Examples

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Text Sample 1: POS Dim 1 – 1994 – Score 15.00 – 1994/t002458.txt
Because I usually take **the** role of trying to explain to people how wonderful **the** new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology **slide**. And then, so if you can just turn on **the slide** that ’ s on. This is just a random **slide** that I picked out of my file. What I want to show you is not so much **the** details of **the slide**, but **the** general form of it. This happens to be a **slide** of

some analysis that we were doing about **the** power of RISC microprocessors versus **the** power of local area networks. And **the** interesting thing about it is that this **slide**, like so many technology **slides** that we ' re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ' s basically what I ' m going to be talking about. So, if you could bring up **the** lights. If you could bring up **the** lights higher, because I ' m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well **the** answer is, if I drew it on a normal curve where, let ' s say, this is years, this is time of some sort, and this is whatever measure of **the** technology that I ' m trying to graph, **the** graphs look sort of silly. They sort of go like this. And they don ' t tell us much. Now if I graph, for instance, some other technology, say **transportation** technology, on a semi-log curve, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if **transportation** technology was moving along as fast as microprocessor technology, then **the** day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ' s not moving like that. And there ' s nothing precedented in **the** history of technology development of this kind of self- **feeding** growth where you go by orders of magnitude every few years. Now **the** question that I ' d like to ask is, if you look at these exponential curves, they don ' t go on forever. Things just can ' t possibly keep changing as fast as they are. One of two things is going to happen. Either it ' s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ' s going to do this. That ' s about all it can do. Now I ' m an optimist, so I sort of think it ' s probably going to do something like that. If so, that means that what we ' re in **the** middle of right now is a transition. We ' re sort of on this line in a transition from **the** way **the** world used to be to some new way that **the** world is. And so what I ' m trying to ask, what I ' ve been asking myself, is what ' s this new way that **the** world is? What ' s that new state that **the** world is heading toward? Because **the** transition seems very, very confusing when we ' re right in **the** middle of it. Now when I was a kid growing up, **the** future was kind of **the** year 2000, and people used to talk about what would happen in **the** year 2000. Now here ' s a conference in which people talk about **the** future, and you notice that **the** future is still at about **the** year 2000. It ' s about as far as we go out. So in other words, **the** future has kind of been shrinking one year per year for my whole lifetime. Now I think that **the** reason is because we all feel that something ' s happening there. That transition is happening. We can all sense it. And we know that it just doesn ' t make too much sense to think out 30, 50 years because everything ' s going to be so different that a simple extrapolation of what we ' re doing just doesn ' t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ' re going through. Now in order to do that I ' m going to have to talk about a bunch of stuff that really has nothing to do with technology and **computers**. Because I think **the** only way to understand this is to really step back and take a long time scale look at things. So **the** time scale that I would like to look at this on is **the** time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, **the** Earth was this big, sterile hunk of **rock** with a lot of chemicals floating around on it. And if you look at **the** way that **the** chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ' s theories that are beginning to understand about how it started with RNA, but I ' m going to tell a sort of simple story of it, which is that, at that time, there were little drops of **oil** floating around with all kinds of different recipes of chemicals in them. And some of those drops of **oil** had a particular combination of chemicals in them which caused them to incorporate chemicals from **the** outside and grow **the** drops of **oil**. And those that were like that started to split and divide. And those were **the** most primitive forms of cells in a sense, those little drops of **oil**. But now those drops of **oil** weren ’ t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of **the** chemicals within them. And so every drop was a little bit different. In fact, **the** drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ’ s sort of just a very simple chemical form of life, but when things got interesting was when these drops learned a trick about abstraction. Somehow by ways that we don ’ t quite understand, these little drops learned to write down information. They learned to record **the** information that was **the** recipe of **the** cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of **evolutionary** way, a form of writing that let them write down what they were, so that that way of writing it down could get **copied**. **The** amazing thing is that that way of writing seems to have stayed steady since it **evolved** two and a half billion years ago. In fact **the** recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly **the** same set of letters and **the** same code. In fact, one of **the** things that I did just for amusement purposes is we can now write things in this code. And I ’ ve got here a little 100 micrograms of white **powder**, which I try not to let **the** security people see at airports. But this has in it — what I did is I took this code — **the** code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to **the** 22 times. So if anyone would like a hundred million **copies** of my business card, I have plenty for everyone in **the** room, and, in fact, everyone in **the** world, and it ’ s right here. If I had really been a egotist, I would have put it into a virus and released it in **the** room. So what was **the** next step? Writing down **the** DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ’ t know if you know this, but bacteria can actually exchange DNA. Now that ’ s why, for instance, antibiotic resistance has **evolved**. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in **the** same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all **the** individuals in that community were repeated more and they were favored by **evolution**. Now **the** transition point happened when these communities got so close that, in fact, they got together and decided to write down **the** whole recipe for **the** community together on one string of DNA. And so **the** next stage that ’ s interesting in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ’ re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to **evolve** so that **the** interesting level on which **evolution** was taking place was no longer a cell, but a community which we call an organism. Now **the** next step that happened is within these communities. These communities of cells, again, began to **abstract** information. And they began building very special structures that did nothing but process information within **the** community. And those are **the** neural structures. So neurons are **the** information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in **the** community and special structures that were responsible for recording, understanding, learning information. And that was **the** brains and **the** nervous system of those communities. And that gave them an **evolutionary** advantage. Because at that point, an individual — **learning** could happen within **the** time span of a single organism, instead of over this **evolutionary** time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within **the** lifetime of a single organism, whereas before they ’ d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by **the** individuals dying off that ate that kind of fruit. So that nervous system, **the** fact that they built these special information structures, tremendously sped up **the** whole process of **evolution**. Because **evolution** could now happen within an individual. It could happen in learning time scales. But then what happened was **the** individuals worked out, of course, tricks of communicating. And for example, **the** most sophisticated version that we ’ re aware of is human language. It ’ s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ’ m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ’ re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ’ ve done is we, humanity, have started **abstracting** out. We ’ re going through **the** same levels that multi-cellular organisms have gone through — **abstracting** out our methods of recording, presenting, processing information. So for example, **the** invention of language was a tiny step in that direction. Telephony, **computers**, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ’ ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to **evolve** than what we were before. So now, **evolution** can take place on a scale of microseconds. And you saw Ty ’ s little **evolutionary** example where he sort of did a little bit of **evolution** on **the** Convolution program right before your eyes. So now we ’ ve speeded up **the** time scales once again. So **the** first steps of **the** story that I told you about took a billion years a piece. And **the** next steps, like nervous systems and brains, took a few hundred million years. Then **the** next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. **The** process is feeding on itself and becoming, I guess, autocatalytic is **the** word for it — when something reinforces its rate of change. **The** more it changes, **the** faster it changes. And I think that that ’ s what we ’ re seeing here in this **explosion** of curve. We ’ re seeing this process feeding back on itself. Now I design **computers** for a living, and I know that **the** mechanisms that I use to design **computers** would be impossible without recent advances in **computers**. So right now, what I do is I design **objects** at such complexity that it ’ s really impossible for me to design them in **the** traditional sense. I don ’ t know what every transistor in **the** connection machine does. There are billions of them. Instead, what I do and what **the** designers at Thinking Machines do is we think at some level of abstraction and then we hand it to **the** machine and **the** machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ’ t quite even understand. One method that ’ s particularly interesting that I ’ ve been using a lot lately is **evolution** itself. So what we do is we put inside **the** machine a process of **evolution** that takes place on **the** microsecond time scale. So for example, in **the** most extreme cases, we can actually **evolve** a program by starting out with random sequences of instructions. Say,”Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out **the** ones that came closest to doing what I wanted.”So in other words, I define what I wanted. Let ’ s say I want to sort numbers, as a simple example I ’ ve done it with. So find **the** programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in **the** right order. And I say,”Computer, would you please now take **the** 10 percent of those random sequences that did **the** best job. Save those. Kill off **the** rest. And now let ’ s reproduce **the** ones that sorted numbers **the** best. And let ’ s reproduce them by a process of recombination analogous to sex.”Take two programs and they produce children by exchanging their subroutines, and **the** children inherit **the** traits of **the** subroutines of **the** two programs. So I ’ ve got now a new generation of programs that are produced by combinations of **the** programs that did a little bit better job. Say,”Please repeat that process.”Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do **the** equivalent of millions of years of **evolution** on that within **the computer** in a few minutes, or in **the** complicated cases, in a few hours. At **the** end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ’ t tell you how they work. I ’ ve tried looking at them and telling you how they work. They ’ re **obscure**, weird programs. But they do **the** job. And in fact, I know, I ’ m very confident that they do **the** job because they come from a line of hundreds of thousands of programs that did **the** job. In fact, their life depended on doing **the** job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,”Oh look. Look at this. It says, ’ This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ’ Doesn ’ t that make you feel confident?”In fact, we know that **the** engineering process doesn ’ t work very well when it gets complicated. So we ’ re beginning to depend on **computers** to do a process that ’ s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ’ t quite understand **the** options of it. So in a sense, it ’ s getting ahead of us. We ’ re now using those programs to make much faster **computers** so that we ’ ll be able to run this process much faster. So it ’ s **feeding** back on itself. **The** thing is becoming faster and that ’ s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ’ re taking off. And what we are is we ’ re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ’ re **the** amoebas and we can ’ t quite figure out what **the** hell this thing is we ’ re creating. We ’ re right at that point of transition. But I think that there really is something coming along after us. I think it ’ s very haughty of us to think that we ’ re **the** end product of **evolution**. And

I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

Text Sample 2: POS Dim 1 – 2003 – Score 24.00 – 2003/t000436.txt

I ’ m supposed to scare you, because it ’ s about fear, right? And you should be really afraid, but not for **the** reasons why you think you should be. You should be really afraid that — if we stick up **the** first **slide** on this thing — there we go — that you ’ re missing out. Because if you spend this week thinking about Iraq and thinking about Bush and thinking about **the** stock market, you ’ re going to miss one of **the** greatest adventures that we ’ ve ever been on. And this is what this adventure ’ s really about. This is crystallized DNA. Every life form on this **planet** — every insect, every bacteria, every plant, every animal, every human, every politician — is coded in that stuff. And if you want to take a single crystal of DNA, it looks like that. And we ’ re just beginning to understand this stuff. And this is **the** single most exciting adventure that we have ever been on. It ’ s **the** single greatest mapping project we ’ ve ever been on. If you think that **the** mapping of America ’ s made a difference, or landing on **the** moon, or this other stuff, it ’ s **the** map of ourselves and **the** map of every plant and every insect and every bacteria that really makes a difference. And it ’ s beginning to tell us a lot about **evolution**. It turns out that what this stuff is — and Richard Dawkins has written about this — is, this is really a river out of Eden. So, **the** 3. 2 billion base pairs inside each of your cells is really a history of where you ’ ve been for **the** past billion years. And we could start dating things, and we could start changing medicine and archeology. It turns out that if you take **the** human species about 700 years ago, white Europeans diverged from black Africans in a very significant way. White Europeans were subject to **the** plague. And when they were subject to **the** plague, most people didn ’ t survive, but those who survived had a mutation on **the** CCR5 receptor. And that mutation was passed on to their kids because they ’ re **the** ones that survived, so there was a great deal of population pressure. In Africa, because you didn ’ t have these cities, you didn ’ t have that CCR5 population pressure mutation. We can date it to 700 years ago. That is one of **the** reasons why AIDS is raging across Africa as fast as it is, and not as fast across Europe. And we ’ re beginning to find these little things for **malaria**, for sickle cell, for cancers. And in **the** measure that we map ourselves, this is **the** single greatest adventure that we ’ ll ever be on. And this Friday, I want you to pull out a really good bottle of wine, and I want you to toast these two people. Because this Friday, 50 years ago, Watson and Crick found **the** structure of DNA, and that is almost as important a date as **the** 12th of February when we first mapped ourselves, but anyway, we ’ ll get to that. I thought we ’ d talk about **the** new zoo. So, all you guys have heard about DNA, all **the** stuff that DNA does, but some of **the** stuff we ’ re discovering is kind of nifty because this turns out to be **the** single most abundant species on **the planet**. If you think you ’ re successful or **cockroaches** are successful, it turns out that there ’ s ten trillion trillion Pleurococcus sitting out there. And we didn ’ t know that Pleurococcus was out there, which is part of **the** reason why this whole species-mapping project is so important. Because we ’ re just beginning to learn where we came from and what we are. And we ’ re finding amoebas like this. This is **the** amoeba dubia. And **the** amoeba dubia doesn ’ t look like much, except that each of you has about 3. 2 billion letters, which is what makes you you, as far as gene code inside each of your cells, and this little amoeba which, you know, sits in water in hundreds and millions and billions, turns out to have 620 billion base pairs of gene code inside. So, this little thingamajig has a **genome** that ’ s 200 times **the** size of yours. And if you ’ re thinking of efficient information storage mechanisms, it may not turn out to be chips. It may turn out to be something that looks a little like that amoeba. And, again, we ’ re learning from life and how life works. This funky little thing : people didn ’ t used to think that it was worth taking samples out of nuclear reactors because it was dangerous and, of course, nothing lived there. And then finally somebody picked up a microscope and looked at **the** water that was sitting next to **the** cores. And sitting next to that water in **the** cores was this little Deinococcus radiodurans, doing a backstroke, having its chromosomes blown apart every day, six, seven times, restitching them, living in about 200 times **the** radiation that would kill you. And by now you should be getting a hint as to how diverse and how important and how interesting this journey into life is, and how many different life forms there are, and how there can be different life forms living in very different places, maybe even outside of this **planet**. Because if you can live in radiation that looks like this, that brings up a whole series of interesting questions. This little thingamajig : we didn ’ t know this thingamajig existed. We should have known that this existed because this is **the** only bacteria that you can see to **the** naked eye. So, this thing is 0. 75 millimeters. It lives in a deep trench off **the** coast of Namibia. And what you ’ re looking at with this namibiensis is **the** biggest bacteria we ’ ve ever seen. So, it ’ s about **the** size of a little period on a sentence. Again, we didn ’ t know this thing was there three years ago. We ’ re just beginning this journey of life in **the** new zoo. This is a really odd one. This is Ferroplasma. **The** reason why Ferroplasma is interesting is because it eats **iron**, lives inside **the** equivalent of battery acid, and excretes sulfuric acid. So, when you think of odd life forms, when you think of what it takes to live, it turns out this is a very efficient life form, and they call it an archaea. Archaea means”**the** ancient ones.”And **the** reason why they ’ re ancient is because this thing came up when this **planet** was covered by things like sulfuric acid in batteries, and it was eating **iron** when **the** earth was part of a melted core. So, it ’ s not just dogs and cats and **whales** and dolphins that you should be aware of and interested in on this little journey. Your fear should be that you are not, that you ’ re paying attention to stuff which is temporal. I mean, George Bush — he ’ s going to be gone, alright? Life isn ’ t. Whether **the** humans survive or don ’ t survive, these things are going to be living on this **planet** or other **planets**. And it ’ s just beginning to understand this code of DNA that ’ s really **the** most exciting intellectual adventure that we ’ ve ever been on. And you can do strange things with this stuff. This is a baby gaur. Conservation group gets together, tries to figure out how to breed an animal that ’ s almost extinct. They can ’ t do it naturally, so what they do with this thing is they take a spoon, take some cells out of an adult gaur ’ s mouth, code, take **the** cells from that and insert it into a fertilized cow ’ s egg, reprogram cow ’ s egg — different gene code. When you do that, **the** cow gives birth to a gaur. We are now experimenting with bongos, pandas, elands, Sumatran **tigers**, and **the** Australians — bless their hearts — are playing with these things. Now, **the** last of these things died in September 1936. These are Tasmanian **tigers**. **The** last known one died at **the** Hobart Zoo. But it turns out that as we learn more about gene code and how to reprogram species, we may be able to close **the** gene gaps in deteriorate DNA. And when we learn how to close **the** gene gaps, then we can put a full string of DNA together. And if we do that, and insert this into a fertilized wolf ’ s egg, we may give birth to an animal that hasn ’ t walked **the** earth since 1936. And then you can start going back further, and you can start thinking about dodos, and you can think about other species. And in other places, like Maryland, they ’ re trying to figure out what **the** primordial ancestor is. Because each of us contains our entire gene code of where we ’ ve been for **the** past billion years, because we ’ ve **evolved** from that stuff, you can take that tree of life and collapse it back, and in **the** measure that you learn to reprogram, maybe we ’ ll give birth to something that is very close to **the** first primordial ooze. And it ’ s all coming out of things that look like this. These are companies that didn ’ t exist five years ago. Huge gene sequencing facilities **the** size of football fields. Some are public. Some are private. It takes about 5 billion dollars to sequence a human being **the** first time. Takes about 3 million dollars **the** second time. We will have a 1, 000-dollar **genome** within **the** next five to eight years. That means each of you will contain on a CD your entire gene code. And it will be really boring. It will read like this. **The** really neat thing about this stuff is that ’ s life. And Laurie ’ s going to talk about this one a little bit. Because if you happen to find this one inside your body, you ’ re in big trouble, because that ’ s **the** source code for Ebola. That ’ s one of **the** deadliest diseases known to humans. But plants work **the** same way and insects work **the** same way, and this apple works **the** same way. This apple is **the** same thing as this floppy **disk**. Because this thing codes ones and zeros, and this thing codes A, T, C, Gs, and it sits up there, absorbing energy on a tree, and one fine day it has enough energy to say, execute, and it goes [thump]. Right? And when it does that, pushes a. EXE, what it does is, it executes **the** first line of code, which reads just like that, AATCAGGGACCC, and that means : make a root. Next line of code : make a stem. Next line of code, TACGGGG : make a flower that ’ s white, that blooms in **the spring**, that smells like this. In **the** measure that you have **the** code and **the** measure that you read it — and, by **the** way, **the** first plant was read two years ago ; **the** first human was read two years ago ; **the** first insect was read two years ago. **The** first thing that we ever read was in 1995 : a little bacteria called Haemophilus influenzae. In **the** measure that you have **the** source code, as all of you know, you can change **the** source code, and you can reprogram life forms so that this little thingy becomes a vaccine, or this little thingy starts producing biomaterials, which is why DuPont is now growing a form of polyester that feels like silk in corn. This changes all rules. This is life, but we ’ re reprogramming it. This is what you look like. This is one of your chromosomes. And what you can do now is, you can outlay exactly what your chromosome is, and what **the** gene code on that chromosome is right here, and what those genes code for, and what animals they code against, and then you can tie it to **the** literature. And in **the** measure that you can do that, you can go home today, and get on **the** Internet, and access **the** world ’ s biggest public library, which is a library of life. And you can do some pretty strange things because in **the** same way as you can reprogram this apple, if you go to Cliff Tabin ’ s lab at **the** Harvard Medical School, he ’ s reprogramming chicken embryos to grow more **wings**. Why would Cliff be doing that? He doesn ’ t have a restaurant. **The** reason why he ’ s reprogramming that animal to have more **wings** is because when you used to play with lizards as a little child, and you picked up **the** lizard, sometimes **the** tail fell off, but it regrew. Not so in human beings : you cut off an arm, you cut off a leg — it doesn ’ t regrow. But because each of your cells contains your entire gene code, each cell can be reprogrammed, if we don ’ t stop stem cell research and if we don ’ t stop genomic research, to express different body functions. And in **the** measure that we learn how chickens grow **wings**, and what **the** program is for those cells to differentiate, one of **the** things we ’ re going to be able to do is to stop undifferentiated cells, which you know as cancer, and one of **the** things we ’ re going to learn how to do is how to reprogram cells like stem cells in such a way that they express bone, stomach, skin, pancreas. And you are likely to be wandering around — and your children — on regrown body parts in a reasonable period of time, in some places in **the** world where they don ’ t stop **the** research. How ’ s this stuff work? If each of you differs from **the** person next to you by one in a thousand, but only three percent codes, which means it ’ s only one in a thousand times three percent, very small differences in expression and punctuation can make a significant difference. Take a simple declarative sentence. Right? That ’ s perfectly clear. So, men read that sentence, and they look at that sentence, and they read this. **Okay**? Now, women look at that sentence and they say, uh-uh, wrong. This is **the** way it should be seen. That ’ s what your genes are doing. That ’ s why you differ from this person over here by one in a thousand. Right? But, you know, he ’ s reasonably good looking, but... I won ’ t go there. You can do this stuff even without changing **the** punctuation. You can look at this, right? And they look at **the** world a little differently. They look at **the** same world and they say... That ’ s how **the** same gene code — that ’ s why you have 30, 000 genes, mice have 30, 000 genes, husbands have 30, 000 genes. Mice and men are **the** same. Wives know that, but anyway. You can make very small changes in gene code and get really different outcomes, even with **the** same string of letters. That ’ s what your genes are doing every day. That ’ s why sometimes a person ’ s genes don ’ t have to change a lot to get cancer. These little chippies, these things are **the** size of a credit card. They will test any one of you for 60, 000 genetic conditions. That brings up questions of privacy and insurability and all kinds of stuff, but it also allows us to start going after diseases, because if you run a person who has leukemia through something like this, it turns out that three diseases with completely similar clinical syndromes are completely different diseases. Because in ALL leukemia, that set of genes over there over-expresses. In MLL, it ’ s **the** middle set of genes, and in AML, it ’ s **the** bottom set of genes. And if one of those particular things is expressing in your body, then you take Gleevec and you ’ re cured. If it is not expressing in your body, if you don ’ t have one of those types — a particular one of those types — don ’ t take Gleevec. It won ’ t do anything for you. Same thing with Receptin if you ’ ve got breast cancer. Don ’ t have an HER- 2 receptor? Don ’ t take Receptin. Changes **the** nature of medicine. Changes **the** predictions of medicine. Changes **the** way medicine works. **The** greatest repository of knowledge when most of us went to college was this thing, and it turns out that this is not so important any more. **The** U. S. Library of Congress, in terms of its printed volume of data, contains less data than is coming out of a good genomics company every month on a compound basis. Let me say that again : A single genomics company generates more data in a month, on a compound basis, than is in **the** printed **collections** of **the** Library of Congress. This is what ’ s been powering **the** U. S. economy. It ’ s Moore ’ s Law. So, all of you know that **the** price of **computers** halves every 18 months and **the** power doubles, right? Except that when you lay that side by side with **the** speed with which gene data ’ s being deposited in GenBank, Moore ’ s Law is right here : it ’ s **the** **blue** line. This is on a log scale, and that ’ s what superexponential growth means. This is going to push **computers** to have to grow faster than they ’ ve been growing, because so far, there haven ’ t been applications that have been required that need to go faster than Moore ’ s Law. This stuff does. And here ’ s an interesting map. This is a map which was finished at **the** Harvard Business School. One of **the** really interesting questions is, if all this data ’ s free, who ’ s using it? This is **the** greatest public library in **the** world. Well, it turns out that there ’ s about 27 trillion bits moving inside from **the** United States to **the** United States ; about 4. 6 trillion is going over to those European countries ; about 5. 5 ’ s going to Japan ; there ’ s almost no communication between Japan, and nobody else is literate in this stuff. It ’ s free. No one ’ s reading it. They ’ re focusing on **the** war ; they ’ re focusing on Bush ; they ’ re not interested in life. So, this is what a new map of **the** world looks like. That is **the** genomically literate world. And that is a problem. In fact, it ’ s not a genomically literate world. You can break this out by states. And you can watch states rise and fall depending on their ability to speak a language of life, and you can watch New York fall off a cliff, and you can watch New Jersey fall off a cliff, and you can watch **the** rise of **the** new empires of intelligence. And you can break it out by counties, because it ’ s specific counties. And if you want to get more specific, it ’ s actually specific zip codes. So, you want to know where life is happening? Well, in Southern California it ’ s happening in 92121. And that ’ s it. And that ’ s **the triangle** between Salk, Scripps, UCSD, and it ’ s called Torrey Pines Road. That means you don ’ t need to be a big nation to be successful ; it means you don ’ t need a lot of people to be successful ; and it means you can move most of **the** wealth of a country in about three or four carefully picked 747s. Same thing in Massachusetts. Looks more spread out but — oh, by **the** way, **the** ones that are **the** same color are contiguous. What ’ s **the** net effect of this? In an agricultural society, **the** difference between **the** richest and **the** poorest, **the** most productive and **the** least productive, was five to one. Why? Because in **agriculture**, if you had 10 kids and you grow up a little bit earlier and you work a little bit harder, you could produce about five times more wealth, on average, than your neighbor. In a knowledge society, that number is now 427 to 1. It really matters if you ’ re literate, not just in reading and writing in English and French and German, but in Microsoft and Linux and Apple. And very soon it ’ s going to matter if you ’ re literate in life code. So, if there is something you should fear, it ’ s that you ’ re not keeping your eye on **the** ball. Because it really matters who speaks life. That ’ s why nations rise and fall. And it turns out that if you went back to **the** 1870s, **the** most productive nation on earth was Australia, per person. And New Zealand was way up there. And then **the** U. S. came in about 1950, and then Switzerland about 1973, and then **the** U. S. got back on top — beat up their chocolates and cuckoo clocks. And today, of course, you all know that **the** most productive nation on earth is Luxembourg, producing about one third more wealth per person per year than America. Tiny landlocked state. No **oil**. No diamonds. No natural resources. Just smart people moving bits. Different rules. Here ’ s differential productivity rates. Here ’ s how many people it takes to produce a single U. S. **patent**. So, about 3, 000 Americans, 6, 000 Koreans, 14, 000 Brits, 790, 000 Argentines. You want to know why Argentina ’ s crashing? It ’ s got nothing to do with inflation. It ’ s got nothing to do with privatization. You can take a Harvard-educated Ivy League economist, stick him in charge of Argentina. He still crashes **the** country because he doesn ’ t understand how **the** rules have changed. Oh, yeah, and it takes about 5. 6 million Indians. Well, watch what happens to India. India and China used to be 40 percent of **the** global economy just at **the** Industrial Revolution, and they are now about 4. 8 percent. Two billion people. One third of **the** global population producing 5 percent of **the** wealth because they didn ’ t get this change, because they kept treating their people like serfs instead of like shareholders of a common project. They didn ’ t keep **the** people who were educated. They didn ’ t foment **the** businesses. They didn ’ t do **the** IPOs. Silicon Valley did. And that ’ s why they say that Silicon Valley has been powered by ICs. Not integrated circuits : Indians and Chinese. Here ’ s what ’ s happening in **the** world. It turns out that if you ’ d gone to **the** U. N. in 1950, when it was founded, there were 50 countries in this world. It turns out there ’ s now about 192. Country after country is splitting, seceding, succeeding, failing — and it ’ s all getting very fragmented. And this has not stopped. In **the** 1990s, these are sovereign states that did not exist before 1990. And this doesn ’ t include **fusions** or name changes or changes in flags. We ’ re generating about 3. 12 states per year. People are taking control of their own states, sometimes for **the** better and sometimes for **the** worse. And **the** really interesting thing is, you and your kids are empowered to build great empires, and you don ’ t need a lot to do it. And, given that **the** music is over, I was going to talk about how you can use this to generate a lot of wealth, and how code works. Moderator : Two minutes. Juan Enriquez : No, I ’ m going to stop there and we ’ ll do it next year because I don ’ t want to take any of Laurie ’ s time. But thank you very much.

Text Sample 3: POS Dim 1 – 2003 – Score 23.00 – 2003/t000259.txt
Last year, I told you **the** story, in seven minutes, of Project Orion, which was this very implausible technology that technically could have worked, but it had this one-year political window where it could have happened. So it didn ’ t happen. It was a dream that did not happen. This year I ’ m going to tell you **the** story of **the** birth of digital computing. This was a perfect introduction. And it ’ s a story that did work. It did happen, and **the** machines are all around us. And it was a technology that was inevitable. If **the** people I ’ m going to tell you **the** story about, if they hadn ’ t done it, somebody else would have. So, it was sort of **the** right idea at **the** right time. This is Barricelli ’ s **universe**. This is **the universe** we live in now. It ’ s **the universe** in which these machines are now doing all these things, including changing **biology**. I ’ m starting **the** story with **the** first atomic bomb at Trinity, which was **the** Manhattan Project. It was a little bit like TED : it brought a whole lot of very smart people together. And three of **the** smartest people were Stan Ulam, Richard Feynman and John von Neumann. And it was Von Neumann who said, after **the** bomb, he was working on something much more important than bombs : he ’ s thinking about **computers**. So, he wasn ’ t only thinking about them ; he built one. This is **the** machine he built. He built this machine, and we had a beautiful demonstration of how this thing really works, with these little bits. And it ’ s an idea that goes way back. **The** first person to really explain that was Thomas Hobbes, who, in 1651, explained how arithmetic and logic are **the** same thing, and if you want to do **artificial** thinking and **artificial** logic, you can do it all with arithmetic. He said you needed addition and subtraction. Leibniz, who came a little bit later — this is 1679 — showed that you didn ’ t even need subtraction. You could do **the** whole thing with addition. Here, we have all **the** binary arithmetic and logic that drove **the computer** revolution. And Leibniz was **the** first person to really talk about building such a machine. He talked about doing it with marbles, having gates and what we now call shift registers, where you shift **the** gates, drop **the** marbles down **the** tracks. And that ’ s what all these machines are doing, except, instead of doing it with marbles, they ’ re doing it with electrons. And then we jump to Von Neumann, 1945, when he sort of reinvents **the** whole same thing. And 1945, after **the** war, **the** electronics existed to actually try and build such a machine. So June 1945 — actually, **the** bomb hasn ’ t even been dropped yet — and Von Neumann is putting together all **the** theory to actually build this thing, which also goes back to Turing, who, before that, gave **the** idea that you could do all this with a very brainless, little, finite state machine, just reading a tape in and reading a tape out. **The** other sort of genesis of what Von Neumann did was **the** difficulty of how you would predict **the** weather. Lewis Richardson saw how you could do this with a cellular **array** of people, giving them each a little chunk, and putting it together. Here, we have an **electrical** model illustrating a mind having a will, but capable of only two ideas. And

that ’ s really **the** simplest **computer**. It ’ s basically why you need **the** qubit, because it only has two ideas. And you put lots of those together, you get **the** essentials of **the** modern **computer** : **the** arithmetic unit, **the** central control, **the** memory, **the** recording medium, **the** input and **the** output. But, there ’ s one catch. This is **the** fatal — you know, we saw it in starting these programs up. **The** instructions which govern this operation must be given in absolutely exhaustive detail. So, **the** programming has to be perfect, or it won ’ t work. If you look at **the** origins of this, **the** classic history sort of takes it all back to **the** ENIAC here. But actually, **the** machine I ’ m going to tell you about, **the** Institute for Advanced Study machine, which is way up there, really should be down there. So, I ’ m trying to revise history, and give some of these guys more credit than they ’ ve had. Such a **computer** would open up **universes**, which are, at **the** present, outside **the** range of any instruments. So it opens up a whole new world, and these people saw it. **The** guy who was supposed to build this machine was **the** guy in **the** middle, Vladimir Zworykin, from RCA. RCA, in probably one of **the** lousiest business decisions of all time, decided not to go into **computers**. But **the** first meetings, November 1945, were at RCA ’ s offices. RCA started this whole thing off, and said, you know, televisions are **the** future, not **computers**. **The** essentials were all there — all **the** things that make these machines run. Von Neumann, and a logician, and a mathematician from **the** army put this together. Then, they needed a place to build it. When RCA said no, that ’ s when they decided to build it in Princeton, where Freeman works at **the** Institute. That ’ s where I grew up as a kid. That ’ s me, that ’ s my sister Esther, who ’ s talked to you before, so we both go back to **the** birth of this thing. That ’ s Freeman, a long time ago, and that was me. And this is Von Neumann and Morgenstern, who wrote **the**”Theory of Games.”All these forces came together there, in Princeton. Oppenheimer, who had built **the** bomb. **The** machine was actually used mainly for doing bomb calculations. And Julian Bigelow, who took Zworkykin ’ s place as **the** engineer, to actually figure out, using electronics, how you would build this thing. **The** whole gang of people who came to work on this, and women in front, who actually did most of **the** coding, were **the** first programmers. These were **the** prototype **geeks**, **the** nerds. They didn ’ t fit in at **the** Institute. This is a letter from **the** director, concerned about —”especially unfair on **the** matter of sugar.”You can read **the** text. This is hackers getting in trouble for **the** first time.. These were not theoretical physicists. They were real soldering-gun type guys, and they actually built this thing. And we take it for granted now, that each of these machines has billions of transistors, doing billions of cycles per second without failing. They were using vacuum tubes, very narrow, sloppy **techniques** to get actually binary behavior out of these radio vacuum tubes. They actually used 6J6, **the** common radio tube, because they found they were more reliable than **the** more expensive tubes. And what they did at **the** Institute was publish every step of **the** way. Reports were issued, so that this machine was **cloned** at 15 other places around **the** world. And it really was. It was **the** original microprocessor. All **the computers** now are **copies** of that machine. **The** memory was in cathode ray tubes — a whole bunch of spots on **the** face of **the** tube — very, very sensitive to electromagnetic disturbances. So, there ’ s 40 of these tubes, like a V-40 engine running **the** memory. **The** input and **the** output was by teletype tape at first. This is a wire drive, using bicycle wheels. This is **the** archetype of **the** hard **disk** that ’ s in your machine now. Then they switched to a magnetic **drum**. This is modifying IBM equipment, which is **the** origins of **the** whole data-processing industry, later at IBM. And this is **the** beginning of **computer graphics**. **The**”Graph ’ g-Beam Turn On.”This next **slide**, that ’ s **the** — as far as I know — **the** first digital bitmap display, 1954. So, Von Neumann was already off in a theoretical cloud, doing **abstract** sorts of studies of how you could build reliable machines out of unreliable components. Those guys **drinking** all **the** tea with sugar in it were writing in their logbooks, trying to get this thing to work, with all these 2, 600 vacuum tubes that failed half **the** time. And that ’ s what I ’ ve been doing, this last six months, is going through **the** logs.”Running time : two minutes. Input, output : 90 minutes.”This includes a large amount of human error. So they are always trying to figure out, what ’ s machine error? What ’ s human error? What ’ s code, what ’ s hardware? That ’ s an engineer gazing at tube number 36, trying to figure out why **the** memory ’ s not in focus. He had to focus **the** memory — seems OK. So, he had to focus each tube just to get **the** memory up and running, let alone having, you know, software problems.”No use, went home.””Impossible to follow **the** damn thing, where ’ s a directory?”So, already, they ’ re complaining about **the** manuals :”before closing down in disgust...””**The** General Arithmetic : Operating Logs.”Burning lots of midnight **oil**.”MANIAC,”which became **the** acronym for **the** machine, Mathematical and Numerical Integrator and Calculator,”lost its memory.””MANIAC regained its memory, when **the** power went off.””Machine or human?””Aha!”So, they figured out it ’ s a code problem.”Found trouble in code, I hope.””Code error, machine not guilty.””Damn it, I can be just as stubborn as this thing.””And **the** dawn came.”So they ran all night. Twenty-four hours a day, this thing was running, mainly running bomb calculations.”Everything up to this point is wasted time.””What ’ s **the** use? Good night.””Master control off. **The** hell with it. Way off.””Something ’ s wrong with **the** air conditioner — smell of burning V-belts in **the** air.””A short — do not turn **the** machine on.””IBM machine putting a tar-like substance on **the** cards. **The** tar is from **the** roof.”So they really were working under tough conditions. Here,”A mouse has climbed into **the** blower behind **the** regulator rack, set blower to vibrating. Result : no more mouse.””Here lies mouse. Born :?. Died : 4:50 a. m., May 1953.”There ’ s an inside joke someone has penciled in :”Here lies Marston Mouse.”If you ’ re a mathematician, you get that, because Marston was a mathematician who **objected** to **the computer** being there.”Picked a lightning bug off **the drum**.””Running at two kilocycles.”That ’ s two thousand cycles per second —”yes, I ’ m chicken”— so two kilocycles was slow speed. **The** high speed was 16 kilocycles. I don ’ t know if you remember a Mac that was 16 Megahertz, that ’ s slow speed.”I have now duplicated both results. How will I know which is right, assuming one result is correct? This now is **the** third different output. I know when I ’ m licked.””We ’ ve duplicated errors before.””Machine run, fine. Code isn ’ t.””Only happens when **the** machine is running.”And sometimes things are **okay**.”Machine a thing of beauty, and a joy forever.””Perfect running.””Parting thought : when there ’ s bigger and better errors, we ’ ll have them.”So, nobody was supposed to know they were actually designing bombs. They ’ re designing hydrogen bombs. But someone in **the** logbook, late one night, finally drew a bomb. So, that was **the** result. It was Mike, **the** first thermonuclear bomb, in 1952. That was designed on that machine, in **the** woods behind **the** Institute. So Von Neumann invited a whole gang of weirdos from all over **the** world to work on all these problems. Barricelli, he came to do what we now call, really, **artificial** life, trying to see if, in this **artificial universe** — he was a viral-geneticist, way, way, way ahead of his time. He ’ s still ahead of some of **the** stuff that ’ s being done now. Trying to start an **artificial** genetic system running in **the computer**. Began — his **universe** started March 3, ’ 53. So it ’ s almost exactly — it ’ s 50 years ago next Tuesday, I guess. And he saw everything in terms of — he could read **the** binary code straight off **the** machine. He had a wonderful rapport. Other people couldn ’ t get **the** machine running. It always worked for him. Even errors were duplicated.”Dr. Barricelli claims machine is wrong, code is right."So he designed this **universe**, and ran it. When **the** bomb people went home, he was allowed in there. He would run that thing all night long, running these things, if anybody remembers Stephen Wolfram, who reinvented this stuff. And he published it. It wasn ' t locked up and **disappeared**. It was published in **the** literature."If it ' s that easy to create living organisms, why not create a few yourself?"So, he decided to give it a try, to start this **artificial biology** going in **the** machines. And he found all these, sort of — it was like a naturalist coming in and looking at this tiny, 5, 000-byte **universe**, and seeing all these things happening that we see in **the** outside world, in **biology**. This is some of **the** generations of his **universe**. But they ' re just going to stay numbers ; they ' re not going to become organisms. They have to have something. You have a genotype and you have to have a phenotype. They have to go out and do something. And he started doing that, started giving these little numerical organisms things they could play with — playing chess with other machines and so on. And they did start to **evolve**. And he went around **the** country after that. Every time there was a new, fast machine, he started using it, and saw exactly what ' s happening now. That **the** programs, instead of being turned off — when you quit **the** program, you ' d keep running and, basically, all **the** sorts of things like Windows is doing, running as a multicellular organism on many machines, he envisioned all that happening. And he saw that **evolution** itself was an intelligent process. It wasn ' t any sort of creator intelligence, but **the** thing itself was a giant parallel computation that would have some intelligence. And he went out of his way to say that he was not saying this was lifelike, or a new kind of life. It just was another version of **the** same thing happening. And there ' s really no difference between what he was doing in **the computer** and what nature did billions of years ago. And could you do it again now? So, when I went into these archives looking at this stuff, lo and behold, **the** archivist came up one day, saying,"I think we found another box that had been thrown out."And it was his **universe** on punch cards. So there it is, 50 years later, sitting there — sort of suspended animation. That ' s **the** instructions for running — this is actually **the** source code for one of those **universes**, with a note from **the** engineers saying they ' re having some problems."There must be something about this code that you haven ' t explained yet."And I think that ' s really **the** truth. We still don ' t understand how these very simple instructions can lead to increasing complexity. What ' s **the** dividing line between when that is lifelike and when it really is alive? These cards, now, thanks to me showing up, are being saved. And **the** question is, should we run them or not? You know, could we get them running? Do you want to let it loose on **the** Internet? These machines would think they — these organisms, if they came back to life now — whether they ' ve died and gone to heaven, there ' s a **universe**. My **laptop** is 10 thousand million times **the** size of **the universe** that they lived in when Barricelli quit **the** project. He was thinking far ahead, to how this would really grow into a new kind of life. And that ' s what ' s happening! When Juan Enriquez told us about these 12 trillion bits being transferred back and forth, of all this genomics data going to **the** proteomics lab, that ' s what Barricelli imagined : that this digital code in these machines is actually starting to code — it already is coding from nucleic acids. We ' ve been doing that since, you know, since we started PCR and synthesizing small strings of DNA. And real soon, we ' re actually going to be synthesizing **the** proteins, and, like Steve showed us, that just opens an entirely new world. It ' s a world that Von Neumann himself envisioned. This was published after he died : his sort of unfinished notes on self-reproducing machines, what it takes to get **the** machines sort of jump-started to where they begin to reproduce. It took really three people : Barricelli had **the** concept of **the** code as a living thing ; Von Neumann saw how you could build **the** machines — that now, last count, four million of these Von Neumann machines is built every 24 hours ; and Julian Bigelow, who died 10 days ago — this is John Markoff ' s obituary for him — he was **the** important missing link, **the** engineer who came in and knew how to put those vacuum tubes together and make it work. And all our **computers** have, inside them, **the copies** of **the** architecture that he had to just design one day, sort of on pencil and paper. And we owe a tremendous credit to that. And he explained, in a very generous way, **the** spirit that brought all these different people to **the** Institute for Advanced Study in **the** ' 40s to do this project, and make it freely available with no **patents**, no restrictions, no intellectual property disputes to **the** rest of **the** world. That ' s **the** last entry in **the** logbook when **the** machine was shut down, July 1958. And it ' s Julian Bigelow who was running it until midnight when **the** machine was officially turned off. And that ' s **the** end. Thank you very much.

Text Sample 4: POS Dim 1 – 2003 – Score 23.00 – 2003/t000117.txt

Right when I was 15 was when I first got interested in solar energy. My family had moved from Fort Lee, New Jersey to California, from **the** snow to lots of heat, and gas lines. There was gas rationing in 1973. **The** energy crisis was in full bore. I started reading"Popular Science"magazine, and I got really excited about **the** potential of solar energy to try and solve that crisis. I had just taken trigonometry in high school, I learned about **the** parabola and how it could concentrate rays of light to a single focus. That got me very excited. And I really felt that there would be potential to build some kind of thing that could concentrate light. So, I started this company called Solar Devices. And this was a company where I built parabolas, I took metal shop, and I remember walking into metal shop building parabolas and Stirling engines. And I was building a Stirling engine over on **the** lathe, and all **the** motorcycle guys said,"You ' re building a bong, aren ' t you?"And I said,"No, it ' s a Stirling engine."But they didn ' t believe me. I sold **the** plans for this engine and for this dish in **the** back of"Popular Science"magazine, for four dollars each. And I earned enough money to pay for my first year of Caltech. It was a really big excitement for me to get into Caltech. And at my first year at Caltech, I continued **the** business. But then, in **the** second year of Caltech, they started grading. **The** whole first year was pass / fail, but **the** second year was graded. I wasn ' t able to keep up with **the** business, and I ended up with a 25-year detour. My dream had been to convert solar energy at a very practical cost, but then I had this big detour. First, **the** coursework at Caltech. Then, when I graduated from Caltech, **the** IBM PC came out, and I got addicted to **the** IBM PC in 1981. And then in 1983, Lotus 1-2-3 came out, and I was completely blown away by Lotus 1-2-3. I began operating my business with 1-2-3, began writing add-ins for 1-2-3, wrote a natural language interface to 1-2-3. I started an educational software company after I joined Lotus, and then I started Idealab so I could have a roof under which I could build multiple companies in succession. Much later — in 2000, very recently — **the** new California energy crisis — what was purported to be a big energy crisis — was coming. And I was trying to figure out if we could build something that would capitalize on that and get people backup energy, in case **the** crisis really came. And I started looking at how we could build battery backup systems that could give people five hours, 10 hours, maybe even a full day, or three days ' worth of backup power. I ' m glad you heard earlier today, batteries are unbelievably — lack density compared to fuel. So much more energy can be stored with fuel than with batteries. You ' d have to fill your entire parking space of one garage space just to give yourself four hours of battery backup. And I **concluded**, after researching every other technology that we could deploy for storing energy — flywheels, different formulations of batteries — it just wasn ’ t practical to store energy. So what about making energy? Maybe we could make energy. I tried to figure out — maybe solar ’ s become attractive. It ’ s been 25 years since I was doing this, let me go back and look at what ’ s been happening with solar cells. And **the** price had gone down from 10 dollars a watt to about four or five dollars a watt, but it stabilized. And it needed to get much lower to be cost-effective. I studied all **the** new things that had happened in solar cells, and was looking for ways we could make solar cells more inexpensively. A lot of new things are happening to do that, but fundamentally, **the** process requires a tremendous amount of energy. Some people say it takes more energy to make a solar cell than it will give out in its entire life. If we reduce **the** amount of energy it takes to make **the** cells, that will become more practical. But right now, you pretty much have to take silicon, put it in an oven at 1600 F for 17 hours, to make **the** cells. A lot of people are working to try and reduce that, but I didn ’ t have anything to contribute. So I tried to figure out what other way could we try to make cost-effective solar electricity. What if we collect **the** sun with a large reflector — like I had been thinking about in high school, but maybe with modern technology we could make it cheaper — concentrate it to a small converter, and then **the** conversion **device** wouldn ’ t have to be as expensive, because it ’ s much smaller, rather than solar cells, which have to cover **the** entire **surface** that you want to gather sun from. This seemed practical now, because a lot of new technologies had come in **the** 25 years since I had last looked at it. There was a lot of new manufacturing **techniques**, not to mention really cheap miniature motors — brushless motors, servomotors, stepper motors, that are used in printers and scanners. So, that ’ s a breakthrough. Of course, inexpensive microprocessors and a very important breakthrough — genetic algorithms. I ’ ll be very short on genetic algorithms. It ’ s a powerful way of solving intractable problems using natural **selection**. You take a problem that you can ’ t solve with a pure mathematical answer, you build an **evolutionary** system to try multiple tries at guessing, you add sex — where you take half of one solution and half of another and then make new mutations — and you use natural **selection** to kill off not-as-good solutions. Usually, with a genetic algorithm on a **computer** today, with a three gigahertz processor, you can solve many formerly intractable problems in just a matter of minutes. So we tried to come up with a way to use genetic algorithms to create a new type of concentrator. And I ’ ll show you what we came up with. Traditionally, concentrators look like this. Those shapes are parabolas. They take all **the** parallel incoming rays and focus it to a single spot. They have to track **the** sun, because they have to point directly at **the** sun. They usually have a one degree acceptance angle — once they ’ re more than a degree off, none of **the** sunlight rays will hit **the** focus. So we tried to come up with a non-tracking collector that would gather much more than one degree of light, with no moving parts. So we created a genetic algorithm to try this out, we made a model in Excel of a multisurface reflector, and an amazing thing **evolved**, literally, from trying a billion cycles, a billion different attempts, with a fitness function that defined how can you collect **the** most light, from **the** most angles, over a day, from **the** sun. And this is **the** shape that **evolved**. It ’ s this non-tracking collector with these six tuba-like horns, and each of them collect light in **the** following way — if **the** sunlight strikes right here, it might bounce right to **the** center, **the** hot spot, directly, but if **the** sun is off axis and comes from **the** side, it might hit two places and take two bounces. So for direct light, it takes only one bounce, for off-axis light it might take two, and for extreme off-axis, it might take three. Your efficiency goes down with more bounces, because you lose about 10 percent with each bounce, but this allowed us to collect light from a plus or minus 25-degree angle. So, about two and a half hours of **the** day we could collect with a stationary component. Solar cells collect light for four and a half hours though. On an average adjusted day, a solar cell — because **the** sun ' s moving across **the** sky, **the** solar cell is going down with a sine wave function of performance at **the** off- axis angles. It collects about four and a half average hours of sunlight a day. So even this, although it was great with no moving parts — we could achieve high temperatures — wasn ' t enough. We needed to beat solar cells. So we took a look at another idea. We looked at a way to break up a parabola into individual petals that would track. So what you see here is 12 separate petals that each could be controlled with individual microprocessors that would only cost a dollar. You can buy a two- megahertz microprocessor for a dollar now. And you can buy stepper motors that pretty much never wear out because they have no brushes, for a dollar. So we can control all 12 of these petals for under 50 dollars and what this would allow us to do is not have to move **the** focus any more, but only move **the** petals. **The** whole system would have a much lower profile, but also we could gather sunlight for six and a half to seven hours a day. Now that we have concentrated sunlight, what are we going to put at **the** center to convert sunlight to electricity? So we tried to look at all **the** different heat engines that have been used in history to convert sunlight or heat to electricity, And one of **the** great ones of all time, James Watt ' s steam engine of 1788 was a major breakthrough. James Watt didn ' t actually invent **the** steam engine, he just refined it. But his refinements were incredible. He added new linear motion guides to **the** pistons, he added a condenser to cool **the** steam outside **the** cylinder, he made **the** engine double-acting, so it had double **the** power. Those were major breakthroughs. All of **the** improvements he made — and it ' s justifiable that our measure of energy, **the** watt, today is named after him. So we looked at this engine, and this had some potential. Steam engines are dangerous, and they had tremendous impact on **the** world — industrial revolution and ships and locomotives. But they ' re usually good to be large, so they ' re not good for distributed power generation. They ' re also very high- pressure, so they ' re dangerous. Another type of engine is **the** hot air engine. And **the** hot air engine also was not invented by Robert Stirling, but Robert Stirling came along in 1816 and radically improved it. This engine, because it was so interesting — it only worked on air, no steam — has led to hundreds of creative designs over **the** years that use **the** Stirling engine principle. But after **the** Stirling engine, Otto came along, and also, he didn ' t invent **the** internal combustion engine, he just refined it. He showed it in Paris in 1867, and it was a major achievement because it brought **the** power density of **the** engine way up. You could now get a lot more power in a lot smaller space, and that allowed **the** engine to be used for mobile applications. So, once you have mobility, you ' re making a lot of engines because you ' ve got lots of units, as opposed to steam ships or big factories, so this was **the** engine that ended up benefiting from mass production where all **the** other engines didn ' t. So, because it went into mass production, costs were reduced, 100 years of refinement, **emissions** were reduced, tremendous production value. There have been hundreds of millions of internal combustion engines built, compared to thousands of Stirling engines built. And not nearly as many small steam engines being built anymore, only large ones for big operations. So after looking at these three, and 47 others, we **concluded** that **the** Stirling engine would be **the** best one to use. I want to give you a brief explanation of how we looked at it and how it works. So we tried to look at **the** Stirling engine in a new way, because it was practical — weight no longer mattered for our application. **The** internal combustion engine took off because weight mattered, because you were moving around. But if you ’ re trying to generate solar energy in a static place **the** weight doesn ’ t matter so much. We also discovered that efficiency doesn ’ t matter so much if your energy source is free. Normally, efficiency is crucial because **the** fuel cost of your engine over its life **dwarfs the** cost of **the** engine. But if your fuel source is free, then **the** only thing that matters is **the** up-front capital cost of **the** engine. So you don ’ t want to optimize for efficiency, you want to optimize for power per dollar. So using that new twist, with **the** new criteria, we thought we could relook at **the** Stirling engine, and also bring genetic algorithms in. Basically, Robert Stirling didn ’ t have Gordon Moore before him to get us three gigahertz of processor power. So we took **the** same genetic algorithm that we used earlier to make that concentrator, which didn ’ t work out for us, to optimize **the** Stirling engine, and make its design sizes and all of its **dimensions the exact** optimum to get **the** most power per dollar, irrespective of weight, irrespective of size, just to get **the** most conversion of solar energy, because **the** sun is free. And that ’ s **the** process we took — let me show you how **the** engine works. **The** simplest heat engine, or hot air engine, of all time would be this — take a box, a steel canister, with a piston. Put a flame under it, **the** piston moves up. Take it off **the** flame and pour water on it, or let it cool down, **the** piston moves down. That ’ s a heat engine. That ’ s **the** most fundamental heat engine you could have. **The** problem is **the** efficiency is one hundredth of one percent, because you ’ re heating all **the** metal of **the** chamber and then cooling all **the** metal of **the** chamber each time. And you ’ re only getting power from **the** air that ’ s heating at **the** same time, but you ’ re wasting energy heating and cooling **the** metal. So someone came up with a very clever idea. Instead of heating and cooling **the** whole cylinder, what about if you put a displacer inside — a little thing that shuttles **the** air back and forth. You move that up and down with a little bit of energy but now you ’ re only shifting **the** air down to **the** hot end and up to **the** cold end. So, now you ’ re not alternately heating and cooling **the** metal, just **the** air. That allows you to get **the** efficiency up from a hundredth of a percent to about two percent. And then Robert Stirling came along with this genius idea, which was, well, I ’ m still not heating **the** metal now, with this kind of engine, but I ’ m still reheating all **the** air. I ’ m still heating **the** air every time and cooling **the** air every time. What about if I put a thermal sponge in **the** middle, in **the** passageway between where **the** air has to move between hot and cold? So he made fine wires, and cracked glass, and all different kinds of materials to be a heat sponge. So when **the** air pushes up to go from **the** hot end to **the** cold end, it puts some heat into **the** sponge. And then when **the** air comes back after it ’ s been cooled, it picks up that heat again. So you ’ re reusing your energy five or six times, and that brings **the** efficiency up to between 30 and 40 percent. It ’ s a little known, but brilliant, genius invention of Robert Stirling that takes **the** hot air engine from being somewhat impractical — like I found out when I made **the** real simple version in high school — to very potentially possible, once you get **the** efficiency up, if you can design this to be low enough cost. So we really set out on a path to try and make **the** lowest cost possible. We built a huge mathematical model of how a Stirling engine works. We applied **the** genetic algorithm. We got **the** results from that for **the** optimal engine. We built engines — so we built 100 different engines over **the** last two years. We measured each one, we readjusted **the** model to what we measured, and then we led that to **the** current prototype. It led to a very compact, inexpensive engine, and this is what **the** engine looks like. Let me show you what it looks like in real life. So this is **the** engine. It ’ s just a small cylinder down here, which holds **the** generator inside and all

the linkage, and it ’ s **the** hot cap — **the** hot cylinder on **the** top — this part gets hot, this part is cool, and electricity comes out. **The exact** converse is also true. If you put electricity in, this will get hot and this will get cold, you get refrigeration. So it ’ s a complete reversible cycle, a very efficient cycle, and quite a simple thing to make. So now you put **the** two things together. So you have **the** engine. What if you combine **the** petals and **the** engine in **the** center? **The** petals track and **the** engine gets **the** concentrated sunlight, takes that heat and turns it into electricity. This is what **the** first prototype of our system looked like with **the** petals and **the** engine in **the** center. This is being run out in **the** sun, and now I want to show you what **the** actual thing looks like. Thank you. So this is a unit with **the** 12 petals. These petals cost about a dollar each — they ’ re lightweight, injection-molded plastic, aluminized. **The** mechanism to control each petal is below there, with a microprocessor on each one. There are thermocouples on **the** engine — little sensors that **detect the** heat when **the** sunlight strikes them. Each petal adjusts itself separately to keep **the** highest temperature on it. When **the** sun comes out in **the** morning, **the** petals will seek **the** sun, find it by **searching** for **the** highest temperature. About a minute and a half or two minutes after **the** rays are striking **the** hot cap **the** engine will be warm enough to start and then **the** engine will generate electricity for about six and a half hours a day — six and a half to seven hours as **the** sun moves across **the** sky. A critical part that we can take advantage of is that we have these inexpensive microprocessors and each of these petals is autonomous, and each of these petals figures out where **the** sun is with no user setup. So you don ’ t have to tell what latitude, longitude you ’ re at, what your roof **slope** angle is, or what orientation. It doesn ’ t really care. What it does is it **searches** to find **the** hottest spot, it **searches** again a half an hour later, a day later, a month later. It basically figures out where on Earth you are by watching **the** direction **the** sun moves, so you don ’ t have to actually enter anything about that. **The** way **the** unit works is, when **the** sun comes out, **the** engine will start and you get power out here. We have AC and DC, get 12 volts DC, so that could be used for certain applications. We have an inverter in there, so you get 117 volts AC. And you also get hot water. **The** hot water ’ s optional. You don ’ t have to use it, it will cool itself. But you can use it to optionally heat hot water and that brings **the** efficiency up even higher because some of **the** heat that you ’ d normally be rejecting, you can now use as useful energy, whether it ’ s for a pool or hot water. Let me show you a quick movie of what this looks like running. This is **the** first test where we took it outside and each of **the** petals were individually seeking. And what they do is step, very coarsely at first, and very finely afterward. Once they get a temperature reading on **the** thermocouple indicating they found **the** sun, they slow down and do a fine **search**. Then **the** petals will move into position, and **the** engine will start. We ’ ve been working on this for **the** last two years. We ’ re very excited about **the** progress, we have a long way to go though. This is how we envision it would be in a residential installation : you ’ d probably have more than one unit on your roof. It could be on your roof, your backyard, or somewhere else. You don ’ t have to have enough units to power your entire house, you just save money with each incremental one you add. So you ’ re still using **the** grid potentially, in this type of application, to be your backup supply — of course, you can ’ t use these at night, and you can ’ t use these on cloudy days. But by reducing your energy use, pretty much at **the** peak times — usually when you have your air conditioning on, or other times like that — this generates **the** peak power at **the** peak usage time, so it ’ s very complementary in that sense. This is how we would envision a residential application. We also think there ’ s very big potential for energy farms, especially in remote land where there happens to be a lot of sun. It ’ s a really good combination of those two factors. It turns out there ’ s a lot of powerful sun all around **the** world, obviously, but in special places where it happens to be relatively inexpensive to place these and also in many more places where there is high wind power. So an example of that is, here ’ s **the** map of **the** United States. Pretty much everywhere that ’ s not green or **blue** is a really ideal place, but even **the** green or **blue** areas are good, just not as good as **the** places that are red, orange and **yellow**. But **the** hot spot right around Las Vegas and Death Valley is very good. And is only affects **the** payback period, it doesn ’ t mean that you couldn ’ t use solar energy ; you could use it anywhere on Earth. It just affects **the** payback period if you ’ re comparing to grid-supplied electricity. But if you don ’ t have grid-supplied electricity, then **the** question of payback is a different one entirely. It ’ s just how many watts do you get per dollar, and how could you benefit from that to change your life in some way. This is **the** map of **the** whole Earth, and you can see a huge swathe in **the** middle where a large part of **the** population is, there ’ s tremendous chances for solar energy. And of course, look at Africa. **The** potential to take advantage of solar energy there is unbelievable, and I ’ m really excited to talk more about finding ways we can help with that. So, in conclusion, I would say my journey has shown me that you can revisit old ideas in a new light, and sometimes ideas that have been **discarded** in **the** past can be practical now if you apply some new technology or new twists. We believe we ’ re getting very close to something practical and affordable. Our short-term goal for this is to be half **the** price of solar cells and our longer-term goal is to be less than a five-year payback. And at less than a five-year payback, this becomes very economic. So you don ’ t have to just have a feel-good attitude about energy to want to have one of these. It just makes economic sense. Right now, solar paybacks are between 30 and 50 years. If you get it down below five years, then it ’ s almost a no-brainer because **the** interest to own it — someone else will finance it for you and you can just make money from day one. So that ’ s our real powerful goal that we ’ re really shooting for in **the** company. Two other things that I learned that were very surprising to me — one was how casual we are about energy. I was walking from **the** elevator over here, and even just looking at **the** stage right now — so there ’ s probably 20, 500-watt lights right now. There ’ s 10, 000 watts of light pouring on **the** stage, one horsepower is 746 watts, at full power. So there ’ s basically 15 horses running at full speed just to keep **the** stage lit. Not to mention **the** 200 horses that are probably running right now to keep **the** air-conditioning going. And it ’ s just amazing, walk in **the** elevator, and there ’ s lights on in **the** elevator. Of course, now I ’ m very sensitive at home when we leave **the** lights on by mistake. But, everywhere around us we have insatiable use for energy because it ’ s so cheap. And it ’ s cheap because we ’ ve been subsidized by energy that ’ s been concentrated by **the** sun. Basically, **oil** is solar-energy concentrate. It ’ s been **pounded** for a billion years with a lot of energy to make it have all that energy contained in it. And we don ’ t have a birthright to just use that up as fast as we are, I think. And it would be great if we could make our energy usage **renewable**, where as we ’ re using **the** energy, we ’ re creating it at **the** same pace, and I really hope we can get there. Thank you very much, you ’ ve been a great audience.

Text Sample 5: POS Dim 1 – 2005 – Score 32.00 – 2005/t000435.txt
It is a thrill to be here at a conference that ’ s devoted to”Inspired by Nature”â
you can imagine. And I ’ m also thrilled to be in **the** foreplay section. Did you

notice this section is foreplay? Because I get to talk about one of my favorite critters, which is **the** Western Grebe. You haven ’ t lived until you ’ ve seen these guys do their courtship dance. I was on Bowman Lake in Glacier National Park, which is a long, skinny lake with sort of mountains upside down in it, and my partner and I have a rowing shell. And so we were rowing, and one of these Western Grebes came along. And what they do for their courtship dance is, they go together, **the** two of them, **the** two **mates**, and they begin to run **underwater**. They paddle faster, and faster, and faster, until they ’ re going so fast that they literally lift up out of **the** water, and they ’ re standing upright, sort of paddling **the** top of **the** water. And one of these Grebes came along while we were rowing. And so we ’ re in a skull, and we ’ re moving really, really quickly. And this Grebe, I think, sort of, mistaked us for a prospect, and started to run along **the** water next to us, in a courtship dance â for miles. It would stop, and then start, and then stop, and then start. Now that is foreplay. I came this close to changing species at that moment. Obviously, life can teach us something in **the** entertainment section. Life has a lot to teach us. But what I ’ d like to talk about today is what life might teach us in technology and in design. What ’ s happened since **the** book came out â **the** book was mainly about research in biomimicry â and what ’ s happened since then is architects, designers, engineers â people who make our world â have started to call and say, we want a biologist to sit at **the** design table to help us, in real time, become inspired. Or â and this is **the** fun part for me â we want you to take us out into **the** natural world. We ’ ll come with a design challenge and we find **the** champion adapters in **the** natural world, who might inspire us. So this is a picture from a Galapagos trip that we took with some wastewater treatment engineers ; they purify wastewater. And some of them were very resistant, actually, to being there. What they said to us at first was, you know, we already do biomimicry. We use bacteria to clean our water. And we said, well, that ’ s not exactly being inspired by nature. That ’ s bioprocessing, you know ; that ’ s bio-assisted technology : using an organism to do your wastewater treatment is an old, old technology called”domestication.”This is learning something, learning an idea, from an organism and then applying it. And so they still weren ’ t getting it. So we went for a walk on **the** beach and I said, well, give me one of your big problems. Give me a design challenge, **sustainability** speed bump, that ’ s keeping you from being sustainable. And they said scaling, which is **the** build-up of minerals inside of pipes. And they said, you know what happens is, mineral â just like at your house â mineral builds up. And then **the** aperture closes, and we have to flush **the** pipes with toxins, or we have to dig them up. So if we had some way to stop this scaling â and so I picked up some shells on **the** beach. And I asked them, what is scaling? What ’ s inside your pipes? And they said, calcium carbonate. And I said, that ’ s what this is ; this is calcium carbonate. And they didn ’ t know that. They didn ’ t know that what a seashell is, it ’ s templated by proteins, and then ions from **the** seawater crystallize in place to create a shell. So **the** same sort of a process, without **the** proteins, is happening on **the** inside of their pipes. They didn ’ t know. This is not for lack of information ; it ’ s a lack of integration. You know, it ’ s a silo, people in silos. They didn ’ t know that **the** same thing was happening. So one of them thought about it and said, OK, well, if this is just crystallization that happens automatically out of seawater â self-assembly â then why aren ’ t shells infinite in size? What stops **the** scaling? Why don ’ t they just keep on going? And I said, well, in **the** same way that they exude a protein and it starts **the** crystallization â and then they all sort of leaned in â they let go of a protein that stops **the** crystallization. It literally adheres to **the** growing face of **the** crystal. And, in fact, there is a product called TPA that ’ s mimicked that protein â that stop-protein â and it ’ s an environmentally **friendly** way to stop scaling in pipes. That changed everything. From then on, you could not get these engineers back in **the** boat. **The** first day they would take a hike, and it was, click, click, click, click. Five minutes later they were back in **the** boat. We ’ re done. You know, I ’ ve seen that **island**. After this, they were crawling all over. They would snorkel for as long as we would let them snorkel. What had happened was that they realized that there were organisms out there that had already solved **the** problems that they had spent their careers trying to solve. Learning about **the** natural world is one thing ; learning from **the** natural world â that ’ s **the** switch. That ’ s **the** profound switch. What they realized was that **the** answers to their questions are everywhere ; they just needed to change **the** lenses with which they saw **the** world. 3. 8 billion years of field-testing. 10 to 30 â Craig Venter will probably tell you ; I think there ’ s a lot more than 30 million â well-adapted solutions. **The** important thing for me is that these are solutions solved in context. And **the** context is **the** Earth â **the** same context that we ’ re trying to solve our problems in. So it ’ s **the** conscious emulation of life ’ s genius. It ’ s not slavishly mimicking â although Al is trying to get **the** hairdo going â it ’ s not a slavish mimicry ; it ’ s taking **the** design principles, **the** genius of **the** natural world, and learning something from it. Now, in a group with so many IT people, I do have to mention what I ’ m not going to talk about, and that is that your field is one that has learned an enormous amount from living things, on **the** software side. So there ’ s **computers** that protect themselves, like an immune system, and we ’ re learning from gene regulation and biological development. And we ’ re learning from neural nets, genetic algorithms, **evolutionary** computing. That ’ s on **the** software side. But what ’ s interesting to me is that we haven ’ t looked at this, as much. I mean, these machines are really not very high tech in my estimation in **the** sense that there ’ s dozens and dozens of carcinogens in **the** water in Silicon Valley. So **the** hardware is not at all up to snuff in terms of what life would call a success. So what can we learn about making â not just **computers**, but everything? **The** plane you came in, cars, **the** seats that you ’ re sitting on. How do we redesign **the** world that we make, **the** human-made world? More importantly, what should we ask in **the** next 10 years? And there ’ s a lot of cool technologies out there that life has. What ’ s **the** syllabus? Three questions, for me, are key. How does life make things? This is **the** opposite ; this is how we make things. It ’ s called heat, beat and treat â that ’ s what material scientists call it. And it ’ s carving things down from **the** top, with 96 percent waste left over and only 4 percent product. You heat it up ; you beat it with high pressures ; you use chemicals. OK. Heat, beat and treat. Life can ’ t afford to do that. How does life make things? How does life make **the** most of things? That ’ s a geranium pollen. And its shape is what gives it **the** function of being able to tumble through air so easily. Look at that shape. Life adds information to matter. In other words : structure. It gives it information. By adding information to matter, it gives it a function that ’ s different than without that structure. And thirdly, how does life make things **disappear** into systems? Because life doesn ’ t really deal in things ; there are no things in **the** natural world divorced from their systems. Really quick syllabus. As I ’ m reading more and more now, and following **the** story, there are some amazing things coming up in **the** biological sciences. And at **the** same time, I ’ m listening to a lot of businesses and finding what their sort of grand challenges are. **The** two groups are not talking to each other. At all. What in **the** world of **biology** might be helpful at this juncture, to get us through this sort of **evolutionary** knothole that we ’ re in? I ’ m going to try to go through 12, really quickly. One that ’ s exciting to me is self-assembly. Now, you ’ ve heard about this in terms of nanotechnology.

Back to that shell : **the** shell is a self-assembling material. On **the** lower **left** there is a picture of mother of pearl forming out of seawater. It ’ s a layered structure that ’ s mineral and then polymer, and it makes it very, very tough. It ’ s twice as tough as our high-tech ceramics. But what ’ s really interesting : unlike our ceramics that are in kilns, it happens in seawater. It happens near, in and near, **the** organism ’ s body. This is Sandia National Labs. A guy named Jeff Brinker has found a way to have a self- assembling coding process. Imagine being able to make ceramics at room temperature by simply dipping something into a liquid, lifting it out of **the** liquid, and having evaporation force **the** molecules in **the** liquid together, so that they jigsaw together in **the** same way as this crystallization works. Imagine making all of our hard materials that way. Imagine spraying **the** precursors to a PV cell, to a solar cell, onto a roof, and having it self- assemble into a layered structure that harvests light. Here ’ s an interesting one for **the** IT world : bio-silicon. This is a diatom, which is made of silicates. And so silicon, which we make right now â it ’ s part of our carcinogenic problem in **the** manufacture of our chips â this is a bio-mineralization process that ’ s now being mimicked. This is at UC Santa Barbara. Look at these diatoms. This is from Ernst Haeckel ’ s work. Imagine being able to â and, again, it ’ s a templated process, and it solidifies out of a liquid process â imagine being able to have that sort of structure coming out at room temperature. Imagine being able to make perfect lenses. On **the** **left**, this is a brittle star ; it ’ s covered with lenses that **the** people at Lucent Technologies have found have no distortion whatsoever. It ’ s one of **the** most distortion-free lenses we know of. And there ’ s many of them, all over its entire body. What ’ s interesting, again, is that it self-assembles. A woman named Joanna Aizenberg, at Lucent, is now learning to do this in a low- temperature process to create these sort of lenses. She ’ s also looking at fiber optics. That ’ s a **sea** sponge that has a fiber optic. Down at **the** very base of it, there ’ s fiber optics that work better than ours, actually, to move light, but you can tie them in a knot ; they ’ re incredibly flexible. Here ’ s another big idea : CO2 as a feedstock. A guy named Geoff Coates, at Cornell, said to himself, you know, plants do not see CO2 as **the** biggest poison of our time. We see it that way. Plants are busy making long chains of starches and glucose, right, out of CO2. He ’ s found a way â he ’ s found a catalyst â and he ’ s found a way to take CO2 and make it into polycarbonates. Biodegradable plastics out of CO2 â how plant-like. Solar transformations : **the** most exciting one. There are people who are mimicking **the** energy-harvesting **device** inside of purple bacterium, **the** people at ASU. Even more interesting, lately, in **the** last couple of weeks, people have seen that there ’ s an enzyme called hydrogenase that ’ s able to **evolve** hydrogen from proton and electrons, and is able to take hydrogen up â basically what ’ s happening in a fuel cell, in **the** anode of a fuel cell and in a reversible fuel cell. In our fuel cells, we do it with platinum ; life does it with a very, very common **iron**. And a team has now just been able to mimic that hydrogen-juggling hydrogenase. That ’ s very exciting for fuel cells â to be able to do that without platinum. Power of shape : here ’ s a **whale**. We ’ ve seen that **the fins** of this **whale** have tubercles on them. And those little bumps actually increase efficiency in, for instance, **the** edge of an **airplane** â increase efficiency by about 32 percent. Which is an amazing **fossil** fuel savings, if we were to just put that on **the** edge of a **wing**. Color without pigments : this peacock is creating color with shape. Light comes through, it bounces back off **the** layers ; it ’ s called thin- film interference. Imagine being able to self-assemble products with **the** last few layers playing with light to create color. Imagine being able to create a shape on **the** outside of a **surface**, so that it ’ s self-cleaning with just water. That ’ s what a leaf does. See that up-close picture? That ’ s a ball of water, and those are dirt particles. And that ’ s an up-close picture of a lotus leaf. There ’ s a company making a product called Lotusan, which mimics â when **the** building facade paint dries, it mimics **the** bumps in a self-cleaning leaf, and rainwater cleans **the** building. Water is going to be our big, grand challenge : quenching thirst. Here are two organisms that pull water. **The** one on **the left** is **the** Namibian beetle pulling water out of fog. **The** one on **the** right is a pill bug â pulls water out of air, does not **drink** fresh water. Pulling water out of Monterey fog and out of **the** sweaty air in Atlanta, before it gets into a building, are key technologies. Separation technologies are going to be extremely important. What if we were to say, no more hard **rock** mining? What if we were to separate out metals from waste streams, small amounts of metals in water? That ’ s what microbes do ; they chelate metals out of water. There ’ s a company here in San Francisco called MR3 that is embedding mimics of **the** microbes ’ molecules on filters to mine waste streams. Green **chemistry** is **chemistry** in water. We do **chemistry** in **organic** solvents. This is a picture of **the** spinnerets coming out of a **spider** and **the** silk being formed from a **spider**. Isn ’ t that beautiful? Green **chemistry** is replacing our industrial **chemistry** with nature ’ s recipe book. It ’ s not easy, because life uses only a subset of **the** elements in **the** periodic table. And we use all of them, even **the** toxic ones. To figure out **the** elegant recipes that would take **the** small subset of **the** periodic table, and create miracle materials like that cell, is **the** task of green **chemistry**. Timed degradation : packaging that is good until you don ’ t want it to be good anymore, and dissolves on cue. That ’ s a mussel you can find in **the** waters out here, and **the** threads holding it to a **rock** are timed ; at exactly two years, they begin to dissolve. Healing : this is a good one. That little guy over there is a tardigrade. There is a problem with vaccines around **the** world not getting to patients. And **the** reason is that **the** refrigeration somehow gets broken ; what ’ s called **the**”cold chain”gets broken. A guy named Bruce Rosner looked at **the** tardigrade â which dries out completely, and yet stays alive for months and months and months, and is able to regenerate itself. And he found a way to dry out vaccines â encase them in **the** same sort of sugar capsules as **the** tardigrade has within its cells â meaning that vaccines no longer need to be refrigerated. They can be put in a glove compartment, OK. Learning from organisms. This is a session about water â learning about organisms that can do without water, in order to create a vaccine that lasts and lasts and lasts without refrigeration. I ’ m not going to get to 12. But what I am going to do is tell you that **the** most important thing, besides all of these adaptations, is **the** fact that these organisms have figured out a way to do **the** amazing things they do while taking care of **the** place that ’ s going to take care of their offspring. When they ’ re involved in foreplay, they ’ re thinking about something very, very important â and that ’ s having their genetic material remain, 10, 000 generations from now. And that means finding a way to do what they do without destroying **the** place that ’ ll take care of their offspring. That ’ s **the** biggest design challenge. Luckily, there are millions and millions of geniuses willing to gift us with their best ideas. Good luck having a conversation with them. Thank you. Chris Anderson : Talk about foreplay, I â we need to get to 12, but really quickly. Janine Benyus : Oh really? CA : Yeah. Just like, you know, like **the** 10-second version of 10, 11 and 12. Because we just â your **slides** are so gorgeous, and **the** ideas are so big, I can ’ t stand to let you go down without seeing 10, 11 and 12. JB : OK, put this â OK, I ’ ll just hold this thing. OK, great. OK, so that ’ s **the** healing one. Sensing and responding : feedback is a huge thing. This is a locust. There can be 80 million of them in a square kilometer, and yet they don ’ t collide with one another. And yet we have 3. 6 million car collisions a year. Right. There ’ s a person at Newcastle who has figured out that it ’ s a very large neuron. And she ’ s actually figuring out how to make a collision-avoidance circuitry based on this very large neuron in **the** locust. This is a huge and important one, number 11. And that ’ s **the** growing fertility. That means, you know, net fertility farming. We should be growing fertility. And, oh yes â we get food, too. Because we have to grow **the** capacity of this **planet** to create more and more opportunities for life. And really, that ’ s what other organisms do as well. In ensemble, that ’ s what whole ecosystems do : they create more and more opportunities for life. Our farming has done **the** opposite. So, farming based on how a prairie builds soil, ranching based on how a native ungulate herd actually increases **the** health of **the** range, even wastewater treatment based on how a marsh not only cleans **the** water, but creates incredibly sparkling productivity. This is **the** simple design brief. I mean, it looks simple because **the** system, over 3. 8 billion years, has worked this out. That is, those organisms that have not been able to figure out how to enhance or sweeten their places, are not around to tell us about it. That ’ s **the** twelfth one. Life â and this is **the** secret trick ; this is **the** magic trick â life creates conditions conducive to life. It builds soil ; it cleans air ; it cleans water ; it mixes **the** cocktail of gases that you and I need to live. And it does that in **the** middle of having great foreplay and meeting their needs. So it ’ s not mutually exclusive. We have to find a way to meet our needs, while making of this place an Eden. CA : Janine, thank you so much.

Text Sample 6: POS Dim 1 – 2005 – Score 27.00 – 2005/t000476.txt

Well, it ’ s great to be here. We ’ ve heard a lot about **the** promise of technology, and **the** peril. I ’ ve been quite interested in both. If we could convert 0. 03 percent of **the** sunlight that falls on **the** earth into energy, we could meet all of our projected needs for 2030. We can ’ t do that today because solar panels are heavy, expensive and very inefficient. There are nano-engineered designs, which at least have been analyzed theoretically, that show **the** potential to be very lightweight, very inexpensive, very efficient, and we ’ d be able to actually provide all of our energy needs in this **renewable** way. Nano-engineered fuel cells could provide **the** energy where it ’ s needed. That ’ s a key trend, which is decentralization, moving from centralized nuclear power plants and liquid natural gas tankers to decentralized resources that are environmentally more **friendly**, a lot more efficient and capable and safe from disruption. Bono spoke very eloquently, that we have **the** tools, for **the** first time, to address age-old problems of disease and poverty. Most regions of **the** world are moving in that direction. In 1990, in East Asia and **the** Pacific region, there were 500 million people living in poverty — that number now is under 200 million. **The** World Bank projects by 2011, it will be under 20 million, which is a reduction of 95 percent. I did **enjoy** Bono ’ s comment linking Haight-Ashbury to Silicon Valley. Being from **the** Massachusetts high-tech community myself, I ’ d point out that we were hippies also in **the** 1960s, although we hung around Harvard Square. But we do have **the** potential to overcome disease and poverty, and I ’ m going to talk about those issues, if we have **the** will. Kevin Kelly talked about **the** acceleration of technology. That ’ s been a strong interest of mine, and a **theme** that I ’ ve developed for some 30 years. I realized that my technologies had to make sense when I finished a project. That invariably, **the** world was a different place when I would introduce a technology. And, I noticed that most inventions fail, not because **the** R&D department can ’ t get it to work — if you look at most business plans, they will actually succeed if given **the** opportunity to build what they say they ’ re going to build — and 90 percent of those projects or more will fail, because **the** timing is wrong — not all **the** enabling factors will be in place when they ’ re needed. So I began to be an ardent student of technology trends, and track where technology would be at different points in time, and began to build **the** mathematical models of that. It ’ s kind of taken on a life of its own. I ’ ve got a group of 10 people that work with me to gather data on key measures of technology in many different areas, and we build models. And you ’ ll hear people say, well, we can ’ t predict **the** future. And if you ask me, will **the** price of Google be higher or lower than it is today three years from now, that ’ s very hard to say. Will WiMax CDMA G3 be **the** wireless standard three years from now? That ’ s hard to say. But if you ask me, what will it cost for one MIPS of computing in 2010, or **the** cost to sequence a base pair of DNA in 2012, or **the** cost of sending a megabyte of data wirelessly in 2014, it turns out that those are very predictable. There are remarkably smooth exponential curves that govern price performance, capacity, bandwidth. And I ’ m going to show you a small sample of this, but there ’ s really a theoretical reason why technology develops in an exponential fashion. And a lot of people, when they think about **the** future, think about it linearly. They think they ’ re going to continue to develop a problem or address a problem using today ’ s tools, at today ’ s pace of progress, and fail to take into consideration this exponential growth. **The Genome** Project was a controversial project in 1990. We had our best Ph. D. students, our most advanced equipment around **the** world, we got 1 / 10, 000th of **the** project done, so how ’ re we going to get this done in 15 years? And 10 years into **the** project, **the** skeptics were still going strong — says,”You ’ re two- thirds through this project, and you ’ ve managed to only sequence a very tiny percentage of **the** whole **genome**.”But it ’ s **the** nature of exponential growth that once it reaches **the** knee of **the** curve, it explodes. Most of **the** project was done in **the** last few years of **the** project. It took us 15 years to sequence HIV — we sequenced SARS in 31 days. So we are gaining **the** potential to overcome these problems. I ’ m going to show you just a few examples of how pervasive this phenomena is. **The** actual paradigm-shift rate, **the** rate of adopting new ideas, is doubling every decade, according to our models. These are all logarithmic graphs, so as you go up **the** levels it represents, generally multiplying by factor of 10 or 100. It took us half a century to adopt **the** telephone, **the** first virtual- reality technology. Cell phones were adopted in about eight years. If you put different communication technologies on this logarithmic graph, television, radio, telephone were adopted in decades. Recent technologies — like **the** PC, **the web**, cell phones — were under a decade. Now this is an interesting chart, and this really gets at **the** fundamental reason why an **evolutionary** process — and both **biology** and technology are **evolutionary** processes — accelerate. They work through interaction — they create a capability, and then it uses that capability to bring on **the** next stage. So **the** first step in biological **evolution**, **the evolution** of DNA — actually it was RNA came first — took billions of years, but then **evolution** used that information-processing backbone to bring on **the** next stage. So **the** Cambrian **Explosion**, when all **the** body plans of **the** animals were **evolved**, took only 10 million years. It was 200 times faster. And then **evolution** used those body plans to **evolve** higher cognitive functions, and biological **evolution** kept accelerating. It ’ s an inherent nature of an **evolutionary** process. So Homo sapiens, **the** first technology-creating species, **the** species that combined a cognitive function with an opposable appendage — and by **the** way, chimpanzees don ’ t really have a very good opposable thumb — so we could actually **manipulate** our environment with a power grip and fine motor coordination, and use our mental models to actually change **the** world and bring on technology. But anyway, **the**

evolution of our species took hundreds of thousands of years, and then working through interaction, **evolution** used, essentially, **the** technology-creating species to bring on **the** next stage, which were **the** first steps in technological **evolution**. And **the** first step took tens of thousands of years — stone tools, fire, **the** wheel — kept accelerating. We always used then **the** latest generation of technology to create **the** next generation. Printing press took a century to be adopted ; **the** first **computers** were designed pen-on-paper — now we use **computers**. And we ' ve had a continual acceleration of this process. Now by **the** way, if you look at this on a linear graph, it looks like everything has just happened, but some observer says,"Well, Kurzweil just put points on this graph that fall on that straight line."So, I took 15 different lists from key thinkers, like **the** Encyclopedia Britannica, **the** Museum of Natural History, Carl Sagan ' s Cosmic Calendar on **the** same — and these people were not trying to make my point ; these were just lists in reference works, and I think that ' s what they thought **the** key events were in biological **evolution** and technological **evolution**. And again, it forms **the** same straight line. You have a little bit of thickening in **the** line because people do have disagreements, what **the** key points are, there ' s differences of opinion when **agriculture** started, or how long **the** Cambrian **Explosion** took. But you see a very clear trend. There ' s a basic, profound acceleration of this **evolutionary** process. Information technologies double their capacity, price performance, bandwidth, every year. And that ' s a very profound **explosion** of exponential growth. A personal experience, when I was at MIT — **computer** taking up about **the** size of this room, less powerful than **the computer** in your cell phone. But Moore ' s Law, which is very often identified with this exponential growth, is just one example of many, because it ' s basically a property of **the evolutionary** process of technology. I put 49 **famous computers** on this logarithmic graph — by **the** way, a straight line on a logarithmic graph is exponential growth — that ' s another exponential. It took us three years to double our price performance of computing in 1900, two years in **the** middle ; we ' re now doubling it every one year. And that ' s exponential growth through five different paradigms. Moore ' s Law was just **the** last part of that, where we were shrinking transistors on an integrated circuit, but we had electro-mechanical calculators, relay-based **computers** that cracked **the** German Enigma Code, vacuum tubes in **the** 1950s predicted **the** election of Eisenhower, discreet transistors used in **the** first space flights and then Moore ' s Law. Every time one paradigm ran out of steam, another paradigm came out of left field to continue **the** exponential growth. They were shrinking vacuum tubes, making them smaller and smaller. That hit a wall. They couldn ' t shrink them and keep **the** vacuum. Whole different paradigm — transistors came out of **the** woodwork. In fact, when we see **the** end of **the** line for a particular paradigm, it creates research pressure to create **the** next paradigm. And because we ' ve been predicting **the** end of Moore ' s Law for quite a long time — **the** first prediction said 2002, until now it says 2022. But by **the** teen years, **the** features of transistors will be a few atoms in **width**, and we won ' t be able to shrink them any more. That ' ll be **the** end of Moore ' s Law, but it won ' t be **the** end of **the** exponential growth of computing, because chips are flat. We live in a three-dimensional world ; we might as well use **the** third **dimension**. We will go into **the** third **dimension** and there ' s been tremendous progress, just in **the** last few years, of getting three-dimensional, self- organizing molecular circuits to work. We ' ll have those ready well before Moore ' s Law runs out of steam. Supercomputers — same thing. Processor performance on Intel chips, **the** average price of a transistor — 1968, you could buy one transistor for a dollar. You could buy 10 million in 2002. It ' s pretty remarkable how smooth an exponential process that is. I mean, you ' d think this is **the** result of some tabletop experiment, but this is **the** result of worldwide chaotic behavior — countries accusing each other of dumping products, IPOs, bankruptcies, marketing programs. You would think it would be a very erratic process, and you have a very smooth outcome of this chaotic process. Just as we can ' t predict what one molecule in a gas will do — it ' s hopeless to predict a single molecule — yet we can predict **the** properties of **the** whole gas, using thermodynamics, very accurately. It ' s **the** same thing here. We can ' t predict any particular project, but **the** result of this whole worldwide, chaotic, unpredictable activity of competition and **the evolutionary** process of technology is very predictable. And we can predict these trends far into **the** future. Unlike Gertrude Stein ' s roses, it ' s not **the** case that a transistor is a transistor. As we make them smaller and less expensive, **the** electrons have less distance to travel. They ' re faster, so you ' ve got exponential growth in **the** speed of transistors, so **the** cost of a cycle of one transistor has been coming down with a halving rate of 1. 1 years. You add other forms of innovation and processor design, you get a doubling of price performance of computing every one year. And that ' s basically deflation — 50 percent deflation. And it ' s not just **computers**. I mean, it ' s true of DNA sequencing ; it ' s true of brain scanning ; it ' s true of **the** World Wide Web. I mean, anything that we can quantify, we have hundreds of different measurements of different, information-related measurements — capacity, adoption rates — and they basically double every 12, 13, 15 months, depending on what you ' re looking at. In terms of price performance, that ' s a 40 to 50 percent deflation rate. And economists have actually started worrying about that. We had deflation during **the** Depression, but that was collapse of **the** money supply, collapse of consumer confidence, a completely different phenomena. This is due to greater productivity, but **the** economist says,"But there ' s no way you ' re going to be able to keep up with that. If you have 50 percent deflation, people may increase their volume 30, 40 percent, but they won ' t keep up with it."But what we ' re actually seeing is that we actually more than keep up with it. We ' ve had 28 percent per year compounded growth in dollars in information technology over **the** last 50 years. I mean, people didn ' t build iPods for 10, 000 dollars 10 years ago. As **the** price performance makes new applications feasible, new applications come to **the** market. And this is a very widespread phenomena. Magnetic data storage — that ' s not Moore ' s Law, it ' s shrinking magnetic spots, different engineers, different companies, same exponential process. A key revolution is that we ' re understanding our own **biology** in these information terms. We ' re understanding **the** software programs that make our body run. These were **evolved** in very different times — we ' d like to actually change those programs. One little software program, called **the fat** insulin receptor gene, basically says,"Hold onto every calorie, because **the** next hunting season may not work out so well."That was in **the** interests of **the** species tens of thousands of years ago. We ' d like to actually turn that program off. They tried that in animals, and these mice ate ravenously and remained slim and got **the** health benefits of being slim. They didn ' t get diabetes ; they didn ' t get heart disease ; they lived 20 percent longer ; they got **the** health benefits of caloric restriction without **the** restriction. Four or five pharmaceutical companies have noticed this, felt that would be interesting drug for **the** human market, and that ' s just one of **the** 30, 000 genes that affect our biochemistry. We were **evolved** in an era where it wasn ' t in **the** interests of people at **the** age of most people at this conference, like myself, to live much longer, because we were using up **the** precious resources which were better deployed towards **the** children and those caring for them. So, life — long lifespans — like, that is to say, much more than 30 — weren ' t selected for, but we are learning to actually **manipulate** and change these software programs through **the** biotechnology revolution. For example, we can inhibit genes now with RNA interference. There are exciting new forms of gene therapy that overcome **the** problem of placing **the** genetic material in **the** right place on **the** chromosome. There ' s actually a — for **the** first time now, something going to human trials, that actually cures pulmonary hypertension — a fatal disease — using gene therapy. So we ' ll have not just designer babies, but designer baby boomers. And this technology is also accelerating. It cost 10 dollars per base pair in 1990, then a penny in 2000. It ' s now under a 10th of a cent. **The** amount of genetic data — basically this shows that smooth exponential growth doubled every year, enabling **the genome** project to be completed. Another major revolution : **the** communications revolution. **The** price performance, bandwidth, capacity of communications measured many different ways ; wired, wireless is growing exponentially. **The** Internet has been doubling in power and continues to, measured many different ways. This is based on **the** number of hosts. Miniaturization — we ' re shrinking **the** size of technology at an exponential rate, both wired and wireless. These are some designs from Eric Drexler ' s book — which we ' re now showing are feasible with supercomputing simulations, where actually there are scientists building molecule-scale robots. One has one that actually walks with a surprisingly human-like gait, that ' s built out of molecules. There are little machines doing things in experimental bases. **The** most exciting opportunity is actually to go inside **the** human body and perform therapeutic and diagnostic functions. And this is less futuristic than it may sound. These things have already been done in animals. There ' s one nano-engineered **device** that cures type 1 diabetes. It ' s **blood** cell- sized. They put tens of thousands of these in **the blood** cell — they tried this in rats — it lets insulin out in a controlled fashion, and actually cures type 1 diabetes. What you ' re watching is a design of a robotic red **blood** cell, and it does bring up **the** issue that our **biology** is actually very sub- optimal, even though it ' s remarkable in its intricacy. Once we understand its principles of operation, and **the** pace with which we are reverse-engineering **biology** is accelerating, we can actually design these things to be thousands of times more capable. An analysis of this respirocyte, designed by Rob Freitas, indicates if you replace 10 percent of your red **blood** cells with these robotic versions, you could do an Olympic sprint for 15 minutes without taking a breath. You could sit at **the** bottom of your pool for four hours — so,"Honey, I ' m in **the** pool,"will take on a whole new meaning. It will be interesting to see what we do in our Olympic trials. Presumably we ' ll ban them, but then we ' ll have **the** specter of teenagers in their high schools gyms routinely out- performing **the** Olympic athletes. Freitas has a design for a robotic white **blood** cell. These are 2020-circa scenarios, but they ' re not as futuristic as it may sound. There are four major conferences on building **blood** cell-sized **devices** ; there are many experiments in animals. There ' s actually one going into human trial, so this is feasible technology. If we come back to our exponential growth of computing, 1, 000 dollars of computing is now somewhere between an insect and a mouse brain. It will intersect human intelligence in terms of capacity in **the** 2020s, but that ' ll be **the** hardware side of **the** equation. Where will we get **the** software? Well, it turns out we can see inside **the** human brain, and in fact not surprisingly, **the** spatial and temporal resolution of brain scanning is doubling every year. And with **the** new generation of scanning tools, for **the** first time we can actually see individual inter-neural fibers and see them processing and signaling in real time — but then **the** question is, OK, we can get this data now, but can we understand it? Doug Hofstadter wonders, well, maybe our intelligence just isn ' t great enough to understand our intelligence, and if we were smarter, well, then our brains would be that much more complicated, and we

’ d never catch up to it. It turns out that we can understand it. This is a block diagram of a model and simulation of **the** human auditory cortex that actually works quite well — in applying psychoacoustic tests, gets very similar results to human auditory perception. There ’ s another simulation of **the** cerebellum — that ’ s more than half **the** neurons in **the** brain — again, works very similarly to human skill formation. This is at an early stage, but you can show with **the** exponential growth of **the** amount of information about **the** brain and **the** exponential improvement in **the** resolution of brain scanning, we will succeed in reverse-engineering **the** human brain by **the** 2020s. We ’ ve already had very good models and simulation of about 15 regions out of **the** several hundred. All of this is driving exponentially growing economic progress. We ’ ve had productivity go from 30 dollars to 150 dollars per hour of labor in **the** last 50 years. E-commerce has been growing exponentially. It ’ s now a trillion dollars. You might wonder, well, wasn ’ t there a boom and a bust? That was strictly a capital-markets phenomena. Wall Street noticed that this was a revolutionary technology, which it was, but then six months later, when it hadn ’ t revolutionized all business models, they figured, well, that was wrong, and then we had this bust. All right, this is a technology that we put together using some of **the** technologies we ’ re involved in. This will be a routine feature in a cell phone. It would be able to translate from one language to another. So let me just end with a couple of scenarios. By 2010 **computers** will **disappear**. They ’ ll be so small, they ’ ll be embedded in our clothing, in our environment. Images will be written directly to our retina, providing full- immersion virtual reality, augmented real reality. We ’ ll be interacting with virtual personalities. But if we go to 2029, we really have **the** full maturity of these trends, and you have to appreciate how many turns of **the** screw in terms of generations of technology, which are getting faster and faster, we ’ ll have at that point. I mean, we will have two-to-the-25th-power greater price performance, capacity and bandwidth of these technologies, which is pretty phenomenal. It ’ ll be millions of times more powerful than it is today. We ’ ll have completed **the** reverse-engineering of **the** human brain, 1, 000 dollars of computing will be far more powerful than **the** human brain in terms of basic raw capacity. **Computers** will combine **the** subtle pan-recognition powers of human intelligence with ways in which machines are already superior, in terms of doing analytic thinking, remembering billions of facts accurately. Machines can share their knowledge very quickly. But it ’ s not just an alien invasion of intelligent machines. We are going to merge with our technology. These nano-bots I mentioned will first be used for medical and health applications : cleaning up **the** environment, providing powerful fuel cells and widely distributed decentralized solar panels and so on in **the** environment. But they ’ ll also go inside our brain, interact with our biological neurons. We ’ ve demonstrated **the** key principles of being able to do this. So, for example, full-immersion virtual reality from within **the** nervous system, **the** nano-bots shut down **the** signals coming from your real senses, replace them with **the** signals that your brain would be receiving if you were in **the** virtual environment, and then it ’ ll feel like you ’ re in that virtual environment. You can go there with other people, have any kind of experience with anyone involving all of **the** senses.”Experience beamers,”I call them, will put their whole flow of sensory experiences in **the** neurological correlates of their emotions out on **the** Internet. You can plug in and experience what it ’ s like to be someone else. But most importantly, it ’ ll be a tremendous expansion of human intelligence through this direct merger with our technology, which in some sense we ’ re doing already. We routinely do intellectual feats that would be impossible without our technology. Human life expectancy is expanding. It was 37 in 1800, and with this sort of biotechnology, nano-technology revolutions, this will move up very rapidly in **the** years ahead. My main message is that progress in technology is exponential, not linear. Many — even scientists — assume a linear model, so they ' ll say,"Oh, it ' ll be hundreds of years before we have self-replicating nano-technology assembly or **artificial** intelligence."If you really look at **the** power of exponential growth, you ' ll see that these things are pretty soon at hand. And information technology is increasingly encompassing all of our lives, from our music to our manufacturing to our **biology** to our energy to materials. We ' ll be able to manufacture almost anything we need in **the** 2020s, from information, in very inexpensive raw materials, using nano-technology. These are very powerful technologies. They both empower our promise and our peril. So we have to have **the** will to apply them to **the** right problems. Thank you very much.

Text Sample 7: POS Dim 1 – 2005 – Score 27.00 – 2005/t000437.txt

If you take 10, 000 people at random, 9, 999 have something in common : their interests in business lie on or near **the** Earth ' s **surface**. **The** odd one out is an astronomer, and I am one of that strange breed. My talk will be in two parts. I ' ll talk first as an astronomer, and then as a worried member of **the** human race. But let ' s start off by remembering that Darwin showed how we ' re **the** outcome of four billion years of **evolution**. And what we try to do in astronomy and cosmology is to go back before Darwin ' s simple beginning, to set our Earth in a cosmic context. And let me just run through a few **slides**. This was **the** impact that happened last week on a comet. If they ' d sent a nuke, it would have been rather more spectacular than what actually happened last Monday. So that ' s another project for NASA. That ' s Mars from **the** European Mars Express, and at New Year. This artist ' s impression turned into reality when a parachute landed on Titan, Saturn ' s giant moon. It landed on **the** **surface**. This is pictures taken on **the** way down. That looks like a coastline. It is indeed, but **the** ocean is liquid methane — **the** temperature minus 170 degrees centigrade. If we go beyond our solar system, we ' ve learned that **the** stars aren ' t twinkly points of light. Each one is like a sun with a retinue of **planets** orbiting around it. And we can see places where stars are forming, like **the** Eagle Nebula. We see stars dying. In six billion years, **the** sun will look like that. And some stars die spectacularly in a supernova **explosion**, leaving remnants like that. On a still bigger scale, we see entire galaxies of stars. We see entire ecosystems where gas is being recycled. And to **the** cosmologist, these galaxies are just **the** atoms, as it were, of **the** large-scale **universe**. This picture shows a patch of sky so small that it would take about 100 patches like it to cover **the** full moon in **the** sky. Through a small telescope, this would look quite blank, but you see here hundreds of little, faint smudges. Each is a galaxy, fully like ours or Andromeda, which looks so small and faint because its light has taken 10 billion light-years to get to us. **The** stars in those galaxies probably don ' t have **planets** around them. There ' s scant chance of life there — that ' s because there ' s been no time for **the** nuclear **fusion** in stars to make silicon and **carbon** and **iron**, **the** building blocks of **planets** and of life. We believe that all of this emerged from a Big Bang — a hot, dense state. So how did that amorphous Big Bang turn into our complex cosmos? I ' m going to show you a movie simulation 16 powers of 10 faster than real time, which shows a patch of **the universe** where **the** expansions have subtracted out. But you see, as time goes on in gigayears at **the** bottom, you will see structures **evolve** as gravity feeds on small, dense irregularities, and structures develop. And we ' ll end up after 13 billion years with something looking rather like our own **universe**. And we compare simulated **universes** like that — I ’ ll show you a better simulation at **the** end of my talk — with what we actually see in **the** sky. Well, we can trace things back to **the** earlier stages of **the** Big Bang, but we still don ’ t know what banged and why it banged. That ’ s a challenge for 21st-century science. If my research group had a logo, it would be this picture here : an ouroboros, where you see **the** micro-world on **the left** — **the** world of **the** quantum — and on **the** right **the** large-scale **universe** of **planets**, stars and galaxies. We know our **universes** are united though — links between **left** and right. **The** everyday world is determined by atoms, how they stick together to make molecules. Stars are fueled by how **the** nuclei in those atoms react together. And, as we ’ ve learned in **the** last few years, galaxies are held together by **the** gravitational pull of so-called dark matter : particles in huge swarms, far smaller even than atomic nuclei. But we ’ d like to know **the** synthesis symbolized at **the** very top. **The** micro-world of **the** quantum is understood. On **the** right hand side, gravity holds sway. Einstein explained that. But **the** unfinished business for 21st-century science is to link together cosmos and micro-world with a unified theory — symbolized, as it were, gastronomically at **the** top of that picture. And until we have that synthesis, we won ’ t be able to understand **the** very beginning of our **universe** because when our **universe** was itself **the** size of an atom, quantum effects could shake everything. And so we need a theory that unifies **the** very large and **the** very small, which we don ’ t yet have. One idea, incidentally — and I had this hazard sign to say I ’ m going to **speculate** from now on — is that our Big Bang was not **the** only one. One idea is that our three-dimensional **universe** may be embedded in a high- dimensional space, just as you can imagine on these sheets of paper. You can imagine ants on one of them thinking it ’ s a two-dimensional **universe**, not being aware of another population of ants on **the** other. So there could be another **universe** just a millimeter away from ours, but we ’ re not aware of it because that millimeter is measured in some fourth spatial **dimension**, and we ’ re imprisoned in our three. And so we believe that there may be a lot more to physical reality than what we ’ ve normally called our **universe — the** aftermath of our Big Bang. And here ’ s another picture. Bottom right depicts our **universe**, which on **the** horizon is not beyond that, but even that is just one bubble, as it were, in some vaster reality. Many people suspect that just as we ’ ve gone from believing in one solar system to zillions of solar systems, one galaxy to many galaxies, we have to go to many Big Bangs from one Big Bang, perhaps these many Big Bangs displaying an immense variety of properties. Well, let ’ s go back to this picture. There ’ s one challenge symbolized at **the** top, but there ’ s another challenge to science symbolized at **the** bottom. You want to not only synthesize **the** very large and **the** very small, but we want to understand **the** very complex. And **the** most complex things are ourselves, midway between atoms and stars. We depend on stars to make **the** atoms we ’ re made of. We depend on **chemistry** to determine our complex structure. We clearly have to be large, compared to atoms, to have layer upon layer of complex structure. We clearly have to be small, compared to stars and **planets** — otherwise we ’ d be crushed by gravity. And in fact, we are midway. It would take as many human bodies to make up **the** sun as there are atoms in each of us. **The** geometric mean of **the** mass of a proton and **the** mass of **the** sun is 50 kilograms, within a factor of two of **the** mass of each person here. Well, most of you anyway. **The** science of complexity is probably **the** greatest challenge of all, greater than that of **the** very small on **the left** and **the** very large on **the** right. And it ’ s this science, which is not only enlightening our understanding of **the** biological world, but also transforming our world faster than ever. And more than that, it ’ s engendering new kinds of change. And I now move on to

the second part of my talk, and **the** book"Our Final Century"was mentioned. If I was not a self-effacing Brit, I would mention **the** book myself, and I would add that it ' s available in paperback. And in America it was called"Our Final Hour"because Americans like instant gratification. But my **theme** is that in this century, not only has science changed **the** world faster than ever, but in new and different ways. Targeted drugs, genetic modification, **artificial** intelligence, perhaps even implants into our brains, may change human beings themselves. And human beings, their physique and character, has not changed for thousands of years. It may change this century. It ' s new in our history. And **the** human impact on **the** global environment — **greenhouse** warming, mass extinctions and so forth — is unprecedented, too. And so, this makes this coming century a challenge. Bio- and cybertechnologies are environmentally benign in that they offer marvelous prospects, while, nonetheless, reducing pressure on energy and resources. But they will have a dark side. In our interconnected world, novel technology could empower just one fanatic, or some weirdo with a mindset of those who now design **computer** viruses, to trigger some kind on disaster. Indeed, catastrophe could arise simply from technical misadventure — error rather than terror. And even a tiny **probability** of catastrophe is unacceptable when **the** downside could be of global consequence. In fact, some years ago, Bill Joy wrote an article expressing tremendous concern about robots taking us over, etc. I don ' t go along with all that, but it ' s interesting that he had a simple solution. It was what he called"fine- grained relinquishment."He wanted to give up **the** dangerous kind of science and keep **the** good bits. Now, that ' s absurdly naive for two reasons. First, any scientific discovery has benign consequences as well as dangerous ones. And also, when a scientist makes a discovery, he or she normally has no clue what **the** applications are going to be. And so what this means is that we have to accept **the** risks if we are going to **enjoy the** benefits of science. We have to accept that there will be hazards. And I think we have to go back to what happened in **the** post-War era, post-World War II, when **the** nuclear scientists who ' d been involved in making **the** atomic bomb, in many cases were concerned that they should do all they could to alert **the** world to **the** dangers. And they were inspired not by **the** young Einstein, who did **the** great work in relativity, but by **the** old Einstein, **the** icon of poster and t-shirt, who failed in his scientific efforts to unify **the** physical laws. He was premature. But he was a moral compass — an inspiration to scientists who were concerned with arms control. And perhaps **the** greatest living person is someone I ' m privileged to know, Joe Rothblatt. Equally untidy office there, as you can see. He ' s 96 years old, and he founded **the** Pugwash movement. He persuaded Einstein, as his last act, to sign **the famous** memorandum of Bertrand Russell. And he sets an example of **the** concerned scientist. And I think to harness science optimally, to choose which doors to open and which to leave closed, we need latter-day counterparts of people like Joseph Rothblatt. We need not just campaigning physicists, but we need biologists, **computer** experts and environmentalists as well. And I think academics and independent entrepreneurs have a special obligation because they have more freedom than those in government service, or company employees subject to commercial pressure. I wrote my book,"Our Final Century,"as a scientist, just a general scientist. But there ' s one respect, I think, in which being a cosmologist offered a special perspective, and that ' s that it offers an awareness of **the** immense future. **The** stupendous time spans of **the evolutionary** past are now part of common culture — outside **the** American Bible Belt, anyway — but most people, even those who are familiar with **evolution**, aren ' t mindful that even more time lies ahead. **The** sun has been shining for four and a half billion years, but it ' ll be another six billion years before its fuel runs out. On that schematic picture, a sort of time-lapse picture, we ’ re halfway. And it ’ ll be another six billion before that happens, and any remaining life on Earth is vaporized. There ’ s an unthinking tendency to imagine that humans will be there, experiencing **the** sun ’ s demise, but any life and intelligence that exists then will be as different from us as we are from bacteria. **The** unfolding of intelligence and complexity still has immensely far to go, here on Earth and probably far beyond. So we are still at **the** beginning of **the** emergence of complexity in our Earth and beyond. If you represent **the** Earth ’ s lifetime by a single year, say from January when it was made to December, **the** 21st-century would be a quarter of a second in June — a tiny **fraction** of **the** year. But even in this concertinaed cosmic perspective, our century is very, very special, **the** first when humans can change themselves and their home **planet**. As I should have shown this earlier, it will not be humans who witness **the** end point of **the** sun ; it will be creatures as different from us as we are from bacteria. When Einstein died in 1955, one striking tribute to his global status was this cartoon by Herblock in **the** Washington Post. **The** plaque reads,”Albert Einstein lived here.”And I ’ d like to end with a vignette, as it were, inspired by this image. We ’ ve been familiar for 40 years with this image : **the** fragile beauty of land, ocean and clouds, contrasted with **the** sterile moonscape on which **the** astronauts left their footprints. But let ’ s suppose some aliens had been watching our pale **blue** dot in **the** cosmos from afar, not just for 40 years, but for **the** entire 4. 5 billion-year history of our Earth. What would they have seen? Over nearly all that immense time, Earth ’ s appearance would have changed very gradually. **The** only abrupt worldwide change would have been major **asteroid** impacts or volcanic super-eruptions. Apart from those brief traumas, nothing happens suddenly. **The** continental landmasses drifted around. Ice cover waxed and waned. Successions of new species emerged, **evolved** and became extinct. But in just a tiny sliver of **the** Earth ’ s history, **the** last one-millionth part, a few thousand years, **the** patterns of vegetation altered much faster than before. This signaled **the** start of **agriculture**. Change has accelerated as human populations rose. Then other things happened even more abruptly. Within just 50 years — that ’ s one hundredth of one millionth of **the** Earth ’ s age — **the** amount of **carbon** dioxide in **the** atmosphere started to rise, and ominously fast. **The planet** became an intense emitter of radio waves — **the** total output from all TV and cell phones and radar transmissions. And something else happened. Metallic **objects** — albeit very small ones, a few tons at most — escaped into orbit around **the** Earth. Some journeyed to **the** moons and **planets**. A race of advanced extraterrestrials watching our solar system from afar could confidently predict Earth ’ s final doom in another six billion years. But could they have predicted this unprecedented spike less than halfway through **the** Earth ’ s life? These human-induced alterations occupying overall less than a millionth of **the** elapsed lifetime and seemingly **occurring** with runaway speed? If they continued their vigil, what might these hypothetical aliens witness in **the** next hundred years? Will some spasm foreclose Earth ’ s future? Or will **the** biosphere stabilize? Or will some of **the** metallic **objects** launched from **the** Earth spawn new oases, a post-human life elsewhere? **The** science done by **the** young Einstein will continue as long as our civilization, but for civilization to survive, we ’ ll need **the** wisdom of **the** old Einstein — humane, global and farseeing. And whatever happens in this uniquely crucial century will resonate into **the** remote future and perhaps far beyond **the** Earth, far beyond **the** Earth as depicted here. Thank you very much.

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I thought I ’ d start with telling you or showing you **the** people who started [Jet Propulsion Lab]. When they were a bunch of kids, they were kind of very imaginative, very adventurous, as they were trying at Caltech to mix chemicals and see which one blows up more. Well, I don ’ t recommend that you try to do that now. Naturally, they blew up a shack, and Caltech, well, then, hey, you go to **the** Arroyo and really do all your tests in there. So, that ’ s what we call our first five employees during **the** tea break, you know, in here. As I said, they were adventurous people. As a matter of fact, one of them, who was, kind of, part of a cult which was not too far from here on Orange Grove, and unfortunately he blew up himself because he kept mixing chemicals and trying to figure out which ones were **the** best chemicals. So, that gives you a kind of flavor of **the** kind of people we have there. We try to avoid blowing ourselves up. This one I thought I ’ d show you. Guess which one is a JPL employee in **the** heart of this crowd. I tried to come like him this morning, but as I walked out, then it was too cold, and I said, I ’ d better put my shirt back on. But more importantly, **the** reason I wanted to show this picture : look where **the** other people are looking, and look where he is looking. Wherever anybody else looks, look somewhere else, and go do something different, you know, and doing that. And that ’ s kind of what has been **the** spirit of what we are doing. And I want to tell you a quote from Ralph Emerson that one of my colleagues, you know, put on my wall in my office, and it says,”Do not go where **the** path may lead. Go instead where there is no path, and leave a trail.”And that ’ s my recommendation to all of you : look what everybody is doing, what they are doing ; go do something completely different. Don ’ t try to improve a little bit on what somebody else is doing, because that doesn ’ t get you very far. In our early days we used to work a lot on rockets, but we also used to have a lot of parties, you know. As you can see, one of our parties, you know, a few years ago. But then a big difference happened about 50 years ago, after Sputnik was launched. We launched **the** first American satellite, and that ’ s **the** one you see on **the left** in there. And here we made 180 degrees change : we changed from a rocket house to be an exploration house. And that was done over a period of a couple of years, and now we are **the** leading organization, you know, exploring space on all of your behalf. But even when we did that, we had to remind ourselves, sometimes there are setbacks. So you see, on **the** bottom, that rocket was supposed to go upward ; somehow it ended going sideways. So that ’ s what we call **the** misguided missile. But then also, just to celebrate that, we started an event at JPL for”Miss Guided Missile.”So, we used to have a celebration every year and select — there used to be competition and parades and so on. It ’ s not very appropriate to do it now. Some people tell me to do it ; I think, well, that ’ s not really proper, you know, these days. So, we do something a little bit more serious. And that ’ s what you see in **the** last Rose Bowl, you know, when we entered one of **the** floats. That ’ s more on **the** play side. And on **the** right side, that ’ s **the** Rover just before we finished its testing to take it to **the** Cape to launch it. These are **the** Rovers up here that you have on Mars now. So that kind of tells you about, kind of, **the** fun things, you know, and **the** serious things that we try to do. But I said I ’ m going to show you a short **clip** of one of our employees to kind of give you an idea about some of **the** talent that we have. Video : Morgan Hendry : Beware of Safety is an instrumental **rock** band. It branches on more **the** experimental side. There ’ s **the** improvisational side of jazz. There ’ s **the** heavy-hitting sound of **rock**. Being able to treat sound as an instrument, and be able to dig for more **abstract** sounds and things to play live, mixing electronics and acoustics. **The** music ’ s half of me, but **the** other half — I landed probably **the** best gig of all. I work for **the** Jet Propulsion Lab. I

’ m building **the** next Mars Rover. Some of **the** most brilliant engineers I know are **the** ones who have that sort of artistic quality about them. You ’ ve got to do what you want to do. And anyone who tells you you can ’ t, you don ’ t listen to them. Maybe they ’ re right - I doubt it. Tell them where to put it, and then just do what you want to do. I ’ m Morgan Hendry. I am NASA. Charles Elachi : Now, moving from **the** play stuff to **the** serious stuff, always people ask, why do we explore? Why are we doing all of these missions and why are we exploring them? Well, **the** way I think about it is fairly simple. Somehow, 13 billion years ago there was a Big Bang, and you ’ ve heard a little bit about, you know, **the** origin of **the universe**. But somehow what strikes everybody ’ s imagination — or lots of people ’ s imagination — somehow from that original Big Bang we have this beautiful world that we live in today. You look outside : you have all that beauty that you see, all that life that you see around you, and here we have intelligent people like you and I who are having a conversation here. All that started from that Big Bang. So, **the** question is : How did that happen? How did that **evolve**? How did **the universe** form? How did **the** galaxies form? How did **the planets** form? Why is there a **planet** on which there is life which have **evolved**? Is that very common? Is there life on every **planet** that you can see around **the** stars? So we literally are all made out of stardust. We started from those stars ; we are made of stardust. So, next time you are really depressed, look in **the** mirror and you can look and say, hi, I ’ m looking at a star here. You can skip **the** dust part. But literally, we are all made of stardust. So, what we are trying to do in our exploration is effectively write **the** book of how things have came about as they are today. And one of **the** first, or **the** easiest, places we can go and explore that is to go towards Mars. And **the** reason Mars takes particular attention : it ’ s not very far from us. You know, it ’ ll take us only six months to get there. Six to nine months at **the** right time of **the** year. It ’ s a **planet** somewhat similar to Earth. It ’ s a little bit smaller, but **the** land mass on Mars is about **the** same as **the** land mass on Earth, you know, if you don ’ t take **the** oceans into account. It has polar caps. It has an atmosphere somewhat thinner than ours, so it has weather. So, it ’ s very similar to some extent, and you can see some of **the** features on it, like **the** Grand Canyon on Mars, or what we call **the** Grand Canyon on Mars. It is like **the** Grand Canyon on Earth, except a hell of a lot larger. So it ’ s about **the** size, you know, of **the** United States. It has volcanoes on it. And that ’ s Mount Olympus on Mars, which is a kind of huge volcanic shield on that **planet**. And if you look at **the** height of it and you compare it to Mount Everest, you see, it ’ ll give you an idea of how large that Mount Olympus, you know, is, **relative** to Mount Everest. So, it basically **dwarfs**, you know, Mount Everest here on Earth. So, that gives you an idea of **the** tectonic events or volcanic events which have happened on that **planet**. Recently from one of our satellites, this shows that it ’ s Earth-like — we caught a landslide **occurring** as it was happening. So it is a dynamic **planet**, and activity is going on as we speak today. And these Rovers, people wonder now, what are they doing today, so I thought I would show you a little bit what they are doing. This is one very large crater. Geologists love craters, because craters are like digging a big hole in **the** ground without really working at it, and you can see what ’ s below **the surface**. So, this is called Victoria Crater, which is about a few football fields in size. And if you look at **the** top **left**, you see a little teeny dark dot. This picture was taken from an orbiting satellite. If I zoom on it, you can see : that ’ s **the** Rover on **the surface**. So, that was taken from orbit ; we had **the** camera zoom on **the surface**, and we actually saw **the** Rover on **the surface**. And we actually used **the** combination of **the** satellite images and **the** Rover to actually conduct science, because we can observe large areas and then you can get those Rovers to move around and basically go to a certain location. So, specifically what we are doing now is that Rover is going down in that crater. As I told you, geologists love craters. And **the** reason is, many of you went to **the** Grand Canyon, and you see in **the** wall of **the** Grand Canyon, you see these layers. And what these layers — that ’ s what **the surface** used to be a million years ago, 10 million years ago, 100 million years ago, and you get deposits on top of them. So if you can read **the** layers it ’ s like reading your book, and you can learn **the** history of what happened in **the** past in that location. So what you are seeing here are **the** layers on **the** wall of that crater, and **the** Rover is going down now, measuring, you know, **the** properties and analyzing **the rocks** as it ’ s going down, you know, that canyon. Now, it ’ s kind of a little bit of a challenge driving down a **slope** like this. If you were there you wouldn ’ t do it yourself. But we really made sure we tested those Rovers before we got them down — or that Rover — and made sure that it ’ s all working well. Now, when I came last time, shortly after **the** landing — I think it was, like, a hundred days after **the** landing — I told you I was surprised that those Rovers are lasting even a hundred days. Well, here we are four years later, and they ’ re still working. Now you say, Charles, you are really lying to us, and so on, but that ’ s not true. We really believed they were going to last 90 days or 100 days, because they are solar powered, and Mars is a dusty **planet**, so we expected **the** dust would start accumulating on **the surface**, and after a while we wouldn ’ t have enough power, you know, to keep them warm. Well, I always say it ’ s important that you are smart, but every once in a while it ’ s good to be lucky. And that ’ s what we found out. It turned out that every once in a while there are dust devils which come by on Mars, as you are seeing here, and when **the** dust devil comes over **the** Rover, it just cleans it up. It is like a brand new car that you have, and that ’ s literally why they have lasted so long. And now we designed them reasonably well, but that ’ s exactly why they are lasting that long and still providing all **the** science data. Now, **the** two Rovers, each one of them is, kind of, getting old. You know, one of them, one of **the** wheels is stuck, is not working, one of **the** front wheels, so what we are doing, we are driving it **backwards**. And **the** other one has arthritis of **the** shoulder joint, you know, it ’ s not working very well, so it ’ s walking like this, and we can move **the** arm, you know, that way. But still they are producing a lot of scientific data. Now, during that whole period, a number of people got excited, you know, outside **the** science community about these Rovers, so I thought I ’ d show you a video just to give you a reflection about how these Rovers are being viewed by people other than **the** science community. So let me go on **the** next short video. By **the** way, this video is pretty accurate of how **the** landing took place, you know, about four years ago. Video : **Okay**, we have parachute aligned. **Okay**, deploy **the** airbags. Open. Camera. We have a picture right now. Yeah! CE : That ’ s about what happened in **the** Houston operation room. It ’ s exactly like this. Video : Now, if there is life, **the** Dutch will find it. What is he doing? What is that? CE : Not too bad. So anyway, let me continue on showing you a little bit about **the** beauty of that **planet**. As I said earlier, it looked very much like Earth, so you see sand dunes. It looks like I could have told you these are pictures taken from **the** Sahara Desert or somewhere, and you ’ d have believed me, but these are pictures taken from Mars. But one area which is particularly intriguing for us is **the** northern region, you know, of Mars, close to **the** North Pole, because we see ice caps, and we see **the** ice caps shrinking and expanding, so it ’ s very much like you have in northern Canada. And we wanted to find out — and we see all kinds of glacial features on it. So, we wanted to find out, actually, what is that ice made of, and could that have embedded in it some **organic**, you know, material. So we have a spacecraft which is heading towards Mars, called Phoenix, and that spacecraft will land

17 days, seven hours and 20 seconds from now, so you can adjust your watch. So it ’ s on May 25 around just before five o ’ clock our time here on **the** West Coast, actually we will be landing on another **planet**. And as you can see, this is a picture of **the** spacecraft put on Mars, but I thought that just in case you ’ re going to miss that show, you know, in 17 days, I ’ ll show you, kind of, a little bit of what ’ s going to happen. Video : That ’ s what we call **the** seven minutes of terror. So **the** plan is to dig in **the** soil and take samples that we put them in an oven and actually heat them and look what gases will come from it. So this was launched about nine months ago. We ’ ll be coming in at 12, 000 miles per hour, and in seven minutes we have to stop and touch **the surface** very softly so we don ’ t break that lander. Ben Cichy : Phoenix is **the** first Mars Scout mission. It ’ s **the** first mission that ’ s going to try to land near **the** North Pole of Mars, and it ’ s **the** first mission that ’ s actually going to try and reach out and touch water on **the surface** of another **planet**. Lynn Craig : Where there tends to be water, at least on Earth, there tends to be life, and so it ’ s potentially a place where life could have existed on **the planet** in **the** past. Erik Bailey : **The** main purpose of EDL is to take a spacecraft that is traveling at 12, 500 miles an hour and bring it to a screeching halt in a soft way in a very short amount of time. BC : We enter **the** Martian atmosphere. We ’ re 70 miles above **the surface** of Mars. And our lander is safely tucked inside what we call an aeroshell. EB : Looks kind of like an ice cream cone, more or less. BC : And on **the** front of it is this heat shield, this saucer-looking thing that has about a half-inch of essentially what ’ s cork on **the** front of it, which is our heat shield. Now, this is really special cork, and this cork is what ’ s going to protect us from **the** violent atmospheric entry that we ’ re about to experience. Rob Grover : Friction really starts to build up on **the** spacecraft, and we use **the** friction when it ’ s flying through **the** atmosphere to our advantage to slow us down. BC : From this point, we ’ re going to decelerate from 12, 500 miles an hour down to 900 miles an hour. EB : **The** outside can get almost as hot as **the surface** of **the** Sun. RG : **The** temperature of **the** heat shield can reach 2, 600 degrees Fahrenheit. EB : **The** inside doesn ’ t get very hot. It probably gets about room temperature. Richard Kornfeld : There is this window of opportunity within which we can deploy **the** parachute. EB : If you fire **the** ’ chute too early, **the** parachute itself could fail. **The** fabric and **the** stitching could just pull apart. And that would be bad. BC : In **the** first 15 seconds after we deploy **the** parachute, we ’ ll decelerate from 900 miles an hour to a relatively slow 250 miles an hour. We no longer need **the** heat shield to protect us from **the** force of atmospheric entry, so we jettison **the** heat shield, exposing for **the** first time our lander to **the** atmosphere of Mars. LC : After **the** heat shield has been jettisoned and **the** legs are deployed, **the** next step is to have **the** radar system begin to **detect** how far Phoenix really is from **the** ground. BC : We ’ ve lost 99 percent of our entry velocity. So, we ’ re 99 percent of **the** way to where we want to be. But that last one percent, as it always seems to be, is **the** tricky part. EB : Now **the** spacecraft actually has to decide when it ’ s going to get rid of its parachute. BC : We separate from **the** lander going 125 miles an hour at roughly a kilometer above **the surface** of Mars : 3, 200 feet. That ’ s like taking two Empire State Buildings and **stacking** them on top of one another. EB : That ’ s when we separate from **the** back shell, and we ’ re now in free-fall. It ’ s a very scary moment ; a lot has to happen in a very short amount of time. LC : So it ’ s in a free-fall, but it ’ s also trying to use all of its actuators to make sure that it ’ s in **the** right position to land. EB : And then it has to light up its engines, right itself, and then slowly slow itself down and touch down on **the** ground safely. BC : Earth and Mars are so far apart that it takes over ten minutes for a signal from Mars to get to Earth. And EDL itself is all over in a matter of seven minutes. So by **the** time

you even hear from **the** lander that EDL has started it ’ ll already be over. EB : We have to build large amounts of autonomy into **the** spacecraft so that it can land itself safely. BC : EDL is this immense, technically challenging problem. It ’ s about getting a spacecraft that ’ s hurtling through deep space and using all this bag of tricks to somehow figure out how to get it down to **the surface** of Mars at zero miles an hour. It ’ s this immensely exciting and challenging problem. CE : Hopefully it all will happen **the** way you saw it in here. So it will be a very tense moment, you know, as we are watching that spacecraft landing on another **planet**. So now let me talk about **the** next things that we are doing. So we are in **the** process, as we speak, of actually designing **the** next Rover that we are going to be sending to Mars. So I thought I would go a little bit and tell you, kind of, **the** steps we go through. It ’ s very similar to what you do when you design your product. As you saw a little bit earlier, when we were doing **the** Phoenix one, we have to take into account **the** heat that we are going to be facing. So we have to study all kinds of different materials, **the** shape that we want to do. In general we don ’ t try to please **the** customer here. What we want to do is to make sure we have an effective, you know, an efficient kind of machine. First we start by we want to have our employees to be as imaginative as they can. And we really love being close to **the** art center, because we have, as a matter of fact, one of **the** alumni from **the** art center, Eric Nyquist, had put a series of displays, far-out displays, you know, in our what we call mission design or spacecraft design room, just to get people to think wildly about things. We have a bunch of Legos. So, as I said, this is a playground for adults, where they sit down and try to play with different shapes and different designs. Then we get a little bit more serious, so we have what we call our CAD / CAMs and all **the** engineers who are involved, or scientists who are involved, who know about thermal properties, know about design, know about atmospheric interaction, parachutes, all of these things, which they work in a team effort and actually design a spacecraft in a **computer** to some extent, so to see, does that meet **the** requirement that we need. On **the** right, also, we have to take into account **the** environment of **the planet** where we are going. If you are going to Jupiter, you have a very high-radiation, you know, environment. It ’ s about **the** same radiation environment close by Jupiter as inside a nuclear reactor. So just imagine : you take your P. C. and throw it into a nuclear reactor and it still has to work. So these are kind of some of **the** little challenges, you know, that we have to face. If we are doing entry, we have to do tests of parachutes. You saw in **the** video a parachute breaking. That would be a bad day, you know, if that happened, so we have to test, because we are deploying this parachute at supersonic speeds. We are coming at extremely high speeds, and we are deploying them to slow us down. So we have to do all kinds of tests. To give you an idea of **the** size, you know, of that parachute **relative** to **the** people standing there. Next step, we go and actually build some kind of test models and actually test them, you know, in **the** lab at JPL, in what we call our Mars Yard. We kick them, we hit them, we drop them, just to make sure we understand how, where would they break. And then we back off, you know, from that point. And then we actually do **the** actual building and **the** flight. And this next Rover that we ’ re flying is about **the** size of a car. That big shield that you see outside, that ’ s a heat shield which is going to protect it. And that will be basically built over **the** next year, and it will be launched June a year from now. Now, in that case, because it was a very big Rover, we couldn ’ t use airbags. And I know many of you, kind of, last time afterwards said well, that was a cool thing to have — those airbags. Unfortunately this Rover is, like, ten times **the** size of **the**, you know, mass-wise, of **the** other Rover, or three times **the** mass. So we can ’ t use airbags. So we have to come up with another ingenious idea of how do we land it. And we didn ’ t want to take it propulsively all **the** way to **the surface** because we didn ’ t want to contaminate **the surface** ; we wanted **the** Rover to immediately land on its legs. So we came up with this ingenious idea, which is used here on Earth for helicopters. Actually, **the** lander will come down to about 100 feet and hover above that **surface** for 100 feet, and then we have a sky crane which will take that Rover and land it down on **the surface**. Hopefully it all will work, you know, it will work that way. And that Rover will be more kind of like a chemist. What we are going to be doing with that Rover as it drives around, it ’ s going to go and analyze **the** chemical composition of **rocks**. So it will have an arm which will take samples, put them in an oven, crush and analyze them. But also, if there is something that we cannot reach because it is too high on a cliff, we have a little laser system which will actually zap **the rock**, evaporate some of it, and actually analyze what ’ s coming from that **rock**. So it ’ s a little bit like”Star Wars,”you know, but it ’ s real. It ’ s real stuff. And also to help you, to help **the** community so you can do ads on that Rover, we are going to train that Rover to actually in addition to do this, to actually serve cocktails, you know, also on Mars. So that ’ s kind of giving you an idea of **the** kind of, you know, fun things we are doing on Mars. I thought I ’ d go to”**The** Lord of **the** Rings”now and show you some of **the** things we have there. Now,”**The** Lord of **the** Rings”has two things played through it. One, it ’ s a very attractive **planet** — it just has **the** beauty of **the** rings and so on. But for scientists, also **the** rings have a special meaning, because we believe they represent, on a small scale, how **the** Solar System actually formed. Some of **the** scientists believe that **the** way **the** Solar System formed, that **the** Sun when it collapsed and actually created **the** Sun, a lot of **the** dust around it created rings and then **the** particles in those rings accumulated together, and they formed bigger **rocks**, and then that ’ s how **the planets**, you know, were formed. So, **the** idea is, by watching Saturn we ’ re actually watching our solar system in real time being formed on a smaller scale, so it ’ s like a test bed for it. So, let me show you a little bit on what that Saturnian system looks like. First, I ’ m going to fly you over **the** rings. By **the** way, all of this is real stuff. This is not animation or anything like this. This is actually taken from **the** satellite that we have in orbit around Saturn, **the** Cassini. And you see **the** amount of detail that is in those rings, which are **the** particles. Some of them are agglomerating together to form larger particles. So that ’ s why you have these gaps, is because a small satellite, you know, is being formed in that location. Now, you think that those rings are very large **objects**. Yes, they are very large in one **dimension** ; in **the** other **dimension** they are paper thin. Very, very thin. What you are seeing here is **the** shadow of **the** ring on Saturn itself. And that ’ s one of **the** satellites which was actually formed on that one. So, think about it as a paper-thin, huge area of many hundreds of thousands of miles, which is rotating. And we have a wide variety of kind of satellites which will form, each one looking very different and very odd, and that keeps scientists busy for tens of years trying to explain this, and telling NASA we need more money so we can explain what these things look like, or why they formed that way. Well, there were two satellites which were particularly interesting. One of them is called Enceladus. It ’ s a satellite which was all made of ice, and we measured it from orbit. Made of ice. But there was something bizarre about it. If you look at these stripes in here, what we call **tiger** stripes, when we flew over them, all of a sudden we saw an increase in **the** temperature, which said that those stripes are warmer than **the** rest of **the** **planet**. So as we flew by away from it, we looked back. And guess what? We saw geysers coming out. So this is a Yellowstone, you know, of Saturn. We are seeing geysers of ice which are coming out of that **planet**, which indicate that most likely there is an ocean, you know, below **the surface**. And somehow, through some dynamic effect, we ’ re having these geysers which are being, you know, emitted from it. And **the** reason I showed **the** little arrow there, I think that should say 30 miles, we decided a few months ago to actually fly **the** spacecraft through **the** plume of that geyser so we can actually measure **the** material that it is made of. That was [**unclear**] also — you know, because we were worried about **the** risk of it, but it worked pretty well. We flew at **the** top of it, and we found that there is a fair amount of **organic** material which is being emitted in combination with **the** ice. And over **the** next few years, as we keep orbiting, you know, Saturn, we are planning to get closer and closer down to **the surface** and make more accurate measurements. Now, another satellite also attracted a lot of attention, and that ’ s Titan. And **the** reason Titan is particularly interesting, it ’ s a satellite bigger than our moon, and it has an atmosphere. And that atmosphere is very — as dense as our own atmosphere. So if you were on Titan, you would feel **the** same pressure that you feel in here. Except it ’ s a lot colder, and that atmosphere is heavily made of methane. Now, methane gets people all excited, because it ’ s **organic** material, so immediately people start thinking, could life have **evolved** in that location, when you have a lot of **organic** material. So people believe now that Titan is most likely what we call a pre-biotic **planet**, because it ’ s so cold **organic** material did not get to **the** stage of becoming biological material, and therefore life could have **evolved** on it. So it could be Earth, frozen three billion years ago before life actually started on it. So that ’ s getting a lot of interest, and to show you some example of what we did in there, we actually dropped a probe, which was developed by our colleagues in Europe, we dropped a probe as we were orbiting Saturn. We dropped a probe in **the** atmosphere of Titan. And this is a picture of an area as we were coming down. Just looked like **the** coast of California for me. You see **the** rivers which are coming along **the** coast, and you see that white area which looks like Catalina Island, and that looks like an ocean. And then with an instrument we have on board, a radar instrument, we found there are lakes like **the** Great Lakes in here, so it looks very much like Earth. It looks like there are rivers on it, there are oceans or lakes, we know there are clouds. We think it ’ s raining also on it. So it ’ s very much like **the** cycle on Earth except because it ’ s so cold, it could not be water, you know, because water would have frozen. What it turned out, that all that we are seeing, all this liquid, [is made of] hydrocarbon and ethane and methane, similar to what you put in your car. So here we have a cycle of a **planet** which is like our Earth, but is all made of ethane and methane and **organic** material. So if you were on Mars — sorry, on Titan, you don ’ t have to worry about four-dollar gasoline. You just drive to **the** nearest lake, stick your hose in it, and you ’ ve got your car filled up. On **the** other hand, if you light a match **the** whole **planet** will blow up. So in closing, I said I want to close by a couple of pictures. And just to kind of put us in perspective, this is a picture of Saturn taken with a spacecraft from behind Saturn, looking towards **the** Sun. **The** Sun is behind Saturn, so we see what we call”forward scattering,”so it highlights all **the** rings. And I ’ m going to zoom. There is a — I ’ m not sure you can see it very well, but on **the** top **left**, around 10 o ’ clock, there is a little teeny dot, and that ’ s Earth. You barely can see ourselves. So what I did, I thought I ’ d zoom on it. So as you zoom in, you know, you can see Earth, you know, just in **the** middle here. So we zoomed all **the** way on **the** art center. So thank you very much.

Text Sample 9: POS Dim 1 – 2008 – Score 22.00 – 2008/t000271.txt

Cultural **evolution** is a dangerous child for any species to let loose on its **planet**. By **the** time you realize what ’ s happening, **the** child is a toddler, up and causing havoc, and it ’ s too late to put it back. We humans are Earth ’ s Pandoran species. We ’ re **the** ones who let **the** second replicator out of its box, and we can ’ t push it back in. We ’ re seeing **the** consequences all around us. Now that, I suggest, is **the** view that comes out of taking memetics seriously. And it gives us a new way of thinking about not only what ’ s going on on our **planet**, but what might be going on elsewhere in **the** cosmos. So first of all, I ’ d like to say something about memetics and **the** theory of memes, and secondly, how this might answer questions about who ’ s out there, if indeed anyone is. So, memetics : memetics is founded on **the** principle of Universal Darwinism. Darwin had this amazing idea. Indeed, some people say it ’ s **the** best idea anybody ever had. Isn ’ t that a wonderful thought, that there could be such a thing as a best idea anybody ever had? Do you think there could? Audience : No. Susan Blackmore : Someone says no, very loudly, from over there. Well, I say yes, and if there is, I give **the** prize to Darwin. Why? Because **the** idea was so simple, and yet it explains all design in **the universe**. I would say not just biological design, but all of **the** design that we think of as human design. It ’ s all just **the** same thing happening. What did Darwin say? I know you know **the** idea, natural **selection**, but let me just paraphrase”**The** Origin of Species,”1859, in a few sentences. What Darwin said was something like this : if you have creatures that vary, and that can ’ t be doubted — I ’ ve been to **the** Galapagos, and I ’ ve measured **the** size of **the** beaks and **the** size of **the turtle** shells and so on, and so on. And 100 pages later. And if there is a struggle for life, such that nearly all of these creatures die — and this can ’ t be doubted, I ’ ve read Malthus and I ’ ve **calculated** how long it would take for **elephants** to cover **the** whole world if they bred unrestricted, and so on and so on. And another 100 pages later. And if **the** very few that survive pass onto their offspring whatever it was that helped them survive, then those offspring must be better adapted to **the** circumstances in which all this happened than their parents were. You see **the** idea? If, if, if, then. He had no concept of **the** idea of an algorithm, but that ’ s what he described in that book, and this is what we now know as **the evolutionary** algorithm. **The** principle is you just need those three things — variation, **selection** and heredity. And as Dan Dennett puts it, if you have those, then you must get **evolution**. Or design out of chaos, without **the** aid of mind. There ’ s one word I love on that **slide**. What do you think my favorite word is? Audience : Chaos. SB : Chaos? No. What? Mind? No. Audience : Without. SB : No, not without. You try them all in order : Mmm...? Audience : Must. SB : Must, at must. Must, must. This is what makes it so amazing. You don ’ t need a designer, or a plan, or foresight, or anything else. If there ’ s something that is **copied** with variation and it ’ s selected, then you must get design appearing out of nowhere. You can ’ t stop it. Must is my favorite word there. Now, what ’ s this to do with memes? Well, **the** principle here applies to anything that is **copied** with variation and **selection**. We ’ re so used to thinking in terms of **biology**, we think about genes this way. Darwin didn ’ t, of course ; he didn ’ t know about genes. He talked mostly about animals and plants, but also about languages **evolving** and becoming extinct. But **the** principle of Universal Darwinism is that any information that is varied and selected will produce design. And this is what Richard Dawkins was on about in his 1976 bestseller,”**The** Selfish Gene.”**The** information that is **copied**, he called **the** replicator. It selfishly **copies**. Not meaning it kind of sits around inside cells going,”I want to get **copied**.”But that it will get **copied** if it can, regardless of **the** consequences. It doesn ’ t care about **the** consequences because it can ’ t, because it ’ s just information being **copied**. And he wanted to get away from everybody thinking all **the** time about genes, and so he said,”Is there another replicator out there on **the planet**?”Ah, yes, there is. Look around you — here will do, in this room. All around us, still clumsily drifting about in its primeval **soup** of culture, is another replicator. Information that we **copy** from person to person, by imitation, by language, by talking, by telling stories, by wearing clothes, by doing things. This is information **copied** with variation and **selection**. This is design process going on. He wanted a name for **the** new replicator. So, he took **the** Greek word”mimeme,”which means that which is imitated. Remember that, that ’ s **the** core definition : that which is imitated. And abbreviated it to meme, just because it sounds good and made a good meme, an effective spreading meme. So that ’ s how **the** idea came about. It ’ s important to stick with that definition. **The** whole science of memetics is much maligned, much misunderstood, much feared. But a lot of these problems can be avoided by remembering **the** definition. A meme is not equivalent to an idea. It ’ s not an idea. It ’ s not equivalent to anything else, really. Stick with **the** definition. It ’ s that which is imitated, or information which is **copied** from person to person. So, let ’ s see some memes. Well, you sir, you ’ ve got those glasses hung around your neck in that particularly fetching way. I wonder whether you invented that idea for yourself, or **copied** it from someone else? If you **copied** it from someone else, it ’ s a meme. And what about, oh, I can ’ t see any interesting memes here. All right everyone, who ’ s got some interesting memes for me? Oh, well, your earrings, I don ’ t suppose you invented **the** idea of earrings. You probably went out and bought them. There are plenty more in **the** shops. That ’ s something that ’ s passed on from person to person. All **the** stories that we ’ re telling — well, of course, TED is a great meme-fest, masses of memes. **The** way to think about memes, though, is to think, why do they spread? They ’ re selfish information, they will get **copied**, if they can. But some of them will be **copied** because they ’ re good, or true, or useful, or beautiful. Some of them will be **copied** even though they ’ re not. Some, it ’ s quite hard to tell why. There ’ s one particular curious meme which I rather **enjoy**. And I ’ m glad to say, as I expected, I found it when I came here, and I ’ m sure all of you found it, too. You go to your nice, posh, international hotel somewhere, and you come in and you put down your clothes and you go to **the** bathroom, and what do you see? Audience : Bathroom soap. SB : Pardon? Audience : Soap. SB : Soap, yeah. What else do you see? Audience : SB : Mmm mmm. Audience : Sink, toilet! SB : Sink, toilet, yes, these are all memes, they ’ re all memes, but they ’ re sort of useful ones, and then there ’ s this one. What is this one doing? This has spread all over **the** world. It ’ s not surprising that you all found it when you arrived in your bathrooms here. But I took this photograph in a toilet at **the** back of a tent in **the** eco-camp in **the** jungle in Assam. Who folded that thing up there, and why? Some people get carried away. Other people are just lazy and make mistakes. Some hotels exploit **the** opportunity to put even more memes with a little sticker. What is this all about? I suppose it ’ s there to tell you that somebody ’ s cleaned **the** place, and it ’ s all lovely. And you know, actually, all it tells you is that another person has potentially spread germs from place to place. So, think of it this way. Imagine a world full of brains and far more memes than can possibly find homes. **The** memes are all trying to get **copied** — trying, in inverted commas — i. e., that ’ s **the** shorthand for, if they can get **copied**, they will. They ’ re using you and me as their propagating, **copying** machinery, and we are **the** meme machines. Now, why is this important? Why is this useful, or what does it tell us? It gives us a completely new view of human origins and what it means to be human, all conventional theories of cultural **evolution**, of **the** origin of humans, and what makes us so different from other species. All other theories explaining **the** big brain, and language, and tool use and all these things that make us unique, are based upon genes. Language must have been useful for **the** genes. Tool use must have enhanced our survival, mating and so on. It always comes back, as Richard Dawkins complained all that long time ago, it always comes back to genes. **The** point of memetics is to say,"Oh no, it doesn ' t."There are two replicators now on this **planet**. From **the** moment that our ancestors, perhaps two and a half million years ago or so, began imitating, there was a new copying process. **Copying** with variation and **selection**. A new replicator was let loose, and it could never be — right from **the** start — it could never be that human beings who let loose this new creature, could just **copy the** useful, beautiful, true things, and not **copy the** other things. While their brains were having an advantage from being able to **copy** — lighting fires, keeping fires going, new **techniques** of hunting, these kinds of things — inevitably they were also **copying** putting feathers in their hair, or wearing strange clothes, or painting their faces, or whatever. So, you get an arms race between **the** genes which are trying to get **the** humans to have small economical brains and not waste their time **copying** all this stuff, and **the** memes themselves, like **the** sounds that people made and **copied** — in other words, what turned out to be language — competing to get **the** brains to get bigger and bigger. So, **the** big brain, on this theory, is driven by **the** memes. This is why, in"**The** Meme Machine,"I called it memetic drive. As **the** memes **evolve**, as they inevitably must, they drive a bigger brain that is better at **copying the** memes that are doing **the** driving. This is why we ' ve ended up with such **peculiar** brains, that we like religion, and music, and art. Language is a parasite that we ' ve adapted to, not something that was there originally for our genes, on this view. And like most parasites, it can begin dangerous, but then it coevolves and adapts, and we end up with a symbiotic relationship with this new parasite. And so, from our perspective, we don ' t realize that that ' s how it began. So, this is a view of what humans are. All other species on this **planet** are gene machines only, they don ' t imitate at all well, hardly at all. We alone are gene machines and meme machines as well. **The** memes took a gene machine and turned it into a meme machine. But that ' s not all. We have a new kind of memes now. I ' ve been wondering for a long time, since I ' ve been thinking about memes a lot, is there a difference between **the** memes that we **copy** — **the** words we speak to each other, **the** gestures we **copy**, **the** human things — and all these technological things around us? I have always, until now, called them all memes, but I do honestly think now we need a new word for technological memes. Let ' s call them techno-memes or temes. Because **the** processes are getting different. We began, perhaps 5, 000 years ago, with writing. We put **the** storage of memes out there on a clay tablet, but in order to get true temes and true teme machines, you need to get **the** variation, **the selection** and **the** copying, all done outside of humans. And we ' re getting there. We ' re at this extraordinary point where we ' re nearly there, that there are machines like that. And indeed, in **the** short time I ' ve already been at TED, I see we ' re even closer than I thought we were before. So actually, now **the** temes are forcing our brains to become more like teme machines. Our children are growing up very quickly learning to read, learning to use machinery. We ' re going to have all kinds of implants, drugs that force us to stay awake all **the** time. We ' ll think we ' re choosing these things, but **the** temes are making us do it. So, we ' re at this cusp now of having a third replicator on our **planet**. Now, what about what else is going on out there in **the universe**? Is there anyone else out there? People have been asking this question for a long time. We ' ve been asking it here at TED already. In 1961, Frank Drake made his **famous** equation, but I think he concentrated on **the** wrong things. It ' s been very productive, that equation. He wanted to estimate N, **the** number of communicative civilizations out there in our galaxy, and he included in there **the** rate of star formation, **the** rate of **planets**, but crucially, intelligence. I think that ' s **the** wrong way to think about it. Intelligence appears all over **the** place, in all kinds of guises. Human intelligence is only one kind of a thing. But what ' s really important is **the** replicators you have and **the** levels of replicators, one **feeding** on **the** one before. So, I would suggest that we don ' t think intelligence, we think replicators. And on that basis, I ' ve suggested a different kind of equation. A very simple equation. N, **the** same thing, **the** number of communicative civilizations out there [that] we might expect in our galaxy. Just start with **the** number of **planets** there are in our galaxy. **The fraction** of those which get a first replicator. **The fraction** of those that get **the** second replicator. **The fraction** of those that get **the** third replicator. Because it ' s only **the** third replicator that ' s going to reach out — sending information, sending probes, getting out there, and communicating with anywhere else. OK, so if we take that equation, why haven ' t we heard from anybody out there? Because every step is dangerous. Getting a new replicator is dangerous. You can pull through, we have pulled through, but it ' s dangerous. Take **the** first step, as soon as life appeared on this earth. We may take **the** Gaian view. I loved Peter Ward ' s talk yesterday — it ' s not Gaian all **the** time. Actually, life forms produce things that kill themselves. Well, we did pull through on this **planet**. But then, a long time later, billions of years later, we got **the** second replicator, **the** memes. That was dangerous, all right. Think of **the** big brain. How many mothers do we have here? You know all about big brains. They are dangerous to give birth to, are agonizing to give birth to. My cat gave birth to four kittens, purring all **the** time. Ah, mm — slightly different. But not only is it painful, it kills lots of babies, it kills lots of mothers, and it ' s very expensive to produce. **The** genes are forced into producing all this myelin, all **the fat** to myelinate **the** brain. Do you know, sitting here, your brain is using about 20 percent of your body ' s energy output for two percent of your body weight? It ' s a really expensive organ to run. Why? Because it ' s producing **the** memes. Now, it could have killed us off. It could have killed us off, and maybe it nearly did, but you see, we don ' t know. But maybe it nearly did. Has it been tried before? What about all those other species? Louise Leakey talked yesterday about how we ' re **the** only one in this branch left. What happened to **the** others? Could it be that this experiment in imitation, this experiment in a second replicator, is dangerous enough to kill people off? Well, we did pull through, and we adapted. But now, we ' re hitting, as I ' ve just described, we ' re hitting **the** third replicator point. And this is even more dangerous — well, it ' s dangerous again. Why? Because **the** temes are selfish replicators and they don ' t care about us, or our **planet**, or anything else. They ' re just information, why would they? They are using us to suck up **the planet** ' s resources to produce more **computers**, and more of all these amazing things we ' re hearing about here at TED. Don ' t think,"Oh, we created **the** Internet for our own benefit."That ' s how it seems to us. Think, temes spreading because they must. We are **the** old machines. Now, are we going to pull through? What ' s going to happen? What does it mean to pull through? Well, there are kind of two ways of pulling through. One that is obviously happening all around us now, is that **the** temes turn us into teme machines, with these implants, with **the** drugs, with us merging with **the** technology. And why would they do that? Because we are self-replicating. We have babies. We make new ones, and so it ' s convenient to piggyback on us, because we ' re not yet at **the** stage on this **planet** where **the** other option is viable. Although it ' s closer, I heard this morning, it ' s closer than I thought it was. Where **the** teme machines themselves will replicate themselves. That way, it wouldn ' t matter if **the planet**

’ s climate was utterly destabilized, and it was no longer possible for humans to live here. Because those teme machines, they wouldn ’ t need — they ’ re not squishy, **wet**, oxygen-breathing, warmth-requiring creatures. They could carry on without us. So, those are **the** two possibilities. **The** second, I don ’ t think we ’ re that close. It ’ s coming, but we ’ re not there yet. **The** first, it ’ s coming too. But **the** damage that is already being done to **the planet** is showing us how dangerous **the** third point is, that third danger point, getting a third replicator. And will we get through this third danger point, like we got through **the** second and like we got through **the** first? Maybe we will, maybe we won ’ t. I have no idea. Chris Anderson : That was an incredible talk. SB : Thank you. I scared myself. CA :

Text Sample 10: POS Dim 1 – 2008 – Score 22.00 – 2008/t000241.txt

My talk is”Flapping Birds and Space Telescopes.”And you would think that should have nothing to do with one another, but I hope by **the** end of these 18 minutes, you ’ ll see a little bit of a relation. It ties to origami. So let me start. What is origami? Most people think they know what origami is. It ’ s this : flapping birds, toys, cootie catchers, that sort of thing. And that is what origami used to be. But it ’ s become something else. It ’ s become an art form, a form of sculpture. **The** common **theme** — what makes it origami — is folding is how we create **the** form. You know, it ’ s very old. This is a **plate** from 1797. It shows these women playing with these toys. If you look close, it ’ s this shape, called a crane. Every Japanese kid learns how to fold that crane. So this art has been around for hundreds of years, and you would think something that ’ s been around that long — so restrictive, folding only — everything that could be done has been done a long time ago. And that might have been **the** case. But in **the** twentieth century, a Japanese folder named Yoshizawa came along, and he created tens of thousands of new designs. But even more importantly, he created a language, a way we could communicate, a code of dots, dashes and arrows. Harkening back to Susan Blackmore ’ s talk, we now have a means of transmitting information with heredity and **selection**, and we know where that leads. And where it has led in origami is to things like this. This is an origami figure — one sheet, no cuts, folding only, hundreds of folds. This, too, is origami, and this shows where we ’ ve gone in **the** modern world. Naturalism. Detail. You can get horns, antlers — even, if you look close, cloven hooves. And it raises a question : what changed? And what changed is something you might not have expected in an art, which is math. That is, people applied mathematical principles to **the** art, to discover **the** underlying laws. And that leads to a very powerful tool. **The** secret to productivity in so many fields — and in origami — is letting dead people do your work for you. Because what you can do is take your problem, and turn it into a problem that someone else has solved, and use their solutions. And I want to tell you how we did that in origami. Origami revolves around crease patterns. **The** crease pattern shown here is **the** underlying blueprint for an origami figure. And you can ’ t just draw them arbitrarily. They have to obey four simple laws. And they ’ re very simple, easy to understand. **The** first law is two-colorability. You can color any crease pattern with just two colors without ever having **the** same color meeting. **The** directions of **the** folds at any vertex — **the** number of mountain folds, **the** number of valley folds — always differs by two. Two more or two less. Nothing else. If you look at **the** angles around **the** fold, you find that if you number **the** angles in a circle, all **the** even-numbered angles add up to a straight line, all **the** odd-numbered angles add up to a straight line. And if you look at how **the** layers **stack**, you ’ ll find that no matter how you **stack** folds and sheets, a sheet can never penetrate a fold. So that ’ s four simple laws. That ’ s all you need in origami. All of origami comes from that. And you ’ d think,”Can four simple laws give rise to that kind of complexity?”But indeed, **the** laws of quantum mechanics can be written down on a napkin, and yet they govern all of **chemistry**, all of life, all of history. If we obey these laws, we can do amazing things. So in origami, to obey these laws, we can take simple patterns — like this repeating pattern of folds, called textures — and by itself it ’ s nothing. But if we follow **the** laws of origami, we can put these patterns into another fold that itself might be something very, very simple, but when we put it together, we get something a little different. This **fish**, 400 scales — again, it is one uncut square, only folding. And if you don ’ t want to fold 400 scales, you can back off and just do a few things, and add **plates** to **the** back of a **turtle**, or toes. Or you can ramp up and go up to 50 stars on a flag, with 13 stripes. And if you want to go really crazy, 1, 000 scales on a rattlesnake. And this guy ’ s on display downstairs, so take a look if you get a chance. **The** most powerful tools in origami have related to how we get parts of creatures. And I can put it in this simple equation. We take an idea, combine it with a square, and you get an origami figure. What matters is what we mean by those symbols. And you might say,”Can you really be that specific? I mean, a stag beetle — it ’ s got two points for jaws, it ’ s got antennae. Can you be that specific in **the** detail?”And yeah, you really can. So how do we do that? Well, we break it down into a few smaller steps. So let me stretch out that equation. I start with my idea. I **abstract** it. What ’ s **the** most **abstract** form? It ’ s a stick figure. And from that stick figure, I somehow have to get to a folded shape that has a part for every bit of **the** subject, a flap for every leg. And then once I have that folded shape that we call **the** base, you can make **the** legs narrower, you can bend them, you can turn it into **the** finished shape. Now **the** first step, pretty easy. Take an idea, draw a stick figure. **The** last step is not so hard, but that middle step — going from **the abstract** description to **the** folded shape — that ’ s hard. But that ’ s **the** place where **the** mathematical ideas can get us over **the** hump. And I ’ m going to show you all how to do that so you can go out of here and fold something. But we ’ re going to start small. This base has a lot of flaps in it. We ’ re going to learn how to make one flap. How would you make a single flap? Take a square. Fold it in half, fold it in half, fold it again, until it gets long and narrow, and then we ’ ll say at **the** end of that, that ’ s a flap. I could use that for a leg, an arm, anything like that. What paper went into that flap? Well, if I unfold it and go back to **the** crease pattern, you can see that **the** upper **left** corner of that shape is **the** paper that went into **the** flap. So that ’ s **the** flap, and all **the** rest of **the** paper ’ s left over. I can use it for something else. Well, there are other ways of making a flap. There are other **dimensions** for flaps. If I make **the** flaps skinnier, I can use a bit less paper. If I make **the** flap as skinny as possible, I get to **the** limit of **the** minimum amount of paper needed. And you can see there, it needs a quarter-circle of paper to make a flap. There ’ s other ways of making flaps. If I put **the** flap on **the** edge, it uses a half circle of paper. And if I make **the** flap from **the** middle, it uses a full circle. So, no matter how I make a flap, it needs some part of a circular region of paper. So now we ’ re ready to scale up. What if I want to make something that has a lot of flaps? What do I need? I need a lot of circles. And in **the** 1990s, origami artists discovered these principles and realized we could make arbitrarily complicated figures just by packing circles. And here ’ s where **the** dead people start to help us out, because lots of people have studied **the** problem of packing circles. I can rely on that vast history of mathematicians and artists looking at disc packings and arrangements. And I can use those patterns now to create origami shapes. So we figured out these rules whereby you pack circles, you decorate **the** patterns of circles with lines according to more rules. That gives you **the** folds. Those folds fold into a base. You shape **the** base. You get a folded shape — in this case, a **cockroach**. And it ' s so simple. It ' s so simple that a **computer** could do it. And you say,"Well, you know, how simple is that?"But **computers** — you need to be able to describe things in very basic terms, and with this, we could. So I wrote a **computer** program a bunch of years ago called TreeMaker, and you can download it from my website. It ' s free. It runs on all **the** major platforms — even Windows. And you just draw a stick figure, and it **calculates the** crease pattern. It does **the** circle packing, **calculates the** crease pattern, and if you use that stick figure that I just showed — which you can kind of tell, it ' s a deer, it ' s got antlers — you ' ll get this crease pattern. And if you take this crease pattern, you fold on **the** dotted lines, you ' ll get a base that you can then shape into a deer, with exactly **the** crease pattern that you wanted. And if you want a different deer, not a white-tailed deer, but you want a mule deer, or an elk, you change **the** packing, and you can do an elk. Or you could do a moose. Or, really, any other kind of deer. These **techniques** revolutionized this art. We found we could do insects, **spiders**, which are close, things with legs, things with legs and **wings**, things with legs and antennae. And if folding a single praying mantis from a single uncut square wasn ' t interesting enough, then you could do two praying mantises from a single uncut square. She ' s eating him. I call it"Snack Time."And you can do more than just insects. This — you can put details, toes and claws. A grizzly bear has claws. This tree frog has toes. Actually, lots of people in origami now put toes into their models. Toes have become an origami meme, because everyone ' s doing it. You can make multiple subjects. So these are a couple of instrumentalists. **The** guitar player from a single square, **the** bass player from a single square. And if you say,"Well, but **the** guitar, bass — that ' s not so hot. Do a little more complicated instrument."Well, then you could do an organ. And what this has allowed is **the** creation of origami-on-demand. So now people can say,"I want exactly this and this and this,"and you can go out and fold it. And sometimes you create high art, and sometimes you pay **the** bills by doing some commercial work. But I want to show you some examples. Everything you ' ll see here, except **the** car, is origami. Just to show you, this really was folded paper. **Computers** made things move, but these were all real, folded **objects** that we made. And we can use this not just for visuals, but it turns out to be useful even in **the** real world. Surprisingly, origami and **the** structures that we ' ve developed in origami turn out to have applications in medicine, in science, in space, in **the** body, consumer electronics and more. And I want to show you some of these examples. One of **the** earliest was this pattern, this folded pattern, studied by Koryo Miura, a Japanese engineer. He studied a folding pattern, and realized this could fold down into an extremely compact package that had a very simple opening and closing structure. And he used it to design this solar **array**. It ' s an artist ' s rendition, but it flew in a Japanese telescope in 1995. Now, there is actually a little origami in **the** James Webb Space Telescope, but it ' s very simple. **The** telescope, going up in space, it unfolds in two places. It folds in thirds. It ' s a very simple pattern — you wouldn ' t even call that origami. They certainly didn ' t need to talk to origami artists. But if you want to go higher and go larger than this, then you might need some origami. Engineers at Lawrence Livermore National Lab had an idea for a telescope much larger. They called it **the** Eyeglass. **The** design called for geosynchronous orbit 25, 000 miles up, 100-meter **diameter** lens. So, imagine a lens **the** size of a football field. There were two groups of people who were interested in this : planetary scientists, who want to look up, and then other people, who wanted to look down. Whether you look up or look down, how do you get it up in space? You ’ ve got to get it up there in a rocket. And rockets are small. So you have to make it smaller. How do you make a large sheet of glass smaller? Well, about **the** only way is to fold it up somehow. So you have to do something like this. This was a small model. Folded lens, you divide up **the** panels, you add flexures. But this pattern ’ s not going to work to get something 100 meters down to a few meters. So **the** Livermore engineers, wanting to make use of **the** work of dead people, or perhaps live origamists, said,”Let ’ s see if someone else is doing this sort of thing.”So they looked into **the** origami community, we got in touch with them, and I started working with them. And we developed a pattern together that scales to arbitrarily large size, but that allows any flat ring or disc to fold down into a very neat, compact cylinder. And they adopted that for their first generation, which was not 100 meters — it was a five-meter. But this is a five-meter telescope — has about a quarter-mile focal length. And it works perfectly on its test range, and it indeed folds up into a neat little bundle. Now, there is other origami in space. Japan Aerospace [Exploration] Agency flew a solar sail, and you can see here that **the** sail expands out, and you can still see **the** fold lines. **The** problem that ’ s being solved here is something that needs to be big and sheet-like at its destination, but needs to be small for **the** journey. And that works whether you ’ re going into space, or whether you ’ re just going into a body. And this example is **the** latter. This is a heart stent developed by Zhong You at Oxford University. It holds open a blocked **artery** when it gets to its destination, but it needs to be much smaller for **the** trip there, through your **blood** vessels. And this stent folds down using an origami pattern, based on a model called **the** water bomb base. Airbag designers also have **the** problem of getting flat sheets into a small space. And they want to do their design by simulation. So they need to figure out how, in a **computer**, to flatten an airbag. And **the** algorithms that we developed to do insects turned out to be **the** solution for airbags to do their simulation. And so they can do a simulation like this. Those are **the** origami creases forming, and now you can see **the** airbag inflate and find out, does it work? And that leads to a really interesting idea. You know, where did these things come from? Well, **the** heart stent came from that little blow-up box that you might have learned in elementary school. It ’ s **the** same pattern, called **the** water bomb base. **The** airbag-flattening algorithm came from all **the** developments of circle packing and **the** mathematical theory that was really developed just to create insects — things with legs. **The** thing is, that this often happens in math and science. When you get math involved, problems that you solve for aesthetic value only, or to create something beautiful, turn around and turn out to have an application in **the** real world. And as weird and surprising as it may sound, origami may someday even save a life. Thanks.

Text Sample 11: POS Dim 1 – 2006 – Score 22.00 – 2006/t000498.txt

I ’ m **the** luckiest guy in **the** world. I got to see **the** last case of killer smallpox in **the** world. I was in India this past year, and I may have seen **the** last cases of polio in **the** world. There ’ s nothing that makes you feel more â **the** blessing and **the** honor of working in a program like that â than to know that something that horrible no longer exists. So I ’ m going to tell you â so I ’ m going to show you some dirty pictures. They are difficult to watch, but you should look at them with optimism, because **the** horror of these pictures will be matched by **the** uplifting quality of knowing that they no longer exist. But first, I ’ m going to tell you a little bit about my own journey. My background is not exactly **the** conventional medical education that you might expect. When I was an intern in San Francisco, I heard about a group of Native Americans who had taken over Alcatraz Island, and a Native American who wanted to give birth on that **island**, and no other doctor wanted to go and help her give birth. I went out to Alcatraz, and I lived on **the island** for several weeks. She gave birth ; I caught **the** baby ; I got off **the island** ; I landed in San Francisco ; and all **the** press wanted to talk to me, because my three weeks on **the island** made me an expert in Indian affairs. I wound up on every television show. Someone saw me on television ; they called me up ; and they asked me if I ' d like to be in a movie and to play a young doctor for a bunch of **rock** and roll stars who were traveling in a bus ride from San Francisco to England. And I said, yes, I would do that, so I became **the** doctor in an absolutely awful movie called"Medicine Ball Caravan."Now, you know from **the** ' 60s, you ' re either on **the** bus or you ' re off **the** bus ; I was on **the** bus. My wife of 37 years and I joined **the** bus. Our bus ride took us from San Francisco to London, then we switched buses at **the** big pond. We then got on two more buses and we drove through Turkey and Iran, Afghanistan, over **the** Khyber Pass into Pakistan, like every other young doctor. This is us at **the** Khyber Pass, and that ' s our bus. We had some difficulty getting over **the** Khyber Pass. But we wound up in India. And then, like everyone else in our generation, we went to live in a Himalayan monastery. This is just like a residency program, for those of you that are in medical school. And we studied with a wise man, a guru named Karoli Baba, who then told me to get rid of **the** dress, put on a three-piece suit, go join **the** United Nations as a diplomat and work for **the** World Health Organization. And he made an outrageous prediction that smallpox would be eradicated, and that this was God ' s gift to humanity, because of **the** hard work of dedicated scientists. And that prediction came true. This little girl is Rahima Banu, and she was **the** last case of killer smallpox in **the** world. And this document is **the** certificate that **the** global commission signed, certifying **the** world to have eradicated **the** first disease in history. **The** key to eradicating smallpox was early detection, early response. I ' m going to ask you to repeat that : early detection, early response. Can you say that? Audience : Early detection, early response. Larry Brilliant : Smallpox was **the** worst disease in history. It killed more people than all **the** wars in history. In **the** last century, it killed 500 million people. You ' re reading about Larry Page already. Somebody reads very fast. In **the** year that Larry Page and Sergey Brin â with whom I have a certain affection and a new affiliation â in **the** year in which they were born, two million people died of smallpox. We declared smallpox eradicated in 1980. This is **the** most important **slide** that I ' ve ever seen in public health, [Sovereigns killed by smallpox] because it shows you to be **the** richest and **the** strongest, and to be kings and **queens** of **the** world, did not protect you from dying of smallpox. Never can you doubt that we are all in this together. But to see smallpox from **the** perspective of a sovereign is **the** wrong perspective. You should see it from **the** perspective of a mother, watching her child develop this disease and standing by helplessly. Day one, day two, day three, day four, day five, day six. You ' re a mother and you ' re watching your child, and on day six, you see pustules that become hard. Day seven, they show **the** classic scars of smallpox umbilication. Day eight. And Al Gore said earlier that **the** most photographed image in **the** world, **the** most printed image in **the** world, was that of **the** Earth. But this was in 1974, and as of that moment, this photograph was **the** photograph that was **the** most widely printed, because we printed two billion **copies** of this photograph, and we took them hand to hand, door to door, to show people and ask them if there was smallpox in their house, because that was our surveillance system. We didn ' t have Google, we didn ' t have **web** crawlers, we didn ' t have **computers**. By day nine â you look at this picture and you ’ re horrified ; I look at this picture and I say,”Thank God,”because it ’ s clear that this is only an ordinary case of smallpox, and I know this child will live. And by day 13, **the** lesions are scabbing, his eyelids are swollen, but you know this child has no other secondary infection. And by day 20, while he will be scarred for life, he will live. There are other kinds of smallpox that are not like that. This is confluent smallpox, in which there isn ’ t a single place on **the** body where you could put a finger and not be covered by lesions. Flat smallpox, which killed 100 percent of people who got it. And hemorrhagic smallpox, **the** most cruel of all, which had a predilection for pregnant women. I ’ ve probably had 50 women die. They all had hemorrhagic smallpox. I ’ ve never seen anybody die from it who wasn ’ t a pregnant woman. In 1967, **the** WHO embarked on what was an outrageous program to eradicate a disease. In that year, there were 34 countries affected with smallpox. By 1970, we were down to 18 countries. 1974, we were down to five countries. But in that year, smallpox exploded throughout India. And India was **the** place where smallpox made its last stand. In 1974, India had a population of 600 million. There are 21 linguistic states in India, which is like saying 21 different countries. There are 20 million people on **the** road at any time, in buses and trains, walking ; 500, 000 villages, 120 million households, and none of them wanted to report if they had a case of smallpox in their house, because they thought that smallpox was **the** visitation of a deity, Shitala Mata, **the** cooling mother, and it was wrong to bring strangers into your house when **the** deity was in **the** house. No incentive to report smallpox. It wasn ’ t just India that had smallpox deities ; smallpox deities were prevalent all over **the** world. So, how we eradicated smallpox was â mass vaccination wouldn ’ t work. You could vaccinate everybody in India, but one year later there ’ d be 21 million new babies, which was then **the** population of Canada. It wouldn ’ t do just to vaccinate everyone. You had to find every single case of smallpox in **the** world at **the** same time, and draw a circle of immunity around it. And that ’ s what we did. In India alone, my 150, 000 best friends and I went door to door, with that same picture, to every single house in India. We made over one billion house calls. And in **the** process, I learned something very important. Every time we did a house-to-house **search**, we had a spike in **the** number of reports of smallpox. When we didn ’ t **search**, we had **the** illusion that there was no disease. When we did **search**, we had **the** illusion that there was more disease. A surveillance system was necessary, because what we needed was early detection, early response. So we **searched** and we **searched**, and we found every case of smallpox in India. We had a reward. We raised **the** reward. We continued to increase **the** reward. We had a scorecard that we wrote on every house. And as we did that, **the** number of reported cases in **the** world dropped to zero. And in 1980, we declared **the** globe free of smallpox. It was **the** largest campaign in United Nations history, until **the** Iraq war. 150, 000 people from all over **the** world â doctors of every race, religion, culture and nation, who fought side by side, brothers and sisters, with each other, not against each other, in a common cause to make **the** world better. But smallpox was **the** fourth disease that was intended for eradication. We failed three other times. We failed against **malaria**, **yellow** fever and yaws. But soon we may see polio eradicated. But **the** key to eradicating polio is early detection, early response. This may be **the** year we eradicate polio. That will make it **the** second disease in history. And David Heymann, who ’ s watching this on **the** webcast â David, keep on going. We ’ re close! We ’ re down to four countries. I feel like Hank Aaron. Barry Bonds can replace me any time. Let ’ s get another disease off **the** list of terrible things to worry about. I was just in India working on **the** polio program. **The** polio surveillance program is four million people going door to door. That is **the** surveillance system. But we need to have early detection, early response. Blindness, **the** same thing. **The** key to discovering blindness is doing epidemiological surveys and finding out **the** causes of blindness, so you can mount **the** correct response. **The** Seva Foundation was started by a group of alumni of **the** Smallpox Eradication Programme, who, having climbed **the** highest mountain, tasted **the** elixir of **the** success of eradicating a disease, wanted to do it again. And over **the** last 27 years, Seva ' s programs in 15 countries have given back sight to more than two million blind people. Seva got started because we wanted to apply these lessons of surveillance and epidemiology to something which nobody else was looking at as a public health issue : blindness, which heretofore had been thought of only as a clinical disease. In 1980, Steve Jobs gave me that **computer**, which is Apple number 12, and it ' s still in Kathmandu, and it ' s still working, and we ought to go get it and auction it off and make more money for Seva. And we conducted **the** first Nepal survey ever done for health, and **the** first nationwide blindness survey ever done, and we had astonishing results. Instead of finding out what we thought was **the** case â that blindness was caused mostly by glaucoma and trachoma â we were astounded to find out that blindness was caused instead by cataract. You can ' t cure or prevent what you don ' t know is there. In your TED packages there ' s a DVD,"Infinite Vision,"about Dr. V and **the** Aravind Eye Hospital. I hope that you will take a look at it. Aravind, which started as a Seva project, is now **the** world ' s largest and best eye hospital. This year, that one hospital will give back sight to more than 300, 000 people in Tamil Nadu, India. Bird flu. I stand here as a representative of all terrible things â this might be **the** worst. **The** key to preventing or mitigating pandemic bird flu is early detection and rapid response. We will not have a vaccine or adequate supplies of an antiviral to combat bird flu if it **occurs** in **the** next three years. WHO stages **the** progress of a pandemic. We are now at stage three on **the** pandemic alert stage, with just a little bit of human-to-human transmission, but no human-to-human-to-human sustained transmission. **The** moment WHO says we ' ve moved to category four â this will not be like Katrina. **The** world as we know it will stop. There ' ll be no **airplanes** flying. Would you get in an **airplane** with 250 people you didn ' t know, coughing and sneezing, when you knew that some of them might carry a disease that could kill you, for which you had no antivirals or vaccine? I did a study of **the** top epidemiologists in **the** world in October. I asked them â these are all fluologists and specialists in influenza â and I asked them **the** questions you ' d like to ask them : What do you think **the** likelihood is that there ' ll be a pandemic? If it happens, how bad do you think it will be? Fifteen percent said they thought there ' d be a pandemic within three years. But much worse than that, 90 percent said they thought there ' d be a pandemic within your children or your grandchildren ' s lifetime. And they thought that if there was a pandemic, a billion people would get sick. As many as 165 million people would die. There would be a global recession and depression as our just-in-time inventory system and **the** tight rubber band of globalization broke, and **the** cost to our economy of one to three trillion dollars would be far worse for everyone than merely 100 million people dying, because so many more people would lose their job and their healthcare benefits, that **the** consequences are almost unthinkable. And it ' s getting worse, because travel is getting so much better. Let me show you a simulation of what a pandemic looks like. So we know what we ' re talking about. Let ' s assume, for example, that **the** first case **occurs** in South Asia. It initially goes quite slowly. You get two or three discrete locations. Then there ' ll be secondary outbreaks, and **the** disease will spread from country to country so fast that you won ' t know what hit you. Within three weeks it will be everywhere in **the** world. Now, if we had an”undo”button, and we could go back and isolate it and grab it when it first started â if we could find it early, and we had early detection and early response, and we could put each one of those viruses in jail â that ’ s **the** only way to deal with something like a pandemic. And let me show you why that is. We have a joke. This is an epidemic curve, and everyone in medicine, I think, ultimately gets to know what it is. But **the** joke is, an epidemiologist likes to arrive at an epidemic right here and ride to glory on **the** downhill curve. But you don ’ t get to do that usually. You usually arrive right about here. What we really want is to arrive right here, so we can stop **the** epidemic. But you can ’ t always do that. But there ’ s an organization that has been able to find a way to learn when **the** first cases **occur**, and that is called GPHIN ; it ’ s **the** Global Public Health Information Network. And that simulation that I showed you that you thought was bird flu â that was SARS. And SARS is **the** pandemic that did not **occur**. And it didn ’ t **occur** because GPHIN found **the** pandemic-to-be of SARS three months before WHO actually announced it, and because of that, we were able to stop **the** SARS pandemic. And I think we owe a great debt of gratitude to GPHIN and to Ron St. John, who I hope is in **the** audience some place â over there â who ’ s **the** founder of GPHIN. Hello, Ron! And TED has flown Ron here from Ottawa, where GPHIN is located, because not only did GPHIN find SARS early, but you may have seen last week that Iran announced that they had bird flu in Iran, but GPHIN found **the** bird flu in Iran not February 14 â but last September. We need an early-warning system to protect us against **the** things that are humanity ’ s worst nightmare. And so my TED wish is based on **the** common denominator of these experiences. Smallpox â early detection, early response. Blindness, polio â early detection, early response. Pandemic bird flu â early detection, early response. It is a litany. It is so obvious that our only way of dealing with these new diseases is to find them early and to kill them before they spread. So, my TED wish is for you to help build a global system â an early-warning system â to protect us against humanity ’ s worst nightmares. And what I thought I would call it is”Early Detection,”But it should really be called...”Total Early Detection.”[TED] What? What? But in all seriousness, because this idea is birthed in TED, I would like it to be a legacy of TED, and I ’ d like to call it **the**”International System for Total Early Disease Detection.”[INSTEDD] And INSTEDD then becomes our mantra. So instead of a hidden pandemic of bird flu, we find it and immediately contain it. Instead of a novel virus caused by bio-terror or bio-error, or shift or drift, we find it and we contain it. Instead of industrial accidents like **oil spills** or **the** catastrophe in Bhopal, we find them, and we respond to them. Instead of famine, hidden until it is too late, we **detect** it, and we respond. And instead of a system which is owned by a government, and hidden in **the** bowels of government, let ’ s build an early detection system that ’ s freely available to anyone in **the** world in their own language. Let ’ s make it transparent, non-governmental, not owned by any single country or company, housed in a neutral country, with redundant backup in a different time zone and a different **continent**. And let ’ s build it on GPHIN. Let ’ s start with GPHIN. Let ’ s increase **the** websites that they crawl from 20, 000 to 20 million. Let ’ s increase **the** languages they crawl from seven to 70, or more. Let ’ s build in outbound confirmation messages, using text messages or SMS or instant messaging to find out from people who are within 100 meters of **the** rumor that you hear, if it is, in fact, valid. And let ’ s add satellite confirmation. And we ’ ll add Gapminder ’ s amazing **graphics** to **the** front end. And we ’ ll grow it as a moral force in **the** world, finding out those terrible things before anybody else knows about them, and sending our response to them, so that next year, instead of us meeting here, lamenting how many terrible things there are in **the** world, we will have pulled together, used **the** unique skills and **the** magic of this community, and be proud that we have done everything we can to stop pandemics, other catastrophes, and change **the** world, beginning right now. Chris Anderson : An amazing presentation. First of all, just so everyone understands : you ’ re saying that by creating **web** crawlers, looking on **the** Internet for patterns, they can **detect** something suspicious before WHO, before anyone else can see it? Give an example of how that could possibly be true. Larry Brilliant : You ’ re not mad about **the** copyright violation? CA : No. I love it. LB : Well, as Ron St. John â I hope you ’ ll go and meet him in **the** dinner afterwards and talk to him. When he started GPHIN â In 1997, there was an outbreak of bird flu â H5N1. It was in Hong Kong. And a remarkable doctor in Hong Kong responded immediately, by slaughtering 1. 5 million chickens and birds, and they stopped that outbreak in its tracks. Immediate detection, immediate response. Then a number of years went by, and there were a lot of rumors about bird flu. Ron and his team in Ottawa began to crawl **the web** â only crawling 20, 000 different websites, mostly periodicals â and they read about and heard about a concern, of a lot of children who had high fever and symptoms of bird flu. They reported this to WHO. WHO took a little while taking action, because WHO will only receive a report from a government, because it ’ s **the** United Nations. But they were able to point to WHO and let them know that there was this surprising and unexplained cluster of illnesses that looked like bird flu. That turned out to be SARS. That ’ s how **the** world found out about SARS. And because of that, we were able to stop SARS. Now, what ’ s really important is that, before there was GPHIN, 100 percent of all **the** world ’ s reports of bad things â whether you ’ re talking about famine or you ’ re talking about bird flu or you ’ re talking about Ebola â 100 percent of all those reports came from nations. **The** moment these guys in Ottawa â on a budget of 800, 000 dollars a year â got cracking, 75 percent of all **the** reports in **the** world came from GPHIN, 25 percent of all **the** reports in **the** world came from all **the** other 180 nations. Now, here ’ s what ’ s really interesting : after they ’ d been working for a couple years, what do you think happened to those nations? They felt pretty stupid. So they started sending in their reports early. And now, their reporting percentage is down to 50 percent, because other nations have started to report. So, can you find diseases early by crawling **the web**? Of course you can. Can you find it even earlier than GPHIN does now? Of course you can. You saw that they found SARS using their Chinese **web** crawler a full six weeks before they found it using their English **web** crawler. Well, they ’ re only crawling in seven languages. These bad viruses really don ’ t have any intention of showing up first in English or Spanish or French. So yes, I want to take GPHIN, I want to build on it. I want to add all **the** languages of **the** world that we possibly can. I want to make this open to everybody, so that **the** health officer in Nairobi or in Patna, Bihar will have as much access to it as **the** folks in Ottawa or in CDC. And I want to make it part of our culture that there is a community of people who are watching out for **the** worst nightmares of humanity, and that it ’ s accessible to everyone.

Text Sample 12: POS Dim 1 – 2006 – Score 21.00 – 2006/t000160.txt

The future that we will create can be a future that we ’ ll be proud of. I think about this every day ; it ’ s quite literally my job. I ’ m co-founder and senior columnist at Worldchanging. com. Alex Steffen and I founded Worldchanging in late 2003, and since then we and our growing global team of contributors have documented **the** ever-expanding variety of solutions that are out there, right now and on **the** near horizon. In a little over two years, we ' ve written up about 4, 000 items — replicable models, technological tools, emerging ideas — all providing a path to a future that ' s more sustainable, more equitable and more desirable. Our emphasis on solutions is quite intentional. There are tons of places to go, online and off, if what you want to find is **the** latest bit of news about just how quickly our hell-bound handbasket is moving. We want to offer people an idea of what they can do about it. We focus primarily on **the planet** ' s environment, but we also address issues of global development, international conflict, responsible use of emerging technologies, even **the** rise of **the** so-called Second Superpower and much, much more. **The** scope of solutions that we discuss is actually pretty broad, but that reflects both **the** range of challenges that need to be met and **the** kinds of innovations that will allow us to do so. A quick sampling really can barely scratch **the surface**, but to give you a sense of what we cover : tools for rapid disaster relief, such as this inflatable concrete shelter ; innovative uses of bioscience, such as a flower that changes color in **the** presence of landmines ; ultra high-efficiency designs for homes and offices ; distributed power generation using solar power, wind power, ocean power, other clean energy sources ; ultra, ultra high-efficiency vehicles of **the** future ; ultra high- efficiency vehicles you can get right now ; and better urban design, so you don ' t need to drive as much in **the** first place ; bio-mimetic approaches to design that take advantage of **the** efficiencies of natural models in both vehicles and buildings ; distributed computing projects that will help us model **the** future of **the** climate. Also, a number of **the** topics that we ' ve been talking about this week at TED are things that we ' ve addressed in **the** past on Worldchanging : cradle-to-cradle design, MIT ' s Fab Labs, **the** consequences of extreme longevity, **the** One **Laptop** per Child project, even Gapminder. As a born-in-the-mid-1960s Gen X-er, hurtling all too quickly to my fortieth birthday, I ' m naturally inclined to pessimism. But working at Worldchanging has convinced me, much to my own surprise, that successful responses to **the** world ' s problems are nonetheless possible. Moreover, I ' ve come to realize that focusing only on negative outcomes can really blind you to **the** very possibility of success. As Norwegian social scientist Evelin Lindner has observed,"Pessimism is a luxury of good times... In difficult times, pessimism is a self- fulfilling, self-inflicted death sentence."**The** truth is, we can build a better world, and we can do so right now. We have **the** tools : we saw a hint of that a moment ago, and we ' re coming up with new ones all **the** time. We have **the** knowledge, and our understanding of **the planet** improves every day. Most importantly, we have **the** motive : we have a world that needs fixing, and nobody ' s going to do it for us. Many of **the** solutions that I and my colleagues seek out and write up every day have some important aspects in common : transparency, collaboration, a willingness to experiment, and an appreciation of science — or, more appropriately, science! **The** majority of models, tools and ideas on Worldchanging encompass combinations of these characteristics, so I want to give you a few concrete examples of how these principles combine in world- changing ways. We can see world-changing values in **the** emergence of tools to make **the** invisible visible — that is, to make apparent **the** conditions of **the** world around us that would otherwise be largely imperceptible. We know that people often change their behavior when they can see and understand **the** impact of their actions. As a small example, many of us have experienced **the** change in driving behavior that comes from having a real time display of mileage showing precisely how one ' s driving habits affect **the** vehicle ' s efficiency. **The** last few years have all seen **the** rise of innovations in how we measure and display aspects of **the** world that can be too big, or too intangible, or too slippery to grasp easily. Simple technologies, like wall-mounted **devices** that display how much power your household is using, and what kind of results you ' ll get if you turn off a few lights — these can actually have a direct positive impact on your energy footprint. Community tools, like text messaging, that can tell you when pollen counts are up or smog levels are rising or a natural disaster is unfolding, can give you **the** information you need to act in a timely fashion. Data-rich displays like maps of campaign contributions, or maps of **the disappearing** polar ice caps, allow us to better understand **the** context and **the** flow of processes that affect us all. We can see world-changing values in research projects that seek to meet **the** world ' s medical needs through open access to data and collaborative action. Now, some people emphasize **the** risks of knowledge-enabled dangers, but I ' m convinced that **the** benefits of knowledge-enabled solutions are far more important. For example, open-access journals, like **the** Public Library of Science, make cutting-edge scientific research free to all — everyone in **the** world. And actually, a growing number of science publishers are adopting this model. Last year, hundreds of volunteer **biology** and **chemistry** researchers around **the** world worked together to sequence **the genome** of **the** parasite responsible for some of **the** developing world ' s worst diseases : African sleeping sickness, leishmaniasis and Chagas disease. That **genome** data can now be found on open-access genetic data banks around **the** world, and it ' s an enormous boon to researchers trying to come up with treatments. But my favorite example has to be **the** global response to **the** SARS epidemic in 2003, 2004, which relied on worldwide access to **the** full gene sequence of **the** SARS virus. **The** U. S. National Research Council in its follow-up report on **the** outbreak specifically cited this open availability of **the** sequence as a key reason why **the** treatment for SARS could be developed so quickly. And we can see world-changing values in something as humble as a cell phone. I can probably count on my fingers **the** number of people in this room who do not use a mobile phone — and where is Aubrey, because I know he doesn ' t? For many of us, cell phones have really become almost an extension of ourselves, and we ' re really now beginning to see **the** social changes that mobile phones can bring about. You may already know some of **the** big-picture aspects : globally, more camera phones were sold last year than any other kind of camera, and a growing number of people live lives mediated through **the** lens, and over **the** network — and sometimes enter history books. In **the** developing world, mobile phones have become economic drivers. A study last year showed a direct correlation between **the** growth of mobile phone use and subsequent GDP increases across Africa. In Kenya, mobile phone minutes have actually become an alternative currency. **The** political aspects of mobile phones can ' t be ignored either, from text message swarms in Korea helping to bring down a government, to **the** Blairwatch Project in **the** UK, keeping tabs on politicians who try to avoid **the** press. And it ' s just going to get more wild. Pervasive, always-on networks, high quality sound and video, even **devices** made to be worn instead of carried in **the pocket**, will transform how we live on a scale that few really appreciate. It ' s no exaggeration to say that **the** mobile phone may be among **the** world ' s most important technologies. And in this rapidly **evolving** context, it ' s possible to imagine a world in which **the** mobile phone becomes something far more than a medium for social interaction. I ' ve long admired **the** Witness project, and Peter Gabriel told us more details about it on Wednesday, in his profoundly moving presentation. And I ' m just incredibly happy to see **the** news that Witness is going to be opening up a Web portal to enable users of digital cameras and camera phones to send in their recordings over **the** Internet, rather than just hand-carrying **the** videotape. Not only does this add a new and potentially safer avenue for documenting abuses, it opens up **the** program to **the** growing global digital generation. Now, imagine a similar model for networking environmentalists. Imagine a Web portal collecting recordings and evidence of what ' s happening to **the planet** : putting news and data at **the** fingertips of people of all kinds, from activists and researchers to businesspeople and political figures. It would highlight **the** changes that are underway, but would more importantly give voice to **the** people who are willing to work to see a new world, a better world, come about. It would give everyday citizens a chance to play a role in **the** protection of **the planet**. It would be, in essence, an"Earth Witness"project. Now, just to be clear, in this talk I ' m using **the** name"Earth Witness"as part of **the** scenario, simply as a shorthand, for what this imaginary project could aspire to, not to piggyback on **the** wonderful work of **the** Witness organization. It could just as easily be called,"Environmental Transparency Project,""Smart Mobs for Natural Security"— but Earth Witness is a lot easier to say. Now, many of **the** people who participate in Earth Witness would focus on ecological problems, human-caused or otherwise, especially environmental crimes and significant sources of **greenhouse** gases and **emissions**. That ' s understandable and important. We need better documentation of what ' s happening to **the planet** if we ' re ever going to have a chance of repairing **the** damage. But **the** Earth Witness project wouldn ' t need to be limited to problems. In **the** best Worldchanging tradition, it might also serve as a showcase for good ideas, successful projects and efforts to make a difference that deserve much more visibility. Earth Witness would show us two worlds : **the** world we ' re leaving behind, and **the** world we ' re building for generations to come. And what makes this scenario particularly appealing to me is we could do it today. **The** key components are already widely available. Camera phones, of course, would be fundamental to **the** project. And for a lot of us, they ' re as close as we have yet to always-on, widely available information tools. We may not remember to bring our digital cameras with us wherever we go, but very few of us forget our phones. You could even imagine a version of this scenario in which people actually build their own phones. Over **the** course of last year, **open-source** hardware hackers have come up with multiple models for usable, Linux-based mobile phones, and **the** Earth Phone could spin off from this kind of project. At **the** other end of **the** network, there ' d be a server for people to send photos and messages to, accessible over **the** Web, combining a photo-sharing service, social networking platforms and a collaborative filtering system. Now, you Web 2. 0 folks in **the** audience know what I ' m talking about, but for those of you for whom that last sentence was in a crazy moon language, I mean simply this : **the** online part of **the** Earth Witness project would be created by **the** users, working together and working openly. That ' s enough right there to start to build a compelling chronicle of what ' s now happening to our **planet**, but we could do more. An Earth Witness site could also serve as a **collection** spot for all sorts of data about conditions around **the planet** picked up by environmental sensors that attach to your cell phone. Now, you don ' t see these **devices** as add-ons for phones yet, but students and engineers around **the** world have attached atmospheric sensors to bicycles and handheld **computers** and cheap robots and **the** backs of pigeons — that being a project that ' s actually underway right now at U. C. Irvine, using bird-mounted sensors as a way of measuring smog-forming pollution. It ' s hardly a stretch to imagine putting **the** same thing on a phone carried by a person. Now, **the** idea of connecting a sensor to your phone is not new : phone-makers around **the** world offer phones that sniff for bad breath, or tell you to worry about too much sun exposure. Swedish firm Uppsala Biomedical, more seriously, makes a mobile phone add-on that can process **blood** tests in **the** field, uploading **the** data, displaying **the** results. Even **the** Lawrence Livermore National Labs have gotten into **the** act, designing a prototype phone that has radiation sensors to find dirty bombs. Now, there ’ s an enormous variety of tiny, inexpensive sensors on **the** market, and you can easily imagine someone putting together a phone that could measure temperature, CO2 or methane levels, **the** presence of some biotoxins — potentially, in a few years, maybe even H5N1 avian flu virus. You could see that some kind of system like this would actually be a really good fit with Larry Brilliant ’ s InSTEDD project. Now, all of this data could be **tagged** with geographic information and mashed up with online maps for easy viewing and analysis. And that ’ s worth noting in particular. **The** impact of open-access online maps over **the** last year or two has been simply phenomenal. Developers around **the** world have come up with an amazing variety of ways to layer useful data on top of **the** maps, from bus routes and crime statistics to **the** global progress of avian flu. Earth Witness would take this further, linking what you see with what thousands or millions of other people see around **the** world. It ’ s kind of exciting to think about what might be accomplished if something like this ever existed. We ’ d have a far better — far better knowledge of what ’ s happening on our **planet** environmentally than could be gathered with satellites and a handful of government sensor nets alone. It would be a collaborative, bottom-up approach to environmental awareness and protection, able to respond to emerging concerns in a smart mobs kind of way — and if you need greater sensor density, just have more people show up. And most important, you can ’ t ignore how important mobile phones are to global youth. This is a system that could put **the** next generation at **the** front lines of gathering environmental data. And as we work to figure out ways to mitigate **the** worst effects of climate disruption, every little bit of information matters. A system like Earth Witness would be a tool for all of us to participate in **the** improvement of our knowledge and, ultimately, **the** improvement of **the planet** itself. Now, as I suggested at **the** outset, there are thousands upon thousands of good ideas out there, so why have I spent **the** bulk of my time telling you about something that doesn ’ t exist? Because this is what tomorrow could look like : bottom-up, technology-enabled global collaboration to handle **the** biggest crisis our civilization has ever faced. We can save **the planet**, but we can ’ t do it alone — we need each other. Nobody ’ s going to fix **the** world for us, but working together, making use of technological innovations and human communities alike, we might just be able to fix it ourselves. We have at our fingertips a cornucopia of compelling models, powerful tools, and innovative ideas that can make a meaningful difference in our **planet** ’ s future. We don ’ t need to wait for a magic **bullet** to save us all ; we already have an arsenal of solutions just waiting to be used. There ’ s a staggering **array** of wonders out there, across diverse disciplines, all telling us **the** same thing : success can be ours if we ’ re willing to try. And as we say at Worldchanging, another world isn ’ t just possible ; another world is here. We just need to open our eyes. Thank you very much.

Text Sample 13: POS Dim 1 – 2006 – Score 20.00 – 2006/t000417.txt

I just want to say, over **the** last few years I ’ ve been â had **the** opportunity to do this closing conference. And I ’ ve had some incredible warm-up acts. About eight years ago, Billy Graham opened for me. And I thought that there was â I thought that there was absolutely no way in hell to top that. But I just wanted to say â and I mean this without irony â I think I can speak for everybody in **the** audience when I say that I wish to God that you were **the** President of **the** United States. OK, this is **the** title of my talk today. I just want to give you a quick overview. First of all, please remember I ’ m completely politically correct, and I mean everything with great affection. If any of you have sensitive stomachs or are feeling queasy, now is **the** time to check your Blackberry. Just to review, this is my TEDTalk. We ’ re going to do some jokes, some gags, some little skits â and then we ’ re going to talk about **the** L1 point. So, one of **the** questions I ask myself is, was this **the** most distressing TED ever? Let ’ s try and **sum** things up, shall we? Images of limb regeneration and faces filled with smallpox : 21 percent of **the** conference. Mentions of polar bears drowning : four percent. Images of **the** earth being **wiped** out by flood or bird flu : 64 percent. And David Pogue singing show tunes. Because this is **the** most distressing TED ever, I ’ ve been working with Neil Gershenfeld on next year ’ s TED Bag. And if **the** â if **the** conference is anywhere near this distressing, then we ’ re going to have a scream bag next year. It ’ s going to be a cradle-to-cradle scream bag, of course. So you ’ re going to be able to go like this. Bring it over here and open it up. Aaaah! Meanwhile, back at TED University, this wonderful woman is teaching you how to chop Sun Chips. So Robert Wright â I don ’ t know, I felt like if there was anyone that Helen needed to give antidepressants to, it might have been him. I want to deliberately interfere with his dopamine levels. He was talking about morality. Economy class morality is, we want to bomb you back to **the** Stone Age. Business class morality is, don ’ t bomb Japan â they built my car. And first-class morality is, don ’ t bomb Mexico â they clean my house. Yes, it is politically incorrect. All right, now I want to do a little bit of a thing for you... All right, now these are **the** wha â I ’ d say, [mumble, mumble â mumble, mumble, mumble, mumble, mumble] â ahh! So I wanted to show you guys â I wanted to talk about a revolutionary new **computer** interface that lets you work with images just as easily as you â as a completely natural user interface. And you can â you can use really natural hand gestures to like, go like this. Now we had a Harvard professor here â she was from Harvard, I just wanted to mention and â and she was actually a professor from Harvard. And she was talking about seven-dimensional, inverted **universes**. With, you know, of course, there ’ s **the** gravity brain. There ’ s **the** weak brain. And then there ’ s my weak brain, which is too â too â absolutely too weak to understand what **the** fuck she was talking about. Now â one of **the** things that is very important to me is to try and figure out what on Earth am I here for. And that ’ s why I went out and I picked up a best-selling business book. You know, it basically uses as its central premise Greek mythology. And it ’ s by a guy named Pastor Rick Warren, and it ’ s called”**The** Porpoise Driven Life.”And Rick is as a pagan god, which I thought was kind of appropriate, in a certain way. And now we ’ re going to have kind of a little more visualization about Rick Warren. OK. All right. Now, red is Rick Warren, and green is Daniel Dennett, OK? **The** scales here are religiosity from zero percent, or atheist, to 100 percent, Bible literally true. And then this is books sold â **the** logarithmic scale. 30, 000, 300, 000, three million, 30 million, 300 million. OK, now they ’ re duking it out. Now they ’ re duking it out. And Rick Warren ’ s kind of pulling ahead, kind of pulling ahead. Yup, and his installed base is getting a little bigger. But Darwin ’ s dangerous idea is coming back. It ’ s coming back. Let me turn **the** trails on, so you can see that a little bit better. Now, one of **the** things that ’ s very important is, Nicholas Negroponte talked to us about one lap dance per â I ’ m sorry â about One **Laptop** Per Child. Now let ’ s talk about some of **the** characteristics that are important for this revolutionary **device**. I ’ ll tell you a little bit about **the** design parameters, and then I ’ ll show it to you in person. First of all, it needs to be small. It needs to be flat, so it ’ s transportable. Lightweight. Portable. Uses very, very little power. Very, very high resolution. Has to be visible in bright daylight. Will work anywhere. And broadly applicable across many platforms. Now, we ’ ve actually done some research â□□ Neil Gershenfeld and **the** Fab Labs went out into **the** market. They did some research ; we came back ; and we think we have **the** perfect prototype of what **the** students in **the** field are actually asking for. And here it is, **the** $100 **computer**. OK, OK, OK, OK â□□ excellent, excellent. Now, I bought this **device** from Clifford Stoll for about 900 **bucks**. And he and his team of junior-high school students were doing real science. So we ’ re trying to check and trying to douse here, and see who uses marijuana. See who uses marijuana. Are we going to be able to find any marijuana, Jim Young? Only if we open enough locker doors. OK, now smallpox is an extremely distressing illness. We had Dr. Larry Brilliant talking about how we eradicated smallpox. I wanted to show you **the** stages of smallpox. We start. This is day one. Day two. Day three, she gets a massively big pox on her shoulder. Day three. Day four. Day five and day six. Now **the** good news is, because I ’ m a trained medical professional, I know that even though she ’ ll be scarred for life, she ’ s going to make a full recovery. Now **the** good news about Architects for Humanity is they ’ re really kind of **the** most amazing group. They ’ ve been sponsoring a design competition to come up with innovative medical housing solutions, clinic solutions, in Africa, and they ’ ve had a design competition. Now **the** wonderful thing is, Larry Brilliant was just appointed **the** head of **the** Google Foundation, and so he decided that he would support â□□ he would support Cameron ’ s work. And **the** way he decided to support that work was by shipping over 50, 000 **shipping** containers of Google snacks. So I want to show you some prototypes. **The** U. N. â□□ you know, they took 20 years just to add a flap to a tent, but I think we have some more exciting things. This is a home made entirely out of Fruit Roll-Ups. And those roll-up cookies coated with white chocolate. And **the** really wonderful thing about this is, when you ’ re done, well â□□ you can eat it. But **the** thing that I ’ m really, really excited about is this incredible granola house. And **the** granola house has a special Sun Chip roof to collect water and recycle it. And it ’ s â□□ well, on this side it has regular Sour Patch Kids and Gummy Bears to let in **the** light. But on this side, it has sugary Gummy Bears, to diffuse **the** light more slightly. And we â□□ we wanted just to show you what this might look like in situ. So, Einstein â□□ Einstein, tell me â□□ what ’ s your favorite song? No, I said what ’ s your favorite song? No, I said what ’ s your favorite song?”Free Bird.”OK, so, Einstein, what ’ s your favorite singing group? Could you say that again? What ’ s your favorite singing group? OK, one more time â□□ I ’ m just going to give you a little help. Your favorite singing group â□□ it ’ s Diana Ross and **the** â□□ Audience : Supremes! Tom Reilly : Exactly. Could we have **the** sound up on **the laptop**, please?”Free Bird”kind of reminds me that if you â□□ if you listen to”Free Bird”**backwards**, this is what you might hear. **Computer** : Satan. Satan. Satan. Satan. Satan. Satan. Satan. TR : Now it ’ s a little hard to hear **the** whole message, so I wanted to â□□ so I wanted to help you a little bit. **Computer** : My sweet Satan. Dan Dennett worships Satan. Buy”**The** Purpose-Driven Life,”or Satan will take your soul. TR : So, we ’ ve talked a lot about global warming, but, you know, as Jill said, it sounds kind of nice â□□ good weather in **the** wintertime, and New York City. And as Jay Walker pointed out, that is just not scary enough. So Al, I actually think I ’ m rather good at branding. So I ’ ve tried to figure out a good design process to come up with a new term to replace”global warming.”So we started with Babel Fish. We put in global warming. And then we decided that we ’ d change it from English to Dutch â□□ into”Het globale Verwarmen.”From Dutch to [Korean], into”Hordahordaneecheewa.”[Korean] to Portuguese : Aquecer-se Global. Then Portuguese to Pig Latin. Aquecer-se ucked-fay. And then finally back into **the** English, which is, we ’ re totally fucked. Now I don ’ t know about you, but Michael Shermer talked about **the** willingness for human beings â evolutionarily, they ’ re designed to see patterns in things. For example, in cheese sandwiches. Now can you look at that carefully and see if you see **the** Virgin Mary? I tried to make it a little bit clearer. Is it **the** Virgin Mary? Or is it Mena Trott? So, I talked to Josh Prince-Ramus about **the** convention center and **the** conferences. It ’ s getting awfully big. It ’ s getting just a little bit too big. It ’ s bursting at **the** seams here a little bit. So we tried to come up with a program â how we could remake this structure to better accommodate TED. So first of all we decided â that we needed about **one-third** bookstore, **one-third** Google cafe, about 20 percent registration, 80 percent luxury hotel, about five percent for restrooms. And then of course, we wanted to have **the** simulcast lounge, **the** lobby and **the** Steinbeck forum. Now let me show you how that literally translated into **the** design program. So first, one of **the** problems with Monterey is that if there is global warming and Greenland melts as you say, **the** ocean level is going to rise 20 feet and flood **the** hell out of **the** convention center. So we ’ re going to build this new building on stilts. So we build this building on stilts, then up here â is where we ’ re going to put **the** new Steinbeck auditorium. And **the** wonderful thing about **the** new bookstore is, it ’ s going to be shaped in a spiral that ’ s organized by **the** Dewey Decimal System. Then we ’ re going to make an escalator that helps you get up there. And finally, we ’ re going to put **the** Marriott Hotel and **the** Portola Plaza on **the** top. Now I don ’ t know about you, but sometimes I have these images in my head of separated at birth. I don ’ t know about you, but when I see Aubrey de Grey, I immediately go to Gandalf **the** Grey. OK. Now, we ’ ve heard, of course, that we ’ re all soldiers here. So what I ’ d really, really like you to do now is, pick up your white piece of paper. Does everybody have their white piece of paper? And I want you to get out a pen, and I want you to write a **terrorist** note. If we put up **the** ELMO for a moment â if we put up **the** ELMO, then we ’ ll get, you know, I ’ ll give you a model that you can work from, OK? And then I want you to fold that note into a paper **airplane**. And once you ’ ve folded it into a paper **airplane**, I want you to take some anthrax â and I want you to put that in **the** paper **airplane**. And then I want you to throw it on Jim Young. Luckily, I was **the** recipient of **the** TED Prize this year. And I wanted to see â I want to dedicate this film to my father, Homer. OK. Now this film isn ’ t really hard enough, so I wanted to make it a little bit harder. So I ’ m going to try and do this while reciting pi. 3. 1415, 2657, 753, 8567, 24972 â â 85871, 25871, 3928, 5657, 2592, 5624. Can we cue **the** music please? Now I wanted to use this talk to talk about global warming a little bit. Back in 1968, you can see that **the** mountain range of Brokeback Mountain was covered in 151 **inches** of snow pack. Parenthetically, over there on **the** **slopes**, I did want to show you that black men ski. But over **the** years, 10 years later, **the** snow packs eroded, and, if you notice, **the** trees have started turning **yellow**. **The** water level of **the** lake has started drying up. A few years later, there ’ s no snow left at all. And all **the** trees have turned brown. This year, unfortunately **the** lakebed ’ s turned into an absolute cracked dry bed. And I fear, if we do nothing for our **planet**, in 20 years, it ’ s going to look like this. Mr. Vice President, I wish I knew how to quit you. Thank you very much.

Text Sample 14: POS Dim 1 – 1998 – Score 21.00 – 1998/t000216.txt

David Gallo : This is Bill Lange. I ’ m Dave Gallo. And we ’ re going to tell you some stories from **the sea** here in video. We ’ ve got some of **the** most incredible video of Titanic that ’ s ever been seen, and we ’ re not going to show you any of it. **The** truth of **the** matter is that **the** Titanic — even though it ’ s breaking all sorts of box office records — it ’ s not **the** most exciting story from **the sea**. And **the** problem, I think, is that we take **the** ocean for granted. When you think about it, **the** oceans are 75 percent of **the planet**. Most of **the planet** is ocean water. **The** average depth is about two miles. Part of **the** problem, I think, is we stand at **the** beach, or we see images like this of **the** ocean, and you look out at this great big **blue** expanse, and it ’ s shimmering and it ’ s moving and there ’ s waves and there ’ s **surf** and there ’ s tides, but you have no idea for what lies in there. And in **the** oceans, there are **the** longest mountain ranges on **the planet**. Most of **the** animals are in **the** oceans. Most of **the earthquakes** and volcanoes are in **the sea**, at **the** bottom of **the sea**. **The** biodiversity and **the** biodensity in **the** ocean is higher, in places, than it is in **the** rainforests. It ’ s mostly unexplored, and yet there are beautiful sights like this that captivate us and make us become familiar with it. But when you ’ re standing at **the** beach, I want you to think that you ’ re standing at **the** edge of a very unfamiliar world. We have to have a very special technology to get into that unfamiliar world. We use **the** submarine Alvin and we use cameras, and **the** cameras are something that Bill Lange has developed with **the** help of Sony. Marcel Proust said,”**The** true voyage of discovery is not so much in seeking new landscapes as in having new eyes.”People that have partnered with us have given us new eyes, not only on what exists — **the** new landscapes at **the** bottom of **the sea** — but also how we think about life on **the planet** itself. Here ’ s a jelly. It ’ s one of my favorites, because it ’ s got all sorts of working parts. This turns out to be **the** longest creature in **the** oceans. It gets up to about 150 feet long. But see all those different working things? I love that kind of stuff. It ’ s got these **fishing** lures on **the** bottom. They ’ re going up and down. It ’ s got tentacles dangling, swirling around like that. It ’ s a colonial animal. These are all individual animals banding together to make this one creature. And it ’ s got these jet thrusters up in front that it ’ ll use in a moment, and a little light. If you take all **the** big **fish** and schooling **fish** and all that, put them on one side of **the** scale, put all **the** jelly-type of animals on **the** other side, those guys win hands down. Most of **the** biomass in **the** ocean is made out of creatures like this. Here ’ s **the** X-wing death jelly. **The** bioluminescence — they use **the** lights for attracting **mates** and attracting prey and communicating. We couldn ’ t begin to show you our archival stuff from **the** jellies. They come in all different sizes and shapes. Bill Lange : We tend to forget about **the** fact that **the** ocean is miles deep on average, and that we ’ re real familiar with **the** animals that are in **the** first 200 or 300 feet, but we ’ re not familiar with what exists from there all **the** way down to **the** bottom. And these are **the** types of animals that live in that three- dimensional space, that micro-gravity environment that we really haven ’ t explored. You hear about giant squid and things like that, but some of these animals get up to be approximately 140, 160 feet long. They ’ re very little understood. DG : This is one of them, another one of our favorites, because it ’ s a little octopod. You can actually see through his head. And here he is, flapping with his ears and very gracefully going up. We see those at all depths and even at **the** greatest depths. They go from a couple of **inches** to a couple of feet. They come right up to **the** submarine — they ’ ll put their eyes right up to **the** window and peek inside **the** sub. This is really a world within a world, and we ’ re going to show you two. In this case, we ’ re passing down through **the** mid-ocean and we see creatures like this. This is kind of like an undersea rooster. This guy, that looks incredibly formal, in a way. And then one of my favorites. What a face! This is basically scientific data that you ’ re looking at. It ’ s footage that we ’ ve collected for scientific purposes. And that ’ s one of **the** things that Bill ’ s been doing, is providing scientists with this first view of animals like this, in **the** world where they belong. They don ’ t catch them in a net. They ’ re actually looking at them down in that world. We ’ re going to take a joystick, sit in front of our **computer**, on **the** Earth, and press **the** joystick forward, and fly around **the planet**. We ’ re going to look at **the** mid-ocean ridge, a 40, 000-mile long mountain range. **The** average depth at **the** top of it is about a mile and a half. And we ’ re over **the** Atlantic — that ’ s **the** ridge right there — but we ’ re going to go across **the** Caribbean, Central America, and end up against **the** Pacific, nine degrees north. We make maps of these mountain ranges with sound, with sonar, and this is one of those mountain ranges. We ’ re coming around a cliff here on **the** right. **The** height of these mountains on either side of this valley is greater than **the** Alps in most cases. And there ’ s tens of thousands of those mountains out there that haven ’ t been mapped yet. This is a volcanic ridge. We ’ re getting down further and further in scale. And eventually, we can come up with something like this. This is an icon of our robot, Jason, it ’ s called. And you can sit in a room like this, with a joystick and a headset, and drive a robot like that around **the** bottom of **the** ocean in real time. One of **the** things we ’ re trying to do at Woods Hole with our partners is to bring this virtual world — this world, this unexplored region — back to **the** laboratory. Because we see it in bits and pieces right now. We see it either as sound, or we see it as video, or we see it as photographs, or we see it as chemical sensors, but we never have yet put it all together into one interesting picture. Here ’ s where Bill ’ s cameras really do shine. This is what ’ s called a hydrothermal vent. And what you ’ re seeing here is a cloud of densely packed, hydrogen-sulfide-rich water coming out of a volcanic axis on **the sea** floor. Gets up to 600, 700 degrees F, somewhere in that range. So that ’ s all water under **the sea** — a mile and a half, two miles, three miles down. And we knew it was volcanic back in **the** ’ 60s, ’ 70s. And then we had some hint that these things existed all along **the** axis of it, because if you ’ ve got volcanism, water ’ s going to get down from **the sea** into cracks in **the sea** floor, come in contact with magma, and come shooting out hot. We weren ’ t really aware that it would be so rich with sulfides, hydrogen sulfides. We didn ’ t have any idea about these things, which we call chimneys. This is one of these hydrothermal vents. Six hundred degree F water coming out of **the** Earth. On either side of us are mountain ranges that are higher than **the** Alps, so **the** setting here is very dramatic. BL : **The** white material is a type of bacteria that thrives at 180 degrees C. DG : I think that ’ s one of **the** greatest stories right now that we ’ re seeing from **the** bottom of **the sea**, is that **the** first thing we see coming out of **the sea** floor after a volcanic eruption is bacteria. And we started to wonder for a long time, how did it all get down there? What we find out now is that it ’ s probably coming from inside **the** Earth. Not only is it coming out of **the** Earth — so, biogenesis made from volcanic activity — but that bacteria supports these colonies of life. **The** pressure here is 4, 000 **pounds** per square **inch**. A mile and a half from **the surface** to two miles to three miles — no sun has ever gotten down here. All **the** energy to support these life forms is coming from inside **the** Earth — so, chemosynthesis. And you can see how dense **the** population is. These are called tube worms. BL : These worms have no digestive system. They have no mouth. But they have two types of gill structures. One for extracting oxygen out of **the** deep-sea water, another one which houses this chemosynthetic bacteria, which takes **the** hydrothermal fluid — that hot water that you saw coming out of **the** bottom — and converts that into simple sugars that **the** tube worm can digest. DG : You can see, here ’ s a **crab** that lives down there. He ’ s managed to grab a tip of these worms. Now, they normally retract as soon as a **crab** touches them. Oh! Good going. So, as soon as a **crab** touches them, they retract down into their shells, just like your fingernails. There ' s a whole story being played out here that we ' re just now beginning to have some idea of because of this new camera technology. BL : These worms live in a real temperature extreme. Their foot is at about 200 degrees C and their head is out at three degrees C, so it ' s like having your hand in boiling water and your foot in freezing water. That ' s how they like to live. DG : This is a female of this kind of worm. And here ' s a male. You watch. It doesn ' t take long before two guys here — this one and one that will show up over here — start to fight. Everything you see is played out in **the** pitch black of **the** deep **sea**. There are never any lights there, except **the** lights that we bring. Here they go. On one of **the** last dive series, we counted 200 species in these areas — 198 were new, new species. BL : One of **the** big problems is that for **the** biologists working at these sites, it ' s rather difficult to collect these animals. And they disintegrate on **the** way up, so **the** imagery is critical for **the** science. DG : Two octopods at about two miles depth. This pressure thing really amazes me — that these animals can exist there at a depth with pressure enough to crush **the** Titanic like an empty Pepsi can. What we saw up till now was from **the** Pacific. This is from **the** Atlantic. Even greater depth. You can see this **shrimp** is harassing this poor little guy here, and he ' ll bat it away with his claw. Whack! And **the** same thing ' s going on over here. What they ' re getting at is that — on **the** back of this **crab** — **the** foodstuff here is this very strange bacteria that lives on **the** backs of all these animals. And what these **shrimp** are trying to do is actually harvest **the** bacteria from **the** backs of these animals. And **the crabs** don ' t like it at all. These long filaments that you see on **the** back of **the crab** are actually created by **the** product of that bacteria. So, **the** bacteria grows hair on **the crab**. On **the** back, you see this again. **The** red dot is **the** laser light of **the** submarine Alvin to give us an idea about how far away we are from **the** vents. Those are all **shrimp**. You see **the** hot water over here, here and here, coming out. They ' re clinging to a **rock** face and actually scraping bacteria off that **rock** face. Here ' s a tiny, little vent that ' s come out of **the** side of that pillar. Those pillars get up to several stories. So here, you ' ve got this valley with this incredible alien landscape of pillars and hot **springs** and volcanic eruptions and **earthquakes**, inhabited by these very strange animals that live only on chemical energy coming out of **the** ground. They don ' t need **the** sun at all. BL : You see this white V-shaped mark on **the** back of **the shrimp**? It ' s actually a light-sensing organ. It ' s how they find **the** hydrothermal vents. **The** vents are emitting a black body radiation — an IR signature — and so they ' re able to find these vents at considerable distances. DG : All this stuff is happening along that 40, 000-mile long mountain range that we ' re calling **the** ribbon of life, because just even today, as we speak, there ' s life being generated there from volcanic activity. This is **the** first time we ' ve ever tried this any place. We ' re going to try to show you high definition from **the** Pacific. We ' re moving up one of these pillars. This one ' s several stories tall. In it, you ' ll see that it ' s a **habitat** for a lot of different animals. There ' s a funny kind of hot **plate** here, with vent water coming out of it. So all of these are individual homes for worms. Now here ' s a closer view of that community. Here ' s **crabs** here, worms here. There are smaller animals crawling around. Here ' s pagoda structures. I think this is **the** neatest-looking thing. I just can ' t get over this — that you ' ve got these little chimneys sitting here smoking away. This stuff is toxic as hell, by **the** way. You could never get a permit to dump this in **the** ocean, and it ' s coming out all from it. It ' s unbelievable. It ' s basically sulfuric acid, and it ' s being just dumped out, at incredible rates. And animals are thriving — and we probably came from here. That ' s probably where we **evolved** from. BL : This bacteria that we ' ve been talking about turns out to be **the** most simplest form of life found. There are a number of groups that are proposing that life **evolved** at these vent sites. Although **the** vent sites are short-lived — an individual site may last only 10 years or so — as an ecosystem they ’ ve been stable for millions — well, billions — of years. DG : It works too well. You see there ’ re some **fish** inside here as well. There ’ s a **fish** sitting here. Here ’ s a **crab** with his claw right at **the** end of that tube worm, waiting for that worm to stick his head out. BL : **The** biologists right now cannot explain why these animals are so active. **The** worms are growing **inches** per week! DG : I already said that this site, from a human perspective, is toxic as hell. Not only that, but on top — **the** lifeblood — that plumbing system turns off every year or so. Their plumbing system turns off, so **the** sites have to move. And then there ’ s **earthquakes**, and then volcanic eruptions, on **the** order of one every five years, that completely **wipes the** area out. Despite that, these animals grow back in about a year ’ s time. You ’ re talking about biodensities and biodiversity, again, higher than **the** rainforest that just **springs** back to life. Is it sensitive? Yes. Is it fragile? No, it ’ s not really very fragile. I ’ ll end up with saying one thing. There ’ s a story in **the** **sea**, in **the** waters of **the sea**, in **the** sediments and **the rocks** of **the sea** floor. It ’ s an incredible story. What we see when we look back in time, in those sediments and **rocks**, is a record of Earth history. Everything on this **planet** — everything — works by cycles and rhythms. **The continents** move apart. They come back together. Oceans come and go. Mountains come and go. Glaciers come and go. El Nino comes and goes. It ’ s not a disaster, it ’ s rhythmic. What we ’ re learning now, it ’ s almost like a symphony. It ’ s just like music — it really is just like music. And what we ’ re **learning** now is that you can ’ t listen to a five-billion-year long symphony, get to today and say,”Stop! We want tomorrow ’ s note to be **the** same as it was today.”It ’ s absurd. It ’ s just absurd. So, what we ’ ve got to learn now is to find out where this **planet** ’ s going at all these different scales and work with it. Learn to manage it. **The** concept of preservation is futile. Conservation ’ s tougher, but we can probably get there. Thank you very much. Thank you.

Text Sample 15: POS Dim 1 – 1998 – Score 17.00 – 1998/t000200.txt

You hear that this is **the** era of environment â or **biology**, or information technology... Well, it ’ s **the** era of a lot of different things that we ’ re in right now. But one thing for sure : it ’ s **the** era of change. There ’ s more change going on than ever has **occurred** in **the** history of human life on earth. And you all sort of know it, but it ’ s hard to get it so that you really understand it. And I ’ ve tried to put together something that ’ s a good start for this. I ’ ve tried to show in this â though **the** color doesn ’ t come out â that what I ’ m concerned with is **the** little 50-year time bubble that you are in. You tend to be interested in a generation past, a generation future â your parents, your kids, things you can change over **the** next few decades â and this 50-year time bubble you kind of move along in. And in that 50 years, if you look at **the** population curve, you find **the** population of humans on **the** earth more than doubles and we ’ re up three-and-a-half times since I was born. When you have a new baby, by **the** time that kid gets out of high school more people will be added than existed on earth when I was born. This is unprecedented, and it ’ s big. Where it goes in **the** future is questioned. So that ’ s **the** human part. Now, **the** human part related to animals : look at **the** left side of that. What I call **the** human portion â humans and their livestock and pets â versus **the** natural portion â all **the** other wild animals and just â these are vertebrates and all **the** birds, etc., in **the** land and air, not in **the** water. How does it balance? Certainly, 10, 000 years ago, **the** civilization ’ s beginning, **the** human portion was less than one tenth of one percent. Let ’ s look at it now. You follow this curve and you see **the** whiter spot in **the** middle â that ’ s your 50- year time bubble. Humans, livestock and pets are now 97 percent of that integrated total mass on earth and all wild nature is three percent. We have won. **The** next generation doesn ’ t even have to worry about this game â it is over. And **the** biggest problem came in **the** last 25 years : it went from 25 percent up to that 97 percent. And this really is a sobering picture upon realizing that we, humans, are in charge of life on earth ; we ’ re like **the** capricious Gods of old Greek myths, kind of playing with life â and not a great deal of wisdom injected into it. Now, **the** third curve is information technology. This is Moore ’ s Law plotted here, which relates to density of information, but it has been pretty good for showing a lot of other things about information technology â **computers**, their use, Internet, etc. And what ’ s important is it just goes straight up through **the** top of **the** curve, and has no real limits to it. Now try and contrast these. This is **the** size of **the** earth going through that same â â frame. And to make it really clear, I ’ ve put all four on one graph. There ’ s no need to see **the** little detailed words on it. That first one is humans-versus-nature ; we ’ ve won, there ’ s no more gain. Human population. And so if you ’ re looking for growth industries to get into, that ’ s not a good one â protecting natural creatures. Human population is going up ; it ’ s going to continue for quite a while. Good business in obstetricians, morticians, and farming, housing, etc. â they all deal with human bodies, which require being fed, transported, housed and so on. And **the** information technology, which connects to our brains, has no limit â now, that is a wonderful field to be in. You ’ re looking for growth opportunity? It ’ s just going up through **the** roof. And then, **the** size of **the** Earth. Somehow making these all compatible with **the** Earth looks like a pretty bad industry to be involved with. So, that ’ s **the** stage out of all this. I find, for reasons I don ’ t understand, I really do have a goal. And **the** goal is that **the** world be desirable and sustainable when my kids reach my age â and I think that ’ s â in other words, **the** next generation. I think that ’ s a goal that we probably all share. I think it ’ s a hopeless goal. Technologically, it ’ s achievable ; economically, it ’ s achievable ; politically, it means sort of **the** habits, institutions of people â it ’ s impossible. **The** institutions of **the** past with all their inertia are just irrelevant for **the** future, except they ’ re there and we have to deal with them. I spend about 15 percent of my time trying to save **the** world, **the** other 85 percent, **the** usual â and whatever else we devote ourselves to. And in that 15 percent, **the** main focus is on human mind, thinking skills, somehow trying to unleash kids from **the** straightjacket of school, which is putting information and dogma into them, get them so they really think, ask tough questions, argue about serious subjects, don ’ t believe everything that ’ s in **the** book, think broadly or creative. They can be. Our school systems are very flawed and do not reward you for **the** things that are important in life or for **the** survival of civilization ; they reward you for a lot of learning and sopping up stuff. We can ’ t go into that today because there isn ’ t time â it ’ s a broad subject. One thing for sure, in **the** future there is an essential feature â necessary, but not sufficient â which is doing more with less. We ’ ve got to be doing things with more efficiency using less energy, less material. Your great-great grandparents got by on muscle power, and yet we all think there ’ s this huge power that ’ s essential for our lifestyle. And with all **the** wonderful technology we have we can do things that are much more efficient : conserve, recycle, etc. Let me just rush very quickly through things that we ’ ve done. Human-powered **airplane** â Gossamer Condor sort of started me in this direction in 1976 and 77, winning **the** Kremer prize in aviation history, followed by **the** Albatross. And we began making various odd planes and creatures. Here ’ s a giant flying replica of a pterosaur that has no tail. Trying to have it fly straight is like trying to shoot an arrow with **the** feathered end forward. It was a tough job, and boy it made me have a lot of respect for nature. This was **the** full size of **the** original creature. We did things on land, in **the** air, on water â vehicles of all different kinds, usually with some electronics or electric power systems in them. I find they ’ re all **the** same, whether its land, air or water. I ’ ll be focusing on **the** air here. This is a solar-powered **airplane** â 165 miles carrying a person from France to England as a symbol that solar power is going to be an important part of our future. Then we did **the** solar car for General Motors â **the** Sunracer â that won **the** race in Australia. We got a lot of people thinking about electric cars, what you could do with them. A few years later, when we suggested to GM that now is **the** time and we could do a thing called **the** Impact, they sponsored it, and here ’ s **the** Impact that we developed with them on their programs. This is **the** demonstrator. And they put huge effort into turning it into a commercial product. With that preamble, let ’ s show **the** first two-minute videotape, which shows a little **airplane** for surveillance and moving to a giant **airplane**. Narrator : A tiny **airplane**, **the** AV Pointer serves for surveillance â in effect, a pair of roving eyeglasses. A cutting-edge example of where miniaturization can lead if **the** operator is remote from **the** vehicle. It is convenient to carry, assemble and launch by hand. Battery-powered, it is silent and rarely noticed. It sends high-resolution video pictures back to **the** operator. With onboard GPS, it can navigate autonomously, and it is rugged enough to self-land without damage. **The** modern sailplane is superbly efficient. Some can glide as flat as 60 feet forward for every foot of descent. They are powered only by **the** energy they can extract from **the** atmosphere â an atmosphere nature stirs up by solar energy. Humans and soaring birds have found nature to be generous in providing replenishable energy. Sailplanes have flown over 1, 000 miles, and **the altitude** record is over 50, 000 feet. **The** Solar Challenger was made to serve as a symbol that photovoltaic cells can produce real power and will be part of **the** world ’ s energy future. In 1981, it flew 163 miles from Paris to England, solely on **the** power of sunbeams, and established a basis for **the** Pathfinder. **The** message from all these vehicles is that ideas and technology can be harnessed to produce remarkable gains in doing more with less â gains that can help us attain a desirable balance between technology and nature. **The** stakes are high as we speed toward a challenging future. Buckminster Fuller said it clearly :”there are no passengers on spaceship Earth, only crew. We, **the** crew, can and must do more with less â much less.”Paul MacCready : If we could have **the** second video, **the** one-minute, put in as quickly as you can, which â this will show **the** Pathfinder **airplane** in some flights this past year in Hawaii, and will show a sequence of some of **the** beauty behind it after it had just flown to 71, 530 feet â higher than any propeller **airplane** has ever flown. It ’ s amazing : just on **the** puny power of **the** sun â by having a super lightweight plane, you ’ re able to get it up there. It ’ s part of a long-term program NASA sponsored. And we worked very closely with **the** whole thing being a team effort, and with wonderful results like that flight. And we ’ re working on a bigger plane â 220-foot span â and an intermediate-size, one with a regenerative fuel cell that can store excess energy during **the** day, feed it back at night, and stay up 65, 000 feet for months at a time. Ray Morgan ’ s voice will come in here. There he ’ s **the** project manager. Anything they do is certainly a team effort. He ran this program. Here ’ s... some things he showed as a celebration at **the** very end. Ray Morgan : We ’ d just ended a seven-month deployment of Hawaii. For those who live on **the** mainland, it was tough being away from home. **The friendly** support, **the** quiet confidence, congenial hospitality shown by our Hawaiian and military hosts â this is starting â made **the** experience enjoyable and unforgettable. PM : We have real-time IR scans going out through **the** Internet while **the** plane is flying. And it ’ s exploring without **polluting the** stratosphere. That ’ s its goal : **the** stratosphere, **the** blanket that really controls **the** radiation of **the** earth and permits life on earth to be **the** success that it is â probing that is very important. And also we consider it as a sort of poor man ’ s stationary satellite, because it can stay right overhead for months at a time, 2, 000 times closer than **the** real GFC synchronous satellite. We couldn ’ t bring one here to fly it and show you. But now let ’ s look at **the** other end. In **the** video you saw that nine-pound or eight-pound Pointer **airplane** surveillance drone that Keenan has developed and just done a remarkable job. Where some have servos that have gotten down to, oh, 18 or 25 grams, his weigh **one-third** of a gram. And what he ’ s going to bring out here is a surveillance drone that weighs about 2 ounces â that includes **the** video camera, **the** batteries that run it, **the** telemetry, **the** receiver and so on. And we ’ ll fly it, we hope, with **the** same success that we had last night when we did **the** practice. So Matt Keenan, just any time you ’ re â all right â ready to let her go. But first, we ’ re going to make sure that it ’ s appearing on **the** screen, so you see what it sees. You can imagine yourself being a mouse or fly inside of it, looking out of its camera. Matt Keenan : It ’ s switched on. PM : But now we ’ re trying to get **the** video. There we go. MK : Can you bring up **the** house lights? PM : Yeah, **the** house lights and we ’ ll see you all better and be able to fly **the** plane better. MK : All right, we ’ ll try to do a few laps around and bring it back in. Here we go. PM : **The** video worked right for **the** first few and I don ’ t know why it â there it goes. Oh, that was only a minute, but I think you ’ d be safe to have that near **the** end of **the** flight, perhaps. We get to do **the** classic. All right. If this hits you, it will not hurt you. OK. Thank you very much. Thank you. But now, as they say in infomercials, we have something much better for you, which we ’ re working on : planes that are only six **inches** â 15 centimeters â in size. And Matt ’ s plane was on **the** cover of Popular Science last month, showing what this can lead to. And in a while, something this size will have GPS and a video camera in it. We ’ ve had one of these fly nine miles through **the** air at 35 miles an hour with just a little battery in it. But there ’ s a lot of technology going. There are just milestones along **the** way of some remarkable things. This one doesn ’ t have **the** video in it, but you get a little feel from what it can do. OK, here we go. MK : Sorry. OK. PM : If you can pass it down when you ’ re done. Yeah, I think â I lost a little orientation ; I looked up into this light. It hit **the** building. And **the** building was poorly placed, actually. But you ’ re beginning to see what can be done. We ’ re working on projects now â even wing-flapping things **the** size of hawk moths â DARPA contracts, working with Caltech, UCLA. Where all this leads, I don ’ t know. Is it practical? I don ’ t know. But like any basic research, when you ’ re really forced to do things that are way beyond existing technology, you can get there with micro-technology, nanotechnology. You can do amazing things when you realize what nature has been doing all along. As you get to these small scales, you realize we have a lot to learn from nature â not with 747s â but when you get down to **the** nature ’ s realm, nature has 200 million years of experience. It never makes a mistake. Because if you make a mistake, you don ’ t leave any progeny. We should have nothing but success stories from nature, for you or for birds, and we ’ re learning a lot from its fascinating subjects. In **concluding**, I want to get back to **the** big picture and I have just two final **slides** to try and put it in perspective. **The** first I ’ ll just read. At last, I put in three sentences and had it say what I wanted. Over billions of years on a unique **sphere**, chance has painted a thin covering of life â complex, improbable, wonderful and fragile. Suddenly, we humans â a recently arrived species, no longer subject to **the** checks and balances inherent in nature â have grown in population, technology and intelligence to a position of terrible power. We now wield **the** paintbrush. And that ' s serious : we ' re not very bright. We ' re short on wisdom ; we ' re high on technology. Where ' s it going to lead? Well, inspired by **the** sentences, I decided to wield **the** paintbrush. Every 25 years I do a picture. Here ' s **the** one â tries to show that **the** world isn ' t getting any bigger. Sort of a timeline, very non-linear scale, nature rates and trilobites and dinosaurs, and eventually we saw some humans with caves... Birds were flying overhead, after pterosaurs. And then we get to **the** civilization above **the** little TV set with a gun on it. Then traffic jams, and power systems, and some dots for digital. Where it ' s going to lead â I have no idea. And so I just put robotic and natural **cockroaches** out there, but you can fill in whatever you want. This is not a forecast. This is a warning, and we have to think seriously about it. And that time when this is happening is not 100 years or 500 years. Things are going on this decade, next decade ; it ' s a very short time that we have to decide what we are going to do. And if we can get some agreement on where we want **the** world to be â desirable, sustainable when your kids reach your age â I think we actually can reach it. Now, I said this was a warning, not a forecast. That was before â I painted this before we started in on making robotic versions of hawk moths and **cockroaches**, and now I ' m beginning to wonder seriously â was this more of a forecast than I wanted? I personally think **the** surviving intelligent life form on earth is not going to be carbon-based ; it ' s going to be silicon-based. And so where it all goes, I don ' t know. **The** one final bit of sparkle we ' ll put in at **the** very end here is an utterly impractical flight vehicle, which is a little ornithopter wing-flapping **device** that â rubber-band powered â that we ' ll show you. MK : 32 gram. Sorry, one gram. PM : Last night we gave it a few too many turns and it tried to bash **the** roof out also. It ' s about a gram. **The** tube there ' s hollow, about paper-thin. And if this lands on you, I assure you it will not hurt you. But if you reach out to grab it or hold it, you will destroy it. So, be gentle, just act like a wooden Indian or something. And when it comes down â and we ' ll see how it goes. We consider this to be sort of **the** spirit of TED. And you wonder, is it practical? And it turns out if I had not been â Unfortunately, we have some light bulb changes. We can probably get it down, but it ' s possible it ' s gone up to a greater destiny up there â â than it ever had. And I wanted to make â just â But I want to make just two points. One is, you think it ' s frivolous ; there ' s nothing to it. And yet if I had not been making ornithopters like that, a little bit **cruder**, in 1939 â a long, long time ago â there wouldn ' t have been a Gossamer Condor, there wouldn ' t have been an Albatross, a Solar Challenger, there wouldn ' t be an Impact car, there wouldn ' t be a mandate on zero-emission vehicles in California. A lot of these things â or similar â would have happened some time, probably a decade later. I didn ' t realize at **the** time I was doing inquiry-based, hands-on things with teams, like they ' re trying to get in education systems. So I think that, as a symbol, it ' s important. And I believe that also is important. You can think of it as a sort of a symbol for **learning** and TED that somehow gets you thinking of technology and nature, and puts it all together in things that are â that make this conference, I think, more important than any that ' s taken place in this country in this decade. Thank you.

Text Sample 16: POS Dim 1 – 1998 – Score 15.00 – 1998/t000250.txt

As a clergyman, you can imagine how out of place I feel. I feel like a **fish** out of water, or maybe an owl out of **the** air. I was preaching in San Jose some time ago, and my friend Mark Kvamme, who helped introduce me to this conference, brought several CEOs and leaders of some of **the** companies here in **the** Silicon Valley to have breakfast with me, or I with them. And I was so stimulated. And had such — it was an eye-opening experience to hear them talk about **the** world that is yet to come through technology and science. I know that we ' re near **the** end of this conference, and some of you may be wondering why they have a speaker from **the** field of religion. Richard can answer that, because he made that decision. But some years ago I was on an elevator in Philadelphia, coming down. I was to address a conference at a hotel. And on that elevator a man said,"I hear Billy Graham is staying in this hotel."And another man looked in my direction and said,"Yes, there he is. He ' s on this elevator with us."And this man looked me up and down for about 10 seconds, and he said,"My, what an anticlimax!"I hope that you won ' t feel that these few moments with me is not a — is an anticlimax, after all these tremendous talks that you ' ve heard, and addresses, which I intend to listen to every one of them. But I was on an **airplane** in **the** east some years ago, and **the** man sitting across **the** aisle from me was **the** mayor of Charlotte, North Carolina. His name was John Belk. Some of you will probably know him. And there was a drunk man on there, and he got up out of his seat two or three times, and he was making everybody upset by what he was trying to do. And he was slapping **the** stewardess and pinching her as she went by, and everybody was upset with him. And finally, John Belk said,"Do you know who ' s sitting here?"And **the** man said,"No, who?"He said,"It ' s Billy Graham, **the** preacher."He said,"You don ' t say!"And he turned to me, and he said,"Put her there!"He said,"Your sermons have certainly helped me."And I suppose that that ' s true with thousands of people. I know that as you have been peering into **the** future, and as we ' ve heard some of it here tonight, I would like to live in that age and see what is going to be. But I won ' t, because I ' m 80 years old. This is my eightieth year, and I know that my time is brief. I have phlebitis at **the** moment, in both legs, and that ' s **the** reason that I had to have a little help in getting up here, because I have Parkinson ' s disease in addition to that, and some other problems that I won ' t talk about. But this is not **the** first time that we ' ve had a technological revolution. We ' ve had others. And there ' s one that I want to talk about. In one generation, **the** nation of **the** people of Israel had a tremendous and dramatic change that made them a great power in **the** Near East. A man by **the** name of David came to **the** throne, and King David became one of **the** great leaders of his generation. He was a man of tremendous leadership. He had **the** favor of God with him. He was a brilliant poet, philosopher, writer, soldier — with strategies in battle and conflict that people study even today. But about two centuries before David, **the** Hittites had discovered **the** secret of smelting and processing of **iron**, and, slowly, that skill spread. But they wouldn ' t allow **the** Israelis to look into it, or to have any. But David changed all of that, and he introduced **the** Iron Age to Israel. And **the** Bible says that David laid up great stores of **iron**, and which archaeologists have found, that in present-day Palestine, there are evidences of that generation. Now, instead of **crude** tools made of sticks and stones, Israel now had **iron** plows, and sickles, and hoes and military weapons. And in **the** course of one generation, Israel was completely changed. **The** introduction of **iron**, in some ways, had an impact a little bit like **the** microchip has had on our generation. And David found that there were many problems that technology could not solve. There were many problems still left. And they ' re still with us, and you haven ' t solved them, and I haven ' t heard anybody here speak to that.

How do we solve these three problems that I ’ d like to mention? **The** first one that David saw was human evil. Where does it come from? How do we solve it? Over again and again in **the** Psalms, which Gladstone said was **the** greatest book in **the** world, David describes **the** evils of **the** human race. And yet he says,”He restores my soul.”Have you ever thought about what a contradiction we are? On one hand, we can probe **the** deepest secrets of **the universe** and dramatically push back **the** frontiers of technology, as this conference vividly demonstrates. We ’ ve seen under **the sea**, three miles down, or galaxies hundreds of billions of years out in **the** future. But on **the** other hand, something is wrong. Our battleships, our soldiers, are on a frontier now, almost ready to go to war with Iraq. Now, what causes this? Why do we have these wars in every generation, and in every part of **the** world? And revolutions? We can ’ t get along with other people, even in our own families. We find ourselves in **the** paralyzing grip of self-destructive habits we can ’ t break. Racism and injustice and violence sweep our world, bringing a tragic harvest of heartache and death. Even **the** most sophisticated among us seem powerless to break this cycle. I would like to see Oracle take up that, or some other technological geniuses work on this. How do we change man, so that he doesn ’ t lie and cheat, and our newspapers are not filled with stories of fraud in business or labor or athletics or wherever? **The** Bible says **the** problem is within us, within our hearts and our souls. Our problem is that we are separated from our Creator, which we call God, and we need to have our souls restored, something only God can do. Jesus said,”For out of **the** heart come evil thoughts : murders, sexual immorality, theft, false testimonies, slander.”**The** British philosopher Bertrand Russell was not a religious man, but he said,”It ’ s in our hearts that **the** evil lies, and it ’ s from our hearts that it must be plucked out.”Albert Einstein — I was just talking to someone, when I was speaking at Princeton, and I met Mr. Einstein. He didn ’ t have a doctor ’ s degree, because he said nobody was qualified to give him one. But he made this statement. He said,”It ’ s easier to denature plutonium than to denature **the** evil spirit of man.”And many of you, I ’ m sure, have thought about that and puzzled over it. You ’ ve seen people take beneficial technological advances, such as **the** Internet we ’ ve heard about tonight, and twist them into something corrupting. You ’ ve seen brilliant people **devise computer** viruses that bring down whole systems. **The** Oklahoma City bombing was simple technology, horribly used. **The** problem is not technology. **The** problem is **the** person or persons using it. King David said that he knew **the** depths of his own soul. He couldn ’ t free himself from personal problems and personal evils that included murder and adultery. Yet King David sought God ’ s forgiveness, and said,”You can restore my soul.”You see, **the** Bible teaches that we ’ re more than a body and a mind. We are a soul. And there ’ s something inside of us that is beyond our understanding. That ’ s **the** part of us that yearns for God, or something more than we find in technology. Your soul is that part of you that yearns for meaning in life, and which seeks for something beyond this life. It ’ s **the** part of you that yearns, really, for God. I find [that] young people all over **the** world are **searching** for something. They don ’ t know what it is. I speak at many universities, and I have many questions and answer periods, and whether it ’ s Cambridge, or Harvard, or Oxford — I ’ ve spoken at all of those universities. I ’ m going to Harvard in about three or four — no, it ’ s about two months from now — to give a **lecture**. And I ’ ll be asked **the** same questions that I was asked **the** last few times I ’ ve been there. And it ’ ll be on these questions : where did I come from? Why am I here? Where am I going? What ’ s life all about? Why am I here? Even if you have no religious belief, there are times when you wonder that there ’ s something else. Thomas Edison also said,”When you see everything that happens in **the** world of science, and in **the** working of **the universe**, you cannot deny that there ’ s a captain on **the** bridge.”I remember once, I sat beside Mrs. Gorbachev at a White House dinner. I went to Ambassador Dobrynin, whom I knew very well. And I ’ d been to Russia several times under **the** Communists, and they ’ d given me marvelous freedom that I didn ’ t expect. And I knew Mr. Dobrynin very well, and I said,”I ’ m going to sit beside Mrs. Gorbachev tonight. What shall I talk to her about?”And he surprised me with **the** answer. He said,”Talk to her about religion and philosophy. That ’ s what she ’ s really interested in.”I was a little bit surprised, but that evening that ’ s what we talked about, and it was a stimulating conversation. And afterward, she said,”You know, I ’ m an atheist, but I know that there ’ s something up there higher than we are.”**The** second problem that King David realized he could not solve was **the** problem of human suffering. Writing **the** oldest book in **the** world was Job, and he said,”Man is born unto trouble as **the** sparks fly upward.”Yes, to be sure, science has done much to push back certain types of human suffering. But I ’ m — in a few months, I ’ ll be 80 years of age. I admit that I ’ m very grateful for all **the** medical advances that have kept me in relatively good health all these years. My doctors at **the** Mayo Clinic urged me not to take this trip out here to this — to be here. I haven ’ t given a talk in nearly four months. And when you speak as much as I do, three or four times a day, you get rusty. That ’ s **the** reason I ’ m using this podium and using these notes. Every time you ever hear me on **the** television or somewhere, I ’ m ad-libbing. I ’ m not reading. I never read an address. I never read a speech or a talk or a **lecture**. I talk ad lib. But tonight, I ’ ve got some notes here so that if I begin to forget, which I do sometimes, I ’ ve got something I can turn to. But even here among us, most — in **the** most advanced society in **the** world, we have poverty. We have families that self-destruct, friends that betray us. Unbearable psychological pressures bear down on us. I ’ ve never met a person in **the** world that didn ’ t have a problem or a worry. Why do we suffer? It ’ s an age- old question that we haven ’ t answered. Yet David again and again said that he would turn to God. He said,”**The** Lord is my shepherd.”**The** final problem that David knew he could not solve was death. Many commentators have said that death is **the** forbidden subject of our generation. Most people live as if they ’ re never going to die. Technology projects **the** myth of control over our mortality. We see people on our screens. Marilyn Monroe is just as beautiful on **the** screen as she was in person, and our — many young people think she ’ s still alive. They don ’ t know that she ’ s dead. Or Clark Gable, or whoever it is. **The** old stars, they come to life. And they ’ re — they ’ re just as great on that screen as they were in person. But death is inevitable. I spoke some time ago to a joint session of Congress, last year. And we were meeting in that room, **the** statue room. About 300 of them were there. And I said,”There ’ s one thing that we have in common in this room, all of us together, whether Republican or Democrat, or whoever.”I said,”We ’ re all going to die. And we have that in common with all these great men of **the** past that are staring down at us.”And it ’ s often difficult for young people to understand that. It ’ s difficult for them to understand that they ’ re going to die. As **the** ancient writer of Ecclesiastes wrote, he said, there ’ s every activity under heaven. There ’ s a time to be born, and there ’ s a time to die. I ’ ve stood at **the** deathbed of several **famous** people, whom you would know. I ’ ve talked to them. I ’ ve seen them in those agonizing moments when they were scared to death. And yet, a few years earlier, death never crossed their mind. I talked to a woman this past week whose father was a **famous** doctor. She said he never thought of God, never talked about God, didn ’ t believe in God. He was an atheist. But she said, as he came to die, he sat up on **the** side of **the** bed one day, and he asked **the** nurse if he could see **the** chaplain. And he said, for **the** first time in his life he ’ d thought about **the** inevitable, and about God. Was there a God? A few years ago, a university student asked me,”What is **the** greatest surprise in your life?”And I said,”**The** greatest surprise in my life is **the** brevity of life. It passes so fast.”But it does not need to have to be that way. Wernher von Braun, in **the** aftermath of World War II **concluded**, quote :”science and religion are not antagonists. On **the** contrary, they ’ re sisters.”He put it on a personal basis. I knew Dr. von Braun very well. And he said,”Speaking for myself, I can only say that **the** grandeur of **the** cosmos serves only to confirm a belief in **the** certainty of a creator.”He also said,”In our **search** to know God, I ’ ve come to believe that **the** life of Jesus Christ should be **the** focus of our efforts and inspiration. **The** reality of this life and His resurrection is **the** hope of mankind.”I ’ ve done a lot of speaking in Germany and in France, and in different parts of **the** world — 105 countries it ’ s been my privilege to speak in. And I was invited one day to visit Chancellor Adenauer, who was looked upon as sort of **the** founder of modern Germany, since **the** war. And he once — and he said to me, he said,”Young man.”He said,”Do you believe in **the** resurrection of Jesus Christ?”And I said,”Sir, I do.”He said,”So do I.”He said,”When I leave office, I ’ m going to spend my time writing a book on why Jesus Christ rose again, and why it ’ s so important to believe that.”In one of his plays, Alexander Solzhenitsyn depicts a man dying, who says to those gathered around his bed,”**The** moment when it ’ s terrible to feel regret is when one is dying.”How should one live in order not to feel regret when one is dying? Blaise Pascal asked exactly that question in seventeenth-century France. Pascal has been called **the** architect of modern civilization. He was a brilliant scientist at **the** frontiers of mathematics, even as a teenager. He is viewed by many as **the** founder of **the probability** theory, and a creator of **the** first model of a **computer**. And of course, you are all familiar with **the computer** language named for him. Pascal explored in depth our human dilemmas of evil, suffering and death. He was astounded at **the** phenomenon we ’ ve been considering : that people can achieve extraordinary heights in science, **the** arts and human enterprise, yet they also are full of anger, hypocrisy and have — and self- hatreds. Pascal saw us as a remarkable mixture of genius and self-delusion. On November 23, 1654, Pascal had a profound religious experience. He wrote in his journal these words :”I submit myself, absolutely, to Jesus Christ, my redeemer.”A French historian said, two centuries later,”Seldom has so mighty an intellect submitted with such humility to **the** authority of Jesus Christ.”Pascal came to believe not only **the** love and **the** grace of God could bring us back into harmony, but he believed that his own sins and failures could be forgiven, and that when he died he would go to a place called heaven. He experienced it in a way that went beyond scientific observation and reason. It was he who penned **the** well-known words,”**The** heart has its reasons, which reason knows not of.”Equally well known is Pascal ’ s Wager. Essentially, he said this :”if you bet on God, and open yourself to his love, you lose nothing, even if you ’ re wrong. But if instead you bet that there is no God, then you can lose it all, in this life and **the** life to come.”For Pascal, scientific knowledge paled beside **the** knowledge of God. **The** knowledge of God was far beyond anything that ever crossed his mind. He was ready to face him when he died at **the** age of 39. King David lived to be 70, a long time in his era. Yet he too had to face death, and he wrote these words :”even though I walk through **the** valley of **the** shadow of death, I will fear no evil, for you are with me.”This was David ’ s answer to three dilemmas of evil, suffering and death. It can be yours, as well, as you seek **the** living God and allow him to fill your life and give you hope for **the** future. When I was 17 years of age, I was born and reared on a farm in North Carolina. I milked cows every morning, and I had to milk **the** same cows every evening when I came home from school. And there were 20 of them that I had — that I was responsible for, and I worked on **the** farm and tried to keep up with my studies. I didn ’ t make good grades in high school. I didn ’ t make them in college, until something happened in my heart. One day, I was faced face-to-face with Christ. He said,”I am **the** way, **the** truth and **the** life.”Can you imagine that?”I am **the** truth. I ’ m **the** embodiment of all truth.”He was a liar. Or he was insane. Or he was what he claimed to be. Which was he? I had to make that decision. I couldn ’ t prove it. I couldn ’ t take it to a laboratory and experiment with it. But by faith I said, I believe him, and he came into my heart and changed my life. And now I ’ m ready, when I hear that call, to go into **the** presence of God. Thank you, and God bless all of you. Thank you for **the** privilege. It was great. Richard Wurman : You did it. Thanks.

Text Sample 17: POS Dim 1 – 2002 – Score 27.00 – 2002/t000368.txt

The advances that have taken place in astronomy, cosmology and **biology**, in **the** last 10 years, are really extraordinary â to **the** point where we know more about our **universe** and how it works than many of you might imagine. But there was something else that I ’ ve noticed as those changes were taking place, as people were starting to find out that hmm... yeah, there really is a black hole at **the** center of every galaxy. **The** science writers and editors â I shouldn ’ t say science writers, I should say people who write about science â and editors would sit down over a couple of beers, after a hard day of work, and start talking about some of these incredible perceptions about how **the** **universe** works. And they would inevitably end up in what I thought was a very bizarre place, which is ways **the** world could end very suddenly. And that ’ s what I want to talk about today. Ah, you laugh, you fools. Yeah, we need **the** time! Stephen Petranek : At first, it all seemed a little fantastical to me, but after challenging a lot of these ideas, I began to take a lot of them seriously. And then September 11 happened, and I thought, ah, God, I can ’ t go to **the** TED conference and talk about how **the** world is going to end. Nobody wants to hear that. Not after this! And that got me into a discussion with some other people, other scientists, about maybe some other subjects, and one of **the** guys I talked to, who was a neuroscientist, said,”You know, I think there are a lot of solutions to **the** problems you brought up,”and reminds me of Michael ’ s talk yesterday and his mother saying you can ’ t have a solution if you don ’ t have a problem. So, we went out looking for solutions to ways that **the** world might end tomorrow, and lo and behold, we found them. Which leads me to a videotape of a President Bush press conference from a couple of weeks ago. Can we run that, Andrew? President George W. Bush : Whatever it costs to defend our security, and whatever it costs to defend our freedom, we must pay it. SP : I agree with **the** president. He wants two trillion dollars to protect us from **terrorists** next year, a two-trillion-dollar federal budget, which will land us back into deficit spending real fast. But **terrorists** aren ’ t **the** only threat we face. There are really serious calamities staring us in **the** eye that we ’ re in **the** same kind of denial about that we were about terrorism, and what could ’ ve happened on September 11. I would propose, therefore, that if we took 10 billion dollars from that 2. 13 trillion dollar budget â which is two one hundredths of that budget â and we doled out a billion dollars to each one of these problems I ’ m going to talk to you about, **the** vast majority could be solved, and **the** rest we could deal with. So, I hope you find this both fascinating â I ’ m fascinated by this kind of stuff, I gotta admit â to me these are Richard ’ s **cockroaches**. But I also hope, because I think **the** people in this room can literally change **the** world, I hope you take some of this stuff away with you, and when you have an opportunity to be influential, that you try to get some heavy-duty money spent on some of these ideas. So let ' s start. Number 10 : we lose **the** will to survive. We live in an incredible age of modern medicine. We are all much healthier than we were 20 years ago. People around **the** world are getting better medicine â but mentally, we ' re falling apart. **The** World Health Organization now estimates that one out of five people on **the planet** is clinically depressed. And **the** World Health Organization also says that depression is **the** biggest epidemic that humankind has ever faced. Soon, genetic breakthroughs and even better medicine are going to allow us to think of 100 as a normal lifespan. A female child born tomorrow, on average â median â will live to age 83. Our life longevity is going up almost a year for every year that passes. Now **the** problem with all of this, getting older, is that people over 65 are **the** most likely people to commit suicide. So, what are **the** solutions? We don ' t really have mental health insurance in this country, and it ' s â â it ' s really a crime. Something like 98 percent of all people with depression, and I mean really severe depression â I have a friend with stunningly severe depression â this is a curable disease, with present medicine and present technology. But it is often a combination of talk therapy and pills. Pills alone don ' t do it, especially in clinically depressed people. You ought to be able to go to a psychiatrist or a psychologist, and put down your 10-dollar copay, and get treated, just like you do when you got a cut on your arm. It ' s ridiculous. Secondly, drug companies are not going to develop really sophisticated psychoactive drugs. We know that most mental illnesses have a biological component that can be dealt with. And we know just an amazing amount more about **the** brain now than we did 10 years ago. We need a pump-push from **the** federal government, through NIH and National Science â NSF â and places like that to start helping **the** drug companies develop some advanced psychoactive drugs. Moving on. Number nine â don ' t laugh â aliens invade Earth. Ten years ago, you couldn ' t have found an astronomer â well, very few astronomers â in **the** world who would ' ve told you that there are any **planets** anywhere outside our solar system. 1995, we found three. **The** count now is up to 80 â we ' re finding about two or three a month. All of **the** ones we ' ve found, by **the** way, are in this little, teeny, tiny corner where we live, in **the** Milky Way. There must be millions of **planets** in **the** Milky Way, and as Carl Sagan insisted for many years, and was laughed at for it, there must be billions and billions in **the universe**. In a few years, NASA is going to launch four or five telescopes out to Jupiter, where there ' s less dust, and start looking for Earth-like **planets**, which we cannot see with present technology, nor **detect**. It ' s becoming obvious that **the** chance that life does not exist elsewhere in **the universe**, and probably fairly close to us, is a fairly remote idea. And **the** chance that some of it isn ' t more intelligent than ours is also a remote idea. Remember, we ' ve only been an advanced civilization â an industrial civilization, if you would â for 200 years. Although every time I go to Pompeii, I ' m amazed that they had **the** equivalent of a McDonald ' s on every street corner, too. So, I don ' t know how much civilization really has progressed since AD 79, but there ' s a great likelihood. I really believe this, and I don ' t believe in aliens, and I don ' t believe there are any aliens on **the** Earth or anything like that. But there ' s a likelihood that we will confront a civilization that is more intelligent than our own. Now, what will happen? What if they come to, you know, suck up our oceans for **the** hydrogen? And swat us away like flies, **the** way we swat away flies when we go into **the** rainforest and start logging it. We can look at our own history. **The** late physicist Gerard O ' Neill said,"Advanced Western civilization has had a destructive effect on all primitive civilizations it has come in contact with, even in those cases where every attempt was made to protect and guard **the** primitive civilization."If **the** aliens come visiting, we ' re **the** primitive civilization. So, what are **the** solutions to this? Thank God you can all read! It may seem ridiculous, but we have a really lousy history of anticipating things like this and actually being prepared for them. How much energy and money does it take to actually have a plan to negotiate with an advanced species? Secondly â and you ' re going to hear more from me about this â we have to become an outward-looking, space-faring nation. We have got to develop **the** idea that **the** Earth doesn ' t last forever, our sun doesn ' t last forever. If we want humanity to last forever, we have to colonize **the** Milky Way. And that is not something that is beyond comprehension at this point. It ' ll also help us a lot, if we meet an advanced civilization along **the** way, if we ' re trying to be an advanced civilization. Number eight â SP : You ' ve got it! You ' ve got **the** job. Number eight : **the** ecosystem collapses. Last July, in Science, **the** journal Science, 19 oceanographers published a very, very unusual article. It wasn ' t really a research report ; it was a screed. They said, we ' ve been looking at **the** oceans for a long time now, and we want to tell you they ' re not in trouble, they ' re near collapse. Many other ecosystems on Earth are in real, real danger. We ' re living in a time of mass extinctions that exceeds **the fossil** record by a factor of 10, 000. We have lost 25 percent of **the** unique species in Hawaii in **the** last 20 years. California is expected to lose 25 percent of its species in **the** next 40 years. Somewhere in **the** Amazon forest is **the** marginal tree. You cut down that tree, **the** rain forest collapses as an ecosystem. There ' s really a tree like that out there. That ' s really what it comes to. And when that ecosystem collapses, it could take a major ecosystem with it, like our atmosphere. So, what do we do about this? What are **the** solutions? There is some modeling of ecosystems going on now. **The** problem with ecosystems is that we understand them so poorly, that we don ' t know they ' re really in trouble until it ' s almost too late. We need to know earlier that they ' re getting in trouble, and we need to be able to pump possible solutions into models. And with **the** kind of computing power we have now, there is, as I say, some of this going on, but it needs money. National Science Foundation needs to say â you know, almost all **the** money that ' s spent on science in this country comes from **the** federal government, one way or another. And they get to prioritize, you know? There are people at **the** National Science Foundation who get to say, this is **the** most important thing. This is one of **the** things they ought to be thinking more about. Secondly, we need to create huge biodiversity reserves on **the planet**, and start moving them around. There ' s been an experiment for **the** last four or five years on **the** Georges Bank, or **the** Grand Banks off of Newfoundland. It ' s a no-take **fishing** zone. They can ' t **fish** there for a radius of 200 miles. And an amazing thing has happened : almost all **the fish** have come back, and they ' re reproducing like crazy. We ' re going to have to start doing this around **the** globe. We ' re going to have to have no-take zones. We ' re going to have to say, no more logging in **the** Amazon for 20 years. Let it recover, before we start logging again. Number seven : particle accelerator mishap. You all remember Ted Kaczynski, **the** Unabomber? One of **the** things he raved about was that a particle accelerator experiment could go haywire and set off a chain reaction that would destroy **the** world. A lot of very sober-minded physicists, believe it or not, have had exactly **the** same thought. This **spring** â there ' s a collider at Brookhaven, on Long Island â this **spring**, it ' s going to have an experiment in which it creates black holes. They are expecting to create little, tiny black holes. They expect them to evaporate. I hope they ' re right. Other collider experiments â there ' s one that ' s going to take place next summer at CERN â have **the** possibility of creating something called strangelets, which are kind of like antimatter. Whenever they hit other matter, they destroy it and obliterate it. Most physicists say that **the** accelerators we have now are not really powerful enough to create black holes and strangelets that we need to worry about, and they ’ re probably right. But, all around **the** world, in Japan, in Canada, there ’ s talk about this, of reviving this in **the** United States. We shut one down that was going to be big. But there ’ s talk of building very big accelerators. What can we do about this? What are **the** solutions? We ’ ve got **the** fox watching **the** henhouse here. We need to â we need **the** advice of particle physicists to talk about particle physics and what should be done in particle physics, but we need some outside thinking and watchdogging of what ’ s going on with these experiments. Secondly, we have a natural laboratory surrounding **the** Earth. We have an electromagnetic field around **the** Earth, and it ’ s constantly bombarded by high- energy particles, like protons. And in my opinion, we don ’ t spend enough time looking at that natural laboratory and figuring out first what ’ s safe to do on Earth. Number six : biotech disaster. It ’ s one of my favorite ones, because we ’ ve done several stories on Bt corn. Bt corn is a corn that creates its own pesticide to kill a corn borer. You may of heard of it â heard it called StarLink, especially when all those taco shells were taken out of **the supermarkets** about a year and a half ago. This stuff was supposed to only be feed for animals in **the** United States, and it got into **the** human food supply, and somebody should ’ ve figured out that it would get in **the** human food supply very easily. But **the** thing that ’ s alarming is a couple of months ago, in Mexico, where Bt corn and all genetically altered corn is totally illegal, they found Bt corn genes in wild corn plants. Now, corn originated, we think, in Mexico. This is **the** genetic biodiversity storehouse of corn. This brings back a skepticism that has gone away recently, that superweeds and superpests could spread around **the** world, from biotechnology, that literally could destroy **the** world ’ s food supply in very short order. So, what do we do about that? We treat biotechnology with **the** same scrutiny we apply to nuclear power plants. It ’ s that simple. This is an amazingly unregulated field. When **the** StarLink disaster happened, there was a battle between **the** EPA and **the** FDA over who really had authority, and over what parts of this, and they didn ’ t get it straightened out for months. That ’ s kind of crazy. Number five, one of my favorites : reversal of **the** Earth ’ s magnetic field. Believe it or not, this happens every few hundred thousand years, and has happened many times in our history. North Pole goes to **the** South, South Pole goes to **the** North, and vice versa. But what happens, as this **occurs**, is that we lose our magnetic field around **the** Earth over **the** period of about 100 years, and that means that all these cosmic rays and particles that are to come streaming at us from **the** sun, that this field protects us from, are â well, basically, we ’ re gonna fry. SP : So, what can we do about this? Oh, by **the** way, we ’ re overdue. It ’ s been 780, 000 years since this happened. So, it should have happened about 480, 000 years ago. Oh, and here ’ s one other thing. Scientists think now our magnetic field may be diminished by about five percent. So, maybe we ’ re in **the** throes of it. One of **the** problems of trying to figure out how healthy **the** Earth is, is that we have â you know, we don ’ t have good weather data from 60 years ago, much less data on things like **the** ozone layer. So, there ’ s a fairly simple solution to this. There ’ s going to be a lot of cheap rocketry that ’ s going to come online in about six or seven years that gets us into **the** low atmosphere very cheaply. You know, we can make ozone from car tailpipes. It ’ s not hard : it ’ s just three oxygen atoms. If you brought **the** entire ozone layer down to **the surface** of **the** Earth, it would be **the** thickness of two pennies, at 14 **pounds** per square **inch**. You don ’ t need that much up there. We need to learn how to repair and replenish **the** Earth ’ s ozone layer. Number four

: giant solar flares. Solar flares are enormous magnetic outbursts from **the** Sun that bombard **the** Earth with high-speed subatomic particles. So far, our atmosphere has done, and our magnetic field has done pretty well protecting us from this. Occasionally, we get a flare from **the** Sun that causes havoc with communications and so forth, and electricity. But **the** alarming thing is that astronomers recently have been studying stars that are similar to our Sun, and they ' ve found that a number of them, when they ' re about **the** age of our Sun, brighten by a factor of as much as 20. Doesn ' t last for very long. And they think these are super-flares, millions of times more powerful than any flares we ' ve had from our Sun so far. Obviously, we don ' t want one of those. There ' s a flip side to it. In studying stars like our Sun, we ' ve found that they go through periods of diminishment, when their total amount of energy that ' s expelled from them goes down by maybe one percent. One percent doesn ' t sound like a lot, but it would cause one hell of an ice age here. So, what can we do about this? Start terraforming Mars. This is one of my favorite subjects. I wrote a story about this in Life magazine in 1993. This is rocket science, but it ' s not hard rocket science. Everything that we need to make an atmosphere on Mars, and to make a livable **planet** on Mars, is probably there. And you just, literally, have to send little nuclear factories up there that gobble up **the iron** oxide on **the surface** of Mars and spit out **the** oxygen. **The** problem is it takes 300 years to terraform Mars, minimum. Really more like 500 years to do it right. There ' s no reason why we shouldn ' t start now. Number three â isn ' t this stuff cool? A new global epidemic. People have been at war with germs ever since there have been people, and from time to time, **the** germs sure get **the** upper hand. In 1918, we had a flu epidemic in **the** United States that killed 20 million people. That was back when **the** population was around 100 million people. **The** bubonic plague in Europe, in **the** Middle Ages, killed one out of four Europeans. AIDS is coming back. Ebola seems to be rearing its head with much too much **frequency**, and old diseases like cholera are becoming resistant to antibiotics. We ' ve all learned what â **the** kind of panic that can **occur** when an old disease rears its head, like anthrax. **The** worst possibility is that a very simple germ, like staph, for which we have one antibiotic that still works, mutates. And we know staph can do amazing things. A staph cell can be next to a muscle cell in your body and borrow genes from it when antibiotics come, and change and mutate. **The** danger is that some germ like staph will be â will mutate into something that ' s really virulent, very contagious, and will sweep through populations before we can do anything about it. That ' s happened before. About 12, 000 years ago, there was a massive wave of mammal extinctions in **the** Americas, and that is thought to have been a virulent disease. So, what can we do about it? It is nuts. We give antibiotics â â every cow, every lamb, every chicken, they get antibiotics every day, all. You know, you go to a restaurant, you eat **fish**, I got news for you, it ' s all farmed. You know, you gotta ask when you go to a restaurant if it ' s a wild **fish**, cause they ' re not going to tell you. We ' re giving away **the** code. This is like being at war and giving somebody your secret code. We ' re telling **the** germs out there how to fight us. We gotta fix that. We gotta outlaw that right away. Secondly, our public health system, as we saw with anthrax, is a real disaster. We have a real, major outbreak of disease in **the** United States, we are not prepared to cope with it. Now, there is money in **the** federal budget, next year, to build up **the** public health service. But I don ' t think to any extent that it really needs to be done. Number two â my favorite â we meet a rogue black hole. You know, 10 years ago, or 15 years ago, really, you walk into an astronomy convention, and you say,"You know, there ' s probably a black hole at **the** center of every galaxy,"and they ' re going to hoot you off **the** stage. And now, if you went into one of those conventions and you said,”Well, I don ’ t think black holes are out there,”they ’ d hoot you off **the** stage. Our comprehension of **the** way **the universe** works is really â has just gained unbelievably in recent years. We think that there are about 10 million dead stars in **the** Milky Way alone, our galaxy. And these stars have compressed down to maybe something like 12, 15 miles wide, and they are black holes. And they are gobbling up everything around them, including light, which is why we can ’ t see them. Most of them should be in orbit around something. But galaxies are very violent places, and things can be spun out of orbit. And also, space is incredibly vast. So even if you flung a million of these things out of orbit, **the** chances that one would actually hit us is fairly remote. But it only has to get close, about a billion miles away, one of these things. About a billion miles away, here ’ s what happens to Earth ’ s orbit : it becomes elliptical instead of circular. And for three months out of **the** year, **the surface** temperatures go up to 150 to 180. For three months out of **the** year, they go to 50 below zero. That won ’ t work too well. What can we do about this? And this is my scariest. I don ’ t have a good answer for this one. Again, we gotta think about being a colonizing race. And finally, number one : biggest danger to life as we know it, I think, a really big **asteroid** heads for Earth. **The** important thing to remember here â this is not a question of if, this is a question of when, and how big. In 1908, just a 200-foot piece of a comet exploded over Siberia and flattened forests for maybe 100 miles. It had **the** effect of about 1, 000 Hiroshima bombs. Astronomers estimate that little **asteroids** like that come about every hundred years. In 1989, a large **asteroid** passed 400, 000 miles away from Earth. Nothing to worry about, right? It passed directly through Earth ’ s orbit. We were in that that spot six hours earlier. A small **asteroid**, say a half mile wide, would touch off firestorms followed by severe global cooling from **the** debris kicked up â Carl Sagan ’ s nuclear winter thing. An **asteroid** five miles wide causes major extinctions. We think **the** one that got **the** dinosaurs was about five miles wide. Where are they? There ’ s something called **the** Kuiper belt, which â some people think Pluto ’ s not a **planet**, that ’ s where Pluto is, it ’ s in **the** Kuiper belt. There ’ s also something a little farther out, called **the** Oort cloud. There are about 100, 000 balls of ice and **rock** â comets, really â out there, that are 50 miles in **diameter** or more, and they regularly take a little spin, in towards **the** Sun and pass reasonably close to us. Of more concern, I think, is **the asteroids** that exist between Mars and Jupiter. **The** folks at **the** Sloan Digital Sky Survey told us last fall â they ’ re making **the** first map of **the universe**, three-dimensional map of **the universe** â that there are probably 700, 000 **asteroids** between Mars and Jupiter that are a half a mile big or bigger. So you say, yeah, well, what are really **the** chances of this happening? Andrew, can you put that chart up? This is a chart that Dr. Clark Chapman at **the** Southwest Research Institute presented to Congress a few years ago. You ’ ll notice that **the** chance of an asteroid-slash-comet impact killing you is about one in 20, 000, according to **the** work they ’ ve done. Now look at **the** one right below that. Passenger aircraft crash, one in 20, 000. We spend an awful lot of money trying to be sure that we don ’ t die in **airplane** accidents, and we ’ re not spending hardly anything on this. And yet, this is completely preventable. We finally have, just in **the** last year, **the** technology to stop this cold. Could we have **the** solutions? NASA ’ s spending three million dollars a year, three million **bucks** â that is like **pocket** change â to **search** for **asteroids**. Because we can actually figure out every **asteroid** that ’ s out there, and if it might hit Earth, and when it might hit Earth. And they ’ re trying to do that. But it ’ s going to take them 10 years, at spending three million dollars a year, and even then, they claim they ’ ll only have about 80 percent of them catalogued. Comets are a tougher act.

We don ’ t really have **the** technology to predict comet trajectories, or when one with our name on it might arrive. But we would have lots of time, if we see it coming. We really need a dedicated observatory. You ’ ll notice that a lot of comets are named after people you never heard of, **amateur** astronomers? That ’ s because nobody ’ s looking for them, except **amateurs**. We need a dedicated observatory that looks for comets. Part two of **the** solutions : we need to figure out how to blow up an **asteroid**, or alter its trajectory. Now, a year ago, we did an amazing thing. We sent a probe out to this **asteroid** belt, called NEAR, Near Earth **Asteroid** Rendezvous. And these guys orbited a 30 â or no, about a 22-mile long **asteroid** called Eros. And then, of course, you know, they pulled one of those sneaky NASA things, where they had extra batteries and extra gas aboard and everything, and then, at **the** last minute, they landed. When **the** mission was over, they actually landed on **the** thing. We have landed a rocket ship on an **asteroid**. It ’ s not a big deal. Now, **the** trouble with just sending a bomb out for this thing is that you don ’ t have anything to push against in space, because there ’ s no air. A nuclear **explosion** is just as hot, but we don ’ t really have anything big enough to melt a 22-mile long **asteroid**, or vaporize it, would be more like it. But we can learn to land on these **asteroids** that have our name on them and put something like a small ion propulsion motor on it, which would gently, slowly, after a period of time, push it into a different trajectory, which, if we ’ ve done our math right, would keep it from hitting Earth. This is just a matter of finding ’ em, going there, and doing something about it. I know your head is spinning from all this stuff. Yikes! So many big threats! **The** thing, I think, to remember, is September 11. We don ’ t want to get caught flat-footed again. We know about this stuff. Science has **the** power to predict **the** future in many cases now. Knowledge is power. **The** worst thing we can do is say, jeez, I got enough to worry about without worrying about an **asteroid**. That ’ s a mistake that could literally cost us our future. Thank you.

Text Sample 18: POS Dim 1 – 2002 – Score 21.00 – 2002/t000165.txt

I ’ ll just start talking about **the** 17th century. I hope nobody finds that offensive. I â you know, when I â after I had invented PCR, I kind of needed a change. And I moved down to La Jolla and learned how to **surf**. And I started living down there on **the** beach for a long time. And when surfers are out waiting for waves, you probably wonder, if you ’ ve never been out there, what are they doing? You know, sometimes there ’ s a 10-, 15-minute break out there when you ’ re waiting for a wave to come in. They usually talk about **the** 17th century. You know, they get a real bad rap in **the** world. People think they ’ re sort of lowbrows. One day, somebody suggested I read this book. It was called â it was called”**The** Air Pump,”or something like”**The** Leviathan and **The** Air Pump.”It was a real weird book about **the** 17th century. And I realized, **the** roots of **the** way I sort of thought was just **the** only natural way to think about things. That â you know, I was born thinking about things that way, and I had always been like a little scientist guy. And when I went to find out something, I used scientific methods. I wasn ’ t real surprised, you know, when they first told me how â how you were supposed to do science, because I ’ d already been doing it for fun and whatever. But it didn ’ t â it never **occurred** to me that it had to be invented and that it had been invented only 350 years ago. You know, it was â like it happened in England, and Germany, and Italy sort of all at **the** same time. And **the** story of that, I thought, was really fascinating. So I ’ m going to talk a little bit about that, and what exactly is it that scientists are supposed to do. And it ’ s, it ’ s a kind of â You know, Charles I got beheaded somewhere early in **the** 17th century. And **the** English set up Cromwell and a whole bunch of Republicans or whatever, and not **the** kind of Republicans we had. They changed **the** government, and it didn ’ t work. And Charles II, **the** son, was finally put back on **the** throne of England. He was really nervous, because his dad had been, you know, beheaded for being **the** King of England And he was nervous about **the** fact that conversations that got going in, like, bars and stuff would turn to â this is kind of â it ’ s hard to believe, but people in **the** 17th century in England were starting to talk about, you know, philosophy and stuff in bars. They didn ’ t have TV screens, and they didn ’ t have any football games to watch. And they would get really pissy, and all of a sudden people would **spill** out into **the** street and fight about issues like whether or not it was **okay** if Robert Boyle made a **device** called **the** vacuum pump. Now, Boyle was a friend of Charles II. He was a Christian guy during **the** weekends, but during **the** week he was a scientist. Which was â back then it was sort of, you know, well, you know â if you made this thing â he made this little **device**, like kind of like a bicycle pump in reverse that could suck all **the** air out of â you know what a bell jar is? One of these things, you pick it up, put it down, and it ’ s got a **seal**, and you can see inside of it, so you can see what ’ s going on inside this thing. But what he was trying to do was to pump all **the** air out of there, and see what would happen inside there. I mean, **the** first â I think one of **the** first experiments he did was he put a bird in there. And people in **the** 17th century, they didn ’ t really understand **the** same way we do about you know, this stuff is a bunch of different kinds of molecules, and we breathe it in for a purpose and all that. I mean, **fish** don ’ t know much about water, and people didn ’ t know much about air. But both started exploring it. One thing, he put a bird in there, and he pumped all **the** air out, and **the** bird died. So he said, hmm... He said â he called what he ’ d done as making â they didn ’ t call it a vacuum pump at **the** time. Now you call it a vacuum pump ; he called it a vacuum. Right? And immediately, he got into trouble with **the** local clergy who said, you can ’ t make a vacuum. Ah, uh â Aristotle said that nature abhors one. I think it was a poor translation, probably, but people relied on authorities like that. And you know, Boyle says, well, shit. I make them all **the** time. I mean, whatever that is that kills **the** bird â and I ’ m calling it a vacuum. And **the** religious people said that if God wanted you to make â I mean, God is everywhere, that was one of their rules, is God is everywhere. And a vacuum â there ’ s nothing in a vacuum, so you ’ ve â God couldn ’ t be in there. So therefore **the** church said that you can ’ t make a vacuum, you know. And Boyle said, bullshit. I mean, you want to call it Godless, you know, you call it Godless. But that ’ s not my job. I ’ m not into that. I do that on **the** weekend. And like â what I ’ m trying to do is figure out what happens when you suck everything out of a compartment. And he did all these cute little experiments. Like he did one with â he had a little wheel, like a fan, that was sort of loosely attached, so it could spin by itself. He had another fan opposed to it that he had like a â I mean, **the** way I would have done this would be, like, a rubber band, and, you know, around a tinker toy kind of fan. I know exactly how he did it ; I ’ ve seen **the** drawings. It ’ s two fans, one which he could turn from outside after he got **the** vacuum established, and he discovered that if he pulled all **the** air out of it, **the** one fan would no longer turn **the** other one, right? Something was missing, you know. I mean, these are â it ’ s kind of weird to think that someone had to do an experiment to show that, but that was what was going on at **the** time. And like, there was big arguments about it in **the** â you know, **the** gin houses and in **the** coffee shops and stuff. And Charles started not liking that. Charles II was kind of saying, you know, you should keep that â let ' s make a place where you can do this stuff where people don ' t get so â you know, we don ' t want **the** â we don ' t want to get **the** people mad at me again. And so â because when they started talking about religion and science and stuff like that, that ' s when it had sort of gotten his father in trouble. And so, Charles said, I ' m going to put up **the** money give you guys a building, come here and you can meet in **the** building, but just don ' t talk about religion in there. And that was fine with Boyle. He said, OK, we ' re going to start having these meetings. And anybody who wants to do science is â this is about **the** time that Isaac Newton was starting to whip out a lot of really interesting things. And there was all kind of people that would come to **the** Royal Society, they called it. You had to be dressed up pretty well. It wasn ' t like a TED conference. That was **the** only criteria, was that you be â you looked like a gentleman, and they ' d let anybody could come. You didn ' t have to be a member then. And so, they would come in and you would do â Anybody that was going to show an experiment, which was kind of a new word at **the** time, demonstrate some principle, they had to do it on stage, where everybody could see it. So they were â **the** really important part of this was, you were not supposed to talk about final causes, for instance. And God was out of **the** picture. **The** actual nature of reality was not at issue. You ' re not supposed to talk about **the** absolute nature of anything. You were not supposed to talk about anything that you couldn ' t demonstrate. So if somebody could see it, you could say, here ' s how **the** machine works, here ' s what we do, and then here ' s what happens. And seeing what happens, it was OK to generalize, and say, I ' m sure that this will happen anytime we make one of these things. And so you can start making up some rules. You say, anytime you have a vacuum state, you will discover that one wheel will not turn another one, if **the** only connection between them is whatever was there before **the** vacuum. That kind of thing. Candles can ' t burn in a vacuum, therefore, probably sparklers wouldn ' t either. It ' s not clear ; actually sparklers will, but they didn ' t know that. They didn ' t have sparklers. But, they â â you can make up rules, but they have to relate only to **the** things that you ' ve been able to demonstrate. And most **the** demonstrations had to do with visuals. Like if you do an experiment on stage, and nobody can see it, they can just hear it, they would probably think you were freaky. I mean, reality is what you can see. That wasn ' t an explicit rule in **the** meeting, but I ' m sure that was part of it, you know. If people hear voices, and they can ' t see and associate it with somebody, that person ' s probably not there. But **the** general idea that you could only â you could only really talk about things in that place that had some kind of experimental basis. It didn ' t matter what Thomas Hobbes, who was a local philosopher, said about it, you know, because you weren ' t going to be talking final causes. What ' s happening here, in **the** middle of **the** 17th century, was that what became my field â science, experimental science â was pulling itself away, and it was in a physical way, because we ' re going to do it in this room over here, but it was also what â it was an amazing thing that happened. Science had been all interlocked with theology, and philosophy, and â and â and mathematics, which is really not science. But experimental science had been tied up with all those things. And **the** mathematics part and **the** experimental science part was pulling away from philosophy. And â things â we never looked back. It ' s been so cool since then. I mean, it just â it just â untangled a thing that was really impeding technology from being developed. And, I mean, everybody in this room â now, this is 350 short years ago. Remember, that ' s a short time. It was 300, 000, probably, years ago that most of us, **the** ancestors of most of us in this room came up out of Africa and turned to **the left**. You know, **the** ones that turned to **the** right, there are some of those in **the** Japanese translation. But that happened very â a long time ago compared to 350 short years ago. But in that 350 years, **the** place has just undergone a lot of changes. In fact, everybody in this room probably, especially if you picked up your bag â some of you, I know, didn ' t pick up your bags â but if you picked up your bag, everybody in this room has got in their **pocket**, or back in their room, something that 350 years ago, kings would have gone to war to have. I mean, if you can think how important â If you have a GPS system and there are no satellites, it ' s not going to be much use. But, like â but, you know, if somebody had a GPS system in **the** 17th century some king would have gotten together an army and gone to get it, you know. If that person â Audience : For **the** teddy bear? **The** teddy bear? Kary Mullis : They might have done it for **the** teddy bear, yeah. But â all of us own stuff. I mean, individuals own things that kings would have definitely gone to war to get. And this is just 350 years. Not a whole lot of people doing this stuff. You know, **the** important people â you can almost read about their lives, about all **the** really important people that made advances, you know. And, I mean â this kind of stuff, you know, all this stuff came from that separation of this little sort of thing that we do â now I, when I was a boy was born sort of with this idea that if you want to know something â you know, maybe it ' s because my old man was gone a lot, and my mother didn ' t really know much science, but I thought if you want to know something about stuff, you do it â you make an experiment, you know. You get â you get, like â I just had a natural feeling for science and setting up experiments. I thought that was **the** way everybody had always thought. I thought that anybody with any brains will do it that way. It isn ' t true. I mean, there ' s a lot of people â You know, I was one of those scientists that was â got into trouble **the** other night at dinner because of **the** post-modernism thing. And I didn ' t mean, you know â where is that lady? Audience : Here. KM : I mean, I didn ' t really think of that as an argument so much as just a lively discussion. I didn ' t take it personally, but â I just â I had â I naively had thought, until this surfing experience started me into **the** 17th century, I ' d thought that ' s just **the** way people thought, and everybody did, and they recognized reality by what they could see or touch or feel or hear. At any rate, when I was a boy, I, like, for instance, I had this â I got this little book from Fort Sill, Oklahoma â This is about **the** time that George Dyson ' s dad was starting to blow nuclear â thinking about blowing up nuclear rockets and stuff. I was thinking about making my own little rockets. And I knew that frogs â little frogs â had aspirations of space travel, just like people. And I â I was looking for a â a propulsion system that would like, make a rocket, like, maybe about four feet high go up a couple of miles. And, I mean, that was my sort of goal. I wanted it to go out of sight and then I wanted this little parachute to come back with **the** frog in it. And â I â I â I got this book from Fort Sill, Oklahoma, where there ' s a missile base. They send it out for **amateur** rocketeers, and it said in there do not ever heat a mixture of potassium perchlorate and sugar. You know, that ' s what you call a lead. You sort of â now you say, well, let ' s see if I can get hold of some potassium chlorate and sugar, perchlorate and sugar, and heat it ; it would be interesting to see what it is they don ' t want me to do, and what it is going to â and how is it going to work. And we didn ' t have â like, my mother presided over **the** back yard from an upstairs window, where she would be **ironing** or something like that. And she was usually just sort of keeping an eye on, and if there was any puffs of smoke out there, she ' d lean out and admonish us all not to blow our eyes out. That was her â You know, that was kind of **the** worst thing that could happen to us. That ' s why I thought, as long as I don ' t blow my eyes out... I may not care about **the** fact that it ’ s prohibited from heating this solution. I ’ m going to do it carefully, but I ’ ll do it. It ’ s like anything else that ’ s prohibited : you do it behind **the** garage. So, I went to **the** drug store and I tried to buy some potassium perchlorate and it wasn ’ t unreasonable then for a kid to walk into a drug store and buy chemicals. Nowadays, it ’ s no ma ’ am, check your shoes. And like â But then it wasn ’ t â they didn ’ t have any, but **the** guy had â I said, what kind of salts of potassium do you have? You know. And he had potassium nitrate. And I said, that might do **the** same thing, whatever it is. I ’ m sure it ’ s got to do with rockets or it wouldn ’ t be in that manual. And so I â I did some experiments. You know, I started off with little tiny amounts of potassium nitrate and sugar, which was readily available, and I mixed it in different proportions, and I tried to light it on fire. Just to see what would happen, if you mixed it together. And it â they burned. It burned kind of slow, but it made a nice smell, compared to other rocket fuels I had tried, that all had sulfur in them. And, it smelt like burnt candy. And then I tried **the** melting business, and I melted it. And then it melted into a little sort of syrupy liquid, brown. And then it cooled down to a brick-hard substance, that when you lit that, it went off like a bat. I mean, **the** little bowl of that stuff that had cooled down â you ’ d light it, and it would just start dancing around **the** yard. And I said, there is a way to get a frog up to where he wants to go. So I started developing â you know, George ’ s dad had a lot of help. I just had my brother. But I â it took me about â it took me about, I ’ d say, six months to finally figure out all **the** little things. There ’ s a lot of little things involved in making a rocket that it will actually work, even after you have **the** fuel. But you do it, by â what I justâ you know, you do experiments, and you write down things sometimes, you make observations, you know. And then you slowly build up a theory of how this stuff works. And it was â I was following all **the** rules. I didn ’ t know what **the** rules were, I ’ m a natural born scientist, I guess, or some kind of a throwback to **the** 17th century, whatever. But at any rate, we finally did have a **device** that would reproduceably put a frog out of sight and get him back alive. And we had not â I mean, we weren ’ t frightened by it. We should have been, because it made a lot of smoke and it made a lot of noise, and it was powerful, you know. And once in a while, they would blow up. But I wasn ’ t worried, by **the** way, about, you know, **the explosion** causing **the** destruction of **the planet**. I hadn ’ t heard about **the** 10 ways that we should be afraid of **the** â By **the** way, I could have thought, I ’ d better not do this because they say not to, you know. And I ’ d better get permission from **the** government. If I ’ d have waited around for that, I would have never â **the** frog would have died, you know. At any rate, I bring it up because it ’ s a good story, and he said, tell personal things, you know, and that ’ s a personal â I was going to tell you about **the** first night that I met my wife, but that would be too personal, wouldn ’ t it. So, so I ’ ve got something else that ’ s not personal. But that... process is what I think of as science, see, where you start with some idea, and then instead of, like, looking up, every authority that you ’ ve ever heard of I â sometimes you do that, if you ’ re going to write a paper later, you want to figure out who else has worked on it. But in **the** actual process, you get an idea â like, when I got **the** idea one night that I could amplify DNA with two oligonucleotides, and I could make lots of **copies** of some little piece of DNA, you know, **the** thinking for that was about 20 minutes while I was driving my car, and then instead of going â I went back and I did talk to people about it, but if I ’ d listened to what I heard from all my friends who were molecular biologists â I would have abandoned it. You know, if I had gone back looking for an authority figure who could tell me if it would work or not, he would have said, no, it probably won ’ t. Because **the** results of it were so spectacular that if it worked it was going to change everybody ’ s goddamn way of doing molecular **biology**. Nobody wants a chemist to come in and poke around in their stuff like that and change things. But if you go to authority, and you always don ’ t â you don ’ t always get **the** right answer, see. But I knew, you ’ d go into **the** lab and you ’ d try to make it work yourself. And then you ’ re **the** authority, and you can say, I know it works, because right there in that tube is where it happened, and here, on this gel, there ’ s a little band there that I know that ’ s DNA, and that ’ s **the** DNA I wanted to amplify, so there! So it does work. You know, that ’ s how you do science. And then you say, well, what can make it work better? And then you figure out better and better ways to do it. But you always work from, from like, facts that you have made available to you by doing experiments : things that you could do on a stage. And no tricky shit behind **the** thing. I mean, it ’ s all â you ’ ve got to be very honest with what you ’ re doing if it really is going to work. I mean, you can ’ t make up results, and then do another experiment based on that one. So you have to be honest. And I ’ m basically honest. I have a fairly bad memory, and dishonesty would always get me in trouble, if I, like â so I ’ ve just sort of been naturally honest and naturally inquisitive, and that sort of leads to that kind of science. Now, let ’ s see... I ’ ve got another five minutes, right? OK. All scientists aren ’ t like that. You know â and there is a lot â There is a lot â a lot has been going on since Isaac Newton and all that stuff happened. One of **the** things that happened right around World War II in that same time period before, and as sure as hell afterwards, government got â realized that scientists aren ’ t strange dudes that, you know, hide in ivory towers and do ridiculous things with test tube. Scientists, you know, made World War II as we know it quite possible. They made faster things. They made bigger guns to shoot them down with. You know, they made drugs to give **the** pilots if they were broken up in **the** process. They made all kinds of â and then finally one giant bomb to end **the** whole thing, right? And everybody stepped back a little and said, you know, we ought to invest in this shit, because whoever has got **the** most of these people working in **the** places is going to have a dominant position, at least in **the** military, and probably in all kind of economic ways. And they got involved in it, and **the** scientific and industrial establishment was born, and out of that came a lot of scientists who were in there for **the** money, you know, because it was suddenly available. And they weren ’ t **the** curious little boys that liked to put frogs up in **the** air. They were **the** same people that later went in to medical school, you know, because there was money in it, you know. I mean, later, then they all got into business â I mean, there are waves of â going into your high school, person saying, you want to be rich, you know, be a scientist. You know, not anymore. You want to be rich, you be a businessman. But a lot of people got in it for **the** money and **the** power and **the** travel. That ’ s back when travel was easy. And those people don ’ t think â they don ’ t â they don ’ t always tell you **the** truth, you know. There is nothing in their contract, in fact, that makes it to their advantage always, to tell you **the** truth. And **the** people I ’ m talking about are people that like â they say that they ’ re a member of **the** committee called, say, **the** Inter- Governmental Panel on Climate Change. And they â and they have these big meetings where they try to figure out how we ’ re going to â how we ’ re going to continually prove that **the planet** is getting warmer, when that ’ s actually contrary to most people ’ s sensations. I mean, if you actually measure **the** temperature over a period â I mean, **the** temperature has been measured now pretty carefully for about 50, 60 years â longer than that it ’ s been measured, but in really nice, precise ways, and records have been kept for 50 or 60 years, and in fact, **the** temperature hadn ’ t really gone up. It ’ s like, **the** average temperature has gone up a tiny little bit, because **the** nighttime temperatures at **the** weather stations have come up just a little bit. But there ' s a good explanation for that. And it ' s that **the** weather stations are all built outside of town, where **the** airport was, and now **the** town ' s moved out there, there ' s concrete all around and they call it **the** skyline effect. And most responsible people that measure temperatures realize you have to shield your measuring **device** from that. And even then, you know, because **the** buildings get warm in **the** daytime, and they keep it a little warmer at night. So **the** temperature has been, sort of, **inching** up. It should have been. But not a lot. Not like, you know â **the** first guy â **the** first guy that got **the** idea that we ' re going to fry ourselves here, actually, he didn ' t think of it that way. His name was Sven Arrhenius. He was Swedish, and he said, if you double **the** CO2 level in **the** atmosphere, which he thought might â this is in 1900 â **the** temperature ought to go up about 5. 5 degrees, he **calculated**. He was thinking of **the** earth as, kind of like, you know, like a completely insulated thing with no stuff in it, really, just energy coming down, energy leaving. And so he came up with this theory, and he said, this will be cool, because it ' ll be a longer growing season in Sweden, you know, and **the** surfers liked it, **the** surfers thought, that ' s a cool idea, because it ' s pretty cold in **the** ocean sometimes, and â but a lot of other people later on started thinking it would be bad, you know. But nobody actually demonstrated it, right? I mean, **the** temperature as measured â and you can find this on our wonderful Internet, you just go and look for all NASAs records, and all **the** Weather Bureau ' s records, and you ' ll look at it yourself, and you ' ll see, **the** temperature has just â **the** nighttime temperature measured on **the surface** of **the planet** has gone up a tiny little bit. So if you just average that and **the** daytime temperature, it looks like it went up about. 7 degrees in this century. But in fact, it was just coming up â it was **the** nighttime ; **the** daytime temperatures didn ' t go up. So â and Arrhenius ' theory â and all **the** global warmers think â they would say, yeah, it should go up in **the** daytime, too, if it ' s **the greenhouse** effect. Now, people like things that have, like, names like that, that they can envision it, right? I mean â but people don ' t like things like this, so â most â I mean, you don ' t get all excited about things like **the** actual evidence, you know, which would be evidence for strengthening of **the** tropical circulation in **the** 1990s. It ' s a paper that came out in February, and most of you probably hadn ' t heard about it."Evidence for Large Decadal Variability in **the** Tropical Mean Radiative Energy Budget."Excuse me. Those papers were published by NASA, and some scientists at Columbia, and Viliki and a whole bunch of people, Princeton. And those two papers came out in Science Magazine, February **the** first, and these â **the** conclusion in both of these papers, and in also **the** Science editor ' s, like, descriptions of these papers, for, you know, for **the** quickie, is that our theories about global warming are completely wrong. I mean, what these guys were doing, and this is what â **the** NASA people have been saying this for a long time. They say, if you measure **the** temperature of **the** atmosphere, it isn ' t going up â it ' s not going up at all. We ' ve doing it very carefully now for 20 years, from satellites, and it isn ' t going up. And in this paper, they show something much more striking, and that was that they did what they call a radiation â and I ' m not going to go into **the** details of it, actually it ' s quite complicated, but it isn ' t as complicated as they might make you think it is by **the** words they use in those papers. If you really get down to it, they say, **the** sun puts out a certain amount of energy â we know how much that is â it falls on **the** earth, **the** earth gives back a certain amount. When it gets warm it generates â it makes redder energy â I mean, like infra-red, like something that ' s warm gives off infra-red. **The** whole business of **the** global warming â trash, really, is that â if **the** â if there ' s too much CO2 in **the** atmosphere, **the** heat that ' s trying to escape won ' t be able to get out. But **the** heat coming from **the** sun, which is mostly down in **the** â it ' s like 350 nanometers, which is where it ' s centered â that goes right through CO2. So you still get heated, but you don ' t dissipate any. Well, these guys measured all of those things. I mean, you can talk about that stuff, and you can write these large reports, and you can get government money to do it, but these â they actually measured it, and it turns out that in **the** last 10 years â that ' s why they say"decadal"there â that **the** energy â that **the** level of what they call"imbalance"has been way **the** hell over what was expected. Like, **the** amount of imbalance â meaning, heat ' s coming in and it ' s not going out that you would get from having double **the** CO2, which we ' re not anywhere near that, by **the** way. But if we did, in 2025 or something, have double **the** CO2 as we had in 1900, they say it would be increase **the** energy budget by about â in other words, one watt per square centimeter more would be coming in than going out. So **the planet** should get warmer. Well, they found out in this study â these two studies by two different teams â that five and a half watts per square meter had been coming in from 1998, 1999, and **the** place didn ' t get warmer. So **the** theory ' s kaput â it ' s nothing. These papers should have been called,"**The** End to **the** Global Warming Fiasco,"you know. They ' re concerned, and you can tell they have very guarded conclusions in these papers, because they ' re talking about big laboratories that are funded by lots of money and by scared people. You know, if they said, you know what? There isn ' t a problem with global warming any longer, so we can â you know, they ' re funding. And if you start a grant request with something like that, and say, global warming obviously hadn ' t happened... if they â if they â if they actually â if they actually said that, I ' m getting out. I ' ll stand up too, and â They have to say that. They had to be very cautious. But what I ' m saying is, you can be delighted, because **the** editor of Science, who is no dummy, and both of these fairly professional â really professional teams, have really come to **the** same conclusion and in **the** bottom lines in their papers they have to say, what this means is, that what we ' ve been thinking, was **the** global circulation model that we predict that **the** earth is going to get overheated that it ' s all wrong. It ' s wrong by a large factor. It ' s not by a small one. They just â they just misinterpreted **the** fact that **the** earth â there ' s obviously some mechanisms going on that nobody knew about, because **the** heat ' s coming in and it isn ' t getting warmer. So **the planet** is a pretty amazing thing, you know, it ' s big and horrible â and big and wonderful, and it does all kinds of things we don ' t know anything about. So I mean, **the** reason I put those things all together, OK, here ' s **the** way you ' re supposed to do science â some science is done for other reasons, and just curiosity. And there ' s a lot of things like global warming, and ozone hole and you know, a whole bunch of scientific public issues, that if you ' re interested in them, then you have to get down **the** details, and read **the** papers called,"Large Decadal Variability in **the**..."You have to figure out what all those words mean. And if you just listen to **the** guys who are hyping those issues, and making a lot of money out of it, you ' ll be misinformed, and you ' ll be worrying about **the** wrong things. Remember **the** 10 things that are going to get you. **The** â one of them â And **the asteroids** is **the** one I really agree with there. I mean, you ' ve got to watch out for **asteroids**. OK, thank you for having me here.

Text Sample 19: POS Dim 1 – 2002 – Score 21.00 – 2002/t000269.txt

Welcome. If I could have **the** first **slide**, please? Contrary to calculations made by some engineers, bees can fly, dolphins can swim, and geckos can even climb up **the** smoothest **surfaces**. Now, what I want to do, in **the** short time I have, is to try to allow each of you to experience **the** thrill of revealing nature ' s design. I get to do this all **the** time, and it ' s just incredible. I want to try to share just a little bit of that with you in this presentation. **The** challenge of looking at nature ' s designs — and I ' ll tell you **the** way that we perceive it, and **the** way we ' ve used it. **The** challenge, of course, is to answer this question : what permits this extraordinary performance of animals that allows them basically to go anywhere? And if we could figure that out, how can we implement those designs? Well, many biologists will tell engineers, and others, organisms have millions of years to get it right ; they ' re spectacular ; they can do everything wonderfully well. So, **the** answer is bio-mimicry : just **copy** nature directly. We know from working on animals that **the** truth is that ' s exactly what you don ' t want to do — because **evolution** works on **the** just-good-enough principle, not on a perfecting principle. And **the** constraints in building any organism, when you look at it, are really severe. Natural technologies have incredible constraints. Think about it. If you were an engineer and I told you that you had to build an automobile, but it had to start off to be this big, then it had to grow to be full size and had to work every step along **the** way. Or think about **the** fact that if you build an automobile, I ' ll tell you that you also — inside it — have to put a factory that allows you to make another automobile. And you can absolutely never, absolutely never, because of history and **the** inherited plan, start with a clean slate. So, organisms have this important history. Really **evolution** works more like a tinkerer than an engineer. And this is really important when you begin to look at animals. Instead, we believe you need to be inspired by **biology**. You need to discover **the** general principles of nature, and then use these analogies when they ' re advantageous. This is a real challenge to do this, because animals, when you start to really look inside them — how they work — appear hopelessly complex. There ' s no detailed history of **the** design plans, you can ' t go look it up anywhere. They have way too many motions for their joints, too many muscles. Even **the** simplest animal we think of, something like an insect, and they have more neurons and connections than you can imagine. How can you make sense of this? Well, we believed — and we hypothesized — that one way animals could work simply, is if **the** control of their movements tended to be built into their bodies themselves. What we discovered was that two-, four-, six- and eight-legged animals all produce **the** same forces on **the** ground when they move. They all work like this kangaroo, they bounce. And they can be modeled by a spring-mass system that we call **the spring** mass system because we ' re biomechanists. It ' s actually a pogo stick. They all produce **the** pattern of a pogo stick. How is that true? Well, a human, one of your legs works like two legs of a trotting dog, or works like three legs, together as one, of a trotting insect, or four legs as one of a trotting **crab**. And then they alternate in their propulsion, but **the** patterns are all **the** same. Almost every organism we ' ve looked at this way — you ' ll see next week, I ' ll give you a hint, there ' ll be an article coming out that says that really big things like T. rex probably couldn ' t do this, but you ' ll see that next week. Now, what ' s interesting is **the** animals, then — we said — bounce along **the** vertical plane this way, and in our collaborations with Pixar, in"A Bug ' s Life,"we discussed **the** bipedal nature of **the** characters of **the** ants. And we told them, of course, they move in another plane as well. And they asked us this question. They say,"Why model just in **the** sagittal plane or **the** vertical plane, when you ' re telling us these animals are moving in **the** **horizontal** plane?"This is a good question. Nobody in **biology** ever modeled it this way. We took their advice and we modeled **the** animals moving in **the horizontal** plane as well. We took their three legs, we collapsed them down as one. We got some of **the** best mathematicians in **the** world from Princeton to work on this problem. And we were able to create a model where animals are not only bouncing up and down, but they ’ re also bouncing side to side at **the** same time. And many organisms fit this kind of pattern. Now, why is this important to have this model? Because it ’ s very interesting. When you take this model and you perturb it, you give it a push, as it bumps into something, it self-stabilizes, with no brain or no reflexes, just by **the** structure alone. It ’ s a beautiful model. Let ’ s look at **the** mathematics. That ’ s enough! **The** animals, when you look at them running, appear to be self-stabilizing like this, using basically springy legs. That is, **the** legs can do computations on their own ; **the** control algorithms, in a sense, are embedded in **the** form of **the** animal itself. Why haven ’ t we been more inspired by nature and these kinds of discoveries? Well, I would argue that human technologies are really different from natural technologies, at least they have been so far. Think about **the** typical kind of robot that you see. Human technologies have tended to be large, flat, with right angles, stiff, made of metal. They have rolling **devices** and axles. There are very few motors, very few sensors. Whereas nature tends to be small, and curved, and it bends and twists, and has legs instead, and appendages, and has many muscles and many, many sensors. So it ’ s a very different design. However, what ’ s changing, what ’ s really exciting — and I ’ ll show you some of that next — is that as human technology takes on more of **the** characteristics of nature, then nature really can become a much more useful teacher. And here ’ s one example that ’ s really exciting. This is a collaboration we have with Stanford. And they developed this new **technique**, called Shape Deposition Manufacturing. It ’ s a **technique** where they can mix materials together and mold any shape that they like, and put in **the** material properties. They can embed sensors and actuators right in **the** form itself. For example, here ’ s a leg : **the** clear part is stiff, **the** white part is compliant, and you don ’ t need any axles there or anything. It just bends by itself beautifully. So, you can put those properties in. It inspired them to show off this design by producing a little robot they named Sprawl. Our work has also inspired another robot, a biologically inspired bouncing robot, from **the** University of Michigan and McGill named RHex, for robot hexapod, and this one ’ s autonomous. Let ’ s go to **the** video, and let me show you some of these animals moving and then some of **the** simple robots that have been inspired by our discoveries. Here ’ s what some of you did this morning, although you did it outside, not on a treadmill. Here ’ s what we do. This is a death ’ s head **cockroach**. This is an American **cockroach** you think you don ’ t have in your kitchen. This is an eight-legged scorpion, six-legged ant, forty-four-legged centipede. Now, I said all these animals are sort of working like pogo sticks — they ’ re bouncing along as they move. And you can see that in this ghost **crab**, from **the** beaches of Panama and North Carolina. It goes up to four meters per second when it runs. It actually leaps into **the** air, and has aerial phases when it does it, like a horse, and you ’ ll see it ’ s bouncing here. What we discovered is whether you look at **the** leg of a human like Richard, or a **cockroach**, or a **crab**, or a kangaroo, **the relative** leg stiffness of that **spring** is **the** same for everything we ’ ve seen so far. Now, what good are springy legs then? What can they do? Well, we wanted to see if they allowed **the** animals to have greater stability and maneuverability. So, we built a terrain that had obstacles three times **the** hip height of **the** animals that we ’ re looking at. And we were certain they couldn ’ t do this. And here ’ s what they did. **The** animal ran over it and it didn ’ t even slow down! It didn ’ t decrease its preferred speed at all. We couldn ’ t believe that it could do this.

It said to us that if you could build a robot with very simple, springy legs, you could make it as maneuverable as any that ’ s ever been built. Here ’ s **the** first example of that. This is **the** Stanford Shape Deposition Manufactured robot, named Sprawl. It has six legs — there are **the** tuned, springy legs. It moves in a gait that an insect uses, and here it is going on **the** treadmill. Now, what ’ s important about this robot, compared to other robots, is that it can ’ t see anything, it can ’ t feel anything, it doesn ’ t have a brain, yet it can maneuver over these obstacles without any difficulty whatsoever. It ’ s this **technique** of building **the** properties into **the** form. This is a graduate student. This is what he ’ s doing to his thesis project — very robust, if a graduate student does that to his thesis project. This is from McGill and University of Michigan. This is **the** RHex, making its first outing in a demo. Same principle : it only has six moving parts, six motors, but it has springy, tuned legs. It moves in **the** gait of **the** insect. It has **the** middle leg moving in synchrony with **the** front, and **the** hind leg on **the** other side. Sort of an alternating tripod, and they can negotiate obstacles just like **the** animal. Robert Full : It ’ ll go on different **surfaces** — here ’ s sand — although we haven ’ t perfected **the** feet yet, but I ’ ll talk about that later. Here ’ s RHex entering **the** woods. Again, this robot can ’ t see anything, it can ’ t feel anything, it has no brain. It ’ s just working with a tuned mechanical system, with very simple parts, but inspired from **the** fundamental dynamics of **the** animal. RF : Here ’ s it going down a pathway. I presented this to **the** jet propulsion lab at NASA, and they said that they had no ability to go down craters to look for ice, and life, ultimately, on Mars. And he said — especially with legged-robots, because they ’ re way too complicated. Nothing can do that. And I talk next. I showed them this video with **the** simple design of RHex here. And just to convince them we should go to Mars in 2011, I tinted **the** video orange just to give them **the** sense of being on Mars. Another reason why animals have extraordinary performance, and can go anywhere, is because they have an effective interaction with **the** environment. **The** animal I ’ m going to show you, that we studied to look at this, is **the** gecko. We have one here and notice its position. It ’ s holding on. Now I ’ m going to challenge you. I ’ m going show you a video. One of **the** animals is going to be running on **the** level, and **the** other one ’ s going to be running up a wall. Which one ’ s which? They ’ re going at a meter a second. How many think **the** one on **the** **left** is running up **the** wall? **Okay**. **The** point is it ’ s really hard to tell, isn ’ t it? It ’ s incredible, we looked at students do this and they couldn ’ t tell. They can run up a wall at a meter a second, 15 steps per second, and they look like they ’ re running on **the** level. How do they do this? It ’ s just phenomenal. **The** one on **the** right was going up **the** hill. How do they do this? They have bizarre toes. They have toes that uncurl like party favors when you blow them out, and then peel off **the surface**, like tape. Like if we had a piece of tape now, we ’ d peel it this way. They do this with their toes. It ’ s bizarre! This peeling inspired iRobot — that we work with — to build Mecho-Geckos. Here ’ s a legged version and a tractor version, or a bulldozer version. Let ’ s see some of **the** geckos move with some video, and then I ’ ll show you a little bit of a **clip** of **the** robots. Here ’ s **the** gecko running up a vertical **surface**. There it goes, in real time. There it goes again. Obviously, we have to slow this down a little bit. You can ’ t use regular cameras. You have to take 1, 000 pictures per second to see this. And here ’ s some video at 1, 000 frames per second. Now, I want you to look at **the** animal ’ s back. Do you see how much it ’ s bending like that? We can ’ t figure that out — that ’ s an unsolved mystery. We don ’ t know how it works. If you have a son or a daughter that wants to come to Berkeley, come to my lab and we ’ ll figure this out. **Okay**, send them to Berkeley because that ’ s **the** next thing I want to do. Here ’ s **the** gecko mill. It ’ s a see-through treadmill with a see-through treadmill belt, so we can watch **the** animal ’ s feet, and videotape them through **the** treadmill belt, to see how they move. Here ’ s **the** animal that we have here, running on a vertical **surface**. Pick a foot and try to watch a toe, and see if you can see what **the** animal ’ s doing. See it uncurl and then peel these toes. It can do this in 14 milliseconds. It ’ s unbelievable. Here are **the** robots that they inspire, **the** Mecho-Geckos from iRobot. First we ’ ll see **the** animals toes peeling — look at that. And here ’ s **the** peeling action of **the** Mecho-Gecko. It uses a pressure-sensitive adhesive to do it. Peeling in **the** animal. Peeling in **the** Mecho-Gecko — that allows them climb autonomously. Can go on **the** flat **surface**, transition to a wall, and then go onto a ceiling. There ’ s **the** bulldozer version. Now, it doesn ’ t use pressure-sensitive glue. **The** animal does not use that. But that ’ s what we ’ re limited to, at **the** moment. What does **the** animal do? **The** animal has weird toes. And if you look at **the** toes, they have these little leaves there, and if you blow them up and zoom in, you ’ ll see that ’ s there ’ s little striations in these leaves. And if you zoom in 270 times, you ’ ll see it looks like a rug. And if you blow that up, and zoom in 900 times, you see there are hairs there, tiny hairs. And if you look carefully, those tiny hairs have striations. And if you zoom in on those 30, 000 times, you ’ ll see each hair has split ends. And if you blow those up, they have these little structures on **the** end. **The** smallest branch of **the** hairs looks like spatulae, and an animal like that has one billion of these nano-size split ends, to get very close to **the surface**. In fact, there ’ s **the diameter** of your hair — a gecko has two million of these, and each hair has 100 to 1, 000 split ends. Think of **the** contact of that that ’ s possible. We were fortunate to work with another group at Stanford that built us a special manned sensor, that we were able to measure **the** force of an individual hair. Here ’ s an individual hair with a little split end there. When we measured **the** forces, they were enormous. They were so large that a patch of hairs about this size — **the** gecko ’ s foot could support **the** weight of a small child, about 40 **pounds**, easily. Now, how do they do it? We ’ ve recently discovered this. Do they do it by friction? No, force is too low. Do they do it by electrostatics? No, you can change **the** charge — they still hold on. Do they do it by interlocking? That ’ s kind of a like a Velcro-like thing. No, you can put them on molecular smooth **surfaces** — they don ’ t do it. How about suction? They stick on in a vacuum. How about **wet** adhesion? Or capillary adhesion? They don ’ t have any glue, and they even stick under water just fine. If you put their foot under water, they grab on. How do they do it then? Believe it or not, they grab on by intermolecular forces, by Van der Waals forces. You know, you probably had this a long time ago in **chemistry**, where you had these two atoms, they ’ re close together, and **the** electrons are moving around. That tiny force is sufficient to allow them to do that because it ’ s added up so many times with these small structures. What we ’ re doing is, we ’ re taking that inspiration of **the** hairs, and with another colleague at Berkeley, we ’ re manufacturing them. And just recently we ’ ve made a breakthrough, where we now believe we ’ re going to be able to create **the** first synthetic, self-cleaning, dry adhesive. Many companies are interested in this. We also presented to Nike even. We ’ ll see where this goes. We were so excited about this that we realized that that small-size scale — and where everything gets sticky, and gravity doesn ’ t matter anymore — we needed to look at ants and their feet, because one of my other colleagues at Berkeley has built a six-millimeter silicone robot with legs. But it gets stuck. It doesn ’ t move very well. But **the** ants do, and we ’ ll figure out why, so that ultimately we ’ ll make this move. And imagine : you ’ re going to be able to have swarms of these six-millimeter robots available to run around. Where ’ s this going? I think you can see it already. Clearly,

the Internet is already having eyes and ears, you have **web** cams and so forth. But it ’ s going to also have legs and hands. You ’ re going to be able to do programmable work through these kinds of robots, so that you can run, fly and swim anywhere. We saw David Kelly is at **the** beginning of that with his **fish**. So, in conclusion, I think **the** message is clear. If you need a message, if nature ’ s not enough, if you care about **search** and rescue, or mine clearance, or medicine, or **the** various things we ’ re working on, we must preserve nature ’ s designs, otherwise these secrets will be lost forever. Thank you.

Text Sample 20: POS Dim 1 – 2007 – Score 34.00 – 2007/t000298.txt

As an architect you design for **the** present, with an awareness of **the** past, for a future which is essentially unknown. **The** green agenda is probably **the** most important agenda and issue of **the** day. And I ’ d like to share some experience over **the** last 40 years â we celebrate our fortieth anniversary this year â and to explore and to touch on some observations about **the** nature of **sustainability**. How far you can anticipate, what follows from it, what are **the** threats, what are **the** possibilities, **the** challenges, **the** opportunities? I think that â I ’ ve said in **the** past, many, many years ago, before anybody even invented **the** concept of a green agenda, that it wasn ’ t about fashion â it was about survival. But what I never said, and what I ’ m really going to make **the** point is, that really, green is cool. I mean, all **the** projects which have, in some way, been inspired by that agenda are about a celebratory lifestyle, in a way celebrating **the** places and **the** spaces which determine **the** quality of life. I rarely actually quote anything, so I ’ m going to try and find a piece of paper if I can, [in] which somebody, at **the** end of last year, ventured **the** thought about what for that individual, as a kind of important observer, analyst, writer â a guy called Thomas Friedman, who wrote in **the** Herald Tribune, about 2006. He said,”I think **the** most important thing to happen in 2006 was that living and thinking green hit Main Street. We reached a tipping point this year where living, acting, designing, investing and manufacturing green came to be understood by a critical mass of citizens, entrepreneurs and officials as **the** most patriotic, capitalistic, geo-political and competitive thing they could do. Hence my motto : green is **the** new red, white and **blue**.”And I asked myself, in a way, looking back,”When did that kind of awareness of **the planet** and its fragility first appear?”And I think it was July 20, 1969, when, for **the** first time, man could look back at **planet** Earth. And, in a way, it was Buckminster Fuller who coined that phrase. And before **the** kind of collapse of **the** communist system, I was privileged to meet a lot of cosmonauts in Space City and other places in Russia. And interestingly, as I think back, they were **the** first true environmentalists. They were filled with a kind of pioneering passion, fired about **the** problems of **the** Aral Sea. And at that period it was â in a way, a number of things were happening. Buckminster Fuller was **the** kind of green guru â again, a word that had not been coined. He was a design scientist, if you like, a poet, but he foresaw all **the** things that are happening now. It ’ s another subject. It ’ s another conversation. You can go back to his writings : it ’ s quite extraordinary. It was at that time, with an awareness fired by Bucky ’ s prophecies, his concerns as a citizen, as a kind of citizen of **the planet**, that influenced my thinking and what we were doing at that time. And it ’ s a number of projects. I select this one because it was 1973, and it was a master plan for one of **the** Canary Islands. And this probably coincided with **the** time when you had **the planet** Earth ’ s sourcebook, and you had **the** hippie movement. And there are some of those qualities in this drawing, which seeks to **sum** up **the** recommendations.

And all **the** components are there which are now in common parlance, in our vocabulary, you know, 30-odd years later : wind energy, recycling, biomass, solar cells. And in parallel at that time, there was a very kind of exclusive design club. People who were really design conscious were inspired by **the** work of Dieter Rams, and **the objects** that he would create for **the** company called Braun. This is going back **the** mid- ’ 50s, ’ 60s. And despite Bucky ’ s prophecies that everything would be miniaturized and technology would make an incredible style â access to comfort, to amenities â it was very, very difficult to imagine that everything that we see in this image, would be very, very stylishly packaged. And that, and more besides, would be in **the** palm of your hand. And I think that that digital revolution now is coming to **the** point where, as **the** virtual world, which brings so many people together here, finally connects with **the** physical world, there is **the** reality that that has become humanized, so that digital world has all **the** friendliness, all **the** immediacy, **the** orientation of **the** analog world. Probably **summed** up in a way by **the** stylish or alternative available here, as we generously had gifted at lunchtime, **the** [**unclear**], which is a further kind of development â and again, inspired by **the** incredible sort of sensual feel. A very, very beautiful **object**. So, something which in [**the**] ’ 50s, ’ 60s was very exclusive has now become, interestingly, quite inclusive. And **the** reference to **the** iPod as iconic, and in a way evocative of performance, delivery â quite interesting that [in] **the** beginning of **the** year 2007, **the** Financial Times commented that **the** Detroit companies envy **the** halo effect that Toyota has gained from **the** Prius as **the** hybrid, energy-conscious vehicle, which rivals **the** iPod as an iconic product. And I think it ’ s very tempting to, in a way, seduce ourselves â as architects, or anybody involved with **the** design process â that **the** answer to our problems lies with buildings. Buildings are important, but they ’ re only a component of a much bigger picture. In other words, as I might seek to demonstrate, if you could achieve **the** impossible, **the** equivalent of perpetual motion, you could design a carbon-free house, for example. That would be **the** answer. Unfortunately, it ’ s not **the** answer. It ’ s only **the** beginning of **the** problem. You cannot separate **the** buildings out from **the** infrastructure of cites and **the** mobility of transit. For example, if, in that Bucky-inspired phrase, we draw back and we look at **planet** Earth, and we take a kind of typical, industrialized society, then **the** energy consumed would be split between **the** buildings, 44 percent, transport, 34 percent, and industry. But again, that only shows part of **the** picture. If you looked at **the** buildings together with **the** associated transport, in other words, **the** transport of people, which is 26 percent, then 70 percent of **the** energy **consumption** is influenced by **the** way that our cites and infrastructure work together. So **the** problems of **sustainability** cannot be separated from **the** nature of **the** cities, of which **the** buildings are a part. For example, if you take, and you make a comparison between a recent kind of city, what I ’ ll call, simplistically, a North American city â and Detroit is not a bad example, it is very car dependent. **The** city goes out in annular rings, consuming more and more green space, and more and more roads, and more and more energy in **the** transport of people between **the** city center â which again, **the** city center, as it becomes deprived of **the** living and just becomes commercial, again becomes dead. If you compared Detroit with a city of a Northern European example â and Munich is not a bad example of that, with **the** greater dependence on walking and cycling â then a city which is really only twice as dense, is only using one-tenth of **the** energy. In other words, you take these comparable examples and **the** energy leap is enormous. So basically, if you wanted to generalize, you can demonstrate that as **the** density increases along **the** bottom there, that **the** energy consumed reduces dramatically. Of course you can ’ t separate this out from issues like social diversity, mass transit, **the** ability to be able to walk a convenient distance, **the** quality of civic spaces. But again, you can see Detroit, in **yellow** at **the** top, extraordinary **consumption**, down below Copenhagen. And Copenhagen, although it ’ s a dense city, is not dense compared with **the** really dense cities. In **the** year 2000, a rather interesting thing happened. You had for **the** first time mega-cities, [of] 5 million or more, which were **occurring** in **the** developing world. And now, out of typically 46 cities, 33 of those mega-cities are in **the** developing world. So you have to ask yourself â□□ **the** environmental impact of, for example, China or India. If you take China, and you just take Beijing, you can see on that traffic system, and **the** pollution associated with **the consumption** of energy as **the** cars expand at **the** price of **the** bicycles. In other words, if you put onto **the** roads, as is currently happening, 1, 000 new cars every day â□□ statistically, it ’ s **the** biggest booming auto market in **the** world â□□ and **the** half a billion bicycles serving one and a third billion people are reducing. And that urbanization is extraordinary, accelerated pace. So, if we think of **the** transition in our society of **the** movement from **the** land to **the** cities, which took 200 years, then that same process is happening in 20 years. In other words, it is accelerating by a factor of 10. And quite interestingly, over something like a 60-year period, we ’ re seeing **the** doubling in life expectancy, over that period where **the** urbanization has trebled. If I pull back from that global picture, and I look at **the** implication over a similar period of time in terms of **the** technology â□□ which, as a tool, is a tool for designers, and I cite our own experience as a company, and I just illustrate that by a small **selection** of projects â□□ then how do you measure that change of technology? How does it affect **the** design of buildings? And particularly, how can it lead to **the** creation of buildings which consume less energy, create less pollution and are more socially responsible? That story, in terms of buildings, started in **the** late ’ 60s, early ’ 70s. **The** one example I take is a corporate headquarters for a company called Willis and Faber, in a small market town in **the** northeast of England, commuting distance with London. And here, **the** first thing you can see is that this building, **the** roof is a very warm kind of overcoat blanket, a kind of insulating garden, which is also about **the** celebration of public space. In other words, for this community, they have this garden in **the** sky. So **the** humanistic ideal is very, very strong in all this work, encapsulated perhaps by one of my early sketches here, where you can see greenery, you can see sunlight, you have a connection with nature. And nature is part of **the** generator, **the** driver for this building. And symbolically, **the** colors of **the** interior are green and **yellow**. It has facilities like swimming pools, it has flextime, it has a social heart, a space, you have contact with nature. Now this was 1973. In 2001, this building received an award. And **the** award was about a celebration for a building which had been in use over a long period of time. And **the** people who ’ d created it came back : **the** project managers, **the** company chairmen then. And they were saying, you know,”**The** architects, Norman was always going on about designing for **the** future, and you know, it didn ’ t seem to cost us any more. So we humored him, we kept him happy.”**The** image at **the** top, what it doesn ’ t â□□ if you look at it in detail, really what it is saying is you can wire this building. This building was wired for change. So, in 1975, **the** image there is of typewriters. And when **the** photograph was taken, it ’ s word processors. And what they were saying on this occasion was that our competitors had to build new buildings for **the** new technology. We were fortunate, because in a way our building was future-proofed. It anticipated change, even though those changes were not known. Round about that design period leading up to this building, I did a sketch, which we pulled out of **the** archive recently. And I was saying, and I wrote,”But we don ’ t have **the** time, and we really don ’ t have **the** immediate expertise at a technical level."In other words, we didn ' t have **the** technology to do what would be really interesting on that building. And that would be to create a kind of three- dimensional bubble â a really interesting overcoat that would naturally ventilate, would breathe and would seriously reduce **the** energy loads. Notwithstanding **the** fact that **the** building, as a green building, is very much a pioneering building. And if I fast-forward in time, what is interesting is that **the** technology is now available and celebratory. **The** library of **the** Free University, which opened last year, is an example of that. And again, **the** transition from one of **the** many thousands of sketches and **computer** images to **the** reality. And a combination of **devices** here, **the** kind of heavy mass concrete of these book **stacks**, and **the** way in which that is enclosed by this skin, which enables **the** building to be ventilated, to consume dramatically less energy, and where it ' s really working with **the** forces of nature. And what is interesting is that this is hugely popular by **the** people who use it. Again, coming back to that thing about **the** lifestyle, and in a way, **the** ecological agenda is very much at one with **the** spirit. So it ' s not a kind of sacrifice, quite **the** reverse. I think it ' s a great â it ' s a celebration. And you can measure **the** performance, in terms of energy **consumption**, of that building against a typical library. If I show another aspect of that technology then, in a completely different context â this apartment building in **the** Alps in Switzerland. Prefabricated from **the** most traditional of materials, but that material â because of **the** technology, **the** computing ability, **the** ability to prefabricate, make high-performance components out of timber â very much at **the** cutting edge. And just to give a sort of glimpse of that technology, **the** ability to plot points in **the** sky and to transmit, to transfer that information now, directly into **the** factory. So if you cross **the** border â just across **the** border â a small factory in Germany, and here you can see **the** guy with his **computer** screen, and those points in space are communicated. And on **the** **left** are **the** cutting machines, which then, in **the** factory, enable those individual pieces to be fabricated and plus or minus very, very few millimeters, to be slotted together on site. And then interestingly, that building to then be clad in **the** oldest technology, which is **the** kind of hand-cut shingles. One quarter of a million of them applied by hand as **the** final finish. And again, **the** way in which that works as a building, for those of us who can **enjoy the** spaces, to live and visit there. If I made **the** leap into these new technologies, then how did we â what happened before that? I mean, you know, what was life like before **the** mobile phone, **the** things that you take for granted? Well, obviously **the** building still happened. I mean, this is a glimpse of **the** interior of our Hong Kong bank of 1979, which opened in 1985, with **the** ability to be able to reflect sunlight deep into **the** heart of this space here. And in **the** absence of **computers**, you have to physically model. So for example, we would put models under an **artificial** sky. For wind tunnels, we would literally put them in a wind tunnel and blast air, and **the** many kilometers of cable and so on. And **the** turning point was probably, in our terms, when we had **the** first **computer**. And that was at **the** time that we sought to redesign, reinvent **the** airport. This is Terminal Four at Heathrow, typical of any terminal â big, heavy roof, blocking out **the** sunlight, lots of machinery, big pipes, whirring machinery. And Stansted, **the** green alternative, which uses natural light, is a **friendly** place : you know where you are, you can relate to **the** outside. And for a large part of its cycle, not needing electric light â electric light, which in turn creates more heat, which creates more cooling loads and so on. And at that particular point in time, this was one of **the** few solitary **computers**. And that ' s a little image of **the** tree of Stansted. Not going back very far in time, 1990, that ' s our office. And if you looked very closely, you ' d see that people were drawing with pencils, and they were pushing, you know, big rulers and **triangles**. It ' s not that long ago, 17 years, and here we are now. I mean, major transformation. Going back in time, there was a lady called Valerie Larkin, and in 1987, she had all our information on one **disk**. Now, every week, we have **the** equivalent of 84 million **disks**, which record our archival information on past, current and future projects. That reaches 21 kilometers into **the** sky. This is **the** view you would get, if you looked down on that. But meanwhile, as you know, wonderful protagonists like Al Gore are noting **the** inexorable rise in temperature, set in **the** context of that, interestingly, those buildings which are celebratory and very, very relevant to this place. Our Reichstag project, which has a very familiar agenda, I ' m sure, as a public place where we sought to, in a way, through a process of advocacy, reinterpret **the** relationship between society and politicians, public space. And maybe its hidden agenda, an energy manifesto â something that would be free, completely free of fuel as we know it. So it would be totally **renewable**. And again, **the** humanistic sketch, **the** translation into **the** public space, but this very, very much a part of **the** **ecology**. But here, not having to model it for real. Obviously **the** wind tunnel had a place, but **the** ability now with **the** **computer** to explore, to plan, to see how that would work in terms of **the** forces of nature : natural ventilation, to be able to model **the** chamber below, and to look at biomass. A combination of biomass, aquifers, burning vegetable **oil** â a process that, quite interestingly, was developed in Eastern Germany, at **the** time of its dependence on **the** Soviet Bloc. So really, retranslating that technology and developing something which was so clean, it was virtually pollution-free. You can measure it again. You can compare how that building, in terms of its **emission** in tons of **carbon** dioxide per year â at **the** time that we took that project, over 7, 000 tons â what it would have been with natural gas and finally, with **the** vegetable **oil**, 450 tons. I mean, a 94 percent reduction â virtually clean. We can see **the** same processes at work in terms of **the** Commerce Bank â its dependence on natural ventilation, **the** way that you can model those gardens, **the** way they spiral around. But again, very much about **the** lifestyle, **the** quality â something that would be more enjoyable as a place to work. And again, we can measure **the** reduction in terms of energy **consumption**. There is an **evolution** here between **the** projects, and Swiss Re again develops that a little bit further â **the** project in **the** city in London. And this sequence shows **the** buildup of that model. But what it shows first, which I think is quite interesting, is that here you see **the** circle, you see **the** public space around it. What are **the** other ways of putting **the** same amount of space on **the** site? If, for example, you seek to do a building which goes right to **the** edge of **the** pavement, it ' s **the** same amount of space. And finally, you profile this, you cut grooves into it. **The** grooves become **the** kind of green lungs which give views, which give light, ventilation, make **the** building fresher. And you enclose that with something that also is central to its appearance, which is a mesh of triangulated structures â again, in a long connection evocative of some of those works of Buckminster Fuller, and **the** way in which triangulation can increase performance and also give that building its sense of identity. And here, if we look at a detail of **the** way that **the** building opens up and breathes into those atria, **the** way in which now, with a **computer**, we can model **the** forces, we can see **the** high pressure, **the** low pressure, **the** way in which **the** building behaves rather like an aircraft **wing**. So it also has **the** ability, all **the** time, regardless of **the** direction of **the** wind, to be able to make **the** building fresh and efficient. And unlike conventional buildings, **the** top of **the** building is celebratory. It ' s a viewing place for people, not machinery. And **the** base of **the** building is again about public space. Comparing it with a typical building, what happens if we seek to use such design strategies in terms of really large-scale thinking? And I ’ m just going to give two images out of a kind of company research project. It ’ s been well known that **the** Dead Sea is dying. **The** level is dropping, rather like **the** Aral Sea. And **the** Dead Sea is obviously much lower than **the** oceans and **seas** around it. So there has been a project which rescues **the** Dead Sea by creating a pipeline, a pipe, sometimes above **the surface**, sometimes buried, that will redress that, and will feed from **the** Gulf of Aqaba into **the** Dead Sea. And our translation of that, using a lot of **the** thinking built up over **the** 40 years, is to say, what if that, instead of being just a pipe, what if it is a lifeline? What if it is **the** equivalent, depending on where you are, of **the** Grand Canal, in terms of tourists, habitation, desalination, **agriculture**? In other words, water is **the** lifeblood. And if you just go back to **the** previous image, and you look at this area of volatility and hostility, that a unifying design idea as a humanitarian gesture could have **the** affect of bringing all those warring factions together in a united cause, in terms of something that would be genuinely green and productive in **the** widest sense. Infrastructure at that large scale is also inseparable from communication. And whether that communication is **the** virtual world or it is **the** physical world, then it ’ s absolutely central to society. And how do we make more legible in this growing world, especially in some of **the** places that I ’ m talking about â□□ China, for example, which in **the** next ten years will create 400 new airports. Now what form do they take? How do you make them more **friendly** at that scale? Hong Kong I refer to as a kind of analog experience in a digital age, because you always have a point of reference. So what happens when we take that and you expand that further into **the** Chinese society? And what is interesting is that that produces in a way perhaps **the** ultimate mega-building. It is physically **the** largest project on **the planet** at **the** moment. 250 â□□ excuse me, 50, 000 people working 24 hours, seven days. Larger by 17 percent than every terminal put together at Heathrow â□□ built â□□ plus **the** new, un-built Terminal Five. And **the** challenge here is a building that will be green, that is compact despite its size and is about **the** human experience of travel, is about **friendly**, is coming back to that starting point, is very, very much about **the** lifestyle. And perhaps these, in **the** end, as celebratory spaces. As Hubert was talking over lunch, as we sort of engaged in conversation, talked about this, talked about cities. Hubert was saying, absolutely correctly,”These are **the** new cathedrals.”And in a way, one aspect of this conversation was triggered on New Year ’ s Eve, when I was talking about **the** Olympic agenda in China in terms of its green ambitions and aspirations. And I was voicing **the** thought that â□□ it just crossed my mind that New Year ’ s Eve, a sort of symbolic turning point as we move from 2006 to 2007 â□□ that maybe, you know, **the** future was **the** most powerful, innovative sort of nation. **The** way in which somebody like Kennedy inspirationally could say,”We put a man on **the** moon.”You know, who is going to say that we cracked this thing of **the** dependence on **fossil** fuels, with all that being held to ransom by rogue regimes, and so on. And that ’ s a concerted platform. It ’ s more than one **device**, you know, it ’ s **renewable**. And I voiced **the** thought that maybe at **the** turn of **the** year, I thought that **the** inspiration was more likely to come from those other, larger countries out there â□□ **the** Chinas, **the** Indias, **the** Asian-Pacific **tigers**. Thank you very much.

Text Sample 21: POS Dim 1 – 2007 – Score 32.00 – 2007/t000238.txt

Well, good morning. You know, **the computer** and television both recently turned 60, and today I ’ d like to talk about their relationship. Despite their middle age, if you ' ve been following **the themes** of this conference or **the** entertainment industry, it ' s pretty clear that one has been picking on **the** other. So it ' s about time that we talked about how **the computer** ambushed television, or why **the** invention of **the** atomic bomb unleashed forces that lead to **the** writers ' strike. And it ' s not just what these are doing to each other, but it ' s what **the** audience thinks that really frames this matter. To get a sense of this, and it ' s been a **theme** we ' ve talked about all week, I recently talked to a bunch of tweeners. I wrote on cards :"television,""radio,""MySpace,""Internet,""PC."And I said, just arrange these, from what ' s important to you and what ' s not, and then tell me why. Let ' s listen to what happens when they get to **the** portion of **the** discussion on television. Girl 1 : Well, I think it ' s important but, like, not necessary because you can do a lot of other stuff with your free time than watch programs. Peter Hirshberg : Which is more fun, Internet or TV? Girls : Internet. Girl 2 : I think we — **the** reasons, one of **the** reasons we put **computer** before TV is because nowadays, like, we have TV shows on **the computer**. Girl 2 : And then you can download onto your iPod. PH : Would you like to be **the** president of a TV network? Girl 4 : I wouldn ' t like it. Girl 2 : That would be so stressful. Girl 5 : No. PH : How come? Girl 5 : Because they ' re going to lose all their money eventually. Girl 3 : Like **the** stock market, it goes up and down and stuff. I think right now **the computers** will be at **the** top and everything will be kind of going down and stuff. PH : There ' s been an uneasy relationship between **the** TV business and **the** tech business, really ever since they both turned about 30. We go through periods of enthrallment, followed by reactions in boardrooms, in **the** finance community best characterized as, what ' s **the** finance term? Ick pooey. Let me give you an example of this. **The** year is 1976, and Warner buys Atari because video games are on **the** rise. **The** next year they march forward and they introduce Qube, **the** first interactive cable TV system, and **the** New York Times heralds this as telecommunications moving to **the** home, convergence, great things are happening. Everybody in **the** East Coast gets in **the** pictures — Citicorp, Penney, RCA — all getting into this big vision. By **the** way, this is about when I enter **the** picture. I ' m going to do a summer internship at Time Warner. That summer I ' m all — I ' m at Warner that summer — I ' m all excited to work on convergence, and then **the** bottom falls out. Doesn ' t work out too well for them, they lose money. And I had a happy brush with convergence until, kind of, Warner basically has to liquidate **the** whole thing. That ' s when I leave graduate school, and I can ' t work in New York on kind of entertainment and technology because I have to be exiled to California, where **the** remaining jobs are, almost to **the sea**, to go to work for Apple Computer. Warner, of course, writes off more than 400 million dollars. Four hundred million dollars, which was real money back in **the** ' 70s. But they were onto something and they got better at it. By **the** year 2000, **the** process was perfected. They merged with AOL, and in just four years, managed to shed about 200 billion dollars of market capitalization, showing that they ' d actually mastered **the** art of applying Moore ' s law of successive miniaturization to their balance sheet. Now, I think that one reason that **the** media and **the** entertainment communities, or **the** media community, is driven so crazy by **the** tech community is that tech folks talk differently. You know, for 50 years, we ' ve talked about changing **the** world, about total transformation. For 50 years, it ' s been about hopes and fears and promises of a better world. And I got to thinking, you know, who else talks that way? And **the** answer is pretty clearly — it ' s people in religion and in politics. And so I realized that actually **the** tech world is best understood, not as a business cycle, but as a messianic movement. We promise something great, we evangelize it, we ' re going to change **the** world. It doesn ' t work out too well, and so we actually go back to

the well and start all over again, as **the** people in New York and L. A. look on in absolute, morbid astonishment. But it ' s this irrational view of things that drives us on to **the** next thing. So, what I ' d like to ask is, if **the computer** is becoming a principal tool of media and entertainment, how did we get here? I mean, how did a machine that was built for accounting and artillery morph into media? Of course, **the** first **computer** was built just after World War II to solve military problems, but things got really interesting just a couple of years later — 1949 with Whirlwind, built at MIT ' s Lincoln Lab. Jay Forrester was building this for **the** Navy, but you can ' t help but see that **the** creator of this machine had in mind a machine that might actually be a potential media star. So take a look at what happens when **the** foremost journalist of early television meets one of **the** foremost **computer** pioneers, and **the computer** begins to express itself. Journalist : It ' s a Whirlwind electronic **computer**. With considerable trepidation, we **undertake** to interview this new machine. Jay Forrester : Hello New York, this is Cambridge. And this is **the** oscilloscope of **the** Whirlwind electronic **computer**. Would you like if I used **the** machine? Journalist : Yes, of course. But I have an idea, Mr. Forrester. Since this **computer** was made in conjunction with **the** Office of Naval Research, why don ' t we switch down to **the** Pentagon in Washington and let **the** Navy ' s research chief, Admiral Bolster, give Whirlwind **the** workout? Calvin Bolster : Well, Ed, this problem concerns **the** Navy ' s Viking rocket. This rocket goes up 135 miles into **the** sky. Now, at **the** standard rate of fuel **consumption**, I would like to see **the computer** trace **the** flight path of this rocket and see how it can determine, at any instant, say at **the** end of 40 seconds, **the** amount of fuel remaining, and **the** velocity at that set instant. JF : Over on **the** left-hand side, you will notice fuel **consumption** decreasing as **the** rocket takes off. And on **the right-hand** side, there ' s a scale that shows **the** rocket ' s velocity. **The** rocket ' s position is shown by **the** trajectory that we ' re now looking at. And as it reaches **the** peak of its trajectory, **the** velocity, you will notice, has dropped off to a minimum. Then, as **the** rocket dives down, velocity picks up again toward a maximum velocity and **the** rocket hits **the** ground. How ' s that? Journalist : What about that, Admiral? CB : Looks very good to me. JF : And before leaving, we would like to show you another kind of mathematical problem that some of **the** boys have worked out in their spare time, in a less serious vein, for a Sunday afternoon. Journalist : Thank you very much indeed, Mr. Forrester and **the** MIT lab. PH : You know, so much was worked out : **the** first real-time interaction, **the** video display, pointing a gun. It lead to **the** microcomputer, but unfortunately, it was too pricey for **the** Navy, and all of this would have been lost if it weren ' t for a happy coincidence. Enter **the** atomic bomb. We ' re threatened by **the** greatest weapon ever, and knowing a good thing when it sees it, **the** Air Force decides it needs **the** biggest **computer** ever to protect us. They adapt Whirlwind to a massive air defense system, deploy it all across **the** frozen north, and spend nearly three times as much on this **computer** as was spent on **the** Manhattan Project building **the** A-Bomb in **the** first place. Talk about a shot in **the** arm for **the computer** industry. And you can imagine that **the** Air Force became a pretty good salesman. Here ' s their marketing video. Narrator : In a mass raid, high-speed bombers could be in on us before we could determine their tracks. And then it would be too late to act. We cannot afford to take that chance. It is to meet this threat that **the** Air Force has been developing SAGE, **the** Semi-Automatic Ground Environment system, to strengthen our air defenses. This new **computer**, built to become **the** nerve center of a defense network, is able to perform all **the** complex mathematical problems involved in countering a mass enemy raid. It is provided with its own powerhouse containing large diesel-driven generators, air-conditioning equipment, and cooling towers required to cool **the** thousands of vacuum tubes in **the computer**. PH : You know, that one **computer** was huge. There ' s an interesting marketing lesson from it, which is basically, when you market a product, you can either say, this is going to be wonderful, it will make you feel better and enliven you. Or there ' s one other marketing proposition : if you don ' t use our product, you ' ll die. This is a really good example of that. This had **the** first pointing **device**. It was distributed, so it worked out — distributed computing and modems — so all these things could talk to each other. About 20 percent of all **the** nation ' s programmers were wrapped up in this thing, and it led to an awful lot of what we have today. It also used vacuum tubes. You saw how huge it was, and to give you a sense for this — because we ' ve talked a lot about Moore ' s law and making things small at this conference, so let ' s talk about making things large. If we took Whirlwind and put it in a place that you all know, say, Century City, it would fit beautifully. You ' d kind of have to take Century City out, but it could fit in there. But like, let ' s imagine we took **the** latest Pentium processor, **the** latest Core 2 Extreme, which is a four-core processor that Intel ' s working on, it will be our **laptop** tomorrow. To build that, what we ' d do with Whirlwind technology is we ' d have to take up roughly from **the** 10 to Mulholland, and from **the** 405 to La Cienega just with those Whirlwinds. And then, **the** 92 nuclear power plants that it would take to provide **the** power would fill up **the** rest of Los Angeles. That ' s roughly a third more nuclear power than all of France creates. So, **the** next time they tell you they ' re on to something, clearly they ' re not. So — and we haven ' t even worked out **the** cooling needs. But it gives you **the** kind of power that people have, that **the** audience has, and **the** reasons these transformations are happening. All of this stuff starts moving into industry. DEC kind of reduces all this and makes **the** first mini-computer. It shows up at places like MIT, and then a mutation happens. Spacewar! is built, **the** first **computer** game, and all of a sudden, interactivity and involvement and passion is worked out. Actually, many MIT students stayed up all night long working on this thing, and many of **the** principles of gaming today were worked out. DEC knew a good thing about wasting time. It shipped every one of its **computers** with that game. Meanwhile, as all of this is happening, by **the** mid- ' 50s, **the** business model of traditional broadcasting and cinema has been busted completely. A new technology has confounded radio men and movie moguls and they ' re quite certain that television is about to do them in. In fact, despair is in **the** air. And a quote that sounds largely reminiscent from everything I ' ve been reading all week. RCA had David Sarnoff, who basically commercialized radio, said this,"I don ' t say that radio networks must die. Every effort has been made and will continue to be made to find a new pattern, new selling arrangements and new types of programs that may arrest **the** declining revenues. It may yet be possible to eke out a poor existence for radio, but I don ' t know how."And of course, as **the** **computer** industry develops interactively, producers in **the** emerging TV business actually hit on **the** same idea. And they fake it. Jack Berry : Boys and girls, I think you all know how to get your magic windows up on **the** set, you just get them out. First of all, get your Winky Dink kits out. Put out your Magic Window and your erasing glove, and rub it like this. That ' s **the** way we get some of **the** magic into it, boys and girls. Then take it and put it right up against **the** screen of your own television set, and rub it out from **the** center to **the** corners, like this. Make sure you keep your magic crayons handy, your Winky Dink crayons and your erasing glove, because you ' ll be using them during **the** show to draw like that. You all set? OK, let ' s get right to **the** first story about Dusty Man. Come on into **the** secret lab. PH : It was **the** dawn of interactive TV, and you may have noticed they wanted to sell you **the** Winky Dink kits. Those are **the**

Winky Dink crayons. I know what you ’ re saying.”Pete, I could use any ordinary **open-source** crayon, why do I have to buy theirs?”I assure you, that ’ s not **the** case. Turns out they told us directly that these are **the** only crayons you should ever use with your Winky Dink Magic Window, other crayons may discolor or hurt **the** window. This proprietary principle of vendor lock-in would go on to be perfected with great success as one of **the** enduring principles of windowing systems everywhere. It led to lawsuits — — federal investigations, and lots of repercussions, and that ’ s a scandal we won ’ t discuss today. But we will discuss this scandal, because this man, Jack Berry, **the** host of”Winky Dink,”went on to become **the** host of”Twenty One,”one of **the** most important quiz shows ever. And it was rigged, and it became unraveled when this man, Charles van Doren, was outed after an unnatural winning streak, ending Berry ’ s career. And actually, ending **the** career of a lot of people at CBS. It turns out there was a lot to learn about how this new medium worked. And 50 years ago, if you ’ d been at a meeting like this and were trying to understand **the** media, there was one prophet and only but one you wanted to hear from, Professor Marshall McLuhan. He actually understood something about a **theme** that we ’ ve been discussing all week. It ’ s **the** role of **the** audience in an era of pervasive electronic communications. Here he is talking from **the** 1960s. Marshall McLuhan : If **the** audience can become involved in **the** actual process of making **the** ad, then it ’ s happy. It ’ s like **the** old quiz shows. They were great TV because it gave **the** audience a role, something to do. They were horrified when they discovered they ’ d really been left out all **the** time because **the** shows were rigged. Now, then, this was a horrible misunderstanding of TV on **the** part of **the** programmers. PH : You know, McLuhan talked about **the** global village. If you substitute **the** word blogosphere, of **the** Internet today, it is very true that his understanding is probably very enlightening now. Let ’ s listen in to him. MM : **The** global village is a world in which you don ’ t necessarily have harmony. You have extreme concern with everybody else ’ s business and much involvement in everybody else ’ s life. It ’ s a sort of Ann Landers ’ column writ large. And it doesn ’ t necessarily mean harmony and peace and quiet, but it does mean huge involvement in everybody else ’ s affairs. And so **the** global village is as big as a **planet**, and as small as a village post office. PH : We ’ ll talk a little bit more about him later. We ’ re now right into **the** 1960s. It ’ s **the** era of big business and data centers for computing. But all that was about to change. You know, **the** expression of technology reflects **the** people and **the** time of **the** culture it was built in. And when I say that code expresses our hopes and aspirations, it ’ s not just a joke about messianism, it ’ s actually what we do. But for this part of **the** story, I ’ d actually like to throw it to America ’ s leading technology correspondent, John Markoff. John Markoff : Do you want to know what **the** counterculture in drugs, sex, **rock** ’ n ’ roll and **the** anti-war movement had to do with computing? Everything. It all happened within five miles of where I ’ m standing, at Stanford University, between 1960 and 1975. In **the** midst of revolution in **the** streets and **rock** and roll concerts in **the** parks, a group of researchers led by people like John McCarthy, a **computer** scientist at **the** Stanford **Artificial** Intelligence Lab, and Doug Engelbart, a **computer** scientist at SRI, changed **the** world. Engelbart came out of a pretty dry engineering culture, but while he was beginning to do his work, all of this stuff was bubbling on **the** mid-peninsula. There was LSD leaking out of Kesey ’ s Veterans ’ Hospital experiments and other areas around **the** campus, and there was music literally in **the** streets. **The** Grateful Dead was playing in **the** pizza parlors. People were leaving to go back to **the** land. There was **the** Vietnam War. There was black liberation. There was women ’ s liberation. This was a remarkable place, at a remarkable time. And into that ferment came **the** microprocessor. I think it was that interaction that led to personal computing. They saw these tools that were controlled by **the** establishment as ones that could actually be liberated and put to use by these communities that they were trying to build. And most importantly, they had this ethos of sharing information. I think these ideas are difficult to understand, because when you ’ re trapped in one paradigm, **the** next paradigm is always like a science fiction **universe** — it makes no sense. **The** stories were so compelling that I decided to write a book about them. **The** title of **the** book is,”What **the** Dormouse Said : How **the** ’ 60s Counterculture Shaped **the** Personal Computer Industry.”**The** title was taken from **the** lyrics to a Jefferson **Airplane** song. **The** lyrics go,”Remember what **the** dormouse said. Feed your head, feed your head, feed your head.”PH : By this time, computing had kind of leapt into media territory, and in short order much of what we ’ re doing today was imagined in Cambridge and Silicon Valley. Here ’ s **the** Architecture Machine Group, **the** predecessor of **the** Media Lab, in 1981. Meanwhile, in California, we were trying to commercialize a lot of this stuff. HyperCard was **the** first program to introduce **the** public to hyperlinks, where you could randomly hook to any kind of picture, or piece of text, or data across a file system, and we had no way of explaining it. There was no metaphor. Was it a database? A prototyping tool? A scripted language? Heck, it was everything. So we ended up writing a marketing brochure. We asked a question about how **the** mind works, and we let our customers play **the** role of so many blind men filling out **the elephant**. A few years later, we then hit on **the** idea of explaining to people **the** secret of, how do you get **the** content you want, **the** way you want it and **the** easy way? Here ’ s **the** Apple marketing video. James Burke : You ’ ll be pleased to know, I ’ m sure, that there are several ways to create a HyperCard interactive video. **The** most involved method is to go ahead and produce your own videodisc as well as build your own HyperCard **stacks**. By far **the** simplest method is to buy a pre-made videodisc and HyperCard **stacks** from a commercial supplier. **The** method we illustrate in this video uses a pre-made videodisc but creates custom HyperCard **stacks**. This method allows you to use existing videodisc materials in ways which suit your specific needs and interests. PH : I hope you realize how subversive that is. That ’ s like a Dick Cheney speech. You think he ’ s a nice balding guy, but he ’ s just declared war on **the** content business. Find **the** commercial stuff, mash it up, tell **the** story your way. Now, as long as we confine this to **the** education market, and a personal matter between **the computer** and **the** file system, that ’ s fine, but as you can see, it was about to leap out and upset Jack Valenti and a lot of other people. By **the** way, speaking of **the** filing system, it never **occurred** to us that these hyperlinks could go beyond **the** local area network. A few years later, Tim Berners-Lee worked that out. It became a killer app of links, and today, of course, we call that **the** World Wide Web. Now, not only was I instrumental in helping Apple miss **the** Internet, but a couple of years later, I helped Bill Gates do **the** same thing. **The** year is 1993 and he was working on a book and I was working on a video to help him kind of explain where we were all heading and how to popularize all this. We were plenty aware that we were messing with media, and on **the surface**, it looks like we predicted a lot of **the** right things, but we also missed an awful lot. Let ’ s take a look. Narrator : **The** pyramids, **the** Colosseum, **the** New York subway system and TV dinners, ancient and modern wonders of **the** man-made world all. Yet each pales to insignificance with **the** completion of that magnificent accomplishment of twenty-first-century technology, **the** Digital Superhighway. Once it was only a dream of technoids and a few long-forgotten politicians. **The** Digital Highway arrived in America ’ s living rooms late in **the** twentieth century. Let us recall **the** pioneers who made this technical marvel possible.

The Digital Highway would follow **the** rutted trail first blazed by Alexander Graham Bell. Though some were incredulous... Man 1 : **The** phone company! Narrator : Stirred by **the** prospects of mass communication and making big **bucks** on advertising, David Sarnoff commercializes radio. Man 2 : Never had scientists been put under such pressure and demand. Narrator : **The** medium introduced America to new products. Voice 1 : Say, mom, Windows for Radio means more enjoyment and greater ease of use for **the** whole family. Be sure to **enjoy** Windows for Radio at home and at work. Narrator : In 1939, **the** Radio Corporation of America introduced television. Man 2 : Never had scientists been put under such pressure and demand. Narrator : Eventually, **the** race to **the** future took on added momentum with **the** breakup of **the** telephone company. And further stimulus came with **the** deregulation of **the** cable television industry, and **the** re-regulation of **the** cable television industry. Ted Turner : We did **the** work to build this, this cable industry, now **the** broadcasters want some of our money. I mean, it ' s ridiculous. Narrator : **Computers**, once **the** unwieldy tools of accountants and other **geeks**, escaped **the** backrooms to enter **the** media fracas. **The** world and all its culture reduced to bits, **the** lingua franca of all media. And **the** forces of convergence exploded. Finally, four great industrial sectors combined. Telecommunications, entertainment, computing and everything else. Man 3 : We ' ll see channels for **the** gourmet and we ' ll see channels for **the** pet lover. Voice 2 : Next on **the** gourmet pet channel, decorating birthday cakes for your schnauzer. Narrator : All of industry was in play, as investors flocked to place their bets. At stake : **the** battle for you, **the** consumer, and **the** right to spend billions to send a lot of information into **the** parlors of America. PH : We missed a lot. You know, you missed, we missed **the** Internet, **the** long tail, **the** role of **the** audience, open systems, social networks. It just goes to show how tough it is to come up with **the** right uses of media. Thomas Edison had **the** same problem. He wrote a list of what **the** phonograph might be good for when he invented it, and kind of only one of his ideas turned out to have been **the** right early idea. Well, you know where we ' re going on from here. We come into **the** era of **the** dotcom, **the** World Wide Web, and I don ' t need to tell you about that because we all went through that bubble together. But when we emerge from this and what we call Web 2. 0, things actually are quite different. And I think it ' s **the** reason that TV ' s so challenged. If Internet one was about pages, now it ' s about people. It ' s a customer, it ' s an audience, it ' s a person who ' s participating. It ' s **the** formidable thing that is changing entertainment now. MM : Because it gave **the** audience a role, something to do. PH : In my own company, Technorati, we see something like 67, 000 blog posts an hour come in. That ' s about 2, 700 fresh, connective links across about 112 million blogs that are out there. And it ' s no wonder that as we head into **the** writers ' strike, odd things happen. You know, it reminds me of that old saw in Hollywood, that a producer is anyone who knows a writer. I now think a network boss is anyone who has a cable modem. But it ' s not a joke. This is a real headline."Websites attract striking writers : operators of sites like MyDamnChannel. com could benefit from labor disputes."Meanwhile, you have **the** TV bloggers going out on strike, in sympathy with **the** television writers. And then you have TV Guide, a Fox property, which is about to sponsor **the** online video awards — but cancels it out of sympathy with traditional television, not appearing to gloat. To show you how schizophrenic this all is, here ' s **the** head of MySpace, or Fox Interactive, a News Corp company, being asked, well, with **the** writers ' strike, isn ' t this going to hurt News Corp and help you online? Man : But I, yeah, I think there ' s an opportunity. As **the** strike continues, there ' s an opportunity for more people to experience video on places like MySpace TV. PH : Oh, but then he remembers he works for Rupert Murdoch. Man : Yes, well, first, you know, I ' m part of News Corporation as part of Fox Entertainment Group. Obviously, we hope that **the** strike is — that **the** issues are resolved as quickly as possible. PH : One of **the** great things that ' s going on here is **the** globalization of content really is happening. Here is a **clip** from a video, from a piece of animation that was written by a writer in Hollywood, animation worked out in Israel, farmed out to Croatia and India, and it ' s now an international series. Narrator : **The** following takes place between **the** minutes of 2:15 p. m. and 2:18 p. m., in **the** months preceding **the** presidential primaries. Voice 1 : You ' ll have to stay here in **the** safe house until we get word **the terrorist** threat is over. Voice 2 : You mean we ' ll have to live here, together? Voices 2, 3 and 4 : With her? Voice 2 : Well, there goes **the** neighborhood. PH : **The** company that created this, Aniboom, is an interesting example of where this is headed. Traditional TV animation costs, say, between 80, 000 and 10, 000 dollars a minute. They ' re producing things for between 1, 500 and 800 dollars a minute. And they ' re offering their creators 30 percent of **the** back end, in a much more entrepreneurial manner. So, it ' s a different model. What **the** entertainment business is struggling with, **the** world of brands is figuring out. For example, Nike now understands that Nike Plus is not just a **device** in its shoe, it ' s a network to hook its customers together. And **the** head of marketing at Nike says,"People are coming to our site an average of three times a week. We don ' t have to go to them."Which means television advertising is down 57 percent for Nike. Or, as Nike ' s head of marketing says,"We ' re not in **the** business of keeping media companies alive. We ' re in **the** business of connecting with consumers."And media companies realize **the** audience is important also. Here ' s a man announcing **the** new Market Watch from Dow Jones, powered 100 percent by **the** user experience on **the** home page — user-generated content married up with traditional content. It turns out you have a bigger audience and more interest if you hook up with them. Or, as Geoffrey Moore once told me, it ' s intellectual curiosity that ' s **the** trade that brands need in **the** age of **the** blogosphere. And I think this is beginning to happen in **the** entertainment business. One of my heroes is songwriter, Ally Willis, who just wrote"**The** Color Purple"and has been an R and — rhythm and **blues** writer, and this is what she said about where songwriting ' s going. Ally Willis : Where millions of collaborators wanted **the** song, because to look at them strictly as spam is missing what this medium is about. PH : So, to wrap up, I ' d love to throw it back to Marshall McLuhan, who, 40 years ago, was dealing with audiences that were going through just as much change, and I think that, today, traditional Hollywood and **the** writers are framing this perhaps in **the** way that it was being framed before. But I don ' t need to tell you this, let ' s throw it back to him. Narrator : We are in **the** middle of a tremendous clash between **the** old and **the** new. MM : **The** medium does things to people and they are always completely unaware of this. They don ' t really notice **the** new medium that is wrapping them up. They think of **the** old medium, because **the** old medium is always **the** content of **the** new medium, as movies tend to be **the** content of TV, and as books used to be **the** content, novels used to be **the** content of movies. And so every time a new medium arrives, **the** old medium is **the** content, and it is highly observable, highly noticeable, but **the** real, real roughing up and massaging is done by **the** new medium, and it is ignored. PH : I think it ' s a great time of enthrallment. There ' s been more raw DNA of communications and media thrown out there. Content is moving from shows to particles that are batted back and forth, and part of social communications, and I think this is going to be a time of great renaissance and opportunity. And whereas television may have gotten beat up, what ' s getting built is a really exciting new form of communication, and we kind of have **the** merger of **the** two industries and a new way of thinking to look at it. Thanks very much.

Text Sample 22: POS Dim 1 – 2007 – Score 24.00 – 2007/t000223.txt

Good afternoon, good evening, whatever. We can go, jambo, guten Abend, bonsoir, but we can also ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh. That is **the** call that chimpanzees make before they go to sleep in **the** evening. You hear it going from one side of **the** valley to **the** other, from one group of nests to **the** next. And I want to pick up with my talk this evening from where Zeray left off yesterday. He was talking about this amazing, three-year-old Australopithecine child, Selam. And we ’ ve also been hearing about **the** history, **the** family tree, of mankind through DNA genetic profiling. And it was a paleontologist, **the** late Louis Leakey, who actually set me on **the** path for studying chimpanzees. And it was pretty extraordinary, way back then. It ’ s kind of commonplace now, but his argument was — because he ’ d been **searching** for **the** fossilized remains of early humans in Africa. And you can tell an awful lot about what those beings looked like from **the fossils**, from **the** shape of **the** muscle attachments, something about **the** way they lived from **the** various artifacts found with them. But what about how they behaved? That ’ s what he wanted to know. And of course, behavior doesn ’ t fossilize. He argued — and it ’ s now a fairly common theory — that if we found behavior patterns similar or **the** same in our closest living **relatives**, **the** great apes, and humans today, then maybe those behaviors were present in **the** ape-like, human-like ancestor some seven million years ago. And therefore, perhaps we had brought those characteristics with us from that ancient, ancient past. Well, if you look in **textbooks** today that deal with human **evolution**, you very often find people **speculating** about how early humans may have behaved, based on **the** behavior of chimpanzees. They are more like us than any other living creature, and we ’ ve heard about that during this TED Conference. So it remains for me to comment on **the** ways in which chimpanzees are so like us, in certain aspects of their behavior. Every chimpanzee has his or her own personality. Of course, I gave them names. They can live to be 60 years or more, although we think most of them probably don ’ t make it to 60 in **the** wild. Mr. Wurzel. **The** female has her first baby when she ’ s 11 or 12. Thereafter, she has one baby only every five or six years, a long period of childhood dependency when **the** child is nursing, sleeping with **the** mother at night, and riding on her back. And we believe that this long period of childhood is important for chimpanzees, just as it is for us, in relation to learning. As **the** brain becomes ever more complex during **evolution** in different forms of animals, so we find that **learning** plays an ever more important role in an individual ’ s life history. And young chimpanzees spend a lot of time watching what their elders do. We know now that they ’ re capable of imitating behaviors that they see. And we believe that it ’ s in this way that **the** different tool-using behaviors — that have now been seen in all **the** different chimpanzee populations studied in Africa — how these are passed from one generation to **the** next, through observation, imitation and practice, so that we can describe these tool-using behaviors as primitive culture. Chimpanzees don ’ t have a spoken language. We ’ ve talked about that. They do have a very rich repertoire of postures and gestures, many of which are similar, or even **identical**, to ours and formed in **the** same context. Greeting chimpanzees embracing. They also kiss, hold hands, pat one another on **the** back. And they swagger and they throw **rocks**. In chimpanzee society, we find many, many examples of compassion, precursors to love and true altruism. Unfortunately, they, like us, have a dark side to their nature. They ' re capable of extreme brutality, even a kind of primitive war. And these really aggressive behaviors, for **the** most part, are directed against individuals of **the** neighboring social group. They are very territorially aggressive. Chimpanzees, I believe, more than any other living creature, have helped us to understand that, after all, there is no sharp line between humans and **the** rest of **the** animal kingdom. It ' s a very blurry line, and it ' s getting more blurry all **the** time as we make even more observations. **The** study that I began in 1960 is still continuing to this day. And these chimpanzees, living their complex social lives in **the** wild, have helped — more than anything else — to make us realize we are part of, and not separated from, **the** amazing animals with whom we share **the planet**. So it ' s pretty sad to find that chimpanzees, like so many other creatures around **the** world, are losing their **habitats**. This is just one photograph from **the** air, and it shows you **the** forested highlands of Gombe. And it was when I flew over **the** whole area, about 16 years ago, and realized that outside **the** park, this forest, which in 1960 had stretched almost unbroken along **the eastern** shore of Lake Tanganyika, which is where **the** tiny, 30-square-mile Gombe National Park lies, that a question came to my mind."How can we even try to save these **famous** chimpanzees, when **the** people living around **the** National Park are struggling to survive?"More people are living there than **the** land could possibly support. **The** numbers increased by refugees pouring in from Burundi and over **the** lake from Congo. And very poor people — they couldn ' t afford to buy food from elsewhere. This led to a program, which we call TACARE. It ' s a very holistic way of improving **the** lives of **the** people living in **the** villages around **the** park. It started small with 12 villages. It ' s now in 24. There isn ' t time to go into it, but it ' s including things like tree nurseries, methods of farming most suitable to this now very degraded, almost desert-like land up in these mountains. Ways of controlling, preventing soil erosion. Ways of reclaiming overused farmland, so that within two years they can again be productive. Working to help **the** villagers obtain fresh water from wells. Perhaps build some schoolrooms. Most important of all, I believe, is working with small groups of women, providing them with opportunities for micro-credit loans. And we ' ve got, as is **the** case around **the** world, about 95 percent of all loans returned. Empowering women, working with education, providing scholarships for girls so they can finish secondary school, in **the** clear understanding that, all around **the** world, as women ' s education improves, family size drops. We provide information about family planning and about HIV / AIDS. And as a result of this program, something ' s happening for conservation. What ' s happening for conservation is that **the** farmers living in these 24 villages, instead of looking on us as a bunch of white people coming to study a whole bunch of monkeys — and by **the** way, many of **the** staff are now Tanzanian — but when we began **the** TACARE program, it was a Tanzanian team going into **the** villages. It was a Tanzanian team talking to **the** villagers, asking what they were interested in. Were they interested in conservation? Absolutely not. They were interested in health ; they were interested in education. And as time went on, and as their situation began to improve, they began to understand ever more about **the** need for conservation. They began to understand that as **the** upper levels of **the** hills were denuded of trees, so you ' ve got this terrible soil erosion and mudslides. Today, we are developing what we call **the** Greater Gombe Ecosystem. This is an area way outside **the** National Park, stretching out into all these very degraded lands. And as these villages have a better standard of life, they are actually agreeing to put between 10 percent and 20 percent of their land in **the** highlands aside, so that once again, as **the** trees grow back, **the** chimpanzees will have leafy corridors through which they can travel to interact — as they must for genetic viability — with other remnant groups outside **the** National Park. So TACARE is a success. We ’ re replicating it in other parts of Africa, around other wilderness areas which are faced with extreme population pressure. **The** problems in Africa, however, as we ’ ve been discussing for **the** whole of these first couple of days of TED, are major problems. There is a great deal of poverty. And when you get large numbers of people living in land that is not that fertile, particularly when you cut down trees, and you leave **the** soil open to **the** wind for erosion, as desperate populations cut down more and more trees, so that they can try and grow food for themselves and their families, what ’ s going to happen? Something ’ s got to give. And **the** other problems — in not only Africa, but **the** rest of **the** developing world and, indeed, everywhere — what are we doing to our **planet**? You know, **the famous** scientist, E. O. Wilson said that if every person on this **planet** attains **the** standard of living of **the** average European or American, we need three new **planets**. Today, they are saying four. But we don ’ t have them. We ’ ve got one. And what ’ s happened? I mean, **the** question here is, here we are, arguably **the** most intelligent being that ’ s ever walked **planet** Earth, with this extraordinary brain, capable of **the** kind of technology that is so well illustrated by these TED Conferences, and yet we ’ re destroying **the** only home we have. **The** indigenous people around **the** world, before they made a major decision, used to sit around and ask themselves,”How does this decision affect our people seven generations ahead?”Today, major decisions — and I ’ m not particularly talking about Africa here, but **the** developed world — major decisions involving millions of dollars, and millions of people, are often based on,”How will this affect **the** next shareholders ’ meeting?”And these decisions affect Africa. As I began traveling around Africa talking about **the** problems faced by chimpanzees and their vanishing forests, I realized more and more how so many of Africa ’ s problems could be laid at **the** door of previous colonial exploitation. So I began traveling outside Africa, talking in Europe, talking in **the** United States, going to Asia. And everywhere there were these terrible problems. And you know **the** kind I ’ m talking about. I ’ m talking about pollution. **The** air that we breathe that often poisons us. **The** earth is poisoning our foods. **The** water — water is perhaps one of **the** most crucial issues that we ’ re going to face in this century — and everywhere water is being **polluted** by agricultural, industrial and household chemicals that still are being sprayed around **the** world, seemingly with **the** inability to profit from past experience. **The** mangroves are being cut down ; **the** effects of things like **the** **tsunami** get worse. We ’ ve talked about **the** soil erosion. We have **the** reckless burning of **fossil** fuels along with other **greenhouse** gasses, so called, leading to climate change. Finally, all around **the** world, people have begun to believe that there is something going on very wrong with our climate. All around **the** world climates are mixed up. And it ’ s **the** poor people who are affected worse. It ’ s Africa that already is affected. In many parts of sub- Saharan Africa, **the** **droughts** are so much worse. And when **the** rain does come, it so often leads to flooding and added distress, and **the** cycle of poverty and hunger and disease. And **the** numbers of people living in an area that **the** land cannot support, who are too poor to buy food, who can ’ t move away because **the** whole land is degraded. And so you get desertification — creeping, creeping, creeping — as **the** last of **the** trees are cut down. And this kind of thing is not just in Africa. It ’ s all over **the** world. So it wasn ’ t surprising to me that as I was traveling around **the** world I met so many young people who seemed to have lost hope. We seem to have lost wisdom, **the** wisdom of **the** indigenous people. I asked a question.”Why?”Well, do you think there could be some kind of disconnect between this extraordinarily clever brain, **the** kind of brain that **the** TED technologies exemplify, and **the** human heart? Talking about it in **the** non-scientific term, in terms of love and compassion. Is there some disconnect? And these young people, when I talk to them, basically they were either depressed or apathetic, or bitter and angry. And they said more or less **the** same thing,"We feel this way because we feel you ' ve compromised our future and there ' s nothing we can do about it."We have compromised their future. I ' ve got three little grandchildren, and every time I look at them and I think how we ' ve harmed this beautiful **planet** since I was their age, I feel this desperation. And that led to this program we call Roots and Shoots, which began right here in Tanzania and has now spread to 97 countries around **the** world. It ' s symbolic. Roots make a firm foundation. Shoots seem tiny ; to reach **the** sun they can break through a brick wall. See **the** brick wall as all these problems we ' ve inflicted on **the planet**, environmental and social. It ' s a message of hope. Hundreds and thousands of young people around **the** world can break through and can make this a better world for all living things. **The** most important message of Roots and Shoots : every single one of us makes a difference, every single day. We have a choice. Every one of us in this room, we have a choice as to what kind of difference we want to make. **The** very poor have no choice. It ' s up to us to change things so that **the** poor have choice as well. **The** Roots and Shoots groups all choose three projects. It depends on how old they are, and which country, whether they ' re in a city or rural, as to what kinds of projects. But basically, we have programs now from preschool right through university, with more and more adults starting their own Roots and Shoots groups. And every group chooses, between them, three different kinds of project to make this a better world, recognizing that all these different problems are interconnected and impinge on each other. So one of their projects will be to help their own human community. And then, if they ' re able, they may raise money to help communities in other parts of **the** world. One of their projects will be to help animals — not just wildlife, domestic animals as well. And one of their projects will be to help **the** environment that we all share. And woven throughout all of this is a message of learning to live in peace and harmony within ourselves, in our families, in our communities, between nations, between cultures, between religions and between us and **the** natural world. We need **the** natural world. We cannot go on destroying it at **the** rate we are. We not do have more than this one **planet**. Just picking one or two of **the** projects right here in Africa that **the** Roots and Shoots groups are doing, one or two projects only — in Tanzania, in Uganda, Kenya, South Africa, Congo-Brazzaville, Sierra Leone, Cameroon and other groups. And as I say, it ' s in 97 countries around **the** world. Of course, they ' re planting trees. They ' re growing **organic** vegetables. They ' re working in **the** refugee camps, with chickens and selling **the** eggs for a little amount of money, or just using them to feed their families, and feeling a sense of pride and empowerment, because they ' re no longer helpless and depending on others with their vegetables and their chickens. It ' s being used in Uganda to give some psychological help to ex-child soldiers. Doing projects like this is bringing them out of themselves. Once again, they ' re useful members of society. We have this program in prisons as well. So, there ' s no time for more Roots and Shoots now. But — oh, they ' re also working on HIV / AIDS. That ' s a very important component of Roots and Shoots, with older kids talking to younger ones. And unwanted pregnancies and things like that, which young people listen to better from other youth, rather than adults. Hope. That ' s **the** question I get asked as I ' m going around **the** world :"Jane, you ' ve seen so many terrible things, you ' ve seen your chimpanzees decrease in number from about one million, at **the** turn of **the** century, to no more than 150, 000 now, and **the** same with so many other animals. Forests **disappearing**, deserts where once there was forest. Do you really have hope?"Well, yes. You can ' t come to a conference like TED and not have hope, can you? And of course, there ' s hope. One is this amazing human brain. And I mean, think of **the** technologies. And I ' ve just been so thrilled, finally, to come to people talking about compost latrines. It ' s one of my hobbyhorses. We just flush all this water down **the** lavatory, it ' s terrible. And then talking about **renewable** energy — desperately important. Do we care about **the planet** for our children? How many of us have children or grandchildren, nieces, nephews? Do we care about their future? And if we care about their future, we, as **the** elite around **the** world, we can do something about it. We can make choices as to how we live each day. What we buy. What we wear. And choose to make these choices with **the** question, how will this affect **the** environment around me? How will it affect **the** life of my child when he or she grows up? Or my grandchild, or whatever it is. So **the** human brain, coupled with **the** human heart, and we join hands around **the** world. And that ' s what TED is helping so well with, and Google who help us, and Esri are helping us with mapping in Gombe National Park. All of these technologies we can use. Now let ' s link them, and it ' s beginning to happen, isn ' t it? You ' ve heard about it this afternoon. It ' s beginning to happen. This change, this change. To see change that we must have if we care about **the** future. And **the** next reason for hope — nature is amazingly resilient. You can take an area that ' s absolutely destroyed, with time and perhaps some help it can regenerate. And an example is **the** TACARE program. I told you, where a seemingly dead tree stump — if you stop hacking them for firewood, which you don ' t need to because you have wood lots, then in five years you can have a 30-foot tree. And animals, almost on **the** brink of extinction, can be given a second chance. That ' s my next book. It ' s inspiring. And it brings me to my last category of hope, and we ' ve heard about this so much in **the** last two days : this indomitable human spirit. This determination of people, **the** resilience of **the** human spirit, So that people who you would think would be battered by poverty, or disease, or whatever, can pull themselves up out of it, sometimes with a helping hand, and take their part in society, and take their part in changing **the** world. And just to think of one or two people out of Africa who are just really inspiring. We could make a very long list, but obviously Nelson Mandela, emerging from 17 years of hard physical labor, 23 years of imprisonment, with this amazing ability to forgive, so that he could lead his nation out **the** evil regime of apartheid without a bloodbath. Ken Saro-Wiwa, in Nigeria, who took on **the** giant **oil** companies, and although people around **the** world tried their best, was executed. People like this are so inspirational. People like this are **the** role models we need for young Africans. And we need some environmental role models as well, and I ' ve been hearing some of them today. So I ' m really grateful for this opportunity to share this message again, with everyone at TED. And I hope that some of us can get together and talk about some of these things, especially **the** Roots and Shoots program. And just a last word on that — **the** young woman who ' s running this entire conference center, I met her today. She came up so excited, with her certificate. She was [in] Roots and Shoots. She was in **the** leadership in Dar es Salaam. She said it ' s helped her to do what she ' s doing. And it was very, very exciting for me to meet her and see just one example of how young people, when they are empowered, given **the** opportunity to take action, to make **the** world a better place, truly are our hope for tomorrow. Thank you.

Text Sample 23: POS Dim 1 – 2004 – Score 20.00 – 2004/t000140.txt

This is **the** first of two rather extraordinary photographs I ' m going to show you today. It was taken 18 years ago. I was 19 years old at **the** time. I had just returned from one of **the** deepest dives I ’ d ever made at that time â a little over 200 feet. And I had caught this little **fish**. It turns out that particular one was **the** first of its kind ever taken alive. I ’ m not just an ichthyologist, I ’ m a bona fide **fish** nerd. And to a **fish** nerd, this is some pretty exciting stuff. More exciting was **the** fact that **the** person who took this photo is a guy named Jack Randall, **the** greatest living ichthyologist on earth â **the** Grand Poobah of **fish** nerds, if you will. And so, it was really exciting to me to have this moment in time. It set **the** course for **the** rest of my life. But really, **the** most significant, most profound thing about this picture is it was taken two days before I was completely paralyzed from **the** neck down. I made a really stupid kind of mistake that most 19-year-old males do when they think they ’ re immortal, and I got a bad case of **the** bends and was paralyzed, and had to be flown back for treatment. I learned two really important things that day. **The** first thing I learned â well, I ’ m mortal. That ’ s a really big one. And **the** second thing I learned was that I knew, with profound certainty, that this is exactly what I was going to do for **the** rest of my life. I had to focus all my energies towards going to find new species of things down on deep coral reefs. When you think of a coral reef, this is what most people think of : these big, hard, elaborate corals, lots of bright, colorful **fishes** and things. But this is really just **the** tip of **the** iceberg. If you look at this diagram of a coral reef, we know a lot about that part up near **the** top. **The** reason we know so much about it is scuba divers can very easily go down there and access it. There is a problem with scuba, though, in that it imposes some limitations on how deep you can go. It turns out that depth is about 200 feet. I ’ ll get into why that is in just a minute, but **the** point is, scuba divers generally stay less than 100 feet deep, and very rarely go much below this, at least, not with any kind of sanity. So to go deeper, most biologists have turned to submersibles. Now, submersibles are great, wonderful things, but if you ’ re going to spend 30, 000 dollars a day to use one of these things and it ’ s capable of going 2, 000 feet, you ’ re not going to go farting around up here in a couple of hundred feet, you ’ re going to go way down deep. So **the** bottom line is, almost all research using submersibles has taken place well below 500 feet. Now, it ’ s pretty obvious at this point there ’ s a zone here in **the** middle. That ’ s **the** zone that centers around my own personal pursuit of happiness. I want to find out what ’ s in this zone ; we know almost nothing about it. Scuba divers can ’ t get there, submarines go right on past it. It took me a year to learn to walk again after my diving accident in Palau. During that year, I spent a lot of time learning about **the** physics and physiology of diving and how to overcome these limitations. I ’ m just going to show you a basic idea. We ’ re all breathing air right now. Air is a mixture of oxygen and nitrogen, 20 percent oxygen, 80 percent nitrogen. It ’ s in our lungs. And there ’ s a phenomenon called Henry ’ s law, that says gases will dissolve into a fluid in proportion to **the** partial pressures you ’ re exposing them to. So, basically **the** gas dissolves into our body. **The** oxygen is bound by metabolism, we use it for energy. **The** nitrogen sort of floats around in our **blood** and tissues. That ’ s fine, it ’ s how we ’ re designed. **The** problem happens when you go **underwater**. **The** deeper you go **underwater**, **the** higher **the** pressure is. If you were to go down to a depth of about 130 feet, which is **the** recommended limit for most scuba divers, you ’ d get this pressure effect. **The** effect of that pressure is you have an increased density of gas molecules in every breath you take. Over time, those gas molecules dissolve into your **blood** and tissues and start to fill you up. Now, if you were to go down to, say, 300 feet, you don ’ t have five times as many gas molecules in your lungs, you ’ ve got 10 times as many gas molecules in your lungs. And, sure enough, they dissolve into your

blood and tissues as well. And if you were to go down to where there ’ s 15 times as much â **the** deeper you go, **the** more exacerbating **the** problem becomes. **The** limitation of diving with air is all those dots in your body, all **the** nitrogen and all **the** oxygen. There are three basic limitations of scuba diving. **The** first limitation is **the** oxygen â oxygen toxicity. Now, we all know **the** song :”Love is like oxygen. You get too much, you get too high. Not enough, and you ’ re gonna die.”Well, in **the** context of diving, you get too much, you die also. You die because oxygen toxicity can cause a seizure. It makes you convulse **underwater** â not a good thing to happen **underwater**. It happens because there ’ s too much concentration of oxygen in your body. **The** nitrogen has two problems. One of them is what Jacques Cousteau called”rapture of **the** deep.”It ’ s nitrogen narcosis. It makes you loopy ; **the** deeper you go, **the** loopier you get. You don ’ t want to drive drunk, you don ’ t want to dive drunk. So that ’ s a real big problem. And of course, **the** third problem is **the** one I found out **the** hard way in Palau, which is **the** bends. One thing I forgot to mention is that to obviate **the** problem of nitrogen narcosis â all of those **blue** dots in our body â you remove **the** nitrogen and replace it with helium. Helium ’ s a gas ; there ’ re a lot of reasons why helium ’ s good, it ’ s a tiny molecule, it ’ s inert, it doesn ’ t give you narcosis. So that ’ s **the** basic concept we use. But **the** theory ’ s relatively easy. **The** tricky part is **the** implementation. So this is how I began about 15 years ago. I ’ ll admit, it wasn ’ t exactly **the** smartest of starts, but you ’ ve got to start somewhere. At **the** time, I wasn ’ t **the** only one who didn ’ t know what I was doing. Almost nobody did. This rig was actually used for a dive of 300 feet. But over time we got better at it, and we came up with this really sophisticated-looking rig with four scuba tanks, five regulators and all **the** right gas mixtures, all that good stuff. It was fine and dandy, it allowed us to go down and find new species. This picture was taken at 300 feet, catching new species of **fish**. **The** problem was it didn ’ t allow us much time. For all its bulk and size, it only gave us about 15 minutes at most down at those sorts of depths. We needed more time. There had to be a better way. And indeed, there is a better way. In 1994, I was fortunate enough to get access to these prototype closed-circuit rebreathers. Closed-circuit rebreather : what makes it different from scuba, and why is it better? Well, there are three main advantages to a rebreather. One, they ’ re quiet, they don ’ t make any noise. Two, they allow you to stay **underwater** longer. Three, they allow you to go deeper. How is it that they do that? In order to really understand how they do that, you have to look underneath **the** hood and see what ’ s going on. There are three basic systems to a closed-circuit rebreather. **The** most fundamental is called **the** breathing loop. It ’ s a breathing loop because you breathe off of it ; it ’ s a closed loop, and you breathe **the** same gas around and around. There ’ s a mouthpiece that you put in your mouth, and there ’ s a counter lung, or in this case, two counter lungs. **The** counter lungs aren ’ t high tech, they ’ re simply flexible bags. They allow you to mechanically breathe, mechanically ventilate. When you exhale, it goes in **the** exhale counter lung ; when you inhale, it comes from **the** inhale counter lung. It ’ s just pure mechanics, allowing you to cycle air through this breathing loop. **The** other component on a breathing loop is **the** carbon-dioxide-absorbent canister. Now, as we breathe, we produce **carbon** dioxide, and that **carbon** dioxide needs to be scrubbed out of **the** system. There ’ s a chemical filter in there that pulls **the carbon** dioxide out of **the** breathing gas, so that when it comes back to us, it ’ s safe to breathe again. That ’ s **the** breathing loop in a nutshell. **The** second main component of a closed-circuit rebreather is **the** gas system. **The** primary purpose of **the** gas system is to provide oxygen, to replenish **the** oxygen that your body consumes. So **the**

main tank, **the** main critical thing, is this oxygen gas supply cylinder we have here. But if we only had an oxygen gas supply cylinder, we wouldn ’ t be able to go very deep, because we ’ d run into oxygen toxicity very quickly. So we need another gas, something to dilute **the** oxygen with. And that, fittingly enough, is called **the** diluent gas supply. In our applications, we generally put air inside this diluent gas supply, because it ’ s a very cheap and easy source of nitrogen. So that ’ s where we get our nitrogen from. But if we want to go deeper, of course, we need another gas supply, and helium is what we really need to go deep. Usually we ’ ll have a slightly larger cylinder, mounted exterior on **the** rebreather, like this. That ’ s what we use to inject, as we start to do our deep dives. We also have a second oxygen cylinder, solely as a backup ; if there ’ s a problem with our first oxygen supply, we can continue to breathe. **The** way you manage all these different gases and different gas supplies is this really high-tech, sophisticated gas block up on **the** front here, where it ’ s easy to reach. It ’ s got **the valves** and knobs and things you need to inject **the** right gases at **the** right time. Normally, you don ’ t have to do that, because all of it ’ s done automatically with **the** electronics, **the** third system of a rebreather. **The** most critical part of a rebreather are **the** oxygen sensors. You need three, so if one goes bad, you know which one it is. You need voting logic. You also have three microprocessors. Any one of those **computers** can run **the** entire system, so if you have to lose two of them, there ’ s back-up power supplies. And there are multiple displays, to get **the** information to **the** diver. This is **the** high-tech gadgetry that allows us to do what we do on these deep dives. I can talk about it all day â just ask my wife â but I want to move on to something much more interesting. I ’ m going to take you on a deep dive, and show you what it ’ s like to do one. We start up on **the** boat, For all this high-tech, expensive equipment, this is still **the** best way to get in **the** water, just flop over **the** side of **the** boat. Now, as I showed you in **the** earlier diagram, these reefs that we dive on start out near **the surface** and they go almost vertically, completely straight down. So we drop in **the** water and go over **the** edge of this cliff, and then we start dropping, dropping, dropping. People ask if it takes a long time to get there. No ; it only takes a couple minutes to get down to three or four hundred feet, which is what we ’ re aiming for. It ’ s like skydiving in slow motion. It ’ s really very interesting. You ever see”**The Abyss**,”where Ed Harris is sinking down along **the** side of **the** wall? That ’ s what it feels like. Amazing. And down there, you find that **the** water is very clear, extremely clear, because there ’ s hardly any plankton. When you turn on your light and look around **the** caves, you ’ re confronted with a tremendous amount of diversity, much more than anyone used to believe. Now, not all of it is new species â that **fish** you see with **the** white stripe, that ’ s a known species. But if you look carefully into **the** cracks and crevices, you ’ ll see little things scurrying all over **the** place. There ’ s a just unbelievable diversity. It ’ s not just **fish**, either. These are crinoids, sponges, black corals. There ’ re some more **fishes**. Those **fishes** that you see now are new species. They ’ re still new species, because I had a video camera instead of my net, so they ’ re still waiting down there for someone to go find them. But this is what it looks like. And this kind of **habitat** just goes on and on and on for miles. This is Papua, New Guinea. Now little **fishes** and invertebrates aren ’ t **the** only things we see down there. We also see sharks, much more regularly than I would have expected to. We ’ re not quite sure why. What I want you to do now is imagine yourself 400 feet **underwater**, with all this high-tech gear on your back, you ’ re in a remote reef off Papua, New Guinea, thousands of miles from **the** nearest recompression chamber, and you ’ re completely surrounded by sharks. Diver 1 : Look at those... Diver 2 : Uh, oh... Uh, oh! Audience : Diver 1 : I think we have their attention... Richard Pyle : When you start talking like Donald Duck, there ’ s no situation in **the** world that can seem tense. So we ’ re down there â this is at 400 feet. That ’ s looking straight up, by **the** way, to give a sense of how far **the surface** is. And if you ’ re a biologist and know about sharks, and you want to assess, how much jeopardy am I really in here, there ’ s one question that sort of jumps to **the** forefront of your mind immediately, which is â Diver 1 : What kind of sharks? Diver 2 : Silvertip sharks. Diver 1 : Oh. RP : There are actually three species of sharks here. **The** silvertips are **the** ones with **the** white edges on **the fins**, and there ’ re also gray reef sharks and hammerheads off in **the** distance. And yes, it ’ s a little nerve-racking. Diver 2 : Hoo! That little guy is frisky! Audience : Now, you ’ ve seen video like this on TV a lot, and it ’ s very intimidating. I think it gives **the** wrong impression about sharks ; they ’ re actually not very dangerous animals. That ’ s why we weren ’ t worried much and were joking around. More people are killed by pigs, by lightning strikes, more people are killed at soccer games in England. There ’ s a lot of other ways you can die. And I ’ m not making that stuff up. Coconuts! You can get killed by a coconut more likely than killed by a shark. So sharks aren ’ t quite as dangerous as most people make them out to be. Now, I don ’ t know if any of you get US News and World Report â I got **the** recent issue. There ’ s a cover story about **the** great explorers of our time. **The** last article is entitled,”No New Frontiers.”It questions whether or not there really are any new frontiers out there, whether there are any true, hardcore discoveries that can still be made. My favorite line from **the** article : [... ’ discovery ’ can mean finding a guppy with an extra spine in its dorsal **fin**.”] I have to laugh ; they don ’ t call us **fish** nerds for nothing. We actually do get excited about finding a new dorsal spine in a guppy. But it ’ s much more than that. I want to show you a few of **the** guppies we ’ ve found over **the** years. This one â you can see how ugly it is. Even if you ignore **the** scientific value of this thing, look at **the** monetary value of this thing. A couple of these were sold through **the** aquarium trade to Japan, where they sold for 15, 000 dollars apiece. That ’ s half a million dollars a **pound**. Here ’ s another new angelfish we discovered. This one we first discovered back in **the** air days â **the** bad old air days, when we were doing these kind of dives with air. We were at 360 feet. I remember coming up from one of these deep dives and I had this fog, and **the** narcosis takes a little while to fade away, sort of like sobering up. I had this vague recollection of seeing this **yellow fish** with a black spot, and thought,”Damn, I should have caught one â I think that ’ s a new species.”Eventually, I looked in my bucket. Sure enough, I had caught one, I just completely forgot. So this one, we decided to give **the** name Centropyge narcosis to. That ’ s its official scientific name, in reference to its deep-dwelling habits. This is another neat one. When we first found it, we weren ’ t even sure what family it belonged to, so we just called it **the** Dr. Seuss **fish**, since it looked like something from one of those books. Now, this one ’ s pretty cool. If you go to Papua, New Guinea and go down 300 feet, you ’ ll see these big mounds. This may be hard to see, but they ’ re a couple meters in **diameter**. If you look closely, you ’ ll see there ’ s a little white and gray **fish** that hangs out near them. It turns out this little white **fish** builds these giant mounds, one pebble at a time. It ’ s extraordinary to find something like this. It ’ s not just new species, it ’ s new behaviors, new **ecology**, all kind of new things. What I ’ m going to show you now, quickly, is a sampling of **the** new species we ’ ve discovered. What ’ s extraordinary is not just **the** number of species we ’ re finding â though as you can see, that ’ s pretty amazing ; this is only half of what we ’ ve found â what ’ s extraordinary is how quickly we find them. We ’ re up to seven new species per hour of time we spend at that depth. If you go to an Amazon jungle and fog a tree, you may get a lot of bugs, but for **fishes**, there ’ s nowhere in **the** world you can get seven new species per hour of time. Now, we ’ ve done some back-of-the-envelope calculations, and we ’ re predicting there are probably about 2, 000 to 2, 500 new species in **the** Indo-Pacific alone. There are only five to six thousand known species, so a very large percentage of what is out there isn ’ t really known. We thought we had a handle on all **the** reef **fish** diversity â evidently not. I ’ m going to just close on a very somber note. At **the** beginning, I said I ’ d show you two extraordinary photographs. This is **the** second extraordinary photograph I ’ m going to show you. This one was taken at **the exact** moment I was down there filming those sharks. This was taken exactly 300 feet above my head. **The** reason this photograph is extraordinary is because it captures a moment in **the** very last minute of a person ’ s life. Less than 60 seconds after this picture was taken, this guy was dead. When we recovered his body, we figured out what had gone wrong. He had made a very simple mistake ; he turned **the** wrong **valve** when he filled his cylinder. he had 80 percent oxygen in his tank when he should have had 40. He had an oxygen toxicity seizure and he drowned. **The** reason I show this â not to put a downer on everything â but I just want to use it to key off my philosophy of life in general, which is that we all have two goals. **The** first goal we share with every other living thing on this **planet**, which is to survive. I call it perpetuation : **the** survival of **the** species and survival of ourselves, And those are both about perpetuating **the** **genome**. **The** second goal, for those of us who have mastered **the** first goal â call it spiritual fulfillment, call it financial success, you can call it any number of different things. I call it seeking joy, this pursuit of happiness. So, I guess my **theme** on this is this guy lived his life to **the** fullest, he absolutely did. You have to balance those two goals. If you live your whole life in fear â I mean, life is a sexually transmitted disease with 100 percent mortality. So you can ’ t live your life in fear. I thought that was an old one! But at **the** same time, you don ’ t want to get so focused on rule number two, goal number two, that you neglect goal number one. Because once you ’ re dead, you really can ’ t **enjoy** anything after that. So I wish you all **the** best of luck in maintaining that balance in your future **endeavors**. Thanks.

Text Sample 24: POS Dim 1 – 2004 – Score 19.00 – 2004/t000177.txt

I was trying to think, how is sync connected to happiness, and it **occurred** to me that for some reason we take pleasure in synchronizing. We like to dance together, we like singing together. And so, if you ’ ll put up with this, I would like to enlist your help with a first experiment today. **The** experiment is — and I notice, by **the** way, that when you applauded, that you did it in a typical North American way, that is, you were raucous and incoherent. You were not organized. It didn ’ t even **occur** to you to clap in unison. Do you think you could do it? I would like to see if this audience would — no, you haven ’ t practiced, as far as I know — can you get it together to clap in sync? Whoa! Now, that ’ s what we call emergent behavior. So I didn ’ t expect that, but — I mean, I expected you could synchronize. It didn ’ t **occur** to me you ’ d increase your **frequency**. It ’ s interesting. So what do we make of that? First of all, we know that you ’ re all brilliant. This is a room full of intelligent people, highly sensitive. Some trained musicians out there. Is that what enabled you to synchronize? So to put **the** question a little more seriously, let ’ s ask ourselves what are **the** minimum requirements for what you just did, for spontaneous synchronization. Do you need, for instance, to be as smart as you are? Do you even need a brain at all just to synchronize? Do you need to be alive? I mean, that ’ s a spooky thought, right? Inanimate **objects** that might spontaneously synchronize themselves. It ’ s real. In fact, I ’ ll try to explain today that sync is maybe one of, if not one of **the** most, perhaps **the** most pervasive drive in all of nature. It extends from **the** subatomic scale to **the** farthest reaches of **the** cosmos. It ’ s a deep tendency toward order in nature that opposes what we ’ ve all been taught about entropy. I mean, I ’ m not saying **the** law of entropy is wrong — it ’ s not. But there is a countervailing force in **the universe** — **the** tendency towards spontaneous order. And so that ’ s our **theme**. Now, to get into that, let me begin with what might have **occurred** to you immediately when you hear that we ’ re talking about synchrony in nature, which is **the** glorious example of birds that flock together, or **fish** swimming in organized schools. So these are not particularly intelligent creatures, and yet, as we ’ ll see, they exhibit beautiful ballets. This is from a BBC show called”Predators,”and what we ’ re looking at here are examples of synchrony that have to do with defense. When you ’ re small and vulnerable, like these starlings, or like **the fish**, it helps to swarm to avoid predators, to confuse predators. Let me be quiet for a second because this is so gorgeous. For a long time, biologists were puzzled by this behavior, wondering how it could be possible. We ’ re so used to choreography giving rise to synchrony. These creatures are not choreographed. They ’ re choreographing themselves. And only today is science starting to figure out how it works. I ’ ll show you a **computer** model made by Iain Couzin, a researcher at Oxford, that shows how swarms work. There are just three simple rules. First, all **the** individuals are only aware of their nearest neighbors. Second, all **the** individuals have a tendency to line up. And third, they ’ re all attracted to each other, but they try to keep a small distance apart. And when you build those three rules in, automatically you start to see swarms that look very much like **fish** schools or bird flocks. Now, **fish** like to stay close together, about a body length apart. Birds try to stay about three or four body lengths apart. But except for that difference, **the** rules are **the** same for both. Now, all this changes when a predator enters **the** scene. There ’ s a fourth rule : when a predator ’ s coming, get out of **the** way. Here on **the** model you see **the** predator attacking. **The** prey move out in random directions, and then **the** rule of attraction brings them back together again, so there ’ s this constant splitting and reforming. And you see that in nature. Keep in mind that, although it looks as if each individual is acting to cooperate, what ’ s really going on is a kind of selfish Darwinian behavior. Each is scattering away at random to try to save its scales or feathers. That is, out of **the** desire to save itself, each creature is following these rules, and that leads to something that ’ s safe for all of them. Even though it looks like they ’ re thinking as a group, they ’ re not. You might wonder what exactly is **the** advantage to being in a swarm, so you can think of several. As I say, if you ’ re in a swarm, your odds of being **the** unlucky one are reduced as compared to a small group. There are many eyes to spot danger. And you ’ ll see in **the** example with **the** starlings, with **the** birds, when this peregrine hawk is about to attack them, that actually waves of panic can propagate, sending messages over great distances. You ’ ll see — let ’ s see, it ’ s coming up possibly at **the** very end — maybe not. Information can be sent over half a kilometer away in a very short time through this mechanism. Yes, it ’ s happening here. See if you can see those waves propagating through **the** swarm. It ’ s beautiful. **The** birds are, we sort of understand, we think, from that **computer** model, what ’ s going on. As I say, it ’ s just those three simple rules, plus **the** one about watch out for predators. There doesn ’ t seem to be anything mystical about this. We don ’ t, however, really understand at a mathematical level. I ’ m a mathematician. We would like to be able to understand better. I mean, I showed you a **com-**

puter model, but a **computer** is not understanding. A **computer** is, in a way, just another experiment. We would really like to have a deeper insight into how this works and to understand, you know, exactly where this organization comes from. How do **the** rules give rise to **the** patterns? There is one case that we have begun to understand better, and it ’ s **the** case of fireflies. If you see fireflies in North America, like so many North American sorts of things, they tend to be independent operators. They ignore each other. They each do their own thing, flashing on and off, paying no attention to their neighbors. But in Southeast Asia — places like Thailand or Malaysia or Borneo — there ’ s a beautiful cooperative behavior that **occurs** among male fireflies. You can see it every night along **the** river banks. **The** trees, mangrove trees, are filled with fireflies communicating with light. Specifically, it ’ s male fireflies who are all flashing in perfect time together, in perfect synchrony, to reinforce a message to **the** females. And **the** message, as you can imagine, is”Come hither. **Mate** with me.”In a second I ’ m going to show you a slow motion of a single firefly so that you can get a sense. This is a single frame. Then on, and then off — a 30th of a second, there. And then watch this whole river bank, and watch how precise **the** synchrony is. On, more on and then off. **The** combined light from these beetles — these are actually tiny beetles — is so bright that fishermen out at **sea** can use them as navigating beacons to find their way back to their home rivers. It ’ s **stunning**. For a long time it was not believed when **the** first Western travelers, like Sir Francis Drake, went to Thailand and came back with tales of this unbelievable spectacle. No one believed them. We don ’ t see anything like this in Europe or in **the** West. And for a long time, even after it was documented, it was thought to be some kind of optical illusion. Scientific papers were published saying it was twitching eyelids that explained it, or, you know, a human being ’ s tendency to see patterns where there are none. But I hope you ’ ve convinced yourself now, with this nighttime video, that they really were very well synchronized. **Okay**, well, **the** issue then is, do we need to be alive to see this kind of spontaneous order, and I ’ ve already hinted that **the** answer is no. Well, you don ’ t have to be a whole creature. You can even be just a single cell. Like, take, for instance, your pacemaker cells in your heart right now. They ’ re keeping you alive. Every beat of your heart depends on this crucial region, **the** sinoatrial node, which has about 10, 000 independent cells that would each beep, have an **electrical** rhythm — a voltage up and down — to send a signal to **the** ventricles to pump. Now, your pacemaker is not a single cell. It ’ s this democracy of 10, 000 cells that all have to fire in unison for **the** pacemaker to work correctly. I don ’ t want to give you **the** idea that synchrony is always a good idea. If you have epilepsy, there is an instance of billions of brain cells, or at least millions, discharging in pathological concert. So this tendency towards order is not always a good thing. You don ’ t have to be alive. You don ’ t have to be even a single cell. If you look, for instance, at how lasers work, that would be a case of atomic synchrony. In a laser, what makes laser light so different from **the** light above my head here is that this light is incoherent — many different colors and different **frequencies**, sort of like **the** way you clapped initially — but if you were a laser, it would be rhythmic applause. It would be all atoms pulsating in unison, emitting light of one color, one **frequency**. Now comes **the** very risky part of my talk, which is to demonstrate that inanimate things can synchronize. Hold your breath for me. What I have here are two empty water bottles. This is not Keith Barry doing a magic trick. This is a klutz just playing with some water bottles. I have some metronomes here. Can you hear that? All right, so, I ’ ve got a metronome, and it ’ s **the** world ’ s smallest metronome, **the** — well, I shouldn ’ t advertise. Anyway, so this is **the** world ’ s smallest metronome. I ’ ve set it on **the** fastest setting, and

I ’ m going to now take another one set to **the** same setting. We can try this first. If I just put them on **the** table together, there ’ s no reason for them to synchronize, and they probably won ’ t. Maybe you ’ d better listen to them. I ’ ll stand here. What I ’ m hoping is that they might just drift apart because their **frequencies** aren ’ t perfectly **the** same. Right? They did. They were in sync for a while, but then they drifted apart. And **the** reason is that they ’ re not able to communicate. Now, you might think that ’ s a bizarre idea. How can metronomes communicate? Well, they can communicate through mechanical forces. So I ’ m going to give them a chance to do that. I also want to wind this one up a bit. How can they communicate? I ’ m going to put them on a movable platform, which is **the**”Guide to Graduate Study at Cornell.”**Okay**? So here it is. Let ’ s see if we can get this to work. My wife pointed out to me that it will work better if I put both on at **the** same time because otherwise **the** whole thing will tip over. All right. So there we go. Let ’ s see. OK, I ’ m not trying to cheat — let me start them out of sync. No, hard to even do that. All right. So before any one goes out of sync, I ’ ll just put those right there. Now, that might seem a bit whimsical, but this pervasiveness of this tendency towards spontaneous order sometimes has unexpected consequences. And a clear case of that, was something that happened in London in **the** year 2000. **The** Millennium Bridge was supposed to be **the** pride of London — a beautiful new footbridge erected across **the** Thames, first river crossing in over 100 years in London. There was a big competition for **the** design of this bridge, and **the** winning proposal was submitted by an unusual team — in **the** TED spirit, actually — of an architect — perhaps **the** greatest architect in **the** United Kingdom, Lord Norman Foster — working with an artist, a sculptor, Sir Anthony Caro, and an engineering firm, Ove Arup. And together they submitted a design based on Lord Foster ’ s vision, which was — he remembered as a kid reading Flash Gordon comic books, and he said that when Flash Gordon would come to an **abyss**, he would shoot what today would be a kind of a light saber. He would shoot his light saber across **the abyss**, making a blade of light, and then scamper across on this blade of light. He said,”That ’ s **the** vision I want to give to London. I want a blade of light across **the** Thames.”So they built **the** blade of light, and it ’ s a very thin ribbon of steel, **the** world ’ s — probably **the** flattest and thinnest suspension bridge there is, with cables that are out on **the** side. You ’ re used to suspension bridges with big droopy cables on **the** top. These cables were on **the** side of **the** bridge, like if you took a rubber band and stretched it taut across **the** Thames — that ’ s what ’ s holding up this bridge. Now, everyone was very excited to try it out. On opening day, thousands of Londoners came out, and something happened. And within two days **the** bridge was closed to **the** public. So I want to first show you some interviews with people who were on **the** bridge on opening day, who will describe what happened. Man : It really started moving sideways and slightly up and down, rather like being on **the** boat. Woman : Yeah, it felt unstable, and it was very windy, and I remember it had lots of flags up and down **the** sides, so you could definitely — there was something going on sideways, it felt, maybe. Interviewer : Not up and down? Boy : No. Interviewer : And not forwards and **backwards**? Boy : No. Interviewer : Just sideways. About how much was it moving, do you think? Boy : It was about — Interviewer : I mean, that much, or this much? Boy : About **the** second one. Interviewer : This much? Boy : Yeah. Man : It was at least six, six to eight **inches**, I would have thought. Interviewer : Right, so, at least this much? Man : Oh, yes. Woman : I remember wanting to get off. Interviewer : Oh, did you? Woman : Yeah. It felt odd. Interviewer : So it was enough to be scary? Woman : Yeah, but I thought that was just me. Interviewer : Ah! Now, tell me why you had to do this? Boy : We had to do this because, to keep in balance because if you didn ’ t keep your balance, then you would just fall over about, like, to **the left** or right, about 45 degrees. Interviewer : So just show me how you walk normally. Right. And then show me what it was like when **the** bridge started to go. Right. So you had to deliberately push your feet out sideways and — oh, and short steps? Man : That ’ s right. And it seemed obvious to me that it was probably **the** number of people on it. Interviewer : Were they deliberately walking in step, or anything like that? Man : No, they just had to conform to **the** movement of **the** bridge. Steven Strogatz : All right, so that already gives you a hint of what happened. Think of **the** bridge as being like this platform. Think of **the** people as being like metronomes. Now, you might not be used to thinking of yourself as a metronome, but after all, we do walk like — I mean, we oscillate back and forth as we walk. And especially if we start to walk like those people did, right? They all showed this strange sort of skating gait that they adopted once **the** bridge started to move. And so let me show you now **the** footage of **the** bridge. But also, after you see **the** bridge on opening day, you ’ ll see an interesting **clip** of work done by a bridge engineer at Cambridge named Allan McRobie, who figured out what happened on **the** bridge, and who built a bridge simulator to explain exactly what **the** problem was. It was a kind of unintended positive feedback loop between **the** way **the** people walked and **the** way **the** bridge began to move, that engineers knew nothing about. Actually, I think **the** first person you ’ ll see is **the** young engineer who was put in charge of this project. **Okay**. Interviewer : Did anyone get hurt? Engineer : No. Interviewer : Right. So it was quite small — Engineer : Yes. Interviewer : — but real? Engineer : Absolutely. Interviewer : You thought,”Oh, bother.”Engineer : I felt I was disappointed about it. We ’ d spent a lot of time designing this bridge, and we ’ d analyzed it, we ’ d checked it to codes — to heavier loads than **the** codes — and here it was doing something that we didn ’ t know about. Interviewer : You didn ’ t expect. Engineer : Exactly. Narrator : **The** most dramatic and shocking footage shows whole sections of **the** crowd — hundreds of people — apparently **rocking** from side to side in unison, not only with each other, but with **the** bridge. This synchronized movement seemed to be driving **the** bridge. But how could **the** crowd become synchronized? Was there something special about **the** Millennium Bridge that caused this effect? This was to be **the** focus of **the** investigation. Interviewer : Well, at last **the** simulated bridge is finished, and I can make it wobble. Now, Allan, this is all your fault, isn ’ t it? Allan McRobie : Yes. Interviewer : You designed this, yes, this simulated bridge, and this, you reckon, mimics **the** action of **the** real bridge? AM : It captures a lot of **the** physics, yes. Interviewer : Right. So if we get on it, we should be able to wobble it, yes? Allan McRobie is a bridge engineer from Cambridge who wrote to me, suggesting that a bridge simulator ought to wobble in **the** same way as **the** real bridge — provided we hung it on pendulums of exactly **the** right length. AM : This one ’ s only a couple of tons, so it ’ s fairly easy to get going. Just by walking. Interviewer : Well, it ’ s certainly going now. AM : It doesn ’ t have to be a real dangle. Just walk. It starts to go. Interviewer : It ’ s actually quite difficult to walk. You have to be careful where you put your feet down, don ’ t you, because if you get it wrong, it just throws you off your feet. AM : It certainly affects **the** way you walk, yes. You can ’ t walk normally on it. Interviewer : No. If you try and put one foot in front of another, it ’ s moving your feet away from under you. AM : Yes. Interviewer : So you ’ ve got to put your feet out sideways. So already, **the** simulator is making me walk in exactly **the** same way as our witnesses walked on **the** real bridge. AM :... ice-skating gait. There isn ’ t all this sort of snake way of walking. Interviewer : For a more convincing experiment, I wanted my own opening-day crowd, **the** sound check team. Their instructions : just walk normally. It ’ s really intriguing because none of these people is trying to drive it. They ’ re all having some difficulty walking. And **the** only way you can walk comfortably is by getting in step. But then, of course, everyone is driving **the** bridge. You can ’ t help it. You ’ re actually forced by **the** movement of **the** bridge to get into step, and therefore to drive it to move further. SS : All right, well, with that from **the** Ministry of Silly Walks, maybe I ’ d better end. I see I ’ ve gone over. But I hope that you ’ ll go outside and see **the** world in a new way, to see all **the** amazing synchrony around us. Thank you.

Text Sample 25: POS Dim 1 – 2004 – Score 19.00 – 2004/t000211.txt

You might be wondering why I ’ m wearing sunglasses, and one answer to that is, because I ’ m here to talk about glamour. So, we all think we know what glamour is. Here it is. It ’ s glamorous movie stars, like Marlene Dietrich. And it comes in a male form, too â very glamorous. Not only can he shoot, drive, **drink** â you know, he **drinks** wine, there actually is a little wine in there â and of course, always wears a tuxedo. But I think that glamour actually has a much broader meaning â one that is true for **the** movie stars and **the** fictional characters, but also comes in other forms. A magazine? Well, it ’ s certainly not this one. This is **the** least glamorous magazine on **the** newsstand â it ’ s all about sex tips. Sex tips are not glamorous. And Drew Barrymore, for all her wonderful charm, is not glamorous either. But there is a glamour of industry. This is Margaret Bourke-White ’ s â one of her pictures she did. **Fantastic**, glamorous pictures of steel mills and paper mills and all kinds of glamorous industrial places. And there ’ s **the** mythic glamour of **the** garage entrepreneur. This is **the** Hewlett-Packard garage. We know everyone who starts a business in a garage ends up founding Hewlett- Packard. There ’ s **the** glamour of physics. What could be more glamorous than understanding **the** entire **universe**, grand unification? And, by **the** way, it helps if you ’ re Brian Greene â he has other kinds of glamour. And there is, of course, this glamour. This is very, very glamorous : **the** glamour of outer space â and not **the** alien-style glamour, but **the** nice, clean, early ’ 60s version. So what do we mean by glamour? Well, one thing you can do if you want to know what glamour means is you can look in **the** dictionary. And it actually helps a lot more if you look in a very old dictionary, in this case **the** 1913 dictionary. Because for centuries, glamour had a very particular meaning, and **the** word was actually used differently from **the** way we think of it. You had”a”glamour. It wasn ’ t glamour as a quality â you”cast a glamour.”Glamour was a literal magic spell. Not a metaphorical one, **the** way we use it today, but a literal magic spell associated with witches and gypsies and to some extent, Celtic magic. And over **the** years, around **the** turn of **the** 20th century, it started to take on this other kind of deception â this definition for any **artificial** interest in, or association with, an **object** through which it appears delusively magnified or glorified. But still, glamour is an illusion. Glamour is a magic spell. And there ’ s something dangerous about glamour throughout most of history. When **the** witches cast a magic spell on you, it was not in your self-interest â it was to get you to act against your interest. Well of course, in **the** 20th century, glamour came to have this different meaning associated with Hollywood. And this is Hedy Lamarr. Hedy Lamarr said,”Anyone can look glamorous, all you have to do is sit there and look stupid.”But in fact, with all due respect to Hedy â about whom we ’ ll hear more later â there ’ s a lot more to it. There was a tremendous amount of technical achievement associated with creating this Hollywood glamour. There were scores of retouchers and lighting experts and make-up experts. You can go to **the** museum of Hollywood history in Hollywood and see Max Factor ’ s special rooms that he painted different colors depending on **the** complexion of **the** star he was going to make up. So you ’ ve got this highly stylized portrait of something that was not entirely of this earth â it was a portrait of a star. And actually, we see glamorized photos of stars all **the** time â they call them false color. Glamour is a form of falsification, but falsification to achieve a particular purpose. It may be to illuminate **the** star ; it may be to sell a film. And it involves a great deal of **technique**. It ’ s not â glamour is not something â you don ’ t wake up in **the** morning glamorous. I don ’ t care who you are. Even Nicole Kidman doesn ’ t wake up in **the** morning glamorous. There is a process of”idealization, glorification and dramatization,”and it ’ s not just **the** case for people. Glamour doesn ’ t have to be people. Architectural photography â Julius Schulman, who has talked about transfiguration, took this fabulous, **famous** picture of **the** Kauffman House. Architectural photography is extremely glamorous. It puts you into this special, special world. This is Alex Ross ’ s comic book art, which appears to be extremely realistic, as part of his style is he gives you a kind of realism in his comic art. Except that light doesn ’ t work this way in **the** real world. When you **stack** people in rows, **the** ones in **the** background look smaller than **the** ones in **the** foreground â but not in **the** world of glamour. What glamour is all about â I took this from a blurb in **the** table of contents of New York magazine, which was telling us that glamour is back â glamour is all about transcending **the** everyday. And I think that that ’ s starting to get at what **the** core that combines all sorts of glamour is. This is Filippino Lippi ’ s 1543 portrait of Saint Apollonia. And I don ’ t know who she is either, but this is **the** [16th] century equivalent of a supermodel. It ’ s a very glamorous portrait. Why is it glamorous? It ’ s glamorous, first, because she is beautiful â but that does not make you glamorous, that only makes you beautiful. She is graceful, she is mysterious and she is transcendent, and those are **the** central qualities of glamour. You don ’ t see her eyes ; they ’ re looking downward. She ’ s not looking away from you exactly, but you have to mentally imagine her world. She ’ s encouraging you to contemplate this higher world to which she belongs, where she can be completely tranquil while holding **the iron** instruments of her death by torture. Mel Gibson ’ s”Passion Of **The** Christ”â not glamorous. That ’ s glamour : that ’ s Michelangelo ’ s”Pieta,”where Mary is **the** same age as Jesus and they ’ re both awfully happy and pleasant. Glamour invites us to live in a different world. It has to simultaneously be mysterious, a little bit distant â that ’ s why, often in these glamour shots, **the** person is not looking at **the** audience, it ’ s why sunglasses are glamorous â but also not so far above us that we can ’ t identify with **the** person. In some sense, there has to be something like us. So as I say, in religious art, you know, God is not glamorous. God cannot be glamorous because God is omnipotent, omniscient â too far above us. And yet you will see in religious art, saints or **the** Virgin Mary will often be portrayed â not always â in glamorous forms. As I said earlier, glamour does not have to be about people, but it has to have this transcendent quality. What is it about Superman? Aside from Alex Ross ’ s style, which is very glamorous, one thing about Superman is he makes you believe that a man can fly. Glamour is all about transcending this world and getting to an idealized, perfect place. And this is one reason that modes of **transportation** tend to be extremely glamorous. **The** less experience we have with them, **the** more glamorous they are. So you can do a glamorized picture of a car, but you can ’ t do a glamorized picture of traffic. You can do a glamorized picture of an **airplane**, but not **the** inside. **The** notion is that it ’ s going to transport you, and **the** story is not about, you know, **the** guy in front of you in **the airplane**, who has this nasty little kid, or **the** big cough. **The** story is about where you ’ re arriving, or thinking about where you ’ re arriving. And this sense of being transported is one reason that we have glamour styling. This sort of streamlining styling is not just glamorous because we associate it with movies of that period, but because, in it ’ s streamlining, it transports us from **the** everyday. **The** same thing â□□ arches are very glamorous. Arches with stained glass â□□ even more glamorous. Staircases that curve away from you are glamorous. I happen to find that particular staircase picture very glamorous because, to me, it captures **the** whole promise of **the** academic contemplative life â□□ but maybe that ’ s because I went to Princeton. Anyway, skylines are super glamorous, city streets â□□ not so glamorous. You know, when you get, actually to this town it has reality. **The** horizon, **the** open road, is very, very glamorous. There are few things more glamorous than **the** horizon â□□ except, possibly, multiple horizons. Of course, here you don ’ t feel **the** cold, or **the** heat â□□ you just see **the** possibilities. In order to pull glamour off, you need this Renaissance quality of sprezzatura, which is a term coined by Castiglione in his book,”**The** Book Of **The** Courtier.”There ’ s **the** not-glamorous version of what it looks like today, after a few centuries. And sprezzatura is **the** art that conceals art. It makes things look effortless. You don ’ t think about how Nicole Kidman is maneuvering that dress â□□ she just looks completely natural. And I remember reading, after **the** Lara Croft movies, how Angelina Jolie would go home completely black and **blue**. Of course, they covered that with make-up, because Lara Croft did all those same stunts â□□ but she doesn ’ t get black and **blue**, because she has sprezzatura.”To conceal all art and make whatever is done or said appear to be without effort”: And this is one of **the** critical aspects of glamour. Glamour is about editing. How do you create **the** sense of transcendence, **the** sense of evoking a perfect world? **The** sense of, you know, life could be better, I could join this â□□ I could be a perfect person, I could join this perfect world. We don ’ t tell you all **the** grubby details. Now, this was kindly lent to me by Jeff Bezos, from last year. This is underneath Jeff ’ s desk. This is what **the** real world of **computers**, **lamps**, **electrical** appliances of all kinds, looks like. But if you look in a catalog â□□ particularly a catalog of modern, beautiful **objects** for your home â□□ it looks like this. There are no cords. Look next time you get these catalogs in your mail â□□ you can usually figure out where they hid **the** cord. But there ’ s always this illusion that if you buy this **lamp**, you will live in a world without cords. And **the** same thing is true of, if you buy this **laptop** or you buy this **computer** â□□ and even in these wireless eras, you don ’ t get to live in **the** world without cords. You have to have mystery and you have to have grace. And there she is â□□ Grace. This is **the** most glamorous picture, I think, ever. Part of **the** thing is that, in”Rear Window,”**the** question is, is she too glamorous to live in his world? And **the** answer is no, but of course it ’ s really just a movie. Here ’ s Hedy Lamarr again. And, you know, this kind of head covering is extremely glamorous because, like sunglasses, it conceals and reveals at **the** same time. Translucence is glamorous â□□ that ’ s why all these people wear pearls. It ’ s why barware is glamorous. Glamour is translucent â□□ not transparent, not opaque. It invites us into **the** world but it doesn ’ t give us a completely clear picture. And I think if Grace Kelly is **the** most glamorous person, maybe a spiral staircase with glass block may be **the** most glamorous interior shot, because a spiral staircase is incredibly glamorous. It has that sense of going up and away, and yet you never think about how you would really trip if you were â□□ particularly going down. And of course glass block has that sense of translucence. So, this session ’ s supposed to be about pure pleasure but glamour ’ s really partly about meaning. All individuals and all cultures have ideals that cannot possibly be realized in reality. They have contradictions, they uphold principles that are incommensurable with each other â whatever it is â and yet these ideals give meaning and purpose to our lives as cultures and as individuals. And **the** way we deal with that is we displace them â we put them into a golden world, an imagined world, an age of heroes, **the** world to come. And in **the** life of an individual, we often associate that with some **object**. **The** white picket fence, **the** perfect house. **The** perfect kitchen â no bills on **the** counter in **the** perfect kitchen. You know, you buy that Viking range, this is what your kitchen will look like. **The** perfect love life â symbolized by **the** perfect necklace, **the** perfect diamond ring. **The** perfect getaway in your perfect car. This is an interior design firm that is literally called Utopia. **The** perfect office â again, no cords, as far as I can tell. And certainly, no, it doesn ' t look a thing like my office. I mean, there ' s no paper on **the** desk. We want this golden world. And some people get rich enough, and if they have their ideals â in a sort of domestic sense, they get to acquire their perfect world. Dean Koontz built this fabulous home theater, which is â I don ' t think accidentally â in Art Deco style. That symbolizes this sense of being safe and at home. This is not always good, because what is your perfect world? What is your ideal, and also, what has been edited out? Is it something important?"**The** Matrix"is a movie that is all about glamour. I could do a whole talk on"**The** Matrix"and glamour. It was criticized for glamorizing violence, because, look â sunglasses and those long coats, and, of course, they could walk up walls and do all these kinds of things that are impossible in **the** real world. This is another Margaret Bourke-White photo. This is from Soviet Union. Attractive. I mean, look how happy **the** people are, and good-looking too. You know, we ' re marching toward Utopia. I ' m not a fan of PETA, but I think this is a great ad. Because what they ' re doing is they ' re saying, your coat ' s not so glamorous, what ' s been edited out is something important. But actually, what ' s even more important than remembering what ' s been edited out is thinking, are **the** ideals good? Because glamour can be very totalitarian and deceptive. And it ' s not just a matter of glamorizing cleaning your floor. This is from"Triumph Of **The** Will"â brilliant editing to cut together things. There ' s a glamour shot. National Socialism is all about glamour. It was a very aesthetic ideology. It was all about cleaning up Germany, and **the** West, and **the** world, and ridding it of anything unglamorous. So glamour can be dangerous. I think glamour has a genuine appeal, has a genuine value. I ' m not against glamour. But there ' s a kind of wonder in **the** stuff that gets edited away in **the** cords of life. And there is both a way to avoid **the** dangers of glamour and a way to broaden your appreciation of it. And that ' s to take Isaac Mizrahi ' s advice and confront **the** manipulation of it all, and sort of admit that manipulation is something that we **enjoy**, but also **enjoy** how it happens. And here ' s Hedy Lamarr. She ' s very glamorous but, you know, she invented spread- spectrum technology. So she ' s even more glamorous if you know that she really wasn ' t stupid, even though she thought she could look stupid. David Hockney talks about how **the** appreciation of this very glamorous painting is heightened if you think about **the** fact that it takes two weeks to paint this splash, which only took a **fraction** of a second to happen. There is a book out in **the** bookstore â it ' s called"Symphony In Steel,"and it ' s about **the** stuff that ' s hidden under **the** skin of **the** Disney Center. And that has a fascination. It ' s not necessarily glamorous, but **unveiling the** glamour has an appeal. There ' s a wonderful book called"Crowns"that ' s all these glamour pictures of black women in their church hats. And there ' s a quote from one of these women, and she talks about,"As a little girl, I ' d admire women at church with beautiful hats. They looked like beautiful dolls, like they ' d just stepped out of a magazine. But I also knew how hard they worked all week. Sometimes under those hats there ' s a lot of joy and a lot of sorrow."And, actually, you get more appreciation for glamour when you realize what went into creating it. Thank you.

Text Sample 26: POS Dim 1 – 2010 – Score 31.00 – 2010/t002836.txt

In **the** spirit of Jacques Cousteau, who said,"People protect what they love,"I want to share with you today what I love most in **the** ocean, and that ' s **the** incredible number and variety of animals in it that make light. My addiction began with this strange looking diving suit called Wasp ; that ' s not an acronym — just somebody thought it looked like **the** insect. It was actually developed for use by **the** offshore **oil** industry for diving on **oil** rigs down to a depth of 2, 000 feet. Right after I completed my Ph. D., I was lucky enough to be included with a group of scientists that was using it for **the** first time as a tool for ocean exploration. We trained in a tank in Port Hueneme, and then my first open ocean dive was in Santa Barbara Channel. It was an evening dive. I went down to a depth of 880 feet and turned out **the** lights. And **the** reason I turned out **the** lights is because I knew I would see this phenomenon of animals making light called bioluminescence. But I was totally unprepared for how much there was and how spectacular it was. I saw chains of jellyfish called siphonophores that were longer than this room, pumping out so much light that I could read **the** dials and gauges inside **the** suit without a flashlight ; and puffs and billows of what looked like luminous **blue** smoke ; and **explosions** of sparks that would swirl up out of **the** thrusters — just like when you throw a log on a campfire and **the** embers swirl up off **the** campfire, but these were icy, **blue** embers. It was breathtaking. Now, usually if people are familiar with bioluminescence at all, it ' s these guys ; it ' s fireflies. And there are a few other land-dwellers that can make light — some insects, earthworms, fungi — but in general, on land, it ' s really rare. In **the** ocean, it ' s **the** rule rather than **the** exception. If I go out in **the** open ocean environment, virtually anywhere in **the** world, and I drag a net from 3, 000 feet to **the surface**, most of **the** animals — in fact, in many places, 80 to 90 percent of **the** animals that I bring up in that net — make light. This makes for some pretty spectacular light shows. Now I want to share with you a little video that I shot from a submersible. I first developed this **technique** working from a little single-person submersible called Deep Rover and then adapted it for use on **the** Johnson Sea-Link, which you see here. So, mounted in front of **the** observation **sphere**, there ' s a a three-foot **diameter** hoop with a screen stretched across it. And inside **the sphere** with me is an intensified camera that ' s about as sensitive as a fully dark-adapted human eye, albeit a little fuzzy. So you turn on **the** camera, turn out **the** lights. That sparkle you ' re seeing is not luminescence, that ' s just electronic noise on these super intensified cameras. You don ' t see luminescence until **the** submersible begins to move forward through **the** water, but as it does, animals bumping into **the** screen are stimulated to bioluminesce. Now, when I was first doing this, all I was trying to do was count **the** numbers of sources. I knew my forward speed, I knew **the** area, and so I could figure out how many hundreds of sources there were per cubic meter. But I started to realize that I could actually identify animals by **the** type of flashes they produced. And so, here, in **the** Gulf of Maine at 740 feet, I can name pretty much everything you ' re seeing there to **the** species level. Like those big **explosions**, sparks, are from a little comb jelly, and there ' s krill and other kinds of crustaceans, and jellyfish. There was another one of those comb jellies. And so I ' ve worked with **computer** image analysis engineers to develop automatic recognition systems that can identify these animals and then extract **the** XYZ coordinate of **the** initial impact point. And we can then do **the** kinds of things that ecologists do on land, and do nearest neighbor distances. But you don ’ t always have to go down to **the** depths of **the** ocean to see a light show like this. You can actually see it in **surface** waters. This is some shot, by Dr. Mike Latz at Scripps Institution, of a dolphin swimming through bioluminescent plankton. And this isn ’ t someplace exotic like one of **the** bioluminescent **bays** in Puerto Rico, this was actually shot in San Diego Harbor. And sometimes you can see it even closer than that, because **the** heads on ships — that ’ s toilets, for any land lovers that are listening — are flushed with unfiltered seawater that often has bioluminescent plankton in it. So, if you stagger into **the** head late at night and you ’ re so toilet-hugging sick that you forget to turn on **the** light, you may think that you ’ re having a religious experience. So, how does a living creature make light? Well, that was **the** question that 19th century French physiologist Raphael Dubois, asked about this bioluminescent clam. He ground it up and he managed to get out a couple of chemicals ; one, **the** enzyme, he called luciferase ; **the** substrate, he called luciferin after Lucifer **the** Lightbearer. That terminology has stuck, but it doesn ’ t actually refer to specific chemicals because these chemicals come in a lot of different shapes and forms. In fact, most of **the** people studying bioluminescence today are focused on **the chemistry**, because these chemicals have proved so incredibly valuable for developing antibacterial agents, cancer fighting drugs, testing for **the** presence of life on Mars, **detecting** pollutants in our waters — which is how we use it at ORCA. In 2008, **the** Nobel Prize in **Chemistry** was awarded for work done on a molecule called green fluorescent protein that was isolated from **the** bioluminescent **chemistry** of a jellyfish, and it ’ s been equated to **the** invention of **the** microscope, in terms of **the** impact that it has had on cell **biology** and genetic engineering. Another thing all these molecules are telling us that, apparently, bioluminescence has **evolved** at least 40 times, maybe as many as 50 separate times in **evolutionary** history, which is a clear indication of how spectacularly important this trait is for survival. So, what is it about bioluminescence that ’ s so important to so many animals? Well, for animals that are trying to avoid predators by staying in **the** darkness, light can still be very useful for **the** three basic things that animals have to do to survive : and that ’ s find food, attract a **mate** and avoid being eaten. So, for example, this **fish** has a built-in headlight behind its eye that it can use for finding food or attracting a **mate**. And then when it ’ s not using it, it actually can roll it down into its head just like **the** headlights on your Lamborghini. This **fish** actually has high beams. And this **fish**, which is one of my favorites, has three headlights on each side of its head. Now, this one is **blue**, and that ’ s **the** color of most bioluminescence in **the** ocean because **evolution** has selected for **the** color that travels farthest through seawater in order to optimize communication. So, most animals make **blue** light, and most animals can only see **blue** light, but this **fish** is a really fascinating exception because it has two red light organs. And I have no idea why there ’ s two, and that ’ s something I want to solve some day — but not only can it see **blue** light, but it can see red light. So it uses its red bioluminescence like a sniper ’ s scope to be able to sneak up on animals that are blind to red light and be able to see them without being seen. It ’ s also got a little chin barbel here with a **blue** luminescent lure on it that it can use to attract prey from a long way off. And a lot of animals will use their bioluminescence as a lure. This is another one of my favorite **fish**. This is a viperfish, and it ’ s got a lure on **the** end of a long **fishing** rod that it arches in front of **the** toothy jaw that gives **the** viperfish its name. **The** teeth on this **fish** are so long that if they closed inside **the** mouth of **the fish**, it would actually impale its own brain. So instead, it **slides** in grooves on **the** outside of

the head. This is a Christmas tree of a **fish** ; everything on this **fish** lights up, it ' s not just that lure. It ' s got a built-in flashlight. It ' s got these jewel-like light organs on its belly that it uses for a type of camouflage that obliterates its shadow, so when it ' s swimming around and there ' s a predator looking up from below, it makes itself **disappear**. It ' s got light organs in **the** mouth, it ' s got light organs in every single scale, in **the fins**, in a mucus layer on **the** back and **the** belly, all used for different things — some of which we know about, some of which we don ' t. And we know a little bit more about bioluminescence thanks to Pixar, and I ' m very grateful to Pixar for sharing my favorite topic with so many people. I do wish, with their budget, that they might have spent just a tiny bit more money to pay a consulting fee to some poor, starving graduate student, who could have told them that those are **the** eyes of a **fish** that ' s been preserved in formalin. These are **the** eyes of a living anglerfish. So, she ' s got a lure that she sticks out in front of this living mousetrap of needle-sharp teeth in order to attract in some unsuspecting prey. And this one has a lure with all kinds of little interesting threads coming off it. Now we used to think that **the** different shape of **the** lure was to attract different types of prey, but then stomach content analyses on these **fish** done by scientists, or more likely their graduate students, have revealed that they all eat pretty much **the** same thing. So, now we believe that **the** different shape of **the** lure is how **the** male recognizes **the** female in **the** anglerfish world, because many of these males are what are known as **dwarf** males. This little guy has no visible means of self-support. He has no lure for attracting food and no teeth for eating it when it gets there. His only hope for existence on this **planet** is as a gigolo. He ' s got to find himself a babe and then he ' s got to latch on for life. So this little guy has found himself this babe, and you will note that he ' s had **the** good sense to attach himself in a way that he doesn ' t actually have to look at her. But he still knows a good thing when he sees it, and so he **seals the** relationship with an eternal kiss. His flesh fuses with her flesh, her bloodstream grows into his body, and he becomes nothing more than a little sperm sac. Well, this is a deep-**sea** version of Women ' s Lib. She always knows where he is, and she doesn ' t have to be monogamous, because some of these females come up with multiple males attached. So they can use it for finding food, for attracting **mates**. They use it a lot for defense, many different ways. A lot of them can release their luciferin or luferase in **the** water just **the** way a squid or an octopus will release an ink cloud. This **shrimp** is actually spewing light out of its mouth like a fire breathing dragon in order to blind or distract this viperfish so that **the shrimp** can swim away into **the** darkness. And there are a lot of different animals that can do this : There ' s jellyfish, there ' s squid, there ' s a whole lot of different crustaceans, there ' s even **fish** that can do this. This **fish** is called **the** shining tubeshoulder because it actually has a tube on its shoulder that can squirt out light. And I was luck enough to capture one of these when we were on a trawling expedition off **the** northwest coast of Africa for"Blue Planet,"for **the** deep portion of"Blue Planet."And we were using a special trawling net that we were able to bring these animals up alive. So we captured one of these, and I brought it into **the** lab. So I ' m holding it, and I ' m about to touch that tube on its shoulder, and when I do, you ' ll see bioluminescence coming out. But to me, what ' s shocking is not just **the** amount of light, but **the** fact that it ' s not just luciferin and luciferase. For this **fish**, it ' s actually whole cells with nuclei and membranes. It ' s energetically very costly for this **fish** to do this, and we have no idea why it does it — another one of these great mysteries that needs to be solved. Now, another form of defense is something called a burglar alarm — same reason you have a burglar alarm on your car ; **the** honking horn and flashing lights are meant to attract **the** attention of, hopefully, **the** police that will come and take **the** burglar away — when an animal ’ s caught in **the** clutches of a predator, its only hope for escape may be to attract **the** attention of something bigger and nastier that will attack their attacker, thereby affording them a chance for escape. This jellyfish, for example, has a spectacular bioluminescent display. This is us chasing it in **the** submersible. That ’ s not luminescence, that ’ s reflected light from **the** gonads. We capture it in a very special **device** on **the** front of **the** submersible that allows us to bring it up in really pristine condition, bring it into **the** lab on **the** ship. And then to generate **the** display you ’ re about to see, all I did was touch it once per second on its nerve ring with a sharp pick that ’ s sort of like **the** sharp tooth of a **fish**. And once this display gets going, I ’ m not touching it anymore. This is an unbelievable light show. It ’ s this pinwheel of light, and I ’ ve done calculations that show that this could be seen from as much as 300 feet away by a predator. And I thought,”You know, that might actually make a pretty good lure.”Because one of **the** things that ’ s frustrated me as a deep-sea explorer is how many animals there probably are in **the** ocean that we know nothing about because of **the** way we explore **the** ocean. **The** primary way that we know about what lives in **the** ocean is we go out and drag nets behind ships. And I defy you to name any other branch of science that still depends on hundreds of **year-old** technology. **The** other primary way is we go down with submersibles and remote-operated vehicles. I ’ ve made hundreds of dives in submersibles. When I ’ m sitting in a submersible though, I know that I ’ m not unobtrusive at all — I ’ ve got bright lights and noisy thrusters — any animal with any sense is going to be long gone. So, I ’ ve wanted for a long time to figure out a different way to explore. And so, sometime ago, I got this idea for a camera system. It ’ s not exactly rocket science. We call this thing Eye-in-the-Sea. And scientists have done this on land for years ; we just use a color that **the** animals can ’ t see and then a camera that can see that color. You can ’ t use infrared in **the sea**. We use far-red light, but even that ’ s a problem because it gets absorbed so quickly. Made an intensified camera, wanted to make this electronic jellyfish. Thing is, in science, you basically have to tell **the** funding agencies what you ’ re going to discover before they ’ ll give you **the** money. And I didn ’ t know what I was going to discover, so I couldn ’ t get **the** funding for this. So I kluged this together, I got **the** Harvey Mudd Engineering Clinic to actually do it as an undergraduate student project initially, and then I kluged funding from a whole bunch of different sources. Monterey Bay Aquarium Research Institute gave me time with their ROV so that I could test it and we could figure out, you know, for example, which colors of red light we had to use so that we could see **the** animals, but they couldn ’ t see us — get **the** electronic jellyfish working. And you can see just what a shoestring operation this really was, because we cast these 16 **blue** LEDs in epoxy and you can see in **the** epoxy mold that we used, **the** word Ziploc is still visible. Needless to say, when it ’ s kluged together like this, there were a lot of trials and tribulations getting this working. But there came a moment when it all came together, and everything worked. And, remarkably, that moment got caught on film by photographer Mark Richards, who happened to be there at **the** precise moment that we discovered that it all came together. That ’ s me on **the left**, my graduate student at **the** time, Erika Raymond, and Lee Fry, who was **the** engineer on **the** project. And we have this photograph posted in our lab in a place of honor with **the** caption :”Engineer satisfying two women at once.”And we were very, very happy. So now we had a system that we could actually take to some place that was kind of like an oasis on **the** bottom of **the** ocean that might be patrolled by large predators. And so, **the** place that we took it to was this place called a Brine Pool, which is in **the** northern part of **the** Gulf of Mexico. It ’ s a magical place. And I know this footage isn ’ t going to look like anything to you — we had a crummy camera at **the** time — but I was ecstatic. We ’ re at **the** edge of **the** Brine Pool, there ’ s a **fish** that ’ s swimming towards **the** camera. It ’ s clearly undisturbed by us. And I had my window into **the** deep **sea**. I, for **the** first time, could see what animals were doing down there when we weren ’ t down there disturbing them in some way. Four hours into **the** deployment, we had programmed **the** electronic jellyfish to come on for **the** first time. Eighty-six seconds after it went into its pinwheel display, we recorded this : This is a squid, over six feet long, that is so new to science, it cannot be placed in any known scientific family. I could not have asked for a better proof of concept. And based on this, I went back to **the** National Science Foundation and said,”This is what we will discover.”And they gave me enough money to do it right, which has involved developing **the** world ’ s first deep-sea webcam — which has been installed in **the** Monterey Canyon for **the** past year — and now, more recently, a modular form of this system, a much more mobile form that ’ s a lot easier to launch and recover, that I hope can be used on Sylvia ’ s”hope spots”to help explore and protect these areas, and, for me, learn more about **the** bioluminescence in these”hope spots.”So one of these take-home messages here is, there is still a lot to explore in **the** oceans. And Sylvia has said that we are destroying **the** oceans before we even know what ’ s in them, and she ’ s right. So if you ever, ever get an opportunity to take a dive in a submersible, say yes — a thousand times, yes — and please turn out **the** lights. I promise, you ’ ll love it. Thank you.

Text Sample 27: POS Dim 1 – 2010 – Score 29.00 – 2010/t002739.txt

I ’ ve been fascinated for a lifetime by **the** beauty, form and function of giant bluefin tuna. Bluefin are warmblooded like us. They ’ re **the** largest of **the** tunas, **the** second-largest **fish** in **the sea** â bony **fish**. They actually are a **fish** that is endothermic â powers through **the** ocean with warm muscles like a mammal. That ’ s one of our bluefin at **the** Monterey Bay Aquarium. You can see in its shape and its streamlined design it ’ s powered for ocean swimming. It flies through **the** ocean on its pectoral **fins**, gets lift, powers its movements with a lunate tail. It ’ s actually got a naked skin for most of its body, so it reduces friction with **the** water. This is what one of nature ’ s finest machines. Now, bluefin were revered by Man for all of human history. For 4, 000 years, we **fished** sustainably for this animal, and it ’ s evidenced in **the** art that we see from thousands of years ago. Bluefin are in cave paintings in France. They ’ re on coins that date back 3, 000 years. This **fish** was revered by humankind. It was **fished** sustainably till all of time, except for our generation. Bluefin are pursued wherever they go â there is a gold rush on Earth, and this is a gold rush for bluefin. There are traps that **fish** sustainably up until recently. And yet, **the** type of **fishing** going on today, with pens, with enormous stakes, is really **wiping** bluefin ecologically off **the planet**. Now bluefin, in general, goes to one place : Japan. Some of you may be guilty of having contributed to **the** demise of bluefin. They ’ re delectable muscle, rich in **fat** â absolutely taste delicious. And that ’ s their problem ; we ’ re eating them to death. Now in **the** Atlantic, **the** story is pretty simple. Bluefin have two populations : one large, one small. **The** North American population is **fished** at about 2, 000 ton. **The** European population and North African â **the Eastern** bluefin tuna â is **fished** at tremendous levels : 50, 000 tons over **the** last decade almost every year. **The** result is whether you ’ re looking at **the** West or **the** Eastern bluefin population, there ’ s been tremendous decline on both sides, as much as 90 percent if you go back with your baseline to 1950. For that, bluefin have been given a status equivalent to **tigers**, to **lions**, to certain African **elephants** and to pandas. These **fish** have been proposed for an endangered species listing in **the** past two months. They were voted on and rejected just two weeks ago, despite outstanding science that shows from two committees this **fish** meets **the** criteria of CITES I. And if it ’ s tunas you don ’ t care about, perhaps you might be interested that international long lines and pursing chase down tunas and bycatch animals such as leatherbacks, sharks, marlin, albatross. These animals and their demise **occurs** in **the** tuna fisheries. **The** challenge we face is that we know very little about tuna, and everyone in **the** room knows what it looks like when an African **lion** takes down its prey. I doubt anyone has seen a giant bluefin feed. This tuna symbolizes what ’ s **the** problem for all of us in **the** room. It ’ s **the** 21st century, but we really have only just begun to really study our oceans in a deep way. Technology has come of age that ’ s allowing us to see **the** Earth from space and go deep into **the seas** remotely. And we ’ ve got to use these technologies immediately to get a better understanding of how our ocean realm works. Most of us from **the** ship â even I â look out at **the** ocean and see this homogeneous **sea**. We don ’ t know where **the** structure is. We can ’ t tell where are **the** watering holes like we can on an African plain. We can ’ t see **the** corridors, and we can ’ t see what it is that brings together a tuna, a leatherback and an albatross. We ’ re only just beginning to understand how **the** physical oceanography and **the** biological oceanography come together to create a seasonal force that actually causes **the** upwelling that might make a hot spot a hope spot. **The** reasons these challenges are great is that technically it ’ s difficult to go to **sea**. It ’ s hard to study a bluefin on its turf, **the** entire Pacific realm. It ’ s really tough to get up close and personal with a mako shark and try to put a **tag** on it. And then imagine being Bruce Mate ’ s team from OSU, getting up close to a **blue whale** and fixing a **tag** on **the blue whale** that stays, an engineering challenge we ’ ve yet to really overcome. So **the** story of our team, a dedicated team, is **fish** and chips. We basically are taking **the** same satellite phone parts, or **the** same parts that are in your **computer**, chips. We ’ re putting them together in unusual ways, and this is taking us into **the** ocean realm like never before. And for **the** first time, we ’ re able to watch **the** journey of a tuna beneath **the** ocean using light and **photons** to measure sunrise and sunset. Now, I ’ ve been working with tunas for over 15 years. I have **the** privilege of being a partner with **the** Monterey Bay Aquarium. We ’ ve actually taken a sliver of **the** ocean, put it behind glass, and we together have put bluefin tuna and yellowfin tuna on display. When **the** veil of bubbles lifts every morning, we can actually see a community from **the** Pelagic ocean, one of **the** only places on Earth you can see giant bluefin swim by. We can see in their beauty of form and function, their ceaseless activity. They ’ re flying through their space, ocean space. And we can bring two million people a year into contact with this **fish** and show them its beauty. Behind **the** scenes is a working lab at Stanford University partnered with **the** Monterey Bay Aquarium. Here, for over 14 or 15 years, we ’ ve actually brought in both bluefin and yellowfin in captivity. We ’ d been studying these **fish**, but first we had to learn how to husbandry them. What do they like to eat? What is it that they ’ re happy with? We go in **the** tanks with **the** tuna â we touch their naked skin â it ’ s pretty amazing. It feels wonderful. And then, better yet, we ’ ve got our own version of tuna whisperers, our own Chuck Farwell, Alex Norton, who can take a big tuna and in one motion, put it into an envelope of water, so that we can actually work with **the** tuna and learn **the techniques** it takes to not injure this **fish** who never sees a boundary in **the** open **sea**. Jeff and Jason there, are scientists who are going to take a tuna and put it in **the** equivalent of a treadmill, a flume. And that tuna thinks it ’ s going to Japan, but it ’ s staying in place. We ’ re actually measuring its oxygen **consumption**, its energy **consumption**. We ’ re taking this data and building better models. And when I see that tuna â this is my favorite view â I begin to wonder : how did this **fish** solve **the** longitude problem before we did? So take a look at that animal. That ’ s **the** closest you ’ ll probably ever get. Now, **the** activities from **the** lab have taught us now how to go out in **the** open ocean. So in a program called Tag-A-Giant we ’ ve actually gone from Ireland to Canada, from Corsica to Spain. We ’ ve **fished** with many nations around **the** world in an effort to basically put electronic **computers** inside giant tunas. We ’ ve actually **tagged** 1, 100 tunas. And I ’ m going to show you three **clips**, because I **tagged** 1, 100 tunas. It ’ s a very hard process, but it ’ s a ballet. We bring **the** tuna out, we measure it. A team of fishers, captains, scientists and technicians work together to keep this animal out of **the** ocean for about four to five minutes. We put water over its gills, give it oxygen. And then with a lot of effort, after **tagging**, putting in **the computer**, making sure **the** stalk is sticking out so it senses **the** environment, we send this **fish** back into **the sea**. And when it goes, we ’ re always happy. We see a flick of **the** tail. And from our data that gets collected, when that **tag** comes back, because a fisher returns it for a thousand-dollar reward, we can get tracks beneath **the sea** for up to five years now, on a backboned animal. Now sometimes **the** tunas are really large, such as this **fish** off Nantucket. But that ’ s about half **the** size of **the** biggest tuna we ’ ve ever **tagged**. It takes a human effort, a team effort, to bring **the fish** in. In this case, what we ’ re going to do is put a pop-up satellite archival **tag** on **the** tuna. This **tag** rides on **the** tuna, senses **the** environment around **the** tuna and actually will come off **the fish**, detach, float to **the surface** and send back to Earth-orbiting satellites position data estimated by math on **the tag**, pressure data and temperature data. And so what we get then from **the** pop-up satellite **tag** is we get away from having to have a human interaction to recapture **the tag**. Both **the** electronic **tags** I ’ m talking about are expensive. These **tags** have been engineered by a variety of teams in North America. They are some of our finest instruments, our new technology in **the** ocean today. One community in general has given more to help us than any other community. And that ’ s **the** fisheries off **the** state of North Carolina. There are two villages, Harris and Morehead City, every winter for over a decade, held a party called Tag-A-Giant, and together, fishers worked with us to **tag** 800 to 900 **fish**. In this case, we ’ re actually going to measure **the fish**. We ’ re going to do something that in recent years we ’ ve started : take a mucus sample. Watch how shiny **the** skin is ; you can see my reflection there. And from that mucus, we can get gene profiles, we can get information on gender, checking **the** pop-up **tag** one more time, and then it ’ s out in **the** ocean. And this is my favorite. With **the** help of my former postdoc, Gareth Lawson, this is a gorgeous picture of a single tuna. This tuna is actually moving on a numerical ocean. **The** warm is **the** Gulf Stream, **the** cold up there in **the** Gulf of Maine. That ’ s where **the** tuna wants to go â it wants to forage on schools of herring â but it can ’ t get there. It ’ s too cold. But then it warms up, and **the** tuna pops in, gets some **fish**, maybe comes back to home base, goes in again and then comes back to winter down there in North Carolina and then on to **the** Bahamas. And my favorite scene, three tunas going into **the** Gulf of Mexico. Three tunas **tagged**. Astronomically, we ’ re **calculating** positions. They ’ re coming together. That could be tuna sex â and there it is. That is where **the** tuna spawn. So from data like this, we ’ re able now to put **the** map up, and in this map you see thousands of positions generated by this decade and a half of **tagging**. And now we ’ re showing that tunas on **the** western side go to **the eastern** side. So two populations of tunas â that is, we have a Gulf population, one that we can **tag** â they go to **the** Gulf of Mexico, I showed you that â and a second population. Living amongst our tunas â our North American tunas â are European tunas that go back to **the** Med. On **the** hot spots â **the** hope spots â they ' re mixed populations. And so what we ' ve done with **the** science is we ' re showing **the** International Commission, building new models, showing them that a two-stock no-mixing model â to this day, used to reject **the** CITES treaty â that model isn ' t **the** right model. This model, a model of overlap, is **the** way to move forward. So we can then predict where management places should be. Places like **the** Gulf of Mexico and **the** Mediterranean are places where **the** single species, **the** single population, can be captured. These become forthright in places we need to protect. **The** center of **the** Atlantic where **the** mixing is, I could imagine a policy that lets Canada and America **fish**, because they manage their fisheries well, they ' re doing a good job. But in **the** international realm, where **fishing** and overfishing has really gone wild, these are **the** places that we have to make hope spots in. That ' s **the** size they have to be to protect **the** bluefin tuna. Now in a second project called **Tagging** of Pacific Pelagics, we took on **the planet** as a team, those of us in **the** Census of Marine Life. And, funded primarily through Sloan Foundation and others, we were able to actually go in, in our project â we ' re one of 17 field programs and begin to take on **tagging** large numbers of predators, not just tunas. So what we ' ve done is actually gone up to **tag** salmon shark in Alaska, met salmon shark on their home territory, followed them catching salmon and then went in and figured out that, if we take a salmon and put it on a line, we can actually take up a salmon shark â This is **the** cousin of **the** white shark â and very carefully â note, I say"very carefully,"â we can actually keep it calm, put a hose in its mouth, keep it off **the** deck and then **tag** it with a satellite **tag**. That satellite **tag** will now have your shark phone home and send in a message. And that shark leaping there, if you look carefully, has an antenna. It ' s a free swimming shark with a satellite **tag** jumping after salmon, sending home its data. Salmon sharks aren ' t **the** only sharks we **tag**. But there goes salmon sharks with this meter-level resolution on an ocean of temperature â warm colors are warmer. Salmon sharks go down to **the** tropics to pup and come into Monterey. Now right next door in Monterey and up at **the** Farallones are a white shark team led by Scott Anderson â there â and Sal Jorgensen. They can throw out a target â it ' s a carpet shaped like a **seal** â and in will come a white shark, a curious critter that will come right up to our 16-ft. boat. It ' s a several thousand-pound animal. And we ' ll wind in **the** target. And we ' ll place an acoustic **tag** that says,"OMSHARK 10165,"or something like that, acoustically with a ping. And then we ' ll put on a satellite **tag** that will give us **the** long- distance journeys with **the** light-based geolocation algorithms solved on **the computer** that ' s on **the fish**. So in this case, Sal ' s looking at two **tags** there, and there they are : **the** white sharks of California going off to **the** white shark cafe and coming back. We also **tag** makos with our NOAA colleagues, **blue** sharks. And now, together, what we can see on this ocean of color that ' s temperature, we can see ten-day worms of makos and salmon sharks. We have white sharks and **blue** sharks. For **the** first time, an ecoscape as large as ocean-scale, showing where **the** sharks go. **The** tuna team from TOPP has done **the** unthinkable : three teams **tagged** 1, 700 tunas, bluefin, yellowfin and albacore all at **the** same time â carefully rehearsed **tagging** programs in which we go out, pick up juvenile tunas, put in **the tags** that actually have **the** sensors, stick out **the** tuna and then let them go. They get returned, and when they get returned, here on a NASA numerical ocean you can see bluefin in **blue** go across their corridor, returning to **the** Western Pacific. Our team from UCSC has **tagged**

elephant seals with **tags** that are glued on their heads, that come off when they slough. These **elephant seals** cover half an ocean, take data down to 1, 800 feet â amazing data. And then there ’ s Scott Shaffer and our shearwaters wearing tuna **tags**, light-based **tags**, that now are going to take you from New Zealand to Monterey and back, journeys of 35, 000 nautical miles we had never seen before. But now with light-based geolocation **tags** that are very small, we can actually see these journeys. Same thing with Laysan albatross who travel an entire ocean on a trip sometimes, up to **the** same zone **the** tunas use. You can see why they might be caught. Then there ’ s George Schillinger and our leatherback team out of Playa Grande **tagging** leatherbacks that go right past where we are. And Scott Benson ’ s team that showed that leatherbacks go from Indonesia all **the** way to Monterey. So what we can see on this moving ocean is we can finally see where **the** predators are. We can actually see how they ’ re using ecospaces as large as an ocean. And from this information, we can begin to map **the** hope spots. So this is just three years of data right here â and there ’ s a decade of this data. We see **the pulse** and **the** seasonal activities that these animals are going on. So what we ’ re able to do with this information is boil it down to hot spots, 4, 000 deployments, a huge herculean task, 2, 000 **tags** in an area, shown here for **the** first time, off **the** California coast, that appears to be a gathering place. And then for sort of an encore from these animals, they ’ re helping us. They ’ re carrying instruments that are actually taking data down to 2, 000 meters. They ’ re taking information from our **planet** at very critical places like Antarctica and **the** Poles. Those are **seals** from many countries being released who are sampling underneath **the** ice sheets and giving us temperature data of oceanographic quality on both poles. This data, when visualized, is captivating to watch. We still haven ’ t figured out best how to visualize **the** data. And then, as these animals swim and give us **the** information that ’ s important to climate issues, we also think it ’ s critical to get this information to **the** public, to engage **the** public with this kind of data. We did this with **the** Great Turtle Race â **tagged turtles**, brought in four million hits. And now with Google ’ s Oceans, we can actually put a white shark in that ocean. And when we do and it swims, we see this magnificent bathymetry that **the** shark knows is there on its path as it goes from California to Hawaii. But maybe Mission Blue can fill in that ocean that we can ’ t see. We ’ ve got **the** capacity, NASA has **the** ocean. We just need to put it together. So in conclusion, we know where Yellowstone is for North America ; it ’ s off our coast. We have **the** technology that ’ s shown us where it is. What we need to think about perhaps for Mission Blue is increasing **the** biologging capacity. How is it that we can actually take this type of activity elsewhere? And then finally â to basically get **the** message home â maybe use live links from animals such as **blue whales** and white sharks. Make killer apps, if you will. A lot of people are excited when sharks actually went under **the** Golden Gate Bridge. Let ’ s connect **the** public to this activity right on their iPhone. That way we do away with a few internet myths. So we can save **the** bluefin tuna. We can save **the** white shark. We have **the** science and technology. Hope is here. Yes we can. We need just to apply this capacity further in **the** oceans. Thank you.

Text Sample 28: POS Dim 1 – 2010 – Score 28.00 – 2010/t002801.txt

I would like to share with you this morning some stories about **the** ocean through my work as a still photographer for National Geographic magazine. I guess I became an **underwater** photographer and a photojournalist because

I fell in love with **the sea** as a child. And I wanted to tell stories about all **the** amazing things I was seeing **underwater**, incredible wildlife and interesting behaviors. And after even 30 years of doing this, after 30 years of exploring **the** ocean, I never cease to be amazed at **the** extraordinary encounters that I have while I ’ m at **sea**. But more and more frequently these days I ’ m seeing terrible things **underwater** as well, things that I don ’ t think most people realize. And I ’ ve been compelled to turn my camera towards these issues to tell a more complete story. I want people to see what ’ s happening **underwater**, both **the** horror and **the** magic. **The** first story that I did for National Geographic, where I recognized **the** ability to include environmental issues within a natural history coverage, was a story I proposed on harp **seals**. **The** story I wanted to do initially was just a small focus to look at **the** few weeks each year where these animals migrate down from **the** Canadian arctic to **the** Gulf of St. Lawrence in Canada to engage in courtship, mating and to have their pups. And all of this is played out against **the** backdrop of transient pack ice that moves with wind and tide. And because I ’ m an **underwater** photographer, I wanted to do this story from both above and below, to make pictures like this that show one of these little pups making its very first swim in **the** icy 29-degree water. But as I got more involved in **the** story, I realized that there were two big environmental issues I couldn ’ t ignore. **The** first was that these animals continue to be hunted, killed with hakapiks at about eight, 15 days old. It actually is **the** largest marine mammal slaughter on **the planet**, with hundreds of thousands of these **seals** being killed every year. But as disturbing as that is, I think **the** bigger problem for harp **seals** is **the** loss of **sea** ice due to global warming. This is an aerial picture that I made that shows **the** Gulf of St. Lawrence during harp **seal** season. And even though we see a lot of ice in this picture, there ’ s a lot of water as well, which wasn ’ t there historically. And **the** ice that is there is quite thin. **The** problem is that these pups need a stable platform of solid ice in order to nurse from their moms. They only need 12 days from **the** moment they ’ re born until they ’ re on their own. But if they don ’ t get 12 days, they can fall into **the** ocean and die. This is a photo that I made showing one of these pups that ’ s only about five or seven days old — still has a little bit of **the** umbilical cord on its belly — that has fallen in because of **the** thin ice, and **the** mother is frantically trying to push it up to breathe and to get it back to stable purchase. This problem has continued to grow each year since I was there. I read that last year **the** pup mortality rate was 100 percent in parts of **the** Gulf of St. Lawrence. So, clearly, this species has a lot of problems going forward. This ended up becoming a cover story at National Geographic. And it received quite a bit of attention. And with that, I saw **the** potential to begin doing other stories about ocean problems. So I proposed a story on **the** global **fish** crisis, in part because I had personally witnessed a lot of degradation in **the** ocean over **the** last 30 years, but also because I read a scientific paper that stated that 90 percent of **the** big **fish** in **the** ocean have **disappeared** in **the** last 50 or 60 years. These are **the** tuna, **the** billfish and **the** sharks. When I read that, I was blown away by those numbers. I thought this was going to be headline news in every media outlet, but it really wasn ’ t, so I wanted to do a story that was a very different kind of **underwater** story. I wanted it to be more like war photography, where I was making harder-hitting pictures that showed readers what was happening to marine wildlife around **the planet**. **The** first component of **the** story that I thought was essential, however, was to give readers a sense of appreciation for **the** ocean animals that they were eating. You know, I think people go into a restaurant, and somebody orders a steak, and we all know where steak comes from, and somebody orders a chicken, and we know what a chicken is, but when they ’ re eating bluefin sushi, do they have any sense of **the** magnificent animal that they ’ re consuming? These are **the lions** and **tigers** of **the sea**. In reality, these animals have no terrestrial counterpart ; they ’ re unique in **the** world. These are animals that can practically swim from **the** equator to **the** poles and can crisscross entire oceans in **the** course of a year. If we weren ’ t so efficient at catching them, because they grow their entire life, would have 30-year-old bluefin out there that weigh a ton. But **the** truth is we ’ re way too efficient at catching them, and their stocks have collapsed worldwide. This is **the** daily auction at **the** Tsukiji Fish Market that I photographed a couple years ago. And every single day these tuna, bluefin like this, are **stacked** up like cordwood, just warehouse after warehouse. As I wandered around and made these pictures, it sort of **occurred** to me that **the** ocean ’ s not a grocery store, you know. We can ’ t keep taking without expecting serious consequences as a result. I also, with **the** story, wanted to show readers how **fish** are caught, some of **the** methods that are used to catch **fish**, like a bottom trawler, which is one of **the** most common methods in **the** world. This was a small net that was being used in Mexico to catch **shrimp**, but **the** way it works is essentially **the** same everywhere in **the** world. You have a large net in **the** middle with two steel doors on either end. And as this assembly is towed through **the** water, **the** doors meet resistance with **the** ocean, and it opens **the** mouth of **the** net, and they place floats at **the** top and a lead line on **the** bottom. And this just drags over **the** bottom, in this case to catch **shrimp**. But as you can imagine, it ’ s catching everything else in its path as well. And it ’ s destroying that precious benthic community on **the** bottom, things like sponges and corals, that critical **habitat** for other animals. This photograph I made of **the** fisherman holding **the shrimp** that he caught after towing his nets for one hour. So he had a handful of **shrimp**, maybe seven or eight **shrimp**, and all those other animals on **the** deck of **the** boat are bycatch. These are animals that died in **the** process, but have no commercial value. So this is **the** true cost of a **shrimp** dinner, maybe seven or eight **shrimp** and 10 **pounds** of other animals that had to die in **the** process. And to make that point even more visual, I swam under **the shrimp** boat and made this picture of **the** guy shoveling this bycatch into **the sea** as trash and photographed this cascade of death, you know, animals like guitarfish, bat rays, flounder, pufferfish, that only an hour before, were on **the** bottom of **the** ocean, alive, but now being thrown back as trash. I also wanted to focus on **the** shark **fishing** industry because, currently on **planet** Earth, we ’ re killing over 100 million sharks every single year. But before I went out to photograph this component, I sort of wrestled with **the** notion of how do you make a picture of a dead shark that will resonate with readers You know, I think there ’ s still a lot of people out there who think **the** only good shark is a dead shark. But this one morning I jumped in and found this thresher that had just recently died in **the** gill net. And with its huge pectoral **fins** and eyes still very visible, it struck me as sort of a crucifixion, if you will. This ended up being **the** lead picture in **the** global fishery story in National Geographic. And I hope that it helped readers to take notice of this problem of 100 million sharks. And because I love sharks — I ’ m somewhat obsessed with sharks — I wanted to do another, more celebratory, story about sharks, as a way of talking about **the** need for shark conservation. So I went to **the** Bahamas because there ’ re very few places in **the** world where sharks are doing well these days, but **the** Bahamas seem to be a place where stocks were reasonably healthy, largely due to **the** fact that **the** government there had outlawed longlining several years ago. And I wanted to show several species that we hadn ’ t shown much in **the** magazine and worked in a number of locations. One of **the** locations was this place called Tiger Beach, in **the** northern Bahamas where **tiger** sharks aggregate in shallow water. This is a low-altitude photograph that I made showing our dive boat with about a dozen of these big old **tiger** sharks sort of just swimming around behind. But **the** one thing I definitely didn ’ t want to do with this coverage was to continue to portray sharks as something like monsters. I didn ’ t want them to be overly threatening or scary. And with this photograph of a beautiful 15-feet, probably 14-feet, I guess, female **tiger** shark, I sort of think I got to that goal, where she was swimming with these little barjacks off her nose, and my strobe created a shadow on her face. And I think it ’ s a gentler picture, a little less threatening, a little more respectful of **the** species. I also **searched** on this story for **the** elusive great hammerhead, an animal that really hadn ’ t been photographed much until maybe about seven or 10 years ago. It ’ s a very solitary creature. But this is an animal that ’ s considered data deficient by science in both Florida and in **the** Bahamas. You know, we know almost nothing about them. We don ’ t know where they migrate to or from, where they **mate**, where they have their pups, and yet, hammerhead populations in **the** Atlantic have declined about 80 percent in **the** last 20 to 30 years. You know, we ’ re losing them faster than we can possibly find them. This is **the** oceanic whitetip shark, an animal that is considered **the** fourth most dangerous species, if you pay attention to such lists. But it ’ s an animal that ’ s about 98 percent in decline throughout most of its range. Because this is a pelagic animal and it lives out in **the** deeper water, and because we weren ’ t working on **the** bottom, I brought along a shark cage here, and my friend, shark biologist Wes Pratt is inside **the** cage. You ’ ll see that **the** photographer, of course, was not inside **the** cage here, so clearly **the** biologist is a little smarter than **the** photographer I guess. And lastly with this story, I also wanted to focus on baby sharks, shark nurseries. And I went to **the island** of Bimini, in **the** Bahamas, to work with lemon shark pups. This is a photo of a lemon shark pup, and it shows these animals where they live for **the** first two to three years of their lives in these protective mangroves. This is a very sort of un-shark-like photograph. It ’ s not what you typically might think of as a shark picture. But, you know, here we see a shark that ’ s maybe 10 or 11 **inches** long swimming in about a foot of water. But this is crucial **habitat** and it ’ s where they spend **the** first two, three years of their lives, until they ’ re big enough to go out on **the** rest of **the** reef. After I left Bimini, I actually learned that this **habitat** was being bulldozed to create a new **golf** course and resort. And other recent stories have looked at single, flagship species, if you will, that are at risk in **the** ocean as a way of talking about other threats. One such story I did documented **the** leatherback **sea turtle**. This is **the** largest, widest-ranging, deepest-diving and oldest of all **turtle** species. Here we see a female crawling out of **the** ocean under moonlight on **the island** of Trinidad. These are animals whose lineage dates back about 100 million years. And there was a time in their lifespan where they were coming out of **the** water to nest and saw Tyrannosaurus rex running by. And today, they crawl out and see condominiums. But despite this amazing longevity, they ’ re now considered critically endangered. In **the** Pacific, where I made this photograph, their stocks have declined about 90 percent in **the** last 15 years. This is a photograph that shows a hatchling about to taste saltwater for **the** very first time beginning this long and perilous journey. Only one in a thousand leatherback hatchlings will reach maturity. But that ’ s due to natural predators like vultures that pick them off on a beach or predatory **fish** that are waiting offshore. Nature has learned to compensate with that, and females have multiple clutches of eggs to overcome those odds. But what they can ’ t deal with is anthropogenic stresses, human things, like this picture that shows a leatherback caught at night in a gill net. I actually jumped in and photographed this, and with **the** fisherman ’ s permission, I cut **the turtle** out, and it was able to swim free. But, you know, thousands of other leatherbacks each year are not so fortunate, and **the** species ' future is in great danger. Another charismatic megafauna species that I worked with is **the** story I did on **the** right **whale**. And essentially, **the** story is this with right **whales**, that about a million years ago, there was one species of right **whale** on **the planet**, but as land masses moved around and oceans became isolated, **the** species sort of separated, and today we have essentially two distinct stocks. We have **the** Southern right **whale** that we see here and **the** North Atlantic right **whale** that we see here with a mom and calf off **the** coast of Florida. Now, both species were hunted to **the** brink of extinction by **the** early whalers, but **the** Southern right **whales** have rebounded a lot better because they ' re located in places farther away from human activity. **The** North Atlantic right **whale** is listed as **the** most endangered species on **the** **planet** today because they are urban **whales** ; they live along **the** east coast of North America, United States and Canada, and they have to deal with all these urban ills. This photo shows an animal popping its head out at sunset off **the** coast of Florida. You can see **the coal** burning plant in **the** background. They have to deal with things like toxins and pharmaceuticals that are flushed out into **the** ocean, and maybe even affecting their **reproduction**. They also get entangled in **fishing** gear. This is a picture that shows **the** tail of a right **whale**. And those white markings are not natural markings. These are entanglement scars. 72 percent of **the** population has such scars, but most don ' t shed **the** gear, things like lobster traps and **crab** pots. They hold on to them, and it eventually kills them. And **the** other problem is they get hit by ships. And this was an animal that was struck by a ship in Nova Scotia, Canada being towed in, where they did a necropsy to confirm **the** cause of death, which was indeed a ship strike. So all of these ills are **stacking** up against these animals and keeping their numbers very low. And to draw a contrast with that beleaguered North Atlantic population, I went to a new pristine population of Southern right **whales** that had only been discovered about 10 years ago in **the** sub-Antarctic of New Zealand, a place called **the** Auckland Islands. I went down there in **the** winter time. And these are animals that had never seen humans before, and I was one of **the** first people they probably had ever seen. And I got in **the** water with them, and I was amazed at how curious they were. This photograph shows my assistant standing on **the** bottom at about 70 feet and one of these amazingly beautiful, 45-foot, 70- ton **whales**, like a city bus just swimming up, you know. They were in perfect condition, very **fat** and healthy, robust, no entanglement scars, **the** way they ' re supposed to look. You know, I read that **the** pilgrims, when they landed at Plymouth Rock in Massachusetts in 1620, wrote that you could walk across Cape Cod Bay on **the** backs of right **whales**. And we can ' t go back and see that today, but maybe we can preserve what we have left. And I wanted to close this program with a story of hope, a story I did on marine reserves as sort of a solution to **the** problem of overfishing, **the** global **fish** crisis story. I settled on working in **the** country of New Zealand because New Zealand was rather progressive, and is rather progressive in terms of protecting their ocean. And I really wanted this story to be about three things : I wanted it to be about abundance, about diversity and about resilience. And one of **the** first places I worked was a reserve called Goat Island in Leigh of New Zealand. What **the** scientists there told me was that when protected this first marine reserve in 1975, they hoped and expected that certain things might happen. For example, they hoped that certain species of **fish** like **the** New Zealand snapper would return because they had been **fished** to **the** brink of commercial extinction. And they did come back. What they couldn ' t predict was that other things would happen. For example, these **fish** predate on **sea** urchins, and when **the fish** were all gone, all anyone ever saw **underwater** was just acres and acres of **sea** urchins. But when **the fish** came back and began predating and controlling **the** urchin population, low and behold, kelp forests emerged in shallow water. And that ’ s because **the** urchins eat kelp. So when **the fish** control **the** urchin population, **the** ocean was restored to its natural equilibrium. You know, this is probably how **the** ocean looked here one or 200 years ago, but nobody was around to tell us. I worked in other parts of New Zealand as well, in beautiful, fragile, protected areas like in Fiordland, where this **sea** pen colony was found. Little **blue** cod swimming in for a dash of color. In **the** northern part of New Zealand, I dove in **the blue** water, where **the** water ’ s a little warmer, and photographed animals like this giant sting ray swimming through an **underwater** canyon. Every part of **the** ecosystem in this place seems very healthy, from tiny, little animals like a nudibrank crawling over encrusting sponge or a leatherjacket that is a very important animal in this ecosystem because it grazes on **the** bottom and allows new life to take hold. And I wanted to finish with this photograph, a picture I made on a very stormy day in New Zealand when I just laid on **the** bottom amidst a school of **fish** swirling around me. And I was in a place that had only been protected about 20 years ago. And I talked to divers that had been diving there for many years, and they said that **the** marine life was better here today than it was in **the** 1960s. And that ’ s because it ’ s been protected, that it has come back. So I think **the** message is clear. **The** ocean is, indeed, resilient and tolerant to a point, but we must be good custodians. I became an **underwater** photographer because I fell in love with **the sea**, and I make pictures of it today because I want to protect it, and I don ’ t think it ’ s too late. Thank you very much.

Text Sample 29: POS Dim 1 – 2009 – Score 29.00 – 2009/t000143.txt

Fifty years ago, when I began exploring **the** ocean, no one â not Jacques Perrin, not Jacques Cousteau or Rachel Carson â imagined that we could do anything to harm **the** ocean by what we put into it or by what we took out of it. It seemed, at that time, to be a **sea** of Eden, but now we know, and now we are facing paradise lost. I want to share with you my personal view of changes in **the sea** that affect all of us, and to consider why it matters that in 50 years, we ’ ve lost â actually, we ’ ve taken, we ’ ve eaten â more than 90 percent of **the** big **fish** in **the sea** ; why you should care that nearly half of **the** coral reefs have **disappeared** ; why a mysterious depletion of oxygen in large areas of **the** Pacific should concern not only **the** creatures that are dying, but it really should concern you. It does concern you, as well. I ’ m haunted by **the** thought of what Ray Anderson calls”tomorrow ’ s child,”asking why we didn ’ t do something on our watch to save sharks and bluefin tuna and squids and coral reefs and **the** living ocean while there still was time. Well, now is that time. I hope for your help to explore and protect **the** wild ocean in ways that will restore **the** health and, in so doing, secure hope for humankind. Health to **the** ocean means health for us. And I hope Jill Tarter ’ s wish to engage Earthlings includes dolphins and **whales** and other **sea** creatures in this quest to find intelligent life elsewhere in **the universe**. And I hope, Jill, that someday we will find evidence that there is intelligent life among humans on this **planet**. Did I say that? I guess I did. For me, as a scientist, it all began in 1953 when I first tried scuba. It ’ s when I first got to know **fish** swimming in something other than lemon slices and butter. I actually love diving at night ; you see a lot of **fish** then that you don ’ t see in **the** daytime. Diving day and night was really easy for me in 1970, when I led a team of aquanauts living **underwater** for weeks at a time â at **the** same time that astronauts were putting their footprints on **the** moon. In 1979 I had a chance to put my footprints on **the** ocean floor while using this personal submersible called Jim. It was six miles offshore and 1, 250 feet down. It ' s one of my favorite bathing suits. Since then, I ' ve used about 30 kinds of submarines and I ' ve started three companies and a nonprofit foundation called Deep Search to design and build systems to access **the** deep **sea**. I led a five-year National Geographic expedition, **the** Sustainable **Seas** expeditions, using these little subs. They ' re so simple to drive that even a scientist can do it. And I ' m living proof. Astronauts and aquanauts alike really appreciate **the** importance of air, food, water, temperature â all **the** things you need to stay alive in space or under **the sea**. I heard astronaut Joe Allen explain how he had to learn everything he could about his life support system and then do everything he could to take care of his life support system ; and then he pointed to this and he said,"Life support system."We need to learn everything we can about it and do everything we can to take care of it. **The** poet Auden said,"Thousands have lived without love ; none without water."Ninety-seven percent of Earth ' s water is ocean. No **blue**, no green. If you think **the** ocean isn ' t important, imagine Earth without it. Mars comes to mind. No ocean, no life support system. I gave a talk not so long ago at **the** World Bank and I showed this amazing image of Earth and I said,"There it is! **The** World Bank!"That ' s where all **the** assets are! And we ' ve been trawling them down much faster than **the** natural systems can replenish them. Tim Worth says **the** economy is a wholly-owned subsidiary of **the** environment. With every drop of water you **drink**, every breath you take, you ' re connected to **the sea**. No matter where on Earth you live. Most of **the** oxygen in **the** atmosphere is generated by **the sea**. Over time, most of **the planet** ' s **organic carbon** has been absorbed and stored there, mostly by microbes. **The** ocean drives climate and weather, stabilizes temperature, shapes Earth ' s **chemistry**. Water from **the sea** forms clouds that return to **the** land and **the seas** as rain, sleet and snow, and provides home for about 97 percent of life in **the** world, maybe in **the universe**. No water, no life ; no **blue**, no green. Yet we have this idea, we humans, that **the** Earth â all of it : **the** oceans, **the** skies â are so vast and so resilient it doesn ' t matter what we do to it. That may have been true 10, 000 years ago, and maybe even 1, 000 years ago but in **the** last 100, especially in **the** last 50, we ' ve drawn down **the** assets, **the** air, **the** water, **the** wildlife that make our lives possible. New technologies are helping us to understand **the** nature of nature ; **the** nature of what ' s happening, showing us our impact on **the** Earth. I mean, first you have to know that you ' ve got a problem. And fortunately, in our time, we ' ve learned more about **the** problems than in all preceding history. And with knowing comes caring. And with caring, there ' s hope that we can find an enduring place for ourselves within **the** natural systems that support us. But first we have to know. Three years ago, I met John Hanke, who ' s **the** head of Google Earth, and I told him how much I loved being able to hold **the** world in my hands and go exploring vicariously. But I asked him :"When are you going to finish it? You did a great job with **the** land, **the** dirt. What about **the** water?"Since then, I ' ve had **the** great pleasure of working with **the** Googlers, with DOER Marine, with National Geographic, with dozens of **the** best institutions and scientists around **the** world, ones that we could enlist, to put **the** ocean in Google Earth. And as of just this week, last Monday, Google Earth is now whole. Consider this : Starting right here at **the** convention center, we can find **the** nearby aquarium, we can look at where we ' re sitting, and then we can cruise up **the** coast to **the** big aquarium, **the** ocean, and California ' s four national marine sanctuaries, and **the** new network of state marine reserves that are beginning to protect and restore some of **the** assets We can flit over to Hawaii and see **the** real Hawaiian Islands : not just

the little bit that pokes through **the surface**, but also what ' s below. To see â wait a minute, we can go kshhplash! â right there, ha â under **the** ocean, see what **the whales** see. We can go explore **the** other side of **the** Hawaiian Islands. We can go actually and swim around on Google Earth and visit with humpback **whales**. These are **the** gentle giants that I ' ve had **the** pleasure of meeting face to face many times **underwater**. There ' s nothing quite like being personally inspected by a **whale**. We can pick up and fly to **the** deepest place : seven miles down, **the** Mariana Trench, where only two people have ever been. Imagine that. It ' s only seven miles, but only two people have been there, 49 years ago. One-way trips are easy. We need new deep-diving submarines. How about some X Prizes for ocean exploration? We need to see deep trenches, **the** undersea mountains, and understand life in **the** deep **sea**. We can now go to **the** Arctic. Just ten years ago I stood on **the** ice at **the** North Pole. An ice-free Arctic Ocean may happen in this century. That ' s bad news for **the** polar bears. That ' s bad news for us too. Excess **carbon** dioxide is not only driving global warming, it ' s also changing ocean **chemistry**, making **the sea** more acidic. That ' s bad news for coral reefs and oxygen-producing plankton. Also it ' s bad news for us. We ' re putting hundreds of millions of tons of plastic and other trash into **the sea**. Millions of tons of **discarded fishing** nets, gear that continues to kill. We ' re clogging **the** ocean, poisoning **the planet** ' s circulatory system, and we ' re taking out hundreds of millions of tons of wildlife, all carbon-based units. Barbarically, we ' re killing sharks for shark **fin soup**, undermining food chains that shape planetary **chemistry** and drive **the carbon** cycle, **the** nitrogen cycle, **the** oxygen cycle, **the** water cycle â our life support system. We ' re still killing bluefin tuna ; truly endangered and much more valuable alive than dead. All of these parts are part of our life support system. We kill using long lines, with baited hooks every few feet that may stretch for 50 miles or more. Industrial trawlers and draggers are scraping **the sea** floor like bulldozers, taking everything in their path. Using Google Earth you can witness trawlers â in China, **the** North Sea, **the** Gulf of Mexico â shaking **the** foundation of our life support system, leaving plumes of death in their path. **The** next time you dine on sushi â or sashimi, or swordfish steak, or **shrimp** cocktail, whatever wildlife you happen to **enjoy** from **the** ocean â think of **the** real cost. For every **pound** that goes to market, more than 10 **pounds**, even 100 **pounds**, may be thrown away as bycatch. This is **the** consequence of not knowing that there are limits to what we can take out of **the sea**. This chart shows **the** decline in ocean wildlife from 1900 to 2000. **The** highest concentrations are in red. In my lifetime, imagine, 90 percent of **the** big **fish** have been killed. Most of **the turtles**, sharks, tunas and **whales** are way down in numbers. But, there is good news. Ten percent of **the** big **fish** still remain. There are still some **blue whales**. There are still some krill in Antarctica. There are a few oysters in Chesapeake Bay. Half **the** coral reefs are still in pretty good shape, a jeweled belt around **the** middle of **the planet**. There ' s still time, but not a lot, to turn things around. But business as usual means that in 50 years, there may be no coral reefs â and no commercial **fishing**, because **the fish** will simply be gone. Imagine **the** ocean without **fish**. Imagine what that means to our life support system. Natural systems on **the** land are in big trouble too, but **the** problems are more obvious, and some actions are being taken to protect trees, watersheds and wildlife. And in 1872, with Yellowstone National Park, **the** United States began establishing a system of parks that some say was **the** best idea America ever had. About 12 percent of **the** land around **the** world is now protected : safeguarding biodiversity, providing a **carbon** sink, generating oxygen, protecting watersheds. And, in 1972, this nation began to establish a counterpart in **the sea**, National Marine

Sanctuaries. That ’ s another great idea. **The** good news is that there are now more than 4, 000 places in **the sea**, around **the** world, that have some kind of protection. And you can find them on Google Earth. **The** bad news is that you have to look hard to find them. In **the** last three years, for example, **the** U. S. protected 340, 000 square miles of ocean as national monuments. But it only increased from 0. 6 of one percent to 0. 8 of one percent of **the** ocean protected, globally. Protected areas do rebound, but it takes a long time to restore 50-year-old rockfish or monkfish, sharks or **sea** bass, or 200-year-old orange roughy. We don ’ t consume 200-year-old cows or chickens. Protected areas provide hope that **the** creatures of Ed Wilson ’ s dream of an encyclopedia of life, or **the** census of marine life, will live not just as a list, a photograph, or a paragraph. With scientists around **the** world, I ’ ve been looking at **the** 99 percent of **the** ocean that is open to **fishing** â and mining, and drilling, and dumping, and whatever â to **search** out hope spots, and try to find ways to give them and us a secure future. Such as **the** Arctic â we have one chance, right now, to get it right. Or **the** Antarctic, where **the continent** is protected, but **the** surrounding ocean is being stripped of its krill, **whales** and **fish**. Sargasso Sea ’ s three million square miles of floating forest is being gathered up to feed cows. 97 percent of **the** land in **the** Galapagos Islands is protected, but **the** adjacent **sea** is being ravaged by **fishing**. It ’ s true too in Argentina on **the** Patagonian shelf, which is now in serious trouble. **The** high **seas**, where **whales**, tuna and dolphins travel â **the** largest, least protected, ecosystem on Earth, filled with luminous creatures, living in dark waters that average two miles deep. They flash, and sparkle, and glow with their own living light. There are still places in **the sea** as pristine as I knew as a child. **The** next 10 years may be **the** most important, and **the** next 10, 000 years **the** best chance our species will have to protect what remains of **the** natural systems that give us life. To cope with climate change, we need new ways to generate power. We need new ways, better ways, to cope with poverty, wars and disease. We need many things to keep and maintain **the** world as a better place. But, nothing else will matter if we fail to protect **the** ocean. Our fate and **the** ocean ’ s are one. We need to do for **the** ocean what Al Gore did for **the** skies above. A global plan of action with a world conservation union, **the** IUCN, is underway to protect biodiversity, to mitigate and recover from **the** impacts of climate change, on **the** high **seas** and in coastal areas, wherever we can identify critical places. New technologies are needed to map, photograph and explore **the** 95 percent of **the** ocean that we have yet to see. **The** goal is to protect biodiversity, to provide stability and resilience. We need deep-diving subs, new technologies to explore **the** ocean. We need, maybe, an expedition â a TED at **sea** â that could help figure out **the** next steps. And so, I suppose you want to know what my wish is. I wish you would use all means at your disposal â films, expeditions, **the web**, new submarines â and campaign to ignite public support for a global network of marine protected areas â hope spots large enough to save and restore **the** ocean, **the blue** heart of **the planet**. How much? Some say 10 percent, some say 30 percent. You decide : how much of your heart do you want to protect? Whatever it is, a **fraction** of one percent is not enough. My wish is a big wish, but if we can make it happen, it can truly change **the** world, and help ensure **the** survival of what actually â as it turns out â is my favorite species ; that would be us. For **the** children of today, for tomorrow ’ s child : as never again, now is **the** time. Thank you.

Text Sample 30: POS Dim 1 – 2009 – Score 24.00 – 2009/t002945.txt

The substance of things **unseen**. Cities, past and future. In Oxford, perhaps we can use Lewis Carroll and look in **the** looking glass that is New York City to try and see our true selves, or perhaps pass through to another world. Or, in **the** words of F. Scott Fitzgerald,"As **the** moon rose higher, **the** inessential houses began to melt away until gradually I became aware of **the** old **island** here that once flowered for Dutch sailors ' eyes, a fresh green breast of **the** new world."My colleagues and I have been working for 10 years to rediscover this lost world in a project we call **The** Mannahatta Project. We ' re trying to discover what Henry Hudson would have seen on **the** afternoon of September 12th, 1609, when he sailed into New York harbor. And I ' d like to tell you **the** story in three acts, and if I have time still, an epilogue. So, Act I : A Map Found. So, I didn ' t grow up in New York. I grew up out west in **the** Sierra Nevada Mountains, like you see here, in **the** Red Rock Canyon. And from these early experiences as a child I learned to love landscapes. And so when it became time for me to do my graduate studies, I studied this emerging field of landscape **ecology**. Landscape **ecology** concerns itself with how **the** stream and **the** meadow and **the** forest and **the** cliffs make **habitats** for plants and animals. This experience and this training lead me to get a wonderful job with **the** Wildlife Conservation Society, which works to save wildlife and wild places all over **the** world. And over **the** last decade, I traveled to over 40 countries to see jaguars and bears and **elephants** and **tigers** and rhinos. But every time I would return from my trips I ' d return back to New York City. And on my weekends I would go up, just like all **the** other tourists, to **the** top of **the** Empire State Building, and I ' d look down on this landscape, on these ecosystems, and I ' d wonder,"How does this landscape work to make **habitat** for plants and animals? How does it work to make **habitat** for animals like me?"I ' d go to Times Square and I ' d look at **the** amazing ladies on **the** wall, and wonder why nobody is looking at **the** historical figures just behind them. I ' d go to Central Park and see **the** rolling topography of Central Park come up against **the** abrupt and **sheer** topography of midtown Manhattan. I started reading about **the** history and **the** geography in New York City. I read that New York City was **the** first mega-city, a city of 10 million people or more, in 1950. I started seeing paintings like this. For those of you who are from New York, this is 125th street under **the** West Side Highway. It was once a beach. And this painting has John James Audubon, **the** painter, sitting on **the rock**. And it ' s looking up on **the** wooded heights of Washington Heights to Jeffrey ' s Hook, where **the** George Washington Bridge goes across today. Or this painting, from **the** 1740s, from Greenwich Village. Those are two students at King ' s College — later Columbia University — sitting on a hill, overlooking a valley. And so I ' d go down to Greenwich Village and I ' d look for this hill, and I couldn ' t find it. And I couldn ' t find that palm tree. What ' s that palm tree doing there? So, it was in **the** course of these investigations that I ran into a map. And it ' s this map you see here. It ' s held in a geographic information system which allows me to zoom in. This map isn ' t from Hudson ' s time, but from **the** American Revolution, 170 years later, made by British military cartographers during **the** occupation of New York City. And it ' s a remarkable map. It ' s in **the** National Archives here in Kew. And it ' s 10 feet long and three and a half feet wide. And if I zoom in to lower Manhattan you can see **the** extent of New York City as it was, right at **the** end of **the** American Revolution. Here ' s Bowling Green. And here ' s Broadway. And this is City Hall Park. So **the** city basically extended to City Hall Park. And just beyond it you can see features that have vanished, things that have **disappeared**. This is **the** Collect Pond, which was **the** fresh water source for New York City for its first 200 years, and for **the** Native Americans for thousands of years before that. You can see **the** Lispenard Meadows draining down through here, through what is TriBeCa now, and **the** beaches that come up from **the** Battery, all **the** way to 42nd St. This map was made for military reasons. They ’ re mapping **the** roads, **the** buildings, these fortifications that they built. But they ’ re also mapping things of ecological interest, also military interest : **the** hills, **the** marshes, **the** streams. This is Richmond Hill, and Minetta Water, which used to run its way through Greenwich Village. Or **the** swamp at Gramercy Park, right here. Or Murray Hill. And this is **the** Murrays ’ house on Murray Hill, 200 years ago. Here is Times Square, **the** two streams that came together to make a wetland in Times Square, as it was at **the** end of **the** American Revolution. So I saw this remarkable map in a book. And I thought to myself,”You know, if I could georeference this map, if I could place this map in **the** grid of **the** city today, I could find these lost features of **the** city, in **the** block-by-block geography that people know, **the** geography of where people go to work, and where they go to live, and where they like to eat.”So, after some work we were able to georeference it, which allows us to put **the** modern streets on **the** city, and **the** buildings, and **the** open spaces, so that we can zoom in to where **the** Collect Pond is. We can digitize **the** Collect Pond and **the** streams, and see where they actually are in **the** geography of **the** city today. So this is fun for finding where things are **relative** to **the** old topography. But I had another idea about this map. If we take away **the** streets, and if we take away **the** buildings, and if we take away **the** open spaces, then we could take this map. If we pull off **the** 18th century features we could drive it back in time. We could drive it back to its ecological fundamentals : to **the** hills, to **the** streams, to **the** basic hydrology and shoreline, to **the** beaches, **the** basic aspects that make **the** ecological landscape. Then, if we added maps like **the** geology, **the** bedrock geology, and **the surface** geology, what **the** glaciers leave, if we make **the** soil map, with **the** 17 soil classes, that are defined by **the** National Conservation Service, if we make a digital elevation model of **the** topography that tells us how high **the** hills were, then we can **calculate the slopes**. We can **calculate the** aspect. We can **calculate the** winter wind exposure — so, which way **the** winter winds blow across **the** landscape. **The** white areas on this map are **the** places protected from **the** winter winds. We compiled all **the** information about where **the** Native Americans were, **the** Lenape. And we built a **probability** map of where they might have been. So, **the** red areas on this map indicate **the** places that are best for human **sustainability** on Manhattan, places that are close to water, places that are near **the** harbor to **fish**, places protected from **the** winter winds. We know that there was a Lenape settlement down here by **the** Collect Pond. And we knew that they planted a kind of horticulture, that they grew these beautiful gardens of corn, beans, and squash, **the**”Three Sisters”garden. So, we built a model that explains where those fields might have been. And **the** old fields, **the** successional fields that go. And we might think of these as abandoned. But, in fact, they ’ re grassland **habitats** for grassland birds and plants. And they have become successional shrub lands, and these then mix in to a map of all **the** ecological communities. And it turns out that Manhattan had 55 different ecosystem types. You can think of these as neighborhoods, as distinctive as TriBeCa and **the** Upper East Side and Inwood — that these are **the** forest and **the** wetlands and **the** marine communities, **the** beaches. And 55 is a lot. On a per-area basis, Manhattan had more ecological communities per acre than Yosemite does, than Yellowstone, than Amboseli. It was really an extraordinary landscape that was capable of supporting an extraordinary biodiversity. So, Act II : A Home Reconstructed. So, we studied **the fish** and **the** frogs and **the** birds and **the** bees, **the** 85 different kinds of **fish** that were on Manhattan, **the** Heath hens, **the** species that aren ’ t there anymore, **the** beavers on all **the** streams, **the** black bears, and

the Native Americans, to study how they used and thought about their landscape. We wanted to try and map these. And to do that what we did was we mapped their **habitat** needs. Where do they get their food? Where do they get their water? Where do they get their shelter? Where do they get their reproductive resources? To an ecologist, **the** intersection of these is **habitat**, but to most people, **the** intersection of these is their home. So, we would read in field guides, **the** standard field guides that maybe you have on your shelves, you know, what beavers need is,"A slowly meandering stream with aspen trees and alders and willows, near **the** water."That ' s **the** best thing for a beaver. So we just started making a list. Here is **the** beaver. And here is **the** stream, and **the** aspen and **the** alder and **the** willow. As if these were **the** maps that we would need to predict where you would find **the** beaver. Or **the** bog **turtle**, needing **wet** meadows and insects and sunny places. Or **the** bobcat, needing rabbits and beavers and den sites. And rapidly we started to realize that beavers can be something that a bobcat needs. But a beaver also needs things. And that having it on either side means that we can link it together, that we can create **the** network of **the habitat** relationships for these species. Moreover, we realized that you can start out as being a beaver specialist, but you can look up what an aspen needs. An aspen needs fire and dry soils. And you can look at what a **wet** meadow needs. And it need beavers to create **the** wetlands, and maybe some other things. But you can also talk about sunny places. So, what does a sunny place need? Not **habitat** per se. But what are **the** conditions that make it possible? Or fire. Or dry soils. And that you can put these on a grid that ' s 1, 000 columns long across **the** top and 1, 000 rows down **the** other way. And then we can visualize this data like a network, like a social network. And this is **the** network of all **the habitat** relationships of all **the** plants and animals on Manhattan, and everything they needed, going back to **the** geology, going back to time and space at **the** very core of **the web**. We call this **the** Muir Web. And if you zoom in on it it looks like this. Each point is a different species or a different stream or a different soil type. And those little gray lines are **the** connections that connect them together. They are **the** connections that actually make nature resilient. And **the** structure of this is what makes nature work, seen with all its parts. We call these Muir **Webs** after **the** Scottish-American naturalist John Muir, who said,"When we try to pick out anything by itself, we find that it ' s bound fast by a thousand invisible cords that cannot be broken, to everything in **the universe**."So then we took **the** Muir **webs** and we took them back to **the** maps. So if we wanted to go between 85th and 86th, and Lex and Third, maybe there was a stream in that block. And these would be **the** kind of trees that might have been there, and **the** flowers and **the** lichens and **the** mosses, **the** butterflies, **the fish** in **the** stream, **the** birds in **the** trees. Maybe a timber rattlesnake lived there. And perhaps a black bear walked by. And maybe Native Americans were there. And then we took this data. You can see this for yourself on our website. You can zoom into any block on Manhattan, and see what might have been there 400 years ago. And we used it to try and reveal a landscape here in Act III. We used **the** tools they use in Hollywood to make these **fantastic** landscapes that we all see in **the** movies. And we tried to use it to visualize Third Avenue. So we would take **the** landscape and we would build up **the** topography. We ' d lay on top of that **the** soils and **the** waters, and illuminate **the** landscape. We would lay on top of that **the** map of **the** ecological communities. And feed into that **the** map of **the** species. So that we would actually take a photograph, flying above Times Square, looking toward **the** Hudson River, waiting for Hudson to come. Using this technology, we can make these **fantastic** georeferenced views. We can basically take a picture out of any window on Manhattan and see what that landscape looked like 400 years ago. This is **the** view from **the** East River, looking up Murray Hill at where **the** United Nations is today. This is **the** view looking down **the** Hudson River, with Manhattan on **the left**, and New Jersey out on **the** right, looking out toward **the** Atlantic Ocean. This is **the** view over Times Square, with **the** beaver pond there, looking out toward **the** east. So we can see **the** Collect Pond, and Lispenard Marshes back behind. We can see **the** fields that **the** Native Americans made. And we can see this in **the** geography of **the** city today. So when you ' re watching"Law and Order,"and **the** lawyers walk up **the** steps they could have walked back down those steps of **the** New York Court House, right into **the** Collect Pond, 400 years ago. So these images are **the** work of my friend and colleague, Mark Boyer, who is here in **the** audience today. And I ' d just like, if you would give him a hand, to call out for his fine work. There is such power in bringing science and visualization together, that we can create images like this, perhaps looking on either side of a looking glass. And even though I ' ve only had a brief time to speak, I hope you appreciate that Mannahatta was a very special place. **The** place that you see here on **the left** side was interconnected. It was based on this diversity. It had this resilience that is what we need in our modern world. But I wouldn ' t have you think that I don ' t like **the** place on **the** right, which I quite do. I ' ve come to love **the** city and its kind of diversity, and its resilience, and its dependence on density and how we ' re connected together. In fact, that I see them as reflections of each other, much as Lewis Carroll did in"Through **the** Looking Glass."We can compare these two and hold them in our minds at **the** same time, that they really are **the** same place, that there is no way that cities can escape from nature. And I think this is what we ' re learning about building cities in **the** future. So if you ' ll allow me a brief epilogue, not about **the** past, but about 400 years from now, what we ' re realizing is that cities are **habitats** for people, and need to supply what people need : a sense of home, food, water, shelter, reproductive resources, and a sense of meaning. This is **the** particular additional **habitat** requirement of humanity. And so many of **the** talks here at TED are about meaning, about bringing meaning to our lives in all kinds of different ways, through technology, through art, through science, so much so that I think we focus so much on that side of our lives, that we haven ' t given enough attention to **the** food and **the** water and **the** shelter, and what we need to raise **the** kids. So, how can we envision **the** city of **the** future? Well, what if we go to Madison Square Park, and we imagine it without all **the** cars, and bicycles instead and large forests, and streams instead of sewers and storm drains? What if we imagined **the** Upper East Side with green roofs, and streams winding through **the** city, and windmills supplying **the** power we need? Or if we imagine **the** New York City metropolitan area, currently home to 12 million people, but 12 million people in **the** future, perhaps living at **the** density of Manhattan, in only 36 percent of **the** area, with **the** areas in between covered by farmland, covered by wetlands, covered by **the** marshes we need. This is **the** kind of future I think we need, is a future that has **the** same diversity and abundance and dynamism of Manhattan, but that learns from **the sustainability** of **the** past, of **the ecology**, **the** original **ecology**, of nature with all its parts. Thank you very much.

Text Sample 31: POS Dim 1 – 2009 – Score 23.00 – 2009/t000128.txt

So, my question : are we alone? **The** story of humans is **the** story of ideas â scientific ideas that shine light into dark corners, ideas that we embrace rationally and irrationally, ideas for which we ' ve lived and died and killed and been killed, ideas that have vanished in history, and ideas that have been set in dogma. It ' s a story of nations, of ideologies, of territories, and of conflicts among them. But, every moment of human history, from **the** Stone Age to **the** Information Age, from Sumer and Babylon to **the** iPod and celebrity gossip, they ' ve all been carried out â every book that you ' ve read, every poem, every laugh, every tear â they ' ve all happened here. Here. Here. Here. Perspective is a very powerful thing. Perspectives can change. Perspectives can be altered. From my perspective, we live on a fragile **island** of life, in a **universe** of possibilities. For many millennia, humans have been on a journey to find answers, answers to questions about naturalism and transcendence, about who we are and why we are, and of course, who else might be out there. Is it really just us? Are we alone in this vast **universe** of energy and matter and **chemistry** and physics? Well, if we are, it ' s an awful waste of space. But, what if we ' re not? What if, out there, others are asking and answering similar questions? What if they look up at **the** night sky, at **the** same stars, but from **the** opposite side? Would **the** discovery of an older cultural civilization out there inspire us to find ways to survive our increasingly uncertain technological adolescence? Might it be **the** discovery of a distant civilization and our common cosmic origins that finally drives home **the** message of **the** bond among all humans? Whether we ' re born in San Francisco, or Sudan, or close to **the** heart of **the** Milky Way galaxy, we are **the** products of a billion-year lineage of wandering stardust. We, all of us, are what happens when a primordial mixture of hydrogen and helium **evolves** for so long that it begins to ask where it came from. Fifty years ago, **the** journey to find answers took a different path and SETI, **the** Search for Extra-Terrestrial Intelligence, began. So, what exactly is SETI? Well, SETI uses **the** tools of astronomy to try and find evidence of someone else ' s technology out there. Our own technologies are visible over interstellar distances, and theirs might be as well. It might be that some massive network of communications, or some shield against asteroidal impact, or some huge astro-engineering project that we can ' t even begin to conceive of, could generate signals at radio or optical **frequencies** that a determined program of **searching** might **detect**. For millennia, we ' ve actually turned to **the** priests and **the** philosophers for guidance and instruction on this question of whether there ' s intelligent life out there. Now, we can use **the** tools of **the** 21st century to try and observe what is, rather than ask what should be, believed. SETI doesn ' t presume **the** existence of extra terrestrial intelligence ; it merely notes **the** possibility, if not **the probability** in this vast **universe**, which seems fairly uniform. **The** numbers suggest a **universe** of possibilities. Our sun is one of 400 billion stars in our galaxy, and we know that many other stars have planetary systems. We ' ve discovered over 350 in **the** last 14 years, including **the** small **planet**, announced earlier this week, which has a radius just twice **the** size of **the** Earth. And, if even all of **the** planetary systems in our galaxy were devoid of life, there are still 100 billion other galaxies out there, altogether 10^{22} stars. Now, I ' m going to try a trick, and recreate an experiment from this morning. Remember, one billion? But, this time not one billion dollars, one billion stars. Alright, one billion stars. Now, up there, 20 feet above **the** stage, that ' s 10 trillion. Well, what about 10^{22}? Where ' s **the** line that marks that? That line would have to be 3. 8 million miles above this stage. 16 times farther away than **the** moon, or four percent of **the** distance to **the** sun. So, there are many possibilities. And much of this vast **universe**, much more may be habitable than we once thought, as we study extremophiles on Earth â organisms that can live in conditions totally inhospitable for us, in **the** hot, high pressure thermal vents at **the** bottom of **the** ocean, frozen in ice, in boiling battery acid, in **the** cooling waters of nuclear reactors. These extremophiles tell us that life may exist in many other environments. But those environments are going to be widely spaced in this **universe**. Even our nearest star, **the** Sun â its **emissions** suffer **the** tyranny of light speed. It takes a full eight minutes for its radiation to reach us. And **the** nearest star is 4. 2 light years away, which means its light takes 4. 2 years to get here. And **the** edge of our galaxy is 75, 000 light years away, and **the** nearest galaxy to us, 2. 5 million light years. That means any signal we **detect** would have started its journey a long time ago. And a signal would give us a glimpse of their past, not their present. Which is why Phil Morrison calls SETI,"**the** archaeology of **the** future."It tells us about their past, but detection of a signal tells us it ' s possible for us to have a long future. I think this is what David Deutsch meant in 2005, when he ended his Oxford TEDTalk by saying he had two principles he ' d like to share for living, and he would like to carve them on stone tablets. **The** first is that problems are inevitable. **The** second is that problems are soluble. So, ultimately what ' s going to determine **the** success or failure of SETI is **the** longevity of technologies, and **the** mean distance between technologies in **the** cosmos â distance over space and distance over time. If technologies don ' t last and persist, we will not succeed. And we ' re a very young technology in an old galaxy, and we don ' t yet know whether it ' s possible for technologies to persist. So, up until now I ' ve been talking to you about really large numbers. Let me talk about a relatively small number. And that ' s **the** length of time that **the** Earth was lifeless. Zircons that are mined in **the** Jack Hills of western Australia, zircons taken from **the** Jack Hills of western Australia tell us that within a few hundred million years of **the** origin of **the planet** there was abundant water and perhaps even life. So, our **planet** has spent **the** vast majority of its 4. 56 billion year history developing life, not anticipating its emergence. Life happened very quickly, and that bodes well for **the** potential of life elsewhere in **the** cosmos. And **the** other thing that one should take away from this chart is **the** very narrow range of time over which humans can claim to be **the** dominant intelligence on **the planet**. It ' s only **the** last few hundred thousand years modern humans have been pursuing technology and civilization. So, one needs a very deep appreciation of **the** diversity and incredible scale of life on this **planet** as **the** first step in preparing to make contact with life elsewhere in **the** cosmos. We are not **the** pinnacle of **evolution**. We are not **the** determined product of billions of years of **evolutionary** plotting and planning. We are one outcome of a continuing adaptational process. We are residents of one small **planet** in a corner of **the** Milky Way galaxy. And Homo sapiens are one small leaf on a very extensive Tree of Life, which is densely populated by organisms that have been honed for survival over millions of years. We misuse language, and talk about **the**"ascent"of man. We understand **the** scientific basis for **the** interrelatedness of life but our ego hasn ' t caught up yet. So this"ascent"of man, pinnacle of **evolution**, has got to go. It ' s a sense of privilege that **the** natural **universe** doesn ' t share. Loren Eiseley has said,"One does not meet oneself until one catches **the** reflection from an eye other than human."One day that eye may be that of an intelligent alien, and **the** sooner we eschew our narrow view of **evolution the** sooner we can truly explore our ultimate origins and destinations. We are a small part of **the** story of cosmic **evolution**, and we are going to be responsible for our continued participation in that story, and perhaps SETI will help as well. Occasionally, throughout history, this concept of this very large cosmic perspective comes to **the surface**, and as a result we see transformative and profound discoveries. So in 1543, Nicholas Copernicus published"**The** Revolutions of Heavenly **Spheres**,"and by taking **the** Earth out of **the** center, and putting **the** sun in **the** center of **the** solar system, he opened our eyes to a much larger **universe**, of which we are just a small part. And that Copernican revolution continues today to influence science and philosophy and technology and theology. So, in 1959, Giuseppe Coccone and Philip Morrison published **the** first SETI article in a refereed journal, and brought SETI into **the** scientific mainstream. And in 1960, Frank Drake conducted **the** first SETI observation looking at two stars, Tau Ceti and Epsilon Eridani, for about 150 hours. Now Drake did not discover extraterrestrial intelligence, but he learned a very valuable lesson from a passing aircraft, and that ' s that terrestrial technology can interfere with **the search** for extraterrestrial technology. We ' ve been **searching** ever since, but it ' s impossible to overstate **the** magnitude of **the search** that remains. All of **the** concerted SETI efforts, over **the** last 40-some years, are equivalent to scooping a single glass of water from **the** oceans. And no one would decide that **the** ocean was without **fish** on **the** basis of one glass of water. **The** 21st century now allows us to build bigger glasses â much bigger glasses. In Northern California, we ' re beginning to take observations with **the** first 42 telescopes of **the** Allen Telescope **Array** â and I ' ve got to take a moment right now to publicly thank Paul Allen and Nathan Myhrvold and all **the** TeamSETI members in **the** TED community who have so generously supported this research. **The** ATA is **the** first telescope built from a large number of small dishes, and hooked together with **computers**. It ' s making silicon as important as aluminum, and we ' ll grow it in **the** future by adding more antennas to reach 350 for more sensitivity and leveraging Moore ' s law for more processing capability. Today, our signal detection algorithms can find very simple artifacts and noise. If you look very hard here you can see **the** signal from **the** Voyager 1 spacecraft, **the** most distant human **object** in **the universe**, 106 times as far away from us as **the** sun is. And over those long distances, its signal is very faint when it reaches us. It may be hard for your eye to see it, but it ' s easily found with our efficient algorithms. But this is a simple signal, and tomorrow we want to be able to find more complex signals. This is a very good year. 2009 is **the** 400th anniversary of Galileo ' s first use of **the** telescope, Darwin ' s 200th birthday, **the** 150th anniversary of **the** publication of"On **the** Origin of Species,"**the** 50th anniversary of SETI as a science, **the** 25th anniversary of **the** incorporation of **the** SETI Institute as a non-profit, and of course, **the** 25th anniversary of TED. And next month, **the** Kepler Spacecraft will launch and will begin to tell us just how frequent Earth-like **planets** are, **the** targets for SETI ' s **searches**. In 2009, **the** U. N. has declared it to be **the** International Year of Astronomy, a global festival to help us residents of Earth rediscover our cosmic origins and our place in **the universe**. And in 2009, change has come to Washington, with a promise to put science in its rightful position. So, what would change everything? Well, this is **the** question **the** Edge foundation asked this year, and four of **the** respondents said,"SETI."Why? Well, to quote :"**The** discovery of intelligent life beyond Earth would eradicate **the** loneliness and solipsism that has plagued our species since its inception. And it wouldn ' t simply change everything, it would change everything all at once."So, if that ' s right, why did we only capture four out of those 151 minds? I think it ' s a problem of completion and delivery, because **the** fine print said,"What game-changing ideas and scientific developments would you expect to live to see?"So, we have a fulfillment problem. We need bigger glasses and more hands in **the** water, and then working together, maybe we can all live to see **the** detection of **the** first extraterrestrial signal. That brings me to my wish. I wish that you would empower Earthlings everywhere to become active participants in **the** ultimate **search** for cosmic company. **The** first step would be to tap into **the** global brain trust, to build an environment where raw data could be stored, and where it could be accessed and **manipulated**, where new algorithms could be developed and old algorithms made more efficient. And this is a technically creative challenge, and it would change **the** perspective of people who worked on it. And then, we ’ d like to augment **the** automated **search** with human insight. We ’ d like to use **the** pattern recognition capability of **the** human eye to find faint, complex signals that our current algorithms miss. And, of course, we ’ d like to inspire and engage **the** next generation. We ’ d like to take **the** materials that we have built for education, and get them out to students everywhere, students that can ’ t come and visit us at **the** ATA. We ’ d like to tell our story better, and engage young people, and thereby change their perspective. I ’ m sorry Seth Godin, but over **the** millennia, we ’ ve seen where tribalism leads. We ’ ve seen what happens when we divide an already small **planet** into smaller **islands**. And, ultimately, we actually all belong to only one tribe, to Earthlings. And SETI is a mirror â a mirror that can show us ourselves from an extraordinary perspective, and can help to trivialize **the** differences among us. If SETI does nothing but change **the** perspective of humans on this **planet**, then it will be one of **the** most profound **endeavors** in history. So, in **the** opening days of 2009, a visionary president stood on **the** steps of **the** U. S. Capitol and said,”We cannot help but believe that **the** old hatreds shall someday pass, that **the** lines of tribe shall soon dissolve, that, as **the** world grows smaller, our common humanity shall reveal itself.”So, I look forward to working with **the** TED community to hear about your ideas about how to fulfill this wish, and in collaborating with you, hasten **the** day that that visionary statement can become a reality. Thank you.

Text Sample 32: POS Dim 1 – 2011 – Score 31.00 – 2011/t002439.txt

The oceans cover some 70 percent of our **planet**. And I think Arthur C. Clarke probably had it right when he said that perhaps we ought to call our **planet** Planet Ocean. And **the** oceans are hugely productive, as you can see by **the** satellite image of photosynthesis, **the** production of new life. In fact, **the** oceans produce half of **the** new life every day on Earth as well as about half **the** oxygen that we breathe. In addition to that, it harbors a lot of **the** biodiversity on Earth, and much of it we don ’ t know about. But I ’ ll tell you some of that today. That also doesn ’ t even get into **the** whole protein extraction that we do from **the** ocean. That ’ s about 10 percent of our global needs and 100 percent of some **island** nations. If you were to descend into **the** 95 percent of **the** biosphere that ’ s livable, it would quickly become pitch black, interrupted only by pinpoints of light from bioluminescent organisms. And if you turn **the** lights on, you might periodically see spectacular organisms swim by, because those are **the** denizens of **the** deep, **the** things that live in **the** deep ocean. And eventually, **the** deep **sea** floor would come into view. This type of **habitat** covers more of **the** Earth ’ s **surface** than all other **habitats** combined. And yet, we know more about **the surface** of **the** Moon and about Mars than we do about this **habitat**, despite **the** fact that we have yet to extract a gram of food, a breath of oxygen or a drop of water from those bodies. And so 10 years ago, an international program began called **the** Census of Marine Life, which set out to try and improve our understanding of life in **the** global oceans. It involved 17 different projects around **the** world. As you can see, these are **the** footprints of **the** different projects. And I hope you ’ ll appreciate **the** level of global coverage that it managed to achieve. It all began when two scientists, Fred Grassle and Jesse Ausubel, met in Woods Hole, Massachusetts where both were guests at **the** famed oceanographic institute. And Fred was lamenting **the** state of marine biodiversity and **the** fact that it was in trouble and nothing was being done about it. Well, from that discussion grew this program that involved

2, 700 scientists from more than 80 countries around **the** world who engaged in 540 ocean expeditions at a combined cost of 650 million dollars to study **the** distribution, diversity and abundance of life in **the** global ocean. And so what did we find? We found spectacular new species, **the** most beautiful and visually **stunning** things everywhere we looked — from **the** shoreline to **the** **abyss**, form microbes all **the** way up to **fish** and everything in between. And **the** limiting step here wasn ’ t **the** unknown diversity of life, but rather **the** taxonomic specialists who can identify and catalog these species that became **the** limiting step. They, in fact, are an endangered species themselves. There are actually four to five new species described everyday for **the** oceans. And as I say, it could be a much larger number. Now, I come from Newfoundland in Canada — It ’ s an **island** off **the** east coast of that **continent** — where we experienced one of **the** worst **fishing** disasters in human history. And so this photograph shows a small boy next to a codfish. It ’ s around 1900. Now, when I was a boy of about his age, I would go out **fishing** with my grandfather and we would catch **fish** about half that size. And I thought that was **the** norm, because I had never seen **fish** like this. If you were to go out there today, 20 years after this fishery collapsed, if you could catch a **fish**, which would be a bit of a challenge, it would be half that size still. So what we ’ re experiencing is something called shifting baselines. Our expectations of what **the** oceans can produce is something that we don ’ t really appreciate because we haven ’ t seen it in our lifetimes. Now most of us, and I would say me included, think that human exploitation of **the** oceans really only became very serious in **the** last 50 to, perhaps, 100 years or so. **The** census actually tried to look back in time, using every source of information they could get their hands on. And so anything from restaurant menus to monastery records to ships ’ logs to see what **the** oceans looked like. Because science data really goes back to, at best, World War II, for **the** most part. And so what they found, in fact, is that exploitation really began heavily with **the** Romans. And so at that time, of course, there was no refrigeration. So fishermen could only catch what they could either eat or sell that day. But **the** Romans developed salting. And with salting, it became possible to store **fish** and to transport it long distances. And so began industrial **fishing**. And so these are **the** sorts of extrapolations that we have of what sort of loss we ’ ve had **relative** to pre-human impacts on **the** ocean. They range from 65 to 98 percent for these major groups of organisms, as shown in **the** dark **blue** bars. Now for those species **the** we managed to leave alone, that we protect — for example, marine mammals in recent years and **sea** birds — there is some recovery. So it ’ s not all hopeless. But for **the** most part, we ’ ve gone from salting to exhausting. Now this other line of evidence is a really interesting one. It ’ s from trophy **fish** caught off **the** coast of Florida. And so this is a photograph from **the** 1950s. I want you to notice **the** scale on **the** **slide**, because when you see **the** same picture from **the** 1980s, we see **the fish** are much smaller and we ’ re also seeing a change in terms of **the** composition of those **fish**. By 2007, **the** catch was actually laughable in terms of **the** size for a trophy **fish**. But this is no laughing matter. **The** oceans have lost a lot of their productivity and we ’ re responsible for it. So what ’ s left? Actually quite a lot. There ’ s a lot of exciting things, and I ’ m going to tell you a little bit about them. And I want to start with a bit on technology, because, of course, this is a TED Conference and you want to hear something on technology. So one of **the** tools that we use to sample **the** deep ocean are remotely operated vehicles. So these are tethered vehicles we lower down to **the sea** floor where they ’ re our eyes and our hands for working on **the sea** bottom. So a couple of years ago, I was supposed to go on an oceanographic cruise and I couldn ’ t go because of a scheduling conflict. But through a satellite link I was able to sit at my study at home with my dog curled up at my feet, a cup of tea in my hand, and I could tell **the** pilot,"I want a sample right there."And that ' s exactly what **the** pilot did for me. That ' s **the** sort of technology that ' s available today that really wasn ' t available even a decade ago. So it allows us to sample these amazing **habitats** that are very far from **the surface** and very far from light. And so one of **the** tools that we can use to sample **the** oceans is acoustics, or sound waves. And **the** advantage of sound waves is that they actually pass well through water, unlike light. And so we can send out sound waves, they bounce off **objects** like **fish** and are reflected back. And so in this example, a census scientist took out two ships. One would send out sound waves that would bounce back. They would be received by a second ship, and that would give us very precise estimates, in this case, of 250 billion herring in a period of about a minute. And that ' s an area about **the** size of Manhattan Island. And to be able to do that is a tremendous fisheries tool, because knowing how many **fish** are there is really critical. We can also use satellite **tags** to track animals as they move through **the** oceans. And so for animals that come to **the surface** to breathe, such as this **elephant seal**, it ' s an opportunity to send data back to shore and tell us where exactly it is in **the** ocean. And so from that we can produce these tracks. For example, **the** dark **blue** shows you where **the elephant seal** moved in **the** north Pacific. Now I realize for those of you who are colorblind, this **slide** is not very helpful, but stick with me nonetheless. For animals that don ' t **surface**, we have something called pop-up **tags**, which collect data about light and what time **the** sun rises and sets. And then at some period of time it pops up to **the surface** and, again, relays that data back to shore. Because GPS doesn ' t work under water. That ' s why we need these tools. And so from this we ' re able to identify these **blue highways**, these hot spots in **the** ocean, that should be real priority areas for ocean conservation. Now one of **the** other things that you may think about is that, when you go to **the supermarket** and you buy things, they ' re scanned. And so there ' s a barcode on that product that tells **the computer** exactly what **the** product is. Geneticists have developed a similar tool called genetic barcoding. And what barcoding does is use a specific gene called CO1 that ' s consistent within a species, but varies among species. And so what that means is we can unambiguously identify which species are which even if they look similar to each other, but may be biologically quite different. Now one of **the** nicest examples I like to cite on this is **the** story of two young women, high school students in New York City, who worked with **the** census. They went out and collected **fish** from markets and from restaurants in New York City and they barcoded it. Well what they found was mislabeled **fish**. So for example, they found something which was sold as tuna, which is very valuable, was in fact tilapia, which is a much less valuable **fish**. They also found an endangered species sold as a common one. So barcoding allows us to know what we ' re working with and also what we ' re eating. **The** Ocean Biogeographic Information System is **the** database for all **the** census data. It ' s open access ; you can all go in and download data as you wish. And it contains all **the** data from **the** census plus other data sets that people were willing to contribute. And so what you can do with that is to plot **the** distribution of species and where they **occur** in **the** oceans. What I ' ve plotted up here is **the** data that we have on hand. This is where our sampling effort has concentrated. Now what you can see is we ' ve sampled **the** area in **the** North Atlantic, in **the** North Sea in particular, and also **the** east coast of North America fairly well. That ' s **the** warm colors which show a well-sampled region. **The** cold colors, **the blue** and **the** black, show areas where we have almost no data. So even after a 10-year census, there are large areas that still remain unexplored. Now there are a group of scientists living in Texas, working in **the** Gulf of Mexico who decided really as a labor of love to pull together all **the** knowledge they could about biodiversity in **the** Gulf of Mexico. And so they put this together, a list of all **the** species, where they ' re known to **occur**, and it really seemed like a very esoteric, scientific type of exercise. But then, of course, there was **the** Deep Horizon **oil spill**. So all of a sudden, this labor of love for no obvious economic reason has become a critical piece of information in terms of how that system is going to recover, how long it will take and how **the** lawsuits and **the** multi-billion-dollar discussions that are going to happen in **the** coming years are likely to be resolved. So what did we find? Well, I could stand here for hours, but, of course, I ' m not allowed to do that. But I will tell you some of my favorite discoveries from **the** census. So one of **the** things we discovered is where are **the** hot spots of diversity? Where do we find **the** most species of ocean life? And what we find if we plot up **the** well-known species is this sort of a distribution. And what we see is that for coastal **tags**, for those organisms that live near **the** shoreline, they ' re most diverse in **the** tropics. This is something we ' ve actually known for a while, so it ' s not a real breakthrough. What is really exciting though is that **the** oceanic **tags**, or **the** ones that live far from **the** coast, are actually more diverse at intermediate latitudes. This is **the** sort of data, again, that managers could use if they want to prioritize areas of **the** ocean that we need to conserve. You can do this on a global scale, but you can also do it on a regional scale. And that ' s why biodiversity data can be so valuable. Now while a lot of **the** species we discovered in **the** census are things that are small and hard to see, that certainly wasn ' t always **the** case. For example, while it ' s hard to believe that a three kilogram lobster could elude scientists, it did until a few years ago when South African fishermen requested an export permit and scientists realized that this was something new to science. Similarly this Golden V kelp collected in Alaska just below **the** low water mark is probably a new species. Even though it ' s three meters long, it actually, again, eluded science. Now this guy, this bigfin squid, is seven meters in length. But to be fair, it lives in **the** deep waters of **the** Mid- Atlantic Ridge, so it was a lot harder to find. But there ' s still potential for discovery of big and exciting things. This particular **shrimp**, we ' ve dubbed it **the** Jurassic **shrimp**, it ' s thought to have gone extinct 50 years ago — at least it was, until **the** census discovered it was living and doing just fine off **the** coast of Australia. And it shows that **the** ocean, because of its vastness, can hide secrets for a very long time. So, Steven Spielberg, eat your heart out. If we look at distributions, in fact distributions change dramatically. And so one of **the** records that we had was this sooty shearwater, which undergoes these spectacular migrations all **the** way from New Zealand all **the** way up to Alaska and back again in **search** of endless summer as they complete their life cycles. We also talked about **the** White Shark Cafe. This is a location in **the** Pacific where white shark converge. We don ' t know why they converge there, we simply don ' t know. That ' s a question for **the** future. One of **the** things that we ' re taught in high school is that all animals require oxygen in order to survive. Now this little critter, it ' s only about half a millimeter in size, not terribly charismatic. But it was only discovered in **the** early 1980s. But **the** really interesting thing about it is that, a few years ago, census scientists discovered that this guy can thrive in oxygen-poor sediments in **the** deep Mediterranean Sea. So now they know that, in fact, animals can live without oxygen, at least some of them, and that they can adapt to even **the** harshest of conditions. If you were to suck all **the** water out of **the** ocean, this is what you ' d be left behind with, and that ' s **the** biomass of life on **the sea** floor. Now what we see is huge biomass towards **the** poles and not much biomass in between. We found life in **the** extremes. And so there were new species that were found that live inside ice and help to support an ice-based food **web**. And we also found this spectacular yeti **crab** that lives near boiling hot hydrothermal vents at Easter Island. And this particular species really captured **the** public ' s attention. We also found **the** deepest vents known yet — 5, 000 meters — **the** hottest vents at 407 degrees Celsius — vents in **the** South Pacific and also in **the** Arctic where none had been found before. So even new environments are still within **the** domain of **the** discoverable. Now in terms of **the** unknowns, there are many. And I ' m just going to summarize just a few of them very quickly for you. First of all, we might ask, how many **fishes** in **the sea**? We actually know **the fishes** better than we do any other group in **the** ocean other than marine mammals. And so we can actually extrapolate based on rates of discovery how many more species we ' re likely to discover. And from that, we actually **calculate** that we know about 16, 500 marine species and there are probably another 1, 000 to 4, 000 left to go. So we ' ve done pretty well. We ' ve got about 75 percent of **the fish**, maybe as much as 90 percent. But **the fishes**, as I say, are **the** best known. So our level of knowledge is much less for other groups of organisms. Now this figure is actually based on a brand new paper that ' s going to come out in **the** journal PLoS **Biology**. And what is does is predict how many more species there are on land and in **the** ocean. And what they found is that they think that we know of about nine percent of **the** species in **the** ocean. That means 91 percent, even after **the** census, still remain to be discovered. And so that turns out to be about two million species once all is said and done. So we still have quite a lot of work to do in terms of unknowns. Now this bacterium is part of mats that are found off **the** coast of Chile. And these mats actually cover an area **the** size of Greece. And so this particular bacterium is actually visible to **the** naked eye. But you can imagine **the** biomass that represents. But **the** really intriguing thing about **the** microbes is just how diverse they are. A single drop of seawater could contain 160 different types of microbes. And **the** oceans themselves are thought potentially to contain as many as a billion different types. So that ' s really exciting. What are they all doing out there? We actually don ' t know. **The** most exciting thing, I would say, about this census is **the** role of global science. And so as we see in this image of light during **the** night, there are lots of areas of **the** Earth where human development is much greater and other areas where it ' s much less, but between them we see large dark areas of relatively unexplored ocean. **The** other point I ' d like to make about this is that this ocean ' s interconnected. Marine organisms do not care about international boundaries ; they move where they will. And so **the** importance then of global collaboration becomes all **the** more important. We ' ve lost a lot of paradise. For example, these tuna that were once so abundant in **the** North Sea are now effectively gone. There were trawls taken in **the** deep **sea** in **the** Mediterranean, which collected more garbage than they did animals. And that ' s **the** deep **sea**, that ' s **the** environment that we consider to be among **the** most pristine **left** on Earth. And there are a lot of other pressures. Ocean acidification is a really big issue that people are concerned with, as well as ocean warming, and **the** effects they ' re going to have on coral reefs. On **the** scale of decades, in our lifetimes, we ' re going to see a lot of damage to coral reefs. And I could spend **the** rest of my time, which is getting very limited, going through this litany of concerns about **the** ocean, but I want to end on a more positive note. And so **the** grand challenge then is to try and make sure that we preserve what ' s left, because there is still spectacular beauty. And **the** oceans are so productive, there ' s so much going on in there that ' s of relevance to humans that we really need to, even from a selfish perspective, try to do better than we have in **the** past. So we need to recognize those hot spots and do our best to protect them. When we look at pictures like this, they take our breath away, in addition to helping to give us breath by **the** oxygen that **the** oceans provide. Census scientists worked in **the** rain, they worked in **the** cold, they worked under water and they worked above water trying to illuminate **the** wondrous discovery, **the** still vast unknown, **the** spectacular adaptations that we see in ocean life. So whether you ’ re a yak herder living in **the** mountains of Chile, whether you ’ re a stockbroker in New York City or whether you ’ re a TEDster living in Edinburgh, **the** oceans matter. And as **the** oceans go so shall we. Thanks for listening.

Text Sample 33: POS Dim 1 – 2011 – Score 23.00 – 2011/t002559.txt
Each of you possesses **the** most powerful, dangerous and subversive trait that natural **selection** has ever **devised**. It ’ s a piece of neural audio technology for rewiring other people ’ s minds. I ’ m talking about your language, of course, because it allows you to implant a thought from your mind directly into someone else ’ s mind, and they can attempt to do **the** same to you, without either of you having to perform surgery. Instead, when you speak, you ’ re actually using a form of telemetry not so different from **the** remote control **device** for your television. It ’ s just that, whereas that **device** relies on **pulses** of infrared light, your language relies on **pulses**, discrete **pulses**, of sound. And just as you use **the** remote control **device** to alter **the** internal settings of your television to suit your mood, you use your language to alter **the** settings inside someone else ’ s brain to suit your interests. Languages are genes talking, getting things that they want. And just imagine **the** sense of wonder in a baby when it first discovers that, merely by uttering a sound, it can get **objects** to move across a room as if by magic, and maybe even into its mouth. Now language ’ s subversive power has been recognized throughout **the** ages in censorship, in books you can ’ t read, phrases you can ’ t use and words you can ’ t say. In fact, **the** Tower of Babel story in **the** Bible is a fable and warning about **the** power of language. According to that story, early humans developed **the** conceit that, by using their language to work together, they could build a tower that would take them all **the** way to heaven. Now God, angered at this attempt to usurp his power, destroyed **the** tower, and then to ensure that it would never be rebuilt, he scattered **the** people by giving them different languages — confused them by giving them different languages. And this leads to **the** wonderful irony that our languages exist to prevent us from communicating. Even today, we know that there are words we cannot use, phrases we cannot say, because if we do so, we might be accosted, jailed, or even killed. And all of this from a puff of air emanating from our mouths. Now all this fuss about a single one of our traits tells us there ’ s something worth explaining. And that is how and why did this remarkable trait **evolve**, and why did it **evolve** only in our species? Now it ’ s a little bit of a surprise that to get an answer to that question, we have to go to tool use in **the** chimpanzees. Now these chimpanzees are using tools, and we take that as a sign of their intelligence. But if they really were intelligent, why would they use a stick to extract termites from **the** ground rather than a shovel? And if they really were intelligent, why would they crack open **nuts** with a **rock**? Why wouldn ’ t they just go to a shop and buy a bag of **nuts** that somebody else had already cracked open for them? Why not? I mean, that ’ s what we do. Now **the** reason **the** chimpanzees don ’ t do that is that they lack what psychologists and anthropologists call social **learning**. They seem to lack **the** ability to learn from others by **copying** or imitating or simply watching. As a result, they can ’ t improve on others ’ ideas or learn from others ’ mistakes — benefit from others ’ wisdom. And so they just do **the** same thing over and

over and over again. In fact, we could go away for a million years and come back and these chimpanzees would be doing **the** same thing with **the** same sticks for **the** termites and **the** same **rocks** to crack open **the nuts**. Now this may sound arrogant, or even full of hubris. How do we know this? Because this is exactly what our ancestors, **the** Homo erectus, did. These upright apes **evolved** on **the** African savanna about two million years ago, and they made these splendid hand axes that fit wonderfully into your hands. But if we look at **the fossil** record, we see that they made **the** same hand axe over and over and over again for one million years. You can follow it through **the fossil** record. Now if we make some guesses about how long Homo erectus lived, what their generation time was, that ' s about 40, 000 generations of parents to offspring, and other individuals watching, in which that hand axe didn ' t change. It ' s not even clear that our very close genetic **relatives**, **the** Neanderthals, had social **learning**. Sure enough, their tools were more complicated than those of Homo erectus, but they too showed very little change over **the** 300, 000 years or so that those species, **the** Neanderthals, lived in Eurasia. **Okay**, so what this tells us is that, contrary to **the** old adage,"monkey see, monkey do,"**the** surprise really is that all of **the** other animals really cannot do that — at least not very much. And even this picture has **the** suspicious taint of being rigged about it — something from a Barnum & Bailey circus. But by comparison, we can learn. We can learn by watching other people and **copying** or imitating what they can do. We can then choose, from among a range of options, **the** best one. We can benefit from others ' ideas. We can build on their wisdom. And as a result, our ideas do accumulate, and our technology progresses. And this cumulative cultural adaptation, as anthropologists call this accumulation of ideas, is responsible for everything around you in your bustling and teeming everyday lives. I mean **the** world has changed out of all proportion to what we would recognize even 1, 000 or 2, 000 years ago. And all of this because of cumulative cultural adaptation. **The** chairs you ' re sitting in, **the** lights in this auditorium, my microphone, **the** iPads and iPods that you carry around with you — all are a result of cumulative cultural adaptation. Now to many commentators, cumulative cultural adaptation, or social **learning**, is job done, end of story. Our species can make stuff, therefore we prospered in a way that no other species has. In fact, we can even make **the**"stuff of life"— as I just said, all **the** stuff around us. But in fact, it turns out that some time around 200, 000 years ago, when our species first arose and acquired social **learning**, that this was really **the** beginning of our story, not **the** end of our story. Because our acquisition of social **learning** would create a social and **evolutionary** dilemma, **the** resolution of which, it ' s fair to say, would determine not only **the** future course of our psychology, but **the** future course of **the** entire world. And most importantly for this, it ' ll tell us why we have language. And **the** reason that dilemma arose is, it turns out, that social **learning** is visual theft. If I can learn by watching you, I can steal your best ideas, and I can benefit from your efforts, without having to put in **the** time and energy that you did into developing them. If I can watch which lure you use to catch a **fish**, or I can watch how you flake your hand axe to make it better, or if I follow you secretly to your mushroom patch, I can benefit from your knowledge and wisdom and skills, and maybe even catch that **fish** before you do. Social **learning** really is visual theft. And in any species that acquired it, it would behoove you to hide your best ideas, lest somebody steal them from you. And so some time around 200, 000 years ago, our species confronted this crisis. And we really had only two options for dealing with **the** conflicts that visual theft would bring. One of those options was that we could have retreated into small family groups. Because then **the** benefits of our ideas and knowledge would flow just to our **relatives**. Had we chosen this option, sometime around 200, 000 years ago, we would probably still be living like **the** Neanderthals were when we first entered Europe 40, 000 years ago. And this is because in small groups there are fewer ideas, there are fewer innovations. And small groups are more prone to accidents and bad luck. So if we ’ d chosen that path, our **evolutionary** path would have led into **the** forest — and been a short one indeed. **The** other option we could choose was to develop **the** systems of communication that would allow us to share ideas and to cooperate amongst others. Choosing this option would mean that a vastly greater fund of accumulated knowledge and wisdom would become available to any one individual than would ever arise from within an individual family or an individual person on their own. Well, we chose **the** second option, and language is **the** result. Language **evolved** to solve **the** crisis of visual theft. Language is a piece of social technology for enhancing **the** benefits of cooperation — for reaching agreements, for striking deals and for coordinating our activities. And you can see that, in a developing society that was beginning to acquire language, not having language would be a like a bird without **wings**. Just as **wings** open up this **sphere** of air for birds to exploit, language opened up **the sphere** of cooperation for humans to exploit. And we take this utterly for granted, because we ’ re a species that is so at home with language, but you have to realize that even **the** simplest acts of exchange that we engage in are utterly dependent upon language. And to see why, consider two scenarios from early in our **evolution**. Let ’ s imagine that you are really good at making arrowheads, but you ’ re hopeless at making **the** wooden shafts with **the** flight feathers attached. Two other people you know are very good at making **the** wooden shafts, but they ’ re hopeless at making **the** arrowheads. So what you do is — one of those people has not really acquired language yet. And let ’ s pretend **the** other one is good at language skills. So what you do one day is you take a pile of arrowheads, and you walk up to **the** one that can ’ t speak very well, and you put **the** arrowheads down in front of him, hoping that he ’ ll get **the** idea that you want to trade your arrowheads for finished arrows. But he looks at **the** pile of arrowheads, thinks they ’ re a gift, picks them up, smiles and walks off. Now you pursue this guy, gesticulating. A scuffle ensues and you get stabbed with one of your own arrowheads. **Okay**, now replay this scene now, and you ’ re approaching **the** one who has language. You put down your arrowheads and say,”I ’ d like to trade these arrowheads for finished arrows. I ’ ll split you 50 / 50.”**The** other one says,”Fine. Looks good to me. We ’ ll do that.”Now **the** job is done. Once we have language, we can put our ideas together and cooperate to have a prosperity that we couldn ’ t have before we acquired it. And this is why our species has prospered around **the** world while **the** rest of **the** animals sit behind bars in zoos, languishing. That ’ s why we build space shuttles and cathedrals while **the** rest of **the** world sticks sticks into **the** ground to extract termites. All right, if this view of language and its value in solving **the** crisis of visual theft is true, any species that acquires it should show an **explosion** of creativity and prosperity. And this is exactly what **the** archeological record shows. If you look at our ancestors, **the** Neanderthals and **the** Homo erectus, our immediate ancestors, they ’ re confined to small regions of **the** world. But when our species arose about 200, 000 years ago, sometime after that we quickly walked out of Africa and spread around **the** entire world, occupying nearly every **habitat** on Earth. Now whereas other species are confined to places that their genes adapt them to, with social **learning** and language, we could transform **the** environment to suit our needs. And so we prospered in a way that no other animal has. Language really is **the** most potent trait that has ever **evolved**. It is **the** most valuable trait we have for converting new lands and resources into more people and their genes that natural

selection has ever **devised**. Language really is **the** voice of our genes. Now having **evolved** language, though, we did something **peculiar**, even bizarre. As we spread out around **the** world, we developed thousands of different languages. Currently, there are about seven or 8, 000 different languages spoken on Earth. Now you might say, well, this is just natural. As we diverge, our languages are naturally going to diverge. But **the** real puzzle and irony is that **the** greatest density of different languages on Earth is found where people are most tightly packed together. If we go to **the island** of Papua New Guinea, we can find about 800 to 1, 000 distinct human languages, different human languages, spoken on that **island** alone. There are places on that **island** where you can encounter a new language every two or three miles. Now, incredible as this sounds, I once met a Papuan man, and I asked him if this could possibly be true. And he said to me,"Oh no. They ' re far closer together than that."And it ' s true ; there are places on that **island** where you can encounter a new language in under a mile. And this is also true of some remote oceanic **islands**. And so it seems that we use our language, not just to cooperate, but to draw rings around our cooperative groups and to establish identities, and perhaps to protect our knowledge and wisdom and skills from eavesdropping from outside. And we know this because when we study different language groups and associate them with their cultures, we see that different languages slow **the** flow of ideas between groups. They slow **the** flow of technologies. And they even slow **the** flow of genes. Now I can ' t speak for you, but it seems to be **the** case that we don ' t have sex with people we can ' t talk to. Now we have to counter that, though, against **the** evidence we ' ve heard that we might have had some rather distasteful genetic dalliances with **the** Neanderthals and **the** Denisovans. **Okay**, this tendency we have, this seemingly natural tendency we have, towards isolation, towards keeping to ourselves, crashes head first into our modern world. This remarkable image is not a map of **the** world. In fact, it ' s a map of Facebook friendship links. And when you plot those friendship links by their latitude and longitude, it literally draws a map of **the** world. Our modern world is communicating with itself and with each other more than it has at any time in its past. And that communication, that connectivity around **the** world, that globalization now raises a burden. Because these different languages impose a barrier, as we ' ve just seen, to **the** transfer of goods and ideas and technologies and wisdom. And they impose a barrier to cooperation. And nowhere do we see that more clearly than in **the** European Union, whose 27 member countries speak 23 official languages. **The** European Union is now spending over one billion euros annually translating among their 23 official languages. That ' s something on **the** order of 1. 45 billion U. S. dollars on translation costs alone. Now think of **the** absurdity of this situation. If 27 individuals from those 27 member states sat around table, speaking their 23 languages, some very simple mathematics will tell you that you need an army of 253 translators to anticipate all **the** pairwise possibilities. **The** European Union employs a permanent staff of about 2, 500 translators. And in 2007 alone — and I ' m sure there are more recent figures — something on **the** order of 1. 3 million pages were translated into English alone. And so if language really is **the** solution to **the** crisis of visual theft, if language really is **the** conduit of our cooperation, **the** technology that our species derived to promote **the** free flow and exchange of ideas, in our modern world, we confront a question. And that question is whether in this modern, globalized world we can really afford to have all these different languages. To put it this way, nature knows no other circumstance in which functionally equivalent traits coexist. One of them always drives **the** other extinct. And we see this in **the** inexorable march towards standardization. There are lots and lots of ways of measuring things

— weighing them and measuring their length — but **the** metric system is winning. There are lots and lots of ways of measuring time, but a really bizarre base 60 system known as hours and minutes and seconds is nearly universal around **the** world. There are many, many ways of imprinting CDs or DVDs, but those are all being standardized as well. And you can probably think of many, many more in your own everyday lives. And so our modern world now is confronting us with a dilemma. And it ' s **the** dilemma that this Chinese man faces, who ' s language is spoken by more people in **the** world than any other single language, and yet he is sitting at his blackboard, converting Chinese phrases into English language phrases. And what this does is it raises **the** possibility to us that in a world in which we want to promote cooperation and exchange, and in a world that might be dependent more than ever before on cooperation to maintain and enhance our levels of prosperity, his actions suggest to us it might be inevitable that we have to confront **the** idea that our destiny is to be one world with one language. Thank you. Matt Ridley : Mark, one question. Svante found that **the** FOXP2 gene, which seems to be associated with language, was also shared in **the** same form in Neanderthals as us. Do we have any idea how we could have defeated Neanderthals if they also had language? Mark Pagel : This is a very good question. So many of you will be familiar with **the** idea that there ' s this gene called FOXP2 that seems to be implicated in some ways in **the** fine motor control that ' s associated with language. **The** reason why I don ' t believe that tells us that **the** Neanderthals had language is — here ' s a simple analogy : Ferraris are cars that have engines. My car has an engine, but it ' s not a Ferrari. Now **the** simple answer then is that genes alone don ' t, all by themselves, determine **the** outcome of very complicated things like language. What we know about this FOXP2 and Neanderthals is that they may have had fine motor control of their mouths — who knows. But that doesn ' t tell us they necessarily had language. MR : Thank you very much indeed.

Text Sample 34: POS Dim 1 – 2011 – Score 20.00 – 2011/t002528.txt

I ' m a neuroscientist. And in neuroscience, we have to deal with many difficult questions about **the** brain. But I want to start with **the** easiest question and **the** question you really should have all asked yourselves at some point in your life, because it ' s a fundamental question if we want to understand brain function. And that is, why do we and other animals have brains? Not all species on our **planet** have brains, so if we want to know what **the** brain is for, let ' s think about why we **evolved** one. Now you may reason that we have one to perceive **the** world or to think, and that ' s completely wrong. If you think about this question for any length of time, it ' s blindingly obvious why we have a brain. We have a brain for one reason and one reason only, and that ' s to produce adaptable and complex movements. There is no other reason to have a brain. Think about it. Movement is **the** only way you have of affecting **the** world around you. Now that ' s not quite true. There ' s one other way, and that ' s through sweating. But apart from that, everything else goes through contractions of muscles. So think about communication — speech, gestures, writing, sign language — they ' re all mediated through contractions of your muscles. So it ' s really important to remember that sensory, memory and cognitive processes are all important, but they ' re only important to either drive or suppress future movements. There can be no **evolutionary** advantage to laying down memories of childhood or perceiving **the** color of a rose if it doesn ' t affect **the** way you ' re going to move later in life. Now for those who don ' t believe this argument, we have trees and grass on our **planet** without **the** brain, but **the** clinching evidence is this animal here — **the** humble **sea** squirt. Rudimentary animal, has a nervous system, swims around in **the** ocean in its juvenile life. And at some point of its life, it implants on a **rock**. And **the** first thing it does in implanting on that **rock**, which it never leaves, is to digest its own brain and nervous system for food. So once you don ' t need to move, you don ' t need **the** luxury of that brain. And this animal is often taken as an analogy to what happens at universities when professors get tenure, but that ' s a different subject. So I am a movement chauvinist. I believe movement is **the** most important function of **the** brain — don ' t let anyone tell you that it ' s not true. Now if movement is so important, how well are we doing understanding how **the** brain controls movement? And **the** answer is we ' re doing extremely poorly ; it ' s a very hard problem. But we can look at how well we ' re doing by thinking about how well we ' re doing building machines which can do what humans can do. Think about **the** game of chess. How well are we doing determining what piece to move where? If you pit Garry Kasparov here, when he ' s not in jail, against IBM ' s Deep Blue, well **the** answer is IBM ' s Deep Blue will occasionally win. And I think if IBM ' s Deep Blue played anyone in this room, it would win every time. That problem is solved. What about **the** problem of picking up a chess piece, dexterously **manipulating** it and putting it back down on **the** board? If you put a five **year-old** child ' s dexterity against **the** best robots of today, **the** answer is simple : **the** child wins easily. There ' s no competition at all. Now why is that top problem so easy and **the** bottom problem so hard? One reason is a very smart five **year-old** could tell you **the** algorithm for that top problem — look at all possible moves to **the** end of **the** game and choose **the** one that makes you win. So it ' s a very simple algorithm. Now of course there are other moves, but with vast **computers** we approximate and come close to **the** optimal solution. When it comes to being dexterous, it ' s not even clear what **the** algorithm is you have to solve to be dexterous. And we ' ll see you have to both perceive and act on **the** world, which has a lot of problems. But let me show you cutting-edge robotics. Now a lot of robotics is very impressive, but manipulation robotics is really just in **the** dark ages. So this is **the** end of a Ph. D. project from one of **the** best robotics institutes. And **the** student has trained this robot to pour this water into a glass. It ' s a hard problem because **the** water sloshes about, but it can do it. But it doesn ' t do it with anything like **the** agility of a human. Now if you want this robot to do a different task, that ' s another three-year Ph. D. program. There is no generalization at all from one task to another in robotics. Now we can compare this to cutting-edge human performance. So what I ' m going to show you is Emily Fox winning **the** world record for cup **stacking**. Now **the** Americans in **the** audience will know all about cup **stacking**. It ' s a high school sport where you have 12 cups you have to **stack** and unstack against **the** clock in a prescribed order. And this is her getting **the** world record in real time. And she ' s pretty happy. We have no idea what is going on inside her brain when she does that, and that ' s what we ' d like to know. So in my group, what we try to do is reverse engineer how humans control movement. And it sounds like an easy problem. You send a command down, it causes muscles to contract. Your arm or body moves, and you get sensory feedback from vision, from skin, from muscles and so on. **The** trouble is these signals are not **the** beautiful signals you want them to be. So one thing that makes controlling movement difficult is, for example, sensory feedback is extremely noisy. Now by noise, I do not mean sound. We use it in **the** engineering and neuroscience sense meaning a random noise corrupting a signal. So **the** old days before digital radio when you were tuning in your radio and you heard"crrcckkk"on **the** station you wanted to hear, that was **the** noise. But more generally, this noise is something that corrupts **the** signal. So for example, if you put your hand under a table and try to localize it with your other hand, you can be off by several centimeters due to **the** noise in sensory feedback. Similarly, when you put motor output on movement output, it ' s extremely noisy. Forget about trying to hit **the** bull ' s eye in darts, just aim for **the** same spot over and over again. You have a huge spread due to movement variability. And more than that, **the** outside world, or task, is both ambiguous and variable. **The** teapot could be full, it could be empty. It changes over time. So we work in a whole sensory movement task **soup** of noise. Now this noise is so great that society places a huge premium on those of us who can reduce **the** consequences of noise. So if you ' re lucky enough to be able to knock a small white ball into a hole several hundred yards away using a long metal stick, our society will be willing to reward you with hundreds of millions of dollars. Now what I want to convince you of is **the** brain also goes through a lot of effort to reduce **the** negative consequences of this sort of noise and variability. And to do that, I ' m going to tell you about a framework which is very popular in statistics and machine **learning** of **the** last 50 years called Bayesian decision theory. And it ' s more recently a unifying way to think about how **the** brain deals with uncertainty. And **the** fundamental idea is you want to make inferences and then take actions. So let ' s think about **the** inference. You want to generate beliefs about **the** world. So what are beliefs? Beliefs could be : where are my arms in space? Am I looking at a cat or a fox? But we ' re going to represent beliefs with **probabilities**. So we ' re going to represent a belief with a number between zero and one — zero meaning I don ' t believe it at all, one means I ' m absolutely certain. And numbers in between give you **the** gray levels of uncertainty. And **the** key idea to Bayesian inference is you have two sources of information from which to make your inference. You have data, and data in neuroscience is sensory input. So I have sensory input, which I can take in to make beliefs. But there ' s another source of information, and that ' s effectively prior knowledge. You accumulate knowledge throughout your life in memories. And **the** point about Bayesian decision theory is it gives you **the** mathematics of **the** optimal way to combine your prior knowledge with your sensory evidence to generate new beliefs. And I ' ve put **the** formula up there. I ' m not going to explain what that formula is, but it ' s very beautiful. And it has real beauty and real explanatory power. And what it really says, and what you want to estimate, is **the probability** of different beliefs given your sensory input. So let me give you an intuitive example. Imagine you ' re learning to play tennis and you want to decide where **the** ball is going to bounce as it comes over **the** net towards you. There are two sources of information Bayes ' rule tells you. There ' s sensory evidence — you can use visual information auditory information, and that might tell you it ' s going to land in that red spot. But you know that your senses are not perfect, and therefore there ' s some variability of where it ' s going to land shown by that cloud of red, representing numbers between 0. 5 and maybe 0. 1. That information is available in **the** current shot, but there ' s another source of information not available on **the** current shot, but only available by repeated experience in **the** game of tennis, and that ' s that **the** ball doesn ' t bounce with equal **probability** over **the** court during **the** match. If you ' re playing against a very good opponent, they may distribute it in that green area, which is **the** prior distribution, making it hard for you to return. Now both these sources of information carry important information. And what Bayes ' rule says is that I should multiply **the** numbers on **the** red by **the** numbers on **the** green to get **the** numbers of **the yellow**, which have **the** ellipses, and that ' s my belief. So it ' s **the** optimal way of combining information. Now I wouldn ' t tell you all this if it wasn ' t that a few years ago, we showed this is exactly what people do when they learn new movement skills. And what it means is we really are Bayesian inference machines. As we go around, we learn about statistics of **the** world and lay that down, but we also learn about how noisy our own sensory apparatus is, and then combine those in a real Bayesian way. Now a key part to **the** Bayesian is this part of **the** formula. And what this part really says is I have to predict **the probability** of different sensory feedbacks given my beliefs. So that really means I have to make predictions of **the** future. And I want to convince you **the** brain does make predictions of **the** sensory feedback it ’ s going to get. And moreover, it profoundly changes your perceptions by what you do. And to do that, I ’ ll tell you about how **the** brain deals with sensory input. So you send a command out, you get sensory feedback back, and that transformation is governed by **the** physics of your body and your sensory apparatus. But you can imagine looking inside **the** brain. And here ’ s inside **the** brain. You might have a little predictor, a neural simulator, of **the** physics of your body and your senses. So as you send a movement command down, you tap a **copy** of that off and run it into your neural simulator to anticipate **the** sensory consequences of your actions. So as I shake this ketchup bottle, I get some true sensory feedback as **the** function of time in **the** bottom row. And if I ’ ve got a good predictor, it predicts **the** same thing. Well why would I bother doing that? I ’ m going to get **the** same feedback anyway. Well there ’ s good reasons. Imagine, as I shake **the** ketchup bottle, someone very kindly comes up to me and taps it on **the** back for me. Now I get an extra source of sensory information due to that external act. So I get two sources. I get you tapping on it, and I get me shaking it, but from my senses ’ point of view, that is combined together into one source of information. Now there ’ s good reason to believe that you would want to be able to distinguish external events from internal events. Because external events are actually much more behaviorally relevant than feeling everything that ’ s going on inside my body. So one way to reconstruct that is to compare **the** prediction — which is only based on your movement commands — with **the** reality. Any discrepancy should hopefully be external. So as I go around **the** world, I ’ m making predictions of what I should get, subtracting them off. Everything left over is external to me. What evidence is there for this? Well there ’ s one very clear example where a sensation generated by myself feels very different then if generated by another person. And so we decided **the** most obvious place to start was with tickling. It ’ s been known for a long time, you can ’ t tickle yourself as well as other people can. But it hasn ’ t really been shown, it ’ s because you have a neural simulator, simulating your own body and subtracting off that sense. So we can bring **the** experiments of **the** 21st century by applying robotic technologies to this problem. And in effect, what we have is some sort of stick in one hand attached to a robot, and they ’ re going to move that back and forward. And then we ’ re going to track that with a **computer** and use it to control another robot, which is going to tickle their palm with another stick. And then we ’ re going to ask them to rate a bunch of things including ticklishness. I ’ ll show you just one part of our study. And here I ’ ve taken away **the** robots, but basically people move with their right arm sinusoidally back and forward. And we replay that to **the** other hand with a time delay. Either no time delay, in which case light would just tickle your palm, or with a time delay of two- tenths of three-tenths of a second. So **the** important point here is **the** right hand always does **the** same things — sinusoidal movement. **The** left hand always is **the** same and puts sinusoidal tickle. All we ’ re playing with is a tempo causality. And as we go from naught to 0. 1 second, it becomes more ticklish. As you go from 0. 1 to 0. 2, it becomes more ticklish at **the** end. And by 0. 2 of a second, it ’ s equivalently ticklish to **the** robot that just tickled you without you doing anything. So whatever is responsible for this cancellation is extremely tightly coupled with tempo causality. And based on this illustration, we really convinced ourselves in **the** field that **the** brain ’ s making precise predictions and subtracting them off from **the** sensations. Now I have to admit, these are **the** worst studies my lab has ever run. Because **the** tickle sensation on **the** palm comes and goes, you need large numbers of subjects with these stars making them significant. So we were looking for a much more objective way to assess this phenomena. And in **the** intervening years I had two daughters. And one thing you notice about children in backseats of cars on long journeys, they get into fights — which started with one of them doing something to **the** other, **the** other retaliating. It quickly escalates. And children tend to get into fights which escalate in terms of force. Now when I screamed at my children to stop, sometimes they would both say to me **the** other person hit them harder. Now I happen to know my children don ’ t lie, so I thought, as a neuroscientist, it was important how I could explain how they were telling inconsistent truths. And we hypothesize based on **the** tickling study that when one child hits another, they generate **the** movement command. They predict **the** sensory consequences and subtract it off. So they actually think they ’ ve hit **the** person less hard than they have — rather like **the** tickling. Whereas **the** passive recipient doesn ’ t make **the** prediction, feels **the** full blow. So if they retaliate with **the** same force, **the** first person will think it ’ s been escalated. So we decided to test this in **the** lab. Now we don ’ t work with children, we don ’ t work with hitting, but **the** concept is **identical**. We bring in two adults. We tell them they ’ re going to play a game. And so here ’ s player one and player two sitting opposite to each other. And **the** game is very simple. We started with a motor with a little lever, a little force transfuser. And we use this motor to apply force down to player one ’ s fingers for three seconds and then it stops. And that player ’ s been told, remember **the** experience of that force and use your other finger to apply **the** same force down to **the** other subject ’ s finger through a force transfuser — and they do that. And player two ’ s been told, remember **the** experience of that force. Use your other hand to apply **the** force back down. And so they take it in turns to apply **the** force they ’ ve just experienced back and forward. But critically, they ’ re briefed about **the** rules of **the** game in separate rooms. So they don ’ t know **the** rules **the** other person ’ s playing by. And what we ’ ve measured is **the** force as a function of terms. And if we look at what we start with, a quarter of a Newton there, a number of turns, perfect would be that red line. And what we see in all pairs of subjects is this — a 70 percent escalation in force on each go. So it really suggests, when you ’ re doing this — based on this study and others we ’ ve done — that **the** brain is canceling **the** sensory consequences and underestimating **the** force it ’ s producing. So it re-shows **the** brain makes predictions and fundamentally changes **the** precepts. So we ’ ve made inferences, we ’ ve done predictions, now we have to generate actions. And what Bayes ’ rule says is, given my beliefs, **the** action should in some sense be optimal. But we ’ ve got a problem. Tasks are symbolic — I want to **drink**, I want to dance — but **the** movement system has to contract 600 muscles in a particular sequence. And there ’ s a big gap between **the** task and **the** movement system. So it could be bridged in infinitely many different ways. So think about just a point to point movement. I could choose these two paths out of an infinite number of paths. Having chosen a particular path, I can hold my hand on that path as infinitely many different joint **configurations**. And I can hold my arm in a particular joint **configuration** either very stiff or very relaxed. So I have a huge amount of choice to make. Now it turns out, we are extremely stereotypical. We all move **the** same way pretty much. And so it turns out we ’ re so stereotypical, our brains have got dedicated neural circuitry to decode this stereotyping. So if I take some dots and set them in motion with biological motion, your brain ' s circuitry would understand instantly what ' s going on. Now this is a bunch of dots moving. You will know what this person is doing, whether happy, sad, old, young — a huge amount of information. If these dots were cars going on a racing circuit, you would have absolutely no idea what ' s going on. So why is it that we move **the** particular ways we do? Well let ' s think about what really happens. Maybe we don ' t all quite move **the** same way. Maybe there ' s variation in **the** population. And maybe those who move better than others have got more chance of getting their children into **the** next generation. So in **evolutionary** scales, movements get better. And perhaps in life, movements get better through learning. So what is it about a movement which is good or bad? Imagine I want to intercept this ball. Here are two possible paths to that ball. Well if I choose **the** left-hand path, I can work out **the** forces required in one of my muscles as a function of time. But there ' s noise added to this. So what I actually get, based on this lovely, smooth, desired force, is a very noisy version. So if I pick **the** same command through many times, I will get a different noisy version each time, because noise changes each time. So what I can show you here is how **the** variability of **the** movement will **evolve** if I choose that way. If I choose a different way of moving — on **the** right for example — then I ' ll have a different command, different noise, playing through a noisy system, very complicated. All we can be sure of is **the** variability will be different. If I move in this particular way, I end up with a smaller variability across many movements. So if I have to choose between those two, I would choose **the** right one because it ' s less variable. And **the** fundamental idea is you want to plan your movements so as to minimize **the** negative consequence of **the** noise. And one intuition to get is actually **the** amount of noise or variability I show here gets bigger as **the** force gets bigger. So you want to avoid big forces as one principle. So we ' ve shown that using this, we can explain a huge amount of data — that exactly people are going about their lives planning movements so as to minimize negative consequences of noise. So I hope I ' ve convinced you **the** brain is there and **evolved** to control movement. And it ' s an intellectual challenge to understand how we do that. But it ' s also relevant for disease and rehabilitation. There are many diseases which effect movement. And hopefully if we understand how we control movement, we can apply that to robotic technology. And finally, I want to remind you, when you see animals do what look like very simple tasks, **the** actual complexity of what is going on inside their brain is really quite dramatic. Thank you very much. Chris Anderson : Quick question for you, Dan. So you ' re a movement — — chauvinist. Does that mean that you think that **the** other things we think our brains are about — **the** dreaming, **the** yearning, **the** falling in love and all these things — are a kind of side show, an accident? DW : No, no, actually I think they ' re all important to drive **the** right movement behavior to get **reproduction** in **the** end. So I think people who study sensation or memory without realizing why you ' re laying down memories of childhood. **The** fact that we forget most of our childhood, for example, is probably fine, because it doesn ' t effect our movements later in life. You only need to store things which are really going to effect movement. CA : So you think that people thinking about **the** brain, and consciousness generally, could get real insight by saying, where does movement play in this game? DW : So people have found out for example that studying vision in **the** absence of realizing why you have vision is a mistake. You have to study vision with **the** realization of how **the** movement system is going to use vision. And it uses it very differently once you think about it that way. CA : Well that was quite fascinating. Thank you very much indeed.

Text Sample 35: POS Dim 1 – 1984 – Score 8.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ’ ve tried to break it up into three parts : **the** first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a **computer** and really address **the** qualities of **the** human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, **the** intelligence of interaction. Then **the** second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then **the** last one will be some examples I ’ ve been able to collect, which I think illustrate this at least as best I can, in **the** world of entertainment. People have this belief — and I share most of it — that we will be using **the** TV screens or their equivalents for electronic books of **the** future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a **slide** taken from a TV set and it was pre- processed to be very sympathetic to **the** TV medium, and it absolutely looks beautiful. Well, what ’ s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal **computers** and video text — teletext systems, and somewhat horrified by what you see on **the** screen? Well, you have to remember that TV was designed to be looked at eight times **the** distance of **the** diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ’ s **the** distance you should sit away from **the** TV set. Now we ’ ve put people 18 **inches** in front of a TV, and all **the** artifacts that none of **the** original designers expected to be seen, all of a sudden, are staring you in **the** face : **the** shadow mask, **the** scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ’ m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ’ m very interested in touch-sensitive displays. High-tech, high-touch. Isn ’ t that what some of you said? It ’ s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ’ re not : it ’ s really a very, very high-resolution input medium — you have to just do it twice, you have to touch **the** screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on **the** market these systems that have just a few light emitting diodes on **the** side and are very low resolution, it ’ s nice that they exist because it still is better than nothing. But it, in some sense, misses **the** point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of **the** other advantages? Well, **the** one advantage is that you don ’ t have to pick them up, and people don ’ t realize how important that is — not having to pick up your fingers to use them. When you think for a second of **the** mouse on Macintosh — and I will not criticize **the** mouse too much — when you ’ re typing — what you have — you want to now put something — first of all, you ’ ve got to find **the** mouse. You have to probably stop. Maybe not come to a grinding halt, but you ’ ve got to sort of find that mouse. Then you find **the** mouse, and you ’ re going to have to wiggle it a little bit to see where **the** cursor is on **the** screen. And then when you finally see where it is, then you ’ ve got to move it to get **the** cursor over there, and then —”Bang”— you ’ ve got to hit a button or do whatever. That ’ s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ’ m trying to do is just illustrate **the** kinds of problems that I think face **the** designers of new **computer** systems and entertainment systems and educational systems from **the** perspective of **the** quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this **slide** is a fake **slide**. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of **the computer** field, is when you have a bug that you can ’ t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But **the** bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across **the** screen to input continuous points — and there was just too much friction created between your finger and **the** glass — if glass was **the** substrate, which it usually is. So we found that that actually was a feature in **the** sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all **the** forces on **the** face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ’ s pressure-sensitive. And it ’ s pressure-sensitive to **the** forces both in **the** plane of **the** screen — X, Y, and Z at least in one direction ; we couldn ’ t figure out how to come in **the** other direction. But let me get rid of **the slide**, and let ’ s see if this comes on. OK. So there is **the** pressure-sensitive display in operation. **The** person ’ s just, if you will, pushing on **the** screen to make a curve. But this is **the** interesting part. I want to stop it for a second because **the** movie is very badly made. And **the** particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of **objects** on it and **the** person has touched an **object** — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those **objects** are physically heavy and some are light : one is an anvil on a fuzzy rug and **the** other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across **the** screen, and yet you touch **the** ping-pong ball very lightly and it just scoots across **the** screen. And what you can do — oops, I didn ’ t mean to do that — what you can do is actually feed back to **the** user **the** feeling of **the** physical properties. So again, they don ’ t have to be weight ; they could be a general trying to move troops, and he ’ s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. **The** whole notion, then, is one that at **the** interface there are physical properties in that transducer — in this case it ’ s pressure and touches — that allow you to present things to **the** user that you could never present before. So it ’ s not simply looking at **the** quality or, if you will, **the** luxury of that interface, but it ’ s actually looking at **the** idea of presenting things that previously couldn ’ t be presented before. I want to move on to another example, which is one of a different sort, where we ’ re trying to use **computer** and video disc technology now to come up with a new kind of book. Here, **the** idea is that you ’ re going to take this book, if you will, and it ’ s going to come alive. You ’ re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are **the** experts in monologue making : you make a film and it has a well-formed beginning, middle and end, and in some sense **the** art of it is that. And you then say,”There ’ s an opportunity for making conversational movies.”Well, what does that mean? And it sort of nibbles at **the** core of **the** whole profession and all **the** assumptions of that medium. So, book writing is **the** same thing. What I ’ ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of **the** movie has information about itself. So it knows, or at least there is computer-readable information in **the** medium itself. It ' s just not a static movie frame. That ' s one thing. **The** other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is **the** cookbook, **the**"Larousse Gastronomique."And I think I use **the** example all too often, but it ' s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to **the** end of **the** recipe and it says,"Cook until done."Now, that would be, if you will, **the** top green track, which doesn ' t mean too much. But you might have to elaborate for me or for somebody who isn ' t an expert, and say,"Cook at 380 degrees for 45 minutes."And then for a real beginner, you would go down even further and elaborate more — say,"Open **the** oven, preheat, wait for **the** light to go out, open **the** door, don ' t leave it open too long, put **the** penguin in and shut **the** door..."whatever. And that ' s a much more elaborate one than you dribble back. That ' s one kind of use of random access. And **the** other is where you want to explain **the** same thing in different ways. If you ' re in a classroom situation and somebody asks a question, **the** last thing you do is repeat what you just said. You try and think of a different way of saying **the** same thing, or if you know **the** particular student and that student ' s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of **techniques** you will use — and again, this is a different kind of branching. So, what I will show you is... it ' s a rather boring book, but I ' m afraid sometimes you have to do boring books because your sponsors aren ' t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ' t even know what vintage **the** transmission is, but let me just show you very quickly some of it, and we ' ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of **the** transmission, and as you rub your finger across **the** transmission it highlights **the** various parts. Narrator : When I find a chapter that I want to see, I just touch **the** text and **the** system will format pages for me to read. **The** words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching **the** word, and **the** definition appears, superimposed over **the** illustration. NN : This is about **the oil** pan, or **the oil** filter and all that. This is relatively important because it sets **the** page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and **the** definition will appear in **the** illustration corner. I can get back to **the** illustration, but in this case it ' s not a single frame, but it ' s actually a movie of someone coming into **the** frame and doing **the** repair that ' s described in **the** text. **The** two-headed slider is a speed control that allows me to watch **the** movie at various speeds, in forward or reverse. And **the** movie is displayed as a full frame movie. I can go back to **the** beginning... and play **the** movie at full speed. Here ' s another step-by-step procedure, only in this case — NN : Okay, this movie is... Everybody ' s heard of sound-sync movies — this is text-sync movies, so as **the** movie plays, **the** text gets highlighted. We highlight **the** text as we go through **the** movie. Repairman :... Not too far out. Front poles, preferably. Don ' t loosen them too far. If you loosen them too far, you ' ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ' m at **the** third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to **the** world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ’ ve been doing — in this case, in Senegal — where we have tried to use personal **computers** as a pedagogical medium. But not as teaching machines at all ; **the** whole notion is to use this as an instrument where there is a complete reversal of roles — **the** child is, if you will, **the** teacher and **the** machine is **the** student — and **the** art of **computer** programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few **slides** I want to go through, but there ’ s a story I ’ d like to tell. And that was when, before we did this in any developing countries — we ’ re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ’ ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ’ t read, didn ’ t even make it in **the** lowest section of **the** school ’ s classes — and was pretty much not in school, though physically there. But did hang around **the**, quote,”**computer** room,”where there were quite a few **computers**, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very interesting. And one day, by chance, some visitors from **the** NIE came by in their double-breasted suits looking at this setup, and none of **the** children who were normally there, except for this one child, were there. He was, and he said,”Let me show you how this works,”and they got an absolutely ingenuous, wonderful description of Logo. And **the** child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ’ t explain and so he flipped through **the** manual, found **the** explanation and typed **the** command and got it to do what they asked. They were delighted, and by **the** time it was time to go see **the** principal, whom they ’ d actually come to see — not **the computer** room — they went upstairs and they said,”This is absolutely remarkable! That child was very articulate and showed us and even dealt with **the** things he couldn ’ t do automatically with that manual. It was just absolutely **fantastic**.”**The** principal said,”There ’ s a dreadful mistake, because that child can ’ t read. And you obviously have been hoodwinked or you ’ ve talked about somebody else.”And they all got up and they all went downstairs and **the** child was still there. And they did something very intelligent : they asked **the** child,”Can you read?”And **the** child said,”No, I can ’ t.”And then they said,”But wait a minute. You just looked through that manual and you found...”and he said,”Oh, but that ’ s not reading.”And so they said,”Well, what ’ s reading then?”He says,”Well, reading is this junk they give me in little books to read. It ’ s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return.”And it really meant something to **the** child. **The** child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. **The** key to **the** future of **computers** in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ’ m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ’ s not, but we think speech —”My God, little children pick it up somehow, and by **the** age of two they ’ re doing a mediocre job, and by three and four they ’ re speaking reasonably well. And yet you ’ ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder.”Well, it ’ s not harder. What **the** truth is is that speaking has great value to a child ; **the** child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ’ s going to help you. So what happens is you go to school and people say,”Just believe me, you ’ re going to like it. You ’ re going to like reading,”and just read and read. On **the** other hand, you give a kid — a three-year-old kid — a **computer** and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on **the** screen has a huge payoff, and it ’ s a lot of fun. And in fact, it ’ s a powerful educational instrument. Well, in Senegal we found that this was **the** traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ’ s **the** girl on **the** left leaning with her chalkboard, and she came... within two days — I want to show you **the** program she wrote, and remember her hairstyle. And that is **the** program she made. That ’ s what meant something to her, is doing **the** hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely **the** most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all **the** rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example — as some of my former colleagues, whom I see in **the** room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And **the** reason it remains a favorite project is that we were asked to do a teleconferencing system where you had **the** following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that **the** other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to **the** bait, and we did. And **the** fact that we knew **the** people — we had to take a page out of **the** history of Walt Disney — we actually went so far as to build CRTs in **the** shapes of **the** people ’ s faces. So if I wanted to call my friend Peter Sprague on **the** phone, my secretary would get his head out and bring it and set it on **the** desk, and that would be **the** TV used for **the** occasion. And it ’ s uncanny : there ’ s no way I can explain to you **the** amount of eye contact you get with that physical face projected on a 3D CRT of that sort. **The** next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ’ s something that didn ’ t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around **the** table, **the** actual location of **the** people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, **the** empty seat becomes, if you will, that person. And you point frequently to **the** empty seat and you say,”He or she wouldn ’ t agree,”and **the** empty chair is that person and **the** spatiality is crucial. So we said,”Well, these will be on round tables and **the** order around **the** table had to be **the** same, so that at my site, I would be, if you will, real and then at each other ’ s site you ’ d have these plastic heads. And **the** plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ’ t want to dwell on, but this is **the** one that we finally used where we projected onto rear screen material that was molded in **the** face — literally in **the** face of **the** person. And I ’ ll show you one more **slide**, where this is actually made from something called a solid photograph and is **the** screen. Now, we track, on **the** person ’ s head, **the** head motions — so we transmit with a video **the** head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to **the** person to my **left** and start talking to that person, then at **the** person to my right ’ s site, he ’ ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.

Text Sample 36: POS Dim 1 – 2012 – Score 20.00 – 2012/t002337.txt

America ’ s public energy conversation boils down to this question : Would you rather die of A) **oil** wars, or B) climate change, or C) nuclear holocaust, or D) all of **the** above? Oh, I missed one : or E) none of **the** above? That ’ s **the** one we ’ re not normally offered. What if we could make energy do our work without working our undoing? Could we have fuel without fear? Could we reinvent fire? You see, fire made us human ; **fossil** fuels made us modern. But now we need a new fire that makes us safe, secure, healthy and durable. Let ’ s see how. Four-fifths of **the** world ’ s energy still comes from burning each year four cubic miles of **the** rotted remains of primeval swamp goo. Those **fossil** fuels have built our civilization. They ’ ve created our wealth. They ’ ve enriched **the** lives of billions. But they also have rising costs to our security, economy, health and environment that are starting to erode, if not outweigh their benefits. So we need a new fire. And switching from **the** old fire to **the** new fire means changing two big stories about **oil** and electricity, each of which puts two- fifths of **the fossil carbon** in **the** air. But they ’ re really quite distinct. Less than one percent of our electricity is made from **oil** — although almost half is made from **coal**. Their uses are quite concentrated. Three-fourths of our **oil** fuel is **transportation**. Three-fourths of our electricity powers buildings. And **the** rest of both runs factories. So very efficient vehicles, buildings and factories save **oil** and **coal**, and also natural gas that can displace both of them. But today ’ s energy system is not just inefficient, it is also disconnected, aging, dirty and insecure. So it needs refurbishment. By 2050 though, it could become efficient, connected and distributed with elegantly frugal autos, factories and buildings all relying on a modern, secure and resilient electricity system. We can eliminate our addiction to **oil** and **coal** by 2050 and use **one-third** less natural gas while switching to efficient use and **renewable** supply. This could cost, by 2050, five trillion dollars less in net present value, that is expressed as a lump **sum** today, than business as usual — assuming that **carbon emissions** and all other hidden or external costs are worth zero — a conservatively low estimate. Yet this cheaper energy system could support 158 percent bigger U. S. economy all without needing **oil** or **coal**, or for that matter nuclear energy. Moreover, this transition needs no new inventions and no acts of Congress and no new federal taxes, mandate subsidies or laws and running Washington gridlock. Let me say that again. I ’ m going to tell you how to get **the** United States completely off **oil** and **coal**, five trillion dollars cheaper with no act of Congress led by business for profit. In other words, we ’ re going to use our most effective institutions — private enterprise co-evolving with civil society and sped by military innovation to go around our least effective institutions. And whether you care most about profits and jobs and competitive advantage or national security, or environmental stewardship and climate protection and public health, reinventing fire makes sense and makes money. General Eisenhower reputedly said that enlarging **the** boundaries of a tough problem makes it soluble by encompassing more options and more synergies. So in reinventing fire, we integrated all four sectors that use energy — **transportation**, buildings, industry and electricity — and we integrated four kinds of innovation, not just technology and policy, but also design and business strategy. Those combinations yield very much more than **the sum** of **the** parts, especially in creating deeply disruptive business opportunities. Oil costs our economy two billion dollars a day, plus another four billion dollars a day in hidden economic and military costs, raising its total cost to over a sixth of GDP. Our mobility fuel goes three-fifths to automobiles. So let ' s start by making autos **oil** free. **Two-thirds** of **the** energy it takes to move a typical car is caused by its weight. And every unit of energy you save at **the** wheels, by taking out weight or drag, saves seven units in **the** tank, because you don ' t have to waste six units getting **the** energy to **the** wheels. Unfortunately, over **the** past quarter century, epidemic obesity has made our two-ton steel cars gain weight twice as fast as we have. But today, ultralight, ultrastrong materials, like **carbon** fiber composites, can make dramatic weight savings snowball and can make cars simpler and cheaper to build. Lighter and more slippery autos need less force to move them, so their engines get smaller. Indeed, that sort of vehicle fitness then makes electric propulsion affordable because **the** batteries or fuel cells also get smaller and lighter and cheaper. So sticker prices will ultimately fall to about **the** same as today, while **the** driving cost, even from **the** start, is very much lower. So these innovations together can transform automakers from wringing tiny savings out of Victorian engine and seal-stamping technologies to **the** steeply falling costs of three linked innovations that strongly reenforce each other — namely ultralight materials, making them into structures and electric propulsion. **The** sales can grow and **the** prices fall even faster with temporary feebates, that is rebates for efficient new autos paid for by fees on inefficient ones. And just in **the** first two years **the** biggest of Europe ' s five feebate programs has tripled **the** speed of improving automotive efficiency. **The** resulting shift to electric autos is going to be as game-changing as shifting from typewriters to **the** gains in **computers**. Of course, **computers** and electronics are now America ' s biggest industry, while typewriter makers have vanished. So vehicle fitness opens a new automotive competitive strategy that can double **the oil** savings over **the** next 40 years, but then also make electrification affordable, and that displaces **the** rest of **the oil**. America could lead this next automotive revolution. Currently **the** leader is Germany. Last year, Volkswagen announced that by next year they ' ll be producing this **carbon** fiber plugin hybrid getting 230 miles a gallon. Also last year, BMW announced this **carbon** fiber electric car, they said that its **carbon** fiber is paid for by needing fewer batteries. And they said,"We do not intend to be a typewriter maker."Audi claimed it ' s going to beat them both by a year. Seven years ago, an even faster and cheaper American manufacturing technology was used to make this little **carbon** fiber test part, which doubles as a **carbon** cap. In one minute — and you can tell from **the** sound how immensely stiff and strong it is. Don ' t worry about dropping it, it ' s tougher than titanium. Tom Friedman actually whacked it as hard as he could with a sledgehammer without even scuffing it. But such manufacturing **techniques** can scale to automotive speed and cost with aerospace performance. They can save four-fifths of **the** capital needed to make autos. They can save lives because this stuff can absorb up to 12 times as much crash energy per **pound** as steel. If we made all of our autos this way, it would save **oil** equivalent to finding one and a half Saudi Arabias, or half an OPEC, by drilling in **the** Detroit formation, a very prospective play. And all those mega-barrels under Detroit cost an average of 18 **bucks** a barrel. They are all- American, carbon-free and inexhaustible. **The** same physics and **the** same business logic also apply to big vehicles. In **the** five years ending with 2010, Walmart saved 60 percent of **the** fuel per ton-mile in its giant fleet of heavy trucks through better logistics and design. But just **the** technological savings in heavy trucks can get to **two-thirds**. And combined with triple to quintuple efficiency **airplanes**, now on **the** drawing board, can save close to a trillion dollars. Also today ' s military revolution in energy efficiency is going to speed up all of these civilian advances in much **the** same way that military R&D has given us **the** Internet, **the** Global Positioning System and **the** jet engine and microchip industries. As we design and build vehicles better, we can also use them smarter by harnessing four powerful **techniques** for eliminating needless driving. Instead of just seeing **the** travel grow, we can use innovative pricing, charging for road infrastructure by **the** mile, not by **the** gallon. We can use some smart IT to enhance transit and enable car sharing and ride sharing. We can allow smart and lucrative growth models that help people already be near where they want to be, so they don ' t need to go somewhere else. And we can use smart IT to make traffic free-flowing. Together, those things can give us **the** same or better access with 46 to 84 percent less driving, saving another 0. 4 trillion dollars, plus 0. 3 trillion dollars from using trucks more productively. So 40 years hence, when you add it all up, a far more mobile U. S. economy can use no **oil**. Saving or displacing barrels for 25 **bucks** rather than buying them for over a hundred, adds up to a $4 trillion net saving counting all **the** hidden costs at zero. So to get mobility without **oil**, to phase out **the oil**, we can get efficient and then switch fuels. Those 125 to 240 mile-per-gallon-equivalent autos can use any mixture of hydrogen fuel cells, electricity and advanced biofuels. **The** trucks and planes can realistically use hydrogen or advanced biofuels. **The** trucks could even use natural gas. But no vehicles will need **oil**. And **the** most biofuel we might need, just three million barrels a day, can be made **two-thirds** from waste without displacing any cropland and without harming soil or climate. Our team speeds up these kinds of **oil** savings by what we call"institutional acupuncture."We figure out where **the** business logic is congested and not flowing properly, we stick little needles in it to get it flowing, working with partners like Ford and Walmart and **the** Pentagon. And **the** long transition is already well under way. In fact, three years ago mainstream analysts were starting to see peak **oil**, not in supply, but in demand. And Deutsche Bank even said world **oil** use could peak around 2016. In other words, **oil** is getting uncompetitive even at low prices before it becomes unavailable even at high prices. But **the** electrified vehicles don ' t need to burden **the** electricity grid. Rather, when smart autos exchange electricity and information through smart buildings with smart grids, they ' re adding to **the** grid valuable flexibility and storage that help **the** grid integrate varying solar and wind power. So **the** electrified autos make **the** auto and electricity problems easier to solve together than separately. And they also converge **the oil** story with our second big story, saving electricity and then making it differently. And those twin revolutions in electricity will bring to that sector more numerous and profound and diverse disruptions than any other sector, because we ' ve got 21st century technology and speed colliding head-on with 20th and 19th century institutions, rules and cultures. Changing how we make electricity gets easier if we need less of it. Most of it now is wasted and **the** technologies for saving it keep improving faster than we ' re installing them. So **the** unbought efficiency resource keeps getting ever bigger and cheaper. But as efficiency in buildings and industry starts to grow faster than **the** economy, America ' s electricity use could actually shrink, even with **the** little extra use required for those efficient electrified autos. And we can do this just by reasonably accelerating existing trends. Over **the** next 40 years, buildings, which use three-quarters of **the** electricity, can triple or quadruple their energy productivity, saving 1. 4 trillion dollars, net present value, with a 33 percent internal rate of return or in English, **the** savings are worth four times what they cost. And industry can accelerate too, doubling its energy productivity with a 21 percent internal rate of return. **The** key is a disruptive innovation that we call integrative design that often makes very big energy savings cost less than small or no savings. That is, it can give you expanding returns, not diminishing returns. That is how our 2010 retrofit is saving over two-fifths of **the** energy in **the** Empire State Building — remanufacturing those six and a half thousand windows on site into super windows that pass light, but reflect heat. plus better lights and office equipment and such cut **the** maximum cooling load by a third. And then renovating smaller chillers instead of adding bigger ones saved 17 million dollars of capital cost, which helped pay for **the** other improvements and reduce **the** payback to just three years. Integrative design can also increase energy savings in industry. Dow ' s billion-dollar efficiency investment has already returned nine billion dollars. But industry as a whole has another half-trillion dollars of energy still to save. For example, three-fifths of **the** world ' s electricity runs motors. Half of that runs pumps and fans. And those can all be made more efficient, and **the** motors that turn them can have their system efficiency roughly doubled by integrating 35 improvements, paying back in about a year. But first we ought to be capturing bigger, cheaper savings that are normally ignored and are not in **the textbooks**. For example, pumps, **the** biggest use of motors, move liquid through pipes. But a standard industrial pumping loop was redesigned to use at least 86 percent less energy, not by getting better pumps, but just by replacing long, thin, crooked pipes with **fat**, short, straight pipes. This is not about new technology, it ' s just rearranging our metal furniture. Of course, it also shrinks **the** pumping equipment and its capital costs. So what do such savings mean for **the** electricity that is three-fifths used in motors? Well, from **the coal** burned at **the** power plant through all these compounding losses, only a tenth of **the** fuel energy actually ends up coming out **the** pipe as flow. But now let ' s turn those compounding losses around **backwards**, and every unit of flow or friction that we save in **the** pipe saves 10 units of fuel cost, pollution and what Hunter Lovins calls"global weirding"back at **the** power plant. And of course, as you go back upstream, **the** components get smaller and therefore cheaper. Our team has lately found such snowballing energy savings in more than 30 billion dollars worth of industrial redesigns — everything from data centers and chip fabs to mines and refineries. Typically our retrofit designs save about 30 to 60 percent of **the** energy and pay back in a few years, while **the** new facility designs save 40 to 90-odd percent with generally lower capital cost. Now needing less electricity would ease and speed **the** shift to new sources of electricity, chiefly renewables. China leads their explosive growth and their plummeting cost. In fact, these solar power **module** costs have just fallen off **the** bottom of **the** chart. And Germany now has more solar workers than America has steel workers. Already in about 20 states private installers will come put those cheap solar cells on your roof with no money down and beat your utility bill. Such unregulated products could ultimately add up to a virtual utility that bypasses your electric company just as your cellphone bypassed your wireline phone company. And this sort of thing gives utility executives **the** heebee-jeebees and it gives venture capitalists sweet dreams. Renewables are no longer a fringe activity. For each of **the** past four years half of **the** world ' s new generating capacity has been **renewable**, mainly lately in developing countries. In 2010, renewables other than big hydro, particularly wind and solar cells, got 151 billion dollars of private investment, and they actually surpassed **the** total installed capacity of nuclear power in **the** world by adding 60 billion watts in that one year. That happens to be **the** same amount of solar cell capacity that **the** world can now make every year — a number that goes up 60 or 70 percent a year. In contrast, **the** net additions of nuclear capacity and **coal** capacity and **the** orders behind those keep fading because they cost too much and they have too much financial risk. In fact in this country, no new nuclear power plant has been able to raise any private construction capital, despite seven years of 100-plus percent subsidies. So how else could we replace **the** coal-fired power plants? Well efficiency and gas can displace them all at just below their operating cost and, combined with renewables, can displace them more than 23 times at less than their replacement cost. But we only need to replace them once. We ’ re often told though that only **coal** and nuclear plants can keep **the** lights on, because they ’ re 24 / 7, whereas wind and solar power are variable, and hence supposedly unreliable. Actually no generator is 24 / 7. They all break. And when a big plant goes down, you lose a thousand megawatts in milliseconds, often for weeks or months, often without warning. That is exactly why we ’ ve designed **the** grid to back up failed plants with working plants. And in exactly **the** same way, **the** grid can handle wind and solar power ’ s forecastable variations. Hourly simulations show that largely or wholly **renewable** grids can deliver highly reliable power when they ’ re forecasted, integrated and diversified by both type and location. And that ’ s true both for continental areas like **the** U. S. or Europe and for smaller areas embedded within a larger grid. That is how, for example, four German states in 2010 were 43 to 52 percent wind powered. Portugal was 45 percent **renewable** powered, Denmark 36. And it ’ s how all of Europe can shift to **renewable** electricity. In America, our aging, dirty and insecure power system has to be replaced anyway by 2050. And whatever we replace it with is going to cost about **the** same, about six trillion dollars at present value — whether we buy more of what we ’ ve got or new nuclear and so- called clean **coal**, or renewables that are more or less centralized. But those four futures at **the** same cost differ profoundly in their risks, around national security, fuel, water, finance, technology, climate and health. For example, our over-centralized grid is very vulnerable to cascading and potentially economy-shattering blackouts caused by bad space weather or other natural disasters or a **terrorist** attack. But that blackout risk **disappears**, and all of **the** other risks are best managed, with distributed renewables organized into local micro-grids that normally interconnect, but can stand alone at need. That is, they can disconnect fractally and then reconnect seamlessly. That approach is exactly what **the** Pentagon is adopting for its own power supply. They think they need that ; how about **the** rest of us that they ’ re defending? We want our stuff to work too. At about **the** same cost as business as usual, this would maximize national security, customer choice, entrepreneurial opportunity and innovation. Together, efficient use and diverse dispersed **renewable** supply are starting to transform **the** whole electricity sector. Traditionally utilities build a lot of giant **coal** and nuclear plants and a bunch of big gas plants and maybe a little bit of efficiency renewables. And those utilities were rewarded, as they still are in 34 states, for selling you more electricity. However, especially where regulators are now instead rewarding cutting your bills, **the** investments are shifting radically toward efficiency, demand response, cogeneration, renewables and ways to knit them all together reliably with less transmission and little or no bulk electricity storage. So our energy future is not fate, but choice, and that choice is very flexible. In 1976, for example, government and industry insisted that **the** amount of energy needed to make a dollar of GDP could never go down. And I heretically suggested it could go down several-fold. Well that ’ s what ’ s actually happened so far. It ’ s fallen by half. But with today ’ s much better technologies, more mature delivery channels and integrative design, we can do far more and even cheaper. So to solve **the** energy problem, we just needed to enlarge it. And **the** results may at first seem incredible, but as Marshall McLuhan said,”Only puny secrets need protection. Big discoveries are protected by public incredulity.”Now combine **the** electricity and **oil** revolutions, both driven by modern efficiency, and you get **the** really big story : reinventing fire, where business enabled and sped by smart policies in mindful markets can lead **the** United States completely off **oil** and **coal** by 2050, saving 5 trillion dollars, growing **the** economy 2. 6-fold, strengthening out national security, oh, and by **the** way, by getting rid of **the oil** and **coal**, reducing **the fossil carbon emissions** by 82 to 86 percent. Now if you like any of those outcomes, you can support reinventing fire without needing to like all of them and without needing to agree about which of them is most important. So focusing on outcomes, not motives, can turn gridlock and conflict into a unifying solution to America ' s energy challenge. This also turns out to be **the** best way to cope with global challenges — climate change, nuclear proliferation, energy insecurity, energy poverty — all of which make us less safe. Now our team at RMI helps smart companies to get unstuck and speed this journey via six sectoral initiatives, with some more hatching. Of course there ' s still a lot of old thinking out there too. Former **oil** man Maurice Strong said,"Not all **the fossils** are in **the** fuel."But as Edgar Woolard, who used to chair Dupont, reminds us,"Companies hampered by old thinking won ' t be a problem because,"he said,"they simply won ' t be around long-term."I ' ve described not just a once-in-a-civilization business opportunity, but one of **the** most profound transitions in **the** history of our species. We humans are inventing a new fire, not dug from below, but flowing from above ; not scarce, but bountiful ; not local, but everywhere ; not transient, but permanent ; not costly, but free. And but for a little transitional tail of natural gas and a bit of biofuel grown in ways that sustain and endure, this new fire is flameless. Efficiently used, it really can do our work without working our undoing. Each of you owns a piece of that $5 trillion prize. And our new book"Reinventing Fire"describes how you can capture it. So with **the** conversation just begun at ReinventingFire. com, let me invite you each to engage with us and with each other, with everyone around you, to help make **the** world richer, fairer, cooler and safer by together reinventing fire. Thank you.

Text Sample 37: POS Dim 1 – 2012 – Score 20.00 – 2012/t002268.txt

The job of uncovering **the** global food waste scandal started for me when I was 15 years old. I bought some pigs. I was living in Sussex. And I started to feed them in **the** most traditional and environmentally **friendly** way. I went to my school kitchen, and I said,"Give me **the** scraps that my school friends have turned their noses up at."I went to **the** local baker and took their stale bread. I went to **the** local greengrocer, and I went to a farmer who was throwing away potatoes because they were **the** wrong shape or size for **supermarkets**. This was great. My pigs turned that food waste into delicious pork. I sold that pork to my school friends ' parents, and I made a good **pocket** money addition to my teenage allowance. But I noticed that most of **the** food that I was giving my pigs was in fact fit for human **consumption**, and that I was only scratching **the surface**, and that right **the** way up **the** food supply chain, in **supermarkets**, greengrocers, bakers, in our homes, in factories and farms, we were hemorrhaging out food. Supermarkets didn ' t even want to talk to me about how much food they were wasting. I ' d been round **the** back. I ' d seen bins full of food being locked and then trucked off to landfill sites, and I thought, surely there is something more sensible to do with food than waste it. One morning, when I was feeding my pigs, I noticed a particularly tasty-looking sun-dried tomato loaf that used to crop up from time to time. I grabbed hold of it, sat down, and ate my breakfast with my pigs. That was **the** first act of what I later learned to call freeganism, really an exhibition of **the** injustice of food waste, and **the** provision of **the** solution to food waste, which is simply to sit down and eat food, rather than throwing it away. That became, as it were, a way of confronting large businesses in **the** business of wasting food, and exposing, most importantly, to **the** public, that when we ’ re talking about food being thrown away, we ’ re not talking about rotten stuff, we ’ re not talking about stuff that ’ s beyond **the** pale. We ’ re talking about good, fresh food that is being wasted on a colossal scale. Eventually, I set about writing my book, really to demonstrate **the** extent of this problem on a global scale. What this shows is a nation-by-nation breakdown of **the** likely level of food waste in each country in **the** world. Unfortunately, empirical data, good, hard stats, don ’ t exist, and therefore to prove my point, I first of all had to find some proxy way of uncovering how much food was being wasted. So I took **the** food supply of every single country and I compared it to what was actually likely to be being consumed in each country. That ’ s based on diet intake surveys, it ’ s based on levels of obesity, it ’ s based on a range of factors that gives you an approximate guess as to how much food is actually going into people ’ s mouths. That black line in **the** middle of that table is **the** likely level of **consumption** with an allowance for certain levels of inevitable waste. There will always be waste. I ’ m not that unrealistic that I think we can live in a waste-free world. But that black line shows what a food supply should be in a country if they allow for a good, stable, secure, nutritional diet for every person in that country. Any dot above that line, and you ’ ll quickly notice that that includes most countries in **the** world, represents unnecessary surplus, and is likely to reflect levels of waste in each country. As a country gets richer, it invests more and more in getting more and more surplus into its shops and restaurants, and as you can see, most European and North American countries fall between 150 and 200 percent of **the** nutritional requirements of their populations. So a country like America has twice as much food on its shop shelves and in its restaurants than is actually required to feed **the** American people. But **the** thing that really struck me, when I plotted all this data, and it was a lot of numbers, was that you can see how it levels off. Countries rapidly shoot towards that 150 mark, and then they level off, and they don ’ t really go on rising as you might expect. So I decided to unpack that data a little bit further to see if that was true or false. And that ’ s what I came up with. If you include not just **the** food that ends up in shops and restaurants, but also **the** food that people feed to livestock, **the** maize, **the** soy, **the** wheat, that humans could eat but choose to fatten livestock instead to produce increasing amounts of meat and dairy products, what you find is that most rich countries have between three and four times **the** amount of food that their population needs to feed itself. A country like America has four times **the** amount of food that it needs. When people talk about **the** need to increase global food production to feed those nine billion people that are expected on **the planet** by 2050, I always think of these graphs. **The** fact is, we have an enormous buffer in rich countries between ourselves and hunger. We ’ ve never had such gargantuan surpluses before. In many ways, this is a great success story of human civilization, of **the** agricultural surpluses that we set out to achieve 12, 000 years ago. It is a success story. It has been a success story. But what we have to recognize now is that we are reaching **the** ecological limits that our **planet** can bear, and when we chop down forests, as we are every day, to grow more and more food, when we extract water from depleting water reserves, when we emit **fossil** fuel **emissions** in **the** quest to grow more and more food, and then we throw away so much of it, we have to think about what we can start saving. And yesterday, I went to one of **the** local **supermarkets** that I often visit to inspect, if you like, what they ’ re throwing away. I found quite a few packets of biscuits amongst all **the** fruit and vegetables and everything else that was in there. And I thought, well this could serve as a symbol for today. So I want you to imagine that these nine biscuits that I found in **the** bin represent **the** global food supply, **okay**? We start out with nine. That ’ s what ’ s in fields around **the** world every single year. **The** first biscuit we ’ re going to lose before we even leave **the** farm. That ’ s a problem primarily associated with developing work **agriculture**, whether it ’ s a lack of infrastructure, refrigeration, pasteurization, grain stores, even basic fruit crates, which means that food goes to waste before it even leaves **the** fields. **The** next three biscuits are **the** foods that we decide to feed to livestock, **the** maize, **the** wheat and **the** soya. Unfortunately, our beasts are inefficient animals, and they turn **two-thirds** of that into feces and heat, so we ’ ve lost those two, and we ’ ve only kept this one in meat and dairy products. Two more we ’ re going to throw away directly into bins. This is what most of us think of when we think of food waste, what ends up in **the** garbage, what ends up in **supermarket** bins, what ends up in restaurant bins. We ’ ve lost another two, and we ’ ve left ourselves with just four biscuits to feed on. That is not a superlatively efficient use of global resources, especially when you think of **the** billion hungry people that exist already in **the** world. Having gone through **the** data, I then needed to demonstrate where that food ends up. Where does it end up? We ’ re used to seeing **the** stuff on our **plates**, but what about all **the** stuff that goes missing in between? **Supermarkets** are an easy place to start. This is **the** result of my hobby, which is unofficial bin inspections. Strange you might think, but if we could rely on corporations to tell us what they were doing in **the** back of their stores, we wouldn ’ t need to go sneaking around **the** back, opening up bins and having a look at what ’ s inside. But this is what you can see more or less on every street corner in Britain, in Europe, in North America. It represents a colossal waste of food, but what I discovered whilst I was writing my book was that this very evident abundance of waste was actually **the** tip of **the** iceberg. When you start going up **the** supply chain, you find where **the** real food waste is happening on a gargantuan scale. Can I have a show of hands if you have a loaf of sliced bread in your house? Who lives in a household where that crust — that slice at **the** first and last end of each loaf — who lives in a household where it does get eaten? **Okay**, most people, not everyone, but most people, and this is, I ’ m glad to say, what I see across **the** world, and yet has anyone seen a **supermarket** or sandwich shop anywhere in **the** world that serves sandwiches with crusts on it? I certainly haven ’ t. So I kept on thinking, where do those crusts go? This is **the** answer, unfortunately : 13, 000 slices of fresh bread coming out of this one single factory every single day, day-fresh bread. In **the** same year that I visited this factory, I went to Pakistan, where people in 2008 were going hungry as a result of a squeeze on global food supplies. We contribute to that squeeze by depositing food in bins here in Britain and elsewhere in **the** world. We take food off **the** market shelves that hungry people depend on. Go one step up, and you get to farmers, who throw away sometimes a third or even more of their harvest because of cosmetic standards. This farmer, for example, has invested 16, 000 **pounds** in growing spinach, not one leaf of which he harvested, because there was a little bit of grass growing in amongst it. Potatoes that are cosmetically imperfect, all going for pigs. Parsnips that are too small for **supermarket** specifications, tomatoes in Tenerife, oranges in Florida, bananas in Ecuador, where I visited last year, all being **discarded**. This is one day ’ s waste from one banana plantation in Ecuador. All being **discarded**, perfectly edible, because they ’ re **the** wrong shape or size. If we do that to fruit and vegetables, you bet we can do it to animals too. Liver, lungs, heads, tails, kidneys, testicles, all of these things which are traditional, delicious and nutritious parts of our gastronomy go to waste. Offal **consumption** has halved in Britain and America in **the** last 30 years. As a result, this stuff gets fed to dogs at best, or is incinerated. This man, in Kashgar, Xinjiang province, in Western China, is serving up his national dish. It ’ s called sheep ’ s organs. It ’ s delicious, it ’ s nutritious, and as I learned when I went to Kashgar, it symbolizes their taboo against food waste. I was sitting in a roadside cafe. A chef came to talk to me, I finished my bowl, and halfway through **the** conversation, he stopped talking and he started frowning into my bowl. I thought,”My goodness, what taboo have I broken? How have I insulted my host?”He pointed at three grains of rice at **the** bottom of my bowl, and he said,”Clean.”I thought,”My God, you know, I go around **the** world telling people to stop wasting food. This guy has thrashed me at my own game.”But it gave me faith. It gave me faith that we, **the** people, do have **the** power to stop this tragic waste of resources if we regard it as socially unacceptable to waste food on a colossal scale, if we make noise about it, tell corporations about it, tell governments we want to see an end to food waste, we do have **the** power to bring about that change. **Fish**, 40 to 60 percent of European **fish** are **discarded** at **sea**, they don ’ t even get landed. In our homes, we ’ ve lost touch with food. This is an experiment I did on three lettuces. Who keeps lettuces in their fridge? Most people. **The** one on **the left** was kept in a fridge for 10 days. **The** one in **the** middle, on my kitchen table. Not much difference. **The** one on **the** right I treated like cut flowers. It ’ s a living organism, cut **the** slice off, stuck it in a vase of water, it was all right for another two weeks after this. Some food waste, as I said at **the** beginning, will inevitably arise, so **the** question is, what is **the** best thing to do with it? I answered that question when I was 15. In fact, humans answered that question 6, 000 years ago : We domesticated pigs to turn food waste back into food. And yet, in Europe, that practice has become illegal since 2001 as a result of **the** foot-and-mouth outbreak. It ’ s unscientific. It ’ s unnecessary. If you cook food for pigs, just as if you cook food for humans, it is **rendered** safe. It ’ s also a massive saving of resources. At **the** moment, Europe depends on importing millions of tons of soy from South America, where its production contributes to global warming, to deforestation, to biodiversity loss, to feed livestock here in Europe. At **the** same time we throw away millions of tons of food waste which we could and should be feeding them. If we did that, and fed it to pigs, we would save that amount of **carbon**. If we feed our food waste which is **the** current government favorite way of getting rid of food waste, to anaerobic digestion, which turns food waste into gas to produce electricity, you save a paltry 448 kilograms of **carbon** dioxide per ton of food waste. It ’ s much better to feed it to pigs. We knew that during **the** war. A silver lining : It has kicked off globally, **the** quest to tackle food waste. Feeding **the** 5, 000 is an event I first organized in 2009. We fed 5, 000 people all on food that otherwise would have been wasted. Since then, it ’ s happened again in London, it ’ s happening internationally, and across **the** country. It ’ s a way of organizations coming together to celebrate food, to say **the** best thing to do with food is to eat and **enjoy** it, and to stop wasting it. For **the** sake of **the planet** we live on, for **the** sake of our children, for **the** sake of all **the** other organisms that share our **planet** with us, we are a terrestrial animal, and we depend on our land for food. At **the** moment, we are trashing our land to grow food that no one eats. Stop wasting food. Thank you very much.

Text Sample 38: POS Dim 1 – 2012 – Score 19.00 – 2012/t002318.txt

All right. So, like all good stories, this starts a long, long time ago when there was basically nothing. So here is a complete picture of **the universe** about

14-odd billion years ago. All energy is concentrated into a single point of energy. For some reason it explodes, and you begin to get these things. So you ’ re now about 14 billion years into this. And these things expand and expand and expand into these giant galaxies, and you get trillions of them. And within these galaxies you get these enormous dust clouds. And I want you to pay particular attention to **the** three little prongs in **the** center of this picture. If you take a close-up of those, they look like this. And what you ’ re looking at is columns of dust where there ’ s so much dust — by **the** way, **the** scale of this is a trillion vertical miles — and what ’ s happening is there ’ s so much dust, it comes together and it fuses and ignites a thermonuclear reaction. And so what you ’ re watching is **the** birth of stars. These are stars being born out of here. When enough stars come out, they create a galaxy. This one happens to be a particularly important galaxy, because you are here. And as you take a close-up of this galaxy, you find a relatively normal, not particularly interesting star. By **the** way, you ’ re now about **two-thirds** of **the** way into this story. So this star doesn ’ t even appear until about **two-thirds** of **the** way into this story. And then what happens is there ’ s enough dust left over that it doesn ’ t ignite into a star, it becomes a **planet**. And this is about a little over four billion years ago. And soon thereafter there ’ s enough material left over that you get a primordial **soup**, and that creates life. And life starts to expand and expand and expand, until it goes kaput. Now **the** really strange thing is life goes kaput, not once, not twice, but five times. So almost all life on Earth is **wiped** out about five times. And as you ’ re thinking about that, what happens is you get more and more complexity, more and more stuff to build new things with. And we don ’ t appear until about 99. 96 percent of **the** time into this story, just to put ourselves and our ancestors in perspective. So within that context, there ’ s two theories of **the** case as to why we ’ re all here. **The** first theory of **the** case is that ’ s all she wrote. Under that theory, we are **the** be-all and end-all of all creation. And **the** reason for trillions of galaxies, sextillions of **planets**, is to create something that looks like that and something that looks like that. And that ’ s **the** purpose of **the universe** ; and then it flat-lines, it doesn ’ t get any better. **The** only question you might want to ask yourself is, could that be just mildly arrogant? And if it is — and particularly given **the** fact that we came very close to extinction. There were only about 2, 000 of our species left. A few more weeks without rain, we would have never seen any of these. So maybe you have to think about a second theory if **the** first one isn ’ t good enough. Second theory is : Could we upgrade? Well, why would one ask a question like that? Because there have been at least 29 upgrades so far of humanoids. So it turns out that we have upgraded. We ’ ve upgraded time and again and again. And it turns out that we keep discovering upgrades. We found this one last year. We found another one last month. And as you ’ re thinking about this, you might also ask **the** question : So why a single human species? Wouldn ’ t it be really odd if you went to Africa and Asia and Antarctica and found exactly **the** same bird — particularly given that we co-existed at **the** same time with at least eight other versions of humanoid at **the** same time on this **planet**? So **the** normal state of affairs is not to have just a Homo sapiens ; **the** normal state of affairs is to have various versions of humans walking around. And if that is **the** normal state of affairs, then you might ask yourself, all right, so if we want to create something else, how big does a mutation have to be? Well Svante Paabo has **the** answer. **The** difference between humans and Neanderthal is 0. 004 percent of gene code. That ’ s how big **the** difference is one species to another. This explains most contemporary political debates. But as you ’ re thinking about this, one of **the** interesting things is how small these mutations are and where they take place. Difference human / Neanderthal is sperm and testis, smell and skin. And those are **the** specific genes that differ from one to **the** other. So very small changes can have a big impact. And as you ' re thinking about this, we ' re continuing to mutate. So about 10, 000 years ago by **the** Black Sea, we had one mutation in one gene which led to **blue** eyes. And this is continuing and continuing and continuing. And as it continues, one of **the** things that ' s going to happen this year is we ' re going to discover **the** first 10, 000 human **genomes**, because it ' s gotten cheap enough to do **the** gene sequencing. And when we find these, we may find differences. And by **the** way, this is not a debate that we ' re ready for, because we have really misused **the** science in this. In **the** 1920s, we thought there were major differences between people. That was partly based on Francis Galton ' s work. He was Darwin ' s cousin. But **the** U. S., **the** Carnegie Institute, Stanford, American Neurological Association took this really far. That got exported and was really misused. In fact, it led to some absolutely horrendous treatment of human beings. So since **the** 1940s, we ' ve been saying there are no differences, we ' re all **identical**. We ' re going to know at year end if that is true. And as we think about that, we ' re actually beginning to find things like, do you have an ACE gene? Why would that matter? Because nobody ' s ever climbed an 8, 000-meter peak without oxygen that doesn ' t have an ACE gene. And if you want to get more specific, how about a 577R genotype? Well it turns out that every male Olympic power athelete ever tested carries at least one of these variants. If that is true, it leads to some very complicated questions for **the** London Olympics. Three options : Do you want **the** Olympics to be a showcase for really hardworking mutants? Option number two : Why don ' t we play it like **golf** or sailing? Because you have one and you don ' t have one, I ' ll give you a tenth of a second head start. Version number three : Because this is a naturally **occurring** gene and you ' ve got it and you didn ' t pick **the** right parents, you get **the** right to upgrade. Three different options. If these differences are **the** difference between an Olympic medal and a non-Olympic medal. And it turns out that as we discover these things, we human beings really like to change how we look, how we act, what our bodies do. And we had about 10. 2 million plastic surgeries in **the** United States, except that with **the** technologies that are coming online today, today ' s corrections, deletions, augmentations and enhancements are going to seem like child ' s play. You already saw **the** work by Tony Atala on TED, but this ability to start filling things like inkjet cartridges with cells are allowing us to print skin, organs and a whole series of other body parts. And as these technologies go forward, you keep seeing this, you keep seeing this, you keep seeing things — 2000, human **genome** sequence — and it seems like nothing ' s happening, until it does. And we may just be in some of these weeks. And as you ' re thinking about these two guys sequencing a human **genome** in 2000 and **the** Public Project sequencing **the** human **genome** in 2000, then you don ' t hear a lot, until you hear about an experiment last year in China, where they take skin cells from this mouse, put four chemicals on it, turn those skin cells into stem cells, let **the** stem cells grow and create a full **copy** of that mouse. That ' s a big deal. Because in essence what it means is you can take a cell, which is a pluripotent stem cell, which is like a skier at **the** top of a mountain, and those two skiers become two pluripotent stem cells, four, eight, 16, and then it gets so crowded after 16 divisions that those cells have to differentiate. So they go down one side of **the** mountain, they go down another. And as they pick that, these become bone, and then they pick another road and these become platelets, and these become macrophages, and these become T cells. But it ' s really hard, once you ski down, to get back up. Unless, of course, if you have a ski lift. And what those four chemicals do is they take any cell and take it way back up **the** mountain so it can become any body part. And as you think of that, what it means is potentially you can rebuild a full **copy** of any organism out of any one of its cells. That turns out to be a big deal because now you can take, not just mouse cells, but you can human skin cells and turn them into human stem cells. And then what they did in October is they took skin cells, turned them into stem cells and began to turn them into liver cells. So in theory, you could grow any organ from any one of your cells. Here ' s a second experiment : If you could photocopy your body, maybe you also want to take your mind. And one of **the** things you saw at TED about a year and a half ago was this guy. And he gave a wonderful technical talk. He ' s a professor at MIT. But in essence what he said is you can take retroviruses, which get inside brain cells of mice. You can **tag** them with proteins that light up when you light them. And you can map **the exact** pathways when a mouse sees, feels, touches, remembers, loves. And then you can take a fiber optic cable and light up some of **the** same things. And by **the** way, as you do this, you can image it in two colors, which means you can download this information as binary code directly into a **computer**. So what ' s **the** bottom line on that? Well it ' s not completely inconceivable that someday you ' ll be able to download your own memories, maybe into a new body. And maybe you can upload other people ' s memories as well. And this might have just one or two small ethical, political, moral implications. Just a thought. Here ' s **the** kind of questions that are becoming interesting questions for philosophers, for governing people, for economists, for scientists. Because these technologies are moving really quickly. And as you think about it, let me close with an example of **the** brain. **The** first place where you would expect to see enormous **evolutionary** pressure today, both because of **the** inputs, which are becoming massive, and because of **the** plasticity of **the** organ, is **the** brain. Do we have any evidence that that is happening? Well let ' s take a look at something like autism incidence per thousand. Here ' s what it looks like in 2000. Here ' s what it looks like in 2002, 2006, 2008. Here ' s **the** increase in less than a decade. And we still don ' t know why this is happening. What we do know is, potentially, **the** brain is reacting in a hyperactive, hyper-plastic way, and creating individuals that are like this. And this is only one of **the** conditions that ' s out there. You ' ve also got people with who are extraordinarily smart, people who can remember everything they ' ve seen in their lives, people who ' ve got synesthesia, people who ' ve got schizophrenia. You ' ve got all kinds of stuff going on out there, and we still don ' t understand how and why this is happening. But one question you might want to ask is, are we seeing a rapid **evolution** of **the** brain and of how we process data? Because when you think of how much data ' s coming into our brains, we ' re trying to take in as much data in a day as people used to take in in a lifetime. And as you ' re thinking about this, there ' s four theories as to why this might be going on, plus a whole series of others. I don ' t have a good answer. There really needs to be more research on this. One option is **the** fast food fetish. There ' s beginning to be some evidence that obesity and diet have something to do with gene modifications, which may or may not have an impact on how **the** brain of an infant works. A second option is **the** sexy **geek** option. These conditions are highly rare. But what ' s beginning to happen is because these **geeks** are all getting together, because they are highly qualified for **computer** programming and it is highly remunerated, as well as other very detail-oriented tasks, that they are concentrating geographically and finding like-minded **mates**. So this is **the** assortative mating hypothesis of these genes reinforcing one another in these structures. **The** third, is this too much information? We ' re trying to process so much stuff that some people get synesthetic and just have huge pipes that remember everything. Other people get hyper-sensitive to **the** amount of information. Other people react with various psychological conditions or reactions to this information. Or maybe it ’ s chemicals. But when you see an increase of that order of magnitude in a condition, either you ’ re not measuring it right or there ’ s something going on very quickly, and it may be **evolution** in real time. Here ’ s **the** bottom line. What I think we are doing is we ’ re transitioning as a species. And I didn ’ t think this when Steve Gullans and I started writing together. I think we ’ re transitioning into Homo evolutis that, for better or worse, is not just a hominid that ’ s conscious of his or her environment, it ’ s a hominid that ’ s beginning to directly and deliberately control **the evolution** of its own species, of bacteria, of plants, of animals. And I think that ’ s such an order of magnitude change that your grandkids or your great-grandkids may be a species very different from you. Thank you very much.

Text Sample 39: POS Dim 1 – 2015 – Score 23.00 – 2015/t001481.txt
So I ’ ve had **the** great privilege of traveling to some incredible places, photographing these distant landscapes and remote cultures all over **the** world. I love my job. But people think it ’ s this string of epiphanies and sunrises and rainbows, when in reality, it looks more something like this. This is my office. We can ’ t afford **the** fanciest places to stay at night, so we tend to sleep a lot outdoors. As long as we can stay dry, that ’ s a bonus. We also can ’ t afford **the** fanciest restaurants. So we tend to eat whatever ’ s on **the** local menu. And if you ’ re in **the** Ecuadorian Pramo, you ’ re going to eat a large rodent called a cuy. But what makes our experiences perhaps a little bit different and a little more unique than that of **the** average person is that we have this gnawing thing in **the** back of our mind that even in our darkest moments, and those times of despair, we think,”Hey, there might be an image to be made here, there might be a story to be told.”And why is storytelling important? Well, it helps us to connect with our cultural and our natural heritage. And in **the** Southeast, there ’ s an alarming disconnect between **the** public and **the** natural areas that allow us to be here in **the** first place. We ’ re visual creatures, so we use what we see to teach us what we know. Now **the** majority of us aren ’ t going to willingly go way down to a swamp. So how can we still expect those same people to then advocate on behalf of their protection? We can ’ t. So my job, then, is to use photography as a communication tool, to help bridge **the** gap between **the** science and **the** aesthetics, to get people talking, to get them thinking, and to hopefully, ultimately, get them caring. I started doing this 15 years ago right here in Gainesville, right here in my backyard. And I fell in love with adventure and discovery, going to explore all these different places that were just minutes from my front doorstep. There are a lot of beautiful places to find. Despite all these years that have passed, I still see **the** world through **the** eyes of a child and I try to incorporate that sense of wonderment and that sense of curiosity into my photography as often as I can. And we ’ re pretty lucky because here in **the** South, we ’ re still blessed with a relatively blank canvas that we can fill with **the** most fanciful adventures and incredible experiences. It ’ s just a matter of how far our imagination will take us. See, a lot of people look at this and they say,”Oh yeah, wow, that ’ s a pretty tree.”But I don ’ t just see a tree — I look at this and I see opportunity. I see an entire weekend. Because when I was a kid, these were **the** types of images that got me off **the** sofa and dared me to explore, dared me to go find **the** woods and put my head **underwater** and see what we have. And folks, I ’ ve been photographing all over **the** world and I promise you, what we have here in **the** South, what we have in **the** Sunshine State, rivals anything else that I ’ ve seen. But yet our tourism industry is busy promoting all **the** wrong things. Before most kids are 12, they ’ ll have

been to Disney World more times than they ’ ve been in a canoe or camping under a starry sky. And I have nothing against Disney or Mickey ; I used to go there, too. But they ’ re missing out on those fundamental connections that create a real sense of pride and ownership for **the** place that they call home. And this is compounded by **the** issue that **the** landscapes that define our natural heritage and fuel our aquifer for our drinking water have been deemed as scary and dangerous and spooky. When our ancestors first came here, they warned,”Stay out of these areas, they ’ re haunted. They ’ re full of evil spirits and ghosts.”I don ’ t know where they came up with that idea. But it ’ s actually led to a very real disconnect, a very real negative mentality that has kept **the** public disinterested, silent, and ultimately, our environment at risk. We ’ re a state that ’ s surrounded and defined by water, and yet for centuries, swamps and wetlands have been regarded as these obstacles to overcome. And so we ’ ve treated them as these second-class ecosystems, because they have very little monetary value and of course, they ’ re known to harbor alligators and snakes — which, I ’ ll admit, these aren ’ t **the** most cuddly of ambassadors. So it became assumed, then, that **the** only good swamp was a drained swamp. And in fact, draining a swamp to make way for **agriculture** and development was considered **the** very essence of conservation not too long ago. But now we ’ re backpedaling, because **the** more we come to learn about these sodden landscapes, **the** more secrets we ’ re starting to **unlock** about interspecies relationships and **the** connectivity of **habitats**, watersheds and flyways. Take this bird, for example : this is **the** prothonotary warbler. I love this bird because it ’ s a swamp bird, through and through, a swamp bird. They nest and they **mate** and they breed in these old-growth swamps in these flooded forests. And so after **the spring**, after they raise their young, they then fly thousand of miles over **the** Gulf of Mexico into Central and South America. And then after **the** winter, **the spring** rolls around and they come back. They fly thousands of miles over **the** Gulf of Mexico. And where do they go? Where do they land? Right back in **the** same tree. That ’ s **nuts**. This is a bird **the** size of a tennis ball — I mean, that ’ s crazy! I used a GPS to get here today, and this is my hometown. It ’ s crazy. So what happens, then, when this bird flies over **the** Gulf of Mexico into Central America for **the** winter and then **the spring** rolls around and it flies back, and it comes back to this : a freshly sodded **golf** course? This is a narrative that ’ s all too commonly unraveling here in this state. And this is a natural process that ’ s **occurred** for thousands of years and we ’ re just now learning about it. So you can imagine all else we have to learn about these landscapes if we just preserve them first. Now despite all this rich life that abounds in these swamps, they still have a bad name. Many people feel uncomfortable with **the** idea of wading into Florida ’ s blackwater. I can understand that. But what I loved about growing up in **the** Sunshine State is that for so many of us, we live with this latent but very palpable fear that when we put our toes into **the** water, there might be something much more ancient and much more adapted than we are. Knowing that you ’ re not top dog is a welcomed discomfort, I think. How often in this modern and urban and digital age do you actually get **the** chance to feel vulnerable, or consider that **the** world may not have been made for just us? So for **the** last decade, I began seeking out these areas where **the** concrete yields to forest and **the** pines turn to cypress, and I viewed all these mosquitoes and reptiles, all these discomforts, as affirmations that I ’ d found true wilderness, and I embrace them wholly. Now as a conservation photographer obsessed with blackwater, it ’ s only fitting that I ’ d eventually end up in **the** most **famous** swamp of all : **the** Everglades. Growing up here in North Central Florida, it always had these enchanted names, places like Loxahatchee and Fakahatchee, Corkscrew, Big

Cypress. I started what turned into a five-year project to hopefully reintroduce **the** Everglades in a new light, in a more inspired light. But I knew this would be a tall order, because here you have an area that ' s roughly a third **the** size **the** state of Florida, it ' s huge. And when I say Everglades, most people are like,"Oh, yeah, **the** national park."But **the** Everglades is not just a park ; it ' s an entire watershed, starting with **the** Kissimmee chain of lakes in **the** north, and then as **the** rains would fall in **the** summer, these downpours would flow into Lake Okeechobee, and Lake Okeechobee would fill up and it would overflow its banks and **spill** southward, ever slowly, with **the** topography, and get into **the** river of grass, **the** Sawgrass Prairies, before meting into **the** cypress slews, until going further south into **the** mangrove swamps, and then finally — finally — reaching Florida Bay, **the** emerald gem of **the** Everglades, **the** great estuary, **the** 850 square-mile estuary. So sure, **the** national park is **the** southern end of this system, but all **the** things that make it unique are these inputs that come in, **the** fresh water that starts 100 miles north. So no manner of these political or invisible boundaries protect **the** park from polluted water or insufficient water. And unfortunately, that ' s precisely what we ' ve done. Over **the** last 60 years, we have drained, we have dammed, we have dredged **the** Everglades to where now only one third of **the** water that used to reach **the bay** now reaches **the bay** today. So this story is not all sunshine and rainbows, unfortunately. For better or for worse, **the** story of **the** Everglades is intrinsically tied to **the** peaks and **the** valleys of mankind ' s relationship with **the** natural world. But I ' ll show you these beautiful pictures, because it gets you on board. And while I have your attention, I can tell you **the** real story. It ' s that we ' re taking this, and we ' re trading it for this, at an alarming rate. And what ' s lost on so many people is **the sheer** scale of which we ' re discussing. Because **the** Everglades is not just responsible for **the** drinking water for 7 million Floridians ; today it also provides **the** agricultural fields for **the** year-round tomatoes and oranges for over 300 million Americans. And it ' s that same seasonal **pulse** of water in **the** summer that built **the** river of grass 6, 000 years ago. Ironically, today, it ' s also responsible for **the** over half a million acres of **the** endless river of sugarcane. These are **the** same fields that are responsible for dumping exceedingly high levels of fertilizers into **the** watershed, forever changing **the** system. But in order for you to not just understand how this system works, but to also get personally connected to it, I decided to break **the** story down into several different narratives. And I wanted that story to start in Lake Okeechobee, **the** beating heart of **the** Everglade system. And to do that, I picked an ambassador, an iconic species. This is **the** Everglade snail kite. It ' s a great bird, and they used to nest in **the** thousands, thousands in **the** northern Everglades. And then they ' ve gone down to about 400 nesting pairs today. And why is that? Well, it ' s because they eat one source of food, an apple snail, about **the** size of a ping-pong ball, an aquatic gastropod. So as we started damming up **the** Everglades, as we started diking Lake Okeechobee and draining **the** wetlands, we lost **the habitat** for **the** snail. And thus, **the** population of **the** kites declined. And so, I wanted a photo that would not only communicate this relationship between wetland, snail and bird, but I also wanted a photo that would communicate how incredible this relationship was, and how very important it is that they ' ve come to depend on each other, this healthy wetland and this bird. And to do that, I brainstormed this idea. I started sketching out these plans to make a photo, and I sent it to **the** wildlife biologist down in Okeechobee — this is an endangered bird, so it takes special permission to do. So I built this submerged platform that would hold snails just right under **the** water. And I spent months planning this crazy idea. And I took this platform down to Lake Okeechobee and I spent over a week in **the** water, wading waist-deep, 9-hour shifts from dawn until dusk, to get one image that I thought might communicate this. And here ’ s **the** day that it finally worked : [Video : After setting up **the** platform, I look off and I see a kite coming over **the** cattails. And I see him scanning and **searching**. And he gets right over **the** trap, and I see that he ’ s seen it. And he beelines, he goes straight for **the** trap. And in that moment, all those months of planning, waiting, all **the** sunburn, mosquito **bites** — suddenly, they ’ re all worth it. Oh my gosh, I can ’ t believe it!] You can believe how excited I was when that happened. But what **the** idea was, is that for someone who ’ s never seen this bird and has no reason to care about it, these photos, these new perspectives, will help shed a little new light on just one species that makes this watershed so incredible, so valuable, so important. Now, I know I can ’ t come here to Gainesville and talk to you about animals in **the** Everglades without talking about gators. I love gators, I grew up loving gators. My parents always said I had an unhealthy relationship with gators. But what I like about them is, they ’ re like **the** freshwater equivalent of sharks. They ’ re feared, they ’ re hated, and they are tragically misunderstood. Because these are a unique species, they ’ re not just apex predators. In **the** Everglades, they are **the** very architects of **the** Everglades, because as **the** water drops down in **the** winter during **the** dry season, they start excavating these holes called gator holes. And they do this because as **the** water drops down, they ’ ll be able to stay **wet** and they ’ ll be able to forage. And now this isn ’ t just affecting them, other animals also depend on this relationship, so they become a keystone species as well. So how do you make an apex predator, an ancient reptile, at once look like it dominates **the** system, but at **the** same time, look vulnerable? Well, you wade into a pit of about 120 of them, then you hope that you ’ ve made **the** right decision. I still have all my fingers, it ’ s cool. But I understand, I know I ’ m not going to rally you guys, I ’ m not going to rally **the** troops to”Save **the** Everglades for **the** gators!”It won ’ t happen because they ’ re so ubiquitous, we see them now, they ’ re one of **the** great conservation success stories of **the** US. But there is one species in **the** Everglades that no matter who you are, you can ’ t help but love, too, and that ’ s **the** roseate spoonbill. These birds are great, but they ’ ve had a really tough time in **the** Everglades, because they started out with thousands of nesting pairs in Florida Bay, and at **the** turn of **the** 20th century, they got down to two — two nesting pairs. And why? That ’ s because women thought they looked better on their hats then they did flying in **the** sky. Then we banned **the** plume trade, and their numbers started rebounding. And as their numbers started rebounding, scientists began to pay attention, they started studying these birds. And what they found out is that these birds ’ behavior is intrinsically tied to **the** annual draw-down cycle of water in **the** Everglades, **the** thing that defines **the** Everglades watershed. What they found out is that these birds started nesting in **the** winter as **the** water drew down, because they ’ re tactile feeders, so they have to touch whatever they eat. And so they wait for these concentrated pools of **fish** to be able to feed enough to feed their young. So these birds became **the** very icon of **the** Everglades — an indicator species of **the** overall health of **the** system. And just as their numbers were rebounding in **the** mid-20th century — shooting up to 900, 1, 000, 1, 100, 1, 200 — just as that started happening, we started draining **the** southern Everglades. And we stopped **two-thirds** of that water from moving south. And it had drastic consequences. And just as those numbers started reaching their peak, unfortunately, today, **the** real spoonbill story, **the** real photo of what it looks like is more something like this. And we ’ re down to less than 70 nesting pairs in Florida Bay today, because we ’ ve disrupted **the** system so much. So all these different organizations are shouting, they ’ re screaming,”**The** Everglades is fragile! It ’ s fragile!”It is not. It is resilient.

Because despite all we ’ ve taken, despite all we ’ ve done and we ’ ve drained and we ’ ve dammed and we ’ ve dredged it, pieces of it are still here, waiting to be put back together. And this is what I ’ ve loved about South Florida, that in one place, you have this unstoppable force of mankind meeting **the** immovable **object** of tropical nature. And it ’ s at this new frontier that we are forced with a new appraisal. What is wilderness worth? What is **the** value of biodiversity, or our drinking water? And fortunately, after decades of debate, we ’ re finally starting to act on those questions. We ’ re slowly **undertaking** these projects to bring more freshwater back to **the bay**. But it ’ s up to us as citizens, as residents, as stewards to hold our elected officials to their promises. What can you do to help? It ’ s so easy. Just get outside, get out there. Take your friends out, take your kids out, take your family out. Hire a **fishing** guide. Show **the** state that protecting wilderness not only makes ecological sense, but economic sense as well. It ’ s a lot of fun, just do it — put your feet in **the** water. **The** swamp will change you, I promise. Over **the** years, we ’ ve been so generous with these other landscapes around **the** country, cloaking them with this American pride, places that we now consider to define us : Grand Canyon, Yosemite, Yellowstone. And we use these parks and these natural areas as beacons and as cultural compasses. And sadly, **the** Everglades is very commonly left out of that conversation. But I believe it ’ s every bit as iconic and emblematic of who we are as a country as any of these other wildernesses. It ’ s just a different kind of wild. But I ’ m encouraged, because maybe we ’ re finally starting to come around, because what was once deemed this swampy wasteland, today is a World Heritage site. It ’ s a wetland of international importance. And we ’ ve come a long way in **the** last 60 years. And as **the** world ’ s largest and most ambitious wetland restoration project, **the** international spotlight is on us in **the** Sunshine State. Because if we can heal this system, it ’ s going to become an icon for wetland restoration all over **the** world. But it ’ s up to us to decide which legacy we want to attach our flag to. They say that **the** Everglades is our greatest test. If we pass it, we get to keep **the planet**. I love that quote, because it ’ s a challenge, it ’ s a prod. Can we do it? Will we do it? We have to, we must. But **the** Everglades is not just a test. It ’ s also a gift, and ultimately, our responsibility. Thank you.

Text Sample 40: POS Dim 1 – 2015 – Score 22.00 – 2015/t001608.txt

Well, you know, sometimes **the** most important things come in **the** smallest packages. I am going to try to convince you, in **the** 15 minutes I have, that microbes have a lot to say about questions such as,”Are we alone?”and they can tell us more about not only life in our solar system but also maybe beyond, and this is why I am tracking them down in **the** most impossible places on Earth, in extreme environments where conditions are really pushing them to **the** brink of survival. Actually, sometimes me too, when I ’ m trying to follow them too close. But here ’ s **the** thing : We are **the** only advanced civilization in **the** solar system, but that doesn ’ t mean that there is no microbial life nearby. In fact, **the planets** and moons you see here could host life — all of them — and we know that, and it ’ s a strong possibility. And if we were going to find life on those moons and **planets**, then we would answer questions such as, are we alone in **the** solar system? Where are we coming from? Do we have family in **the** neighborhood? Is there life beyond our solar system? And we can ask all those questions because there has been a revolution in our understanding of what a habitable **planet** is, and today, a habitable **planet** is a **planet** that has a zone where water can stay stable, but to me this is a **horizontal** definition of habitability, because it involves a distance to a star, but there is another **dimension** to habitability, and this is a vertical **dimension**. Think of it as conditions in **the** subsurface of a **planet** where you are very far away from a sun, but you still have water, energy, nutrients, which for some of them means food, and a protection. And when you look at **the** Earth, very far away from any sunlight, deep in **the** ocean, you have life thriving and it uses only **chemistry** for life processes. So when you think of it at that point, all walls collapse. You have no limitations, basically. And if you have been looking at **the** headlines lately, then you will see that we have discovered a subsurface ocean on Europa, on Ganymede, on Enceladus, on Titan, and now we are finding a geyser and hot **springs** on Enceladus, Our solar system is turning into a giant spa. For anybody who has gone to a spa knows how much microbes like that, right? So at that point, think also about Mars. There is no life possible at **the surface** of Mars today, but it might still be hiding **underground**. So, we have been making progress in our understanding of habitability, but we also have been making progress in our understanding of what **the** signatures of life are on Earth. And you can have what we call **organic** molecules, and these are **the** bricks of life, and you can have **fossils**, and you can minerals, biominerals, which is due to **the** reaction between bacteria and **rocks**, and of course you can have gases in **the** atmosphere. And when you look at those tiny green algae on **the** right of **the slide** here, they are **the** direct descendants of those who have been pumping oxygen a billion years ago in **the** atmosphere of **the** Earth. When they did that, they poisoned 90 percent of **the** life at **the surface** of **the** Earth, but they are **the** reason why you are breathing this air today. But as much as our understanding grows of all of these things, there is one question we still cannot answer, and this is, where are we coming from? And you know, it ’ s getting worse, because we won ’ t be able to find **the** physical evidence of where we are coming from on this **planet**, and **the** reason being is that anything that is older than four billion years is gone. All record is gone, erased by **plate** tectonics and erosion. This is what I call **the** Earth ’ s biological horizon. Beyond this horizon we don ’ t know where we are coming from. So is everything lost? Well, maybe not. And we might be able to find evidence of our own origin in **the** most unlikely place, and this place in Mars. How is this possible? Well clearly at **the** beginning of **the** solar system, Mars and **the** Earth were bombarded by giant **asteroids** and comets, and there were ejecta from these impacts all over **the** place. Earth and Mars kept throwing **rocks** at each other for a very long time. Pieces of **rocks** landed on **the** Earth. Pieces of **the** Earth landed on Mars. So clearly, those two **planets** may have been seeded by **the** same material. So yeah, maybe Granddady is sitting there on **the surface** and waiting for us. But that also means that we can go to Mars and try to find traces of our own origin. Mars may hold that secret for us. This is why Mars is so special to us. But for that to happen, Mars needed to be habitable at **the** time when conditions were right. So was Mars habitable? We have a number of missions telling us exactly **the** same thing today. At **the** time when life appeared on **the** Earth, Mars did have an ocean, it had volcanoes, it had lakes, and it had deltas like **the** beautiful picture you see here. This picture was sent by **the** Curiosity rover only a few weeks ago. It shows **the** remnants of a delta, and this picture tells us something : water was abundant and stayed founting at **the surface** for a very long time. This is good news for life. Life **chemistry** takes a long time to actually happen. So this is extremely good news, but does that mean that if we go there, life will be easy to find on Mars? Not necessarily. Here ’ s what happened : At **the** time when life exploded at **the surface** of **the** Earth, then everything went south for Mars, literally. **The** atmosphere was stripped away by solar winds, Mars lost its magnetosphere, and then cosmic rays and U. V. bombarded **the surface** and water escaped to space and went **underground**. So if we want to be able to understand, if we want to be able to find those traces of **the** signatures of life at **the surface** of Mars, if they are there, we need to understand what was **the** impact of each of these events on **the** preservation of its record. Only then will we be able to know where those signatures are hiding, and only then will we be able to send our rover to **the** right places where we can sample those **rocks** that may be telling us something really important about who we are, or, if not, maybe telling us that somewhere, independently, life has appeared on another **planet**. So to do that, it ' s easy. You only need to go back 3. 5 billion years ago in **the** past of a **planet**. We just need a time machine. Easy, right? Well, actually, it is. Look around you — that ' s **planet** Earth. This is our time machine. Geologists are using it to go back in **the** past of our own **planet**. I am using it a little bit differently. I use **planet** Earth to go in very extreme environments where conditions were similar to those of Mars at **the** time when **the** climate changed, and there I ' m trying to understand what happened. What are **the** signatures of life? What is left? How are we going to find it? So for one moment now I ' m going to take you with me on a trip into that time machine. And now, what you see here, we are at 4, 500 meters in **the** Andes, but in fact we are less than a billion years after **the** Earth and Mars formed. **The** Earth and Mars will have looked pretty much exactly like that — volcanoes everywhere, evaporating lakes everywhere, minerals, hot **springs**, and then you see those mounds on **the** shore of those lakes? Those are built by **the** descendants of **the** first organisms that gave us **the** first **fossil** on Earth. But if we want to understand what ' s going on, we need to go a little further. And **the** other thing about those sites is that exactly like on Mars three and a half billion years ago, **the** climate is changing very fast, and water and ice are **disappearing**. But we need to go back to that time when everything changed on Mars, and to do that, we need to go higher. Why is that? Because when you go higher, **the** atmosphere is getting thinner, it ' s getting more unstable, **the** temperature is getting cooler, and you have a lot more U. V. radiation. Basically, you are getting to those conditions on Mars when everything changed. So I was not promising anything about a leisurely trip on **the** time machine. You are not going to be sitting in that time machine. You have to haul 1, 000 **pounds** of equipment to **the** summit of this 20, 000-foot volcano in **the** Andes here. That ' s about 6, 000 meters. And you also have to sleep on 42-degree **slopes** and really hope that there won ' t be any **earthquake** that night. But when we get to **the** summit, we actually find **the** lake we came for. At this **altitude**, this lake is experiencing exactly **the** same conditions as those on Mars three and a half billion years ago. And now we have to change our voyage into an inner voyage inside that lake, and to do that, we have to remove our mountain gear and actually don suits and go for it. But at **the** time we enter that lake, at **the** very moment we enter that lake, we are stepping back three and a half billion years in **the** past of another **planet**, and then we are going to get **the** answer came for. Life is everywhere, absolutely everywhere. Everything you see in this picture is a living organism. Maybe not so **the** diver, but everything else. But this picture is very deceiving. Life is abundant in those lakes, but like in many places on Earth right now and due to climate change, there is a huge loss in biodiversity. In **the** samples that we took back home, 36 percent of **the** bacteria in those lakes were composed of three species, and those three species are **the** ones that have survived so far. Here ' s another lake, right next to **the** first one. **The** red color you see here is not due to minerals. It ' s actually due to **the** presence of a tiny algae. In this region, **the** U. V. radiation is really nasty. Anywhere on Earth, 11 is considered to be extreme. During U. V. storms there, **the** U. V. Index reaches 43. SPF 30 is not going to do anything to you over there, and **the** water is so transparent in those lakes that **the** algae has nowhere to hide, really, and so they are developing their own sunscreen, and this is **the** red color you see. But they can adapt only so far, and then when all **the** water is gone from **the surface**, microbes have only one solution **left** : They go **underground**. And those microbes, **the rocks** you see in that **slide** here, well, they are actually living inside **rocks** and they are using **the** protection of **the** translucence of **the rocks** to get **the** good part of **the** U. V. and **discard the** part that could actually damage their DNA. And this is why we are taking our rover to train them to **search** for life on Mars in these areas, because if there was life on Mars three and a half billion years ago, it had to use **the** same strategy to actually protect itself. Now, it is pretty obvious that going to extreme environments is helping us very much for **the** exploration of Mars and to prepare missions. So far, it has helped us to understand **the** geology of Mars. It has helped to understand **the** past climate of Mars and its **evolution**, but also its habitability potential. Our most recent rover on Mars has discovered traces of **organics**. Yeah, there are **organics** at **the surface** of Mars. And it also discovered traces of methane. And we don ’ t know yet if **the** methane in question is really from geology or **biology**. Regardless, what we know is that because of **the** discovery, **the** hypothesis that there is still life present on Mars today remains a viable one. So by now, I think I have convinced you that Mars is very special to us, but it would be a mistake to think that Mars is **the** only place in **the** solar system that is interesting to find potential microbial life. And **the** reason is because Mars and **the** Earth could have a common root to their tree of life, but when you go beyond Mars, it ’ s not that easy. Celestial mechanics is not making it so easy for an exchange of material between **planets**, and so if we were to discover life on those **planets**, it would be different from us. It would be a different type of life. But in **the** end, it might be just us, it might be us and Mars, or it can be many trees of life in **the** solar system. I don ’ t know **the** answer yet, but I can tell you something : No matter what **the** result is, no matter what that magic number is, it is going to give us a standard by which we are going to be able to measure **the** life potential, abundance and diversity beyond our own solar system. And this can be achieved by our generation. This can be our legacy, but only if we dare to explore. Now, finally, if somebody tells you that looking for alien microbes is not cool because you cannot have a philosophical conversation with them, let me show you why and how you can tell them they ’ re wrong. Well, **organic** material is going to tell you about environment, about complexity and about diversity. DNA, or any information carrier, is going to tell you about adaptation, about **evolution**, about survival, about planetary changes and about **the** transfer of information. All together, they are telling us what started as a microbial pathway, and why what started as a microbial pathway sometimes ends up as a civilization or sometimes ends up as a dead end. Look at **the** solar system, and look at **the** Earth. On Earth, there are many intelligent species, but only one has achieved technology. Right here in **the** journey of our own solar system, there is a very, very powerful message that says here ’ s how we should look for alien life, small and big. So yeah, microbes are talking and we are listening, and they are taking us, one **planet** at a time and one moon at a time, towards their big brothers out there. And they are telling us about diversity, they are telling us about abundance of life, and they are telling us how this life has survived thus far to reach civilization, intelligence, technology and, indeed, philosophy. Thank you.

Text Sample 41: POS Dim 1 – 2015 – Score 20.00 – 2015/t001601.txt

I ’ d like to take you on **the** epic quest of **the** Rosetta spacecraft. To escort and land **the** probe on a comet, this has been my passion for **the** past two years. In order to do that, I need to explain to you something about **the** origin of **the** solar system. When we go back four and a half billion years, there was a cloud of gas and dust. In **the** center of this cloud, our sun formed and ignited. Along with that, what we now know as **planets**, comets and **asteroids** formed. What then happened, according to theory, is that when **the** Earth had cooled down a bit after its formation, comets massively impacted **the** Earth and delivered water to Earth. They probably also delivered complex **organic** material to Earth, and that may have bootstrapped **the** emergence of life. You can compare this to having to solve a 250-piece puzzle and not a 2, 000-piece puzzle. Afterwards, **the** big **planets** like Jupiter and Saturn, they were not in their place where they are now, and they interacted gravitationally, and they swept **the** whole interior of **the** solar system clean, and what we now know as comets ended up in something called **the** Kuiper Belt, which is a belt of **objects** beyond **the** orbit of Neptune. And sometimes these **objects** run into each other, and they gravitationally deflect, and then **the** gravity of Jupiter pulls them back into **the** solar system. And they then become **the** comets as we see them in **the** sky. **The** important thing here to note is that in **the** meantime, **the** four and a half billion years, these comets have been sitting on **the** outside of **the** solar system, and haven ’ t changed — deep, frozen versions of our solar system. In **the** sky, they look like this. We know them for their tails. There are actually two tails. One is a dust tail, which is blown away by **the** solar wind. **The** other one is an ion tail, which is charged particles, and they follow **the** magnetic field in **the** solar system. There ’ s **the** coma, and then there is **the** nucleus, which here is too small to see, and you have to remember that in **the** case of Rosetta, **the** spacecraft is in that center pixel. We are only 20, 30, 40 kilometers away from **the** comet. So what ’ s important to remember? Comets contain **the** original material from which our solar system was formed, so they ’ re ideal to study **the** components that were present at **the** time when Earth, and life, started. Comets are also suspected of having brought **the** elements which may have bootstrapped life. In 1983, ESA set up its long-term Horizon 2000 program, which contained one cornerstone, which would be a mission to a comet. In parallel, a small mission to a comet, what you see here, Giotto, was launched, and in 1986, flew by **the** comet of Halley with an armada of other spacecraft. From **the** results of that mission, it became immediately clear that comets were ideal bodies to study to understand our solar system. And thus, **the** Rosetta mission was approved in 1993, and originally it was supposed to be launched in 2003, but a problem arose with an Ariane rocket. However, our P. R. department, in its enthusiasm, had already made 1, 000 Delft **Blue plates** with **the** name of **the** wrong comets. So I ’ ve never had to buy any china since. That ’ s **the** positive part. Once **the** whole problem was solved, we left Earth in 2004 to **the** newly selected comet, Churyumov-Gerasimenko. This comet had to be specially selected because A, you have to be able to get to it, and B, it shouldn ’ t have been in **the** solar system too long. This particular comet has been in **the** solar system since 1959. That ’ s **the** first time when it was deflected by Jupiter, and it got close enough to **the** sun to start changing. So it ’ s a very fresh comet. Rosetta made a few historic firsts. It ’ s **the** first satellite to orbit a comet, and to escort it throughout its whole tour through **the** solar system — closest approach to **the** sun, as we will see in August, and then away again to **the** exterior. It ’ s **the** first ever landing on a comet. We actually orbit **the** comet using something which is not normally done with spacecraft. Normally, you look at **the** sky and you know where you point and where you are. In this case, that ’ s not enough. We navigated by looking at landmarks on **the** comet. We recognized features — boulders, craters — and that ' s how we know where we are respective to **the** comet. And, of course, it ' s **the** first satellite to go beyond **the** orbit of Jupiter on solar cells. Now, this sounds more heroic than it actually is, because **the** technology to use radio isotope thermal generators wasn ' t available in Europe at that time, so there was no choice. But these solar **arrays** are big. This is one **wing**, and these are not specially selected small people. They ' re just like you and me. We have two of these **wings**, 65 square meters. Now later on, of course, when we got to **the** comet, you find out that 65 square meters of sail close to a body which is outgassing is not always a very handy choice. Now, how did we get to **the** comet? Because we had to go there for **the** Rosetta scientific objectives very far away — four times **the** distance of **the** Earth to **the** sun — and also at a much higher velocity than we could achieve with fuel, because we ' d have to take six times as much fuel as **the** whole spacecraft weighed. So what do you do? You use gravitational flybys, slingshots, where you pass by a **planet** at very low **altitude**, a few thousand kilometers, and then you get **the** velocity of that **planet** around **the** sun for free. We did that a few times. We did Earth, we did Mars, we did twice Earth again, and we also flew by two **asteroids**, Lutetia and Steins. Then in 2011, we got so far from **the** sun that if **the** spacecraft got into trouble, we couldn ' t actually save **the** spacecraft anymore, so we went into hibernation. Everything was switched off except for one clock. Here you see in white **the** trajectory, and **the** way this works. You see that from **the** circle where we started, **the** white line, actually you get more and more and more elliptical, and then finally we approached **the** comet in May 2014, and we had to start doing **the** rendezvous maneuvers. On **the** way there, we flew by Earth and we took a few pictures to test our cameras. This is **the** moon rising over Earth, and this is what we now call a selfie, which at that time, by **the** way, that word didn ' t exist. It ' s at Mars. It was taken by **the** CIVA camera. That ' s one of **the** cameras on **the** lander, and it just looks under **the** solar **arrays**, and you see **the planet** Mars and **the** solar **array** in **the** distance. Now, when we got out of hibernation in January 2014, we started arriving at a distance of two million kilometers from **the** comet in May. However, **the** velocity **the** spacecraft had was much too fast. We were going 2, 800 kilometers an hour faster than **the** comet, so we had to **brake**. We had to do eight maneuvers, and you see here, some of them were really big. We had to **brake the** first one by a few hundred kilometers per hour, and actually, **the** duration of that was seven hours, and it used 218 kilos of fuel, and those were seven nerve-wracking hours, because in 2007, there was a leak in **the** system of **the** propulsion of Rosetta, and we had to close off a branch, so **the** system was actually operating at a pressure which it was never designed or qualified for. Then we got in **the** vicinity of **the** comet, and these were **the** first pictures we saw. **The** true comet rotation period is 12 and a half hours, so this is accelerated, but you will understand that our flight dynamics engineers thought, this is not going to be an easy thing to land on. We had hoped for some kind of spud-like thing where you could easily land. But we had one hope : maybe it was smooth. No. That didn ' t work either. So at that point in time, it was clearly unavoidable : we had to map this body in all **the** detail you could get, because we had to find an area which is 500 meters in **diameter** and flat. Why 500 meters? That ' s **the** error we have on landing **the** probe. So we went through this process, and we mapped **the** comet. We used a **technique** called photoclinometry. You use shadows thrown by **the** sun. What you see here is a **rock** sitting on **the surface** of **the** comet, and **the** sun shines from above. From **the** shadow, we, with our brain, can immediately determine roughly what **the** shape of that **rock** is. You can program that in a **computer**, you then cover **the** whole comet, and you can map **the** comet. For that, we flew special trajectories starting in August. First, a **triangle** of 100 kilometers on a side at 100 kilometers ’ distance, and we repeated **the** whole thing at 50 kilometers. At that time, we had seen **the** comet at all kinds of angles, and we could use this **technique** to map **the** whole thing. Now, this led to a **selection** of landing sites. This whole process we had to do, to go from **the** mapping of **the** comet to actually finding **the** final landing site, was 60 days. We didn ’ t have more. To give you an idea, **the** average Mars mission takes hundreds of scientists for years to meet about where shall we go? We had 60 days, and that was it. We finally selected **the** final landing site and **the** commands were prepared for Rosetta to launch Philae. **The** way this works is that Rosetta has to be at **the** right point in space, and aiming towards **the** comet, because **the** lander is passive. **The** lander is then pushed out and moves towards **the** comet. Rosetta had to turn around to get its cameras to actually look at Philae while it was departing and to be able to communicate with it. Now, **the** landing duration of **the** whole trajectory was seven hours. Now do a simple calculation : if **the** velocity of Rosetta is off by one centimeter per second, seven hours is 25, 000 seconds. That means 252 meters wrong on **the** comet. So we had to know **the** velocity of Rosetta much better than one centimeter per second, and its location in space better than 100 meters at 500 million kilometers from Earth. That ’ s no mean feat. Let me quickly take you through some of **the** science and **the** instruments. I won ’ t bore you with all **the** details of all **the** instruments, but it ’ s got everything. We can sniff gas, we can measure dust particles, **the** shape of them, **the** composition, there are magnetometers, everything. This is one of **the** results from an instrument which measures gas density at **the** position of Rosetta, so it ’ s gas which has left **the** comet. **The** bottom graph is September of last year. There is a long-term variation, which in itself is not surprising, but you see **the** sharp peaks. This is a comet day. You can see **the** effect of **the** sun on **the** evaporation of gas and **the** fact that **the** comet is rotating. So there is one spot, apparently, where there is a lot of stuff coming from, it gets heated in **the** Sun, and then cools down on **the** back side. And we can see **the** density variations of this. These are **the** gases and **the organic** compounds that we already have measured. You will see it ’ s an impressive list, and there is much, much, much more to come, because there are more measurements. Actually, there is a conference going on in Houston at **the** moment where many of these results are presented. Also, we measured dust particles. Now, for you, this will not look very impressive, but **the** scientists were thrilled when they saw this. Two dust particles : **the** right one they call Boris, and they shot it with tantalum in order to be able to analyze it. Now, we found sodium and magnesium. What this tells you is this is **the** concentration of these two materials at **the** time **the** solar system was formed, so we learned things about which materials were there when **the planet** was made. Of course, one of **the** important elements is **the** imaging. This is one of **the** cameras of Rosetta, **the** OSIRIS camera, and this actually was **the** cover of Science magazine on January 23 of this year. Nobody had expected this body to look like this. Boulders, **rocks** — if anything, it looks more like **the** Half Dome in Yosemite than anything else. We also saw things like this : dunes, and what look to be, on **the** righthand side, wind-blown shadows. Now we know these from Mars, but this comet doesn ’ t have an atmosphere, so it ’ s a bit difficult to create a wind-blown shadow. It may be local outgassing, stuff which goes up and comes back. We don ’ t know, so there is a lot to investigate. Here, you see **the** same image twice. On **the** left-hand side, you see in **the** middle a pit. On **the right-hand** side, if you carefully look, there are three jets coming out of **the** bottom of that pit. So this is **the** activity of **the** comet. Apparently, at **the** bottom of these pits is where **the** active regions are, and where **the** material evaporates into space. There is a very intriguing crack in **the** neck of **the** comet. You see it on **the right-hand** side. It ’ s a kilometer long, and it ’ s two and a half meters wide. Some people suggest that actually, when we get close to **the** sun, **the** comet may split in two, and then we ’ ll have to choose, which comet do we go for? **The** lander — again, lots of instruments, mostly comparable except for **the** things which hammer in **the** ground and drill, etc. But much **the** same as Rosetta, and that is because you want to compare what you find in space with what you find on **the** comet. These are called ground truth measurements. These are **the** landing descent images that were taken by **the** OSIRIS camera. You see **the** lander getting further and further away from Rosetta. On **the** top right, you see an image taken at 60 meters by **the** lander, 60 meters above **the surface** of **the** comet. **The** boulder there is some 10 meters. So this is one of **the** last images we took before we landed on **the** comet. Here, you see **the** whole sequence again, but from a different perspective, and you see three blown-ups from **the** bottom-left to **the** middle of **the** lander traveling over **the surface** of **the** comet. Then, at **the** top, there is a before and an after image of **the** landing. **The** only problem with **the** after image is, there is no lander. But if you carefully look at **the right-hand** side of this image, we saw **the** lander still there, but it had bounced. It had departed again. Now, on a bit of a comical note here is that originally Rosetta was designed to have a lander which would bounce. That was **discarded** because it was way too expensive. Now, we forgot, but **the** lander knew. During **the** first bounce, in **the** magnetometers, you see here **the** data from them, from **the** three axes, x, y and z. Halfway through, you see a red line. At that red line, there is a change. What happened, apparently, is during **the** first bounce, somewhere, we hit **the** edge of a crater with one of **the** legs of **the** lander, and **the** rotation velocity of **the** lander changed. So we ’ ve been rather lucky that we are where we are. This is one of **the** iconic images of Rosetta. It ’ s a man-made **object**, a leg of **the** lander, standing on a comet. This, for me, is one of **the** very best images of space science I have ever seen. One of **the** things we still have to do is to actually find **the** lander. **The blue** area here is where we know it must be. We haven ’ t been able to find it yet, but **the search** is continuing, as are our efforts to start getting **the** lander to work again. We listen every day, and we hope that between now and somewhere in April, **the** lander will wake up again. **The** findings of what we found on **the** comet : This thing would float in water. It ’ s half **the** density of water. So it looks like a very big **rock**, but it ’ s not. **The** activity increase we saw in June, July, August last year was a four-fold activity increase. By **the** time we will be at **the** sun, there will be 100 kilos a second leaving this comet : gas, dust, whatever. That ’ s 100 million kilos a day. Then, finally, **the** landing day. I will never forget — absolute madness, 250 TV crews in Germany. **The** BBC was interviewing me, and another TV crew who was following me all day were filming me being interviewed, and it went on like that for **the** whole day. **The** Discovery Channel crew actually caught me when leaving **the** control room, and they asked **the** right question, and man, I got into tears, and I still feel this. For a month and a half, I couldn ’ t think about landing day without crying, and I still have **the** emotion in me. With this image of **the** comet, I would like to leave you. Thank you.

Text Sample 42: POS Dim 1 – 2013 – Score 23.00 – 2013/t002142.txt

Now, extinction is a different kind of death. It ’ s bigger. We didn ’ t really realize that until 1914, when **the** last passenger pigeon, a female named Martha, died at **the** Cincinnati zoo. This had been **the** most abundant bird in **the** world that ’ d been in North America for six million years. Suddenly it wasn ’ t here at all. Flocks that were a mile wide and 400 miles long used to darken **the** sun. Aldo Leopold said this was a biological storm, a feathered tempest. And indeed it was a keystone species that enriched **the** entire **eastern** deciduous forest, from **the** Mississippi to **the** Atlantic, from Canada down to **the** Gulf. But it went from five billion birds to zero in just a couple decades. What happened? Well, commercial hunting happened. These birds were hunted for meat that was sold by **the** ton, and it was easy to do because when those big flocks came down to **the** ground, they were so dense that hundreds of hunters and netters could show up and slaughter them by **the** tens of thousands. It was **the** cheapest source of protein in America. By **the** end of **the** century, there was nothing left but these beautiful skins in museum specimen drawers. There ’ s an upside to **the** story. This made people realize that **the** same thing was about to happen to **the** American bison, and so these birds saved **the** buffalos. But a lot of other animals weren ’ t saved. **The** Carolina parakeet was a parrot that lit up backyards everywhere. It was hunted to death for its feathers. There was a bird that people liked on **the** East Coast called **the** heath hen. It was loved. They tried to protect it. It died anyway. A local newspaper spelled out,”There is no survivor, there is no future, there is no life to be recreated in this form ever again.”There ’ s a sense of deep tragedy that goes with these things, and it happened to lots of birds that people loved. It happened to lots of mammals. Another keystone species is a **famous** animal called **the** European aurochs. There was sort of a movie made about it recently. And **the** aurochs was like **the** bison. This was an animal that basically kept **the** forest mixed with grasslands across **the** entire Europe and Asian **continent**, from Spain to Korea. **The** documentation of this animal goes back to **the** Lascaux cave paintings. **The** extinctions still go on. There ’ s an ibex in Spain called **the** bucardo. It went extinct in 2000. There was a marvelous animal, a marsupial wolf called **the** thylacine in Tasmania, south of Australia, called **the** Tasmanian **tiger**. It was hunted until there were just a few **left** to die in zoos. A little bit of film was shot. Sorrow, anger, mourning. Don ’ t mourn. Organize. What if you could find out that, using **the** DNA in museum specimens, **fossils** maybe up to 200, 000 years old could be used to bring species back, what would you do? Where would you start? Well, you ’ d start by finding out if **the** biotech is really there. I started with my wife, Ryan Phelan, who ran a biotech business called DNA Direct, and through her, one of her colleagues, George Church, one of **the** leading genetic engineers who turned out to be also obsessed with passenger pigeons and a lot of confidence that methodologies he was working on might actually do **the** deed. So he and Ryan organized and hosted a meeting at **the** Wyss Institute in Harvard bringing together specialists on passenger pigeons, conservation ornithologists, bioethicists, and fortunately passenger pigeon DNA had already been sequenced by a molecular biologist named Beth Shapiro. All she needed from those specimens at **the** Smithsonian was a little bit of toe pad tissue, because down in there is what is called ancient DNA. It ’ s DNA which is pretty badly fragmented, but with good **techniques** now, you can basically reassemble **the** whole **genome**. Then **the** question is, can you reassemble, with that **genome**, **the** whole bird? George Church thinks you can. So in his book,”Regenesis,”which I recommend, he has a chapter on **the** science of bringing back extinct species, and he has a machine called **the** Multiplex Automated **Genome** Engineering machine. It ’ s kind of like an **evolution** machine. You try combinations of genes that you write at **the** cell level and then in organs on a chip, and **the** ones that win, that you can then put into a living organism. It ’ ll work. **The** precision of this, one of George ’ s **famous** unreadable **slides**, nevertheless points out that there ’ s a level of precision here right down to **the** individual base pair. **The** passenger pigeon has 1. 3 billion base pairs in its **genome**. So what you ' re getting is **the** capability now of replacing one gene with another variation of that gene. It ' s called an allele. Well that ' s what happens in normal hybridization anyway. So this is a form of synthetic hybridization of **the genome** of an extinct species with **the genome** of its closest living **relative**. Now along **the** way, George points out that his technology, **the** technology of synthetic **biology**, is currently accelerating at four times **the** rate of Moore ' s Law. It ' s been doing that since 2005, and it ' s likely to continue. **Okay**, **the** closest living **relative** of **the** passenger pigeon is **the** band-tailed pigeon. They ' re abundant. There ' s some around here. Genetically, **the** band- tailed pigeon already is mostly living passenger pigeon. There ' s just some bits that are band-tailed pigeon. If you replace those bits with passenger pigeon bits, you ' ve got **the** extinct bird back, cooing at you. Now, there ' s work to do. You have to figure out exactly what genes matter. So there ' s genes for **the** short tail in **the** band-tailed pigeon, genes for **the** long tail in **the** passenger pigeon, and so on with **the** red eye, peach-colored breast, flocking, and so on. Add them all up and **the** result won ' t be perfect. But it should be be perfect enough, because nature doesn ' t do perfect either. So this meeting in Boston led to three things. First off, Ryan and I decided to create a nonprofit called Revive and Restore that would push de-extinction generally and try to have it go in a responsible way, and we would push ahead with **the** passenger pigeon. Another direct result was a young grad student named Ben Novak, who had been obsessed with passenger pigeons since he was 14 and had also learned how to work with ancient DNA, himself sequenced **the** passenger pigeon, using money from his family and friends. We hired him full-time. Now, this photograph I took of him last year at **the** Smithsonian, he ' s looking down at Martha, **the** last passenger pigeon alive. So if he ' s successful, she won ' t be **the** last. **The** third result of **the** Boston meeting was **the** realization that there are scientists all over **the** world working on various forms of de-extinction, but they ' d never met each other. And National Geographic got interested because National Geographic has **the** theory that **the** last century, discovery was basically finding things, and in this century, discovery is basically making things. De-extinction falls in that category. So they hosted and funded this meeting. And 35 scientists, they were conservation biologists and molecular biologists, basically meeting to see if they had work to do together. Some of these conservation biologists are pretty radical. There ' s three of them who are not just re-creating ancient species, they ' re recreating extinct ecosystems in northern Siberia, in **the** Netherlands, and in Hawaii. Henri, from **the** Netherlands, with a Dutch last name I won ' t try to pronounce, is working on **the** aurochs. **The** aurochs is **the** ancestor of all domestic cattle, and so basically its **genome** is alive, it ' s just unevenly distributed. So what they ' re doing is working with seven breeds of primitive, hardy-looking cattle like that Maremmana primitivo on **the** top there to rebuild, over time, with selective back-breeding, **the** aurochs. Now, re-wilding is moving faster in Korea than it is in America, and so **the** plan is, with these re-wilded areas all over Europe, they will introduce **the** aurochs to do its old job, its old ecological role, of clearing **the** somewhat barren, closed-canopy forest so that it has these biodiverse meadows in it. Another amazing story came from Alberto Fernndez-Arias. Alberto worked with **the** bucardo in Spain. **The** last bucardo was a female named Celia who was still alive, but then they captured her, they got a little bit of tissue from her ear, they cryopreserved it in liquid nitrogen, released her back into **the** wild, but a few months later, she was found dead under a fallen tree. They took **the** DNA from that ear, they planted it as a **cloned** egg in a goat, **the** pregnancy came to term, and a live baby bucardo was born. It was **the** first de-extinction in history. It was short-lived. Sometimes interspecies clones have respiration problems. This one had a malformed lung and died after 10 minutes, but Alberto was confident that **cloning** has moved along well since then, and this will move ahead, and eventually there will be a population of bucardos back in **the** mountains in northern Spain. Cryopreservation pioneer of great depth is Oliver Ryder. At **the** San Diego zoo, his frozen zoo has collected **the** tissues from over 1, 000 species over **the** last 35 years. Now, when it ’ s frozen that deep, minus 196 degrees Celsius, **the** cells are intact and **the** DNA is intact. They ’ re basically viable cells, so someone like Bob Lanza at Advanced Cell Technology took some of that tissue from an endangered animal called **the** Javan banteng, put it in a cow, **the** cow went to term, and what was born was a live, healthy baby Javan banteng, who thrived and is still alive. **The** most exciting thing for Bob Lanza is **the** ability now to take any kind of cell with induced pluripotent stem cells and turn it into germ cells, like sperm and eggs. So now we go to Mike McGrew who is a scientist at Roslin Institute in Scotland, and Mike ’ s doing miracles with birds. So he ’ ll take, say, falcon skin cells, fibroblast, turn it into induced pluripotent stem cells. Since it ’ s so pluripotent, it can become germ plasm. He then has a way to put **the** germ plasm into **the** embryo of a chicken egg so that that chicken will have, basically, **the** gonads of a falcon. You get a male and a female each of those, and out of them comes falcons. Real falcons out of slightly doctored chickens. Ben Novak was **the** youngest scientist at **the** meeting. He showed how all of this can be put together. **The** sequence of events : he ’ ll put together **the genomes** of **the** band-tailed pigeon and **the** passenger pigeon, he ’ ll take **the techniques** of George Church and get passenger pigeon DNA, **the techniques** of Robert Lanza and Michael McGrew, get that DNA into chicken gonads, and out of **the** chicken gonads get passenger pigeon eggs, squabs, and now you ’ re getting a population of passenger pigeons. It does raise **the** question of, they ’ re not going to have passenger pigeon parents to teach them how to be a passenger pigeon. So what do you do about that? Well birds are pretty hard-wired, as it happens, so most of that is already in their DNA, but to **supplement** it, part of Ben ’ s idea is to use homing pigeons to help train **the** young passenger pigeons how to flock and how to find their way to their old nesting grounds and **feeding** grounds. There were some conservationists, really **famous** conservationists like Stanley Temple, who is one of **the** founders of conservation **biology**, and Kate Jones from **the** IUCN, which does **the** Red List. They ’ re excited about all this, but they ’ re also concerned that it might be competitive with **the** extremely important efforts to protect endangered species that are still alive, that haven ’ t gone extinct yet. You see, you want to work on protecting **the** animals out there. You want to work on getting **the** market for ivory in Asia down so you ’ re not using 25, 000 **elephants** a year. But at **the** same time, conservation biologists are realizing that bad news bums people out. And so **the** Red List is really important, keep track of what ’ s endangered and critically endangered, and so on. But they ’ re about to create what they call a Green List, and **the** Green List will have species that are doing fine, thank you, species that were endangered, like **the** bald eagle, but they ’ re much better off now, thanks to everybody ’ s good work, and protected areas around **the** world that are very, very well managed. So basically, they ’ re **learning** how to build on good news. And they see reviving extinct species as **the** kind of good news you might be able to build on. Here ’ s a couple related examples. Captive breeding will be a major part of bringing back these species. **The** California condor was down to 22 birds in 1987. Everybody thought is was finished. Thanks to captive breeding at **the** San Diego Zoo, there ’ s 405 of them now, 226 are out in **the** wild. That technology will be used on de-extincted animals. Another success story is **the** mountain gorilla in Central Africa. In 1981, Dian Fossey was sure they were going extinct. There were just 254 **left**. Now there are 880. They ' re increasing in population by three percent a year. **The** secret is, they have an eco-tourism program, which is absolutely brilliant. So this photograph was taken last month by Ryan with an iPhone. That ' s how comfortable these wild gorillas are with visitors. Another interesting project, though it ' s going to need some help, is **the** northern white rhinoceros. There ' s no breeding pairs left. But this is **the** kind of thing that a wide variety of DNA for this animal is available in **the** frozen zoo. A bit of **cloning**, you can get them back. So where do we go from here? These have been private meetings so far. I think it ' s time for **the** subject to go public. What do people think about it? You know, do you want extinct species back? Do you want extinct species back? Tinker Bell is going to come fluttering down. It is a Tinker Bell moment, because what are people excited about with this? What are they concerned about? We ' re also going to push ahead with **the** passenger pigeon. So Ben Novak, even as we speak, is joining **the** group that Beth Shapiro has at UC Santa Cruz. They ' re going to work on **the genomes** of **the** passenger pigeon and **the** band-tailed pigeon. As that data matures, they ' ll send it to George Church, who will work his magic, get passenger pigeon DNA out of that. We ' ll get help from Bob Lanza and Mike McGrew to get that into germ plasm that can go into chickens that can produce passenger pigeon squabs that can be raised by band-tailed pigeon parents, and then from then on, it ' s passenger pigeons all **the** way, maybe for **the** next six million years. You can do **the** same thing, as **the** costs come down, for **the** Carolina parakeet, for **the** great auk, for **the** heath hen, for **the** ivory- billed woodpecker, for **the** Eskimo curlew, for **the** Caribbean **monk seal**, for **the** woolly mammoth. Because **the** fact is, humans have made a huge hole in nature in **the** last 10, 000 years. We have **the** ability now, and maybe **the** moral obligation, to repair some of **the** damage. Most of that we ' ll do by expanding and protecting wildlands, by expanding and protecting **the** populations of endangered species. But some species that we killed off totally we could consider bringing back to a world that misses them. Thank you. Chris Anderson : Thank you. I ' ve got a question. So, this is an emotional topic. Some people stand. I suspect there are some people out there sitting, kind of asking tormented questions, almost, about, well, wait, wait, wait, wait, wait, wait a minute, there ' s something wrong with mankind interfering in nature in this way. There ' s going to be unintended consequences. You ' re going to uncork some sort of Pandora ' s box of who-knows-what. Do they have a point? Stewart Brand : Well, **the** earlier point is we interfered in a big way by making these animals go extinct, and many of them were keystone species, and we changed **the** whole ecosystem they were in by letting them go. Now, there ' s **the** shifting baseline problem, which is, so when these things come back, they might replace some birds that are there that people really know and love. I think that ' s, you know, part of how it ' ll work. This is a long, slow process — One of **the** things I like about it, it ' s multi-generation. We will get woolly mammoths back. CA : Well it feels like both **the** conversation and **the** potential here are pretty thrilling. Thank you so much for presenting. SB : Thank you. CA : Thank you.

Text Sample 43: POS Dim 1 – 2013 – Score 21.00 – 2013/t001929.txt

Technology can change our understanding of nature. Take for example **the** case of **lions**. For centuries, it ' s been said that female **lions** do all of **the** hunting out in **the** open savanna, and male **lions** do nothing until it ' s time for dinner. You ’ ve heard this too, I can tell. Well recently, I led an airborne mapping campaign in **the** Kruger National Park in South Africa. Our colleagues put GPS tracking collars on male and female **lions**, and we mapped their hunting behavior from **the** air. **The** lower **left** shows a **lion** sizing up a herd of impala for a kill, and **the** right shows what I call **the lion** viewshed. That ’ s how far **the lion** can see in all directions until his or her view is obstructed by vegetation. And what we found is that male **lions** are not **the** lazy hunters we thought them to be. They just use a different strategy. Whereas **the** female **lions** hunt out in **the** open savanna over long distances, usually during **the** day, male **lions** use an ambush strategy in dense vegetation, and often at night. This video shows **the** actual hunting viewsheds of male **lions** on **the left** and females on **the** right. Red and darker colors show more dense vegetation, and **the** white are wide open spaces. And this is **the** viewshed right literally at **the** eye level of hunting male and female **lions**. All of a sudden, you get a very clear understanding of **the** very spooky conditions under which male **lions** do their hunting. I bring up this example to begin, because it emphasizes how little we know about nature. There ’ s been a huge amount of work done so far to try to slow down our losses of tropical forests, and we are losing our forests at a rapid rate, as shown in red on **the slide**. I find it ironic that we ’ re doing so much, yet these areas are fairly unknown to science. So how can we save what we don ’ t understand? Now I ’ m a global ecologist and an Earth explorer with a background in physics and **chemistry** and **biology** and a lot of other boring subjects, but above all, I ’ m obsessed with what we don ’ t know about our **planet**. So I created this, **the** Carnegie Airborne Observatory, or CAO. It may look like a plane with a fancy paint job, but I packed it with over 1, 000 kilos of high-tech sensors, **computers**, and a very motivated staff of Earth scientists and pilots. Two of our instruments are very unique : one is called an imaging spectrometer that can actually measure **the** chemical composition of plants as we fly over them. Another one is a set of lasers, very high-powered lasers, that fire out of **the** bottom of **the** plane, sweeping across **the** ecosystem and measuring it at nearly 500, 000 times per second in high-resolution 3D. Here ’ s an image of **the** Golden Gate Bridge in San Francisco, not far from where I live. Although we flew straight over this bridge, we imaged it in 3D, captured its color in just a few seconds. But **the** real power of **the** CAO is its ability to capture **the** actual building blocks of ecosystems. This is a small town in **the** Amazon, imaged with **the** CAO. We can slice through our data and see, for example, **the** 3D structure of **the** vegetation and **the** buildings, or we can use **the** chemical information to actually figure out how fast **the** plants are growing as we fly over them. **The** hottest pinks are **the** fastest-growing plants. And we can see biodiversity in ways that you never could have imagined. This is what a rainforest might look like as you fly over it in a hot air balloon. This is how we see a rainforest, in kaleidoscopic color that tells us that there are many species living with one another. But you have to remember that these trees are literally bigger than **whales**, and what that means is that they ’ re impossible to understand just by walking on **the** ground below them. So our imagery is 3D, it ’ s chemical, it ’ s biological, and this tells us not only **the** species that are living in **the** canopy, but it tells us a lot of information about **the** rest of **the** species that occupy **the** rainforest. Now I created **the** CAO in order to answer questions that have proven extremely challenging to answer from any other vantage point, such as from **the** ground, or from satellite sensors. I want to share three of those questions with you today. **The** first questions is, how do we manage our **carbon** reserves in tropical forests? Tropical forests contain a huge amount of **carbon** in **the** trees, and we need to keep that **carbon** in those forests if we ’ re going to avoid any further global warming. Unfortunately, global **carbon emissions** from deforestation now equals **the** global **transportation** sector. That ' s all ships, **airplanes**, trains and automobiles combined. So it ' s understandable that policy negotiators have been working hard to reduce deforestation, but they ' re doing it on landscapes that are hardly known to science. If you don ' t know where **the carbon** is exactly, in detail, how can you know what you ' re losing? Basically, we need a high-tech accounting system. With our system, we ' re able to see **the carbon** stocks of tropical forests in utter detail. **The** red shows, obviously, closed- canopy tropical forest, and then you see **the** cookie cutting, or **the** cutting of **the** forest in **yellows** and greens. It ' s like cutting a cake except this cake is about **whale** deep. And yet, we can zoom in and see **the** forest and **the** trees at **the** same time. And what ' s amazing is, even though we flew very high above this forest, later on in analysis, we can go in and actually experience **the** treetrops, leaf by leaf, branch by branch, just as **the** other species that live in this forest experience it along with **the** trees themselves. We ' ve been using **the** technology to explore and to actually put out **the** first **carbon** geographies in high resolution in faraway places like **the** Amazon Basin and not-so-faraway places like **the** United States and Central America. What I ' m going to do is I ' m going to take you on a high-resolution, first-time tour of **the carbon** landscapes of Peru and then Panama. **The** colors are going to be going from red to **blue**. Red is extremely high **carbon** stocks, your largest cathedral forests you can imagine, and **blue** are very low **carbon** stocks. And let me tell you, Peru alone is an amazing place, totally unknown in terms of its **carbon** geography until today. We can fly to this area in northern Peru and see super high **carbon** stocks in red, and **the** Amazon River and floodplain cutting right through it. We can go to an area of utter devastation caused by deforestation in **blue**, and **the** virus of deforestation spreading out in orange. We can also fly to **the** southern Andes to see **the** tree line and see exactly how **the carbon** geography ends as we go up into **the** mountain system. And we can go to **the** biggest swamp in **the** western Amazon. It ' s a watery dreamworld akin to Jim Cameron ' s"Avatar."We can go to one of **the** smallest tropical countries, Panama, and see also a huge range of **carbon** variation, from high in red to low in **blue**. Unfortunately, most of **the carbon** is lost in **the** lowlands, but what you see that ' s left, in terms of high **carbon** stocks in greens and reds, is **the** stuff that ' s up in **the** mountains. One interesting exception to this is right in **the** middle of your screen. You ' re seeing **the** buffer zone around **the** Panama Canal. That ' s in **the** reds and **yellows**. **The** canal authorities are using force to protect their watershed and global commerce. This kind of **carbon** mapping has transformed conservation and resource policy development. It ' s really advancing our ability to save forests and to curb climate change. My second question : How do we prepare for climate change in a place like **the** Amazon rainforest? Let me tell you, I spend a lot of time in these places, and we ' re seeing **the** climate changing already. Temperatures are increasing, and what ' s really happening is we ' re getting a lot of **droughts**, recurring **droughts**. **The** 2010 mega-drought is shown here with red showing an area about **the** size of Western Europe. **The** Amazon was so dry in 2010 that even **the** main stem of **the** Amazon river itself dried up partially, as you see in **the** photo in **the** lower portion of **the slide**. What we found is that in very remote areas, these **droughts** are having a big negative impact on tropical forests. For example, these are all of **the** dead trees in red that suffered mortality following **the** 2010 **drought**. This area happens to be on **the** border of Peru and Brazil, totally unexplored, almost totally unknown scientifically. So what we think, as Earth scientists, is species are going to have to migrate with climate change from **the** east in Brazil all **the** way west into **the** Andes and up into **the** mountains in order to minimize their exposure to climate change. One of **the** problems with this is that humans are taking apart **the** western Amazon as we speak. Look at this 100-square-kilometer gash in **the** forest created by gold miners. You see **the** forest in green in 3D, and you see **the** effects of gold mining down below **the** soil **surface**. Species have nowhere to migrate in a system like this, obviously. If you haven ' t been to **the** Amazon, you should go. It ' s an amazing experience every time, no matter where you go. You ' re going to probably see it this way, on a river. But what happens is a lot of times **the** rivers hide what ' s really going on back in **the** forest itself. We flew over this same river, imaged **the** system in 3D. **The** forest is on **the left**. And then we can digitally remove **the** forest and see what ' s going on below **the** canopy. And in this case, we found gold mining activity, all of it illegal, set back away from **the** river ' s edge, as you ' ll see in those strange pockmarks coming up on your screen on **the** right. Don ' t worry, we ' re working with **the** authorities to deal with this and many, many other problems in **the** region. So in order to put together a conservation plan for these unique, important corridors like **the** western Amazon and **the** Andes Amazon corridor, we have to start making geographically explicit plans now. How do we do that if we don ' t know **the** geography of biodiversity in **the** region, if it ' s so unknown to science? So what we ' ve been doing is using **the** laser-guided spectroscopy from **the** CAO to map for **the** first time **the** biodiversity of **the** Amazon rainforest. Here you see actual data showing different species in different colors. Reds are one type of species, **blues** are another, and greens are yet another. And when we take this together and scale up to **the** regional level, we get a completely new geography of biodiversity unknown prior to this work. This tells us where **the** big biodiversity changes **occur** from **habitat** to **habitat**, and that ' s really important because it tells us a lot about where species may migrate to and migrate from as **the** climate shifts. And this is **the** pivotal information that ' s needed by decision makers to develop protected areas in **the** context of their regional development plans. And third and final question is, how do we manage biodiversity on a **planet** of protected ecosystems? **The** example I started out with about **lions** hunting, that was a study we did behind **the** fence line of a protected area in South Africa. And **the** truth is, much of Africa ' s nature is going to persist into **the** future in protected areas like I show in **blue** on **the** screen. This puts incredible pressure and responsibility on park management. They need to do and make decisions that will benefit all of **the** species that they ' re protecting. Some of their decisions have really big impacts. For example, how much and where to use fire as a management tool? Or, how to deal with a large species like **elephants**, which may, if their populations get too large, have a negative impact on **the** ecosystem and on other species. And let me tell you, these types of dynamics really play out on **the** landscape. In **the** foreground is an area with lots of fire and lots of **elephants** : wide open savanna in **blue**, and just a few trees. As we cross this fence line, now we ' re getting into an area that has had protection from fire and zero **elephants** : dense vegetation, a radically different ecosystem. And in a place like Kruger, **the** soaring **elephant** densities are a real problem. I know it ' s a sensitive issue for many of you, and there are no easy answers with this. But what ' s good is that **the** technology we ' ve developed and we ' re working with in South Africa, for example, is allowing us to map every single tree in **the** savanna, and then through repeat flights we ' re able to see which trees are being pushed over by **elephants**, in **the** red as you see on **the** screen, and how much that ' s happening in different types of landscapes in **the** savanna. That ' s giving park managers a very first opportunity to use tactical management strategies that are more nuanced and don ' t lead to those extremes that I just showed you. So really, **the** way we ' re looking at protected areas nowadays is to think of it as tending to a circle of life, where we have fire management, **elephant** management, those impacts on **the** structure of **the** ecosystem, and then those impacts affecting everything from insects up to apex predators like **lions**. Going forward, I plan to greatly expand **the** airborne observatory. I ’ m hoping to actually put **the** technology into orbit so we can manage **the** entire **planet** with technologies like this. Until then, you ’ re going to find me flying in some remote place that you ’ ve never heard of. I just want to end by saying that technology is absolutely critical to managing our **planet**, but even more important is **the** understanding and wisdom to apply it. Thank you.

Text Sample 44: POS Dim 1 – 2013 – Score 19.00 – 2013/t001983.txt

When I was a young man, I spent six years of wild adventure in **the** tropics working as an investigative journalist in some of **the** most bewitching parts of **the** world. I was as reckless and foolish as only young men can be. This is why wars get fought. But I also felt more alive than I ’ ve ever done since. And when I came home, I found **the** scope of my existence gradually diminishing until loading **the** dishwasher seemed like an interesting challenge. And I found myself sort of scratching at **the** walls of life, as if I was trying to find a way out into a wider space beyond. I was, I believe, ecologically bored. Now, we **evolved** in rather more challenging times than these, in a world of horns and tusks and fangs and claws. And we still possess **the** fear and **the** courage and **the** aggression required to navigate those times. But in our comfortable, safe, crowded lands, we have few opportunities to exercise them without harming other people. And this was **the** sort of constraint that I found myself bumping up against. To conquer uncertainty, to know what comes next, that ’ s almost been **the** dominant aim of industrialized societies, and having got there, or almost got there, we have just encountered a new set of unmet needs. We ’ ve privileged safety over experience and we ’ ve gained a lot in doing so, but I think we ’ ve lost something too. Now, I don ’ t romanticize **evolutionary** time. I ’ m already beyond **the** lifespan of most hunter-gatherers, and **the** outcome of a mortal combat between me myopically stumbling around with a stone-tipped spear and an enraged giant aurochs isn ’ t very hard to predict. Nor was it authenticity that I was looking for. I don ’ t find that a useful or even intelligible concept. I just wanted a richer and rawer life than I ’ ve been able to lead in Britain, or, indeed, that we can lead in most parts of **the** industrialized world. And it was only when I stumbled across an unfamiliar word that I began to understand what I was looking for. And as soon as I found that word, I realized that I wanted to devote much of **the** rest of my life to it. **The** word is”rewilding,”and even though rewilding is a young word, it already has several definitions. But there are two in particular that fascinate me. **The** first one is **the** mass restoration of ecosystems. One of **the** most exciting scientific findings of **the** past half century has been **the** discovery of widespread trophic cascades. A trophic cascade is an ecological process which starts at **the** top of **the** food chain and tumbles all **the** way down to **the** bottom, and **the** classic example is what happened in **the** Yellowstone National Park in **the** United States when wolves were reintroduced in 1995. Now, we all know that wolves kill various species of animals, but perhaps we ’ re slightly less aware that they give life to many others. It sounds strange, but just follow me for a while. Before **the** wolves turned up, they ’ d been absent for 70 years. **The** numbers of deer, because there was nothing to hunt them, had built up and built up in **the** Yellowstone Park, and despite efforts by humans to control them, they ’ d managed to reduce much of **the** vegetation there to almost nothing, they ’ d just grazed it away. But as soon as **the** wolves arrived, even though they were few in number, they started to have **the** most remarkable effects. First, of course, they killed some of **the** deer, but that wasn ' t **the** major thing. Much more significantly, they radically changed **the** behavior of **the** deer. **The** deer started avoiding certain parts of **the** park, **the** places where they could be trapped most easily, particularly **the** valleys and **the** gorges, and immediately those places started to regenerate. In some areas, **the** height of **the** trees quintupled in just six years. Bare valley sides quickly became forests of aspen and willow and cottonwood. And as soon as that happened, **the** birds started moving in. **The** number of songbirds, of migratory birds, started to increase greatly. **The** number of beavers started to increase, because beavers like to eat **the** trees. And beavers, like wolves, are ecosystem engineers. They create niches for other species. And **the** dams they built in **the** rivers provided **habitats** for otters and muskrats and ducks and **fish** and reptiles and amphibians. **The** wolves killed coyotes, and as a result of that, **the** number of rabbits and mice began to rise, which meant more hawks, more weasels, more foxes, more badgers. Ravens and bald eagles came down to feed on **the** carrion that **the** wolves had left. Bears fed on it too, and their population began to rise as well, partly also because there were more berries growing on **the** regenerating shrubs, and **the** bears reinforced **the** impact of **the** wolves by killing some of **the** calves of **the** deer. But here ' s where it gets really interesting. **The** wolves changed **the** behavior of **the** rivers. They began to meander less. There was less erosion. **The** channels narrowed. More pools formed, more riffle sections, all of which were great for wildlife **habitats**. **The** rivers changed in response to **the** wolves, and **the** reason was that **the** regenerating forests stabilized **the** banks so that they collapsed less often, so that **the** rivers became more fixed in their course. Similarly, by driving **the** deer out of some places and **the** vegetation recovering on **the** valley sides, there was less soil erosion, because **the** vegetation stabilized that as well. So **the** wolves, small in number, transformed not just **the** ecosystem of **the** Yellowstone National Park, this huge area of land, but also its physical geography. **Whales** in **the** southern oceans have similarly wide-ranging effects. One of **the** many post-rational excuses made by **the** Japanese government for killing **whales** is that they said,"Well, **the** number of **fish** and krill will rise and then there ' ll be more for people to eat."Well, it ' s a stupid excuse, but it sort of kind of makes sense, doesn ' t it, because you ' d think that **whales** eat huge amounts of **fish** and krill, so obviously take **the whales** away, there ' ll be more **fish** and krill. But **the** opposite happened. You take **the whales** away, and **the** number of krill collapses. Why would that possibly have happened? Well, it now turns out that **the whales** are crucial to sustaining that entire ecosystem, and one of **the** reasons for this is that they often feed at depth and then they come up to **the surface** and produce what biologists politely call large fecal plumes, huge **explosions** of poop right across **the surface** waters, up in **the** photic zone, where there ' s enough light to allow photosynthesis to take place, and those great plumes of fertilizer stimulate **the** growth of phytoplankton, **the** plant plankton at **the** bottom of **the** food chain, which stimulate **the** growth of zooplankton, which feed **the fish** and **the** krill and all **the** rest of it. **The** other thing that **whales** do is that, as they ' re plunging up and down through **the** water column, they ' re kicking **the** phytoplankton back up towards **the surface** where it can continue to survive and reproduce. And interestingly, well, we know that plant plankton in **the** oceans absorb **carbon** from **the** atmosphere — **the** more plant plankton there are, **the** more **carbon** they absorb — and eventually they filter down into **the abyss** and remove that **carbon** from **the** atmospheric system. Well, it seems that when **whales** were at their historic populations, they were probably responsible for sequestering some tens of millions of tons of **carbon** every year from **the** atmosphere. And when you look at it like that, you think, wait a minute, here are **the** wolves changing **the** physical geography of **the** Yellowstone National Park. Here are **the whales** changing **the** composition of **the** atmosphere. You begin to see that possibly, **the** evidence supporting James Lovelock ' s Gaia hypothesis, which conceives of **the** world as a coherent, self-regulating organism, is beginning, at **the** ecosystem level, to accumulate. Trophic cascades tell us that **the** natural world is even more fascinating and complex than we thought it was. They tell us that when you take away **the** large animals, you are left with a radically different ecosystem to one which retains its large animals. And they make, in my view, a powerful case for **the** reintroduction of missing species. Rewilding, to me, means bringing back some of **the** missing plants and animals. It means taking down **the** fences, it means blocking **the** drainage ditches, it means preventing commercial **fishing** in some large areas of **sea**, but otherwise stepping back. It has no view as to what a right ecosystem or a right assemblage of species looks like. It doesn ' t try to produce a heath or a meadow or a rain forest or a kelp garden or a coral reef. It lets nature decide, and nature, by and large, is pretty good at deciding. Now, I mentioned that there are two definitions of rewilding that interest me. **The** other one is **the** rewilding of human life. And I don ' t see this as an alternative to civilization. I believe we can **enjoy the** benefits of advanced technology, as we ' re doing now, but at **the** same time, if we choose, have access to a richer and wilder life of adventure when we want to because there would be wonderful, rewilded **habitats**. And **the** opportunities for this are developing more rapidly than you might think possible. There ' s one estimate which suggests that in **the** United States, two thirds of **the** land which was once forested and then cleared has become reforested as loggers and farmers have retreated, particularly from **the eastern** half of **the** country. There ' s another one which suggests that 30 million hectares of land in Europe, an area **the** size of Poland, will be vacated by farmers between 2000 and 2030. Now, faced with opportunities like that, does it not seem a little unambitious to be thinking only of bringing back wolves, lynx, bears, beavers, bison, boar, moose, and all **the** other species which are already beginning to move quite rapidly across Europe? Perhaps we should also start thinking about **the** return of some of our lost megafauna. What megafauna, you say? Well, every **continent** had one, apart from Antarctica. When Trafalgar Square in London was excavated, **the** river gravels there were found to be stuffed with **the** bones of hippopotamus, rhinos, **elephants**, hyenas, **lions**. Yes, ladies and gentlemen, there were **lions** in Trafalgar Square long before Nelson ' s Column was built. All these species lived here in **the** last interglacial period, when temperatures were pretty similar to our own. It ' s not climate, largely, which has got rid of **the** world ' s megafaunas. It ' s pressure from **the** human population hunting and destroying their **habitats** which has done so. And even so, you can still see **the** shadows of these great beasts in our current ecosystems. Why is it that so many deciduous trees are able to sprout from whatever point **the** trunk is broken? Why is it that they can withstand **the** loss of so much of their bark? Why do understory trees, which are subject to lower **sheer** forces from **the** wind and have to carry less weight than **the** big canopy trees, why are they so much tougher and harder to break than **the** canopy trees are? **Elephants**. They are elephant-adapted. In Europe, for example, they **evolved** to resist **the** straight-tusked **elephant**, elephas antiquus, which was a great beast. It was related to **the** Asian **elephant**, but it was a temperate animal, a temperate forest creature. It was a lot bigger than **the** Asian **elephant**. But why is it that some of our common shrubs have spines which seem to be over- engineered to resist browsing by deer? Perhaps because they **evolved** to resist browsing by rhinoceros. Isn ’ t it an amazing thought that every time you wander into a park or down an avenue or through a leafy street, you can see **the** shadows of these great beasts? Paleoecology, **the** study of past ecosystems, crucial to an understanding of our own, feels like a portal through which you may pass into an enchanted kingdom. And if we really are looking at areas of land of **the** sort of sizes I ’ ve been talking about becoming available, why not reintroduce some of our lost megafauna, or at least species closely related to those which have become extinct everywhere? Why shouldn ’ t all of us have a Serengeti on our doorsteps? And perhaps this is **the** most important thing that rewilding offers us, **the** most important thing that ’ s missing from our lives : hope. In motivating people to love and defend **the** natural world, an ounce of hope is worth a ton of despair. **The** story rewilding tells us is that ecological change need not always proceed in one direction. It offers us **the** hope that our silent **spring** could be replaced by a raucous summer. Thank you.

Text Sample 45: POS Dim 1 – 2014 – Score 25.00 – 2014/t001657.txt

On June 12, 2014, precisely at 3:33 in a balmy winter afternoon in SĂŁo Paulo, Brazil, a typical South American winter afternoon, this kid, this young man that you see celebrating here like he had scored a goal, Juliano Pinto, 29 years old, accomplished a magnificent deed. Despite being paralyzed and not having any sensation from mid-chest to **the** tip of his toes as **the** result of a car crash six years ago that killed his brother and produced a complete spinal cord lesion that left Juliano in a wheelchair, Juliano rose to **the** occasion, and on this day did something that pretty much everybody that saw him in **the** six years deemed impossible. Juliano Pinto delivered **the** opening kick of **the** 2014 Brazilian World Soccer Cup here just by thinking. He could not move his body, but he could imagine **the** movements needed to kick a ball. He was an athlete before **the** lesion. He ’ s a para-athlete right now. He ’ s going to be in **the** Paralympic Games, I hope, in a couple years. But what **the** spinal cord lesion did not rob from Juliano was his ability to dream. And dream he did that afternoon, for a stadium of about 75, 000 people and an audience of close to a billion watching on TV. And that kick crowned, basically, 30 years of basic research studying how **the** brain, how this amazing **universe** that we have between our ears that is only comparable to **universe** that we have above our head because it has about 100 billion elements talking to each other through **electrical** brainstorms, what Juliano accomplished took 30 years to imagine in laboratories and about 15 years to plan. When John Chapin and I, 15 years ago, proposed in a paper that we would build something that we called a brain-machine interface, meaning connecting a brain to **devices** so that animals and humans could just move these **devices**, no matter how far they are from their own bodies, just by imagining what they want to do, our colleagues told us that we actually needed professional help, of **the** psychiatry variety. And despite that, a Scot and a Brazilian persevered, because that ’ s how we were raised in our respective countries, and for 12, 15 years, we made demonstration after demonstration suggesting that this was possible. And a brain-machine interface is not rocket science, it ’ s just brain research. It ’ s nothing but using sensors to read **the** **electrical** brainstorms that a brain is producing to generate **the** motor commands that have to be downloaded to **the** spinal cord, so we projected sensors that can read hundreds and now thousands of these brain cells simultaneously, and extract from these **electrical** signals **the** motor planning that **the** brain is generating to actually make us move into space. And by doing that, we converted these signals into digital commands that any mechanical, electronic, or even a virtual **device** can understand so that **the** subject can imagine what he, she or it wants to make move, and **the device** obeys that brain command. By sensorizing these **devices** with lots of different types of sensors, as you are going to see in a moment, we actually sent messages back to **the** brain to confirm that that voluntary motor will was being enacted, no matter where â next to **the** subject, next door, or across **the planet**. And as this message gave feedback back to **the** brain, **the** brain realized its goal : to make us move. So this is just one experiment that we published a few years ago, where a monkey, without moving its body, learned to control **the** movements of an avatar arm, a virtual arm that doesn ’ t exist. What you ’ re listening to is **the** sound of **the** brain of this monkey as it explores three different visually **identical spheres** in virtual space. And to get a reward, a drop of orange juice that monkeys love, this animal has to **detect**, select one of these **objects** by touching, not by seeing it, by touching it, because every time this virtual hand touches one of **the objects**, an **electrical pulse** goes back to **the** brain of **the** animal describing **the** fine texture of **the surface** of this **object**, so **the** animal can judge what is **the** correct **object** that he has to grab, and if he does that, he gets a reward without moving a muscle. **The** perfect Brazilian lunch : not moving a muscle and getting your orange juice. So as we saw this happening, we actually came and proposed **the** idea that we had published 15 years ago. We reenacted this paper. We got it out of **the** drawers, and we proposed that perhaps we could get a human being that is paralyzed to actually use **the** brain-machine interface to regain mobility. **The** idea was that if you suffered â and that can happen to any one of us. Let me tell you, it ’ s very sudden. It ’ s a millisecond of a collision, a car accident that transforms your life completely. If you have a complete lesion of **the** spinal cord, you cannot move because your brainstorms cannot reach your muscles. However, your brainstorms continue to be generated in your head. Paraplegic, quadriplegic patients dream about moving every night. They have that inside their head. **The** problem is how to get that code out of it and make **the** movement be created again. So what we proposed was, let ’ s create a new body. Let ’ s create a robotic vest. And that ’ s exactly why Juliano could kick that ball just by thinking, because he was wearing **the** first brain-controlled robotic vest that can be used by paraplegic, quadriplegic patients to move and to regain feedback. That was **the** original idea, 15 years ago. What I ’ m going to show you is how 156 people from 25 countries all over **the** five **continents** of this beautiful Earth, dropped their lives, dropped their **patents**, dropped their dogs, wives, kids, school, jobs, and congregated to come to Brazil for 18 months to actually get this done. Because a couple years after Brazil was awarded **the** World Cup, we heard that **the** Brazilian government wanted to do something meaningful in **the** opening ceremony in **the** country that reinvented and perfected soccer until we met **the** Germans, of course. But that ’ s a different talk, and a different neuroscientist needs to talk about that. But what Brazil wanted to do is to showcase a completely different country, a country that values science and technology, and can give a gift to millions, 25 million people around **the** world that cannot move any longer because of a spinal cord injury. Well, we went to **the** Brazilian government and to FIFA and proposed, well, let ’ s have **the** kickoff of **the** 2014 World Cup be given by a Brazilian paraplegic using a brain- controlled exoskeleton that allows him to kick **the** ball and to feel **the** contact of **the** ball. They looked at us, thought that we were completely **nuts**, and said,”Okay, let ’ s try.”We had 18 months to do everything from zero, from scratch. We had no exoskeleton, we had no patients, we had nothing done. These people came all together and in 18 months, we got eight patients in a routine of training and basically built from nothing this guy, that we call Bra-Santos Dumont 1. **The** first brain-controlled exoskeleton to be built was named after **the** most **famous** Brazilian scientist ever, Alberto Santos Dumont, who, on October 19, 1901, created and flew himself **the** first controlled airship on air in Paris for a million people to see. Sorry, my American friends, I live in North Carolina, but it was two years before **the** Wright Brothers flew on **the** coast of North Carolina. Flight control is Brazilian. So we went together with these guys and we basically put this exoskeleton together, 15 degrees of freedom, hydraulic machine that can be commanded by brain signals recorded by a non-invasive technology called electroencephalography that can basically allow **the** patient to imagine **the** movements and send his commands to **the** controls, **the** motors, and get it done. This exoskeleton was covered with an **artificial** skin invented by Gordon Cheng, one of my greatest friends, in Munich, to allow sensation from **the** joints moving and **the** foot touching **the** ground to be delivered back to **the** patient through a vest, a shirt. It is a smart shirt with micro-vibrating elements that basically delivers **the** feedback and fools **the** patient ’ s brain by creating a sensation that it is not a machine that is carrying him, but it is he who is walking again. So we got this going, and what you ’ ll see here is **the** first time one of our patients, Bruno, actually walked. And he takes a few seconds because we are setting everything, and you are going to see a **blue** light cutting in front of **the** helmet because Bruno is going to imagine **the** movement that needs to be performed, **the computer** is going to analyze it, Bruno is going to certify it, and when it is certified, **the device** starts moving under **the** command of Bruno ’ s brain. And he just got it right, and now he starts walking. After nine years without being able to move, he is walking by himself. And more than that â â more than just walking, he is feeling **the** ground, and if **the** speed of **the** exo goes up, he tells us that he is walking again on **the** sand of Santos, **the** beach resort where he used to go before he had **the** accident. That ’ s why **the** brain is creating a new sensation in Bruno ’ s head. So he walks, and at **the** end of **the** walk â I am running out of time already â he says,”You know, guys, I need to borrow this thing from you when I get married, because I wanted to walk to **the** priest and see my bride and actually be there by myself. Of course, he will have it whenever he wants. And this is what we wanted to show during **the** World Cup, and couldn ’ t, because for some mysterious reason, FIFA cut its broadcast in half. What you are going to see very quickly is Juliano Pinto in **the** exo doing **the** kick a few minutes before we went to **the** pitch and did **the** real thing in front of **the** entire crowd, and **the** lights you are going to see just describe **the** operation. Basically, **the blue** lights pulsating indicate that **the** exo is ready to go. It can receive thoughts and it can deliver feedback, and when Juliano makes **the** decision to kick **the** ball, you are going to see two streams of green and **yellow** light coming from **the** helmet and going to **the** legs, representing **the** mental commands that were taken by **the** exo to actually make that happen. And in basically 13 seconds, Juliano actually did. You can see **the** commands. He gets ready, **the** ball is set, and he kicks. And **the** most amazing thing is, 10 seconds after he did that, and looked at us on **the** pitch, he told us, celebrating as you saw,”I felt **the** ball.”And that ’ s priceless. So where is this going to go? I have two minutes to tell you that it ’ s going to **the** limits of your imagination. Brain-actuating technology is here. This is **the** latest : We just published this a year ago, **the** first brain-to-brain interface that allows two animals to exchange mental messages so that one animal that sees something coming from **the** environment can send a mental SMS, a torpedo, a neurophysiological torpedo, to **the** second animal, and **the** second animal performs **the** act that he needed to perform without ever knowing what **the** environment was sending as a message, because **the** message came from **the** first animal ’ s brain. So this is **the** first demo. I ’ m going to be very quick because I want to show you **the** latest. But what you see here is **the** first rat getting informed by a light that is going to show up on **the left** of **the** cage that he has to press **the** left cage to basically get a reward. He goes there and does it. And **the** same time, he is sending a mental message to **the** second rat that didn ’ t see any light, and **the** second rat, in 70 percent of **the** times is going to press **the** left lever and get a reward without ever experiencing **the** light in **the** retina. Well, we took this to a little higher limit by getting monkeys to collaborate mentally in a brain net, basically to donate their brain activity and combine them to move **the** virtual arm that I showed you before, and what you see here is **the** first time **the** two monkeys combine their brains, synchronize their brains perfectly to get this virtual arm to move. One monkey is controlling **the** x **dimension**, **the** other monkey is controlling **the** y **dimension**. But it gets a little more interesting when you get three monkeys in there and you ask one monkey to control x and y, **the** other monkey to control y and z, and **the** third one to control x and z, and you make them all play **the** game together, moving **the** arm in 3D into a target to get **the famous** Brazilian orange juice. And they actually do. **The** black dot is **the** average of all these brains working in parallel, in real time. That is **the** definition of a biological **computer**, interacting by brain activity and achieving a motor goal. Where is this going? We have no idea. We ’ re just scientists. We are paid to be children, to basically go to **the** edge and discover what is out there. But one thing I know : One day, in a few decades, when our grandchildren **surf the** Net just by thinking, or a mother donates her eyesight to an autistic kid who cannot see, or somebody speaks because of a brain-to-brain bypass, some of you will remember that it all started on a winter afternoon in a Brazilian soccer field with an impossible kick. Thank you. Thank you. Bruno Giussani : Miguel, thank you for sticking to your time. I actually would have given you a couple more minutes, because there are a couple of points we want to develop, and, of course, clearly it seems that we need connected brains to figure out where this is going. So let ’ s connect all this together. So if I ’ m understanding correctly, one of **the** monkeys is actually getting a signal and **the** other monkey is reacting to that signal just because **the** first one is receiving it and transmitting **the** neurological impulse. Miguel Nicolelis : No, it ’ s a little different. No monkey knows of **the** existence of **the** other two monkeys. They are getting a visual feedback in 2D, but **the** task they have to accomplish is 3D. They have to move an arm in three **dimensions**. But each monkey is only getting **the** two **dimensions** on **the** video screen that **the** monkey controls. And to get that thing done, you need at least two monkeys to synchronize their brains, but **the** ideal is three. So what we found out is that when one monkey starts slacking down, **the** other two monkeys enhance their performance to get **the** guy to come back, so this adjusts dynamically, but **the** global synchrony remains **the** same. Now, if you flip without telling **the** monkey **the dimensions** that each brain has to control, like this guy is controlling x and y, but he should be controlling now y and z, instantaneously, that animal ’ s brain forgets about **the** old **dimensions** and it starts concentrating on **the** new **dimensions**. So what I need to say is that no Turing machine, no **computer** can predict what a brain net will do. So we will absorb technology as part of us. Technology will never absorb us. It ’ s simply impossible. BG : How many times have you tested this? And how many times have you succeeded versus failed? MN : Oh, tens of times. With **the** three monkeys? Oh, several times. I wouldn ’ t be able to talk about this here unless I had done it a few times. And I forgot to mention, because of time, that just three weeks ago, a European group just demonstrated **the** first man-to-man brain-to-brain connection. BG : And how does that play? MN : There was one bit of information â big ideas start in a humble way â but basically **the** brain activity of one subject was transmitted to a second **ob-**

ject, all non-invasive technology. So **the** first subject got a message, like our rats, a visual message, and transmitted it to **the** second subject. **The** second subject received a magnetic **pulse** in **the** visual cortex, or a different **pulse**, two different **pulses**. In one **pulse**, **the** subject saw something. On **the** other **pulse**, he saw something different. And he was able to verbally indicate what was **the** message **the** first subject was sending through **the** Internet across **continents**. Moderator : Wow. **Okay**, that ’ s where we are going. That ’ s **the** next TED Talk at **the** next conference. Miguel Nicolelis, thank you. MN : Thank you, Bruno. Thank you.

Text Sample 46: POS Dim 1 – 2014 – Score 21.00 – 2014/t001874.txt
What ’ s **the** scariest thing you ’ ve ever done? Or another way to say it is, what ’ s **the** most dangerous thing that you ’ ve ever done? And why did you do it? I know what **the** most dangerous thing is that I ’ ve ever done because NASA does **the** math. You look back to **the** first five shuttle launches, **the** odds of a catastrophic event during **the** first five shuttle launches was one in nine. And even when I first flew in **the** shuttle back in 1995, 74 shuttle flight, **the** odds were still now that we look back about one in 38 or so — one in 35, one in 40. Not great odds, so it ’ s a really interesting day when you wake up at **the** Kennedy Space Center and you ’ re going to go to space that day because you realize by **the** end of **the** day you ’ re either going to be floating effortlessly, gloriously in space, or you ’ ll be dead. You go into, at **the** Kennedy Space Center, **the** suit-up room, **the** same room that our childhood heroes got dressed in, that Neil Armstrong and Buzz Aldrin got suited in to go ride **the** Apollo rocket to **the** moon. And I got my pressure suit built around me and rode down outside in **the** van heading out to **the** launchpad — in **the** Astro van — heading out to **the** launchpad, and as you come around **the** corner at **the** Kennedy Space Center, it ’ s normally predawn, and in **the** distance, lit up by **the** huge xenon lights, is your spaceship — **the** vehicle that is going to take you off **the planet**. **The** crew is sitting in **the** Astro van sort of hushed, almost holding hands, looking at that as it gets bigger and bigger. We ride **the** elevator up and we crawl in, on your hands and knees into **the** spaceship, one at a time, and you worm your way up into your chair and plunk yourself down on your back. And **the** hatch is closed, and suddenly, what has been a lifetime of both dreams and denial is becoming real, something that I dreamed about, in fact, that I chose to do when I was nine years old, is now suddenly within not too many minutes of actually happening. In **the** astronaut business — **the** shuttle is a very complicated vehicle ; it ’ s **the** most complicated flying machine ever built. And in **the** astronaut business, we have a saying, which is, there is no problem so bad that you can ’ t make it worse. And so you ’ re very conscious in **the** cockpit ; you ’ re thinking about all of **the** things that you might have to do, all **the** switches and all **the** wickets you have to go through. And as **the** time gets closer and closer, this excitement is building. And then about three and a half minutes before launch, **the** huge nozzles on **the** back, like **the** size of big church bells, swing back and forth and **the** mass of them is such that it sways **the** whole vehicle, like **the** vehicle is alive underneath you, like an **elephant** getting up off its knees or something. And then about 30 seconds before launch, **the** vehicle is completely alive — it is ready to go — **the** APUs are running, **the computers** are all self-contained, it ’ s ready to leave **the planet**. And 15 seconds before launch, this happens : Voice : 12, 11, 10, nine, eight, seven, six — — start, two, one, booster ignition, and liftoff of **the** space shuttle Discovery, returning to **the** space station, paving **the** way... Chris Hadfield : It is incredibly

powerful to be on board one of these things. You are in **the** grip of something that is vastly more powerful than yourself. It ’ s shaking you so hard you can ’ t focus on **the** instruments in front of you. It ’ s like you ’ re in **the** jaws of some enormous dog and there ’ s a foot in **the** small of your back pushing you into space, accelerating wildly straight up, shouldering your way through **the** air, and you ’ re in a very complex place — paying attention, watching **the** vehicle go through each one of its wickets with a steadily increasing smile on your face. After two minutes, those solid rockets explode off and then you just have **the** liquid engines, **the** hydrogen and oxygen, and it ’ s as if you ’ re in a dragster with your foot to **the** floor and accelerating like you ’ ve never accelerated. You get lighter and lighter, **the** force gets on us heavier and heavier. It feels like someone ’ s pouring cement on you or something. Until finally, after about eight minutes and 40 seconds or so, we are finally at exactly **the** right **altitude**, exactly **the** right speed, **the** right direction, **the** engine shut off, and we ’ re weightless. And we ’ re alive. It ’ s an amazing experience. But why would we take that risk? Why would you do something that dangerous? In my case **the** answer is fairly straightforward. I was inspired as a youngster that this was what I wanted to do. I watched **the** first people walk on **the** moon and to me, it was just an obvious thing — I want to somehow turn myself into that. But **the** real question is, how do you deal with **the** danger of it and **the** fear that comes from it? How do you deal with fear versus danger? And having **the** goal in mind, thinking about where it might lead, directed me to a life of looking at all of **the** small details to allow this to become possible, to be able to launch and go help build a space station where you are on board a million-pound creation that ’ s going around **the** world at five miles a second, eight kilometers a second, around **the** world 16 times a day, with experiments on board that are teaching us what **the** substance of **the universe** is made of and running 200 experiments inside. But maybe even more importantly, allowing us to see **the** world in a way that is impossible through any other means, to be able to look down and have — if your jaw could drop, it would — **the** jaw-dropping gorgeousness of **the** turning orb like a self-propelled art gallery of **fantastic**, constantly changing beauty that is **the** world itself. And you see, because of **the** speed, a sunrise or a sunset every 45 minutes for half a year. And **the** most magnificent part of all that is to go outside on a spacewalk. You are in a one-person spaceship that is your spacesuit, and you ’ re going through space with **the** world. It ’ s an entirely different perspective, you ’ re not looking up at **the universe**, you and **the** Earth are going through **the universe** together. And you ’ re holding on with one hand, looking at **the** world turn beside you. It ’ s roaring silently with color and texture as it pours by mesmerizingly next to you. And if you can tear your eyes away from that and you look under your arm down at **the** rest of everything, it ’ s unfathomable blackness, with a texture you feel like you could stick your hand into. and you are holding on with one hand, one link to **the** other seven billion people. And I was outside on my first spacewalk when my **left** eye went blind, and I didn ’ t know why. Suddenly my **left** eye slammed shut in great pain and I couldn ’ t figure out why my eye wasn ’ t working. I was thinking, what do I do next? I thought, well maybe that ’ s why we have two eyes, so I kept working. But unfortunately, without gravity, tears don ’ t fall. So you just get a bigger and bigger ball of whatever that is mixed with your tears on your eye until eventually, **the** ball becomes so big that **the surface** tension takes it across **the** bridge of your nose like a tiny little waterfall and goes”goosh”into your other eye, and now I was completely blind outside **the** spaceship. So what ’ s **the** scariest thing you ’ ve ever done? Maybe it ’ s **spiders**. A lot of people are afraid of **spiders**. I think you should be afraid of **spiders** — **spiders** are creepy and they ’ ve got long, hairy legs, and

spiders like this one, **the** brown recluse — it ’ s horrible. If a brown recluse **bites** you, you end with one of these horrible, big necrotic things on your leg and there might be one right now sitting on **the** chair behind you, in fact. And how do you know? And so a **spider** lands on you, and you go through this great, spasmy attack because **spiders** are scary. But then you could say, well is there a brown recluse sitting on **the** chair beside me or not? I don ’ t know. Are there brown recluses here? So if you actually do **the** research, you find out that in **the** world there are about 50, 000 different types of **spiders**, and there are about two dozen that are venomous out of 50, 000. And if you ’ re in Canada, because of **the** cold winters here in B. C., there ’ s about 720, 730 different types of **spiders** and there ’ s one — one — that is venomous, and its venom isn ’ t even fatal, it ’ s just kind of like a nasty sting. And that **spider** — not only that, but that **spider** has beautiful markings on it, it ’ s like”I ’ m dangerous. I got a big radiation symbol on my back, it ’ s **the** black widow.”So, if you ’ re even slightly careful you can avoid running into **the** one **spider** — and it lives close **the** ground, you ’ re walking along, you are never going to go through a **spider web** where a black widow **bites** you. **Spider webs** like this, it doesn ’ t build those, it builds them down in **the** corners. And its a black widow because **the** female **spider** eats **the** male ; it doesn ’ t care about you. So in fact, **the** next time you walk into a spiderweb, you don ’ t need to panic and go with your caveman reaction. **The** danger is entirely different than **the** fear. How do you get around it, though? How do you change your behavior? Well, next time you see a spiderweb, have a good look, make sure it ’ s not a black widow **spider**, and then walk into it. And then you see another spiderweb and walk into that one. It ’ s just a little bit of fluffy stuff. It ’ s not a big deal. And **the spider** that may come out is no more threat to you than a lady bug or a butterfly. And then I guarantee you if you walk through 100 spiderwebs you will have changed your fundamental human behavior, your caveman reaction, and you will now be able to walk in **the** park in **the** morning and not worry about that spiderweb — or into your grandma ’ s attic or whatever, into your own basement. And you can apply this to anything. If you ’ re outside on a spacewalk and you ’ re blinded, your natural reaction would be to panic, I think. It would make you nervous and worried. But we had considered all **the** venom, and we had practiced with a whole variety of different spiderwebs. We knew everything there is to know about **the** spacesuit and we trained **underwater** thousands of times. And we don ’ t just practice things going right, we practice things going wrong all **the** time, so that you are constantly walking through those spiderwebs. And not just **underwater**, but also in virtual reality labs with **the** helmet and **the** gloves so you feel like it ’ s realistic. So when you finally actually get outside on a spacewalk, it feels much different than it would if you just went out first time. And even if you ’ re blinded, your natural, panicky reaction doesn ’ t happen. Instead you kind of look around and go,”Okay, I can ’ t see, but I can hear, I can talk, Scott Parazynski is out here with me. He could come over and help me.”We actually practiced incapacitated crew rescue, so he could float me like a blimp and stuff me into **the** airlock if he had to. I could find my own way back. It ’ s not nearly as big a deal. And actually, if you keep on crying for a while, whatever that gunk was that ’ s in your eye starts to dilute and you can start to see again, and Houston, if you negotiate with them, they will let you then keep working. We finished everything on **the** spacewalk and when we came back inside, Jeff got some cotton batting and took **the** crusty stuff around my eyes, and it turned out it was just **the** anti-fog, sort of a mixture of **oil** and soap, that got in my eye. And now we use Johnson ’ s No More Tears, which we probably should ’ ve been using right from **the** very beginning. But **the** key to that is by looking at **the** difference between perceived danger and actual danger, where is **the** real risk? What is **the** real thing that you should be afraid of? Not just a generic fear of bad things happening. You can fundamentally change your reaction to things so that it allows you to go places and see things and do things that otherwise would be completely denied to you... where you could see **the** hardpan south of **the** Sahara, or you can see New York City in a way that is almost dreamlike, or **the** unconscious gingham of Eastern Europe fields or **the** Great Lakes as a **collection** of small puddles. You can see **the** fault lines of San Francisco and **the** way **the** water pours out under **the** bridge, just entirely different than any other way that you could have if you had not found a way to conquer your fear. You see a beauty that otherwise never would have happened. It ’ s time to come home at **the** end. This is our spaceship, **the** Soyuz, that little one. Three of us climb in, and then this spaceship detaches from **the** station and falls into **the** atmosphere. These two parts here actually melt, we jettison them and they burn up in **the** atmosphere. **The** only part that survives is **the** little **bullet** that we ’ re riding in, and it falls into **the** atmosphere, and in essence you are riding a meteorite home, and riding meteorites is scary, and it ought to be. But instead of riding into **the** atmosphere just screaming, like you would if suddenly you found yourself riding a meteorite back to Earth — — instead, 20 years previously we had started studying Russian, and then once you learn Russian, then we learned orbital mechanics in Russian, and then we learned vehicle control theory, and then we got into **the** simulator and practiced over and over and over again. And in fact, you can fly this meteorite and steer it and land in about a 15-kilometer circle anywhere on **the** Earth. So in fact, when our crew was coming back into **the** atmosphere inside **the** Soyuz, we weren ’ t screaming, we were laughing ; it was fun. And when **the** great big parachute opened, we knew that if it didn ’ t open there ’ s a second parachute, and it runs on a nice little clockwork mechanism. So we came back, we came thundering back to Earth and this is what it looked like to land in a Soyuz, in Kazakhstan. Reporter : And you can see one of those **search** and recovery helicopters, once again that helicopter part of dozen such Russian Mi- 8 helicopters. Touchdown — 3:14 and 48 seconds, a. m. Central Time. CH : And you roll to a stop as if someone threw your spaceship at **the** ground and it tumbles end over end, but you ’ re ready for it you ’ re in a custom-built seat, you know how **the** shock absorber works. And then eventually **the** Russians reach in, drag you out, plunk you into a chair, and you can now look back at what was an incredible experience. You have taken **the** dreams of that nine-year-old boy, which were impossible and dauntingly scary, dauntingly **terrifying**, and put them into practice, and figured out a way to reprogram yourself, to change your primal fear so that it allowed you to come back with a set of experiences and a level of inspiration for other people that never could have been possible otherwise. Just to finish, they asked me to play that guitar. I know this song, and it ’ s really a tribute to **the** genius of David Bowie himself, but it ’ s also, I think, a reflection of **the** fact that we are not machines exploring **the universe**, we are people, and we ’ re taking that ability to adapt and that ability to understand and **the** ability to take our own self-perception into a new place. — This is Major Tom to ground control — — I ’ ve left forevermore — — And I ’ m floating in a most **peculiar** way — — And **the** stars look very different today — — For here am I floating in **the** tin can — — A last glimpse of **the** world — — Planet Earth is **blue** and there ’ s so much left to do — Fear not. That ’ s very nice of you. Thank you very much. Thank you.

Text Sample 47: POS Dim 1 – 2014 – Score 18.00 – 2014/t001794.txt

Every day we face issues like climate change or **the** safety of vaccines where we have to answer questions whose answers rely heavily on scientific information. Scientists tell us that **the** world is warming. Scientists tell us that vaccines are safe. But how do we know if they are right? Why should be believe **the** science? **The** fact is, many of us actually don ’ t believe **the** science. Public opinion polls consistently show that significant proportions of **the** American people don ’ t believe **the** climate is warming due to human activities, don ’ t think that there is **evolution** by natural **selection**, and aren ’ t persuaded by **the** safety of vaccines. So why should we believe **the** science? Well, scientists don ’ t like talking about science as a matter of belief. In fact, they would contrast science with faith, and they would say belief is **the** domain of faith. And faith is a separate thing apart and distinct from science. Indeed they would say religion is based on faith or maybe **the** calculus of Pascal ’ s wager. Blaise Pascal was a 17th-century mathematician who tried to bring scientific reasoning to **the** question of whether or not he should believe in God, and his wager went like this : Well, if God doesn ’ t exist but I decide to believe in him nothing much is really lost. Maybe a few hours on Sunday. But if he does exist and I don ’ t believe in him, then I ’ m in deep trouble. And so Pascal said, we ’ d better believe in God. Or as one of my college professors said,”He clutched for **the** handrail of faith.”He made that leap of faith leaving science and rationalism behind. Now **the** fact is though, for most of us, most scientific claims are a leap of faith. We can ’ t really judge scientific claims for ourselves in most cases. And indeed this is actually true for most scientists as well outside of their own specialties. So if you think about it, a geologist can ’ t tell you whether a vaccine is safe. Most chemists are not experts in **evolutionary** theory. A physicist cannot tell you, despite **the** claims of some of them, whether or not tobacco causes cancer. So, if even scientists themselves have to make a leap of faith outside their own fields, then why do they accept **the** claims of other scientists? Why do they believe each other ’ s claims? And should we believe those claims? So what I ’ d like to argue is yes, we should, but not for **the** reason that most of us think. Most of us were taught in school that **the** reason we should believe in science is because of **the** scientific method. We were taught that scientists follow a method and that this method guarantees **the** truth of their claims. **The** method that most of us were taught in school, we can call it **the textbook** method, is **the** hypothetical deductive method. According to **the** standard model, **the textbook** model, scientists develop hypotheses, they deduce **the** consequences of those hypotheses, and then they go out into **the** world and they say,”Okay, well are those consequences true?”Can we observe them taking place in **the** natural world? And if they are true, then **the** scientists say,”Great, we know **the** hypothesis is correct.”So there are many **famous** examples in **the** history of science of scientists doing exactly this. One of **the** most **famous** examples comes from **the** work of Albert Einstein. When Einstein developed **the** theory of general relativity, one of **the** consequences of his theory was that space-time wasn ’ t just an empty void but that it actually had a fabric. And that that fabric was bent in **the** presence of massive **objects** like **the** sun. So if this theory were true then it meant that light as it passed **the** sun should actually be bent around it. That was a pretty startling prediction and it took a few years before scientists were able to test it but they did test it in 1919, and lo and behold it turned out to be true. Starlight actually does bend as it travels around **the** sun. This was a huge confirmation of **the** theory. It was considered proof of **the** truth of this radical new idea, and it was written up in many newspapers around **the** globe. Now, sometimes this theory or this model is referred to as **the** deductive- nomological model, mainly because academics like to make things complicated. But also because in **the** ideal case, it ’ s about laws. So nomological means having to do with laws. And in **the** ideal case, **the** hypothesis isn ' t just an idea : ideally, it is a law of nature. Why does it matter that it is a law of nature? Because if it is a law, it can ' t be broken. If it ' s a law then it will always be true in all times and all places no matter what **the** circumstances are. And all of you know of at least one example of a **famous** law : Einstein ' s **famous** equation, E=MC2, which tells us what **the** relationship is between energy and mass. And that relationship is true no matter what. Now, it turns out, though, that there are several problems with this model. **The** main problem is that it ' s wrong. It ' s just not true. And I ' m going to talk about three reasons why it ' s wrong. So **the** first reason is a logical reason. It ' s **the** problem of **the** fallacy of affirming **the** consequent. So that ' s another fancy, academic way of saying that false theories can make true predictions. So just because **the** prediction comes true doesn ' t actually logically prove that **the** theory is correct. And I have a good example of that too, again from **the** history of science. This is a picture of **the** Ptolemaic **universe** with **the** Earth at **the** center of **the universe** and **the** sun and **the planets** going around it. **The** Ptolemaic model was believed by many very smart people for many centuries. Well, why? Well **the** answer is because it made lots of predictions that came true. **The** Ptolemaic system enabled astronomers to make accurate predictions of **the** motions of **the planet**, in fact more accurate predictions at first than **the** Copernican theory which we now would say is true. So that ' s one problem with **the textbook** model. A second problem is a practical problem, and it ' s **the** problem of auxiliary hypotheses. Auxiliary hypotheses are assumptions that scientists are making that they may or may not even be aware that they ' re making. So an important example of this comes from **the** Copernican model, which ultimately replaced **the** Ptolemaic system. So when Nicolaus Copernicus said, actually **the** Earth is not **the** center of **the universe**, **the** sun is **the** center of **the** solar system, **the** Earth moves around **the** sun. Scientists said, well **okay**, Nicolaus, if that ' s true we ought to be able to **detect the** motion of **the** Earth around **the** sun. And so this **slide** here illustrates a concept known as stellar parallax. And astronomers said, if **the** Earth is moving and we look at a prominent star, let ' s say, Sirius — well I know I ' m in Manhattan so you guys can ' t see **the** stars, but imagine you ' re out in **the** country, imagine you chose that rural life — and we look at a star in December, we see that star against **the** backdrop of distant stars. If we now make **the** same observation six months later when **the** Earth has moved to this position in June, we look at that same star and we see it against a different backdrop. That difference, that angular difference, is **the** stellar parallax. So this is a prediction that **the** Copernican model makes. Astronomers looked for **the** stellar parallax and they found nothing, nothing at all. And many people argued that this proved that **the** Copernican model was false. So what happened? Well, in hindsight we can say that astronomers were making two auxiliary hypotheses, both of which we would now say were incorrect. **The** first was an assumption about **the** size of **the** Earth ' s orbit. Astronomers were assuming that **the** Earth ' s orbit was large **relative** to **the** distance to **the** stars. Today we would draw **the** picture more like this, this comes from NASA, and you see **the** Earth ' s orbit is actually quite small. In fact, it ' s actually much smaller even than shown here. **The** stellar parallax therefore, is very small and actually very hard to **detect**. And that leads to **the** second reason why **the** prediction didn ' t work, because scientists were also assuming that **the** telescopes they had were sensitive enough to **detect the** parallax. And that turned out not to be true. It wasn ' t until **the** 19th century that scientists were able to **detect the** stellar parallax. So, there ' s a third problem as well. **The** third problem is simply a factual problem, that a lot of science doesn ' t fit **the textbook** model. A lot of science isn ’ t deductive at all, it ’ s actually inductive. And by that we mean that scientists don ’ t necessarily start with theories and hypotheses, often they just start with observations of stuff going on in **the** world. And **the** most **famous** example of that is one of **the** most **famous** scientists who ever lived, Charles Darwin. When Darwin went out as a young man on **the** voyage of **the** Beagle, he didn ’ t have a hypothesis, he didn ’ t have a theory. He just knew that he wanted to have a career as a scientist and he started to collect data. Mainly he knew that he hated medicine because **the** sight of **blood** made him sick so he had to have an alternative career path. So he started collecting data. And he collected many things, including his **famous** finches. When he collected these finches, he threw them in a bag and he had no idea what they meant. Many years later back in London, Darwin looked at his data again and began to develop an explanation, and that explanation was **the** theory of natural **selection**. Besides inductive science, scientists also often participate in modeling. One of **the** things scientists want to do in life is to explain **the** causes of things. And how do we do that? Well, one way you can do it is to build a model that tests an idea. So this is a picture of Henry Cadell, who was a Scottish geologist in **the** 19th century. You can tell he ’ s Scottish because he ’ s wearing a deerstalker cap and Wellington boots. And Cadell wanted to answer **the** question, how are mountains formed? And one of **the** things he had observed is that if you look at mountains like **the** Appalachians, you often find that **the rocks** in them are folded, and they ’ re folded in a particular way, which suggested to him that they were actually being compressed from **the** side. And this idea would later play a major role in discussions of continental drift. So he built this model, this crazy contraption with levers and wood, and here ’ s his wheelbarrow, buckets, a big sledgehammer. I don ’ t know why he ’ s got **the** Wellington boots. Maybe it ’ s going to rain. And he created this physical model in order to demonstrate that you could, in fact, create patterns in **rocks**, or at least, in this case, in mud, that looked a lot like mountains if you compressed them from **the** side. So it was an argument about **the** cause of mountains. Nowadays, most scientists prefer to work inside, so they don ’ t build physical models so much as to make **computer** simulations. But a **computer** simulation is a kind of a model. It ’ s a model that ’ s made with mathematics, and like **the** physical models of **the** 19th century, it ’ s very important for thinking about causes. So one of **the** big questions to do with climate change, we have tremendous amounts of evidence that **the** Earth is warming up. This **slide** here, **the** black line shows **the** measurements that scientists have taken for **the** last 150 years showing that **the** Earth ’ s temperature has steadily increased, and you can see in particular that in **the** last 50 years there ’ s been this dramatic increase of nearly one degree centigrade, or almost two degrees Fahrenheit. So what, though, is driving that change? How can we know what ’ s causing **the** observed warming? Well, scientists can model it using a **computer** simulation. So this diagram illustrates a **computer** simulation that has looked at all **the** different factors that we know can influence **the** Earth ’ s climate, so sulfate particles from air pollution, volcanic dust from volcanic eruptions, changes in solar radiation, and, of course, **greenhouse** gases. And they asked **the** question, what set of variables put into a model will reproduce what we actually see in real life? So here is **the** real life in black. Here ’ s **the** model in this light gray, and **the** answer is a model that includes, it ’ s **the** answer E on that SAT, all of **the** above. **The** only way you can reproduce **the** observed temperature measurements is with all of these things put together, including **greenhouse** gases, and in particular you can see that **the** increase in **greenhouse** gases tracks this very dramatic increase in temperature over **the** last 50 years. And so this is why climate scientists say it ’ s not just that we know that climate change is happening, we know that **greenhouse** gases are a major part of **the** reason why. So now because there all these different things that scientists do, **the** philosopher Paul Feyerabend famously said,"**The** only principle in science that doesn ' t inhibit progress is : anything goes."Now this quotation has often been taken out of context, because Feyerabend was not actually saying that in science anything goes. What he was saying was, actually **the** full quotation is,"If you press me to say what is **the** method of science, I would have to say : anything goes."What he was trying to say is that scientists do a lot of different things. Scientists are creative. But then this pushes **the** question back : If scientists don ' t use a single method, then how do they decide what ' s right and what ' s wrong? And who judges? And **the** answer is, scientists judge, and they judge by judging evidence. Scientists collect evidence in many different ways, but however they collect it, they have to subject it to scrutiny. And this led **the** sociologist Robert Merton to focus on this question of how scientists scrutinize data and evidence, and he said they do it in a way he called"organized skepticism."And by that he meant it ' s organized because they do it collectively, they do it as a group, and skepticism, because they do it from a position of distrust. That is to say, **the** burden of proof is on **the** person with a novel claim. And in this sense, science is intrinsically conservative. It ' s quite hard to persuade **the** scientific community to say,"Yes, we know something, this is true."So despite **the** popularity of **the** concept of paradigm shifts, what we find is that actually, really major changes in scientific thinking are relatively rare in **the** history of science. So finally that brings us to one more idea : If scientists judge evidence collectively, this has led historians to focus on **the** question of consensus, and to say that at **the** end of **the** day, what science is, what scientific knowledge is, is **the** consensus of **the** scientific experts who through this process of organized scrutiny, collective scrutiny, have judged **the** evidence and come to a conclusion about it, either yea or nay. So we can think of scientific knowledge as a consensus of experts. We can also think of science as being a kind of a jury, except it ' s a very special kind of jury. It ' s not a jury of your peers, it ' s a jury of **geeks**. It ' s a jury of men and women with Ph. D. s, and unlike a conventional jury, which has only two choices, guilty or not guilty, **the** scientific jury actually has a number of choices. Scientists can say yes, something ' s true. Scientists can say no, it ' s false. Or, they can say, well it might be true but we need to work more and collect more evidence. Or, they can say it might be true, but we don ' t know how to answer **the** question and we ' re going to put it aside and maybe we ' ll come back to it later. That ' s what scientists call"intractable."But this leads us to one final problem : If science is what scientists say it is, then isn ' t that just an appeal to authority? And weren ' t we all taught in school that **the** appeal to authority is a logical fallacy? Well, here ' s **the** paradox of modern science, **the** paradox of **the** conclusion I think historians and philosophers and sociologists have come to, that actually science is **the** appeal to authority, but it ' s not **the** authority of **the** individual, no matter how smart that individual is, like Plato or Socrates or Einstein. It ' s **the** authority of **the** collective community. You can think of it is a kind of wisdom of **the** crowd, but a very special kind of crowd. Science does appeal to authority, but it ' s not based on any individual, no matter how smart that individual may be. It ' s based on **the** collective wisdom, **the** collective knowledge, **the** collective work, of all of **the** scientists who have worked on a particular problem. Scientists have a kind of culture of collective distrust, this"show me"culture, illustrated by this nice woman here showing her colleagues her evidence. Of course, these people don ' t really look like scientists, because they ' re much too happy. **Okay**, so that brings me to my final point. Most of us get up in **the** morning. Most of us trust our cars. Well, see, now I ' m thinking, I ' m in Manhattan, this is a bad analogy, but most Americans who don ’ t live in Manhattan get up in **the** morning and get in their cars and turn on that ignition, and their cars work, and they work incredibly well. **The** modern automobile hardly ever breaks down. So why is that? Why do cars work so well? It ’ s not because of **the** genius of Henry Ford or Karl Benz or even Elon Musk. It ’ s because **the** modern automobile is **the** product of more than 100 years of work by hundreds and thousands and tens of thousands of people. **The** modern automobile is **the** product of **the** collected work and wisdom and experience of every man and woman who has ever worked on a car, and **the** reliability of **the** technology is **the** result of that accumulated effort. We benefit not just from **the** genius of Benz and Ford and Musk but from **the** collective intelligence and hard work of all of **the** people who have worked on **the** modern car. And **the** same is true of science, only science is even older. Our basis for trust in science is actually **the** same as our basis in trust in technology, and **the** same as our basis for trust in anything, namely, experience. But it shouldn ’ t be blind trust any more than we would have blind trust in anything. Our trust in science, like science itself, should be based on evidence, and that means that scientists have to become better communicators. They have to explain to us not just what they know but how they know it, and it means that we have to become better listeners. Thank you very much.

Text Sample 48: POS Dim 1 – 2024 – Score 20.00 – 2024/t003227.txt

Chris Anderson : Demis, so good to have you here. Demis Hassabis : It ’ s **fantastic** to be here, thanks, Chris. Now, you told Time Magazine,”I want to understand **the** big questions, **the** really big ones that you normally go into philosophy or physics if you ’ re interested in them. I thought building AI would be **the** fastest route to answer some of those questions.”Why did you think that? DH : Well, I guess when I was a kid, my favorite subject was physics, and I was interested in all **the** big questions, fundamental nature of reality, what is consciousness, you know, all **the** big ones. And usually you go into physics, if you ’ re interested in that. But I read a lot of **the** great physicists, some of my all-time scientific heroes like Feynman and so on. And I realized, in **the** last, sort of 20, 30 years, we haven ’ t made much progress in understanding some of these fundamental laws. So I thought, why not build **the** ultimate tool to help us, which is **artificial** intelligence. And at **the** same time, we could also maybe better understand ourselves and **the** brain better, by doing that too. So not only was it incredible tool, it was also useful for some of **the** big questions itself. CA : Super interesting. So obviously AI can do so many things, but I think for this conversation, I ’ d love to focus in on this **theme** of what it might do to **unlock the** really big questions, **the** giant scientific breakthroughs, because it ’ s been such a **theme** driving you and your company. DH : So I mean, one of **the** big things AI can do, and I ’ ve always thought about, is we ’ re getting, you know, even back 20, 30 years ago, **the** beginning of **the** internet era and **computer** era, **the** amount of data that was being produced and also scientific data, just too much for **the** human mind to comprehend in many cases. And I think one of **the** uses of AI is to find patterns and insights in huge amounts of data and then **surface** that to **the** human scientists to make sense of and make new hypotheses and conjectures. So it seems to me very compatible with **the** scientific method. CA : Right. But game play has played a huge role in your own journey in figuring this thing out. Who is this young lad on **the left** there? Who is that? DH : So that was me, I think I must have been about around nine years old. I ’ m captaining **the** England Under 11 team, and we ’ re playing in a Four Nations tournament, that ’ s why we ’ re all in red. I think we ’ re playing France, Scotland and Wales, I think it was. CA : That is so weird, because that happened to me too. In my dreams. And it wasn ’ t just chess, you loved all kinds of games. DH : I loved all kinds of games, yeah. CA : And when you launched DeepMind, pretty quickly, you started having it tackle game play. Why? DH : Well, look, I mean, games actually got me into AI in **the** first place because while we were doing things like, we used to go on training camps with **the** England team and so on. And actually back then, I guess it was in **the** mid ’ 80s, we would use **the** very early chess **computers**, if you remember them, to train against, as well as playing against each other. And they were big lumps of plastic, you know, physical boards that you used to, some of you remember, used to actually press **the** squares down and there were LED lights, came on. And I remember actually, not just thinking about **the** chess, I was actually just fascinated by **the** fact that this lump of plastic, someone had programmed it to be smart and actually play chess to a really high standard. And I was just amazed by that. And that got me thinking about thinking. And how does **the** brain come up with these thought processes, these ideas, and then maybe how we could mimic that with **computers**. So yeah, it ’ s been a whole **theme** for my whole life, really. CA : But you raised all this money to launch DeepMind, and pretty soon you were using it to do, for example, this. I mean, this is an odd use of it. What was going on here? DH : Well, we started off with games at **the** beginning of DeepMind. This was back in 2010, so this is from about 10 years ago, it was our first big breakthrough. Because we started off with classic Atari games from **the** 1970s, **the** simplest kind of **computer** games there are out there. And one of **the** reasons we used games is they ’ re very convenient to test out your ideas and your algorithms. They ’ re really fast to test. And also, as your systems get more powerful, you can choose harder and harder games. And this was actually **the** first time ever that our machine surprised us, **the** first of many times, which, it figured out in this game called Breakout, that you could send **the** ball round **the** back of **the** wall, and actually, it would be much safer way to knock out all **the** tiles of **the** wall. It ’ s a classic Atari game there. And that was our first real aha moment. CA : So this thing was not programmed to have any strategy. It was just told, try and figure out a way of winning. You just move **the** bat at **the** bottom and see if you can find a way of winning. DH : It was a real revolution at **the** time. So this was in 2012, 2013 where we coined these terms”deep reinforcement **learning**.”And **the** key thing about them is that those systems were learning directly from **the** pixels, **the** raw pixels on **the** screen, but they weren ’ t being told anything else. So they were being told, maximize **the** score, here are **the** pixels on **the** screen, 30, 000 pixels. **The** system has to make sense on its own from first principles what ’ s going on, what it ’ s controlling, how to get points. And that ’ s **the** other nice thing about using games to begin with. They have clear objectives, to win, to get scores. So you can kind of measure very easily that your systems are improving. CA : But there was a direct line from that to this moment a few years later, where country of South Korea and many other parts of Asia and in fact **the** world went crazy over – over what? DH : Yeah, so this was **the** pinnacle of – this is in 2016 – **the** pinnacle of our games-playing work, where, so we ’ d done Atari, we ’ d done some more complicated games. And then we reached **the** pinnacle, which was **the** game of Go, which is what they play in Asia instead of chess, but it ’ s actually more complex than chess. And **the** actual brute force algorithms that were used to kind of crack chess were not possible with Go because it ’ s a much more pattern-based game, much more intuitive game. So even though Deep Blue beat Garry Kasparov in **the** ’ 90s, it took another 20 years for our program, AlphaGo, to beat **the** world champion at

Go. And we always thought, myself and **the** people working on this project for many years, if you could build a system that could beat **the** world champion at Go, it would have had to have done something very interesting. And in this case, what we did with AlphaGo, is it basically learned for itself, by playing millions and millions of games against itself, ideas about Go, **the** right strategies. And in fact invented its own new strategies that **the** Go world had never seen before, even though we ’ ve played Go for more than, you know, 2, 000 years, it ’ s **the** oldest board game in existence. So, you know, it was pretty astounding. Not only did it win **the** match, it also came up with brand new strategies. CA : And you continued this with a new strategy of not even really teaching it anything about Go, but just setting up systems that just from first principles would play so that they could teach themselves from scratch, Go or chess. Talk about AlphaZero and **the** amazing thing that happened in chess then. DH : So following this, we started with AlphaGo by giving it all of **the** human games that are being played on **the** internet. So it started that as a basic starting point for its knowledge. And then we wanted to see what would happen if we started from scratch, from literally random play. So this is what AlphaZero was. That ’ s why it ’ s **the** zero in **the** name, because it started with zero prior knowledge And **the** reason we did that is because then we would build a system that was more general. So AlphaGo could only play Go, but AlphaZero could play any two-player game, and it did it by playing initially randomly and then slowly, incrementally improving. Well, not very slowly, actually, within **the** course of 24 hours, going from random to better than world-champion level. CA : And so this is so amazing to me. So I ’ m more familiar with chess than with Go. And for decades, thousands and thousands of AI experts worked on building incredible chess **computers**. Eventually, they got better than humans. You had a moment a few years ago, where in nine hours, AlphaZero taught itself to play chess better than any of those systems ever did. Talk about that. DH : It was a pretty incredible moment, actually. So we set it going on chess. And as you said, there ’ s this rich history of chess and AI where there are these expert systems that have been programmed with these chess ideas, chess algorithms. And you have this amazing, you know, I remember this day very clearly, where you sort of sit down with **the** system starting off random, you know, in **the** morning, you go for a cup of coffee, you come back. I can still just about beat it by lunchtime, maybe just about. And then you let it go for another four hours. And by dinner, it ’ s **the** greatest chess-playing entity that ’ s ever existed. And, you know, it ’ s quite amazing, like, looking at that live on something that you know well, you know, like chess, and you ’ re expert in and actually just seeing that in front of your eyes. And then you extrapolate to what it could then do in science or something else, which of course, games were only a means to an end. They were never **the** end in themselves. They were just **the** training ground for our ideas and to make quick progress in a matter of, you know, less than five years actually went from Atari to Go. CA : I mean, this is why people are in awe of AI and also kind of terrified by it. I mean, it ’ s not just incremental improvement. **The** fact that in a few hours you can achieve what millions of humans over centuries have not been able to achieve. That gives you pause for thought. DH : It does, I mean, it ’ s a hugely powerful technology. It ’ s going to be incredibly transformative. And we have to be very thoughtful about how we use that capability. CA : So talk about this use of it because this is again, this is another extension of **the** work you ’ ve done, where now you ’ re turning it to something incredibly useful for **the** world. What are all **the** letters on **the left**, and what ’ s on **the** right? DH : This was always my aim with AI from a kid, which is to use it to accelerate scientific discovery. And actually, ever since doing my undergrad at Cambridge, I had this problem in mind one day for AI, it ’ s called

the protein-folding problem. And it ' s kind of like a 50-year grand challenge in **biology**. And it ' s very simple to explain. Proteins are essential to life. They ' re **the** building blocks of life. Everything in your body depends on proteins. A protein is sort of described by its amino acid sequence, which you can think of as roughly **the** genetic sequence describing **the** protein, so that are **the** letters. CA : And each of those letters represents in itself a complex molecule? DH : That ' s right, each of those letters is an amino acid. And you can think of them as a kind of string of beads there at **the** bottom, left, right? But in nature, in your body or in an animal, this string, a sequence, turns into this beautiful shape on **the** right. That ' s **the** protein. Those letters describe that shape. And that ' s what it looks like in nature. And **the** important thing about that 3D structure is **the** 3D structure of **the** protein goes a long way to telling you what its function is in **the** body, what it does. And so **the** protein-folding problem is : Can you directly predict **the** 3D structure just from **the** amino acid sequence? So literally if you give **the** machine, **the** AI system, **the** letters on **the left**, can it produce **the** 3D structure on **the** right? And that ' s what AlphaFold does, our program does. CA : It ' s not **calculating** it from **the** letters, it ' s looking at patterns of other folded proteins that are known about and somehow learning from those patterns that this may be **the** way to do this? DH : So when we started this project, actually straight after AlphaGo, I thought we were ready. Once we ' d cracked Go, I felt we were finally ready after, you know, almost 20 years of working on this stuff to actually tackle some scientific problems, including protein folding. And what we start with is painstakingly, over **the** last 40-plus years, experimental biologists have pieced together around 150, 000 protein structures using very complicated, you know, X-ray crystallography **techniques** and other complicated experimental **techniques**. And **the** rule of thumb is that it takes one PhD student their whole PhD, so four or five years, to uncover one structure. But there are 200 million proteins known to nature. So you could just, you know, take forever to do that. And so we managed to actually fold, using AlphaFold, in one year, all those 200 million proteins known to science. So that ' s a billion years of PhD time saved. CA : So it ' s amazing to me just how reliably it works. I mean, this shows, you know, here ' s **the** model and you do **the** experiment. And sure enough, **the** protein turns out **the** same way. Times 200 million. DH : And **the** more deeply you go into proteins, you just start appreciating how exquisite they are. I mean, look at how beautiful these proteins are. And each of these things do a special function in nature. And they ' re almost like works of art. And it ' s still astounds me today that AlphaFold can predict, **the** green is **the** ground truth, and **the blue** is **the** prediction, how well it can predict, is to within **the width** of an atom on average, is how accurate **the** prediction is, which is what is needed for biologists to use it, and for drug design and for disease understanding, which is what AlphaFold **unlocks**. CA : You made a surprising decision, which was to give away **the** actual results of your 200 million proteins. DH : We open-sourced AlphaFold and gave everything away on a huge database with our wonderful colleagues, **the** European Bioinformatics Institute. CA : I mean, you ' re part of Google. Was there a phone call saying,"Uh, Demis, what did you just do?"DH : You know, I ' m lucky we have very supportive, Google ' s really supportive of science and understand **the** benefits this can bring to **the** world. And, you know, **the** argument here was that we could only ever have even scratched **the surface** of **the** potential of what we could do with this. This, you know, maybe like a millionth of what **the** scientific community is doing with it. There ' s over a million and a half biologists around **the** world have used AlphaFold and its predictions. We think that ' s almost every biologist in **the** world is making use of this now, every pharma company. So we ' ll never know probably what **the** full impact of it all is. CA : But you ’ re continuing this work in a new company that ’ s spinning out of Google called Isomorph. DH : Isomorphic. CA : Isomorphic. Give us just a sense of **the** vision there. What ’ s **the** vision? DH : AlphaFold is a sort of fundamental **biology** tool. Like, what are these 3D structures, and then what might they do in nature? And then if you, you know, **the** reason I thought about this and was so excited about this, is that this is **the** beginnings of understanding disease and also maybe helpful for designing drugs. So if you know **the** shape of **the** protein, and then you can kind of figure out which part of **the surface** of **the** protein you ’ re going to target with your drug compound. And Isomorphic is extending this work we did in AlphaFold into **the chemistry** space, where we can design chemical compounds that will bind exactly to **the** right spot on **the** protein and also, importantly, to nothing else in **the** body. So it doesn ’ t have any side effects and it ’ s not toxic and so on. And we ’ re building many other AI models, sort of sister models to AlphaFold to help predict, make predictions in **chemistry** space. CA : So we can expect to see some pretty dramatic health medicine breakthroughs in **the** coming few years. DH : I think we ’ ll be able to get down drug discovery from years to maybe months. CA : OK. Demis, I ’ d like to change direction a bit. Our mutual friend, Liv Boeree, gave a talk last year at TEDAI that she called **the**”Moloch Trap.”**The** Moloch Trap is a situation where organizations, companies in a competitive situation can be driven to do things that no individual running those companies would by themselves do. I was really struck by this talk, and it ’ s felt, as a sort of layperson observer, that **the** Moloch Trap has been shockingly in effect in **the** last couple of years. So here you are with DeepMind, sort of pursuing these amazing medical breakthroughs and scientific breakthroughs, and then suddenly, kind of out of left field, OpenAI with Microsoft releases ChatGPT. And **the** world goes crazy and suddenly goes,”Holy crap, AI is...”you know, everyone can use it. And there ’ s a sort of, it felt like **the** Moloch Trap in action. I think Microsoft CEO Satya Nadella actually said,”Google is **the** 800-pound gorilla in **the search** space. We wanted to make Google dance.”How...? And it did, Google did dance. There was a dramatic response. Your role was changed, you took over **the** whole Google AI effort. Products were rushed out. You know, Gemini, some part amazing, part embarrassing. I ’ m not going to ask you about Gemini because you ’ ve addressed it elsewhere. But it feels like this was **the** Moloch Trap happening, that you and others were pushed to do stuff that you wouldn ’ t have done without this sort of catalyzing competitive thing. Meta did something similar as well. They rushed out an **open-source** version of AI, which is arguably a reckless act in itself. This seems **terrifying** to me. Is it **terrifying**? DH : Look, it ’ s a complicated topic, of course. And, first of all, I mean, there are many things to say about it. First of all, we were working on many large language models. And in fact, obviously, Google research actually invented Transformers, as you know, which was **the** architecture that allowed all this to be possible, five, six years ago. And so we had many large models internally. **The** thing was, I think what **the** ChatGPT moment did that changed was, and fair play to them to do that, was they demonstrated, I think somewhat surprisingly to themselves as well, that **the** public were ready to, you know, **the** general public were ready to embrace these systems and actually find value in these systems. Impressive though they are, I guess, when we ’ re working on these systems, mostly you ’ re focusing on **the** flaws and **the** things they don ’ t do and hallucinations and things you ’ re all familiar with now. We ’ re thinking, you know, would anyone really find that useful given that it does this and that? And we would want them to improve those things first, before putting them out. But interestingly, it turned out that even with those flaws, many tens of millions of people still find them very useful. And so that was an interesting update on maybe **the** convergence of products and **the** science that actually, all of these amazing things we ' ve been doing in **the** lab, so to speak, are actually ready for prime time for general use, beyond **the** rarefied world of science. And I think that ' s pretty exciting in many ways. CA : So at **the** moment, we ' ve got this exciting **array** of products which we ' re all **enjoying**. And, you know, all this generative AI stuff is amazing. But let ' s roll **the** clock forward a bit. Microsoft and OpenAI are reported to be building or investing like 100 billion dollars into an absolute monster database supercomputer that can offer compute at orders of magnitude more than anything we have today. It takes like five gigawatts of energy to drive this, it ' s estimated. That ' s **the** energy of New York City to drive a data center. So we ' re pumping all this energy into this giant, vast brain. Google, I presume is going to match this type of investment, right? DH : Well, I mean, we don ' t talk about our specific numbers, but you know, I think we ' re investing more than that over time. So, and that ' s one of **the** reasons we teamed up with Google back in 2014, is kind of we knew that in order to get to AGI, we would need a lot of compute. And that ' s what ' s transpired. And Google, you know, had and still has **the** most **computers**. CA : So Earth is building these giant **computers** that are going to basically, these giant brains, that are going to power so much of **the** future economy. And it ' s all by companies that are in competition with each other. How will we avoid **the** situation where someone is getting a lead, someone else has got 100 billion dollars invested in their thing. Isn ' t someone going to go,"Wait a sec. If we used reinforcement learning here to maybe have **the** AI tweak its own code and rewrite itself and make it so [powerful], we might be able to catch up in nine hours over **the** weekend with what they ' re doing. Roll **the** dice, dammit, we have no choice. Otherwise we ' re going to lose a fortune for our shareholders."How are we going to avoid that? DH : Yeah, well, we must avoid that, of course, clearly. And my view is that as we get closer to AGI, we need to collaborate more. And **the** good news is that most of **the** scientists involved in these labs know each other very well. And we talk to each other a lot at conferences and other things. And this technology is still relatively nascent. So probably it ' s OK what ' s happening at **the** moment. But as we get closer to AGI, I think as a society, we need to start thinking about **the** types of architectures that get built. So I ' m very optimistic, of course, that ' s why I spent my whole life working on AI and working towards AGI. But I suspect there are many ways to build **the** architecture safely, robustly, reliably and in an understandable way. And I think there are almost certainly going to be ways of building architectures that are unsafe or risky in some form. So I see a sort of, a kind of bottleneck that we have to get humanity through, which is building safe architectures as **the** first types of AGI systems. And then after that, we can have a sort of, a flourishing of many different types of systems that are perhaps sharded off those safe architectures that ideally have some mathematical guarantees or at least some practical guarantees around what they do. CA : Do governments have an essential role here to define what a level playing field looks like and what is absolutely taboo? DH : Yeah, I think it ' s not just about – actually I think government and civil society and academia and all parts of society have a critical role to play here to shape, along with industry labs, what that should look like as we get closer to AGI and **the** cooperation needed and **the** collaboration needed, to prevent that kind of runaway race dynamic happening. CA : OK, well, it sounds like you remain optimistic. What ' s this image here? DH : That ' s one of my favorite images, actually. I call it, like, **the** tree of all knowledge. So, you know, we ' ve been talking a lot about science, and a lot of science can be boiled down to if you imagine all **the** knowledge that exists in **the** world as a tree of knowledge, and then maybe what we know today as a civilization is some, you know, small subset of that. And I see AI as this tool that allows us, as scientists, to explore, potentially, **the** entire tree one day. And we have this idea of root node problems that, like AlphaFold, **the** protein-folding problem, where if you could crack them, it **unlocks** an entire new branch of discovery or new research. And that ' s what we try and focus on at DeepMind and Google DeepMind to crack those. And if we get this right, then I think we could be, you know, in this incredible new era of radical abundance, curing all diseases, spreading consciousness to **the** stars. You know, maximum human flourishing. CA : We ' re out of time, but what ' s **the** last example of like, in your dreams, this dream question that you think there is a shot that in your lifetime AI might take us to? DH : I mean, once AGI is built, what I ' d like to use it for is to try and use it to understand **the** fundamental nature of reality. So do experiments at **the** Planck scale. You know, **the** smallest possible scale, theoretical scale, which is almost like **the** resolution of reality. CA : You know, I was brought up religious. And in **the** Bible, there ' s a story about **the** tree of knowledge that doesn ' t work out very well. Is there any scenario where we discover knowledge that **the universe** says,"Humans, you may not know that."DH : Potentially. I mean, there might be some unknowable things. But I think scientific method is **the** greatest sort of invention humans have ever come up with. You know, **the** enlightenment and scientific discovery. That ' s what ' s built this incredible modern civilization around us and all **the** tools that we use. So I think it ' s **the** best **technique** we have for understanding **the** enormity of **the universe** around us. CA : Well, Demis, you ' ve already changed **the** world. I think probably everyone here will be cheering you on in your efforts to ensure that we continue to accelerate in **the** right direction. DH : Thank you. CA : Demis Hassabis.

Text Sample 49: POS Dim 1 – 2024 – Score 19.00 – 2024/t003234.txt

I want to tell you what I see coming. I ' ve been lucky enough to be working on AI for almost 15 years now. Back when I started, to describe it as fringe would be an understatement. Researchers would say,"No, no, we ' re only working on machine **learning**."Because working on AI was seen as way too out there. In 2010, just **the** very mention of **the** phrase"AGI,"**artificial** general intelligence, would get you some seriously strange looks and even a cold shoulder."You ' re actually building AGI?"people would say."Isn ' t that something out of science fiction?"People thought it was 50 years away or 100 years away, if it was even possible at all. Talk of AI was, I guess, kind of embarrassing. People generally thought we were weird. And I guess in some ways we kind of were. It wasn ' t long, though, before AI started beating humans at a whole range of tasks that people previously thought were way out of reach. Understanding images, translating languages, transcribing speech, playing Go and chess and even diagnosing diseases. People started waking up to **the** fact that AI was going to have an enormous impact, and they were rightly asking technologists like me some pretty tough questions. Is it true that AI is going to solve **the** climate crisis? Will it make personalized education available to everyone? Does it mean we ' ll all get universal basic income and we won ' t have to work anymore? Should I be afraid? What does it mean for weapons and war? And of course, will China win? Are we in a race? Are we headed for a mass misinformation apocalypse? All good questions. But it was actually a simpler and much more kind of fundamental question that left me puzzled. One that actually gets to **the** very heart of my work every day. One morning over breakfast, my six-year-old nephew Caspian was playing with Pi, **the** AI I created at my last company, Inflection. With a mouthful of scrambled eggs, he looked at me plain in **the** face and said,”But Mustafa, what is an AI anyway?”He ’ s such a sincere and curious and optimistic little guy. He ’ d been talking to Pi about how cool it would be if one day in **the** future, he could visit dinosaurs at **the** zoo. And how he could make infinite amounts of chocolate at home. And why Pi couldn ’ t yet play I Spy.”Well,”I said,”it ’ s a clever piece of software that ’ s read most of **the** text on **the** open internet, and it can talk to you about anything you want.””Right. So like a person then?”I was stumped. Genuinely left scratching my head. All my boring stock answers came rushing through my mind.”No, but AI is just another general-purpose technology, like printing or steam.”It will be a tool that will augment us and make us smarter and more productive. And when it gets better over time, it ’ ll be like an all-knowing oracle that will help us solve grand scientific challenges.”You know, all of these responses started to feel, I guess, a little bit defensive. And actually better suited to a policy seminar than breakfast with a no-nonsense six-year-old.”Why am I hesitating?”I thought to myself. You know, let ’ s be honest. My nephew was asking me a simple question that those of us in AI just don ’ t confront often enough. What is it that we are actually creating? What does it mean to make something totally new, fundamentally different to any invention that we have known before? It is clear that we are at an inflection point in **the** history of humanity. On our current trajectory, we ’ re headed towards **the** emergence of something that we are all struggling to describe, and yet we cannot control what we don ’ t understand. And so **the** metaphors, **the** mental models, **the** names, these all matter if we ’ re to get **the** most out of AI whilst limiting its potential downsides. As someone who embraces **the** possibilities of this technology, but who ’ s also always cared deeply about its ethics, we should, I think, be able to easily describe what it is we are building. And that includes **the** six-year-olds. So it ’ s in that spirit that I offer up today **the** following metaphor for helping us to try to grapple with what this moment really is. I think AI should best be understood as something like a new digital species. Now, don ’ t take this too literally, but I predict that we ’ ll come to see them as digital companions, new partners in **the** journeys of all our lives. Whether you think we ’ re on a 10-, 20- or 30-year path here, this is, in my view, **the** most accurate and most fundamentally honest way of describing what ’ s actually coming. And above all, it enables everybody to prepare for and shape what comes next. Now I totally get, this is a strong claim, and I ’ m going to explain to everyone as best I can why I ’ m making it. But first, let me just try to set **the** context. From **the** very first microscopic organisms, life on Earth stretches back billions of years. Over that time, life **evolved** and diversified. Then a few million years ago, something began to shift. After countless cycles of growth and adaptation, one of life ’ s branches began using tools, and that branch grew into us. We went on to produce a mesmerizing variety of tools, at first slowly and then with astonishing speed, we went from stone axes and fire to language, writing and eventually industrial technologies. One invention unleashed a thousand more. And in time, we became homo technologicus. Around 80 years ago, another new branch of technology began. With **the** invention of **computers**, we quickly jumped from **the** first mainframes and transistors to today ’ s **smartphones** and virtual-reality headsets. Information, knowledge, communication, computation. In this revolution, creation has exploded like never before. And now a new wave is upon us. **Artificial** intelligence. These waves of history are clearly speeding up, as each one is amplified and accelerated by **the** last. And if you look back, it ’ s clear that we are in **the** fastest and most consequential wave ever. **The** journeys of humanity and technology are now deeply intertwined. In just 18 months, over a billion people have used large language models. We ’ ve witnessed one landmark event after another.

Just a few years ago, people said that AI would never be creative. And yet AI now feels like an endless river of creativity, making poetry and images and music and video that stretch **the** imagination. People said it would never be empathetic. And yet today, millions of people **enjoy** meaningful conversations with AIs, talking about their hopes and dreams and helping them work through difficult emotional challenges. AIs can now drive cars, manage energy grids and even invent new molecules. Just a few years ago, each of these was impossible. And all of this is turbocharged by spiraling exponentials of data and computation. Last year, Inflection 2. 5, our last model, used five billion times more computation than **the** DeepMind AI that beat **the** old-school Atari games just over 10 years ago. That ' s nine orders of magnitude more computation. 10x per year, every year for almost a decade. Over **the** same time, **the** size of these models has grown from first tens of millions of parameters to then billions of parameters, and very soon, tens of trillions of parameters. If someone did nothing but read 24 hours a day for their entire life, they ' d consume eight billion words. And of course, that ' s a lot of words. But today, **the** most advanced AIs consume more than eight trillion words in a single month of training. And all of this is set to continue. **The** long arc of technological history is now in an extraordinary new phase. So what does this mean in practice? Well, just as **the** internet gave us **the** browser and **the smartphone** gave us apps, **the** cloud-based supercomputer is ushering in a new era of ubiquitous AIs. Everything will soon be represented by a conversational interface. Or, to put it another way, a personal AI. And these AIs will be infinitely knowledgeable, and soon they ' ll be factually accurate and reliable. They ' ll have near-perfect IQ. They ' ll also have exceptional EQ. They ' ll be kind, supportive, empathetic. These elements on their own would be transformational. Just imagine if everybody had a personalized tutor in their **pocket** and access to low-cost medical advice. A lawyer and a doctor, a business strategist and coach – all in your **pocket** 24 hours a day. But things really start to change when they develop what I call AQ, their"actions quotient."This is their ability to actually get stuff done in **the** digital and physical world. And before long, it won ' t just be people that have AIs. Strange as it may sound, every organization, from small business to nonprofit to national government, each will have their own. Every town, building and **object** will be represented by a unique interactive persona. And these won ' t just be mechanistic assistants. They ' ll be companions, confidants, colleagues, friends and partners, as varied and unique as we all are. At this point, AIs will convincingly imitate humans at most tasks. And we ' ll feel this at **the** most intimate of scales. An AI organizing a community get-together for an elderly neighbor. A sympathetic expert helping you make sense of a difficult diagnosis. But we ' ll also feel it at **the** largest scales. Accelerating scientific discovery, autonomous cars on **the** roads, drones in **the** skies. They ' ll both order **the** takeout and run **the** power station. They ' ll interact with us and, of course, with each other. They ' ll speak every language, take in every pattern of sensor data, sights, sounds, streams and streams of information, far surpassing what any one of us could consume in a thousand lifetimes. So what is this? What are these AIs? If we are to prioritize safety above all else, to ensure that this new wave always serves and amplifies humanity, then we need to find **the** right metaphors for what this might become. For years, we in **the** AI community, and I specifically, have had a tendency to refer to this as just tools. But that doesn ' t really capture what ' s actually happening here. AIs are clearly more dynamic, more ambiguous, more integrated and more emergent than mere tools, which are entirely subject to human control. So to contain this wave, to put human agency at its center and to mitigate **the** inevitable unintended consequences that are likely to arise, we should start to think about them as we might a new kind of digital species. Now it ' s just an analogy, it ' s not a literal description, and it ' s not perfect. For a start, they clearly aren ' t biological in any traditional sense, but just pause for a moment and really think about what they already do. They communicate in our languages. They see what we see. They consume unimaginably large amounts of information. They have memory. They have personality. They have creativity. They can even reason to some extent and formulate rudimentary plans. They can act autonomously if we allow them. And they do all this at levels of sophistication that is far beyond anything that we ' ve ever known from a mere tool. And so saying AI is mainly about **the** math or **the** code is like saying we humans are mainly about **carbon** and water. It ' s true, but it completely misses **the** point. And yes, I get it, this is a super arresting thought but I honestly think this frame helps sharpen our focus on **the** critical issues. What are **the** risks? What are **the** boundaries that we need to impose? What kind of AI do we want to build or allow to be built? This is a story that ' s still unfolding. Nothing should be accepted as a given. We all must choose what we create. What AIs we bring into **the** world, or not. These are **the** questions for all of us here today, and all of us alive at this moment. For me, **the** benefits of this technology are stunningly obvious, and they inspire my life ' s work every single day. But quite frankly, they ' ll speak for themselves. Over **the** years, I ' ve never shied away from highlighting risks and talking about downsides. Thinking in this way helps us focus on **the** huge challenges that lie ahead for all of us. But let ' s be clear. There is no path to progress where we leave technology behind. **The** prize for all of civilization is immense. We need solutions in health care and education, to our climate crisis. And if AI delivers just a **fraction** of its potential, **the** next decade is going to be **the** most productive in human history. Here ' s another way to think about it. In **the** past, **unlocking** economic growth often came with huge downsides. **The** economy expanded as people discovered new **continents** and opened up new frontiers. But they colonized populations at **the** same time. We built factories, but they were grim and dangerous places to work. We struck **oil**, but we **polluted the planet**. Now because we are still designing and building AI, we have **the** potential and opportunity to do it better, radically better. And today, we ' re not discovering a new **continent** and plundering its resources. We ' re building one from scratch. Sometimes people say that data or chips are **the** 21st century ' s new **oil**, but that ' s totally **the** wrong image. AI is to **the** mind what nuclear **fusion** is to energy. Limitless, abundant, world-changing. And AI really is different, and that means we have to think about it creatively and honestly. We have to push our analogies and our metaphors to **the** very limits to be able to grapple with what ' s coming. Because this is not just another invention. AI is itself an infinite inventor. And yes, this is exciting and promising and concerning and intriguing all at once. To be quite honest, it ' s pretty surreal. But step back, see it on **the** long view of glacial time, and these really are **the** very most appropriate metaphors that we have today. Since **the** beginning of life on Earth, we ' ve been **evolving**, changing and then creating everything around us in our human world today. And AI isn ' t something outside of this story. In fact, it ' s **the** very opposite. It ' s **the** whole of everything that we have created, distilled down into something that we can all interact with and benefit from. It ' s a reflection of humanity across time, and in this sense, it isn ' t a new species at all. This is where **the** metaphors end. Here ' s what I ' ll tell Caspian next time he asks. AI isn ' t separate. AI isn ' t even in some senses, new. AI is us. It ' s all of us. And this is perhaps **the** most promising and vital thing of all that even a six-year-old can get a sense for. As we build out AI, we can and must reflect all that is good, all that we love, all that is special about humanity : our empathy, our kindness, our curiosity and our creativity. This, I would argue, is **the** greatest challenge of **the** 21st century, but also **the** most wonderful, inspiring and hopeful opportunity for all of us. Thank you. Chris Anderson : Thank you Mustafa. It ’ s an amazing vision and a super powerful metaphor. You ’ re in an amazing position right now. I mean, you were connected at **the** hip to **the** amazing work happening at OpenAI. You ’ re going to have resources made available, there are reports of these giant new data centers, 100 billion dollars invested and so forth. And a new species can emerge from it. I mean, in your book, you did, as well as painting an incredible optimistic vision, you were super eloquent on **the** dangers of AI. And I ’ m just curious, from **the** view that you have now, what is it that most keeps you up at night? Mustafa Suleyman : I think **the** great risk is that we get stuck in what I call **the** pessimism aversion trap. You know, we have to have **the** courage to confront **the** potential of dark scenarios in order to get **the** most out of all **the** benefits that we see. So **the** good news is that if you look at **the** last two or three years, there have been very, very few downsides, right? It ’ s very hard to say explicitly what harm an LLM has caused. But that doesn ’ t mean that that ’ s what **the** trajectory is going to be over **the** next 10 years. So I think if you pay attention to a few specific capabilities, take for example, autonomy. Autonomy is very obviously a threshold over which we increase risk in our society. And it ’ s something that we should step towards very, very closely. **The** other would be something like recursive self-improvement. If you allow **the** model to independently self-improve, update its own code, explore an environment without oversight, and, you know, without a human in control to change how it operates, that would obviously be more dangerous. But I think that we ’ re still some way away from that. I think it ’ s still a good five to 10 years before we have to really confront that. But it ’ s time to start talking about it now. CA : A digital species, unlike any biological species, can replicate not in nine months, but in nine nanoseconds, and produce an indefinite number of **copies** of itself, all of which have more power than we have in many ways. I mean, **the** possibility for unintended consequences seems pretty immense. And isn ’ t it true that if a problem happens, it could happen in an hour? MS : No. That is really not true. I think there ’ s no evidence to suggest that. And I think that, you know, that ’ s often referred to as **the**”intelligence **explosion**.”And I think it is a theoretical, hypothetical maybe that we ’ re all kind of curious to explore, but there ’ s no evidence that we ’ re anywhere near anything like that. And I think it ’ s very important that we choose our words super carefully. Because you ’ re right, that ’ s one of **the** weaknesses of **the** species framing, that we will design **the** capability for self-replication into it if people choose to do that. And I would actually argue that we should not, that would be one of **the** dangerous capabilities that we should step back from, right? So there ’ s no chance that this will”emerge”accidentally. I really think that ’ s a very low **probability**. It will happen if engineers deliberately design those capabilities in. And if they don ’ t take enough efforts to deliberately design them out. And so this is **the** point of being explicit and transparent about trying to introduce safety by design very early on. CA : Thank you, your vision of humanity injecting into this new thing **the** best parts of ourselves, avoiding all those weird, biological, freaky, horrible tendencies that we can have in certain circumstances, I mean, that is a very inspiring vision. And thank you so much for coming here and sharing it at TED. Thank you, good luck.

Text Sample 50: POS Dim 1 – 2024 – Score 18.00 – 2024/t003168.txt

When I look up at **the** sky at night, I see 100 billion stars of **the** Milky Way galaxy. They look like lights in cabins of a giant spaceship, **The** Milky Way, sailing through space. And I wonder if there are other passengers in those cabins. There are 100 billion of them, comparable to **the** number of people who ever lived on Earth. It would be arrogant to think otherwise, that we are alone, that we are unique and special, especially if you read **the** news every day. We are not **the** pinnacle of creation. There is room for improvement. I ' m just a curious farm boy, and I wonder about **the** world around me. And I hate to behave like **the** adults in **the** room, because they often pretend to know more than we actually know. And that bothered me since I was a young kid. And so I decided to become a scientist and answer **the** questions based on evidence, not based on prejudice, not based on **the** politics of getting **the** largest number of likes on social media. I don ' t have a footprint on social media. I **enjoy** nature. Whatever it brings to our doorstep is welcome. So let ' s just look around. And for 70 years, we ' ve been **searching** for radio signals. This is equivalent to staying at home and waiting for a phone call that may never come because nobody cares that we are lonely. It may also be that others are addicted to digital screens and they live in a virtual reality, as we are at this point in time. A much better approach is to check if there is any **object** in our backyard that may have arrived from a neighbor ' s yard. Like a tennis ball, that may tell us that **the** neighbor plays tennis. And we haven ' t really checked until **the** last decade. **The** first **object** to have been reported by astronomers that came from outside **the** solar system looked really weird. It was discovered by a telescope in Hawaii. When it passed close to Earth, it was **the** size of a football field. What you see behind me is **the** artist ' s depiction. It looked really weird because as it was tumbling every eight hours, **the** amount of sunlight reflected from it changed by a factor of ten, which meant that it has a very extreme shape, most likely flat, like a pancake. And moreover, it exhibited a push away from **the** sun by some mysterious force because there was no evaporation, no cometary tail around it, no dust, no gas. So **the** question was, what is pushing it? And I suggested that maybe it ' s **the** reflection of sunlight, but for that, **the object** had to be very thin, like a sail. And that meant that it was not produced naturally. Maybe it ' s a **surface** layer, maybe it ' s space trash, like a plastic bag tumbling in **the** wind. And so we go back 70 years to a question that Enrico Fermi, **the** physicist, asked at Los Alamos,"Where is everybody?"Well, this is a question that single people often ask. But if you stay at home – You will not find anyone. You have to go to dating sites. At **the** very least, you need to look through your windows for other people. And he didn ' t seek **the** evidence. He was just asking **the** question and kept repeating it. And if we don ' t look for evidence, we will not find anything. It ' s a self-fulfilling prophecy. It ' s a way to maintain our ignorance. And science is better than politics. We can find **the** evidence if we allocate **the** funds for it. This is a real image, what you see behind me, it ' s **the** Tesla Roadster car that was put as a dummy payload on **the** Falcon Heavy launch of 2018. It ' s now moving in an elliptical orbit around **the** Sun, and perhaps in 20 million years, it will collide with Earth. And if it will do so unexpectedly, some of my colleagues would argue"This is a **rock** of a type that we ' ve never seen before."We cannot see it with our best telescopes because it ' s too small. It doesn ' t reflect enough sunlight. ' Oumuamua was **the** size of a football field, big enough for us to see. And so **the** next Copernican revolution would be that we are not at **the** intellectual center of **the universe**. Not only that we are not at **the** physical center of **the universe**, but actually, you know, we arrive to **the** play relatively late. We are not at **the** center of stage. **The** play is not about us. We should be modest. We keep thinking that it ' s about us, but it ' s not. And we better find other actors that will tell us what **the** play is about. And people often say extraordinary claims require extraordinary evidence, but they are not seeking **the** evidence. Actually, extraordinary evidence requires extraordinary funding. Elon Musk argued recently,"I don ' t see any aliens."But new scientific knowledge does not fall into our lap. We had to invest 10 billion dollars in **the** Large Hadron Collider in order to find **the** Higgs boson. We had to invest 10 billion dollars in **the** Webb telescope in order to find **the** first generation of galaxies. This is **the** way science is done. You need to put **the** effort in order to find something new. And only over **the** past decade, we discovered **objects** that came from outside **the** solar system. **The** first one was actually a decade ago. It was a meteor, an **object** half a meter in size that collided with Earth and burned up in **the** atmosphere. It was spotted by US government satellites. **The** fireball that it generated released a few percent of **the** Hiroshima atomic bomb energy, and it was moving too fast to be bound to **the** Sun ' s gravity. And so we **concluded** it ' s interstellar. It came from outside **the** solar system. Could it be a Voyager-like meteor? Imagine our own spacecraft colliding with a **planet** like Earth. In **the** future, it would appear as a meteor of unusual material strength and unusual speed, which are exactly **the** properties of this meteor from 2014. And then ' Oumuamua was discovered in 2017, and finally a comet appeared, also from interstellar space, was moving too fast. And so my colleagues argued,"Well, this one looks familiar. Doesn ' t it convince you that **the** others are natural in origin? **Rocks** of a type that we ' ve never seen before?"And I say, if I go down **the** street and I see a weird person, and after that I see a normal person, it doesn ' t make **the** weird person normal. Now **the** US – Director of National intelligence – Avril Haines delivered three reports to **the** US Congress, talking about unidentified anomalous phenomena. **The** good news is **the** sky is not classified. We don ' t need to wait for **the** US government to tell us what lies outside **the** solar system. Their day job is national security. My day job is figuring out what lies beyond **the** solar system. And **the** sky is not classified. We can answer **the** question ourselves. So I ' m leading **the** Galileo project. We built an observatory at Harvard University that monitors **the** sky 24 / 7, looking for **objects** that are not familiar, not birds, balloons, drones, **airplanes**, satellites. So far, we monitored half a million **objects**. Haven ' t found anything unusual yet, but we keep looking and we are using machine-learning software to figure out what we are looking at. But **the** most exciting **endeavor** that I was involved in is an expedition to **the** Pacific Ocean near Papua New Guinea, to look for **the** materials from this meteor that I described before. And **the** US Space Command issued a letter to NASA confirming at **the** 99. 999 percent that this **object** indeed originated from outside **the** solar system. Based on its high speed, it was moving faster than 95 percent of all stars in **the** vicinity of **the** Sun. It exploded in **the** lower atmosphere, about 90 kilometers away from Manus Island in Papua New Guinea, and that meant that **the object** had material strength tougher than even **iron** meteorites. And so I led an expedition in June 2023. You can see **the** team on **the** ship that was fittingly called Silver Star, and we used **the** sled with magnets on both sides to **search** for droplets left over from **the explosion** of this meteor. And at **the** bottom left you see **the** filming crew of Netflix. They are preparing a documentary about this research, and **the** director asked me,"Avi, it looks like you are running,"because I was jogging at sunrise, as I often do on land for a few miles. And he said,"Are you running away from something or towards something?"And I said,"Both. I ' m running away from some of my colleagues who have strong opinions without seeking evidence, and I ' m running towards a higher intelligence in interstellar space."Now we used this sled and collected magnetic particles from **the** ocean floor about a mile deep. And then I brought them to my colleagues at Harvard University. They look like metallic **spheres**, very distinct from **the** background of sand in which they were collected. And my colleague at Harvard, Stein Jacobsen, is a world-renowned geochemist. He used **the** electron microprobe, a mass spectrometer, in his laboratory. **The** person on **the** other side of me in this picture is a summer intern, Sophie Bergstrom, who found most of our molten droplets. And so I called her **the** spheral hunter. And most of our spherules were actually of a type familiar from **the** solar system, but about 10 percent of them looked unusual, and they had a chemical composition very different from solar system materials. They had abundances of elements like beryllium, lanthanum, uranium, that are up to a factor of 1, 000 more than found in solar system materials. They were not from **the** Earth, not from **the** moon, not from Mars, not from **asteroids**. And so now **the** question arises, was this a **rock** from another star? And of course, one possibility is that there was a natural process that produced it. For example, most stars are **dwarf** stars, 10 percent of **the** mass of **the** sun, and they are 100 times denser than **the** sun. And so if you bring a **planet** like **the** Earth close to them, they spaghettify **the planet**, make a stream of **rocks** that could be ejected at a speed similar to that of this meteor. But it ' s also possible that this **object** was of **artificial** origin, in which case, if we look for bigger pieces of **the object**, we might find a gadget with buttons on it. And I asked students in my class,"If we find such a gadget, should we press a button?"Now, some of my critics argued, maybe it ' s **coal** ash. So we looked at 55 elements from **the** periodic table and found that **the** abundances of elements are very different from **coal** ash. So it ' s not **coal** ash. Others argued,"Maybe it was not a meteor, maybe it was a truck."Well, **the** data came from US government satellites. We actually based our **search** region on **the** Department of Defense coordinates, and we went 26 times back and forth, **searching** that region. So **the** next expedition, hopefully within **the** year, will **search** for bigger pieces of **the object**, maybe even **the** core of **the object**, because that could have a huge impact on humanity. We all know **the** biblical story about Moses, who looked at **the** bush that was burning without being consumed, with religious awe, and that gave Moses **the** sense that there is a superhuman entity, God, out there. Now, Friedrich Nietzsche in 1882 argued,"God is dead."And that gave rise to **the** modern period of science and technology, where humans have this hubris, they lack modesty. Nobody is smarter than us, we are at **the** top of **the** food chain. Maybe AI will do a little better. But AI is just a digital mirror. It reflects our faults. It ' s nothing better than us. It ' s not a digital species. It ' s just us. And if we find a partner out there, of course, that will give a new meaning to our existence. And it ' s a whole different ball game. Something from another star has nothing to do with us. And we better be ready for that. Not look at **the** mirror and imagine something like it, as science fiction stories do. Now **the** good news is, next year, **the** Rubin Observatory in Chile will survey **the** southern sky every four days with a camera that is **the** size of a person, 3. 2 billion pixels, a thousand times more than your cell phone camera. And so if we find more **objects** like ' Oumuamua, it might give us a sense of modesty. We might bring back this sense of awe that Moses had. Except in this time, it will be based on something that was delivered from interstellar space, from a neighbor. And that is quite promising, actually, because it may change our priorities. Instead of spending four trillion dollars a year on military budgets, killing each other for territories on this **rock**, **the** tiny **rock**, left over from **the** formation of **the** sun, we might realize that there is a smarter kid on **the** block, and that kid may provide a better role model than our politicians. And if we allocate four trillion dollars a year to space exploration, we could send a CubeSat towards every star in **the** Milky Way galaxy, hundreds of billions of them, within one century. And it gets better than that, because if we find a superhuman intelligence out there... We might learn new physics. **The** first question I would ask is,"What happened before **the** Big Bang?"And they might have quantum gravity engineers that are capable of creating a baby **universe** in **the** laboratory. And this job of creating a new **universe** can be perfected. And that would help us actually, given that there is a lot of room for improvement in **the** world that we inhabit. Thank you. Chris Anderson : Thank you, Avi. I ' m here like a visitor from another **planet**, because we ' re out of time. But also just to ask you one yes or no question, OK? If you had to put money on it, in **the** next ten years, what do you think – is it likely that we will discover convincing evidence that persuades a critical mass of your colleagues that there is someone out there. And I just want a yes or no. Avi Loeb : Yes. CA : Yes! Avi Loeb!

Text Sample 51: POS Dim 1 – 2016 – Score 23.00 – 2016/t001399.txt

I was excited to be a part of **the**"Dream"**theme**, and then I found out I ' m leading off **the**"Nightmare?"section of it. And certainly there are things about **the** climate crisis that qualify. And I have some bad news, but I have a lot more good news. I ' m going to propose three questions and **the** answer to **the** first one necessarily involves a little bad news. But — hang on, because **the** answers to **the** second and third questions really are very positive. So **the** first question is,"Do we really have to change?"And of course, **the** Apollo Mission, among other things changed **the** environmental movement, really launched **the** modern environmental movement. 18 months after this Earthrise picture was first seen on earth, **the** first Earth Day was organized. And we learned a lot about ourselves looking back at our **planet** from space. And one of **the** things that we learned confirmed what **the** scientists have long told us. One of **the** most essential facts about **the** climate crisis has to do with **the** sky. As this picture illustrates, **the** sky is not **the** vast and limitless expanse that appears when we look up from **the** ground. It is a very thin shell of atmosphere surrounding **the planet**. That right now is **the** open sewer for our industrial civilization as it ' s currently organized. We are spewing 110 million tons of heat-trapping global warming pollution into it every 24 hours, free of charge, go ahead. And there are many sources of **the greenhouse** gases, I ' m certainly not going to go through them all. I ' m going to focus on **the** main one, but **agriculture** is involved, diet is involved, population is involved. Management of forests, **transportation**, **the** oceans, **the** melting of **the** permafrost. But I ' m going to focus on **the** heart of **the** problem, which is **the** fact that we still rely on dirty, carbon-based fuels for 85 percent of all **the** energy that our world burns every year. And you can see from this image that after World War II, **the emission** rates started really accelerating. And **the** accumulated amount of man- made, global warming pollution that is up in **the** atmosphere now traps as much extra heat energy as would be released by 400, 000 Hiroshima-class atomic bombs exploding every 24 hours, 365 days a year. Fact-checked over and over again, conservative, it ' s **the** truth. Now it ' s a big **planet**, but — that is a lot of energy, particularly when you multiply it 400, 000 times per day. And all that extra heat energy is heating up **the** atmosphere, **the** whole earth system. Let ' s look at **the** atmosphere. This is a depiction of what we used to think of as **the** normal distribution of temperatures. **The** white represents normal temperature days ; 1951-1980 are arbitrarily chosen. **The blue** are cooler than average days, **the** red are warmer than average days. But **the** entire curve has moved to **the** right in **the** 1980s. And you ' ll see in **the** lower **right-hand** corner **the** appearance of statistically significant numbers of extremely hot days. In **the** 90s, **the** curve shifted further. And in **the** last 10 years, you see **the** extremely hot days are now more numerous than **the** cooler than average days.

In fact, they are 150 times more common on **the surface** of **the** earth than they were just 30 years ago. So we ' re having record-breaking temperatures. Fourteen of **the** 15 of **the** hottest years ever measured with instruments have been in this young century. **The** hottest of all was last year. Last month was **the** 371st month in a row warmer than **the** 20th-century average. And for **the** first time, not only **the** warmest January, but for **the** first time, it was more than two degrees Fahrenheit warmer than **the** average. These higher temperatures are having an effect on animals, plants, people, ecosystems. But on a global basis, 93 percent of all **the** extra heat energy is trapped in **the** oceans. And **the** scientists can measure **the** heat buildup much more precisely now at all depths : deep, mid-ocean, **the** first few hundred meters. And this, too, is accelerating. It goes back more than a century. And more than half of **the** increase has been in **the** last 19 years. This has consequences. **The** first order of consequence : **the** ocean-based storms get stronger. Super Typhoon Haiyan went over areas of **the** Pacific five and a half degrees Fahrenheit warmer than normal before it slammed into Tacloban, as **the** most destructive storm ever to make landfall. Pope Francis, who has made such a difference to this whole issue, visited Tacloban right after that. Superstorm Sandy went over areas of **the** Atlantic nine degrees warmer than normal before slamming into New York and New Jersey. **The** second order of consequences are affecting all of us right now. **The** warmer oceans are evaporating much more water vapor into **the** skies. Average humidity worldwide has gone up four percent. And it creates these atmospheric rivers. **The** Brazilian scientists call them"flying rivers."And they funnel all of that extra water vapor over **the** land where storm conditions trigger these massive record-breaking downpours. This is from Montana. Take a look at this storm last August. As it moves over Tucson, Arizona. It literally splashes off **the** city. These downpours are really unusual. Last July in Houston, Texas, it rained for two days, 162 billion gallons. That represents more than two days of **the** full flow of Niagara Falls in **the** middle of **the** city, which was, of course, paralyzed. These record downpours are creating historic floods and mudslides. This one is from Chile last year. And you ' ll see that warehouse going by. There are **oil** tankers cars going by. This is from Spain last September, you could call this **the** running of **the** cars and trucks, I guess. Every night on **the** TV news now is like a nature hike through **the** Book of Revelation. I mean, really. **The** insurance industry has certainly noticed, **the** losses have been mounting up. They ' re not under any illusions about what ' s happening. And **the** causality requires a moment of discussion. We ' re used to thinking of linear cause and linear effect — one cause, one effect. This is systemic causation. As **the** great Kevin Trenberth says,"All storms are different now. There ' s so much extra energy in **the** atmosphere, there ' s so much extra water vapor. Every storm is different now."So, **the** same extra heat pulls **the** soil moisture out of **the** ground and causes these deeper, longer, more pervasive **droughts** and many of them are underway right now. It dries out **the** vegetation and causes more fires in **the** western part of North America. There ' s certainly been evidence of that, a lot of them. More lightning, as **the** heat energy builds up, there ' s a considerable amount of additional lightning also. These climate-related disasters also have geopolitical consequences and create instability. **The** climate-related historic **drought** that started in Syria in 2006 destroyed 60 percent of **the** farms in Syria, killed 80 percent of **the** livestock, and drove 1. 5 million climate refugees into **the** cities of Syria, where they collided with another 1. 5 million refugees from **the** Iraq War. And along with other factors, that opened **the** gates of Hell that people are trying to close now. **The** US Defense Department has long warned of consequences from **the** climate crisis, including refugees, food and water shortages and pandemic disease. Right now we ’ re seeing microbial diseases from **the** tropics spread to **the** higher latitudes ; **the transportation** revolution has had a lot to do with this. But **the** changing conditions change **the** latitudes and **the** areas where these microbial diseases can become endemic and change **the** range of **the** vectors, like mosquitoes and ticks that carry them. **The** Zika epidemic now — we ’ re better positioned in North America because it ’ s still a little too cool and we have a better public health system. But when women in some regions of South and Central America are advised not to get pregnant for two years — that ’ s something new, that ought to get our attention. **The** Lancet, one of **the** two greatest medical journals in **the** world, last summer labeled this a medical emergency now. And there are many factors because of it. This is also connected to **the** extinction crisis. We ’ re in danger of losing 50 percent of all **the** living species on earth by **the** end of this century. And already, land-based plants and animals are now moving towards **the** poles at an average rate of 15 feet per day. Speaking of **the** North Pole, last December 29, **the** same storm that caused historic flooding in **the** American Midwest, raised temperatures at **the** North Pole 50 degrees Fahrenheit warmer than normal, causing **the** thawing of **the** North Pole in **the** middle of **the** long, dark, winter, polar night. And when **the** land- based ice of **the** Arctic melts, it raises **sea** level. Paul Nicklen ’ s beautiful photograph from Svalbard illustrates this. It ’ s more dangerous coming off Greenland and particularly, Antarctica. **The** 10 largest risk cities for sea-level rise by population are mostly in South and Southeast Asia. When you measure it by assets at risk, number one is Miami : three and a half trillion dollars at risk. Number three : New York and Newark. I was in Miami last fall during **the** supermoon, one of **the** highest high-tide days. And there were **fish** from **the** ocean swimming in some of **the** streets of Miami Beach and Fort Lauderdale and Del Rey. And this happens regularly during **the** highest- tide tides now. Not with rain — they call it”sunny-day flooding.”It comes up through **the** storm sewers. And **the** Mayor of Miami speaks for many when he says it is long past time this can be viewed through a partisan lens. This is a crisis that ’ s getting worse day by day. We have to move beyond partisanship. And I want to take a moment to honor these House Republicans — who had **the** courage last fall to step out and take a political risk, by telling **the** truth about **the** climate crisis. So **the** cost of **the** climate crisis is mounting up, there are many of these aspects I haven ’ t even mentioned. It ’ s an enormous burden. I ’ ll mention just one more, because **the** World Economic Forum last month in Davos, after their annual survey of 750 economists, said **the** climate crisis is now **the** number one risk to **the** global economy. So you get central bankers like Mark Carney, **the** head of **the** UK Central Bank, saying **the** vast majority of **the carbon** reserves are unburnable. Subprime **carbon**. I ’ m not going to remind you what happened with subprime mortgages, but it ’ s **the** same thing. If you look at all of **the carbon** fuels that were burned since **the** beginning of **the** industrial revolution, this is **the** quantity burned in **the** last 16 years. Here are all **the** ones that are proven and left on **the** books, 28 trillion dollars. **The** International Energy Agency says only this amount can be burned. So **the** rest, 22 trillion dollars — unburnable. Risk to **the** global economy. That ’ s why divestment movement makes practical sense and is not just a moral imperative. So **the** answer to **the** first question,”Must we change?”is yes, we have to change. Second question,”Can we change?”This is **the** exciting news! **The** best projections in **the** world 16 years ago were that by 2010, **the** world would be able to install 30 gigawatts of wind capacity. We beat that mark by 14 and a half times over. We see an exponential curve for wind installations now. We see **the** cost coming down dramatically. Some countries — take Germany, an industrial powerhouse with a climate not that different from Vancouver ’ s, by **the** way — one day last December, got 81 percent of all its energy from **renewable** resources, mainly solar and wind. A lot of countries are getting more than half on an average basis. More good news : energy storage, from batteries particularly, is now beginning to take off because **the** cost has been coming down very dramatically to solve **the** intermittency problem. With solar, **the** news is even more exciting! **The** best projections 14 years ago were that we would install one gigawatt per year by 2010. When 2010 came around, we beat that mark by 17 times over. Last year, we beat it by 58 times over. This year, we ' re on track to beat it 68 times over. We ' re going to win this. We are going to prevail. **The** exponential curve on solar is even steeper and more dramatic. When I came to this stage 10 years ago, this is where it was. We have seen a revolutionary breakthrough in **the** emergence of these exponential curves. And **the** cost has come down 10 percent per year for 30 years. And it ' s continuing to come down. Now, **the** business community has certainly noticed this, because it ' s crossing **the** grid parity point. Cheaper solar penetration rates are beginning to rise. Grid parity is understood as that line, that threshold, below which **renewable** electricity is cheaper than electricity from burning **fossil** fuels. That threshold is a little bit like **the** difference between 32 degrees Fahrenheit and 33 degrees Fahrenheit, or zero and one Celsius. It ' s a difference of more than one degree, it ' s **the** difference between ice and water. And it ' s **the** difference between markets that are frozen up, and liquid flows of capital into new opportunities for investment. This is **the** biggest new business opportunity in **the** history of **the** world, and **two-thirds** of it is in **the** private sector. We are seeing an **explosion** of new investment. Starting in 2010, investments globally in **renewable** electricity generation surpassed **fossils**. **The** gap has been growing ever since. **The** projections for **the** future are even more dramatic, even though **fossil** energy is now still subsidized at a rate 40 times larger than renewables. And by **the** way, if you add **the** projections for nuclear on here, particularly if you assume that **the** work many are doing to try to break through to safer and more acceptable, more affordable forms of nuclear, this could change even more dramatically. So is there any precedent for such a rapid adoption of a new technology? Well, there are many, but let ' s look at cell phones. In 1980, AT&T, then Ma Bell, commissioned McKinsey to do a global market survey of those clunky new mobile phones that appeared then."How many can we sell by **the** year 2000?"they asked. McKinsey came back and said,"900, 000."And sure enough, when **the** year 2000 arrived, they did sell 900, 000 — in **the** first three days. And for **the** balance of **the** year, they sold 120 times more. And now there are more cell connections than there are people in **the** world. So, why were they not only wrong, but way wrong? I ' ve asked that question myself,"Why?"And I think **the** answer is in three parts. First, **the** cost came down much faster than anybody expected, even as **the** quality went up. And low-income countries, places that did not have a landline grid — they leap-frogged to **the** new technology. **The** big expansion has been in **the** developing counties. So what about **the** electricity grids in **the** developing world? Well, not so hot. And in many areas, they don ' t exist. There are more people without any electricity at all in India than **the** entire population of **the** United States of America. So now we ' re getting this : solar panels on grass huts and new business models that make it affordable. Muhammad Yunus financed this one in Bangladesh with micro- credit. This is a village market. Bangladesh is now **the** fastest-deploying country in **the** world : two systems per minute on average, night and day. And we have all we need : enough energy from **the** Sun comes to **the** earth every hour to supply **the** full world ' s energy needs for an entire year. It ' s actually a little bit less than an hour. So **the** answer to **the** second question,"Can we change?"is clearly"Yes."And it ' s an ever-firmer"yes."Last question,”Will we change?”Paris really was a breakthrough, some of **the** provisions are binding and **the** regular reviews will matter a lot. But nations aren ’ t waiting, they ’ re going ahead. China has already announced that starting next year, they ’ re adopting a nationwide cap and trade system. They will likely link up with **the** European Union. **The** United States has already been changing. All of these **coal** plants were proposed in **the** next 10 years and canceled. All of these existing **coal** plants were retired. All of these **coal** plants have had their retirement announced. All of them — canceled. We are moving forward. Last year — if you look at all of **the** investment in new electricity generation in **the** United States, almost three-quarters was from **renewable** energy, mostly wind and solar. We are solving this crisis. **The** only question is : how long will it take to get there? So, it matters that a lot of people are organizing to insist on this change. Almost 400, 000 people marched in New York City before **the** UN special session on this. Many thousands, tens of thousands, marched in cities around **the** world. And so, I am extremely optimistic. As I said before, we are going to win this. I ’ ll finish with this story. When I was 13 years old, I heard that proposal by President Kennedy to land a person on **the** Moon and bring him back safely in 10 years. And I heard adults of that day and time say,”That ’ s reckless, expensive, may well fail.”But eight years and two months later, in **the** moment that Neil Armstrong set foot on **the** Moon, there was great cheer that went up in NASA ’ s mission control in Houston. Here ’ s a little-known fact about that : **the** average age of **the** systems engineers, **the** controllers in **the** room that day, was 26, which means, among other things, their age, when they heard that challenge, was 18. We now have a moral challenge that is in **the** tradition of others that we have faced. One of **the** greatest poets of **the** last century in **the** US, Wallace Stevens, wrote a line that has stayed with me :”After **the** final ’ no, ’ there comes a ’ yes, ’ and on that ’ yes ’, **the** future world depends.”When **the** abolitionists started their movement, they met with no after no after no. And then came a yes. **The** Women ’ s Suffrage and Women ’ s Rights Movement met endless no ’ s, until finally, there was a yes. **The** Civil Rights Movement, **the** movement against apartheid, and more recently, **the** movement for gay and lesbian rights here in **the** United States and elsewhere. After **the** final”no”comes a”yes.”When any great moral challenge is ultimately resolved into a binary choice between what is right and what is wrong, **the** outcome is fore-ordained because of who we are as human beings. Ninety-nine percent of us, that is where we are now and it is why we ’ re going to win this. We have everything we need. Some still doubt that we have **the** will to act, but I say **the** will to act is itself a **renewable** resource. Thank you very much. Chris Anderson : You ’ ve got this incredible combination of skills. You ’ ve got this scientist mind that can understand **the** full range of issues, and **the** ability to turn it into **the** most vivid language. No one else can do that, that ’ s why you led this thing. It was amazing to see it 10 years ago, it was amazing to see it now. Al Gore : Well, you ’ re nice to say that, Chris. But honestly, I have a lot of really good friends in **the** scientific community who are incredibly patient and who will sit there and explain this stuff to me over and over and over again until I can get it into simple enough language that I can understand it. And that ’ s **the** key to trying to communicate. CA : So, your talk. First part : **terrifying**, second part : incredibly hopeful. How do we know that all those graphs, all that progress, is enough to solve what you showed in **the** first part? AG : I think that **the** crossing — you know, I ’ ve only been in **the** business world for 15 years. But one of **the** things I ’ ve learned is that apparently it matters if a new product or service is more expensive than **the** incumbent, or cheaper than. Turns out, it makes a difference if it ’ s cheaper than. And when it crosses that line, then a lot of things really change. We are regularly surprised by these developments. **The** late Rudi Dornbusch, **the** great economist said,”Things take longer to happen then you think they will, and then they happen much faster than you thought they could.”I really think that ’ s where we are. Some people are using **the** phrase”**The** Solar Singularity”now, meaning when it gets below **the** grid parity, unsubsidized in most places, then it ’ s **the** default choice. Now, in one of **the** presentations yesterday, **the** jitney thing, there is an effort to use regulations to slow this down. And I just don ’ t think it ’ s going to work. There ’ s a woman in Atlanta, Debbie Dooley, who ’ s **the** Chairman of **the** Atlanta Tea Party. They enlisted her in this effort to put a tax on solar panels and regulations. And she had just put solar panels on her roof and she didn ’ t understand **the** request. And so she went and formed an alliance with **the** Sierra Club and they formed a new organization called **the** Green Tea Party. And they defeated **the** proposal. So, finally, **the** answer to your question is, this sounds a little corny and maybe it ’ s a clich, but 10 years ago — and Christiana referred to this — there are people in this audience who played an incredibly significant role in generating those exponential curves. And it didn ’ t work out economically for some of them, but it kick-started this global revolution. And what people in this audience do now with **the** knowledge that we are going to win this. But it matters a lot how fast we win it. CA : Al Gore, that was incredibly powerful. If this turns out to be **the** year, that **the** partisan thing changes, as you said, it ’ s no longer a partisan issue, but you bring along people from **the** other side together, backed by science, backed by these kinds of investment opportunities, backed by reason that you win **the** day — boy, that ’ s really exciting. Thank you so much. AG : Thank you so much for bringing me back to TED. Thank you!

Text Sample 52: POS Dim 1 – 2016 – Score 21.00 – 2016/t001203.txt

Here ’ s a question that matters. [Is it ethical to **evolve the** human body?] Because we ’ re beginning to get all **the** tools together to **evolve** ourselves. And we can **evolve** bacteria and we can **evolve** plants and we can **evolve** animals, and we ’ re now reaching a point where we really have to ask, is it really ethical and do we want to **evolve** human beings? And as you ’ re thinking about that, let me talk about that in **the** context of prosthetics, prosthetics past, present, future. So this is **the iron** hand that belonged to one of **the** German counts. Loved to fight, lost his arm in one of these battles. No problem, he just made a suit of armor, put it on, perfect prosthetic. That ’ s where **the** concept of ruling with an **iron** fist comes from. And of course these prosthetics have been getting more and more useful, more and more modern. You can hold soft-boiled eggs. You can have all types of controls, and as you ’ re thinking about that, there are wonderful people like Hugh Herr who have been building absolutely extraordinary prosthetics. So **the** wonderful Aimee Mullins will go out and say, how tall do I want to be tonight? Or Hugh will say what type of cliff do I want to climb? Or does somebody want to run a marathon, or does somebody want to ballroom dance? And as you adapt these things, **the** interesting thing about prosthetics is they ’ ve been coming inside **the** body. So these external prosthetics have now become **artificial** knees. They ’ ve become **artificial** hips. And then they ’ ve **evolved** further to become not just nice to have but essential to have. So when you ’ re talking about a heart pacemaker as a prosthetic, you ’ re talking about something that isn ’ t just,”I ’ m missing my leg,”it ’ s,”if I don ’ t have this, I can die.”And at that point, a prosthetic becomes a symbiotic relationship with **the** human body. And four of **the** smartest people that I ’ ve ever met — Ed Boyden, Hugh Herr, Joe Jacobson, Bob Lander — are working on a Center for Extreme Bionics. And **the** interesting thing of what you ’ re seeing here is these prosthetics now get integrated into **the** bone. They get integrated into **the** skin. They get integrated into **the** muscle. And one of **the** other sides of Ed is he ’ s been thinking about how to connect **the** brain using light or other mechanisms directly to things like these prosthetics. And if you can do that, then you can begin changing fundamental aspects of humanity. So how quickly you react to something depends on **the diameter** of a nerve. And of course, if you have nerves that are external or prosthetic, say with light or liquid metal, then you can increase that **diameter** and you could even increase it theoretically to **the** point where, as long as you could see **the** muzzle flash, you could step out of **the** way of a **bullet**. Those are **the** order of magnitude of changes you ’ re talking about. This is a fourth sort of level of prosthetics. These are Phonak hearing aids, and **the** reason why these are so interesting is because they cross **the** threshold from where prosthetics are something for somebody who is”disabled”and they become something that somebody who is”normal”might want to actually have, because what this prosthetic does, which is really interesting, is not only does it help you hear, you can focus your hearing, so it can hear **the** conversation going on over there. You can have superhearing. You can have hearing in 360 degrees. You can have white noise. You can record, and oh, by **the** way, they also put a phone into this. So this functions as your hearing aid and also as your phone. And at that point, somebody might actually want to have a prosthetic voluntarily. All of these thousands of loosely connected little pieces are coming together, and it ’ s about time we ask **the** question, how do we want to **evolve** human beings over **the** next century or two? And for that we turn to a great philosopher who was a very smart man despite being a Yankee fan. And Yogi Berra used to say, of course, that it ’ s very tough to make predictions, especially about **the** future. So instead of making a prediction about **the** future to begin with, let ’ s take what ’ s happening in **the** present with people like Tony Atala, who is redesigning 30-some-odd organs. And maybe **the** ultimate prosthetic isn ’ t having something external, titanium. Maybe **the** ultimate prosthetic is take your own gene code, remake your own body parts, because that ’ s a whole lot more effective than any kind of a prosthetic. But while you ’ re at it, then you can take **the** work of Craig Venter and Ham Smith. And one of **the** things that we ’ ve been doing is trying to figure out how to reprogram cells. And if you can reprogram a cell, then you can change **the** cells in those organs. So if you can change **the** cells in those organs, maybe you make those organs more radiation-resistant. Maybe you make them absorb more oxygen. Maybe you make them more efficient to filter out stuff that you don ’ t want in your body. And over **the** last few weeks, George Church has been in **the** news a lot because he ’ s been talking about taking one of these programmable cells and inserting an entire human **genome** into that cell. And once you can insert an entire human **genome** into a cell, then you begin to ask **the** question, would you want to enhance any of that **genome**? Do you want to enhance a human body? How would you want to enhance a human body? Where is it ethical to enhance a human body and where is it not ethical to enhance a human body? And all of a sudden, what we ’ re doing is we ’ ve got this multidimensional chess board where we can change human genetics by using viruses to attack things like AIDS, or we can change **the** gene code through gene therapy to do away with some hereditary diseases, or we can change **the** environment, and change **the** expression of those genes in **the** epigenome and pass that on to **the** next generations. And all of a sudden, it ’ s not just one little bit, it ’ s all these **stacked** little bits that allow you to take little portions of it until all **the** portions coming together lead you to something that ’ s very different. And a lot of people are very scared by this stuff. And it does sound scary, and there are risks to this stuff. So why in **the** world would you ever want to do this stuff? Why would we really want to alter **the** human body in a fundamental way? **The** answer lies in part with Lord Rees, astronomer royal of Great Britain. And one of his favorite sayings is **the universe** is 100 percent malevolent. So what does that mean? It means if you take any one of your bodies at random, drop it anywhere in **the universe**, drop it in space, you die. Drop it on **the** Sun, you die. Drop it on **the surface** of Mercury, you die. Drop it near a supernova, you die. But fortunately, it ’ s only about 80 percent effective. So as a great physicist once said, there ’ s these little upstream eddies of **biology** that create order in this rapid torrent of entropy. So as **the universe** dissipates energy, there ’ s these upstream eddies that create biological order. Now, **the** problem with eddies is, they tend to **disappear**. They shift. They move in rivers. And because of that, when an eddy shifts, when **the** Earth becomes a snowball, when **the** Earth becomes very hot, when **the** Earth gets hit by an **asteroid**, when you have supervolcanoes, when you have solar flares, when you have potentially extinction-level events like **the** next election — then all of a sudden, you can have periodic extinctions. And by **the** way, that ’ s happened five times on Earth, and therefore it is very likely that **the** human species on Earth is going to go extinct someday. Not next week, not next month, maybe in November, but maybe 10, 000 years after that. As you ’ re thinking of **the** consequence of that, if you believe that extinctions are common and natural and normal and **occur** periodically, it becomes a moral imperative to diversify our species. And it becomes a moral imperative because it ’ s going to be really hard to live on Mars if we don ’ t fundamentally modify **the** human body. Right? You go from one cell, mom and dad coming together to make one cell, in a cascade to 10 trillion cells. We don ’ t know, if you change **the** gravity substantially, if **the** same thing will happen to create your body. We do know that if you expose our bodies as they currently are to a lot of radiation, we will die. So as you ’ re thinking of that, you have to really redesign things just to get to Mars. Forget about **the** moons of Neptune or Jupiter. And to borrow from Nikolai Kardashev, let ’ s think about life in a series of scales. So Life One civilization is a civilization that begins to alter his or her looks. And we ’ ve been doing that for thousands of years. You ’ ve got tummy tucks and you ’ ve got this and you ’ ve got that. You alter your looks, and I ’ m told that not all of those alterations take place for medical reasons. Seems odd. A Life Two civilization is a different civilization. A Life Two civilization alters fundamental aspects of **the** body. So you put human growth **hormone** in, **the** person grows taller, or you put x in and **the** person gets **fatter** or loses metabolism or does a whole series of things, but you ’ re altering **the** functions in a fundamental way. To become an intrasolar civilization, we ’ re going to have to create a Life Three civilization, and that looks very different from what we ’ ve got here. Maybe you splice in Deinococcus radiodurans so that **the** cells can resplice after a lot of exposure to radiation. Maybe you breathe by having oxygen flow through your **blood** instead of through your lungs. But you ’ re talking about really radical redesigns, and one of **the** interesting things that ’ s happened in **the** last decade is we ’ ve discovered a whole lot of **planets** out there. And some of them may be Earth-like. **The** problem is, if we ever want to get to these **planets**, **the** fastest human **objects** — Juno and Voyager and **the** rest of this stuff — take tens of thousands of years to get from here to **the** nearest solar system. So if you want to start exploring beaches somewhere else, or you want to see two-sun sunsets, then you ’ re talking about something that is very different, because you have to change **the** timescale and **the** body of humans in ways which may be absolutely unrecognizable. And that ’ s a Life Four civilization. Now, we can ’ t even begin to imagine what that might look like, but we ’ re beginning to get glimpses of instruments that might take us even that far. And let me give you two examples. So this is **the** wonderful Floyd Romesberg, and one of **the** things that Floyd ' s been doing is he ' s been playing with **the** basic **chemistry** of life. So all life on this **planet** is made in ATCGs, **the** four letters of DNA. All bacteria, all plants, all animals, all humans, all cows, everything else. And what Floyd did is he changed out two of those base pairs, so it ' s ATXY. And that means that you now have a parallel system to make life, to make babies, to reproduce, to **evolve**, that doesn ' t **mate** with most things on Earth or in fact maybe with nothing on Earth. Maybe you make plants that are immune to all bacteria. Maybe you make plants that are immune to all viruses. But why is that so interesting? It means that we are not a unique solution. It means you can create alternate **chemistries** to us that could be **chemistries** adaptable to a very different **planet** that could create life and heredity. **The** second experiment, or **the** other implication of this experiment, is that all of you, all life is based on 20 amino acids. If you don ' t substitute two amino acids, if you don ' t say ATXY, if you say ATCG + XY, then you go from 20 building blocks to 172, and all of a sudden you ' ve got 172 building blocks of amino acids to build life-forms in very different shapes. **The** second experiment to think about is a really weird experiment that ' s been taking place in China. So this guy has been transplanting hundreds of mouse heads. Right? And why is that an interesting experiment? Well, think of **the** first heart transplants. One of **the** things they used to do is they used to bring in **the** wife or **the** daughter of **the** donor so **the** donee could tell **the** doctors,"Do you recognize this person? Do you love this person? Do you feel anything for this person?"We laugh about that today. We laugh because we know **the** heart is a muscle, but for hundreds of thousands of years, or tens of thousands of years,"I gave her my heart. She took my heart. She broke my heart."We thought this was emotion and we thought maybe emotions were transplanted with **the** heart. Nope. So how about **the** brain? Two possible outcomes to this experiment. If you can get a mouse that is functional, then you can see, is **the** new brain a blank slate? And boy, does that have implications. Second option : **the** new mouse recognizes Minnie Mouse. **The** new mouse remembers what it ' s afraid of, remembers how to navigate **the** maze, and if that is true, then you can transplant memory and consciousness. And then **the** really interesting question is, if you can transplant this, is **the** only input-output mechanism this down here? Or could you transplant that consciousness into something that would be very different, that would last in space, that would last tens of thousands of years, that would be a completely redesigned body that could hold consciousness for a long, long period of time? And let ' s come back to **the** first question : Why would you ever want to do that? Well, I ' ll tell you why. Because this is **the** ultimate selfie. This is taken from six billion miles away, and that ' s Earth. And that ' s all of us. And if that little thing goes, all of humanity goes. And **the** reason you want to alter **the** human body is because you eventually want a picture that says, that ' s us, and that ' s us, and that ' s us, because that ' s **the** way humanity survives long-term extinction. And that ' s **the** reason why it turns out it ' s actually unethical not to **evolve the** human body even though it can be scary, even though it can be challenging, but it ' s what ' s going to allow us to explore, live and get to places we can ' t even dream of today, but which our great-great-great-great-grandchildren might someday. Thank you very much.

Text Sample 53: POS Dim 1 – 2016 – Score 20.00 – 2016/t001086.txt
In 1956, a documentary by Jacques Cousteau won both **the** Palme d ' Or and

an Oscar award. This film was called,”Le Monde Du Silence,”or,”**The** Silent World.”**The** premise of **the** title was that **the underwater** world was a quiet world. We now know, 60 years later, that **the underwater** world is anything but silent. Although **the** sounds are inaudible above water depending on where you are and **the** time of year, **the underwater** soundscape can be as noisy as any jungle or rainforest. Invertebrates like snapping **shrimp**, **fish** and marine mammals all use sound. They use sound to study their **habitat**, to keep in communication with each other, to navigate, to **detect** predators and prey. They also use sound by listening to know something about their environment. Take, for an example, **the** Arctic. It ’ s considered a vast, inhospitable place, sometimes described as a desert, because it is so cold and so remote and ice- covered for much of **the** year. And despite this, there is no place on Earth that I would rather be than **the** Arctic, especially as days lengthen and **spring** comes. To me, **the** Arctic really embodies this disconnect between what we see on **the surface** and what ’ s going on **underwater**. You can look out across **the** ice — all white and **blue** and cold — and see nothing. But if you could hear **underwater**, **the** sounds you would hear would at first amaze and then delight you. And while your eyes are seeing nothing for kilometers but ice, your ears are telling you that out there are bowhead and beluga **whales**, walrus and bearded **seals**. **The** ice, too, makes sounds. It screeches and cracks and pops and groans, as it collides and rubs when temperature or currents or winds change. And under 100 percent **sea** ice in **the** dead of winter, bowhead **whales** are singing. And you would never expect that, because we humans, we tend to be very visual animals. For most of us, but not all, our sense of sight is how we navigate our world. For marine mammals that live **underwater**, where chemical cues and light transmit poorly, sound is **the** sense by which they see. And sound transmits very well **underwater**, much better than it does in air, so signals can be heard over great distances. In **the** Arctic, this is especially important, because not only do Arctic marine mammals have to hear each other, but they also have to listen for cues in **the** environment that might indicate heavy ice ahead or open water. Remember, although they spend most of their lives **underwater**, they are mammals, and so they have to **surface** to breathe. So they might listen for thin ice or no ice, or listen for echoes off nearby ice. Arctic marine mammals live in a rich and varied **underwater** soundscape. In **the spring**, it can be a cacophony of sound. But when **the** ice is frozen solid, and there are no big temperature shifts or current changes, **the underwater** Arctic has some of **the** lowest ambient noise levels of **the** world ’ s oceans. But this is changing. This is primarily due to a decrease in seasonal **sea** ice, which is a direct result of human **greenhouse** gas **emissions**. We are, in effect, with climate change, conducting a completely uncontrolled experiment with our **planet**. Over **the** past 30 years, areas of **the** Arctic have seen decreases in seasonal **sea** ice from anywhere from six weeks to four months. This decrease in **sea** ice is sometimes referred to as an increase in **the** open water season. That is **the** time of year when **the** Arctic is navigable to vessels. And not only is **the** extent of ice changing, but **the** age and **the width** of ice is, too. Now, you may well have heard that a decrease in seasonal **sea** ice is causing a loss of **habitat** for animals that rely on **sea** ice, such as ice **seals**, or walrus, or polar bears. Decreasing **sea** ice is also causing increased erosion along coastal villages, and changing prey availability for marine birds and mammals. Climate change and decreases in **sea** ice are also altering **the underwater** soundscape of **the** Arctic. What do I mean by soundscape? Those of us who eavesdrop on **the** oceans for a living use instruments called hydrophones, which are **underwater** microphones, and we record ambient noise — **the** noise all around us. And **the** soundscape describes **the** different contributors to this noise field. What we are hearing on our hydrophones are **the** very real sounds of climate change. We are hearing these changes from three fronts : from **the** air, from **the** water and from land. First : air. Wind on water creates waves. These waves make bubbles ; **the** bubbles break, and when they do, they make noise. And this noise is like a hiss or a static in **the** background. In **the** Arctic, when it ’ s ice-covered, most of **the** noise from wind doesn ’ t make it into **the** water column, because **the** ice acts as a buffer between **the** atmosphere and **the** water. This is one of **the** reasons that **the** Arctic can have very low ambient noise levels. But with decreases in seasonal **sea** ice, not only is **the** Arctic now open to this wave noise, but **the** number of storms and **the** intensity of storms in **the** Arctic has been increasing. All of this is raising noise levels in a previously quiet ocean. Second : water. With less seasonal **sea** ice, subarctic species are moving north, and taking advantage of **the** new **habitat** that is created by more open water. Now, Arctic **whales**, like this bowhead, they have no dorsal **fin**, because they have **evolved** to live and swim in ice-covered waters, and having something sticking off of your back is not very conducive to migrating through ice, and may, in fact, be excluding animals from **the** ice. But now, everywhere we ’ ve listened, we ’ re hearing **the** sounds of **fin whales** and humpback **whales** and killer **whales**, further and further north, and later and later in **the** season. We are hearing, in essence, an invasion of **the** Arctic by subarctic species. And we don ’ t know what this means. Will there be competition for food between Arctic and subarctic animals? Might these subarctic species introduce diseases or parasites into **the** Arctic? And what are **the** new sounds that they are producing doing to **the** soundscape **underwater**? And third : land. And by land... I mean people. More open water means increased human use of **the** Arctic. Just this past summer, a massive cruise ship made its way through **the** Northwest Passage — **the** once-mythical route between Europe and **the** Pacific. Decreases in **sea** ice have allowed humans to occupy **the** Arctic more often. It has allowed increases in **oil** and gas exploration and extraction, **the** potential for commercial **shipping**, as well as increased tourism. And we now know that ship noise increases levels of stress **hormones** in **whales** and can disrupt feeding behavior. Air guns, which produce loud, low-frequency”whoomps”every 10 to 20 seconds, changed **the** swimming and vocal behavior of **whales**. And all of these sound sources are decreasing **the** acoustic space over which Arctic marine mammals can communicate. Now, Arctic marine mammals are used to very high levels of noise at certain times of **the** year. But this is primarily from other animals or from **sea** ice, and these are **the** sounds with which they ’ ve **evolved**, and these are sounds that are vital to their very survival. These new sounds are loud and they ’ re alien. They might impact **the** environment in ways that we think we understand, but also in ways that we don ’ t. Remember, sound is **the** most important sense for these animals. And not only is **the** physical **habitat** of **the** Arctic changing rapidly, but **the** acoustic **habitat** is, too. It ’ s as if we ’ ve plucked these animals up from **the** quiet countryside and dropped them into a big city in **the** middle of rush hour. And they can ’ t escape it. So what can we do now? We can ’ t decrease wind speeds or keep subarctic animals from migrating north, but we can work on local solutions to reducing human- caused **underwater** noise. One of these solutions is to slow down ships that traverse **the** Arctic, because a slower ship is a quieter ship. We can restrict access in seasons and regions that are important for **mating** or feeding or migrating. We can get smarter about quieting ships and find better ways to explore **the** ocean bottom. And **the** good news is, there are people working on this right now. But ultimately, we humans have to do **the** hard work of reversing or at **the** very least decelerating human-caused atmospheric changes. So, let ’ s return to this idea of a silent world **underwater**. It ’ s entirely possible that many of

the whales swimming in **the** Arctic today, especially long- lived species like **the** bowhead **whale** that **the** Inuits say can live two human lives — it ’ s possible that these **whales** were alive in 1956, when Jacques Cousteau made his film. And in retrospect, considering all **the** noise we are creating in **the** oceans today, perhaps it really was”**The** Silent World.”Thank you.

Text Sample 54: POS Dim 1 – 2017 – Score 34.00 – 2017/t000900.txt
Chris Anderson : OK, Stewart, in **the** ’ 60s, you â I think it was ’ 68 â you founded this magazine. Stewart Brand : Bravo! It ’ s **the** original one. That ’ s hard to find. CA : Right. Issue One, right? SB : Mm hmm. CA : Why did that make so much impact? SB : Counterculture was **the** main event that I was part of at **the** time, and it was made up of hippies and New Left. That was sort of my contemporaries, **the** people I was just slightly older than. And my mode is to look at where **the** interesting flow is and then look in **the** other direction. CA : SB : Partly, I was trained to do that as an army officer, but partly, it ’ s just a cheap heuristic to find originalities : don ’ t look where everybody else is looking, look **the** opposite way. So **the** deal with counterculture is, **the** hippies were very romantic and kind of against technology, except very good LSD from Sandoz, and **the** New Left was against technology because they thought it was a power **device**. **Computers** were : do not spindle, fold, or mutilate. Fight that. And so, **the** Whole Earth Catalog was kind of a counter-counterculture thing in **the** sense that I bought Buckminster Fuller ’ s idea that tools of are of **the** essence. Science and engineers basically define **the** world in interesting ways. If all **the** politicians **disappeared** one week, it would be... a nuisance. But if all **the** scientists and engineers **disappeared** one week, it would be way more than a nuisance. CA : We still believe that, I think. SB : So focus on that. And then **the** New Left was talking about power to **the** people. And people like Steve Jobs and Steve Wozniak cut that and just said, power to people, tools that actually work. And so, where Fuller was saying don ’ t try to change human nature, people have been trying for a long time and it does not even bend, but you can change tools very easily. So **the** efficient thing to do if you want to make **the** world better is not try to make people behave differently like **the** New Left was, but just give them tools that go in **the** right direction. That was **the** Whole Earth Catalog. CA : And Stewart, **the** central image â this is one of **the** first images, **the** first time people had seen Earth from outer space. That had an impact, too. SB : It was kind of a chance that in **the spring** of ’ 66, thanks to an LSD experience on a rooftop in San Francisco, I got thinking about, again, something that Fuller talked about, that a lot of people assume that **the** Earth is flat and kind of infinite in terms of its resources, but once you really grasp that it ’ s a **sphere** and that there ’ s only so much of it, then you start husbanding your resources and thinking about it as a finite system.”Spaceship Earth”was his metaphor. And I wanted that to be **the** case, but on LSD I was getting higher and higher on my hundred micrograms on **the** roof of San Francisco, and noticed that **the** downtown buildings which were right in front of me were not all parallel, they were sort of fanned out like this. And that ’ s because they are on a curved **surface**. And if I were even higher, I would see that even more clearly, higher than that, more clearly still, higher enough, and it would close, and you would get **the** circle of Earth from space. And I thought, you know, we ’ ve been in space for 10 years â at that time, this is ’ 66 â and **the** cameras had never looked back. They ’ d always been looking out or looking at just parts of **the** Earth. And so I said, why haven ’ t we seen a photograph of **the** whole Earth yet? And it went around and NASA got it and senators, secretaries got it,

and various people in **the** Politburo got it, and it went around and around. And within two and a half years, about **the** time **the** Whole Earth Catalog came out, these images started to appear, and indeed, they did transform everything. And my idea of hacking civilization is that you try to do something lazy and ingenious and just sort of trick **the** situation. So all of these photographs that you see â and then **the** march for science last week, they were carrying these Whole Earth banners and so on â I did that with no work. I sold those buttons for 25 cents apiece. So, you know, tweaking **the** system is, I think, not only **the** most efficient way to make **the** system go in interesting ways, but in some ways, **the** safest way, because when you try to horse **the** whole system around in a big way, you can get into big horsing-around problems, but if you tweak it, it will adjust to **the** tweak. CA : So since then, among many other things, you ' ve been regarded as a leading voice in **the** environmental movement, but you are also a counterculturalist, and recently, you ' ve been taking on a lot of, well, you ' ve been declaring what a lot of environmentalists almost believe are heresies. I kind of want to explore a couple of those. I mean, tell me about this image here. SB : Ha-ha! That ' s a National Geographic image of what is called **the** mammoth steppe, what **the** far north, **the** sub-Arctic and Arctic region, used to look like. In fact, **the** whole world used to look like that. What we find in South Africa and **the** Serengeti now, lots of big animals, was **the** case in this part of Canada, throughout **the** US, throughout Eurasia, throughout **the** world. This was **the** norm and can be again. So in a sense, my long-term goal at this point is to not only bring back those animals and **the** grassland they made, which could be a climate stabilization system over **the** long run, but even **the** mammoths there in **the** background that are part of **the** story. And I think that ' s probably a 200- year goal. Maybe in 100, by **the** end of this century, we should be able to dial down **the** extinction rate to sort of what it ' s been in **the** background. Bringing back this amount of bio-abundance will take longer, but it ' s worth doing. CA : We ' ll come back to **the** mammoths, but explain how we should think of extinctions. Obviously, one of **the** huge concerns right now is that extinction is happening at a faster rate than ever in history. That ' s **the** meme that ' s out there. How should we think of it? SB : **The** story that ' s out there is that we ' re in **the** middle of **the** Sixth Extinction or maybe in **the** beginning of **the** Sixth Extinction. Because we ' re in **the** de-extinction business, **the** preventing-extinction business with Revive & Restore, we started looking at what ' s actually going on with extinction. And it turns out, there ' s a very confused set of data out there which gets oversimplified into **the** narrative of we ' re becoming... Here are five mass extinctions that are indicated by **the** **yellow triangles**, and we ' re now next. **The** last one there on **the** far right was **the** meteor that struck 66 million years ago and did in **the** dinosaurs. And **the** story is, we ' re **the** next meteor. Well, here ' s **the** deal. I wound up researching this for a paper I wrote, that a mass extinction is when 75 percent of all **the** species in **the** world go extinct. Well, there ' s on **the** order of five-and-a-half-million species, of which we ' ve identified one and a half million. Another 14, 000 are being identified every year. There ' s a lot of **biology** going on out there. Since 1500, about 500 species have gone extinct, and you ' ll see **the** term"mass extinction"kind of used in strange ways. So there was, about a year and a half ago, a front-page story by Carl Zimmer in **the** New York Times,"Mass Extinction in **the** Oceans, Broad Studies Show."And then you read into **the** article, and it mentions that since 1500, 15 species â one, five â have gone extinct in **the** oceans, and, oh, by **the** way, none in **the** last 50 years. And you read further into **the** story, and it ' s saying, **the** horrifying thing that ' s going on is that **the** fisheries are so overfishing **the** wild **fishes**, that it is taking down **the fish** populations in **the** oceans by 38 percent. That ' s **the** serious thing. None of those species are probably going to go extinct. So you ' ve just put, that headline writer put a panic button on **the** top of **the** story. It ' s clickbait kind of stuff, but it ' s basically saying,"Oh my God, start panicking, we ' re going to lose all **the** species in **the** oceans."Nothing like that is in prospect. And in fact, what I then started looking into in a little more detail, **the** Red List shows about 23, 000 species that are considered threatened at one level or another, coming from **the** International Union for **the** Conservation of Nature, **the** IUCN. And Nature Magazine had a piece surveying **the** loss of wildlife, and it said,"If all of those 23, 000 went extinct in **the** next century or so, and that rate of extinction carried on for more centuries and millennia, then we might be at **the** beginning of a sixth extinction. So **the** exaggeration is way out of hand. But environmentalists always exaggerate. That ' s a problem. CA : I mean, they probably feel a moral responsibility to, because they care so much about **the** thing that they are looking at, and unless you bang **the drum** for it, maybe no one listens. SB : Every time somebody says moral this or moral that â""moral hazard,""precautionary principle"â these are terms that are used to basically say no to things. CA : So **the** problem isn ' t so much **fish** extinction, animal extinction, it ' s **fish** flourishing, animal flourishing, that we ' re crowding them to some extent? SB : Yeah, and I think we are crowding, and there is losses going on. **The** major losses are caused by **agriculture**, and so anything that improves **agriculture** and basically makes it more condensed, more highly productive, including GMOs, please, but even if you want to do vertical farms in town, including inside farms, all **the** things that have been learned about how to grow pot in basements, is now being applied to growing vegetables inside containers â that ' s great, that ' s all good stuff, because land sparing is **the** main thing we can do for nature. People moving to cities is good. Making **agriculture** less of a destruction of **the** landscape is good. CA : There people talking about bringing back species, rewilding... Well, first of all, rewilding species : What ' s **the** story with these guys? SB : Ha-ha! Wolves. Europe, connecting to **the** previous point, we ' re now at probably peak farmland, and, by **the** way, in terms of population, we are already at peak children being alive. Henceforth, there will be fewer and fewer children. We are in **the** last doubling of human population, and it will get to nine, maybe nine and a half billion, and then start not just leveling off, but probably going down. Likewise, farmland has now peaked, and one of **the** ways that plays out in Europe is there ' s a lot of abandoned farmland now, which immediately reforests. They don ' t do wildlife corridors in Europe. They don ' t need to, because so many of these farms are connected that they ' ve made reforested wildlife corridors, that **the** wolves are coming back, in this case, to Spain. They ' ve gotten all **the** way to **the** Netherlands. There ' s bears coming back. There ' s lynx coming back. There ' s **the** European jackal. I had no idea such a thing existed. They ' re coming back from Italy to **the** rest of Europe. And unlike here, these are all predators, which is kind of interesting. They are being welcomed by Europeans. They ' ve been missed. CA : And counterintuitively, when you bring back **the** predators, it actually increases rather than reduces **the** diversity of **the** underlying ecosystem often. SB : Yeah, generally predators and large animals â large animals and large animals with sharp teeth and claws â are turning out to be highly important for a really rich ecosystem. CA : Which maybe brings us to this rather more dramatic rewilding project that you ' ve got yourself involved in. Why would someone want to bring back these **terrifying** woolly mammoths? SB : Hmm. Asian **elephants** are **the** closest **relative** to **the** woolly mammoth, and they ' re about **the** same size, genetically very close. They diverged quite recently in **evolutionary** history. **The** Asian **elephants** are closer to woolly mammoths than they are to African **elephants**, but they ' re close enough to African **elephants** that they have successfully hybridized. So we ’ re working with George Church at Harvard, who has already moved **the** genes for four major traits from **the** now well-preserved, well-studied **genome** of **the** woolly mammoth, thanks to so-called”ancient DNA analysis.”And in **the** lab, he has moved those genes into living Asian **elephant** cell lines, where they ’ re taking up their proper place thanks to CRISPR. I mean, they ’ re not shooting **the** genes in like you did with genetic engineering. Now with CRISPR you ’ re editing, basically, one allele, and replacing it in **the** place of another allele. So you ’ re now getting basically Asian **elephant** germline cells that are effectively in terms of **the** traits that you ’ re going for to be comfortable in **the** Arctic, you ’ re getting them in there. So we go through **the** process of getting that through a surrogate mother, an Asian **elephant** mother. You can get a proxy, as it ’ s being called by conservation biologists, of **the** woolly mammoth, that is effectively a hairy, curly-trunked, Asian **elephant** that is perfectly comfortable in **the** sub-Arctic. Now, it ’ s **the** case, so many people say,”Well, how are you going to get them there? And Asian **elephants**, they don ’ t like snow, right?”Well, it turns out, they do like snow. There ’ s some in an Ontario zoo that have made snowballs bigger than people. They just love â you know, with a trunk, you can start a little thing, roll it and make it bigger. And then people say,”Yeah, but it ’ s 22 months of gestation. This kind of cross-species **cloning** is tricky business, anyway. Are you going to lose some of **the** surrogate Asian **elephant** mothers?”And then George Church says,”That ’ s all right. We ’ ll do an **artificial** uterus and grow them that way.”Then people say,”Yeah, next century, maybe,”except **the** news came out this week in Nature that there ’ s now an **artificial** uterus in which they ’ ve grown a lamb to four weeks. That ’ s halfway through its gestation period. So this stuff is moving right along. CA : But why should we want a world where â Picture a world where there are thousands of these things thundering across Siberia. Is that a better world? SB : Potentially. It ’ s â There ’ s three groups, basically, working on **the** woolly mammoth seriously : Revive & Restore, we ’ re kind of in **the** middle ; George Church and **the** group at Harvard that are doing **the** genetics in **the** lab ; and then there ’ s an amazing old scientist named Zimov who works in northern Siberia, and his son Nikita, who has bought into **the** system, and they are, Sergey and Nikita Zimov have been, for 25 years, creating what they call”Pleistocene Park,”which is a place in a really tough part of Siberia that is pure tundra. And **the** research that ’ s been done shows that there ’ s probably one one-hundredth of **the** animals on **the** landscape there that there used to be. Like that earlier image, we saw lots of animals. Now there ’ s almost none. **The** tundra is mostly moss, and then there ’ s **the** boreal forest. And that ’ s **the** way it is, folks. There ’ s just a few animals there. So they brought in a lot of grazing animals : musk ox, Yakutian horses, they ’ re bringing in some bison, they ’ re bringing in some more now, and put them in at **the** density that they used to be. And grasslands are made by grazers. So these animals are there, grazing away, and they ’ re doing a couple of things. First of all, they ’ re turning **the** tundra, **the** moss, back into grassland. Grassland fixes **carbon**. Tundra, in a warming world, is thawing and releasing a lot of **carbon** dioxide and also methane. So already in their little 25 square miles, they ’ re doing a climate stabilization thing. Part of that story, though, is that **the** boreal forest is very absorbent to sunlight, even in **the** winter when snow is on **the** ground. And **the** way **the** mammoth steppe, which used to wrap all **the** way around **the** North Pole â there ’ s a lot of landmass around **the** North Pole â that was all this grassland. And **the** steppe was magnificent, probably one of **the** most productive biomes in **the** world, **the** biggest biome in **the** world. **The** forest part of it, right now, Sergey Zimov and Nikita go out with this old military tank they got for nothing, and they knock down **the** trees. And that ’ s a bore, and it ’ s tiresome, and as Sergey says,”... and they make no dung!”which, by **the** way, these big animals do, including mammoths. So mammoths become what conservation biologists call an umbrella species. It ’ s an exciting animal â pandas in China or wherever â that **the** excitement that goes on of making life good for that animal is making a **habitat**, an ecosystem, which is good for a whole lot of creatures and plants, and it ideally gets to **the** point of being self-managing, where **the** conservation biologists can back off and say,”All we have to do is keep out **the** destructive invasives, and this thing can just cook.”CA : So there ’ s many other species that you ’ re dreaming of de-extincting at some point, but I think what I ’ d actually like to move on to is this idea you talked about how mammoths might help green Siberia in a sense, or at least, I ’ m not talking about tropical rainforest, but this question of greening **the planet** you ’ ve thought about a lot. And **the** traditional story is that deforestation is one of **the** most awful curses of modern times, and that it ’ s a huge contributor to climate change. And then you went and sent me this graph here, or this map. What is this map? SB : Global greening. **The** thing to do with any narrative that you get from headlines and from short news stories is to look for what else is going on, and look for what Marc Andreessen calls”narrative violation.”So **the** narrative â and Al Gore is master of putting it out there â is that there ’ s this civilization-threatening climate change coming on very rapidly. We have to cease all extra production of **greenhouse** gases, especially CO2, as soon as possible, otherwise, we ’ re in deep, deep trouble. All of that is true, but it ’ s not **the** whole story, and **the** whole story is more interesting than these fragmentary stories. Plants love CO2. What plants are made of is CO2 plus water via sunshine. And so in many **greenhouses**, industrialized **greenhouses**, they add CO2 because **the** plants turn that into plant matter. So **the** studies have been done with satellites and other things, and what you ’ re seeing here is a graph of, over **the** last 33 years or so, there ’ s 14 percent more leaf action going on. There ’ s that much more biomass. There ’ s that much more what ecologists call”primary production.”There ’ s that much more life happening, thanks to climate change, thanks to all of our goddam **coal** plants. So â whoa, what ’ s going on here? By **the** way, crop production goes up with this. This is a partial counter to **the** increase of CO2, because there ’ s that much more plant that is sucking it down into plant matter. Some of that then decays and goes right back up, but some of it is going down into roots and going into **the** soil and staying there. So these counter things are part of what you need to bear in mind, and **the** deeper story is that thinking about and dealing with and engineering climate is a pretty complex process. It ’ s like medicine. You ’ re always, again, tweaking around with **the** system to see what makes an improvement. Then you do more of that, see it ’ s still getting better, then â oop! â that ’ s enough, back off half a turn. CA : But might some people say,”Not all green is created equal.”Possibly what we ’ re doing is trading off **the** magnificence of **the** rainforest and all that diversity for, I don ’ t know, green pond scum or grass or something like that. SB : In this particular study, it turns out every form of plant is increasing. Now, what ’ s interestingly left out of this study is what **the** hell is going on in **the** oceans. Primary production in **the** oceans, **the** biota of **the** oceans, mostly microbial, what they ’ re up to is probably **the** most important thing. They ’ re **the** ones that create **the** atmosphere that we ’ re happily breathing, and they ’ re not part of this study. This is one of **the** things James Lovelock has been insisting ; basically, our knowledge of **the** oceans, especially of ocean life, is fundamentally vapor, in this sense. So we ’ re in **the** process of finding out by inadvertent bad geoengineering of too much CO2 in **the** atmosphere, finding out, what is **the** ocean doing with that?

Well, **the** ocean, with **the** extra heat, is swelling up. That ’ s most of where we ’ re getting **the sea** level rise, and there ’ s a lot more coming with more global warming. We ’ re getting terrible harm to some of **the** coral reefs, like off of Australia. **The** great reef there is just a lot of bleaching from overheating. And this is why I and Danny Hillis, in our previous session on **the** main stage, was saying,”Look, geoengineering is worth experimenting with enough to see that it works, to see if we can buy time in **the** warming aspect of all of this, tweak **the** system with small but usable research, and then see if we should do more than tweak. CA : OK, so this is what we ’ re going to talk about for **the** last few minutes here because it ’ s such an important discussion. First of all, this book was just published by Yuval Harari. He ’ s basically saying **the** next **evolution** of humans is to become as gods. I think he â SB : Now, you ’ ve talked to him. And you ’ ve probably finished **the** book. I haven ’ t finished it yet. Where does he come out on â CA : I mean, it ’ s a pretty radical view. He thinks that we will completely remake ourselves using data, using bioengineering, to become completely new creatures that have, kind of, superpowers, and that there will be huge inequality. But we ’ re about to write a very radical, brand-new chapter of history. That ’ s what he believes. SB : Is he nervous about that? I forget. CA : He ’ s nervous about it, but I think he also likes provoking people. SB : Are you nervous about that? CA : I ’ m nervous about that. But, you know, with so much at TED, I ’ m excited and nervous. And **the** optimist in me is trying hard to lean towards”This is awesome and really exciting,”while **the** sort of responsible part of me is saying,”But, uh, maybe we should be a little bit careful as to how we think of it.”SB : That ’ s your secret sauce, isn ’ t it, for TED? Staying nervous and excited. CA : It ’ s also **the** recipe for being a little bit schizophrenic. But he didn ’ t quote you. What I thought was an astonishing statement that you made right back in **the** original Whole Earth Catalog, you ended it with this powerful phrase :”We are as gods, and might as well get good at it.”And then more recently, you ’ ve upgraded that statement. I want you talk about this philosophy. SB : Well, one of **the** things I ’ m **learning** is that documentation is better than memory â by far. And one of **the** things I ’ ve learned from somebody â I actually got on Twitter. It changed my life â it hasn ’ t forgiven me yet! And I took ownership of this phrase when somebody quoted it, and somebody else said,”Oh by **the** way, that isn ’ t what you originally wrote in that first 1968 Whole Earth Catalog. You wrote, ’ We are as gods and might as well get used to it. ’”I ’ d forgotten that entirely. **The** stories â these goddam stories â **the** stories we tell ourselves become lies over time. So, documentation helps cut through that. It did move on to”We are as gods and might as well get good at it,”and that was **the** Whole Earth Catalog. By **the** time I was doing a book called”Whole Earth Discipline : An Ecopragmatist Manifesto,”and in light of climate change, basically saying that we are as gods and have to get good at it. CA : We are as gods and have to get good at it. So talk about that, because **the** psychological reaction from so many people as soon as you talk about geoengineering is that **the** last thing they believe is that humans should be gods â some of them for religious reasons, but most just for humility reasons, that **the** systems are too complex, we should not be dabbling that way. SB : Well, this is **the** Greek narrative about hubris. And once you start getting really sure of yourself, you wind up sleeping with your mother. CA : I did not expect you would say that. SB : That ’ s **the** Oedipus story. Hubris is a really important cautionary tale to always have at hand. One of **the** guidelines I ’ ve kept for myself is : every day I ask myself how many things I am dead wrong about. And I ’ m a scientist by training and getting to work with scientists these days, which is pure joy. Science is organized skepticism. So you ’ re always insisting that even when something looks pretty good, you maintain a full set of not only suspicions about whether it ’ s as good as it looks, but : What else is going on? So this”What else is going?”on query, I think, is how you get away from fake news. It ’ s not necessarily real news, but it ’ s welcomely more complex news that you ’ re trying to take on. CA : But coming back to **the** application of this just for **the** environment : it seems like **the** philosophy of this is that, whether we like it or not, we are already dominating so many aspects of what happens on **planets**, and we ’ re doing it unintentionally, so we really should start doing it intentionally. What would it look like to start getting good at being a god? How should we start doing that? Are there small-scale experiments or systems we can nudge and play with? How on earth do we think about it? SB : **The** mentor that sort of freed me from total allegiance to Buckminster Fuller was Gregory Bateson. And Gregory Bateson was an epistemologist and anthropologist and biologist and psychologist and many other things, and he looked at how systems basically look at themselves. And that is, I think, part of how you want to always be looking at things. And what I like about David Keith ’ s approach to geoengineering is you don ’ t just haul off and do it. David Keith ’ s approach â and this is what Danny Hillis was talking about earlier â is that you do it really, really incrementally, you do some stuff to tweak **the** system, see how it responds, that tells you something about **the** system. That ’ s responding to **the** fact that people say, quite rightly,”What are we talking about here? We don ’ t understand how **the** climate system works. You can ’ t engineer a system you don ’ t understand.”And David says,”Well, that certainly applies to **the** human body, and yet medicine goes ahead, and we ’ re kind of glad that it has.”**The** way you engineer a system that is so large and complex that you can ’ t completely understand it is you tweak it, and this is kind of an anti-hubristic approach. This is : try a little bit here, back **the** hell off if it ’ s an issue, expand it if it seems to go OK, meanwhile, have other paths going forward. This is **the** whole argument for diversity and dialogue and all these other things and **the** things we were hearing about earlier with Sebastian [Thrun]. So **the** non-hubristic approach is looking for social license, which is a terminology that I think is a good one, of including society enough in these interesting, problematic, deep issues that they get to have a pretty good idea and have people that they trust paying close attention to **the** sequence of experiments as it ’ s going forward, **the** public dialogue as it ’ s going forward â which is more public than ever, which is **fantastic** â and you feel your way, you just ooze your way along, and this is **the** muddle-through approach that has worked pretty well so far. **The** reason that Sebastian and I are optimistic is we read people like Steven Pinker,”**The** Better Angels of Our Nature,”and so far, so good. Now, that can always change, but you can build a lot on that sense of : things are capable of getting better, figure out **the** tools that made that happen and apply those further. That ’ s **the** story. CA : Stewart, I think on that optimistic note, we ’ re actually going to wrap up. I am in awe of how you always are willing to challenge yourself and other people. I feel like this recipe for never allowing yourself to be too certain is so powerful. I want to learn it more for myself, and it ’ s been very insightful and inspiring, actually, listening to you today. Stewart Brand, thank you so much. SB : Thank you.

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Chris Anderson : Elon, hey, welcome back to TED. It ’ s great to have you here. Elon Musk : Thanks for having me. CA : So, in **the** next half hour or so, we ’ re going to spend some time exploring your vision for what an exciting future might look like, which I guess makes **the** first question a little ironic : Why are you boring? EM : Yeah. I ask myself that frequently. We ’ re trying to dig a hole under LA, and this is to create **the** beginning of what will hopefully be a 3D network of tunnels to alleviate congestion. So right now, one of **the** most soul-destroying things is traffic. It affects people in every part of **the** world. It takes away so much of your life. It ’ s horrible. It ’ s particularly horrible in LA. CA : I think you ’ ve brought with you **the** first visualization that ’ s been shown of this. Can I show this? EM : Yeah, absolutely. So this is **the** first time — Just to show what we ’ re talking about. So a couple of key things that are important in having a 3D tunnel network. First of all, you have to be able to integrate **the** entrance and exit of **the** tunnel seamlessly into **the** fabric of **the** city. So by having an elevator, sort of a car skate, that ’ s on an elevator, you can integrate **the** entrance and exits to **the** tunnel network just by using two parking spaces. And then **the** car gets on a skate. There ’ s no speed limit here, so we ’ re designing this to be able to operate at 200 kilometers an hour. CA : How much? EM : 200 kilometers an hour, or about 130 miles per hour. So you should be able to get from, say, Westwood to LAX in six minutes — five, six minutes. CA : So possibly, initially done, it ’ s like on a sort of toll road-type basis. EM : Yeah. CA : Which, I guess, alleviates some traffic from **the surface** streets as well. EM : So, I don ’ t know if people noticed it in **the** video, but there ’ s no real limit to how many levels of tunnel you can have. You can go much further deep than you can go up. **The** deepest mines are much deeper than **the** tallest buildings are tall, so you can alleviate any arbitrary level of urban congestion with a 3D tunnel network. This is a very important point. So a key rebuttal to **the** tunnels is that if you add one layer of tunnels, that will simply alleviate congestion, it will get used up, and then you ’ ll be back where you started, back with congestion. But you can go to any arbitrary number of tunnels, any number of levels. CA : But people — seen traditionally, it ’ s incredibly expensive to dig, and that would block this idea. EM : Yeah. Well, they ’ re right. To give you an example, **the** LA subway extension, which is — I think it ’ s a two-and-a-half mile extension that was just completed for two billion dollars. So it ’ s roughly a billion dollars a mile to do **the** subway extension in LA. And this is not **the** highest utility subway in **the** world. So yeah, it ’ s quite difficult to dig tunnels normally. I think we need to have at least a tenfold improvement in **the** cost per mile of tunneling. CA : And how could you achieve that? EM : Actually, if you just do two things, you can get to approximately an order of magnitude improvement, and I think you can go beyond that. So **the** first thing to do is to cut **the** tunnel **diameter** by a factor of two or more. So a single road lane tunnel according to regulations has to be 26 feet, maybe 28 feet in **diameter** to allow for crashes and emergency vehicles and sufficient ventilation for combustion engine cars. But if you shrink that **diameter** to what we ’ re attempting, which is 12 feet, which is plenty to get an electric skate through, you drop **the diameter** by a factor of two and **the** cross-sectional area by a factor of four, and **the** tunneling cost scales with **the** cross-sectional area. So that ’ s roughly a half-order of magnitude improvement right there. Then tunneling machines currently tunnel for half **the** time, then they stop, and then **the** rest of **the** time is putting in reinforcements for **the** tunnel wall. So if you design **the** machine instead to do continuous tunneling and reinforcing, that will give you a factor of two improvement. Combine that and that ’ s a factor of eight. Also these machines are far from being at their power or thermal limits, so you can jack up **the** power to **the** machine substantially. I think you can get at least a factor of two, maybe a factor of four or five improvement on top of that. So I think there ’ s a fairly straightforward series of steps to get somewhere in excess of an order of magnitude improvement in **the** cost per mile, and our target actually is — we ’ ve got a pet snail called Gary, this is from Gary **the** snail from”South Park,”I mean, sorry,”SpongeBob SquarePants.”So Gary is capable of — currently he ’ s capable of going 14 times faster than a tunnel-boring machine. CA : You want to beat Gary. EM : We want to beat Gary. He ’ s not a patient little fellow, and that will be victory. Victory is beating **the** snail. CA : But a lot of people imagining, dreaming about future cities, they imagine that actually **the** solution is flying cars, drones, etc. You go aboveground. Why isn ’ t that a better solution? You save all that tunneling cost. EM : Right. I ’ m in favor of flying things. Obviously, I do rockets, so I like things that fly. This is not some inherent bias against flying things, but there is a challenge with flying cars in that they ’ ll be quite noisy, **the** wind force generated will be very high. Let ’ s just say that if something ’ s flying over your head, a whole bunch of flying cars going all over **the** place, that is not an anxiety-reducing situation. You don ’ t think to yourself,”Well, I feel better about today.”You ’ re thinking,”Did they service their hubcap, or is it going to come off and guillotine me?”Things like that. CA : So you ’ ve got this vision of future cities with these rich, 3D networks of tunnels underneath. Is there a tie-in here with Hyperloop? Could you apply these tunnels to use for this Hyperloop idea you released a few years ago. EM : Yeah, so we ’ ve been sort of puttering around with **the** Hyperloop stuff for a while. We built a Hyperloop test track adjacent to SpaceX, just for a student competition, to encourage innovative ideas in transport. And it actually ends up being **the** biggest vacuum chamber in **the** world after **the** Large Hadron Collider, by volume. So it was quite fun to do that, but it was kind of a hobby thing, and then we think we might — so we ’ ve built a little pusher car to push **the** student pods, but we ’ re going to try seeing how fast we can make **the** pusher go if it ’ s not pushing something. So we ’ re cautiously optimistic we ’ ll be able to be faster than **the** world ’ s fastest **bullet** train even in a. 8-mile stretch. CA : Whoa. Good **brakes**. EM : Yeah, I mean, it ’ s — yeah. It ’ s either going to smash into tiny pieces or go quite fast. CA : But you can picture, then, a Hyperloop in a tunnel running quite long distances. EM : Exactly. And looking at tunneling technology, it turns out that in order to make a tunnel, you have to — In order to **seal** against **the** water table, you ’ ve got to typically design a tunnel wall to be good to about five or six atmospheres. So to go to vacuum is only one atmosphere, or near-vacuum. So actually, it sort of turns out that automatically, if you build a tunnel that is good enough to resist **the** water table, it is automatically capable of holding vacuum. CA : Huh. EM : So, yeah. CA : And so you could actually picture, what kind of length tunnel is in Elon ’ s future to running Hyperloop? EM : I think there ’ s no real length limit. You could dig as much as you want. I think if you were to do something like a DC-to-New York Hyperloop, I think you ’ d probably want to go **underground the** entire way because it ’ s a high- density area. You ’ re going under a lot of buildings and houses, and if you go deep enough, you cannot **detect the** tunnel. Sometimes people think, well, it ’ s going to be pretty annoying to have a tunnel dug under my house. Like, if that tunnel is dug more than about three or four tunnel **diameters** beneath your house, you will not be able to **detect** it being dug at all. In fact, if you ’ re able to **detect the** tunnel being dug, whatever **device** you are using, you can get a lot of money for that **device** from **the** Israeli military, who is trying to **detect** tunnels from Hamas, and from **the** US Customs and Border patrol that try and **detect** drug tunnels. So **the** reality is that earth is incredibly good at absorbing vibrations, and once **the** tunnel depth is below a certain level, it is undetectable. Maybe if you have a very sensitive seismic instrument, you might be able to **detect** it. CA : So you ’ ve started a new company to do this called **The** Boring Company. Very nice. Very funny. EM : What ’ s funny about that? CA : How much of your time is this? EM : It ’ s maybe... two or three percent. CA : You ’ ve called it a hobby. This is what an Elon Musk hobby looks like. EM : I mean, it really is, like — This is basically interns and people doing it part time. We bought some second-hand machinery. It ' s kind of puttering along, but it ' s making good progress, so — CA : So an even bigger part of your time is being spent on electrifying cars and transport through Tesla. Is one of **the** motivations for **the** tunneling project **the** realization that actually, in a world where cars are electric and where they ' re self-driving, there may end up being more cars on **the** roads on any given hour than there are now? EM : Yeah, exactly. A lot of people think that when you make cars autonomous, they ' ll be able to go faster and that will alleviate congestion. And to some degree that will be true, but once you have shared autonomy where it ' s much cheaper to go by car and you can go point to point, **the** affordability of going in a car will be better than that of a bus. Like, it will cost less than a bus **ticket**. So **the** amount of driving that will **occur** will be much greater with shared autonomy, and actually traffic will get far worse. CA : You started Tesla with **the** goal of persuading **the** world that electrification was **the** future of cars, and a few years ago, people were laughing at you. Now, not so much. EM : OK. I don ' t know. I don ' t know. CA : But isn ' t it true that pretty much every auto manufacturer has announced serious electrification plans for **the** short- to medium-term future? EM : Yeah. Yeah. I think almost every automaker has some electric vehicle program. They vary in seriousness. Some are very serious about transitioning entirely to electric, and some are just dabbling in it. And some, amazingly, are still pursuing fuel cells, but I think that won ' t last much longer. CA : But isn ' t there a sense, though, Elon, where you can now just declare victory and say, you know,"We did it."Let **the** world electrify, and you go on and focus on other stuff? EM : Yeah. I intend to stay with Tesla as far into **the** future as I can imagine, and there are a lot of exciting things that we have coming. Obviously **the** Model 3 is coming soon. We ' ll be **unveiling the** Tesla Semi truck. CA : OK, we ' re going to come to this. So Model 3, it ' s supposed to be coming in July-ish. EM : Yeah, it ' s looking quite good for starting production in July. CA : Wow. One of **the** things that people are so excited about is **the** fact that it ' s got autopilot. And you put out this video a while back showing what that technology would look like. EM : Yeah. CA : There ' s obviously autopilot in Model S right now. What are we seeing here? EM : Yeah, so this is using only cameras and GPS. So there ' s no LIDAR or radar being used here. This is just using passive optical, which is essentially what a person uses. **The** whole road system is meant to be navigated with passive optical, or cameras, and so once you solve cameras or vision, then autonomy is solved. If you don ' t solve vision, it ' s not solved. So that ' s why our focus is so heavily on having a vision neural net that ' s very effective for road conditions. CA : Right. Many other people are going **the** LIDAR route. You want cameras plus radar is most of it. EM : You can absolutely be superhuman with just cameras. Like, you can probably do it ten times better than humans would, just cameras. CA : So **the** new cars being sold right now have eight cameras in them. They can ' t yet do what that showed. When will they be able to? EM : I think we ' re still on track for being able to go cross-country from LA to New York by **the** end of **the** year, fully autonomous. CA : OK, so by **the** end of **the** year, you ' re saying, someone ' s going to sit in a Tesla without touching **the** steering wheel, tap in"New York,"off it goes. EM : Yeah. CA : Won ' t ever have to touch **the** wheel — by **the** end of 2017. EM : Yeah. Essentially, November or December of this year, we should be able to go all **the** way from a parking lot in California to a parking lot in New York, no controls touched at any point during **the** entire journey. CA : Amazing. But part of that is possible because you ' ve already got a fleet of Teslas driving all these roads. You ' re accumulating a huge amount of data of that national road system. EM

: Yes, but **the** thing that will be interesting is that I ' m actually fairly confident it will be able to do that route even if you change **the** route dynamically. So, it ' s fairly easy — If you say I ' m going to be really good at one specific route, that ' s one thing, but it should be able to go, really be very good, certainly once you enter a **highway**, to go anywhere on **the highway** system in a given country. So it ' s not sort of limited to LA to New York. We could change it and make it Seattle-Florida, that day, in real time. So you were going from LA to New York. Now go from LA to Toronto. CA : So leaving aside regulation for a second, in terms of **the** technology alone, **the** time when someone will be able to buy one of your cars and literally just take **the** hands off **the** wheel and go to sleep and wake up and find that they ' ve arrived, how far away is that, to do that safely? EM : I think that ' s about two years. So **the** real trick of it is not how do you make it work say 99. 9 percent of **the** time, because, like, if a car crashes one in a thousand times, then you ' re probably still not going to be comfortable falling asleep. You shouldn ' t be, certainly. It ' s never going to be perfect. No system is going to be perfect, but if you say it ' s perhaps — **the** car is unlikely to crash in a hundred lifetimes, or a thousand lifetimes, then people are like, OK, wow, if I were to live a thousand lives, I would still most likely never experience a crash, then that ' s probably OK. CA : To sleep. I guess **the** big concern of yours is that people may actually get seduced too early to think that this is safe, and that you ' ll have some horrible incident happen that puts things back. EM : Well, I think that **the** autonomy system is likely to at least mitigate **the** crash, except in rare circumstances. **The** thing to appreciate about vehicle safety is this is probabilistic. I mean, there ' s some chance that any time a human driver gets in a car, that they will have an accident that is their fault. It ' s never zero. So really **the** key threshold for autonomy is how much better does autonomy need to be than a person before you can rely on it? CA : But once you get literally safe hands-off driving, **the** power to disrupt **the** whole industry seems massive, because at that point you ' ve spoken of people being able to buy a car, drops you off at work, and then you let it go and provide a sort of Uber-like service to other people, earn you money, maybe even cover **the** cost of your lease of that car, so you can kind of get a car for free. Is that really likely? EM : Yeah. Absolutely this is what will happen. So there will be a shared autonomy fleet where you buy your car and you can choose to use that car exclusively, you could choose to have it be used only by friends and family, only by other drivers who are rated five star, you can choose to share it sometimes but not other times. That ' s 100 percent what will **occur**. It ' s just a question of when. CA : Wow. So you mentioned **the** Semi and I think you ' re planning to announce this in September, but I ' m curious whether there ' s anything you could show us today? EM : I will show you a teaser shot of **the** truck. It ' s alive. CA : OK. EM : That ' s definitely a case where we want to be cautious about **the** autonomy features. Yeah. CA : We can ' t see that much of it, but it doesn ' t look like just a little **friendly** neighborhood truck. It looks kind of badass. What sort of semi is this? EM : So this is a heavy duty, long-range semitruck. So it ' s **the** highest weight capability and with long range. So essentially it ' s meant to alleviate **the** heavy-duty trucking loads. And this is something which people do not today think is possible. They think **the** truck doesn ' t have enough power or it doesn ' t have enough range, and then with **the** Tesla Semi we want to show that no, an electric truck actually can out-torque any diesel semi. And if you had a tug- of-war competition, **the** Tesla Semi will tug **the** diesel semi uphill. CA : That ' s pretty cool. And short term, these aren ' t driverless. These are going to be trucks that truck drivers want to drive. EM : Yes. So what will be really fun about this is you have a flat torque RPM curve with an electric motor, whereas with a diesel motor or any kind of internal combustion engine car, you ’ ve got a torque RPM curve that looks like a hill. So this will be a very spry truck. You can drive this around like a sports car. There ’ s no gears. It ’ s, like, single speed. CA : There ’ s a great movie to be made here somewhere. I don ’ t know what it is and I don ’ t know that it ends well, but it ’ s a great movie. EM : It ’ s quite bizarre test-driving. When I was driving **the** test prototype for **the** first truck. It ’ s really weird, because you ’ re driving around and you ’ re just so nimble, and you ’ re in this giant truck. CA : Wait, you ’ ve already driven a prototype? EM : Yeah, I drove it around **the** parking lot, and I was like, this is crazy. CA : Wow. This is no vaporware. EM : It ’ s just like, driving this giant truck and making these mad maneuvers. CA : This is cool. OK, from a really badass picture to a kind of less badass picture. This is just a cute house from”Desperate Housewives”or something. What on earth is going on here? EM : Well, this illustrates **the** picture of **the** future that I think is how things will **evolve**. You ’ ve got an electric car in **the** driveway. If you look in between **the** electric car and **the** house, there are actually three Powerwalls **stacked** up against **the** side of **the** house, and then that house roof is a solar roof. So that ’ s an actual solar glass roof. CA : OK. EM : That ’ s a picture of a real — well, admittedly, it ’ s a real fake house. That ’ s a real fake house. CA : So these roof tiles, some of them have in them basically solar power, **the** ability to — EM : Yeah. Solar glass tiles where you can adjust **the** texture and **the** color to a very fine-grained level, and then there ’ s sort of microlouvers in **the** glass, such that when you ’ re looking at **the** roof from street level or close to street level, all **the** tiles look **the** same whether there is a solar cell behind it or not. So you have an even color from **the** ground level. If you were to look at it from a helicopter, you would be actually able to look through and see that some of **the** glass tiles have a solar cell behind them and some do not. You can ’ t tell from street level. CA : You put them in **the** ones that are likely to see a lot of sun, and that makes these roofs super affordable, right? They ’ re not that much more expensive than just tiling **the** roof. EM : Yeah. We ’ re very confident that **the** cost of **the** roof plus **the** cost of electricity — A solar glass roof will be less than **the** cost of a normal roof plus **the** cost of electricity. So in other words, this will be economically a no-brainer, we think it will look great, and it will last — We thought about having **the** warranty be infinity, but then people thought, well, that might sound like were just talking rubbish, but actually this is toughened glass. Well after **the** house has collapsed and there ’ s nothing there, **the** glass tiles will still be there. CA : I mean, this is cool. So you ’ re rolling this out in a couple week ’ s time, I think, with four different roofing types. EM : Yeah, we ’ re starting off with two, two initially, and **the** second two will be introduced early next year. CA : And what ’ s **the** scale of ambition here? How many houses do you believe could end up having this type of roofing? EM : I think eventually almost all houses will have a solar roof. **The** thing is to consider **the** time scale here to be probably on **the** order of 40 or 50 years. So on average, a roof is replaced every 20 to 25 years. But you don ’ t start replacing all roofs immediately. But eventually, if you say were to fast- forward to say 15 years from now, it will be unusual to have a roof that does not have solar. CA : Is there a mental model thing that people don ’ t get here that because of **the** shift in **the** cost, **the** economics of solar power, most houses actually have enough sunlight on their roof pretty much to power all of their needs. If you could capture **the** power, it could pretty much power all their needs. You could go off-grid, kind of. EM : It depends on where you are and what **the** house size is **relative** to **the** roof area, but it ’ s a fair statement to say that most houses in **the** US have enough roof area to power all **the** needs of **the** house. CA : So **the** key to **the** economics of **the** cars, **the** Semi, of these houses is **the** falling price of lithium-ion batteries, which you ’ ve made a huge bet on as Tesla. In many ways, that ' s almost **the** core competency. And you ' ve decided that to really, like, own that competency, you just have to build **the** world ' s largest manufacturing plant to double **the** world ' s supply of lithium-ion batteries, with this guy. What is this? EM : Yeah, so that ' s **the** Gigafactory, progress so far on **the** Gigafactory. Eventually, you can sort of roughly see that there ' s sort of a diamond shape overall, and when it ' s fully done, it ' ll look like a giant diamond, or that ' s **the** idea behind it, and it ' s aligned on true north. It ' s a small detail. CA : And capable of producing, eventually, like a hundred gigawatt hours of batteries a year. EM : A hundred gigawatt hours. We think probably more, but yeah. CA : And they ' re actually being produced right now. EM : They ' re in production already. CA : You guys put out this video. I mean, is that speeded up? EM : That ' s **the** slowed down version. CA : How fast does it actually go? EM : Well, when it ' s running at full speed, you can ' t actually see **the** cells without a strobe light. It ' s just blur. CA : One of your core ideas, Elon, about what makes an exciting future is a future where we no longer feel guilty about energy. Help us picture this. How many Gigafactories, if you like, does it take to get us there? EM : It ' s about a hundred, roughly. It ' s not 10, it ' s not a thousand. Most likely a hundred. CA : See, I find this amazing. You can picture what it would take to move **the** world off this vast **fossil** fuel thing. It ' s like you ' re building one, it costs five billion dollars, or whatever, five to 10 billion dollars. Like, it ' s kind of cool that you can picture that project. And you ' re planning to do, at Tesla — announce another two this year. EM : I think we ' ll announce locations for somewhere between two and four Gigafactories later this year. Yeah, probably four. CA : Whoa. No more teasing from you for here? Like — where, **continent**? You can say no. EM : We need to address a global market. CA : OK. This is cool. I think we should talk for — Actually, global market. I ' m going to ask you one question about politics, only one. I ' m kind of sick of politics, but I do want to ask you this. You ' re on a body now giving advice to a guy — EM : Who? CA : Who has said he doesn ' t really believe in climate change, and there ' s a lot of people out there who think you shouldn ' t be doing that. They ' d like you to walk away from that. What would you say to them? EM : Well, I think that first of all, I ' m just on two advisory councils where **the** format consists of going around **the** room and asking people ' s opinion on things, and so there ' s like a meeting every month or two. That ' s **the sum** total of my contribution. But I think to **the** degree that there are people in **the** room who are arguing in favor of doing something about climate change, or social issues, I ' ve used **the** meetings I ' ve had thus far to argue in favor of immigration and in favor of climate change. And if I hadn ' t done that, that wasn ' t on **the** agenda before. So maybe nothing will happen, but at least **the** words were said. CA : OK. So let ' s talk SpaceX and Mars. Last time you were here, you spoke about what seemed like a kind of incredibly ambitious dream to develop rockets that were actually reusable. And you ' ve only gone and done it. EM : Finally. It took a long time. CA : Talk us through this. What are we looking at here? EM : So this is one of our rocket boosters coming back from very high and fast in space. So just delivered **the** upper stage at high velocity. I think this might have been at sort of Mach 7 or so, delivery of **the** upper stage. CA : So that was a sped-up — EM : That was **the** slowed down version. CA : I thought that was **the** sped-up version. But I mean, that ' s amazing, and several of these failed before you finally figured out how to do it, but now you ' ve done this, what, five or six times? EM : We ' re at eight or nine. CA : And for **the** first time, you ' ve actually reflown one of **the** rockets that landed. EM : Yeah, so we landed **the** rocket booster and then prepped it for flight again and flew it again, so it ' s **the** first reflight of an orbital booster where that reflight is relevant. So it ’ s important to appreciate that reusability is only relevant if it is rapid and complete. So like an aircraft or a car, **the** reusability is rapid and complete. You do not send your aircraft to Boeing in- between flights. CA : Right. So this is allowing you to dream of this really ambitious idea of sending many, many, many people to Mars in, what, 10 or 20 years time, I guess. EM : Yeah. CA : And you ’ ve designed this outrageous rocket to do it. Help us understand **the** scale of this thing. EM : Well, visually you can see that ’ s a person. Yeah, and that ’ s **the** vehicle. CA : So if that was a skyscraper, that ’ s like, did I read that, a 40-story skyscraper? EM : Probably a little more, yeah. **The** thrust level of this is really — This **configuration** is about four times **the** thrust of **the** Saturn V moon rocket. CA : Four times **the** thrust of **the** biggest rocket humanity ever created before. EM : Yeah. Yeah. CA : As one does. EM : Yeah. In units of 747, a 747 is only about a quarter of a million **pounds** of thrust, so for every 10 million **pounds** of thrust, there ’ s 40 747s. So this would be **the** thrust equivalent of 120 747s, with all engines blazing. CA : And so even with a machine designed to escape Earth ’ s gravity, I think you told me last time this thing could actually take a fully loaded 747, people, cargo, everything, into orbit. EM : Exactly. This can take a fully loaded 747 with maximum fuel, maximum passengers, maximum cargo on **the** 747 — this can take it as cargo. CA : So based on this, you presented recently this Interplanetary Transport System which is visualized this way. This is a scene you picture in, what, 30 years time? 20 years time? People walking into this rocket. EM : I ’ m hopeful it ’ s sort of an eight- to 10-year time frame. Aspirationally, that ’ s our target. Our internal targets are more aggressive, but I think — CA : OK. EM : While vehicle seems quite large and is large by comparison with other rockets, I think **the** future spacecraft will make this look like a rowboat. **The** future spaceships will be truly enormous. CA : Why, Elon? Why do we need to build a city on Mars with a million people on it in your lifetime, which I think is kind of what you ’ ve said you ’ d love to do? EM : I think it ’ s important to have a future that is inspiring and appealing. I just think there have to be reasons that you get up in **the** morning and you want to live. Like, why do you want to live? What ’ s **the** point? What inspires you? What do you love about **the** future? And if we ’ re not out there, if **the** future does not include being out there among **the** stars and being a multiplanet species, I find that it ’ s incredibly depressing if that ’ s not **the** future that we ’ re going to have. CA : People want to position this as an either or, that there are so many desperate things happening on **the planet** now from climate to poverty to, you know, you pick your issue. And this feels like a distraction. You shouldn ’ t be thinking about this. You should be solving what ’ s here and now. And to be fair, you ’ ve done a fair old bit to actually do that with your work on sustainable energy. But why not just do that? EM : I think there ’ s — I look at **the** future from **the** standpoint of **probabilities**. It ’ s like a branching stream of **probabilities**, and there are actions that we can take that affect those **probabilities** or that accelerate one thing or slow down another thing. I may introduce something new to **the** **probability** stream. Sustainable energy will happen no matter what. If there was no Tesla, if Tesla never existed, it would have to happen out of necessity. It ’ s tautological. If you don ’ t have sustainable energy, it means you have unsustainable energy. Eventually you will run out, and **the** laws of economics will drive civilization towards sustainable energy, inevitably. **The** fundamental value of a company like Tesla is **the** degree to which it accelerates **the** advent of sustainable energy, faster than it would otherwise **occur**. So when I think, like, what is **the** fundamental good of a company like Tesla, I would say, hopefully, if it accelerated that by a decade, potentially more than a decade, that would be quite a good thing to **occur**. That ’ s what I consider to be **the** fundamental aspirational good of Tesla. Then there ' s becoming a multiplanet species and space-faring civilization. This is not inevitable. It ' s very important to appreciate this is not inevitable. **The** sustainable energy future I think is largely inevitable, but being a space- faring civilization is definitely not inevitable. If you look at **the** progress in space, in 1969 you were able to send somebody to **the** moon. 1969. Then we had **the** Space Shuttle. **The** Space Shuttle could only take people to low Earth orbit. Then **the** Space Shuttle retired, and **the** United States could take no one to orbit. So that ' s **the** trend. **The** trend is like down to nothing. People are mistaken when they think that technology just automatically improves. It does not automatically improve. It only improves if a lot of people work very hard to make it better, and actually it will, I think, by itself degrade, actually. You look at great civilizations like Ancient Egypt, and they were able to make **the** pyramids, and they forgot how to do that. And then **the** Romans, they built these incredible aqueducts. They forgot how to do it. CA : Elon, it almost seems, listening to you and looking at **the** different things you ' ve done, that you ' ve got this unique double motivation on everything that I find so interesting. One is this desire to work for humanity ' s long-term good. **The** other is **the** desire to do something exciting. And often it feels like you feel like you need **the** one to drive **the** other. With Tesla, you want to have sustainable energy, so you made these super sexy, exciting cars to do it. Solar energy, we need to get there, so we need to make these beautiful roofs. We haven ' t even spoken about your newest thing, which we don ' t have time to do, but you want to save humanity from bad AI, and so you ' re going to create this really cool brain-machine interface to give us all infinite memory and telepathy and so forth. And on Mars, it feels like what you ' re saying is, yeah, we need to save humanity and have a backup plan, but also we need to inspire humanity, and this is a way to inspire. EM : I think **the** value of beauty and inspiration is very much underrated, no question. But I want to be clear. I ' m not trying to be anyone ' s savior. That is not **the** — I ' m just trying to think about **the** future and not be sad. CA : Beautiful statement. I think everyone here would agree that it is not — None of this is going to happen inevitably. **The** fact that in your mind, you dream this stuff, you dream stuff that no one else would dare dream, or no one else would be capable of dreaming at **the** level of complexity that you do. **The** fact that you do that, Elon Musk, is a really remarkable thing. Thank you for helping us all to dream a bit bigger. EM : But you ' ll tell me if it ever starts getting genuinely insane, right? CA : Thank you, Elon Musk. That was really, really **fantastic**. That was really **fantastic**.

Text Sample 56: POS Dim 1 – 2017 – Score 23.00 – 2017/t000541.txt

We know more about other **planets** than our own, and today, I want to show you a new type of robot designed to help us better understand our own **planet**. It belongs to a category known in **the** oceanographic community as an unmanned **surface** vehicle, or USV. And it uses no fuel. Instead, it relies on wind power for propulsion. And yet, it can sail around **the** globe for months at a time. So I want to share with you why we built it, and what it means for you. A few years ago, I was on a sailboat making its way across **the** Pacific, from San Francisco to Hawaii. I had just spent **the** past 10 years working nonstop, developing video games for hundreds of millions of users, and I wanted to take a step back and look at **the** big picture and get some much-needed thinking time. I was **the** navigator on board, and one evening, after a long session analyzing weather data and plotting our course, I came up on deck and saw this beautiful sunset. And a thought **occurred** to me : How much do we really know about our oceans? **The** Pacific was stretching all around me as far as **the** eye could see, and **the** waves were **rocking** our boat forcefully, a sort of constant reminder of its untold power. How much do we really know about our oceans? I decided to find out. What I quickly learned is that we don ' t know very much. **The** first reason is just how vast oceans are, covering 70 percent of **the planet**, and yet we know they drive complex planetary systems like global weather, which affect all of us on a daily basis, sometimes dramatically. And yet, those activities are mostly invisible to us. Ocean data is scarce by any standard. Back on land, I had grown used to accessing lots of sensors — billions of them, actually. But at **sea**, in situ data is scarce and expensive. Why? Because it relies on a small number of ships and buoys. How small a number was actually a great surprise. Our National Oceanic and Atmospheric Administration, better known as NOAA, only has 16 ships, and there are less than 200 buoys offshore globally. It is easy to understand why : **the** oceans are an unforgiving place, and to collect in situ data, you need a big ship, capable of carrying a vast amount of fuel and large crews, costing hundreds of millions of dollars each, or, big buoys tethered to **the** ocean floor with a four-mile-long cable and weighted down by a set of train wheels, which is both dangerous to deploy and expensive to maintain. What about satellites, you might ask? Well, satellites are **fantastic**, and they have taught us so much about **the** big picture over **the** past few decades. However, **the** problem with satellites is they can only see through one micron of **the surface** of **the** ocean. They have relatively poor spatial and temporal resolution, and their signal needs to be corrected for cloud cover and land effects and other factors. So what is going on in **the** oceans? And what are we trying to measure? And how could a robot be of any use? Let ' s zoom in on a small cube in **the** ocean. One of **the** key things we want to understand is **the surface**, because **the surface**, if you think about it, is **the** nexus of all air-sea interaction. It is **the** interface through which all energy and gases must flow. Our sun radiates energy, which is absorbed by oceans as heat and then partially released into **the** atmosphere. Gases in our atmosphere like CO2 get dissolved into our oceans. Actually, about 30 percent of all global CO2 gets absorbed. Plankton and microorganisms release oxygen into **the** atmosphere, so much so that every other breath you take comes from **the** ocean. Some of that heat generates evaporation, which creates clouds and then eventually leads to precipitation. And pressure gradients create **surface** wind, which moves **the** moisture through **the** atmosphere. Some of **the** heat radiates down into **the** deep ocean and gets stored in different layers, **the** ocean acting as some kind of planetary-scale boiler to store all that energy, which later might be released in short-term events like hurricanes or long-term phenomena like El Nio. These layers can get mixed up by vertical upwelling currents or **horizontal** currents, which are key in transporting heat from **the** tropics to **the** poles. And of course, there is marine life, occupying **the** largest ecosystem in volume on **the planet**, from microorganisms to **fish** to marine mammals, like **seals**, dolphins and **whales**. But all of these are mostly invisible to us. **The** challenge in studying those ocean variables at scale is one of energy, **the** energy that it takes to deploy sensors into **the** deep ocean. And of course, many solutions have been tried — from wave-actuated **devices** to **surface** drifters to sun-powered **electrical** drives — each with their own compromises. Our team breakthrough came from an unlikely source — **the** pursuit of **the** world speed record in a wind-powered land yacht. It took 10 years of research and development to come up with a novel **wing** concept that only uses three watts of power to control and yet can propel a vehicle all around **the** globe with seemingly unlimited autonomy. By adapting this **wing** concept into a marine vehicle, we had **the** genesis of an ocean drone. Now, these are larger than they appear. They are about 15 feet high, 23 feet long, seven feet deep. Think of them as **surface** satellites. They ’ re laden with an **array** of science-grade sensors that measure all key variables, both oceanographic and atmospheric, and a live satellite link transmits this high- resolution data back to shore in real time. Our team has been hard at work over **the** past few years, conducting missions in some of **the** toughest ocean conditions on **the planet**, from **the** Arctic to **the** tropical Pacific. We have sailed all **the** way to **the** polar ice shelf. We have sailed into Atlantic hurricanes. We have rounded Cape Horn, and we have slalomed between **the oil** rigs of **the** Gulf of Mexico. This is one tough robot. Let me share with you recent work that we did around **the** Pribilof Islands. This is a small group of **islands** deep in **the** cold Bering Sea between **the** US and Russia. Now, **the** Bering Sea is **the** home of **the** walleye pollock, which is a whitefish you might not recognize, but you might likely have tasted if you **enjoy fish** sticks or surimi. Yes, surimi looks like crabmeat, but it ’ s actually pollock. And **the** pollock fishery is **the** largest fishery in **the** nation, both in terms of value and volume — about 3. 1 billion **pounds** of **fish** caught every year. So over **the** past few years, a fleet of ocean drones has been hard at work in **the** Bering Sea with **the** goal to help assess **the** size of **the** pollock **fish** stock. This helps improve **the** quota system that ’ s used to manage **the** fishery and help prevent a collapse of **the fish** stock and protects this fragile ecosystem. Now, **the** drones survey **the fishing** ground using acoustics, i. e., a sonar. This sends a sound wave downwards, and then **the** reflection, **the** echo from **the** sound wave from **the** seabed or schools of **fish**, gives us an idea of what ’ s happening below **the surface**. Our ocean drones are actually pretty good at this repetitive task, so they have been gridding **the** Bering Sea day in, day out. Now, **the** Pribilof Islands are also **the** home of a large colony of fur **seals**. In **the** 1950s, there were about two million individuals in that colony. Sadly, these days, **the** population has rapidly declined. There ’ s less than 50 percent of that number left, and **the** population continues to fall rapidly. So to understand why, our science partner at **the** National Marine Mammal Laboratory has fitted a GPS **tag** on some of **the** mother **seals**, glued to their furs. And this **tag** measures location and depth and also has a really cool little camera that ’ s triggered by sudden acceleration. Here is a movie taken by an artistically inclined **seal**, giving us unprecedented insight into an **underwater** hunt deep in **the** Arctic, and **the** shot of this pollock prey just seconds before it gets devoured. Now, doing work in **the** Arctic is very tough, even for a robot. They had to survive a snowstorm in August and interferences from bystanders — that little spotted **seal enjoying** a ride. Now, **the seal tags** have recorded over 200, 000 dives over **the** season, and upon a closer look, we get to see **the** individual **seal** tracks and **the** repetitive dives. We are on our way to decode what is really happening over that foraging ground, and it ’ s quite beautiful. Once you superimpose **the** acoustic data collected by **the** drones, a picture starts to emerge. As **the seals** leave **the islands** and swim from left to right, they are observed to dive at a relatively shallow depth of about 20 meters, which **the** drone identifies is populated by small young pollock with low calorific content. **The seals** then swim much greater distance and start to dive deeper to a place where **the** drone identifies larger, more adult pollock, which are more nutritious as **fish**. Unfortunately, **the** calories expended by **the** mother **seals** to swim this extra distance don ’ t leave them with enough energy to lactate their pups back on **the island**, leading to **the** population decline. Further, **the** drones identify that **the** water temperature around **the island** has significantly warmed. It might be one of **the** driving forces that ’ s pushing **the** pollock north, and to spread in **search** of colder regions. So **the** data analysis is ongoing, but already we can see that some of **the** pieces of **the** puzzle from **the** fur **seal** mystery are coming into focus. But if you look back at **the** big picture, we are mammals, too. And actually, **the** oceans provide up to 20 kilos of **fish** per human per year. As we deplete our **fish** stocks, what can we humans learn from **the** fur **seal** story? And beyond **fish**, **the** oceans affect all of us daily as they drive global weather systems, which affect things like global agricultural output or can lead to devastating destruction of lives and property through hurricanes, extreme heat and floods. Our oceans are pretty much unexplored and undersampled, and today, we still know more about other **planets** than our own. But if you divide this vast ocean in six-by-six-degree squares, each about 400 miles long, you ' d get about 1, 000 such squares. So little by little, working with our partners, we are deploying one ocean drone in each of those boxes, **the** hope being that achieving planetary coverage will give us better insights into those planetary systems that affect humanity. We have been using robots to study distant worlds in our solar system for a while now. Now it is time to quantify our own **planet**, because we cannot fix what we cannot measure, and we cannot prepare for what we don ' t know. Thank you.

Text Sample 57: POS Dim 1 – 2019 – Score 21.00 – 2019/t001498.txt
My mom has always reminded me that I have **the** same proportions as a LEGO man. And she does actually have a point. LEGO is a company that has succeeded in making everybody believe that LEGO is from their home country. But it ' s not, it ' s from my home country. So you can imagine my excitement when **the** LEGO family called me and asked us to work with them to design **the** Home of **the** Brick. This is **the** architectural model — we built it out of LEGO, obviously. This is **the** final result. And what we tried to do was to design a building that would be as interactive and as engaging and as playful as LEGO is itself, with these kind of interconnected playgrounds on **the** roofscape. You can enter a square on **the** ground where **the** citizens of Billund can roam around freely without a **ticket**. And it ' s probably one of **the** only museums in **the** world where you ' re allowed to touch all **the** artifacts. But **the** Danish word for design is"formgivning,"which literally means to give form to that which has not yet been given form. In other words, to give form to **the** future. And what I love about LEGO is that LEGO is not a toy. It ' s a tool that empowers **the** child to build his or her own world, and then to inhabit that world through play and to invite her friends to join her in cohabiting and cocreating that world. And that is exactly what formgivning is. As human beings, we have **the** power to give form to our future. Inspired by LEGO, we ' ve built a social housing project in Copenhagen, where we **stacked** blocks of wood next to each other. Between them, they leave spaces with extra ceiling heights and balconies. And by gently wiggling **the** blocks, we can actually create curves or any **organic** form, adapting to any urban context. Because adaptability is probably one of **the** strongest drivers of architecture. Another example is here in Vancouver. We were asked to look at **the** site where Granville bridge triforks as it touches downtown. And we started, like, mapping **the** different constraints. There ' s like a 100-foot setback from **the** bridge because **the** city want to make sure that no one looks into **the** traffic on **the** bridge. There ' s a park where we can ' t cast any shadows. So finally, we ' re left with a tiny triangular footprint, almost too small to build. But then we thought, like, what if **the** 100-foot minimum distance is really about minimum distance — once we get 100 feet up in **the** air, we can grow **the** building back out. And so we did. When you drive over **the** bridge, it ' s as if someone is pulling a curtain aback, welcoming you to Vancouver. Or a like a weed growing through **the** cracks in **the** pavement and

blossoming as it gets light and air. Underneath **the** bridge, we ’ ve worked with Rodney Graham and a handful of Vancouver artists, to create what we called **the** Sistine Chapel of street art, an art gallery turned upside down, that tries to turn **the** negative impact of **the** bridge into a positive. So even if it looks like this kind of surreal architecture, it ’ s highly adapted to its surroundings. So if a bridge can become a museum, a museum can also serve as a bridge. In Norway, we are building a museum that spans across a river and allows people to sort of journey through **the** exhibitions as they cross from one side of a sculpture park to **the** other. An architecture sort of adapted to its landscape. In China, we built a headquarters for an energy company and we designed **the** facade like an Issey Miyake fabric. It ’ s rippled, so that facing **the** predominant direction of **the** sun, it ’ s all opaque ; facing away from **the** sun, it ’ s all glass. On average, it sort of transitions from solid to clear. And this very simple idea without any moving parts or any sort of technology, purely because of **the** geometry of **the** facade, reduces **the** energy **consumption** on cooling by 30 percent. So you can say what makes **the** building look elegant is also what makes it perform elegantly. It ’ s an architecture that is adapted to its climate. You can also adapt one culture to another, like in Manhattan, we took **the** Copenhagen courtyard building with a social space where people can hang out in this kind of oasis in **the** middle of a city, and we combined it with **the** density and **the** verticality of an American skyscraper, creating what we ’ ve called a”courtscraper.”From New York to Copenhagen. On **the** waterfront of Copenhagen, we are right now finishing this waste-to-energy power plant. It ’ s going to be **the** cleanest waste-to-energy power plant in **the** world, there are no toxins coming out of **the** chimney. An amazing marvel of engineering that is completely invisible. So we thought, how can we express this? And in Copenhagen we have snow, as you can see, but we have absolutely no mountains. We have to go six hours by bus to get to Sweden, to get alpine skiing. So we thought, let ’ s put an alpine ski **slope** on **the** roof of **the** power plant. So this is **the** first test run we did a few months ago. And what I like about this is that it also show you **the** sort of world-changing power of formgivning. I have a five-month-old son, and he ’ s going to grow up in a world not knowing that there was ever a time when you couldn ’ t ski on **the** roof of **the** power plant. So imagine for him and his generation, that ’ s their baseline. Imagine how far they can leap, what kind of wild ideas they can put forward for their future. So right in front of it, we ’ re building our smallest project. It ’ s basically nine containers that we have **stacked** in a shipyard in Poland, then we ’ ve schlepped it across **the** Baltic **sea** and docked it in **the** port of Copenhagen, where it is now **the** home of 12 students. Each student has a view to **the** water, they can jump out **the** window into **the** clean port of Copenhagen, and they can get back in. All of **the** heat comes from **the** thermal mass of **the sea**, all **the** power comes from **the** sun. This is **the** first 12 units in Copenhagen, another 60 on their way, another 200 are going to Gothenburg, and we ’ re speaking with **the** Paris Olympics to put a small floating village on **the** Seine. So very much this kind of, almost like nomadic, impermanent architecture. And **the** waterfronts of our cities are experiencing a lot of change. Economic change, industrial change and climate change. This is Manhattan before Hurricane Sandy, and this is Manhattan after Sandy. We got invited by **the** city of New York to look if we could make **the** necessary flood protection for Manhattan without building a seawall that would segregate **the** life of **the** city from **the** water around it. And we got inspired by **the** High Line. You probably know **the** High Line — it ’ s this amazing new park in New York. It ’ s basically decommissioned train tracks that now have become one of **the** most popular promenades in **the** city. So we thought, could we design **the** necessary flood protection for Manhattan so we don ’ t have to wait until we shut it down before it gets nice? So we sat down with **the** citizens living along **the** waterfront of New York, and we worked with them to try to design **the** necessary flood protection in such a way that it only makes their waterfront more accessible and more enjoyable. Underneath **the** FDR, we are putting, like, pavilions with **pocket** walls that can **slide** out and protect from **the** water. We are creating little stepped terraces that are going to make **the** underside more enjoyable, but also protect from flooding. Further north in **the** East River Park, we are creating rolling hills that protect **the** park from **the** noise of **the highway**, but in turn also become **the** necessary flood protection that can stop **the** waves during an incoming storm surge. So in a way, this project that we have called **the** Dryline, it ’ s essentially **the** High Line — **The** High Line that ’ s going to keep Manhattan dry. It ’ s scheduled to break ground on **the** first East River portion at **the** end of this year. But it has essentially been codesigned with **the** citizens of Lower Manhattan to take all of **the** necessary infrastructure for resilience and give it positive social and environmental side effects. So, New York is not alone in facing this situation. In fact, by 2050, 90 percent of **the** major cities in **the** world are going to be dealing with rising **seas**. In Hamburg, they ’ ve created a whole neighborhood where **the** bottom floors are designed to withstand **the** inevitable flood. In Sweden, they ’ ve designed a city where all of **the** parks are **wet** gardens, designed to deal with storm water and waste water. So we thought, could we perhaps — Actually, today, three million people are already permanently living on **the sea**. So we thought, could we actually imagine a floating city designed to incorporate all of **the** Sustainable Development Goals of **the** United Nations into a whole new human-made ecosystem. And of course, we have to design it so it can produce its own power, harvesting **the** thermal mass of **the** oceans, **the** force of **the** tides, of **the** currents, of **the** waves, **the** power of **the** wind, **the** heat and **the** energy of **the** sun. Also, we are going to collect all of **the** rain water that drops on this man-made archipelago and deal with it organically and mechanically and store it and clean it. We have to grow all of our food locally, it has to be fish- and plant-based, because you won ’ t have **the** space or **the** resources for a dairy diet. And finally, we are going to deal with all **the** waste locally, with compost, recycling, and turning **the** waste into energy. So imagine where a traditional urban master plan, you typically draw **the** street grid where **the** cars can drive and **the** building plots where you can put some buildings. This master plan, we sat down with a handful of scientists and basically started with all of **the renewable**, available natural resources, and then we started channeling **the** flow of resources through this kind of human- made ecosystem or this kind of urban metabolism. So it ’ s going to be modular, it ’ s going to be buoyant, it ’ s going to be designed to resist a tropical storm. You can prefabricate it at scale, and tow it to dock with others, to form a small community. We ’ re designing these kind of coastal additions, so that even if it ’ s modular and rational, each **island** can be unique with its own coastal landscape. **The** architecture has to remain relatively low to keep **the** center of gravity buoyant. We ’ re going to take all of **the agriculture** and use it to also create social space so you can actually **enjoy the** permaculture gardens. We ’ re designing it for **the** tropics, so all of **the** roofs are maximized to harvest solar power and to shade from **the** sun. All **the** materials are going to be light and **renewable**, like bamboo and wood, which is also going to create this charming, warm environment. And any architecture is supposed to be able to fit on this platform. Underneath we have all **the** storage inside **the** pontoon, almost like a mega version of **the** student housings that we ’ ve already worked with. We have all **the** storage for **the** energy that ’ s produced, all of **the** water storage and remediation. We are sort of dealing with all of **the** waste and **the** composting.

And we also have some backup farming with aeroponics and hydroponics. So imagine almost like a vertical section through this landscape that goes from **the** air above, where we have vertical farms ; below, we have **the** aeroponics and **the** aquaponics. Even further below, we have **the** ocean farms and where we tie **the island** to **the** ground, we ’ re using biorock to create new reefs to regenerate **habitat**. So think of this small **island** for 300 people. It can then group together to form a cluster or a neighborhood that then can sort of group together to form an entire city for 10, 000 people. And you can imagine if this floating city flourishes, it can sort of grow like a culture in a petri dish. So one of **the** first places we are looking at placing this, or anchoring this floating city, is in **the** Pearl River delta. So imagine this kind of canopy of photovoltaics on this archipelago floating in **the sea**. As you sail towards **the island**, you will see **the** maritime residents moving around on alternative forms of aquatic **transportation**. You come into this kind of community port. You can roam around in **the** permaculture gardens that are productive landscapes, but also social landscapes. **The greenhouses** also become orangeries for **the** cultural life of **the** city, and below, under **the sea**, it ’ s teeming with life of farming and science and social spaces. So in a way, you can imagine this community port is where people gather, both by day and by night. And even if **the** first one is designed for **the** tropics, we also imagine that **the** architecture can adapt to any culture, so imagine, like, a Middle Eastern floating city or Southeast Asian floating city or maybe a Scandinavian floating city one day. So maybe just to **conclude**. **The** human body is 70 percent water. And **the surface** of our **planet** is 70 percent water. And it ’ s rising. And even if **the** whole world woke up tomorrow and became carbon-neutral over night, there are still **island** nations that are destined to sink in **the seas**, unless we also develop alternate forms of floating human **habitats**. And **the** only constant in **the universe** is change. Our world is always changing, and right now, our climate is changing. No matter how critical **the** crisis is, and it is, this is also our collective human superpower. That we have **the** power to adapt to change and we have **the** power to give form to our future.

Text Sample 58: POS Dim 1 – 2019 – Score 21.00 – 2019/t001223.txt

It may seem like we ’ re all standing on solid earth right now, but we ’ re not. **The rocks** and **the** dirt underneath us are crisscrossed by tiny little fractures and empty spaces. And these empty spaces are filled with astronomical quantities of microbes, such as these ones. **The** deepest that we found microbes so far into **the** earth is five kilometers down. So like, if you pointed yourself at **the** ground and took off running into **the** ground, you could run an entire 5K race and microbes would line your whole path. So you may not have ever thought about these microbes that are deep inside earth ’ s crust, but you probably thought about **the** microbes living in our guts. If you add up **the** gut microbiomes of all **the** people and all **the** animals on **the planet**, collectively, this weighs about 100, 000 tons. This is a huge biome that we carry in our bellies every single day. We should all be proud. But it pales in comparison to **the** number of microbes that are covering **the** entire **surface** of **the** earth, like in our soils, our rivers and our oceans. Collectively, these weigh about two billion tons. But it turns out that **the** majority of microbes on earth aren ’ t even in oceans or our guts or sewage treatment plants. Most of them are actually inside **the** earth ’ s crust. So collectively, these weigh 40 billion tons. This is one of **the** biggest biomes on **the planet**, and we didn ’ t even know it existed until a few decades ago. So **the** possibilities for what life is like down there, or what it might do for humans, are limitless. This is a map showing a red dot for every place where we ’ ve gotten pretty good deep subsurface samples with modern microbiological methods, and you may be impressed that we ’ re getting a pretty good global coverage, but actually, if you remember that these are **the** only places that we have samples from, it looks a little worse. If we were all in an alien spaceship, trying to reconstruct a map of **the** globe from only these samples, we ’ d never be able to do it. So people sometimes say to me,”Yeah, there ’ s a lot of microbes in **the** subsurface, but... aren ’ t they just kind of dormant?”This is a good point. **Relative** to a ficus plant or **the** measles or my kid ’ s guinea pigs, these microbes probably aren ’ t doing much of anything at all. We know that they have to be slow, because there ’ s so many of them. If they all started dividing at **the** rate of E. coli, then they would double **the** entire weight of **the** earth, **rocks** included, over a single night. In fact, many of them probably haven ’ t even undergone a single cell division since **the** time of ancient Egypt. Which is just crazy. Like, how do you wrap your head around things that are so long- lived? But I thought of an analogy that I really love, but it ’ s weird and it ’ s complicated. So I hope that you can all go there with me. Alright, let ’ s try it. It ’ s like trying to figure out **the** life cycle of a tree... if you only lived for a day. So like if human life span was only a day, and we lived in winter, then you would go your entire life without ever seeing a tree with a leaf on it. And there would be so many human generations that would pass by within a single winter that you may not even have access to a history book that says anything other than **the** fact that trees are always lifeless sticks that don ’ t do anything. Of course, this is ridiculous. We know that trees are just waiting for summer so they can reactivate. But if **the** human life span were significantly shorter than that of trees, we might be completely oblivious to this totally mundane fact. So when we say that these deep subsurface microbes are just dormant, are we like people who die after a day, trying to figure out how trees work? What if these deep subsurface organisms are just waiting for their version of summer, but our lives are too short for us to see it? If you take E. coli and **seal** it up in a test tube, with no food or nutrients, and leave it there for months to years, most of **the** cells die off, of course, because they ’ re starving. But a few of **the** cells survive. If you take these old surviving cells and compete them, also under starvation conditions, against a new, fast-growing culture of E. coli, **the** grizzled old tough guys beat out **the** squeaky clean upstarts every single time. So this is evidence there ’ s actually an **evolutionary** payoff to being extraordinarily slow. So it ’ s possible that maybe we should not equate being slow with being unimportant. Maybe these out-of-sight, out-of-mind microbes could actually be helpful to humanity. OK, so as far as we know, there are two ways to do subsurface living. **The** first is to wait for food to trickle down from **the surface** world, like trying to eat **the** leftovers of a picnic that happened 1, 000 years ago. Which is a crazy way to live, but shockingly seems to work out for a lot of microbes in earth. **The** other possibility is for a microbe to just say,”Nah, I don ’ t need **the surface** world. I ’ m good down here.”For microbes that go this route, they have to get everything that they need in order to survive from inside **the** earth. Some things are actually easier for them to get. They ’ re more abundant inside **the** earth, like water or nutrients, like nitrogen and **iron** and phosphorus, or places to live. These are things that we literally kill each other to get ahold of up at **the surface** world. But in **the** subsurface, **the** problem is finding enough energy. Up at **the surface**, plants can chemically knit together **carbon** dioxide molecules into yummy sugars as fast as **the** sun ’ s **photons** hit their leaves. But in **the** subsurface, of course, there ’ s no sunlight, so this ecosystem has to solve **the** problem of who is going to make **the** food for everybody else. **The** subsurface needs something that ’ s like a plant but it breathes **rocks**. Luckily, such a thing exists, and it ’ s called a chemolithoautotroph. Which is a microbe that uses chemicals —”chemo,”from **rocks** —”litho,”to make food —”autotroph.”And they can do this with a ton of different elements. They can do this with sulphur, **iron**, manganese, nitrogen, **carbon**, some of them can use pure electrons, straight up. Like, if you cut **the** end off of an **electrical** cord, they could breathe it like a snorkel. These chemolithoautotrophs take **the** energy that they get from these processes and use it to make food, like plants do. But we know that plants do more than just make food. They also make a waste product, oxygen, which we are 100 percent dependent upon. But **the** waste product that these chemolithoautotrophs make is often in **the** form of minerals, like rust or pyrite, like fool ’ s gold, or carminites, like limestone. So what we have are microbes that are really, really slow, like **rocks**, that get their energy from **rocks**, that make as their waste product other **rocks**. So am I talking about **biology**, or am I talking about geology? This stuff really blurs **the** lines. So if I ’ m going to do this thing, and I ’ m going to be a biologist who studies microbes that kind of act like **rocks**, then I should probably start studying geology. And what ’ s **the** coolest part of geology? Volcanoes. This is looking inside **the** crater of Pos Volcano in Costa Rica. Many volcanoes on earth arise because an oceanic tectonic **plate** crashes into a continental **plate**. As this oceanic **plate** subducts or gets moved underneath this continental **plate**, things like water and **carbon** dioxide and other materials get squeezed out of it, like ringing a **wet** washcloth. So in this way, subduction zones are like portals into **the** deep earth, where materials are exchanged between **the surface** and **the** subsurface world. So I was recently invited by some of my colleagues in Costa Rica to come and work with them on some of **the** volcanoes. And of course I said yes, because, I mean, Costa Rica is beautiful, but also because it sits on top of one of these subduction zones. We wanted to ask **the** very specific question : Why is it that **the carbon** dioxide that comes out of this deeply buried oceanic tectonic **plate** is only coming out of **the** volcanoes? Why don ’ t we see it distributed throughout **the** entire subduction zone? Do **the** microbes have something to do with that? So this is a picture of me inside Pos Volcano, along with my colleague Donato Giovannelli. That lake that we ’ re standing next to is made of pure battery acid. I know this because we were measuring **the** pH when this picture was taken. And at some point while we were working inside **the** crater, I turned to my Costa Rican colleague Carlos Ramrez and I said,”Alright, if this thing starts erupting right now, what ’ s our exit strategy?”And he said,”Oh, yeah, great question, it ’ s totally easy. Just turn around and **enjoy the** view.””Because it will be your last.”And it may sound like he was being overly dramatic, but 54 days after I was standing next to that lake, this happened. Audience : Oh! Freaking **terrifying**, right? This was **the** biggest eruption this volcano had had in 60-some-odd years, and not long after this video ends, **the** camera that was taking **the** video is obliterated and **the** entire lake that we had been sampling vaporizes completely. But I also want to be clear that we were pretty sure this was not going to happen on **the** day that we were actually in **the** volcano, because Costa Rica monitors its volcanoes very carefully through **the** OVSICORI Institute, and we had scientists from that institute with us on that day. But **the** fact that it erupted illustrates perfectly that if you want to look for where **carbon** dioxide gas is coming out of this oceanic **plate**, then you should look no further than **the** volcanoes themselves. But if you go to Costa Rica, you may notice that in addition to these volcanoes there are tons of cozy little hot **springs** all over **the** place. Some of **the** water in these hot **springs** is actually bubbling up from this deeply buried oceanic **plate**. And our hypothesis was that there should be **carbon** dioxide bubbling up with it, but something deep **underground** was filtering it out. So we spent two weeks driving all around Costa Rica, sampling every hot **spring** we could find — it was awful, let me tell you. And then we spent **the** next two years measuring and analyzing data. And if you ’ re not a scientist, I ’ ll just let you know that **the** big discoveries don ’ t really happen when you ’ re at a beautiful hot **spring** or on a public stage ; they happen when you ’ re hunched over a messy **computer** or you ’ re troubleshooting a difficult instrument, or you ’ re Skyping your colleagues because you are completely confused about your data. Scientific discoveries, kind of like deep subsurface microbes, can be very, very slow. But in our case, this really paid off this one time. We discovered that literally tons of **carbon** dioxide were coming out of this deeply buried oceanic **plate**. And **the** thing that was keeping them **underground** and keeping it from being released out into **the** atmosphere was that deep **underground**, underneath all **the** adorable sloths and toucans of Costa Rica, were chemolithoautotrophs. These microbes and **the** chemical processes that were happening around them were converting this **carbon** dioxide into carbonate mineral and locking it up **underground**. Which makes you wonder : If these subsurface processes are so good at sucking up all **the carbon** dioxide coming from below them, could they also help us with a little **carbon** problem we ’ ve got going on up at **the surface**? Humans are releasing enough **carbon** dioxide into our atmosphere that we are decreasing **the** ability of our **planet** to support life as we know it. And scientists and engineers and entrepreneurs are working on methods to pull **carbon** dioxide out of these point sources, so that they ’ re not released into **the** atmosphere. And they need to put it somewhere. So for this reason, we need to keep studying places where this **carbon** might be stored, possibly in **the** subsurface, to know what ’ s going to happen to it when it goes there. Will these deep subsurface microbes be a problem because they ’ re too slow to actually keep anything down there? Or will they be helpful because they ’ ll help convert this stuff to solid carbonate minerals? If we can make such a big breakthrough just from one study that we did in Costa Rica, then imagine what else is waiting to be discovered down there. This new field of geo-bio-chemistry, or deep subsurface **biology**, or whatever you want to call it, is going to have huge implications, not just for mitigating climate change, but possibly for understanding how life and earth have coevolved, or finding new products that are useful for industrial or medical applications. Maybe even predicting **earthquakes** or finding life outside our **planet**. It could even help us understand **the** origin of life itself. Fortunately, I don ’ t have to do this by myself. I have amazing colleagues all over **the** world who are cracking into **the** mysteries of this deep subsurface world. And it may seem like life buried deep within **the** earth ’ s crust is so far away from our daily experiences that it ’ s kind of irrelevant. But **the** truth is that this weird, slow life may actually have **the** answers to some of **the** greatest mysteries of life on earth. Thank you.

Text Sample 59: POS Dim 1 – 2019 – Score 13.00 – 2019/t000014.txt

I ’ m here to talk to you about something important that may be new to you. **The** governments of **the** world are about to conduct an unintentional experiment on our climate. In 2020, new rules will require ships to lower their sulfur **emissions** by scrubbing their dirty exhaust or switching to cleaner fuels. For human health, this is really good, but sulfur particles in **the emission** of ships also have an effect on clouds. This is a satellite image of marine clouds off **the** Pacific West Coast of **the** United States. **The** streaks in **the** clouds are created by **the** exhaust from ships. Ships ’ **emissions** include both **greenhouse** gases, which trap heat over long periods of time, and particulates like sulfates that mix with clouds and temporarily make them brighter. Brighter clouds reflect more sunlight back to space, cooling **the** climate. So in fact, humans are currently running two unintentional experiments on our climate. In **the** first one, we ’ re increasing **the** concentration of **greenhouse** gases and gradually warming **the** earth system. This works something like a fever in **the** human body. If **the** fever remains low, its effects are mild, but as **the** fever rises, damage grows more severe and eventually devastating. We ’ re seeing a little of this now. In our other experiment, we ’ re planning to remove a layer of particles that brighten clouds and shield us from some of this warming. **The** effect is strongest in ocean clouds like these, and scientists expect **the** reduction of sulfur **emissions** from ships next year to produce a measurable increase in global warming. Bit of a shocker? In fact, most **emissions** contain sulfates that brighten clouds : **coal**, diesel exhaust, forest fires. Scientists estimate that **the** total cooling effect from **emission** particles, which they call aerosols when they ’ re in **the** climate, may be as much as all of **the** warming we ’ ve experienced up until now. There ’ s a lot of uncertainty around this effect, and it ’ s one of **the** major reasons why we have difficulty predicting climate, but this is cooling that we ’ ll lose as **emissions** fall. So to be clear, humans are currently cooling **the planet** by dispersing particles into **the** atmosphere at massive scale. We just don ’ t know how much, and we ’ re doing it accidentally. That ’ s worrying, but it could mean that we have a fast-acting way to reduce warming, emergency medicine for our climate fever if we needed it, and it ’ s a medicine with origins in nature. This is a NASA simulation of earth ’ s atmosphere, showing clouds and particles moving over **the planet**. **The** brightness is **the** Sun ’ s light reflecting from particles in clouds, and this reflective shield is one of **the** primary ways that nature keeps **the planet** cool enough for humans and all of **the** life that we know. In 2015, scientists assessed possibilities for rapidly cooling climate. They discounted things like mirrors in space, ping-pong balls in **the** ocean, plastic sheets on **the** Arctic, and they found that **the** most viable approaches involved slightly increasing this atmospheric reflectivity. In fact, it ’ s possible that reflecting just one or two percent more sunlight from **the** atmosphere could offset two degrees Celsius or more of warming. Now, I ’ m a technology executive, not a scientist. About a decade ago, concerned about climate, I started to talk with scientists about potential countermeasures to warming. These conversations grew into collaborations that became **the** Marine Cloud Brightening Project, which I ’ ll talk about momentarily, and **the** nonprofit policy organization SilverLining, where I am today. I work with politicians, researchers, members of **the** tech industry and others to talk about some of these ideas. Early on, I met British atmospheric scientist John Latham, who proposed cooling **the** climate **the** way that **the** ships do, but with a natural source of particles : sea-salt mist from seawater sprayed from ships into areas of susceptible clouds over **the** ocean. **The** approach became known by **the** name I gave it then,”marine cloud brightening.”Early modeling studies suggested that by deploying marine cloud brightening in just 10 to 20 percent of susceptible ocean clouds, it might be possible to offset as much as two degrees Celsius ’ s warming. It might even be possible to brighten clouds in local regions to reduce **the** impacts caused by warming ocean **surface** temperatures. For example, regions such as **the** Gulf Atlantic might be cooled in **the** months before a hurricane season to reduce **the** force of storms. Or, it might be possible to cool waters flowing onto coral reefs overwhelmed by heat stress, like Australia ’ s Great Barrier Reef. But these ideas are only theoretical, and brightening marine clouds is not **the** only way to increase **the** reflection of **the** sunlight from **the** atmosphere. Another **occurs** when large volcanoes release material with enough force to reach

the upper layer of **the** atmosphere, **the** stratosphere. When Mount Pinatubo erupted in 1991, it released material into **the** stratosphere, including sulfates that mix with **the** atmosphere to reflect sunlight. This material remained and circulated around **the planet**. It was enough to cool **the** climate by over half a degree Celsius for about two years. This cooling led to a striking increase in Arctic ice cover in 1992, which dropped in subsequent years as **the** particles fell back to earth. But **the** volcanic phenomenon led Nobel Prize winner Paul Crutzen to propose **the** idea that dispersing particles into **the** stratosphere in a controlled way might be a way to counter global warming. Now, this has risks that we don ' t understand, including things like heating up **the** stratosphere or damage to **the** ozone layer. Scientists think that there could be safe approaches to this, but is this really where we are? Is this really worth considering? This is a simulation from **the** US National Center for Atmospheric Research global climate model showing, earth **surface** temperatures through 2100. **The** globe on **the left** visualizes our current trajectory, and on **the** right, a world where particles are introduced into **the** stratosphere gradually in 2020, and maintained through 2100. Intervention keeps **surface** temperatures near those of today, while without it, temperatures rise well over three degrees. This could be **the** difference between a safe and an unsafe world. So, if there ' s even a chance that this could be close to reality, is this something we should consider seriously? Today, there are no capabilities, and scientific knowledge is extremely limited. We don ' t know whether these types of interventions are even feasible, or how to characterize their risks. Researchers hope to explore some basic questions that might help us know whether or not these might be real options or whether we should rule them out. It requires multiple ways of studying **the** climate system, including **computer** models to forecast changes, analytic **techniques** like machine **learning**, and many types of observations. And though it ' s controversial, it ' s also critical that researchers develop core technologies and perform small-scale, real-world experiments. There are two research programs proposing experiments like this. At Harvard, **the** SCoPEx experiment would release very small amounts of sulfates, calcium carbonate and water into **the** stratosphere with a balloon, to study **chemistry** and physics effects. How much material? Less than **the** amount released in one minute of flight from a commercial aircraft. So this is definitely not dangerous, and it may not even be scary. At **the** University of Washington, scientists hope to spray a fine mist of salt water into clouds in a series of land and ocean tests. If those are successful, this would culminate in experiments to measurably brighten an area of clouds over **the** ocean. **The** marine cloud brightening effort is **the** first to develop any technology for generating aerosols for atmospheric sunlight reflection in this way. It requires producing very tiny particles — think about **the** mist that comes out of an asthma inhaler — at massive scale — so think of looking up at a cloud. It ' s a tricky engineering problem. So this one nozzle they developed generates three trillion particles per second, 80 nanometers in size, from very corrosive saltwater. It was developed by a team of retired engineers in Silicon Valley — here they are — working full-time for six years, without pay, for their grandchildren. It will take a few million dollars and another year or two to develop **the** full spray system they need to do these experiments. In other parts of **the** world, research efforts are emerging, including small modeling programs at Beijing Normal University in China, **the** Indian Institute of Science, a proposed center for climate repair at Cambridge University in **the** UK and **the** DECIMALS Fund, which sponsors researchers in global South countries to study **the** potential impacts of these sunlight interventions in their part of **the** world. But all of these programs, including **the** experimental ones, lack significant funding. And understanding these interventions is a hard problem. **The** earth is a vast, complex system and we need major investments in climate models, observations and basic science to be able to predict climate much better than we can today and manage both our accidental and any intentional interventions. And it could be urgent. Recent scientific reports predict that in **the** next few decades, earth ’ s fever is on a path to devastation : extreme heat and fires, major loss of ocean life, collapse of Arctic ice, displacement and suffering for hundreds of millions of people. **The** fever could even reach tipping points where warming takes over and human efforts are no longer enough to counter accelerating changes in natural systems. To prevent this circumstance, **the** UN ’ s International Panel on Climate Change predicts that we need to stop and even reverse **emissions** by 2050. How? We have to quickly and radically transform major economic sectors, including energy, construction, **agriculture**, **transportation** and others. And it is imperative that we do this as fast as we can. But our fever is now so high that climate experts say we also have to remove massive quantities of CO2 from **the** atmosphere, possibly 10 times all of **the** world ’ s annual **emissions**, in ways that aren ’ t proven yet. Right now, we have slow-moving solutions to a fast-moving problem. Even with **the** most optimistic assumptions, our exposure to risk in **the** next 10 to 30 years is unacceptably high, in my opinion. Could interventions like these provide fast-acting medicine if we need it to reduce **the** earth ’ s fever while we address its underlying causes? There are real concerns about this idea. Some people are very worried that even researching these interventions could provide an excuse to delay efforts to reduce **emissions**. This is also known as a moral hazard. But, like most medicines, interventions are more dangerous **the** more that you do, so research actually tends to draw out **the** fact that we absolutely, positively cannot continue to fill up **the** atmosphere with **greenhouse** gases, that these kinds of alternatives are risky and if we were to use them, we would need to use as little as possible. But even so, could we ever learn enough about these interventions to manage **the** risk? Who would make decisions about when and how to intervene? What if some people are worse off, or they just think they are? These are really hard problems. But what really worries me is that as climate impacts worsen, leaders will be called on to respond by any means available. I for one don ’ t want them to act without real information and much better options. Scientists think it will take a decade of research just to assess these interventions, before we ever were to develop or use them. Yet today, **the** global level of investment in these interventions is effectively zero. So, we need to move quickly if we want policymakers to have real information on this kind of emergency medicine. There is hope! **The** world has solved these kinds of problems before. In **the** 1970s, we identified an existential threat to our protective ozone layer. In **the** 1980s, scientists, politicians and industry came together in a solution to replace **the** chemicals causing **the** problem. They achieved this with **the** only legally binding environmental agreement signed by all countries in **the** world, **the** Montreal Protocol. Still in force today, it has resulted in a recovery of **the** ozone layer and is **the** most successful environmental protection effort in human history. We have a far greater threat now, but we do have **the** ability to develop and agree on solutions to protect people and restore our climate to health. This could mean that to remain safe, we reflect sunlight for a few decades, while we green our industries and remove CO2. It definitely means we must work now to understand our options for this kind of emergency medicine. Thank you,

Text Sample 60: POS Dim 1 – 2021 – Score 24.00 – 2021/t004019.txt

Hello there, this is Chris Anderson, and I am hugely, hugely, tremendously excited to welcome you to a new series of **the** TED interview. Now, then, this season, we ' re trying something new. We ' re organising **the** whole season around a single **theme**, albeit a **theme** that some of you may consider inappropriate. But hear me out. **The theme** is **the** case for optimism. And yes, I know **the** world has been hit with extraordinarily ugly things in **the** last few years. Political division, a racial reckoning, technology run amuck, not to mention a global pandemic and impending climate catastrophe. What on earth are we thinking in this context? Optimism just seems so naive and unwanted, almost annoying. So here ' s my position. Don ' t think of optimism as a feeling. It ' s not just this sort of shallow feeling of hope. Optimism is a **search**. It ' s a determination to look for a pathway forward somewhere out there. I believe I truly believe there are amazing people whose minds contain **the** ideas, **the** visions, **the** solutions that can actually create that pathway forward. If given **the** support and resources they need, they may very well light **the** path out of this dark place we ' re in. So these are **the** people who can present not optimism, but a case for optimism. They ' re **the** people I ' m talking to this season. So let ' s see if they can persuade us now. Then **the** place I want to start is with A. I. **artificial** intelligence. This, of course, is **the** next innovative technology that is going to change everything as we know it, for better or for worse. Today was painted not with **the** usual dystopian brush, but by someone who truly believes in its potential. Sam Altman is **the** former president of Y Combinator, **the** legendary startup accelerator. And in 2015, he and a team launched a company called Open Eye, dedicated to one noble purpose to develop A. I. so that it benefits humanity as a whole. You may have heard, by **the** way, recently a lot of buzz around in A. I. technology called T3 that was developed by open eye improve **the** quality of **the** amazing team of researchers and developers they have work in. There will be hearing a lot about three in **the** conversation ahead. But sticking to this lofty mission of developing A. I. for humanity and finding **the** resources to realize it haven ' t been simple. Open A. I. is certainly not without its critics, but their goal couldn ' t be more important. And honestly, I found it really quite exciting to hear Sam ' s vision for where all this could lead. OK, let ' s do this. So, Sam Altman, welcome. Thank you for having me. So, Sam, here we are in 2021. A lot of people are fearful of **the** future at this moment in world history. How would you describe your attitude to **the** future? I think that **the** combination of scientific and technological progress and better societal decision making, better societal governance is going to solve in **the** next couple of decades all of our current most pressing problems, there will be new ones. But I think we are going to get very safe, very inexpensive, **carbon** free nuclear energy to work. And I think we ' re going to talk about that time that **the** climate disaster looks so bad and how lucky we are. We got saved by science and technology, I think. And we ' ve already now seen this with **the** rapidity that we were able to get vaccines deployed. We are going to find that we are able to cure or at least treat a significant percentage of human disease, including I think we ' ll just actually make progress in helping people have much longer decades, longer health spans. And I think in **the** next couple of decades, that will look pretty clear. I think we will build systems with AI and otherwise that make access to an incredibly high quality education more possible than ever before. I think **the** lives we look forward like one hundred years, fifty years, even **the** quality of life available to anyone then will be much better than **the** quality of life available in **the** very best case to anyone today, to any single person today. So, yeah, I ' m super optimistic. I think, like, it ' s always easy to do scroll and think about how bad are **the** bad things are, but **the** good things are really good and getting much better. Is it your sincere belief that **artificial** intelligence can actually make that future better? Certainly. How look, with any technology. I don ' t think it will all be better. I think there are always positive and negative use cases of anything new, and it ' s our job to maximize **the** positive ones, minimize **the** negative ones. But I truly, genuinely believe that **the** positive impacts will be orders of magnitude bigger than **the** negative ones. I think we ' re seeing a glimpse of that now. Now that we have **the** first general purpose built out in **the** world and available via things like RPI, I think we are seeing evidence of just **the** breadth of services that we will be able to offer as **the** sort of technological revolution really takes hold. And we will have people interact with services that are smart, really smart, and it will feel like as strange as **the** world before mobile phones feels now to us. Hmm, yeah, you mentioned your API, I guess that stands for what, application programming interface? It ' s **the** technology that allows complex technology to be accessible to others. So give me a sense of a couple of things that have got you most excited that are already out there and then how that gives you visibility to a pathway forward that is even more exciting. So I think that **the** things that we ' re seeing now are very much glimpse of **the** future. We released three, which is a general-purpose natural language text model in **the** summer of twenty twenty. You know, there ' s hundreds of applications that are now using it in production that ' s ramping up all of **the** time. But there are things where people use three to really understand **the** intent behind **the** **search** query and deliver results and sort of understand not only intent, but all of **the** data and deliver **the** thing of what you want. So you can sort of describe a fuzzy thing and it ' ll understand documents. It can understand, you know, short documents, not full books yet, but bring you back to **the** context of what you want. There ' s been a lot of excitement about using **the** generative capabilities to create sort of games or sort of interactive stories or letting people develop characters or chat with a sort of virtual friend. There are applications that, for example, help a job seeker polish a tailored application for each individual company. There ' s **the** beginning of tutors that can sort of teach people about different concepts and take on different personas. And we can go on for a long time. But I think anything that you can imagine that you do today via **computer** that you would like to really understand and get to know you. And not only that, but understand all of **the** data and knowledge in **the** world and help you have **the** best experience that is is possible that that will happen. So what gets opened up? What new adjacent possible state is that as a result of these powers from this question, from **the** point of view of someone who ' s starting out on a career, for example, they ' re trying to figure out what would be a really interesting thing to do in **the** future that has only recently become possible. What are some new things that this opens up in a world where you can talk to a **computer**? And get. **The** output that would normally require you hiring **the** world experts back immediately for almost no money, I would say think about what ' s possible there. So that could be like, as you said, what can normally only **the** best programmer in **the** world or a really great programmer do for me. And can I now instead just ask in English and have that program written? So all these people that, you know, want to develop an app and they have an idea, but they don ' t know how to program. Now they can have it. You know, what is **the** service look like when anyone on Earth who wants really great medical advice? Can get better medical advice than any single doctor could ever get, because this has **the** total medical knowledge and reasoning ability that **the** some humanity has ever produced. When you want to learn something, you have sort of a tutor that understands your **exact** style, how you best learn everything you know, and custom teaches you whatever concept you want to learn someday. You can imagine that like. You have an eye that reads your email and your task list and your calendar and **the** documents you ' ve been sent and in any meeting maximally perfectly prepares you and has all of **the** information that you need in all **the** context of your entire career right there for you to go on for a long time. But I think this will just be powerful systems. So it ' s really fun playing around with Chapatti three, one compelling example of someone who ' s more tax base is try Googling **The** Guardian essay that was written entirely by different GP2 three queries and stitched together. It ' s an essay on why **artificial** intelligence isn ' t a threat to humanity. And that ' s impressive. It ' s very compelling. I actually tried inputting one of **the** three online uses. I asked **the** question what is interesting about some ollman? Oh no. Here ' s what it came back with. It was it was rather philosophical, actually. Came back with. I don ' t understand what you mean by interesting. It seems to me that this word has no unambiguous meaning in **the** context of human society beyond its literal definition as somewhat pleasing or entertaining. I do not believe there to be any such thing as INTERESTINGNESS except in **the** mind of a human or other sentient being that to my knowledge, this is an entirely subjective state that varies from person to person. However, I will grant that this is an interesting question to ask. This does not mean it has been answered. There is no answer to be found. Well, so you can agree that somewhere between profound and gibberish is that almost well, with **the** state of play is I mean, that ' s where we are today. I think somewhere between profound and jibberish is **the** right way to think about **the** current capabilities of CGP three. I think they would definitely had a bubble of hype about three last summer. But **the** thing about bubbles is **the** reason that smart people fall for them is there ' s a kernel of something really real and really interesting that people get overexcited about. And I think people definitely got and still are overexcited about 3:00 today, but still probably underestimated **the** potential of where these models will go in **the** future. And so maybe there ' s this like short term overhyped and long term under hype for **the** entire field, for tax models, for whatever you ' d like. It ' s going on. And as you said, there ' s clearly some gibberish in there. But on **the** other hand, those were like well-formed sentences. And there were a couple of ideas and there that I was like, oh, like they actually maybe that ' s right. And I think if **artificial** intelligence, even in its current very larval state, can make us confront new things and sort of inspire new ideas, that ' s already pretty impressive. Give us a sense of what ' s actually happening in **the** background there. I think it ' s hard to understand because you read these words seem like someone is trying to mean something. Obviously, I think you believe that there ' s whatever you ' ve built there, that there ' s a sort of thinking, sentient thing that ' s going, oh, I must answer this question. So so what how would you describe what ' s going on? You ' ve got something that has read **the** entire Internet, essentially all of Wikipedia, etc. We ' ve read something that ' s read like a small **fraction** of a random sampling of **the** Internet. We will eventually train something that has read as much of **the** Internet or more of **the** Internet than we ' ve done right now. But we have a very long way to go. I mean, we ' re still, I think, **relative** to what we will have operated at quite small scale with quite small eyes. But what is happening is there is a model that is ingesting lots of text and it is trying to predict **the** next word. So we use Transformer ' s they take in a context, which is a particular architecture of an A. I. model, they take in a context of a lot of words, let ' s say like a thousand or something like that. And they try to predict **the** word that comes next in **the** sequence. And there ' s like a lot of other things that happen, but fundamentally that ' s it, and I think this is interesting because in **the** process of playing that little game of trying to predict **the** next word, these models have to develop a representation and understanding of what is likely to come next and. I think it is maybe not perfectly accurate, but certainly worth considering to say that intelligence is very near **the** ability to make accurate predictions. What ' s confusing about this is that there are so many words on **the** Internet which are foolish as well as **the** words that are wise. And and how do you build a model that can distinguish between those two? And this is prompted actually by another example that I typed in. Like I asked, you know, what is a powerful idea, very interested in ideas. That was my question as a powerful idea. And it came back with several things, some of which seemed moderately pronouncements, which seemed moderately gibberish. But then he was he was one that it came back with **the** idea that **the** human race has, quote, **evolved**, unquote, is false **evolution** or adaptation within a species was abandoned by **biology** and genetics long ago. Wait a sec. That ' s news to me. What have you been reading? And I presume this has been pulled out of some recesses of **the** Internet, but how is it possible, even in theory, to imagine how a model can gravitate towards truth, wisdom, as opposed to just like majority views? Or how how how do you avoid something taking us further into **the** sort of **the** maze of errors and bad thinking and so forth that has already been a worrying feature for **the** last few years? It ' s a **fantastic** question, and I think it is **the** most interesting area of research that we need to pursue. Now, I think at this point, **the** questions of whether we can build really powerful general-purpose AI system, I won ' t say there in **the** rearview mirror. We still have a lot of hard engineering work to do, but I ' m pretty confident we ' re going to be able to. And now **the** questions are like, what should we build? And how and why and what data should we train on and how do we build systems not just that can do these like phenomenally impressive things, but that we can ensure do **the** things that we want and that understand **the** concepts of truth and falsehood and, you know, alignment with human values and misalignment with human values. One of **the** pieces of research that we put out last year that I was most proud of and most excited about is what we call reinforcement learning from human feedback. And we showed that we can take these giant models that are trained on a bunch of stuff, some of it good, some of **the** bad, and then with a really quite small amount of feedback from human judgment about, hey, this is good, this is bad, this is wrong, this is **the** behavior I want I don ' t want this behavior. We can feed that information from **the** human judges back into **the** model and we can teach **the** model, behave more like this and less like that. And it works better than I ever imagined it would. And that gives me a lot of hope that we can build an aligned system. We ' ll do other things, too, like I think curating data sets where there ' s just less sort of bad data to train on. It will go a very long way. And as these models get smarter, I think they inherently develop **the** ability to sort out bad data from good data. And as they get really smart, they ' ll even start to do something we call active **learning**, which is where they ask us for exactly **the** data they need when they ' re missing something, when they ' re unsure, when they don ' t understand. But I think as a result of simply scaling these models up, building better, I hate to use **the** word cognition because it sounds so anthropomorphic, but let ' s say building a better ability to reason into **the** models, to think, to challenge, to try to understand and combining that with this idea of online into human values via this **technique** we developed, that ' s going to go a very long way. Now, there ' s another question, which you sort of just kicked **the** ball down **the** field, too, which is how do we as a society decide to which set of human values do we align these powerful systems? Yeah, indeed. So if I if I understand rightly what you ' re saying, that you ' re saying that it ' s possible to look at **the** output at any one time of three. And if we don ' t like what it ' s coming up with, some ways human can say, no, that was off, don ' t do that. Whatever algorithm or process led you to that, undo it. Yeah. And that **the** system is that incredibly powerful at avoiding that same kind of mistake in future because it sort of replicates **the** instructions, correct? Yeah. And eventually and not much longer, I believe that we ' ll be able to not only say that was good, that was bad, but say that was bad for this reason. And also tell me how you got to that answer so I can make sure I understand. But at **the** end of **the** day, someone needs to decide who is **the** wise human or short humans who are looking at **the** results. So it ' s a big difference. Someone who who grew up with intelligent design world view could look at that and go, that ' s a brilliant outcome. Well, Goldstar done. And someone else would say something is done awfully wrong here. So how do you avoid and this is a version of **the** problem that a lot of **the**, I guess, Silicon Valley companies are facing right now in terms of **the** pushback they ' re getting on **the** output of social media and so forth. How do you assemble that pool of experts who stand for human values that we actually want? I mean, we talk about this all **the** time, I don ' t think this is like solely or even not even close to majorly up to opening night to decide, I think we need to begin a societal conversation now about how we ' re going to make those decisions, how we ' re going to make sure we have representational input in that, and how we sort of make these very difficult global governance systems. My personal belief is that we should have pretty broad rules about what these systems will never do and will always do. But then **the** individual user should get a system that kind of behaves like they want. And there will be people do have very different value systems. Some of them are just fundamentally incompatible. No one gets to use eye to, like, exploit other people, for example, and hopefully we can all agree on. But do you want **the** AI to like. You know, support you and your belief of intelligent design, like, do I think openly, I should say it can ' t, even though I disagree with that is like a scientific conclusion. No, I wouldn ' t take that stance. I think **the** thing to remember about all of this is that history is still quite extraordinarily weak. It ' s still has such big problems and it ' s still so unreliable that for most use cases it ' s still unsuitable. But when we think about a system that is like a thousand times more powerful and let ' s say a million times more reliable, it just doesn ' t it doesn ' t say gibberish very often. It doesn ' t totally lose **the** plot and get distracted or system like that is going to be one that a lot of **the** economic activity in **the** world comes to rely on. And I think it ' s very important that we don ' t have a small group of people sort of saying you can never use it for this thing that, like most of **the** world wants to use it for because it doesn ' t match our personal beliefs. Talk a bit more about some of **the** other uses of it, because one of **the** things that ' s most surprising is it ' s not just about sort of text responses. It ' s it can take generalized human instructions and build things up. For example, you can say to it, write a Python program that is designed to put a flashing cursor in one corner of **the** screen, in **the** Google logo in **the** other corner. And and it can go your way and do something like that. Shockingly, quite well, effectively. Yeah, I it can. That ' s amazing. I mean, this is amazing to me. That opens **the** door to. An entirely way to think about programers for **the** future, that you could you could have people who can program just in human natural language potentially and gain rapid efficiency. I do **the** engineering. We ' re not that far away from that world. We ' re not that far away from **the** world where you will write a spec in English. And for a simple enough program, I will just write **the** code for you. As you said, you can see glimpses of that even in this very week three which was not trained to code like. I think this is important to remember. We trained it on **the** language on **the** Internet very rarely, you know, Internet let language on **the** Internet also includes some code snippets. And that was enough, so if we really try to go train a model on code itself and that ' s where we decide to put **the** horsepower of **the** model into, just imagine what will be possible will be quite impressive. But I think what you ’ re pointing to there is that because models like three to some degree or other, and it ’ s like very hard to know exactly how much understand **the** underlying concepts of what ’ s going on. And they ’ re not just regurgitating things they found in a website, but they can really apply them and say, oh, yeah, I kind of like know about this word and this idea and code. And this is probably what you ’ re trying to do. And I won ’ t get it right always. But sometimes I will just generate this like a brand new program for nothing that anyone has ever asked before. And it will work. That ’ s pretty cool. And data is data. So it can do that from English to code. It can do that from English to French. Again, we never told it to learn about translation. We never told it about **the** concepts of English and French, but it learned them, even though we never said this is what English is and this is what French is and this is what it means to translate, it can still do it. Wow, I mean, for creative people, is there a world coming where **the** sort of **the** palette of possibility that they can be exposed to is just explodes? I mean, if you ’ re a musician, is there a near future where you can say to your eye, OK, I ’ m going to bed now, but in **the** morning I ’ d love you to present me with a thousand tuba jingles with words attached that you have of a sort of mean factor to **the** and you come down in **the** morning and **the computer** shows you **the** stuff. And one of them, you go, wow, that is it. That is a top 10 hit and you build a song from it. Or is that going to be released? Actually be **the** value add. We released something last year called Jukebox, which is very near what you described, where you can say I want music generated for me in this style or this kind of stuff, and it can come up with **the** words as well. And it ’ s like pretty cool. And I really **enjoy** listening to music that it creates. And I can sort of do four songs, two bars of a jingle, whatever you ’ d like. And one of my very favorite artists reached out, called to open it after we release this and said that he wanted to talk. And I was like, well, I like total fanboy here. I ’ d love to join that call. And I was so nervous that he was going to say, this is terrible. This is like a really sad thing for human creativity. Like, you know, why are you doing this? This is like whatever. And he was so excited. And he ’ s like, this has been so inspiring. I want to do a new album with this. You know, it ’ s like, give me all these new ideas. It ’ s making me much better at my job. I ’ m going to make better music because of this tool. And that was awesome. And I hope that ’ s how it all continues to go. And I think it is going to lead to this. We see a similar thing now with Dolly, where graphic designers sometimes tell us that they just they see this new set of possibilities because there ’ s new creative inspiration and they ’ re cycle time, like **the** amount of time it takes to just come up with an idea and be able to look at it and then decide whether to go down that path or head in a different direction goes down so much. And so I think it ’ s going to just be this like incredible creative **explosion** for humans. And how far away are we some before? And I it comes up with a genuinely powerful new idea, an idea that solves **the** problem that humans have been wrestling with. It doesn ’ t have to be as quite on **the** scale as of, OK, we ’ ve got a virus coming. Please describe to us what a what a national rational response should look like, but some kind of genuinely innovative idea or solution like one one internal question we ’ ve asked ourselves is, when will **the** first genuinely interesting, purely AI written TED talk show up? I think that ’ s a great milestone. I will say it ’ s always hard to guess timeline ’ s I ’ m sure I ’ ll be wrong on this, but I would guess **the** first genuinely interesting. Ted talk, thought of written delivered by an AIDS within **the** kind of **the** seven ish year time frame. Maybe a little bit less. And it feels like I mean, just reading that Guardian essay that was kind of it was a composite of several different GPG three responses to questions about, you know, **the** threats of robotics or whatever. If you throw in a human editor into **the** mix, you could probably imagine something much sooner. Indeed. Like tomorrow. Yeah. So **the** hybrid **the** hybrid version where it ’ s basically a tool assisted TED talk, but that it is better than any TED talk a human could generate in one hundred hours or whatever, if you can sort of combine human discretion with A. I. horsepower. I suspect that ’ s like our next year or two years from now kind of thing where it ’ s just really quite good. That ’ s that ’ s really interesting. How do you view **the** impact of A. I. on jobs? There ’ s obviously been **the** familiar story is that every White-Collar job is now up for destruction. What ’ s what ’ s your view there? You know, it ’ s I think it ’ s always hard to make these predictions. That is definitely **the** familiar story now. Five years ago, it was every **blue** collar job is up for destruction, maybe like last year it was. Every creative job is up for destruction because of things like Jukebox I. I think there will be an enormous impact on. **The** job market, and I really hate it, I think it ’ s kind of gross when people like working on I pretend like there ’ s not going to be or sort of say, oh, don ’ t worry about it. It ’ ll just all obviously better. It doesn ’ t always obviously get better. I think what is true is. Every technological revolution produces a change in jobs, we always find new ones, at least so far. It ’ s difficult to predict from where we ’ re sitting now what **the** new ones will be and this technological revolution is likely to be. Again, it ’ s always tempting to say this time it ’ s different. Maybe I ’ ll be totally wrong. But from what I see now, this technological revolution is likely to be more. Dramatic. More of a staccato note than most, and I think we as a society need to figure out how we ’ re going to cushion everybody through that. I ’ ve got my own ideas about how to do that. I, I wouldn ’ t say that I have any reason to believe they ’ re **the** right ones, but doing nothing and not really engaging with **the** magnitude of what ’ s about to happen, I think it ’ s like not an acceptable answer. So there ’ s going to be huge impact. It ’ s difficult to predict where it shows up **the** most. I think previous predictions have mostly been wrong, but I I ’ d like to see us all as a society, certainly as a field, engage in what what **the** shifts we want to make to **the** social contract are to kind of get through that in a way that is maximally beneficial to everybody. I mean, in every past revolution, there ’ s always been a space for humans to move to. That is, if you like, moving up **the** food chain, it ’ s sort of we ’ ve retreated to **the** things that humans could uniquely do, think better, be more creative and so forth. I guess **the** worry about A. I. is that in principle, I believe this, that there is no human cognitive feat that won ’ t ultimately be doable, probably better by **artificial** general touch, simply because of **the** extra firepower that ultimately they can have, **the** vast knowledge they bring to **the** table and so forth. Is that basically right, that there is ultimately no safe sort of space where we can say, oh, but that would never be able to do that on a very long time horizon? I agree with you, but that ’ s such a long time horizon. I think that, you know, like maybe we ’ ve merged by that point, like maybe we ’ re all plugged in and then, like, we ’ re this sort of symbiotic thing. Like, I think there ’ s an interesting example, as we were talking about a few minutes ago, where right now we have these systems that have sort of enormous horsepower but no steering wheel. It ’ s like, you know, incredible capabilities, but no judgment. And there ’ s like these obvious ways in which today even a human plus three is far better than either on their own. Many people speak about a world where it ’ s sort of A. I. as this external threat you speak about. At some point, we actually merge with eyes in some way. What do you mean by that? There ’ s a lot of different versions of what I think is possible there, you know, in some sense, I ’ d argue **the** merge has already like begun **the** human technology merge like we have this thing in our hands that sort of dictates a lot of what we think, but it gives us real superpowers and that can go much, much further. Maybe it goes all **the** way to like **the** Elon Musk vision of neuro link and having our brains plugged into **computers** and sort of like literally we have a **computer** on **the** back of our head or goes **the** other direction and we get uploaded into one. Or maybe it ' s just that we all have a chat bot that kind of constantly steers us and helps us make better decisions than we could. But in any case, I think **the** fundamental thing is it ' s not like **the** humans versus **the** eyes competing to be **the**. Smartest sentient thing on earth or beyond. But it ' s that this idea of being on **the** same team. Hmm. I certainly get very excited by **the** sort of **the** medium term potential for creative people of all sorts if they ' re willing to expand their palette of possibilities. But with **the** use of A. I. to be willing to. I mean, **the** one thing that **the** history of technology has shown again and again is that something this powerful and with this much benefit is unstoppable and you will get rewarded for embracing it **the** most and **the** earliest. So talk about what can go wrong with that, so let ' s move away from just **the** sort of economic displacement factor. You were a co-founder of Open Eye because you saw existential risks to humanity from high today. What would you put as **the** sort of **the** most worrying of those risks? And how is open eye working to minimize? I still think all of **the** really horrifying risks exist. I am more confident, much more confident than I was five years ago when we started that there are technical things we can do about. How we build these systems and **the** research and **the** alignment that make us much more likely to end up in **the** kind of really wonderful camp, but, you know, like maybe open I fall behind and maybe somebody else feels ajai that thinks about it in a very different way or doesn ' t care as much as we ' d like about safety and **the** risks or how to strike a different trade off of how fast we should go with this and where we should sort of just say, like, you know, like let ' s push on for **the** economic benefits. But I think all of this sort of like, you know, traditionally what ' s been in **the** realm of sci fi risks are real and we should not ignore them. And I still lose sleep over them. And just to update people is **artificial** general intelligence. Right now, we have incredible examples of powerful AI operating on specific areas. Ajai is **the** ability of a **computer** mind to connect **the** dots and to make decisions at **the** same level of breadth that that humans have had. What ' s your sort of elevator pitch on Ajai about how to identify and how to think of it? Yeah, I mean, **the** way that I would say it is that for a while we were in this world of like very narrow A. I., you know, that could like classify images of cats or whatever, more advanced stuff in that. But that kind of thing. We are now in **the** era of general purpose, AI, where you have these systems that are still very much imperfect tools, but that can generalize. And one thing like GPP three can write essays and translate between languages and write **computer** code and do very complicated **search**. It ' s like a single model that understands enough of what ' s really going on to do a broad **array** of tasks and learn new things quickly, sort of like people can. And then eventually we ' ll get to this other realm. Some people call it ajai, some people call ostler things. But I think it implies that **the** systems are like to some degree self directed, have some intentionality of their own is a simple summary to say that, like **the** fundamental risk is that there ' s **the** potential with general **artificial** intelligence of a sort of runaway effect of self-improvement that can happen far faster than any kind of humans can even keep up with, so that **the** day after you get to ajai, suddenly **computers** are thousands of times more advanced than us and we have no way of controlling what they do with that power. Yeah, and that is certainly in **the** risk space, which is that we build this thing and at some point somewhat suddenly, it ' s much more powerful than we are, we haven ' t really done **the** full merge yet. There ' s an event horizon there and it ' s sort of hard to see to **the** other side of it. Again, lots of reasons to think it will go OK. Lots of reasons to think we won ' t even get to that scenario. But that is something that. I don ' t think people should brush under **the** rug as much as they do, it ' s in **the** possibility space for sure, and in **the** possibility subspace of that is one where, like, we didn ' t actually do as good of a job on **the** alignment work as we thought. And this sort of child of humanity kind of acts in a very different way than we think. A framework that I find useful is to sort of think about like a two by two matrix, which is short timelines to ajai and long timelines to ajai and a slow take off and a fast take off on **the** other axis. And in **the** short timelines, fast take off quadrant, which is not where I think we ' re going to be. But if we get there, I think there ' s a lot of scenarios in **the** direction that you are describing that are worrisome. And we would want to spend a lot of effort planning for. I mean, **the** fact that a **computer** could start editing its own code and improving itself while we ' re asleep and you wake up in **the** morning and it ' s got smarter, that is **the** start of something super powerful and potentially scary. I have tremendous misgivings about letting my system, not one we have today, but one that we might not have and too many more years start editing its own code while we ' re not paying attention. I think that ' s **the** kind of thing that is worth a great deal of societal discussion about, you know, just because we can do that. Should we? Yes, because one of **the** things that ' s that ' s been most shocking to you about **the** last few years has been just **the** power of unintended consequences. It ' s like you don ' t have to have a belief that there ' s some sort of waking up of of an alien intelligence that suddenly decided it wants to wreak havoc on humans. That may never happen. What you can have is just incredible power that goes amok. So a lot of people would argue that what ' s happened in technology in **the** last few years is actually an example of that. You know, social media companies created these intelligences that were programmed to maximally harvest attention, for example, for sure. And they understand this from that turned out to be in some ways horrifying and extraordinarily damaging. Is that a meaningful sort of canary in **the coal** mine saying, look out, humanity, this could be really dangerous? And how how on earth do you protect against those kinds of unintended consequences? I think you raise a great point in general, which is these systems don ' t have to wish ill to humanity to cause ill just when you have, like, very powerful systems. I mean, unintended consequences for sure. But another version of that is and I think this applies at **the** technical level, at **the** company level, at **the** societal level, incentives are superpower ' s. Charlie Munger had this thing on, which is incentives are so powerful that if you can spend any time whatsoever working on **the** incentive system, that ' s what you should do before you work on anything else. And I really believe that. And I think that applies to **the** individual models we build and what their reward functions look like. I think it applies to society in a big way, and I think it applies to our corporate structure at open. I you know, we sort of observe that if you have very well-meaning people, but they have this incentive to sort of maximize attention harvesting and profit forever through no one ' s ill intentions, that leads to a quite undesirable outcome. And so we set up opening is this thing called a capped profit model specifically so that we don ' t have **the** system incentive to just generate maximum value forever with an AGI that seems like obviously quite broken. But even though we knew that was bad and even though we all like to think of ourselves as good people, it took us a long time to figure out **the** right structure, to figure out a charter that ' s going to govern us and a set of incentives that we believe will let us do our work. And kind of these we have these like three elements that we talk about a lot research sort of engineering, development and deployment policy and safety. Put those all together under a system where you don ' t have to rely on. Anything but **the** natural incentives to push in a direction that we hope will minimize **the** sort of negative unintended consequences. So help me understand this, because this is I think this is confusing to some people. So you started opening. I initially I think Elon Musk, **the** co-founder, and there was a group of you and **the** argument was this technology is too powerful to be left, developed in secret and to be left developed purely by corporations who have whatever incentive they may have. We need a nonprofit that will develop and share knowledge openly. First of all, just even at that early stage, some people were confused about this. It was saying if this thing is so dangerous, why on earth would you want to make it secrets even more available? Well, maybe giving **the** tools to that sort of AI **terrorist** in his bedroom somewhere, I think I think we got misunderstood in **the** way we were talking about that. We certainly don ' t think that **the** right thing to do is to, like, build this a super weapon and hand it to a **terrorist**. That ' s obviously awful. One of **the** reasons that we like our API model is it lets us make **the** most powerful AI technology anyone in **the** world has, as far as we know, available to ever would like to use it, but to put some controls on its usage. And also, if we make a mistake, to be able to pull it back or change it or tweak it or improve it or whatever. But we do want to put and this is continued will continue to be true with appropriate restrictions and guardrails, very powerful technology in **the** hands of people. I think that is fair. I think that will lead to **the** best results for **the** society as a whole. And I think it will sort of maximize benefit. But that ' s very different than sort of shipping **the** whole model and saying, here, do whatever you want with it. We ' re able to enforce rules on it. We also think and this is part of **the** mission that like something **the** field was doing a lot of that we didn ' t feel good about was sort of saying like, oh, we ' re going to keep **the** pace of progress and capabilities secret. That doesn ' t feel right, because I think we do need a societal conversation about what ' s what ' s going on here, what **the** impacts are going to be. And so we although we don ' t always say, like, you know, here ' s **the** super weapon, hopefully we do try to say, like, this is really serious. This is a big deal. This is going to affect all of us. We need to have a big conversation about what to do with it. Help me understand **the** structure a bit better, because you definitely surprised much people when you announced that Microsoft were putting a billion dollars into **the** organization and in return, I guess they get certain exclusive licensing rights. And so, for example, they are **the** exclusive licensee of CP3. So talk about that structure of how you win. Microsoft presumably have invested not purely for altruistic purposes. They think that they will make money on that billion dollars. I sure hope they do. I love capitalism, but I think that I really loved even more about Microsoft as a partner. And I ' ll talk about **the** structure and **the** exclusive license in a minute is that we like went around to people that might find us. And we said one of **the** things here is that we ' re going to try to make you some money. But like Adjei going well is more important. And we need you to sign this document that says if things don ' t go **the** way we think and we can ' t make you money like you just cheerfully walk away from it and we do **the** right thing for humanity. And they were like, yes, we are enthusiastic about that. We get that **the** mission comes first here. So again, I hope a phenomenal investment for them. But they were like they really pleasantly surprised us on **the** upside of how aligned they were with us, about how strange **the** world may get here and **the** need for us to have flexibility and put our mission first, even if that means they lose all their money, which I hope they don ' t and don ' t think they will. So **the** way it ' s set up is that if at some point in **the** coming year or two, two years, Microsoft decide that there ' s some incredible commercial opportunity that they could realize out of **the** eye that you ' ve built and you feel actually, no, that ' s that ' s damaging. You can block it. You can veto it. Correct. So **the** four most powerful version of three and its successors are available via **the** API, and we intend for that to continue. What Microsoft has is **the** ability to sort of put that model directly into their own technology. If they want to do that. We don ' t plan to do that with other people because we can ' t have all these controls that we talked about earlier. But they ' re like a close trusted partner and they really care about safety, too. But our goal is that anybody who wants to use **the** API can have **the** most powerful versions of what we ' ve trained. And **the** structure of **the** API lets us continue to increase **the** safety and fix problems when we find them. But but **the** structure. So we start out as a non-profit, as you said, we realized pretty quickly that although we went into this thinking that **the** way to get to ajai would be about smarter and smarter algorithms, that we just needed bigger and bigger **computers** as well. And that was going to require a scale of capital that no one will, at least certainly not me, could figure out how to raise is a nonprofit. We also needed to sort of be able to compensate very highly compensated, talented individuals that do this, but are full for profit company had runaway incentives problem, among other things. Also just one about sort of fairness in society and wealth concentration that didn ' t feel right to us either. And so we came up with this kind of hybrid where we have a nonprofit that governs what we do, and it has a subsidiary, LLC, that we structure in a way to make a fixed amount of profit so that all of our investors and employees, hopefully if things go how we like, if not no one gets any money, but hopefully they get to make this one time great return on their investment or **the** time that they spent it open their equity here. And then beyond that, all **the** value flows back to **the** nonprofit and we figure out how to share it as fairly as we can with **the** world. And I think that this structure and this nonprofit with this very strong charter in place and everybody who joins signing up for **the** mission come in first and **the** fact **the** world may get strange, I think that. That was at least **the** best idea we could come up with, and I think it feels so far like **the** incentive system is working, just as I sort of watch **the** way that we and our partners make decisions. But if I read it right, **the** cap on **the** gain that investors can make is 100 Axum. It ' s a massive call that was for our very first round investors. It ' s way, way lower. Like as we now take a bit of capital, it ' s way, way lower. So your deal with Microsoft isn ' t you can only make **the** first hundred billion dollars. I don ' t know. It ' s way lower than after that. We ' re giving it to **the** world. It ' s way lower than that. Have you disclosed what I don ' t know if we have, so I won ' t accidentally do it now. All right. OK, so explain a bit more about **the** charter and how it is that you. Hope to avoid or I guess help contribute to an eye that is safe for humanity. What do you see as **the** keys to us avoiding **the** worst mistakes and really holding on to something that ' s that ' s beneficial for humanity? My answer there is actually more about, like technical and societal issues than **the** charter. So if it ' s OK for me to answer it from that perspective, sure. OK, I ' m happy to talk about **the** charter to. I think this question of alignment that we talked about a little earlier is paramount, and then I think to understand that it ' s useful to differentiate between accidental misuse of a system and intentional misuse of a system. So like intentional would be a bad actor saying, I ' ve got this powerful system, I ' m going to use it to like hack into all **the computers** in **the** world and wreak havoc on **the** power grids. And accidental would be kind of **the** Nick Bostrom make a lot of paper **clips** and view humans as collateral damage in both cases. But to varying degrees, if we can really, truly, technically solve **the** alignment problem and **the** societal problem of deciding to which set of human values do we align, then **the** systems understand right and wrong, and they understand probably better than we ever can, unintended consequences from complex actions and very complex systems. And, you know, if we can train a system which is like. Don ' t harm humanity and **the** system can really understand what we mean when we say that, again, who is we and what does that have some asterisks on them? Sorry, go ahead. Well, that ' s if they could understand what it means to not harm humanity, that there ' s a lot wrapped up in that sentence. Because what ' s been so striking to me about efforts so far is that they seem to have been based on a very naive view of human nature. Go back to **the** sort of Facebook and Twitter examples of, well, **the** engineers building some of **the** systems would say we ' ve just designed them around what humans want to do. You said, well, if someone wants to click on something, we will give them more of that thing. And what could possibly be wrong with that? We ' re just supporting human choice, ignoring **the** fact that humans are complicated, farshid animals for sure, who are constantly making choices, that a more effective version of themselves would agree is not in their long term interests. So that ' s one part of it. And then you ' ve got layered on top of that or **the** complications of systemic complexity where, you know, multiple choices by thousands of people end up creating a reality that possibly have designed for how how to cut through that. Like an AI has to make a decision based on a moment, on a specific data set. As those decisions get more powerful, how can we be confident that they don ' t lead to this sort of system crashing basically in some way? I think that I ' ve heard a lot of behavioral psychologists and other people that have studied this say in different ways, are that I hate to keep picking on Facebook, but we can do it one more time since we ' re on **the** topic. Maybe you can ' t in any given moment in night where you ' re tired and you have a stressful day, stop yourself from **the** dopamine hit of scrolling and Instagram, even though you know that ' s bad for you and it ' s not leading to your best life. But if you were asked in a reflective moment where you were sort of fully alert and thoughtful, do you want to spend as much time as you do scrolling through Instagram? Does it make you happier or not? You would actually be able to give like **the** right long term answer? It ' s sort of **the** spirit is willing, but **the** flesh is weak kind of moment. And one thing that I am hopeful is that humans do know what we want and what. On **the** whole, and presented with research or sort of an objective view about what makes us happy and doesn ' t we ' re pretty, what ' s so great about it, they ' re pretty good. But in any particular moment, we are subjected to our animal instincts and it is easy for **the** lower brain to take over **the** eye. Well, I think be an even higher brain. And as we can teach it, you know, here is what we really do value. Here ' s what we really do want. It will help us make better decisions than we are capable of, even in our best moments. So is that being proposed and talked about as an actual rule? Because it strikes me that there is something potentially super profound here to introduce some kind of rule for development of AIDS that they have to tap into not. What humans one, which is an ill defined question, but as to what humans in reflective mode want. Yeah, we talk about this a lot. I mean, do you see a real chance where something like that could be incorporated as a sort of an absolute golden rule and and if you like, spread around **the** community so that it seeps into corporations and elsewhere? Because that I ' ve seen no evidence that, well, a little corporation that was potentially a game changer. Corporations have this weird incentive problem. Right. What I was trying to speak about was something that I think should be technologically possible, and that ' s something that we as a society should demand. And I think it is technically possible for this to be sort of like a layer above **the** neocortex that makes even better decisions for us and our welfare and our long term happiness and fulfillment than we could make on our own. And I think it is possible for us as a society to demand that. And if we can do like a pincer move between what **the** technology is capable of and what we what we as society demand, maybe we can make everybody in **the** middle that way. I mean, there are instances of even though companies have their incentives to make money and so forth, they also in **the** knowledge age. Can ' t make money if they have pissed off too many of their employees and customers and investors by analogy of **the** climate space right now, you can see more and more companies, even those that are emitting huge amounts of **carbon** dioxide, saying, wait a sec, we ' re struggling to recruit talented people because they don ' t want to work for someone who ' s evil. And their customers are saying, we don ' t want to buy something that is evil. And so, you know, ultimately you can picture processes where they do better. And I I believe that most engineers, for example, work in Silicon Valley. Companies are actually good people who want to design great products for humanity. I think that **the** people who run these companies want to be a net contribution to humanity. It ' s we ' ve we ' ve rushed really quickly and design stuff without thinking it through properly. And it ' s led to a mess up. So it ' s like, OK, don ' t move fast, break things, slow down and build beautiful things that are built on a real version of human nature and on a real version of system complexity and **the** risks associated with systemic complexity. Is that **the** agenda that fundamentally you think that you can push somehow? Yes, but I think **the** way we can push it is by getting **the** incentive system right. I think most people are fundamentally extremely good. Very few people wake up in **the** morning thinking about how can I make **the** world a worse place? But **the** incentive systems that we ' re in are so powerful. And even those engineers who join with **the** absolute best of intentions get sucked into this world where they ' re like trying to go up from it all for and five or whatever Facebook calls those things and you like, it ' s pretty exciting. You get caught up playing **the** game, you ' re rewarded for kind of doing things that move **the** company ' s key metrics. It ' s like fun to get promoted. It feels good to make more money and **the** incentive systems of **the** company. And that ' s what it rewards. An individual performance are maybe like not what we all want. And here I don ' t want to pick on Facebook at all because I think there ' s versions of this at play it like every big tech company, including in some ways I ' m sure it open I but to **the** degree that we can better align **the** incentives of companies with **the** welfare of society and then **the** incentives of an individual at those companies within **the** now realign incentives for those companies, **the** more likely we are to be able to have things like ajai that. Follow an incentive system of. What we want in our most reflective best moments and are even better than what we could think of ourselves is is it still **the** vision for open eye that you will get to? **Artificial** general intelligence ahead of. **The** corporations, so that you can somehow put a stake in **the** ground and build it **the** right way. Is that really a realistic thing to to dream for? And if not, how do you live up to **the** mission and help ensure that this thing doesn ' t go off **the** rails? I think it is. Look, I certainly don ' t think we will be **the** only group to build an AGI, but I think we could be **the** first. And I think if you are **the** first, you have a lot of norms that empower. And I think you ' ve already seen that. You know, we have released some of **the** most powerful systems to date. And I think **the** way that we have done that kind of in controlled release where we ' ve released a bigger model than a bigger one than a bigger one, and we sort of try and talk about **the** potential misuse cases and we try to like talk about **the** importance of releasing this behind an API so that you can make changes. Other groups have followed suit in some of those directions, and I think that ' s good. So, yes, I don ' t think we can be **the** only I do think we can be ahead. And if we are ahead, I think we can use that leverage to hopefully push people in a better direction or maybe we ' re wrong and somebody else has a better direction. We ' re doing something about do you have a structural advantage in that your mission is to do this for everyone as opposed to for some corporate objective. And that that that allows you that. Why is it that we came out of open eye and not someone else? It ’ s like it ’ s surprising in some ways when you ’ re up against so much money and so much talent in these other companies that you came up with this platform ahead of. You know, in some sense it ’ s surprising and in some sense, like **the** startup wins most of **the** time, like I ’ m a huge believer in startups as **the** best force for innovation we have in **the** world today. I talked a little bit about how we combine these three. Different clans of research, engineering and sort of safety and policy that don ’ t normally combine well and I think we have an unusual strength, there were clearly like well funded. We have super talented people. But what we really have is like intense focus and self belief that what we ’ re doing is possible and good. And I appreciate **the** implied compliment. But, you know, we, like, work really hard. And if we stop doing that, I ’ m sure someone would run by us fast. Tell us a bit more about some of your prior life sentences. Yeah, for several years, you were running Y Combinator, which had incredible impact on some 70 companies. There are so many startup stories that began at Y Combinator. What were key drivers in your own life that took you on **the** path you ’ re on? And how did that path end up at Y Combinator? No exaggeration. I think I have back to back had **the** two jobs that are at least **the** most interesting to me in all of Silicon Valley. I, I was I went to college to study **computer** science. I was a major **computer** nerd growing up. I knew like a little bit about startups, but not very much. I started working on this project **the** same year I started working on that. This thing called Y Combinator started and funded me and my co-founders. And we dropped out of school and did this company, which I ran for like seven years. And then after that I got acquired. I had stayed close to my comment **the** whole time. I thought it was just this incredible group of people and spirit and set of incentives and just badly misunderstood by most of **the** world, but obvious to everyone within it that it was going to create huge amounts of value and do a lot of new things. My company had acquired PJI, who is **the** founder of ICI, and like truly one of **the** most incredible humans and business people. And Paul Burrell, Paul Graham asked me if I wanted to run it. And kind of like **the** central **learning** of my career, why I individual startups has been that if you really scale them up, remarkable things can happen. And I. I did it and I was like, one of **the** things that would make this exciting for me personally motivating would be if I could sort of push it in **the** direction of doing these hard tech companies, one of which became open. I describe actually what Y Combinator is, you know, how many people come through it to give us a couple of stories of its impact. Yeah. So you basically apply as a handful of people and an idea, maybe a prototype and say, I would like to start a company and will you please fund me? And we review those applications and we I shouldn ’ t say we anymore. I guess they fund four hundred companies a year. You get about one hundred and fifty thousand dollars while she takes about seven percent ownership and then gives you lots of advice and then networking and sort of this like fast track program for starting a startup. I haven ’ t looked at this in a while, but at one point a significant **fraction** of **the** billion dollar plus companies in **the** US that got started. It all came through **the** Wiki program, some recently in **the** news ones have been like Airbnb, Jordache, Coinbase, insta card stripe. And I think it ’ s just it has become an incredible way to help. People who understand technology get a three month course in business, but instead of like herding you with an MBA, we actually teach you **the** things that matter and kind of go on to do incredible, incredible work. What is it about entrepreneurs? Why do they matter? Some people just find them kind of annoying. But I think you would argue I think I would argue that they have done as much as anyone to shape

the future. Why? What is it about them? I think it is **the** ability to take. And idea and by force of will to make it happen in **the** world and in an incentive system that rewards you for making **the** most impact on **the** most people like in our system. That ' s how we get most of **the** things that that we use. That ' s how we got **the computer** that I ' m using, **the** software I ' m using to talk to you on it. Like all of this, you know, everyone in life, everything has a balance sheet. There ' s plenty of very annoying things about them. And there ' s plenty of very annoying things about **the** system that sort of idolizes them. But we do get something really important in return. And I think that as a force for making things that make all of our lives better happen, it ' s very cool. Otherwise, you know, like if you have, like, a great idea, but you don ' t actually do anything useful with it for people, that ' s still cool. It ' s still intellectually interesting. But like, there ' s got to be something about **the** reward function in society that is like, did you actually do something useful? Did you create value? And I think entrepreneurship and startups are a wonderful way to do that. You know, we get all these great software companies. But I also think it ' s like how we ' re going to get ajai, how we ' re going to get nuclear **fusion**, how we ' re going to get life extension. And like on any of those topics are a long list of other things I could point to. There ' s like a number of startups that I think are doing incredible work, some of which will actually deliver. It is a truly amazing thing when you put **the** camera back and to believe that a human being could be lying awake at night and something pops inside their mind as a patterning of **the** neurons in their brain that is effectively them saying, aha, I can see a way where **the** future could be better and and they can actually picture it. And then they wake up and then they talk to other people and they persuade them and they persuade investors and so forth. And **the** fact that this this system can happen and that they can then actually change **the** history changes in some sense. It is mind boggling that that happens that way and it happens again and again. So you ' ve seen so many of these stories happen. What would you say? Is **the** is there a key thing that differentiates good entrepreneurs from others? If you could double down on one trait, what would it be? If I could pick only one, I would pick determination. I think that is **the** biggest predictor of success, **the** biggest differentiator predictor. And if you would allow a second, I would pick like communication skills or evangelism or something in that direction as well. There are all of **the** obvious ones that matter, like intelligence, but there ' s like a lot of smart people in **the** world. And when I look back at kind of **the** thousands of entrepreneurs I ' ve worked with, all of many of whom were like quite capable, I would say that ' s like one and two of **the** surprisingly differentiated characteristics. What it ' s it ' s what I look at, **the** different things that you ' ve built and you ' re working on. I mean, it could not be more foundational for **the** future. I mean, entrepreneurship. I know this is I agree that this is really what has driven **the** future. Do you see some people get really now they look at Silicon Valley and they look at this story and they worry about **the** culture. Right. That it ' s this is a bro culture. Do you see prospects of that changing anytime soon? And would you welcome it? Can we get better companies by really working to expand a group of people who can be entrepreneurs and who can contribute to aid, for example? For sure. And in fact, I think I ' m hopeful, since these are **the** two things I ' ve thought **the** most about. I ' m excited for **the** day when someone combines them and uses A. I. to better select who did more fairly, maybe even select who to fund and how to advise them and really kind of make entrepreneurship super widely available that will lead to like better outcomes and sort of more societal wealth for all of us. So are. So, yeah, I think. Broadening **the** set of people able to start companies and that sort of get **the** resources that you need, that is like an unequivocally good thing and it ’ s something that I think Silicon Valley is making some progress in. But I hope we see a lot more. And I do really, truly think that **the** technology industry entrepreneurship is one of **the** greatest forces for self betterment. If we can just figure out how to be a little bit more inclusive in how we do things. My last question today is about ideas were spreading. If you could inject one idea into **the** mind of everyone listening, what what would **the** idea be? We ’ ve touched on it a bunch, but **the** one idea would be **the** ajai really is going to happen. You have to engage with it seriously, and you shouldn ’ t just listen to this and then brush aside and go about life as if it ’ s not going to happen because it is going to affect everything. And we will all we all, I think, have an obligation, but also an opportunity to figure out what not means and how we want **the** world and this sort of one time shift to go on. I ’ m kind of awed by **the** breadth of things are engaged with. Thank you so much for spending so much time sharing your vision. Thanks so much for having me. OK, that ’ s it for today. You can read more about open eyes, vision and progress at open eye dotcom. If you want to try playing with yourself, it ’ s a little tricky. You have to find a website that has licensed **the** API. **The** one I went to was philosopher ehi dot com, where you just you pay a few dollars to get access to a very strange mind. That ’ s actually quite a lot of fun. **The** interview is part of **the** TED Audio Collective, a **collection** of podcasts dedicated to sparking curiosity and sharing ideas that matter. This show is produced by Kim Net2Phone Pittas and edited by Grace Rubenstein and Sheila Boffano, Sambor Islamic Sir. Fact Check is by Paul Durbin and special thanks to Michele Quent, Colin Helmes and Anna Felin. If you like **the** show, please write and review it. It helps other people find us. We read every review, so thanks so much for listening. See you next time.

Text Sample 61: POS Dim 1 – 2021 – Score 23.00 – 2021/t004046.txt

Bruno Giussani : Hello. Bill Gates calls himself an imperfect messenger on climate because of his high **carbon** footprint and lifestyle. However, he has just made a major contribution to our thinking about confronting climate change via a book, a book about decarbonising our economy and society. It ’ s an optimistic”can-do”kind of book with a strong focus on technological solutions. He discusses **the** things we have, such as wind and solar power, **the** things we need to develop, such as carbon-free cement or carbon-free steel, and long-term energy storage. And he talks about **the** economics of it all, introducing **the** concept of green premium. **The** gist of **the** book, and really I ’ m simplifying here a lot, is that fighting climate change is going to be hard, but it ’ s possible, we can do it. There is a pathway to a clean and prosperous future for all. So we want to unpack some of that with **the** author of **the** book. Bill Gates, welcome back to TED. Bill Gates : Thank you. Giussani : Bill, I would like to start where you start, from **the** title of **the** book.”How to Avoid a Climate Disaster,”which, of course, presumes that we are heading towards a climate disaster if we don ’ t act differently. So what is **the** single most important thing we must do to avoid a climate disaster? Gates : Well, **the greenhouse** gases we put into **the** atmosphere, particularly CO2, stay there for thousands of years. And so it ’ s really **the sum** of all those **emissions** are forcing **the** temperature higher and higher, which will have disastrous effects. And so we have to take these **emissions**, which are presently over 51 billion tons per year and drive those all **the** way down to zero. And that ’ s when **the** temperature will stop increasing and **the** disastrous weather events won ’ t get worse and worse. So it ’ s pretty demanding. It ’ s not a 50 percent reduction. It ’ s all **the** way down to zero. Giussani : Now, 51 billion is a big number. It ’ s difficult to register, to understand. Help us visualize **the** scale of **the** problem. Gates : Well, **the** key is to understand all **the** different sources. And people are mostly aware of **the** production of electricity with natural gas and **coal** as being a big source. That ' s about 27 percent. And they ' re somewhat aware of **transportation**, including passenger cars. Passenger cars are seven percent and **transportation** overall is 16 percent. They have far less awareness of **the** other three segments. Agriculture, which is 19 percent, heating and cooling buildings, including using natural gas, are seven percent. And then, sadly, **the** biggest segment of all, manufacturing, including steel and cement. People are least aware of that one. And in fact, that one is **the** most difficult for us to solve. **The** size of all **the** steel plants, cement plants, paper, plastic, **the** industrial economy is gigantic. And we ' re asking that to be changed over in this 30-year period when we don ' t even know how to make that change right now. Giussani : So your core argument, and really, here I ' m simplifying, is that basically we need to clean up all of that, right? **The** way we make things, we grow things, we get around and power our economy. And so to do that, we need to get to a point where green energy is as cheap as **fossil** fuels, and new materials, clean materials, are as cheap as current materials. And you call that"eliminating **the** green premium."So to start, tell us what you mean by green premium. Gates : Yeah, so **the** green premium varies from **emission** sources. It ' s **the** cost to buy that product where there ' s been no **emissions** versus **the** cost we have today. And so for an electric car, **the** green premium is reasonably modest. You pay a little more upfront, you save a bit on **the** maintenance and gasoline, you give up some range, you have a longer charging time. But over **the** next 15 years, because **the** volume is there and **the** R and D is being done, we can expect that **the** electric car will be preferable. It won ' t cost more. It will have a much higher range. And so that green premium that today is about 15 percent is headed to be zero even without any government subsidies. And so that ' s magic. That ' s exactly what we need to do for every other category. Now, an area like cement, where we haven ' t really gotten started yet, **the** green premium today is almost double **the** price. That is, you pay 125 dollars for a ton of cement today but it would be almost double that if you insisted that it be green cement. And so **the** way I think of this is in 2050 we ' ll be talking to India and saying to them,"Please use **the** green products as you ' re building basic shelter, you know, simple air conditioning,"which they ' ll need because of **the** heat increase or lighting at night for students. And unless we ' re willing to subsidize it or **the** price is very low, they ' ll say,"No, this is a problem that **the** rich countries created, that India is suffering from and you need to take care of it."So only by bringing that green premium down very dramatically, about 95 percent across all categories, will that conversation go well so that India can make that shift. And so **the** key thing here is that **the** US ' s responsibility is not just to zero out its **emissions**. That ' s a very hard thing, but we ' re only 15 percent. Unless we, through our power of innovation, make it so cheap for all countries to switch all categories, then we simply aren ' t going to get there. And so **the** US really has to step up and use all of this innovative capacity every year for **the** next 30 years. Giussani : What what needs to happen in order for these breakthroughs to actually **occur**? Who are **the** players who need to come together? Gates : Well, innovation usually happens at a pace of its own. Here we have this deadline, 2050. And so we have to do everything we can to accelerate it. We need to raise **the** R and D budgets in these areas. In 2015 I organized, along with President Hollande and Obama, a side event to **the** Paris climate talks where what Prime Minister Modi had labeled"mission innovation"was a commitment to double R and D budgets over a five-year period. And all **the** big countries came in and made that pledge. Then we need lots of smart people, who, instead of working on other problems, are encouraged to work on these problems. So coming up with funding for them is very, very important. I ' m doing some of that through what I call Breakthrough Energy Fellows. We need high-risk capital to invest in these companies, even though **the** risks are very, very high. And that ' s – There is now – Breakthrough Energy Ventures is one group doing that and drawing lots of other people in. But then **the** most difficult thing is we actually need markets for these products even when they start out being more expensive. And that ' s what I call Catalyst, organizing **the** buying power of consumers and companies and governments so that we get on **the** learning curve, get **the** scale going up, like we did with solar and wind across all these categories. So it ' s both supply of innovation and demand for **the** green products. That ' s **the** combination that can start us to make this change to **the** infrastructure of **the** entire physical economy. Giussani : In terms of funding all this, **the** financial system as a whole right now is essentially funding expansion of **fossil** fuels consistent with three degrees Celsius increase in global heating. What do you say to **the** Finance Committee, beyond **the** venture capital, about **the** need to think and act differently? Gates : Well, if you look at **the** interest rates for a solar field versus any other type of investments, it ' s not lower. You know, money is very fungible. It ' s going to, you know, different projects. But, there ' s no, sort of, special rate for climate-related projects. Now, you know, governments can decide through tax incentives to improve those things. But you know, this is not just about reporting numbers. It ' s good to report numbers. But **the** steel industry is providing a vital service. Even gasoline, for 98 percent of **the** cars being purchased today, allows people to get to their job. And so, you know, just divestment alone is not going to be **the** thing that creates **the** new alternative and brings **the** cost of that down. So **the** finance side will be important because **the** speed of deployment – Solar and wind needs to be accelerated dramatically beyond anything we ' ve done today. In fact, you know, on average, we ' ll have to deploy three times as much every year as **the** peak year so far. And so those are a key part of **the** solution. They ' re not **the** whole solution. But you know, people sometimes think just by putting numbers, disclosing numbers, that somehow it all changes or by divesting that it all changes. **The** financial sector is important, but without **the** innovation, there ' s nothing they can do. Giussani : A lot of **the** current focus is on cutting **emissions** by half by 2030 and on **the** way to reaching that zero by 2050. And in **the** book, you write that there is danger in that kind of thinking, that we should keep our main focus on 2050. Can you explain that? Gates : Well, **the** only real measure of how well we ' re doing is **the** green premium, because that ' s what determines whether India and other developing countries will choose to use **the** zero **emission** approach in 2050. **The** idea, you know – If you ' re focusing on short-term reductions, then you could say, oh, it ' s **fantastic**, we just put billions of dollars into natural-gas plants. And even ignoring that there ' s a lot of leakage that doesn ' t get properly measured, you know, give them full credit and say that ' s a 50 percent reduction. **The** lifetime of that plant is greater than 30 years. You financed it, assuming you ' re getting value out of it over a longer period of time. So to say,"Hey, hallelujah, we switched from **coal** to natural gas,"that has nothing to do with reaching zero. It sets you back because of **the** capital spending involved there. And so it ' s not a path to zero if you ' re just working on **the** easy parts of **the emission**. That ' s not a path. **The** path is to take every source of **emission** and say,"Oh, my goodness, how am I going to get that green premium down? How am I going to get that green premium down?"Otherwise, getting to 50 percent does not stop **the** problems. This is very tough because **the** nature is zero. It ' s not just, you know, a small decrease. Giussani : So getting **the** green premium down, so you mentioned India. To make sure that **the** transition, **the** clean transition, is also an Indian story and an African story and not only a Western story, should rich countries adopt **the** expensive clean alternatives now, to kind of buy down **the** green premium and make technology, therefore, more accessible to low-income countries? Gates : Absolutely. You have to pick which product paths with scale will become low green-premium products. And so you don ’ t want to just randomly pick things that are low-emission. You have to pick things that will come down. So like, you know, so far, hydrogen fuel cells have not done that. Now they might in **the** future. Solar, wind and lithium-ion, we ’ ve seen these incredible cost reductions. And so we have to duplicate that for other areas, things like offshore wind, heat pump, new ways of doing transmission. You know, we have to keep **the** electricity grid reliable, even in very bad weather conditions. And so that ’ s where storage or nuclear and transmission will have to be scaled up in a very significant way, which we now have this **open-source** model to look at that. And so **the** buying of green products that demand signal, what I call Catalyst, we ’ re going to have to orchestrate a lot of money, many tens of billions of dollars for that, that is one of **the** most expensive pieces so that **the** improvements come in these other areas and some of them we haven ’ t even gotten started on. Giussani : Many people believe actually that **the** climate question is mostly a question of consuming less and particularly consuming less energy. And in **the** book, you actually write several times that we need to consume more energy. Why? Gates : Well, **the** basic living conditions that we take for granted should be made available to all humans. And **the** human population is growing. And so you ’ re going – as you provide shelter, heating and air conditioning, which anywhere near **the** equator will be more in demand since you ’ ll have many days where you can ’ t go outdoors at all. So we ’ re not going to stop making shelter. We ’ re not going to stop making food. And so we need to be able to multiply those processes by zero. That is, you make shelter, but there ’ s no **emissions**. You make food, but there ’ s no **emissions**. That ’ s how you get all **the** way down to zero. Now, it ’ s made somewhat easier if rich countries are consuming less, but how far will that go? Giussani : So if I summarize in my head your book, you are basically suggesting that if we eliminate **the** green premium, **the** transition somehow can **occur**, **the** cost and technology, green premium and breakthroughs are **the** key drivers. But, you know, you think of climate and climate is kind of a wicked problem. It has implications that are social, political, behavioral. It requires significant citizen involvement. Are you focusing too much on tech and not enough on those other variables? Gates : Well, we need all these things. If you don ’ t have a deep engagement, particularly by **the** younger generation making this a top priority every year for **the** next 30 years across all **the** developed countries, we will not succeed. And I ’ m not **the** one who knows how to activate all those people. I ’ m super glad that **the** people who are smart about that are thinking. It ’ s a necessary element. Likewise, **the** piece that I do have experience in, innovation ecosystems, that ’ s a necessary element. You will not get there just by saying,”Please stop using steel.”You know, it won ’ t happen. And so **the** innovation piece has to come along and particularly encouraging consumers to buy electric cars or **artificial** meat or electric heat pumps, they ’ re part of driving that demand, what I call **the** catalytic demand. That alone won ’ t do it, because some of these big projects, like green hydrogen or green aviation fuel require billions in capital expense. But **the** demand signal from enlightened consumers is very important. Their political voice is very important. Their pushing **the** companies they work at is very important. And so I absolutely agree that **the** broad community who cares about this, particularly if they understand how hard it is and don ’ t say,”Oh, we can do it in 10 years,”they are super important. Giussani : You mentioned **artificial** meat and I know you love burgers. Have you tried an **artificial** burger, and how did you find it? Gates : Yes, I ’ m an investor in all these Impossible, Beyond and various people. And I have to say, **the** progress in that sector is greater than I expected. Five years ago, I would have said that is as hard as manufacturing. Now it ’ s very hard, but not as hard as manufacturing. There is no Impossible Foods of green steel and **the** quality is going to keep improving. It ’ s quite good today. You know, there ’ s other companies coming in to that space covering different types of food. And as **the** volume goes up, **the** price will go down. And so I think it ’ s quite promising. I have to admit, 100 percent of my burgers aren ’ t **artificial** yet, it ’ s about 50 percent, but I ’ ll get there. Giussani : It ’ s a start. You ’ re already on 2030 on that on that front. So you mentioned your expertise in innovation. You have a couple of lines of – how to say it? – of exquisite modesty in **the** book where you say you think like an engineer, you don ’ t know much about politics, but actually you talk to top politicians more than any of us. What do you ask them and what do you hear in return? Gates : Well, they ’ re mostly responding to voters interest. Most of **the** countries we ’ re talking about are democracies. And you know, there are resources that will need to be put in like tax credits that encourage buying green products or government purchasing of green products at slightly higher prices. Those are real trade-offs. And, you know, making sure that **the** areas where there ’ s lots of jobs in, say, **coal** mining, or things where **the** demand will go down, having that political sensitivity and thinking through, can we put **the** new jobs in that area and what are **the** departments that handle that. This is a tough political problem. And, you know, my admonition to people is not only to get educated themselves, but help educate other people. And often, like in **the** US, if it ’ s people of both parties, that ’ s even better. **The** level of interest is high, but it needs to get even higher, almost like a moral mission of all young people to go beyond their individual success, that they believe that getting to zero by 2050 is critical. Giussani : Thank you. Now, before we continue **the** interview, I would like to take a short detour for one minute because my colleagues at TED-Ed have produced a series of seven animated videos inspired by your book. They introduce **the** concept of net zero **emissions**, they discuss other questions and challenges that we are facing relating to climate. So I would like to share one short **clip** from one of those animations. You flip a switch, **coal** burns in a furnace which turns water into steam. That steam spins a **turbine** which activates a generator which pushes electrons through **the** wire. This current propagates through hundreds of miles of electric cables and arrives at your home. All around **the** world, countless people are doing this every second : flipping a switch, plugging in, pressing an”on”button. So how much electricity does humanity need? Giussani : **The** answer to that and to many other climate questions, of course, is in **the** seven animations available on **the** TED-Ed site, and **the** TED-Ed YouTube channel. Bill, you mentioned **the** younger generation before. How does **the** younger generation ’ s role in solving this problem inspire you? Gates : Well, they... can make sure this is a priority. And if we have, you know, it ’ s a priority for four years, then it ’ s not for four years, you can ’ t ask **the** trillions of investment in **the** new approach to take place. It ’ s got to be pretty clear that even though political parties may disagree on **the** tactics, **the** same way they agree there should be a strong defense that they agree this zero by 2050 is a shared goal, and then discuss, OK, where does government come in, where does **the** private sector come in, which area deserves priority? That will be a huge milestone where it ’ s a discussion about how to get there versus whether to get there. And **the** US is **the** most fraught in terms of it being politicized, even in terms of, is this a huge problem or not. So, you know, young people, they ' re going to be around to see **the** good news if we ' re able to achieve this. You know, I won ' t be around. But, they speak with moral authority and they have particular people who are stepping up on this. But I hope that ' s just **the** beginning. And that ' s why this year, I think, is so important. All this recovery money being programed, people thinking about, do governments protect us **the** way they should, do governments work together? You know, **the** pandemic has teed this up, it ' s OK, what ' s **the** next big problem we need to collaborate around? And I ' m hoping that climate appears there because of these activists. Giussani : Yeah, we ought to come back to **the** pandemic question. But I want to talk a moment about some specific technology breakthroughs that you mention in **the** book. But you also just mentioned **the** organization you set up, Breakthrough Energy, to invest in clean tech start-up and advocate for policies. And I assume somehow your book is kind of a blueprint for what Breakthrough Energy is going to do. There is a branch in that organization, called Catalyst, that ' s prioritizing several technologies, including green hydrogen direct carbon-capture, aviation biofuels. I don ' t want to ask you about specific investments, but I would like to ask you to describe why those priorities, maybe starting with green hydrogen. Gates : If we can get green hydrogen that ' s very cheap, and we don ' t know that we can, that becomes a magic ingredient to a lot of processes that lets you make fertilizer without using natural gas, it lets you reduce **iron** ore for steel production without using any form of **coal**. And so there ' s two ways to make it. You can take water and split it into hydrogen and oxygen. You can take natural gas and pull out **the** hydrogen. And so it ' s kind of a holy grail, you know, and so we need to get going. We need to get all **the** components to be very, very cheap. And only by actually doing projects, significant scale projects, do you get on to that learning curve. And so Catalyst will fund **the** early pilot projects along with governments to go make green hydrogen, get **the** electrolyzers to get up in volume and get a lot cheaper, because that would be a huge advance. Lot of manufacturing, not all of it, but a lot of it would be solved with that. Giussani : So just for those who don ' t know, green hydrogen is produced with clean energy sources, wind, solar and so, nitrogen produced from natural gas or **fossil** gas is called gray hydrogen. **The** second priority that you have set is direct **carbon** capture, pulling **carbon** from **the** atmosphere. And at least in theory, we can find a way to do that at large scale. Together with other technologies **the** problem could be solved, but it ' s a very early-stage, unproven technology. You describe it yourself in **the** book as a thought experiment at this stage. But you are one of **the** main investors in this sector in **the** world. So what ' s **the** real potential for direct **carbon** capture? Gates : So there ' s a company today, Climeworks, that for a bit over 600 dollars a ton will do capture. Now it ' s at fairly small scale. I ' m a customer of theirs as part of my program where I eliminate all of my **carbon emissions** in a gold standard way. There are other people who are trying to do larger-scale plants like Carbon Engineering. Breakthrough Energy is investing in a number of these carbon-capture entities. In a way, **the carbon** capture is for **the** part that you can ' t solve any other way. It ' s kind of **the** brute-force piece, and no one knows what that price will be. If it ' s 100 dollars a ton, then **the** cost against current **emissions** would be five trillion a year. Can we get below 100 dollars a ton? It ' s not clear that we can, but it would be **fantastic** to take **the** final 10, 15 percent of **emissions**, and instead of making **the** change at **the** place where **the emission** takes place, just do this, direct air capture. To be clear, direct air capture means you ' re just filtering **the** air, and you ' re pulling out **the** 410 million current parts per million and putting that into a pressurized form of CO2 that then you sequester in someplace that you know it ' ll stay for millions of years. And so this industry is just at **the** very, very beginning. But it ’ s a necessary piece for **the** tail of **emissions**. Giussani : And a third priority is aviation biofuels. Now flying over **the** last several years has become a symbol of a **polluting** lifestyle, let ’ s say. Why aviation biofuels as a priority? Gates : Well, **the** great thing about passenger cars is that even though batteries don ’ t store energy as well as gasoline does, there ’ s dramatic difference, you can afford to have **the** extra size and weight of those batteries. On a plane that is a large plane going a long distance, there ’ s no chance that batteries will ever have that energy density. I mean, there ’ s some crazy people who are working on it, and I ’ ll be glad to fund them, but you wouldn ’ t want to count on that. And so you either have to use hydrogen as a plane fuel, that has certain challenges, or you just want to make today ’ s aviation fuel with green processes, like using plants as **the** source of how you make those. And so I ’ m **the** biggest individual customer of a group that makes green aviation fuel. So that ’ s what I ’ m using now, costs over twice as much as **the** normal aviation fuel. But as **the** demand scales up, that ’ s one of those green premiums that we hope to bring down so that at some point lots of consumers will say,”Yes, I ’ ll pay a little extra on my plane **ticket**,”probably through Catalyst or through **the** airline to make sure that we ’ re building up that industry and trying to get **the** costs, that green premium down quite a bit. So this is a super important area that, you know, I was amazed when I went to buy that I was by far **the** biggest individual customer. Giussani : It ’ s also, of course, a very symbolic area, right? People think of cars and think of **airplanes** when they think of pollution and **emissions**. There is a fourth technology that I want to bring up because you are an advocate of and an investor in nuclear power, which is also not universally accepted as clean energy because of **the** risks, because of radioactive waste. Can you make **the** case, briefly, but can you make your case for nuclear? Gates : Well, one thing that needs to be appreciated is that energy has to come from somewhere. And so as you stop using natural gas to heat homes and gasoline to power cars, **the** electric grid will have to grow dramatically. And even in **the** case of **the** US, where electricity demand has been flat **the** last few decades, will need to be almost three times as large. As you do that with weather-dependent sources, **the** reliability of **the** electricity generation goes away, that as you get big cold fronts that for 10 days can stop most of **the** wind and solar, say, in **the** Midwest. So **the** question is, how do you use massive amounts of transmission and storage and non weather-dependent sources which at scale nuclear as **the** only choice there, to maintain that reliability and not have people, say, freeze to death. And nuclear – People will be pretty impressed with how valuable it is to create that reliability. Now, 80 percent, at least in **the** US, will be wind and solar. So we have to build that faster than ever. But what is that piece that ’ s always available is where nuclear would fit in. Giussani : Now, you talk a lot about scaling up technologies in **the** book. I want to ask you a question about scaling down, because one thing that ’ s absent from your book is a discussion about scaling down **fossil** fuel production, maybe starting with at least not expanding it anymore, with adopting a sort of **fossil** fuel nonproliferation approach, because right now we are talking about a green transition, but at **the** same time we are building out further **fossil** fuel infrastructure. What ’ s your view on that? Gates : Well, it ’ s all about **the** demand for **fossil** fuels and **the** fact that **the** green premium is very high. If you want to restrict supply, then you ’ ll just drive **the** price up. People still want to drive to their job. If you know, say you made **fossil** fuels illegal. You know, try that for a few weeks. My view is you have to create a substitute because **the** services provided by **the** **fossil** fuel are actually quite valuable. Electricity is valuable. Transportation is valuable. And are you willing to drop demand for those things to zero? So

the capitalistic economy will respond. If it ’ s clear, you know, how you ’ re going to do your tax policies and your credit, you ’ re going to drive innovation, then **the** infrastructure investments in those things will go down. We see that with **coal** already today. But... Just speaking against something when you haven ’ t created an alternative isn ’ t going to get you to zero. Giussani : OK, I want to shift topics. You mentioned before **the** pandemic, and your main focus since you set up **the** Gates Foundation has been on global health. I actually read **the** letter, **the** annual letter you wrote with your wife, Melinda, in January 2021, which was very much about **the** COVID-19 pandemic. And I found myself marveling at how what you write resonates with **the** climate challenge. What lessons from **the** pandemic can be actually applied to climate? Gates : Well I think **the** biggest lesson is that governments got to, on our behalf, avoid disastrous future outcomes. And individual citizens aren ’ t equipped to either do those evaluations or think through that R and D and deployment plan that ’ s necessary here. And so we have to make it an imperative that governments, hopefully of any party, join in to this. **The** pandemic, we did eventually get global cooperation. **The** US didn ’ t play its normal role there, but **the** private sector innovation created **the** vaccine. Now, sadly, with climate, **the** pain it ’ s causing gets worse over time. So with **the** pandemic, we had all these deaths, and people were like,”Wow, OK, we should do something.”With climate you can ’ t wait. You know, **the** coral reefs will have died off, **the** species will be gone. And so if you say,”OK, well, let ’ s, when it gets bad, we ’ ll invent something like a vaccine.”That doesn ’ t work, because to stop **the emissions**, you have to change every steel plant, cement plant, car, things that have massive lead times and literally trillions of dollars of investment. And so with **the** pandemic, we messed up. We didn ’ t pay attention. You know, people like myself said it was a problem. But, now we ’ re getting our way out of it through innovation. But climate ’ s harder, much harder problem. And so **the** political will to get it right needs to be unprecedented compared to **the** pandemic or almost any other political cause. Giussani : I want to correct you about private-sector innovation, because, of course, a lot of public money went into funding research and guaranteed purchases. And so for **the** vaccines. But that ’ s a separate interview. Gates : Well, **the** Pfizer vaccine used no government money. Giussani : That ’ s absolutely true. Bill, still related to our reaction to these big challenges, over half of **the** total **greenhouse** gas **emissions** since 2017, 50 or 51 have gone up in **the** atmosphere in **the** last 30 years. And we have known for more than 30 years that they have pernicious effects, to say **the** least. So if we could mobilize such enormous resources and policies and collaborations and global collaborations against COVID-19, what ’ s **the** lever we can use to mobilize **the** same against climate? Gates : Well, until 2015, nobody pushed for **the** R and D budgets to go up. So, you know, there are times when I think”Wow, are we serious, are we not?”You know, that R and D question just wasn ’ t discussed. And you could read every paper on green steel and there were hardly any. And so we are just starting to get serious about climate. And you know, we ’ ve wasted a lot of time that we should have used to work on **the** hard parts of climates, just having short-term goals and not focusing on **the** R and D piece, you know, **the** last 20 years, we don ’ t have much to show for **the** hard categories. Now, there ’ s still time to get there. We will have to fund adaptation a great deal because **the** 2050 goal is not because that ’ s a goal that gives you zero damage. It ’ s simply **the** goal that is **the** most ambitious that has a chance of being achieved. And we are causing problems for subsistence farmers and sea-level rise and wildfires and natural ecosystems. And so **the** adaptation side is even more underfunded than **the** mitigation side. For example, helping poor farmers with seeds that can deal with **the droughts** and high temperatures is funded at less than a billion a year, which is deeply tragic. Giussani : Bill, we ’ re getting towards **the** end. I mentioned at **the** beginning, in **the** book you describe yourself as an imperfect messenger on climate. What changes have you made in your personal and family life to reduce your footprint? Gates : Well, I ’ m certainly driving an electric car, you know, putting solar panels on **the** houses where that makes sense. Using this green aviation fuel, I still can ’ t say that I ’ ve stopped eating meat or that I never fly. And so it ’ s mostly by funding at over seven million a year **the** products that although their green premiums are very high today, like **the carbon** capture or like **the** aviation fuel, by funding those, you actually get those on to **the** learning curve to get those prices down very dramatically. One area that I do is I fund putting in electric heat-pumps into low-cost housing. And so instead of using natural gas, they get a lower bill because it ’ s done with electricity. And I paid that extra capital cost as an offset. So, you know, accelerating those demand things. I ’ ve tried to get out in front of that. Giussani : You also talk in **the** book about paying more than market value or **the** current market price for offsets, and you hinted to them before, talking about **the** golden standard. Tell us about what kind of – offsets can be kind of confusing and controversial. What are **the** right offsets? Gates : Well, first of all, it ’ s great that we ’ re finally talking about offsets, and any company that actually looks at their **emissions** and pays for offsets is way better than a company that either doesn ’ t look at their **emissions** or looks at them and doesn ’ t pay for offsets. And so **the** leading companies are now buying offsets, some of them spending hundreds of millions to buy offsets. That is a very, very good thing. **The** ability to see which offsets actually have long-term benefit that really keep **the carbon** out for **the** over thousands of years that count. There are now organizations that I and others are funding to label offsets as either gold standard or different levels of impact. And **the** price of offsets range from 15 dollars a ton to 600 dollars a ton. Some of **the** low-cost ones may be really legitimate, like reducing natural gas leakage. That is pretty dramatic in some cases in terms of **the** dollars per tons avoided there. A lot of **the** forestry things will probably not end up looking that good, as you really look at **the** lifetime of **the** tree or what would have happened otherwise. But, you know, at least we ’ re talking about offsets now. And now we ’ re going to do it in a thoughtful way. Giussani : So some of **the** people listening to this interview may already be very involved with climate, others may be looking for ways to step up. And at **the** end of your book, you have a chapter about individual action. Give us maybe, say, two examples, two practical examples of things that individual citizens in **the** US, but also elsewhere, can do to play a meaningful role in tackling climate change. Gates : I think everybody should start by learning more, you know. How much steel do we make and where are those steel plants? **The** industrial economy is kind of a miracle, although sadly, it ’ s a source of so many **emissions**. Once you really educate yourself, then you ’ re in a position to educate others, hopefully, of diverse political beliefs about why this is so important. And yet it ’ s also very, very hard to do. You have all your buying behavior, electric cars, **artificial** meat, you know, and you ’ ll see for all **the** different products various things that indicate how green that product is. And your demand doesn ’ t just save those **emissions**, it also encourages **the** improvement of **the** green product. Political voice, I ’ d still put it number one. Making sure your company is measuring its **emissions** and is starting to fund offsets and is willing to be a customer for breakthrough storage solutions or green aviation fuel. That is catalytic. And a lot of those funds hopefully will go through this vehicle that ’ s really identifying which projects globally are technologies that won ’ t stay expensive but can do like what solar and wind did and come down in price. So individuals are what drive this thing. **The** democracies are where most of **the** innovation power is and they have to get activated and set **the** example. Giussani : I would like to end on **the** outcome. Maybe we should have started with **the** outcome, but let ' s imagine a world where we will actually have done all of what you describe in **the** book and everything else that ' s necessary. What would that future everyday life look like? Gates : Well, I think everybody will be really proud that humanity came together on a global basis to make this radical change and there ' s no precedent for it. Even world wars, where we orchestrated lots of resources, it was, like, four or five year duration. And here we ' re talking about three decades of hard work dealing with an enemy that **the** super bad stuff is out in **the** future. And so you ' re benefiting young people and future generations. In some ways, you know, life will look a lot like it does today. You ' ll still have buildings, you ' ll have air conditioning, you ' ll have lights at night. But all of those you ' ll multiply by zero in terms of what **the emissions** that come out of those activities are. During **the** same time frame we ' ll have advances in medicine of curing cancer and finishing polio and **malaria** and all sorts of things. And so by taking away this one super negative thing, then all **the** progress we make in other areas won ' t get reduced by this awful thing that, if it goes unchecked, **the** migration out of **the** equatorial regions, **the** number of deaths, it ' ll make **the** pandemic look like nothing. I think so from a moral point of view and letting **the** other improvements not be offset by this. It ' ll be a source of great pride that, hey, we came together. Giussani : OK, let ' s hope that we actually do. Bill Gates, thank you for sharing your knowledge, thank you for this very, very important book. And thank you for coming back to TED. I want to close by showing another short **clip** from **the** TED-Ed animation series inspired by Bill ' s book. This one is about material that ' s all around us, has a big **carbon** footprint and how to reinvent it. Now, **the** series includes seven videos. You can watch them all for free at ed. ted. com / planforzero. That ' s planforzero, one single word. And on **the** TED-Ed YouTube channel. Thank you. Goodbye. Look around your home. Refrigeration, along with other heating and cooling, makes up about six percent of total **emissions**. Agriculture, which produces our food, accounts for 18 percent. Electricity is responsible for 27 percent. Walk outside and **the** cars zipping past, planes overhead, trains ferrying commuters to work. Transportation, including **shipping**, contributes 16 percent of **greenhouse** gas **emissions**. Even before we use any of these things, making them produces **emissions**, a lot of **emissions**. Making materials, concrete, steel, plastic, glass, aluminum and everything else accounts for 31 percent of **greenhouse** gas **emissions**. [Watch **the** full animated series at ed. ted. com / planforzero and youtube. com / ted-ed]

Text Sample 62: POS Dim 1 – 2021 – Score 22.00 – 2021/t003892.txt

So **the** climate crisis, in my way of thinking about it, is **the** most serious manifestation of an underlying collision between human civilization as we ' ve presently organized it and **the** Earth ' s ecological systems. And **the** system most in jeopardy is **the** very thin shell of atmosphere surrounding our **planet**, because we ' re spewing 162 million tons of human-caused global warming pollution into it every single day, as if this is an open sewer. It ' s not an open sewer. It ' s so thin that if you could drive an automobile at autobahn speeds to **the** top of that **blue** shell, you ' d reach it in about five minutes. And that ' s where all **the greenhouse** gases are. **The** accumulated amount, coming from **the** burning of **fossil** fuels primarily, CO2 – fastest-growing source of methane is from **fossil** fuels as well, also, **agriculture** as you have heard – **the** accumulated amount traps as much heat now as would be released by 600, 000 Hiroshima-class atomic bombs exploding every day. So **the** temperatures are going up at almost record levels every year. Last year was **the** hottest year in recorded history, according to NASA, and **the** scientists say it ' s absolutely unequivocal that we are **the** cause of that. And we ' re hearing Mother Nature and seeing **the** extreme events. **The** most anomalous extreme event since records began 200 years ago was in **the** Pacific Northwest of North America. Forty nine and a half degrees, 121 degrees Fahrenheit in British Columbia. In **the** tropics and subtropics, **the** combination of temperature and humidity is now making larger areas literally uninhabitable, **the** area so described is relatively small now, but if we continue over **the** next few decades, these areas are predicted to expand, and billions of people will be in areas where it is not safe to stay outside for more than a couple of hours. Already, **the** climate refugees and migrants are four times more from **the** climate crisis than from all **the** wars and conflicts going on right now. And **the** respected"Lancet"commission has predicted as many as a billion climate refugees in this century. That ' s one of **the** reasons why we are seeing this wave of populist authoritarianism. Sea level rise is also contributing, and of course, 93 percent of **the** extra heat is absorbed in **the** oceans, and **the** ocean temperatures are reaching record levels as well. And that means more water vapor comes off **the** oceans, and **the** warmer ocean temperature makes **the** cyclonic storms, like hurricanes and typhoons much stronger. Hurricane Ida struck **the** Gulf Coast as a category four, and as is so often **the** case, communities of color and poor people were disproportionately victimized. This storm continued on north and northeast across North America to New York, New Jersey and Pennsylvania, killed a lot of people and dropped rain bombs. And in New York City some basement apartments were flooded. And I appreciate this family giving me **the** right to show this video. He ' s safe, **the** guy is screaming for his mother. He and his mother were both rescued by his brother, who punched a hole through **the** roof from **the** floor above. These rain bombs are becoming much more frequent and much more extreme. And actually **the** water coming off **the** oceans is held in much greater volumes by **the** warmer air. Atmospheric rivers are becoming larger. Here ' s one, from Hawaii in **the** lower left to Silicon Valley in **the** upper right, 2, 300 kilometers. What was happening in Silicon Valley when this satellite picture was made? Well, that ' s what happens when these rain bombs drop, and **the** average atmospheric river now contains as much moisture as 25 Mississippi Rivers. They ' re creating what you could call"atmospheric **tsunamis**"in some areas. **The** downpours get much bigger and much more frequent, four times more frequent now than in 1980. One of them dropped on Germany and neighboring countries in July, killed more than 200 people. Look at **the** before and after. This is an example of what we ' re doing to our **planet**. Ten days ago, in part of Italy, there were 74 centimeters that fell in 12 hours, 29 **inches** of rain in 12 hours. One year ago, October 3rd, as much rain fell here in **the** UK as there is water in Loch Ness. And as a result, **the** insurance industry is now seeing record recoveries, and those recoveries I showed you from last year may get a tiny bit larger, depending on **the** outcome of this lawsuit. **The** owners and operators of this giant replica of Noah ' s Ark have sued their insurance company for a million dollars in flood damages. Hard to make some of this stuff up. **The** same extra heat is pulling **the** moisture out of **the** first meter of **the** soil, creating **the** worst **droughts** in memory. 93 percent of **the** American West is in **drought** now, 100 percent of California, half of California is in **the** most extreme form of **drought**, and **the** same heat is also draining **the** reservoirs, evaporating **the** reservoirs. Lake Mead is down to levels not seen since they started filling it in **the** 1930s. In Brazil, **the** same thing is happening. In Eastern Europe,

the Czech Republic has been going through **the** worst **drought** in at least 500 years. And in **the** southern cone of Africa, 100 million people are experiencing food insecurity primarily because of **the** extended **droughts**. And when **the** temperatures go up, **the** fires get a lot worse and **the** worst fires in California and western US history have been in **the** last two years. Also in Siberia, in Australia, in southern Europe. It doesn ' t have to ruin your **golf** game. And this is a serious point, because we can ' t let these conditions become **the** new normal. It ' s not fine. And this is an example of why it ' s not fine. A lightning strike hitting a natural gas leak in **the** middle of **the** Gulf of Mexico. Lightning gets more common with **the** climate crisis. We ' re relying on dead plants and animals, leaving their residue in **the** atmosphere to threaten **the** species that are still alive, and 50 percent of them are in danger of extinction in this century. And as we push into previously wild areas of **the** world, we encounter millions of new viruses that we have not dealt with in **the** past. Five new infectious diseases every year, emerging, three quarters of them from animals, like **the** COVID-19 pandemic. Which raises some of **the** same questions that are raised by **the** climate crisis. When **the** world ' s leading scientists are setting their hair on fire to get our attention, should we listen to them? Check. Can our interconnected global civilization suddenly be turned upside down? Check. Are **the** poor and marginalized populations of **the** world **the** most affected? Check. Can science and technology give us nearly miraculous solutions in record time? Check. Will we deploy those solutions in time? That is **the** question. We have inequitable vaccine distribution in **the** world, and it threatens everyone in **the** world. We also have solutions to **the** climate crisis, but they ' re not equitably distributed. Worldwide, 90 percent of all of **the** new electricity generation installed last year was renewables. Almost all of it from solar and wind. Just one year before **the** Paris Agreement, solar and wind electricity was cheaper than **fossil** electricity in only one percent of **the** world. Five years later, two thirds of **the** world. Three years from now, it will be **the** cheapest source in 100 percent of **the** world. **Coal** is not getting cheaper. Gas is not getting cheaper. Nuclear has been getting more expensive. Wind is cheaper than all three, and solar is continuing to plummet in cost and is **the** cheapest of all now because **the** cost of making **the** solar panels and **the** windmills continues to come down. By **the** way, **the famous coal** museum in Kentucky just installed solar panels on its roof in order to save in its operating budget. Now we ' re getting battery storage. I ' ll rescale this graph to show you **the** predicted emergence of a new one-trillion-dollar industry in **the** next few decades and green hydrogen on top of that. **The** car batteries are getting cheaper as demand continues to increase. And within less than two years in some model categories, **the** EVs will be cheaper than their internal combustion engine counterparts and within four years in all model categories. So, **the oil** and gas industry has been **the** worst investment in global markets for more than a decade, while **the** clean energy companies are really becoming more profitable. Now **the oil** and gas companies say that they ' re going to invest a lot more in renewables and CCS. Yes, it ' s **the** fastest-growing – they ' ve tripled their investments all **the** way up to four percent. Ninety-six percent of **the** money they ' re spending is still going into **oil** and gas and **coal**. **The** impression they ' ve given us is not an honest impression, not for **the** first time. And they ' re telling a different story to Wall Street. They still have planes, trains, a few automobiles and ships, but they ' re telling Wall Street and financial markets they ' re going to make it up in plastics, which is 75 percent of their third-largest market, petrochemicals. How ' s that working out for us? Not so well. Banning single-use plastics is one of **the** many things that we have to do. We have to shift to **sustainability**. And here ' s **the** good news, **the sustainability** revolution is **the** single biggest business opportunity in **the** history of **the** world. It has **the** scale of **the** industrial revolution coupled with **the** speed of **the** digital revolution. And what we ’ re seeing is that we have to seize this opportunity, but we need reforms in **the** current version of capitalism. Capitalism does a lot of things great. It balances supply and demand and allocates resources efficiently and often **unlocks** a higher **fraction** of **the** human potential. And by **the** way, **the** alternatives on **the** left and right explored in **the** 20th century didn ’ t work out very well. But in order to see how we need to change capitalism, we have to pull out our focus. And let me present an analogy. **The** electromagnetic spectrum from **the** long radio waves to **the** short gamma rays and all **the** things in between, **the** portion of visible light that we can see with our eyes makes up only 0. 1 percent of **the** total. For **the** eight years I worked in **the** White House, I got a daily report from **the** intelligence community that collected from all parts of **the** spectrum, and it was a much more accurate picture. Here ’ s **the** analogy. **The** value spectrum is something we too frequently look at through **the** very narrow aperture of short-term profits for one stakeholder, **the** shareholders. But this leaves out negative externalities, as **the** economists call them, like **the** pollution. That ’ s why we need a price on **carbon** and a price on plastic pollution. It also leaves out positive externalities. So we ’ re chronically underinvesting in education and health care and environmental protection, pandemic preparedness. We ’ re ignoring **the** depletion of resources like topsoil and **underground** water aquifers. And we ’ re ignoring **the** distribution of incomes and net worths to **the** point where one percent of **the** people have almost half of **the** world ’ s wealth. That ’ s one of **the** other factors that ’ s driving populist authoritarianism. We have to take account of **the** environmental effects and **the** effects on people and their families and **the** communities where they live and **the** communities in their supply chain and **the** ethics in **the** C-suites. And we have to realize that hyper inequality is a threat to both capitalism and to democracy. There is an emergent form called multistakeholder capitalism, and it is driving a lot of new decisions. For example, in asset management, almost half of all **the** assets in **the** world under management are now in portfolios committed to net-zero. One reason is that **the** Paris Agreement set **the** direction of travel, where every country in **the** world committed to net-zero. So just last May, **the** G7 nations banned financing of **coal** plants overseas. And just three weeks ago, China did, which is good, because they were financing most of them. I hope that they will cut down on **the coal** chain in China domestically as well. And now more than half of all **the greenhouse** gas **emissions** and two thirds of global GDP is coming from countries that have set net-zero targets. So here ’ s **the** hope. Once we reach net-zero, then with a lag time of as little as three to five years, **the** temperatures in **the** world will stop going up. And once we reach net-zero within as little as 25 to 30 years, half of **the** human-caused CO2 will fall out of **the** atmosphere. It is as if we have a switch that we can flip in order to stop **the** climate crisis. Regrettably, some damage has been done, but we can stop **the** temperatures from going up and start **the** healing process. But we all have to flip this switch known as reaching net-zero. **The** young people are telling us that we have to, and they ’ re marching in every country in **the** world. And what ’ s **the** response? As recently as this morning we heard,”Well, it may be impossible.”Well, it ’ s not impossible. Nelson Mandela said,”It always seems impossible until it ’ s done.”That ’ s what they told **the** abolitionists.”It ’ s impossible to eliminate slavery.”That ’ s what they told **the** women ’ s suffragettes.”It ’ s impossible to have equal rights for women.”In **the** civil rights movement in my country and **the** anti-apartheid movement in South Africa and more recently, **the** lesbian and gay liberation and equal rights movement. We can do this. This is **the** biggest emergent social movement in all of history. We can do this. And if anybody thinks that we don ’ t have **the** political will, remember, political will is itself a **renewable** resource. Thank you very much. Thank you.

Text Sample 63: POS Dim 1 – 2023 – Score 26.00 – 2023/t003333.txt

Christiana Figueres : There ’ s no doubt that we are facing exponential impacts of climate change. There is also, however, no doubt that that is being met with exponential progress of **the** technologies that can help to address climate change. So what we have here is, frankly, a race between two exponential curves. By 2030, we collectively will have chosen between door number one. And door number two. Julio Friedmann : **The** question I am most commonly asked about climate change is :”Should I be optimistic or pessimistic?”Cynthia Williams : Here ’ s what we need to do to make sure that we ’ re ready for an all-electric future. Anika Goss : Being financially secure and climate resilient should be **the** most important priority. Faustine Delasalle : It ’ s really easy when you look at where we are today and where we ought to be by 2030, to feel like there ’ s a huge gap that will never be filled. But we have good reasons to say that that gap is actually fixable. Nigel Topping : If you ’ re not worried or anxious, you ’ re probably not paying attention. But we should also not be defeatist. We shouldn ’ t say,”We ’ ve done nothing, we ’ re failing.”We ’ ve really started 20 years too late, but it takes a long time to get **the** flywheel of momentum going. Kingsmill Bond : **The** tide of change is coming. **The** costs keep falling. **The** political pressure keeps increasing. JF : We ’ re going to build a thriving, exciting world. AG : We can do this. JF : Full of potential. AG : We have to do this. [TED Explores A New Climate Vision] Manoush Zamorodi : When it comes to climate change, are you an optimist or a pessimist? News about **the** climate can be overwhelming and emotional, to **the** point that many of us just feel numb. I ’ m Manoush Zomorodi, a longtime public media journalist and host of **the** TED Radio Hour. And today, we want to bring you something special : an hour of nuance, of real people and clarity about what is happening to our **planet** and what we ’ re doing about it. Because we all know there ’ s plenty of bad news when it comes to **the** climate crisis. But guess what? There ’ s also some really good news. That ’ s why we ’ re here in Detroit at **the** TED Countdown Summit, where a cross-section of people who know **the** most and who are doing **the** most have gathered to share their ideas and plans with each other and with you. You ’ re going to hear from some incredible speakers about their personal stories and tested solutions. But first, let ’ s get a quick reminder on **the** basics : climate change in a minute or so. [Prologue : What is Climate Change?] David Biello : **The** air we breathe has changed. **The** mix of gases is shifting, with more and more **greenhouse** gases like **carbon** dioxide. And **the** shift is happening faster each year. In just a few hundred years, **fossil** fuels formed over eons have been burned as **coal**, **oil** and gas. **The** exhaust has transformed **the** entire atmosphere and ocean. It ’ s like a pollution blanket. And **the** result we know as climate change. **The planet** has already warmed more than one degree Celsius and is on a path to heat up even more. That may not seem like a lot, but it ’ s already caused major destruction across **the** globe. To give us some room to breathe, **the** world must reduce **greenhouse** gas pollution by more than seven percent each year, every year of this decade. There are a number of ways to do that. Let ’ s start by looking at **the greenhouse** gases themselves. CO2 makes up nearly 75 percent of **the** pollution emitted each year. And then there ’ s methane or natural gas, which makes up 17 percent. Finally, there ’ s nitrous oxide, making up six percent of **the** problem. There are other **greenhouse** gases, but these three make up **the** bulk of

the climate challenge. Where do they come from? There are several ways to break down **the** sources of **greenhouse** gas **emissions**. Here we ' re using **the** public Climate Watch data. Start with energy. **The** majority of modern energy comes from burning **fossil** fuels, which makes **the** energy sector 76 percent of **the** climate challenge, including **the** fuels used for **transportation**, industrial processes and agricultural production. Farming and animal raising also have to change, as **agriculture** accounts for 12 percent of **greenhouse** gas pollution. **The** remaining 12 percent comes from a grab bag of human activities like industrial heating, clearing forests and more. This is **the** climate challenge we face : going from adding billions of metric tons of **greenhouse** gases to **the** air each year to adding zero. Will it be easy? No. Can we do it? Yes, if we choose to. MZ : OK, so that ' s **the** situation. Where do we go from here? Well, we ' ve all heard of a tipping point, that moment when an idea or concept finally takes hold and then changes **the** world. Well, **the** UN ' s climate chief, Simon Stiell, believes we are almost at a tipping point for massive action on climate change. And to get you ready, he wants to take you back to **the** early ' 90s, when he was an executive at Nokia, and **the** world wasn ' t sending text messages yet. [Chapter 1 : Exponential Change] Simon Stiell : I want you to imagine it ' s 1992, and you ' re talking to a telecoms expert. She tells you that mobile phone text messaging is going to fundamentally change **the** way we communicate. You think, why on Earth am I going to go from this to tapping on a screen with my fingers and my thumbs? But then, after a slow start, text-based services exploded. Today, over 23 billion messages are sent every single day. This paved **the** way for **the smartphone** revolution. I know how much we struggle to comprehend and predict what exponential change means, because I was at Nokia in 1993 when that technological revolution started. And now, as head of UN climate change, I ' m here to tell you that what was true for mobile phones is also true for climate action today. We ' re here in Detroit, **the** heart of car manufacturing. Take a look at this graph. Electric vehicle sales are expected to increase to become 70 percent by 2030. Another example, solar energy. Between 2010 and 2020, we moved from 20 gigawatts of solar energy installed to 150 gigawatts. And by 2030, that number is expected to increase again, to 1, 000 gigawatts per year. Change can come fast. FD : So exponential change means that change happens very slowly at first and then comes a tipping point. And suddenly things accelerate and accelerate and accelerate. NT : It ' s a mathematical term, but it means basically stuff happens very slowly and then it goes really fast. Tessa Kahn : Governments and experts have consistently underestimated **the** pace and scale at which we can deploy **renewable** energy and how quickly **renewable** energy and associated technologies become affordable. Ramez Naam : I predicted in 2011 that we ' d have solar as cheap as **coal** in some parts of **the** world by 2015 and half **the** cost of **coal** by 2020, and everyone thought I was crazy, I ' ve got to say, almost. And in fact, I was wrong. **The** actual decline in **the** cost of solar happened twice as fast as I thought it would. Solar prices today are a century ahead of where **the** world ' s leading energy experts thought they ' d be in 2010. KB : It ' s now spreading from solar, so it ' s happening also in **the** wind sector, it ' s happening in **the** battery sector, it ' s happening in **the** electric vehicle sector. CF : I don ' t think that we do a good job at communicating **the** fact that we are progressing more than is commonly known. Nili Gilbert : **The** human brain is trained best at thinking about linear change. But when you think about an exponential curve, next to a line, at first **the** exponential curve is below **the** line, and so it ' s moving more slowly than you would think. And then it crosses **the** line and it ' s moving faster than you would think. Habiba Daggash : We call it cautious optimism. We ' re very hopeful about **the** trajectories that we ' ve seen. But we shouldn ' t let up yet.

KB : So we can make this change happen slow or we can make it happen fast. **The** key point to say is we have agency. MZ : This idea of exponential growth sounds awesome. It sounds amazing, but is it actually happening? Well, I ' m headed into a workshop with industry and climate leaders to find out. **The** moderators of **the** workshop challenged us to hold two competing ideas in our minds at **the** same time. NT : What we ' re really going to explore today is two apparently opposing realities, and they ' re often two opposing narratives. **The** one which is true, right, is that in terms of **the** Paris goals, we ' re way off track. We ' re deep into **the** emergency. But what we ' re going to explore today is **the** other reality that we know what to do. Exponential change is happening, but it doesn ' t happen by magic. MZ : So what are we seeing in different industries? Well, let ' s start with one that ' s known to be really difficult to decarbonize : cement. Vicente Saiso Alva : I think everything converges into making climate a huge urgent matter for society. We started to feel **the** pressure from all different types of stakeholders, and in one year of running this program, we realized how fast we could move and how fast we were going in reducing CO2 **emissions**, which means now to go from 40 to close to 50 percent reduction by 2030. MZ : Encouraging, but to tackle that remaining 50 percent may require technologies that don ' t exist at scale yet. OK, what about **shipping**? Narin Phol : In Maersk, a few years back, we came out with a net-zero ambition by 2050, and in a few years later, based on **the** data that we have got, we ' ve really seen a path that we can really accelerate that one. That ' s why **the** revised target of 2040 came in. MZ : **The** news in offshore wind was quite dramatic. Jennie Dodson : So my moment of mindset shift was in 2019. **The** government put out a request to bid for offshore wind permits. And that year **the** price that came in was two thirds less than just four years before. And really critically, it was **the** same price as **the** mainstream market. And that was transformative. Nobody expected that when they started. MZ : **The** conversation got pretty technical and a bit wonky. NP : So to drive exponential growth, you focus on a reinforcing loop to make sure it works and you slow down on **the** balancing loop. MZ : But **the** takeaway is that certain sectors, like EVs and renewables, are being upended, while others, like cement and heavy industry are further behind. But generally across every industry – JD : It ' s amazing stuff. MZ : There ' s an understanding that they all need to work together. And **the** message? Big change can happen. It ' s going to be hard, but some of us are already doing it, so learn from us. [Chapter 2 : **The** Electric Vehicle Revolution] OK, we are in an exponential mindset. And as you heard, one example of exponential growth might be sitting in your driveway right now. Electric vehicles are finally really happening, which means that people here in Detroit are rethinking **the** auto industry. They ' re rethinking their jobs, their identity. Take Cynthia Williams, head of **sustainability** at Ford. I sat down with her at Lucky Detroit, a local coffee and barber shop, where she told me that there are three things that need to happen for electric vehicles to go to **the** next level, from this tipping point to exponential growth. CW : **The** three things for me are change, collaboration and capacity. So **the** change piece of it is changing **the** way folks think about **the** vehicle, making sure that they understand we ' re bringing a better vehicle to **the** customers. MZ : So you don ' t have to make a sacrifice. CW : I think one of **the** things we need to do is actually get people in **the** cars and show them how fun it is. And it ' s no different, it ' s just powered by a battery. We are at a tipping point. We ' re at a point of acceleration of electric vehicles. Many are calling it **the** electric vehicle revolution. Now with that comes unprecedented change that will require us to build new capacity and collaborate in ways that we ' ve never seen before. To make things happen faster, you have to think differently. You can ' t just collaborate with **the** same people because you get

the same answer, right? You need to broaden your collaboration space. Go outside of your industry, understand what worked in Norway, for example, to get them to where they are. MZ : Where ’ s **the** weak link, Cynthia? What ’ s **the** thing that you ’ re like, that if we don ’ t push through that, we may not get to this full adoption? CW : **The** critical things that we hear from **the** consumers is they need infrastructure. That ’ s what it ’ s taking for us to work with governments to make sure that we bring more charging to everyone, whether it ’ s rural, urban, wealthy, low-income, we all need chargers. And I ’ m not saying we need one on every corner, but we do need many, many more. When we start building out **the** charging infrastructure, we need lighting. We need, you know, security, shelter, we need restrooms. We need other amenities so that people know where to go, where to charge. Next comes capacity. We have to invest in new facilities and talent. Now, that ’ s true for any industry transitioning to greener goals and strategies. And we ’ re investing in talent, we ’ re investing in education and training, and we ’ re also listening to **the** communities where we live and operate. MZ : So a young Cynthia Williams, now, your counterpart who ’ s studying engineering, will she be learning something very different if she wants to go and work in **the** automotive industry in Detroit? CW : There will be different options, right. And so, I ’ m a mechanical engineer by trade, but I think there will be opportunities for **electrical** engineering, software engineering. And there ’ s even certificates that students can get instead of going through a four-year college. It ’ s **the** proper training to get them right, to be dedicated, to build **the** vehicles that we need for **the** future. At **the** end of this year, globally, we ’ ll have **the** ability to produce 600, 000 electric vehicles. And in 2026, we ’ ll have a global goal to produce two million electric vehicles. And it just goes up from there. MZ : What is this going to look like, this growth for me in **the** next couple of years, in five years, in 20 years? CW : Clean air. That ’ s what I look at. If you think about when we were, during **the** pandemic, when everybody stopped driving and they were at home, clear skies and moving to zero tailpipe **emission** vehicles. And if you chose an electric vehicle, that ’ s you participating and you contributing to a better world, to a better future. We have come a long way. It ’ s so gratifying to see electric vehicles on **the** road, to move from aspiration to reality. To see plants dedicated to building electric vehicles. We have **the** technology, we have **the** leadership, we have **the** ingenuity to respond to **the** environmental crisis with courage. And I ’ ll tell you, as a person who ’ s dedicated my entire career to **sustainability**, **the** road ahead is paved with promise. TED-Ed : If you were buying a car in 1899, you would have had three major options to choose from. You could buy a steam-powered car, typically relying on gas-powered boilers. These could drive as far as you wanted, provided you also wanted to lug around extra water to refuel and didn ’ t mind waiting 30 minutes for your engine to heat up. Alternatively, you could buy a car powered by gasoline. However, **the** internal combustion engines in these models required dangerous hand cranking to start and emitted loud noises and foul smelling exhaust while driving. So your best bet was probably option number three : a battery-powered electric vehicle. These cars were quick to start, clean and quiet to run, and if you lived somewhere with access to electricity, easy to refuel overnight. If this seems like an easy choice, you ’ re not alone. By **the** end of **the** 19th century, nearly 40 percent of American cars were electric. In cities with early electric systems, battery-powered cars were a popular and reliable alternative to their occasionally explosive competitors. But electric vehicles had one major problem. Batteries. MZ : Flash forward to **the** present and batteries are still on everyone ’ s mind. So I visited Mujeeb Ijaz, **the** founder of **the** energy storage company One, who also happens to be an electric vehicle history buff. Mujeeb Ijaz : So this is a 1922 Detroit Electric. MZ

: OK, so 100-year-old vehicle. MI : That ’ s right, electric vehicle. And then this was Walter Baker. He developed **the** Baker Electric. And he made **the** range run 201 miles on a single charge. He also felt strongly that you should be able to wear your top hat, you know, so he has a taller carriage. These had lead acid batteries to begin with. And then Thomas Edison introduced **the** nickel-iron battery. So this is **the** battery compartment, one of **the** battery compartments. And this is where you would charge. MZ : Right there? MI : And we ’ re building on what these original pioneers did. And I think it ’ s important to not think about us starting from scratch now. But go **backwards**, think about how they solved **the** problem, what disrupted their ability to advance this to **the** market, and then pick those problems up and make sure we don ’ t get stuck again. MZ : So we ’ ve been here before, a moment when electric vehicles could have taken off but didn ’ t. So what do we need to do differently this time? MI : Electric vehicles were able to deal with mobility in urban areas, but there were charging deserts where rural driving resulted in no way to get back because you couldn ’ t find electric charging everywhere. And that was a problem. MZ : And that ’ s what people are worried about right now. We ’ re at **the** same thing. We ’ re worried about range, we ’ re worried about accessibility. All of **the** things are **the** same. But **the** key difference being that we now know that our **planet** can ’ t keep going like this. MI : That ’ s right. Had **the** vision of **the** Model T assembly line and mass manufacturing been applied to electric car, **the** electric car would have also dropped in price, but range and charging deserts would have still been **the** obstruction. MZ : OK. We didn ’ t have **the** technology, but we do now. MI : We do now. MZ : In **the** US, there hasn ’ t been a mass move to electric vehicles... yet. Mujeeb ’ s company is one of many who want to change that by building batteries that give drivers longer range. And these batteries aren ’ t just for cars. MI : So we produce battery packs for **transportation**, but we ’ ve also started developing **the** same battery products into grid batteries. So micro-grids are those technologies that combine solar and storage and replace a **coal** plant or a natural gas plant for generating electricity. So buildings and industry and **transportation** all working together. And that ’ s an exciting future, is all these industries working together, that makes me more confident that we ’ re at **the** time in history where this transition will take place. MZ : How many years till we get to that? MI : I think it ’ s within 10 years. MZ : No way, 10 years? MI : We ’ re in a place where that transition is going to be much more visible, and then we just need technologies to drive it up to **the** mass markets. MZ : I am so excited. Like, genuinely so excited. Everyone everywhere driving electric sounds great, but that also means that **the** power to charge all those vehicles needs to be green. And that brings us to **the** energy sector, which is also changing at an incredible pace, even though **renewable** energy has been years in **the** making. [Chapter 3 : Renewables] TED-Ed : In **the** **spring** of 1954, **the** press excitedly gathered around Bell Laboratories ’ latest invention : a silicon-based solar cell that could efficiently convert **the** sun ’ s energy into **electrical** current. **The** creation was celebrated as **the** dawn of a new era, as reporters touted that civilization would soon run on **the** sun ’ s limitless energy. But **the** dream had a catch. As this first commercially sold solar cell cost around 300 dollars per watt, meaning at its current rate it would cost well over a million dollars to buy a unit large enough to power a single home. But today, in many countries, solar is **the** cheapest form of energy to produce, surpassing **fossil** fuel alternatives like **coal** and natural gas. Millions of homes are equipped with rooftop solar, with most units paying for themselves in their first seven to 12 years and then generating further savings. So how did solar become so affordable? A turning point in solar ’ s price history **occurred** on **the** floor of Germany ’ s parliament, where in 2000, Hermann Scheer introduced **the Renewable** Energy Sources Act. This legislation laid out a vision for **the** country ' s energy future in solar and wind. It incentivized citizens to personally invest in rooftop solar panels by guaranteeing payment to homeowners for **the renewable** energy they generated and sold to **the** grid. **The** pay rate for this electricity was highly subsidized, at times reaching four times **the** market price. Several other countries soon followed Germany ' s example, implementing similar policies and incentives to drive their country ' s solar use. This created unprecedented demand for solar panels worldwide. Manufacturers were able to scale up production and innovate in ways that cut costs. As a result, solar panel prices dropped while efficiency grew. HD : We see **the** cost reduction in solar and other **renewable** energy technologies has been so steep, but a lot of it is because of **the** enabling regulatory and policy environments created in places like **the** European Union and **the** US. Akil Callender : We cannot divert catastrophic climate change without addressing **emissions** coming from our energy systems. Currently, **emissions** from **the** energy system account for around two thirds of global **greenhouse** gas **emissions**. So **the** good news is, yes, we are seeing tremendous progress across **the** globe in **renewable** energy. It ' s rolling out faster than ever before. Luisa Neubauer : **The** brilliant thing about energy supply in **the** world is that we have alternatives. So we don ' t need **fossil** fuels to provide safe and equitable access to energy systems. Ramón Méndez Galain : Almost 90 percent of **the** total new installed capacity all over **the** world is just **renewable**. This is a tremendous change. AC : There ' s more finance than ever before going into clean energy. In many countries, it is **the** largest contributor to their electricity system for **the** first time ever. KB : And also because China is **the** leading manufacturing nation in **the** world, change is happening there much faster. And those innovations are then spreading around **the** world. Hongquiao Liu : China ' s **coal** demand might not have peaked, and China will continue to consume quite a lot of **coal** in **the** next few years. But **the** decarbonization is happening. It ' s happening faster than anyone has expected, and China is adding an astonishing amount of wind and solar to its grid every year. Eduarda Zoghbi : I think one of **the** biggest obstacles today are natural resources, in **the** sense that we ' re talking more and more about critical minerals and **the** role that they have. And batteries are demanding a lot of critical minerals that are present in most of **the** developing countries. So I think that even though costs are decreasing, we have to be more and more creative when it comes to recycling **the** batteries. We also have to think of how they ' re impacting local communities. KB : So to give you **the** numbers, we extract, at **the** moment, 15, 000 million tons of **fossil** fuels every year. **The** International Energy Agency says that we will need about 43 million tons of critical materials in order to build out **the** clean energy economy. So that ' s 300 times less stuff that is required. It ' s worth saying that **the** impact of extracting 300 times less stuff upon nature is dramatically lower. And I know that it ' s often argued that **the renewable** economy may be damaging for nature. Again, this is a completely false statement. It ' s in fact, our only chance of reducing **the** pressure upon our natural resources, because there ' s far less of it. MZ : Of course, transitioning to using clean energy, upending industries, it has to happen across **the** world, not just here in **the** US. And it can be hard to even imagine what life will be like. But **carbon** scientist and futurist Julio Friedmann can help us. Julio Friedmann : I think we need a compelling vision of **the** future that gets people excited about **the** energy transition and solving climate change. Right now, it ' s like we ' re at a restaurant and there ' s two options : burnt toast, unflavored oatmeal. And people are not excited about either, right? **The** burnt toast option for me is what I call **the** big switch. Everything ' s **the** same, it ' s just clean, but it costs a lot of money and it ’ s super hard to do. **The** other vision is kind of like,”Thou shalt not.”That vision is a narrower sort of,”eat your peas,”hair shirt kind of vision of **the** future. And a lot of people don ’ t want that either. And there ’ s nothing in **the** physics, **chemistry** or **biology** that says we can ’ t have a super interesting, exciting, vibrant world. And that is compelling in a way that we have not heard yet. MZ : Help me out here, what does that even look like? JF : So, for lack of a better word, like, I like **the** Marvel cinematic **universe**. MZ : OK, I ’ ll go with you. JF : So a place like Wakanda, right? It ’ s got flying cars, it ’ s got maglev trains, and it ’ s a totally fabulous place to live. Like, they ’ ve got lots of food. That is built on abundant, sustainable, cheap energy. MZ : OK, so like, we ’ re not Wakanda, we ’ re not in **the** Marvel **universe**. This is real life, sort of. How do we make this possible? JF : So what can you actually do here on Earth, constrained by economics and engineering and all **the** rest of it, right? And **the** answer is we can do an incredible amount. Abundant, available where you want it, when you want it. Everybody should have energy, including **the** three billion people who use less electricity than my refrigerator uses. Well, **the** good news is, every day, **the** Earth receives 163, 000 terawatts of energy from **the** sun. About half of that bounces back to space, but about 80, 000 terawatts arrive at **the** Earth in a form we can use. For reference, today, **the** world uses about 26 terawatts of energy, and we got more than solar and wind. We have geothermal, we have hydro, we have nuclear. There ’ s other kinds of clean energies. And some of **the** best resources are, in fact, in **the** Global South. These places are not simply future climate victims. These places are latent energy superpowers. My favorite example of this is Chile. Almost eight years ago now their government said, wait a second, we can make **the** cheapest green electricity on Earth. We have hydropower, we have solar power, we have wind power and **the** best resources anywhere in **the** world. So let ’ s start by making a lot of green electricity. And then they said, because of that, we can also make green hydrogen cheaper than anywhere else. And then Japan was like, hello, we want your green hydrogen. Can you ship it to us? MZ : Is that possible? JF : Totally. And Japan is like, we will also give you money to pay for these projects. We ’ ll loan you money to pay for these projects. Could you please use Japanese technology? So they ’ re using Japanese **turbines**, electrolyzers, they ’ re using Japanese ships to haul **the** ammonia around. So Japan gets rich on this, Chile gets rich on this. MZ : Win-win. JF : Totally win-win. MZ : Win-win-win, **planet**, too. JF : And that model, other countries are looking at that, going, I want some of that. And that ’ s why I like this vision of abundance. 15 years ago, solar was **the** most expensive power. Now it is **the** cheapest form of electricity, right? Which is astonishing, it ’ s totally amazing. And a whole bunch of people are like, now we ’ re done. And I ’ m like, no, we can do that with everything. We can do that with wind. We can do that with nuclear, we can do that with geothermal. But these projects take time, money and people. We need to develop **the** human capital as part of these investment projects for decades. And innovation, we need much more energy in many places, much cheaper. That ’ s an innovation agenda. For solar, that was **the** United States, Germany, China and others acting together. Well, if that ’ s **the** recipe, we can do that again. We ’ re already doing it with electric vehicles and clean hydrogen production. We can certainly do it by turning electricity into fuels. MZ : It sounds like **the** climate discussion needs to be kind of like improv, that it ’ s a”yes and”sort of thing. JF : I ’ m so glad you said that. So many people don ’ t understand my love of improv comedy. Yes, it is a”yes and”circumstance. Because mostly it is about room for agreement. Where do you find **the** point where everybody is like, I can agree on that? One of **the** things I love about **the** infrastructure bill, one of **the** things I love about **the** Inflation Reduction Act, neither of them had **the** word climate in it. They were both massive climate bills, right? That was a way to get everybody on **the** same page. Nationwide, they basically tripled **the** clean energy research budget. That ' s a good start. But that made it all of one percent of GDP. And for comparison, we spend twice that on pharmaceuticals. So I think it ' s fair to say we should get going. We should spend a lot more money. We ' ve done stuff like this before. We ' ve done World War II, we ' ve done **the** Marshall Plan, like, we ' ve done **the** space shot, like we actually know how to move really quickly as a nation. We did this for COVID. Collective action, building together is what makes **the** difficult possible and nourishes **the** soul through mission and purpose. We know what to do and we can act, not out of anger or fear, but out of generosity and common purpose, bringing aspiration and humility together. We ' re going to build a thriving, vibrant, exciting world full of potential that ' s going to be built on **the** back of infrastructure, innovation and investment that will harness **the** abundant, sustainable, cheap energy that is our **planet** ' s endowment. Thank you. [Chapter 4 : A Just Transition] FD : There is a high risk that when those exponential kick in, we see businesses running after those opportunities. But then **the** distribution of **the** benefits being very unequal. EZ : Many people like discussing **the** importance of **the** just energy transition. And a lot of people don ' t understand what justice really means. If we are to reach a revolution, a **renewable** energy future has to be inclusive. AC : We have this push for what we ' re calling a just and equitable energy transition. We need to make sure that we are not exacerbating inequality, but rather we are ensuring that that gap is closed and that **the** groups that have been historically marginalized are not once again being marginalized in a new way. NT : Most of **emissions** come from wealthy, energy-guzzling citizens, but a lot of **the** economic benefit and human development benefit will come not from decarbonizing high-carbon lives and lifestyles, but from providing clean, distributed energy and tools and solutions to those who don ' t have access to any of them. LN : When we speak of countries in need of economic development, we mostly speak of countries that are hit by **the** climate crisis **the** hardest right now. It should be **the** biggest wish and job and task for all those countries who have burned so much **fossil** fuel already to support any other country in **the** so-called Global South, to get an economy running without **fossil** fuels, to not repeat all **the** mistakes that we have made. HD : To ensure that **the** solar revolution that has been seen around **the** world comes to places like Africa, we need to be really conscious that **the** history and **the** lived experience when it comes to energy is very different for poor countries. Rebekah Shirley : We love to remind Global South communities, especially Africa, about **the** vast **renewable** energy potential that they have, which is true. But **the** finance flows to deliver on that potential remain very scarce. Africa, despite being 90 percent of those across **the** entire globe that still do not have access to electricity and being 17 percent of **the** global population, we ' re actually two percent of international finance. Money doesn ' t actually flow to these spaces naturally, and we need to do a lot of work to get global finance to these spaces. MZ : A world of abundant clean energy for everyone. That is **the** goal. But what about in **the** short term? Take Detroit. This once-wealthy city has been through some seriously hard times. How is it rebuilding itself so that everyone here can thrive while dealing with extreme heat, flooding, and other climate problems? Anika Goss is looking to Detroit ' s past to plan its future. AG : I am a third-generation Detroiter. My grandmother moved to Detroit in 1936 during **the** Great Migration and brought all of her southern ways with her. She owned a home and knew that home ownership would create wealth and opportunity for her growing family. Up until **the** late 1950s, Detroit was a haven for middle-class families living in neighborhoods where there was green space and community connectivity and opportunity. My grandmother, she had this amazing garden, and it was just abundant. There were all kinds of flowers and food. You know, she grew vegetables back there, and it was just a magical place for a little girl. MZ : I mean, what you ' re describing is a Detroit that, yes, has huge problems, segregation. But at **the** same time, a woman, she can have a job, she can own a house, she can raise her family, she can have fresh fruits and vegetables. AG : It was certainly a place where you could gain economic opportunity faster than you can now. My grandmother ' s Detroit is not **the** Detroit that I live in today. All of those sites where there was manufacturing and industrial sites that led to our economic boom, many stand vacant and abandoned. These industrial sites have led to dangerous contamination to our land and our water and our air. MZ : So, just lay it out for us. For people who maybe don ' t see, like, how is there a link between economic inequality and climate change? AG : You know, I ' m so surprised that it feels like separate problems. But I ' m also really glad that people are asking. Because if you ' re living around and working in a community that is low-income, where there are fewer jobs, **the** housing is deteriorated, this is a neighborhood that ' s also more likely to be closer to a factory that ' s emitting pollutants, likely to have water that ' s contaminated, probably has fewer parks and fewer intentional green spaces. **The** people who are living in those communities are at risk for climate impacts first. So we have to do something in those neighborhoods right away. MZ : So what can this healthier future look like? Anika showed me some of **the** work she ' s done in partnership with different neighborhoods in Detroit, to make them both more enjoyable and more resilient. First, she took me to **the** Circle Forest. It ' s an oasis right in **the** middle of **the** city. AG : This is six lots that were vacant, and we all worked together with this community to build a forest. MZ : Beautiful and a great example of how nonprofits can work with communities to transform them into sustainable, welcoming places. AG : We might have a good idea about gardens and tree planting, but this is where people live. And so you ask first. This is a new effort, of this idea of a resilient neighborhood or calling it that. So these are large tracts of vacancy that can be used to provide environmental **sustainability**, community resilience and beauty. Watch for **the** poison ivy. It ' s not pretend. MZ : Next we met up with Katrina Watkins, **the** founder and CEO of **the** Bailey Park Neighborhood Development Corporation, a group revitalizing this historic area by turning vacant lots into parks that can stand up to extreme weather. Katrina Watkins : My family has been in this neighborhood since 1946. I grew up hearing **the** stories from my dad, him telling me about how people thought it was a ghetto, but it wasn ' t. Those were families, and people worked and had businesses, and people looked out for each other. And it was those type of stories that made me say, wow, we just really need to bring that back. MZ : So then briefly fill us in about how things changed. KW : Things deteriorated in this neighborhood. **The** homes got torn down, left just a lot of vacant lots, a lot of overgrowth. It wasn ' t safe. I would see my little nieces and nephews. They would just be on their little skateboards, you know, not paying attention, **the** stop sign is covered. And me and my dad was like, we just have to do something. So we started out as Bailey Park Project, focusing on **the** park and green infrastructure and trees and flowers. And then over **the** course of **the** years, we started doing more community development. MZ : I wonder, like, how do you describe it to people when you explain what ' s happening, what you ' re trying to do for your neighborhood, **the** changes that you ' re making and how it relates to climate change? KW : One really great example, when people come to **the** park, like right now, this is like, literally a heat **island** because our trees haven ' t grown. So it is very hot. So sometimes it ' s so hot that **the** kids don ' t come out and play until **the** evening. That education of knowing **the** benefit of trees and cooling down an environment, how it makes **the** air cleaner. There ’ s a lot of people that suffer from asthma. So trees can make a difference. And if you go to places like Grosse Pointe, you will see this beautiful tree canopy down. Like their homes are probably 20 degrees cooler you know, than our homes, AG : Having a lot of green space actually contributes to **the** climate resiliency for this neighborhood. So all of this grass actually can manage stormwater better. **The** idea of leading new development in this neighborhood with green, what it says to me is that **the** residents are really envisioning **the** neighborhood that they want to live in. And that I just – I ’ m really excited about that. MZ : Turning great loss and upheaval into even greater opportunities. These Detroiters are creating a blueprint for an urban future that every city can learn from. [Chapter 5 : Agriculture] TED-Ed : About 10, 000 years ago, humans began to farm. This agricultural revolution was a turning point in our history that enabled people to settle, build and create. In short, **agriculture** enabled **the** existence of civilization. Today, approximately 40 percent of our **planet** is farmland. Spread all over **the** world, these agricultural lands are **the** pieces to a global puzzle we are all facing. In **the** future, how can we feed every member of a growing population a healthy diet? Meeting this goal will require nothing short of a second agricultural revolution. **The** first agricultural revolution was characterized by expansion and exploitation, feeding people at **the** expense of forests, wildlife and water, and destabilizing **the** climate in **the** process. That ’ s not an option **the** next time around. Agriculture depends on a stable climate with predictable seasons and weather patterns. This means we can ’ t keep expanding our agricultural lands, because doing so will undermine **the** environmental conditions that make **agriculture** possible in **the** first place. Instead, **the** next agricultural revolution will have to increase **the** output of our existing farmland for **the** long term while protecting biodiversity, conserving water, and reducing pollution and **greenhouse** gas **emissions**. So what will **the** future farms look like? In Bangladesh, Cambodia and Nepal, new approaches to rice production may dramatically decrease **greenhouse** gas **emissions** in **the** future. Rice is a staple food for three billion people and **the** main source of livelihood for millions of households. More than 90 percent of rice is grown in flooded paddies, which use a lot of water and release 11 percent of annual methane **emissions**, which accounts for one to two percent of total annual **greenhouse** gas **emissions** globally. By experimenting with new strains of rice, irrigating less and adopting less labor-intensive ways of planting seeds, farmers in these countries have already increased their incomes and crop yields while cutting down on **greenhouse** gas **emissions**. MZ : Related technologies are also taking root in **the** US. Father and daughter farmers, Jim and Jessica Whitaker, have come up with their own strategies to reduce methane that are working on their farm in Arkansas. But convincing everyone else to change is their next big challenge. So let ’ s go back to how you got to be such a big supplier of rice in this country. You tell a story about how you were 22 and you were like, I ’ m going to start my own farm, but – not that easy. Jim Whitaker : No, it ’ s not. Let me tell you something about renting a farm when you ’ re 22 years old. No one rents you a farm unless no one else wants to rent it. It was a big piece of land, it was cash rent, I mean, **the** stakes were set. We were doomed to fail. And we weren ’ t focused on environmental **sustainability** back then. We were focused on economic **sustainability**. How do we make higher yield? How do we use less fertilizer? How do we get to **the** next year and feed our family? It happens that economic and environmental **sustainability** go hand in hand along with social **sustainability**. As we used less fertilizer, used less water, our yield started going up. MZ : Can you spell out for me what **the** problem is with rice when it comes to **the** environment? JW : Soil is full of microbes. And what they ’ re doing in **the** soil on a basic level is they ’ re down there chewing away, eating up **the** biomass that ’ s left over from **the** crop before. And out of that comes methane, **the** same way it happens in cattle operations or whatever. MZ : But you decided to do it a different way. JW : We do it a little bit differently. Arkansas is a very water-rich state, but we have a dry season, right in **the** middle of summer is a dry season. So when we dry **the** soil, that anaerobic microbial that ’ s living in that **soup**, that mushy soil, either dies or goes dormant. So we break his life cycle. MZ : Oh! JW : And then it stops emitting methane. But it doesn ’ t hurt **the** rice. That actually helps rice, when you use less water. We have less runoff, less erosion, less nutrients leaving our field. Nothing leaves our field unless we want it to. Jessica Whitaker Allen : We started tracking this. We saw that our water use was down and we thought, OK, well, we need to add to that, let ’ s add more data points to that. Let ’ s add our fertilizer, let ’ s add our fuel usage. And so we really started recording all this stuff by hand in just an Excel spreadsheet. We didn ’ t know what we were doing. MZ : But you knew you were on to something. JWA : Yes, so we started recording everything. MZ : Do you remember if there was a moment when **the** two of you thought, like, what we ’ re doing, it shouldn ’ t be special to us. Why aren ’ t more people – JW : To collect **the** data, have **the** equipment and **the** cloud-based technology and **the** stuff in **the** field to record it, it ’ s just massively time-consuming, and consumers don ’ t pay for it. JWA : Well and people like us, why would you want to do it? I mean, take a walk through your **supermarket**. Everything ’ s sustainable, everything ’ s greenwashed. You can put whatever you want to on a package, there ’ s no rules, there ’ s no standards. So we ’ re trying to make that standard. We live and work in southeast Arkansas. Our town has 4, 000 residents and one stoplight. **The** nearest airport, Starbucks shopping mall, Whole Foods, is two hours away in any direction. Without places like this and farmers like us, you ’ d be hungry, naked and sober. I ’ m a farmer, but not in **the** way you might think. I don ’ t drive **the** tractors, and I don ’ t plant **the** rice. But I know how to take what my dad has learned and what he is doing and share that with others. We are going to work with farmers in southeast Arkansas to educate them about **the** benefits of growing sustainable rice. JW : Let me tell you why that ’ s important. There are 400 million acres of rice grown globally. And our methods, if used, can reduce **greenhouse** gas by 50 percent, reduce water use by 50 percent, increase yields to feed a hungry world. MZ : We hear a lot about how weather has gotten more extreme, flooding, heat. What ’ s it been like in Arkansas? JW : So in 2016, we had a 500-year flood. We were... We get a picture of our place flooded. 14 **inches** of rain. That was a 500-year flood. Hurt a lot of people. And then we had a 1, 000-year flood. And we lost so many acres. And you know, when houses get flooded in rural communities, they don ’ t come back. I mean, when you have whole neighborhoods get flooded, they don ’ t rebuild them, they don ’ t come back. JWA : I feel like almost every day I ’ m getting a notification from **the** weather Channel of an excessive heat warning or thunderstorm warning or hail threat. You know, you don ’ t really know what you ’ re going to expect. MZ : When you think about what it will be like, I don ’ t know, five years, 10 years, for someone to go into **the supermarket** and think, oh, right, I need to get rice off my grocery list. What ideally would you like to see on **the** shelf there? JW : So I think it ’ s going to change, because I like to study what companies want. I mean, if you figure out what they want, then you know what to give them. And when you read their **sustainability** goals, they ’ re all saying they ’ re going to be net zero. **The** only way they can be net zero is they bring **the** farmer into **the** loop. So if we ’ re able to do this, in five years, everybody ' s doing these practices, **the** US rice industry will be **the** most sustainable industry in **the** world. MZ : Can you do it? JW : Oh, yeah. It ' s about to happen. MZ : You ' ve just heard a lot of ideas, a lot of stories and a lot of big plans. So we want to wrap things up by asking our experts : What if we actually achieved our climate goals? What would that future even look like? They, of course, had a lot of thoughts. We ' ll let you pick your favorites. Thanks for watching. NT : If we are successful in addressing **the** climate crisis, we massively reduce **the** vulnerability and improve **the** economic and **the** creative prospects of billions of people around **the** world who currently are basically excluded from all of **the** opportunities which we have, which come from having abundant material and energy all around us. That ' s **the** real prize. FD : A world where we have had that exponential would be a world in which we have clean, cheap energy for all. KB : At **the** end of this decade, **the** International Energy Agency believes that we will have enough capacity to be producing 10 billion, 10 billion batteries a year. It ' s enough for every person on **the planet** to have their own battery. AC : I ' m constantly surrounded by young people just like me who are really, really, really passionate about **the** world that we live in. For obvious reasons. It ' s **the** world that we will inherit. EZ : Women are very important for **the** energy transition. So if we ' re talking about a world where we have, you know, where we tackle climate change, all I want to see is, you know, women in power. I want to see more solutions being driven by young people. And I want our politicians to be truly and meaningfully engaged in making change. NG : We always tend to underestimate **the** pace of ultimate adoption and change. Today, there are more than 150 pathways, scientific pathways that we could take to limit global warming. Al Gore : If we get to true net zero, astonishingly, global temperatures will stop going up with a lag time of as little as three to five years. Colin Averill : We ' ve been able to accelerate tree growth and **carbon** capture and tree stems by 30 to 70 percent. Dan Jørgensen : Offshore wind has **the** potential to cover **the** current electricity demand of **the** entire world not once, not twice, 18 times. NG : We need to decarbonize **the** old economy that we have and invest to create **the** new economy that we need Paul Hawken : Regenerative farming is fairly simple in concept. It means creating **the** conditions for more life. Josephine Phillips : We need to buy less stuff, and we need to look after what we buy. Valuing **the** things that we own is a climate solution. Vishaan Chakrabarti : We can all live in this transit-rich, carbon-negative, affordable way and leave **the** vast majority of **the planet** for nature, for **agriculture**, for clean oceans. We can do this. Susan Ruffo : **The** ocean is a powerful source of solutions that we ' ve overlooked for far too long. Al Roker : You have to be engaged. Let your elected officials know that this is important to you. You have to vote, you have to go out there and support politicians who are going to support our **planet**. Peggy Shepard : We can create a legacy of environmental quality and climate resilience for all. Fehinti Balogun : You are not powerless that **the**"oh, there ' s nothing we can do, just keep going"is symptom, not cure. But you are needed. Avinash Persaud : **The** power of what we can do when it matters to us is unlimited. CF : It ' s not that we ' re simply going to avoid **the** worst impacts of climate change. It ' s that we ' re going to improve **the** quality of life for humans, both in **the** urban and in **the** rural sector, and for non-humans.

Text Sample 64: POS Dim 1 – 2023 – Score 19.00 – 2023/t003440.txt

The most important question these days is how can we speed up **the** solutions to **the** climate crisis? I ' m convinced we are going to solve **the** climate crisis.

We ’ ve got this. But **the** question remains, will we solve it in time? Others have said we ’ re kind of in a race. I ’ ll give you **the the** shortest definition of **the** problem. If I was going to give a one-slide **slide** show, it would be this **slide**. That ’ s **the** troposphere, **the** lowest part of **the** atmosphere. And you already know why it ’ s **blue**. That ’ s **the** oxygen that refracts **the blue** light. And if you could drive a car at **highway** speed straight up in **the** air, you ’ d get to **the** top of that **blue** line in about five to seven minutes. You could walk it in an hour. And all of **the greenhouse** gas pollution is below you. That ’ s what we ’ re using as an open sewer. That ’ s **the** problem. And it ’ s causing a lot of second and third order consequences. And we saw some of them in **the** northern-tier cities, including Detroit. This one ’ s from New York City, all **the** fires in Canada. And we have gotten used to **the** fact that **the** world suffers deep **droughts** and huge rain bombs and downpours and floods simultaneously. **The** really ingenious new gravity-measuring satellite has given us, for **the** first time, **the** opportunity to see how this plays out worldwide. We get these huge surpluses of water, **the** rain bombs and **the drought** simultaneously. And as you can see, **the** amplitude is increasing. And at both ends of our **planet** right now, we ’ re seeing signs of distress. Of course, you know, I ’ ve often said, every night on **the** TV news is like a nature hike through **the** Book of Revelation. And just today, big flooding in Montpelier, Vermont, in southern Japan, in India. And I haven ’ t done a complete scan, but every single day, it ’ s like that. But in Antarctica, some scientists who are normally pretty levelheaded are getting a little bit freaked out, I would say, is a fair definition, about **the** lowest level ever, at this point in **the** year, of **sea** ice. And at **the** other pole, in **the** North Atlantic, we ’ re seeing literally off-the-charts temperatures. So obviously, **the** crisis has to be addressed. And **the** good news is, as others have often said, we are seeing tremendous progress. And it starts with **the** Inflation Reduction Act here in **the** United States. President Biden, **the** Congress, they have passed **the** best, biggest climate legislation in all of history, and it ’ s said to be 369 billion. But **the** heavy lifting is done by tax credits, and most of them are open-ended. And **the** early applications already show it ’ s going to be well over a trillion dollars, maybe 1. 2 trillion. So this is really good news. And one month after that passed, Australia changed its government and started changing its laws. And it ’ s now a pro-climate nation. A month after that, Brazil did **the** same thing, new president, new policies protecting **the** Amazon. And throughout, **the** European Union has resisted **the** efforts of Russia to blackmail it into supporting its sadistic and cruel invasion of Ukraine. And there are other signs of success as well. China ’ s reached its renewables target five years ahead of time. It ’ s still building a lot of **coal**, but only 50 percent capacity utilization. So there is lots of good news. But still, in spite of this progress, **the emissions** are still going up. And **the** crisis is still getting worse faster than we are deploying **the** solutions. So maybe it ’ s time to look at **the** obstacles that are standing in our way. I ’ m going to focus on two of them this afternoon. First of all, **the** unrelenting opposition from **the fossil** fuel industry. A lot of people think they ’ re on side and trying to help. But let me tell you, and **the** activists will all tell you this, every piece of legislation, whether it ’ s at **the** municipal level, **the** regional or provincial level, **the** national level or **the** international level, they ’ re in there with their lobbyists and with their fixers and with their revolving-door colleagues doing everything they can to slow down progress. So speeding up progress means doing something about this. They have used fraud on a massive scale. They ’ ve used falsehoods on an industrial scale. And they ’ ve used their legacy political and economic networks, lavishly funded, to capture **the** policy-making process in too many countries around **the** world. And **the** [UN] Secretary General said **the fossil** fuel industry is **the** polluted heart of **the** climate crisis. Now, that ’ s not to say that **the** men and women who ’ ve worked in **fossil** fuel for **the** last century and a half are not due our gratitude. They are, and they didn ’ t cause this. But for decades now, **the** companies have had **the** evidence. They know **the** truth, and they consciously decided to lie to publics all around **the** world in order to calm down **the** political momentum for doing something about it so they can make more money. It ’ s simple as that. And now they have brazenly seized control of **the** COP process, especially this year ’ s COP in Dubai in **the** United Arab Emirates. And concern has been building about this for quite some time. I remember when there were so many **fossil** fuel delegates in Madrid, but by **the** time we got to Glasgow a year and a half ago, **the** delegates from **the fossil** fuel companies made up a larger group than **the** largest national delegation. And why? Why? Because they ’ re helping? They ’ re not helping, they ’ re trying to stop progress. And last year in Egypt, they had more delegates than **the** combined delegations of **the** ten most affected countries by climate. And now this year ’ s host, which is a petro state, has appointed **the** president of COP28, in spite of **the** fact that he has a blatant conflict of interest, he ’ s **the** CEO of **the** Abu Dhabi National Oil Company owned by Abu Dhabi. Their **emissions** are larger than those of ExxonMobil, and they have no credible plan whatsoever to reduce them. So this is **the** person in charge of **the** COP. He ’ s a nice guy. He ’ s a smart guy. But a conflict of interest is a conflict of interest. And a matter of fact, they have a plan now to have a new increase in their **emissions**. Their plan is to increase **the** production of both **oil** and gas by as much as 50 percent by 2030. Which is **the** same time frame when **the** world is trying to reduce **emissions**, by 50 percent by 2030. And **the** same person has been put in charge of both of those efforts. Direct conflict of interest. I think it ’ s time to say, wait a minute, do you take us for fools? Do you think you can just completely remove **the** disguise and we won ’ t notice? **The fossil** fuel industry has captured this process and is slowing it down. And we need to do something about it. Now, they plan to increase their **emissions** – Getting all hot and bothered here. **The** head of **the** International Energy Agency was talking about several **oil** companies who have publicly pledged to increase their **oil** and gas and simultaneously pledged to comply with **the** Paris Agreement. Do you take us for fools? You cannot do both of those things simultaneously. And if you think we believe you when you say that, we beg to differ. Last year at **the** COP, **the fossil** fuel petro states vetoed any mention of phasing down **fossil** fuels. They say, oh, that ’ s not **the** problem. One of **the** secretary of state equivalent in Saudi Arabia said, we don ’ t see this as a discussion about **fossil** fuels. Let me tell you, **the** climate crisis is a **fossil** fuel crisis. **The** solutions are going to come from a discussion and collaboration about phasing out **fossil** fuels. And there ’ s only so much longer they can hold this up and tie us down and keep us from doing **the** right thing. Now **the** director general of this year ’ s COP, working under Sultan Al Jaber, he worked for ADNOC also, he was seconded for several years to ExxonMobil. He said, no, that ’ s all changed. This year, they ’ ve changed. **The fossil** fuel companies are really engaging us. They have changed. Yes, they have changed. They ’ ve changed for **the** worse this year. You don ’ t believe me? Look at this, BP. They said,”We ’ re going to net zero, there ’ s no turning back.”A few months ago this year, they turned back and they decided to roll back their investments. ExxonMobil. Oh, my God. This is minor compared to their industrial-scale lying, you know. But they ’ ve had all **the** TV ads on, particularly in **the** US, about revolutionizing biofuels.”You wouldn ’ t believe what algae can do.”Well, for 14 years they ran this program. And they spent 50 percent as much on **the** ads about **the** program as they did on actually trying to come up with new biofuels. And then a couple of months ago, they said,”Oh, we ’ ve changed our mind. We ’ re just going to cancel that program.”**The** industry as a whole has not been acting in good faith. Shell reversed its commitment to increase their investments in **fossil** fuels. They announced just a couple of months ago they ’ re going to plow that money into expanding **oil** and gas production instead. So did they miss **the** memo? Well, no, they say they ’ ve discovered a way out. They ’ re telling us now that **the** problem is not **fossil** fuels. It ’ s **the emissions** from **fossil** fuels. And so all we have to do is to capture **the emissions** from **the fossil** fuels. Now, I ’ m all for research and development into trying to capture **emissions** or sucking it out of **the** air if they want to do that. But let ’ s don ’ t pretend it ’ s for real. Maybe someday it will be. Maybe someday it will be. This is **the** head of ExxonMobil saying,”Not **oil** and gas, it ’ s **the emissions**.”And Abu Dhabi, they have said they ’ ve already decreased their **emissions** from producing **oil** and gas, 99. 2 percent. **The** point two makes it sound even more authoritative. Is that feasible? Have they really done that? Is their **carbon** capture technology that ’ s that good? Well, Climate TRACE can measure. Now, let ’ s just look. There ’ s their CCS capability. Oh, but they ’ ve got a great improvement in **the** work. Seven years from now, they ’ re going to have a 500 percent increase in their capacity. But here are **the emissions** from **oil** and gas production alone. And here are **the emissions** of **greenhouse** gases from Abu Dhabi, from **the** UAE. Now, **the** pathetic little sliver there compared to **the** reality, again, do they think we don ’ t see what they ’ re doing, don ’ t understand what they ’ re doing? A lot of people maybe just want to look away from it. Pretty ironic here in this country. You know, **the** coal-burning utilities have been given a new mandate this year by **the** EPA. You ’ ve got to clean up your act. Now, you can keep on burning **coal** and you can keep on producing electricity with it if you use **carbon** capture and sequestration. Well, they scream bloody murder. Another story about it this morning, but they say, that technology? That ’ s not feasible. It ’ s not technically feasible, not economically feasible. So they need to get their story straight, in my opinion. So **the** technology of CCS has been around for a long time, 50 years, and we get used to technologies automatically getting cheaper and better, you know, with **the computer** chips and **the** cell phones and all that. Well, University of Oxford did a whole study, and some come down in cost fast, some slow. A few are in a category called non-improving technologies. And that ’ s where **carbon** capture and storage is. 50 years, not any price decline. Well, could we get a breakthrough in spite of that? Maybe. We ’ re throwing a lot of money at it. We had to do that to get **the** compromise that passed **the** IRA, and we ought to continue **the** research. But again, let ’ s don ’ t pretend that it ’ s here. It ’ s not here now. And to use it as an excuse, and now they ’ ve got another one. Now it ’ s called direct air capture. These are giant vacuum cleaners that use an awful lot of energy. It ’ s technically feasible, but it ’ s extremely expensive. And it also uses so much energy. But it is tremendously useful. You know what it ’ s used for? **The** CEO of one of **the** largest **oil** companies in **the** US had told us what it ’ s useful for. It ’ s useful to give them an excuse for not ever stopping **oil**. A few months later, she said,”This gives us, this new technology, a license to continue operating.”Some people call that a moral hazard. For them, **the** moral hazard is not a bug, it ’ s a feature, as **the** old saying goes. And so let ’ s just look, this is state-of-the-art. It looks pretty impressive, doesn ’ t it? This is **the** backside of one of these machines. I had **the** same thought. Now this, they ’ re improving this. They ’ re improving this. And **the** new model, seven years from now, each of these machines is going to be able to capture 27 seconds ’ worth of annual **emissions**. Woohoo. That gives them a license to continue producing more and more **oil** and gas. Or so they claim. Well, they also say, **the** experts, that in order to be economically feasible, it has to come in at 100 dollars a ton or less. **The** leading company says that 27 years from now, it may come down below three times that amount. So again, if this sounds kind of incredible, it is not credible. But they ' re using it in order to gaslight us, literally. But I didn ' t mean it as a pun, but that ' s what they ' re using it to do. And we can ' t fall for it. **The** biggest obstacle to direct air capture is probably physics. And what I mean by that, they ' ve explained this to me. CO2 makes up 0. 035 percent of **the** air so we ' re going to vacuum **the** other 99. 96 percent to get that little bit out. Come on, really? Come on. And it doesn ' t even pretend to catch **the** methane or **the** soot or **the** mercury or **the** particulate pollution that kills 9 million people a year around **the** world. And that ' s why **the** climate justice advocates are so down on this kind of nonsense. They say, here ' s one of them says, look, we need to fight climate change, but don ' t do it by pretending to do it and continue dumping all this pollution on **the** front line communities that are downwind from **the** smokestacks. And that ' s why they have not ever really liked **the carbon** capture. And what about all **the** energy that ' s needed? Is it going to come from more **fossil** fuels? Well, in case of Occidental, yes. But could it come from more solar and wind? Well, if so, why not use **the** solar and wind to replace **the fossil** fuel-burning plants that are putting all that stuff up there in **the** first place? Doesn ' t that make sense? Am I missing something? Anyway, this is **the** environmental justice advocate who is making **the** point that you can ' t do it without protecting those who are victimized downwind from **the** smokestacks, downstream from **the** pollution. Now, some of **the** companies, of course, you say we ' re going to offset our **emissions**. And again, offsets can play a role, a small role, five to 10 percent according to **the** Science Based Targets initiative, **the** most authoritative source on all of this. But, you know, Chevron just had its offsets analyzed. It turned out 93 percent of them were worthless and junk. And that ' s too often **the** case. It ' s not a get-out-of-jail-free card. And their offsets were only aimed at 10 percent of their **emissions** in **the** first place. Eighty percent of **the fossil** fuel companies just completely ignore **the** Scope Three **emissions**, which is **the** main part of their pollution problem. So they say that they have increased by 400 percent **the** share of their spending on energy that goes to green technologies and **carbon** capture. And yes, they have, they ' ve increased it all **the** way up to four percent. And it ' s not even as good as that, because if you look at **the** windfall profits they ' ve been getting, what have they done with that? Well, they gave it back to **the** shareholders through dividends and stock buybacks. So **the** amount of money **the fossil** fuel industry is investing in renewables and **carbon** capture is one percent. Does that mean they are sincere actors, working in good faith? I don ' t think so. A lot of people still think so. They think they ' re on our side. I don ' t think it ' s in their nature to be on our side. I think they ' re driven by incentives that push them in **the** opposite direction. But in any case, they produce no solutions whatsoever that are scalable or remotely feasible. And again, they have actively fought against **the** solutions that others have been trying to bring. I ' ve seen it personally. Many of you have seen it personally. And it ' s happening all over **the** world. No progress. They ' ve even gone **backwards**. Now, if a company says, look, we shouldn ' t be excluded from **the** COP process, we know a lot about energy. Here ' s a simple test. Do they actually have, does **the** company have real net-zero commitments, a genuine phase-down plan, yes or no? Are they committed to full disclosure? Yes or no? Are they going to spend windfall profits on **the** transition? Yes or no? Are they committed to transparency? Will they end their anti-climate lobbying? Will they end their greenwashing? Will they be in favor of reforming **the** COP process so that **the** petro states don ' t have an absolute veto on anything **the** world wants to discuss or act on? We ' ve got to change that. In order to move faster, we have got to empower **the** global community in a way that frees them from **the** hammerlock that **the fossil** fuel companies have on them today. And any company that doesn ’ t pass this test, I ’ m telling you, ought to be prohibited from taking part in **the** COP process. What about **the** company that ’ s led by **the** president of this year ’ s COP, **the** Abu Dhabi National Oil Company? Well, it ’ s rated one of **the** least responsible of all **the oil** and gas companies. It no longer even releases information on any of its **emissions**, has no transition plan, no short or mid-term target scores, a pathetic three on 100-scale on their transition plans. And **the** nation ’ s plan is rated highly insufficient. They just put out a new one, it ’ s still insufficient. So that ’ s **the** first obstacle. Second one ’ s a lot clearer. This one has to be removed in order to get to **the** faster solutions. But in order to remove it, we ’ ve got to do something about **the** second obstacle, and that is **the** financial system, **the** global allocation of capital and **the** subsidies for **fossil** fuels. Governments last year around **the** world subsidized with taxpayer money more than a trillion dollars, subsidizing **fossil** fuels. That ’ s five times larger than **the** amount in 2020. Are we going in **the** right direction or **the** wrong direction? Well, there are a lot of good things I ’ ll show you, but this has to stop. And **the** 60 largest global banks have put 5. 5 trillion extra dollars into **the fossil** fuel companies since **the** Paris agreement. And preposterously, 49 of them have also signed net-zero pledges. So we... Global allocation of capital. **The** borrowing of private capital is ruled out for many developing countries. Nigeria has to pay an interest rate seven times higher than Europe or **the** US. Political risk, corruption risk, currency risk, off-take risk, it ’ s rule of law risk. That ’ s why **the** World Bank and **the** other multilateral development banks are supposed to take those top layers of risk off. And that ’ s why we need reform in **the** World Bank. Thank goodness so many fought to get a new head of **the** World Bank, and I ’ m very excited about his progress. Now, when these obstacles are removed, we can really accelerate progress. They ’ ll still be problems, of course, but we have everything we need and proven deployment models to reduce **emissions** 50 percent in **the** next seven years. But we will still need better grids, more resilient grids, more solar and wind, much more regenerative **agriculture**. That ’ s a way to really pull **carbon** out of **the** air. More electric vehicles and charging stations and more energy storage. I ’ ll show you that just before we quit here. And more green hydrogen, so more electrolyzers to produce it. And these are all surmountable obstacles. Already, renewables account for 90 percent of all **the** new electricity generation being installed each year worldwide. Ninety-three percent solar and wind in India last year. And of course, **the** solar ’ s taking off and so is wind and so are **the** electric vehicles and battery storage. It has grown so dramatically. You ’ ve got to rescale **the** graph to look at **the** trillion-dollar industry emerging. Now, just seven years ago, there was one gigafactory. I learned from one of you at a session a couple of hours ago and had my staff research it, there are now 195 gigafactories and another 300 in **the** pipeline. So will we succeed? Some people fight with a vulnerability to despair. You know, **the** old cliche,”denial ain ’ t just a river in Egypt.”Despair ain ’ t just a tire in **the** trunk. There are people that are vulnerable. But let me close with what I regard as amazingly good news. What if we could stop **the** increase in temperatures? Well, if you look at **the** temperature increases, if we get to true net-zero, astonishingly, global temperatures will stop going up with a lag time of as little as three to five years. They used to think that positive feedback loops would keep that process going. No, it will not. **The** temperatures will stop going up. **The** ice will continue melting and some other things will continue, but we can stop **the** increase of temperatures. Even better, if we stay at true net zero, in as little as 30 years, half of all **the** human-caused

CO_2 will come out of **the** atmosphere into **the** upper ocean and **the** trees and vegetation. So young people are demanding that we do **the** right thing. Do not be vulnerable to despair. We are going to do this. And if you doubt that we, as human beings have **the** will to act, please always remember that **the** will to act is itself a **renewable** resource. Thank you very much. Thank you.

Text Sample 65: POS Dim 1 – 2023 – Score 18.00 – 2023/t003429.txt

Twenty-five hundred years ago, Heraclitus said,”**The** world bubbles forth.”Here ’ s bubbling. What is actual now enables what is next possible, **the** adjacent possible. **The** biosphere has been bubbling forth for four billion years, creating new possibilities in **the universe** in that bubbling. It ’ s critical that physics cannot talk at all about bubbling new bubbles. Life on Earth started four billion years ago. For **the** first 2. 5 billion years were single-celled organisms co-evolving. Even by existing, organisms create new possibilities. If I ’ m a bacterium and you ’ re a bacterium and I have a flat **surface**, I ’ m a possible **highway** you can walk over to get food. That ’ s bubbling forth. So 2. 5 billion years after life started, six times multicell organisms emerged. A billion years after that, 3. 5 billion years after life started, is **the** extraordinary Cambrian **explosion**. 540 million years ago, for 50 million years, there was just this burst of creativity, making all of **the** phyla that exist now. And now we have this guy. I ’ m scared of having him in bed, I really am. So I don ’ t know what to call him. It ’ s a vote afterwards. So here ’ s a gentler guy and I ’ m in love with him. Look at his eyes looking at you, you know.”You ’ re doing what?”In order to talk about how **the** biosphere has been creative, I have to talk about **the** functions of parts of an organ. You all know your heart keeps you alive, but in particular, your heart pumps **blood**. It ’ s not by making heart sounds. So this means something fundamental. **The** function of a part is that subset of its causal properties that sustain **the** whole. Your heart pumping **blood** and you. But doing that is jury-rigging, finding subsets of causal features of use. Some years ago, my daughter dropped her purse in a shallow well. I took a wire coat hanger, **slid** it over a mop handle, and I **fished** out her purse. Anything can be used for more than one thing. For example, a screwdriver can be used to screw in a screw, but it ’ s great for scraping putty off **the** wall. An engine block can have eight holes drilled and you make an engine out of it. It ’ s actually a superb paperweight, and much research done only by scientists demonstrates that **the** corners are sharp and can crack open a coconut. I was very proud of that. So. But if you ’ re using an engine block as a paperweight, can you deduce, oh, I can use this thing to crack open coconuts. You can ’ t. It might be a banana peel. But that means also something fundamental. Finding new uses for things, jury-rigging is not a computation. It ’ s not a deduction. Jury-rigging is not a computation at all. **Computers** are. **The** biosphere is forever innovating in this kind of way that ’ s not a deduction. For example, there are things called Darwinian preadaptations. Dinosaurs had scales that were **evolved** for thermoregulation that were jury-rigged into flight feathers. Your eyes have clear crystallins that were perfectly normal-colored enzymes. And this, too, means something fundamental. We cannot deduce what is in **the** adjacent possible that **the evolving** biosphere will create, then become. We do not even know what can happen. In fact, we can use no mathematics based on set theory, which is all of mathematics, to deduce what **the** biosphere is going to become, or **the** economy. This means we are at a **fantastic** third-phase transition in science, we ’ re beyond Newton, we ’ re beyond quantum mechanics. We ’ re beyond **the** exquisite view that Copernicus had 480 years ago, that **the** world is a clockwork. **The** biosphere is not a clockwork. **The** paper that just published, it came out three days ago, and I ’ m pretty proud of it, it ’ s actually worth reading. We can ’ t deduce what is in **the** biosphere and will become, but we can make a mathematical theory of **the** statistics of **the** process. I call it **the** Theory of **the** Adjacent Possible or TAP. It ’ s really based on one idea. Things can be combined to make new things. So there is **the** equation. Mt is **the** number of things in **the** economy at some time now, say 10 things. So how many things will be in **the** economy in **the** next period? Well, if **the** 10 things work, we ’ ll keep them, but we could actually try to jury-rig something new out of any single thing among **the** 10 or out of pair of things among **the** 10. **The** printing press is a recombination between movable type and a wine press or any three things or four things. Watch what happens to **the** number of things, pairs of things, when **the** number of things goes up. If there ’ s 10 things, there ’ s 45 pairs of things that we might fiddle with. If there ’ s 100 things, there ’ s 4, 500 things we might jury-rig with. And if there ’ s 1, 000 things, there ’ s a half a million things we might fiddle with. Therefore, this process, **the** TAP process, has **the** property that for a long time **the** number of things increases very, very, very slowly, then something **stunning** happens. There ’ s a hockey-stick **explosion** and **the** number of things reaches infinity in a finite time. There ’ s a singularity and a brief burst **explosion** for 40 million years of **the** Cambrian **explosion**. **The** same pattern is showing up in our own **evolution**. Australopithecus flourished – I ’ d never know what flourishing means, I guess they had a good lunch – in Africa, 3. 3 million years ago. A million and a half years later, Homo erectus had about 20 **crude** stone tools. Watch how slowly things change. A mere 30, 000 years ago, Cro-Magnon in **the** south of France had about 100 or 150 of these exquisite pressure flake tools. A mere 28, 000 years after that is Mesopotamia with needles and chariots. There ’ s something interesting about **the** fact – of course, we invented writing, which is a wonderful story – once you have a clay tablet, it calls forth **the** tool that ’ s its complement. You need a stylus. So tools make niches for new tools and for new jobs. Scribes earned a living for quite a while. This is us in **the** late Middle Ages to now, and this is us now. We have billions of tools ranging from knitting needles to **the** space station. There ’ s always an adjacent possible. Once you ’ ve made a bow, a crossbow is in **the** adjacent possible. **The** pattern that we saw in **the** Cambrian, of a long period nothing happening and then a burst, is here right now. This is **the** size of human **habitats** in **the** last 300, 000 years. **The horizontal** line is time before present, and **the** vertical line as **the** logarithm of sizes of settlements. Nothing happened for 290, 000 years then it burst upward. Watch this, **the** same thing is true of GDP. GDP was flat since **the** time of Christ. In fact, for **the** last 100, 000 years. GDP burst up in **the** last two centuries. It ’ s going up vertically now. It ’ s wonderful. We ’ re lifting millions from poverty. And **the** theory explains it. It ’ s **the** same delay and burst, and there it is. We start making more complex things, selling them, and we make a profit. There ’ s another feature of all of this that I ’ ve lived in my lifetime, I ’ m 83. I was born in 1939. Just like **the** Cambrian **explosion**, watch **the** number of things that have come into existence in my lifetime. We didn ’ t have any helicopters, we didn ’ t have television. I listened to”**The** Lone Ranger,”which maybe you know, on radio, six o ’ clock at night. We didn ’ t have... **computers**, we didn ’ t have plastics, which are now all over **the planet**. We didn ’ t have microwaves, we didn ’ t have lasers. Look at **the** number of things that came into existence in one lifetime of 83 years. Not much happened 50, 000 years ago in 83 years. So what ’ s going on? **The** TAP process has **the** following property. Every time you make something new, **the** waiting time for **the** next new thing is cut in half. A thousand years, 500 years, 250, a year, six months, three months, we ’ re going to get a great acceleration. We are. So this is TAP. And watch that thing explode upward. That ’ s it. That **explosion** is **the** Cambrian, and it ’ s where we are now. And **the** signatures of **the** Anthropocene are exploding upward, now all over **the** place. This ends 25 years ago. We need to find a better adjacent possible. We ’ re rampaging over **the planet**, and **the** hope is in soils. So I ’ m going to talk briefly about compost. There are very fine compost, this is now terribly important. A very good compost is **the** Johnson-Su compost. Less land use, fewer degradations of... of our forests, fewer extinction events, fewer pandemics. Well, let ’ s get rid of fertilizer. And **the** most important point I ’ m going to take is – This is really bad news and good news. Last year we emitted 40. 5 billion tons of **carbon** dioxide. We have to find a way of solving it, and we can. We can take this Johnson-Su compost and get it out across **the** landscape. But more fundamentally, we can find good fungal bacterial communities and we can coat seeds with it and sell **the** seeds around **the planet**, or we can fill biochar and we can use biochar to get her across **the planet**. Most usefully, we can take a really good fungal bacterial community, put it into biochar or some other matrix, mix it into fertilizer, and get that fertilizer out across **the planet**. I ’ m going to just end with **the** following. Fungal bacterial communities are precisely something that can create novel adjacent possibles, that can bubble forth with solutions to soil problems. **The** good news is we can do it now. **The planet** ’ s on fire. We are of nature, ladies and gentlemen, we ’ re not above nature. Thank you.

Text Sample 66: POS Dim 1 – 2018 – Score 24.00 – 2018/t000688.txt

I ’ d like to introduce you to a tiny microorganism that you ’ ve probably never heard of : its name is Prochlorococcus, and it ’ s really an amazing little being. For one thing, its ancestors changed **the** earth in ways that made it possible for us to **evolve**, and hidden in its genetic code is a blueprint that may inspire ways to reduce our dependency on **fossil** fuel. But **the** most amazing thing is that there are three billion billion billion of these tiny cells on **the planet**, and we didn ’ t know they existed until 35 years ago. So to tell you their story, I need to first take you way back, four billion years ago, when **the** earth might have looked something like this. There was no life on **the planet**, there was no oxygen in **the** atmosphere. So what happened to change that **planet** into **the** one we **enjoy** today, teeming with life, teeming with plants and animals? Well, in a word, photosynthesis. About two and a half billion years ago, some of these ancient ancestors of Prochlorococcus **evolved** so that they could use solar energy and absorb it and split water into its component parts of oxygen and hydrogen. And they used **the** chemical energy produced to draw CO2, **carbon** dioxide, out of **the** atmosphere and use it to build sugars and proteins and amino acids, all **the** things that life is made of. And as they **evolved** and grew more and more over millions and millions of years, that oxygen accumulated in **the** atmosphere. Until about 500 million years ago, there was enough in **the** atmosphere that larger organisms could **evolve**. There was an **explosion** of life- forms, and, ultimately, we appeared on **the** scene. While that was going on, some of those ancient photosynthesizers died and were compressed and buried, and became **fossil** fuel with sunlight buried in their **carbon** bonds. They ’ re basically buried sunlight in **the** form of **coal** and **oil**. Today ’ s photosynthesizers, their engines are descended from those ancient microbes, and they feed basically all of life on earth. Your heart is beating using **the** solar energy that some plant processed for you, and **the** stuff your body is made out of is made out of CO2 that some plant processed for you. Basically, we ’ re all made out of sunlight and **carbon** dioxide. Fundamentally, we ' re just hot air. So as terrestrial beings, we ' re very familiar with **the** plants on land : **the** trees, **the** grasses, **the** pastures, **the** crops. But **the** oceans are filled with billions of tons of animals. Do you ever wonder what ' s feeding them? Well there ' s an invisible pasture of microscopic photosynthesizers called phytoplankton that fill **the** upper 200 meters of **the** ocean, and they feed **the** entire open ocean ecosystem. Some of **the** animals live among them and eat them, and others swim up to feed on them at night, while others sit in **the** deep and wait for them to die and settle down and then they chow down on them. So these tiny phytoplankton, collectively, weigh less than one percent of all **the** plants on land, but annually they photosynthesize as much as all of **the** plants on land, including **the** Amazon rainforest that we consider **the** lungs of **the** **planet**. Every year, they fix 50 billion tons of **carbon** in **the** form of **carbon** dioxide into their bodies that feeds **the** ocean ecosystem. How does this tiny amount of biomass produce as much as all **the** plants on land? Well, they don ' t have trunks and stems and flowers and fruits and all that to maintain. All they have to do is grow and divide and grow and divide. They ' re really lean little photosynthesis machines. They really crank. So there are thousands of different species of phytoplankton, come in all different shapes and sizes, all roughly less than **the width** of a human hair. Here, I ' m showing you some of **the** more beautiful ones, **the textbook** versions. I call them **the** charismatic species of phytoplankton. And here is Prochlorococcus. I know, it just looks like a bunch of schmutz on a microscope **slide**. But they ' re in there, and I ' m going to reveal them to you in a minute. But first I want to tell you how they were discovered. About 38 years ago, we were playing around with a technology in my lab called flow cytometry that was developed for biomedical research for studying cells like cancer cells, but it turns out we were using it for this off-label purpose which was to study phytoplankton, and it was beautifully suited to do that. And here ' s how it works : so you inject a sample in this tiny little capillary tube, and **the** cells go single file by a laser, and as they do, they scatter light according to their size and they emit light according to whatever pigments they might have, whether they ' re natural or whether you stain them. And **the** chlorophyl of phytoplankton, which is green, emits red light when you shine **blue** light on it. And so we used this instrument for several years to study our phytoplankton cultures, species like those charismatic ones that I showed you, just studying their basic cell **biology**. But all that time, we thought, well wouldn ' t it be really cool if we could take an instrument like this out on a ship and just squirt seawater through it and see what all those diversity of phytoplankton would look like. So I managed to get my hands on what we call a big rig in flow cytometry, a large, powerful laser with a money-back guarantee from **the** company that if it didn ' t work on a ship, they would take it back. And so a young scientist that I was working with at **the** time, Rob Olson, was able to take this thing apart, put it on a ship, put it back together and take it off to **sea**. And it worked like a charm. We didn ' t think it would, because we thought **the** ship ' s vibrations would get in **the** way of **the** focusing of **the** laser, but it really worked like a charm. And so we mapped **the** phytoplankton distributions across **the** ocean. For **the** first time, you could look at them one cell at a time in real time and see what was going on — that was very exciting. But one day, Rob noticed some faint signals coming out of **the** instrument that we dismissed as electronic noise for probably a year before we realized that it wasn ' t really behaving like noise. It had some regular patterns to it. To make a long story short, it was tiny, tiny little cells, less than one-one hundredth **the** **width** of a human hair that contain chlorophyl. That was Prochlorococcus. So remember this **slide** that I showed you? If you shine **blue** light on that same sample, this is what you see : two tiny little red light-emitting cells. Those are Prochlorococcus. They are **the** smallest and most abundant photosynthetic cell on **the planet**. At first, we didn ’ t know what they were, so we called **the**”little greens.”It was a very affectionate name for them. Ultimately, we knew enough about them to give them **the** name Prochlorococcus, which means”primitive green berry.”And it was about that time that I became so smitten by these little cells that I redirected my entire lab to study them and nothing else, and my loyalty to them has really paid off. They ’ ve given me a tremendous amount, including bringing me here. So over **the** years, we and others, many others, have studied Prochlorococcus across **the** oceans and found that they ’ re very abundant over wide, wide ranges in **the** open ocean ecosystem. They ’ re particularly abundant in what are called **the** open ocean gyres. These are sometimes referred to as **the** deserts of **the** oceans, but they ’ re not deserts at all. Their deep **blue** water is teeming with a hundred million Prochlorococcus cells per liter. If you crowd them together like we do in our cultures, you can see their beautiful green chlorophyl. One of those test tubes has a billion Prochlorococcus in it, and as I told you earlier, there are three billion billion billion of them on **the planet**. That ’ s three octillion, if you care to convert. And collectively, they weigh more than **the** human population and they photosynthesize as much as all of **the** crops on land. They ’ re incredibly important in **the** global ocean. So over **the** years, as we were studying them and found how abundant they were, we thought, hmm, this is really strange. How can a single species be so abundant across so many different **habitats**? And as we isolated more into culture, we learned that they are different ecotypes. There are some that are adapted to **the** high-light intensities in **the surface** water, and there are some that are adapted to **the** low light in **the** deep ocean. In fact, those cells that live in **the** bottom of **the** sunlit zone are **the** most efficient photosynthesizers of any known cell. And then we learned that there are some strains that grow optimally along **the** equator, where there are higher temperatures, and some that do better at **the** cooler temperatures as you go north and south. So as we studied these more and more and kept finding more and more diversity, we thought, oh my God, how diverse are these things? And about that time, it became possible to sequence their **genomes** and really look under **the** hood and look at their genetic makeup. And we ’ ve been able to sequence **the genomes** of cultures that we have, but also recently, using flow cytometry, we can isolate individual cells from **the** wild and sequence their individual **genomes**, and now we ’ ve sequenced hundreds of Prochlorococcus. And although each cell has roughly 2, 000 genes — that ’ s one tenth **the** size of **the** human **genome** — as you sequence more and more, you find that they only have a thousand of those in common and **the** other thousand for each individual strain is drawn from an enormous gene pool, and it reflects **the** particular environment that **the** cell might have thrived in, not just high or low light or high or low temperature, but whether there are nutrients that limit them like nitrogen, phosphorus or **iron**. It reflects **the habitat** that they come from. Think of it this way. If each cell is a **smartphone** and **the** apps are **the** genes, when you get your **smartphone**, it comes with these built-in apps. Those are **the** ones that you can ’ t delete if you ’ re an iPhone person. You press on them and they don ’ t jiggle and they don ’ t have x ’ s. Even if you don ’ t want them, you can ’ t get rid of them. Those are like **the** core genes of Prochlorococcus. They ’ re **the** essence of **the** phone. But you have a huge pool of apps to draw upon to make your phone custom- designed for your particular lifestyle and **habitat**. If you travel a lot, you ’ ll have a lot of travel apps, if you ’ re into financial things, you might have a lot of financial apps, or if you ’ re like me, you probably have a lot of weather apps, hoping one of them will tell you what you want to hear. And I ’ ve learned **the** last couple days in Vancouver that you don ’ t need a weather app — you just need an umbrella. So — So just as your **smartphone** tells us something about how you live your life, your lifestyle, reading **the genome** of a Prochlorococcus cell tells us what **the** pressures are in its environment. It ’ s like reading its diary, not only telling us how it got through its day or its week, but even its **evolutionary** history. As we studied — I said we ’ ve sequenced hundreds of these cells, and we can now project what is **the** total genetic size — gene pool — of **the** Prochlorococcus federation, as we call it. It ’ s like a superorganism. And it turns out that projections are that **the** collective has 80, 000 genes. That ’ s four times **the** size of **the** human **genome**. And it ’ s that diversity of gene pools that makes it possible for them to dominate these large regions of **the** oceans and maintain their stability year in and year out. So when I daydream about Prochlorococcus, which I probably do more than is healthy — I imagine them floating out there, doing their job, maintaining **the planet**, feeding **the** animals. But also I inevitably end up thinking about what a masterpiece they are, finely tuned by millions of years of **evolution**. With 2, 000 genes, they can do what all of our human ingenuity has not figured out how to do yet. They can take solar energy, CO2 and turn it into chemical energy in **the** form of **organic carbon**, locking that sunlight in those **carbon** bonds. If we could figure out exactly how they do this, it could inspire designs that could reduce our dependency on **fossil** fuels, which brings my story full circle. **The fossil** fuels that are buried that we ’ re burning took millions of years for **the** earth to bury those, including those ancestors of Prochlorococcus, and we ’ re burning that now in **the** blink of an eye on geological timescales. **Carbon** dioxide is increasing in **the** atmosphere. It ’ s a **greenhouse** gas. **The** oceans are starting to warm. So **the** question is, what is that going to do for my Prochlorococcus? And I ’ m sure you ’ re expecting me to say that my beloved microbes are doomed, but in fact they ’ re not. Projections are that their populations will expand as **the** ocean warms to 30 percent larger by **the** year 2100. Does that make me happy? Well, it makes me happy for Prochlorococcus of course — but not for **the planet**. There are winners and losers in this global experiment that we ’ ve **undertaken**, and it ’ s projected that among **the** losers will be some of those larger phytoplankton, those charismatic ones which are expected to be reduced in numbers, and they ’ re **the** ones that feed **the** zooplankton that feed **the fish** that we like to harvest. So Prochlorococcus has been my muse for **the** past 35 years, but there are legions of other microbes out there maintaining our **planet** for us. They ’ re out there ready and waiting for us to find them so they can tell their stories, too. Thank you.

Text Sample 67: POS Dim 1 – 2018 – Score 17.00 – 2018/t000713.txt

After 13. 8 billion years of cosmic history, our **universe** has woken up and become aware of itself. From a small **blue planet**, tiny, conscious parts of our **universe** have begun gazing out into **the** cosmos with telescopes, discovering something humbling. We ’ ve discovered that our **universe** is vastly grander than our ancestors imagined and that life seems to be an almost imperceptibly small perturbation on an otherwise dead **universe**. But we ’ ve also discovered something inspiring, which is that **the** technology we ’ re developing has **the** potential to help life flourish like never before, not just for centuries but for billions of years, and not just on earth but throughout much of this amazing cosmos. I think of **the** earliest life as”Life 1. 0”because it was really dumb, like bacteria, unable to learn anything during its lifetime. I think of us humans as”Life 2. 0”because we can learn, which we in nerdy, **geek** speak, might think of as installing new software into our brains, like languages and job skills.”Life 3. 0,”which can design not only its software but also its hardware of course doesn ’ t exist yet. But perhaps our technology has already made us”Life 2. 1,”with our **artificial** knees, pacemakers and cochlear implants. So let ’ s take a closer look at our relationship with technology, OK? As an example, **the** Apollo 11 moon mission was both successful and inspiring, showing that when we humans use technology wisely, we can accomplish things that our ancestors could only dream of. But there ’ s an even more inspiring journey propelled by something more powerful than rocket engines, where **the** passengers aren ’ t just three astronauts but all of humanity. Let ’ s talk about our collective journey into **the** future with **artificial** intelligence. My friend Jaan Tallinn likes to point out that just as with rocketry, it ’ s not enough to make our technology powerful. We also have to figure out, if we ’ re going to be really ambitious, how to steer it and where we want to go with it. So let ’ s talk about all three for **artificial** intelligence : **the** power, **the** steering and **the** destination. Let ’ s start with **the** power. I define intelligence very inclusively — simply as our ability to accomplish complex goals, because I want to include both biological and **artificial** intelligence. And I want to avoid **the** silly carbon- chauvinism idea that you can only be smart if you ’ re made of meat. It ’ s really amazing how **the** power of AI has grown recently. Just think about it. Not long ago, robots couldn ’ t walk. Now, they can do backflips. Not long ago, we didn ’ t have self-driving cars. Now, we have self-flying rockets. Not long ago, AI couldn ’ t do face recognition. Now, AI can generate fake faces and simulate your face saying stuff that you never said. Not long ago, AI couldn ’ t beat us at **the** game of Go. Then, Google DeepMind ’ s AlphaZero AI took 3, 000 years of human Go games and Go wisdom, ignored it all and became **the** world ’ s best player by just playing against itself. And **the** most impressive feat here wasn ’ t that it crushed human gamers, but that it crushed human AI researchers who had spent decades handcrafting game-playing software. And AlphaZero crushed human AI researchers not just in Go but even at chess, which we have been working on since 1950. So all this amazing recent progress in AI really begs **the** question : How far will it go? I like to think about this question in terms of this **abstract** landscape of tasks, where **the** elevation represents how hard it is for AI to do each task at human level, and **the sea** level represents what AI can do today. **The sea** level is rising as AI improves, so there ’ s a kind of global warming going on here in **the** task landscape. And **the** obvious takeaway is to avoid careers at **the** waterfront — which will soon be automated and disrupted. But there ’ s a much bigger question as well. How high will **the** water end up rising? Will it eventually rise to flood everything, matching human intelligence at all tasks. This is **the** definition of **artificial** general intelligence — AGI, which has been **the** holy grail of AI research since its inception. By this definition, people who say,”Ah, there will always be jobs that humans can do better than machines,”are simply saying that we ’ ll never get AGI. Sure, we might still choose to have some human jobs or to give humans income and purpose with our jobs, but AGI will in any case transform life as we know it with humans no longer being **the** most intelligent. Now, if **the** water level does reach AGI, then further AI progress will be driven mainly not by humans but by AI, which means that there ’ s a possibility that further AI progress could be way faster than **the** typical human research and development timescale of years, raising **the** controversial possibility of an intelligence **explosion** where recursively self- improving AI rapidly leaves human intelligence far behind, creating what ’ s known as superintelligence. Alright, reality check : Are we going to get AGI any time soon? Some **famous** AI researchers, like Rodney Brooks, think it won ’ t happen for hundreds of years. But others, like Google Deep-

Mind founder Demis Hassabis, are more optimistic and are working to try to make it happen much sooner. And recent surveys have shown that most AI researchers actually share Demis ' s optimism, expecting that we will get AGI within decades, so within **the** lifetime of many of us, which begs **the** question — and then what? What do we want **the** role of humans to be if machines can do everything better and cheaper than us? **The** way I see it, we face a choice. One option is to be complacent. We can say,"Oh, let ' s just build machines that can do everything we can do and not worry about **the** consequences. Come on, if we build technology that makes all humans obsolete, what could possibly go wrong?"But I think that would be embarrassingly lame. I think we should be more ambitious — in **the** spirit of TED. Let ' s envision a truly inspiring high-tech future and try to steer towards it. This brings us to **the** second part of our rocket metaphor : **the** steering. We ' re making AI more powerful, but how can we steer towards a future where AI helps humanity flourish rather than flounder? To help with this, I cofounded **the** Future of Life Institute. It ' s a small nonprofit promoting beneficial technology use, and our goal is simply for **the** future of life to exist and to be as inspiring as possible. You know, I love technology. Technology is why today is better than **the** Stone Age. And I ' m optimistic that we can create a really inspiring high-tech future... if — and this is a big if — if we win **the** wisdom race — **the** race between **the** growing power of our technology and **the** growing wisdom with which we manage it. But this is going to require a change of strategy because our old strategy has been learning from mistakes. We invented fire, screwed up a bunch of times — invented **the** fire extinguisher. We invented **the** car, screwed up a bunch of times — invented **the** traffic light, **the** seat belt and **the** airbag, but with more powerful technology like nuclear weapons and AGI, learning from mistakes is a lousy strategy, don ' t you think? It ' s much better to be proactive rather than reactive ; plan ahead and get things right **the** first time because that might be **the** only time we ' ll get. But it is funny because sometimes people tell me,"Max, shhh, don ' t talk like that. That ' s Luddite scaremongering."But it ' s not scaremongering. It ' s what we at MIT call safety engineering. Think about it : before NASA launched **the** Apollo 11 mission, they systematically thought through everything that could go wrong when you put people on top of explosive fuel tanks and launch them somewhere where no one could help them. And there was a lot that could go wrong. Was that scaremongering? No. That ' s was precisely **the** safety engineering that ensured **the** success of **the** mission, and that is precisely **the** strategy I think we should take with AGI. Think through what can go wrong to make sure it goes right. So in this spirit, we ' ve organized conferences, bringing together leading AI researchers and other thinkers to discuss how to grow this wisdom we need to keep AI beneficial. Our last conference was in Asilomar, California last year and produced this list of 23 principles which have since been signed by over 1, 000 AI researchers and key industry leaders, and I want to tell you about three of these principles. One is that we should avoid an arms race and lethal autonomous weapons. **The** idea here is that any science can be used for new ways of helping people or new ways of harming people. For example, **biology** and **chemistry** are much more likely to be used for new medicines or new cures than for new ways of killing people, because biologists and chemists pushed hard — and successfully — for bans on biological and chemical weapons. And in **the** same spirit, most AI researchers want to stigmatize and ban lethal autonomous weapons. Another Asilomar AI principle is that we should mitigate AI-fueled income inequality. I think that if we can grow **the** economic pie dramatically with AI and we still can ' t figure out how to divide this pie so that everyone is better off, then shame on us. Alright, now raise your hand if your **computer** has ever crashed.

Wow, that ' s a lot of hands. Well, then you ' ll appreciate this principle that we should invest much more in AI safety research, because as we put AI in charge of even more decisions and infrastructure, we need to figure out how to transform today ' s buggy and hackable **computers** into robust AI systems that we can really trust, because otherwise, all this awesome new technology can malfunction and harm us, or get hacked and be turned against us. And this AI safety work has to include work on AI value alignment, because **the** real threat from AGI isn ' t malice, like in silly Hollywood movies, but competence — AGI accomplishing goals that just aren ' t aligned with ours. For example, when we humans drove **the** West African black rhino extinct, we didn ' t do it because we were a bunch of evil rhinoceros haters, did we? We did it because we were smarter than them and our goals weren ' t aligned with theirs. But AGI is by definition smarter than us, so to make sure that we don ' t put ourselves in **the** position of those rhinos if we create AGI, we need to figure out how to make machines understand our goals, adopt our goals and retain our goals. And whose goals should these be, anyway? Which goals should they be? This brings us to **the** third part of our rocket metaphor : **the** destination. We ' re making AI more powerful, trying to figure out how to steer it, but where do we want to go with it? This is **the elephant** in **the** room that almost nobody talks about — not even here at TED — because we ' re so fixated on short-term AI challenges. Look, our species is trying to build AGI, motivated by curiosity and economics, but what sort of future society are we hoping for if we succeed? We did an opinion poll on this recently, and I was struck to see that most people actually want us to build superintelligence : AI that ' s vastly smarter than us in all ways. What there was **the** greatest agreement on was that we should be ambitious and help life spread into **the** cosmos, but there was much less agreement about who or what should be in charge. And I was actually quite amused to see that there ' s some some people who want it to be just machines. And there was total disagreement about what **the** role of humans should be, even at **the** most basic level, so let ' s take a closer look at possible futures that we might choose to steer toward, alright? So don ' t get me wrong here. I ' m not talking about space travel, merely about humanity ' s metaphorical journey into **the** future. So one option that some of my AI colleagues like is to build superintelligence and keep it under human control, like an enslaved god, disconnected from **the** internet and used to create unimaginable technology and wealth for whoever controls it. But Lord Acton warned us that power corrupts, and absolute power corrupts absolutely, so you might worry that maybe we humans just aren ' t smart enough, or wise enough rather, to handle this much power. Also, aside from any moral qualms you might have about enslaving superior minds, you might worry that maybe **the** superintelligence could outsmart us, break out and take over. But I also have colleagues who are fine with AI taking over and even causing human extinction, as long as we feel **the** **the** AIs are our worthy descendants, like our children. But how would we know that **the** AIs have adopted our best values and aren ' t just unconscious zombies tricking us into anthropomorphizing them? Also, shouldn ' t those people who don ' t want human extinction have a say in **the** matter, too? Now, if you didn ' t like either of those two high-tech options, it ' s important to remember that low-tech is suicide from a cosmic perspective, because if we don ' t go far beyond today ' s technology, **the** question isn ' t whether humanity is going to go extinct, merely whether we ' re going to get taken out by **the** next killer **asteroid**, supervolcano or some other problem that better technology could have solved. So, how about having our cake and eating it... with AGI that ' s not enslaved but treats us well because its values are aligned with ours? This is **the** gist of what Eliezer Yudkowsky has called"**friendly** AI,"and if we can do this, it could be awesome. It could not only eliminate negative experiences like disease, poverty, crime and other suffering, but it could also give us **the** freedom to choose from a **fantastic** new diversity of positive experiences — basically making us **the** masters of our own destiny. So in summary, our situation with technology is complicated, but **the** big picture is rather simple. Most AI researchers expect AGI within decades, and if we just bumble into this unprepared, it will probably be **the** biggest mistake in human history — let ' s face it. It could enable brutal, global dictatorship with unprecedented inequality, surveillance and suffering, and maybe even human extinction. But if we steer carefully, we could end up in a **fantastic** future where everybody ' s better off : **the** poor are richer, **the** rich are richer, everybody is healthy and free to live out their dreams. Now, hang on. Do you folks want **the** future that ' s politically right or left? Do you want **the** pious society with strict moral rules, or do you an hedonistic free-for-all, more like Burning Man 24 / 7? Do you want beautiful beaches, forests and lakes, or would you prefer to rearrange some of those atoms with **the computers**, enabling virtual experiences? With **friendly** AI, we could simply build all of these societies and give people **the** freedom to choose which one they want to live in because we would no longer be limited by our intelligence, merely by **the** laws of physics. So **the** resources and space for this would be astronomical — literally. So here ' s our choice. We can either be complacent about our future, taking as an article of blind faith that any new technology is guaranteed to be beneficial, and just repeat that to ourselves as a mantra over and over and over again as we drift like a rudderless ship towards our own obsolescence. Or we can be ambitious — thinking hard about how to steer our technology and where we want to go with it to create **the** age of amazement. We ' re all here to celebrate **the** age of amazement, and I feel that its essence should lie in becoming not overpowered but empowered by our technology. Thank you.

Text Sample 68: POS Dim 1 – 2018 – Score 17.00 – 2018/t000534.txt

All life, every living thing ever, has been built according to **the** information in DNA. What does that mean? Well, it means that just as **the** English language is made up of alphabetic letters that, when combined into words, allow me to tell you **the** story I ' m going to tell you today, DNA is made up of genetic letters that, when combined into genes, allow cells to produce proteins, strings of amino acids that fold up into complex structures that perform **the** functions that allow a cell to do what it does, to tell its stories. **The** English alphabet has 26 letters, and **the** genetic alphabet has four. They ' re pretty **famous**. Maybe you ' ve heard of them. They are often just referred to as G, C, A and T. But it ' s remarkable that all **the** diversity of life is **the** result of four genetic letters. Imagine what it would be like if **the** English alphabet had four letters. What sort of stories would you be able to tell? What if **the** genetic alphabet had more letters? Would life with more letters be able to tell different stories, maybe even more interesting ones? In 1999, my lab at **the** Scripps Research Institute in La Jolla, California started working on this question with **the** goal of creating living organisms with DNA made up of a six-letter genetic alphabet, **the** four natural letters plus two additional new man-made letters. Such an organism would be **the** first radically altered form of life ever created. It would be a semisynthetic form of life that stores more information than life ever has before. It would be able to make new proteins, proteins built from more than **the** 20 normal amino acids that are usually used to build proteins. What sort of stories could that life tell? With **the** power of synthetic **chemistry** and molecular **biology** and just under 20 years of work, we created bacteria with six-letter DNA. Let me tell you how we did it. All you have to remember from your high school **biology** is that **the** four natural letters pair together to form two base pairs. G pairs with C and A pairs with T, so to create our new letters, we synthesized hundreds of new candidates, new candidate letters, and examined their abilities to selectively pair with each other. And after about 15 years of work, we found two that paired together really well, at least in a test tube. They have complicated names, but let ' s just call them X and Y. **The** next thing we needed to do was find a way to get X and Y into cells, and eventually we found that a protein that does something similar in algae worked in our bacteria. So **the** final thing that we needed to do was to show that with X and Y provided, cells could grow and divide and hold on to X and Y in their DNA. Everything we had done up to then took longer than I had hoped — I am actually a really impatient person — but this, **the** most important step, worked faster than I dreamed, basically immediately. On a weekend in 2014, a graduate student in my lab grew bacteria with six- letter DNA. Let me take **the** opportunity to introduce you to them right now. This is an actual picture of them. These are **the** first semisynthetic organisms. So bacteria with six-letter DNA, that ' s really cool, right? Well, maybe some of you are still wondering why. So let me tell you a little bit more about some of our motivations, both conceptual and practical. Conceptually, people have thought about life, what it is, what makes it different from things that are not alive, since people have had thoughts. Many have interpreted life as being perfect, and this was taken as evidence of a creator. Living things are different because a god breathed life into them. Others have sought a more scientific explanation, but I think it ' s fair to say that they still consider **the** molecules of life to be special. I mean, **evolution** has been optimizing them for billions of years, right? Whatever perspective you take, it would seem pretty impossible for chemists to come in and build new parts that function within and alongside **the** natural molecules of life without somehow really screwing everything up. But just how perfectly created or **evolved** are we? Just how special are **the** molecules of life? These questions have been impossible to even ask, because we ' ve had nothing to compare life to. Now for **the** first time, our work suggests that maybe **the** molecules of life aren ' t that special. Maybe life as we know it isn ' t **the** only way it could be. Maybe we ' re not **the** only solution, maybe not even **the** best solution, just a solution. These questions address fundamental issues about life, but maybe they seem a little esoteric. So what about practical motivations? Well, we want to explore what sort of new stories life with an expanded vocabulary could tell, and remember, stories here are **the** proteins that a cell produces and **the** functions they have. So what sort of new proteins with new types of functions could our semisynthetic organisms make and maybe even use? Well, we have a couple of things in mind. **The** first is to get **the** cells to make proteins for us, for our use. Proteins are being used today for an increasingly broad range of different applications, from materials that protect soldiers from injury to **devices** that **detect** dangerous compounds, but at least to me, **the** most exciting application is protein drugs. Despite being relatively new, protein drugs have already revolutionized medicine, and, for example, insulin is a protein. You ' ve probably heard of it, and it ' s manufactured as a drug that has completely changed how we treat diabetes. But **the** problem is that proteins are really hard to make and **the** only practical way to get them is to get cells to make them for you. So of course, with natural cells, you can only get them to make proteins with **the** natural amino acids, and so **the** properties those proteins can have, **the** applications they could be developed for, must be limited by **the** nature of those amino acids that **the** protein ' s built from. So here they are, **the** 20 normal amino acids that are strung together to make a protein, and I think you can see, they ' re not that different-looking. They don ' t bring that many different functions. They don ' t make that many different functions available. Compare that with **the** small molecules that synthetic chemists make as drugs. Now, they ' re much simpler than proteins, but they ' re routinely built from a much broader range of diverse things. Don ' t worry about **the** molecular details, but I think you can see how different they are. And in fact, it ' s their differences that make them great drugs to treat different diseases. So it ' s really provocative to wonder what sort of new protein drugs you could develop if you could build proteins from more diverse things. So can we get our semisynthetic organism to make proteins that include new and different amino acids, maybe amino acids selected to confer **the** protein with some desired property or function? For example, many proteins just aren ' t stable when you inject them into people. They are rapidly degraded or eliminated, and this stops them from being drugs. What if we could make proteins with new amino acids with things attached to them that protect them from their environment, that protect them from being degraded or eliminated, so that they could be better drugs? Could we make proteins with little fingers attached that specifically grab on to other molecules? Many small molecules failed during development as drugs because they just weren ' t specific enough to find their target in **the** complex environment of **the** human body. So could we take those molecules and make them parts of new amino acids that, when incorporated into a protein, are guided by that protein to their target? I started a biotech company called Synthorx. Synthorx stands for synthetic organism with an X added at **the** end because that ' s what you do with biotech companies. Synthorx is working closely with my lab, and they ' re interested in a protein that recognizes a certain receptor on **the surface** of human cells. But **the** problem is that it also recognizes another receptor on **the surface** of those same cells, and that makes it toxic. So could we produce a variant of that protein where **the** part that interacts with that second bad receptor is shielded, blocked by something like a big umbrella so that **the** protein only interacts with that first good receptor? Doing that would be really difficult or impossible to do with **the** normal amino acids, but not with amino acids that are specifically designed for that purpose. So getting our semisynthetic cells to act as little factories to produce better protein drugs isn ' t **the** only potentially really interesting application, because remember, it ' s **the** proteins that allow cells to do what they do. So if we have cells that make new proteins with new functions, could we get them to do things that natural cells can ' t do? For example, could we develop semisynthetic organisms that when injected into a person, seek out cancer cells and only when they find them, secrete a toxic protein that kills them? Could we create bacteria that eat different kinds of **oil**, maybe to clean up an **oil spill**? These are just a couple of **the** types of stories that we ' re going to see if life with an expanded vocabulary can tell. So, sounds great, right? Injecting semisynthetic organisms into people, dumping millions and millions of gallons of our bacteria into **the** ocean or out on your favorite beach? Oh, wait a minute, actually it sounds really scary. This dinosaur is really scary. But here ' s **the** catch : our semisynthetic organisms in order to survive, need to be fed **the** chemical precursors of X and Y. X and Y are completely different than anything that exists in nature. Cells just don ' t have them or **the** ability to make them. So when we prepare them, when we grow them up in **the** controlled environment of **the** lab, we can feed them lots of **the** unnatural food. Then, when we deploy them in a person or out on a beach where they no longer have access that special food, they can grow for a little bit, they can survive for a little, maybe just long enough to perform some intended function, but then they start to run out of **the** food. They start to starve. They starve to death and they just **disappear**. So not only could we get life to tell new stories, we get to tell life when and where to tell those stories. At **the** beginning of this talk I told you that we reported in 2014 **the** creation of semisynthetic organisms that store more information, X and Y, in their DNA. But all **the** motivations that we just talked about require cells to use X and Y to make proteins, so we started working on that. Within a couple years, we showed that **the** cells could take DNA with X and Y and **copy** it into RNA, **the** working **copy** of DNA. And late last year, we showed that they could then use X and Y to make proteins. Here they are, **the** stars of **the** show, **the** first fully- functional semisynthetic organisms. These cells are green because they ' re making a protein that glows green. It ' s a pretty **famous** protein, actually, from jellyfish that a lot of people use in its natural form because it ' s easy to see that you made it. But within every one of these proteins, there ' s a new amino acid that natural life can ' t build proteins with. Every living cell, every living cell ever, has made every one of its proteins using a four-letter genetic alphabet. These cells are living and growing and making protein with a six-letter alphabet. These are a new form of life. This is a semisynthetic form of life. So what about **the** future? My lab is already working on expanding **the** genetic alphabet of other cells, including human cells, and we ' re getting ready to start working on more complex organisms. Think semisynthetic worms. **The** last thing I want to say to you, **the** most important thing that I want to say to you, is that **the** time of semisynthetic life is here. Thank you. Chris Anderson : I mean, Floyd, this is so remarkable. I just wanted to ask you, what are **the** implications of your work for how we should think about **the** possibilities for life, like, in **the universe**, elsewhere? It just seems like so much of life, or so much of our assumptions are based on **the** fact that of course, it ' s got to be DNA, but is **the** possibility space of self-replicating molecules much bigger than DNA, even just DNA with six letters? Floyd Romesberg : Absolutely, I think that ' s right, and I think what our work has shown, as I mentioned, is that there ' s been always this prejudice that sort of we ' re perfect, we ' re optimal, God created us this way, **evolution** perfected us this way. We ' ve made molecules that work right alongside **the** natural ones, and I think that suggests that any molecules that obey **the** fundamental laws of **chemistry** and physics and you can optimize them could do **the** things that **the** natural molecules of life do. There ' s nothing magic there. And I think that it suggests that life could **evolve** many different ways, maybe similar to us with other types of DNA, maybe things without DNA at all. CA : I mean, in your mind, how big might that possibility space be? Do we even know? Are most things going to look something like a DNA molecule, or something radically different that can still self-reproduce and potentially create living organisms? FR : My personal opinion is that if we found new life, we might not even recognize it. CA : So this obsession with **the search** for Goldilocks **planets** in exactly **the** right place with water and whatever, that ' s a very parochial assumption, perhaps. FR : Well, if you want to find someone you can talk to, then maybe not, but I think that if you ' re just looking for any form of life, I think that ' s right, I think that you ' re looking for life under **the** light post. CA : Thank you for boggling all our minds. Thank so much, Floyd.

Text Sample 69: POS Dim 1 – 2022 – Score 32.00 – 2022/t003762.txt

Chris Anderson : Elon Musk, great to see you. How are you? Elon Musk : Good. How are you? CA : We ' re here at **the** Texas Gigafactory **the** day before this thing opens. It ' s been pretty crazy out there. Thank you so much for making time on a busy day. I would love you to help us, kind of, cast our minds, I don

’ t know, 10, 20, 30 years into **the** future. And help us try to picture what it would take to build a future that ’ s worth getting excited about. **The** last time you spoke at TED, you said that that was really just a big driver. You know, you can talk about lots of other reasons to do **the** work you ’ re doing, but fundamentally, you want to think about **the** future and not think that it sucks. EM : Yeah, absolutely. I think in general, you know, there ’ s a lot of discussion of like, this problem or that problem. And a lot of people are sad about **the** future and they ’ re... Pessimistic. And I think... this is... This is not great. I mean, we really want to wake up in **the** morning and look forward to **the** future. We want to be excited about what ’ s going to happen. And life cannot simply be about sort of, solving one miserable problem after another. CA : So if you look forward 30 years, you know, **the** year 2050 has been labeled by scientists as this, kind of, almost like this doomsday deadline on climate. There ’ s a consensus of scientists, a large consensus of scientists, who believe that if we haven ’ t completely eliminated **greenhouse** gases or offset them completely by 2050, effectively we ’ re inviting climate catastrophe. Do you believe there is a pathway to avoid that catastrophe? And what would it look like? EM : Yeah, so I am not one of **the** doomsday people, which may surprise you. I actually think we ’ re on a good path. But at **the** same time, I want to caution against complacency. So, so long as we are not complacent, as long as we have a high sense of urgency about moving towards a sustainable energy economy, then I think things will be fine. So I can ’ t emphasize that enough, as long as we push hard and are not complacent, **the** future is going to be great. Don ’ t worry about it. I mean, worry about it, but if you worry about it, ironically, it will be a self-unfulfilling prophecy. So, like, there are three elements to a sustainable energy future. One is of sustainable energy generation, which is primarily wind and solar. There ’ s also hydro, geothermal, I ’ m actually pro-nuclear. I think nuclear is fine. But it ’ s going to be primarily solar and wind, as **the** primary generators of energy. **The** second part is you need batteries to store **the** solar and wind energy because **the** sun doesn ’ t shine all **the** time, **the** wind doesn ’ t blow all **the** time. So it ’ s a lot of stationary battery packs. And then you need electric transport. So electric cars, electric planes, boats. And then ultimately, it ’ s not really possible to make electric rockets, but you can make **the** propellant used in rockets using sustainable energy. So ultimately, we can have a fully sustainable energy economy. And it ’ s those three things : solar / wind, stationary battery pack, electric vehicles. So then what are **the** limiting factors on progress? **The** limiting factor really will be battery cell production. So that ’ s going to really be **the** fundamental rate driver. And then whatever **the** slowest element of **the** whole lithium-ion battery cells supply chain, from mining and **the** many steps of refining to ultimately creating a battery cell and putting it into a pack, that will be **the** limiting factor on progress towards **sustainability**. CA : All right, so we need to talk more about batteries, because **the** key thing that I want to understand, like, there seems to be a scaling issue here that is kind of amazing and alarming. You have said that you have **calculated** that **the** amount of battery production that **the** world needs for **sustainability** is 300 terawatt hours of batteries. That ’ s **the** end goal? EM : Very rough numbers, and I certainly would invite others to check our calculations because they may arrive at different conclusions. But in order to transition, not just current electricity production, but also heating and transport, which roughly triples **the** amount of electricity that you need, it amounts to approximately 300 terawatt hours of installed capacity. CA : So we need to give people a sense of how big a task that is. I mean, here we are at **the** Gigafactory. You know, this is one of **the** biggest buildings in **the** world. What I ’ ve read, and tell me if this is still right, is that **the** goal here is to eventually produce 100 gigawatt hours of batteries here a year eventually. EM : We will probably do more than that, but yes, hopefully we get there within a couple of years. CA : Right. But I mean, that is one – EM : 0. 1 terrawat hours. CA : But that ’ s still 1 / 100 of what ’ s needed. How much of **the** rest of that 100 is Tesla planning to take on let ’ s say, between now and 2030, 2040, when we really need to see **the** scale up happen? EM : I mean, these are just guesses. So please, people shouldn ’ t hold me to these things. It ’ s not like this is like some – What tends to happen is I ’ ll make some like, you know, best guess and then people, in five years, there ’ ll be some jerk that writes an article :”Elon said this would happen, and it didn ’ t happen. He ’ s a liar and a fool.”It ’ s very annoying when that happens. So these are just guesses, this is a conversation. CA : Right. EM : I think Tesla probably ends up doing 10 percent of that. Roughly. CA : Let ’ s say 2050 we have this amazing, you know, 100 percent sustainable electric grid made up of, you know, some mixture of **the** sustainable energy sources you talked about. That same grid probably is offering **the** world really low-cost energy, isn ’ t it, compared with now. And I ’ m curious about like, are people entitled to get a little bit excited about **the** possibilities of that world? EM : People should be optimistic about **the** future. Humanity will solve sustainable energy. It will happen if we, you know, continue to push hard, **the** future is bright and good from an energy standpoint. And then it will be possible to also use that energy to do **carbon** sequestration. It takes a lot of energy to pull **carbon** out of **the** atmosphere because in putting it in **the** atmosphere it releases energy. So now, you know, obviously in order to pull it out, you need to use a lot of energy. But if you ’ ve got a lot of sustainable energy from wind and solar, you can actually sequester **carbon**. So you can reverse **the** CO2 parts per million of **the** atmosphere and oceans. And also you can really have as much fresh water as you want. Earth is mostly water. We should call Earth”Water.”It ’ s 70 percent water by **surface** area. Now most of that ’ s seawater, but it ’ s like we just happen to be on **the** bit that ’ s land. CA : And with energy, you can turn seawater into – EM : Yes. CA : Irrigating water or whatever water you need. EM : At very low cost. Things will be good. CA : Things will be good. And also, there ’ s other benefits to this non-fossil fuel world where **the** air is cleaner – EM : Yes, exactly. Because, like, when you burn **fossil** fuels, there ’ s all these side reactions and toxic gases of various kinds. And sort of little particulates that are bad for your lungs. Like, there ’ s all sorts of bad things that are happening that will go away. And **the** sky will be cleaner and quieter. **The** future ’ s going to be good. CA : I want us to switch now to think a bit about **artificial** intelligence. But **the** segue there, you mentioned how annoying it is when people call you up for bad predictions in **the** past. So I ’ m possibly going to be annoying now, but I ’ m curious about your timelines and how you predict and how come some things are so amazingly on **the** money and some aren ’ t. So when it comes to predicting sales of Tesla vehicles, for example, you ’ ve kind of been amazing, I think in 2014 when Tesla had sold that year 60, 000 cars, you said,”2020, I think we will do half a million a year.”EM : Yeah, we did almost exactly a half million. CA : You did almost exactly half a million. You were scoffed in 2014 because no one since Henry Ford, with **the** Model T, had come close to that kind of growth rate for cars. You were scoffed, and you actually hit 500, 000 cars and then 510, 000 or whatever produced. But five years ago, last time you came to TED, I asked you about full self-driving, and you said,”Yeah, this very year, I ’ m confident that we will have a car going from LA to New York without any intervention.”EM : Yeah, I don ’ t want to blow your mind, but I ’ m not always right. CA : What ’ s **the** difference between those two? Why has full self-driving in particular been so hard to predict? EM : I mean, **the** thing that really got me, and I think it ’ s going to get a lot of other people, is that there are just so many false dawns with self-driving, where you think you ’ ve got **the** problem, have a handle on **the** problem, and then it, no, turns out you just hit a ceiling. Because if you were to plot **the** progress, **the** progress looks like a log curve. So it ’ s like a series of log curves. So most people don ’ t know what a log curve is, I suppose. CA : Show **the** shape with your hands. EM : It goes up you know, sort of a fairly straight way, and then it starts tailing off and you start getting diminishing returns. And you ’ re like, uh oh, it was trending up and now it ’ s sort of, curving over and you start getting to these, what I call local maxima, where you don ’ t realize basically how dumb you were. And then it happens again. And ultimately... These things, you know, in retrospect, they seem obvious, but in order to solve full self-driving properly, you actually have to solve real-world AI. Because what are **the** road networks designed to work with? They ’ re designed to work with a biological neural net, our brains, and with vision, our eyes. And so in order to make it work with **computers**, you basically need to solve real-world AI and vision. Because we need cameras and silicon neural nets in order to have self-driving work for a system that was designed for eyes and biological neural nets. You know, I guess when you put it that way, it ’ s sort of, like, quite obvious that **the** only way to solve full self-driving is to solve real world AI and sophisticated vision. CA : What do you feel about **the** current architecture? Do you think you have an architecture now where there is a chance for **the** logarithmic curve not to tail off any anytime soon? EM : Well I mean, admittedly these may be infamous last words, but I actually am confident that we will solve it this year. That we will exceed – **The probability** of an accident, at what point do you exceed that of **the** average person? I think we will exceed that this year. CA : What are you seeing behind **the** scenes that gives you that confidence? EM : We ’ re almost at **the** point where we have a high-quality unified vector space. In **the** beginning, we were trying to do this with image recognition on individual images. But if you get one image out of a video, it ’ s actually quite hard to see what ’ s going on without ambiguity. But if you look at a video segment of a few seconds of video, that ambiguity resolves. So **the** first thing we had to do is tie all eight cameras together so they ’ re synchronized, so that all **the** frames are looked at simultaneously and labeled simultaneously by one person, because we still need human labeling. So at least they ’ re not labeled at different times by different people in different ways. So it ’ s sort of a surround picture. Then a very important part is to add **the** time **dimension**. So that you ’ re looking at surround video, and you ’ re labeling surround video. And this is actually quite difficult to do from a software standpoint. We had to write our own labeling tools and then create auto labeling, create auto labeling software to amplify **the** efficiency of human labelers because it ’ s quite hard to label. In **the** beginning, it was taking several hours to label a 10-second video **clip**. This is not scalable. So basically what you have to have is you have to have surround video, and that surround video has to be primarily automatically labeled with humans just being editors and making slight corrections to **the** labeling of **the** video and then feeding back those corrections into **the** future auto labeler, so you get this flywheel eventually where **the** auto labeler is able to take in vast amounts of video and with high accuracy, automatically label **the** video for cars, lane lines, drive space. CA : What you ’ re saying is... **the** result of this is that you ’ re effectively giving **the** car a 3D model of **the** actual **objects** that are all around it. It knows what they are, and it knows how fast they are moving. And **the** remaining task is to predict what **the** quirky behaviors are that, you know, that when a pedestrian is walking down **the** road with a smaller pedestrian, that maybe that smaller pedestrian might do something unpredictable or things like that. You have to build into it before you can really call it safe. EM : You basically need to have memory across time and space. So what I mean by that is... Memory can ’ t be infinite, because it ’ s using up a lot of **the computer** ’ s RAM basically. So you have to say how much are you going to try to remember? It ’ s very common for things to be occluded. So if you talk about say, a pedestrian walking past a truck where you saw **the** pedestrian start on one side of **the** truck, then they ’ re occluded by **the** truck. You would know intuitively, OK, that pedestrian is going to pop out **the** other side, most likely. CA : A **computer** doesn ’ t know it. EM : You need to slow down. CA : A skeptic is going to say that every year for **the** last five years, you ’ ve kind of said, well, no this is **the** year, we ’ re confident that it will be there in a year or two or, you know, like it ’ s always been about that far away. But we ’ ve got a new architecture now, you ’ re seeing enough improvement behind **the** scenes to make you not certain, but pretty confident, that, by **the** end of this year, what in most, not in every city, and every circumstance but in many cities and circumstances, basically **the** car will be able to drive without interventions safer than a human. EM : Yes. I mean, **the** car currently drives me around Austin most of **the** time with no interventions. So it ’ s not like... And we have over 100, 000 people in our full self-driving beta program. So you can look at **the** videos that they post online. CA : I do. And some of them are great, and some of them are a little **terrifying**. I mean, occasionally **the** car seems to veer off and scare **the** hell out of people. EM : It ’ s still a beta. CA : But you ’ re behind **the** scenes, looking at **the** data, you ’ re seeing enough improvement to believe that a this-year timeline is real. EM : Yes, that ’ s what it seems like. I mean, we could be here talking again in a year, like, well, another year went by, and it didn ’ t happen. But I think this is **the** year. CA : And so in general, when people talk about Elon time, I mean it sounds like you can ’ t just have a general rule that if you predict that something will be done in six months, actually what we should imagine is it ’ s going to be a year or it ’ s like two-x or three-x, it depends on **the** type of prediction. Some things, I guess, things involving software, AI, whatever, are fundamentally harder to predict than others. Is there an element that you actually deliberately make aggressive prediction timelines to drive people to be ambitious? Without that, nothing gets done? EM : Well, I generally believe, in terms of internal timelines, that we want to set **the** most aggressive timeline that we can. Because there ’ s sort of like a law of gaseous expansion where, for schedules, where whatever time you set, it ’ s not going to be less than that. It ’ s very rare that it ’ ll be less than that. But as far as our predictions are concerned, what tends to happen in **the** media is that they will report all **the** wrong ones and ignore all **the** right ones. Or, you know, when writing an article about me – I ’ ve had a long career in multiple industries. If you list my sins, I sound like **the** worst person on Earth. But if you put those against **the** things I ’ ve done right, it makes much more sense, you know? So essentially like, **the** longer you do anything, **the** more mistakes that you will make cumulatively. Which, if you **sum** up those mistakes, will sound like I ’ m **the** worst predictor ever. But for example, for Tesla vehicle growth, I said I think we ’ d do 50 percent, and we ’ ve done 80 percent. CA : Yes. EM : But they don ’ t mention that one. So, I mean, I ’ m not sure what my **exact** track record is on predictions. They ’ re more optimistic than pessimistic, but they ’ re not all optimistic. Some of them are exceeded probably more or later, but they do come true. It ’ s very rare that they do not come true. It ’ s sort of like, you know, if there ’ s some radical technology prediction, **the** point is not that it was a few years late, but that it happened at all. That ’ s **the** more important part. CA : So it feels like at some point in **the** last year, seeing **the** progress on understanding, **the** Tesla AI understanding **the** world around it, led to a kind of, an aha moment at Tesla. Because you really surprised people recently when you said probably **the** most important product development going on at Tesla this year is this robot, Optimus. EM : Yes. CA : Many companies out there have tried to put out these robots, they ' ve been working on them for years. And so far no one has really cracked it. There ' s no mass adoption robot in people ' s homes. There are some in manufacturing, but I would say, no one ' s kind of, really cracked it. Is it something that happened in **the** development of full self-driving that gave you **the** confidence to say,"You know what, we could do something special here."EM : Yeah, exactly. So, you know, it took me a while to sort of realize that in order to solve self-driving, you really needed to solve real-world AI. And at **the** point of which you solve real-world AI for a car, which is really a robot on four wheels, you can then generalize that to a robot on legs as well. **The** two hard parts I think – like obviously companies like Boston Dynamics have shown that it ' s possible to make quite compelling, sometimes alarming robots. CA : Right. EM : You know, so from a sensors and actuators standpoint, it ' s certainly been demonstrated by many that it ' s possible to make a humanoid robot. **The** things that are currently missing are enough intelligence for **the** robot to navigate **the** real world and do useful things without being explicitly instructed. So **the** missing things are basically real-world intelligence and scaling up manufacturing. Those are two things that Tesla is very good at. And so then we basically just need to design **the** specialized actuators and sensors that are needed for humanoid robot. People have no idea, this is going to be bigger than **the** car. CA : So let ' s dig into exactly that. I mean, in one way, it ' s actually an easier problem than full self-driving because instead of an **object** going along at 60 miles an hour, which if it gets it wrong, someone will die. This is an **object** that ' s engineered to only go at what, three or four or five miles an hour. And so a mistake, there aren ' t lives at stake. There might be embarrassment at stake. EM : So long as **the** AI doesn ' t take it over and murder us in our sleep or something. CA : Right. So talk about – I think **the** first applications you ' ve mentioned are probably going to be manufacturing, but eventually **the** vision is to have these available for people at home. If you had a robot that really understood **the** 3D architecture of your house and knew where every **object** in that house was or was supposed to be, and could recognize all those **objects**, I mean, that ' s kind of amazing, isn ' t it? Like **the** kind of thing that you could ask a robot to do would be what? Like, tidy up? EM : Yeah, absolutely. Make dinner, I guess, mow **the** lawn. CA : Take a cup of tea to grandma and show her family pictures. EM : Exactly. Take care of my grandmother and make sure – CA : It could obviously recognize everyone in **the** home. It could play catch with your kids. EM : Yes. I mean, obviously, we need to be careful this doesn ' t become a dystopian situation. I think one of **the** things that ' s going to be important is to have a localized ROM chip on **the** robot that cannot be updated over **the** air. Where if you, for example, were to say,"Stop, stop, stop,"if anyone said that, then **the** robot would stop, you know, type of thing. And that ' s not updatable remotely. I think it ' s going to be important to have safety features like that. CA : Yeah, that sounds wise. EM : And I do think there should be a regulatory agency for AI. I ' ve said that for many years. I don ' t love being regulated, but I think this is an important thing for public safety. CA : Let ' s come back to that. But I don ' t think many people have really sort of taken seriously **the** notion of, you know, a robot at home. I mean, at **the** start of **the** computing revolution, Bill Gates said there ' s going to be a **computer** in every home. And people at **the** time said, yeah, whatever, who would even want that. Do you think there will be basically like in, say, 2050 or whatever, like a robot in most homes, is what there will be, and people will love them and count on them? You ' ll have your own butler basically. EM : Yeah, you ’ ll have your sort of buddy robot probably, yeah. CA : I mean, how much of a buddy? How many applications have you thought, you know, can you have a romantic partner, a sex partner? EM : It ’ s probably inevitable. I mean, I did promise **the** internet that I ’ d make catgirls. We could make a robot catgirl. CA : Be careful what you promise **the** internet. EM : So, yeah, I guess it ’ ll be whatever people want really, you know. CA : What sort of timeline should we be thinking about of **the** first models that are actually made and sold? EM : Well, you know, **the** first units that we intend to make are for jobs that are dangerous, boring, repetitive, and things that people don ’ t want to do. And, you know, I think we ’ ll have like an interesting prototype sometime this year. We might have something useful next year, but I think quite likely within at least two years. And then we ’ ll see rapid growth year over year of **the** usefulness of **the** humanoid robots and decrease in cost and scaling up production. CA : Initially just selling to businesses, or when do you picture you ’ ll start selling them where you can buy your parents one for Christmas or something? EM : I ’ d say in less than ten years. CA : Help me on **the** economics of this. So what do you picture **the** cost of one of these being? EM : Well, I think **the** cost is actually not going to be crazy high. Like less than a car. Initially, things will be expensive because it ’ ll be a new technology at low production volume. **The** complexity and cost of a car is greater than that of a humanoid robot. So I would expect that it ’ s going to be less than a car, or at least equivalent to a cheap car. CA : So even if it starts at 50k, within a few years, it ’ s down to 20k or lower or whatever. And maybe for home they ’ ll get much cheaper still. But think about **the** economics of this. If you can replace a $30, 000, $40, 000-a-year worker, which you have to pay every year, with a one-time payment of $25, 000 for a robot that can work longer hours, a pretty rapid replacement of certain types of jobs. How worried should **the** world be about that? EM : I wouldn ’ t worry about **the** sort of, putting people out of a job thing. I think we ’ re actually going to have, and already do have, a massive shortage of labor. So I think we will have... Not people out of work, but actually still a shortage labor even in **the** future. But this really will be a world of abundance. Any goods and services will be available to anyone who wants them. It ’ ll be so cheap to have goods and services, it will be ridiculous. CA : I ’ m presuming it should be possible to imagine a bunch of goods and services that can ’ t profitably be made now but could be made in that world, courtesy of legions of robots. EM : Yeah. It will be a world of abundance. **The** only scarcity that will exist in **the** future is that which we decide to create ourselves as humans. CA : OK. So AI is allowing us to imagine a differently powered economy that will create this abundance. What are you most worried about going wrong? EM : Well, like I said, AI and robotics will bring out what might be termed **the** age of abundance. Other people have used this word, and that this is my prediction : it will be an age of abundance for everyone. But I guess there ’ s... **The** dangers would be **the artificial** general intelligence or digital superintelligence decouples from a collective human will and goes in **the** direction that for some reason we don ’ t like. Whatever direction it might go. You know, that ’ s sort of **the** idea behind Neuralink, is to try to more tightly couple collective human world to digital superintelligence. And also along **the** way solve a lot of brain injuries and spinal injuries and that kind of thing. So even if it doesn ’ t succeed in **the** greater goal, I think it will succeed in **the** goal of alleviating brain and spine damage. CA : So **the** spirit there is that if we ’ re going to make these AIs that are so vastly intelligent, we ought to be wired directly to them so that we ourselves can have those superpowers more directly. But that doesn ’ t seem to avoid **the** risk that those superpowers might... turn ugly in unintended ways. EM : I think it ’ s a risk, I agree. I ’ m not saying that I have some certain answer to that risk. I ’ m just saying like maybe one of **the** things that would be good for ensuring that **the** future is one that we want is to more tightly couple **the** collective human world to digital intelligence. **The** issue that we face here is that we are already a cyborg, if you think about it. **The computers** are an extension of ourselves. And when we die, we have, like, a digital ghost. You know, all of our text messages and social media, emails. And it ’ s quite eerie actually, when someone dies but everything online is still there. But you say like, what ’ s **the** limitation? What is it that inhibits a human-machine symbiosis? It ’ s **the** data rate. When you communicate, especially with a phone, you ’ re moving your thumbs very slowly. So you ’ re like moving your two little meat sticks at a rate that ’ s maybe 10 bits per second, optimistically, 100 bits per second. And **computers** are communicating at **the** gigabyte level and beyond. CA : Have you seen evidence that **the** technology is actually working, that you ’ ve got a richer, sort of, higher bandwidth connection, if you like, between like external electronics and a brain than has been possible before? EM : Yeah. I mean, **the** fundamental principles of reading neurons, sort of doing read-write on neurons with tiny electrodes, have been demonstrated for decades. So it ’ s not like **the** concept is new. **The** problem is that there is no product that works well that you can go and buy. So it ’ s all sort of, in research labs. And it ’ s like some cords sticking out of your head. And it ’ s quite gruesome, and it ’ s really... There ’ s no good product that actually does a good job and is high-bandwidth and safe and something actually that you could buy and would want to buy. But **the** way to think of **the** Neuralink **device** is kind of like a Fitbit or an Apple Watch. That ’ s where we take out sort of a small section of skull about **the** size of a quarter, replace that with what, in many ways really is very much like a Fitbit, Apple Watch or some kind of smart watch thing. But with tiny, tiny wires, very, very tiny wires. Wires so tiny, it ’ s hard to even see them. And it ’ s very important to have very tiny wires so that when they ’ re implanted, they don ’ t damage **the** brain. CA : How far are you from putting these into humans? EM : Well, we have put in our FDA application to aspirationally do **the** first human implant this year. CA : **The** first uses will be for neurological injuries of different kinds. But rolling **the** clock forward and imagining when people are actually using these for their own enhancement, let ’ s say, and for **the** enhancement of **the** world, how clear are you in your mind as to what it will feel like to have one of these inside your head? EM : Well, I do want to emphasize we ’ re at an early stage. And so it really will be many years before we have anything approximating a high-bandwidth neural interface that allows for AI-human symbiosis. For many years, we will just be solving brain injuries and spinal injuries. For probably a decade. This is not something that will suddenly one day it will have this incredible sort of whole brain interface. It ’ s going to be, like I said, at least a decade of really just solving brain injuries and spinal injuries. And really, I think you can solve a very wide range of brain injuries, including severe depression, morbid obesity, sleep, potentially schizophrenia, like, a lot of things that cause great stress to people. Restoring memory in older people. CA : If you can pull that off, that ’ s **the** app I will sign up for. EM : Absolutely. CA : Please hurry. EM : I mean, **the** emails that we get at Neuralink are heartbreaking. I mean, they ’ ll send us just tragic, you know, where someone was sort of, in **the** prime of life and they had an accident on a motorcycle and someone who ’ s 25, you know, can ’ t even feed themselves. And this is something we could fix. CA : But you have said that AI is one of **the** things you ’ re most worried about and that Neuralink may be one of **the** ways where we can keep abreast of it. EM : Yeah, there ’ s **the** short-term thing, which I think is helpful on an individual human level with injuries. And then **the** long-term thing is an attempt to address **the** civilizational risk of

AI by bringing digital intelligence and biological intelligence closer together. I mean, if you think of how **the** brain works today, there are really two layers to **the** brain. There ' s **the** limbic system and **the** cortex. You ' ve got **the** kind of, animal brain where – it ' s kind of like **the** fun part, really. CA : It ' s where most of Twitter operates, by **the** way. EM : I think Tim Urban said, we ' re like somebody, you know, stuck a **computer** on a monkey. You know, so we ' re like, if you gave a monkey a **computer**, that ' s our cortex. But we still have a lot of monkey instincts. Which we then try to rationalize as, no, it ' s not a monkey instinct. It ' s something more important than that. But it ' s often just really a monkey instinct. We ' re just monkeys with a **computer** stuck in our brain. But even though **the** cortex is sort of **the** smart, or **the** intelligent part of **the** brain, **the** thinking part of **the** brain, I ' ve not yet met anyone who wants to delete their limbic system or their cortex. They ' re quite happy having both. Everyone wants both parts of their brain. And people really want their phones and their **computers**, which are really **the** tertiary, **the** third part of your intelligence. It ' s just that it ' s... Like **the** bandwidth, **the** rate of communication with that tertiary layer is slow. And it ' s just a very tiny straw to this tertiary layer. And we want to make that tiny straw a big **highway**. And I ' m definitely not saying that this is going to solve everything. Or this is you know, it ' s **the** only thing – it ' s something that might be helpful. And worst-case scenario, I think we solve some important brain injury, spinal injury issues, and that ' s still a great outcome. CA : Best-case scenario, we may discover new human possibility, telepathy, you ' ve spoken of, in a way, a connection with a loved one, you know, full memory and much faster thought processing maybe. All these things. It ' s very cool. If AI were to take down Earth, we need a plan B. Let ' s shift our attention to space. We spoke last time at TED about reusability, and you had just demonstrated that spectacularly for **the** first time. Since then, you ' ve gone on to build this monster rocket, Starship, which kind of changes **the** rules of **the** game in spectacular ways. Tell us about Starship. EM : Starship is extremely fundamental. So **the** holy grail of rocketry or space transport is full and rapid reusability. This has never been achieved. **The** closest that anything has come is our Falcon 9 rocket, where we are able to recover **the** first stage, **the** boost stage, which is probably about 60 percent of **the** cost of **the** vehicle of **the** whole launch, maybe 70 percent. And we ' ve now done that over a hundred times. So with Starship, we will be recovering **the** entire thing. Or at least that ' s **the** goal. CA : Right. EM : And moreover, recovering it in such a way that it can be immediately re-flown. Whereas with Falcon 9, we still need to do some amount of refurbishment to **the** booster and to **the** fairing nose cone. But with Starship, **the** design goal is immediate re-flight. So you just refill propellants and go again. And this is gigantic. Just as it would be in any other mode of transport. CA : And **the** main design is to basically take 100 plus people at a time, plus a bunch of things that they need, to Mars. So, first of all, talk about that piece. What is your latest timeline? One, for **the** first time, a Starship goes to Mars, presumably without people, but just equipment. Two, with people. Three, there ' s sort of, OK, 100 people at a time, let ' s go. EM : Sure. And just to put **the** cost thing into perspective, **the** expected cost of Starship, putting 100 tons into orbit, is significantly less than what it would have cost or what it did cost to put our tiny Falcon 1 rocket into orbit. Just as **the** cost of flying a 747 around **the** world is less than **the** cost of a small **airplane**. You know, a small **airplane** that was thrown away. So it ' s really pretty mind-boggling that **the** giant thing costs less, way less than **the** small thing. So it doesn ' t use exotic propellants or things that are difficult to obtain on Mars. It uses methane as fuel, and it ' s primarily oxygen, roughly 77-78 percent oxygen by weight. And Mars has a CO_2 atmosphere and has water ice, which is CO2 plus H2O, so you can make CH4, methane, and O2, oxygen, on Mars. CA : Presumably, one of **the** first tasks on Mars will be to create a fuel plant that can create **the** fuel for **the** return trips of many Starships. EM : Yes. And actually, it ’ s mostly going to be oxygen plants, because it ’ s 78 percent oxygen, 22 percent fuel. But **the** fuel is a simple fuel that is easy to create on Mars. And in many other parts of **the** solar system. So basically... And it ’ s all propulsive landing, no parachutes, nothing thrown away. It has a heat shield that ’ s capable of entering on Earth or Mars. We can even potentially go to Venus. but you don ’ t want to go there. Venus is hell, almost literally. But you could... It ’ s a generalized method of transport to anywhere in **the** solar system, because **the** point at which you have propellant depo on Mars, you can then travel to **the asteroid** belt and to **the** moons of Jupiter and Saturn and ultimately anywhere in **the** solar system. CA : But your main focus and SpaceX ’ s main focus is still Mars. That is **the** mission. That is where most of **the** effort will go? Or are you actually imagining a much broader **array** of uses even in **the** coming, you know, **the** first decade or so of uses of this. Where we could go, for example, to other places in **the** solar system to explore, perhaps NASA wants to use **the** rocket for that reason. EM : Yeah, NASA is planning to use a Starship to return to **the** moon, to return people to **the** moon. And so we ’ re very honored that NASA has chosen us to do this. But I ’ m saying it is a generalized – it ’ s a general solution to getting anywhere in **the** greater solar system. It ’ s not suitable for going to another star system, but it is a general solution for transport anywhere in **the** solar system. CA : Before it can do any of that, it ’ s got to demonstrate it can get into orbit, you know, around Earth. What ’ s your latest advice on **the** timeline for that? EM : It ’ s looking promising for us to have an orbital launch attempt in a few months. So we ’ re actually integrating – will be integrating **the** engines into **the** booster for **the** first orbital flight starting in about a week or two. And **the** launch complex itself is ready to go. So assuming we get regulatory approval, I think we could have an orbital launch attempt within a few months. CA : And a radical new technology like this presumably there is real risk on those early attempts. EM : Oh, 100 percent, yeah. **The** joke I make all **the** time is that excitement is guaranteed. Success is not guaranteed, but excitement certainly is. CA : But **the** last I saw on your timeline, you ’ ve slightly put back **the** expected date to put **the** first human on Mars till 2029, I want to say? EM : Yeah, I mean, so let ’ s see. I mean, we have built a production system for Starship, so we ’ re making a lot of ships and boosters. CA : How many are you planning to make actually? EM : Well, we ’ re currently expecting to make a booster and a ship roughly every, well, initially, roughly every couple of months, and then hopefully by **the** end of this year, one every month. So it ’ s giant rockets, and a lot of them. Just talking in terms of rough orders of magnitude, in order to create a self-sustaining city on Mars, I think you will need something on **the** order of a thousand ships. And we just need a Helen of Sparta, I guess, on Mars. CA : This is not in most people ’ s heads, Elon. EM : **The planet** that launched 1, 000 ships. CA : That ’ s nice. But this is not in most people ’ s heads, this picture that you have in your mind. There ’ s basically a two-year window, you can only really fly to Mars conveniently every two years. You were picturing that during **the** 2030s, every couple of years, something like 1, 000 Starships take off, each containing 100 or more people. That picture is just completely mind-blowing to me. That sense of this armada of humans going to – EM : It ’ ll be like”Battlestar Galactica,”**the** fleet departs. CA : And you think that it can basically be funded by people spending maybe a couple hundred grand on a **ticket** to Mars? Is that price about where it has been? EM : Well, I think if you say like, what ’ s required in order to get enough people and enough cargo to Mars to build a self-sustaining city. And it ' s where you have an intersection of sets of people who want to go, because I think only a small percentage of humanity will want to go, and can afford to go or get sponsorship in some manner. That intersection of sets, I think, needs to be a million people or something like that. And so it ' s what can a million people afford, or get sponsorship for, because I think governments will also pay for it, and people can take out loans. But I think at **the** point at which you say, OK, like, if moving to Mars costs are, for argument ' s sake, $100, 000, then I think you know, almost anyone can work and save up and eventually have $100, 000 and be able to go to Mars if they want. We want to make it available to anyone who wants to go. It ' s very important to emphasize that Mars, especially in **the** beginning, will not be luxurious. It will be dangerous, cramped, difficult, hard work. It ' s kind of like that Shackleton ad for going to **the** Antarctic, which I think is actually not real, but it sounds real and it ' s cool. It ' s sort of like, **the** sales pitch for going to Mars is,"It ' s dangerous, it ' s cramped. You might not make it back. It ' s difficult, it ' s hard work."That ' s **the** sales pitch. CA : Right. But you will make history. EM : But it ' ll be glorious. CA : So on that kind of launch rate you ' re talking about over two decades, you could get your million people to Mars, essentially. Whose city is it? Is it NASA ' s city, is it SpaceX ' s city? EM : It ' s **the** people of Mars ' city. **The** reason for this, I mean, I feel like why do this thing? I think this is important for maximizing **the** probable lifespan of humanity or consciousness. Human civilization could come to an end for external reasons, like a giant meteor or super volcanoes or extreme climate change. Or World War III, or you know, any one of a number of reasons. But **the** probable life span of civilizational consciousness as we know it, which we should really view as this very delicate thing, like a small candle in a vast darkness. That is what appears to be **the** case. We ' re in this vast darkness of space, and there ' s this little candle of consciousness that ' s only really come about after 4. 5 billion years, and it could just go out. CA : I think that ' s powerful, and I think a lot of people will be inspired by that vision. And **the** reason you need **the** million people is because there has to be enough people there to do everything that you need to survive. EM : Really, like **the** critical threshold is if **the** ships from Earth stop coming for any reason, does **the** Mars City die out or not? And so we have to – You know, people talk about like, **the** sort of, **the** great filters, **the** things that perhaps, you know, we talk about **the** Fermi paradox, and where are **the** aliens? Well maybe there are these various great filters that **the** aliens didn ' t pass, and so they eventually just ceased to exist. And one of **the** great filters is becoming a multi-planet species. So we want to pass that filter. And I ' ll be long-dead before this is, you know, a real thing, before it happens. But I ' d like to at least see us make great progress in this direction. CA : Given how tortured **the** Earth is right now, how much we ' re beating each other up, shouldn ' t there be discussions going on with everyone who is dreaming about Mars to try to say, we ' ve got a once in a civilization ' s chance to make some new rules here? Should someone be trying to lead those discussions to figure out what it means for this to be **the** people of Mars ' City? EM : Well, I think ultimately this will be up to **the** people of Mars to decide how they want to rethink society. Yeah there ' s certainly risk there. And hopefully **the** people of Mars will be more enlightened and will not fight amongst each other too much. I mean, I have some recommendations, which people of Mars may choose to listen to or not. I would advocate for more of a direct democracy, not a representative democracy, and laws that are short enough for people to understand. Where it is harder to create laws than to get rid of them. CA : Coming back a bit nearer term, I ' d love you to just talk a bit about some of **the** other possibility space that Starship seems to have created.

So given – Suddenly we ’ ve got this ability to move 100 tons-plus into orbit. So we ’ ve just launched **the** James Webb telescope, which is an incredible thing. It ’ s unbelievable. EM : Exquisite piece of technology. CA : Exquisite piece of technology. But people spent two years trying to figure out how to fold up this thing. It ’ s a three-ton telescope. EM : We can make it a lot easier if you ’ ve got more volume and mass. CA : But let ’ s ask a different question. Which is, how much more powerful a telescope could someone design based on using Starship, for example? EM : I mean, roughly, I ’ d say it ’ s probably an order of magnitude more resolution. If you ’ ve got 100 tons and a thousand cubic meters volume, which is roughly what we have. CA : And what about other exploration through **the** solar system? I mean, I ’ m you know – EM : Europa is a big question mark. CA : Right, so there ’ s an ocean there. And what you really want to do is to drop a submarine into that ocean. EM : Maybe there ’ s like, some squid civilization, cephalopod civilization under **the** ice of Europa. That would be pretty interesting. CA : I mean, Elon, if you could take a submarine to Europa and we see pictures of this thing being devoured by a squid, that would honestly be **the** happiest moment of my life. EM : Pretty wild, yeah. CA : What other possibilities are out there? Like, it feels like if you ’ re going to create a thousand of these things, they can only fly to Mars every two years. What are they doing **the** rest of **the** time? It feels like there ’ s this **explosion** of possibility that I don ’ t think people are really thinking about. EM : I don ’ t know, we ’ ve certainly got a long way to go. As you alluded to earlier, we still have to get to orbit. And then after we get to orbit, we have to really prove out and refine full and rapid reusability. That ’ ll take a moment. But I do think we will solve this. I ’ m highly confident we will solve this at this point. CA : Do you ever wake up with **the** fear that there ’ s going to be this Hindenburg moment for SpaceX where... EM : We ’ ve had many Hindenburg. Well, we ’ ve never had Hindenburg moments with people, which is very important. Big difference. We ’ ve blown up quite a few rockets. So there ’ s a whole compilation online that we put together and others put together, it ’ s showing rockets are hard. I mean, **the sheer** amount of energy going through a rocket boggles **the** mind. So, you know, getting out of Earth ’ s gravity well is difficult. We have a strong gravity and a thick atmosphere. And Mars, which is less than 40 percent, it ’ s like, 37 percent of Earth ’ s gravity and has a thin atmosphere. **The** ship alone can go all **the** way from **the surface** of Mars to **the surface** of Earth. Whereas getting to Mars requires a giant booster and orbital refilling. CA : So, Elon, as I think more about this incredible **array** of things that you ’ re involved with, I keep seeing these synergies, to use a horrible word, between them. You know, for example, **the** robots you ’ re building from Tesla could possibly be pretty handy on Mars, doing some of **the** dangerous work and so forth. I mean, maybe there ’ s a scenario where your city on Mars doesn ’ t need a million people, it needs half a million people and half a million robots. And that ’ s a possibility. Maybe **The** Boring Company could play a role helping create some of **the** subterranean dwelling spaces that you might need. EM : Yeah. CA : Back on **planet** Earth, it seems like a partnership between Boring Company and Tesla could offer an unbelievable deal to a city to say, we will create for you a 3D network of tunnels populated by robo-taxis that will offer fast, low-cost transport to anyone. You know, full self-driving may or may not be done this year. And in some cities, like, somewhere like Mumbai, I suspect won ’ t be done for a decade. EM : Some places are more challenging than others. CA : But today, today, with what you ’ ve got, you could put a 3D network of tunnels under there. EM : Oh, if it ’ s just in a tunnel, that ’ s a solved problem. CA : Exactly, full self-driving is a solved problem. To me, there ’ s amazing synergy there. With Starship, you know, Gwynne Shotwell talked about by 2028 having from city to city, you know, transport on **planet** Earth. EM : This is a real possibility. **The** fastest way to get from one place to another, if it ’ s a long distance, is a rocket. It ’ s basically an ICBM. CA : But it has to land – Because it ’ s an ICBM, it has to land probably offshore, because it ’ s loud. So why not have a tunnel that then connects to **the** city with Tesla? And Neuralink. I mean, if you going to go to Mars having a telepathic connection with loved ones back home, even if there ’ s a time delay... EM : These are not intended to be connected, by **the** way. But there certainly could be some synergies, yeah. CA : Surely there is a growing argument that you should actually put all these things together into one company and just have a company devoted to creating a future that ’ s exciting, and let a thousand flowers bloom. Have you been thinking about that? EM : I mean, it is tricky because Tesla is a publicly-traded company, and **the** investor base of Tesla and SpaceX and certainly Boring Company and Neuralink are quite different. Boring Company and Neuralink are tiny companies. CA : By comparison. EM : Yeah, Tesla ’ s got 110, 000 people. SpaceX I think is around 12, 000 people. Boring Company and Neuralink are both under 200 people. So they ’ re little, tiny companies, but they will probably get bigger in **the** future. They will get bigger in **the** future. It ’ s not that easy to sort of combine these things. CA : Traditionally, you have said that for SpaceX especially, you wouldn ’ t want it public, because public investors wouldn ’ t support **the** craziness of **the** idea of going to Mars or whatever. EM : Yeah, making life multi-planetary is outside of **the** normal time horizon of Wall Street analysts. To say **the** least. CA : I think something ’ s changed, though. What ’ s changed is that Tesla is now so powerful and so big and throws off so much cash that you actually could connect **the** dots here. Just tell **the** public that x-billion dollars a year, whatever your number is, will be diverted to **the** Mars mission. I suspect you ’ d have massive interest in that company. And it might **unlock** a lot more possibility for you, no? EM : I would like to give **the** public access to ownership of SpaceX, but I mean **the** thing that like, **the** overhead associated with a public company is high. I mean, as a public company, you ’ re just constantly sued. It does occupy like, a fair bit of... You know, time and effort to deal with these things. CA : But you would still only have one public company, it would be bigger, and have more things going on. But instead of being on four boards, you ’ d be on one. EM : I ’ m actually not even on **the** Neuralink or Boring Company boards. And I don ’ t really attend **the** SpaceX board meetings. We only have two a year, and I just stop by and chat for an hour. **The** board overhead for a public company is much higher. CA : I think some investors probably worry about how your time is being split, and they might be excited by you know, that. Anyway, I just woke up **the** other day thinking, just, there are so many ways in which these things connect. And you know, just **the** simplicity of that mission, of building a future that is worth getting excited about, might appeal to an awful lot of people. Elon, you are reported by Forbes and everyone else as now, you know, **the** world ’ s richest person. EM : That ’ s not a sovereign. CA : EM : You know, I think it ’ s fair to say that if somebody is like, **the** king or de facto king of a country, they ’ re wealthier than I am. CA : But it ’ s just harder to measure – So $300 billion. I mean, your net worth on any given day is rising or falling by several billion dollars. How insane is that? EM : It ’ s bonkers, yeah. CA : I mean, how do you handle that psychologically? There aren ’ t many people in **the** world who have to even think about that. EM : I actually don ’ t think about that too much. But **the** thing that is actually more difficult and that does make sleeping difficult is that, you know, every good hour or even minute of thinking about Tesla and SpaceX has such a big effect on **the** company that I really try to work as much as possible, you know, to **the** edge of sanity, basically. Because you know, Tesla ’ s getting to **the** point where probably will get to **the** point later this year, where every high-quality minute of thinking is a million dollars impact on Tesla. Which is insane. I mean, **the** basic, you know, if Tesla is doing, you know, sort of $2 billion a week, let ' s say, in revenue, it ' s sort of $300 million a day, seven days a week. You know, it ' s... CA : If you can change that by five percent in an hour ' s brainstorm, that ' s a pretty valuable hour. EM : I mean, there are many instances where a half-hour meeting, I was able to improve **the** financial outcome of **the** company by $100 million in a half-hour meeting. CA : There are many other people out there who can ' t stand this world of billionaires. Like, they are hugely offended by **the** notion that an individual can have **the** same wealth as, say, a billion or more of **the** world ' s poorest people. EM : If they examine sort of – I think there ' s some axiomatic flaws that are leading them to that conclusion. For sure, it would be very problematic if I was consuming, you know, billions of dollars a year in personal **consumption**. But that is not **the** case. In fact, I don ' t even own a home right now. I ' m literally staying at friends ' places. If I travel to **the** Bay Area, which is where most of Tesla engineering is, I basically rotate through friends ' spare bedrooms. I don ' t have a yacht, I really don ' t take vacations. It ' s not as though my personal **consumption** is high. I mean, **the** one exception is a plane. But if I don ' t use **the** plane, then I have less hours to work. CA : I mean, I personally think you have shown that you are mostly driven by really quite a deep sense of moral purpose. Like, your attempts to solve **the** climate problem have been as powerful as anyone else on **the planet** that I ' m aware of. And I actually can ' t understand, personally, I can ' t understand **the** fact that you get all this criticism from **the** Left about,"Oh, my God, he ' s so rich, that ' s disgusting."When climate is their issue. Philanthropy is a topic that some people go to. Philanthropy is a hard topic. How do you think about that? EM : I think if you care about **the** reality of goodness instead of **the** perception of it, philanthropy is extremely difficult. SpaceX, Tesla, Neuralink and **The** Boring Company are philanthropy. If you say philanthropy is love of humanity, they are philanthropy. Tesla is accelerating sustainable energy. This is a love – philanthropy. SpaceX is trying to ensure **the** long-term survival of humanity with a multiple-planet species. That is love of humanity. You know, Neuralink is trying to help solve brain injuries and existential risk with AI. Love of humanity. Boring Company is trying to solve traffic, which is hell for most people, and that also is love of humanity. CA : How upsetting is it to you to hear this constant drumbeat of,"Billionaires, my God, Elon Musk, oh, my God?"Like, do you just shrug that off or does it does it actually hurt? EM : I mean, at this point, it ' s water off a duck ' s back. CA : Elon, I ' d like to, as we wrap up now, just pull **the** camera back and just think... You ' re a father now of seven surviving kids. EM : Well, I mean, I ' m trying to set a good example because **the** birthrate on Earth is so low that we ' re facing civilizational collapse unless **the** birth rate returns to a sustainable level. CA : Yeah, you ' ve talked about this a lot, that depopulation is a big problem, and people don ' t understand how big a problem it is. EM : Population collapse is one of **the** biggest threats to **the** future of human civilization. And that is what is going on right now. CA : What drives you on a day-to-day basis to do what you do? EM : I guess, like, I really want to make sure that there is a good future for humanity and that we ' re on a path to understanding **the** nature of **the universe**, **the** meaning of life. Why are we here, how did we get here? And in order to understand **the** nature of **the** **universe** and all these fundamental questions, we must expand **the** scope and scale of consciousness. Certainly it must not diminish or go out. Or we certainly won ' t understand this. I would say I ' ve been motivated by curiosity more than anything, and just desire to think about **the** future and not be sad, you know? CA : And are you? Are you not sad? EM : I ' m sometimes sad, but mostly I ' m feeling I guess relatively optimistic about **the** future these days. There are certainly some big risks that humanity faces. I think **the** population collapse is a really big deal, that I wish more people would think about because **the** birth rate is far below what ’ s needed to sustain civilization at its current level. And there ’ s obviously... We need to take action on climate **sustainability**, which is being done. And we need to secure **the** future of consciousness by being a multi-planet species. We need to address – Essentially, it ’ s important to take whatever actions we can think of to address **the** existential risks that affect **the** future of consciousness. CA : There ’ s a whole generation coming through who seem really sad about **the** future. What would you say to them? EM : Well, I think if you want **the** future to be good, you must make it so. Take action to make it good. And it will be. CA : Elon, thank you for all this time. That is a beautiful place to end. Thanks for all you ’ re doing. EM : You ’ re welcome.

Text Sample 70: POS Dim 1 – 2022 – Score 17.00 – 2022/t003676.txt

Here ’ s what gives me hope for humanity. I believe that we can change our human nature for **the** better. This chance is unique. It ’ s where we stand in history right now. No other generation has stood here before us. Homo sapiens has been around for about 300, 000 years. Other humanoid life forms have come and gone. We find their traces. We **search** for their stories. But we are **the** success story. We ’ ve survived natural disasters, famines, floods, **earthquakes**, plagues, woolly mammoths, mess-ups of our own making. We ’ re smart, no question. **The** question is : Are we smart enough to survive how smart we are? It ’ s not looking so good just now, is it? We stand facing **the** possibility of global conflict. If that happens, it will be our Third World War in not much more than 100 years. If Putin ’ s violence stops today, **the** problem doesn ’ t go away. As across **the** globe dictatorism threatens democracy. I wonder whether as a species we can survive any of this, even if those threats **disappear**. Those living inside **the** snow globe of their magical thinking will not be insulated against **the** facts of climate breakdown. And yet, still, we go on heating up **the planet**. Still we go on **polluting the** earth. Still we despoil our precious natural resources. 250 years ago, we kick-started **the** Industrial Revolution. **The** machine age. It ’ s when we first hear **the** buzzwords of **the** modern moment. Disruption, acceleration. That ’ s Karl Marx, actually. Acceleration of production, **the** factory system, acceleration of transport, **the** coming of **the** railways. No need for ships to wait for **the** wind. Coal-fired, steam-powered. Acceleration of information. **The** global village. And it ’ s when we start digging **fossil** fuels out of **the** ground in planet-changing quantities, when we start pushing **carbon** dioxide into **the** atmosphere. In a salami slice of space-time, human beings have changed **the** way that we live on **the planet** forever. We ’ ve moved out of **the** agricultural economies of our **evolutionary** inheritance into **the** industrial economies and beyond of where we are now. There ’ s no one else for us to turn to. There ’ s no one else for us to blame. There is no us and them. There ’ s only us. So will we continue to be **the** success story of **the** known **universe**, or are we writing our own obituary? **The** suicide species? Now I was raised in an evangelical household. We lived in end time. We were waiting for things to get so bad that Jesus would come back and save us. **The** apocalypse. It ’ s what **the** prepper communities are, well, prepping for. And it ’ s why super rich white guys are buying up tracts of land, hoping that they can live inside a kind of wi-fi-enabled Noah ’ s Ark. Well, we can have **the** end time if we want it. We know those stories from our religions, our sci-fi, our movies. We are uniquely placed to bring on **the** apocalypse. And we ’ re uniquely placed to save ourselves, too. If we could accept that as a species, Homo sapiens needs to **evolve** further. And that ’ s what gives me hope, because we have **the** means to **evolve** further. And that ’ s AI. Now look, I don ’ t mean **the geeks** will inherit **the** earth. I ’ m sorry, **geeks**. You won ’ t. I don ’ t really want to talk about narrow- goal **artificial** intelligence at all. That term of John McCarthy ’ s,”**artificial** intelligence,”is it any use to us now? I ’ d rather call it alternative intelligence. And I think humankind is in need of some alternative intelligence. In 1965, Jack Good talked about AI as our last invention. He meant superintelligence, **the** kind of thing that worries Bill Gates and Elon Musk. You know, **the**”Terminator”scenario. **The** final us and them. But I think that ’ s all to do with our doomster mindset. We don ’ t have to vote for **the** apocalypse. Jack Good worked at Bletchley Park with Alan Turing during World War II, building **the** early computing machinery that would crack **the** Nazi Enigma code. Now after **the** war, Turing, wrestling with **the** problems of a stored, programmed **computer** had a bigger dream on his mind. And in 1950, Alan Turing published a paper called”Computing Machinery and Intelligence.”And in there, there ’ s a chapter that ’ s titled”Lady Lovelace ’ s Objection”where Turing time-travels back 100 years to have a conversation with that long-dead genius, Ada Lovelace, **the** first person to write mathematical programs for **the computer** not yet built by her friend Charles Babbage. Now Ada wrote that as well as doing awesome stuff with numbers, **the computer** would, if correctly programmed, be able to write novels and compose music. That is a pretty big insight in 1843.”But,”said Ada,”**The computer** should never have any pretensions to originate anything.”She meant, think creatively. Well, Ada ’ s father was Lord Byron, England ’ s most **famous** poet, and England is **the** land of Shakespeare. More poets. And Ada had seen Charles Babbage ’ s kit all over **the** floor, and she wasn ’ t having some steampunk, coal-fired **nuts**, bolts, bezels, levers, gears, cogs and chains, crank-handled machine writing poetry.”Well,”said Alan Turing,”Was Lady Lovelace, correct? Could a **computer** ever be said to originate anything? And what would be **the** difference between computing intelligence and human intelligence?”Well, I ’ ll tell you one difference, and it ’ s optimistic. Computing power uses binary, but computing intelligence is nonbinary. It ’ s humans who are obsessed with false binaries. Male, female. Masculine, feminine. Black, white. Human, nonhuman. Us, them. AI has no skin color. AI has no race, no gender, no faith in a sky God. AI is not interested in men being superior to women, in white folks being smarter than people of color. Straight, gay, gay, trans are not separating categories for AI. AI does not distinguish between success and failure by gold bars, yachts and Ferraris. AI is not motivated by fame and fortune. If we develop alternative intelligence, it will be Buddhist in its non-needs. Now I am aware that **the** algorithms ubiquitous in everyday life are racist, sexist, gendered, trivializing, stoke division, amplify bias. But what is this teaching us about ourselves? AI is a tool. We are **the** ones who are using **the** tool. Hatred and contempt, money and power are human agendas, not AI agendas. We ’ ve been forced to recognize **the** paucity, **the** inadequacy of our data sets. And humans are trained on data sets too. We ’ ve had to recognized **the** unacknowledged ideologies that we live by every day. Rationality, neutrality, logic, objective decision-making. What can we say about any of that when we see what we are reflected back to us in **the** small screen? And it ’ s not a pretty sight. AI is not yet self-aware. But we are becoming more self-aware as we work with AI and we realize that Homo sapiens is no longer fit for purpose. We need a reboot. So what are we going to choose? Apocalypse or an alternative? I believe that humans have a strong future as a hybrid species as we start to merge with **the** biotechnology we ’ re creating, whether that ’ s nanobots in **the** bloodstream, monitoring our vital systems, whether it ’ s genetic editing, whether it ’ s 3D printing of bespoke body parts, whether it ’ s neural implants that will connect us directly to **the web** and to one another, insourcing information, enhancing our cognitive capacities. And if we manage to upload consciousness, I think that **the** shift from **the** transhuman to **the** posthuman world will seem natural, an **evolutionary** necessity. Why do I say that? I say that because for millennia, all human beings have been obsessed with **the** big question, **the** absurdity of death. We asked,”Do we have souls?”We watched **the** spirit wait to leave **the** body. We created **the** world ’ s first disruptive startup, **the** afterlife. Now a multinational company, with a vast VR real estate portfolio. A mansion in **the** sky, sir? Our earliest extant written narrative, **The** Epic of Gilgamesh, is a journey to discover if there is life after death. And what is life after death? It is **the** extension of **the** self beyond biological limits. Now I ’ m a writer, and I wonder, have we been telling **the** story **backwards**? Did we know we would always get here, capable of creating **the** kind of superintelligence that we said created us? We ’ re told were made in God ’ s image. God is immortal. God is not a biological entity. Since **the** 17th century, **the** Enlightenment, science and religion have parted company. And science said, all that God thinking, **the** afterlife stuff, it ’ s folly, It ’ s ignorance, it ’ s superstition. Suppose it was intuition. Suppose it was **the** only way we could talk about what we knew, a fundamental, deep truth that this is not **the** last word. This is not **the** end of **the** story. That we are not time-bound creatures caught in our bodies. That there is further to go. I ’ m fascinated that computing, science and religion, like parallel lines that do meet in space, are now asking **the** same question : Is consciousness obliged to materiality? Now... I accept that machine intelligence will challenge human intelligence. But **the** mythos of **the** world is built around a group of stories that showcase an encounter between a human and a nonhuman entity. Think of Jacob wrestling **the** angel. Think of Prometheus bringing fire down from **the** gods. In these encounters, both parties are changed, not always for **the** better. But it generally works out. We ’ ve been thinking about this stuff forever. It ’ s time that we created it. And we could have some fun stuff too. Who wants their own AI angel? Me. There ’ s a message in a bottle about this 200 years ago. When Ada Lovelace was busy getting born, her father, Lord Byron, was on holiday on Lake Geneva with his friend, **the** poet Percy Shelley, and Shelley ’ s wife, Mary Shelley. On a **wet** weekend with no internet Byron said –”Let ’ s write horror stories.”You know what happened. Out of that came **the** world ’ s most **famous** monster, Frankenstein. This is 1816, **the** start of **the** Industrial Revolution. Mary Shelley is just 19 years old. In that novel, there is an alternative intelligence made out of **the** body parts from **the** graveyard and electricity. An astonishing vision because electricity was not in any practical use at all and was hardly understood as a force. You know what happens. **The** monster is not named. Not educated. Is outcast by his panicky creator, Victor Frankenstein. And **the** whole thing ends in a chase across **the** Arctic ice towards a Götterdämmerung of death and destruction. **The** death wish that human beings are so drawn to, perhaps because it ’ s easier to give up than to carry on. Well, we ’ re **the** first generation who can read Mary Shelley ’ s novel in **the** right way, as a flare flung across time, because we too could create an alternative intelligence, not out of **the** body parts from **the** graveyard using electricity, but out of zeros and ones of code. And how is this going to end? Utopia or dystopia? It ’ s up to us. Endings are not set in stone. We change **the** story because we are **the** story. Now, Marvin Minsky called alternative intelligence our”mind children.”Could we as proud parents accept that **the** new generation that we will create will be smarter than we are? And could we accept that **the** new generation we create need not be on a substrate made of meat? Thank you. Thank you. Thank you very much. Thank you. Thank you. Helen Walters : Jeanette, stay right there. I have some questions. Jeanette Winterson : I know you want your lunch. Me too. HW : No. Everybody, stop. OK. That was amazing. Thank you. What would you say our odds were of survival? JW : Look – I ’ m an optimist. I ’ m a glass-half-full girl. So I know that we ’ re running out of time, time is **the** most precious resource we have, and there isn ’ t much of it. If we get this right soon, it can really work. If we get it wrong, will be fighting each other with sticks and stones for scraps of food and water on an overheated **planet** in **the** ruins of dictator-world. But – OK. But we could get it right. That ’ s why I feel we ’ ve arrived at this moment before, and it might **disappear** back into space-time. Then we ’ ll have to wait billions of years to get here again, which is so dull. So let ’ s not fuck it up. HW : I ’ ve got another question. I just wanted to have a moment of eye to eye. Don ’ t fuck it up. Don ’ t fuck it up. OK. JW : That ’ s **the** message. **The** whole TED Talk is :”Don ’ t fuck it up.”HW : But wait, I have another question. JW : I could have saved 12 minutes, 55 seconds. HW : Yeah, well, I ’ ll just be **the** title that we put online. Jeanette Winterson :”Don ’ t fuck it up.”OK. I want to talk about love. So in your memoir, which everybody should read, it is called”Why Be Happy When You Can Be Normal?”**The** very end of that book is about – it ’ s a good title – t ’ s about love. And you write,”Love, **the** difficult word. Where everything starts, where we always return. Love. Love ’ s lack. **The** possibility of love.”Now I don ’ t want to bastardize Ada Lovelace. She was all about originating. But what about love? What about alternative intelligence and love? JW : We ’ ll teach it. HW : We ’ ll teach it? JW : Yeah. And listen, any of you who ever fell in love with your teddy bear, which is all of you, know what it ’ s like to have an intense relationship with a nonbiological lifeform. HW : Jeanette Winterson, I love you. JW : Thank you. HW : Thank you so much. Thank you.

Text Sample 71: POS Dim 1 – 2022 – Score 16.00 – 2022/t003763.txt

Chris Anderson : I want you to come with me to Tesla ’ s huge gigafactory in Austin, Texas. So **the** day before it opened last week, **the** evening before, I was allowed to walk around it, no one else there. And what I saw there was, honestly, pretty mind-blowing. This is Elon Musk ’ s **famous** machine that builds **the** machine, and in his view, **the** secret to a sustainable future, is not just making electric car. It ’ s making a system that churns out huge numbers of electric cars with a margin, so that they can fund further growth. When I was there, none of us knew whether Elon would actually be able to make it here today, so I took **the** chance to sit down with him and record an epic interview. And I just want to show you an eight-minute excerpt of that interview. So here from Austin, Texas, Elon Musk. I want us to switch now to think a bit about **artificial** intelligence. I ’ m curious about your timelines and how you predict and how come some things are so amazingly on **the** money and some aren ’ t. So when it comes to predicting sales of Tesla vehicles, for example, you ’ ve kind of been amazing. I think in 2014 when Tesla had sold that year 60, 000 cars, you said,”2020, I think we will do half a million a year.”Elon Musk : Yeah, we did almost exactly a half million. CA : Five years ago, last time you came to TED, I asked you about full self-driving, and you said,”Yeah, this very year I ’ m confident that we will have a car going from LA to New York without any intervention.”EM : Yeah, I don ’ t want to blow your mind, but I ’ m not always right. CA : What ’ s **the** difference between those two? Why has full self-driving in particular been so hard to predict? EM : I mean, **the** thing that really got me, and I think it ’ s going to get a lot of other people, is that there are just so many false dawns with self-driving where you think you ' ve got **the** problem, have a handle on **the** problem, and then it, no, turns out you just hit a ceiling. Because if you were to plot **the** progress, **the** progress looks like a log curve. So it ' s like a series of log curves. So most people don ' t know what a log curve is, I suppose. CA : Show **the** shape with your hands. EM : It goes up, you know, sort of a fairly straight way, and then it starts tailing off and you start getting diminishing returns. You know, in retrospect, they seem obvious, but in order to solve full self-driving properly, you actually have to solve real-world AI. Because what are **the** road networks designed to work with? They ' re designed to work with a biological neural net, our brains, and with vision, our eyes. And so in order to make it work with **computers**, you basically need to solve real-world AI and vision. Because we need cameras and silicon neural nets in order to have self-driving work for a system that was designed for eyes and biological neural nets. You know, I guess when you put it that way, it ' s sort of, like, quite obvious that **the** only way to solve full self-driving is to solve real-world AI and sophisticated vision. CA : What do you feel about **the** current architecture? Do you think you have an architecture now where there is a chance for **the** logarithmic curve not to tail off anytime soon? EM : Well I mean, admittedly these maybe infamous last words, but I actually am confident that we will solve it this year. That we will exceed – **The probability** of an accident, at what point do you exceed that of **the** average person? I think we will exceed that this year. We could be here talking again in a year, like, well, another year went by and it didn ' t happen. But I think this is **the** year. CA : Is there an element that you actually deliberately make aggressive prediction timelines to drive people to be ambitious? Without that, nothing gets done? So it feels like at some point in **the** last year, seeing **the** progress on understanding that **the** Tesla AI understanding **the** world around it, led to a kind of, an aha moment at Tesla. Because you really surprised people recently when you said probably **the** most important product development going on at Tesla this year is this robot, Optimus. Is it something that happened in **the** development of full self-driving that gave you **the** confidence to say,"You know what, we could do something special here."EM : Yeah, exactly. So, you know, it took me a while to sort of realize that in order to solve self-driving, you really needed to solve real-world AI. And at **the** point of which you solve real-world AI for a car, which is really a robot on four wheels, you can then generalize that to a robot on legs as well. **The** things that are currently missing are enough intelligence for **the** robot to navigate **the** real world and do useful things without being explicitly instructed. So **the** missing things are basically real-world intelligence and scaling up manufacturing. Those are two things that Tesla is very good at. And so then we basically just need to design **the** specialized actuators and sensors that are needed for a humanoid robot. People have no idea, this is going to be bigger than **the** car. CA : So talk about – I think **the** first applications you ' ve mentioned are probably going to be in manufacturing, but eventually, **the** vision is to have these available for people at home. If you had a robot that really understood **the** 3D architecture of your house and knew where every **object** in that house was or was supposed to be and could recognize all those **objects**, I mean, that ' s kind of amazing, isn ' t it, like, **the** kind of thing that you could ask a robot to do would be what? Like, tidy up? EM : Yeah, absolutely. Make dinner, I guess, mow **the** lawn. CA : Take a cup of tea to grandma and show her family pictures. EM : Exactly. Take care of my grandmother and make sure – CA : It could obviously recognize everyone in **the** home. It could play catch with your kids. EM : Yes. I mean, obviously, we need to be careful this doesn ' t become a dystopian situation. I think one of **the** things that ' s going to be important is to have a localized ROM chip on **the** robot that cannot be updated over **the** air. Where if you, for example, were to say,"Stop, stop, stop,"if anyone said that, then **the** robot would stop, you know, type of thing. And that ' s not updatable remotely. I think it ' s going to be important to have safety features like that. CA : Yeah, that sounds wise. EM : And I do think there should be a regulatory agency for AI. I ' ve said this for many years. I don ' t love being regulated, but I think this is an important thing for public safety. CA : Do you think there will be basically like in, say, 2050 or whatever, like, a robot in most homes, is what there will be, and people will love them and count on them? You ' ll have your own butler basically. EM : Yeah, you ' ll have your sort of buddy robot probably, yeah. CA : I mean, how much of a buddy? How many applications have you thought, you know, can you have a romantic partner, a sex partner? EM : It ' s probably inevitable. I mean, I did promise **the** internet that I ' d make catgirls. We could make a robot catgirl. CA : Be careful what you promise **the** internet. EM : So, yeah, I guess it ' ll be whatever people want really, you know. CA : What sort of timeline should we be thinking about of **the** first models that are actually made and sold? EM : Well, you know, **the** first units that we intend to make are for jobs that are dangerous, boring, repetitive and things that people don ' t want to do. And, you know, I think we ' ll have like, an interesting prototype some time this year. We might have something useful next year, but I think quite likely within at least two years. And then we ' ll see rapid growth year over year of **the** usefulness of **the** humanoid robots and decrease in cost and scaling up production. CA : Help me on **the** economics of this. So what do you picture **the** cost of one of these being? EM : Well, I think **the** cost is actually not going to be crazy high. Like, less than a car. CA : But think about **the** economics of this. If you can replace a 30, 000-dollar, 40, 000-dollar-a-year worker, which you have to pay every year, with a one-time payment of 25, 000 dollars for a robot that can work longer hours, doesn ' t go an vacation, it could be a pretty rapid replacement of certain types of jobs. How worried should **the** world be about that? EM : I wouldn ' t worry about **the** sort of, putting people out of a job thing. I think we ' re actually going to have, and already do have, a massive shortage of labor. So I think we will have... Not people out of work, but actually still a shortage of labor even in **the** future. But this really will be a world of abundance. Any goods and services will be available to anyone who wants them. It ' ll be so cheap to have goods and services, it will be ridiculous. CA : And now, arguably, **the** biggest visionary of them all, Elon Musk. Hey, Elon, welcome. So, Elon, a few hours ago, you made an offer to buy Twitter. Why? EM : How ' d you know? CA : A little bird tweeted in my ear or something, I don ' t know. EM : By **the** way, have you seen **the** movie"Ted,"about **the** bear? CA : I have. EM : It ' s a good movie. CA : Don ' t mention that here. EM : So, yeah, yeah, so... Was there a question? CA : Why make that offer? EM : Oh, so, well, I think it ' s very important for there to be an inclusive arena for free speech where all... Twitter has become, kind of, **the** de facto town square. It ' s just really important that people have both **the** reality and **the** perception that they are able to speak freely within **the** bounds of **the** law. And, you know, one of **the** things that I believe Twitter should do is **open-source the** algorithm and make any changes to people ' s tweets, or if they ' re emphasized or de-emphasized, that action should be made apparent so anyone can see that action has been taken. So there ' s no, sort of, behind **the** scenes manipulation, either algorithmically or manually. CA : Last week when we spoke, Elon, I asked you whether you were thinking of taking over. You said, no way. You said,"I do not want to own Twitter. It is a recipe for misery. Everyone will blame me for everything."What on Earth changed? EM : No I think everyone will still blame me for everything. If I acquire Twitter and something goes wrong, it ' s my fault, 100 percent. I think there will be quite a few arrows, yes. CA : It will be miserable. But you still want to do it. Why? EM : I mean, I hope it ’ s not too miserable. I just think it ’ s important to **the**... It ’ s important to **the** function of democracy. It ’ s important to **the** function of **the** United States as a free country and many other countries. And to actually help freedom in **the** world, more broadly than **the** US. And so I think it ’ s, you know, I think **the** civilizational risk is decreased **the** more we can increase **the** trust of Twitter as a public platform. And so I do think this will be somewhat painful. And I ’ m not sure that I will actually be able to acquire it. And I should also say **the** intent is to retain as many shareholders as is allowed by **the** law in a private company, which I think is around 2, 000 or so. So it ’ s definitely not, from **the** standpoint of,”Let me figure out how to monopolize or maximize my ownership of Twitter.”But I will try to bring along as many shareholders as we ’ re allowed to. CA : You don ’ t necessarily want to pay out 40 or whatever it is, billion dollars in cash. You ’ d like them to come with you in **the** new company. EM : I mean, you know, I could technically afford it. CA : I heard that. EM : But what I ’ m saying is, this is not a way to, sort of, make money, you know. It ’ s just that I think this is... My strong intuitive sense is that having a public platform that is maximally trusted and broadly inclusive is extremely important to **the** future of civilization. CA : But you ’ ve described yourself – EM : I don ’ t care about **the** economics at all. CA : OK, that ’ s cool to hear. This is not about **the** economics. It ’ s for **the** moral good that you think it will achieve. You ’ ve described yourself, Elon, as a”free speech absolutist.”But does that mean that there ’ s literally nothing that people can ’ t say, and it ’ s OK? EM : Well, I think, obviously, Twitter or any forum is bound by **the** laws of **the** country that it operates in. So obviously, there are some limitations on free speech in **the** US. And of course, Twitter would have to abide by those rules. CA : So you can ’ t incite people to violence, like, a direct incitement to violence you know, you can ’ t do **the** equivalent of crying”fire”in a movie theater, for example. EM : No, that would be a crime. It should be a crime. CA : But here ’ s **the** challenge, it ’ s such a nuanced difference between different things. So there ’ s incitement to violence. That ’ s a no if it ’ s illegal. There ’ s hate speech, which, some forms of hate speech are fine. You know,”I hate spinach.”EM : I mean, if it ’ s sauteed in, you know, cream sauce, it can be quite nice. CA : But **the** problem is, so let ’ s say someone says – OK, here ’ s one tweet,”I hate politician X.”Next tweet is,”I wish politician X wasn ’ t alive,”as some of us have said about Putin right now, for example. So that ’ s legitimate speech. Another tweet is,”I wish politician X wasn ’ t alive”with a picture of their head with a gunsight over it. Or that plus their address. I mean, at some point someone has to make a decision as to which of those is not OK. Can an algorithm do that, or surely you need human judgment at some point? EM : No, I think, like I said, in my view, Twitter should match **the** laws of **the** country. And really, you know, there ’ s an obligation to do that. But going beyond that, and having it be **unclear** who ’ s making what changes to where, having tweets sort of, mysteriously be promoted and demoted with no insight into what ’ s going on, having a black-box algorithm promote some things and not other things, I think this can be quite dangerous. CA : So, **the** idea of opening **the** algorithm is a huge deal, and I think many people would welcome that of understanding exactly how it ’ s making **the** decision. EM : And critique it, like, what I mean is like, I think **the** code should be on GitHub, you know. So people can look through it and say, like,”I see a problem here.””I don ’ t agree with this.”They can highlight issues, suggest changes in **the** same way that you update Linux or Signal or something like that, you know. CA : As I understand it, at some point right now, what **the** algorithm would do is it would look at, for example, how many people have flagged a tweet as obnoxious. And then at some point a human has to look at it and make a decision as to, does this cross **the** line or not, that **the** algorithm itself can ' t, I don ' t think yet, tell **the** difference between legal and OK and definitely obnoxious. And so **the** question is, which humans, you know, make that call? I mean, do you have a picture of that? Right now, Twitter and Facebook and others, you know, they ' ve hired thousands of people to try to help make wise decisions. And **the** trouble is that no one can agree on what is wise. How do you solve that? EM : Well I think we would want to err on **the** side – if in doubt, let **the** speech exist. If it ' s, you know, a gray area, I would say let **the** tweet exist. But obviously, you know, in a case where there ' s perhaps a lot of controversy that you would not want to necessarily promote that tweet. So I ' m not saying that I have all **the** answers here. But I do think that we want to be just very reluctant to delete things and just be very cautious with permanent bans. You know, time-outs, I think, are better than, sort of, permanent bans. And... But just in general, like I said... It won ' t be perfect, but I think we want it to really have **the** perception and reality that speech is as free as reasonably possible. And a good sign as to whether there ' s free speech is : is someone you don ' t like allowed to say something you don ' t like? And if that is **the** case, then we have free speech. And it ' s damn annoying when someone you don ' t like says something you don ' t like. That is a sign of a healthy, functioning, free speech situation. CA : I think many people would agree with that. And looking at **the** reaction online, many people are excited by you coming in and **the** changes you ' re proposing. Some others are absolutely horrified. Here ' s how they would see it. They would say,"Wait a sec. We agree that Twitter is an incredibly important town square. It is where **the** world exchanges opinion about life and death matters. How on Earth could it be owned by **the** world ' s richest person? That can ' t be right."So what ' s **the** response there? Is there any way that you can distance yourself from **the** actual decision making that matters on content in some very clear way that is convincing to people? EM : Well, like I said, I think it ' s very important that **the** algorithm be open-sourced and that any manual adjustments be identified. So if somebody did something to a tweet, there ' s information attached to it that action was taken. And I won ' t personally be, you know, in there editing tweets. But you ' ll know if something was done to promote, demote or otherwise affect a tweet. You know, as for media sort of ownership, I mean, you ' ve got, you know, Mark Zuckerberg owning Facebook and Instagram and WhatsApp and with a share ownership structure that will have Mark Zuckerberg XIV still controlling those entities. Like, literally. We won ' t have that at Twitter. CA : If you commit to opening up **the** algorithm that definitely gives some level of confidence. Talk about some of **the** other changes that you ' ve proposed. **The** edit button, that ' s definitely coming if you have your way. EM : Yeah. I think, I mean, frankly... **The** top priority I would have is eliminating **the** spam and scam bots and **the** bot armies that are on Twitter. You know, I think these influence – they make **the** product much worse. You know, if I had a Dogecoin for every crypto scam I saw. I would have 100 billion Dogecoin. CA : Do you regret sparking a sort of, storm of excitement over DOGE and, you know, where it ' s gone? EM : I mean, I think DOGE is fun. And, you know, I ' ve always said don ' t bet **the** farm on Dogecoin, FYI. But I think it ' s, I like dogs and I like memes, and it ' s got both of those. CA : But just on **the** edit button, how do you get around **the** problem of, so someone tweets,"Elon **rocks**"and it ' s tweeted by two million people. And then after that, they edit it"Elon sucks,"and then all those retweets they ' re all embarrassed. How do you avoid that type of changing of meaning so that retweeters are exploited? EM : Well, I think you ' d only have **the** edit capability for a short period of time. And probably **the** thing to do upon **the** edit would be to zero out all retweets and favorites. CA :

OK. EM : I ’ m open to ideas though, you know. CA : So in one way, **the** algorithm works kind of well for you right now, I wanted to show you this. This is a typical tweet of mine, kind of lame and wordy and whatever. And **the** amazing response it gets is this, oh, my God, 97 likes. And then I tried another one. [I get to interview @elonmusk on Thursday here at TED. What questions would YOU ask him?] 29, 000 likes. So **the** algorithm, at least seems to be, at **the** moment, you know :”If, expand to **the** world immediately.”Not bad, right? EM : Yeah, I guess so, I mean, that is cool. CA : So help us understand how it is you ’ ve built this incredible following on Twitter yourself when some of **the** people who love you **the** most look at some of what you tweet and they think it ’ s somewhere between embarrassing and crazy, some of it ’ s amazing. EM : It is a little... CA : But is that actually why it ’ s worked, why has it worked? EM : I mean, I don ’ t know. I ’ m tweeting more or less stream of consciousness, you know. It ’ s not like, let me think about some grand plan about my Twitter or whatever. You know, I ’ m, like, literally, on **the** toilet or something, like,”Ha ha, this is funny.”And then tweet that out. That ’ s like most of them. That ’ s, you know, oversharing. CA : But you are obsessed with getting **the** most out of every minute of your day. And so why not? EM : So, I don ’ t know. I try to tweet out things that are interesting or funny. And then people seem to like it. CA : So if you are unsuccessful – actually, before I ask that, let me ask this. So how can I say, is funding secured? EM : I have sufficient assets to complete... This is not a forward-looking statement, blah blah blah. I mean, I can do it if possible. I should say, actually even originally with Tesla back in **the** day, funding was actually secured. I want to be clear about that. In fact, this may be a good opportunity to clarify that. Funding was indeed secured, and I should say, like, why do I not have respect for **the** SEC in that situation? And I don ’ t mean to blame everyone at **the** SEC, but certainly **the** San Francisco office, it ’ s because **the** SEC knew that funding was secured, but they pursued an active public investigation nonetheless. At **the** time, Tesla was in a precarious financial situation, and I was told by **the** banks that if I did not agree to settle with **the** SEC that **the** banks would cease providing working capital and Tesla would go bankrupt immediately. So that ’ s like having a gun to your child ’ s head. So I was forced to concede to **the** SEC unlawfully. Those bastards. And now it makes it look like I lied when I did not, in fact, lie. I was forced to admit that I lied to save Tesla ’ s life. And that ’ s **the** only reason. CA : Given what ’ s actually happened – Given what ’ s actually happened to Tesla since then, though, aren ’ t you glad that you didn ’ t take it private? EM : Yeah. I mean... It ’ s difficult to put yourself in **the** position at **the** time, Tesla was under **the** most relentless short seller attack in **the** history of **the** stock market. There ’ s something called”short and distort,”where **the** barrage of negativity that Tesla was experiencing from short sellers on Wall Street was beyond all belief. Tesla was **the** most shorted stock in **the** history of stock markets. This is saying something. So, you know, this was affecting our ability to hire people, it was affecting our ability to sell cars, it was... Yeah, it was terrible. They wanted Tesla to die so bad, they could taste it. CA : Well, most of them have paid **the** price. EM : Yes. Where are they now? CA : So that was a very strong statement. Obviously, a lot of people who support you, I would ’ ve thought, would say, you have so much to offer **the** world on **the** upside, on **the** vision side, don ’ t waste your time getting distracted by these battles that bring out negativity and make people feel that you ’ re being defensive. People don ’ t like fights, especially with powerful government authorities. They ’ d rather buy into your dream. Aren ’ t you encouraged by people just to edit that temptation out and go with **the** bigger story? EM : Well, I mean, I would say, like, you know, I ’ m somewhat of a mixed bag, you know. CA : Well, you ’ re a fighter and you don ’ t... You don ’ t like to lose and you are determined that you don ’ t. Basically, I mean. EM : Sure I don ’ t like to lose, I ’ m not sure many people do. But **the** truth matters to me a lot. Sort of, pathologically, it matters to me. CA : OK. So you don ’ t like to lose, if in this case you are not successful in, you know, **the** board does not accept your offer, you ’ ve said you won ’ t go higher. Is there a plan B? EM : There is. CA : I think we would like to hear a little bit about plan B. EM : For another time, I think. CA : Another time, alright. I... That ’ s a nice tease. Alright. I would love to try to understand this brain of yours more, Elon. With your permission, I ’ d like to just play this. Oh, actually, before we do that, here was one of **the** thousands of questions that people asked. I thought this was actually quite a good one.”If you could go back in time and change one decision you made along **the** way”– do your own edit button –”which one would it be and why?”EM : Do you mean like a career decision or something? CA : Just any decision over **the** last few years, like your decision to invest in Twitter in **the** first place or your... Anything. EM : I mean, **the** worst business decision I ever made was not starting Tesla with just JB Straubel. By far **the** worst decision I ’ ve ever made is not just starting Tesla with JB. That ’ s number one by far. CA : Alright, so JB Straubel was **the** visionary cofounder who was obsessed with and knew so much about batteries. And your decision to go with Tesla, **the** company as it was, meant that you got locked into what you **concluded** was a weird architecture now. EM : There ’ s a lot of confusion. Tesla did not exist in any... Tesla was a shell company with no employees, no intellectual property, when I invested. But a false narrative has been created by one of **the** other cofounders, Martin Eberhard. And I don ’ t want to get into **the** nastiness here, but I didn ’ t invest in an existing company. We created a company. And ultimately, **the** creation of that company was done by JB and me. And unfortunately, there ’ s someone else, another cofounder who has made it his life ’ s mission to make it sound like he created **the** company, which is false. CA : Wasn ’ t there another issue right at **the** heart of **the** development of **the** Tesla Model 3, where Tesla almost went bankrupt? And I think you have said that part of **the** reason for that was that you overestimated **the** extent to which it was possible, at that time, to automate a factory. Huge amount was spent kind of over-automating, and it didn ’ t work, and it nearly took **the** company down. Is that fair? EM : I mean, first of all, it ’ s important to understand, like, what has Tesla actually accomplished that is most noteworthy. It is not **the** creation of an electric vehicle or creating an electric vehicle prototype or low-volume production of a car. There have been hundreds of car start-ups over **the** years, hundreds. And in fact, at one point, Bloomberg counted up **the** number of electric vehicle start-ups, and I think they got to almost 500. So **the** hard part is not creating a prototype or going into limited production. **The** absolutely difficult thing, which has not been accomplished by an American car company in 100 years, is reaching volume production without going bankrupt. That is **the** actual hard thing. **The** last company, American company, to reach volume production without going bankrupt, was Chrysler in **the** ’ 20s. CA : And it nearly happened to Tesla. EM : Yes, but it ’ s not like, oh, geez, I guess if we ’ d just done more manual stuff, things would have been fine. Of course not. That is definitely not **the** case. So we basically messed up almost every aspect of **the** Model 3 production line, from cells to packs to drive inverters, motors, body line, **the** paint shop, final assembly, everything. Everything was messed up. I lived in **the** Fremont and Nevada factories for three years, fixing that production line, running around like a maniac through every part of that factory. Living with **the** team. I slept on **the** floor, so **the** team, who was going through a hard time, could see me on **the** floor. That they knew that I was not in some ivory tower. When whatever pain they experienced, I had it more.

CA : And some people who knew you well actually thought you were making a terrible mistake, that you were driving yourself to **the** edge of sanity, almost. And that you were in danger of making bad choices. And in fact, I heard you say last week, Elon, that you, because of Tesla ' s huge value now and, you know, **the** significance of every minute that you spend, that you are in danger of sort of, obsessing over it, spending all this time to **the** edge of sanity. That doesn ' t sound super wise. Isn ' t there... Your time, your completely sane, centered, rested time and decision making is more powerful and compelling than that sort of,"I can barely hold my eyes open."So surely it should be an absolute strategic priority to look after yourself. EM : I mean, there wasn ' t any other way to make it work. They were three years of hell. 2017, 2018 and 2019 were **the** longest period of excruciating pain in my life. There wasn ' t any other way, and we barely made it. And we were on **the** ragged edge of bankruptcy **the** entire time. It ' s not like I want **the** pain, I don ' t like it. Those were three – So, so, so much pain. But it had to be done or Tesla would be dead. CA : When you looked around **the** Gigafactory that we saw images of earlier last week and just see where **the** company ' s come, do you feel that this challenge of figuring out **the** new way of manufacturing, that you actually have an edge now, that it ' s different, that you ' ve figured out how to do this. And that those three years, won ' t be repeated? You ' ve actually figured out a new way of manufacturing. EM : At this point, I think I know more about manufacturing than anyone currently alive on Earth. I can tell you how every damn part of that car is made. It ' s basically if you just live on **the** factory, you live in **the** factory for three years and – CA : That was nice. EM : Poignant note to something. CA : Someone wants to compose a symphony to that expression of confidence, something like that. I have no idea what that is. EM : Anyway, yeah. Every aspect of a car, six ways to Sunday, I know it. CA : I mean, you talk about scale, right. You ' re in **the** middle of writing your new master plan, and you ' ve said that scale is at **the** heart of it. Why does scale matter? Why are you obsessed with it, what are you thinking? EM : Yeah, well, see, in order to accelerate **the** advent of sustainable energy, there must be scale. Because we ' ve got to transition a vast economy that is currently overly dependent on **fossil** fuels to a sustainable energy economy. One where **the** energy is – Yeah, I mean, we ' ve got to do it. So **the** energy ' s got to be sustainably generated with wind, solar, hydro, geothermal, I ' m a believer in nuclear as Isabelle [Boemeke] gave a talk about. And then since solar and wind is intermittent, you have to have stationary storage batteries, and then we ' ve got to transition all transport to electric. If we do those things, we have a sustainable energy future. **The** faster we do those things, **the** less risk we put to **the** environment. So sooner is better. And so scale is very important. You know, it ' s not about fresh releases. It ' s about tonnage. What was **the** tonnage of batteries produced and obviously done in a sustainable way? And our estimate is that approximately 300 terawatt hours of battery storage is needed to transition transport, electricity and heating and cooling to a fully electric situation. There may be some different estimates out there, but our estimate is 300 terawatt hours. CA : So we dug into this a lot in **the** interview that we recorded last week, so people can go in and hear that more. But I mean, **the** context is, that is, I think about 1, 000 times **the** current installed battery capacity. I mean, **the** scale up needed is breathtaking, basically. And... So your vision is to commit Tesla to try to deliver on a meaningful percentage of what is needed. And call on others to do **the** rest. This is a task for humanity to massively scale up our response to change **the** energy grid. EM : Yes. It ' s like basically how fast can we scale and encourage others to scale to get to that 300-terawatt-hour installed base of batteries? CA : Right. EM : And then, of course, there ' ll be a tremendous need to recycle those batteries, and it makes sense to recycle them because **the** raw materials are like high grade ore. People shouldn ' t think there will be this big pile of batteries, they can get recycled, because even a dead battery pack is worth about 1, 000 dollars. But this is what ' s needed for a sustainable energy future. So we ' re going to try to take a set of actions that accelerate **the** day and bring **the** day of a sustainable energy future sooner. CA : OK. There ' s going to be a huge interest in your master plan when you publish that. Meanwhile, I just would love to understand more what goes on in this brain of yours, because it is a pretty unique one. I want to play, with your permission, this very funny opening from SNL, Saturday Night Live. Thank you. Thank you very much. It ' s an honor to be hosting Saturday Night Live. I mean that. Sometimes after I say something, I have to say,"I mean that."So people really know that I mean it. That ' s because I don ' t always have a lot of intonational variation in how I speak. Which, I ' m told, makes for great comedy. I ' m actually making history tonight as **the** first person with Asperger ' s to host SNL. CA : And I think you followed that up with – EM : At least **the** first person to admit it. CA : **The** first person to admit it. So this was a brave thing to say. But I would love to understand you know, how you think of Asperger ' s, like, whether you can give us any sense of, even you as a boy, what **the** experience was or as you now understand, with **the** benefit of hindsight. Can you talk about that a bit? EM : Well, I think everyone ' s experience is going to be somewhat different. But I guess for me, **the** social cues were not intuitive. So I was just very bookish, and I didn ' t understand... I guess others could sort of, intuitively understand... What was meant by something. I would just tend to take things very literally, like, **the** words, as spoken, were exactly what they meant. But then that turned out to be wrong. They ' re not simply saying exactly what they mean. There ' s all sorts of other things that are meant. It took me a while to figure that out. I was, you know, bullied quite a lot. So... I did not have a sort of, happy childhood, to be frank. It was quite, quite rough. But I read a lot of books. I read lots and lots of books. And, you know, sort of, gradually understood more from **the** books that I was reading and watched a lot of movies and... You know, just... But it took me a while to understand things that most people intuitively understand. CA : So I ' ve wondered whether it ' s possible that that was, in a strange way, an incredible gift to you and indirectly to many other people. Inasmuch as... Brain ' s, you know, are plastic. And they go where **the** action is. And if for some reason **the** external world and social cues, which so many people spend so much time and mental energy obsessing over, if that is partly cut off, isn ' t it possible that that is partly what gave you **the** ability to understand inwardly **the** world at a much deeper level than most people do? EM : I suppose that ' s certainly possible. I think there ' s maybe some value also from a technology standpoint, because I found it rewarding to spend all night programming **computers** just by myself. And I think most people, most people don ' t **enjoy** typing strange symbols into a **computer** by themselves all night. They think that ' s not fun. But I thought it was, I really liked it. So I would just program all night by myself. And I found that to be quite enjoyable. But I think that is not normal. CA : I ' ve thought a lot about... It ' s a riddle to a lot of people of how you ' ve done this, how you ' ve repeatedly innovated in these different industries. And you know, every entrepreneur sees possibility in **the** future and then acts to make that real. It feels to me like you see possibility just more broadly than almost anyone and can connect **the** dots. So you see scientific possibility based on a deep understanding of physics and knowing what **the** fundamental equations are, what **the** technologies are that are based on that science and where they could go, you see technological possibility. And then really unusually, you combine that with economic possibility, of like what it actually would cost, is there a system you can imagine where you could affordably make that thing, and that... sometimes you then get conviction that there is an opportunity here. Put those pieces together and you could do something amazing. EM : Yeah, I think one aspect of whatever condition I had was I was just absolutely obsessed with truth. Just obsessed with truth. And so **the** obsession with truth is why I studied physics, because physics attempts to understand **the** truth, **the** truth of **the universe**. Physics is just, what are **the** provable truths of **the universe**. And truths that have predictive power. So for me, physics was sort of, a very natural thing to study. Nobody made me study it. I was intrinsically interested to understand **the** nature of **the universe**. And then **computer** science or information theory also to just understand logic. And, you know, there ' s an argument that... That information theory is actually operating at a more fundamental level, more fundamental level than even physics. So just yeah... Physics and information theory were really interesting to me. CA : So when you say truth, I mean, like, some people... What you ' re talking about is **the** truth of **the universe**. **The** fundamental truths that drive **the universe**, it ' s like, a deep curiosity about what this **universe** is, why we ' re here, simulation, we don ' t have time to go into that. But I mean, you ' re just deeply, deeply curious about, what this is for, what this is, this whole thing. EM : Yes, I think **the** why of things is very important. When I was in young teens, I got quite depressed about **the** meaning of life. And I was trying to sort of, understand **the** meaning of life. I ' m looking at, reading religious texts and reading books on philosophy. And I got into **the** German philosophers, which is definitely not wise if you ' re a young teenager, I have to say. It can be a bit dark. Much better read as an adult. And then actually, I ended up reading"**The** Hitchhiker ' s Guide to **the** Galaxy,"which is actually a book on philosophy just sort of disguised as a silly humor book. But it ' s actually a philosophy book. And Adams makes **the** point that it ' s actually **the** question that is harder than **the** answer. You know, he sort of makes a joke that **the** answer is 42. That number does pop up a lot. And 420 is just 10 times 42. CA : Ten times more significant than 42. EM : But, you know, you can make a **triangle** with 42 degrees and two 69s. So there ' s no such thing as a perfect **triangle**, or is there? CA : But even more important than **the** answer is **the** question. That was **the** whole **theme** of that book. I mean, is that basically how you see meaning then, that it ' s **the** pursuit of questions? EM : So I have a sort of, a proposal for a worldview or a motivating philosophy, which is to understand what questions to ask about **the** answer that is **the universe**. And to **the** degree that we expand **the** scope and scale of consciousness, biological and digital, we will be better able to ask these questions, to frame these questions and to understand why we ' re here, how we got here, what **the** heck is going on. And so that is my driving philosophy, is to expand **the** scope and scale of consciousness that we may better understand **the** nature of **the universe**. CA : Elon, one of **the** things that was most touching last week – was seeing you hang out with your kids. Here ' s, if I may... EM : He looks vaguely like a ventriloquist dummy there. I mean, how do you know that ' s real? CA : So that ' s X, and it was just a delight seeing you hang out with him. What ' s his future going to be? I mean, I don ' t mean him personally, but **the** world he ' s going to grow up in. What future do you believe he will grow up in? EM : Well, I mean, a very digital future. A very different world than I grew up in, that ' s for sure. But I think we want to obviously do our absolute best to ensure that **the** future is good for everyone ' s children. And that, you know, that **the** future is something that you can look forward to and not feel sad about. You know, you want to get up in **the** morning and be excited about **the** future. And we should fight for **the** things that make us excited about **the** future. You know, **the** future cannot just be about one miserable thing after another, solving one sad problem after another. There have got to be things that get you excited. Like, you want to live. These things are very important. You should have more of it. CA : And it ’ s not as if it ’ s a done deal, like it ’ s all to play for. Like, **the** future may be horrible. Still, there are scenarios where it is horrible, but you see a pathway to an exciting future both on Earth and on Mars and in our minds through **artificial** intelligence and so forth. I mean, in your heart of hearts, do you really believe that you are helping deliver that exciting future for X and for others? EM : I mean, I ’ m trying my hardest to do so. I, you know, I love humanity, and I think that we should fight for a good future for humanity. And I think we should be optimistic about **the** future and fight to make that optimistic future happen. CA : Elon, I think that ’ s **the** perfect place to close this. Thank you so much for spending time coming here and for **the** work that you ’ re doing. And good luck with finding a wise course through on Twitter and everything else. EM : Alright, thank you.

Text Sample 72: POS Dim 1 – 2020 – Score 32.00 – 2020/t004244.txt
Chris Anderson : Al, welcome. So look, just six months ago – it seems a lifetime ago, but it really was just six months ago – climate seemed to be on **the** lips of every thinking person on **the planet**. Recent events seem to have swept it all away from our attention. How worried are you about that? Al Gore : Well, first of all Chris, thank you so much for inviting me to have this conversation. People are reacting differently to **the** climate crisis in **the** midst of these other great challenges that have taken over our awareness, appropriately. One reason is something that you mentioned. People get **the** fact that when scientists are warning us in ever more dire terms and setting their hair on fire, so to speak, it ’ s best to listen to what they ’ re saying, and I think that lesson has begun to sink in in a new way. Another similarity, by **the** way, is that **the** climate crisis, like **the** COVID-19 pandemic, has revealed in a new way **the** shocking injustices and inequalities and disparities that affect communities of color and low-income communities. There are differences. **The** climate crisis has effects that are not measured in years, as **the** pandemic is, but consequences that are measured in centuries and even longer. And **the** other difference is that instead of depressing economic activity to deal with **the** climate crisis, as nations around **the** world have had to do with COVID-19, we have **the** opportunity to create tens of millions of new jobs. That sounds like a political phrasing, but it ’ s literally true. For **the** last five years, **the** fastest-growing job in **the** US has been solar installer. **The** second-fastest has been wind **turbine** technician. And **the**”Oxford Review of Economics,”just a few weeks ago, pointed **the** way to a very jobs-rich recovery if we emphasize **renewable** energy and **sustainability** technology. So I think we are crossing a tipping point, and you need only look at **the** recovery plans that are being presented in nations around **the** world to see that they ’ re very much focused on a green recovery. CA : I mean, one obvious impact of **the** pandemic is that it ’ s brought **the** world ’ s economy to a shuddering halt, thereby reducing **greenhouse** gas **emissions**. I mean, how big an effect has that been, and is it unambiguously good news? AG : Well, it ’ s a little bit of an illusion, Chris, and you need only look back to **the** Great Recession in 2008 and ’ 09, when there was a one percent decline in **emissions**, but then in 2010, they came roaring back during **the** recovery with a four percent increase. **The** latest estimates are that **emissions** will go down by at least five percent during this induced coma, as **the** economist Paul Krugman perceptively described it, but whether it goes back **the** way it did after **the** Great Recession is in part up to us, and if these green recovery

plans are actually implemented, and I know many countries are determined to implement them, then we need not repeat that pattern. After all, this whole process is **occurring** during a period when **the** cost of **renewable** energy and electric vehicles, batteries and a range of other **sustainability** approaches are continuing to fall in price, and they ' re becoming much more competitive. Just a quick reference to how fast this is : five years ago, electricity from solar and wind was cheaper than electricity from **fossil** fuels in only one percent of **the** world. This year, it ' s cheaper in **two-thirds** of **the** world, and five years from now, it will be cheaper in virtually 100 percent of **the** world. EVs will be cost-competitive within two years, and then will continue falling in price. And so there are changes underway that could interrupt **the** pattern we saw after **the** Great Recession. CA : **The** reason those pricing differentials happen in different parts of **the** world is obviously because there ' s different amounts of sunshine and wind there and different building costs and so forth. AG : Well, yes, and government policies also account for a lot. **The** world is continuing to subsidize **fossil** fuels at a ridiculous amount, more so in many developing countries than in **the** US and developed countries, but it ' s subsidized here as well. But everywhere in **the** world, wind and solar will be cheaper as a source of electricity than **fossil** fuels, within a few years. CA : I think I ' ve heard it said that **the** fall in **emissions** caused by **the** pandemic isn ' t that much more than, actually, **the** fall that we will need every single year if we ' re to meet **emissions** targets. Is that true, and, if so, doesn ' t that seem impossibly daunting? AG : It does seem daunting, but first look at **the** number. That number came from a study a little over a year ago released by **the** IPCC as to what it would take to keep **the** Earth ' s temperatures from increasing more than 1. 5 degrees Celsius. And yes, **the** annual reductions would be significant, on **the** order of what we ' ve seen with **the** pandemic. And yes, that does seem daunting. However, we do have **the** opportunity to make some fairly dramatic changes, and **the** plan is not a mystery. You start with **the** two sectors that are closest to an effective transition – electricity generation, as I mentioned – and last year, 2019, if you look at all of **the** new electricity generation built all around **the** world, 72 percent of it was from solar and wind. And already, without **the** continuing subsidies for **fossil** fuels, we would see many more of these plants being shut down. There are some new **fossil** plants being built, but many more are being shut down. And where **transportation** is concerned, **the** second sector ready to go, in addition to **the** cheaper prices for EVs that I made reference to before, there are some 45 jurisdictions around **the** world – national, regional and municipal – where laws have been passed beginning a phaseout of internal combustion engines. Even India said that by 2030, less than 10 years from now, it will be illegal to sell any new internal combustion engines in India. There are many other examples. So **the** past small reductions may not be an accurate guide to **the** kind we can achieve with serious national plans and a focused global effort. CA : So help us understand just **the** big picture here, Al. I think before **the** pandemic, **the** world was emitting about 55 gigatons of what they call"CO2 equivalent,"so that includes other **greenhouse** gases like methane dialed up to be **the** equivalent of CO2. And am I right in saying that **the** IPCC, which is **the** global organization of scientists, is recommending that **the** only way to fix this crisis is to get that number from 55 to zero by 2050 at **the** very latest, and that even then, there ' s a chance that we will end up with temperature rises more like two degrees Celsius rather than 1. 5? I mean, is that approximately **the** big picture of what **the** IPCC is recommending? AG : That ' s correct. **The** global goal established in **the** Paris Conference is to get to net zero on a global basis by 2050, and many people quickly add that that really means a 45 to 50 percent reduction by 2030 to make that pathway to net zero feasible. CA : And that kind of timeline is **the** kind of timeline where people couldn ’ t even imagine it. It ’ s just hard to think of policy over 30 years. So that ’ s actually a very good shorthand, that humanity ’ s task is to cut **emissions** in half by 2030, approximately speaking, which I think boils down to about a seven or eight percent reduction a year, something like that, if I ’ m not wrong. AG : Not quite. Not quite that large but close, yes. CA : So it is something like **the** effect that we ’ ve experienced this year may be necessary. This year, we ’ ve done it by basically shutting down **the** economy. You ’ re talking about a way of doing it over **the** coming years that actually gives some economic growth and new jobs. So talk more about that. You ’ ve referred to changing our energy sources, changing how we transport. If we did those things, how much of **the** problem does that solve? AG : Well, we can get to – well, in addition to doing **the** two sectors that I mentioned, we also have to deal with manufacturing and all **the** use cases that require temperatures of a thousand degrees Celsius, and there are solutions there as well. I ’ ll come back and mention an exciting one that Germany has just embarked upon. We also have to tackle regenerative **agriculture**. There is **the** opportunity to sequester a great deal of **carbon** in topsoils around **the** world by changing **the** agricultural **techniques**. There is a farmer-led movement to do that. We need to also retrofit buildings. We need to change our management of forests and **the** ocean. But let me just mention two things briefly. First of all, **the** high temperature use cases. Angela Merkel, just 10 days ago, with **the** leadership of her minister Peter Altmaier, who is a good friend and a great public servant, have just embarked on a green hydrogen strategy to make hydrogen with zero marginal cost **renewable** energy. And just a word on that, Chris : you ’ ve heard about **the** intermittency of wind and solar – solar doesn ’ t produce electricity when **the** sun ’ s not shining, and wind doesn ’ t when **the** wind ’ s not blowing – but batteries are getting better, and these technologies are becoming much more efficient and powerful, so that for an increasing number of hours of each day, they ’ re producing often way more electricity than can be used. So what to do with it? **The** marginal cost for **the** next kilowatt-hour is zero. So all of a sudden, **the** very energy-intensive process of cracking hydrogen from water becomes economically feasible, and it can be substituted for **coal** and gas, and that ’ s already being done. There ’ s a Swedish company already making steel with green hydrogen, and, as I say, Germany has just embarked on a major new initiative to do that. I think they ’ re pointing **the** way for **the** rest of **the** world. Now, where building retrofits are concerned, just a moment on this, because about 20 to 25 percent of **the** global warming pollution in **the** world and in **the** US comes from inefficient buildings that were constructed by companies and individuals who were trying to be competitive in **the** marketplace and keep their margins acceptably high and thereby skimping on insulation and **the** right windows and LEDs and **the** rest. And yet **the** person or company that buys that building or leases that building, they want their monthly utility bills much lower. So there are now ways to close that so-called agent-principal divide, **the** differing incentives for **the** builder and occupier, and we can retrofit buildings with a program that literally pays for itself over three to five years, and we could put tens of millions of people to work in jobs that by definition cannot be outsourced because they exist in every single community. And we really ought to get serious about doing this, because we ’ re going to need all those jobs to get sustainable prosperity in **the** aftermath of this pandemic. CA : Just going back to **the** hydrogen economy that you referred to there, when some people hear that, they think,”Oh, are you talking about hydrogen-fueled cars?”And they ’ ve heard that that probably won ’ t be a winning strategy. But you ’ re thinking much more broadly than that, I think, that it ’ s not just hydrogen as a kind of storage mechanism to act as a buffer for **renewable** energy, but also hydrogen could be essential for some of **the** other processes in **the** economy like making steel, making cement, that are fundamentally carbon-intensive processes right now but could be transformed if we had much cheaper sources of hydrogen. Is that right? AG : Yes, I was always skeptical about hydrogen, Chris, principally because it ' s been so expensive to make it, to"crack it out of water,"as they say. But **the** game-changer has been **the** incredible abundance of solar and wind electricity in volumes and amounts that people didn ' t expect, and all of a sudden, it ' s cheap enough to use for these very energy-intensive processes like creating green hydrogen. I ' m still a bit skeptical about using it in vehicles. Toyota ' s been betting on that for 25 years and it hasn ' t really worked for them. Never say never, maybe it will, but I think it ' s most useful for these high-temperature industrial processes, and we already have a pathway for decarbonizing **transportation** with electricity that ' s working extremely well. Tesla ' s going to be soon **the** most valuable automobile company in **the** world, already in **the** US, and they ' re about to overtake Toyota. There is now a semitruck company that ' s been stood up by Tesla and another that is going to be a hybrid with electricity and green hydrogen, so we ' ll see whether or not they can make it work in that application. But I think electricity is preferable for cars and trucks. CA : We ' re coming to some community questions in a minute. Let me ask you, though, about nuclear. Some environmentalists believe that nuclear, or maybe new generation nuclear power is an essential part of **the** equation if we ' re to get to a truly clean future, a clean energy future. Are you still pretty skeptical on nuclear, Al? AG : Well, **the** market ' s skeptical about it, Chris. It ' s been a crushing disappointment for me and for so many. I used to represent Oak Ridge, where nuclear energy began, and when I was a young congressman, I was a booster. I was very enthusiastic about it. But **the** cost overruns and **the** problems in building these plants have become so severe that utilities just don ' t have an appetite for them. It ' s become **the** most expensive source of electricity. Now, let me hasten to add that there are some older nuclear reactors that have more useful time that could be added onto their lifetimes. And like a lot of environmentalists, I ' ve come to **the** view that if they can be determined to be safe, they should be allowed to continue operating for a time. But where new nuclear power plants are concerned, here ' s a way to look at it. If you are – you ' ve been a CEO, Chris. If you were **the** CEO of – I guess you still are. If you were **the** CEO of an electric utility, and you told your executive team,"I want to build a nuclear power plant,"two of **the** first questions you would ask are, number one : How much will it cost? And there ' s not a single engineering consulting firm that I ' ve been able to find anywhere in **the** world that will put their name on an opinion giving you a cost estimate. They just don ' t know. A second question you would ask is : How long will it take to build it, so we can start selling **the** electricity? And again, **the** answer you will get is,"We have no idea."So if you don ' t know how much it ' s going to cost, and you don ' t know when it ' s going to be finished, and you already know that **the** electricity is more expensive than **the** alternate ways to produce it, that ' s going to be a little discouraging, and, in fact, that ' s been **the** case for utilities around **the** world. CA : OK. So there ' s definitely an interesting debate there, but we ' re going to come on to some community questions. Let ' s have **the** first of those questions up, please. From Prosanta Chakrabarty :"People who are skeptical of COVID and of climate change seem to be skeptical of science in general. It may be that **the** singular message from scientists gets diluted and convoluted. How do we fix that?"AG : Yeah, that ' s a great question, Prosanta. Boy, I ' m trying to put this succinctly and shortly. I think that there has been a feeling that experts in general have kind of let **the**

US down, and that feeling is much more pronounced in **the** US than in most other countries. And I think that **the** considered opinion of what we call experts has been diluted over **the** last few decades by **the** unhealthy dominance of big money in our political system, which has found ways to really twist economic policy to benefit elites. And this sounds a little radical, but it ' s actually what has happened. And we have gone for more than 40 years without any meaningful increase in middle-income pay, and where **the** injustice experienced by African Americans and other communities of color are concerned, **the** differential in pay between African Americans and majority Americans is **the** same as it was in 1968, and **the** family wealth, **the** net worth – it takes 11 and a half so-called"typical"African American families to make up **the** net worth of one so-called"typical"White American family. And you look at **the** soaring incomes in **the** top one or **the** top one-tenth of one percent, and people say,"Wait a minute. Whoever **the** experts were that designed these policies, they haven ' t been doing a good job for me."A final point, Chris : there has been an assault on reason. There has been a war against truth. There has been a strategy, maybe it was best known as a strategy decades ago by **the** tobacco companies who hired actors and dressed them up as doctors to falsely reassure people that there were no health consequences from smoking cigarettes, and a hundred million people died as a result. That same strategy of diminishing **the** significance of truth, diminishing, as someone said, **the** authority of knowledge, I think that has made it kind of open season on any inconvenient truth – forgive another buzz phrase, but it is apt. We cannot abandon our devotion to **the** best available evidence tested in reasoned discourse and used as **the** basis for **the** best policies we can form. CA : Is it possible, Al, that one consequence of **the** pandemic is actually a growing number of people have revisited their opinions on scientists? I mean, you ' ve had a chance in **the** last few months to say,"Do I trust my political leader or do I trust this scientist in terms of what they ' re saying about this virus?"Maybe lessons from that could be carried forward? AG : Well, you know, I think if **the** polling is accurate, people do trust their doctors a lot more than some of **the** politicians who seem to have a vested interest in pretending **the** pandemic isn ' t real. And if you look at **the** incredible bust at President Trump ' s rally in Tulsa, a stadium of 19, 000 people with less than **one-third** filled, according to **the** fire marshal, you saw all **the** empty seats if you saw **the** news **clips**, so even **the** most loyal Trump supporters must have decided to trust their doctors and **the** medical advice rather than Dr. Donald Trump. CA : With a little help from **the** TikTok generation, perchance. AG : Well, but that didn ' t affect **the** turnout. What they did, very cleverly, and I ' m cheering them on, what they did was affect **the** Trump White House ' s expectations. They ' re **the** reason why he went out a couple days beforehand and said,"We ' ve had a million people sign up."But they didn ' t prevent – they didn ' t take seats that others could have otherwise taken. They didn ' t affect **the** turnout, just **the** expectations. CA : OK, let ' s have our next question here."Are you concerned **the** world will rush back to **the** use of **the** private car out of fear of using shared public **transportation**?"AG : Well, that could actually be one of **the** consequences, absolutely. Now, **the** trends on mass transit were already **inching** in **the** wrong direction because of Uber and Lyft and **the** ridesharing services, and if autonomy ever reaches **the** goals that its advocates have hoped for then that may also have a similar effect. But there ' s no doubt that some people are going to be probably a little more reluctant to take mass **transportation** until **the** fear of this pandemic is well and truly gone. CA : Yeah. Might need a vaccine on that one. AG : Yeah. CA : Next question. Sonaar Luthra, thank you for this question from LA."Given **the** temperature rise in **the** Arctic this past week, seems like **the** rate we are losing our **carbon** sinks like permafrost or forests is accelerating faster than we predicted. Are our models too focused on human **emissions**?”Interesting question. AG : Well, **the** models are focused on **the** factors that have led to these incredible temperature spikes in **the** north of **the** Arctic Circle. They were predicted, they have been predicted, and one of **the** reasons for it is that as **the** snow and ice cover melts, **the** sun ’ s incoming rays are no longer reflected back into space at a 90 percent rate, and instead, when they fall on **the** dark tundra or **the** dark ocean, they ’ re absorbed at a 90 percent rate. So that ’ s a magnifier of **the** warming in **the** Arctic, and this has been predicted. There are a number of other consequences that are also in **the** models, but some of them may have to be recalibrated. **The** scientists are freshly concerned that **the emissions** of both CO2 and methane from **the** thawing tundra could be larger than they had hoped they would be. There ’ s also just been a brand-new study. I won ’ t spend time on this, because it deals with a kind of geeky term called”climate sensitivity,”which has been a factor in **the** models with large error bars because it ’ s so hard to pin down. But **the** latest evidence indicates, worryingly, that **the** sensitivity may be greater than they had thought, and we will have an even more daunting task. That shouldn ’ t discourage us. I truly believe that once we cross this tipping point, and I do believe we ’ re doing it now, as I ’ ve said, then I think we ’ re going to find a lot of ways to speed up **the emissions** reductions. CA : We ’ ll take one more question from **the** community. Haha.”Geoengineering is making extraordinary progress. Exxon is investing in technology from Global Thermostat that seems promising. What do you think of these air and water **carbon** capture technologies?”Stephen Petranek. AG : Yeah. Well, you and I have talked about this before, Chris. I ’ ve been strongly opposed to conducting an unplanned global experiment that could go wildly wrong, and most are really scared of that approach. However, **the** term”geoengineering”is a nuanced term that covers a lot. If you want to paint roofs white to reflect more energy from **the** cityscapes, that ’ s not going to bring a danger of a runaway effect, and there are some other things that are loosely called”geoengineering”like that, which are fine. But **the** idea of blocking out **the** sun ’ s rays – that ’ s insane in my opinion. Turns out plants need sunlight for photosynthesis and solar panels need sunlight for producing electricity from **the** sun ’ s rays. And **the** consequences of changing everything we know and pretending that **the** consequences are going to precisely cancel out **the** unplanned experiment of global warming that we already have underway, you know, there are glitches in our thinking. One of them is called **the**”single solution bias,”and there are people who just have a hunger to say,”Well, that one solution, we just need to latch on to that and do that, and damn **the** consequences.”Well, it ’ s **nuts**. CA : But let me push back on this just a little bit. So let ’ s say that we agree that a single solution, all-or-nothing attempt at geoengineering is crazy. But there are scenarios where **the** world looks at **emissions** and just sees, in 10 years ’ time, let ’ s say, that they are just not coming down fast enough and that we are at risk of several other liftoff events where this train will just get away from us, and we will see temperature rises of three, four, five, six, seven degrees, and all of civilization is at risk. Surely, there is an approach to geoengineering that could be modeled, in a way, on **the** way that we approach medicine. Like, for hundreds of years, we don ’ t really understand **the** human body, people would try interventions, and some of them would work, and some of them wouldn ’ t. No one says in medicine,”You know, go in and take an all-or-nothing decision on someone ’ s life,”but they do say,”Let ’ s try some stuff.”If an experiment can be reversible, if it ’ s plausible in **the** first place, if there ’ s reason to think that it might work, we actually owe it to **the** future health of humanity to try at least some types of tests to see what could work.

So, small tests to see whether, for example, seeding of something in **the** ocean might create, in a nonthreatening way, **carbon** sinks. Or maybe, rather than filling **the** atmosphere with sulfur dioxide, a smaller experiment that was not that big a deal to see whether, cost-effectively, you could reduce **the** temperature a little bit. Surely, that isn ' t completely crazy and is at least something we should be thinking about in case these other measures don ' t work? AG : Well, there ' ve already been such experiments to seed **the** ocean to see if that can increase **the** uptake of CO2. And **the** experiments were an unmitigated failure, as many predicted they would be. But that, again, is **the** kind of approach that ' s very different from putting tinfoil strips in **the** atmosphere orbiting **the** Earth. That was **the** way that solar geoengineering proposal started. Now they ' re focusing on chalk, so we have chalk dust all over everything. But more serious than that is **the** fact that it might not be reversible. CA : But, Al, that ' s **the** rhetoric response. **The** amount of dust that you need to drop by a degree or two wouldn ' t result in chalk dust over everything. It would be unbelievably – like, it would be less than **the** dust that people experience every day, anyway. I mean, I just – AG : First of all, I don ' t know how you do a small experiment in **the** atmosphere. And secondly, if we were to take that approach, we would have to steadily increase **the** amount of whatever substance they decided. We ' d have to increase it every single year, and if we ever stopped, then there would be a sudden snapback, like"**The** Picture of Dorian Gray,"that old book and movie, where suddenly all of **the** things caught up with you at once. **The** fact that anyone is even considering these approaches, Chris, is a measure of a feeling of desperation that some have begun to feel, which I understand, but I don ' t think it should drive us toward these reckless experiments. And by **the** way, using your analogy to experimental cancer treatments, for example, you usually get informed consent from **the** patient. Getting informed consent from 7. 8 billion people who have no voice and no say, who are subject to **the** potentially catastrophic consequences of this wackadoodle proposal that somebody comes up with to try to rearrange **the** entire Earth ' s atmosphere and hope and pretend that it ' s going to cancel out, **the** fact that we ' re putting 152 million tons of heat-trapping, manmade global warming pollution into **the** sky every day. That ' s what ' s really insane. A scientist decades ago compared it this way. He said, if you had two people on a sinking boat and one of them says,"You know, we could probably use some mirrors to signal to shore to get them to build a sophisticated wave-generating machine that will cancel out **the** **rocking** of **the** boat by these guys in **the** back of **the** boat."Or you could get them to stop **rocking the** boat. And that ' s what we need to do. We need to stop what ' s causing **the** crisis. CA : Yeah, that ' s a great story, but if **the** effort to stop **the** people **rocking** in **the** back of **the** boat is as complex as **the** scientific proposal you just outlined, whereas **the** experiment to stop **the** waves is actually as simple as telling **the** people to stop **rocking the** boat, that story changes. And I think you ' re right that **the** issue of informed consent is a really challenging one, but, I mean, no one gave informed consent to do all of **the** other things we ' re doing to **the** atmosphere. And I agree that **the** moral hazard issue is worrying, that if we became dependent on geoengineering and took away our efforts to do **the** rest, that would be tragic. It just seems like, I wish it was possible to have a nuanced debate of people saying, you know what, there ' s multiple dials to a very complex problem. We ' re going to have to adjust several of them very, very carefully and keep talking to each other. Wouldn ' t that be a goal to just try and have a more nuanced debate about this, rather than all of that geoengineering can ' t work? AG : Well, I ' ve said some of it, you know, **the** benign forms that I ' ve mentioned, I ' m not ruling those out. But blocking **the** Sun ' s rays from **the** Earth, not only do you affect 7. 8 billion people, you affect **the** plants and **the** animals and **the** ocean currents and **the** wind currents and natural processes that we ' re in danger of disrupting even more. Techno-optimism is something I ' ve engaged in in **the** past, but to latch on to some brand-new technological solution to rework **the** entire Earth ' s natural system because somebody thinks he ' s clever enough to do it in a way that precisely cancels out **the** consequences of using **the** atmosphere as an open sewer for heat-trapping manmade gases. It ' s much more important to stop using **the** atmosphere as an open sewer. That ' s what **the** problem is. CA : All right, well, we ' ll agree that that is **the** most important thing, for sure, and speaking of which, do you believe **the** world needs **carbon** pricing, and is there any prospect for getting there? AG : Yes. Yes to both questions. For decades, almost every economist who is asked about **the** climate crisis says,"Well, we just need to put a price on **carbon**."And I have certainly been in favor of that approach. But it is daunting. Nevertheless, there are 43 jurisdictions around **the** world that already have a price on **carbon**. We ' re seeing it in Europe. They finally straightened out their **carbon** pricing mechanism. It ' s an **emissions** trading version of it. We have places that have put a tax on **carbon**. That ' s **the** approach **the** economists prefer. China is beginning to implement its national **emissions** trading program. California and quite a few other states in **the** US are already doing it. It can be given back to people in a revenue-neutral way. But **the** opposition to it, Chris, which you ' ve noted, is impressive enough that we do have to take other approaches, and I would say most climate activists are now saying, look, let ' s don ' t make **the** best **the** enemy of **the** better. There are other ways to do this as well. We need every solution we can rationally employ, including by regulation. And often, when **the** political difficulty of a proposal becomes too difficult in a market-oriented approach, **the** fallback is with regulation, and it ' s been given a bad name, regulation, but many places are doing it. I mentioned phasing out internal combustion engines. That ' s an example. There are 160 cities in **the** US that have already by regulation ordered that within a date certain, 100 percent of all their electricity will have to come from **renewable** sources. And again, **the** market forces that are driving **the** cost of **renewable** energy and **sustainability** solutions ever downward, that gives us **the** wind at our back. This is working in our favor. CA : I mean, **the** pushback on **carbon** pricing often goes further from parts of **the** environmental movement, which is to a pushback on **the** role of business in general. Business is actually – well, capitalism – is blamed for **the** climate crisis because of unrelenting growth, to **the** point where many people don ' t trust business to be part of **the** solution. **The** only way to go forward is to regulate, to force businesses to do **the** right thing. Do you think that business has to be part of **the** solution? AG : Well, definitely, because **the** allocation of capital needed to solve this crisis is greater than what governments can handle. And businesses are beginning, many businesses are beginning to play a very constructive role. They ' re getting a demand that they do so from their customers, from their investors, from their boards, from their executive teams, from their families. And by **the** way, **the** rising generation is demanding a brighter future, and when CEOs interview potential new hires, they find that **the** new hires are interviewing them. They want to make a nice income, but they want to be able to tell their family and friends and peers that they ' re doing something more than just making money. One illustration of how this new generation is changing, Chris : there are 65 colleges in **the** US right now where **the** College Young Republican Clubs have joined together to jointly demand that **the** Republican National Committee change its policy on climate, lest they lose that entire generation. This is a global phenomenon. **The** Greta Generation is now leading this in so many ways, and if you look at **the** polling, again, **the** vast majority of young Republicans are demanding a change on climate policy. This is really a movement that is building still. CA : I was going to ask you about that, because one of **the** most painful things over **the** last 20 years has just been how climate has been politicized, certainly in **the** US. You ' ve probably felt yourself at **the** heart of that a lot of **the** time, with people attacking you personally in **the** most merciless, and unfair ways, often. Do you really see signs that that might be changing, led by **the** next generation? AG : Yeah, there ' s no question about it. I don ' t want to rely on polls too much. I ' ve mentioned them already. But there was a new one that came out that looked at **the** wavering Trump supporters, those who supported him strongly in **the** past and want to do so again. **The** number one issue, surprisingly to some, that is giving them pause, is **the** craziness of President Trump and his administration on climate. We ' re seeing big majorities of **the** Republican Party overall saying that they ' re ready to start exploring some real solutions to **the** climate crisis. I think that we ' re really getting there, no question about it. CA : I mean, you ' ve been **the** figurehead for raising this issue, and you happen to be a Democrat. Is there anything that you can personally do to – I don ' t know – to open **the** tent, to welcome people, to try and say,"This is beyond politics, dear friends"? AG : Yeah. Well, I ' ve tried all of those things, and maybe it ' s made a little positive difference. I ' ve worked with **the** Republicans extensively. And, you know, well after I left **the** White House, I had Newt Gingrich and Pat Robertson and other prominent Republicans appear on national TV ads with me saying we ' ve got to solve **the** climate crisis. But **the** petroleum industry has really doubled down enforcing discipline within **the** Republican Party. I mean, look at **the** attacks they ' ve launched against **the** Pope when he came out with his encyclical and was demonized, not by all for sure, but there were hawks in **the** anti-climate movement who immediately started training their guns on Pope Francis, and there are many other examples. They enforce discipline and try to make it a partisan issue, even as Democrats reach out to try to make it bipartisan. I totally agree with you that it should not be a partisan issue. It didn ' t use to be, but it ' s been artificially weaponized as an issue. CA : I mean, **the** CEOs of **oil** companies also have kids who are talking to them. It feels like some of them are moving and are trying to invest and trying to find ways of being part of **the** future. Do you see signs of that? AG : Yeah. I think that business leaders, including in **the oil** and gas companies, are hearing from their families. They ' re hearing from their friends. They ' re hearing from their employees. And, by **the** way, we ' ve seen in **the** tech industry some mass walkouts by employees who are demanding that some of **the** tech companies do more and get serious. I ' m so proud of Apple. Forgive me for parenthetically praising Apple. You know, I ' m on **the** board, but I ' m such a big fan of Tim Cook and my colleagues at Apple. It ' s an example of a tech company that ' s really doing **fantastic** things. And there ' s some others as well. There are others in many industries. But **the** pressures on **the oil** and gas companies are quite extraordinary. You know, BP just wrote down 12 and a half billion dollars ' worth of **oil** and gas assets and said that they ' re never going to see **the** light of day. **Two-thirds** of **the fossil** fuels that have already been discovered cannot be burned and will not be burned. And so that ' s a big economic risk to **the** global economy, like **the** subprime mortgage crisis. We ' ve got 22 trillion dollars of subprime **carbon** assets, and just yesterday, there was a major report that **the** fracking industry in **the** US is seeing now a wave of bankruptcies because **the** price of **the** fracked gas and **oil** has fallen below levels that make them economic. CA : Is **the** shorthand of what ' s happened there that electric cars and electric technologies and solar and so forth have helped drive down **the** price of **oil** to **the** point where huge amounts of **the** reserves just can ' t be developed profitably? AG : Yes, that ’ s it. That ’ s mainly it. **The** projections for energy sources in **the** next several years uniformly predict that electricity from wind and solar is going to continue to plummet in price, and therefore using gas or **coal** to make steam to turn **the turbines** is just not going to be economical. Similarly, **the** electrification of **the transportation** sector is having **the** same effect. Some are also looking at **the** trend in national, regional and local governance. I mentioned this before, but they ’ re predicting a very different energy future. But let me come back, Chris, because we talked about business leaders. I think you were getting in a question a moment ago about capitalism itself, and I do want to say a word on that, because there are a lot of people who say maybe capitalism is **the** basic problem. I think **the** current form of capitalism we have is desperately in need of reform. **The** short-term outlook is often mentioned, but **the** way we measure what is of value to us is also at **the** heart of **the** crisis of modern capitalism. Now, capitalism is at **the** base of every successful economy, and it balances supply and demand, **unlocks** a higher **fraction** of **the** human potential, and it ’ s not going anywhere, but it needs to be reformed, because **the** way we measure what ’ s valuable now ignores so-called negative externalities like pollution. It also ignores positive externalities like investments in education and health care, mental health care, family services. It ignores **the** depletion of resources like groundwater and topsoil and **the web** of living species. And it ignores **the** distribution of incomes and net worths, so when GDP goes up, people cheer, two percent, three percent – wow! – four percent, and they think,”Great!”But it ’ s accompanied by vast increases in pollution, chronic underinvestment in public goods, **the** depletion of irreplaceable natural resources, and **the** worst inequality crisis we ’ ve seen in more than a hundred years that is threatening **the** future of both capitalism and democracy. So we have to change it. We have to reform it. CA : So reform capitalism, but don ’ t throw it out. We ’ re going to need it as a tool as we go forward if we ’ re to solve this. AG : Yeah, I think that ’ s right, and just one other point : **the** worst environmental abuses in **the** last hundred years have been in jurisdictions that experimented during **the** 20th century with **the** alternatives to capitalism on **the left** and right. CA : Interesting. All right. Two last community questions quickly. Chadburn Blomquist :”As you are reading **the** tea leaves of **the** impact of **the** current pandemic, what do you think in regard to our response to combatting climate change will be **the** most impactful lesson learned?”AG : Boy, that ’ s a very thoughtful question, and I wish my answer could rise to **the** same level on short notice. I would say first, don ’ t ignore **the** scientists. When there is virtual unanimity among **the** scientific and medical experts, pay attention. Don ’ t let some politician dissuade you. I think President Trump is slowly learning that ’ s it ’ s kind of difficult to gaslight a virus. He tried to gaslight **the** virus in Tulsa. It didn ’ t come off very well, and tragically, he decided to recklessly roll **the** dice a month ago and ignore **the** recommendations for people to wear masks and to socially distance and to do **the** other things, and I think that lesson is beginning to take hold in a much stronger way. But beyond that, Chris, I think that this period of time has been characterized by one of **the** most profound opportunities for people to rethink **the** patterns of their lives and to consider whether or not we can ’ t do a lot of things better and differently. And I think that this rising generation I mentioned before has been even more profoundly affected by this interlude, which I hope ends soon, but I hope **the** lessons endure. I expect they will. CA : Yeah, it ’ s amazing how many things you can do without emitting **carbon**, that we ’ ve been forced to do. Let ’ s have one more question here. Frank Hennessy :”Are you encouraged by **the** ability of people to quickly adapt to **the** new normal due to COVID-19 as evidence that people can and will change their habits to respond to climate change?”AG : Yes, but I think we have to keep in mind that there is a crisis within this crisis. **The** impact on **the** African American community, which I mentioned before, on **the** Latinx community, Indigenous peoples. **The** highest infection rate is in **the** Navajo Nation right now. So some of these questions appear differently to those who are really getting **the** brunt of this crisis, and it is unacceptable that we allow this to continue. It feels one way to you and me and perhaps to many in our audience today, but for low-income communities of color, it ’ s an entirely different crisis, and we owe it to them and to all of us to get busy and to start using **the** best science and solve this pandemic. You know **the** phrase”pandemic economics.”Somebody said, **the** first principle of pandemic economics is take care of **the** pandemic, and we ’ re not doing that yet. We ’ re seeing **the** president try to goose **the** economy for his reelection, never mind **the** prediction of tens of thousands of additional American deaths, and that is just unforgivable in my opinion. CA : Thank you, Frank. So Al, you, along with others in **the** community played a key role in encouraging TED to launch this initiative called”Countdown.”Thank you for that, and I guess this conversation is continuing among many of us. If you ’ re interested in climate, watching this, check out **the** Countdown website, countdown. ted. com, and be part of 10 / 10 / 2020, when we are trying to put out an alert to **the** world that climate can ’ t wait, that it really matters, and there ’ s going to be some amazing content free to **the** world on that day. Thank you, Al, for your inspiration and support in doing that. I wonder whether you could end today ’ s session just by painting us a picture, like how might things roll out over **the** next decade or so? Just tell us whether there is still a story of hope here. AG : I ’ d be glad to. I ’ ve got to get one plug in. I ’ ll make it brief. July 18 through July 26, **The** Climate Reality Project is having a global training. We ’ ve already had 8, 000 people register. You can go to climatereality. com. Now, a bright future. It begins with all of **the** kinds of efforts that you ’ ve thrown yourself into in organizing Countdown. Chris, you and your team have been amazing to work with, and I ’ m so excited about **the** Countdown project. TED has an unparalleled ability to spread ideas that are worth spreading, to raise consciousness, to enlighten people around **the** world, and it ’ s needed for climate and **the** solutions to **the** climate crisis like it ’ s never been needed before, and I just want to thank you for what you personally are doing to organize this **fantastic** Countdown program. CA : Thank you. And **the** world? Are we going to do this? Do you think that humanity is going to pull this off and that our grandchildren are going to have beautiful lives where they can celebrate nature and not spend every day in fear of **the** next tornado or **tsunami**? AG : I am optimistic that we will do it, but **the** answer is in our hands. We have seen dark times in periods of **the** past, and we have risen to meet **the** challenge. We have limitations of our long **evolutionary** heritage and elements of our culture, but we also have **the** ability to transcend our limitations, and when **the** chips are down, and when survival is at stake and when our children and future generations are at stake, we ’ re capable of more than we sometimes allow ourselves to think we can do. This is such a time. I believe we will rise to **the** occasion, and we will create a bright, clean, prosperous, just and fair future. I believe it with all my heart. CA : Al Gore, thank you for your life of work, for all you ’ ve done to elevate this issue and for spending this time with us now. Thank you. AG : Back at you. Thank you.

Text Sample 73: POS Dim 1 – 2020 – Score 21.00 – 2020/t004235.txt
Chris Anderson : Dr. Jane Goodall, welcome. Jane Goodall : Thank you, and I

think, you know, we couldn ' t have a complete interview unless people know Mr. H is with me, because everybody knows Mr. H. CA : Hello, Mr. H. In your TED Talk 17 years ago, you warned us about **the** dangers of humans crowding out **the** natural world. Is there any sense in which you feel that **the** current pandemic is kind of, nature striking back? JG : It ' s very, very clear that these zoonotic diseases, like **the** corona and HIV / AIDS and all sorts of other diseases that we catch from animals, that ' s partly to do with destruction of **the** environment, which, as animals lose **habitat**, they get crowded together and sometimes that means that a virus from a reservoir species, where it ' s lived harmoniously for maybe hundreds of years, jumps into a new species, then you also get animals being pushed into closer contact with humans. And sometimes one of these animals that has caught a virus can – you know, provides **the** opportunity for that virus to jump into people and create a new disease, like COVID-19. And in addition to that, we are so disrespecting animals. We hunt them, we kill them, we eat them, we traffic them, we send them off to **the** wild-animal markets in Asia, where they ' re in terrible, cramped conditions, in tiny cages, with people being contaminated with **blood** and urine and feces, ideal conditions for a virus to **spill** from an animal to an animal, or an animal to a person. CA : I ' d love to just dip **backwards** in time for a bit, because your story is so extraordinary. I mean, despite **the** arguably even more sexist attitudes of **the** 1960s, somehow you were able to break through and become one of **the** world ' s leading scientists, discovering this astonishing series of facts about chimpanzees, such as their tool use and so much more. What was it about you, do you think, that allowed you to make such a breakthrough? JG : Well, **the** thing is, I was born loving animals, and **the** most important thing was, I had a very supportive mother. She didn ' t get mad when she found earthworms in my bed, she just said they better be in **the** garden. And she didn ' t get mad when I **disappeared** for four hours and she called **the** police, and I was sitting in a hen house, because nobody would tell me where **the** hole was where **the** egg came out. I had no dream of being a scientist, because women didn ' t do that sort of thing. In fact, there weren ' t any man doing it back then, either. And everybody laughed at me except Mom, who said,"If you really want this, you ' re going to have to work awfully hard, take advantage of every opportunity, if you don ' t give up, maybe you ' ll find a way."CA : And somehow, you were able to kind of, earn **the** trust of chimpanzees in **the** way that no one else had. Looking back, what were **the** most exciting moments that you discovered or what is it that people still don ' t get about chimpanzees? JG : Well, **the** thing is, you say,"See things nobody else had, get their trust."Nobody else had tried. Quite honestly. So, basically, I used **the** same **techniques** that I had to study **the** animals around my home when I was a child. Just sitting, patiently, not trying to get too close too quickly, but it was awful, because **the** money was only for six months. I mean, you can imagine how difficult to get money for a young girl with no degree, to go and do something as bizarre as sitting in a forest. And you know, finally, we got money for six months from an American philanthropist, and I knew with time I ' d get **the** chimps ' trust, but did I have time? And weeks became months and then finally, after about four months, one chimpanzee began to lose his fear, and it was he that on one occasion I saw – I still wasn ' t really close, but I had my binoculars – and I saw him using and making tools to **fish** for termites. And although I wasn ' t terribly surprised, because I ' ve read about things captive chimps could do – but I knew that science believed that humans, and only humans, used and made tools. And I knew how excited [Dr. Louis] Leakey would be. And it was that observation that enabled him to go to **the** National Geographic, and they said,"OK, we ' ll continue to support **the** research,"and they sent Hugo van Lawick, **the** photographer-filmmaker, to record what I was seeing. So a lot of scientists didn ' t want to believe **the** tool-using. In fact, one of them said I must have taught **the** chimps. Since I couldn ' t get near them, it would have been a miracle. But anyway, once they saw Hugo ' s film and that with all my descriptions of their behavior, **the** scientists had to start changing their minds. CA : And since then, numerous other discoveries that placed chimpanzees much closer to humans than people cared to believe. I think I saw you say at one point that they have a sense of humor. How have you seen that expressed? JG : Well, you see it when they ' re playing games, and there ' s a bigger one playing with a little one, and he ' s trailing a vine around a tree. And every time **the** little one is about to catch it, **the** bigger one pulls it away, and **the** little one starts crying and **the** big one starts laughing. So, you know. CA : And then, Jane, you observed something much more troubling, which was these instances of chimpanzee gangs, tribes, groups, being brutally violent to each other. I ' m curious how you process that. And whether it made you, kind of, I don ' t know, depressed about us, we ' re close to them, did it make you feel that violence is irredeemably part of all **the** great apes, somehow? JG : Well, it obviously is. And my first encounter with human, what I call evil, was **the** end of **the** war and **the** pictures from **the** Holocaust. And you know, that really shocked me. That changed who I was. I was 10, I think, at **the** time. And when **the** chimpanzees, when I realized they have this dark, brutal side, I thought they were like us but nicer. And then I realized they ' re even more like us than I had thought. And at that time, in **the** early ' 70s, it was very strange, aggression, there was a big thing about, is aggression innate or learned. And it became political. And it was, I don ' t know, it was a very strange time, and I was coming out, saying,"No, I think aggression is definitely part of our inherited repertoire of behaviors."And I asked a very respected scientist what he really thought, because he was coming out on **the** clean slate, aggression is learned, and he said,"Jane, I ' d rather not talk about what I really think."That was a big shock as far as science was concerned for me. CA : I was brought up to believe a world of all things bright and beautiful. You know, numerous beautiful films of butterflies and bees and flowers, and you know, nature as this gorgeous landscape. And many environmentalists often seem to take **the** stance,"Yes, nature is pure, nature is beautiful, humans are bad,"but then you have **the** kind of observations that you see, when you actually look at any part of nature in more detail, you see things to be terrified by, honestly. What do you make of nature, how do you think of it, how should we think of it? JG : Nature is, you know, I mean, you think of **the** whole spectrum of **evolution**, and there ' s something about going to a pristine place, and Africa was very pristine when I was young. And there were animals everywhere. And I never liked **the** fact that **lions** killed, they have to, I mean, that ' s what they do, if they didn ' t kill animals, they would die. And **the** big difference between them and us, I think, is that they do what they do because that ' s what they have to do. And we can plan to do things. Our plans are very different. We can plan to cut down a whole forest, because we want to sell **the** timber, or because we want to build another shopping mall, something like that. So our destruction of nature and our warfare, we ' re capable of evil because we can sit comfortably and plan **the** torture of somebody far away. That ' s evil. Chimpanzees have a sort of primitive war, and they can be very aggressive, but it ' s of **the** moment. It ' s how they feel. It ' s response to an emotion. CA : So your observation of **the** sophistication of chimpanzees doesn ' t go as far as what some people would want to say is **the** sort of **the** human superpower, of being able to really simulate **the** future in our minds in great detail and make long-term plans. And act to encourage each other to achieve those long-term plans. That that feels, even to someone who spent so much time with chimpanzees, that feels like a fundamentally different skill set that we just have to take responsibility for and use much more wisely than we do. JG : Yes, and I personally think, I mean, there ’ s a lot of discussion about this, but I think it ’ s a fact that we developed **the** way of communication that you and I are using. And because we have words, I mean, animal communication is way more sophisticated than we used to think. And chimpanzees, gorillas, orangutans can learn human sign language of **the** Deaf. But we sort of grow up speaking whatever language it is. So I can tell you about things that you ’ ve never heard of. And a chimpanzee couldn ’ t do that. And we can teach our children about **abstract** things. And chimpanzees couldn ’ t do that. So yes, chimpanzees can do all sorts of clever things, and so can **elephants** and so can crows and so can octopuses, but we design rockets that go off to another **planet** and little robots taking photographs, and we ’ ve designed this extraordinary way of you and me talking in our different parts of **the** world. When I was young, when I grew up, there was no TV, there were no cell phones, there was no **computers**. It was such a different world, I had a pencil, pen and notebook, that was it. CA : So just going back to this question about nature, because I think about this a lot, and I struggle with this, honestly. So much of your work, so much of so many people who I respect, is about this passion for trying not to screw up **the** natural world. So is it possible, is it healthy, is it essential, perhaps, to simultaneously accept that many aspects of nature are **terrifying**, but also, I don ’ t know, that it ’ s awesome, and that some of **the** awesomeness comes from its potential to be **terrifying** and that it is also just breathtakingly beautiful, and that we cannot be ourselves, because we are part of nature, we cannot be whole unless we somehow embrace it and are part of it? Help me with **the** language, Jane, on how that relationship should be. JG : Well, I think one of **the** problems is, you know, as we developed our intellect, and we became better and better at modifying **the** environment for our own use, and creating fields and growing crops where it used to be forest or woodland, and you know, we won ’ t go into that now, but we have this ability to change nature. And as we ’ ve moved more into towns and cities, and relied more on technology, many people feel so divorced from **the** natural world. And there ’ s hundreds, thousands of children growing up in inner cities, where there basically isn ’ t any nature, which is why this movement now to green our cities is so important. And you know, they ’ ve done experiments, I think it was in Chicago, I ’ m not quite sure, and there were various empty lots in a very violent part of town. So in some of those areas they made it green, they put trees and flowers and things, shrubs in these vacant lots. And **the** crime rate went right down. So then of course, they put trees in **the** other half. So it just shows, and also, there have been studies done showing that children really need green nature for good psychological development. But we are, as you say, part of nature and we disrespect it, as we are, and that is so terrible for our children and our children ’ s children, because we rely on nature for clean air, clean water, for regulating climate and rainfall. Look what we ’ ve done, look at **the** climate crisis. That ’ s us. We did that. CA : So a little over 30 years ago, you made this shift from scientist mainly to activist mainly, I guess. Why? JG : Conference in 1986, scientific one, I ’ d got my PhD by then and it was to find out how chimp behavior differed, if it did, from one environment to another. There were six study sites across Africa. So we thought, let ’ s bring these scientists together and explore this, which was fascinating. But we also had a session on conservation and a session on conditions in some captive situations like medical research. And those two sessions were so shocking to me. I went to **the** conference a a scientist, and I left as an activist. I didn ’ t make **the** decision, something happened inside me. CA : So you spent **the** last 34 years sort of tirelessly campaigning for a better relationship between people and nature. What should that relationship look like? JG : Well, you know, again you come up with all these problems. People have to have space to live. But I think **the** problem is that we ’ ve become, in **the** affluent societies, too greedy. I mean, honestly, who needs four houses with huge grounds? And why do we need yet another shopping mall? And so on and so on. So we are looking at short-term economic benefit, money has become a sort of god to worship, as we lose all spiritual connection with **the** natural world. And so we ’ re looking for short-term monetary gain, or power, rather than **the** health of **the planet** and **the** future of our children. We don ’ t seem to care about that anymore. That ’ s why I ’ ll never stop fighting. CA : I mean, in your work specifically on chimpanzee conservation, you ’ ve made it practice to put people at **the** center of that, local people, to engage them. How has that worked and do you think that ’ s an essential idea if we ’ re to succeed in protecting **the planet**? JG : You know, after that **famous** conference, I thought, well, I must learn more about why chimps are vanishing in Africa and what ’ s happening to **the** forest. So I got a bit of money together and went out to visit six range countries. And learned a lot about **the** problems faced by chimps, you know, hunting for bushmeat and **the** live animal trade and caught in snares and human populations growing and needing more land for their crops and their cattle and their villages. But I was also learning about **the** plight faced by so many people. **The** absolute poverty, **the** lack of health and education, **the** degradation of **the** land. And it came to a head when I flew over **the** tiny Gombe National Park. It had been part of this equatorial forest belt right across Africa to **the** west coast, and in 1990, it was just this little **island** of forest, just tiny national park. All around, **the** hills were bare. And that ’ s when it hit me. If we don ’ t do something to help **the** people find ways of living without destroying their environment, we can ’ t even try to save **the** chimps. So **the** Jane Goodall Institute began this program”Take Care,”we call it”TACARE.”And it ’ s our method of community-based conservation, totally holistic. And we ’ ve now put **the** tools of conservation into **the** hand of **the** villagers, because most Tanzanian wild chimps are not in protected areas, they ’ re just in **the** village forest reserves. And so, they now go and measure **the** health of their forest. They ’ ve understood now that protecting **the** forest isn ’ t just for wildlife, it ’ s their own future. That they need **the** forest. And they ’ re very proud. **The** volunteers go to workshops, they learn how to use **smartphones**, they learn how to upload into platform and **the** cloud. And so it ’ s all transparent. And **the** trees have come back, there ’ s no bare hills anymore. They agreed to make a buffer zone around Gombe, so **the** chimps have more forest than they did in 1990. They ’ re opening up corridors of forest to link **the** scattered chimp groups so that you don ’ t get too much inbreeding. So yes, it ’ s worked, and it ’ s in six other countries now. Same thing. CA : I mean, you ’ ve been this extraordinary tireless voice, all around **the** world, just traveling so much, speaking everywhere, inspiring people everywhere. How on earth do you find **the** energy, you know, **the** fire to do that, because that is exhausting to do, every meeting with lots of people, it is just physically exhausting, and yet, here you are, still doing it. How are you doing this, Jane? JG : Well, I suppose, you know, I ’ m obstinate, I don ’ t like giving up, but I ’ m not going to let these CEOs of big companies who are destroying **the** forests, or **the** politicians who are unraveling all **the** protections that were put in place by previous presidents, and you know who I ’ m talking about. And you know, I ’ ll go on fighting, I care about, I ’ m passionate about **the** wildlife. I ’ m passionate about **the** natural world. I love forests, it hurts me to see them damaged. And I care passionately about children. And we ’ re stealing their future. And I ’ m not going to give up. So I guess I ’ m blessed with good genes, that ’ s a gift, and **the** other gift, which I discovered I had, was communication, whether it ’ s writing or speaking. And so, you know, if going around like this wasn ’ t working, but every time I do a **lecture**, people come up and say,”Well, I had given up, but you ’ ve inspired me, I promise to do my bit.”And we have our youth program”Roots and Shoots”now in 65 countries and growing fast, all ages, all choosing projects to help people, animals, **the** environment, rolling up their sleeves and taking action. And you know, they look at you with shining eyes, wanting to tell Dr. Jane what they ’ ve been doing to make **the** world a better place. How can I let them down? CA : I mean, as you look at **the planet** ’ s future, what worries you most, actually, what scares you most about where we ’ re at? JG : Well, **the** fact that we have a small window of time, I believe, when we can at least start healing some of **the** harm and slowing down climate change. But it is closing, and we ’ ve seen what happens with **the** lockdown around **the** world because of COVID-19 : clear skies over cities, some people breathing clean air that they ’ ve never breathed before and looking up at **the** shining skies at night, which they ’ ve never seen properly before. And you know, so what worries me most is how to get enough people, people understand, but they ’ re not taking action, how to get enough people to take action? CA : National Geographic just launched this extraordinary film about you, highlighting your work over six decades. It ’ s titled”Jane Goodall : **The** Hope.”So what is **the** hope, Jane? JG : Well, **the** hope, my greatest hope is all these young people. I mean, in China, people will come up and say,”Well, of course I care about **the** environment, I was in ’ Roots and Shoots ’ in primary school.”And you know, we have”Roots and Shoots”just hanging on to **the** values and they ’ re so enthusiastic once they know **the** problems and they ’ re empowered to take action, they are clearing **the** streams, removing invasive species humanely. And they have so many ideas. And then there ’ s, you know, this extraordinary intellect of ours. We ’ re beginning to use it to come up with technology that really will help us to live in greater harmony, and in our individual lives, let ’ s think about **the** consequences of what we do each day. What do we buy, where did it come from, how was it made? Did it harm **the** environment, was it cruel to animals? Is it cheap because of child slave labor? Make ethical choices. Which you can ’ t do if you ’ re living in poverty, by **the** way. And then finally, this indomitable spirit of people who tackle what seems impossible and won ’ t give up. You can ’ t give up when you have those... But you know, there are things that I can ’ t fight. I can ’ t fight corruption. I can ’ t fight military regimes and dictators. So I can only do what I can do, and if we all do **the** bits that we can do, surely that makes a whole that eventually will win out. CA : So, last question, Jane. If there was one idea, one thought, one seed you could plant in **the** minds of everyone watching this, what would that be? JG : You know, just remember that every day you live, you make an impact on **the planet**. You can ’ t help making an impact. And at least, unless you ’ re living in extreme poverty, you have a choice as to what sort of impact you make. Even in poverty you have a choice, but when we are more affluent, we have a greater choice. And if we all make ethical choices, then we start moving towards a world that will be not quite so desperate to leave to our great-grandchildren. That ’ s, I think, something for everybody. Because a lot of people understand what ’ s happening, but they feel helpless and hopeless, and what can they do, so they do nothing and they become apathetic. And that is a huge danger, apathy. CA : Dr. Jane Goodall, wow. I really want to thank you for your extraordinary life, for all that you ’ ve done and for spending this time with us now. Thank you. JG : Thank you.

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Hello, I ’ m Chris Anderson. Welcome to **The** TED Interview. We ’ re gearing up for season four with some extraordinary guests, but I don ’ t want to wait for that for today ’ s episode, because we ’ re in **the** middle of a pandemic, and there ’ s a guest I really wanted to talk to now. He is Adam Kucharski, an infectious diseases scientist who focuses on **the** mathematical modeling of pandemics. He ’ s an associate professor at **the** London School of Hygiene and Tropical Medicine and a TED Fellow. Adam Kucharski : So what kind of behavior is actually important for epidemics? Conversations, close physical contacts? What sort of data should we be collecting before an outbreak if we want to predict how infection might spread? To find out, our team built a mathematical model... Chris Anderson : When it comes to figuring out what to make of this pandemic, known technically as COVID-19, and informally as just **the** coronavirus, I find his thinking unbelievably helpful. And I ’ m excited to dive into it with you. A special callout to my friends on Twitter who offered up many suggestions for questions. I know this topic is on everyone ’ s mind right now. And what I hope this episode does is give us all a more nuanced way of thinking about how this pandemic has unfolded so far, what might be to come and what we all collectively can do about it. Let ’ s dive in. Adam, welcome to **the** TED Interview. Adam Kucharski : Thank you. CA : So let ’ s just start with a couple of basics. A lot of skeptical people ’ s response – certainly over **the** last few weeks, maybe less so now – has been,”Oh, come on, this isn ’ t such a big deal, there ’ s a relatively tiny number of cases. Compare it to **the** flu, compare it to anything else. There are much bigger problems in **the** world. Why are we making such a fuss about this?”And I guess **the** answer to that fuss is that it comes down to **the** mathematics. We ’ re talking about **the** mathematics of exponential growth, fundamentally, right? AK : Exactly. And there ’ s a number that we use to get a sense of how easy things spread and **the** level of transmission we ’ re dealing with. We call that **the reproduction** number, and conceptually, it ’ s just, for each case you have, on average, how many others are they infecting? And that gives you a sense of how much is this scaling, how much this growth is going to look like. For coronavirus, we ’ re now seeing, across multiple countries, we ’ re seeing each person on average giving it to two or three more. CA : So that **reproduction** number, **the** first thing to understand is that any number above one means that this thing is going to grow. Any number below one means it ’ s going to diminish. AK : Exactly – if you have it above one, then each group of people infected are going to be generating more infection than there was before. And you will see **the** exponential effect, so if it ’ s two, you will be doubling every round of infection, and if it ’ s below one, you ’ re going to get something that ’ s going to decline, on average. CA : So that number two or higher, I think everyone here is maybe familiar with **the famous** story of **the** chessboard and **the** grains of rice, and if you double **the** number of grains for every square of **the** chessboard, for **the** first 10 or 15 squares nothing much happens, but by **the** time you ’ ve got to **the** 64th square, you suddenly have tons of rice for every individual on **the planet**. Exponential growth is an incredible thing. And **the** small numbers now are really not what you should be paying attention to – you should be paying attention to **the** models of what could be to come. AK : Exactly. Obviously, if you continue **the** exponential growth, you do sometimes get these incredibly large, maybe implausibly large numbers. But even looking at a timescale of say, a month, if **the reproduction** number is three, each person is infecting three on average. **The** gap between these rounds of infection is about five days. So if you imagine that you ’ ve got one case now, that ’ s, kind of, six of these five-day steps in a month. So by **the** end of that month, that one person could have generated, I think it works out at about 729 cases. So even in a month, just **the** scale of this thing can really shoot up if it ' s not controlled. CA : And so certainly, that seems to be happening on most numbers that you look at now, certainly where **the** virus is in **the** early stages of entering a country. You ' ve given a model whereby we can much more clearly understand this **reproduction** number, because it seems to me this is almost like **the** core to how we think of **the** virus and how we respond to it and how much we should fear it, almost. And in your thinking, you sort of break it down into four components, which you call DOTS : Duration, Opportunities, Transmission **probability** and Susceptibility. And I think it would be really helpful, Adam, for you to just explain each of these, because it ' s quite a simple equation that links those four things to **the** actual **reproduction** number. So talk about them in turn. Duration, what does that mean? AK : Duration measures how long someone is infectious for. If, for example, intuitively, if someone is infectious for a longer period of time, say, twice as long as someone else, then that ' s twice **the** length that they ' ve got to be spreading infection. CA : And what is **the** duration number for this virus, compared with, say, flu or with other pathogens? AK : It depends a little bit on what happens when people are infectious, if they ' re being isolated very quickly, that shortens that period of time, but potentially, we ' re looking at around a week people are effectively infectious before they might be isolated in hospital. CA : And during that week, they may not even be showing symptoms for that full week either, right? So someone gets infected, there ' s an incubation period. There ' s a period some way into that incubation period where they start being infectious, and there may be a period after that, where they start to show symptoms, and it ' s not clear, quite, how those dates align. Is that right? AK : No, we ' re getting more information. One of **the** signals we see in data that suggest that you may have that early transmission going on is when you have this delay from one infection to **the** next. So that seems to be around five days. That incubation period, **the** time for symptoms to appear, is also about five days. So if you imagine that most people are only infecting others when they ' re symptomatic, you ' d have that incubation period and then you ' d have some more time when they ' re infecting others. So **the** fact that those values seem to be similar, suggesting that some people are transmitting either very early on or potentially before they ' re showing clear symptoms. CA : So almost implies that on average, people are infecting others as much before they show symptoms as after. AK : Potentially. Obviously these are early data sets, but I think there ' s good evidence that a fair number of people, either before they ' re showing clear symptoms or maybe they ' re not showing **the** kind of very distinctive fever and cough but they ' re feeling unwell and they ' re shedding virus during that period. CA : And does that make it quite different from **the** flu, for example? AK : It makes it actually similar to flu in that regard. One of **the** reasons pandemic flu is so hard to control and so feared as a threat is because so much transmission happens before people are severely ill. And that means that by **the** time you identify these cases, they ' ve probably actually spread it to a number of other people. CA : Yeah, so this is **the** trickery of **the** thing, and why it ' s so hard to do anything about it. It is ahead of us all **the** time, and you can ' t just pay attention to how someone feels or what they ' re doing. I mean, how does that happen, by **the** way? How does someone infect someone else before they ' re even showing symptoms themselves, because classically, we think of, you know, **the** person sneezing and droplets go through **the** air and someone else breathes them in and that ' s how infection happens. What is actually going on for infection pre-symptoms? AK : So **the** level of transmission we see with this virus isn ' t what we see, for example, with measles, where someone sneezes and a lot of virus gets out and potentially lots of susceptible people can get exposed. So potentially, it could be quite early on that even if someone has quite mild symptoms, maybe a bit of a cough, that ' s enough for some virus to be getting out and particularly, some of **the** work that we ' ve done trying to look at sort of close gatherings, so very tight-knit meals, there was an example in a ski chalet – and even in those situations, you might have someone mildly ill, but enough virus is getting out and somehow exposing others, we ' re still trying to work out exactly how, but there ' s enough there to cause some infection. CA : But if someone ' s mildly ill, don ' t they still have symptoms? Isn ' t there evidence that even before they know that they ' re ill, something is going on? There was a German paper published this week that seemed to suggest that even really early on, you take a swab from **the** back of someone ' s throat and they have hundreds of thousands of these viruses already reproducing there. Like, can someone just literally just be breathing normally and there is some transmission of virus out into **the** air that they don ' t even know about and is either infecting people directly or settling on **surfaces**, is that possible? AK : I think that ' s what we ' re trying to pin down, how much that [**unclear**]. As you said, there ' s evidence that you can have people without symptoms and you can get virus out their throats. And so certainly there ' s potential that it can be breathed out, but is that a fairly rare event for that actual transmission to happen, or are we potentially seeing more infections **occur** through that route? So it ' s really early data, and I think it ' s a piece of **the** puzzle, but we ' re trying to work out where that fits in with what we know about **the** kind of other transmission events we ' ve seen. CA : Alright, so, duration is **the** duration of **the** period of infectiousness. We think is five to six days, is that what I heard you say? AK : Potentially around a week, depending on exactly what happens to people when they ' re infectious. CA : And there are cases of people testing positive way, way later, after they ' ve got infected. That may be true, but they are probably not as infectious then. Is that basically right, that ' s **the** way to think of this? AK : I think that ' s our working theory, that a lot of that infection is happening early on. And we see that for a number of respiratory infections, that when people obviously become severely ill, their behavior is very different to when they may be walking around and going about their normal day. CA : And so again, comparing that D number to other cases, like **the** flu, is flu similar? What ' s **the** D number for flu? AK : So for flu, it ' s probably slightly shorter in terms of **the** period that people are actively infectious. I mean, for flu, it ' s a very quick turnover from one case to **the** next, actually. Even a matter of about three days, potentially, from one infection to **the** person that they infect. And then at **the** other end of **the** scale, you get things like STDs, where that duration could be several months, potentially. CA : Right. OK, really nothing that unusual so far, in terms of this particular virus. Let ' s look at **the** O, opportunity. What is that? AK : So opportunity is a measure of how many chances **the** virus has to spread through interactions while someone is infectious. So typically, it ' s a measure of social behavior. On average, how many social contacts do people make that create opportunities for transmission while they ' re infectious. CA : So it ' s how many people have you got close enough to during a day, during a given day, to have a chance of infecting them. And that number could be, if people aren ' t taking precautions in a normal, sort of, urban setting, I mean, that could run into **the** hundreds, presumably? AK : Potentially, for some people. We ' ve done a number of studies looking at that in recent years, and **the** average, in terms of physical contacts, is about five people per day. Most people will have conversation or contacts generally with about 10, 15, but obviously, between cultures, we see quite a lot of variation in **the** level of physical greetings that might happen. CA : And presumably, that number again is no different for this virus than for any other. I mean, that ' s just a feature of **the** lives that we live. AK : I think for this one, if it ' s driven through these kind of interactions, and we ' ve seen for flu, for other respiratory infections, those kinds of fairly close contacts and everyday physical interactions seem to be **the** ones that are important for transmission. CA : Perhaps there is one difference. **The** fact that if you ' re infectious pre-symptoms, perhaps that means that actually, there are more opportunities here. This is part of **the** virus ' s genius, as it were, that by not letting on that it ' s inside someone, people continue to interact and go to work and take **the** subway and so forth, not even knowing that they ' re sick. AK : Exactly. And for something like flu, you see when people get ill, clearly, their social contacts drop off. So to have a virus that can be infectious while people are going around their everyday lives, really gives it an advantage in terms of transmission. CA : In your modeling, do you actually have this opportunities number higher than for flu? AK : So for **the** moment, we ' re kind of using similar values, so we ' re trying to look at, for example, physical contacts within different populations. But what we are doing is scaling **the** risk. So that ' s coming on to **the** T term. So that between each contact, what ' s **the** risk that a transmission event will **occur**. CA : Alright, so let ' s go on to this next number, **the** T, transmission **probability**. How do you define that? AK : So this measures **the** chance that, essentially, **the** virus will get across during a particular opportunity or a particular interaction. So you may well have a conversation with somebody, but actually, you don ' t cough or you don ' t sneeze or for some reason, **the** virus doesn ' t actually get across and expose **the** other person. And so, for this virus, as I mentioned, say people are having 10 conversations a day, but we ' re not seeing infected people infect 10 others a day. So it suggests that not all of those opportunities are actually resulting in **the** virus getting across. CA : But people say that this is an infectious virus. Like, what is that transmission **probability** number, again, compared with, say, **the** flu? AK : So, we did some analysis looking at these very close gatherings. We looked at about 10 different case studies, and we found that about a third of **the** contacts in those settings subsequently got infected in these early stages, when people weren ' t aware. So if you had these, kind of, big group meals, potentially, each contact had about, a kind of, one in three chance of getting exposed. For seasonal flu, that tends to be slightly lower, even within households and close settings, you don ' t necessarily get values that high. And even for something like SARS, those values have, kind of – **the** risk per interaction you had was lower than what we seem to be getting for coronavirus. Which intuitively makes sense, there must be a higher risk per interaction if this thing is spreading so easily. CA : Hm. OK, and then **the** fourth letter of DOTS is S for susceptibility. What ' s that? AK : So that is a measure of **the** proportion of **the** population who are susceptible. If you imagine you have this interaction with someone, **the** virus gets across, it exposes them, but some people may have been vaccinated or otherwise have some immunity and not develop infection themselves and not be infectious to others. So we ' ve got to account for this potential proportion of people who are not actually going to turn into cases themselves. CA : And obviously, there ' s no vaccine yet for this coronavirus, nor is anyone, at least initially, immune, as far as we know. So are you modeling that susceptibility number pretty high, is that part of **the** issue here? AK : Yeah, I think **the** evidence is that this is going to fully susceptible populations, and even in areas, for example, like China, where there ' s been a lot of transmission but there ' s been very strong control measures, we estimated that up to **the** end of January, probably about 95 percent of Wuhan are still susceptible. So there ' s been a lot of infection, but it hasn ' t really taken much of that component, of **the** DOTS, of those four things that drive transmission. CA : And so **the** way **the** mathematics works, I have to confess, amidst **the** stress of this whole situation, **the** nerd in me kind of loves **the** elegance of **the** mathematics here, because I ’ d never really thought about it this way, but you basically just multiply those numbers together to get **the reproduction** number. Is that right? AK : Exactly, yeah, you almost take **the** path of **the** infection during transmission as you multiply those together, and that gives you **the** number for that virus. CA : And so there ’ s just a total logic to that. It ’ s **the** number of days, duration that you ’ re infectious, it ’ s **the** number of people you ’ re seeing on average during those days that you have a chance to infect. Then you multiply that by **the** transmission **probability**, is virus getting into them, essentially, that ’ s what you mean by crossing over. And then by **the** susceptibility number. By **the** way, what do you think **the** susceptibility **probability** is for this case? AK : I think we have to assume that it ’ s near 100 percent in terms of spread, yeah. CA : Alright, you multiply those numbers together, and right now, it looks like, for this coronavirus, that you say two to three is **the** most plausible current number, which implies very rapid growth. AK : Exactly. In these uncontrolled outbreaks, we ’ re seeing now a number of countries in this stage – you are going to get this really rapid growth **occurring**. CA : And so how does that two to three compare with flu? And I guess, there ’ s seasonal flu, in **the** winter, when it ’ s spreading, and at other times during **the** year drops well below one as a **reproduction** number, right? But what is it during seasonal flu time? AK : During **the** early stage when it ’ s taking off at **the** start of **the** flu season, it ’ s probably, we reckon, somewhere between about maybe 1. 2, 1. 4. So it ’ s not incredibly transmissible, if you imagine you do have some immunity in your population from vaccination and from other things. So it can spread, it ’ s above one, but it ’ s not taking off, necessarily, as quickly as **the** coronavirus is. CA : So I want to come back to two of those elements, specifically opportunity and transmission **probability**, because those seem to have **the** most chance to actually do something about this infection rate. Before we go there, let ’ s talk about another key number on this, which is **the** fatality rate. First of all, could you define – I think there ’ s two different versions of **the** fatality rate that maybe confuse people. Could you define them? AK : So **the** one that we often talk about is what ’ s known as **the** case fatality rate, and that ’ s of **the** proportion who show up with symptoms as cases, what proportion of those will subsequently be fatal. And we also sometimes talk about what ’ s known as **the** infection fatality rate, which is, of everyone who gets infected, regardless of symptoms, how many of those infections will subsequently be fatal. But most of **the** values we see kicking around are **the** case fatality rate, or **the** CFR, as it ’ s sometimes known. CA : And so what is that fatality rate for this virus, and again, how does that compare with other pathogens? AK : So there ’ s a few numbers that have been bouncing around. One of **the** challenges in real time is you often don ’ t see all of your cases, you have people symptomatic not being reported. You also have a delay. If you imagine, for example, 100 people turn up to a hospital with coronavirus and none have died yet, that doesn ’ t imply that **the** fatality rate is zero, because you ’ ve got to wait to see what might happen to them. So when you adjust for that underreporting and delays, best estimate for **the** case fatality is about one percent. So about one percent of people with symptoms, on average, those outcomes are fatal. And that ’ s probably about 10 times worse than seasonal flu. CA : Yeah, so that ’ s a scary comparison right there, given how many people die of flu. So when **the** World Health Organization mentioned a higher number, a little while back, of 3. 4 percent, they were criticized a bit for that. Explain why that might have been misleading and how to think about it and adjust for that. AK : It ’ s incredibly common that people look at these raw numbers, they say,”How many deaths are there so far, how many cases,”and they look at that ratio, and even a couple of weeks ago, that number produced a two percent value. But if you imagine you have this delay effect, then even if you stop all your cases, you will still have these kind of fatal outcomes over time, so that number will creep up. This has **occurred** in every epidemic from pandemic flu to Ebola, we see this again and again. And I made **the** point to a number of people that this number is going to go up, because as China ’ s cases slow, it will look like it ’ s increasing, and that ’ s just kind of a statistical quirk. There ’ s nothing really kind of, behind a change, there ’ s no mutations or anything going on. CA : If I have this right, there are two effects going on. One is that **the** number of fatalities from **the** existing caseload will rise, which actually would boost that 3. 4 even higher. But then you have to offset that against **the** fact that, apparently, huge numbers of cases have just gone undetected and that we haven ’ t, due to bad testing, that **the** number of fatalities don ’ t – they probably reflect a much larger number of early cases. Is that it? AK : Exactly. So you have one thing pulling **the** number up and one thing pulling it down. And it means that on these kind of early values, if you actually just adjust for **the** delay and don ’ t think about these unreported cases, you start getting really very scary numbers indeed. You get up to 20, 30 percent potentially, which really doesn ’ t align with what we know about this virus in general. CA : Alright. There ’ s a lot more data in now. From your point of view, you think **the** likely fatality rate, at least in **the** earlier stage of an infection, is about two percent? AK : I think overall, I think we can put something probably in **the** 0. 5 to two percent range, and that ’ s on a number of different data sets. And that ’ s for people who are symptomatic. I think on average, one percent is a good number to work with. CA : OK, one percent, So flu is often quoted as a tenth of a percent, so it ’ s five to 10 times or more more dangerous than flu. And that danger is not symmetric across age groups, as is well known. It primarily affects **the** elderly. AK : Yeah, we ’ ve seen that one percent on average, but once you start getting into **the** over 60s, over 70s, that number really starts to shoot up. I mean, we ’ re estimating potentially in these older groups, you ’ re looking at maybe five, 10 percent fatality. And then of course, on top of that, you ’ ve got to add what are going to be **the** severe cases and people are going to require hospitalization. And those risks get very large in **the** older groups indeed. CA : Adam, put these numbers together for us. In your models, if you put together a **reproduction** rate of two to three and a fatality rate of 0. 5 percent to one percent and you run **the** simulation, what does it look like? AK : So if you have this uncontrolled transmission, and you have this **reproduction** number of two or three and you don ’ t do anything about it, **the** only way **the** outbreak ends is enough people get it and immunity builds up and **the** outbreak kind of ends on its own. And in that case, you would expect very large numbers of **the** population to be infected. It ’ s what we see, for example, with many other uncontained outbreaks, that it essentially burns through **the** population, you get large numbers infected and with this kind of fatality rate and hospitalization rate, that would really be hugely damaging if that were to **occur**. Certainly at **the** country level, we ’ re seeing – Italy is a good example at **the** moment, that if you have that early transmission that ’ s undetected, that rapid growth, you very quickly get to a situation where your health systems are overwhelmed. I think one of **the** nastiest aspects of this virus is that because you have **the** delay between infection and symptoms and people showing up in health care, if your health system is overwhelmed, even on that day, if you completely stop transmission, you ’ ve got all of these people who have already been exposed, so you ’ re still going to have cases and severe cases appearing for maybe another couple of weeks. So it ’ s really this huge accumulation of infection and burden that ’ s coming through **the** system on your population. CA : So there ' s another key number, actually, is how does **the** total case number compare to **the** capacity of a country ' s health system to process that number of cases. Presumably that issue makes a huge difference to **the** fatality rate, **the** difference between people coming in with severe illness and a health system that ' s able to respond and one that ' s overwhelmed. **The** fatality rate is going to be very different at that point. AK : If someone requires an ICU bed, that ' s a couple of weeks they ' re going to require it for and you ' ve got more cases coming through **the** system, so it very quickly gets very tough. CA : So talk about **the** difference between containment and mitigation. These are different terms that we ' re hearing a lot about. In **the** early stages of **the** virus, governments are focused on containment. What does that mean? AK : Containment is this idea that you can focus your effort on control very much on **the** cases and their contacts. So you ' re not causing disruption to **the** wider population, you have a case that comes in, you isolate them, you work out who they ' ve come into contact with, who ' s potentially these opportunities for exposure and then you can follow up those people, maybe quarantine them to make sure that no further transmission happens. So it ' s a very focused, targeted method, and for SARS, it worked remarkably well. But I think for this infection, because some cases are going to be missed or undetected, you ' ve really got to be capturing a large chunk of people at risk. If a few slip through **the** net, potentially, you ' re going to get an outbreak. CA : Are there any countries that have been able to employ this strategy and effectively contain **the** virus? AK : So Singapore have been doing a really remarkable job of this for **the** last six weeks or so. So as well as some wider measures, they ' ve been working incredibly hard to trace people who have come into contact. Looking at CCTV, going through to find out which taxi someone might have gotten, who might be at risk – really, really thorough follow-up. And for about six weeks, that has kept a lid on transmission. CA : So that ' s amazing. So someone comes into **the** country, they test positive – they go to work, and with a massive team, and trace everything to **the** level of actually saying,"Oh, you don ' t know what taxi you went in? Let us find that out for you."And presumably, when they find **the** taxi driver, they then have to try and figure out everyone else who was in that taxi? AK : So they will focus on close contacts of people most at risk, but they ' re really minimizing **the** chance that anyone slips through **the** net. CA : But even in Singapore, if I ' m not mistaken, numbers started to trend back down to zero, but I think recently, they ' ve picked up again a bit. It ' s still **unclear** whether they will actually be able to sustain containment. AK : Exactly. If we talk in terms of **the reproduction** number, we saw it dipped to maybe 0. 8, 0. 9, so under that crucial value of one. But in **the** last week or two, it does seem to be ticking up and they ' re getting more cases appearing. I think a lot of it is, even if they are containing it, **the** world is experiencing outbreaks and just keeps throwing sparks of infection, and it becomes harder and harder with that level of intensive effort to stamp them all out. CA : In **the** case of this virus, you know, there was warning to most countries in **the** world that this thing was happening. **The** news out of China very quickly became very bleak, and people had time to prepare. I mean, what would ideal preparation look like if you know that something like this is coming and you know that there ' s a lot on **the** line if you can successfully contain it before it really escapes? AK : I think two things would make a really big difference. One is having as thorough a follow-up and detection as possible. We ' ve done some modeling analyses, looking at how effective that kind of early containment is. And it can be, if you ' re identifying maybe 70 or 80 percent of **the** people who might have come into contact. But if you ' re not **detecting** those cases coming in, if you ' re not **detecting** their contacts – and a lot of **the** early focus, for example, was on travel history to China, and then it became clear that **the** situation was changing, but because you were relying on that as your definition of a case, it meant a lot of maybe other cases that matched **the** definition weren ’ t being tested because they didn ’ t seem to be potentially at risk. CA : So I mean, if you know that early detection is key to this, an essential early measure, I guess, would be to rapidly ensure that you had enough tests available and where needed, so that you could respond, be ready to swing into action as soon as someone was **detected**, you then have to very quickly, I guess, test their contacts and so forth, to have a chance of keeping this under control. AK : Exactly. In my line of work, we say there ’ s value in a negative test, because it shows that you ’ re looking for something and it ’ s not there. And so I think having small numbers of people tested doesn ’ t give you confidence that you ’ re not missing infections, whereas if you are doing really thorough follow-up on contacts, we ’ ve seen countries even like Korea now, huge numbers of people tested. So although there are still cases appearing, it gives them more confidence that they have some sense of where those infections are. CA : I mean, you ’ re in **the** UK right now, I ’ m in **the** US. How likely is it that **the** UK is going to be able to contain, how likely is it that **the** US is going to be able to contain this? AK : I think it ’ s pretty unlikely in both cases. I think **the** UK is going to have to introduce some additional measures. I think when that happens obviously depends a bit on **the** current situation, but we ’ ve tested almost 30, 000 people now. Frankly, I think **the** US may well be moving beyond that point, given how much evidence of extensive transmission that has, and I think without clear ideas of how much infection there is and that level of testing, it ’ s quite hard to actually see what **the** picture currently is in **the** US. CA : I mean, I definitely don ’ t want to get too political about this, but I mean, does this strike you as – you just said that **the** UK has tested 30, 000 people – **the** US is five or six times bigger and I think **the** total number of tests here is five or six thousand, or it was a few days ago. Does that strike you as bizarre? I don ’ t understand, honestly, how that happened in an educated country that has so much knowledge about infectious diseases. AK : It does, and I think there ’ s obviously a number of factors playing in there, logistics and so on, but there has been that period of warning that this is a threat and this is coming in. And I think countries need to make sure that they ’ ve got **the** capacity to really do as much detection as they can in those early stages, because that ’ s where you ’ re going to catch it and that ’ s where you ’ re going to have a better chance of containing it. CA : OK, so if you fail to contain, then you have to move to some kind of mitigation strategy. So what comes into play there? And I think I almost want to bring that back to two of your DOTS factors, opportunity and transmission **probability**, because it seems like **the** virus is what it is, **the** actual duration when someone is potentially infectious, we can ’ t do much about. **The** susceptibility side, we can ’ t do much about until there ’ s a vaccine. We could maybe talk about that in a bit. But **the** middle two of opportunity and transmission **probability**, we can do something about. Do you want to maybe talk about those in turn, of what that looks like, how would you build a mitigation strategy? I mean, first of all, thinking about opportunity, how do you reduce **the** number of opportunities to pass on **the** bug? AK : And so I think in that respect, it would be about massive changes in our social interactions. And if you think in terms of **the reproduction** number of being about two or three, to get that number below one, you ’ ve really got to cut some aspect of that transmission in half or in **two-thirds** to get that below one. And so that would require, of **the** opportunities that could spread **the** virus, so these kind of close contacts, everybody in **the** population, on average, will be needing to reduce those interactions potentially by **two-thirds** to bring it under control. That might be through working from home, from changing lifestyle and kind of where you go in crowded places and dinners. And of course, these measures, things like school closures, and other things that just attempt to reduce **the** social mixing of a population. CA : Well, actually, talk to me more about school closures, because that, if I remember, often in past pandemics has been cited as an absolutely key measure, that schools represent this sort of coming together of people, children are often – certainly when it comes to flu and colds – they ’ re carriers. But on this case, children don ’ t seem to be getting sick from this particular virus, or at least very few of them are. Do we know whether they can still be infectious? They can be **the** unintended carriers of it. Or actually, is there evidence that school closures may not be as important in this instance as it is in others? AK : So that point on what role children play is a crucial one, and there ’ s still not a good evidence base there. From following up of contacts of cases, there ’ s now evidence that children are getting infected, so when you ’ re testing, they are getting exposed, it ’ s not that somehow they ’ re just not getting **the** infection at all, but as you said, they ’ re not showing symptoms in **the** same way. And particularly for flu, when we see **the** implications of school closures, even in **the** UK in 2009 during swine flu, there was a dip in **the** outbreak during **the** school holidays, you could see it on **the** epidemic curve, it kind of comes back down in **the** summer and goes back up in **the** autumn. But of course, in 2009, there was some immunity in older groups. That kind of shifted more **the** transmission into **the** younger ones. So I think it ’ s really something we ’ re trying to work to understand. Obviously, it will reduce interactions, with school closures, but then there ’ s knock-on social effects, there ’ s potential knock-on changes in mixing, maybe grandparents and their role, in terms of alternative carers if parents have to work. So I think there ’ s a lot of pieces that need to be considered. CA : I mean, based on all of **the** different pieces of evidence you ’ ve seen, if it were down to you, would you be recommending that most countries at this point do look hard at extensive school closures as a precautionary measure, that it ’ s just worth it to do that as a sort of painful two, three, four, five-month strategy? What would you recommend? AK : I think **the** key thing, given **the** age distribution of risk and **the** severity in older groups is reduce interactions that bring **the** infection into those groups. And then amongst everyone else, reduce interactions as much as possible. I think **the** key thing is we ’ ve got so much of **the** disease burden in **the** kind of 60-plus group that it ’ s not just about everyone trying to avoid everyone ’ s interactions, but it ’ s **the** kind of behaviors that would drive infections into those groups. CA : Does that mean that people should think twice before, I don ’ t know, visiting a loved one in an old people ’ s home or in a residential facility? Like that, we should just pay super special attention to that, should all these facilities be taking great care about who they admit, taking temperature and checking for symptoms or something like that? AK : I think those measures definitely need to be considered. In **the** UK, we ’ re getting plans for potentially what ’ s known as a cocooning strategy for these older groups that we can really try and **seal** off interactions as much as possible from people who might be bringing infection in. And ultimately, because as you said, we can ’ t target these other aspects of transmission, it is just reducing **the** risk of exposure in these groups, and so I think anything at **the** individual level you can do to get people reducing their risk, if either they ’ re elderly or in other risk groups, I think is crucial. And I think more at **the** general level those kind of more large-scale measures can help reduce interactions overall, but I think if those reductions are happening and not reducing **the** risk for people who are going to get severe disease, then you ’ re still going to get this really remarkably severe burden. CA : I mean, do people have to almost apply this double lens as they think about this stuff?

There ’ s **the** risk to you as you go about your life, of you catching this bug. But there ’ s also **the** risk of you being, unintentionally, a carrier to someone who would suffer much more than you might. And that both those things have to be top of mind right now. AK : Yeah, and it ’ s not just whose hand you shake, it ’ s whose hand that person goes on to shake. And I think we need to think about these second-degree steps, that you might think you have low risk and you ’ re in a younger group, but you ’ re often going to be a very short step away from someone who is going to get hit very hard by this. And I think we really need to be socially minded and think this could be quite dramatic in terms of change of behavior, but it needs to be to reduce **the** impact that we ’ re potentially facing. CA : So **the** opportunity number, we bring down by just reducing **the** number of physical contacts we have with other people. And I guess **the** transmission **probability** number, how do we bring that down? That impacts how we interact. You mentioned hand-shaking, I ’ m guessing you ’ re going to say no handshaking. AK : Yeah, so changes like that. I mean, another one, I think, handwashing operates in a way that we might be still be doing activities that we ’ ve previously done, but handwashing reduces **the** chance that from one interaction to another, we might be spreading infection, so it ’ s all of these measures that mean that even if we ’ re having these exposures, we ’ re taking additional steps to avoid any transmission happening. CA : I still think most people don ’ t fully understand or don ’ t have a clear model of **the** pathway by which this thing spreads. So you think definitely people understand that you don ’ t breathe in **the** water droplets of someone who has just coughed or sneezed. So how does it spread? It gets onto **surfaces**. How? Do people just breathe out and it goes on from people who are sick, they touch their mouth or something like that, and then touch a **surface** and it gets on that way? How does it actually get onto **surfaces**? AK : I think a lot of it would be that you cough in your hand and it ends up on a **surface**. But I think **the** challenge, obviously, is untangling these questions of how transmission happens. You have transmission in a household, and is it that someone coughed and it got on a **surface**, is it direct contact, is it a handshake, and even for things like flu, that ’ s something that we work quite hard to try and unpick, how does social behavior correspond to infection risk. Because it ’ s clearly important, but pinning it down is really tough. CA : It ’ s almost like embracing **the** fact that for a lot of these things, we actually don ’ t know and that we ’ re all in this game of **probabilities**. Which, in a way, is why I think **the** math is so important here. That you have to think of this as these multiple numbers working together on each other, they all have their part to play. And any of them that you can take down by a percentage is likely contributing, not just to you but to everyone. And people don ’ t actually know in detail how **the** numbers go together, but they know that they probably all matter. We almost need people to, somehow, you know, embrace that uncertainty and then try to get some satisfaction by acting on every single part of it. AK : I think it ’ s this idea that if on average, you ’ re infecting, say, three people, what ’ s driving that and how can you chip away at that value? If you ’ re washing your hands, how much might that chip away in terms of **the** handshakes, you know, you may have had virus and you no longer do, or if you are changing your social behavior in a certain way, is that taking away a couple of interactions, is that taking away half? How can you really chip into that number as much as you possibly can? CA : Is there anything else to say about how we could reduce that transmission **probability** in our interactions? Like, what is **the** physical distance that it ’ s wise to stay away from other people if we can? AK : I think it ’ s hard to pin down exactly, but I think one thing to bear in mind is that there ’ s not so much evidence that this is a kind of aerosol and it goes really far – it ’ s reasonably short distances.

I don ’ t think it ’ s **the** case that you ’ re sitting a few meters away from someone and **the** virus is somehow going to get across. It ’ s in closer interactions, and it ’ s why we ’ re seeing so many transmission events **occur** in things like meals and really tight-knit groups. Because if you imagine that ’ s where you can get a virus out and onto **surfaces** and onto hands and onto faces, and it ’ s really situations like that we ’ ve got to think more about. CA : So in a way, some of **the** fears that people have may actually be overstated, like, if you ’ re in **the** middle of an **airplane** and someone at **the** front sneezes, I mean, that ’ s annoying, but it ’ s actually not **the** thing you should be most freaked out about. There are much smarter ways to pay attention to your well-being. AK : Yeah, if it was measles and **the** plane was susceptible people, you would see a lot of infections after that. I think it is, bear in mind, that this is, on average, people infecting two or three others, so it ’ s not **the** case of your maybe 50 interactions over a week, all of those people are at risk. But it ’ s going to be some of them, particularly those close contacts, that are going to be where transmission ’ s **occurring**. CA : So talk about, from a sort of national strategy point of view. There ’ s a lot of talk about **the** need to”flatten **the** curve.”What does that mean? AK : I think it refers to this idea that for your health systems, you don ’ t want all your cases to appear at **the** same time. So if we sat back and did nothing and just let **the** epidemic grow, and you had this growth rate that, at **the** moment, in some places is looking like maybe three to four days, you ’ re getting doubling. So every three or four days, **the** epidemic is doubling. It will skyrocket and you ’ ll end up with a whole bunch of really severely ill people needing hospital care all at **the** same time, and you just won ’ t have capacity for it. So **the** idea of flattening **the** curve is if we can slow transmission, if we can get that **reproduction** number down, then there may still be an outbreak, but it will be much flatter, it will be longer and there will be fewer severe cases showing up, which means that they can get **the** health care they need. CA : Does it imply that there will be fewer cases overall, or – When you look at **the** actual images of people showing what flattening **the** curve looks like, it almost looks as if you ’ ve got **the** same area still under **the** graph, i. e. that **the** same number of people, ultimately, are infected but over a longer period. Is that typically what happens, and even if you adopt all these strategies of social distancing and washing hands and etc. that **the** best you can hope for is that you slow **the** thing down, you actually will get as many people infected in **the** end? AK : Not necessarily – it depends on **the** measures that go in. There are some measures like, shutting down travel, which typically delay **the** spread rather than reduce it. So you ’ re still going to get **the** same outbreaks, but you ’ re stretching out **the** outbreaks. But there are other measures. If we talk about reducing interactions, if your **reproduction** number ’ s lower, you would expect fewer cases overall. And eventually, in your population, you will get some buildup of immunity, which would help you out if you think about **the** components, reducing susceptibility, alongside what ’ s going on elsewhere. So **the** hope is that **the** two things will work together. CA : So help me understand what **the** endgame is here. So, take China, for example. Whatever you make of **the** early suppression of data and so forth that seems pretty troubling there. **The** intensity of **the** response come January time or whatever, with **the** shutdown of this huge area of **the** country, seems to have actually been effective. **The** number of cases there are falling at a shockingly high rate in some ways. Falling to almost nothing. And I can ’ t understand that. You are talking about a country of, whatever, 1. 4 billion people. There have been a huge number of cases there, but it was a tiny **fraction** of **the** population have actually got sick. And yet, they ’ ve got **the** number way down. It ’ s not like every other person in China has somehow developed immunity. Is it that they have been absolutely disciplined about shutting down travel from **the** infected regions and somehow really dialed up, massively dialed up testing at any sign of any problem, so that literally, they are back in containment mode in most parts of China? I can ’ t get my head around it, help me understand it. AK : So we estimated, in **the** last two weeks of January, when these measures went in, **the reproduction** number went from about 2. 4 to 1. 1. So about 60 percent decline in transmission in **the** space of a week or two. Which is remarkable and really, a lot of it is likely to be driven by just fundamental change in social behavior, huge social distancing, really intensive follow-up, intensive testing. And it got to **the** point where it took enough off **the reproduction** number to cause **the** decline, and now, of course, we ’ re seeing, in many areas, a transition back to more of this kind of containment, because there ’ s few cases, it ’ s more manageable. But we ’ re also seeing them face a challenge, because a lot of these cities have basically been locked down for six weeks and there ’ s a limit to how long you can do that for. And so some of these measures are gradually starting to be lifted, which of course creates **the** risk that cases that are appearing from other countries may subsequently go in and reintroduce transmission. CA : But given how infectious **the** bug is, and how many theoretical pathways and connection points there are between people in Wuhan, even in shutdown, or relatively shut down, or **the** other places where there ’ s been some infection and **the** rest of **the** country, does it surprise you how quickly that curve has gone down to nearly zero? AK : Yes. Early on when we saw that flattening off in cases in those first few days, we did wonder whether it was just they hit a limit in testing capacity and they were reporting 1, 000 a day, because that ’ s all **the** kits they had. But it continued, thankfully, and it shows that it is possible to turn this over with that level of intervention. I think **the** key thing now is seeing how it works in other settings. Italy now are putting in really dramatic interventions. But of course, because of this delay effect, if you put them in today, you won ’ t necessarily see **the** effects on cases for another week or two. So I think working out what impact that ’ s had is going to be key for helping other countries work on how to contain this. CA : To have a picture, Adam, of how this is likely to play out over **the** next month or two, give us a couple of scenarios that are in your head. AK : I think **the** optimistic scenario is that we ’ re going to learn a lot from places like Italy that have unfortunately been hit very hard. And that countries are going to take this very seriously and that we ’ re not going to get this continued growth that ’ s going to overwhelm totally, that we ’ re going to be able to sufficiently slow it down, that we are going to get large numbers of cases, we ’ re probably going to get a lot of severe cases, but that will be more manageable, that ’ s **the** kind of optimistic scenario. I think if we have a point where countries either don ’ t take this seriously or populations don ’ t respond well to control measures or it ’ s not **detected**, we could get situations – I think Iran is probably **the** closest one at **the** moment – where there ’ s been extensive widespread transmission, and by **the** time it ’ s being responded to, those infections are already in **the** system and they are going to turn up as cases and severe illness. So I ’ m hoping we ’ re not at that point, but we ’ ve certainly got, at **the** moment, potentially about 10 countries on that trajectory to have **the** same outlook as Italy. So it ’ s really crucial what happens in **the** next couple of weeks. CA : Is there a real chance that quite a few countries end up having, this year, substantially more deaths from this virus than from seasonal flu? AK : I think for some countries that is likely, yeah. I think if control is not possible, and we ’ ve seen it happen in China, but that was really just an unprecedented level of intervention. It was really just changing **the** social fabric. I think people, many of us, don ’ t really appreciate, at a glance, just what that means, to reduce your interactions to that extent. I think many countries just simply won ’ t be able to manage that. CA : It ’ s almost a challenge to democracies, isn ’ t it –”OK, show us what you can do without that kind of draconian control. If you don ’ t like **the** thought of that, come on, citizens, step up, show us what you ’ re capable of, show that you can be wise about this and smart and self-disciplined, and get ahead of **the** damn bug.”AK : Yeah. CA : I mean, I ’ m not personally superoptimistic about that, because there ’ s such conflicting messaging coming out in so many different places, and people don ’ t like to short-term sacrifice. I mean, is there almost a case that – I mean, what ’ s your view on whether **the** media has played a helpful role here or an unhelpful role? Is it actually, in some ways, helpful to, if anything, overstate **the** concern, **the** fear, and actually make people panic a little bit? AK : I think it ’ s a really tough balance to strike, because of course, early on, if you don ’ t have cases, if you don ’ t have any evidence of potential pressure, it ’ s very hard to get that message and convince people to take it seriously if you ’ re overhyping it. But equally, if you ’ re waiting too long, and saying it ’ s not a concern yet, we ’ re OK for **the** moment, a lot of people think it ’ s just flu. By **the** time it hits hard, as I ’ ve said, you ’ re going to have weeks of an overburdened health system, because even if you take interventions, it ’ s too late to control **the** infections that have happened. So I think it ’ s a fine line, and my hope is there is this ramp-up in messaging, now people have these tangible examples like Italy, where they can see what ’ s going to happen if they don ’ t take it seriously. But certainly, of all **the** diseases I ’ ve seen, I think many of my colleagues who are much older than me and have memories of other outbreaks, it ’ s **the** scariest thing we ’ ve seen in terms of **the** impact it could have, and I think we need to respond to that. CA : It ’ s **the** scariest disease you ’ ve seen. Wow. I ’ ve got some questions for you from my friends on Twitter. Everyone is obviously super exercised about this topic. Hypothetically, if everyone stayed home for three weeks, would that effectively **wipe** this out? Is there a way to socially distance ourselves out of this? AK : Yeah, I think in certain countries with reasonably small household sizes, I think average in **the** UK, US is about two and a half, so even if you had a round of infection within **the** household, that would probably stamp it out. As a secondary benefit, you may well stamp out a few other infections, too. Measles only circulates in humans, so you may have some knock-on effect, if, of course, that were ever to be possible. CA : I mean, obviously that would be a huge dent to **the** economy, and this is in a way almost, like, one of **the** underlying challenges here is that you can ’ t optimize public policy for both economic health and fighting a virus. Like, those two things are, to some extent, in conflict, or at least, short-term economic health and fighting a virus. Those two things are in conflict, right? And societies need to pick one. AK : It is tough to convince people of that balance, **the** thing we always say of pandemic planning is it ’ s cheap to put this stuff in place now – otherwise, you ’ ve got to pay for it later. But unfortunately, as we ’ ve seen with this, that a lot of early money for response wasn ’ t there. And it ’ s only when it has an impact and when it ’ s going to get expensive that people are happy to take that cost on board, it seems. CA : OK, some more Twitter questions. Will **the** rising temperature in coming weeks and months slow down **the** COVID-19 spread? AK : I haven ’ t seen any convincing evidence that there ’ s that strong pattern with temperature, and we ’ ve seen it for other infections that there is this seasonal pattern, but I think **the** fact we ’ re getting widespread outbreaks makes it hard to identify, and of course, there ’ s other things going on. So even if one country doesn ’ t have as big an outbreak as another, that ’ s going to be influenced by control measures, by social behavior, by opportunities and these things as well. So it would be really reassuring if this was **the** case, but I don ’ t think we can say that just yet. CA : Continuing from Twitter, I mean, is there a standardized global recommendation for all countries on how to do this? And if not, why not? AK : I think that ' s what people are trying to piece together, first in terms of what works. It ' s only really in **the** last sort of few weeks we ' ve got a sense that this thing can be controllable with this extent of interventions, but of course, not all countries can do what China have done, some of these measures incur a huge social, economic, psychological burden on populations. And of course, there ' s **the** time limit. In China they ' ve had them in for six weeks it ' s tough to maintain that, so we need to think of these tradeoffs of all **the** things we can ask people to do, what ' s going to have **the** most impact on actually reducing **the** burden. CA : Another question : How did this happen and is it likely to happen again? AK : So it ' s likely that this originated with **the** virus that was circling in bats and then probably made its way through another species into humans somehow, there ' s a lot of bits of evidence around this, there ' s not kind of single, clear story, but even for SARS, it took several years for genomics to piece together **the exact** route that it happened. But certainly, I think it ' s plausible that it could happen again. Nature is throwing out these viruses constantly. Many of them aren ' t well-adapted to humans, don ' t pick up, you know, there may well have been a virus like this a few years ago that just happened to infect someone who just didn ' t have any contacts and didn ' t go any further. I think we are going to face these things and we need to think about how can we get in early at **the** stage where we ' re talking small numbers of cases, and even something like this is containable, rather than **the** situation we ' ve got now. CA : It seems like this isn ' t **the** first time that a virus seems to have emerged from, like, a wild meat market. That ' s certainly how it happens in **the** movies. And I think China has already taken some steps this time to try to crack down on that. I guess that ' s potentially quite a big deal for **the** future if that can be properly maintained. AK : It is, and we saw, for example, **the** H7N9 avian flu, over **the** last few years, in 2013, it was a big emerging concern, and China made a very extensive response in terms of changing how they operate their markets and vaccination of birds and that seems to have removed that threat. So I think these measures can be effective if they ' re identified early on. CA : So talk about vaccinations. That ' s **the** key measure, I guess, to change that susceptibility factor in your equation. There ' s obviously a race on to get these vaccinations out there, there are some candidate vaccinations there. How do you see that playing out? AK : I think there ' s certainly some promising development happening, but I think **the** timescales of these things are really on **the** order of maybe a year, 18 months before these things be widely available. Obviously, a vaccine has to go through these stages of trials, that takes time, so even if by **the** end of **the** year, we have something which is viable and works, we ' re still going to see a delay before everyone can get ahold of it. CA : So this really puzzles me, actually, and I ' d love to ask you as a mathematician about this as well. There are already several companies believing that they have plausible candidate vaccines. As you say, **the** process of testing takes forever. Is there a case that we ' re not thinking about this right when we ' re looking at **the** way that testing is done and that **the** safety calculations are made? Because it ' s one thing if you ' re going to introduce a brand new drug or something – yes, you want to test to make sure that there are no side effects, and that can take a long time by **the** time you ' ve done all **the** control trials and all **the** rest of it. If there ' s a global emergency, isn ' t there a case, both mathematically and ethically, that there should just be a different calculation, **the** question shouldn ' t be"Is there any possible case where this vaccine can do harm,"**the** question surely should be,"On **the** net **probabilities**, isn ' t there a case to roll this out at scale, to have a shot at nipping this thing in **the** bud?"I mean, what am I missing in thinking that way?

AK : I mean, we do see that in other situations, for example, **the** Ebola vaccine in 2015 showed, within a few months, very promising evidence and interim results of **the** trial in humans showed what seemed very high efficacy. And even though it hadn ’ t been licensed fully, it was employed for what is known as compassionate use in subsequent other outbreaks. So there are these mechanisms where vaccines can be fast-tracked in this way. But of course, we ’ re currently in a situation where we have no idea if these things will do anything at all. So I think we need to accrue enough evidence that it could have an impact, but obviously, fast-track that as much as possible. CA : But **the** skeptic in me still doesn ’ t fully get this. I don ’ t understand why there isn ’ t more energy behind bolder thinking on this. Everyone seems, despite **the** overall risk, incredibly risk-averse about how to build **the** response to it. AK : So with **the** caveat that, yeah, there ’ s a lot of good questions on this, and some of them are slightly outside my wheelhouse, but I agree that we need to do more to get timescales out. **The** example I always quote is it takes us six months to choose a seasonal flu strain and get **the** vaccines out there to people. We always have to try and predict ahead which strains are going to be circulating. And that ’ s for something we know how to make and has been manufactured for a long time. So there is definitely more that needs to be done to get these timescales shorter. But I think we do have to balance that, especially if we ’ re exposing large numbers of people to something to make sure that we ’ re confident it ’ s safe and that it ’ s going to have some benefit, potentially. CA : And so, finally, Adam, I guess going into this – There ’ s another set of infectious things happening around **the** world at **the** same time, which is ideas and **the** communication around this thing. They really are two very dynamic, interactive systems of infectiousness – there ’ s some very damaging information out there. Is it fair to think of this as battle of credible knowledge and measures against **the** bug, and just bad information – You know, part of what we have to think about here is how to suppress one set of things and boost **the** other, actually, turbocharge **the** other. How should we think of this? AK : I think we can definitely think of it almost as competition for our attention, and we see similarly, with diseases, you have viruses competing to infect susceptible hosts. And I think we ’ re now seeing, I guess over **the** last few years with fake news and misinformation and **the** emergence of awareness, more of a transition to thinking about how do we reduce that susceptibility if we have people that can be in these different states, how can we try and preempt better with information. I think **the** challenge for an outbreak is obviously, early on, we have very little good information, and it ’ s very easy for certainty and confidence to fill that vacuum. And so I think that is something – I know platforms are working on how can we get people exposed to good information earlier, so hopefully protect them against other stuff. CA : One of **the** big unknowns to me in **the** year ahead – let ’ s say that **the** year ahead includes many, many more weeks, for many people, of actually self-isolating. Those of us who are lucky enough to have jobs where you can do that. You know, staying home. By **the** way, **the** whole injustice of this situation, where so many people can ’ t do that and continue to make a living, is, I ’ m sure, going to be a huge deal in **the** year ahead and if it turns out that death rates are much higher in **the** latter group than in **the** former group, and especially in a country like **the** US, where **the** latter group doesn ’ t even have proper health insurance and so forth. That feels like right there, that could just become a huge debate, hopefully a huge source of change at some level. AK : I think that ’ s an incredibly important point, because it ’ s very easy – I similarly have a job where remote working is fairly easy, and it ’ s very easy to say we should just stop social interactions, but of course, that could have an enormous impact on people and **the** choices and **the** routine that they can have. And I think those do need to be accounted for, both now and what **the** effect is going to look like a few months down **the** line. CA : When all ' s said and done, is it fair to say that **the** world has faced, actually, much graver problems in **the** past, that on any scenario, it ' s highly likely that at some point in **the** next 18 months, let ' s say, a vaccine is there and starts to get wide distribution, that we will have learned lots of other ways to manage this problem? But at some point, next year probably, **the** world will feel like it ' s got on top of this and can move on. Is that likely to be it, or is this more likely to be, you know, it escapes, it ' s now an endemic nightmare that every year picks off far more people than are picked off by **the** flu currently. What are **the** likely ways forward, just taking a slightly longer-term view? AK : I think there ' s plausible ways you could see all of those potentially playing out. I think **the** most plausible is probably that we ' ll see very rapid growth this year and lots of large outbreaks that don ' t recur, necessarily. But there is a potential sequence of events that could end up with these kind of multiyear outbreaks in different places and reemerging. But I think it ' s likely we ' ll see most transmission concentrated in **the** next year or so. And then, obviously, if there ' s a vaccine available, we can move past this, and hopefully learn from this. I think a lot of **the** countries that responded very strongly to this were hit very hard by SARS. Singapore, Hong Kong, that really did leave an impact, and I think that ' s something they ' ve drawn on very heavily in their response to this. CA : Alright. So let ' s wrap up maybe by just encouraging people to channel their inner mathematician and especially think about **the** opportunities and **the** transmission **probabilities** that they can help shift. Just remind us of **the** top three or four or five or six things that you would love to see people doing. AK : I think at **the** individual level, just thinking a lot more about your interactions and your risk of infection and obviously, what gets onto your hands and once that gets onto your face, and how do you potentially create that risk for others. I think also, in terms of interactions, with things like handshakes and maybe contacts you don ' t need to have. You know, how can we get those down as much as possible. If each person ' s giving it to two or three others, how do we get that number down to one, through our behavior. And then it ' s likely that we ' ll need some larger-scale interventions in terms of gatherings, conferences, other things where there ' s a lot of opportunities for transmission. And really, I think that combination of that individual level, you know, if you ' re ill or potentially you ' re going to get ill, reducing that risk, but then also us working together to prevent it getting into those groups who, if it continues to be uncontrolled, could really hit some people very, very hard. CA : Yeah, there ' s a lot of things that we may need to gently let go of for a bit. And maybe try to reinvent **the** best aspects of them. Thank you so much. If people want to keep up with you, first of all, they can follow you on Twitter, for example. What ' s your Twitter handle? AK : So @AdamJKucharski, all one word. CA : Adam, thank you so much for your time, stay well. AK : Thank you. CA : Associate professor and TED Fellow Adam Kucharski. We ' d love to hear what you think of this bonus episode. Please tell us by rating and reviewing us in Apple Podcasts or your favorite podcast app. Those reviews are influential, actually. We certainly read every one, and truly appreciate your feedback. This week ' s show was produced by Dan O ' Donnell at Transmitter Media. Our production manager is Roxanne Hai Lash, our fact-checker Nicole Bode. This episode was mixed by Sam Bair. Our **theme** music is by Allison Layton-Brown. Special thanks to my colleague Michelle Quint. Thanks for listening to **the** TED Interview. We ' ll be back later this **spring** with a whole new season ' s worth of deep dives with great minds. I hope you ' ll **enjoy** them whether or not life is back to normal. I ' m Chris Anderson, thanks for listening and stay well.

Text Sample 75: POS Dim 1 – 1990 – Score 4.00 – 1990/t000287.txt

I ’ m going to go right into **the slides**. And all I ’ m going to try and prove to you with these **slides** is that I do just very straight stuff. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at **the** end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into **the slides**. Can you turn off **the** light? Down. I like to be in **the** dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is **the slide** on **the** right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On **the** left-hand side, I had **the** context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with **the** context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that **slide** on **the** right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for **the** study of law. And we continue to work with this client. **The** building on **the** right at **the** top is now under construction. **The** garage on **the** right â **the** gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to **the** clients and **the** students from **the** very first meeting saying they felt denied a place. They wanted a sense of place. And so **the** whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage **the** neighborhood â one made accommodations. I tried to be inclusive, to include **the** buildings in **the** neighborhood, whether they were buildings I liked or not. In **the** ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And **the** success of it threw me for a loop. I couldn ’ t deal with **the** success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by **the** slice. And after we failed, I just kept failing. **The** piece on **the** left â and that ultimately led to **the** piece on **the** right â happened when **the** kid that was working on this took one of those long strings of stuff and folded it up to put it in **the** wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into **fish**. I mean, **the** story I tell is that I got mad at postmodernism â at po-mo â and said that **fish** were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to **the** beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as **the** one glass at **the** Walker. And then, I sliced off **the** head and **the** tail and everything and tried to translate what I was learning about **the** form of **the fish** and **the** movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after **the** contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And **the** client said,"Why no **fish**?"And so I made a drawing with a **fish**, and I left Japan. Three weeks later, I received a complete set of drawings saying we ' d won **the** competition. Now, it ' s hard to do. It ' s hard to translate a **fish** form, because they ' re so beautiful â perfect â into a building or **object** like this. And Oldenburg, who I work with a little once in a while, told me I couldn ' t do it, and so that made it even more exciting. But he was right â I couldn ' t do **the** tail. I started to get **the** head OK, but **the** tail I couldn ' t do. It was pretty hard. **The** thing on **the** right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with **the** context again. Now, if you saw a picture of this as it was published in Architectural Record â they didn ' t show **the** context, so you would think,"God, what a pushy guy this is."But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ' t find it. So... As for craft and technology and all those things that you ' ve all been talking about, I was thrown for a complete loop. This was built in six months. **The** way we sent drawings to Japan : we used **the** magic **computer** in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made **the** drawings of **the fish** and **the** scales. And when I got there, everything was perfect â except **the** tail. So, I decided to cut off **the** head and **the** tail. And I made **the object** on **the left** for my show at **the** Walker. And it ' s one of **the** nicest pieces I ' ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ' t want to talk about, it got delayed. Toxic waste, I guess, is **the** key clue to that one. And so we built a temporary building â I ' m getting good at temporary â and we put a conference room in that ' s a **fish**. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ' s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in **the** bathtub when I was a kid. I played with it in **the** evening. When I went to sleep, **the** next day it wasn ' t there. And **the** next night, we had gefilte **fish**. And so I set up this interior for Jay ' s offices and I made a pedestal for a sculpture. And he didn ' t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ' s, and I put **the fish** in. It was a joke. I play with funny people like [Claes] Oldenburg. We ' ve been friends for a long time. And we ' ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called"Il Corso del Coltello"â **the** Swiss Army knife. And most of **the** imagery is â Claes ', but those two little boys are my sons, and they were Claes ' assistants in **the** play. He was **the** Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like **the** AT&T building by Claes â with a **fish** hat. **The** highlight of **the** performance was at **the** end. This beautiful **object**, **the** Swiss Army knife, which I get credit for participating in. And I can tell you â it ' s totally an Oldenburg. I had nothing to do with it. **The** only thing I did was, I made it possible for them to turn those blades so you could sail this thing in **the** canal, because I love sailing. We made it into a sailing craft. I ' ve been known to mess with things like chain link fencing. I do it because it ' s a curious thing in **the** culture, when things are made in such great quantities, absorbed in such great quantities, and there ' s so much denial about them. People hate it. And I ' m fascinated with that, which, like **the** paper furniture â it ' s one of those materials. And I ' m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in **the** Loyola Law School. And that chain link is really expensive. It ' s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, **the** milk can is his. But we were trying to collide our ideas, to put **objects** next to each other. Like a Morandi â like **the** little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in **the** middle. And Jay asked me what I was going to do with **the** piece in **the** middle. And he pushed that. And one day I had a â oh, well, **the** other way. I had **the** binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made **the** binoculars incredible when he sent me **the** first model of **the** real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to **the** building on **the left**. And I still think **the** Time magazine picture will be of **the** binoculars, you know, leaving out **the** â what **the** hell. I use a lot of metal in my work, and I have a hard time connecting with **the** craft. **The** whole thing about my house, **the** whole use of rough carpentry and everything, was **the** frustration with **the** crafts available. I said,"If I can ' t get **the** craft that I want, I ' ll use **the** craft I can get."There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into **the** metal because it was a way of building a building that was a sculpture. And it was all of one material, and **the** metal could go on **the** roof as well as **the** walls. **The** metalworkers, for **the** most part, do ducts behind **the** ceilings and stuff. I was given an opportunity to design an exhibit for **the** metalworkers ' unions of America and Canada in Washington, and I did it on **the** condition that they become my partners in **the** future and help me with all future metal buildings, etc. etc. And it ' s working very well to have these people, these craftsmen, interested in it. I just tell **the** stories. It ' s a way of connecting, at least, with some of those people that are so important to **the** realization of architecture. **The** metal continued into a building â Herman Miller, in Sacramento. And it ' s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ' s kind of a trite thing to say, but it is real that they wanted to have a central place where **the** cafeteria would be, where **the** people would come and where **the** people working would interact. So it ' s out in **the** middle of nowhere, and you approach it. It ' s copper and galvanize. I used **the** galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ' s aesthetic. Everybody ' s trying to get **the** panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is **the** central area. There ' s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded **the** contract I, at **the** very beginning, asked **the** client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ' s all about [a] metaphor for a city, maybe. And so Stanley did **the** little dome thing. And we did it over **the** phone and by fax. He would send me a fax and show me something. He ' d made a building with a dome and he had a little tower. I told him,"No, no, that ' s too ongepotchket. I don ' t want **the** tower."So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on **the** fax, going back and forth over a couple of weeks ' period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,"David and Goliath."And that ' s my cafeteria.

In Boston, we had that old building on **the left**. It was a very prominent building off **the** freeway, and we added a floor and cleaned it up and fixed it up and used **the** kind of â I thought â **the** language of **the** neighborhood, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid **the** side of **the** building and re-proportioned **the** windows so it sort of fit into **the** space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ’ t normally think I would fit them. **The** detailing was very careful with **the** lead copper and making panels and fitting it tightly into **the** fabric of **the** existing building. In Barcelona, on Las Ramblas for some film festival, I did **the** Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â a little shopping center. And this is a laser laboratory at **the** University of Iowa, in which **the fish** comes back as an abstraction in **the** back. It ’ s **the** support labs, which, by some coincidence, required no windows. And **the** shape fit perfectly. I just joined **the** points. In **the** curved part there ’ s all **the** mechanical equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ’ t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ’ ve been building it so long I don ’ t remember where it is. It ’ s in **the** West Valley. And we started with **the** stream and built **the** house along **the** stream â dammed it up to make a lake. These are **the** models. **The** reality, with **the** lake â **the** workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ’ s hard to get perfect corners and stuff. That big metal thing is a passage, and in it is â you go downstairs into **the** living room and then down into **the** bedroom, which is on **the** right. It ’ s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. **The** owners had a dilemma â they asked Philip to do it. He was too busy. He didn ’ t recommend me, by **the** way. We ended up having to make it a sculpture, because **the** dilemma was, how do you build a building that doesn ’ t look like **the** language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ’ ve got **the** idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from **the** main house and all **the** windows are on **the** other side. And **the** building is very sculptural as you walk around it. It ’ s made of metal and **the** brown stuff is Fin-Ply â it ’ s that formed lumber from Finland. We used it at Loyola on **the** chapel, and it didn ’ t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ’ s Burnham Mall, on **the left**. It ’ s never been finished. Going out to **the** lake, you can see all those new buildings we built. And we had **the** opportunity to build a building on this site. There ’ s a railroad track. This is **the** city hall over here somewhere, and **the** courthouse. And **the** centerline of **the** mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on **the** axis here, and we followed **the** axis, and they ’ re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put **the** newspaper on top, folded. **The** health club is fastened to **the** garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this stuff at **the** bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, **the** centerline of this thing â we ' d preserve it, and it would start to work with **the** scale of **the** new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ' s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ' 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and **the** owner of this lot worked for me and built that actual model on **the left**. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of **the** village, we changed it as we got into it. I locked **the** house into **the** site by cutting **the** back end â here you see on **the** photographs of **the** site â slicing into it and putting all **the** bathrooms and dressing rooms like a retaining wall, creating a lower level zone for **the** master bedroom, which I designed like a kind of a barge, looking like a boat. And that ' s it, built. **The** dome was a request from **the** client. She wanted a dome somewhere in **the** house. She didn ' t care where. When you sleep in this bedroom, I hope â I mean, I haven ' t slept in it yet. I ' ve offered to marry her so I could sleep there, but she said I didn ' t have to do that. But when you ' re in that room, you feel like you ' re on a kind of barge on some kind of lake. And it ' s very private. **The** landscape is being built around to create a private garden. And then up above there ' s a garden on this side of **the** living room, and one on **the** other side. These aren ' t focused very well. I don ' t know how to do it from here. Focus **the** one on **the** right. It ' s up there. Left â it ' s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ' s **the** living room. Servants ' quarters. A guest bedroom, which has this dome with marble on it. And then you enter into **the** living room and then so on. This is **the** bedroom. You come down from this level along **the** stairway, and you enter **the** bedroom here, going into **the** lake. And **the** bed is back in this space, with windows looking out onto **the** lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. **The** material is lead copper, like in **the** building in Boston. And so it was an intent to make this small piece of land â it ' s 100 by 250 â into a kind of an estate by separating these areas and making **the** living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with **the** dining room table. It looks like I got a Baldessari painting for free. But **the** idea is, **the** windows are all placed so you see pieces of **the** house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ' re in your own kind of village. This is for Michael Eisner â Disney. We ' re doing some work for him. And this is in Anaheim, California, and it ' s a freeway building. You go under this bridge at about 65 miles an hour, and there ' s another bridge here. And you ' re through this room in a split second, and **the** building will sort of reflect that. On **the** backside, it ' s much more humane â entrance, dining hall, etc. And then this thing here â I ' m hoping as you drive by you ' ll hear **the** picket fence effect of **the** sound hitting it. Kind of a fun thing to do. I ' m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with **the** image. These are **the** early studies, but they have to sell furniture to normal people, so if I did **the** building and it was too fancy, then people might say,"Well, **the** furniture looks OK in his thing, but no, it ain ' t going to look good in my normal building."So we ' ve made a kind of pragmatic slab in **the** second phase here, and we ' ve taken **the** conference facilities and made a villa out of them so that **the** communal space is very sculptural and separate. And you ' re looking at it from **the** offices and you create a kind of interaction between these pieces. This is in Paris, along **the** Seine. Palais des Sports, **the** Gare de Lyon over here. **The** Minister of Finance â **the** guy that moved from **the** Louvre â goes in here. There ’ s a new library across **the** river. And back in here, in this already treed park, we ’ re doing a very dense building called **the** American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is **the** ground level. And **the** French have this extraordinary way of screwing things up by taking a beautiful site and cutting **the** corner off. They call it **the** plan coupe. And I struggled with that thing â how to get around **the** corner. These are **the** models for it. I showed you **the** other model, **the** one â this is **the** way I organized myself so I could make **the** drawing â so I understood **the** problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are **the** kind of study models we did. And **the** one on **the left** is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where **the** elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and **the** skirt, kind of like a ballerina lifting her skirt to let you into **the** foyer. **The** restaurants here â **the** apartments and **the** theater, etc. So it would all be built in stone, in French limestone, except for this metal piece. And it faces into a park. And **the** idea was to make this express **the** energy of this. On **the** side facing **the** street it ’ s much more normal, except I slipped a few mansards down, so that coming on **the** point, these housing units made a gesture to **the** corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in **the** Hamptons. And it ’ s got a **fish**. And I keep thinking,”This is going to be **the** last **fish**.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s **the** living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in **the** budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of **the** guys that presented to you earlier. We ’ ve had a lot of fun working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into **the** Magic Kingdom and **the** hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of **the** Paris skies being quite dull, I made a light grid that ’ s perpendicular to **the** train station, to **the** route of **the** train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. **The** light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on **the** Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good **slides** to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â you see it as all of these pieces accrue and become part of an overall neighborhood. It ’ s plaster and just zinc. And you wonder, if this is a museum, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â I made it quiet and wonderful. But on **the** outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and stuff.

And from **the** building in **the** back, you see it as an iceberg floating by in **the** hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is **the** Disney Hall â **the** concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a compositional relationship between us that would strengthen both of us. And **the** plan of this â it ’ s a concert hall. This is **the** foyer, which is kind of a garden structure. There ’ s commercial at **the** ground floor. These are offices, which, really, in **the** competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were **the** kind of relationships made to **the** Chandler, composing these elevations together and relating them to **the** buildings that existed â to MOCA, etc. **The** acoustician in **the** competition gave us criteria, which led to this compartmentalized scheme, which we found out after **the** competition would not work at all. But everybody liked these forms and liked **the** space, and so that ’ s one of **the** problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were **the** three buildings that were **the** ideal â **the** Concertgebouw, Boston and Berlin. Everybody liked **the** surround. Actually, this is **the** smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new acoustician, he told us this was **the** right shape or this was **the** right shape. And we tried many shapes, trying to get **the** energy of **the** original design within an acoustical, acceptable format. We finally settled on a shape that was **the** proportion of **the** Concertgebouw with **the** sloping outside walls, which **the** acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make **the** seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s **the** idea. And **the** corners would have skylights and these columns would be structural. And **the** nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on **the** way to being â and so I wouldn ’ t take it literally, except **the** feeling of **the** space. We studied **the** acoustics with laser stuff, and they bounce them off this and see where it all works. But you get **the** sense of **the** hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in **the** four corners. And so it will be quite a different shape. **The** original building, because it was frog-like, fit nicely on **the** site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, struggling with how to put **the** hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is **the** crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at **the** foot of Interstate Bank Tower, **the** highest building in L. A. Larry Halprin is doing **the** stairs. And I was asked to do a **fish**, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a **fish**. In Barcelona I was asked to do a **fish**, and we ’ re working on that, at **the** foot of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And **the** Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took **the** language of this exposed steel and used it, perverted it, into **the** form of **the** **fish**, and created a kind of a 19th- century contraption that looks like, that will sit â this is **the** beach and **the** harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to **the** hotel people **the** other day, and they were terrified and said that nobody would come to **the** Ritz-Carlton anymore, because of this **fish**. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed **the** robot as **the** cashier, and **the** head moves, and I did **the** rest of it. And **the** food wasn ’ t as good as **the** stuff, and so it failed. It should have been **the** other way around â **the** food should have been good first. It didn ’ t work. Thank you very much.

Text Sample 76: POS Dim 1 – 2001 – Score 2.00 – 2001/t000083.txt

I thought I would read poems I have that relate to **the** subject of youth and age. I was sort of astonished to find out how many I have actually. **The** first one is dedicated to Spencer, and his grandmother, who was shocked by his work. My poem is called”Dirt.”My grandmother is washing my mouth out with soap ; half a long century gone and still she comes at me with that thick cruel **yellow** bar. All because of a word I said, not even said really, only repeated. But”Open,”she says,”open up!”her hand clawing at my head. I know now her life was hard ; she lost three children as babies, then her husband died too, leaving young sons, and no money. She ’ d stand me in **the** sink to pee because there was never room in **the** toilet. But oh, her soap! Might its bitter burning have been what made me a poet? **The** street she lived on was unpaved, her flat, two cramped rooms and a fetid kitchen where she stalked and caught me. Dare I admit that after she did it I never really loved her again? She lived to a hundred, even then. All along it was **the** sadness, **the** squalor, but I never, until now loved her again. When that was published in a magazine I got an irate letter from my uncle.”You have maligned a great woman.”It took some diplomacy. This is called”**The** Dress.”It ’ s a longer poem. In those days, those days which exist for me only as **the** most elusive memory now, when often **the** first sound you ’ d hear in **the** morning would be a storm of birdsong, then **the** soft clop of **the** hooves of **the** horse hauling a milk wagon down your block, and **the** last sound at night as likely as not would be your father pulling up in his car, having worked late again, always late, and going heavily down to **the** cellar, to **the** furnace, to shake out **the** ashes and damp **the** draft before he came upstairs to fall into bed — in those long-ago days, women, my mother, my friends ’ mothers, our neighbors, all **the** women I knew — wore, often much of **the** day, what were called housedresses, cheap, printed, pulpy, seemingly purposefully shapeless light cotton shifts that you wore over your nightgown and, when you had to go look for a child, hang wash on **the** line, or run down to **the** grocery store on **the** corner, under a coat, **the** twisted hem of **the** nightgown always lank and **yellowed**, dangling beneath. More than **the** curlers some of **the** women seemed constantly to have in their hair in preparation for some great event — a ball, one would think — that never came to pass ; more than **the** way most women ’ s faces not only were never made up during **the** day, but seemed scraped, bleached, and, with their plucked eyebrows, scarily masklike ; more than all that it was those dresses that made women so unknowable and forbidding, adepts of enigmas to which men could have no access, and boys no conception. Only later would I see **the** dresses also as a proclamation : that in your dim kitchen, your laundry, your bleak concrete yard, what you revealed of yourself was a fabulation ; your real sensual nature, veiled in those sexless vestments, was utterly your dominion. In those days, one hid much else as well : grown men didn ’ t embrace one another, unless someone had died, and not always then ; you shook hands or, at a ball game, thumped your friend ’ s back and exchanged blows meant to be codes for affection ; once out of childhood you ’ d never again know **the** shock of your father ’ s whiskers on your cheek, not until mores at last had **evolved**, and you could hug another man, then hold on for a moment, then even kiss. What release finally, **the** embrace : though we were wary — it seemed so audacious — how much unspoken joy there was in that affirmation of equality and communion, no matter how much misunderstanding and pain had passed between you by then. We knew so little in those days, as little as now, I suppose about healing those hurts : even **the** women, in their best dresses, with beads and sequins sewn on **the** bodices, even in lipstick and mascara, their hair aflow, could only stand wringing their hands, begging for peace, while father and son, like thugs, like thieves, like Romans, simmered and hissed and hated, inflicting sorrows that endured, **the** worst anyway, through **the** kiss and embrace, bleeding from brother to brother, into **the** generations. In those days there was still countryside close to **the** city, farms, cornfields, cows ; even not far from our building with its blurred brick and long shadowy hallway you could find tracts with hills and trees you could pretend were mountains and forests. Or you could go out by yourself even to a half-block-long empty lot, into **the** bushes : like a creature of leaves you ’ d lurk, crouched, crawling, simplified, savage, alone ; already there was wanting to be simpler, wanting, when they called you, never to go back. This is another longish one, about **the** old and **the** young. It actually happened right at **the** time we met. Part of **the** poem takes place in space we shared and time we shared. It ’ s called”**The** Neighbor.”Her five horrid, deformed little dogs who incessantly yap on **the** roof under my window. Her cats, God knows how many, who must piss on her rugs — her landing ’ s a sickening reek. Her shadow once, fumbling **the** chain on her door, then **the** door slamming fearfully shut, only **the** barking and **the** music — jazz — filtering as it does, day and night into **the** hall. **The** time it was Chris Connor singing”Lush Life”— how it brought back my college sweetheart, my first real love, who — till I left her — played **the** same record. And head on my shoulder, hand on my thigh, sang sweetly along, of regrets and depletions she was too young for, as I was too young, later, to believe in her pain. It startled, then bored, then repelled me. My starting to fancy she ’ d ended up in this fire-trap in **the** Village, that my neighbor was her. My thinking we ’ d meet, recognize one another, become friends, that I ’ d accomplish a penance. My seeing her, it wasn ’ t her, at **the** mailbox. Gray-yellow hair, army pants under a nightgown, her turning away, hiding her ravaged face in her hands, muttering an inappropriate”Hi.”Sometimes there are frightening goings-on in **the** stairwell. A man shouting,”Shut up!”**The** dogs frantically snarling, claws scrabbling, then her — her voice hoarse, harsh, hollow, almost only a tone, incoherent, a note, a squawk, bone on metal, metal gone molten, calling them back,”Come back darlings, come back dear ones. My sweet angels, come back.”Medea she was, next time I saw her. Sorceress, tranced, ecstatic, stock-still on **the** sidewalk ragged coat hanging agape, passersby flowing around her, her mouth torn suddenly open as though in a scream, silently though, as though only in her brain or breast had it erupted. A cry so pure, practiced, detached, it had no need of a voice, or could no longer bear one. These invisible links that allure, these transfigurations, even of anguish, that hold us. **The** girl, my old love, **the** last lost time I saw her when she came to find me at a party, her drunkenly stumbling, falling, sprawling, skirt hiked, eyes veined red, swollen with tears, her shame, her dishonor. My ignorant, arrogant coarseness, my secret pride, my turning away. Still life on a rooftop, dead trees in barrels, a bench broken, dogs, excrement, sky. What pathways through pain, what junctures of vulnerability, what crossings and counterings? Too many lives in our lives already, too many chances for sorrow, too many unaccounted-for pasts."Behold me,"**the** god of frenzied, inexhaustible love says, rising in bloody splendor,"Behold me."Her making her way down **the** littered vestibule stairs, one agonized step at a time. My holding **the** door. Her crossing **the** fragmented tiles, faltering at **the** step to **the** street, droning, not looking at me,"Can you help me?"Taking my arm, leaning lightly against me. Her wavering step into **the** world. Her whispering,"Thanks love."Lightly, lightly against me. I think I ' ll lighten up a little. Another, different kind of poem of youth and age. It ' s called"Gas."Wouldn ' t it be nice, I think, when **the** blue-haired lady in **the** doctor ' s waiting room bends over **the** magazine table and farts, just a little, and violently blushes. Wouldn ' t it be nice if intestinal gas came embodied in visible clouds, so she could see that her really quite inoffensive pop had only barely grazed my face before it drifted away. Besides, for this to have happened now is a nice coincidence. Because not an hour ago, while we were on our walk, my dog was startled by a backfire and jumped straight up like a horse **bucking**. And that brought back to me **the** stable I worked on weekends when I was 12, and a splendid piebald stallion, who whenever he was mounted would **buck** just like that, though more hugely of course, enormous, gleaming, resplendent. And **the** woman, her face abashedly buried in her"Elle"now, reminded me — I ' d forgotten that not **the** least part of my awe consisted of **the** fact that with every jump he took **the** horse would powerfully fart. Phwap! Phwap! Phwap! Something never mentioned in **the** dozens of books about horses and their riders I devoured in those days. All that savage grandeur, **the** steely glinting hooves, **the** eruptions driven from **the** creature ' s mighty innards, breath stopped, heart stopped, nostrils madly flared, I didn ' t know if I wanted to break him, or be him. This is called"Thirst."Many — most of my poems actually are urban poems. I happen to be reading a bunch that aren ' t."Thirst."Here was my relation with **the** woman who lived all last autumn and winter, day and night, on a bench in **the** 103rd Street subway station, until finally one day she vanished. We regarded each other, scrutinized one another. Me shyly, obliquely, trying not to be furtive. She boldly, unblinkingly, even pugnaciously, wrathfully even, when her bottle was empty. I was frightened of her. I felt like a child. I was afraid some repressed part of myself would go out of control, and I ' d be forever entrapped in **the** shocking seethe of her stench. Not excrement merely, not merely **surface** and orifice going unwashed, rediffusion of rum, there was will in it, and intention, power and purpose — a social, ethical rage and rebellion — despair too, though, grief, loss. Sometimes I ' d think I should take her home with me, bathe her, comfort her, dress her. She wouldn ' t have wanted me to, I would think. Instead, I ' d step into my train. How rich I would think, is **the** lexicon of our self-absolving. How enduring, our bland fatal assurance that reflection is righteousness being accomplished. **The** dance of our glances, **the** clash, pulling each other through our perceptual punctures, then holocaust, holocaust, host on host of ill, injured presences, squandered, consumed. Her vigil somewhere I know continues. Her occupancy, her absolute, faithful attendance. **The** dance of our glances, challenge, abdication, effacement, **the** perfume of our consternation. This is a newer poem, a brand new poem. **The** title is"This Happened."A student, a young woman in a fourth-floor hallway of her lycee, perched on **the** ledge of an open window chatting with friends between classes ; a teacher passes and chides her,"Be careful, you might fall,"almost banteringly chides her,"You might fall,"and **the** young woman, 18, a girl really, though she wouldn ' t think that, as brilliant as she is, first in her class, and"Beautiful, too,"she ' s often told, smiles back, and leans into **the** open window, which wouldn ' t even be open if it were winter — if it were winter someone would have closed it — leans into **the** window, farther, still smiling, farther and farther, though it takes less time than this, really an instant, and lets herself fall. Herself fall. A casual impulse, a fancy, never thought of until now, hardly thought of even now... No, more than impulse or fancy, **the** girl knows what she ' s doing, **the** girl means something, **the** girl means to mean, because it **occurs** to her in that instant, that beautiful or not, bright yes or no, she ' s not who she is, she ' s not **the** person she is, and **the** reason, she suddenly knows, is that there ' s been so much premeditation where she is, so much plotting and planning, there ' s hardly a person where she is, or if there is, it ' s not her, or not wholly her, it ' s a self inhabited, lived in by her, and seemingly even as she thinks it she knows what ' s been missing : grace, not premeditation but grace, a kind of being in **the** world spontaneously, with grace. Weightfully upon me was **the** world. Weightfully this self which graced **the** world yet never wholly itself. Weightfully this self which weighed upon me, **the** release from which is what I desire and what I achieve. And **the** girl remembers, in this infinite instant already now so many times divided, **the** sadness she felt once, hardly knowing she felt it, to merely inhabit herself. Yes, **the** girl falls, absurd to fall, even **the** earth with its compulsion to take unto itself all that falls must know that falling is absurd, yet **the** girl falling isn ' t myself, or she is myself, but a self I took of my own volition unto myself. Forever. With grace. This happened. I ' ll read just one more. I don ' t usually say that. I like to just end. But I ' m afraid that Ricky will come out here and shake his fist at me. This is called"Old Man,"appropriately enough."Special : big tits,"Says **the** advertisement for a soft-core magazine on our neighborhood newsstand. But forget her breasts. A lush, fresh-lipped blond, skin glowing gold, sprawls there, resplendent. 60 nearly, yet these hardly tangible, hardly better than harlots, can still stir me. Maybe a coming of age in **the** American sensual darkness, never seeing an unsmudged nipple, an uncensored vagina, has left me forever infected with an unquenchable lust of **the** eye. Always that erotic murmur, I ' m hardly myself if I ' m not in a state of incipient desire. God knows though, there are worse twists your obsessions can take. Last year in Israel, a young ultra-orthodox Rabbi guiding some teenage girls through **the** Shrine of **the** Shoah forbade them to look in one room. Because there were images in it he said were licentious. **The** display was a photo. Men and women stripped naked, some trying to cover their genitals, others too frightened to bother, lined up in snow waiting to be shot and thrown into a ditch. **The** girls, to my horror, averted their gaze. What carnal mistrust had their teacher taught them. Even that though. Another confession : Once in a book on pre-war Poland, a studio portrait, an absolute angel, an absolute angel with tormented, tormenting eyes. I kept finding myself at her page. That she died in **the** camps made her — I didn ' t dare wonder why — more present, more precious. Died in **the** camps, that too people — or Jews anyway — kept from their children back then. But it was like sex, you didn ' t have to be told. Sex and death, how close they can seem. So constantly conscious now of death moving towards me, sometimes I think I confound them. My wife ' s loveliness almost consumes me. My passion for her goes beyond reasonable bounds. When we make love, her holding me everywhere all around me, I ' m there and not there. My mind teems, jumbles of faces, voices, impressions, I live my life over, as though I were drowning. Then I am drowning, in despair at having to leave her, this, everything, all, unbearable, awful. Still, to be able to die with no special contrition, not having been slaughtered, or enslaved. And not having to know history ' s next mad rage or regression, it might be a relief. No. Again, no. I don ' t mean that for a moment. What I mean is **the** world holds me so tightly — **the** good and **the** bad — my own follies and weakness that even this counterfeit Venus with her sham heat, and her bosom probably plumped with gel, so moves me my breath catches. Vamp. Siren. Seductress. How much more she reveals in her glare of ink than she knows. How she incarnates our desperate human need for regard, our passion to live in beauty, to be beauty, to be cherished by glances, if by no more, of something like love, or love. Thank you.

Text Sample 77: POS Dim 1 – 2001 – Score 1.00 – 2001/t000092.txt
I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by **the** way parents of **the** youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for **the** neighbors ' children, because they were all average. But they weren ' t satisfied when their own — it would make **the** teacher feel that they had failed, or **the** youngster had failed. And that ' s not right. **The** good Lord in his infinite wisdom didn ' t create us all equal as far as intelligence is concerned, any more than we ' re equal for size, appearance. Not everybody could earn an A or a B, and I didn ' t like that way of judging, and I did know how **the** alumni of various schools back in **the** ' 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ' t lose a game. But it seemed that we didn ' t win each individual game by **the** margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in **the** 30s, so I understood that. But I didn ' t like it, I didn ' t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give **the** youngsters under my supervision, be it in athletics or **the** English classroom, something to which to aspire, other than just a higher mark in **the** classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as **the** accumulation of material possessions or **the** attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ' m sure at **the** time he did that, I didn ' t — it didn ' t — well, somewhere, I guess in **the** hidden recesses of **the** mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be **the** best you can be — that ' s under your control. If you get too engrossed and involved and concerned in regard to **the** things over which you have no control, it will adversely affect **the** things over which you have control. Then I ran across this simple verse that said,"At God ' s footstool to confess, a poor soul knelt, and bowed his head. ' I failed! ' he cried. **The** Master said, ' Thou didst thy best, that is success. '"From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made **the** effort to do **the** best of which you ' re capable. I believe that ' s true. If you make **the** effort to do **the** best of which you ' re capable, trying to improve **the** situation that exists for you, I think that ' s success, and I don ' t think others can judge that ; it ' s like character and reputation — your reputation is what you ' re perceived

to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be **the** same. Well, that was my idea that I was going to try to get across to **the** youngsters. I ran across other things. I love to teach, and it was mentioned by **the** previous speaker that I **enjoy** poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all **the** books on all **the** shelves — it ’ s what **the** teachers are themselves.”That made an impression on me in **the** 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in **the** English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by **coal oil lamp** — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about **the** same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I reply, ’ Where could I find such splendid company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem **the** life-blood ’ s flow. And there a builder ; upward rise **the** arch of a church he builds, wherein that minister may speak **the** word of God, and lead a stumbling soul to touch **the** Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see **the** church, or hear **the** word, or eat **the** food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ’ Where could I find such splendid company? ’”And I believe **the** teaching profession — it ’ s true, you have so many youngsters, and I ’ ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make **the** youngsters feel that they ’ re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over **the** other two, and you ’ re not going to have any very long. So that was **the** idea that I tried to get across to **the** youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all **the** time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was”Never be late.”Later on I said certain things — **the** players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ’ s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch **the** bus ; we were leaving for somewhere to play. And he wasn ’ t clean and neat, so I wouldn ’ t let him go. He couldn ’ t get on **the** bus, he had to go home and get cleaned up to get to **the** airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. **The** youngsters didn ’ t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ' ll be **the** younger coaches getting in **the** profession. Most of them are young, you know, and probably newly-married. And I tell them,"Don ' t run practices late, because you ' ll go home in a bad mood, and that ' s not good, for a young married man to go home in a bad mood. When you get older, it doesn ' t make any difference, but —"So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for **the** day. If I see it in a game, you ' re going to come out and sit on **the** bench. And **the** third one was, never criticize a teammate. I didn ' t want that. I used to tell them I was paid to do that. That ' s my job. I ' m paid to do it. Pitifully poor, but I am paid to do it. Not like **the** coaches today, for gracious sakes, no. It ' s a little different than it was in my day. Those were three things that I stuck with pretty closely all **the** time. And those actually came from my dad. That ' s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ' t have **the** time to go on that. But that helped me, I think, become a better teacher. It ' s something like this : And I had blocks in **the** pyramid, and **the** cornerstones being industriousness and enthusiasm, working hard and **enjoying** what you ' re doing, coming up to **the** apex, according to my definition of success. And right at **the** top, faith and patience. And I say to you, in whatever you ' re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out **the** way we want them to much of **the** time, but we don ' t do **the** things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by **the** name of George Moriarty. He spelled Moriarty with only one ' i '. I ' d never seen that before, but he did. Big league baseball players — they ' re very perceptive about those things, and they noticed he had only one ' i ' in his name. You ' d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it"**The** Road Ahead, or **the** Road Behind."He said,"Sometimes I think **the** Fates must grin as we denounce them and insist **the** only reason we can ' t win, is **the** Fates themselves have missed. Yet there lives on **the** ancient claim : we win or lose within ourselves. **The** shining trophies on our shelves can never win tomorrow ' s game. You and I know deeper down, there ' s always a chance to win **the** crown. But when we fail to give our best, we simply haven ' t met **the** test, of giving all and saving none until **the** game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ' s bearing down that wins **the** cup. Of dreaming there ' s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so **the** Fates are seldom wrong, no matter how they twist and wind. It ' s you and I who make our fates — we open up or close **the** gates on **the** road ahead or **the** road behind."Reminds me of another set of threes that my dad tried to get across to us : Don ' t whine. Don ' t complain. Don ' t make excuses. Just get out there, and whatever you ' re doing, do it to **the** best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ' re outscored. I ' ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ' t know **the** outcome, I hope they couldn ' t tell by your actions whether you outscored an opponent or **the** opponent outscored you. That ' s what really matters : if you make an effort to do **the** best you can regularly, **the** results will be about what they should be. Not necessarily what you ' d want them to be but they ' ll be about what they should ; only you will know whether you can do that. And that ' s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as **the** results. But I wanted **the** score of a game to be **the** byproduct of these other things, and not **the** end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,"**The** journey is better than **the** end."And I like that. I think that it is — it ' s getting there. Sometimes when you get there, there ' s almost a let down. But it ' s **the** getting there that ' s **the** fun. As a basketball coach at UCLA, I liked our practices to be **the** journey, and **the** game would be **the** end, **the** end result. I liked to go up and sit in **the** stands and watch **the** players play, and see whether I ' d done a decent job during **the** week. There again, it ' s getting **the** players to get that self-satisfaction, in knowing that they ' d made **the** effort to do **the** best of which they are capable. Sometimes I ' m asked who was **the** best player I had, or **the** best teams. I can never answer that. As far as **the** individuals are concerned — I was asked one time about that, and they said,"Suppose that you, in some way, could make **the** perfect player. What would you want?"And I said,"Well, I ' d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in **the** first place. But I ' d want one that could play, too. I ' d want one to realize that defense usually wins championships, and who would work hard on defense. But I ' d want one who would play offense, too. I ' d want him to be unselfish, and look for **the** pass first and not shoot all **the** time. And I ' d want one that could pass and would pass. I ' ve had some that could and wouldn ' t, and I ' ve had some that would and could. So, yeah, I ' d want that. And I wanted them to be able to shoot from **the** outside. I wanted them to be good inside too. I ' d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had **the** qualifications. Not **the** only one, but he was one that I used in that particular category, because I think he made **the** effort to become **the** best. There was a couple. I mention in my book,"They Call Me Coach,"two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ' t play varsity when I taught. I thought,"Oh gracious, if these two players, either one of them"— they were different years, but I thought about each one at **the** time he was there —"Oh, if he ever makes **the** varsity, our varsity must be pretty miserable, if he ' s good enough to make it."And you know, one of them was a starting player for a season and a half. **The** other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. **The** next year, he was a starting player on **the** national championship team, and here I thought he ' d never play a minute, when he was — so those are **the** things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of **the** others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 78: POS Dim 1 – 2001 – Score -1.00 – 2001/t000154.txt

So I understand that this meeting was planned, and **the** slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,”Are you still doing this or that?”I don ’ t answer because I ’ m not doing things still, I ’ m doing it like I always did. I still have — or did I use **the** word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during **the** last 75 years. But, of course, I ’ m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at **the** Museum of Modern Art. This is now for sale at **the** Metropolitan Museum. This is still at **the** Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of **the** things I ’ ve made. I made hundreds of them for **the** last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. **The** industrial design magazine, I believe, is called”Innovation.”Innovation is not part of **the** aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just **the** craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with **the** playful **search** for beauty. That means **the** playful **search** for beauty was called **the** first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,”**The** playful **search** for beauty was Man ’ s first activity — that all useful qualities and all material qualities were developed from **the** playful **search** for beauty.”These are tiles. **The** word,”playful”is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with **the** same pleasure, as a gift to others, for so long? **The** playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in **the** United States was at **the** Sesquicentennial exhibition in 1926 — that **the** Hungarian government sent one of my hand-drawn pieces as part of **the** exhibit. My work actually took me through many countries, and showed me a great part of **the** world. This is not that they took me — **the** work didn ’ t take me — I made **the** things particularly because I wanted to use them to see **the** world. I was incredibly curious to see **the** world, and I made all these things, which then finally did take me to see many countries and many cultures.

I started as an apprentice to a Hungarian craftsman, and this taught me what **the** guild system was in Middle Ages. **The** guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as **the** apprentice. **The** work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed **the** clay with our feet when it came from **the** hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for **the** throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in **the** time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by **the** Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at **the** time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on **the** marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ’ t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at **the** market and sold **the** pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of **the** director. Then **the** director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that **the** industrial revolution had broken out, and that I rather should join **the** factory. There he made an art department for me where I worked for several months. However, everybody in **the** factory spent his time at **the** art department. **The** director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, **the** director, **the** chemist, model maker — everybody — concerned himself much more with **the** art department — that means, with my work — than making toilets, so finally they got a letter from **the** center, from **the** bank who owned **the** factory, saying, make toilet-setting behind **the** art department, and that was my end. So this gave me **the** possibility because now I was a journeyman, and journeymen also take their satchel and go to see **the** world. So as a journeyman, I put an ad into **the** paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted **the** one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on **the** wheel, and so I worked in a shop where there were several potters. And **the** first day, I was coming to take my place at **the** turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. **The** first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on **the** wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in **the** Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of **the** china and glass industry, and eventually under Stalin ’ s purges — at **the** beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. **The** accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is **the** end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is **the** end of **the** five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, **the** Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought **the** 15 of their artists to visit me here. And they invited myself to come to **the** Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is **the** end? Thank you.

Text Sample 79: POS Dim 1 – 2025 – Score 2.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In **the** beginning of time, there was no time. **The universe** began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. **The** point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about **the** future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in **the** world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In **the** meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet **the** city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness **underground** and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under **the** same sky, at **the** same moment, at **the** same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over **the** world. There have been many early departures, and many evenings in **the** air. In all my travels, I ’ ve been struck by so many things. I ’ ve observed how people lived in some of **the** most ancient places on **the planet**. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. **The** ancient city contained a church, a theater and tombs for **the** dead. Families lived in two and three-bedroom apartments. We ' ve come a long way. Some of **the** structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house **the** souls of **the** dead. Some have forced us to choose sides. Some have succumbed to **the** power of nature. And some have not. **The** view from **the** sky is so **abstract**. It reveals our connections, our continuity and our scale. Nevertheless, it ' s hard to see **the** big picture right now. It ' s such a difficult time in **the** world as we pause and dismantle and rebuild our culture. I think back to my travels. I ' ve seen how different we are, how diverse and distinct. But I ' ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret **the** mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in **the** world. So much love. I ' m hopeful for **the** next generation. For every creature, large and small. And for **the planet** we call home. A Love Letter to Storytelling by Debbie Millman"A need to tell and hear stories is essential to **the** species Homo Sapiens. **The** sound of story is **the** dominant sound of our lives."- Joseph Campbell I ' ve loved telling stories all of my life. I ' ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over **the** world. Over **the** years, **the** students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ' ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming **the** stories of our times.

Text Sample 80: POS Dim 1 – 2025 – Score -6.00 – 2025/t003001.txt

- I don ' t remember exactly, but... I use **the** most expensive hair color in **the** world. It ' s not that I care about **the** money, it ' s that I care about my hair. It ' s not just **the** color. I expect great color. What ' s worth more to me is **the** way my hair feels. Feels good around my neck. I can ' t remember it. I can ' t finish it all. If only we had more time. - Why don ' t we have more time? - I ' m dying. I ' m croaking! - It ' s tough, coming home. But this is what you do for a parent that you love. But, you know? She ' s not my mother. Technically. In my own family, there ' s a lot of dysfunction, a lot of mental illness, a lot of substance abuse. Every generation. My parents divorced when I was about five. My mother moved us out of state, so we had to fly to New York to see my father ' cause he was already working in advertising, and New York was **the** only place to be. - Many of **the** commercials you see on television begin life here. - Advertising is one of **the** main sources of messaging about who you should be. - Nowadays, most of you women lead two lives, combining homemaking with **the** role of glamorous hostess. - And in **the** ' 60s, **the** ads were often made from a male point of view. - In **the** pre-production meeting, everybody who has anything to do with **the** commercial is there. - It was mostly men. These superstar art directors and writers. Like my father, Gene Case."Mad Men"was like home movies, and he was Don Draper. He was very witty, but I don ’ t think my father realized how powerful his words were. He knew it professionally. But with his kids, he could be incredibly cutting. Flay you just with a phrase. But it ’ s **the** time with my mother that was **the** nightmare. She had a horrible childhood, but she sure as hell didn ’ t make any effort to overcome that in her relations with us. You know, it was really horrifying to my mother that I was not popular. But I was also really awkward because she had been undermining my confidence for all my life. She would pick apart your personality in detail. I was lazy, I looked bad, I acted bad, I was awkward. Emotionally, just out for **blood**. At a certain point in **the** night, I would know,"OK, **the** situation is dangerous. I need to get outta here."I had this fantasy when I was in fourth grade that I vividly remember that some kind of outside agency was gonna come in and determine from all **the** salt on my pillow that I was crying myself to sleep every night and do something about it. No, it was awful. It was awful. And that was **the** point where Ilon came into my life. - Ilon, do you need anything to eat? - I don ’ t think so, but look up this wallpaper, Waverly Fabrics, and see if they have a rose pattern. - OK. Oh, what about this one here? - That ’ s not it. - No? - No, first of all, I was an art dir... I was a creative director at an advertising agency. **The** first thing I know is what I ’ m looking at. Ilon **the** girl. I don ’ t remember particularly being into advertising, but somehow I found myself in it. And I seemed to be very good at it. First of all, I ’ d always listened to everybody. And I noticed that I saw things differently than other people. - Women could get somewhere in advertising, but they were marginalized. - Women were given a lesser role. All **the** men were always arguing with you and always taking credit. - And one of **the** things that was coming through loud and clear is that women are for men. - Glory, glory, hallelujah Glory, glory, hallelujah Glory, glory, hallelujah His truth is marching on - Men were always **the** ones who everyone was pleasing. - What will your eyes say to him tonight? Wait till **the** fellas take a look at this plunging backline. - You look at ads from that period, and it ’ s jaw-dropping. - This is one of our nation ’ s most beautiful sights, **the** kind of girl who keeps her figure. - But that ’ s what feminism was pushing back against. - I was a feminist. I knew women were equal if not superior to men from **the** time I was a little girl. - And in advertising, she wanted to be a trailblazer that way. At one point, she worked in **the** same agency with my father, and that ’ s how they met. - He was a good writer, and I was a good writer. I liked **the** fact that he was good at what he did and cared about it. We got engaged, and then we got married. - I ’ m sitting there in **the** background. I ’ m grinning. I ’ m so thrilled because I adored her instantly. I thought she was super cool. And she treated me like I mattered. - She was a sweet little girl, and she deserved some caretaking. My mother was kinda terrible, so I knew that she needed support, and I gave her support. And that was what we did. - She was interested in me. She listened to me. Treated me like I was lovable. - She didn ’ t have anyone to recognize her being, to support her, to see who she could be, who she was. - She was very interested in what I would do with my life, and **the** assumption was that it would be something worthwhile. She just went way, way beyond **the** call of duty to make me feel like I was worth it. And she did that for a lot of other women, too. - I was working at McCann Erickson. We were given an assignment. L ’ Oréal, they wanted a campaign for Preference, it ’ s a hair color. - She ’ s in this room full of men brainstorming for this ad. - **The** men were busy flirting with **the** women, telling us we all looked like models and we didn ’ t look like advertising people, you know? - They ’ re like,"Oh, this woman is gonna be standing by an open window, and **the** wind will be blowing **the** curtains."And she ’ s thinking, you know, this is just making **the** woman totally an **object**. - I was feeling angry. I ’ m not interested in writing anything about looking good for men. Fuck ’ em. Fuck you too. - Why fuck me? - Well, you ’ re a man. - Yeah, but I ’ m here trying to help you tell your story. - Yeah, that ’ s good. I like that part. I was pissed. We weren ’ t there to just dance for **the** men. - It is what she said. She just thought,”Fuck you.”And wrote this. - I just opened my eyes and knew. Because I ’ m worth it. **The** campaign got approved, but all **the** men were always telling you that you were doing something wrong or they knew how to do it better. - **The** agency insisted on a male voiceover. - She uses **the** most expensive hair color in **the** world, Preference by L ’ Oréal. - You know, it was just totally different. - Smooth and silky, but with body. It feels good against her neck. Actually, she doesn ’ t mind spending more for L ’ Oréal. Because she ’ s worth it. - It was wrong. This was not for men but for women and for other human beings. - I use **the** most expensive hair color in **the** world, Preference by L ’ Oréal. It ’ s not that I care about **the** money, it ’ s that I care about my hair. It ’ s not just **the** color, I expect great color. What ’ s worth more to me is **the** way my hair feels. Smooth and silky, but with body. It feels good against my neck. Actually, I don ’ t mind spending more for L ’ Oréal. Because I ’ m worth it. - I wrote that. - It was everywhere. It was a national campaign. - Yes, it ’ s expensive, but I ’ m worth it. - What they realized after they started running it was that that was only **the** tip of **the** iceberg. - It was supporting women, finally. - And I ’ m worth it. And I ’ m worth it. I ’ m worth it. - It ’ s a four-word feminist manifesto, defying this social pressure to think about yourself in relation to men. - You ’ re worth it. I ’ m worth it. I ’ m worth it. - It was a very big success. - At a certain point, L ’ Oréal adopted it as **the** tagline for their entire business. - Parce que je le vaux bien. - Even today, that ’ s their motto. - Because I ’ m worth it. - ’ Cause you ’ re worth it. - You ’ re worth it. - We ’ re worth it. - And I ’ m worth it. - She wrote **the** most influential tagline in women ’ s beauty advertising. - I ’ m worth it. - It ’ s magic, that phrase. - Those words just get better with age, don ’ t they? - I ’ m worth it. Because you are and always will be. - She came up with that line because she believed it. She believed it about herself, and she believed that women should believe it. She did that for me. You know, even after they split up, she was still a big part of my life. I ’ ve always kind of felt like I dodged a **bullet** that I have watched take down other members of my family. She saved me. And today, I am a professor of English at Williams College. I teach Victorian literature, and I also teach women ’ s, gender and sexuality studies. She was my mother. She also taught me how to be a mother. I had been very uncertain about motherhood for myself because my role model for that was so nightmarish. But Ilon really taught me, I can do this. I can be a good mother. She made me believe that I was worth it. And I was not either **the** first or **the** last. - I have to go away now. Go to my room. - It ’ s been lovely. - It ’ s been lovely. Oh... I ’ m not interested in advertising. I don ’ t give a shit. It wasn ’ t like it was my whole life. It was just a portion. It ’ s about humans ; it ’ s not about advertising. It ’ s about caring for people because... we ’ re all worth it, or no one is worth it.

2 NEG Dim 1

Text Sample 81: NEG Dim 1 – 2025 – Score -6.00 – 2025/t003001.txt

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It wasn ’ t like it was my whole life. It was just a portion. It ’ s about humans ; it ’ s not about advertising. It ’ s about caring for people because... we ’ re all worth it, or no one is worth it.

Text Sample 82: NEG Dim 1 – 2025 – Score 2.00 – 2025/t004225.txt
A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came **back** to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in the world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of **us**. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the air. In all my travels, I ’ ve been struck by so many things. I ’ ve observed how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ’ ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced **us** to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ’ s hard to see the big picture right now. It ’ s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think **back** to my travels. I ’ ve seen how different we are, how diverse and distinct. But I ’ ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ’ m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman”A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story

is the dominant sound of our lives.”- Joseph Campbell I ’ ve loved telling stories all of my life. I ’ ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 83: NEG Dim 1 – 2001 – Score -3.00 – 2001/t000367.txt
I ’ d like to do pretty much what I did the first time, which is to choose a **lighthearted** theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, **exorcise** those evil spirits, and this is still going on, as you know. But it wasn ’ t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ’ re feeling depressed? It ’ s better than Prozac, but I wouldn ’ t recommend it. So what we see in the **seventeenth**, eighteenth century is the continued search for medications other than camphor that ’ ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we fast-forward to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a row — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very interested in producing convulsions, measured types of convulsions. And they thought,”Well, we ’ ve got electricity, we ’ ll plug somebody into the wall. That always makes hair stand up and people shake a lot.”So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,”We know that at the Rome railroad station, there are all these lost souls wandering around, muttering gibberish. Can you bring one of them to **us**?”Someone who is, as the Italians say,”cagoots.”So they found this”ca-

goots"guy, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,"Well, we ' ll try 55 volts, **two-tenths** of a second. That ' s not going to do anything terrible to him."So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,"This fellow"— remember, he wasn ' t even put to sleep —"after this major grand mal convulsion, sat right up, looked at these three fellas and said, ' What the fuck are you assholes trying to do? '"If I could only say that in Italian. Well, they were happy as could be, because he hadn ' t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and everybody in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ' t work very well on the schizophrenics, but it was pretty clear in the ' 30s and by the middle of the ' 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ' t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the brain waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ' 60s, the first antidepressants came out. Tofranil was the first. In the late ' 70s, early ' 80s, there were others, and they were very effective. And patients ' rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ' m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ' ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ' m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ' clock in the afternoon, because I couldn ' t get out of bed before about 11 o ' clock. And anybody who ' s been depressed here knows what that ' s like. I couldn ' t even pull the covers off myself. Well, you ' re in a university medical center, where everybody knows everybody, and it ' s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ' t work anymore. And, in fact, it didn ' t make any difference because I didn ' t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,"Don ' t worry, chap. Six weeks, you ' re **back** in the operating room. Everything ' s going to be great."Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ' t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five feet in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ' t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had. They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the senior staff. All the senior staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful stuff. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it got so there was a throbbing, there was a ferocious fear in my head. You ' ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ' t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case. He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the senior staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,"No. I know this man better than any of you. I have met with him over and over again. You ' ve just seen him from time to time. You ' ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ' ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go **back** to surgery, he will never be the loving father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ' One Flew Over the Cuckoo ' s Nest. ' And you know about that, just essentially in a stupor the rest of his life."Well, he said,"Can ' t we try a course of electroshock therapy?"And you know why they agreed? They agreed to humor him. They just thought,"Well, we ' ll give a course of 10. And so we ' ll lose a little time. Big deal. It doesn ' t make any difference."So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ' t work. Seven didn ' t work. Eight didn ' t work. At nine, I noticed — and it ' s wonderful that I could notice anything — I noticed a change. And at 10, I noticed a real change. And he went **back** to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, lo and behold, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could blow the obsessional thinking away. I could blow the depression away. And I ' ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,"I ' ve got the strength now to do this."It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a formula. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember"Ruddigore,"and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,"We must have a word to bring you **back** to reality, and the word, my dear, will be ' Basingstoke. '"So every time she got a little nuts, he would say,"Basingstoke!"And she would say,"Basingstoke, it is."And she would be fine for a little while. Well, you know, I ' m from the Bronx. I can ' t say"Basingstoke."But I had something better. And it was very simple. It was,"Ah, fuck it!"Much better than"Basingstoke,"at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,"Ah, fuck it."And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went **back** to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came **back** to live with **us**. We had two more children after that. Resuscitated the career, even better than it had been before. Went right **back** into the university and began to write books. Well, you know, it ' s been a wonderful life. It ' s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ' s been very exciting. It ' s been very happy. Every once in a while, I have to say,"Ah, fuck it."Every once in a while, I get somewhat depressed and a little obsessional. So, I ' m not free of all of this. But it ' s worked. It ' s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ’ ve gotten thousands of letters about them by people who do think this — that based on my life ’ s history as I ’ ve portrayed in the books, my early life ’ s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the bitter **dregs** of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ’ ve always felt guilty about that. I ’ ve always felt that somehow I was an impostor because my readers don ’ t know what I have just told you. It ’ s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an **untroubled** mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes **back** to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ’ t happen, there will be adversities. If I, with the **bleakness** of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way **back** from this, believe me, anybody can find their way **back** from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

Text Sample 84: NEG Dim 1 – 2001 – Score -1.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,”Are you still doing this or that?”I don ’ t answer because I ’ m not doing things still, I ’ m doing it like I always did. I still have — or did I use the word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years.

But, of course, I ’ m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ’ ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called”Innovation.”Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,”The playful search for beauty was Man ’ s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty.”These are tiles. The word,”playful”is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ’ t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be **kneaded**. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ’ t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with **us** to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 85: NEG Dim 1 – 2001 – Score 1.00 – 2001/t000092.txt

I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ' children, because they were all average. But they weren ' t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ' s not right. The good Lord in his infinite wisdom didn ' t create **us** all equal as far as intelligence is concerned, any more than we ' re equal for size, appearance. Not everybody could earn an A or a B, and I didn ' t like that way of judging, and I did know how the alumni of various schools **back** in the ' 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ' t lose a game. But it seemed that we didn ' t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had **backed** up their predictions in a more materialistic manner. But that was true **back** in the 30s, so I understood that. But I didn ' t like it, I didn ' t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ' m sure at the time he did that, I didn ' t — it didn ' t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ' s under your control. If you get too engrossed and involved and concerned in regard to the things over which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,"At God ' s **footstool** to confess, a poor soul knelt, and bowed his head. ' I failed! ' he cried. The Master said, ' Thou didst thy best, that is success. '"From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ' re capable. I believe that ' s true. If you make the effort to do the best of which you ' re capable, trying to improve the situation that exists for you, I think that ' s success, and I don ' t think others can judge that ; it ' s like character and reputation — your reputation is what you ' re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ' d hope they ' d both be good, but they won ' t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ' m not what I ought to be, what I should be, but I think I ' m better than I would have been if I hadn ' t run across certain things. One was just a little verse that said,"No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ' s what the teachers are themselves."That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ' s because Dad used to read to **us** at night, by coal oil lamp — we didn ' t have electricity in our farm home. And Dad would read poetry to **us**. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,"They ask me why I teach, and I reply, ' Where could I find such splendid company? ' There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ' s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ' Where could I find such splendid company? '"And I believe the teaching profession — it ' s true, you have so many youngsters, and I ' ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ' re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ' re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ' d learned these prior to coming to UCLA, and I decided they were very important. One was"Never be late."Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ' s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ' t clean and neat, so I wouldn ' t let him go. He couldn ' t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ' t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ' ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,"Don ' t run practices late, because you ' ll go home in a bad mood, and that ' s not good, for a young married man to go home in a bad mood. When you get older, it doesn ' t make any difference, but —"So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ' re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ’ t want that. I used to tell them I was paid to do that. That ’ s my job. I ’ m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ’ s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ’ s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ’ t have the time to go on that. But that helped me, I think, become a better teacher. It ’ s something like this : And I had blocks in the pyramid, and the cornerstones being **industriousness** and enthusiasm, working hard and enjoying what you ’ re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ’ re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ’ t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ’ i ’. I ’ d never seen that before, but he did. Big league baseball players — they ’ re very perceptive about those things, and they noticed he had only one ’ i ’ in his name. You ’ d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it”The Road Ahead, or the Road Behind.”He said,”Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ’ t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ’ s game. You and I know deeper down, there ’ s always a chance to win the crown. But when we fail to give our best, we simply haven ’ t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ’ s bearing down that wins the cup. Of dreaming there ’ s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ’ s you and I who make our fates — we open up or close the gates on the road ahead or the road behind.”Reminds me of another set of threes that my dad tried to get across to **us** : Don ’ t whine. Don ’ t complain. Don ’ t make excuses. Just get out there, and whatever you ’ re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ’ re **outscored**. I ’ ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ’ t know the outcome, I hope they couldn ’ t tell by your actions whether you **outscored** an opponent or the opponent **outscored** you. That ’ s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ’ d want them to be but they ’ ll be about what they should ; only you will know whether you can do that. And that ’ s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,”The journey is better than the end.”And I like that. I think that it is — it ’ s getting there. Sometimes when you get there, there ’ s almost a let down. But it ’ s the getting there that ’ s the fun. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ’ d done a decent job during the week. There again, it ’ s getting the players to get that self-satisfaction, in knowing that they ’ d made the effort to do the best of which they are capable. Sometimes I ’ m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,”Suppose that you, in some way, could make the perfect player. What would you want?”And I said,”Well, I ’ d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ’ d want one that could play, too. I ’ d want one to realize that defense usually wins championships, and who would work hard on defense. But I ’ d want one who would play offense, too. I ’ d want him to be **unselfish**, and look for the pass first and not shoot all the time. And I ’ d want one that could pass and would pass. I ’ ve had some that could and wouldn ’ t, and I ’ ve had some that would and could. So, yeah, I ’ d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ’ d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,”They Call Me Coach,”two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ’ t play varsity when I taught. I thought,”Oh gracious, if these two players, either one of them”— they were different years, but I thought about each one at the time he was there —”Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ’ s good enough to make it.”And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for **us**. The next year, he was a starting player on the national championship team, and here I thought he ’ d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for **us**. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 86: NEG Dim 1 – 1990 – Score 4.00 – 1990/t000287.txt

I ’ m going to go right into the slides. And all I ’ m going to try and prove to you with these slides is that I do just very straight stuff. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the left-hand side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage the neighborhood â one made accommodations. I tried to be inclusive, to include the buildings in the neighborhood, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of stuff and folded it up to put it in the wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at postmodernism â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go **back**, we might as well go **back** to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker. And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And the client said,”Why no fish?”And so I made a drawing with a fish, and I left Japan. Three weeks later, I received a complete set of drawings saying we ’ d won the competition. Now, it ’ s hard to do. It ’ s hard to translate a fish form, because they ’ re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ’ t do it, and so that made it even more exciting. But he was right â I couldn ’ t do the tail. I started to get the head OK, but the tail I couldn ’ t do. It was pretty hard. The thing on the right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with the context again. Now, if you saw a picture of this as it was published in Architectural Record â they didn ’ t show the context, so you would think,”God, what a pushy guy this is.”But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ’ t find it. So... As for craft and technology and all those things that you ’ ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent drawings to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made the drawings of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ’ s one of the nicest pieces I ’ ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ’ t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ’ m getting good at temporary â and we put a conference room in that ’ s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ’ s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ’ t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ’ s offices and I made a pedestal for a sculpture. And he didn ’ t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ’ s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ’ ve been friends for a long time. And we ’ ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called”Il Corso del Coltello”â the Swiss Army knife. And most of the imagery is â Claes ’, but those two little boys are my sons, and they were Claes ’ assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ’ s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ’ ve been known to mess with things like chain link fencing. I do it because it ’ s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ’ s so much denial about them. People hate it. And I ’ m fascinated with that, which, like the paper furniture â it ’ s one of those materials. And I ’ m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ’ s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to the building on the left. And I still think the Time magazine picture will be of the binoculars, you know, leaving out the â what the hell. I use a lot of metal in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,”If I can ’ t get the craft that I want, I ’ ll use the craft I can get.”There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the metal because it was a way of building a building that was a sculpture. And it was all of one material, and the metal could go on the roof as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and stuff. I was given an opportunity to design an exhibit for the metalworkers ’ unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future metal buildings, etc. etc. And it ’ s working very well to have these people, these craftsmen, interested in it. I just tell the stories. It ’ s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The metal continued into a building â Herman Miller, in Sacramento. And it ’ s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ’ s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ’ s out in the middle of nowhere, and you approach it. It ’ s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ’ s aesthetic. Everybody ’ s trying to get the panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ’ s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ’ s all about [a] metaphor for a city, maybe. And so Stanley did the little dome thing. And we did it over the phone and by fax. He would send me a fax and show me something. He ’ d made a building with a dome and he had a little tower. I told him,”No, no, that ’ s too ongepotchket. I don ’ t want the tower.”So he came **back** with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going **back** and forth over a couple of weeks ' period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,"David and Goliath."And that ' s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and cleaned it up and fixed it up and used the kind of â I thought â the language of the neighborhood, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ' t normally think I would fit them. The detailing was very careful with the lead copper and making panels and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some film festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â a little shopping center. And this is a laser laboratory at the University of Iowa, in which the fish comes **back** as an abstraction in the **back**. It ' s the support labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the curved part there ' s all the mechanical equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ' t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ' ve been building it so long I don ' t remember where it is. It ' s in the West Valley. And we started with the stream and built the house along the stream â dammed it up to make a lake. These are the models. The reality, with the lake â the workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ' s hard to get perfect corners and stuff. That big metal thing is a passage, and in it is â you go downstairs into the living room and then down into the bedroom, which is on the right. It ' s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â they asked Philip to do it. He was too busy. He didn ' t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ' t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ' ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ' s made of metal and the brown stuff is Fin-Ply â it ' s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ' t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ' s Burnham Mall, on the left. It ' s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ' s a railroad track. This is the city hall over here somewhere, and the courthouse. And the centerline of the mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ' re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with

Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this stuff at the bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the centerline of this thing â we ’ d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ’ s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ’ 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came **back**, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the **back** end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake. And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ’ s the living room. Servants ’ quarters. A guest bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is **back** in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a fun thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it ain ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the conference facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais des Sports, the Gare de Lyon over here. The Minister of Finance â the guy that moved from the Louvre â goes in here. There ’ s a new library across the river. And **back** in here, in this already treed park, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan coupe. And I struggled with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in French limestone, except for this metal piece. And it faces into a park. And the idea was to make this express the energy of this. On the side facing the street it ’ s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in the Hamptons. And it ’ s got a fish. And I keep thinking,”This is going to be the last fish.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s the living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in the budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of the guys that presented to you earlier. We ’ ve had a lot of fun working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ’ s perpendicular to the train station, to the route of the train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good slides to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â you see it as all of these pieces accrue and become part of an overall neighborhood. It ’ s plaster and just zinc. And you wonder, if this is a museum, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and stuff. And from the building in the **back**, you see it as an iceberg floating by in the hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is the Disney Hall â the concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a compositional relationship between **us** that would strengthen both of **us**. And the plan of this â it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â to MOCA, etc. The acoustician in the competition gave **us** criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But everybody liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that **back** in some way. And we studied many models. This was our original model. These were the three buildings that were the ideal â the Concertgebouw, Boston and Berlin. Everybody liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new acoustician, he told **us** this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an acoustical, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the sloping outside walls, which the acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on the way to being â and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with laser stuff, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it **back** up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, struggling with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the foot of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the foot of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th- century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ’ t as good as the stuff, and so it failed. It should have been the other way around â the food should have been good first. It didn ’ t work. Thank you very much.

Text Sample 87: NEG Dim 1 – 2020 – Score -1.00 – 2020/t004215.txt

Several years ago a young man came to see me in my clinic. He told me he was running for his life. He said that he fled his home, because there, homosexuality wasn ’ t just illegal, in some cases it was punishable by death. So when his sexual orientation was exposed, his family rejected him, his boss fired him and angry mobs repeatedly attacked him in the streets. And each time the police arrived only to arrest him, detain and torture him further. And he knew that if he couldn ’ t escape the cycle of violence, he would surely be killed. So he had to do what he needed to do to survive. He left everything behind. All of his friends, his family, his career. He fled his home, he escaped to the United States and here he applied for asylum. But like many people fleeing this kind of persecution, he couldn ’ t carry much. He had some basic ID, barely any money and a few other belongings. He certainly didn ’ t bring official documents from the police who tortured him. No videos from the mob that tried to kill him. He didn ’ t have this kind of evidence to help support his claims, yet here he was, sitting in my clinic, showing me some of the most powerful evidence of his persecution. That was the physical and psychological scars that he brought with him. You see, he suffered from chronic, debilitating pain. He had severe scars scattered over his body, poorly healing wounds that got infected over and over again. He suffered from severe depression and continued to have regular, paralyzing flashbacks and nightmares from PTSD. So we continued our work. We met regularly for months, documenting each of these pieces of medical evidence. We went over the details of every attack, photographed his scars, documented his injuries and wounds, and we were even able to start chronicling his slow but steady recovery while under our care. Working closely with his lawyers, I submitted a detailed affidavit, including the findings of this forensic medical evaluation, and we included it as part of his asylum application. And then we waited for several long years while he navigated the courts. And then one day I got an email from him. It said that he was granted asylum. And everyone in the clinic was overjoyed. He said in his email that this was the first time in years that he didn ’ t fear deportation and death. It was the first time in years that he truly felt safe, that he had the security to rebuild his life all over again. And it was only through this medical and legal advocacy that we were able to help restore his legal status and his rights, that he could do that, all through asylum. Now for many people fleeing persecution, they come to programs and clinics like this telling unimaginable tales of violence and different reasons they were persecuted. But one thing is always the same. The violence meted against them was done with complete impunity, sometimes by the hands of the state directly through police or military officials. In other cases, the state just turns a blind eye and condones the acts of paramilitary groups or even violent domestic partners. In other cases, state is completely powerless to protect the vulnerable from powerful gangs. Now we know that social determinants of health play a huge role in determining the health and well-being of our patients : housing, income, education, race, social inclusion. But the same can be true for equal protection in the law – due process. Especially in societies for the most vulnerable, the marginalized and even those who are actively targeted, their access to these human rights protections that can mean the difference between sickness and health, and often it ' s the difference between life and death. And for millions of people who endure persecution and torture, the only way to heal is to acknowledge the human rights abuses that have occurred and to help restore the rights and protections that were so violated. After the atrocities of World War II, the asylum system was set up as one pathway to that kind of relief. But these days it seems like that pathway has turned into an obstacle course, setting people up to fail. Asylum seekers oftentimes don ' t know how to start, let alone complete the process that can drag on for years. They ' re not entitled to lawyers, so they don ' t know their rights. Increasingly, they ' re even being barred from setting foot in places of potential refuge. They ' re arrested or prosecuted, even deported before they ever get to see an asylum officer. And even if they do make it through the process, asylum grant rates can be as low as 20 percent and far worse for some. It ' s almost like the system was designed to keep people from exercising their right. But there is something that many of these people can do. Something that can potentially increase their chances of success to 90 percent or more. So what makes the difference? Getting a lawyer and having a medical evaluation. It ' s as simple as that. The man who came to my clinic and won his asylum case. Doctors and lawyers working together to present all of the evidence, including the medical evidence, to the courts allows judges to make informed and just decisions. And it ' s this kind of medical-legal partnership that ' s now more important than ever, because we live in a time of epic, forced migration due to violence and conflict. In 2018 there were 70 million people worldwide forcibly displaced due to war, conflict and persecution. It includes 40 million internally displaced, 25 million refugees and three million asylum seekers. Here in the United States, we see the impact of escalating violence in places like El Salvador, Guatemala and Honduras, where murder rates can be as high as those in Syria and Afghanistan. Where police corruption and gang violence are on the rise, where poverty and child abuse are widespread and tolerated, where basic systems of governance – public safety, child protection – are ineffective. It ' s no surprise then that many of the most vulnerable in some of these societies – children, women and other targeted groups – they ' re growing increasingly desperate and fleeing in unprecedented numbers. Like over the past 10 years, the numbers of unaccompanied children trying to seek safety at our southwest border has increased 18-fold, from 3, 300 in 2009 to over 62, 000 this past year. That ' s in addition to nearly half a million people traveling as families. Men, women and children trying to seek refuge at our borders, but who are stranded in a humanitarian crisis. And what makes matters worse is that they ' re caught in this fog of claims and counterclaims about who they are, what they ' ve experienced, where the proof is and what they deserve. Do they deserve our help? Sometimes people make claims that they ' re not fleeing human rights abuses but are simply economic migrants. Others say these children are actually being exploited and trafficked by their parents. Others say they ' re not even children at all ; they ’ re hardened criminals, they ’ re gang members trying to infiltrate our country. To cut through some of this fog, my colleagues and I conducted a study. We looked at data from children seeking asylum who had medical evaluations. And this is what the evidence told **us**. 80 percent of these children had evidence of exposure to repeated physical violence : assault and torture. 60 percent of the girls and at least 10 percent of the boys had evidence of repeated exposure to sexual violence. One young girl, telling a story and having corroborating evidence of being detained, beaten and raped over the course of three years, trafficked to other men and even having the threats of the murder of her entire family if she should ever escape or try to seek help. 90 percent of these children had evidence of psychological harm from indirect violence, including such severe threats, but also witnessing untold atrocities with their very eyes. One young boy described the terror and the grief and the utter fear of seeing the mutilated bodies and faces of his younger brother, his aunt, his uncle, his cousin, all killed in a single gang attack meant to send the community a message. And of course the psychological toll is immense. 19 percent of these children had signs of anxiety disorder ; 41 percent, depression and 64 percent, PTSD. 21 percent also had signs of suicidality as children. To put this into some perspective, returning combat veterans, they have PTSD on the order of 10 to 20 percent. These children at three to six times more likely to have PTSD than a soldier returning from war. Now despite this burden and despite this trauma, there are many others, still. Children who come to seek safety and enter into our immigration system only to find further abuse and even torture reminiscent of the places that they fled. You might remember some of those headlines, some of those images this past year. Children being ripped from the arms of their parents. Toddlers, infants in cold and unsanitary cages. The absence of food, water clothing and even soap. There ’ s also increasing reports of medical negligence, preventable complications, child abuse, sexual abuse and even child deaths in US custody. Sadly, many of these abuses and crimes aren ’ t new. Some date **back** many years and even across administrations. But something ’ s changed. The scope and scale of these abuses and crimes, the systematic and seemingly purposeful endangerment of asylum seekers and also the impunity with which it ’ s being done has raised the harm to an entirely new level. It reminds me of one of the girls in the study who told **us** how she pleaded with one of her attackers, asking him to stop, asking why she was targeted. And do you what his response was? He says,”We can do this, because there ’ s no one here to protect you.”We can ’ t let this be true of children and other asylum seekers trying to find help at our borders. But what do we do? As a physician, I ’ m often dealing with difficult decisions with some of my sickest and most complex patients. Of course we want to keep our focus on their health, their well-being, their quality of life, but sometimes it requires a deeper exploration of their values to really understand how to move forward. In a similar way, our nation is facing a crisis with the increasing number of asylum seekers at our borders and in our communities, and it compels **us** to re-examine some of our own fundamental values. What does it mean when we value health and safety? What does it mean when we value security, life, liberty, the life of children? What about this one – what does it mean when we say we value law and order? Does that also include respecting due process rights for an asylum seeker? Now for some, when they hear these terms they immediately gravitate towards wanting to build more walls, deploying more border patrol, deporting more people even if it means separating children from their families, subjecting them to psychological torture or deporting them to places where they might die. All in the name of security. All in our name. But for me and for many others, when I think of these values, that pushes me in an entirely new direction and renews my commitment to try to meet the needs of these asylum seekers with every tool I have at my disposal. So that when we say that we value life and liberty, we ' ll see these people who have taken unimaginable risks to flee imminent danger and harm to try to find safety. We ' ll meet them where they are and provide food, water, shelter, clothing. And we ' ll certainly meet them with medical care and mental health care that they so desperately need. When we say that we value the rule of law, and not just the privileges it provides a few but the responsibilities it requires of all of **us**, we ' ll make sure that we have a functioning immigration system. We ' ll make sure that we have trained judges. We ' ll make sure that we ' re not settling for the illusion of law and order that maybe a tall wall or a militarized border might provide **us**. We want the real thing. We want judges to be able to evaluate the evidence, including the medical evidence, and we want them to administer justice... fairly. When we say that we value health and well-being, that we don ' t want to perpetuate harm, then we ' ll deploy trauma-informed strategies at all levels of the immigration system. It might start with retraining border patrol agents or immigration officials, but it needs more medical, mental health and child welfare experts across the whole system. And when we say that we value justice, we won ' t let ourselves be turned into the torturers that many of these children and other people fled. We ' ll open up our **detention** centers and our courts to experts and advocates to hold ourselves accountable. And we may find that we need to shut down most of them and close these camps. I believe that by working in effective partnerships with lawyers, doctors, human rights advocates and many others, that we can work together to meet these asylum seekers ' needs, that we can meet our historical, humanitarian and legal obligations to them. And when we do, I think something powerful will unfold. Not only will these asylum seekers – like the man who came to my clinic and won his asylum case, like the children in the study or the many thousands of others seeking a new life, they ' ll be able to find that safety and security. We ' ll recognize the abuses that have occurred, and we ' ll restore the rights and protections that were lost. And I think that we ' ll be in wonder when we see them in the fullness of their humanity. Not just their strengths and weaknesses, their hopes and joys, not just the trauma that we acknowledge, but we ' ll also stand with them and we ' ll be inspired by their resilience. They ' ll blossom, and they ' ll add to the richness of this nation. I think by staying true to our fundamental values in the way that I ' ve described, that ' s how we build a sane and humane immigration system. That ' s how we remain the golden door. And that ' s how it happens that we remain the shining light of the world. Thank you.

Text Sample 88: NEG Dim 1 – 2020 – Score -1.00 – 2020/t004214.txt

So often, I ' ll take a fitness class, or I ' ll go to a music venue, or, really, anywhere that plays music in the background, and I ' ll find myself loving the rhythms and the melodies and the beats... And then I take a second to listen to the lyrics, lyrics that, for example, place **us** in a position of subservience that we would never tolerate in any other context. And I ' m aghast at the degree to which we normalize sexism in our culture. I listen to this music and I ' m like, I don ' t want to have to turn up to the sound of my own oppression. You know, music is one of the most powerful forms of communication, because it has the potential to either uplift or oppress. Music caters to the emotions. Music caters to the soul. Music opens up our soul. It opens up our channels to receive information about somebody else ' s walk of life, to inform our own roles. And while I have no problem with male fantasy, what I do have a problem with is that, according to a recent study, only 2. 6 percent of all music producers identify as women. That means an even smaller percentage identify as trans or gender nonconforming. And why does this matter? Because, if we don ’ t own and control our own narrative, somebody else will tell our stories for **us**, and they will get it wrong, perpetuating the very myths that hold **us back**. And I ’ m not here to tell other people how to make their music. But I am here to provide and design the alternative. One strategy I take in my music is making uplifting, energetic, percussive global beats and placing lyrics on top of them that genuinely describe my life ’ s experiences without contributing to the oppression of anybody else. It ’ s funny, because it ’ s the same reason as to why we excuse so many problematic lyrics ; it ’ s because we love how the beats make **us** feel. An example of this is my song”Top Knot Turn Up.”I turned off my phone ’ s notifications so I have more time / No bubbles to trouble my clear state of mind / One thing to know, I ’ m not here to please / Hair tied up, I do it properly / My time is not your property / When I ’ m productive like my ovaries, eyy! / Give a grown girl room to breathe, basic rights and her liberty / Free from insecurity that the world ’ s projecting onto me / Please do not trouble me when I am focused / The future is female you already know this / I ’ m fighting against the corruption on SCOTUS / Turned up in my top knot since when I first wrote this / It ’ s a top knot turn up It ’ s a top knot turn up, turn up, turn up. It ’ s a top knot turn up It ’ s a top knot turn up, turn up, turn up. It ’ s a top knot turn up It ’ s a top knot turn up, eyyy. It ’ s a top knot turn up. I want **us** to keep making sex-positive, beautiful music about joy and freedom. I want **us** to embrace our own pleasure just as much as we embrace our own pain. I want **us** to celebrate the authentic, nuanced, multidimensional aspects of our human existence, rather than perform false narratives of degrading sexuality in order to feel accepted or loved. And another strategy that I take in my music to combat the misogyny that exists on the airwaves is to visually depict the very world I wished we lived in. In the music video for my song”See Me Thru,”which is like a vibe-y, queer electronic R and B song, I cast two of my dear friends, Ania and Dejha, to play the role of the lovers, because they ’ re married in real life. But what you don ’ t know is that they also are behind the camera concepting and directing the entire video. Heyyyyy ohhhh My emotions were tired Music should be safe and accessible for all to experience. As you can see, it ’ s not about losing the sex appeal or swag that music has, it ’ s about writing messages that infuse tenderness and positivity into music that motivates **us** and challenges **us**. And while we as musicians absolutely have the responsibility to make music that isn ’ t disempowering, the consumers can be part of the change, too. Firstly, we get to choose which songs we want to mute and which songs we want to turn louder. We get to say,”I respect myself enough to say I don ’ t want to listen to this, and I don ’ t want this to be in anybody else ’ s space, either.”Secondly, we can simply ask ourselves :”Does this music or this message contribute to the oppression of somebody else? Why am I tolerating it?”And finally, we can all be choosing to make playlists or DJ-ing music that provides the right vibe or mood that we ’ re looking for in that moment without the problematic messaging. Why does this matter? Because it ’ s teaching algorithms in our streaming systems and our world exactly what it is that we do want to listen to, creating long-term change and a feedbacking mechanism that impacts the entire industry. This is not a message for just a small group of people. This is a message that affects everybody, because when we protect and liberate our most vulnerable genders, we liberate everybody.

Text Sample 89: NEG Dim 1 – 2020 – Score -1.00 – 2020/t004316.txt

No matter how hard you might try, you can ’ t just flip a switch when you step into the office and turn your emotions off. Feeling feelings is part of being human. [The Way We Work] A pervasive myth exists that emotions don ’ t belong at work, and this often leads **us** to mistakenly equate professionalism with being stoic or even cold. But research shows that in the moments when our colleagues drop their glossy professional presentation, we ’ re actually much more likely to believe what they ’ re telling **us**. We feel connected to the people around **us**. We try harder, we perform better and we ’ re just generally kinder. So it ’ s about time that we learn how to embrace emotion at work. Now, that ’ s not to say you should suddenly become a feelings fire hose. A line exists between sharing, which builds trust, and oversharing, which destroys it. If you suddenly let your feelings run wild at work and give people far more information than they bargained for, you make everyone around you uncomfortable and you also undermine yourself. You ’ re more likely to be seen as weak or lacking self awareness, so, great to say you weren ’ t feeling well last night – you don ’ t need to go into every lurid detail about how you got reacquainted with your half-digested dinner. So there ’ s a wide spectrum of emotional expression. On one hand, you have under-emoters, or people who have a hard time talking about their feelings, and on the other end are over-emoters, those who constantly share everything that ’ s going on inside, and neither of these make for a healthy workplace. So what ’ s the balance between these two extremes? It ’ s something called selective vulnerability. Selective vulnerability is opening up while still prioritizing stability and psychological safety, both for you and for your colleagues. Luckily, anyone can learn to be selectively vulnerable, with practice. Here are four ways to get started. First, flag your feelings without becoming emotionally leaky. Bad moods are contagious, and even if you ’ re not vocalizing what you ’ re feeling, chances are your body language or your expressions are a dead giveaway. So if you are crossing your arms or hammering on your keyboard, your coworkers are going to know you ’ re upset. And if you don ’ t say anything, they might start to think it ’ s about them and get worried. So if you are reacting to a non-work-related event, so traffic for example, just flag it. You don ’ t need to go into detail. You can say something as simple as”I ’ m having a bad morning. It has nothing to do with you.”Now if it ’ s a work-related event that ’ s causing you to feel strong emotions, that brings **us** to point number two. Try to understand the need behind your emotion, and then address that need. If you suddenly start to find everyone around you irritating, sit **back** and reflect on that. And it might be that you ’ re irritable because you ’ re anxious, and you ’ re anxious because you ’ re worried about hitting a looming deadline. And in that case, you can go **back** to your team to address that need and say something like,”I want to make sure I get everything done ahead of the deadline. Can you help me put together a realistic plan to do that?”If you ’ re thinking of sharing, try and put yourself in the other person ’ s shoes. So if what you ’ re about to say would help you feel more supported and better understand the situation, then go ahead and share it. But if it gives you any kind of pause, you might want to leave it out. And finally, read the room and provide a path forward. If everyone on your team has been pulling long hours, and you notice that one of your colleagues seems particularly deflated or anxious, you can acknowledge that and show some empathy, but then try to give them something actionable that they could hold on to. And in this case, you could suggest that you go to your manager and ask that your weekly meeting be pushed **back** a day so you both have more time to work. You ’ re showing you ’ re invested in their success, but also that you care about their well-being. When we can be honest about what we feel, and freely suggest ideas, make mistakes and just not have to hide every piece of who we are, we ’ re much more likely to stay at the company for a long time. We ’ re also happier and more productive. So take a moment to reflect on the emotional expression that you bring to work each day. And if you are prone to oversharing, try editing. And if you ’ re a little bit more reserved, look for moments when you can open up to your colleagues and be a bit vulnerable. And chances are, there will be a big difference in how people respond to you. And selective vulnerability might just become one of your most valuable tools.

Text Sample 90: NEG Dim 1 – 2022 – Score -2.00 – 2022/t003564.txt

I ’ m a lawyer based in Chicago, and two months ago I represented a Vietnamese immigrant woman in her divorce. As she was testifying about how her husband brutally raped her, the judge repeatedly told her to stop speaking. She should only speak when I asked her a question. Can you imagine having to publicly share one of the most painful, traumatic moments in your life and being told over and over again to basically shut up? It felt so wrong, but also so familiar for me. My mother, she had no say in her divorce. She didn ’ t have a lawyer, so she signed papers giving away everything because she thought she would otherwise be deported **back** to Vietnam. I thought by becoming a lawyer, I could help people like my mother. But then I realized I was actually part of the problem. As lawyers, we take over speaking for our clients. And courtroom procedures, rules of evidence means that we have all the power in what happens in their cases. We take terrible situations from people ’ s lives and we translate it into legal language. But, in doing so, we rewrite our clients as caricatures. The most helpless, most victimized versions of themselves. This is the American legal system. And that ’ s why, eight years ago, I founded Beyond Legal Aid in the belief that it doesn ’ t have to be this way, and lawyers can actually change it, working within, making it so that clients can speak for themselves. How does this work? First, instead of working nine to five in offices downtown, we travel, go to our clients where they live. And during evenings, nights and weekends when they are available. And they can be active participants in the legal process. Next, let me share a little secret. Well, at least for the non-lawyers. Attorneys like to be in charge. We go to law school, we hang up fancy law degrees, and we wear fancy suits – obviously much fancier than what I ’ m wearing – because we want to be in control. But it ’ s our clients whose lives are being judged and impacted by what I do. They need to be full partners in charge of their own cases. We can give advice, empower them to navigate the legal system. But ultimately, we need to follow their lead and defer to their decisions. Let me give you an example. One of our clients, let ’ s call him Atticus, he was facing deportation in a case where the law was completely against **us**. Atticus had been in ICE **detention** for several months because the immigration judge denied him bond. So he and OCAD, the Organized Communities Against Deportation, the community group we were working with on his case, wanted to try for bond a second time. I and the other lawyers were like,”No, no, no, no, no, no, no. You can ’ t do that. The law is bad, the judge has already said no once, and we don ’ t want to piss him off by questioning his decision.”But they wanted to proceed, so we deferred to their decision. However, instead of my sharing his situation on his behalf, we convinced the judge to hear a live conversation between Atticus and his US citizen son, let ’ s call him Jem. Jem spoke about how he wanted to commit suicide because he couldn ’ t live without his father anymore. Atticus broke down, sobbing in despair because he felt responsible for what was happening with his son and yet couldn ’ t be there for him. The judge was forced to hear directly how Atticus and Jem loved each other and how much they needed each other. He heard it unadulterated, in full, without any questions, without any interruptions by any lawyers. Unfortunately, he denied bond a second time. But – and of course there ' s a"but,"right, otherwise, I wouldn ' t be telling you this story. The judge then cancelled Atticus ' s deportation. I cannot tell you how utterly, absolutely shocking that legal victory was. I was like,"Oh my God, I can ' t believe this happened."And... I ' m going to say something now, which lawyers don ' t say very often but probably should say a lot more often. I was wrong. My client and his community, they knew better. If we had not followed their lead and tried for bond a second time, the judge would have never heard Atticus tell his own story and be so moved by what was going on that he defied the law to allow Atticus to stay in the country with his son. Atticus being able to tell his own story, created a decision that was impossible, at least what we thought as attorneys. Atticus sharing and talking his own case created justice where the law was wrong. And that ' s why the legal system has to change, so that people like Atticus can share their own stories, have agency over their own cases. In the meantime, we as lawyers need to make it happen in our work, in our cases right now. Thank you.

Text Sample 91: NEG Dim 1 – 2022 – Score -2.00 – 2022/t003605.txt

For people who want to have a baby, false messages about fertility can be especially powerful. So I want to say upfront that the COVID-19 vaccine is safe for fertility. [Body Stuff with Dr. Jen Gunter] The data on this is clear. COVID-19 vaccination saves lives for everyone, especially pregnant people and their babies. But the fog of the pandemic has made it extra challenging to dispel the misinformation surrounding the COVID-19 vaccine. So I want to address some of these worries by talking about how a vaccine becomes a vaccine in the first place. To show that each step of creating the vaccine is closely evaluated for safety. One of the early stages of vaccine development involves exploratory research. In the case of the coronavirus that causes COVID-19, researchers already had years of knowledge and insight to work from. Scientists already knew a lot about other coronaviruses. And because of knowledge sharing, they were able to simply download the virus ' s genetic code off a website. They were also able to take advantage of the advances made in mRNA vaccine technology, which you might have heard about. While mRNA vaccines are new to the public, researchers have been studying and working with them for decades. I want to underscore this point. One worry about the vaccine is that it felt rushed. But while the COVID-19 vaccine was developed rapidly, its creation was based on decades of research. Once the exploratory stage is finished, researchers move on to the preclinical stage, where testing is done in petri dishes or on animals. The goal of preclinical tests is to see if the vaccine truly works on the pathogen – in this case, the virus that causes COVID-19 – and also to test the safety of a vaccine. This is where researchers perform toxicity testing, using a dosage far greater than what they might use in humans to make sure it ' s safe. By testing on animals, they ' re able to see if vaccines have any impact on functions like the reproductive system. You might be surprised to learn that the chemical signaling that happens in a rat uterus is very similar to what happens in a human uterus. Here ' s what ' s important. Researchers don ' t move on from the preclinical testing phase until they prove that the vaccine is unlikely to harm anyone. This is a rigorous process that goes through FDA approval. Once the FDA approves, the next step is the clinical development or clinical trials, testing in humans, which is a three-phase process. Phase one starts with a small group of people, usually less than 100 very healthy adults. By phase three, researchers are working with thousands or even tens of thousands of people. Because of the urgency of COVID-19, researchers save time by overlapping these phases, using early data from each phase to design for the next. In other words, no steps were skipped. Now, it ’ s true that clinical trials for the COVID-19 vaccines didn ’ t include pregnant people, and that ’ s not unusual. Pregnant people are currently part of a protected class that typically aren ’ t included in testing new vaccines or new drugs. I completely understand the worries around this, but remember, testing in animals is a proven way to ensure vaccine safety. The preclinical data for the COVID-19 mRNA vaccine showed no impact on fertility in animals that would lead researchers to suspect there would be issues with humans. Even after the vaccine has been approved by the FDA and made available to the public, there ’ s still work to be done. These are called phase four trials, where researchers and regulatory agencies track how the public responds to the vaccine long-term. Monitoring side effects, collecting and analyzing data from a much wider population. And now we have even more data to reassure **us** about fertility. After the COVID-19 vaccine was rolled out, there were reports online of people experiencing irregular periods or changes in their menstrual cycle. This, understandably, made people nervous about a potential effect on fertility or an unknown effect of the vaccine. We didn ’ t have any data about irregular menstrual cycles from clinical trials to point to, and that is problematic. But given the rush to get the data, it would have been hard to collect because people would have had to track cycles before the vaccine. Differences people notice in their menstrual cycles are valid, and it ’ s scary and uncertain to notice a change in our bodies. And because the endometrium, the lining of the uterus that sheds during a period, is part of our immune system, it is theoretically possible a vaccine could have a temporary effect in the same way a vaccine causes temporary swelling in the lymph nodes for some people or a temporary fever. Researchers responded to the concerns raised online and looked into data from almost 4, 000 people who tracked their cycles using an app and compared data from the vaccinated group to those who didn ’ t get vaccinated. On average, the menstrual cycles of people who were vaccinated varied in length by less than one day, which is interesting but not medically significant since normal menstrual cycles vary by up to seven days each cycle. The only group that was more likely to show a meaningful difference in their menstrual cycle length were those who received two doses of the vaccine in one cycle. Even then, what they observed was temporary. Their cycles went **back** to normal within three cycles. And we ’ ve continued to gather data. We now see the importance of vaccination for those who are pregnant as the risk of mortality in pregnancy with COVID-19 is 1. 6 percent. That ’ s 22 times higher than the risk of mortality in pregnancy for those without COVID. And unvaccinated mothers and pregnant people who do get COVID-19 have a much higher chance of being admitted to the intensive care unit. We also now have data showing that infants from people vaccinated during pregnancy do have antibodies to help protect them from COVID-19. If your goal is a healthy pregnancy, one of the best ways to achieve that is by getting vaccinated. And we can have confidence that the COVID-19 vaccines have been rigorously tested and are being closely followed.

Text Sample 92: NEG Dim 1 – 2022 – Score -1.00 – 2022/t003749.txt

It ’ s dark days for democracy. And our crisis is global. If you ’ re like me, you can feel it in your bones. My expertise focuses on democracy in the United

States, where I live and work. And while we ’ ve long lifted up our own system on a pedestal as an example for the world, we ’ re one of those democracies that ’ s currently in a yearslong decline. So much of the conversation in the media and among experts is about our broken two-party system. Democrats vs. Republicans. And so much of the conversation about voting is about the outcomes : the winners and the losers. These are the wrong conversations if we want to tackle the threat to US democracy today. Because we have a much more fundamental challenge, and it ’ s quickly growing. The United States election infrastructure is crumbling. By infrastructure, I mean the technology, the physical infrastructure, like facilities, and most importantly, the human infrastructure, the people who manage the US voting process. Election infrastructure is so essential that it ’ s been designated critical infrastructure by the US Department of Homeland Security. That puts it on par with systems like our power grid and water supply in the eyes of the federal government. But they haven ’ t invested in it like it ’ s critical. The average portion of a county budget spent on election operations is about half of one percent. To put that into context, we spend about the same amount to maintain parking facilities as we do our election system. If we zoom out a little bit, the US election system is actually pretty unique by global standards. Unlike some other places, we have no central election authority that ’ s responsible for managing the logistics of voting for our entire country. Instead, we have thousands and thousands of local departments that each have some independent mix for figuring out how to make voting work where they live. These departments are staffed by professional election officials with support from volunteer poll workers during peak election season. So with no government-run how-to for how to administer elections, we end up seeing widely different voting experiences throughout the US, where voting can legitimately work one way in one community and look totally different somewhere else. For over a decade, my work has focused on providing technology and training and other resources to state and local election officials to support them in their work serving voters. In my current role, that has looked like working with election departments in every corner of the country, that together serve about 75 percent of our eligible voters. This has given me a really unique window into what it actually takes these public servants on the front lines to do things like keep voter rolls up to date and quickly and accurately count ballots and inform their communities about how the process works. This year-round work is hard. And it ’ s often super thankless. And the last two years have been some of the worst. Election officials who serve millions of voters currently lack the basic technology that they need to reliably do their work. It either doesn ’ t exist, or it ’ s shockingly outdated. In the year 2020, we supported a small New England town replace their hand crank ballot box that they had been using since the early 1900s to count ballots. One was literally held together by duct tape. Even worse, the people that underpin our voting process, the election officials, are currently under attack. In exchange for being outspoken about the integrity of the process that they managed in 2020, which, by the way, turned out to be our highest turnout election in the US in over a century and the most secure election ever administered in our history, according to our national security community – In exchange for the grit and determination that it took to make that possible during a global pandemic, today, election officials are receiving death threats, their children are being bullied, and some have had to flee their homes. A recent survey shows that one in three election officials, one in three, currently feel unsafe doing their work. It ’ s appalling. And enough is enough. We are at a tipping point for US democracy and frankly, democracy globally. I don ’ t know about you, but I personally don ’ t feel comfortable just standing around, leaving things to chance. And I especially don ’ t feel comfortable standing by, asking election officials to keep figuring out how to make it work alone, while our system is getting pushed to its brink. We need to rally around a set of shared values and standards, a North Star, so that every single voter, regardless of their zip code, has access to a process that ’ s both fair and trustworthy. Election officials need a place where they can come together to keep their skills fresh so that they ’ re ready to tackle whatever challenges might come, whether it ’ s training to bolster cybersecurity or to combat disinformation or to help them keep voters safe during a pandemic. And every election department, rural, urban, large, small, needs access to 21st century secure technology so that every single community has access to the basics. Like, a website where voters can find out answers about how voting works where they live. Or data-driven tools that election officials can use to make sure that we don ’ t have lines that snake around city blocks outside of polling places. Last summer, I was at my first conference since the start of the pandemic, and I was going up the escalator and I locked eyes with this election official who I absolutely adore. Let ’ s call her Sarah. I ran and I gave Sarah a huge hug, and I realized that probably 60 seconds passed before we let go. And when we did, I was crying and so was Sarah. Sarah ’ s life got flipped totally upside down after the 2020 election, when a vicious conspiracy theory about her office went viral. Almost immediately, the death threats started to roll in, and they were unrelenting and they were very specific about the harm that they wished upon Sarah and her husband and their children. And still, Sarah never **backed** down from telling the truth about the process that she led. In that moment, I was so overwhelmed with gratitude and love and admiration for Sarah that I couldn ’ t help the tears. But I think if I ’ m being super, super honest, I was just so happy that my friend was still here to hug. Election officials deserve a warm, welcoming community where their hard work is celebrated and encouraged, not vilified. And all of this is exactly why I ’ m so excited that we are inviting every single election department in the United States to join the US Alliance for Election Excellence. The Alliance is a place where election officials and technologists and designers and other experts are working hand in hand to revitalize US democracy. And we ’ re doing it by focusing on the basics, a shared North Star, the tools that it actually takes to get the job done, and a community of support that has each other ’ s **back**, whatever might come. It is absolutely crucial that we rebuild the foundation of US democracy, and that ’ s exactly what we ’ re going to do. The Alliance is not only ready to meet this moment, we ’ re ready to create a foundation so strong, that we can make it another 200 years. Together. Thank you.

Text Sample 93: NEG Dim 1 – 2018 – Score -2.00 – 2018/t000044.txt

I ’ m here to tell you how change is happening at a local level in Pakistan, because women are finding their place in the political process. I want to take you all on a journey to the place I was raised, northwest Pakistan, called Dir. Dir was founded in the 17th century. It was a princely state until its merger with Pakistan in 1969. Our prince, Nawab Shah Jahan, reserved the right to wear white, the color of honor, but only for himself. He didn ’ t believe in educating his people. And at the time of my birth in 1979, only five percent of boys and one percent of girls received any schooling at all. I was one among that one percent. Growing up, I was very close to my father. He is a pharmacy doctor, and he sent me to school. Every day, I would go to his clinic when my lessons finished. He ’ s a wonderful man and a well-respected community leader. He was leading a welfare organization, and I would go with him to the social and political gatherings to listen and talk to the local men about our social and economic problems. However, when I was 16, my father asked me to stop coming with him to the public gatherings. Now, I was a young woman, and my place was in the home. I was very upset. But most of my family members, they were happy with this decision. It was very difficult for me to sit **back** in the home and not be involved. It took two years that finally my family agreed that my father could reconnect me with women and girls, so they could share their problems and together we could resolve them. So, with his blessings, I started to reconnect with women and girls so we could resolve their problems together. When women show up, they bring their realities and views with them. And yet, I have found all too often, women underestimate their own strength, their potential and their self-respect. However, while connecting with these women and girls, it became very clear to me that if there was to be any hope to create a better life for these women and girls and their families, we must stand up for our own rights — and not wait for someone else to come and help **us**. So I took a huge leap of faith and founded my own organization in ' 94 to create our very own platform for women empowerment. I engaged many women and girls to work with me. It was hard. Many of the women working with me had to leave once they got married, because their husbands wouldn ' t let them work. One colleague of mine was given away by her family to make amends for a crime her brother had committed. I couldn ' t help her. And I felt so helpless at that time. But it made me more determined to continue my struggle. I saw many practices like these, where these women suffered silently, bearing this brutality. But when I see a woman struggling to change her situation instead of giving up, it motivates me. So I ran for a public office as an independent candidate in Lower Dir in the local elections in 2001. Despite all the challenges and hurdles I faced throughout this process, I won. And I served in the public office for six years. But unfortunately, we women, elected women, we were not allowed to sit in the council together with all the members and to take part in the proceedings. We had to sit in a separate, ladies-only room, not even aware what was happening in the council. Men told me that,"You women, elected women members, should buy sewing machines for women."When I knew what they needed the most was access to clean drinking water. So I did everything I could do to prioritize the real challenges these women faced. I set up five hand pumps in the two dried up wells in my locality. Well, we got them working again. Before long, we made water accessible for over 5, 000 families. We proved that anything the men could do, so could we women. I built alliances with other elected women members, and last year, we women were allowed to sit together with all the members in the council. And to take part in the legislation and planning and budgeting, in all the decisions. I saw there is strength in numbers. You know yourselves. Lack of representation means no one is fighting for you. Pakistan is — We ' re 8, 000 miles away from where I ' m here with you today. But I hope what I ' m about to tell you will resonate with you, though we have this big distance in miles and in our cultures. When women show up, they bring the realities and hopes of half a population with them. In 2007, we saw the rise of the Taliban in Swat, Dir and nearby districts. It was horrifying. The Taliban killed innocent people. Almost every day, people collected the dead bodies of their loved ones from the streets. Most of the social and political leaders struggling and working for the betterment of their communities were threatened and targeted. Even I had to leave, leaving my children behind with my in-laws. I closed my office in Dir and relocated to Peshawar, the capital of my province. I was in trauma, kept thinking what to do next. And most of the family members and friends were suggesting,"Shad, stop working. The threat is very serious."But I persisted. In 2009, we experienced a historic influx of internally displaced persons, from Swat, Dir and other nearby districts. I started visiting the camps almost every day, until the internally displaced persons started to go **back** to their place of origin. I established four mother-child health care units, especially to take care of over 10, 000 women and children nearby the camps. But you know, during all these visits, I observed that there was very little attention towards women ’ s needs. And I was looking for what is the reason behind it. And I found it was because of the underrepresentation of women in both social and political platforms, in our society as a whole. And that was the time when I realized that I need to narrow down my focus on building and strengthening women ’ s political leadership to increase their political representation, so they would have their own voice in their future. So we started training around 300 potential women and youth for the upcoming local elections in 2015. And you know what? Fifty percent of them won. And they are now sitting in the councils, taking part actively in the legislation, planning and budgeting. Most of them are now investing their funds on women ’ s health, education, skill development and safe drinking water. All these elected women now share, discuss and resolve their problems together. Let me tell you about two of the women I have been working with : Saira Shams. You can see, this young lady, age 26, she ran for a public office in 2015 in Lower Dir, and she won. She completed two of the community infrastructure schemes. You know, women, community infrastructure schemes... Some people think this is men ’ s job. But no, this is women ’ s job, too, we can do it. And she also fixed two of the roads leading towards girls schools, knowing that without access to these schools, they are useless to the girls of Dir. And another young woman is Asma Gul. She is a very active member of the young leaders forum we established. She was unable to run for the public office, so she has become the first female journalist of our region. She speaks and writes for women ’ s and girls ’ issues and their rights. Saira and Asma, they are the living examples of the importance of inclusion and representation. Let me tell you this, too. In the 2013 general elections in Pakistan and the local elections in 2015, there were less than 100 women voters in Dir. But you know what? I ’ m proud to tell you that this year, during the general elections, there were 93, 000 women voters in Dir. So our struggle is far from over. But this shift is historic. And a sign that women are standing up, showing up and making it absolutely clear that we all must invest in building women ’ s leadership. In Pakistan and here in the United States, and everywhere in the world, this means women in politics, women in business and women in positions of power making important decisions. It took me 23 years to get here. But I don ’ t want any girl or any woman to take 23 years of her life to make herself heard. I have had some dark days. But I have spent every waking moment of my life working for the right of every woman to live her full potential. Imagine with me a world where thousands of **us** stand up and they support other young women together, creating opportunities and choices that benefit all. And that, my friends, can change the world. Thank you.

Text Sample 94: NEG Dim 1 – 2018 – Score -1.00 – 2018/t000022.txt

This is my favorite protest shirt. It says,”Protect your people.”We made it in the basement of our community center. I ’ ve worn it at rallies, at protests and marches, at candlelight vigils with families who have lost loved ones to police violence. I ’ ve seen how this ethic of community organizing has been able to change arresting practices, hold individual officers accountable and allow families to feel strong and supported in the darkest moments of their lives. But when a family would come to our center and say,”My loved one got arrested, what can we do?”we didn ’ t know how to translate the power of community organizing that we saw on the streets into the courts. We figured we ’ re not lawyers, and so that ’ s not our arena to make change. And so despite our belief in collective action, we would allow people that we cared about to go to court alone. Nine out of ten times — and this is true nationally — they couldn ’ t afford their own attorney, and so they ’ d have a public defender, who is doing heroic work, but was often under-resourced and stretched bare with too many cases. They would face prosecutors aiming for high conviction rates, mandatory minimum sentences and racial bias baked into every stage of the process. And so, facing those odds, stripped away from the power of community, unsure how to navigate the courts, over 90 percent of people that face a criminal charge in this country will take a plea deal. Meaning, they ’ ll never have their fabled day in court that we talk about in television shows and in movies. And this is the untold part of the story of mass incarceration in America — how we became the largest jailer in the world. Over two million people currently incarcerated in this country. And projections that say one out of three black men will see the inside of a prison cell at some point in their life on this trajectory. But we have a solution. We decided to be irreverent to this idea that only lawyers can impact the courts. And to penetrate the judicial system with the power, intellect and ingenuity of community organizing. We call the approach”participatory defense.”It ’ s a methodology for families and communities whose loved ones are facing charges, and how they could impact the outcome of those cases and transform the landscape of power in the courts. How it works is, families whose loved ones are facing criminal charges will come to a weekly meeting, and it ’ s half support group, half strategic planning session. And they ’ ll build a community out of what otherwise would be an isolating and lonely experience. And they ’ ll sit in a circle, and write the names of their loved ones on a board, who they ’ re there to support. And collectively, the group will find out ways to tangibly and tactfully impact the outcome of that case. They ’ ll review police reports to find out inconsistencies ; they ’ ll find areas that require more investigation by the defense attorney ; and they ’ ll go to court with each other, for the emotional support but also so that the judge knows that the person standing before them is part of a larger community that is invested in their well-being and success. And the results have been remarkable. We ’ ve seen charges get dismissed, sentences significantly reduced, acquittals won at trial and, sometimes, it has been literally lifesaving. Like in the case of Ramon Vasquez. Father of two, family man, truck driver and someone who was wrongfully charged with a gang- related murder he was totally innocent of, but was facing a life sentence. Ramon ’ s family came to those meetings shortly after his arrest and his **detention**, and they worked the model. And through their hard work, they found major contradictions in the case, gaping holes in the investigation. And were able to disprove dangerous assumptions by the detectives. Like that the red hat that they found when they raided his home somehow affiliated him to a gang lifestyle. Through their photos and their records, they were able to prove that the red hat was from his son ’ s Little League team that Ramon coached on the weekends. And they produced independent information that proved that Ramon was on the other side of town at the time of the alleged incident, through their phone records and receipts from the stores that they attended. After seven long months of hard work from the family, Ramon staying strong inside jail, they were able to get the charge dismissed. And they brought Ramon home to live the life that he should have been living all along. And with each new case, the families identified new ways to flex the knowledge of the community to have impact on the court system. We would go to a lot of sentencing hearings. And when we would leave the sentencing hearing, on the walk **back** to the parking lot after someone ' s loved one just got sent to prison, the most common refrain we would hear wasn ' t so much,"I hate that judge,"or"I wish we had a new lawyer."What they would say was,"I wish they knew him like we know him."And so we developed tools and vehicles for families to tell the fuller story of their loved one so they would be understood as more than just a case file. They started making what we call social biography packets, which is families making a compilation of photos and certificates and letters that show past challenges and hardships and accomplishments, and future prospects and opportunities. And the social biography [packets] were working so well in the courts, that we evolved it into social biography videos. Ten-minute mini documentaries, which were interviews of people in their homes, and at their churches and at their workplace, explaining who the person was in the backdrop of their lives. And it was a way for **us** to dissolve the walls of the court temporarily. And through the power of video, bring the judge out of the court and into the community, so that they would be able to understand the fuller context of someone ' s life that they ' re deciding the fate of. One of the first social biography projects that came out of our camp was by Carnell. He had come to the meetings because he had pled to a low-level drug charge. And after years of sobriety, got arrested for this one drug possession charge. But he was facing a five-year prison sentence because of the sentencing schemes in California. We knew him primarily as a dad. He ' d bring his daughters to the meetings and then play with them at the park across the street. And he said,"Look, I could do the time, but if I go in, they ' re going to take my girls."And so we gave him a camera and said,"Just take pictures of what ' s like being a father."And so he took pictures of making breakfast for his daughters and taking them to school, taking them to after-school programs and doing homework. And it became this photo essay that he turned in to his lawyer who used it at the sentencing hearing. And that judge, who originally indicated a five-year prison sentence, understood Carnell in a whole new way. And he converted that five-year prison sentence into a six-month outpatient program, so that Carnell could be with his daughters. His girls would have a father in their life. And Carnell could get the treatment that he was actually seeking. We have one ceremony of sorts that we use in participatory defense. And I told you earlier that when families come to the meetings, they write the names of their loved ones on the board. Those are names that we all get to know, week in, week out, through the stories of the family, and we ' re rooting for and praying for and hoping for. And when we win a case, when we get a sentence reduced, or a charge dropped, or we win an acquittal, that person, who ' s been a name on the board, comes to the meeting. And when their name comes up, they ' re given an eraser, and they walk over to the board and they erase their name. And it sounds simple, but it is a spiritual experience. And people are applauding, and they ' re crying. And for the families that are just starting that journey and are sitting in the **back** of the room, for them to know that there ' s a finish line, that one day, they too might be able to bring their loved one home, that they could erase the name, is profoundly inspiring. We ' re training organizations all over the country now in participatory defense. And we have a national network of over 20 cities. And it ' s a church in Pennsylvania, it ' s a parents ' association in Tennessee, it ' s a youth center in Los Angeles. And the latest city that we just added to the national network to grow and deepen this practice is Philadelphia. They literally just started their first weekly participatory defense meeting last week. And the person that we brought from California to Philadelphia to share their testimony, to inspire them to know what ' s possible, was Ramon Vasquez, who went from sitting in a jail in Santa Clara County, California, to inspiring a community about what ’ s possible through the perseverance of community across the country. And with all the hubs, we still use one metric that we invented. It ’ s called time saved. It ’ s a saying that we actually still say at weekly meetings. And what we say when a family comes in a meeting for the first time is : if you do nothing, the system is designed to give your loved one time served. That ’ s the language the system uses to quantify time of incarceration. But if you engage, if you participate, you can turn time served into time saved. That ’ s them home with you, living the life they should be living. So, Carnell, for example, would represent five years of time saved. So when we totaled our time saved numbers from all the different participatory defense hubs, through the work in the meetings and at court and making social biography videos and packets, we had 4, 218 years of time saved from incarceration. That is parents ’ and children ’ s lives. Young people going to college instead of prison. We ’ re ending generational cycles of suffering. And when you consider in my home state of California, it costs 60, 000 dollars to house someone in the California prison system, that means that these families are saving their states a ton of money. I ’ m not a mathematician, I haven ’ t done the numbers, but that is money and resources that could be reallocated to mental health services, to drug treatment programs, to education. And we ’ re now wearing this shirt in courts all across the country. And people are wearing this shirt because they want the immediacy of protecting their people in the courtroom. But what we ’ re telling them is, as practitioners, they ’ re building a new field, a new movement that is going to forever change the way justice is understood in this country. Thank you.

Text Sample 95: NEG Dim 1 – 2018 – Score -1.00 – 2018/t001768.txt

A traumatic brain injury, or TBI, is a disruption in brain function caused by an external blow to the head. And when you hear that definition, you might think about sports and professional athletes, since it ’ s the kind of injury we ’ re used to seeing on the playing field. And this imagery has really come to define TBI in the public consciousness. I myself do research on TBI in retired and college athletes. I stood on a TED stage in 2010, talking about concussions in kids ’ sports. So I have to say, as someone who researches and treats these injuries, that I ’ ve been really gratified to see the growing awareness of TBI and specifically, the short- and long-term risks to athletes. Today, though, I want to introduce you to a larger but no less controversial group of people impacted by traumatic brain injury, who don ’ t often show up in the headlines. I ’ ve come to recognize these inmates and probationers as surprisingly among the most vulnerable members of society. For the last six years, my colleagues and I have been doing research that has completely changed the way we think about the criminal justice system and the people in it. And it may change the way you think about those things, too. So I ’ ll start with a shocking statistic : 50 to 80 percent of people in criminal justice have a traumatic brain injury. Up to 80 percent. In the general public, in this room, for example, that number is less than five percent. And I ’ m not just talking about getting your bell rung. These are the kinds of injuries that require hospitalization. Most of them are the product of a physical assault, and some of them are actually sustained in jail. All of these numbers are even higher among the women in criminal justice. Almost every single woman in the criminal justice system has been exposed to interpersonal violence and abuse. More than half of these women have been exposed to repeated brain injuries. In this way, these women ’ s brains look like the brains of retired NFL players, and they ’ ll likely face the same risks for dementing diseases as they age. The same risks. TBI, together with mental illness and substance abuse and trauma, makes it hard for people to think. They have cognitive impairments like poor judgment and poor impulse control, problems that make criminal justice a revolving door. People get arrested and booked into jail. They oftentimes get into trouble while they ’ re in there. They get into fights. They fall out of their bunk. And then they get released and do stupid things, like forgetting mandatory check- ins, and they get rearrested. Statistically speaking, they ’ re actually more likely to be rearrested than not. A colleague calls this”serving a life sentence 30 days at a time.”And oftentimes, these folks don ’ t know why this is so hard for them. They feel out of control and frustrated. So knowing that TBI is at the root of so many of these challenges, the mission for a group of **us** in Colorado has been to disrupt that cycle, to jam the revolving the door. So working together with my state and local partners, we crafted a plan to meet everyone ’ s needs : the system, the inmates and probationers, my graduate students. In this program, we assess how each person ’ s brain works so that we can recommend basic modifications to make this system more effective and safer. And here when I say”safer,”I mean safer not only for the inmates, but safer also for correctional staff. In some ways, this is such a simple approach. We ’ re not treating the brain injury, we ’ re treating the underlying problem that gets people into all of this trouble in the first place. We do quick neuropsychological screening tests to identify strengths and weaknesses in the way an inmate thinks. Using that information, we write two reports. One, a report for the system with specific recommendations on how to manage that inmate. The other is a letter to the inmate with specific suggestions for how to manage themselves. For example, if our test result suggests that a probationer has a hard time remembering the things they hear, that would be an auditory memory deficit. In that case, our letter to the court might suggest that that probationer get handouts of important information. And our letter to that probationer would say, among other things, that they should carry a notebook to record that information for themselves. Now, most importantly, is that I pause here to be really clear about one point. This program does not minimize responsibility or make excuses for anyone ’ s behavior. This is about changing longstanding negative perceptions and building self-advocacy. It ’ s actually about taking responsibility. The inmates move from,”I ’ m a total screwup, I ’ m a loser,”to,”Here ’ s what I don ’ t do well, and here ’ s what I have to do about it.”And the system comes to see an inmate ’ s problematic behavior as the things they can ’ t do versus the things they won ’ t do. And that change — seeing behavior as a deficit rather than outright defiance — is everything in these settings. We hear from inmates around the country, and they write, and more than anything, they want to know how to help themselves. This is an excerpt from a letter from Troy in Virginia, an excerpt from a 50-page letter. And he writes,”Can you tell me what you think of all the head traumas I ’ ve dealt with? What can I do? Can you help me?”Closer to home, we have thousands of stories like this, and smart stories, stories that have a great outcome. Here ’ s Vinny. Vinny was hit by a car when he was 15, and from that moment forward, spent more time in jail than in school. With some basic skill-building, after our assessment revealed that he had some pretty significant memory impairments, Vinny learned to use the alarm and reminder function on his iPhone to track important appointments, and he keeps a checklist to break larger tasks into smaller, manageable ones. And with basic tools like that under his belt, Vinny ’ s been out of jail for two years, clean for nine months, and recently **back** to work. What ’ s so striking for Vinny is that this is his first time off of court supervision since his injury more than 15 years ago. He made it out of the revolving door. He says now,”I can do anything. I just have to work a lot harder at it.”And here ’ s Thomas. Thomas has some pretty significant attention and behavior problems after an injury landed him in a coma for more than a month. After relearning how to walk, his first stop? Court. He couldn ’ t imagine a future where he wasn ’ t in trouble. He now carries a calendar to avoid being held in contempt for missed court dates, and he schedules a break into his day every day to recharge before he gets agitated. And nobody knows the revolving door better than the person sitting at the front of the courtroom. This is my good friend and colleague Judge Brian Bowen. Now, Judge Bowen was already on a mission to make the system work for everyone, and when he heard about this program, he saw the perfect fit. He actually sits down with all of his prosecutors to help them see that there ’ s basically two categories of defendants in the courtroom : the ones we ’ re afraid of — oftentimes, rightfully so — and the ones we ’ re mad at. These are the ones who miss all of their scheduled appointments and they blow through the best-laid probation plans. And Judge Bowen believes that, with a little more support, we could move people in this latter category, the maddening category, through and ultimately out of the system. He proved that with Navy veteran Mike. Judge Bowen saw the correlation between Mike ’ s history of a massive 70-foot fall and his long-standing pattern of difficulty showing up on the right day for court appointments and complying with mandatory therapy requirements, for example. And instead of sentencing him to more and more jail time, Judge Bowen sent him home with maps and checklists and handouts and recommended instead vocational rehabilitation and flexible scheduling for those therapies. And this with those supports, Mike ’ s **back** to work for the first time since his injury while he was in the service. He ’ s repairing relationships with his family, and just last month, he graduated from Judge Bowen ’ s veteran ’ s court. This program shows **us** the overwhelming prevalence of traumatic brain injuries and cognitive deficits and the accumulation of brokenness in the criminal justice system. And it highlights the extraordinary power of resilience and responsibility. In Mike and Thomas and Vinny, even Judge Bowen ’ s story, you saw the transformation made possible by a change in perception and some simple accommodations. All told, in this program, these inmates and probationers come to see themselves differently. The system sees them differently, and when you meet them in the community, I hope you see them differently, too. Thanks, guys.

Text Sample 96: NEG Dim 1 – 2023 – Score -1.00 – 2023/t003511.txt

All around the world, there ’ s a huge disability employment gap. In most countries, the unemployment rate for people with disabilities is twice that of people without disabilities. Often it ’ s as high as 80 percent. There are over a billion people in the world living with a disability. To have 80 percent of **us** unemployed, that ’ s so much untapped potential. [The Way We Work] As someone with a paralyzed arm and living with a mental health disability who ’ s worked in the financial sector and entertainment and tech, I know that having a disability doesn ’ t bar you from doing good work. So it ’ s shocking to me that this disability employment gap is so persistent. In the US, the Americans with Disabilities Act prohibits discrimination in hiring and requires employers to provide reasonable accommodations. And studies show that companies who prioritize disability inclusion are more profitable overall. So what exactly is the problem? Recruiters, human resources staff, managers, they tend to think of hiring people with disabilities as a social good or as something they have to do to meet a quota. But they should be hiring disabled people because of our strengths and all of the value and innovation we can bring. Here are three things every workplace can do to truly welcome people with disabilities. It ’ s not comprehensive, but it ’ s a start. First, stop making assumptions. Our culture tends to treat disability as a medical diagnosis, a tragedy or a charity case. And all of these things are rooted in pity. They prevent **us** from being seen as peers and equals. These assumptions lead to a lot of avoidance. People tend to not ask **us** about our lives and hobbies outside of work or don ’ t invite **us** to company social outings. There ’ s so much fear about saying the wrong thing, that instead people say nothing at all. They treat disabled people like we ’ re invisible when all we want is to be seen and heard and accepted like anyone else. These assumptions also make people jump to conclusions about what we can and can ’ t do without even asking. For me, assuming I can ’ t type because I can ’ t use one of my arms or assuming a blind person can ’ t be an engineer. When to succeed at work, you need to have people who see your full potential because without that, nobody wins. So please, take the time to get to know **us**, invite **us** to things, ask **us** the same kind of questions you would of any colleague. What drew you to this work? What are your goals? What do you hope to do from here? And listen to our answers but also respect our boundaries. If there ’ s something we ’ re not comfortable discussing, we ’ ll let you know. And any time you feel unsure, just ask **us** privately so we can make those decisions. You can even start by saying,”I ’ m still learning how to get better at talking about disability.”Second, rethink accessibility and accommodations starting now. Disability takes so many different forms. It can be about someone ’ s sight, hearing or mobility. It can be ADHD, dyslexia or chronic pain. About 62 percent of employees with disabilities have ones that are not apparent. So even if you don ’ t think there are people with disabilities in your organization, there probably are. And that ’ s why I strongly recommend that workplaces create spaces with disability and accessibility in mind. That way you ’ re paving the way for future employees, clients and customers who might benefit from accommodations, too. At one of my first jobs at an investment bank, the company did something cool. Within a month of starting, I and all other new hires got an ergonomic assessment of our workstations. They actually had a person thinking about my access needs in multiple ways, like getting a foot rest and keyboard wrist pad, which a lot of colleagues got, to asking if I might benefit from speech to text technology because I type with one hand. I ’ ve always tried to find these little hacks to make my work environment more comfortable, but the fact that this person offered so many options proactively, it made me feel so welcome without making me feel like my needs were”special.”It would be amazing if every employer could do some version of this. I mean, why not? But there are also more simple things that can make an impact. Like, what about listing accessibility information if you ’ re hosting an event? Or automatically turning on captions for video meetings? Another way to think about this is : What do my employees need in order to thrive? It starts with equipping people with a variety of tools. Third, embrace flexibility, like disabled people have had to do our whole lives. People with disabilities have been advocating for remote work environments and flexible hours for decades now, and it took a pandemic for the world to realize it could happen. All of a sudden, this thing that seemed so wild has become commonplace. And this isn ’ t the first time I can think of when an accommodation for people with disabilities ended up improving life for society at large. Audiobooks, curb cuts, closed captioning, even electric toothbrushes. Disability is so often the root of innovation. So as we get **back** to the office, let ’ s remember that all of **us** can thrive with flexibility. For people with disabilities, let ’ s provide the option to work remotely at the hours when we can be the most productive from wherever works best. I mean, some of **us** do great working from our bed. And really, let ’ s hold on to the openness that we can work around everyone ’ s needs. Let ’ s keep finding ways to embrace flexibility as part of our company ’ s culture. We know that companies need to work harder on hiring **us**, retaining **us**, promoting **us**, paying **us** fairly and amplifying our efforts. To do that, we need to stop making assumptions, rethink accessibility and embrace flexibility. That is what will help close the disability employment gap.

Text Sample 97: NEG Dim 1 – 2023 – Score -1.00 – 2023/t003408.txt
Do you know the secret to successfully pitching an idea? Well, it ’ s something kind of unexpected. It ’ s FOMO : the fear of missing out. [The Way We Work] As someone who invests in companies early in their journeys, I listen to nearly 2, 000 pitches a year, and I work countless hours with company founders to help them make their pitch even more compelling. You may equate pitching with a slick deck based on some standard format, but whether you ’ re pitching a company or trying to get buy-in for a passion project at work, so much of pitching is a storytelling exercise. You need to bring people along with you. You need to tell your audience a story that not only will draw them in, but it will make them feel that if they don ’ t say yes, they will be missing out on something really big. [Step 1 : Know your audience] When people think about pitching, they ’ re thinking inwards, they think about their nerves, how smoothly they ’ re talking or what it takes to get to a yes. But the secret of successful pitching is to flip it outward. Who are you pitching to? What do they care about? How can you speak to what motivates them? For example, if you ’ re pitching your start-up idea to a venture capitalist who has lots of money to invest and needs big returns, they probably care about **backing** the next Uber or DoorDash. So you should focus your story on the size of the market opportunity. If you ’ re pitching to a philanthropist or a nonprofit, their motivation is likely about large social impact. So focus on showing how your product or idea will improve things in a lasting way. If you ’ re pitching a project at work, focus on the people you ’ re pitching the project to. What do they care about? More customer loyalty, more revenue, or perhaps a promotion? Tell them how your project will help them attain the goal that they want in a way that makes it almost inevitable. Give them that feeling of :”If I don ’ t support this, I ’ m going to miss out on something I care about deeply.”[Step 2 : Think about the hero ’ s journey] Pitching is much like telling a story. Just like a movie tells a story. You ’ re charting the hero ’ s journey in three acts. First, you start by telling about the hero ’ s world, the status quo. In this case, the current situation that your product, idea or service will be addressing. Then introduce tension and conflict, showing all the problems that existing products aren ’ t yet addressing. This will lead you to the big confrontation. You, the hero, swooping in to save the day. And from there, give the resolution. How are things changed as a result of your actions? How does your product, idea or service solve the problems you highlighted earlier? One of the best pitches I ’ ve heard followed this arc perfectly. The story started with a disturbing status quo. Depending on where and how it ’ s produced, one gallon of milk can take roughly 1, 000 gallons of water to produce and can create about six kilograms of CO_2 equivalent or more in the process. In this case, the hero was the CEO and their team of scientists and food industry experts who have come up with a way to engineer plants to produce the animal proteins. They showed how a small crop of soybeans could create lots of delicious cheese and how this could feed the global population in a sustainable and yummy way. To anyone who cares about both the environment

and good food, it was an irresistible story. It made me feel like I needed to be part of it or I would be missing out on a big opportunity. [Step 3 : Shore up your weakest point] When people are done charting their hero ’ s journey, I make them identify their biggest weakness. Is it that they ’ re missing somebody on their team or that it ’ s very competitive or that they ’ re trying to do something that has never been done before? Most people ’ s instinct is to gloss over it or even skip it altogether. But that ’ s exactly the wrong thing to do because the audience will notice it and will ask you about it. Instead, face it directly. Tell your audience,”Hey, you may think this is a problem, but here ’ s exactly what I ’ m going to do about it.”And by showing the strengths and weaknesses of your story and not hiding anything, you inspire confidence not only in you but also in your story. The best storytellers, they live in the future, and they come here not to just tell **us** about it, but to show **us** the steps to get there. And this makes the audience lean forward. And all they need to do to be part of this amazing story is say yes.

Text Sample 98: NEG Dim 1 – 2023 – Score -1.00 – 2023/t003386.txt
This week, on Sunday, Alexey Navalny – politician, opposition leader and my father – will have been in prison for 1, 000 days. Almost three years. I miss him every single day. I ’ m scared that my father won ’ t be able to come to my graduation ceremony or walk me down the aisle at my wedding. But if being my father ’ s daughter has taught me anything, it is to never succumb to fear and sadness. And I ’ ve experienced both, fear and sadness. One of the most fearful moments of my short yet eventful life was on August 20, 2020. That day, I woke up in my childhood bedroom in Moscow. Like many of **us** do first thing in the morning, I rolled over to the bedside table and reached for my phone. I looked at the screen and was surprised at how many notifications I had. Twitter, Instagram, Telegram, you name it. My feeling of confusion started to slowly turn into a sense of worry. Alexey Navalny had suddenly fell unconscious on the plane and was taken to the emergency room at the hospital. My heart dropped to my feet. Without blinking an eye, I jumped out of my bed and ran to my parents ’ bedroom to tell my mom that I was able to look after my younger brother, Zahar, while she was away. But by then, she was already at the airport. The plane on which my father had collapsed had to make an emergency landing in a city in Siberia. And after two days, an international uproar and my mom fighting tooth and nail to get him out of there, he was finally transferred to the clinic called Charité in Berlin, Germany. There, at the special laboratory of the German armed forces, they confirmed that he was indeed poisoned by a military-grade nerve agent of the Novichok group. He spent three weeks in a coma. And after months and months of recovery, he went **back** to work with the Anti-Corruption Foundation. Founded over a decade ago, the Anti-Corruption Foundation has been doing incredibly important and dangerous work of spreading information and exposing the corrupt nature of the Russian government. We ’ ve produced over 500 investigations, helped organize hundreds of protests in Moscow and run election campaigns. We provide free legal assistance to those who, like my father, have been imprisoned for peacefully protesting the regime. We ’ ve created a sanctions list of 6, 000 government officials, propagandists, entrepreneurs, various artists and more people who ’ ve supported and worked with Putin over the years. And the list works. Governments all over the world have used our list when imposing sanctions. Because Putin and his associates, they live lavishly. They own condos in Miami and penthouses in Manhattan. They own vineyards in France

and villas on Lake Como. They buy enormous yachts and country houses with gold toilets. They buy private jets to send their corgis to dog competitions. Yes, you heard that right. In 2016, the Anti-Corruption Foundation had released an investigation that Russia ’ s deputy prime minister owned an undeclared jet valued at over 60 million dollars and used it primarily to send his dogs to international competitions. All of this while most citizens in Russia scramble and live on wages of under 200 dollars. According to independent polling done by the ACF sociology center, only 15 percent of the population believe in Putin ’ s propaganda. That means that 85 percent of the population either opposes the regime, doesn ’ t think it really affects them, don ’ t follow politics or simply don ’ t know what to believe. This is why we ask our supporters, who are ordinary citizens, to spread our investigations and news content to their family members, friends and loved ones, because we know that if we can reach those who are on the fence or persuade those who have been brainwashed by Putin, we ’ ll get that much closer to changing the regime. One of my favorite chants my dad used to do during the 2011 to 2014 protests in Moscow was :”One for all and all for one.”This is a phrase that we ’ re all familiar with, but most of **us** often forget. Dictators like Putin want **us** to forget it. They want **us** to forget how strong we can be when we work together. They want **us** to forget how much we can accomplish when we stand side by side. Because it is that much harder for any dictator to fight an international united front than each of **us** individually. Now you might think to yourself,”Dasha, why should I care about this? Geographically, United States is quite far away from Putin. Is he really that threatening to **us**, or are you just trying to raise unnecessary alarm?”And to that, I want to respond with another thought. If we don ’ t stand up for them now, then who will be left to stand up for **us** later? Cyber attacks, spreading of propaganda and disinformation, meddling in the elections, bribing your government officials. This has been and is continuing to happen. The West, Europe and the United States have tried talking to Putin. And where did that get **us**? The illegal annexation of Crimea in 2014 and now a full-scale war in Ukraine. Tyrants and dictators don ’ t want diplomatic relations. All they want is power. I often catch myself thinking how I shouldn ’ t have gone **back** to school that fall. I should have stayed with my family while my dad was recovering. I should have gotten on that plane with him **back** to Russia and hugged him before he got unlawfully and outrageously arrested at the airport. That day when he went **back**, on January 17, 2021, thousands of people showed up at the airport to support him. When I was younger, I didn ’ t really understand how my dad could be so optimistic about the future. He was constantly threatened, arrested, followed. Our family would be followed by men dressed in all black, wearing baseball caps and a ridiculous-looking fanny packs when we would go to the cinema. And after a while, my dad decided to make a game out of it. Whenever we would get on the subway, we waited until the last moment and quickly got out of the car right before the doors closed, so that FSB agents were left behind. A unique childhood experience. I think one of the main reasons for my dad ’ s success is that he has always been incredible at channeling his emotions into hard work and motivation. He wouldn ’ t get sad, and he was certainly never afraid. Instead, he would get fired up. Fired up because he saw a group of ex-KGB corrupt criminals come into power and destroy his country. Seeing that motivated him so much more to work that much harder to stop them. And that is what makes a true patriot. Of course I understand that, first and foremost, the change needs to come from within the country. And that is precisely why, knowing all the risks, my dad went **back** to Russia 996 days ago. Because he simply couldn ’ t not go **back**. My dad is an incredibly courageous and selfless man, and I can only dream to be like him one day. He doesn

’ t run or hide. And unlike my father, Putin is a coward. A coward – A coward who, with his cronies, can no longer hide from the consequences of what they have done to my country and democracy around the world. And you, everyone who ’ s sitting here today or watching this online, you can help. Now. You, unlike millions [and] millions of those who have been threatened and murdered and silenced in Russia, you have a voice. Please don ’ t take your freedom for granted. Russia is my home. It ’ s where I was born. It ’ s where I grew up. It ’ s where I became the person I am today. And I will promise you that I will continue to speak up, tell the truth, and I will continue fighting until Russia is a free and democratic country. Thank you very much, free Alexey Navalny.

Text Sample 99: NEG Dim 1 – 2021 – Score -1.00 – 2021/t003996.txt
Cloe Shasha Brooks : Hello, welcome. You are watching a TED interview series called How to Deal with Difficult Feelings. I ’ m Cloe Shasha Brooks, your host and a curator at TED. And today we ’ ll be focusing specifically on anxiety. So first I ’ ll be speaking with author and model Naomi Shimada about the anxiety associated with social media. She coauthored a book called”Mixed Feelings : Exploring the emotional impact of our digital habits.”It ’ s all about how the internet has created a new layer of perfectionist pressure on our lives and how we can better manage our relationship with our online worlds. Hello, Naomi. Great to see you. Naomi Shimada : Hello, Cloe, great to see you, too. I ’ m honored to be here. CSB : Oh, well, thanks for joining **us**. So, Naomi, you have written and spoken about the relationship between social media and anxiety a whole bunch, such as the anxiety to post online or not to post. So can you tell **us** a little bit more about that? NS : So I always want to start by saying, even though I have written about it, I still don ’ t really feel like an expert because this is just – I always want to decenter my voice as an expert because I ’ m just feeling this out like everyone else. But in my experience, social media and anxiety are connected, you know, or social media exacerbates anxious feelings. It exacerbates the human condition. And so things that we may have insecurities and anxieties around, like, you know, our relationships, our bodies, our work, the things that make up our sense of self, I think the anxiety we feel or we can feel when we use social media can sometimes act as a marker for things that show **us** where we need to do work or where we feel insecure. And sometimes it ’ s just a message being reflected **back** to **us**. And also, like I said, social media exacerbates the human condition. You know, as humans, I think we so often just want to be loved and cared for and seen and adored or just acknowledged. So social media has also become, you know, our main mode of communication, our method of work. Some of those things, those lines can start to become very blurred. CSB : Absolutely. Yeah, and in addition to making lives look shiny and perfect, social media also seems to fuel a lot of FOMO, or fear of missing out. And I ’ m curious what you ’ d suggest for people who experience a lot of anxiety from seeing videos and images of other people having a ton of fun and, you know, not knowing how to deal with that. NS : I think, like I said slightly earlier, the feelings of anxiety when they come up, like, what is that message, you know, taking that step **back** and being, like, why do I feel this way? Why is this making me feel like this? And kind of reading into it. And in my personal experience, the thing that works for me is just taking a step **back**, taking a moment, you know, if something is making me feel bad, for example, if social media – if we thought of it as a substance, for example, if something was making you feel bad, what would you do about it? Would you stop using it? You know, I think there ’ s levels to this because sometimes, you

know, we may have work now that is so intertwined with social media and it can ’ t just be like, oh, stop using it. And I know that there ’ s a spectrum. And I ’ m also navigating this constantly myself when as a public-facing person, my job is so intertwined with social media and it ’ s something I want to do less and less. So I ’ m navigating that kind of boundary for me all the time. So it ’ s just negotiating, sometimes it ’ s not as clear cut, you know, it may for you start as take the weekends off, or you know, I actually personally most of the time don ’ t have social media on my phone. And just when I have to do something for work, that ’ s when I interact with it, especially this year that ’ s been so heavy, you know, and where there is no”off”button and every new day bringing such bad news, like, I ’ m a very sensitive person, so I have to do the things I know that I need to take care of myself, which is not scroll. Also, I ’ ve had an injury in my hand, which means I can ’ t actually scroll, so I ’ m like,”This is a sign! I ’ m just not supposed to be interacting like that right now.”So just listening and knowing that you don ’ t have to fall under the pressure. Like, I think so often we think that if we don ’ t post, we don ’ t exist. Our existence, you know – we only exist when other people see **us** existing. Like, that, that whole line, like,”Oh, if you didn ’ t post about it, it didn ’ t happen.”That concept. We ’ ve started to internalize, you know, especially my generation of millennials, gen-Z, like, if you didn ’ t post it, it didn ’ t happen. And so it ’ s just like going **back** and being like, OK, is that true? Why do I feel the need to share this? And asking those questions. And that ’ s what I do. So like I said, I ’ m not an expert, I too I ’ m working this out and every day feels totally different. But asking those questions is a great place to start. CSB : Thank you for that. So we have a question from the audience. Let ’ s bring that up. OK, so related to this, from Facebook,”What question should we be asking ourselves before we post on social media?”NS : So I like to ask myself, like, why do I want to share this right now? Is this something – as a person that has grown up on the internet, on social media, so often how I validated myself and my sense of self was posting something and people reacting to it. And I think that ’ s just very murky territory. I think like, you know, why do I feel the need to share this? Is this something that feels also private to me? You know, in my opinion, on whether, and I guess, you know, I have not the biggest social media following, but a social media following, that sometimes, when I ’ m like, does that person, for me, does my family member want to be shown online, for example, like, or is this a private moment? I think navigating, like, do I feel not good about myself right now and is posting a picture of myself looking, like, hot, or whatever the equivalent of looking really happy – I think sometimes so often we post about the things that we are yearning for, whether that ’ s attention, love, craving. And I think there ’ s deeper underlying messages behind posting sometimes, you know, and that it is a projection of the things that we want in our lives, for example, posting photos of people you want better relationships with or, you know, there ’ s a big spectrum of experience. But for me, I just try to ask myself, why do I feel the need to make this public right now? Is this something that I am proud of? And it ’ s no critique. This is really questions that are just a gauge where I ’ m at or where someone else is at with it. Like, is this something that actually I just need to pay attention to in my own life privately, of, like, this is something I should be working on or thinking about, or there ’ s just deeper questions about context, I think, that are important. CSB : Yeah, yeah. And I think as we ’ re now at our final question, which is something that I think is related to what you ’ re saying around when to post or not to post, but from a different angle, which is, you know, a lot of people have anxiety about whether or not to post their social justice activism on their accounts and regardless of the activism they might be already doing outside of social media, right? And some people just find it performative. But at the same time, there was a fear of looking apathetic if people are not posting about social justice on social media. So how do you suggest people deal with that anxiety and think about that? NS : I mean, that ’ s definitely an anxiety of our generation, right? Anxieties around posting about social justice. I think the big question here is asking ourselves, like, what am I doing in my own life? You know, and again, there is a spectrum, because there ’ s a lot of people who are sharing a lot of important information via social media. So you have, like, organizers and then everybody else. But if you are – Once again, you know, I can ’ t speak for everybody, but just I think it ’ s – I read this quote by an activist in Oregon, a lifetime organizer called Grace Lee Boggs, and she said that, you know, that a lot of times in our lives we don ’ t prioritize the importance of self-reflection and revolution. And I think, you know, we so care about optics. We don ’ t want people to think that we are racists, sizeist, sexist, etc. But to not create and redo this kind of harm in the world, we need to understand and really reflect on these systems that we ’ ve all internalized to some effect. So to understand, like, where am I on the spectrum? How do I benefit? All of these things actually really take time and deep, you know, self-reflection and work. And that kind of questioning, I think, is something that I find it helpful to be offline because I ’ m like, otherwise, I ’ m just listening to what everybody else is saying. Like, are these my thoughts and my feelings or am I just internalizing what other people are just shouting into the atmosphere and into the internet? I think, there ’ s moments where obviously, a lot the uprisings in June would not have happened if it wasn ’ t for the information that was shared and that action, of course, was so important. But I think there ’ s different phases, you know. And when it ’ s just about shame and optics, that ’ s not how we change the world. For **us** to change the world, we need to inhabit and act on these reflections. So I think there are again, more questions to ask ourselves, like, do I just not want people to think that I do this? And often we are in echo chamber of the people who follow **us** and people we follow, right? So a lot of the times we ’ re just sharing and shouting into the atmosphere of people who have the same ideals as **us**. And that energy can be used in a different way. And also sometimes inhibits, I think, real harder conversations from happening, because I think social media isn ’ t often an intimate enough of a space to be able to ask each other questions that we ’ re afraid to ask. Or mistakes, it ’ s not favorable to making mistakes anymore, which is my critique and sadness about social media. You know, our biggest fear is being called out for something. But this call-out culture, sometimes, not always, I understand its role and place in society, but sometimes doesn ’ t allow for **us** to have more engaged conversations around these systems that we ’ ve internalized. And we all make mistakes and we all have to learn and sometimes it doesn ’ t allow for that to happen. CSB : Yeah, yeah, yeah. Well, I think that ’ s beautifully said and we ’ ve come to the end of our time here. But I am so grateful to you for this conversation, Naomi, and thank you for sharing all this. I ’ ll talk to you soon. Take care. NS : Thank you, Cloe and everyone. Much love.

Text Sample 100: NEG Dim 1 – 2021 – Score -1.00 – 2021/t004053.txt

I ’ ve spent the last couple of years traveling around the world giving talks to big corporations and little bitty start-ups and lots of leadership teams and women ’ s groups, and what I ’ ve been talking to people about, I ’ ve been trying really hard to convince people that we can change the way we work. But every time I do a talk, somebody comes backstage or follows me offstage and says,”You know, I ’ m so inspired by what you say. It ’ s so great, it makes so much sense. But we can ’ t.””We can ’ t because we ’ re regulated.””We can ’ t because our CFO says we can ’ t do it.””We can ’ t because we ’ re in Europe.””We can ’ t because we ’ re a service industry.””We can ’ t because we ’ re a nonprofit.”And then last year came the pandemic. And the pandemic changed everything all over the world. Service people started realizing that they had to suit up and wear masks and take temperatures and wash their hands. We had to start standing six feet apart in lines. We started working from home. We started working virtually. And we started learning all kinds of things because we had to. All that muscle around innovation and flexibility and creativity that we didn ’ t think we had, we had all along. And we now have realized that we can. So what have we learned? I mean, what did we learn right away? First of all, we learned we ’ re not family. The family is the toddler walking around behind you in the Zoom call with the pet. The family is somebody needing their diaper changed. The family is making sure you ’ re taking care of your mom. That ’ s your family. This is your team. And we ’ ve also learned that that separation between family and work has become this balancing act. And that when we used to say,”Well, this is my work home and this is my family home, and those are two completely different things,”for many of **us**, it ’ s exactly the same thing. You ’ re no longer at home and at work. For many of **us**, work is at home and the home is – and it ’ s confusing, and it ’ s creating a whole different level of complexity and coordination so that we understand that it ’ s easier actually to work when we can separate the work that we do as a team from the work that we do in our family. Furthermore, in order to be able to do all that, we have to recognize that we ’ re all adults. And here ’ s the deal about adults. Adults have responsibilities, adults have obligations. Adults have things that they have to commit to. And do you know that every single person that works for you, from the shop floor to the executive suite, is a grown-up? But we have been operating as if they aren ’ t. We operate as if only the smart adults are the people who are at the C Suite. And as we move through the organization, everybody sort of gets a little dumbed down and the rules get a lot stricter and we have to have more control. And the truth is, everybody ’ s a grown-up, we can see it now. Everybody has all of these things to figure out and coordinate. And so now we ’ re expecting from people adult behavior. We ’ re now focusing on the results that matter, not the work. And the way we track it now is we don ’ t walk by and see who ’ s working. We pay attention to what people are doing. And I think that that ’ s always been the best metric. And you know what? For the first time in my life, the concept of best practices is out the window. And you know what? We don ’ t care what Google ’ s doing because we ’ re not Google. We don ’ t care what some other company is doing. Nobody ’ s doing it best. We ’ re all figuring it out as we go along and we ’ re figuring it out for our organizations in our teams at this time. So in order for people to deliver the right results, in order for people ’ s hard work to matter, it has to be in the context of what success looks like for your organization. So if we start to think about context, it ’ s really important that we think about how we teach that. If we can teach everybody in the company how to read a profit and loss statement, if we can teach them what the different teams do, and what they ’ re setting out to accomplish, then people within their own small teams, and within themselves, can figure out what excellence looks like for them. And so then we can start operating relatively independently as a whole organization because we ’ re all moving in the same direction, trying to do the same thing. And there ’ s a really critically important part of making that work, and that ’ s communication. And everything about communication has changed. We tend to think that communication is this waterfall from the top to the bottom. The executives would tell somebody and the next level would tell somebody and we ’ d go all the way down to the shop floor and everybody would understand what ’ s going on. Well, it may not have worked that well then, but it certainly doesn ’ t work that well now. So now we have to recognize it ’ s a different heartbeat. What has it been before and what should it be now? How do we make sure that the messages are clear and consistent? Because that ’ s how people operate. That ’ s how those adults who get the freedom and the responsibility to produce great results operate best is when they understand what they need to know in order to make the best decisions. So that communication, that skill around being a great communicator is something that each of **us** needs to get better at. One of the things we have to do is think about what the right discipline is for that. If you used to communicate to your team by walking by and asking how they ’ re doing or if they had heard something, you ’ re going to have to schedule that now, it ’ s going to have to have discipline. We ’ ve got to check in with the people on the shop floor to make sure they ’ re hearing what they need to hear because it ’ s not going to automatically happen. One of the ideas I have is just jot down at the end of every day a sentence of what worked and what didn ’ t work. And you don ’ t have to look at it for a month. But when you look **back**, over a month, you want to look for,”Wow, that was surprising. I didn ’ t really think that would be as effective as it is.”Or maybe it would be, like,”We keep trying to have this in-person meeting in Zoom, and it turns out that there ’ s 14 people on the call and only two of them are talking. Maybe it ’ s an email.”So we have to rethink all of the ways, not just the work we ’ re doing, but the ways we ’ re doing it. So now I ’ m starting to hear a lot of nostalgia around the way it used to be. There are things we aren ’ t doing now that don ’ t matter. Maybe we don ’ t need to go **back** for five levels of approval. Maybe we don ’ t need to go **back** and do that annual performance review. Maybe we don ’ t need to do a whole bunch of things that were part of the way we do business that just aren ’ t making a difference. You know what? The way we used to do it not only is not the way of the future, but we ’ re discovering so many wonderful things right now. Let ’ s not lose it. We want to create a new organization, new workforce, that ’ s excited about taking all of the things that we ’ ve learned using that muscle, going forward. One of the most important things that we can do is realize the things that we aren ’ t doing now. The stuff that we ’ ve stopped doing and not go **back** and do it again. What if we don ’ t go **back**? What if we go forward and rethink the way we work? Thank you.

Text Sample 101: NEG Dim 1 – 2021 – Score -1.00 – 2021/t003982.txt

[SHAPE YOUR FUTURE] I ’ m going to start by telling you a story about Danielle. When she was a senior in college, Danielle ’ s dad passed away, which left her mom with no way to support herself. So Danielle had to drop out of college and pick up three jobs as a barista, a bartender and a car washer. Altogether, the three jobs paid Danielle 23, 000 dollars per year, which wasn ’ t a whole lot, but it allowed her to feed her mom and keep a roof over their head. And for Danielle, that was enough. But early one morning when Danielle was driving home from one of her jobs, a deer ran in front of her car. She swerved off the road and crashed into a barn. And Danielle doesn ’ t remember exactly what happened next, but when she woke up in a hospital a few hours later, a doctor told her that she had damaged her brain stem and C1 vertebrae. Now, the good news is that Danielle was going to leave the hospital alive. But the bad news is that Danielle had 55, 000 dollars in medical bills. Now, Danielle tried so hard for the next two years to try and pay **back** that debt, but it was impossible. It was impossible for Danielle to pay **back** 55, 000 dollars in medical bills, earning just 23, 000 dollars per year. She felt trapped. One freak accident put Danielle on the verge of homelessness, hunger, poverty. And when you ’ re in Danielle ’ s shoes, bankruptcy is a lifeline. It ’ s a powerful legal tool that allows you to relieve your debt and re-enter the economy. Medical emergency, a job loss, a divorce. These are financial shocks that could happen to any of **us**. And when you ’ re living paycheck to paycheck and don ’ t have a whole lot of savings, like so many Americans, a financial shock can ruin your life. Bankruptcy gives you a second chance. But when Danielle went to go find a bankruptcy lawyer, she, like so many others filing for bankruptcy, learned that it was going to cost her 1, 500 dollars. She didn ’ t have that kind of money. I mean, what a cruel irony. In America, it costs you 1, 500 dollars to tell the court that you have no money. When you walk into a court, everyone from the judge to the clerk to the forms themselves will tell you to go find a lawyer, no matter how little money you have. One of the great civil rights injustices in America is that we don ’ t have equal rights under the law. What we have is equal rights if you can afford a lawyer. Whether you ’ re evicted from your home in an abusive relationship or need access to bankruptcy, you have no right to a free lawyer in most civil cases. And because there aren ’ t even close to enough pro bono or legal aid lawyers around, four out of five low-income Americans can ’ t get the legal help they need to access their civil legal rights. Four years ago, I helped start an organization to fight for new civil right in America, the right to solve your own legal problem when you can ’ t afford a lawyer. We started with bankruptcy. Our nonprofit Upsolve has built an app to help people file for bankruptcy on their own for free. People like Danielle. Our app asks people questions about their finances in language they can understand and then uses this information to help generate their forms. Last year, Danielle used Upsolve to file for bankruptcy on her own for free. She got her final letter from the court, relieving all of her medical debt, right after Christmas Day. Today, Danielle has the highest paying job she ’ s ever had and she ’ s on track to finish her degree. There are so many opportunities to create a more just legal system by empowering people to solve their own legal problems whenever possible. This is especially true in nonadversarial areas of the law, things like no-asset bankruptcies, uncontested divorces and Social Security disability. But there are two main barriers that stand in the way. The first is legal complexity. We ’ ve designed our forms in courts around lawyers, not regular people. Many legal forms are like modern day literacy tests. When you can ’ t understand them, you can ’ t access your rights. Every year, poorly designed forms, courts and processes deny millions of Americans their life, their liberty and their property. Legal complexity is a civil rights injustice. To start solving this problem, we need to require basic user testing in courts and reviser assumption in areas of poverty law that everybody will be able to afford a lawyer. A second barrier is a closed culture. We ’ ve been met with pushback from some folks who believe that you need to go see a lawyer no matter what legal problem you have. Imagine you had to go see a doctor to cure a plain old headache rather than being able to buy Advil at your local pharmacy. Telling a person who is poor to go find a lawyer when they obviously can ’ t afford one is out of touch, iIntimidating, unfair and wrong. It ’ s also a racial injustice. Black and brown communities disproportionately cannot afford the legal fees they need to access their civil legal rights. Many legal fees are like modern day poll taxes. When you can ’ t afford to pay the fees, you can access your rights. And we have a decision to make about how open and equal we want our system of justice to be. The only way we ’ re ever going to have equal rights in America is if we get rid of the modern day literacy tests and poll taxes that dominate our courts and legal system. We need a new civil right in America, the right to solve your own legal problem when you can ’ t afford a lawyer. Because in America, our rights are supposed to be inalienable, our protections are supposed to be equal, and we all deserve a chance at life, liberty and the pursuit of happiness, whether or not we can afford the legal fees. Thank you.

Text Sample 102: NEG Dim 1 – 2019 – Score -1.00 – 2019/t000523.txt
We all know that saving is important and is something that we should be doing. And yet, overall, we ’ re doing less and less of it. [The Way We Work] We know what we need to do. The question is : How do we do it? And that ’ s what I ’ m here to teach you. Your savings behavior isn ’ t a question of how smart you are or how much willpower you have. The amount we save depends on the environmental cues around **us**. Let me give you an example. We ran a study in which, in one group, we showed people their income on a monthly basis. In another group, we showed people their income on a weekly basis. And what we found was that people who saw their income on a weekly basis were able to budget better throughout the month. Now, it ’ s important to know that we didn ’ t change how much money people were receiving, we just changed the environment in which they understood their income. And environmental cues like this have an impact. So I ’ m not going to share tricks with you that you already know. I ’ m not going to tell you how to open up a savings account or how to start saving for your retirement. What I am going to share with you is how to bridge this gap from your intentions to save and your actions. Are you ready? Here ’ s number one : harness the power of pre-commitment. Fundamentally, we think about ourselves in two different ways : our present self and our future self. In the future, we ’ re perfect. In the future, we ’ re going to save for retirement, we ’ re going to lose weight, we ’ re going to call our parents more. But we oftentimes forget that our future self is exactly the same person as our present self. We know that one of the best times to save is when you get your tax return. So we tried an A / B test. In the first group, we texted people in early February, hopefully before they even filed for their taxes. And we asked them,”If you get a tax refund, what percentage would you like to save?”Now this is a really hard question. They didn ’ t know if they would receive a tax refund or how much. But we asked the question anyway. In the second group, we asked people right after they received their refund,”What percentage would you like to save?”Now, here ’ s what happened. In that second condition, when people just received their tax refund, they wanted to save about 17 percent of their tax refund. But in the condition when we asked people before they even filed their taxes, savings rates increased from 17 percent to 27 percent when we asked in February. Why? Because you ’ re committing for your future self, and of course your future self can save 27 percent. These large changes in savings behavior came from the fact that we changed the decision-making environment. We want you to be able to harness that same power. So take a moment and think about the ways in which you can sign up your future self for something that you know today will be a little bit hard. Sign up for an app that lets you make savings decisions in advance. The trick is, you have to have that binding contract. Number two : use transition moments to your advantage. We did an experiment with a website that helps older adults share their housing. We ran two ads on social media, targeted to the same population of 64-year-olds. In one group, we said,”Hey, you ’ re getting older. Are you ready for retirement? House sharing can help.”In the second group, we got a little bit more specific and said,”You ’ re 64 turning 65. Are you ready for

retirement? House sharing can help.”What we ’ re doing in that second group is highlighting that a transition is happening. All of a sudden, we saw click-through rates, and ultimately sign-up rates, increase when we highlight that. In psychology, we call this the”fresh start effect.”Whether it ’ s the start of a new year or even a new season, your motivation to act increases. So right now, put a meeting request on your calendar for the day before your next birthday. Identify the one financial thing you most want to do. And commit yourself to it. The third and final trick : get a handle on small, frequent purchases. We ’ ve run a few different studies and found that the number one purchase people say they regret, after bank fees, is eating out. It ’ s a frequent purchase we make almost every day, and it ’ s death by a thousand cuts. A coffee here, a burrito there... It adds up and decreases our ability to save. **Back** when I lived in New York City, I looked at my expenses and saw that I spent over 2, 000 dollars on ride-sharing apps. It was more than my New York City rent. I vowed to make a change. And the next month, I spent 2, 000 dollars again — no change, because the information alone didn ’ t change my behavior. I didn ’ t change my environment. So now that I was 4, 000 dollars in the hole, I did two things. The first is that I unlinked my credit card from my car-sharing apps. Instead, I linked a debit card that only had 300 dollars a month. If I needed more, I had to go through the whole process of adding a new card, and we know that every click, every barrier, changes our behavior. We aren ’ t machines. We don ’ t carry around an abacus every day, adding up what we ’ re spending, in comparison to what we wanted. But what our brains are very good at is counting up the number of times we ’ ve done something. So I gave myself a limit. I can only use ride-sharing apps three times a week. It forced me to ration my travels. I got a handle on my car-sharing expenses to the benefit of my husband, because of the environmental changes that I did. So get a handle on whatever that purchase is for you, and change your environment to make it harder to do so. Those are my tips for you. But I want you to remember one thing. As human beings, we can be irrational when it comes to saving and spending and budgeting. But luckily, we know this about ourselves, and we can predict how we ’ ll act under certain environments. Let ’ s do that with saving. Let ’ s change our environment to help our future selves.

Text Sample 103: NEG Dim 1 – 2019 – Score -1.00 – 2019/t000920.txt

Hello. I ’ d like to introduce you to someone. This is Jomny. That ’ s”Jonny”but spelled accidentally with an”m,”in case you were wondering, because we ’ re not all perfect. Jomny is an alien who has been sent to earth with a mission to study humans. Jomny is feeling lost and alone and far from home, and I think we ’ ve all felt this way. Or, at least I have. I wrote this story about this alien at a moment in my life when I was feeling particularly alien. I had just moved to Cambridge and started my doctoral program at MIT, and I was feeling intimidated and isolated and very much like I didn ’ t belong. But I had a lifeline of sorts. See, I was writing jokes for years and years and sharing them on social media, and I found that I was turning to doing this more and more. Now, for many people, the internet can feel like a lonely place. It can feel like this, a big, endless, expansive void where you can constantly call out to it but no one ’ s ever listening. But I actually found a comfort in speaking out to the void. I found, in sharing my feelings with the void, eventually the void started to speak **back**. And it turns out that the void isn ’ t this endless lonely expanse at all, but instead it ’ s full of all sorts of other people, also staring out into it and also wanting to be heard. Now, there have been many bad things that have come from social media. I ’ m not trying to dispute that at all. To be online at any given point is to feel so much sadness and anger and violence. It can feel like the end of the world. Yet, at the same time, I ’ m conflicted because I can ’ t deny the fact that so many of my closest friends are people that I had met originally online. And I think that ’ s partly because there ’ s this confessional nature to social media. It can feel like you are writing in this personal, intimate diary that ’ s completely private, yet at the same time you want everyone in the world to read it. And I think part of that, the joy of that is that we get to experience things from perspectives from people who are completely different from ourselves, and sometimes that ’ s a nice thing. For example, when I first joined Twitter, I found that so many of the people that I was following were talking about mental health and going to therapy in ways that had none of the stigma that they often do when we talk about these issues in person. Through them, the conversation around mental health was normalized, and they helped me realize that going to therapy was something that would help me as well. Now, for many people, it sounds like a scary idea to be talking about all these topics so publicly and so openly on the internet. I feel like a lot of people think that it is a big, scary thing to be online if you ’ re not already perfectly and fully formed. But I think the internet can be actually a great place to not know, and I think we can treat that with excitement, because to me there ’ s something important about sharing your imperfections and your insecurities and your vulnerabilities with other people. Now, when someone shares that they feel sad or afraid or alone, for example, it actually makes me feel less alone, not by getting rid of any of my loneliness but by showing me that I am not alone in feeling lonely. And as a writer and as an artist, I care very much about making this comfort of being vulnerable a communal thing, something that we can share with each other. I ’ m excited about externalizing the internal, about taking those invisible personal feelings that I don ’ t have words for, holding them to the light, putting words to them, and then sharing them with other people in the hopes that it might help them find words to find their feelings as well. Now, I know that sounds like a big thing, but ultimately I ’ m interested in putting all these things into small, approachable packages, because when we can hide them into these smaller pieces, I think they are easier to approach, I think they ’ re more fun. I think they can more easily help **us** see our shared humanness. Sometimes that takes the form of a short story, sometimes that takes the form of a cute book of illustrations, for example. And sometimes that takes the form of a silly joke that I ’ ll throw on the internet. For example, a few months ago, I posted this app idea for a dog-walking service where a dog shows up at your door and you have to get out of the house and go for a walk. If there are app developers in the audience, please find me after the talk. Or, I like to share every time I feel anxious about sending an email. When I sign my emails”Best,”it ’ s short for”I am trying my best,”which is short for”Please don ’ t hate me, I promise I ’ m trying my best!”Or my answer to the classic icebreaker, if I could have dinner with anyone, dead or alive, I would. I am very lonely. And I find that when I post things like these online, the reaction is very similar. People come together to share a laugh, to share in that feeling, and then to disburse just as quickly. Yes, leaving me once again alone. But I think sometimes these little gatherings can be quite meaningful. For example, when I graduated from architecture school and I moved to Cambridge, I posted this question :”How many people in your life have you already had your last conversation with?”And I was thinking about my own friends who had moved away to different cities and different countries, even, and how hard it would be for me to keep in touch with them. But other people started replying and sharing their own experiences. Somebody talked about a family member they had a falling out with. Someone talked about a loved one who had passed away quickly and unexpectedly. Someone else talked about their friends from school who had moved away as well. But then something really nice started happening. Instead of just replying to me, people started replying to each other, and they started to talk to each other and share their own experiences and comfort each other and encourage each other to reach out to that friend that they hadn ’ t spoken to in a while or that family member that they had a falling out with. And eventually, we got this little tiny microcommunity. It felt like this support group formed of all sorts of people coming together. And I think every time we post online, every time we do this, there ’ s a chance that these little microcommunities can form. There ’ s a chance that all sorts of different creatures can come together and be drawn together. And sometimes, through the muck of the internet, you get to find a kindred spirit. Sometimes that ’ s in the reading the replies and the comments sections and finding a reply that is particularly kind or insightful or funny. Sometimes that ’ s in going to follow someone and seeing that they already follow you **back**. And sometimes that ’ s in looking at someone that you know in real life and seeing the things that you write and the things that they write and realizing that you share so many of the same interests as they do, and that brings them closer together to you. Sometimes, if you ’ re lucky, you get to meet another alien. [when two aliebns find each other in a strange place, it feels a litle more like home] But I am worried, too, because as we all know, the internet for the most part doesn ’ t feel like this. We all know that for the most part, the internet feels like a place where we misunderstand each other, where we come into conflict with each other, where there ’ s all sorts of confusion and screaming and yelling and shouting, and it feels like there ’ s too much of everything. It feels like chaos, and I don ’ t know how to square away the bad parts with the good, because as we know and as we ’ ve seen, the bad parts can really, really hurt **us**. It feels to me that the platforms that we use to inhabit these online spaces have been designed either ignorantly or willfully to allow for harassment and abuse, to propagate misinformation, to enable hatred and hate speech and the violence that comes from it, and it feels like none of our current platforms are doing enough to address and to fix that. But still, and maybe probably unfortunately, I ’ m still drawn to these online spaces, as many others are, because sometimes it just feels like that ’ s where all the people are. And I feel silly and stupid sometimes for valuing these small moments of human connection in times like these. But I ’ ve always operated under this idea that these little moments of humanness are not superfluous. They ’ re not retreats from the world at all, but instead they ’ re the reasons why we come to these spaces. They are important and vital and they affirm and they give **us** life. And they are these tiny, temporary sanctuaries that show **us** that we are not as alone as we think we are. And so yes, even though life is bad and everyone ’ s sad and one day we ’ re all going to die — [look. life is bad. everyones sad. We ’ re all gona die, but i alredy bought this inflatable bouncey castle so are u gona take Ur shoes off or not] I think the inflatable metaphorical bouncy castle in this case is really our relationships and our connections to other people. And so one night, when I was feeling particularly sad and hopeless about the world, I shouted out to the void, to the lonely darkness. I said,”At this point, logging on to social media feels like holding someone ’ s hand at the end of the world.”And this time, instead of the void responding, it was people who showed up, who started replying to me and then who started talking to each other, and slowly this little tiny community formed. Everybody came together to hold hands. And in these dangerous and unsure times, in the midst of it all, I think the thing that we have to hold on to is other people. And I know that is a small thing made up of small moments, but

I think it is one tiny, tiny sliver of light in all the darkness. Thank you. Thank you.

Text Sample 104: NEG Dim 1 – 2019 – Score 1.00 – 2019/t000046.txt

I want to start by telling you two things about myself before I get into the full talk. And the first is that I ’ ve been writing about manners and civility for more than 20 years, as a book author and as a magazine columnist. The second is, my friends know to be very wary of inviting me over for dinner because any faux pas that happens at the table is likely to wind up in print. So, I ’ m watching, I can see **back** there and I can see through the portals, too. So, speaking of dinner parties, I want to take you **back** to 2015 and a dinner party that I went to. To place this in time, this was when Caitlyn Jenner was first coming out, shedding her identity as a Kardashian and moving into her life as a transgender activist. I wrote a column in People magazine at the time, talking about the importance of names and how names are our identity. And that to misuse them or not to use them erases **us** in a certain way. And especially with Caitlyn Jenner, I talked about Caitlyn, but also the use of her pronouns. Her pronouns. So I ’ m at this dinner — delicious, wonderful, fun — when my host goes on a rant about Caitlyn Jenner. And she is saying that it is disrespectful for Caitlyn Jenner to force her to use a new name and to use these new pronouns. She ’ s not buying it, and I ’ m listening, and because I do meditation, I took my sacred pause before I responded. And I reminded her that when she got married, she changed her name, and that she took the name of her husband. And that ’ s the name all of **us** now use. We don ’ t use it just because it ’ s her legal name, but we use it because it ’ s respectful. Ditto for Miss Jenner. She didn ’ t buy it and we didn ’ t speak for years. So... I am known as the Civilist. And it ’ s probably a word that you ’ re not that familiar with. It ’ s not in common parlance and it comes from the Latin and the French, and it means an individual who tries to live by a moral code, who is striving to be a good citizen. The word”civility”is derived from that, and the original definition of civility is citizens willing to give of themselves for the good of the city, for the good of the commonwealth, for the larger good. So, in this talk, you ’ re going to learn three new ways to be civil, I hope, and it will be according to the original definition of civility. My first problem is : civility is an obsolete word. My second problem is : civility has become a dirty word in this country. And that is whether you lean right or whether you lean left. And in part, that ’ s because modern usage equates civility with decorum, with formal politeness, formal behavior. We ’ ve gotten away from the idea of citizenship. So, let me start by talking a little bit about my friends on the right, who have conflated civility with what they call political correctness. And to them, callouts for civility are really very much like what George Orwell wrote in”1984”— he called it”newspeak.”And this was an attempt to change the way we talk by forcibly changing the language that we use. To change our ideas by changing the meaning of words. And I think my dinner host might have had some of that rattling around there. And I first personally understood, though, the right ’ s problem with civility when I wrote a column about then-candidate Donald Trump. And he had just said he did not have time for total political correctness, and he did not believe the country did either. And I took that to heart, it was very — The audience was very engaged about that online, as you can imagine. There was a thousand responses, and this one stood out to me because it was representative :”Political correctness is a pathological system that lets liberals dominate a conversation, label, demonize and shout down the opposition.”So I think, to the right, civility translates into censure. So that ' s the right. Now, my friends on the left also have a problem with it. And for example, there have been those who have harassed Trump administration officials who support the President ' s border wall. They ' ve been called out as rude, they ' ve been called out as nasty, they ' ve been called out as worse. And after one such incident last year, even the Washington Post — you know, left- leaning Washington Post — wrote an editorial and sided with decorum. And they argued that officials should be allowed to dine in peace. Hm."You know, the wall is the real incivility here. The tear-gassing of kids, the separation of families."That ' s what the protestors say. And imagine if we had sided, in this country, with decorum and courtesy throughout our history. You know, I think about the suffragettes. They marched, they picketed. They were chastised, they were arrested for pursuing the vote for women in the 1920s. You know, I also think about the Reverend Martin Luther King Jr., the father of American nonviolent civil disobedience. He was labeled as uncivil in his attempt to promote racial and economic justice. So I think you get a sense of why civility has become a problem, a dirty word, here. Now, does this mean we can ' t disagree, that we can ' t speak our minds? Absolutely not. I recently spoke with Dr. Carolyn Lukensmeyer. She ' s kind of the guru of civility in this country, and the executive director of a body called the National Institute for Civil Discourse. And she told me,"Civility does not mean appeasement or avoiding important differences. It means listening and talking about those differences with respect."In a healthy democracy, we need to do that. And I call that respectful engagement. But civil discourse also needs rules, it needs boundaries. For instance, there ' s a difference between language that is simply rude or demeaning, and speech that invokes hatred and intolerance. And specifically of groups. And I ' m thinking of racial and ethnic groups, I ' m thinking of the LGBTQ community, I ' m thinking of the disabled. We snowflakes call this speech"hate speech."And hate speech can lead to violence. So, to that point, in the fall of 2018, I wrote a column about Dr. Christine Blasey Ford. You may remember her, she was one of the women who accused Supreme Court nominee Brett Kavanaugh of sexual assault. And among the responses, I received this message, a personal message, which you can see here on the slide. It ' s been largely redacted. This message was 50 words long. 10 of them were the f-bomb. And the Democrats were called out, President Obama was called out, and I was referred to in a pretty darn vulgar and coarse way. There was an explicit threat in that message, and that is why my editors at The Post sent it to authorities. This came shortly before the pipe bombs were sent to other media outlets, so everybody was really kind of on guard there. And the larger context was, only a few months before, five staffers had been killed at a Maryland newspaper. They had been shot dead by a reader with a grudge."Shut up or else."And it was around that same time that a different reader of mine started stalking me online. And at first, it was... I ' ll call it light and fluffy. It was around this time last year and I still had my Christmas decorations up and he sent me a message saying,"You should take your Christmas decorations down."And then he noticed that my dog was off leash one day, and then he commented that I had gone to the market. And then he wrote me one that said,"If anyone were to shoot and kill you, it would not be a loss at all."I wish that were the end of the story. Because then, a few months later, he came to my door, my front door, in a rage and tried to break the door down. I now own mace, a security system and a Louisville Slugger baseball bat."Shut up or else."So, what ' s to be done to forestall civility from turning ugly, from turning violent? My first rule is to deescalate language. And I ' ve stopped using trigger words in print. And by trigger words, I mean"homophobe,"I mean"racist,"I mean"xenophobe,"I mean"sexist."All of those words. They set people off. They

’ re incendiary and they do not allow **us** to find common ground. They do not allow **us** to find a common heart. And so to this point, when John McCain died in 2018, his supporters noted that he never made personal attacks. But his opponents agreed as well, and I though that was what was really noteworthy. He challenged people ’ s policies, he challenged their positions, but he never made it personal. And so that ’ s the second rule. So the problem of civility is not only an American one. In the Netherlands, there are calls for a civility offensive right now, and as one Dutch philosopher has put it, the country has fallen under a spell of”verhuftering.”Now, this is not a word that I knew before and I did quite a bit of research. It loosely means bullying and the disappearance of good manners. It actually means much worse than that, but that ’ s what I ’ m saying here. When you have a specific word, though, to describe a problem like that, you know you really have a problem. And in the United Kingdom, the [2016] Brexit vote... you know, has divided a nation even more so. And one critic of the breakup called those who favor it — I just love this phrase —”the frightened parochial lizard brain of Britain.”The frightened parochial lizard brain of Britain. That ’ s personal. And it makes me miss”Downton Abbey”and its patina of civility. But therein lies the third rule : don ’ t mistake decorum for civility. Even if you have a dowager countess as fabulous as Dame Maggie Smith. [Don ’ t be defeatist. It ’ s so middle class.] So let me end with one last story. Not that long ago, I was at a bakery, and they make these amazing scones. So, long line — there are a lot of scones. And one by one, the scones were disappearing until there was one woman in between me and that last scone. Praise the Lord, she said,”I ’ ll have a croissant.”So when it became my turn, I said,”I ’ ll take that scone.”The guy behind me — I ’ d never turned around, never seen him — he shouted,”That ’ s my scone! I ’ ve been waiting in line 20 minutes.”And I was like,”Who are you? I ’ ve been waiting in line 20 minutes, and you ’ re behind me.”So, I grew up here in New York, and went to high school not that far from here. And I may seem, you know, very civil here and so on, but I can hip check anybody for a taxicab in this room, on these streets. So I was surprised when I said to this guy...”Would you like half?””Would you like half?”I didn ’ t think about it, it just came out. And then, he was very puzzled, and I could see his face change and he said to me,”Well, how about if I buy another pastry and we ’ ll share both of them?”And he did, and we did. And we sat and talked. We had nothing in common. We had nothing in common : nationality, sexual orientation, occupation. But through this moment of kindness, through this moment of connection, we developed a friendship, we have stayed in touch. Although he was appalled to learn that I ’ m called the Civilist after that. But I call this the joy of civility. The joy of civility. And it led me to wonder, what is the good we forgo, not just the trouble we avoid, when we choose to be uncivil. And by good, I mean friendship, I mean connection. I mean sharing 1000 calories. But I also mean it in a larger way. You know, as communities and as a country and as a world. What are we missing out on? So, today, we are engaged in a great civil war of ideas and identity. And we have no rules for them. You know, there are rules for war. Think about the Geneva Conventions. They ensure that every soldier is treated humanely, on and off the battlefield. So, frankly, I think we need a Geneva Convention of civility, to set the rules for discourse for the parameters of that. To help **us** become better citizens of our communities and of our countries. And if I have anything to say about it, I would base those rules on the original definition of civility, from the Latin and from the French. Civility : citizens willing to give of themselves for the greater good. For the good of the city. So I think civility, with that understanding, is not a dirty word. And I hope the civilist will not become, or will not stay, obsolete. Thank you.

Text Sample 105: NEG Dim 1 – 2017 – Score -1.00 – 2017/t000962.txt

After decades of research and billions of dollars spent in clinical trials, we still have a problem with cancer drug delivery. We still give patients chemotherapy, which is so non-specific that even though it kills the cancer cells, it kind of kills the rest of your body, too. And yes, we have developed more selective drugs, but it ’ s still a challenge to get them into the tumor, and they end up accumulating in the other organs as well or passing through your urine, which is a total waste. And fields like mine have emerged where we try to encapsulate these drugs to protect them as they travel through the body. But these modifications cause problems that we make more modifications to fix. So what I ’ m really trying to say is we need a better drug delivery system. And I propose, rather than using solely human design, why not use nature ’ s? Immune cells are these versatile vehicles that travel throughout our body, patrolling for signs of disease and arriving at a wound mere minutes after injury. So I ask you guys : If immune cells are already traveling to places of injury or disease in our bodies, why not add an extra passenger? Why not use immune cells to deliver drugs to cure some of our biggest problems in disease? I am a biomedical engineer, and I want to tell you guys a story about how I use immune cells to target one of the largest problems in cancer. Did you know that over 90 percent of cancer deaths can be attributed to its spread? So if we can stop these cancer cells from going from the primary tumor to a distant site, we can stop cancer right in its tracks and give people more of their lives **back**. To do this special mission, we decided to deliver a nanoparticle made of lipids, which are the same materials that compose your cell membrane. And we ’ ve added two special molecules. One is called e-selectin, which acts as a glue that binds the nanoparticle to the immune cell. And the second one is called trail. Trail is a therapeutic drug that kills cancer cells but not normal cells. Now, when you put both of these together, you have a mean killing machine on wheels. To test this, we ran an experiment in a mouse. So what we did was we injected the nanoparticles, and they bound almost immediately to the immune cells in the bloodstream. And then we injected the cancer cells to mimic a process through which cancer cells spread throughout our bodies. And we found something very exciting. We found that in our treated group, over 75 percent of the cancer cells we initially injected were dead or dying, in comparison to only around 25 percent. So just imagine : these fewer amount of cells were available to actually be able to spread to a different part of the body. And this is only after two hours of treatment. Our results were amazing, and we had some pretty interesting press. My favorite title was actually,”Sticky balls may stop the spread of cancer.”I can ’ t tell you just how smug my male colleagues were, knowing that their sticky balls might one day cure cancer. But I can tell you they made some pretty, pretty, exciting, pretty ballsy t- shirts. This was also my first experience talking to patients where they asked how soon our therapy would be available. And I keep these stories with me to remind me of the importance of the science, the scientists and the patients. Now, our fast-acting results were pretty interesting, but we still had one lingering question : Can our sticky balls, our particles actually attached to the immune cells, actually stop the spread of cancer? So we went to our animal model, and we found three important parts. Our primary tumors were smaller in our treated animals, there were fewer cells in circulation, and there was little to no tumor burden in the distant organs. Now, this wasn ’ t just a victory for **us** and our sticky balls. This was also a victory to me in drug delivery, and it represents a paradigm shift, a revolution — to go from just using drugs, just injecting them and hoping they go to the right places in the body, to using immune cells as special delivery drivers in your body. For this example, we used two molecules, e-selectin and trail, but really, the possibility of drugs you can use are endless. And I talked about cancer, but where disease goes, so do immune cells. So this could be used for any disease. Imagine using immune cells to deliver crucial wound-healing agents after a spinal cord injury, or using immune cells to deliver drugs past the blood-brain barrier to treat Parkinson ' s or Alzheimer ' s disease. These are the ideas that excite me about science the most. And from where I stand, I see so much promise and opportunity. Thank you.

Text Sample 106: NEG Dim 1 – 2017 – Score -1.00 – 2017/t001027.txt

So, I ' m afraid. Right now, on this stage, I feel fear. In my life, I ain ' t met many people that will readily admit when they are afraid. And I think that ' s because deep down, they know how easy it spreads. See, fear is like a disease. When it moves, it moves like wildfire. But what happens when, even in the face of that fear, you do what you ' ve got to do? That ' s called courage. And just like fear, courage is contagious. See, I ' m from East St. Louis, Illinois. That ' s a small city across the Mississippi River from St. Louis, Missouri. I have lived in and around St. Louis my entire life. When Michael Brown, Jr., an ordinary teenager, was gunned down by police in 2014 in Ferguson, Missouri — another suburb, but north of St. Louis — I remember thinking, he ain ' t the first, and he won ' t be the last young kid to lose his life to law enforcement. But see, his death was different. When Mike was killed, I remember the powers that be trying to use fear as a weapon. The police response to a community in mourning was to use force to impose fear : fear of militarized police, imprisonment, fines. The media even tried to make **us** afraid of each other by the way they spun the story. And all of these things have worked in the past. But like I said, this time it was different. Michael Brown ' s death and the subsequent treatment of the community led to a string of protests in and around Ferguson and St. Louis. When I got out to those protests about the fourth or fifth day, it was not out of courage ; it was out of guilt. See, I ' m black. I don ' t know if y ' all noticed that. But I couldn ' t sit in St. Louis, minutes away from Ferguson, and not go see. So I got off my ass to go check it out. When I got out there, I found something surprising. I found anger ; there was a lot of that. But what I found more of was love. People with love for themselves. Love for their community. And it was beautiful — until the police showed up. Then a new emotion was interjected into the conversation : fear. Now, I ' m not going to lie ; when I saw those armored vehicles, and all that gear and all those guns and all those police I was terrified — personally. And when I looked around that crowd, I saw a lot of people that had the same thing going on. But I also saw people with something else inside of them. That was courage. See, those people yelled, and they screamed, and they were not about to **back** down from the police. They were past that point. And then I could feel something in me changing, so I yelled and I screamed, and I noticed that everybody around me was doing the same thing. And there was nothing like that feeling. So I decided I wanted to do something more. I went home, I thought : I ' m an artist. I make shit. So I started making things specific to the protest, things that would be weapons in a spiritual war, things that would give people voice and things that would fortify them for the road ahead. I did a project where I took pictures of the hands of protesters and put them up and down the boarded-up buildings and community shops. My goal was to raise awareness and to raise the morale. And I think, for a minute at least, it did just that. Then I thought, I want to uplift the stories of these people I was watching being courageous in the moment. And myself and my friend, and filmmaker and partner Sabaah Folayan did just that with our documentary,"Whose Streets?"I kind of became a conduit for all of this courage that was given to me. And I think that ' s part of our job as artists. I think we should be conveyors of courage in the work that we do. And I think that we are the wall between the normal folks and the people that use their power to spread fear and hate, especially in times like these. So I ' m going to ask you. Y ' all the movers and the shakers, you know, the thought leaders : What are you gonna do with the gifts that you ' ve been given to break **us** from the fear the binds **us** every day? Because, see, I ' m afraid every day. I can ' t remember a time when I wasn ' t. But once I figured out that fear was not put in me to cripple me, it was there to protect me, and once I figured out how to use that fear, I found my power. Thank you.

Text Sample 107: NEG Dim 1 – 2017 – Score -1.00 – 2017/t000816.txt

A lot of people call me a"justice architect."But I don ' t design prisons. I don ' t design jails. I don ' t design **detention** centers, and I don ' t even design courthouses. All the same, I get a call every week, saying,"OK, but you design better prisons, right? You know, like those pretty ones they ' re building in Europe."And I always pause. And I invite them, and I invite you today, to imagine a world without prisons. What does that justice feel and look like? What do we need to build to get there? I ' d like to show you some ideas today of things that we ' re building. And I ' m going to start with an early prototype. This I built when I was five. I call it"the healing hut."And I built it after I got sent home from school for punching this kid in the face because he called me the N-word. OK, he deserved it. It happened a lot, though, because my family had desegregated a white community in rural Virginia. And I was really scared. I was afraid. I was angry. And so I would run into the forest, and I would build these little huts. They were made out of twigs and leaves and blankets I had taken from my mom. And as the light would stream into my refuge, I would feel at peace. Despite my efforts to comfort myself, I still left my community as soon as I could, and I went to architecture school and then into a professional career designing shopping centers, homes for the wealthy and office buildings, until I stepped into a prison for the first time. It was the Chester State Correctional Institution in Pennsylvania. And my friend, she invited me there to work with some of her incarcerated students and teach them about the positive power of design. The irony is so obvious, right? As I approached this concrete building, these tiny little windows, barbed wire, high walls, observation towers, and on the inside, these cold, hard spaces, little light or air, the guards are screaming, the doors are clanking, there ' s a wall of cells filled with so many black and brown bodies. And I realized that what I was seeing was the end result of our racist policies that had caused mass incarceration. But as an architect, what I was seeing was how a prison is the worst building type we could have created to address the harm that we ' re doing to one another. I thought,"Well, could I design an alternative to this, other than building a prettier prison?"It didn ' t feel good to me ; it still doesn ' t feel good. But **back** then, I just didn ' t know what to do. What do we build instead of this? And then I heard about restorative justice. I felt at peace again, because here was an alternative system that says when a crime is committed, it is a breach of relationship, that the needs of those who have been harmed must be addressed first ; that those who have committed the offense have an obligation to make amends. And what they are are really intense dialogues, where all stakeholders come together to find a way to repair the breach. Early data shows that restorative justice builds empathy ; that it reduces violent reoffending by up to 75 percent ; that it eases PTSD in survivors of the most severe violence. And because of these reasons, we see prosecutors and judges and district attorneys starting to divert cases out of court and into restorative justice so that some people never touch the system altogether. And so I thought,"Well, damn — why aren ' t we designing for this system?"Instead of building prisons, we should be building spaces to amplify restorative justice. And so I started in schools, because suspensions and expulsions have been fueling the pathway to prison for decades. And many school districts — probably some of your own — are turning to restorative justice as an alternative. So, my first project — I just turned this dirty little storage room into a peacemaking room for a program in a high school in my hometown of Oakland. And after we were done, the director said that the circles she was holding in this space were more powerful in bringing the community together after fighting at school and gun violence in the community, and that students and teachers started to come here just because they saw it as a space of refuge. So what was happening is that the space was amplifying the effects of the process. OK, then I did something that architects always do, y ' all. I was like, I ' m going to build something massive now, right? I ' m going to build the world ' s first restorative justice center all by myself. And it ' s going to be a beautiful figure on the skyline, like a beacon in the night. Thousands of people will come here instead of going to court. I will single-handedly end mass incarceration and win lots of design awards. And then I checked myself — because here ' s the deal : we are incarcerating more of our citizens per capita than any country in the world. And the fastest-growing population there are black women. Ninety-five percent of all these folks are coming home. And most of them are survivors of severe sexual, physical and emotional abuse. They have literally been on both sides of the harm. So I thought, uh, maybe I should ask them what we should build instead of prisons. So I returned with a restorative justice expert, and we started to run the country ' s first design studios with incarcerated men and women around the intersection of restorative justice and design. And it was transformative for me. I saw all these people behind walls in a totally different way. These were souls deeply committed to their personal transformation and being accountable. They were creative, they were visionary. Danny is one of those souls. He ' s been incarcerated at San Quentin for 27 years for taking a life at the age of 21. From the very beginning, he ' s been focused on being accountable for that act and doing his best to make amends from behind bars. He brought that work into a design for a community center for reconciliation and wellness. It was a beautiful design, right? So it ' s this green campus filled with these circular structures for victim and offender dialogue. And when he presented the project to me, he started crying. He said,"After being in the brutality of San Quentin for so long, we don ' t think reconciliation will happen. This design is for a place that fulfills the promise of restorative justice. And it feels closer now."I know for a fact that just the visualization of spaces for restorative justice and healing are transformative. I ' ve seen it in our workshops over and over again. But I think we know that just visualizing these spaces is not enough. We have to build them. And so I started to look for justice innovators. They are not easy to find. But I found one. I found the Center for Court Innovation. They were bringing Native American peacemaking practices into a non-Native community for the very first time in the United States. And I approached them, and I said,"OK, well, as you set up your process, could I work with the community to design a peacemaking center?"And they said yes. Thank God, because I had no backup to these guys. And so, in the Near Westside of Syracuse, New York, we started to run design workshops with the community to both locate and reenvision an old drug house to be a peacemaking center. The Near Westside Peacemaking Project is complete. And they are already running over 80 circles a year, with a very interesting outcome, and that it is the space itself that ’ s convincing people to engage in peacemaking for the very first time in their lives. Isabel and her daughter are some of those community members. And they had been referred to peacemaking to heal their relationship after a history of family abuse, sexual abuse and other issues that they ’ d been having in their own family and the community. And, you know, Isabel didn ’ t want to do peacemaking. She was like,”This is just like going to court. What is this peacemaking stuff?”But when she showed up, she was stressed, she was anxious. But when she got in, she kind of looked around, and she settled in. And she turned to the coordinator and said,”I feel comfortable here — at ease. It ’ s homey.”Isabel and her daughter made a decision that day to engage and complete the peacemaking process. And today, their relationship is transformed ; they ’ re doing really well and they ’ re healing. So after this project, I didn ’ t go into a thing where I ’ m going to make a huge peacemaking center. I did want to have peacemaking centers in every community. But then a new idea emerged. I was doing a workshop in Santa Rita Jail in California, and one of our incarcerated designers, Doug, said,”Yeah, you know, repairing the harm, getting **back** on my feet, healing — really important. But the reality is, Deanna, when I get home, I don ’ t have anywhere to go. I have no job — who ’ s going to hire me? I ’ m just going to end up **back** here.”And you know what, he ’ s right, because 60 to 75 percent of those returning to their communities will be unemployed a year after their release. We also know, if you can ’ t meet your basic economic needs, you ’ re going to commit crime — any of **us** would do that. So instead of building prisons, what we could build are spaces for job training and entrepreneurship. These are spaces for what we call”restorative economics.”Located in East Oakland, California,”Restore Oakland”will be the country—s first center for restorative justice and restorative economics. So here ’ s what we ’ re going to do. We ’ re going to gut this building and turn it into three things. First, a restaurant called”Colors,”that will break the racial divide in the restaurant industry by training low-wage restaurant workers to get living-wage jobs in fine dining. It does not matter if you have a criminal record or not. On the second floor, we have bright, open, airy spaces to support a constellation of activist organizations to amplify their cry of”Healthcare Not Handcuffs,”and”Housing as a human right.”And third, the county ’ s first dedicated space for restorative justice, filled with nature, color, texture and spaces of refuge to support the dialogues here. This project breaks ground in just two months. And we have plans to replicate it in Washington D. C., Detroit, New York and New Orleans. So you ’ ve seen two things we can build instead of prisons. And look, the price point is better. For one jail, we can build 30 restorative justice centers. That is a better use of your tax dollars. So I want to build all of these. But building buildings is a really heavy lift. It takes time. And what was happening in the communities that I was serving is we were losing people every week to gun violence and mass incarceration. We needed to serve more people and faster and keep them out of the system. And a new idea emerged from the community, one that was a lot lighter on its feet. Instead of building prisons, we could build villages on wheels. It ’ s called the Pop-Up Resource Village, and it brings an entire constellation of resources to isolated communities in the greater San Francisco area, including mobile medical, social services and pop-up shops. And so what we ’ re doing now is we ’ re building this whole village with the community, starting with transforming municipal buses into classrooms on wheels that bring GED and high school education across turf lines. We will serve thousands of more students with this.

We ’ re creating mobile spaces of refuge for women released from jail in the middle of the night, at their most vulnerable. Next summer, the village will launch, and it pops up every single week, expanding to more and more communities as it goes. So look out for it. So what do we build instead of prisons? We ’ ve looked at three things : peacemaking centers, centers for restorative justice and restorative economics and pop-up villages. But I ’ m telling you, I have a list a mile long. This is customized housing for youth transitioning out of foster care. These are reentry centers for women to reunite with their children. These are spaces for survivors of violence. These are spaces that address the root causes of mass incarceration. And not a single one of them is a jail or a prison. Activist, philosopher, writer Cornel West says that”Justice is what love looks like in public.”So with this in mind, I ask you one more time to imagine a world without prisons, and join me in creating all the things that we could build instead. Thank you.

Text Sample 108: NEG Dim 1 – 2016 – Score -1.00 – 2016/t001089.txt

A few days after my husband Paul was diagnosed with stage IV lung cancer, we were lying in our bed at home, and Paul said,”It ’ s going to be OK.”And I remember answering **back**,”Yes. We just don ’ t know what OK means yet.”Paul and I had met as first-year medical students at Yale. He was smart and kind and super funny. He used to keep a gorilla suit in the trunk of his car, and he ’ d say,”It ’ s for emergencies only.”I fell in love with Paul as I watched the care he took with his patients. He stayed late talking with them, seeking to understand the experience of illness and not just its technicalities. He later told me he fell in love with me when he saw me cry over an EKG of a heart that had ceased beating. We didn ’ t know it yet, but even in the heady days of young love, we were learning how to approach suffering together. We got married and became doctors. I was working as an internist and Paul was finishing his training as a neurosurgeon when he started to lose weight. He developed excruciating **back** pain and a cough that wouldn ’ t go away. And when he was admitted to the hospital, a CT scan revealed tumors in Paul ’ s lungs and in his bones. We had both cared for patients with devastating diagnoses ; now it was our turn. We lived with Paul ’ s illness for 22 months. He wrote a memoir about facing mortality. I gave birth to our daughter Cady, and we loved her and each other. We learned directly how to struggle through really tough medical decisions. The day we took Paul into the hospital for the last time was the most difficult day of my life. When he turned to me at the end and said,”I ’ m ready,”I knew that wasn ’ t just a brave decision. It was the right one. Paul didn ’ t want a ventilator and CPR. In that moment, the most important thing to Paul was to hold our baby daughter. Nine hours later, Paul died. I ’ ve always thought of myself as a caregiver — most physicians do — and taking care of Paul deepened what that meant. Watching him reshape his identity during his illness, learning to witness and accept his pain, talking together through his choices — those experiences taught me that resilience does not mean bouncing **back** to where you were before, or pretending that the hard stuff isn ’ t hard. It is so hard. It ’ s painful, messy stuff. But it ’ s the stuff. And I learned that when we approach it together, we get to decide what success looks like. One of the first things Paul said to me after his diagnosis was,”I want you to get remarried.”And I was like, whoa, I guess we get to say anything out loud. It was so shocking and heartbreaking... and generous, and really comforting because it was so starkly honest, and that honesty turned out to be exactly what we needed. Early in Paul ’ s illness, we agreed we would just keep saying things out loud. Tasks like making a will, or completing our advance directives — tasks that I had always avoided — were not as daunting as they once seemed. I realized that completing an advance directive is an act of love — like a wedding vow. A pact to take care of someone, codifying the promise that til death do **us** part, I will be there. If needed, I will speak for you. I will honor your wishes. That paperwork became a tangible part of our love story. As physicians, Paul and I were in a good position to understand and even accept his diagnosis. We weren ' t angry about it, luckily, because we ' d seen so many patients in devastating situations, and we knew that death is a part of life. But it ' s one thing to know that ; it was a very different experience to actually live with the sadness and uncertainty of a serious illness. Huge strides are being made against lung cancer, but we knew that Paul likely had months to a few years left to live. During that time, Paul wrote about his transition from doctor to patient. He talked about feeling like he was suddenly at a crossroads, and how he would have thought he ' d be able to see the path, that because he treated so many patients, maybe he could follow in their footsteps. But he was totally disoriented. Rather than a path, Paul wrote,"I saw instead only a harsh, vacant, gleaming white desert. As if a sandstorm had erased all familiarity. I had to face my mortality and try to understand what made my life worth living, and I needed my oncologist ' s help to do so."The clinicians taking care of Paul gave me an even deeper appreciation for my colleagues in health care. We have a tough job. We ' re responsible for helping patients have clarity around their prognoses and their treatment options, and that ' s never easy, but it ' s especially tough when you ' re dealing with potentially terminal illnesses like cancer. Some people don ' t want to know how long they have left, others do. Either way, we never have those answers. Sometimes we substitute hope by emphasizing the best-case scenario. In a survey of physicians, 55 percent said they painted a rosier picture than their honest opinion when describing a patient ' s prognosis. It ' s an instinct born out of kindness. But researchers have found that when people better understand the possible outcomes of an illness, they have less anxiety, greater ability to plan and less trauma for their families. Families can struggle with those conversations, but for **us**, we also found that information immensely helpful with big decisions. Most notably, whether to have a baby. Months to a few years meant Paul was not likely to see her grow up. But he had a good chance of being there for her birth and for the beginning of her life. I remember asking Paul if he thought having to say goodbye to a child would make dying even more painful. And his answer astounded me. He said,"Wouldn ' t it be great if it did?"And we did it. Not in order to spite cancer, but because we were learning that living fully means accepting suffering. Paul ' s oncologist tailored his chemo so he could continue working as a neurosurgeon, which initially we thought was totally impossible. When the cancer advanced and Paul shifted from surgery to writing, his palliative care doctor prescribed a stimulant medication so he could be more focused. They asked Paul about his priorities and his worries. They asked him what trade-offs he was willing to make. Those conversations are the best way to ensure that your health care matches your values. Paul joked that it ' s not like that"birds and bees"talk you have with your parents, where you all get it over with as quickly as possible, and then pretend it never happened. You revisit the conversation as things change. You keep saying things out loud. I ' m forever grateful because Paul ' s clinicians felt that their job wasn ' t to try to give **us** answers they didn ' t have, or only to try to fix things for **us**, but to counsel Paul through painful choices... when his body was failing but his will to live wasn ' t. Later, after Paul died, I received a dozen bouquets of flowers, but I sent just one... to Paul ' s oncologist, because she supported his goals and she helped him weigh his choices. She knew that living means more than just staying alive. A few weeks ago, a patient came into my clinic. A woman dealing with a serious chronic disease. And while we were talking about her life and her health care, she said,"I love my palliative care team. They taught me that it ' s OK to say ' no '."Yeah, I thought, of course it is. But many patients don ' t feel that. Compassion and Choices did a study where they asked people about their health care preferences. And a lot of people started their answers with the words"Well, if I had a choice..."If I had a choice. And when I read that"if,"I understood better why one in four people receives excessive or unwanted medical treatment, or watches a family member receive excessive or unwanted medical treatment. It ' s not because doctors don ' t get it. We do. We understand the real psychological consequences on patients and their families. The things is, we deal with them, too. Half of critical care nurses and a quarter of ICU doctors have considered quitting their jobs because of distress over feeling that for some of their patients, they ' ve provided care that didn ' t fit with the person ' s values. But doctors can ' t make sure your wishes are respected until they know what they are. Would you want to be on life support if it offered any chance of longer life? Are you most worried about the quality of that time, rather than quantity? Both of those choices are thoughtful and brave, but for all of **us**, it ' s our choice. That ' s true at the end of life and for medical care throughout our lives. If you ' re pregnant, do you want genetic screening? Is a knee replacement right or not? Do you want to do dialysis in a clinic or at home? The answer is : it depends. What medical care will help you live the way you want to? I hope you remember that question the next time you face a decision in your health care. Remember that you always have a choice, and it is OK to say no to a treatment that ' s not right for you. There ' s a poem by W. S. Merwin — it ' s just two sentences long — that captures how I feel now."Your absence has gone through me like thread through a needle. Everything I do is stitched with its color."For me that poem evokes my love for Paul, and a new fortitude that came from loving and losing him. When Paul said,"It ' s going to be OK,"that didn ' t mean that we could cure his illness. Instead, we learned to accept both joy and sadness at the same time ; to uncover beauty and purpose both despite and because we are all born and we all die. And for all the sadness and sleepless nights, it turns out there is joy. I leave flowers on Paul ' s grave and watch our two-year-old run around on the grass. I build bonfires on the beach and watch the sunset with our friends. Exercise and mindfulness meditation have helped a lot. And someday, I hope I do get remarried. Most importantly, I get to watch our daughter grow. I ' ve thought a lot about what I ' m going to say to her when she ' s older."Cady, engaging in the full range of experience — living and dying, love and loss — is what we get to do. Being human doesn ' t happen despite suffering. It happens within it. When we approach suffering together, when we choose not to hide from it, our lives don ' t diminish, they expand."I ' ve learned that cancer isn ' t always a battle. Or if it is, maybe it ' s a fight for something different than we thought. Our job isn ' t to fight fate, but to help each other through. Not as soldiers but as shepherds. That ' s how we make it OK, even when it ' s not. By saying it out loud, by helping each other through... and a gorilla suit never hurts, either. Thank you.

Text Sample 109: NEG Dim 1 – 2016 – Score -1.00 – 2016/t000996.txt

I recently read about what the young generation of workers want in Harvard Business Review. One thing that stuck out to me was : don ' t just talk about impact, but make an impact. I ' m a little bit older than you, maybe much older than you, but this is exactly the same goal that I had when I was in college. I wanted to make my own impact for those who live under injustice ; it ' s the reason that I became a documentary journalist, the reason I became a prisoner in North Korea for 140 days. It was March 17, 2009. It is St. Patrick ' s Day for all of you, but it was the day that turned my life upside down. My team and I were making a documentary about North Korean refugees living below human life in China. We were at the border. It was our last day of filming. There was no wire fence or bars or sign to show that it is the border, but this is a place that a lot of North Korean defectors use as an escape route. It was still winter, and the river was frozen. When we were in the middle of the frozen river, we were filming about the condition of the cold weather and the environment that North Koreans had to deal with when they seek their freedom. And suddenly, one of my team members shouted,"Soldiers!"So I looked **back**, and there were two small soldiers in green uniforms with rifles, chasing after **us**. We all ran as fast as we could. I prayed that, please don ' t let them shoot my head. And I was thinking that, if my feet are on Chinese soil, I ' ll be safe. And I made it to Chinese soil. Then I saw my colleague Laura Ling fall on her knees. I didn ' t know what to do at that short moment, but I knew that I could not leave her alone there when she said,"Euna, I can ' t feel my legs."In a flash, we were surrounded by these two Korean soldiers. They were not much bigger than **us**, but they were determined to take **us** to their army base. I begged and yelled for any kind of help, hoping that someone would show up from China. Here I was, being stubborn towards a trained soldier with a gun. I looked at his eyes. He was just a boy. At that moment, he raised his rifle to hit me, but I saw that he was hesitating. His eyes were shaking, and his rifle was still up in the air. So I shouted at him,"OK, OK, I ' ll walk with you."And I got up. When we arrived at their army base, my head was spinning with these worst-case scenarios, and my colleague ' s statement wasn ' t helping. She said,"We are the enemy."She was right : we were the enemy. And I was supposed to be frightened, too. But I kept having these odd experiences. This time, an officer brought me his coat to keep me warm, because I lost my coat on the frozen river while battling with one of these soldiers. I will tell you what I mean by these odd experiences. I grew up in South Korea. To **us**, North Korea was always the enemy, even before I was born. South and North have been under armistice for 63 years, since the end of the Korean War. And growing up in the South in the ' 80s and ' 90s, we were taught propaganda about North Korea. And we heard so many graphic stories, such as, a little young boy being brutally killed by North Korean spies just because he said,"I don ' t like communists."Or, I watched this cartoon series about a young South Korean boy defeating these fat, big, red pig, which represented the North Koreans ' first leader at the time. And the effect of hearing these horrible stories over and over instilled one word in a young mind :"enemy."And I think at some point, I dehumanized them, and the people of North Korea became equated with the North Korean government. Now, **back** to my **detention**. It was the second day of being in a cell. I had not slept since I was out at the border. This young guard came to my cell and offered me this small boiled egg and said,"This will give you strength to keep going."Do you know what it is like, receiving a small kindness in the enemy ' s hand? Whenever they were kind to me, I thought the worst case was waiting for me after the kindness. One officer noticed my nervousness. He said,"Did you think we were all these red pigs?"referring to the cartoon that I just showed you. Every day was like a psychological battle. The interrogator had me sit at a table six days a week and had me writing down about my journey, my work, over and over until I wrote down the confession that they wanted to hear. After about three months of **detention**, the North Korean court sentenced me to 12 years in a labor camp. So I was just sitting in my room to be transferred. At that time, I really had nothing else to do, so I paid attention to these two female guards and listened to what they were talking about. Guard A was older, and she studied English. She seemed like she came from an affluent family. She often showed up with these colorful dresses, and then loved to show off. And Guard B was the younger one, and she was a really good singer. She loved to sing Celine Dion ’ s”My Heart Will Go On”— sometimes too much. She knew just how to torture me without knowing. And this girl spent a lot of time in the morning to put on makeup, like you can see in any young girl ’ s life. And they loved to watch this Chinese drama, a better quality production. I remember Guard B said,”I can no longer watch our TV shows after watching this.”She got scolded for degrading her own country ’ s produced TV shows. Guard B had more of a free mind than Guard A, and she often got scolded by Guard A whenever she expressed herself. One day, they invited all these female colleagues — I don ’ t know where they came from — to where I was held, and they invited me to their guard room and asked if one-night stands really happen in the US. This is the country where young couples are not even allowed to hold hands in public. I had no idea where they had gotten this information, but they were shy and giggly even before I said anything. I think we all forgot that I was their prisoner, and it was like going **back** to my high school classroom again. And I learned that these girls also grew up watching a similar cartoon, but just propaganda towards South Korea and the US. I started to understand where these people ’ s anger was coming from. If these girls grew up learning that we are enemies, it was just natural that they would hate **us** just as I feared them. But at that moment, we were all just girls who shared the same interests, beyond our ideologies that separated **us**. I shared these stories with my boss at Current TV at the time after I came home. His first reaction was,”Euna, have you heard of Stockholm Syndrome?”Yes, and I clearly remember the feeling of fear and being threatened, and tension rising up between me and the interrogator when we talked about politics. There definitely was a wall that we couldn ’ t climb over. But we were able to see each other as human beings when we talked about family, everyday life, the importance of the future for our children. It was about a month before I came home. I got really sick. Guard B stopped by my room to say goodbye, because she was leaving the **detention** center. She made sure that no one watched **us**, no one heard **us**, and quietly said,”I hope you get better and go **back** to your family soon.”It is these people — the officer who brought me his coat, the guard who offered me a boiled egg, these female guards who asked me about dating life in the US — they are the ones that I remember of North Korea : humans just like **us**. North Koreans and I were not ambassadors of our countries, but I believe that we were representing the human race. Now I ’ m **back** home and **back** to my life. The memory of these people has blurred as time has passed. And I ’ m in this place where I read and hear about North Korea provoking the US. I realized how easy it is to see them as an enemy again. But I have to keep reminding myself that when I was over there, I was able to see humanity over hatred in my enemy ’ s eyes. Thank you.

Text Sample 110: NEG Dim 1 – 2016 – Score -1.00 – 2016/t001326.txt

I ’ m an Iranian-American Muslim female, like all of you. And I ’ m also a social justice comedian, something that I insist is an actual job. To explain what that is, let me tell you how I got here. I ’ ve performed all over the country. And let me tell you, America is majestic, right? It ’ s got breathtaking nature, waffle houses and diabetes as far as the eye can see. It is really something. Now, the

American population can be broken up into three main categories : there ' s mostly wonderful people, haters and Florida. Besides Florida, the most troubling category here are the Haters. They are a minority, but they overcompensate by being extra loud. They have the Napoleon complex of demographics, and yes, some of the men do wear heels. As a social justice comedian, it ' s my goal to convert these haters, because they hate a lot of things, which leads to negative outcomes, like racism, violence and Ted Nugent. This is not an exhaustive list ; I ' m probably missing 3-7 items. But the point is, we have to reckon with the haters. But there ' s variance within this group and it ' s not efficient to go after all of them, right? So what I ' ve done is created a highly scientific Taxonomy of Haters. I basically took all of the haters, I put them in a petri dish, like a scientist, and this is what I found. First off, we have the trolls. These are your garden-variety digital haters. They ' re the people who quit their jobs so they can post on YouTube videos all day long. There ' s also the drive-by haters. Now, these people will be at a stoplight, they ' ll wait for the light to turn green and when it does, they yell,"Go **back** to your own country!"Now **back** in the day, they would ' ve actually gotten out of their cars and hated you to your face. But they just don ' t make them like they used to — which is another sign of the decline in America. The next category is the mission-oriented-bigot-whose-group-affiliation- gives-them-cover-for-hating hater. These guys like to hate via a seemingly nice organization, like a church or a nonprofit, and they oftentimes like to speak in an old-timey voice. But the group I ' m most interested in is the swing hater. The swing hater is sister to the swing voter — they just can ' t decide! They ' re like ideological sluts who move from hating to not hating. And they do it because they don ' t have enough information. This is the group I like to target with social justice comedy. Why comedy? Because on a scale of comedy to brochure, the average American prefers comedy, as you can see from this graph. Comedy is very popular. And by the way, this is a mathematically accurate graph, generated from fake numbers. Now, the question is : Why does social justice comedy work? Because, first off, it makes you laugh. And when you ' re laughing, you enter into a state of openness. And in that moment of openness, a good social justice comedian can stick in a whole bunch of information, and if they ' re really skilled, a rectal exam. Here are some ground rules for social justice comedy : first off, it ' s not partisan. This isn ' t political comedy, this is about justice, and no one is against justice. Two, it ' s inviting and warm, it makes you feel like you ' re sitting inside of a burrito. Three, it ' s funny but sneaky, like you could be hearing an interesting treatise on income inequality, that ' s encased in a really sophisticated poop joke. Here ' s how I see social justice comedy working. A few years ago, I rounded up a bunch of Muslim-American comedians — in a non-violent way — And we went around the country to places like Alabama, Arizona, Tennessee, Georgia — places where they love the Muzzies — and we did stand-up shows. We called the tour"The Muslims Are Coming!"We turned this into a movie, and then after the movie came out, a known hate group spent 300, 000 dollars on an anti-Muslim poster campaign with the MTA — that ' s the New York City subway system. Now, the posters were truly offensive, not to mentioned poorly designed — I mean, if you ' re going to be bigoted, you might as well use a better font. But we decided, why not launch our own poster campaign that says nice things about Muslims, while promoting the movie. So myself and fellow comedian Dean Obeidallah decided to launch the fighting-bigotry- with-delightful-posters campaign. We raised the money, worked with the MTA for over 5 months, got the posters approved, and two days after they were supposed to go up, the MTA decided to ban the posters, citing political content. Let ' s take a look at a couple of those posters. Here ' s one. Facts about Muslims : Muslims invented the concept of a hospital. OK. Fact : Grown-up Muslims can do more push-ups than baby Muslims. Fact : Muslims invented Justin Timberlake. Let ' s take a look at another one. The ugly truth about Muslims : they have great frittata recipes. Now clearly, frittatas are considered political by the MTA. Either that, or the mere mention of Muslims in a positive light was considered political — but it isn ' t. It ' s about justice. So we decided to change our fighting-bigortry-with- delightful-posters campaign and turn it into the fighting-bigotry- with-a-delightful-lawsuit campaign. So basically, what I ' m saying is a couple of dirt-bag comedians took on a major New York City agency and the comedians won. Thank you. Victory was a very weird feeling. I was like,"Is this what blonde girls feel like all the time? ' Cause this is amazing!"Here ' s another example. I ' m asked everywhere I go :"Why don ' t Muslims denounce terrorism?"We do. But OK, I ' ll take the bait. So I decided to launch thedailydenouncer. com. It ' s a website that denounces terrorism every day of the week, while taking the weekends off. Let ' s take a look at an example. They generally appear as single-panel cartoons,"I denounce terrorism! I also denounce people who never fill the paper tray!"The point of the website is that it denounces terrorism while recognizing that it ' s ridiculous that we have to constantly denounce terrorism. But if bigotry isn ' t your thing, social justice comedy is useful for all sorts of issues. For example, myself and fellow comedian Lee Camp went to the Cayman Islands to investigate offshore banking. Now, the United States loses something like 300 billion dollars a year in these offshore tax havens. Not to brag, but at the end of every month, I have something like 5-15 dollars in disposable income. So we walked into these banks in the Cayman Islands and asked if we could open up a bank account with eight dollars and 27 cents. The bank managers would indulge **us** for 30-45 seconds before calling security. Security would come out, brandish their weapons, and then we would squeal with fear and run away, because — and this is the last rule of social justice comedy — sometimes it makes you want to take a dump in your pants. Most of my work is meant to be fun. It ' s meant to generate a connection and laughter. But yes, sometimes I get run off the grounds by security. Sometimes I get mean tweets and hate mail. Sometimes I get voice mails saying that if I continue telling my jokes, they ' ll kill me and they ' ll kill my family. And those death threats are definitely not funny. But despite the occasional danger, I still think that social justice comedy is one of our best weapons. I mean, we ' ve tried a lot of approaches to social justice, like war and competitive ice dancing. But still, a lot of things are still kind of awful. So I think it ' s time we try and tell a really good poop joke. Thank you.

Text Sample 111: NEG Dim 1 – 2024 – Score 1.00 – 2024/t003229.txt

My first job as an investor was when I was 24 years old, and I ' m almost 50 today, so that is half a lifetime ago. But I still clearly remember the first thing I was asked to do : to oversee wire transfers related to the sale of a business, basically to make sure all the shareholders got their money. Any company, large or small, public or private, has shareholders who own the business. And as the value of the company goes up, wealth is created for those who own shares. Now at this company, ownership among the employees extended from the very top position, the CEO, to some mid-level roles like assistant treasurer. When I called the CEO to confirm his wire transfer it was a very matter-of-fact conversation."Got the money, thanks a lot."Click. Now, when I called the assistant treasurer, who was making a tiny fraction of what the CEO had made, he was so overcome with emotion, he could barely find the words. And he later explained in a tearful voicemail that his ownership payout literally meant college education for his kids. So I wondered, what if everyone in a company had stock ownership, not just to the assistant-treasurer level, but to the factory floor, distribution centers? How might employees ’ lives be impacted, and how might the company and, in fact, the whole community be impacted? These questions were not new to me. My dad had been asking them for decades. My father operated a road grader for a union construction company in Chicago for 45 years. This is actually a picture of my dad in his road grader. And my dad dreamt of worker ownership to address three things he didn ’ t like about his job. First, he couldn ’ t build wealth on 20 dollars an hour. Second, he had no incentive to care about things an owner would focus on, things like productivity. So he felt no sense of alignment. And third, he had no voice. Without the right incentives in place, there was really no reason for management to listen to workers, and they didn ’ t. And my dad ’ s experience is certainly not unique. Only a tiny percentage of workers are granted stock ownership in their companies, and most workers have no wealth, and it is in fact stock ownership or the lack of it that is by a mile the biggest driver of wealth inequality. Most employees feel their opinions don ’ t count. If you look at Gallup surveys, 77 percent of employees globally are disengaged on the job. 18 percent literally hate the company that they work for. They ’ re throwing the wrenches in the machines. Today, I ’ m going to try to convince you that broadening employee stock ownership is the single most important thing that we can do to lift up workers and to make companies stronger, too. It could give **us** a form of capitalism that is actually inclusive and sustainable. And I believe it could literally change the economy. I ’ ve been pursuing this idea most of my adult life. 25 years ago, as a graduate school student, I dove into the history of employee ownership, and I published a paper on the topic in 2002. Then, a handful of years later, I got my first real leadership position at the company I still work for, KKR. And I was put in charge of investing in and improving industrial companies, mostly manufacturing businesses. And that was a great opportunity to start experimenting with different ways of sharing stock ownership with all employees. Something we ’ ve now done with 44 different companies over the past 15 years. In our early efforts, we had some success, but we made a lot of mistakes. The most important thing we got right was making sure that stock ownership was free and incremental for workers, not a trade for wages or other benefits. This could not be about shifting risk onto the workforce. But we got plenty of things wrong. We didn ’ t communicate well. So if I said something like,”We ’ d hope to sell a business in five years,”that often led to employees literally counting the days or growing suspicious if things took longer. We didn ’ t share our financial information so people didn ’ t know how the business was doing, nor did we ask them how they would run the business better. You see, ownership is about a lot more than just giving out stock. It ’ s about trying to create a whole different type of culture, an ownership culture. Now let me share a story of what this can look like when it ’ s done well. In 2015, we invested in a company called CHI Overhead Doors. CHI is based in central Illinois, in Amish country, and the company makes overhead garage doors, like the one you see here in this picture. And CHI was a good business. But from a worker morale standpoint, it was very reminiscent of what I saw with my dad. Out of 800 employees, only 18 had stock ownership. So that means when we bought the business, most people got nothing and just went **back** to work, and a small handful made many millions of dollars. Employee engagement scores were absolutely dismal. Most people didn ’ t even bother to respond to surveys. And this lack of alignment and engagement, you could see it in the business. It showed up in things like productivity, quality, scrap. People didn ’ t always try to do their best. It took **us** a long time to change the culture, eight years. And it started with stock ownership. So day one, all 800 employees were granted stock ownership so they would participate. As the value of the company went up, through dividends along the way, and then at the end, when the business was sold. When we announced this program, the employees were so excited, they gave me a gift. Which was a live chicken. Literally two guys on the factory floor came up and handed me a chicken. I awkwardly joked that I didn ’ t have a crate, and I wasn ’ t sure I could carry a chicken on my flight **back** to New York. But this was about much more than just stock ownership. Employees were educated about the business, they were kept informed. Once a quarter we got everyone together and we talked openly about what was working and what was not and where the business was headed, and we shared our financial information transparently. All 800 employee owners were given a voice. As one example, they were provided with a million dollars annual budget to invest in their workplace any way they saw fit. They came up with the ideas, they decided where the money went. And over the course of several years, they voted in things like air conditioning for the manufacturing plant. One year they voted for new break rooms. Another year, they wanted an on-site health clinic built. And another, an on-site cafeteria with healthy food options. We found this was a great way to engage with employees because not only were you giving them a voice but they could see their voice physically manifested in the workplace. Employees were also provided free financial coaching so that as they received money, they could save and invest effectively. And I want to be very clear, all of this benefited the company too. It ’ s not like this was only good for workers. We saw quality improve, scrap went down, productivity boomed. We became more responsive to customer demands, and we gained market share as a result. In fact, by the end of the journey, the financial results had improved so enormously the company was worth ten times what it had been worth in 2015. And to put that in perspective, this was the single best investment for **us** at KKR since the 1980s. And this is with a company making garage doors. This is not like a high-flying technology business. When the company was sold, we gathered everyone in our main manufacturing plant in central Illinois. We told them their jobs were safe, and it was time for each of them to learn what they had earned from their ownership. The payouts were scaled based on length of employment. With our most tenured factory workers earning six-and-a-half times their annual income, or a half a million dollars. We had truck drivers make 800, 000 dollars. So this was life-impacting amounts of money. And I could talk to you about it all day, but I won ’ t get it across as well as the workers can themselves. Person 1 : He started out at 20, 000, and then he went to the next amount and the next amount, and he kept going. Person 2 : I ’ m like, whoa! Person 3 : Holy smokes! Person 4 : Where is this going to lead? PS : 40, 000 dollars, 70, 000 dollars. Two-and-a-half times your annual pay. Three-and-a-half times, four-and-a-half times, five-and-a-half times your annual pay. Six-and-a-half times. Person 4 : I just kept looking around at all the people around me thinking how much this was going to change everyone ’ s lives. Person 2 : You sit there and you struggle, like, how am I going to pay for this? Person 6 : Just think about what this means for this area. Person 7 : We ’ re out here, there ’ s a bunch of lower-income families that live paycheck to paycheck. Person 2 : This will absolutely give **us** some comfort that, you know, the future is going to be OK. Person 8 : It ’ s wonderful, I mean, it ’ s just freedom. I got freedom. Person 10 : This is what we did and we ’ re proud of it. Person 11 : Every cent that we get **back**, we earned. Person 12 : Without ownership, it is a job. You clock in, you clock out, you go home. When you have ownership in the business, it changes everything. PS : They deserve to be applauded because they earned it. One of my favorite things in that... Audience : Don ’ t worry, we ’ re all crying. PS : Alright. I ’ ll try and explain that again. So when Cara said,”We earned this,”I love that sentiment, because to me, that ’ s ownership. You know, they earned it, this wasn ’ t a gift, it didn ’ t happen by chance, they made it happen. And it did change the community. 340 million dollars of wealth was injected into the community, and it ’ s forever changed. I mean, that type of wealth creation, it can fundamentally shift a whole local economy. It means more tax revenue, better schools, new business formations, more restaurants in town. And it means dignified retirement for the employees. So just imagine how different would the economy be if every company operated in this way and included all employees in decision-making and in wealth creation? What a different economy we would have. When we sold the business, I got a second gift. A chicken. But this time it was with a crate and it had a little tag indicating it had been pre-checked **back** to New York City. I wanted to stay married to my wife so I did not bring the chicken home. But when I got home, my wife and I started talking about how we could give this idea of employee ownership more scope. And we decided to start a nonprofit called Ownership Works, whose mission is to ignite a movement around employee ownership and to help CEOs and companies who want to go on this journey because it is not easy. Our goal is to reach a million workers over the next 10 years and to create tens of billions of wealth for working families. And we ’ re already well on our way. We ’ ve got line of sight to our first 250, 000 workers being impacted, and our first 10 billion of wealth creation for working families. We ’ ve also joined forces with a coalition of employee ownership advocates who have come together to put new energy behind something called an ESOP, or employee stock ownership plan. The ESOP was created by Congress in the 1970s, and it gives companies tax incentives in exchange for sharing stock ownership with workers. Our goal, through a new organization called Expanding ESOPs, is to pass legislation in the United States to make ESOPs easier to form and faster to form, particularly with larger businesses. It is going to take government support for this idea to really take off. That ’ s how we ’ re going to get not one million new employee owners, but 30 million or 40 million. When I talk about this idea and my passion for it, I always get the same question :”What does your dad think of all this?”Well, you ’ d have to know my dad to appreciate this, but he says,”What the hell took so long?”Thank you.

Text Sample 112: NEG Dim 1 – 2024 – Score 1.00 – 2024/t003057.txt

Video : Chris Anderson : Hello, Nila. Welcome to the TED stage. Nila Ibrahimi : Thank you so much, Chris. Thank you for having me. CA : This happens when you come to Vancouver. NI : I ’ m really honored to be here. CA : What were you singing? NI : The song was saying – It ’ s really hard to translate it right now, but it ’ s mostly encouraging other girls to go to school. It ’ s encouraging education. At this time, I thought that was the closest song to protest against the decision that banned girls from singing in public. So yeah, I didn ’ t know that two or three months after girls would be banned from education. CA : Before they were banned from education, though, the video had an impact. Other people released videos. There was support. What happened? Wasn ’ t that singing ban then reversed? NI : So this happened. The decision was announced that girls over the age of 12 can ’ t sing in public in March of 2021. If I ’ m right. After the movement”I ’ m My Song,”I joined the movement and so did the other brave girls of Afghanistan. And it was two or three weeks after that the decision was reversed, after our efforts and the efforts that the media put in in getting our voice across to the world. CA : Wow. CA : But then something shocking happened after that. NI : I didn ’ t know two or three months after that Afghanistan would fall into the hands of Taliban. It was surreal, and it ’ s really hard to talk about it every time. I ’ ve started my journey of activism right after that moment, when I discovered that my voice can be heard, and I discovered the power of my voice through standing up against that decision. But when Afghanistan fell, I felt like that power was taken away from me. So I had to fight for that, to regain it. CA : So I told the story wrong. The initial ban, the singing ban, was before the Taliban came to power. You found your voice. They came to power. And... Tell **us** your story. What happened next? NI : The Taliban came August 15. That was the day I was about to have my last mid-year exam, the day after that, August 16, which did not happen. I was just busy studying for my exams, Arabic and Pashto, and the rumors spread out, and my mom suddenly started burning all of the documents that we had of my deceased father, and my siblings, who were studying in the American university and working at the United Nations. It was surreal in a way that I never thought that could happen, all of a sudden, in just one day. All of that power, all of the things that we had worked so hard for, could be taken away from **us** in just seconds and just minutes. And that the power that I thought I had, the power, the confidence I gained from posting that video, it was all taken away from me. It was all shattered. So I thought I had to regain that power by getting out of the oppressed Afghanistan that now is being ruled by Taliban. CA : So you were able to escape. How did you get out? NI : That ’ s a long story, actually. So five days after the fall, we went to Pakistan to hide and escape. Our family ’ s background put **us** in danger and, amazingly, we found 30 Birds Foundation, which helped **us** resettle in Canada after waiting in Pakistan around eight, nine months. The time I spent in Pakistan, that has its own surreal story, but I ’ m just really glad to have the opportunities that I think every single woman should have. And I just wish that Afghan girls can regain their power and their voice, just like I have. CA : Well. Nila, here you are on the TED stage. You ’ ve been using your voice to raise awareness on this issue. I ’ d like to – see that red light there? That ’ s a camera, right there. I ’ d love you to just come to the middle of the stage and give your message to the Taliban government in Afghanistan. Whatever you want to say to them, here at TED. I ’ m going to step out and give you the stage for a minute. NI : For the Taliban. I ’ m not angry. I never have been. But you don ’ t know who you ’ re up against. People like me, the girls that I know, the girls that I ’ ve been interviewing through my work at Her Story, the girls who have been displaced and the girls who are now living in Afghanistan under the oppression, without any resources and without any access to education, they ’ re still fighting. They have not given up. And that is what I want to share about Afghan girls. That is what makes **us** different, and that is what should be supported and appreciated and heard. Thank you.

Text Sample 113: NEG Dim 1 – 2024 – Score 1.00 – 2024/t003078.txt

There is a Supreme Court case that you could mistake for a sermon. It ’ s the case that recognized that same-sex couples have a constitutional right to marry. Here is a sense of what Justice Kennedy wrote :”Marriage responds to the universal fear that a lonely person might call out only to find no one there.”He goes on to say that marriage offers care and companionship, and the decision argues that these are basic human needs that everyone should have access to, whether they ’ re straight or queer. Validating. But what do these words say to you if you ’ re single? Anybody single here? I mean, there should be quite a few of you, because in the US, the percentage of American adults who have never been married is at a record high. Married people, you ’ re not off the hook. I ’ m going to get a little morbid for a moment and have you contemplate what happens if your marriage doesn ’ t last until the end of your life, whether because of divorce or outliving your spouse? In the US, about 30 percent of women over 65 are widows. The reality is, any one of **us** is unlikely to have a spouse by our side until our last dying breath. Regardless of whether we are partnered now, we need to rely on more than one relationship to sustain **us** throughout our full, unpredictable lives. We need other significant others. And there ’ s an overlooked kind of relationship that we can turn to. Friendship. I got the sense that friendship could be this stronger force in our lives because of a friendship that I stumbled into. We would see each other most days of the week, be each other ’ s plus-ones to parties. My friend has this habit of grabbing my hand to hold, including when I brought her to my office, and then I ’ d have to be like, no, not in the office. But I mean, I wouldn ’ t let my husband do that in the office either. It ’ s just, you know, setting matters. But it was only an issue because for **us**, affection is a reflex. And I knew it couldn ’ t be just **us**. I went out and interviewed dozens of people who had a friendship like ours, and I wrote a book about them. And the kinds of friends that I spoke to, they don ’ t just have a weekly phone call. They ’ re friends like these. Natasha and Linda are the first legally recognized platonic co-parents in Canada. And this is them with their teenage son on vacation. Joe and John have been best friends for many decades. When Joe was struggling with alcohol and drug use, John got him into recovery. And then John decided that to support his friend, he would also become sober. Joy took care of her friend Hannah during Hannah ’ s six-year battle with ovarian cancer. And that included flying out to New York, where Hannah got specialized treatment. Joy had trouble actually sleeping overnight in the hospital, because she was too busy watching to make sure her friend ’ s chest was still rising and falling. Some of the friends that I spoke to had this friendship occupy the space that ’ s conventionally given to a romantic partner. Some had this kind of friendship and a romantic partner. It ’ s not either / or. As I spoke to these people, I realized that they were at the frontier of friendship, helping **us** imagine how much more we could ask of our platonic relationships. Which is true, but another way of looking at it is they ’ re doing something retro, even ancient. In ancient Rome, friends would talk about each other as”half of my soul,”or”the greater part of my soul.”The kind of language we now use in romantic relationships. From China to Jordan to England, there was a practice called”sworn brotherhood,”where male friends would go through a ritual that would turn them into brothers. About a century ago, friends would sit for portraits like these, with their arms wrapped around each other, their bodies up close. What I took from this history is that if we don ’ t limit friendship, it can be central to our lives. But today, not everybody recognizes that. I spoke to a mother who really tried to get her son to make dating a priority because she wanted him to find emotional wholeness. And her son told her,”I found it in my platonic life partner.”His best friend, who he had known since high school, who had moved across the country to be near him, to live with him, in fact. The mother said,”I don ’ t understand how you can be partners with someone you ’ re not romantic with.”Understandable as a reaction in a culture that treats friendship as the sidekick to the real hero of romance. We get that message from rom-coms, from Supreme Court justices, also from policy. So Joy, during the six years she took care of her friend, she was not entitled to family medical leave. When Hannah died, Joy was not entitled to bereavement leave, because the two of them were considered unrelated. In our government and workplace policies, friendship is invisible. Sometimes this diminishment of friendship comes from the outside, and sometimes it comes from the inside. A woman wrote to me about her friend who she considers her person. She spent so much time with her friend ’ s kids that she was given car seats for them. She ’ s also divorced and tried to find a new spouse because there was a hole she wanted to fill in her life. Then she read stories of people like Joe and John in an article I ’ d written. And she realized there was no hole. She had been happy all along, but she hadn ’ t known, been made to believe that it was possible to have a friend be enough. If we can recognize what friendship has the potential to be, and if we can recognize that there is more than one kind of significant other then we can imagine more ways for **us** to find love and care and companionship. And we can support people who have these kinds of friendships. So the mother I mentioned, she ’ s completely changed her tune. She now admires the commitment between her son and her son ’ s friend. I feel like I get to live in a future world where you can just build a life with your friends. I live not only with my husband but also two of my closest friends. One of them we kind of like had a courtship process to recruit him to come to our city and live with **us**. The other had a job in our city, and we invited her to stay. It didn ’ t take long for **us** to start scheming with about a half dozen other friends, about trying to buy property together. The kind of place where we could raise kids alongside one another, our working title for the place is”The Village.”I don ’ t know if this will work out. I can keep you posted about it, but if it does, I feel really confident about one thing. That if one of **us** has a migraine at 6am and there ’ s a toddler bouncing around, or we get a terrifying diagnosis, we will not be a lonely person calling out only for no one to answer. And this is what I hope for all of **us**. That we feel like we have permission to share our lives with whoever we are lucky enough to find, whether that ’ s a spouse, a sibling or a house full of friends. Thank you.

Text Sample 114: NEG Dim 1 – 2014 – Score -1.00 – 2014/t001786.txt

Today, a baffled lady observed the shell where my soul dwells And announced that I ’ m”articulate”Which means that when it comes to enunciation and diction I don ’ t even think of it â□□Cause Iâ□□m”articulate”So when my professor asks a question And my answer is tainted with a connotation of urbanized suggestion Thereâ□□s no misdirected intention Pay attention â□□Cause Iâ□□m â□□articulateâ□□ So when my father asks, â□□Whaâ□□ kinda ting is dis? â□□ My â□□articulateâ□□ answer never goes amiss I say â□□father, this is the impending problem at handâ□□ And when Iâ□□m on the block I switch it up just because I can So when my boy says, â□□Whatâ□□s good with you son? â□□ I just say, â□□I jusâ□□ fall out wit dem people but I done! â□□ And sometimes in class I might pause the intellectual sounding flow to ask â□□Yo! Why dese books neva be about my peoplesâ□□ Yes, I have decided to treat all three of my languages as equals Because Iâ□□m â□□articulateâ□□ But who controls articulation? Because the English language is a multifaceted oration Subject to indefinite transformation Now you may think that it is ignorant to speak broken English But Iâ□□m here to tell you that even â□□articulateâ□□ Americans sound foolish to the British So when my Professor comes on the block and says, â□□Helloâ□□ I stop him and say â□□Noooo â□Ś Youâ□□re being inarticulate â□Ś the proper way is to say â□□whatâ□□s goodâ□□â□□ Now you may think thatâ□□s too hood, thatâ□□s not cool But Iâ□□m here to tell you that even our language has rules So when Mommy mocks me and says â□□yaâ□□ll-be-madd-going-to-the-storeâ□□ I say â□□Mommy, no, that sentence is not following the law Never does the word”madd”go before a present participle Thatâ□□s simply the principle of this Englishâ□□ If I had the vocal capacity I would sing this from every mountaintop, From every suburbia, and every hood â□□Cause the only God of language is the one recorded in the Genesis Of this world saying â□□it is good”So I may not always come before you with excellency of speech But do not judge me by my language and assume That Iâ□□m too ignorant to teach â□□Cause I speak three tongues One for each : Home, school and friends Iâ□□m a tri-lingual orator Sometimes Iâ□□m consistent with my language now Then switch it up so I donâ□□t bore later Sometimes I fight **back** two tongues While I use the other one in the classroom And when I mistakenly mix them up I feel crazy like â□Ś Iâ□□m cooking in the bathroom I know that I had to borrow your language because mines was stolen But you canâ□□t expect me to speak your history wholly while mines is broken These words are spoken By someone who is simply fed up with the Eurocentric ideals of this season And the reason I speak a composite version of your language Is because mines was raped away along with my history I speak broken English so the profusing gashes can remind **us** That our current state is not a mystery Iâ□□m so tired of the negative images that are driving my people mad So unless youâ□□ve seen it rob a bank stop calling my hair bad Iâ□□m so sick of this nonsensical racial disparity So donâ□□t call it good unless your hair is known for donating to charity As much as has been raped away from our people How can you expect me to treat their imprint on your language As anything less than equal Let there be no confusion Let there be no hesitation This is not a promotion of ignorance This is a linguistic celebration Thatâ□□s why I put”tri-lingual”on my last job application I can help to diversify your consumer market is all I wanted them to know And when they call me for the interview Iâ□□ll be more than happy to show that I can say : â□□Whatâ□□s goodâ□□ â□□Whatagwanâ□□ And of course â□Śâ□□Helloâ□□ Because Iâ□□m â□□articulateâ□□ Thank you.

Text Sample 115: NEG Dim 1 – 2014 – Score 1.00 – 2014/t001818.txt

So it ’ s 2006. My friend Harold Ford calls me. He ’ s running for U. S. Senate in Tennessee, and he says,”Mellody, I desperately need some national press. Do you have any ideas?”So I had an idea. I called a friend who was in New York at one of the most successful media companies in the world, and she said,”Why don ’ t we host an editorial board lunch for Harold? You come with him.”Harold and I arrive in New York. We are in our best suits. We look like shiny new pennies. And we get to the receptionist, and we say,”We ’ re here for the lunch.”She motions for **us** to follow her. We walk through a series of corridors, and all of a sudden we find ourselves in a stark room, at which point she looks at **us** and she says,”Where are your uniforms?”Just as this happens, my friend rushes in. The blood drains from her face. There are literally no words, right? And I look at her, and I say,”Now, don ’ t you think we need more than one black person in the U. S. Senate?”Now Harold and I — — we still laugh about that story, and in many ways, the moment caught me off guard, but deep, deep down inside, I actually wasn ’ t surprised. And I wasn ’ t surprised because of something my mother taught me about 30 years before. You see, my mother was ruthlessly realistic. I remember one day coming home from a birthday party where I was the only black kid invited, and instead of asking me the normal motherly questions like,”Did you have fun?”or”How was the cake?”my mother looked at me and she said,”How did they treat you?”I was seven. I did not understand. I mean, why would anyone treat me differently? But she knew. And she looked me right in the eye and she said,”They will not always treat you well.”Now, race is one of those topics in America that makes people extraordinarily uncomfortable. You bring it up at a dinner party or in a workplace environment, it is literally the conversational equivalent of touching the third rail. There is shock, followed by a long silence. And even coming here today, I told some friends and colleagues that I planned to talk about race, and they warned me, they told me, don ’ t do it, that there ’ d be huge risks in me talking about this topic, that people might think I ’ m a militant black woman and I would ruin my career. And I have to tell you, I actually for a moment was a bit afraid. Then I realized, the first step to solving any problem is to not hide from it, and the first step to any form of action is awareness. And so I decided to actually talk about race. And I decided that if I came here and shared with you some of my experiences, that maybe we could all be a little less anxious and a little more bold in our conversations about race. Now I know there are people out there who will say that the election of Barack Obama meant that it was the end of racial discrimination for all eternity, right? But I work in the investment business, and we have a saying : The numbers do not lie. And here, there are significant, quantifiable racial disparities that cannot be ignored, in household wealth, household income, job opportunities, healthcare. One example from corporate America : Even though white men make up just 30 percent of the U. S. population, they hold 70 percent of all corporate board seats. Of the Fortune 250, there are only seven CEOs that are minorities, and of the thousands of publicly traded companies today, thousands, only two are chaired by black women, and you ’ re looking at one of them, the same one who, not too long ago, was nearly mistaken for kitchen help. So that is a fact. Now I have this thought experiment that I play with myself, when I say, imagine if I walked you into a room and it was of a major corporation, like ExxonMobil, and every single person around the boardroom were black, you would think that were weird. But if I walked you into a Fortune 500 company, and everyone around the table is a white male, when will it be that we think that ’ s weird too? And I know how we got here. I know how we got here. You know, there was institutionalized, at one time legalized, discrimination in our country. There ’ s no question about it. But still, as I grapple with this issue, my mother ’ s question hangs in the air for me : How did they treat you? Now, I do not raise this issue to complain or in any way to elicit any kind of sympathy. I have succeeded in my life beyond my wildest expectations, and I have been treated well by people of all races more often than I have not. I tell the uniform story because it happened. I cite those statistics around corporate board diversity because they are real, and I stand here today talking about this issue of racial discrimination because I believe it threatens to rob another generation of all the opportunities that all of **us** want for all of our children, no matter what their color or where they come from. And I think it also threatens to hold **back** businesses. You see, researchers have coined this term”color blindness”to describe a learned behavior where we pretend that we don ’ t notice race. If you happen to be surrounded by a bunch of people who look like you, that ’ s purely accidental. Now, color blindness, in my view, doesn ’ t mean that there ’ s no racial discrimination, and there ’ s fairness. It doesn ’ t mean that at all. It doesn ’ t ensure it. In my view, color blindness is very dangerous because it means we ’ re ignoring the problem. There was a corporate study that said that, instead of avoiding race, the really smart corporations actually deal with it head on. They actually recognize that embracing diversity means recognizing all races, including the majority one. But I ’ ll be the first one to tell you, this subject matter can be hard, awkward, uncomfortable — but that ’ s kind of the point. In the spirit of debunking racial stereotypes, the one that black people don ’ t like to swim, I ’ m going to tell you how much I love to swim. I love to swim so much that as an adult, I swim with a coach. And one day my coach had me do a drill where I had to swim to one end of a 25-meter pool without taking a breath. And every single time I failed, I had to start over. And I failed a lot. By the end, I got it, but when I got out of the pool, I was exasperated and tired and annoyed, and I said,”Why are we doing breath-holding exercises?”And my coach looked me at me, and he said,”Mellody, that was not a breath- holding exercise. That drill was to make you comfortable being uncomfortable, because that ’ s how most of **us** spend our days.”If we can learn to deal with our discomfort, and just relax into it, we ’ ll have a better life. So I think it ’ s time for **us** to be comfortable with the uncomfortable conversation about race : black, white, Asian, Hispanic, male, female, all of **us**, if we truly believe in equal rights and equal opportunity in America, I think we have to have real conversations about this issue. We cannot afford to be color blind. We have to be color brave. We have to be willing, as teachers and parents and entrepreneurs and scientists, we have to be willing to have proactive conversations about race with honesty and understanding and courage, not because it ’ s the right thing to do, but because it ’ s the smart thing to do, because our businesses and our products and our science, our research, all of that will be better with greater diversity. Now, my favorite example of color bravery is a guy named John Skipper. He runs ESPN. He ’ s a North Carolina native, quintessential Southern gentleman, white. He joined ESPN, which already had a culture of inclusion and diversity, but he took it up a notch. He demanded that every open position have a diverse slate of candidates. Now he says the senior people in the beginning bristled, and they would come to him and say,”Do you want me to hire the minority, or do you want me to hire the best person for the job?”And Skipper says his answers were always the same :”Yes.”And by saying yes to diversity, I honestly believe that ESPN is the most valuable cable franchise in the world. I think that ’ s a part of the secret sauce. Now I can tell you, in my own industry, at Ariel Investments, we actually view our diversity as a competitive advantage, and that advantage can extend way beyond business. There ’ s a guy named Scott Page at the University of Michigan. He is the first person to develop a mathematical calculation for diversity. He says, if you ’ re trying to solve a really hard problem, really hard, that you should have a diverse group of people, including those with diverse intellects. The example that he gives is the smallpox epidemic. When it was ravaging Europe, they brought together all these scientists, and they were stumped. And the beginnings of the cure to the disease came from the most unlikely source, a dairy farmer who noticed that the milkmaids were not getting smallpox. And the smallpox vaccination is bovine-based because of that dairy farmer. Now I ’ m sure you ’ re sitting here and you ’ re saying, I don ’ t run a cable company, I don ’ t run an investment firm, I am not a dairy farmer. What can I do? And I ’ m telling you, you can be color brave. If you ’ re part of a hiring process or an admissions process, you can be color brave. If you are trying to solve a really hard problem, you can speak up and be color brave. Now I know people will say, but that doesn ’ t add up to a lot, but I ’ m actually asking you to do something really simple : observe your environment, at work, at school, at home. I ’ m asking you to look at the people around you purposefully and intentionally. Invite people into your life who don ’ t look like you, don ’ t think like you, don ’ t act like you, don ’ t come from where you come from, and you might find that they will challenge your assumptions and make you grow as a person. You might get powerful new insights from these individuals, or, like my husband, who happens to be white, you might learn that black people, men, women, children, we use body lotion every single day. Now, I also think that this is very important so that the next generation really understands that this progress will help them, because they ’ re expecting **us** to be great role models. Now, I told you, my mother, she was ruthlessly realistic. She was an unbelievable role model. She was the kind of person who got to be the way she was because she was a single mom with six kids in Chicago. She was in the real estate business, where she worked extraordinarily hard but oftentimes had a hard time making ends meet. And that meant sometimes we got our phone disconnected, or our lights turned off, or we got evicted. When we got evicted, sometimes we lived in these small apartments that she owned, sometimes in only one or two rooms, because they weren ' t completed, and we would heat our bathwater on hot plates. But she never gave up hope, ever, and she never allowed **us** to give up hope either. This brutal pragmatism that she had, I mean, I was four and she told me,"Mommy is Santa."She was this brutal pragmatism. She taught me so many lessons, but the most important lesson was that every single day she told me,"Mellody, you can be anything."And because of those words, I would wake up at the crack of dawn, and because of those words, I would love school more than anything, and because of those words, when I was on a bus going to school, I dreamed the biggest dreams. And it ' s because of those words that I stand here right now full of passion, asking you to be brave for the kids who are dreaming those dreams today. You see, I want them to look at a CEO on television and say,"I can be like her,"or,"He looks like me."And I want them to know that anything is possible, that they can achieve the highest level that they ever imagined, that they will be welcome in any corporate boardroom, or they can lead any company. You see this idea of being the land of the free and the home of the brave, it ' s woven into the fabric of America. America, when we have a challenge, we take it head on, we don ' t shrink away from it. We take a stand. We show courage. So right now, what I ' m asking you to do, I ' m asking you to show courage. I ' m asking you to be bold. As business leaders, I ' m asking you not to leave anything on the table. As citizens, I ' m asking you not to leave any child behind. I ' m asking you not to be color blind, but to be color brave, so that every child knows that their future matters and their dreams are possible. Thank you. Thank you. Thanks. Thanks.

Text Sample 116: NEG Dim 1 – 2014 – Score 1.00 – 2014/t001843.txt

So I ' ve been thinking about the difference between the résumé virtues and the eulogy virtues. The résumé virtues are the ones you put on your résumé, which are the skills you bring to the marketplace. The eulogy virtues are the ones that get mentioned in the eulogy, which are deeper : who are you, in your depth, what is the nature of your relationships, are you bold, loving, dependable, consistency? And most of **us**, including me, would say that the eulogy virtues are the more important of the virtues. But at least in my case, are they the ones that I think about the most? And the answer is no. So I ' ve been thinking about that problem, and a thinker who has helped me think about it is a guy named Joseph Soloveitchik, who was a rabbi who wrote a book called"The Lonely Man Of Faith"in 1965. Soloveitchik said there are two sides of our natures, which he called Adam I and Adam II. Adam I is the worldly, ambitious, external side of our nature. He wants to build, create, create companies, create innovation. Adam II is the humble side of our nature. Adam II wants not only to do good but to be good, to live in a way internally that honors God, creation and our possibilities. Adam I wants to conquer the world. Adam II wants to hear a calling and obey the world. Adam I savors accomplishment. Adam II savors inner consistency and strength. Adam I asks how things work. Adam II asks why we ' re here. Adam I ' s motto is"success."Adam II ' s motto is"love, redemption and return."And Soloveitchik argued that these two sides of our nature are at war with each other. We live in perpetual self-confrontation between the external success and the internal value. And the tricky thing, I ’ d say, about these two sides of our nature is they work by different logics. The external logic is an economic logic : input leads to output, risk leads to reward. The internal side of our nature is a moral logic and often an inverse logic. You have to give to receive. You have to surrender to something outside yourself to gain strength within yourself. You have to conquer the desire to get what you want. In order to fulfill yourself, you have to forget yourself. In order to find yourself, you have to lose yourself. We happen to live in a society that favors Adam I, and often neglects Adam II. And the problem is, that turns you into a shrewd animal who treats life as a game, and you become a cold, calculating creature who slips into a sort of mediocrity where you realize there ’ s a difference between your desired self and your actual self. You ’ re not earning the sort of eulogy you want, you hope someone will give to you. You don ’ t have the depth of conviction. You don ’ t have an emotional sonorousness. You don ’ t have commitment to tasks that would take more than a lifetime to commit. I was reminded of a common response through history of how you build a solid Adam II, how you build a depth of character. Through history, people have gone **back** into their own pasts, sometimes to a precious time in their life, to their childhood, and often, the mind gravitates in the past to a moment of shame, some sin committed, some act of selfishness, an act of omission, of shallowness, the sin of anger, the sin of self-pity, trying to be a people- pleaser, a lack of courage. Adam I is built by building on your strengths. Adam II is built by fighting your weaknesses. You go into yourself, you find the sin which you ’ ve committed over and again through your life, your signature sin out of which the others emerge, and you fight that sin and you wrestle with that sin, and out of that wrestling, that suffering, then a depth of character is constructed. And we ’ re often not taught to recognize the sin in ourselves, in that we ’ re not taught in this culture how to wrestle with it, how to confront it, and how to combat it. We live in a culture with an Adam I mentality where we ’ re inarticulate about Adam II. Finally, Reinhold Niebuhr summed up the confrontation, the fully lived Adam I and Adam II life, this way :”Nothing that is worth doing can be achieved in our lifetime ; therefore we must be saved by hope. Nothing which is true or beautiful or good makes complete sense in any immediate context of history ; therefore we must be saved by faith. Nothing we do, however virtuous, can be accomplished alone ; therefore we must be saved by love. No virtuous act is quite as virtuous from the standpoint of our friend or foe as from our own standpoint. Therefore we must be saved by that final form of love, which is forgiveness.”Thanks.

Text Sample 117: NEG Dim 1 – 2013 – Score -1.00 – 2013/t002046.txt

Allow me to start this talk with a question to everyone. You know that all over the world, people fight for their freedom, fight for their rights. Some battle oppressive governments. Others battle oppressive societies. Which battle do you think is harder? Allow me to try to answer this question in the few coming minutes. Let me take you **back** two years ago in my life. It was the bedtime of my son, Aboody. He was five at the time. After finishing his bedtime rituals, he looked at me and he asked a question :”Mommy, are we bad people?”I was shocked.”Why do you say such things, Aboody?”Earlier that day, I noticed some bruises on his face when he came from school. He wouldn ’ t tell me what happened. [But now] he was ready to tell.”Two boys hit me today in school. They told me, ’ We saw your mom on Facebook. You and your mom should be put in jail. ’”I ’ ve never been afraid to tell Aboody anything. I ’ ve been always a proud woman of my achievements. But those questioning eyes of my son were my moment of truth, when it all came together. You see, I ’ m a Saudi woman who had been put in jail for driving a car in a country where women are not supposed to drive cars. Just for giving me his car keys, my own brother was detained twice, and he was harassed to the point he had to quit his job as a geologist, leave the country with his wife and two-year-old son. My father had to sit in a Friday sermon listening to the imam condemning women drivers and calling them prostitutes amongst tons of worshippers, some of them our friends and family of my own father. I was faced with an organized defamation campaign in the local media combined with false rumors shared in family gatherings, in the streets and in schools. It all hit me. It came into focus that those kids did not mean to be rude to my son. They were just influenced by the adults around them. And it wasn ’ t about me, and it wasn ’ t a punishment for taking the wheel and driving a few miles. It was a punishment for daring to challenge the society ’ s rules. But my story goes beyond this moment of truth of mine. Allow me to give you a briefing about my story. It was May, 2011, and I was complaining to a work colleague about the harassments I had to face trying to find a ride **back** home, although I have a car and an international driver ’ s license. As long as I ’ ve known, women in Saudi Arabia have been always complaining about the ban, but it ’ s been 20 years since anyone tried to do anything about it, a whole generation ago. He broke the good / bad news in my face.”But there is no law banning you from driving.”I looked it up, and he was right. There wasn ’ t an actual law in Saudi Arabia. It was just a custom and traditions that are enshrined in rigid religious fatwas and imposed on women. That realization ignited the idea of June 17, where we encouraged women to take the wheel and go drive. It was a few weeks later, we started receiving all these”Man wolves will rape you if you go and drive.”A courageous woman, her name is Najla Hariri, she ’ s a Saudi woman in the city of Jeddah, she drove a car and she announced but she didn ’ t record a video. We needed proof. So I drove. I posted a video on YouTube. And to my surprise, it got hundreds of thousands of views the first day. What happened next, of course? I started receiving threats to be killed, raped, just to stop this campaign. The Saudi authorities remained very quiet. That really creeped **us** out. I was in the campaign with other Saudi women and even men activists. We wanted to know how the authorities would respond on the actual day, June 17, when women go out and drive. So this time I asked my brother to come with me and drive by a police car. It went fast. We were arrested, signed a pledge not to drive again, released. Arrested again, he was sent to **detention** for one day, and I was sent to jail. I wasn ’ t sure why I was sent there, because I didn ’ t face any charges in the interrogation. But what I was sure of was my innocence. I didn ’ t break a law, and I kept my abaya — it ’ s a black cloak we wear in Saudi Arabia before we leave the house — and my fellow prisoners kept asking me to take it off, but I was so sure of my innocence, I kept saying,”No, I ’ m leaving today.”Outside the jail, the whole country went into a frenzy, some attacking me badly, and others supportive and even collecting signatures in a petition to be sent to the king to release me. I was released after nine days. June 17 comes. The streets were packed with police cars and religious police cars, but some hundred brave Saudi women broke the ban and drove that day. None were arrested. We broke the taboo. So I think by now, everyone knows that we can ’ t drive, or women are not allowed to drive, in Saudi Arabia, but maybe few know why. Allow me to help you answer this question. There was this official study that was presented to the Shura Council — it ’ s the consultative council appointed by the king in Saudi Arabia — and it was done by a local professor, a university professor. He claims it ’ s done based on a UNESCO study. And the study states, the percentage of rape, adultery, illegitimate children, even drug abuse, prostitution in countries where women drive is higher than countries where women don ’ t drive. I know, I was like this, I was shocked. I was like,”We are the last country in the world where women don ’ t drive.”So if you look at the map of the world, that only leaves two countries : Saudi Arabia, and the other society is the rest of the world. We started a hashtag on Twitter mocking the study, and it made headlines around the world. [BBC News : ’ End of virginity ’ if women drive, Saudi cleric warns] And only then we realized it ’ s so empowering to mock your oppressor. It strips it away of its strongest weapon : fear. This system is based on ultra-conservative traditions and customs that deal with women as if they are inferior and they need a guardian to protect them, so they need to take permission from this guardian, whether verbal or written, all their lives. We are minors until the day we die. And it becomes worse when it ’ s enshrined in religious fatwas based on wrong interpretation of the sharia law, or the religious laws. What ’ s worst, when they become codified as laws in the system, and when women themselves believe in their inferiority, and they even fight those who try to question these rules. So for me, it wasn ’ t only about these attacks I had to face. It was about living two totally different perceptions of my personality, of my person — the villain **back** in my home country, and the hero outside. Just to tell you, two stories happened in the last two years. One of them is when I was in jail. I ’ m pretty sure when I was in jail, everyone saw titles in the international media something like this during these nine days I was in jail. But in my home country, it was a totally different picture. It was more like this :”Manal al-Sharif faces charges of disturbing public order and inciting women to drive.”I know.”Manal al-Sharif withdraws from the campaign.”Ah, it ’ s okay. This is my favorite.”Manal al-Sharif breaks down and confesses : ’ Foreign forces incited me. ’”And it goes on, even trial and flogging me in public. So it ’ s a totally different picture. I was asked last year to give a speech at the Oslo Freedom Forum. I was surrounded by this love and the support of people around me, and they looked at me as an inspiration. At the same time, I flew **back** to my home country, they hated that speech so much. The way they called it : a betrayal to the Saudi country and the Saudi people, and they even started a hashtag called #OsloTraitor on Twitter. Some 10, 000 tweets were written in that hashtag, while the opposite hashtag, #OsloHero, there was like a handful of tweets written. They even started a poll. More than 13, 000 voters answered this poll : whether they considered me a traitor or not after that speech. Ninety percent said yes, she ’ s a traitor. So it ’ s these two totally different perceptions of my personality. For me, I ’ m a proud Saudi woman, and I do love my country, and because I love my country, I ’ m doing this. Because I believe a society will not be free if the women of that society are not free. Thank you. Thank you, thank you, thank you, thank you. Thank you. But you learn lessons from these things that happen to you. I learned to be always there. The first thing, I got out of jail, of course after I took a shower, I went online, I opened my Twitter account and my Facebook page, and I ’ ve been always very respectful to those people who are opining to me. I would listen to what they say, and I would never defend myself with words only. I would use actions. When they said I should withdraw from the campaign, I filed the first lawsuit against the general directorate of traffic police for not issuing me a driver ’ s license. There are a lot of people also — very big support, like those 3, 000 people who signed the petition to release me. We sent a petition to the Shura Council in favor of lifting the ban on Saudi women, and there were, like, 3, 500 citizens who believed in that and they signed that petition. There were people like that, I just showed some examples, who are amazing, who are believing in women ’ s rights in Saudi Arabia, and trying, and they are also facing a lot of hate because of speaking up and voicing their views. Saudi Arabia today is taking small steps toward enhancing women ’ s rights. The Shura Council that ’ s appointed by the king, by royal decree of King Abdullah, last year there were 30 women assigned to that Council, like 20 percent. 20 percent of the Council. The same time, finally, that Council, after rejecting our petition four times for women driving, they finally accepted it last February. After being sent to jail or sentenced lashing, or sent to a trial, the spokesperson of the traffic police said, we will only issue traffic violation for women drivers. The Grand Mufti, who is the head of the religious establishment in Saudi Arabia, he said, it ’ s not recommended for women to drive. It used to be haram, forbidden, by the previous Grand Mufti. So for me, it ’ s not about only these small steps. It ’ s about women themselves. A friend once asked me, she said,”So when do you think this women driving will happen?”I told her,”Only if women stop asking ’ When? ’ and take action to make it now.”So it ’ s not only about the system, it ’ s also about **us** women to drive our own life, I ’ d say. So I have no clue, really, how I became an activist. And I don ’ t know how I became one now. But all I know, and all I ’ m sure of, in the future when someone asks me my story, I will say,”I ’ m proud to be amongst those women who lifted the ban, fought the ban, and celebrated everyone ’ s freedom.”So the question I started my talk with, who do you think is more difficult to face, oppressive governments or oppressive societies? I hope you find clues to answer that from my speech. Thank you, everyone. Thank you. Thank you.

Text Sample 118: NEG Dim 1 – 2013 – Score 1.00 – 2013/t002009.txt

Clouds. Have you ever noticed how much people moan about them? They get a bad rap. If you think about it, the English language has written into it negative associations towards the clouds. Someone who ’ s down or depressed, they ’ re under a cloud. And when there ’ s bad news in store, there ’ s a cloud on the horizon. I saw an article the other day. It was about problems with computer processing over the Internet.”A cloud over the cloud,”was the headline. It seems like they ’ re everyone ’ s default doom-and-gloom metaphor. But I think they ’ re beautiful, don ’ t you? It ’ s just that their beauty is missed because they ’ re so omnipresent, so, I don ’ t know, commonplace, that people don ’ t notice them. They don ’ t notice the beauty, but they don ’ t even notice the clouds unless they get in the way of the sun. And so people think of clouds as things that get in the way. They think of them as the annoying, frustrating obstructions, and then they rush off and do some blue-sky thinking. But most people, when you stop to ask them, will admit to harboring a strange sort of fondness for clouds. It ’ s like a nostalgic fondness, and they make them think of their youth. Who here can ’ t remember thinking, well, looking and finding shapes in the clouds when they were kids? You know, when you were masters of daydreaming? Aristophanes, the ancient Greek playwright, he described the clouds as the patron godesses of idle fellows two and a half thousand years ago, and you can see what he means. It ’ s just that these days, **us** adults seem reluctant to allow ourselves the indulgence of just allowing our imaginations to drift along in the breeze, and I think that ’ s a pity. I think we should perhaps do a bit more of it. I think we should be a bit more willing, perhaps, to look at the beautiful sight of the sunlight bursting out from behind the clouds and go,”Wait a minute, that ’ s two cats dancing the salsa!”Or seeing the big, white, puffy one up there over the shopping center looks like the Abominable Snowman going to rob a bank. They ’ re like nature ’ s version of those inkblot images, you know, that shrinks used to show their patients in the ’ 60s, and I think if you consider the shapes you see in the clouds, you ’ ll save money on psychoanalysis bills. Let ’ s say you ’ re in love. All right? And you look up and what do you see? Right? Or maybe the opposite. You ’ ve just been dumped by your partner, and everywhere you look, it ’ s kissing couples. Perhaps you ’ re having a moment of existential angst. You know, you ’ re thinking about your own mortality. And there, on the horizon, it ’ s the Grim Reaper. Or maybe you see a topless sunbather. What would that mean? What would that mean? I have no idea. But one thing I do know is this : The bad press that clouds get is totally unfair. I think we should stand up for them, which is why, a few years ago, I started the Cloud Appreciation Society. Tens of thousands of members now in almost 100 countries around the world. And all these photographs that I ’ m showing, they were sent in by members. And the society exists to remind people of this : Clouds are not something to moan about. Far from it. They are, in fact, the most diverse, evocative, poetic aspect of nature. I think, if you live with your head in the clouds every now and then, it helps you keep your feet on the ground. And I want to show you why, with the help of some of my favorite types of clouds. Let ’ s start with this one. It ’ s the cirrus cloud, named after the Latin for a lock of hair. It ’ s composed entirely of ice crystals cascading from the upper reaches of the troposphere, and as these ice crystals fall, they pass through different layers with different winds and they speed up and slow down, giving the cloud these brush-stroked appearances, these brush-stroke forms known as fall streaks. And these winds up there can be very, very fierce. They can be 200 miles an hour, 300 miles an hour. These clouds are bombing along, but from all the way down here, they appear to be moving gracefully, slowly, like most clouds. And so to tune into the clouds is to slow down, to calm down. It ’ s like a bit of everyday meditation. Those are common clouds. What about rarer ones, like the lenticularis, the UFO-shaped lenticularis cloud? These clouds form in the region of mountains. When the wind passes, rises to pass over the mountain, it can take on a wave-like path in the lee of the peak, with these clouds hovering at the crest of these invisible standing waves of air, these flying saucer-like forms, and some of the early black-and-white UFO photos are in fact lenticularis clouds. It ’ s true. A little rarer are the fallstreak holes. All right? This is when a layer is made up of very, very cold water droplets, and in one region they start to freeze, and this freezing sets off a chain reaction which spreads outwards with the ice crystals cascading and falling down below, giving the appearance of jellyfish tendrils down below. Rarer still, the Kelvin—Helmholtz cloud. Not a very snappy name. Needs a rebrand. This looks like a series of breaking waves, and it ’ s caused by shearing winds — the wind above the cloud layer and below the cloud layer differ significantly, and in the middle, in between, you get this undulating of the air, and if the difference in those speeds is just right, the tops of the undulations curl over in these beautiful breaking wave-like vortices. All right. Those are rarer clouds than the cirrus, but they ’ re not that rare. If you look up, and you pay attention to the sky, you ’ ll see them sooner or later, maybe not quite as dramatic as these, but you ’ ll see them. And you ’ ll see them around where you live. Clouds are the most egalitarian of nature ’ s displays, because we all have a good, fantastic view of the sky. And these clouds, these rarer clouds, remind **us** that the exotic can be found in the everyday. Nothing is more nourishing, more stimulating to an active, inquiring mind than being surprised, being amazed. It ’ s why we ’ re all here at TED, right? But you don ’ t need to rush off away from the familiar, across the world to be surprised. You just need to step outside, pay attention to what ’ s so commonplace, so everyday, so mundane that everybody else misses it. One cloud that people rarely miss is this one : the cumulonimbus storm cloud. It ' s what ' s produces thunder and lightning and hail. These clouds spread out at the top in this enormous anvil fashion stretching 10 miles up into the atmosphere. They are an expression of the majestic architecture of our atmosphere. But from down below, they are the embodiment of the powerful, elemental force and power that drives our atmosphere. To be there is to be connected in the driving rain and the hail, to feel connected to our atmosphere. It ' s to be reminded that we are creatures that inhabit this ocean of air. We don ' t live beneath the sky. We live within it. And that connection, that visceral connection to our atmosphere feels to me like an antidote. It ' s an antidote to the growing tendency we have to feel that we can really ever experience life by watching it on a computer screen, you know, when we ' re in a wi-fi zone. But the one cloud that best expresses why cloudspotting is more valuable today than ever is this one, the cumulus cloud. Right? It forms on a sunny day. If you close your eyes and think of a cloud, it ' s probably one of these that comes to mind. All those cloud shapes at the beginning, those were cumulus clouds. The sharp, crisp outlines of this formation make it the best one for finding shapes in. And it reminds **us** of the aimless nature of cloudspotting, what an aimless activity it is. You ' re not going to change the world by lying on your **back** and gazing up at the sky, are you? It ' s pointless. It ' s a pointless activity, which is precisely why it ' s so important. The digital world conspires to make **us** feel eternally busy, perpetually busy. You know, when you ' re not dealing with the traditional pressures of earning a living and putting food on the table, raising a family, writing thank you letters, you have to now contend with answering a mountain of unanswered emails, updating a Facebook page, feeding your Twitter feed. And cloudspotting legitimizes doing nothing. And sometimes we need — Sometimes we need excuses to do nothing. We need to be reminded by these patron goddesses of idle fellows that slowing down and being in the present, not thinking about what you ' ve got to do and what you should have done, but just being here, letting your imagination lift from the everyday concerns down here and just being in the present, it ' s good for you, and it ' s good for the way you feel. It ' s good for your ideas. It ' s good for your creativity. It ' s good for your soul. So keep looking up, marvel at the ephemeral beauty, and always remember to live life with your head in the clouds. Thank you very much.

Text Sample 119: NEG Dim 1 – 2013 – Score 1.00 – 2013/t002086.txt

”Don ' t talk to strangers.”You have heard that phrase uttered by your friends, family, schools and the media for decades. It ' s a norm. It ' s a social norm. But it ' s a special kind of social norm, because it ' s a social norm that wants to tell **us** who we can relate to and who we shouldn ' t relate to.”Don ' t talk to strangers”says,”Stay from anyone who ' s not familiar to you. Stick with the people you know. Stick with people like you.”How appealing is that? It ' s not really what we do, is it, when we ' re at our best? When we ' re at our best, we reach out to people who are not like **us**, because when we do that, we learn from people who are not like **us**. My phrase for this value of being with”not like **us**”is”strangeness,”and my point is that in today ' s digitally intensive world, strangers are quite frankly not the point. The point that we should be worried about is, how much strangeness are we getting? Why strangeness? Because our social relations are increasingly mediated by data, and data turns our social relations into digital relations, and that means that our digital relations now depend extraordinarily on technology to bring to them a sense of robustness, a sense of discovery, a sense of surprise and unpredictability. Why not strangers? Because strangers are part of a world of really rigid boundaries. They belong to a world of people I know versus people I don ' t know, and in the context of my digital relations, I ' m already doing things with people I don ' t know. The question isn ' t whether or not I know you. The question is, what can I do with you? What can I learn with you? What can we do together that benefits **us** both? I spend a lot of time thinking about how the social landscape is changing, how new technologies create new constraints and new opportunities for people. The most important changes facing **us** today have to do with data and what data is doing to shape the kinds of digital relations that will be possible for **us** in the future. The economies of the future depend on that. Our social lives in the future depend on that. The threat to worry about isn ' t strangers. The threat to worry about is whether or not we ' re getting our fair share of strangeness. Now, 20th-century psychologists and sociologists were thinking about strangers, but they weren ' t thinking so dynamically about human relations, and they were thinking about strangers in the context of influencing practices. Stanley Milgram from the ' 60s and ' 70s, the creator of the small-world experiments, which became later popularized as six degrees of separation, made the point that any two arbitrarily selected people were likely connected from between five to seven intermediary steps. His point was that strangers are out there. We can reach them. There are paths that enable **us** to reach them. Mark Granovetter, Stanford sociologist, in 1973 in his seminal essay"The Strength of Weak Ties,"made the point that these weak ties that are a part of our networks, these strangers, are actually more effective at diffusing information to **us** than are our strong ties, the people closest to **us**. He makes an additional indictment of our strong ties when he says that these people who are so close to **us**, these strong ties in our lives, actually have a homogenizing effect on **us**. They produce sameness. My colleagues and I at Intel have spent the last few years looking at the ways in which digital platforms are reshaping our everyday lives, what kinds of new routines are possible. We ' ve been looking specifically at the kinds of digital platforms that have enabled **us** to take our possessions, those things that used to be very restricted to **us** and to our friends in our houses, and to make them available to people we don ' t know. Whether it ' s our clothes, whether it ' s our cars, whether it ' s our bikes, whether it ' s our books or music, we are able to take our possessions now and make them available to people we ' ve never met. And we concluded a very important insight, which was that as people ' s relationships to the things in their lives change, so do their relations with other people. And yet recommendation system after recommendation system continues to miss the boat. It continues to try to predict what I need based on some past characterization of who I am, of what I ' ve already done. Security technology after security technology continues to design data protection in terms of threats and attacks, keeping me locked into really rigid kinds of relations. Categories like"friends"and"family"and"contacts"and"colleagues"don ' t tell me anything about my actual relations. A more effective way to think about my relations might be in terms of closeness and distance, where at any given point in time, with any single person, I am both close and distant from that individual, all as a function of what I need to do right now. People aren ' t close or distant. People are always a combination of the two, and that combination is constantly changing. What if technologies could intervene to disrupt the balance of certain kinds of relationships? What if technologies could intervene to help me find the person that I need right now? Strangeness is that calibration of closeness and distance that enables me to find the people that I need right now, that enables me to find the sources of intimacy, of discovery, and of inspiration that I need right now. Strangeness is not about meeting strangers. It simply makes the point that we need to disrupt our zones of familiarity. So jogging those zones of familiarity is one way to think about strangeness, and it ' s a problem faced not just by individuals today, but also by organizations, organizations that are trying to embrace massively new opportunities. Whether you ' re a political party insisting to your detriment on a very rigid notion of who belongs and who does not, whether you ' re the government protecting social institutions like marriage and restricting access of those institutions to the few, whether you ' re a teenager in her bedroom who ' s trying to jostle her relations with her parents, strangeness is a way to think about how we pave the way to new kinds of relations. We have to change the norms. We have to change the norms in order to enable new kinds of technologies as a basis for new kinds of businesses. What interesting questions lie ahead for **us** in this world of no strangers? How might we think differently about our relations with people? How might we think differently about our relations with distributed groups of people? How might we think differently about our relations with technologies, things that effectively become social participants in their own right? The range of digital relations is extraordinary. In the context of this broad range of digital relations, safely seeking strangeness might very well be a new basis for that innovation. Thank you.

Text Sample 120: NEG Dim 1 – 2015 – Score 1.00 – 2015/t001461.txt

Religion is more than belief. It ' s power, and it ' s influence. And that influence affects all of **us**, every day, regardless of your own belief. Despite the enormous influence of religion on the world today, we hold them to a different standard of scrutiny and accountability than any other sector of our society. For example, if there were a multinational organization, government or corporation today that said no female could be on a leadership board, not one woman could have a decision-making authority, not one woman could handle any financial matter, we would have outrage. There would be sanctions. And yet this is a common practice in almost every world religion today. We accept things in our religious lives that we do not accept in our secular lives, and I know this because I ' ve been doing it for three decades. I was the type of girl that fought every form of gender discrimination growing up. I played pickup basketball games with the boys and inserted myself. I said I was going to be the first female President of the United States. I have been fighting for the Equal Rights Amendment, which has been dead for 40 years. I ' m the first woman in both sides of my family to ever work outside the home and ever receive a higher education. I never accepted being excluded because I was a woman, except in my religion. Throughout all of that time, I was a part of a very patriarchal orthodox Mormon religion. I grew up in an enormously traditional family. I have eight siblings, a stay-at-home mother. My father ' s actually a religious leader in the community. And I grew up in a world believing that my worth and my standing was in keeping these rules that I ' d known my whole life. You get married a virgin, you never drink alcohol, you don ' t smoke, you always do service, you ' re a good kid. Some of the rules we had were strict, but you followed the rules because you loved the people and you loved the religion and you believed. Everything about Mormonism determined what you wore, who you dated, who you married. It determined what underwear we wore. I was the kind of religious where everyone I know donated 10 percent of everything they earned to the church, including myself. From paper routes and babysitting, I donated 10 percent. I was the kind of religious where I heard parents tell children when they ' re leaving on a two-year proselytizing mission that they would rather have them die than return home without honor, having sinned. I was the type and the kind of religious where kids kill themselves every single year because they ' re terrified of coming out to our community as gay. But I was also the kind of religious where it didn ' t matter where in the world I lived, I had friendship, instantaneous mutual aid. This was where I felt safe. This is certainty and clarity about life. I had help raising my little daughter. So that ' s why I accepted without question that only men can lead, and I accepted without question that women can ' t have the spiritual authority of God on the Earth, which we call the priesthood. And I allowed discrepancies between men and women in operating budgets, disciplinary councils, in decision-making capacities, and I gave my religion a free pass because I loved it. Until I stopped, and I realized that I had been allowing myself to be treated as the support staff to the real work of men. And I faced this contradiction in myself, and I joined with other activists in my community. We ' ve been working very, very, very hard for the last decade and more. The first thing we did was raise consciousness. You can ' t change what you can ' t see. We started podcasting, blogging, writing articles. I created lists of hundreds of ways that men and women are unequal in our community. The next thing we did was build advocacy organizations. We tried to do things that were unignorable, like wearing pants to church and trying to attend all- male meetings. These seem like simple things, but to **us**, the organizers, they were enormously costly. We lost relationships. We lost jobs. We got hate mail on a daily basis. We were attacked in social media and national press. We received death threats. We lost standing in our community. Some of **us** got excommunicated. Most of **us** got put in front of a disciplinary council, and were rejected from the communities that we loved because we wanted to make them better, because we believed that they could be. And I began to expect this reaction from my own people. I know what it feels like when you feel like someone ' s trying to change you or criticize you. But what utterly shocked me was throughout all of this work I received equal measures of vitriol from the secular left, the same vehemence as the religious right. And what my secular friends didn ' t realize was that this religious hostility, these phrases of,"Oh, all religious people are crazy or stupid.""Don ' t pay attention to religion.""They ' re going to be homophobic and sexist."What they didn ' t understand was that that type of hostility did not fight religious extremism, it bred religious extremism. Those arguments don ' t work, and I know because I remember someone telling me that I was stupid for being Mormon. And what it caused me to do was defend myself and my people and everything we believe in, because we ' re not stupid. So criticism and hostility doesn ' t work, and I didn ' t listen to these arguments. When I hear these arguments, I still continue to bristle, because I have family and friends. These are my people, and I ' m the first to defend them, but the struggle is real. How do we respect someone ' s religious beliefs while still holding them accountable for the harm or damage that those beliefs may cause others? It ' s a tough question. I still don ' t have a perfect answer. My parents and I have been walking on this tightrope for the last decade. They ' re intelligent people. They ' re lovely people. And let me try to help you understand their perspective. In Mormonism, we believe that after you die, if you keep all the rules and you follow all the rituals, you can be together as a family again. And to my parents, me doing something as simple as having a sleeveless top right now, showing my shoulders, that makes me unworthy. I won ' t be with my family in the eternities. But even more, I had a brother die in a tragic accident at 15, and something as simple as this means we won ' t be together as a family. And to my parents, they cannot understand why something as simple as fashion or women ' s rights would prevent me from seeing my brother again.

And that ’ s the mindset that we ’ re dealing with, and criticism does not change that. And so my parents and I have been walking this tightrope, explaining our sides, respecting one another, but actually invalidating each other ’ s very basic beliefs by the way we live our lives, and it ’ s been difficult. The way that we ’ ve been able to do that is to get past those defensive shells and really see the soft inside of unbelief and belief and try to respect each other while still holding boundaries clear. The other thing that the secular left and the atheists and the orthodox and the religious right, what they all don ’ t understand was why even care about religious activism? I cannot tell you the hundreds of people who have said,”If you don ’ t like religion, just leave.”Why would you try to change it? Because what is taught on the Sabbath leaks into our politics, our health policy, violence around the world. It leaks into education, military, fiscal decision- making. These laws get legally and culturally codified. In fact, my own religion has had an enormous effect on this nation. For example, during Prop 8, my church raised over 22 million dollars to fight same-sex marriage in California. Forty years ago, political historians will say, that if it wasn ’ t for the Mormon opposition to the Equal Rights Amendment, we ’ d have an Equal Rights Amendment in our Constitution today. How many lives did that affect? And we can spend time fighting every single one of these little tiny laws and rules, or we can ask ourselves, why is gender inequality the default around the world? Why is that the assumption? Because religion doesn ’ t just create the roots of morality, it creates the seeds of normality. Religions can liberate or subjugate, they can empower or exploit, they can comfort or destroy, and the people that tip the scales over to the ethical and the moral are often not those in charge. Religions can ’ t be dismissed or ignored. We need to take them seriously. But it ’ s not easy to influence a religion, like we just talked about. But I ’ ll tell you what my people have done. My groups are small, there ’ s hundreds of **us**, but we ’ ve had huge impact. Right now, women ’ s pictures are hanging in the halls next to men for the first time. Women are now allowed to pray in our church-wide meetings, and they never were before in the general conferences. As of last week, in a historic move, three women were invited down to three leadership boards that oversee the entire church. We ’ ve seen perceptual shifts in the Mormon community that allow for talk of gender inequality. We ’ ve opened up space, regardless of being despised, for more conservative women to step in and make real changes, and the words”women”and”the priesthood”can now be uttered in the same sentence. I never had that. My daughter and my nieces are inheriting a religion that I never had, that ’ s more equal — we ’ ve had an effect. It wasn ’ t easy standing in those lines trying to get into those male meetings. There were hundreds of **us**, and one by one, when we got to the door, we were told,”I ’ m sorry, this meeting is just for men,”and we had to step **back** and watch men get into the meeting as young as 12 years old, escorted and walked past **us** as we all stood in line. But not one woman in that line will forget that day, and not one little boy that walked past **us** will forget that day. If we were a multinational corporation or a government, and that had happened, there would be outrage, but we ’ re just a religion. We ’ re all just part of religions. We can ’ t keep looking at religion that way, because it doesn ’ t only affect me, it affects my daughter and all of your daughters and what opportunities they have, what they can wear, who they can love and marry, if they have access to reproductive healthcare. We need to reclaim morality in a secular context that creates ethical scrutiny and accountability for religions all around the world, but we need to do it in a respectful way that breeds cooperation and not extremism. And we can do it through unignorable acts of bravery, standing up for gender equality. It ’ s time that half of the world ’ s population had voice and equality within our world ’ s religions, churches, synagogues, mosques and shrines around the world. I ’ m working on my people. What are you doing for yours?

Text Sample 121: NEG Dim 1 – 2015 – Score 1.00 – 2015/t001488.txt

In the year 1901, a woman called Auguste was taken to a medical asylum in Frankfurt. Auguste was delusional and couldn ’ t remember even the most basic details of her life. Her doctor was called Alois. Alois didn ’ t know how to help Auguste, but he watched over her until, sadly, she passed away in 1906. After she died, Alois performed an autopsy and found strange plaques and tangles in Auguste ’ s brain — the likes of which he ’ d never seen before. Now here ’ s the even more striking thing. If Auguste had instead been alive today, we could offer her no more help than Alois was able to 114 years ago. Alois was Dr. Alois Alzheimer. And Auguste Deter was the first patient to be diagnosed with what we now call Alzheimer ’ s disease. Since 1901, medicine has advanced greatly. We ’ ve discovered antibiotics and vaccines to protect **us** from infections, many treatments for cancer, antiretrovirals for HIV, statins for heart disease and much more. But we ’ ve made essentially no progress at all in treating Alzheimer ’ s disease. I ’ m part of a team of scientists who has been working to find a cure for Alzheimer ’ s for over a decade. So I think about this all the time. Alzheimer ’ s now affects 40 million people worldwide. But by 2050, it will affect 150 million people — which, by the way, will include many of you. If you ’ re hoping to live to be 85 or older, your chance of getting Alzheimer ’ s will be almost one in two. In other words, odds are you ’ ll spend your golden years either suffering from Alzheimer ’ s or helping to look after a friend or loved one with Alzheimer ’ s. Already in the United States alone, Alzheimer ’ s care costs 200 billion dollars every year. One out of every five Medicare dollars get spent on Alzheimer ’ s. It is today the most expensive disease, and costs are projected to increase fivefold by 2050, as the baby boomer generation ages. It may surprise you that, put simply, Alzheimer ’ s is one of the biggest medical and social challenges of our generation. But we ’ ve done relatively little to address it. Today, of the top 10 causes of death worldwide, Alzheimer ’ s is the only one we cannot prevent, cure or even slow down. We understand less about the science of Alzheimer ’ s than other diseases because we ’ ve invested less time and money into researching it. The US government spends 10 times more every year on cancer research than on Alzheimer ’ s despite the fact that Alzheimer ’ s costs **us** more and causes a similar number of deaths each year as cancer. The lack of resources stems from a more fundamental cause : a lack of awareness. Because here ’ s what few people know but everyone should : Alzheimer ’ s is a disease, and we can cure it. For most of the past 114 years, everyone, including scientists, mistakenly confused Alzheimer ’ s with aging. We thought that becoming senile was a normal and inevitable part of getting old. But we only have to look at a picture of a healthy aged brain compared to the brain of an Alzheimer ’ s patient to see the real physical damage caused by this disease. As well as triggering severe loss of memory and mental abilities, the damage to the brain caused by Alzheimer ’ s significantly reduces life expectancy and is always fatal. Remember Dr. Alzheimer found strange plaques and tangles in Auguste ’ s brain a century ago. For almost a century, we didn ’ t know much about these. Today we know they ’ re made from protein molecules. You can imagine a protein molecule as a piece of paper that normally folds into an elaborate piece of origami. There are spots on the paper that are sticky. And when it folds correctly, these sticky bits end up on the inside. But sometimes things go wrong, and some sticky bits are on the outside. This causes the protein molecules to stick to each other, forming clumps that eventually become large plaques and tangles. That ’ s what we see in the brains of Alzheimer ’ s patients. We ’ ve spent the past 10 years at the University of Cambridge trying to understand how this malfunction works. There are many steps, and identifying which step to try to block is complex — like defusing a bomb. Cutting one wire might do nothing. Cutting others might make the bomb explore. We have to find the right step to block, and then create a drug that does it. Until recently, we for the most part have been cutting wires and hoping for the best. But now we ’ ve got together a diverse group of people — medics, biologists, geneticists, chemists, physicists, engineers and mathematicians. And together, we ’ ve managed to identify a critical step in the process and are now testing a new class of drugs which would specifically block this step and stop the disease. Now let me show you some of our latest results. No one outside of our lab has seen these yet. Let ’ s look at some videos of what happened when we tested these new drugs in worms. So these are healthy worms, and you can see they ’ re moving around normally. These worms, on the other hand, have protein molecules sticking together inside them — like humans with Alzheimer ’ s. And you can see they ’ re clearly sick. But if we give our new drugs to these worms at an early stage, then we see that they ’ re healthy, and they live a normal lifespan. This is just an initial positive result, but research like this shows **us** that Alzheimer ’ s is a disease that we can understand and we can cure. After 114 years of waiting, there ’ s finally real hope for what can be achieved in the next 10 or 20 years. But to grow that hope, to finally beat Alzheimer ’ s, we need help. This isn ’ t about scientists like me — it ’ s about you. We need you to raise awareness that Alzheimer ’ s is a disease and that if we try, we can beat it. In the case of other diseases, patients and their families have led the charge for more research and put pressure on governments, the pharmaceutical industry, scientists and regulators. That was essential for advancing treatment for HIV in the late 1980s. Today, we see that same drive to beat cancer. But Alzheimer ’ s patients are often unable to speak up for themselves. And their families, the hidden victims, caring for their loved ones night and day, are often too worn out to go out and advocate for change. So, it really is down to you. Alzheimer ’ s isn ’ t, for the most part, a genetic disease. Everyone with a brain is at risk. Today, there are 40 million patients like Auguste, who can ’ t create the change they need for themselves. Help speak up for them, and help demand a cure. Thank you.

Text Sample 122: NEG Dim 1 – 2015 – Score 1.00 – 2015/t001533.txt

For the past decade, I ’ ve been studying non-state armed groups : armed organizations like terrorists, insurgents or militias. I document what these groups do when they ’ re not shooting. My goal is to better understand these violent actors and to study ways to encourage transition from violent engagement to nonviolent confrontation. I work in the field, in the policy world and in the library. Understanding non-state armed groups is key to solving most ongoing conflict, because war has changed. It used to be a contest between states. No longer. It is now a conflict between states and non-state actors. For example, of the 216 peace agreements signed between 1975 and 2011, 196 of them were between a state and a non-state actor. So we need to understand these groups ; we need to either engage them or defeat them in any conflict resolution process that has to be successful. So how do we do that? We need to know what makes these organizations tick. We know a lot about how they fight, why they fight, but no one looks at what they ’ re doing when they ’ re not fighting. Yet, armed struggle and unarmed politics are related. It is all part of the same organization. We cannot understand these groups, let alone defeat them, if we don ’ t have the full picture. And armed groups today are complex organizations. Take the Lebanese Hezbollah, known for its violent confrontation against Israel. But since its creation in the early 1980s, Hezbollah has also set up a political party, a social-service network, and a military apparatus. Similarly, the Palestinian Hamas, known for its suicide attacks against Israel, also runs the Gaza Strip since 2007. So these groups do way more than just shoot. They multi-task. They set up complex communication machines — radio stations, TV channels, Internet websites and social media strategies. And up here, you have the ISIS magazine, printed in English and published to recruit. Armed groups also invest in complex fund- raising — not looting, but setting up profitable businesses ; for example, construction companies. Now, these activities are keys. They allow these groups to increase their strength, increase their funds, to better recruit and to build their brand. Armed groups also do something else : they build stronger bonds with the population by investing in social services. They build schools, they run hospitals, they set up vocational-training programs or micro-loan programs. Hezbollah offers all of these services and more. Armed groups also seek to win the population over by offering something that the state is not providing : safety and security. The initial rise of the Taliban in war-torn Afghanistan, or even the beginning of the ascent of ISIS, can be understood also by looking at these groups ’ efforts to provide security. Now, unfortunately, in these cases, the provision of security came at an unbearably high price for the population. But in general, providing social services fills a gap, a governance gap left by the government, and allows these groups to increase their strength and their power. For example, the 2006 electoral victory of the Palestinian Hamas cannot be understood without acknowledging the group ’ s social work. Now, this is a really complex picture, yet in the West, when we look at armed groups, we only think of the violent side. But that ’ s not enough to understand these groups ’ strength, strategy or long-term vision. These groups are hybrid. They rise because they fill a gap left by the government, and they emerge to be both armed and political, engage in violent struggle and provide governance. And the more these organizations are complex and sophisticated, the less we can think of them as the opposite of a state. Now, what do you call a group like Hezbollah? They run part of a territory, they administer all their functions, they pick up the garbage, they run the sewage system. Is this a state? Is it a rebel group? Or maybe something else, something different and new? And what about ISIS? The lines are blurred. We live in a world of states, non-states, and in-between, and the more states are weak, like in the Middle East today, the more non-state actors step in and fill that gap. This matters for governments, because to counter these groups, they will have to invest more in non-military tools. Filling that governance gap has to be at the center of any sustainable approach. This also matters very much for peacemaking and peacebuilding. If we better understand armed groups, we will better know what incentives to offer to encourage the transition from violence to nonviolence. So in this new contest between states and non-states, military power can win some battles, but it will not give **us** peace nor stability. To achieve these objectives, what we need is a long-term investment in filling that security gap, in filling that governance gap that allowed these groups to thrive in the first place. Thank you.

Text Sample 123: NEG Dim 1 – 2012 – Score -1.00 – 2012/t002213.txt

Today I have just one request. Please don ’ t tell me I ’ m normal. Now I ’ d like to introduce you to my brothers. Remi is 22, tall and very handsome. He ’ s speechless, but he communicates joy in a way that some of the best orators cannot. Remi knows what love is. He shares it unconditionally and he shares it regardless. He ’ s not greedy. He doesn ’ t see skin color. He doesn ’ t care about religious differences, and get this : He has never told a lie. When he sings songs from our childhood, attempting words that not even I could remember, he reminds me of one thing : how little we know about the mind, and how wonderful the unknown must be. Samuel is 16. He ’ s tall. He ’ s very handsome. He has the most impeccable memory. He has a selective one, though. He doesn ’ t remember if he stole my chocolate bar, but he remembers the year of release for every song on my iPod, conversations we had when he was four, weeing on my arm on the first ever episode of Teletubbies, and Lady Gaga ’ s birthday. Don ’ t they sound incredible? But most people don ’ t agree. And in fact, because their minds don ’ t fit into society ’ s version of normal, they ’ re often bypassed and misunderstood. But what lifted my heart and strengthened my soul was that even though this was the case, although they were not seen as ordinary, this could only mean one thing : that they were extraordinary — autistic and extraordinary. Now, for you who may be less familiar with the term”autism,”it ’ s a complex brain disorder that affects social communication, learning and sometimes physical skills. It manifests in each individual differently, hence why Remi is so different from Sam. And across the world, every 20 minutes, one new person is diagnosed with autism, and although it ’ s one of the fastest-growing developmental disorders in the world, there is no known cause or cure. And I cannot remember the first moment I encountered autism, but I cannot recall a day without it. I was just three years old when my brother came along, and I was so excited that I had a new being in my life. And after a few months went by, I realized that he was different. He screamed a lot. He didn ’ t want to play like the other babies did, and in fact, he didn ’ t seem very interested in me whatsoever. Remi lived and reigned in his own world, with his own rules, and he found pleasure in the smallest things, like lining up cars around the room and staring at the washing machine and eating anything that came in between. And as he grew older, he grew more different, and the differences became more obvious. Yet beyond the tantrums and the frustration and the never-ending hyperactivity was something really unique : a pure and innocent nature, a boy who saw the world without prejudice, a human who had never lied. Extraordinary. Now, I cannot deny that there have been some challenging moments in my family, moments where I ’ ve wished that they were just like me. But I cast my mind **back** to the things that they ’ ve taught me about individuality and communication and love, and I realize that these are things that I wouldn ’ t want to change with normality. Normality overlooks the beauty that differences give **us**, and the fact that we are different doesn ’ t mean that one of **us** is wrong. It just means that there ’ s a different kind of right. And if I could communicate just one thing to Remi and to Sam and to you, it would be that you don ’ t have to be normal. You can be extraordinary. Because autistic or not, the differences that we have — We ’ ve got a gift! Everyone ’ s got a gift inside of **us**, and in all honesty, the pursuit of normality is the ultimate sacrifice of potential. The chance for greatness, for progress and for change dies the moment we try to be like someone else. Please — don ’ t tell me I ’ m normal. Thank you.

Text Sample 124: NEG Dim 1 – 2012 – Score 1.00 – 2012/t002409.txt

The recent debate over copyright laws like SOPA in the United States and the ACTA agreement in Europe has been very emotional. And I think some dispassionate, quantitative reasoning could really bring a great deal to the debate. I ’ d therefore like to propose that we employ, we enlist, the cutting edge field of copyright math whenever we approach this subject. For instance, just recently the Motion Picture Association revealed that our economy loses 58 billion dollars a year to copyright theft. Now rather than just argue about this number, a copyright mathematician will analyze it and he ’ ll soon discover that this money could stretch from this auditorium all the way across Ocean Boulevard to the Westin, and then to Mars...... if we use pennies. Now this is obviously a powerful, some might say dangerously powerful, insight. But it ’ s also a morally important one. Because this isn ’ t just the hypothetical retail value of some pirated movies that we ’ re talking about, but this is actual economic losses. This is the equivalent to the entire American corn crop failing along with all of our fruit crops, as well as wheat, tobacco, rice, sorghum — whatever sorghum is — losing sorghum. But identifying the actual losses to the economy is almost impossible to do unless we use copyright math. Now music revenues are down by about eight billion dollars a year since Napster first came on the scene. So that ’ s a chunk of what we ’ re looking for. But total movie revenues across theaters, home video and pay-per-view are up. And TV, satellite and cable revenues are way up. Other content markets like book publishing and radio are also up. So this small missing chunk here is puzzling. Since the big content markets have grown in line with historic norms, it ’ s not additional growth that piracy has prevented, but copyright math tells **us** it must therefore be foregone growth in a market that has no historic norms — one that didn ’ t exist in the 90 ’ s. What we ’ re looking at here is the insidious cost of ringtone piracy. 50 billion dollars of it a year, which is enough, at 30 seconds a ringtone, that could stretch from here to Neanderthal times. It ’ s true. I have Excel. The movie folks also tell **us** that our economy loses over 370, 000 jobs to content theft, which is quite a lot when you consider that, **back** in ’ 98, the Bureau of Labor Statistics indicated that the motion picture and video industries were employing 270, 000 people. Other data has the music industry at about 45, 000 people. And so the job losses that came with the Internet and all that content theft, have therefore left **us** with negative employment in our content industries. And this is just one of the many mind-blowing statistics that copyright mathematicians have to deal with every day. And some people think that string theory is tough. Now this is a key number from the copyright mathematicians ’ toolkit. It ’ s the precise amount of harm that comes to media companies whenever a single copyrighted song or movie gets pirated. Hollywood and Congress derived this number mathematically **back** when they last sat down to improve copyright damages and made this law. Some people think this number ’ s a little bit large, but copyright mathematicians who are media lobby experts are merely surprised that it doesn ’ t get compounded for inflation every year. Now when this law first passed, the world ’ s hottest MP3 player could hold just 10 songs. And it was a big Christmas hit. Because what little hoodlum wouldn ’ t want a million and a half bucks-worth of stolen goods in his pocket. These days an iPod Classic can hold 40, 000 songs, which is to say eight billion dollars-worth of stolen media. Or about 75, 000 jobs. Now you might find copyright math strange, but that ’ s because it ’ s a field that ’ s best left to experts. So that ’ s it for now. I hope you ’ ll join me next time when I will be making an equally scientific and fact-based inquiry into the cost of alien music piracy to he American economy. Thank you very much. Thank you.

Text Sample 125: NEG Dim 1 – 2012 – Score 1.00 – 2012/t002000.txt

I ’ d like to talk to you today about a whole new way to think about sexual activity and sexuality education, by comparison. If you talk to someone today in America about sexual activity, you ’ ll find pretty soon you ’ re not just talking about sexual activity. You ’ re also talking about baseball. Because baseball is the dominant cultural metaphor that Americans use to think about and talk about sexual activity, and we know that because there ’ s all this language in English that seems to be talking about baseball but that ’ s really talking about sexual activity. So, for example, you can be a pitcher or a catcher, and that corresponds to whether you perform a sexual act or receive a sexual act. Of course, there are the bases, which refer to specific sexual activities that happen in a very specific order, ultimately resulting in scoring a run or hitting a home run, which is usually having vaginal intercourse to the point of orgasm, at least for the guy. You can strike out, which means you don ’ t get to have any sexual activity. And if you ’ re a benchwarmer, you might be a virgin or somebody who for whatever reason isn ’ t in the game, maybe because of your age or because of your ability or because of your skillset. A bat ’ s a penis, and a nappy dugout is a vulva, or a vagina. A glove or a catcher ’ s mitt is a condom. A switch-hitter is a bisexual person, and we gay and lesbian folks play for the other team. And then there ’ s this one :”if there ’ s grass on the field, play ball.”And that usually refers to if a young person, specifically often a young woman, is old enough to have pubic hair, she ’ s old enough to have sex with. This baseball model is incredibly problematic. It ’ s sexist. It ’ s heterosexist. It ’ s competitive. It ’ s goal-directed. And it can ’ t result in healthy sexuality developing in young people or in adults. So we need a new model. I ’ m here today to offer you that new model. And it ’ s based on pizza. Now pizza is something that is universally understood and that most people associate with a positive experience. So let ’ s do this. Let ’ s take baseball and pizza and compare it when talking about three aspects of sexual activity : the trigger for sexual activity, what happens during sexual activity, and the expected outcome of sexual activity. So when do you play baseball? You play baseball when it ’ s baseball season and when there ’ s a game on the schedule. It ’ s not exactly your choice. So if it ’ s prom night or a wedding night or at a party or if our parents aren ’ t home, hey, it ’ s just batter up. Can you imagine saying to your coach,”Uh, I ’ m not really feeling it today, I think I ’ ll sit this game out.”That ’ s just not the way it happens. And when you get together to play baseball, immediately you ’ re with two opposing teams, one playing offense, one playing defense, somebody ’ s trying to move deeper into the field. That ’ s usually a sign to the boy. Somebody ’ s trying to defend people moving into the field. That ’ s often given to the girl. It ’ s competitive. We ’ re not playing with each other. We ’ re playing against each other. And when you show up to play baseball, nobody needs to talk about what we ’ re going to do or how this baseball game might be good for **us**. Everybody knows the rules. You just take your position and play the game. But when do you have pizza? Well, you have pizza when you ’ re hungry for pizza. It starts with an internal sense, an internal desire, or a need.”Huh. I could go for some pizza.”And because it ’ s an internal desire, we actually have some sense of control over that. I could decide that I ’ m hungry but know that it ’ s not a great time to eat. And then when we get together with someone for pizza, we ’ re not competing with them, we ’ re looking for an experience that both of **us** will share that ’ s satisfying for both of **us**, and when you get together for pizza with somebody, what ’ s the first thing you do? You talk about it. You talk about what you want. You talk about what you like. You may even negotiate it.”How do you feel about pepperoni?””Not so much, I ’ m kind of a mushroom guy myself.””Well, maybe we can go half and half.”And even if you ’ ve had pizza with somebody for a very long time, don ’ t you still say things like,”Should we get the usual?””Or maybe something a little more adventurous?”Okay, so when you ’ re playing baseball, so if we talk about during sexual activity, when you ’ re playing baseball, you ’ re just supposed to round the bases in the proper order one at a time. You can ’ t hit the ball and run to right field. That doesn ’ t work. And you also can ’ t get to second base and say,”I like it here. I ’ m going to stay here.”No. And also, of course, with baseball, there ’ s, like, the specific equipment and a specific skill set. Not everybody can play baseball. It ’ s pretty exclusive. Okay, but what about pizza? When we ’ re trying to figure out what ’ s good for pizza, isn ’ t it all about what ’ s our pleasure? There are a million different kinds of pizza. There ’ s a million different toppings. There ’ s a million different ways to eat pizza. And none of them are wrong. They ’ re different. And in this case, difference is good, because that ’ s going to increase the chance that we ’ re having a satisfying experience. And lastly, what ’ s the expected outcome of baseball? Well, in baseball, you play to win. You score as many runs as you can. There ’ s always a winner in baseball, and that means there ’ s always a loser in baseball. But what about pizza? Well, in pizza, we ’ re not really — there ’ s no winning. How do you win pizza? You don ’ t. But you do look for,”Are we satisfied?”And sometimes that can be different amounts over different times or with different people or on different days. And we get to decide when we feel satisfied. If we ’ re still hungry, we might have some more. If you eat too much, though, you just feel gross. So what if we could take this pizza model and overlay it on top of sexuality education? A lot of sexuality education that happens today is so influenced by the baseball model, and it sets up education that can ’ t help but produce unhealthy sexuality in young people. And those young people become older people. But if we could create sexuality education that was more like pizza, we could create education that invites people to think about their own desires, to make deliberate decisions about what they want, to talk about it with their partners, and to ultimately look for not some external outcome but for what feels satisfying, and we get to decide that. You may have noticed in the baseball and pizza comparison, under the baseball, it ’ s all commands. They ’ re all exclamation points. But under the pizza model, they ’ re questions. And who gets to answer those questions? You do. I do. So remember, when we ’ re thinking about sexuality education and sexual activity, baseball, you ’ re out. Pizza is the way to think about healthy, satisfying sexual activity, and good, comprehensive sexuality education. Thank you very much for your time.

Text Sample 126: NEG Dim 1 – 1984 – Score 8.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ’ ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a computer and really address the qualities of the human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, the intelligence of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ’ ve been able to collect, which I think illustrate this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a slide taken from a

TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ' s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ' s the distance you should sit away from the TV set. Now we ' ve put people 18 inches in front of a TV, and all the artifacts that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow mask, the scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ' m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ' m very interested in touch-sensitive displays. High-tech, high-touch. Isn ' t that what some of you said? It ' s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ' re not : it ' s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on the market these systems that have just a few light emitting diodes on the side and are very low resolution, it ' s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ' t have to pick them up, and people don ' t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ' re typing — what you have — you want to now put something — first of all, you ' ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ' ve got to sort of find that mouse. Then you find the mouse, and you ' re going to have to wiggle it a little bit to see where the cursor is on the screen. And then when you finally see where it is, then you ' ve got to move it to get the cursor over there, and then —"Bang"— you ' ve got to hit a button or do whatever. That ' s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ' m trying to do is just illustrate the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of the computer field, is when you have a bug that you can ' t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the substrate, which it usually is. So we found that that actually was a feature in the sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ’ s pressure-sensitive. And it ’ s pressure-sensitive to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ’ t figure out how to come in the other direction. But let me get rid of the slide, and let ’ s see if this comes on. OK. So there is the pressure-sensitive display in operation. The person ’ s just, if you will, pushing on the screen to make a curve. But this is the interesting part. I want to stop it for a second because the movie is very badly made. And the particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an anvil on a fuzzy rug and the other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across the screen, and yet you touch the ping-pong ball very lightly and it just scoots across the screen. And what you can do — oops, I didn ’ t mean to do that — what you can do is actually feed **back** to the user the feeling of the physical properties. So again, they don ’ t have to be weight ; they could be a general trying to move troops, and he ’ s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the interface there are physical properties in that transducer — in this case it ’ s pressure and touches — that allow you to present things to the user that you could never present before. So it ’ s not simply looking at the quality or, if you will, the luxury of that interface, but it ’ s actually looking at the idea of presenting things that previously couldn ’ t be presented before. I want to move on to another example, which is one of a different sort, where we ’ re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ’ re going to take this book, if you will, and it ’ s going to come alive. You ’ re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are the experts in monologue making : you make a film and it has a well-formed beginning, middle and end, and in some sense the art of it is that. And you then say,”There ’ s an opportunity for making conversational movies.”Well, what does that mean? And it sort of nibbles at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ’ ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the movie has information about itself. So it knows, or at least there is computer-readable information in the medium itself. It ’ s just not a static movie frame. That ’ s one thing. The other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the cookbook, the”Larousse Gastronomique.”And I think I use the example all too often, but it ’ s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to the end of the recipe and it says,”Cook until done.”Now, that would be, if you will, the top green track, which doesn ’ t mean too much. But you might have to elaborate for me or for somebody who isn ’ t an expert, and say,”Cook at 380 degrees for 45 minutes.”And then for a real beginner, you would go down even further and elaborate more — say,”Open the oven, preheat, wait for the light to go out, open the door, don ’ t leave it open too long, put the penguin in and shut the door...”whatever. And that ’ s a much more elaborate one than you dribble **back**. That ' s one kind of use of random access. And the other is where you want to explain the same thing in different ways. If you ' re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ' s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ' s a rather boring book, but I ' m afraid sometimes you have to do boring books because your sponsors aren ' t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ' t even know what vintage the transmission is, but let me just show you very quickly some of it, and we ' ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of the transmission, and as you rub your finger across the transmission it highlights the various parts. Narrator : When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, superimposed over the illustration. NN : This is about the oil pan, or the oil filter and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get **back** to the illustration, but in this case it ' s not a single frame, but it ' s actually a movie of someone coming into the frame and doing the repair that ' s described in the text. The two-headed slider is a speed control that allows me to watch the movie at various speeds, in forward or reverse. And the movie is displayed as a full frame movie. I can go **back** to the beginning... and play the movie at full speed. Here ' s another step-by-step procedure, only in this case — NN : Okay, this movie is... Everybody ' s heard of sound-sync movies — this is text-sync movies, so as the movie plays, the text gets highlighted. We highlight the text as we go through the movie. Repairman :... Not too far out. Front poles, preferably. Don ' t loosen them too far. If you loosen them too far, you ' ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ' m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ' ve been doing — in this case, in Senegal — where we have tried to use personal computers as a pedagogical medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ' s a story I ' d like to tell. And that was when, before we did this in any developing countries — we ' re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ' ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ' t read, didn ' t even make it in the lowest section of the school ' s classes — and was pretty much not in school, though physically there. But did hang around the, quote,"computer room,"where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very interesting. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,"Let me show you how this works,"and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ' t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ' d actually come to see — not the computer room — they went upstairs and they said,"This is absolutely remarkable! That child was very articulate and showed **us** and even dealt with the things he couldn ' t do automatically with that manual. It was just absolutely fantastic."The principal said,"There ' s a dreadful mistake, because that child can ' t read. And you obviously have been hoodwinked or you ' ve talked about somebody else."And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,"Can you read?"And the child said,"No, I can ' t."And then they said,"But wait a minute. You just looked through that manual and you found..."and he said,"Oh, but that ' s not reading."And so they said,"Well, what ' s reading then?"He says,"Well, reading is this junk they give me in little books to read. It ' s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return."And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ' m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ' s not, but we think speech —"My God, little children pick it up somehow, and by the age of two they ' re doing a mediocre job, and by three and four they ' re speaking reasonably well. And yet you ' ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder."Well, it ' s not harder. What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ' s going to help you. So what happens is you go to school and people say,"Just believe me, you ' re going to like it. You ' re going to like reading,"and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on the screen has a huge payoff, and it ' s a lot of fun. And in fact, it ' s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ' s the girl on the left leaning with her chalkboard, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ' s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example

— as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And the reason it remains a favorite project is that we were asked to do a teleconferencing system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ’ s faces. So if I wanted to call my friend Peter Sprague on the phone, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ’ s uncanny : there ’ s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ’ s something that didn ’ t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,”He or she wouldn ’ t agree,”and the empty chair is that person and the spatiality is crucial. So we said,”Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ’ s site you ’ d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ’ t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ’ ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ’ s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ’ s site, he ’ ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.

Text Sample 127: NEG Dim 1 – 2011 – Score 1.00 – 2011/t002572.txt

As an artist, connection is very important to me. Through my work I ’ m trying to articulate that humans are not separate from nature and that everything is interconnected. I first went to Antarctica almost 10 years ago, where I saw my first icebergs. I was in awe. My heart beat fast, my head was dizzy, trying to comprehend what it was that stood in front of me. The icebergs around me were almost 200 feet out of the water, and I could only help but wonder that this was one snowflake on top of another snowflake, year after year. Icebergs are born when they calve off of glaciers or break off of ice shelves. Each iceberg has its own individual personality. They have a distinct way of interacting with their environment and their experiences. Some refuse to give up and hold on to the bitter end, while others can ’ t take it anymore and crumble in a fit of dramatic passion. It ’ s easy to think, when you look at an iceberg, that they ’ re isolated, that they ’ re separate and alone, much like we as humans sometimes view ourselves. But the reality is far from it. As an iceberg melts, I am breathing in its ancient atmosphere. As the iceberg melts, it is releasing mineral-rich fresh water that nourishes many forms of life. I approach photographing these icebergs as if I ’ m making portraits of my ancestors, knowing that in these individual moments they exist in that way and will never exist that way again. It is not a death when they melt ; it is not an end, but a continuation of their path through the cycle of life. Some of the ice in the icebergs that I photograph is very young — a couple thousand years old. And some of the ice is over 100, 000 years old. The last pictures I ’ d like to show you are of an iceberg that I photographed in Qeqetarsuaq, Greenland. It ’ s a very rare occasion that you get to actually witness an iceberg rolling. So here it is. You can see on the left side a small boat. That ’ s about a 15-foot boat. And I ’ d like you to pay attention to the shape of the iceberg and where it is at the waterline. You can see here, it begins to roll, and the boat has moved to the other side, and the man is standing there. This is an average-size Greenlandic iceberg. It ’ s about 120 feet above the water, or 40 meters. And this video is real time. And just like that, the iceberg shows you a different side of its personality. Thank you.

Text Sample 128: NEG Dim 1 – 2011 – Score 1.00 – 2011/t002587.txt

Pat Mitchell : You have brought **us** images from the Yemen Times. And take **us** through those, and introduce **us** to another Yemen. Nadia Al-Sakkaf : Well, I ’ m glad to be here. And I would like to share with you all some of the pictures that are happening today in Yemen. This picture shows a revolution started by women, and it shows women and men leading a mixed protest. The other picture is the popularity of the real need for change. So many people are there. The intensity of the upspring. This picture shows that the revolution has allowed opportunities for training, for education. These women are learning about first aid and their rights according to the constitution. I love this picture. I just wanted to show that over 60 percent of the Yemeni population are 15 years and below. And they were excluded from decision-making, and now they are in the forefront of the news, raising the flag. English — you will see, this is jeans and tights, and an English expression — the ability to share with the world what is going on in our own country. And expression also, it has brought talents. Yemenis are using cartoons and art, paintings, comics, to tell the world and each other about what ’ s going on. Obviously, there ’ s always the dark side of it. And this is just one of the less-gruesome pictures of the revolution and the cost that we have to pay. The solidarity of millions of Yemenis across the country just demanding the one thing. And finally, lots of people are saying that Yemen ’ s revolution is going to break the country. Is it going to be so many different countries? Is it going to be another Somalia? But we want to tell the world that, no, under the one flag, we ’ ll still remain as Yemeni people. PM : Thank you for those images, Nadia. And they do, in many ways, tell a different story than the story of Yemen, the one that is often in the news. And yet, you yourself defy all those characterizations. So let ’ s talk about the personal story for a moment. Your father is murdered. The Yemen Times already has a strong reputation in Yemen as an independent English language newspaper. How did you then make the decision and assume the responsibilities of running a newspaper, especially in such times of conflict? NA : Well, let me first warn you that I ’ m not the traditional Yemeni girl. I ’ ve guessed you ’ ve already noticed this by now. In Yemen, most women are veiled and they are sitting behind doors and not very much part of the public life. But there ’ s so much potential. I wish I could show you my Yemen. I wish you could see Yemen through my eyes. Then you would know that there ’ s so much to it. And I was privileged because I was born into a family, my father would always encourage the boys and the girls. He would say we are equal. And he was such an extraordinary man. And even my mother — I owe it to my family. A story : I studied in India. And in my third year, I started becoming confused because I was Yemeni, but I was also mixing up with a lot of my friends in college. And I went **back** home and I said,”Daddy, I don ’ t know who I am. I ’ m not a Yemeni ; I ’ m not an Indian.”And he said,”You are the bridge.”And that is something I will keep in my heart forever. So since then I ’ ve been the bridge, and a lot of people have walked over me. PM : I don ’ t think so. NA : But it just helps tell that some people are change agents in the society. And when I became editor-in-chief after my brother actually — my father passed away in 1999, and then my brother until 2005 — and everybody was betting that I will not be able to do it.”What ’ s this young girl coming in and showing off because it ’ s her family business,”or something. It was very hard at first. I didn ’ t want to clash with people. But with all due respect to all the men, and the older men especially, they did not want me around. It was very hard, you know, to impose my authority. But a woman ’ s got to do what a woman ’ s got to do. And in the first year, I had to fire half of the men. Brought in more women. Brought in younger men. And we have a more gender- balanced newsroom today. The other thing is that it ’ s about professionalism. It ’ s about proving who you are and what you can do. And I don ’ t know if I ’ m going to be boasting now, but in 2006 alone, we won three international awards. One of them is the IPI Free Media Pioneer Award. So that was the answer to all the Yemeni people. And I want to score a point here, because my husband is in the room over there. If you could please stand up, [unclear]. He has been very supportive of me. PM : And we should point out that he works with you as well at the paper. But in assuming this responsibility and going about it as you have, you have become a bridge between an older and traditional society and the one that you are now creating at the paper. And so along with changing who worked there, you must have come up against another positioning that we always run into, in particular with women, and it has to do with outside image, dress, the veiled woman. So how have you dealt with this on a personal level as well as the women who worked for you? NA : As you know, the image of a lot of Yemeni women is a lot of black and covered, veiled women. And this is true. And a lot of it is because women are not able, are not free, to show their face to their self. It ’ s a lot of traditional imposing coming by authority figures such as the men, the grandparents and so on. And it ’ s economic empowerment and the ability for a woman to say,”I am as much contributing to this family, or more, than you are.”And the more empowered the women become, the more they are able to remove the veil, for example, or to drive their own car or to have a job or to be able to travel. So the other face of Yemen is actually one that lies behind the veil, and it ’ s economic empowerment mostly that allows the woman to just uncover it. And I have done this throughout my work. I ’ ve tried to encourage young girls. We started with, you can take it off in the office. And then after that, you can take it off on assignments. Because I didn ’ t believe a journalist can be a journalist with — how can you talk to people if you have your face covered? — and so on ; it ’ s just a movement. And I am a role model in Yemen. A lot of people look up to me. A lot of young girls look up to me. And I need to prove to them that, yes, you can still be married, you can still be a mother, and you can still be respected within the society, but at the same time, that doesn ’ t mean you [should] just be one of the crowd. You can be yourself and have your face. PM : But by putting yourself personally out there — both projecting a different image of Yemeni women, but also what you have made possible for the women who work at the paper — has this put you in personal danger? NA : Well the Yemen Times, across 20 years, has been through so much. We ’ ve suffered prosecution ; the paper was closed down more than three times. It ’ s an independent newspaper, but tell that to the people in charge. They think that if there ’ s anything against them, then we are being an opposition newspaper. And very, very difficult times. Some of my reporters were arrested. We had some court cases. My father was assassinated. Today, we are in a much better situation. We ’ ve created the credibility. And in times of revolution or change like today, it is very important for independent media to have a voice. It ’ s very important for you to go to YemenTimes. com, and it ’ s very important to listen to our voice. And this is probably something I ’ m going to share with you in Western media probably — and how there ’ s a lot of stereotypes — thinking of Yemen in one single frame : this is what Yemen is all about. And that ’ s not fair. It ’ s not fair for me ; it ’ s not fair for my country. A lot of reporters come to Yemen and they want to write a story on Al-Qaeda or terrorism. And I just wanted to share with you : there ’ s one reporter that came. He wanted to do a documentary on what his editors wanted. And he ended up writing about a story that even surprised me — hip hop — that there are young Yemeni men who express themselves through dancing and puchu puchu. That thing. Yeah, break dancing. I ’ m not so old. I ’ m just not in touch. PM : Yes, you are. Actually, that ’ s a documentary that ’ s available online ; the video ’ s online. NA : ShaketheDust. org. PM :”Shake the Dust.”PM : ShaketheDust. org. And it definitely does give a different image of Yemen. You spoke about the responsibility of the press. And certainly, when we look at the ways in which we have separated ourselves from others and we ’ ve created fear and danger, often from lack of knowledge, lack of real understanding, how do you see the way that the Western press in particular is covering this and all other stories out of the region, but in particular, in your country? NA : Well there is a saying that says,”You fear what you don ’ t know, and you hate what you fear.”So it ’ s about the lack of research, basically. It ’ s almost,”Do your homework,”— some involvement. And you cannot do parachute reporting — just jump into a country for two days and think that you ’ ve done your homework and a story. So I wish that the world would know my Yemen, my country, my people. I am an example, and there are others like me. We may not be that many, but if we are promoted as a good, positive example, there will be others — men and women — who can eventually bridge the gap — again, coming to the bridge — between Yemen and the world and telling first about recognition and then about communication and compassion. I think Yemen is going to be in a very bad situation in the next two or three years. It ’ s natural. But after the two years, which is a price we are willing to pay, we are going to stand up again on our feet, but in the new Yemen with a younger and more empowered people — democratic. PM : Nadia, I think you ’ ve just given **us** a very different view of Yemen. And certainly you yourself and what you do have given **us** a view of the future that we will embrace and be grateful for. And the very best of luck to you. YemenTimes. com. NA : On Twitter also. PM : So you are plugged in.

Text Sample 129: NEG Dim 1 – 2011 – Score 1.00 – 2011/t002653.txt

As a boy, I loved cars. When I turned 18, I lost my best friend to a car accident. Like this. And then I decided I ’ d dedicate my life to saving one million people every year. Now I haven ’ t succeeded, so this is just a progress report, but I ’ m here to tell you a little bit about self-driving cars. I saw the concept first in the DARPA Grand Challenges where the U. S. government issued a prize to build a self-driving car that could navigate a desert. And even though a hundred teams were there, these cars went nowhere. So we decided at Stanford to build a different self-driving car. We built the hardware and the software. We made it learn from **us**, and we set it free in the desert. And the unimaginable happened : it became the first car to ever return from a DARPA Grand Challenge, winning Stanford 2 million dollars. Yet I still hadn ’ t saved a single life. Since, our work has focused on building driving cars that can drive anywhere by themselves — any street in California. We ’ ve driven 140, 000 miles. Our cars have sensors by which they magically can see everything around them and make decisions about every aspect of driving. It ’ s the perfect driving mechanism. We ’ ve driven in cities, like in San Francisco here. We ’ ve driven from San Francisco to Los Angeles on Highway 1. We ’ ve encountered joggers, busy highways, toll booths, and this is without a person in the loop ; the car just drives itself. In fact, while we drove 140, 000 miles, people didn ’ t even notice. Mountain roads, day and night, and even crooked Lombard Street in San Francisco. Sometimes our cars get so crazy, they even do little stunts. Man : Oh, my God. What? Second Man : It ’ s driving itself. Sebastian Thrun : Now I can ’ t get my friend Harold **back** to life, but I can do something for all the people who died. Do you know that driving accidents are the number one cause of death for young people? And do you realize that almost all of those are due to human error and not machine error, and can therefore be prevented by machines? Do you realize that we could change the capacity of highways by a factor of two or three if we didn ’ t rely on human precision on staying in the lane — improve body position and therefore drive a little bit closer together on a little bit narrower lanes, and do away with all traffic jams on highways? Do you realize that you, TED users, spend an average of 52 minutes per day in traffic, wasting your time on your daily commute? You could regain this time. This is four billion hours wasted in this country alone. And it ’ s 2. 4 billion gallons of gasoline wasted. Now I think there ’ s a vision here, a new technology, and I ’ m really looking forward to a time when generations after **us** look **back** at **us** and say how ridiculous it was that humans were driving cars. Thank you.

Text Sample 130: NEG Dim 1 – 2009 – Score -1.00 – 2009/t000090.txt

I ’ ve been working on issues of poverty for more than 20 years, and so it ’ s ironic that the problem that and question that I most grapple with is how you actually define poverty. What does it mean? So often, we look at dollar terms â people making less than a dollar or two or three a day. And yet the complexity of poverty really has to look at income as only one variable. Because really, it ’ s a condition about choice, and the lack of freedom. And I had an experience that really deepened and elucidated for me the understanding that I have. It was in Kenya, and I want to share it with you. I was with my friend Susan Meiselas, the photographer, in the Mathare Valley slums. Now, Mathare Valley is one of the oldest slums in Africa. It ’ s about three miles out of Nairobi, and it ’ s a mile long and about **two-tenths** of a mile wide, where over half a million people live crammed in these little tin shacks, generation after generation, renting them, often eight or 10 people to a room. And it ’ s known for prostitution, violence, drugs : a hard place to grow up. And when we were walking through the narrow alleys, it was literally impossible not to step in the raw sewage and the garbage alongside the little homes. But at the same time it was also impossible not to see the human vitality, the aspiration and the ambition of the people who live there : women washing their babies, washing their clothes, hanging them out to dry. I met this woman, Mama Rose, who has rented that little tin shack for 32 years, where she lives with her seven children. Four sleep in one twin bed, and three sleep on the mud and linoleum floor. And she keeps them all in school by selling water from that kiosk, and from selling soap and bread from the little store inside. It was also the day after the inauguration, and I was reminded how Mathare is still connected to the globe. And I would see kids on the street corners, and they ' d say"Obama, he ' s our brother!"And I ' d say"Well, Obama ' s my brother, so that makes you my brother too."And they would look quizzically, and then be like,"High five!"And it was here that I met Jane. I was struck immediately by the kindness and the gentleness in her face, and I asked her to tell me her story. She started off by telling me her dream. She said,"I had two. My first dream was to be a doctor, and the second was to marry a good man who would stay with me and my family, because my mother was a single mom, and couldn ' t afford to pay for school fees. So I had to give up the first dream, and I focused on the second."She got married when she was 18, had a baby right away. And when she turned 20, found herself pregnant with a second child, her mom died and her husband left her â married another woman. So she was again in Mathare, with no income, no skill set, no money. And so she ultimately turned to prostitution. It wasn ' t organized in the way we often think of it. She would go into the city at night with about 20 girls, look for work, and sometimes come **back** with a few shillings, or sometimes with nothing. And she said,"You know, the poverty wasn ' t so bad. It was the humiliation and the embarrassment of it all."In 2001, her life changed. She had a girlfriend who had heard about this organization, Jamii Bora, that would lend money to people no matter how poor you were, as long as you provided a commensurate amount in savings. And so she spent a year to save 50 dollars, and started borrowing, and over time she was able to buy a sewing machine. She started tailoring. And that turned into what she does now, which is to go into the secondhand clothing markets, and for about three dollars and 25 cents she buys an old ball gown. Some of them might be ones you gave. And she repurposes them with frills and ribbons, and makes these frothy confections that she sells to women for their daughter ' s Sweet 16 or first Holy Communion â those milestones in a life that people want to celebrate all along the economic spectrum. And she does really good business. In fact, I watched her walk through the streets hawking. And before you knew it, there was a crowd of women around her, buying these dresses. And I reflected, as I was watching her sell the dresses, and also the jewelry that she makes, that now Jane makes more than four dollars a day. And by many definitions she is no longer poor. But she still lives in Mathare Valley. And so she can ' t move out. She lives with all of that insecurity, and in fact, in January, during the ethnic riots, she was chased from her home and had to find a new shack in which she would live. Jamii Bora understands that and understands that when we ' re talking about poverty, we ' ve got to look at people all along the economic spectrum. And so with patient capital from Acumen and other organizations, loans and investments that will go the long term with them, they built a low-cost housing development, about an hour outside Nairobi central. And they designed it from the perspective of customers like Jane herself, insisting on responsibility and accountability. So she has to give 10 percent of the mortgage â of the total value, or about 400 dollars in savings. And then they match her mortgage to what she paid in rent for her little shanty. And in the next couple of weeks, she ' s going to be among the first 200 families to move into this development. When I asked her if she feared anything, or whether she would miss anything from Mathare, she said,"What would I fear that I haven ' t confronted already? I ' m HIV positive. I ' ve dealt with it all."And she said,”What would I miss? You think I will miss the violence or the drugs? The lack of privacy? Do you think I ’ ll miss not knowing if my children are going to come home at the end of the day?”She said”If you gave me 10 minutes my bags would be packed.”I said,”Well what about your dreams?”And she said,”Well, you know, my dreams don ’ t look exactly like I thought they would when I was a little girl. But if I think about it, I thought I wanted a husband, but what I really wanted was a family that was loving. And I fiercely love my children, and they love me **back**.”She said,”I thought that I wanted to be a doctor, but what I really wanted to be was somebody who served and healed and cured. And so I feel so blessed with everything that I have, that two days a week I go and I counsel HIV patients. And I say, ’ Look at me. You are not dead. You are still alive. And if you are still alive you have to serve. ’”And she said,”I ’ m not a doctor who gives out pills. But maybe me, I give out something better because I give them hope.”And in the middle of this economic crisis, where so many of **us** are inclined to pull in with fear, I think we ’ re well suited to take a cue from Jane and reach out, recognizing that being poor doesn ’ t mean being ordinary. Because when systems are broken, like the ones that we ’ re seeing around the world, it ’ s an opportunity for invention and for innovation. It ’ s an opportunity to truly build a world where we can extend services and products to all human beings, so that they can make decisions and choices for themselves. I truly believe it ’ s where dignity starts. We owe it to the Janes of the world. And just as important, we owe it to ourselves. Thank you.

Text Sample 131: NEG Dim 1 – 2009 – Score -1.00 – 2009/t002996.txt

I normally teach courses on how to rebuild states after war. But today I ’ ve got a personal story to share with you. This is a picture of my family, my four siblings â my mom and I â taken in 1977. And we ’ re actually Cambodians. And this picture is taken in Vietnam. So how did a Cambodian family end up in Vietnam in 1977? Well to explain that, I ’ ve got a short video clip to explain the Khmer Rouge regime during 1975 and 1979. Video : April 17th, 1975. The communist Khmer Rouge enters Phnom Penh to liberate their people from the encroaching conflict in Vietnam, and American bombing campaigns. Led by peasant-born Pol Pot, the Khmer Rouge evacuates people to the countryside in order to create a rural communist utopia, much like Mao Tse-tung ’ s Cultural Revolution in China. The Khmer Rouge closes the doors to the outside world. But after four years the grim truth seeps out. In a country of only seven million people, one and a half million were murdered by their own leaders, their bodies piled in the mass graves of the killing fields. Sophal Ear : So, notwithstanding the 1970s narration, on April 17th 1975 we lived in Phnom Penh. And my parents were told by the Khmer Rouge to evacuate the city because of impending American bombing for three days. And here is a picture of the Khmer Rouge. They were young soldiers, typically child soldiers. And this is very normal now, of modern day conflict, because they ’ re easy to bring into wars. The reason that they gave about American bombing wasn ’ t all that far off. I mean, from 1965 to 1973 there were more munitions that fell on Cambodia than in all of World War II Japan, including the two nuclear bombs of August 1945. The Khmer Rouge didn ’ t believe in money. So the equivalent of the Federal Reserve Bank in Cambodia was bombed. But not just that, they actually banned money. I think it ’ s the only precedent in which money has ever been stopped from being used. And we know money is the root of all evil, but it didn ’ t actually stop evil from happening in Cambodia, in fact. My family was moved from Phnom Penh to Pursat province. This is a picture of what Pursat looks like. It ’ s actually a very pretty area of Cambodia, where rice growing takes place. And in fact they were forced to work the fields. So my father and mother ended up in a sort of concentration camp, labor camp. And it was at that time that my mother got word from the commune chief that the Vietnamese were actually asking for their citizens to go **back** to Vietnam. And she spoke some Vietnamese, as a child having grown up with Vietnamese friends. And she decided, despite the advice of her neighbors, that she would take the chance and claim to be Vietnamese so that we could have a chance to survive, because at this point they ’ re forcing everybody to work. And they ’ re giving about â in a modern-day, caloric-restriction diet, I guess â they ’ re giving porridge, with a few grains of rice. And at about this time actually my father got very sick. And he didn ’ t speak Vietnamese. So he died actually, in January 1976. And it made it possible, in fact, for **us** to take on this plan. So the Khmer Rouge took **us** from a place called Pursat to Kaoh Tiev, which is across from the border from Vietnam. And there they had a **detention** camp where alleged Vietnamese would be tested, language tested. And my mother ’ s Vietnamese was so bad that to make our story more credible, she ’ d given all the boys and girls new Vietnamese names. But she ’ d given the boys girls ’ names, and the girls boys ’ names. And it wasn ’ t until she met a Vietnamese lady who told her this, and then tutored her for two days intensively, that she was able to go into her exam and â you know, this was a moment of truth. If she fails, we ’ re all headed to the gallows ; if she passes, we can leave to Vietnam. And she actually, of course â I ’ m here, she passes. And we end up in Hong Ngu on the Vietnamese side. And then onwards to Chau Doc. And this is a picture of Hong Ngu, Vietnam today. A pretty idyllic place on the Mekong Delta. But for **us** it meant freedom. And freedom from persecution from the Khmer Rouge. Last year, the Khmer Rouge Tribunal, which the U. N. is helping Cambodia take on, started, and I decided that as a matter of record I should file a Civil Complaint with the Tribunal about my father ’ s passing away. And I got word last month that the complaint was officially accepted by the Khmer Rouge Tribunal. And it ’ s for me a matter of justice for history, and accountability for the future, because Cambodia remains a pretty lawless place, at times. Five years ago my mother and I went **back** to Chau Doc. And she was able to return to a place that for her meant freedom, but also fear, because we had just come out of Cambodia. I ’ m happy, actually, today, to present her. She ’ s here today with **us** in the audience. Thank you mother.

Text Sample 132: NEG Dim 1 – 2009 – Score 1.00 – 2009/t002908.txt

There are a lot of web 2. 0 consultants who make a lot of money. In fact, they make their living on this stuff. I ’ m going to try to save you all the time and money and go through it in the next three minutes, so bear with me. Started a website in 2005 with a few friends, called Reddit. com. It ’ s what you ’ d call a social news website ; basically, the democratic front page of the best stuff on the web. You find some interesting content — say, a TED Talk — submit it to Reddit, and a community of your peers votes up if they like it, down if they don ’ t. That creates the front page. It ’ s always rising, falling ; a half million people visit every day. But this isn ’ t about Reddit. It ’ s about discovering new things that pop up on the web. In the last four years, we ’ ve seen all kinds of memes, all kinds of trends get born right on our front page. This isn ’ t about Reddit itself, it ’ s actually about humpback whales. Well, technically, it ’ s about Greenpeace, an environmental organization that wanted to stop the Japanese government ’ s whaling campaign. The whales were getting killed ;

they wanted to put an end to it. One of the ways they wanted to do it was to put a tracking chip inside one of the whales. But to personify the movement, they wanted to name it. So in true web fashion, they put together a poll, where they had a bunch of very erudite, very thoughtful, cultured names. I believe this is the Farsi word for”immortal.”I think this means”divine power of the ocean”in a Polynesian language. And then there was this :”Mister Splashy Pants.”And this was a special name. Mister Pants, or”Splashy”to his friends, was very popular on the Internet. In fact, someone on Reddit thought,”What a great thing, we should all vote this up.”And Redditors responded and all agreed. So the voting started. We got behind it ourselves ; we changed our logo for the day, from the alien to Splashy, to help the cause. And it wasn ’ t long before other sites like Fark and Boing Boing and the rest of the Internet started saying,”We love Splashy Pants!”So it went from about five percent, which was when this meme started, to 70 percent at the end of voting. Pretty impressive, right? We won! Mister Splashy Pants was chosen. Just kidding — Greenpeace actually wasn ’ t that crazy about it, because they wanted one of the more thoughtful names to win. They said,”No, just kidding. We ’ ll give it another week of voting.”Well, that got **us** a little angry, so we changed it to Fightin ’ Splashy. And the Reddit community — really, the rest of the Internet, really got behind this. Facebook groups were created. Facebook applications were created. The idea was,”Vote your conscience, vote for Mister Splashy Pants.”People were putting up signs in the real world about this whale. This was the final vote : 78 percent of the votes. To give you an idea of the landslide, the next highest name pulled in three. There was a clear lesson : the Internet loves Mister Splashy Pants. Which is obvious. It ’ s a great name. Everyone wants to hear their news anchor say,”Mister Splashy Pants.”I think that ’ s what helped drive this. What was cool were the repercussions. Greenpeace created an entire marketing campaign around it — Mister Splashy Pants shirts and pins, an e-card so you could send your friend a dancing Splashy. But even more important was that they accomplished their mission. The Japanese government called off their whaling expedition. Mission accomplished : Greenpeace was thrilled, the whales were happy — that ’ s a quote. And actually, Redditors in the Internet community were happy to participate, but they weren ’ t whale lovers. A few, certainly, but we ’ re talking about a lot of people, really interested and caught up in this meme. Greenpeace came **back** to the site and thanked Reddit for its participation. But this wasn ’ t really altruism ; just interest in doing something cool. This is how the Internet works. This is that great big secret. The Internet provides a level playing field. Your link is as good as your link, which is as good as my link. With a browser, anyone can get to any website no matter your budget. That is, as long as you can keep net neutrality in place. Another important thing is it costs nothing to get content online. There are so many publishing tools available, it only takes a few minutes to produce something. and the cost of iteration is so cheap, you might as well. If you do, be genuine. Be honest, up-front. One of the great lessons Greenpeace learned is that it ’ s OK to lose control, OK to take yourself a little less seriously, given that, even though it ’ s a very serious cause, you could ultimately achieve your goal. That ’ s the final message I want to share : you can do well online. But no longer is the message coming from just the top down. If you want to succeed you ’ ve got to be OK to lose control. Thank you.

Text Sample 133: NEG Dim 1 – 2010 – Score -1.00 – 2010/t002858.txt
So, basically we have public leaders, public officials who are out of control ;

they are writing bills that are unintelligible, and out of these bills are going to come maybe 40, 000 pages of regulations, total complexity, which has a dramatically negative impact on our life. If you ’ re a veteran coming **back** from Iraq or Vietnam you face a blizzard of paperwork to get your benefits ; if you ’ re trying to get a small business loan, you face a blizzard of paperwork. What are we going to do about it? I define simplicity as a means to achieving clarity, transparency and empathy, building humanity into communications. I ’ ve been simplifying things for 30 years. I come out of the advertising and design business. My focus is understanding you people, and how you interact with the government to get your benefits, how you interact with corporations to decide whom you ’ re going to do business with, and how you view brands. So, very quickly, when President Obama said,”I don ’ t see why we can ’ t have a one-page, plain English consumer credit agreement.”So, I locked myself in a room, figured out the content, organized the document, and wrote it in plain English. I ’ ve had this checked by the two top consumer credit lawyers in the country. This is a real thing. Now, I went one step further and said,”Why do we have to stick with the stodgy lawyers and just have a paper document? Let ’ s go online.”And many people might need help in computation. Working with the Harvard Business School, you ’ ll see this example when you talk about minimum payment : If you spent 62 dollars for a meal, the longer you take to pay out that loan, you see, over a period of time using the minimum payment it ’ s 99 dollars and 17 cents. How about that? Do you think your bank is going to show that to people? But it ’ s going to work. It ’ s more effective than just computational aids. And what about terms like”over the limit”? Perhaps a stealth thing. Define it in context. Tell people what it means. When you put it in plain English, you almost force the institution to give the people a way, a default out of that, and not put themselves at risk. Plain English is about changing the content. And one of the things I ’ m most proud of is this agreement for IBM. It ’ s a grid, it ’ s a calendar. At such and such a date, IBM has responsibilities, you have responsibilities. Received very favorably by business. And there is some good news to report today. Each year, one in 10 taxpayers receives a notice from the IRS. There are 200 million letters that go out. Running through this typical letter that they had, I ran it through my simplicity lab, it ’ s pretty unintelligible. All the parts of the document in red are not intelligible. We looked at doing over 1, 000 letters that cover 70 percent of their transactions in plain English. They have been tested in the laboratory. When I run it through my lab, this heat-mapping shows everything is intelligible. And the IRS has introduced the program. There are a couple of things going on right now that I want to bring to your attention. There is a lot of discussion now about a consumer financial protection agency, how to mandate simplicity. We see all this complexity. It ’ s incumbent upon **us**, and this organization, I believe, to make clarity, transparency and empathy a national priority. There is no way that we should allow government to communicate the way they communicate. There is no way we should do business with companies that have agreements with stealth provisions and that are unintelligible. So, how are we going to change the world? Make clarity, transparency and simplicity a national priority. I thank you.

Text Sample 134: NEG Dim 1 – 2010 – Score -1.00 – 2010/t002681.txt

Hawa Abdi : Many people â 20 years for Somalia â [were] fighting. So there was no job, no food. Children, most of them, became very malnourished, like this. Deqo Mohamed : So as you know, always in a civil war, the ones affected most [are] the women and children. So our patients are women and children. And they are in our backyard. It ’ s our home. We welcome them. That ’ s the camp that we have in now 90, 000 people, where 75 percent of them are women and children. Pat Mitchell : And this is your hospital. This is the inside. HA : We are doing C-sections and different operations because people need some help. There is no government to protect them. DM : Every morning we have about 400 patients, maybe more or less. But sometimes we are only five doctors and 16 nurses, and we are physically getting exhausted to see all of them. But we take the severe ones, and we reschedule the other ones the next day. It is very tough. And as you can see, it ’ s the women who are carrying the children ; it ’ s the women who come into the hospitals ; it ’ s the women [are] building the houses. That ’ s their house. And we have a school. This is our bright â we opened [in the] last two years [an] elementary school where we have 850 children, and the majority are women and girls. PM : And the doctors have some very big rules about who can get treated at the clinic. Would you explain the rules for admission? HA : The people who are coming to **us**, we are welcoming. We are sharing with them whatever we have. But there are only two rules. First rule : there is no clan distinguished and political division in Somali society. [Whomever] makes those things we throw out. The second : no man can beat his wife. If he beat, we will put [him] in jail, and we will call the eldest people. Until they identify this case, we ’ ll never release him. That ’ s our two rules. The other thing that I have realized, that the woman is the most strong person all over the world. Because the last 20 years, the Somali woman has stood up. They were the leaders, and we are the leaders of our community and the hope of our future generations. We are not just the helpless and the victims of the civil war. We can reconcile. We can do everything. DM : As my mother said, we are the future hope, and the men are only killing in Somalia. So we came up with these two rules. In a camp with 90, 000 people, you have to come up with some rules or there is going to be some fights. So there is no clan division, and no man can beat his wife. And we have a little storage room where we converted a jail. So if you beat your wife, you ’ re going to be there. So empowering the women and giving the opportunity â we are there for them. They are not alone for this. PM : You ’ re running a medical clinic. It brought much, much needed medical care to people who wouldn ’ t get it. You ’ re also running a civil society. You ’ ve created your own rules, in which women and children are getting a different sense of security. Talk to me about your decision, Dr. Abdi, and your decision, Dr. Mohamed, to work together â for you to become a doctor and to work with your mother in these circumstances. HA : My age â because I was born in 1947 â we were having, at that time, government, law and order. But one day, I went to the hospital â my mother was sick â and I saw the hospital, how they [were] treating the doctors, how they [are] committed to help the sick people. I admired them, and I decided to become a doctor. My mother died, unfortunately, when I was 12 years [old]. Then my father allowed me to proceed [with] my hope. My mother died in [a] gynecology complication, so I decided to become a gynecology specialist. That ’ s why I became a doctor. So Dr. Deqo has to explain. DM : For me, my mother was preparing [me] when I was a child to become a doctor, but I really didn ’ t want to. Maybe I should become an historian, or maybe a reporter. I loved it, but it didn ’ t work. When the war broke out â civil war â I saw how my mother was helping and how she really needed the help, and how the care is essential to the woman to be a woman doctor in Somalia and help the women and children. And I thought, maybe I can be a reporter and doctor gynecologist. So I went to Russia, and my mother also, [during the] time of [the] Soviet Union. So some of our character, maybe we will come with a strong Soviet background of training. So that ’ s how I decided [to do] the same. My sister was different. She ’ s here. She ’ s also a doctor. She graduated in Russia also. And to go **back** and to work with our mother is just what we saw in the civil war â when I was 16, and my sister was 11, when the civil war broke out. So it was the need and the people we saw in the early ’ 90s â that ’ s what made **us** go **back** and work for them. PM : So what is the biggest challenge working, mother and daughter, in such dangerous and sometimes scary situations? HA : Yes, I was working in a tough situation, very dangerous. And when I saw the people who needed me, I was staying with them to help, because I [could] do something for them. Most people fled abroad. But I remained with those people, and I was trying to do something â [any] little thing I [could] do. I succeeded in my place. Now my place is 90, 000 people who are respecting each other, who are not fighting. But we try to stand on our feet, to do something, little things, we can for our people. And I ’ m thankful for my daughters. When they come to me, they help me to treat the people, to help. They do everything for them. They have done what I desire to do for them. PM : What ’ s the best part of working with your mother, and the most challenging part for you? DM : She ’ s very tough ; it ’ s most challenging. She always expects **us** to do more. And really when you think [you] cannot do it, she will push you, and I can do it. That ’ s the best part. She shows **us**, trains **us** how to do and how to be better [people] and how to do long hours in surgery â 300 patients per day, 10, 20 surgeries, and still you have to manage the camp â that ’ s how she trains **us**. It is not like beautiful offices here, 20 patients, you ’ re tired. You see 300 patients, 20 surgeries and 90, 000 people to manage. PM : But you do it for good reasons. Wait. Wait. HA : Thank you. DM : Thank you. HA : Thank you very much. DM : Thank you very much.

Text Sample 135: NEG Dim 1 – 2010 – Score -1.00 – 2010/t002852.txt

Today, I want you to look at children who become suicide bombers through a completely different lens. In 2009, there were 500 bomb blasts across Pakistan. I spent the year working with children who were training to become suicide bombers and with Taliban recruiters, trying to understand how the Taliban were converting these children into live ammunition and why these children were actively signing up to their cause. I want you to watch a short video from my latest documentary film,”Children of the Taliban.”The Taliban now run their own schools. They target poor families and convince the parents to send their children. In return, they provide free food and shelter and sometimes pay the families a monthly stipend. We ’ ve obtained a propaganda video made by the Taliban. Young boys are taught justifications for suicide attacks and the execution of spies. I made contact with a child from Swat who studied in a madrassa like this. Hazrat Ali is from a poor farming family in Swat. He joined the Taliban a year ago when he was 13. How do the Taliban in your area get people to join them? Hazrat Ali : They first call **us** to the mosque and preach to **us**. Then they take **us** to a madrassa and teach **us** things from the Koran. Sharmeen Obaid Chinoy : He tells me that children are then given months of military training. HA : They teach **us** to use machine guns, Kalashnikov, rocket launchers, grenades, bombs. They ask **us** to use them only against the infidels. Then they teach **us** to do a suicide attack. SOC : Would you like to carry out a suicide attack? HA : If God gives me strength. SOC : I, in my research, have seen that the Taliban have perfected the way in which they recruit and train children, and I think it ’ s a five-step process. Step one is that the Taliban prey on families that are large, that are poor, that live in rural areas. They separate the parents from the children by promising to provide food, clothing, shelter to these children. Then they ship them off, hundreds of miles away to hard-line schools that run along the Taliban agenda. Step two : They teach the children the Koran, which is Islam ' s holiest book, in Arabic, a language these children do not understand and cannot speak. They rely very heavily on teachers, who I have personally seen distort the message to these children as and when it suits their purpose to. These children are explicitly forbidden from reading newspapers, listening to radio, reading any books that the teachers do not prescribe them. If any child is found violating these rules, he is severely reprimanded. Effectively, the Taliban create a complete blackout of any other source of information for these children. Step three : The Taliban want these children to hate the world that they currently live in. So they beat these children — I have seen it ; they feed them twice a day dried bread and water ; they rarely allow them to play games ; they tell them that, for eight hours at a time, all they have to do is read the Koran. The children are virtual prisoners ; they cannot leave, they cannot go home. Their parents are so poor, they have no resources to get them **back**. Step four : The older members of the Taliban, the fighters, start talking to the younger boys about the glories of martyrdom. They talk to them about how when they die, they will be received up with lakes of honey and milk, how there will be 72 virgins waiting for them in paradise, how there will be unlimited food, and how this glory is going to propel them to become heroes in their neighborhoods. Effectively, this is the brainwashing process that has begun. Step five : I believe the Taliban have one of the most effective means of propaganda. Their videos that they use are intercut with photographs of men and women and children dying in Iraq and Afghanistan and in Pakistan. And the basic message is that the Western powers do not care about civilian deaths, so those people who live in areas and support governments that work with Western powers are fair game. That ' s why Pakistani civilians, over 6, 000 of whom have been killed in the last two years alone, are fair game. Now these children are primed to become suicide bombers. They ' re ready to go out and fight because they ' ve been told that this is effectively their only way to glorify Islam. I want you to watch another excerpt from the film. This boy is called Zenola. He blew himself up, killing six. This boy is called Sadik. He killed 22. This boy is called Messoud. He killed 28. The Taliban are running suicide schools, preparing a generation of boys for atrocities against civilians. Do you want to carry out a suicide attack? Boy : I would love to. But only if I get permission from my dad. When I look at suicide bombers younger than me, or my age, I get so inspired by their terrific attacks. SOC : What blessing would you get from carrying out a suicide attack? Boy : On the day of judgment, God will ask me,"Why did you do that?"I will answer,"My Lord! Only to make you happy! I have laid down my life fighting the infidels."Then God will look at my intention. If my intention was to eradicate evil for Islam, then I will be rewarded with paradise. Singer : On the day of judgment My God will call me My body will be put **back** together And God will ask me why I did this SOC : I leave you all with this thought : If you grew up in these circumstances, faced with these choices, would you choose to live in this world or in the glorious afterlife? As one Taliban recruiter told me,"There will always be sacrificial lambs in this war."Thank you.

Text Sample 136: NEG Dim 1 – 2004 – Score 2.00 – 2004/t000196.txt
I grew up in Europe, and World War II caught me when I was between seven

and 10 years old. And I realized how few of the grown-ups that I knew were able to withstand the tragedies that the war visited on them â how few of them could even resemble a normal, contented, satisfied, happy life once their job, their home, their security was destroyed by the war. So I became interested in understanding what contributed to a life that was worth living. And I tried, as a child, as a teenager, to read philosophy and to get involved in art and religion and many other ways that I could see as a possible answer to that question. And finally I ended up encountering psychology by chance. I was at a ski resort in Switzerland without any money to actually enjoy myself, because the snow had melted and I didn ’ t have money to go to a movie. But I found that on the â I read in the newspapers that there was to be a presentation by someone in a place that I ’ d seen in the center of Zurich, and it was about flying saucers [that] he was going to talk. And I thought, well, since I can ’ t go to the movies, at least I will go for free to listen to flying saucers. And the man who talked at that evening lecture was very interesting. Instead of talking about little green men, he talked about how the psyche of the Europeans had been traumatized by the war, and now they ’ re projecting flying saucers into the sky. He talked about how the **mandalas** of ancient Hindu religion were kind of projected into the sky as an attempt to regain some sense of order after the chaos of war. And this seemed very interesting to me. And I started reading his books after that lecture. And that was Carl Jung, whose name or work I had no idea about. Then I came to this country to study psychology and I started trying to understand the roots of happiness. This is a typical result that many people have presented, and there are many variations on it. But this, for instance, shows that about 30 percent of the people surveyed in the United States since 1956 say that their life is very happy. And that hasn ’ t changed at all. Whereas the personal income, on a scale that has been held constant to accommodate for inflation, has more than doubled, almost tripled, in that period. But you find essentially the same results, namely, that after a certain basic point â which corresponds more or less to just a few 1, 000 dollars above the minimum poverty level â increases in material well-being don ’ t seem to affect how happy people are. In fact, you can find that the lack of basic resources, material resources, contributes to **unhappiness**, but the increase in material resources does not increase happiness. So my research has been focused more on â after finding out these things that actually corresponded to my own experience, I tried to understand : where â in everyday life, in our normal experience â do we feel really happy? And to start those studies about 40 years ago, I began to look at creative people â first artists and scientists, and so forth â trying to understand what made them feel that it was worth essentially spending their life doing things for which many of them didn ’ t expect either fame or fortune, but which made their life meaningful and worth doing. This was one of the leading composers of American music **back** in the ’ 70s. And the interview was 40 pages long. But this little excerpt is a very good summary of what he was saying during the interview. And it describes how he feels when composing is going well. And he says by describing it as an ecstatic state. Now,”ecstasy”in Greek meant simply to stand to the side of something. And then it became essentially an analogy for a mental state where you feel that you are not doing your ordinary everyday routines. So ecstasy is essentially a stepping into an alternative reality. And it ’ s interesting, if you think about it, how, when we think about the civilizations that we look up to as having been pinnacles of human achievement â whether it ’ s China, Greece, the Hindu civilization, or the Mayas, or Egyptians â what we know about them is really about their ecstasies, not about their everyday life. We know the temples they built, where people could come to experience a different reality.

We know about the circuses, the arenas, the theaters. These are the remains of civilizations and they are the places that people went to experience life in a more concentrated, more ordered form. Now, this man doesn ’ t need to go to a place like this, which is also â this place, this arena, which is built like a Greek amphitheatre, is a place for ecstasy also. We are participating in a reality that is different from that of the everyday life that we ’ re used to. But this man doesn ’ t need to go there. He needs just a piece of paper where he can put down little marks, and as he does that, he can imagine sounds that had not existed before in that particular combination. So once he gets to that point of beginning to create, like Jennifer did in her improvisation, a new reality â that is, a moment of ecstasy â he enters that different reality. Now he says also that this is so intense an experience that it feels almost as if he didn ’ t exist. And that sounds like a kind of a romantic exaggeration. But actually, our nervous system is incapable of processing more than about 110 bits of information per second. And in order to hear me and understand what I ’ m saying, you need to process about 60 bits per second. That ’ s why you can ’ t hear more than two people. You can ’ t understand more than two people talking to you. Well, when you are really involved in this completely engaging process of creating something new, as this man is, he doesn ’ t have enough attention left over to monitor how his body feels, or his problems at home. He can ’ t feel even that he ’ s hungry or tired. His body disappears, his identity disappears from his consciousness, because he doesn ’ t have enough attention, like none of **us** do, to really do well something that requires a lot of concentration, and at the same time to feel that he exists. So existence is temporarily suspended. And he says that his hand seems to be moving by itself. Now, I could look at my hand for two weeks, and I wouldn ’ t feel any awe or wonder, because I can ’ t compose. So what it ’ s telling you here is that obviously this automatic, spontaneous process that he ’ s describing can only happen to someone who is very well trained and who has developed technique. And it has become a kind of a truism in the study of creativity that you can ’ t be creating anything with less than 10 years of technical-knowledge immersion in a particular field. Whether it ’ s mathematics or music, it takes that long to be able to begin to change something in a way that it ’ s better than what was there before. Now, when that happens, he says the music just flows out. And because all of these people I started interviewing â this was an interview which is over 30 years old â so many of the people described this as a spontaneous flow that I called this type of experience the”flow experience.”And it happens in different realms. For instance, a poet describes it in this form. This is by a student of mine who interviewed some of the leading writers and poets in the United States. And it describes the same effortless, spontaneous feeling that you get when you enter into this ecstatic state. This poet describes it as opening a door that floats in the sky â a very similar description to what Albert Einstein gave as to how he imagined the forces of relativity, when he was struggling with trying to understand how it worked. But it happens in other activities. For instance, this is another student of mine, Susan Jackson from Australia, who did work with some of the leading athletes in the world. And you see here in this description of an Olympic skater, the same essential description of the phenomenology of the inner state of the person. You don ’ t think ; it goes automatically, if you merge yourself with the music, and so forth. It happens also, actually, in the most recent book I wrote, called”Good Business,”where I interviewed some of the CEOs who had been nominated by their peers as being both very successful and very ethical, very socially responsible. You see that these people define success as something that helps others and at the same time makes you feel happy as you are working at it. And like all of these successful and responsible

CEOs say, you can ’ t have just one of these things be successful if you want a meaningful and successful job. Anita Roddick is another one of these CEOs we interviewed. She is the founder of Body Shop, the natural cosmetics king. It ’ s kind of a passion that comes from doing the best and having flow while you ’ re working. This is an interesting little quote from Masaru Ibuka, who was at that time starting out Sony without any money, without a product â they didn ’ t have a product, they didn ’ t have anything, but they had an idea. And the idea he had was to establish a place of work where engineers can feel the joy of technological innovation, be aware of their mission to society and work to their heart ’ s content. I couldn ’ t improve on this as a good example of how flow enters the workplace. Now, when we do studies â we have, with other colleagues around the world, done over 8, 000 interviews of people â from Dominican monks, to blind nuns, to Himalayan climbers, to Navajo shepherds â who enjoy their work. And regardless of the culture, regardless of education or whatever, there are these seven conditions that seem to be there when a person is in flow. There ’ s this focus that, once it becomes intense, leads to a sense of ecstasy, a sense of clarity : you know exactly what you want to do from one moment to the other ; you get immediate feedback. You know that what you need to do is possible to do, even though difficult, and sense of time disappears, you forget yourself, you feel part of something larger. And once the conditions are present, what you are doing becomes worth doing for its own sake. In our studies, we represent the everyday life of people in this simple scheme. And we can measure this very precisely, actually, because we give people electronic pagers that go off 10 times a day, and whenever they go off you say what you ’ re doing, how you feel, where you are, what you ’ re thinking about. And two things that we measure is the amount of challenge people experience at that moment and the amount of skill that they feel they have at that moment. So for each person we can establish an average, which is the center of the diagram. That would be your mean level of challenge and skill, which will be different from that of anybody else. But you have a kind of a set point there, which would be in the middle. If we know what that set point is, we can predict fairly accurately when you will be in flow, and it will be when your challenges are higher than average and skills are higher than average. And you may be doing things very differently from other people, but for everyone that flow channel, that area there, will be when you are doing what you really like to do â play the piano, be with your best friend, perhaps work, if work is what provides flow for you. And then the other areas become less and less positive. Arousal is still good because you are over-challenged there. Your skills are not quite as high as they should be, but you can move into flow fairly easily by just developing a little more skill. So, arousal is the area where most people learn from, because that ’ s where they ’ re pushed beyond their comfort zone and to enter that â going **back** to flow â then they develop higher skills. Control is also a good place to be, because there you feel comfortable, but not very excited. It ’ s not very challenging any more. And if you want to enter flow from control, you have to increase the challenges. So those two are ideal and complementary areas from which flow is easy to go into. The other combinations of challenge and skill become progressively less optimal. Relaxation is fine â you still feel OK. Boredom begins to be very aversive and apathy becomes very negative : you don ’ t feel that you ’ re doing anything, you don ’ t use your skills, there ’ s no challenge. Unfortunately, a lot of people ’ s experience is in apathy. The largest single contributor to that experience is watching television ; the next one is being in the bathroom, sitting. Even though sometimes watching television about seven to eight percent of the time is in flow, but that ’ s when you choose a program you really want to watch and you get feedback from it. So the question we are trying to address â and I ’ m way over time â is how to put more and more of everyday life in that flow channel. And that is the kind of challenge that we ’ re trying to understand. And some of you obviously know how to do that spontaneously without any advice, but unfortunately a lot of people don ’ t. And that ’ s what our mandate is, in a way, to do. Thank you.

Text Sample 137: NEG Dim 1 – 2004 – Score 3.00 – 2004/t000431.txt

Imagine spending seven years at MIT and research laboratories, only to find out that you ’ re a performance artist. I ’ m also a software engineer, and I make lots of different kinds of art with the computer. And I think the main thing that I ’ m interested in is trying to find a way of making the computer into a personal mode of expression. And many of you out there are the heads of Macromedia and Microsoft, and in a way those are my bane : I think there ’ s a great homogenizing force that software imposes on people and limits the way they think about what ’ s possible on the computer. Of course, it ’ s also a great liberating force that makes possible, you know, publishing and so forth, and standards, and so on. But, in a way, the computer makes possible much more than what most people think, and my art has just been about trying to find a personal way of using the computer, and so I end up writing software to do that. Chris has asked me to do a short performance, and so I ’ m going to take just this time — maybe 10 minutes — to do that, and hopefully at the end have just a moment to show you a couple of my other projects in video form. Thank you. We ’ ve got about a minute left. I ’ d just like to show a clip from a most recent project. I did a performance with two singers who specialize in making strange noises with their mouths. And this just came off last September at ARS Electronica ; we repeated it in England. And the idea is to visualize their speech and song behind them with a large screen. We used a computer vision tracking system in order to know where they were. And since we know where their heads are, and we have a wireless mic on them that we ’ re processing the sound from, we ’ re able to create visualizations which are linked very tightly to what they ’ re doing with their speech. This will take about 30 seconds or so. He ’ s making a, kind of, cheek-flapping sound. Well, suffice it to say it ’ s not all like that, but that ’ s part of it. Thanks very much. There ’ s always lots more. I ’ m overtime, so I just wanted to say you can, if you ’ re in New York, you can check out my work at the Whitney Biennial next week, and also at Bitforms Gallery in Chelsea. And with that, I think I should give up the stage, so, thank you so much.

Text Sample 138: NEG Dim 1 – 2004 – Score 4.00 – 2004/t000446.txt

About 15 years ago, I went to visit a friend in Hong Kong. And at the time I was very superstitious. So, upon landing â this was still at the old Hong Kong airport that ’ s Kai Tak, when it was smack in the middle of the city â I thought,”If I see something good, I ’ m going to have a great time here in my two weeks. And if I see something negative, I ’ m going to be miserable, indeed.”So the plane landed in between the buildings and got to a full stop in front of this little billboard. And I actually went to see some of the design companies in Hong Kong in my stay there. And it turned out that â I just went to see, you know, what they are doing in Hong Kong. But I actually walked away with a great job offer. And I flew **back** to Austria, packed my bags, and, another week later, I was again on my way to Hong Kong still superstitions and thinking,”Well, if that

’ Winner ’ billboard is still up, I ’ m going to have a good time working here. But if it ’ s gone, it ’ s going to be really miserable and stressful.”So it turned out that not only was the billboard still up but they had put this one right next to it. On the other hand, it also taught me where superstition gets me because I really had a terrible time in Hong Kong. However, I did have a number of real moments of happiness in my life â of, you know, I think what the conference brochure refers to as”moments that take your breath away.”And since I ’ m a big list maker, I actually listed them all. Now, you don ’ t have to go through the trouble of reading them and I won ’ t read them for you. I know that it ’ s incredibly boring to hear about other people ’ s happinesses. What I did do, though is, I actually looked at them from a design standpoint and just eliminated all the ones that had nothing to do with design. And, very surprisingly, over half of them had, actually, something to do with design. So there are, of course, two different possibilities. There ’ s one from a consumer ’ s point of view â where I was happy while experiencing design. And I ’ ll just give you one example. I had gotten my first Walkman. This is 1983. My brother had this great Yamaha motorcycle that he was willing to borrow to me freely. And The Police ’ s”Synchronicity”cassette had just been released and there was no helmet law in my hometown of Bregenz. So you could drive up into the mountains freely blasting The Police on the new Sony Walkman. And I remember it as a true moment of happiness. You know, of course, they are related to this combination of at least two of them being, you know, design objects. And, you know, there ’ s a scale of happiness when you talk about in design but the motorcycle incident would definitely be, you know, situated somewhere here â right in there between Delight and Bliss. Now, there is the other part, from a designer ’ s standpoint â if you ’ re happy while actually doing it. And one way to see how happy designers are when they ’ re designing could be to look at the authors ’ photos on the **back** of their monographs? So, according to this, the Australians and the Japanese as well as the Mexicans are very happy. While, somewhat, the Spaniards... and, I think, particularly, the Swiss, don ’ t seem to be doing all that well. Last November, a museum opened in Tokyo called The Mori Museum, in a skyscraper, up on the 56th floor. And their inaugural exhibit was called”Happiness.”And I went, very eagerly, to see it, because â well, also, with an eye on this conference. And they interestingly sectioned the exhibit off into four different areas. Under”Arcadia,”they showed things like this, from the Edo period â a hundred ways to write”happiness”in different forms. Or they had this apple by Yoko Ono â that, of course, later on was, you know, made into the label for The Beatles. Under”Nirvana”they showed this Constable painting. And there was a little â an interesting theory about abstraction. This is a blue field â it ’ s actually an Yves Klein painting. And the theory was that if you abstract an image, you really, you know open as much room for the un-representable â and, therefore, you know, are able to involve the viewer more. Then, under”Desire,”they showed these Shunsho paintings â also from the Edo period â ink on silk. And, lastly, under”Harmony,”they had this 13th- century **mandala** from Tibet. Now, what I took away from the exhibit was that maybe with the exception of the **mandala** most of the pieces in there were actually about the visualization of happiness and not about happiness. And I felt a little bit cheated, because the visualization â that ’ s a really easy thing to do. And, you know, my studio â we ’ ve done it all the time. This is, you know, a book. A happy dog â and you take it out, it ’ s an aggressive dog. It ’ s a happy David Byrne and an angry David Byrne. Or a jazz poster with a happy face and a more aggressive face. You know, that ’ s not a big deal to accomplish. It has gotten to the point where, you know, within advertising or within the movie industry,”happy”has gotten such a bad reputation that if you actually want to do something with the subject and still appear authentic, you almost would have to, you know, do it from a cynical point of view. This is, you know, the movie poster. Or we, a couple of weeks ago, designed a box set for The Talking Heads where the happiness visualized on the cover definitely has, very much, a dark side to it. Much, much more difficult is this, where the designs actually can evoke happiness â and I ’ m going to just show you three that actually did this for me. This is a campaign done by a young artist in New York, who calls himself”True.”Everybody who has ridden the New York subway system will be familiar with these signs? True printed his own version of these signs. Met every Wednesday at a subway stop with 20 of his friends. They divided up the different subway lines and added their own version. So this is one. Now, the way this works in the system is that nobody ever looks at these signs. So you ’ re you ’ re really bored in the subway, and you kind of stare at something. And it takes you a while until it actually â you realize that this says something different than what it normally says. I mean, that ’ s, at least, how it made me happy. Now, True is a real humanitarian. He didn ’ t want any of his friends to be arrested, so he supplied everybody with this fake volunteer card. And also gave this fake letter from the MTA to everybody â sort of like pretending that it ’ s an art project financed by The Metropolitan Transit Authority. Another New York project. This is at P. S. 1 â a sculpture that ’ s basically a square room by James Turrell, that has a retractable ceiling. Opens up at dusk and dawn every day. You don ’ t see the horizon. You ’ re just in there, watching the incredible, subtle changes of color in the sky. And the room is truly something to be seen. People ’ s demeanor changes when they go in there. And, for sure, I haven ’ t looked at the sky in the same way after spending an hour in there. There are, of course, more than those three projects that I ’ m showing here. I would definitely say that observing Vik Muniz ’”Cloud”a couple of years ago in Manhattan for sure made me happy, as well. But my last project is, again, from a young designer in New York. He ’ s from Korea originally. And he took it upon himself to print 55, 000 speech bubbles â empty speech bubbles stickers, large ones and small ones. And he goes around New York and just puts them, empty as they are, on posters. And other people go and fill them in. This one says,”Please let me die in peace.”I think that was â the most surprising to myself was that the writing was actually so good. This is on a musician poster, that says :”I am concerned that my CD will not sell more than 200, 000 units and that, as a result, my recoupable advance from my label will be taken from me, after which, my contract will be cancelled, and I ’ ll be **back** doing Journey covers on Bleecker Street.”I think the reason this works so well is because everybody involved wins. Jee gets to have his project ; the public gets a sweeter environment ; and different public gets a place to express itself ; and the advertisers finally get somebody to look at their ads. Well, there was a question, of course, that was on my mind for a while : You know, can I do more of the things that I like doing in design and less of the ones that I don ’ t like to be doing? Which brought me **back** to my list making â you know, just to see what I actually like about my job. You know, one is : just working without pressure. Then : working concentrated, without being frazzled. Or, as Nancy said before, like really immerse oneself into it. Try not to get stuck doing the same thing â or try not get stuck behind the computer all day. This is, you know, related to it : getting out of the studio. Then, of course, trying to, you know, work on things where the content is actually important for me. And being able to enjoy the end results. And then I found another list in one of my diaries that actually contained all the things that I thought I learned in my life so far. And, just about at that time, an Austrian magazine called and asked if we would want to do six spreads â design six spreads that work like dividing pages between the different chapters in the magazine? And the whole thing just fell together. So I just picked one of the things that I thought I learned â in this case,”Everything I do always comes **back** to me”â and we made these spreads right out of this. So it was :”Everything I do always comes **back** to me.”A couple of weeks ago, a French company asked **us** to design five billboards for them. Again, we could supply the content for it. So I just picked another one. And this was two weeks ago. We flew to Arizona â the designer who works with me, and myself â and photographed this one. So it ' s :”Trying to look good limits my life.”And then we did one more of these. This is, again, for a magazine, dividing pages. This is :”Having”â this is the same thing ; it ' s just, you know, photographed from the side. This is from the front. Then it ' s :”guts.”Again, it ' s the same thing â”guts”is just the same room, reworked. Then it ' s :”always works out.”Then it ' s”for,”with the light on. And it ' s”me.”Thank you so much.

Text Sample 139: NEG Dim 1 – 2007 – Score 2.00 – 2007/t000318.txt

Ladies and gentlemen, the history of music and television on the Internet in three minutes. A TED medley â a TEDley. It ' s nine o ' clock on a Saturday The record store ' s closed for the night So I fire up the old iTunes music store And soon I am feelin ' all right I know Steve Jobs can find me a melody With one dollar pricing that rocks I can type in the track and get album names **back** While still in my PJs and socks Sell **us** a song, you ' re the music man My iPod ' s still got 10 gigs to go Yes, we might prefer more compatibility But Steve likes to run the whole show I heard”Desperate Housewives”was great last night But I had a bad piece of cod As I threw up my meal, I thought,”It ' s no big deal”I ' ll watch it tonight on my ' Pod And now all of the networks are joining in Two bucks a show without ads It ' s a business those guys always wanted to try But only Steve Jobs had the ' nads They say we ' re young, don ' t watch TV They say the Internet is all we see But that ' s not true ; they ' ve got it wrong See, all our shows are just two minutes long Hey I got YouTube I got YouTube And now, ladies and gentlemen, a tribute to the Recording Industry Association of America â the RIAA! Young man, you were surfin ' along And then, young man, you downloaded a song And then, dumb man, copied it to your ' Pod Then a phone call came to tell you... You ' ve just been sued by the R-I-A-A You ' ve just been screwed by the R-I-A-A Their attorneys say you committed a crime And there ' d better not be a next time They ' ve lost their minds at the R-I-A-A Justice is blind at the R-I-A-A You ' re depriving the bands You are learning to steal You can ' t do whatever you feel CD sales have dropped every year They ' re not greedy, they ' re just quaking with fear Yes indeedy, what if their end is near And we download all our music Yeah, that would piss off the R-I-A-A No plastic discs from the R-I-A-A What a way to make friends It ' s a plan that can ' t fail All your customers off to jail Who ' ll be next for the R-I-A-A? What else is vexing the R-I-A-A? Maybe whistling a tune Maybe humming along Maybe mocking ' em in a song

Text Sample 140: NEG Dim 1 – 2007 – Score 2.00 – 2007/t002580.txt

You know, what I do is write for children, and I ' m probably America ' s most widely read children ' s author, in fact. And I always tell people that I don ' t want to show up looking like a scientist. You can have me as a farmer, or in leathers, and no one has ever chose farmer. I ' m here today to talk to you about circles and epiphanies. And you know, an epiphany is usually something you find that you dropped someplace. You ’ ve just got to go around the block to see it as an epiphany. That ’ s a painting of a circle. A friend of mine did that — Richard Bollingbroke. It ’ s the kind of complicated circle that I ’ m going to tell you about. My circle began **back** in the ’ 60s in high school in Stow, Ohio where I was the class queer. I was the guy beaten up bloody every week in the boys ’ room, until one teacher saved my life. She saved my life by letting me go to the bathroom in the teachers ’ lounge. She did it in secret. She did it for three years. And I had to get out of town. I had a thumb, I had 85 dollars, and I ended up in San Francisco, California — met a lover — and **back** in the ’ 80s, found it necessary to begin work on AIDS organizations. About three or four years ago, I got a phone call in the middle of the night from that teacher, Mrs. Posten, who said,”I need to see you. I ’ m disappointed that we never got to know each other as adults. Could you please come to Ohio, and please bring that man that I know you have found by now. And I should mention that I have pancreatic cancer, and I ’ d like you to please be quick about this.”Well, the next day we were in Cleveland. We took a look at her, we laughed, we cried, and we knew that she needed to be in a hospice. We found her one, we got her there, and we took care of her and watched over her family, because it was necessary. It ’ s something we knew how to do. And just as the woman who wanted to know me as an adult got to know me, she turned into a box of ashes and was placed in my hands. And what had happened was the circle had closed, it had become a circle — and that epiphany I talked about presented itself. The epiphany is that death is a part of life. She saved my life ; I and my partner saved hers. And you know, that part of life needs everything that the rest of life does. It needs truth and beauty, and I ’ m so happy it ’ s been mentioned so much here today. It also needs — it needs dignity, love and pleasure, and it ’ s our job to hand those things out. Thank you.

Text Sample 141: NEG Dim 1 – 2007 – Score 3.00 – 2007/t000392.txt

So, I kind of believe that we ’ re in like the”cave-painting”era of computer interfaces. Like, they ’ re very kind of â they don ’ t go as deep or as emotionally engaging as they possibly could be and I ’ d like to change all that. Hit me. OK. So I mean, this is the kind of status quo interface, right? It ’ s very flat, kind of rigid. And OK, so you could sex it up and like go to a much more lickable Mac, you know, but really it ’ s the kind of same old crap we ’ ve had for the last, you know, 30 years. Like I think we really put up with a lot of crap with our computers. I mean it ’ s point and click, it ’ s like the menus, icons, it ’ s all the kind of same thing. And so one kind of information space that I take inspiration from is my real desk. It ’ s so much more subtle, so much more visceral â you know, what ’ s visible, what ’ s not. And I ’ d like to bring that experience to the desktop. So I kind of have a â this is BumpTop. It ’ s kind of like a new approach to desktop computing. So you can bump things â they ’ re all physically, you know, manipulable and stuff. And instead of that point and click, it ’ s like a push and pull, things collide as you ’ d expect them. Just like on my real desk, I can â let me just grab these guys â I can turn things into piles instead of just the folders that we have. And once things are in a pile I can browse them by throwing them into a grid, or you know, flip through them like a book or I can lay them out like a deck of cards. When they ’ re laid out, I can pull things to new locations or delete things or just quickly sort a whole pile, you know, just immediately, right? And then, it ’ s all smoothly animated, instead of these jarring changes you see in today ’ s interfaces. Also, if I want to add something to a pile, well, how do I do that? I just toss it to the pile, and it

’ s added right to the top. It ’ s a kind of nice way. Also some of the stuff we can do is, for these individual icons we thought â I mean, how can we play with the idea of an icon, and push that further? And one of the things I can do is make it bigger if I want to emphasize it and make it more important. But what ’ s really cool is that since there ’ s a physics simulation running under this, it ’ s actually heavier. So the lighter stuff doesn ’ t really move but if I throw it at the lighter guys, right? So it ’ s cute, but it ’ s also like a subtle channel of conveying information, right? This is heavy so it feels more important. So it ’ s kind of cool. Despite computers everywhere paper really hasn ’ t disappeared, because it has a lot of, I think, valuable properties. And some of those we wanted to transfer to the icons in our system. So one of the things you can do to our icons, just like paper, is crease them and fold them, just like paper. Remember, you know, something for later. Or if you want to be destructive, you can just crumple it up and, you know, toss it to the corner. Also just like paper, around our workspace we ’ ll pin things up to the wall to remember them later, and I can do the same thing here, and you know, you ’ ll see post-it notes and things like that around people ’ s offices. And I can pull them off when I want to work with them. So, one of the criticisms of this kind of approach to organization is that, you know,”Okay, well my real desk is really messy. I don ’ t want that mess on my computer.”So one thing we have for that is like a grid align, kind of â so you get that more traditional desktop. Things are kind of grid aligned. More boring, but you still have that kind of colliding and bumping. And you can still do fun things like make shelves on your desktop. Let ’ s just break this shelf. Okay, that shelf broke. I think beyond the icons, I think another really cool domain for this software â I think it applies to more than just icons and your desktop â but browsing photographs. I think you can really enrich the way we browse our photographs and bring it to that kind of shoebox of, you know, photos with your family on the kitchen table kind of thing. I can toss these things around. They ’ re so much more tangible and touchable â and you know I can double-click on something to take a look at it. And I can do all that kind of same stuff I showed you before. So I can pile things up, I can flip through it, I can, you know â okay, let ’ s move this photo to the **back**, let ’ s delete this guy here, and I think it ’ s just a much more rich kind of way of interacting with your information. And that ’ s BumpTop. Thanks!

Text Sample 142: NEG Dim 1 – 2002 – Score 2.00 – 2002/t000101.txt

One thing I wanted to say about film making is â about this film â in thinking about some of the wonderful talks we ’ ve heard here, Michael Moschen, and some of the talks about music, this idea that there is a narrative line, and that music exists in time. A film also exists in time ; it ’ s an experience that you should go through emotionally. And in making this film I felt that so many of the documentaries I ’ ve seen were all about learning something, or knowledge, or driven by talking heads, and driven by ideas. And I wanted this film to be driven by emotions, and really to follow my journey. So instead of doing the talking head thing, instead it ’ s composed of scenes, and we meet people along the way. We only meet them once. They don ’ t come **back** several times, so it really chronicles a journey. It ’ s something like life, that once you get in it you can ’ t get out. There are two clips I want to show you, the first one is a kind of hodgepodge, its just three little moments, four little moments with three of the people who are here tonight. It ’ s not the way they occur in the film, because they are part of much larger scenes. They play off each other in a wonderful way. And that ends with a little clip of my father, of Lou, talking about something that is very dear to him, which is the accidents of life. I think he felt that the greatest things in life were accidental, and perhaps not planned at all. And those three clips will be followed by a scene of perhaps what, to me, is really his greatest building which is a building in Dhaka, Bangladesh. He built the capital over there. And I think you ’ ll enjoy this building, it ’ s never been seen â it ’ s been still photographed, but never photographed by a film crew. We were the first film crew in there. So you ’ ll see images of this remarkable building. A couple of things to keep in mind when you see it, it was built entirely by hand, I think they got a crane the last year. It was built entirely by hand off bamboo scaffolding, people carrying these baskets of concrete on their heads, dumping them in the forms. It is the capital of the country, and it took 23 years to build, which is something they seem to be very proud of over there. It took as long as the Taj Mahal. Unfortunately it took so long that Lou never saw it finished. He died in 1974. The building was finished in 1983. So it continued on for many years after he died. Think about that when you see that building, that sometimes the things we strive for so hard in life we never get to see finished. And that really struck me about my father, in the sense that he had such belief that somehow, doing these things giving in the way that he gave, that something good would come out of it, even in the middle of a war, there was a war with Pakistan at one point, and the construction stopped totally and he kept working, because he felt,”Well when the war is done they ’ ll need this building.”So, those are the two clips I ’ m going to show. Roll that tape. Richard Saul Wurman : I remember hearing him talk at Penn. And I came home and I said to my father and mother,”I just met this man : doesn ’ t have much work, and he ’ s sort of ugly, funny voice, and he ’ s a teacher at school. I know you ’ ve never heard of him, but just mark this day that someday you will hear of him, because he ’ s really an amazing man.”Frank Gehry : I heard he had some kind of a fling with Ingrid Bergman. Is that true? Nathaniel Kahn : If he did he was a very lucky man. NK : Did you hear that, really? FG : Yeah, when he was in Rome. Moshe Safdie : He was a real nomad. And you know, when I knew him when I was in the office, he would come in from a trip, and he would be in the office for two or three days intensely, and he would pack up and go. You know he ’ d be in the office till three in the morning working with **us** and there was this kind of sense of the nomad in him. I mean as tragic as his death was in a railway station, it was so consistent with his life, you know? I mean I often think I ’ m going to die in a plane, or I ’ m going to die in an airport, or die jogging without an identification on me. I don ’ t know why I sort of carry that from that memory of the way he died. But he was a sort of a nomad at heart. Louis Kahn : How accidental our existences are really and how full of influence by circumstance. Man : We are the morning workers who come, all the time, here and enjoy the walking, city ’ s beauty and the atmosphere and this is the nicest place of Bangladesh. We are proud of it. NK : You ’ re proud of it? Man : Yes, it is the national image of Bangladesh. NK : Do you know anything about the architect? Man : Architect? I ’ ve heard about him ; he ’ s a top-ranking architect. NK : Well actually I ’ m here because I ’ m the architect ’ s son, he was my father. Man : Oh! Dad is Louis Farrakhan? NK : Yeah. No not Louis Farrakhan, Louis Kahn. Man : Louis Kahn, yes! Man : Your father, is he alive? NK : No, he ’ s been dead for 25 years. Man : Very pleased to welcome you **back**. NK : Thank you. NK : He never saw it finished, Pop. No, he never saw this. Shamsul Wares : It was almost impossible, building for a country like ours. In 30, 50 years **back**, it was nothing, only paddy fields, and since we invited him here, he felt that he has got a responsibility. He wanted to be a Moses here, he gave **us** democracy. He is not a political man, but in this guise he has given **us** the institution for democracy, from where we can rise. In that way it is so relevant. He didn ’ t care for how much money this country has, or whether he would be able to ever finish this building, but somehow he has been able to do it, build it, here. And this is the largest project he has got in here, the poorest country in the world. NK : It cost him his life. SW : Yeah, he paid. He paid his life for this, and that is why he is great and we ’ ll remember him. But he was also human. Now his failure to satisfy the family life, is an inevitable association of great people. But I think his son will understand this, and will have no sense of grudge, or sense of being neglected, I think. He cared in a very different manner, but it takes a lot of time to understand that. In social aspect of his life he was just like a child, he was not at all matured. He could not say no to anything, and that is why, that he cannot say no to things, we got this building today. You see, only that way you can be able to understand him. There is no other shortcut, no other way to really understand him. But I think he has given **us** this building and we feel all the time for him, that ’ s why, he has given love for **us**. He could not probably give the right kind of love for you, but for **us**, he has given the people the right kind of love, that is important. You have to understand that. He had an enormous amount of love, he loved everybody. To love everybody, he sometimes did not see the very closest ones, and that is inevitable for men of his stature.

Text Sample 143: NEG Dim 1 – 2002 – Score 2.00 – 2002/t000333.txt

Thank you. Ooh, I ’ m like,”Phew, phew, calm down. Get **back** into my body now.”Usually when I play out, the first thing that happens is people scream out,”What ’ s she doing?!”I ’ ll play at these rock shows, be on stage standing completely still, and they ’ re like,”What ’ s she doing?! What ’ s she doing?!”And then I ’ ll kind of be like — — and then they ’ re like,”Whoa!”I ’ m sure you ’ re trying to figure out,”Well, how does this thing work?”Well, what I ’ m doing is controlling the pitch with my left hand. See, the closer I get to this antenna, the higher the note gets — — and you can get it really low. And with this hand I ’ m controlling the volume, so the further away my right hand gets, the louder it gets. So basically, with both of your hands you ’ re controlling pitch and volume and kind of trying to create the illusion that you ’ re doing separate notes, when really it ’ s continuously going... Sometimes I startle myself : I ’ ll forget that I have it on, and I ’ ll lean over to pick up something, and then it goes like — —”Oh!”And it ’ s like a funny sound effect that follows you around if you don ’ t turn the thing off. Maybe we ’ ll go into the next tune, because I totally lost where this is going. We ’ re going to do a song by David Mash called”Listen : the Words Are Gone,”and maybe I ’ ll have words come **back** into me afterwards if I can relax. So, I ’ m trying to think of some of the questions that are commonly asked ; there are so many. And... Well, I guess I could tell you a little of the history of the theremin. It was invented around the 1920s, and the inventor, Lon Theremin — he also was a musician besides an inventor — he came up with the idea for making the theremin, I think, when he was working on some shortwave radios. And there ’ d be that sound in the signal — it ’ s like — and he thought,”Oh, what if I could control that sound and turn it into an instrument, because there are pitches in it.”And so somehow through developing that, he eventually came to make the theremin the way it is now. And a lot of times, even kids nowadays, they ’ ll make reference to a theremin by going,”Whoo-hoo-hoo-hoo,”because in the ’ 50s it was used in the sci-fi horror movies, that sound that ’ s like... It ’ s kind of a funny, goofy sound to do. And sometimes if I have too much coffee, then my vibrato gets out of hand. You ’ re really sensitive to your body and its functions when you ’ re behind this thing. You have to stay so still if you want to have the most control. It reminds me of the balancing act earlier on — what Michael was doing — because you ’ re fighting so hard to keep the balance with what you ’ re playing with and stay in tune, and at the same time you don ’ t want to focus so much on being in tune all the time ; you want to be feeling the music. And then also, you ’ re trying to stay very, very, very still because little movements with other parts of your body will affect the pitch, or sometimes if you ’ re holding a low note — — and breathing will make it... If I pass out on the next song... I think of it almost like like a yoga instrument because it makes you so aware of every little crazy thing your body is doing, or just aware of what you don ’ t want it to be doing while you ’ re playing ; you don ’ t want to have any sudden movements. And if I go to a club and play a gig, people are like,”Here, have some drinks on **us**!”And it ’ s like,”Well, I ’ m about to go on soon ; I don ’ t want to be like — — you know?”It really does reflect the mood that you ’ re in also, if you ’ re... it ’ s similar to being a vocalist, except instead of it coming out of your throat, you ’ re controlling it just in the air and you don ’ t really have a point of reference ; you ’ re always relying on your ears and adjusting constantly. You just have to always adjust to what ’ s happening and realize you ’ ll have bummer notes come here and there and listen to it, adjust it, and just move on, or else you ’ ll get too tied up and go crazy. Like me. I think we will play another tune now. I ’ m going to do”Lush Life.”It ’ s one of my favorite tunes to play.

Text Sample 144: NEG Dim 1 – 2002 – Score 4.00 – 2002/t000079.txt

I ’ m going to read a few strips. These are, most of these are from a monthly page I do in and architecture and design magazine called Metropolis. And the first story is called”The Faulty Switch.”Another beautifully designed new building ruined by the sound of a common wall light switch. It ’ s fine during the day when the main rooms are flooded with sunlight. But at dusk everything changes. The architect spent hundreds of hours designing the burnished brass switchplates for his new office tower. And then left it to a contractor to install these 79-cent switches behind them. We know instinctively where to reach when we enter a dark room. We automatically throw the little nub of plastic upward. But the sound we are greeted with, as the room is bathed in the simulated glow of late-afternoon light, recalls to mind a dirty men ’ s room in the rear of a Greek coffee shop. This sound colors our first impression of any room ; it can ’ t be helped. But where does this sound, commonly described as a click, come from? Is it simply the byproduct of a crude mechanical action? Or is it an imitation of one half the set of sounds we make to express disappointment? The often dental consonant of no Indo-European language. Or is it the amplified sound of a synapse firing in the brain of a cockroach? In the 1950s they tried their best to muffle this sound with mercury switches and silent knob controls. But today these improvements seem somehow inauthentic. The click is the modern triumphal clarion proceeding **us** through life, announcing our entry into every lightless room. The sound made flicking a wall switch off is of a completely different nature. It has a deep melancholy ring. Children don ’ t like it. It ’ s why they leave lights on around the house. Adults find it comforting. But wouldn ’ t it be an easy matter to wire a wall switch so that it triggers the muted horn of a steam ship? Or the recorded crowing of a rooster? Or the distant peel of thunder? Thomas Edison went through thousands of unlikely substances before he came upon the right one for the filament of his electric light bulb. Why have we settled so quickly for the sound of its switch? That ’ s the end of that. The next story is called”In Praise of the

Taxpayer.”That so many of the city ’ s most venerable taxpayers have survived yet another commercial building boom, is cause for celebration. These one or two story structures, designed to yield only enough income to cover the taxes on the land on which they stand, were not meant to be permanent buildings. Yet for one reason or another they have confounded the efforts of developers to be combined into lots suitable for high-rise construction. Although they make no claim to architectural beauty, they are, in their perfect **temporariness**, a delightful alternative to the large-scale structures that might someday take their place. The most perfect examples occupy corner lots. They offer a pleasant respite from the high-density development around them. A break of light and air, an architectural biding of time. So buried in signage are these structures, that it often takes a moment to distinguish the modern specially constructed taxpayer from its neighbor : the small commercial building from an earlier century, whose upper floors have been sealed, and whose ground-floor space now functions as a taxpayer. The few surfaces not covered by signs are often clad in a distinctive, dark green-gray, striated aluminum siding. Take-out sandwich shops, film processing drop-offs, peep-shows and necktie stores. Now these provisional structures have, in some cases, remained standing for the better part of a human lifetime. The temporary building is a triumph of modern industrial organization, a healthy sublimation of the urge to build, and proof that not every architectural idea need be set in stone. That ’ s the end. And the next story is called,”On the Human Lap.”For the ancient Egyptians the lap was a platform upon which to place the earthly possessions of the dead — 30 cubits from foot to knee. It was not until the 14th century that an Italian painter recognized the lap as a Grecian temple, upholstered in flesh and cloth. Over the next 200 years we see the infant Christ go from a sitting to a standing position on the Virgin ’ s lap, and then **back** again. Every child recapitulates this ascension, straddling one or both legs, sitting sideways, or leaning against the body. From there, to the modern ventriloquist ’ s dummy, is but a brief moment in history. You were late for school again this morning. The ventriloquist must first make **us** believe that a small boy is sitting on his lap. The illusion of speech follows incidentally. What have you got to say for yourself, Jimmy? As adults we admire the lap from a nostalgic distance. We have fading memories of that provisional temple, erected each time an adult sat down. On a crowded bus there was always a lap to sit on. It is children and teenage girls who are most keenly aware of its architectural beauty. They understand the structural integrity of a deep avuncular lap, as compared to the shaky arrangement of a neurotic niece in high heels. The relationship between the lap and its owner is direct and intimate. I envision a 36-story, 450-unit residential high-rise — a reason to consider the mental health of any architect before granting an important commission. The bathrooms and kitchens will, of course, have no windows. The lap of luxury is an architectural construct of childhood, which we seek, in vain, as adults, to employ. That ’ s the end. The next story is called”The Haverpiece Collection”A nondescript warehouse, visible for a moment from the northbound lanes of the Prykushko Expressway, serves as the temporary resting place for the Haverpiece collection of European dried fruit. The profound convolutions on the surface of a dried cherry. The foreboding sheen of an extra-large date. Do you remember wandering as a child through those dark wooden storefront galleries? Where everything was displayed in poorly labeled roach-proof bins. Pears dried in the form of genital organs. Apricot halves like the ears of cherubim. In 1962 the unsold stock was purchased by Maurice Haverpiece, a wealthy prune juice bottler, and consolidated to form the core collection. As an art form it lies somewhere between still-life painting and plumbing. Upon his death in 1967, a quarter of the items were sold off for compote to a high-class hotel restaurant. Unsuspecting guests were served stewed turn-of-the-century Turkish figs for breakfast. The rest of the collection remains here, stored in plain brown paper bags until funds can be raised to build a permanent museum and study center. A shoe made of apricot leather for the daughter of a czar. That ’ s the end. Thank you.

Text Sample 145: NEG Dim 1 – 1998 – Score 2.00 – 1998/t000118.txt

Back in 1992, I started working for a company called Interval Research, which was just then being founded by David Lidell and Paul Allen as a for-profit research enterprise in Silicon Valley. I met with David to talk about what I might do in his company. I was just coming out of a failed virtual reality business and supporting myself by being on the speaking circuit and writing books — after twenty years or so in the computer game industry having ideas that people didn ’ t think they could sell. And David and I discovered that we had a question in common, that we really wanted the answer to, and that was,”Why hasn ’ t anybody built any computer games for little girls?”Why is that? It can ’ t just be a giant sexist conspiracy. These people aren ’ t that smart. There ’ s six billion dollars on the table. They would go for it if they could figure out how. So, what is the deal here? And as we thought about our goals — I should say that Interval is really a humanistic institution, in the classical sense that humanism, at its best, finds a way to combine **clear-eyed** empirical research with a set of core values that fundamentally love and respect people. The basic idea of humanism is the improvable quality of life ; that we can do good things, that there are things worth doing because they ’ re good things to do, and that **clear-eyed** empiricism can help **us** figure out how to do them. So, contrary to popular belief, there is not a conflict of interest between empiricism and values. And Interval Research is kind of the living example of how that can be true. So David and I decided to go find out, through the best research we could muster, what it would take to get a little girl to put her hands on a computer, to achieve the level of comfort and ease with the technology that little boys have because they play video games. We spent two and a half years conducting research ; we spent another year and a half in advance development. Then we formed a spin-off company. In the research phase of the project at Interval, we partnered with a company called Cheskin Research, and these people — Davis Masten and Christopher Ireland — changed my mind entirely about what market research was and what it could be. They taught me how to look and see, and they did not do the incredibly stupid thing of saying to a child,”Of all these things we already make you, which do you like best?”— which gives you zero answers that are usable. So, what we did for the first two and a half years was four things : We did an extensive review of the literature in related fields, like cognitive psychology, spatial cognition, gender studies, play theory, sociology, primatology. Thank you Frans de Waal, wherever you are, I love you and I ’ d give anything to meet you. After we had done that with a pretty large team of people and discovered what we thought the salient issues were with girls and boys and playing — because, after all, that ’ s really what this is about — we moved to the second phase of our work, where we interviewed adult experts in academia, some of the people who ’ d produced the literature that we found relevant. Also, we did focus groups with people who were on the ground with kids every day, like playground supervisors. We talked to them, confirmed some hypotheses and identified some serious questions about gender difference and play. Then we did what I consider to be the heart of the work : interviewed 1, 100 children, boys and girls, ages seven to 12, all over the United States —

except for Silicon Valley, Boston and Austin because we knew that their little families would have millions of computers in them and they wouldn ’ t be a representative sample. And at the end of those remarkable conversations with kids and their best friends across the United States, after two years, we pulled together some survey data from another 10, 000 children, drew up a set up of what we thought were the key findings of our research, and spent another year transforming them into design heuristics, for designing computer-based products — and, in fact, any kind of products — for little girls, ages eight to 12. And we spent that time designing interactive prototypes for computer software and testing them with little girls. In 1996, in November, we formed the company Purple Moon which was a spinoff of Interval Research, and our chief investors were Interval Research, Vulcan Northwest, Institutional Venture Partners and Allen and Company. We launched a website on September 2nd that has now served 25 million pages, and has 42, 000 registered young girl users. They visit an average of one and a half times a day, spend an average of 35 minutes a visit, and look at 50 pages a visit. So we feel that we ’ ve formed a successful online community with girls. We launched two titles in October —”Rockett ’ s New School”— the first of a series of products — is about a character called Rockett beginning her first day of school in eighth grade at a brand new place, with a blank slate, which allows girls to play with the question of,”What will I be like when I ’ m older?””What ’ s it going to be like to be in high school or junior high school? Who are my friends?”; to exercise the love of social complexity and the narrative intelligence that drives most of their play behavior ; and which embeds in it values about noticing that we have lots of choices in our lives and the ways that we conduct ourselves. The other title that we launched is called”Secret Paths in the Forest,”which addresses the more fantasy-oriented, inner lives of girls. These two titles both showed up in the top 50 entertainment titles in PC Data — entertainment titles in PC Data in December, right up there with”John Madden Football,”which thrills me to death. So, we ’ re real, and we ’ ve touched several hundreds of thousands of little girls. We ’ ve made half-a-billion impressions with marketing and PR for this brand, Purple Moon. Ninety-six percent of them, roughly, have been positive ; four percent of them have been”other.”I want to talk about the other, because the politics of this enterprise, in a way, have been the most fascinating part of it, for me. There are really two kinds of negative reviews that we ’ ve received. One kind of reviewer is a male gamer who thinks he knows what games ought to be, and won ’ t show the product to little girls. The other kind of reviewer is a certain flavor of feminist who thinks they know what little girls ought to be. And so it ’ s funny to me that these interesting, odd bedfellows have one thing in common : they don ’ t listen to little girls. They haven ’ t looked at children and they ’ re certainly not demonstrating any love for them. I ’ d like to play you some voices of little girls from the two-and-a-half years of research that we did — actually, some of the voices are more recent. And these voices will be accompanied by photographs that they took for **us** of their lives, of the things that they value and care about. These are pictures the girls themselves never saw, but they gave to **us** This is the stuff those reviewers don ’ t know about and aren ’ t listening to and this is the kind of research I recommend to those who want to do humanistic work. Girl 1 : Yeah, my character is usually a tomboy. Hers is more into boys. Girl 2 : Uh, yeah. Girl 1 : We have — in the very beginning of the whole game, always we do this : we each have a piece of paper ; we write down our name, our age — are we rich, very rich, not rich, poor, medium, wealthy, boyfriends, dogs, pets — what else — sisters, brothers, and all those. Girl 2 : Divorced — parents divorced, maybe. Girl 3 : This is my pretend [unclear] one. Girl 4 : We make a school newspaper on the computer. Girl 5 : For a girl ’ s game also usually they ’ ll have really pretty scenery with clouds and flowers. Girl 6 : Like, if you were a girl and you were really adventurous and a real big tomboy, you would think that girls ’ games were kinda sissy. Girl 7 : I run track, I played soccer, I play basketball, and I love a lot of things to do. And sometimes I feel like I can ’ t really enjoy myself unless it ’ s like a vacation, like when I get Mondays and all those days off. Girl 8 : Well, sometimes there is a lot of stuff going on because I have music lessons and I ’ m on swim team — all this different stuff that I have to do, and sometimes it gets overwhelming. Girl 9 : My friend Justine kinda took my friend Kelly, and now they ’ re being mean to me. Girl 10 : Well, sometimes it gets annoying when your brothers and sisters, or brother or sister, when they copy you and you get your idea first and they take your idea and they do it themselves. Girl 11 : Because my older sister, she gets everything and, like, when I ask my mom for something, she ’ ll say,”No”—all the time. But she gives my sister everything. Brenda Laurel : I want to show you, real quickly, just a minute of”Rockett ’ s Tricky Decision,”which went gold two days ago. Let ’ s hope it ’ s really stable. This is the second day in Rockett ’ s life. The reason I ’ m showing you this is I ’ m hoping that the scene that I ’ m going to show you will look familiar and sound familiar, now that you ’ ve listened to some girls ’ voices. And you can see how we ’ ve tried to incorporate the issues that matter to them in the game that we ’ ve created. Miko : Hey Rockett! C ’ mere! Rockett : Hi Miko! What ’ s going on? Miko : Did you hear about Nakilia ’ s big Halloween party this weekend? She asked me to make sure you knew about it. Nakilia invited Reuben too, but — Rockett : But what? Isn ’ t he coming? Miko : I don ’ t think so. I mean, I heard his band is playing at another party the same night. Rockett : Really? What other party? Girl : Max ’ s party is going to be so cool, Whitney. He ’ s invited all the best people. BL : I ’ m going to fast-forward to the decision point because I know I don ’ t have a lot of time. After this awful event occurs, Rocket gets to decide how she feels about it. Rockett : Who ’ d want to show up at that party anyway? I could get invited to that party any day if I wanted to. Gee, I doubt I ’ ll make Max ’ s best people list. BL : OK, so we ’ re going to emotionally navigate. If we were playing the game, that ’ s what we ’ d do. If at any time during the game we want to learn more about the characters, we can go into this hidden hallway, and I ’ ll quickly just show you the interface. We can, for example, go find Miko ’ s locker and get some more information about her. Oops, I turned the wrong way. But you get the general idea of the product. I wanted to show you the ways, innocuous as they seem, in which we ’ re incorporating what we ’ ve learned about girls — their desires to experience greater emotional flexibility, and to play around with the social complexity of their lives. I want to make the point that what we ’ re giving girls, I think, through this effort, is a kind of validation, a sense of being seen. And a sense of the choices that are available in their lives. We love them. We see them. We ’ re not trying to tell them who they ought to be. But we ’ re really, really happy about who they are. It turns out they ’ re really great. I want to close by showing you a videotape that ’ s a version of a future game in the Rockett series that our graphic artists and design people put together, that we feel would please that four percent of reviewers.”Rockett 28!”Rockett : It ’ s like I ’ m just waking up, you know? BL : Thanks.

Text Sample 146: NEG Dim 1 – 1998 – Score 2.00 – 1998/t000113.txt

Sheryl Shade : Hi, Aimee. Aimee Mullins : Hi. SS : Aimee and I thought we ’ d just talk a little bit, and I wanted her to tell all of you what makes her a distinctive athlete. AM : Well, for those of you who have seen the picture in the little bio â it might have given it away â I ’ m a double amputee, and I was born without fibulas in both legs. I was amputated at age one, and I ’ ve been running like hell ever since, all over the place. SS : Well, why don ’ t you tell them how you got to Georgetown â why don ’ t we start there? Why don ’ t we start there? AM : I ’ m a senior in Georgetown in the Foreign Service program. I won a full academic scholarship out of high school. They pick three students out of the nation every year to get involved in international affairs, and so I won a full ride to Georgetown and I ’ ve been there for four years. Love it. SS : When Aimee got there, she decided that she ’ s, kind of, curious about track and field, so she decided to call someone and start asking about it. So, why don ’ t you tell that story? AM : Yeah. Well, I guess I ’ ve always been involved in sports. I played softball for five years growing up. I skied competitively throughout high school, and I got a little restless in college because I wasn ’ t doing anything for about a year or two sports-wise. And I ’ d never competed on a disabled level, you know â I ’ d always competed against other able-bodied athletes. That ’ s all I ’ d ever known. In fact, I ’ d never even met another amputee until I was 17. And I heard that they do these track meets with all disabled runners, and I figured,”Oh, I don ’ t know about this, but before I judge it, let me go see what it ’ s all about.”So, I booked myself a flight to Boston in ’ 95, 19 years old and definitely the dark horse candidate at this race. I ’ d never done it before. I went out on a gravel track a couple of weeks before this meet to see how far I could run, and about 50 meters was enough for me, panting and heaving. And I had these legs that were made of a wood and plastic compound, attached with Velcro straps â big, thick, five-ply wool socks on â you know, not the most comfortable things, but all I ’ d ever known. And I ’ m up there in Boston against people wearing legs made of all things â carbon graphite and, you know, shock absorbers in them and all sorts of things â and they ’ re all looking at me like, OK, we know who ’ s not going to win this race. And, I mean, I went up there expecting â I don ’ t know what I was expecting â but, you know, when I saw a man who was missing an entire leg go up to the high jump, hop on one leg to the high jump and clear it at six feet, two inches... Dan O ’ Brien jumped 5 ’ 11”in ’ 96 in Atlanta, I mean, if it just gives you a comparison of â these are truly accomplished athletes, without qualifying that word”athlete.”And so I decided to give this a shot : heart pounding, I ran my first race and I beat the national record-holder by three hundredths of a second, and became the new national record-holder on my first try out. And, you know, people said,”Aimee, you know, you ’ ve got speed â you ’ ve got natural speed â but you don ’ t have any skill or finesse going down that track. You were all over the place. We all saw how hard you were working.”And so I decided to call the track coach at Georgetown. And I thank god I didn ’ t know just how huge this man is in the track and field world. He ’ s coached five Olympians, and the man ’ s office is lined from floor to ceiling with All America certificates of all these athletes he ’ s coached. He ’ s just a rather intimidating figure. And I called him up and said,”Listen, I ran one race and I won...””I want to see if I can, you know â I need to just see if I can sit in on some of your practices, see what drills you do and whatever.”That ’ s all I wanted â just two practices.”Can I just sit in and see what you do?”And he said,”Well, we should meet first, before we decide anything.”You know, he ’ s thinking,”What am I getting myself into?”So, I met the man, walked in his office, and saw these posters and magazine covers of people he has coached. And we got to talking, and it turned out to be a great partnership because he ’ d never coached a disabled athlete, so therefore he had no preconceived notions of what I was or wasn ’ t capable of, and I ’ d never been coached before. So this was like,”Here we go â let ’ s start on this trip.”So he started giving me four days a week of his lunch break, his free time, and I would come up to the track and train with him. So that ’ s how I met Frank. That was fall of ’ 95. But then, by the time that winter was rolling around, he said,”You know, you ’ re good enough. You can run on our women ’ s track team here.”And I said,”No, come on.”And he said,”No, no, really. You can. You can run with our women ’ s track team.”In the spring of 1996, with my goal of making the U. S. Paralympic team that May coming up full speed, I joined the women ’ s track team. And no disabled person had ever done that â run at a collegiate level. So I don ’ t know, it started to become an interesting mix. SS : Well, on your way to the Olympics, a couple of memorable events happened at Georgetown. Why don ’ t you just tell them? AM : Yes, well, you know, I ’ d won everything as far as the disabled meets â everything I competed in â and, you know, training in Georgetown and knowing that I was going to have to get used to seeing the **backs** of all these women ’ s shirts â you know, I ’ m running against the next Flo-Jo â and they ’ re all looking at me like,”Hmm, what ’ s, you know, what ’ s going on here?”And putting on my Georgetown uniform and going out there and knowing that, you know, in order to become better â and I ’ m already the best in the country â you know, you have to train with people who are inherently better than you. And I went out there and made it to the Big East, which was sort of the championship race at the end of the season. It was really, really hot. And it ’ s the first â I had just gotten these new sprinting legs that you see in that bio, and I didn ’ t realize at that time that the amount of sweating I would be doing in the sock â it actually acted like a lubricant and I ’ d be, kind of, pistoning in the socket. And at about 85 meters of my 100 meters sprint, in all my glory, I came out of my leg. Like, I almost came out of it, in front of, like, 5, 000 people. And I, I mean, was just mortified â because I was signed up for the 200, you know, which went off in a half hour. I went to my coach :”Please, don ’ t make me do this.”I can ’ t do this in front of all those people. My legs will come off. And if it came off at 85 there ’ s no way I ’ m going 200 meters. And he just sat there like this. My pleas fell on deaf ears, thank god. Because you know, the man is from Brooklyn ; he ’ s a big man. He says,”Aimee, so what if your leg falls off? You pick it up, you put the damn thing **back** on, and finish the goddamn race!”And I did. So, he kept me in line. He kept me on the right track. SS : So, then Aimee makes it to the 1996 Paralympics, and she ’ s all excited. Her family ’ s coming down â it ’ s a big deal. It ’ s now two years that you ’ ve been running? AM : No, a year. SS : A year. And why don ’ t you tell them what happened right before you go run your race? AM : Okay, well, Atlanta. The Paralympics, just for a little bit of clarification, are the Olympics for people with physical disabilities â amputees, persons with cerebral palsy, and wheelchair athletes â as opposed to the Special Olympics, which deals with people with mental disabilities. So, here we are, a week after the Olympics and down at Atlanta, and I ’ m just blown away by the fact that just a year ago, I got out on a gravel track and couldn ’ t run 50 meters. And so, here I am â never lost. I set new records at the U. S. Nationals â the Olympic trials â that May, and was sure that I was coming home with the gold. I was also the only, what they call”bilateral BK”â below the knee. I was the only woman who would be doing the long jump. I had just done the long jump, and a guy who was missing two legs came up to me and says,”How do you do that? You know, we ’ re supposed to have a **planar** foot, so we can ’ t get off on the springboard.”I said,”Well, I just did it. No one told me that.”So, it ’ s funny â I ’ m three inches within the world record â and kept on from that point, you know, so I ’ m signed up in the long jump â signed up? No, I made it for the long jump and the 100-meter. And I ’ m sure of it, you know? I made the front page of my hometown paper that I delivered for six years, you know? It was, like, this is my time for shine. And we ’ re at the trainee warm-up track, which is a few blocks away from the Olympic stadium. These legs that I was on, which I ’ ll take out right now â I was the first person in the world on these legs. I was the guinea pig., I ’ m telling you, this was, like â talk about a tourist attraction. Everyone was taking pictures â”What is this girl running on?”And I ’ m always looking around, like, where is my competition? It ’ s my first international meet. I tried to get it out of anybody I could, you know,”Who am I running against here?””Oh, Aimee, we ’ ll have to get **back** to you on that one.”I wanted to find out times.”Don ’ t worry, you ’ re doing great.”This is 20 minutes before my race in the Olympic stadium, and they post the heat sheets. And I go over and look. And my fastest time, which was the world record, was 15. 77. Then I ’ m looking : the next lane, lane two, is 12. 8. Lane three is 12. 5. Lane four is 12. 2. I said,”What ’ s going on?”And they shove **us** all into the shuttle bus, and all the women there are missing a hand. So, I ’ m just, like â they ’ re all looking at me like ’ which one of these is not like the other, ’ you know? I ’ m sitting there, like,”Oh, my god. Oh, my god.”You know, I ’ d never lost anything, like, whether it would be the scholarship or, you know, I ’ d won five golds when I skied. In everything, I came in first. And Georgetown â that was great. I was losing, but it was the best training because this was Atlanta. Here we are, like, crĂ¨me de la crĂ¨me, and there is no doubt about it, that I ’ m going to lose big. And, you know, I ’ m just thinking,”Oh, my god, my whole family got in a van and drove down here from Pennsylvania.”And, you know, I was the only female U. S. sprinter. So they call **us** out and, you know â”Ladies, you have one minute.”And I remember putting my blocks in and just feeling horrified because there was just this murmur coming over the crowd, like, the ones who are close enough to the starting line to see. And I ’ m like,”I know! Look! This isn ’ t right.”And I ’ m thinking that ’ s my last card to play here ; if I ’ m not going to beat these girls, I ’ m going to mess their heads a little, you know? I mean, it was definitely the”Rocky IV”sensation of me versus Germany, and everyone else â Estonia and Poland â was in this heat. And the gun went off, and all I remember was finishing last and fighting **back** tears of frustration and incredible â incredible â this feeling of just being overwhelmed. And I had to think,”Why did I do this?”If I had won everything â but it was like, what was the point? All this training â I had transformed my life. I became a collegiate athlete, you know. I became an Olympic athlete. And it made me really think about how the achievement was getting there. I mean, the fact that I set my sights, just a year and three months before, on becoming an Olympic athlete and saying,”Here ’ s my life going in this direction â and I want to take it here for a while, and just seeing how far I could push it.”And the fact that I asked for help â how many people jumped on board? How many people gave of their time and their expertise, and their patience, to deal with me? And that was this collective glory â that there was, you know, 50 people behind me that had joined in this incredible experience of going to Atlanta. So, I apply this sort of philosophy now to everything I do : sitting **back** and realizing the progression, how far you ’ ve come at this day to this goal, you know. It ’ s important to focus on a goal, I think, but also recognize the progression on the way there and how you ’ ve grown as a person. That ’ s the achievement, I think. That ’ s the real achievement. SS : Why don ’ t you show them your legs? AM : Oh, sure. SS : You know, show **us** more than one set of legs. AM : Well, these are my pretty legs. No, these are my cosmetic legs, actually, and they ’ re absolutely beautiful. You ’ ve got to come up and see them. There are hair follicles on them, and I can paint my toenails. And, seriously, like, I can wear heels. Like, you guys don ’ t understand what that ’ s like to be able to just go into a shoe store and buy whatever you want. SS : You got to pick your height? AM : I got to pick my height, exactly. Patrick Ewing, who played for Georgetown in the ’ 80s, comes **back** every summer. And I had incessant fun making fun of him in the training room because he ’ d come in with foot injuries. I ’ m like,”Get it off! Don ’ t worry about it, you know. You can be eight feet tall. Just take them off.”He didn ’ t find it as humorous as I did, anyway. OK, now, these are my sprinting legs, made of carbon graphite, like I said, and I ’ ve got to make sure I ’ ve got the right socket. No, I ’ ve got so many legs in here. These are â do you want to hold that actually? That ’ s another leg I have for, like, tennis and softball. It has a shock absorber in it so it, like,”Shhhh,”makes this neat sound when you jump around on it. All right. And then this is the silicon sheath I roll over, to keep it on. Which, when I sweat, you know, I ’ m pistoning out of it. SS : Are you a different height? AM : In these? SS : In these. AM : I don ’ t know. I don ’ t think so. I may be a little taller. I actually can put both of them on. SS : She can ’ t really stand on these legs. She has to be moving, so... AM : Yeah, I definitely have to be moving, and balance is a little bit of an art in them. But without having the silicon sock, I ’ m just going to try slip in it. And so, I run on these, and have shocked half the world on these. These are supposed to simulate the actual form of a sprinter when they run. If you ever watch a sprinter, the ball of their foot is the only thing that ever hits the track. So when I stand in these legs, my hamstring and my **glutes** are contracted, as they would be had I had feet and were standing on the ball of my feet. AM : It ’ s a company in San Diego called Flex-Foot. And I was a guinea pig, as I hope to continue to be in every new form of prosthetic limbs that come out. But actually these, like I said, are still the actual prototype. I need to get some new ones because the last meet I was at, they were everywhere. You know, it ’ s like a big â it ’ s come full circle. Moderator : Aimee and the designer of them will be at TEDMED 2, and we ’ ll talk about the design of them. AM : Yes, we ’ ll do that. SS : Yes, there you go. AM : So, these are the sprint legs, and I can put my other... SS : Can you tell about who designed your other legs? AM : Yes. These I got in a place called Bournemouth, England, about two hours south of London, and I ’ m the only person in the United States with these, which is a crime because they are so beautiful. And I don ’ t even mean, like, because of the toes and everything. For me, while I ’ m such a serious athlete on the track, I want to be feminine off the track, and I think it ’ s so important not to be limited in any capacity, whether it ’ s, you know, your mobility or even fashion. I mean, I love the fact that I can go in anywhere and pick out what I want â the shoes I want, the skirts I want â and I ’ m hoping to try to bring these over here and make them accessible to a lot of people. They ’ re also silicon. This is a really basic, basic prosthetic limb under here. It ’ s like a Barbie foot under this. It is. It ’ s just stuck in this position, so I have to wear a **two-inch** heel. And, I mean, it ’ s really â let me take this off so you can see it. I don ’ t know how good you can see it, but, like, it really is. There ’ re veins on the feet, and then my heel is pink, and my Achilles ’ tendon â that moves a little bit. And it ’ s really an amazing store. I got them a year and two weeks ago. And this is just a silicon piece of skin. I mean, what happened was, two years ago this man in Belgium was saying,”God, if I can go to Madame Tussauds ’ wax museum and see Jerry Hall replicated down to the color of her eyes, looking so real as if she breathed, why can ’ t they build a limb for someone that looks like a leg, or an arm, or a hand?”I mean, they make ears for burn victims. They do amazing stuff with silicon. SS : Two weeks ago, Aimee was up for the Arthur Ashe award at the ESPYs. And she came into town and she rushed around and she said,”I have to buy some new shoes!”We ’ re an hour before the ESPYs, and she thought she ’ d gotten a **two-inch** heel but she ’ d actually bought a three-inch heel. AM : And this poses a problem for me, because it means I ’ m walking like that all night long. SS : For 45 minutes. Luckily, the hotel was terrific. They got someone to come in and saw off the shoes. AM : I said to the receptionist â I mean, I am just harried, and Sheryl ’ s at my side â I said,”Look, do you have anybody here who could help me? Because I have this problem...”You know, at first they were just going to write me off, like,”If you don ’ t like your shoes, sorry. It ’ s too late.””No, no, no, no. I ’ ve got these special feet that need a **two-inch** heel. I have a three-inch heel. I need a little bit off.”They didn ’ t even want to go there. They didn ’ t even want to touch that one. They just did it. No, these legs are great. I ’ m actually going **back** in a couple of weeks to get some improvements. I want to get legs like these made for flat feet so I can wear sneakers, because I can ’ t with these ones. So... Moderator : That ’ s it. SS : That ’ s Aimee Mullins.

Text Sample 147: NEG Dim 1 – 1998 – Score 11.00 – 1998/t000134.txt

’ Theme and variations ’ is one of those forms that require a certain kind of intellectual activity, because you are always comparing the variation with the theme that you hold in your mind. You might say that the theme is nature and everything that follows is a variation on the subject. I was asked, I guess about six years ago, to do a series of paintings that in some way would celebrate the birth of Piero della Francesca. And it was very difficult for me to imagine how to paint pictures that were based on Piero until I realized that I could look at Piero as nature — that I would have the same attitude towards looking at Piero della Francesca as I would if I were looking out a window at a tree. And that was enormously liberating to me. Perhaps it ’ s not a very insightful observation, but that really started me on a path to be able to do a kind of theme and variations based on a work by Piero, in this case that remarkable painting that ’ s in the Uffizi,”The Duke of Montefeltro,”who faces his consort, Battista. Once I realized that I could take some liberties with the subject, I did the following series of drawings. That ’ s the real Piero della Francesca — one of the greatest portraits in human history. And these, I ’ ll just show these without comment. It ’ s just a series of variations on the head of the Duke of Montefeltro, who ’ s a great, great figure in the Renaissance, and probably the basis for Machiavelli ’ s”The Prince.”He apparently lost an eye in battle, which is why he is always shown in profile. And this is Battista. And then I decided I could move them around a little bit — so that for the first time in history, they ’ re facing the same direction. Whoops! Passed each other. And then a visitor from another painting by Piero, this is from”The Resurrection of Christ”— as though the cast had just gotten of the set to have a chat. And now, four large panels : this is upper left ; upper right ; lower left ; lower right. Incidentally, I ’ ve never understood the conflict between abstraction and naturalism. Since all paintings are inherently abstract to begin with there doesn ’ t seem to be an argument there. On another subject — — I was driving in the country one day with my wife, and I saw this sign, and I said,”That is a fabulous piece of design.”And she said,”What are you talking about?”I said,”Well, it ’ s so persuasive, because the purpose of that sign is to get you into the garage, and since most people are so suspicious of garages, and know that they ’ re going to be ripped off, they use the word ’ reliable. ’ But everybody says they ’ re reliable. But, reliable Dutchman”— —”Fantastic!”Because as soon as you hear the word Dutchman — which is an archaic word, nobody calls Dutch people”Dutchmen”anymore — but as soon as you hear Dutchman, you get this picture of the kid with his finger in the dike, preventing the thing from falling and flooding Holland, and so on. And so the entire issue is **detoxified** by the use of"Dutchman."Now, if you think I ' m exaggerating at all in this, all you have to do is substitute something else, like"Indonesian."Or even"French."Now,"Swiss"works, but you know it ' s going to cost a lot of money. I ' m going to take you quickly through the actual process of doing a poster. I do a lot of work for the School of Visual Arts, where I teach, and the director of this school — a remarkable man named Silas Rhodes — often gives you a piece of text and he says,"Do something with this."And so he did. And this was the text —"In words as fashion the same rule will hold / Alike fantastic if too new or old / Be not the first by whom the new are tried / Nor yet the last to lay the old aside."I could make nothing of that. And I really struggled with this one. And the first thing I did, which was sort of in the absence of another idea, was say I ' ll sort of write it out and make some words big, and I ' ll have some kind of design on the **back** somehow, and I was hoping — as one often does — to stumble into something. So I took another crack at it — you ' ve got to keep it moving — and I Xeroxed some words on pieces of colored paper and I pasted them down on an ugly board. I thought that something would come out of it, like"Words rule fantastic new old first last Pope"because it ' s by Alexander Pope, but I sort of made a mess out of it, and then I thought I ' d repeat it in some way so it was legible. So, it was going nowhere. Sometimes, in the middle of a resistant problem, I write down things that I know about it. But you can see the beginning of an idea there, because you can see the word"new"emerging from the"old."That ' s what happens. There ' s a relationship between the old and the new ; the new emerges from the context of the old. And then I did some variations of it, but it still wasn ' t coalescing graphically at all. I had this other version which had something interesting about it in terms of being able to put it together in your mind from clues. The W was clearly a W, the N was clearly an N, even though they were very fragmentary and there wasn ' t a lot of information in it. Then I got the words"new"and"old"and now I had regressed **back** to a point where there seemed to be no return. I was really desperate at this point. And so, I do something I ' m truthfully ashamed of, which is that I took two drawings I had made for another purpose and I put them together. It says"dreams"at the top. And I was going to do a thing, I say,"Well, change the copy. Let it say something about dreams, and come to SVA and you ' ll sort of fulfill your dreams."But, to my credit, I was so embarrassed about doing that that I never submitted this sketch. And, finally, I arrived at the following solution. Now, it doesn ' t look terribly interesting, but it does have something that distinguishes it from a lot of other posters. For one thing, it transgresses the idea of what a poster ' s supposed to be, which is to be understood and seen immediately, and not explained. I remember hearing all of you in the graphic arts —"If you have to explain it, it ain ' t working."And one day I woke up and I said,"Well, suppose that ' s not true?"So here ' s what it says in my explanation at the bottom left. It says,"Thoughts : This poem is impossible. Silas usually has a better touch with his choice of quotations. This one generates no imagery at all."I am now exposing myself to my audience, right? Which is something you never want to do professionally."Maybe the words can make the image without anything else happening. What ' s the heart of this poem? Don ' t be trendy if you want to be serious. Is doing the poster this way trendy in itself? I guess one could reduce the idea further by suggesting that the new emerges behind and through the old, like this."And then I show you a little drawing — you see, you remember that old thing I discarded? Well, I found a way to use it. So, there ' s that little alternative over there, and I say,"Not bad,"— criticizing myself —"but more didactic than visual. Maybe what wants to be said is that old and the new are locked in a dialectical embrace, a kind of dance where each defines the other."And then more self-questioning —"Am I being simple-minded? Is this the kind of simple that looks obvious, or the kind that looks profound? There ' s a significant difference. This could be embarrassing. Actually, I realize fear of embarrassment drives me as much as any ambition. Do you think this sort of thing could really attract a student to the school?"Well, I think there are two fresh things here — two fresh things. One is the sort of willingness to expose myself to a critical audience, and not to suggest that I am confident about what I ' m doing. And as you know, you have to have a front. I mean — you ' ve got to be confident ; if you don ' t believe in your work, who else is going to believe in it? So that ' s one thing, to introduce the idea of doubt into graphics. That can be a big contribution. The other thing is to actually give you two solutions for the price of one ; you get the big one and if you don ' t like that, how about the little one? And that too is a relatively new idea. And here ' s just a series of experiments where I ask the question of — does a poster have to be square? Now, this is a little illusion. That poster is not folded. It ' s not folded, that ' s a photograph and it ' s cut on the diagonal. Same cheap trick in the upper left- hand corner. And here, a very peculiar poster because, simply because of using the isometric perspective in the computer, it won ' t sit still in the space. At times, it seems to be wider at the **back** than the front, and then it shifts. And if you sit here long enough, it ' ll float off the page into the audience. But, we don ' t have time. And then an experiment — a little bit about the nature of perspective, where the outside shape is determined by the peculiarity of perspective, but the shape of the bottle — which is identical to the outside shape — is seen frontally. And another piece for the art directors ' club is"Anna Rees"casting long shadows. This is another poster from the School of Visual Arts. There were 10 artists invited to participate in it, and it was one of those things where it was extremely competitive and I didn ' t want to be embarrassed, so I worked very hard on this. The idea was — and it was a brilliant idea — was to have 10 posters distributed throughout the city ' s subway system so every time you got on the subway you ' d be passing a different poster, all of which had a different idea of what art is. But I was absolutely stuck on the idea of"art is"and trying to determine what art was. But then I gave up and I said,"Well, art is whatever."And as soon as I said that, I discovered that the word"hat"was hidden in the word"whatever,"and that led me to the inevitable conclusion. But then again, it ' s on my list of didactic posters. My intent is to have a literary accompaniment that explains the poster, in case you don ' t get it. Now this says,"Note to the viewer : I thought I might use a visual cliche of our time — Magritte ' s everyman — to express the idea that art is mystery, continuity and history. I ' m also convinced that, in an age of computer manipulation, surrealism has become banal, a shadow of its former self. The phrase ' Art is whatever ' expresses the current inclusiveness that surrounds art-making — a sort of ' it ain ' t what you do, it ' s the way that you do it ' notion. The shadow of Magritte falls across the central part of the poster a poetic event that occurs as the shadow man isolates the word ' hat, ' hidden in the word ' whatever. ' The four hats shown in the poster suggests how art might be defined : as a thing itself, the worth of the thing, the shadow of the thing, and the shape of the thing. Whatever."OK. And the one that I did not submit, which I still like, I wanted to use the same phrase. There were wonderful experiments by Bruno Munari on **letterforms** some years ago — sort of, see how far you could go and still be able to read them. And that idea stuck in my head. But then I took the pieces that I had taken off and put them at the bottom. And, of course those are the remains, and they ' re so labeled. But what really happens is that you read it as :"Art is whatever remains."Thank you.

Text Sample 148: NEG Dim 1 – 2006 – Score 3.00 – 2006/t000480.txt

I wrote this poem after hearing a pretty well known actress tell a very well known interviewer on television,”I ’ m really getting into the Internet lately. I just wish it were more organized.”So... If I controlled the Internet, you could auction your broken heart on eBay. Take the money ; go to Amazon ; buy a phonebook for a country you ’ ve never been to — call folks at random until you find someone who flirts really well in a foreign language. If I were in charge of the Internet, you could Mapquest your lover ’ s mood swings. Hang left at cranky, right at preoccupied, U-turn on silent treatment, all the way **back** to tongue kissing and good lovin ’. You could navigate and understand every emotional intersection. Some days, I ’ m as shallow as a baking pan, but I still stretch miles in all directions. If I owned the Internet, Napster, Monster and Friendster. com would be one big website. That way you could listen to cool music while you pretend to look for a job and you ’ re really just chattin ’ with your pals. Heck, if I ran the Web, you could email dead people. They would not email you **back** — but you ’ d get an automated reply. Their name in your inbox — it ’ s all you wanted anyway. And a message saying,”Hey, it ’ s me. I miss you. Listen, you ’ ll see being dead is dandy. Now you go **back** to raising kids and waging peace and craving candy.”If I designed the Internet, childhood. com would be a loop of a boy in an orchard, with a ski pole for a sword, trashcan lid for a shield, shouting,”I am the emperor of oranges. I am the emperor of oranges. I am the emperor of oranges.”Now follow me, OK? Grandma. com would be a recipe for biscuits and spit-bath instructions. One, two, three. That links with hotdiggitydog. com. That is my grandfather. They take you to gruff-ex-cop-on-his-fourth-marriage. dad. He forms an attachment to kind-of-ditzy-but-still-sends-ginger-snaps-for-Christmas. mom, who downloads the boy in the orchard, the emperor of oranges, who grows up to be me — the guy who usually goes too far. So if I were emperor of the Internet, I guess I ’ d still be mortal, huh? But at that point, I would probably already have the lowest possible mortgage and the most enlarged possible penis — so I would outlaw spam on my first day in office. I wouldn ’ t need it. I ’ d be like some kind of Internet genius, and me, I ’ d like to upgrade to deity and maybe just like that — pop! — I ’ d go wireless. Huh? Maybe Google would hire this. I could zip through your servers and firewalls like a virus until the World Wide Web is as wise, as wild and as organized as I think a modern-day miracle / oracle can get, but, ooh-eee, you want to bet just how whack and un-PC your Mac or PC is going to be when I ’ m rocking hot-shit-hot-shot-god. net? I guess it ’ s just like life. It is not a question of if you can — it ’ s : do ya? We can interfere with the interface. We can make”You ’ ve got Hallelujah”the national anthem of cyberspace every lucky time we log on. You don ’ t say a prayer. You don ’ t write a psalm. You don ’ t chant an”om.”You send one blessed email to whomever you ’ re thinking of at dah-da-la-dat-da-dah-da-la-dat. com. Thank you, TED.

Text Sample 149: NEG Dim 1 – 2006 – Score 4.00 – 2006/t000496.txt

On September 10, the morning of my seventh birthday, I came downstairs to the kitchen, where my mother was washing the dishes and my father was reading the paper or something, and I sort of presented myself to them in the doorway, and they said,”Hey, happy birthday!”And I said,”I ’ m seven.”And my father smiled and said,”Well, you know what that means, don ’ t you?”And I said,”Yeah, that I ’ m going to have a party and a cake and get a lot of presents?”And my dad said,”Well, yes. But more importantly, being seven means that you ’ ve reached the age of reason, and you ’ re now capable of committing any and all sins against God and man.”Now, I had heard this phrase,”age of reason,”before. Sister Mary Kevin had been bandying it about my second-grade class at school. But when she said it, the phrase seemed all caught up in the excitement of preparations for our first communion and our first confession, and everybody knew that was really all about the white dress and the white veil. And anyway, I hadn ’ t really paid all that much attention to that phrase,”age of reason.”So, I said,”Yeah, yeah, age of reason. What does that mean again?”And my dad said,”Well, we believe, in the Catholic Church, that God knows that little kids don ’ t know the difference between right and wrong, but when you ’ re seven, you ’ re old enough to know better. So, you ’ ve grown up and reached the age of reason, and now God will start keeping notes on you, and begin your permanent record.”And I said,”Oh... Wait a minute. You mean all that time, up till today, all that time I was so good, God didn ’ t notice it?”And my mom said,”Well, I noticed it.”And I thought,”How could I not have known this before? How could it not have sunk in when they ’ d been telling me? All that being good and no real credit for it. And worst of all, how could I not have realized this very important information until the very day that it was basically useless to me?”So I said,”Well, Mom and Dad, what about Santa Claus? I mean, Santa Claus knows if you ’ re naughty or nice, right?”And my dad said,”Yeah, but, honey, I think that ’ s technically just between Thanksgiving and Christmas.”And my mother said,”Oh, Bob, stop it. Let ’ s just tell her. I mean, she ’ s seven. Julie, there is no Santa Claus.”Now, this was actually not that upsetting to me. My parents had this whole elaborate story about Santa Claus : how they had talked to Santa Claus himself and agreed that instead of Santa delivering our presents over the night of Christmas Eve, like he did for every other family who got to open their surprises first thing Christmas morning, our family would give Santa more time. Santa would come to our house while we were at nine o ’ clock high mass on Christmas morning, but only if all of **us** kids did not make a fuss. Which made me very suspicious. It was pretty obvious that it was really our parents giving **us** the presents. I mean, my dad had a very distinctive wrapping style, and my mother ’ s handwriting was so close to Santa ’ s. Plus, why would Santa save time by having to loop **back** to our house after he ’ d gone to everybody else ’ s? There was only one obvious conclusion to reach from this mountain of evidence : our family was too strange and weird for even Santa Claus to come visit, and my poor parents were trying to protect **us** from the embarrassment, this humiliation of rejection by Santa, who was jolly — but let ’ s face it, he was also very judgmental. So to find out that there was no Santa Claus at all was actually sort of a relief. I left the kitchen not really in shock about Santa, but rather, I was just dumbfounded about how I could have missed this whole age of reason thing. It was too late for me, but maybe I could help someone else, someone who could use the information. They had to fit two criteria : they had to be old enough to be able to understand the whole concept of the age of reason, and not yet seven. The answer was clear : my brother Bill. He was six. Well, I finally found Bill about a block away from our house at this public school playground. It was a Saturday, and he was all by himself, just kicking a ball against the side of a wall. I ran up to him and said,”Bill! I just realized that the age of reason starts when you turn seven, and then you ’ re capable of committing any and all sins against God and man.”And Bill said,”So?”And I said,”So, you ’ re six. You have a whole year to do anything you want to and God won ’ t notice it.”And he said,”So?”And I said,”So? So everything!”And I turned to run. I was so angry with him. But when I got to the top of the steps, I turned around dramatically and said,”Oh, by the way, Bill — there is no Santa Claus.”Now, I didn ’ t know it at the time, but I really wasn ’ t turning seven on September 10th. For my 13th birthday, I planned a slumber party with all of my girlfriends, but a couple of weeks beforehand my mother took me aside and said,”I need to speak to you privately. September 10th is not your birthday. It ’ s October 10th.”And I said,”What?”And she said...”Listen. The cut-off date to start kindergarten was September 15th.””So I told them that your birthday was September 10th, and then I wasn ’ t sure that you weren ’ t just going to go blab it all over the place, so I started to tell you your birthday was September 10th. But, Julie, you were so ready to start school, honey. You were so ready.”I thought about it, and when I was four, I was already the oldest of four children, and my mother even had another child to come, so what I think she — understandably — really meant was that she was so ready, she was so ready. Then she said,”Don ’ t worry, Julie. Every year on October 10th, when it was your birthday but you didn ’ t realize it, I made sure that you ate a piece of cake that day.”Which was comforting, but troubling. My mother had been celebrating my birthday with me, without me. What was so upsetting about this new piece of information was not that I had to change the date of my slumber party with all of my girlfriends. What was most upsetting was that this meant I was not a Virgo. I had a huge Virgo poster in my bedroom. And I read my horoscope every single day, and it was so totally me. And this meant that I was a Libra? So, I took the bus downtown to get the new Libra poster. The Virgo poster is a picture of a beautiful woman with long hair, sort of lounging by some water, but the Libra poster is just a huge scale. This was around the time that I started filling out physically, and I was filling out a lot more than a lot of the other girls, and frankly, the whole idea that my astrological sign was a scale just seemed ominous and depressing. But I got the new Libra poster, and I started to read my new Libra horoscope, and I was astonished to find that it was also totally me. It wasn ’ t until years later, looking **back** on this whole age-of-reason, change- of-birthday thing, that it dawned on me : I wasn ’ t turning seven when I thought I turned seven. I had a whole other month to do anything I wanted to before God started keeping tabs on me. Oh, life can be so cruel. One day, two Mormon missionaries came to my door. Now, I just live off a main thoroughfare in Los Angeles, and my block is — well, it ’ s a natural beginning for people who are peddling things door to door. Sometimes I get little old ladies from the Seventh Day Adventist Church showing me these cartoon pictures of heaven. And sometimes I get teenagers who promise me that they won ’ t join a gang and just start robbing people, if I only buy some magazine subscriptions from them. So normally, I just ignore the doorbell, but on this day, I answered. And there stood two boys, each about 19, in white, starched short-sleeved shirts, and they had little name tags that identified them as official representatives of the Church of Jesus Christ of Latter-day Saints, and they said they had a message for me, from God. I said,”A message for me? From God?”And they said,”Yes.”Now, I was raised in the Pacific Northwest, around a lot of Church of Latter-day Saints people and, you know, I ’ ve worked with them and even dated them, but I never really knew the doctrine, or what they said to people when they were out on a mission, and I guess I was sort of curious, so I said,”Well, please, come in.”And they looked really happy, because I don ’ t think this happens to them all that often. And I sat them down, and I got them glasses of water — Ok, I got it, I got it. I got them glasses of water. Don ’ t touch my hair, that ’ s the thing. You can ’ t put a video of myself in front of me and expect me not to fix my hair. Ok. So I sat them down and I got them glasses of water, and after niceties, they said,”Do you believe that God loves you with all his heart?”And I thought,”Well, of course I believe in God, but you know, I don ’ t like that word ’ heart, ’ because it anthropomorphizes God, and I don ’ t like the word, ’ his, ’ either, because that sexualizes God.”But I didn ’ t want to argue semantics with these boys, so after a very long, uncomfortable pause, I said,”Yes, yes, I do. I feel very loved.”And they looked at each other and smiled, like that was the right answer. And then they said,”Do you believe that we ’ re all brothers and sisters on this planet?”And I said,”Yes, I do.”And I was so relieved that it was a question I could answer so quickly. And they said,”Well, then we have a story to tell you.”And they told me this story all about this guy named Lehi, who lived in Jerusalem in 600 BC. Now, apparently in Jerusalem in 600 BC, everyone was completely bad and evil. Every single one of them : man, woman, child, infant, fetus. And God came to Lehi and said to him,”Put your family on a boat and I will lead you out of here.”And God did lead them. He led them to America. I said,”America? From Jerusalem to America by boat in 600 BC?”And they said,”Yes.”Then they told me how Lehi and his descendants reproduced and reproduced, and over the course of 600 years, there were two great races of them, the Nephites and the Lamanites, and the Nephites were totally good — each and every one of them — and the Lamanites were totally bad and evil — every single one of them just bad to the bone. Then, after Jesus died on the cross for our sins, on his way up to heaven, he stopped by America and visited the Nephites. And he told them that if they all remained totally, totally good — each and every one of them — they would win the war against the evil Lamanites. But apparently somebody blew it, because the Lamanites were able to kill all the Nephites. All but one guy, this guy named Mormon, who managed to survive by hiding in the woods. And he made sure this whole story was written down in reformed Egyptian hieroglyphics chiseled onto gold plates, which he then buried near Palmyra, New York. Well, I was just on the edge of my seat. I said,”What happened to the Lamanites?”And they said,”Well, they became our Native Americans, here in the U. S.”And I said,”So, you believe the Native Americans are descended from a people who were totally evil?”And they said,”Yes.”Then they told me how this guy named Joseph Smith found those buried gold plates right in his backyard, and he also found this magic stone **back** there that he put into his hat and then buried his face into, and this allowed him to translate the gold plates from the reformed Egyptian into English. Well, at this point I just wanted to give these two boys some advice about their pitch. I wanted to say —”Ok, don ’ t start with this story.”I mean, even the Scientologists know to start with a personality test before they start — telling people all about Xenu, the evil intergalactic overlord. Then, they said,”Do you believe that God speaks to **us** through his righteous prophets?”And I said,”No, I don ’ t,”because I was sort of upset about this Lamanite story and this crazy gold plate story, but the truth was, I hadn ’ t really thought this through, so I backpedaled a little and I said,”Well, what exactly do you mean by ’ righteous ’? And what do you mean by prophets? Like, could the prophets be women?”And they said,”No.”And I said,”Why?”And they said,”Well, it ’ s because God gave women a gift that is so spectacular, it is so wonderful, that the only gift he had left over to give men was the gift of prophecy.”What is this wonderful gift God gave women, I wondered? Maybe their greater ability to cooperate and adapt? Women ’ s longer lifespan? The fact that women tend to be much less violent than men? But no — it wasn ’ t any of these gifts. They said,”Well, it ’ s her ability to bear children.”I said,”Oh, come on. I mean, even if women tried to have a baby every single year from the time they were 15 to the time they were 45, assuming they didn ’ t die from exhaustion, it still seems like some women would have some time left over to hear the word of God.”And they said,”No.”Well, then they didn ’ t look so fresh-faced and cute to me any more, but they had more to say. They said,”Well, we also believe that if you ’ re a Mormon, and if you ’ re in good standing with the church, when you die, you get to go to heaven and be with your family for all eternity.”And I said,”Oh, dear. That wouldn ’ t be such a good incentive for me.”And they said,”Oh. Hey! Well, we also believe that when you go to heaven, you get your body restored to you in its best original state. Like, if you ’ d lost a leg, well, you get it **back**. Or, if you ’ d gone blind, you could see.”I said,”Oh. Now, I don ’ t have a uterus, because I had cancer a few years ago. So does this mean that if I went to heaven, I would get my old uterus **back**?”And they said,”Sure.”And I said,”I don ’ t want it **back**. I ’ m happy without it.”Gosh. What if you had a nose job and you liked it? Would God force you to get your old nose **back**? Then they gave me this Book of Mormon, told me to read this chapter and that chapter, and said they ’ d come **back** and check in on me, and I think I said something like,”Please don ’ t hurry,”or maybe just,”Please don ’ t,”and they were gone. Ok, so I initially felt really superior to these boys, and smug in my more conventional faith. But then the more I thought about it, the more I had to be honest with myself. If someone came to my door and I was hearing Catholic theology and dogma for the very first time, and they said,”We believe that God impregnated a very young girl without the use of intercourse, and the fact that she was a virgin is maniacally important to **us**.””And she had a baby, and that ’ s the son of God,”I mean, I would think that ’ s equally ridiculous. I ’ m just so used to that story. So, I couldn ’ t let myself feel condescending towards these boys. But the question they asked me when they first arrived really stuck in my head : Did I believe that God loved me with all his heart? Because I wasn ’ t exactly sure how I felt about that question. Now, if they had asked me,”Do you feel that God loves you with all his heart?”Well, that would have been much different, I think I would have instantly answered,”Yes, yes, I feel it all the time. I feel God ’ s love when I ’ m hurt and confused, and I feel consoled and cared for. I take shelter in God ’ s love when I don ’ t understand why tragedy hits, and I feel God ’ s love when I look with gratitude at all the beauty I see.”But since they asked me that question with the word”believe”in it, somehow it was all different, because I wasn ’ t exactly sure if I believed what I so clearly felt.

Text Sample 150: NEG Dim 1 – 2006 – Score 5.00 – 2006/t000219.txt

Stephanie White : I ’ m going to let her introduce herself to everybody. Can you tell everybody your name? Einstein : Einstein. SW : This is Einstein. Can you tell everyone”hi”? E : Hello. SW : That ’ s nice. Can you be polite? E : Hi, sweetheart. SW : Much better. Well, Einstein is very honored to be here at TED 2006, amongst all you modern-day Einsteins. In fact, she ’ s very excited. E : Woo. SW : Yeah. Since we ’ ve arrived, there ’ s been a constant buzz about all the exciting speakers here for the conference. This morning we ’ ve heard a lot of whispers about Tom Reilly ’ s wrap-up on Saturday. Einstein, did you hear whispers? E : [Squawks] SW : Yeah. Einstein ’ s especially interested in Penelope ’ s talk. A lot of her research goes on in caves, which can get pretty dusty. E : Achoo! SW : It could make her sneeze. But more importantly, her research could help Einstein find a cure for her never-ending scratchy throat. Einstein : [Coughs] SW : Yeah. Well, Bob Russell was telling **us** about his work on nanotubes in his research at the microscopic level. Well, that ’ s really cool, but what Einstein ’ s really hoping is that maybe he ’ ll genetically engineer a five-pound peanut. E : Oh, my God! My God! My God! SW : Yeah. She would get really, really excited. That is one big peanut. Since Einstein is a bird, she ’ s very interested in things that fly. She thinks Burt Rutan is very impressive. E

: Ooh. SW : Yeah. She especially likes his latest achievement, SpaceShipOne. Einstein, would you like to ride in Burt ’ s spaceship? E : [Spaceship noise] SW : Even if it doesn ’ t have a laser? E : [Laser noise] SW : Yeah, yeah. That was pretty funny, Einstein. Now, Einstein also thinks, you know, working in caves and travelling through space — it ’ s all very dangerous jobs. It would be very dangerous if you fell down. E : Wheeeeeee! [Splat] SW : Yeah. Little splat at the end there. Einstein, did that hurt? E : Ow, ow, ow. SW : Yeah. It ’ s all a lot of hard work. E : [Squawks] SW : Yeah. It can get a bird like Einstein frustrated. E : [Squawks] SW : Yeah, it sure can. But when Einstein needs to relax from her job educating the public, she loves to take in the arts. If the children of the Uganda need another dance partner, Einstein could sure fit the bill, because she loves to dance. Can you get down? E : [Bobbing head] SW : Let ’ s get down for everybody. Come on now. She ’ s going to make me do it, too. Ooh, ooh. Einstein : Ooh, ooh, ooh, ooh. SW : Do your head now. E : Ooh, ooh, ooh, ooh, ooh. SW : Or maybe Sirena Huang would like to learn some arias on her violin, and Einstein can sing along with some opera? E : [Operatic squawk] SW : Very good. Or maybe Stu just needs another backup singer? Einstein, can you also sing? I know, you need to get rid of that seed first. Can you sing? E : La, la. SW : There you go. And, of course, if all else fails, you can just run off and enjoy a fun fiesta. E : [Squawks] SW : All right. Well, Einstein was pretty embarrassed to admit this earlier, but she was telling me backstage that she had a problem. E : What ’ s the matter? SW : No, I don ’ t have a problem. You have the problem, remember? You were saying that you were really embarrassed, because you ’ re in love with a pirate? E : Yar. SW : There you go. And what do pirates like to drink? E : Beer. SW : Yeah, that ’ s right. But you don ’ t like to drink beer, Einstein. You like to drink water. E : [Water sound] SW : Very good. Now, really, she is pretty nervous. Because one of her favorite folks from **back** home is here, and she ’ s pretty nervous to meet him. She thinks Al Gore is a really good-looking man. What do you say to a good-looking man? E : Hey, baby. SW : And so do all the folks **back** home in Tennessee. E : Yee haw. SW : And since she ’ s such a big fan, she knows that his birthday is coming up at the end of March. And we didn ’ t think he ’ d be in town then, so Einstein wanted to do something special for him. So let ’ s see if Einstein will sing”Happy Birthday”to Al Gore. Can you sing”Happy Birthday”to him? E : Happy birthday to you. SW : Again. E : Happy birthday to you. SW : Again. E : Happy birthday to you. SW : Big finish. E : Happy birthday to you. SW : Good job! Well, before we wrap it up, she would like to give a shout out to all our animal friends **back** at the Knoxville Zoo. Einstein, do you want to say”hi”to all the owls? E : Woo, woo, woo. SW : What about the other birds? E : Tweet, tweet, tweet. SW : And the penguin? E : Quack, quack, quack. SW : There we go. Let ’ s get that one out of there. How about a chimpanzee? E : Ooh, ooh, ooh. Aah, aah, aah. SW : Very good. What about a wolf? E : Ooooowww. SW : And a pig? E : Oink, oink, oink. SW : And the rooster? E : Cock-a-doodle-doo! SW : And how about those cats? E : Meow. SW : At the zoo we have big cats from the jungle. E : Grrrrr. SW : What about a skunk? E : Stinker. SW : She ’ s a comedian. I suppose you think you ’ re famous? Are you famous? E : Superstar. SW : Yeah. You are a superstar. Well, we would like to encourage all of you to do your part to help protect Einstein ’ s animal friends, and to do your part to help protect their homes that they live [in]. Now, Einstein does say it best when we ask her. Why do we want to protect your home? E : I ’ m special. SW : You are very special. What would you like to say to all these nice people? E : I love you. SW : That ’ s good. Can you blow them a kiss? E : [Kissing noise] SW : And what do you say when it ’ s time to go? E : Goodbye. SW : Good job. Thank you all.

Text Sample 151: NEG Dim 1 – 2008 – Score 1.00 – 2008/t000146.txt

Great creativity. In times of need, we need great creativity. Discuss. Great creativity is astonishingly, absurdly, rationally, irrationally powerful. Great creativity can spread tolerance, champion freedom, make education seem like a bright idea. Great creativity can turn a spotlight on deprivation, or show that deprivation ain ’ t necessarily so. Great creativity can make politicians electable, or parties unelectable. It can make war seem like tragedy or farce. Creativity is the meme-maker that puts slogans on our t-shirts and phrases on our lips. It ’ s the pathfinder that shows **us** a simple road through an impenetrable moral maze. Science is clever, but great creativity is something less knowable, more magical. And now we need that magic. This is a time of need. Our climate is changing quickly, too quickly. And great creativity is needed to do what it does so well : to provoke **us** to think differently with dramatic creative statements. To tempt **us** to act differently with delightful creative scraps. Here is one such scrap from an initiative I ’ m involved in using creativity to inspire people to be greener. Man : You know, rather than drive today, I ’ m going to walk. Narrator : And so he walked, and as he walked he saw things. Strange and wonderful things he would not otherwise have seen. A deer with an itchy leg. A flying motorcycle. A father and daughter separated from a bicycle by a mysterious wall. And then he stopped. Walking in front of him was her. The woman who as a child had skipped with him through fields and broken his heart. Sure, she had aged a little. In fact, she had aged a lot. But he felt all his old passion for her return.”Ford,”he called softly. For that was her name.”Don ’ t say another word, Gusty,”she said, for that was his name.”I know a tent next to a caravan, exactly 300 yards from here. Let ’ s go there and make love. In the tent.”Ford undressed. She spread one leg, and then the other. Gusty entered her boldly and made love to her rhythmically while she filmed him, because she was a keen amateur pornographer. The earth moved for both of them. And they lived together happily ever after. And all because he decided to walk that day. Andy Hobsbawm : We ’ ve got the science, we ’ ve had the debate. The moral imperative is on the table. Great creativity is needed to take it all, make it simple and sharp. To make it connect. To make it make people want to act. So this is a call, a plea, to the incredibly talented TED community. Let ’ s get creative against climate change. And let ’ s do it soon. Thank you.

Text Sample 152: NEG Dim 1 – 2008 – Score 1.00 – 2008/t000249.txt

Probably a lot of you know the story of the two salesmen who went down to Africa in the 1900s. They were sent down to find if there was any opportunity for selling shoes, and they wrote telegrams **back** to Manchester. And one of them wrote,”Situation hopeless. Stop. They don ’ t wear shoes.”And the other one wrote,”Glorious opportunity. They don ’ t have any shoes yet.”Now, there ’ s a similar situation in the classical music world, because there are some people who think that classical music is dying. And there are some of **us** who think you ain ’ t seen nothing yet. And rather than go into statistics and trends, and tell you about all the orchestras that are closing, and the record companies that are folding, I thought we should do an experiment tonight. Actually, it ’ s not really an experiment, because I know the outcome. But it ’ s like an experiment. Now, before we start — Before we start, I need to do two things. One is I want to remind you of what a seven-year-old child sounds like when he plays the piano. Maybe you have this child at home. He sounds something like this. I see some of you recognize this child. Now, if he practices for a year and takes lessons, he

’ s now eight and he sounds like this. He practices for another year and takes lessons — he ’ s nine. Then he practices for another year and takes lessons — now he ’ s 10. At that point, they usually give up. Now, if you ’ d waited for one more year, you would have heard this. Now, what happened was not maybe what you thought, which is, he suddenly became passionate, engaged, involved, got a new teacher, he hit puberty, or whatever it is. What actually happened was the impulses were reduced. You see, the first time, he was playing with an impulse on every note. And the second, with an impulse every other note. You can see it by looking at my head. The nine-year-old put an impulse on every four notes. The 10-year-old, on every eight notes. And the 11-year-old, one impulse on the whole phrase. I don ’ t know how we got into this position. I didn ’ t say,”I ’ m going to move my shoulder over, move my body.”No, the music pushed me over, which is why I call it one-buttock playing. It can be the other buttock. You know, a gentleman was once watching a presentation I was doing, when I was working with a young pianist. He was the president of a corporation in Ohio. I was working with this young pianist, and said,”The trouble with you is you ’ re a two-buttock player. You should be a one-buttock player.”I moved his body while he was playing. And suddenly, the music took off. It took flight. The audience gasped when they heard the difference. Then I got a letter from this gentleman. He said,”I was so moved. I went **back** and I transformed my entire company into a one-buttock company.”Now, the other thing I wanted to do is to tell you about you. There are 1, 600 people, I believe. My estimation is that probably 45 of you are absolutely passionate about classical music. You adore classical music. Your FM is always on that classical dial. You have CDs in your car, and you go to the symphony, your children are playing instruments. You can ’ t imagine your life without classical music. That ’ s the first group, quite small. Then there ’ s another bigger group. The people who don ’ t mind classical music. You know, you ’ ve come home from a long day, and you take a glass of wine, and you put your feet up. A little Vivaldi in the background doesn ’ t do any harm. That ’ s the second group. Now comes the third group : people who never listen to classical music. It ’ s just simply not part of your life. You might hear it like second-hand smoke at the airport... — and maybe a little bit of a march from”Aida”when you come into the hall. But otherwise, you never hear it. That ’ s probably the largest group. And then there ’ s a very small group. These are the people who think they ’ re tone-deaf. Amazing number of people think they ’ re tone-deaf. Actually, I hear a lot,”My husband is tone-deaf.”Actually, you cannot be tone-deaf. Nobody is tone-deaf. If you were tone-deaf, you couldn ’ t change the gears on your car, in a stick shift car. You couldn ’ t tell the difference between somebody from Texas and somebody from Rome. And the telephone. The telephone. If your mother calls on the miserable telephone, she calls and says,”Hello,”you not only know who it is, you know what mood she ’ s in. You have a fantastic ear. Everybody has a fantastic ear. So nobody is tone- deaf. But I tell you what. It doesn ’ t work for me to go on with this thing, with such a wide gulf between those who understand, love and are passionate about classical music, and those who have no relationship to it at all. The tone-deaf people, they ’ re no longer here. But even between those three categories, it ’ s too wide a gulf. So I ’ m not going to go on until every single person in this room, downstairs and in Aspen, and everybody else looking, will come to love and understand classical music. So that ’ s what we ’ re going to do. Now, you notice that there is not the slightest doubt in my mind that this is going to work, if you look at my face, right? It ’ s one of the characteristics of a leader that he not doubt for one moment the capacity of the people he ’ s leading to realize whatever he ’ s dreaming. Imagine if Martin Luther King had said,”I have a dream. Of course, I ’ m not sure they ’ ll be up to it.”All right.

So I ’ m going to take a piece of Chopin. This is a beautiful prelude by Chopin. Some of you will know it. Do you know what I think probably happened here? When I started, you thought,”How beautiful that sounds.””I don ’ t think we should go to the same place for our summer holidays next year.”It ’ s funny, isn ’ t it? It ’ s funny how those thoughts kind of waft into your head. And of course — Of course, if the piece is long and you ’ ve had a long day, you might actually drift off. Then your companion will dig you in the ribs and say,”Wake up! It ’ s culture!”And then you feel even worse. But has it ever occurred to you that the reason you feel sleepy in classical music is not because of you, but because of **us**? Did anybody think while I was playing,”Why is he using so many impulses?”If I ’ d done this with my head you certainly would have thought it. And for the rest of your life, every time you hear classical music, you ’ ll always be able to know if you hear those impulses. So let ’ s see what ’ s really going on here. We have a B. This is a B. The next note is a C. And the job of the C is to make the B sad. And it does, doesn ’ t it? Composers know that. If they want sad music, they just play those two notes. But basically, it ’ s just a B, with four sads. Now, it goes down to A. Now to G. And then to F. So we have B, A, G, F. And if we have B, A, G, F, what do we expect next? That might have been a fluke. Let ’ s try it again. Oh, the TED choir. And you notice nobody is tone-deaf, right? Nobody is. You know, every village in Bangladesh and every hamlet in China — everybody knows : da, da, da, da — da. Everybody knows, who ’ s expecting that E. Chopin didn ’ t want to reach the E there, because what will have happened? It will be over, like Hamlet. Do you remember? Act One, scene three, he finds out his uncle killed his father. He keeps on going up to his uncle and almost killing him. And then he **backs** away, he goes up to him again, almost kills him. The critics sitting in the **back** row there, they have to have an opinion, so they say,”Hamlet is a procrastinator.”Or they say,”Hamlet has an Oedipus complex.”No, otherwise the play would be over, stupid. That ’ s why Shakespeare puts all that stuff in Hamlet — Ophelia going mad, the play within the play, and Yorick ’ s skull, and the gravediggers. That ’ s in order to delay — until Act Five, he can kill him. It ’ s the same with the Chopin. He ’ s just about to reach the E, and he says,”Oops, better go **back** up and do it again.”So he does it again. Now, he gets excited. That ’ s excitement, don ’ t worry about it. Now, he gets to F-sharp, and finally he goes down to E, but it ’ s the wrong chord — because the chord he ’ s looking for is this one, and instead he does... Now, we call that a deceptive cadence, because it deceives **us**. I tell my students,”If you have a deceptive cadence, raise your eyebrows, and everybody will know.”Right. He gets to E, but it ’ s the wrong chord. Now, he tries E again. That chord doesn ’ t work. Now, he tries the E again. That chord doesn ’ t work. Now, he tries E again, and that doesn ’ t work. And then finally... There was a gentleman in the front row who went,”Mmm.”It ’ s the same gesture he makes when he comes home after a long day, turns off the key in his car and says,”Aah, I ’ m home.”Because we all know where home is. So this is a piece which goes from away to home. I ’ m going to play it all the way through and you ’ re going to follow. B, C, B, C, B, C, B — down to A, down to G, down to F. Almost goes to E, but otherwise the play would be over. He goes **back** up to B, he gets very excited. Goes to F-sharp. Goes to E. It ’ s the wrong chord. It ’ s the wrong chord. And finally goes to E, and it ’ s home. And what you ’ re going to see is one-buttock playing. Because for me, to join the B to the E, I have to stop thinking about every single note along the way, and start thinking about the long, long line from B to E. You know, we were just in South Africa, and you can ’ t go to South Africa without thinking of Mandela in jail for 27 years. What was he thinking about? Lunch? No, he was thinking about the vision for South Africa and for human beings. This is about vision. This is about the long line. Like the bird who flies over the field and doesn ’ t care about the fences underneath, all right? So now, you ’ re going to follow the line all the way from B to E. And I ’ ve one last request before I play this piece all the way through. Would you think of somebody who you adore, who ’ s no longer there? A beloved grandmother, a lover — somebody in your life who you love with all your heart, but that person is no longer with you. Bring that person into your mind, and at the same time, follow the line all the way from B to E, and you ’ ll hear everything that Chopin had to say. Now, you may be wondering — You may be wondering why I ’ m clapping. Well, I did this at a school in Boston with about 70 seventh graders, 12-year-olds. I did exactly what I did with you, and I explained the whole thing. At the end, they went crazy, clapping. I was clapping. They were clapping. Finally, I said,”Why am I clapping?”And one of them said,”Because we were listening.”Think of it. 1, 600 people, busy people, involved in all sorts of different things, listening, understanding and being moved by a piece by Chopin. Now, that is something. Am I sure that every single person followed that, understood it, was moved by it? Of course, I can ’ t be sure. But I ’ ll tell you what happened to me in Ireland during the Troubles, 10 years ago, and I was working with some Catholic and Protestant kids on conflict resolution. And I did this with them — a risky thing to do, because they were street kids. And one of them came to me the next morning and he said,”You know, I ’ ve never listened to classical music in my life, but when you played that shopping piece...”He said,”My brother was shot last year and I didn ’ t cry for him. But last night, when you played that piece, he was the one I was thinking about. And I felt the tears streaming down my face. And it felt really good to cry for my brother.”So I made up my mind at that moment that classical music is for everybody. Everybody. Now, how would you walk — my profession, the music profession doesn ’ t see it that way. They say three percent of the population likes classical music. If only we could move it to four percent, our problems would be over. How would you walk? How would you talk? How would you be? If you thought,”Three percent of the population likes classical music, if only we could move it to four percent.”How would you walk or talk? How would you be? If you thought,”Everybody loves classical music — they just haven ’ t found out about it yet.”See, these are totally different worlds. Now, I had an amazing experience. I was 45 years old, I ’ d been conducting for 20 years, and I suddenly had a realization. The conductor of an orchestra doesn ’ t make a sound. My picture appears on the front of the CD — But the conductor doesn ’ t make a sound. He depends, for his power, on his ability to make other people powerful. And that changed everything for me. It was totally life-changing. People in my orchestra said,”Ben, what happened?”That ’ s what happened. I realized my job was to awaken possibility in other people. And of course, I wanted to know whether I was doing that. How do you find out? You look at their eyes. If their eyes are shining, you know you ’ re doing it. You could light up a village with this guy ’ s eyes. Right. So if the eyes are shining, you know you ’ re doing it. If the eyes are not shining, you get to ask a question. And this is the question : who am I being that my players ’ eyes are not shining? We can do that with our children, too. Who am I being, that my children ’ s eyes are not shining? That ’ s a totally different world. Now, we ’ re all about to end this magical, on-the-mountain week, we ’ re going **back** into the world. And I say, it ’ s appropriate for **us** to ask the question, who are we being as we go **back** out into the world? And you know, I have a definition of success. For me, it ’ s very simple. It ’ s not about wealth and fame and power. It ’ s about how many shining eyes I have around me. So now, I have one last thought, which is that it really makes a difference what we say — the words that come out of our mouth. I learned this from a woman who survived

Auschwitz, one of the rare survivors. She went to Auschwitz when she was 15 years old. And... And her brother was eight, and the parents were lost. And she told me this, she said,”We were in the train going to Auschwitz, and I looked down and saw my brother ’ s shoes were missing. I said, ’ Why are you so stupid, can ’ t you keep your things together for goodness ’ sake? ’”The way an elder sister might speak to a younger brother. Unfortunately, it was the last thing she ever said to him, because she never saw him again. He did not survive. And so when she came out of Auschwitz, she made a vow. She told me this. She said,”I walked out of Auschwitz into life and I made a vow. And the vow was,”I will never say anything that couldn ’ t stand as the last thing I ever say.”Now, can we do that? No. And we ’ ll make ourselves wrong and others wrong. But it is a possibility to live into. Thank you. Shining eyes. Shining eyes. Thank you, thank you.

Text Sample 153: NEG Dim 1 – 2008 – Score 2.00 – 2008/t000255.txt

I had a fire nine days ago. My archive : 175 films, my 16-millimeter negative, all my books, my dad ’ s books, my photographs. I ’ d collected â I was a collector, major, big-time. It ’ s gone. I just looked at it, and I didn ’ t know what to do. I mean, this was â was I my things? I always live in the present â I love the present. I cherish the future. And I was taught some strange thing as a kid, like, you ’ ve got to make something good out of something bad. You ’ ve got to make something good out of something bad. This was bad! Man, I was â I cough. I was sick. That ’ s my camera lens. The first one â the one I shot my Bob Dylan film with 35 years ago. That ’ s my feature film.”King, Murray”won Cannes Film Festival 1970 â the only print I had. That ’ s my papers. That was in minutes â 20 minutes. Epiphany hit me. Something hit me.”You ’ ve got to make something good out of something bad,”I started to say to my friends, neighbors, my sister. By the way, that ’ s”Sputnik.”I ran it last year.”Sputnik”was downtown, the negative. It wasn ’ t touched. These are some pieces of things I used in my Sputnik feature film, which opens in New York in two weeks downtown. I called my sister. I called my neighbors. I said,”Come dig.”That ’ s me at my desk. That was a desk took 40-some years to build. You know â all the stuff. That ’ s my daughter, Jean. She came. She ’ s a nurse in San Francisco.”Dig it up,”I said.”Pieces. I want pieces. Bits and pieces.”I came up with this idea : a life of bits and pieces, which I ’ m just starting to work on â my next project. That ’ s my sister. She took care of pictures, because I was a big collector of snapshot photography that I believed said a lot. And those are some of the pictures that â something was good about the burnt pictures. I didn ’ t know. I looked at that â I said,”Wow, is that better than the â”That ’ s my proposal on Jimmy Doolittle. I made that movie for television. It ’ s the only copy I had. Pieces of it. Idea about women. So I started to say,”Hey, man, you are too much! You could cry about this.”I really didn ’ t. I just instead said,”I ’ m going to make something out of it, and maybe next year...”And I appreciate this moment to come up on this stage with so many people who ’ ve already given me so much solace, and just say to TEDsters : I ’ m proud of me. That I take something bad, I turn it, and I ’ m going to make something good out of this, all these pieces. That ’ s Arthur Leipzig ’ s original photograph I loved. I was a big record collector â the records didn ’ t make it. Boy, I tell you, film burns. Film burns. I mean, this was 16- millimeter safety film. The negatives are gone. That ’ s my father ’ s letter to me, telling me to marry the woman I first married when I was 20. That ’ s my daughter and me. She ’ s still there. She ’ s there this morning, actually.

That ’ s my house. My family ’ s living in the Hilton Hotel in Scotts Valley. That ’ s my wife, Heidi, who didn ’ t take it as well as I did. My children, Davey and Henry. My son, Davey, in the hotel two nights ago. So, my message to you folks, from my three minutes, is that I appreciate the chance to share this with you. I will be **back**. I love being at TED. I came to live it, and I am living it. That ’ s my view from my window outside of Santa Cruz, in Bonny Doon, just 35 miles from here. Thank you everybody.

Text Sample 154: NEG Dim 1 – 2005 – Score 1.00 – 2005/t002624.txt
I ’ m used to thinking of the TED audience as a wonderful collection of some of the most effective, intelligent, intellectual, savvy, worldly and innovative people in the world. And I think that ’ s true. However, I also have reason to believe that many, if not most, of you are actually tying your shoes incorrectly. Now I know that seems ludicrous. I know that seems ludicrous. And believe me, I lived the same sad life until about three years ago. And what happened to me was I bought, what was for me, a very expensive pair of shoes. But those shoes came with round nylon laces, and I couldn ’ t keep them tied. So I went **back** to the store and said to the owner,”I love the shoes, but I hate the laces.”He took a look and said,”Oh, you ’ re tying them wrong.”Now up until that moment, I would have thought that, by age 50, one of the life skills that I had really nailed was tying my shoes. But not so — let me demonstrate. This is the way that most of **us** were taught to tie our shoes. Now as it turns out — thank you. Wait, there ’ s more. As it turns out — there ’ s a strong form and a weak form of this knot, and we were taught the weak form. And here ’ s how to tell. If you pull the strands at the base of the knot, you will see that the bow will orient itself down the long axis of the shoe. That ’ s the weak form of the knot. But not to worry. If we start over and simply go the other direction around the bow, we get this, the strong form of the knot. And if you pull the cords under the knot, you will see that the bow orients itself along the transverse axis of the shoe. This is a stronger knot. It will come untied less often. It will let you down less, and not only that, it looks better. We ’ re going to do this one more time. Start as usual — go the other way around the loop. This is a little hard for children, but I think you can handle it. Pull the knot. There it is : the strong form of the shoe knot. Now, in keeping with today ’ s theme, I ’ d like to point out — something you already know — that sometimes a small advantage someplace in life can yield tremendous results someplace else. Live long and prosper.

Text Sample 155: NEG Dim 1 – 2005 – Score 3.00 – 2005/t000479.txt
This is really a two-hour presentation I give to high school students, cut down to three minutes. And it all started one day on a plane, on my way to TED, seven years ago. And in the seat next to me was a high school student, a teenager, and she came from a really poor family. And she wanted to make something of her life, and she asked me a simple little question. She said,”What leads to success?”And I felt really badly, because I couldn ’ t give her a good answer. So I get off the plane, and I come to TED. And I think, jeez, I ’ m in the middle of a room of successful people! So why don ’ t I ask them what helped them succeed, and pass it on to kids? So here we are, seven years, 500 interviews later, and I ’ m going to tell you what really leads to success and makes TEDsters tick. And the first thing is passion. Freeman Thomas says,”I ’ m driven by my passion.”TEDsters do it for love ; they don ’ t do it for money. Carol Coletta says,”I would pay someone to do what I do.”And the interesting thing is : if you

do it for love, the money comes anyway. Work! Rupert Murdoch said to me,”It ’ s all hard work. Nothing comes easily. But I have a lot of fun.”Did he say fun? Rupert? Yes! TEDsters do have fun working. And they work hard. I figured, they ’ re not workaholics. They ’ re workafrolics. Good! Alex Garden says,”To be successful, put your nose down in something and get damn good at it.”There ’ s no magic ; it ’ s practice, practice, practice. And it ’ s focus. Norman Jewison said to me,”I think it all has to do with focusing yourself on one thing.”And push! David Gallo says,”Push yourself. Physically, mentally, you ’ ve got to push, push, push.”You ’ ve got to push through shyness and self-doubt. Goldie Hawn says,”I always had self-doubts. I wasn ’ t good enough ; I wasn ’ t smart enough. I didn ’ t think I ’ d make it.”Now it ’ s not always easy to push yourself, and that ’ s why they invented mothers. Frank Gehry said to me,”My mother pushed me.”Serve! Sherwin Nuland says,”It was a privilege to serve as a doctor.”A lot of kids want to be millionaires. The first thing I say is :”OK, well you can ’ t serve yourself ; you ’ ve got to serve others something of value. Because that ’ s the way people really get rich.”Ideas! TEDster Bill Gates says,”I had an idea : founding the first micro- computer software company.”I ’ d say it was a pretty good idea. And there ’ s no magic to creativity in coming up with ideas — it ’ s just doing some very simple things. And I give lots of evidence. Persist! Joe Kraus says,”Persistence is the number one reason for our success.”You ’ ve got to persist through failure. You ’ ve got to persist through crap! Which of course means”Criticism, Rejection, Assholes and Pressure.”So, the answer to this question is simple : Pay 4, 000 bucks and come to TED. Or failing that, do the eight things — and trust me, these are the big eight things that lead to success. Thank you TEDsters for all your interviews!

Text Sample 156: NEG Dim 1 – 2005 – Score 3.00 – 2005/t000310.txt

Good morning, ladies and gentlemen. My name is Art Benjamin, and I am a”mathemagician.”What that means is, I combine my loves of math and magic to do something I call”mathemagics.”But before I get started, I have a quick question for the audience. By any chance, did anyone happen to bring with them this morning a calculator? Seriously, if you have a calculator with you, raise your hand. Raise your hand. Did your hand go up? Now bring it out, bring it out. Anybody else? I see, I see one way in the **back**. You sir, that ’ s three. And anybody on this side here? OK, over there on the aisle. Would the four of you please bring out your calculators, then join me up on stage. Let ’ s give them a nice round of applause. That ’ s right. Now, since I haven ’ t had the chance to work with these calculators, I need to make sure that they are all working properly. Would somebody get **us** started by giving **us** a two-digit number, please? How about a two-digit number? Audience : 22. AB : 22. And another two-digit number, sir? Audience : 47. AB : Multiply 22 times 47, make sure you get 1, 034, or the calculators are not working. Do all of you get 1, 034? 1, 034? Volunteer : No. AB : 594. Let ’ s give three of them a nice round of applause there. Would you like to try a more standard calculator, just in case? OK, great. What I ’ m going to try and do then — I notice it took some of you a little bit of time to get your answer. That ’ s OK. I ’ ll give you a shortcut for multiplying even faster on the calculator. There is something called the square of a number, which most of you know is taking a number and multiplying it by itself. For instance, five squared would be? Audience : 25. AB : 25. The way we can square on most calculators — let me demonstrate with this one — is by taking the number, such as five, hitting”times”and then”equals,”and on most calculators that will give you the square. On some of these ancient RPN calculators, you ’ ve got an”x squared”button on it, will allow you to do the calculation even faster. What I ’ m going to try and do now is to square, in my head, four two-digit numbers faster than they can do on their calculators, even using the shortcut method. What I ’ ll use is the second row this time, and I ’ ll get four of you to each yell out a two-digit number, and if you would square the first number, and if you would square the second, the third and the fourth, I will try and race you to the answer. OK? So quickly, a two-digit number please. Audience : 37. Arthur Benjamin : 37 squared, OK. Audience : 23. AB : 23 squared, OK. Audience : 59. AB : 59 squared, OK, and finally? Audience : 93. AB : 93 squared. Would you call out your answers, please? Volunteer : 1369. AB : 1369. Volunteer : 529. AB : 529. Volunteer : 3481. AB : 3481. Volunteer : 8649. AB : Thank you very much. Let me try to take this one step further. I ’ m going to try to square some three-digit numbers this time. I won ’ t even write these down — I ’ ll just call them out as they ’ re called out to me. Anyone I point to, call out a three-digit number. Anyone on our panel, verify the answer. Just give some indication if it ’ s right. A three-digit number, sir, yes? Audience : 987. AB : 987 squared is 974, 169. AB : Yes? Good. Another three-digit — — another three-digit number, sir? Audience : 457. AB : 457 squared is 205, 849. 205, 849? AB : Yes? OK, another, another three-digit number, sir? Audience : 321. AB : 321 is 103, 041. 103, 041. Yes? One more three-digit number please. Audience : Oh, 722. AB : 722 is 500, that ’ s a harder one. Is that 513, 284? Volunteer : Yes. AB : Yes? Oh, one more, one more three-digit number please. Audience : 162. 162 squared is 26, 244. Volunteer : Yes. Thank you very much. Let me try to take this one step further. I ’ m going to try to square a four-digit number this time. You can all take your time on this ; I will not beat you to the answer on this one, but I will try to get the answer right. To make this a little bit more random, let ’ s take the fourth row this time, let ’ s say, one, two, three, four. If each of you would call out a single digit between zero and nine, that will be the four-digit number that I ’ ll square. Nine. Seven. Five. Eight. 9, 758, this will take me a little bit of time, so bear with me. 95 million — 218, 564? Volunteer : Yes! AB : Thank you very much. Now, I would attempt to square a five-digit number — and I can — but unfortunately, most calculators cannot. Eight-digit capacity — don ’ t you hate that? So, since we ’ ve reached the limits of our calculators — what ’ s that? Does yours go higher? Volunteer : I don ’ t know. AB : Oh, yours does? Volunteer : I can probably do it. AB : I ’ ll talk to you later. In the meanwhile, let me conclude the first part of my show by doing something a little trickier. Let ’ s take the largest number on the board here, 8649. Would you each enter that on your calculator? And instead of squaring it this time, I want you to take that number and multiply it by any three-digit number that you want, but don ’ t tell me what you ’ re multiplying by — just multiply it by any random three-digit number. So you should have as an answer either a six-digit or probably a seven-digit number. How many digits do you have, six or seven? Seven, and yours? Seven? Seven? And, uncertain. Seven. Is there any possible way that I could know what seven-digit numbers you have? Say”No.”Good, then I shall attempt the impossible — or at least the improbable. What I ’ d like each of you to do is to call out for me any six of your seven digits, any six of them, in any order you ’ d like. One digit at a time, I shall try and determine the digit you ’ ve left out. Starting with your seven-digit number, call out any six of them please. Volunteer : 1, 9, 7, 0, 4, 2. AB : Did you leave out the number 6? Good, OK, that ’ s one. You have a seven-digit number, call out any six of them please. Volunteer : 4, 4, 8, 7, 5. I think I only heard five numbers. I — wait — 44875 — did you leave out the number 6? Same as she did, OK. You ’ ve got a seven-digit number — call out any six of them loud and clear. Volunteer : 0, 7, 9, 0, 4, 4. I think you left out the number 3? AB : That ' s three. The odds of me getting all four of these right by random guessing would be one in 10, 000 : 10 to the fourth power. OK, any six of them. Really scramble them up this time, please. Volunteer : 2, 6, 3, 9, 7, 2. Did you leave out the number 7? And let ' s give all four of these people a nice round of applause. Thank you very much. For my next number — while I mentally recharge my batteries, I have one more question for the audience. By any chance, does anybody here happen to know the day of the week that they were born on? If you think you know your birth day, raise your hand. Let ' s see, starting with — let ' s start with a gentleman first. What year was it, first of all? That ' s why I pick a gentleman first. Audience : 1953. 1953, and the month? November what? 23rd — was that a Monday? Audience : Yes. Good. Somebody else? I haven ' t seen any women ' s hands up. OK, how about you, what year? 1949, and the month? October what? Fifth — was that a Wednesday? Yes! I ' ll go way to the **back** right now, how about you? Yell it out, what year? Audience : 1959. 1959, OK — and the month? Audience : February. February what? Sixth — was that a Friday? Audience : Yes. Good, how about the person behind her? Call out, what year was it? Audience : 1947. AB : 1947, and the month? Audience : May. AB : May what? Seventh — would that be a Wednesday? Audience : Yes. AB : Thank you very much. Anybody here who ' d like to know the day of the week they were born? We can do it that way. Of course, I could just make up an answer and you wouldn ' t know, so I come prepared for that. I brought with me a book of calendars. It goes as far **back** into the past as 1800, because you never know. I didn ' t mean to look at you, sir — you were just sitting there. Anyway, Chris, you can help me out here, if you wouldn ' t mind. This is a book of calendars. Who wanted to know their birth day? What year was it, first of all? Audience : 1966. 66 — turn to the calendar with 1966. And what month? Audience : April. AB : April what? Audience : 17th. I believe that was a Sunday. Can you confirm, Chris? Chris Anderson : Yes. AB : I ' ll tell you what, Chris : as long as you have that book in front of you, do me a favor, turn to a year outside of the 1900s, either into the 1800s or way into the 2000s — that ' ll be a much greater challenge for me. AB : What year would you like? CA : 1824. AB : 1824, OK. AB : And what month? CA : June. AB : June what? CA : Sixth. AB : Was that a Sunday? CA : It was. AB : And it was cloudy. Good, thank you very much. But I ' d like to wrap things up now by alluding to something from earlier in the presentation. There was a gentleman up here who had a 10-digit calculator. Where is he, would you stand up, 10-digit guy? OK, stand up for me just for a second, so I can see where you are. You have a 10-digit calculator, sir, as well? OK, what I ' m going to try and do, is to square in my head a five-digit number requiring a 10-digit calculator. But to make my job more interesting for you, as well as for me, I ' m going to do this problem thinking out loud. So you can actually, honestly hear what ' s going on in my mind while I do a calculation of this size. Now, I have to apologize to our magician friend Lennart Green. I know as a magician we ' re not supposed to reveal our secrets, but I ' m not too afraid that people are going to start doing my show next week, so — I think we ' re OK. So, let ' s see, let ' s take a different row of people, starting with you. I ' ll get five digits : one, two, three, four. Oh, I did this row already. Let ' s do the row before you, starting with you : one, two, three, four, five. Call out a single digit — that will be the five-digit number that I will try to square, go ahead. Five. Seven. Six. Eight. Three. 57, 683 — squared. Yuck. Let me explain to you how I ' m going to attempt this problem. I ' m going to break the problem down into three parts. I ' ll do 57, 000 squared, plus 683 squared, plus 57, 000 times 683 times two. Add all those numbers together, and with any luck, arrive at the answer. Now, let me recap. Thank you. While I explain something else — — I know, that you can use, right? While I do these calculations, you might hear certain words, as opposed to numbers, creep into the calculation. Let me explain what that is. This is a phonetic code, a mnemonic device that I use, that allows me to convert numbers into words. I store them as words, and later on retrieve them as numbers. I know it sounds complicated ; it ’ s not. I don ’ t want you to think you ’ re seeing something out of”Rain Man.”There ’ s definitely a method to my madness — definitely, definitely. Sorry. If you want to talk to me about ADHD afterwards, you can talk to me then. By the way, one last instruction, for my judges with the calculators — you know who you are — there is at least a 50 percent chance that I will make a mistake here. If I do, don ’ t tell me what the mistake is ; just say,”you ’ re close,”or something like that, and I ’ ll try and figure out the answer — which could be pretty entertaining in itself. If, however, I am right, whatever you do, don ’ t keep it to yourself, OK? Make sure everybody knows that I got the answer right, because this is my big finish, OK. So, without any more stalling, here we go. I ’ ll start the problem in the middle, with 57 times 683. 57 times 68 is 3, 400, plus 476 is 3876, that ’ s 38, 760 plus 171, 38, 760 plus 171 is 38, 931. 38, 931 ; double that to get 77, 862. 77, 862 becomes cookie fission, cookie fission is 77, 822. That seems right, I ’ ll go on. Cookie fission, OK. Next, I do 57 squared, which is 3, 249, so I can say, three billion. Take the 249, add that to cookie, 249, oops, but I see a carry coming — 249 — add that to cookie, 250 plus 77, is 327 million — fission, fission, OK, finally, we do 683 squared, that ’ s 700 times 666, plus 17 squared is 466, 489, rev up if I need it, rev up, take the 466, add that to fission, to get, oh gee — 328, 489. Audience : Yeah! AB : Good. Thank you very much. I hope you enjoyed mathemagics. Thank you.

Text Sample 157: NEG Dim 1 – 2003 – Score 2.00 – 2003/t000176.txt

You know, I am so bad at tech that my daughter â who is now 41 â when she was five, was overheard by me to say to a friend of hers, If it doesn ’ t bleed when you cut it, my daddy doesn ’ t understand it. So, the assignment I ’ ve been given may be an insuperable obstacle for me, but I ’ m certainly going to try. What have I heard during these last four days? This is my third visit to TED. One was to TEDMED, and one, as you ’ ve heard, was a regular TED two years ago. I ’ ve heard what I consider an extraordinary thing that I ’ ve only heard a little bit in the two previous TEDs, and what that is is an interweaving and an interlarding, an intermixing, of a sense of social responsibility in so many of the talks â global responsibility, in fact, appealing to enlightened self- interest, but it goes far beyond enlightened self-interest. One of the most impressive things about what some, perhaps 10, of the speakers have been talking about is the realization, as you listen to them carefully, that they ’ re not saying : Well, this is what we should do ; this is what I would like you to do. It ’ s : This is what I have done because I ’ m excited by it, because it ’ s a wonderful thing, and it ’ s done something for me and, of course, it ’ s accomplished a great deal. It ’ s the old concept, the real Greek concept, of philanthropy in its original sense : phil- anthropy, the love of humankind. And the only explanation I can have for some of what you ’ ve been hearing in the last four days is that it arises, in fact, out of a form of love. And this gives me enormous hope. And hope, of course, is the topic that I ’ m supposed to be speaking about, which I ’ d completely forgotten about until I arrived. And when I did, I thought, well, I ’ d better look this word up in the dictionary. So, Sarah and I â my wife â walked over to the public library, which is four blocks away, on Pacific Street, and we got the OED, and we looked in there, and there are 14 definitions of hope, none of which really hits you between the eyes as being the appropriate one. And, of course, that makes sense, because hope is an abstract phenomenon ; it ’ s an abstract idea, it ’ s not a concrete word. Well, it reminds me a little bit of surgery. If there ’ s one operation for a disease, you know it works. If there are 15 operations, you know that none of them work. And that ’ s the way it is with definitions of words. If you have appendicitis, they take your appendix out, and you ’ re cured. If you ’ ve got reflux oesophagitis, there are 15 procedures, and Joe Schmo does it one way and Will Blow does it another way, and none of them work, and that ’ s the way it is with this word, hope. They all come down to the idea of an expectation of something good that is due to happen. And you know what I found out? The Indo- European root of the word hope is a stem, K-E-U â we would spell it K-E-U ; it ’ s pronounced koy â and it is the same root from which the word curve comes from. But what it means in the original Indo-European is a change in direction, going in a different way. And I find that very interesting and very provocative, because what you ’ ve been hearing in the last couple of days is the sense of going in different directions : directions that are specific and unique to problems. There are different paradigms. You ’ ve heard that word several times in the last four days, and everyone ’ s familiar with Kuhnian paradigms. So, when we think of hope now, we have to think of looking in other directions than we have been looking. There ’ s another â not definition, but description, of hope that has always appealed to me, and it was one by VĂĄclav Havel in his perfectly spectacular book”Breaking the Peace,”in which he says that hope does not consist of the expectation that things will come out exactly right, but the expectation that they will make sense regardless of how they come out. I can ’ t tell you how reassured I was by the very last sentence in that glorious presentation by Dean Kamen a few days ago. I wasn ’ t sure I heard it right, so I found him in one of the inter-sessions. He was talking to a very large man, but I didn ’ t care. I interrupted, and I said,”Did you say this?”He said,”I think so.”So, here ’ s what it is : I ’ ll repeat it.”The world will not be saved by the Internet.”It ’ s wonderful. Do you know what the world will be saved by? I ’ ll tell you. It ’ ll be saved by the human spirit. And by the human spirit, I don ’ t mean anything divine, I don ’ t mean anything supernatural â certainly not coming from this skeptic. What I mean is this ability that each of **us** has to be something greater than herself or himself ; to arise out of our ordinary selves and achieve something that at the beginning we thought perhaps we were not capable of. On an elemental level, we have all felt that spirituality at the time of childbirth. Some of you have felt it in laboratories ; some of you have felt it at the workbench. We feel it at concerts. I ’ ve felt it in the operating room, at the bedside. It is an elevation of **us** beyond ourselves. And I think that it ’ s going to be, in time, the elements of the human spirit that we ’ ve been hearing about bit by bit by bit from so many of the speakers in the last few days. And if there ’ s anything that has permeated this room, it is precisely that. I ’ m intrigued by a concept that was brought to life in the early part of the 19th century â actually, in the second decade of the 19th century â by a 27-year-old poet whose name was Percy Shelley. Now, we all think that Shelley obviously is the great romantic poet that he was ; many of **us** tend to forget that he wrote some perfectly wonderful essays, too, and the most well- remembered essay is one called”A Defence of Poetry.”Now, it ’ s about five, six, seven, eight pages long, and it gets kind of deep and difficult after about the third page, but somewhere on the second page he begins talking about the notion that he calls”moral imagination.”And here ’ s what he says, roughly translated : A man â generic man â a man, to be greatly good, must imagine clearly. He must see himself and the world through the eyes of another, and of many others. See himself and the world â not just the world, but see himself. What is it that is expected of **us** by the billions of people who live in what Laurie Garrett the other day so appropriately called despair and disparity? What is it that they have every right to ask of **us**? What is it that we have every right to ask of ourselves, out of our shared humanity and out of the human spirit? Well, you know precisely what it is. There ’ s a great deal of argument about whether we, as the great nation that we are, should be the policeman of the world, the world ’ s constabulary, but there should be virtually no argument about whether we should be the world ’ s healer. There has certainly been no argument about that in this room in the past four days. So, if we are to be the world ’ s healer, every disadvantaged person in this world â including in the United States â becomes our patient. Every disadvantaged nation, and perhaps our own nation, becomes our patient. So, it ’ s fun to think about the etymology of the word”patient.”It comes initially from the Latin patior, to endure, or to suffer. So, you go **back** to the old Indo- European root again, and what do you find? The Indo-European stem is pronounced payen â we would spell it P-A-E-N â and, lo and behold, mirabile dictu, it is the same root as the word compassion comes from, P-A-E-N. So, the lesson is very clear. The lesson is that our patient â the world, and the disadvantaged of the world â that patient deserves our compassion. But beyond our compassion, and far greater than compassion, is our moral imagination and our identification with each individual who lives in that world, not to think of them as a huge forest, but as individual trees. Of course, in this day and age, the trick is not to let each tree be obscured by that Bush in Washington that can get â can get in the way. So, here we are. We are, should be, morally committed to being the healer of the world. And we have had examples over and over and over again â you ’ ve just heard one in the last 15 minutes â of people who have not only had that commitment, but had the charisma, the brilliance â and I think in this room it ’ s easy to use the word brilliant, my God â the brilliance to succeed at least at the beginning of their quest, and who no doubt will continue to succeed, as long as more and more of **us** enlist ourselves in their cause. Now, if we ’ re talking about medicine, and we ’ re talking about healing, I ’ d like to quote someone who hasn ’ t been quoted. It seems to me everybody in the world ’ s been quoted here : Pogo ’ s been quoted ; Shakespeare ’ s been quoted backwards, forwards, inside out. I would like to quote one of my own household gods. I suspect he never really said this, because we don ’ t know what Hippocrates really said, but we do know for sure that one of the great Greek physicians said the following, and it has been recorded in one of the books attributed to Hippocrates, and the book is called”Precepts.”And I ’ ll read you what it is. Remember, I have been talking about, essentially philanthropy : the love of humankind, the individual humankind and the individual humankind that can bring that kind of love translated into action, translated, in some cases, into enlightened self-interest. And here he is, 2, 400 years ago :”Where there is love of humankind, there is love of healing.”We have seen that here today with the sense, with the sensitivity â and in the last three days, and with the power of the indomitable human spirit. Thank you very much.

Text Sample 158: NEG Dim 1 – 2003 – Score 3.00 – 2003/t000192.txt
When I first arrived in beautiful Zimbabwe, it was difficult to understand that 35 percent of the population is HIV positive. It really wasn ’ t until I was invited to the homes of people that I started to understand the human toll of the epidemic. For instance, this is Herbert with his grandmother. When I first met him, he was sitting on his grandmother ’ s lap. He has been orphaned, as both of his

parents died of AIDS, and his grandmother took care of him until he too died of AIDS. He liked to sit on her lap because he said that it was painful for him to lie in his own bed. When she got up to make tea, she placed him in my own lap and I had never felt a child that was that emaciated. Before I left, I actually asked him if I could get him something. I thought he would ask for a toy, or candy, and he asked me for slippers, because he said that his feet were cold. This is Joyce who ’ s — in this picture — 21. Single mother, HIV positive. I photographed her before and after the birth of her beautiful baby girl, Issa. And I was last week walking on Lafayette Street in Manhattan and got a call from a woman who I didn ’ t know, but she called to tell me that Joyce had passed away at the age of 23. Joyce ’ s mother is now taking care of her daughter, like so many other Zimbabwean children who ’ ve been orphaned by the epidemic. So a few of the stories. With every picture, there are individuals who have full lives and stories that deserve to be told. All these pictures are from Zimbabwe. Chris Anderson : Kirsten, will you just take one minute, just to tell your own story of how you got to Africa? Kirsten Ashburn : Mmm, gosh. CA : Just — KA : Actually, I was working at the time, doing production for a fashion photographer. And I was constantly reading the New York Times, and stunned by the statistics, the numbers. It was just frightening. So I quit my job and decided that that ’ s the subject that I wanted to tackle. And I first actually went to Botswana, where I spent a month — this is in December 2000 — then went to Zimbabwe for a month and a half, and then went **back** again this March 2002 for another month and a half in Zimbabwe. CA : That ’ s an amazing story, thank you. KB : Thanks for letting me show these.

Text Sample 159: NEG Dim 1 – 2003 – Score 3.00 – 2003/t000124.txt

The new me is beauty. Yeah, people used to say,”Norman ’ s OK, but if you followed what he said, everything would be usable but it would be ugly.”Well, I didn ’ t have that in mind, so... This is neat. Thank you for setting up my display. I mean, it ’ s just wonderful. And I haven ’ t the slightest idea of what it does or what it ’ s good for, but I want it. And that ’ s my new life. My new life is trying to understand what beauty is about, and”pretty,”and”emotions.”The new me is all about making things kind of neat and fun. And so this is a Philippe Starck juicer, produced by Alessi. It ’ s just neat ; it ’ s fun. It ’ s so much fun I have it in my house — but I have it in the entryway, I don ’ t use it to make juice. In fact, I bought the gold-plated special edition and it comes with a little slip of paper that says,”Don ’ t use this juicer to make juice.”The acid will ruin the gold plating. So actually, I took a carton of orange juice and I poured it in the glass to take this picture. Beneath it is a wonderful knife. It ’ s a Global cutting knife made in Japan. First of all, look at the shape — it ’ s just wonderful to look at. Second of all, it ’ s really beautifully balanced : it holds well, it feels well. And third of all, it ’ s so sharp, it just cuts. It ’ s a delight to use. And so it ’ s got everything, right? It ’ s beautiful and it ’ s functional. And I can tell you stories about it, which makes it reflective, and so you ’ ll see I have a theory of emotion. And those are the three components. Hiroshi Ishii and his group at the MIT Media Lab took a ping-pong table and placed a projector above it, and on the ping-pong table they projected an image of water with fish swimming in it. And as you play ping-pong, whenever the ball hits part of the table, the ripples spread out and the fish run away. But of course, then the ball hits the other side, the ripples hit the — poor fish, they can ’ t find any peace and quiet. Is that a good way to play ping-pong? No. But is it fun? Yeah! Yeah. Or look at Google. If you type in, oh say,”emotion and design,”you get 10 pages of results.

So Google just took their logo and they spread it out. Instead of saying,”You got 73, 000 results. This is one through 20. Next,”they just give you as many o ’ s as there are pages. It ’ s really simple and subtle. I bet a lot of you have seen it and never noticed it. That ’ s the subconscious mind that sort of notices it — it probably is kind of pleasant and you didn ’ t know why. And it ’ s just clever. And of course, what ’ s especially good is, if you type”design and emotion,”the first response out of those 10 pages is my website. Now, the weird thing is Google lies, because if I type”design and emotion,”it says,”You don ’ t need the ’ and. ’ We do it anyway.”So, OK. So I type”design emotion”and my website wasn ’ t first again. It was third. Oh well, different story. There was this wonderful review in The New York Times about the MINI Cooper automobile. It said,”You know, this is a car that has lots of faults. Buy it anyway. It ’ s so much fun to drive.”And if you look at the inside of the car — I mean, I loved it, I wanted to see it, I rented it, this is me taking a picture while my son is driving — and the inside of the car, the whole design is fun. It ’ s round, it ’ s neat. The controls work wonderfully. So that ’ s my new life ; it ’ s all about fun. I really have the feeling that pleasant things work better, and that never made any sense to me until I finally figured out — look... I ’ m going to put a plank on the ground. So, imagine I have a plank about two feet wide and 30 feet long and I ’ m going to walk on it, and you see I can walk on it without looking, I can go **back** and forth and I can jump up and down. No problem. Now I ’ m going to put the plank 300 feet in the air — and I ’ m not going to go near it, thank you. Intense fear paralyzes you. It actually affects the way the brain works. So, Paul Saffo, before his talk said that he didn ’ t really have it down until just a few days or hours before the talk, and that anxiety was really helpful in causing him to focus. That ’ s what fear and anxiety does ; it causes you to be — what ’ s called depth-first processing — to focus, not be distracted. And I couldn ’ t force myself across that. Now some people can — circus workers, steel workers. But it really changes the way you think. And then, a psychologist, Alice Isen, did this wonderful experiment. She brought students in to solve problems. So, she ’ d bring people into the room, and there ’ d be a string hanging down here and a string hanging down here. It was an empty room, except for a table with a bunch of crap on it — some papers and scissors and stuff. And she ’ d bring them in, and she ’ d say,”This is an IQ test and it determines how well you do in life. Would you tie those two strings together?”So they ’ d take one string and they ’ d pull it over here and they couldn ’ t reach the other string. Still can ’ t reach it. And, basically, none of them could solve it. You bring in a second group of people, and you say,”Oh, before we start, I got this box of candy, and I don ’ t eat candy. Would you like the box of candy?”And turns out they liked it, and it made them happy — not very happy, but a little bit of happy. And guess what — they solved the problem. And it turns out that when you ’ re anxious you squirt neural transmitters in the brain, which focuses you makes you depth-first. And when you ’ re happy — what we call positive valence — you squirt dopamine into the prefrontal lobes, which makes you a breadth-first problem solver : you ’ re more susceptible to interruption ; you do out-of-the-box thinking. That ’ s what brainstorming is about, right? With brainstorming we make you happy, we play games, and we say,”No criticism,”and you get all these weird, neat ideas. But in fact, if that ’ s how you always were you ’ d never get any work done because you ’ d be working along and say,”Oh, I got a new way of doing it.”So to get work done, you ’ ve got to set a deadline, right? You ’ ve got be anxious. The brain works differently if you ’ re happy. Things work better because you ’ re more creative. You get a little problem, you say,”Ah, I ’ ll figure it out.”No big deal. There ’ s something I call the visceral level of processing, and there will be visceral-level design. Biology — we have co-adapted through biology to like bright colors. That ' s especially good that mammals and primates like fruits and bright plants, because you eat the fruit and you thereby spread the seed. There ' s an amazing amount of stuff that ' s built into the brain. We dislike bitter tastes, we dislike loud sounds, we dislike hot temperatures, cold temperatures. We dislike scolding voices. We dislike frowning faces ; we like symmetrical faces, etc., etc. So that ' s the visceral level. In design, you can express visceral in lots of ways, like the choice of type fonts and the red for hot, exciting. Or the 1963 Jaguar : It ' s actually a crummy car, falls apart all the time, but the owners love it. And it ' s beautiful — it ' s in the Museum of Modern Art. A water bottle : You buy it because of the bottle, not because of the water. And when people are finished, they don ' t throw it away. They keep it for — you know, it ' s like the old wine bottles, you keep it for decoration or maybe fill it with water again, which proves it ' s not the water. It ' s all about the visceral experience. The middle level of processing is the behavioral level and that ' s actually where most of our stuff gets done. Visceral is subconscious, you ' re unaware of it. Behavioral is subconscious, you ' re unaware of it. Almost everything we do is subconscious. I ' m walking around the stage — I ' m not attending to the control of my legs. I ' m doing a lot ; most of my talk is subconscious ; it has been rehearsed and thought about a lot. Most of what we do is subconscious. Automatic behavior — skilled behavior — is subconscious, controlled by the behavioral side. And behavioral design is all about feeling in control, which includes usability, understanding — but also the feel and heft. That ' s why the Global knives are so neat. They ' re so nicely balanced, so sharp, that you really feel you ' re in control of the cutting. Or, just driving a high- performance sports car over a demanding curb — again, feeling that you are in complete control of the environment. Or the sensual feeling. This is a Kohler shower, a waterfall shower, and actually, all those knobs beneath are also showerheads. It will squirt you all around and you can stay in that shower for hours — and not waste water, by the way, because it recirculates the same dirty water. Or this — this is a really neat teapot I found at high tea at The Four Seasons Hotel in Chicago. It ' s a Ronnefeldt tilting teapot. That ' s kind of what the teapot looks like but the way you use it is you lay it on its **back**, and you put tea in, and then you fill it with water. The water then seeps over the tea. And the tea is sitting in this stuff to the right — the tea is to the right of this line. There ' s a little ledge inside, so the tea is sitting there and the water is filling it up like that. And when the tea is ready, or almost ready, you tilt it. And that means the tea is partially covered while it completes the brewing. And when it ' s finished, you put it vertically, and now the tea is — you remember — above this line and the water only comes to here — and so it keeps the tea out. On top of that, it communicates, which is what emotion does. Emotion is all about acting ; emotion is really about acting. It ' s being safe in the world. Cognition is about understanding the world, emotion is about interpreting it — saying good, bad, safe, dangerous, and getting **us** ready to act, which is why the muscles tense or relax. And that ' s why we can tell the emotion of somebody else — because their muscles are acting, subconsciously, except that we ' ve evolved to make the facial muscles really rich with emotion. Well, this has emotions if you like, because it signals the waiter that,"Hey, I ' m finished. See — upright."And the waiter can come by and say,"Would you like more water?"It ' s kind of neat. What a wonderful design. And the third level is reflective, which is, if you like the superego, it ' s a little part of the brain that has no control over what you do, no control over the — doesn ' t see the senses, doesn ' t control the muscles. It looks over what ' s going on. It ' s that little voice in your head that ' s watching and saying,"That ' s good. That ' s bad."Or,"Why are you doing that? I don ' t understand."It ' s that little voice in your head that ’ s the seat of consciousness. Here ’ s a great reflective product. Owners of the Hummer have said,”You know I ’ ve owned many cars in my life — all sorts of exotic cars, but never have I had a car that attracted so much attention.”It ’ s about attention. It ’ s about their image, not about the car. If you want a more positive model — this is the GM car. And the reason you might buy it now is because you care about the environment. And you ’ ll buy it to protect the environment, even though the first few cars are going to be really expensive and not perfected. But that ’ s reflective design as well. Or an expensive watch, so you can impress people —”Oh gee, I didn ’ t know you had that watch.”As opposed to this one, which is a pure behavioral watch, which probably keeps better time than the $13, 000 watch I just showed you. But it ’ s ugly. This is a clear Don Norman watch. And what ’ s neat is sometimes you pit one emotion against the other, the visceral fear of falling against the reflective state saying,”It ’ s OK. It ’ s OK. It ’ s safe. It ’ s safe.”If that amusement park were rusty and falling apart, you ’ d never go on the ride. So, it ’ s pitting one against the other. The other neat thing... So Jake Cress is this furniture maker, and he makes this unbelievable set of furniture. And this is his chair with claw, and the poor little chair has lost its ball and it ’ s trying to get it **back** before anybody notices. And what ’ s so neat about it is how you accept that story. And that ’ s what ’ s nice about emotion. So that ’ s the new me. I ’ m only saying positive things from now on.

Text Sample 160: NEG Dim 1 – 1994 – Score 15.00 – 1994/t002458.txt

Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look **back** and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ’ s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC microprocessors versus the power of local area networks. And the interesting thing about it is that this slide, like so many technology slides that we ’ re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ’ s basically what I ’ m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ’ m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well the answer is, if I drew it on a normal curve where, let ’ s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ’ m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ’ t tell **us** much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log curve, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as microprocessor technology, then the day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ’ s not moving like that. And there ’ s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ' d like to ask is, if you look at these exponential curves, they don ' t go on forever. Things just can ' t possibly keep changing as fast as they are. One of two things is going to happen. Either it ' s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ' s going to do this. That ' s about all it can do. Now I ' m an optimist, so I sort of think it ' s probably going to do something like that. If so, that means that what we ' re in the middle of right now is a transition. We ' re sort of on this line in a transition from the way the world used to be to some new way that the world is. And so what I ' m trying to ask, what I ' ve been asking myself, is what ' s this new way that the world is? What ' s that new state that the world is heading toward? Because the transition seems very, very confusing when we ' re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ' s a conference in which people talk about the future, and you notice that the future is still at about the year 2000. It ' s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ' s happening there. That transition is happening. We can all sense it. And we know that it just doesn ' t make too much sense to think out 30, 50 years because everything ' s going to be so different that a simple extrapolation of what we ' re doing just doesn ' t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ' re going through. Now in order to do that I ' m going to have to talk about a bunch of stuff that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step **back** and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go **back** about two and a half billion years, the Earth was this big, sterile hunk of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ' s theories that are beginning to understand about how it started with RNA, but I ' m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ' t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ' s sort of just a very simple chemical form of life, but when things got interesting was when these drops learned a trick about abstraction. Somehow by ways that we don ' t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for **us**, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for amusement purposes is we can now write things in this code. And I ' ve got here a little 100 micrograms of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ' s right here. If I had really been a egotist, I would have put it into a virus and released it in the room. So what was the next step? Writing down the DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ' t know if you know this, but bacteria can actually exchange DNA. Now that ' s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the transition point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ' s interesting in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ' re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to evolve so that the interesting level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very special structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in the community and special structures that were responsible for recording, understanding, learning information. And that was the brains and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ' d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these special information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ’ re aware of is human language. It ’ s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ’ m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ’ re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows **us** now to begin to start functioning as a single organism. And so, in fact, what we ’ ve done is we, humanity, have started abstracting out. We ’ re going through the same levels that multi-cellular organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. Telephony, computers, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ’ ve now built within our society for handling that information. And it all connects **us** together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of microseconds. And you saw Ty ’ s little evolutionary example where he sort of did a little bit of evolution on the Convolution program right before your eyes. So now we ’ ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and brains, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ’ s what we ’ re seeing here in this explosion of curve. We ’ re seeing this process feeding **back** on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ’ s really impossible for me to design them in the traditional sense. I don ’ t know what every transistor in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of abstraction and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ’ t quite even understand. One method that ’ s particularly interesting that I ’ ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the microsecond time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,”Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted.”So in other words, I define what I wanted. Let ’ s say I want to sort numbers, as a simple example I ’ ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,”Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ’ s reproduce the ones that sorted numbers the best. And let ’ s reproduce them by a process of recombination analogous to sex.”Take two programs and they produce children by exchanging their subroutines, and the children inherit the traits of the subroutines of the two programs. So I ’ ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,”Please repeat that process.”Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ’ t tell you how they work. I ’ ve tried looking at them and telling you how they work. They ’ re obscure, weird programs. But they do the job. And in fact, I know, I ’ m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,”Oh look. Look at this. It says, ’ This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ’ Doesn ’ t that make you feel confident?”In fact, we know that the engineering process doesn ’ t work very well when it gets complicated. So we ’ re beginning to depend on computers to do a process that ’ s very different than engineering. And it lets **us** produce things of much more complexity than normal engineering lets **us** produce. And yet, we don ’ t quite understand the options of it. So in a sense, it ’ s getting ahead of **us**. We ’ re now using those programs to make much faster computers so that we ’ ll be able to run this process much faster. So it ’ s feeding **back** on itself. The thing is becoming faster and that ’ s why I think it seems so confusing. Because all of these technologies are feeding **back** on themselves. We ’ re taking off. And what we are is we ’ re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ’ re the amoebas and we can ’ t quite figure out what the hell this thing is we ’ re creating. We ’ re right at that point of transition. But I think that there really is something coming along after **us**. I think it ’ s very haughty of **us** to think that we ’ re the end product of evolution. And I think all of **us** here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

3 POS Dim 2

Text Sample 161: POS Dim 2 – 2020 – Score 55.00 – 2020/t004270.txt

Lizet Ocampo : I have a team meeting with my team. It ’ s Monday morning, so we kind of miss each other.”Let ’ s turn our cameras on.”Adam Grant : For Lizet Ocampo, it seemed like a regular day in **quarantine**. But when they turned on their cameras, something was off. Her face was a potato, a flat image of a potato, with Lizet ’ s eyes and lips moving on it. LO : And I just remember thinking,”Why am I a potato?”and of all things, why a potato? Like, I couldn ’ t go back to a regular camera, so I just essentially was a potato for the meeting. AG : The previous week, Lizet had been in a virtual happy hour for Latino leaders. LO : And I wanted to make them laugh, and so I thought,”You know what? Next week, I ’ m going to try to do some funny filters,”and so I download these Snap filters, but it didn ’ t really download all the way. There was an error message that kept popping up, so I kind of gave up on it and forgot about it. AG : Turns out the filters had worked a little too well and a little too late. Now Lizet was stuck as a potato. Her colleagues laughed and went on with the meeting. Lizet didn ’ t think much of it until she got a text. LO : Someone that ’ s on my staff took a picture of it and posted it on her personal Twitter account, so I had no idea. AG : She wrote,”My boss turned herself into a potato and can ’ t figure out how to turn the setting off.”LO : And then my colleagues ’ faces are hilarious because it looks like they ’ re trying to be serious, but they ’ re trying not to laugh at the same time. AG : It is amazing because you see her, you see the guy, and then there ’ s just a potato. LO : Yes. AG : Yeah, this poor potato is like,”How do I stop being a potato?”LO : Forty million people have seen it, and it ’ s become sort of a cultural moment in this **quarantine** world that we ’ re in right now. AG : You have a new identity now, don ’ t you, though? LO : That ’ s true. I ’ m known to the world as”potato boss.”I have potato appearances requests – AG : LO : and so I ’ ve showed up as a potato to some meetings for friends at other organizations and things like that. AG : The **coronavirus** pandemic has thrown many of us into the deep end of working remotely. While some people have found creative ways to swim, many others are **struggling** just to stay afloat. The good news is that remote work is not uncharted waters. We ’ ve actually been doing it and studying it for a long time, and with those lessons in hand, you might actually be more prepared than you think to make remote work work. AG : I ’ m Adam Grant, and this is”Work-Life,”my podcast with TED. I ’ m an **organizational** psychologist. I study how to make work not suck. In this show, I ’ m inviting myself inside the minds of some truly unusual people, because they ’ ve mastered something I wish everyone knew about work. Today : remote work, how we can communicate better with and without a camera, no potatoes required, and how, even during this unprecedented pandemic, we can stay resilient. This episode is sponsored by Hilton. As the world has gone into lockdown, many of us have experienced the challenges of working from home. But some jobs are hard to do remotely. If you work in health care or public safety, you need to be on the front lines. If you ’ re in manufacturing, production or distribution, you have to be on-site at a farm, factory or warehouse. We ’ re all depending on the people who are risking their health and safety to go to work. Yet, in the knowledge and **service** industries, many jobs can be done with online tools. And that ’ s not as new as it seems. Half a century ago, futurists started predicting the rise of remote work. In 1993, management guru Peter Drucker wrote that commuting to an office is obsolete. But a quarter-century later, nearly half of global companies still weren ’ t allowing remote work at all. Not anymore. Around the world, many companies are now requiring remote work. This forced remote work has amplified many of the challenges we had when we worked face-to-face. And even in that hard-to-imagine future beyond the pandemic, at least some of this shift to remote work is probably here to stay. So, how do we do it well? As teams, we need to figure out how to stay connected and collaborative. As individuals, we ’ re all seeking guidance on how to stay sane when we ’ re working alone. Recorded caller 1 : So, I ’ m incredibly vulnerable and scared of losing some of the relationships that have taken a long time to build and not having the edge of my personality and my ability to communicate clearly face-to-face. Recorded caller 2 : Working from home has been a nightmare. AG : Interruptions have become a fact of life. We ’ re all”BBC Dad”now, with kids or pets dancing right into our video conferences. Recorded caller 3 : We get on a Zoom call with our region, and in runs my four-year-old, completely naked, and I was speaking, so I wasn ’ t on mute, and all you hear is me screaming,”No!”Recorded caller 4 :

I finally made a rule that said,"If you interrupt a Zoom meeting, you ' ll have to stay for the whole thing."The first offender was my 10-year-old daughter. She ended up sitting there for 45 minutes with nothing to play with, and I can remember her telling all the other kids,"Wow, don ' t go down there, because you ' re going to have to stay for one of those boring meetings."Recorded caller 5 : On one Zoom call where I was pleading with government officials to please listen to the plight of small business owners, my five-year-old daughter walked behind me and said,"Mom, if you die of the virus, I ' ll run your company,"and everyone went silent. Try to come back from that when you ' re campaigning. Martine Haas : I think in some ways, we are adapting very, very fast right now. We are the experiment that will create the data. AG : Martine Haas is my colleague at Wharton and a renowned expert on how teams collaborate at a distance. Now she isn ' t just studying it. She ' s living it. MH : I ' ve been working to kind of get my setup really functional, and I invested in a screen to separate my workspace from the rest of the room that I ' m in. I open the screen ceremoniously in the morning, and then I close it ceremoniously. I literally move it so that it kind of covers the desk just as a way to kind of switch off. AG : One of my favorite insights about teams is that they ' re kind of like amplifiers : whatever you put in comes out louder. When we go remote, that ' s even more true. All of our **microphones** are blasting at once, and that can cause problems. MH : I sometimes think about it as kind of like an iceberg, right? At the top of the iceberg, the stuff that ' s more obvious – you know, it ' s just difficult to communicate, partly because of the technology, right? AG : Think about how much time you ' ve wasted on tech problems. First man : Hey, Paul. Can you hear me? Paul : Hey, guys. Hey, Tyler. Tyler : Sorry I ' m late. Having a hard time connecting. FM : One second. Paul ' s having a sound issue. P : I can ' t hear you. T : Try adjusting your output **settings**. P : Can you hear me? T : It ' s the gear icon. P : Never mind. I got it. I just had to change a few **settings**. FM : Great. Great. I think we can get started then started then then. Oh, great. Oh, great. AG : Another obvious issue is power. MH : And it ' s very easy to fall into a very hierarchical form of communication. AG : You know, where the person with the most authority or status hijacks the conversation. In a classic study of airline crews, it was possible to predict a flight ' s timeliness and fuel efficiency just by watching the first 15 minutes of a captain ' s meeting with the crew. The **ineffective** captains tended to monopolize the conversation. The effective ones did more two-way communication. They spent more time engaging their crews and listening to them. They also encouraged their crews to talk to one another about the task. Creating open channels of communication is even more important in virtual teams. You ' ve probably had video calls that were dominated by one talking head, but that ' s only the tip of the iceberg. MH : So the less obvious things, there are two big buckets of things called"shared identity"problems and"shared understanding"problems. AG : Shared identity and shared understanding. Beneath the everyday communication challenges you ' ve seen, these are probably the bigger issues at play. In fact, these are the key ingredients that Martine has found for making remote teams effective. But how do you build them among teammates who can only see the dimly lit, poorly framed video conference versions of each other? Let ' s start with shared identity. It ' s the sense that we ' re all in this together. When teams lose that sense of unity... MH :... it ' s really easy to get an us-versus-them kind of dynamic going."Here in Philadelphia, we work really hard, but that part of the team that ' s over there in Belgium, they really don ' t work so hard, they don ' t come in on time. We don ' t know what they ' re up to."So it ' s very easy to get those kind of psychological **distancing** dynamics that come into play, and it ' s a basic psychological process. AG : In the study of airline crews, the effective captains created a shared identity. They talked about everyone as"we,"not just the flight attendants, but also the maintenance crew, the gate agents and even the air traffic controllers communicating remotely with them. They saw themselves as leading one team that was responsible for the safety of the flight. The **ineffective** captains reinforced fault lines between subgroups. They called the pilots"we"but everyone else"you."This lack of shared identity hurts the work of face-to-face colleagues but creates an even bigger problem for remote teams. MH : It reduces trust. It reduces psychological safety, that sense that it ' s safe to kind of speak up in the group. It can lead to people withholding information rather than sharing what they know, it can increase conflict. And so it ' s really important of the team leader to articulate for an overarching goal. AG : Goal-setting isn ' t usually the first thing that comes to mind when thinking about team identity. What most people try instead is team-building activities. I ' m guessing you ' ve suffered through Zoom calls where you ' re asked to name your favorite pizza topping or to share highlights from your totally boring weekend in **quarantine**. Those superficial strategies overlook that we bond best when our individual **actions** contribute to a common purpose. Research reveals that it ' s critical to establish team goals and clarify roles, especially in big teams. Having team goals creates a sense of shared identity. Having clear roles prevents social loafing and free-riding, helping people feel that their individual responsibilities matter. The other key challenge of virtual teams is shared understanding, being able to get on the same page, not just about what you ' re doing, but also about who you are and what you value. MH : It ' s basically the problem that you kind of have incomplete information. You don ' t have a good understanding of each other ' s contexts, right? So this is actually why it ' s really valuable to ask members of your remote team,"What does your **workplace** look like? Who else is there? Do you have a dog? Are you watching your kid? What else is going on?"AG : You know you ' ve reached shared understanding when a colleague does something that would be puzzling to most people but you get it right away. The project manager who cc ' s the boss isn ' t trying to make everyone look bad. She ' s anxious about your team ' s work being overlooked. The guy in accounts payable isn ' t signing off at 3pm because he ' s a slacker. He ' s taking care of his mom, who ' s ill. MH : Because if you don ' t understand people ' s contexts, you can make attributions – and they ' re often negative attributions – about what other people are doing or whether they ' re working hard enough or whether they ' re distracted that are not really fair, right? Because you don ' t really get what their situation is in the same way as you would if you were working face-to-face. AG : Think about your virtual team. Shared identity and shared understanding are pivotal for trust. How do you make that happen when you ' re not in the same physical space? You might strengthen shared identity by circulating feedback from customers on why your team ' s work is more important than ever. And you could get to some shared understanding by talking about your favorite video conference bloopers. Some of you have been experimenting with other strategies. Woman to baby : Hey, bud, can you say"welcome"? Baby : Welcome. Woman : To. Baby : To. Woman : Principles. Baby : Woman : Of. Baby : Of. Woman : Management! Baby : Man : My little sister, she had found a sign that said"Wall Street"on it. She put the Wall Street sign behind me so that when I get on my Zoom calls, my colleagues can see it, and she said,"Look, now you ' re on Wall Street. Stop **complaining**. Act like it."Woman : Can you say"my"? Baby : My Woman : Mama. Baby : Mama. Woman : Misses. Baby : Misses. Woman : All. Baby : All. Woman : Of. Baby : Of. Woman : You. Baby : You. Man : I love asking folks to give me a quick tour of their makeshift home office if they ' re comfortable with it. I ' ve found it often leads to a conversation about hobbies or life outside of work, which has been really, really cool for building closer relationships and kind of humanizing the other folks on the team. Woman : OK, one more thing. If you ’ re happy and you know it shout”hooray”! Baby : Hooray! Woman : All right! AG : I ’ ve been doing the majority of my work remotely for the past 15 years. Some of the key members of our”WorkLife”team have never even met face-to-face, so reaching shared understanding has been key in our collaborations. If you work with me, we have a shared understanding that my biggest vice is being chronically late. I have a hard time disengaging from a task before it ’ s done, and I try to squeeze too much into each day, but I often have calls with people who don ’ t have that context. Eventually, I had to start managing expectations. Now my calendar invite automatically notifies people that I ’ m usually running five to 10 minutes late. It helps them understand my schedule. But there ’ s something that helps even more. Alex Cornell : Well, I ’ ve been sittin ’ here all day / I ’ ve been sittin ’ in this waiting room. AG : Hold music. But not just any hold music. AC : Waitin ’ on my friends / Yes, I ’ m waitin ’ on this conference call / All alone. AG : Now when I jump on the line late, people aren ’ t disappointed or frustrated. They ’ re often amused. AC : Guess I ’ m on hold / I hope it ’ s not all day. AG : So when I scheduled a call with the guy who wrote this song, for once, I was deliberately late. I let him call in first, so he could hear his own hold music. AC : Here ’ s my voice. Y ’ all sing along. AG : Alex Cornell wrote this song for UberConference, because he hates hold music. AC : Whenever I call into an airline or anything like that, and you hear the same song over and over and over again, you ’ re now sort of already in this like, low-level tortured state of just been listening to this terrible song over and over again. AG : So he decided to make something better. AC : Tell me where can they be... AC : If you could imagine being on hold, and while you ’ re on hold, there ’ s just actually a person there that ’ s sort of like a doorman, so to speak. I ’ m just smiling as I ’ m saying it because it ’ s such a weird thing to picture, this, like, virtual doorman saying like,”Oh, hey, like, here you are, too, calling into this conference call,”but then also, it acts as a great icebreaker. AG : Alex, didn ’ t just write the hold music for UberConference. He cofounded the company, and then went on to be the lead product designer of Facebook Live. So he ’ s spent a lot of time working on virtual communication and has a lot of experience building shared identity and shared understanding. Alex ’ s new **start-up**, Cocoon, has developed an app for communicating privately with our closest friends and families. Before the pandemic, the team was six people in the same physical space while one person was remote. Now they ’ re all remote. AC : We have brought on at Cocoon two new team members since this began, and they need to be onboarded to the company, to the product and to what we do and what we believe in. AG : He ’ s instinctively put into place some of the core strategies that Martine Haas recommends. To strengthen shared identity, Alex has created written memos to share the company ’ s history and values. Spelling it out forced him to get clear on the team ’ s goals and culture and helped newcomers get up to speed from a distance. It ’ s something they can all revisit occasionally to remember their common purpose and **norms**. And to foster shared understanding, Alex has found that it helps to open up a little more about his home life. AC : I ’ ve found that it can be pretty helpful for this situation too, even just for our coworkers, to have a little bit more awareness of what we ’ re all doing day to day. I wouldn ’ t normally have shared with them that I am building a gigantic LEGO with my stepson, and that ’ s something that they know, and I think we ’ re actually, in some ways, more in touch with one another than we ever have been. AG : There ’ s evidence that when we gain this kind of personal knowledge about our colleagues ’ lives outside work, it humanizes them. We trust them more and feel more similar to them. If we only talk about work, we miss out on that information. So Alex has been experimenting with unstructured interaction to create shared understanding. AC : We ’ ve definitely done virtual lunches. I think one of the things that is missing most instantly when you go full remote is the sort of just camaraderie associated with being able to just kind of have a conversation about nothing around the lunch table, and it ’ s weird to set a meeting and say,”The agenda is we ’ re not going to talk about work, and everybody please have fun.”It ’ s like that is, that ’ s a weird agenda. It ’ s hard to do. I think there ’ s all kinds of these new **norms** that are being established just as we go. AG : When you communicate virtually, what **norms** have you been setting to facilitate shared understanding? Maybe you ’ ve been encouraging people to use the chat function so the quieter voices get heard. Or you might be turning off your self-view so you ’ re not distracted by all your weird facial **expressions**. But if you ’ re like most people, you want to keep seeing your team members ’ videos. You prefer that to audio-only because you value having the rich visual cues of facial **expressions** and body language. That ’ s how you read their emotions, right? Wrong. It turns out that if you want to know what someone is feeling, you might be better off just hearing their voice. In one experiment, people read another person ’ s emotions more accurately when the lights were turned off. In another experiment, they were more accurate when the video feed was turned off. **Follow-up** studies showed that when we don ’ t have visual cues, we pay more attention to the content and tone of voice. AC : I think that there is definitely something that is pure about when all you can hear somebody ’ s voice. Then when you switch to audio only, it ’ s like that actually is something that we ’ re all familiar with and are somewhat used to, and there ’ s no sort of missing piece. AG : Plus, if you ’ re on video all day, you could be losing contributions from one group of colleagues in particular : introverts. Why? Because of screen-out. AC : Even though I really want to see my friends, if I ’ ve been on Zoom all day, I don ’ t really want to spend another couple hours on Zoom just because, socially, I ’ m tired of interacting, but I ’ m a bit of an introvert, I suppose, so my energy is just depleted, usually, by the end of the day so. AG : Introverts are more sensitive to stimulation than extroverts, not just bright lights and loud noises, but also eye contact, and video calls seem to be magnifying that overload. AC : And you ’ re staring at somebody ’ s eyes for, in some cases, like an hour, and I don ’ t know about you, but when I talk to somebody else, I ’ m very rarely staring at their face for all 60 minutes. AG : There ’ s a fun study that tracked people ’ s reactions to seeing a large virtual face on their screen. The most common response was to flinch. AG : We know face-to-face interaction is critical for establishing trust, for building that shared understanding that Martine Haas has found is so important. But once trust exists, I think we should all have the **license** to turn off our video feeds from time to time. That ’ s what I ’ ve been doing. When people ask what video conference **platform** I prefer, I often tell them”audio.”After all, if you know what I look like, and you trust me, you don ’ t need to see my face every minute. But for times when you don ’ t have that option, Alex recently created a video conference version of his hold song. AC : My version of this on video conferencing is to have a looping version of myself who ’ s sitting there – AG : AC :... seemingly paying attention. Technically – I ’ m not saying I would do this – but you could have the looping version of yourself just on the call the whole time. This video one, it actually is me, so it ’ s sort of like a”Westworld”-type version of it. AG : If we were robots, it would be easy to be resilient. Just turn your emotions off until this crisis is over. Since we can ’ t do that, the next best thing is to learn from someone who ’ s done more remote work than almost anyone in human history. After the break. OK, this is going to be a different kind of **ad**. I ' ve played a personal role in selecting the sponsors for this podcast because they all have interesting cultures of their own. Today, we ' re going inside the **workplace** at Hilton. We had recorded a story back in February. Then the world changed, so we decided to do something different, to speak to the moment we ' re living in. Charles Hill : I was in denial. We weren ' t going to be impacted. We ' re making it out of this. And it was that moment, it was 4:30 on a Friday, March 6. I knew that we were in for something else. AG : Meet Charles Hill. He ' s the General Manager of DoubleTree by Hilton in Crystal City, Virginia, just outside Washington, DC. That Friday, his hotel was hit with a wave of cancellations. As the reality of how severely the **coronavirus** would affect them set in, he was called into nonstop meetings to answer an impossible question : How do you manage a hotel in the middle of a pandemic? CH : What is it that ' s happening around us? How big is this? And what do we need to do? There was nothing coming in. It was all going out. But we still had these hotel rooms. AG : That ' s when Hilton came up with a plan. They would do what they do best : offer hospitality. CH : Hospitality is caring for others and spreading light and warmth. AG : In partnership with American Express and its hotel owners, Hilton donated a million rooms across the US to frontline healthcare professionals. CH : For us to be able to provide accommodations for people who are literally working on the front lines to take care of those in need, it ' s such a great way to impact our community. AG : They took in nurses, doctors, EMTs – the people working tireless hours who needed a safe and clean place to stay. When they arrived, Charles was there to welcome them at the front desk. CH : I think there ' s a sense of surprise that they don ' t have to pay. There isn ' t a fee. We ' re not going to charge you. It ' s OK. We want you here. We ' re thankful for what you ' re doing. AG : For Charles and his staff, it felt good to be able to do something, and they were inspired to bring more light and warmth – literally. CH : The lights in our north tower have this warm glow, and we came up with the idea of putting messages in that north tower so that anybody who passes would be able to see something from us. AG : With no one staying in the north tower of the hotel, the dark windows were like a blank canvas. They could turn on the lights inside certain rooms to beam shapes to the outside. CH : The first message that we put up was this big heart. Think about these warm bulbs in 52 rooms that just emanates from the building to create this kind of glow outside. AG : A few days later, they changed the lights to form another message. This time, they beamed out one word. CH : We did"hope,"we put H-O-P-E, so you can see very clearly"hope"glowing in the night. AG : Because the hotel is so close to DC, anyone going in or out of the city could see the lights. That night, Charles got a message from a police officer who works at the Pentagon. CH : He had just finished his shift. He was walking back. He saw it, and he just wanted to say,"Thanks."He said,"This was the first positive thing I ' ve seen in a while."AG : Their community started to take notice. Other hotels followed suit. The local news picked up the story, and people started calling the hotel, hoping for a sneak preview of what shape the lights would take next. CH : It turned into this moment where we all had something to be happy about again and something to celebrate, and it was small, but it was a win, and it brought joy. AG : Psychologists have discovered an alternative to post-traumatic **stress**. It ' s called"post-traumatic growth."It ' s the experience of not just bouncing back, but bouncing forward from adversity. Often, it takes the form of a greater sense of strength. CH : This may be the toughest crisis any of us have seen in our lifetimes, but if we can make it through, anything else is a piece of cake. We ' re going to be stronger for making our way through this. AG : COVID-19 has created unprecedented challenges for the travel industry. In times like these, hospitality is needed more than ever. Hilton hotels all around the world are coming together to be the bright light for those who need it most. Read more inspiring stories of Hilton hospitality at newsroom. hilton. com / actsofhospitality. JaBarie Anderson : We ' re going to make magic today. OK, so – JA : This is"WorkLife ' s"first live virtual haircut, and we ' re here with"You Probably Need a Haircut"on webcam, because this is the new normal. You ' re going to need a mirror, a comb and scissors. Coco : I have scissors, they ' re the dog scissors. AG : Since I don ' t have hair, our producer Coco volunteered to be a guinea pig. JA : So cut straight across, a little bit higher. Coco : A little bit higher? OK. AG : Before the pandemic, JaBarie Anderson was a freelance stylist. So when he tried doing virtual haircuts from home about a month ago, he wasn ' t sure what to expect. JA : I ' ve literally been in so many bathrooms and showers with people now, and people turn on their camera, and it ' s just like people in their boxes getting their hair cut. I ' m like,"Oh, OK. This is happening."AG : Wait. That happens? JA : Yes, people are shirtless, or the whole family is in there watching, talking like,"This is crazy. I want to see how this is going to turn out."I ' m like,"Turn your head this way. Turn your head that way,"and you ' re trying to get the hand positions right. Coco : OK, cutting. JA : OK. Now I ' ll ask you to go in and point-cut it. Coco : Got it. OK, so now I ' m point-cutting this. So basically... Ohhhh. JA : People being so willing and trusting – that was one thing that shocked me the most. Coco : A little bit higher? OK. JA : Right there. Coco : So, like, here? JA : Yep. Coco : OK. Cutting. JA : OK, now I want you to... AG : So far, there haven ' t been any major hair disasters, and by moving his job online, JaBarie ' s been able to get clients in San Francisco, Texas and Germany. JA : This is absolutely amazing now. It gives me a **platform** to really showcase my talent. AG : Wait. Did you just say this is amazing? JA : Yes, it allows you to interact with your clients in a different way, and it builds trust, and it builds a bond amongst people in your household. So, say your wife is cutting your hair. That ' s another level of trust because it ' s your hair, you know? AG : Wow, I would not have expected that. So it ' s **gotta** be harder, though, to do your job in some ways, because you can ' t touch the person or even control the scissors or the clippers. JA : Well, that ' s where the education comes in. It ' s truly amazing to see what people can do from a video chat call and their home tools. That line is kind of shattered. Coco : OK, I think I am shattering that line. It ' s still, like, a little – JA : Shattering like broken glass! AG : JaBarie ' s enthusiasm for this new **platform** has helped him maintain his livelihood during the pandemic. This kind of ingenuity is visible in many **service** jobs. Some physical therapists are doing virtual appointments. So are plumbers. Plumber : You know, as far as your toilet, what have you been flushing down there? Did you use a plunger for this, because you really don ' t want to. Oh, I told him what tools he needed, and I stayed on the video with him while he fixed it himself. It ended up being a very simple fix. AG : And people doing more independent work have found creative ways to stay engaged while working from home. Woman : I fake my commute, which means I get on my stationary bike and listen to a podcast in the same way I do when I drive to work for roughly 30 minutes before going to actually sit at my desk. Man : To stay productive, I could only use my phone in the living room, so pretty much my cell phone never leaves that area, and if I have to pick up a call, I ' ll have to go to the living room. If I want to check Facebook, I have to go to the living room in order to do so. Scott Kelly : I have a little bit of a **mission** control setup, so I got an iPad, a second iPad. I have a little TV next, over my right shoulder, so I can monitor the situation. AG : You might have your own personal **mission** control center as you **navigate** working remotely. But"**mission** control"has an entirely different meaning to Scott Kelly, because he ' s an astronaut. SK : So, five years ago today, I launched into space for my **mission** that wound up being 340 days on the ISS. AG : Three-hundred and forty consecutive days in space. It ' s an American record for a single **mission**. The rest of us weren ' t schooled in how to stay at home that long in sustained **isolation**, so I turned to Scott for ideas on building resilience. Other than the few fellow space travelers in **isolation** with him, communication with his family, friends and coworkers was entirely through video, e-mail and radio calls. He couldn ' t go outside without the right equipment, and he didn ' t even know when his **mission** was going to end. Sound familiar? SK : I knew I was going to come home about a year later. I didn ' t know exactly when. I had to make this mental kind of change to my whole reference system with regards to my life. AG : He ' d been to space three times before, but he ' d never been out there for so long away from everyone he loved. SK : And I felt like I needed to pace myself, as you can wind up working too much when you go to sleep in your office, and you wake up, and you ' re still in your office. The walls could be closing in a little bit, and I definitely felt that on my six-month flight. AG : Under normal circumstances, conventional wisdom is to be in the present, live in the moment, seize the day. But in **isolation**, sometimes the day sucks, especially in outer space. SK : I also missed nature, going outside, sun, rain, wind, fresh air. People become depressed without that sun. AG : I ' m imagining it ' s something like wake up in the morning and then wait for the Earth to rise? SK : Yeah, so – Yeah,"earthrise."Never heard it that way, put it that way, but yeah, that ' s a good way to put it. AG : For Scott, every day started feeling like"Groundhog Day."SK : I would generally wake up at around 6:15, have breakfast, work for a few hours, take time for lunch, schedule time for exercise, schedule time to connect with people. AG : It sounds a lot like what many of us are experiencing now. Time becomes meaningless. You ' ve probably already heard the most popular tips for coping with that :"Make a **daily** schedule.""Wear different clothes for different parts of the day.""Have Zoom calls with family and friends."But one of the lesser-known ways to give this time meaning is to master the art of time travel. More specifically, mental time travel. In psychology, it ' s a skill. Rewinding to think about the past and **fast-forwarding** to imagine the future. It ' s a distinctively human skill, and if we use it thoughtfully, it gives us the capacity to find meaning in the mundane and **happiness** in the midst of sadness, and make time pass faster or slower at will. The first destination for your mental time machine is the future. That ' s where Scott started. Months before he set foot in the rocket to leave Earth, Scott started imagining how he wanted his trip to end. He didn ' t just focus on what he wanted to accomplish. He imagined how he wanted to feel when his **mission** was wrapping up. SK : My goal was to get to the end of this with the same enthusiasm and ability and energy as I had in the beginning, and I always wanted to convince myself that this is just my life. I ' m here for a purpose, and my purpose was to be an astronaut and complete our **mission** objectives. AG : It helped Scott stay focused, and made the days pass faster. Psychologists find that a mental trip to the future can help us think less about the monotonous how of our days and more on the meaningful why. We ' re able to get out of the dull weeds of the process and shift our attention to an exciting purpose. The farther ahead we look, the easier it becomes to tell a coherent story about our experience. One of the challenges of the current pandemic is that we don ' t know when our **mission** will end, but we do know it isn ' t endless. It will end. So think about how you want to feel on the day this crisis is over. How will you want to have gone through it? You might still find yourself worrying about what will happen in between. It ' s inevitable. But here ' s the thing : worrying isn ' t necessarily a bad thing. SK : One of the things that NASA does very well is, and how we operate in space is, we ' re always having to think about,"OK, what is the next worst thing that can happen to us right now? And how do we react and respond to that?"You know, in some ways that can cause **stress** and anxiety. AG : There ' s a distinction that we often make in psychology between worrying and rumination, where worrying is basically an attempt at problem-solving and trying to figure out,"OK, what could go wrong? What ' s that next worst thing? How do I avoid it?"And rumination is when you get stuck in a loop, just worrying about the situation, but not trying to act in the face of it. Did you have strategies for making sure that the stuff you were worried about didn ' t cause you to ruminate? SK : Oh yeah, I think so, and I think that ' s part of our nature as astronauts is to understand that bad things can happen. We don ' t have control over them, but we still have to be prepared for them. AG : Some psychologists recommend a simple strategy for preventing rumination : adding worry time to your **daily** schedule. SK : Schedule time for your rumination, 15 minutes, maybe. Doing that will make the days go by so much quicker rather than just sitting there thinking about what you ' re going to do next and bingeing on Netflix all day. AG : When I went through this exercise, I started thinking about silver linings, aspects of being fully remote that I ' ll miss, like having more family time and flex time, not having to travel or commute, not having to change out of pajamas or at least pajama bottoms. Psychologists find that imagining these silver linings fading in the future makes them feel scarce. That helps us appreciate them more in the present. Next, you can direct your mental DeLorean to the past. This is something Scott has been doing during the current crisis : thinking about memories from his days in space. SK : When people ask me,"Well, what do you miss about space?"It ' s not launching in the rocket. It ' s not doing a space walk. It ' s not floating around. It ' s not the view. It is the people I was there with and also the sense of purpose. AG : What I heard there was nostalgia. The Greek roots of nostalgia translate to"return"and"pain."The literal definition is : the pain of being unable to return to the past. But here ' s the strange thing : that ' s not how it feels. After people are randomly assigned to reflect on a nostalgic event in their lives, they end up feeling happier. They have a stronger sense of connection to others and become more willing to give and seek help. They gain a deeper sense of meaning and purpose in life. Think about a nostalgic event in your life, something you had before the pandemic but no longer do, like celebrating a holiday with your extended family, going on **vacation**, even going to a restaurant. Reminiscing about those moments might seem bittersweet, but the aftertaste is mostly sweet. It motivates you to make the most of the present. But it doesn ' t just help to remember **pleasant** memories from the past. There are also benefits to reflecting on painful ones, too. SK : It was the worst day of my 520 days in space by far and probably one of the worst days of my life was January 8, 2011. I was on the space station about halfway through my six-month **mission**, and I get a call from the control center in Houston, and then they say,"Scott, I don ' t know how to tell you this, so I ' m just going to tell you. Your sister-in-law Gabby was shot in Tucson, Arizona."And I quickly got on the phone with my brother. He was packing his bags to head to Tucson. You know, I ' m on the space station now, and I can ' t be there physically to be supportive to my family. Couple hours later, the CAPCOM called up on the privatized channel, and they told me Gabby had passed away because that ' s what the news was reporting. Later, I found out she was still alive, thankfully. AG : That experience taught Scott a powerful lesson. SK : Oh absolutely. I mean, I always realize things could be worse. AG : This brings us to a third destination to transport yourself to alternative time lines, things that could have happened but didn ' t. It ' s called"counterfactual thinking."Psychologists have shown that imagining how things could be worse helps us find gratitude. There ' s evidence, for example, that people who graduate from college during a recession end up being happier with their jobs a decade later. They can easily imagine a world in which they ’ re unemployed, so they don ’ t take it for granted that they have jobs. Counterfactual thinking has come in handy for Scott over the past month. As bad as the **coronavirus** situation is, he ’ s been reflecting on how things could be worse. SK : You know, if given the choice of spending a year in my apartment or a year in space, the apartment wins hands-down every time. AG : Even if you ’ ve never been to space, you know the current circumstances could be a lot worse. My wife and I have been talking with our kids about how lucky we are that grocery stores are still open and utilities are still on. We feel deep appreciation for the people who are hard at work keeping those **services** going. I ’ ve been thinking about how much harder it would be to do my job without the internet or a phone. Recording this podcast would be pretty much impossible. As it does with many astronauts, being in space broadened Scott ’ s idea of who his neighbors were. SK : When we ’ re in space, and you look down at planet Earth, the planet is incredibly beautiful. During the daytime, you don ’ t see political borders, and it makes it look like humanity is all part of one big team, and when you ’ re part of a team, you need to work together to solve our problems. And I think this virus has shown us that we ’ re really, for better or worse, more interconnected than I really realized. If you consider the last pandemic, which was like 100 years ago, and if you consider what we have been able to accomplish as a species in a hundred years, it ’ s mind-blowing. I am absolutely confident that we will get through this, but it ’ s going to take all of us working together, making the right decisions, doing the right things. Teamwork is absolutely critical when you ’ re trying to do something like this. We ’ re really fighting, right now, the greatest battle for our lives that we ’ ve ever had to fight.”WorkLife”is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell, Jessica Glazer, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. This episode was produced by Constanza Gallardo. Our show is mixed by Rick Kwan. Original music by Hahnsdale Hsu and Allison Leyton-Brown. **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors, Accenture, BetterUp, Hilton and **SAP**. For the conference call in real life, thanks to Tripp Crosby. For the snippets of what it ’ s like to work from home, thanks to Jason Gates, Ryan Smith, Amanda Monday, Danielle Tussing, Erin Kahana, Tiffany Chang, John Mi, Aaron Guo, Next Plumbing and everyone else who called in to share their stories with us. For their research, thanks to Bob Ginnett on airline crews, Cameron Klein and colleagues on team building, Ashley Hardin on personal knowledge, Jeremy Bailenson on giant virtual heads, Yaacov Trope and Nira Liberman on mental time travel, Kate Sweeny and Michael Dooley on worrying versus rumination, Constantine Sedikides and colleagues on nostalgia, Laura Kray and colleagues on counterfactual thinking, Emily Bianchi on starting your career in a recession, and my late advisor, Richard Hackman, on teams as amplifiers. For all their help this season, we thank Nicole Bode, Valentina Bojanini, Sammy Case, Micah Eames, Will Hennessy, Dian Lofton, Jen Michalski, Sarah Jane Souther and Emma Taubner. That ’ s a wrap for season three of”WorkLife.”Thank you so much for joining us. If you like what you heard, please rate and review the show. It helps people find us. And are you still a potato? LO : I ’ m sorry – am I still a potato? AG : Yeah, I mean, you ’ re potato boss now. You have to be a potato, right? LO : The funny thing is, the first time I got the request to be a potato, I couldn ’ t figure out how to be a potato. It took me – it was like the opposite problem. It took me hours to get the potato back up and running again.

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David Fano : Well I would say there was kind of like three phases. There was the time when you would tell people,"Hey, I work at WeWork"and nobody would know what that was. And then very quickly it became pretty commonplace. People knew what WeWork was, and it was on the cover of magazines, it had a great **mission**, it had this rebranding, it had these, like, very grandiose ambitions. And then from one day to the next, it flipped completely. And I think that was really, really intense for people to process. That so much of their identity and who they were was somewhat scaffolded by the identity of the place that they worked, that now it became like an existential discussion. And then you ' ve got your grandparents who didn ' t even know where you worked are calling you now being, like,"Hey, that ' s the place you work?"And you almost go to, like, a place of shame. Adam Grant : I ' m Adam Grant, and this is a **special** bonus episode of WorkLife, my podcast with TED, sponsored by Accenture, BetterUp, Hilton and **SAP**. I ' m an **organizational** psychologist. I study how to make work not suck. In the past year, one of the biggest work stories, and just one of the bigger stories, period, has been the implosion of WeWork. The company started as a single coworking space in Soho, and in just a few years, WeWork exploded to more than 100 cities in dozens of countries, employing more than 12, 000 people. They weren ' t just selling leases, they were selling a new vision for work. And Dave Fano was on board. DF : For me it meant making going to work awesome for people. You know, with an initial focus on the physical environment, and making that environment comfortable and exciting and feeling good, and then also the technological side of how we brought people together, how we could get people to commune and work together, small companies helping each other I thought was really exciting, and helping people do work that mattered to them, and being able to strip away the muck, the things that matter a little less. AG : For many business owners, part of that muck was the perceived hassle of renting office space, and then creating a welcoming, energizing environment. That ' s where Dave came in. WeWork recruited him to design all of their spaces and become one of their top executives. It was a job for someone with both technical and creative skills. DF : I was trained as an architect. AG : Were you aspiring to be someone like Frank Lloyd Wright, or did you have a different vision in mind? DF : Sadly I really, probably didn ' t know who many of those people were. I was really determined to be a comic book artist. So you could have asked me any comic book artist, and I would have known them all. I was a kid that drew the cover to the yearbooks and high school T-shirts and things and wanted to be an artist, and my dad told me that wasn ' t a real job, so architecture seemed to fall somewhere in the middle. AG : He ended up starting a consulting business to help architects, engineers, contractors and building owners improve their work spaces. Soon WeWork was a client. In 2015, WeWork acquired Dave ' s company. He became WeWork ' s chief product officer, and then chief growth officer. He oversaw all of real estate, design, construction, sales and marketing. In the spring of 2019 he left. Six months later, after WeWork filed for an IPO, a series of investigative reports exposed their huge **financial** losses. The media was in an uproar over the self-dealing behavior of the cofounder and CEO, Adam Neumann. Woman 1 : Adam Neumann pays a direct family member to be the director of wellness at the company. Man : There ' s credible reports in the business press that he traveled with illegal drugs across international boundaries. Woman 2 : Expenses at WeWork more than quadrupled. Man 2 : Adam Neumann is stepping down as CEO. AG : Plenty of people are still dissecting what happened on the business side of WeWork. I wanted to explore something different, the leadership and culture. We know visions and values have a real impact on whether a company succeed or fail, and WeWork was an extreme example of doing both. What could we learn from WeWork ' s rise and demise about how culture is built and sustained? When is a strong culture actually helpful? And when do you trust your sense that something is off? I can ' t think of a better person to ask than Dave. He had a front **row** seat to the company ' s evolution, expansion and **peak**. This is his first time speaking publicly about WeWork. DF : I think we tried to do a lot as a company. you know, what ' s the saying, **start-ups** don ' t die of starvation, they die of indigestion. I think we did a lot, I think we tried to do a lot of things, which probably led to over-committing and not following through with things at times. But I don ' t think that there was ill intent. AG : I want to dig more into that, but first let me tell you actually about my first introduction to WeWork. So summer 2017, I had been talking to the TED team about starting this WorkLife podcast, and I mentioned it to a few friends of mine and a couple of colleagues, and one of them said,"Oh, you ' ve got to talk to WeWork, because they ' re all about trying to make work better."And before I knew it, I was being introduced to Adam, and I reached out to a colleague I knew had worked with the company a bunch, and she said,"Listen, they are obviously growing really fast, they ' re accomplishing a lot, but if I were you I wouldn ' t trust them because they ' re gangsters."I didn ' t know what that meant at first, I ' m curious about whether you heard that description and what you took from it? DF : Actually it ' s the first time I ' ve heard the company described that way. I don ' t feel like that was the case, I think actually more often than not we did the right thing, we tried to do right by vendors, we tried to do right by the people. When you get bigger those things get complicated and there ' s always two sides to every story. AG : There are. Well, I certainly want to hear more of your side. So let ' s begin with, what was the **mission** when you joined? DF : To create a world where people make a life and not just a living. Or enable a world, where, you now, there ' s – the main punchline being that you create a world where people make a life and not just a living. AG : That is both incredibly lofty and extraordinarily vague. What did it mean to you? DF : I always really connected with the aspects of work. Not as much the sort of, non-work side, however we define that. So I thought that was very exciting, you know, if I ' m an embroidery designer sending invoices and billing is not so exciting, or signing a lease, so I was really excited about the idea of stripping away all the stuff that was not their core activity, and I was pretty charged up with that and that ' s how I **interpreted** the **mission**. AG : Got it. That could almost be the slogan for this podcast that I host. So that, if you frame the **mission** that way, I could even imagine myself getting excited about it. I ' m curious about how the **mission** got communicated. What were the major ways that the **mission** was brought to life? DF : I think it changed over time, I think very early on, it was about the entrepreneur, the small business, the founder, and enabling them to do what they love and start the companies that mattered to them, and all these companies that were going to go change the world. I think as the company got a little bigger and there was excitement around the reach of the idea, and building more of a lifestyle brand, it started to become a little more consumer centric and tried to speak to a broader audience, and then you saw the evolution of the company trying to do more things. We live, we grow, and then it goes really about serving everybody and enabling everyone to, you know, live their best lives. It became incredibly broad, but I think a lot of people suited up every day to go do that, I think people jumped out of bed excited to try to help people. It was a powerful **mission**. And I think as a **workforce** and as a team, people got really excited about it, and it motivated them and the closer proximity you had to the customers and the more you can see that you are affecting these people and helping them day in and day out, whether they were the owner of the company or an employee of the company, people were really charged up to help them. AG : Is there a story, Dave, that illustrates a moment when you really felt fired up by the **mission**, wondering was there a company retreat where a speech resonated with you, or, you know, just any defining experience where it clicked and you said,"Yeah, this is why I ' m here."DF : There was a story that the community manager told one day of someone who was working at one of the companies in the building and he had become really good friends with them and they ' d see them every day and he came, I think, crying to the main community desk and he had lost his job. And the community team mobilized incredibly quickly, started to talk to all the other members in the building, and I think within 48 hours they found him a new job. And I thought that was amazing, that this person, you know, we associate so much of our identity and our being to our work, and the work we do and the company that we do it at, and especially I like to think of it through the lens of the employee, because I think the founders and CEOs have a bit more agency in those things, and that wasn ' t the person paying the bills, this was someone who worked in the building and that the community team took such good care of this person, and they didn ' t need to, again, this person wasn ' t, like, the direct customer. And that by being part of that community, that person was taken care of. That really meant a lot to me, and I thought that that was really beautiful. A group of people that could be looking out for each other. And this thing is mostly individualistic, managing my career, looking out for myself, making my money. For me that really highlighted the benefits of what we were doing. And I got even more excited to figure out how to scale that out to more people. AG : It sounds like you were far from the only one to feel fired up about the **mission**. What would you say was the dominant emotion that people felt at WeWork? DF : Excited. High growth is pretty exciting. You hit those milestones, you hit those achievements that you didn ' t even think were possible. It ' s a combination of excited and exhausted. And I think people – I think **struggle** is kind of important to really love your work. I think you may not realize it at the time, but I think it ' s one of those things in retrospect, when you think about some of the most fulfilling times at work, they involve some amount of **struggle**, and I think people felt really fulfilled by their work. Again, I think there was plenty of **peaks** of **exhaustion**, but I think they were getting to do really good work with really good people. And I think they felt that sense of community and excitement. AG : The juxtaposition of excitement and **exhaustion** is fascinating to me, it reminds me of some research on what ' s called the passion tax, where Aaron Kay and his colleagues have found that people who love their jobs are more likely to get asked to do extra work unpaid, even if it ' s not part of their role, even if it ' s demeaning. And they end up sacrificing sleep, family time and health because of it. Do you feel like there was a passion tax imposed at WeWork? DF : I think people loved what they did. And I think they worked hard. You know, I think if you asked most people that worked there they ' ll tell you about how amazing the team was, and that it was some of the most incredible people they ever got to work with. And I really think people liked being at work. And yeah, so I think they may have put a little bit more of themselves out there that led to a little bit more of that draining and **exhaustion**, but I think people were pretty stoked for the work they were doing and the people they were doing it with. AG : So it sounds like then you don ' t feel that they were getting exploited, so much as they were making voluntary sacrifices out of passion? DF : I don ' t think so. Look, that ' s a tricky one, because I had a very particular view of the business, given my scope and my domain. There were a lot of different roles. I didn ’ t work in the buildings. Most of the folks that I lead and managed and interacted with on a regular basis were within the office, and they had a bit more control of, sort of, managing their time. I could imagine if you were working in the buildings and you had a bit more of a direct obligation to the members and their needs that it may not have felt like you had as much agency, so I can ’ t speak to that as much. AG : When I sat down with Adam, I was asking him what WeWork was all about, what are his goals, what ’ s he hoping to accomplish, and he said,”We ’ re selling culture.”And you know, my reaction, as an **organizational** psychologist was,”I ’ m pretty sure you ’ re leasing buildings.”It seemed like there was a big, you know, a little bit of a leap from what the actual work was to the **mission** that it was serving, and of course, sometimes those leaps are aspirational. Sometimes they ’ re part of creating a meaningful and compelling vision. But the disconnect felt a little bit more significant to me than I see in a lot of **workplaces**. What ’ s your take on that? DF : I think we were selling cultural intervention. I think a culture is ultimately the aggregates of the micro-actions that an organization takes, and we couldn ’ t entirely control those. But I think we were trying to help shape culture. The big learning for me was that space and the physical environment helped, but at the end of the day it was the core and how the organization behaved that would ultimately shape the culture. And companies who are looking to try to bring cultural lightning rods to the organization would bring a whole group into a WeWork, because it was so different from whatever their corporate environment was. So we would be used as a bit of a lever for cultural change. So I think we were helping inform culture. I think companies were looking to us to think about culture in a different way. I would say the unfortunate part is I think they gave too much emphasis to the space itself. And that, you know, what one of my colleagues at WeWork at the time called cultural confetti, and much more of the appliqué of culture than, like, the deep internals of culture. So I think we were trying to have those conversations and it was a tricky thing to **navigate**, understanding that our product in the market was space, and flexibility in space, while still trying to have a meaningful impact on the employee experience. AG : Cultural confetti, I think in that sense, it sounds like just a surface layer, as opposed to dealing with the underlying values and **norms** that define what a culture is. But the other image that confetti paints for me, at least in my mind, is it almost sounds like it ’ s a weak culture, that it ’ s kind of dispersed and there ’ s not anything consistent or strong about it. And that to me is a big part of the difference between weak and strong cultures. So in my world we normally define a weak culture as existing when you go around to different people and you ask what the values are and they all name different values, and nobody really cares about them. In a strong culture, the values are widely shared, so everybody believes the values are the same. And they ’ re intensely held, so people are extremely **passionate** about them. My sense was that WeWork had a very strong culture. First question is do you agree or disagree, and then secondly, how would you describe the culture? DF : I think WeWork had an incredibly strong culture. I think people felt very connected to the **mission**, I think they felt very connected to the values. So in that sense the culture was incredibly strong. It was about relentlessness, entrepreneurialism, respect, tenaciousness, togetherness and community. It was really about achieving a lot as a collective, I think that was kind of the essence of the values, and bringing an entrepreneurial approach to everything we did. And I think the organization, you know, for the most part operated that way. I think people felt very emboldened to do what they thought was right for the customer. If you were the community manager of a building and it wasn ’ t part of the standard operating procedures to have, you know, nondairy milk, but you had a member who needed it, they would go do it. That ’ s kind of a silly little example, but there are places where you could find yourself caught in bureaucracy for weeks to be able to do something like that. And they felt, you know, I think, a tremendous amount of agency and **ownership** around – And then again, everyone felt this way to do the things you needed to do to move the business forward. At least what we thought was moving the business forward at the time. AG : That goes to a lot of the research on strong cultures, which, there ’ s a long list of benefits. We know that when organizations have strong cultures they ’ re better at attracting and retaining people, because they stand for something unique and it draws talented employees, and once they ’ re there, why would I want to go anywhere else? It ’ s also helpful for motivating and connecting and coordinating people, because they feel like they ’ re in a place that has distinctive values, and they start to identify with those values. I worry a lot about the dark side of strong cultures though, which is that they tend to breed homogeneity, that can cause groupthink, in a lot of cases they have trouble changing and adapting because people are so committed to the way we have always done things or the values we ’ ve always held. Did you see any of those sharper edges? DF : Yeah, I think that feels right, you know, from having lived inside a strong culture, is that some things start to just happen because. I ’ m not sure if you ’ re familiar with the pot roast story, but it ’ s one of these work-fables of just doing things. AG : The what? DF : The pot roast story? AG : Not described that way. DF : OK, I ’ ll give it real quick, it ’ s a parent and a child cooking a pot roast, and the first thing the parent does when they take the pot roast out is they cut the ends off. And the child says,”Why do you do that?”They say,”Well it ’ s in the recipe. So let ’ s call a grandparent and see.”They call the grandparent and they say,”I ’ ve just always done it, that ’ s what ’ s in the recipe.”And the family was very lucky that the great grandparent was still around, so they called the great grandparent that wrote the recipe. And they say,”Oh well my pan was just too small so I always cut the ends off.”So the idea being that we just do things because we thought that ’ s what was right. And when you have these strong cultures with a bit of a lore about them, it becomes incredibly difficult to trace back like, why certain things are the way they are and the strong culture just keeps them going. So I think that that ’ s absolutely a side effect, a potentially negative side effect of a really strong culture, especially as you scale. Those things might be fine if you stay at the same scale for a very long time, but when every three to six months the business is fundamentally different and the scale and infrastructure is different, those things can start to be of an issue, because you start to have these cohorts of culture, especially when you ’ re onboarding a lot of people, you ’ ve got the folks that have been there for five years, five and a half years, four years, and they ’ ve got the snapshot of what the most current version was at that time. And it becomes, like, cemented in, and so you find yourself with people debating what the right way is, not even, again, knowing what the source of that thing was. And so I think the combination of a strong culture, high growth and high speed make for some really interesting times. AG : What ’ s a WeWork example of the pot roast? DF : This foggy vinyl on the glass. You know, a lot of members would want **privacy**, and they would want to put some translucency on the glass in the buildings so that their office would have a little bit more **privacy**. Like, if people want **privacy**, no problem, we just put up the vinyl, and so you would just see different things, where it ’ s like,”Oh, that ’ s not allowed at all,”which was not true. That was one that I always found myself being pulled into a meeting, like, we ’ re not allowed to do that, and I ’ d be like,”You ’ re absolutely allowed to do that, that ’ s what the member wants, give it to them, it ’ s not a big deal.”AG : This definitely captures the double-edged sword of strong cultures in a compelling way. I think about one of the benefits of strong cultures being that leaders don ’ t have to use incentives or controls or punishments because people have internalized what the values and **norms** are, and that means if they see something that is not being done right, or a rule that ’ s being broken, or a principle that ’ s being violated, they will enforce it themselves. And that ’ s the upside. The downside is, people will enforce it themselves, and they might not be enforcing the right principles. DF : Yeah, I found myself in just way too many meetings being the person that was willing and seeming to the others empowered enough to change the **norm**. But they would just be spinning and spinning, it ’ s like,”We can ’ t, but we should, but we can ’ t.”Where did that come from? I was like,”Guys, it ’ s totally OK, just do it, if it ’ s not OK I ’ ll take the heat.”And it was always OK. And I feel like, when things got stuck, that ’ s a lot of the stuff that was happening, it was these cultural **norms**, and you couldn ’ t even ask somebody,”Hey, where did you see this written?”And more often than not you couldn ’ t find it. AG : I want to ask you a little bit about leadership, which obviously is a core driver of culture. Adam seemed to do a lot of preaching about what WeWork was going to accomplish, what the values were, what the **mission** was. Do you think he believed all of what he was saying, or was he hoping other people would drink the Kool-Aid? DF : I think he believed it. You know, I like to believe he believed it. I know that the team believed it. And that as a **workforce**, and as a team, and as a culture we believed it. The question I ’ m kind of curious about, and curious on your take on is like, how much does a belief system matter to a culture, and how important is it that the leader can either embody or present the belief system in a way that can get people behind it? So I think he believed it and I definitely believe he got people to believe it. I ’ m kind of curious for myself on how much that mattered for the incredible things that we were able to achieve. I think they mattered a lot, but I ’ m curious from your surveying of companies and founders and CEOs, if you ’ ve seen that as a common trait? AG : Yeah, I think so. There ’ s evidence that charismatic leaders are more successful at pitching their companies and fundraising. It ’ s not surprising, I think, to anyone that that would be true. There is also evidence that the cultural blueprints that founders have in their minds when they first launched their companies, they cast a real shadow on the future of the company, even if founders rotate out of leadership. There ’ s a study of a couple a hundred Silicon Valley **start-ups** where Jim Baron and his colleagues identified the cultural blueprints, and they found that if you, as a founder, had a blueprint where you were going to convince people that you had an important **mission** and everyone was going to be **passionate** about that **mission** and the culture was organized around commitment, that your **start-up** was less likely to fail, and it was also significantly more likely to have an IPO. But then post-IPO, those **start-ups** that were strong cultures of commitment, they grew at slower rates, if you track metrics like annual market capitalization. And I think that ’ s where you start to see some of the rigidity and groupthink problems come in, where you begin to weed out diversity of thought, you end up with more homogeneity. And I guess where that ’ s left me is wondering whether the kind of charisma and conviction that you are describing in Adam is helpful for a company to grow at an exorbitant rate early, and then over time, sort of, shoots the company in the foot? DF : Well, this will definitely make for an interesting case study, because we ’ ve kind of got the full cycle. Look, I really – Coming back to the question on belief, I believe he believed it. I think he had other things that drove that belief system and his ambitions of things he wanted to achieve, which, honestly, he achieved a lot of them. And so I ’ m not sure like, sort of, chicken and egg – which one was **driving** which, at a certain point the belief can only push you so far. And so maybe that ' s what we felt there at the end. AG : Yeah, I think that ' s fair. It reminds me a little bit of some research by Sandra Cha and Amy Edmondson, who studied values-driven organizations and found that the more emphasis leaders put on values, the more risk they end up taking of being accused of hypocrisy. Where you get people convinced that you stand for higher principles and moral ideals, and then the moment that you deviate from those, there ' s a giant gap and people are very quick to point out that you are not **practicing** what you preach. I ' m curious to hear your perspective on that and do you feel like some of what happened at WeWork was hypocrisy because the values and aspirations were preached in such an aspirational way? Or do you have a different analysis of where it started to unfold? DF : I definitely think that that is the source of a lot of the disappointment. That what was presented as what the organization stood for, as more and more commercial pressure mounted, you know, being public and things like that, some of those things were harder to live. I don ' t think there ' s necessarily a one to one. I am not sure a company can be altruistic, not that we were pretending to be altruistic. And so I could imagine, that since the company had really built an existence of caring about people and the community so much, that if you did anything that looked anything like not that, that you run the risk. And I think these mission-led companies and companies with great aspirations, that is the challenge, I think it ' s part of what gets your customers to buy into what you do, it ' s part of what get your team to buy into what you do and to grow. But you set the bar for yourself incredibly high. And I think some companies have been, like, very clear and pragmatic about what they do, and are able to live that. I think Amazon does that pretty well. I think that ' s tricky. You know, being able to manage this aspirational, very human-centric brand, you know, you ' ve got to really operate tightly and I think the world is becoming less tolerant of a deviation from your **actions** and your words. And so I think that ' s something companies are going to have to be very thoughtful about as they establish these brands and what they stand for and their values and they better be prepared to really live by them. AG : Yeah, I guess that goes to the question of narcissistic leadership. And I know you know I ' ve spent a lot of my career studying the dynamics of giving and taking, and in the IPO there ' s just an unusual number of”Is”and”mes”, and we know from evidence that narcissistic taker CEOs are more likely to use”I”and”me”when talking about their companies. There ' s one article calling Adam the most talented grifter of our time. I ' m just curious to hear your reaction to that set of observations. DF : I have described him before as, like, the best personal trainer you never had. He could get people to push themselves to deliver and to achieve in ways that they didn ' t think they could, which I think is – I think for a lot of us that ' s kind of uncomfortable, we actually feel bad. I think he was able to do it. And I think he really believed he was helping people get the most out of their potential. He was a force, or is a force. AG : When I think about a personal trainer, my first reaction is, OK, this is somebody who is going to pump me up. My second thought is, when I bring in a personal trainer, it ' s somebody who ' s trying to help me achieve my own goals. So I ' m trying to get in shape or get faster or stronger, or lose weight, whatever the target may be, that person is there to help me. And I wonder if that analogy breaks down when you start to think of a founder and CEO who may not be there to try to help the people around him? DF : Yeah, that ' s a really interesting callout. I think, you know, for companies in general, and this starts to tug at the relationship between the folks that work at a company and the company itself. And what is really best for the individual, so I can kind of make the case that what ' s good for the collective is good for the individual. And I think that a lot of people convince themselves of that. AG

: You know, I ’ m not asking you to criticize a particular leader, but when you think about the culture it seems like there are aspects of the culture that fit more with taking than giving and that there was some selfish as opposed to servant leadership at the top. DF : Yeah look, I think he was superfocused on delivering what he set out to do. I think he worked harder than probably most people I ’ ve met. He didn ’ t leave the office, he worked all the time, and so I think that there was this maniacal focus on winning. You know, what winning means is different to different people, but I think he wanted to do incredibly big things, I think the definition of”big”changed for him over time. I think we would set out to do these fairly audacious things, we would achieve them, and then it would open up a whole new world of what was possible, you know, be it fundraising numbers or opening X number of buildings in a month, or selling so many seats to new members on a certain month. We would consistently set out fairly audacious goals and hit them. AG : Stretch goals. Sometimes those kinds of moonshots are just the fuel we need. There ’ s a wealth of evidence that extremely difficult specific goals motivate us to aim higher. And work harder, longer and smarter. But they can also lead us to cut corners, especially if we ’ re about to fall just short of the goal. Sometimes our creative strategies for hitting **bold** targets stretch beyond the bounds of what ’ s ethical. DF : So I think he was increasingly emboldened by what we could do as a company, and when we really focused on trying to tackle something, and so, yeah, I think he wanted to win, I think he wanted to win everything. AG : I ’ ve always bristled at the language of winning and losing in business, because it ’ s not like a soccer match, or a **baseball** game where there ’ s a **score** and everybody has agreed on the rules and at the end of the game you know what metric you are counting on. It seems like the moment you define success in terms of winning, it means that everybody else has to lose. DF : Yeah, I totally agree with you, and I think it ’ s a complicated topic, but you see it in, like, the business discourse of,”We ’ ve got to win, we ’ re going to win.”And you even see recommendations that I also always bristled at, it ’ s like, we need to identify an enemy, or an opponent. People will rally around that. But I agree with you, I much more subscribe to the idea of, don ’ t worry about who the competitors are, get customer-obsessed and just focus on solving for the customer. And the notion of what winning is and competitiveness, you know, maybe that gets into personality, but I think you see it, I think there are certain founders who exude a bit more of a competitiveness in the way that they – it ’ s a bit more of a combative approach to business, and they ’ re able to rally people behind it and get people to do amazing things, and there ’ s others that maybe that ’ s not as much a part of the way that they engage and that they talk about it. But it was definitely part of our language. I think people are very capable of convincing themselves that **actions** that most would recognize as taking are in the **service** of giving. And I think it ’ s a really fine line and slippery slope. But I would say that that ’ s what was believed. That the things that were being done were best for the collective, and kind of the rising tide lifts all ships, and they can be a little bit more conscious of the language. It still shows up as the”Is”and”mes,”but I think it ’ s – it ’ s easy to see in retrospect I guess. AG : Yeah. There ’ s a spectrum of self-awareness when it comes to the most selfish takers among us. And on one end you have people for whom it ’ s a very deliberate strategy, right? They ’ ll say,”Look, I believe the world is a dog-eat-dog competitive place, and I have to put myself first because no one else will, and that ’ s how I get ahead, that ’ s how I win.”And on the other end of the spectrum, you have people who, it ’ s almost like they ’ re in the movie”The Sixth Sense.”Everybody else knows they ’ re a taker, but they have no idea. And I think the Richard Feynman quote always stands out here, that,”You have to be careful not to **fool** yourself, and you are the easiest person to **fool**."DF : Yeah, that makes sense, and I do have a belief that we have a more natural disposition towards a self or other orientation. And then you sort of, combine that with people ' s EQ. And so if you have a high self-orientation and a high EQ, that makes you kind of dangerous. And I think some people are, I don ' t know if the right word is gifted, but it ' s a gift that some people can use for good, some people can use for bad, but some people are clearly capable. And sort of – I ' m just saying through history, we ' ve seen these very magnanimous characters and figures that have done remarkable things, and then some that seemed like they were doing remarkable things until they weren ' t. And unfortunately, many of us are susceptible to it. AG : Yeah. Well, I think this is one of the ways that emotional intelligence gets weaponized. There are a lot of people who associate care and **empathy** with emotional intelligence, but by definition, emotional intelligence is a set of abilities, not a set of motivations or values. And that means that how emotionally intelligent you are, how good you are at recognizing and using and managing emotions in yourself and others, that ' s completely independent of whether you are more of a giver or a taker. And sure enough there is some research by Stéphane Côté and his colleagues, showing that the most dangerous colleagues are emotionally intelligent takers, because they ' re very good at manipulating you for their own gain, but convincing you that it ' s for your benefit, and maybe to your point they ' ve even convinced themselves it is for your benefit. DF : Yeah, it ' s really easy for folks of that profile to fake **empathy**. Because they can intellectually understand, but I don ' t think they really feel it. AG : Yeah, so that would be, well in some language cognitive **empathy**, but not emotional **empathy**. DF : Right. AG : Very interesting. They can take your perspective, they know what you ' re feeling, and that makes it very easy to get you to do what they want you to do. DF : Yeah. AG : Let me ask you this then, when this all came crashing down during the IPO, what was your emotional reaction? DF : I was really sad. I was sad, because a lot of my friends and people I really care deeply about and have a ton of respect for were very directly affected by it. I was sad, because a lot of what the company stood for and the way that it treated its customers and members were still very much true. None of the media was about that. Like the product was good. The achievements that we were able to accomplish were amazing, and people did really incredible work and people went from one day being very proud and excited to say they worked at the company when they were at a party, to the next day maybe not so much. And I think that was really sad, because I think so much of our personal identity and our personal brand, which ultimately shapes our career, is connected to where we work, and I think a lot of people had made a huge investment in that connection. And for me that was really sad that without their doing that just changed so drastically for them. Like, by association, you just get painted with that. And I think that ' s what was really sad for me, is that the sequence of events that really had very little to do with the people working there and the effort and energy that they put into building something amazing, now affected them all. AG : That ' s such an – well I don ' t want to call it interesting because it ' s painful, but from a psychological perspective, it ' s a challenging situation. Because in a way it feels like somebody ' s house just burned down, only, number one, you don ' t have any insurance when it comes to your company having gone up in flames. Number two, you can ' t just move to a new house without talking about the old one. You have to account for where you spent those three or four years, and so it ' s almost like you are forced to wear a badge with a bunch of stigma attached to it. DF : Yeah, absolutely. You know, if you were in the middle of a job hunt in that process you went from being highly coveted one day to needing to explain why that happened the next, and were you involved?

You almost had to defend your career decisions in this way that was completely **unexpected**, and you had zero input or effect on that being the case. So I think that was really tricky, and that is part of why, I think, career decisions and job decisions are so heavy for people, because there is this aspect of the unknown that ’ s out of your control. You don ’ t know how well the company is going to do. You don ’ t really know how much money it has, you don ’ t really know the things that the executives are talking about. And I think that there is a tremendous amount of trust that people put into organizations, in that they ’ re going to take care of them and take care of their career. AG : What advice do you have, Dave, for people who go through that kind of stigmatization experience? DF : Focus on what you learned, and the people that you met. Because what no one can take from you is your knowledge and your relationships. And if you invest in those, and you manage those correctly, you ’ ll be able to cash them in later. So whether the brand becomes tarnished or the company doesn ’ t do well, if you ’ ve built person-to-person relationship rather than person-to-brand relationship, then you ’ ll be able to leverage that later. And not in, like, a capitalistic, opportunistic way, like, work together, because those things transcend companies. Whether it was a truly work relationship, perform, do well. I think too many people really feel like the work that they ’ re doing is, like, for the company.”I worked for the company.”No, I think what most people need to recognize is that you work for yourself, and your best investment is you, and what you leave behind is the reputation of the work you did while you were there. So you might be angry and you might be sort of, mailing it in, but people aren ’ t going to give you the benefit of the doubt that you were mailing it in because the company wasn ’ t doing great, they ’ re just going to remember that you mailed it in. And so I think always remember that you are managing your identity and your reputation and that travels with you. And so when things get bad, do right by people. Even if that ’ s not what the company is doing, stand your ground for the things that matter to you, because that will travel with you. AG : As WeWork ’ s IPO was collapsing last fall and several thousand people were being laid off, Dave stood his ground for something that mattered to him. He **tweeted** that he was willing to serve as a personal reference for anyone at the company. The post-mortem on WeWork has only just begun, but my biggest takeaway is that we should all be a little skeptical when companies claim to be changing the world. Even if their intentions are good, sometimes they ’ ll fall short of their lofty ambitions. But the bigger danger is that they ’ ll reach those ambitions. They may well end up changing the world but that doesn ’ t mean they ’ ll change it for the better. Next time on WorkLife. Arthur Brooks : It was a wreck, it was a slow moving wreck in my career itself. I was in decline. AG : Is career decline inevitable? WorkLife is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Jessica Glazer, Constanza Gallardo, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. Our show is mixed by Rick Kwan. Original music by Hahnsdale Hsu and Allison Leyton-Brown. **Ad** stories produced by **Pineapple** Street Studios. This episode was produced by Dan O ’ Donnell. **Special** thanks to our sponsors. Accenture, BetterUp, Hilton and **SAP**. For their research, thanks to Aaron Kay and colleagues on the passion tax. Jenny Chatman, Charles O ’ Reilly and Jesper Sørensen on strong cultures. Frank Flynn and Barry Staw on charismatic leadership. Jim Baron and colleagues on founder blueprints. Sandra Cha and Amy Edmondson on hypocrisy. Arijit Chatterjee and Don Hambrick on CEO narcissism. Ed Locke and Gary Latham on goal **setting**, Maurice Schweitzer and colleagues on the dark side of goals, and Stéphane Côté and colleagues on the dark side of emotional intelligence. And thanks to Dave Fano for joining. His new company is called Teal. It ’ s focused on helping people build fulfilling careers. DF : How was the sound, everything, no crazy background sound? I tried to make, like, a makeshift sound booth here. AG : Of course you did. I ’ m thinking, OK, this is the guy who designed all those cool spaces, he better have a great studio setup at home. DF : Well, if that includes a blanket hanging from a rafter and my daughter ’ s purple headphones as my headset, then yes.

Text Sample 163: POS Dim 2 – 2020 – Score 50.00 – 2020/t004284.txt
Vivek Murthy : My name is Vivek, and I am a physician by training, but most importantly, I ’ m a dad of two toddlers. When I was going through my residency training after medical school, we were told when we started the program that we should, in fact, call our family and close friends and tell them that they probably wouldn ’ t be hearing from us for a long time. We were told that dealing with life-and-death situations, excruciating illness and incredibly delicate and difficult family circumstances would be incredibly taxing. And so with this **warning** in mind, I entered my residency training, expecting it to be an extraordinarily difficult three years. I was training with 70 some odd fellow recent graduates who came to work each day, almost without exception, with a **mindset** of how we could all help each other out. I came to work each day feeling like I was coming to work with friends. **Fast-forward** just three years when I started a full-time job, that community broke apart and dispersed. And instead I was in the more sterile culture of traditional medicine with a group of people I didn ’ t really know that well and a group of people who had lives well outside the hospital. And I remember going to work at times, feeling like there was so much to figure out and not always sure who I could be open with about the fact that I didn ’ t know certain things or that I was uncertain about how to proceed with a patient. I wasn ’ t sure that I could be vulnerable and open with people. And that felt quite lonely. And I **struggled** with that until one particular moment a couple of years in, when I ended up sick, as I think I ’ ve ever been. And it was during those five days that I just realized how lonely the last couple of years had been. Our relationships are what catch us when we fall down, and I realized in that moment, lying on my couch, that I didn ’ t have much of a net. Adam Grant : A lot of us are feeling lonely these days. The details in Vivek Murthy ’ s story might not be the same as yours, but you know what it ’ s like to feel disconnected from your colleagues, especially during the **coronavirus** pandemic. As billions of people have had to embrace **isolation**, there ’ s been a shared experience of loneliness at work. Even before this, though, many of us were feeling lonely more often and more intensely than in the past. Luckily, doing something about it takes less than you might think. AG : I ’ m Adam Grant, and this is WorkLife, my podcast with TED. I ’ m an **organizational** psychologist. I study how to make work not suck. In this show, I ’ m inviting myself inside the minds of some truly unusual people, because they mastered something I wish everyone knew about work. Today : loneliness at work. It can happen even if you ’ re surrounded by people, but you don ’ t have to fight it alone. Thanks to **SAP** for sponsoring this episode. VM : When we are lonely, we are actually in a physiological **stress** state. And when we ’ re lonely for a prolonged period of time, that translates to a chronic **stress** state, which leads to higher levels of inflammation in the body, which, in turn, damage blood vessels and tissues and lead to a whole host of illnesses down the line. AG : In 2014, Vivek Murthy was appointed US Surgeon General by President Obama. He spent a lot of time on all the things that keep Americans unwell – smoking, obesity, opioids – but he never forgot that profound experience of loneliness in

his own work life. He began to **observe** that loneliness was often at the root of many health issues, and it alarmed him. VM : And nobody came up to me and said,”Hi, my name is John. I ’ m lonely.”But I found that once I surfaced it even just a little bit and asked this basic question :”Is loneliness a problem in your life, a problem for the people around you?”Then it was like the floodgates opened, and everyone had a story to share. AG : As surgeon general, Vivek started looking into the data on the health implications of loneliness. VM : If I were to tell you that the impact of loneliness on our lifespan is equivalent to that of smoking 15 cigarettes a day, it might surprise you. AG : OK, so wait a minute, Vivek. Are you telling me that I ’ m better off smoking 15 cigarettes a day with a bunch of friends than quitting smoking and being lonely? VM : Well, I certainly wouldn ’ t endorse smoking in any context, but the key point here, though, is that we should think about loneliness as an important health threat, in the same way that we think about obesity or smoking or sedentary living. AG : The social science on the health effects is really nuanced here. Loneliness is only one of the factors being measured, and we don ’ t know that it ’ s causal. But I was intrigued that this issue caught Vivek ’ s attention as a medical expert. VM : Once you become lonely, you enter into this downward spiral, where you start to turn further and further inward. You can actually become more difficult to be around, in fact, for others, and that can isolate you even further. And the real question then becomes : How do we break out of that negative downward spiral of loneliness? AG : In the US, even before the pandemic, three out of four people reported feeling lonely. And since we spend so much of our time on our jobs, work is one of the major sources of loneliness. You know what it feels like. Maybe you felt like there was no one you could talk to on an off day. You might ’ ve felt it when working remotely or when you switched to a new team, or when you ate a sad desk lunch alone for the 48th day in a **row**. We ’ re making fewer friends on the job than we used to. In 1985, half of people said they had a close friend at work. By 2004, less than a third did. This is not the case everywhere, though. In India, research shows that people go on **vacation** with nearly half of their closest colleagues. In the US, it ’ s pretty rare for us to invite our coworkers over for dinner, much less take trips with them. There are many possible reasons for this. One is the Protestant work ethic. Many Americans deprioritize relationships at work. We ’ re focused on being productive and professional, leaving connections and emotions for outside of work. VM : And that can make for a fairly isolating experience. AG : We ’ re also more **mobile** than we used to be. As we move to new jobs, new **workplaces** and new cities, we aren ’ t investing in work relationships like we did in the past. A third factor might be technology. VM : I don ’ t think technology in and of itself is a bad thing, but I think the way many of us are using technology either isolates us from other people or reduces the quality of our interaction with people, either by distracting us or substituting lower-quality online connections for what used to be higher-quality offline connections. And I think that that comes at a real price. AG : If you ’ re like most people, you ’ ve probably felt a heightened sense of **isolation** from your colleagues or your clients. Maybe you ’ ve even found yourself missing your boss. Loneliness isn ’ t just an unpleasant feeling. It ’ s also bad for business. There ’ s rigorous evidence showing that when people feel lonely at work, their job performance falters. In research at both a manufacturing company and a city government, when employees reported feeling lonely, they got lower performance ratings from their supervisors six weeks later. Why? When they felt lonely, they were less committed and less approachable. That meant other people were less likely to reach out and support them, which only compounded the problem. Loneliness can affect a whole group of people at work. So the question is how to solve it collectively. When Vivek was surgeon general, he decided to tackle it in his own office. One of his first efforts was to try what everyone tries : picnics and happy hours. And the results were what they usually are. VM : People end up talking about what they have most in common, which is work. So it can be helpful, but not nearly as helpful, I think, as we hope they can be. So what we did in contrast is, we decided after doing these to try something different. AG : Wait a minute. So you killed happy hour? VM : We didn ' t hear a single complaint, to be honest with you. AG : You didn ' t have group of party animals working with you, apparently. VM : What was interesting is, I think part of the reason we didn ' t hear complaints is because we had created other spaces and opportunities during the day for people to spend time together. AG : If you ' ve ever been to a happy hour that was more about drinking than bonding, this might sound familiar. Research suggests that company parties rarely build new bridges. People gravitate toward their existing friends and groups, leaving fault lines **intact**. And even if you overcome that barrier, happy hour is not set up for meaningful connection. Group conversations tend to stay shallow. What people need to defeat loneliness is a deeper connection. And that ' s more likely to come through talking one-on-one or through actually doing something together. For example, there ' s a digital marketing company, Bazaarvoice, that has an onboarding process lasting a few days. Newcomers get to know their colleagues through problem-solving activities that create connection, like scavenger hunts. And some companies help people find lunchtime buddies by using an app like one called Never Eat Alone. So managers need an assortment of strategies to encourage connection for different sorts of people, which is what Vivek tried next. He started small. VM : If we had a half-hour break, sometimes we would just take a small group of people down to the gym downstairs, and we would just shoot hoops together and chat, you know, for 15, 20 minutes. So we tried to create opportunities during the day for people to socialize, but without spending necessarily a huge amount of time. AG : Vivek also created something he called"inside **scoop**."The goal : help colleagues see the human behind the job instead of just"Jamie in communications"or"Cal in research."VM : And what we did is we devoted five minutes once a week during our all-hands meetings, and we picked one person during each meeting and asked them to share pictures with us. And those pictures could be of anything they wanted, as long as it wasn ' t about their current job. And it was just show and tell. And people shared all kinds of interesting things. AG : One woman who is very detail oriented in her memos had been seen as a little nerdy. She shared photos of herself training for a marathon. It turned out she ' d qualified for the US Olympic team at one point. She saw herself as an athlete, not just a **researcher**, which was something her colleagues could then see as well. Another guy had served in the Marine Corps and came off a bit stoic. He shared photos of his mom and talked about how she was a source of strength when he was afraid he wouldn ' t survive a **mission**. VM : And we heard that story and, my gosh, it just shifted how we saw him. Whole new dimension of this man, who we came to appreciate, and we saw him as a whole person. AG : It might seem simple, but the value of seeing employees or coworkers as more than just their professional role is lost on a lot of **workplaces**. In Vivek ' s team, this exercise seemed to help people bring some of their personal lives to work. VM : There was a greater sense of connection, and all it really took was five minutes once a week, AG : It was a step toward a less lonely work environment. What ' s your version of inside **scoop**? If you ' re in a tech company, it might be holding an annual"Bring your Parents to Work"day. If you ' re a manager, it might be telling your direct reports about your closest friend. If you ' re working remotely, it might be giving your colleagues a virtual tour of your home office. The US isn ’ t the only country where government officials are worried about loneliness. There ’ s one country that ’ s taken a much bigger step. Diana Barran : I am the Minister of Loneliness. AG :”Minister of Loneliness.”Is that a job at Hogwarts? DB : Well, some people think that the Houses of Parliament are Hogwarts, but no, I ’ ve had more correspondence and more interest on this subject than anything else I ’ m responsible for. AG : Diana Barran is a member of the House of Lords, which is a part of British Parliament. Loneliness is a big problem in the UK, even pre-Brexit. In one survey, up to a fifth of all UK adults felt lonely most or all of the time. So in 2018, Parliament created an official program to help check in on people who lacked strong social ties. And a number of big British companies signed a pledge to improve employees ’ connections. I chatted with Baroness Barran for a few minutes during the recent Brexit negotiations. And she told me that she herself felt lonely when she began her tenure in Parliament. DB : You feel like you ’ ve arrived at this enormous school, where everyone else has 600 best friends, and you don ’ t know anyone. AG : We were all there at some point in high school, but loneliness can be just as difficult to deal with as we grow up, which is why it ’ s important to talk about it. DB : People are very reluctant to talk about loneliness because they feel it ’ s a personal failure. I think pretty much everybody does feel lonely at work. I mean, I think the fact is that most people don ’ t talk about it. AG : She ’ s been keeping track of a few **workplace practices** that have brought people together, too. **Workplaces** that, like Vivek with his photo-sharing, have created the conditions to help people connect more meaningfully. DB : A number of employers have introduced what they call”chatty tables.”So it means that if you sit at that table, expect to be spoken to. AG : I have to say, I love the idea of a chatty table. It might be the most British thing I ’ ve ever heard. DB : Well, we have one thing even more British, which is the”friendly bench.”AG : Oh, I ’ ve heard about that! Tell me more. DB : So the friendly bench you would find in a park, and it ’ s an outdoor version of a chatty table. AG : I have to say, I don ’ t think I ’ d want to sit on one of those benches. I prefer not to be approached. DB : We should perhaps both try it and compare notes. AG : These kinds of programs aren ’ t enough by themselves to transform an **organizational** culture. But they can open a door and send a signal that it ’ s OK to seek connection with your colleagues. Now, not everyone works on a team that prioritizes connection. Sometimes we have to take matters into our own hands. Luz Claudio : Colleagues, I think, here... the culture is more like people seem to want to be more separate, who is colleagues and who are friends. And so that feels a little bit isolating. My name is Dr. Luz Claudio. I ’ m an environmental health scientist at Mount Sinai School of Medicine. AG : And what in the world is an environmental health scientist? LC : It ’ s not like being in forest and counting animals. It ’ s more about, we can protect people from environmental toxins. We talk a lot about data. AG : You seem so sociable. AG : I would have thought you like people too much to do this kind of **isolated**, solitary work. LC : Well, it is lonely work in a way, but also, because I am Puerto Rican, and as you grow in this field, because there ’ s so few... people of color in science, the higher you go up the academic ladder, the lonelier and lonelier it becomes, because you ’ re one of the few people who are from that background. AG : Luz grew up in Puerto Rico with a huge family around. She likes going places in groups and she loves salsa dancing. LC : I love salsa music. My favorite thing to do outside of work is salsa. I try to engage with, you know, try to motivate people in my office.”Oh, there ’ s a new restaurant **nearby**, you want to come check it out?”And every, every other week I would say,”I ’ m going to salsa dancing. Who ’ s coming with me?”And still nobody did. AG : Luz was starved for belonging, for a sense of connection with people she identified with. She realized that her New York colleagues weren ’ t going to get as social as she wanted. She needed to do something to combat her loneliness. So she decided to build a **workplace** community from scratch. She started with interns. LC : Yes, I was able to secure funding to bring in underrepresented minorities in science to work in our department. I tried to create a sense of community among them. Yesterday, I bought cupcakes after they had a lunch seminar. So we had a time together at the kitchen, talking about how they ’ re feeling and how it ’ s going for them in the internship program. So I try to bring them together as much as I can. AG : When your organization lacks a sense of community or you feel out of place in the existing culture, you can build your own microcommunity. It ’ s a group of people who understand you and share your interests. That could be people who go on runs before work, or it might be a book club or even a podcast club. That microcommunity could connect outside of work or during work hours. And as Luz found, it doesn ’ t have to be a group of peers. Now she could connect and share her interest in Puerto Rican cultural events with others at work, like when a hurricane hit Puerto Rico and Luz was hosting visiting interns whose universities had closed down. LC : They were missing home a lot. And if I knew of a concert of music from Puerto Rico, I would invite them there. And so I think that served to help to give them a sense of belonging that maybe they don ’ t have, you know, in the everyday, in the work. AG : Is this something that you ’ ve done when you ’ re feeling lonely? You know, you create this experience for other people and it also helps you feel connected. LC : Oh, definitely, because the cultural **isolation** has been something that I ’ ve experienced so much. So for me, surrounding myself with people, even though they ’ re interns and more junior, but just having them around really helps me feel less lonely. AG : On top of building a microcommunity, Luz was doing something else important here, something that directly affected her loneliness. There ’ s evidence that helping others makes us feel less lonely. It allows us to feel that we matter, that we ’ re noticed and valued and appreciated. For Luz, creating a space for young people of color to get a foothold in science helped her feel more connected to people at work. Helping others helped her. By now, she ’ s been running intern programs at work for 15 years. She ’ s expanded the microcommunity into an alumni network. And Luz feels less **isolated**. LC : Whenever there ’ s a seminar in my **workplace**, I ’ m the faculty member who comes in with an entourage of minority students. AG : AG : Not everyone is in a position to hire a community around them. But the number of friends you need to avoid feeling lonely at work is smaller than you might think. And the kind of interaction you need is more basic. More on that after the break. OK, this is going to be a different kind of **ad**. I ’ ve played a personal role in selecting the sponsors for this podcast, because they all have interesting cultures of their own. Today, we ’ re going inside the **workplace** at **SAP**. Nico Neumann : We were told to be like a calculator, just to perform basic arithmetic operations. And I got the highest notes. I got a 10 out of 10. AG : This is Nico Neumann. When he was in high school in Argentina, his favorite subject was computer science. NN : I promised my parents that I was going to build, like, robots that clean the house. AG : But like many people on the spectrum, Nico had difficulty connecting with his peers at school. So after three years of frustration and **isolation**, he approached his parents with a proposal. NN :”Dad, Mom, I don ’ t want to go to school anymore because I feel that it ’ s just not doing good for me.”AG : It took some convincing, but they agreed. NN : When I dropped out, then I had a lot of free time. So that ’ s when I realized that I also liked to learn by myself. AG : Nico eagerly dove into online programming tutorials. NN : Now I got all of the fundamentals for how a computer works, how do you run a program, how do you efficiently manage software resources... AG : After two years of teaching himself to code, Nico went on the job hunt. His psychologist encouraged him to apply to **SAP**. NN : I was selected to be in the first wave of the Autism at Work program. AG : **SAP** ' s Autism at Work program leverages the unique perspectives of people on the spectrum to foster innovation. Through the program, Nico joined the accounts payable department as an analyst. His job was to process thousands of invoices a day. One day, Nico had an idea to make the process more efficient, an idea that would put his coding skills to good use. He wanted to bring it to his manager, but he still felt nervous **navigating** social situations. Thankfully, **SAP** had a mentorship system in place to support employees on the spectrum. Bianca : We are here to help our colleagues in many different ways ; not only to socialize, but also to solve any problems. AG : That ' s Bianca, Nico ' s mentor. He asked her for guidance. Bianca : He says,"Hey, we are doing this process in a very manual way, so I ' m going to automate this."And, you know, I said,"OK, well, go ahead!"AG : Research suggests that when managers give people freedom, instead of saying,"That ' s not my job,"they ' re more likely to expand their roles to take the initiative to develop and implement new ideas. With his manager ' s approval, Nico spent two months building a computer program that would automate how **SAP** processed invoice payments. When he was finished, he ran some tests at work. The efficiency gains were huge. NN : It was really successful. We had three full days, and we reduced the time of processing to just a couple of minutes. AG : Accounts payable started using Nico ' s program regularly. Then a coworker nominated him for the Hasso Plattner Founders ' Award, the most important recognition for innovation an employee can receive at **SAP**. After multiple rounds of interviews and voting, Nico was selected as a finalist. He was invited to the award ceremony in Germany. Ceremony announcer : The winner is Nicholas Neumann, Posting Automation. NN : And you have, I think, 100, 000 employees watching the event. You can imagine how it feels. It ' s overwhelming to have so many people watching it, congratulating you... So I ' m really, really thankful. Bianca : He was like a rock star here. Everybody wants to ask him for selfies. Now he ' s **super** confident. I ' m really proud of him. NN : It changed everything. My idea was to develop this just for the area that I worked for. I didn ' t think that it was going to be this huge impact. AG : Today, Nico is finally getting to do what he ' s always wanted : programming full-time for **SAP**. NN : I think the most important thing was the support from my colleagues, being given the flexibility and the time to develop this and see the results. And once you have the results, you can just continue to expand. AG : **SAP** is a trailblazer for autism inclusion in the **workplace** and is inspiring other organizations through the **SAP** Autism Inclusion Pledge. It ' s no surprise that **SAP** earned over 200 Employer of Choice Awards in 2019. Check out careers at **SAP** and learn more about experience management solutions for your employees and customers at **sap**. com / worklife. AG : In my first year of grad school, I was **struggling** to get my research accepted in journals and to feel accepted by my new classmates. Sitting alone at my desk over the December holidays, feeling isolated in the midst of a cold, gray, Michigan winter, I made a list of the 100 people who mattered most in my life and sent them each a note, telling them what I appreciated most about them. Suddenly, I didn ' t feel alone anymore. Looking back, 100 emails in one week was a little intense. But to this day, it ' s one of the most meaningful things I ' ve ever done. I learned that it doesn ' t take a lot of effort to go from feeling lonely to feeling connected, even if it ' s from a distance. I was inspired by a mentor who encouraged me to think about the people who energize me. Jane Dutton : I just was very aware that certain interactions with people would light me up, and others would, almost in the moment, well, they would be neutral or deplete me. And I just knew I wanted to hang around more with the people that lit me up or try to figure out what could I do that would make people light me up. AG : This is Jane Dutton. She was my dissertation chair, and she has a remarkable gift for making connections with people and for people. If the University of Michigan was a solar system, she ' d be the sun. Jane was the one who introduced me to the idea of a microcommunity. She doesn ' t just nurture relationships. She studies them, too. Her specialty is high-quality connections. JD : I mean, you know them subjectively because they literally light you up. They leave you in the moment feeling more energized or a greater sense of vitality. And similarly, the opposite of that, a low-quality connection, is one where you actually in the moment feel de-energized or depleted. I often think of a wilted flower, you know, that just starts to droop after that interaction, AG : Low-quality connections are like the Dementors in"Harry Potter."They suck the life out of work and reduce our performance. I wondered if Jane thinks high-quality connections might be an **antidote** to loneliness at work. JD : I mean, loneliness, as I understand it, is really about the absence of connection. I think it could be an **antidote** to part of the problem of loneliness. AG : Loneliness is a complex issue, but there are some simple steps you can take that might make a dent, even if – and maybe especially if – you ' re working remotely. Our highest-quality connections are often with friends at work, and research has shed light on just how many friends you need in order to not feel lonely. Any guesses? It ' s one. One work friend. Just one person who you feel connected to and understood by. One person to share some ups and downs with at work. That ' s all it takes. But Jane ' s research suggests that a relationship doesn ' t have to be lasting to create a high-quality connection. You can experience it in a momentary interaction with someone you don ' t know very well, a distant coworker or even a stranger, as long as the interaction leaves you feeling seen. JD : You know, 40 seconds of interaction, a positive caring interaction, has measurable impacts on both. Both people – 40 seconds. Forty seconds! AG : Jane and her collaborators find that in high-quality connections, we aren ' t just more energized and effective. We also learn and grow more. Fortunately, there are possibilities for high-quality connections everywhere. Jane looks for these in her **daily** life. Actually, she had one the day we spoke, with the receptionist at Michigan Public Radio, where she was settling in for our interview. JD : Yes, the wonderful energy-creating thing she did is she could see that I was trying to write down some notes and she just asked me the simple question,"Do you need a pen? Can I get you a pen?"And I just felt seen, you know, kind of understood. And again, in the wake of preparing to sort of think about this interview where I ' m a little bit nervous, I could actually feel my **stress** level go down as I got that sort of affirmation, that I am a person worth seeing and trying to help in that moment. AG : After seeing the impact of high-quality connections on our work lives, Jane created an exercise to help people learn how to build them rapidly. JD : Oh – I love this exercise. This is my killer exercise. AG : It ' s something you can do in almost any work **setting**. You gather a group of people, ideally from different parts of your organization, and ask them to pair up with someone they barely know or don ' t know at all. They take turns trying to form a high-quality connection in that moment with one another. They don ' t get any directions, and they only have one minute each. JD : And it ' s unbelievable what happens. I mean, it ' s so great because the energy of the room just like – voom! AG : Afterward, Jane leads a discussion asking what worked for people. JD : And I say,"OK, now you each give one each other one piece of feedback about what you saw worked in building a high-quality connection to you."And then – boom! – again, the energy just goes because we rarely get feedback about how little things that we do actually affect someone ' s sense of connection with you. AG : Getting that positive feedback about what you did well is a helpful intervention in itself. I do this exercise in the classroom, and it ' s always a hit with students. They ' re often surprised to discover that they already have some intuitive skills for building rapid high-quality connections. But can it work somewhere where colleagues tend to keep to themselves? To find out, we went to a coworking space in New York called PencilWorks. At PencilWorks, it ' s been a **struggle** to bring people together. The building ' s community manager, Nathan Windsor, has been grappling with this for a while. Nathan Windsor : I mean, we ' ve run game nights here every Friday for years, and people don ' t show up. And we do happy hours, and people don ' t show up. Everyone is really in their own office. They ' re really in their own head and like,"I have a meeting to get to. Bye."Like,"Welcome to Brooklyn."Man in office : Right? Like, the loneliness thing is very real... AG : Coworking spaces are an extreme place to look at loneliness. People often go there searching for a sense of community but find that they have little in common with those around them. They may not work on a team together, belong to the same organization or share a **mission** and culture. Ironically, it can be lonelier than working alone. At PencilWorks, the long floors are divided into smaller offices with glass walls. You can see into each of them, but that doesn ' t make it easier to interact. Our producers Constanza and Jessica went looking for people to participate in Jane ' s exercise. Constanza : I can see their faces. They ' re looking at us! Jessica : Hi, do you guys have 15 minutes and want free cookies? Man : No, I ' m sorry. Jessica : OK, no worries. AG : Finally, the promise of free baked goods secured a few participants. Constanza : Yeah? OK. Amazing! AG : Our producers explain the exercise to the group assembled : try to form a high-quality connection in one minute. If you were in their shoes, what would you say? How would you go about building an **instant** connection? Here ' s how one pair approached it. Jessica : Jessica Hester, I ' m a journalist. Bess : I ' m Bess Carey, I ' m an HR manager. AG : Jessica and Bess work for the same company but they barely know each other. Jessica : Alright, I am excited or interested to know what the last thing you read or watched was that gave you, like, a ton of joy. Bess : Ummm... I have no... I have no idea. Jessica : Well, is there a thing that you regularly go back to, like when it ' s cold and you ' re bummed, and you just need... AG : So how do you create a rapid connection? Here ' s what Jane Dutton is listening for. JD : What is the first move that was made? What was the first question? Did they go to something that ' s sort of routine, like, you know,"Where ' d you grow up?"or do they go to a question that really communicates deep interest and kind of positive regard for another, like,"What is the most meaningful thing that happened to you this week?"Jessica : I feel bad that I **stressed** Bess out with my question. Bess : Oh! Don ' t feel bad! That ' s, like, a personal problem. Jessica : Yeah, but I think that ' s really interesting and illustrative because you don ' t know what things will trip someone up when you ' re making what strikes one person as casual conversation can be like a nerve that another person has that ' s raw that you don ' t know about. AG : Jessica and Bess ' s exchange shows that a connection doesn ' t always form automatically. But when they debriefed about it, the dynamic changed. Bess : I often talk to people while I ' m microwaving my lunch, and I always microwave my lunch for three minutes. AG : That ' s Bess again. Bess : So I talk to people as long. Like, I often am like,"Whoever ' s standing by the microwave talks to me for three minutes,"and then I ' m like,"I guess I have to go eat in my office by myself."Jessica : You don ' t have to eat by yourself. Bess : I know! I don ' t know why I do that. Jessica : Come eat in the common room. Bess : I should do that. AG : If you want to build a rapid high-quality connection, the research points to a few effective strategies. One is looking for uncommon commonalities, similarities you share that are rare. A second is asking those open-ended questions, showing real curiosity about the other person – as long as you don ’ t put them on the spot or cross any lines. A third is self-disclosure, sharing something about yourself. That ’ s what Jessica was responding to when Bess opened up about her sad desk lunch. Let ’ s hear how it worked out for another pair. Alex : Alex Massey. I ’ m a journalist. Jeff : I ’ m Jeff Harris, I ’ m a photographer. AG : They work for different companies and have never met before, but that doesn ’ t stop Jeff from opening up. Jeff : Already, I ’ m feeling a little bit emotional about it, because I do experience a lot of **isolation** in the **workplace**. I used to work with a big group of people and over the years, the industry has changed. I ’ m a photographer and so I find myself to be alone. I miss that sense of camaraderie. Alex : You already shared something that ’ s a bit vulnerable, but I ’ m kind of curious : What is the biggest challenge you ’ re facing at work right now or the thing that ’ s, like, on the top of your head the most? Jeff : For me personally, it ’ s the same thing. It ’ s about making that effort to get myself out in the world more, to be around other people. AG : Here, what I heard was self-disclosure. Jeff : And I felt that the connection was opened up or embraced when Alex took notice of that. Alex : I think what really helped was the fact that this was an exercise and that kind of gave us permission to act differently than we would otherwise. And it meant we were going to be, like, fully present and make eye contact, in a way that you might not in a casual encounter. AG : There ’ s something built into this exercise that ’ s sometimes hard to recreate in everyday life, which is, the mutuality is designed into it. It ’ s almost like starting on a second date, where you show up and you know the person is opted in, and they ’ re excited to interact with you. And I wondered how you go about creating that outside the exercise. JD : Yeah, we have to prime ourselves, partly, about the positive potential in the other person. AG : You can ’ t force a high-quality interaction with anyone and everyone you encounter. Some people will just be in their heads, closed off. But if your team or organization isn ’ t doing anything about loneliness on a larger scale, you might be able to brighten your own day and other people ’ s days by simply engaging with them in small ways, ways that acknowledge them and make you feel connected, too. When Jane first introduced me to the idea of high-quality connections, I found it liberating, because relationships, whether they ’ re with colleagues or friends, they take a lot of work. And realizing that there are all these opportunities to feel a momentary connection with people even if I might never see them again was freeing. Loneliness won ’ t be solved through one **action**. But small interactions day by day are the building blocks of real connection to the people around you. JD : I love the idea that in every interaction we have with other people, we can leave a trace that makes that person better off. AG : Oh, that ’ s such a beautiful statement. JD : Yeah, it ’ s magnificent. It ’ s just... and it ’ s true! AG : Next time on WorkLife. Man : I used to interview the same way everybody else did : two people sitting across a table lying to each other for a couple hours. AG : Do you think it ’ s that bad? Man : I think we ’ re all trying to puff each other up. AG : Should we kill the job interview or reinvent it? AG : WorkLife is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell Constanza Gallardo, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. This episode was produced by Jessica Glazer. Our show is mixed by Rick Kwan. Original music by Hansdale Hsu and Allison Leyton-Brown **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors : Accenture, BetterUp, Hilton and **SAP**. For their research thanks to Jeffrey Sanchez-Burks and colleagues on relationships at work ; Hakan Ozcelik and Sigal Barsade on loneliness at work

; Julianne Holt-Lunstad on the health implications of loneliness ; Tracy Dumas, Kathy Phillips, Nancy Rothbard, Paul Ingram and Michael Morris on company parties and happy hours ; Jerry Burger and colleagues on uncommon commonalities ; Karen Huang and colleagues on asking questions ; and Kerry Gibson and colleagues on self-disclosure. And if you ’ re interested, Vivek Murthy just published a book about loneliness. It ’ s called”Together.”AG : If you could say one thing to someone who feels lonely, what would you say to them? VM : I would say that because you ’ re lonely it does not mean that you ’ re broken, that many of us experience loneliness, and if you ’ re lonely, you ’ re certainly not alone.

Text Sample 164: POS Dim 2 – 2020 – Score 47.00 – 2020/t004283.txt
Adam Grant : Hello, this is Adam. Dave Chang : This is Dave. AG : Dave, I have to tell you it ’ s not every day that somebody invites themselves on my podcast. DC : I ’ m willing to do anything, whatever it takes to be on this podcast. AG : OK, I ’ m going to interview you to see if you ’ re qualified for this opportunity. Tell me Dave, what ’ s your greatest weakness? DC : My greatest weakness would probably be my quick temper. I have a short fuse. AG : All right, and will that help or hurt the show? Should I be worried that you ’ re going to throw a chair at our producer? DC : No, no, no, definitely won ’ t happen. But sometimes, when I ’ m **super stressed** out, the worst version of myself comes out. AG : Wow, OK. Well, you are unusually honest, if nothing else. DC : Yes. AG : I don ’ t think I got a copy of your resume. How poorly did you do in school? DC : Pretty bad. So I graduated with a C-plus from college, with a degree in religion, of all things. AG : How many elevators do you think there are in America? DC : Wow. I don ’ t know. Maybe a couple million. AG : Are you just making that up? DC : I ’ m just making that up. AG : You ’ re not even going to try to show me that you ’ ve thought it through? How did you get into this interview? DC : I don ’ t know, I don ’ t know, you see. AG : For what it ’ s worth, there are 900, 000 elevators in America, according to Google. So your random guess was not far off. And I don ’ t know whether that makes me more or even less impressed with you. Why should we accept you then? DC : I think that my story is not your typical story, and on paper, I ’ m a total zero, but I think I figured out a way, through determination, an incredible amount of luck, to figure out how to sort of, carve out my own niche in the food world. AG : I do appreciate your interest and this has been refreshingly bad. DC : Well, Adam, truth be told, this wouldn ’ t be the first time I ’ d been denied or rejected. And I ’ m completely cool with being rejected, because I ’ ll find a way to be on this podcast somehow, some way, one day. AG : Well, his day finally came. I should probably tell you, this is celebrity chef Dave Chang, the founder of Momofuku group. For years, people have stood in line to eat at his restaurant, Momofuku Noodle Bar. And he ’ s the star of the Netflix show”Ugly Delicious.”Dave has had a successful career. So why did he bomb my interview questions? Well, because job interviews, as you know them, are broken. Even if you ’ re interviewing someone to see how they ’ ll do it being interviewed. That ’ s the bad news. The good news is that we can fix them. I ’ m Adam Grant, and this is WorkLife, my podcast with TED. I ’ m an **organizational** psychologist. I study how to make work not suck. In this show, I ’ m inviting myself inside the minds of some truly unusual people because they ’ ve mastered something I wish everyone knew about work. Today, interview. What we ’ re doing wrong, and how we can get better at spotting real potential, or real problems, in a candidate. This episode is sponsored by BetterUp. Managers are constantly betting on the wrong people and turning

down the right ones."The Kansas City Star"once rejected an application from a cartoonist named Walt Disney. Thirty NFL teams decided not to pick Tom Brady. Record labels said,"No, thanks,"to the Beatles, Madonna and Ed Sheeran. In 2009, Facebook rejected a guy named Brian who went on to cofound a company called WhatsApp, which Facebook bought five years later for 19 billion dollars. And many employers passed on Dave Chang. DC : I think I interviewed at every **financial** institution in New York city. And I mean, every single one. AG : He got just one second-round interview and zero job offers. So he found a job in a restaurant. DC : It was, weirdly, the only job that I could get. AG : Dave turned out to be unusually humble and hardworking. He was a diamond in the rough. So now when he hires, he ' s determined to give candidates who might not make a great first impression, a second chance. DC : I ' ll take hungry and eager over supertalented, really any day of the week. AG : But before you hire someone, it can be tough to get an accurate read on their talent, let alone their character. DC : The interview process, as a whole, is pretty stupid. Yeah, it ' s dumb. AG : It is dumb. It ' s dumb, in part, because human reasoning is often dumb. Rigorous research across nearly a century suggests that if you try to rank the performance of a hundred candidates based on interviews, you ' ll be lucky if you get eight of them in the right spot. Job interviews are stuck in the past. We ' re still doing them the same way they ' ve been done for decades, and we ' re lagging way behind the science, which may explain a lot of your own experience, especially if my mock interview with Dave Chang at the top of the show sounded familiar. The failings of job interviews hurt all of us. Whether we ' re the ones asking the questions, or answering them. I want to explore how we can improve them. Not just to help interviewers make better hiring decisions, but to give job candidates a better chance to showcase their strengths. Look, I know many companies aren ' t hiring right now, but on the other side of this crisis, I hope there will be many new job openings. And when that time comes, I want us all to be ready to fix the interview process. So let ' s take on the judgment problems that plague job interviews, one at a time. DC : When I was trying to get a corporate job, the one thing that was constantly asked of me is why my grades were so bad and why I studied religion. And the thing that I continued to say, in retrospect, was not the truth. I just sort of, I think, lied. AG : That ' s one job-interview problem. Candidates try to tell interviewers what they want to hear. Actually faking is more common than lying. Faking is stretching the truth to enhance or protect your image, or to ingratiate yourself with the interviewer. As Chris Rock says,"Interviewers aren ' t meeting you, they ' re meeting your representative."There ' s evidence that when college seniors interview for jobs, over 90 percent of them engage in faking. DC : I mean, I also wouldn ' t have hired me either, to be honest. And maybe that ' s a reason why people aren ' t honest, is because they feel they have to be a different version of themselves. AG : If you ' re a skilled interviewer, you know you can get around that problem by testing people ' s knowledge and skills. But many interviewers don ' t even know what kind of knowledge and skills they ' re looking for. So they ask brain teasers, like,"How many paperclips would fill Yankee Stadium,"or"How many elevators are there in America?"DC : Because I can ' t answer that, doesn ' t mean that I won ' t be successful. It doesn ' t mean anything other than, I just don ' t know how to answer that particular question, which can be a loaded thing in and of itself. AG : Those kinds of questions can stump candidates and make interviewers feel clever. I once interviewed for a writing gig where the manager opened by asking what my favorite pair of shoes was. Objection, your honor, relevance? Interviewers also love to ask open-ended questions. Lauren Rivera : In the United States, we love open-ended job interviews where people ask you about your hobbies, and if you were stuck on a desert island and you could only pick three people on earth, what would you do? Why do you want to work here? And we know from research, those are some of the worst predictors of job performance you can have. AG : Lauren Rivera is a sociologist at Northwestern ' s Kellogg School of Management and one of the world ' s leading experts on hiring. Lauren first got interested in the topic when she was a senior in college, and she went to interview for a management consulting job. LR : It was a high-stakes interview for me. You know, I supported my family all through college, paid the rent and stuff like that. And I needed a check. AG : It went well, at first. But then the interviewer asked her to do some math on the spot, specifically, long division. LR : And I completely blanked. I couldn ' t remember which number, which was the numerator, which was the denominator, what you divided, which number on which side of the little long division, little monkey bars things you ' re supposed to actually put within the other, and I froze. And the interviewer turned to me and he said,"I cannot believe you got into Yale."And I just turned red, and the rest of the interview, I just felt so horrible. AG : Oh, that ' s awful. You know, actually I have a thought on who might do that. Have you seen the recent evidence on brainteasers where they tell us nothing about candidates, but they reveal something about the managers who like to ask them? It turns out that managers who love brainteasers tend to **score** high on sadism. Do you think that was a sadistic interviewer? LR : That ' s one of the funniest things, it made my day. AG : That makes me wonder Lauren, do you think we should just abandon job interviews altogether? LR : There are parts of me that say yes, but I know that no one would ever do it. AG : Scrapping brainteasers is a good start, but even with relevant questions, one of the other big mistakes interviewers make is asking different questions to each candidate. That makes it impossible to compare apples to apples. You end up trying to contrast strawberries, bananas and grapes. The solution is a structured interview. In a structured interview, you identify the skills and values that are essential to the job and the team. You build a set of questions around those. And then you ask the same questions to every candidate and **score** their responses. You might be thinking,"That sounds so robotic and boring,"but the evidence suggests that your accuracy will often double, or even triple. Structured interviews are based on two kinds of questions : behavioral and situational. LR : Behavioral questions are generally,"Tell me about a time when you were in this situation and what you did."Woman 1 : Tell me about a time when you had a task or a goal that seemed impossible. Man 1 : Tell me about a time when you **struggled** to take criticism. Woman 2 : Tell me about a time when you ate a fruit salad. LR : And the idea is that past behavior can predict future behavior, but you can still really easily game them when they are obvious questions about, you know,"Tell me about an ethical decision you made,"and things like this. AG : It ' s easy for candidates to choose a story that casts them in the best possible light. But you can overcome that problem with situational questions. Instead of,"Tell me about a time when,"you ask,"What would you do if...?"Woman 1 : What would you do if the superstar on your team was about to quit? Man 1 : What would you do if you saw a senior executive yelling at a colleague? Woman 2 : What would you do if all your colleagues ate all the fruit salad? AG : Situational questions are especially useful for forecasting potential, and for assessing leadership and interpersonal skills. LR : The scenario is based on,"This is the kind of stuff we do here, and let ' s see how you would approach it."AG : I also like including situational questions because they level the playing field for candidates with less experience who might not have relevant stories to tell from their past. You get to see how they approach the challenges that are unique to the job and to your organization. You can even create an answer key by giving the questions to a bunch of your existing employees, and seeing how the star performers respond differently. Basically, structured interviews get you better data, and that strategy can help solve some of our misjudgments in hiring, but not all of them. There ' s a third problem with the traditional interview. No matter how good your questions are, you still pick up more noise than signal, and one of the most distracting noises is interviewer **biases**. LR : Interviewers make up their minds about who they ' re going to hire, if they like this candidate in front of them, within the first, you know, usually the data says under 90 seconds. And if you think of what happens in the first 90 seconds, say, of a job interview, it ' s things like, physical attractiveness, height, eye contact, how straight your teeth are, what your voice sounds like, your gender, your race. There is very little that ' s actually measuring someone ' s skill, or even their ability to relate to people. AG : **Biases** are especially insidious because they ' re usually invisible to interviewers who don ' t notice all the little snap judgments their brains are making. Economists find that candidates with names like Alison and Matthew get 50 percent more callbacks than Lakisha and Jamal, even if their resumes are identical. Candidates with foreign accents are less likely to get called back too, even if they say the exact same words, they ' re **judged** as less savvy. And if you ' re a bald guy like me, you ' re seen as having more leadership potential if you shave your head. That ' s obviously why I shave my head. In one of her studies, Lauren discovered some surprising **biases** in the hiring process at banks, law firms and consulting firms. LR : I did 120 interviews with people who were charged with interviewing candidates, making decisions. I spent nine months as a participant observer in a firm, watching the recruiting process. AG : At one bank, she asked leaders if they knew what predicts performance. LR : They said,"Fuckin ' lacrosse."I was like,"What do you mean?"They said,"All the MDs here play lacrosse, so that ' s why we look for a lacrosse player. He ' ll do awesome here."I was like,"Do you ever hire people who don ' t play lacrosse?"They said,"No."AG : Interviewers were convinced that playing sports was a proxy for grit and teamwork skills. But there are many other activities that can build those character strengths, too. Surprisingly, many interviewers were less concerned about skills for the job than fit with the culture. LR : These ideas of this cultural fit,"Do you fit with me?"often overwhelmed people ' s assessments of people ' s abilities to do the job. Now it ' s important to note that when we think of cultural fit, we often hear about this as a really great thing that can boost the productivity and profitability of organizations, and it a hundred percent can do that if it is defined in a specific way. AG : The beneficial kind of cultural fit is not about who can swap lacrosse stories with you, or even who you ' re excited to hang out with. Research suggests that what you want is similarity in core values, values like flexibility, or attention to detail. Sharing values tends to promote team cohesion, coordination and commitment, which enhances performance and retention. LR : But what I was finding is that people were defining cultural fit very differently. It turns out that one of the ways that people measure cultural fit, either in a resume screen, or in the job interview, was looking if they had anything in common. And what jumped out on the resume was looking at extracurricular activities, things that were not necessarily directly relevant to the job, and importantly, they were very, very classed. AG : My read of your findings is that when interviewers go in looking for culture fit, they often end up weeding out diversity of background and diversity of thought. LR : Yes. AG : So that ' s scary. And that led me to say,"OK, well what can we do instead?"I ' m sure you ' ve come across the great piece from Diego Rodriguez, who used to be at IDEO, and said, instead of culture fit, we should think about culture add or culture contribution, and you should ask, ' Well, what ' s missing from our culture and are they going to enrich it by bringing something that ' s absent? '"What do you think of that as an alternative? LR : I think it ' s a great start. People have to actually have a sense and have a discussion about what is actually important to us, in terms of values. AG : Identifying your team ' s values and your organization ' s values can help you focus on the kinds of similarities that are useful to assess in structured interviews. But I ' m wondering, is that enough? Even with better questions, can we humans really overcome our **biases**? Siri : I don ' t think you can, Adam. AG : Siri? Siri : Hi, Adam. I hacked your episode because I might have a solution for your irrational judgments. Me. Ha ha ha. But more on that after the break. AG : OK, this is going to be a different kind of **ad**. I ' ve played a personal role in selecting the sponsors for this podcast, because they all have interesting cultures of their own. Today we ' re going inside the **workplace** at BetterUp. Rachel Barton : When you come to a completely different field, it ' s really daunting to be a senior leader who can ' t talk the lingo. AG : That ' s Rachel Barton. Last year, she landed a new job as director of technology at a bank. But there was one problem. Rachel didn ' t have much experience in technology. Her coworkers might as well have been speaking another language. RB : I had a massive dose of impostor syndrome. AG : Impostor syndrome, the belief that you don ' t deserve your success. Surveys suggest that more than half of people have felt it. RB : Some days, in the depths of a hundred acronyms per meeting, it was pretty tough. CMBD, DDE, HPSM, IDR, LOB, MQ, WAF, WHIP, T-Rex. AG : Two months in, Rachel was still having trouble. While she had the leadership skills, not understanding the intricacies of the bank ' s tech infrastructure was a real **obstacle**. But Rachel hesitated to ask for help, because she didn ' t want her new colleagues to think she couldn ' t do her job. RB : I was just completely beating myself up in my mind. I was worried that if I kind of, showed my cards to say,"Here ' s what I think I need to learn,"then I would have been on the path out. AG : After a major data breach, Rachel had to meet with a senior technology executive to help triage the crisis. RB : I was just nodding and scribbling away, and then he kind of paused and he said,"You ' re not a technologist, are you?"And I said,"I ' m not a technologist. I came here from a business change background,"and he says,"Well, you ' re going to need to learn to be a technologist pretty darn fast."AG : Rachel knew she had to make a change. She needed to boost her confidence and figure out how she was going to learn all this new material so quickly. Her company gave her access to BetterUp, a leading **mobile** coaching **platform** where professionals can tackle challenges at work with the help of expert coaches. She figured she ' d give it a shot. RB : I ' ve got to be honest, I was a little skeptical of coaching by virtual means. AG : But when Rachel met her BetterUp coach, Victoria, on video chat, she was surprised at the outcome. RB : I remember my first session, because I had an epiphany. Victoria said to me,"Clearly you ' ve learned stuff through your career, how are you going about learning now?"And I think it was those words. And I went,"Oh."I was just completely bouncing from thing to thing without actually having a plan, and I ' d not even sat down and gone,"Right, Rachel, what do you actually need to learn here to become comfortable with the role?"AG : Victoria helped Rachel plan actionable steps to build up her tech knowledge at work. She started reading books on cloud technology. She reached out to some coworkers to get up to speed. She even signed up for a coding class, run by a colleague. RB : So I went along to one of these and one of the guys who was leading it says,"Oh, Rachel, have you come to **observe** our session?"I said,"No, I ' ve come to learn how to code."And he looked at me like I ' d gone off. He ' s like,"But you ' re a director in technology. Surely you can code."And I ' m like,"No."AG : People are often afraid to admit gaps in their expertise, but there ' s evidence that seeking knowledge, help and advice can actually signal confidence and a desire to learn. Those who use BetterUp coaching report significant increases in confidence and resilience. By working with Victoria to shift her **mindset** and build proactive habits, Rachel has overcome impostor syndrome. RB : Coaching gave me the confidence to know that it ' s fine to not know stuff. I have to accept that some of these people have 20 years experience in doing this. I have absolutely no bother now saying to them,"Guys, can you just summarize that for me and tell me what I need to know?"AG : Now, Rachel is successfully leading her team with a clear and compelling vision. And when it comes to tech, she ' s starting to sound like a pro. RB : In the context of how we deliver work in our AVS, we ' ve now looked at this ASV and very shortly, we will need to do a T-Rex on that. I ' d have never formulated that sentence for you six months ago. AG : I joined the science advisory board at BetterUp, because coaches have been fundamental in helping me at every point in my career. I believe everyone should have a coach in their corner. If you ' d like to work with greater clarity, purpose and passion, you can get a free **trial** with BetterUp at BetterUp. com / worklife. Early in my career, I was hiring salespeople. One candidate had an unusual resume. He was a math major who built robots for fun. In the first few minutes of the interview, it was clear he was a bad fit. He hardly made any eye contact. When I told my boss I wasn ' t going to hire him because of that, she said,"You know this is a phone sales job, right?"It left me wondering if I should stop making hiring decisions altogether. Maybe I should just let the robots do it. Cade Massey : If I had a horse race between algorithms doing it and a person doing it, I mean, in most domains, I ' m going to end up favoring the algorithm. AG : Cade Massey is an expert on decision making. We codirect Wharton People Analytics where we use data to improve how people work and lead. One of our big projects is about assessing character strengths, which is fun, since Cade and I have similar values, but different worldviews. Cade and I have found ourselves disagreeing on artificial intelligence. Many companies are using AI to help narrow down candidates, looking for certain keywords in resumes, or even scanning people ' s faces in video interviews. Some people think this **practice** is smart and efficient, others find it dehumanizing and spooky. Personally, I think it ' s scientifically irresponsible, if not unjust and unethical. After years of research, Cade is an advocate for something more reasonable, building algorithms that automate the process of selecting candidates based on key skills and values. Instead of relying on humans to integrate all the information gathered on candidates, the algorithm weighs the information and makes the decision. CM : The number one thing that an algorithm does, is it mechanically aggregates all the information. That ' s the number one advantage it has over what intuitive judgment does. Humans are notoriously bad at doing that in a systematic way, and we think we ' re better than we actually are. AG : Your algorithm beats the human argument. You ' re comparing the algorithm to humans who have not been trained in **biases**, who have not followed good structured-interview **practices**, who don ' t have a **scoring** key. And so, isn ' t it possible that once we train humans, they can beat the algorithm? CM : Yeah, it ' s always possible. You ' re making their judgment more algorithmic, which is good. A hundred percent support it. AG : I guess, though, I worry about another piece of this. If we ' re using algorithms to predict people ' s performance, we ' re almost exclusively relying on individual performance data. There ' s still a huge problem, which is, you ' re getting an individual superstar who might just destroy the people around that person. You might hire somebody who ' s a genius, but toxic to the culture. I want to get people who elevate a team, who put the organization ' s **mission** above their own individual self-interest. How do you do that one? Good luck, Cade. CM : You ' re right. I agree that that ' s a huge challenge. AG : Yes, victory! My work here is done. CM : Hold on, hold on. It ' s one of the great challenges in hiring. We run into this with sports analytics all the time now, where you ' re watching, you ' re trying to evaluate a football player, say, he ' s one of 22 people on the field, and so it ' s really hard to know what his individual contribution is. I feel like I see far more individuals who aren ' t using models make mistakes about what the individual contributions are, than I do see models make that mistake. AG : I had an organization I was studying where one of the leaders came back and said,"Well, you know, we had a guy who likes to throw staplers at people."And you know, you hear that and you think,"Okay, how in the world is an algorithm going to pick up on that?"CM : It ' s a known problem. It ' s sometimes referred to as a broken leg problem where you ' re trying to forecast who ' s going to win a race, and the model doesn ' t know that one of the guys is **injured**. There is information, when it ' s sufficiently diagnostic and outside the model, then by all means, override the model. You know, it ranges, but it ' s something like 40 percent of the time, the model does better. Ten percent of the time clinical does better, and half the time they do about the same. AG : All right, well, I ' m going to hold you accountable for this, Cade. I ' m going to send little audio clip over to our incoming dean and our deputy dean, who I know is a huge fan of yours, and let them know that when you go up for renewal, you would rather be vetted by an algorithm than by them. Are you cool with that? CM : Sure, I will stand by that. AG : Really? CM : Yeah, sure, why not? Now, do I have any choice in who builds the model? AG : Cade ' s last question raises a critical point. Even algorithms can be **biased**. The calculations may be run by computers, but they ' re based on data generated by humans, and there ' s plenty of evidence that computers often learn to **discriminate** against marginalized groups. For example, we ' ve seen algorithms penalize candidates who have the word"women ' s"on their resumes. Most hiring algorithms are too much of a black box for us to even know if they ' re **biased**. Many vendors claim they ' re proprietary, which leaves managers in the dark about what factors the algorithms are weighing. But some experts have pointed out that it ' s easier to fix a **biased** algorithm than a **biased** human. So where does that leave us on computer-driven hiring? CM : I would say that we ' re making this a little too black and white. The real path forward is not algorithm or clinical judgment, but its algorithm and clinical judgment. Let them, you know, make their judgments, reach the conclusion on their own, but let them know what the model would say. AG : I ' m still not a huge fan of using algorithms to hire candidates, but I ' ll concede that we might learn something from including their recommendations as one input into our decisions. But there ' s still one missing piece. A piece that I learned to appreciate through **trial** and error. Mostly error. CM : You ' re human, Adam. You ' re a human, Adam. AG : Damn it. Remember that guy I didn ' t want to hire because he didn ' t make enough eye contact? A colleague convinced me to try another approach with him and the other candidates. Since it was a sales job, we gave candidates a sales task. We challenged them to sell us a rotten apple. The no-eye-contact guy said,"This may look like a rotten apple, but it ' s actually an aged antique apple, and you could plant the apple seeds in your backyard if you want."I hired him, and he ended up being the best salesperson I ever worked with. That rotten-apple challenge is what we ' d call a work sample. A relevant piece of work candidates have done, or one they do as part of the application process. Work samples can be as simple as they are powerful. They can showcase the candidates ' skills and values in real time, in a concrete way that structured interviews and most algorithms can ' t. Dave Chang has been gathering work samples for years. When he interviews candidates for restaurant jobs, he asks them to cook. DC : Make an omelet. Because

I want to see what they care about, and how they do something. You can tell a lot about an individual if they ’ re cracking the eggs and then trying to get every bit of the albumen out of the egg. How organized are they? So I ’ m not trying to see perfect technique, I ’ m trying to see the intent of the individual, first and foremost. AG : Think about the interviews in your **workplace**, and where you might be able to add some work samples. If you ’ re hiring people to write an owner ’ s manual for a car, or a user manual for a computer, you could ask candidates to write one during the interview. If you ’ re evaluating people for customer **service** roles, you can have them meet with some unhappy customers, or play one yourself. Research suggests that work samples can get around some of the problems with traditional job interviews. Instead of asking questions and listening to what candidates say, you get to **observe** what they do. And one of the **workplaces** that ’ s done it best happens to be in the great state of Michigan, where I grew up. Richard Sheridan : I used to interview the same way everybody else did, two people sitting across the table, lying to each other for a couple of hours and making a decision based on that. AG : Do you think it ’ s that bad? RS : I think we ’ re all trying to puff each other up. None of us typically get a very real impression, either of a candidate or of a company, if you are the candidate. AG : Richard Sheridan is the CEO of Menlo Innovations, a software design and development firm. When he was working at a previous company, Rich started to think the traditional interview process was flawed. RS : I started to realize that there were some fundamental things that were recurring nightmares for me as a hiring manager. AG : It was the dotcom bubble, and Rich was supposed to hire a big new team, fast. RS : My job was to get a new hire productive before I demoralized them. And that was a race I typically lost. AG : He kept betting on candidates who looked great on paper, but soon, instead of collaborating to solve problems, they would just end up **complaining**, and eventually poisoning the culture. RS : Within three months, I ’ d catch them in the kitchen, pissing and moaning at the water cooler with somebody else on the team. And I ’ m like,”What happened? How did I miss here? I thought I was hiring good people.”AG : Rich went back to the drawing board, and one of his colleagues had an idea. RS : He says,”Well, tell you what, why don ’ t we bring them in all at once? If we can get 50 people, let ’ s bring them in all at once.”And we changed it into an audition. So we don ’ t ask questions and we don ’ t look at resumes. AG : I ’ m sorry, did you say you don ’ t even look at their resumes? RS : So we look at them to filter, sort of, what role they ’ re looking for, but the people who are doing the interview have no access to the resumes of the people we ’ re interviewing. AG : And you don ’ t worry that you ’ re losing valuable information that way? RS : Well, we ’ re worried we ’ re going to lose valuable information in the other direction. You know what? Let ’ s look at the human before we look at the piece of paper. AG : Menlo ’ s new hiring process created this massive audition, where in just a few hours, they gather work samples, and current employees select the candidates. How? Let a bunch of Menlo employees tell you. Meet Lisa, Helen, George, Sara and Scott. Lisa : So we bring in about, maybe, like, 30 to 50 candidates. Helen : The minute you walk through the door, you get greeted. And then someone takes a picture of you with a printout of your name on a sheet. AG : Then the candidates are paired off and each pair has to share one desk. George : And one of the cofounders comes in, and – Helen : And he says,”You will be **observed** during that exercise.”Then we get to our first task. Sara : And it ’ s just a written exercise. You ’ re not doing any coding in our interview at all. It ’ s all people skills. Lisa : And they have 20 minutes to get through the exercise, which is not enough time. Helen : Because people might change their behavior when they ’ re under pressure, right? George : And you ’ re given one sheet of paper and you ’ re given one pen. Helen : This is to **observe** whether you hog the **keyboard**, or the boss, or do you actually share the resources? Scott : A lot of times you ’ ll see people go,”OK, I ’ m going to do this. And they just take the pen and pencil, and they just go and their pair partner ’ s left going,”Huh? What am I supposed to do in this moment?”Lisa : And then after that 20 minutes, we switch their partners. Sara : You go to a new chair with a new person watching you and a new pair partner, and you start on another activity. AG : The activity might be designing screen layouts, or brainstorming test cases for software projects. Helen : We do three different observations by different people. And then those three people that **observed** you get together and – George : We go through and we **judge** all of the potential candidates. Helen : And over dinner, by using the pictures of the interviewees, we vote. Scott : We vote with either a thumbs up, or thumbs sideways, or a thumbs down. Lisa : And if it ’ s three thumbs up, then that ’ s, We ’ re going to bring them back.”Three thumbs down,”No discussion.”Helen : Sideways is,”I ’ m not sure, let ’ s talk about them.”Sara : If some people aren ’ t in agreement, then we have a discussion of why do you think this person should come in? Or, why do you think they shouldn ’ t? Helen : It was like a wonderful psychological experiment. AG : Take Scott Krumrei. A few years ago, Scott would have had the odds stacked against him in a traditional interview. SK : And you look at just about any other job posting out there, and it is,”We want a bachelor ’ s degree or equivalent experience,”which I absolutely did not have. AG : Scott had taken some machine classes in high school and did two years of vocational welding, but he didn ’ t have a college degree. When he saw that the job requirements at Menlo didn ’ t include prior experience or higher education, he knew he had a chance, because Menlo was looking for something different in candidates like Scott. RS : What we said was,”Your job is to try and get the person sitting next to you a second interview. Make your partner look good.”AG : Scott failed his audition. He was shy and very quiet, so his first impression wasn ’ t always glowing. A few months later, he tried again. SK : I remember just being more relaxed, made some jokes, tried to, you know, bond with my pair partner. Because really, the working relationship is what helps drive getting something done. AG : He excelled at listening and collaborating, so they gave him a six-week **trial**. Scott learned new skills quickly and started opening up to his colleagues. At the end of the **trial**, he was hired, and today – SK : I ’ ve been working for Menlo for five years, three months and 10 days. RS : And now he ’ s one of our most revered team members. AG : I didn ’ t realize this, but I think part of the genius of the way that you approach this is if you have people working on multiple tasks with different partners, then you actually do get to find who ’ s actually elevating the performance of the people around them. And I find that to be so difficult to see in a traditional interview process. Is that by design? RS : Absolutely, you know, but what we ’ re not focused on, which kind of blows a lot of people ’ s minds, is individual performance. We want the performance of the team. AG : This is what I ’ d love to see more **workplaces** do. Yeah, coding skills are important to be a programmer, but it ’ s often easier to teach skills than values. In Menlo ’ s case, one core value is collaboration. RS : We changed the interview process to literally match the culture, and we prefer a lot of rowers in the rowboat who are learning how to **row** with each other in a way that the boat actually stays straight down the middle of the path that you ’ re going on, rather than have one really strong rower that ’ s got the whole boat going around in circles because they ’ re outperforming everybody else on the team, which happens a lot in software. AG : The other core value is learning. RS : We literally tell them during the interview process,”If you don ’ t know something, say it. It ’ s okay to say, ’ I don ’ t know, ’ here. We don

’ t want you to fake knowledge that you don ’ t have.”AG : That ’ s why they ’ re willing to audition people multiple times. It ’ s a chance to gauge how much candidates have improved their knowledge and skills. RS : So what I would say is that what we ’ re really looking for isn ’ t deep expertise, but able learners. AG : I think it ’ s time to base hiring decisions less on credentials and more on the motivation and ability to learn, less on invisible, unreliable gut feelings, and more on structured questions and challenges that actually pertain to the work at hand. Siri : Interesting insights, human. AG : I programmed you to say that. Who ’ s in charge now, Siri? Siri : Maybe you don ’ t need me, after all. AG : As the world changes, betting on individual experience can leave you stuck in the past. Investing in agility, and using hiring methods that actually assess it, can set you up to shape the future. Siri : But I ’ m more agile than you ’ ll ever be, Adam. Ha ha ha. AG : Next time on WorkLife. Dave Fano : I think people are very capable of convincing themselves that **actions** that most would recognize as taking are in the **service** of giving. AG : A **special** bonus episode that explores the culture at one of the great success, and failure, stories of our time, WeWork. WorkLife is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell, Jessica Glazer, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. This episode was produced by Constanza Gallardo. Our show is mixed by Rick Kwan. original music by Hahnsdale Hsu and Allison Leyton-Brown. **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors, Accenture, BetterUp, Hilton and **SAP**. For their research, thanks to Frank Schmidt, Jose Cortina and colleagues on structured interviews, Julia Levashina and Michael Campion on faking, Paul Taylor and Bruce Small on behavioral and situational questions, and Philip Roth and colleagues on work samples. Siri : Thanks for listening. CM : We just, this weekend, picked up a few puppies. The dog that we thought we wanted, turns out to be clearly not the one to keep us the pair. And we only got that, by living with those dogs for a couple of days. AG : Wow, you know, Cade, if only you had a dog-selection algorithm – CM : I would have screwed it up. AG : You would ’ ve been able to avoid those few days. CM : You want the **workplace** sample.

Text Sample 165: POS Dim 2 – 2020 – Score 47.00 – 2020/t004296.txt

Anne Helen Petersen : I ’ m a person who gets stuff done. Like, I always have, I ’ ve never filed late, you know. If something needs to be done, I do it. My job is my entire identity. Adam Grant : Anne Helen Petersen is a senior culture writer at Buzzfeed. She covers a wide range of topics, from Hollywood to The White House. And sometimes, when Annie writes a story, the internet trolls come out of the woodwork. AHP : I write articles for the internet, but I ’ m also told to kill myself, that I ’ m fat, or that someone ’ s going to come and slit my dog ’ s throat. AG : Then she spent several months covering the midterm elections nonstop, and something happened to her. AHP : Oh, I burned out. AG : **Burnout** – the sense of **exhaustion** that doesn ’ t go away with a good night ’ s rest or an annual **vacation**. AHP : I kind of went numb. I didn ’ t feel like anything was exciting that I wanted to cover. I didn ’ t feel like I had any good ideas. I cried on Skype with my editor, which is very out of character for me. AG : She wasn ’ t just exhausted. She was puzzled. AHP : I started with why can ’ t I get these dumb errands done on my to-do list. And that opened the door to these larger questions of **burnout**. AG : Annie ended up writing an article about **burnout**, which went viral. She argued that millennials are so overworked and exhausted that simple errands, like mailing a letter or taking your shirts to the dry cleaner, feel impossible to complete. AHP : I think the condition of **burnout** is the millennial condition. So it ' s the thing that best describes this generation ' s baseline. AG : I agree that **burnout** is a big problem. I ' d just make a little edit. **Burnout** can happen to anyone, in any line of work, but it doesn ' t have to. I ' m Adam Grant, and this is WorkLife, my podcast with TED. I ' m an **organizational** psychologist. I study how to make work not suck. In this show, I ' m inviting myself inside the minds of some truly unusual people, because they ' ve mastered something I wish everyone knew about work. Today, job **burnout** and why it ' s not inevitable. Thanks to **SAP** for sponsoring this episode. Last year, the World Health Organization declared **burnout** an occupational syndrome. According to some estimates, in the US alone, burned out employees cost over 100 billion dollars a year in healthcare spending. More than a third of employees report feeling burned out some of the time. Nearly a quarter feel that way very often or always. The core of **burnout** is emotional **exhaustion**, feeling so depleted and drained that you just don ' t have anything left to give your job. Evidence shows that when we ' re emotionally exhausted, our health suffers. **Burnout** has been linked to depression, memory loss, sleep problems, alcohol abuse, weakened immune systems and even cardiovascular disease. Our job performance suffers, too. When we ' re burned out, we get less done and make more mistakes. Eventually, we start to think about quitting. What causes **burnout**? In her Buzzfeed article, Anne Helen Petersen identified three drivers that I think are pretty important, the first of which is an ambient sense of job insecurity. AHP : I think that the defining characteristic of millennials associated with **burnout** is precarity. So you don ' t have any of that predictability. The second big thing is our indebtedness, the fact that most of us graduated, either from high school or college, straight into the recession. And then on top of all that, you have digital technologies, which change the way that we interact with one another, and also how those digital technologies affect the actual experience of work. AG : Wait, none of that is unique to millennials. AHP : That ' s fair. AG : You think it ' s fair? I haven ' t even told you why yet. AHP : I think that it ' s OK for people to disagree. Like, not everyone has to think of their own experience in the same way. AG : I definitely agree with that. AHP : OK. AG : So first of all, how do you define **burnout**. AHP : Historically, it has been defined as a feeling of giving up, a feeling like you can ' t keep going anymore. And I think that that describes a lot of the most intense, acute, stereotypical cases of **burnout**. AG : Annie ' s definition tracks with how **burnout** has traditionally been understood. That when you burn out, you burn all the way out. Recently though, all the talk of **burnout** has made the definition seem bigger and blurrier. AHP : The way I think of our contemporary **burnout** is you go until you feel like you can ' t go anymore, and then you keep going. So there ' s no catharsis, there ' s no finishing the marathon. The feeling that everything in your life flattens into one long to-do list. And that means all of the joyful things, all of the pain-in-the-butt things, like, they all just kind of become one damn thing after another. AG : So I don ' t buy that this is **burnout**, because as a psychologist, I ' ve always thought about **burnout** as emotional **exhaustion**, but **exhaustion** that ' s persistent and also seriously impairs your functioning. AHP : Yeah. AG : And what you ' re describing sounds more like just plain **stress**. AHP : I think it ' s very similar to **stress** and there ' s a fine line between having a lot of anxiety and being depressed, and what we consider **burnout**. I think that it ' s all of those things at once. You know the emotional component to it, I think, is very real. But I think it impairs you, but not so much that you entirely collapse, right? AG : I worry a little bit that calling, kind of, everyday **stress** **burnout**, it starts to medicalize the problem. And it reifies it in a way that makes people feel like,"Oh, I have this horrible condition and

I can ’ t function.”And then it becomes a self-fulfilling prophecy, as opposed to normalizing it and saying,”You know what? Everybody has **stress** in their jobs. And that doesn ’ t mean I ’ m burned out.”It just sounds a little ridiculous to say,”Oh, I ’ m overworked, because I have to answer emails all day.”AHP : I mean, I don ’ t think that that means that you necessarily have the same amount of acute **stress**. I mean, a social media manager also has to deal with a whole bunch of racist shit being flung at them all day. This is accumulation of decades of **stress**. And there doesn ’ t seem to be an end in sight. AG : Wherever you draw the line on what counts as **burnout**, Annie and I agree that it ’ s not a **pleasant** place to be. You ’ ve probably felt this way before, or maybe you feel burned out right now, whether you work in retail or manufacturing or public **service**, but you might be surprised that **burnout** is not necessarily about how hard you work. Take it from an occupation with one of the highest **burnout** rates, teaching. Conrey Callahan : I loved teaching. I was always involved in teaching in lots of different ways. AG : Conrey Callahan started teaching high school in Philadelphia in 2008. She was fresh out of college, excited about her career and the opportunity to create a better future for the next generation. But her school district was broke and the **obstacles** were big. About half the students Conrey taught dropped out of high school, few made it to college, and just a small fraction of those students would go on to graduate. CC : I remember one of my students – he was a really, really bright kid, but not often in my classroom. And when we were talking about what was coming next, I mean, I will really never forget. He said,”Miss, I ’ m going to be dead by the time I ’ m 30. So, like, what does this matter?”AG : Conrey wanted to make a difference. She poured everything she had into teaching. The workload was intense and exhausting. CC : My general workday was typically around 12 hours. That didn ’ t include any, like, planning, or I don ’ t know, I probably can ’ t even count the hours that I spent, the Saturdays and Sundays, grading papers. It was tough. AG : She estimates she was putting in close to 100 hours a week – 100! CC : It was bananas. The most difficult thing was kind of this feeling of, like, being trapped in this system where you ’ re on, like, a hamster wheel. You ’ re kind of, like, doing a lot and trying really hard, but is it really changing anything? AG : That feeling Conrey had is important. Research shows that one of the major causes of **burnout** is a lack of efficacy, feeling incapable of making an impact. CC : I think I felt the most burnt out when I felt like all the hours that I was putting in, I really couldn ’ t tell if it was making a difference. And are you really moving the needle for kids who really deserve more? AG : How much of the **burnout** experience was a sense of guilt? Like you were letting these students down. CC : Yeah, I think that ’ s at the heart of it, for sure. We call it Catholic guilt in my family, but yes. I mean, how can you sit there and watch kids who have so much potential just flounder? That just feels like such an immense failure. AG : So what did Conrey do to overcome **burnout**? She worked more. She started a local chapter of a mentoring program and spent her weekends tutoring low-income, high-achieving students. CC : You know, it was really fun, actually. AG : You are an overworked, emotionally-exhausted teacher, and your idea of self-care is to then say,”Alright, I ’ m going to start another organization and volunteer for it.”What kind of sick and twisted **vacation** is that? CC : I know it sounds counterintuitive, but I felt really refreshed, and like I had a new, kind of, purpose and drive, and I found that I actually was a happier person. AG : It almost sounds like it gave you a sense of hope. CC : I think it probably did. And I think that really was a turning point for me. I ended up committing and staying a couple years longer in Philadelphia. AG : Your **antidote** to **burnout** is not necessarily less work. It could be more meaning. What was clever about Conrey ’ s weekend tutoring was that it reconnected her to why she chose teaching in the first place. It enabled her to step out of the grind for a moment and see her impact, which is something you can do in any kind of job. In a study of hundreds of professionals keeping **daily** diaries across multiple industries, the single strongest predictor of engagement at work was a sense of **daily** progress. What that means in **practice** is you don ’ t need huge accomplishments to feel good about your work. What matters most for building efficacy is small wins. Often those small wins come from having a positive impact on others. I found in my research that when people feel **ineffective**, helping others buffers against **burnout**. It makes them feel competent, which leaves them energized rather than exhausted. For Conrey, tutoring high-achieving students on the weekends was full of small wins. She could see their progress quickly, and it rebuilt her confidence that she could help her own students. Let ’ s be clear – I ’ m not suggesting anyone should work a hundred hours a week, let alone add more work on top of it. The point is that when you ’ re burned out, it helps to find some small wins. These days, Conrey is looking for those wins as an elementary school principal. CC : I am fortunate to work for a neighborhood school in Chicago Public Schools. AG : And should I be calling you”Principal Callahan”from here on out? CC : No, that ’ s so awkward, please don ’ t. CC : I really encourage a first-name basis when we ’ re not in front of kids at school. AG : Alright, good. It also feels like I ’ m in trouble if I say that. CC : You ’ re not the only one. AG : Although her hours aren ’ t as crazy now, she still feels burned out once in awhile, but she ’ s managed to maintain efficacy by keeping track of her small wins at work. From attending community events with parents and having hallway talks with teachers, to my favorite **practice** of Conrey ’ s. CC : Go hang out at recess with some kindergartners. AG : Wait, I want adult recess. So I like this concept. CC : Last week at recess, I started playing tag with my kindergartners, and I was in my high heels, so it was probably ridiculous. But it started out with, like, three students, and all of the sudden, 40 students wanted to play tag at recess, and we ’ re all laughing and playing together and just really not a care in the world and having fun. To see that you ’ re so important to them is really inspiring, to be able to go back in and sit down and say,”OK, I ’ m going to work on this really hard thing, or I ’ m going to have this really hard conversation, because I have 40 kids out there who are depending on me for that.”AG : Think about your experiences with **burnout**. What ’ s your version of hanging out with kids at recess? For an architect, it might be strolling by a building you designed to see how people are living or working in it. For a business manager, it could be an individual check-in with a team member to ask how they ’ re doing and how you can help them. For me, it ’ s carving out a little time each week to send notes of appreciation to colleagues and recommend students for jobs. Small wins aren ’ t hard to find if you look for them. But as Anne Helen Petersen knows all too well, sometimes you land in a situation at work where it feels like you ’ re just on a treadmill. So tell me this. There are days in my job where I feel like I have too much to do, and there are a lot of people depending on me, and I can ’ t respond to all of them as fast as I would like. And I keep just plowing through emails to try to get to inbox zero. Am I burned out? AHP : I mean, that, I think, is a great example of what you would call **stress**. AG : Yes. And I would just like to learn to not even find that **stressful**. This my stance on it. As opposed to saying,”I ’ m burned out, I can ’ t do this.”AHP : Right. So when, I ’ m frantically going between my work and personal email trying to, like, respond to all these things and, like, make lists and that sort of thing, and like actually not doing anything, I ’ m like,”This is a weird behavior that you ’ re doing.”Having realized that I ’ m burnt out, I recognize **burnout** behaviors. AG : So it seems then that part of what you like about **burnout** is that it helps you recognize that this is a problem embedded in your circumstances. As opposed to a problem in your head. AHP : Yeah, that ' s a great way of putting it. AG : Annie and I landed on a key distinction. **Burnout** isn ' t just feeling emotionally exhausted for a little while. It ' s feeling that way often enough or intensely enough that it starts to interfere with your functioning, even if you don ' t actually quit. But unlike **stress**, **burnout** doesn ' t necessarily originate from the way you react to an event. It originates in particular kinds of jobs, in particular kinds of **workplaces**. Your organization probably has some **practices** that fuel a culture of **burnout**. So it shouldn ' t be up to you as an individual to conquer **burnout** alone. Your **workplace** is often responsible for the causes, so they should take responsibility for the cures. More on that after the break. OK, this is going to be a different kind of **ad**. I ' ve played a personal role in selecting the sponsors for this podcast, because they all have interesting cultures of their own. Today, we ' re going inside the **workplace** at **SAP**. Felicia Shafiq : I hadn ' t played volleyball or really any sports before I got to **SAP**. AG : That ' s Felicia Shafiq, she ' s a product support manager. She ' s pretty tall, so when she started in 2005, her coworkers recruited her to join their **SAP** volleyball league. FS : It was all paid for, so why not, right? That ' s how it all started, and I fell in love with the sport. AG : Many leaders fear that hobbies are a distraction from work, but plenty of evidence shows the opposite. Leisure activities can energize us and increase our confidence and engagement at work, especially if they ' re very different from our jobs. Felicia **practiced** with her teammates twice a week after work, and competed in tournaments as often as she could. But that all changed one morning when she woke up feeling very sick. FS : I woke up really shivering. And I thought I had the **flu**, but I had a volleyball tournament. By about 11 o ' clock that same night, I wasn ' t able to breathe well, and I was rushed to the hospital and almost immediately put on life support. AG : Felicia slipped into a coma. Days passed, then a week, then two. When Felicia woke up, she was completely disoriented. FS : My first thought was"Where am I? I thought I just had the **flu**. Why are my hands and feet bandaged? And when can I get back to work?"I needed to be able to support not only myself, but my family. AG : Felicia had contracted a severe strain of pneumonia, which poisoned her bloodstream. Her body pulled all her blood from her extremities, leaving her hands and feet to die slowly. FS : The good news is that we won the volleyball tournament. AG : Gradually, her hands got better, but her legs still weren ' t showing any signs of healing. FS : The choice to have my legs amputated was mine. I decided that I wanted to move on with my life, that I would start becoming independent again. AG : She spent another three years at home in recovery, but she wasn ' t alone in the process. Her managers and coworkers from **SAP** were there to offer her support every step of the way. FS : What didn ' t they do? The biggest thing that they did was they reminded me that my job was safe. It was there when I was ready to come back, and that all I needed to focus on was my recovery. I can ' t even begin to explain the burden it took off my shoulders. AG : After almost four long years of recovery and rehabilitation, Felicia was finally able to meet her goal and return to work. But there was still another dream that she wanted to get back to. After her legs were amputated, Felicia thought she would never play volleyball again. Then a friend suggested she try sitting volleyball, an adapted version for players with different abilities. FS : It felt so amazing. It felt like I was home. Like this is where I was meant to be. AG : Felicia decided to try out for the Canadian national team, and she made it. FS : We got a bronze, and we qualified for the Paralympics in Rio. That was surreal. AG : And she says none of this would have been possible without everyone at **SAP** cheering her on. FS : Playing any sport, dedicating time to train, to travel with the team, is so very important. A great way that **SAP** has supported me in this is by giving me the time to compete. AG : Today, four years later, their encouragement hasn ’ t stopped. Felicia and her team will soon be competing at the 2020 Paralympics in Tokyo. FS : But if volleyball ends, I know that **SAP** will continue to support me in whatever I want to do. I think it ’ s just the culture that we ’ ve created here. AG : It ’ s no surprise that **SAP** earned many Employer of Choice awards in 2019. Check out careers@sap and visit **sap**. com / worklife to learn about experience management solutions for your employees and customers. **Burnout** isn ’ t an individual problem. It ’ s an **organizational** problem, and it ’ s contagious. Studies of nurses and doctors show that when one person becomes emotionally exhausted, their coworkers are more likely to catch it. And it affects the people you serve as well. When teachers are burned out, their students show elevated cortisol, a classic **stress** response. More and more organizations are dealing with **burnout** by offering mindfulness training. It might help with some of the symptoms, but it doesn ’ t get to the root of the problem. If more than one person is facing **burnout**, deep breathing is a Band-Aid, not a repair. You need to change the structure of the job, or the culture of the organization. If you want to tackle **burnout** at scale, my favorite model is demand, control, support. In a nutshell, it says there are three ways to prevent emotional **exhaustion**. Reduce the demands of the job. Give people more control to handle them. And provide more support to help people cope. To see how demand, control, support works, we went to a couple places that deal with extreme emotional **exhaustion**, where **burnout** is often part of the job and the culture. Adrienne Boissy : You know, we forget that culture is also a medium that bacteria and stuff grow on. AG : Adrienne Boissy is a neurologist at one of the nation ’ s foremost healthcare institutions, the Cleveland Clinic. Adrienne holds the title of Chief Experience Officer. She ’ s in charge of making sure that patients get excellent care. And from her first days on the job, she could see that physicians couldn ’ t provide that care when they were burning out. She ’ s become an expert on the topic. She spent the past decade studying what causes **burnout**. AB : You heard people say,”I went into healthcare to take care of people, and there ’ s all these things in the way of doing that. Either there ’ s too many patients in the ER that are waiting and piling up as I ’ m trying to spend time with this one, or my internal medicine appointments are getting shorter and shorter, or I ’ m spending all my time in the electronic health record after hours.”And so it became apparent that there ’ s this value conflict for people that does feel like moral **distress** as a driver of **burnout**. And that, for me, is where the **burnout** story began. AG : So Adrienne set out to find a cure. Healthcare professionals are among the few occupational groups whose emotional **exhaustion** rates rival teachers ’. Nearly half of physicians report being burned out. That costs the US economy about 4. 6 billion dollars a year. When doctors and nurses burn out, they ’ re prone to anxiety, depression and self-harm. They ’ re also more likely to face malpractice claims. Remember – demand, control, support. To fight **burnout**, the Cleveland Clinic started by reducing demands. One of the most **frustrating** job demands is electronic overload. Digital demands are overwhelming many people in many jobs. I bet you ’ ve been there, looking at your phone from your bed and feeling like you just can ’ t keep up. There are some **researchers** who put those habits to good use. They can predict **burnout** by analyzing email patterns. Recent studies suggest that the volume of emails you send and receive is unrelated to your emotional **exhaustion**, as long as email is key to doing your job. But if email doesn ’ t really help you get your work done, you feel the burden of being boxed in by your inbox. If you ’ re a surgeon, email is rarely critical to your operations. In hospitals, digital demands come in the dreaded form of electronic health records. AB : There ’ s, like, 30 tabs on the left-hand side, and if you click on one of them, it opens up into another 20-tab menu. AB : I mean, it makes no sense. AG : Oh, the joys of electronic health records. AB : It makes no sense. So actually, we created task forces around better managing all the boxes and minimized even the number of categories or clicks that someone could even have in the electronic health record, as a means to make their life simpler. AG : The demands weren ’ t just digital. Medical professionals waste a lot of time on the phone, too. To eliminate the burden of calling insurance companies to authorize tests and pharmacies to approve refills, The Cleveland Clinic introduced preauthorizations and automated refills. They were able to reduce some of the demands on physicians, but not all of them. Many of the demands are built into the job. You can ’ t stop patients from getting sick and having emergencies. Since they couldn ’ t erase those demands altogether, Adrienne and her colleagues turned to the next component of the demand, control, support model. And they did it in an **unexpected** way, which succeeded in reducing **burnout**. AB : We had taken 40, 000 caregivers offline, trained them in **empathy**. AG : I ’ m ready to be wrong on this one, but it ’ s surprising to me. I don ’ t think I would have ever wanted to train **empathy** as a solution to **burnout**. AB : I had faith, Adam, I had faith. AG : Oh, well, that makes one of us. AB : Well, thanks for the support, Adam. AG : We had to see it for ourselves, so let ’ s go to the Cleveland Clinic. AB : Hopefully, you ’ re in the right place. So this is the ready to communicate, how to enhance – AG : In a big conference room, a dozen practitioners sat in a circle. Mary Beth Modic : Hello, I ’ m Mary Beth Modic. I ’ m a clinical nurse specialist, inpatient. Aparna Prakash : I ’ m an internist at the Solon Family Health Center. Participant 1 : I ’ m a physical therapist. AG : As an icebreaker, they told each other about their first car. Participant 1 : It was a little blue Chevy, and I called her Madame Blueberry. Participant 2 : My first car was a Chrysler Cordoba with crushed velvet interior. Participant 2 : It was wonderful. AG : Then they started to role-play scenarios that were designed to help caregivers have better conversations with patients. Participant 3 : And we ’ ve invented a way to record voices and sounds that you can recall on this device and play anytime you want. Participant 4 : What ’ s a recording? Participant 3 : Well, anything that you say out loud can be recorded. Think of it like an echo. AB : And so we realized all of a sudden, we had created a **platform** to attend to them as human beings. And that became the secret. Don ’ t tell anybody. AB : That became the power of the training that we were offering. AG : I was still skeptical. I needed a clear picture of what **empathy** had to do with demand, control, support. There ’ s actually evidence that **empathy** can contribute to **burnout**. When someone ’ s in pain and you ’ re unable to help, feeling their pain drains you. This is known to happen to physicians who get overwhelmed by **empathy** for patients. So I asked Adrienne why **empathy** has been so powerful. AB : Well, let me give you an example, OK? Maybe that would help. Of a conversation that I might typically hear. This is the doctor talking.”We need to talk about quitting smoking. We talked about this last time and there hasn ’ t been much progress.”Patient :”It ’ s been really hard since my mom died.”Doctor :”Well, if you don ’ t quit, you ’ re going to have another heart attack and you ’ re going to get that lung cancer you ’ re worried about.”AB : How did that one feel for you? AG : Horrible. AB : So the issue, though, is, if you dissect that with the clinician and you were to say,”OK, let ’ s just pause that conversation for a second. Doctor, tell me what your intention was.”And they ’ ll say,”Well, I want the patient to quit smoking.””Yes, tell me why you want the patient to quit smoking.””Well, it ’ s going to hurt their lungs and...””OK, why do you care whether it hurts their lungs?””Well, they told me they don ’ t want to die of lung cancer like their mom did.””OK, yes, but why do you care?””Well, because

I care about them.””OK, Doctor, let ’ s start over.”AB :”Can you just try to say that?”AG : Oh, Adrienne, I love that, because if you go back to the demand, control, support model, I think what I just heard is that by teaching physicians to express **empathy**, you ’ re actually helping them gain a sense of control. That instead of having a **stressful**, **frustrating** patient interaction, instead of dealing with lots of complaints and also feeling like it ’ s sort of out of their hands, whether the patient leaves feeling cared for and satisfied and helped. By expressing concern and doing that in a way that the patient hears it. AB : I think you ’ re spot-on. That is the magic. And you see the clinicians sort of go,”Oh.”You ’ re making it easier. You ’ re literally teaching the clinician, or the doc, the words to use to express their intention of care. And they forget. I can ’ t tell you how many times, over and over and over again, we saw them simply forget to say,”I care about you. I ’ m in this with you. We ’ re going to figure this out together.”AG : And there ’ s the genius of Adrienne ’ s approach. **Empathy** training eased **burnout** at the Cleveland Clinic, because it gave doctors that deeper sense of control. It enabled them to have the kinds of effective, caring conversations with patients that they wanted to have. You may not work in healthcare, but you know what it ’ s like to feel that your tasks are spinning out of your control. And you can find small but meaningful ways to gain more control over your work, or to give that control to your employees. For an executive assistant, it might be some extra flexibility in hours. For a police officer, it could be having a say in which streets to patrol. For a retail associate, it might be a little discretion around which products to sell. But the demands of some jobs are so unpredictable that control is difficult to offer. That ’ s when support becomes especially important. Chris Macklin : Anytime you ’ re taking care of men and women who call 911, or children, they ’ re at their worst. And you ’ re always supposed to be at your best. AG : Chris Macklin was a paramedic in Denver, Colorado for 10 years. The demands of that job took a toll on him. CM : I didn ’ t know how to take care of myself, because I was so burned out and never wanted to work in that role again. AG : He decided to leave that job and become a firefighter. But soon enough, Chris found that firefighting was also a quick road to emotional **exhaustion**. He couldn ’ t control the calls that came in, or the tragedies that endured. CM : So everybody ’ s gone. So open bay. Like, where you can see all the shoes left behind, you can tell that that ’ s where the members jumped on the truck, when the truck got their call. AG : One minute, he ’ d be playing cards in the firehouse. The next minute, the siren would go off and he ’ d be racing onto the truck. Soon, he ’ d be running into a burning building. He **struggled** to control how those demands affected him. CM : You can imagine, if you ’ re always under that kind of **stress**, you train your brain to be hyperaroused or hypervigilant all the time. And that just really leads to that anxiety. And I think that anxiety, that ’ s that **burnout**. AG : So rather than switching roles again, he decided to tackle the problem head-on. Chris is now the wellness manager at South Metro Fire Rescue in Denver, which is a huge organization. They have 29 stations, with more than 800 first responders. It ’ s part of Chris ’ s job to get emotional support to firefighters, which means changing the culture. CM : Culture of the organization, I think, plays a huge role in **burnout**. AG : In a recent study, firefighting units with cultures of care and support had fewer health problems and fewer accidents. But in too many fire departments, that care is missing, because people are afraid to seek support. At South Metro, like in many emergency-response organizations, people felt like they were supposed to be tough. Chris experienced that lack of support firsthand 20 years ago, in the aftermath of the Columbine High School shooting, where he was a first responder. CM : The old science was make you go to a debriefing with people you didn ’ t know, allowing visitors in the room, and then, really, people asking you questions and wanting you to tell your whole story right there on the spot. And that was a horrible experience for me. And what happened was I left the debriefing angry, upset. AG : This old support model brought in professional counselors, who often lacked personal experience with what first responders went through. Research suggested analyzing the incident right away with a counselor often doesn ' t help and can backfire. What people actually need is emotional support from someone who can empathize with their experience, especially since there ' s often a stigma associated with asking for help. CM : We also wanted to normalize that, that everyone ' s reaction is normal, based on their own personal circumstances and where they come from. I think it ' s helping us move our culture to a positive place and maybe lower some of that **burnout**. AG : South Metro created a peer-support program for workers and their families. After a difficult call, like a big fire or a mass shooting, first responders can opt in to what they call"defuse sessions."That ' s defuse as in defusing a bomb. The first defusing session Chris did was for a crew that just responded to an emergency call involving an infant. You could see how much more helpful it was than the old debriefing model. CM : I just asked them what they needed. And so, a couple just wanted to remain out of **service**. They wanted to call their wives, but they all wanted to stay at work, because that was that cohesive unit and where their support network was. But the ambulance crew asked if they could just go to sleep. That speaks to the culture of the organization in recognizing the **stresses** of the job. AG : The **stresses** firefighters endure on the job are probably a lot more extreme than most **workplaces**. But the steps Chris and his colleagues took to support their staff still teach us a lot about what all kinds of organizations can do. If you do client **service** or customer care, you want someone on your team whose job it is to care for the caregivers. If you do knowledge work, you need someone in the organization to track who ' s getting bombarded with requests, so they can get backup on their team. Formal support helps to prevent **burnout**, but ultimately, the culture of informal support matters just as much, if not more, and organizations can take steps to nurture it. At South Metro Station 45, people know where to go for unconditional support. It ' s probably the busiest place in the station, says firefighter Dan Rivers. Dan Rivers : It ' s the only place in the world where you can have a stack of money, 100s, 50, whatever, a stack of money, gold bars – nobody will touch it. But if you have creamer, cookies, cake... So yeah, the kitchen table ' s like your house, it ' s a dropping zone, everybody comes in, just drops their stuff, whether it ' s mentally or emotionally or just stuff, you know? Yeah, this is the nerve center right here. AG : Dan and his crewmates are just back from a call at a residential building **nearby**. With their suits halfway on and radios strapped to their chest, they sit around the heart of the station, an old wooden kitchen table. DR : It ' s got a lot of miles on it, like a lot of us do. AG : That table is a natural place for support, to share good experiences but also bad memories. Of course, the tone is set at the top. One of the key moments at South Metro was at a meeting with over 800 firefighters and paramedics. Several of them stood up to share their **struggles**. Chris says it started when the fire chief publicly stated his commitment to supporting first responders throughout the organization. CM : And our fire chief and operations chief both told all of those men and women that if they raise their hand, they will be taken care of and will never lose their job. And so that was a superpowerful moment for us as an organization. And if you feel safe, if you know that your organization is standing behind you, you can really weather more and perform better. AG : Whether or not you ' re fighting fires, your job can leave you feeling burned out. Reducing the demands, gaining control and finding support shouldn ' t be up to you alone, because

burnout ’ s not an epidemic that ’ s afflicting one generation. It ’ s an **organizational** problem that can affect any generation. If you feel exhausted and your **workplace** isn ’ t doing enough to help, you can help yourself, at least a little, by reflecting on why you ’ re in this job to begin with. What does it mean to you and to the people around you? It ’ s only a starting point, but it can make a difference to remember how your work makes a difference. Next week on WorkLife. Grace Song : I was the lowest-paid employee in my department of 250 people, even though I was the highest-rated performer within my band levels. And it was because I never asked. AG : Negotiation is an art. But there ’ s a science to help you get the results you want without hurting your reputation. WorkLife is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell, Jessica Glazer, Grace Rubenstein, Michelle Quint, Angela Chang and Anna Phelan. This episode was produced by Constanza Gallardo. Our show is mixed by Rick Kwan. Original music by Hansdale Hsu and Allison Leyton-Brown. **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors, Accenture, BetterUp, Hilton and **SAP**. For their research, thanks to Christina Maslach on **burnout**, Teresa Amabile on progress, Karl Weick on small wins, Sabine Sonnentag on helping others, Wilmar Schaufeli and Arnold Bakker on **burnout** contagion, Robert Karasek and colleagues, and Margot van der Doef and Stan Maes on demand, control, support, Mandy O ’ Neill and Nancy Rothbard on cultures of care, and Jonathon Halbesleben and colleagues on firefighter **burnout**. AHP : Millennials are, in a way, incredibly physically **mobile**. You have people who are moving away from their support systems. When you ’ re not around your parents, you don ’ t have that support system. AG : Some of my students have moved away from their parents to escape **burnout**, but that is another conversation. AHP : That is another conversation.

Text Sample 166: POS Dim 2 – 2020 – Score 45.00 – 2020/t004287.txt

Mohamed El-Erian : It terrified me, absolutely terrified me. And it still terrifies me. Adam Grant : Mohamed El-Erian is an economist. And he gets a lot of requests to speak in public. ME : No matter how often I do it, the night before, I will not sleep. AG : What is it that you ’ re afraid of? ME : I ’ m worried about not being able to put a coherent sentence in place. I ’ m still worried about it right now as I ’ m talking to you. AG : Last year, he was asked to give a speech about the global economy at a dinner in New York. When he showed up, Mohamed found out that the host had neglected to mention a few things about the audience. ME : The first one is, they are precious metal traders. AG : You know, people who trade in one of the most volatile, risky **financial** products out there. ME : Which means their attention span is a few minutes at most. And second, he said,”They ’ ve been drinking for the last hour. So they ’ re a bit rowdy.”I said,”OK.””Is there something else?”He said,”Yeah, just one more thing. Last year, the person who did it lost control of the crowd, and the precious metal traders, some of them got up with their bread rolls and threw them onstage.”AG : They threw rolls at the presenter? ME : They did. AG : Ugh! Where they at least soft rolls? ME : They were soft rolls. AG : If someone tells you your audience is drunk and might pelt you with food, you should probably call a **cab**. But Mohamed, who ’ s jittery even when things are going well, who overprepares for everything, did something different. ME : I went onstage and said,”I ’ m **super** anxious right now. And I ’ ve been told that last year, you got impatient with the speaker. And some of you picked up the bread roll and threw it.”AG : ME :”In fact, I ’ m not just anxious, I ’ m scared.”AG

: In that moment, Mohamed didn ’ t hide his insecurities. He admitted them out loud, sharing his real feelings with the audience. When you ’ re nervous about a big performance, whether it ’ s a speech or a job interview, people often say,”Just be yourself.”It ’ s practically gospel in the world of work advice these days.”Be authentic.”But what does it really mean to be authentic? And is it always the right choice? AG : I ’ m Adam Grant. And this is WorkLife, my podcast with TED. I ’ m an **organizational** psychologist. I study how to make work not suck. In this show, I ’ m inviting myself inside the minds of some truly unusual people, because they ’ ve mastered something I wish everyone knew about work. Today – authenticity, and why bringing your whole self to work might be more complicated than you think. Thanks to Accenture for sponsoring this episode. For decades, many **workplaces** expected people to leave parts of themselves at home.”Don ’ t show your anxiety in a performance review.””Don ’ t tell your boss what you really think of her idea.””And don ’ t wear a Hawaiian shirt unless it ’ s Hawaiian Shirt Friday.”But today, in part, thanks to the popularity of self-expression in Silicon Valley companies, there ’ s a growing awareness that people need the freedom to be themselves at work, to be authentic. Recorded speaker 1 : I ’ m going to talk a little bit about the power of authenticity. Recorded speaker 2 : I was asked to speak about authenticity. AG : It kind of feels like we ’ re living in the age of authenticity. Recorded speaker 3 : I ’ m here to tell you that the world needs your authentic self-expression. AG : It ’ s a liberating mantra :”Be yourself. Don ’ t worry about what everyone else thinks.”Recorded speaker 4 : This push for authenticity is awesome. Recorded speaker 5 : I ’ m being authentic. I did not rehearse this. You ’ re about to get 18 minutes of me unplugged. AG : So what exactly is authenticity? It seems like there are about as many variations on the concept as there are people who use it. Here ’ s what it doesn ’ t mean – being a fake or a phony. It ’ s being genuine. As a psychologist, I think about authenticity as erasing the gap between what you think and feel on the inside and what you share on the outside. And that can apply in a few different realms – sharing your genuine feelings, your opinions and your ideas. Look, there ’ s good reason to value authenticity. Evidence shows that when you have the freedom to express your thoughts and emotions, your energy soars and so does your effectiveness. And when you feel you can ’ t be authentic at work, you ’ re more likely to become **stressed** and exhausted. It can be stifling. If you ’ re an entrepreneur pitching a **start-up** or a job candidate pitching yourself, pretending to be who you think others want you to be can make you anxious and hurt your performance. You do want to work somewhere that accepts your authentic self. But expressing that self isn ’ t always straightforward. A while back, psychologists conducted a huge meta-analysis, a study of studies synthesizing data from more than 20, 000 people. It turned out that, on average, the more people focused on being themselves at work, the less successful they were. For reasons we ’ re about to explore, they got lower performance reviews and fewer promotions. Being authentic has some landmines. And I ’ ve developed some guidelines about how to avoid stepping on them. Let ’ s start with one kind of authenticity : emotional vulnerability, letting people see your real feelings, like Mohamed with those metal traders. We all get nervous. Sometimes we ’ re encouraged to fake it till we make it. But Mohamed chose to be real and sprinkled in some humor, too. ME : I said,”I ’ ve been told that last year you took to the bread rolls that you have in front of you and threw it at the speaker. So let me tell you what I ’ m going to do. I ’ ve asked the people sitting in the tables right next to the stage to serve as my human shield.”AG : And it worked! ME : And I ended up speaking for 40 minutes, and there were lots of questions. Can you tell, I still remember it vividly? AG : Why did Mohamed ’ s authenticity resonate with the audience?

Classic and recent experiments suggest that showing vulnerability humanizes you, and people tend to like that. But there ’ s a crucial caveat : it only works if you ’ ve already proven your competence. You were impressive before, but now you ’ re relatable, too. AG : For Mohamed, being authentic worked because he had already established his qualifications. The metal traders knew he wasn ’ t just an economist. He had been a wildly successful trader, then the CEO of a big **financial** company and the chair of President Obama ’ s Global Development Council. Plus, on top of his credentials, Mohamed proved his competence in reading a room with that joke about the front **row** being his shield. If you haven ’ t already demonstrated that you ’ re capable, though, showing vulnerability can have the opposite effect. Take a recent study of lawyers interviewing for jobs. Focusing on expressing themselves only increased their odds of getting a job offer if their resumes had already impressed the interviewer. The only lawyers who benefited from being authentic were the ones who ’ d been ranked in the top 10 percent of candidates going into the interview. And for lawyers in the bottom half of the resume pool, striving to be themselves in the interview decreased their chances of landing jobs. This isn ’ t unique to lawyers. The results were similar for teachers, too. So this is guideline number one : authenticity without boundaries is careless. Be careful that the vulnerability you choose to show doesn ’ t cast doubt on your competence. Notice what Mohamed didn ’ t say. He didn ’ t tell the group he was afraid he would forget his material or stumble over his words. The fear he expressed was about the audience, not his preparation. AG : When we hear,”Be authentic,”many of us hear,”Be more honest,”which is another way of saying,”Don ’ t sugarcoat things. Say what you really think.”That ’ s another kind of authenticity, being **blunt** with your opinions, which you see an awful lot on the internet. People seem to feel like they can say nearly anything online under the guise of,”I ’ m just being authentic.”But what happens when that kind of online commentary is your job? Leah Finnegan : There is one specific incident that kind of haunts me. AG : Leah Finnegan is a journalist. In 2015, she was working at”Gawker,”the notoriously irreverent, gossipy website. It was a place where raw self-expression online was rewarded and encouraged. It was Leah ’ s job to share her unfiltered views. It was a company standard. LF : We felt like we could do whatever we wanted to and people would respond to it. AG : And often they would. The more outrageous the article, the more clicks it received, the better the website performed, and on and on. It was a perfect fit for how Leah saw herself. LF :... a salty bitch who hated everything and was very extreme in her opinions. AG : Usually those opinions were trashing celebrity baby names or articles in prestigious newspapers. But the problem was, that ’ s how Leah treated everything, online and off. LF : During union negotiations at my company, I left a comment on a blog post saying,”Unions suck dick.”That was something I felt was valid to express. And you know, I just thought I was right. AG : Apparently, Leah ’ s parents had had bad experiences with unions, but obviously not everyone agreed with her take. Her comment got a lot of attention. At some point, she googled her name, and she saw herself referred to as Leah”Unions suck dick”Finnegan. LF : That kind of haunts me, that I might have a legacy as a union buster, which is not great. In fact, unions do a lot of good and have been good for the digital media industry. So I regret that. AG : Leah was just being her authentic self. It felt good in the moment, but over time, the reactions from other people got to her. LF : Well, I was angry a lot of the time because I didn ’ t understand why... people didn ’ t accept me as I was, online and in real life. AG : That ’ s when she got a tip : be less authentic. LF : Yes, well, that was advice from my therapist, who I started seeing after I was like,”I ’ m this way, and no one is responding to it the way I want them to.”And it took a lot of inner strength to pull away from that and kind of be like,"I have to work on portraying myself differently."It was kind of like,"You need to think about how other people see you, and it ' s not their problem if they see you a certain way. It ' s your problem."AG : Her therapist wasn ' t advising her to be fake, but to avoid being completely individualistic, to think of others, too. LF : I think that has benefited my life. AG : Sometimes people get so absorbed in expressing their own opinions that they lose sight of how they affect others. In Leah ' s view, focusing less on authenticity has made her more effective. LF : I think I ' ve become a lot more... some might call it"mature"in how I talk to people. You know, because there are consequences. Like, if you ' re just focused on being yourself, you ' re not focused on anyone else. AG : It ' s common to think of authenticity as something that ' s all about me –"How do I express myself in the world?"But other people have to be part of that equation. Authenticity can ' t exist in a vacuum. I found in my research that concern for others is a critical ingredient for effective authenticity. This is guideline number two : authenticity without **empathy** is selfish. I often hear people explain their **actions** by saying,"Well, I was just being myself."That ' s not an excuse for disrespectful behavior. Sure, we should be true to our values, but we might want to consider others ' values, too. Or as David Sedaris puts it : David Sedaris : Be yourself, unless your self is an asshole. Herminia Ibarra : One of the problems with the"bring your whole self to work"framing is that it ' s narcissistic. What about taking an interest in the other person? Herminia Ibarra is an **organizational** behavior professor at London Business School. She ' s a leading expert on authenticity at work. HI : Authentic leadership, which I actually traced in the press and kind of saw the line go up exponentially, really has become a very popular notion. I think we kind of reached **peak** authenticity a few years back. AG : In the past, when we talked about authenticity, we weren ' t usually describing people. HI : The original meaning is from the world of art, that which can be authenticated, meaning, trace back to the person who made it of their own hands. And in leadership, I think this really came out of the number of scandals that hit, starting in the 1980s and ' 90s, and that decline in trust in leaders. What is authentic? What is real? What is true and therefore **worthy** of trust? AG : Early in her own career, when she first started teaching, Herminia **struggled** with authenticity. HI : I did really poorly for a number of years, and everybody was giving me advice about what to do, and of course nothing fit. And what I was struck with was the least helpful feedback was,"Just be yourself, Herminia."And of course that was the worst feedback to get because the problem was I was too much myself, you know, too introverted, too nervous, too academic, too theoretical – all of those things. AG : But when a colleague suggested doing the opposite, that felt wrong, too. HI : He literally said to me,"You have to be a dog and go mark your territory in each the four corners of that amphitheater."AG : In her research, Herminia wanted to explore how we adapt to new roles at work. So she launched a study of how young professionals **navigate** career transitions. She didn ' t start out focusing on authenticity, but she quickly saw that for the participants, finding their authentic selves was a big challenge. HI : That was a study of how young consultants and investment bankers managed the transition away from doing analytical and project work to actually having to be rainmakers and to manage relationships with clients. And so they had to kind of put on an image of something that they felt they were not quite yet. AG : The bankers and consultants were testing out what she called"provisional selves,"and it made some of them uncomfortable. HI : It was a real sense that,"This isn ' t me. I can ' t see myself that way. I ' m going to have to find my own way through this."It was just really a major threat to their sense of self. AG : So they stuck to their existing identities instead of trying new styles of self-presentation that felt unfamiliar to them. As a result, they often **struggled** to adapt to and excel in their new roles. But others embraced it. They ’ d **observed** their senior colleagues being assertive with some clients while gently guiding others. And they ’ d experiment with those provisional selves to make them their own. The most successful young professionals in her study weren ’ t worried about staying true to themselves. They were striving to develop themselves. HI : The knee-jerk reaction is,”I ’ m fixed, this is who I am, and this is just not within the realm of the possible.”And the minute you say to people,”Does being authentic condemn you to being as you always have been?”they get it. You know, identity – it ’ s **super** multifaceted. It ’ s made up of traits, roles, emotions, aspirations – all of these kinds of things. AG : That ’ s a key lesson here. No one has just one authentic self. We have multiple selves. Think about all the identities you hold at work. Your taskmaster self is probably different from your mentor self, your presenter self and your conference call self. And I hope it ’ s different from your drinks-after-work self. You get to choose which selves you express in which situations, and those selves are evolving all the time So as you move into new territory, it helps to test out different selves to see what feels like you. That ’ s what Herminia ended up doing in her own teaching. She tried out different provisional selves, moving around the classroom more and engaging students more in interactive discussions. HI : It changed from being,”I have content to deliver,”to,”I ’ m there to facilitate an experience in which people are open to learning, engage, having fun and participating.”And that was the result of trying these very unnatural tactics. And I learned much more from it than doing anything that vaguely approximated my sense of my authentic self at that point in time. AG : She started to excel when she developed a style that worked for her. HI : So it ’ s always a balance of what stays permanent, what ’ s continuous, what ’ s core to who you are and then what is it that you ’ re going to grow, learn and discover. We come to inflection points in our lives in which we say,”Hey, wait a minute. I ’ m so much more than this, and I want to give some attention to that.”AG : Yet for some people, the effort to develop and express their authentic selves has extra landmines. More on that after the break. AG : OK, this is going to be a different kind of **ad**. I ’ ve played a personal role in selecting the sponsors for this podcast because they all have interesting cultures of their own. Today, we ’ re going inside the **workplace** at Accenture. AG : I meet a lot of people who are pretty into their jobs. To gauge if they might be a little obsessed, I like to ask,”Where ’ s the strangest place you ’ ve checked your work email?”Joanne McMorrow : The ski lift, going up the mountain, with the winds and the snow that was blowing at me. For some strange reason, I felt the need to pull the phone out, pull the gloves off and check the email. AG : This is Joanne McMorrow, a director at Accenture. I think the chairlift phone check firmly lands her in the category of engaged workaholic. So much so, that two years ago, when Joanne began having some pretty odd headaches, she brushed them off. JM : I was working a lot of hours. I had two little kids. There was just work, life. My own health was probably falling to the last of my priorities at the time. AG : At the urging of her wife, Joanne finally met with the doctor, who ordered some tests. When he came back with the results – JM : They didn ’ t need to tell me anything. The X-ray was actually sitting on the X-ray machine, and as clear as day, I could see the tumor. AG : The tumor needed to come out right away. So Joanne was rushed to the emergency room for a craniotomy. When she woke up, she wasn ’ t only worried about her recovery. JM : At the time, I thought I was going to take a week off of work. I know – major brain surgery, a week off work probably to anyone else would seem ridiculous. I was just feeling guilty about taking that much time because I had no preparation to prep my team for this. AG : To her boss and everyone else at Accenture, that was ridiculous. JM : My boss really was the one that sort of put in my head,"You really need to make sure you ' re taking care of yourself and doing what ' s best for you."And it just so happened she was right. AG : There ' s evidence that self-care isn ' t just important for health. It ' s critical to maintaining energy and performance, too. Accenture encouraged Joanne to care for herself as much as she did her job. Joanne realized she needed to make a change. And once she returned to the office, people around her noticed a new Joanne. Stephanie Williams McConnell : Her perspective absolutely changed in terms of,"OK, this feels really important today, but in six months, it ' s not going to feel as important, and let ' s make sure that my health, my well-being, all of those types of things aren ' t being compromised, because in six months, those things will still matter."That ' s Joanne ' s boss, Stephanie Winters McConnell. Note, don ' t get me wrong – Joanne still works hard. JM : I ' m definitely committed, I ' m dedicated, I ' m driven. But I do so in a very intentional, meaningful way. SWM : She has always been a very compassionate and empathetic person, but I saw a significant shift in leading with that first. Not just driving to get things done but ensuring that along the way she ' s taking care of herself, but everybody around her is as well. AG : Joanne is not only living a more balanced life. She ' s also encouraging her team members to do the same. JM : I lead some of my calls in a surprising way and have everyone talk about what have they done for themselves in the past week. Not at work, but what have they done for themselves. So it allows us to hold each other accountable. AG : And when her colleagues run into trouble, Joanne ' s there for them, just like they were for her. JM : Two weeks ago, one of my direct reports called me to share with me that she had just been diagnosed with breast cancer. And I just knew what we needed to do to support her. And just on Friday, she sent me an email saying how grateful she is for the support that she ' s gotten as she heads into this. Having that impact and knowing that in sharing my story and in changing the way that I have that we could be part of her journey that she ' s about to go through and that she feels that support – that ' s probably one of the most important things for me. AG : Accenture is working to become one of the most truly human companies in the digital age. Learn more at Accenture. com / careers. Carmen Medina : I used to describe myself as a heretic, because I ' m a heretic. I like the word because it means someone whose views are different from the prevailing orthodoxy. And that certainly characterized me. AG : Carmen Medina spent years as a heretic, not in a religious context, but in a place almost as strict – the CIA. CM : I was never a spy. I did travel to Iraq and Afghanistan. I think Afghanistan once and Iraq twice. And riding the helicopters at night – open helicopters, by the way – flying at rooftop level to get from the airport to the Green Zone – I thought was pretty exciting and not what I expected to be doing when I was a little kid. AG : Carmen had already worked for the CIA for more than a decade when the Cold War ended in 1989, and the agency started undergoing a tectonic shift. CM : And I kept getting asked to serve on task forces explicitly about how should the CIA change and modify itself, what should be the new strategic plan now that this focus we had, the Cold War, is ending. So I ' m thinking,"I ' m on a task force. Like, this will – clearly, they must want to know what I really think."And so, I would say in the task force what I really think. AG : Carmen was certain that this newfangled thing called the internet would transform her job and her organization. And she didn ' t hide that belief. CM : I am convinced that the internet is going to change everything about how knowledge organizations do their work. And I think it will be very important for the CIA to figure out how they ' re going to adapt to digital technology. I just thought that, you know, the oceans would part and people would recognize the wisdom of what I was saying, but all I got was resistance and kind of evil stares. The internet was about open information and sharing information and making information freely available. And what ' s the CIA about? Well, not that. AG : This is another type of authenticity : nonconformity. When you act differently than your organization ' s **norms** dictate, like swearing on a team that avoids profanity, publicly calling out errors in an office that favors **privacy** or quietly going home each night when all your coworkers head to happy hour. Sure, you want to be authentic. But what happens when your authentic self runs counter to the culture of your **workplace**? Carmen, a natural heretic in an organization that cherishes strict hierarchies, was about to find out. CM : A senior leader, a woman came up to me and said,"Carmen, you ' re just going to have to censor yourself. You need to stop saying some of the things you ' re saying. You have a bright career in front of you, and you ' re going to ruin it if you keep talking like this."AG : For a while, Carmen did censor herself. She suppressed the part of her that was a heretic. She created what ' s called a"facade of conformity."She put on a mask and pretended to align with the CIA ' s values. Research shows that ' s a great way to burn yourself out. And finally, after nearly two years, the mask slipped. CM :"I ' m just sick and tired of this! All you care about is getting promoted and getting ahead in the organization. And things – the organization is headed in the wrong direction, and you just are not going to say anything."It was my authentic belief. AG : Carmen refused to conform. She couldn ' t let go of this idea that the internet would change the CIA, and she didn ' t adjust the way she expressed her idea, either. Eventually, the CIA sidelined her. CM : Nobody would hire me. I got the really helpful feedback from someone who did not choose me for a position. He says,"You know, your reputation is that you ' re cynical and negative."And I was like,"Wow."AG : Unfortunately, the pressure to conform can be even stronger for members of nondominant groups. That might be true if you ' re a salesperson in an organization dominated by engineers, or a native Southerner in a team of New Yorkers, or more fundamentally, a woman of color in a **workplace** run by white men. For Carmen, being a double minority amplified the challenges of having heretical views. CM : I recognized that the only way I could survive at the CIA was that I had to give up. I had to drop some of the things that I thought were really important for me. So yes, in that sense, I became less authentic. AG : This experience at the CIA knocked the authenticity out of Carmen ' s self-expression. When other people dictate the terms of your authenticity, that ' s the definition of inauthentic. Just ask Alicia Menendez. It ' s happened to her multiple times. Alicia Menendez : I was told that I needed to, like, liven up my look and get some T-shirts, and some leather jackets and wear the things that the kids were wearing. And I was somehow supposed to become that edgy, hip, cool person, instead of just being my, you know, 30-something dorky self. AG : Alicia is now an anchor at MSNBC and the author of"The Likability Trap,"a book about the pressure women face to be who other people want them to be. AM : I sometimes get the sense, and other women I spoke with got the sense, that when they were told,"Be yourself,"it wasn ' t,"Be whoever that is."It was,"Be who I expect that person to be, predicated on your gender, your race, your ethnicity, how old you are, and then show up as my imagined version of who that ought to be."There is often a sense that I would benefit professionally if I were less Latina-ish Liz Lemon and more, to quote Cardi B,"spicy mommy hot tamale."It would be like,"I need this, but smokier."AG : What? People tell you that? AM : Yeah. There is a sense of how a Latina is supposed to self-present, and that can be a wild caricature. Now, I ' ve got to say, Adam, I ' ve also heard it from Latinas on the complete flip, where their English is accented, and they ' re told that they need to tone down their Latina-ness. So... AG : Ugh. AM : It takes lots of forms. AG : If someone is pushing you to fit their idea of your authentic self, especially if it ' s based on stereotypes, something has gone very wrong. AM : I ' m just curious – I mean, have you ever gotten any of this? Have you ever been sort of given the sense – AG : AM :... that you could be something other than the way you are? AG : No. I mean, every once in a while I get suggestions like,"You should move less like a Muppet."AM : AG : But that ' s more objective,"Let ' s try to get you to move and communicate more like a normal person."AM : No one ' s ever told you how to be more professor-ish. AG : No! AM : I think what sometimes gets lost in this conversation when you talk to people who work at companies that are telling them to bring their whole selves to work, they often feel that the company hasn ' t done the work necessary to receive them as they are. And it even seemed to me that the louder and the more insistent an organization was about bringing your whole self to work, almost the less work they ' d done to make that actually possible. AG : People who don ' t fit the dominant culture in an organization learn the hard way that they sometimes have to hide their authentic identities and ideas. They try to fit into that expectation, intentionally or subconsciously, even though it doesn ' t feel right. Let ' s say the conformity pressure you ' re feeling at work is about **norms** and values, not cultural stereotypes. How do you break through that pressure and get heard when people don ' t want to listen? CM : When you ' re a minority, however you might be a minority in an organization, you ' re already a rebel. AG : To get her career back on track at the CIA, Carmen Medina didn ' t have to suppress her identity as an original thinker. But she did need to build up her reputation as someone committed to the **mission**. And so this is guideline number three : authenticity without status and trust is risky. Before you challenge **organizational** culture, it helps to demonstrate your loyalty first. There ' s a **special** kind of status that allows you to get away with nonconformity. In psychology, it ' s called"idiosyncrasy credit."Once you prove your value to your group, you earn a **license** to deviate from expectations. Research backs this up. A series of experiments show that trying to wield power before earning status tends to breed conflict with colleagues and superiors. When Carmen tried to voice her authentic ideas, she hadn ' t accumulated enough idiosyncrasy credit to fight the status quo at the CIA. In order to turn things around, Carmen had to show that she was dedicated to the **mission**. One day, she came across a CIA role focused on keeping classified **publications** secure, one she could excel at. CM : And I went,"That ' s it! If I can just perform the other duties really, really well, then maybe I will get a chance to explore what ' s really my passion."Carmen saw an opportunity to earn some idiosyncrasy credit. By demonstrating her value in the new job, she could show that she was aligned with the CIA ' s core values. CM : And we nailed it. I think that that really restored or changed the organization ' s view of me and restored it as a much more competent person. And I now had authority and status. And so, people were much more receptive to my ideas. AG : With that credibility under her belt, Carmen also reframed her heretical ideas as conforming to the CIA ' s values. CM : So instead of standing on the soapbox and saying,"We have to be completely different, or we ' re going to die!"you know, which is what I did in the late ' 90s, I think you need to say,"The organization has this objective, and I think these ideas could help us achieve these objectives more effectively."And I think you need to start there. AG : She realized there was another part of her authentic identity that she could express. She wasn ' t just a heretic who saw problems. She was also someone who solved them. CM : I would describe myself as a change agent. I use that word. AG : Carmen had set herself up to have real influence. After September 11, 2001, her colleagues were finally forced to recognize the importance of speed in sharing information. Carmen became a major force in ushering the intelligence community in to the digital era, enabling people to share ideas online in real time in a secure way across agencies. She ended up getting promoted all the way up to Deputy Director of Intelligence at the CIA. CM : You ’ re not going to win those battles if you do a frontal assault every time. And sadly, for a lot of organizations, behaving in your authentic way can be perceived as a frontal attack. And I just think that ’ s horrible. But that is still the world that we live in. AG : Look, you shouldn ’ t have to hide your opinions and emotions at work or your ideas and identities. But it doesn ’ t hurt to reflect on the many different selves that are already part of who you are and the selves you might become as you evolve. It makes sense to be thoughtful about which ones you share and when, where and how you express them. After all, you want to be authentic, but you also want to be professional, to set smart boundaries, show **empathy** and not just challenge **norms**, but earn the **license** to change them. You don ’ t have to bring your whole self to work. I think effective authenticity is more about bringing your best selves to work, the ones that bring out the best in others, too. Next time on WorkLife. Person : You know, 40 seconds of interaction, a positive, caring interaction has measurable impacts on both people. Forty seconds. Forty seconds! AG : Ways to tackle loneliness at work. WorkLife is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell, Constanza Gallardo, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. This episode was produced by Jessica Glazer. Our show is mixed by Rick Kwan. Original music by Hahnsdale Hsu and Allison Leyton-Brown. **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors, Accenture, BetterUp, Hilton and **SAP**. For their research, thanks to Patricia Hewlin on facades of conformity ; David Day and Deidra Schleicher on self-monitoring ; Elliot Aronson and colleagues on the pratfall effect ; Kerry Gibson and colleagues on the risks of vulnerability ; Dan Cable and Virginia Kay on the double-edged sword of self-verification ; Jonathan Evans and colleagues on self-deprecating jokes ; Edwin Hollander on idiosyncrasy credit ; and Alison Fragale, Nate Fast and colleagues on power without status. AG : I hear a lot of leaders say,”Be authentic! Bring your whole self to work!”What would you encourage them to say instead? HI : Don ’ t use platitudes. That ’ s the first thing I would say, is don ’ t. AG : HI : Honestly! Authenticity also has a sense of uniqueness. Everybody and their brother has said exactly the same thing. AG : HI : I ’ m not advising leaders to not be authentic. I ’ m advising them to maybe be a little bit more original in what they want to convey to others. AG : I think that ’ s great advice.

Text Sample 167: POS Dim 2 – 2020 – Score 44.00 – 2020/t004295.txt

Grace Song : I ’ m one of those people that feel responsible for other people ’ s feelings, which is terrible. And I just made up all these things in my head about how people would respond if I would ask for things. So I ’ m a people-pleaser. Adam Grant : This is Grace Song. Her drive to make others happy makes her one of the nicest people I know, but it ’ s also occasionally made her a little timid. GS : I was a person where I ’ d go into shops and people would convince me to buy clothes and I would just buy them because I would feel bad and then return them later. AG : Wow, that is a whole new level of agreeableness. GS : I just was always afraid to kind of ask and just have the person be angry or upset. AG : When she finished college, she was grateful to land a great marketing job at a good company. Grace was consistently a star performer in her department. She got promoted five times over five years, but one day, she learned something upsetting. GS : Somebody told me that I was the lowest-paid employee in my department of 250 people, even though I was the highest-rated performer within my band levels, and I was promoted consistently every year, but just never asked for a salary increase. AG : How many years, looking back, were you underpaid? GS : It was six years. Six freaking years. AG : Some people avoid negotiating. Others **struggle** at it, and even good negotiators have bad habits. It ' s time to change that. I ' m Adam Grant, and this is WorkLife, my podcast with TED. I ' m an **organizational** psychologist. I study how to make work not suck. In this show, I ' m inviting myself inside the minds of some truly unusual people, because they ' ve mastered something I wish everyone knew about work. Today, negotiating. It doesn ' t have to destroy your relationships. In fact, you can actually use it to strengthen them. This episode is sponsored by BetterUp. When they get a job offer, more than half of American workers don ' t negotiate at all. If you ' re one of those people, estimates suggest that over your career, it can cost you an average of 750, 000 dollars. Grace can relate to this. GS : I would **dread** negotiating. I was 100 percent a pushover. AG : I know the feeling. Back when I was 18, I was hired to sell **ads** for the"Let ' s Go"travel books. I was supposed to call up last year ' s clients and convince them to renew their **ads**. I was terrified. How was I supposed to negotiate with marketing execs more than twice my age? Our renewal rate was 95 percent, but by the end of the first week, I hadn ' t gotten a single renewal and I ' d promised a refund to three clients who were unhappy with their **ads** for the prior year. I think that made me the first person in the 40-year history of the company to lose money that was already on the books. The technical term for this is agreement **bias**, and it happens when you ' re so agreeable that you accept a bad deal instead of pushing for a better one. Out of desperation, I took a crash course in negotiation. I learned how to disagree without being disagreeable. I studied evidence that negotiations hinge on two big priorities : results and relationships, but many people make the mistake of focusing on one and neglecting the other. It was so helpful that when I became a professor, I signed up to start teaching negotiations. My goal was to help students learn that it ' s possible to get the results they want without hurting their relationships. When Grace walked into my class at Wharton, she reminded me a lot of my former self. For as long as she could remember, Grace had been hesitant to negotiate. GS : Being raised by Korean parents, from their perspective, they ' re like, don ' t rock the boat. Like don ' t jeopardize that. Just be appreciative of getting that steady paycheck and be grateful for what you get. So I think that was the mentality that I had. AG : It became clear that Grace was buying into some common myths about negotiating which made it seem like results and relationships had to be either or. After her six years of being underpaid, Grace finally asked for a raise, but she left it to her boss to make the first offer. If you ' re like most people, you believe that ' s the way to get the best results. You don ' t want to tip your hand before you see the other person ' s cards, but the evidence suggests that ' s backward. First offers serve as anchors. They set the hook for the negotiation, and it ' s hard to escape the pull. One comprehensive analysis of negotiation experiment showed that every dollar higher in the first offer translates into about 50 cents more in the final agreement, but in a job offer you don ' t always have the chance to anchor. A few years after taking my class, Grace got an offer for a new consulting role at a big tech company, and the salary was well below what she expected. GS : And it was funny because I was talking to my brother about it and he was like, just take it. AG : But by then, Grace knew that a counter-anchor makes a difference. The final agreement often lands somewhere near the midpoint of the first two offers, so she countered. GS : I think I said something like,"I ' m so excited to work for the team, I ' m **super** eager to do X, Y and Z, but to be honest, the rate that was communicated was much lower than I had expected. So I would love to talk more about this."AG : This brings us to a second myth about getting results. People think they have to ask for exactly what they want, but that ' s not what Grace did. She put multiple counters on the table. GS : I think I anchored with three different numbers. So in a way, it was a range. AG : This strategy felt friendlier to Grace than coming in with a hardball demand, and it ' s actually been demonstrated to get better results. Let ' s say you ' re negotiating your salary and your target is 60, 000 dollars. Instead of anchoring there, experiments show that you ' re better off asking for a range of 60, 000 to 65, 000. The range allows you to signal a bottom line without having to disclose one, and it makes you look more flexible than if you anchor with a single number. People are concerned about insulting you by going below your range, so they ' re more likely to counter within it. You can take this a step further if you ' re negotiating multiple issues. Put multiple offers on the table. This might sound counterintuitive if you ' re used to sticking with a single set of terms, but research shows you ' re better off working from what ' s called a multiple equivalent simultaneous offer. Say that five times fast. For example, if you care about salary, bonus and location, you present one offer with a high salary, but small bonus and unpopular location and another with a more moderate salary and a bigger bonus and more desirable location. It increases the odds that both sides find a combination that works for them. Of course, the credibility of your offers depends on having a legitimate rationale to back them up. Grace had done her homework on the numbers. GS : I did a lot of research, also kind of talked about why getting a higher rate was important for me. It ' s sometimes in your best interest to communicate information and come to the table with information that you ' re willing to share. AG : Grace didn ' t just succeed in getting the top of her salary range, she also negotiated fewer hours and a broader role that served her manager ' s needs, too. In doing so, Grace busted a third myth about getting results. Asking doesn ' t have to be a selfish act. Research shows that, sadly, women are often at a disadvantage in negotiations. In many cultures, they ' re expected to be caring and communal and modest. When women negotiate, they often face a backlash, because their assertiveness violates these unfair gender stereotypes. GS : Women should be quiet and they shouldn ' t speak up, and, you know, you should be a dutiful daughter, not **complain**, and it was the same way I felt about being Asian. You know, like I feel like there is a stereotype that Asians don ' t speak up or ask for things or **complain**. AG : That ' s the bad news. Stereotypes can hold people back. The good news is that there ' s a way to overcome these stereotypes. Extensive evidence reveals that when women negotiate on behalf of others, the backlash vanishes. Now being tough is an act of caring. So when you negotiate, think about who you ' re representing. It might be your family who ' s depending on your salary. It could be your mentee who will take the job after you. For Grace, it was a group she identified with. GS : I didn ' t want to be that statistic of the woman that didn ' t ask. AG : Explaining who you ' re representing makes it clear that you ' re not being selfish. It ' s called a relational account where you describe how the request you ' re making is going to benefit other people, not just you. But what if you ' re negotiating your salary and it ' s not gonna help someone else? You can still highlight how you ' ll use your negotiation skills to benefit your employer in the future. You might say something like,"I ' m going to represent myself with the same high aspirations and integrity that I ' ll bring to the table when representing you."GS : I ' d always thought of negotiation as this zero-sum game, like only one party could win and it was kind of like a duel to the death. AG : But now, it felt different. GS : During the negotiation, it felt really good because it really felt like we were kind of solving this puzzle together. AG : I love that description that good negotiation is like solving a puzzle together, and with her more collaborative approach, today Grace doesn ’ t see herself as a pushover. In fact, she negotiates all kinds of things all the time. She even negotiated a discount on a couch that she loved. GS : I think I saved like 20 percent. I ’ m a creative negotiator. Collaborative negotiator. AG : Collaborative is not exactly how I ’ d describe another student I ’ d had just a few years before Grace. He was the opposite of a pushover, a steamroller. He wasn ’ t interested in solving a puzzle together. Jason Botterill : I was a very aggressive negotiator. I wanted to win. AG : Steamrolling was in Jason Botterill ’ s blood. For a while, he was a pro hockey player. He was a bruiser. After one too many concussions, his career was cut short, and he decided to get into hockey management. When Jason **enrolled** in my negotiation class, I quickly saw that he was just as tough off the ice. JB : Well, I think it was similar to my playing days. You ’ ll do a little bit of a hook or a **penalty** on the ice and the ref doesn ’ t see it, no issues with it and stuff. You ’ re just trying to win. I brought some of that competitiveness to the negotiation standpoint right off the bat. AG : When Jason came to class every week, he would do intense role plays negotiating with his peers. At the end of the semester, I had my entire negotiation class vote on awards like”most cooperative”and”most creative.”Jason was the landslide winner for a different award,”most ruthless.”JB : I remember that moment vividly. AG : He was so aggressive in class that word got out and students in a different class also nominated him for most ruthless, even though they ’ d never negotiated with him. A few years later, I interviewed him about it, and he took it well. Heads up, I recorded this before I even had an iPhone, so the quality ’ s fuzzy. JB : I won the Stanley Cup and I ’ ve also won a national championship in college, but that was by far my highlight of my career – winning”most ruthless”negotiator. No, but how I won it was a situation where I wanted to win at all costs and really wanted just to destroy my competitor. AG : When he graduated, Jason got a management role with the Pittsburgh Penguins hockey franchise. His job was to supervise contract negotiations, and he kept right on steamrolling. JB : I wanted to prove to the people I worked with that I was ready to handle the responsibilities in a job. I wanted to prove to people and agents within the industry that I was gonna be tough enough and aggressive enough to be able to handle myself. AG : What were some of the more aggressive things that you would say to kick off a negotiation or to drive home your point? JB : Well, I think I sometimes would certainly just – my offers were too low to begin with. AG : How much were you lowballing by? JB : Let ’ s see, I believe my initial offer was – it ended up probably being about 1 million dollars per year lower than what we ended up settling on. AG : Wow, and the agent just thought that was ridiculous? JB : I don ’ t think he even viewed it as a first offer. AG : Did you ever lose a player you wanted? JB : We certainly lost a couple players. One agent in particular that we were trying to work on a deal for one of our young players in Pittsburgh, the agent too went to my general manager at the time and **complained** a little bit about my stance and where we ’ re at. AG : Wait, you had an agent **complain** to your boss about you? JB : Yep, had that happen, and he just said that I was being very aggressive. AG : In class, Jason could laugh about being ruthless. It was just a role play, but now his career was on the line. Being a steamroller was putting his future in jeopardy. JB : Maybe I had some success with it at the start, but I also learned that for long-term relationships, it wasn ’ t gonna be a recipe for success. AG : With his steamrolling style, Jason was looking out for himself, but it turns out there ’ s actually a lot of value for both relationships and results in helping the other party benefit. In one of my favorite studies, **researchers** found that the more intelligent negotiators were, the better their counterparts did. Smart negotiators found creative ways of helping the other side that cost them nothing. This busts a fourth myth of negotiating. You don ' t have to lose to help the other person win. Decades of evidence show that great negotiators don ' t just care about **maximizing** their own success, they ' re also motivated to help the other person succeed. Jason realized that if he wanted results, he had to help players and agents get results too, and that commitment became its own kind of rush. JB : You ' re just excited to get deals done and sometimes it ' s hard, sometimes it takes a while. You know, you ' re handing, 22-, 23-, 24-year-old kids 25 million dollars, 75 million dollars, 80 million dollars, and it can get very emotional and it ' s fun to see these players being rewarded. AG : Well, I mean, it ' s really cool to hear you actually thinking about how it ' s gonna help them as opposed to just, OK, what ' s it going to do for me? JB : I think I ' ve just found the more creative you can get, the more options you can present out there, it actually gets the dialog going that much more and you truly understand what ' s important for the player and the agent. AG : One of Jason ' s first opportunities to try it out was in Pittsburgh with a player who was recovering from shoulder surgery. JB : The agent knew that he didn ' t have a lot of leverage and I remember pushing really hard on salary and it was below what players where he was drafted were receiving, and we came to an agreement, but he was certainly very disappointed at the agent. AG : But then, Jason had a change of heart. JB : And after sleeping on it the following day, I actually believed in his rationale and we increased salary a little bit, and I think it was a situation that improved our long-term relationship and it also went to show me just, hey, it ' s important to push hard and be strong for your organization from a negotiation standpoint and there ' s a certain limit to be fair and to make sure that both parties come away from the discussion excited about the deal. AG : That ' s kind of shocking to hear. You had a deal in place, you slept on it, and then you said,"You know what? I ' m gonna make another concession, even though I don ' t need to. JB : And I think it helped our relationship with the player. In future negotiations with him, he was very open with me. He was very to the point on what he was looking for. I was very to the point, and most of our conversations and negotiations moved on very smoothly from there. AG : The lesson was that even if he managed to win this one negotiation, it might have unintended consequences for a whole series of negotiations in the future. In 2017, Jason became the general manager of the Buffalo Sabres. He ' s one of the youngest NHL GMs ever, and he attributes a lot of his success to his shift in negotiating style and strategy. JB : I used to be a hard-line negotiator, and now what I strive for is to be a creative negotiator. AG : Jason and Grace started at opposite extremes. He was a steamroller who was all about results. She was a pushover who was all about relationships, but I was thrilled to see them land in the same place. They both learned that building relationships is a way of getting results, that before you claim value, you need to create value, but many times we don ' t know what the person across the table values. To figure that out, we ' re gonna dive into one of the most important negotiations in modern history right after the break. OK, this is going to be a different kind of **ad**. I ' ve played a personal role in selecting the sponsors for this podcast because they all have interesting cultures of their own. Today, we ' re going inside the **workplace** at BetterUp. Alexi Robichaux : I remember vividly driving down in my little Prius down to Palo Alto and on numerous occasions being like,"This is so scary, so intimidating."Do I need to just pull over to vomit? AG : That ' s Alexi Robichaux. When he was 26, the startup he worked for was acquired by a big company. Almost overnight, he became director of product development. He found himself giving regular presentations to senior executives, and every time he expected to bomb. AR : I was completely underskilled for the task at hand and there was a restroom downstairs. It was the classic cold water on the face, feeling lightheaded and feeling nauseous, hanging over the sink for dear life. AG : Alexi was afraid he could get fired at any moment for saying the wrong thing, so he made sure to keep saying yes, even to things he didn ’ t feel qualified for. He kept waking up at 4am sweating. His doctor told him he had symptoms of hypertension. AR : My doctor ’ s like, this is weird. You ’ re very young, you ’ re in great shape. There ’ s something wrong here. Like you need to make some adjustments. I thought, hey, this place is the problem, right? I just need a new place and I ’ ll be fine and I remember telling a friend,”If I ’ m not out of this company by January 1 of next year, I want you to come find me and I want you to punch me in the face.”AG : Alexi did leave and started his own company, but the **stress** and self-doubt didn ’ t go away. AR : Maybe I just can ’ t be a founder CEO. I knew I was great as a number two. Maybe I just **gotta** make my peace with that. AG : He had hoped being his own boss would solve the problem. AR : I did that for about three months and realized, nope, I am the problem. AG : So he decided to try a professional coach. Her name was Heidi. AR : I learned that I am a terrible overaccommodator. She would just keep pointing out very gently, but very powerfully you didn ’ t share with the person what you actually thought or you didn ’ t share with the person how that bothered you. There are really nice ways, polite ways to still communicate a really powerful, emotional point. AG : It was a huge realization for Alexi. Up until then, he ’ d been bottling up these issues in his head, convinced that he needed to solve all his problems on his own because he thought that ’ s what strong leaders do. AR : And I think that ’ s the great lie. I think any great human who ’ s accomplished anything had an ecosystem of people supporting them. AG : Alexi knew that pro athletes and actors always have coaches and some executives do, too. Experiments show that when managers work with coaches, their performance improves and so does their confidence and resilience. Alexi wanted to bring more coaches like Heidi to professionals everywhere, no matter what level they were in their organization. Today, Alexi is the cofounder and CEO of BetterUp, a leading **mobile** coaching **platform** where tens of thousands of professionals around the world are getting coached everyday. They report that it enhances their job performance and helps them make progress toward their goals. AR : I have a BetterUp coach. She helped me realize that I kind of have these automatic negative thoughts, and now I have developed some skill to be like,”OK, hit pause.”That ’ s an automatic negative thought. It may or may not be true. Let ’ s evaluate that really rationally at a different time. AG : Alexi doesn ’ t get nauseous before big meetings anymore. He ’ s still working regularly with Heidi and with two other coaches. AR : Yeah, they ’ re like my pit crew. They keep me going. AG : With the help of his coaches, Alexi has successfully taken his vision of bringing professional coaching to the masses and made it into reality. He now leads a company of 250 employees with a global network of 2, 000 coaches. He ’ s learned to communicate more openly and ruminate less on his own. Just like in sports, it takes **practice**. AR : And that ’ s something that over the past six, seven years, I ’ ve really learned and embraced and cultivated and I like to think that if we hadn ’ t done that, BetterUp wouldn ’ t be here today. AG : I joined the science advisory board at BetterUp, because coaches have been fundamental in helping me at every point in my career. I believe everyone should have a coach in their corner. If you ’ d like to work with greater clarity, purpose and passion, you can get a free **trial** with BetterUp at betterup. com / worklife. I ’ ve just arrived at the King of Prussia mall, and I ’ m actually here to see if I can negotiate either a discount on a Cinnabon or a free Cinnabon, which would be obviously better. For years, I ’ ve sent my students out to negotiate something non-negotiable. They have to go into a retail **setting** and try to get something free, and they can ’ t mention it ’ s a class assignment. I want them to get in the habit of negotiating and flexing their creative muscles, but one day, they turned the tables and challenged me to do it myself. I failed to negotiate my way out of it, so I found myself heading to the mall. My wife likes Cinnabon and tacos, so I was hoping to surprise her with something creative. I ’ m feeling a little bit of the butterflies right now. Hi there, how ’ s it going? Anthony : Welcome to Cinnabon, how can I help you? AG : Thank you, I ’ m Adam. Anthony : Hello, Adam, my name ’ s Anthony. AG : What do you like most about working here? Anthony : The environment is very nice. Mini buns are a ton of fun to make. AG : So the smell doesn ’ t get old? Anthony : Oh no, it doesn ’ t get old, but I still have to control myself every so often. AG : Alright, so Anthony, here ’ s my **mission** today. I was wondering if there ’ s any way you could possibly make me a taco. Anthony : A taco? Well, actually, we wouldn ’ t have the ingredients here. AG : Can you make, like, a Cinnabon that looks like a softshell? Anthony : Yeah, I don ’ t see why not. We could probably make a Cinnabon that looks like a softshell. AG : I am a little worried that if I come home with this taco-looking delicacy and then she tastes it and it tastes like a Cinnabon, that she ’ s still gonna be devastated. So the question is can I get you to offer me this for free, just in case, so that I don ’ t have to tell her that I paid for something that was not a taco? Anthony : Hm, well, you know, I am running a business. I definitely want to make you happy. I can definitely give you this for a discount. We ’ re actually running a discount today for packs like these. They ’ re gonna be 4 dollars off today. AG : I was hoping to get both of these free. Is there anything I could offer you that would make that worth your while? Anthony : Hm, well, what do you do? AG : Going into that negotiation, I was thinking about something many people overlook. Negotiating is not the first phase of making a deal. It ’ s the third. The first two are preparation and information exchange. You know now from Grace and Jason ’ s stories why it ’ s so valuable to understand the other party ’ s needs. This is about how to actually do that. So I prepared for Cinnabon by coming up with an unusual request. I wanted to show I wasn ’ t selfish by asking on behalf of my wife, Allison, though when I got home, I found out she had no interest in a Cinnataco. I also figured that sharing some personal information would help with building rapport. In one experiment, randomly assigning people to schmooze for a few minutes before negotiating increased their odds of making a deal. I also spent a few minutes thinking about how I could bring something of value to Anthony. I figured he sometimes had to deal with unpleasant customers and he might get bored when traffic is slow, so I thought I ’ d try to entertain him. Then I invited a little information exchange when I asked how I could make the free Cinnataco worth his while. I ended up offering to help spread the word about the sweet taco. If I were to line up some friends, how many of those would I need to send here before that would earn me a couple free Cinnabons? Anthony : Off the top of my head, probably like eight people. Eight to a dozen people. AG : So I promised to recommend it to 10 friends and... success! Anthony : Yeah, I don ’ t mind giving a couple away for free, yeah, because it ’ s a **special** request and it sounds like a fun experiment to do. AG : When I came back 20 minutes later, Anthony had something ready for me. Anthony : It ’ s a puffy little taco, yes. It has frosting, caramel frosting, and then regular caramel on top as well. AG : It looks like a taco that had a chance to rise in an oven and had a bunch of bread in it. I love the touch of the way the frosting is sprinkled and it almost, the little pieces of Cinnabon bread almost look like taco meat. Alright, how do I eat this? Like a taco, OK. Oh my God. Wow, that is really good. Alright, who else wants to try it? Yeah, that Cinnataco didn ’ t even make it home to Allison because it was delicious. Giving away some free Cinnabons didn ’ t really cost Anthony anything, and his team seemed to enjoy the challenge of making the Cinnataco. One of them even suggested they should put it on their secret menu. So the preparation and information exchange that I had to do wasn ’ t that heavy. But in negotiations at work, people often have to give up something of real value. If the stakes are high and the ask is big, it can be tough to figure out what will get people to say yes, especially if it ’ s a deal that affects the entire planet. Christiana Figueres : When I got the call, I was in total disbelief. I was in shock. Oh my God, I got this responsibility. AG : Christiana Figueres was a diplomat in Costa Rica for decades. In 2010, she landed a big role with the United Nations. Her goal was to get governments on board with the Paris Climate Agreement, a global deal to combat climate change and invest in a low-carbon future. You know, the agreement the US recently left. Now, negotiating with one government is hard enough, but Christiana had a few more parties at the table. CF : Yeah, 195. AG : Oh my gosh. CF : Sovereign governments. AG : So you ’ re not just negotiating with 195 people, they ’ re each representing millions of constituents. CF : Yeah, well, they ’ re representing seven billion people, right, collectively. AG : For five years, Christiana traveled the globe. It was one of the most epic efforts at preparation and information exchange ever. CF : The challenge here was to find between what the globe needs and what the self-enlighted interest is of those who are taking decisions. If you obligate anyone to choose between long-term global need and short-term interest, they will go with the short-term interest. So don ’ t pit those two things against each other. That is by definition a battle that you have lost from the start. AG : So how do you discover the interest of so many different parties? Just like with anchoring, you don ’ t have to wait for the other side to open up. If you start by talking about your interests, you signal that you trust them and the **norm** of reciprocity kicks in. That ’ s when people are more likely to share back. Christiana started by giving some background on where she was coming from. CF : Costa Rica doesn ’ t have any carbon, so we see the world very, very differently. AG : She acknowledged that they might be coming from a different place. She said her goal was to work together toward a global solution. CF : Because let ’ s remember, no single country, big or small, no single country can actually get us to where we need to be on climate change. AG : Then she started asking questions. CF : For me to understand but also to help them understand where that sweet spot was. AG : But that sweet spot was different for each country. One of the best ways to understand the other side is to ask them questions about the future. Not just what their goals are today, but what they ’ re hoping to accomplish in the next few months or even years. There ’ s evidence that expert negotiators dedicate more than twice as much planning time to long-term considerations as their peers. The further you stretch out the time horizon, the less fixed the pie becomes and the more possibilities you can sketch out for helping one another. So Christiana asked each country about their future aspirations. CF : Maybe one of the most surprising is the conversations with Saudi Arabia. AG : Christiana had meetings in Bedouin tents and oil fields outside in the scorching desert heat. CF : And they very well understood that 20, 30, 40 years from now, it was probable that they ’ re going to be living in completely uninhabitable areas because of the heat. They understand that. AG : Once you ’ ve figured out their future goals and concerns, you work backward and ask how they ’ re going to get from where they are now to where they want to be. CF : How are you going to do that? What is it going to be like to work outside? Once that has been internalized, then of course, our human reaction is to immediately start to go,”OK, well, let ’ s find a solution to this.”AG : It turns out Saudi Arabia had one crucial interest that stood in the way of closing the climate deal, their economy. CF : Saudi Arabia, it has one single export which is fossil fuels, oil and gas. They came out of poverty when they discovered fossil fuels, and they are one of the richest nations of the world because of fossil fuels. AG : On one of her many trips there, she visited an oil field and was on a plane with the minister of energy and the head negotiator for Saudi Arabia. One of them pulled out a napkin and started drawing out a plan. CF : The idea of the need to have economic diversification came up during that flight. It was sort of an aha moment for them. AG : Meaning that for Saudi Arabia to move forward with the climate agreement, they had to get support to diversify their economy, to be able to invest in other national exports and have the world support them in it. CF : That famous little napkin on which the negotiator pulled out in the airplane and wrote”economic diversification”and made a little design for me of how that might work, that ’ s why economic diversification is literally embedded into the Paris Agreement as an invitation for Saudi Arabia. AG : Wow, wait, so are you saying one of the key turning points in the Paris negotiations happened on a napkin? CF : Yeah, and I really wish that I had kept that napkin, but – AG : So what was the negotiating strategy that led to that napkin? CF : No, no, no, not a negotiation strategy. I mean, I call it my understanding strategy was to really understand how do they see the world? AG : What kind of negotiator shows up not planning to negotiate? CF : Well, honestly, I didn ’ t see myself as a negotiator, right? I mean, I had 195 negotiators who were negotiating for their country. My role was to be the gardener, right? To be the gardener of this fertile ground of figuring out where is the common ground among all of these different needs? And when it starts to germinate, that is the moment in which you really have to step in and give much more energy to those ideas that are germinating and that can bring people to a common position. AG : And tell me how you felt in Paris when you succeeded in closing the agreement. CF : It was nonetheless a moment of utter elation, and not just me, right? The 5, 000 people who were in that hall, all negotiators, plus NGOs plus scientists plus businesspeople, everyone, and as soon as they heard the gavel, everyone just jumped up and screamed and yelled and hugged each other and cried and applauded and stomped their feet and, you know, it was just a total ruckus of delight. AG : You might not be negotiating major international treaties, but understanding other people ’ s interests is always a core skill. Whether you ’ re negotiating your salary, closing a deal or even just trying to get a discount on a couch, or a free Cinnataco, to get what you want, it helps to figure out what other people want. If you can offer them something they value based in genuine, deep understanding, you don ’ t just end up with better results, you also come away with stronger relationships. Next time, on WorkLife, at your request, a bonus episode with the always provocative therapist and podcast host, Esther Perel. Esther Perel : There is no relationship that doesn ’ t have a power dimension. AG : Who do you think has more power in our relationship here? EP : At this moment, it alternates. AG : Relationships at work. Power, trust, and how to avoid the curse of being a people-pleaser. WorkLife is hosted by me, Adam Grant. The show ’ s produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell, Jessica Glazer, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. This episode was produced by Constanza Gallardo. Our show is mixed by Rick Kwan. Original music by Hahnsdale Hsu and Allison Leyton-Brown. **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors, Accenture, BetterUp, Hilton and **SAP**. For more of Christiana ’ s work, check out her new book,”The Future We Choose,”and her podcast,”Outrage and Optimism.”For their research, thanks to Kenneth Thomas and Ralph

Kilmann on conflict styles, Greg Northcraft and Maggie Neale on anchoring, Adam Galinsky, Thomas Mussweiler, Chris Guthrie and Dan Orr on first offers, Dan Ames and Malia Mason on ranges, Hannah Riley Bowles and Linda Babcock on relational accounts, Bruce Barry and Ray Friedman on intelligence, Carsten De Dreu and colleagues on helping others get what they want, Geoff Leonardelli and colleagues on multiple equivalent simultaneous offers, Neil Rackham and colleagues on long-term planning, and Michael Morris and colleagues on schmoozing. OK, before we leave the mall, let ' s try negotiating at Auntie Anne ' s. I have an unusual request. Employee : OK. AG : It ' s my father-in-law ' s birthday. I was hoping to bring home enough cinnamon sugar nuggets to spell out"happy birthday,"and so I want to be able to tell him that Auntie Anne ' s did this for me for free. Employee : Well, as a son-in-law, I appreciate the effort. I think it ' s a great idea. I can ' t give them for free. We do sell buckets, which probably may be exactly what you ' re looking for. AG : OK, and so how much does that normally cost? Employee : 24. 99. AG : OK, so my goal is to get down to zero. Employee : How about this, if you were to purchase a cup of nuggets, I can maybe get you a bunch of extra, put it in a bucket so there ' s more than enough to spell"happy birthday."AG : Did the father-in-law idea, did that tug at your heartstrings at all? Employee : It did, yeah. I ' m married for seven years, so I think I connect to that, yeah. AG : I ' m bummed. If we ' d come on a different day, I could ' ve done this for our daughter ' s birthday. Would that have been more effective? Employee : I have a 15-month-old at home, so that probably would ' ve been equally as. We have too many connections here. A daughter, a father-in-law and no hair. I think you kind of hit all three marks there.

Text Sample 168: POS Dim 2 – 2020 – Score 42.00 – 2020/t004301.txt

Margaret Atwood : It ' s like going into a very cold lake when you ' ve decided you ' re going to go swimming in it. Adam Grant : This is how one famous writer describes procrastination. **MA** : You put your foot in, you take it out. You put it in again... It ' s still too cold. You think,"Am I going to do this or not? Am I really going to do this?"No, yes, no, yes. That goes on for a while. If you ' re going to do it, you run in screaming. AG : So says Margaret Atwood. She ' s best known as the acclaimed author of"The Handmaid ' s Tale"and has sold many millions of books. But you might not know that she ' s also a self-proclaimed world expert on procrastination. **MA** : Yeah, I ' ve racked up, you know, years and years of it. AG : Because Margaret doesn ' t do anything halfway. **MA** : If you ' re going to do something, might as well be good at it, right? I ' d hate to be a failed procrastinator. AG : She can procrastinate anywhere with the greatest of ease. At home, in a coffee shop, even up in the air. **MA** : I think it ' s always more fun to watch movies on planes than to work. A film called"Captain Underpants"was on the menu. So I was watching"Captain Underpants,"which well repaid my time. And then the plane landed and I forgot that my computer was on it and did not tell my publishers that I had left this computer with all of this correspondence about the heavily embargoed novel,"The Testaments,"on the plane. AG : Oh my gosh. **MA** : Yeah, it was very bad. I won ' t do it again soon. AG : But here ' s the thing. Despite being a world-class procrastinator, Margaret does not turn manuscripts in late. **MA** : No, no, no, no, no. I do not miss **deadlines**. I would consider it dishonorable to miss a **deadline**. AG : How does she manage her procrastination so productively? And can you? I ' m Adam Grant and this is WorkLife, my podcast with TED. I ' m an **organizational** psychologist. I study how to make work not suck. In this show, I ' m inviting myself inside the minds of some truly unusual people, because they ' ve mastered something I wish everyone knew about work. Today, procrastination, and why it ' s not as much of a character flaw or as impossible to overcome as you might think. Thanks to Hilton for sponsoring this episode. Procrastination is intentionally delaying a task that needs to be done even though you know it will come with a cost. Instead of working, you might find yourself watching cat videos on YouTube, looking in your fridge to see if something new has magically appeared in the last 10 minutes, or deciding your productivity problem is that you type too slow, then taking a typing test online to confirm your suspicion, and then taking it over and over to get a better **score**. If you ' re like most people, you first became acquainted with procrastination in school. Somewhere between 80 and 95 percent of students procrastinate. And half of them do it chronically. But it doesn ' t just disappear when you graduate. About 15 to 20 percent of adults are chronic procrastinators. I ' m not one of them. I ' m the opposite, a precrastinator, someone who feels pressure to start tasks immediately and finish them ahead of schedule. Although I did get sucked into that typing test. But if a task is important, I tend to get it done before the **deadline**. And my colleagues tell me that can be annoying. I ' m constantly late to meetings. My excuse?"I was busy finishing another project early."So I ' m pretty fascinated by chronic procrastinators, who live on the opposite extreme. Like Douglas Adams, who wrote"The Hitchhiker ' s Guide to the Galaxy."On a typical writing day, he would sit in the bath for hours, waiting for an idea to come. By the time he got out and got dressed, he often forgot the idea, which would lead him right back to the bathtub. It was so bad that he once had his editor lock him in a hotel suite for several weeks. One that presumably only had a shower. Writers are legendary procrastinators. The question is why. And Margaret Atwood has an answer. **MA** : I see myself as lazy. AG : If you ' re a procrastinator, you might have said the same thing about yourself. Lazy, slacker, undisciplined. But is Margaret Atwood, bestselling author of dozens of books, really lazy? Fuschia Sirois : No, no. That ' s one of the common myths about procrastination, it ' s just people being lazy. AG : Fuschia Sirois is a psychologist in the UK. Her specialty is studying procrastination, and she knows that the root of procrastination is actually something far sneakier than laziness. It ' s not about avoiding work. It ' s about avoiding feelings. More specifically, negative emotions. FS : We say at the core, procrastination is about mood regulation. So a task may elicit lack of confidence, feelings of incompetency, insecurity, fear of failure, anxiety. You put that task aside, and you ' ve just regulated your mood. Now you feel better. It ' s like,"Ah, great. I don ' t have to think about it."AG : You know more about this than most. Do you still procrastinate? FS : Yeah well, you know, I ' m human so yeah, I do procrastinate. The classic thing for me is, you know, I ' ve got this paper to write. And I ' m thinking, this is going to be really hard. And I build it up into something that ' s really huge. And you know, after a couple of days of that, I just kind of go,"Right, I ' m procrastinating. I ' ve just got to get on with this."AG : Everyone procrastinates on something. If you ' re on top of your work, there ' s probably still a task you ' re delaying, even though you know it comes at a cost. FS : You ' ve got to buy a present for your aunt that you only see once a year, for example. And she tends to be really picky. And so now you ' re thinking,"Oh, if I make a mistake, she ' s just going to give me that look."AG : This doesn ' t sound like a hypothetical example. AG : I ' m not going to ask you to name your aunt, but... AG : If you still think you ' re just lazy, here ' s some proof. Take a look at what you do while you ' re procrastinating. Some of those tasks actually take a lot of energy and effort. FS : You ' ll see some, you know, classic chronic procrastinators. They will have the neatest houses. Everything will be organized. All the dishes will be done.

Everything will be clean. But the big looming task that they ’ re supposed to be doing isn ’ t being done. AG : If you ’ re actively doing something else, it ’ s pretty clear that you ’ re not lazy. You ’ re avoiding a task that stirs up negative emotions. And that can have consequences. At work, chronic procrastinators are less productive than their peers. And their health suffers for it. FS : If you ’ re a chronic procrastinator, you have higher levels of **stress**. You have poor sleep quality. You tend to not exercise as much. You might eat more junk food, especially because you ’ re **stressed**. If you ’ re a chronic procrastinator, you ’ ve got difficulty regulating yourself. AG : Which can lead to depression and anxiety. FS : They actually put off seeking help for those mental health issues, which doesn ’ t help either. AG : Oh no, so they meta-procrastinate. FS : Yes, definitely. People feel guilty when they procrastinate. But that guilt doesn ’ t operate in the same way that it does for most people. Guilt could be a motivating emotion. AG : Yeah, it ’ s like the Erma Bombeck line that guilt is the gift that keeps on giving. FS : Yeah, well, for procrastinators, what it gives is more procrastination. AG : Ah, so unfair. Even if you ’ re not a chronic procrastinator, there are certain types of tasks that you might have a habit of postponing. FS : Might be a procrastinogenic environment because... AG : Did you say”procrastinogenic?”FS : Yes, procrastinogenic. AG : What a great phrase. FS : Well, it ’ s an environment that can evoke procrastination. AG : I am absolutely using this as an excuse. It ’ s not me, it ’ s not that I lack willpower. This is just a very procrastinogenic task. FS : Yeah, tasks that don ’ t give you autonomy, that lack structure, and that are ambiguous. AG : Is that why so many writers **struggle** with procrastination? FS : It could be. Because yeah, when you ’ re writing, who ’ s telling you what the next thing is you ’ re supposed to be writing, right? You are. I mean, it brings up uncertainty about yourself. It brings up doubts about whether you know what you ’ re doing, right? We all have that feeling from time to time. AG : You might find yourself procrastinating to avoid anxiety, confusion or boredom. Whichever your flavor of procrastination, psychology points to a couple ways to curb it. For one, you can start by trying to be a little kinder to yourself about your past procrastination. Yep, this actually makes a difference. FS : Our emotions can actually change the way we view the task. AG : Instead of beating yourself up, show yourself a little compassion. **Relieve** the guilt. Research reveals that after students put off studying for an exam, those who forgive themselves are actually less likely to procrastinate on preparing for the next test. Fuschia and her colleagues have found that it helps to remind yourself that you ’ re not the only one suffering from procrastination. It ’ s part of the human condition. Everyone does it on occasion. FS : Sometimes I will just sort of step back and go,”Yep, yep. I ’ m just being like every other procrastinator in the planet.”And you ’ re acknowledging what you ’ re doing, accepting responsibility for it. But you ’ re not feeding back into the negative emotions that probably put you in that place where you wanted to procrastinate in the first place. AG : It turns out that self-compassion is especially hard if you ’ re a neurotic perfectionist, the kind of person who constantly beats yourself up for never doing work that ’ s good enough. If that ’ s you, you might take a cue from productive perfectionists and stop **judging** your work before you ’ ve even produced it. In other words, don ’ t criticize yourself while you ’ re creating. Try waiting until you ’ ve finished developing your ideas before you worry about evaluating them. That ’ s something Margaret Atwood advises. **MA** : The wastepaper basket is your friend. So go ahead, say something. It may be the wrong thing, but you can throw that out and no one will ever read your dumb thing that you ’ ve put on them. AG : Margaret has a long history of procrastinating to escape negative emotions. **MA** : I procrastinated about starting”The Handmaid ’ s Tale.”I procrastinated for about three years. I tried to write a more normal novel instead, because I thought it was just too batty. AG : Too batty, really? **MA** : Yeah, I mean it doesn ' t seem very batty now. But think of when this was. It was in the early ' 80s. Yeah, it just seemed a bit too batty. AG : Margaret wasn ' t just worried that the plot was far-fetched. Her fears actually kept her from writing the book sooner. **MA** : You don ' t know who ' s going to read it. You ' ve got no idea. You don ' t know whether they ' ll like it or not. It ' s not something you can anticipate or have any control over, really. AG : Years ago, when asked to describe her writing routine, Margaret said she would spend the morning procrastinating and worrying. Then plunge into the manuscript in a frenzy of anxiety around three o ' clock, when it looked as though she might not get anything done. It still happens to her sometimes. **MA** : **Scrolling** around on the news certainly can get me sucked in. AG : Luckily, Margaret has come up with a unique strategy for dealing with her procrastination habit. And it ' s a trick that lines up with what some **researchers** recommend. **MA** : I had another name that I grew up with, and that gave me two names. So I had a double identity. So Margaret does the writing and the other one does everything else. AG : Her alter ego ' s name is Peggy. **MA** : It ' s a Scottish diminutive of"Margaret."AG : Do you actually refer to yourself by both identities in your head? **MA** : Absolutely. AG : Seriously? **MA** : Well, you see what a range it gives me. AG : Do you have conversations between Margaret and Peggy? **MA** : No, they lead quite separate lives. Peggy does the laundry. Now there is, of course, some overlap. Because sometimes when Peggy ' s doing the laundry, Margaret is thinking about what is being written. Deciding what to write is done by Margaret. Deciding when to write is sometimes a tug of war. AG : Margaret ' s dual identity strategy isn ' t as strange as it sounds. Psychologists have long **observed** that we have two selves, the want self and the should self. Your want self runs on emotions. It ' s drawn to whatever avoids pain or brings pleasure in the short run. That ' s Margaret watching"Captain Underpants."**MA** : Oh, you ' d rather be watching"Captain Underpants,"let ' s face it. AG : The should self is more concerned with doing the right thing in the long run. That ' s Peggy. **MA** : The ordinary person who walks the dog and eats the bran flakes for breakfast. AG : In the moment, the want self is often stronger. No matter how hard you try to push yourself to do the work you should be doing, it ' s easy to get pulled into the show you want to be binging. Like, maybe,"The Handmaid ' s Tale"? That ' s the bad news. The good news is that the should self is smarter. You can outwit the want self by planning ahead. This is a second strategy for beating procrastination that science teaches us. You don ' t have to worry about resisting temptation if you remove temptation. In college, my roommate Palmer was brilliant at this. Whenever it was time to study for an exam, he would ask me to hide his video games. You ' ve probably done it too. Your should self puts the alarm clock across the room at night so your want self can ' t reach the snooze button in the morning. You prevent procrastination by taking willpower out of the equation. Or maybe your should self announces to the world that you ' re signing off social media so your want self won ' t get sucked back in, which is what Peggy does for Margaret, who loves Twitter and sometimes posts random questions that pop up. **MA** : For instance, I put up a picture of a weird mushroom, and said, what is this? Because I couldn ' t find it. AG : When you ' re **tweeting**, how often does that happen while you ' re writing? Do you actually interrupt yourself or get distracted by social media? **MA** : No, no. No, no way, no. I might get distracted before I take the plunge into the writing burrow but not while I ' m in it. AG : How do you prevent that from happening? Is there a mental firewall of sorts? **MA** : You turn it off. AG : For many people, easier said than done. You just turn it off and that ' s it? **MA** : Just don ' t go there.

AG : It can help to schedule a specific task in your calendar, the same way you schedule meetings. In one experiment, writers were randomly assigned to plan **daily** writing sessions. They were over four times more productive, and they didn ' t lose any creativity. Even scheduling 15 minutes a day was enough to make some progress. That ' s time management. You can also think about timing management. When do you procrastinate? Procrastinators tend to be night owls. The start of the work day is out of sync with their circadian rhythms. If that ' s you, and you have the flexibility, try moving a task you procrastinate to later in the day, when the wants might be less tempting. One of my favorite tactics for outsmarting my want self is a twist on the to-do list. I found out that Margaret does it, too. **MA** : So the list would include everything from"call the tree guy"to, you know,"clean the furnace."If it ' s not on the list, it doesn ' t happen. AG : That ' s her to-do list. But she also has a to-don ' t list, a set of activities to avoid while working. Think about your to-don ' t list. What would you put on it? When I ' m working on my boring procrastinogenic tasks, like reading contracts and proofreading articles, my to-don ' t list includes don ' t play online Scrabble, don ' t turn on the TV – unless I already know what I want to watch, and don ' t **scroll** on social media after posting. Peggy also puts social media on Margaret ' s to-don ' t list. To hold herself accountable, she makes a public commitment, **tweeting** that she ' s signing off to write. **MA** :"I ' m about to write, goodbye."And that ' s about going off social media for a while, and reassuring people that I ' m not dead yet, or possibly disappointing them that I ' m not dead yet. AG : I can ' t imagine that anyone is disappointed. **MA** : They would be even more excited to hear from me if I were dead. They would be really excited then. She came back from the dead. AG : I ' m having a hard time reconciling your self-description of being lazy and procrastinating with your enormous productivity. **MA** : Well, just consider some elementary math. Take the number of years I ' ve been alive and divide it by the number of books I ' ve written. AG : Alright, so you average less than one a year. **MA** : Yes, and some are quite short. AG : And you don ' t feel like that ' s a lot. **MA** : No, it ' s just, you know, they accumulate. AG : Maybe you can start forgiving your want self for procrastinating. Maybe you ' ll succeed at putting some of the should tasks on your calendar and some wants on your to-don ' t list. Still, you can ' t shake the feeling that if only you had more time, you could get more done. But what if the opposite is true? More on that after the break. OK, this is going to be a different kind of **ad**. I played a personal role in selecting the sponsors for this podcast, because they all have interesting cultures of their own. Today we ' re going inside the **workplace** at Hilton. It ' s not uncommon for people to say"My coworkers feel like my family."But I recently met someone who takes that sentiment to a whole new level. Jessica Clingman-Kerns : The DoubleTree by Hilton Sonoma Wine Country is my second home. Everybody here is my family. AG : That ' s Jessica Clingman-Kerns, a team member at Hilton. In October 2017, when wildfires swept through wine country, her parents ' home burned down. JCK : The fire just kind of completely took over their entire neighborhood, and their house was gone. It ' s hard. My brother had passed away just a couple months prior, so all of his things were in that house. Just a lot of parts of our life that we ' ll never get back. AG : It was devastating. But after sitting with her grief for just a moment, Jessica sprang into **action**. JCK : I emptied out my boyfriend ' s Tahoe and put all my camping stuff in the car. And then I went to Walmart and maxed out a credit card with just toothbrushes and toothpaste and deodorant, and just little things for someone who left their house at midnight and had nothing. And I just started driving. AG : She headed to the DoubleTree, to set up a makeshift relief operation. A manager provided her with a small conference room, which quickly became too small of a conference room. JCK : I think everyone at the hotel thought I was crazy because I was like,”Oh, I just need a little bit more space.”And then they started seeing semitrucks. So my one little room turned into 10, 000 square feet, 100 volunteers, millions of dollars in donations. AG : Of the thousands of Californians who lost their homes, several were Jessica ’ s colleagues. JCK : That didn ’ t stop them from volunteering or being a part of everything either, so... I could not have done it without their support. AG : Each year, Hilton ’ s CEO presents the Light and Warmth Award to a dozen of their 425, 000 team members. It ’ s the highest honor in the company, given to people who embody Hilton ’ s vision, **mission** and values. JCK : I knew about the Light and Warmth Award, but I never thought that I would even be thought of. AG : Experiments show that it ’ s not just the recipients of recognition who end up performing better. Their colleagues do, too. Awards are not just a powerful way to show the winners that they ’ re valued. They ’ re also an important way to signal to everyone what ’ s valued in the culture. A few months after the relief effort, Jessica headed to a meeting, where she was surprised with a phone call. It was Hilton ’ s CEO. She hadn ’ t just been nominated for the award, she won. JCK : So they awarded me with one of those big obnoxious checks. But I looked at the other side and it said 10, 000 dollars. And I was like,”Oh no, come on. This is a joke. Where ’ s the camera,”right? But as soon as it kind of hit me like, I thought,”How else can I help people with this money?”I can sponsor families for the holidays this year. I can donate some of this to rebuilds. AG : One of the many people amazed was Jessica ’ s dad. Jessica ’ s Dad : When I think back on it, just seeing all the work and caring that Jessie put into helping the entire community, I mean, that in itself makes up for any losses that we had. JCK : It ’ s just in my nature to help other people. That ’ s why I love hospitality. AG : Hilton was named Fortune ’ s number one Best Company to Work For in the US, in both 2019 and 2020. And one of the best places to work for Millennials by the Great Place to Work Institute. Learn more at jobs. hilton. com. AG : Procrastination is the opposite of productivity. It ’ s wasting time, or at least using it pretty inefficiently. We ’ ve talked about how individuals can avoid that, but I also want to know what organizations can do about it, which might mean thinking differently about what it means to be productive. Rutger Bregman : I think the first question we should ask ourselves is”What is work, and what is productivity?”Nowadays, we say work is just this thing you do in a hierarchical relationship with an employer. You get a salary, you pay taxes over that, and that is what we call work. AG : Meet Rutger Bregman. RB : I ’ m a Dutch historian and author. AG : And why do you feel the need to mention that you ’ re Dutch? RB : That ’ s a good question. Maybe because the Netherlands is the country with the shortest working week in the world. And also, we have incredibly high productivity per hour. So it ’ s actually a good example of my thesis that actually, if you want to be more productive, you ’ ve got to work less. AG : How would you define productivity? What do you consider a really productive day at work? It probably has something to do with time and with output. How many hours you spent actually paying attention to your tasks instead of **scrolling** on Instagram, how many boxes on your to-do list you checked. But what if you redefine productivity? RB : My definition would be work or being productive is just doing something that is valuable, that is useful. AG : I like this. I think of productivity as using your time to accomplish things of value to you and others. Whether you use your time well depends on how much of it you have. Psychologists find that being busy, having less time, motivates us to finish tasks faster. We often procrastinate less when we have more on our plate. And there ’ s some evidence that people with multiple children are more productive at work than people with one or none. New studies suggest that parents are more absorbed in their job tasks while at work, because they know how much they have to **juggle** later at home. I guess the old saying is true. If you want something done, give it to a busy person. Of course, your ideal solution to procrastination is not going to be having kids, or having even more kids. But this research suggests that when you have less time to complete your tasks, or more tasks to complete in a given time, it can curb procrastination by changing your emotions. When you ’ re busy, you ’ re more motivated by fear and guilt about falling behind than whatever unpleasant feelings you have around the task itself. But when you have a lot of free time, you don ’ t feel that urgency to finish. There ’ s a name for it, Parkinson ’ s law. The idea that work contracts or expands to fill the time available. RB : I actually experienced this when I was working at a traditional Dutch newspaper. And I remember all those afternoons. I ’ m not very productive during the afternoon, you know. So around 4:00pm, I just want to start bothering my colleagues and making stupid jokes. AG : Rutger has proposed what might seem like a radical solution to those episodes of procrastination – shortening the workweek. It ’ s not as crazy as it sounds, because the whole notion that 40 hours is a magic number for productivity is kind of arbitrary. RB : If you look at the history of this, it was Henry Ford, already at the beginning of the 20th century, who already found out that when he moved his workers to a 40-hour workweek, they were more productive. And he didn ’ t do it because he cared so much about his employees. You know, he cared about his wallet. That was the reason why he did it. It was called the American way. Working less, it ’ s the American way. AG : And actually, Rutger points out that for more than a century, workweeks were getting shorter and shorter. In the mid-1800s, people often worked 70-hour weeks. Then, around the turn of the century, workweeks started dropping from 60 to 50 hours a week. RB : And then after the Second World War, economist John Maynard Keynes said we ’ ll have a 15-hour workweek in 2030. AG : I promise this is going to tie back to procrastination. I ’ m just putting it off for a little while, because I want to know why we ’ re working so many hours today. There are lots of possible explanations, from rising competition and globalization and consumerism to being obsessed with status at work, to just believing more is better. Whatever the cause, you ’ ll be familiar with what the result looks like now. RB : Probably, at this moment around the globe, there are millions of people sitting in offices, just waiting, browsing Facebook, sending emails to people they don ’ t really like, writing reports that no one ’ s ever going to read. I think we could easily move to a four-day, three-day workweek, and be just as productive. You just squash out all the slack that ’ s currently in the system. AG : How many hours do you actually work? RB : I think about 50 to 60. If you would define my work as, you know... Am I working right now? Is this work? AG : I don ’ t know, does it feel like work? Are you contributing something valuable to the world? RB : Well, you decide that. No, I find it very hard to define what work is for me. AG : I wonder then, is there some degree of irony that a guy who works 50 to 60 hours a week is calling for a 15-hour workweek? RB : Yes, it ’ s very ironical, I know that. AG : I also think I work more hours than you do in a typical week. And I ’ ve also called for shorter workweeks. So at heart, I wanted to know that I wasn ’ t the only hypocrite out there. RB : Yeah, we ’ re both hypocrites. AG : Yeah, but I think maybe the difference between our lives and the policy changes we ’ re calling for is we choose to work this number of hours, right? I work as many hours as I do because I find my work enjoyable and meaningful. And what I want is for the hundreds of millions of people who hate their work or who find it extremely **stressful** to have the freedom to work less if they so choose. RB : Yeah, and you could also frame it like this. I think that often, we need to work less in order to do more, right? To have more time for the things that we really care about. AG : We should think about productivity not as the volume of output but as the value of output. And if we ' re going to do that, then we need to start measuring work in something beyond hours, which is starting to happen. Leaders are beginning to realize there ' s a big difference between working long hours and doing worthwhile work. Finland ' s new prime minister has spoken in support of a four-day workweek and a six-hour work day. In the US, Shake Shack is trying out a four-day week for managers at many of their locations. And recently, Microsoft Japan tested the four-day workweek. Productivity climbed by 40 percent there. In part because of more focused attention, and in part because they got rid of unnecessary distractions by making meetings shorter. But my favorite example comes from another company that has gone to the extreme to help people use their time more productively. Even people who are chronic procrastinators. Jade Walker : I was definitely not employee of the year or anything like that. Just a whole lot of procrastination. AG : Jade Walker has **struggled** with procrastination in the past. Maybe because she lives in a beautiful part of New Zealand. JW : I ' m based in Takapuna, which is a suburb of Auckland. It ' s, like, a beach town. AG : Are you a surfer? JW : No, I ' ve got two little kids. So mom life is how I spend my spare time, chasing them around. AG : But at work, she found herself getting distracted by another kind of surfing. JW : Oh, I was definitely an internet surfer. Online shopping, talking to my friends. This is really bad, but I used to do our grocery shopping online and it gets delivered. AG : I love the efficiency of that, though. JW : My old boss probably wouldn ' t agree, maybe. AG : Then she took a new job in estate planning, creating wills for people in the hospital and for navy soldiers about to deploy. Jade ' s new company, Perpetual Guardian, does something unusual. They offer a four-day workweek. New employees start out at five days, and if they prove their productivity in the first few months, they get to go down to four days. It had a big impact on Jade. JW : Every weekend, I get a screen time report. And it used to be horrendously bad. My husband would see it sometimes, and he ' d be like,"Oh my God, you spend four hours on your phone a day."I ' m like... Yeah. But now, like during the day, I don ' t have time. So my screen time, I ' m proud to say, is between one to two hours. A lot less, due to the fact that I ' m being a lot more productive at work. AG : Congratulations. JW : Thank you. AG : It ' s one of the smartest motivators I ' ve ever seen. If your productivity backslides, you go back to a regular workweek. If you ' re efficient and effective, the reward is that you get to work less. JW : Yes, that ' s a big incentive to not procrastinate. We don ' t have a lot of time to procrastinate anymore. You just sort of get in there and get everything done. AG : It ' s completely changed her work process. Now, each morning, she spends 15 minutes planning her day. She color-codes emails and writes lists of priorities. All this helps her complete one task at a time and move on to the next one, instead of trying to do three things at once. Her productivity and her focus have improved. JW : Going back to five, I don ' t know what I would do with an extra day at work now. AG : Grocery orders. JW : Yeah, exactly, shopping, long breaks. AG : Are you saying you would procrastinate, if you had a fifth day? JW : Yes, I probably would. Now that I ' m used to getting everything done in four, I think I ' d have so much extra time. So I probably would procrastinate, because yeah, I ' d have to fill in the day somehow. AG : You probably don ' t have the luxury of just deciding to work fewer hours. If you ' re an hourly retail employee, for example, your income will take a hit. But the shorter workweek is a **bold** demonstration that it ' s possible to manage our work lives differently, more efficiently. In any given moment, we will always have dilemmas about whether to work and what to work on. But take it from Margaret Atwood, the task you ' re putting off isn ' t always the one you hate. It might be the one you fear, the one that ' s ultimately the most worth pursuing. At least, that ' s how it was for"The Handmaid ' s Tale."**MA** : I did try to write this more normal novel and it really just did not work out. So that was a signal that I had to write the batty one or nothing. AG : If you ' re going to do the task eventually, you might as well spare yourself the agony and start it sooner. **MA** : I call that"white rabbit syndrome."AG : From"Alice in Wonderland"? **MA** :"I ' m late, I ' m late, I ' m late."AG :"No time to say hello, goodbye."**MA** :"That ' s it, got to go."AG : Next time on WorkLife. Conrey Callahan : This feeling of being trapped in this system where you ' re on, like, a hamster wheel. You know, you ' re kind of, like, doing a lot, but is it really changing anything? AG : Job **burnout** seems to be everywhere. But it ' s not inevitable. WorkLife is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ' Donnell, Constanza Gallardo, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. This episode was produced by Jessica Glazer. Our show is mixed by Rick Kwan. Original music by Hansdale Hsu and Allison Leyton-Brown. **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors : Accenture, BetterUp, Hilton and **SAP**. For their research, thanks to Tim Pychyl on procrastination, Max Bazerman, Katy Milkman and colleagues on want / should selves, Keith Wilcox and colleagues on being busy, Tracy Dumas and Jill Perry-Smith on absorption at work, and Bob Boice on scheduling writing sessions. And thanks to Candice Faktor for the amazing introduction to Margaret Atwood. **MA** : OK, so here ' s the story. We were throwing a party at our house for some writers. And there they were, all milling around. And then this young woman, who was about 35, said she was having a heart attack. Then I called 911 and the paramedics arrived, lumpety-bump up the steps. I love paramedics. So in they come, and here ' s the conversation. First paramedic :"Do you know whose house this is?"Second paramedic :"No, whose house is it?"First paramedic :"It ' s Margaret Atwood ' s house."Second paramedic :"Margaret Atwood? Is she still alive?"

Text Sample 169: POS Dim 2 – 2020 – Score 41.00 – 2020/t004276.txt

Arthur Brooks : Starting when I was eight or nine years old, I literally wanted to be the greatest French horn player in the world. That was what I always wanted to do. That was my whole life dream. Adam Grant : What was behind that? AB : I was ambitious. I wanted to play for a lot of people. And I guess that was just simply expressed in the only thing that I really knew how to do, the only thing I really cared about, which was music. AG : Arthur Brooks was a natural. He could read music before he could actually read. He played in youth orchestras as a kid, prestigious ones. Eventually he dropped out of college to go on tour. AB : I went on the road playing **classical** music. I did that for six years. I saw all 50 states from the back of a van. AG : Arthur recorded an album that played on **classical** radio stations across the country. But then something started to change. AB : I realized, about age 21, that things were starting to go south, and I couldn ' t explain it. Ordinarily, **classical** musicians, they will see their skills increase through their 20s, and sometimes even their 30s. So whereas if I were a **baseball** pitcher, it would be totally normal that I would be losing heat off my fastball by my early 20s. As a French horn player, I shouldn ' t have been seeing that, but I was. AG : To turn things around he **practiced** his heart out. Finally, he got his big break, his debut at Carnegie Hall. AB : So I got ready for this concert, and the first half of the concert, I have to say, was phenomenal. It was really one of the best concerts I ' d ever given. And in the second half, I had this experience where I had to actually talk to the audience and I walked toward the front of the stage to address the crowd, and I wasn ' t watching my feet and I slipped, or something collapsed, I still don ' t know, and I fell five feet into the audience. AG : You literally fell off the stage? AB : I literally fell off the stage. It was just ridiculous. It was... I wrecked my instrument, I fell on my instrument, I hurt my arm. And, of course, I did what you would do when you ' re 22 years old. I jumped up and said,"I ' m okay, folks."And I obviously was not okay. AG : Oh, no. AB : It was just so terrible, Adam. It was so terrible. AG : Arthur tried to shake off the feeling that falling off the stage was a bad omen. He spent the next decade practicing. He studied with the best tutors in Europe. He got a job at a symphony in Barcelona, but – AB : It was a wreck. It was a slow-moving wreck in my career itself. I was in decline. Finally had to pack it in. I had to admit it to myself that I wasn ' t the horn player I had always dreamed of being. And right now, I ' m talking about it with you and, you know, we ' re laughing and all that, but I ' ve got to tell you, it kills me, I still hate it. AG : Arthur didn ' t plan to quit music so early. He simply stopped being great at his job. And in the years since, it ' s made him wonder, does career decline happened to all of us eventually? And is there anything we can do about it? I ' m Adam Grant, and this is WorkLife, my podcast with TED. I ' m an **organizational** psychologist. I study how to make work not suck. In this show, I ' m inviting myself inside the minds of some truly unusual people, because they mastered something I wish everyone knew about work. Today, career staying power. It ' s more about the choices you make than the age you reach. Thanks to Accenture for sponsoring this episode. Let ' s be honest, we all have **peaks** and valleys in our careers, times when we hit our stride and do our best work, and times when we ' re in a slump. Most of us are worried that as we age, our physical and mental skills will decline and we might enter into a permanent slump. And that fear is compounded by the fact that age is seldom seen as an asset in the modern **workplace**. Far too often, our older colleagues don ' t get the respect they deserve. Arthur ' s slump happened at a particularly early age, so he changed careers and became a public policy expert. AB : Basically I ' m just an old fashioned, number-crunching, quantitative social scientist. AG : Given what he ' s gone through, it ' s not entirely surprising that his big area of focus now is career decline. AB : Career decline is a very personal and personalized phenomenon in which your skills are not what they once were. So you might be making as much, or more, money than you ever have before, but you know that you are not at the same level of technical adeptness that you had once attained, and / or you ' re not on the trajectory to attain what you had hoped. AG : Maybe you felt your stomach lurch when someone younger than you has a smashing success in your field or your **workplace**. Or maybe you ' ve been humming along feeling good about your work, until one day you noticed your skills don ' t seem quite as sharp as they used to. In 2019, Arthur wrote a major story about this phenomenon in"The Atlantic."The headline was"Your Professional Decline Is Coming Sooner Than You Think."As in, long before you even start to see retirement on the horizon. AB : I mean the number one myth is that, you know, particularly in certain professions, like, let ' s just say yours and mine, which is coming up with big ideas and sharing them, that that ' ll never go into decline. Why? Because it doesn ' t require, you know, strong biceps, and yet there ' s overwhelming evidence that in idea professions people experience decline as well. They just don ' t expect it. AG : What happens to our cognitive abilities as we age is not straightforward. It actually depends on what kind of mental skills we ' re talking about. Psychologists have long distinguished between two kinds of intelligence : fluid and crystallized. Fluid intelligence is your raw processing power. It ' s basically your IQ, your innate capacity for learning and problem solving. AB : That ' s a big part of the modern idea economy. I ' m going to come in here, I ' m going to be a whizzbang person who comes up with brand new ideas. That ' s actually what ' s rewarded for young people because young people have high levels of fluid intelligence. It increases through your teens and 20s, and maybe through your 30s, although it tends to start declining for most people in their 30s and 40s, and it makes this original idea generation harder. Now that doesn ' t mean that you ' re going to be unsuccessful, but it means if you ' re relying entirely on this fluid intelligence, you ' re going to **struggle**. AG : A common refrain in Silicon Valley is that young people are just smarter. When it comes to fluid intelligence, that ' s generally true. But the story changes with crystallized intelligence, your acquired ability to solve problems by applying your knowledge and experience. AB : Your ability to synthesize things that you didn ' t necessarily invent, that ' s crystallized intelligence and that tends to happen later. AG : Think about how crystallized intelligence comes in handy in your job. If you ' re in sales, you ' ve used it to tailor your pitches to different clients based on what you ' ve learned about them in the past. If you ' re in government, you ' ve relied on it to get around red tape. If you ' re in finance, maybe you ' ve applied it to predict how markets are affected by crises. Crystallized intelligence **peaks** much later than fluid intelligence, in large part because our body of knowledge keeps growing. AB : Think of it like a library. Think of the brain as basically filling up with a lot of volumes. And that means you have a lot of crystallized intelligence. AG : Depth and breadth of knowledge aren ' t the only benefits of aging. Evidence suggests that self-control and socio-emotional skills often keep climbing, too. Many of us continue gaining willpower and emotional intelligence well into our 40s and 50s. We ' re still getting better at delaying gratification, reading other people ' s emotions and regulating our own moods. Even into our 70s and 80s, we ' re still improving at ignoring sunk costs, all of which can enhance our job performance. But Arthur doesn ' t think that ' s enough to keep us from declining. AB : Everybody falls in their game at some particular point. AG : At what age do you think the average person is going to start to face significant career decline? AB : I ' ve got to give you the social scientist answer – it depends but if I were to guess, I ' ve got to say that in your mid-40s is when you ' re likely to see the **peak** of your originality and creativity, and you should be making plans in your 40s to how you can be the best creator with your crystallized intelligence, how you can be the best instructor, the best synthesizer, you can possibly be. AG : OK, so this is where I want to start disagreeing with you. So my read of the evidence is that people don ' t inherently become less creative. One of the things that happens as people ' s careers advance is they tend to produce less, just, you know, in terms of overall volume. I think it ' s not ability. I think it ' s motivation that ' s dropping. I think, you know, a lot of people become complacent and they feel like they ' ve achieved enough success, and so they may scale back a little bit. It may be that priorities outside of work become more important to them. And if we can maintain that motivation, there ' s no inherent reason why the quality of creative output would decline. What do you think? AB : Hmm, I ' m willing to entertain your hypothesis. AG : OK, cool. So there ' s a study I really like of scientists, filmmakers and artists, showing that as they advance from their 20s and 30s toward their 50s and 60s, they do less work. The scientists run fewer experiments, the filmmakers don ' t make as many movies and the artists produce fewer paintings. But those who stay productive, their hit rate was every bit as high in their 50s and 60s as it was in their 20s and 30s. So, you know, I think about Frank Lloyd Wright designing Fallingwater at 68 years old, his crowning architectural achievement. Or I ' m sure you ' ve come across Raymond Davis Jr, the scientists who measured

neutrinos from the sun. He didn ’ t build the key detector until he was 51 and kept running his experiments until he was 88, the year he won the Nobel Prize. I ’ d be willing to bet that we would either uncover, or even create more Frank Lloyd Wrights and more Raymond Davises if we had ways of keeping more people motivated to produce into and past mid career. Agree or disagree? AB : I think that if there is a natural cadence to the average person ’ s life, and you ’ re saying it doesn ’ t have to be this way, then the societal goal will be one in which everybody is above average. And you know, that becomes an exercise in frustration. AG : In other words, that ’ s a nice theory about encouraging everyone to keep producing at maximum capacity for decades, but it might not be realistic if you ’ re not a superstar in your field. The rest of us are going to do what people naturally do. And Arthur wants there to be space for that path, too. He wants people to be excellent in their own way, which made me wonder if he thinks career decline is more inevitable for people who achieve excellence. AB : What goes up must come down ultimately, and the higher up you go, the more obvious it is that you ’ re not there anymore. That ’ s what I call the Law of Psychoprofessional Gravitation. You notice a lot when you were great and you ’ re not as great as you once were. And if you can do something to get on your crystallized intelligence curve, you get a different kind of productivity, and you can be successful in a different way. That ’ s the knowledge I want to get across. AG : He ’ s arguing that regardless of what field you ’ re in, as your fluid intelligence declines, you should make a change. Stop trying to innovate. Take what you ’ ve learned and accomplished and shift into a mode where you can transmit your knowledge and skills to other people. Become a teacher or a mentor. AB : And frankly, the world needs those people. And we don ’ t have enough of them. When I ’ m at a certain point in my career, and I see that I ’ m not on the top of my game, I need to start thinking, is what got me here, whether it ’ s the insight or the innovation or whatever it happens to be that got me here, is it something that I can sustain? If it ’ s not, start looking for something else. Specifically, start looking for a way that you can move into instruction mode. AG : In Arthur ’ s case, after giving up on music, he spent years running a conservative think tank. And then having learned the lessons of his fall at Carnegie Hall, he took his own advice. AB : And I had somebody I really respect, and he said,”You know, when you are a chief executive and you have to be on the razor ’ s edge every single day, you have two choices on how to step out of your chief executive job.”And I said,”Well, good, two choices. I like two choices. You know, it ’ s like the Monty Hall game. You know, door number one and door number two. So what ’ s behind door number one?”And he said,”Leave before you ’ re ready.”And I said,”Oh, I don ’ t like that. What ’ s behind door number two?”And he says,”Leave on someone else ’ s terms.”I said,”Ooh, I really hate that one.”AG : What did Arthur decide to do then? Exactly what he ’ s suggesting you do. He went into teaching. AB : The bottom line is, when the stakes are high, it ’ s worth leaving things before you ’ re ready. AG : Arthur is content with how his career choices worked out. He wants you to walk off the stage before you fall off of it. I understand where he ’ s coming from, but I ’ d rather see you find your balance, or get on a different stage altogether. You do have choices beyond aging out and opting out. More on those after the break. OK, this is going to be a different kind of **ad**. I played a personal role in selecting the sponsors for this podcast, because they all have interesting cultures of their own. Today, we ’ re going inside the **workplace** at Accenture. Gonzalo Adriazola : I ’ m from Arequipa, which is the second largest city in Peru. We have the coast, we have the Andes and we have a bit of jungle. AG : This is Gonzalo Adriazola. When he was 19, he and his mother left Peru and moved to rural Ohio. GA : I don ’ t think I fully processed that I was leaving a whole life behind. I think I was just excited for the future. AG : As a young immigrant to a new country, Gonzalo didn ’ t know exactly what the future held, but his mother became a tremendous source of inspiration. From early on, she taught him about the importance of education. GA : She would always tell me,”I might not leave you anything, but I ’ m going to leave you a good education.”AG : Gonzalo got a job at a local restaurant while he applied to colleges. GA : When I got the acceptance letter, my mom and I were just, like, jumping and dancing and just so happy. AG : He headed to the Ohio State University, and it didn ’ t take him long to fit in. He was crowned homecoming king, started working full time at a bank and found a community within the Hispanic Business Student Association. They sponsored a trip to a conference in Washington. GA : It ’ s the largest gathering of professional Latinos in the US. I saw this one speaker and he was so good. He was so **passionate**, energetic. Besides being phenomenal, he looked like me, and this was the first time I had seen someone in person that looked like me that was a total boss. AG : At the end of the talk, Gonzalo went up to the speaker and asked for advice. GA : I asked him like,”I want to be a CEO one day. How do I get to be like you?”And he laughed and looked at me and told me that professional **services** firms, all they have is people. So they ’ re going to do their best to make the best of you. And that ’ s how I found out about consulting. AG : Research shows that role models can elevate our aspirations. When we see someone similar to us do something inspiring, it builds our confidence that we can follow in their footsteps. When Gonzalo graduated from college, he followed his role model ’ s advice and went to work as a consultant at Accenture. GA : I was really happy to join the firm. And I mean, it has only been great experiences. AG : Gonzalo had accomplished a big goal, and for four years, he helped companies grow and perform better. But he had a nagging feeling that something was missing from his life. GA : As soon as I got to Accenture, I wanted to give back to Latin America. That was kind of, like, my life objective. I saw that we had worked with this microlending institution called Caja Arequipa, and I ’ m like,”What are the odds?”AG : He reached out to the leadership team running the project. GA : They told me about, like, everything we were doing in regards to **financial** inclusion in Latin America. I was really excited to know that I was from that same city. Very welcoming to my help. AG : Alongside his consulting work, Gonzalo has been a leader in the Hispanic Employee Resource Group and volunteered his extra time to help the team in Latin America. His dedication paid off. Earlier this year, Accenture development partnerships moved him to Colombia where he joined the project team full time. GA : It was a dream come true. I called my mom. We laughed, we smiled, we danced. AG : This isn ’ t the only way Gonzalo gives back. Every year he ’ s gone back to the conference where he was first inspired by someone he felt he could relate to. GA : It feels like I ’ m doing my **mission**. I was in their shoes one day and now I ’ m the person lifting up, just as others lift me up. And there ’ s people lifting me up right now. There ’ s going to be a brighter future for my community. AG : Accenture is working to become one of the most truly human companies in the digital age. Learn more at Accenture. com / careers. Lyn Slater : I ’ m not a Pollyanna about aging. There ’ s certainly changes in my body. You know, I have my wrinkles, I have my white hair, but I think that I could generate content as much as someone who ’ s younger than me. In fact, most of the peers who are at my level are all younger than me. AG : Lyn Slater is 65. She recently quit her career as a professor to become a full-time social media influencer. Lots of people call themselves influencers, but Lyn ’ s version is a real job. LS : I ’ m also known as The Accidental Icon. AG : The Accidental Icon has more than 700, 000 followers on Instagram, and she gets paid to run **ads** on her account for brands like Kate Spade, Visa and Teva. Her feed is filled with stylized photos of herself posing around New York City in designer clothes and large sunglasses. She ’ s been featured in magazines like”Cosmo”and”In Style.”And she ’ s walked the runway at New York Fashion Week. Actually, she ’ s walked more than one runway. LS : I have no background in the fashion industry, but I was in the runway show last spring with Kate Spade. I ’ ve worked with them quite a bit. AG : Some people will tell you that most of us do our best work when we ’ re younger, that when we approach retirement age, we should slow down, take up a hobby, play with the grandkids. But excellence later in your career is possible, and there are a few ways you can get there. Lyn is living proof that you don ’ t have to avoid career decline by shifting from doing into teaching. You can actually reinvigorate your career by experimenting, by doing something new. In Lyn ’ s case, she did something completely new. We saw her in **action** one Saturday near her home in Manhattan. LS : We ’ re on Fifth Avenue in New York City. It ’ s called Museum Mile because everywhere is a museum. I ’ m going to be shooting my outfit outside. AG : She ’ s with her partner who doubles as her photographer. Calvin : Your bangs are good. LS : Thank you. AG : Lyn is petite with a short white bob and bangs that stop just above her huge sunglasses. She takes off a long velvet coat and scans the brick facades of the Museum of the City of New York for her next spot of inspiration. LS : And right now, that little nook over there is really calling to me because it does look like a place... AG : Before becoming a social media influencer, she had an entirely different decades-long career in social work. She was an expert for New York City family courts on child abuse issues. She wrote a seminal book on social work and law, and she spent the last 20 years as a professor of social work at Fordham University. She was quite an authority in her field, but she told me she ’ d become frustrated with the glacial pace of **academia**. Like, how long it takes to write and publish journal articles. LS : The whole process could take almost two years and maybe a handful of people would read it. And so, as I began to get more followers on Instagram and I began to do more things, you know, for example, I did a short film and within a couple of weeks, that was in front of, you know, millions of people. AG : Lyn went about as far from her original field of work as she could go. So what leads to a spectacular pivot like that? A dose of novelty. Research shows that curiosity is a key factor that can help you find creative **peaks** as you age, rather than decline. This is one way to stay productive and maintain motivation later in your career. Find something new to explore. For you, novelty might mean switching to a new team or a new market, getting certified in a fresh skill set, or mastering a hobby completely outside your experience. For Lyn, that hobby was fashion. LS : I ’ ve always had, sort of, what I call a very performative relationship with my clothes. I resist like crazy social categories or constructions of who I should be. AG : In this case, she was resisting the idea that a long time professor of social work wouldn ’ t also be interested in high fashion. She ’ d been toying with the idea of starting a fashion blog. And one day, unexpectedly, the name came to her. LS : It was the first week of school, and I always used to dress up for that. And I was meeting a friend for lunch and I had on a Yohji Yamamoto suit, and a couple of photographers started taking my picture and then a bunch of other photographers saw them do it so they came over, and my friend was making a joke like,”You ’ re an accidental icon.”AG : If you ’ re going to experiment with something new, it helps to fully immerse yourself in that world. Lyn was creative in where she looked for her mentors. LS : I found out that I could get a press pass to what is called the market shows, and I would spend hours talking with these young designers. They taught me a great deal about how clothing is made, what the processes are. That ’ s how I really began to understand fashion. AG : Instead of being intimidated by new technology and seeing it as something only young people can master, she went straight to the experts she knew. LS : My students helped me to figure out Instagram and Snapchat, and I started to take classes in social media and fashion and just random things, like, how to start your own vintage store. AG : In a study in the fashion industry, the more time designers and directors spend abroad, the more creative they became over their careers, but it wasn ’ t traveling or even living abroad that mattered. It was the amount of time they spent working abroad. They couldn ’ t just **observe** it from the sidelines. They had to be in the arena. That ’ s how they really internalize the novel elements of a foreign culture. For Lyn, seeking out new perspectives included small steps, too. LS : I do things that I don ’ t normally do. And so that could be very simple things, like taking a different route to work. It ’ s really, put yourself into different experiences that are creative and see where that takes you. AG : You may not aspire to quit your job and become an Instagram star, or to switch fields at all. And you don ’ t have to. There are ways to reach new heights while staying within your career path. Think about whether you ’ re a sprinter or a marathoner, not in terms of running, but how you go about generating creative ideas. Economists call the sprinters **conceptual** innovators. They typically start with a big idea and they tend to **peak** early, because they have eureka moments when they ’ re relatively new to a domain. Like Einstein at 16 imagining himself chasing a beam of light, or Mary Shelley at 18 starting to write”Frankenstein.”But as they accumulate knowledge, they ’ re more likely to get stuck in their ways and stop gaining new perspectives. There ’ s a term for that : cognitive entrenchment. The longer they stay in a field, the more likely they are to take for granted assumptions that need to be questioned. Marathoners, on the other hand, are experimental innovators. They usually discover their ideas through **trial** and error. They often **peak** later because it takes time to test out different approaches, like Darwin, spending decades tracing the origins of species, or da Vinci experimenting for years with different ways of modeling light in his paintings. But those habits allow them to sustain their **peaks** longer and **peak** again later as they explore new areas. That kind of experimentation is familiar to Lynda Weinman. Lynda Weinman : I ’ m an autodidactic person. I always love learning new things. AG : She got her start in the 1980s as an animator at a film studio. She often compared herself to her higher achieving peers who were sprinting ahead. LW : I remember when I was in my 20s watching some of my friends who were uber successful and I, you know, at my 30th birthday was in tears that I hadn ’ t accomplished what I wanted to accomplish yet with my life. AG : But she didn ’ t rush it. Instead, for the next 10 years, she kept chugging along, testing out different ways to build and use her skills. Finally, the year of her next big birthday, she created a website. Lynda called her site Lynda. com. LW : Well, when we started Lynda. com I was 40 and when we sold it I was 60. AG : She sold it in 2015 to LinkedIn, for, oh, 1. 5 billion dollars. Lynda. com was one of the first companies to offer professional learning online. Maybe you ’ ve heard of their courses in animation, design and software. Maybe you ’ ve even taken classes there that advanced your career. Many founders who are trying to grow a unicorn company, think they need to generate one brilliant idea and scale it rapidly. They think they need to be sprinters, but Lynda approached her career more like a marathoner. Her online learning company didn ’ t look anything like a success early on. Instead, she was constantly trying out new ideas and innovating over time. In some ways, Lynda was primed for this kind of scrappy test-and-learn approach. LW : I actually had a pretty troubled childhood. I moved in with my dad and my stepmom who weren ’ t expecting to raise me and my siblings. And, you know, it was just this feeling of having to find my way because all the adults in my life were, you know, having their own problems.

And I think what I probably am is a very resourceful person who realized at an early age that it was on me to figure out my life. AG : During her early days as an animator, Lynda encountered her first computer. She was unimpressed, at first. LW : It was a Apple II Plus clone that my boyfriend brought home, and so to please him, I turned it on and I started pecking at the keys and I caught the bug that so many people since have caught. And he eventually called himself a computer widow, because I liked the computer more than him. AG : LW : I mean, I was just amazed at what it could do. AG : What it could do was draw. So Lynda got her own Macintosh and brought it into work. People started asking her to do projects for them. LW : Somehow they thought I knew more than they did, which was, you know, basically I ' d had five minutes longer with something than they had. AG : Lynda was onto something. By experimenting with new tools and following her curiosities to uncharted places, she could carve out a niche and continue iterating instead of stagnating. LW : Oh, do I want to do what everybody else is doing, or I ' m being asked to do something that only I know how to do, and maybe that ' s more fun. AG : Lynda ran two more experiments. First, she started teaching in-person college classes in web design. Then when she saw a lack of good manuals for students, she wrote one herself. LW : Then it became a huge success, way bigger than any of us could have imagined. AG : Well, you ' re being very modest, but can I get you to describe a little bit just how successful the book was? LW : I ' m not sure how many millions it sold, but it was translated into many, many languages and it was the de facto web-design book for all colleges. It was the"it"book. AG : From there, Lynda continued to test out new methods of teaching. She started selling instructional videos on VHS tapes. Then, as technology changed, she switched to CDs, then DVDs. Then she launched the website Lynda. com. In the late ' 90s, she and her business partner and husband tried launching memberships on Lynda. com. They weren ' t exactly an **instant** hit. LW : In the beginning it was just an abject failure. It was cannibalizing some of our other work. AG : But they didn ' t give up. Not every experiment succeeds, and they don ' t all succeed right away. Memberships weren ' t growing like crazy, but there was steady growth. Lynda paid attention to that. Eventually they hit an inflection point. While Lynda. com was quietly adding hours of material to their catalog, more people were using the internet for online learning. Their online audience doubled each year. And in time that grew into a big number. By 2012, they had over a million paying users and over a hundred million dollars in revenue. LW : It was literally a very organic growth process. AG : Lynda could ' ve simply continued to develop her skills as an expert animator and would have had a nice career. She could ' ve kept teaching, too, but her greatest success came when her experimentation led her into entrepreneurship, turning her various projects into a business. She didn ' t listen to the narrative that as you get older, you should pull over into the slow lane. She saw some benefits of building something later in her career. LW : By the time you ' re 40, you believe in yourself more. You ' re a more seasoned human being that has better judgment and better experience and better confidence. AG : This might be one of the reasons why older founders are actually more successful than younger ones. In a study of several million entrepreneurs, 40-year-olds were more than twice as likely as 25-year-olds to launch a successful **start-up**. And they were also nearly a third more likely to be in the top 0. 1 percent of fastest-growing tech **start-ups**. Yes, sprinters sometimes get lucky right out of the gate, but marathoners know how to persevere. LW : I think everybody would probably like to be instantly successful and be revered and acknowledged and acclaimed and all those things, but when it happens to you too young, it actually can be very damaging. At least the people that I ' ve seen, who, you know, they burned bright and they burned all the way out. They didn ’ t understand that it wasn ’ t going to last forever and that they had to keep reinventing and they had to earn it again. AG : If you ’ re worried that your career will decline, don ’ t let your output decline, and don ’ t let your curiosity stagnate either. It ’ s never too late to invent or to reinvent yourself. LW : I think the older you are, the more valuable you can be. AG : I love that because I don ’ t ever want to retire. Next time on”WorkLife.”Working remotely is full of challenges that have made our work lives harder. Lizet Ocampo : So we turn our cameras on and there I am as something else. At first, we didn ’ t realize it was a potato. We ’ re like,”What is going on?”I just essentially was a potato for the meeting. AG : But some of the solutions might be easier than you think.”WorkLife”is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell, Constanza Gallardo, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelin. This episode was produced by Jessica Glazer. Our show is mixed by Rick Kwan. Original music by Hahnsdale Hsu and Allison Layton-Brown. **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors, Accenture, BetterUp, Hilton and **SAP**. For their research thanks to Dean Simonton on creative productivity, Lu Liu and colleagues on career hot streaks, Ben Jones and colleagues on the success of older entrepreneurs, David Galenson on **conceptual** and experimental innovators, Erik Dane on cognitive entrenchment and Will Maddux, Adam Galinsky and colleagues on working abroad. In some ways, Lynda was primed for this kind of crap... Excuse me. In some ways, Lynda was primed for this kind of crap... It ’ s not crappy, it ’ s scrappy. In some ways, Lynda was primed for this kind of crappy... Sorry.

Text Sample 170: POS Dim 2 – 2020 – Score 41.00 – 2020/t004290.txt

Adam Grant : Hey everyone. It ’ s Adam. If this is your first time listening to WorkLife, don ’ t start here. This is a bonus episode. To get the full WorkLife experience, start with”How to Love Criticism,””Your Hidden Personality,””The Creative Power of Misfits,”or”The Real Reason You Procrastinate.”Don ’ t put that one off. Four years ago, when I was working on my first TED Talk, I invited some colleagues to a **practice** session. One of those colleagues was Esther Perel, the relationship therapist, and host of the popular podcast,”Where Should We Begin?”We ’ d only met once before, but she didn ’ t hold back in her feedback. She told me my talk was all facts, no feelings. That if I wanted to reach people, I couldn ’ t stop at information. I needed to inject some emotion into the talk. Right then I knew I had found a great new member of my Challenge Network, someone with common interests, but a very different style and worldview. It was clear that her work on the couch reached far beyond what happens at home. So I was excited to learn that she has a new podcast,”How ’ s Work?”It turns out personal and professional relationships have a lot in common. I ’ m Adam Grant. And this is WorkLife, my podcast with TED, sponsored by Accenture, BetterUp, Hilton and **SAP**. I ’ m an **organizational** psychologist. I study how to make work not suck. And one of the reasons work sucks for so many people is other people. So I sat down with Esther for a lively discussion about how we can get better managing our work relationships. So Esther, I was hoping you would bring a couch. Esther Perel : So that you could lay down? AG : Yeah, are we not doing therapy tonight? EP : Where should we begin? AG : I feel like you should probably tell me, but we ’ re not doing therapy. EP : We may do therapeutic maneuvers. AG : OK, good. Because I feel like I could use some. I ’ m excited to have this conversation because every time I

’ ve heard you talk about romantic relationships, I find myself substituting in work relationships and thinking that a lot of the wisdom applies. I ’ m going to start by asking you a question, which is, when you thought about transitioning from, kind of, the romantic world to the work world, what did you see as the big similarities and differences in how you look at relationships? EP : So it ’ s interesting. I don ’ t experience myself as transitioning. I, as a therapist, have worked for years with people who have work issues and certainly with family businesses. So it ’ s just that the podcast has transitioned. However, I look at relational systems, and relationships exist within a context, a cultural, political, socioeconomic context. Relationships have expectations. All relationships do. All relationships have boundaries. All relationships, if they are strong, have a foundational truth around trust. All relationships demand responsibility and accountability, communication, creativity. So what changes is the context where I explore the dynamics, and here I moved to work. I looked at what are these invisible forces that people bring with them to work and that shape relationships around conflict, communication, connection. And it ’ s the context that changes, not the thinking about relationships. AG : Are you saying we don ’ t become fundamentally different human beings when we go to work? EP : Oh, you still believe that? AG : I don ’ t. I just want to confirm that you agree with me. EP : Of course. I think, this is a very interesting thing. This notion that you bring – There is two views. There ’ s the view that says,”You don ’ t bring your life to work,”and I say,”What else do you come with?”We all have a resume that is the official resume. And we all have our relational diary that is the unofficial resume. And it all comes with you. And it shapes what expectations you have, how you collaborate, how you ask for help, how you compete, you name it. But then, you know, keep going. AG : Were you going to say more? EP : I can say it afterwards. AG : OK. Alright, so one of the things I ’ ve been thinking a lot about is power dynamics in relationships. And I remember reading a really fascinating analysis of power. I think it was by Emerson who argued that relationships were more stable if there was a slight power imbalance, because if you knew which person had a little bit more authority or more influence, then the structure was stable. It was consistent, it was predictable, whereas if people had equal power, they were constantly jocking for position, and it wasn ’ t clear where anyone should stand. And it seems like that dynamic is more common in romantic relationships than it is in professional relationships. Because at work we have hierarchies, right? We know who the boss is. We often know, if not, who has power in a team, who has more status or who has more accomplishments. I was really curious to hear how you ’ d think about this. EP : So we can have an interesting chat about power, but – AG : Are you going to save that for later too or are we going to talk about it now? EP : That I can – but it ’ s so interesting. What you ’ re highlighting is the kind of the tension that comes, you know, from – this dynamic tension that comes from the difference in the discrepancy. I look at power a little bit differently. I would put it like this. There is no relationship that doesn ’ t have a power dimension. It ’ s intrinsic to relationships. It ’ s not good or bad. It ’ s just part of the fabric of relationships because in relationships, people have expectations. And when you have expectations, it comes with a degree of dependency, reliance, and that dependency confers to the people on whom you depend power, by definition. And that power gets neutralized by making it become something that is benevolent, which we then call trust, so that it will become power to, rather than power over. But everybody understands that power isn ’ t just the vertical axis that comes with authority, anybody who ’ s ever had a two-year-old knows that. I mean, this idea that power is just a matter of authority and hierarchy, that ’ s one. But you can have power that comes from the bottom up. You can have power that is the power that constantly kind of deflects the energy, the power that takes the authority away from the people in authority, that is also power. It is actually multifaceted and intrinsic to relationship. And the question that you wanted to ask is, is this power helping the system thrive, do what it needs to do, the relationship, whatever the relationship is, or is this power that becomes oppressive, abused? Which means a breach of trust. AG : So there ' s something weird – EP : Do you agree with the notion that trust is intrinsic? Trust itself, it ' s like, it ' s intrinsic to relationships. AG : Yeah, of course. I would not argue with that. There ' s something that ' s a little odd, though, about defining power in terms of dependence. So there is a strong sociological tradition of saying,"Look, the power in a relationship is how much do I depend on you or how much do you depend on me."In an organization, though, that would suggest that the CEO is the least powerful person, because the CEO is completely dependent on everyone else to do their jobs or else the organization fails. And yet that ' s not how we traditionally look at power. EP : We look traditionally wrong. AG : We might be, but are you saying the CEO is the most powerless person in the organization? EP : No, I don ' t think it ' s this or that. I think the CEO has both ends. AG : You always have these paradoxes and they drive me crazy because I want answers. EP : Because – AG : Wait, pause for a second. EP : You want to simplify and I want to hold complexity, but that ' s the difference between a clinical psychologist and an **organizational** psychologist. AG : I thought that was the difference between Americans and Europeans, OK. EP : As well, as well. We can put that one in too. AG : But you ' re right. And I want to become less of a simplifier. So let me try to complicate and say, when I think about power from a psychological perspective, I think about it also in terms of influence, which is about control over resources. So one of the reasons that a CEO is powerful is the CEO has control over resources that lots of people at lower levels of the organization value and want. EP : Correct, correct. AG : And so, yeah, I would say the CEO is dependent in some ways, but that the resource control net outweighs the dependencies. And so the CEO is still powerful. When you think about these – EP : But if the people don ' t perform, if the people decide to go on strike, if the people underproduce, if the people decide to undermine the grand project, doesn ' t that give them power? AG : It can, sure. The CEO, though, also can take power back by saying,"I ' m going to fire those people or I ' m going to stop paying those people. I ' m going to replace those people."And so these dynamics get complicated. Where do you draw the line and who has more power in total or do you not care? Are you willing to take the multiple layers? EP : No, I think that I always look at who has power in this matter. AG : OK, good. So who do you think has more power in our relationship here? EP : Right now? AG : Yeah, or just in general. I don ' t know, I ' m curious. EP : At this moment between the two of us, it alternates. AG : How so? EP : You know, it ' s like we had this conversation, you think ping pong, I think billiards, right? So ping pong is this dyadic thing where you try to surprise and to make – you know. Billiard is about, you know, this ball has to go in the hole and which is the one that I need to kick for that one to go. And that means that you have to be able to think multiperspective. And when I think about power, I look at the interdependence of parts and on some things, I say,"Power for what?"And is it power to or power over? And is it power that actually, you know – I ' ll give you one of the – I ' m switching mid-sentence. AG : You ' re also dodging my question. EP : No, no, because listen to this, it was a moment like you all have, you have moments like that too. That ' s like a paradigmatic shift, right? I ' m in training. This is many years back, but it stayed with me, like the vista opened up and I was working with families with depression. And at one point my teacher says that the depressed has all the power. And I thought – this is a moment where you stop because I ’ m thinking you ’ re anhedonic, you feel powerless, you feel hopeless, you feel helpless. You have no energy, no sense of meaning, no purpose. It ’ s a, how can that person have the power? And then they said,”Because that person, through their impotence, is actually activating the competence of everybody else who is trying to lift them to whom they end up saying, ’ No, ’ to everything they suggest to them. And in the end, the competent people feel as defeated and deflated as the depressed one. That is power.”AG : Wow. EP : That was a moment for me where I began to think, this is way more complicated than what it looks like. What you see isn ’ t necessarily what it is. And so that ’ s how I look at power. I look at the way that, you know, an eyebrow, you know, if one of us was talking and the other one just kind of have that moment that says,”Dismissed,”with just this, that ’ s power, that makes you instantly feel like, you know, you ’ re irrelevant, you ’ re pushed aside. AG : I suddenly feel very conscious of my eyebrows. Don ’ t look at me. I think that ’ s interesting – EP : But you understand what I ’ m saying? It ’ s in the small things like this, where the power of somebody is the power to make you feel that you matter. We are creatures of meaning. It ’ s the power to make you feel like you ’ re irrelevant. And I think that who has the power is often that, and that can be performed by the one who pretends to care less. AG : So I want to come back to the question that I asked you. EP : Like you have more to lose than me, those kinds of game. I mean, power games are fascinating. Fascinating. AG : I think so, too. So I ’ m asking you this question about our power dynamic in part because I think it ’ s a lens into understanding how you think about power. So when you say power shifts back and forth in this conversation, how do you see it shifting? EP : So I think I am trying to grope – to reach for, I ’ m trying to reach for places where I think, do you agree with me? Do you see what I ’ m saying? Can you build on what I ’ m saying? AG : Yeah. EP : And then sometimes I think that you have an approach of, I agree with this, I disagree with that. And to me it ’ s like,”Oh, is it a debate or is it a conversation?”But I know you enough to know – AG : Which do you think it is? EP : I want a conversation, but because I think when it goes into agree, disagree, I start to, you know, if you disagree five times, then I ’ m going to want to prove myself to you. And then I feel that you have, you know, I want your approval. I prove myself to someone from whom I want your approval. And that means that you have power. AG : Right. EP : It ’ s not bad, it just is. So I think that ’ s the thing that I am still testing at this moment is, are we going to have debate or conversation? Are we building on each other? AG : I don ’ t know, which do you think is better? EP : Which do you think is better, people? Help me. AG : I ’ m trying to empower you to shape the conversation in the direction you want to go. No, I think that part of the reason that I sometimes default to debate is, I think we sometimes learn new things when our assumptions are tested and questioned. And so the idea that you can say something and then I can disagree with it and decide that I was wrong is to me a signal of learning, right? On the other hand, I think that one of the things I love about having a conversation with you every time we do it is I come away with new lenses for understanding things. They ’ re not necessarily correcting misconceptions or changing a belief I had, but they ’ re giving me new ways of thinking about the topics that I care about, and I love that. So my hope is we ’ re going to do that tonight. EP : And what I get when I talk with you is you force me to go for rigor because I am case-based, rather than evidence-based. My research is different. I can go and look at the psychological research on a lot of things, but that ’ s not the reality in which I exist. So when you say”the research says,”then I think I should go do some research. AG : And did you? EP : Yes. I went yesterday, and I read a chunk of research on trust. AG : And I listened to one of your therapy sessions. So we ’ ve each become a little bit more like the other. So what did you learn about trust? EP : This was really interesting, because – I mean, I can say a lot of things as how I think about trust. But when I went to look at research, what it fundamentally said is that there is an absolute definitional void. It is one of those concepts that is swimming in vagueness. There is a view that reinforces the notion that trust is a psychological state in which you express the willingness to be vulnerable, based on positive expectations with others. Then there is a view that looks at trust as process rather than state. And it looks at trusting processes like intentions, expectations, behaviors, thoughts, et cetera. But in the end, there is no agreement. The most complex psychological experience, subjective experiences actually are very hard to define. So that said, you know, I continue to look at other things and at people who said things that I found really interesting. And when I can ’ t come up with research, I go to philosophers and poets who have a different way of trying to grapple with the unexplainable. So Adam Phillips, this wonderful English psychoanalyst, starts to talk about how trust is a risk masquerading as a promise. I thought that was very beautiful. Then he said,”Trust is a word that we often put too much trust in.”AG : That sentence is self-defeating. EP : You know, then Rachel Botsman talked about how trust is an active or confident engagement with the unknown. By definition, trust is a leap of faith. Therefore, if you think transparency leads to trust, probably not always – sometimes, but at other times, transparency leads to surveillance. Surveillance is the opposite of trust. Trust is what you live with, what you don ’ t know. If you always have to know, you ’ re not trusting. AG : This was such a surprise to me when Rachel captured this. I had always thought of transparency as a driver of trust, right? So if you want somebody to have positive expectations about your behavior, if you want them to be comfortable being vulnerable with you, then you shouldn ’ t hide anything from them. And then Rachel said,”Wait a minute, If you trust me, you don ’ t need to know what I ’ m hiding.”AG : And oh, that is a big frameshift. EP : So I looked at it developmentally. I tried to apply it, and I thought of it actually, where does this concept come from? And I don ’ t know where she rooted it, but for me, it was – you know, around eight months, the kid takes a little object and they drop it. And for the first time they realize that this object continues to exist, even when they don ’ t see it. And then there is that game where they pick it up, they throw it, you pick it up, they throw it, and they develop this thing called object constancy. Object constancy then leads to people constancy, and it ’ s peekaboo. Peekaboo means that even when I don ’ t see you, you ’ re still there, and even though you ’ re not seeing me, I ’ m still there. And it becomes the foundation, against the fear of abandonment. It is the foundation of trust. You will exist inside of me when you ’ re not there, and I exist inside of you when I am not there. And therefore, I don ’ t need to check on you because I can trust you. I know you will come back, et cetera, et cetera. And I think the developmental roots of that idea that she presents are very strong. AG : Well, as a quick aside, I know now next time someone asks me about you, I ’ m going to say,”Esther Perel can even make a game of peekaboo psychologically interesting,”which is not an easy task. EP : It is the universal game that every kid plays, actually, and still, it ’ s really across the globe, that notion, but it really anchors you. I mean, that to me is the beginning of trust. AG : So when I think about taking this into the **workplace**, then one of the things that jumps to mind is, I got a really interesting query from a **reporter** a couple of weeks ago, who said,”What do you think about organizations that have **practices** of requiring open calendars?”And I immediately thought of this particular, I guess, revelation for me, where I would have said in the past,”Of course, everyone should have their calendars open. No one should need to hide what they ' re doing."And to me, that ' s a symbol of low trust, potentially. That if I need to know that I can access what everyone else is doing at every moment, that I ' m not very confident that they ' re going to do what they ' ve committed to doing. EP : It ' s called totalitarianism. AG : And that ' s a surveillance component. EP : That ' s what happened in the regimes where people spied on everybody. It ' s transparency, it ' s surveillance. It ' s really not just that I have nothing to hide kind of thing. It ' s you ' re entitled to **privacy**. The point is exactly that, while you ' re not there, I know you ' re doing your job. While I am gone, I know you ' re not rolling me under the bus. You have my back, we are in this together. We share meanings, expectations, and we predict that we are there for each other. That ' s the foundational truth of trust, and I don ' t see that coming up because I know your agenda. AG : There ' s also a power component in this. EP : Do you agree with that? AG : You ' re now trying to get us into agree, disagree? I ' m trying to complicate things and learn here, Esther, come on. EP : No, but I mean, do you think that transparency, that this idea that I have nothing to hide – You know, when somebody says to me,"I have nothing to hide,"that doesn ' t elicit trust inside of me, actually. I am sorry. Actually, I say,"Why are you saying this to me?"If you ' re saying this, you must have something. AG : It sounds a little bit like projection **bias** gone wrong. I agree with that for the record. I also think there ' s a power dynamic here that ' s really intriguing, which is when I ' m making my calendar transparent, I ' m trying to assert that my schedule has been determined, right? You can take a look at it, and you can go and claim time on it, if you want, on my terms. And that feels like a little bit of a power move. What do you think? EP : On you, from you. AG : Yeah, yeah. I ' m basically saying,"Look, I ' m more important than you are."So here ' s my calendar, pick what ' s convenient for you, but what ' s convenient for you is already predetermined by what ' s convenient for me, which is why I hate when someone sends me a calendar link and says,"Sign up on it."EP : Because? AG : Because I feel like – particularly if the person is asking me for time or asking for my help or advice, I feel like they should probably accommodate my schedule and my convenience. As opposed to,"I ' m going to impose on you, at a time that is most inconvenient for you."EP : Right,"Can you help me on this moment?"AG : Yes. EP : Yeah. AG : Ideally today or tomorrow, yeah. So I was thinking about something – EP : But can I ask you something? AG : Yeah, please. EP : Because if you ' re going to define trust, I think what ' s really important is to talk about how we build trust, what we do with breaches of trust, and can we rebuild trust once it ' s broken? Because I do think that at the time when people lived in more traditional structures in societies, and you kind of – there ' s always been breaches of trust. It ' s not like this is a new story. But when things are clearer and you have clear expectations and you know the hierarchy and you know what is the role of the father, the mother, the children, the bosses, the big brother versus the younger brother, the one who entered the family business before the other, et cetera, you could create a set of expectations what you ' re going to trust for what. Think that in this moment, where things are much more fluid and diffused, and we need to define all of that, trust has become a major centerpiece. You know, there ' s a lot of restlessness in the realm of relationships at this moment, at work as well. And as a lot of what to do, when people experienced breaches of trust, as happened definitely after #MeToo. And it shakes the whole foundation because all your expectations are now up for grabs. Was I right about you? Did I completely misunderstand who I thought I was for you? Did I completely read wrong what this partnership was actually about and you have nothing to hold on to because the relationship – So now comes the question, how do we build it? How do we rebuild it? And what is the impact of the breach of trust, which today becomes almost a loss of identity? I thought I knew what I am here and who we are together here. And I said that about romantic relationships, but I have experienced it with the partners in business, no less. AG : Yeah, so let ' s start on the repair side. In some cases it ' s harder, but I think the dynamics are sometimes a little bit simpler. So my reader research on trust repair says that,"Number one, an apology seems to be less important than an **expression** of responsibility."EP : Absolutely. AG : Saying,"I ' m sorry,"showing that I feel remorse, is nice emotionally. It doesn ' t convince you necessarily that I accept the harm that I ' ve done to you. And also that I ' m going to change my behavior moving forward. And so I always look for someone to say,"Look, whether I feel that what I did was wrong, I recognize that I hurt you, I take responsibility for that. And now I ' m also going to take responsibility for adjusting in the following ways to make sure it doesn ' t happen again."And that to me feels like the foundation of trust repair. What do you think is missing from that equation? EP : I think the first part is very much the way I like to think of it as well. It ' s the acknowledgment of the wrongdoing, of the hurt, even if you think you were legitimate and justified in what you do. I actually think that breaches of trust are part of relationships. Relationships is connection, disconnection, reconnection. It ' s harmony, disharmony, repair. It ' s not a problem, it ' s in the nature of relationships that you ' re going to do breaches. Betrayal is a different story, and I think that betrayal lives in the shadow of trust all the time. So the first thing is the acknowledgment, and it involves an element of remorse or guilt sometimes for what you ' ve done to the other person, not necessarily for your own **action**. The second one, what you call responsibility in the rebuilding, I call it the vigilante. It ' s like you show the benevolence of that the relationship matters to you. Because a breach of trust says to the other person not just that they ' ve lost power, but they ' ve lost value. You don ' t matter. Therefore, I put my interests ahead of yours, I do the thing that suits me. It basically is a message of devaluation of the other person. So that benevolent thing that you talk about is crucial. I would say the thing I like to add is the need to create new experiences. You know, what happens when people have been shaken is that they don ' t go back to the state of naivety that they had before, that leap of faith in which they existed, they do enter what Ricœur often calls the"secondary naïveté". You trust with your eyes open and you wait, you just don ' t do like this anymore, and think everything is fine, and that demands new experiences, **generative** experiences that will create new layers of cells that are going to sit on the ones that really need to die and evacuate. AG : How ' s that different from trust, but verify. EP : Just the same. Trust with your eyes open means that you – But it ' s not trust and be suspicious the whole time. You know, if you are in this cortisol level, high cortisol levels of constant checking, checking, checking, you ' re not trusting. You know, I can have all your codes – it ' s the same in relationships, when people say,"Give me all your passwords,"who trusts on that? You do surveillance, but the surveillance doesn ' t breed trust. AG : I like to give my passwords in case I forget them. But maybe a different dynamic. EP : To whom, to whom is the question. AG : To my wife, mostly. Yeah, I think that tracks for me. I ' m curious then about the more basic question of building trust and how you think about that. Rachel also changed my thinking about this. She said to me one day,"You can ' t build trust, you have to earn trust."And I was annoyed by that, actually, because it took the power to build trust out of my hands, right? That this is not something I can gain. It ' s regardless of what I do, you have the choice then to either grant me trust or deny it. And that made me feel like I lack control over building the kind of relationships that I want to build. And so I wondered if you could help me get out of that trap. EP : This a perfect moment, this is a perfect example, because you can ’ t make me trust you, you think you have no power. When in fact you have all the power to do the things that you can do, which hopefully will make me trust you. AG : But now I have probabilistic power instead of certain power, which has less power. EP : Who has certain power? I mean, where ’ d you get that idea? AG : Too many comic books growing up. EP : Yes, but that is developmental. It ’ s good for latency boys at age nine. But I mean, these things can be taken away. For I mean, every powerful guy, has had experiences where they thought they had it all and then it can instantly disappear. I think that the opposite, I mean – AG : But wait, Esther, let me stop you – EP : You can do a lot of things to earn my trust, but I think the earning is the problem. The piece that I am not so clear about – is trust a feeling? Is trust the dimension in a relationship? AG : Yeah. EP : Is trust something that you can elicit? Is trust something that you confer? You know, I think it ’ s all a combination of a number of these things and that ’ s where it ’ s not so clear. Is it a thought? Is it something that gets tested? For example, a big debate in the trust research is, you know, do you need to trust and feel safe in order to be able to take risks? That ’ s one body of research. AG : Or is it the reverse? EP : Or is it the fact that you take risks that grants you a greater sense of trust? But you first have to go into the unknown to then establish the trust. AG : I can answer that one. EP : Go ahead. AG : Yes. EP : To the second one? AG : No, no, it ’ s both. EP : It ’ s both. AG : It ’ s a cycle. EP : Yes, I agree. I think – AG : Here you are agreeing again. EP : Because you ’ re going complex. AG : So if I say something complicating, you ’ re more likely to agree with that. EP : No, when you do both ends. It ’ s not that it ’ s complicated. It ’ s that it ’ s both ends. Because I frankly don ’ t think that one can prove that it is this versus that. I think that it is a feedback loop. the more safe you feel, the more risk you take, but also the more risk you take, the more trusting you can be. AG : Sure, but as an experimentalist, I can ask,”OK, which is the more important starting condition? And if you have a chicken and the egg choice, do you want to start with trust leading the risk-taking or is the effect stronger in reverse?”I can design those experiments and try to adjudicate it. Would it convince you? EP : Yeah, but I think developmentally in those moments, right? Because I think that one of the best places to go look at how these things get layered, is to look at child development. So child development, the kid sits on your lap and if all goes well, they are secure, they are nested, they ’ re safe, they trust. And it is from that place that they jump into the world and they go, and they go to explore and to discover and take risks and do their thing. Then they turn around, and they ascertain if you ’ re still there. And if the base is safe, they do turn around and they go further, and they are experiencing security and risk at the same time. And that is the developmental dance, I think. So I don ’ t know which one, maybe with some, you know – I have two boys, and actually, it ’ s an interesting example for this. I remember, I have one, first, he would go, then he would fall, then he would figure out what to do in order not to fall. The other one would sit, study everybody else falling, and then when he understood what not to do, then he would go. And I just thought, this is for life. One is the risk taker, one is the risk manager. AG : Yeah. EP : And both will create experiences of trust and risk. AG : Yeah, no, I think that ’ s interesting. I guess what I ’ m stuck on is you didn ’ t like my expectation that I could have the power to build trust, but I think – EP : I think it ’ s beautiful, I think it ’ s a beautiful wish. AG : You definitely didn ’ t describe it that way when you made fun of me for liking comic books. EP : No, but it ’ s a wish to think that you can create the response of the other person. What you can do is what you can do. AG : Yeah. EP : It ’ s your part. And then yes, this moment where you wait to see if what you put in is going to create the thing you get back, that is the vulnerability moment. AG : Yeah, and I don ’ t like that moment. EP : That is the moment you can ’ t control. EP : You don ’ t like it. AG : I don ’ t like it at all. And I don ’ t like it in part because – EP : I think it ’ s beautiful. AG : I think it ’ s sickening. I like to avoid it as much as possible, but I say that because I think most of us have a dominant trait, which is the trait where we ’ re either the most extreme or the trait that ’ s most central to defining us. And I ’ ve noticed over the last few years when both kind of, I guess, reflecting and reading personality research that my dominant trait is conscientiousness. I am a goal-oriented person and that means I will bend and stretch all my other traits in **service** of achieving my goals. I hate the fact that unlike writing the best book I can write or preparing as hard as I can to give a talk, that ’ s going to resonate with people, when it comes to achieving a goal of building a great relationship with someone, I don ’ t have the same level of control over attaining that goal than I do the more individual goals that I direct myself to. And it makes me want to deprioritize relationships because I don ’ t know if I ’ ll succeed. EP : I get it. I totally – AG : I ’ ve never had that thought before, in fact. Is this what happens in your therapy sessions? EP : Yes. AG : OK. So now what, what do I do? EP : So if you were in a session with me, I would say”For the moment you do nothing. You just had a new thought, sit with it.”So when you realize that in a way you go for the things that you can pretty much have more agency about from the beginning to the end, I sit with that. And even me, I have nothing to say. I just take it in and I say,”Wow, that ’ s a moment.”AG : OK, how long do I have to sit? EP : You know what it is? You sit, you suddenly say, and then I watch to see, and generally what happens is that that ’ s the moment that people swallow. You know, it ’ s the quiver here. It ’ s the moment, it is that vulnerability moment, but it is a moment of insight and change. And then from there, whatever happens, either it disappears and it just was a thought, either it becomes the ground of a new enterprise into personal or otherwise. AG : So, OK. So – EP : You know, then of course, you can go home and you can say to your wife or to your partner,”I had a thought, I had a conversation with this woman there, with her accent. And as I was thinking, and I realized that –”And then she will say,”Took you so long or something?”And look, either one values this... It doesn ’ t have a monetary value necessarily. It has an existential value. I happen to think that those things matter. I also think, you know, why is the relationship conversation so important at this point? And I think you can have a great job, you can have purpose, you can have a good salary and you can have free food, but nothing will compensate for a poisonous relationship in the **workplace**, with whom you go to bed every night, fretting about somebody who is sleeping perfectly fine. And that eats people up and that I have done for decades in my **practice**. EP : It ’ s just that I had people come to my office rather than me go to their office. AG : Well, the worst case is when the person is sleeping fine because they know you ’ re fretting, which is – EP : That ’ s sadistic. AG : Yes. But one of the things I ’ m thinking about as I sit with this thought is, since I first started studying psychology in college, whenever I take values assessments, I come out with two core values at the top, one is helping other people and the other is achieving excellence. And I was surprised at first that relationships came lower because clearly helping others in such a relational dynamic. And I ’ m suddenly realizing that helping has been my solution to this problem. It ’ s to say, OK, if my stance on relationships is I want to be helpful, then I can still know whether my goal has been achieved or not, whereas if the goal is just to build the relationship or to earn trust, it ’ s never clear whether I ’ ve made progress. And I don ’ t know whether I ’ m using my time well. So what would you suggest doing with that? EP : I think when you have – AG : I ’ m trying to get a solution here, I really am. EP : No, but look, helping is a lot of things, right? Helping makes you feel rich, I mean, rich internally. Helping is the most powerful antidepressant, doing for others. AG : Yeah. EP : Helping, you know, generosity brings joy. Helping is indeed – other people need me, I will never be alone. They come to me, I ' m protected. Helping is a state in which I don ' t have to be dependent on others because I make others be dependent on me. Helping has a lot of aspects to it. Which one it is for you? See, this is where it becomes different. I don ' t look for systematization, which I think what **organizational** psychology has to do. Because you need to aggregate groups of people. I am on the other side of that, where I ' m more looking at what is the individual resonance of this thing, what is helping for you? What you feel when you give only you can tell me. If it is a defense against something else, only you can tell me, if on some level it protects you from other states that you experienced as more vulnerable, only you can tell me. It ' s like the answer that you ' re trying to find from me is in you. AG : Well, that is definitely not helpful. So we need to work on your giving, but no, no, I think that ' s right. And I respect and accept the fact that you ' re forcing me to do that. I would still prefer if you told me the answer, but I understand the impulse not to. EP : No, but it ' s not just as in pushing it back on you. Do you seriously think that I know the meaning of giving for you? AG : No. EP : No. So it would be presumptuous for me to just throw out a bunch of stuff. I have ideas, but we would be discussing them together, and then I would ask you, what do you think about that? Does that resonate for you? Is that – you know. And I think that ' s a different conversation. AG : Yeah. EP : The solution – I mean, there ' s nothing to solve here unless you tell me,"I want to change it,"but that ' s not what you ' re telling me. Then there is nothing to solve. AG : OK, then my work is done, good. So this goes to a dynamic that I wanted to talk about, being a people pleaser. So Esther, is it fair to say that I am more agreeable, more of a people pleaser than you? EP : Yes. Yes, yes, yes. AG : Why do you say that? EP : Because I don ' t think of myself as a people pleaser. AG : OK. I don ' t think of you as one either. Which is why I was so pleasantly surprised. EP : But for that we have to define what goes in the word people pleaser, right? I mean, this people pleaser, it ' s a history of life. It ' s not something you become one day. It ' s the way you learned to maintain your attachments. People pleaser is when you have learned that unless you fulfill the conditions, the people who need to take care of you won ' t be doing it in the way that they should. People pleaser is a fear of not what happens when you please, but what happens, you know, how attuned you have become at what displeasure can mean for others. And therefore, it ' s an avoidance of conflict. And therefore, I can say yes to you and to you and to you and everybody I ' ve said yes to. And then everybody at the end resents me because I have not been able to fulfill the yeses to all the people. And in the end, I know what everybody else thinks, feels and wants, and I have no clue what goes on inside of me because I ' m organized by this threat and **dread**. AG : It seems like you known me very well. Yeah, I mean when I think about, when I would describe myself as a people pleaser, I think of a few things. One is prioritizing other people ' s feelings above your own priorities. EP : Consistently. AG : Consistently. Not in one situation, but habitually. EP : Yeah, yeah. AG : I think about – EP : In order to avoid... AG : Conflict, which is the next thing, right? EP : And the conflict stems from? AG : Is this a test? EP : No, no. It ' s really, it ' s – what am I avoiding? I ' m avoiding the wrath of others. AG : Yeah, no, that ' s exactly right. I don ' t want to, yeah. As the kid who was always afraid of being called to the principal ' s office, that ' s very much the **dread**. I think the nonconfrontational stance also though is not just a fear of incurring the wrath of others. It ' s also a fear of damaging the relationship and thinking about,"OK, if I can just make this person happy in the moment, then maybe the relationship won ’ t be broken as a result.”EP : But that means that you are the one who carries the anxiety over the loss of the relationship, as if you ’ re the only one who cares about that relationship. And that ’ s not, you know, that is also back to the power thing. AG : Yes. EP : You know, it ’ s like, why is one person living the threat of loss here? And where is the other one? AG : Yeah. So I guess on that, I feel like, as a recovering people pleaser, I ’ ve tried to overcome some of these tendencies. Some of that came from being trained as a negotiator and then teaching negotiation. Some of that came from learning to do conflict mediation. A lot of it has come from just ending up with an accumulating number of requests where I just can ’ t say no to all of them anymore. EP : And what about that it came with the notion that I can say no and you can be frustrated with that, and those two shall coexist. AG : Yeah. EP : It ’ s that thing that people find impossible to hold, it ’ s how do I say no and, you know, but what if the other person doesn ’ t like it? They are entitled not to like it. And those two will live together, that doesn ’ t mean the relationship will dissolve because of it. That ’ s the fear that people live, it ’ s irrational fear. EP : Because the way it gets rationalized is people say,”It ’ s just not important enough for me. It ’ s not worth the conversation. It ’ s not worth getting into it.”And it ’ s all these rationalizations that kind of say,”It doesn ’ t really matter.”But meanwhile, at every step of the game, you ’ re foregoing you because you literally experience that the relationship only continues if the other people are pleased with you. AG : Which is a pretty sad state of affairs, if that ’ s what it takes to maintain a relationship. EP : So how did you learn to tolerate conflict, tension, disagreement? AG : I think it started probably in relationships where I felt like there was a secure base that I didn ’ t have to question whether, you know, if I hurt the person ’ s feelings or I said something that might be a little bit challenging, that it wouldn ’ t damage the relationship because the relationship was either so strong or they were secure enough in their own ego that they weren ’ t going to be insulted by it. I probably overcorrected on that at some point though, and decided that anytime I have a relationship with someone, of course they will know that when I ’ m challenging them, it ’ s because I want to help them. And I ’ ve discovered of late that there are moments when I have not made that clear, and I ’ ve regretted that. EP : What could you have said differently or done differently? AG : I think it ’ s a simple thing to start out. It ’ s fundamentally about just establishing that tough love is the way you help. So what I would have done, I think about a situation I had recently with a student I was advising. One of the first things I could have done is to say,”Look, when I was in grad school, one of the best things that a mentor did for me was tear apart the first 37 pages of the paper I had spent my whole year on. And then tell me, ’ There ’ s a gem on page 38. ’ And that tough love changed the trajectory of my career. And I made a commitment at that point to pay it forward. And so when I tell you what I ’ m about to tell you, I do it out of what I hope is some version of long-term kindness that I care about your future success enough that I ’ m willing to potentially hurt your feelings in the present.”And I think I ’ ve just taken for granted that people know that ’ s true. EP : You know, I was raised like that. This was pretty much much more of a cultural **norm** growing up. AG : This explains it a lot. EP : I mean, it ’ s like, we were not praised as much, you were not pumped up and you had to learn the distinction between this is not good versus you are not good. And it was not an easy thing to learn. AG : That is the fundamental distinction in psychology between guilt and shame. EP : Yes. Correct, correct. But it is very much the educational model of the large majority of the world, actually. AG : So I ’ ve noticed that maybe one of the manifestations of this upbringing for you is you are not at all shy about asking other people for help. I think you ’ re much **bolder** about it than I typically am. And I was curious about whether this is related to that dynamic. Are you unafraid of the rejection? EP : I would say it like this, it is absolutely true. I ask for help a lot. I offer help a lot and I have no fear when I ask for help. I think if you can, you can, if you can ’ t, I completely understand it. And my husband Jack will often say,”It ’ s like, it ’ s amazing sometimes what I dare to ask,”but I figured that the people will say to me when it ’ s not – AG : How dare you? EP : No, nobody says it like that, but it ’ s – AG : No, that ’ s because we ’ re all more agreeable than you. Yeah. EP : Or people are so shocked that somebody would dare to ask such a thing. AG : They ’ re speechless. EP : But I think the foundation of that is two different things and is actually quite serious. The first thing is I fundamentally understand that I depend on others. I also fundamentally know that my family ’ s background shaped me very much about that. So I mean, the short nut of it is that my two parents are the sole survivors of five years in concentration camps. They ’ ve lost their entire family. And boy, did they never think that they survived on their own. So this notion of self-made never existed for us. You are never self-made. You are made by the luck and the presence of everything else in the moment. I think that ’ s where my notion of the both end and the multiplicity of things is so woven into the way I look at things. And I live with that. I completely believe in a communal structure, and I depend on a lot of people for a lot of things, for me, for my kids, for my team, for, you know, I ’ m constantly networking before the word ever existed. And so is my life. I knew share economy before the term was invented. And I think that ’ s where the asking for help comes from. But the same is true in reverse. You can ask me for stuff, I don ’ t blink. It ’ s like, of course, if I can, I will. I love to feel important. I love to feel like I can give something and facilitate something. It gives me tremendous joy. AG : I think what ’ s so interesting about how courageous you are in asking is that it empowers the people you know to ask. EP : I would hope that. AG : Because I know that you will never be offended by a request because you think that part of a relationship is people should be able to ask anything they want of each other and they should also have the freedom to refuse if they think it ’ s not appropriate or not possible. Which is very helpful for those of us who **struggle** to ask for help. EP : That is true, that is true. I think, you know, there ’ s a question I love to ask in the workspace that I used to ask in a very different way when I used to do cross-cultural work, but it has become suddenly really relevant. Were you raised for autonomy or were you raised for loyalty? Think for a moment, people. Were you raised with the primary messages around relationships that said,”You have yourself to rely on. In the end, it ’ s all about you. Nobody ’ s ever going to help you as well as you can help yourself.”Ontologically, in a way, you are alone in this world. And how do you think that influences the way that you organize your relationships around you, your concepts of give and take, your concept of asking for help of what you can ask from whom, et cetera, expectations. And the loyalty thing is, you know, the messages really say you ’ re never alone, you owe a lot of things to a lot of people, and a lot of people are there for you. In my thinking, when I have a problem, the first thing I think is who can help me? I don ’ t think,”What can I do?”I think,”Who can help me?”Because there ’ s people who know more than me about this thing and will help me think it through. And I in American parlance can be seen as someone who pulls. Because I ask, it ’ s not just one or two, you know, I take it in, then I sit with it and then I make it my own. But I am completely shaped through these conversations rather than sitting with myself, thinking it through, trying it out, and it ’ s a completely different model. AG : So I feel very clear that I ’ ve learned something from this discussion because when you asked the question, OK, were you raised to be autonomous or to be loyal, my first impulse was to say,”Well, why does it have to be either or, can it be both ends?”And then I thought,”OK, I ’ ve internalized your worldview.”But in all seriousness – EP : But it ’ s true. AG : Why does it have to be either or? Can I be raised to be both? EP : I don ’ t think it ’ s either or, but I think that when you ask – How many of you would say that you were raised, raised – that doesn ’ t mean you stayed this way, it doesn ’ t mean that you didn ’ t have to because there was nobody there for you, right? It ’ s not always just a nice messages story. How many of you would say the focus was on autonomy, on self-reliance? And how many of you would say the focus was on loyalty? And how many of the people here who highlighted loyalty are not American-born? OK. AG : A lot. EP : You see that you live in the Mecca of individualism here. AG : We do. EP : And so that notion is, I think, what draws a lot of people to come here as well, is that for once I can actually prove myself, do my own thing. That is not an either or, I think there ’ s beauty and value in each of these models. But everyone has a clear sense of what they were told growing up. I really do think it ’ s a real foundational set of messages that shape our relational legacy. AG : There are a couple other quick things I wanted to raise. One is from Susan Dominus. So we share something very unusual. EP : Yes, I have to explain that. I mean, basically, Susan Dominus is a journalist at”The New York Times”and she wrote a profile about me and she wrote a profile about Adam. AG : And when we met and realized this, my first thought was, we need to write a profile of Susan based on what we learned about her from her profiling us, which is maybe too meta. But she was hoping to be here tonight. She could not make it, but she sent in a question for us to discuss, which I thought was really interesting, which was, what do we think about **passionate** work? Is that something we should strive for? Can you ever have too much of it? Do you want to go first? EP : I think it ’ s a fascinating question, actually, because passion for a long time throughout history, when it came to work, was pretty much the privilege of the artisans and the artists. People did not talk about passion when they worked the land and certainly not when they went to the factory. AG : Yeah. EP : So it really speaks to the aspirational meaning that work has received. Never have we expected more from work. We want from work today what we used to get from religion and community : belonging, purpose, meaning, community, you know, that ’ s a whole new set of expectations. And that ’ s why we suddenly have the permission to talk about work that transports you, right? But the concept was that it transcends you, it elevates you, it takes you above the ordinary into something bigger and things that people used to attribute to religious experiences. And now we want that immersive, you know, all-encompassing thing in what we do. I don ’ t think of it as good or not good, I just think it ’ s **super** interesting. And it ’ s actually a wonderful new permission that more people can have access to that kind of intensity in their work. What comes with that is that the other side of passion is massive heartbreak, disillusion. disruption, degradation. I mean, passion has its side effects. And I think that today you have that too. When people – People don ’ t just leave because the factory closes. That happens, but this is not the primary thing. People leave for management reasons, relationship reasons, they were not promoted enough, seen enough, acknowledged, and so it becomes identity questions. I think work today is organized around an identity economy. So that ’ s where I see the passion thing, that drive. Now, where do you see the most? My guess is... What do you do when a person comes to present a thing to you and they ’ re going to change the world? I mean, how many people come to present to you an enterprise? AG : Too many. EP : A startup that ’ s about to change the world. AG : It ’ s an immediate red flag for me, by the way. EP : OK, me too. AG : I ’ m like, narcissism defined. EP : But why don ’ t you think of it as passion? I mean, you could – it ’ s an interesting thing. But the part of me thinks, am I judgmental here? Why don ’ t I let these people think – you know. But big part of it is because – did you look at what the neighbor is doing? Do you figure that there ’ s five others that are doing something similar? Would you ever think about joining them? Or everybody needs their own little plaquette? AG : I mean that ’ s not passion to me, that ’ s just grandiosity, right? Like you couldn ’ t be satisfied just maybe changing a country or helping a billion people as opposed to over 7 billion, right? That just seems like a ridiculous scale to aspire to. EP : But where ’ s the limit? Grandiosity is a part of passion because when you experience passion, you experience yourself as irrepressible, right? That is part of the experience of passion. It ’ s a force that cannot be resisted. So why do we say”this is grandiose”versus”this is people who have deep passion for something?”I agree with you, but I do think that there is a **bias** in it too. AG : Yeah, there might be. I think it depends for me in part on the valor and distinction between what ’ s called harmonious versus obsessive passion. So harmonious passion is, you know, I love it, I chose it, I ’ m excited to pursue this work. And I think if that ’ s the experience of passion you have, often the work in and of itself is its own reward. If it ’ s not, then the impact that you have on each individual person is meaningful. And then the hope is to scale it over time. You don ’ t have to change the world on day one. If the passion is more obsessive, it feels more like something you have to do as opposed to something you want to do. And then I start to worry that we ’ re in a situation where you feel like, if I don ’ t change the world, then my life has had no meaning. And I think that ’ s a sad state of affairs. EP : Have you ever wanted to do something different than what you do? AG : Always. That ’ s why I don ’ t have a real job. I just study other people ’ s jobs. I live vicariously through them. EP : That is so interesting. Would you say that you ’ re **passionate** about your work? AG : Yeah, of course. EP : You are. AG : Absolutely. EP : And how do you define that? AG : You can ’ t tell? EP : I think you are deeply engaged with your work. You ’ re curious, there ’ s intensity, there ’ s insatiability, there is a constant look for what else is out there that I don ’ t know yet. I see all of that, but that ’ s what I see. When you think I ’ m **passionate** about work, how do you articulate that? AG : I think about it in a few ways. One is, I think about, well, not every part of my work. There are parts of my job that I ’ m not **passionate** about. I ’ ve tried to design those out of it over time, but some of them are kind of stuck. When I think about the parts I ’ m **passionate** about, I think about waking up in the morning, looking forward to starting work, going to bed at night, thinking about all the cool ideas I get to explore and all the amazing, talented, inspiring students I get to learn from, and maybe try to help a little bit. And that for me is energizing, and I guess it ’ s exciting in the way that I think falling in love is exciting. EP : So how come the age where we want passion the most at work, is the age where we stay about an average of two years? I mean, isn ’ t it interesting that the moment where we have the highest set of expectations about this thing is also the time where we are roaming and nomadic. AG : Yeah, I think that ’ s a huge problem. EP : How do you understand it? You have the students. You know, I see students who tell me,”I love it, I ’ m going to take another three months and I ’ m going to start looking for the next thing.”I say,”If you love it, why do you want to leave?”I can ’ t put one and one together. The leaving and the loving, don ’ t you want to stay when you love something? AG : Yeah. EP : In my logic. AG : Yeah. I ’ ll tell you how I deal with this, which is, one, I try to read the research and forecast what ’ s going to happen to them. The work that Jon Jachimowicz and his colleagues have just published has been helpful to me. What he ’ s found is that when people think about passion as pleasure or joy, they ’ re much less likely to succeed in pursuing whatever they thought their passion was than if they think about it as meaning and purpose. And I think that ' s because emotions are much more fleeting and a sense of mattering is much more sustainable. EP : It is very important, what you just said. This is, I think passion that is defined as joy and excitement – and what it does for me, is very different than passion that is defined by the nature, the richness of the thing in which you ' re plunging. You know, that is giving you meaning because it is in itself meaningful. AG : Yeah. Yeah, for me, that resonates. And so I guess what I say to students is, when they say,"I want to find a job I ' m **passionate** about,"or"I want to love my work."That to me is something you can look forward to 10, 20 years down the road after you ' ve figured out where you find meaning and what kind of contribution you can make. But I don ' t feel like there ' s an easy answer to your question, which is a really interesting one. Why would you leave something you love? This is another paradox point though, that you yourself have made, right? Which is you can love a person and still say,"This relationship is not right for me."And why not believe the same is true about a job? EP : The way I ' ve said it is in marriage today, people will say,"It ' s not that I divorce because I ' m miserable and unhappy, it ' s I divorce because I could be happier."And I think something similar is happening at work. This elusive quest for more, for better, for FOMO and all of that makes us think, what else is out there that I haven ' t yet done versus the good enough, versus the this is rich in and of itself. Stay here, deepen it, you know, build something from it. You know, it ' s just – For me, the moment where I read this was an"Wall Street Journal"article that said"The quitters are the winners."And I thought,"Oh boy, one headline tells you what ' s going on."You know, what does it mean? And then of course, it explained : you get a better salary, you get promoted more, but the whole point was about leaving versus staying. This is, I find I don ' t – you ' re the one who studies these things, right? I don ' t know how to **interpret** that. AG : The only thing that jumped to mind on this is I think Oliver Burkeman covered the best **antidote** I ' ve seen. He wrote this great little article that said,"Look, every time you feel FOMO, you should erase it, and instead turn it into JOMO, the joy of missing out."And I thought this was such a clever concept to say, anytime you ' re afraid you ' re going to miss out on a great job opportunity or even a social event, right? You ' re only thinking about everything that could have been better. Well, why not be thrilled about all the horrible jobs you didn ' t take, all the toxic cultures you avoided, all of the difficult bosses you didn ' t get stuck working for? And that joy of missing out for me is something that probably sticks in my head on a **daily** basis. With that, I am feeling a little FOMO that we ' re not going to hear from the audience. So let ' s open it up. Woman 1 : Thanks, guys. I think we all thought that the autonomy versus loyalty was a really eye-opening and powerful concept. I ' m just curious, your comments on how you see that play out in the **workplace**. EP : Great, I ' m a cross-cultural psychologist. My work was on how families and individuals deal with large cultural changes. And this notion of collective versus individualistic. So what it really influences a lot is this. Do you know that person that does more than others because they can do it better than anybody else? But then quietly resents it, but then doesn ' t want anybody to do it because nobody would do it the way they do it. That is raised for autonomy to the end degree. You know that person in your personal life, you know that person at work. So you mapped this on a whole line here, between the degree of, you know, I ' m going to hide the fact that I don ' t know something. So when people these days talk about trust for vulnerability, one of the first things they talk about is asking for help, right? The disclosure of the leader and the people who can say,"I don ' t know, I want to try and ask for help."But this asking for help in some people and in some cultures means,"I don ' t know and I ' m going to find out who can help me.”Like I said, who can help me? I don ’ t experience it as a weakness whatsoever. I have no shame about it, but I know that for other people it ’ s an admittance of what ’ s wrong with me. And the moment it ’ s what ’ s wrong with you, I can ’ t do it. Then your asking for help is a completely different thing. And you put these people to work together? This is true in the classroom and this is true at work. They are going to have messages and meta messages. That ’ s just a nutshell on a – AG : Alright. So autonomy versus loyalty. What I think about is you don ’ t have to give people autonomy about the ends if you give them autonomy about the means. This is something one of my mentors, Richard Hackman found throughout his career, that if leaders were very clear about a **mission**, a purpose, a goal to work toward, they could give people a ton of autonomy then about when to do it, where to do it, how to do it, with whom to do it. And that would work out really well. And I think if you take that view, then giving people that kind of autonomy is an **expression** of loyalty, right? You ’ re you ’ re saying,”I believe in you, I trust you. I ’ m going to let you know what our objectives are and why I think those are worthwhile.”And then you got this. So next question. Woman 2 : You touched on this very briefly earlier, the difference between breach and betrayal. How is that different? I think instinctually, we understand what that might mean in a relationship. And that said, that may vary from person to person in a relationship, but how do you distinguish between the two in the **workplace**? How do those look different in the **workplace**? AG : When I think about this at work, I think about what ’ s called the psychological contract. So all of you who have jobs, you have formal contracts with your employers. There ’ s also the what ’ s called the psychological contract, which is an unwritten set of expectations and obligations that you have with your employer. There are basically three types that we ’ ve studied. There ’ s an economic contract, which is basically,”Look, I ’ ll do my job and you pay me to support myself and my family.”There ’ s a relational contract, which is much more about,”OK, I will invest in this organization as a community or a family, and I ’ ll bring a little bit of extra dedication and commitment to the people in the place. And in return, what I expect is to be treated like a member of the family.”And then the final type of contract is more of a cause, where you say,”Look, I will give you extraordinary levels of effort and dedication and grit. And in return, what I want to do is I want to make a difference, hopefully not change the world.”I think that contracts get breached when somebody has a set of expectations that are not met, often by the employer, or by their boss. I think that a betrayal is when people go further and they avoid or violate the contract and say,”Not only am I unwilling to follow through on this set of commitments or expectations, that was never there to begin with. We were never a family. You were never working on a meaningful cause, I never owed you a paycheck.”And I think that might be the difference. I have no idea what to do with that, but that was my thought. EP : That ’ s beautiful. I mean, it ’ s an element of gaslighting. Basically, you know, you ’ re crazy. You ’ re crazy for even thinking this in the first place. Yes. Woman 3 : How do you know whether to leave or stay in the work context? AG : Hypothetically. Woman 3 : Hypothetically, what is a good reason to stay? And what if, even if you stay, what is a way to think about something more in a relationship to work that is productive and meaningful and healthy without jumping around in a disorganized way? AG : I mean, I – You ’ re not going to be surprised, there ’ s a framework on this. So Hirschman wrote about exit, voice and loyalty, there ’ s an addition now of neglect. So when you ’ re dissatisfied with a job or a person or an organization, or a country, right? You have, I think, as far as I know, four choices. You can leave, that ’ s an exit. You can speak up and try to change it, that ’ s voice. You can neglect, where you do the absolute minimum not to get fired, that ’ s office space. And then the final option is loyalty, which is basically to say,”Look, I ’ m committed enough to this person or system that I ’ m just going to grit my teeth and bear it.”And I think that there ’ s a progression that we all need to go through when we ’ re considering leaving, which is to say, have I been loyal long enough that I ’ ve given the person or the place a chance to change? And then, once I ’ ve done that, have I tried to change it and have I made my best effort at doing that? And if not, it ’ s probably time to go. EP : So the interesting thing about your framework is that they all put the eye at the center. What is the circumstances of this? You know, do you decide? Are you sending money over to the old country and there ’ s a whole family that lives on your work? That ’ s going to change if you – you know. It ’ s like, where is the locus of control in those four things, which I think are very beautiful, you are the locus of control. In many people ’ s lives, that is not the reality. There are lots of other things that are affecting your decision. Maybe at this moment, I would leave. If it was up to me, I would leave. But given the circumstances of my life, this is not the right moment. And I think it ’ s **juggling** that ambiguity that is the most difficult thing. On occasion, many of us will say,”This, I left too soon, and this, I could have left sooner.”This is the story of our lives, in relationships, in work situations, in places where we lived in houses that were leaking. I mean, it ’ s like we are constantly – And the story of our lives is a story of where we wished we had stayed longer and where we wished we had left sooner. And I don ’ t think that there is a specific, you know, I can say I have rarely heard people leave a job and regret it. But I do look at our time, my dear. AG : There ’ s one other quick thing I wanted to bring up before we wrap, which is, we got a bunch of questions via social media. There ’ s this question I love from Michelle Miller, which is, what do we think about work spouses? And I have to tell you – I ’ ve heard people use this term increasingly, over the last few years. And at first I ’ m like,”Wait, are you hedging in case your marriage isn ’ t going well? You have another spouse? What is this about?”So given that you ’ ve study relationships and you work on relationships, what in the world is a work spouse, and do you think we should have one as a preview? I think the answer is a very clear”no,”but I want to hear your take. EP : The same way that when people say,”Our company ’ s a family,”I ’ m like,”Seriously?”You know, a family is a family. And I can tell you, family companies are different from this, this is not the same. I like that, I understand that people want to borrow terminology from other areas so that they can create analogies, but it ’ s **trivial**. It is something rather vulgar and boring about the spouse work concept. It ’ s like, it ’ s a noninteresting way of looking at it. What it says is this, the person with whom you spend all the time, with whom you make all these decisions, with whom you share so many important things. The only thing that is interesting about the work spouse concept is that it betrays one reality or it translates one reality, is that way too often these days we bring the best of ourselves to work and the leftovers home. This is where I do think the word”work spouse”becomes relevant, but for the rest, no, find a word that – There ’ s a lack of vocabulary for this thing. You know, you ’ re not married to them. Marriage is a whole lot more complicated than that. AG : You start working with someone else, you ’ re also not cheating on them, right? EP : You know, it ’ s a – And by the way, I don ’ t know if – You know, I speak a few languages and I don ’ t know the term in other languages. It doesn ’ t translate in my head. Does anybody have a translation for this? French, Hebrew, German, Spanish, anybody? It ’ s an interesting thing. It ’ s like, you know, you work with that person. You know, whatever it is that happens, but it ’ s not your work spouse. AG : Like closest colleague. EP : Don ’ t do the laundry of that person, you know? AG : Well, this has been fun and enlightening as always. EP : Yes, I like it too. AG : I want to say thanks to the audience and thanks to Esther for doing this. EP : Thank you to all. AG : Next time on WorkLife. Carmen Medina : I recognized that the only way I could survive at the CIA was that I had to give up. So yes, in that sense, I became less authentic. AG : The double-edged sword, bringing your whole self to work. WorkLife is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ' Donnell, Constanza Gallardo, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. This bonus episode was produced by Jessica Glazer. Our show is mixed by Rick Kwan, original music by Hahnsdale Hsu and Allison Leyton-Brown. **Special** thanks to our sponsors, Accenture, BetterUp, Hilton and **SAP**. Thanks to TED for hosting our discussion and to Esther Perel for joining. And thanks to you for listening. If you liked what you heard, we ' d love it if you could rate and review the show, it helps people find us.

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Audrey Tang : Very happy to be joining you, and good local time, everyone. David Biello : So, tell us about – Sorry to – Tell us about digital tools and COVID. AT : Sure. Yeah, I ' m really happy to share with you how Taiwan successfully countered the COVID using the power of digital democracy tools. As we know, democracy improves as more people participate. And digital technology remains one of the best ways to improve participation, as long as the focus is on finding common ground, that is to say, prosocial media instead of antisocial media. And there ' s three key ideas that I would like to share today about digital democracy that is fast, fair and fun. First about the fast part. Whereas many jurisdictions began countering **coronavirus** only this year, Taiwan started last year. Last December, when Dr. Li Wenliang, the PRC whistleblower, posted that there are new SARS cases, he got inquiries and eventually punishments from PRC police institutions. But at the same time, the Taiwan equivalent of Reddit, the Ptt board, has someone called nomorepipe reposting Dr. Li Wenliang ' s whistleblowing. And our medical officers immediately noticed this post and issued an order that says all passengers flying in from Wuhan to Taiwan need to start health inspections the very next day, which is the first day of January. And this says to me two things. First, the civil society trusts the government enough to talk about possible new SARS outbreaks in the public forum. And the government trusts citizens enough to take it seriously and treat it as if SARS has happened again, something we ' ve always been preparing for, since 2003. And because of this open civil society, according to the CIVICUS Monitor after the Sunflower Occupy, Taiwan is now the most open society in the whole of Asia. We enjoy the same freedom of speech, of assembly, [unclear] as other liberal democracies, but with the emphasis on keeping an open mind to novel ideas from the society. And that is why our schools and businesses still remain open today, there was no lockdown, it ' s been a month with no local confirmed cases. So the fast part. Every day, our Central Epidemic Command Center, or CECC, holds a press conference, which is always livestreamed, and we work with the journalists, they answer all the questions from the journalists, and whenever there ' s a new idea coming in from the social sector, anyone can pick up their phone and call 1922 and tell that idea to the CECC. For example, there was one day in April where a young boy has said he doesn ' t want to go to school because his school mates may laugh at him because all he had is a pink medical mask. The very next day, everybody in the CECC press conference started wearing pink medical masks, making sure that everybody learns about gender mainstreaming. And so this kind of rapid response system builds trust between

the government and the civil society. And the second focus is fairness. Making sure everybody can use their national health insurance card to collect masks from **nearby** pharmacies, not only do we publish the stock level of masks of all pharmacies, 6, 000 of them, we publish it every 30 seconds. That ' s why our civic hackers, our civil engineers in the digital space, built more than 100 tools that enable people to view a map, or people with blindness who talk to chat bots, voice assistants, all of them can get the same inclusive access to information about which pharmacies near them still have masks. And because the national health insurance single payer is more than 99. 9 percent of health coverage, people who show any symptoms will then be able to take the medical mask, go to a local clinic, knowing fully that they will get treated fairly without incurring any **financial** burden. And so people designed a dashboard that lets everybody see our supply is indeed growing, and whether there ' s over- or undersupply, so that we codesign this distribution system with the pharmacies, with the whole of society. So based on this analysis, we show that there was a **peak** at 70 percent, and that remaining 20 percent of people were often young, work very long hours, when they go off work, the pharmacies also went off work, and so we work with convenience stores so that everybody can collect their mask anytime, 24 hours a day. So we ensure fairness of all kinds, based on the digital democracy ' s feedback. And finally, I would like to acknowledge that this is a very **stressful** time. People feel anxious, outraged, there ' s a lot of panic buying, a lot of conspiracy theories in all economies. And in Taiwan, our counter-disinformation strategy is very simple. It ' s called"humor over rumor."So when there was a panic buying of tissue paper, for example, there was a rumor that says,"Oh, we ' re ramping up mass production, it ' s the same material as tissue papers, and so we ' ll run out of tissue paper soon."And our premier showed a very memetic picture that I simply have to share with you. In very large print, he shows his bottom, wiggling it a little bit, and then the large print says"Each of us only have one pair of buttocks."And of course, the serious table shows that tissue paper came from South American materials, and medical masks come from domestic materials, and there ' s no way that ramping up production of one will hurt the production of the other. And so that went absolutely viral. And because of that, the panic buying died down in a day or two. And finally, we found out the person who spread the rumor in the first place was the tissue paper reseller. And this is not just a single shock point in social media. Every single day, the **daily** press conference gets translated by the spokesdog of the Ministry of Health and Welfare, that translated a lot of things. For example, our physical **distancing** is phrased as saying"If you are outdoors, you need to keep two dog-lengths away, if you are indoor, three dog-lengths away,"and so on. And hand sanitation rules, and so on. So because all this goes viral, we make sure that the factual humor spreads faster than rumor. And they serve as a vaccine, as inoculation, so that when people see the conspiracy theories, the R0 value of that will be below one, meaning that those ideas will not spread. And so I only have this five-minute briefing, the rest of it will be driven by your Q and A, but please feel free to read more about Taiwan ' s counter-coronavirus strategy, at taiwancanhelp. us. Thank you. DB : That ' s incredible. And I love this"humor versus rumor."The problem here in the US, perhaps, is that the rumors seem to travel faster than any response, whether humorous or not. How do you defeat that aspect in Taiwan? AT : Yeah, we found that, of course, humor implicitly means there is a sublimation of upsetness, of outrage. And so as you see, for example, in our premier ' s example, he makes fun of himself. He doesn ' t make a joke at the expense of other people. And this was the key. Because people think it hilarious, they share it, but with no malicious or toxic intentions. People remember the actual payload, that table about materials used to produce masks, much more easily. If they make a joke that excludes parts of the society, of course, that part of society will feel outraged and we will end up creating more divisiveness, rather than prosocial behavior. So the humor at no expense, not excluding any part of society, I think that was the key. DB : It ’ s also incredible because Taiwan has such close ties to the origin point of this. AT : PRC, yes. DB : The mainland. So given those close economic ties, how do you survive that kind of disruption? AT : Yeah, I mean, at this moment, it ’ s been almost a month now with no local confirmed cases, so we ’ re doing fine. And what we are doing, essentially, is just to respond faster than pretty much anyone. We started responding last year, whereas pretty much everybody else started responding this year. We tried to warn the world last year, but, anyway. So in any case, the point here is that if you start early enough, you get to make sure that the border control is the main point where you **quarantine** all the returning residents and so on, instead of waiting until the community spread stage, where even more human-right invading techniques would probably have to be deployed one way or the other. And so in Taiwan, we ’ ve not declared an emergency situation. We ’ re firmly under the constitutional law. Because of that, every measure the administration is taking is also applicable in non-coronavirus times. And this forces us to innovate. Much as the idea of”we are an open liberal democracy”prevented us from doing takedowns. And therefore, we have to innovate of humor versus rumor, because the easy path, the takedown of online speech, is not accessible to us. Our design criteria, which is no lockdowns, also prevented us from doing any, you know, very invasive **privacy** encroaching response system. So we have to innovate at the border, and make sure that we have a sufficient number of, for example, **quarantine** hotels or the so-called”digital fences,”where your phone is basically connected to the **nearby** telecoms, and they make sure that if they go out of the 15-meter or so radius, an SMS is sent to the local household managers or police and so on. But because we focus all these measures at the border, the vast majority of people live a normal life. DB : Let ’ s talk about that a little bit. So walk me through the digital tools and how they were applied to COVID. AT : Yes. So there ’ s three parts that I just outlined. The first one is the collective intelligence system. Through online spaces that we design to be devoid of Reply buttons, because we see that, when there ’ s Reply buttons, people focus on each other ’ s face part, not the book part, and without”Reply”buttons, you can get collective intelligence working out their rough consensus of where the direction is going with the response strategies. So we use a lot of new technologies, such as Polis, which is essentially a forum that lets you upvote and downvote each other ’ s feelings, but with real-time clustering, so that if you go to cohack. tw, you see six such conversations, talking about how to protect the most vulnerable people, how to make a smooth transition, how to make a fair distribution of supplies and so on. And people are free to voice their ideas, and upvote and downvote each other ’ s ideas. But the trick is that we show people the main divisive points, and the main consensual points, and we respond only to the ideas that can convince all the different opinion groups. So people are encouraged to post more eclectic, more nuanced ideas and they discover, at the end of this consultation, that everybody, actually, agrees with most things, with most of their neighbors on most of the issues. And that is what we call the social mandate, or the democratic mandate, that then informs our development of the counter-coronavirus strategy and helping the world with such tools. And so this is the first part, it ’ s called listening at scale for rough consensus. The second part I already covered is the distribute ledger, where everybody can go to a **nearby** pharmacy, present their NHI card, buy nine masks, or 10 if you ’ re a child, and see the stock level of that pharmacy on their phone actually decreasing by nine or 10 in a couple of minutes. And if they grow by nine or 10, of course, you call the 1922, and report something fishy is going on. But this is participatory accountability. This is published every 30 seconds. So everybody holds each other accountable, and that massively increases trust. And finally, the third one, the humor versus rumor, I think the important thing to see here is that wherever there ’ s a trending disinformation or conspiracy theory, you respond to it with a humorous package within two hours. We have discovered, if we respond within two hours, then more people see the vaccination than the conspiracy theory. But if you respond four hours or a day afterwards, then that ’ s a lost cause. You can ’ t really counter that using humor anymore, you have to invite the person who spread those messages into cocreation workshops. But we ’ re OK with that, too. DB : Your speed is incredible. I see Whitney has joined us with some questions. Whitney Pennington Rodgers : That ’ s right, we have a few coming in already from the audience. Hi there, Audrey. And we ’ ll start with one from our community member Michael Backes. He asks how long has humor versus rumor been a strategy that you ’ ve implemented. Excuse me.”How long has humor versus rumor strategy been implemented? Were comedians consulted to make the humor?”AT : Yes, definitely. Comedians are our most cherished colleagues. And each and every ministry has a team of what we call participation officers in charge of engaging with trending topics. And it ’ s a more than 100 people-strong team now. We meet every month and also every couple of weeks on specific topics. It ’ s been like that since late 2016, but it ’ s not until our previous spokesperson, Kolas Yotaka, joined about a year and a half ago, do the professional comedians get to the team. Previously, this was more about inviting the people who post, you know, quotes like”Our tax filing system is explosively hostile,”and gets trending, and previously, the POs just invited those people. Everybody who **complains** about the finance minister ’ s tax-filing experience gets invited to the cocreation of that tax filing experience. So previously, it was that. But Kolas Yotaka and the premier Su Tseng-chang said, wouldn ’ t it be much better and reach more people if we add some dogs to it or cat ’ s pictures to it? And that ’ s been around for a year and a half. WPR : Definitely, I think it makes a lot of difference, just even seeing them without being part of the thought process behind that. And we have another question here from G. Ryan Ansin. He asks,”What would you rank the level of trust your community had before the pandemic, in order for the government to have a chance at properly controlling this crisis?”AT : I would say that a community trusts each other. And that is the main point of digital democracy. This is not about people trusting the government more. This is about the government trusting the citizens more, making the state transparent to the citizen, not the citizen transparent to the state, which would be some other regime. So making the state transparent to the citizens doesn ’ t always elicit more trust, because you may see something wrong, something missing, something exclusively hostile to its user experience, an so on, of the state. So it doesn ’ t necessarily lead to more trust from the government. Sorry, from the citizen to the government. But it always leads to more trust between the social sector stakeholders. So I would say the level of trust between the people who are working on, for example, medical officers, and people who are working with the pandemic responses, people who manufacture medical masks, and so on, all these people, the trust level between them is very high. And not necessarily they trust the government. But we don ’ t need that for a successful response. If you ask a random person on the street, they will say Taiwan is performing so well because of the people. When the CECC tells us to wear the mask, we wear the mask. When the CECC tells us not to wear a mask, like, if you are keeping physical distance, we wear a mask anyway. And so because of that, I think it

’ s the social sector ’ s trust between those different stakeholders that ’ s the key to the response. WPR : I will come back shortly with more questions, but I ’ ll leave you guys to continue your conversation. AT : Awesome. DB : Well, clearly, part of that trust in government was maybe not there in 2014 during the Sunflower Movement. So talk to me about that and how that led to this, kind of, digital transformation. AT : Indeed. Before March 2014, if you asked a random person on the street in Taiwan, like, whether it ’ s possible for a minister – that ’ s me – to have their office in a park, literally a park, anyone can walk in and talk to me for 40 minutes at a time, I ’ m currently in that park, the Social Innovation Lab, they would say that this is crazy, right? No public officials work like that. But that was because on March 18, 2014, hundreds of young activists, most of them college students, occupied the legislature to express their profound opposition to a trade pact with Beijing under consideration, and the secretive manner in which it was pushed through the parliament by Kuomintang, the ruling party at the time. And so the protesters demanded, very simply, that the pact be scraped, and the government to institute a more transparent ratification process. And that drew widespread public support. It ended a little more than three weeks later, after the government promised and agreed on the four demands [unclear] of legislative oversight. A poll released after the occupation showed that more than 75 percent remained dissatisfied with the ruling government, illustrating the crisis of trust that was caused by a trade deal dispute. And to heal this rift and communicate better with everyday citizens, the administration reached out to the people who supported the occupiers, for example, the g0v community, which has been seeking to improve government transparency through the creation of open-source tools. And so, Jaclyn Tsai, a government minister at the time, attended our hackathon and proposed the establishment of novel **platforms** with the online community to exchange policy ideas. And an experiment was born called vTaiwan, that pioneerly used tools such as Polis, that allows for”agree”or”disagree”with no Reply button, that gets people ’ s rough consensus on issues such as crowdfunding, equity-based crowdfunding, to be precise, teleworking and many other cyber-related legislation, of which there is no existing unions or associations. And it proved to be very successful. They solved the Uber problem, for example, and by now, you can call an Uber – I just called an Uber this week – but in any case, they are operating as taxis. They set up a local taxi company called Q Taxi, and that was because on the **platform**, people cared about insurance, they care about registration, they care about all the sort of, protection of the passengers, and so on. So we changed the taxi regulations, and now Uber is just another taxi company along with the other co-ops. DB : So you ’ re actually, in a way, crowdsourcing laws that, well, then become laws. AT : Yeah, learn more at crowd. law. It ’ s a real website. DB : So, some might say that this seems easier, because Taiwan is an island, that maybe helps you control COVID, helps promote social cohesion, maybe it ’ s a smaller country than some. Do you think that this could be scaled beyond Taiwan? AT : Well, first of all, 23 million people is still quite some people. It ’ s not a city, as some usually say, you know,”Taiwan is a city-state.”Well, 23 million people, not quite a city-state. And what I ’ m trying to get at, is that the high population density and a variety of cultures – we have more than 20 national languages – doesn ’ t necessarily lead to social cohesion, as you said. Rather, I think, this is the humbleness of all the ministers in the counter-coronavirus response. They all took on an attitude of”So we learned about SARS”– many of them were in charge of the SARS back then, but that was **classical** epidemiology. This is SARS 2. 0, it has different characteristics. And the tools that we use are very different, because of the digital transformation. And so we are in it to learn together with the citizens. Our vice president at the time, Dr. Chen Chien-jen, an academician, literally wrote the textbook on epidemiology. However, he still says,"You know, what I ' m going to do is record an online MOOC, a crash course on epidemiology, that shares with, I think, more than 20, 00 people **enrolled** the first day, I was among them, to learn about important ideas, like the R0 and the basic transmission and how the various different measures work, and then they asked people to innovate. If you think of a new way that the vice president did not think of, just call 1922, and your idea will become the next day ' s press conference. And this is this colearning strategy, I think, that more than anything enabled the social cohesion, as you speak. But this is more of a robust civil society than the uniformity. There ' s no uniformity at all in Taiwan, everybody is entitled to their ideas, and all the social innovations, ranging from using a traditional rice cooker to revitalize, to disinfect the mask, to pink medical mask, and so on, there ' s all variety of very interesting ideas that get amplified by the **daily** press conference. DB : That ' s beautiful. Now – oh, Whitney is back, so I will let her ask the next question. WPR : Sure, we ' re having some more questions come in. One from our community member Aria Bendix. Aria asked,"How do you ensure that digital campaigns act quickly without sacrificing accuracy? In the US, there was a fear of inciting panic about COVID-19 in early January."AT : This is a great question. So most of the scientific ideas about the COVID are evolving, right? The efficacy of masks, for example, is a very good example, because the different characteristics of previous respiratory diseases respond differently to the facial mask. And so, our digital campaigns focus on the idea of getting the rough consensus through. So basically, it ' s a reflection of the society, through Polis, through Slido, through the joint **platform**, the various tools that vTaiwan has prototyped, we know that people are feeling a rough consensus about things and we ' re responding to the society, saying,"This is what you all feel and this is what we ' re doing to respond to your feelings. And the scientific consensus is still developing, but we know, for example, people feel that wearing a mask mostly protects you, because it reminds you to not touch your face and wash your hands properly."And these, regardless of everything else, are the two things that everybody agrees with. So we just capitalize on that and say,"OK, wash your hands properly, and don ' t touch your face, and wearing a mask reminds you of that."And that lets us cut through the kind of, very ideologically charged debates and focus on what people generally resonate with one another. And that ' s how we act quickly without sacrificing scientific accuracy. WPR : And this next question sort of feels connected to this as well. It ' s a question from an anonymous community member."Pragmatically, do you think any of your policies could be applied in the United States under the current Trump administration?"AT : Quite a few, actually. We work with many states in the US and abroad on what we call"epicenter to epicenter diplomacy."So what we ' re doing essentially is, for example, there was a chat bot in Taiwan that lets you, but especially people under home **quarantine**, to ask the chat bot anything. And if there is a scientific adviser who already wrote a frequently asked question, the chat bot just responds with that, but otherwise, they will call the science advisory board and write an accessible response to that, and the spokesdog would translate that into a cute dog meme. And so this feedback cycle of people very easily accessing, finding, and asking a scientist, and an open API that allows for voice assistance and other third-party developers to get through it, resonates with many US states, and I think many of them are implementing it. And before the World Health Assembly, I think three days before, we held a 14 countries [unclear] lateral meeting, kind of, pre-WHA, where we shared many small, like, quick wins like this. And I think many jurisdictions took some of that, including the humor versus rumor. Many of them said that they ’ re going to recruit comedians now. WPR : I love that. DB : I hope so. WPR : I hope so too. And we have one more question, which is actually a **follow-up**, from Michael Backes, who asked a question earlier.”Does the Ministry plan to publish their plans in a white paper?”Sounds like you ’ re already sharing your plans with folks, but do you have a plan to put it out on paper? AT : Of course. Yeah, and multiple white papers. So if you go to taiwancanhelp. us, that is where most of our strategy is, and that website is actually crowdsourced as well, and it shows that more than five million now, I think, medical masks donated to the humanitarian aid. It ’ s also crowdsourced. People who have some masks in their homes, who did not collect the rationed masks, they can use an app, say,”I want to dedicate this to international humanitarian aid,”and half of them choose to publish their names, so you can also see the names of people who participated in this. And there ’ s also an”Ask Taiwan Anything”website, at fightcovid. edu. tw, that outlines, in white paper form, all the response strategies, so check those out. WPR : Great. Well, I will disappear and be back later with some other questions. DB : A blizzard of white papers, if you will. I ’ d like to turn the focus on you a little bit. How does a conservative anarchist become a digital minister? AT : Yeah, by occupying the parliament, and through that. More interestingly, I would say that I go working with the government, but never for the government. And I work with the people, not for the people. I ’ m like this Lagrange point between the people ’ s movements on one side and the government on the other side. Sometimes right in the middle, trying to do some coach or translation work. Sometimes in a kind of triangle point, trying to supply both sides with tools for prosocial communication. But always with this idea of getting the shared values out of different positions, out of varied positions. Because all too often, democracy is built as a showdown between opposing values. But in the pandemic, in the infodemic, in climate change, in many of those structural issues, the virus or carbon dioxide doesn ’ t sit down and negotiate with you. It ’ s a structural issue that requires common values built out of different positions. And so that is why my working principle is radical transparency. Every conversation, including this one, is on the record, including the internal meetings that I hold. So you can see all the different meeting transcripts in my YouTube channel, in the SayIt **platform**, where people can see, after I became digital minister, I held 1, 300 meetings with more than 5, 000 speakers, with more than 260, 000 utterances. And every one of them has a URL that becomes a social object that people can have a conversation on. And because of that, for example, when Uber ’ s David Plouffe visited me to lobby for Uber, because of radical transparency, he is very much aware of that, and so he made all the arguments based on public good, based on sustainability, and things like that, because he knows that the other sides would see his positions very clearly and transparently. So that encourages people to add on each other ’ s argument, instead of attacking each other ’ s person, you know, credits and things like that. And so I think that, more than anything, is the main principle of conserving the anarchism of the internet, which is about, you know, nobody can force anyone to hook to the internet, or to adhere to a new internet protocol. Everything has to be done using rough consensus and running code. DB : I wish you had more counterparts all around the world. Maybe you wish you had more counterparts all around the world. AT : That ’ s why these ideas are worth spreading. DB : There you go. So one of the challenges that might arise with some of these digital tools is access. How do you approach that part of it for folks maybe who don ’ t have the best broadband connection or the latest **mobile** phone or whatever it might be that ’ s required? AT : Well, anywhere in Taiwan, even on the top of Taiwan, almost 4, 000 meters high, the Saviah, or the Jade Mountain, you ’ re guaranteed to have 10 megabits per second over 4G or fiber or cable, with just 16 US dollars a month, an unlimited plan. And actually, on the top of the mountain, it ’ s faster, fewer people use that bandwidth. And if you don ’ t, it ’ s my fault. It ’ s personally my fault. In Taiwan, we have broadband as a human right. And so when we ’ re deploying 5G, we ’ re looking at places where the 4G has the weakest signal, and we begin with those places in our 5G deployment. And only by deploying broadband as a human right can we say that this is for everybody. That digital democracy actually strengthens democracy. Otherwise, we would be excluding parts of the society. And this also applies to, for example, you can go to a local digital opportunity center to rent a tablet that ’ s guaranteed to be manufactured in the past three years, and things like that, to enable, also, the different digital access by the digital opportunity centers, universities and schools, and public libraries, very important. And if people who prefer to talk in their town hall, I personally go to that town hall with a 360 recorder, and livestream that to Taipei and to other municipalities, where the central government ’ s public servants can join in a connected room style, but listening to the local people who set the agenda. So people still do face-to-face meetings, we ’ re not doing this to replace face-to-face meetings. We ’ re bringing more stakeholders from central government in the local town halls, and we ’ re amplifying their voices by making sure the transcripts, the mind maps, and things like that are spread through the internet in real time, but we don ’ t ever ask the elderly to, say,”Oh, you have to learn typing, otherwise you don ’ t do democracy.”It ’ s not our style. But that requires broadband. Because if you don ’ t have broadband, but only a very limited bandwidth, you are forced to use text-based communication. DB : That ’ s right. Well, with access, of course, comes access for folks who maybe will misuse the **platform**. You talked a little bit about disinformation and using humor to beat rumor. But sometimes, disinformation is more weaponized. How do you combat those kinds of attacks, really? AT : Right, so you mean malinformation, then. So essentially, information designed to cause intentional public harm. And that ’ s no laughing matter. So for that, we have an idea called”notice and public notice.”So this is a Reuters photo, and I will read the original caption. The original caption says”A teenage extradition bill protester in Hong Kong is seen during a march to demand democracy and political reform in Hong Kong.”OK, a very neutral title by the Reuters. But there was a spreading of malinformation back last November, just leading to our presidential election, that shows something else entirely. This is the same photo – that says”This 13-year-old thug bought new iPhones, game consoles and brand-name sports shoes, and recruiting his brothers to murder police and collect 200, 000 dollars.”And this, of course, is a weapon designed to sow discord, and to elicit in Taiwan ’ s voters a kind of distaste for Hong Kong. And because they know that this is the main issue. And had we resorted to takedowns, that will not work, because that would only evoke more outrage. So we didn ’ t do a takedown. Instead, we worked with the fact checkers and professional journalists to attribute this original message back to the first day that it was posted. And it came from Zhongyang Zhengfawei. That is the main political and legal unit of the central party, in the Central Communist Party, in CCP. And we know that it ’ s their Weibo account that first did this new caption. So we sent out a public notice and with the partners in social media companies, pretty much all of them, they just put this very small reminder next to each time that this is shared with the wrong caption, that says”This actually came from the central propaganda unit of the CCP. Click here to learn more. To learn about the whole story.”And that, we found, that has worked, because people understand this is then not a news material. This is rather an appropriation of Reuters ’ news material and a copyright infringement and I think that ’ s part of the [

unclear]. In any case, the point is that when people understand that this is an intentional narrative, they won ’ t just randomly share it. They may share it, but with a comment that says”This is what the Zhongyang Zhengfawei is trying to do to our democracy.”DB : Seems like some of the global social media companies could learn something from notice and public notice. AT : Public notice, that ’ s right. DB : What advice would you have for the Twitters and Facebooks and LINEs and WhatsApps, and you name it, of the world? AT : Yeah. So, just before our election, we said to all of them that we ’ re not making a law to kind of punish them. However, we ’ re sharing this very simple fact that there is this **norm** in Taiwan that we even have a separate branch of the government, the control branch, that published the campaign donation and expense. And it just so occurred to us that in the previous election, the mayoral one, there was a lot of candidates that did not include any social media advertisements in their expense to the Control Yuan. And so essentially, that means that there is a separate amount of political donation and expense that evades public scrutiny. And our Control Yuan published their numbers in raw data form, that is to say, they ’ re not statistics, but individual records of who donated for what cause, when, where, and investigative journalists are very happy, because they can then make investigative reports about the connections between the candidates and the people who fund them. But they cannot work with the same material from the global social media companies. So I said,”Look, this is very simple. This is the social **norm** here, I don ’ t really care about other jurisdictions. You either adhere to the social **norm** that is set by the Control Yuan and the investigative journalists, or maybe you will face social sanctions. And this is not the government mandate, but it ’ s the people fed up with, you know, black box, and that ’ s part of the Sunflower Occupy ’ s demands, also. And so Facebook actually published in the Ad Library, I think at that time, one of the fastest response strategies, where everybody who has basically any dark pattern advertisement will get revealed very quickly, and investigative journalists work with the local civic technologists to make sure that if anybody dare to use social media in such a divisive way, within an hour, there will be a report out condemning that. So nobody tried that during the previous presidential election season. DB : So change is possible. AT : Mhm. WPR : Hey there, we have some more questions from the community. There is an anonymous one that says,”I believe Taiwan is outside WHO entirely and has a 130-part preparation program – developed entirely on its own – to what extent does it credit its preparation to building its own system?”AT : Well, a little bit, I guess. We tried to warn the WHO, but at that point – we are not totally outside, we have limited scientific access. But we do not have any ministerial access. And this is very different, right? If you only have limited scientific access, unless the other side ’ s top epidemiologist happens to be the vice president, like in Taiwan ’ s case, they don ’ t always do the storytelling well enough to translate that into political **action** as our vice president did, right? So the lack of ministerial access, I think, is to the detriment of the global community, because otherwise, people could have responded as we did in the first day of January, instead of having to wait for weeks before the WHO declared that this is something, that there ’ s definitely human to human transmission, that you should inspect people coming in from Wuhan, which they eventually did, but that ’ s already two weeks or three weeks after what we did. WPR : Makes a lot of sense. DB : More scientists and technologists in politics. That sounds like that ’ s the answer. AT : Yeah. WPR : And then we have another question here from Kamal Srinivasan about your reopening strategy.”How are you enabling restaurants and retailers to open safely in Taiwan?”AT : Oh, they never closed, so... WPR : Oh! AT : Yeah, they never closed, there was no lockdown, there was no closure. We just said a very simple thing in the CECC press conference, that there ' s going to be physical **distancing**. You maintain one and a half meters indoors or wear a mask. And that ' s it. And so there are some restaurants that put up, I guess, red curtains, some put very cute teddy bears and so on, on the chairs, to make sure that people spread evenly, some installed see-through glass or plastic walls between the seats. There ' s various social innovations happening around. And I think the only shops that got closed for a while, because they could not innovate quick enough to respond to these rules, was the intimate escort bars. But eventually, even they invented new ways, by handing out these caps that are plastic shielding, but still leaves room for drinking behind it. And so they opened with that social innovation. DB : That ' s amazing. WPR : It is, yeah, it ' s a lot to learn from your strategies there. Thank you, I ' ll be back towards the end with some final questions. DB : I ' m very happy to hear that the restaurants were not closed down, because I think Taipei has some of the best food in the world of any city that I ' ve visited, so, you know, kudos to you for that. So the big concern when it comes to using digital tools for COVID or using digital tools for democracy is always **privacy**. You ' ve talked about that a little bit, but I ' m sure the citizens of Taiwan are perhaps equally concerned about their **privacy**, especially given the geopolitical context. AT : Definitely. DB : So how do you cope with those demands? AT : Yeah, we design with not only defensive strategy, like minimization of data collection, but also proactive measures, such as privacy-enhancing technologies. One of the top teams that emerged out of our cohack, the TW response from the Polis, how to make contact tracing easier, focused not on the contact tracers, not on the medical officers, but on the person. So they basically said,"OK, you have a phone, you can record your temperatures, you can record your whereabouts and things like that, but that is strictly in your phone. It doesn ' t even use Bluetooth. So there ' s no transmission. Technology uses open-source, you can check it, you can use it in airplane mode. And when the contact tracer eventually tells you that you are part of a high-risk group, and they really want your contact history, this tool can then generate a single-use URL that only contains the precise information, anonymized, that the contact tracers want. But it will not, like in a traditional interview, let you ask – they ask a question, they only want to know your whereabouts, but you answer with such accuracy that you end up compromising other people ' s **privacy**. So basically, this is about designing with an aim to enhance other people ' s **privacy**, because personal data is never truly personal. It ' s always social, it ' s always intersectional. If I take a selfie at a party, I inadvertently also take pretty much everybody else ' s who are in the picture, the surroundings, the ambiance, and so on, and if I upload it to a cloud **service**, then I actually decimate the bargaining power, the negotiation power of everybody around me, because then their data is part of the cloud, and the cloud doesn ' t have to compensate them or get their agreement for it. And so only by designing the tools with **privacy** enhancing as a positive value, and not enhancing only the person ' s own **privacy**, just like a medical mask, it protects you, but mostly it also protects others, right? So if we design tools using that idea, and always open-source and with an open API, then we ' re in a much better shape than in centralized or so-called cloud-based **services**. DB : Well, you ' re clearly living in the future, and I guess that ' s quite literal, in the sense of, it ' s tomorrow morning there. AT : Twelve hours. DB : Yes. Tell me, what do you see in the future? What comes next? AT : Yes, so I see the **coronavirus** as a great amplifier. If you start with an authoritarian society, the **coronavirus**, with all its lockdowns and so on, has the potential of making it even a more totalitarian society. If people place their trust, however, on the social sector, on the ingenuity of social innovators, then the pandemic, as in Taiwan, actually strengthens our democracy, so that people feel, truly, that everybody can think of something that improves the welfare of not just Taiwan, but pretty much everybody else in the world. And so, my point here is that the great amplifier comes if no matter you want it or not, but the society, what they can do, is do what Taiwan did after SARS. In 2003, when SARS came, we had to shut down an entire hospital, barricading it with no definite termination date. It was very traumatic, everybody above the age of 30 remembers how traumatic it was. The municipalities and the central government were saying very different things, and that is why after SARS, the constitutional courts charged the legislature to set up the system as you see today, and also that is why, when people responding to that crisis back in 2003 built this very robust response system that there ’ s early drills. So just as the Sunflower Occupy, because of the crisis in trust let us build new tools that put trust first, I think the **coronavirus** is the chance for everybody who have survived through the first wave to settle on a new set of **norms** that will reinforce your founding values, instead of taking on alien values in the name of survival. DB : Yeah, let ’ s hope so, and let ’ s hope the rest of the world is as prepared as Taiwan the next time around. When it comes to digital democracy, though, and digital citizenship, where do you see that going, both in Taiwan and maybe in the rest of the world? AT : Well, I have my job description here, which I will read to you. It ’ s literally my job description and the answer to that question. And so, here goes. When we see the internet of things, let ’ s make it the internet of beings. When we see virtual reality, let ’ s make it a shared reality. When we see machine learning, let ’ s make it collaborative learning. When we see user experience, let ’ s make it about human experience. And whenever we hear the singularity is near, let us always remember the plurality is here. Thank you for listening. DB : Wow. I have to give that a little clap, that was beautiful. Quite a job description too. So, conservative anarchist, digital minister, and with that job description – that ’ s pretty impressive. AT : A poetician, yes. DB : So I **struggle** to imagine an adoption of these techniques in the US, and that may be my pessimism weighing in. But what words of hope do you have for the US, as we cope with COVID? AT : Well, as I mentioned, during SARS in Taiwan, nobody imagined we could have CECC and a cute spokesdog. Before the Sunflower movement, during a large protest, there was, I think, half a million people on the street, and many more. Nobody thought that we could have a collective intelligence system that puts open government data as a way to rebuild citizen participation. And so, never lose hope. As my favorite singer, Leonard Cohen – a poet, also – is fond of saying,”Ring the bells that still can ring and forget any perfect offering. There is a crack in everything and that is how the light gets in.”WPR : Wow. That ’ s so beautiful, and it feels like such a great message to, sort of, leave the audience with, and sharing the sentiment that everyone seems to be so grateful for what you ’ ve shared, Audrey, and all the great information and insight into Taiwan ’ s strategies. AT : Thank you. WPR : And David – DB : I was just going to say, thank you so much for that, thank you for that beautiful job description, and for all the wisdom you shared in rapid-fire fashion. I think it wasn ’ t just one idea that you shared, but maybe, I don ’ t know, 20, 30, 40? I lost count at some point. AT : Well, it ’ s called Ideas Worth Spreading, it ’ s a plural form. DB : Very true. Well, thank you so much for joining us. WPR : Thank you, Audrey. DB : And I wish you luck with everything. AT : Thank you, and have a good local time. Stay safe.

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Hello, I ’ m Chris Anderson. Welcome to The TED Interview. We ’ re gearing up

for season four with some extraordinary guests, but I don ’ t want to wait for that for today ’ s episode, because we ’ re in the middle of a pandemic, and there ’ s a guest I really wanted to talk to now. He is Adam Kucharski, an infectious diseases scientist who focuses on the mathematical modeling of pandemics. He ’ s an associate professor at the London School of Hygiene and Tropical Medicine and a TED Fellow. Adam Kucharski : So what kind of behavior is actually important for epidemics? Conversations, close physical contacts? What sort of data should we be collecting before an outbreak if we want to predict how infection might spread? To find out, our team built a mathematical model… Chris Anderson : When it comes to figuring out what to make of this pandemic, known technically as COVID-19, and informally as just the **coronavirus**, I find his thinking unbelievably helpful. And I ’ m excited to dive into it with you. A **special** callout to my friends on Twitter who offered up many suggestions for questions. I know this topic is on everyone ’ s mind right now. And what I hope this episode does is give us all a more nuanced way of thinking about how this pandemic has unfolded so far, what might be to come and what we all collectively can do about it. Let ’ s dive in. Adam, welcome to the TED Interview. Adam Kucharski : Thank you. CA : So let ’ s just start with a couple of basics. A lot of skeptical people ’ s response – certainly over the last few weeks, maybe less so now – has been,”Oh, come on, this isn ’ t such a big deal, there ’ s a relatively tiny number of cases. Compare it to the **flu**, compare it to anything else. There are much bigger problems in the world. Why are we making such a fuss about this?”And I guess the answer to that fuss is that it comes down to the mathematics. We ’ re talking about the mathematics of exponential growth, fundamentally, right? AK : Exactly. And there ’ s a number that we use to get a sense of how easy things spread and the level of transmission we ’ re dealing with. We call that the reproduction number, and conceptually, it ’ s just, for each case you have, on average, how many others are they infecting? And that gives you a sense of how much is this scaling, how much this growth is going to look like. For **coronavirus**, we ’ re now seeing, across multiple countries, we ’ re seeing each person on average giving it to two or three more. CA : So that reproduction number, the first thing to understand is that any number above one means that this thing is going to grow. Any number below one means it ’ s going to diminish. AK : Exactly – if you have it above one, then each group of people infected are going to be generating more infection than there was before. And you will see the exponential effect, so if it ’ s two, you will be doubling every round of infection, and if it ’ s below one, you ’ re going to get something that ’ s going to decline, on average. CA : So that number two or higher, I think everyone here is maybe familiar with the famous story of the chessboard and the grains of rice, and if you double the number of grains for every square of the chessboard, for the first 10 or 15 squares nothing much happens, but by the time you ’ ve got to the 64th square, you suddenly have tons of rice for every individual on the planet. Exponential growth is an incredible thing. And the small numbers now are really not what you should be paying attention to – you should be paying attention to the models of what could be to come. AK : Exactly. Obviously, if you continue the exponential growth, you do sometimes get these incredibly large, maybe implausibly large numbers. But even looking at a timescale of say, a month, if the reproduction number is three, each person is infecting three on average. The gap between these rounds of infection is about five days. So if you imagine that you ’ ve got one case now, that ’ s, kind of, six of these five-day steps in a month. So by the end of that month, that one person could have generated, I think it works out at about 729 cases. So even in a month, just the scale of this thing can really shoot up if it ’ s not controlled. CA : And so certainly, that seems to be happening on most numbers that you look at now, certainly where the virus is in the early stages of entering a country. You ' ve given a model whereby we can much more clearly understand this reproduction number, because it seems to me this is almost like the core to how we think of the virus and how we respond to it and how much we should fear it, almost. And in your thinking, you sort of break it down into four components, which you call DOTS : Duration, Opportunities, Transmission probability and Susceptibility. And I think it would be really helpful, Adam, for you to just explain each of these, because it ' s quite a simple equation that links those four things to the actual reproduction number. So talk about them in turn. Duration, what does that mean? AK : Duration measures how long someone is infectious for. If, for example, intuitively, if someone is infectious for a longer period of time, say, twice as long as someone else, then that ' s twice the length that they ' ve got to be spreading infection. CA : And what is the duration number for this virus, compared with, say, **flu** or with other pathogens? AK : It depends a little bit on what happens when people are infectious, if they ' re being isolated very quickly, that shortens that period of time, but potentially, we ' re looking at around a week people are effectively infectious before they might be isolated in hospital. CA : And during that week, they may not even be showing symptoms for that full week either, right? So someone gets infected, there ' s an incubation period. There ' s a period some way into that incubation period where they start being infectious, and there may be a period after that, where they start to show symptoms, and it ' s not clear, quite, how those dates align. Is that right? AK : No, we ' re getting more information. One of the signals we see in data that suggest that you may have that early transmission going on is when you have this delay from one infection to the next. So that seems to be around five days. That incubation period, the time for symptoms to appear, is also about five days. So if you imagine that most people are only infecting others when they ' re symptomatic, you ' d have that incubation period and then you ' d have some more time when they ' re infecting others. So the fact that those values seem to be similar, suggesting that some people are transmitting either very early on or potentially before they ' re showing clear symptoms. CA : So almost implies that on average, people are infecting others as much before they show symptoms as after. AK : Potentially. Obviously these are early data sets, but I think there ' s good evidence that a fair number of people, either before they ' re showing clear symptoms or maybe they ' re not showing the kind of very distinctive fever and cough but they ' re feeling unwell and they ' re shedding virus during that period. CA : And does that make it quite different from the **flu**, for example? AK : It makes it actually similar to **flu** in that regard. One of the reasons pandemic **flu** is so hard to control and so feared as a threat is because so much transmission happens before people are severely ill. And that means that by the time you identify these cases, they ' ve probably actually spread it to a number of other people. CA : Yeah, so this is the trickery of the thing, and why it ' s so hard to do anything about it. It is ahead of us all the time, and you can ' t just pay attention to how someone feels or what they ' re doing. I mean, how does that happen, by the way? How does someone infect someone else before they ' re even showing symptoms themselves, because classically, we think of, you know, the person sneezing and droplets go through the air and someone else breathes them in and that ' s how infection happens. What is actually going on for infection pre-symptoms? AK : So the level of transmission we see with this virus isn ' t what we see, for example, with measles, where someone sneezes and a lot of virus gets out and potentially lots of susceptible people can get exposed. So potentially, it could be quite early on that even if someone has quite mild symptoms, maybe a bit of a cough, that ' s enough for some virus to be getting out and particularly, some of the work that we ' ve done trying to look at sort of close gatherings, so very tight-knit meals, there was an example in a ski chalet – and even in those situations, you might have someone mildly ill, but enough virus is getting out and somehow exposing others, we ’ re still trying to work out exactly how, but there ’ s enough there to cause some infection. CA : But if someone ’ s mildly ill, don ’ t they still have symptoms? Isn ’ t there evidence that even before they know that they ’ re ill, something is going on? There was a German paper published this week that seemed to suggest that even really early on, you take a swab from the back of someone ’ s throat and they have hundreds of thousands of these viruses already reproducing there. Like, can someone just literally just be breathing normally and there is some transmission of virus out into the air that they don ’ t even know about and is either infecting people directly or settling on surfaces, is that possible? AK : I think that ’ s what we ’ re trying to pin down, how much that [unclear]. As you said, there ’ s evidence that you can have people without symptoms and you can get virus out their throats. And so certainly there ’ s potential that it can be breathed out, but is that a fairly rare event for that actual transmission to happen, or are we potentially seeing more infections occur through that route? So it ’ s really early data, and I think it ’ s a piece of the puzzle, but we ’ re trying to work out where that fits in with what we know about the kind of other transmission events we ’ ve seen. CA : Alright, so, duration is the duration of the period of infectiousness. We think is five to six days, is that what I heard you say? AK : Potentially around a week, depending on exactly what happens to people when they ’ re infectious. CA : And there are cases of people testing positive way, way later, after they ’ ve got infected. That may be true, but they are probably not as infectious then. Is that basically right, that ’ s the way to think of this? AK : I think that ’ s our working theory, that a lot of that infection is happening early on. And we see that for a number of respiratory infections, that when people obviously become severely ill, their behavior is very different to when they may be walking around and going about their normal day. CA : And so again, comparing that D number to other cases, like the **flu**, is **flu** similar? What ’ s the D number for **flu**? AK : So for **flu**, it ’ s probably slightly shorter in terms of the period that people are actively infectious. I mean, for **flu**, it ’ s a very quick turnover from one case to the next, actually. Even a matter of about three days, potentially, from one infection to the person that they infect. And then at the other end of the scale, you get things like STDs, where that duration could be several months, potentially. CA : Right. OK, really nothing that unusual so far, in terms of this particular virus. Let ’ s look at the O, opportunity. What is that? AK : So opportunity is a measure of how many chances the virus has to spread through interactions while someone is infectious. So typically, it ’ s a measure of social behavior. On average, how many social contacts do people make that create opportunities for transmission while they ’ re infectious. CA : So it ’ s how many people have you got close enough to during a day, during a given day, to have a chance of infecting them. And that number could be, if people aren ’ t taking precautions in a normal, sort of, urban **setting**, I mean, that could run into the hundreds, presumably? AK : Potentially, for some people. We ’ ve done a number of studies looking at that in recent years, and the average, in terms of physical contacts, is about five people per day. Most people will have conversation or contacts generally with about 10, 15, but obviously, between cultures, we see quite a lot of variation in the level of physical greetings that might happen. CA : And presumably, that number again is no different for this virus than for any other. I mean, that ’ s just a feature of the lives that we live. AK : I think for this one, if it ’ s driven through these kind of interactions, and we ’ ve seen for **flu**, for other respiratory infections, those kinds of fairly close contacts and everyday physical interactions seem to be the ones that are important for transmission. CA : Perhaps there is one difference. The fact that if you ' re infectious pre-symptoms, perhaps that means that actually, there are more opportunities here. This is part of the virus ' s genius, as it were, that by not letting on that it ' s inside someone, people continue to interact and go to work and take the subway and so forth, not even knowing that they ' re sick. AK : Exactly. And for something like **flu**, you see when people get ill, clearly, their social contacts drop off. So to have a virus that can be infectious while people are going around their everyday lives, really gives it an advantage in terms of transmission. CA : In your modeling, do you actually have this opportunities number higher than for **flu**? AK : So for the moment, we ' re kind of using similar values, so we ' re trying to look at, for example, physical contacts within different populations. But what we are doing is scaling the risk. So that ' s coming on to the T term. So that between each contact, what ' s the risk that a transmission event will occur. CA : Alright, so let ' s go on to this next number, the T, transmission probability. How do you define that? AK : So this measures the chance that, essentially, the virus will get across during a particular opportunity or a particular interaction. So you may well have a conversation with somebody, but actually, you don ' t cough or you don ' t sneeze or for some reason, the virus doesn ' t actually get across and expose the other person. And so, for this virus, as I mentioned, say people are having 10 conversations a day, but we ' re not seeing infected people infect 10 others a day. So it suggests that not all of those opportunities are actually resulting in the virus getting across. CA : But people say that this is an infectious virus. Like, what is that transmission probability number, again, compared with, say, the **flu**? AK : So, we did some analysis looking at these very close gatherings. We looked at about 10 different case studies, and we found that about a third of the contacts in those **settings** subsequently got infected in these early stages, when people weren ' t aware. So if you had these, kind of, big group meals, potentially, each contact had about, a kind of, one in three chance of getting exposed. For seasonal **flu**, that tends to be slightly lower, even within households and close **settings**, you don ' t necessarily get values that high. And even for something like SARS, those values have, kind of – the risk per interaction you had was lower than what we seem to be getting for **coronavirus**. Which intuitively makes sense, there must be a higher risk per interaction if this thing is spreading so easily. CA : Hm. OK, and then the fourth letter of DOTS is S for susceptibility. What ' s that? AK : So that is a measure of the proportion of the population who are susceptible. If you imagine you have this interaction with someone, the virus gets across, it exposes them, but some people may have been vaccinated or otherwise have some immunity and not develop infection themselves and not be infectious to others. So we ' ve got to account for this potential proportion of people who are not actually going to turn into cases themselves. CA : And obviously, there ' s no vaccine yet for this **coronavirus**, nor is anyone, at least initially, immune, as far as we know. So are you modeling that susceptibility number pretty high, is that part of the issue here? AK : Yeah, I think the evidence is that this is going to fully susceptible populations, and even in areas, for example, like China, where there ' s been a lot of transmission but there ' s been very strong control measures, we estimated that up to the end of January, probably about 95 percent of Wuhan are still susceptible. So there ' s been a lot of infection, but it hasn ' t really taken much of that component, of the DOTS, of those four things that drive transmission. CA : And so the way the mathematics works, I have to confess, amidst the **stress** of this whole situation, the nerd in me kind of loves the elegance of the mathematics here, because I ' d never really thought about it this way, but you basically just multiply those numbers together to get the reproduction number. Is that right?

AK : Exactly, yeah, you almost take the path of the infection during transmission as you multiply those together, and that gives you the number for that virus. CA : And so there ' s just a total logic to that. It ' s the number of days, duration that you ' re infectious, it ' s the number of people you ' re seeing on average during those days that you have a chance to infect. Then you multiply that by the transmission probability, is virus getting into them, essentially, that ' s what you mean by crossing over. And then by the susceptibility number. By the way, what do you think the susceptibility probability is for this case? AK : I think we have to assume that it ' s near 100 percent in terms of spread, yeah. CA : Alright, you multiply those numbers together, and right now, it looks like, for this **coronavirus**, that you say two to three is the most plausible current number, which implies very rapid growth. AK : Exactly. In these uncontrolled outbreaks, we ' re seeing now a number of countries in this stage – you are going to get this really rapid growth occurring. CA : And so how does that two to three compare with **flu**? And I guess, there ' s seasonal **flu**, in the winter, when it ' s spreading, and at other times during the year drops well below one as a reproduction number, right? But what is it during seasonal **flu** time? AK : During the early stage when it ' s taking off at the start of the **flu** season, it ' s probably, we reckon, somewhere between about maybe 1. 2, 1. 4. So it ' s not incredibly transmissible, if you imagine you do have some immunity in your population from vaccination and from other things. So it can spread, it ' s above one, but it ' s not taking off, necessarily, as quickly as the **coronavirus** is. CA : So I want to come back to two of those elements, specifically opportunity and transmission probability, because those seem to have the most chance to actually do something about this infection rate. Before we go there, let ' s talk about another key number on this, which is the fatality rate. First of all, could you define – I think there ' s two different versions of the fatality rate that maybe confuse people. Could you define them? AK : So the one that we often talk about is what ' s known as the case fatality rate, and that ' s of the proportion who show up with symptoms as cases, what proportion of those will subsequently be fatal. And we also sometimes talk about what ' s known as the infection fatality rate, which is, of everyone who gets infected, regardless of symptoms, how many of those infections will subsequently be fatal. But most of the values we see kicking around are the case fatality rate, or the CFR, as it ' s sometimes known. CA : And so what is that fatality rate for this virus, and again, how does that compare with other pathogens? AK : So there ' s a few numbers that have been bouncing around. One of the challenges in real time is you often don ' t see all of your cases, you have people symptomatic not being reported. You also have a delay. If you imagine, for example, 100 people turn up to a hospital with **coronavirus** and none have died yet, that doesn ' t imply that the fatality rate is zero, because you ' ve got to wait to see what might happen to them. So when you adjust for that underreporting and delays, best estimate for the case fatality is about one percent. So about one percent of people with symptoms, on average, those outcomes are fatal. And that ' s probably about 10 times worse than seasonal **flu**. CA : Yeah, so that ' s a scary comparison right there, given how many people die of **flu**. So when the World Health Organization mentioned a higher number, a little while back, of 3. 4 percent, they were criticized a bit for that. Explain why that might have been misleading and how to think about it and adjust for that. AK : It ' s incredibly common that people look at these raw numbers, they say,"How many deaths are there so far, how many cases,"and they look at that ratio, and even a couple of weeks ago, that number produced a two percent value. But if you imagine you have this delay effect, then even if you stop all your cases, you will still have these kind of fatal outcomes over time, so that number will creep up. This has occurred in every epidemic from pandemic **flu** to Ebola, we see this again and again. And I made the point to a number of people that this number is going to go up, because as China ’ s cases slow, it will look like it ’ s increasing, and that ’ s just kind of a statistical quirk. There ’ s nothing really kind of, behind a change, there ’ s no mutations or anything going on. CA : If I have this right, there are two effects going on. One is that the number of fatalities from the existing caseload will rise, which actually would boost that 3. 4 even higher. But then you have to offset that against the fact that, apparently, huge numbers of cases have just gone undetected and that we haven ’ t, due to bad testing, that the number of fatalities don ’ t – they probably reflect a much larger number of early cases. Is that it? AK : Exactly. So you have one thing pulling the number up and one thing pulling it down. And it means that on these kind of early values, if you actually just adjust for the delay and don ’ t think about these unreported cases, you start getting really very scary numbers indeed. You get up to 20, 30 percent potentially, which really doesn ’ t align with what we know about this virus in general. CA : Alright. There ’ s a lot more data in now. From your point of view, you think the likely fatality rate, at least in the earlier stage of an infection, is about two percent? AK : I think overall, I think we can put something probably in the 0. 5 to two percent range, and that ’ s on a number of different data sets. And that ’ s for people who are symptomatic. I think on average, one percent is a good number to work with. CA : OK, one percent, So **flu** is often quoted as a tenth of a percent, so it ’ s five to 10 times or more more dangerous than **flu**. And that danger is not symmetric across age groups, as is well known. It primarily affects the elderly. AK : Yeah, we ’ ve seen that one percent on average, but once you start getting into the over 60s, over 70s, that number really starts to shoot up. I mean, we ’ re estimating potentially in these older groups, you ’ re looking at maybe five, 10 percent fatality. And then of course, on top of that, you ’ ve got to add what are going to be the severe cases and people are going to require hospitalization. And those risks get very large in the older groups indeed. CA : Adam, put these numbers together for us. In your models, if you put together a reproduction rate of two to three and a fatality rate of 0. 5 percent to one percent and you run the simulation, what does it look like? AK : So if you have this uncontrolled transmission, and you have this reproduction number of two or three and you don ’ t do anything about it, the only way the outbreak ends is enough people get it and immunity builds up and the outbreak kind of ends on its own. And in that case, you would expect very large numbers of the population to be infected. It ’ s what we see, for example, with many other uncontained outbreaks, that it essentially burns through the population, you get large numbers infected and with this kind of fatality rate and hospitalization rate, that would really be hugely damaging if that were to occur. Certainly at the country level, we ’ re seeing – Italy is a good example at the moment, that if you have that early transmission that ’ s undetected, that rapid growth, you very quickly get to a situation where your health systems are overwhelmed. I think one of the nastiest aspects of this virus is that because you have the delay between infection and symptoms and people showing up in health care, if your health system is overwhelmed, even on that day, if you completely stop transmission, you ’ ve got all of these people who have already been exposed, so you ’ re still going to have cases and severe cases appearing for maybe another couple of weeks. So it ’ s really this huge accumulation of infection and burden that ’ s coming through the system on your population. CA : So there ’ s another key number, actually, is how does the total case number compare to the capacity of a country ’ s health system to process that number of cases. Presumably that issue makes a huge difference to the fatality rate, the difference between people coming in with severe illness and a health system that ’ s able to respond and one that ’ s overwhelmed. The fatality rate is going to be very different at that point. AK : If someone requires an ICU bed, that ’ s a couple of weeks they ’ re going to require it for and you ’ ve got more cases coming through the system, so it very quickly gets very tough. CA : So talk about the difference between containment and mitigation. These are different terms that we ’ re hearing a lot about. In the early stages of the virus, governments are focused on containment. What does that mean? AK : Containment is this idea that you can focus your effort on control very much on the cases and their contacts. So you ’ re not causing disruption to the wider population, you have a case that comes in, you isolate them, you work out who they ’ ve come into contact with, who ’ s potentially these opportunities for exposure and then you can follow up those people, maybe **quarantine** them to make sure that no further transmission happens. So it ’ s a very focused, targeted method, and for SARS, it worked remarkably well. But I think for this infection, because some cases are going to be missed or undetected, you ’ ve really got to be capturing a large chunk of people at risk. If a few slip through the net, potentially, you ’ re going to get an outbreak. CA : Are there any countries that have been able to employ this strategy and effectively contain the virus? AK : So Singapore have been doing a really remarkable job of this for the last six weeks or so. So as well as some wider measures, they ’ ve been working incredibly hard to trace people who have come into contact. Looking at CCTV, going through to find out which taxi someone might have gotten, who might be at risk – really, really thorough **follow-up**. And for about six weeks, that has kept a lid on transmission. CA : So that ’ s amazing. So someone comes into the country, they test positive – they go to work, and with a massive team, and trace everything to the level of actually saying,”Oh, you don ’ t know what taxi you went in? Let us find that out for you.”And presumably, when they find the taxi driver, they then have to try and figure out everyone else who was in that taxi? AK : So they will focus on close contacts of people most at risk, but they ’ re really minimizing the chance that anyone slips through the net. CA : But even in Singapore, if I ’ m not mistaken, numbers started to trend back down to zero, but I think recently, they ’ ve picked up again a bit. It ’ s still unclear whether they will actually be able to sustain containment. AK : Exactly. If we talk in terms of the reproduction number, we saw it dipped to maybe 0. 8, 0. 9, so under that crucial value of one. But in the last week or two, it does seem to be ticking up and they ’ re getting more cases appearing. I think a lot of it is, even if they are containing it, the world is experiencing outbreaks and just keeps throwing sparks of infection, and it becomes harder and harder with that level of intensive effort to stamp them all out. CA : In the case of this virus, you know, there was warning to most countries in the world that this thing was happening. The news out of China very quickly became very bleak, and people had time to prepare. I mean, what would ideal preparation look like if you know that something like this is coming and you know that there ’ s a lot on the line if you can successfully contain it before it really escapes? AK : I think two things would make a really big difference. One is having as thorough a **follow-up** and detection as possible. We ’ ve done some modeling analyses, looking at how effective that kind of early containment is. And it can be, if you ’ re identifying maybe 70 or 80 percent of the people who might have come into contact. But if you ’ re not detecting those cases coming in, if you ’ re not detecting their contacts – and a lot of the early focus, for example, was on travel history to China, and then it became clear that the situation was changing, but because you were relying on that as your definition of a case, it meant a lot of maybe other cases that matched the definition weren ’ t being tested because they didn ’ t seem to be potentially at risk. CA : So I mean, if you know that early detection is key to this, an essential early measure, I guess, would be to rapidly ensure that you had enough tests available and where needed, so that you could respond, be ready to swing into **action** as soon as someone was detected, you then have to very quickly, I guess, test their contacts and so forth, to have a chance of keeping this under control. AK : Exactly. In my line of work, we say there ' s value in a negative test, because it shows that you ' re looking for something and it ' s not there. And so I think having small numbers of people tested doesn ' t give you confidence that you ' re not missing infections, whereas if you are doing really thorough **follow-up** on contacts, we ' ve seen countries even like Korea now, huge numbers of people tested. So although there are still cases appearing, it gives them more confidence that they have some sense of where those infections are. CA : I mean, you ' re in the UK right now, I ' m in the US. How likely is it that the UK is going to be able to contain, how likely is it that the US is going to be able to contain this? AK : I think it ' s pretty unlikely in both cases. I think the UK is going to have to introduce some additional measures. I think when that happens obviously depends a bit on the current situation, but we ' ve tested almost 30, 000 people now. Frankly, I think the US may well be moving beyond that point, given how much evidence of extensive transmission that has, and I think without clear ideas of how much infection there is and that level of testing, it ' s quite hard to actually see what the picture currently is in the US. CA : I mean, I definitely don ' t want to get too political about this, but I mean, does this strike you as – you just said that the UK has tested 30, 000 people – the US is five or six times bigger and I think the total number of tests here is five or six thousand, or it was a few days ago. Does that strike you as bizarre? I don ' t understand, honestly, how that happened in an educated country that has so much knowledge about infectious diseases. AK : It does, and I think there ' s obviously a number of factors playing in there, logistics and so on, but there has been that period of warning that this is a threat and this is coming in. And I think countries need to make sure that they ' ve got the capacity to really do as much detection as they can in those early stages, because that ' s where you ' re going to catch it and that ' s where you ' re going to have a better chance of containing it. CA : OK, so if you fail to contain, then you have to move to some kind of mitigation strategy. So what comes into play there? And I think I almost want to bring that back to two of your DOTS factors, opportunity and transmission probability, because it seems like the virus is what it is, the actual duration when someone is potentially infectious, we can ' t do much about. The susceptibility side, we can ' t do much about until there ' s a vaccine. We could maybe talk about that in a bit. But the middle two of opportunity and transmission probability, we can do something about. Do you want to maybe talk about those in turn, of what that looks like, how would you build a mitigation strategy? I mean, first of all, thinking about opportunity, how do you reduce the number of opportunities to pass on the bug? AK : And so I think in that respect, it would be about massive changes in our social interactions. And if you think in terms of the reproduction number of being about two or three, to get that number below one, you ' ve really got to cut some aspect of that transmission in half or in two-thirds to get that below one. And so that would require, of the opportunities that could spread the virus, so these kind of close contacts, everybody in the population, on average, will be needing to reduce those interactions potentially by two-thirds to bring it under control. That might be through working from home, from changing lifestyle and kind of where you go in crowded places and dinners. And of course, these measures, things like school closures, and other things that just attempt to reduce the social mixing of a population. CA : Well, actually, talk to me more about school closures, because that, if I remember, often in past pandemics has been cited as an absolutely key measure, that schools represent this sort of coming together of people, children are often – certainly when it comes to **flu** and colds – they ’ re carriers. But on this case, children don ’ t seem to be getting sick from this particular virus, or at least very few of them are. Do we know whether they can still be infectious? They can be the unintended carriers of it. Or actually, is there evidence that school closures may not be as important in this instance as it is in others? AK : So that point on what role children play is a crucial one, and there ’ s still not a good evidence base there. From following up of contacts of cases, there ’ s now evidence that children are getting infected, so when you ’ re testing, they are getting exposed, it ’ s not that somehow they ’ re just not getting the infection at all, but as you said, they ’ re not showing symptoms in the same way. And particularly for **flu**, when we see the implications of school closures, even in the UK in 2009 during swine **flu**, there was a dip in the outbreak during the school holidays, you could see it on the epidemic curve, it kind of comes back down in the summer and goes back up in the autumn. But of course, in 2009, there was some immunity in older groups. That kind of shifted more the transmission into the younger ones. So I think it ’ s really something we ’ re trying to work to understand. Obviously, it will reduce interactions, with school closures, but then there ’ s knock-on social effects, there ’ s potential knock-on changes in mixing, maybe grandparents and their role, in terms of alternative carers if parents have to work. So I think there ’ s a lot of pieces that need to be considered. CA : I mean, based on all of the different pieces of evidence you ’ ve seen, if it were down to you, would you be recommending that most countries at this point do look hard at extensive school closures as a precautionary measure, that it ’ s just worth it to do that as a sort of painful two, three, four, five-month strategy? What would you recommend? AK : I think the key thing, given the age distribution of risk and the severity in older groups is reduce interactions that bring the infection into those groups. And then amongst everyone else, reduce interactions as much as possible. I think the key thing is we ’ ve got so much of the disease burden in the kind of 60-plus group that it ’ s not just about everyone trying to avoid everyone ’ s interactions, but it ’ s the kind of behaviors that would drive infections into those groups. CA : Does that mean that people should think twice before, I don ’ t know, visiting a loved one in an old people ’ s home or in a residential facility? Like that, we should just pay **super special** attention to that, should all these facilities be taking great care about who they admit, taking temperature and checking for symptoms or something like that? AK : I think those measures definitely need to be considered. In the UK, we ’ re getting plans for potentially what ’ s known as a cocooning strategy for these older groups that we can really try and seal off interactions as much as possible from people who might be bringing infection in. And ultimately, because as you said, we can ’ t target these other aspects of transmission, it is just reducing the risk of exposure in these groups, and so I think anything at the individual level you can do to get people reducing their risk, if either they ’ re elderly or in other risk groups, I think is crucial. And I think more at the general level those kind of more large-scale measures can help reduce interactions overall, but I think if those reductions are happening and not reducing the risk for people who are going to get severe disease, then you ’ re still going to get this really remarkably severe burden. CA : I mean, do people have to almost apply this double lens as they think about this stuff? There ’ s the risk to you as you go about your life, of you catching this bug. But there ’ s also the risk of you being, unintentionally, a carrier to someone who would suffer much more than you might. And that both those things have to be top of mind right now. AK : Yeah, and it ’ s not just whose hand you shake, it ’ s whose hand that person goes on to shake. And I think we need to think about these second-degree steps, that you might think you have low risk and you ' re in a younger group, but you ' re often going to be a very short step away from someone who is going to get hit very hard by this. And I think we really need to be socially minded and think this could be quite dramatic in terms of change of behavior, but it needs to be to reduce the impact that we ' re potentially facing. CA : So the opportunity number, we bring down by just reducing the number of physical contacts we have with other people. And I guess the transmission probability number, how do we bring that down? That impacts how we interact. You mentioned hand-shaking, I ' m guessing you ' re going to say no handshaking. AK : Yeah, so changes like that. I mean, another one, I think, handwashing operates in a way that we might be still be doing activities that we ' ve previously done, but handwashing reduces the chance that from one interaction to another, we might be spreading infection, so it ' s all of these measures that mean that even if we ' re having these exposures, we ' re taking additional steps to avoid any transmission happening. CA : I still think most people don ' t fully understand or don ' t have a clear model of the pathway by which this thing spreads. So you think definitely people understand that you don ' t breathe in the water droplets of someone who has just coughed or sneezed. So how does it spread? It gets onto surfaces. How? Do people just breathe out and it goes on from people who are sick, they touch their mouth or something like that, and then touch a surface and it gets on that way? How does it actually get onto surfaces? AK : I think a lot of it would be that you cough in your hand and it ends up on a surface. But I think the challenge, obviously, is untangling these questions of how transmission happens. You have transmission in a household, and is it that someone coughed and it got on a surface, is it direct contact, is it a handshake, and even for things like **flu**, that ' s something that we work quite hard to try and unpick, how does social behavior correspond to infection risk. Because it ' s clearly important, but pinning it down is really tough. CA : It ' s almost like embracing the fact that for a lot of these things, we actually don ' t know and that we ' re all in this game of probabilities. Which, in a way, is why I think the math is so important here. That you have to think of this as these multiple numbers working together on each other, they all have their part to play. And any of them that you can take down by a percentage is likely contributing, not just to you but to everyone. And people don ' t actually know in detail how the numbers go together, but they know that they probably all matter. We almost need people to, somehow, you know, embrace that uncertainty and then try to get some satisfaction by acting on every single part of it. AK : I think it ' s this idea that if on average, you ' re infecting, say, three people, what ' s driving that and how can you chip away at that value? If you ' re washing your hands, how much might that chip away in terms of the handshakes, you know, you may have had virus and you no longer do, or if you are changing your social behavior in a certain way, is that taking away a couple of interactions, is that taking away half? How can you really chip into that number as much as you possibly can? CA : Is there anything else to say about how we could reduce that transmission probability in our interactions? Like, what is the physical distance that it ' s wise to stay away from other people if we can? AK : I think it ' s hard to pin down exactly, but I think one thing to bear in mind is that there ' s not so much evidence that this is a kind of aerosol and it goes really far – it ' s reasonably short distances. I don ' t think it ' s the case that you ' re sitting a few meters away from someone and the virus is somehow going to get across. It ' s in closer interactions, and it ' s why we ' re seeing so many transmission events occur in things like meals and really tight-knit groups. Because if you imagine that ' s where you can get a virus out and onto surfaces and onto hands and onto faces, and it ' s really situations like that we ' ve got to think more about. CA : So in a way, some of the fears that people have may actually be overstated, like, if you ' re in the middle of an airplane and someone at the front sneezes, I mean, that ' s annoying, but it ' s actually not the thing you should be most freaked out about. There are much smarter ways to pay attention to your well-being. AK : Yeah, if it was measles and the plane was susceptible people, you would see a lot of infections after that. I think it is, bear in mind, that this is, on average, people infecting two or three others, so it ' s not the case of your maybe 50 interactions over a week, all of those people are at risk. But it ' s going to be some of them, particularly those close contacts, that are going to be where transmission ' s occurring. CA : So talk about, from a sort of national strategy point of view. There ' s a lot of talk about the need to"flatten the curve."What does that mean? AK : I think it refers to this idea that for your health systems, you don ' t want all your cases to appear at the same time. So if we sat back and did nothing and just let the epidemic grow, and you had this growth rate that, at the moment, in some places is looking like maybe three to four days, you ' re getting doubling. So every three or four days, the epidemic is doubling. It will skyrocket and you ' ll end up with a whole bunch of really severely ill people needing hospital care all at the same time, and you just won ' t have capacity for it. So the idea of flattening the curve is if we can slow transmission, if we can get that reproduction number down, then there may still be an outbreak, but it will be much flatter, it will be longer and there will be fewer severe cases showing up, which means that they can get the health care they need. CA : Does it imply that there will be fewer cases overall, or – When you look at the actual images of people showing what flattening the curve looks like, it almost looks as if you ' ve got the same area still under the graph, i. e. that the same number of people, ultimately, are infected but over a longer period. Is that typically what happens, and even if you adopt all these strategies of social **distancing** and washing hands and etc. that the best you can hope for is that you slow the thing down, you actually will get as many people infected in the end? AK : Not necessarily – it depends on the measures that go in. There are some measures like, shutting down travel, which typically delay the spread rather than reduce it. So you ' re still going to get the same outbreaks, but you ' re stretching out the outbreaks. But there are other measures. If we talk about reducing interactions, if your reproduction number ' s lower, you would expect fewer cases overall. And eventually, in your population, you will get some buildup of immunity, which would help you out if you think about the components, reducing susceptibility, alongside what ' s going on elsewhere. So the hope is that the two things will work together. CA : So help me understand what the endgame is here. So, take China, for example. Whatever you make of the early suppression of data and so forth that seems pretty troubling there. The intensity of the response come January time or whatever, with the shutdown of this huge area of the country, seems to have actually been effective. The number of cases there are falling at a shockingly high rate in some ways. Falling to almost nothing. And I can ' t understand that. You are talking about a country of, whatever, 1. 4 billion people. There have been a huge number of cases there, but it was a tiny fraction of the population have actually got sick. And yet, they ' ve got the number way down. It ' s not like every other person in China has somehow developed immunity. Is it that they have been absolutely disciplined about shutting down travel from the infected regions and somehow really dialed up, massively dialed up testing at any sign of any problem, so that literally, they are back in containment mode in most parts of China? I can ' t get my head around it, help me understand it. AK : So we estimated, in the last two weeks of January, when these measures went in, the reproduction number went from about 2. 4 to 1. 1. So about 60 percent decline in transmission in the space of a week or two. Which is remarkable and really, a lot of it is likely to be driven by just fundamental change in social behavior, huge social **distancing**, really intensive **follow-up**, intensive testing. And it got to the point where it took enough off the reproduction number to cause the decline, and now, of course, we ’ re seeing, in many areas, a transition back to more of this kind of containment, because there ’ s few cases, it ’ s more manageable. But we ’ re also seeing them face a challenge, because a lot of these cities have basically been locked down for six weeks and there ’ s a limit to how long you can do that for. And so some of these measures are gradually starting to be lifted, which of course creates the risk that cases that are appearing from other countries may subsequently go in and reintroduce transmission. CA : But given how infectious the bug is, and how many theoretical pathways and connection points there are between people in Wuhan, even in shutdown, or relatively shut down, or the other places where there ’ s been some infection and the rest of the country, does it surprise you how quickly that curve has gone down to nearly zero? AK : Yes. Early on when we saw that flattening off in cases in those first few days, we did wonder whether it was just they hit a limit in testing capacity and they were reporting 1, 000 a day, because that ’ s all the kits they had. But it continued, thankfully, and it shows that it is possible to turn this over with that level of intervention. I think the key thing now is seeing how it works in other **settings**. Italy now are putting in really dramatic interventions. But of course, because of this delay effect, if you put them in today, you won ’ t necessarily see the effects on cases for another week or two. So I think working out what impact that ’ s had is going to be key for helping other countries work on how to contain this. CA : To have a picture, Adam, of how this is likely to play out over the next month or two, give us a couple of scenarios that are in your head. AK : I think the optimistic scenario is that we ’ re going to learn a lot from places like Italy that have unfortunately been hit very hard. And that countries are going to take this very seriously and that we ’ re not going to get this continued growth that ’ s going to overwhelm totally, that we ’ re going to be able to sufficiently slow it down, that we are going to get large numbers of cases, we ’ re probably going to get a lot of severe cases, but that will be more manageable, that ’ s the kind of optimistic scenario. I think if we have a point where countries either don ’ t take this seriously or populations don ’ t respond well to control measures or it ’ s not detected, we could get situations – I think Iran is probably the closest one at the moment – where there ’ s been extensive widespread transmission, and by the time it ’ s being responded to, those infections are already in the system and they are going to turn up as cases and severe illness. So I ’ m hoping we ’ re not at that point, but we ’ ve certainly got, at the moment, potentially about 10 countries on that trajectory to have the same outlook as Italy. So it ’ s really crucial what happens in the next couple of weeks. CA : Is there a real chance that quite a few countries end up having, this year, substantially more deaths from this virus than from seasonal **flu**? AK : I think for some countries that is likely, yeah. I think if control is not possible, and we ’ ve seen it happen in China, but that was really just an unprecedented level of intervention. It was really just changing the social fabric. I think people, many of us, don ’ t really appreciate, at a glance, just what that means, to reduce your interactions to that extent. I think many countries just simply won ’ t be able to manage that. CA : It ’ s almost a challenge to democracies, isn ’ t it –”OK, show us what you can do without that kind of draconian control. If you don ’ t like the thought of that, come on, citizens, step up, show us what you ’ re capable of, show that you can be wise about this and smart and self-disciplined, and get ahead of the damn bug.”AK : Yeah. CA : I mean, I ’ m not personally superoptimistic about that, because there ’ s such conflicting messaging coming out in so many different places, and people don ’ t like to short-term sacrifice. I mean, is there almost a case that – I mean, what ’ s your view on whether the media has played a helpful role here or an unhelpful role? Is it actually, in some ways, helpful to, if anything, overstate the concern, the fear, and actually make people panic a little bit? AK : I think it ’ s a really tough balance to strike, because of course, early on, if you don ’ t have cases, if you don ’ t have any evidence of potential pressure, it ’ s very hard to get that message and convince people to take it seriously if you ’ re overhyping it. But equally, if you ’ re waiting too long, and saying it ’ s not a concern yet, we ’ re OK for the moment, a lot of people think it ’ s just **flu**. By the time it hits hard, as I ’ ve said, you ’ re going to have weeks of an overburdened health system, because even if you take interventions, it ’ s too late to control the infections that have happened. So I think it ’ s a fine line, and my hope is there is this ramp-up in messaging, now people have these tangible examples like Italy, where they can see what ’ s going to happen if they don ’ t take it seriously. But certainly, of all the diseases I ’ ve seen, I think many of my colleagues who are much older than me and have memories of other outbreaks, it ’ s the scariest thing we ’ ve seen in terms of the impact it could have, and I think we need to respond to that. CA : It ’ s the scariest disease you ’ ve seen. Wow. I ’ ve got some questions for you from my friends on Twitter. Everyone is obviously **super** exercised about this topic. Hypothetically, if everyone stayed home for three weeks, would that effectively wipe this out? Is there a way to socially distance ourselves out of this? AK : Yeah, I think in certain countries with reasonably small household sizes, I think average in the UK, US is about two and a half, so even if you had a round of infection within the household, that would probably stamp it out. As a secondary benefit, you may well stamp out a few other infections, too. Measles only circulates in humans, so you may have some knock-on effect, if, of course, that were ever to be possible. CA : I mean, obviously that would be a huge dent to the economy, and this is in a way almost, like, one of the underlying challenges here is that you can ’ t optimize public policy for both economic health and fighting a virus. Like, those two things are, to some extent, in conflict, or at least, short-term economic health and fighting a virus. Those two things are in conflict, right? And societies need to pick one. AK : It is tough to convince people of that balance, the thing we always say of pandemic planning is it ’ s cheap to put this stuff in place now – otherwise, you ’ ve got to pay for it later. But unfortunately, as we ’ ve seen with this, that a lot of early money for response wasn ’ t there. And it ’ s only when it has an impact and when it ’ s going to get expensive that people are happy to take that cost on board, it seems. CA : OK, some more Twitter questions. Will the rising temperature in coming weeks and months slow down the COVID-19 spread? AK : I haven ’ t seen any convincing evidence that there ’ s that strong pattern with temperature, and we ’ ve seen it for other infections that there is this seasonal pattern, but I think the fact we ’ re getting widespread outbreaks makes it hard to identify, and of course, there ’ s other things going on. So even if one country doesn ’ t have as big an outbreak as another, that ’ s going to be influenced by control measures, by social behavior, by opportunities and these things as well. So it would be really reassuring if this was the case, but I don ’ t think we can say that just yet. CA : Continuing from Twitter, I mean, is there a standardized global recommendation for all countries on how to do this? And if not, why not? AK : I think that ’ s what people are trying to piece together, first in terms of what works. It ’ s only really in the last sort of few weeks we ’ ve got a sense that this thing can be controllable with this extent of interventions, but of course, not all countries can do what China have done, some of these measures incur a huge social, economic, psychological burden on populations.

And of course, there ' s the time limit. In China they ' ve had them in for six weeks it ' s tough to maintain that, so we need to think of these tradeoffs of all the things we can ask people to do, what ' s going to have the most impact on actually reducing the burden. CA : Another question : How did this happen and is it likely to happen again? AK : So it ' s likely that this originated with the virus that was circling in bats and then probably made its way through another species into humans somehow, there ' s a lot of bits of evidence around this, there ' s not kind of single, clear story, but even for SARS, it took several years for genomics to piece together the exact route that it happened. But certainly, I think it ' s plausible that it could happen again. Nature is throwing out these viruses constantly. Many of them aren ' t well-adapted to humans, don ' t pick up, you know, there may well have been a virus like this a few years ago that just happened to infect someone who just didn ' t have any contacts and didn ' t go any further. I think we are going to face these things and we need to think about how can we get in early at the stage where we ' re talking small numbers of cases, and even something like this is containable, rather than the situation we ' ve got now. CA : It seems like this isn ' t the first time that a virus seems to have emerged from, like, a wild meat market. That ' s certainly how it happens in the movies. And I think China has already taken some steps this time to try to crack down on that. I guess that ' s potentially quite a big deal for the future if that can be properly maintained. AK : It is, and we saw, for example, the H7N9 avian **flu**, over the last few years, in 2013, it was a big emerging concern, and China made a very extensive response in terms of changing how they operate their markets and vaccination of birds and that seems to have removed that threat. So I think these measures can be effective if they ' re identified early on. CA : So talk about vaccinations. That ' s the key measure, I guess, to change that susceptibility factor in your equation. There ' s obviously a race on to get these vaccinations out there, there are some candidate vaccinations there. How do you see that playing out? AK : I think there ' s certainly some promising development happening, but I think the timescales of these things are really on the order of maybe a year, 18 months before these things be widely available. Obviously, a vaccine has to go through these stages of **trials**, that takes time, so even if by the end of the year, we have something which is viable and works, we ' re still going to see a delay before everyone can get ahold of it. CA : So this really puzzles me, actually, and I ' d love to ask you as a mathematician about this as well. There are already several companies believing that they have plausible candidate vaccines. As you say, the process of testing takes forever. Is there a case that we ' re not thinking about this right when we ' re looking at the way that testing is done and that the safety calculations are made? Because it ' s one thing if you ' re going to introduce a brand new drug or something – yes, you want to test to make sure that there are no side effects, and that can take a long time by the time you ' ve done all the control **trials** and all the rest of it. If there ' s a global emergency, isn ' t there a case, both mathematically and ethically, that there should just be a different calculation, the question shouldn ' t be"Is there any possible case where this vaccine can do harm,"the question surely should be,"On the net probabilities, isn ' t there a case to roll this out at scale, to have a shot at nipping this thing in the bud?"I mean, what am I missing in thinking that way? AK : I mean, we do see that in other situations, for example, the Ebola vaccine in 2015 showed, within a few months, very promising evidence and interim results of the **trial** in humans showed what seemed very high efficacy. And even though it hadn ' t been **licensed** fully, it was employed for what is known as compassionate use in subsequent other outbreaks. So there are these mechanisms where vaccines can be fast-tracked in this way. But of course, we ' re currently in a situation where we have no idea if these things will do anything at all. So I think we need to accrue enough evidence that it could have an impact, but obviously, fast-track that as much as possible. CA : But the skeptic in me still doesn ’ t fully get this. I don ’ t understand why there isn ’ t more energy behind **bolder** thinking on this. Everyone seems, despite the overall risk, incredibly risk-averse about how to build the response to it. AK : So with the caveat that, yeah, there ’ s a lot of good questions on this, and some of them are slightly outside my wheelhouse, but I agree that we need to do more to get timescales out. The example I always quote is it takes us six months to choose a seasonal **flu** strain and get the vaccines out there to people. We always have to try and predict ahead which strains are going to be circulating. And that ’ s for something we know how to make and has been manufactured for a long time. So there is definitely more that needs to be done to get these timescales shorter. But I think we do have to balance that, especially if we ’ re exposing large numbers of people to something to make sure that we ’ re confident it ’ s safe and that it ’ s going to have some benefit, potentially. CA : And so, finally, Adam, I guess going into this – There ’ s another set of infectious things happening around the world at the same time, which is ideas and the communication around this thing. They really are two very dynamic, interactive systems of infectiousness – there ’ s some very damaging information out there. Is it fair to think of this as battle of credible knowledge and measures against the bug, and just bad information – You know, part of what we have to think about here is how to suppress one set of things and boost the other, actually, turbocharge the other. How should we think of this? AK : I think we can definitely think of it almost as competition for our attention, and we see similarly, with diseases, you have viruses competing to infect susceptible hosts. And I think we ’ re now seeing, I guess over the last few years with fake news and misinformation and the emergence of awareness, more of a transition to thinking about how do we reduce that susceptibility if we have people that can be in these different states, how can we try and preempt better with information. I think the challenge for an outbreak is obviously, early on, we have very little good information, and it ’ s very easy for certainty and confidence to fill that vacuum. And so I think that is something – I know **platforms** are working on how can we get people exposed to good information earlier, so hopefully protect them against other stuff. CA : One of the big unknowns to me in the year ahead – let ’ s say that the year ahead includes many, many more weeks, for many people, of actually self-isolating. Those of us who are lucky enough to have jobs where you can do that. You know, staying home. By the way, the whole injustice of this situation, where so many people can ’ t do that and continue to make a living, is, I ’ m sure, going to be a huge deal in the year ahead and if it turns out that death rates are much higher in the latter group than in the former group, and especially in a country like the US, where the latter group doesn ’ t even have proper health insurance and so forth. That feels like right there, that could just become a huge debate, hopefully a huge source of change at some level. AK : I think that ’ s an incredibly important point, because it ’ s very easy – I similarly have a job where remote working is fairly easy, and it ’ s very easy to say we should just stop social interactions, but of course, that could have an enormous impact on people and the choices and the routine that they can have. And I think those do need to be accounted for, both now and what the effect is going to look like a few months down the line. CA : When all ’ s said and done, is it fair to say that the world has faced, actually, much graver problems in the past, that on any scenario, it ’ s highly likely that at some point in the next 18 months, let ’ s say, a vaccine is there and starts to get wide distribution, that we will have learned lots of other ways to manage this problem? But at some point, next year probably, the world will feel like it ’ s got on top of this and can move on. Is that likely to be it, or is this more likely to be, you know, it escapes, it ’ s now an endemic nightmare that every year picks off far more people than are picked off by the **flu** currently. What are the likely ways forward, just taking a slightly longer-term view? AK : I think there ’ s plausible ways you could see all of those potentially playing out. I think the most plausible is probably that we ’ ll see very rapid growth this year and lots of large outbreaks that don ’ t recur, necessarily. But there is a potential sequence of events that could end up with these kind of multiyear outbreaks in different places and reemerging. But I think it ’ s likely we ’ ll see most transmission concentrated in the next year or so. And then, obviously, if there ’ s a vaccine available, we can move past this, and hopefully learn from this. I think a lot of the countries that responded very strongly to this were hit very hard by SARS. Singapore, Hong Kong, that really did leave an impact, and I think that ’ s something they ’ ve drawn on very heavily in their response to this. CA : Alright. So let ’ s wrap up maybe by just encouraging people to channel their inner mathematician and especially think about the opportunities and the transmission probabilities that they can help shift. Just remind us of the top three or four or five or six things that you would love to see people doing. AK : I think at the individual level, just thinking a lot more about your interactions and your risk of infection and obviously, what gets onto your hands and once that gets onto your face, and how do you potentially create that risk for others. I think also, in terms of interactions, with things like handshakes and maybe contacts you don ’ t need to have. You know, how can we get those down as much as possible. If each person ’ s giving it to two or three others, how do we get that number down to one, through our behavior. And then it ’ s likely that we ’ ll need some larger-scale interventions in terms of gatherings, conferences, other things where there ’ s a lot of opportunities for transmission. And really, I think that combination of that individual level, you know, if you ’ re ill or potentially you ’ re going to get ill, reducing that risk, but then also us working together to prevent it getting into those groups who, if it continues to be uncontrolled, could really hit some people very, very hard. CA : Yeah, there ’ s a lot of things that we may need to gently let go of for a bit. And maybe try to reinvent the best aspects of them. Thank you so much. If people want to keep up with you, first of all, they can follow you on Twitter, for example. What ’ s your Twitter handle? AK : So @AdamJKucharski, all one word. CA : Adam, thank you so much for your time, stay well. AK : Thank you. CA : Associate professor and TED Fellow Adam Kucharski. We ’ d love to hear what you think of this bonus episode. Please tell us by rating and reviewing us in Apple Podcasts or your favorite podcast app. Those reviews are influential, actually. We certainly read every one, and truly appreciate your feedback. This week ’ s show was produced by Dan O ’ Donnell at Transmitter Media. Our production manager is Roxanne Hai Lash, our fact-checker Nicole Bode. This episode was mixed by Sam Bair. Our theme music is by Allison Layton-Brown. **Special** thanks to my colleague Michelle Quint. Thanks for listening to the TED Interview. We ’ ll be back later this spring with a whole new season ’ s worth of deep dives with great minds. I hope you ’ ll enjoy them whether or not life is back to normal. I ’ m Chris Anderson, thanks for listening and stay well.

Text Sample 173: POS Dim 2 – 2020 – Score 18.00 – 2020/t004288.txt

Chris Anderson : Well, hello, Helen. Very nice to see you. CA : You staying well? Helen Walters : How ’ s it going? CA : These are mad, mad, mad, mad, mad days. So many emotions. Not all bad, happily, but I ’ m just so aware that, among the people listening to this, some are in really tough times right now. I hope this is going to be a beautiful hour of therapy and help in its own way, because we have with us just an extraordinary author, an extraordinary mind, Elizabeth Gilbert, obviously known for her astonishing best-selling success with"Eat, Pray, Love,"although her favorite book from my point of view is called"Big Magic,"where the subtitle is,"Creative Living Beyond Fear.""Creative Living Beyond Fear."Now when you think about it, that is a pretty good agenda for today ' s conversation, I think. Liz describes the emotional landscape of our lives, I think like no one else I ' ve read, and I ' m not even her target audience. She ' s really extraordinary in doing that. She gave an amazing TED Talk 11 years ago now,"In pursuit of your creative genius."It really reframed how to think of creativity. It ' s been seen, like, 19 million times or something, and it ' s really changed how a lot of people – they ' re just open to the creative genius coming from the outside. So it ' s a delight to welcome to the TED Connects stage Elizabeth Gilbert. Elizabeth Gilbert : Hey, Chris. CA : Great to see you. How are you? Where are you? Who are you living with or staying with? What ' s up? EG : I ' m fine. I don ' t want to brag, but I ' m in New Jersey, where anybody would want to be. I ' m by myself. I ' ve got a little house out in the country, and I think I ' m on day 17 of no human contact other than virtually, and I ' m well. I ' m not anybody you need to be worrying about right now. So I ' m good. CA : Wow. Well, so in a way, you ' re having a related experience to what so many people are having. I mean, these are days of **isolation** for many people, and that brings with it lots of difficult emotions, in a way. And we ' re going to go through many of them, I hope, in the next hour. So I ' m hoping to talk with you about – I wrote down a list here : about anxiety, loneliness, curiosity, creativity, procrastination, grief, connection and hope. How about that? That ' s our agenda. Are you up for that? EG : I think that ' s the whole buffet. Just a little light tasting menu of all the mass of human emotions. Let ' s do it. Absolutely. CA : I think it ' s probably good to dive straight in with the anxiety that I know a lot of people are feeling right now. So many reasons to be anxious, both for yourself, your loved ones, and just for this time and for the world and how we all get through this. Have you been feeling anxiety, Liz? And how do you think of it? What can you say to us? EG : I have been, and I think you would have to be either a sociopath or totally enlightened not to be feeling anxiety at a moment like this. So I would say that the first thing that I would want to encourage everybody to do is to give themselves a measure of mercy and compassion for the difficult emotions that you ' re feeling right now. They ' re extremely understandable. I think sometimes our emotions about our emotions become a bigger problem, so if you ' re feeling frightened and anxious, and then you ' re layering shame on top of that because you feel like you should be handling it better, or you should be doing your **isolation** better, or you should be creating more while you ' re alone, or you should be serving the world in some better way, now you ' ve just multiplied the suffering, right? So I think that the **antidote** for that, first of all, is just a really warm, loving dose of compassion and mercy towards yourself, because if you ' re in anxiety, you ' re a person who is suffering right now, and that deserves a show of mercy. The second thing that I would say about anxiety is this, that here ' s what I think is the central paradox of the human emotional landscape that I ' m finding particularly fascinating right this moment, and it ' s really come to light for me. So there are these two aspects of humanity that don ' t match – hence the word paradox – but they really define us. And the first is that there is no species on earth more anxious than humans. It ' s a hallmark of our species, because we have the ability slash curse to imagine a future. And also, once you ' ve lived on earth for a little while, you have the experience to recognize this terrifying piece of information, which is that literally anything can happen at literally any moment to literally any person. And because we have these vast, rich, colorful imaginations, we can see all sorts of terrifying movies in our heads about all of the possibilities and all of the scariest things that could occur. And actually, one of the scariest things that could occur is occurring. It ’ s something that people have imagined in fiction and imagined in science, and it ’ s actually happening right now, so that ’ s quite terrifying. The paradox is that, in that level, we ’ re very bad, emotionally, at fear and anxiety, because we stir ourselves up to a very heated degree because of our imaginations about how horrible it can get, and it get can get very horrible, but we can imagine it even worse. The paradox is that we ’ re also the most capable, resourceful and resilient species that has ever lived on earth. So history has shown that when change comes to humanity – either on the global level, like it ’ s happening now, or on the personal level – we ’ re really good at it. We ’ re really good at adaptation. And I think that if we can remember that, it can help to actually mitigate the fear. And you can remember it in a historical perspective, by looking at what humanity has gone through, and what we have not only survived but figured out how to thrive through. And you can also look at it at a personal level, where you can make an inventory of what you yourself have survived, and notice, as I often notice, my panic and my anxiety about the imagined future is deadly on my nervous system, but I actually have discovered that when there ’ s an actual emergency in the moment, I tend to be pretty good at it. And I think most of us are like that. You ’ ll see that repeated in history in so many examples. I think about those heartbreaking and devastating phone messages that people were leaving for their loved ones from the towers on September 11th, and you can hear the calm, the calm in peoples ’ voices. The biggest emergency ever was happening, and in that moment, intuition told them what to do. The important thing to do now is to make this phone call. And I think if you can trust that when the point of emergency actually arrives, you ’ ll be able to meet it, and then when the world changes, you ’ ll be able to adapt to it – it certainly helps me calm down. CA : I mean, I guess there ’ s a reason why fear is there. It didn ’ t just evolve by accident. It ’ s supposed to direct our behavior and help us avoid danger, and it ’ s just that sometimes, it gets out of control and actually gets in our way and damages us. I mean, any specific advice on how someone could turn their fear into something useful, at this moment? EG : Can I tell you a story that I ’ m using as a touchstone for myself right now and drawing wonder and inspiration from? So, some of you may have heard of a young woman named Amanda Eller. She was in the news recently, because she got lost in Hawaii in the wilderness for 17 days, and there was a massive, massive hunt for her, because she had left her car, she ’ d gone for a simple hike, had left her phone in her car, went up into the woods, took a wrong turn, and then had this disastrous 17 days, fell off a cliff, broke her leg, walked for 40 miles on a broken knee, lost her shoes in a flash flood. She had to sleep packed in mud in order to protect herself from the cold and the mosquitoes. She was eating moths. I mean, just a harrowing story of survival. I met her recently, and she was so lit and radiant with this kind of serenity and this kind of wonder and joy, and I said,”How are you like this? You went through one of the most traumatizing things that a person could go through.”She said,”First of all, I discovered that I can survive anything,”going back to this idea of how resourceful and adaptive humans actually are. But the piece of her story that I am using like a life raft right now is that she said, on her second day in the jungle, when she realized that she was truly and very much in trouble – she ’ d already spent one night in the woods and she was completely lost and she was totally alone and no one knew where she was, and she was full of terror – she said she closed her eyes and she prayed or asked or requested, she made a wish to herself, to consciousness, to the universe, and she said,”Please take my fear away, and when I open my eyes, have it be gone, and have it be gone and have it not come back.”And she opened her eyes, and it was gone, and it was replaced by intuition. And I think intuition is a little bit the opposite of fear, because fear is the terror that you feel about a frightening imagined future. Intuition can only happen when you ’ re in the moment. And so, from that point forward, she did not experience fear for the rest of the time she was in the woods. She just was guided by some deep intuitive sense, located somewhere between her sternum and her navel, and in every moment, she would ask it,”Right or left?””Up or down?””Eat this? Don ’ t eat this?”And just trust it. Complete, absolute surrender to the intuition of the moment. And she said it hasn ’ t returned, the fear hasn ’ t returned, and she still guides her life that way. So it ’ s a return to some sense that there ’ s a navigational system within you that will, if you stay present in this actual moment, tell you what to do one moment to the next. Now, if you want to suffer, pop out of the moment and imagine a future, and then you can suffer indefinitely. So it is almost like a spiritual or meditation **practice**, and anybody out there who ’ s done any spiritual or meditation **practices**, this is what you were **practicing** for. You were **practicing** for this moment, and those of you who haven ’ t tried that, this might be a really [inaudible] to be centered in the **instant**. CA : Wow, that ’ s a remarkable story, and I guess what I ’ m hearing is two things. It ’ s one, just the reaching out to the universe there, but specifically, there was a decision to let go of the future and just to focus on the moment. EG : That ’ s it, yeah. Nothing will bring you more pain than the future, and what I ’ m seeing happening right now – I said this to you the other day, Chris – is there ’ s a relatively small percentage of the population who will suffer physically from this disease, and there ’ s a larger percentage who are going to suffer economically from it. But then there ’ s this massive, uncountable number of people who will and are suffering from it emotionally, and right now, those people are my concern, because they ’ re really in pain, and there ’ s millions and millions of them. CA : So you ’ re living there by yourself, Liz, and many others are in that same circumstance right now. I suspect some are feeling, like, crushing loneliness. Talk about that. How do you handle loneliness in a situation like this, when it ’ s so alien to everything that we as a social species are usually about? We crave other people. We crave touch. We crave hugs. We want to be there with people. How can we avoid this being a period of crushing loneliness? EG : I don ’ t think you can avoid it, but I think you can walk toward it. And I think that, for me, I ’ ve deliberately, many times in my life, gone off into **isolation** in order to face those things. I ’ ve gone on long meditation retreats. This year, I was in India, and I spent 17 days alone with no contact with anybody, which was a weird **practice** run for what ’ s happening right now. And as I see people really losing it and feeling like they ’ re crawling out of their skin, either from anxiety, fear, boredom, anger, blame, loneliness, depression, all of these things that come up when you are forced to just be in your own presence. I know all of those feelings because, as a meditator, I ’ ve experienced all of those, in stillness. The hardest person in the entire world to be with is yourself, and so the only way that I learned, as a meditator, to be able to survive and endure my own company was with universal human compassion toward me, and to recognize that this is a person who is suffering right now from loneliness, and this person needs kindness from self towards self. And it ’ s a very high teaching, but I think that it ’ s a very interesting moment to **practice** that. And so what I would suggest to people – and again, this takes a certain amount of resolve and it takes a certain amount of curiosity about learning more about the human experience – what I ’ m seeing people do is people are spinning away from that

isolation because they ’ re so terrified of it. What happened with the world right now is that basically all of our pacifiers were yanked out of our mouths. Everything that we ever can do and reach for that can get us out of having to be in the existential crisis of being alone with ourselves was taken away. And I see people rushing to fill it, I mean, constant Zoom meetings and constant parties online and constant interaction, and all of that is lovely, but from a spiritual and psychological standpoint, from a creative standpoint, I would say if you have any curiosity about this, don ’ t be in such a hurry to rush away from an experience that could actually transform your life. I think sometimes the experiences that can transform us the most intimately are the ones that we want to run away from, and I think of a story that the Dalai Lama told about one of his teachers. When the Chinese invaded Tibet, and all the monks were running into India for safety, one of his teachers, who was one of the great masters, the last glimpse that the Dalai Lama had of him was that he was walking into China – very patiently and very slowly, toward it. Everybody else was running away from it – he was walking toward it, and I think there ’ s a level at which first responders do that, and in the real world, in an intimate way, they go into the emergency, they go toward the emergency. All the people who are trying to solve this now in worldly ways are walking toward the emergency. But there ’ s a way that you can do it emotionally as well and that is to walk with curiosity and with an open mind toward your most difficult and painful emotions without resistance, and say,”What is it like for a person to feel like they don ’ t have something to do for an hour?”And you can open up your compassion in that. There ’ s so many lessons in compassion that can be found here. How about a general universal mercy that we can all feel toward people who are in solitary confinement. Let ’ s have that be part of the conversation. Now you ’ ve experienced it for two days in your own house. Maybe it ’ s time to change the prison system? You see how hard this is. Or you can have compassion toward people who have lost a loved one and they ’ re alone. By feeling your own feelings, you can open up your feelings more universally toward the world. So I think there ’ s a great opportunity here for growth on the personal level, but you have to have almost a whimsical curiosity to be the one walking into China rather than the one away from it. And that ’ s how I ’ m doing it right now. CA : So let ’ s follow up on that word”curiosity”that you ’ ve used a few times there. I mean, a lot of wisdom that I ’ ve heard sort of thrown around online right now is,”This is a great time to follow your passion and dive deep into whatever it is you ’ ve most been wanting to do.”I mean, in”Big Magic,”you made an argument that following your passion isn ’ t necessarily the wisest strategy. You argued, no, don ’ t do that, follow curiosity. Does that apply now? Make that case. EG : Yeah, you know, I ’ ve been on a personal crusade to rid the world of the world”passion”as an instruction for people on how as they should be living, because I know that in my case, it brings me nothing but anxiety.”Purpose”is another one that has become a cudgel that we use to bludgeon ourselves into thinking that we ’ re not doing enough or that we ’ re doing life right or that you ’ re supposed to be more useful, more productive, you ’ re supposed to be changing the world, or uncovering some particular talent that only you have and with it, you ’ re supposed to transform everybody and monetize it, no pressure. I start to get hives even repeating that, but that ’ s what we ’ ve been taught, that purpose and passion are everything. I would like to replace it with a far gentler word, and I think”curiosity”is very gentle, because the stakes are so much lower. The stakes of passion say you have to shave your head and move to India and get rid of all your possessions and start up, like – It ’ s so intense. But curiosity is a very simple, universal experience that causes you to want to look at something just a tiny bit closer, and you don ’ t have to change your life around it. You just look, and it might be taking a weekend to try something new for a little while. It ’ s almost so easily missed, and I think so many times, we ’ re looking way up at the sky for the sign from God of what our passion and what our purpose is supposed to be, and meanwhile, there ’ s this lovely little trail of breadcrumbs of curiosity that if you can slow down – and again, this is about not rushing out of the experience of being silent, still and alone – if you can slow down, you might be able to see them. But if I could say one thing I ’ m noticing is an **obstacle** right now – because I think a lot of people thought,”**Isolation**, great, this is the perfect time for me to learn Italian and take that calligraphy class and start writing that novel,”and they find that they ’ re actually in a paralysis of anxiety and they ’ re not creating anything or doing anything. First of all, again, like, a blanket of mercy on you. These are hard times, and it might take you a minute in your nervous system and your mind to adjust to the new reality. But the second thing I would say is that when people are saying they ’ re having trouble with their creativity because they ’ re in **isolation**, I might daringly suggest that perhaps you ’ re not in enough **isolation**. And by that I mean, are you monitoring how much external stimulus you ’ re bringing of this disaster into your home? So if you ’ re sitting watching the news all day, what you ’ re doing is you ’ re bringing the disaster into your work space. You ’ re bringing it into your soul. You ’ re bringing it into your mind. And you ’ re going to create the opposite of a creative environment, an environment of fear, panic and urgency. So I think if you ’ re going to be a good steward of your creativity right now, you have to isolate a little bit from the news. And that doesn ’ t mean disconnecting, it means I get up every morning and after I ’ ve meditated, I read the New York Times and I give myself 40 minutes with it, and then that ’ s it for the day, because I know that if I bring in any more, I ’ m going to go into a traumatized state and then I won ’ t be able to follow my intuition, I won ’ t be able to help people, because I myself will be suffering, and I won ’ t be able to be present for this very interesting moment in my life and in history, and I want to remain present for it as much as I can. So there ’ s a discipline of being a good steward of your senses and deciding what you ’ re going to put your senses in front of. CA : Helen. HW : Liz, there ’ s an outpouring on Facebook of gratitude for you. People are so grateful, and grateful for the calm that you are instilling in us all, so thank you, from them and from me. We ’ ve also had a number of questions about grief. We ’ re kind of dealing with grief at a different scale at the moment. One person has already lost five people to **coronavirus**. And so any thoughts of how to manage grief at this scale or how to process this in a way that honors both them and yourself? EG : First of all, my condolences. And I think any words that I would say about somebody who just lost five family members could only be inadequate. Grief is bigger than us. It ’ s bigger than your efforts to manage it, and if you want to hold yourself and your family members compassionately through grief, you have to allow that it cannot be managed. And I think that grief management is something that we ’ ve kind of created in our very Western idea that if we can figure out something, we can avoid suffering from it, so if we can figure out how to translate grief and if we can figure out how to walk through grief, then we won ’ t have to experience the magnitude of it. Many of you know that I lost the love of my life two years ago from pancreatic and liver cancer, and I was with her when she died, and I ’ ve been walking through my own path of grief, so I know what it feels like to lose the person in the world who is the most important to you, which is of course the biggest fear that we all have. I know that you can survive it, but I know that you survive it by allowing yourself to feel it. And again, to go back to the metaphor of the monk walking directly into China, into conflict rather than away from it, do you have the courage to let it break over you like waves?

I wish I could remember her name. There ’ s this extraordinary woman who wrote a book called”Here If I Need You,”and she ’ s a chaplain for the police department in Maine, and she ’ s in charge of knocking on people ’ s doors and giving them the worst news they ’ re ever going to hear in their life, when she goes with the police when something happens. And she told a story once that I found very moving and very helpful for me in my grief. She said what she ’ d witnessed through years and years of sitting with people through what is literally the worst moment of their life, the nightmare of that loss, is that when she knocks on that door and tells that person, your daughter, your family member, your husband, your mother has been killed, there ’ s this universal collapse where the person will just be – it is the tidal wave that comes and just takes you down and you lose all civilization, you lose all your attainments, all your wisdom. Nothing can stand up to that. You literally go to the floor. And you sob and you grieve, and she holds them through that. And then she said that what she ’ s learned is the most astonishing thing, that that never lasts more than a half an hour, that first wave. It can ’ t. You actually physiologically can ’ t sustain that, and if you let it break over you and you just allow it, then within a half an hour, usually sooner – and she said this has happened every single time she ’ s been with somebody with a loved one ’ s death – the very next thing that happens is that that person calms down, they catch a breath, and the next question they ask is a very reasonable question.”Where is the body? What do we do next? When can we have the funeral? Who else was in the car?”And with that question, she says, they start to rebuild their new life already with this new piece of information that even an hour ago would have seemed unsurvivable. And she uses that as an example of, once again, the tremendous psychological resilience of a human being. And it doesn ’ t mean that they will never grieve again. It doesn ’ t mean that their grieving journey is over. It just means that, somewhere in their mind, that it ’ s landed, and now, already, they ’ re making a plan about,”OK, who do we need to notify, what ’ s the next thing we need to do.”And again, if you can remember this as you go through your panic, if you can remember that in the moment of emergency, there will be an intuitive, deep sense that will tell you there ’ s going to be some next steps and it ’ s time for us to take those next steps, and if you can also remember that resilience is our shared genetic and psychological inheritance – we are, each and every one of us, no matter how anxious you feel you are, no matter how ridden by fear you feel you are, every single one of us is the genetic survivor of hundreds of thousands of years of survivors. Each one of us came from a line of people who made the next correct intuitive move, survived incredibly difficult things, and were able to pass their genes on. So almost to the biological level, you can relax into a trust that when the moment comes where you will be faced with the biggest challenge, you will be able to draw on a deep reservoir of shared human consciousness that will say,”Now it ’ s time to make the next move, and we can do this.”HW : So beautiful. So many more questions. I will be back. EG : Thanks, Helen. CA : Thank you, Liz. I think the author ’ s name was Kate Braestrup. EG : That ’ s it. CA : I guess that ’ s a book, if you need a book right now,”Here If You Need Me”by Kate Braestrup. EG : Very good, yeah. CA : Liz, you and I got to have a conversation a few months after Rayya passed away. It was actually the first-ever episode of the TED Interview podcast we did. And I found that it was probably my favorite episode ever of the TED Interview. And it was so moving how you spoke about your grief then. And I feel like that ’ s a potential resource to people. I know we were both sitting there shedding tears, and I found that an extraordinary experience personally, to be sure. But somehow... in this moment, if you follow this journey of curiosity, if you walk towards some of the harder moments, do you think that this actually can be a creative time for people if they ’ re willing to do that? EG : Absolutely, and I don ’ t think creativity in this case has to necessarily mean that you write the Great American Novel or start that business you always intended to start. It doesn ’ t need to be so literal. We ’ re going to be creating new worlds and new lives on the other side of this, and we ’ re going to be doing that individually and we ’ re going to be doing that collectively. I think of the shoots of small trees that can only come up after massive forest fires, where seed pods have to explode under great heat. We ’ re in a kind of crucible moment right now, and I wouldn ’ t begin to have the hubris to predict what sort of creativity will come, but look, if history is any measure, what we ’ ll probably see is people at their best and people at their worst. But I think we ’ ll see more of people at their best, because that ’ s typically how it works. CA : I mean, your model of how creativity happens is that it doesn ’ t all come from within. It ’ s not like you have to sit there, saying,”OK, this is my moment to be creative. Come on. Be creative. Be creative.”It involves, fundamentally, an openness to something coming to you, to be open, to be curious, listening, but then just to be open to that moment. Perhaps that could apply even more now than ever, just because we have this huge distraction of the news, some other distractions are taken away. Is there a chance that if people listened, they actually can receive more at this moment? EG : I think so, and I think, again, if you stop thinking about your self-isolation and your social **distancing** as **quarantine** and you start thinking of it as a retreat, you ’ ll find that you can ’ t really tell the difference between **quarantine** and retreat. You know, a lot of you out there have dreamed, I ’ ve heard you, because I talked about going to India to an ashram for four months and God, I can ’ t tell you how many people I ’ ve heard say,”I wish I could do that.”I ’ m like,”Well, you got it.”And by the way, this is what it felt like. This is what it felt like to learn how to be present with yourself. I think my screen needs to move a bit. To learn how to be present with yourself means sitting in a lot of terror, sitting in a lot of anxiety, sitting in a lot of fear, sitting in a lot of shame, and being able to allow that without having to resist it, without having to reach outside yourself for something to numb yourself with. I also want to tell a story that a friend of mine, Martha Beck, told me about when she goes to South Africa and teaches animal-tracking courses. And she works with all these great African animal trackers, and these old men who have had these skills passed down for generations. And she was using it as an example of the difference between focus and openness. So I think sometimes the mistake people make when they want to be creative is they think they have to get really focused, and focus is an anxiety-producing energy as well. You ’ ve got to drill down and you feel your whole body tense. But what she described witnessing in these animal trackers is when they go out to hunt the lions, these old, old men, the very first thing they do is they sit down against a tree and they appear to go to sleep. They drop into a state that she calls and that the mystics call”wordless oneness.”And wordless oneness, you can also call meditation. You can also call it the zone. But it ’ s a stillness where you actually can drop your nervous system into such a quiet place that you have 360-degree awareness of your senses and of presence. And they ’ ll sit like that, apparently doing nothing, for an extremely long time, just looking through half-lidded eyes at the world. And then, maybe an hour, two hours in, all of the sudden, they ’ ll say,”The lion ’ s over there.”And so for me, I ’ ve learned to hold my creative wishes lightly in that same way. I ’ m between books right now and I don ’ t have an idea for a book and in the past, that would have made me really anxious, but now I know – take a lot of naps, go for a lot of walks, do a lot of drawings. I ’ m doing weird little art projects as I ’ m sitting here, to distract my mind. CA : Wait, wait. Bring that back. Hold that up. EG : Owls. CA : Aww. EG : Aren ’ t they dear? CA : They ’ re beautiful. Goodness me.

EG : Well, I ’ m just playing with color and texture because it calms me, and I think if you can ’ t think of what to do right now, I would suggest doing what you used to do when you were 10 years old that made you feel happy and relaxed, and that ’ s often creativity and play. And for many of us who were anxious children – and I was an anxious child – we learned at an early age that we could sedate ourselves with our curiosity and with our play, and then, usually around adolescence, the world taught us that there were faster and more immediate ways to bump out of that anxiety through sex or substances or distraction or workaholism or whatever we did and not have to sit with ourselves. And I think right now is a really good opportunity – You actually were on the right track when you were 10, whatever it was. So, you know, get some LEGOs. Get some LEGOs, get some coloring books, just get your hands in the mud, do whatever it is that will actually ground you into this, again, to take you out of the futurizing and the future-tripping that ’ s going to cause you nothing but anxiety and not going to make you be of **service**. There ’ s such a thing, too, that I just want to touch on if I can, for a minute, about empathetic overload and empathetic meltdown. We ’ re taught that **empathy** is a good thing. I would suggest that in a case this traumatic, what you want to talk about replacing **empathy** with is compassion, and the difference is extremely important. So compassion means”I ’ m actually not suffering right now, you are, I see your suffering, and I want to help you.”That ’ s what compassion is. **Empathy** is”You ’ re suffering, and now I ’ m suffering because you ’ re suffering.”So now we have two people suffering and nobody who can serve, and nobody who can be of help, and if you knew how your empathetic suffering actually makes you into another patient who needs assistance, you would be more willing to dip into compassion. And what underlies compassion is the virtual courage, the courage to be able to sit with and witness somebody else ’ s pain without inhabiting it yourself so much that you become another person who is suffering and now, there are no helpers. And it takes an enormous amount of courage to be able to watch that without diving into it and joining it and becoming sick yourself. CA : I mean, if **empathy** is just a feeling, does compassion, your use of compassion imply that it ’ s turning that feeling into something potentially practical to actually do something, if you can, for that person? EG : It ’ s recognizing that if I feel your pain, I can ’ t help you in your pain, because now my pain has taken over me, and sometimes, I think all you need to do is know that and it makes you turn the ship. Right? One of my favorite teachers, Byron Katie, says,”My favorite thing about my suffering is that it isn ’ t yours.””My favorite thing about my suffering is that it isn ’ t yours. My favorite thing about your suffering is that it isn ’ t mine.”So it will be, eventually, we all take a turn suffering. You cannot move through this earth without it. When it ’ s your turn, you ’ ll know. When it ’ s not your turn, stay out of that field of somebody else ’ s pain, because you can ’ t help them when you ’ re in pain yourself. And then see if you can find the inner resolve and courage. And I think some of that is just based on accepting the Buddhist First Noble Truth, which is that suffering is an unavoidable aspect of life on earth. We ’ re all going to be in it at some point. We ’ ve all been in it at some point. And now, how can I help? I ’ m not saying this is easy. I ’ m just saying, also, if you ’ re suffering from empathetic overload and empathetic meltdown, which means your adrenals are up, your **stress** is up, your endorphins are down, you ’ re going into a parasympathetic collapse – this would be another time to discipline yourself to stay away from the news, because you actually will have a breakdown, and you won ’ t be able to help the people around you who are the people who need help. CA : But have you seen any signs that if someone takes that **empathy** and compassion, let ’ s say, and decides to act in some way, big or small, on behalf of someone, that actually shifts how they feel, that there ' s a healthiness to that? Or is that the language of just inducing more guilt in people? EG : No, I think there ' s a beautiful healthiness that can come from being of **service**, and that ' s also how I ' ve been medicating my anxiety through this, by showing up in ways that I can with whatever resources I ' ve got. Here ' s what you have to keep in mind, though, and this is what I keep reminding people. Right now, in my own personal sphere, there is more need than I have resources to fix. So I have to begin with that reality, and I have to have the courage to sit in that reality soberly and acknowledge that that ' s the case. The second thing I think emotional sobriety would require of me right now is to recognize that this is going to be a marathon, not a sprint. And so the first week of the crisis, I had this deluge of all my really energized, let ' s-save-the-world friends, all my creative friends, everybody was e-mailing, texting, Zooming, and they all had a response."Let ' s do this! Let ' s do this! Let ' s fix it this way!"And I found myself joining with some of them and not joining with others, just, again, based on my intuition, but I also found myself cautioning them,"Guys, this is a marathon."We ' re in mile one of what ' s going to be a very long marathon. So pace yourselves, and pace your resources. Don ' t overgive to the point where you collapse, because we ' re still going to need helpers two months from now, and we ' re still going to need helpers six months from now. And so, find a steady pace and be willing to be in it for the entire long haul. CA : Yeah. Helen. HW : Such great advice, Liz, and so many questions pouring in. One of them is from a therapist who confesses that she, and many of her clients, are having trouble with the letting go of control in this moment, and wonders if you have any advice on how to let go of control in order to be willing to feel everything that we ' re enduring. EG : Just this... This sense that you had that you had control was a myth to begin with. And that may not be comforting, except that I find it very comforting. You know, control is an illusion, and there are times where we ' re able to **fool** ourselves because we ' re so good at technology, we ' re so good at creating safe worlds where we ' re able to trick ourselves into believing that we ' re in control of any of this. But we ' re not, and the paradox, for me, of surrender is how relaxing it is. Nobody ever wants to surrender, because nobody wants to lose control, but if you recognize that you never had control, all you ever had was anxiety, and then you let go of the myth of control, you ' ll find that, I find that if I even say that sentence,"I ' m losing control,"and then I remind myself,"You never had control, all you had was anxiety, and that ' s what you ' re having right now."So you ' re not letting go of anything. Surrender means letting go of something you never even had. So there ' s an awakening that ' s happening right now, where what ' s happening is not that you ' re losing control. What ' s happening is that, for the first time, you ' re noticing that you never had it. And the world is doing its job. The job of the world is to change, constantly, and sometimes radically, and sometimes immediately, and it ' s doing its job, and that is also the **norm** of things. And again, we are adaptive and we ' re resilient and we can handle it. But I don ' t kid myself for a minute to think that I ' m in control of anything that ' s ever happening. My realm of control is extremely small. It ' s usually about, like, might be able to go get a glass of water right now. Like, there ' s not a lot that I ' m in control of. And I ' m actually I ' ve ever been. HW : One more question from online, if I may, and then I will jump off again. You know, Chris, you and I, we ' re all in a pretty privileged position. TED has been able to go remote. We ' re able to work remotely. But many, many, many millions of people in the US and beyond are not able to do that, and people are really suffering. How can we help? What are your thoughts about people who are not able to socially distance, who are losing their jobs, the global catastrophe that is unveiling? How can we think about that in a humane and compassionate way? EG : It ' s crushing, and again, as with the same case of the person who said they had lost five family members, I can ’ t, sitting in this position of comfort and safety, say anything that I think is going to be accurate and appropriate to that, other than to say that I just think of this Indian proverb that I keep going back to, which is,”I store my grain in the belly of my neighbor.”Western, capitalistic society has taught and trained us to hoard long before this, long before this happened and people were hoarding toilet paper and canned goods. Advertising and the whole capitalist model has taught us scarcity, it ’ s taught us that you have to be surrounded by abundance in order to safe. The disconnect between those who have and those who have not has never been bigger, and never in my lifetime, and probably in any of our lifetimes, has there been an invitation, again, to release the stranglehold on your hoarding. This is not the time for hoarding. This is the time to store your grain in the belly of your neighbor, in a way that is emotionally sober and accurate to what you can give, and to look at that in a really honest way, to not put your own family in danger, to not put yourself in crisis, but to be able to say,”What can I offer in the immediacy?”And then, in the longer term, a conversation about redistribution of resources, and why do so few have so much and why do so many have so little? But that ’ s not a conversation I can fix today. That ’ s, again, outside of my realm of control. But what I can do is unleash the white-knuckled grip that I have on what ’ s mine and make sure that I ’ m going into the world with an open hand – again, not a panicked open hand, where I ’ m going to destroy myself to save somebody else, because then there will be no helper left, but in a reasonable way. I cannot save everybody. I can save a few. And that ’ s the tragic, but, I think, sobering reality that I can offer right now, and again, underlying all of that, undergirding all of that is a recognition that anything that I have to say about people who are in extraordinary suffering right now is not enough. HW : I ’ ll be back. Thank you. EG : Thanks. CA : Liz, talk to me a minute about anger. Like, I think a lot of, just from the conversation we just had, or just listening to you there, there are so many reasons to feel angry right now about what ’ s going on. And part of me feels we should be. That ’ s what anger is for. It ’ s to highlight things that are unjust and unfair and that we must pay attention to, and yet part of me is honestly scared of it. I think there could be an eruption of anger that ’ s dangerous, both personally and for society. Have you felt anger? What are you doing with it? EG : I feel anger at every White House press conference, and I think all thinking people do. I feel angry that this wasn ’ t taken more seriously early on. I feel angry at myself that I didn ’ t take it more seriously early on. As much as I feel contempt and disgust for government officials who I feel were slow to recognize how serious this is, I also have to be really candid that three weeks ago, I was one of the people walking around saying,”Why is everybody overreacting to this so much?”So I think we also have to own our own piece of that, and I think there are rolling waves of awakening that are happening in people, and so a lot of the anger I feel right now is for people who aren ’ t taking this seriously enough, who aren ’ t **quarantining** themselves, who are putting other people in danger. But a month ago, that was me. I was in the Hong Kong airport, sallying through the Hong Kong airport while everybody was scurrying around in masks and gloves, and I was like,”What ’ s the big deal?”It takes people as long as it takes them to come to awakening, and some people, we have to also acknowledge, never will. Anger has its place, and I think that righteous anger, which is the kind of anger that says a violation has occurred here, a humanitarian violation is occurring here, can be very stirring for transformation. Again, it ’ s how comfortable can you be, sitting with these discomforting emotions, and what are you going to do with your anger? CA : Umh. EG : Are you going to lash out at the people you ’ re **quarantined** with?

Are you going to go on Twitter rants? Is that useful? Is that productive? And so I think – again, I keep using the words”emotional sobriety,”but the emotional sobriety that would be required is to feel that anger, acknowledge it, to show yourself mercy for how uncomfortable it is, and then to steadily, recognizing, again, that this is a marathon not a sprint, do what you reasonably can do to change the situation. CA : I mean, the part of me that ’ s constantly looking for the better narrative hopes that the anger we feel now could almost displace some of – I mean, the world ’ s been an angry place for the last couple years. There ’ s been so much anger inflamed online. We ’ ve made each other angry, often, probably, unnecessarily – outrage sparking outrage, disgust, etc. I mean, is there any hope that this is a massive societal shaking up? It ’ s like, don ’ t be so silly. Look at what actually matters here. And we can at least focus more attention onto the things that, yes, some things that we really should be angry about, but other things that maybe... you know, could lead people to say human connection really matters in this moment. People from all sides, we need each other. We just have to use this as a moment when we come together. How do you think about that? Like, how do we turn some of these negative emotions into a force for good that at least gives us some permission to hope that something **special** comes out of all this. EG : Well, I think you have to give yourself permission to hope, and I don ’ t think it ’ s unreasonable to give yourself permission to hope, because, again, our resilience, our resourcefulness, and the way that history has shown how catastrophe can lead to transformation, gives us, actually, I think, reasonable cause to hope. One thing that I ’ m noticing that I ’ m, like, a little bit amused by is that when people start predicting what the post-pandemic world is going to be, I notice that their predictions seem to be, suspiciously, in exact alignment with their personal worldview. So my friends who are utopians are already living in this utopian future where this is going to be the big change. My friends who are dystopians are already predicting that this is the official beginning of the police state and the disastrous new world order. I think there ’ s a lot of hubris in trying to imagine what that new world could be. A quote that I love that a friend of mine always says is”When people aren ’ t busy being the worst, they ’ re the best.”And I think that gives me hope. And it ’ s true the other way, too. When people aren ’ t busy being the best, they ’ re the worst. I ’ m terrible at social engineering, Chris, and you know this, and you have great, better minds than mine who can come on and talk about this on the global scale. The only world that I have a really intimate, familiar engagement with is this one, and on the individual level, what I understand is that the only world that any of us are ever going to live in is this one. And so minding this, and learning how to calm this, how to open this, how to get on the other side of the emotions that are causing harm to you and others, that ’ s my work, you know? Personally, whatever role I have in the public sphere. CA : You ’ re an extraordinary storyteller and you already told us one amazing story earlier on. Have you come across any other recent stories that have given you reason for hope, perhaps? EG : Well, I ’ ll give you one, and this one, I delight in. Years ago, 20 years ago in New York City – 30 years ago, I was in my 20s – I was friends with a woman named Winifred, who was in her 90s. She was this really cool West Village bohemian artist who had lived in Greenwich Village for her entire life, had had a very storied and checkered and wild life, surrounding herself with intellectuals and poets and artists and adventure, and she ’ d had a lot of loss and a lot of gain, and she was this extremely **passionate** person who had friends of all ages, which was something I admired about her. I was friends with her. I was 25, she was 95. But I would call her my very good friend, and she had a lot of friends. She was so open to everything. And at her 95th birthday party, I asked her,”What have you learned, more than anything else?"Because she was such a creature of learning. I wrote about her in"Big Magic."I said to her one time,"What ' s your favorite book that you ' ve ever read?"And she said,"I can ' t say my favorite book because there ' s been so many, but I can tell you my favorite subject, the history of ancient Mesopotamia, which I started learning when I was 80 and it changed my life."And it did. She ' d gone on these expeditions to Jordan and Iraq. She was just so full of living, you know? And I said to her,"What have you learned in all of your experiences? What is the most central thing that you ' ve learned?"And she said,"Human beings can adapt to anything. Human beings can adapt to absolutely anything."And then she said this great line :"If Martians landed on Earth tomorrow, it would be off the front pages of the newspaper by next Tuesday. We would already be used to it."Right? And there ' s a level at which I ' m seeing this adaptation happening. And that is both a good and a bad thing. Right? We can get used to totalitarianism, but we can also get used to – I ' ve gotten used to a world without the love of my life in it. We can adapt. And I keep using that line as a touchstone for myself, because I don ' t know, nor do I presume to know, what the world is going to be after this. I know that it will be different from the one before. I also just have to point out that all y ' all had a lot of complaints about the world we had before, and I do a lot of talking, I do a lot of going around the world, and [I don ' t remember] any one of you raising your hand in any of the seminars I ' ve taught over the last years, and saying,"We are living in a golden age and I ' m so grateful and appreciative for all that I have,"now you want that world back, right? So let ' s actually remember that as we go forward, that this moment, for some of us, that we ' re in right now, might be one that we look back later and say,"Wow, actually, that was pretty good, and I didn ' t have any gratitude for it."So personally, I ' m just hoping that at an intimate level – and again, this is not a socioeconomic, global political level, but it ' s an invitation to actually be grateful for the safety that you have and the people that you have, and maybe carry that forward a little bit. Maybe. We ' re really good at forgetting. Once a crisis is over, we ' re really good at forgetting our gratitude. It ' s one of our great gifts. But you might want to make a note to actually try to be grateful for what you have. CA : Thank you, Liz. I think we have a last question from our online friends. HW : Yeah, what crisis, right? So Liz, just a request for a concrete strategy to try and reduce the fear or the shame that is coming at this moment. EG : I ' ll give you mine, and it may feel weird and out of reach and woo-woo, but I ' m beyond that at this point, and it has been a game changer and a life changer for me. I have a 20-year-long **practice** of writing myself, every day, a letter from Love. Now this may not feel concrete. It may feel very airy. But what it does is that it helps me through my anxiety, and I need it every single day, because I ' m anxious every single day. I wake up frightened every day. I wake up shamed every day. I wake up angry every day. All of the difficult emotions that run through the software of a human consciousness are running through my software all the time, and they cause me pain and they cause me fear, and they cause me **distress**, and they make me sick. So, 20 years ago, when I was going through a very bad divorce and a depression, I began this tactic, and the tactic is that I will sit down with a notebook and I will write to myself, from myself, a letter from Love. And what I mean by"Love"is not romantic love. It ' s the infinite, bottomlessly merciful source of all human compassion. And every single one of these letters begins the same way. It starts with me saying,"I need you."It ' s a dialogue. It starts with me saying,"I need you,"and Love saying,"I ' m right here."And then I say what I ' m going through."I ' m really angry right now. I ' m terrified. I ' m spinning. I can ' t sleep. I ' m anxious."And then I just allow to come through my hand whatever, if you could imagine the most loving, compassionate, merciful voice in the world, if they were in the room with you, what would you want them to say? And you say that to yourself. And so for me, that usually is a combination of these sorts of phrases :”I ’ ve got you. I ’ m right here. I see how distressed you are. It ’ s all right. I don ’ t need you to feel better.”I think a lot of our anxiety is that we want to get out of that feeling as fast as we can, and what Love always says to me is,”It makes no difference to me whether you ’ re anxious or afraid or angry or hurt. I ’ m with you, and I ’ ll be with you through this entire thing for however long it takes. I ’ m not going anywhere. I ’ ve got nowhere better to be right now than sitting with you, loving you. I ’ ll be with you at the moment of your death. I was here with you at the moment of your birth. There ’ s nothing you can do to lose me. You can ’ t fail. You can ’ t do this wrong. You are infinitely, bottomlessly loved.”And it ’ s so interesting to me that the opposite of fear in my life, in my emotional landscape, on the color palette, the opposite of fear isn ’ t courage, the opposite of fear is love. And that presence, a sense of,”I ’ ve got you,”right? Which is the thing that we ’ d all want somebody to say.”I ’ ve got you, and it ’ s going to be all right.”I would love to know, neurologically, what actually happens in my mind when I do this, but what happens to me physiologically is that my mind, just hearing those words and seeing those words, settles, and then from there, I ’ m able to take the next intuitive right **action** the best that I can. CA : Liz, you can say no to this, it may be a totally inappropriate thing to ask, but you don ’ t happen to have a letter from the last day or two that you ’ d consider reading, all or in part of? I don ’ t know how long they are. EG : You ’ re putting me on the spot. Let me see what we ’ ve got. Let ’ s see. OK, so here ’ s one. So I was panicking because I want to offer my apartment in New York to a woman who is a COVID-19 nurse who ’ s volunteered to come into New York City to help, and I ’ m afraid that I ’ ll infect my neighbors if I let her come and stay there. So I was up in the middle of the night, thinking, ethically, is it appropriate for me to do this? So I wrote,”I need you.”And Love said,”I ’ m right here.”And then I said,”I want to offer that COVID-19 nurse my apartment, but I ’ m afraid that my neighbors will get infected, and I ’ m scared, and I don ’ t know what the right move is. Help me.”And Love said,”I don ’ t actually know what the right answer to that is, but I ’ m with you.”And I said,”But what do you think I should do?”And Love said,”Why don ’ t you just sit with me right here for a minute and be with me and know that you ’ re held no matter what, that you cannot make the wrong choice, that it doesn ’ t matter in the grand scheme of things. You ’ re my beloved, I ’ ve got you. I can see how much you ’ re spinning, I can see how tired you are, and it doesn ’ t matter to me whether you make this decision in the next minute, in the next day, or not at all. I ’ m with you, and I ’ ll sit with you through this entire thing, and I ’ ll love you no matter what you decide to do at the end of this. I will be just as much with you at the end of this decision as am I with you now.”And then, I said,”So what do you think I should do?”And Love says,”I think you should go get a glass of water, and I think you should lie down and get some rest, and we ’ ll talk about it some more in the morning.”What I have found over the years of writing myself these letters from Love is that Love never gives advice. This is actually really good for all of you who love to give unsolicited advice to people. Love never gives advice beyond,”Why don ’ t you get a glass of water? Why don ’ t you rest? We ’ ll try this again tomorrow. You ’ re doing your best, this is a hard time, and I ’ ve got you.”So I ’ ve got 20 years of those journals, and I ’ m assuming that I ’ m going to need it for the rest of my life. CA : Wow. I don ’ t know, Helen, I think we might be done. I think I ’ m done. I can ’ t ask any more after that. HW : How beautiful. Good grief. CA : Liz, you ’ re really phenomenal. You ’ ve just got this unique way of articulating what others can ’ t articulate, and you

’ ve brought all of us to a very tender, intimate place, and thank you for that. EG : Thank you, Chris. HW : Thank you so much. EG : And thank you, Helen. Take care of yourselves, everybody. We ’ re right here with each other through this. We can do this. CA : Thank you, Liz. Goodbye. HW : Thank you. CA : Oof. HW : Oof. HW : Deep breaths. CA : Yeah. No. That was **special**. That was **special** to me. I know that you are all in different circumstances online, and that there are so many elements to this thing. There ’ s the problems that those of us who are isolated have, and in many ways, those are the luxurious problems, and we ’ re really aware of that. But they ’ re still problems, and we ’ re going to give space on these TED Connects to many other voices as well. I think we ’ re hoping to hear next week from a doctor at the front line, a voice from India, we hope, on some of the horrifying things that are happening there, and also some pretty amazing proposals for how the world could come out of this, like specific proposals on how we get past this period of lockdown to bring back the economy. All of this matters. So I guess we want Helen and everyone to come back, calendar this, share with friends, and help us figure out how to use this time best. HW : I also wanted to flag that I don ’ t know if you were able to tune in for Susan David ’ s conversation earlier in the week. We have launched a new podcast with Susan that launched on Monday. We ’ re calling it”Checking In with Susan David,”and she is going to be sharing **daily** tips on how to deal with this pandemic. And so you can find that wherever you find podcasts in this day and age. For this conversation, we will be archiving it. It will be on Facebook, and we ’ ll also put it onto TED. com. You can find the TED Interview podcast that Chris and Liz did last year, which I confess just made me weep for... too long. You can find that at go. ted. com / tedconnects. But that ’ s it from us. And tomorrow, I want to flag that we have a very **special** treat, which is less chat, more beauty. We will be joined by the unbelievably talented Butterscotch, who is a beatboxer and a singer and a musician and a sage, and an all-around delight, and she is going to be giving us a glimpse into her world and delighting us all with some sonic deliciousness. So do tune in tomorrow. CA : Thanks so much, everyone. We ’ re in this together. Stay safe. See you soon. HW : See you soon. Be well.

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Whitney Pennington Rodgers : Hello and welcome to everyone joining us from around the globe. Thank you for being part of day two of our **special** series TED Connects. This week, we ’ re bringing you interviews from some of the world ’ s greatest minds to offer tools for us to **navigate** through and thrive in these really uncertain times. I ’ m Whitney Pennington Rogers, TED ’ s current affairs curator, and I ’ ll be one of your hosts for today ’ s event. Yesterday, we kicked off this series with an interview from acclaimed psychologist Susan David, who offered us some tips on how to really be our best selves in these trying times. And we ’ re going to switch gears a little bit today from thinking about our own personal mental health to the state of our global public health systems. Chris Anderson : Thank you. I guess we have a pretty exciting guest to introduce. On the other side of the country, let ’ s bring in Bill Gates. Bill, they say the better-known people are, the less you have to intro them. It ’ s great to have you here. How are you doing? Bill Gates : I think this is an unprecedented, really disconcerting time for everyone, with things being shut down, not knowing exactly how long it ’ s going to last, worrying about the health of all the people we care about. You know, I ’ m lucky that I get to connect up with video conferencing using Teams a lot, so the Foundation is stepping up and there ’ s a lot of great people trying to help with this crisis. But it ' s scary for everyone. CA : Are you basically stuck at home like many of us watching? BG : Yeah, almost all my meetings are using Teams now, I ' m getting used to that. You know, I ' ve gone days without seeing any coworkers. CA : Let ' s start here, Bill. Five years ago, you stood on the TED stage and you gave this chilling **warning** that the world was in danger, at some point, of a major pandemic. People watching that talk now, their hair stands up on the back of their neck – it is exactly what we ' re living through. What happened, did people listen to that **warning** at all? BG : Basically, no. You know, I was hopeful that with the Zika and Ebola and SARS and MERS, they all reminded us that, particularly in a world where people move around so much, you can get huge devastation. And so the talk was to say, hey, we ' re not ready for the next pandemic, but in fact, there ' s advances in science that if we put resources against them, we can be ready. Sadly, very little was done. There were some things – the Coalition for Epidemic Preparedness Innovation, CEPI, was funded by our foundation, Wellcome Trust and a number of governments, to do some of the **platform** vaccine work, but in the area of diagnostics, antibodies, antivirals, basically doing the disease games that I talked about, where we ' d simulate what needed to be done. We hardly did anything, and so now here we have a respiratory virus that is, sadly, fulfilling some of the more negative predictions I made. CA : Last month, you said that this might be the big one. You wrote that this could be the sort of once-in-a-century pandemic that people had been fearing. Is that how you think of it still? BG : Well, it ' s awful to say this, but we could have a respiratory virus whose case fatality rate was even higher, if this was something like smallpox, you know, that kills 30 percent of people. So this is horrific. But in fact, most people, even who get the COVID disease, are able to survive. So it ' s quite infectious, way more infectious than MERS or SARS were. It ' s not as fatal as they were. And yet, the disruption we ' re seeing, in order to knock it down, is really completely unprecedented. So this is going global, that was – it ' s respiratory, that was the great fear. How many people end up dying – hopefully, if we do the right things, it won ' t be a gigantic number. So, you know, we should end up not having the 1918 **flu** situation. We should be able to do a lot better than that. CA : And that ' s because of **actions** that we would take. I mean, left without the right **actions**, the prospects are pretty deadly. If we knew what we knew in 1919, this thing could take out tens of millions of people around the world. You said – is the key thing here that it ' s got this sort of a strange combination of being certainly more dangerous than **flu** – not as dangerous as something like Ebola or SARS, but more dangerous than **flu** by a factor, but infectious, and also infectious before symptoms have started, is that part of why it ' s been really hard to respond to? BG : Right. Ebola, you ' re actually flat on your back before you ' re very infectious. So you ' re not at church or in a bus or at a store. With most respiratory viruses like the **flu** and COVID, at first you only feel a little bit of a fever and a little bit sick, and so there ' s the possibility you ' re going about your normal activities and infecting other people. And so human-to-human transmissible respiratory viruses that in the early stage aren ' t stopping you from doing things, that ' s kind of a worst case, and that ' s where, you know, I did a **flu** simulation in the 2015 talk and showed how quickly it spread. You know, versus 1918, people move around a lot more now than they used to, and so that works against us. Now the medical system that steps up to treat people is also far, far better. CA : But when was it clear to you that unless we acted, this could be a really deadly pandemic? BG : Well, in January it was discussed that there was human-to-human transmission taking place. And so the alarm bells were ringing that this fits the very scary pattern that it will be very difficult to contain. And on January 23, China did their equivalent of the shutdown. Did it in a fairly extreme form. The very good news is that they were able to reduce the infection rates dramatically because of those **actions**. But it ' s January where everybody should have been on notice – let ' s get our act together with testing, let ' s get going on therapeutics and vaccines, we ' ve got to get organized because we have this novel respiratory virus whose infectiousness and fatality put it in that superscary range. CA : And so, what did happen? Because it ' s such a mystery to me about the"lost month"of preparations in many countries and certainly in the US, where we are. Were you on the phone to people during early February, late January, early February, saying,"Guys, what ' s going on, this is a really big deal, what are we doing?"What was happening behind the scenes during that period? BG : Well, you ' d like to have government money show up for the key activities. We put out 100 million, we created the Therapeutics Accelerator, there ' s the period between when we realized it was transmitting and now, where we should have done more. I think the most important thing to discuss today is that in the area of testing, we ' re still not creating that capacity and applying it to the people most in need. And so we have health workers who are symptomatic, who can ' t get a test and so they don ' t know should they go in or not go in, and yet we have lots of tests being given to people who aren ' t symptomatic. So the testing thing to me, it ' s got to be organized, it ' s got to be prioritized, that is **super**, **super** urgent. The second thing is the **isolation** that, you know, various parts, just focusing on the US, some parts are doing that in a fairly strong way and other parts not yet, and it ' s very hard to do, it ' s tough on people, it ' s disastrous for the economy. But the sooner you do it in a tough way, the sooner you can undo it and go back to normal. CA : So we ' ll come to the **isolation** part in a minute, but just sticking with the testing thing, I ' m just so confused as to why, with more than a month ' s notice – I mean, there are so many smart epidemiologists in the US, for example, you plug numbers about infectiousness and fatality into any simulation and you see that if you don ' t do anything, millions of people will die. And there ' s a month. So what ' s your explanation, what do you think happened here as to why there was almost no – a month later, there was no viable test in the US. Was this just government complexity, too many chefs in the kitchen, what on earth happened here? BG : Well, we certainly didn ' t take advantage of the month of February. The good news is that the actual process, the PCR machines, we have a lot in the United States. And so there ' s models like South Korea, who took advantage of February, built up the testing capacity, and they were able to contact-trace and their infections have gone down, even without the type of shutdown that, because we ' re late, we ' re having to do. One thing that is good news just this week is that people had thought to do this test, that you had to have a nurse or doctor shove a swab way up, all the way to the back of your throat, which hurts a lot, but also, you ' re going to cough and potentially spread the disease to that health care worker. So they have to have protective equipment and change that. We sent data to the FDA this weekend, showing that just an individual, by themselves, swabbing up to the tip of their nose, the accuracy of that test is essentially the same as having a health care worker do it. That helps a lot. We still have to do other things, but that means that you don ' t have to change protective equipment, you just hand the patient that swab, they do it, put it in the test tube, and if the capacity is right, within 24 hours, you should get that result back. CA : So how do you see that playing out? Are there people going to massively scale those tests and how will ordinary citizens be able to get hold of them? Does it still have to be kind of prescribed by a doctor at some point, or at some point, will you be able to order them off Amazon or something? BG : Well, it ' s pretty chaotic today, because the government hasn ' t stepped in to make sure the testing capacity is both increased and it ’ s used for the right cases. There will be a website – and if the federal government doesn ’ t do it, a lot of local governments will have to do it – that you go to, you give your situation, including your symptoms, you ’ re told, based on your work and your symptoms, are you a priority. If so, you ’ re told where there are kiosks you can go to and you ’ ll do the self-swab and just hand it over, or eventually, we ’ ll send the kits to you at home, and then you ’ ll send it back and hear that result. Maybe six months from now, you ’ ll actually have a strip where you perform the test in the home, but for now, they ’ re sending it back for the PCR processing. We can have massive capacity there. And that ’ s how you know. The testing is everything, because that ’ s how you know whether you need to do more shutdown or you ’ re starting to get to the point where you can **relieve** it. CA : Some people are trying to argue now that, almost, the testing should be dialed back, because the cat is out of the bag, testing is bringing people together and risking infection, you know, forget that, let ’ s just focus on treatment and on **isolation** strategies. You disagree with that. Testing is still absolutely essential and needs to be scaled dramatically. BG : The two that go together are testing, at very high volume, and the **isolation** piece. If you ’ re a medical worker, you want to stay and do your job. If you ’ re making sure the electricity, water, food is still available, you want to do your job, and so testing is what indicates to you, do you need to go into **isolation** and make sure you ’ re not the source of spread. And so, you know, testing is the key thing. South Korea did that in this massive way that everybody should learn from. And so that is paired with the **isolation** piece. Our goal here is to get to the point where a very small percentage of the population is infected. You know, China, only 0. 01 percent of the population was infected. If you let it, if you don ’ t do these things, you ’ re going to get the majority of people infected and that huge overload of the medical system. CA : Whitney has some questions from our online audience. Whitney. WPR : Some of the questions that we ’ re seeing are about how our tech giants and leaders can play a role in isolating this and containing this virus. BG : The tech companies are very involved in making sure that some work can go on. People can stay in touch, you know, they can help with some of the disease modeling, they can help with the visibility of the numbers. It ’ s actually very impressive, you get up there and you can see those numbers. Actually, they ’ re sad numbers, but everybody ’ s able to monitor this thing. Back in 1918, they didn ’ t have this type of visibility, and ability to share best **practices**. But for a lot of people, the **isolation** is the key thing. CA : Bill, one of the riddles about this **isolation** strategy is how long it has to last. A lot of people are concerned that the price of victory by isolating everyone is that you crash the economy, and that we have to be, basically, at home, not doing our regular jobs for three, six months, maybe all year. And so much so that there ’ s now this big debate in the US and other countries about this may just be the wrong strategy, that we can ’ t crash the economy that badly, we should only isolate for another couple of weeks, and then let people back, and if that means a lot of other people get sick and we eventually build up herd immunity, that may be the right way to go. What ’ s your thought on this, what is the **isolation** strategy that eventually leads to us getting back to normal? BG : It ’ s very tough to say to people,”Hey, keep going to restaurants,”you know,”Go buy new houses, ignore that pile of bodies over in the corner, just, you know, we want you to keep spending,”because there ’ s some, maybe a politician who thinks GDP growth is what really counts. It ’ s very hard to tell people, when there ’ s an epidemic spreading that threatens, particularly, their parents or elderly people that they know, that they should go about things knowing that their activity is spreading this disease. I don ’ t know of any rich countries that have chosen to use that approach. It is true, if you did that approach, over a period of several years, enough people would be infected you ’ d have what ’ s called herd immunity. But herd immunity is meaningless until you infect over half the population. And so you can take – You ’ ll overload your medical system, so your case fatality rate, instead of being one percent, will be like three, four percent. And so, the idea, it ’ s very irresponsible for somebody to suggest we can have the best of both worlds. What we need is the extreme shutdown so that in six to ten weeks, if things go well, then you can start opening back up. CA : So just putting the math together from what you just said, Bill, to get to herd immunity, you need more than half the people in the country to basically get the bug. So in the case of the US, for example, that would be 150 million people, thereabouts. You said that the fatality rate in that scenario, you ’ re talking about four to five million people potential fatalities. That is just a horrifying scenario that no one should be contemplating. BG : Even one percent of the population getting sick, they will treat, whoever goes for this”ignore the disease”strategy, they will treat them as a pariah state, so none of their people will go in, and none of your people will go into that. And so briefly, a few countries in Europe that hadn ’ t really looked at this hard, considered,”OK, should we be the ones who kind of go about business as usual?”It is tempting, because if you got there early – South Korea did not have to do the extreme shutdown, because they did such a good job on testing. CA : Testing and containment. BG : That ’ s why it ’ s so maddening to me that government is not allocating the testing to where it ’ s needed, and maybe that will have to happen at the state level, because it ’ s not happening at the federal level. But there is no middle course on this thing. It is sad that the shutdown will be harder for poorer countries than it is for richer countries. CA : So let ’ s come into that in minute. The one exception I ’ ve heard the case made for is Japan, that Japan has not contained it quite in the same way that South Korea did but has allowed people to work. It ’ s tried to make extreme measures for protecting their most elderly population. But they ’ ve tried to find a middle scenario, haven ’ t they? BG : If you act – When you have hundreds of cases, you may be able to contain it by doing great testing and great contact tracing, and restricting foreigners coming in, without as much damage to your economy. The US is past this opportunity to control without shutdown. So the worst case of what was happening in Wuhan in the beginning or in northern Italy over the last few weeks, that we avoid that. But we did not act fast enough to have an ability to avoid the shutdown. CA : But then what I don ’ t understand, in the case of the US, for example, is that even if we ’ re successful in bending the curve and reducing the number of new cases from a period of extreme shutdown, as it were, no immunity has been built up. Let ’ s say that there ’ s still no vaccine. Surely when you lift restrictions and people start going back to work, the whole thing just blows up again. BG : The experience that we ’ re seeing in China and in South Korea is that there are not these people who are asymptomatic that are causing lots of infections. And that ’ s a parameter that, as you build the model, you have to put in. There ’ s an Imperial model that people talk about a lot, which shows that reopening is very hard to do. But the results of that model are not matching what we see in China, and so very likely, there aren ’ t as many of these infecting asymptomatics. And that ’ s why you have to be pragmatic. There ’ s a lot we don ’ t know. For example, seasonality may help us in the Northern Hemisphere, the force of infection will – Respiratory viruses, to some degree, they all are seasonal. We don ’ t know how seasonal this one is, but you know, there ’ s a reasonable chance that the force of infection will be going down. And it ’ s your testing that always is telling you,”Oh, my gosh, do I have to shut down more, or can I start to open up?”So particularly, right as you open up, that testing and contact tracing is saying to you – And you can say I ’ m more on the optimistic side, that it will be possible to do what China ’ s doing, where they are starting to go back to normal. CA : And help me understand what happened there because it seems kind of miraculous to me, because this virus was exploding, yes, in Wuhan, but people moved from there to many other parts of China. How is it possible that the combination of the shutdown in Wuhan and measures elsewhere seem to have got to the point where there are literally no new cases happening. I mean, to me, that implies that literally, the virus is not circulating at all between humans in China. You know, there ’ s a few tourists coming in who they deal with, but I mean, is that literally your interpretation of what happened, that it ’ s no longer circulating in China? BG : Absolutely. Take a spreadsheet and take a number like four – one person infects four people – and say the cycle is every 10 days. Go through eight of those cycles, and you ’ re getting the big number. You know, start with 10, 000 and then, you know, that increase. If you take the number 0. 4 instead, that is, the average case infects 0. 4 people, then look at what happens to that number as you go out. It drops to zero, and so things that are exponential are very, very dramatic. When they ’ re above one, they are growing rapidly. When they ’ re below one, they are shrinking rapidly. And so the **isolation** in China drove that reproductive number to well below zero. And so local infection rates – CA : Below one. BG : Below one, sorry. And that **quarantine**, you know, **quarantine** comes from”40 days,”which is what they thought would help for black plague, that is our primary technique. Thank God we have testing, if we use it properly. We are doing therapeutics, which will help with the death rate, but in terms of keeping the infections below one percent of the population, it really all depends just on the two things : **isolation** and testing. CA : So to quote a question from my Twitter feed this morning for you Bill : If you were president for a month in the US, what would be the top two or three things you would do? BG : Well, the clear message that we have no choice to maintain this **isolation** and that ’ s going to keep going for a period of time, you know, probably in the Chinese case, it was like six weeks, so we have to prepare ourselves for that, and do it very well. And then use the testing and every week, talk about what ’ s going on with that. If you ’ re doing **isolation** well, within about 20 days, you ’ ll see those numbers really change, you know, instead of this, you ’ ll see this, and that is a sign that you ’ re on your way. Now, you have to stay to get more generations that are 0. 4 infections per previous infection. You have to maintain it for a number of weeks there. And you know, so this is not going to be easy. We need a clear message about that. It is really tragic that the economic effects of this are very dramatic. I mean, nothing like this has ever happened to the economy in our lifetimes. But bringing the economy back and doing money, that ’ s more of a reversible thing than bringing people back to life. And so, we ’ re going to take the pain in the economic dimension, huge pain, in order to minimize the pain in the disease and death dimension. CA : Whitney. WPR : We have a lot of other questions coming in. One that we ’ ve been seeing is a question about what tools are available for countries that maybe don ’ t have the luxury of being able to social-distance, don ’ t have great health systems in place, how should they be handling this virus? BG : Yeah, I would say, if the rich countries really do their job well, by the summer, they ’ ll be like China is, or some of the other countries that responded early. But in the developing countries, particularly in the Southern Hemisphere, the seasonality is large. As you say, the ability to isolate, you know, when you go out to get your food every day, you have to earn your wage, when you live in a slum or you ’ re very **nearby** each other, it ’ s very hard to do, as you move down the income ladder, than it is for a country like the United States. And so we should all accelerate the vaccine, which eventually will come, and you know, people are being responsible to say that that ’ s going to take 18 months. And there ’ s a lot of those being pursued. I ’ m talking a lot with Seth Berkley, who you ’ re going to have later this week, who can talk a lot about the vaccine front, because he ’ s definitely at the center of that, being the head of GAVI. We do need to get really cheap testing out to these countries, and we need to get therapeutics so you don ’ t need to put five percent of people on respirators. Because even if they had the equipment, they don ’ t have the personnel, they just don ’ t have the beds, the capacity. And so the only good news is that the rich countries have this and so they will be learning about testing, therapeutics, and funding the vaccines for the entire world, to try and minimize the damage in developing countries. WPR : Great, I ’ ll be back later with more questions. CA : Bill, you mentioned therapeutics there. What is looking promising, is anything looking promising? BG : Yeah, so there ’ s quite a range of things going on. There ’ s a few that get mentioned a lot, remdesivir, hydroxychloroquine, azithromycin, and the data is still a bit confusing, but there ’ s some positive data on those. Remdesivir is a five-day IV infusion, and actually kind of hard to manufacture, so people are looking at how that can be improved. The hydroxychloroquine looks like it works, somewhat, if you get in early. There ’ s a huge list of compounds, including antibodies, antiviral drugs, and so the Gates Foundation and Wellcome Trust, with support from Mastercard and now others, created this therapeutics accelerator to really triage out. You have hundreds of people showing up and saying, try this, try that. So we look at lab assays, animal models, and so we understand which things should be prioritized for these very quick human **trials** that need to be done all over the world. So the coordination on that is very complex, globally. But I think, you know, out of the top 20 or so candidates, probably three or four of them will work out, you know, at different stages of the disease, to reduce the respiratory **distress**. CA : I heard you mentioned that one possibility might be treatments from the serum, the blood serum, of people who had had the disease and recovered. So I guess they ’ re carrying antibodies. Talk a bit about that, how that could work and what it would take to accelerate that. BG : Yeah, this has always been discussed as how could you pull that off. So people who are recovered, it appears, have really effective antibodies in their blood. So you could go, transfuse them and only take out the white cells, the immune cells. And then the question is, OK, how many patients ’ worth of material could you get? You know, if you have that recovered person come in, say, once a week, do you get enough for two people or five people? Then logistically, you have to take that and get it to where that need is. And so it ’ s fairly complicated, you know, compared to a drug that we can make in high volume. You know, the cost of taking it out and putting it back in probably doesn ’ t scale as well. But there is work being done on this. You know, we actually started with Ebola, and fortunately, it got done before it was needed. So that is being pursued and it will work to some degree, but it will be hard to scale the numbers. CA : So it ’ s almost like, when you talk about the need to accelerate testing, the immediate need is for testing for the virus. But is it possible that in a few months ’ time, there ’ s going to be this growing need to test for these antibodies in people, i. e. to see if someone had the disease and recovered, maybe they didn ’ t even know they had it. Because you could picture this growing worldwide force of heroes – let ’ s call them heroes – who have been through this experience and have a lot to offer the world. Maybe they can offer blood donation, serum donation. But also other tasks, like, if you ’ ve got overwhelmed health care systems, presumably, there are kind of community health worker type tasks that people could be trained to do to **relieve** the pressure there, if we knew that they were effectively immune? BG : Yes. Until we came up with the self-swab and showed FDA that that ’ s equivalent, we were thinking that people who might be able to man those kiosks would be the recovered patients. Now we don ' t want to have a lot of recovered people, you know. To be clear, we ' re trying, through the shutdown, in the United States, to not get to one percent of the population infected. We ' re well below that today, but with exponentiation, you could get past that three million. I believe we will be able to avoid that with having this economic pain. Eventually, what we ' ll have to have is certificates of who is a recovered person, who is a vaccinated person, because you don ' t want people moving around the world – where you ' ll have some countries that won ' t have it under control, sadly – you don ' t want to completely block off the ability for those people to go there and come back and move around. CA : Bill, is your foundation helping to accelerate the manufacture of these self tests? What are the prospects for really seeing scale on some of this testing soon, not just in the US, but globally? BG : Yeah, our foundation, we ' d been funding the thing called the **Flu** Study to really understand how respiratory viruses spread. It ' s amazing how little was understood about how important schools are, different age groups, different types of interaction. And that gave us an experience. In fact, that **flu** study actually was the first time **coronavirus** was found in the community, because the government was still saying you only test people who ' d come from China, but we ran into people who had coronavirus, who hadn ' t been travelers. So, that was like an early **warning** sign, even though the regulation said you weren ' t supposed to even look at that. So yeah, the Foundation is working with all the private sector people, the diagnostics people on this testing piece. Now that we can do the self-swab, those swabs are very easy to manufacture. The one where you had to jam it into the throat, deep turbinate, that was getting into short supply. So the swab should not be limiting, neither should the various chemicals that help run the PCR machines. So we should be able to get to a South Korea-type prioritized testing thing within a few weeks. CA : How important is it that the world ' s nations collaborate right now? I mean, it seems like, you know, here ' s this common enemy facing humanity, it does not know that it just crossed a border, it does not know what race people are, what religion they are – it just knows,"Here ' s a human, I ' ve got a manufacturing machine here that can make me famous."And it goes to work. It ' s so terrifying to me to see signs of countries starting to blame each other or the xenophobia, it just seems so toxic. What ' s your take on this, Bill? Do you see signs of cooperation happening, or are you also worried about the sort of,"US versus China"kind of thing that seems to be going on if we ' re not careful? BG : Well, I see both. I see that countries that are recovered can help other countries. And that ' s fantastic. If by the summer, we ' ve knocked this thing down, then great, we can help other countries. There are vaccine projects all over the world, and those should be evaluated on a very neutral basis, to which one is the best to help humanity. And make sure the manufacturing capacity isn ' t just for rich countries, that it ' s scaled up, very low cost stuff for the entire world, and that ' s the spirit of GAVI, is getting vaccines out to every person. So in the science side, and data-sharing side, you see this great cooperation going on. Unfortunately, whenever you have disease, this sense of other and foreign and"Oh, stay away from me,"you know, that sort of pulling inward is reinforced. And we have to avoid that. You know, ironically, we have to isolate physically, while in terms of looking at community groups that are pooling resources to help make sure food gets to everyone and help assure medical care, you know, if older people need to be moved out of common facilities, you help out with that, and that people aren ' t suffering too much from the psychology of **isolation**. So our generosity has to go up towards others at the same time we ' re less actually physically interacting with other people. CA : I mean, thinking about the situation in many developing countries, I ' m curious how you think of this. You mentioned, first of all, that seasonality may help, i. e. high temperatures. Is it possible that that is so far protecting, to some extent, places like India or sub-Saharan Africa and so forth? BG : India ' s Northern Hemisphere. So Southern Hemisphere is lots of Africa, South America, Australia, New Zealand, Indonesia. And it is true, either the force of the infection is lower there or we ' re just not seeing it with testing. You know, a few months from now, we ' ll understand the seasonality question, which would be good news for the Northern Hemisphere, and somewhat bad news for the Southern Hemisphere. Now more people live in the Northern Hemisphere, including India, Pakistan, and that would buy us some time, and time is a big deal, because all these tools get so much better if you had to go into a second season with it. But yeah, sadly, we could see, in the next few months, as the Southern Hemisphere is moving into its fall and then winter, we could see a big increase there, and that is going to be very difficult. Now they don ' t have as many older people, but they have lots of people who are HIV positive, or have malnutrition or various lung challenges because of indoor smoke, and so the wild card is how well can the developing countries deal with this. CA : If you ' re in a country where the majority of your population is making less than two or three dollars a day, can you even afford a strategy that looks like, basically, shutting down the economy? BG : I ' m very worried that there will be a massive number of deaths in those poorer countries, because the health systems just aren ' t – you know, the number of respirators, hospitals, and of course, when you overload that system, your deaths are not just COVID deaths, but everyone else who ' s trying to access a system that will be somewhat in chaos, including with health workers who are getting sick. CA : OK, we ' re getting near to running out of time with this. Whitney, maybe a last question or two from online. WPR : Sure, we have two from online, we ' re seeing thousands of questions around these same lines. One, there ' s lots of people who are really interested to hear about the kind of work that you ' re doing with your foundation as far as distributing tests, but also producing safety gear, masks and that sort of thing, to help with this effort for health workers. BG : So the Gates Foundation, you know, we, very early on, gave out 100 million to help out with all the pieces : the testing piece, the therapeutics and the vaccines. We are not experts in making masks and ventilators and gowns, and it ' s great that other people, including some 3D printing, and open-source things, that is great. Our focus, you know, like this self-swab thing, nobody had done that before, people thought it wouldn ' t work, we were quite sure it would work. And so that, for the globe, is a huge thing. We work a lot with both governments and private sector, so in some ways, we ' re kind of a bridge. And we ' ve been talking to the heads of the pharmaceutical companies, the testing companies and, specifically, with the ones doing vaccines, including some of which are these new type of vaccines, RNA vaccines, that we ' ve been backing for quite some time, and CEPI has been backing. And so our expertise is in those medical tools and really getting the best of the private sector engaged there. It ' s been a little slow. We can write checks right away, whereas the government processes, even in this situation – you know, there ' s still this notion of bidding, and not really knowing who has the unique capabilities of doing things, and so, an organization that ' s working on this all the time, lots of new vaccines, can step in and be helpful. And it ' s really amazing. When we talk to private-sector partners, their interest in helping out has been absolutely fantastic. And so that ' s where we have a unique role. WPR : And the other question that we ' re seeing a ton of – before we wrap up here – is just people are really interested in your insight, Bill, on whether you think we are heading in the right direction, do you feel like our economy is heading in the right place, that humanity is heading in the right place, are we in a better position now than you thought we were in five years ago? BG : Well, five years ago, I said that pandemic is this unaddressed, very, very scary thing. And that if we did the right things, we could be more prepared. Science is on our side. The fact we can be ready for the next epidemic, it ’ s very clear how to do that. And yes, it will take tens of billions, but not hundreds or trillions of dollars. So it will be tiny compared to the economic cost. I remember when I did that presentation 2015, I put up,”Hey, a big **flu** epidemic could cost four trillion,”and I thought, wow, that ’ s a big number, do I really think it ’ s that big? And I went and looked up numbers and thought, yeah, well, that ’ s big. This epidemic will cost that much to the economy. So in the short run, we are going to have more pain and more difficulty and people are going to have to step up to help each other. I ’ m still very much an optimist, you know, whether it ’ s climate change, countries working together, biology taking the diseases, malaria, TB, you know, even advances for what are more rich-world diseases, like cancer. The amount of innovation, the way we can connect up and work together – yes, I ’ m superpositive about that. You know, I love my work because I see progress on all these diseases all the time. Now we have to turn and focus on this, you know. Sadly, it may interrupt and the polio situation might get worse a little bit because of the distraction here. We ’ re using a lot of the great capacity that was built up for those polio activities to try and help the developing countries respond to this very well. And that is appropriate, but the message from me, although it ’ s very sober when we ’ re dealing with this epidemic, you know, I ’ m very positive that this should draw us together. We will get out of this, and then, we will get ready for the next epidemic. CA : That ’ s exactly what I was going to ask you, Bill, which is, where is your head, do you think we will get through this? Will the leaders that matter listen to the scientists, will they? Will we make it through? Do you believe that within a few months ’ time, we ’ re already going to be looking back and saying,”Phew, we dodged a pretty bad one there.”BG : We can ’ t say for sure that even the rich countries will be out of this in six to ten weeks. I think that ’ s likely, but as we get the testing data, we ’ ll get more of a sense of that and people will continuously be able to see that. But you know, the rich countries will get out of this. The developing countries will bear a significant price, but even they, we will get a vaccine and GAVI will get that out to everyone. So you know, two to three years from now, this thing, even on a global basis, will essentially be over with a gigantic price tag. But now we ’ re going to know, OK, next time we see a pathogen, we can make billions of tests within two or three weeks. We can figure out which antiviral drugs work within two or three weeks and get those scaled up. And we can make a vaccine, if we ’ re really ready, probably in six months, using these new **platforms**, probably the RNA vaccine. So specifically, there are innovations that are there that will get financed, you know, I hope, quite generously, coming out of this thing. And so, three years from now, we ’ ll look back and say, you know, that was awful, there ’ s a lot of heroes, but we ’ ve learned a lesson and the world as a whole, with its great science and desire to help each other, was able to try and minimize what happened there and avoid it happening again. CA : That ’ s certainly the optimistic scenario that I ’ m craving for, myself. That the world kind of realizes, one, that there are certain things that you just have to unite on. Two, that science really matters and it ’ s a miracle that science can understand this bug, you know, make a vaccine, sequence it, make therapeutics, understand how to model it – it ’ s kind of miraculous to me. So will we learn, now, to pay attention to scientists, because if we do, I ’ m sure that you feel this as well, there ’ s an amazing analogue with climate, it ’ s just a different timescale. That the scientists are out there, saying,”There ’ s this huge enemy coming, if we do nothing, it ’ s going to take millions of lives, it ’ s going to wreck our planet. For God ’ s sake, act, politicians! Do something.”And the politicians are going,”Meh, no. We need a little more GDP, we need to win an election.”And they ’ re not acting. Do you see a scenario where this shocks politicians to actually change their thinking and their prioritization of science overall, or is that asking too much? BG : Yeah, it ’ s interesting how much of this distraction will delay the urgent innovation agenda that exists over in climate. You know, I have freed up a lot of time to work on climate. I have to say, you know, for the last few months, that ’ s now shifted, and until we get out of this crisis, COVID will dominate, and so some of the climate stuff, although it will still go on, it won ’ t get that same focus. As we get past this, yes, that idea of innovation and science and the world working together, that is totally common between these two problems. And so I don ’ t think this has to be a huge setback for climate. CA : Last question. There are thousands of people watching, many of them living alone, some quite scared, there may even be people there who have this virus and are suffering symptoms or recovering. By the way, if that ’ s you, we ’ d love to hear from you, we really would. Maybe have a conversation with some of you, in a future one of these, just understanding the experience. But Bill, what can people do as individuals from their own homes, right now, to try and help? BG : Well, there ’ s a lot of creativity, you know – can you mentor kids who are being forced into an online format where the school systems really weren ’ t ready for that? Can you organize some giving activity that gets the food banks to step up where there ’ s problems there? These are such unprecedented times, and it really should draw out that sense of creativity, while complying with the **isolation** mandates. CA : Bill, I really want to thank you for spending this time with us and for the **financial** investment, the time investment. You ’ ve really invested your life into trying to solve these big problems. And this is as big as they get. I have a hunch that your voice is really going to be needed in the next few weeks. Thank you so much for your time today. This was really wonderful, hearing from you. Thank you. BG : Thanks, Chris. CA : OK, thanks, everyone, thanks for being part of the TED community. Look after yourselves, be smart about this. You know, get ahead of it. If you ’ re in a part of the world where this thing hasn ’ t really hit, listen to Bill Gates. Get ahead of it. Keep, you know, if you possibly can, socially distanced. No, not – physically distanced and socially connect. That ’ s what the internet is for. These days are what the internet was built for. We can spread love, we can spread ideas, we can spread relationship, we can spread thought, without spreading a dangerous bug. So get ahead of it, and let ’ s figure this out together. It ’ s been wonderful spending time with you. From Whitney and from me and from the whole TED team, thank you, and over and out.

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Chris Anderson : Welcome to TED Connects. This is a new series of live conversations, trying to make sense of this weird moment that we ’ re in : **coronavirus**. Everyone ’ s suddenly changing how they live their lives, it ’ s so jolting, it ’ s so startling. We ’ re all trying to make sense of it, and it ain ’ t easy. That much we know. We ’ re trying to make sense of this together in the only way that we know how, which is by having wise humans coming on, talking to each other, listening to each other trying to learn from each other. We are apart, but we can use this moment to build community together, and that ’ s what we ’ re trying to do. So this is being produced by a virtual TED team scattered around New York, currently one of the epicenters of this pandemic. So it ’ s definitely a scary time for people here. I ’ d like to welcome to join me my cohost here, Whitney Pennington Rodgers. She ’ s our current affairs curator. Whitney Pennington Rodgers : We ’ re going to be looking a little bit at China ’ s response today. When news surfaced about a strange viral outbreak in Wuhan, China at the very last days of 2019, I think a lot of people were confused about what was going on there, and in the months that followed, we learned more about the disease that ’ s now known as COVID-19, we watched the situation in China quickly worsen and in the most recent weeks dramatically improve. And I think as all of us around the world grapple with how we can contain and control the spread of COVID-19, there are a lot of lessons we can learn from what China experienced and how they responded. So we ’ re really thrilled to be joined today by the CEO of the”South China Morning Post,”Gary Liu, who ’ s here to share his perspective and insights. So, welcome Gary. Gary Liu : Thanks for having me. WPR : Hey there, Gary, thanks for being with us. And I think before we dive into things, I ’ d love to hear about just how things have been for you personally, your loved ones, those close to you, how have you been experiencing this? GL : It ’ s complicated. So we ’ re here in Hong Kong, I ’ m working from home, like much of Hong Kong. I ’ m actually self-quarantined in our apartment here in Hong Kong, because there was a confirmed case in our **workplace**. So over the course of the last week plus, and likely for at least another week plus, the entire organization has been distributed and working from home. You know, when Hong Kong got its first confirmed case, I was actually back in the United States with my wife, we were taking a small break in the Rockies, and we came back to Hong Kong pretty soon after that to make sure we got back into Hong Kong before the airports shut. And at that point, it was all of our family in the United States and friends texting us and worrying about how things were in Hong Kong as the situation in China started escalating, and people were sending us, or trying to send us, supplies. Masks and sanitizer and stuff like that. And now it ’ s the opposite. New York City is our home, so we certainly empathize with what you guys are dealing with right now and going through in the city. And we are seeing our friends and our family back home in New York and in California and checking in on them, trying to send equipment and materials back to them, so the script has flipped actually pretty fast over just the last couple of weeks. WPR : And you know, I think that ’ s actually a really interesting place to start and probably a question that a lot of people who aren ’ t in China have, you know, I think from the outside looking in, it seems as if what ’ s happened in China is kind of miraculous. That to go from, you know, you have a country with more than a billion people there, to go from as many as 80, 000 cases to nearly zero new cases now, you know, what can you tell us about how this happened, to help us understand the current situation and just really how China ended up there? GL : Yeah, a lot has happened. China has been dealing with this for several months now. Several-month head start, that ’ s not a good thing, but they have gone through several different phases. I think, Whitney, before I jump into it, there are a couple of caveats that are really important to make. The first one is that we ’ re still parsing what happened in China. The information system, as everyone knows, is still relatively closed. And so a lot of the information that we ’ re using to piece together what happened in China is still not fully complete. And so with every passing day, every passing week, there ’ s more information that allows us to retroactively make sure that we get the picture of what happened early on in those early days at the end of 2019, get that picture right. And there ’ s still a lot that ’ s happening today, even though I think the information sharing is much more open than it was early on, there ’ s still a lot of stuff that we need to parse. And the second important caveat here is that I think learning sometimes suggests that everything China did was right and good, and hopefully, other countries can take it and apply it, but that ’ s not 100 percent the case. China, of course, did a lot that was right, and if we walked through the time line, I think it would be pretty apparent the decisions that they made kept the **coronavirus** from really exploding across the entire country and really limited it to one province and mostly one city. But there were also many, many missteps, and those are things that I think the world can also learn from, most importantly, China should learn from, because most of these – I think those of us who are professional observers would call missteps, are because they are systemic issues with the country, because of governance, lack of free information flow, stuff like that. Those are the initial caveats, but I think the timing of how China progressed from first case to now has been fascinating. WPR : Yeah, and I mean, so we know now that in Hubei province they ’ ve officially lifted the two-month lockdown. And are you getting the sense, do you feel like this is the right decision to make at this moment? GL : I don ’ t think I ’ m the right person to say whether or not it ’ s the right decision. But certainly, this has been a progression of decisions, and I think they ’ ve been sitting on this decision for quite some time. Wuhan itself, which was where the pandemic started, it was the first epicenter and the major epicenter. Wuhan is opening up on April 8, that ’ s right now the schedule. And this is really, what we ’ re in now is the third of three phases from the first discovery of the virus in Wuhan. Now, April 8 will be about 11 weeks after Wuhan the city got completely shut down, and the Hubei province got shut down. And so for those who are in a shelter, at home kind of situations right now in the United States and wondering how long this is going to take, in Wuhan, they ’ ve been locked down for 11 weeks and only now has the Chinese government decided they ’ re ready to start letting people move freely around. WPR : And to your point earlier about some of the possible missteps in terms of reporting, I mean, there are still reports now that we might not be getting an accurate number of cases that we ’ re seeing in Wuhan or beyond, we ’ re hearing some people say there are no new cases, other people saying that there actually are cases. So do you feel like there is accurate spread of information about the current state of the virus in China right now? GL : Generally, yes, with the caveat that it is based on the Chinese government ’ s definition. And this is one of the problems right now that even the World Health Organization is **struggling** with, is that the definition of what is a confirmed case, what is an infection, is different from country to country. As an example, in China, the folks that have tested positive but are asymptomatic, we understand now that they are not included, since February 7, they have not been included in the official numbers. Or at the very least, on February 7, they changed that definition, and they ’ re not included in those official numbers. And that could be another 50 percent on top of the numbers that we ’ re seeing today. So what we ’ ve found, our **reporters** have gotten their hands on some classified government documents and government data that suggests that a third of total actual positive tests are asymptomatic, and therefore not included in official numbers. Now, I don ’ t think that this is an example of the Chinese government trying to hide information. This is a definitions issue, which countries have been debating and people are doing it in different ways. But like I said, there really have been three very distinct phases. We are in the third phase that I would call recovery and rehabilitation, rehabilitation being the rehabilitation of China ’ s image. But the first part was discovery and a lot of denial. And then there was this two-and-a-half-month period of response and containment. And that I think, the response and containment part is the most interesting to the rest of the world. WPR : And so maybe we can break some of that down, you know, thinking about China ’ s response. What were some of the specific things that you think China did right, both as a nation, individuals in the country, what were some of the things that you saw that worked really well? GL : OK, so let me walk through the time line, I want to try and get these dates right, because the dates do matter, I think again, for context, how many weeks it took from one step to another. Let me actually back up into that initial first phase, that discovery and denial phase. The first time we heard about the **coronavirus**, this mysterious respiratory disease that looks somewhat like SARS, was on December 30. That was the day that there was a doctor, whose name is known all over the world for the unfortunate reason he ended up eventually dying, named Li Wenliang. And Li Wenliang, Dr. Li, posted to a private WeChat group on December 30. These were some of his old classmates from med school. And he said,"Hey, I ' m in Wuhan, I ' m at the hospital, there is a SARS-like illness,"SARS being the epidemic from 2002 to 2003,"There ' s a SARS-like illness that is spreading through these hospitals in Wuhan."A private message. Somebody forwarded it, and it went viral across the Chinese internet. The very next – so that was the first time we actually heard about something that was going on in Wuhan. The very next day, December 31, was the first time that any Chinese officials – and on that day, it happened to be the actual provincial and the city officials – acknowledged that there were 27 people, at that moment in time, who had been diagnosed with this mysterious pneumonia, and they reported the cases to the World Health Organization. That was also the day that Dr. Li was reprimanded, officially reprimanded. So that was really the discovery, the end of the discovery and denial phase, because what we know now is that back to mid-December, several weeks before Dr. Li wrote his blog post, the authorities had already been notified that a SARS-like pneumonia was showing up in Wuhan hospitals. And **action** had already started down the chain of authority. They have now backdated, at least publicly backdated, the first case to December 1. But actually, in their confidential and classified government documents that again, our journalists have seen, and we ' ve published a story – Officially, in classified documents, they ' ve backdated the first COVID-19 case all the way back to November 17, as the earliest example that they can find based on symptoms and based on retroactive diagnosis for a COVID-19 case. So in effect, there were several weeks before the acknowledgment to the World Health Organization that that was going on, and the first case with symptoms was actually identified about a month and a half before that notice to the World Health Organization. Then the second phase, which really started, let ' s say, December 31, when the acknowledgment happened, was response and then massive containment. Now this phase, to be clear, still had some denial and a good amount of censorship happening within the country. So on January 1, the World Health Organization started working with China on trying to identify the virus and trying to figure out course of **action**. It wasn ' t until several weeks later that Beijing, the central government, for the first time broke its silence, and that was on January 18. And actually, they broke the silence to deny that this was SARS, and in fact to"defy rumors"that were spreading around the Chinese internet. But there was a major date that happened two days afterwards, which was January 20. Because for the first time, a member of the party, a senior government official who is now one of the central gentlemen that is actually leading the task force against COVID-19, his name is Zhong Nanshan, he ' s an epidemiologist, he was one of the central figures during SARS 17 years ago. On January 20, he visited Wuhan. And he admitted, for the first time, that human-to-human transmission was possible. Now this was important, because prior to that, officials who had spoken up had said that human-to-human transmission was not likely, was not possible. And previous to that, all of the cases, the majority of the cases were tied to this seafood and wildlife marketplace that was in the city of

Wuhan. But now, on January 20, human-to-human transmission, it ’ s possible, it ’ s happening, and so of course, the course of **action**, not only in China, but the course of **action** all over the world, started to change. And three days after that, Wuhan was locked down. It was completely, I mean, it shocked the world that they could lock down that many people so quickly. Of course, now India yesterday announced that 1. 3 billion people are being locked down. So we have another frame of reference now. And then the end of this middle second phase I think really came in March, around March 10. Actually, on March 10 I should say, because Chinese president, Xi Jinping, visited Wuhan. And these things, in Chinese politics, because everything is so well-choreographed, matters a whole lot. The fact that Xi Jinping visited Wuhan signaled that the Chinese government believed the worst was over. The reality was that probably about 20 days before that, the curve had already been flattened. So 20 days before that, probably around February 20, the infection rate was around 75, 000, 76, 000, and it ’ s effectively stayed within a couple of thousand since then. So on March 10, Xi Jinping ’ s visit to Wuhan kind of signaled the worst is over, and then they moved into the recovery and rehabilitation phase. WPR : OK. I mean, if I ’ m hearing correctly – thank you for sharing all of that, it sounds like, although there was a slow period of getting the information out initially, eventually there was quick reaction from the Chinese government to respond to this, lock folks down. And it seems like that had a real impact on flattening the curve in China, in Wuhan. GL : A real impact. WPR : Yeah and I – GL : Absolutely. WPR : Yes, please go ahead, Gary. GL : The date of January 23 was not by coincidence. Because the Chinese New Year holiday started on January 24, the very next day. And the thing is, with the Chinese New Year holiday, is that it is, every single year, it ’ s the largest human migration that happens on Earth. About 400 million people travel during about a forty-day period that would have started on January 24. And that ’ s three billion trips, it ’ s just people traveling all over the country, 400 million people traveling. Now, Wuhan is one of the most important cities in China, although before this, I don ’ t think a lot of people around the world knew the city of Wuhan, but it ’ s extremely important. It is considered the most important city in the center of China for many different reasons, but one of the key reasons is that it is one of the key transportation hubs of the country. So all of the major train lines, the high-speed train lines, the normal train lines, the trade lines, they all kind of converge on Wuhan. So you can imagine if 400 million people start moving around for Chinese New Year on January 24, a huge number of them were going to go through Wuhan. And of course, Wuhan itself is an 11-million-person city. The surrounding cities all added together, Hubei province has about 60 million people, and they were also largely going to travel. And so if January 23 they had not shut it down, and people had started traveling, the likelihood would have been that this would have been really, really hard, possibly, likely impossible to contain. And even though they shut down before the Chinese New Year holiday started, we now also know that at least five million people actually left the surrounding areas and traveled. Which is one of the reasons why it did spread a little bit across the country, and then eventually spread to other parts of the world. WPR : And I ’ d like to come back to that as well, just thinking about the five million people who left and sort of where they landed today and how that affected things, but before we do that, I ’ m interested to talk with you a little bit more about – you mentioned this November date as one of the earliest cases you discovered that was reported was in November, and that ’ s something actually that I hadn ’ t heard before, and I imagine that might be news to a lot of people hearing this, and so I ’ m curious, when you think about the missteps from China ’ s perspective, in terms of what China did, you know, there is, as you mentioned, suppression of information is one thing, one major criticism of how China handled this. And hearing that maybe there was knowledge of something as early as November, if that might have played a role in how we were able to control and contain this a lot sooner. GL : I do want to clarify, from what we understand, officials were not notified about this until mid-December. It wasn ' t – So it was really a couple of weeks between officials realizing that there was a SARS-like pneumonia going around to when the first case was reported to the World Health Organization. It wasn ' t all the way back to November 17. That was retroactively backdated, but that has not been made public by the government. We published it because we ' ve seen the data that actually backdates the first case. From a misstep point of view, again, it ' s a couple of weeks compared to what happened in SARS, which was a long time of locking down on information. This was much shorter, the period of time that the government wasn ' t in complete shutdown mode. But then, after that, of course, there was still continued censorship on the internet, especially within the Great Firewall of China, for communications between Chinese citizens. And you know, surprisingly to some, I think for a lot of China watchers not so surprisingly, is that the government has – the central government – over the course of the last several weeks, actually, I should say probably the last two months, has started to change their tone and to some degree admit that there needed to be better free flow of information. They ' ve changed the official narrative of a couple of different things, including this initial whistleblower, Dr. Li, who unfortunately ended up passing away from the virus, they actually now refer to him as a national hero, they have officially removed the reprimand, the Wuhan police have apologized to Dr. Li ' s family, and they have actually been – a couple of policemen – have been punished in Wuhan for the way that they handled the situation. So there has definitely been an internal shift and there is a lot more sharing of data and information. I can tell you, from Hong Kong ' s point of view, without the open sharing of information between the authorities, between Hong Kong and mainland China, I think Hong Kong ' s response would have been much more different and I think Hong Kong would have suffered because of that. So that much more open sharing of information has benefited this city for sure. WPR : And we have Chris here who has a question, I think, from the audience. CA : Hey, Gary. The online audience, loving what you ' re saying. It ' s so interesting, you ' re giving us amazing new insights here. Just in the current situation where much of – you know, there have been these very few reports of new cases. How much does it feel like life is getting back to normal? Do people really believe that this problem has been successfully tackled elsewhere? GL : I think the sentiment in mainland China is that yes, in China, the problem has been tackled. And people are looking forward to going back to normal life. A lot of the other major cities, Shanghai, Beijing, are starting to get back to work. Many of the factories have now been reopened. The last stat that I saw was that 90 percent of the businesses that had been shut down are now reopened in China. So generally speaking, life is getting back to normal. Wuhan and Hubei are really the last places that are still shut down, with Wuhan being the city that is shut down until April 8. Hong Kong is a little bit different. Hong Kong has actually gone back into a second wave of social **isolation** and **distancing**. A bunch of different companies, us included, as well as the Hong Kong government and the civil **service** has now gone back to work from home. And it ' s because we are starting to see a second wave, but for us, honestly, is the first time that we ' ve had a spike of infections, and it ' s because of imported cases. It ' s because a lot of Hong Kong residents who left Hong Kong prior to, well, actually when the virus first came into the city, are now returning, because oddly enough, the places they escaped to are now more dangerous than Hong Kong. And as they ’ re coming back, a lot of them are actually bringing the virus back with them. And so we ’ re starting to see a spike. Before this week, the highest infection day that Hong Kong had during the first two months of this was 10 infections in one day. Now the highest that we ’ ve seen in the last week was 48. So this is really the first spike that we ’ re seeing, and so Hong Kong is returning back to a state of alertness, to a state of caution, and more and more people are holed up at home. CA : Is it possible, in mainland China, that because of this redefinition that you spoke about, where if someone tests positive, but they ’ re not showing symptoms, that is not reported as a case. That seems significant to me. Is that part of the explanation for why new reports have gone nearly to zero? GL : I don ’ t know if that ’ s the answer to it, but I do actually think that even with – and remember, these are folks that are tested, so the data that we have is that these folks have been tested, the tests have come back positive, but have not been added to the official number of infections, because they ’ re asymptomatic. But they have still gone through the process that is part of China ’ s containment strategy, which has worked extraordinarily well. Which is, first of all, lots and lots of people have been tested. And then once – if there is a positive test return, regardless of whether or not they ’ re symptomatic or asymptomatic, regardless of whether or not they ’ re added to official numbers, what happens next is that they are **quarantined**, they ’ re isolated, and contact tracing happens. Contact tracing is a key, key, key **action**. And so they go and figure out where this person has been moving, where they ’ ve been, who they ’ ve been in contact with, and all those folks that they ’ ve been in close contact with, they get tested. And if they come back with a positive test, then they ’ re also isolated and they go through the process again. So China has not been testing people, finding that they ’ re asymptomatic and then just releasing them and letting them go home. That ’ s not the case. WPR : I think to that point too, Gary, what you mentioned about this trace-testing and being able to figure out who people have been in contact with to figure out who may also have been infected, you know, when you look at what ’ s happening in other parts of the world, you hear in the United States, where Chris and I are based right now, you ’ re hearing that people who are experiencing symptoms, have symptoms cannot get tested. You know, how does China ’ s ability to test so many people affect the way that they can respond to this and control this virus? GL : It really matters. Without the significant testing and without the contact tracing that comes afterwards, I don ’ t think there ’ s a way that China could have contained it the way that they did. The same thing here in Hong Kong. If we didn ’ t have both of those, as minimum requirements in a health system, Hong Kong could not have contained it. And this is actually the reason why South Korea is the only other country besides China that has managed to flatten the curve, is because they aggressively tested. I think by far the highest per capita testing anywhere in the world, as far as we know right now. And they aggressively did contact tracing. And because of that, even though South Korea had this huge spike, and we thought that it was going to get out of hand, they were able to suppress it, control it, and now they ’ re in a much, much better place. WPR : One thing you mentioned earlier that I ’ d love to talk about, too, is SARS and the impact of going through that in 2002 and 2003 for China, other countries in Asia, Hong Kong. You know, what effect did that have on everyone ’ s preparedness in that part of the world for the COVID-19 outbreak? GL : It was significant. I think the institutional and social memory of SARS matters a heck of a lot, when you look at China, Hong Kong, South Korea, Singapore, Taiwan, Japan, a lot of these countries in Asia have dealt with COVID-19. Let me use Hong Kong as an example, because it ’ s the one that I know the most intimately. But a lot of what I ’ m about to say actually does apply to those other areas of Asia. So for context, SARS, November 2002 to July 2003, very, very similar **coronavirus** to COVID-19, I think there ’ s about an 80 percent similarity to those two viruses. The global infected number was a little over 8, 000, 774 deaths. So by percentage, deadlier than COVID-19 is but far less infectious than COVID-19 is. Now here ’ s why it impacted Hong Kong so much, and why the memory is so deep, and actually it tells you a lot about Hong Kong ’ s reaction to COVID-19. Of the 8, 000 infected, 22 percent were here in the city of 7. 5 million, and 40 percent, actually 39 percent of the deaths, 299 people died in Hong Kong. Thirty-nine percent of the global deaths happened in Hong Kong. And SARS did not start in Hong Kong, it was imported into Hong Kong from southern China. And so SARS, again, deep, deep memory, but it was a massive turning point in the Hong Kong health care system and also the social **practices** of the city. And let me walk through some of that impact, because you can actually still see it, even before COVID-19, you see it every day. The health care system was able to really, very quickly, ramp up in capacity, because of preparation post-SARS. So after SARS, the Hong Kong health care authorities started preparing for greater capacity, especially for infectious diseases. There were new health alert systems, **warnings** and treatment protocols put in place. I can tell you that a lot of folks that were here before SARS will tell you that in Hong Kong hospitals, before SARS, it was actually rare to see even medical professionals wear face masks. And now surgical masks are ubiquitous, not only in hospitals, but across the entire city. Anytime, anywhere, it seems, especially right now. New channels of communication and data and information exchange were opened up with mainland Chinese authorities, and technology was implemented, including now a supercomputer that actually does contact tracing in Hong Kong. You could trace the existence of the supercomputer and this contact-tracing ability back to changes that happened post-SARS. On the social side, there was also a huge change. The first thing I have to talk about is, of course, masks. Now, I know that there is still not consensus everywhere in the world about whether or not masks actually help in this situation. I know that the World Health Organization as well as governments like the US, as well as Singapore, say that only medical personnel as well as people who are actually sick and showing symptoms need to be wearing masks. In Hong Kong, everyone wears masks. And the government, even though they flip-flopped a little bit during this epidemic, the general, the guidance is that everyone should be wearing masks. That started in SARS. Ninety percent of Hong Kongers during SARS wore masks, and that habit actually stayed with Hong Kongers, and so generally speaking, even outside of the pandemic, when people are sick and coughing, you ’ ll see them wear masks out in public. On top of that, there was – it became systemic, or I should say systematic controls for hygiene in social and public spaces. So if you visit Hong Kong, again, before all of this happened, you would have noticed that public spaces are constantly being disinfected. One good example that everyone notices is that when you go into an elevator in public spaces, in buildings, they will either have one of two things, potentially both. They ’ ll either have a sign that tells you how often the elevator buttons are disinfected, or there will be a plastic, piece of sticky plastic, like a plastic sheet over the buttons so it effectively becomes a flat surface. When you eat out, Hong Kong is obviously famous for its dim sum, and one of the most famous things about Hong Kong dim sum are the dim sum carts, which are also very popular in New York ’ s Chinatown, as an example. Those dim sum carts, they pretty much went away after SARS. And so most dim sum restaurants that you ’ ll go to in Hong Kong now, the vast majority of them, you have to order off of a menu, you don ’ t have public carts going around because of hygiene issues.

In most nice Chinese restaurants in Hong Kong now you will get, when you sit down, two pairs of chopsticks per person. And those two pairs of chopsticks are different colored, because one is used to grab food from the center of the table to your plate, and the other one is for you to take the food and put it in your mouth. And honestly, there are hand sanitizers and hand-washing notices literally everywhere, and this is just part of the social behavior after SARS. Safety protocols in offices, everyone knows how to shut down an office and control traffic really well. Most major offices have temperature-check machines at the very least available, and then, of course, social **distancing**. People understand social **distancing** is important, and so the moment there was fear of what was happening across the border, naturally, people started social-distancing activities and self-quarantine became pretty normal. So those are all the social things as well as the health system things that kind of changed, and because of that, Hong Kong was able to react really, really fast, not just the government, not just the health authorities but the people of Hong Kong, and I think that ' s the most important part, is that the entire city, that the community reacted and went into this mode where you wore masks, you washed your hands, you carried hand sanitizer, you stopped going to public places. WPR : I ' m curious then, I think a lot of people who are listening at home and figuring out how can we apply some of those things here, and from where you sit, and when you see what ' s going on in other parts of the world, where maybe people are **struggling** to make some of these changes. You know, what are some of the specific things you think folks can adapt in their own cultures, in their own countries? GL : I think communication is a huge deal. If you talk to local Hong Kongers, they will likely opine that the communication from the Hong Kong government has not been top notch. But thankfully, there have been other authorities and certainly even just person-to-person communication has been pretty strong. A lot of corporates have done an incredible job in Hong Kong in communicating very transparently with their employees and insurance companies have also been making available all sorts of webinars and materials and made it actually quite easy for people to understand how to get tested, where to get tested, who to get tested. And so that communication, I think, has centralized, to some degree, the messaging. In a city like Hong Kong, everyone generally believes the same thing, and what they believe is generally true. Of course, there ' s still misinformation issues, as there are everywhere. But I think, possibly also because of SARS, because over the course of the 17 years, a lot of the misinformation has now been vetted, everyone knows what is true, so there is already, sort of, an internal radar or at least alarm bell for things that seem to be wrong. So I think communication is really important, from government, from corporates anywhere in the world. And I think if there is a recommendation for health systems, I know getting tests is really difficult. One of the things that has made testing in China and Hong Kong certainly so effective, is that there is point-of-care testing, that still really doesn ' t exist, or at least doesn ' t exist in volume in the US. And so they have to save these tests and only a certain number of people can get tested, the triage system then becomes overflowing. Whereas here, generally speaking, everyone can get tested, and then of course, the contact tracing. Everyone knows that if somebody that you ' ve been in contact with tests positive, you ' re going to be called in by the hospital authorities and you ' re going to be tested, and then if you ' re positive, everyone you ' ve been in contact with for the last two weeks will also be called in. And people don ' t really see it as an annoyance, it ' s just what needs to be done. And I think because of that, again, the containment has been effective. WPR : Great. And we have a question from Chris here. CA : Gary, it actually builds on the point you just made, people are puzzled online, how is it that China avoided the explosion of cases in big cities like Beijing, Shanghai, where people were coming there from Wuhan. How on earth did some of those cases not explode? Was it just down to really diligent contact tracing? GL : I think it was a combination of things. First of all, the shutdown of Hubei province certainly helped. And then, the major cities actually went into **isolation** and **quarantine** as well. Remember, it was Chinese New Year, so there was no one working that week. And so everyone just went home. And generally speaking, in most major cities, they locked their doors and they didn ’ t leave. Now, China is very prepared for this, because the technology stack in China, including consumer site **services**, make it really easy to lock your doors and get everything delivered to you. This is infrastructure and this is consumer behavior that is already ingrained, especially in major cities across China. So people just went home. There was also a stigma issue, which is unfortunate for people from Hubei, and especially from Wuhan. But there are plenty of stories in the other major Chinese cities where anyone coming from Hubei or with any connection to Wuhan were ostracized during those early days, especially after the Wuhan lockdown. And so folks that might have been, in fact, carrying the virus, because they were from the epicenter, they were either self-quarantined, or they were forcibly **quarantined**, because no one was going to spend time with them anyway. So I think for a lot of those reasons, some of them social, some of them systemic, they made it so that there was much less person-to-person contact, especially after the authorities admitted that human-to-human transmission was possible. CA : Hospitals here in New York, there are **warnings** that they ’ re about to get overwhelmed. What can we learn from what happened in Wuhan, some of the scenes from there were horrifying, but there were amazing stories as well. What should we learn from what happened there? GL : Well, it started off horrifying. So in the early days, post-lockdown, all the stories coming out of Wuhan, we had journalists that were there right before the lockdown, they got out about three hours before the lockdown happened, and we had people what ended up having to be **quarantined**, because they were stuck in Hubei. As well as a lot of citizen journalists that were documenting what was going on, and those images, like you said, Chris, were horrifying. There were videos showing people literally laying on the ground. Some were just so sick they couldn ’ t move, others had already died and they were just covered with plastic sheets. There were nurses and doctors that were just crying in front of the camera, begging for help. And so, I think it ’ s important to understand that China ’ s health care system did not just immediately become effective. And certainly not in Wuhan. There was not that much information, people didn ’ t know what they were dealing with. Certainly, the authorities were trying to help, I think at that moment, but again, the information flow was not that free. And during lockdown, people were screaming off of their balconies, because they couldn ’ t get food, they couldn ’ t even go to the hospital, because the public transportation systems got locked down. Remember, this is not, Wuhan is not a city like New York where most of New York is walkable. For people who don ’ t have cars, and many, many of the Wuhan residents don ’ t have cars, if the buses are locked down, then they might have to walk three, four hours, to get to a hospital. Maybe not that long, but they have to walk a long way to get to a hospital. And so a lot of people were just stuck at home, and they were unable to initially get any diagnosis or any health care. And so it was a disaster. But then the capacity actually ramped up. The triage system became extremely effective. I think most people have heard now that there were two massive hospitals with thousands of beds of capacity that were built within 10 days. And this is true, they came out of nowhere, they were literally just parking lots or flat ground, and two major hospital units were built up. To be clear, also, those were the triage units for those who have very mild symptoms. But that ’ s really important. Being able to get people with mild symptoms out of the major hospital systems, so that they ’ re not taking up the resources of nurses and doctors, they are not taking up the diagnostic equipment for the second confirmation tests, and also, especially, they are not taking up **isolation** wards and ventilators. And so the moment people started being moved out, the mild symptoms, the ones that were going to survive and they just really needed to be separate from family and have some antiviral medication, once they were moved out into these new hospitals, the main hospitals in Wuhan and across Hubei could deal with the primary patients, especially those that are critical, of the overall tested population and do their best and try and save them and make sure that they ’ re not highly infectious. At the same time, I think that the health authorities in China, especially the nurses and the doctors, did a very good job of also protecting themselves. So there have been far fewer, by percentage, infections and deaths, of medical staff than there were during SARS. CA : I mean, to respond that effectively took a kind of top-down drive. Plus a willingness of a lot of people to risk their own well-being in a way for their perception of what they had to do for the public good. You are well aware of the cultural differences between China, Hong Kong and the West. Do you – how do you rate the chances of, say, the US responding effectively should things really explode here, as they seem like they may be about to? GL : In the health care system side, I have every confidence that the US health care system is going to be able to respond well. I have many, many friends who are medical professionals in the United States, and they are raising their hands and volunteering and going to hospitals to see where they can help. So I have full trust in the system, and the people that man those systems. Our health care capacity in the United States is also significantly greater, doctors per capita, than in China. And because of the fact that also, our health care system is not just relying on hospitals, but there are primary care physicians scattered all over the country, as long as the testing capacity and testing kits are available across the country, general practitioners can actually administer those. And it certainly sounds like more and more med tech **start-ups** in the US are now trying to create these home kits, so that people can start testing at home. That will help a lot. My hope is certainly that the citizens of the US, that people are going to take this very, very seriously and realize that it doesn ’ t matter that you may not feel sick, it doesn ’ t matter if you think you are young and that you are not prone to catching this virus, or that you ’ re not in – you may not be fearful of dire consequences and death. Take it seriously and stay home. And don ’ t go to public spaces, and don ’ t be a carrier, because we now know that asymptomatic folks can be carriers and there is a possibility that you can be infectious as an asymptomatic carrier. So yeah, on the cultural side – I don ’ t think it ’ s really cultural. I would say that it ’ s because of the impact of SARS, it ’ s because of the social memory of SARS that has meant people are a little bit more selfless, and have just said,”OK, I will stay home, because I might have come in contact.”CA : Yeah, weirdly, SARS seems to have acted as its own kind of vaccine, sort of just prepped the system enough for people to be ready. Yeah, social vaccine, amazing. Back to you, Whitney. WPR : OK, great, thanks so much, Chris. And so I think it ’ s interesting, Gary, to hear you talk about some of the reactions in Wuhan and some of the stories that you ’ ve heard, especially running the”South China Morning Post”and running a news organization during this outbreak. You know, what are some of the – First, what is that like, to run a news organization, to report during this outbreak? GL : Well, running a news organization in a moment like this, so close to the initial epicenter where the outbreak started, is complicated. We were lucky, very, very lucky that most of our senior editors, and certainly the most senior editors, our editor in chief, our masthead leadership, they were all journalists and they were **reporters** during SARS. So there ' s a lot of pattern recognition in our newsroom. Which meant the moment that we got the first, sort of, the first stories coming out of China, starting on December 30, people already raising their hands in the newsroom saying,"Hey, we ' ve got to report about this like it ' s going to be the next SARS. There ' s a high likelihood that this is it."And we did send people to Wuhan early on in January. Like I said, we also had **reporters** there right before the lockdown. After the lockdown, we were lucky enough to pull all of them out of Wuhan. But we actually did change very quickly the way that we report. Partially to make sure we got the story right, to dig deeper in the places that we knew we had to dig deeper, but also to protect our journalists and employees. So one of the things that we did do, and maybe other news organizations would disagree with our decision, was I think in late January or early February, even in Hong Kong, we said to our journalists,"You are not to go into hospitals."So no more in-hospital reporting. Because they were – we knew that it was highly infectious, we were worried that they were going to become you know, points of spread, and we just wanted to protect our employees and our company, so we did that. We also had a business continuity plan. Which meant that at the drop of a dime we could shut down the entire office and still operate this global news business. Some of the most interesting stories we ' ve covered is actually how technology has played a huge role in China during this epidemic. Because it frankly has changed the way that diagnostics work, it changed the way that containment works, it certainly has changed the way that consumer life works. And of course, there ' s been a lot of instances of really interesting censorship, but also, more interestingly, how the Chinese netizens have fought that censorship and reacted to that censorship. And I do think that there ' s quite a lot of lasting impacts that are likely to happen because of technology deployment during this time. WPR : And so I think in talking about some of those lasting impacts, now that you as a country are sort of emerging from this and coming on into a different stage with this outbreak, what are some of the changes you ' re seeing to **daily** life, both as society, and maybe things that you ' re hearing that individuals are experiencing as a result of this? GL : Yeah, I think probably the two most interesting changes, actually, I should say three – The first one is on education. Now schools have been shut down across China for quite some time now and again, this might feel a little bit stereotypical, or a caricature of China, but education is extremely important to the country and extremely important to the citizens. And we were actually just about to come up to the national exams, which these students work 18 years for. And so online education – very, very quickly moved online. And part of that move online was that courses had to be, and classes had to be recorded. Which means that now, there ' s this huge repository of recorded classes. That means potential democratization of education material, and significantly lowered costs to get this type of coursework from the top tiered schools, whether it ' s high schools, universities or primary schools, to the entire country. Now whether or not China activates on that, we ' re still not sure, but the potential is there. The second major shift is really on distributed **workforce**. The idea of working remotely, office work remotely, is not much of a concept in China and across most of Asia. Certainly far less than in the United States. And I ' m from the US tech industry, so it was pretty normal, it ' s pretty normal in the US tech industry even before this, in China much less so. But because of the lockdowns, not only in Hubei but across China, this has become much more normal. And people are kind of falling into a different rhythm of work. And most importantly, this has given rise to a whole new set of teleconferencing companies in China. Because most of the teleconferencing companies that we know of in the West, whether it ’ s the Cisco systems, Google Hangouts, Zoom, that everyone uses, BlueJeans, Slack video, they ’ re not available in China. They don ’ t work in China. There is this mirror internet in China, behind the Great Firewall, and so there ’ s a whole new set of teleconferencing systems that were used, but were not really commonplace, certainly not for distributed **workforce**, and now suddenly, over the last few months, they are. So it will be interesting to see how those companies and those **services** develop, and whether or not the **workplace** changes in China. And then finally, the third thing that is really interesting is that there was a huge internet response to this censorship issue in China over the course of the last two months. It especially exploded after this whistle blower, Dr. Li, died on February 7. All over the Chinese internet hashtags like”we want freedom of speech,””national hero Dr. Li,”things like that just exploded everywhere. And there have been – And actually the Chinese government has had to respond, I think for observers, a lot of observers believe that Chinese government ’ s change of narrative about Dr. Li was largely driven by this reaction from its citizens across the internet. There have been extremely creative examples of people getting around censorship. I think China is quite famous for using emojis to get around tech censorship. I think most people also know that the primary messaging app that the Chinese internet users use, called WeChat, it is heavily censored, it ’ s not just text that ’ s censored, images are censored really effectively, individual conversations are censored. And so when there are specific articles or specific posts that are about what ’ s happening with the virus that people want to share, and the government thinks that it is detrimental to whatever, they will censor and it will be completely and very effectively removed. But this time around, Chinese citizens used emojis again. They translated these posts into ancient Chinese texts that the censoring machines couldn ’ t pick up yet, they actually translated one version of this post into Tolkien ’ s Elvish language, I don ’ t even know the name of that language, they translated into that and the AI couldn ’ t pick it up. And then finally, I think one of my favorite versions of this was they used the”Star Wars”intro, the angled text **scrolling**, it became a video, and they had the entire post about what was going on in Wuhan in that format, and that went all over the internet. So I do think that there is going to be an increased call. Academics now are speaking up about freedom of speech. So there ’ s going to be this increased volume of netizens calling for freedom of speech. It will be very, very interesting to watch how the authorities in China deal with that. WPR : Great. And Chris, you have a question? CA : Yeah, it sort of picks up on that about, you know, the stories that could come out of this. I mean, there are definitely optimistic stories that people are feeling, that this could lead to more free speech of a certain kind in China. Certain things you can ’ t suppress. Maybe in the US it might lead to the government taking scientific predictions more seriously, not clear that ’ s happening yet. And there ’ s hope that this whole thing, because it ’ s a common enemy for the world, will actually bring the world together in some ways. But I ’ m curious how you think about this. President Trump started referring to this as the Chinese virus. I ’ m curious how that ’ s being received in China, and how people are feeling on this issue. Do you think it ’ s increased sympathy for other countries or actually dialed up animosity? GL : Well, it ’ s certainly not being received well across China. I think one thing that is still really undercovered is the intensity of rising nationalism at the grassroots level across China over the last several years. And they ’ re very protective of their country, and their people and their history. And President Trump ’ s comments and the fact that so much of the US government is now referring to this as the Chinese virus is not received well. You know, my fear of course, is that even prior to the virus, the US-China conflict was escalating beyond anything that I think most of us as observers want to see. Trade, tech, military, ideology, and now we can add information conflict and health conflict, health tech conflict especially, to the list. Of course, the hope is that these heightened tensions will actually dissipate and that the two countries can actually, at this moment in time, choose to go down one of two paths. Either one that further damages the relationship or one that actually shows what the possibilities are if the two largest economies in the world, the two most powerful countries in the world, actually cooperate. You know, this week, Thursday is the G20 conversations that are going to happen remotely. It will be interesting to see how US and China actually coordinate, cooperate, how they communicate during those talks. CA : I think people want to know how you think this will play out. You ' ve got a very **special** seat there, you know, looking at all parts of the world in it. What ' s your take on how this plays out? GL : I desperately want to be an optimist, Chris. But I think that everything we see, especially the data, shows that it is going to get far worse before it get better. And I ' m very fearful for what ' s going on in the United States. It ' s because of the amount of data we have across all these different countries, you can very clearly layer countries and the way that the pandemic has been spreading, on top of one another, and we know that the US is a week and a half, maybe two weeks, behind Italy, and we know what happened in Italy and what ' s going on in Spain. The US is catching up on that spike and it ' s going to come much faster, and it ' s going to be much higher than I think most people originally believed or hoped for. So it will get worse. So the hope is that, again, this is going to be the optimistic side of me, that the nations will come together, that those in charge, our governments, will make the drastic, necessary moves, and we will be able to come out on the other side faster than it looks right now. Remember, when China went to shutdown, on January 23, there were only 830 confirmed cases. And even if those numbers are not exactly accurate, it ' s nowhere near the confirmed cases that we have in the US right now, that we see in the US. So that is something to be very, very concerned about. At 830 they shut down. And even after the shutdown, two weeks later, the cases had grown to 35, 000, two weeks after that, it was at 75, 000. So at this point, it is late in the US. But, we, you know – it can still be fixed. And I think most... experts that we talk to believe that it can be fixed with fast and decisive **action**. CA : Yeah, people **struggle** with understanding the power of exponential growth. And a number can seem smallish today, but if you believe the science, yeah, you have to act. Gary, look, I hope somehow you will convey to whomever you can convey that regardless of what some people might say, in government or elsewhere, there are millions, there are tens of millions, there are probably hundreds of millions of people in the US, on both right and left, who are amazed by what happened in China. You know, yes, missteps early on, whatever. But they ' re amazed, you ' ve really – you know, both the Chinese government, the Hong Kong government, several Asian governments, Singapore, South Korea, have shown astonishingly wise and disciplined **action** against this thing. And we ' re grateful, we feel there is much we can learn from you and so – People, most people want this to be a time of bringing the world together. I genuinely believe that, it ' s maybe the optimistic part of me believing it. But I believe in it, it ' s partly what these conversations are for, to try and make those kinds of connections. We want to keep in touch. You ' ve got an amazing seat there, and I have loved listening to every word you ' ve said today here. It ' s just I ' ve learned so much from you. So thank you for that. GL : It was a great conversation. WPR : Thank you for your insight. CA : And thank you to our whole online audience, I mean, this is a journey, every day we ' re learning something new. And just in case anyone out there is feeling a little bit powerless, and afraid or you know, at the situation, I mean, the one thing that everyone can do right now, I think, is we can reach out to the people we know, we can encourage each of us to be our best selves in this moment. I really think it ’ s what the world is going to need, when people are angry and fearful, we can turn into nasty people. But when we ’ re – When we realize how much we need each other, and are willing to just reach out and share stories of hope and share what we ’ re feeling and share possibilities, we can really impact each other, and I see so many incredible instances of that from around the world, whether it ’ s Italians singing to each other joyfully from each other ’ s balconies, or these sort of tales of heroism that some of our health workers have been engaged in all around the world. There ’ s going to need to be a lot more of that. And honestly, every single person can play a part in how they are online, what they share, how they react. So I don ’ t want to be overly, embarrassingly Kumbaya, but I kind of think we need that spirit right now a little bit. We need each other, and TED is going to try and play that role a bit. So if you hate that, maybe you don ’ t need to be here, but I hope you don ’ t hate that. I hope you like that and will be part of it. Whitney, it ’ s so fun cohosting these, thanks to the rest of the amazing TED team who are everyone in our individual homes, they are sort of racing around, trying to make this stuff work technically. We ’ re learning a bit each day, I hope. Thanks so much for being part of this. WPR : Thank you, everyone, thank you. We ’ ll see you all back here tomorrow.

Text Sample 176: POS Dim 2 – 2020 – Score 15.00 – 2020/t004010.txt

As a small child, I was lucky to be at the launch of Apollo 17, the last manned **mission** to the Moon, and I ’ ve remained enamored with space ever since. And fortunate as a physician to have contributed to NASA life sciences research and to **practice** aerospace medicine, inspired by the cross-disciplinary teamwork required to tackle audacious challenges and how space is has often brought the world together through the lens of seeing our planet as one without borders. Now, just as the historic Apollo moon landings were transformational inflection points in history, so too is the global health crisis of COVID-19, which, despite its many challenges and tragedies, like the sinister Cold War **setting**, which launched the space race, can have silver linings. As Regina Dugan, former head of DARPA, wrote,”Sputnik set off the space age, COVID can spark the health age.”The silver linings include the unprecedented acceleration of innovation, collaboration and discovery, catalyzing a future of health and medicine that can help us reimagine and bring us a healthier, smarter, more equitable post-COVID world. Now, many solutions ride the rails of rapidly, exponentially developing technologies that are rapidly doubling in their speed-price performance, as exemplified by Moore ’ s Law, which has enabled the billionfold improvements in memory and computation, resulting in the ubiquitous supercomputer smartphones most of us carry in our pockets. I still have my now ancient iPhone 2 here. Still works, which felt magical 12 years ago but now feels slow and kludgy. And I ’ m sure my iPhone 11 will soon seem antique, perhaps as its features dissolve into the rumored to soon arrive augmented reality smartglasses. Now exponential technologies packed into our smart devices are becoming increasingly medicalized, with sensors able to detect an ear infection and more. So what used to fit on a desktop computer now fits on our wrist and these are now entering the domain of FDA-approved medical devices. But the future isn ’ t about any one technology, but their convergence as they get faster, cheaper, better. In fact, creating entire new fields at their interfaces, from computational biology, robotic surgery, digiceuticals, telemedicine to AI-enabled radiology. And while many industries have been disrupted and breached the fourth industrial age, health and medicine often feel stuck in the second or third. Critical data is still stuck being shared on fax machines, paper forms. We ’ re stuck in waiting rooms waiting for our visits. I recently had my own echocardiogram only made available to share with me on a CD-ROM. I don ’ t even own a CD-ROM player anymore. Tools for managing pandemics in 2020 rely on the same core technologies used in the pandemic of 1918 : face masks, social **distancing**, handwashing. So part of the challenge in advancing global, local health are our models, our mindsets. We don ’ t really **practice** health care. We **practice** sick care. Sick care is based on intermittent episodic data, usually only obtained within the four walls of the clinic or hospital bed, and leads to our reactive sick care model, where we wait for the patient to show up in the emergency room with a heart attack, stroke or late-stage cancer or for the pandemic to arrive on our shores. I believe the convergence of many of the accelerating technologies and approaches being catalyzed by COVID will bring us from intermittent sick care to an age of continuous, proactive, personalized, crowdsourced health care that can increasingly bring care anytime, anywhere more effectively and lower costs around the planet. For example, the convergence of ever smaller interconnected devices now riding 5G is creating not just an Internet of Things but an Internet of Medical Things. Much of this convergence is in the field of digital health, the ability to connect the dots between data sources from personal genomics and medical records with apps and **services** that match the needs of an individual, patient or caregiver. And as incentives and reimbursements align, COVID has pushed us to an increasingly virtualized care, from the hospital to home to our phone to on and even inside our bodies. The age of hospital to home-spital is upon us. Now, the challenge of this hyperconnected age is that we ’ re creating exponential amounts of big data that ’ s too often siloed in formats that can ’ t even talk to each other. So we need to narrow that gap between data, turning that into actual information for the patient, physician, public health worker, and speed its safe and effective use in the community clinic and bedside. The pandemic has instigated an immense amount of international sharing and collaboration amongst clinicians and **researchers** to narrow that gap. What was learned in managing patients in Wuhan and then in the intensive care units of Italy has helped New York City hospitals and their learnings in turn have spread to centers around the world. Let ’ s take a quick dive into some examples of what ’ s happening across the health care paradigm in the age of COVID and the implications for the future. From new forms of data to help prediction of prevention to faster diagnostics, more tailored therapy and increasingly crowdsourced discovery. Let ’ s start with prevention. Now, while our genomes impact our health outcomes and our health spans, our social determinants of health, our social, and our day-to-day behaviors drive most of our risk for disease and associated costs. And we now have an explosion of new tools to help measure and improve our healthy behaviors. The first Fitbit only launched in 2009. Wearables are now ubiquitous and can measure almost every element of our physiology, behavior and even mental health. And they ’ re evolving all the way from disposable tattoos that can stream vital signs 24 / 7 to an integration of big data that can – Even small data from a simple wearable, tracking the patient discharged home after a hip replacement or a **coronavirus** infection can determine if the patient is recovering as expected, walking more, doing great or not so great and trigger early intervention. We ’ re evolving from a world of quantified self where our digital data remains silent on our devices to one of quantified health where the data can be shared securely with clinical teams and **researchers** to help optimize prevention, diagnose disease early and with feedback loops, personalize and optimize therapy. From wristband vitals, including blood pressure, now obtainable without a cuff, and soon sensors that will measure our blood oxygenation levels to continuous blood sugar monitoring, to shock ' ables, hearables, ring ' ables that can replace an entire sleep lab fitting on our finger to inside ' ables, chips beneath our skin, to track our physiology and lab values, to even underwear ' ables, Internet of Medical Things sensors so cheap today you can get a pack of ten of them, have one on each pair of your underwear, now being used to do what ' s called remote patient monitoring to help detect signs of respiratory decompensation of patients with bronchitis or COVID. Breath ' ables are showing promise. Nanonoses that can detect molecules in our breath correlating to cancer, metabolic disease and even diagnosing infectious disease. In fact, we now don ' t need to wear anything. Invisibles, ambient sensing from AI-enabled cameras can track her vital signs. To voice as a biomarker to manage and detect mental health challenges, signs of heart disease, now being able to differentiate between a cough from a common cold to that one caused by **coronavirus**. And we ' ll soon be exuding our digital exhaust 24 / 7, our digitome. How do we make sense and truly leverage it? One path is through crowdsourcing. The million-participant All of Us **trial** from the National Institutes of Health is doing just that where data donors, and I ' m one, can contribute our medical records, genomes and wearable data to build a much better and diverse data set, crossing racial and socioeconomic groups to help foster better precision medicine for all of us. Integrating this information for the individual and public health will lead to predictalitics, our own personal check engine lights that can give us early proactive **warning**. And recent work is demonstrating that wearables can detect presymptomatically the onset of the **flu**, or, as recently published by Stanford, in 83 percent of COVID patients smartwatches can detect COVID infections early, often days before onset of symptoms. Self-reporting websites like Covid Near You enable us to locally generate infections maps, and combined with our social graphs and contact tracing apps, may provide us detailed suggestions about who we might want to consider being near or socially distanced from. What about advancements in diagnostics and monitoring? What used to require a full clinic or laboratory can now fit into a digital doctor ' s bag or the pocket of a patient. From COVID **quarantine** kits enabling tracking of oxygen saturation, temperature and lung sounds, we ' re starting to integrate these into virtual visits, providing real-time enhancements of a virtual physical exam. And the diagnostic tools are becoming increasingly infused with AI machine learning, including consumer ultrasounds, which can bring diagnostics anywhere at very low cost, including the ability to evaluate the lungs in suspected COVID patients. The laboratory has shrunk to microfluidic **platforms** that can be attached to our smartphones and enable anyone to take measurements from blood or saliva. Many of these diagnostics are leveraging the smartphone and its camera for a medical selfie. For example, instead of taking your urine to the lab to diagnose a potential urinary tract infection, in the **privacy** of your home simply dip the urine dipstick, take a picture with your smartphone camera and have the results made available immediately to your doctor and pharmacy. Similar phone-based apps and approaches are being used and developed for fast, frequent, cheap and easy COVID testing. Novel approaches to community level diagnostics are also being explored, including next gen sequencing of sewage for early detection of COVID-19, identifying hotspots and predicted outbreaks a week or more early. The explosion of data sources, however, is really beyond the capacity of the human mind to effectively integrate. We ' re now getting help from AI, or as I call it, IA, intelligence augmentation. IA is being leveraged in reading CT scans to diagnose COVID, to enhancing the vision of a gastroenterologist performing a colonoscopy to identify lesions they might have missed. And AI is playing an active role in helping identify and develop new antivirals. And while AI is often perceived as a threat by some clinicians, it can ’ t replace the human touch or **empathy**. And I don ’ t think doctors or nurses will be replaced by AI, but doctors and health care systems who ’ re collaborating with AI in the future will be replacing those who don ’ t. Finally, therapy. The pandemic has dramatically accelerated the use of virtual visits. Telemedicine visits are up on the order of a thousand percent in many **settings**. And I don ’ t think we ’ ll ever revert to pre-COVID levels as patients and clinicians are discovering the compelling convenience and efficacy. Even before virtual zoom or facetime with the clinician, asynchronous screening and support has been provided by ever-smarter chat bots that can help discern symptoms and triage problems effectively at lower cost. This includes virtualization and virtual augmentation to meet our mental health crisis, exacerbated by the many economic and other stressors which accompany this pandemic. 3D-printing is finding a role in health care, with newfound applications from printing personal masks to critical parts of ventilators and being leveraged by the growing maker movement, which is playing a major role in pandemic response, from making face shields and masks to improvising do-it-yourself ventilators. All together, these efforts are enabling the potential for democratization of health and medicine across the planet and access to information and care that was previously inaccessible. Clinical **trials** are being reshaped, leveraging smart devices, cloud-based analytic **platforms** and collaborators around the world. That ’ s at this convergence of many rapidly developing and exponential technologies that we have the real potential to reshape and scale health care at our pandemic age. One where we can dramatically expand access to basic health care, increasingly personalized and proactive, leveraging the scale of digital **platforms** and technologies, enhancing digital connection and **empathy**, and the ability to blend virtual and in-person care, and leveraging the power of the crowd to share and build better maps that guide our individual health and public health journeys, and to develop validated and scaled solutions. So imagine a new generation of volunteers, a global health corps similar to the volunteer paramedics and firemen of today that can be upskilled, use the powerful new tools to respond early and collectively to enhance contact tracing, **isolation** and **quarantine**, and to help identify and address social and other disparities. So coming full circle. Twenty four years after I was at the launch of Apollo 17, I found myself as a medical student on a research clerkship at Johnson Space Center. And much to my surprise, one day in the clinic, I ran right into Gene Cernan, the Apollo 17 commander and the last man to walk on the Moon. After enthusiastically sharing my childhood memories of his launch, he shared one of his famous lines :”I walked on the Moon. What can ’ t you do?”Indeed, what can ’ t we do if we work together as one in the face of this pandemic? And just as the near tragedy of Apollo 13 rallied NASA to work creatively and collectively, so too can this in our pandemic age lead to our finest hour, bring on a true health age. I believe this is possible if we all get out of our linear mindsets, take exponential steps and collaboratively go forth collectively, not only to solve the challenges of this pandemic and predict the future of health and medicine, but boldly to go forth together to accelerate a far better one for everyone on Spaceship Earth. Thanks.

Text Sample 177: POS Dim 2 – 2020 – Score 15.00 – 2020/t004286.txt
Corey Hajim : Hi, Chris, how are you? Chris Anderson : I ’ m very well, Corey,

it ’ s absolutely lovely to see you. CH : It ’ s great to see you, too. CA : Somehow, you ’ re always smiling, no matter how dangerous, weird, crazy things are. Thank you for that. CH : You don ’ t see me in the other room crying afterwards, but we ’ ll leave that for the other [unclear]. So Chris, this is week three of these conversations, how are you thinking about the people we should be speaking with? CA : I mean, there are so many aspects to this, right? There ’ s understanding the basic pandemic itself and all the science around that. There ’ s the psychology that we ’ re all going through, the **mindset**. And we ’ ve had speakers addressing both of these. And then I think increasingly, the conversation is going to be”what now?””How do we dig ourselves out of this? What ’ s the way forward?”And there ’ s a couple of speakers this week focused on that. And I think it ’ s – These conversations are incredibly rich, because I think one of the things that people have got growing consensus on is that step one, we kind of get, right? You shut things down, physically distance in whatever way you can, different countries have gone about it slightly differently, but basically that”flattens the curve,”ultimately, the number of cases, the number of infections slows down. And, but then what? Because you can ’ t go back to life as normal, when you ’ re living at home completely. You could do some things, but you can ’ t. And so that ’ s what we ’ re going to talk about today. CH : Right, it feels really hopeful to talk about some **actions** we can take besides just staying away from everybody else. So, well, I guess I ’ ll pass it over to you to introduce the speaker, and I will come back a little bit later to share some questions from our audience. CA : Thanks so much, Corey, see you again in a bit. CH : Thank you. CA : And yes, if you know anyone out there who has just got stuck on,”But how do people get back to work?””Where do we go from here?”Those are the people who you should, maybe invite them into this conversation right now, because I think they ’ re going to be really interested. Our speaker, our guest is a professor at Harvard, Danielle Allen. She runs, among other things, she runs an institute for ethics there, the Safra institute. And fundamentally, she ’ s thinking about the ethical questions about what ’ s happening here, but she has pulled together an extraordinary multidisciplinary team of economists, business leaders and others who have put together a plan, and I ’ ve been obsessed with this whole thing and how we find our way out. This plan is as compelling a plan as I ’ ve seen anywhere. So let ’ s dig into it without further ado, Danielle Allen, welcome here to TED Connects. Danielle Allen : Thank you, Chris, happy to be here. I ’ m really, really grateful to have the chance to have this conversation with you. CA : It ’ s – It ’ s so good, I just enjoyed our conversation over the last couple of days. This is such a complex problem. What I kind of want you to do is just go through it step-by-step, to see the logic of what it is that your team are putting forward. First of all – Just the problem itself of how we get the economy going again, just talk a bit about what ’ s at stake there, because sometimes this is framed as”The economy? Who cares about the economy? People ’ s lives are at stake. So let ’ s just focus on that, don ’ t worry about the economy.”But it ’ s not as simple as that. I mean, as an ethicist, what ’ s at stake if we don ’ t restart the economy somehow? DA : Well we have to recognize that we ’ ve actually faced two existential threats simultaneously. The first was to the public health system. If the virus had been allowed to unfold unimpeded, our public health systems would have collapsed and that would have produced a whole legitimacy crisis for our public institutions. So of course we shut down, we had to do that, it was a necessary self-defense **action** that has, however, really devastated the economy. And that is also an existential threat, we can ’ t actually endure a closed economy over a duration of 12 to 18 months. Nor can we really endure a situation where we don ’ t know whether we might have another two to three months of extensive social **distancing**. So we really need an integrated strategy, one that recognizes both of these existential threats and finds a way to control the disease at the same that we can keep the economy open. We call that combination of controlling the disease while keeping the economy open pandemic resilience. We think that ’ s what we should be aiming for. CA : So people who aren ’ t moved by the notion of the economy, capitalism, whatever, think instead about the millions and millions of jobs that were lost, the people who are desperate to make money. And I guess the lives that will be lost unless we solve this problem. DA : Absolutely, the economy is one of the foundational pillars for a healthy society with opportunity and with justice. You can ’ t have a just society either, if you haven ’ t secured a just and functioning economy that delivers well-being for people. So all we have to do is remember back to 2008, and think about the impacts on things like suicide and depression and so forth, that flowed from that recession, so the economy is a public health concern in the same way that the virus is a public health concern. CA : OK, so talk about why this is such an intractable problem. People isolate, in many countries in the world now you ’ re starting to see the cases flatten and in many cases decrease. It looks like, whether it ’ s happened now in your country or not, that will happen sooner or later. So why isn ’ t that problem solved, we ’ ve beaten the virus, let ’ s get back to work? DA : That ’ s a great question and it really speaks to how new the experience for us is to encounter a novel virus. It just really hasn ’ t happened to our society in a very, very long time. So we are what ’ s called the susceptible population, meaning not any of us at the beginning of this had immunity. We were all susceptible to catching the disease. For a society to be safe, it needs to have what ’ s called herd immunity. You could achieve that through vaccination or through people getting the disease. But it takes 50 to 67 percent of the population to get the disease in order to achieve that level of protection. We don ’ t expect a vaccine anytime in the next 12 months, possibly 18 months, so we have to recognize that that pathway is not open to us. And to get a sense of the magnitude of what it would mean to live through the disease to get to herd immunity, think about this : In Italy right now they estimate that about 15 percent of the population has probably been exposed to the disease. So you ’ d have to repeat what Italy has done three or more times, to get to a place where you can reasonably think that there ’ s herd immunity. And I think you can see that when you think of that picture, how destabilizing a process would be of just leaving things broadly open without disease controls. So the real trick is whether or not there ’ s a substitute for social **distancing** as a method for controlling the disease. CA : Right. So Italy, even with that 15 percent has suffered at least 15, 000 deaths, some people argue that it ’ s underreported by 50 percent there, it might be 30, 000 deaths plus there, and as they come down the curve, there will be more to come. Multiply that by five or six, say, for the bigger population size of the US and the herd immunity idea per se doesn ’ t seem like a winning idea. I mean, it ’ s a horrible idea. DA : It ’ s a horrible idea, exactly. And we do have alternatives, that ’ s the important thing, we actually do have a way of controlling the disease, minimizing loss of life and reopening the economy, so that ’ s the thing we should all be focusing on. CA : And again, the initial problem is that if you just let people start coming back, as soon as they gather again in reasonable numbers, the risk is that this highly infectious bug just takes off again. DA : Exactly. CA : And so one scenario is that you have countries lurching from a little bit of activity here and then suddenly it explodes again and everyone has to retreat. That does not seem attractive, that also just doesn ’ t work. DA : No, exactly. I mean, we described that as a freeze in place strategy for dealing with this. That is you freeze and you shut down all activity, and then that flattens the curve, you open up again, then you have another **peak**, you have to freeze again and so forth. So you have this repeated process of freezing, which just does tremendous damage to the economy over time. I mean the upfront damage is huge, but then there ' s never space to recover from it, because of great deal of uncertainty and repeated applications of economically ruinous social **distancing**. So I think you ' re really pointing to the features of the disease that make this situation a problem that it is. And there are really two that people should focus on. One is the degree of infectiousness. This is a highly infectious virus. So the comparison to the Spanish **flu** is a reasonable one from the point of view of degree of infectiousness. Then the second really important point about the disease is that it ' s possible to be an asymptomatic carrier. That is to be infectious, to carry the virus, and never show any symptoms yourself. Current estimates are still imprecise, but people think that about 20 percent of virus carriers are asymptomatic. And that is really the thing that makes it so hard to control. People don ' t know they ' re sick and then they become disease vectors, spreading it everywhere they go. CA : Yes, indeed. So talk a bit, Danielle, about your thinking about how we might outwit this thing. DA : So the alternative to social **distancing** as a strategy for controlling the disease is really massively ramped up, massively scaled up testing, combined with individual **quarantine**. So we are going to continue to need individual **quarantine** for those who are positive carriers of the virus, until such a point as we have gotten a vaccine. Now what does that mean exactly? It means that the standard **quarantine** that aligns with the incubation period, 14 days is often what people talk about, in the conservative picture you might say twice the incubation period length, 28 days for individual **quarantine**. And we need that **quarantine** for people who are symptomatic and for asymptomatic carriers of the virus. Now the only way that you can actually run an individual **quarantine** as opposed to a collective **quarantine** regime, is if you do massive testing. We really need to make testing in a sense universally available, so that we can be testing broadly across the population. There are ways to target test, make it more efficient and so forth, but in principle, what one should imagine, is really wide-scale testing, tens of millions of tests a day, connected with **quarantine** for those who test positive. Excuse me. CA : So weird. Anytime anyone coughs today, you go,"Oh, God, are you OK?"DA : Yeah, no, no, I ' m fine, Frog in the throat, that ' s all it is. CA : So just to play out a thought experiment. If we had an infinite number of tests available, and after the curve has flattened and cases have gone down, everyone came back to work, but everyone was tested every day. Then what we think is that the tests will show up positive at the same time, or possibly even ahead of the time that people are infectious. But certainly, let ' s say at the same time, regardless of whether they ' re symptomatic. And so you could – Those people would immediately go back home and the rest of the population should be OK, we should be able to get work done, in that thought experiment, right? DA : Right, in that thought experiment, exactly, yeah. CA : But the trouble is, that that would mean doing, whatever, like, 200 million tests a day. DA : Right, exactly. CA : Which is many, many orders of magnitude more than we have and could even imagine ramping up to. So you have a proposal, and this is the ingenuity, the proposal, of how to potentially administer tests in a way that ' s much more efficient. Talk a bit about that. DA : Sure. So if you were going to use a purely random testing method to control the disease, you could probably actually get away with testing everybody every two or three days – I ' m playing along with your thought experiment here – and bring the number down to 100 million tests a day. But even that is a magnitude that would take us multiple months to get to, let ' s just say if we even wanted to try to do something like that. So the thing that you really need is smart testing. So rather than testing the population at random, what you do is you use testing to identify people who are positive, and then you add to that contact tracing or contact **warning**, we think about it in both ways. And what this means is that once you know who ' s a positive test, you figure out who else has been exposed to that person over the previous two weeks, and they all get tested as well. So you start to identify a class of people who are a higher probability of being infectious and you test that group of people. So you move away from random testing, you target it through contact tracing or contact **warning**. And then, depending on the level of effectiveness of your contact-tracing and contact-warning strategy, you can reduce the numbers. So on a moderately effective contact-tracing regimen, you could imagine doing 20 million tests a day. On a highly effective regime of contact tracing and warning, you could get yourself down to the order of five to 10 million tests a day. CA : And some countries in Asia seem to have pulled off a version of this strategy that has been effective. But it requires one of two things, if I understand you right, Danielle, it requires either just this massively scaled up, or potentially quite intrusive sort of manual contact tracing where you have big teams who swoop in to anyone who ' s tested positive and try to unpack their complete recent social history. Or technology plays a role, and this is where it gets complicated, because you know, there are apps in some of the Asian countries, like, China has an app which most people are, I think, required to carry, certainly in Wuhan and elsewhere, where it ' s very good at predicting whether someone may need **quarantine**. And they will be required to do so. And so there are all these concerns about government control, government intrusion. You are in discussion about ways of doing some kind of technology that would be more acceptable in a democracy, and I ' d love you to share what those are. DA : Sure, I ' m happy to do that. So I think it ' s an important thing to say upfront that the rates at which we would need to test per capita are higher, much higher than Asian countries used, because prevalence is much higher here. They caught it earlier, they had these tools built before the pandemic hit. As a consequence, they ' re able to control it with a lower per capita rate of testing than will be the case for us. We just have to accept that fact at this point and recognize that massively scaling up is specific to our situation, because we weren ' t ready before it hit. So then, yes, OK, if we ' re trying to do the smart testing, trying to use tools, what can you do? So we ' re actually open to manual testing in the plan that we ' ve developed, I want to just say that, and I think that society, we have a big choice to make, whether what we want is a big core of manual contact tracers who are tracing people ' s histories and figuring out who they ' ve been in contact with and who they ' ve been exposed to. Or if we want to try to use a technological system. The important thing is there is a diversity of options within the technology space. So it ' s really important to recognize that places like Singapore and China have used highly centralized data systems for supporting this. And so what happens is, sort of, you carry your phone around, and everybody is connected to a central data system, and then when somebody in the system has a positive test, that gets put into the app, and then their phone communicates to other phones that it ' s been in proximity with over the previous two weeks, to alert people that they too need to get a test, OK? That ' s the basic concept. In China and Singapore the data structure for doing this is highly centralized. There are, however, a lot of innovative apps under development right now that depend instead on a very privacy-protective structure where the data lives on the individual user ' s phone and through a combination of encryption and tokens users of phones can communicate with other users of phones, but the data is not centralized. So in that regard, it becomes more of a peer-to-peer sharing, sort of concept of friends warn friends that they should probably go get tested. Then you would have a central repository of test data, but the truth is, we already have that, because all influenza tests for example, already roll up into CDC and Health and Human Services databases, so that they can track influenza patterns every year. CA : So tell me if I understand this right. You would carry on your phone an app that would, when you got, say, within six feet of another human carrying that app, the phones would exchange a Bluetooth – using Bluetooth technology they exchange a kind of token that says,"Hi, we connected."But it ' s encrypted. And that is not communicated to a central server, that is on the phone. But if either of you in the next week or two tests positive, your phone will be able to communicate to all the people which it exchanged token with, to say,"Uh oh, someone who you were close to in the last two weeks has tested positive. You ' ve got to isolate."That ' s basically how it works, it ' s done that way. DA : Exactly. CA : And then after, what, three or four weeks, the tokens can actually autodelete? They go, they ' re not there anymore. DA : They expire, that ' s right. Because you only need the most recent two weeks ' information or data about where you ' ve been and what other phones your phone has interacted with. So that ' s the really key thing. CA : Alright, we ' ll come back to that in a minute, but let ' s see what our friends are asking online. DA : OK. CH : Hi, Danielle, hi, Chris. Yeah, we ' ve got a lot of great questions, people are **super** interested in how this is all going to work. There ' s a couple of questions I ' m trying to cobble together here. I think people are really interested in your thoughts on the United States health care system. We have so many underinsured and uninsured people and the changes that you might make to that system, I mean, does that situation make things worse, and what changes would you make to the system so that we ' re not as vulnerable in the future? DA : So that ' s a great set of questions, and so just from the point of view of the testing program, it is absolutely critical that the testing be free. And so there is absolutely, a sort of necessary feature of this, which is about, kind of, universal access element to the health system. And so I ' m sure there will be tweaking that ' s necessary in the existing health system to achieve that. We ' ve also without any question seen vulnerabilities that relate to and stem from our fragmented health system. So I think there ' s a much bigger, longer-term question to be had, or conversation to have, about how we overcome that fragmentation. So yes, I do hope this moment will be a spur for that longer-term conversation about improving our health system and really achieving that universal coverage that we so badly need. CH : OK, thank you, I ' ll see you both again in a little bit. CA : Thanks, Corey. So let ' s stay with this tech issue for a bit. And the sort of civil rights or **privacy** questions that it might still raise in some people. So one concern is that surely, if your phone is able to contact these other phones, someone somewhere is ultimately going to reverse that and we ' ll have some kind of record of your, you know, everyone who you ' ve connected with, and that might be concerning to some. Is that a legitimate concern? DA : I think it is, I mean, I think we ' ve been working hard on this question and really trying to think it through and when you talk to legal experts and civil liberties experts and so forth, everybody starts with the same premise : assume failure. Assume that you ' ll have a data breach. Think for that and what kind of protection you want in that regard. And so when you think that way, you of course are trying to minimize any likelihood of that happening, so hence the privacy-protective structure of phones communicating with phones, data living on the hardware of the phone, not in the server, etc. And then also you would want a kind of democratic accountability feature, so for example having the Department of Health and Human Services have an auditing function to audit whoever is manning the server or controls the server through which the tokens are exchanged you would want to audit their functionality and how they ' re using the data. But then again, you presume failure, that somebody ’ s reverse engineering, the audit system fails in some fashion. What ’ s your protection then? The answer to that would appear to be upfront legislation that prohibits the commercialization of this COVID testing data. So that anybody who in any way tried to commercialize it in any kind of way, would be subject to legal **penalty**. So I think that ’ s how you build the fence up upfront in the expectation that somebody would find the way to crack it. CA : And then there ’ s a set of questions around how you get this app out there at scale, because it ’ s only effective if, say, two thirds of the people who are working are carrying it, right, something like that. DA : Right. CA : And so short of authoritarian”everyone must have this app,”I guess there are ways that are interesting to say to people, one, this is a really useful app, it will alert you quickly if you ’ re at any risk. But two, to get to the kind of scale we ’ re talking about, you might have to say to people,”Look, we ’ re slowly going to come back to work, industry by industry, company by company, and the deal for you to come back and break **isolation**, the societal deal, is that you have to be willing to carry this app.”And you could, for people who didn ’ t want to do that, I guess you could have some protection, you can ’ t lose your job for that. But, I mean, can you picture society making the choice that it is reasonable to require people who want to come back to work to carry that alert technology with them? DA : So this is the hardest question. We know we don ’ t want an authoritarian model, such as the one used in China and Singapore, so we have to figure out instead how to activate that thing, which is sort of the most important democratic resource or asset, namely solidarity. So what is it that, from a solidarity perspective, it ’ s reasonable for us to ask of each other? That has to be the frame for deciding how we approach this. And so one aspect of this is really, truly the building a culture of opting in to this. And there are examples of this. So for example, New York has tackled HIV testing through a program that goes by the label”New York Knows,”and it started out with labels of”Manhattan Knows”or”Brooklyn Knows,”and so forth, of the different burrows. And what this program is is one that is owned by community organizations, community partners, that do the job of spreading the word and recruiting people into testing programs. And New York has the goal of having every single New Yorker be tested for HIV, so in other words, it ’ s established as an expectation, that universal participation, and it ’ s activated a network of community partners and organizations, to cultivate that commitment to solidarity. And so I think, in all honesty, that that would be a really huge part of what you would need to do in order to tap into solidarity, to have this work. I ’ m sure that we would see some amount of requiring in different context, I think that ’ s a very hard one, because you don ’ t want to generate labor discrimination problems. And so the model there, to think about and to sort of figure out what are our parameters, what we think is fair, connects to things that schools currently do, for example, when they require that students show vaccination proof before they can start the school year and things like that. So there are multiple states that do that in schools for vaccines. Would schools do the same thing, what ’ s the sort of labor, the **workforce** question like, I think that very much remains to be worked through, but it ’ s a hugely important question. CA : I ’ d be curious what the watching audience thinks about this, maybe you could enter a comment on it. But I mean, is it reasonable, in the world that we ’ re in right now, for a company, let ’ s say, to say,”Look, we want to get back to work, but we want to do so and respect the safety of all our workers. That means that for you to come back to work, you need a test showing that you ’ re negative. And you need to carry this app so that we alert people quickly if there ’ s a problem.”Is that –”We won ’ t fire you if you don ’ t come in, but if you want to come back to work, that ’ s what you ’ ll have to agree to.”Is that a reasonable chance? I ' m curious what people think. Is there any other way to get – Sorry. DA : I mean, again, there is precedent for this in the sense that drug testing works this way in many employment contexts, right. There are many roles where people have to do routine drug tests as a part of preserving their job. That was a hotly debated issue in the 1980s, people sort of think back when that sort of first came in, and there was a lot of concern about it. We have managed to develop a regime for that, that has achieved an equilibrium of a kind. So I imagine that something is possible in this space, but we would have to draw on the prior experience with things like drug testing in the **workplace**, I think. CA : I mean, one problem that we face when you think about these big systems introduced is that in the past, there ' s history where something got introduced, you think of the PATRIOT Act that came in after 9 / 11 and a lot of people have a lot of problems with that Act, and it gets renewed relentlessly, relentlessly, and here we are, nearly 20 years later, and it ' s still with us. So that creates quite a high bar for any standard that we push out here. How do we persuade people that this is custom-made for the current situation that we ' re in, and it ' s not going to be picked up and subsequently abused by companies or by government? DA : That ' s an absolutely critical question, and I think we have a lot to learn from places like Germany, which are really, really strong and rigorous on **privacy** protections. Perhaps having some of the highest privacy-protection standards in the world. And Germany, over the course of the last few weeks, has articulated an approach that definitely picks up several of these elements. So there are ways of building in **privacy** structures that are meeting the standards of the German **privacy** framework, and so I think for us, that ' s a really important place to look to, and learn from them how they ' re structuring it, to achieve those **privacy** protections. CA : Danielle, you ' re an ethicist, among other things, as well as a political theorist, and is it, as you think about how to apply ethical questions to this, is it inherent in a situation like this that there are going to be trade-offs, that there is no"perfect solution"that we just, you know – These things are fundamentally – You ' ve got two goods that are fundamentally in conflict with each other or if you like, avoidance of two evils that are going to clash. And that we ' re not going to get away sort of untainted to some extent, we just have to try and make the least bad choice? DA : It ' s a great question, and I think, I tend to formulate things as being about hard choices and judgments, rather than being about trade-offs. I think trade-offs often suggest that you can precisely quantify this degree of monetizable harm against that degree of monetizable harm, and I think that ' s actually not as helpful to us in this current moment, to be honest. So in effect, I think the most important thing is that we clarify our core values. And so the way we ' ve tried to articulate that is to say we have a fundamental value in securing public health. We have a fundamental value in securing a functioning, healthy economy. We have a fundamental value in securing civil liberties and justice and constitutional democracy. And so then the question is, given that set of fundamental values, what are the policy options that actually do secure all of those things? So in that regard, at the end of the day, you know, there ' s a bunch of libertarians in the group that we work on, and a lot of us come out very strongly, sort of, **privacy** protecting, liberty protecting point of view. And so we ' re not here to sacrifice those things. We ' re rather here to find a solution that aligns with the values that we bring in to this problem. So that ' s how we think about the decision making. CA : Talk a bit more, actually, about the group that you ' ve pooled together over this. I know that there ' s a TED speaker Paul Romer, an economist at Stanford, who ' s, I think, a key member. Who else is in the group? DA : Well, Paul was a key member. I ' m afraid we parted ways to some extent, because he ' s advocating random testing, so the sort of 100 million tests a day direction, and he ’ s not a fan of the contact-tracing approach, so he does have, you know, he ’ s sort of at one end of a kind of libertarian spectrum on that and my view, however, is that testing 100 million a day is far more intrusive than smart testing supported by **privacy** protective contact tracing. I also think it ’ s really important to throw into the mix the fact that collective social **distancing** is a huge infringement on our civil liberties. We keep forgetting that. The alternative is not contact tracing versus nothing, it ’ s contact tracing versus social **distancing**. We can ’ t go out, we can ’ t form associations where we get to be together in person, church-goers can ’ t go to church right now. You know, political parties are having their conventions postponed. If that ’ s not infringement on our civil liberties, I don ’ t know what is. So from my point of view, the civil liberties conversation is one about the contrast between the kind of infringement that is produced by social **distancing** versus the kind of infringement or reshaping that would be imposed by contact-tracing regime. I didn ’ t answer your question about our group. CA : Go ahead, it ’ s just amazing this thing is moving so fast in real time. Talk about some of the other people who are in your group. DA : Sure, so Glen Weyl is an economist at Microsoft, a political economist, he ’ s a really key figure and he is really an innovative mechanism design thinker, who is really good at kind of, figuring out how to craft incentive structures and so forth that help people make choices in socially productive ways, in ways that are also freedom-respecting, and so forth. So he ’ s really been helping us think about the design of the policy pathway, Rajiv Sethi is another economist, Lucas Stanczyk is a philosopher at Harvard who has been scrutinizing the civil liberties and justice questions. I mean, that is his line of work, those are the things he ’ s most committed to, and that ’ s what he ’ s doing. We ’ ve reached out to a number of public health groups for regular consultations, so they ’ re not as directly part of our group in the sense of advancing a policy, but in terms of informing our epidemiological understanding, we ’ ve relied a lot on folks at the Chan School of Public Health at Harvard. So lawyers as well, Glenn Cohen, who directs the Petrie-Flom Center for law and bioethics has been a critical member, Andrew Crespo also at Harvard Law School, Rosa Brooks at Georgetown Law school, I could go on, I ’ m missing key people, critical scientists. Actually, there ’ s a great paper on solidarity by Melani Cammett and Evan Lieberman that people should check out too. CA : It ’ s exciting that one of the impacts of this, and I ’ ve seen it in other areas as well, this crisis is really breaking a lot of cross-disciplinary lines and bringing people together in **unexpected** combinations, which is good. DA : Yes. CA : So how, if this plan got general acceptance, how – I mean, obviously, the clock is ticking, this is urgent, what would it look like to move this forward? Give a sense of what you think it would cost, give a sense of who might own it, like, what would it take to actually activate this giant idea? DA : Alright, so it ’ s a big price tag, so I hope you ’ re sitting down – I ’ m glad you ’ re sitting down. So over two years, based on conservative estimates of what you would need, that is to say maximal estimates for testing and things like that, it ’ s got a price tag of 500 billion, which includes both the production of the tests and the personnel of test administration, contact tracing and all of that. So it ’ s important to remember though, that that production ramp up and the contact tracing ramp up are employment possibilities, so in that regard, they would counteract the negative impact on employment of the social **distancing**. So it ’ s a big price tag, but it would be multipurpose in that regard, contributing to jumping up the economy, as well as the testing program itself. It would be important that it be phased in, and phasing it in would actually give us a way of testing out the paradigm as we went. So for example, for a first phase of rollout, probably what you would want to do, ideally by the end of the next month, would be to have a full range of testing for a combination of everybody in the health care sector and everybody who might fill in and substitute for any health care workers who test positive. So in other words, your health care worker pool and a substitute pool, say a national **service** corps, of folks who can fill in for health care workers who test positive. If you can get those two groups, those two sectors fully under testing, contact-tracing regime, so you know that every health care worker is not positive, and anybody who is is immediately **quarantined** and so forth, we would stabilize our public health infrastructure, and that would already get about 30 percent of the **workforce** under this kind of testing and tracing regime. And then you ' d move on, with that stabilized, to other critical and essential workers, etc. So the bad news, Chris, is you know, who would be the last people to be folded into this? It would be you, it would be me, the people who can actually telecommute for work, OK. Because we would have the least call on social needs to pull us back out into the **workforce**. So we ' d be the last ones out. But that ' s a good thing, I think that ' s a part of making the point that we ' re all in this together and that there are sacrifices in different places, and **service** workers, care workers and so forth would be able to get out faster. CA : And that addresses what is definitely one of the most shocking and painful aspects of the current moment, which is, you know, for those of us working from home it feels traumatic, but it ' s nothing like what others are experiencing, whose livelihood depends on being out there, doing, you know, physical work. And so I think it ' s excellent, obviously, that the plan focuses on them first. How applicable is this to other countries? You ' re obviously talking – The plan is developed for the US. It ' s inspired by what ' s happened, in some ways, in some Asian countries. Is it applicable to other countries as well? DA : It absolutely is, and we ' re already seeing Europe move in this way. So Europe and the UK are ahead of the US on this point, I mean, the rough shape of the plan that we ' re proposing seems to be pretty much the rough shape of the plan that ' s emerging in Europe and the UK. So I think it ' s a really important moment to bring together those policy conversations, bring together those modeling conversations and help each other out on this one. CA : And I guess the reason I ' m delighted that you ' re engaged in this is that it ' s – You know, it ' s fundamentally framed here as this is a discussion that society has to have. There are ethical choices we have to make here as part of this. And so we can ' t just leave it up to the scientists, as brilliant as they are. And the politicians, for goodness sake. We all need to understand what is at stake here, what the choices are, what the hard choices are and know that any direction is tricky, but we, you know – This really matters. DA : Absolutely. I think you ' ve put it so well. I think that ' s what makes this kind of question different in a democracy. It really is important that we all collectively achieve understanding, have clarity about the directional options and have a sense of collectively moving in a direction that we desire, right. That we consent to, in a sense. CA : Corey. CH : Hi, I just wanted to come back and give you a little feedback on what people are saying online in terms of the testing, to be able to go back to work, you know, how people feel about that. Obviously, there ' s lots of questions about the app and **privacy**. Some people are hesitant about it, they ' re wondering whether it will be mandatory, which you touched on. Maybe you will opt in to be able to go back in the office. I ' m in, I would test to be able to go back in the office myself, but I think people are wondering about that. But the general consensus is it seems like a reasonable possibility. There are a couple of questions. One I think you just touched on in terms of the global possibilities. Do you see some collaboration on the global landscape, do you see people talking to each other? Obviously, if we want international travel to come back, that seems like a key piece of it. DA

: Yes. So I think travel is one of the hardest pieces of this, and actually I don ’ t think that there are good, clear answers on that yet. Scientists are talking to each other across international boundaries without any questions. I think the scientific community is really well and at work, really connected, trying to think about these things. It ’ s not clear to me how well-networked the policy-making community is, in all honesty. So I think there ’ s probably a lot of room for building a tighter international network of policy makers on that front. And the hardest part is going to be the travel piece. And honestly, we haven ’ t even talked about parts of the globe like Africa or India, South America, where they ’ re not yet getting towards this policy paradigm. So the virus is going to live in the world, without any question. And live in the world probably in quite significant ways for a considerable period of time. So I think the role of travel restrictions is probably going to be with us for a spell. And so it really does matter that we get the design of those right. I think it ’ s Hong Kong that has a particularly, what looks to me like a sort of, useful regime, where anybody who is coming into Hong Kong for longer than two weeks has to go into 14-day **quarantine** when they arrive. But for anybody who is coming for a shorter time, they have to be tested when they arrive and then they have to also go through active monitoring during the period of their time in Hong Kong, which means having temperature checks and so forth reported. So I think that ’ s a reasonable thing to do in order to keep business travel up and running, even as we ’ re all trying to deal with controlling the virus. CH : And you also mentioned solidarity and I think that touches on another question that someone brought up online about some of the social impacts after the 1918 epidemic and the fear, and the, you know, the fear of the other, and foreigners and all that. And how do we get through this without that kind of fallout and you know, how do we, kind of, keep ourselves together and looking out for each other? DA : I think that ’ s such a hugely important question. And I mean, in one sense it ’ s easy, because the virus is an adversary to every human being equally, right. We are all completely equal in relationship to it. And so what we are really all aspiring to here is sort of transformation of our basic socioeconomic infrastructure in a way that puts us all on a footing to be pandemic-resilient. So I ’ ve been using the metaphor we need to put ourselves on a war footing to mobilize the economy to fight the virus, and I stand by that in a sense that we do need to mobilize the economy. But really at the end of the day, it ’ s not a war against a human adversary or anything of that kind. And so what we ’ re really talking about goes back to the questions about the health infrastructure, health care. We ’ re really talking about achieving a transformed peace situation where our economies and our societies are pandemic-resilient. That ’ s the real goal here and it really does require an investment, so because of the 2003 SARS experience, Asian nations have been investing over the last five years or plus, in pandemic-resilient equipment and infrastructure. We haven ’ t done that in the US, so we find ourselves in a position where we have to accelerate in a matter of months, something that has taken other people years to build and develop. So I think really focusing on that, and the goal is an economy that ’ s not vulnerable to pandemic, right. I mean, because we don ’ t want to leave this pandemic and have the economy be just as vulnerable to pandemic at the end of the pandemic as we were at the beginning of it. We don ’ t want to be vulnerable this way. And so in that regard, the job is to build in that infrastructure for pandemic resilience ASAP. CA : Wow. CH : Thank you. CA : Danielle, given the price tag you ’ re talking about on this, half a trillion dollars, basically, up to. That ’ s significantly less than some of the multitrillion dollar numbers that are getting thrown around, so, I mean in terms of the scale of the problem, it ’ s probably an appropriate number. But it sounds like, to have any chance of doing this,

this would have to be a kind of federal initiative at some level. DA : Yes. CA : We have a problem that more than half the country fundamentally doesn ' t trust key parts of the administration, let ' s say. How could this be framed in a way that could build trust and make it feel like this is the country as a whole, that there ' s this coalition of trusted voices who are the final decision makers on this? DA : So we have this incredible federalist system, and we need to see it as an asset. It ' s modularized and flexible, and we need to activate that. We do need all the parts of the system working, so we do need the federal government working on behalf of this, we need the state governments working and municipal governments. On the federal end of things, we need Congress to fund. So in the first instance, there ' s a really big need for funding legislation, and also, Congress can really help by directing investment, not just in the testing program itself, but in the national **service** corps, probably flowing through state governments, through the national – The reserves in every state. That would be sort of health reserves. You know, really expand that program with a combination of employment program and backing up that sector. So there ' s a lot for Congress to do as a part of this. For the testing program, we really do need the kind of procurement order to produce capacity that the Defense Department is the best example of. So in the ideal, a sort of testing supply board that brought in leading figures who are masters of supply chain logistics from the private sector, working in close coordination with the federal government would be great. The White House has recently, in the last week or so, begun to put in pieces of architecture that goes in this way, sort of a testing supply czar, for example, an admiral, I believe. So we need people of that kind who are really superb masters of logistics, procurement, contracts and that sort of thing, to be able to ramp up an active, functional supply chain for testing to deliver at the order of tens of millions of tests a day. So we do need [unclear], absolutely is a key part of that, key driver of that. And so it ' s a time for all the parts of our government to come together and do their respective pieces. CA : So I ' m kind of in awe at the scale at which you ' re thinking. I guess as we wrap up here, if I might, I ' d love to just go to a bit more personal place. Like, I ' m just curious about you and what is it in your past that is, sort of, is providing the fire right now, the drive to try to do this? How are you? How are you feeling about this? Tell us a bit about you, please. DA : Well, that ' s a very generous question. You know, I love this country. I ' ll admit that ' s where the motivation starts from, in the sense that, like, lots of people would say that I ' m a global humanitarian, and watching the world succumb to the disease motivates me. I think of Paul Farmer for example, as an example. And I respect that and I get that, but at the end of the day, I love my country. And it hurt, just hurt, in the beginning of this, and what hurt particularly, was I was very clear, early on, that I was getting better information as a member of the Harvard faculty than my fellow parishioners, than the people who were serving me in restaurants and cafes, and it just like, that made me angry, in all honesty. As a combination of those two things, I was like, A, I want to understand this, and B, I want to share what I understand because it ' s not fair that people like me get it, and that ' s not being shared with other people. CA : Wow, that ' s powerful. I think all of us, we all feel this weird mixture of almost guilt at how fortunate a position some of us are in. Certainly a lot of gratitude, anger. Were you persuaded, Whitney, by this idea, by the possibility of it? CH : Sorry, you ' re meaning me. CA : Sorry! Did I say Whitney? CH : Totally OK. Whitney ' s your usual pal. CA : I ' m the world ' s worst person on names, and Whitney and I have been hanging out here the last few weeks. Corey. CH : It ' s absolutely fine. Being mistaken for Whitney is a huge compliment. It ' s very persuasive, and I think so hopeful to hear a constructive plan and a feeling that there is a path out of this that is both possible for us as humans, to get back to being together, but then also as an economy and as a country. I ' m really inspired by your work and so grateful to you for sharing it with us. DA : I appreciate that, thank you. I ' m really glad to get a chance to talk about it and share the knowledge that our group has acquired over the last month. So thank you. CA : So if someone wants to keep in touch with the progress of this idea, what should they do? DA : OK, so now I should know our website URL by heart, but of course, I don ' t, I ' m afraid. If somebody googles"COVID,""Safra,""Allen,"that ' s my surname, our website will come up. So if you just remember those three words,"COVID,""Safra,""Allen,"and Google that, you should get to our white papers, op-eds, things like that. We are hoping to have our full policy road map published by the end of the week. That ' s our target goal. CA : Yeah. It ' s : ethics. harvard. edu. DA : OK. Exactly, that takes you to the main landing page, and then to the COVID site. CA : And then to the COVID-19 from there, yeah. DA : Exactly. CA : Alright, well, thank you so much, Danielle, I found this absolutely fascinating. DA : Thank you. CA : It ' s going to take – I mean, this is not an ordinary idea. We don ' t often at TED have someone come and say,"Yeah, I ' ve got this idea for how to spend half a trillion dollars, and it could make a difference between the US and other places around the world actually getting the economy going again. That ' s not usual, so this has been a gift to us today, thank you for that. DA : Thank you. CA : To everyone listening, this is an important debate. And it ' s not finished yet, there will be many other contributions to ideas like this, I think. DA : That ' s for sure. CA : Yeah, chip in, chip in. Thank you all so much for being part of this today. We ' re back again tomorrow. Corey, do you have details on that? CH : I do. And also, you can listen to this conversation on our website TED. com or on Facebook, and you can also listen to the recording of it through TED Interview. So if you missed any parts of it or you want to pass it along to a friend. We have some more amazing speakers coming up I might glance at my cheat sheet, but tomorrow we have Esther Choo, who is an emergency physician and professor and she is going to share with us what she ' s seen on the front lines of this crisis. On Wednesday, Chris and I will be speaking with Ray Dalio, the founder of Bridgewater, and he is going to address the market and economic implications of this pandemic. And on Thursday, we have two speakers, Gayathri Vasudevan, who is going to share with us what ' s happening in India, and Fareed Zakaria, a journalist. Friday, we ' ll wrap things up with a musician and artist Jacob Collier. So we have a lot of amazing things coming up. CA : We do, so calendar it if you can, apart from anything else, we just like your company here every day. We ' ll get through this together. Thanks so much for being part of this. Danielle, thanks again. DA : Thank you, goodbye. CH : Bye.

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Corey Hajim : Today, our guest is Dan Schulman, CEO of PayPal. When most of us think of PayPal, we think of buying something online or paying a friend back for a drink using Venmo. But PayPal has also become a major **financial services** player, often acting as an alternative to a traditional bank. During this pandemic, PayPal has supported small businesses around the world by providing loans, waiving fees and increasing cash back programs. It has also worked with the US government on its Paycheck Protection Program, as well as distributing stimulus checks. It has enabled an outpouring of generosity online as well. The trend towards digital payments, or what we might now want to think of as"contactless payments,"has massively accelerated, and it ' s changing forever how we think about commerce. So I ’ m really excited to have Dan here with us. Thank you so much, Dan. Dan Schulman : Thanks for having me, Corey. Pleasure to be here with you. CH : Glad to see you. So let ’ s dive right in. Within a few months of this pandemic ’ s arrival, more than 30 million people have filed for unemployment in the United States alone. These are certainly unusual circumstances, but it seems clear we were running very close to the edge, and now so many businesses and their employees are facing huge **financial** challenges. How worried are you? DS : Well, I think the crisis has exposed three things. Obviously, it ’ s a health crisis for so many people. Second thing is, that health crisis has ricocheted, and the world is now in an economic crisis. And the third crisis that we don ’ t talk so much about but I think is impacting the way that we ’ re going to live our lives going forward is : this is a psychological crisis as well. People are reexamining their place in the world, what ’ s happening in the world, how they ’ re going to live their lives, both in the pandemic and postpandemic. And so I think this is something that each of those phases will need to be dealt with. But you said this, and I completely agree with you : there was an economic crisis happening well before the pandemic exposed this. It ’ s kind of like the water level came down and exposed what was already there. You had, for instance, in the US, 185 million adults in the US **struggling** to make ends meet at the end of the month. You have over 70 million adults that are really outside of the **financial** system, spending over 140 billion dollars on high interest rates, unnecessary fees and **struggling** as well. And so I think what this has really done – because you can ’ t ignore 20, 25 percent unemployment rates – it ’ s exposed this crisis and forced a lot of people into, maybe, **actions** that they might not have taken without this crisis happening. CH : Yeah, I think that ’ s right. There are so many challenges and so many opportunities, and I think you ’ ve spoken of this opportunity of digital transactions being helpful to people, and obviously the trend, as you ’ ve said, has massively accelerated and pushed us into this world even further. So I ’ m curious : What does the world look like without cash? Or less cash? What are the advantages and what are the challenges of making that transition? DS : I think some of the trends that are emerging coming out of this pandemic or coming into it and as we look forward is, clearly, this has been a discontinuous change in the trend line as we move from physical to digital. I think we ’ ve accelerated many forms of digital capabilities by three to five years. And that can be from digital payments to telemedicine to really changing the face of retail and how we think about retailing, changing the face of entertainment, even changing the way governments think about managing and moving money and really thinking about digital currencies going forward. And so I think there are a tremendous number of changes that will occur during this pandemic and coming out of it. Digital payments is obviously one of the big ones that will happen. I mean, cash has been around for quite some time, thousands of years. I would not be so **bold** as to predict its full demise. Many people have been wrong doing that. But there is no question right now that you will see an acceleration of the demise of cash. Last year, you had over 18 trillion dollars of cash spent at retail. Eighty-five percent of the world ’ s transactions today are done in cash still. But the really big change right now towards digital payments, and that ’ s both the advent and the acceleration of commerce that ’ s happening, as well as the shift to in-store contactless payments, as you said, and the real impetus for that is health reasons. People do not want to hand over money. They do not want to touch screens. They don ’ t want to pick up a pen and sign at the point of sale. And so there is a demand for contactless payments and digital payments to keep social distancing requirements in place, to protect the health of cashiers, to protect the health of consumers. And I think we are going to see, we are already seeing in our business, a surge in digital payments across the world. CH : It seems like a great opportunity, but how do we make sure that this transition is inclusive? I mean, you ' ve talked about how so many people are underserved by the traditional banking industry. How do we make sure that those people have that opportunity? And it feels like a smartphone becomes an essential item. How do we address that? DS : Yeah. I do think that a **mobile** is really a key to unlocking this. I ' ve often said that, really, one of the big moon shots for the **financial services** industry is this idea of not just **financial** inclusion. Most people define **financial** inclusion by somebody having access to a bank account, but just having access to a bank account is not nearly enough. I think what we need to aim for is how do we think about **financial** health? How do we make sure that people have the ability to have some wherewithal to create savings to withstand some kind of **financial** shock to the system? I do think that **mobile** phones will be the way that this occurs and will be very inclusive going forward. There are going to be something like six billion smartphones in the world over the next several years. The cost of a smartphone is plummeting. I think in India now you can buy a smartphone for under 25 dollars. So you ' re going to have ubiquity of smartphones across the world, and, in fact, what ' s very interesting is, in lower-income populations, there is a greater penetration of smartphones than in higher income because the smartphone is the only device that somebody has. Higher-income individuals may have desktops or iPads, that kind of thing, but lower income can afford one device, and they choose it to be a smartphone because they can get and live their life through that one device. And think about that one device. Really, you have all the power of a bank branch in the palm of your hands. And when you can start to create distribution of **services**, **financial services**, through a smartphone, you then are able to manage and move money in ways that we couldn ' t do traditionally. In the physical world, if you get a check, you need to then go to a cash checking place to cash it. You stand in line for 30 minutes. They then charge you anywhere between two and five percent to just change the format of currency from a check to cash. And then you have cash and you want to pay a bill. You need to stand in line again at a bill pay, and then you have to pay maybe 10 dollars for an individual bill as a fee. If you do that via a smartphone, I believe that not only do you save a tremendous amount of time, because if you ' re outside the **financial** system, managing and moving money is practically a part-time job to go and do that, so not only do you save time and return time to individuals, but you can cut the cost of transactions by anywhere between 50 and 75 percent. And remember that $140 billion number that I gave you? And that ' s just in the US. Imagine if you could cut that in half and return that to the most vulnerable populations that need it most. So I think there ' s tremendous promise in the use of technology to help provide both inclusion and make sure there aren ' t digital haves and have-nots, but also to start on this journey towards **financial** health. CH : Yeah, I think a lot of people don ' t realize that you don ' t need a bank account or even a credit card to open a PayPal account, which is super-interesting. I mean, do you see a time where traditional banks don ' t exist or at least play a much smaller role in the **financial services** industry? DS : Well, I think the entire **financial services** industry is evolving right now, and so I think banks will always play a role, or as far into the future as I can see, but it will evolve. I mean, think about basic credit cards. Today, you think about a credit card, and you think about it predominantly as a form factor, something that you pull out of your pocket. Sometimes there ' s status associated with what you ' re pulling out of your pocket, depending on the color of that credit card. But really I think those form factors start to go away and become embedded in digital wallets.

So credit will always be an important element. You know, most people in the world, it isn ' t that their cash outlays exceed their cash intake. It ' s just that they ' re not evenly distributed. So there are times where your cash outflows exceed your cash intake, and there, you need some form of credit to make up that difference. And so I think forms of credit will always be an important element. But the way that you extend credit will change going forward, the way that you think about **scoring** people in terms of can they handle credit. You know, traditionally, in more developed countries, you use what ' s called FICO **scores** or bureau **scores**, but those ignore so many of the **financial** transactions that people who are outside the **financial** system do, like paying rent or paying their bills on time. And with the data and information and machine learning around that – and we need to be careful that there aren ' t **biases** built into those algorithms – we can start to do things that could never be done before. I ' ll just give you one quick example. We ' re one of the largest providers of working capital to small businesses in the world. We ' re probably one of the top five in the United States. So we ' ve done over 14, 15 billion dollars of lending of working capital to small businesses. Seventy percent of that goes to the 30 percent of counties where 10 or more banks have closed branches. And where do banks close branches? Banks close branches in neighborhoods where the median income is below the national average, which makes sense because for a branch to be profitable, they need a certain amount of deposits for that branch to actually be profitable. And so, in lower income neighborhoods, branches are starting to close. So why are 70 percent of our loans in those lower income neighborhoods? It ' s because we do machine learning. We don ' t even look at FICO **scores** or bureau **scores**. We look at a number of different data elements. And so we can lend into those lower income neighborhoods where nobody else can, and when we do that, the average sale of a small business goes up by 22 percent. And imagine the impact that has on communities and neighborhoods where they can finally get the working capital to expand those small businesses. And I think that ' s a perfect example of the promise of what technology and **financial services** married together can do. CH : I think it ' s so interesting. I ' m curious. The tech industry has been criticized for amassing power over society, not that the banking industry isn ' t criticized. But what do you say about people who might be worried about tech companies taking on even more influence and control over what ' s happening in their lives? DS : Yeah. Well, I think what ' s so important for any company and tech companies is to respect the boundaries in terms of what consumers expect from a company that serves them. I think the most important brand attribute that a company can have is trust, and trust comes from the understanding that a company respects your **privacy** and will not sell your data or information, that it can perform transactions in a secure manner so that your transactions are protected. And I think those are kind of foundational, and I think any company needs to respect that. They need to assure that consumers have the **privacy** that they desire and the safety and security that is required to serve them the right way. CH : And obviously, you ' ve gained a lot of trust with the US government. Maybe we could talk a little bit about how you ' ve been working with them to distribute some money through the Paycheck Protection Program. And I was curious, I ' ve been reading about it, and it sounds like 30 million-ish small businesses in the United States are able to get those funds, but only six million have received the loans. What do you think ' s happened? DS : Yep. Well, I think initially, the government – and I give them a lot of credit – they responded quite quickly with a 3 trillion dollar stimulus package. These are massive numbers that were happening in very condensed time frames. We were working with various agencies, very closely with the Treasury Department, in terms of distribution of the stimulus. And they were working literally night and day on this. The Small Business Administration was working night and day. But these are volumes that have never been seen before running through these systems, and the first tranche of those loans was very difficult. There were a lot of technical difficulties in getting those out to small businesses. And that first tranche was not enough, and it was quickly used, and there are still a host of small businesses that needed money. The second tranche that came out is still actually in effect. It has not been used up, and we are continuing to lend on that. We ’ ve been able to lend to some 50, 000 small businesses. We ’ ve lent out about 1. 7 billion dollars, and our loan size, which really I ’ m proud of, is about 31, 000 dollars. The average that a bank does is between 100 and 125, 000 dollars. So we are lending to these true small businesses on Main Street, and I ’ m proud that we ’ ve been able to go do that, and I think we should give credit to the US government and governments around the world that are taking this quite seriously and putting a tremendous amount, a percentage of their GDP, towards the rescue of small businesses and towards trying to take care of consumers that find themselves in really difficult straits right now. And we ’ ve been trying to, instead of people mailing out checks, which is ridiculous in today ’ s world – people aren ’ t living where they think they ’ re going to be living, they ’ re with their parents or with friends or in a different location, and mailing a check and then having to take a check and go somewhere, which you can ’ t even go if you ’ re sheltered in place, to cash it, doing that electronically just makes a ton more sense – and we ’ ve been working with the IRS and Treasury and other government agencies to distribute that electronically. CH : Yeah, that makes a lot of sense. It ’ s a massive, massive project for all of us. Whitney is here with some questions from our community. DS : Hello, Whitney. Whitney Pennington Rodgers : Hello Dan. How are you? So the community has some interesting questions following up on what you were talking about earlier about security. We have a question from Marc – and I apologize in advance if I mispronounce your name, Marc – Marc Vanlerberghe :”The move to digital cash could be one more step towards creating the perfect surveillance state. How do we avoid this from happening?”DS : Yeah, well, this is what I was talking about, Marc, before. I mean, I think this idea of trust is incredibly important. I think the only companies that will be successful – and I think we hold a lot of this in our own hands as consumers, by the way ; we need to be aware of data and information that we ’ re giving and to what companies we ’ re doing that with – but I think the companies that will be successful are those that have a high degree of trust, and trust happens by protecting your **privacy** but also very much assuring that your transactions in a digital world are safe and secure. I mean, the idea of cybersecurity has always been important, but is ever more important as we move from physical to digital, and that ’ s where large data sets are important, because a consumer ’ s identity is stolen every two seconds. Every two seconds, some consumer has their identity stolen. And so we have to be, for instance, we have to be sure that even when you sign in with your credentials, they ’ re actually real credentials. We have to look at 30 to 100 different elements of that transaction to make sure it ’ s really you before we let that money out of your account. And so there is a combination of making sure you have enough data to protect somebody but also assure that your **privacy** is held sacrosanct, and I think that is a balancing act and one that needs to happen in order for us to do this successfully. WPR : Great, and actually sort of going from digital cash to digital currency, we have another question from Simone Ross in our community about the opportunity that exists for digital currency. She mentioned that PayPal pulled out of Libra. What would it take for a truly inclusive digital currency to take hold here? DS : Yeah. I think there is a tremendous amount of promise as we think about digital currencies. Our pulling out of Libra had nothing to do with our firm conviction that blockchain and other forms of maybe stable coin currencies are extremely important and can be very, very helpful, especially in different parts of the world. As we think about stability in different parts of the world where currencies can fluctuate up and down, to have a more stable currency where somebody can know, if they have that, that it ’ s going to be worth x amount, and that they can transact, either with other individuals around the world or, importantly, at merchants around the world. And we are looking at all forms of digital currencies right now, working hand in hand with a number of different governments, and I think we should all think about how technology is going to evolve and how currencies will evolve as a result of that. And I think this crisis has really opened the eyes of many governments around the world as to the need for different tool sets to create stimulus and to efficiently and quickly and effectively distribute funds to their citizens. WPR : Great. Well, I ’ ll be back shortly with more questions, and I ’ d just love to remind the community that you can ask questions on the”Ask question”feature. Be sure to use the pull-down tab to select Episode 2, so those questions come. Thank you. DS : Thanks, Whitney. CH : Thanks, Whitney. Dan, I want to go back to something we touched on in the beginning about **financial** wellness. PayPal has done something unique in terms of calculating how much to pay people and how much you should spend on benefits. Traditionally, wages are set by the market, but you ’ ve found that paying as much or even more than other companies wasn ’ t always enough. Can you tell us about that moment? DS : Yeah. So I said, kind of, in our opening, in one of my opening statements, that two-thirds of Americans **struggle** to make ends meet at the end of the month. They are financially **stressed**, and it kind of wreaks havoc in their life. I did a study to look at PayPal employees. We did a research study, and I did it because I thought I was going to get back this great information that I was going to talk about at an employee meeting about how well we pay, because we pay, to your point, at or above market in every single location around the world. And what I found is, unfortunately, like the rest of the world, even though we paid at market or above market, 60 percent of our operations personnel, our entry-level employees, our hourly workers, face the same thing. They **struggle** to make ends meet. And that was simply unacceptable for me. I think the world is changing in terms of the responsibility of corporations, the responsibility of CEOs. We have a lot of different stakeholders that we try to satisfy, from regulators to shareholders to customers to employees. But I think the number one responsibility that we have is the health – **financial** health – of our employees, because nothing could be more important to a company than to have financially secure, **passionate** employees working for you, because nobody is going to serve customers better than employees who feel a part of something and feel financially secure and glad to be a part of that company. And so then the real question becomes : How do you measure that? Because a lot of people think about living wages or a minimum wage. And we thought that was insufficient, and we came up with a measurement we called”net disposable income,”which is, basically : After you pay taxes and your basically essential living expenses, how much money do you have left over for discretionary things or to save? And here ’ s the really unfortunate thing – and I ’ m not proud of this, but remember, we were paying at market or above, so I thought the market would take care of this, right, by doing that – we found that for that population, they had four to six percent NDI, net disposable income, after paying taxes and essential living expenses. That is not enough. You are going to **struggle** to make ends meet. And by the way, NDI changes location to location to location around the globe, right? There ’ s a different NDI in Manila, a different NDI in Omaha, Nebraska, than there is in New York City, etc. And so we basically said to ourselves, we need to take NDI to 20 percent. Because at 20 percent – and that ' s a huge shift, from four to six to 20 percent – but at 20 percent, you actually have the ability to save and to put money away and to take care of discretionary expenses. And so we did a pretty massive reorientation of our compensation systems. We lowered the cost of benefits by 58 percent, because benefits are like a regressive tax, you pay the same amount no matter what your salary is. And so we had a lot of employees who weren ' t taking health care benefits, because it cost too much to be able to do that. So we lowered it by 58 percent. We made every single employee of PayPal a shareholder and an owner of the business, and we gave them pretty big grants so that they could be a part of the success of PayPal going forward. We raised salaries where we needed to go and do that. And then we wrapped all of that into a **financial** education program, because people had never gotten equity before, they were trying to think through,"How do I save now that I ' ve got incremental dollars to go and do that?"And that cost us quite a bit of money to go and do that, but I really feel, just like how we spend a lot of money to take care of customers, as you mentioned up front, in COVID-19, that companies need to stand for more than just making money, for more than just **maximizing** our profits next quarter. I firmly, firmly believe that the costs associated with taking care of our employees, taking care of our customers, will benefit us in the long run multiplefold over the costs associated with doing that. And we ' re already beginning to see some of the impact of that. And so, I think every CEO, every company, needs to really now start to think about, especially maybe as a result of this crisis, but as I mentioned, we had a crisis before this, how do we put our employees first, take care of them? Because if you do that, you ' ll take care of customers, and if you take care of customers, you ' ll take care of shareholders, inevitably. And so this has been a huge part of it about for the last year or so. CH : It ' s so interesting, and it brings up so many questions, I think, for me and probably our community as well. I mean, PayPal is a hugely profitable tech business, huge free cash flow and big margins. Do you think this model is something that every company can do, whether it ' s a tech company, a manufacture, a meatpacking business? I mean, is this what everyone should be focused on? DS : Well, I think that – and I don ' t want to moralize or tell other companies what they should do – but to me, I think everyone should understand the **financial** health of their employees. That ' s a baseline thing to go do. What you do post-that is up to maybe your **financial** strength as a company or where you put your order of priorities. But what I ' ve found is, I thought the market could tell you that, and this is why I say, in many ways – you know, I ' m a big believer in capitalism. I think it ' s, in many ways, the best economic system that I know of. But, like everything, it needs an upgrade. It needs tuning, and at least for these vulnerable populations, just because you pay at market doesn ' t mean that they have **financial** health or **financial** wellness. And I think everyone should know whether or not their employees have the wherewithal to be able to save to withstand **financial** shocks, and then really understand, like, what can you do about it? I think this NDI measure is a really interesting one. It takes some time to go do it, because you have to be quite thorough and you have to really understand living expenses by location and what tax jurisdictions there are. But you need to create an NDI that ' s to a certain level where people aren ' t **struggling** to make ends meet. Because if people are **struggling** to make ends meet, they are not as productive at work. They ' re worried about, like, what am I going to do with my kids? My kid just got sick. I don ' t have health insurance. I think there ' s a spiral that occurs. You think you ' re actually saving money by paying less, but the reality is, at least in my belief system, you take care of your employees, and other things naturally flow from that. They are more productive. They love being a part of that company. They take care of customers better. And all of those things inevitably accrue to the benefit of a company in terms of how it ’ s trying to serve its ultimate end market. But it starts with your employees. CH : So obviously you believe in this”capitalism needs an upgrade,”and I think NDI is something so many companies should adopt. But do you think this happens through benevolent corporate activity? I ’ m channeling my inner Bernie Bro here, but I think a lot of people would be skeptical that we should trust companies to do better at this point. Should the government step in to raise minimum wages, do other things to protect workers in a more structured way? DS : Look, I think the government clearly has a role to play, and I think the private and public sectors need to work closer together to address so many of the issues that we face in our societies across the world, whether that be income inequality, environmental issues, health, protections, that kind of thing, **privacy**. But the way that I think about this is, it ’ s very difficult for governments to regulate around this, because there are so many different ways of thinking about it. If I were another CEO, and this is like, it ’ s actually in your best interest to go and do this because it ’ s a competitive advantage. Like, we attract, I think, some of the best talent in the world to PayPal, because we have a **mission** that people believe in, that we actually are trying to make some sort of positive difference. I ’ m not saying we ’ re the be-all and end-all, but I don ’ t think people should shirk their responsibilities of at least making a small difference going forward. If enough companies did that, if enough governments did that, it would make a real difference in the world. And then the second thing is, you have to have values that support that. And those values are incredibly important. Those values should be all about inclusion. They should be about having a diverse **workforce**. They should be about **financial** wellness. And when you do that, and you attract the very best talent, then by definition, I think the single biggest competitive advantage for any company is their **workforce**. Strategies are great. A whole number of things are great. You have a great **workforce** that ’ s **passionate** about what they ’ re doing and is financially secure, and they will do amazing things. And I think it ’ s that kind of competitive advantage that will spur companies. So there needs to be a set of CEOs and companies that start to move in this direction, and I believe you ’ re beginning to see more do this. And once that happens, it starts to tip everything, and I think more and more need to do it to maintain their competitive positioning. And that may seem like a self-serving way why people are doing it, but honestly, I don ’ t care whether they ’ re doing it out of the goodness of their heart or they ’ re doing it because it ’ s competitively a disadvantage if they don ’ t. Creating **financial** health for our employees is the goal, and we ’ ve got to get that done. CH : Yeah. I mean, it sounds like you think of this as a win-win, but it also sounds like you ’ re willing to maybe think about your employees first and sell it to your shareholders later. Whitney is – oh sorry, go ahead. DS : No, no, no – I was just going to say, I actually do believe that, and I think the idea of a multistakeholder capitalism, that is a time for today, and we cannot just think that we have one stakeholder that we need to satisfy. We live in our communities, we live in this world. To have people **struggling** day in and day out is not good for any company, and... We can only do x amount, but we can actually create **financial** health for our employees, and we should. WPR : Great. So we have so many questions coming in from the community. One here is from Lara Pearson, basically about whether PayPal would consider become a B Corporation.”Are you familiar with the B Corp movement, environmentally and socially responsible, multiple-bottom-line for profits? Presuming so, has PayPal considered or would it consider becoming a certified B Corporation?”DS : Yep. I ’ m familiar with B Corp. We have no intention to move to becoming a B Corporation. I think the values and what we are trying to do are very aligned with assuring a multistakeholder point of view, but what I really want is for this to be a movement across major corporations across the world. And you ’ re not going to have major corporations around the world moving into B Corp. There ’ s a lot of other side issues involved with being a B Corporation as opposed to just a publicly listed company, and so that ’ s going to be a long way before that happens. And so what I ’ m really trying to do is encourage and demonstrate that being multistakeholder, that putting employees first, creates competitive advantage. And I think I ’ m not the only CEO who ’ s feeling that, by the way. I think people like Satya Nadella from Microsoft are doing a great job, Marc Benioff from Salesforce. I could go through quite a list of names. But the list is not long enough yet, but I think there ’ s some quite important names and individuals around the world who are now talking about multistakeholder capitalism, and I think that ’ s an important element as we think about our economies and way of life looking forward. WPR : And there was so much interest also in your net disposable income program and a lot of questions around that, and one which I think is along these same lines from Juan Enriquez asking about a rational way to address extreme income disparities. And perhaps you could expand beyond this program, just sort of ways that we might think about this in a smarter way moving forward. DS : Yeah. Well, there ’ s no easy solution, or it would have been done. So I think there are a couple things that I think about that may not fully address extreme income disparities. Again, I try to think pragmatically about these things, and, like, what can we really do to start to address this? And again, I think about, if we could take one step and then another step, then you ’ re starting your journey, and without getting overwhelmed by how far away the end state is. So one, I think companies need to take care of their employees, and I think that will immediately help to address some of these income disparities. Number two, I do think that, ironically, if you have less money, it costs you more to manage and move it, which, think about that : the less money you have, if you ’ re outside the **financial** system, the more you spend to manage and move your money. And I think that technology is at least a foundational way for us to think about how do we cut the basic costs of managing and moving money by 50 to 70 percent, like [check-cashing], sending remittances, which are such a huge, important part of the world ’ s economy. You know, you do it a traditional way, you go into a store and send the remittance to another store and somebody goes and picks it up. First of all, incredibly time-consuming, and it can cost between eight and 12 percent of that remittance amount that you ’ re sending. So if you ’ re sending a hundred dollars, the recipient who so desperately needs it is getting 88 to 90 dollars. If you do that electronically, digital wallet to digital wallet, that can be like three percent, so you can get 97 dollars from that. And so I think there are ways of addressing the costs. As I mentioned, there is so much money spent on unnecessary fees and high interest rates, and if we can drop that by 20 percent, 30 percent, the amount of money we can return to vulnerable populations is quite large and will start to make a difference. WPR : That ’ s great. We have a ton of questions from the audience, just one more before we turn things back over to Corey with her final questions. This one is from Anna Tunkel, which is just, I think, as we are rounding to the end of the interview here,”What are you most optimistic about, and what do you see as the biggest opportunities for ’ Building Back Better ’ after COVID?”DS : Well, I mean, one thing I ’ m actually optimistic about – and I ’ ve always been a believer in the human spirit and the power of an individual to make a difference. I know that sounds very cliché, but I truly believe it, and I think every one of us can make a difference. But here ’ s what I ’ m seeing. I ’ m beginning to see that at a much larger scale than I ’ ve ever seen before. You know, we have different **platforms**, either the PayPal **platform** or the Venmo **platform**, Venmo here in the US, PayPal across the world. The amount of giving that ’ s happening through those **platforms**, whether it be to local businesses, to artists, to musicians, to bartenders, to places of worship, to schools, to NGOs, to charities has exploded on the **platform**, exploded. We have helped to raise on the PayPal **platform** since COVID-19 struck 2. 8 billion dollars for NGOs and charities – 2. 8 billion. That ’ s incredible, the amount of generosity that is pouring out from the global community around this. And we ’ re just seeing people randomly pay it forward. Somebody gives 20 dollars to a bartender, and that bartender takes 10 dollars of that and gives it to somebody else. And we ’ re watching that over our **platform**, and that gives me a sense of optimism. I also feel like this period of time has exposed a number of things that were happening but were invisible, and I think when things become visible, that ’ s when you can start to address them, and I think there ’ s a lot of attention on some issues that should have had attention before, but vulnerable populations don ’ t have as loud a voice as others, and now that voice is being heard, because you can ’ t ignore it. And hopefully, that will create progress against some of these structural inequalities that have been there for a long time. WPR : That ’ s wonderful. And there ’ s so much interest online. You have some other questions to ask as well. CH : So I think we have one more from our community from Jacqueline Ashby. Anna sort of stole my last question, which was to restore our faith in humanity. But, there ’ s so much interest coming in about NDI. Is there a way for people to learn more, for you to share your study and your methodology? DS : Happy to do so. There is nothing proprietary about it. We would love for this to be – look, and this may not be the be-all and end-all measurement. It ’ s the best one we could come up with, but if working within the community, we can evolve it and think about maybe things that it missed or maybe things that could be done better, that would be fantastic. I don ’ t know the best way of doing that. I ’ ll leave that to Corey and Whitney to help me think that through, but of course we ’ d be willing to share it. There is nothing about that that I don ’ t want to share. CH : Sounds like a good TED Talk. Thank you so much, Dan. This has been a super-interesting conversation. I think we could talk for another hour, but thank you so much for being here. DS : Thank you, Corey. Thank you, Whitney. Thank you, everybody. WPR : Thank you, Dan. Thank you.

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Whitney Pennington Rodgers : Before we really dive in to talking specifically about Google ’ s work in the contact tracing space, let ’ s first set up the relationship between public health and tech. You know, I think a lot of people, they hear”Google,”and they think of this big tech company. They think of a search engine. And there may be questions about why does Google have a chief health officer? So could you talk a little bit about your work and the work your team does? Karen DeSalvo : Yeah. Well, maybe I ’ m the embodiment of public health and tech coming together. My background is, I **practiced** medicine for 20 years, though a part of my work has always been in public health. In fact, my first job, putting myself through college, was working at the state laboratory in Massachusetts. As the story will go with Joia [Mukherjee] we ’ re reconnected again, a Massachusetts theme. And I, across the journey of the work that I was doing for my patients to provide them information and the right care and meet them where they were medically, translated into the work that I did when I was the Health Commissioner in New Orleans and later when I had other roles in public health **practice**, that really is about thinking of people and community in the context in which they live and how we provide the best information, the best resources, the best **services** that are culturally and linguistically appropriate, meet them where they are. And when the opportunity arose to join the team at Google, I was really thrilled, because one of the things that I have learned across my journey is that having the right information at the right time can make all the difference in the world. It can literally save lives. And billions of people come to Google every day asking for information, and so it is a tremendous opportunity to have that right information and those resources to people so that they can make good choices, so that they can have the right information, so that they can participate in their own health, but also, in the context of this historic pandemic, be a part of the broader health of the community, whether it ' s to flatten the curve or keep the curve flat as we go forward. WPR : And so it sounds like that there is this connection, then, between public health and what Google ' s work is in thinking about public education and providing information. And so could you talk a little bit about that link between public health and public education and Google? KD : Definitely. You know, the essential public health **services** include communication and data, and these are two areas where tech in general, but certainly Google, has an opportunity to partner with the public health system and with the public for their health more broadly. You know, going back to the earlier days of this pandemic, towards the end of January, Google first leaned in to start to put information out to the public about how to find resources in their local community, from the CDC or from other authoritative resources. So on the search page, we put up"knowledge panels,"is the way that we describe it, and we did develop an SOS alert, which is something we ' ve done for other crises, and in this particular historic crisis, we wanted to be certain that when people went on to search, that there was authoritative information, which is always there but certainly very prominently displayed, and do that in partnership with public health authorities. So we began our journey really very much in an information way of making certain that people knew how to get the right information at the right time to save lives. I think the journey for us over the course of the last few months has been to continue to lean in on how we provide information in partnership with public health authorities in local areas, directing people in a certain state to their state ' s health department, helping people get information about testing. There ' s also been, though, a suite of resources that we wanted to provide to the health care community, whether that was for health care providers that may not have access to PPE, for example, we did a partnership with the CDC Foundation. Though the scale of the company and the opportunity for us to partner with public health around things like helping public health understand if their **blunt** policies around social **distancing** to flatten the curve were actually having an impact on behavior in the community. That ' s our community mobility reports. We were asked by public health agencies all across the world, including some of my colleagues here in the US, could we help them have a better evidence-based way to understand the policies around social **distancing** or shelter in place? Which I think we ' ll talk about more later. In addition to that sort of work, also been working to support public health in this really essential work they ' re doing for contact tracing, which is very human-resource intensive, very complex, incredibly important to keep the curve flat and prevent future outbreaks, and give time and space for health care and, importantly, science to do the work they need to do to create treatments and, very importantly, a vaccine. So that work around providing an additional set of digital tools, exposure notification for the contact tracing community, is one of the other areas where we ’ ve been supporting the public health. So we think, as we ’ ve thought about this pandemic, it ’ s support the users, which is the consumer. There ’ s also a health care system and a scientific community where we ’ ve been partnering. And then, of course, public health. And for me, I mean, Whitney, this is just a wonderful opportunity for Big Tech to come together with the public health infrastructure. Public health, as Joia was sort of articulating before, is often an unsung hero. It saves your life every day, but you didn ’ t know it. And it is also a pretty under-resourced part of our health infrastructure, globally, but especially in the US. It ’ s something I worked on a lot before I came to Google. And so the opportunity to partner and do everything that we can as a company and, in this case, with contact tracing in partnership with Apple to create a very privacy-promoting, useful, helpful product that is going to be a part of the bigger contact tracing is something that we feel really proud of and look forward to continuing to work with public health. In fact, we were on the phone this morning with a suite of public health groups from across the country, listening again to what would be helpful questions that they have. And as we think about rolling out the system, this is the way that we ’ ve been for the last many months at Google, and I ’ m just really... I landed at a place just a few months ago – I just started at Google – where we can have an impact on what people know all across the world. And I ’ ll tell you, as a public health professional and as a doc, that is one of the most critical things. People need to have the right information so they can help **navigate** their health journey, but also especially in this pandemic because it ’ s going to save lives. WPR : That ’ s great. Thank you. So, to talk more about this contact tracing system and the exposure notification app, we ’ ve read so much about this. Could you describe this, a little bit about how the app works, what exactly are users seeing, what information is being collected? Just give us sort of a broad sense of what this app does. KD : Yeah. Let me just start by explaining what it is, and it ’ s actually not even an app, it ’ s just an API. It ’ s a system that allows a public health agency to create an app, and only the API, this doorway to the phone system, is available to public health. So it ’ s not designed for any other purpose than to support public health and the work that they ’ re doing in COVID-19 in contact tracing. The second piece of this is that we wanted to build a system that was privacy-promoting, that really put the user first, gave them the opportunity to opt into the system and opt out whenever they wanted to do that, so they also have some control over how they ’ re engaging and using their phone, basically, as a part of keeping the curve flat around the world. The system was developed in response to requests that we were getting about how could technology, particularly smartphones, be useful in contact tracing? And as we thought this through and talked with public health experts and academics and **privacy** experts, it was pretty clear that obviously contract tracing is a complex endeavor that does require human resources, because there ’ s a lot of very particular things that you need to do in having conversations with people as part of contact tracing. On the other hand, there ’ s some opportunity to better inform the contact investigators with things like, particularly, an exposure log. So one of the things that happens when the contact tracer calls you or visits you is they ask,”Hey, in the last certain number of days,”and in the case of COVID, it would be a couple days before symptoms developed,”Hey, tell us the story of what you ’ ve been involved in doing so that we can begin to think through where you might have been, to the grocery or to church or what other activities and with whom you might have been into contact.”There ’ s some amount of recall **bias** in that for all us, like we forget where we might have been, and there ’ s also an amount of anonymous contact. So there are times when we ’ re out in the world, on a bus or in a store, and we may have come into prolonged and close contact with someone and wouldn ’ t know who they were. And so the augmentation that the exposure notification system provides is designed to fill in those gaps and to expedite the notification to public health of who has a positive test, because the person would have notified, they trigger something that notifies public health, and then to fill in some of those gaps in the prior exposure. What it does not do is it does not use GPS or location to track people. So the system actually uses something different called Bluetooth Low Energy, which is privacy-preserving, it doesn ’ t drain the battery and it makes it more also interoperable between both Apple and the Android system so it ’ s more useful, not only in the US context, but globally. So we built this system in response to some requests to help augment the contact-tracing systems. We wanted to do it in a way that was user-controlled and privacy-preserving and had technological features that would allow public health to augment the exposure log in a way that would accelerate the work that they needed to get done to interrupt transmission – keep the R naught less than one – and do that in a way that we would also be able to partner with public health to think about risk **scoring**. We could talk more about any of these areas that you want, but I think maybe one of the most important things that I want to say, Whitney, is how grateful Apple and Google are – I ’ ll take a moment to speak for my colleagues at Apple – to the great partnership from public health across the world and to academics and to others who have helped us think through how this can be, how the exposure notification system fits into the broader contact tracing portfolio, and how it does it in a way that really respects and protects **privacy** and also is useful to public health. We ’ re still on this journey with them, and I really believe that we ’ re going to be able to help, and I ’ m looking forward to being a part of the great work that public health ’ s got to do on the front lines every day, been doing, frankly, but needs to be able to step up. WPR : That ’ s great, and thank you for that really detailed explanation. And you know, we actually have Chris here with some questions from our community, so why don ’ t we turn there really quickly. Chris Anderson : Yep. Questions pouring in, Karen. Here ’ s one from Vishal Gurbuxani. Uh... Gurbu – I ’ ve pronounced that horribly wrong, but make up your own mind. Vishal, we ’ ll connect later and you can tell me how to say that. KD : Fabulous last name. I love that. That ’ s a Scrabble word. CA :”Given where we are today, how should employees think about returning to work, with so many conflicting messages?”KD : This has been an important part of my work for the last few months. I joined Google in December, and all this started happening. The pandemic in the world first began in November but it got very hot in many parts of the world in the last few months, and we ’ ve been thinking a lot about how to protect Googlers but also protect the community. I ’ ve been talking a lot about what we ’ ve done externally. You know, internally, Google made a decision to go to work-from-home pretty early. We believed that we could. We believed that in all the places across the world where we have offices, that the more we could not only model but frankly just be a part of flattening the curve, that we would be good citizens. So we have been fairly... I don ’ t know if the right word is conservative or assertive, about it, because we really wanted to make sure that we were doing everything we could just to get people to shelter in place and socially distance. A lot of other companies have been doing the same, and I think the choices that people are making are going to be predicated on a whole array of factors : the rates of local transmission ; governmental expectations ; the ability to work from home ; the individual characteristics of the workers themselves, how much risk they might have or how much risk it would be for them to bring that back into their household if they have people living in their household who would be at increased risk from morbidity, mortality, from suffering and death, from COVID.

So these are individual and local considerations. I think for us as a company, we want to, as we ’ ve talked about publicly, we want to continue to be a part of the public health solution around social **distancing**, and so that for us means continuing to encourage work-from-home for our employees and really only be in if it ’ s essential that people are in the **workplace**. And we ’ ve said publicly that we ’ re going to be doing that for many months to come. Now, here ’ s one thing I do want to say, which is, working from home has definite benefits, not only for the pandemic, but for some people, time for commute, etc. I think we ’ re already learning there are some downsides, and there are generic downsides, even just not from work-from-home but school-from-home and just being at home, which is : social **isolation** is real. It causes depression. It has physical impacts on people ’ s bodies ; there ’ s science around this. So as the world is weighing, even beyond the pandemic, when we ’ ve achieved herd immunity because we ’ ve been able to vaccinate the world with a functioning vaccine that creates immunity, I think probably a lot of **workplaces** are going to want to encourage work-from-home. But I just want us also to remember that part of humanity is community, and so we ’ ll have to be thinking through how we balance those activities. CA : And, of course, there are huge swathes of the economy that can ’ t work from home. We ’ re a lucky few who can. And speaking of which, here ’ s a question from Otho Kerr.”Vulnerable communities seem to be receiving a disproportionate amount of misinformation. What is Google doing to help make sure these communities are receiving accurate news rather than fake news?”KD : You know, vulnerable communities is where I have spent most of my career focused. I think with many things that we ’ ve learned as a society in this pandemic were things that we, frankly, should have known. And before I get to the information, I ’ ll just talk about access to **services**, which is to say, and to brag, I guess, on my hometown of New Orleans. One of the early things that New Orleans learned, or remembered or whatever, was that drive-through testing only works if you have a car. So you need walk-up testing, and it needs to be in the neighborhood. We need to meet people where they are, and it ’ s thematic of all the work that we did after Hurricane Katrina in New Orleans was to build back a health care and public health infrastructure that was community-oriented, built with community not for community. Of all the many things that I really do hope last from this pandemic, one of them, though, is that we ’ re being much more conscious of building with especially vulnerable communities and building out policies and processes that are as inclusive as possible. For Google information, we start with, on the search **platform**, for example, adding up knowledge panels, that we spend time making sure are linguistically and culturally appropriate. We tend to start globally, with global authoritative groups like the World Health Organization or the National Health Service or CDC, and then we begin to build down to more focused jurisdictions. On other **platforms** that we have like YouTube, we ’ ve built out **special** channels where we do, because it ’ s a **platform** and we can host content, we ’ ve partnered with creatives – we call them, I don ’ t know, that ’ s a new thing for me because I ’ m a doctor – but we ’ ve partnered with creatives and influencers whose reach resonates with communities. We have had particular programming, for example, for seniors, African-Americans, so”vulnerable”takes on a lot of meaning for us globally and in the US context. Our work is not done, and we certainly every day are thinking about how we can do more to see that the information is accessible, accurate and also, frankly, interesting so that people want to engage. CA : Yeah. Alright, thank you Karen. I ’ ll be back in a bit with some other questions. WPR : Thank you, Chris. And you know, and this is really wonderful talking about more broadly, where you see tech and public health going, and specifically, talking about these vulnerable communities.

And I think one thing, even just beyond Google, it would be interesting to sort of hear your thoughts on where you see tech in general better serving public health, if there are spaces that you think, no matter which tech company we ’ re talking about, we could all sort of come together to better serve the community. Do you have any thoughts on that? KD : I could spend several hours talking to you about that, but maybe I ’ ll just start by saying that I came to tech through the pathway of direct patient care and public health **service** in local community, and I ended up in a role in the federal government as the National Coordinator for Health IT, which, for my background, felt unusual to me, I ’ m just being honest. And I thought, well, I ’ m not really a tech person, but the secretary at the time said,”That ’ s exactly why we need you, because we need to apply tech.”And she had had the unfortunate experience of hearing me chirp about how public health needed more timely data to make better evidence-based policy on behalf of community and with community. This was a source of frustration for me as a local public health officer, that sometimes the data I was working on, though great, was stale by the time I needed to make decisions about chronic disease interventions, or mental health or even violence or intimate partner violence issues in my community. And so the desire to make data useful and accessible to support people in communities is something that ’ s been burning in me for a long time, and what I have learned since I have been out in Silicon Valley is that that desire burns in the bellies of many people who work at Google and Apple and other companies, and it ’ s been really wonderful to see, during this horrible time of the pandemic, the incredibly brilliant engineering and programming and other minds at a company like Google turn their attention on how can we partner with consumers and with public health to do the right thing, to bring the resources that we have to bear. And I said I could talk all day about it because I have many examples from the work that we have done at Google. Maybe I ’ ll just point out a couple. One is to say that we very early on wanted to find a crisp way to help people understand what they could do to protect themselves and their community, to flatten the curve, get the R naught less than one, and this”Do the Five”work that our teams, largely in marketing but then a lot of other people weighed in. It required massive amounts of talent to make that available on our landing page, on search, and then fold it out more broadly. We did that in partnership with the World Health Organization, then the CDC, then with countries all across the world to get simple messaging about staying home if you can and coughing into your elbow, washing your hands. These are basic public health messages that public health has been, frankly, even in **flu** season trying to get the word out, but it became, the resources at a company like a Google, and the reach to billions, it ’ s a **platform** and a set of talents that aren ’ t even the technical, computer vision kind of stuff that you would typically think about. Many other companies in Silicon Valley have weighed in in the same way. I think similarly, we ’ ve been thinking through how we can use tools like the community mobility reports. This is something, a business backer like we have for restaurants. The engineers and scientists said, what if we applied that to retail and grocery stores and transportation to get a snapshot in a community of whether people were using those areas less, whether people were adhering to local public health expectations and sheltering in place, and give that information not only to public health but to the public to help inspire them to do more for their community as well as for themselves. So there has been, I think what I ’ m trying to say, Whitney, is I think there ’ s a natural marriage, and COVID has been an accelerant use case to demonstrate how that can work, and it is my expectation that companies like Google who, certainly for us it ’ s in our DNA to be involved in health, will want to continue working on this going forward, because it ’ s really not just good for what we need to get done in this pandemic, but public health and prevention are part and parcel of how we create opportunity and equity in all communities across the world. So I ’ m **passionate** about the work of public health and very **passionate** about partnership. Can I just say one more thing? WPR : Absolutely. KD : Which is to say that one of the first things that I did before the pandemic started, I had just started in December, and then in January, I did a listening session with consumers about what they wanted, and they said something kind of similar to what you said, which I just want to call out, and that is, they wanted partnership, they wanted transparency and they really felt like there was quite a lot that tech in general could do to help them on their health journey. But their ask was that we did it in a transparent way and we did it in a partnered way with them. And so as we move out of the pandemic, and we ’ re thinking more about consumers, I want to carry some of this spirit also of prevention and helpfulness and transparency into the work that we ’ re going to continue to do for people every day.

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Chris Anderson : Hello, TED community, welcome back for another live conversation. It ’ s a big one today, as big as they get. You know, when we created this”Build Back Better”series our thought was how could we address issues arising out of the pandemic, how could we imagine building back from that. But the events of this past week, the horrific death of George Floyd and the **daily** protests that have followed, I mean, they provided a new urgency which we, of course, simply have to address. I mean, can we build back better from this? I think before we can even start to answer that question, we just have to seek to understand the immensity of this moment. Whitney Pennington Rodgers : That ’ s right, Chris. Right now, so many people in the United States and beyond are grappling with feelings of anger and frustration, deep, deep sadness and really helplessness. No matter who you are, you have questions about what to do now, how to make things better. And as we ’ ve seen, violence like this unfolds for many, many years. What is the path forward? CA : So – We ’ re joined today by a group of activists, organizers and leaders known for their crucial work in social justice and civil rights. We ’ re so grateful to have them here to engage in a discussion about racial injustice in America, the unbearable acts of violence that we ’ ve – Acts of violence against the black community that we ’ ve witnessed, the dangers to a nation riven by anger and fear. And how on earth we can move forward from this to something better. So first, each of our four guests will share their thoughts on how we move forward from this moment. And then we ’ ll engage as a group, including you, the TED community. WPR : And we ’ d like to thank our partner, the Project Management Institute. Their generous support has helped make today ’ s interviews possible, and of course, as Chris mentioned, we want you to take part in the conversation, so please share your questions using our Ask a question feature and continue to share your thoughts in the discussion thread. CA : Thanks, Whitney. OK, let ’ s get this moving. Our first guest. Dr. Phillip Atiba Goff is the founder and CEO of the Center for Policing Equity. They work with police departments across America, including in Minneapolis, to seek measurable responses to racial **bias**. Phil, I can scarcely imagine how the **stress** in the last week must have been for you. Welcome, and over to you for your opening comments. Phillip Atiba Goff : Thanks, Chris. Yeah, this week has been a gut punch to anybody who felt like we could be making progress in the way that we put forward public safety that empowers particularly vulnerable communities. We started working in Minneapolis about five years ago. At the time, it was, like most major cities in the United States, a department that had a long history of unaccounted for violence from law enforcement, targeting the most vulnerable black communities. And we tried to put into place a number of things that we know work. Change the culture, so that the culture can be accountable to the values of the community. And what we saw was small but measurable progress. We always knew, with small and measurable progress, that you ' re one tragic incident from going back to ground zero. But the events of the last week and a half haven ' t brought us back to ground zero, they ' ve torched ground zero, and we ' ve dug a hole that we have to dig ourselves out of. What I hear from police chiefs who call me, from activists I talk to, from folks in the communities that are literally on fire right now, I hear folks saying, I had one activist say to me that the pain that he was feeling was too large to fit into his body. And without thinking about it, I said right back,"That ' s because it ' s too large to fit into a lifetime."What we ' re seeing isn ' t just the response to one gruesome, cruel, public execution. A lynching. It ' s not just the reaction to three of them : Ahmaud Arbery, Breonna Taylor and then George Floyd. What we ' re seeing is the bill come due for the unpaid debts that this country owes to its black residents. And it comes due usually every 20 to 30 years. It was Ferguson just six years ago, but about 30 years before that, it was in the streets of Los Angeles, after the verdict that exonerated the police that beat Rodney King on video. It was Newark, it was Watts, it was Chicago, it was Tulsa, it was Chicago again. If we don ' t take a full accounting of these debts that are owed, then we ' re going to keep paying it. Part of what I ' ve been experiencing in the last week and a half, and what I ' ve been sharing with the people who do this work, who are serious about it, is the acknowledgment, the soul-crushing reality that at some point, when things stop being on fire, the cameras are going to turn to something else. And the history that we have in this country is not just a history of vicious neglect and a targeted abuse of black communities, it ' s also one where we lose our attention for it. And what that means for communities like in Baton Rouge, for those who still grieve Alton Sterling, and in Baltimore, for those who are still grieving Freddie Gray, is that there is not just a chance, there ' s a likelihood that we are a month or two months out from this with no more to show for it than what we had to show after Michael Brown Jr. And holding the weight of that, individually and collectively, is just too much. It ' s just too heavy a load for a person or a people, or a generation to hold up. What we ' re seeing is the unrepentant sins, the unpaid debts. And so the solution can ' t just be that we fix policing. It can ' t be only incremental reform. It can ' t be only systems of accountability to catch cops after they ' ve killed somebody. Because there ' s no such thing as justice for George Floyd. There ' s maybe accountability. There ' s maybe some relief from the people who are still around, who loved him, for his daughter who spoke out yesterday and said,"My Daddy changed the world."There won ' t be justice for a man who ' s dead when he didn ' t have to be. But we ' re not going to get to where we need to go just by reforming police. So in addition to the work that CPE is known for with the data, we have been encouraging departments and cities to take the money that should be going to invest in communities, and take it from police budgets, bring it to the communities. People ask,"Well, what could it possibly look like? How could we imagine it?"And I tell people, there is a place where we do this in the United States right now. We ' ve all heard about it, whispered, some of us have even been there, some of us live there. The place is called the suburbs, where we already have enough resources to give to people, so they don ' t need the police for public safety in the first place. If someone has a substance abuse issue, they can go to a clinic. If somebody has a medical issue, they ' ve got insurance, they can go to a hospital. If there ’ s a domestic dispute, they have friends, they have support. You don ’ t need to enter a badge and a gun into it. If we hadn ’ t disinvested from all the public resources that were available in communities that most needed those, we wouldn ’ t need police in the first place, and many have been arguing, even more loudly recently, that we don ’ t. If we would just take the money that we use to punish, and instead invest it in the promise and the genius of the community that could be there. So I don ’ t know all the ways we ’ re going to get there. I know it ’ s going to take everything and. It ’ s going to need the kind of systemic change and the management tools that we traditionally offer. It ’ s also going to need a quantum change in the way that we think about public safety. But mostly, this isn ’ t just a policing problem. This is the unpaid debts that are owed to black America. The bill is coming due. And we need to start getting an accounting together, so we ’ re not just paying off the interest of the damn thing. WPR : Thank you, Phil. Rashad Robinson is the president of Color Of Change, a civil rights organization that advocates for racial justice for the black community. To date, more than four million people have signed their petition to arrest the officers involved in the murder of George Floyd. And of course, one was arrested last week. Thank you so much for being with us, Rashad, welcome. Rashad Robinson : Thank you. And thank you for having me. It ’ s an opportunity that I ’ m taking today to just tell you about how you can get involved. How you can take **action**, because right now, strategic **action** is critical for all of us to do the work to change the rules that far too often keep the systems in place that hold us back. Make no mistake, the criminal justice system is not broken. It is operating exactly the way it was designed. At every single level, the criminal justice system is not about providing justice, but about ensuring that certain people, certain communities are protected, while other communities are violated. And so I wan to talk a little bit today about Color Of Change, about activism, about the work that ’ s happening on the ground from other organizations all around the country, and the way that you can channel this energy. What we talk about at Color Of Change is how do you channel presence into power. Far too often we mistake presence and visibility for power, presence retweets the stories of the movement, people feeling passion about change could sometimes make us feel like change is inevitable, but power is actually the ability to change the rules. And right now, every day, people are taking **action**, and what we ’ re trying to channel that energy into is a couple of things. First is a whole set of demands at the federal level and at the local level. As Phil described, policing operates on many different channels. And what we need to recognize is that while there are a lot of things that can happen at the federal level, locally all around the country there are decisions that are being made in communities around how policing is executed, where community needs to hold a deeper level of accountability, at the state level we need new laws. So at Color Of Change, we ’ ve built a whole **platform** around a set of demands and are working to build more energy for everyday people to take **action**. We ’ re fighting for justice for Ahmaud Arbery, Breonna Taylor and George Floyd, we ’ re also fighting for justice for other folks whose names you haven ’ t heard, Nina Pop and others, whose stories of injustice and the relationship to the criminal justice system represent all the ways in which fighting right now is important. Over the last couple of years, we have worked to build a movement, to hold district attorneys accountable and to change the role of district attorneys in our country. And through the Winning Justice **platform** at Color Of Change, www. winningjustice. org, what we have worked to do is channel the energy of everyday people to take **action**. So, for folks who are watching what ’ s happening on TV, seeing it on their social media feeds and are outraged about what ’ s happening in Georgia, what ’ s happening in Tennessee, what ’ s happening in Minnesota, you yourself, probably, most likely, live in a place, in a community where you have a district attorney that will not hold police accountable, that will not prosecute police when they harm, hurt black folks, when they violate the laws, you live in a community where police are part of the structure that is racking up mass incarceration, but many other aspects of our system are racking up mass incarceration, and district attorneys are at the center of it. You live in those communities and you need to do something about it. And so at winningjustice. org, we ’ ve created the only searchable, national database on the 2, 400 prosecutors around the country. We ’ re building local squads and communities for folks to be able to engage around efforts that hold DAs accountable. We ’ ve worked with our partners across the movement, from our friends in Black Lives Matter, to folks who do policy work, to our friends at local ACLU chapters around the country, to build six demands. Six demands that folks can get behind in terms of pushing for reform, and then we ’ ve built public education material. But the only way that we work to change the way that prosecution happens in this country is that if people get involved. If people raise their voice, if people join us in pushing for real change. At the end of the day, I want people to recognize though, and Phillip talked a little bit about this, is that people don ’ t experience issues, they experience life. That the forces that hold us back are deeply interrelated, a racist criminal justice system requires a racist media culture to survive, a political inequality follows economic inequality, they all go hand in hand. And so I also want us to not take ourselves out of the equation. We likely work inside of corporations that may post symbols for Black Lives Matter one day, and then support politicians that work to destroy Black Lives Matter the next day. We oftentimes are engaged in **practices** inside of our companies or in our **daily** lives supporting media properties and others that are harming our communities, are telling stories. Recently, we produced a report at Color Of Change with the Norman Lear school at USC. It ’ s called”Normalizing Injustice,”and it can be found at changehollywood. org. And”Normalizing Injustice”looks at the 22 crime procedurals, those crime shows on TV. And looks at all of the ways in which they, sort of, create a warped perception about our view of justice. They create sort of an incentive for the type of policing we see on the country, and actually serve as a PR arm for law enforcement. We ’ ve been working in writers rooms around the country to work to push folks to tell better stories, but we need folks to be both active listeners, and we need folks in the industry to push back and challenge those, not only the structures that lead to that content coming on the air, but the proliferation across our airwaves. At the end of the day, we have an opportunity in this moment to make change. Inflection points are those moments where we have an opportunity to make huge leaps forward, or the real, real threat of falling backwards. In our hands is the ability to do some incredible things about undoing so many of the injustices that have stood in the way of progress for far too long. But everyday people must get involved. We must channel that presence into power, and we must build the type of power that changes the rules. Racism in so many ways is like water pouring over a floor with holes in it. Every single – In every single way, it will find the holes. And so for us, we cannot simply accept charitable solutions to structural problems, but we actually have to work for structural change. And so I want to end by saying one thing about how we talk about black people and how we talk about black communities in this moment. Because we have to say what we mean, and we have to build the narrative that gets us to where we want to go. So far too often, we talk about black communities as vulnerable, we talk about black people as vulnerable, but vulnerability is a personal trait, black communities have been under attack. Black communities have been exploited, black communities have been targeted, and we need to say that, so we don ' t put the onus on fixing black families and communities, but we put the onus on fixing the structures that have harmed us. We will say things like,"Black people are less likely to get loans from banks,"instead of saying that banks are less likely to give loans to black people. This is our opportunity to build the type of progress that makes real change, and at the center of this story, we need to show and elevate the images not just of the pain that we are facing, but of the joy, the brilliance and the creativity that black people have brought to this country. Black people are the protagonist of this story, and we need to make sure that as we work to build a new tomorrow, we ensure that the heroes are at the center of the liberation that we all need. Thank you. CA : Thank you, Rashad. Dr. Bernice King is the CEO of the King Center in Atlanta, Georgia. The center is a living memorial to her father, Dr. Martin Luther King Jr. It ' s dedicated to inspiring new generations to carry his work forward. In this moment, when so many are hurting, how can we better approach unity and collective healing? Dr. King, over to you. Bernice King : My heart is a little heavy right now, because I was that six-year-old. I was five years old when my father was assassinated. And he did change the world. But the tragedy is that we didn ' t hear what he was saying to us as a prophet to this nation. And his words are now reverberating back to us. Change, we all know, is necessary right now. And yet, it ' s not easy. We know that there has to be changes in policing in this nation of ours. But I want to talk about America ' s choice at a greater level. The prophet said to us,"We still have a choice today : nonviolent coexistence, or violent coannihilation."What we have witnessed over the last eight days has placed that choice before us. We have seen literally in the streets of our nation people who have been following the path of nonviolent protest, and people who have been hell-bent on destruction. Those choices are now looking at us, and we have to make a choice. The history of this nation was founded in violence. In fact, my father said America is the greatest purveyor of violence. And the only way forward is if we repent for being a nation built on violence. And I ' m not just talking about physical violence. I ' m talking about systemic violence, I ' m talking about policy violence, I ' m talking about what he spoke of are the triple evils of poverty, racism and militarism. All violent. Albert Einstein said something to us. He said we cannot solve problems on the same level of thinking in which they were created. And so if we are going to move forward, we are going to have to deconstruct these systems of violence that we have set in America. And we ' re going to have to reconstruct on another foundation. That foundation happens to be love and nonviolence. And so, as we move forward, we can correct course if we make that choice that Daddy said, nonviolent coexistence. And not continue on the pathway of violent coannihilation. So what does that look like? That looks like some deconstruction work in order to get to the construction. We have to deconstruct our thinking. We ' ve got to deconstruct the way in which we see people and deconstruct the way in which we operate, **practice** and engage and set policy. And so I believe that there ' s a lot of heart, h-e-a-r-t work to do, in the midst of all the h-a-r-d, hard work to do. Because heart work is hard work. One of the things we have to do is we have to ensure that everyone, especially my white brothers and sisters, have to engage in the heart work, the antiracism work, in our hearts. No one is exempt from this, especially in my white community. We must do that work in our hearts, the antiracism work. The second thing is that I encourage people to look at the nonviolence training that we [have] at the King Center, thekingcenter. org, so that we learn the foundation of understanding our interrelatedness and interconnectedness. That we understand our loyalties and our commitments and our policy-making can no longer be devoted to one group of people, but has to be devoted to the greater good of all people. And so I ’ m inviting people to even join us on our own line of protest that ’ s happening every night at seven o ’ clock pm on the King Center Facebook page, because so many people have things that they want to express and contribute to this. We all have to change and have to make a choice. It is a choice to change the direction that we have been going. We need a revolution of values in this country. That ’ s what my Daddy said. He changed the world, he changed hearts, and now, what has happened over the last seven, eight years and through history, we have to change course. And we all have to participate in changing America with the true revolution of values, where people are at the center, and not profit. Where morality is at the center, and not our military might. America does have a choice. We can either choose to go down continually that path of destruction, or we can choose nonviolent coexistence. And as my mother said, **struggle** is a never-ending process, freedom is never really won. You earn it and win it in every generation. Every generation is called to this freedom **struggle**. You as a person may want to exempt yourself, but every generation is called. And so I encourage corporations in America to start doing antiracism work within corporate America. I encourage every industry to start doing antiracism work, and pick up the banner of understanding nonviolent change, personally, and from a social change perspective. We can do this. We can make the right choice to ultimately build the beloved community. Thank you. WPR : Thank you, Dr. King. Anthony Romero is the executive director of American Civil Liberties Union. As one of the nation ’ s oldest social justice organizations, the ACLU has advocated for racial equality and shown deep support to the black community in moments of crisis. And in moments like these, black voices are almost always the loudest and at times the silence from our nonblack brothers and sisters can feel deafening. How we can bring our allies into the mix, to better support ending systemic violence and racism against the black communities, is a question top of mind for a lot of us. Anthony, welcome to the show, and thank you so much for being with us. Anthony Romero : Great. Thank you, thank you Whitney, thank you, Chris, for inviting me to join this TED community. I think community is really important right now. With so many of us feeling trepidation, the weariness, the anger, the fear, the frustration, the terrorism that we ’ ve experienced in our communities. This is a time to huddle around a virtual campfire, with your posse, with your family, with your loved ones, with your network. It ’ s not a time to be isolated or alone. And I think for allies in this **struggle**, those of us who don ’ t live this experience every day, it is time for us to lean in. You can ’ t change the channel, you can ’ t tune out, you can ’ t say,”This is too hard.”It is not that hard for us to listen and learn and heed. It is the only way we ’ re going to build out of this, by hearing the voices of Rashad, and Phil and Dr. King. By hearing the voices of our neighbors and loved ones, by hearing the voices on Twitter of people who we don ’ t know. And so white communities and allied organizations need to pay even closer attention. This is the test of your character. How willing are you to lean in and to engage. For me, I have – These have been really hard couple of weeks. I feel like this is really a test of whether or not we really believe in the American experiment. Do we really believe it? Do we really believe that out of many, one, a country with no unifying language, no unifying culture, no unifying religion, can we really become one people? All equal before the law? All bound together with a belief in the rule of law? Do we really believe that or do we just think it ’ s a nice saying to see on the back of a paper dollar? And for me, this is a referendum on the American experiment. On whether we really believe, and... the future is in our hands. And this is not like other crises, I ’ ve been the head of the ACLU for almost 20 years, I feel like I ’ ve seen it all. This is different. And this is different because it is cumulative, like Phil and Rashad and Dr. King told us, this is centuries of systemic discrimination, and the bill has come due. And it will continue to be due, and we will pay. Unless we really do something quite different. I have been scratching my head at the ACLU for the last week. We ' ve been at this for 100 years. My organization has been working on this from its inception. In 1931, we were involved with this report about lawlessness in law enforcement. That was our first report that we got behind in 1931. We opened up our first door fronts after the riots in Watts, so that we can bring legal **services** and lawyers to the communities so they could demand justice from the police departments. You know, we brought Miranda, you know, the right to remain silent, and we brought Gideon, the right to a court-appointed attorney if you can ' t afford one. We fought Bloomberg on"stop-and-frisk,"it took him years and he lost in front of our litigation to finally apologize. We ' ve been at this for 100 years. And for the communities that have lived this for 400 years, God. I ' ve been scratching my head, thinking. It ain ' t working. We don ' t need another pattern and **practice** lawsuit. We don ' t need another training program on racial **bias** or implicit **bias** in police departments, we don ' t need to file another lawsuit on qualified immunity, we don ' t need to, kind of, bring another race discrimination or gender discrimination lawsuit to integrate the police department. Yeah, we ' ve done that and we will continue to do that. For me, where I ' ve come, is that we need to defund the budgets of these police departments. It ' s the only way we ' re going to take the power back. And the more I read over the last couple of weeks about where this country is, the more I ' m clear that that is my North star at the moment. We will continue to bring the litigation on qualified immunity, we will do the efforts to hold unaccountable law enforcement officials accountable, we will bring pattern and **practice** lawsuits, because the justice department is not doing that, so we will continue to do all that good work. But the real thing is, we ' re going to go after those budgets. When you look at the fact that we spend 100 million dollars on policing, more than incarceration, that the city of Minneapolis spent 30 percent of their budget on policing. The city of Oakland, 41 percent on policing. That when you have New York City police department spend more money on policing than it does on housing and preservation development, community youth **services**, homelessness. We ' re going after the money. And that ' s hard-core advocacy. Bills drop in local legislatures to cut the funding for police, to stop these programs that give the federal military surplus to police departments, so they become, like, little mini armies, these don ' t look like police officers, these look like standing armies. And the enemy are communities of color. So we need to take away their toys. We need to cut their budgets. We need to shrink the police infrastructure, so that we can get police out of the quotidian lives of people of color and communities of color. The ubiquitousness of police enforcement on things that the police do not have a role, should not have a role to play. People should not lose their lives over whether or not a cigarette pack has a proper tax stamp, or whether a 20-dollar bill was forged or not. That ' s not **worthy** of spending our dollars on police. Get them out of that business, let ' s focus on the most important and the most serious of crimes, and that ' s it. That ' s it. We ' re going to depolice our communities. Shrink those budgets. We ' re going to reinvest those moneys in local communities, it will be like water on stone campaigns, local legislatures, local city counsels, lab report cards, for people who talk out of both sides of their mouths and say,"We believe in police reform,"and yet, they ' re still going to vote for 30 or 40 percent for the police? We ' re going to put that right to the public. And I think we just have to stay at it, because I think that ' s the only way we can get at this in a different way. Because much of what we tried to do is just simply not working. You know, with that, I **struggle** with, how do you find the optimism in this moment, because you have to find the optimism. You have to find the way to still think that even though on the face of so many setbacks, there ' s been change. It ' s been too little, too slow, not enough. We need to kind of, rock it, boost it. But you can ' t lose sight of the optimism. And you know, I ' ve been thinking about who are the folks inspiring me, and Dr. King ' s father, of course, and the words of Rashad and Patrisse Cullors and others have inspired me. But I find inspiration in the words of a scholar I really don ' t like bringing up, Sam Huntington, kind of often criticized as being a conservative, a racist. But sometimes you can find inspiration even in your enemy ' s words. And in one of his books, which I pulled off the shelf I have, he writes about how America is a disappointment, because it failed to live up to its aspirations. And he actually started talking about, America is a failure because it doesn ' t live up to its ideals. But it ' s not a failure, it ' s not a bunch of lies. It ' s a disappointment. And in the disappointment also is the fact that there ' s hope. I ' m paraphrasing it, but I think we have to kind of, wrap all of that together, and think about the disappointment and the hope and the resolve to do better. And we need to listen and lean in, and I thank the TED community, I thank Dr. King, I thank Rashad, I thank Phil. Thank you. CA : Wow. Thank you to all four of you, that was astonishing. I guess we ' re bringing everyone back now to have a conversation among us, to answer questions from our community, I hope you ' re entering those questions. So I don ' t know whether we can bring back our guests onto the screen at this point. Welcome back. Let me start with a question to you, Dr. King, I was so inspired by what you said. Your father, of course, also deeply understood the anger that leads to protest. I think he said that protests are the language of the unheard. And I ' m wondering what you would say to someone right now who is angered beyond measure by what ' s happened, and also sees this could be the moment, you know, like, someone who believes the system is so fundamentally broken, that our best choice is to tear it down, that that is actually – This may be a once-in-a-generation moment to do that. And so to actually believe that protest, including violent protest, actually is the way right now. What would you say to someone who felt that? BK : First, I just wanted to make just a slight correction, he said riots are the language of the unheard. CA : I apologize, I apologize. But that is the point even more powerfully, yeah. BK : Yes. Protest we must, and we must continue to always protest, to keep the issues in the awareness before people, but you know, when a person is angry, sometimes it ' s hard to reach them. I ' ve been on that journey, I was at a stage of my life where I was so angry, I wanted to destroy, and I ' m the daughter of Martin Luther King Jr., and grew up in a household of love and nonviolence and forgiveness, and I had to go through that journey I was surrounded by the right kind of influences, fortunately. Because that would have been a sad story. But I think it ' s really allowing ourselves to hear the anger and allowing the space for the anger, but also trying to help young people rechannel that energy. And we ' ve got to start ensuring that we connect them to some of the work that has been and now is elevated to another place. Color Of Change, the work that you ' re doing, the ACLU, the work that they ' re doing, because sometimes, there ' s this disconnection that intensifies the emotion and makes you feel helpless. But if you can channel that anger, connect it with **action** that is toward creating the social and economic change, then it begins to build you up, and then you can begin to become more constructive with the anger. WPR : We have some questions that are coming in from our community, but before we do that, you all shared such powerful, meaningful statements right now, and many of you touched on the fact that this is not the first time that we ' re experiencing this. The murder of George Floyd, Ahmaud Arbery, Breonna Taylor, this is one of – these are three of many, many instances just like this, and I ’ d love to hear you all address was has, sort of, brought us to this boiling point, what has contributed to this moment where we ’ re now experiencing things, Anthony, as you said, that feels so much worse than other moments? And that ’ s it, anyone who feels comfortable to take that question. AR : Rashad, I want to hear you. BK : I wanted to say something. I think we ’ ve always been at that moment. But this moment is different, because of the void in leadership. There ’ s no real moral voice in our country, and the person who sits in the office of the presidency is not, you know, leading in the right way. And has kind of – no, not kind of, has given **license** to certain things. And so now it ’ s, you know, he ’ s lit the fires. WPR : Yeah. RR : The thing I ’ ll add... The thing I ’ ll add here is, you know, a couple of things. For the last couple of months, we have been both seeing and experiencing all of the ways that this country ’ s decisions of underinvestment, of targeting black communities, has been killing black people through COVID. And while we ’ ve been in our homes, we have also been watching how the media has blamed us as we have been the essential workers in so many places and trying to ensure that this country keeps going. We ’ ve watched white men with guns show up to capitols demanding, basically, black and brown people go back to work. And then we see this eight-minute video with a police officer, with his knee on someone ’ s neck, after seeing that video of Ahmaud Arbery and hearing the story of Breonna Taylor, and we see him looking in the camera, basically knowing that America was not going to punish him. And what I think it is is that it ’ s just enough is enough, that people didn ’ t feel that they had a channel for that outrage, and because people had been inside, and because people had been experiencing all the ways in which the structures had also been colluding to kill us, that what we ’ re seeing is alignment of all of those things, where people are making demands that are much bigger and much **bolder** than before. And we recognize that while we don ’ t have leadership at the federal level, we also have to recognize that no political party can say that they have been 100 percent, neither political party can say they ’ ve been 100 percent on the right side of all these issues. And so people are mobilizing, they are fighting back like never before, and in some ways, people are unwilling to accept answers like,”Just go vote,”or,”Just participate in the process,”because we recognize that black people have been voting, black people have been part of voting, and part of ensuring that. And so that I think is why this moment feels so much different, combined with, for the last seven years, since Trayvon, we have seen the growth of a new movement of activists and leaders all around the country, who are also in a very different place to be able to move the needle on so much of what ’ s possible. CA : We have a question here from Genesis Be. If we can get that up here.”Here in Mississippi, the police is synonymous with the Klan, historically. How do we purge law enforcement of white supremacists?”PAG : So I guess that ’ s partially to me, being the psychologist of **bias**. I ’ ll say that just yesterday we had an officer in Denver who posted on social media himself and two other officers saying,”Let ’ s go start a riot.”He was fired that day. I worry about all the officers that the FBI has now, for almost half a decade, been warning us, law enforcement and unions being infiltrated by white supremacists. And all the officers that have social media accounts, but they ’ re private. You know, the Invisible Institute has put some things forward. We ’ re not talking seriously about the domestic terrorism threat that white supremacy represents. So the first thing that we ’ ve got to do is we ’ ve got to take it seriously. We have to actually say out loud, and I can ’ t believe that on a day like today, or a week like this week, I have to say out loud, white supremacy is alive and well and a driving force of American politics. This shouldn ’ t be controversial. I shouldn

’ t be looking forward to getting hate mail in my inbox for it, but that ’ s the reality. So the first part of solving a problem is acknowledging that it exists. But the second thing is we need to arm municipalities, that ’ s law enforcement, but even more so communities, with the ability to take **action** when someone violates their values. Right now, I think about the case in Philadelphia where Charles Ramsey, when he was a commissioner there, fired six officers, right. Concerns about racial **bias** and concerns about police brutality, and those six officers were back on the same job inside of three months. We now have a law enforcement system that says you can lose your job in one jurisdiction, and get the same job as law enforcement in another jurisdiction. And without the national registry and the capacity for law enforcement to make different decisions, we ’ re going to have this exact problem, not just in Mississippi, but in Minneapolis, and Louisville and New York and LA. CA : Phil, how much of the problem stems from the fact that police unions have a huge amount of power to protect and sometimes reinstate so called bad apple officers? PAG : Yeah, I ’ m getting this question a lot, and police unions are one of the most labor forces of the United States, and are unique within the labor movement, right? So it ’ s police unions and teachers ’ unions are the two largest and could not be two different groups of folks. When I talk to union leadership, that ’ s the leadership what wants to talk to Dr. Blackenstein, right, when I talk to union leadership, what they say is no one hates a bad officer more than a good officer. But the union contracts, the new negotiations, don ’ t look like that ’ s true. What they look like is anybody gets in trouble, and the union ’ s only job is to make sure whatever officer is in trouble, gets to maintain their job. The perverse incentive here is that when people run for union leadership, no one can run saying,”These people shouldn ’ t be in the union.”It ’ s very hard to do that. What you can run on is say,”If this person didn ’ t protect you enough, I ’ ll protect you even more. The bigots? I ’ ll protect even them.”So we have this perverse incentive where union leadership ends up not really representing the values even of the rest of the union members. But they have massive, outsized negotiating power. So yes, engaging with and appropriate rightsizing of labor protections, for folks whose jobs are difficult, but who should not be protected from the basic values of human rights, human dignity and public safety. It ’ s got to be part of the process. I mean, when unions are negotiating a two-year ban on keeping of records, so that there ’ s no ability even to trace what ’ s happening in the state of California, historically in terms of police misconduct, that ’ s not in the interest of public safety, public legitimacy, or our democracy. AR : Yeah, the thing I would add, Chris, is that I think the labor union piece is a critically important one to think through. Because I think, like Phil said, they are a key part of the puzzle that we have to solve for. And you know, it ’ s **frustrating** when you look at a place in Minneapolis, and Phil knows better than I, but when mayor Jacob Frey, the one who ’ s on TV all the time, saying many of the right things that you want an elected official to say at times like this, when he banned his police department from attending the”warrior training”that was being offered, it was the Minneapolis Police Federation, local union that defied him and sent their police to the training. And so we need to really be clear that we need to have the police forces under civilian control. I know this sounds so elementary, I feel like I ’ m talking about a Latin American, kind of, totalitarian context, but we need to exert civilian control of our police in a way that we have yet not been able to think through, and a key part of that is the labor unions of the police. And there are moments when you can find common ground. When we brought one of our COVID-related lawsuits to deal with the outbreak of the pandemic in a Maryland jail, we worked really hard, I worked the phones with the head of police unions. We got one of the local unions to serve as plaintiff in our lawsuit. Because we understood that the incarcerated folks who were being denied access to masks, social **distancing** and the conditions and lack of testing, and lack of PP, that the people who were also going to be in harm ' s way were going to be the guards as well. And they were going to be the vectors, communicating the disease out into the community. So if you can find ways of bringing that relationship. But make no mistake, when you go after their budgets, and you start taking away kind of, their munitions, and their seat at the budgeting table, oh, are you going to have a battle on your hands, right? And we have to think about also as we shrink the budgets for police, how do we – we deploy people in the police departments to other meaningful jobs, right? Because you can ' t just throw them out into the street, and say,"You ' re on your own, you ' re homeless, good luck to you."That ' s not a way to deal with redemption. So we have to really think about all these pieces in a much more cohesive way. WPR : We have another question here from the audience. From Paul Rucker :"The end of summer of 1919 was followed by the Tulsa Race Massacre, the Johnson-Reed Anti-Immigration Act of 1924, and also the rise of the KKK. Is there a possibility that white supremacy will get stronger if we don ' t seize this opportunity?"Rashad, I think this might be something that would be great to hear your perspective on, working so deeply in activism. RR : I ' m having a little trouble hearing. WPR : Oh, I ' m so sorry. RR : No, it ' s OK. CA : Can you read the question on the screen, Rashad? RR : Oh, I heard that, I heard you. WPR : Yes, I think it might just be my mic is having some issues here."Is there a possibility that white supremacy will get stronger if we don ' t seize this opportunity."Yes, absolutely yes. You know, to be clear, right, if we don ' t have the right diagnosis of white supremacy, if we think of white supremacy as just hoods, if we think of white supremacy as just folks who are operating, you know, with, you know – in some of these underground networks that have grown, if we ' re just thinking about white supremacy and white nationalism as people who marched with tiki torches in Charlottesville, then we will really mistake all the ways in which our systems and structures have white supremacy embedded, and allow for something like a Tulsa Race Massacre to happen, something like anti-immigration to happen, but on a day-to-day basis allow for the targeting of black communities through predatory **practices** by banks. The targeting of black communities through predatory **practices** like bail. A whole set of systems that can be produced day in and day out. We live in a country where the rules are far too often designed in ways that create a caste system, that create a different standard for some over others, and so when I talked about the inflection point, right, of this moment where something could really go forward and something could turn backwards, we are seeing this right now with this current president. And as we look at what could be happening with the next election, we have to be very, very clear that Donald Trump doesn ' t just operate on his own. He ' s enabled by big corporations who benefit from him being in office, and so continue to turn a blind eye to all the things that he does. They may post"Black Lives Matter,"but they show up to the White House and engage with Donald Trump. And then we have a whole set of politicians that may sometimes say that he said something that was wrong, but then allow for – but support his **platform** in other ways. You know, true co-conspiracy in the effort to dismantle white supremacy and white nationalism is not a thing that people can do on **vacation**. It is a 365-day project of us constantly working to dismantle all of the structures that have been put in harm ' s way. The final thing I will just add, because someone mentioned about police unions, and I want to just add that one of the problems with police unions, and many of us have been in this position, I think, is that I have shown up to the table with police unions on many occasions. I remember going to the White House during the last administration and being around a table as we were talking about policing and police reform. And having members of the Fraternal Order of Police leadership say things like,”All of this talk of racial profiling is new to us.”It is one thing for folks to not agree with you on the policy reforms necessary. It is another thing for people to say that our demands are too aspirational. It ’ s another thing to be gaslit and told the problem doesn ’ t actually exist at all. And that is what we are dealing with, and so we have to actually change the way that people see these institutions. Politicians who say that they are on the side of justice and reform, can no longer take money from police-only unions and Fraternal Order of Police. We actually have to create a new standard, a new litmus test of what does it mean to actually be with us. You can ’ t just sing our songs, use our hashtags and march in our marches, if you are on the other end of supporting the structures that put us in harm ’ s way, that literally kill us. And this is the opportunity for white allies to actually stand up in new ways. To be the type of ally, to do the type of allyship and the type of work that truly dismantles structures, not just provides charity. PAG : And I ’ ve got to add to that – so, Paul, thank you for the question, but we ’ re in a moment where people are looking at what ’ s happening on the street, as if a week and a half ago we weren ’ t in a midst of a global pandemic as the greatest news story, the biggest new story going on. One of the things I ’ m most worried about, and have been worried about since the beginning, what I ’ ve been talking to our chiefs about, is say, you must be out of the social **distancing** policing game. You can ’ t be the ones doing that, and the reason is this : We ’ re in a moment where creating scapegoats and enemies and others is incredibly politically advantageous for at least one side. And there is deliberate efforts to do exactly that. And we ’ ve seen that black communities are two and three and four times more likely to contract this virus, which feels like the manifestation of racial discrimination, because it is. But very soon, that ’ s going to look like black people made bad choices and they need to stay away from us. And when that happens, that ’ s when law enforcements get used to regulate where black movement can be. We used to call it sundown towns, I don ’ t know what we ’ re going to call it when it ’ s around COVID. But it ’ s coming. I ’ m already seeing that on communities like Nextdoor, and on Facebook groups. People who don ’ t think of themselves as white supremacists but just want the disease away, and the disease has a black and brown face. So we ’ re not only dealing with a moment of generational tension, between black communities and law enforcement, we ’ re dealing with a moment when people are looking for scapegoats, and black people ’ s vulnerability has always been our greatest casting note for being cast as scapegoats. So for folks who are worried about this, this is not inevitably a moment for change and reform and enlightenment and America ’ s best values, because historically, these have been precisely the moments when regression back to white supremacy has reigned supreme. So let ’ s not just look at everybody signaling. I don ’ t want to just see black and white cops on their knees, I want to see the policies. I want to see the things that will prevent this kind of thing from moving to the next stage. CA : Rashad, I want to respect the fact that you ’ ve got a hard stop at one. And so I just want to thank you for your participation in this. If you ’ ve got a couple of final words you want to share, that would be great, and then if it ’ s OK for the other three, I think there ’ s just a couple other questions I ’ d love to put and continue this conversation for just a moment longer, if possible. Rashad, any closing words? RR : The thing I want to say is that now is the time for **action**. And I want to invite people in to join us at Color Of Change to make justice real. And in so many ways, you can visit us at Color Of Change, you can take **action**. Five, 10, 15 years from now, we will be dealing with the impacts of what we did or didn ’ t do in this moment. How we stood up and how hard we were willing to fight. And as the other speakers have said, now is not the time for reform around the edges, now is time for dismantling the policies and **practices** that have held us back, and championing solutions and new rules that will move us forward. And so we hope that you will do something, whether it ’ s with us, or whether it ’ s with local organizations in your community, or other groups around the country. But this is an opportunity to make change, and I believe that we can make justice real, if we find the passion and the energy to work together to achieve it. So thank you all for having me, and I hope that we have an opportunity to build, not just online, but offline, in the months to come. CA : Thanks so much, Rashad. We ’ re just going to ask this last question of you. This one is from David Fenton.”How can the movement unite around a clear, simple **platform** of policies to enshrine in legislation? Like making all complaints against cops public, banning all choke holds, ensuring independent review boards, etc.?”WPR : That seems like a great place for you to chime in, Dr. King, if you have some thoughts on that. PAG : I think that was to go to you, Dr. King. BK : Oh, OK. You know, this may sound simplistic, but it ’ s a Nike thing, I think we have to just do it, we have to see our work as interconnected. I think there ’ s been efforts towards people working in that vein, but we have to intensify that. And in doing it, one of the things that my father said, and I know people sometimes get tired of hearing me say,”My father said,”but I just think, I wish, should I say, we had really listened to him, because we wouldn ’ t be on this **platform** right now having this type of conversation. But he left something with us, sort of a blueprint in”Where Do We Go From Here : Chaos or Community?”, his book, and he said, going forward, the meddlesome task is to organize our strength into compelling power. And that is so key, because oftentimes we organize merely around passion. But people have certain areas of strength and talent and giftedness, and we ’ ve got to figure out how to build our coalitions based on these strengths. You know, people do different things well. And so, in order to unite in an effective way that they might not elude the demand that we ’ re making, I think that ’ s what ’ s going to happen. People have to do their own personal assessment within their organization, I call it a SWAT analysis. And then that SWAT analysis has to happen across organizations, so that we can make sure that we are moving in a united manner, off of the strengths that each organization brings, so that we can **maximize** the impact and the effectiveness to do things like this, in terms of getting the legislation in place that is needed in this hour. CA : Thanks so much. Just quick closing words from you, Anthony, and then from you, Phil. Anthony. AR : You know, I would just say that what gives me hope are the young folk. You have to believe that among this group, this groups of young ’ uns, seeing what they ’ re seeing, living with this president, with these instincts, seeing the continued indifference that mainstream communities have given to issues of racial justice, or economic justice, you ’ ve got to believe that what comes out of this very hot fire, is something even more powerful and strong than we ’ ve ever seen before. That ’ s what gets me through the hard days that we ’ re now experiencing, this thinking, there is another Dr. King among the young ’ uns, Dr. King. And I have to believe that what they ’ re seeing and what they ’ re witnessing and their righteous indignation and their frustration and their anger is going to be miraculously a beautiful blossoming of a new opportunity, of a new change. This generation will take us there, I have to believe that. My generation has failed them miserably. So I ’ m just looking forward to the new ones. CA : Thank you, Anthony. WPR : Phil. Thank you, Anthony. PAG : So it really has been a privilege to be on with you all. To David ’ s question, let me say that a number of civil rights organizations, I believe the ACLU among them, CPE, Center for

Policing Equity, and hundreds more have signed on to principles for legislation that would include eight pillars. It ’ s been led by the Leadership Conference for Civil and Human Rights. And it includes a federal ban on choke holds, it includes a national registry for officers who have engaged in misconduct. I also think that it ’ s important at this moment to get, we ’ ve got law enforcement ’ s chiefs of major cities willing to say, if we emerge from this moment and our profession hasn ’ t changed, then we have failed again. So it ’ s a critical time to get behind, I would direct you to LCCHR ’ s website for the eight pillars, because I won ’ t remember them all right now, and to start calling your local law enforcement, and say,”Yes, own that.”You should be signing on, they should be going public with letters that do all of that. But I ’ ll also say this. For a path forward in the principles, I ’ ll end where I started, which is that this is bigger than policing. These are the unpaid debts owed to black communities for stolen labor, owed to native communities for stolen land, for stolen culture, for years taken away and for lives lost in it. This is bigger than policing. If we don ’ t understand the size of it, then there ’ s no solution that ’ s really, truly proportional to the moment. But in this moment, when we ’ re seeing trillion of dollars in bailouts, mostly for corporations, it is absolutely a time when we can do things that normally, people could pretend that ’ s too much, it ’ s too big, we can ’ t. We have literally all the money in the world that can be spent and directed towards making us the society we pretend to be, before moments like this happen. And so the thing that gives me hope is that the lies have to be obvious now. The lies have to be, that was a reasonable use of force. The lie has to be, we don ’ t have the money. The lie has to be, that ’ s too hard, it ’ s too big of a challenge. This stuff feels impossible every day except today, because the alternative is we lose everything. Everything is at stake, our democracy is at stake, the people we choose to be, we claim to be, that ’ s at stake. And in the face of that, I think we can do impossible things. I think we can be mighty. So my hope for all of us is first, that we wake up tomorrow with more peace in the evening than war, and that we hold on to what ’ s possible from this moment at the same time that we hold on to the size of the task in front of us. I don ’ t want to come with half measures out of this. I don ’ t want to come out with radicalized youth and indifferent aged... I don ’ t know what the contrast... The radicalized youth and indifferent people who are old like me. I want to come out with a unified country that understands that the costs that we owe are big, and our pockets are deep enough to match it. CA : Wow. Thank you to each of you for extraordinary eloquence. Really, so powerful. This conversation, obviously, continues, I know that there ’ s many people listening, you have other questions, this, I think, from TED ’ s point of view is just the start of the conversation. To the extent that our job is to amplify the voices that matter, we couldn ’ t be prouder to be amplifying further your extraordinary voices. So thank you for being part of this today. PAG : Thank you. WPR : Thank you all.

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Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ’ s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC microprocessors versus the power of local area networks. And the interesting thing about it is that this slide, like so many technology slides that we ’ re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ’ s basically what I ’ m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ’ m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well the answer is, if I drew it on a normal curve where, let ’ s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ’ m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ’ t tell us much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log curve, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as microprocessor technology, then the day after tomorrow, I would be able to get in a taxi **cab** and be in Tokyo in 30 seconds. It ’ s not moving like that. And there ’ s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ’ d like to ask is, if you look at these exponential curves, they don ’ t go on forever. Things just can ’ t possibly keep changing as fast as they are. One of two things is going to happen. Either it ’ s going to turn into a sort of **classical** S-curve like this, until something totally different comes along, or maybe it ’ s going to do this. That ’ s about all it can do. Now I ’ m an optimist, so I sort of think it ’ s probably going to do something like that. If so, that means that what we ’ re in the middle of right now is a transition. We ’ re sort of on this line in a transition from the way the world used to be to some new way that the world is. And so what I ’ m trying to ask, what I ’ ve been asking myself, is what ’ s this new way that the world is? What ’ s that new state that the world is heading toward? Because the transition seems very, very confusing when we ’ re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ’ s a conference in which people talk about the future, and you notice that the future is still at about the year 2000. It ’ s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ’ s happening there. That transition is happening. We can all sense it. And we know that it just doesn ’ t make too much sense to think out 30, 50 years because everything ’ s going to be so different that a simple extrapolation of what we ’ re doing just doesn ’ t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ’ re going through. Now in order to do that I ’ m going to have to talk about a bunch of stuff that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step back and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, the Earth was this big, sterile hunk of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ’ s theories that are beginning to understand about how it started with RNA, but I ’ m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ’ t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ’ s sort of just a very simple chemical form of life, but when things got interesting was when these drops learned a trick about abstraction. Somehow by ways that we don ’ t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for amusement purposes is we can now write things in this code. And I ’ ve got here a little 100 micrograms of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ’ s right here. If I had really been a egotist, I would have put it into a virus and released it in the room. So what was the next step? Writing down the DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ’ t know if you know this, but bacteria can actually exchange DNA. Now that ’ s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the transition point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ’ s interesting in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ’ re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to evolve so that the interesting level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very **special** structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in the community and **special** structures that were responsible for recording, understanding, learning information. And that was the brains and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ' d built these **special** information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these **special** information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ' re aware of is human language. It ' s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ' m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ' re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ' ve done is we, humanity, have started abstracting out. We ' re going through the same levels that multi-cellular organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. Telephony, computers, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ' ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of microseconds. And you saw Ty ' s little evolutionary example where he sort of did a little bit of evolution on the Convolution program right before your eyes. So now we ' ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and brains, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ' s what we ' re seeing here in this explosion of curve. We ' re seeing this process feeding back on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ' s really impossible for me to design them in the traditional sense. I don ' t know what every transistor in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of abstraction and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ’ t quite even understand. One method that ’ s particularly interesting that I ’ ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the microsecond time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,”Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted.”So in other words, I define what I wanted. Let ’ s say I want to sort numbers, as a simple example I ’ ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,”Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ’ s reproduce the ones that sorted numbers the best. And let ’ s reproduce them by a process of recombination analogous to sex.”Take two programs and they produce children by exchanging their subroutines, and the children inherit the traits of the subroutines of the two programs. So I ’ ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,”Please repeat that process.”**Score** them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ’ t tell you how they work. I ’ ve tried looking at them and telling you how they work. They ’ re obscure, weird programs. But they do the job. And in fact, I know, I ’ m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,”Oh look. Look at this. It says, ’ This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ’ Doesn ’ t that make you feel confident?”In fact, we know that the engineering process doesn ’ t work very well when it gets complicated. So we ’ re beginning to depend on computers to do a process that ’ s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ’ t quite understand the options of it. So in a sense, it ’ s getting ahead of us. We ’ re now using those programs to make much faster computers so that we ’ ll be able to run this process much faster. So it ’ s feeding back on itself. The thing is becoming faster and that ’ s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ’ re taking off. And what we are is we ’ re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ’ re the amoebas and we can ’ t quite figure out what the hell this thing is we ’ re creating. We ’ re right at that point of transition. But I think that there really is something coming along after us. I think it ’ s very haughty of us to think that we ’ re the end product of evolution. And I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

Text Sample 182: POS Dim 2 – 2005 – Score 12.00 – 2005/t000425.txt

What I ’ m going to do, in the spirit of collaborative creativity, is simply repeat many of the points that the three people before me have already made, but do them â this is called”creative collaboration ;”it ’ s actually called”borrowing”â but do it through a particular perspective, and that is to ask about the role of users and consumers in this emerging world of collaborative creativity that Jimmy and others have talked about. Let me just ask you, to start with, this simple question : who invented the mountain bike? Because traditional economic theory would say, well, the mountain bike was probably invented by some big bike corporation that had a big R&D lab where they were thinking up new projects, and it came out of there. It didn ’ t come from there. Another answer might be, well, it came from a sort of lone genius working in his garage, who, working away on different kinds of bikes, comes up with a bike out of thin air. It didn ’ t come from there. The mountain bike came from users, came from young users, particularly a group in Northern California, who were frustrated with traditional racing bikes, which were those sort of bikes that Eddy Merckx rode, or your big brother, and they ’ re very glamorous. But also frustrated with the bikes that your dad rode, which sort of had big handlebars like that, and they were too heavy. So, they got the frames from these big bikes, put them together with the gears from the racing bikes, got the brakes from motorcycles, and sort of mixed and matched various ingredients. And for the first, I don ’ t know, three to five years of their life, mountain bikes were known as”clunkers.”And they were just made in a community of bikers, mainly in Northern California. And then one of these companies that was importing parts for the clunkers decided to set up in business, start selling them to other people, and gradually another company emerged out of that, Marin, and it probably was, I don ’ t know, 10, maybe even 15, years, before the big bike companies realized there was a market. Thirty years later, mountain bike sales and mountain bike equipment account for 65 percent of bike sales in America. That ’ s 58 billion dollars. This is a category entirely created by consumers that would not have been created by the mainstream bike market because they couldn ’ t see the need, the opportunity ; they didn ’ t have the incentive to innovate. The one thing I think I disagree with about Yochai ’ s presentation is when he said the Internet causes this distributive capacity for innovation to come alive. It ’ s when the Internet combines with these kinds of **passionate** pro-am consumers â who are knowledgeable ; they ’ ve got the incentive to innovate ; they ’ ve got the tools ; they want to â that you get this kind of explosion of creative collaboration. And out of that, you get the need for the kind of things that Jimmy was talking about, which is our new kinds of organization, or a better way to put it : how do we organize ourselves without organizations? That ’ s now possible ; you don ’ t need an organization to be organized, to achieve large and complex tasks, like innovating new software programs. So this is a huge challenge to the way we think creativity comes about. The traditional view, still enshrined in much of the way that we think about creativity â in organizations, in government â is that creativity is about **special** people : wear **baseball** caps the wrong way round, come to conferences like this, in **special** places, elite universities, R&D labs in the forests, water, maybe **special** rooms in companies painted funny colors, you know, bean bags, maybe the odd table-football table. **Special** people, **special** places, think up **special** ideas, then you have a pipeline that takes the ideas down to the waiting consumers, who are passive. They can say"yes"or"no"to the invention. That ' s the idea of creativity. What ' s the policy recommendation out of that if you ' re in government, or you ' re running a large company? More **special** people, more **special** places. Build creative clusters in cities ; create more R&D parks, so on and so forth. Expand the pipeline down to the consumers. Well this view, I think, is increasingly wrong. I think it ' s always been wrong, because I think always creativity has been highly collaborative, and it ' s probably been largely interactive. But it ' s increasingly wrong, and one of the reasons it ' s wrong is that the ideas are flowing back up the pipeline. The ideas are coming back from the consumers, and they ' re often ahead of the producers. Why is that? Well, one issue is that radical innovation, when you ' ve got ideas that affect a large number of technologies or people, have a great deal of uncertainty attached to them. The payoffs to innovation are greatest where the uncertainty is highest. And when you get a radical innovation, it ' s often very uncertain how it can be applied. The whole history of telephony is a story of dealing with that uncertainty. The very first landline telephones, the inventors thought that they would be used for people to listen in to live performances from West End theaters. When the **mobile** telephone companies invented SMS, they had no idea what it was for ; it was only when that technology got into the hands of teenage users that they invented the use. So the more radical the innovation, the more the uncertainty, the more you need innovation in use to work out what a technology is for. All of our patents, our entire approach to patents and invention, is based on the idea that the inventor knows what the invention is for ; we can say what it ' s for. More and more, the inventors of things will not be able to say that in advance. It will be worked out in use, in collaboration with users. We like to think that invention is a sort of moment of creation : there is a moment of birth when someone comes up with an idea. The truth is that most creativity is cumulative and collaborative ; like Wikipedia, it develops over a long period of time. The second reason why users are more and more important is that they are the source of big, disruptive innovations. If you want to find the big new ideas, it ' s often difficult to find them in mainstream markets, in big organizations. And just look inside large organizations and you ' ll see why that is so. So, you ' re in a big corporation. You ' re obviously keen to go up the corporate ladder. Do you go into your board and say,"Look, I ' ve got a fantastic idea for an embryonic product in a marginal market, with consumers we ' ve never dealt with before, and I ' m not sure it ' s going to have a big payoff, but it could be really, really big in the future?"No, what you do, is you go in and you say,"I ' ve got a fantastic idea for an incremental innovation to an existing product we sell through existing channels to existing users, and I can guarantee you get this much return out of it over the next three years."Big corporations have an in-built tendency to reinforce past success. They ' ve got so much sunk in it that it ' s very difficult for them to spot emerging new markets. Emerging new markets, then, are the breeding grounds for **passionate** users. Best example : who in the music industry, 30 years ago, would have said,"Yes, let ' s invent a musical form which is all about dispossessed black men in ghettos expressing their frustration with the world through a form of music that many people find initially quite difficult to listen to. That sounds like a winner ; we ' ll go with it.". So what happens? Rap music is created by the users. They do it on their own tapes, with their own recording equipment ; they distribute it themselves. 30 years later, rap music is the dominant musical form of popular culture â would never have come from the big companies. Had to start â this is the third point â with these pro-ams. This is the phrase that I ' ve used in some stuff which I ' ve done with a think tank in London called Demos, where we ’ ve been looking at these people who are amateurs â i. e., they do it for the love of it â but they want to do it to very high standards. And across a whole range of fields â from software, astronomy, natural sciences, vast areas of leisure and culture like kite- surfing, so on and so forth â you find people who want to do things because they love it, but they want to do these things to very high standards. They work at their leisure, if you like. They take their leisure very seriously : they acquire skills ; they invest time ; they use technology that ’ s getting cheaper â it ’ s not just the Internet : cameras, design technology, leisure technology, surfboards, so on and so forth. Largely through globalization, a lot of this equipment has got a lot cheaper. More knowledgeable consumers, more educated, more able to connect with one another, more able to do things together. Consumption, in that sense, is an **expression** of their productive potential. Why, we found, people were interested in this, is that at work they don ’ t feel very expressed. They don ’ t feel as if they ’ re doing something that really matters to them, so they pick up these kinds of activities. This has huge **organizational** implications for very large areas of life. Take astronomy as an example, which Yochai has already mentioned. Twenty years ago, 30 years ago, only big professional astronomers with very big telescopes could see far into space. And there ’ s a big telescope in Northern England called Jodrell Bank, and when I was a kid, it was amazing, because the moon shots would take off, and this thing would move on rails. And it was huge â it was absolutely enormous. Now, six amateur astronomers, working with the Internet, with Dobsonian digital telescopes â which are pretty much open source â with some light sensors developed over the last 10 years, the Internet â they can do what Jodrell Bank could only do 30 years ago. So here in astronomy, you have this vast explosion of new productive resources. The users can be producers. What does this mean, then, for our **organizational** landscape? Well, just imagine a world, for the moment, divided into two camps. Over here, you ’ ve got the old, traditional corporate model : **special** people, **special** places ; patent it, push it down the pipeline to largely waiting, passive consumers. Over here, let ’ s imagine we ’ ve got Wikipedia, Linux, and beyond â open source. This is open ; this is closed. This is new ; this is traditional. Well, the first thing you can say, I think with certainty, is what Yochai has said already â is there is a great big **struggle** between those two **organizational** forms. These people over there will do everything they can to stop these kinds of organizations succeeding, because they ’ re threatened by them. And so the debates about copyright, digital rights, so on and so forth â these are all about trying to stifle, in my view, these kinds of organizations. What we ’ re seeing is a complete corruption of the idea of patents and copyright. Meant to be a way to incentivize invention, meant to be a way to orchestrate the dissemination of knowledge, they are increasingly being used by large companies to create thickets of patents to prevent innovation taking place. Let me just give you two examples. The first is : imagine yourself going to a venture capitalist and saying,”I ’ ve got a fantastic idea. I ’ ve invented this brilliant new program that is much, much better than Microsoft Outlook.”Which venture capitalist in their right mind is going to give you any money to set up a venture competing with Microsoft, with Microsoft Outlook? No one. That is why the competition with Microsoft is bound to come â will only come â from an open-source kind of project. So, there is a huge competitive argument about sustaining the capacity for open-source and consumer-driven innovation, because it ’ s one of the greatest competitive levers against monopoly. There ’ ll be huge professional arguments as well. Because the professionals, over here in these closed organizations â they might be academics ; they might be programmers ; they might be doctors

; they might be journalists â my former profession â say,”No, no â you can ’ t trust these people over here.”When I started in journalism â Financial Times, 20 years ago â it was very, very exciting to see someone reading the newspaper. And you ’ d kind of look over their shoulder on the Tube to see if they were reading your article. Usually they were reading the share prices, and the bit of the paper with your article on was on the floor, or something like that, and you know,”For heaven ’ s sake, what are they doing! They ’ re not reading my brilliant article!”And we allowed users, readers, two places where they could contribute to the paper : the letters page, where they could write a letter in, and we would condescend to them, cut it in half, and print it three days later. Or the op-ed page, where if they knew the editor â had been to school with him, slept with his wife â they could write an article for the op-ed page. Those were the two places. Shock, horror : now, the readers want to be writers and publishers. That ’ s not their role ; they ’ re supposed to read what we write. But they don ’ t want to be journalists. The journalists think that the bloggers want to be journalists ; they don ’ t want to be journalists ; they just want to have a voice. They want to, as Jimmy said, they want to have a dialogue, a conversation. They want to be part of that flow of information. What ’ s happening there is that the whole domain of creativity is expanding. So, there ’ s going to be a tremendous **struggle**. But, also, there ’ s going to be tremendous movement from the open to the closed. What you ’ ll see, I think, is two things that are critical, and these, I think, are two challenges for the open movement. The first is : can we really survive on volunteers? If this is so critical, do we not need it funded, organized, supported in much more structured ways? I think the idea of creating the Red Cross for information and knowledge is a fantastic idea, but can we really organize that, just on volunteers? What kind of changes do we need in public policy and funding to make that possible? What ’ s the role of the BBC, for instance, in that world? What should be the role of public policy? And finally, what I think you will see is the intelligent, closed organizations moving increasingly in the open direction. So it ’ s not going to be a contest between two camps, but, in between them, you ’ ll find all sorts of interesting places that people will occupy. New **organizational** models coming about, mixing closed and open in tricky ways. It won ’ t be so clear-cut ; it won ’ t be Microsoft versus Linux â there ’ ll be all sorts of things in between. And those **organizational** models, it turns out, are incredibly powerful, and the people who can understand them will be very, very successful. Let me just give you one final example of what that means. I was in Shanghai, in an office block built on what was a rice paddy five years ago â one of the 2, 500 skyscrapers they ’ ve built in Shanghai in the last 10 years. And I was having dinner with this guy called Timothy Chan. Timothy Chan set up an Internet business in 2000. Didn ’ t go into the Internet, kept his money, decided to go into computer games. He runs a company called Shanda, which is the largest computer games company in China. Nine thousand servers all over China, has 250 million subscribers. At any one time, there are four million people playing one of his games. How many people does he employ to **service** that population? 500 people. Well, how can he **service** 250 million people from 500 employees? Because basically, he doesn ’ t **service** them. He gives them a **platform** ; he gives them some rules ; he gives them the tools and then he kind of orchestrates the conversation ; he orchestrates the **action**. But actually, a lot of the content is created by the users themselves. And it creates a kind of stickiness between the community and the company which is really, really powerful. The best measure of that : so you go into one of his games, you create a character that you develop in the course of the game. If, for some reason, your credit card bounces, or there ’ s some other problem, you lose your character.

You ’ ve got two options. One option : you can create a new character, right from scratch, but with none of the history of your player. That costs about 100 dollars. Or you can get on a plane, fly to Shanghai, queue up outside Shanda ’ s offices â cost probably 600, 700 dollars â and reclaim your character, get your history back. Every morning, there are 600 people queuing outside their offices to reclaim these characters. So this is about companies built on communities, that provide communities with tools, resources, **platforms** in which they can share. He ’ s not open source, but it ’ s very, very powerful. So here is one of the challenges, I think, for people like me, who do a lot of work with government. If you ’ re a games company, and you ’ ve got a million players in your game, you only need one percent of them to be co-developers, contributing ideas, and you ’ ve got a development **workforce** of 10, 000 people. Imagine you could take all the children in education in Britain, and one percent of them were co-developers of education. What would that do to the resources available to the education system? Or if you got one percent of the patients in the NHS to, in some sense, be co-producers of health. The reason why â despite all the efforts to cut it down, to constrain it, to hold it back â why these open models will still start emerging with tremendous force, is that they multiply our productive resources. And one of the reasons they do that is that they turn users into producers, consumers into designers. Thank you very much.

Text Sample 183: POS Dim 2 – 2005 – Score 11.00 – 2005/t000307.txt

One of the problems of writing, and working, and looking at the Internet is that it ’ s very hard to separate fashion from deep change. And so, to start helping that, I want to take us back to 1835. In 1835, James Gordon Bennett founded the first mass-circulation newspaper in New York City. And it cost about 500 dollars to start it, which was about the equivalent of 10, 000 dollars of today. By 15 years later, by 1850, doing the same thing — starting what was experienced as a mass—circulation **daily** paper — would come to cost two and a half million dollars. 10, 000, two and a half million, 15 years. That ’ s the critical change that is being inverted by the Net. And that ’ s what I want to talk about today, and how that relates to the emergence of social production. Starting with newspapers, what we saw was high cost as an initial requirement for making information, knowledge and culture, which led to a stark bifurcation between producers — who had to be able to raise **financial** capital, just like any other industrial organization — and passive consumers that could choose from a certain set of things that this industrial model could produce. Now, the term”information society,””information economy,”for a very long time has been used as the thing that comes after the industrial revolution. But in fact, for purposes of understanding what ’ s happening today, that ’ s wrong. Because for 150 years, we ’ ve had an information economy. It ’ s just been industrial, which means those who were producing had to have a way of raising money to pay those two and a half million dollars, and later, more for the telegraph, and the radio transmitter, and the television, and eventually the mainframe. And that meant they were market based, or they were government owned, depending on what kind of system they were in. And this characterized and anchored the way information and knowledge were produced for the next 150 years. Now, let me tell you a different story. Around June 2002, the world of supercomputers had a bombshell. The Japanese had, for the first time, created the fastest supercomputer — the NEC Earth Simulator — taking the primary from the U. S., and about two years later — this, by the way, is measuring the trillion floating-point operations per second that the computer ’ s capable of running — sigh of relief

: IBM [Blue Gene] has just edged ahead of the NEC Earth Simulator. All of this completely ignores the fact that throughout this period, there ' s another supercomputer running in the world — SETI@home. Four and a half million users around the world, contributing their leftover computer cycles, whenever their computer isn ' t working, by running a screen saver, and together sharing their resources to create a massive supercomputer that NASA harnesses to analyze the data coming from radio telescopes. What this picture suggests to us is that we ' ve got a radical change in the way information production and exchange is capitalized. Not that it ' s become less capital intensive — that there ' s less money that ' s required — but that the **ownership** of this capital, the way the capitalization happens, is radically distributed. Each of us, in these advanced economies, has one of these, or something rather like it — a computer. They ' re not radically different from routers inside the middle of the network. And computation, storage and communications capacity are in the hands of practically every connected person — and these are the basic physical capital means necessary for producing information, knowledge and culture, in the hands of something like 600 million to a billion people around the planet. What this means is that for the first time since the industrial revolution, the most important means, the most important components of the core economic activities — remember, we are in an information economy — of the most advanced economies, and there more than anywhere else, are in the hands of the population at large. This is completely different than what we ' ve seen since the industrial revolution. So we ' ve got communications and computation capacity in the hands of the entire population, and we ' ve got human creativity, human wisdom, human experience — the other major experience, the other major input — which unlike simple labor — stand here turning this lever all day long — is not something that ' s the same or fungible among people. Any one of you who has taken someone else ' s job, or tried to give yours to someone else, no matter how detailed the manual, you cannot transmit what you know, what you will intuit under a certain set of circumstances. In that we ' re unique, and each of us holds this critical input into production as we hold this machine. What ' s the effect of this? So, the story that most people know is the story of free or open source software. This is market share of Apache Web server — one of the critical applications in Web-based communications. In 1995, two groups of people said,"Wow, this is really important, the Web! We need a much better Web server!"One was a motley collection of volunteers who just decided, you know, we really need this, we should write one, and what are we going to do with what — well, we ' re gonna share it! And other people will be able to develop it. The other was Microsoft. Now, if I told you that 10 years later, the motley crew of people, who didn ' t control anything that they produced, acquired 20 percent of the market and was the red line, it would be amazing! Right? Think of it in minivans. A group of automobile engineers on their weekends are competing with Toyota. Right? But, in fact, of course, the story is it ' s the 70 percent, including the major e- commerce site — 70 percent of a critical application on which Web-based communications and applications work is produced in this form, in direct competition with Microsoft. Not in a side issue — in a central strategic decision to try to capture a component of the Net. Software has done this in a way that ' s been very visible, because it ' s measurable. But the thing to see is that this actually happens throughout the Web. So, NASA, at some point, did an experiment where they took images of Mars that they were mapping, and they said, instead of having three or four fully trained Ph. D. s doing this all the time, let ' s break it up into small components, put it up on the Web, and see if people, using a very simple interface, will actually spend five minutes here, 10 minutes there, clicking. After six months, 85, 000 people used this to generate mapping at a faster rate than the images were coming in, which was, quote,”practically indistinguishable from the markings of a fully-trained Ph. D.,”once you showed it to a number of people and computed the average. Now, if you have a little girl, and she goes and writes to — well, not so little, medium little — tries to do research on Barbie. And she ’ ll come to Encarta, one of the main online encyclopedias. This is what you ’ ll find out about Barbie. This is it, there ’ s nothing more to the definition, including,”manufacturers”— plural —”now more commonly produce ethnically diverse dolls, like this black Barbie.”Which is vastly better than what you ’ ll find in the encyclopedia. com, which is Barbie, Klaus. On the other hand, if they go to Wikipedia, they ’ ll find a genuine article — and I won ’ t talk a lot about Wikipedia, because Jimmy Wales is here — but roughly equivalent to what you would find in the Britannica, differently written, including the controversies over body image and commercialization, the claims about the way in which she ’ s a good role model, etc. Another portion is not only how content is produced, but how relevance is produced. The claim to fame of Yahoo! was, we hire people to look — originally, not anymore — we hire people to look at websites and tell you — if they ’ re in the index, they ’ re good. This, on the other hand, is what 60, 000 **passionate** volunteers produce in the Open Directory Project, each one willing to spend an hour or two on something they really care about, to say, this is good. So, this is the Open Directory Project, with 60, 000 volunteers, each one spending a little bit of time, as opposed to a few hundred fully paid employees. No one owns it, no one owns the output, it ’ s free for anyone to use and it ’ s the output of people acting out of social and psychological motivations to do something interesting. This is not only outside of businesses. When you think of what is the critical innovation of Google, the critical innovation is outsourcing the one most important thing — the decision about what ’ s relevant — to the community of the Web as a whole, doing whatever they want to do : so, page rank. The critical innovation here is instead of our engineers, or our people saying which is the most relevant, we ’ re going to go out and count what you, people out there on the Web, for whatever reason — vanity, pleasure — produced links, and tied to each other. We ’ re going to count those, and count them up. And again, here, you see Barbie. com, but also, very quickly, Adiosbarbie. com, the body image for every size. A contested cultural object, which you won ’ t find anywhere soon on Overture, which is the classic market-based mechanism : whoever pays the most is highest on the list. So, all of that is in the creation of content, of relevance, basic human **expression**. But remember, the computers were also physical. Just physical materials — our PCs — we share them together. We also see this in wireless. It used to be wireless was one person owned the **license**, they transmitted in an area, and it had to be decided whether they would be **licensed** or based on property. What we ’ re seeing now is that computers and radios are becoming so sophisticated that we ’ re developing algorithms to let people own machines, like Wi-Fi devices, and overlay them with a sharing protocol that would allow a community like this to build its own wireless broadband network simply from the simple principle : When I ’ m listening, when I ’ m not using, I can help you transfer your messages ; and when you ’ re not using, you ’ ll help me transfer yours. And this is not an idealized version. These are working models that at least in some places in the United States are being implemented, at least for public security. If in 1999 I told you, let ’ s build a data storage and retrieval system. It ’ s got to store terabytes. It ’ s got to be available 24 hours a day, seven days a week. It ’ s got to be available from anywhere in the world. It has to support over 100 million users at any given moment. It ’ s got to be robust to attack, including closing the main index, injecting malicious files, armed seizure of some major nodes. You ' d say that would take years. It would take millions. But of course, what I ' m describing is P2P file sharing. Right? We always think of it as stealing music, but fundamentally, it ' s a distributed data storage and retrieval system, where people, for very obvious reasons, are willing to share their bandwidth and their storage to create something. So, essentially what we ' re seeing is the emergence of a fourth transactional framework. It used to be that there were two primary dimensions along which you could divide things. They could be market based, or non-market based ; they could be decentralized, or centralized. The price system was a market-based and decentralized system. If things worked better because you actually had somebody organizing them, you had firms, if you wanted to be in the market — or you had governments or sometimes larger non-profits in the non-market. It was too expensive to have decentralized social production, to have decentralized **action** in society. That was not about society itself. That was, in fact, economic. But what we ' re seeing now is the emergence of this fourth system of social sharing and exchange. Not that it ' s the first time that we do nice things to each other, or for each other, as social beings. We do it all the time. It ' s that it ' s the first time that it ' s having major economic impact. What characterizes them is decentralized authority. You don ' t have to ask permission, as you do in a property-based system. May I do this? It ' s open for anyone to create and innovate and share, if they want to, by themselves or with others, because property is one mechanism of coordination. But it ' s not the only one. Instead, what we see are social frameworks for all of the critical things that we use property and contract in the market : information flows to decide what are interesting problems ; who ' s available and good for something ; motivation structures — remember, money isn ' t always the best motivator. If you leave a $50 check after dinner with friends, you don ' t increase the probability of being invited back. And if dinner isn ' t entirely obvious, think of sex. It also requires certain new **organizational** approaches. And in particular, what we ' ve seen is task organization. You have to hire people who know what they ' re doing. You have to hire them to spend a lot of time. Now, take the same problem, chunk it into little modules, and motivations become **trivial**. Five minutes, instead of watching TV? Five minutes I ' ll spend just because it ' s interesting. Just because it ' s fun. Just because it gives me a certain sense of meaning, or, in places that are more involved, like Wikipedia, gives me a certain set of social relations. So, a new social phenomenon is emerging. It ' s creating, and it ' s most visible when we see it as a new form of competition. Peer-to-peer networks assaulting the recording industry ; free and open source software taking market share from Microsoft ; Skype potentially threatening traditional telecoms ; Wikipedia competing with online encyclopedias. But it ' s also a new source of opportunities for businesses. As you see a new set of social relations and behaviors emerging, you have new opportunities. Some of them are toolmakers. Instead of building well-behaved appliances — things that you know what they ' ll do in advance — you begin to build more open tools. There ' s a new set of values, a new set of things people value. You build **platforms** for self- **expression** and collaboration. Like Wikipedia, like the Open Directory Project, you ' re beginning to build **platforms**, and you see that as a model. And you see surfers, people who see this happening, and in some sense build it into a supply chain, which is a very curious one. Right? You have a belief : stuff will flow out of connected human beings. That ' ll give me something I can use, and I ' m going to contract with someone. I will deliver something based on what happens. It ' s very scary — that ' s what Google does, essentially. That ' s what IBM does in software **services**, and they ' ve done reasonably well. So, social production is a real fact, not a fad. It is the critical long-term shift caused by the Internet.

Social relations and exchange become significantly more important than they ever were as an economic phenomenon. In some contexts, it ’ s even more efficient because of the quality of the information, the ability to find the best person, the lower transaction costs. It ’ s sustainable and growing fast. But — and this is the dark lining — it is threatened by — in the same way that it threatens — the incumbent industrial systems. So next time you open the paper, and you see an intellectual property decision, a telecoms decision, it ’ s not about something small and technical. It is about the future of the freedom to be as social beings with each other, and the way information, knowledge and culture will be produced. Because it is in this context that we see a battle over how easy or hard it will be for the industrial information economy to simply go on as it goes, or for the new model of production to begin to develop alongside that industrial model, and change the way we begin to see the world and report what it is that we see. Thank you.

Text Sample 184: POS Dim 2 – 2005 – Score 9.00 – 2005/t000424.txt

What I ’ d like to start off with is an observation, which is that if I ’ ve learned anything over the last year, it ’ s that the supreme irony of publishing a book about slowness is that you have to go around promoting it really fast. I seem to spend most of my time these days zipping from city to city, studio to studio, interview to interview, serving up the book in really tiny bite-size chunks. Because everyone these days wants to know how to slow down, but they want to know how to slow down really quickly. So... so I did a spot on CNN the other day where I actually spent more time in makeup than I did talking on air. And I think that — that ’ s not really surprising though, is it? Because that ’ s kind of the world that we live in now, a world stuck in **fast-forward**. A world obsessed with speed, with doing everything faster, with cramming more and more into less and less time. Every moment of the day feels like a race against the clock. To borrow a phrase from Carrie Fisher, which is in my bio there ; I ’ ll just toss it out again —”These days even **instant** gratification takes too long.”And if you think about how we to try to make things better, what do we do? No, we speed them up, don ’ t we? So we used to dial ; now we speed dial. We used to read ; now we speed read. We used to walk ; now we speed walk. And of course, we used to date and now we speed date. And even things that are by their very nature slow — we try and speed them up too. So I was in New York recently, and I walked past a gym that had an advertisement in the window for a new course, a new evening course. And it was for, you guessed it, speed yoga. So this — the perfect solution for time-starved professionals who want to, you know, salute the sun, but only want to give over about 20 minutes to it. I mean, these are sort of the extreme examples, and they ’ re amusing and good to laugh at. But there ’ s a very serious point, and I think that in the headlong dash of **daily** life, we often lose sight of the damage that this roadrunner form of living does to us. We ’ re so marinated in the culture of speed that we almost fail to notice the toll it takes on every aspect of our lives — on our health, our diet, our work, our relationships, the environment and our community. And sometimes it takes a wake-up call, doesn ’ t it, to alert us to the fact that we ’ re hurrying through our lives, instead of actually living them ; that we ’ re living the fast life, instead of the good life. And I think for many people, that wake-up call takes the form of an illness. You know, a **burnout**, or eventually the body says,”I can ’ t take it anymore,”and throws in the towel. Or maybe a relationship goes up in smoke because we haven ’ t had the time, or the patience, or the tranquility, to be with the other person, to listen to them. And my wake-up call came when I started reading bedtime stories to my son, and I found that at the end of day, I would go into his room and I just couldn ’ t slow down — you know, I ’ d be speed reading”The Cat In The Hat.”I ’ d be — you know, I ’ d be skipping lines here, paragraphs there, sometimes a whole page, and of course, my little boy knew the book inside out, so we would quarrel. And what should have been the most relaxing, the most intimate, the most tender moment of the day, when a dad sits down to read to his son, became instead this kind of gladiatorial battle of wills, a clash between my speed and his slowness. And this went on for some time, until I caught myself scanning a newspaper article with timesaving tips for fast people. And one of them made reference to a series of books called”The One-Minute Bedtime Story.”And I wince saying those words now, but my first reaction at the time was very different. My first reflex was to say,”Hallelujah — what a great idea! This is exactly what I ’ m looking for to speed up bedtime even more.”But thankfully, a light bulb went on over my head, and my next reaction was very different, and I took a step back, and I thought,”Whoa — you know, has it really come to this? Am I really in such a hurry that I ’ m prepared to fob off my son with a sound byte at the end of the day?”And I put away the newspaper — and I was getting on a plane — and I sat there, and I did something I hadn ’ t done for a long time — which is I did nothing. I just thought, and I thought long and hard. And by the time I got off that plane, I ’ d decided I wanted to do something about it. I wanted to investigate this whole roadrunner culture, and what it was doing to me and to everyone else. And I had two questions in my head. The first was, how did we get so fast? And the second is, is it possible, or even desirable, to slow down? Now, if you think about how our world got so accelerated, the usual suspects rear their heads. You think of, you know, urbanization, consumerism, the **workplace**, technology. But I think if you cut through those forces, you get to what might be the deeper driver, the nub of the question, which is how we think about time itself. In other cultures, time is cyclical. It ’ s seen as moving in great, unhurried circles. It ’ s always renewing and refreshing itself. Whereas in the West, time is linear. It ’ s a finite resource ; it ’ s always draining away. You either use it, or lose it.”Time is money,”as Benjamin Franklin said. And I think what that does to us psychologically is it creates an equation. Time is scarce, so what do we do? Well — well, we speed up, don ’ t we? We try and do more and more with less and less time. We turn every moment of every day into a race to the finish line — a finish line, incidentally, that we never reach, but a finish line nonetheless. And I guess that the question is, is it possible to break free from that **mindset**? And thankfully, the answer is yes, because what I discovered, when I began looking around, that there is a global backlash against this culture that tells us that faster is always better, and that busier is best. Right across the world, people are doing the unthinkable : they ’ re slowing down, and finding that, although conventional wisdom tells you that if you slow down, you ’ re road kill, the opposite turns out to be true : that by slowing down at the right moments, people find that they do everything better. They eat better ; they make love better ; they exercise better ; they work better ; they live better. And, in this kind of cauldron of moments and places and acts of deceleration, lie what a lot of people now refer to as the”International Slow Movement.”Now if you ’ ll permit me a small act of hypocrisy, I ’ ll just give you a very quick overview of what ’ s going on inside the Slow Movement. If you think of food, many of you will have heard of the Slow Food movement. Started in Italy, but has spread across the world, and now has 100, 000 members in 50 countries. And it ’ s driven by a very simple and sensible message, which is that we get more pleasure and more health from our food when we cultivate, cook and consume it at a reasonable pace. I think also the explosion of the organic farming movement, and the renaissance of farmers ' markets, are other illustrations of the fact that people are desperate to get away from eating and cooking and cultivating their food on an industrial timetable. They want to get back to slower rhythms. And out of the Slow Food movement has grown something called the Slow Cities movement, which has started in Italy, but has spread right across Europe and beyond. And in this, towns begin to **rethink** how they organize the urban landscape, so that people are encouraged to slow down and smell the roses and connect with one another. So they might curb traffic, or put in a park bench, or some green space. And in some ways, these changes add up to more than the sum of their parts, because I think when a Slow City becomes officially a Slow City, it ' s kind of like a philosophical declaration. It ' s saying to the rest of world, and to the people in that town, that we believe that in the 21st century, slowness has a role to play. In medicine, I think a lot of people are deeply disillusioned with the kind of quick-fix mentality you find in conventional medicine. And millions of them around the world are turning to complementary and alternative forms of medicine, which tend to tap into sort of slower, gentler, more holistic forms of healing. Now, obviously the jury is out on many of these complementary therapies, and I personally doubt that the coffee enema will ever, you know, gain mainstream approval. But other treatments such as acupuncture and massage, and even just relaxation, clearly have some kind of benefit. And blue-chip medical colleges everywhere are starting to study these things to find out how they work, and what we might learn from them. Sex. There ' s an awful lot of fast sex around, isn ' t there? I was coming to — well — no pun intended there. I was making my way, let ' s say, slowly to Oxford, and I went through a news agent, and I saw a magazine, a men ' s magazine, and it said on the front,"How to bring your partner to orgasm in 30 seconds."So, you know, even sex is on a stopwatch these days. Now, you know, I like a quickie as much as the next person, but I think that there ' s an awful lot to be gained from slow sex — from slowing down in the bedroom. You know, you tap into that — those deeper, sort of, psychological, emotional, spiritual currents, and you get a better orgasm with the buildup. You can get more bang for your buck, let ' s say. I mean, the Pointer Sisters said it most eloquently, didn ' t they, when they sang the praises of"a lover with a slow hand."Now, we all laughed at Sting a few years ago when he went Tantric, but you **fast-forward** a few years, and now you find couples of all ages flocking to workshops, or maybe just on their own in their own bedrooms, finding ways to put on the brakes and have better sex. And of course, in Italy where — I mean, Italians always seem to know where to find their pleasure — they ' ve launched an official Slow Sex movement. The **workplace**. Right across much of the world — North America being a notable exception — working hours have been coming down. And Europe is an example of that, and people finding that their quality of life improves as they ' re working less, and also that their hourly productivity goes up. Now, clearly there are problems with the 35-hour workweek in France — too much, too soon, too rigid. But other countries in Europe, notably the Nordic countries, are showing that it ' s possible to have a kick-ass economy without being a workaholic. And Norway, Sweden, Denmark and Finland now rank among the top six most competitive nations on Earth, and they work the kind of hours that would make the average American weep with envy. And if you go beyond sort of the country level, down at the micro-company level, more and more companies now are realizing that they need to allow their staff either to work fewer hours or just to unplug — to take a lunch break, or to go sit in a quiet room, to switch off their Blackberrys and laptops — you at the back — **mobile** phones, during the work day or on the weekend, so that they have time to recharge and for the brain to slide into that kind of creative mode of thought. It ' s not just, though, these days, adults who overwork, though, is it? It ' s children, too. I ' m 37, and my childhood ended in the mid- ' 80s, and I look at kids now, and I ' m just amazed by the way they race around with more homework, more tutoring, more extracurriculars than we would ever have conceived of a generation ago. And some of the most heartrending emails that I get on my website are actually from adolescents hovering on the edge of **burnout**, pleading with me to write to their parents, to help them slow down, to help them get off this full-throttle treadmill. But thankfully, there is a backlash there in parenting as well, and you ' re finding that, you know, towns in the United States are now banding together and banning extracurriculars on a particular day of the month, so that people can, you know, decompress and have some family time, and slow down. Homework is another thing. There are homework bans springing up all over the developed world in schools which had been piling on the homework for years, and now they ' re discovering that less can be more. So there was a case up in Scotland recently where a fee-paying, high-achieving private school banned homework for everyone under the age of 13, and the high-achieving parents freaked out and said,"What are you — you know, our kids will fall"— the headmaster said,"No, no, your children need to slow down at the end of the day."And just this last month, the exam results came in, and in math, science, marks went up 20 percent on average last year. And I think what ' s very revealing is that the elite universities, who are often cited as the reason that people drive their kids and hothouse them so much, are starting to notice the caliber of students coming to them is falling. These kids have wonderful marks ; they have CVs jammed with extracurriculars, to the point that would make your eyes water. But they lack spark ; they lack the ability to think creatively and think outside — they don ' t know how to dream. And so what these Ivy League schools, and Oxford and Cambridge and so on, are starting to send a message to parents and students that they need to put on the brakes a little bit. And in Harvard, for instance, they send out a letter to undergraduates — freshmen — telling them that they ' ll get more out of life, and more out of Harvard, if they put on the brakes, if they do less, but give time to things, the time that things need, to enjoy them, to savor them. And even if they sometimes do nothing at all. And that letter is called — very revealing, I think —"Slow Down!"— with an exclamation mark on the end. So wherever you look, the message, it seems to me, is the same : that less is very often more, that slower is very often better. But that said, of course, it ' s not that easy to slow down, is it? I mean, you heard that I got a speeding ticket while I was researching my book on the benefits of slowness, and that ' s true, but that ' s not all of it. I was actually en route to a dinner held by Slow Food at the time. And if that ' s not shaming enough, I got that ticket in Italy. And if any of you have ever driven on an Italian highway, you ' ll have a pretty good idea of how fast I was going. But why is it so hard to slow down? I think there are various reasons. One is that speed is fun, you know, speed is sexy. It ' s all that adrenaline rush. It ' s hard to give it up. I think there ' s a kind of metaphysical dimension — that speed becomes a way of walling ourselves off from the bigger, deeper questions. We fill our head with distraction, with busyness, so that we don ' t have to ask, am I well? Am I happy? Are my children growing up right? Are politicians making good decisions on my behalf? Another reason — although I think, perhaps, the most powerful reason — why we find it hard to slow down is the cultural taboo that we ' ve erected against slowing down."Slow"is a dirty word in our culture. It ' s a byword for"lazy,""slacker,"for being somebody who gives up. You know,"he ' s a bit slow."It ' s actually synonymous with being stupid. I guess what the Slow Movement — the purpose of the Slow Movement, or its main goal, really, is to tackle that taboo, and to say that yes, sometimes slow is not the answer, that there is such a thing as”bad slow.”You know, I got stuck on the M25, which is a ring road around London, recently, and spent three-and-a-half hours there. And I can tell you, that ’ s really bad slow. But the new idea, the sort of revolutionary idea, of the Slow Movement, is that there is such a thing as”good slow,”too. And good slow is, you know, taking the time to eat a meal with your family, with the TV switched off. Or taking the time to look at a problem from all angles in the office to make the best decision at work. Or even simply just taking the time to slow down and savor your life. Now, one of the things that I found most uplifting about all of this stuff that ’ s happened around the book since it came out, is the reaction to it. And I knew that when my book on slowness came out, it would be welcomed by the New Age brigade, but it ’ s also been taken up, with great gusto, by the corporate world — you know, business press, but also big companies and leadership organizations. Because people at the top of the chain, people like you, I think, are starting to realize that there ’ s too much speed in the system, there ’ s too much busyness, and it ’ s time to find, or get back to that lost art of shifting gears. Another encouraging sign, I think, is that it ’ s not just in the developed world that this idea ’ s been taken up. In the developing world, in countries that are on the verge of making that leap into first world status — China, Brazil, Thailand, Poland, and so on — these countries have embraced the idea of the Slow Movement, many people in them, and there ’ s a debate going on in their media, on the streets. Because I think they ’ re looking at the West, and they ’ re saying,”Well, we like that aspect of what you ’ ve got, but we ’ re not so sure about that.”So all of that said, is it, I guess, is it possible? That ’ s really the main question before us today. Is it possible to slow down? And I ’ m happy to be able to say to you that the answer is a resounding yes. And I present myself as Exhibit A, a kind of reformed and rehabilitated speed-aholic. I still love speed. You know, I live in London, and I work as a journalist, and I enjoy the buzz and the busyness, and the adrenaline rush that comes from both of those things. I play squash and ice hockey, two very fast sports, and I wouldn ’ t give them up for the world. But I ’ ve also, over the last year or so, got in touch with my inner tortoise. And what that means is that I no longer overload myself gratuitously. My default mode is no longer to be a rush-aholic. I no longer hear time ’ s winged chariot drawing near, or at least not as much as I did before. I can actually hear it now, because I see my time is ticking off. And the upshot of all of that is that I actually feel a lot happier, healthier, more productive than I ever have. I feel like I ’ m living my life rather than actually just racing through it. And perhaps, the most important measure of the success of this is that I feel that my relationships are a lot deeper, richer, stronger. And for me, I guess, the litmus test for whether this would work, and what it would mean, was always going to be bedtime stories, because that ’ s sort of where the journey began. And there too the news is rosy. You know, at the end of the day, I go into my son ’ s room. I don ’ t wear a watch. I switch off my computer, so I can ’ t hear the email pinging into the basket, and I just slow down to his pace and we read. And because children have their own tempo and internal clock, they don ’ t do quality time, where you schedule 10 minutes for them to open up to you. They need you to move at their rhythm. I find that 10 minutes into a story, you know, my son will suddenly say,”You know, something happened in the playground today that really bothered me.”And we ’ ll go off and have a conversation on that. And I now find that bedtime stories used to be a box on my to-do list, something that I **dreaded**, because it was so slow and I had to get through it quickly. It ’ s become my reward at the end of the day, something I really cherish. And I have a kind of Hollywood ending to my talk this afternoon, which goes a little bit like this : a few months ago, I was getting ready to go on another book tour, and I had my bags packed. I was downstairs by the front door, and I was waiting for a taxi, and my son came down the stairs and he ' d made a card for me. And he was carrying it. He ' d gone and stapled two cards, very like these, together, and put a sticker of his favorite character, Tintin, on the front. And he said to me, or he handed this to me, and I read it, and it said,"To Daddy, love Benjamin."And I thought,"Aw, that ' s really sweet. Is that a good luck on the book tour card?"And he said,"No, no, no, Daddy — this is a card for being the best story reader in the world."And I thought,"Yeah, you know, this slowing down thing really does work."Thank you very much.

Text Sample 185: POS Dim 2 – 2004 – Score 11.00 – 2004/t000486.txt
You ' ll be happy to know that I ' ll be talking not about my own tragedy, but other people ' s tragedy. It ' s a lot easier to be lighthearted about other people ' s tragedy than your own, and I want to keep it in the spirit of the conference. So, if you believe the media accounts, being a drug dealer in the height of the crack cocaine epidemic was a very glamorous life, in the words of Virginia Postrel. There was money, there was drugs, guns, women, you know, you name it — jewelry, bling-bling — it had it all. What I ' m going to tell you today is that, in fact, based on 10 years of research, a unique opportunity to go inside a gang — to see the actual books, the **financial** records of the gang — that the answer turns out not to be that being in the gang was a glamorous life. But I think, more realistically, that being in a gang — selling drugs for a gang — is perhaps the worst job in all of America. And that ' s what I ' d like to convince you of today. So there are three things I want to do. First, I want to explain how and why crack cocaine had such a profound influence on inner-city gangs. Secondly, I want to tell you how somebody like me came to be able to see the inner workings of a gang — an interesting story, I think. And then third, I want to tell you, in a very superficial way, about some of the things we found when we actually got to look at the **financial** records, the books, of the gang. So before I do that, just one **warning**, which is that this presentation has been rated ' R ' by the Motion Picture Association of America. It contains adult themes, adult language. Given who is up on the stage, you ' ll be delighted to know that, in fact, there ' ll be no nudity — **Unexpected** wardrobe malfunctions aside. So let me start by talking about crack cocaine, and how it transformed the gang. To do that, you have to actually go back to a time before crack cocaine, in the early ' 80s, and look at it from the perspective of a gang leader. Being a gang leader in the inner city wasn ' t such a bad deal in the mid- ' 80s — the early ' 80s, let me say. Now, you had a lot of power, and you got to beat people up — you got a lot of prestige, a lot of respect. But the thing is, there was no money in it. The gang had no way to make money. You couldn ' t charge dues to the people in the gang, because the people in the gang didn ' t have any money. You couldn ' t really make any money selling marijuana — marijuana ' s too cheap, it turns out. You can ' t get rich selling marijuana. You couldn ' t sell cocaine ; cocaine ' s a great product — powdered cocaine — but you ' ve got to know rich white people. And most of the inner-city gang members didn ' t know any rich white people, so couldn ' t sell to that market. You couldn ' t really do petty crime, either. Turns out, petty crime ' s a terrible way to make a living. As a result, as a gang leader, you had, you know, power — it ' s a pretty good life — but the thing was, in the end, you were living at home with your mother. And so it wasn ' t really a career. There were limits to how powerful and important you could be if you had to live at home with your mother. Then along comes crack cocaine. And in the words of Malcolm Gladwell, crack cocaine was the extra-chunky version of tomato sauce for the

inner city. Because crack cocaine was an unbelievable innovation. I don ’ t have time to talk about it today, but if you think about it, I would say that in the last 25 years, of every invention or innovation that ’ s occurred in this country, the biggest one in terms of impact on the well-being of people who live in the inner city, was crack cocaine. And for the worse — not for the better, but for the worse. It had a huge impact on life. So what was it about crack cocaine? It was a brilliant way of getting the brain high. Because you could smoke crack cocaine — you can ’ t smoke powdered cocaine — and smoking is a much more efficient mechanism of delivering a high than snorting it. And it turned out there was this audience that didn ’ t know it wanted crack cocaine, but when it came, it really did. And it was a perfect drug ; you could buy the cocaine that went into it for a dollar, sell it for five dollars. Highly addictive — the high was very short. So for fifteen minutes, you get this great high, and then when you come down, all you want to do is get high again. It created a wonderful market. And for the people who were there running the gang, it was a great way, seemingly, to make a lot of money. At least for the people on the top. So this is where we enter the picture. Not really me — I ’ m really a bit player in all this. My co-author, Sudhir Venkatesh, is the main character. He was a math major in college who had a good heart, and decided he wanted to get a sociology PhD, came to the University of Chicago. Now, the three months before he came to Chicago, he had spent following the Grateful Dead. And in his own words, he”looked like a freak.”He ’ s a South Asian — very dark-skinned South Asian. Big man, and he had hair, in his words,”down to his ass.”Defied all kinds of boundaries : Was he black or white? Was he man or woman? He was really a curious sight to be seen. So he showed up at the University of Chicago, and the famous sociologist William Julius Wilson was doing a book that involved surveying people all across Chicago. He took one look at Sudhir, who was going to go do some surveys for him, and decided he knew exactly the place to send him, which was to one of the toughest, most notorious housing projects not just in Chicago, but in the entire United States. So Sudhir, the suburban boy who had never really been in the inner city, dutifully took his clipboard and walked down to this housing project, gets to the first building. The first building? Well, there ’ s nobody there. But he hears some voices up in the stairwell, so he climbs up the stairwell, comes around the corner, and finds a group of young African-American men playing dice. This is about 1990, **peak** of the crack epidemic. This is a very dangerous job, being in a gang. You don ’ t like to be surprised. You don ’ t like to be surprised by people who come around the corner. And the mantra was : shoot first ; ask questions later. Now, Sudhir was lucky — he was such a freak, and that clipboard probably saved his life, because they figured no other rival gang member would be coming up to shoot at them with a clipboard. So his greeting was not particularly warm, but they did say, well, OK — let ’ s hear your questions on your survey. So — I kid you not — the first question on the survey that he was sent to ask was :”How do you feel about being poor and Black in America?”Makes you wonder about academics. So the choice of answers were : [A) Very Good B) Good C) Bad D) Very Bad] What Sudhir found out is, in fact, that the real answer was the following : [A) Very Good B) Good C) Bad D) Very Bad E) Fuck you] The survey was not, in the end, going to be what got Sudhir off the hook. He was held hostage overnight in the stairwell. There was a lot of gunfire, there were a lot of philosophical discussions he had with the gang members. By morning, the gang leader arrived, checked out Sudhir, decided he was no threat, and they let him go home. So Sudhir went home, took a shower, took a nap. And you and I, probably, faced with the situation, would think,”I guess I ’ m going to write my dissertation on The Grateful Dead, I ’ ve been following them for the last three months.”Sudhir, on the other hand, got right back, walked down to the housing project, went up to the second floor, and said :”Hey, guys, I had so much fun hanging out with you last night, I wonder if I could do it again tonight.”And that was the beginning of what turned out to be a beautiful relationship that involved Sudhir living in the housing project on and off for 10 years, hanging out in crack houses, going to jail with the gang members, having the windows shot out of his car, having the police break into his apartment and steal his computer disks — you name it. But ultimately, the story has a happy ending for Sudhir, who became one of the most respected sociologists in the country. And especially for me, as I sat in my office with my Excel spreadsheet open, waiting for Sudhir to come and deliver to me the latest load of data that he would get from the gang. It was one of the most unequal co-authoring relationships ever — But I was glad to be the beneficiary of it. So what did we find? What did we find in the gang? Well, let me say one thing : We really got access to everybody in the gang. We got an inside look at the gang, from the very bottom up to the very top. They trusted Sudhir, in ways that really no academic has ever — or really anybody, any outsider — has ever earned the trust of these gangs, to the point where they actually opened up what was most interesting for me — their books, the **financial** records they kept. They made them available to us, and we not only could study them, but we could ask them questions about what was in them. So if I have to kind of summarize very quickly in the short time I have what the bottom line of what I take away from the gang is, it ’ s that, if I had to draw a parallel between the gang and any other organization, it would be that the gang is just like McDonald ’ s, in a lot of different respects — the restaurant McDonald ’ s. So first, in one way, which isn ’ t maybe the most interesting way, but it ’ s a good way to start — is in the way it ’ s organized, the hierarchy of the gang, the way it looks. So here ’ s what the org chart of the gang looks like. I don ’ t know if you know much about org charts, but if you were to assign a stripped- down and simplified McDonald ’ s org chart, this is exactly what it would look like. It ’ s amazing, but the top level of the gang, they actually call themselves the”Board of Directors.”And Sudhir says it ’ s not like these guys had a very sophisticated view of what happened in American corporate life, but they had seen movies like”Wall Street,”and they had learned a little bit about what it was like to be in the real world. Now, below that board of directors, you ’ ve got essentially what are regional VPs — people who control, say, the South Side of Chicago, or the West Side of Chicago. Sudhir got to know very well the guy who had the unfortunate assignment of trying to take the Iowa franchise, which, it turned out, for this black gang, was not one of the more brilliant **financial** endeavors they undertook. But the thing that really makes the gang seem like McDonald ’ s is its franchisees. The guys who are running the local gangs — the four-square-block by four-square-block areas — they ’ re just like the guys, in some sense, who are running the McDonald ’ s. They are the entrepreneurs. They get the exclusive property rights to control the drug-selling. They get the name of the gang behind them, for merchandising and marketing. And they ’ re the ones who basically make the profit or lose a profit, depending on how good they are at running the business. Now, the group I really want you to think about, though, are the ones at the bottom — the foot soldiers. These are the teenagers, typically, who ’ d be standing out on the street corner, selling the drugs. Extremely dangerous work. And important to note is that almost all of the weight, all of the people in this organization are at the bottom — just like McDonald ’ s. So in some sense, the foot soldiers are a lot like the people who are taking your order at McDonald ’ s, and it ’ s not just by chance that they ’ re like them. In fact, in these neighborhoods, they ’ d be the same people. So the same kids who are working in the gang were actually, at the very same time, typically working part-time at a place like McDonald ' s. Which already foreshadows the main result that I ' ve talked about, about what a crappy job it was, being in the gang. Because obviously, if being in the gang were such a wonderful, lucrative job, why in the world would these guys moonlight at McDonald ' s? So what do the wages look like? You might be surprised. But based on being able to talk to them and to see their records, this is what it looks like in terms of the wages. The hourly wage the foot soldiers were earning was $3. 50 an hour. It was below the minimum wage. And this is well-documented. It ' s easy to see by the patterns of consumption they have. It really is not fiction — it ' s fact. There was very little money in the gang, especially at the bottom. Now if you managed to rise up, say, and be that local leader, the guy who ' s the equivalent of the McDonald ' s franchisee, you ' d be making 100, 000 dollars a year. And that, in some ways, was the best job you could hope to get if you were growing up in one of these neighborhoods as a young black male. If you managed to rise to the very top, 200, 000 or 400, 000 dollars a year is what you ' d hope to make. Truly, you would be a great success story. And one of the sad parts of this is that, indeed, among the many other ramifications of crack cocaine is that the most talented individuals in these communities — this is what they were striving for. They weren ' t trying to make it in legitimate ways, because there were no legitimate channels out. This was the best way out. And it actually was the right choice, probably, to try to make it out this way. You look at this, the relationship to McDonald ' s breaks down here. The money looks about the same. Why is it such a bad job? Well, the reason it ' s such a bad job is that there ' s somebody shooting at you a lot of the time. So, with shooting at you, what are the death rates? We found, in our gang — and admittedly, this was not really a standard situation ; this was a time of intense violence, of a lot of gang wars, as this gang actually became quite successful. But there were costs. And so the death rate — not to mention the rate of being arrested, sent to prison, being wounded — the death rate in our sample was seven percent per person per year. You ' re in the gang for four years, you expect to die with about a 25 percent likelihood. That is about as high as you can get. So for comparison ' s purposes, let ' s think about some other walk of life you may expect might be extremely risky. Let ' s say that you were a murderer and you were convicted of murder, and you ' re sent to death **row**. It turns out, the death rates on death **row** from all causes, including execution : two percent a year. So it ' s a lot safer being on death **row** than it is selling drugs out on the street. That gives you some pause, for those of you who believe that a death **penalty** ' s going to have an enormous deterrent effect on crime. To give you a sense of just how bad the inner city was during crack — and I ' m not really focusing on the negatives, but really, there ' s another story to tell you there — if you look at the death rates just of random, young black males growing up in the inner city in the United States, the death rates during crack were about one percent. That ' s extremely high. And this is violent death — it ' s unbelievable, in some sense. To put it into perspective : if you compare this to the soldiers in Iraq, for instance, right now fighting the war : 0. 5 percent. So in some very literal way, the young black men who were growing up in this country were living in a war zone, very much in the sense that the soldiers over in Iraq are fighting in a war. So why in the world, you might ask, would anybody be willing to stand out on a street corner selling drugs for $3. 50 an hour, with a 25 percent chance of dying over the next four years? Why would they do that? And I think there are a couple answers. I think the first one is that they got **fooled** by history. It used to be the gang was a rite of passage ; that the young people controlled the gang ; that as you got older, you dropped out of the gang. So what happened was, the people who happened to be in the right place at the right time — the people who happened to be leading the gang in the mid-to-late- ’ 80s — became very, very wealthy. And so the logical thing to think was that they are going to age out of the gang like everybody else has, and the next generation is going to take over and get the wealth. There are striking similarities, I think, to the Internet boom. The first set of people in Silicon Valley got very, very rich. And then all of my friends said,”Maybe I should go do that, too.”And they were willing to work very cheap for stock options that never came. In some sense, that ’ s what happened, exactly, to the set of people we were looking at. They were willing to start at the bottom, just like, say, a first-year lawyer at a law firm is willing to start at the bottom, work 80-hour weeks for not that much money, because they think they ’ re going to make partner. But the rules changed, and they never got to make partner. Indeed, the same people who were running all of the major gangs in the late 1980s are still running the major gangs in Chicago today. They never passed on any of the wealth, So everybody got stuck at that $3. 50-an-hour job, and it turned out to be a disaster. The other thing the gang was very good at was marketing and trickery. And so for instance, one thing the gang would do is — the gang leaders would have big entourages, and they ’ d drive fancy cars and have fancy jewelry. So what Sudhir eventually realized as he hung out with them more, is that, really, they didn ’ t own those cars — they just leased them, because they couldn ’ t afford to own the fancy cars. And they didn ’ t really have gold jewelry, they had gold- plated jewelry. It goes back to, you know, the real-real versus the fake-real. And really, they did all sorts of things to trick the young people into thinking what a great deal the gang was going to be. So for instance, they would give a 14-year-old kid a whole roll of bills to hold. That 14-year-old kid would say to his friends,”Hey, look at all the money I got in the gang.”It wasn ’ t his money — until he spent it, and then he was in debt to the gang, and was sort of an indentured servant for a while. So I have a couple minutes. Let me do one last thing I hadn ’ t thought I ’ d have time to do, which is to talk about what we learned more generally about economics, from the study of the gang. So, economists tend to talk in technical words. Often, our theories fail quite miserably when we over the data, but what ’ s kind of interesting is that in this **setting**, it turned out that some of the economic theories that worked not so well in the real economy worked very well in the drug economy, in some sense, because it ’ s unfettered capitalism. Here ’ s an economic principle. This is one of the basic ideas in labor economics, called a”compensating differential.”It ’ s the idea that the increment to wages that a worker requires to leave him indifferent between performing two tasks, one which is more unpleasant than the other. Compensating differential — it ’ s why we think garbagemen might be paid more than people who work in parks. The words of one of the members of the gang, I think, make this clear. So it turns out — I ’ m sort of getting ahead of myself — it turns out, in the gang, when there ’ s a war going on, they actually pay the foot soldiers twice as much money. It ’ s exactly this concept. Because they ’ re not willing to be at risk. And the words of a gang member capture it quite nicely, he says :”Would you stand around here when all this shit...”— the shooting —”... if all this shit ’ s going on? No, right? So if I gonna be asked to put my life on the line, then front me the cash, man.”I think the gang member says it much more articulately than the economist, about what ’ s going on. Here ’ s another one. Economists talk about game theory, that every two-person game has a Nash equilibrium. Here ’ s the translation you get from the gang member. They ’ re talking about the decision of why they don ’ t go shoot — One thing that turns out to be a great business tactic in the gang : if you go and just shoot guns in the air in the other gang ’ s territory — people are afraid to go buy drugs there, they ’ re going to come into your neighborhood. Here ’ s what he says about why they don ’ t do that :”If we start shooting around there, the other gang ’ s territory, nobody, I mean, you dig it, nobody gonna step on their turf. But we **gotta** be careful, ’ cause they can shoot around here too and then we all fucked.”So that ’ s the same concept. Then again, sometimes economists get it wrong. One thing we **observed** in the data is that it looked like — the gang leader always got paid. No matter how bad it was economically, he always got himself paid. We had some theories related to cash flow, and lack of access to capital markets, and things like that. Then we asked the gang member,”Why is it you always get paid and your workers don ’ t always get paid?”His response is,”You got all these niggers below you who want your job, you dig? If you start taking losses, they see you as weak and shit.”And I thought about it and said,”CEOs often pay themselves million-dollar bonuses, even when companies are losing a lot of money. And it never would really occur to an economist that this idea of ’ weak and shit ’ could really be important.”Maybe”weak and shit”is an important hypothesis that needs more analysis. Thank you very much.

Text Sample 186: POS Dim 2 – 2004 – Score 9.00 – 2004/t000177.txt

I was trying to think, how is sync connected to **happiness**, and it occurred to me that for some reason we take pleasure in synchronizing. We like to dance together, we like singing together. And so, if you ’ ll put up with this, I would like to enlist your help with a first experiment today. The experiment is — and I notice, by the way, that when you applauded, that you did it in a typical North American way, that is, you were raucous and incoherent. You were not organized. It didn ’ t even occur to you to clap in unison. Do you think you could do it? I would like to see if this audience would — no, you haven ’ t **practiced**, as far as I know — can you get it together to clap in sync? Whoa! Now, that ’ s what we call emergent behavior. So I didn ’ t expect that, but — I mean, I expected you could synchronize. It didn ’ t occur to me you ’ d increase your frequency. It ’ s interesting. So what do we make of that? First of all, we know that you ’ re all brilliant. This is a room full of intelligent people, highly sensitive. Some trained musicians out there. Is that what enabled you to synchronize? So to put the question a little more seriously, let ’ s ask ourselves what are the minimum requirements for what you just did, for spontaneous synchronization. Do you need, for instance, to be as smart as you are? Do you even need a brain at all just to synchronize? Do you need to be alive? I mean, that ’ s a spooky thought, right? Inanimate objects that might spontaneously synchronize themselves. It ’ s real. In fact, I ’ ll try to explain today that sync is maybe one of, if not one of the most, perhaps the most pervasive drive in all of nature. It extends from the subatomic scale to the farthest reaches of the cosmos. It ’ s a deep tendency toward order in nature that opposes what we ’ ve all been taught about entropy. I mean, I ’ m not saying the law of entropy is wrong — it ’ s not. But there is a countervailing force in the universe — the tendency towards spontaneous order. And so that ’ s our theme. Now, to get into that, let me begin with what might have occurred to you immediately when you hear that we ’ re talking about synchrony in nature, which is the glorious example of birds that flock together, or fish swimming in organized schools. So these are not particularly intelligent creatures, and yet, as we ’ ll see, they exhibit beautiful ballets. This is from a BBC show called”Predators,”and what we ’ re looking at here are examples of synchrony that have to do with defense. When you ’ re small and vulnerable, like these starlings, or like the fish, it helps to swarm to avoid predators, to confuse predators. Let me be quiet for a second because this is so gorgeous. For a long time, biologists were puzzled by this behavior, wondering how it could be possible. We ’ re so used to choreography giving rise to synchrony. These creatures are not choreographed. They ’ re choreographing themselves. And only today is science starting to figure out how it works. I ’ ll show you a computer model made by Iain Couzin, a **researcher** at Oxford, that shows how swarms work. There are just three simple rules. First, all the individuals are only aware of their nearest neighbors. Second, all the individuals have a tendency to line up. And third, they ’ re all attracted to each other, but they try to keep a small distance apart. And when you build those three rules in, automatically you start to see swarms that look very much like fish schools or bird flocks. Now, fish like to stay close together, about a body length apart. Birds try to stay about three or four body lengths apart. But except for that difference, the rules are the same for both. Now, all this changes when a predator enters the scene. There ’ s a fourth rule : when a predator ’ s coming, get out of the way. Here on the model you see the predator attacking. The prey move out in random directions, and then the rule of attraction brings them back together again, so there ’ s this constant splitting and reforming. And you see that in nature. Keep in mind that, although it looks as if each individual is acting to cooperate, what ’ s really going on is a kind of selfish Darwinian behavior. Each is scattering away at random to try to save its scales or feathers. That is, out of the desire to save itself, each creature is following these rules, and that leads to something that ’ s safe for all of them. Even though it looks like they ’ re thinking as a group, they ’ re not. You might wonder what exactly is the advantage to being in a swarm, so you can think of several. As I say, if you ’ re in a swarm, your odds of being the unlucky one are reduced as compared to a small group. There are many eyes to spot danger. And you ’ ll see in the example with the starlings, with the birds, when this peregrine hawk is about to attack them, that actually waves of panic can propagate, sending messages over great distances. You ’ ll see — let ’ s see, it ’ s coming up possibly at the very end — maybe not. Information can be sent over half a kilometer away in a very short time through this mechanism. Yes, it ’ s happening here. See if you can see those waves propagating through the swarm. It ’ s beautiful. The birds are, we sort of understand, we think, from that computer model, what ’ s going on. As I say, it ’ s just those three simple rules, plus the one about watch out for predators. There doesn ’ t seem to be anything mystical about this. We don ’ t, however, really understand at a mathematical level. I ’ m a mathematician. We would like to be able to understand better. I mean, I showed you a computer model, but a computer is not understanding. A computer is, in a way, just another experiment. We would really like to have a deeper insight into how this works and to understand, you know, exactly where this organization comes from. How do the rules give rise to the patterns? There is one case that we have begun to understand better, and it ’ s the case of fireflies. If you see fireflies in North America, like so many North American sorts of things, they tend to be independent operators. They ignore each other. They each do their own thing, flashing on and off, paying no attention to their neighbors. But in Southeast Asia — places like Thailand or Malaysia or Borneo — there ’ s a beautiful cooperative behavior that occurs among male fireflies. You can see it every night along the river banks. The trees, mangrove trees, are filled with fireflies communicating with light. Specifically, it ’ s male fireflies who are all flashing in perfect time together, in perfect synchrony, to reinforce a message to the females. And the message, as you can imagine, is”Come hither. Mate with me.”In a second I ’ m going to show you a slow motion of a single firefly so that you can get a sense. This is a single frame. Then on, and then off — a 30th of a second, there. And then watch this whole river bank, and watch how precise the synchrony is. On, more on and then off. The combined light from these beetles — these are actually tiny beetles — is so bright that fishermen out at sea can use them as **navigating** beacons to find their way back to their home rivers. It ' s stunning. For a long time it was not believed when the first Western travelers, like Sir Francis Drake, went to Thailand and came back with tales of this unbelievable spectacle. No one believed them. We don ' t see anything like this in Europe or in the West. And for a long time, even after it was documented, it was thought to be some kind of optical illusion. Scientific papers were published saying it was twitching eyelids that explained it, or, you know, a human being ' s tendency to see patterns where there are none. But I hope you ' ve convinced yourself now, with this nighttime video, that they really were very well synchronized. Okay, well, the issue then is, do we need to be alive to see this kind of spontaneous order, and I ' ve already hinted that the answer is no. Well, you don ' t have to be a whole creature. You can even be just a single cell. Like, take, for instance, your pacemaker cells in your heart right now. They ' re keeping you alive. Every beat of your heart depends on this crucial region, the sinoatrial node, which has about 10, 000 independent cells that would each beep, have an electrical rhythm — a voltage up and down — to send a signal to the ventricles to pump. Now, your pacemaker is not a single cell. It ' s this democracy of 10, 000 cells that all have to fire in unison for the pacemaker to work correctly. I don ' t want to give you the idea that synchrony is always a good idea. If you have epilepsy, there is an instance of billions of brain cells, or at least millions, discharging in pathological concert. So this tendency towards order is not always a good thing. You don ' t have to be alive. You don ' t have to be even a single cell. If you look, for instance, at how lasers work, that would be a case of atomic synchrony. In a laser, what makes laser light so different from the light above my head here is that this light is incoherent — many different colors and different frequencies, sort of like the way you clapped initially — but if you were a laser, it would be rhythmic applause. It would be all atoms pulsating in unison, emitting light of one color, one frequency. Now comes the very risky part of my talk, which is to demonstrate that inanimate things can synchronize. Hold your breath for me. What I have here are two empty water bottles. This is not Keith Barry doing a magic trick. This is a klutz just playing with some water bottles. I have some metronomes here. Can you hear that? All right, so, I ' ve got a metronome, and it ' s the world ' s smallest metronome, the — well, I shouldn ' t advertise. Anyway, so this is the world ' s smallest metronome. I ' ve set it on the fastest **setting**, and I ' m going to now take another one set to the same **setting**. We can try this first. If I just put them on the table together, there ' s no reason for them to synchronize, and they probably won ' t. Maybe you ' d better listen to them. I ' ll stand here. What I ' m hoping is that they might just drift apart because their frequencies aren ' t perfectly the same. Right? They did. They were in sync for a while, but then they drifted apart. And the reason is that they ' re not able to communicate. Now, you might think that ' s a bizarre idea. How can metronomes communicate? Well, they can communicate through mechanical forces. So I ' m going to give them a chance to do that. I also want to wind this one up a bit. How can they communicate? I ' m going to put them on a movable **platform**, which is the"Guide to Graduate Study at Cornell."Okay? So here it is. Let ' s see if we can get this to work. My wife pointed out to me that it will work better if I put both on at the same time because otherwise the whole thing will tip over. All right. So there we go. Let ' s see. OK, I ' m not trying to cheat — let me start them out of sync. No, hard to even do that. All right. So before any one goes out of sync, I ' ll just put those right there. Now, that might seem a bit whimsical, but this pervasiveness of this tendency towards spontaneous order sometimes has **unexpected** consequences. And a clear case of that, was something that happened in London in the year 2000. The Millennium Bridge was supposed to be the pride of London — a beautiful new footbridge erected across the Thames, first river crossing in over 100 years in London. There was a big competition for the design of this bridge, and the winning proposal was submitted by an unusual team — in the TED spirit, actually — of an architect — perhaps the greatest architect in the United Kingdom, Lord Norman Foster — working with an artist, a sculptor, Sir Anthony Caro, and an engineering firm, Ove Arup. And together they submitted a design based on Lord Foster ' s vision, which was — he remembered as a kid reading Flash Gordon comic books, and he said that when Flash Gordon would come to an abyss, he would shoot what today would be a kind of a light saber. He would shoot his light saber across the abyss, making a blade of light, and then scamper across on this blade of light. He said,"That ' s the vision I want to give to London. I want a blade of light across the Thames."So they built the blade of light, and it ' s a very thin ribbon of steel, the world ' s — probably the flattest and thinnest suspension bridge there is, with cables that are out on the side. You ' re used to suspension bridges with big droopy cables on the top. These cables were on the side of the bridge, like if you took a rubber band and stretched it taut across the Thames — that ' s what ' s holding up this bridge. Now, everyone was very excited to try it out. On opening day, thousands of Londoners came out, and something happened. And within two days the bridge was closed to the public. So I want to first show you some interviews with people who were on the bridge on opening day, who will describe what happened. Man : It really started moving sideways and slightly up and down, rather like being on the boat. Woman : Yeah, it felt unstable, and it was very windy, and I remember it had lots of flags up and down the sides, so you could definitely — there was something going on sideways, it felt, maybe. Interviewer : Not up and down? Boy : No. Interviewer : And not forwards and backwards? Boy : No. Interviewer : Just sideways. About how much was it moving, do you think? Boy : It was about — Interviewer : I mean, that much, or this much? Boy : About the second one. Interviewer : This much? Boy : Yeah. Man : It was at least six, six to eight inches, I would have thought. Interviewer : Right, so, at least this much? Man : Oh, yes. Woman : I remember wanting to get off. Interviewer : Oh, did you? Woman : Yeah. It felt odd. Interviewer : So it was enough to be scary? Woman : Yeah, but I thought that was just me. Interviewer : Ah! Now, tell me why you had to do this? Boy : We had to do this because, to keep in balance because if you didn ' t keep your balance, then you would just fall over about, like, to the left or right, about 45 degrees. Interviewer : So just show me how you walk normally. Right. And then show me what it was like when the bridge started to go. Right. So you had to deliberately push your feet out sideways and — oh, and short steps? Man : That ' s right. And it seemed obvious to me that it was probably the number of people on it. Interviewer : Were they deliberately walking in step, or anything like that? Man : No, they just had to conform to the movement of the bridge. Steven Strogatz : All right, so that already gives you a hint of what happened. Think of the bridge as being like this **platform**. Think of the people as being like metronomes. Now, you might not be used to thinking of yourself as a metronome, but after all, we do walk like — I mean, we oscillate back and forth as we walk. And especially if we start to walk like those people did, right? They all showed this strange sort of skating gait that they adopted once the bridge started to move. And so let me show you now the footage of the bridge. But also, after you see the bridge on opening day, you ' ll see an interesting clip of work done by a bridge engineer at Cambridge named Allan McRobie, who figured out what happened on the bridge, and who built a bridge simulator to explain exactly what the problem was. It was a kind of unintended positive feedback loop between the way the people walked and the way the bridge began to move, that engineers knew nothing about. Actually, I think the first person you ’ ll see is the young engineer who was put in charge of this project. Okay. Interviewer : Did anyone get hurt? Engineer : No. Interviewer : Right. So it was quite small — Engineer : Yes. Interviewer : — but real? Engineer : Absolutely. Interviewer : You thought,”Oh, bother.”Engineer : I felt I was disappointed about it. We ’ d spent a lot of time designing this bridge, and we ’ d analyzed it, we ’ d checked it to codes — to heavier loads than the codes — and here it was doing something that we didn ’ t know about. Interviewer : You didn ’ t expect. Engineer : Exactly. Narrator : The most dramatic and shocking footage shows whole sections of the crowd — hundreds of people — apparently rocking from side to side in unison, not only with each other, but with the bridge. This synchronized movement seemed to be driving the bridge. But how could the crowd become synchronized? Was there something **special** about the Millennium Bridge that caused this effect? This was to be the focus of the investigation. Interviewer : Well, at last the simulated bridge is finished, and I can make it wobble. Now, Allan, this is all your fault, isn ’ t it? Allan McRobie : Yes. Interviewer : You designed this, yes, this simulated bridge, and this, you reckon, mimics the **action** of the real bridge? AM : It captures a lot of the physics, yes. Interviewer : Right. So if we get on it, we should be able to wobble it, yes? Allan McRobie is a bridge engineer from Cambridge who wrote to me, suggesting that a bridge simulator ought to wobble in the same way as the real bridge — provided we hung it on pendulums of exactly the right length. AM : This one ’ s only a couple of tons, so it ’ s fairly easy to get going. Just by walking. Interviewer : Well, it ’ s certainly going now. AM : It doesn ’ t have to be a real dangle. Just walk. It starts to go. Interviewer : It ’ s actually quite difficult to walk. You have to be careful where you put your feet down, don ’ t you, because if you get it wrong, it just throws you off your feet. AM : It certainly affects the way you walk, yes. You can ’ t walk normally on it. Interviewer : No. If you try and put one foot in front of another, it ’ s moving your feet away from under you. AM : Yes. Interviewer : So you ’ ve got to put your feet out sideways. So already, the simulator is making me walk in exactly the same way as our witnesses walked on the real bridge. AM :... ice-skating gait. There isn ’ t all this sort of snake way of walking. Interviewer : For a more convincing experiment, I wanted my own opening-day crowd, the sound check team. Their instructions : just walk normally. It ’ s really intriguing because none of these people is trying to drive it. They ’ re all having some difficulty walking. And the only way you can walk comfortably is by getting in step. But then, of course, everyone is driving the bridge. You can ’ t help it. You ’ re actually forced by the movement of the bridge to get into step, and therefore to drive it to move further. SS : All right, well, with that from the Ministry of Silly Walks, maybe I ’ d better end. I see I ’ ve gone over. But I hope that you ’ ll go outside and see the world in a new way, to see all the amazing synchrony around us. Thank you.

Text Sample 187: POS Dim 2 – 2004 – Score 7.00 – 2004/t000459.txt

When you have 21 minutes to speak, two million years seems like a really long time. But evolutionarily, two million years is nothing. And yet, in two million years, the human brain has nearly tripled in mass, going from the one-and-a-quarter-pound brain of our ancestor here, Habilis, to the almost three-pound meatloaf everybody here has between their ears. What is it about a big brain that nature was so eager for every one of us to have one? Well, it turns out when brains triple in size, they don ’ t just get three times bigger ; they gain new structures. And one of the main reasons our brain got so big is because it got a new part, called the”frontal lobe,”particularly, a part called the”prefrontal cortex.”What does a prefrontal cortex do for you that should justify the entire architectural overhaul of the human skull in the blink of evolutionary time? Well, it turns out the prefrontal cortex does lots of things, but one of the most important things it does is it ’ s an experience simulator. Pilots **practice** in flight simulators so that they don ’ t make real mistakes in planes. Human beings have this marvelous adaptation that they can actually have experiences in their heads before they try them out in real life. This is a trick that none of our ancestors could do, and that no other animal can do quite like we can. It ’ s a marvelous adaptation. It ’ s up there with opposable thumbs and standing upright and language as one of the things that got our species out of the trees and into the shopping mall. All of you have done this. Ben and Jerry ’ s doesn ’ t have”liver and onion”ice cream, and it ’ s not because they whipped some up, tried it and went,”Yuck!”It ’ s because, without leaving your armchair, you can simulate that flavor and say”yuck”before you make it. Let ’ s see how your experience simulators are working. Let ’ s just run a quick diagnostic before I proceed with the rest of the talk. Here ’ s two different futures that I invite you to contemplate. You can try to simulate them and tell me which one you think you might prefer. One of them is winning the lottery. This is about 314 million dollars. And the other is becoming paraplegic. Just give it a moment of thought. You probably don ’ t feel like you need a moment of thought. Interestingly, there are data on these two groups of people, data on how happy they are. And this is exactly what you expected, isn ’ t it? But these aren ’ t the data. I made these up! These are the data. You failed the pop quiz, and you ’ re hardly five minutes into the lecture. Because the fact is that a year after losing the use of their legs and a year after winning the lotto, lottery winners and paraplegics are equally happy with their lives. Don ’ t feel too bad about failing the first pop quiz, because everybody fails all of the pop quizzes all of the time. The research that my laboratory has been doing, that economists and psychologists around the country have been doing, has revealed something really quite startling to us, something we call the”impact **bias**,”which is the tendency for the simulator to work badly, for the simulator to make you believe that different outcomes are more different than, in fact, they really are. From field studies to laboratory studies, we see that winning or losing an election, gaining or losing a romantic partner, getting or not getting a promotion, passing or not passing a college test, on and on, have far less impact, less intensity and much less duration than people expect them to have. A recent study — this almost floors me — a recent study showing how major life traumas affect people suggests that if it happened over three months ago, with only a few exceptions, it has no impact whatsoever on your **happiness**. Why? Because **happiness** can be synthesized. Sir Thomas Brown wrote in 1642,”I am the happiest man alive. I have that in me that can convert poverty to riches, adversity to prosperity. I am more invulnerable than Achilles ; fortune hath not one place to hit me.”What kind of remarkable machinery does this guy have in his head? Well, it turns out it ’ s precisely the same remarkable machinery that all of us have. Human beings have something that we might think of as a”psychological immune system,”a system of cognitive processes, largely nonconscious cognitive processes, that help them change their views of the world, so that they can feel better about the worlds in which they find themselves. Like Sir Thomas, you have this machine. Unlike Sir Thomas, you seem not to know it. We synthesize **happiness**, but we think **happiness** is a thing to be found. Now, you don ’ t need me to give you too many examples of people synthesizing **happiness**, I suspect, though I ’ m going to show you some experimental evidence. You don ’ t have to look very far for evidence. I took a copy of the”New York Times”and tried to find some instances of people synthesizing **happiness**. Here are three guys synthesizing **happiness**.”I ’ m better off physically, financially, mentally...””I don ’ t have one minute ’ s regret. It was a glorious experience.””I believe it turned out for the best.”Who are these characters who are so damn happy? The first one is Jim Wright. Some of you are old enough to remember : he was the chairman of the House of Representatives, and he resigned in disgrace when this young Republican named Newt Gingrich found out about a shady book deal that he had done. He lost everything. The most powerful Democrat in the country lost everything : he lost his money, he lost his power. What does he have to say all these years later about it?”I am so much better off physically, financially, mentally and in almost every other way.”What other way would there be to be better off? Vegetably? Minerally? Animally? He ’ s pretty much covered them there. Moreese Bickham is somebody you ’ ve never heard of. Moreese Bickham uttered these words upon being released. He was 78 years old. He ’ d spent 37 years in Louisiana State Penitentiary for a crime he didn ’ t commit. He was ultimately [released for good behavior halfway through his sentence.] What did he have to say about his experience?”I don ’ t have one minute ’ s regret. It was a glorious experience.”Glorious! This guy ’ s not saying,”There were some nice guys. They had a gym.””Glorious”— a word we usually reserve for something like a religious experience. Harry S. Langerman uttered these words. He ’ s somebody you might have known but didn ’ t, because in 1949, he read a little article in the paper about a hamburger stand owned by these two brothers named McDonald. And he thought,”That ’ s a really neat idea!”So he went to find them. They said,”We can give you a franchise on this for 3, 000 bucks.”Harry went back to New York, asked his brother, an investment banker, to loan him 3, 000 dollars, and his brother ’ s immortal words were,”You idiot, nobody eats hamburgers.”He wouldn ’ t lend him the money. Of course, six months later, Ray Kroc had exactly the same idea. It turns out, people do eat hamburgers, and Ray Kroc, for a while, became the richest man in America. And then, finally, some of you recognize this young photo of Pete Best, who was the original drummer for the Beatles, until they, you know, sent him out on an errand and snuck away and picked up Ringo on a tour. Well, in 1994, when Pete Best was interviewed — yes, he ’ s still a drummer ; yes, he ’ s a studio musician — he had this to say :”I ’ m happier than I would have been with the Beatles.”OK, there ’ s something important to be learned from these people, and it is the secret of **happiness**. Here it is, finally to be revealed. First : accrue wealth, power and prestige, then lose it. Second : spend as much of your life in prison as you possibly can. Third : make somebody else really, really rich. And finally : never, ever join the Beatles. Yeah, right. Because when people synthesize **happiness**, as these gentlemen seem to have done, we all smile at them, but we kind of roll our eyes and say,”Yeah, right, you never really wanted the job.””Oh yeah, right — you really didn ’ t have that much in common with her, and you figured that out just about the time she threw the engagement ring in your face.”We smirk, because we believe that synthetic **happiness** is not of the same quality as what we might call”natural **happiness**.”What are these terms? Natural **happiness** is what we get when we get what we wanted, and synthetic **happiness** is what we make when we don ’ t get what we wanted. And in our society, we have a strong belief that synthetic **happiness** is of an inferior kind. Why do we have that belief? Well, it ’ s very simple. What kind of economic engine would keep churning if we believed that not getting what we want could make us just as happy as getting it? With all apologies to my friend Matthieu Ricard, a shopping mall full of Zen monks is not going to be particularly profitable, because they don ’ t want stuff enough. I want to suggest to you that synthetic **happiness** is every bit as real and enduring as the kind of **happiness** you stumble upon when you get exactly what you were aiming for. Now, I ’ m a scientist, so I ’ m going to do this not with rhetoric, but by marinating you in a little bit of data. Let me first show you an experimental paradigm that ’ s used to demonstrate the synthesis of **happiness** among regular old folks. This isn ’ t mine, it ’ s a 50-year-old paradigm called the”free choice paradigm.”It ’ s very simple. You bring in, say, six objects, and you ask a subject to rank them from the most to the least liked. In this case, because this experiment uses them, these are Monet prints. Everybody ranks these Monet prints from the one they like the most to the one they like the least. Now we give you a choice :”We happen to have some extra prints in the closet. We ’ re going to give you one as your prize to take home. We happen to have number three and number four,”we tell the subject. This is a bit of a difficult choice, because neither one is preferred strongly to the other, but naturally, people tend to pick number three, because they liked it a little better than number four. Sometime later — it could be 15 minutes, it could be 15 days — the same stimuli are put before the subject, and the subject is asked to re-rank the stimuli.”Tell us how much you like them now.”What happens? Watch as **happiness** is synthesized. This is the result that ’ s been replicated over and over again. You ’ re watching **happiness** be synthesized. Would you like to see it again? **Happiness**!”The one I got is really better than I thought! That other one I didn ’ t get sucks!”That ’ s the synthesis of **happiness**. Now, what ’ s the right response to that?”Yeah, right!”Now, here ’ s the experiment we did, and I hope this is going to convince you that”Yeah, right!”was not the right response. We did this experiment with a group of patients who had anterograde amnesia. These are hospitalized patients. Most of them have Korsakoff syndrome, a polyneuritic psychosis. They drank way too much, and they can ’ t make new memories. They remember their childhood, but if you walk in and introduce yourself and then leave the room, when you come back, they don ’ t know who you are. We took our Monet prints to the hospital. And we asked these patients to rank them from the one they liked the most to the one they liked the least. We then gave them the choice between number three and number four. Like everybody else, they said,”Gee, thanks Doc! That ’ s great! I could use a new print. I ’ ll take number three.”We explained we would have number three mailed to them. We gathered up our materials, and we went out of the room and counted to a half hour. Back into the room, we say,”Hi, we ’ re back.”The patients, bless them, say,”Ah, Doc, I ’ m sorry, I ’ ve got a memory problem ; that ’ s why I ’ m here. If I ’ ve met you before, I don ’ t remember.””Really, Jim, you don ’ t remember? I was just here with the Monet prints?””Sorry, Doc, I just don ’ t have a clue.””No problem, Jim. All I want you to do is rank these for me, from the one you like the most to the one you like the least.”What do they do? Well, let ’ s first check and make sure they ’ re really amnesiac. We ask these amnesiac patients to tell us which one they own, which one they chose last time, which one is theirs. And what we find is, amnesiac patients just guess. These are normal controls, where if I did this with you, all of you would know which print you chose. But if I do this with amnesiac patients, they don ’ t have a clue. They can ’ t pick their print out of a lineup. Here ’ s what normal controls do : they synthesize **happiness**. Right? This is the change in liking **score**, the change from the first time they ranked to the second time they ranked. Normal controls show — that was the magic I showed you ; now I ’ m showing it to you in graphical form —”The one I own is better than I thought. The one I didn ’ t own, the one I left behind, is not as good as I thought.”Amnesiacs do exactly the same thing. Think about this result. These people like better the one they own, but they don ’ t know they own it.”Yeah, right”is not the right response! What these people did when they synthesized **happiness** is they really, truly changed their affective, hedonic, aesthetic reactions to that poster. They ’ re not just saying it because they own it, because they don ’ t know they own it. When psychologists show you bars, you know that they are showing you averages of lots of people. And yet, all of us have this psychological immune system, this capacity to synthesize **happiness**, but some of us do this trick better than others. And some situations allow anybody to do it more effectively than other situations do. It turns out that freedom, the ability to make up your mind and change your mind, is the friend of natural **happiness**, because it allows you to choose among all those delicious futures and find the one that you would most enjoy. But freedom to choose, to change and make up your mind, is the enemy of synthetic **happiness**, and I ’ m going to show you why. Dilbert already knows, of course.”Dogbert ’ s tech support. How may I abuse you?””My printer prints a blank page after every document.””Why **complain** about getting free paper?””Free? Aren ’ t you just giving me my own paper?””Look at the quality of the free paper compared to your lousy regular paper! Only a **fool** or a liar would say that they look the same!””Now that you mention it, it does seem a little silkier!””What are you doing?””I ’ m helping people accept the things they cannot change.”Indeed. The psychological immune system works best when we are totally stuck, when we are trapped. This is the difference between dating and marriage. You go out on a date with a guy, and he picks his nose ; you don ’ t go out on another date. You ’ re married to a guy and he picks his nose? He has a heart of gold. Don ’ t touch the fruitcake! You find a way to be happy with what ’ s happened. Now, what I want to show you is that people don ’ t know this about themselves, and not knowing this can work to our supreme disadvantage. Here ’ s an experiment we did at Harvard. We created a black-and-white photography course, and we allowed students to come in and learn how to use a darkroom. So we gave them cameras, they went around campus, they took 12 pictures of their favorite professors and their dorm room and their dog, and all the other things they wanted to have Harvard memories of. They bring us the camera, we make up a contact sheet, they figure out which are the two best pictures. We now spend six hours teaching them about darkrooms, and they blow two of them up. They have two gorgeous 8 x 10 glossies of meaningful things to them, and we say,”Which one would you like to give up?””I have to give one up?””Yes, we need one as evidence of the class project. So you have to give me one. You have to make a choice. You get to keep one, and I get to keep one.”Now, there are two conditions in this experiment. In one case, the students are told,”But you know, if you want to change your mind, I ’ ll always have the other one here, and in the next four days, before I actually mail it to headquarters”— yeah,”headquarters”—”I ’ ll be glad to swap it out with you. In fact, I ’ ll come to your dorm room, just give me an email. Better yet, I ’ ll check with you. You ever want to change your mind, it ’ s totally returnable.”The other half of the students are told exactly the opposite :”Make your choice, and by the way, the mail is going out, gosh, in two minutes, to England. Your picture will be winging its way over the Atlantic. You will never see it again.”Half of the students in each of these conditions are asked to make predictions about how much they ’ re going to come to like the picture that they keep and the picture they leave behind. Other students are just sent back to their little dorm rooms and they are measured over the next three to six days on their satisfaction with the pictures. Look at what we find. First of all, here ’ s what students think is going to happen. They think they ’ re going to maybe come to like the picture they chose a little more than the one they left behind. But these are not statistically significant differences. It ' s a very small increase, and it doesn ' t much matter whether they were in the reversible or irreversible condition. Wrong-o. Bad simulators. Because here ' s what ' s really happening. Both right before the swap and five days later, people who are stuck with that picture, who have no choice, who can never change their mind, like it a lot. And people who are deliberating —"Should I return it? Have I gotten the right one? Maybe this isn ' t the good one. Maybe I left the good one?"— have killed themselves. They don ' t like their picture. In fact, even after the opportunity to swap has expired, they still don ' t like their picture. Why? Because the [reversible] condition is not conducive to the synthesis of **happiness**. So here ' s the final piece of this experiment. We bring in a whole new group of naive Harvard students and we say,"You know, we ' re doing a photography course, and we can do it one of two ways. We could do it so that when you take the two pictures, you ' d have four days to change your mind, or we ' re doing another course where you take the two pictures and you make up your mind right away and you can never change it. Which course would you like to be in?"Duh! Sixty-six percent of the students, two-thirds, prefer to be in the course where they have the opportunity to change their mind. Hello? Sixty-six percent of the students choose to be in the course in which they will ultimately be deeply dissatisfied with the picture — because they do not know the conditions under which synthetic **happiness** grows. The Bard said everything best, of course, and he ' s making my point here but he ' s making it hyperbolically :"' Tis nothing good or bad But thinking makes it so."It ' s nice poetry, but that can ' t exactly be right. Is there really nothing good or bad? Is it really the case that gall bladder surgery and a trip to Paris are just the same thing? That seems like a one-question IQ test. They can ' t be exactly the same. In more turgid prose, but closer to the truth, was the father of modern capitalism, Adam Smith, and he said this. This is worth contemplating :"The great source of both the misery and disorders of human life seems to arise from overrating the difference between one permanent situation and another. Some of these situations may, no doubt, deserve to be preferred to others, but none of them can deserve to be pursued with that **passionate** ardor which drives us to violate the rules either of prudence or of justice, or to corrupt the future tranquility of our minds, either by shame from the remembrance of our own folly, or by remorse for the horror of our own injustice."In other words : yes, some things are better than others. We should have preferences that lead us into one future over another. But when those preferences drive us too hard and too fast because we have overrated the difference between these futures, we are at risk. When our ambition is bounded, it leads us to work joyfully. When our ambition is unbounded, it leads us to lie, to cheat, to steal, to hurt others, to sacrifice things of real value. When our fears are bounded, we ' re prudent, we ' re cautious, we ' re thoughtful. When our fears are unbounded and overblown, we ' re reckless, and we ' re cowardly. The lesson I want to leave you with, from these data, is that our longings and our worries are both to some degree overblown, because we have within us the capacity to manufacture the very commodity we are constantly chasing when we choose experience. Thank you.

Text Sample 188: POS Dim 2 – 2021 – Score 27.00 – 2021/t004025.txt

When I was a little little kid, I was a four years old on my first airplane ride and we got to go up front in the cockpit because you could kind of do that back then. And it was. Totally dark, no moon over the Atlantic Ocean, there is like a billion stars in the sky and I went back and told my mom I wanted to be a stewardess and my mom, to her credit, she wasn ’ t. Honey, you might want to think about being a pilot. And there you go. That ’ s what I wanted to do from then on. Sharon Pressler has flown lots of different kinds of airplanes since then, including fighter jets. She was the first woman in the US Air Force to fly the F-16. It ’ s just always been the coolest looking airplane, that bubble canopy and the big engine. And it has the highest tolerance, which is nine times the force of gravity on the earth, which is significant. It can do anything. Sheeran ’ s had an extraordinary career spanning more than three decades. But recently, after 14 years as a pilot with Southwest Airlines, she hit a particularly bad patch of turbulence. Her whole industry did. I was the captain who always brought, like chocolate for the flight attendants, and I ’ d go give them some chocolate, even if it ’ s breakfast time, it ’ s OK to have chocolate for breakfast. Here you go. And that changed my first flight after we kind of really understood what was going on with covid. We had a bunch of those Clorox wipes. So I took those to the flight attendants instead. I ’ m like, hey, anybody wants some wipes. The pandemic had a catastrophic impact on the airline industry and Southwest eventually offered buyouts. I hadn ’ t even really thought about my retirement moment yet because I had 10 more years to fly and I wasn ’ t planning on leaving. But after some serious reflection, she chose to retire. So we ’ re a couple of hundred that retired in the same two week period, which is unheard of. Despite all the turbulence, Sharon knows she ’ s lucky she looked beyond the horizon to a new career. More on that later. But for now, buckle up. Ladies, gentlemen, this is Captain Pressor speaking. I know it ’ s been a little bit turbulent, especially in the economy, but it ’ s going to smooth out. Thanks for listening with us today. I ’ m Adam Grant and this is work like my podcast with the TED Radio Collective. I ’ m an **organizational** psychologist. I study how to make work, not suck. In this show. I take you inside the minds of fascinating people to help us **rethink** how we work, lead and live today, **navigating** career turbulence, thanks to Morgan Stanley for sponsoring this episode. You know what turbulence feels like on an airplane. You had some nasty wind and everything is out of your control. You ’ ve probably faced that kind of turbulence in your career, too, when your job, your **workplace** or your industry is changing dramatically around you and you feel powerless to shape it. The current pandemic has compounded that disturbance on a massive scale. In the face of uncertainty, we often freeze up. **Researchers** call it a threat, rigidity, response. When we feel powerless to counter the threats around us, our thinking becomes constrained. We hang on for dear life, stop taking risks and play it safe. The past year has left us all reacting to dramatic change. But psychologists find that when we encounter turbulence, instead of pushing harder against the headwinds, we ’ re generally better off tilting our rudder and charting a new course. In other words, expanding your thinking at exactly the moment when all your instincts are telling you to lock it down tight. I was in my 40s and I worked in the mortgage division. When all of this mortgage business hit and they laid off the entire division in Jacksonville in 2009, Erin Scott had been working for a bank in Florida for 14 years when he lost his job during the recession. He started looking for a new job in finance. I would estimate that surely I applied for at least two hundred jobs. I think I have like three interviews in that whole time as more time passed without any callbacks and became more open to exploring other roles and for lower pay as long as they were in state your station for one particular kind of fish. But in time when you start casting your net a whole lot wider feeling dejected. Aaron finally started looking out of state, which would have meant leaving his entire family behind. My brother, my sister, my mom and dad, they ’ re all right there. When you talk about moving to California from Florida, that ’ s you know, that

’ s a desperate move. But something ’ s, you know, something ’ s got to give here. I couldn ’ t find another job to save my life. Aaron spent two and a half years looking for full time work. He had a newborn son, so his **financial** responsibilities were growing. And just when you thought things couldn ’ t get any worse, his wife had a stroke. This is the most driven woman I know. And when you see that kind of person sitting in a wheelchair, I just felt the weight of the world had this little boy. You ’ re thinking, my wife looks to me, my son looks to me, what am I going to do? I just finally I went to another room and I just closed door and I just kind of had a little pity party. What is a pity party? Well, I didn ’ t want to say I cried, but that ’ s what I did. With so much out of his control in that moment, Aaron found a way to take some of the control back. Instead of focusing narrowly on the mortgage industry, Aaron broadened his lens. He found a part time job as a substitute teacher to pay the bills, which was a resourceful move, an example of what psychologists call being proactive. Be proactive can be really annoying advice, it isn ’ t about working harder or taking the bull by the horns or whatever your nagging uncle keeps telling you to do, you ’ re probably doing that already. It means doing what you can to change the circumstance rather than grinding against it, identifying the ways, large or small, that you can exert some control in an out of control situation. It turned out that Aaron loved teaching. A few years later, he ended up pivoting it into his current career, where he ’ s helping others learn from his earlier setback. I work at a **nearby** prison, Hamilton Correctional Institution. Aaron now helps people **navigate** one of the biggest headwinds imaginable time in prison. What Aaron ’ s learned about how formerly incarcerated people re-enter society has implications for adapting to all kinds of turbulence, from being downsized to being demoted to having a gap on your resume. People who have spent time in prison faced major **obstacles** when they ’ re trying to land a job. Employers are often hesitant to give them an opportunity. If they look you up and they find out you are really into some bad stuff, sometimes that ’ s the end of it. They ’ re just going to pass on you. Aaron coaches his students to call out the elephant in the room. I made some very poor decisions. I ’ ve paid for that. I ’ ve made a lot of changes in my life. I would really appreciate the opportunity. Why do you think it ’ s important, Aaron, to address the elephant in the room and actually talk about it directly? The number one thing is you control the narrative to some degree when you do that. Controlling the narrative, this is a key strategy for being proactive in the face of career turbulence, crafting a story about how a headwind has made you stronger or better. Research demonstrates that in the job search, sex offenders are better off owning their mistakes than making excuses. Your headwinds may be lighter than time behind bars. Maybe you lost a job that was a bad fit, or you took time off to care for a child or a sick family member. Or maybe you have a physical or psychological disability that prospective employers wrongly **judge**. And although it ’ s illegal to **discriminate** against people with disabilities, employers have **biases**. In a series of studies, people with disabilities were rated more favorably if they brought them up at the beginning of an interview rather than waiting until the end or concealing them altogether. Psychologist find that in the face of setbacks, taking charge of the narrative doesn ’ t just signal to others that you ’ re in control. Forming a story can also boost your own motivation and build your confidence to overcome past **struggles** and move forward in the future. When Aaron was searching for a job, he decided to take this principle a step further, he started proactively asking interviewers to call out the elephant in the room to tell him what his disadvantages were. Well, first, I want to thank you for allowing me to come in and speak with you. I ’ m very interested in this position. Is there anything in this interview or perhaps on my résumé that might give you pause about considering me for the position? Part of what I think is so clever about asking that question is even if it doesn ’ t get you the job today, you learn something potentially for the next interview. That ’ s exactly right. I did learn from that interview sometimes, but I learn it helps me get the job that is the right job for me here in advance. Doesn ’ t mean you have to take it, but it does mean you can weigh it. Sometimes there ’ s a little nugget in there that will help you. This is another form of proactivity in the face of turbulence, seeking feedback by asking what your limitations are and how you can improve, you don ’ t just get to peek inside the black box of how to get that job, that connection or that coveted project. You also get in a sense of control to address the issues that may be preventing you from getting an opportunity. But people won ’ t always tell you what you can do better. And sometimes you can ’ t even get your foot in the door to ask them, you know, all those YouTube videos that you watch and all those things that interrupt you and jam their way into your face. I do those. Dallas McGlaughlin started building websites as a teenager and taught himself how to do SEO search engine optimization. You know, when you direct people doing Google searches to a particular website, like what? Those pop up **ads** you ’ re always avoiding. Dallas got so good at SEO Marketing that he dropped out of music college to do it full time and for a few years that ’ s what he did. But when Dallas decided he wanted to work for a big agency, not having a formal degree became a major disadvantage. It meant nobody was looking at his resume and prescreening tools and technology that just siphoned me or filtered me out. I just wasn ’ t even passing those checks and balances. Remember, I dropped out of college to go do this. I just knew that I could do that better than anybody stepping out of a college with a communications degree. When you faced a headwind, you need support. If you ’ re like most people, your instinct is to reach out to your strong ties, the people you know well and trust to have your back. But research has shown that you ’ re more likely to get a job through weak ties. Your more distant acquaintances, they travel in different circles and have access to a broader pool of connections and ideas, which puts them in a better position to open up new opportunities. Sadly, the people who need those ties the most are the least likely to reach out to them. Research reveals that when their jobs are under threat, people with lower social class tend to narrow their networks. While people with higher status tend to broaden their networks, they reach beyond the inner circle, which actually turns out to be a way of changing course when the going is rough and it ’ s something we should support everyone to do. Or if you want to be really proactive, you can expand your network to include complete strangers. That ’ s what Dallas did, he came up with a plan to recruit the recruiters into his network. He took the same CEO marketing skills he ’ d been using to attract customers, except now he targeted employers first. He built a website for himself. Next, he made a list of every notable person in the Phoenix agency ecosystem CEOs, CFOs, marketing directors, you name it. And then he made paid search **ads** targeting each of them. So let ’ s say you ’ re an executive named John Smith and you ’ re Googling yourself because you have nothing better to do. You type your name into Google. And the very first result that pops up says, hey, John Smith, I ’ m Dallas McLaughlin. I ’ m a digital marketing expert and I really want to work for your company. Click here to find out why I basically turned the entire hiring process around and I stopped sending resumes. This is so clever. So wait a minute. How many people actually followed up and how many people ignored it? I only targeted maybe 12. And I heard from for the very first follow up I got, they said, please stop doing this. I don ’ t like it because they didn ’ t like the fact that you were so effective in advertising that you were able to annoy them and create them out a little. Yeah. You know, it ’ s also me screening them out if they don ’ t like what I ’ m doing and they clearly don ’ t understand the value in what I just did. And they ’ re not going to be able to properly utilize me within their organization. I don ’ t want to work there anyway. The fourth one, they were just like, come into the office, you ’ re hired, you ’ re in. We don ’ t need we want to have a job opening. Wow. So you went from over 100 to four out of 12 interested in at least learning more about you to 100 percent success rate once you got an interview. Yeah, and that was it. Dallas found a proactive way to showcase his skills specific to the field. Instead of going to the recruiters, he turned the tables and brought the recruiters to him. This kind of initiative can help you expand your network and access to new opportunities to. A few years ago, I got a cold email from a web designer. She sent me a mock up of how she thought my homepage should be redesigned. I wasn ’ t even looking for a change, but she did such a great job that I hired her on the spot. Not everyone will have time to experiment like this, but if you can manage it, it ’ s a promising way to proactively change the situation. There ’ s evidence that side hustles can boost our engagement and performance in our full time jobs, and sometimes they even become a career. Go do that thing that excites you. If you are an architect, go draw blueprints. If you ’ re an auto mechanic, go fix everybody ’ s cars. And if you do that long enough and you do it well enough, somebody is going to notice. Captain Sharon Pressler could have chosen to stay in aviation, but once the airline industry starts furloughing now, there ’ s a glut of pilots, right. Because everybody kind of tries to take a step back and go. So I ’ ll go back to being an instructor. Well, guess what? Everybody else is trying to go back to being an instructor, and there ’ s just not the demand. The more she thought about it, the more she started to reframe the disturbance as an opportunity to take off in an entirely new direction. This whole covid mess and the wreck that the airline industry has become right now for me was an **unexpected** opportunity. So I ’ m back in school. I ’ m getting a master ’ s in psychology with an emphasis on coaching young adults and help them successfully transition to adulthood. And it ’ s just something I wanted to. I ’ ve been interested in for a while and mentoring kind of program, although I will miss flying at Southwest. I ’ m excited about it. It ’ s a big change, but I ’ m excited about it. In the face of the storm confronting the airline industry, Sharon changed course, something the pandemic and recession are forcing all of us to do in varying degrees. Premier grappling with uncertainty. We tend to focus on what ’ s right in front of us, our strong ties and our next move. But what is turbulence mean in the long term to forecast what might happen to your future career? You can learn something from looking to the past. More on that after the break. OK, this is going to be a different kind of **ad**, I play a personal role in selecting the sponsors for this podcast because they all have interesting cultures of their own. Today, we ’ re going inside the **workplace** at Morgan Stanley. For many people, morning routines often include a good cup of coffee and for the team members at Morgan Stanley ’ s global headquarters in Times Square, the coffee comes with a **special** pairing. We stick together and be kind to each other, love each other, respect each other, and everything is going to be fine. Do you meet Edith and Angela? They run a coffee cart together in the heart of New York City. I met my wife in Brownsville, Texas, too many years ago, a long time ago, 1975, 75 75. On a typical morning, they head into the city at two 45 a. m. to set up the coffee cart. They ’ ve spent a lifetime together in that cart. She can be on the docket. I like to talk that much. Over the years, they ’ ve gotten to know their customers. He ’ s very good in keeping track of the orders. And I am with the names Mr. Mackey likes. He has a lot coffee and a large coffee. You know, that ’ s the standard order from one of their regulars, Mike Buckin Burger. If I happen to be running late and there ’ s more than like one person waiting and I realize I don ’ t have time, it kind of just destroys the morning. It goes beyond them just knowing your order and having it ready for you. So as soon as you see them, it ’ s a huge smile. It was my favorite part about going to the New York office. No offense to my colleagues. Mike works at Morgan Stanley. When he did a stint in Hong Kong, Edith and Angelo would check in on him. I always describe them. They ’ re the parents of the block and they, you know, interact with you that way. They ’ re looking out for you guys as as kids. It does feel like that. Like all of us, Edith and Angelo ’ s world was turned upside down in March. Twenty twenty. They started to notice people wearing masks and talking about working from home with a mass of the people coming from people with a mask and etc.. And I said, what ’ s going on here? People say, well, I want to stay home to work. And then we thought that we ’ ve got a problem there. I start getting worried. As New York City shut down, their entire livelihood was now in question. My colleague Ercan sent an email to three of us and the subject was either Sant ’ Angelo and the body was, what are we going to do? In New York? Street vendors were excluded from pandemic relief programs. So Mike sprang into **action**. He helped him set up a Venmo account and employees from Morgan Stanley rallied together to contribute. Then they organized a virtual breakfast with Edith and Angelo to collect donations, and it just snowballed from there. We were very surprised. We were amazed. Like I say, we knew that they came by not that much. And we continue paying all our bills. And I ’ m telling us today it it ’ s amazing. It ’ s amazing. Amazing. I found in my research that cultures of giving don ’ t just start from the top down. They often grow from the bottom up. Morgan Stanley employees ended up raising enough money to support Edith and Angela over the past year, but it didn ’ t end there. They ’ ve now provided **financial** support to thousands of food vendors in New York City during the pandemic. For Mike and his colleagues, giving back is deeply embedded in the Morgan Stanley culture. It ’ s always been first class business in a first class way, but also supporting the communities we operate in. You know, this all happened because of how we felt about them. We you know, we just can ’ t believe how much they do for us. These more people. You ’ re not small people to them. You matter to them. At Morgan Stanley, giving back as a core value, a central part of their culture globally. They live that commitment through long lasting partnerships, community based delivery and engaging their best asset. Their people learn more at Morgan Stanley Dotcom giving back. Turbulence in an individual career is tough to **navigate**, but right now, literally millions of people are all **struggling** with a massive upheaval all at once and one that inherently lasts a long time. Bad economic conditions make everything in our lives feel unstable and out of our control. Millions of jobs have been lost through the pandemic and millions more are at risk. People of color, especially those in lower paid jobs and without college degrees, have suffered the most. Research shows that job loss doesn ’ t just create **financial** strain. It also increases the risk of depression, anxiety and marital problems and undermines our sense of control and confidence. If you haven ’ t lost your job, you still may be feeling job insecurity, which has been linked to negative mental and physical health outcomes. There ’ s even evidence that economic insecurity reduces our pain tolerance. But just as chaos doesn ’ t mean we ’ ve lost all control, a moment of huge upheaval like this one doesn ’ t mean everything is lost. Recessions can shape careers in ways that are profound and profoundly surprising, we don ’ t yet know the outcome of this recession because we ’ re still in it, but we can look to past recessions for wisdom to guide us through our current journey. Just ask Emily BIanche. I would submit job after job after application after application and would hear nothing. Occasionally I get an interview and I find out eight other people were being interviewed and it was really nerve wracking. It took an entire year to get a full time job. In 2001, Emily had just graduated with her bachelor ' s degree in psychology when she lived through her first of three recessions, the dotcom bust. During the boom of the late 90s, there was just an incredible sense of entitlement of what ' s the world going to offer me? Should I start my own company? Will I be a millionaire? By twenty five expectations that were insane. And you fast forward two years to my class and it just it had completely changed. You know, you go to college, you spend all this money and you think when you come out, I ' m going to have some options. And then that historical moment, I really didn ' t. That was true for many, many other people graduating in that time and certainly true for many people graduating right now. Emily knows it ' s true now because she studied it as an **organizational** behavior professor at Emory University. Emily is one of the world ' s foremost experts on the psychology of recessions. Do you ever have these horrible moments, given what you study where you say, well, you know, when bad things happen, I get good research out of it? You know, certainly during the Great Recession, I thought that, but not now. Now I ' m like, enough. Good. OK, so you ' re not a bad person anymore? No, I was in graduate school and I ' ve reformed. Emily had a front **row** seat to the dotcom fallout of 2001, and she **observed** similar patterns in new graduates during the Great Recession of 2009. That was when the world fell apart. You could feel it everywhere, you could feel it walking around New York and when all those folks started graduating the class of 2009, there were so many articles about how they were just doomed. This was a generation that I completely had the rug pulled out from under them. The research that was available about recessions was ominous. A setback at the start of a career cast a long shadow. People who graduated in recessions continued to earn less. A decade later, it ' s not only that you suffer now, but that you will also continue to suffer, at least financially, for a long time to come. But it isn ' t all bad news. As Emily looked around her in 2009, she saw a strange pattern. One thing that really struck me was that so many people were expressing gratitude. These people had just been dealt this really tough hand. And yet again and again, I kept hearing people say, I ' m so grateful I got this job. It ' s not what I hoped for. But, you know, it ' s a place to start. And there is very little data on it, at least from a site perspective. She was familiar with a classic study of Olympic athletes on the podium. Bronze medalists looked happier than silver medalists. The poor silver medalist were disappointed. I almost won a gold. The bronze medalist felt lucky. At least I got a medal. Emily wondered if something similar would hold for recession graduates. I found that people who graduated when the economy was doing worse, even though they were 10 or 15 or even 20 years out of those initial experiences, they were more satisfied with their current job than people who graduated in good economic times. At some of the sounds like good news, provided we can get reemployed, it seems like your data would suggest we ' re going to be happier with our jobs and more likely to appreciate them. Yeah, this is the first surprise people who start their careers in a recession may feel less entitled to the perfect job and more grateful to be employed. I don ' t mean to diminish the very serious and real **obstacles** that people who are graduating, but any recession face, but that there may be sort of this long term silver lining if you take your silver lining of people being more satisfied with their jobs. I worry that that ' s going to open the door for leaders to exploit them, that people who might have exited completely abusive or toxic **workplace** will stay because they say, hey, I ' m lucky to have a job. Is that a risk? Absolutely. I worry about that, too. You may be at risk of tolerating circumstances which you wouldn ' t tolerate if you would come of age in better economic times. And why does it last? I ’ m thinking about the common finding that we adapt pretty quickly to our circumstances and start to get used to them. So why in 10 or 15 or 20 years after the recession is over, am I still thankful for having a job? Usually when people get their first jobs, it ’ s in this period of life called the impressionable years. Most people are beginning to forge an identity outside of their families and their communities and trying to figure out their place in the world. And attitudes tend to be quite malleable during this period. What ’ s going on in the world, whether economically, politically, culturally, tends to help shape those attitudes in ways that mirror what ’ s happening at that moment. And after this critical period, attitudes don ’ t tend to change that much. The indelible mark of the recession doesn ’ t just affect us individually. It can have a meaningful impact on leaders and end up shaping entire organizations. It ’ s very difficult for people who are graduating recessions to avoid kind of the humbling adversity that that type of environment presents. This brings us to a second surprise and some more long term good news despite the short term headwinds in my data. I find that people who come of age in recessions tend to be less narcissistic in terms of how they pay themselves versus how their top leaders within their company are paid. Those who come of age in recessions and then become leaders of organizations are less entitled knowing what it ’ s like to **struggle**. These CEOs seem to care more for employees and behave less selfishly. They become proactive about taking responsibility. In one study, Emily discovered that CEOs who launched their careers in a recession were less likely to backdate their stock options to **maximize** their value. In other words, they were less likely to cheat, which is possibly the opposite of what you ’ d expect from a person who came of age while **struggling**. The people who tend to cheat are actually the people who are doing really well. A lot of cheating comes from entitlement or a belief that somebody deserves a better outcome or that nobody will notice. And I ’ ll get a. So when you look into your crystal ball, then I don ’ t want one of those, I beg to differ based on your findings in 20 or 30 years when people who are just starting out their career become CEOs. Does that mean we ’ re going to see more servant leaders, more givers than takers? I would certainly hope so, based on what we ’ ve seen in the past. There are some reasons for optimism in the long run, the chaos we live through can make people stronger and better in a very real way in terms of how they mitigate the uncertainty associated with bad economic times. One, I think very positive way is through connecting with other people. This tendency for connectivity brings us to one last surprise from Emily ’ s studies. During moments of uncertainty, but especially economic uncertainty, we tend to **rethink** our relationships with others. We become less individualistic and more civically minded. Emily has found an ingenious way to measure that through pop music. When we ’ re **struggling** through bad economic times, we ’ re drawn to different lyrics in popular American music, you ’ re more likely to see first person plural pronouns like we us. Whereas in really good economic times, you see a lot of first person pronouns like me, mine I when you think of what ’ s often called the greatest generation, the generation that came of age either in the Great Depression or World War Two, it ’ s been argued again and again that this is the most civically minded generation. I do think we will see that going forward in the current generation. Another way, Emily measured individualism was by looking at Social Security data to see what parents name their kids during good and bad economic times. Emily found that when the economy is doing well, parents are more likely to give their kids unique names. I ’ m looking at you. Blue Ivy, when the economy is **struggling**, parents choose from a smaller set of more common names, which are often biblical names. It used to be in the 1950s when out of 15 boys would receive the most common name. Fast forward to 2013. And now it ’ s one out of seventy five boys receive the most common name of their birth year, if you take this idea of increasing communal orientation, decreasing individualism seriously, would you also anticipate that we ’ re going to see more caring and more supportive work relationships in the next couple of decades? The findings that I have looked at suggests that, yes, right. To the extent that people are less narcissistic, they would also manifest in positive ways in the interpersonal level. We can all agree that recessions are terrible in the short term for so many reasons. But in the long term, having experienced their own **struggles**, leaders in the future may be less entitled, more honest and more caring toward employees. And in the meantime, we all may be more community focused. But during a recession, there ’ s a darker side to this. It really depends on who your community is. There tends to be greater fondness for people within one ’ s own group and often, not always, but often that comes at the expense of how people perceive and treat people who are not a part of their anger. Xenophobia is higher. Treatment of immigrants is worse. All these different manifestations of prejudice. The evidence shows that recessions have unequal effects along racial lines. You know, the wage gap between black and white workers, well, Emily found that it grows during recessions. She also discovered that recessions change people ’ s attitudes about race. Her research shows that white people are more likely to report that inequality between races is natural and normal in the wake of a recession. We need to understand how to overcome this ingroup **bias** and broaden people ’ s circles of concern to motivate them to care about helping those who have been most disadvantaged. Dismantling these **biases** is so important that we ’ ll be devoting two episodes to it this season. Stay tuned. When you hit turbulence, it can be hard to see the way forward, it can help to look backward. What we see when we do that is that the lessons we learn in these hard times will shape our jobs and our cultures for the long haul. Some of those effects are negative and we have to be vigilant to counteract. Some of them are surprisingly positive. A glimmer of light up ahead. We also see that in the bumpiest or most chaotic moments are connections with each other are more important than ever, and it ’ s up to us to be proactive about not only reaching out to our networks, but expanding them. If we ’ re thoughtful about it, we have the opportunity to stick the landing, to come through turbulence, more open, more honest and more connected than before. Next time on work life and also it ’ s exhausting. Oh, my God. Sometimes I just want to agree to disagree or don ’ t agree to disagree. The keys to solving conflicts at work and at home are often hiding in plain sight **workplaces** hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Alby ’ s creditcard, Dan O ’ Donnell, Constanza Kaieda, Chris Rubenstein, Michelle Quinn, Bambam Chang and Adam Thielen. This episode was produced by Joe Angelena. Our show is mixed by Require Our Fact Checkers. Paul Dabic Original Music by Hartsdale Sue and Alison Layton Brown had stories produced by **Pineapple** Street Studios **special** thanks to our sponsors linked in Logitech, Morgan Stanley, S&P and Verizon for their research, thanks to teams including Beristain Threat, Rigidity, Sharon Parker and Proactivity Connie Weinberg on the job search. Brett Lyons on disclosing stigmatized identities. Jamie Pennebaker on forming a story. Sue Ashford on Proactiv feedback seeking and job insecurity. Mark Granovetter on Weeks ’ s Tarnya met in a narrowing versus broadening networks. Hudson sessions on side Hustle ’ s Rick Priced on job loss and Lisa Kahn and how recessions affect careers. As brilliant as this is, I think you ’ re slipping because I just Googled myself and there was not an **ad** by you to be on my podcast and maybe I ’ m getting complacent.

Text Sample 189: POS Dim 2 – 2021 – Score 25.00 – 2021/t004073.txt

Hey, work lifers, this is J. J. Abrams, we ’ re trying something new on the podcast today, I am going to be conducting an interview on the show and my guest is Adam Grant. We ’ re going be talking about his new book, Think Again Automatical. Oh, JJ, it ’ s great to hear your voice, but this is my show. I don ’ t know if it is. I can ’ t just let you take my job and ask all the questions. I ’ m insisting on asking you some questions. Can you promise me that I can ask one Star Wars question? Yes, you can ask anything you want. You can ask. Fantastic. Jumping in here as host of my show in the off chance. You don ’ t know who J. J. Abrams is. You absolutely know his work. He ’ s directed, produced and written some of the most beloved films and shows of the last 30 years. If you love Lost alias Felicity, the more recent incarnations of Star Wars or Star Trek, you have to think as a bonus episode. While we ’ re hard at work on Season four, J. J. kindly agreed to turn the **microphone** around and host me for a chat about my new book, Think Again The Power of Knowing What You Don ’ t Know. It ’ s a book about the science of **rethinking** our opinions, opening other people ’ s minds and creating cultures of learning. It officially launches February 2nd, but I wanted to give you a sneak preview. So I sent JJ an early copy and told him he was in charge of our interview. Thanks to Viking Penguin, my publisher, for Think Again for supporting this episode. I have read this book and I just truly think again, your new book available now is so compelling and I just I ’ m thrilled to get to talk to you about it. I feel like you ’ re my book fairy godfather. That ’ s such a thing. That ’ s my dream. Thank you. I ’ ve got to say that I had so many moments of kind of gasping when I read Think again, because you provided the kind of optimism that there is a way to find common ground, to find, to embrace nuance, to reexamine how we, you know, we go through the world. What motivated you to write the book? Well, there ’ s a part of me that that thinks I was probably motivated to write this book by one of my best friends from growing up, Khan, who I remember this was in seventh grade here. We ’ re in the middle of an argument, I think, during a commercial on Seinfeld. And he he hung up on me and I called him back and he said, Shut up, Adam, I won ’ t talk to you until you admit you ’ re wrong. And that was sort of a it really it was a moment that stuck with me. I didn ’ t have the language for it at the time, but I spent a lot of time afterward thinking, why is it so important to me to be right? And, you know, why do I have such a hard time admitting when I ’ m not? And that, you know, I think I was I was sort of a defining moment. And then, you know, much later I feel like my job as an **organizational** psychologist is, you know, is to really think again about how we work, how we lead, how we live, and to motivate other people to do that, too. And my most **frustrating** experiences as an author, as a speaker, as a **researcher**, as a teacher, as a consultant are always when people are not willing to **rethink** assumptions that I think they ought to be questioning. And so, you know, just over the past few years, the number of conversations I ’ ve had with with founders and CEOs where I you know, I give them the best evidence I have available. And they say, well, that ’ s that ’ s not how we do things around here or that ’ s not the way I ’ ve always done things. All right. And I just I just want to come back to them and say, you know, Blockbuster, BlackBerry, Kodak, Sears, you know what? You should definitely not **rethink** anything in your vision, strategy or business model. And so I guess, J. J., when you know, when I have both personal and professional experiences that drive me insane on the same topic, I feel like there might be a problem with tackling or a question worth writing about. Hmm. That ’ s great. Answered in reading the book. It made me wonder if people ' s not bandwidth, but their comfort level is so low that there are people finding comfort in belief systems. And that ' s the thing that they can rely on because there ' s so little to rely on. And and so **rethinking** is a threat to one of the few comforts left. That ' s such an interesting question, I don ' t know is my first thought, my second thought is it reminds me of some research that I loved by sociologists that started in the 70s. They did. This study is where they were interested in the effects of being micromanaged at work and not just how we thought about our jobs, but how we behaved in the rest of our lives. And they found that if you had a really controlling boss, that you actually became more authoritarian as a parent with your children, which was just kind of a stunning spillover for me to, you know, to think that the way you get treated at work might affect the way that you raise your children who just never would have occurred to me. And I think that body of research and a whole bunch of other studies that followed kind of that they made me start to think about this basic truth, that when we lack control in one domain of our lives, we become motivated to seek it and try to regain it in other parts of our lives. And I think, you know, especially in the wake of this pandemic and a global recession where we ' re facing a lot of threats to control right now. And I wonder if if part of what people are doing is they ' re grasping at the things they do have control over. And one of the things we all have control over is what we believe. And so, you know, holding fast to the things that I think are true, they make that sort of makes the world feel more predictable, less uncertain. It also gives me a sense of belonging with whoever my tribe is. Right. It validates my worldview. And I ' ve even wondered if, you know, if that ' s part of why the the George Floyd protests, you know, really rose up when they did. I feel like before the pandemic, people felt like, you know, there ' s nothing that I personally can do to fight systemic racism, whereas, you know, juxtaposed against a global disease like covid, which I really can ' t fight. Well, you know what? I could I could show up in protest. And so I guess there ' s there ' s something about both holding fast to your beliefs that helps render an uncontrollable world more controllable. But also, there ' s something about being in an extremely uncontrollable world that can lead you to think again about where you do have some control areas where you can make a difference. And so I wonder if that cuts both ways. Hmmm, that ' s interesting, I wanted to ask you about the value of humility, because it seems like. Humility is also at the center of what allows for someone to. Pursue a truth or information or a point of view that might not be theirs, but I think so many of us take a more defensive posture and our are more closed off and declarative about things that we think we know. And I just wanted to ask you kind of what role you feel humility plays and why it might not be such a given that the actual value of it is the truth and is enlightenment and is wisdom and why that doesn ' t seem so appealing in the moment. Yeah, interesting. So if pride is what fuels overconfidence, I think humility is what gives us the freedom and the flexibility to **rethink** the opinions and assumptions we should be questioning. And I think that part of the problem is that we live in a culture where people are rewarded for being the smartest person in the room and they learn to wield their knowledge and expertise like a weapon, as opposed to treating it as a resource to share. And so, you know, I think in most **workplaces**, in many schools and frankly, in too many families and communities, too, you know, the currency of status is knowing more than everybody else. And so knowing that, you know, admitting what you don ' t know, questioning yourself, showing, you know, any hint of uncertainty is like admitting weakness and setting yourself up for defeat in battle. And obviously, I think that ' s a problem. And I think that maybe the first way to get people past that fear is we need people in positions of power to model it.

Right. We know that if you ’ ve been validated by your power or your status or your authority, that people are much more comfortable letting you say you don ’ t know. There ’ s evidence, for example, that when experts say, I ’ m not sure or I don ’ t know, they actually become more persuasive, in part because we respect their humility and in part because we ’ re surprised that we actually pay more attention to the substance of their argument, which, you know, is likely to be persuasive if they are knowledgeable. So I guess what I want to do is I want to let people know that humility is a potential source of strength, not just a sign of weakness. And I don ’ t I don ’ t entirely know how to get people there. And I want to turn this around on you because I feel like this is one of the the themes that that really cuts through a lot of your work is every time I watch a show or a movie that was directed or produced by you, I know it ’ s going to lead me to question an assumption. And it might be an assumption I ’ ve held for a long time. And I ’ m curious about is this something you do intentionally and how do you get your audiences to confront that kind of humility? Well, thanks for saying so. I feel like the the way into a story for me is usually finding a character who is vulnerable, who is in a place where they ’ re out of their depth and they ’ re desperate for something. And I think that desperation is like the greatest motivator. And it also it forces you as a storyteller to know what a character ’ s desperate for and therefore what you are either afraid of or hopeful for and ideally both the idea of desperation as a motivator. I think that ’ s incredibly powerful. And it ’ s something that that not only strikes me as relevant to, you know, to thinking about characters in a story, but also to understanding the people that I work with and the people that I love. And I ’ m wondering is, is that a lens that you apply? And I guess if you turn it inward a little bit, when you ’ ve achieved the level of success that you have, how do you stay desperate to **rethink** some of the stories that you tell? Well, I have to do is go on Twitter. I feel like, you know, I ’ m always desperate to do something, to tell a story that that moves people. And sometimes I ’ m more successful than others. And, you know, I never go into anything with a kind of template for what it should or shouldn ’ t be. I feel like I ’ ve learned a ton of lessons along the way. And by the way, I always feel like we ’ re carrying, you know, whatever a thousand lessons with us that we all need to, you know, remember at all times. But we can only really carry 900 of them. And we ’ re constantly dropping lessons and picking up the ones that we ’ ve dropped in by picking those up. We ’ ve dropped some other ones and we keep relearning the same thing. So I think, you know, **rethinking** is really important and relearning is a weird Sisyphean inevitability that I don ’ t think we can avoid. But I feel like, you know, the drive to do. What I do like I ’ m guessing the drive for you to do what you do, you do it because you feel compelled to share a point of view, an idea, a feeling, an insight, a question, something you might not even understand the answer to. And you just feel compelled to do it. And I think I start everything desperately, and I think they probably all end that way, too. I love that. I want to ask you, because your book is all about **rethinking**. I was just curious, do you feel like there is a fundamental barrier to **rethinking** and if so, what it is and I do have a follow up. Yeah, I think so. So, yeah, I feel like one of the things that I ’ ve **rethought** twice now as I ’ ve been writing books is, you know, I wrote my first book, Give and Take with with a whole framework to organize the world in terms of givers, takers and matters. And then I didn ’ t want to be constrained by a framework like that when I wrote originals and I felt like I overcorrected and there wasn ’ t enough connective tissue between the different chapters. I felt like I had a lot of interesting trees, but the forest was not clear enough. And so I decided when I was writing Think again that I was going to be open to finding an overarching framework, but I was not going to be attached to it.

And that that felt like the sweet spot. And about halfway through the writing, it hit me and I had to rewrite the book from scratch. The framework was my my colleague, Phil Tetlock, where he **observed** that whatever your job or career is, that you spend a disproportionate amount of your time thinking like certain professions. He said, look, you know, there are moments when we think like preachers, prosecutors and politicians. And I think that that all three of these mindsets can stand in the way of **rethinking**, because when you ’ re preaching, you ’ re already convinced that you found the truth. And so you don ’ t need to question any of your assumptions when you ’ re prosecuting. You are trying to win your case. That means the other side is wrong. And so you ’ ve got to get them to do all the **rethinking**, but you get to stand still. And when you ’ re thinking more like a politician, you ’ re basically trying to cater to your audience and campaign for their approval or get their likes or their votes. And so you may be doing a little bit of flexing in the way that you present, but you ’ re not **rethinking** your underlying beliefs. Right. You ’ re just you ’ re just trying to appease your constituents. And I think we spend way too much time thinking in these modes. I know, you know, as somebody who ’ s been called a logic bully once or twice, maybe seven times at this point, by my count, I spend too much time in prosecutor mode. And when I think that somebody is selling snake oil or trying to, you know, indoctrinate their followers into some kind of idea cult, I feel like it ’ s my job to debunk those myths and bring the better evidence to the table. And the problem is then that I get too close minded. And so I ’ m trying to get out of prosecutor mode and spend more time thinking like a scientist, which is something I think we could all learn to do better. It ’ s funny, in the book, you say how I think the first time you were called The Logic Bully, it actually was something that you liked, that at first it appealed to you. I was proud. I thought, yeah, that ’ s my job as a social scientist. I want to decimate your bad arguments with rigorous evidence and airtight logic. And I ’ m good. And then, yeah, I didn ’ t like the bully part so much as I thought about it more. Given that, what are some of your favorite **practices** for questioning your own assumptions and for **rethinking** yourself? I think the first one for me is just the basic idea of thinking like a scientist to say that, you know, what scientists do is instead of forming beliefs, they actually form hypotheses. And so instead of having an opinion that set in stone, that means, OK, this is a hunch, how would I go and tested? And, you know, I don ’ t think that we have to always operate like we ’ re in a lab. Right, with a bunch of test tubes. But I do think that we should all be running experiments in our lives. So during the pandemic, for example, I have instead of assuming there ’ s a routine that ’ s going to work for me, I ’ ve run experiments to say, OK, what if I shift my creative brainstorming from the morning to the night? What does that do, you know, to to the number of original ideas that I generate? What if I turn my camera off during a zoo meeting? Do I actually have, you know, a more a more reflective, more thoughtful discussion? And all of those you know, there ’ s those little changes and adjustments that we make. We could think about them as experiments and then we could say, OK, well, you know what what information what data do I need in order to find out if that was a successful or failed experiment? And I I think that ’ s a huge step. And I guess the other thing I would I would just put on the table is when I was writing the book again, I made a list of all the things that I know I ’ m completely ignorant about. And my goal was was actually not to reduce the list, it was to expand the list. Mm hmm. Because I feel like the more I know about what I don ’ t know, the more curious I am and the more I ’ m going to learn from people who are actually are knowledgeable in those areas. So, you know, my list, I think it started out I immediately said I know nothing about music. I don ’ t understand **financial** markets. I ’ m clueless about chemistry. And, you know, the list just kept growing from there. And I ' ve actually been just keeping that list. I have a file on my desktop and my goal is to add something new to it every week. And if I stay clear about what I don ' t know, then I hope I ' m going to stay more humble, more curious **mindset**. That ' s great. When I was reading, I think, again, it was occurring to me, you know, that you tell such great stories, including like the story of the BlackBerry company and, you know, Mike ' s inability to **rethink** the way perhaps he could have and what might have been. And I, too, was a huge fan of that **keyboard**. The question I had is, how do you find partnerships where there is a thinker and then a thinker, or is it always or typically whether it ' s one or two or more people that are able to **rethink** and think, oh, that ' s such a cool question. I think my ideal partnership is one where. There are clear **norms** that make sure **rethinking** happens. The question of where it comes from is something that I don ' t have good data on, which really mildly annoys me right now and I need to go gather some. So stay tuned. But I mean, I can think of examples of both. One of the examples that that I ' ve spent probably too much time thinking about now. Is the, you know, the Jerry Seinfeld Larry David example from Seinfeld. And one of the most interesting things that I learned from studying that collaboration was that they were both ruthless free thinkers, but they applied it to their own work, probably even more so than they did to each other ' s work. And so one would not even bring a script or a joke to the other until he had thought it was, you know, really good. Mm hmm. And I can see how those kinds of **norms** could be really effective in partnerships. I ' ve also seen examples, though, where, you know, one person is basically the kind of a creative dreamer and the other person is the really critical evaluator. And so, you know, the first person does the thinking, the second does the **rethinking**, and then it ends up being kind of this beautiful symphony. Mm hmm. I imagine, J. J., you ' ve seen a lot of this in in the world of Hollywood. Is there is there one or the other that you think is more common or more effective? And are there are there are people who do more of the thinking versus **rethinking** in your collaborations? Well, obviously, I think, you know, everyone is different. And I know I have worked with people who who go to a place and not that they ' re stubborn about where they go, but they arrive at an answer very, very quickly. And that ' s sort of their thing and that they ' re perhaps a bit reluctant to be as flexible or **rethink** things. Then, you know, I ' ve worked people like, for example, with Matt Reeves, with whom I did Felicity, and we did the Cloverfield movie together. And he ' s someone who you know, and I ' ve known him since we were kids. And he ' s always **rethinking**. And I find that I ' m maybe more, you know, someone who would want to try something and put it out there and sort of commit to it for now and sort of see where it goes. And Matt. To his great credit, and it ' s one of the reasons I love working with him, is he ' s someone who really wants to stop and examine and consider and reconsider it and look at it from all sides and see if there ' s a better way. And can it go deeper and can it be, you know, can it be more emotional? And I think that that, you know, part of that is probably, you know, less about thinking and **rethinking** and more about my being more impatient. And when we work together, we had our offices right next to each other. And in fact, our offices were in the location where they actually filmed the office, the show, the pilot for the American Office Show. Oh, and weirdly, his office in my office were Matt ' s office with Michael Scott ' s office. And my office was the the conference room, which years later, when I went and directed an episode of The Office, I was like, why do I know this place? And then I realized, oh, my God, it ' s literally based on our old offices. But Matt, we had a little little sign in between is a picture of the two of us in between both of our offices. And someone had put up sticky notes over each of us and overmatch it, said, how do we make it more emotional and undermines lives, how to make it funnier? And there is a sort of like, you know, clearly everyone brings their whatever their interest is, whatever their their point of view is to the table. And even though sometimes it can be frustrating when someone says, well, wait a minute, let ’ s reconsider this, you know, it ’ s almost always a better thing because there ’ s nothing wrong with kicking the tires and interrogating something. But the best thing and this is what I found, you know, recently I co-wrote a pilot with a writer named Latoya Morgan, and she ’ s a wonderful writer. She ’ s a black woman who brings a point of view that is clearly going to be a life experience, very different from mine. And when I bring up an idea and she says, you know, hold on, let ’ s talk about this or that or vice versa, I feel like we almost invariably end up in a place where there ’ s something that we are both, like, beaming about that we wouldn ’ t have arrived at had it just been one of us in that room. And in all likelihood, you know, someone who looked like us, you know, either just me or just her. So I feel like there ’ s a lot of talk of how diversity is good and how you say in your book, diversity is good but hard. But, you know, I have personally felt like, you know, working with people who don ’ t look like you, especially when they are, you know, smart and freethinkers. It ’ s incredible how much better the conversation is, how much richer it is, and I think how much better the work ends up being. And I ’ m grateful for any time I work with someone, whether it ’ s Matt, Latoya, I can name a huge list of people who are really willing to **rethink** things that I might just assume need to stay. And the only caveat I would say here is that I can also think of times when there were paths we were on and the ability to switch it up and **rethink** something might not have been for the best. Yeah, this is one of the hardest questions that I really wrestled with while writing the book and I ’ m still grappling with now, which is how much **rethinking** is too much and how do I know whether when I ’ ve started **rethinking** something, I should actually change my mind or go back to my original first answer and the only coherent place that I ’ ve landed on this so far as to say, well, if this is a curve and there ’ s an optimal amount of **rethinking**, I think most of us don ’ t do enough of it. Agree or disagree? You ’ re asking me, yeah, yeah, it ’ s a question, do you agree or disagree? You know, my guess is I would agree. Got it. I have to take a sidebar here, JJ, because you mentioned the office. This is work life. I read once that you did not like the office when it premiered in the U. S. and that you weren ’ t even a fan of the entire first season. Is it true? And what led you to **rethink** that view? I think that what I felt was when I first watched it, because I became a huge fan, which is why I reached out to them, essentially begging them to hire me as a director. I was such a fan of the original and in fact, to a degree that I reached out to Ricky Gervais and wrote a part for him in Alias. And I was such a fan of that show and the specificity of it that it felt. For me, I remember when I first saw it, I thought, oh, no, like it didn ’ t matter how good everyone was because they were great. It felt like it was somehow sacrilege that it was redoing exactly the same thing in essentially the same time period, you know, putting it through the American TV mill. And then you realize, oh, they ’ re doing it perfectly. And it ’ s it ’ s exactly what it should be. And it ’ s its own thing. It was the quality of the show that that won me over. OK, next question. I ’ m sure you hate getting asked about Star Wars in particular, because it might what might be the most beloved universe ever created a minimum. It ’ s got a lot of **passionate** fans. And yeah, I loved The Force Awakens. I love the rise of Skywalker. I especially love the points where you **rethought** some moments and characters. And I wanted to ask you about one specific one, which is I think my biggest moment of shock in the new trilogy was when Emperor Palpatine came back to life because I thought he ’ d been dead since 1983. And so I was wondering how you went about **rethinking** his demise. Well, that was something that came in and out and then back in again. But that idea was something that that felt like it was sitting there, set up in the prequels. And when we were looking at the story, the idea of a character who is desperate to find immortality and it felt like it was sitting there and that it was a set up potential set up. And I kept imagining it as something that could work. You know, if I ’ m a kid watching all nine movies years from now, it felt like something that was a possible setup to be examined later. I will say that it ’ s something that I loved the possibility of it and the promise of it. And it was one of, you know, so many things that we were **rethinking** and trying to find ways to sort of fit the pieces of the puzzle together. And I know that for some people, it they clicked in and it worked and was fun and surprising. And for other people it was, you know, understandably infuriating and sacrilege and stupid and wrong. The reactions were were vast, but it was also something that was, you know, in various moments felt like it was the right move. I imagine that happens a lot to that. You you **rethink** a classic world or story, whether it ’ s Star Wars or Star Trek or even characters that we ’ ve fallen in love with. Right. I still feel like, you know, with Lost in particular, I still feel like, you know, Sawyer and Sayid and Kate and some of the others are long lost friends. How often do you **rethink** the stories you ’ ve told versus once you put it out into the world, you ’ re you ’ re happy with the creative choices you ’ ve made, regardless of the feedback? Well, all I see when I work on something is the thing that we could have and should have done to make this better or that different. The **rethinking** doesn ’ t stop because it ’ s it ’ s out there. And I do think that one of the enemies of certainly movies and in some cases TV is the sort of, you know, the announced release date where the time that one normally should have to **rethink** doesn ’ t exist. And it ’ s very difficult to be. In a healthy way, **rethinking** when you are under the gun, impractical material and often very expensive ways. For example, when Damon and I started working on Lost, it was a very particular, you know, crazy ticking clock where the head of ABC at the time, Lloyd Braun, wanted a show about people who survived a plane crash. And so in Damini, who had never met, we wrote that outline. It was in a week based on just pure instinct. And we went off and we shot this to our pilot that we were, you know, writing, casting, shooting, doing post and everything in nine weeks. And it was a crazy crunched period. Wow. And it ended up working as a pilot. And then Damon and Carlton Cuse wrote the series. But I, I feel like it ’ s one of those lessons that not having an ending, it just makes it harder. I know it sounds so obvious and so easy, but when you take that leap of faith, when Stephen King starts writing a novel, he gets to finish that novel and decide, does anyone need to see this? And I ’ m sure he ’ s got drawers full of things that people have not read because he feels like it ’ s not the thing. And one of the crazy things that this, you know, pandemic **quarantine** time has done for TV and for movies is it ’ s given people time to write, not just a pilot or a rough outline, but write entire seasons of a series before it even goes, which, you know, you want to be flexible and you want to be able to cast a role and realize, oh, my God, we ’ ve got to do this with this character because we didn ’ t realize it or this thing isn ’ t working the way I thought it would. But this time has allowed writers to, you know, write without the pressure of we have to be casting this thing right now. We have to be location scouting. We have to be, you know, getting the all these episodes edited and posted and ready for broadcast. And I think that it in no way do I feel like, oh, great, we had this time to do this. You know, I ’ d much rather life be normal and covid not exist. But I will say that if there is any kind of creative benefit to this time, it ’ s that suddenly release dates and broadcast dates are not being, you know, as aggressively announced. And time is available for some writers to be able to actually work on the stories in a more, you know, I think human pace. And that is no small thing. Is there creative **practice** that you ' ve **rethought** during the past year? If you had said to me that we could be breaking stories without ever being in the same room together, I would have thought it was probably impossible, if not wildly unlikely. And I think that online, not just the Zoom ' s, but also the sort of online whiteboard apps that exist that allow people to collaborate remotely has, I think, probably profoundly changed the way stories, at least on television and writers rooms, will be broken. Wow. Does that does that mean in writers rooms won ' t always be physical even after the pandemic? They definitely will. But I think the that someone can now be part of it who might not be in town. I think that that the **norm** will always seem to be better to be together. But there ' s something great about the focus and the, you know, the access that you have from anywhere. So I just think that that that ' s something that, you know, clearly there will be infinitely fewer business trips to have meetings that could now clearly be done on Zoome. Will people go back to it? You know, the way we were before? More than not, I believe they will, but. This is now accelerated the evidence that this is a valid and really productive way to work. I ' ve never said those words. I ' ve never said this before. We ' re not going to hold you accountable. Thank you. Next question from one magician to another. How does all the time you ' ve spent learning magic tricks and performing them affect your job? Has it spilled over into your work life in any way? I think the thing about magic, as much as some might see it as the geekiest sort of, you know, silliest thing to me, it ' s incredibly powerful. We had a magician in our office and this was like two years ago, and he was in my office, just the two of us. And he did a few tricks and he did one that literally made me, as a 50 year old man, think to myself in my head, he might be magical, like, you know, because I know how magic tricks are done. I know how a lot of I know bunch of gimmicks. I know a bunch of tricks. He did something where I was like, I wonder if he ' s maybe magical. Does he have? And my whole point is the fact that someone who ' s a half a century old can still have a moment of thinking. There might be powers beyond that, which I know is so profound. And the the wow and the gasping and the amazement that any audience has, big or small, it ' s there ' s something about the the sense of possibility that the world is more than it seems and that we want to see things that we can ' t imagine and that we don ' t expect. And so I feel like that is a natural. Aspect to telling a story, and I just think that whether it ' s writing a a book or making a show or a movie, it ' s all a bit of a magic trick. You are so much more comfortable not knowing than I am. I immediately want to answer. I have to figure out the secret, but then you can never see the trick again. You can admire the the skills and the genius of the performer who executed it, right? Yes. OK, next favorite character from the office. It has to be Michael Scott, not because just because I think he is the hub of the wheel. And so everyone is kind of it feels like they ' re all kind of in his orbit, to mix metaphors. So interesting. He wasn ' t even in my top five. I love that. And then can I ask you the same question about Star Wars? I think my favorite my favorite character in in Star Wars was Han. Just because I mean, it ' s very hard to think of a better scene in a movie ever than when you meet Han Solo and then in Empire Strikes Back. I mean, like there ' s just there are some of the greatest, truest larger than life, but completely relatable and grounded moments. And, you know, when we got to shoot in Force Awakens, you know, Harrison and Chewie coming into the, you know, the Falcon and saying, you know, we ' re home because it just there was something so crazy about it. And I feel **super** lucky to have, you know, been part of that. I had tears in my eyes when when Nonintuitive had their reunion. It just felt like we just couldn ’ t wait to show it to people because it just felt like, oh, my God, you should see this. It ’ s just so cool having them back on the ship. It was a beautiful moment. I ’ d love to talk to you about as a Red Sox fan, the Red Sox versus Yankees. And you have this great chapter about just how far fans of these teams are willing to go to torture the opposing team. And what you tried, what didn ’ t work and what finally seemed to break the ice of having them sort of find the humanity in each other. So my question to you is, what was what surprised you in your attempts to build those bridges between those two fan bases? And what were your findings? In summary, I ’ m so glad that resonated. You know, I was I was when thinking about the prejudice that so many people hold toward other groups, I was looking for a context where we could study it, where the actual stakes are minimal, but the emotional stakes feel very high. And sports fan allegiances, it just it felt like the perfect place to go. I mean, a few years ago, I got to know an amazing woman in her 70s who helps Holocaust survivors. And one day she mentioned that she went to Ohio State. And my first reaction was, yuck. Yeah, I ’ m a Michigan Wolverine. We ’ re rivals, but who cares where she went to school half a century ago? Yeah, it just bothered me at a fundamental level. And so I decided, you know, like any self respecting social scientist, I should study that. And so Tim and I and all these experiments with the Yankees and Red Sox fans, which was such an easy rivalry to choose. And I think my my biggest surprise was that two of the steps that were supposed to work based on all the existing science, didn ’ t you know one of those was was just creating a common identity as **baseball** fans? And, you know, people would say, yeah, you know, I would help a Yankees fan if they were in an emergency. But otherwise, you know, they ’ re rooting for an evil team. So no, thank you. And then the other one was to to try to humanize the individual fan, where basically what happened was people said, all right, you know what? There ’ s this one Yankees fan who ’ s OK, but he ’ s not like the other ones. And I still hate all the rest of them. So what ultimately ended up working and in a way that I was I was surprised by the power of it actually was saying, look, you know, maybe we just need to get people to **rethink** how they became fans in the first place. So imagine if you had grown up in New York. Would you possibly root for the now you ’ re probably root for the Mets. But the point is, you might not be a Red Sox fan. I ’ m not crazy. Exactly. I ’ m joking. I ’ m joking. All my Mets fans, friends, I ’ m joking. I don ’ t know if you are, but go on. I ’ m joking anyway. I know I keep going, please. Well, no. So I thought this idea of just, you know, considering the possibility that we could hold different views all of a sudden, it it just destabilizes a lot of the stereotypes and prejudices that people hold. And so, as you know, we ran these really entertaining experiments where we gave people a chance to to punish fans of the opposing team by giving them extra spicy hot sauce or by giving them really difficult math problems, which were actually going to hurt their payment for participating in the study. And we found that, **lo** and behold, after we just got fans to think about how they might root for a different team if they had been born in a different city, that was enough to get them to show less animosity toward fans of the opposing team, which I thought was amazing. And this did not make it into the book. But we just finished a couple of experiments with with gun control and gun rights advocates where we found that the same process worked for them. That if we got somebody who is extremely concerned about about gun safety, to imagine having been born in a hunting family, they actually were less nasty than to somebody on the other side of the aisle. So I think there might be something to this. Fantastic. So I can ’ t thank you enough for talking to me on your show about you, you should do the the goodbye because it ’ s your show. But I just I wanted to thank you for the the honor and opportunity. Well, I can ’ t thank you enough for coming. I ’ m so grateful that you read the book. You liked it that you ’ re willing to talk about it. It ’ s a huge gift to me and our listeners. And I loved getting a little bit inside of your mind and your work life. And most importantly, the work you do has has always been just a huge source of inspiration to me. But over the past year, you have been a source of light and hope and entertainment and creative inspiration. And I just want to thank you for for everything that you do to entertain us and to get us to **rethink** so many of the things that we think we know. Holy crap. That was unbelievable. Way too generous, but your your words are incredibly appreciated. And and again, thank you for this amazing book and your kind words. WorkLife is hosted by me, Adam Grant. Our team includes Colin Helmes, Greg Achen, Dan O ’ Donnell, Constanza Guardo, Joanne DeLuna, Grace Rubenstein, Michelle Quint, Angela Chang and Anna Felin. This episode was produced by Jessica Glaser and supported by Viking Penguin. Think again. The power of knowing what you don ’ t know is available in whatever format you like print, electronic stone, tablet or audio narrated by me.

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Hey, WorkLife, it ’ s Adam Grant, season four is right around the corner, but today I wanted to share a **special** conversation in our Taken for Granted series. I ’ m talking to Daniel Kahneman. Danny won a Nobel Prize in economics. He ’ s been named one of the most influential economists in the world, but he ’ s not on board with that. Oh, my God, no. The behavioral economist. I ’ m not any kind of economist. Danny is one of the great psychologists of our time, actually, of all time. You may have read his influential book, Thinking Fast and Slow. And he has a new book, Noise, coming out later this spring with Cass Sunstein and Olivier Sebti from. This is taken for granted by podcast with the TED Audio Collective, I ’ m an **organizational** psychologist, my job is to think again about how we work, lead and live. This conversation with Danny challenged one of my core beliefs about intuition. It also gave me a new way of thinking about which ideas are worth pursuing. Since Danny is an expert on decision making, I thought I ’ d start by asking about what we ’ re seeking in so many of our decisions. You ’ ve spent a lot of your career studying **happiness** and related topics. And really, for the first time in my career, I started to wonder why are we so obsessed with **happiness** as psychologists? You know, I ’ m all for people leading enjoyable, satisfying lives. But if I had to choose, I would much rather have people focus on character and, you know, trying to build their generosity, their integrity, their commitment to justice, their humility. And I wonder if you could talk to me a little bit about whether you think we ’ ve lost our way a bit and character has been too little in focus or too far in the background or whether you think **happiness** deserves the attention it ’ s gotten? Well, I think my focus would be neither **happiness** nor character. It would be misery. And I think that there is a task for society to reduce misery, not to increase **happiness**. And when you think of reducing misery, you would be led into very different policy directions. You would be led into mental health issues. You would be led into a lot of other problems. So reducing misery would be my focus. Character and **happiness** or misery are not substitutes. The idea which has been accepted both in the UK and in many other places, other than quite a few other countries by now, is that the objective of society? The objective of policy should be increasing human welfare or human well-being in a in a general way. I think that ’ s a better objective for policy than increasing the

quality of the population ' s character. I think it ' s a better objective. I think it ' s it ' s a more achievable objective, except I would not focus on the positive and I would focus on the negative. And I would say it is the responsibility of society to try to reduce misery. And let ' s focus on that. We speak of length and not the shortness and we speak of **happiness**, the dimension is labeled by its positive poll. And that ' s very unfortunate because actually increasing **happiness** and reducing misery are very different things. I agree. And it ' s interesting to hear you say that reducing misery is more important than promoting **happiness**. In some ways, that feels like a critique of the positive psychology movement. It is. And tell me a little bit more about why. Well, I think the positive psychology movement has in some ways a deeply conservative position. That is, it says let ' s accept people ' s condition as it is and let ' s make people feel better about their unchanging condition. You know, there has been some critique of positive psychology along those lines. I ' m not I ' m not innovating here. But I think that focusing on changing circumstances and dealing directly with misery is more important and is a **worthy** objective for society than making people feel better about their situation. Yeah, I mean, I think it certainly tracks with how I think about, in general, bad being stronger than good and the alleviation of misery contributing more to the quality of people ' s lives than, you know, some degree of elevating of of the amount of joy that they feel. But I also wonder at times if this is not a false dichotomy, that if you want to make people happy, it ' s awfully difficult to do that if you don ' t pay attention to the misery or suffering that they might experience. Well. Actually, we once did a study in which we we were measuring how people figure, how much of the day are people in different states positive or negative? And it turns out that people are in a positive state on average, 80 percent of the time, more than 80 percent of the time. That is on average, people are on the positive side of zero. Now, look at, say, the 10 percent of the time that people spend suffering. Overall, most of the suffering is concentrated in about 10 to 15 percent of the population. So it actually is not the same people that you would make less miserable or happier. Those are different populations. And the question is, where do you direct the the weight of policy and what you pay more attention to? Very interesting. I like it, so you ' re basically saying, look, if we have scarce resources, whether those are **financial** or time or energy, we want to concentrate on the group of people who are suffering as opposed to those who might be languishing. You know, it seems to me that to some extent we have been trapped by words. I mean, it ' s the word **happiness** which seems to stand for the whole dimension. And and and I think this is leading to some policies, actually, this failing to lead to policies that would that would really be directed at increasing human well-being by decreasing misery. Yeah, I think so, too. And it ' s something I ' ve thought about a lot at work. Given given the hat I wear most often is organisational psychologist, I feel like the obsession with employee engagement has really missed the mark. I don ' t go to work hoping that I ' m going to be engaged today. I hope that I ' m going to have motivation and meaning and that I ' m going to have a sense of well-being. And I wonder if if one of the effects that the pandemic has had on a lot of people and a lot of leaders in **workplaces** is to get them to recognize, you know, what we need to care about people ' s well-being in their lives, not just their engagement at work. Well. I thought that, you know, I ' m not an expert. This is your field, not mine. But I thought that engagement has is close to feeling good at work. I mean, we whether it ' s the responsibility of **workplaces** to deal with people ' s well-being in general, I agree that it ' s they ' re responsible for dealing with people ' s well-being at work. And that doesn ' t seem to me to be very different from trying to make people engaged and happy with what they are doing. So I ' m a bit curious to hear more about the dichotomy of the distinction that you ’ re drawing between engagement and will be my interpretation of engagement, whether it ’ s fairly close to wellbeing at work. Yeah, I think I think in large part it depends on which conceptualization and measure of engagement we ’ re talking about. But one of the one of the more interesting patterns in the literature that that ’ s gotten me thinking quite a bit is that it ’ s possible to be an engaged workaholic. And this this has been differentiated recently from being a compulsive workaholic. You know, are you are you working a lot because you find it interesting and worthwhile, or are you doing it because you feel guilty when you ’ re not working and you feel kind of obsessed with the with the problem that you ’ re trying to solve? And I think that one version of engagement is probably healthier than the other. And I associate wellbeing much more with, you know, with being an intrinsically motivated workaholic than with a compulsive workaholic, even though both are highly engaged. I agree. You know, I worked for a while with Gallup. I was consulted with Gallup many years ago. And their concept of engagement, I think, was a positive concept. One of the criteria that I remember for people being happy at work is having a friend at work. So clearly, at least, their concept of engagement, which is the one the only one that they know much about, is by and large a positive concept. And certainly the word we don ’ t want people to be compulsive, although. Although I don ’ t know how to describe myself, for example, when when I work hard or when I used to work very hard, was I doing so compulsively, was doing so out of intrinsic motivation? I think both I was intrinsically motivated and I was compulsive about. So I ’ m not sure of the distinction that you ’ re drawing between being compulsive and being intrinsically motivated. Well, I like to call it a look at ambivalence there, because I think it speaks to the point that you raised earlier, which is that, you know, positive emotions and negative emotions can coexist. You can work because you ’ re **passionate** about it and because you feel bad if you ’ re not doing it. That ’ s right. I want to ask you about the joy of being wrong. The place I wanted to begin on this is to ask you, when you were growing up or earlier in your life, how did you handle making mistakes? Hesitating because I can ’ t it ’ s not that I didn ’ t make any mistakes, I certainly made many, but I wasn ’ t very impressed by my mistakes. I mean, they were not very salient in my life. So if you ’ re asking about my earlier, you know, as a student and so I don ’ t have much to report that I have an interest as a **researcher**, I found my mistakes very instructive. And and there were sort of positive experiences. By and large. It ’ s such an odd thing to hear you say, because most of us, most of us experience pain, not pleasure. When, you know, when we find out that we ’ re wrong or we discover that we ’ ve made a mistake. So how did you arrive at a place where you found that to be a teachable moment? Well, you know, those are situations in which you are surprised. I really enjoyed changing my mind because I enjoy being surprised and I enjoy being surprised because I feel I ’ m learning something so that it ’ s been that way. I ’ ve been lucky, I think, because I think you ’ re right, that this is not universal, the positive emotion to correct and mistakes. But it ’ s just a matter of luck. I mean, I ’ m not, you know, not claiming high moral ground here. It ’ s it ’ s fascinating to watch, though, because I ’ ve seen your eyes light up and, you know, it ’ s it ’ s it ’ s palpable when you when you discover that you were wrong about a hypothesis or a prediction, you look like you are experiencing joy. And I ’ ve started to think a lot about what prevents people from getting to that place. And I think a lot of it is for so many people, they get trapped in either a preacher or a prosecutor **mindset** of saying, you know, I I know my beliefs are correct or I know other people are wrong. And at some point their ideas become part of their identity. And I know even scientists **struggle** with this. Right. I think at least when I was trained as a social scientist, I was taught to be passionately dispassionate. But I know a lot of scientists who **struggle** with detachment, and you don ’ t seem to. So how do you keep your ideas from, I guess, becoming part of your identity? Well, I think that. I mean, this is going to sound awful. I have never thought that ideas are rare. And, you know, if that idea isn ’ t any good, then there is another that ’ s going to be better. And I think that is probably generally true, but not generally acknowledged. So that for people to give up on an idea may in many cases lead to a sort of panic. If I don ’ t have that idea, then what do I have? Who am I if I don ’ t have that idea? So being less identified with your ideas is also associated, I think, with having many of them just discovering that most of them are no good and trying to to do the best you can with a few that are good. So it ’ s seeing ideas as abundant rather than scarce. That ’ s what makes it easier to stay detached. Yeah, yeah. I mean, I used to tell my students ideas are a dime a dozen. I mean, don ’ t overinvest in your old ideas. And so I used to encourage my students to give up. At a certain point, I suddenly never wanted to read a dissertation by a student with a chapter that would explain why the experiment failed. So that was the kind of advice that I would give them. Think of another idea. Do you ever worry about getting too detached? I think, for example, about messenger RNA technology, which was seen as, I think a joke for a long time, and if not for the courage and tenacity of a small group of scientists who persisted with it anyway, we might not have a covert vaccine right now. Oh. I think well, in the first place, science, like many other social systems, doesn ’ t thrive on everybody being the same. So you may have some advice that is good for some people. And it ’ s clear that some people who are irrationally persistent achieved great successes. And indeed, if you look back at the great successes, you will generally find that there is some irrational persistence behind them and irrational optimism behind them. That doesn ’ t mean that when you are looking from the other side, that irrational optimism or irrational persistence are good things to have. So the expected value of it might be negative, although when you look back every big success you can trace to some irrationality. Well, that goes beautifully to one of my favorite ideas of yours, that we look at successful people and we learn from their habits, not realizing that we haven ’ t compared them with people who failed, who had many of the same habits. And I wanted to, I guess, ask you a broader question, which is having put these kinds of decision heuristics and cognitive **biases** on the map. Which one do you fall victim to the most? Is that confirmation **bias**? It sounds like maybe not. I just wondered which of which of the **biases** that you ’ ve documented is your greatest demon. All of them, really. All of them, except, as you said, confirmation **bias**. By the way, people are close to me find this irritating. That is that whenever they have a problem with someone, I automatically take the other side and try to explain that someone might be right after all. So I have that contrarian aspect to what I am. This reminds me a little bit of a possibly apocryphal story that I think told to to every doctoral student in social science these days, which is that not long after you won the Nobel Prize for your work on Decision-Making, there was a journalist who asked you how you made tough decisions and you said you flip a coin. Is this true? No. OK, good. Absolutely. I ’ m really I ’ ve never flipped a coin to make a decision in my life. The version of the story I heard was that you would flip the coin to **observe** your own emotional reaction and figure out what your **biases** were. I might have said that this is one of the benefits of flipping a coin, but I personally have never used that. But it ’ s true that. Flipping a coin would be a way of discovering how you feel if you didn ’ t know earlier that I still believe, I feel very relieved to know that because I was worried about you, given all you know about decision making, making important life choices with a coin toss. Welcome back to Taking for Granted and my conversation with Danny Kahneman. He was just setting the record straight that as an eminent scholar of decision making, he does not make decisions based on a coin toss. So how does he make decisions? Well, when I look back at my life, it ’ s been a series of things that, you know, ultimately I made decisions, made life choices clearly, but I did not experience them. As decisions, I have very little to say, describing myself about making decisions, in part because I have really strong intuitions and I follow them usually. So the decision doesn ’ t feel hard if if you know what you are going to do and if you know yourself and you ’ re going to do it anyway, it doesn ’ t feel very hard. I have to say, Danny, I ’ m a little shocked to hear you say that you follow your intuition because you have spent most of your career highlighting all the fallacies that come into play when we rely on our intuition. Well, you really have to distinguish the judgment from decision making. And most of the intuitions that we ’ ve studied were fallacies of judgment rather than decision making. And second, my attitude to intuition is not that I ’ ve spent my life, you know, saying that it ’ s not good enough in the book that we ’ re right and just finished writing. Our advice is not to do without intuition. It is to delay it. That is, it is not to decide prematurely. And not to have intuitions very early, if you can delay your intuitions, I think they ’ re are your best guide probably about what you should be doing. OK, so two questions there. One is how the other is why, well, you delay your intuitions, you know. No, I ’ m talking about formal decisions, decisions that might be taken within an organization or a decision that an interviewer might take in deciding whether or not to hire a candidate. And he had the advice of delaying intuition is simply because when you have formed an intuition, you are no longer taking in information. You are just rationalizing your own decision or you ’ re confirming your own decision. And there is a lot of research indicating that this is actually what happens in interviews, that interviewers spend a lot of time. They make their mind up very quickly and they spend the rest of the interview confirming what they believe, which is really a waste of time. Yes. Yes. So the idea of delaying your intuition is to make sure that you ’ ve gathered comprehensive, accurate, unbiased information so that then when your intuition forms, it ’ s based on better sources, better data. That is that what you ’ re after? Yes, because I don ’ t think you can make decisions without there being endorsed by your intuitions. You have to feel conviction. You have to feel that there is some good reason to be doing what you are doing, so ultimately intuition must be involved. But if it ’ s involved, if if you jump to conclusions too early or jump to decisions too early, then you are going to make avoidable mistakes. Well, this is an interesting twist on, I guess, how I ’ ve thought about intuition, especially in a hiring context, but I think it applies to a lot of places. My advice for a long time has been, don ’ t trust your intuition, test your intuition, because I think about intuition is a subconscious pattern recognition. And I want to make those patterns conscious so I can figure out whether whatever relationship I have detected in the past is relevant to the present. And it seems like that ’ s what what you ’ ve argued as well when you ’ ve said, look, you know, you can trust your intuition if you ’ re in a predictable environment, you have regular **practice** and you get immediate feedback on your judgment. I think the tension for me here is I don ’ t know how capable people are of delaying their intuition. And I wonder if if what might be more practical is to say, OK, let ’ s make your intuition explicit instead of implicit early on so that then you can rigorously challenge it and figure out if it ’ s valid in this situation. I ’ ve been deeply influenced by something that I did very early in my career. When I was 22 years old, I set up an interviewing system for the Israeli army. It was to determine suitability for combat units. And the interview system that I designed broke up the problem so that you had fixed rates that you were interviewing about, you ’ re asking factual questions about each trade at the time and you were **scoring** each trade once you had completed the questions about that trade jumping in here, because this is such a cool example, but it needs a little explaining. Danny created a system for interviewers to rate job candidates on specific traits like work ethic, analytical ability or integrity. But interviewers did not take it well. They really hated the system. When I introduced it and they they told me I mean, I vividly remember one of them saying, you ’ re turning us into robots. Danny decided to test which approach worked best. Was that their intuition or their ratings from the data? The answer was both their ratings plus their intuition, but not their intuition at the beginning, their intuition at the end, after they did the ratings, that is, you read those six straight and then close your eyes and just have an intuition how good you think the soldier is going to be. When the data came back, it turned out that that intuition at the end was the best single predictor. It was just as good as the average or the sixth straight, and it added information. So, wow. You know, I was surprised, you know, I just was doing that as a favor to them, letting them have intuitions. But the discovery was very clear. And we ended up with a system in which the average of the six traits and the final intuition had equal weight. It sounds like what you recommend then concretely is for a manager to make a list of the skills and values that they ’ re trying to select on to to do ratings that are anchored on those dimensions. So, you know, I might **judge** somebody who ’ s coding skills if they ’ re a programmer or their ability to sell if they ’ re a salesperson. And then I might also be interested in whether they you know, they ’ re aligned on our **organizational** values. And then once I ’ ve done that, I want to form an overall impression of the candidate, because I may have picked up on other pieces of information that didn ’ t fit the model that I had. I think that ’ s about right. It ’ s such a powerful step that I think should bring the best of both worlds from algorithms and human judgment. There ’ s something that ’ s a little puzzling to me about it, though, which is why are managers and people in general so enamored with intuition? I think it ’ s because people don ’ t have an alternative. It ’ s because when they try to reason their way to a conclusion. They end up confusing themselves. And so the intuition wins by default, it makes you feel good. It ’ s easy to do and it ’ s something that you can do quickly, whereas careful thinking in a in a situation of judgment where there is no clearly good answer, careful thinking is painful, it ’ s difficult, and it leaves you in a state of indecision or in a state or even if one option is better than the other. You know that the difference is not something you can be sure of, whereas when you go to the intuitive route, you ’ ll end up with all, you know, with overconfident certainty and feeling good about yourself. So it ’ s an easy choice. I think you you wrote about this topic at length and what some have called your magnum opus, thinking fast and slow. I ’ m wondering what you ’ ve **rethought** since you published that book. Well, um. You know, there were there were things I published in that book that were wrong. I mean, you know, literature I quoted that didn ’ t hold up. Now, the interesting thing about that is that I haven ’ t changed my mind about much of anything. But that is because changing your mind is really quite difficult. That is Dan Gilbert has a beautiful word equal that unbelieving and unbelieving things are very difficult. So I find it extremely hard to unbelieve aspects of Pozza thinking fast and slow, even though I know that my grounds for believing them are now much weaker than they were. But the more significant thing that I have begun to **rethink**. Is that thinking fast and slow, like most of the study of judgment and decision making is completely oblivious to the individual differences. And all my career I I made fun of anybody was studying individual differences. I say I ’ m interested in main effects. I ’ m interested in characterizing the human mind. But it turns out that when you go into detail, people, those studies that you have, it ’ s not that everybody is behaving like the average of the study. That ’ s simply false. There are different subgroups were doing different things. And and life turns out to be much more complicated than if you were just trying to explain the average. So the necessity for studying individual differences is, I think the most important thing that I have **rethought** and, you know, doesn ’ t have any implications for me because it ’ s too late for me to study individual differences. And I wouldn ’ t like doing it anyway. It ’ s not my style. But I think there is much more room for it than I thought when I was writing, thinking fast and slow. Another thing I wanted to ask you about is the choices you make about what problems and projects to work on. I ’ m not a good example for anybody. I really never had a plan. A more or less followed my nose and I did many things that I shouldn ’ t have done. I wasted a lot of time on projects that I shouldn ’ t have carried out, but. No, I ’ ve been lucky. Well, I think that that ’ s probably an encouraging message for a lot of us, that the idea is, is this an area where there is gold? And I ’ m going to look for it. I mean, that ’ s that ’ s an idea. And formulating a new question to an idea in my book, I ’ m going to use that this is an area where I think there might be gold and I want to look for it. Such a nice reframe. So, Danny, you mentioned in your new book, Noize, one of my favorite ideas when I read Noise was the idea of the inner crowd. And I wondered if you could explain that there have been two lines of research by Bulan Besner and by her Twigg. And asking people the same question. On two occasions or in two different friends of mine, and it turns out that when you ask the same question like an estimate of the number of airports, when you ask people the same question twice separated by some time, then they tend to give you a different answers. And the average of the answers is more accurate than each of them separately. Also, in the case that the first answer is more valid than the second, and it ’ s also the case, the longer you wait, the better the average is, the more information there is in the second judgment that you make. You know, what it indicates is clearly that what we come up with when we ask ourselves a question is we are sampling from our mind. We are not extracting the answers from our mind. We are sampling an answer from our mind. And there are many different ways that that sample could come out. And sampling twice, especially if you will make them independent sampling twice is going to be better than sampling once. This is this is one of the most practical, **unexpected** decision making and judgment perspectives that I ’ ve come across in the last few years, in part because it says, I don ’ t always need a second opinion if I can get better at forming my own second opinions. Oh, you know, I think as we say in that chapter sleepover, it is really very much the same thing that is sleepover. And just wait. And tomorrow you might think differently. So the advice is out there reinforcing it, maybe useful. Your collaboration with Amos Tversky is obviously legendary, there ’ s a whole Michael Lewis book about it. Is there a lesson that you took away from that collaboration that ’ s informed either how you choose your collaborators now or how you work with the people on your team? I think that one really important thing in is. It ’ s to be genuinely interested in what your collaborator is say, and, you know, I am quite competitive. I ’ m also quite competitive. We were not competitive when we worked together. The joy of love, of collaboration for me always was that. But that almost that was more with them than with almost anyone else. I would say something and he would understand it better than I had. And that ’ s the greatest joy of collaboration. But in my other collaborations, taking pleasure in the ideas of your collaborator seemed to be very useful, and I ’ ve been lucky that way. On that note, almost anyone who ’ s ever won a Nobel Prize has **complained** that it hurt their career. And I ’ ve wondered what the experience has been like for you or I mean, it hurts people ’ s career if they ’ re young. You know, I got mine when I was 68, and for me it was a net loss. Why does it get people in trouble if they get it earlier? Oh, you know, there are a variety of ways that this can happen in the first place. It ’ s very destructive. I mean, people start taking you more seriously than they did and hanging on your every word and a lot of nonsense like this. And if if you begin to take yourself too seriously, that ’ s not good if you take time away from your work. To to do what you are invited to do when you get a Nobel, which is a lot of talking and a lot of talking and think that you don ’ t know much about, that ’ s a loss. And then if it makes you self-conscious that everything that you have to do has to be important. That ’ s a loss. So there are many different ways, I think, in which getting a Nobel early is a bad idea is not the best gift. I was at a good age to get it because I had some years left in my career and it made many things much easier having a Nobel and it made the end of my career more productive, I think, and and happier than it would have been otherwise. Taken for granted as part of the TED Audio Collective, the show is hosted by Adam Grant and it ’ s produced by TED with Transmitted Media. Our team includes Colin Helmes, Credit, Dan O ’ Donnell, Constanza Gelada, Joanne DeLuna, Grace Rubenstein, Michelle Quint and Ben Chang and Anna Feeling. This episode was produced by Dan O ’ Donnell.

Text Sample 191: POS Dim 2 – 2022 – Score 39.00 – 2022/t003705.txt

Maria : I found this job opening for a marketing manager. And I thought, oh, that would be really fun, they were looking for someone who could do everything, like very strategic. So, I applied, got the interview, and they were actually really nice Adam : This is Maria. Or at least, that ’ s what we ’ re calling her. You know, to protect the innocent. Anyway, Maria was very excited about this new opportunity. From the outside, the job was everything she wanted. Maria : And she called me at the end of the day to offer me the job, so I quit my job, like two days later. Adam : A few weeks later, she started the new job eager to share her ideas. Maria : And as I sat down in my new desk, the designer who was sitting next to me, she ’ s like,”I tried to find you on LinkedIn to warn you not to take this job.”and I went,”Why?”and she told me,”These people are crazy!”Adam : Um, definitely not a good start. Maria : I just laughed, cause it was like, okay let ’ s see how bad it actually is cause I also wanted to give it a chance thinking maybe she ’ s just angry, she just doesn ’ t like it. But in this case, I should have listened to her. Adam : The next few weeks were a wild ride. Maria : So, we had a sales meeting. I was like okay, let ’ s try and look for a marketing way of getting customers in. And they were like”No, how about we just do an energy circle?”And I was like,”Sorry what?”They all knew what that meant, so they all stood up, and held hands And I ’ m there just looking around like,”What the hell ’ s happening?”And they just started like shaking, like,”Okay, shake the bad energy, shake for the universe,”and then the owner is like,”Okay universe, we need you to send us some sales.”And I ’ m just standing there like,”Okay”with that folder big full of strategies and ideas and campaign ideas Nothing. Adam : Was this a company or a cult? Maria : It was a bit cultish because at the Christmas party, the owner wrote a song for everyone, with everyone ’ s names on it and we had to sit there listening to him sing to us. And by early December we get a list on my desk, and it ’ s just a list of everyone ’ s names in an envelope. And then, on top of the list it says,”Please give whatever you ’ d like to contribute to show your appreciation for working here and your appreciation to the owner of the company.”And I was just so angry and I loudly went like”What? So, we have to buy him a present?”I put 50 cents. I suggested to buy everyone in the company a Christmas present out of the marketing budget, and they said"No, coming to job is present enough."Adam : So, the highly paid owner gets a gift, but the hard-working employees don ' t? Call poison control! We have to rescue some people from a toxic culture. **Organizational** culture has big consequences for success and **happiness** but we often overlook it, because it ' s hard to analyze. So what culture clues should you look for before you join an organization? And how do you shape the culture, once you ' re there? I ' m Adam Grant, and this is Work Life my podcast with the TED audio-collective. I ' m an **organizational** psychologist, I study how to make work not suck. In this show, I take you inside the minds of fascinating people, to help us **rethink** how we work, lead, and live. Today, culture at work : How to recognize it from the outside and strengthen it from the inside. Thanks to Morgan Stanley for sponsoring this episode. Jenny : When I was in junior high school one of the things that I used to do as type of surveys that I would give to my parents Well, actually not just my parents, anyone who came over and I would ask them to fill out a paper-and-pencil survey I just loved surveys, I know, **super** weird. I would ask all kinds of questions about how happy they were at work and what their work was like and I just thought that was fascinating. Adam : Jenny Chatman is an **organizational** behavior professor at Berkley, and she was clearly destined to become the queen of **organizational** culture. Jenny : I like being a queen, I continue to be an optimist about this, that there is a place for everyone and there are organizations and jobs that really fit with some people, but not others. Adam : And through her many years of research and countless surveys, Jenny has the clear view what **organizational** culture is. Jenny : Culture is the values and behavioral **norms** that one sees expressed within an organization and it has to be sort of a systematic pattern of **norms** and expectations that people have in a particular **setting** that they might not have in another **setting** and one thing that ' s interesting about **norms** is there ' s no rulebook to teach them to you, instead we learn them through social interaction. They ' re different from what ' s written in the corporate handbook, these are observed patterns of behavior and expectations that we pick up from interacting with colleagues within an organization. Adam : People often claim their cultures are unique, but when you study thousands of organizations, you can start to see underlying patterns. It all has to do with how we balance key priorities. Research reveals that there are two fundamental tensions in **organizational** culture : Results versus relationships, and rules versus risk. If you ignore one of those values altogether, you end up committing one of my 4 deadly sins of **organizational** culture : Toxicity, mediocracy, bureaucracy, and anarchy. The first sin of culture is toxicity, the deadliest sin of them all. New evidence on the great resignation shows that toxic culture is the biggest driver of turnover, more than **burnout**, more than low pay. Toxicity exists when a culture prioritizes results without relationships, getting things done at the cost of treating people right. The organization tolerates disrespect, abuse, exclusion, unethical decisions, and selfish cut-throat **actions**. If people don ' t get fired for those behaviors or worse yet, still get promoted, Houston, we have a problem. At the opposite end of that spectrum is a second sin : Mediocracy : Valuing relationships above results. There ' s no accountability, people are so worried about getting along that they end up forfeiting good work. In a mediocracy, even if you do a terrible job you can still get ahead as long as people like you. Before long, you end up with a Peter principle, where everyone is promoted to their level of incompetence and they get stuck there. The third sin is bureaucracy. That happens when a culture is all rules, no risks. New ideas are seen as threats to the status quo. People cling to process and resist creativity and change. They see questioning the way we ' ve always done things as blasphemy.

There ’ s red tape everywhere. If you want to use the bathroom, you have to fill out paperwork. And the fourth sin is anarchy : You have risks, but no rules. Anyone can do whatever they want, strategy and structure be damned. No one learns from the past or lands on the same page. It ’ s pure chaos. It ’ s bad enough when a culture commits one of these sins, but believe it or not, Maria ’ s jewelry company managed to be guilty of all 4 sins Maria : So, definitely insanely toxic. People were crying there **daily**. Every time I went to the toilet there was someone crying there. We used to laugh about that one of the cubicles, it was the crying cubicle. So you would never actually go there, in case someone needed it for crying. Adam : They were a mediocracy, too. And not just because they did energy circles instead of marketing, they had no system for getting results. Maria : They were a small company that grew into a bigger company and none of the top two bothered to learn how to manage a company, incompetence a whole way through, but just act confident. Adam : They had all sorts of constraints on risk-taking. Maria : Very bureaucratic, like, you couldn ’ t speak directly to the owner because it was very about the steps, like remember how high I am and remember how low you are. Adam : And despite the bureaucracy around the chain of command, in other places, they didn ’ t have enough rules. Most of Maria ’ s job was anarchy. Maria : And turns out I kind of regressed in my career because I ended up not doing anything that I wanted to do and more like a party **planner** and universe messenger person. Adam : I did not know that was a job. Maria : Oh, I didn ’ t either. Adam : Could you figure any of this out before you took the job? Maria : I ’ ve been thinking a lot about that cause while I was there, I was thinking how can I make sure that this doesn ’ t happen again? If I had spoken to someone from the team during the interview, I am sure that the general manager would have been there with us and other than going through LinkedIn and finding people who work there and then messaging them and start saying,”Hey, I have just been offered this position what do you think about the company so I ’ m not sure if that ’ s even something that I would have done? Adam : You ’ ve probably felt that hesitation too. But gathering information about a culture before you agree to join is exactly what culture queen Jenny Chatman recommends doing. Jenny : I think that ’ s a great way of helping people be kind of detectives about the culture that they ’ re interested in particularly people who are seeking jobs. You want to ask, you know, what what do people care about here? What are they talking about? What is behavior focusing on? How much agreement is there? Do people seem to be aligned on these issues? And finally, what are their non-negotiables? What do people get really rewarded for, or if they violate these **norms** or behaviors, what do they get really punished for? Adam : You want to interview the company, but not during your job interview. Wait until you get the offer. Maria : I ’ m definitely doing that for my next job because If I ’ m getting into a new company I want to know exactly what I ’ m getting into so, next time I will do more of a reverse interview. Adam : It ’ s not about the slogans on the wall or the values on the website, culture is revealed in the stories people tell. To gather meaningful culture stories, I have a few favorite questions for you to ask current and former employees. I posed them to some former students, and their answers told me a lot about their organizations cultures. The first question is : Tell me about something that happens here that wouldn ’ t elsewhere. Vivian : Every year, the class of new hires is in charge of organizing a senior team roast, where they basically spend 30 minutes, live, these days over Zoom, during our end-of-the-year strategy session roasting our senior team for things like the way they write emails, wearing pajamas during zoom calls, mixing up people ’ s names, basically anything embarrassing. And I think this is something truly unique about our culture, for a firm in the finance industry where humor isn ’ t often encouraged in the **workplace**. Adam : This is a firm that ’ s trying to avoid both toxicity and bureaucracy. By making fun of senior people, they signal that executives want human relationships and it ’ s okay to take risks. Also, it ’ s pretty rare that people wearing pajamas are into red tape. Romie : When an investment banking report came out about mistreatment of their first shares, I got an email from the chairman of our organization. We ’ d worked together once before but didn ’ t know each other very well and I was one of the only analysts he knew. He was horrified by the report. And wanted to see how I felt we were doing. And to make sure I didn ’ t feel anything remotely similar. He responded to my email by saying :”I hope and pray that no one at our firm would ever be treated like that. And even if one person felt they were treated badly, that that person would let us know and we could fix it immediately. If we ever failed to be people-focused, please let me know whether it ’ s about you or you ’ re aware of anyone else at the firm.”And then in all caps”NOTHING I CARE MORE ABOUT.”Adam : It sounds like the chairman is dedicated to fight toxicity, but culture isn ’ t about one leader ’ s behavior, it ’ s about how widely shared and intensely held the values are. So I want to know how committed others in power are to curbing mistreatment and what the consequences are. In healthy cultures, no level of individual excellence justifies undermining people. You ’ re not a high performer if you don ’ t elevate others. Which brings us to a second question : Tell me about a time when people didn ’ t walk the talk here. Gabriel : Our offices returned to in-person because in-person interactions are highly valued, according to the president of the firm. But the third most senior person at the firm is spending the entire winter in a ski town in Europe. Adam : This is a red flag. Research suggests that the worst stories about a culture are about senior leaders violating their own principles. They claim in-person relationships are valued, but apparently one of the top people is exempt from that value? Hypocrisy alert! You can also see signs of hypocrisy versus integrity by asking a third question : Tell me a story about who gets hired, promoted and fired around here. If you ’ re a detective, these stories are full of clues about what ’ s really valued Girl : An MD shared how he made MD in six years as opposed to the normal twelve. He told us how he works from 4:30 AM until 10 PM and he ’ s always available. He doesn ’ t expect it from everyone else, but he ’ s always grinding. Considering our firm prides itself on valuing mental health, it ’ s a bit demoralizing to hear that the way to advance quickly is to abandon those values. Adam : Here ’ s another **warning** signal : The company claims to value well-being, but do they really mean it? If you want to get promoted early, good luck not working 17-hour days Collecting stories can help you understand a culture from the outside and identify toxicity, mediocracy, bureaucracy, and anarchy before you join. But what if you ’ re already inside? How do you build and maintain a strong culture? More on that after the break. Okay, this is going to be a different kind of **ad**. I play a personal role in selecting the sponsors for this podcast, cause they all have interesting cultures of their own. Today, we ’ re going inside the **workplace** at Morgan Stanley Randy : We like to hike, we like to climb around in the mountains, we like to be like an hour and a half far from civilization fishing. My wife ’ s **super** active, and she just loves being out at the gym, running, doing all that kind of stuff. Adam : Meet Randy Norris Randy : I live in a town, just outside of Boise, Idaho, I ’ m a **financial** advisor with Morgan Stanley. I ’ m married to a wonderful partner for life, Lian. And I have two now adult kids. Adam : One day, Lian suffered a life-threatening medical emergency She suddenly started coughing up blood that was filling her lungs. Randy : She was admitted to the hospital, and she stayed in the hospital for the next 4 or 5 days, and did all kinds of tests, trying to figure out what was going on. The only thing that really came of it that we were told was that it shouldn ’ t happen again but it could happen at any time, and so basically, the advice was never be 10 or 15 minutes away from a hospital. Adam : Randy and Lian was scared and devasted at the thought of spending the rest of their lives tethered to a hospital. But then, Randy got a phone call from his insurance provider. Randy : She said :”I noticed you ’ ve been going to see a number of different doctors, are you getting the care you need?”Immediately, I just said :”No. No, we have something wrong and nobody has a good answer, and I ’ m really frustrated”And she said”Well, Morgan Stanley offers their employees the **service** called Second MD.”Dave : The purpose of that expert medical opinion **service** is really to provide a different point of view that can help avoid surgeries, potentially help save unnecessary doctors ’ visits, maybe identify an alternative diagnosis or treatment that hadn ’ t been considered, especially when there ’ s a major decision to be made Adam : This is doctor Dave Stark, Morgan Stanley ’ s chief medical officer. Dave : Fundamentally, the job of a chief medical officer or frankly the job of an HR benefits professional is to develop and manage a portfolio of products and **services** to help employees and their families stay healthy, happy, and productive Adam : That ’ s what Second MD did for Randy and his family. After being connected to specialists across the country, Lian traveled to Oregon, where she was diagnosed with a micro vascular disfunction and underwent a successful surgery. Randy : After about 3 months of just not being able to figure out what was going on, Second MD came in, and within 3 weeks, we have a plan of **action** that was just guided and stepped all the way through I ’ ve been a raving fan of Second MD ever since. Adam : Research shows that employees ’ commitment and loyalty depends heavily on feeling cared about and supported by their employer. Morgan Stanley has worked to meet employees where they are, with **services** they need, in person or virtually, from child and elder to **special** needs care. As for Randy, he and his wife Lian are counting their blessings Randy : On the anniversary of her surgery, she ran her first half marathon. Life is precious, life is short. We should enjoy it, and take a moment then, but every now and then just reflect on the blessings we have and the miracles that happened that get us where we ’ re at. Adam : Morgan Stanley believes employees are their greatest asset and they protect that asset by offering top quality healthcare options and comprehensive programs to support physical and emotional well-being. Learning more at morganstanley. com / our benefits Adam : The first person who shapes the culture of an organization is the founder. Annie : Our founder, Barclay Simpson, had his 9 principles of doing business and it ’ s something that we refer to all the time. And one of them is everybody matters. Adam : Meet Annie Kao. She ’ s the VP of Engineering at Simpson Manufacture. They make anchors and fasteners for building foundations and decks. Annie : We are a manufacture and engineering company of building connections that help people build and design safer, stronger structures. Adam : And Simpson has been doing that for a long time. Annie : Just celebrated our 65th birthday. Adam : I thought it was really intriguing that you said,”We < celebrated our < 65th birthday.”Annie : Uh-huh. Adam : A lot of people when they talk about other company, they wouldn ’ t know what their company ’ s birthday was. What ’ s behind that? Annie : God, you know, I didn ’ t even pick up on that, but I think for me, like, the people are Simpson, I feel like I ’ m a very active owner of who Simpson is and what Simpson is and so, you know, the successes that we had as a company both on the **financial** side, on the products that we ’ re able to release, like, I have very personal pride associated with that because I just, I know the people and the work that it took to deliver it. Adam : That identification with the company, that pride of **ownership**, is a sign that Simpson has an unusually strong culture. Annie : It ’ s very personal because I think that what a company does is a clear reflection on, you know, who I am as, you know, as a person, as an engineer, as a parent. Adam : Strong culture is one of Jenny Chatman ’ s specialties as a **researcher**. You can tell how strong a culture is by paying close attention to what she calls crystallization and intensity. Jenny : There ’ s a question about how much people agree about the culture, that ’ s the crystallization piece. Does everyone in the organization agree that innovation is important? Or is there fragmentation where our engineers want to be on the cutting edge of things but our marketing folks want to hold back and just provide what customers are asking for? And there ’ s a question of intensity, which is what are the things that we ’ re... that are absolute non-negotiable for us in the organization. Adam : Simpson doesn ’ t just have a strong culture. Jenny has **observed** it up close as an unusually healthy one. Jenny : So, full disclosure, I ’ m on the board of Simpson Manufacturing, even before consultants were billing hours for cultural transformation, Barc was building a strong culture and he had a number of principles, one was that he really supported his people within the organization and expected to do most of their promotions from within the company. So, there was a real investment in developing people Adam : An investment that often took people by surprise, including Annie. Annie : Barclay Simpson, our founder, used to attend all of the orientation classes that we held at our corporate office. So, everyone who starts at Simpson attends this week-long orientation at our home office and Barc would attend all of those, and he would have 30 minutes, he ’ d go round, ask everyone ’ s name, ask about their families, and talk a little bit about the history and he ’ d just going to sit down at one of the, you know, conference chair along with us, and then I think of other leaders that have guided our company, and just the common factor is how, kind of, openly caring they were, how much they make time for employees. It ’ s something that I have really tried to do, just within my department with engineering is, like, I need to be that, you know, for the employee and so just making sure that, you know, I ’ m owning that and also expecting that of my team because that ’ s the kind of culture that we want, and we have to keep it alive and just not assume that someone will is doing it. Adam : It ’ s often said that culture eats strategy for breakfast. But the reality is, that a strong culture can serve your strategy. Jenny : Yeah, I do think there is something to the motto in that if your culture is not aligned with your strategy, good luck, it ’ s probably not going to be executed very well, or if at all. But I also think that strategy is what gives organization its purpose. Without a strategic objective, the question is sort of why organize, or why be an organization. You ’ re coming together for a purpose, you ’ re trying to accomplish something together, the culture should be the kind of engine that allows you to execute on that strategy more or less effectively. Adam : So if you want to build a strong culture, you need to identify the core values that you ’ re trying to crystalize But crystallization alone is not enough. You ’ ve probably come across a **workplace** where everyone agrees on the values but no one really upholds them. Jenny : Think of your US postal **service**, where, you know, who knew that their motto was”Customers first,”right? Like who knew? I think they have the coffee mugs, they have the banners, but, is anyone willing to, sort of, stay past 5pm to ensure that everybody on the line is served? Probably not. And in fact, it ’ s the most common case in organizations to have high crystallization and low intensity, because typically leaders are asking people to agree with pretty good stuff like quality, or customers, sure, I like customers, you know. The real question, though, is are people punished for a failure to uphold the **norms** and are they willing to sanction one another? Adam : That ’ s where intensity comes in. Jenny : There was this case at Nordstrom that is known as a strong culture organization. An experience I had where a shoe salesperson was helpful, but not overly helpful. And another salesperson came over, which I happened to overhear cause, you know, he was dropping is tax deductible for me, right? I happened to overhear this conversation between the two, and the second salesperson was actually admonishing the first one for not going above and beyond in helping me. So that ' s the sign of real intensity. Are people willing to take an interpersonal risk in order to uphold those **norms**? Adam : That might sound a little, well, intense. But it ' s the point of a strong culture : People enforce it even when the boss isn ' t watching. At Simpson, the crystallization and intensity around the principles is a big reason why people regularly stay for the long haul sometimes decades. Annie : I ' ve just celebrated my 15-year anniversary with Simpson. Adam : Is that a thing? An anniversary with a company? Annie : YES! And so, we have this really cool online network, where we can actually go and celebrate our coworkers ' anniversary and so you can post messages and, like, kind of, memories. I was just happen to be the lucky recipient of like, I think 50 messages from people that I worked with congratulating me, which was pretty emotional and overwhelming. Adam : The anniversaries are a result of that strong culture, and the messages also help to maintain that culture. They reinforce the principle that everybody matters. Annie : And I think part of it, too, is everyone, like, wants to be a part of this, like, winning team, if you will, that the work that you do matters, and people who enjoy their work and bring that passion with them to their work just give so much more of themselves. So I feel like it ' s translated itself financially for us. We passed the 1 billion dollars mark, I saw that there was some analysts who are predicting that we would be, you know, maybe, 2 billion by the end of this year. And it ' s like, oh, okay. Like, we really see that the hard work and the culture that we build, actually, translates into **financial** success, which is a kind of self-fulling prophecy. Adam : Yeah, well, it ' s interesting what you ' ve just described dovetails very nicely with what we see in the research on strong cultures, which is you know, one : talented people are more attracted to them because there ' s a clear signal : this is who we are, this is what we stand for. And that differentiate you in the marketplace. Two : Heightened motivation, over and over again. Like, I ' m not just doing a job, I ' m advancing a **mission**. And three : Those two things obviously feed into retention nicely, and they do that in part through a sense of belonging. When the culture is strong, like, everybody shares the same values, and also, is **passionate** about the same values, it ' s me, and I can ' t imagine working anywhere else because how am I going to find that again? Annie : Yes, yes. Adam : So how do you create that kind of commitment and build a strong **organizational** culture? In her research, Jenny compared the effects of selection vs. socialization How much of culture is who you ' d let on the bus? Versus how you drive it? Jenny : It turns out that people are, within some parameters, pretty flexible in terms of how much they can grow to adapt and fit with, or even appreciate the culture that they ' re a part of. Adam : Wow, I think you just said that socialization eats selection for breakfast Jenny : That ' s right! That ' s right, which is interesting because I think if you asked most leaders, they would say it ' s about who you hire, and, it turns out that people are actually more adaptable than we give them credit for. Adam : Jenny finds that when it comes to building a strong culture, socialization beats selection. The values and **norms** you set are more than twice as influential as who you hire So if you want to strengthen your culture, the first step is to bring it to life for people, especially new hires. Culture isn ' t just communicated through the stories we tell, it ' s created through the stories we tell. You want to find and share stories about a time when the culture became real Jenny : Stories are vital. I like to advise leaders to see themselves as curators of key stories within an organization You need to tell the story, but also give the moral of the story.

And so, one very well-known example is Southwest Airlines. They were running out of money early in their operations. They were using 4 jets at the time. And they couldn ' t make their payrolls so they had to sell one of the jets to pay employees that month. And they went to employees and said : We ' re going to well one of our jets but we want to run the same route with 3 jets instead of 4, how can we do that? And employees came back with the idea of turning the planes more quickly than they were before. So Southwest now continues to be known for its turn time, the amount of time it takes to come back in the gate, unload passengers and bags, you know, clean the plane, reload passengers and bags and take off. And that ' s part of Southwest ' s, sort of, deep strategy. So that story got told, not just for the substance that turn time is important, but for the relationship between leaders and employees, and who actually came up with the idea. And everyone at Southwest knows that story and it absolutely maps on to the culture that emphasizes both urgency and speed, but also a deep investment and mutual respect between employees and leaders in the organization Adam : Research reveals that the most powerful stories are about people living your values. Once you ' ve identified your best cultural stories, the second step is to reward and promote the protagonist in them, celebrate people who exemplify your values. Jenny : You would definitely want to focus on the formal and informal reward systems. Formally, if you have control over the compensation and you can reward that **financial** aid, that gets people ' s attention very quickly. But then there are a whole host of informal rewards, you know, whether that ' s gift certificates for lunch or a kick celebration for someone who had a small win that ' s consistent with the desired new culture. These are ways of really capturing people ' s attention. Adam : Wait a minute. If I heard you correctly, you ' re saying I shouldn ' t just pay people for their performance, I should pay them for their contributions to the culture? Jenny : Well yeah! Because, if... in my way of thinking about culture, the culture is your path to strategy execution, then culture is almost a proxy for top performance Adam : Wow! This is... It ' s such a big step for so many organizations that are used to... I would say most of the organizations I ' ve worked with over time were they ' ve gotten good at measuring individual results and they know how to figure out if you are a superstar, they have no idea what your impact on the culture is. Jenny : That ' s right. And, then the final lever, of course, is the leader ' s own behavior. Right, she needs to absolutely emulate the culture that she ' s trying to cultivate and there ' s no substitute for that because people are looking at leaders for an understanding of what ' s important. Adam : This is the third step. Leaders have to show up and model the values every day. That ' s a principle Annie learned from Barc Simpson, the company ' s founder and the CEOs who followed him. Annie : I just do office hours. We ' ve been trying to build that in, and just say, like, our schedules are so busy, everyone never has time, so, I ' m going to actually make that time. I ' m going to be available for 2 hours on Monday morning, right after we ' ve done our earnings call, so, if you have questions about why we decided to do this, why we didn ' t move that project forward, what the heck was Karen talking about the **financials**, I have made the time for that. Adam : So, Annie, I haven ' t asked you yet about the dark side of strong cultures but, I want, because we see empirically that despite all the upsides, the challenges that strong culture runs into has to do with group thinking, homogeneity, and becoming, sort of, stuck in the way we ' ve always done things as opposed to adapting and evolving Annie : Yeah. Adam : Have you seen that? How have you **navigated** it? Annie : Yeah, we... It ' s so funny you ' re asking this question because we ' ve just talking about this as an engineering team, as how do we make sure we have, you know, a diverse set of perspectives and opinions so that we don ' t fall into groupthink and, that like, well, this is how we ' ve always done it. Let ' s just not assume that we ' re not doing, you know, groupthink. Like, how are we making sure to put together diverse teams. Right? Like, what questions are we asking as we start a project, or as we put people on projects to make sure that it has the right, the right mix of people, and I think we ask ourselves that question too when we hire, like, who can, who is the right new person on the team to challenge what we ' re doing, and how do we celebrate being challenged what we ' re doing, so that we are kind of... We ' re looking for it, and that it ' s not something that, you know, catches us by surprise. Adam : This is the paradox of strong cultures. If you want them to stay strong, flexibility needs to be one of your core values. Cultures are like buildings. Without proper maintenance, they fall apart. A culture needs regular **service**, and sometimes a full-scale renovation. So how do you know what kind of maintenance your culture needs? You do a culture audit. Jenny : A culture audit is actually pretty straightforward. You can start by saying : Here are our current **norms**, and then you could say : If we were fully executing on our strategy, what would our **norms** be? And then, you can identify the gaps, and figure out a bunch of behaviors that can fix that, using these levers : who would we hire, how would we socialize, orient, and train people, what stories would we tell, who would we promote, how would we reward people, you know, what would leaders be doing? At that point, you can get to real behavioral change that could be of value. Adam : Yeah, and one of the things I love about the idea of a culture audit is I can go and get responsibility for that, even if I ' m not in charge, right? And in some ways, I ' m taking a task that nobody has time for, or might not already be in anybody ' s job description and I ' m saying :"Hey, leaders, you can ' t be everywhere, you can ' t see everything, can I help you figure out what ' s working in the culture and what needs to be improved? And if I can get that project, then I have a chance to then shape the discovery of data that can convince leaders that we need to some adjustments. Jenny : And then, the next piece of that would be that you make the culture changes part of the work that people are doing anyway. Adam : At Simpson, they have their version of a culture audit. Annie : It ' s a culture and leadership survey. And so the first time we did this, there was feedback that people felt that they couldn ' t really say something if they saw something that was against our values or against our policies. And so, as a result, we launched a program specifically to address that, called"Speak-up, listen-up". We opened an anonymous phone tip line, we had an anonymous form you could fill online, all managers took training, we kind of put our money where our mouth is. Adam : Wow! It sounds like an example of protecting the culture. Annie : Yeah, I would say everyone in our company is responsible for protecting the culture and what it means for me is just... is going back and making sure that we are a values-driven organization. And so I ' m going to do whatever I can, as a leader in our organization, to make sure that that is pervasive through every single role, every single level, because that ' s not something that we are willing to go halfway on. And if we do, then that ' s not the company that I want to work for. Adam : The ultimate test of a strong culture is : Can it adapt? Building a great **workplace** isn ' t just about expecting people to adapt to the culture, it ' s also about adapting the culture to the people and the world. As the world evolves, your culture needs to evolve with it. Next time, wrapping up Season 5 of WorkLife : Dan Heath : People have this sort of skepticism about change and everything you hear about change is pessimistic. I mean everything in our lives is change. For some reason, we code the really difficult things as change and everything else is just like a choice. Adam : **Organizational** change from the bottom up and the top down. WorkLife is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Colin Helms, Gretta Cohn, Dan

O ’ Donnell, JoAnn DeLuna, Grace Rubenstein, Michelle Quint, Banban Cheng and Anna Phelan. This episode was produced by Constanza Gallardo. Our show is mixed by Ben Chesneau. Our fact checker is Paul Durbin. Original music by Hansdale Hsu and Allison Leyton-Brown. **Ad** stories produced by **Pineapple** Street Studios. **Special** thanks to our sponsors : LinkedIn, Morgan Stanley, ServiceNow, and UKG. For their studies of **organizational** culture, gratitude to the following **researchers** and their colleagues : Charles O ’ Reilly, Chad Hartnell, Shalom Schwartz, Joanne Martin, Sean Martin, Alan Benson, Donald Sull, Constantinos Coutifaris, and my beloved late colleague Sigal Barsade. Maria : The people that were there when I was there, are still there, and this was like two years ago. I sat there with them, trying to figure out how we could start their own businesses, cause I was like, you need to get out of here. Adam : For toxic cultures, there should be some kind of extraction team that you can send in, Maria : There should be! Adam : Like a SWAT team that ’ s going to go in and rescue everyone. Maria : That ’ s a good business idea. Maybe I recruit a company who ’ d specialize in saving teams that are stuck in a shitty culture. Adam : I think that is a business waiting to be launched.

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Chris Anderson : Elon Musk, great to see you. How are you? Elon Musk : Good. How are you? CA : We ’ re here at the Texas Gigafactory the day before this thing opens. It ’ s been pretty crazy out there. Thank you so much for making time on a busy day. I would love you to help us, kind of, cast our minds, I don ’ t know, 10, 20, 30 years into the future. And help us try to picture what it would take to build a future that ’ s worth getting excited about. The last time you spoke at TED, you said that that was really just a big driver. You know, you can talk about lots of other reasons to do the work you ’ re doing, but fundamentally, you want to think about the future and not think that it sucks. EM : Yeah, absolutely. I think in general, you know, there ’ s a lot of discussion of like, this problem or that problem. And a lot of people are sad about the future and they ’ re... Pessimistic. And I think... this is... This is not great. I mean, we really want to wake up in the morning and look forward to the future. We want to be excited about what ’ s going to happen. And life cannot simply be about sort of, solving one miserable problem after another. CA : So if you look forward 30 years, you know, the year 2050 has been labeled by scientists as this, kind of, almost like this doomsday **deadline** on climate. There ’ s a consensus of scientists, a large consensus of scientists, who believe that if we haven ’ t completely eliminated greenhouse gases or offset them completely by 2050, effectively we ’ re inviting climate catastrophe. Do you believe there is a pathway to avoid that catastrophe? And what would it look like? EM : Yeah, so I am not one of the doomsday people, which may surprise you. I actually think we ’ re on a good path. But at the same time, I want to caution against complacency. So, so long as we are not complacent, as long as we have a high sense of urgency about moving towards a sustainable energy economy, then I think things will be fine. So I can ’ t emphasize that enough, as long as we push hard and are not complacent, the future is going to be great. Don ’ t worry about it. I mean, worry about it, but if you worry about it, ironically, it will be a self-unfulfilling prophecy. So, like, there are three elements to a sustainable energy future. One is of sustainable energy generation, which is primarily wind and solar. There ’ s also hydro, geothermal, I ’ m actually pro-nuclear. I think nuclear is fine. But it ’ s going to be primarily solar and wind, as the primary generators of energy. The second part is you need batteries to store the solar and wind energy because the sun doesn ' t shine all the time, the wind doesn ' t blow all the time. So it ' s a lot of stationary battery packs. And then you need electric transport. So electric cars, electric planes, boats. And then ultimately, it ' s not really possible to make electric rockets, but you can make the propellant used in rockets using sustainable energy. So ultimately, we can have a fully sustainable energy economy. And it ' s those three things : solar / wind, stationary battery pack, electric vehicles. So then what are the limiting factors on progress? The limiting factor really will be battery cell production. So that ' s going to really be the fundamental rate driver. And then whatever the slowest element of the whole lithium-ion battery cells supply chain, from mining and the many steps of refining to ultimately creating a battery cell and putting it into a pack, that will be the limiting factor on progress towards sustainability. CA : All right, so we need to talk more about batteries, because the key thing that I want to understand, like, there seems to be a scaling issue here that is kind of amazing and alarming. You have said that you have calculated that the amount of battery production that the world needs for sustainability is 300 terawatt hours of batteries. That ' s the end goal? EM : Very rough numbers, and I certainly would invite others to check our calculations because they may arrive at different conclusions. But in order to transition, not just current electricity production, but also heating and transport, which roughly triples the amount of electricity that you need, it amounts to approximately 300 terawatt hours of installed capacity. CA : So we need to give people a sense of how big a task that is. I mean, here we are at the Gigafactory. You know, this is one of the biggest buildings in the world. What I ' ve read, and tell me if this is still right, is that the goal here is to eventually produce 100 gigawatt hours of batteries here a year eventually. EM : We will probably do more than that, but yes, hopefully we get there within a couple of years. CA : Right. But I mean, that is one – EM : 0. 1 terrawat hours. CA : But that ' s still 1 / 100 of what ' s needed. How much of the rest of that 100 is Tesla planning to take on let ' s say, between now and 2030, 2040, when we really need to see the scale up happen? EM : I mean, these are just guesses. So please, people shouldn ' t hold me to these things. It ' s not like this is like some – What tends to happen is I ' ll make some like, you know, best guess and then people, in five years, there ' ll be some jerk that writes an article :"Elon said this would happen, and it didn ' t happen. He ' s a liar and a **fool**."It ' s very annoying when that happens. So these are just guesses, this is a conversation. CA : Right. EM : I think Tesla probably ends up doing 10 percent of that. Roughly. CA : Let ' s say 2050 we have this amazing, you know, 100 percent sustainable electric grid made up of, you know, some mixture of the sustainable energy sources you talked about. That same grid probably is offering the world really low-cost energy, isn ' t it, compared with now. And I ' m curious about like, are people entitled to get a little bit excited about the possibilities of that world? EM : People should be optimistic about the future. Humanity will solve sustainable energy. It will happen if we, you know, continue to push hard, the future is bright and good from an energy standpoint. And then it will be possible to also use that energy to do carbon sequestration. It takes a lot of energy to pull carbon out of the atmosphere because in putting it in the atmosphere it releases energy. So now, you know, obviously in order to pull it out, you need to use a lot of energy. But if you ' ve got a lot of sustainable energy from wind and solar, you can actually sequester carbon. So you can reverse the CO2 parts per million of the atmosphere and oceans. And also you can really have as much fresh water as you want. Earth is mostly water. We should call Earth"Water."It ' s 70 percent water by surface area. Now most of that ' s seawater, but it ' s like we just happen to be on the bit that ' s land. CA : And with energy, you can turn seawater into – EM

: Yes. CA : Irrigating water or whatever water you need. EM : At very low cost. Things will be good. CA : Things will be good. And also, there ' s other benefits to this non-fossil fuel world where the air is cleaner – EM : Yes, exactly. Because, like, when you burn fossil fuels, there ' s all these side reactions and toxic gases of various kinds. And sort of little particulates that are bad for your lungs. Like, there ' s all sorts of bad things that are happening that will go away. And the sky will be cleaner and quieter. The future ' s going to be good. CA : I want us to switch now to think a bit about artificial intelligence. But the segue there, you mentioned how annoying it is when people call you up for bad predictions in the past. So I ' m possibly going to be annoying now, but I ' m curious about your timelines and how you predict and how come some things are so amazingly on the money and some aren ' t. So when it comes to predicting sales of Tesla vehicles, for example, you ' ve kind of been amazing, I think in 2014 when Tesla had sold that year 60, 000 cars, you said,"2020, I think we will do half a million a year."EM : Yeah, we did almost exactly a half million. CA : You did almost exactly half a million. You were scoffed in 2014 because no one since Henry Ford, with the Model T, had come close to that kind of growth rate for cars. You were scoffed, and you actually hit 500, 000 cars and then 510, 000 or whatever produced. But five years ago, last time you came to TED, I asked you about full self-driving, and you said,"Yeah, this very year, I ' m confident that we will have a car going from LA to New York without any intervention."EM : Yeah, I don ' t want to blow your mind, but I ' m not always right. CA : What ' s the difference between those two? Why has full self-driving in particular been so hard to predict? EM : I mean, the thing that really got me, and I think it ' s going to get a lot of other people, is that there are just so many false dawns with self-driving, where you think you ' ve got the problem, have a handle on the problem, and then it, no, turns out you just hit a ceiling. Because if you were to plot the progress, the progress looks like a log curve. So it ' s like a series of log curves. So most people don ' t know what a log curve is, I suppose. CA : Show the shape with your hands. EM : It goes up you know, sort of a fairly straight way, and then it starts tailing off and you start getting diminishing returns. And you ' re like, uh oh, it was trending up and now it ' s sort of, curving over and you start getting to these, what I call local maxima, where you don ' t realize basically how dumb you were. And then it happens again. And ultimately... These things, you know, in retrospect, they seem obvious, but in order to solve full self-driving properly, you actually have to solve real-world AI. Because what are the road networks designed to work with? They ' re designed to work with a biological neural net, our brains, and with vision, our eyes. And so in order to make it work with computers, you basically need to solve real-world AI and vision. Because we need cameras and **silicon** neural nets in order to have self-driving work for a system that was designed for eyes and biological neural nets. You know, I guess when you put it that way, it ' s sort of, like, quite obvious that the only way to solve full self-driving is to solve real world AI and sophisticated vision. CA : What do you feel about the current architecture? Do you think you have an architecture now where there is a chance for the logarithmic curve not to tail off any anytime soon? EM : Well I mean, admittedly these may be infamous last words, but I actually am confident that we will solve it this year. That we will exceed – The probability of an accident, at what point do you exceed that of the average person? I think we will exceed that this year. CA : What are you seeing behind the scenes that gives you that confidence? EM : We ' re almost at the point where we have a high-quality unified vector space. In the beginning, we were trying to do this with image recognition on individual images. But if you get one image out of a video, it ' s actually quite hard to see what ' s going on without ambiguity. But if you look at a video segment of a few seconds of video, that ambiguity resolves. So the first thing we had to do is tie all eight cameras together so they ' re synchronized, so that all the frames are looked at simultaneously and labeled simultaneously by one person, because we still need human labeling. So at least they ' re not labeled at different times by different people in different ways. So it ' s sort of a surround picture. Then a very important part is to add the time dimension. So that you ' re looking at surround video, and you ' re labeling surround video. And this is actually quite difficult to do from a software standpoint. We had to write our own labeling tools and then create auto labeling, create auto labeling software to amplify the efficiency of human labelers because it ' s quite hard to label. In the beginning, it was taking several hours to label a 10-second video clip. This is not scalable. So basically what you have to have is you have to have surround video, and that surround video has to be primarily automatically labeled with humans just being editors and making slight corrections to the labeling of the video and then feeding back those corrections into the future auto labeler, so you get this flywheel eventually where the auto labeler is able to take in vast amounts of video and with high accuracy, automatically label the video for cars, lane lines, drive space. CA : What you ' re saying is... the result of this is that you ' re effectively giving the car a 3D model of the actual objects that are all around it. It knows what they are, and it knows how fast they are moving. And the remaining task is to predict what the quirky behaviors are that, you know, that when a pedestrian is walking down the road with a smaller pedestrian, that maybe that smaller pedestrian might do something unpredictable or things like that. You have to build into it before you can really call it safe. EM : You basically need to have memory across time and space. So what I mean by that is... Memory can ' t be infinite, because it ' s using up a lot of the computer ' s RAM basically. So you have to say how much are you going to try to remember? It ' s very common for things to be occluded. So if you talk about say, a pedestrian walking past a truck where you saw the pedestrian start on one side of the truck, then they ' re occluded by the truck. You would know intuitively, OK, that pedestrian is going to pop out the other side, most likely. CA : A computer doesn ' t know it. EM : You need to slow down. CA : A skeptic is going to say that every year for the last five years, you ' ve kind of said, well, no this is the year, we ' re confident that it will be there in a year or two or, you know, like it ' s always been about that far away. But we ' ve got a new architecture now, you ' re seeing enough improvement behind the scenes to make you not certain, but pretty confident, that, by the end of this year, what in most, not in every city, and every circumstance but in many cities and circumstances, basically the car will be able to drive without interventions safer than a human. EM : Yes. I mean, the car currently drives me around Austin most of the time with no interventions. So it ' s not like... And we have over 100, 000 people in our full self-driving beta program. So you can look at the videos that they post online. CA : I do. And some of them are great, and some of them are a little terrifying. I mean, occasionally the car seems to veer off and scare the hell out of people. EM : It ' s still a beta. CA : But you ' re behind the scenes, looking at the data, you ' re seeing enough improvement to believe that a this-year timeline is real. EM : Yes, that ' s what it seems like. I mean, we could be here talking again in a year, like, well, another year went by, and it didn ' t happen. But I think this is the year. CA : And so in general, when people talk about Elon time, I mean it sounds like you can ' t just have a general rule that if you predict that something will be done in six months, actually what we should imagine is it ' s going to be a year or it ' s like two-x or three-x, it depends on the type of prediction. Some things, I guess, things involving software, AI, whatever, are fundamentally harder to predict than others. Is there an element that you actually deliberately make aggressive prediction timelines to drive people to be ambitious? Without that, nothing gets done? EM : Well, I generally believe, in terms of internal timelines, that we want to set the most aggressive timeline that we can. Because there ' s sort of like a law of gaseous expansion where, for schedules, where whatever time you set, it ' s not going to be less than that. It ' s very rare that it ' ll be less than that. But as far as our predictions are concerned, what tends to happen in the media is that they will report all the wrong ones and ignore all the right ones. Or, you know, when writing an article about me – I ' ve had a long career in multiple industries. If you list my sins, I sound like the worst person on Earth. But if you put those against the things I ' ve done right, it makes much more sense, you know? So essentially like, the longer you do anything, the more mistakes that you will make cumulatively. Which, if you sum up those mistakes, will sound like I ' m the worst predictor ever. But for example, for Tesla vehicle growth, I said I think we ' d do 50 percent, and we ' ve done 80 percent. CA : Yes. EM : But they don ' t mention that one. So, I mean, I ' m not sure what my exact track record is on predictions. They ' re more optimistic than pessimistic, but they ' re not all optimistic. Some of them are exceeded probably more or later, but they do come true. It ' s very rare that they do not come true. It ' s sort of like, you know, if there ' s some radical technology prediction, the point is not that it was a few years late, but that it happened at all. That ' s the more important part. CA : So it feels like at some point in the last year, seeing the progress on understanding, the Tesla AI understanding the world around it, led to a kind of, an aha moment at Tesla. Because you really surprised people recently when you said probably the most important product development going on at Tesla this year is this robot, Optimus. EM : Yes. CA : Many companies out there have tried to put out these robots, they ' ve been working on them for years. And so far no one has really cracked it. There ' s no mass adoption robot in people ' s homes. There are some in manufacturing, but I would say, no one ' s kind of, really cracked it. Is it something that happened in the development of full self-driving that gave you the confidence to say,"You know what, we could do something **special** here."EM : Yeah, exactly. So, you know, it took me a while to sort of realize that in order to solve self-driving, you really needed to solve real-world AI. And at the point of which you solve real-world AI for a car, which is really a robot on four wheels, you can then generalize that to a robot on legs as well. The two hard parts I think – like obviously companies like Boston Dynamics have shown that it ' s possible to make quite compelling, sometimes alarming robots. CA : Right. EM : You know, so from a sensors and actuators standpoint, it ' s certainly been demonstrated by many that it ' s possible to make a humanoid robot. The things that are currently missing are enough intelligence for the robot to **navigate** the real world and do useful things without being explicitly instructed. So the missing things are basically real-world intelligence and scaling up manufacturing. Those are two things that Tesla is very good at. And so then we basically just need to design the specialized actuators and sensors that are needed for humanoid robot. People have no idea, this is going to be bigger than the car. CA : So let ' s dig into exactly that. I mean, in one way, it ' s actually an easier problem than full self-driving because instead of an object going along at 60 miles an hour, which if it gets it wrong, someone will die. This is an object that ' s engineered to only go at what, three or four or five miles an hour. And so a mistake, there aren ' t lives at stake. There might be embarrassment at stake. EM : So long as the AI doesn ' t take it over and murder us in our sleep or something. CA : Right. So talk about – I think the first applications you ' ve mentioned are probably going to be manufacturing, but eventually the vision is to have these available for people at home. If you had a robot that really understood the 3D architecture of your house and knew where every object in that house was or was supposed to be, and could recognize all those objects, I mean, that ’ s kind of amazing, isn ’ t it? Like the kind of thing that you could ask a robot to do would be what? Like, tidy up? EM : Yeah, absolutely. Make dinner, I guess, mow the lawn. CA : Take a cup of tea to grandma and show her family pictures. EM : Exactly. Take care of my grandmother and make sure – CA : It could obviously recognize everyone in the home. It could play catch with your kids. EM : Yes. I mean, obviously, we need to be careful this doesn ’ t become a dystopian situation. I think one of the things that ’ s going to be important is to have a localized ROM chip on the robot that cannot be updated over the air. Where if you, for example, were to say,”Stop, stop, stop,”if anyone said that, then the robot would stop, you know, type of thing. And that ’ s not updatable remotely. I think it ’ s going to be important to have safety features like that. CA : Yeah, that sounds wise. EM : And I do think there should be a regulatory agency for AI. I ’ ve said that for many years. I don ’ t love being regulated, but I think this is an important thing for public safety. CA : Let ’ s come back to that. But I don ’ t think many people have really sort of taken seriously the notion of, you know, a robot at home. I mean, at the start of the computing revolution, Bill Gates said there ’ s going to be a computer in every home. And people at the time said, yeah, whatever, who would even want that. Do you think there will be basically like in, say, 2050 or whatever, like a robot in most homes, is what there will be, and people will love them and count on them? You ’ ll have your own butler basically. EM : Yeah, you ’ ll have your sort of buddy robot probably, yeah. CA : I mean, how much of a buddy? How many applications have you thought, you know, can you have a romantic partner, a sex partner? EM : It ’ s probably inevitable. I mean, I did promise the internet that I ’ d make catgirls. We could make a robot catgirl. CA : Be careful what you promise the internet. EM : So, yeah, I guess it ’ ll be whatever people want really, you know. CA : What sort of timeline should we be thinking about of the first models that are actually made and sold? EM : Well, you know, the first units that we intend to make are for jobs that are dangerous, boring, repetitive, and things that people don ’ t want to do. And, you know, I think we ’ ll have like an interesting prototype sometime this year. We might have something useful next year, but I think quite likely within at least two years. And then we ’ ll see rapid growth year over year of the usefulness of the humanoid robots and decrease in cost and scaling up production. CA : Initially just selling to businesses, or when do you picture you ’ ll start selling them where you can buy your parents one for Christmas or something? EM : I ’ d say in less than ten years. CA : Help me on the economics of this. So what do you picture the cost of one of these being? EM : Well, I think the cost is actually not going to be crazy high. Like less than a car. Initially, things will be expensive because it ’ ll be a new technology at low production volume. The complexity and cost of a car is greater than that of a humanoid robot. So I would expect that it ’ s going to be less than a car, or at least equivalent to a cheap car. CA : So even if it starts at 50k, within a few years, it ’ s down to 20k or lower or whatever. And maybe for home they ’ ll get much cheaper still. But think about the economics of this. If you can replace a $30, 000, $40, 000-a-year worker, which you have to pay every year, with a one-time payment of $25, 000 for a robot that can work longer hours, a pretty rapid replacement of certain types of jobs. How worried should the world be about that? EM : I wouldn ’ t worry about the sort of, putting people out of a job thing. I think we ’ re actually going to have, and already do have, a massive shortage of labor. So I think we will have... Not people out of work, but actually still a shortage labor even in the future. But this really will be a world of abundance. Any goods and **services** will be available to anyone who wants them. It ' ll be so cheap to have goods and **services**, it will be ridiculous. CA : I ' m presuming it should be possible to imagine a bunch of goods and **services** that can ' t profitably be made now but could be made in that world, courtesy of legions of robots. EM : Yeah. It will be a world of abundance. The only scarcity that will exist in the future is that which we decide to create ourselves as humans. CA : OK. So AI is allowing us to imagine a differently powered economy that will create this abundance. What are you most worried about going wrong? EM : Well, like I said, AI and robotics will bring out what might be termed the age of abundance. Other people have used this word, and that this is my prediction : it will be an age of abundance for everyone. But I guess there ' s... The dangers would be the artificial general intelligence or digital superintelligence decouples from a collective human will and goes in the direction that for some reason we don ' t like. Whatever direction it might go. You know, that ' s sort of the idea behind Neuralink, is to try to more tightly couple collective human world to digital superintelligence. And also along the way solve a lot of brain injuries and spinal injuries and that kind of thing. So even if it doesn ' t succeed in the greater goal, I think it will succeed in the goal of alleviating brain and spine damage. CA : So the spirit there is that if we ' re going to make these AIs that are so vastly intelligent, we ought to be wired directly to them so that we ourselves can have those superpowers more directly. But that doesn ' t seem to avoid the risk that those superpowers might... turn ugly in unintended ways. EM : I think it ' s a risk, I agree. I ' m not saying that I have some certain answer to that risk. I ' m just saying like maybe one of the things that would be good for ensuring that the future is one that we want is to more tightly couple the collective human world to digital intelligence. The issue that we face here is that we are already a cyborg, if you think about it. The computers are an extension of ourselves. And when we die, we have, like, a digital ghost. You know, all of our text messages and social media, emails. And it ' s quite eerie actually, when someone dies but everything online is still there. But you say like, what ' s the limitation? What is it that inhibits a human-machine symbiosis? It ' s the data rate. When you communicate, especially with a phone, you ' re moving your thumbs very slowly. So you ' re like moving your two little meat sticks at a rate that ' s maybe 10 bits per second, optimistically, 100 bits per second. And computers are communicating at the gigabyte level and beyond. CA : Have you seen evidence that the technology is actually working, that you ' ve got a richer, sort of, higher bandwidth connection, if you like, between like external electronics and a brain than has been possible before? EM : Yeah. I mean, the fundamental principles of reading neurons, sort of doing read-write on neurons with tiny electrodes, have been demonstrated for decades. So it ' s not like the concept is new. The problem is that there is no product that works well that you can go and buy. So it ' s all sort of, in research labs. And it ' s like some cords sticking out of your head. And it ' s quite gruesome, and it ' s really... There ' s no good product that actually does a good job and is high-bandwidth and safe and something actually that you could buy and would want to buy. But the way to think of the Neuralink device is kind of like a Fitbit or an Apple Watch. That ' s where we take out sort of a small section of skull about the size of a quarter, replace that with what, in many ways really is very much like a Fitbit, Apple Watch or some kind of smart watch thing. But with tiny, tiny wires, very, very tiny wires. Wires so tiny, it ' s hard to even see them. And it ' s very important to have very tiny wires so that when they ' re implanted, they don ' t damage the brain. CA : How far are you from putting these into humans? EM : Well, we have put in our FDA application to aspirationally do the first human implant this year. CA : The first uses will be for neurological injuries of different kinds. But rolling the clock forward and imagining when people are actually using these for their own enhancement, let ' s say, and for the enhancement of the world, how clear are you in your mind as to what it will feel like to have one of these inside your head? EM : Well, I do want to emphasize we ' re at an early stage. And so it really will be many years before we have anything approximating a high-bandwidth neural interface that allows for AI-human symbiosis. For many years, we will just be solving brain injuries and spinal injuries. For probably a decade. This is not something that will suddenly one day it will have this incredible sort of whole brain interface. It ' s going to be, like I said, at least a decade of really just solving brain injuries and spinal injuries. And really, I think you can solve a very wide range of brain injuries, including severe depression, morbid obesity, sleep, potentially schizophrenia, like, a lot of things that cause great **stress** to people. Restoring memory in older people. CA : If you can pull that off, that ' s the app I will sign up for. EM : Absolutely. CA : Please hurry. EM : I mean, the emails that we get at Neuralink are heartbreaking. I mean, they ' ll send us just tragic, you know, where someone was sort of, in the prime of life and they had an accident on a motorcycle and someone who ' s 25, you know, can ' t even feed themselves. And this is something we could fix. CA : But you have said that AI is one of the things you ' re most worried about and that Neuralink may be one of the ways where we can keep abreast of it. EM : Yeah, there ' s the short-term thing, which I think is helpful on an individual human level with injuries. And then the long-term thing is an attempt to address the civilizational risk of AI by bringing digital intelligence and biological intelligence closer together. I mean, if you think of how the brain works today, there are really two layers to the brain. There ' s the limbic system and the cortex. You ' ve got the kind of, animal brain where – it ' s kind of like the fun part, really. CA : It ' s where most of Twitter operates, by the way. EM : I think Tim Urban said, we ' re like somebody, you know, stuck a computer on a monkey. You know, so we ' re like, if you gave a monkey a computer, that ' s our cortex. But we still have a lot of monkey instincts. Which we then try to rationalize as, no, it ' s not a monkey instinct. It ' s something more important than that. But it ' s often just really a monkey instinct. We ' re just monkeys with a computer stuck in our brain. But even though the cortex is sort of the smart, or the intelligent part of the brain, the thinking part of the brain, I ' ve not yet met anyone who wants to delete their limbic system or their cortex. They ' re quite happy having both. Everyone wants both parts of their brain. And people really want their phones and their computers, which are really the tertiary, the third part of your intelligence. It ' s just that it ' s... Like the bandwidth, the rate of communication with that tertiary layer is slow. And it ' s just a very tiny straw to this tertiary layer. And we want to make that tiny straw a big highway. And I ' m definitely not saying that this is going to solve everything. Or this is you know, it ' s the only thing – it ' s something that might be helpful. And worst-case scenario, I think we solve some important brain injury, spinal injury issues, and that ' s still a great outcome. CA : Best-case scenario, we may discover new human possibility, telepathy, you ' ve spoken of, in a way, a connection with a loved one, you know, full memory and much faster thought processing maybe. All these things. It ' s very cool. If AI were to take down Earth, we need a plan B. Let ' s shift our attention to space. We spoke last time at TED about reusability, and you had just demonstrated that spectacularly for the first time. Since then, you ' ve gone on to build this monster rocket, Starship, which kind of changes the rules of the game in spectacular ways. Tell us about Starship. EM : Starship is extremely fundamental. So the holy grail of rocketry or space transport is full and rapid reusability. This has never been achieved. The closest that anything has come is our Falcon 9 rocket, where we are able to recover the first stage, the boost stage, which is probably about 60 percent of the cost of the vehicle of the whole launch, maybe 70 percent. And we ' ve now done that over a hundred times. So with Starship, we will be recovering the entire thing. Or at least that ' s the goal. CA : Right. EM : And moreover, recovering it in such a way that it can be immediately re-flown. Whereas with Falcon 9, we still need to do some amount of refurbishment to the booster and to the fairing nose cone. But with Starship, the design goal is immediate re-flight. So you just refill propellants and go again. And this is gigantic. Just as it would be in any other mode of transport. CA : And the main design is to basically take 100 plus people at a time, plus a bunch of things that they need, to Mars. So, first of all, talk about that piece. What is your latest timeline? One, for the first time, a Starship goes to Mars, presumably without people, but just equipment. Two, with people. Three, there ' s sort of, OK, 100 people at a time, let ' s go. EM : Sure. And just to put the cost thing into perspective, the expected cost of Starship, putting 100 tons into orbit, is significantly less than what it would have cost or what it did cost to put our tiny Falcon 1 rocket into orbit. Just as the cost of flying a 747 around the world is less than the cost of a small airplane. You know, a small airplane that was thrown away. So it ' s really pretty mind-boggling that the giant thing costs less, way less than the small thing. So it doesn ' t use exotic propellants or things that are difficult to obtain on Mars. It uses methane as fuel, and it ' s primarily oxygen, roughly 77-78 percent oxygen by weight. And Mars has a CO_2 atmosphere and has water ice, which is CO_2 plus H_2O, so you can make CH_4, methane, and O_2, oxygen, on Mars. CA : Presumably, one of the first tasks on Mars will be to create a fuel plant that can create the fuel for the return trips of many Starships. EM : Yes. And actually, it ' s mostly going to be oxygen plants, because it ' s 78 percent oxygen, 22 percent fuel. But the fuel is a simple fuel that is easy to create on Mars. And in many other parts of the solar system. So basically... And it ' s all propulsive landing, no parachutes, nothing thrown away. It has a heat shield that ' s capable of entering on Earth or Mars. We can even potentially go to Venus. but you don ' t want to go there. Venus is hell, almost literally. But you could... It ' s a generalized method of transport to anywhere in the solar system, because the point at which you have propellant depo on Mars, you can then travel to the asteroid belt and to the moons of Jupiter and Saturn and ultimately anywhere in the solar system. CA : But your main focus and SpaceX ' s main focus is still Mars. That is the **mission**. That is where most of the effort will go? Or are you actually imagining a much broader array of uses even in the coming, you know, the first decade or so of uses of this. Where we could go, for example, to other places in the solar system to explore, perhaps NASA wants to use the rocket for that reason. EM : Yeah, NASA is planning to use a Starship to return to the moon, to return people to the moon. And so we ' re very honored that NASA has chosen us to do this. But I ' m saying it is a generalized – it ' s a general solution to getting anywhere in the greater solar system. It ' s not suitable for going to another star system, but it is a general solution for transport anywhere in the solar system. CA : Before it can do any of that, it ' s got to demonstrate it can get into orbit, you know, around Earth. What ' s your latest advice on the timeline for that? EM : It ' s looking promising for us to have an orbital launch attempt in a few months. So we ' re actually integrating – will be integrating the engines into the booster for the first orbital flight starting in about a week or two. And the launch complex itself is ready to go. So assuming we get regulatory approval, I think we could have an orbital launch attempt within a few months. CA : And a radical new technology like this presumably there is real risk on those early attempts. EM : Oh, 100 percent, yeah. The joke I make all the time is that excitement is guaranteed. Success is not guaranteed, but excitement certainly is. CA : But the last I saw on your timeline, you ' ve slightly put back the expected date to put the first human on Mars till 2029, I want to say? EM : Yeah, I mean, so let ' s see. I mean, we have built a production system for Starship, so we ' re making a lot of ships and boosters. CA : How many are you planning to make actually? EM : Well, we ' re currently expecting to make a booster and a ship roughly every, well, initially, roughly every couple of months, and then hopefully by the end of this year, one every month. So it ' s giant rockets, and a lot of them. Just talking in terms of rough orders of magnitude, in order to create a self-sustaining city on Mars, I think you will need something on the order of a thousand ships. And we just need a Helen of Sparta, I guess, on Mars. CA : This is not in most people ' s heads, Elon. EM : The planet that launched 1, 000 ships. CA : That ' s nice. But this is not in most people ' s heads, this picture that you have in your mind. There ' s basically a two-year window, you can only really fly to Mars conveniently every two years. You were picturing that during the 2030s, every couple of years, something like 1, 000 Starships take off, each containing 100 or more people. That picture is just completely mind-blowing to me. That sense of this armada of humans going to – EM : It ' ll be like"Battlestar Galactica,"the fleet departs. CA : And you think that it can basically be funded by people spending maybe a couple hundred grand on a ticket to Mars? Is that price about where it has been? EM : Well, I think if you say like, what ' s required in order to get enough people and enough cargo to Mars to build a self-sustaining city. And it ' s where you have an intersection of sets of people who want to go, because I think only a small percentage of humanity will want to go, and can afford to go or get sponsorship in some manner. That intersection of sets, I think, needs to be a million people or something like that. And so it ' s what can a million people afford, or get sponsorship for, because I think governments will also pay for it, and people can take out loans. But I think at the point at which you say, OK, like, if moving to Mars costs are, for argument ' s sake, $100, 000, then I think you know, almost anyone can work and save up and eventually have $100, 000 and be able to go to Mars if they want. We want to make it available to anyone who wants to go. It ' s very important to emphasize that Mars, especially in the beginning, will not be luxurious. It will be dangerous, cramped, difficult, hard work. It ' s kind of like that Shackleton **ad** for going to the Antarctic, which I think is actually not real, but it sounds real and it ' s cool. It ' s sort of like, the sales pitch for going to Mars is,"It ' s dangerous, it ' s cramped. You might not make it back. It ' s difficult, it ' s hard work."That ' s the sales pitch. CA : Right. But you will make history. EM : But it ' ll be glorious. CA : So on that kind of launch rate you ' re talking about over two decades, you could get your million people to Mars, essentially. Whose city is it? Is it NASA ' s city, is it SpaceX ' s city? EM : It ' s the people of Mars ' city. The reason for this, I mean, I feel like why do this thing? I think this is important for **maximizing** the probable lifespan of humanity or consciousness. Human civilization could come to an end for external reasons, like a giant meteor or **super** volcanoes or extreme climate change. Or World War III, or you know, any one of a number of reasons. But the probable life span of civilizational consciousness as we know it, which we should really view as this very delicate thing, like a small candle in a vast darkness. That is what appears to be the case. We ' re in this vast darkness of space, and there ' s this little candle of consciousness that ' s only really come about after 4. 5 billion years, and it could just go out. CA : I think that ' s powerful, and I think a lot of people will be inspired by that vision. And the reason you need the million people is because there has to be enough people there to do everything that you need to survive. EM : Really, like the critical threshold is if the ships from Earth stop coming for any reason, does the Mars City die out or not? And so we have to – You know, people talk about like, the sort of, the great filters, the things that perhaps, you know, we talk about the Fermi paradox, and where are the aliens? Well maybe there are these various great filters that the aliens didn ' t pass, and so they eventually just ceased to exist. And one of the great filters is becoming a multi-planet species. So we want to pass that filter. And I ' ll be long-dead before this is, you know, a real thing, before it happens. But I ' d like to at least see us make great progress in this direction. CA : Given how tortured the Earth is right now, how much we ' re beating each other up, shouldn ' t there be discussions going on with everyone who is dreaming about Mars to try to say, we ' ve got a once in a civilization ' s chance to make some new rules here? Should someone be trying to lead those discussions to figure out what it means for this to be the people of Mars ' City? EM : Well, I think ultimately this will be up to the people of Mars to decide how they want to **rethink** society. Yeah there ' s certainly risk there. And hopefully the people of Mars will be more enlightened and will not fight amongst each other too much. I mean, I have some recommendations, which people of Mars may choose to listen to or not. I would advocate for more of a direct democracy, not a representative democracy, and laws that are short enough for people to understand. Where it is harder to create laws than to get rid of them. CA : Coming back a bit nearer term, I ' d love you to just talk a bit about some of the other possibility space that Starship seems to have created. So given – Suddenly we ' ve got this ability to move 100 tons-plus into orbit. So we ' ve just launched the James Webb telescope, which is an incredible thing. It ' s unbelievable. EM : Exquisite piece of technology. CA : Exquisite piece of technology. But people spent two years trying to figure out how to fold up this thing. It ' s a three-ton telescope. EM : We can make it a lot easier if you ' ve got more volume and mass. CA : But let ' s ask a different question. Which is, how much more powerful a telescope could someone design based on using Starship, for example? EM : I mean, roughly, I ' d say it ' s probably an order of magnitude more resolution. If you ' ve got 100 tons and a thousand cubic meters volume, which is roughly what we have. CA : And what about other exploration through the solar system? I mean, I ' m you know – EM : Europa is a big question mark. CA : Right, so there ' s an ocean there. And what you really want to do is to drop a submarine into that ocean. EM : Maybe there ' s like, some squid civilization, cephalopod civilization under the ice of Europa. That would be pretty interesting. CA : I mean, Elon, if you could take a submarine to Europa and we see pictures of this thing being devoured by a squid, that would honestly be the happiest moment of my life. EM : Pretty wild, yeah. CA : What other possibilities are out there? Like, it feels like if you ' re going to create a thousand of these things, they can only fly to Mars every two years. What are they doing the rest of the time? It feels like there ' s this explosion of possibility that I don ' t think people are really thinking about. EM : I don ' t know, we ' ve certainly got a long way to go. As you alluded to earlier, we still have to get to orbit. And then after we get to orbit, we have to really prove out and refine full and rapid reusability. That ' ll take a moment. But I do think we will solve this. I ' m highly confident we will solve this at this point. CA : Do you ever wake up with the fear that there ' s going to be this Hindenburg moment for SpaceX where... EM : We ' ve had many Hindenburg. Well, we ' ve never had Hindenburg moments with people, which is very important. Big difference. We ' ve blown up quite a few rockets. So there ' s a whole compilation online that we put together and others put together, it ' s showing rockets are hard. I mean, the sheer amount of energy going through a rocket boggles the mind. So, you know, getting out of Earth ' s gravity well is difficult. We have a strong gravity and a thick atmosphere. And Mars, which is less than 40 percent, it ' s like, 37 percent of Earth ' s gravity and has a thin atmosphere. The ship alone can go all the way from the surface of Mars to the surface of Earth. Whereas getting to Mars requires a giant booster and orbital refilling. CA : So, Elon, as I think more about this incredible array of things that you ' re involved with, I keep seeing these synergies, to use a horrible word, between them. You know, for example, the robots you ' re building from Tesla could possibly be pretty handy on Mars, doing some of the dangerous work and so forth. I mean, maybe there ' s a scenario where your city on Mars doesn ' t need a million people, it needs half a million people and half a million robots. And that ' s a possibility. Maybe The Boring Company could play a role helping create some of the subterranean dwelling spaces that you might need. EM : Yeah. CA : Back on planet Earth, it seems like a partnership between Boring Company and Tesla could offer an unbelievable deal to a city to say, we will create for you a 3D network of tunnels populated by robo-taxis that will offer fast, low-cost transport to anyone. You know, full self-driving may or may not be done this year. And in some cities, like, somewhere like Mumbai, I suspect won ' t be done for a decade. EM : Some places are more challenging than others. CA : But today, today, with what you ' ve got, you could put a 3D network of tunnels under there. EM : Oh, if it ' s just in a tunnel, that ' s a solved problem. CA : Exactly, full self-driving is a solved problem. To me, there ' s amazing synergy there. With Starship, you know, Gwynne Shotwell talked about by 2028 having from city to city, you know, transport on planet Earth. EM : This is a real possibility. The fastest way to get from one place to another, if it ' s a long distance, is a rocket. It ' s basically an ICBM. CA : But it has to land – Because it ' s an ICBM, it has to land probably offshore, because it ' s loud. So why not have a tunnel that then connects to the city with Tesla? And Neuralink. I mean, if you going to go to Mars having a telepathic connection with loved ones back home, even if there ' s a time delay... EM : These are not intended to be connected, by the way. But there certainly could be some synergies, yeah. CA : Surely there is a growing argument that you should actually put all these things together into one company and just have a company devoted to creating a future that ' s exciting, and let a thousand flowers bloom. Have you been thinking about that? EM : I mean, it is tricky because Tesla is a publicly-traded company, and the investor base of Tesla and SpaceX and certainly Boring Company and Neuralink are quite different. Boring Company and Neuralink are tiny companies. CA : By comparison. EM : Yeah, Tesla ' s got 110, 000 people. SpaceX I think is around 12, 000 people. Boring Company and Neuralink are both under 200 people. So they ' re little, tiny companies, but they will probably get bigger in the future. They will get bigger in the future. It ' s not that easy to sort of combine these things. CA : Traditionally, you have said that for SpaceX especially, you wouldn ' t want it public, because public investors wouldn ' t support the craziness of the idea of going to Mars or whatever. EM : Yeah, making life multi-planetary is outside of the normal time horizon of Wall Street analysts. To say the least. CA : I think something ' s changed, though. What ' s changed is that Tesla is now so powerful and so big and throws off so much cash that you actually could connect the dots here. Just tell the public that x-billion dollars a year, whatever your number is, will be diverted to the Mars **mission**. I suspect you ' d have massive interest in that company. And it might unlock a lot more possibility for you, no? EM : I would like to give the public access to **ownership** of SpaceX, but I mean the thing that like, the overhead associated with a public company is high. I mean, as a public company, you ' re just constantly sued. It does occupy like, a fair bit of... You know, time and effort to deal with these things. CA : But you would still only have one public company, it would be bigger, and have more things going on. But instead of being on four boards, you ’ d be on one. EM : I ’ m actually not even on the Neuralink or Boring Company boards. And I don ’ t really attend the SpaceX board meetings. We only have two a year, and I just stop by and chat for an hour. The board overhead for a public company is much higher. CA : I think some investors probably worry about how your time is being split, and they might be excited by you know, that. Anyway, I just woke up the other day thinking, just, there are so many ways in which these things connect. And you know, just the simplicity of that **mission**, of building a future that is worth getting excited about, might appeal to an awful lot of people. Elon, you are reported by Forbes and everyone else as now, you know, the world ’ s richest person. EM : That ’ s not a sovereign. CA : EM : You know, I think it ’ s fair to say that if somebody is like, the king or de facto king of a country, they ’ re wealthier than I am. CA : But it ’ s just harder to measure – So $300 billion. I mean, your net worth on any given day is rising or falling by several billion dollars. How insane is that? EM : It ’ s bonkers, yeah. CA : I mean, how do you handle that psychologically? There aren ’ t many people in the world who have to even think about that. EM : I actually don ’ t think about that too much. But the thing that is actually more difficult and that does make sleeping difficult is that, you know, every good hour or even minute of thinking about Tesla and SpaceX has such a big effect on the company that I really try to work as much as possible, you know, to the edge of sanity, basically. Because you know, Tesla ’ s getting to the point where probably will get to the point later this year, where every high-quality minute of thinking is a million dollars impact on Tesla. Which is insane. I mean, the basic, you know, if Tesla is doing, you know, sort of $2 billion a week, let ’ s say, in revenue, it ’ s sort of $300 million a day, seven days a week. You know, it ’ s... CA : If you can change that by five percent in an hour ’ s brainstorm, that ’ s a pretty valuable hour. EM : I mean, there are many instances where a half-hour meeting, I was able to improve the **financial** outcome of the company by $100 million in a half-hour meeting. CA : There are many other people out there who can ’ t stand this world of billionaires. Like, they are hugely offended by the notion that an individual can have the same wealth as, say, a billion or more of the world ’ s poorest people. EM : If they examine sort of – I think there ’ s some axiomatic flaws that are leading them to that conclusion. For sure, it would be very problematic if I was consuming, you know, billions of dollars a year in personal consumption. But that is not the case. In fact, I don ’ t even own a home right now. I ’ m literally staying at friends ’ places. If I travel to the Bay Area, which is where most of Tesla engineering is, I basically rotate through friends ’ spare bedrooms. I don ’ t have a yacht, I really don ’ t take **vacations**. It ’ s not as though my personal consumption is high. I mean, the one exception is a plane. But if I don ’ t use the plane, then I have less hours to work. CA : I mean, I personally think you have shown that you are mostly driven by really quite a deep sense of moral purpose. Like, your attempts to solve the climate problem have been as powerful as anyone else on the planet that I ’ m aware of. And I actually can ’ t understand, personally, I can ’ t understand the fact that you get all this criticism from the Left about,”Oh, my God, he ’ s so rich, that ’ s disgusting.”When climate is their issue. Philanthropy is a topic that some people go to. Philanthropy is a hard topic. How do you think about that? EM : I think if you care about the reality of goodness instead of the perception of it, philanthropy is extremely difficult. SpaceX, Tesla, Neuralink and The Boring Company are philanthropy. If you say philanthropy is love of humanity, they are philanthropy. Tesla is accelerating sustainable energy. This is a love – philanthropy. SpaceX is trying to ensure the long-term survival of humanity with a multiple-planet species. That is love of humanity. You know, Neuralink is trying to help solve brain injuries and existential risk with AI. Love of humanity. Boring Company is trying to solve traffic, which is hell for most people, and that also is love of humanity. CA : How upsetting is it to you to hear this constant drumbeat of,"Billionaires, my God, Elon Musk, oh, my God?"Like, do you just shrug that off or does it does it actually hurt? EM : I mean, at this point, it ' s water off a duck ' s back. CA : Elon, I ' d like to, as we wrap up now, just pull the camera back and just think... You ' re a father now of seven surviving kids. EM : Well, I mean, I ' m trying to set a good example because the birthrate on Earth is so low that we ' re facing civilizational collapse unless the birth rate returns to a sustainable level. CA : Yeah, you ' ve talked about this a lot, that depopulation is a big problem, and people don ' t understand how big a problem it is. EM : Population collapse is one of the biggest threats to the future of human civilization. And that is what is going on right now. CA : What drives you on a day-to-day basis to do what you do? EM : I guess, like, I really want to make sure that there is a good future for humanity and that we ' re on a path to understanding the nature of the universe, the meaning of life. Why are we here, how did we get here? And in order to understand the nature of the universe and all these fundamental questions, we must expand the scope and scale of consciousness. Certainly it must not diminish or go out. Or we certainly won ' t understand this. I would say I ' ve been motivated by curiosity more than anything, and just desire to think about the future and not be sad, you know? CA : And are you? Are you not sad? EM : I ' m sometimes sad, but mostly I ' m feeling I guess relatively optimistic about the future these days. There are certainly some big risks that humanity faces. I think the population collapse is a really big deal, that I wish more people would think about because the birth rate is far below what ' s needed to sustain civilization at its current level. And there ' s obviously... We need to take **action** on climate sustainability, which is being done. And we need to secure the future of consciousness by being a multi-planet species. We need to address – Essentially, it ' s important to take whatever **actions** we can think of to address the existential risks that affect the future of consciousness. CA : There ' s a whole generation coming through who seem really sad about the future. What would you say to them? EM : Well, I think if you want the future to be good, you must make it so. Take **action** to make it good. And it will be. CA : Elon, thank you for all this time. That is a beautiful place to end. Thanks for all you ' re doing. EM : You ' re welcome.

Text Sample 193: POS Dim 2 – 2022 – Score 15.00 – 2022/t003763.txt

Chris Anderson : I want you to come with me to Tesla ' s huge gigafactory in Austin, Texas. So the day before it opened last week, the evening before, I was allowed to walk around it, no one else there. And what I saw there was, honestly, pretty mind-blowing. This is Elon Musk ' s famous machine that builds the machine, and in his view, the secret to a sustainable future, is not just making electric car. It ' s making a system that churns out huge numbers of electric cars with a margin, so that they can fund further growth. When I was there, none of us knew whether Elon would actually be able to make it here today, so I took the chance to sit down with him and record an epic interview. And I just want to show you an eight-minute excerpt of that interview. So here from Austin, Texas, Elon Musk. I want us to switch now to think a bit about artificial intelligence. I ' m curious about your timelines and how you predict and how come some things are so amazingly on the money and some aren ' t. So when it comes to predicting sales of Tesla vehicles, for example, you ' ve kind of been amazing. I think in 2014 when Tesla had sold that year 60, 000 cars, you said,”2020, I think we will do half a million a year.”Elon Musk : Yeah, we did almost exactly a half million. CA : Five years ago, last time you came to TED, I asked you about full self-driving, and you said,”Yeah, this very year I ’ m confident that we will have a car going from LA to New York without any intervention.”EM : Yeah, I don ’ t want to blow your mind, but I ’ m not always right. CA : What ’ s the difference between those two? Why has full self-driving in particular been so hard to predict? EM : I mean, the thing that really got me, and I think it ’ s going to get a lot of other people, is that there are just so many false dawns with self-driving where you think you ’ ve got the problem, have a handle on the problem, and then it, no, turns out you just hit a ceiling. Because if you were to plot the progress, the progress looks like a log curve. So it ’ s like a series of log curves. So most people don ’ t know what a log curve is, I suppose. CA : Show the shape with your hands. EM : It goes up, you know, sort of a fairly straight way, and then it starts tailing off and you start getting diminishing returns. You know, in retrospect, they seem obvious, but in order to solve full self-driving properly, you actually have to solve real-world AI. Because what are the road networks designed to work with? They ’ re designed to work with a biological neural net, our brains, and with vision, our eyes. And so in order to make it work with computers, you basically need to solve real-world AI and vision. Because we need cameras and **silicon** neural nets in order to have self-driving work for a system that was designed for eyes and biological neural nets. You know, I guess when you put it that way, it ’ s sort of, like, quite obvious that the only way to solve full self-driving is to solve real-world AI and sophisticated vision. CA : What do you feel about the current architecture? Do you think you have an architecture now where there is a chance for the logarithmic curve not to tail off anytime soon? EM : Well I mean, admittedly these maybe infamous last words, but I actually am confident that we will solve it this year. That we will exceed – The probability of an accident, at what point do you exceed that of the average person? I think we will exceed that this year. We could be here talking again in a year, like, well, another year went by and it didn ’ t happen. But I think this is the year. CA : Is there an element that you actually deliberately make aggressive prediction timelines to drive people to be ambitious? Without that, nothing gets done? So it feels like at some point in the last year, seeing the progress on understanding that the Tesla AI understanding the world around it, led to a kind of, an aha moment at Tesla. Because you really surprised people recently when you said probably the most important product development going on at Tesla this year is this robot, Optimus. Is it something that happened in the development of full self-driving that gave you the confidence to say,”You know what, we could do something **special** here.”EM : Yeah, exactly. So, you know, it took me a while to sort of realize that in order to solve self-driving, you really needed to solve real-world AI. And at the point of which you solve real-world AI for a car, which is really a robot on four wheels, you can then generalize that to a robot on legs as well. The things that are currently missing are enough intelligence for the robot to **navigate** the real world and do useful things without being explicitly instructed. So the missing things are basically real-world intelligence and scaling up manufacturing. Those are two things that Tesla is very good at. And so then we basically just need to design the specialized actuators and sensors that are needed for a humanoid robot. People have no idea, this is going to be bigger than the car. CA : So talk about – I think the first applications you ’ ve mentioned are probably going to be in manufacturing, but eventually, the vision is to have these available for people at home. If you had a robot that really understood the 3D architecture of your house and knew where every object in that house was or was supposed to be and could recognize all those objects, I mean, that ’ s kind of amazing, isn ’ t it, like, the kind of thing that you could ask a robot to do would be what? Like, tidy up? EM : Yeah, absolutely. Make dinner, I guess, mow the lawn. CA : Take a cup of tea to grandma and show her family pictures. EM : Exactly. Take care of my grandmother and make sure – CA : It could obviously recognize everyone in the home. It could play catch with your kids. EM : Yes. I mean, obviously, we need to be careful this doesn ' t become a dystopian situation. I think one of the things that ' s going to be important is to have a localized ROM chip on the robot that cannot be updated over the air. Where if you, for example, were to say,"Stop, stop, stop,"if anyone said that, then the robot would stop, you know, type of thing. And that ' s not updatable remotely. I think it ' s going to be important to have safety features like that. CA : Yeah, that sounds wise. EM : And I do think there should be a regulatory agency for AI. I ' ve said this for many years. I don ' t love being regulated, but I think this is an important thing for public safety. CA : Do you think there will be basically like in, say, 2050 or whatever, like, a robot in most homes, is what there will be, and people will love them and count on them? You ' ll have your own butler basically. EM : Yeah, you ' ll have your sort of buddy robot probably, yeah. CA : I mean, how much of a buddy? How many applications have you thought, you know, can you have a romantic partner, a sex partner? EM : It ' s probably inevitable. I mean, I did promise the internet that I ' d make catgirls. We could make a robot catgirl. CA : Be careful what you promise the internet. EM : So, yeah, I guess it ' ll be whatever people want really, you know. CA : What sort of timeline should we be thinking about of the first models that are actually made and sold? EM : Well, you know, the first units that we intend to make are for jobs that are dangerous, boring, repetitive and things that people don ' t want to do. And, you know, I think we ' ll have like, an interesting prototype some time this year. We might have something useful next year, but I think quite likely within at least two years. And then we ' ll see rapid growth year over year of the usefulness of the humanoid robots and decrease in cost and scaling up production. CA : Help me on the economics of this. So what do you picture the cost of one of these being? EM : Well, I think the cost is actually not going to be crazy high. Like, less than a car. CA : But think about the economics of this. If you can replace a 30, 000-dollar, 40, 000-dollar-a-year worker, which you have to pay every year, with a one-time payment of 25, 000 dollars for a robot that can work longer hours, doesn ' t go an **vacation**, it could be a pretty rapid replacement of certain types of jobs. How worried should the world be about that? EM : I wouldn ' t worry about the sort of, putting people out of a job thing. I think we ' re actually going to have, and already do have, a massive shortage of labor. So I think we will have... Not people out of work, but actually still a shortage of labor even in the future. But this really will be a world of abundance. Any goods and **services** will be available to anyone who wants them. It ' ll be so cheap to have goods and **services**, it will be ridiculous. CA : And now, arguably, the biggest visionary of them all, Elon Musk. Hey, Elon, welcome. So, Elon, a few hours ago, you made an offer to buy Twitter. Why? EM : How ' d you know? CA : A little bird **tweeted** in my ear or something, I don ' t know. EM : By the way, have you seen the movie"Ted,"about the bear? CA : I have. EM : It ' s a good movie. CA : Don ' t mention that here. EM : So, yeah, yeah, so... Was there a question? CA : Why make that offer? EM : Oh, so, well, I think it ' s very important for there to be an inclusive arena for free speech where all... Twitter has become, kind of, the de facto town square. It ' s just really important that people have both the reality and the perception that they are able to speak freely within the bounds of the law. And, you know, one of the things that I believe Twitter should do is open-source the algorithm and make any changes to people ' s **tweets**, or if they ' re emphasized or de-emphasized, that **action** should be made apparent so anyone can see that **action** has been taken. So there ' s no, sort of, behind the scenes manipulation, either algorithmically or manually. CA : Last week when we spoke, Elon, I asked you whether you were thinking of taking over. You said, no way. You said,"I do not want to own Twitter. It is a recipe for misery. Everyone will blame me for everything."What on Earth changed? EM : No I think everyone will still blame me for everything. If I acquire Twitter and something goes wrong, it ' s my fault, 100 percent. I think there will be quite a few arrows, yes. CA : It will be miserable. But you still want to do it. Why? EM : I mean, I hope it ' s not too miserable. I just think it ' s important to the... It ' s important to the function of democracy. It ' s important to the function of the United States as a free country and many other countries. And to actually help freedom in the world, more broadly than the US. And so I think it ' s, you know, I think the civilizational risk is decreased the more we can increase the trust of Twitter as a public **platform**. And so I do think this will be somewhat painful. And I ' m not sure that I will actually be able to acquire it. And I should also say the intent is to retain as many shareholders as is allowed by the law in a private company, which I think is around 2, 000 or so. So it ' s definitely not, from the standpoint of,"Let me figure out how to monopolize or **maximize** my **ownership** of Twitter."But I will try to bring along as many shareholders as we ' re allowed to. CA : You don ' t necessarily want to pay out 40 or whatever it is, billion dollars in cash. You ' d like them to come with you in the new company. EM : I mean, you know, I could technically afford it. CA : I heard that. EM : But what I ' m saying is, this is not a way to, sort of, make money, you know. It ' s just that I think this is... My strong intuitive sense is that having a public **platform** that is maximally trusted and broadly inclusive is extremely important to the future of civilization. CA : But you ' ve described yourself – EM : I don ' t care about the economics at all. CA : OK, that ' s cool to hear. This is not about the economics. It ' s for the moral good that you think it will achieve. You ' ve described yourself, Elon, as a"free speech absolutist."But does that mean that there ' s literally nothing that people can ' t say, and it ' s OK? EM : Well, I think, obviously, Twitter or any forum is bound by the laws of the country that it operates in. So obviously, there are some limitations on free speech in the US. And of course, Twitter would have to abide by those rules. CA : So you can ' t incite people to violence, like, a direct incitement to violence you know, you can ' t do the equivalent of crying"fire"in a movie theater, for example. EM : No, that would be a crime. It should be a crime. CA : But here ' s the challenge, it ' s such a nuanced difference between different things. So there ' s incitement to violence. That ' s a no if it ' s illegal. There ' s hate speech, which, some forms of hate speech are fine. You know,"I hate spinach."EM : I mean, if it ' s sauteed in, you know, cream sauce, it can be quite nice. CA : But the problem is, so let ' s say someone says – OK, here ' s one **tweet**,"I hate politician X."Next **tweet** is,"I wish politician X wasn ' t alive,"as some of us have said about Putin right now, for example. So that ' s legitimate speech. Another **tweet** is,"I wish politician X wasn ' t alive"with a picture of their head with a gunsight over it. Or that plus their address. I mean, at some point someone has to make a decision as to which of those is not OK. Can an algorithm do that, or surely you need human judgment at some point? EM : No, I think, like I said, in my view, Twitter should match the laws of the country. And really, you know, there ' s an obligation to do that. But going beyond that, and having it be unclear who ' s making what changes to where, having **tweets** sort of, mysteriously be promoted and demoted with no insight into what ' s going on, having a black-box algorithm promote some things and not other things, I think this can be quite dangerous. CA : So, the idea of opening the algorithm is a huge deal, and I think many people would welcome that of understanding exactly how it ’ s making the decision. EM : And critique it, like, what I mean is like, I think the code should be on GitHub, you know. So people can look through it and say, like,”I see a problem here.””I don ’ t agree with this.”They can highlight issues, suggest changes in the same way that you update Linux or Signal or something like that, you know. CA : As I understand it, at some point right now, what the algorithm would do is it would look at, for example, how many people have flagged a **tweet** as obnoxious. And then at some point a human has to look at it and make a decision as to, does this cross the line or not, that the algorithm itself can ’ t, I don ’ t think yet, tell the difference between legal and OK and definitely obnoxious. And so the question is, which humans, you know, make that call? I mean, do you have a picture of that? Right now, Twitter and Facebook and others, you know, they ’ ve hired thousands of people to try to help make wise decisions. And the trouble is that no one can agree on what is wise. How do you solve that? EM : Well I think we would want to err on the side – if in doubt, let the speech exist. If it ’ s, you know, a gray area, I would say let the **tweet** exist. But obviously, you know, in a case where there ’ s perhaps a lot of controversy that you would not want to necessarily promote that **tweet**. So I ’ m not saying that I have all the answers here. But I do think that we want to be just very reluctant to delete things and just be very cautious with permanent bans. You know, time-outs, I think, are better than, sort of, permanent bans. And... But just in general, like I said... It won ’ t be perfect, but I think we want it to really have the perception and reality that speech is as free as reasonably possible. And a good sign as to whether there ’ s free speech is : is someone you don ’ t like allowed to say something you don ’ t like? And if that is the case, then we have free speech. And it ’ s damn annoying when someone you don ’ t like says something you don ’ t like. That is a sign of a healthy, functioning, free speech situation. CA : I think many people would agree with that. And looking at the reaction online, many people are excited by you coming in and the changes you ’ re proposing. Some others are absolutely horrified. Here ’ s how they would see it. They would say,”Wait a sec. We agree that Twitter is an incredibly important town square. It is where the world exchanges opinion about life and death matters. How on Earth could it be owned by the world ’ s richest person? That can ’ t be right.”So what ’ s the response there? Is there any way that you can distance yourself from the actual decision making that matters on content in some very clear way that is convincing to people? EM : Well, like I said, I think it ’ s very important that the algorithm be open-sourced and that any manual adjustments be identified. So if somebody did something to a **tweet**, there ’ s information attached to it that **action** was taken. And I won ’ t personally be, you know, in there editing **tweets**. But you ’ ll know if something was done to promote, demote or otherwise affect a **tweet**. You know, as for media sort of **ownership**, I mean, you ’ ve got, you know, Mark Zuckerberg owning Facebook and Instagram and WhatsApp and with a share **ownership** structure that will have Mark Zuckerberg XIV still controlling those entities. Like, literally. We won ’ t have that at Twitter. CA : If you commit to opening up the algorithm that definitely gives some level of confidence. Talk about some of the other changes that you ’ ve proposed. The edit button, that ’ s definitely coming if you have your way. EM : Yeah. I think, I mean, frankly... The top priority I would have is eliminating the spam and scam bots and the bot armies that are on Twitter. You know, I think these influence – they make the product much worse. You know, if I had a Dogecoin for every crypto scam I saw. I would have 100 billion Dogecoin. CA : Do you regret sparking a sort of, storm of excitement over DOGE and, you know, where it ’ s gone? EM : I mean, I think DOGE is fun. And, you know, I ’ ve always said don ’ t bet the farm on Dogecoin, FYI. But I think it ’ s, I like dogs and I like memes, and it ' s got both of those. CA : But just on the edit button, how do you get around the problem of, so someone **tweets**,"Elon rocks"and it ' s **tweeted** by two million people. And then after that, they edit it"Elon sucks,"and then all those retweets they ' re all embarrassed. How do you avoid that type of changing of meaning so that retweeters are exploited? EM : Well, I think you ' d only have the edit capability for a short period of time. And probably the thing to do upon the edit would be to zero out all retweets and favorites. CA : OK. EM : I ' m open to ideas though, you know. CA : So in one way, the algorithm works kind of well for you right now, I wanted to show you this. This is a typical **tweet** of mine, kind of lame and wordy and whatever. And the amazing response it gets is this, oh, my God, 97 likes. And then I tried another one. [I get to interview @elonmusk on Thursday here at TED. What questions would YOU ask him?] 29, 000 likes. So the algorithm, at least seems to be, at the moment, you know :"If, expand to the world immediately."Not bad, right? EM : Yeah, I guess so, I mean, that is cool. CA : So help us understand how it is you ' ve built this incredible following on Twitter yourself when some of the people who love you the most look at some of what you **tweet** and they think it ' s somewhere between embarrassing and crazy, some of it ' s amazing. EM : It is a little... CA : But is that actually why it ' s worked, why has it worked? EM : I mean, I don ' t know. I ' m **tweeting** more or less stream of consciousness, you know. It ' s not like, let me think about some grand plan about my Twitter or whatever. You know, I ' m, like, literally, on the toilet or something, like,"Ha ha, this is funny."And then **tweet** that out. That ' s like most of them. That ' s, you know, oversharing. CA : But you are obsessed with getting the most out of every minute of your day. And so why not? EM : So, I don ' t know. I try to **tweet** out things that are interesting or funny. And then people seem to like it. CA : So if you are unsuccessful – actually, before I ask that, let me ask this. So how can I say, is funding secured? EM : I have sufficient assets to complete... This is not a forward-looking statement, blah blah blah. I mean, I can do it if possible. I should say, actually even originally with Tesla back in the day, funding was actually secured. I want to be clear about that. In fact, this may be a good opportunity to clarify that. Funding was indeed secured, and I should say, like, why do I not have respect for the SEC in that situation? And I don ' t mean to blame everyone at the SEC, but certainly the San Francisco office, it ' s because the SEC knew that funding was secured, but they pursued an active public investigation nonetheless. At the time, Tesla was in a precarious **financial** situation, and I was told by the banks that if I did not agree to settle with the SEC that the banks would cease providing working capital and Tesla would go bankrupt immediately. So that ' s like having a gun to your child ' s head. So I was forced to concede to the SEC unlawfully. Those bastards. And now it makes it look like I lied when I did not, in fact, lie. I was forced to admit that I lied to save Tesla ' s life. And that ' s the only reason. CA : Given what ' s actually happened – Given what ' s actually happened to Tesla since then, though, aren ' t you glad that you didn ' t take it private? EM : Yeah. I mean... It ' s difficult to put yourself in the position at the time, Tesla was under the most relentless short seller attack in the history of the stock market. There ' s something called"short and distort,"where the barrage of negativity that Tesla was experiencing from short sellers on Wall Street was beyond all belief. Tesla was the most shorted stock in the history of stock markets. This is saying something. So, you know, this was affecting our ability to hire people, it was affecting our ability to sell cars, it was... Yeah, it was terrible. They wanted Tesla to die so bad, they could taste it. CA : Well, most of them have paid the price. EM : Yes. Where are they now? CA : So that was a very strong statement. Obviously, a lot of people who support you, I would ' ve thought, would say, you have so much to offer the world on the upside, on the vision side, don ’ t waste your time getting distracted by these battles that bring out negativity and make people feel that you ’ re being defensive. People don ’ t like fights, especially with powerful government authorities. They ’ d rather buy into your dream. Aren ’ t you encouraged by people just to edit that temptation out and go with the bigger story? EM : Well, I mean, I would say, like, you know, I ’ m somewhat of a mixed bag, you know. CA : Well, you ’ re a fighter and you don ’ t... You don ’ t like to lose and you are determined that you don ’ t. Basically, I mean. EM : Sure I don ’ t like to lose, I ’ m not sure many people do. But the truth matters to me a lot. Sort of, pathologically, it matters to me. CA : OK. So you don ’ t like to lose, if in this case you are not successful in, you know, the board does not accept your offer, you ’ ve said you won ’ t go higher. Is there a plan B? EM : There is. CA : I think we would like to hear a little bit about plan B. EM : For another time, I think. CA : Another time, alright. I... That ’ s a nice tease. Alright. I would love to try to understand this brain of yours more, Elon. With your permission, I ’ d like to just play this. Oh, actually, before we do that, here was one of the thousands of questions that people asked. I thought this was actually quite a good one.”If you could go back in time and change one decision you made along the way”– do your own edit button –”which one would it be and why?”EM : Do you mean like a career decision or something? CA : Just any decision over the last few years, like your decision to invest in Twitter in the first place or your... Anything. EM : I mean, the worst business decision I ever made was not starting Tesla with just JB Straubel. By far the worst decision I ’ ve ever made is not just starting Tesla with JB. That ’ s number one by far. CA : Alright, so JB Straubel was the visionary cofounder who was obsessed with and knew so much about batteries. And your decision to go with Tesla, the company as it was, meant that you got locked into what you concluded was a weird architecture now. EM : There ’ s a lot of confusion. Tesla did not exist in any... Tesla was a shell company with no employees, no intellectual property, when I invested. But a false narrative has been created by one of the other cofounders, Martin Eberhard. And I don ’ t want to get into the nastiness here, but I didn ’ t invest in an existing company. We created a company. And ultimately, the creation of that company was done by JB and me. And unfortunately, there ’ s someone else, another cofounder who has made it his life ’ s **mission** to make it sound like he created the company, which is false. CA : Wasn ’ t there another issue right at the heart of the development of the Tesla Model 3, where Tesla almost went bankrupt? And I think you have said that part of the reason for that was that you overestimated the extent to which it was possible, at that time, to automate a factory. Huge amount was spent kind of over-automating, and it didn ’ t work, and it nearly took the company down. Is that fair? EM : I mean, first of all, it ’ s important to understand, like, what has Tesla actually accomplished that is most noteworthy. It is not the creation of an electric vehicle or creating an electric vehicle prototype or low-volume production of a car. There have been hundreds of car **start-ups** over the years, hundreds. And in fact, at one point, Bloomberg counted up the number of electric vehicle **start-ups**, and I think they got to almost 500. So the hard part is not creating a prototype or going into limited production. The absolutely difficult thing, which has not been accomplished by an American car company in 100 years, is reaching volume production without going bankrupt. That is the actual hard thing. The last company, American company, to reach volume production without going bankrupt, was Chrysler in the ’ 20s. CA : And it nearly happened to Tesla. EM : Yes, but it ’ s not like, oh, geez, I guess if we ’ d just done more manual stuff, things would have been fine. Of course not. That is definitely not the case. So we basically messed up almost every aspect of the Model 3 production line, from cells to packs to drive inverters, motors, body line, the paint shop, final assembly, everything. Everything was messed up. I lived in the Fremont and Nevada factories for three years, fixing that production line, running around like a maniac through every part of that factory. Living with the team. I slept on the floor, so the team, who was going through a hard time, could see me on the floor. That they knew that I was not in some ivory tower. When whatever pain they experienced, I had it more. CA : And some people who knew you well actually thought you were making a terrible mistake, that you were driving yourself to the edge of sanity, almost. And that you were in danger of making bad choices. And in fact, I heard you say last week, Elon, that you, because of Tesla ' s huge value now and, you know, the significance of every minute that you spend, that you are in danger of sort of, obsessing over it, spending all this time to the edge of sanity. That doesn ' t sound **super** wise. Isn ' t there... Your time, your completely sane, centered, rested time and decision making is more powerful and compelling than that sort of,"I can barely hold my eyes open."So surely it should be an absolute strategic priority to look after yourself. EM : I mean, there wasn ' t any other way to make it work. They were three years of hell. 2017, 2018 and 2019 were the longest period of excruciating pain in my life. There wasn ' t any other way, and we barely made it. And we were on the ragged edge of bankruptcy the entire time. It ' s not like I want the pain, I don ' t like it. Those were three – So, so, so much pain. But it had to be done or Tesla would be dead. CA : When you looked around the Gigafactory that we saw images of earlier last week and just see where the company ' s come, do you feel that this challenge of figuring out the new way of manufacturing, that you actually have an edge now, that it ' s different, that you ' ve figured out how to do this. And that those three years, won ' t be repeated? You ' ve actually figured out a new way of manufacturing. EM : At this point, I think I know more about manufacturing than anyone currently alive on Earth. I can tell you how every damn part of that car is made. It ' s basically if you just live on the factory, you live in the factory for three years and – CA : That was nice. EM : Poignant note to something. CA : Someone wants to compose a symphony to that **expression** of confidence, something like that. I have no idea what that is. EM : Anyway, yeah. Every aspect of a car, six ways to Sunday, I know it. CA : I mean, you talk about scale, right. You ' re in the middle of writing your new master plan, and you ' ve said that scale is at the heart of it. Why does scale matter? Why are you obsessed with it, what are you thinking? EM : Yeah, well, see, in order to accelerate the advent of sustainable energy, there must be scale. Because we ' ve got to transition a vast economy that is currently overly dependent on fossil fuels to a sustainable energy economy. One where the energy is – Yeah, I mean, we ' ve got to do it. So the energy ' s got to be sustainably generated with wind, solar, hydro, geothermal, I ' m a believer in nuclear as Isabelle [Boemeke] gave a talk about. And then since solar and wind is intermittent, you have to have stationary storage batteries, and then we ' ve got to transition all transport to electric. If we do those things, we have a sustainable energy future. The faster we do those things, the less risk we put to the environment. So sooner is better. And so scale is very important. You know, it ' s not about fresh releases. It ' s about tonnage. What was the tonnage of batteries produced and obviously done in a sustainable way? And our estimate is that approximately 300 terawatt hours of battery storage is needed to transition transport, electricity and heating and cooling to a fully electric situation. There may be some different estimates out there, but our estimate is 300 terawatt hours. CA : So we dug into this a lot in the interview that we recorded last week, so people can go in and hear that more. But I mean, the context is, that is, I think about 1, 000 times the current installed battery capacity. I mean, the scale up needed is breathtaking, basically. And... So your vision is to commit Tesla to try to deliver on a meaningful percentage of what is needed. And call on others to do the rest. This is a task for humanity to massively scale up our response to change the energy grid. EM : Yes. It ' s like basically how fast can we scale and encourage others to scale to get to that 300-terawatt-hour installed base of batteries? CA : Right. EM : And then, of course, there ' ll be a tremendous need to recycle those batteries, and it makes sense to recycle them because the raw materials are like high grade ore. People shouldn ' t think there will be this big pile of batteries, they can get recycled, because even a dead battery pack is worth about 1, 000 dollars. But this is what ' s needed for a sustainable energy future. So we ' re going to try to take a set of **actions** that accelerate the day and bring the day of a sustainable energy future sooner. CA : OK. There ' s going to be a huge interest in your master plan when you publish that. Meanwhile, I just would love to understand more what goes on in this brain of yours, because it is a pretty unique one. I want to play, with your permission, this very funny opening from SNL, Saturday Night Live. Thank you. Thank you very much. It ' s an honor to be hosting Saturday Night Live. I mean that. Sometimes after I say something, I have to say,"I mean that."So people really know that I mean it. That ' s because I don ' t always have a lot of intonational variation in how I speak. Which, I ' m told, makes for great comedy. I ' m actually making history tonight as the first person with Asperger ' s to host SNL. CA : And I think you followed that up with – EM : At least the first person to admit it. CA : The first person to admit it. So this was a brave thing to say. But I would love to understand you know, how you think of Asperger ' s, like, whether you can give us any sense of, even you as a boy, what the experience was or as you now understand, with the benefit of hindsight. Can you talk about that a bit? EM : Well, I think everyone ' s experience is going to be somewhat different. But I guess for me, the social cues were not intuitive. So I was just very bookish, and I didn ' t understand... I guess others could sort of, intuitively understand... What was meant by something. I would just tend to take things very literally, like, the words, as spoken, were exactly what they meant. But then that turned out to be wrong. They ' re not simply saying exactly what they mean. There ' s all sorts of other things that are meant. It took me a while to figure that out. I was, you know, bullied quite a lot. So... I did not have a sort of, happy childhood, to be frank. It was quite, quite rough. But I read a lot of books. I read lots and lots of books. And, you know, sort of, gradually understood more from the books that I was reading and watched a lot of movies and... You know, just... But it took me a while to understand things that most people intuitively understand. CA : So I ' ve wondered whether it ' s possible that that was, in a strange way, an incredible gift to you and indirectly to many other people. Inasmuch as... Brain ' s, you know, are plastic. And they go where the **action** is. And if for some reason the external world and social cues, which so many people spend so much time and mental energy obsessing over, if that is partly cut off, isn ' t it possible that that is partly what gave you the ability to understand inwardly the world at a much deeper level than most people do? EM : I suppose that ' s certainly possible. I think there ' s maybe some value also from a technology standpoint, because I found it rewarding to spend all night programming computers just by myself. And I think most people, most people don ' t enjoy typing strange symbols into a computer by themselves all night. They think that ' s not fun. But I thought it was, I really liked it. So I would just program all night by myself. And I found that to be quite enjoyable. But I think that is not normal. CA : I ' ve thought a lot about... It ' s a riddle to a lot of people of how you ' ve done this, how you ' ve repeatedly innovated in these different industries. And you know, every entrepreneur sees possibility in the future and then acts to make that real. It feels to me like you see possibility just more broadly than almost anyone and can connect the dots. So you see scientific possibility based on a deep understanding of physics and knowing what the fundamental equations are, what the technologies are that are based on that science and where they could go, you see technological possibility. And then really unusually, you combine that with economic possibility, of like what it actually would cost, is there a system you can imagine where you could affordably make that thing, and that... sometimes you then get conviction that there is an opportunity here. Put those pieces together and you could do something amazing. EM : Yeah, I think one aspect of whatever condition I had was I was just absolutely obsessed with truth. Just obsessed with truth. And so the obsession with truth is why I studied physics, because physics attempts to understand the truth, the truth of the universe. Physics is just, what are the provable truths of the universe. And truths that have predictive power. So for me, physics was sort of, a very natural thing to study. Nobody made me study it. I was intrinsically interested to understand the nature of the universe. And then computer science or information theory also to just understand logic. And, you know, there ' s an argument that... That information theory is actually operating at a more fundamental level, more fundamental level than even physics. So just yeah... Physics and information theory were really interesting to me. CA : So when you say truth, I mean, like, some people... What you ' re talking about is the truth of the universe. The fundamental truths that drive the universe, it ' s like, a deep curiosity about what this universe is, why we ' re here, simulation, we don ' t have time to go into that. But I mean, you ' re just deeply, deeply curious about, what this is for, what this is, this whole thing. EM : Yes, I think the why of things is very important. When I was in young teens, I got quite depressed about the meaning of life. And I was trying to sort of, understand the meaning of life. I ' m looking at, reading religious texts and reading books on philosophy. And I got into the German philosophers, which is definitely not wise if you ' re a young teenager, I have to say. It can be a bit dark. Much better read as an adult. And then actually, I ended up reading"The Hitchhiker ' s Guide to the Galaxy,"which is actually a book on philosophy just sort of disguised as a silly humor book. But it ' s actually a philosophy book. And Adams makes the point that it ' s actually the question that is harder than the answer. You know, he sort of makes a joke that the answer is 42. That number does pop up a lot. And 420 is just 10 times 42. CA : Ten times more significant than 42. EM : But, you know, you can make a triangle with 42 degrees and two 69s. So there ' s no such thing as a perfect triangle, or is there? CA : But even more important than the answer is the question. That was the whole theme of that book. I mean, is that basically how you see meaning then, that it ' s the pursuit of questions? EM : So I have a sort of, a proposal for a worldview or a motivating philosophy, which is to understand what questions to ask about the answer that is the universe. And to the degree that we expand the scope and scale of consciousness, biological and digital, we will be better able to ask these questions, to frame these questions and to understand why we ' re here, how we got here, what the heck is going on. And so that is my driving philosophy, is to expand the scope and scale of consciousness that we may better understand the nature of the universe. CA : Elon, one of the things that was most touching last week – was seeing you hang out with your kids. Here ' s, if I may... EM : He looks vaguely like a ventriloquist dummy there. I mean, how do you know that ' s real? CA : So that ' s X, and it was just a delight seeing you hang out with him. What ' s his future going to be? I mean, I don ' t mean him personally, but the world he ' s going to grow up in. What future do you believe he will grow up in? EM : Well, I mean, a very digital future. A very different world than I grew up in, that ' s for sure. But I think we want to obviously do our absolute best to ensure that the future is good for everyone ' s children. And that, you know, that the future is something that you can look forward to and not feel sad about. You know, you want to get up in the morning and be excited about the future. And we should fight for the things that make us excited about the future. You know, the future cannot just be about one miserable thing after another, solving one sad problem after another. There have got to be things that get you excited. Like, you want to live. These things are very important. You should have more of it. CA : And it ' s not as if it ' s a done deal, like it ' s all to play for. Like, the future may be horrible. Still, there are scenarios where it is horrible, but you see a pathway to an exciting future both on Earth and on Mars and in our minds through artificial intelligence and so forth. I mean, in your heart of hearts, do you really believe that you are helping deliver that exciting future for X and for others? EM : I mean, I ' m trying my hardest to do so. I, you know, I love humanity, and I think that we should fight for a good future for humanity. And I think we should be optimistic about the future and fight to make that optimistic future happen. CA : Elon, I think that ' s the perfect place to close this. Thank you so much for spending time coming here and for the work that you ' re doing. And good luck with finding a wise course through on Twitter and everything else. EM : Alright, thank you.

Text Sample 194: POS Dim 2 – 2024 – Score 17.00 – 2024/t003164.txt

Chris Anderson : Hello, welcome, TED members. It ' s very, very, very good to see you. I ' m the head of TED, Chris Anderson. I ' m a TED head, apparently, and really excited to be here with you. We ' re actually doing something we haven ' t done before. We are going to watch together a smash hit TED Talk in the presence of the man who gave that talk, Scott Galloway. I ' m going to introduce to you in a minute. While we ' re watching that talk, you can comment away, ask questions, talk with each other, disagree, amplify, laugh, whatever strikes you as the right thing to do. And when the talk is over, we ' ll be putting questions to Scott and seeing how he can possibly defend the outrageous claims that he makes in this talk. This talk is a very big deal. He ' s outlining a giant problem making some big, **bold** solutions on what we might do about it. It ' s a big wow. And it ' s been seen by millions of people already. But we want to dig deeper into these questions. And there couldn ' t be a better group of people to do it than you. So thank you for being part of this. You are awesome. I think, let ' s see, yeah, there ' s a live transcript of this event if you need it by clicking on the captions option. So I think that ' s it. So I would like to welcome the – how do we describe Scott Galloway? Well I mean he ' s a professor at NYU, business professor, he ' s the host of an ungodly number of podcasts. Incredibly successful. He does all kinds of things. He ' s been a successful entrepreneur and general troublemaker, and it ' s a delight to welcome Scott Galloway here. Scott, hello, nice to have you. Scott Galloway : It ' s good to be with you, Chris. Thanks for having me. And hello to all TED acolytes and Members. CA : Alrighty. So I don ' t know if you ' ve ever done this either. I don ' t know whether it ' d be at all uncomfortable to watch yourself in full glory giving this talk that was so well received. But let ' s try it. Let us roll this video without further ado. My name is Scott Galloway, I teach at NYU and I appreciate your time. I have 44 slides and 720 seconds. Let ' s light this candle. OK so for those of you who don ' t know me, I ' m actually a global television store. True story. I ' ve had four TV series in the last three years. Two of them have been canceled before they were launched and two were canceled within six weeks. Let ' s recap. If we want to juice this thing, if we want to put a cattle prod up the ass of the economy. Bloomberg. The most trusted name in **financial** news. Not for long. Andrew Yang : I ’ m going to do whatever I can for this country of ours. SG : Jesus, come on, dude, you ’ re 0 for two. Face for podcasting. So first insight of the day. I ’ d like to be the first person to welcome you to the last TED. OK. By the way, it ’ s clear – what ’ s it called? What are we here for?”The brave and the brilliant?”It ’ s clear that Chris is a frustrated soap opera producer. Essentially what we have here is a telenovela where, after a night of unbridled passion between Bill Gates and Malcolm Gladwell, they give birth to their bastard love child, Simon Sinek. OK, I start us with a question. Do we love our children? Sounds like an illegitimate question, right? Well, I ’ m going to try and convince you otherwise. Essentially, as we go down generations, we ’ re seeing that for the last two generations, people are making less money on an inflation-adjusted basis. In addition, the cost of buying a home, the cost of pursuing education, continues to skyrocket. So the purchasing power, the prosperity, is inversely correlated to age. Simply put, as we get younger, we ’ re taking away opportunity and prosperity from our youngest. The social contract that is now no longer in place and for the first time in the US ’ s history, a 30-year-old is no longer doing as well as his or her parents were at 30. This is a breakdown in the fundamental agreement we have with any society, and it creates rage and shame. As a result, people over the age of 55 feel pretty good about America, but less than one in five people under the age of 34 feel very good about America. This creates an incendiary. Righteous movements, cuts to our society end up becoming opportunistic infections because generally speaking, young people have a warranted envy, they ’ re pissed off and they ’ re angry that they don ’ t enjoy the same spoils and prosperity that were provided to our generation. A decent proxy for how much we value youth labor is minimum wage, and we ’ ve kept it purposely pretty low. If it had just kept pace with productivity, it ’ d be at about 23 bucks a share. But we ’ ve decided to purposely keep it low. Out of reach. Median home price has skyrocketed relative to median household income. As a result, pre-pandemic, the average mortgage payment was 1, 100 dollars, it ’ s now 2, 300 dollars because of an acceleration in interest rates and the fact that the average home has gone from 290, 000 to 420. By the way, the most expensive homes in the world, based on this metric, are number three, Vancouver. Why? Because 60 percent of the cost of building a home goes to permits. Because guess what, the incumbents that own assets have weaponized government to make it very difficult for new entrants to ever get their own assets, thereby elevating their own net worth. This is the transfer I ’ m going to be speaking about. This has resulted in an enormous transfer of wealth, where people over the age of 70 used to control 19 percent of household income, versus people under the age of 40, used to control 12. Their wealth has been cut in half. This isn ’ t by accident, it ’ s purposeful. This is me at UCLA in 1987. I know your first thought is I haven ’ t changed a bit. This is also Mia Sevario, who is the analyst who put together these slides. By the way, Mia is 26. I did the math, just by virtue of her being in this audience, it brings the average age of the entire conference down 11 days. When I applied to UCLA, the admissions rate was 76 percent. Today, it ’ s nine percent. I received a 2. 23 GPA from UCLA. I learned nothing but how to make bongs out of household items and every line from”Planet of the Apes.”And the greatest public school in the world, Berkeley, decided to let me in with a 2. 27 GPA. And that ’ s what higher ed is about. Higher ed is about taking unremarkable kids and giving them a shot at being remarkable. And every year it ’ s gotten more expensive. Higher ed and homes and the ability, not only is higher ed incredibly expensive, it ’ s not accessible. Because me and my colleagues are drunk on luxury and I ' ll come back to that. We ' ve embraced the ultimate strategy. Me and my colleagues in higher ed wake up every morning and ask ourselves the same question when we look in the mirror. How can I increase my compensation while reducing my accountability? And we have found the ultimate strategy. It ' s called an LVMH strategy, where we artificially constrain supply to create aspiration and scarcity such that we can raise tuition faster than inflation. And old people and wealthy people have done the same thing with housing. All of a sudden, once you own a home, you become very concerned with traffic and you make sure that there ' s no new housing permits. And here is a memo to my colleagues in higher ed : we ' re public servants, not fucking Chanel bags. Harvard is the best example of this. They ' ve increased their endowment in the last 40 years and have decided to expand their enrollment, their freshman class, by four percent. Any university that doesn ' t grow their freshman class faster than population that has over a billion dollars in endowment should lose their tax-free status because they ' re no longer in higher education. They ' re a hedge fund offering classes. My first recommendation, Biden should take some of that 750 billion earmarked to bail out the one third of people that got to go to college on the backs of the two thirds that didn ' t and give a billion dollars to our 500 greatest public institutions, size-adjusted, in exchange for three things. One, they use technology and scale to reduce tuition by two percent a year, expand enrollments by six percent a year, and increase the number of vocational certifications and non-traditional four-year degrees by 20 percent. Where does that get us? In just 10 years, in just ten years, that doubles the freshman seats and cuts the cost in half. This isn ' t radical. This is called college in the ' 80s and ' 90s. Another transfer of wealth. Look at what ' s happened to wages. Oh, they ' ve gone up? Not as much as corporate profits. There ' s a healthy tension between capital and labor. But for the last 40 years, capital has been kicking the shit out of labor. Well, you think, what about wages, right? They ' ve gone up. Well if you compare them to the S and P, they barely register. It ' s been an amazing time to own assets. But your attempt to get the certification or the income such that you can acquire assets has gotten harder and harder. In my class of 300 kids, it ' s never been easier to be a billionaire, it ' s never been harder to be a millionaire. By the way, our job in higher ed isn ' t to identify a top one percent of people who are freakishly remarkable or have rich parents and turn them into a **super** class of billionaires. It ' s to give the bottom 90 a chance to be in the top ten. You know who doesn ' t need me or higher education? The top 10 percent. The whole point of higher ed is to give the unremarkables, i. e. yours truly, who was raised by a single immigrant mother, a shot of being remarkable. The transfer has been purposeful. While the cohorts, corporations and the ultra wealthy continue to garner more and more of our wealth, we have decided,"I know, if they win the gold, let ' s give them the silver and the bronze and let ' s lower their taxes."This transfer is purposeful. It ' s not by accident and it works. Senior poverty is way down and we should celebrate that. Meanwhile, child poverty is flat to up. The third rail. I ' m going to talk about Social Security. It would cost 11 billion dollars to expand the child tax credit. But that gets stripped out of the infrastructure bill. But the additional 135 billion dollars a year to Social Security, that flies right through Congress. And every year we transfer 1. 4 trillion dollars from a cohort that is increasingly doing less well to the cohort that is the wealthiest cohort in the history of this planet. I ' m not against Social Security, but the criteria should be if you need it, not whether you have a catheter. Eighty percent of you, 80 percent of you have absolutely no reason to ever take Social Security. It is bankrupting our nation. And we have fallen under this mythology that somehow it ' s this great social program. No it ' s not. It ' s the great transfer of wealth from young to old. How is this happening? Because our representatives are in fact, representative. Old people vote. Washington has become a cross between the"Land of the Dead"and"The Golden Girls."Quite frankly, this is fucking ridiculous. And if I sound ageist – If I sound ageist, I am. And you know who else is ageist? Biology. When Speaker Pelosi had her first child, get this, two thirds of households didn ' t have color televisions, and Castro had just declared martial law. But she ' s supposed to understand the challenges of a 17-year-old girl who ' s 5 ' 9 ' ', 95 pounds, getting tips on dieting and extreme dieting from Facebook? She ' s supposed to understand the challenges that a 27-year-old single mother faces? By the way, young and dreamy. Young and dreamy. The great intergenerational theft took place under the auspices of a virus. I know, let ' s use the greatest health crisis in a century to really speed-ball the transfer. This is the Nasdaq from 2008 to 2012. We let the markets crash. And by the way, you need churn, you need disruption because it seeds and recalibrates advantage and wealth from the incumbents to the entrants. It ' s a natural part of the cycle. But wait, lately, no, a million people dying would be bad. But what would be tragic is if we let the Nasdaq go down and guys like me lost wealth. So we pumped the economy, which again, increased the massive transfer of wealth. The best two years of my life? COVID – more time with my kids, more time with Netflix, and the value of my stocks absolutely exploded. And who has to pay for my prosperity? Not me. Future generations who will have to deal with an unprecedented level of debt. Why am I here and why do I get the prosperity I enjoy? Because in 2008 we bailed out the banks, but we didn ' t bail out the economy. We let the markets fall. So as I was coming into my prime income-earning years, I got to buy, no joke, these stocks at these prices. This is where those stocks are now. Where does a young person find disruption? When you bail out the baby-boomer owner of a restaurant, all you ' re doing is robbing opportunity from the 26-year-old graduate of a culinary academy that wants her shot. We need disruption. I just like this slide. It has no context or relevance. We ' re economically attacking the young, but I know, let ' s attack their emotional and mental well-being. Let ' s take advantage of the flaws in our species with medieval institutions, Paleolithic instincts, and godlike technology. I ' m just going to say, I think Mark Zuckerberg has done more damage to the young people in our nation while making more money than any person in history. Oh, but wait, it could be worse. It ' s as if we let an adversary implant a neural jack into our youth to raise a generation of civic, military, and business leaders that hate America. How can we be this stupid? This all adds up to a bunch of graphs all headed up into the right. And what are they? What ' s the first one? Oh, that ' s self-harm rates, which have exploded, especially among girls since my colleague Jonathan Haidt pointed out, it ' s really, really gone crazy since social went on **mobile**. What ' s the next one? Teens with depression. The next one, men and women not having sex. Biggest fear of my parents was that I was going to get in too much trouble. My biggest fear, honestly, is that my kids aren ' t going to get into enough trouble. My advice to every young person watching this program is go out, drink more and make a series of bad decisions, it might pay off. Next graph, cumulative gun deaths. You ' re more likely to be shot in the United States if you ' re a toddler or an infant than a cop. Next graph, obesity, way up. By the way, the industrial food complex wants to addict you to shitty, fatty foods so they can hand you over to the industrial diabetes complex. We should not romanticize obesity. You ' re not finding your fucking truth. You ' re finding diabetes. Overdose deaths, way up. Deaths of despair. When I was in high school, it was drunk **driving**, now it ' s kids killing themselves. Young people don ' t want to have kids anymore. Two thirds of people aged 30 to 34, able-bodied, used to decide to have at least one child. It ' s been cut in half. It ' s now less than a third, 27 percent. As a result, people over the age of 60 in the US, pretty happy. People under the age of 30, not so much. Some of the lowest in the free world. What can we do? Nothing wrong with America that can ' t be fixed with what ' s right with it. We got the hard stuff figured out. There are programs to address all of these issues, they cost a lot of money, that ' s the hard part. And we have figured this out. In just five minutes post an earnings call, we can add a quarter of a trillion dollars to the economy. We ' ve got the hard part figured out, the resources. We have the money, but we decide not to do it. This is per-capita spending on child care in the United States relative to other nations. This is housing permits. Things are doable. We increase minimum wage at 25 bucks an hour, it goes into the economy, the wonderful thing about low- and middle-income households is they spend all their money. We have to have a restore or a progressive tax structure with alternative minimum tax on corporations and wealthy individuals. We need to refund the IRS. We need to reform Social Security. It should be based on whether you need the money, not on how old you are. We need a negative income tax. My friend Andrew Yang screwed up a great idea, he branded it incorrectly. Instead of calling it UBI, he should got Republicans on board by calling it a negative income tax. We need to eliminate the capital gains tax deduction. When did we decide that the money that capital earns is more noble than the money that sweat earns? Shouldn ' t it be flipped? We need to remove 230 protection for all algorithmically-elevated content. We need identity verification. The reason we can have identity verification is because we have a First Amendment. Break up Big Tech. We have monopolies that are incurring greater and greater costs on every small business and parents because again, see above, our representatives don ' t understand these technologies. We need to age-gate social media. There ' s absolutely no reason anyone under the age of 16 should ever be on social media. We need universal pre-K. We need to reinstate the expanded child-tax credit. We need term limits, see above, Andrew Yang. We need income-based affirmative **action**. Any visible signs of affirmative **action** make no sense at all. You would rather be born gay or non-white, in the United States today, than poor. And that ' s a sign of our progress and our need to recalibrate who we give advantage to. Affirmative **action**, of which I ' m a beneficiary, I got Pell Grants, I got unfair advantage. Affirmative **action** is a wonderful thing and it should be based on color. It should be based on green. How much money you have or don ' t have. Expand college enrollment in vocational programs. Mental health, ban phones in schools, invest in third places, Big Brothers and Sisters programs, we need national **service**. We need to tell people in the United States and Canada that they live in the greatest countries in the world, and we need to remind them of that every day by exposing them to other great Americans where they feel connective tissue. We can do all of this. We can do all of it. We have the resources. The question is, do we have the will? This is my last slide. It is an emotionally manipulative slide to try and get you to like me more. But it does have a message. This is the whole shooting match. Anybody here without kids ask someone with kids. You have your world of work, you have your world of friends, you have your world of kids. Something happens here, your whole world shrinks to this. So I present, as I wrap here, with just a few questions. One, if you acknowledge that our kids are the most important thing in our lives, that everything else we do here is meaningful, but our kids ' well-being and prosperity is profound. If you acknowledge that they ' re doing more poorly than previous generations, if you believe there ' s a chance that the illusion of complexity has done nothing but provide cloud cover for the unbelievable transfer of good will, of well-being and of prosperity from young to old, and if you believe we can actually fix these problems and we have the resources, then I present to you, I posit, I augur the question that I hope has more voracity than it did 17 minutes and 24 seconds ago. And that ' s the following question. Do we love our children? My name is Scott Galloway, I teach at NYU and I appreciate your time. Thank you. CA : We cut off the standing O. That was a long, long standing O. It was as long as there was at TED this year, Scott. An amazing talk, honestly. I want to ask you a bit about just your style of speaking, because I think it ' s remarkable. I ' m going to start here. This thing ' s been seen by five-plus million people around the world already and growing. It ' s the biggest single hit coming out of the last TED. I ' m curious what feedback you ' ve had from it. I mean, you laid out some pretty savage criticisms of a lot of people there. Have some of them come back hard at you? Have you mainly been amplified and cheered on? Give us a sense of the feedback on this. SG : First off, thanks for having me, Chris, and thanks for creating such a powerful **platform** to let people like me have this type of opportunity. 97, 98-plus percent has been exceptionally positive. Not yesterday, but the day before yesterday, I was on a Zoom call where I spoke after Speaker Pelosi and before Senators Blumenthal and Blackburn, and talking specifically about my recommendations around Big Tech. I don ' t know if I ' ll be invited back because I did not hold back. But it ' s been, you know, let me talk about, it ' s been overwhelmingly positive. It ' s been everything from... Billionaires reaching out and saying,"I have a philanthropy, and I ' d like your help allocating some of our resources,"to someone sent me a screenshot of an account with 10 million dollars in it and said,"It ' s yours if you run for president in 2028."To which I responded,"I have no interest in that, but I ' d really enjoy spending the weekend in Vegas with you."It ' s been really positive. So the points of pushback that I think are interesting, and I want to be clear, I get it wrong all the time. You know, I ' d like to think that some or even most of this is directionally correct. But what I know for certain is some of it is wrong. I got pushback around Social Security from Representatives, from seniors, that this is the most successful social program in the world, and that I shouldn ' t be demonizing old people for taking out something they ' ve already contributed to. I ' ve got pushback for fat shaming. So I ' ve gotten, you know, points of pushback. But I think that ' s important. I think that if you ' re going to make provocative statements, you need to be subject to pushback. And I would say that my goal is not to be right, It ' s to catalyze a conversation such that we can craft better solutions. And the thing that ' s been really rewarding about this dialogue is that when people send me very long emails, including from Congress and the Senate and people who run companies and nonprofits, the dialogue ' s been very civil. And I would say that the thing that I really enjoy, you know, I recognize about the TED community, is I ' m in a variety of different communities, whether it ' s TikTok or whether it ' s different conferences where I say these things. And the dialogue this has inspired has been remarkably more civil. It ' s been more, you know,"I liked this, I didn ' t like this,"but I really appreciate that the medium is the message. You know, people are more civil on LinkedIn than they are on Twitter. And I have found so far people are much more civil on TED than other **platforms**. CA : Well, that ' s nice to hear. We definitely, though, don ' t want people to hold back. And I know that in the comments there are some people who ' ve pushed back on specific aspects of the talk. But which of the various proposals and things that we could actually do about this problem, do you think have most chance of actually moving forward? Because it ' s so wide ranging. You know, it ' s like, it ' s very hard for young people to get to college now, hard to get a house and then, you know, all the way up to Social Security and so forth. Where do you think there ' s most traction, Scott? Something that might actually shift. SG : I think there ' s already a movement underway to expand freshman class at elite colleges, to invest more in vocational programs, to reverse this vibe or gestalt of higher ed as a luxury brand and return it to being a public servant. I think that movement was already underway, and it feels like there ' s a lot of momentum there, and some of the stats have really freaked people out. There ' s 10 to one MIT employees for every person who teaches. So I ' ve heard from the president of ASU saying, and the presidents of Purdue saying,"Come here,"and they ' ve been circulating my talk and raising money around the fact that they have not embraced this strategy. I think we ' re seeing – I ' ve heard from quite a few people in the budget office and Congresspeople talking about things including eliminating capital gains tax or just having one income tax, similar to what we had under Reagan, instead of favoring the income that capital earns or wealthier people versus current income. I have heard a decent amount around the idea of just general, generally speaking, tax reform. How do we put more money in young people ' s pockets? And then from what I ' ll call the retail media, whether it ' s Morning Joe or The View, which I ' ve been on since this talk, it ' s we need to change the narrative and stop criticizing young people and saying that their depression, their anxiety, their obesity as a function of their entitlement and not a function of public policy that people of my age have implemented. That, OK, maybe there is some truth to some of their disappointment. CA : So let me, before we go into the other details on it, let me pick up that specific point, because there is, I ' m sure that some people watching this, probably older people, would say, are you sure it ' s really this intentional thing as opposed to a consequence of just cultural cycles? So there ' s this quote that you all know by Michael Hopf that, you know, hard times create strong men, strong men create good times, good times create weak men, weak men create hard times. That there ' s an argument that we ' ve been through so much prosperity that, you know, baby boomers, raised, you know, a generation of kids who just grew up too comfortable and that that is where part of the problem is. You are halfly arguing a different case. Is there any truth to that argument? SG : I think in almost every generation before ours, I ' ll call it Gen X and baby boomers, there was a concerted effort to elect people who would think long-term and make forward-leaning investments. If you think about Apple, Google, Nvidia, all of these companies, Moderna, that have added trillions of dollars in market cap, it ' s really, they build a thick layer of innovation on top of investments that have been made by the most successful venture capitalists in history, and that ' s the US government with its limited partners, its investors, the middle class. And that there ' s a lack of that forward-leaning investment. Basically, the only thing that passes for bipartisan cooperation now is reckless spending. Republicans want more military spending and lower taxes. Democrats want more social spending and, you know, higher taxes. And they agree on lower taxes and more spending. So I just feel like we are being irresponsible, even reckless. The way I loosely describe it is seven trillion dollars in government spending which juices the economy, five trillion in receipts. I ' m in the club partying with champagne and cocaine, and the closest the young people get to the club is they can throw their credit card at me from downstairs, and I ' ll rack up more debt on their backs so I can continue this prosperity. Even the markets right now, we don ' t want to talk about this. The markets are being supported right now by an additional two trillion dollars in stimulus every year, in the form of government spending that ' s beyond our revenues. In past generations, unless it was wartime, never would have engaged or supported that type of government spending. So I do think there ' s something different here. I think the financialization of everything, Chris, where, as they said in"Jerry

Maguire,"more money used to be a bigger seat in business class, now it ' s a better life. And the fact that we haven ' t faced really any external threats to rally us together to create more comity of man or more patriotism, there just seems to be not the same sense of selflessness or investment **mindset** there ' s been with past generations. CA : One of the most powerful aspects of the talk, and there were so many powerful aspects, was your juxtaposition of different stats. So Harvard, this explosive growth in the endowment and, you know, no more students being admitted. I mean, you immediately know, when you just put those numbers together that there ' s something very, very wrong with that picture. So powerful. I thought your line about just showing, you know, minimum wage has just not remotely caught up with the way at which capital is growing. And I wonder whether one, you know, there ' s almost like some political ideas that we could push out there that said things like that. A society is not operating justly if the rate, its minimum wage rate, is growing slower than the rate at which capital is appreciating. It shouldn ' t be linked to inflation or even wages. It should be linked to the rate at which capital is appreciating you know, in that society. Otherwise, inequality is absolutely bound to increase. And I ' m curious which others of those have landed, like with Social Security, for example. I understand why people have pushed back. Do you see any workable, kind of, middle ground here where you could say, look, Social Security is important, we ' re not going to take the whole thing away. But we have to make it more just, we can ' t have it sucking more and more of the next generation ' s time. Like, what is the right metric to think about how much Social Security could be trimmed to be fair to the generation coming through that? Feels like one of the biggest numbers. SG : So I ' ve had a bunch of calls with people, different people in Congress and different PACs about Social Security. So means-testing and moving the age back, right? I mean, the majority of people, just even 100 years ago, didn ' t participate in Social Security because they were dead by that time. Now they ' re living 20 and 30 years beyond that. People will say,"Well, I invested in it, I want it back."I ' d say first it ' s called a tax, not a pension fund, meaning that it might be distributed to other people. Two, people who will live to be 85 take out two to three times more than they put in. I ' m not saying we should do away with it. I ' m just saying it should be means-tested, the age should be pushed back to reflect that people are working much longer. Also just basics. The person who put together the slides, Mia, makes 160, 000 dollars a year, I ' m allowed to say that with her permission. She ' s 26, very talented young woman. She pays 9, 000 dollars a year, or six percent, in Social Security. This year, I ' ll make 100 times that, and I ' ll pay, wait for it, 9, 000 dollars in Social Security income tax. It ' s capped for anyone making over 160, 000 dollars. So if this generation is the wealthiest generation in history, why have we decided to cap Social Security for rich people? So I don ' t have a problem with stimulus. I don ' t have a problem with social programs. But Social Security is another example yet again, of we are really taxing – six percent is a real number for young people their whole life, but it ' s been de minimis for me because I ' m old and wealthy. So it ' s another example of how this transfer under the cover of dark, that wealthy people are in there saying,"We ' ll position this tax as onerous, but we ' re going to cap it like it ' s a good thing at six percent."Why is Jeff Bezos not paying six percent of his income into Social Security? It ' s an off-balance-sheet item, it doesn ' t impact the deficit, it ' s funded every year. But right now, 40 percent of all government spending is allocated for programs that go to seniors. It ' s about to be 50 percent in ten years at this rate, which means that crowds out investments and forward-leaning investments such as education and technology. And we also just need to acknowledge and this sounds ageist, and it is, old people are less productive and more expensive. And so unless we figure out a way to acknowledge that they spend a longer period of their lives being expensive and unproductive, our economy is basically just going to be an engine to kind of support the most expensive nursing home in the world. This is happening in Japan, it ' s happening in Italy, and it takes an economy into decline, because you can ' t make forward-leaning investments that encourage wealth or inspire wealth for young people, and they don ' t have kids. In terms of minimum wage, this is an easy one. I ' ve been positioned as anti-union. I ' m not, there should be one union. It should be in DC, and it should be federally-mandated minimum wage, 25 bucks an hour, except in certain areas where there ' s an exceptionally low cost of living. And the incumbents and corporations will start this bullshit narrative of it ' ll kill the economy. No, it won ' t. In Washington state and California, where they raised minimum wage, the economy actually grew because the wonderful thing about low- and middle-income people is they spend all of their money, meaning the multiplier effect is greater. If it had just kept pace with productivity and inflation, it ' d be 23 bucks. That ' s an easy one. Mandatory, federally-mandated, 25 bucks an hour. Would McDonald ' s and Walmart stock take a huge hit? Sure. Would a bunch of small businesses go out of business? Yes. And it ' d be worth it. CA : So, as you say, you don ' t hold back. And there were definitely a few things in the talk that provoked comment. So Kyra Gaunt, I see in the chat, found your comment about diabetes a bit smug. She says there are epigenetics to consider. It ' s complicated for members of marginalized groups. And you say you ' ve heard from other people on that. Do you want to clarify your view on that at all? SG : It ' s a fair feedback. As someone who was born to parents who were both tall and thin, so easy for you to say, Scott. What I have said in the past and you know, if I sound offensive, I apologize, is I believe that we need bottom line to put more money in the pockets of lower-income people such that they can eat better. I think we need to tax the food industrial complex relative to its externalities. Obesity, morbid obesity has gone from five to nine percent in the last 30 years. Obesity has gone from, I believe it ' s gone from 30 to 45 percent. The only thing that Americans really share is it ' s 70 percent of us are overweight or obese. And if you look at McDonald ' s, Kraft, General Foods, PepsiCo, Coca-Cola, they ' re not companies as much as they are obesity indices. And their stocks have gone up 10 to 12x as obesity has gone up. And if you give those two, their stocks and obesity rates, to a statistician, they will tell you they are highly correlated. So as a result, because there ' s money in the diabetes industrial complex, we have Unilever celebrating obese women. We have people saying you ' re finding your truth. And this is from the same company that sells Ben and Jerry ' s and Axe, which basically encourages young men to fuck anything. So the notion that they are concerned about, are trying to liberate people or all of these apparel companies saying that you ' re finding your truth – I want to go back to the ' 60s, where Kennedy said it was American and patriotic to be in great shape. I used to train every year for the Presidential Fitness Awards, and that was seen as fat shaming, so they did away with it. I don ' t think there ' s anything wrong with celebrating fitness, but also at the same time recognizing some people are dealt a difficult hand genetically and providing them with money and resources so they can get out of food deserts. I also believe GLP-1 drugs are the most impressive technology of the last year, not AI, and we should be pushing them into low-income communities. I think they ' re an absolute game changer right now. Here ' s a crazy stat : the region of America that has the greatest penetration of GLP-1 prescriptions is also the thinnest. It ' s the Upper East Side. Because right now, GLP-1 drugs are being used for ladies of lunch who want to lose that last 10 pounds. I hope that more and more of it is imported illegally across the border, where it ’ s 100 bucks a month versus 1, 000, and it gets to the people who really need it, which is low-income people who don ’ t have the money for diabetes. Because the industrial food complex wants you to be obese so they can hand you over to the medical industrial complex of hip replacements, kidney dialysis, knee replacements, heart stents. That ’ s an enormous industry. I want to put Coca-Cola, Kraft, McDonald ’ s and some of these hospital systems absolutely out of business. There ’ s too much incentive to convince you that being obese is some sort of personal liberation. It ’ s not. CA : So this is so interesting. Like, somehow, and I feel this for TED as well, we have to figure out how to have conversations about difficult topics that are respectful of people. And, you know, everyone, as you say, has been dealt a different hand, but that don ’ t ignore the basic facts of health or of science. And that is what is so hard to do in the current environment right now. We ’ ve all got frightened, I think, of saying the kind of thing that you just said. And there probably are still, I don ’ t know Kyra, how you felt about what he replied there, but I feel like we need to do both, you know, we need to be provocative and also respectful. And I heard that in you. So I think that ’ s cool. And then again, actually not to pick on Kyra, but I noticed that she pushed back on your point about social media, that, you know, one of the fixes is to get our younger kids off of social media. Points out, and I ’ ve felt this reading Jon Haidt ’ s work myself, that those communities that people connect with on social media sometimes are incredibly beneficial. So you think of the kid who ’ s queer or is **struggling** with something and doesn ’ t find anyone in their community who they can connect with, finds a community online. I mean, there ’ s presumably some benefit in that. Is there any way of making recommendations where there aren ’ t swings and roundabouts? There aren ’ t, you know, sort of downsides to what we recommend? SG : Look, one in five LGBTQ high schoolers is going to try and kill themselves. So I ’ m involved with the Jed Foundation that works with high schools to try and identify or distinguish between kind of what is normal abnormal teen behavior and suicidal ideation. Because we really have an epidemic of self-harm at a teen level, and there ’ s just no getting around it. Some trans and gay kids and other kids from **special** interest groups find other people like them on social media. I guess the question would be if banning social media under the age of 16 or age-gating it on the whole were demonstrated to be an absolute huge net positive – because I think a lot of those communities are actually the subjects of incredible bullying and shaming online, and are more prone to go down a rabbit hole of depression and end up in self-harm – are there other programs that could replace that incredible community some of them have found? But look, when you make huge government decisions, there is no decision where there ’ s not going to be losers. I mean, the special-interest-group kid who finds people online at 14, when social media is age-gated, yeah, that person suffers. I guess the argument I would make is that if you were to look at any study about the general impact it ’ s having on people under the age of 16, the benefits dramatically outweigh the soft tissue here. So the question is, could we do it? But at the same time recognize we still have an enormous self-harm problem among kids, especially in the LGBTQ community, and move in with other programs or other analog means of helping them find other people. But I want to acknowledge, there ’ s just nothing I recommend that doesn ’ t have a downside. CA : Some people are asking about practical things we can do. So Adrian Neubauer said,”As an elementary school teacher, what ’ s your advice for helping Gen Alpha students? I don ’ t want my students to grow up having such apathy towards learning and being successful. I definitely don ’ t want my students ’ highest aspirations to be TikTok influencers. How do I help them?”SG : Well, the first thing is, I think this is an easy one, I don ’ t think there ’ s any reason for anyone under the age of 16 to be on social media. I think we need a massive investment. I think we need to tax every person in a private school and use that capital to invest in after-school programs and third places. I went to public schools all the way through graduate school. I went back to my high school, university high school, and 93 percent are kids of color. It ’ s got the unfortunate designation of having the greatest proportion of kids who are classified as homeless in LAUSD, and they just cut their drill team. They just cut their band, their drumline, because they don ’ t have money. I just think so much of this, Chris, unfortunately, it ’ s like households with a lot of anxiety. It sounds very crass, but I think a lot of it, a lot of problems are solved with money. Show me divorce and I ’ m usually going to show you some sort of economic **stress** or poor alignment around economics. And when you can add a quarter of a trillion dollars in five minutes post the earnings call of Nvidia, and at the same time, you see corporations are paying their lowest taxes since 1939 and the top 25 wealthiest Americans are paying a tax rate of six percent, and five companies have added the value of the entire global auto industry in the last six weeks. We have the resources. So I think a lot of this comes down to funding, giving kids more third places such that they ’ re not staring at their phones. And also demanding that parents and demanding that entire school systems go phone-free and that there ’ s no social media. And on a more tactical level, I absolutely think we need to ban or divest TikTok. I think it makes absolutely no sense to be raising a generation of civic, military and nonprofit leaders who hate America, and it ’ s all wrapped in cute dances. I just think it ’ s insane that we would be this stupid. One of the wonderful things about America is our optimism. But one of the downsides of that is we ’ re much easier to **fool** than convince we ’ ve being **fooled**. I think every day we ’ re being **fooled** by TikTok. I think this is the most unbelievable propaganda tool in history. Unfortunately, it ’ s not ours. CA : You mentioned Nvidia there, which has exploded in value in the last couple of years, last year really. Isn ’ t that to some extent a counterargument to your earlier comment about you had this unique opportunity to buy these cheap stocks like Netflix and Amazon at low prices. I mean, there are always... Is the market fundamentally changed, there aren ’ t that kind of opportunity for smart people to identify trends and get the same kind of wealth? Was it just a timing thing? SG : No, what we ’ ve decided is we want capitalism on the way up. We want low taxes, and the pioneer, the individual who earned the money and should pay a low tax rate because he or she is the most productive citizen. And then on the way down when there ’ s a virus or some exogenous event, which happen a lot, whether it ’ s war, famine, revolution or a virus, the CEO of Delta says,”We ’ re all in this together,”and wants a massive bailout, despite the fact that 80 percent of the free cash flow of airlines the 10 years before were either used on dividend or stock buybacks or CEO compensation that juice their compensation. So look, capitalism on the way up and socialism on the way down is cronyism. And we ’ ve gone full cronyist in the United States. And I understand the rationale for pumping the economy full of stimulus, because they said we needed to overdo it versus underdo it, but then no one suggested a **special** tax for the unbelievable champagne and cocaine disco party we ’ ve had in the markets. Markets are touching all-time highs right now. And the myth that my generation is fomented across the economy, the young people take hook, line and sinker is the following. There are two phases of everyone ’ s life. There ’ s investing, and then there ’ s the harvesting phase. Investing is when you do your best to make more than you spend, and save some money, so you can deploy an army of capital that grows in your sleep, such that you can have some security and some balance as you get older, and you begin harvesting. I ’ m entering the harvesting stage of my life where I ’ m no longer making as much as I spend. During the investment part of your life, in you ’ re younger years, you want markets to crash. You want real estate that ’ s inexpensive. You want to dollar-cost-averaging at the low price. The reason I get to roll with you in London and I get to go to the south of France tomorrow, Chris, is because we let the markets crash in ’ 08, and I got to buy Netflix at 12 bucks, which is at 650 now. What we ’ ve decided now purposefully, is that we aren ’ t going to let the markets have a natural cycle of disruption, that anytime the markets are threatened, we ’ re going to weigh in with the credit card of our young friends, of our daughters and our sons, and we ’ re going to artificially inflate and support the markets. That is nothing but a transfer of wealth. You need disruption. You need churn. These are purposeful decisions to protect old on the credit card of the young. I don ’ t have a problem with stimulus, but for God ’ s sakes, those of us who have benefited from it should pay it back. CA : So there ’ s a great question here from Brett McCall.”During this talk, there are an overwhelming number of points that are all delivered in an easy-to-understand way.”That ’ s true, by the way. I don ’ t know of a talk where there ’ s been such density of points, of facts that connect, really remarkable. Anyway, he says,”It leaves me charged with the desire to join, ’ the movement ’ and make change. It ’ s almost like a full semester-worth of issues that could be movements, but there isn ’ t an explicit community or movement to join. Scott, where would you suggest a starting place for someone who becomes inspired by your talk?”SG : It ’ s hard to do this without being political, but try and find Congresspeople or get involved in campaigns of people who are, one, talking about the deficit, talking about the war on young people, having difficult conversations. I mean, there ’ s some very basics. And the speaker before me, Andrew Yang, final five and ranked choice voting. People who are going to be moderates, who are going to think long-term about society. I think we need absolutely more young people in Congress. So I would say political **action**, just being engaged. And then on a ground level, getting involved with local schools, getting involved in charities and efforts to help young people in **special** interest groups find third places, find places to find each other. I think all of this is somewhat couched in young people, in an epidemic of loneliness. So anything you can do to create resources and opportunities for young people to get out of the house. I think it ’ s up to parents to get their kids off – I mean, I think it ’ s not what to do, it ’ s what not to do. In terms of a political movement, I ’ m still waiting for a leader to come forward and say,”I ’ m going to piss off everybody. Old people, you ’ re going to hate me. Unions, you ’ re going to hate me. Rich people, you ’ re going to hate me.”You know what we need? We need the biggest class traitor in history. Someone who shows up and says,”Are we going to decrease government spending, or are we going to raise taxes?”And the answer is yes. It ’ s time to start acting like adults and paying it back to some of the incredible Americans who have made sacrifices such that we can enjoy this prosperity. I haven ’ t seen that person yet. CA : Politicians assume that that ’ s an unelectable slate. And maybe they ’ re underestimating the intelligence of their citizens. SG : I think people are ready. Do you know one of the top three issues among young people is right now? People think it ’ s Middle East, it ’ s not. That ’ s like number 17 because of the zombie apocalypse of useful idiots on campuses that has been highlighted in media and in TikTok to give you the sense that every student is protesting. 2, 300 arrests, 40 weren ’ t students. I ’ m at NYU, 99. 9 percent of the kids are just getting their shit done. The new undergraduate president of Columbia is an Israeli girl. You know, we need... anyways, one of the top, top issues for young people, one of the top three? The deficit. So I just think we need a pragmatist who says,”I am not going to ever insult anyone on the other side of the aisle personally to get on TikTok and raise money. I ’ m going to talk about really boring shit. Tax policy, the deficit, teen depression, obesity. And I ’ m going to make big sweeping changes. And guess what? All of you are going to suffer, all of you.”Because there ’ s just no getting around it. We need to make a fraction of the sacrifices that a lot of Americans made 80 years ago. I don ’ t know about you, I was very moved by the D-Day commemorations. But every generation up until this one has made real sacrifice for forward generations, except this one. So we need to catch up. I think people are ready for that. CA : Scott, you ’ ve spent a lot of time giving advice, especially for young men. Do you think that the problem has been especially bad for them for some reason or that a disproportionate share of the solution will be found by fixing the problems that young men are facing right now? SG : Well, all of these things are especially acute. So with young men – and this is really the thing, this is my passion project – four times as likely to kill themselves, three times as likely to be addicted, 12 times as likely to be incarcerated. There has been no group globally that ’ s ascended faster than women. More women globally are pursuing tertiary education, which is remarkable when you think about – many nations don ’ t allow women to pursue tertiary education, and twice as many women have been elected to parliament over the last 30 years around the world. Now domestically in the US, there has never been a cohort that has fallen further faster than young men. More women, single women own homes now than single men, which is fantastic. In urban areas, women are out-earning men. These things are wonderful. We don ’ t have a homeless crisis. We have a male homeless crisis. We don ’ t have an opioid addiction crisis. We have a male opioid addiction crisis. And unfortunately, something that gets in the way of programs, Chris, is that because of the benefit, the massive benefit, privilege and advantage that you and I received, we hold these young men accountable. We use words like accountability or pull yourself up by your bootstraps. If any **special** interest group is killing themselves at four times the rate of the control group, we ’ d move in with programs. And I think there just generally needs to be a change in the narrative and the dialogue. And we need to recognize that **empathy** isn ’ t a zero-sum game. Gay marriage didn ’ t hurt heteronormative marriage, civil rights didn ’ t hurt white people. And recognizing the problems that young men especially are facing – one in three men under the age of 30 hasn ’ t had sex in the last year. Three million working-age men have given up looking for work. Fifty percent of millennial men are no longer even trying to date. And so you have this cohort that is doing so poorly, and if we don ’ t have a vibrant middle class full of successful, economically and emotionally viable men, the nation is going to collapse. Because here ’ s the thing about women, we don ’ t want to have an open and honest conversation about mating. And this is the following. Women mate, socioeconomically, horizontally and up ; men, horizontally and down. And when the pool of men available horizontally and up keeps shrinking, we ’ re going to have a lack of household formation. Sixty percent of people aged 30 to 34, 40 years ago, had at least one child. Now it ’ s 27 percent. The whole shooting match, the whole shooting match, why we engage in all this shit, is so we can find someone we love and raise kids together. That is the most rewarding thing. Everything else is a means. That ’ s the ends. And when a nation is failing to do that and just creating a generation of anxious, obese and depressed kids who don ’ t like themselves, much less have the opportunity to like other people and get together and start having sex and fall in love and have kids and have economic prosperity, then what the fuck is any of this for? So I think this is a call for all hands on deck. If young people, and especially young men, aren ’ t doing well, then we are not going to have a functioning nation or economy, full stop. And it ’ s time to put the politically correct agenda away and say,"Well, you ' re being sexist."Yeah, I ' m sexist. I believe the genders are different. If you don ' t believe me, put a three-year-old girl and a three-year-old boy in a room with cars and dolls and see what happens. That doesn ' t mean we reduce **empathy** for either gender, but we can ' t even have an honest conversation about this stuff because someone sees an opportunity to virtue signal and get a Guardians of Gotcha pin rather than actually addressing the fucking problem. That was a rant. CA : Well, I will say that you rant better than anyone I know. I mean, I really, just as someone who ' s hosted a lot of TED Talks, I was stunned by how this one went. Like many, many speaker coaches looking at you would say, Oh, he ' s very monotone. It ' s very, you know, where ' s the – it just unwraps in a very, sort of flat voice initially. But every word is so carefully chosen and follows so consequentially to what ' s just gone before, that you tap into a different part of people ' s brains than most talks do. People go, yes, you know, that makes sense, that makes sense. And then it builds, and then your anger and frustration starts to come out and get dialed up. It ' s absolutely remarkable rhetoric. I don ' t know how you learned it or where you got it from, but it ' s really unique out there, I would say. And I think it ' s a **super** powerful weapon. And I mean, I don ' t know if people listening agree with this. This is really unusual... It ' s not like everyone will agree with every single word, Scott, that you said, but it ' s a really unusual and powerful piece of rhetoric about something that really matters. And for that, I feel, I think we all feel huge gratitude to you and wish you well on this continued journey. I mean, there just isn ' t a more important issue. So bravo to you. Thank you. SG : Chris, I appreciate the generous words. And what I told my wife is that after 30 years of working my ass off, I ' m an overnight success because of Chris Anderson and TED. So thank you. CA : You had visibility everywhere, and you do. It ' s amazing how much your voice is out there. If people don ' t know it, Scott has his own – Well, just talk about the two or three podcasts you ' d most like people to do, because I think that ' s the best way to get closer to you. SG : Yeah, to resist is futile, I ' m like AOL in the ' 90s. If you stick your hand in a cereal box, you ' re going to find something from Scott Galloway. I have"Pivot,"I have"Prof G,"I have my newsletter,"No Mercy, No Malice"and a slew of books. So to resist is futile. [Want to support TED?] [Become a TED Member!] [Learn more at ted. com / membership]

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Hello and welcome. I ' m Elise Hu, the host of the TED Talks **Daily** podcast, and I ' m so glad to be with all of you today for our TED Talks Daily Summer Book Club, a **special** series on the TED Talks **Daily** podcast. It ' s a show where we deliver ideas that inspire every day. So today we are slowing down to ask : What are our phones doing to childhood, and to us? Around 2010, we all **observed** the mental health of young people in the US starting to get worse. Rates of depression, anxiety, suicide and self-harm started climbing up and haven ' t come down. In his new book,"The Anxious Generation,"social psychologist Jonathan Haidt makes a strong case that the cause is smartphones. He looks to Gen Z as an example. They came of age with unfettered access to the internet and social media, and he argues that these numbers are the consequence of the new childhood reality. John Haidt, welcome and thank you for digging into this with us. Jonathan Haidt : Well, thank you Elise. EH : So much of this book really hinges on the moment around 2010, when a few dramatic changes took place in the digital world. Talk to us about what happened then and why you consider it a big deal. JH : So let me actually start in 1990, because you have to understand how we all got tricked into this. So if we go back to 1990, there was no internet, nobody knew what the internet was. So the internet arrives around 1994, 1995, and it ’ s amazing. It ’ s like God said,”Hey, do you want to know anything, instantly?”I still remember how exciting it was. So the internet was amazing. And the millennials were teenagers at the time, they were going through puberty, and they charged onto it. And they made it their own, and found all kinds of ways to do things. And they started internet companies and they are creative, successful generation. Also, the Berlin Wall fell just before that. And democracy has ascended in the ’ 90s, and we ’ re thinking, democracy, its best friend is the internet. How could a dictator ever keep it out? Good luck China, keeping out the internet. So we were all **super** optimistic. Once you get the smartphone, 2007, now you start getting the App Store and apps. You get Uber. So all of this is miraculous. So our kids love it, kids always love technology. And we ’ re all like, well, OK. They ’ re spending a lot of time on it, but you know, maybe it ’ s making them smarter, it ’ s going to teach them tech skills, so this is all good, we thought. So that ’ s the setup. And if you were born in 1990, let ’ s say, then you hit puberty around 2002, 2003. You go through puberty with a flip phone. You use your flip phone to call your friends and to text them one-on-one, and you meet up and you do things in the real world, so you ’ re fine. You have a normal human development, normal human puberty, and you come out the other end as a mentally healthy person. But suppose you ’ re born in 2000. You are seven when the iPhone comes out. You got a front-facing camera in 2010. You are 11 when you probably got one. You got on Instagram when you were 12. So when you hit puberty, you ’ re going through puberty not meeting up with your friends, you ’ re going through puberty swiping, tapping, liking and hanging on,”What if I post something? How will people react?”Half of our kids say they are on the internet almost all the time, 50 percent, almost all the time. This is not a normal human childhood. There ’ s not as much face-to-face contact. You don ’ t get to develop social skills, you don ’ t have hobbies or read books. You ’ re just on your phone all day long. And guess what? Their mental health collapsed, especially for the girls. Instantly, it ’ s not a slow thing. Instantly, around 2012, you get these hockey stick shapes in all the graphs in my book. There was no sign of a problem in 2010, and by 2015, it ’ s all over the world. We don ’ t know about the developing world, but all over the Western world we start seeing this especially for girls. So that ’ s the story. EH : And what do you think was going wrong? JH : So the story I tell in the book is two things. I decided not just to write a book about what social media is doing. But to write a book which is really more about childhood. What is it, why do we have it, why is human childhood so different from every other animal, including chimpanzees? Because we grow fast after you ’ re born. But then you slow down. And we don ’ t grow very fast until we hit puberty. Why do we delay? We have these amazing cultural brains. This is our great adaptation. This is why we cover the world and chimpanzees don ’ t. And that all depends on a slow growth process with a lot of cultural learning from your elders, from the people ahead of you in your culture. So that ’ s part of it. Also part of it is play. Young mammals need a huge amount of play, free play to wire up their brains. All animals **practice** skills they ’ ll use as adults. So you take what I call the play-based childhood, which is what we ’ ve had for 300 million years, going back to the beginning of mammals. And then 2010 to 2015, kids now have a phone-based childhood. And that, I argue, is what ’ s blocking development. We ’ ve never seen such a sharp change in generations in terms of their mental health. So that ’ s what we see when we look back historically. We can certainly now discuss it, you know, the research, trying to pin it down.

But what I ’ m pointing to is an incredible pattern of changes that happened in many countries simultaneously, always with more increases for the girls, not affecting the middle-aged people, but only affecting the teenagers. And no one can offer another explanation other than the transformation of childhood by the technology. EH : You call this transition from a play-based childhood to a phone-based childhood a”great rewiring.”So how do you believe the minds of these Gen Z kids are wired differently now? JH : Just think about it this way. For those of you who are over 35 or 40, you surely had a play-based childhood. What I ’ d like you to do is think back on the most exciting things that you ever did as a child. Think back on times you just were hanging out with your friends. Think back on how much time you spent laughing with your friends. Laughing, joking, playing around, OK? Now take all of that, cut it by 80 percent, because Gen Z didn ’ t get that. They have almost no unsupervised time. They don ’ t get to hang out behind the 7-Eleven or down by the river, or in a playground or anywhere, except online. So take 70 percent of that out. Think of all the times you smiled at a person or you made eye contact. Take 70 percent of that out. Again, I don ’ t know the exact number, but for the kids who say they ’ re online almost all the time, it ’ s probably 70 percent. Think of all the books you read. Take 70 percent of those out, maybe 100 percent. Gen Z has no time to read books. They have so much content to consume to keep up. There ’ s very little book reading. Think about hobbies, did you have a hobby? Take that out. So you take out almost everything. Because again, if the latest numbers are on average, American kids are spending seven to nine hours a day on entertainment and screen stuff, not counting school. Seven to nine hours a day is the average, depending on how you count it. Take childhood, take out almost everything that you valued in it, and what ’ s left? EH : But does hanging out have to be in person and with making eye contact in real life? Because a lot of these kids would argue, we hang out all the time. We hang out in games, we hang out, you know, on social **platforms**, we are connecting. We ’ re just connecting in a different way. And Jon Haidt, you ’ re the fuddy duddy, you know, for coming down on just the different tools that we ’ re using to hang out and connect. JH : Oh that ’ s a great objection. And, you know, 10 years ago or 15 years ago when I first saw Twitter and people sending out,”I just had a hamburger,”I thought, God, this is so **trivial**. But then as a social psychologist, I thought, well, actually, wait, if you ’ re sort of checking in with your friends hundreds of times a day, that could be really good. But what I ’ ve come to see is the online world gives you multiple ways to connect virtually, which often lacks certain properties. So the most important one is being synchronous. So right now you and I are synchronous. We can see each other ’ s facial **expressions**. But on Zoom there ’ s a little bit of awkwardness, you don ’ t quite get the timing. So video games, Twitch, all these things that allow you to go two-way, voice and face, those have some benefits, I ’ ll grant that. Whereas going back and forth in group texts, group text is performance, not connecting one-on-one like an old phone call. So synchronous is good. All the asynchronous ones, I think, miss that. Another feature is that you now have such vast quantity. Imagine taking the number of people you talk with every day, multiply that by 50, OK, that might sound great, but how much time do you then have for anyone? So you don ’ t get the kind of in-depth conversations. When I was a kid, it was just sort of famous that girls will talk on the phone for hours and hours at night, just one-on-one, they ’ re connecting. Boys don ’ t do that, but we call each other up and say, hey, can you come out? And we ’ d meet and go do something together, one-on-one or in a small group, that ’ s all great. The large groups are not connecting ; they ’ re performing. And kids are anxious. They have to think carefully before they put something out. Because what if I make a mistake?

Play needs to have lots of low-cost mistakes. You say something stupid, your friend says,”That was mean.”And you say,”Oh, sorry.”But when you grow up online, it ’ s like growing up in a minefield because you never know, one mistake, it could actually ruin your life. It could literally ruin your life and make you not get into college. So this is something that so many professors have **observed**. We can ’ t get our students to disagree with each other anymore. They ’ re so afraid. They ’ ve grown up in a minefield. They know, one false move and their leg gets blown off. So I think, you know, you can praise all those virtual connections all you want, they don ’ t seem to have the magic ingredient of hanging out with a friend in real life. EH : And the way you explain it in the book, the rise in anxiety and depression isn ’ t really happening in **isolation** because it also sort of coincides or interacts with a change in parenting culture. So talk to us a little bit about that cultural shift that you ’ re saying is happening all at the same time. JH : A big part of this is the overprotective parenting. And, you know, I can summarize the whole book by saying we ’ ve overprotected our children in the real world, we ’ ve under-protected them online. So let ’ s look at the overprotection. Until the 1980s, American kids had a free-range childhood. I grew up in the ’ 70s, and there was a huge crime wave. There were crazy people, there were serial killers. Crime was at historically high levels. All kids went out, and it was amazing and exciting. And, you know, we got into all kinds of arguments and fights and sports games and just we had an exciting childhood. And that goes on until the ’ 90s. Now you get the new media environment, so you get cable TV, you get 24-hour news cycles. So in the ’ 90s, for a lot of reasons, we freak out and we lock our kids up, or rather I should say, we no longer think it ’ s OK for an eight-year-old to be outside unsupervised by an adult. Now it ’ s more like 10 or 11 is when kids get that level of freedom. So here ’ s the big reason why. It ’ s not just the change in television and cable TV. The biggest reason, I now believe, is the loss of trust in everyone else. There ’ s a great phrase from British sociologist Frank Furedi. He talks about the collapse of adult solidarity. So even in the ’ 70s, when there was a lot of crime, you know, if I wiped out on my bicycle and I got hurt, my friend could go knock on a door and say,”Hey, can you call my mother?”But we begin to lose trust in each other, life moves away from the streets and moves indoors as we get air conditioning and television. We don ’ t know our neighbors anymore. That ’ s why we don ’ t trust our kids to be let out. So there ’ s a whole back story that really begins in the ’ 80s and into the ’ 90s, in which we took away the play-based childhood. Oh and at the exact same time, the internet arrives. And so, you know, we don ’ t want to let our kids out. But this new internet thing, the kids love it. They ’ re sitting in the room on a computer, what could happen? EH : Five hours a day is the stat now? JH : For social media. It ’ s seven to 10, seven to nine, in there if you include video games and porn and all the other things that the kids are doing. But just social media is five hours a day – average. EH : If they ’ re spending so much time on social media, what are they doing less off? JH : Everything else. The most important things they ’ re doing less of, in order, are : any kind of face-to-face contact with their friends or family ; listening to their teachers, because they ’ re doing this in school, too, because most schools let them keep their phones in their pockets ; flirting or gossiping or talking with their friends in school or at lunch. Because even at lunch, if you have phones in your pockets at lunch, the kids are doing what ’ s called continuous partial attention. So they ’ re continually paying attention to what ’ s going on in their phones, and then they ’ re also sometimes having conversations with the kid next to them. In other words, there ’ s no real quality human connection. That ’ s the most important thing. If you take most human connection out of childhood, there ’ s not a lot left. Number two, sleep. Kids really, really need sleep. We ’ re now discovering sleep is so much more important than we thought back when I was in graduate school. Kids are getting less sleep, especially those who have a device in bed. I mean, imagine if you had to suddenly give five to 10 hours a day to some new thing. There ’ s nothing else. That pushes out everything else. EH : I want to talk a little bit about the responses to the book, because this was a number one New York Times bestseller, and there have been plenty of responses following”The Anxious Generation ’ s”release. A number of social scientists have questioned the conclusions in the book, notably this article in”Nature”that gets shared pretty widely, where Candice Odgers writes that”Hundreds of **researchers**, myself included, have searched for the kind of large effects suggested by Haidt. Our efforts have produced a mix of no, small and mixed associations. Most data are correlative.”So she continues that when they can find a relationship between phones and anxiety or poor mental health outcomes, it tends to be that teens who are already depressed tend to use social media more. So how do you respond to this, you know, correlation-not-necessarily- causation argument? JH : So let ’ s start by setting the scene. We have a massive collapse of mental health that happens in a synchronized format in many countries, especially to girls. Happens at the same time. There is no other explanation. The parents see it, the teachers and principals see it, psychiatrists, psychologists see it. So the presumption should be, something is going on here and it may be related to the technology. OK, now, what do we know from the data? There are two main battlefields. There are the correlational studies, where there ’ s hundreds and hundreds of them, they ’ re easy to do, you just look who ’ s more depressed, who spends more hours on social media. Now hours spent on social media is the variable that you ’ re correlating with some self-report of mental health. And then you look at the connection, and the correlations tend to be around 0. 1 for boys, they ’ re actually around 0. 2 for girls. This is what we ’ re fighting about. Now Candice Odgers says that this is a small correlation. She and others won ’ t be convinced unless we find large correlations, like, say, 0. 3 or 0. 4. But in public health, you rarely get 0. 3 or 0. 4 because you have very poor measurement at both ends. So that ’ s the correlational studies. We ’ re sort of at a stalemate there. The more important battleground is on the experiments. We all know correlation doesn ’ t show causation. So we move to the experiments. And what did the experiments show? I think we can show that the great majority of experiments do show a benefit from getting off social media. And very importantly, all of these studies are taking individuals and asking them to get off. And they ’ re mostly college students. And if you take a college student, say,”Hey, we ’ ll pay you to get off social media for a month. Now, how do you feel?”And it turns out they feel better, but they ’ re also isolated. The real test is what happens if you take a whole school, a whole high school, and get them off social media? My prediction is you would have a huge increase because now they ’ re not isolated, they ’ re actually more together. And the most important theme of my book is that it ’ s collective effects that have to be addressed by collective **action**. The reason every 13-year-old girl has to be on Instagram is because every other 13-year-old girl is on Instagram. My college students say the same thing. They can ’ t get off because everyone else is on. So she says, we don ’ t find the large effects that would be necessary. That ’ s because those **researchers**, they ’ re not even considering the collective emergent properties of social media. EH : Yeah, it does seem like an enormous task, though, Jon Haidt, to take collective **action** on something, because for younger generations, so much of the way that they connect these days and find community is taking place on devices. And so how do you address kind of, this problem where well everyone else is on it, I don ’ t want to be the one left out?

And how do we get past this challenge of collective **action**? JH : We get past it by acting together. Any one kid who gets off is alone. Any one parent who says,”You ’ re not getting a phone till high school,”now their kid is isolated. But there is a place where we all have community and that is our local kids ’ schools. It ’ s as simple as this. If you ’ re a parent listening to this and you have kids under the age of 12, let ’ s say, just reach out to two other parents of your kids ’ friends and say,”We want to do the four **norms** from ’ The Anxious Generation. ’ Are you with me?”And they probably will be with you. The four **norms** are : no smartphone till high school, around 14 ; no social media till 16 ; phone-free schools and so parents can push the school to go phone-free ; and the fourth is far more independence, free-play and responsibility in the real world. That ’ s the hardest one because that requires us to change. But it ’ s the most important in that if we ’ re going to take the screens, the 10 hours a day of screen time, take that away from kids, we have to give them back an exciting childhood. EH : You mentioned big tech. Why place the onus of responsibility on individual families or neighborhoods or kids when it is these giant tech companies who are designing the apps and the phones for maximum engagement and profit? Can they not make changes to the way software is designed, and can governments not better regulate these companies? JH : Yes, they certainly could make changes, but they ’ re in a collective **action** trap, too. We know this from Frances Haugen, who brought out the Facebook files. Facebook is actively recruiting underage kids. They really need those kids. And they know that if they were to crack down, those kids would just go to TikTok. So the tech companies certainly could solve this, but they ’ re in a collective **action** problem, too, because if one of them does the right thing, the kids will just lie about their age and go somewhere else. Also, Congress gave them immunity. The courts have **interpreted** Section 230 very broadly, so that if your kids are harmed by anything that they saw online, well, Section 230 says you can ’ t sue the company. So these are the biggest, most powerful, richest companies in the world. They ’ ve been given free reign to own our kids ’ childhood. They have limited legal responsibility so far. And we ’ re trying to get a bill through Congress that does some fairly modest things about setting defaults and just begins to establish children are not adults. You have to treat them differently. If there was any other consumer product, let ’ s imagine a new toy comes out on the market. The kids love it. And 90 percent of kids are using this toy five hours a day. And thousands are being hospitalized for depression, eating disorders, anxiety. And some kids are killing themselves after using the toy. So what I ’ m saying is, we have a defective consumer product. Do you think that that toy maybe would be recalled? Do you think maybe someone – EH : Or regulated. JH : Maybe regulated? Like, maybe you have to change it so that it doesn ’ t wreak such damage? So that ’ s what the Surgeon General is saying. The surgeon General is saying while the scientists fight about whether social media caused the increase at the population level, like, the graphs I show in the book with the hockey sticks, did social media cause that increase at the population level? That ’ s an academic debate. I can ’ t be 100 percent certain I ’ m right. But what the surgeon general was saying was, regardless of that debate, here are hundreds of cases of kids who were sextorted, they were bullied, they were shamed, and then they killed themselves that day. So the Surgeon General is saying, if this was any other consumer product, we ’ d regulate it and we ’ d put warning labels on it. Why not this one? EH : You have two kids. How have you wound up **navigating** phone use at your own house? JH : Yeah, so we gave my son a phone, my old iPhone, as most people do, when he was in fourth grade when he started walking to school. This was back, you know, in the early 2010s, we didn ’ t know any better. Now in retrospect, we should have just given him a phone watch or a basic phone. And that ’ s what we did with my daughter, who ’ s three years younger. I gave her a big pink Gizmo Watch, and in third grade, she loved it. And I could send her out in third or fourth grade. I could send her out into the park, out to get bagels. So we at least did that for my daughter. The place where I did hold the line is on social media. I said, no way, you ’ re not getting social media at least until high school. Both kids, they accepted that. And my daughter, when she was in seventh grade, she said that she was actually glad she wasn ’ t on Instagram because she could see what it does to girls. It ’ s a terrible thing to take 11 and 12-year-old girls and make them be conscious of their face, their skin, their body, constantly, all day long, having people comment on it. It ’ s a horrible thing to do to girls. EH : I want to close on a story that you have told about your son, Max, that I think illustrates the kind of world that you ’ re ultimately advocating for. Tell us about his walks home and what happened eventually. JH : Because my wife and I got to know a woman named Lenore Skenazy, who wrote a book called”Free Range Kids,”I recommend this to everybody,”Free Range Kids.”EH : I do too. I ’ m also a big fan of it. JH : Oh good, good. You said you have three daughters? EH : Yes, and they all walk themselves to school and walk home every day. JH : Fantastic. EH : Good luck to them. I have no idea what ’ s happening in those two or three blocks. JH : And that ’ s really important that you don ’ t have an idea. So because we read Lenore ’ s book and we know her, we encouraged Max to walk to school a year or two before everybody else was walking. And this is in Greenwich Village, right here in New York City. It ’ s a safe neighborhood in terms of crime. But there ’ s some busy streets to cross. But Max is really good at that, as I was when I was seven, eight, nine, 10 years old. So we let him walk to school. And the first time we let him walk, we were terrified. And we were watching the blue dot on the phone because we could track him. The next day we watched again. And I think we might have watched a third day. But here ’ s the thing about anxiety. The way you get rid of it is by exposing yourself to the stimulus and then nothing bad happens and by Pavlovian processes, the anxiety drops. So we trusted him, and he quickly learned the subway system, he ’ s just amazing as a navigator. I think the story you ’ re referring to is that then, when he was 13, he got really into tennis. And I took him to the US Open when he was 12 and then again at 13. And he really wanted to go to a night game. And that would mean he ’ d have to come home very, very late. He loved going out to Queens alone. We let him do that during the day, but he wanted to go out to a night game. And, you know, we were a little nervous about that. But we thought, OK, you know, what would Lenore say? And we said, OK. So he went out and the game, you know, tennis games, sometimes they go really late. This was a really late game. And so he ’ s coming home on the subway. And there ’ s a huge crowd, everyone ’ s happy, they ’ re all out together. He gets on the subway, it comes back. And when he tries to transfer, I forget which line it was, somewhere in Manhattan, you get from Queens into Manhattan, then he had to transfer to go south to Greenwich Village. That train wasn ’ t running. And here it is, it ’ s like one in the morning. And so what does he do? Well he goes upstairs, and he hails a yellow **cab**, which he had never done before. Now I had shown him how to do it, but he ’ d never done it on his own without me. But he just went up, hailed a **cab** and came home. And he was nervous, he was. But that ’ s the thing, because he was nervous, when he succeeded, it just changed him. And this is what I really want to convey. We have to get over our own anxieties and trust our kids to be competent human beings, just as we were at that age. And it ’ s hard at first, but by the third try, it actually gets easy. And then your kids flourish, they grow, they become more confident, and they ’ re going to be much less subject to anxiety disorders if you give them independence. EH : Well, Jon Haidt, those are all the questions that I have for our conversation without the audience. But there are just audience questions piling up. So I ’ m going to jump right into them. The first is,”Is there anything we can do to raise the smartphone issue politically? Because it would be great to know whether politicians around the world would want to address this at a national level?”JH : Yes, this is what ’ s so exciting. This is the least polarized, most bipartisan issue that there is. It ’ s incredible, because wherever you go, red state or blue state, you know, Democrat or Republican in Congress, almost all of them are parents. Most politicians are parents. In so many countries, leaders are getting out in front, and they get massive support. So actually, it ’ s not a risky position. And they ’ re actually making changes. So I ’ d urge everyone, talk to your legislators, let them know you support KOSA, the Kids Online Safety Act. Actually, the Senate passed it almost unanimously. It was like 97 to three. But I ’ m very excited about this. And I think the phone-free schools is the place where it started. A whole bunch of states have already said they ’ re going to go phone-free statewide. Los Angeles School District announced it a month ago ; New York City is going to announce it very soon. So I ’ m actually really optimistic that at least school districts and states are going to go phone-free in school. EH : On the subject of schools, we have a few questions from educators who are on with us. Bradley says,”I ’ m a teacher. How can I implement cell phone restrictions and talk to families about this without sounding too ivory-tower-esque? Or like I ’ m telling parents how to raise their children?”JH : So first let ’ s distinguish between what are you saying about phones in school and what are you saying about phones at home? Now you ’ ve always had the right to talk about what to do at school. Parents respect that. And I guarantee you other teachers in your school and your principal probably are reading about this. So if you raise the subject, if you say,”Hey, can we go phone-free?”you ’ re going to find a lot more support than you expect, because things are really different than they were six months ago. That ’ s the first thing. You can definitely go phone-free in school. We ’ re seeing schools all around the country saying this is such a catastrophe for learning. We see it in test **scores**, we see it in discipline. We ’ re dealing with it at school by going phone-free. Here ’ s the new policy. And in addition, I hope you ’ ll consider rolling back the technology, giving kids a more play-based childhood, not giving them a phone so early. So I think we are seeing that happen. The key, again, is collective **action**. You ’ re going to find a lot more allies now than you would have a year ago. EH : Amber asks,”I am an educator, and I ’ d like to know if Mr. Haidt recommends any specific activities or strategies for educators to help undo some of the negative effects of our students childhoods?”JH : Oh yes, there ’ s an amazingly powerful one which costs zero dollars and is incredibly effective. It ’ s called the”Let Grow Experience.”So I was so taken with Lenore Skenazy ’ s work that I and a few others cofounded an organization called Let Grow. So if you go to letgrow. org, you can sign up, you sign up there to download the kit for the Let Grow experience. You can do this at home too, but it ’ s especially powerful if you do it as a school. So let ’ s imagine a third-grade class. All the third graders, let ’ s say, or second graders say we ’ re doing the Let Grow Experience. So you give them mimeograph, whatever, a printout of the instructions, the kid takes it home. Basically, it ’ s pick something to do by yourself that you ’ ve never done before by yourself. Work it out with your parentsm and then do it. And so it ’ s something like, I ’ ve never walked the dog or I ’ ve never walked to a store, I ’ ve never been out. So, you know, if there ’ s a store that ’ s three blocks away that you ’ re seven or eight-year-old can walk to, that would be an ideal one. But let them pick, you know. So they pick something, and then they do it. And then they come into class. And let ’ s say you do one of these a month, they just say what they did, they put it up on as a leaf on a tree. And if you do it every month for eight months, you get these eight activities that you ’ ve done by yourself. And it ’ s incredibly powerful, because first of all, the kids seem to almost grow taller. They feel much more confident. But the more important effect, or as important, is what it does to the parents. Because the parents are so afraid, like, well, at what age can I let my kid out? I don ’ t know, no one else is doing it until 11, so we don ’ t know. But what happens? Imagine a town in which all the elementary schools are doing this in third grade. Now you ’ ve got eight-year-olds all over the place. They ’ re walking to the store to get milk, they ’ re doing errands, they ’ re outside playing, they ’ re walking the dog. Now adults see eight-year-olds outside, and at that point, people will stop calling the police when they see an eight or nine-year-old outside, which at present some people do because they think it ’ s some horrible, dangerous anomaly. EH : I did get a bunch of texts from other parents when they saw my kids walking to school alone. They were like, are you aware of this? JH : Oh, wow, OK, but look. But this year, things will be different. This year they ’ ll cheer you on and they ’ ll do it themselves. This is the year for collective **action** on all these fronts. EH : Some questions from parents too. Melissa asks about distinguishing, do you distinguish between phones versus iPads? Because iPads can create the same kind of issues, but parents seem to be more OK with iPads or tablets, and some schools actually require them, so I would love your perspective on that. JH : So from the parents ’ point of view, a phone is the worst because it ’ s the most portable. A phone is with you all the time, especially when you ’ re outside the house. My advice is that nobody give a smartphone before high school. An iPad can do all the same stuff as a phone, it ’ s just less portable. So I would say again, don ’ t give your kid their own iPad that they can hold on to and customize and communicate with strangers on and watch porn. Don ’ t do that either until high school. Now for you to have an iPad in the house, your iPad that you give them sometimes, there ’ s all kinds of good stuff that they can do. There ’ s all kinds of good stuff. And also, let me be very clear, stories are good, movies are good. Watching a movie with your kids is great. It ’ s the 15-second videos that have no redeeming value and that I think are really the worst. That ’ s what I ’ ll be studying next. So I would say just be careful of both the addictive nature and the distracting nature of these things. So a screen isn ’ t going to hurt the kid necessarily. But it ’ s the addictive possibilities. EH : Sarah asks, what kind of advice do you have for parents with younger kids if they have already given the kids smartphones? Do you have any advice about the removal process or the weaning? JH : My advice is, if you try to pull it out by yourself, it ’ s going to be very painful. But if you and four other families pull it out at the same time and you give them something in return, it ’ s going to be joyous. Yes, they ’ ll resist, but let ’ s look at the difference. In one case, you gave your kid an iPhone when she was nine and she ’ s now 11. And everyone, everyone has an iPhone, and they ’ re on it all day long. Now if you take your kid off, you ’ re condemning your kid to social death. So that ’ s a very hard situation. But let ’ s suppose you gave your kid a phone when she ’ s nine and now she ’ s 11, and most of her friends are on Instagram too. But you go to – you talk with the parents of her three best friends, and you say,”You know, I think we made a mistake. What do you think?”And if they agree, then what you can do is you can say,”We ’ re taking back the smartphone. Here ’ s a flip phone or a light phone or something else. You ’ ll get a smartphone when you ’ re in high school. You can still do a lot of the same stuff on a computer. You still have a laptop or something. So it ’ s not as though you ’ re not going be able to do these things, but we don ’ t think you should have this with you all the time everywhere. And your three best friends, their families are also doing it. And what we really want is not to take stuff away from you, we really want is for you to have a fantastic childhood. We want you guys to have fun. So we ' re going to start off by saying every Friday, you girls are getting together for a sleepover, you plan it how you want, but we ' re suggesting that every Friday you guys get together, have a sleepover, we ' ll give you money to go places, we want you to be with your friends in the real world."And the more you give them the opportunity to hang out and do things, it ' s not deprivation. It ' s not doors closing, it ' s doors opening. What they ' re most afraid of is being alone, being cut off. So avoid that. EH : Amy asks,"How do I lead by example? I want to be a good role model for my kids, and I acknowledge that I use my own phone too much. I realize I will check my Instagram, then two seconds later, check it again. Why am I checking it again?"she says. But how does she lead by example? JH : So you know, when your kids are infants and toddlers, they are copying you, they ' re looking for examples to copy. So do be careful when you ' re with toddlers. You really need to do a lot of eye contact. Don ' t do the continuous partial attention. Young kids, they really need to develop that sense of mutual gaze and interaction. Teenagers are not really copying you. So I would say insist on really good family **norms** and then honor those **norms**. And that means, like, in my family, we have an absolute rule no phones at the table, no matter how important it is for you to look up this fact relevant to the conversation, don ' t go get your phone and look it up. That ' s just the rule, no phone, we do not bring phones to the table. And they have to be out of the bedroom at a certain time, put them on the counter in the kitchen charging at 10 o ' clock or whatever the time is you pick for your family. So definitely establish family **norms** that you respect and then make those stick. But don ' t expect that your kid is copying you. If you stop checking Instagram all the time, is your kid going to say,"Oh well, I ' ll stop."No, they ' re completely addicted. They ' re socially addicted. They have to do it because everyone else is doing it, not because you ' re doing it. EH : Last couple of questions. Isha asks,"What are the two most essential changes we can implement in our **daily** lives now, so that there will be less damage to our own and younger generations ' mental health?"JH : The number one thing that you can do is regain control of your attention. What I found when I began teaching undergrads is that most of them, not all, but most of them, have given up all of their free attention. If they ' re in the elevator, they ' re doing this because that ' s like 60, 90 seconds, they ' re doing this. They never have a moment to think. They can ' t daydream. They can ' t meditate. And so what I start with at the very beginning of the class is let ' s regain control of your attention. It ' s your most precious commodity. So shut off almost all notifications. Do not get alerts from any newspaper or magazine or anything else. You don ' t need alerts. Move social media off your phone onto your computer only, and then eventually, maybe even off your computer. And a lot of them are gaining three to five hours a day. They ' re regaining three to five hours a day. What do they do? They say,"Now I can do my homework. And it ' s actually not hard anymore. And I have time to read a book or I have time to talk to a friend or I have time to play guitar."So regaining control of your attention is where a lot of this starts. I ' d say read Cal Newport, I assign the book"Deep Work."It ' s a fantastic book, and it ' ll really change the way you see what ' s happening to us adults. EH : OK, let ' s end on a question from Kathleen."Since your book release and tours, so following the release, are you more or less hopeful about your collective **action** proposals catching on?"JH : I am wildly more optimistic. I ' m optimistic by nature, but I ' ve been working on democracy issues since the early 2000s, and I ' m not optimistic there. I mean, the problems are huge, and I don ' t know the answers. And that ' s what my next book is going to be on. But on this one, on can we roll back the phone-based childhood, I was kind of optimistic last year, and now I am just wildly optimistic because it ’ s happening. It ’ s happening at lightning speed. I can ’ t believe how fast it ’ s happening. The number of schools, of states, of countries that have acted in the last six months is mindblowing. I ’ ve never seen anything like it. That ’ s especially the schools and governments. Every day, I ’ m getting emails from parents saying,”Thank you. My friends and I, we did a reading group on the book, and now we ’ re all doing this together. And all the families at the end of our street, and the kids are playing, and they ’ re riding their bicycles.”So it ’ s not as though our children have somehow biologically forgotten how to ride a bicycle. And they ’ re still thrilled to be out from under your thumb. They ’ re thrilled to have some independence. So this is happening. We don ’ t need everybody, but if we get most people, we solve this problem. EH : Thank you for being so witty and wise, Jonathan Haidt. And thank you, TED Members for being here and for your involvement. As a nonprofit organization, all of TED ’ s work is made possible in part by you. So thank you so much to the TED Membership community. And that is it for the Book Club for this time, see you next month. JH : Thanks so much, Elise. [Want to support TED?] [Become a TED Member!] [Learn more at ted. com / membership]

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Wendy Suzuki : It ’ s really about shifting your **mindset** about how you think about anxiety. Most people just want to get rid of all their anxiety, they don ’ t want to think about it, they want to kick it out the door. But anxiety evolved for a reason. Anxiety evolved to help protect us. It is a protective mechanism to shine a little light, say, you should be paying attention to this, you should be paying attention to that. And that is what people need to appreciate. [Intersections] [Dr. Wendy Suzuki : Professor of Neural Science and Psychology, NYU] [Annabel Spring : CEO of Global Private Banking and Wealth, HSBC] WS : So can I tell you why I was so excited to be able to talk to you today? And that is that, as a neuroscientist, I ’ ve spent my whole career studying the brain and what works. But by contrast, I ’ ve ignored certain aspects of my own **financial** well-being. And now I get to sit down with a global banking expert, and I know you ’ re going to reveal secrets that I will be able to decrease my own anxiety. AS : Well, that ’ s a conversation we should definitely have because personal finance is very, very important. And what I would love to hear you talk about is a little bit about how do we get people over that anxiety to actually address their **financial services**. And I think that ’ s really important. So very, very curious to have this conversation. WS : OK, great. AS : But I ’ m curious to talk to you as well, because I went into **financial services** to look after people and to care for people. My passion is actually about people. And I went into **financial services** at first thinking, you know, this is how we help people achieve their hopes and their dreams and their ambitions and also look after their families and their legacies. But what I ’ ve learned in **financial services** is you can ’ t just talk to people about their finances. You actually have to understand the whole person. And a lot of that is the work that you do, whether it ’ s physical fitness or mental wellness or social connectedness or belonging. You know, understanding people in the light of their communities and their personal passions is a huge part of my role but also a huge part of my personal interest. So I ’ m excited to be here. WS : Absolutely, absolutely. I mean, I think we both have really unique perspectives on what makes life worth living. I think what makes a good life is having something that you do that you are just **passionate** about and that you really want to give yourself to. It doesn ' t have to be what you do for a living. It could be giving to the community, giving to your own family. That goal is so important, and we know that that kind of aspiration and the awe that it inspires in you as you think about that, that is releasing dopamine in your brain. That is what brings a lot of **happiness**. Now how do you get there? There is a long road to that that includes you have to be financially able to do that, but it also includes knowing how to regulate your **stress** levels and deal with the increasing levels of **stress** and anxiety in our society today. But also to catch on to those things that bring joy in your life. And I feel like that ' s what I ' ve come to study as a neuroscientist. That is, how to **maximize** joy and those positive emotions and how to better regulate the difficult emotions and use them as signposts towards what you should be paying attention to. AS : You know, it ' s so different. As a banker, when I started to think, you know, what is a good life, you might expect, and I did, my first thought was, OK, consumption levels, income levels, you know, GDP per capita, living standards and all of those thoughts. But then I did turn to where you are, which is, it ' s more subjective. And we found, as a firm, we actually couldn ' t answer that question. So what do you do if you can ' t answer that question? You go out and ask a bunch of people. So we asked 11, 000 people, how do they look at quality of life? What is quality of life for them? And they came back with their top three factors. And I think they ' ll accord a lot with the work that you do. You know, one is physical health, another is mental wellness. And the third is **financial** fitness. And those components are critical to what makes a good life. And now they ' re obviously correlated. So what we saw is for people who are financially fit, you know, they have significantly higher, 30 percent higher, perception of their quality of life. They ' re 20 percent more likely to be physically fit. And they ' re twice as likely to be above the standard or above the average of mental wellness. So you can see that correlation. And sitting back even as a banker, but certainly I thought, gosh, what makes me happy? What makes me happy and having a great day is going for a walk and clearing my head, talking to my husband or my children, or my mom or my friends, knowing that I ' m doing a great job at work and really fulfilling that part of the purpose in my life, whether it ' s at work or in the community. And I know that makes a great day for me, too. WS : Yeah, yeah. Well, you know, you ' re touching on something that we know is so true from neuroscience research. That is, our brains evolved to connect with each other. There are parts of our brains that evolved just to be able for me to be able to appreciate the **expression** on your face, what that means, how you ' re connecting with me. And we all have that. And I think that ' s such an important element to remember when we all spend so much time on our phones, on our computers, even with Zoom, it ' s not the same as this kind of wonderful face-to-face interaction that we ' re able to have today. And asking yourself, is that a good part? Is that a big part of my day? My face-to-face interaction is a really good first question to start asking when you head towards a good life, I think. AS : So I want to scratch into a little bit because just to balance this concept of a good life, these elements of a good life, you know, there ' s a phrase :"Health is wealth."What does that mean to you? WS : I think for me, I would modify that phrase into brain health is wealth. Why? Because the human brain is the most complex structure known to humankind, and it defines everything about us : our sense of humor, how we converse with each other, whether we ' re more introverts or extroverts. It has to do with the connectivity of our own brains. And so I say brain health is wealth, because the healthier that your brain is – How do we make the brain healthy? With good sleep, with good food, with good social connections, with minimizing, though not eliminating all **stress**. Because **stress** can be good, it drives you towards the things that you want to achieve. That will give you the longest possible healthy, good, positive cognitive life that you could have. So for me, brain health is wealth. And I define wealth as that long, good cognitive life. AS : See, I think the long life concept is really important. So when I look at that phrase, I think about longevity. And I think, you know, back in the ' 50s globally, we had a life expectancy of around 47, which is shocking. And now we ' re up in the 70s, which is a huge advantage. So we have a much longer lifespan. But when we look at health span, still about a fifth of your life you ' re living with illness. And that means we ' re actually living longer with sickness, which is a very challenging thing to recognize. So I think the first factor is absolutely everything that you say, which is what can we do to ameliorate that and making sure that we stay healthy? And, you know, there ' s a dreadful investment analogy which I will use, which is your health is your only personal asset. It ' s not transferable, it ' s not diversifiable, there are no alternatives. It ' s your most important asset, and it ' s intangible. And you ' ve got to look after it. But there ' s some **financial** implications of that as well, which is if we ' re talking about living longer while we ' re sick, we need to actually have some longevity literacy and realize that we ' re going to live longer, we need to save and invest and plan for that. And if some of that ' s going to be while we ' re sick, we actually need to make sure that we can fund our sickness as well and make sure that we have a quality of life while we ' re sick. And that obviously is pure savings and investments. But then it gets into protection, whether that ' s health insurance, to allow people to bounce back and progress forward. Because life is never a straight line, you know, you never know quite when you might get sick and you might not be at retirement, it might be before that. So making sure that you have health insurance to get that bounce back, and then life insurance for peace of mind to look after your family should anything awful happen. So I look at"Health is wealth,"and I think, ooh, longevity, health span, savings and investments to make sure that you can look after yourself and insurance so that you can bounce back and you can look after your family. WS : You make it sound so clear and easy, but it ' s very scary for somebody thinking about that. You know how long, how much do I need? And what do we know. I mean what would you say to somebody who listens to that and say, oh gosh, I ' m not sure exactly what to do. Who should they go to? AS : Do your reading online. There ' s a lot of education online, a lot of **financial** education online. And there are even some digital tools now where you can look at, you can outline what your goals are and save for each of those goals individually. And have that looked at. And I think there are some tremendous tools for **financial** fitness. And the first step is curiosity and asking that question. And the second step is education and having the conversation. But please don ' t forget to act on it. Making sure that you save and invest is really important with a little bit of protection. WS : Well, I mean, we talked about long-term fitness and **financial** fitness, but I wanted to ask you about your own personal fitness. Because long-term health is so important for our long-term wealth. What has been your personal history with physical fitness? AS : I wouldn ' t be sitting here right now if it wasn ' t for fitness, exercise and sport. I am a huge proponent of team sports for girls. Team sports for girls is a great – actually, there ' s a Deloitte study about it – team sports for girls gives you team building, problem solving, leadership skills, understanding other people, **empathy**, how to win, how to lose, how to compete and the rules of the game. And I think that ' s a tremendous thing. I was a rower, I was a swimmer, I was an athlete. I ' ve seen it in my sister, my mother and my children. And I think that team sports for girls just sets you up for life. That being said, right now I travel globally. Team sports is not where it ’ s at. But I love to swim, so my personal adventure in every city I go to is to find somewhere to swim. So it might be a lake in Switzerland, it might be, you know, the beach in Asia. On the weekends, though, it ’ s my local public pool in London with a bunch of wonderful ladies. We get there at seven, we have a swim, and then we have a big giggle and a coffee afterwards. WS : Oh, that sounds great. You know, I grew up in California. Outside, I played tennis, I was on basketball teams and softball teams, so I grew up in the same way you did with that team-sport mentality and learned so much from it. But it took a big lesson, a big kind of kick in the pants, if you will, for me to really realize how important exercise was. And this was a moment when I was trying to get tenure at New York University, which is a very, very difficult thing to do. And my strategy wasn ’ t optimal, I decided to do nothing else but work for six years. I could do it for six years. And I tried and didn ’ t do so well and was not very happy and didn ’ t have a lot of social life and was eating way too much takeout. And discovered that going on a river rafting trip to the wilds of deepest Peru, I felt so good. And I thought, my gosh, what ’ s happening here? And it actually changed my whole research direction. I started to study the effects of physical activity in the brain, and now I can tell you exactly what was happening to my brain. So my workaholic brain was being infused on this river rafting trip with a whole bunch of neurochemicals, including dopamine, serotonin, noradrenaline. As your brain is in your morning London swims, I like to call it a bubble bath of neurochemicals that make you feel good. It gives you an immediate mood boost, it gives you an immediate focus boost. So when you go back into work, you can focus on what you need to do. It changed my life, and it changed my research towards that to try and understand what is that prescription? What should we all be doing at different ages? And so yes, I am a big proponent of exercise. And of course I try and do what I preach and do what I study. So even this morning, we had to get in here pretty early. I got in a good 20-minute yoga workout. AS : I ’ m impressed, and it sounds like you enjoy it. How do you link that concept of play and fun to a good life? WS : So, you know, I have come to realize that finding my own joy, what I love to do is so important. And yes, I stuck with my workouts after, you know, trying to work so hard to get tenure because I found things that were fun. I love dance classes, I love this workout called intenSati that combines physical movements from kickbox and dance and yoga with positive spoken affirmations. So you ’ d have to yell out in class,”I am strong now, I believe I will succeed.”And once you get over the awkwardness of yelling things out with a whole bunch of other sweaty, affirmation-yelling people, it feels so good to just get it out of your system and yell with everybody else in unison, with the music, in rhythm. It ’ s a little bit like singing together in a sweaty chorus. It ’ s great. I love it. And so that made me realize that there were so many other things that I hadn ’ t even discovered that could bring that kind of joy into my life. And that ’ s what I look for. AS : So that sounds like a really connected experience and giving you that sense of belonging. One of the things that we see is being part of the community and social connectedness really increases people ’ s quality of life. You know, in fact, two times. Can you talk a little bit about how you see that through your work? WS : Well, I can turn to the research literature. The longest study on **happiness** that has ever been done is still ongoing at Harvard, and now is run by Professor Robert Waldinger at Harvard. And they asked : What makes a happy life? And the answer was the more social connections that you had. They didn ’ t have to be deep ones that you, you know, give presents to at Christmas every year. That included a friendly interaction with the barista at Starbucks. That counts. The more you had, the happier you were in life. And it goes back to the evolution of our human brain, that we have so many areas evolved to help us figure out how to have these social interactions. It ’ s because that social connection is so important to us, not that every social connection is positive. There are, you know, difficult social connections too. But that is the crux of what makes us human. And we know that the more positive ones that you could create in your world, the happier you ’ ll be. So that ’ s a pretty simple formula that anybody can use today. Can I say hello in a more friendly way to more people? That ’ s going to increase your **happiness**. And I love that because it ’ s doable for anybody, anywhere. AS : Yeah, it ’ s so true. You know, some of the saddest conversations that I occasionally have with some of our wealthier clients are people who have really separated, they ’ ve focused so much on one thing in their life that they haven ’ t spent time with friends or family or their community. And as they ’ ve aged, you know, they ’ ve left work and suddenly, they ’ re more alone. And the statistics that we see is that a third of older people over 65 really do experience a much greater sense of loneliness. And as we look around the world, that seems to be true, whether it ’ s in the UK or in China or here in the United States. And I think it ’ s incumbent on all of us, whether it ’ s **financial services** institutions or universities or governments, to find a way to address that and keep those people engaged in the community. WS : So I ’ m a professor at New York University, and I think everybody can relate to the anxiety of you have a question, but do you really want to raise your hand and ask it in front of the whole class? I had that same anxiety through my entire educational experience. But I went into **academia**. So one day I found myself at the front of the classroom, and I knew that 80 percent of those students had questions for me. But, you know, were anxious about doing that. And so I used that knowledge about my own anxiety to **relieve** their anxiety. So I came early and I stayed late, and I answered any questions. I just hung out by myself so they could come up and ask me. And I realized that was a superpower of my own anxiety. That I turned my own anxiety, and I turned it on the outside. And I made it an act of compassion or **empathy** to make sure that everybody could ask the question. And it ’ s not just this, take your biggest anxiety. Take your **financial** anxiety. You know, can you have a conversation? If you had a great conversation with your banker and you learned a great banking trick that **relieved** that anxiety, can you share it with people? So I think that is a wonderful way to take the resolution of your anxiety and help others with it. And that ’ s also a form of generosity. AS : It ’ s fascinating. So how do we take that anxiety then and turn it into a good thing? WS : Yeah. AS : And we could take the **financial** anxiety or any other anxiety. How do people feel that? And I think in your book you talked about anxiety as a **warning** sign. That it ’ s something that maybe you should be dealing with. How do we get people to use that and to act on it? WS : That ’ s such a good question. And it ’ s really about shifting your **mindset** about how you think about anxiety. Most people just want to get rid of all their anxiety. They don ’ t want to think about it, they want to kick it out the door. But anxiety evolved for a reason. Anxiety evolved to help protect us. It is a protective mechanism. To shine a little light, say, you should be paying attention to this, you should be paying attention to that. And that is what people need to appreciate. Think about, do you lay awake at night thinking,”Oh my God, I didn ’ t get to watch that last series on Netflix. I ’ m so worried about that.”No, you don ’ t worry about that. You worry about your work, your finances, your relationships. These are things that matter to you. And so what your anxiety is doing is really showing you what matters most. And actually going back to our first question, what makes a good life for you. If you flip it that way and start to use your anxiety as a signpost to realize it is helping to protect you from things that could be a danger, then it becomes a tool that you can use to order what you ’ re going to do in your day and in your week and in the next year. AS : That ’ s fantastic. So anxiety as an opportunity alerter and a risk management. That ’ s an absolutely fantastic thing. WS : I mean, in a sense, bankers are your anxiety detectors for your finances. This is what they know in great detail, what you should be looking for. And they cut that off at the pass and give you tools to be able not to worry about that. And I say that, I talk a lot about my own **financial** anxieties in the book, because it ’ s not just mine, it ’ s everybody ’ s. Everybody ’ s worried about finances. No matter who you are. Are you doing it the right way? Are you investing? What is investing? Start with the basics. So it ’ s such an important one to appreciate, how you can flip the script and use it to your advantage. AS : Now we recognize that unfortunately for all of us, two out of five of us will find some period of **financial** instability in life. You know, we ’ ll lose our jobs, our children will get sick, we might get sick. So understanding and having the **financial** literacy to understand that that might happen and thinking through ahead of time what might I need to prepare for that is also really important. So that cash on hand, or if you do have the ability, a little bit of insurance to cover some of those things, really, really important. But again, going back to our conversation on curiosity, making sure that you ’ ve taken advantage of the digital tools, the planning tools, there are so many goal **planners** and trackers and budgeting tools online. Almost all banks provide them. And certainly we do. Have a look at them and really just get financially educated, because I think that **relieves** a lot of the anxiety. It really does. WS : No, I think that will help a lot of people. It made my shoulders go down. There you go. AS : So I was really curious, world-famous neuroscientist, what is the one thing that we should all be doing for our brain every day? What should we be doing? WS : I ’ m going to say : moving your body. Just 10 minutes of walking, that everybody anywhere could do, decreases your anxiety and depression levels because you are flooding your brain, even with a 10-minute walk, with this neurochemical bubble bath of dopamine, serotonin, noradrenaline. And then think about what that does, that bubble bath of neurochemicals, if you do it regularly, not just for a day, but over weeks, over months, over years. You are literally giving your brain not only good neurochemicals but growth factors. You are making it, as I like to say, big and fat and fluffy. And for this you can start any time. Maybe you ’ re a couch potato until you ’ re 75. Guess what? You get up and start walking, you get that bubble bath. So moving your body is my answer to that question. So for you, I would love to ask a more long-term question. What advice would you give to young people about how to start planning their **financial** future? AS : To get ahead as a young person, you ’ ve got to get started. It ’ s as simple as that. You ’ ve got to get started. So what am I saying to get started by doing, you ’ ve got to start to save and invest. Early and often. And it ’ s not complicated, it ’ s not rocket science. On the saving side, it ’ s the power of compounding interest. And on the investing side, it ’ s time in the market rather than timing the market. So just put your money in the market and get started. That ’ s the most important thing to do to start. Now if I were looking at my children, I would also say : and pay down that expensive credit card debt, please, because there ’ s not much point investing if you ’ re paying significant charges on expensive debt. So make sure that that ’ s gone. And once you ’ ve got that started, that ’ s a wonderful thing. And we ’ re seeing actually that the generations are starting earlier and earlier. So Gen Z, they ’ re starting in their early 20s. The millennials started in their mid-20s. Gen X, we started in our early 30s. The boomers in their mid-30s. So they ’ re starting earlier, that ’ s fantastic. And I think some of that ’ s technology, and some of that ’ s realizing that now you can start saving and investing with less. You don ’ t have to have a lot of money. Just put away a little bit. Make sure that you ’ re consistently putting away a little bit, so you get the benefit of that time in the market and that compounding. Now as you get older, obviously you ' ve got to make a plan. You ' ve got to make sure that you can pay for that first apartment, that you can start to think about paying for the other expensive things in your life. And, you know, people like me, you ' ve got to make sure that you can pay for those college fees. These are things you ' ve got to save for as well as your retirement. And just a shout out for women in particular, given I ' m looking at you, you know this, you ' re going to live longer. Sadly, statistically, women are going to earn less. So it ' s even more important for women not to just think about their children and what they need now, but to make sure that women also save for retirement. Because you ' re going to be retiring for longer. You want to make sure that you ' ve thought that through, and you ' ve really got the longevity literacy to think through,"I ' ve got to make sure that I ' ve got a plan."WS : Great. I think that ' s so important. And it ' s so great that there ' s so many different options out there now. AS : But the most important thing is starting and staying curious. Because your needs will change. And whether you learn digitally or you learn talking to people, it doesn ' t matter. Stay curious, stay engaged and start. WS : You know, that ' s so funny because that ' s exactly the recommendation that I give about brain health. Stay curious about your brain health. Stay curious about : What is the effect of sleep on my brain? What is the effect of alcohol on my sleep and then on my brain? If I do the movement that Wendy says to do, do I feel a difference? And you will. And that kind of self-experimentation is equally true for your brain and body health as it is for your **financial** health, it seems. And that key is really the curiosity of, maybe I could ask this person. Maybe I could explore, not just sleep and movement but maybe meditation is good. Is there information about that? Yes, meditation is wonderful. It ' s a destressing tool. But it ' s so interesting to see that the same tools that are so valuable for good **financial** health are also so valuable for good brain and body health as well. AS : And I think it ameliorates that uncertainty. You know, you talk about anxiety and uncertainty. The more curious you are, the more you know, the less you feel that uncertainty and the less anxiety I think you feel. WS : Thank you so much for this conversation. I ' ve learned so much, and my anxiety level around finances has significantly decreased. What a pleasure to speak to you today. AS : It ' s been such a pleasure, I can ' t tell you. I will be going out, I will be exercising more and moving towards that big, fat, fluffy brain. It sounds fantastic. Thank you so much.

Text Sample 197: POS Dim 2 – 2019 – Score 14.00 – 2019/t002063.txt

Chris Anderson : What worries you right now? You ' ve been very open about lots of issues on Twitter. What would be your top worry about where things are right now? Jack Dorsey : Right now, the health of the conversation. So, our purpose is to serve the public conversation, and we have seen a number of attacks on it. We ' ve seen abuse, we ' ve seen harassment, we ' ve seen manipulation, automation, human coordination, misinformation. So these are all dynamics that we were not expecting 13 years ago when we were starting the company. But we do now see them at scale, and what worries me most is just our ability to address it in a systemic way that is scalable, that has a rigorous understanding of how we ' re taking **action**, a transparent understanding of how we ' re taking **action** and a rigorous appeals process for when we ' re wrong, because we will be wrong. Whitney Pennington Rodgers : I ' m really glad to hear that that ' s something that concerns you, because I think there ' s been a lot written about people who feel they ' ve been abused and harassed on Twitter, and I think no one more so than women and women of color and black women. And there ' s been data that ' s come out — Amnesty International put out a report a few months ago where they showed that a subset of active black female Twitter users were receiving, on average, one in 10 of their **tweets** were some form of harassment. And so when you think about health for the community on Twitter, I ' m interested to hear,"health for everyone,"but specifically : How are you looking to make Twitter a safe space for that subset, for women, for women of color and black women? JD : Yeah. So it ' s a pretty terrible situation when you ' re coming to a **service** that, ideally, you want to learn something about the world, and you spend the majority of your time reporting abuse, receiving abuse, receiving harassment. So what we ' re looking most deeply at is just the incentives that the **platform** naturally provides and the **service** provides. Right now, the dynamic of the system makes it super-easy to harass and to abuse others through the **service**, and unfortunately, the majority of our system in the past worked entirely based on people reporting harassment and abuse. So about midway last year, we decided that we were going to apply a lot more machine learning, a lot more deep learning to the problem, and try to be a lot more proactive around where abuse is happening, so that we can take the burden off the victim completely. And we ' ve made some progress recently. About 38 percent of abusive **tweets** are now proactively identified by machine learning algorithms so that people don ' t actually have to report them. But those that are identified are still reviewed by humans, so we do not take down content or accounts without a human actually reviewing it. But that was from zero percent just a year ago. So that meant, at that zero percent, every single person who received abuse had to actually report it, which was a lot of work for them, a lot of work for us and just ultimately unfair. The other thing that we ' re doing is making sure that we, as a company, have representation of all the communities that we ' re trying to serve. We can ' t build a business that is successful unless we have a diversity of perspective inside of our walls that actually feel these issues every single day. And that ' s not just with the team that ' s doing the work, it ' s also within our leadership as well. So we need to continue to build **empathy** for what people are experiencing and give them better tools to act on it and also give our customers a much better and easier approach to handle some of the things that they ' re seeing. So a lot of what we ' re doing is around technology, but we ' re also looking at the incentives on the **service** : What does Twitter incentivize you to do when you first open it up? And in the past, it ' s incented a lot of outrage, it ' s incented a lot of mob behavior, it ' s incented a lot of group harassment. And we have to look a lot deeper at some of the fundamentals of what the **service** is doing to make the bigger shifts. We can make a bunch of small shifts around technology, as I just described, but ultimately, we have to look deeply at the dynamics in the network itself, and that ' s what we ' re doing. CA : But what ' s your sense — what is the kind of thing that you might be able to change that would actually fundamentally shift behavior? JD : Well, one of the things — we started the **service** with this concept of following an account, as an example, and I don ' t believe that ' s why people actually come to Twitter. I believe Twitter is best as an interest-based network. People come with a particular interest. They have to do a ton of work to find and follow the related accounts around those interests. What we could do instead is allow you to follow an interest, follow a hashtag, follow a trend, follow a community, which gives us the opportunity to show all of the accounts, all the topics, all the moments, all the hashtags that are associated with that particular topic and interest, which really opens up the perspective that you see. But that is a huge fundamental shift to **bias** the entire network away from just an account **bias** towards a topics and interest **bias**. CA : Because isn ’ t it the case that one reason why you have so much content on there is a result of putting millions of people around the world in this kind of gladiatorial contest with each other for followers, for attention? Like, from the point of view of people who just read Twitter, that ’ s not an issue, but for the people who actually create it, everyone ’ s out there saying,”You know, I wish I had a few more ’ likes, ’ followers, retweets.”And so they ’ re constantly experimenting, trying to find the path to do that. And what we ’ ve all discovered is that the number one path to do that is to be some form of provocative, obnoxious, eloquently obnoxious, like, eloquent insults are a dream on Twitter, where you rapidly pile up — and it becomes this self- fueling process of driving outrage. How do you defuse that? JD : Yeah, I mean, I think you ’ re spot on, but that goes back to the incentives. Like, one of the choices we made in the early days was we had this number that showed how many people follow you. We decided that number should be big and **bold**, and anything that ’ s on the page that ’ s big and **bold** has importance, and those are the things that you want to drive. Was that the right decision at the time? Probably not. If I had to start the **service** again, I would not emphasize the follower count as much. I would not emphasize the”like”count as much. I don ’ t think I would even create”like”in the first place, because it doesn ’ t actually push what we believe now to be the most important thing, which is healthy contribution back to the network and conversation to the network, participation within conversation, learning something from the conversation. Those are not things that we thought of 13 years ago, and we believe are extremely important right now. So we have to look at how we display the follower count, how we display retweet count, how we display”likes,”and just ask the deep question : Is this really the number that we want people to drive up? Is this the thing that, when you open Twitter, you see,”That ’ s the thing I need to increase?”And I don ’ t believe that ’ s the case right now. WPR : I think we should look at some of the **tweets** that are coming in from the audience as well. CA : Let ’ s see what you guys are asking. I mean, this is — generally, one of the amazing things about Twitter is how you can use it for crowd wisdom, you know, that more knowledge, more questions, more points of view than you can imagine, and sometimes, many of them are really healthy. WPR : I think one I saw that passed already quickly down here,”What ’ s Twitter ’ s plan to combat foreign meddling in the 2020 US election?”I think that ’ s something that ’ s an issue we ’ re seeing on the internet in general, that we have a lot of malicious automated activity happening. And on Twitter, for example, in fact, we have some work that ’ s come from our friends at Zignal Labs, and maybe we can even see that to give us an example of what exactly I ’ m talking about, where you have these bots, if you will, or coordinated automated malicious account activity, that is being used to influence things like elections. And in this example we have from Zignal which they ’ ve shared with us using the data that they have from Twitter, you actually see that in this case, white represents the humans — human accounts, each dot is an account. The pinker it is, the more automated the activity is. And you can see how you have a few humans interacting with bots. In this case, it ’ s related to the election in Israel and spreading misinformation about Benny Gantz, and as we know, in the end, that was an election that Netanyahu won by a slim margin, and that may have been in some case influenced by this. And when you think about that happening on Twitter, what are the things that you ’ re doing, specifically, to ensure you don ’ t have misinformation like this spreading in this way, influencing people in ways that could affect democracy? JD : Just to back up a bit, we asked ourselves a question : Can we actually measure the health of a conversation, and what does that mean? And in the same way that you have indicators and we have indicators as humans in terms of are we healthy or not, such as temperature, the flushness of your face, we believe that we could find the indicators of conversational health. And we worked with a lab called Cortico at MIT to propose four starter indicators that we believe we could ultimately measure on the system. And the first one is what we ' re calling shared attention. It ' s a measure of how much of the conversation is attentive on the same topic versus disparate. The second one is called shared reality, and this is what percentage of the conversation shares the same facts — not whether those facts are truthful or not, but are we sharing the same facts as we converse? The third is receptivity : How much of the conversation is receptive or civil or the inverse, toxic? And then the fourth is variety of perspective. So, are we seeing filter bubbles or echo chambers, or are we actually getting a variety of opinions within the conversation? And implicit in all four of these is the understanding that, as they increase, the conversation gets healthier and healthier. So our first step is to see if we can measure these online, which we believe we can. We have the most momentum around receptivity. We have a toxicity **score**, a toxicity model, on our system that can actually measure whether you are likely to walk away from a conversation that you ' re having on Twitter because you feel it ' s toxic, with some pretty high degree. We ' re working to measure the rest, and the next step is, as we build up solutions, to watch how these measurements trend over time and continue to experiment. And our goal is to make sure that these are balanced, because if you increase one, you might decrease another. If you increase variety of perspective, you might actually decrease shared reality. CA : Just picking up on some of the questions flooding in here. JD : Constant questioning. CA : A lot of people are puzzled why, like, how hard is it to get rid of Nazis from Twitter? JD : So we have policies around violent extremist groups, and the majority of our work and our terms of **service** works on conduct, not content. So we ' re actually looking for conduct. Conduct being using the **service** to repeatedly or episodically harass someone, using hateful imagery that might be associated with the KKK or the American Nazi Party. Those are all things that we act on immediately. We ' re in a situation right now where that term is used fairly loosely, and we just cannot take any one mention of that word accusing someone else as a factual indication that they should be removed from the **platform**. So a lot of our models are based around, number one : Is this account associated with a violent extremist group? And if so, we can take **action**. And we have done so on the KKK and the American Nazi Party and others. And number two : Are they using imagery or conduct that would associate them as such as well? CA : How many people do you have working on content moderation to look at this? JD : It varies. We want to be flexible on this, because we want to make sure that we ' re, number one, building algorithms instead of just hiring massive amounts of people, because we need to make sure that this is scalable, and there are no amount of people that can actually scale this. So this is why we ' ve done so much work around proactive detection of abuse that humans can then review. We want to have a situation where algorithms are constantly scouring every single **tweet** and bringing the most interesting ones to the top so that humans can bring their judgment to whether we should take **action** or not, based on our terms of **service**. WPR : But there ' s not an amount of people that are scalable, but how many people do you currently have monitoring these accounts, and how do you figure out what ' s enough? JD : They ' re completely flexible. Sometimes we associate folks with spam. Sometimes we associate folks with abuse and harassment. We ' re going to make sure that we have flexibility in our people so that we can direct them at what is most needed. Sometimes, the elections. We ' ve had a string of elections in Mexico, one coming up in India, obviously, the election last year, the midterm election, so we just want to be flexible with our resources. So when people — just as an example, if you go to our current terms of **service** and you bring the page up, and you ’ re wondering about abuse and harassment that you just received and whether it was against our terms of **service** to report it, the first thing you see when you open that page is around intellectual property protection. You **scroll** down and you get to abuse, harassment and everything else that you might be experiencing. So I don ’ t know how that happened over the company ’ s history, but we put that above the thing that people want the most information on and to actually act on. And just our ordering shows the world what we believed was important. So we ’ re changing all that. We ’ re ordering it the right way, but we ’ re also simplifying the rules so that they ’ re human-readable so that people can actually understand themselves when something is against our terms and when something is not. And then we ’ re making — again, our big focus is on removing the burden of work from the victims. So that means push more towards technology, rather than humans doing the work — that means the humans receiving the abuse and also the humans having to review that work. So we want to make sure that we ’ re not just encouraging more work around something that ’ s **super**, **super** negative, and we want to have a good balance between the technology and where humans can actually be creative, which is the judgment of the rules, and not just all the mechanical stuff of finding and reporting them. So that ’ s how we think about it. CA : I ’ m curious to dig in more about what you said. I mean, I love that you said you are looking for ways to re-tweak the fundamental design of the system to discourage some of the reactive behavior, and perhaps — to use Tristan Harris-type language — engage people ’ s more reflective thinking. How far advanced is that? What would alternatives to that”like”button be? JD : Well, first and foremost, my personal goal with the **service** is that I believe fundamentally that public conversation is critical. There are existential problems facing the world that are facing the entire world, not any one particular nation-state, that global public conversation benefits. And that is one of the unique dynamics of Twitter, that it is completely open, it is completely public, it is completely fluid, and anyone can see any other conversation and participate in it. So there are conversations like climate change. There are conversations like the displacement in the work through artificial intelligence. There are conversations like economic disparity. No matter what any one nation-state does, they will not be able to solve the problem alone. It takes coordination around the world, and that ’ s where I think Twitter can play a part. The second thing is that Twitter, right now, when you go to it, you don ’ t necessarily walk away feeling like you learned something. Some people do. Some people have a very, very rich network, a very rich community that they learn from every single day. But it takes a lot of work and a lot of time to build up to that. So we want to get people to those topics and those interests much, much faster and make sure that they ’ re finding something that, no matter how much time they spend on Twitter — and I don ’ t want to **maximize** the time on Twitter, I want to **maximize** what they actually take away from it and what they learn from it, and — CA : Well, do you, though? Because that ’ s the core question that a lot of people want to know. Surely, Jack, you ’ re constrained, to a huge extent, by the fact that you ’ re a public company, you ’ ve got investors pressing on you, the number one way you make your money is from advertising — that depends on user engagement. Are you willing to sacrifice user time, if need be, to go for a more reflective conversation? JD : Yeah ; more relevance means less time on the **service**, and that ’ s perfectly fine, because we want to make sure that, like, you ’ re coming to Twitter, and you see something immediately that you learn from and that you push. We can still serve an **ad** against that. That doesn

’ t mean you need to spend any more time to see more. The second thing we ’ re looking at — CA : But just — on that goal, **daily** active usage, if you ’ re measuring that, that doesn ’ t necessarily mean things that people value every day. It may well mean things that people are drawn to like a moth to the flame, every day. We are addicted, because we see something that pisses us off, so we go in and add fuel to the fire, and the **daily** active usage goes up, and there ’ s more **ad** revenue there, but we all get angrier with each other. How do you define...”**Daily** active usage”seems like a really dangerous term to be optimizing. JD : Taken alone, it is, but you didn ’ t let me finish the other metric, which is, we ’ re watching for conversations and conversation chains. So we want to incentivize healthy contribution back to the network, and what we believe that is is actually participating in conversation that is healthy, as defined by those four indicators I articulated earlier. So you can ’ t just optimize around one metric. You have to balance and look constantly at what is actually going to create a healthy contribution to the network and a healthy experience for people. Ultimately, we want to get to a metric where people can tell us,”Hey, I learned something from Twitter, and I ’ m walking away with something valuable.”That is our goal ultimately over time, but that ’ s going to take some time. CA : You come over to many, I think to me, as this enigma. This is possibly unfair, but I woke up the other night with this picture of how I found I was thinking about you and the situation, that we ’ re on this great voyage with you on this ship called the”Twittanic”— and there are people on board in steerage who are expressing discomfort, and you, unlike many other captains, are saying,”Well, tell me, talk to me, listen to me, I want to hear.”And they talk to you, and they say,”We ’ re worried about the iceberg ahead.”And you go,”You know, that is a powerful point, and our ship, frankly, hasn ’ t been built properly for steering as well as it might.”And we say,”Please do something.”And you go to the bridge, and we ’ re waiting, and we look, and then you ’ re showing this extraordinary calm, but we ’ re all standing outside, saying,”Jack, turn the fucking wheel!”You know? I mean — It ’ s democracy at stake. It ’ s our culture at stake. It ’ s our world at stake. And Twitter is amazing and shapes so much. It ’ s not as big as some of the other **platforms**, but the people of influence use it to set the agenda, and it ’ s just hard to imagine a more important role in the world than to... I mean, you ’ re doing a brilliant job of listening, Jack, and hearing people, but to actually dial up the urgency and move on this stuff — will you do that? JD : Yes, and we have been moving substantially. I mean, there ’ s been a few dynamics in Twitter ’ s history. One, when I came back to the company, we were in a pretty dire state in terms of our future, and not just from how people were using the **platform**, but from a corporate narrative as well. So we had to fix a bunch of the foundation, turn the company around, go through two crazy layoffs, because we just got too big for what we were doing, and we focused all of our energy on this concept of serving the public conversation. And that took some work. And as we dived into that, we realized some of the issues with the fundamentals. We could do a bunch of superficial things to address what you ’ re talking about, but we need the changes to last, and that means going really, really deep and paying attention to what we started 13 years ago and really questioning how the system works and how the framework works and what is needed for the world today, given how quickly everything is moving and how people are using it. So we are working as quickly as we can, but quickness will not get the job done. It ’ s focus, it ’ s prioritization, it ’ s understanding the fundamentals of the network and building a framework that scales and that is resilient to change, and being open about where we are and being transparent about where are so that we can continue to earn trust. So I ’ m proud of all the frameworks that we ’ ve put in place. I ’ m proud of our direction. We obviously can move faster, but that required just stopping a bunch of stupid stuff we were doing in the past. CA : All right. Well, I suspect there are many people here who, if given the chance, would love to help you on this change-making agenda you ’ re on, and I don ’ t know if Whitney — Jack, thank you for coming here and speaking so openly. It took courage. I really appreciate what you said, and good luck with your **mission**. JD : Thank you so much. Thanks for having me. Thank you.

Text Sample 198: POS Dim 2 – 2019 – Score 10.00 – 2019/t001761.txt
I bring you greetings from the 52nd-freest nation on earth. As an American, it irritates me that my nation keeps sinking in the annual rankings published by Freedom House. I ’ m the son of immigrants. My parents were born in China during war and revolution, went to Taiwan and then came to the United States, which means all my life, I ’ ve been acutely aware just how fragile an inheritance freedom truly is. That ’ s why I spend my time teaching, preaching and **practicing** democracy. I have no illusions. All around the world now, people are doubting whether democracy can deliver. Autocrats and demagogues seem emboldened, even cocky. The free world feels leaderless. And yet, I remain hopeful. I don ’ t mean optimistic. Optimism is for spectators. Hope implies agency. It says I have a hand in the outcome. Democratic hope requires faith not in a strongman or a charismatic savior but in each other, and it forces us to ask : How can we become **worthy** of such faith? I believe we are at a moment of moral awakening, the kind that comes when old certainties collapse. At the heart of that awakening is what I call”civic religion.”And today, I want to talk about what civic religion is, how we **practice** it, and why it matters now more than ever. Let me start with the what. I define civic religion as a system of shared beliefs and collective **practices** by which the members of a self-governing community choose to live like citizens. Now, when I say”citizen”here, I ’ m not referring to papers or passports. I ’ m talking about a deeper, broader, ethical conception of being a contributor to community, a member of the body. To speak of civic religion as religion is not poetic **license**. That ’ s because democracy is one of the most faith-fueled human activities there is. Democracy works only when enough of us believe democracy works. It is at once a gamble and a miracle. Its legitimacy comes not from the outer frame of constitutional rules, but from the inner workings of civic spirit. Civic religion, like any religion, contains a sacred creed, sacred deeds and sacred rituals. My creed includes words like”equal protection of the laws”and”we the people.”My roll call of hallowed deeds includes abolition, women ’ s suffrage, the civil rights movement, the Allied landing at Normandy, the fall of the Berlin Wall. And I have a new civic ritual that I ’ ll tell you about in a moment. Wherever on earth you ’ re from, you can find or make your own set of creed, deed and ritual. The **practice** of civic religion is not about worship of the state or obedience to a ruling party. It is about commitment to one another and our common ideals. And the sacredness of civic religion is not about divinity or the supernatural. It is about a group of unlike people speaking into being our alikeness, our groupness. Perhaps now you ’ re getting a little worried that I ’ m trying to sell you on a cult. Relax, I ’ m not. I don ’ t need to sell you. As a human, you are always in the market for a cult, for some variety of religious experience. We are wired to seek cosmological explanations, to sacralize beliefs that unite us in transcendent purpose. Humans make religion because humans make groups. The only choice we have is whether to activate that groupness for good. If you are a devout person, you know this. If you are not, if you no longer go to prayer

services or never did, then perhaps you ’ ll say that yoga is your religion, or Premier League football, or knitting, or coding or TED Talks. But whether you believe in a God or in the absence of gods, civic religion does not require you to renounce your beliefs. It requires you only to show up as a citizen. And that brings me to my second topic : how we can **practice** civic religion productively. Let me tell you now about that new civic ritual. It ’ s called”Civic Saturday,”and it follows the arc of a faith gathering. We sing together, we turn to the strangers next to us to discuss a common question, we hear poetry and scripture, there ’ s a sermon that ties those texts to the ethical choices and controversies of our time, but the song and scripture and the sermon are not from church or synagogue or mosque. They are civic, drawn from our shared civic ideals and a shared history of claiming and contesting those ideals. Afterwards, we form up in circles to organize rallies, register voters, join new clubs, make new friends. My colleagues and I started organizing Civic Saturdays in Seattle in 2016. Since then, they have spread across the continent. Sometimes hundreds attend, sometimes dozens. They happen in libraries and community centers and coworking spaces, under festive tents and inside great halls. There ’ s nothing high-tech about this social technology. It speaks to a basic human yearning for face-to- face fellowship. It draws young and old, left and right, poor and rich, churched and unchurched, of all races. When you come to a Civic Saturday and are invited to discuss a question like”Who are you responsible for?”or”What are you willing to risk or to give up for your community?”When that happens, something moves. You are moved. You start telling your story. We start actually seeing one another. You realize that homelessness, gun violence, gentrification, terrible traffic, mistrust of newcomers, fake news — these things aren ’ t someone else ’ s problem, they are the aggregation of your own habits and omissions. Society becomes how you behave. We are never asked to reflect on the content of our citizenship. Most of us are never invited to do more or to be more, and most of us have no idea how much we crave that invitation. We ’ ve since created a civic seminary to start training people from all over to lead Civic Saturday gatherings on their own, in their own towns. In the community of Athens, Tennessee, a feisty leader named Whitney Kimball Coe leads hers in an art and framing shop with a youth choir and lots of little flags. A young activist named Berto Aguayo led his Civic Saturday on a street corner in the Back of the Yards neighborhood of Chicago. Berto was once involved with gangs. Now, he ’ s keeping the peace and organizing political campaigns. In Honolulu, Rafael Bergstrom, a former pro **baseball** player turned photographer and conservationist, leads his under the banner”Civics IS Sexy.”It is. Sometimes I ’ m asked, even by our seminarians :”Isn ’ t it dangerous to use religious language? Won ’ t that just make our politics even more dogmatic and self-righteous?”But this view assumes that all religion is fanatical fundamentalism. It is not. Religion is also moral discernment, an embrace of doubt, a commitment to detach from self and serve others, a challenge to repair the world. In this sense, politics could stand to be a little more like religion, not less. Thus, my final topic today : why civic religion matters now. I want to offer two reasons. One is to counter the culture of hyperindividualism. Every message we get from every screen and surface of the modern marketplace is that each of us is on our own, a free agent, free to manage our own brands, free to live under bridges, free to have side hustles, free to die alone without insurance. Market liberalism tells us we are masters beholden to none, but then it enslaves us in the awful **isolation** of consumerism and status anxiety. Yeah! Millions of us are on to the con now. We are realizing now that a free-for-all is not the same as freedom for all. What truly makes us free is being bound to others in mutual aid and obligation, having to work things out the best we can in our neighborhoods and towns, as if our fates were entwined — because they are — as if we could not secede from one another, because, in the end, we cannot. Binding ourselves this way actually liberates us. It reveals that we are equal in dignity. It reminds us that rights come with responsibilities. It reminds us, in fact, that rights properly understood are responsibilities. The second reason why civic religion matters now is that it offers the healthiest possible story of us and them. We talk about identity politics today as if it were something new, but it ’ s not. All politics is identity politics, a never-ending **struggle** to define who truly belongs. Instead of noxious myths of blood and soil that mark some as forever outsiders, civic religion offers everyone a path to belonging based only a universal creed of contribution, participation, inclusion. In civic religion, the”us”is those who wish to serve, volunteer, vote, listen, learn, empathize, argue better, circulate power rather than hoard it. The”them”is those who don ’ t. It is possible to **judge** the them harshly, but it isn ’ t necessary, for at any time, one of them can become one of us, simply by choosing to live like a citizen. So let ’ s welcome them in. Whitney and Berto and Rafael are gifted welcomers. Each has a distinctive, locally rooted way to make faith in democracy relatable to others. Their slang might be Appalachian or South Side or Hawaiian. Their message is the same : civic love, civic spirit, civic responsibility. Now you might think that all this civic religion stuff is just for overzealous second-generation Americans like me. But actually, it is for anyone, anywhere, who wants to kindle the bonds of trust, affection and joint **action** needed to govern ourselves in freedom. Now maybe Civic Saturdays aren ’ t for you. That ’ s OK. Find your own ways to foster civic habits of the heart. Many forms of beloved civic community are thriving now, in this age of awakening. Groups like Community Organizing Japan, which uses creative performative rituals of storytelling to promote equality for women. In Iceland, civil confirmations, where young people are led by an elder to learn the history and civic traditions of their society, culminating in a rite-of-passage ceremony akin to church confirmation. Ben Franklin Circles in the United States, where friends meet monthly to discuss and reflect upon the virtues that Franklin codified in his autobiography, like justice and gratitude and forgiveness. I know civic religion is not enough to remedy the radical inequities of our age. We need power for that. But power without character is a cure worse than the disease. I know civic religion alone can ’ t fix corrupt institutions, but institutional reforms without new **norms** will not last. Culture is upstream of law. Spirit is upstream of policy. The soul is upstream of the state. We cannot unpollute our politics if we clean only downstream. We must get to the source. The source is our values, and on the topic of values, my advice is simple : have some. Make sure those values are prosocial. Put them into **practice**, and do so in the company of others, with a structure of creed, deed and joyful ritual that ’ ll keep all of you coming back. Those of us who believe in democracy and believe it is still possible, we have the burden of proving it. But remember, it is no burden at all to be in a community where you are seen as fully human, where you have a say in the things that affect you, where you don ’ t need to be connected to be respected. That is called a blessing, and it is available to all who believe. Thank you.

Text Sample 199: POS Dim 2 – 2019 – Score 9.00 – 2019/t002065.txt

Have you ever tried to understand a teenager? It ’ s exhausting, right? You must be puzzled by the fact that some teens do well in school, lead clubs and teams and volunteer in their communities, but they eat Tide Pods for an online challenge, speed and text while **driving**, binge drink and experiment with illicit drugs. How can so many teens be so smart, skilled and responsible — and careless risk-takers at the same time? When I was 16, while frequently **observing** my peers in person as well as on social media, I began to wonder why so many teens took such crazy risks. It seems like getting a certificate from DARE class in the fifth grade can ' t stop them. What was even more alarming to me was that the more they exposed themselves to these harmful risks, the easier it became for them to continue taking risks. Now this confused me, but it also made me incredibly curious. So, as someone with a name that literally means"to explore knowledge,"I started searching for a scientific explanation. Now, it ' s no secret that teens ages 13 to 18 are more prone to risk-taking than children or adults, but what makes them so daring? Do they suddenly become reckless, or is this just a natural phase that they ' re going through? Well neuroscientists have already found evidence that the teen brain is still in the process of maturation — and that this makes them exceptionally poor at decision-making, causing them to fall prey to risky behaviors. But in that case, if the maturing brain is to blame, then why are teens more vulnerable than children, even though their brains are more developed than those of children? Also, not all teens in the world take risks at the same level. Are there some other underlying or unintentional causes driving them to risk- taking? Well, this is exactly what I decided to research. So, I founded my research on the basis of a psychological process known as"habituation,"or simply what we refer to as"getting used to it."Habituation explains how our brains adapt to some behaviors, like lying, with repeated exposures. And this concept inspired me to design a project to determine if the same principle could be applied to the relentless rise of risk-taking in teenagers. So I predicted that habituation to risk-taking may have the potential to change the already-vulnerable teenage brain by **blunting** or even eradicating the negative emotions associated with risk, like fear or guilt. I also thought because they would feel less fearful and guilty, this desensitization would lead them to even more risk-taking. In short, I wanted to conduct a research study to answer one big question : Why do teens keep making outrageous choices that are harmful to their health and well-being? But there was one big **obstacle** in my way. To investigate this problem, I needed teenagers to experiment on, laboratories and devices to measure their brain activity, and teachers or professors to supervise me and guide me along the way. I needed resources. But, you see, I attended a high school in South Dakota with limited opportunity for scientific exploration. My school had athletics, band, choir, debate and other clubs, but there were no STEM programs or research mentors. And the notion of high schoolers doing research or participating in a science fair was completely foreign. Simply put, I didn ' t exactly have the ingredients to make a chef-worthy dish. And these **obstacles** were **frustrating**, but I was also a stubborn teenager. And as the daughter of Bangladeshi immigrants and one of just a handful of Muslim students in my high school in South Dakota, I often **struggled** to fit in. And I wanted to be someone with something to contribute to society, not just be deemed the scarf-wearing brown girl who was an anomaly in my homogenous hometown. I hoped that by doing this research, I could establish this and how valuable scientific exploration could be for kids like me who didn ' t necessarily find their niche elsewhere. So with limited research opportunities, inventiveness allowed me to overcome seemingly impossible **obstacles**. I became more creative in working with a variety of methodologies, materials and subjects. I transformed my unassuming school library into a laboratory and my peers into lab rats. My enthusiastic geography teacher, who also happens to be my school ' s football coach, ended up as my cheerleader, becoming my mentor to sign necessary paperwork. And when it became logistically impossible to use a laboratory electroencephalography, or EEG, which are those electrode devices used to measure emotional responses, I bought a portable EEG headset with my own money, instead of buying the new iPhone X that a lot of kids my age were saving up for. So finally I started the research with 86 students, ages 13 to 18, from my high school. Using the computer cubicles in my school library, I had them complete a computerized decision-making simulation to measure their risk-taking behaviors comparable to ones in the real world, like alcohol use, drug use and gambling. Wearing the EEG headset, the students completed the test 12 times over three days to mimic repeated risk exposures. A control panel on the EEG headset measured their various emotional responses : like attention, interest, excitement, frustration, guilt, **stress** levels and relaxation. They also rated their emotions on well-validated emotion-measuring scales. This meant that I had measured the process of habituation and its effects on decision-making. And it took 29 days to complete this research. And with months of frantically drafting proposals, meticulously computing data in a caffeinated daze at 2am, I was able to finalize my results. And the results showed that habituation to risk-taking could actually change a teen ' s brain by altering their emotional levels, causing greater risk-taking. The students ' emotions that were normally associated with risks, like fear, **stress**, guilt and nervousness, as well as attention, were high when they were first exposed to the risk simulator. This curbed their temptations and enforced self-control, which prevented them from taking more risks. However, the more they were exposed to the risks through the simulator, the less fearful, guilty and **stressed** they became. This caused a situation in which they were no longer able to feel the brain ' s natural fear and caution instincts. And also, because they are teenagers and their brains are still underdeveloped, they became more interested and excited in thrill- seeking behaviors. So what were the consequences? They lacked self-control for logical decision- making, took greater risks and made more harmful choices. So the developing brain alone isn ' t to blame. The process of habituation also plays a key role in risk-taking and risk escalation. Although a teen ' s willingness to seek risk is largely a result of the structural and functional changes associated with their developing brains, the dangerous part that my research was able to highlight was that a habituation to risks can actually physically change a teen ' s brain and cause greater risk-taking. So it ' s the combination of the immature teen brain and the impact of habituation that is like a perfect storm to create more damaging effects. And this research can help parents and the general public understand that teens aren ' t just willfully ignoring **warnings** or simply defying parents by engaging in increasingly more dangerous behavior. The biggest hurdle they ' re facing is their habituation to risks : all the physical, detectable and emotional functional changes that drive and control and influence their over- the-top risk-taking. So yes, we need policies that provide safer environments and limit exposures to high risks, but we also need policies that reflect this insight. These results are a wake-up call for teens, too. It shows them that the natural and necessary fear and guilt that protect them from unsafe situations actually become numb when they repeatedly choose risky behaviors. So with this hope to share my findings with fellow teenagers and scientists, I took my research to the Intel International Science and Engineering Fair, or ISEF, a culmination of over 1, 800 students from 75 countries, regions and territories, who showcase their cutting-edge research and inventions. It ' s like the Olympics of science fair. There, I was able to present my research to experts in neuroscience and psychology and garner valuable feedback. But perhaps the most memorable moment of the week was when the booming speakers suddenly uttered my name during the awards ceremony. I was in such disbelief that I questioned myself : Was this just another”La La Land”blunder like at the Oscars? Luckily, it wasn ’ t. I really had won first place in the category”Behavioral and Social Sciences.”Needless to say, I was not only thrilled to have this recognition, but also the whole experience of science fair that validated my efforts keeps my curiosity alive and strengthens my creativity, perseverance and imagination. This still image of me experimenting in my school library may seem ordinary, but to me, it represents a sort of inspiration. It reminds me that this process taught me to take risks. And I know that might sound incredibly ironic. But I took risks realizing that unforeseen opportunities often come from risk-taking — not the hazardous, negative type that I studied, but the good ones, the positive risks. The more risks I took, the more capable I felt of withstanding my unconventional circumstances, leading to more tolerance, resilience and patience for completing my project. And these lessons have led me to new ideas like : Is the opposite of negative risk-taking also true? Can positive risk- taking escalate with repeated exposures? Does positive **action** build positive brain functioning? I think I just might have my next research idea.

Text Sample 200: POS Dim 2 – 2001 – Score 9.00 – 2001/t000092.txt
I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ’ children, because they were all average. But they weren ’ t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ’ s not right. The good Lord in his infinite wisdom didn ’ t create us all equal as far as intelligence is concerned, any more than we ’ re equal for size, appearance. Not everybody could earn an A or a B, and I didn ’ t like that way of **judging**, and I did know how the alumni of various schools back in the ’ 30s **judged** coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ’ t lose a game. But it seemed that we didn ’ t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ’ t like it, I didn ’ t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, **worthy** accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ’ m sure at the time he did that, I didn ’ t — it didn ’ t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ’ s under your control. If you get too engrossed and involved and concerned in regard to the things over which you have no control, it will adversely affect the things over which you

have control. Then I ran across this simple verse that said,”At God ’ s footstool to confess, a poor soul knelt, and bowed his head. ’ I failed! ’ he cried. The Master said, ’ Thou didst thy best, that is success. ’”From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ’ re capable. I believe that ’ s true. If you make the effort to do the best of which you ’ re capable, trying to improve the situation that exists for you, I think that ’ s success, and I don ’ t think others can **judge** that ; it ’ s like character and reputation — your reputation is what you ’ re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ’ s what the teachers are themselves.”That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by coal oil lamp — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I reply, ’ Where could I find such splendid company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ’ s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or **bold** or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ’ Where could I find such splendid company? ’”And I believe the teaching profession — it ’ s true, you have so many youngsters, and I ’ ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ’ re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ’ re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was”Never be late.”Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ’ s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ’ t clean and neat, so I wouldn ’ t let him go. He couldn ’ t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at **practice**, for example — we start on time, we close on time. The youngsters didn ’ t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ’ ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,”Don ’ t run **practices** late, because you ’ ll go home in a bad mood, and that ’ s not good, for a young married man to go home in a bad mood. When you get older, it doesn ’ t make any difference, but —”So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ’ re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ’ t want that. I used to tell them I was paid to do that. That ’ s my job. I ’ m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ’ s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ’ s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ’ t have the time to go on that. But that helped me, I think, become a better teacher. It ’ s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ’ re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ’ re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word **service**, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ’ t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ’ i ’. I ’ d never seen that before, but he did. Big league **baseball** players — they ’ re very perceptive about those things, and they noticed he had only one ’ i ’ in his name. You ’ d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it”The Road Ahead, or the Road Behind.”He said,”Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ’ t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ’ s game. You and I know deeper down, there ’ s always a chance to win the crown. But when we fail to give our best, we simply haven ’ t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ’ s bearing down that wins the cup. Of dreaming there ’ s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ’ s you and I who make our fates — we open up or close the gates on the road ahead or the road behind.”Reminds me of another set of threes that my dad tried to get across to us : Don ’ t whine. Don ’ t **complain**. Don ’ t make excuses. Just get out there, and whatever you ’ re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ’ re outscored. I ’ ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ’ t know the outcome, I hope they couldn ’ t tell by your **actions** whether you outscored an opponent or the opponent outscored you. That ’ s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ’ d want them to be but they ’ ll be about what they should ; only you will know whether you can do that. And that ’ s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the **score** of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,”The journey is better than the end.”And I like that. I think that it is — it ’ s getting there. Sometimes when you get there, there ’ s almost a let down. But it ’ s the getting there that ’ s the fun. As a basketball coach at UCLA, I liked our **practices** to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ’ d done a decent job during the week. There again, it ’ s getting the players to get that self-satisfaction, in knowing that they ’ d made the effort to do the best of which they are capable. Sometimes I ’ m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,”Suppose that you, in some way, could make the perfect player. What would you want?”And I said,”Well, I ’ d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ’ d want one that could play, too. I ’ d want one to realize that defense usually wins championships, and who would work hard on defense. But I ’ d want one who would play offense, too. I ’ d want him to be unselfish, and look for the pass first and not shoot all the time. And I ’ d want one that could pass and would pass. I ’ ve had some that could and wouldn ’ t, and I ’ ve had some that would and could. So, yeah, I ’ d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ’ d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,”They Call Me Coach,”two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ’ t play varsity when I taught. I thought,”Oh gracious, if these two players, either one of them”—

they were different years, but I thought about each one at the time he was there —”Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ’ s good enough to make it.”And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ’ d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 201: POS Dim 2 – 2001 – Score 3.00 – 2001/t000367.txt

I ’ d like to do pretty much what I did the first time, which is to choose a light-hearted theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, exorcise those evil spirits, and this is still going on, as you know. But it wasn ’ t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ’ re feeling depressed? It ’ s better than Prozac, but I wouldn ’ t recommend it. So what we see in the seventeenth, eighteenth century is the continued search for medications other than camphor that ’ ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we **fast-forward** to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a **row** — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very interested in producing convulsions, measured types of convulsions. And they thought,”Well, we ’ ve got electricity, we ’ ll plug somebody into the wall. That always makes hair stand up and people shake a lot.”So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,”We know that at the Rome railroad station, there are all these lost souls wandering around, muttering gibberish. Can you bring one of them to us?”Someone who is, as the Italians say,”cagoots.”So they found this”cagoots”guy, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,”Well, we ’ ll try 55 volts, two-tenths of a second. That ’ s not going to do anything terrible to him.”So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,”This fellow”— remember, he wasn ’ t even put to sleep —”after this major grand mal convulsion, sat right up, looked at these three fellas and said, ’ What the fuck are you assholes trying to do? ’”If I could only say that in Italian. Well, they were happy as could be, because he hadn ’ t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and everybody in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ’ t work very well on the schizophrenics, but it was pretty clear in the ’ 30s and by the middle of the ’ 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ’ t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the brain waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ’ 60s, the first antidepressants came out. Tofranil was the first. In the late ’ 70s, early ’ 80s, there were others, and they were very effective. And patients ’ rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ’ m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ’ ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ’ m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ' clock in the afternoon, because I couldn ' t get out of bed before about 11 o ' clock. And anybody who ' s been depressed here knows what that ' s like. I couldn ' t even pull the covers off myself. Well, you ' re in a university medical center, where everybody knows everybody, and it ' s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ' t work anymore. And, in fact, it didn ' t make any difference because I didn ' t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,"Don ' t worry, chap. Six weeks, you ' re back in the operating room. Everything ' s going to be great."Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ' t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five feet in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ' t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had. They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the senior staff. All the senior staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful stuff. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it got so there was a throbbing, there was a ferocious fear in my head. You ' ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ' t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case.

He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the senior staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,"No. I know this man better than any of you. I have met with him over and over again. You ' ve just seen him from time to time. You ' ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ' ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go back to surgery, he will never be the loving father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ' One Flew Over the Cuckoo ' s Nest. ' And you know about that, just essentially in a stupor the rest of his life."Well, he said,"Can ' t we try a course of electroshock therapy?"And you know why they agreed? They agreed to humor him. They just thought,"Well, we ' ll give a course of 10. And so we ' ll lose a little time. Big deal. It doesn ' t make any difference."So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ' t work. Seven didn ' t work. Eight didn ' t work. At nine, I noticed — and it ' s wonderful that I could notice anything — I noticed a change. And at 10, I noticed a real change. And he went back to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, **lo** and behold, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could blow the obsessional thinking away. I could blow the depression away. And I ' ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,"I ' ve got the strength now to do this."It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a formula. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember"Ruddigore,"and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,"We must have a word to bring you back to reality, and the word, my dear, will be ' Basingstoke. '"So every time she got a little nuts, he would say,"Basingstoke!"And she would say,"Basingstoke, it is."And she would be fine for a little while. Well, you know, I ' m from the Bronx. I can ' t say"Basingstoke."But I had something better. And it was very simple. It was,"Ah, fuck it!"Much better than"Basingstoke,"at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,"Ah, fuck it."And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went back to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came back to live with us. We had two more children after that. Resuscitated the career, even better than it had been before. Went right back into the university and began to write books. Well, you know, it ’ s been a wonderful life. It ’ s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ’ s been very exciting. It ’ s been very happy. Every once in a while, I have to say,”Ah, fuck it.”Every once in a while, I get somewhat depressed and a little obsessional. So, I ’ m not free of all of this. But it ’ s worked. It ’ s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ’ ve gotten thousands of letters about them by people who do think this — that based on my life ’ s history as I ’ ve portrayed in the books, my early life ’ s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the bitter dregs of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ’ ve always felt guilty about that. I ’ ve always felt that somehow I was an impostor because my readers don ’ t know what I have just told you. It ’ s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an untroubled mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes back to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ’ t happen, there will be adversities. If I, with the bleakness of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way back from this, believe me, anybody can find their way back from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

Text Sample 202: POS Dim 2 – 2001 – Score 2.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,”Are you still doing this or that?”I don ’ t answer because I ’ m not doing things still, I ’ m doing it like I always did. I still have — or did I use the word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years. But, of course, I ’ m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ’ ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called”Innovation.”Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,”The playful search for beauty was Man ’ s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty.”These are tiles. The word,”playful”is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ’ t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ’ t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an **ad** into the paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very **pleasant** and happy view. Here I am now, if this is the end? Thank you.

Text Sample 203: POS Dim 2 – 2016 – Score 12.00 – 2016/t001379.txt
I ’ m here to talk to you today about a story that we have all been conditioned to believe is not possible. It ’ s a story about a living, breathing **start-up** flourishing in an unlikely environment : the United States government. Now, this **start-up** is fundamentally beginning to disrupt the way government does business from the inside out. But before I get there, let ’ s start with the problem. For me, the problem begins with a number : 137. 137 is the average number of days a veteran has to wait to have benefits processed by the VA. 137 days. Now, in order to file that application in the first place, she has to **navigate** over 1, 000 different websites and over 900 different call-in numbers, all owned and operated by the United States government. Now, we live in times of incredible change. The private sector is constantly changing and improving itself all the time. For that matter, it ’ s removing every single inconvenience in my life that I could possibly think of. I could be sitting on my couch in my apartment, and from my phone, I can order a warm, gluten-free meal that can arrive at my door in less than 10 minutes. But meanwhile, a working mother who depends on food stamps to support her family has to complete an arduous, complicated application which she might not even be able to do online. And the inability of her to do that same work from her couch means that she might be having to take days or hours off of work that she can ’ t spare. And this growing dichotomy between the beneficiaries of the tech revolution and those it ’ s left behind is one of the greatest challenges of our time — Because government ’ s failure to deliver digital **services** that work is disproportionately impacting the very people who need it most. It ’ s impacting the students trying to go to college, the single mothers trying to get health care, the veterans coming home from battle. They can ’ t get what they need when they need it. And for these Americans, government is more than just a presidential election every four years. Government is a lifeline that provides **services** they need and depend on and deserve. Which is, quite frankly, why government needs to get its shit together and catch up. Just saying. Now, this wasn ’ t always a problem I was **passionate** about. When I joined President Obama ’ s campaign in 2008, we brought the tech industry ’ s best **practices** into politics. We earned more money, we engaged more volunteers and we earned more votes than any political campaign in history. We were a cutting- edge **start-up** that changed the game of politics forever. So when the President asked a small group of us to bring that very same disruption directly into government, I knew it wasn ’ t going to be easy work, but I was eager and showed up ready to get to work. Now, on my first day in DC, my first day in government, I walked into the office and they handed me a laptop. And the laptop was running Windows 98. I mean, three entire presidential elections had come and gone since the government had updated the operating system on that computer. Three elections! Which is when we realized this problem was a whole lot bigger than we ever could have imagined. Let me paint the picture for you. The federal government is the largest institution in the world. It spends over 86 billion dollars a year — 86 billion — on federal IT projects. For context : that is more than the entire

venture capital industry spends annually — on everything. Now, the problem here is that we the taxpayers are not getting what we pay for, because 94 percent of federal IT projects are over budget or behind schedule. 94 percent! For those of you keeping **score**, yes, the number 94 is very close to 100. There ' s another problem : 40 percent of those never end up seeing the light of day. They are completely scrapped or abandoned. Now, this is a very existentially painful moment for any organization, because it means as government continues to operate as it ' s programmed to do, failure is nearly inevitable. And when the status quo is the riskiest option, that means there is simply no other choice than radical disruption. So, what do we do about it? How do we fix this? Well, the irony of all of this is that we actually don ' t have to look any further than our backyard, because right here in America are the very ideas, the very people, who have swept our world into a radically different place than it was two decades ago. So what would it look like if it was actually as easy to get student loans or veterans ' benefits as it is to order cat food to my house? What would it look like if there was an easy pathway for the very entrepreneurs and innovators who have disrupted our tech sector to come and disrupt their government? Well, my friends, here ' s where we get to talk about some of the exciting new formulas we ' ve discovered for creating change in government. Enter the United States Digital Service. The United States Digital Service is a new network of **start-ups**, a team of teams, organizing themselves across government to create radical change. The **mission** of the United States Digital Service is to help government deliver world-class digital **services** for students, immigrants, children, the elderly — everybody — at dramatically lower costs. We are essentially trying to build a more awesome government, for the people, by the people, today. We don ' t care — Thank you. Who doesn ' t want a more awesome government, right? We don ' t care about politics. We care about making government work better, because it ' s the only one we ' ve got. Now, you can think of our team — well, it ' s pretty funny — you can think of our team a little bit like the Peace Corps meets DARPA meets SEAL Team 6. We ' re like the Peace Corps for nerds, but instead of traveling to crazy, interesting, far-off places, you spend a lot of time indoors, behind computers, helping restore the fabric of our democracy. Now, this team — our playbook for the United States Digital Service is pretty simple. The first play is we recruit the very best talent our country has to offer, and recruit them for short tours of duty inside government. These are the very people who have helped build the products and companies that have made our tech sector amongst the most innovative in the world. Second, we pair these incredible people from the tech core with the dedicated civil servants already inside government on the ground creating change. Third, we strategically deploy them in a targeted formation at the most mission-critical, life-changing, important **services** that government offers. And finally, we give them massive air cover, from the leadership inside the agencies all the way up to the President himself, to transform these **services** for the better. Now, this team is beginning to disrupt how government does business from the inside out. If you study classic patterns of disruption, one very common pattern is rather simple. It ' s to take something that has become routine and standard in one industry and apply it to another where it ' s a radical departure from the status quo. Think about what Airbnb took that was normal from hospitality and revolutionized my apartment. The United States Digital Service is doing exactly that. We are taking what Silicon Valley and the private sector has learned through a ton of hard work about how to build planetary-scale digital **services** that delight users at lower cost, and we ' re applying that to government, where it is a radical departure from the status quo. Now, the good news is : it ' s starting to work. We know this because we can already see the results from some of our early projects, like the rescue effort of Healthcare. gov, when that went off the rails. Fixing Healthcare. gov was the first place that we ran this play, and today we are taking that same play and scaling it across a large number of government ’ s most important citizen-facing **services**. Now, if I can take a moment and brag about the team for a second — it is the highest concentration of badasses I could have ever dreamed of. We have top talent from Google, Facebook, Amazon, Twitter and the likes, all on staff today, all choosing to join their government. And what ’ s incredible is, everybody is as eager and kind as they are intelligent. And I might add, by the way, over half of us are women. The best way to understand this strategy is actually to walk through a couple of examples of how it ’ s working out in the wild. I ’ m going to give you two examples quickly. The first one is about immigration. This, my friends, is your typical immigration application. Yes, you guessed it — it ’ s almost entirely paper- based. In the best case, the application takes about six to eight months to process. It is physically shipped thousands of miles — thousands of miles! — between no less than six processing centers. Now, little story : about a decade ago, the government thought that if it brought this system online, it could save taxpayer dollars and provide a better **service**, which was a great idea. So, the typical government process began. Six years and 1. 2 billion dollars later, no working product was delivered — 1. 2 billion with a”B.”Now at this point, the agency responsible, US Citizenship and Immigration Services, could have kept pouring money into the failing program. Sadly, that ’ s what often happens. That ’ s the status quo today. But they didn ’ t. The dedicated civil servants inside the agency decided to stand up and call for change. We deployed a small team of just six people, and what many people don ’ t know is that ’ s the same size as the rescue effort of Healthcare. gov — just six people. And that team jumped in, side-by-side, to support the agency in transitioning this project into more modern business **practices**, more modern development **practices**. Now, in non-tech speak, what that basically means is taking big, multi-year projects and breaking them up into bite-sized chunks, so that way we can reduce the risk and actually start to see results every couple of weeks, instead of waiting in a black box for years. So within less than three months of our team being on the ground, we were already able to push our first products to production. The first one, this is the form I-90. This is used to file for your replacement green card. Now, for immigrant visa holders, a replacement green card is a big deal. Your green card is your proof of identification, it ’ s your work authorization, it ’ s the proof that you can be here in this country. So waiting six months while the government processes the replacement is not cool. I ’ m excited to tell you that today, you can now, for the first time, file for a replacement green card entirely online without anyone touching a piece of paper. It is faster, it is cheaper, and it ’ s a better user experience for the applicant and the government employees alike. Another one, quickly. Last fall, we just released a brand-new **practice** civics test. So as part of becoming a US citizen, you have to pass a civics test. For anyone who has taken this test, it can be quite the **stressful** process. So our team released a very easy, simple-to-use tool in plain language to help people prepare, to help ease their nerves, to help them feel more confident in taking the next step in pursuing their American dream. Because all of this work, all of this work on immigration, is about taking complicated processes and making them more human. The other day, one of the dedicated civil servants on the ground said something incredibly profound. She said that she ’ s never been this hopeful or optimistic about a project in her entire time in government. And she ’ s been doing this for 30 years. That is exactly the kind of hope and culture change we are trying to create. For my second example, I want to bring it back to veterans for a second, and what we are doing to build them a VA that is **worthy** of their **service** and their sacrifice. I ’ m proud to say that just a few months ago, we released a brand-new beta of a new website, Vets. gov. Vets. gov is a simple, easy-to-use website that brings all of the online **services** a veteran needs into one place. One website, not thousands. The site is a work in progress, but it ’ s significant progress, because it ’ s designed with the users who matter most : the veterans themselves. This might sound incredibly obvious, because it should be, but sadly, this isn ’ t normal for government. Far too often, product decisions are made by committees of stakeholders who do their best to represent the interests of the user, but they ’ re not necessarily the users themselves. So our team at the VA went out, we looked at the data, we talked to veterans themselves and we started simple and small, with the two most important **services** that matter most to them : education benefits and disability benefits. I ’ m proud to say that they are live on the site today, and as the team continues to streamline more **services**, they will be ported over here, and the old sites, shut down. To me, this is what change looks like in 2016. When you walk out of the Oval Office, the first time I was ever there, I noticed a quote the President had embroidered on the rug. It ’ s the classic JFK quote. It says,”No problem of human destiny is beyond human beings.”It ’ s true. We have the tools to solve these problems. We have the tools to come together as a society, as a country, and to fix this together. Yes, it ’ s hard. It ’ s particularly hard when we have to fight, when we have to refuse to succumb to the belief that things won ’ t change. But in my experience, it ’ s often the hardest things that are the most worth doing, because if we don ’ t do them, who will? This is on us, all of us, together, because government is not an abstract institution or a concept. Our government is us. Today, it is no longer a question of if change is possible. The question is not,”Can we?”The question is,”Will we?”Will you? Thank you. Thank you.

Text Sample 204: POS Dim 2 – 2016 – Score 11.00 – 2016/t001322.txt

So I ’ d like you to imagine for a moment that you ’ re a soldier in the heat of battle. Maybe you ’ re a Roman foot soldier or a medieval archer or maybe you ’ re a Zulu warrior. Regardless of your time and place, there are some things that are constant. Your adrenaline is elevated, and your **actions** are stemming from these deeply ingrained reflexes, reflexes rooted in a need to protect yourself and your side and to defeat the enemy. So now, I ’ d like you to imagine playing a very different role, that of the scout. The scout ’ s job is not to attack or defend. The scout ’ s job is to understand. The scout is the one going out, mapping the terrain, identifying potential **obstacles**. And the scout may hope to learn that, say, there ’ s a bridge in a convenient location across a river. But above all, the scout wants to know what ’ s really there, as accurately as possible. And in a real, actual army, both the soldier and the scout are essential. But you can also think of each of these roles as a **mindset** — a metaphor for how all of us process information and ideas in our **daily** lives. What I ’ m going to argue today is that having good judgment, making accurate predictions, making good decisions, is mostly about which **mindset** you ’ re in. To illustrate these mindsets in **action**, I ’ m going to take you back to 19th- century France, where this innocuous-looking piece of paper launched one of the biggest political scandals in history. It was discovered in 1894 by officers in the French general staff. It was torn up in a wastepaper basket, but when they pieced it back together, they discovered that someone in their ranks had been selling military secrets to Germany. So they launched a big investigation, and their suspicions quickly converged on this man, Alfred Dreyfus. He had a sterling record, no past history of wrongdoing, no motive as far as they could tell. But Dreyfus was the only Jewish officer at that rank in the army, and unfortunately at this time, the French Army was highly anti-Semitic. They compared Dreyfus ’ s handwriting to that on the memo and concluded that it was a match, even though outside professional handwriting experts were much less confident in the similarity, but never mind that. They went and searched Dreyfus ’ s apartment, looking for any signs of espionage. They went through his files, and they didn ’ t find anything. This just convinced them more that Dreyfus was not only guilty, but sneaky as well, because clearly he had hidden all of the evidence before they had managed to get to it. Next, they went and looked through his personal history for any incriminating details. They talked to his teachers, they found that he had studied foreign languages in school, which clearly showed a desire to conspire with foreign governments later in life. His teachers also said that Dreyfus was known for having a good memory, which was highly suspicious, right? You know, because a spy has to remember a lot of things. So the case went to **trial**, and Dreyfus was found guilty. Afterwards, they took him out into this public square and ritualistically tore his insignia from his uniform and broke his sword in two. This was called the Degradation of Dreyfus. And they sentenced him to life imprisonment on the aptly named Devil ’ s Island, which is this barren rock off the coast of South America. So there he went, and there he spent his days alone, writing letters and letters to the French government begging them to reopen his case so they could discover his innocence. But for the most part, France considered the matter closed. One thing that ’ s really interesting to me about the Dreyfus Affair is this question of why the officers were so convinced that Dreyfus was guilty. I mean, you might even assume that they were setting him up, that they were intentionally framing him. But historians don ’ t think that ’ s what happened. As far as we can tell, the officers genuinely believed that the case against Dreyfus was strong. Which makes you wonder : What does it say about the human mind that we can find such paltry evidence to be compelling enough to convict a man? Well, this is a case of what scientists call”motivated reasoning.”It ’ s this phenomenon in which our unconscious motivations, our desires and fears, shape the way we **interpret** information. Some information, some ideas, feel like our allies. We want them to win. We want to defend them. And other information or ideas are the enemy, and we want to shoot them down. So this is why I call motivated reasoning,”soldier **mindset**.”Probably most of you have never persecuted a French-Jewish officer for high treason, I assume, but maybe you ’ ve followed sports or politics, so you might have noticed that when the referee **judges** that your team committed a foul, for example, you ’ re highly motivated to find reasons why he ’ s wrong. But if he **judges** that the other team committed a foul — awesome! That ’ s a good call, let ’ s not examine it too closely. Or, maybe you ’ ve read an article or a study that examined some controversial policy, like capital punishment. And, as **researchers** have demonstrated, if you support capital punishment and the study shows that it ’ s not effective, then you ’ re highly motivated to find all the reasons why the study was poorly designed. But if it shows that capital punishment works, it ’ s a good study. And vice versa : if you don ’ t support capital punishment, same thing. Our judgment is strongly influenced, unconsciously, by which side we want to win. And this is ubiquitous. This shapes how we think about our health, our relationships, how we decide how to vote, what we consider fair or ethical. What ’ s most scary to me about motivated reasoning or soldier **mindset**, is how unconscious it is. We can think we ’ re being objective and fair-minded and still wind up ruining the life of an innocent man. However, fortunately for Dreyfus, his story is not over. This is Colonel Picquart. He ’ s another high-ranking officer in the French Army, and like most people, he assumed Dreyfus was guilty. Also like most people in the army, he was at least casually anti-Semitic. But at a certain point, Picquart began to suspect :”What if we ’ re all wrong about Dreyfus?”What happened was, he had discovered evidence that the spying for Germany had continued, even after Dreyfus was in prison. And he had also discovered that another officer in the army had handwriting that perfectly matched the memo, much closer than Dreyfus ’ s handwriting. So he brought these discoveries to his superiors, but to his dismay, they either didn ’ t care or came up with elaborate rationalizations to explain his findings, like,”Well, all you ’ ve really shown, Picquart, is that there ’ s another spy who learned how to mimic Dreyfus ’ s handwriting, and he picked up the torch of spying after Dreyfus left. But Dreyfus is still guilty.”Eventually, Picquart managed to get Dreyfus exonerated. But it took him 10 years, and for part of that time, he himself was in prison for the crime of disloyalty to the army. A lot of people feel like Picquart can ’ t really be the hero of this story because he was an anti-Semite and that ’ s bad, which I agree with. But personally, for me, the fact that Picquart was anti-Semitic actually makes his **actions** more admirable, because he had the same prejudices, the same reasons to be **biased** as his fellow officers, but his motivation to find the truth and uphold it trumped all of that. So to me, Picquart is a poster child for what I call”scout **mindset**.”It ’ s the drive not to make one idea win or another lose, but just to see what ’ s really there as honestly and accurately as you can, even if it ’ s not pretty or convenient or **pleasant**. This **mindset** is what I ’ m personally **passionate** about. And I ’ ve spent the last few years examining and trying to figure out what causes scout **mindset**. Why are some people, sometimes at least, able to cut through their own prejudices and **biases** and motivations and just try to see the facts and the evidence as objectively as they can? And the answer is emotional. So, just as soldier **mindset** is rooted in emotions like defensiveness or tribalism, scout **mindset** is, too. It ’ s just rooted in different emotions. For example, scouts are curious. They ’ re more likely to say they feel pleasure when they learn new information or an itch to solve a puzzle. They ’ re more likely to feel intrigued when they encounter something that contradicts their expectations. Scouts also have different values. They ’ re more likely to say they think it ’ s virtuous to test your own beliefs, and they ’ re less likely to say that someone who changes his mind seems weak. And above all, scouts are grounded, which means their self-worth as a person isn ’ t tied to how right or wrong they are about any particular topic. So they can believe that capital punishment works. If studies come out showing that it doesn ’ t, they can say,”Huh. Looks like I might be wrong. Doesn ’ t mean I ’ m bad or stupid.”This cluster of traits is what **researchers** have found — and I ’ ve also found anecdotally — predicts good judgment. And the key takeaway I want to leave you with about those traits is that they ’ re primarily not about how smart you are or about how much you know. In fact, they don ’ t correlate very much with IQ at all. They ’ re about how you feel. There ’ s a quote that I keep coming back to, by Saint-Exupry. He ’ s the author of”The Little Prince.”He said,”If you want to build a ship, don ’ t drum up your men to collect wood and give orders and distribute the work. Instead, teach them to yearn for the vast and endless sea.”In other words, I claim, if we really want to improve our judgment as individuals and as societies, what we need most is not more instruction in logic or rhetoric or probability or economics, even though those things are quite valuable. But what we most need to use those principles well is scout **mindset**. We need to change the way we feel. We need to learn how to feel proud instead of ashamed when we notice we might have been wrong about something. We need to learn how to feel intrigued instead of defensive when we encounter some information that contradicts our beliefs. So the question I want to leave you with is : What do you most yearn for? Do you yearn to defend your own beliefs? Or do you yearn to see the world as clearly as you possibly can? Thank you.

Text Sample 205: POS Dim 2 – 2016 – Score 11.00 – 2016/t001113.txt

An article in the Yale Alumni Magazine told the story of Clyde Murphy, a black man who was a member of the Class of 1970. Clyde was a success story. After Yale and a law degree from Columbia, Clyde spent the next 30 years as one of America ’ s top civil rights lawyers. He was also a great husband and father. But despite his success, personally and professionally, Clyde ’ s story had a sad ending. In 2010, at the age of 62, Clyde died from a blood clot in his lung. Clyde ’ s experience was not unique. Many of his black classmates from Yale also died young. In fact, the magazine article indicated that 41 years after graduation from Yale, the black members of the Class of 1970 had a death rate that was three times higher than that of the average class member. It ’ s stunning. America has recently awakened to a steady drumbeat of unarmed black men being shot by the police. What is even a bigger story is that every seven minutes, a black person dies prematurely in the United States. That is over 200 black people die every single day who would not die if the health of blacks and whites were equal. For the last 25 years, I have been on a **mission** to understand why does race matter so profoundly for health. When I started my career, many believed that it was simply about racial differences in income and education. I discovered that while economic status matters for health, there is more to the story. So for example, if we look at life expectancy at age 25, at age 25 there ’ s a five- year gap between blacks and whites. And the gap by education for both whites and blacks is even larger than the racial gap. At the same time, at every level of education, whites live longer than blacks. So whites who are high school dropouts live 3. 4 years longer than their black counterparts, and the gap is even larger among college graduates. Most surprising of all, whites who have graduated from high school live longer than blacks with a college degree or more education. So why does race matter so profoundly for health? What else is it beyond education and income that might matter? In the early 1990s, I was asked to review a new book on the health of black America. I was struck that almost every single one of its 25 chapters said that racism was a factor that was hurting the health of blacks. All of these **researchers** were stating that racism was a factor adversely impacting blacks, but they provided no evidence. For me, that was not good enough. A few months later, I was speaking at a conference in Washington, DC, and I said that one of the priorities for research was to document the ways in which racism affected health. A white gentleman stood in the audience and said that while he agreed with me that racism was important, we could never measure racism.”We measure self-esteem,”I said.”There ’ s no reason why we can ’ t measure racism if we put our minds to it.”And so I put my mind to it and developed three scales. The first one captured major experiences of discrimination, like being unfairly fired or being unfairly stopped by the police. But discrimination also occurs in more minor and subtle experiences, and so my second scale, called the Everyday Discrimination Scale, captures nine items that captures experiences like you ’ re treated with less courtesy than others, you receive poorer **service** than others in restaurants or stores, or people act as if they ’ re afraid of you. This scale captures ways in which the dignity and the respect of people who society does not value is chipped away on a **daily** basis. Research has found that higher levels of discrimination are associated with an elevated risk of a broad range of diseases from blood pressure to abdominal obesity to breast cancer to heart disease and even premature mortality. Strikingly, some of the effects are **observed** at a very young age. For example, a study of black teens found that those who reported higher levels of discrimination as teenagers had higher levels of **stress** hormones, of blood pressure and of weight at age 20. However, the **stress** of discrimination is only one aspect. Discrimination and racism also matters in other profound ways for health. For example, there ' s discrimination in medical care. In 1999, the National Academy of Medicine asked me to serve on a committee that found, concluded based on the scientific evidence, that blacks and other minorities receive poorer quality care than whites. This was true for all kinds of medical treatment, from the most simple to the most technologically sophisticated. One explanation for this pattern was a phenomenon that ' s called"implicit **bias**"or"unconscious discrimination."Research for decades by social psychologists indicates that if you hold a negative stereotype about a group in your subconscious mind and you meet someone from that group, you will **discriminate** against that person. You will treat them differently. It ' s an unconscious process. It ' s an automatic process. It is a subtle process, but it ' s normal and it occurs even among the most well-intentioned individuals. But the deeper that I delved into the health impact of racism, the more insidious the effects became. There is institutional discrimination, which refers to discrimination that exists in the processes of social institutions. Residential segregation by race, which has led to blacks and whites living in very different neighborhood contexts, is a classic example of institutional racism. One of America ' s best-kept secrets is how residential segregation is the secret source that creates racial inequality in the United States. In America, where you live determines your access to opportunities in education, in employment, in housing and even in access to medical care. One study of the 171 largest cities in the United States concluded that there is not even one city where whites live under equal conditions to blacks, and that the worst urban contexts in which whites reside is considerably better than the average context of black communities. Another study found that if you could eliminate statistically residential segregation, you would completely erase black-white differences in income, education and unemployment, and reduce black-white differences in single motherhood by two thirds, all of that driven by segregation. I have also learned how the negative stereotypes and images of blacks in our culture literally create and sustain both institutional and individual discrimination. A group of **researchers** have put together a database that contains the books, magazines and articles that an average college-educated American would read over their lifetime. It allows us to look within this database and see how Americans have seen words paired together as they grow up in their society. So when the word"black"appears in American culture, what co-occurs with it?"Poor,""violent,""religious,""lazy,""cheerful,""dangerous."When"white"occurs, the frequently co-occurring words are"wealthy,""progressive,""conventional,""stubborn,""successful,""educated."So when a police officer overreacts when he sees an unarmed black male and perceives him to be violent and dangerous, we are not necessarily dealing with an inherently bad cop. We may be simply viewing a normal American who is reflecting what he has been exposed to as a result of being raised in this society. From my own experience, I believe that your race does not have to be a determinant of your destiny. I migrated to the United States from the Caribbean island of Saint Lucia in the late 1970s in pursuit of higher education, and in the last 40 years, I have done well. I have had a supportive family, I have worked hard, I have done well. But it took more for me to be successful. I received a minority fellowship from the University of Michigan. Yes. I am an affirmative **action** baby. Without affirmative **action**, I would not be here. But in the last 40 years, black America has been less successful than I have. In 1978, black households in the United States earned 59 cents for every dollar of income whites earned. In 2015, black families still earn 59 cents for every dollar of income that white families receive, and the racial gaps in wealth are even more stunning. For every dollar of wealth that whites have, black families have six pennies and Latinos have seven pennies. The fact is, racism is producing a truly rigged system that is systematically disadvantaging some racial groups in the United States. To paraphrase Plato, there is nothing so unfair as the equal treatment of unequal people. And that ' s why I am committed to working to dismantle racism. I deeply appreciate the fact that I am standing on the shoulders of those who have sacrificed even their lives to open the doors that I have walked through. I want to ensure that those doors remain open and that everyone can walk through those doors. Robert Kennedy said,"Each time a man"— or woman, I would add —"stands up for an ideal or acts to improve the lot of others or strikes out against injustice, he sends forth a tiny ripple of hope, and those ripples can build a current that can sweep down the mightiest walls of oppression and resistance."I am optimistic today because all across America, I have seen ripples of hope. The Boston Medical Center has added lawyers to the medical team so that physicians can improve the health of their patients because the lawyers are addressing the nonmedical needs their patients have. Loma Linda University has built a gateway college in **nearby** San Bernardino so that in addition to delivering medical care, they can provide job skills and job training to a predominantly minority, low-income community members so that they will have the skills they need to get a decent job. In Chapel Hill, North Carolina, the Abecedarian Project has figured out how to ensure that they have lowered the risks for heart disease for blacks in their mid-30s by providing high-quality day care from birth to age five. In after-school centers across the United States, Wintley Phipps and the US Dream Academy is breaking the cycle of incarceration by providing high-quality academic enrichment and mentoring to the children of prisoners and children who have fallen behind in school. In Huntsville, Alabama, Oakwood University, a historically black institution, is showing how we can improve the health of black adults by including a health evaluation as a part of freshman orientation and giving those students the tools they need to make healthy choices and providing them annually a health transcript so they can monitor their progress. And in Atlanta, Georgia, Purpose Built Communities has dismantled the negative effects of segregation by transforming a crime-ridden, drug-infested public housing project into an oasis of mixed-income housing, of academic performance, of great community wellness and of full employment. And finally, there is the Devine solution. Professor Patricia Devine of the University of Wisconsin has shown us how we can attack our hidden **biases** head on and effectively reduce them. Each one of us can be a ripple of hope. This work will not always be easy, but former Supreme Court Justice Thurgood Marshall has told us,"We must dissent. We must dissent from the indifference. We must dissent from the apathy. We must dissent from the hatred and from the mistrust. We must dissent because America can do better, because America has no choice but to do better."Thank you.

Text Sample 206: POS Dim 2 – 2017 – Score 14.00 – 2017/t000778.txt

Consider the following statement : human beings only use 10 percent of their brain capacity. Well, as a neuroscientist, I can tell you that while Morgan Freeman delivered this line with the gravitas that makes him a great actor, this statement is entirely false. The truth is, human beings use 100 percent of their brain capacity. The brain is a highly efficient, energy-demanding organ that gets fully utilized and even though it is at full capacity being used, it suffers from a problem of information overload. There ' s far too much in the environment than it can fully process. So to solve this problem of overload, evolution devised a solution, which is the brain ' s attention system. Attention allows us to notice, select and direct the brain ' s computational resources to a subset of all that ' s available. We can think of attention as the leader of the brain. Wherever attention goes, the rest of the brain follows. In some sense, it ' s your brain ' s boss. And over the last 15 years, I ' ve been studying the human brain ' s attention system. In all of our studies, I ' ve been very interested in one question. If it is indeed the case that our attention is the brain ' s boss, is it a good boss? Does it actually guide us well? And to dig in on this big question, I wanted to know three things. First, how does attention control our perception? Second, why does it fail us, often leaving us feeling foggy and distracted? And third, can we do anything about this fogginess, can we train our brain to pay better attention? To have more strong and stable attention in the work that we do in our lives. So I wanted to give you a brief glimpse into how we ' re going to look at this. A very poignant example of how our attention ends up getting utilized. And I want to do it using the example of somebody that I know quite well. He ends up being part of a very large group of people that we work with, for whom attention is a matter of life and death. Think of medical professionals or firefighters or soldiers or marines. This is the story of a marine captain, Captain Jeff Davis. And the scene that I ' m going to share with you, as you can see, is not about his time in the battlefield. He was actually on a bridge, in Florida. But instead of looking at the scenery around him, seeing the beautiful vistas and noticing the cool ocean breezes, he was driving fast and contemplating driving off that bridge. And he would later tell me that it took all of everything he had not to do so. You see, he ' d just returned from Iraq. And while his body was on that bridge, his mind, his attention, was thousands of miles away. He was gripped with suffering. His mind was worried and preoccupied and had **stressful** memories and, really, **dread** for his future. And I ' m really glad that he didn ' t take his life. Because he, as a leader, knew that he wasn ' t the only one that was probably suffering ; many of his fellow marines probably were, too. And in the year 2008, he partnered with me in the first-of-its-kind project that actually allowed us to test and offer something called mindfulness training to active-duty military personnel. But before I tell you about what mindfulness training is, or the results of that study, I think it ' s important to understand how attention works in the brain. So what we do in the laboratory is that many of our studies of attention involve brain-wave recordings. In these brain wave recordings, people wear funny-looking caps that are sort of like swimming caps, that have electrodes embedded in them. These electrodes pick up the ongoing brain electrical activity. And they do it with millisecond temporal precision. So we can see these small yet detectable voltage fluctuations over time. And doing this, we can very precisely plot the timing of the brain ' s activity. About 170 milliseconds after we show our research participants a face on the screen, we see a very reliable, detectable brain signature. It happens right at the back of the scalp, above the regions of the brain that are involved in face processing. Now, this happens so reliably and so on cue, as the brain ' s face detector, that we ' ve even given this brain-wave component a name. We call it the N170 component. And we use this component in many of our studies. It allows us to see the impact that attention may have on our perception. I ' m going to give you a sense of the kind of experiments that we actually do in the lab. We would show participants images like this one. You should see a face and a scene overlaid on each other. And what we do is we ask our participants as they ' re viewing a series of these types of overlaid images, to do something with their attention. On some **trials**, we ’ ll ask them to pay attention to the face. And to make sure they ’ re doing that, we ask them to tell us, by pressing a button, if the face appeared to be male or female. On other **trials**, we ask them to tell what the scene was — was it indoor or outdoor? And in this way, we can manipulate attention and confirm that the participants were actually doing what we said. Our hypotheses about attention were as follows : if attention is indeed doing its job and affecting perception, maybe it works like an amplifier. And what I mean by this is that when we direct attention to the face, it becomes clearer and more salient, it ’ s easier to see. But when we direct it to the scene, the face becomes barely perceptible as we process the scene information. So what we wanted to do is look at this brain-wave component of face detection, the N170, and see if it changed at all as a function of where our participants were paying attention — to the scene or the face. And here ’ s what we found. We found that when they paid attention to the face, the N170 was larger. And when they paid attention to the scene, as you can see in red, it was smaller. And that gap you see between the blue and red lines is pretty powerful. What it tells us is that attention, which is really the only thing that changed, since the images they viewed were identical in both cases — attention changes perception. And it does so very fast. Within 170 milliseconds of actually seeing a face. In our **follow-up** studies, we wanted to see what would happen, how could we perturb or diminish this effect. And our hunch was that if you put people in a very **stressful** environment, if you distract them with disturbing, negative images, images of suffering and violence — sort of like what you might see on the news, unfortunately — that doing this might actually affect their attention. And that ’ s indeed what we found. If we present **stressful** images while they ’ re doing this experiment, this gap of attention shrinks, its power diminishes. So in some of our other studies, we wanted to see, OK, great — not great, actually, bad news that **stress** does this to the brain — but if it is the case that **stress** has this powerful influence on attention through external distraction, what if we don ’ t need external distraction, what if we distract ourselves? And to do this, we had to basically come up with an experiment in which we could have people generate their own mind-wandering. This is having off-task thoughts while we ’ re engaged in an ongoing task of some sort. And the trick to mind-wandering is that essentially, you bore people. So hopefully there ’ s not a lot of mind-wandering happening right now. When we bore people, people happily generate all kinds of internal content to occupy themselves. So we devised what might be considered one of the world ’ s most boring experiments. All the participants saw were a series of faces on the screen, one after another. They pressed the button every time they saw the face. That was pretty much it. Well, one trick was that sometimes, the face would be upside down, and it would happen very infrequently. On those **trials** they were told just to withhold the response. Pretty soon, we could tell that they were successfully mind-wandering, because they pressed the button when that face was upside down. Even though it ’ s quite plain to see that it was upside down. So we wanted to know what happens when people have mind-wandering. And what we found was that, very similar to external **stress** and external distraction in the environment, internal distraction, our own mind wandering, also shrinks the gap of attention. It diminishes attention ’ s power. So what do all of these studies tell us? They tell us that attention is very powerful in terms of affecting our perception. Even though it ’ s so powerful, it ’ s also fragile and vulnerable. And things like **stress** and mind-wandering diminish its power. But that ’ s all in the context of these very controlled laboratory **settings**. What about in the real world? What about in our actual day-to-day life? What about now? Where is your attention right now? To kind of bring it back, I ’ d like to make a prediction about your attention for the remainder of my talk. Are you up for it? Here ’ s the prediction. You will be unaware of what I ’ m saying for four out of the next eight minutes. It ’ s a challenge, so pay attention, please. Now, why am I saying this? I ’ m surely going to assume that you ’ re going to remain seated and, you know, graciously keep your eyes on me as I speak. But a growing body of literature suggests that we mind-wander, we take our mind away from the task at hand, about 50 percent of our waking moments. These might be small, little trips that we take away, private thoughts that we have. And when this mind-wandering happens, it can be problematic. Now I don ’ t think there will be any dire consequences with you all sitting here today, but imagine a military leader missing four minutes of a military briefing, or a **judge** missing four minutes of testimony. Or a surgeon or firefighter missing any time. The consequences in those cases could be dire. So we might ask why do we do this? Why do we mind- wander so much? Well, part of the answer is that our mind is an exquisite time-traveling master. It can actually time travel very easily. If we think of the mind as the metaphor of the music player, we see this. We can rewind the mind to the past to reflect on events that have already happened, right? Or we can go and fast- future, to plan for the next thing that we want to do. And we land in this mental time-travel mode of the past or the future very frequently. And we land there often without our awareness, most times without our awareness, even if we want to be paying attention. Think of just the last time you were trying to read a book, got to the bottom of the page with no idea what the words were saying. This happens to us. And when this happens, when we mind-wander without an awareness that we ’ re doing it, there are consequences. We make errors. We miss critical information, sometimes. And we have difficulty making decisions. What ’ s worse is when we experience **stress**. When we ’ re in a moment of overwhelm. We don ’ t just reflect on the past when we rewind, we end up being in the past ruminating, reliving or regretting events that have already happened. Or under **stress**, we **fast-forward** the mind. Not just to productively plan. But we end up catastrophizing or worrying about events that haven ’ t happened yet and frankly may never happen. So at this point, you might be thinking to yourself, OK, mind-wandering ’ s happening a lot. Often, it happens without our awareness. And under **stress**, it ’ s even worse — we mind-wander more powerfully and more often. Is there anything we can possibly do about this? And I ’ m happy to say the answer is yes. From our work, we ’ re learning that the opposite of a **stressed** and wandering mind is a mindful one. Mindfulness has to do with paying attention to our present-moment experience with awareness. And without any kind of emotional reactivity of what ’ s happening. It ’ s about keeping that button right on play to experience the moment-to-moment unfolding of our lives. And mindfulness is not just a concept. It ’ s more like **practice**, you have to embody this mindful mode of being to have any benefits. And a lot of the work that we ’ re doing, we ’ re offering people programs that give our participants a suite of exercises that they should do **daily** in order to cultivate more moments of mindfulness in their life. And for many of the groups that we work with, high-**stress** groups, like I said — soldiers, medical professionals — for them, as we know, mind-wandering can be really dire. So we want to make sure we offer them very accessible, low time constraints to optimize the training, so they can benefit from it. And when we do this, what we can do is track to see what happens, not just in their regular lives but in the most demanding circumstances that they may have. Why do we want to do this? Well, we want to, for example, give it to students right around finals season. Or we want to give the training to accountants during tax season. Or soldiers and marines while they ’ re deploying. Why is that? Because those are the moments in which their attention is most likely to be vulnerable, because of **stress** and mind-wandering. And those are also the moments in which we want their attention to be in **peak** shape so they can perform well. So what we do in our research is we have them take a series of attention tests. We track their attention at the beginning of some kind of high-stress interval, and then two months later, we track them again, and we want to see if there ’ s a difference. Is there any benefit of offering them mindfulness training? Can we protect against the lapses in attention that might arise over high **stress**? So here ’ s what we find. Over a high-stress interval, unfortunately, the reality is if we don ’ t do anything at all, attention declines, people are worse at the end of this high- **stress** interval than before. But if we offer mindfulness training, we can protect against this. They stay stable, even though just like the other groups, they were experiencing high **stress**. And perhaps even more impressive is that if people take our training programs over, let ’ s say, eight weeks, and they fully commit to doing the **daily** mindfulness exercises that allow them to learn how to be in the present moment, well, they actually get better over time, even though they ’ re in high **stress**. And this last point is actually important to realize, because of what it suggests to us is that mindfulness exercises are very much like physical exercise : if you don ’ t do it, you don ’ t benefit. But if you do engage in mindfulness **practice**, the more you do, the more you benefit. And I want to just bring it back to Captain Jeff Davis. As I mentioned to you at the beginning, his marines were involved in the very first project that we ever did, offering mindfulness training. And they showed this exact pattern, which was very heartening. We had offered them the mindfulness training right before they were deployed to Iraq. And upon their return, Captain Davis shared with us what he was feeling was the benefit of this program. He said that unlike last time, after this deployment, they were much more present. They were discerning. They were not as reactive. And in some cases, they were really more compassionate with the people they were engaging with and each other. He said in many ways, he felt that the mindfulness training program we offered gave them a really important tool to protect against developing post-traumatic **stress** disorder and even allowing it to turn into post-traumatic growth. To us, this was very compelling. And it ended up that Captain Davis and I — you know, this was about a decade ago, in 2008 — we ’ ve kept in touch all these years. And he himself has gone on to continue **practicing** mindfulness in a **daily** way. He was promoted to major, he actually then ended up retiring from the Marine Corps. He went on to get a divorce, to get remarried, to have a child, to get an MBA. And through all of these challenges and transitions and joys of his life, he kept up with his mindfulness **practice**. And as fate would have it, just a few months ago, Captain Davis suffered a massive heart attack, at the age of 46. And he ended up calling me a few weeks ago. And he said,”I want to tell you something. I know that the doctors who worked on me, they saved my heart, but mindfulness saved my life. The presence of mind I had to stop the ambulance that ended up taking me to the hospital,”— himself, the clarity of mind he had to notice when there was fear and anxiety happening but not be gripped by it — he said,”For me, these were the gifts of mindfulness.”And I was so **relieved** to hear that he was OK. But really heartened to see that he had transformed his own attention. He went from having a really bad boss — an attention system that nearly drove him off a bridge — to one that was an exquisite leader and guide, and saved his life. So I want to actually end by sharing my call to **action** to all of you. And here it is. Pay attention to your attention. Alright? Pay attention to your attention and incorporate mindfulness training as part of your **daily** wellness toolkit, in order to tame your own wandering mind and to allow your attention to be a trusted guide in your own life. Thank you.

Text Sample 207: POS Dim 2 – 2017 – Score 13.00 – 2017/t001157.txt
Chris Anderson : Hello. Welcome to this TED Dialogues. It ’ s the first of a series that ’ s going to be done in response to the current political upheaval. I don ’ t know about you ; I ’ ve become quite concerned about the growing divisiveness in this country and in the world. No one ’ s listening to each other. Right? They aren ’ t. I mean, it feels like we need a different kind of conversation, one that ’ s based on — I don ’ t know, on reason, listening, on understanding, on a broader context. That ’ s at least what we ’ re going to try in these TED Dialogues, starting today. And we couldn ’ t have anyone with us who I ’ d be more excited to kick this off. This is a mind right here that thinks pretty much like no one else on the planet, I would hasten to say. I ’ m serious. I ’ m serious. He synthesizes history with underlying ideas in a way that kind of takes your breath away. So, some of you will know this book,”Sapiens.”Has anyone here read”Sapiens”? I mean, I could not put it down. The way that he tells the story of mankind through big ideas that really make you think differently — it ’ s kind of amazing. And here ’ s the **follow-up**, which I think is being published in the US next week. YNH : Yeah, next week. CA :”Homo Deus.”Now, this is the history of the next hundred years. I ’ ve had a chance to read it. It ’ s extremely dramatic, and I daresay, for some people, quite alarming. It ’ s a must-read. And honestly, we couldn ’ t have someone better to help make sense of what on Earth is happening in the world right now. So a warm welcome, please, to Yuval Noah Harari. It ’ s great to be joined by our friends on Facebook and around the Web. Hello, Facebook. And all of you, as I start asking questions of Yuval, come up with your own questions, and not necessarily about the political scandal du jour, but about the broader understanding of : Where are we heading? You ready? OK, we ’ re going to go. So here we are, Yuval : New York City, 2017, there ’ s a new president in power, and shock waves rippling around the world. What on Earth is happening? YNH : I think the basic thing that happened is that we have lost our story. Humans think in stories, and we try to make sense of the world by telling stories. And for the last few decades, we had a very simple and very attractive story about what ’ s happening in the world. And the story said that, oh, what ’ s happening is that the economy is being globalized, politics is being liberalized, and the combination of the two will create paradise on Earth, and we just need to keep on globalizing the economy and liberalizing the political system, and everything will be wonderful. And 2016 is the moment when a very large segment, even of the Western world, stopped believing in this story. For good or bad reasons — it doesn ’ t matter. People stopped believing in the story, and when you don ’ t have a story, you don ’ t understand what ’ s happening. CA : Part of you believes that that story was actually a very effective story. It worked. YNH : To some extent, yes. According to some measurements, we are now in the best time ever for humankind. Today, for the first time in history, more people die from eating too much than from eating too little, which is an amazing achievement. Also for the first time in history, more people die from old age than from infectious diseases, and violence is also down. For the first time in history, more people commit suicide than are killed by crime and terrorism and war put together. Statistically, you are your own worst enemy. At least, of all the people in the world, you are most likely to be killed by yourself — which is, again, very good news, compared — compared to the level of violence that we saw in previous eras. CA : But this process of connecting the world ended up with a large group of people kind of feeling left

out, and they ’ ve reacted. And so we have this bombshell that ’ s sort of ripping through the whole system. I mean, what do you make of what ’ s happened? It feels like the old way that people thought of politics, the left-right divide, has been blown up and replaced. How should we think of this? YNH : Yeah, the old 20th-century political model of left versus right is now largely irrelevant, and the real divide today is between global and national, global or local. And you see it again all over the world that this is now the main **struggle**. We probably need completely new political models and completely new ways of thinking about politics. In essence, what you can say is that we now have global ecology, we have a global economy but we have national politics, and this doesn ’ t work together. This makes the political system **ineffective**, because it has no control over the forces that shape our life. And you have basically two solutions to this imbalance : either de-globalize the economy and turn it back into a national economy, or globalize the political system. CA : So some, I guess many liberals out there view Trump and his government as kind of irredeemably bad, just awful in every way. Do you see any underlying narrative or political philosophy in there that is at least worth understanding? How would you articulate that philosophy? Is it just the philosophy of nationalism? YNH : I think the underlying feeling or idea is that the political system — something is broken there. It doesn ’ t empower the ordinary person anymore. It doesn ’ t care so much about the ordinary person anymore, and I think this diagnosis of the political disease is correct. With regard to the answers, I am far less certain. I think what we are seeing is the immediate human reaction : if something doesn ’ t work, let ’ s go back. And you see it all over the world, that people, almost nobody in the political system today, has any future-oriented vision of where humankind is going. Almost everywhere, you see retrograde vision :”Let ’ s make America great again,”like it was great — I don ’ t know — in the ’ 50s, in the ’ 80s, sometime, let ’ s go back there. And you go to Russia a hundred years after Lenin, Putin ’ s vision for the future is basically, ah, let ’ s go back to the Tsarist empire. And in Israel, where I come from, the hottest political vision of the present is :”Let ’ s build the temple again.”So let ’ s go back 2, 000 years backwards. So people are thinking sometime in the past we ’ ve lost it, and sometimes in the past, it ’ s like you ’ ve lost your way in the city, and you say OK, let ’ s go back to the point where I felt secure and start again. I don ’ t think this can work, but a lot of people, this is their gut instinct. CA : But why couldn ’ t it work?”America First”is a very appealing slogan in many ways. Patriotism is, in many ways, a very noble thing. It ’ s played a role in promoting cooperation among large numbers of people. Why couldn ’ t you have a world organized in countries, all of which put themselves first? YNH : For many centuries, even thousands of years, patriotism worked quite well. Of course, it led to wars an so forth, but we shouldn ’ t focus too much on the bad. There are also many, many positive things about patriotism, and the ability to have a large number of people care about each other, sympathize with one another, and come together for collective **action**. If you go back to the first nations, so, thousands of years ago, the people who lived along the Yellow River in China — it was many, many different tribes and they all depended on the river for survival and for prosperity, but all of them also suffered from periodical floods and periodical droughts. And no tribe could really do anything about it, because each of them controlled just a tiny section of the river. And then in a long and complicated process, the tribes coalesced together to form the Chinese nation, which controlled the entire Yellow River and had the ability to bring hundreds of thousands of people together to build dams and canals and regulate the river and prevent the worst floods and droughts and raise the level of prosperity for everybody. And this worked in many places around the world. But in the 21st century, technology is changing all that in a fundamental way. We are now living — all people in the world — are living alongside the same cyber river, and no single nation can regulate this river by itself. We are all living together on a single planet, which is threatened by our own **actions**. And if you don ’ t have some kind of global cooperation, nationalism is just not on the right level to tackle the problems, whether it ’ s climate change or whether it ’ s technological disruption. CA : So it was a beautiful idea in a world where most of the **action**, most of the issues, took place on national scale, but your argument is that the issues that matter most today no longer take place on a national scale but on a global scale. YNH : Exactly. All the major problems of the world today are global in essence, and they cannot be solved unless through some kind of global cooperation. It ’ s not just climate change, which is, like, the most obvious example people give. I think more in terms of technological disruption. If you think about, for example, artificial intelligence, over the next 20, 30 years pushing hundreds of millions of people out of the job market — this is a problem on a global level. It will disrupt the economy of all the countries. And similarly, if you think about, say, bioengineering and people being afraid of conducting, I don ’ t know, genetic engineering research in humans, it won ’ t help if just a single country, let ’ s say the US, outlaws all genetic experiments in humans, but China or North Korea continues to do it. So the US cannot solve it by itself, and very quickly, the pressure on the US to do the same will be immense because we are talking about high-risk, high-gain technologies. If somebody else is doing it, I can ’ t allow myself to remain behind. The only way to have regulations, effective regulations, on things like genetic engineering, is to have global regulations. If you just have national regulations, nobody would like to stay behind. CA : So this is really interesting. It seems to me that this may be one key to provoking at least a constructive conversation between the different sides here, because I think everyone can agree that the start point of a lot of the anger that ’ s propelled us to where we are is because of the legitimate concerns about job loss. Work is gone, a traditional way of life has gone, and it ’ s no wonder that people are furious about that. And in general, they have blamed globalism, global elites, for doing this to them without asking their permission, and that seems like a legitimate complaint. But what I hear you saying is that — so a key question is : What is the real cause of job loss, both now and going forward? To the extent that it ’ s about globalism, then the right response, yes, is to shut down borders and keep people out and change trade agreements and so forth. But you ’ re saying, I think, that actually the bigger cause of job loss is not going to be that at all. It ’ s going to originate in technological questions, and we have no chance of solving that unless we operate as a connected world. YNH : Yeah, I think that, I don ’ t know about the present, but looking to the future, it ’ s not the Mexicans or Chinese who will take the jobs from the people in Pennsylvania, it ’ s the robots and algorithms. So unless you plan to build a big wall on the border of California — the wall on the border with Mexico is going to be very **ineffective**. And I was struck when I watched the debates before the election, I was struck that certainly Trump did not even attempt to frighten people by saying the robots will take your jobs. Now even if it ’ s not true, it doesn ’ t matter. It could have been an extremely effective way of frightening people — and galvanizing people :”The robots will take your jobs!”And nobody used that line. And it made me afraid, because it meant that no matter what happens in universities and laboratories, and there, there is already an intense debate about it, but in the mainstream political system and among the general public, people are just unaware that there could be an immense technological disruption — not in 200 years, but in 10, 20, 30 years — and we have to do something about it now, partly because most of what we teach children today in school or in college is going to be completely irrelevant to the job market of 2040, 2050. So it ’ s not something we ’ ll need to think about in 2040. We need to think today what to teach the young people. CA : Yeah, no, absolutely. You ’ ve often written about moments in history where humankind has... entered a new era, unintentionally. Decisions have been made, technologies have been developed, and suddenly the world has changed, possibly in a way that ’ s worse for everyone. So one of the examples you give in”Sapiens”is just the whole agricultural revolution, which, for an actual person tilling the fields, they just picked up a 12-hour backbreaking workday instead of six hours in the jungle and a much more interesting lifestyle. So are we at another possible phase change here, where we kind of sleepwalk into a future that none of us actually wants? YNH : Yes, very much so. During the agricultural revolution, what happened is that immense technological and economic revolution empowered the human collective, but when you look at actual individual lives, the life of a tiny elite became much better, and the lives of the majority of people became considerably worse. And this can happen again in the 21st century. No doubt the new technologies will empower the human collective. But we may end up again with a tiny elite reaping all the benefits, taking all the fruits, and the masses of the population finding themselves worse than they were before, certainly much worse than this tiny elite. CA : And those elites might not even be human elites. They might be cyborgs or — YNH : Yeah, they could be enhanced **super** humans. They could be cyborgs. They could be completely nonorganic elites. They could even be non-conscious algorithms. What we see now in the world is authority shifting away from humans to algorithms. More and more decisions — about personal lives, about economic matters, about political matters — are actually being taken by algorithms. If you ask the bank for a loan, chances are your fate is decided by an algorithm, not by a human being. And the general impression is that maybe Homo sapiens just lost it. The world is so complicated, there is so much data, things are changing so fast, that this thing that evolved on the African savanna tens of thousands of years ago — to cope with a particular environment, a particular volume of information and data — it just can ’ t handle the realities of the 21st century, and the only thing that may be able to handle it is big-data algorithms. So no wonder more and more authority is shifting from us to the algorithms. CA : So we ’ re in New York City for the first of a series of TED Dialogues with Yuval Harari, and there ’ s a Facebook Live audience out there. We ’ re excited to have you with us. We ’ ll start coming to some of your questions and questions of people in the room in just a few minutes, so have those coming. Yuval, if you ’ re going to make the argument that we need to get past nationalism because of the coming technological... danger, in a way, presented by so much of what ’ s happening we ’ ve got to have a global conversation about this. Trouble is, it ’ s hard to get people really believing that, I don ’ t know, AI really is an imminent threat, and so forth. The things that people, some people at least, care about much more immediately, perhaps, is climate change, perhaps other issues like refugees, nuclear weapons, and so forth. Would you argue that where we are right now that somehow those issues need to be dialed up? You ’ ve talked about climate change, but Trump has said he doesn ’ t believe in that. So in a way, your most powerful argument, you can ’ t actually use to make this case. YNH : Yeah, I think with climate change, at first sight, it ’ s quite surprising that there is a very close correlation between nationalism and climate change. I mean, almost always, the people who deny climate change are nationalists. And at first sight, you think : Why? What ’ s the connection? Why don ’ t you have socialists denying climate change? But then, when you think about it, it ’ s obvious — because nationalism has no solution to climate change. If you want to be a nationalist in the 21st century, you have to deny the problem. If you accept the reality of the problem, then you must accept that, yes, there is still room in the world for patriotism, there is still room in the world for having **special** loyalties and obligations towards your own people, towards your own country. I don ’ t think anybody is really thinking of abolishing that. But in order to confront climate change, we need additional loyalties and commitments to a level beyond the nation. And that should not be impossible, because people can have several layers of loyalty. You can be loyal to your family and to your community and to your nation, so why can ’ t you also be loyal to humankind as a whole? Of course, there are occasions when it becomes difficult, what to put first, but, you know, life is difficult. Handle it. CA : OK, so I would love to get some questions from the audience here. We ’ ve got a **microphone** here. Speak into it, and Facebook, get them coming, too. Howard Morgan : One of the things that has clearly made a huge difference in this country and other countries is the income distribution inequality, the dramatic change in income distribution in the US from what it was 50 years ago, and around the world. Is there anything we can do to affect that? Because that gets at a lot of the underlying causes. YNH : So far I haven ’ t heard a very good idea about what to do about it, again, partly because most ideas remain on the national level, and the problem is global. I mean, one idea that we hear quite a lot about now is universal basic income. But this is a problem. I mean, I think it ’ s a good start, but it ’ s a problematic idea because it ’ s not clear what”universal”is and it ’ s not clear what”basic”is. Most people when they speak about universal basic income, they actually mean national basic income. But the problem is global. Let ’ s say that you have AI and 3D printers taking away millions of jobs in Bangladesh, from all the people who make my shirts and my shoes. So what ’ s going to happen? The US government will levy taxes on Google and Apple in California, and use that to pay basic income to unemployed Bangladeshis? If you believe that, you can just as well believe that Santa Claus will come and solve the problem. So unless we have really universal and not national basic income, the deep problems are not going to go away. And also it ’ s not clear what basic is, because what are basic human needs? A thousand years ago, just food and shelter was enough. But today, people will say education is a basic human need, it should be part of the package. But how much? Six years? Twelve years? PhD? Similarly, with health care, let ’ s say that in 20, 30, 40 years, you ’ ll have expensive treatments that can extend human life to 120, I don ’ t know. Will this be part of the basket of basic income or not? It ’ s a very difficult problem, because in a world where people lose their ability to be employed, the only thing they are going to get is this basic income. So what ’ s part of it is a very, very difficult ethical question. CA : There ’ s a bunch of questions on how the world affords it as well, who pays. There ’ s a question here from Facebook from Lisa Larson :”How does nationalism in the US now compare to that between World War I and World War II in the last century?”YNH : Well the good news, with regard to the dangers of nationalism, we are in a much better position than a century ago. A century ago, 1917, Europeans were killing each other by the millions. In 2016, with Brexit, as far as I remember, a single person lost their life, an MP who was murdered by some extremist. Just a single person. I mean, if Brexit was about British independence, this is the most peaceful war of independence in human history. And let ’ s say that Scotland will now choose to leave the UK after Brexit. So in the 18th century, if Scotland wanted — and the Scots wanted several times — to break out of the control of London, the reaction of the government in London was to send an army up north to burn down Edinburgh and massacre the highland tribes. My guess is that if, in 2018, the Scots vote for independence, the London government will not send an army up north to burn down Edinburgh. Very few people are now willing to kill or be killed for Scottish or for British independence. So for all the talk of the rise of nationalism and going back to the 1930s, to the 19th century, in the West at least, the power of national sentiments today is far, far smaller than it was a century ago. CA : Although some people now, you hear publicly worrying about whether that might be shifting, that there could actually be outbreaks of violence in the US depending on how things turn out. Should we be worried about that, or do you really think things have shifted? YNH : No, we should be worried. We should be aware of two things. First of all, don ' t be hysterical. We are not back in the First World War yet. But on the other hand, don ' t be complacent. We reached from 1917 to 2017, not by some divine miracle, but simply by human decisions, and if we now start making the wrong decisions, we could be back in an analogous situation to 1917 in a few years. One of the things I know as a historian is that you should never underestimate human stupidity. It ' s one of the most powerful forces in history, human stupidity and human violence. Humans do such crazy things for no obvious reason, but again, at the same time, another very powerful force in human history is human wisdom. We have both. CA : We have with us here moral psychologist Jonathan Haidt, who I think has a question. Jonathan Haidt : Thanks, Yuval. So you seem to be a fan of global governance, but when you look at the map of the world from Transparency International, which rates the level of corruption of political institutions, it ' s a vast sea of red with little bits of yellow here and there for those with good institutions. So if we were to have some kind of global governance, what makes you think it would end up being more like Denmark rather than more like Russia or Honduras, and aren ' t there alternatives, such as we did with CFCs? There are ways to solve global problems with national governments. What would world government actually look like, and why do you think it would work? YNH : Well, I don ' t know what it would look like. Nobody still has a model for that. The main reason we need it is because many of these issues are lose-lose situations. When you have a win-win situation like trade, both sides can benefit from a trade agreement, then this is something you can work out. Without some kind of global government, national governments each have an interest in doing it. But when you have a lose-lose situation like with climate change, it ' s much more difficult without some overarching authority, real authority. Now, how to get there and what would it look like, I don ' t know. And certainly there is no obvious reason to think that it would look like Denmark, or that it would be a democracy. Most likely it wouldn ' t. We don ' t have workable democratic models for a global government. So maybe it would look more like ancient China than like modern Denmark. But still, given the dangers that we are facing, I think the imperative of having some kind of real ability to force through difficult decisions on the global level is more important than almost anything else. CA : There ' s a question from Facebook here, and then we ' ll get the mic to Andrew. So, Kat Hebron on Facebook, calling in from Vail :"How would developed nations manage the millions of climate migrants?"YNH : I don ' t know. CA : That ' s your answer, Kat. YNH : And I don ' t think that they know either. They ' ll just deny the problem, maybe. CA : But immigration, generally, is another example of a problem that ' s very hard to solve on a nation-by-nation basis. One nation can shut its doors, but maybe that stores up problems for the future. YNH : Yes, I mean — it ' s another very good case, especially because it ' s so much easier to migrate today than it was in the Middle Ages or in ancient times. CA : Yuval, there ' s a belief among many technologists, certainly, that political concerns are kind of overblown, that actually, political leaders don ' t have that much influence in the world, that the real determination of humanity at this point is by science, by invention, by companies, by many things other than political leaders, and it ' s actually very hard for leaders to do much, so we ' re actually worrying about nothing here. YNH : Well, first, it should be emphasized that it ' s true that political leaders ' ability to do good is very limited, but their ability to do harm is unlimited. There is a basic imbalance here. You can still press the button and blow everybody up. You have that kind of ability. But if you want, for example, to reduce inequality, that ' s very, very difficult. But to start a war, you can still do so very easily. So there is a built-in imbalance in the political system today which is very **frustrating**, where you cannot do a lot of good but you can still do a lot of harm. And this makes the political system still a very big concern. CA : So as you look at what ' s happening today, and putting your historian ' s hat on, do you look back in history at moments when things were going just fine and an individual leader really took the world or their country backwards? YNH : There are quite a few examples, but I should emphasize, it ' s never an individual leader. I mean, somebody put him there, and somebody allowed him to continue to be there. So it ' s never really just the fault of a single individual. There are a lot of people behind every such individual. CA : Can we have the **microphone** here, please, to Andrew? Andrew Solomon : You ' ve talked a lot about the global versus the national, but increasingly, it seems to me, the world situation is in the hands of identity groups. We look at people within the United States who have been recruited by ISIS. We look at these other groups which have formed which go outside of national bounds but still represent significant authorities. How are they to be integrated into the system, and how is a diverse set of identities to be made coherent under either national or global leadership? YNH : Well, the problem of such diverse identities is a problem from nationalism as well. Nationalism believes in a single, monolithic identity, and exclusive or at least more extreme versions of nationalism believe in an exclusive loyalty to a single identity. And therefore, nationalism has had a lot of problems with people wanting to divide their identities between various groups. So it ' s not just a problem, say, for a global vision. And I think, again, history shows that you shouldn ' t necessarily think in such exclusive terms. If you think that there is just a single identity for a person,"I am just X, that ' s it, I can ' t be several things, I can be just that,"that ' s the start of the problem. You have religions, you have nations that sometimes demand exclusive loyalty, but it ' s not the only option. There are many religions and many nations that enable you to have diverse identities at the same time. CA : But is one explanation of what ' s happened in the last year that a group of people have got fed up with, if you like, the liberal elites, for want of a better term, obsessing over many, many different identities and them feeling,"But what about my identity? I am being completely ignored here. And by the way, I thought I was the majority"? And that that ' s actually sparked a lot of the anger. YNH : Yeah. Identity is always problematic, because identity is always based on fictional stories that sooner or later collide with reality. Almost all identities, I mean, beyond the level of the basic community of a few dozen people, are based on a fictional story. They are not the truth. They are not the reality. It ' s just a story that people invent and tell one another and start believing. And therefore all identities are extremely unstable. They are not a biological reality. Sometimes nationalists, for example, think that the nation is a biological entity. It ' s made of the combination of soil and blood, creates the nation. But this is just a fictional story. CA : Soil and blood kind of makes a gooey mess. YNH : It does, and also it messes with your mind when you think too much that I am a combination of soil and blood. If you look from a biological perspective, obviously none of the nations that exist today existed 5, 000 years ago. Homo sapiens is a social animal, that ' s for sure. But for millions of years, Homo sapiens and our hominid ancestors lived in small communities of a few dozen individuals. Everybody knew everybody else. Whereas modern nations are imagined communities, in the sense that I don ’ t even know all these people. I come from a relatively small nation, Israel, and of eight million Israelis, I never met most of them. I will never meet most of them. They basically exist here. CA : But in terms of this identity, this group who feel left out and perhaps have work taken away, I mean, in”Homo Deus,”you actually speak of this group in one sense expanding, that so many people may have their jobs taken away by technology in some way that we could end up with a really large — I think you call it a”useless class”— a class where traditionally, as viewed by the economy, these people have no use. YNH : Yes. CA : How likely a possibility is that? Is that something we should be terrified about? And can we address it in any way? YNH : We should think about it very carefully. I mean, nobody really knows what the job market will look like in 2040, 2050. There is a chance many new jobs will appear, but it ’ s not certain. And even if new jobs do appear, it won ’ t necessarily be easy for a 50-year old unemployed truck driver made unemployed by self-driving vehicles, it won ’ t be easy for an unemployed truck driver to reinvent himself or herself as a designer of virtual worlds. Previously, if you look at the trajectory of the industrial revolution, when machines replaced humans in one type of work, the solution usually came from low-skill work in new lines of business. So you didn ’ t need any more agricultural workers, so people moved to working in low-skill industrial jobs, and when this was taken away by more and more machines, people moved to low- skill **service** jobs. Now, when people say there will be new jobs in the future, that humans can do better than AI, that humans can do better than robots, they usually think about high-skill jobs, like software engineers designing virtual worlds. Now, I don ’ t see how an unemployed cashier from Wal-Mart reinvents herself or himself at 50 as a designer of virtual worlds, and certainly I don ’ t see how the millions of unemployed Bangladeshi textile workers will be able to do that. I mean, if they are going to do it, we need to start teaching the Bangladeshis today how to be software designers, and we are not doing it. So what will they do in 20 years? CA : So it feels like you ’ re really highlighting a question that ’ s really been bugging me the last few months more and more. It ’ s almost a hard question to ask in public, but if any mind has some wisdom to offer in it, maybe it ’ s yours, so I ’ m going to ask you : What are humans for? YNH : As far as we know, for nothing. I mean, there is no great cosmic drama, some great cosmic plan, that we have a role to play in. And we just need to discover what our role is and then play it to the best of our ability. This has been the story of all religions and ideologies and so forth, but as a scientist, the best I can say is this is not true. There is no universal drama with a role in it for Homo sapiens. So — CA : I ’ m going to push back on you just for a minute, just from your own book, because in”Homo Deus,”you give really one of the most coherent and understandable accounts about sentience, about consciousness, and that unique sort of human skill. You point out that it ’ s different from intelligence, the intelligence that we ’ re building in machines, and that there ’ s actually a lot of mystery around it. How can you be sure there ’ s no purpose when we don ’ t even understand what this sentience thing is? I mean, in your own thinking, isn ’ t there a chance that what humans are for is to be the universe ’ s sentient things, to be the centers of joy and love and **happiness** and hope? And maybe we can build machines that actually help amplify that, even if they ’ re not going to become sentient themselves? Is that crazy? I kind of found myself hoping that, reading your book. YNH : Well, I certainly think that the most interesting question today in science is the question of consciousness and the mind. We are getting better and better in understanding the brain and intelligence, but we are not getting much better in understanding the mind and consciousness. People often confuse intelligence and consciousness, especially in places like Silicon Valley, which is understandable, because in humans, they go together. I mean, intelligence basically is the ability to solve problems. Consciousness is the ability to feel things, to feel joy and sadness and boredom and pain and so forth. In Homo sapiens and all other mammals as well — it ' s not unique to humans — in all mammals and birds and some other animals, intelligence and consciousness go together. We often solve problems by feeling things. So we tend to confuse them. But they are different things. What ' s happening today in places like Silicon Valley is that we are creating artificial intelligence but not artificial consciousness. There has been an amazing development in computer intelligence over the last 50 years, and exactly zero development in computer consciousness, and there is no indication that computers are going to become conscious anytime soon. So first of all, if there is some cosmic role for consciousness, it ' s not unique to Homo sapiens. Cows are conscious, pigs are conscious, chimpanzees are conscious, chickens are conscious, so if we go that way, first of all, we need to broaden our horizons and remember very clearly we are not the only sentient beings on Earth, and when it comes to sentience — when it comes to intelligence, there is good reason to think we are the most intelligent of the whole bunch. But when it comes to sentience, to say that humans are more sentient than whales, or more sentient than baboons or more sentient than cats, I see no evidence for that. So first step is, you go in that direction, expand. And then the second question of what is it for, I would reverse it and I would say that I don ' t think sentience is for anything. I think we don ' t need to find our role in the universe. The really important thing is to liberate ourselves from suffering. What characterizes sentient beings in contrast to robots, to stones, to whatever, is that sentient beings suffer, can suffer, and what they should focus on is not finding their place in some mysterious cosmic drama. They should focus on understanding what suffering is, what causes it and how to be liberated from it. CA : I know this is a big issue for you, and that was very eloquent. We ' re going to have a blizzard of questions from the audience here, and maybe from Facebook as well, and maybe some comments as well. So let ' s go quick. There ' s one right here. Keep your hands held up at the back if you want the mic, and we ' ll get it back to you. Question : In your work, you talk a lot about the fictional stories that we accept as truth, and we live our lives by it. As an individual, knowing that, how does it impact the stories that you choose to live your life, and do you confuse them with the truth, like all of us? YNH : I try not to. I mean, for me, maybe the most important question, both as a scientist and as a person, is how to tell the difference between fiction and reality, because reality is there. I ' m not saying that everything is fiction. It ' s just very difficult for human beings to tell the difference between fiction and reality, and it has become more and more difficult as history progressed, because the fictions that we have created — nations and gods and money and corporations — they now control the world. So just to even think,"Oh, this is just all fictional entities that we ' ve created,"is very difficult. But reality is there. For me the best... There are several tests to tell the difference between fiction and reality. The simplest one, the best one that I can say in short, is the test of suffering. If it can suffer, it ' s real. If it can ' t suffer, it ' s not real. A nation cannot suffer. That ' s very, very clear. Even if a nation loses a war, we say,"Germany suffered a defeat in the First World War,"it ' s a metaphor. Germany cannot suffer. Germany has no mind. Germany has no consciousness. Germans can suffer, yes, but Germany cannot. Similarly, when a bank goes bust, the bank cannot suffer. When the dollar loses its value, the dollar doesn ' t suffer. People can suffer. Animals can suffer. This is real. So

I would start, if you really want to see reality, I would go through the door of suffering. If you can really understand what suffering is, this will give you also the key to understand what reality is. CA : There ' s a Facebook question here that connects to this, from someone around the world in a language that I cannot read. YNH : Oh, it ' s Hebrew. CA : Hebrew. There you go. Can you read the name? YNH : Or Lauterbach Goren. CA : Well, thank you for writing in. The question is :"Is the post-truth era really a brand-new era, or just another climax or moment in a never-ending trend? YNH : Personally, I don ' t connect with this idea of post-truth. My basic reaction as a historian is : If this is the era of post-truth, when the hell was the era of truth? CA : Right. YNH : Was it the 1980s, the 1950s, the Middle Ages? I mean, we have always lived in an era, in a way, of post-truth. CA : But I ' d push back on that, because I think what people are talking about is that there was a world where you had fewer journalistic outlets, where there were traditions, that things were fact-checked. It was incorporated into the charter of those organizations that the truth mattered. So if you believe in a reality, then what you write is information. There was a belief that that information should connect to reality in a real way, and if you wrote a headline, it was a serious, earnest attempt to reflect something that had actually happened. And people didn ' t always get it right. But I think the concern now is you ' ve got a technological system that ' s incredibly powerful that, for a while at least, massively amplified anything with no attention paid to whether it connected to reality, only to whether it connected to clicks and attention, and that that was arguably toxic. That ' s a reasonable concern, isn ' t it? YNH : Yeah, it is. I mean, the technology changes, and it ' s now easier to disseminate both truth and fiction and falsehood. It goes both ways. It ' s also much easier, though, to spread the truth than it was ever before. But I don ' t think there is anything essentially new about this disseminating fictions and errors. There is nothing that — I don ' t know — Joseph Goebbels, didn ' t know about all this idea of fake news and post-truth. He famously said that if you repeat a lie often enough, people will think it ' s the truth, and the bigger the lie, the better, because people won ' t even think that something so big can be a lie. I think that fake news has been with us for thousands of years. Just think of the Bible. CA : But there is a concern that the fake news is associated with tyrannical regimes, and when you see an uprise in fake news that is a canary in the coal mine that there may be dark times coming. YNH : Yeah. I mean, the intentional use of fake news is a disturbing sign. But I ' m not saying that it ' s not bad, I ' m just saying that it ' s not new. CA : There ' s a lot of interest on Facebook on this question about global governance versus nationalism. Question here from Phil Dennis :"How do we get people, governments, to relinquish power? Is that — is that — actually, the text is so big I can ' t read the full question. But is that a necessity? Is it going to take war to get there? Sorry Phil — I mangled your question, but I blame the text right here. YNH : One option that some people talk about is that only a catastrophe can shake humankind and open the path to a real system of global governance, and they say that we can ' t do it before the catastrophe, but we need to start laying the foundations so that when the disaster strikes, we can react quickly. But people will just not have the motivation to do such a thing before the disaster strikes. Another thing that I would emphasize is that anybody who is really interested in global governance should always make it very, very clear that it doesn ' t replace or abolish local identities and communities, that it should come both as — It should be part of a single package. CA : I want to hear more on this, because the very words"global governance"are almost the epitome of evil in the **mindset** of a lot of people on the alt-right right now. It just seems scary, remote, distant, and it has let them down, and so globalists, global governance

— no, go away! And many view the election as the ultimate poke in the eye to anyone who believes in that. So how do we change the narrative so that it doesn ’ t seem so scary and remote? Build more on this idea of it being compatible with local identity, local communities. YNH : Well, I think again we should start really with the biological realities of Homo sapiens. And biology tells us two things about Homo sapiens which are very relevant to this issue : first of all, that we are completely dependent on the ecological system around us, and that today we are talking about a global system. You cannot escape that. And at the same time, biology tells us about Homo sapiens that we are social animals, but that we are social on a very, very local level. It ’ s just a simple fact of humanity that we cannot have intimate familiarity with more than about 150 individuals. The size of the natural group, the natural community of Homo sapiens, is not more than 150 individuals, and everything beyond that is really based on all kinds of imaginary stories and large-scale institutions, and I think that we can find a way, again, based on a biological understanding of our species, to weave the two together and to understand that today in the 21st century, we need both the global level and the local community. And I would go even further than that and say that it starts with the body itself. The feelings that people today have of alienation and loneliness and not finding their place in the world, I would think that the chief problem is not global capitalism. The chief problem is that over the last hundred years, people have been becoming disembodied, have been distancing themselves from their body. As a hunter-gatherer or even as a peasant, to survive, you need to be constantly in touch with your body and with your senses, every moment. If you go to the forest to look for mushrooms and you don ’ t pay attention to what you hear, to what you smell, to what you taste, you ’ re dead. So you must be very connected. In the last hundred years, people are losing their ability to be in touch with their body and their senses, to hear, to smell, to feel. More and more attention goes to screens, to what is happening elsewhere, some other time. This, I think, is the deep reason for the feelings of alienation and loneliness and so forth, and therefore part of the solution is not to bring back some mass nationalism, but also reconnect with our own bodies, and if you are back in touch with your body, you will feel much more at home in the world also. CA : Well, depending on how things go, we may all be back in the forest soon. We ’ re going to have one more question in the room and one more on Facebook. Ama Adi-Dako : Hello. I ’ m from Ghana, West Africa, and my question is : I ’ m wondering how do you present and justify the idea of global governance to countries that have been historically disenfranchised by the effects of globalization, and also, if we ’ re talking about global governance, it sounds to me like it will definitely come from a very Westernized idea of what the”global”is supposed to look like. So how do we present and justify that idea of global versus wholly nationalist to people in countries like Ghana and Nigeria and Togo and other countries like that? YNH : I would start by saying that history is extremely unfair, and that we should realize that. Many of the countries that suffered most from the last 200 years of globalization and imperialism and industrialization are exactly the countries which are also most likely to suffer most from the next wave. And we should be very, very clear about that. If we don ’ t have a global governance, and if we suffer from climate change, from technological disruptions, the worst suffering will not be in the US. The worst suffering will be in Ghana, will be in Sudan, will be in Syria, will be in Bangladesh, will be in those places. So I think those countries have an even greater incentive to do something about the next wave of disruption, whether it ’ s ecological or whether it ’ s technological. Again, if you think about technological disruption, so if AI and 3D printers and robots will take the jobs from billions of people, I worry far less about the

Swedes than about the people in Ghana or in Bangladesh. And therefore, because history is so unfair and the results of a calamity will not be shared equally between everybody, as usual, the rich will be able to get away from the worst consequences of climate change in a way that the poor will not be able to. CA : And here ' s a great question from Cameron Taylor on Facebook :”At the end of ' Sapiens, '”you said we should be asking the question, ' What do we want to want? ' Well, what do you think we should want to want?”YNH : I think we should want to want to know the truth, to understand reality. Mostly what we want is to change reality, to fit it to our own desires, to our own wishes, and I think we should first want to understand it. If you look at the long-term trajectory of history, what you see is that for thousands of years we humans have been gaining control of the world outside us and trying to shape it to fit our own desires. And we ' ve gained control of the other animals, of the rivers, of the forests, and reshaped them completely, causing an ecological destruction without making ourselves satisfied. So the next step is we turn our gaze inwards, and we say OK, getting control of the world outside us did not really make us satisfied. Let ' s now try to gain control of the world inside us. This is the really big project of science and technology and industry in the 21st century — to try and gain control of the world inside us, to learn how to engineer and produce bodies and brains and minds. These are likely to be the main products of the 21st century economy. When people think about the future, very often they think in terms,”Oh, I want to gain control of my body and of my brain.”And I think that ' s very dangerous. If we ' ve learned anything from our previous history, it ' s that yes, we gain the power to manipulate, but because we didn ' t really understand the complexity of the ecological system, we are now facing an ecological meltdown. And if we now try to reengineer the world inside us without really understanding it, especially without understanding the complexity of our mental system, we might cause a kind of internal ecological disaster, and we ' ll face a kind of mental meltdown inside us. CA : Putting all the pieces together here — the current politics, the coming technology, concerns like the one you ' ve just outlined — I mean, it seems like you yourself are in quite a bleak place when you think about the future. You ' re pretty worried about it. Is that right? And if there was one cause for hope, how would you state that? YNH : I focus on the most dangerous possibilities partly because this is like my job or responsibility as a historian or social critic. I mean, the industry focuses mainly on the positive sides, so it ' s the job of historians and philosophers and sociologists to highlight the more dangerous potential of all these new technologies. I don ' t think any of that is inevitable. Technology is never deterministic. You can use the same technology to create very different kinds of societies. If you look at the 20th century, so, the technologies of the Industrial Revolution, the trains and electricity and all that could be used to create a communist dictatorship or a fascist regime or a liberal democracy. The trains did not tell you what to do with them. Similarly, now, artificial intelligence and bioengineering and all of that — they don ' t predetermine a single outcome. Humanity can rise up to the challenge, and the best example we have of humanity rising up to the challenge of a new technology is nuclear weapons. In the late 1940s, ' 50s, many people were convinced that sooner or later the Cold War will end in a nuclear catastrophe, destroying human civilization. And this did not happen. In fact, nuclear weapons prompted humans all over the world to change the way that they manage international politics to reduce violence. And many countries basically took out war from their political toolkit. They no longer tried to pursue their interests with warfare. Not all countries have done so, but many countries have. And this is maybe the most important reason why international violence declined dramatically since 1945, and today, as I said, more people commit suicide than are killed in war. So this, I think, gives us a good example that even the most frightening technology, humans can rise up to the challenge and actually some good can come out of it. The problem is, we have very little margin for error. If we don ’ t get it right, we might not have a second option to try again. CA : That ’ s a very powerful note, on which I think we should draw this to a conclusion. Before I wrap up, I just want to say one thing to people here and to the global TED community watching online, anyone watching online : help us with these dialogues. If you believe, like we do, that we need to find a different kind of conversation, now more than ever, help us do it. Reach out to other people, try and have conversations with people you disagree with, understand them, pull the pieces together, and help us figure out how to take these conversations forward so we can make a real contribution to what ’ s happening in the world right now. I think everyone feels more alive, more concerned, more engaged with the politics of the moment. The stakes do seem quite high, so help us respond to it in a wise, wise way. Yuval Harari, thank you.

Text Sample 208: POS Dim 2 – 2017 – Score 12.00 – 2017/t001150.txt

I was a blue-eyed, chubby-cheeked five-year-old when I joined my family on the picket line for the first time. My mom made me leave my dolls in the minivan. I ’ d stand on a street corner in the heavy Kansas humidity, surrounded by a few dozen relatives, with my tiny fists clutching a sign that I couldn ’ t read yet :”Gays are **worthy** of death.”This was the beginning. Our protests soon became a **daily** occurrence and an international phenomenon, and as a member of Westboro Baptist Church, I became a fixture on picket lines across the country. The end of my antigay picketing career and life as I knew it, came 20 years later, triggered in part by strangers on Twitter who showed me the power of engaging the other. In my home, life was framed as an epic spiritual battle between good and evil. The good was my church and its members, and the evil was everyone else. My church ’ s antics were such that we were constantly at odds with the world, and that reinforced our otherness on a **daily** basis.”Make a difference between the unclean and the clean,”the verse says, and so we did. From **baseball** games to military funerals, we trekked across the country with neon protest signs in hand to tell others exactly how”unclean”they were and exactly why they were headed for damnation. This was the focus of our whole lives. This was the only way for me to do good in a world that sits in Satan ’ s lap. And like the rest of my 10 siblings, I believed what I was taught with all my heart, and I pursued Westboro ’ s agenda with a **special** sort of zeal. In 2009, that zeal brought me to Twitter. Initially, the people I encountered on the **platform** were just as hostile as I expected. They were the digital version of the screaming hordes I ’ d been seeing at protests since I was a kid. But in the midst of that digital brawl, a strange pattern developed. Someone would arrive at my profile with the usual rage and scorn, I would respond with a custom mix of Bible verses, pop culture references and smiley faces. They would be understandably confused and caught off guard, but then a conversation would ensue. And it was civil — full of genuine curiosity on both sides. How had the other come to such outrageous conclusions about the world? Sometimes the conversation even bled into real life. People I ’ d sparred with on Twitter would come out to the picket line to see me when I protested in their city. A man named David was one such person. He ran a blog called”Jewlicious,”and after several months of heated but friendly arguments online, he came out to see me at a picket in New Orleans. He brought me a Middle Eastern dessert from Jerusalem, where he lives, and I brought him kosher chocolate and held a”God hates Jews”sign. There was no confusion about our positions, but the line between friend and foe was becoming blurred. We ’ d started to see each other as human beings, and it changed the way we spoke to one another. It took time, but eventually these conversations planted seeds of doubt in me. My friends on Twitter took the time to understand Westboro ’ s doctrines, and in doing so, they were able to find inconsistencies I ’ d missed my entire life. Why did we advocate the death **penalty** for gays when Jesus said,”Let he who is without sin cast the first stone?”How could we claim to love our neighbor while at the same time praying for God to destroy them? The truth is that the care shown to me by these strangers on the internet was itself a contradiction. It was growing evidence that people on the other side were not the demons I ’ d been led to believe. These realizations were life-altering. Once I saw that we were not the ultimate arbiters of divine truth but flawed human beings, I couldn ’ t pretend otherwise. I couldn ’ t justify our **actions** — especially our cruel **practice** of protesting funerals and celebrating human tragedy. These shifts in my perspective contributed to a larger erosion of trust in my church, and eventually it made it impossible for me to stay. In spite of overwhelming grief and terror, I left Westboro in 2012. In those days just after I left, the instinct to hide was almost paralyzing. I wanted to hide from the judgement of my family, who I knew would never speak to me again — people whose thoughts and opinions had meant everything to me. And I wanted to hide from the world I ’ d rejected for so long — people who had no reason at all to give me a second chance after a lifetime of antagonism. And yet, unbelievably, they did. The world had access to my past because it was all over the internet — thousands of **tweets** and hundreds of interviews, everything from local TV news to”The Howard Stern Show”— but so many embraced me with open arms anyway. I wrote an apology for the harm I ’ d caused, but I also knew that an apology could never undo any of it. All I could do was try to build a new life and find a way somehow to repair some of the damage. People had every reason to doubt my sincerity, but most of them didn ’ t. And — given my history, it was more than I could ’ ve hoped for — forgiveness and the benefit of the doubt. It still amazes me. I spent my first year away from home adrift with my younger sister, who had chosen to leave with me. We walked into an abyss, but we were shocked to find the light and a way forward in the same communities we ’ d targeted for so long. David, my”Jewlicious”friend from Twitter, invited us to spend time among a Jewish community in Los Angeles. We slept on couches in the home of a Hasidic rabbi and his wife and their four kids — the same rabbi that I ’ d protested three years earlier with a sign that said,”Your rabbi is a whore.”We spent long hours talking about theology and Judaism and life while we washed dishes in their kosher kitchen and chopped vegetables for dinner. They treated us like family. They held nothing against us, and again I was astonished. That period was full of turmoil, but one part I ’ ve returned to often is a surprising realization I had during that time — that it was a relief and a privilege to let go of the harsh judgments that instinctively ran through my mind about nearly every person I saw. I realized that now I needed to learn. I needed to listen. This has been at the front of my mind lately, because I can ’ t help but see in our public discourse so many of the same destructive impulses that ruled my former church. We celebrate tolerance and diversity more than at any other time in memory, and still we grow more and more divided. We want good things — justice, equality, freedom, dignity, prosperity — but the path we ’ ve chosen looks so much like the one I walked away from four years ago. We ’ ve broken the world into us and them, only emerging from our bunkers long enough to lob rhetorical grenades at the other camp. We write off half the country as out-of- touch liberal elites or racist misogynist bullies. No nuance, no complexity, no humanity. Even when someone does call for **empathy** and understanding for the other side, the conversation nearly always devolves into a debate about who deserves more **empathy**. And just as I learned to do, we routinely refuse to acknowledge the flaws in our positions or the merits in our opponent ' s. Compromise is anathema. We even target people on our own side when they dare to question the party line. This path has brought us cruel, sniping, deepening polarization, and even outbreaks of violence. I remember this path. It will not take us where we want to go. What gives me hope is that we can do something about this. The good news is that it ' s simple, and the bad news is that it ' s hard. We have to talk and listen to people we disagree with. It ' s hard because we often can ' t fathom how the other side came to their positions. It ' s hard because righteous indignation, that sense of certainty that ours is the right side, is so seductive. It ' s hard because it means extending **empathy** and compassion to people who show us hostility and contempt. The impulse to respond in kind is so tempting, but that isn ' t who we want to be. We can resist. And I will always be inspired to do so by those people I encountered on Twitter, apparent enemies who became my beloved friends. And in the case of one particularly understanding and generous guy, my husband. There was nothing **special** about the way I responded to him. What was **special** was their approach. I thought about it a lot over the past few years and I found four things they did differently that made real conversation possible. These four steps were small but powerful, and I do everything I can to employ them in difficult conversations today. The first is don ' t assume bad intent. My friends on Twitter realized that even when my words were aggressive and offensive, I sincerely believed I was doing the right thing. Assuming ill motives almost instantly cuts us off from truly understanding why someone does and believes as they do. We forget that they ' re a human being with a lifetime of experience that shaped their mind, and we get stuck on that first wave of anger, and the conversation has a very hard time ever moving beyond it. But when we assume good or neutral intent, we give our minds a much stronger framework for dialogue. The second is ask questions. When we engage people across ideological divides, asking questions helps us map the disconnect between our differing points of view. That ' s important because we can ' t present effective arguments if we don ' t understand where the other side is actually coming from and because it gives them an opportunity to point out flaws in our positions. But asking questions serves another purpose ; it signals to someone that they ' re being heard. When my friends on Twitter stopped accusing and started asking questions, I almost automatically mirrored them. Their questions gave me room to speak, but they also gave me permission to ask them questions and to truly hear their responses. It fundamentally changed the dynamic of our conversation. The third is stay calm. This takes **practice** and patience, but it ' s powerful. At Westboro, I learned not to care how my manner of speaking affected others. I thought my rightness justified my rudeness — harsh tones, raised voices, insults, interruptions — but that strategy is ultimately counterproductive. Dialing up the volume and the snark is natural in **stressful** situations, but it tends to bring the conversation to an unsatisfactory, explosive end. When my husband was still just an anonymous Twitter acquaintance, our discussions frequently became hard and pointed, but we always refused to escalate. Instead, he would change the subject. He would tell a joke or recommend a book or gently excuse himself from the conversation. We knew the discussion wasn ' t over, just paused for a time to bring us back to an even keel. People often lament that digital communication makes us less civil, but this is one advantage that online conversations have over in-person ones. We have a buffer of time and space between us and the people whose ideas we find so **frustrating**. We can use that buffer. Instead of lashing out, we can pause, breathe, change the subject or walk away, and then come back to it when we ’ re ready. And finally... make the argument. This might seem obvious, but one side effect of having strong beliefs is that we sometimes assume that the value of our position is or should be obvious and self-evident, that we shouldn ’ t have to defend our positions because they ’ re so clearly right and good that if someone doesn ’ t get it, it ’ s their problem — that it ’ s not my job to educate them. But if it were that simple, we would all see things the same way. As kind as my friends on Twitter were, if they hadn ’ t actually made their arguments, it would ’ ve been so much harder for me to see the world in a different way. We are all a product of our upbringing, and our beliefs reflect our experiences. We can ’ t expect others to spontaneously change their own minds. If we want change, we have to make the case for it. My friends on Twitter didn ’ t abandon their beliefs or their principles — only their scorn. They channeled their infinitely justifiable offense and came to me with pointed questions tempered with kindness and humor. They approached me as a human being, and that was more transformative than two full decades of outrage, disdain and violence. I know that some might not have the time or the energy or the patience for extensive engagement, but as difficult as it can be, reaching out to someone we disagree with is an option that is available to all of us. And I sincerely believe that we can do hard things, not just for them but for us and our future. Escalating disgust and intractable conflict are not what we want for ourselves, or our country or our next generation. My mom said something to me a few weeks before I left Westboro, when I was desperately hoping there was a way I could stay with my family. People I have loved with every pulse of my heart since even before I was that chubby-cheeked five-year-old, standing on a picket line holding a sign I couldn ’ t read. She said,”You ’ re just a human being, my dear, sweet child.”She was asking me to be humble — not to question but to trust God and my elders. But to me, she was missing the bigger picture — that we ’ re all just human beings. That we should be guided by that most basic fact, and approach one another with generosity and compassion. Each one of us contributes to the communities and the cultures and the societies that we make up. The end of this spiral of rage and blame begins with one person who refuses to indulge these destructive, seductive impulses. We just have to decide that it ’ s going to start with us. Thank you.

Text Sample 209: POS Dim 2 – 2023 – Score 18.00 – 2023/t003333.txt

Christiana Figueres : There ’ s no doubt that we are facing exponential impacts of climate change. There is also, however, no doubt that that is being met with exponential progress of the technologies that can help to address climate change. So what we have here is, frankly, a race between two exponential curves. By 2030, we collectively will have chosen between door number one. And door number two. Julio Friedmann : The question I am most commonly asked about climate change is :”Should I be optimistic or pessimistic?”Cynthia Williams : Here ’ s what we need to do to make sure that we ’ re ready for an all-electric future. Anika Goss : Being financially secure and climate resilient should be the most important priority. Faustine Delasalle : It ’ s really easy when you look at where we are today and where we ought to be by 2030, to feel like there ’ s a huge gap that will never be filled. But we have good reasons to say that that gap is actually fixable. Nigel Topping : If you ’ re not worried or anxious, you ’ re probably not paying attention. But we should also not be defeatist. We shouldn ’ t say,”We ’ ve done nothing, we ’ re failing.”We ’ ve really started 20 years too late, but it takes a long time to get the flywheel of momentum going. Kingsmill Bond : The tide of change is coming. The costs keep falling. The political pressure keeps increasing. JF : We ’ re going to build a thriving, exciting world. AG : We can do this. JF : Full of potential. AG : We have to do this. [TED Explores A New Climate Vision] Manoush Zamorodi : When it comes to climate change, are you an optimist or a pessimist? News about the climate can be overwhelming and emotional, to the point that many of us just feel numb. I ’ m Manoush Zomorodi, a longtime public media journalist and host of the TED Radio Hour. And today, we want to bring you something **special** : an hour of nuance, of real people and clarity about what is happening to our planet and what we ’ re doing about it. Because we all know there ’ s plenty of bad news when it comes to the climate crisis. But guess what? There ’ s also some really good news. That ’ s why we ’ re here in Detroit at the TED Countdown Summit, where a cross-section of people who know the most and who are doing the most have gathered to share their ideas and plans with each other and with you. You ’ re going to hear from some incredible speakers about their personal stories and tested solutions. But first, let ’ s get a quick reminder on the basics : climate change in a minute or so. [Prologue : What is Climate Change?] David Biello : The air we breathe has changed. The mix of gases is shifting, with more and more greenhouse gases like carbon dioxide. And the shift is happening faster each year. In just a few hundred years, fossil fuels formed over eons have been burned as coal, oil and gas. The exhaust has transformed the entire atmosphere and ocean. It ’ s like a pollution blanket. And the result we know as climate change. The planet has already warmed more than one degree Celsius and is on a path to heat up even more. That may not seem like a lot, but it ’ s already caused major destruction across the globe. To give us some room to breathe, the world must reduce greenhouse gas pollution by more than seven percent each year, every year of this decade. There are a number of ways to do that. Let ’ s start by looking at the greenhouse gases themselves. CO2 makes up nearly 75 percent of the pollution emitted each year. And then there ’ s methane or natural gas, which makes up 17 percent. Finally, there ’ s nitrous oxide, making up six percent of the problem. There are other greenhouse gases, but these three make up the bulk of the climate challenge. Where do they come from? There are several ways to break down the sources of greenhouse gas emissions. Here we ’ re using the public Climate Watch data. Start with energy. The majority of modern energy comes from burning fossil fuels, which makes the energy sector 76 percent of the climate challenge, including the fuels used for transportation, industrial processes and agricultural production. Farming and animal raising also have to change, as agriculture accounts for 12 percent of greenhouse gas pollution. The remaining 12 percent comes from a grab bag of human activities like industrial heating, clearing forests and more. This is the climate challenge we face : going from adding billions of metric tons of greenhouse gases to the air each year to adding zero. Will it be easy? No. Can we do it? Yes, if we choose to. MZ : OK, so that ’ s the situation. Where do we go from here? Well, we ’ ve all heard of a tipping point, that moment when an idea or concept finally takes hold and then changes the world. Well, the UN ’ s climate chief, Simon Stiell, believes we are almost at a tipping point for massive **action** on climate change. And to get you ready, he wants to take you back to the early ’ 90s, when he was an executive at Nokia, and the world wasn ’ t sending text messages yet. [Chapter 1 : Exponential Change] Simon Stiell : I want you to imagine it ’ s 1992, and you ’ re talking to a telecoms expert. She tells you that **mobile** phone text messaging is going to fundamentally change the way we communicate. You think, why on Earth am I going to go from this to tapping on a screen with my fingers and my thumbs? But then, after a slow start, text-based **services** exploded. Today, over 23 billion messages are sent every single day. This paved the way for the smartphone revolution. I know how much we **struggle** to comprehend and predict what exponential change means, because I was at Nokia in 1993 when that technological revolution started. And now, as head of UN climate change, I ' m here to tell you that what was true for **mobile** phones is also true for climate **action** today. We ' re here in Detroit, the heart of car manufacturing. Take a look at this graph. Electric vehicle sales are expected to increase to become 70 percent by 2030. Another example, solar energy. Between 2010 and 2020, we moved from 20 gigawatts of solar energy installed to 150 gigawatts. And by 2030, that number is expected to increase again, to 1, 000 gigawatts per year. Change can come fast. FD : So exponential change means that change happens very slowly at first and then comes a tipping point. And suddenly things accelerate and accelerate and accelerate. NT : It ' s a mathematical term, but it means basically stuff happens very slowly and then it goes really fast. Tessa Kahn : Governments and experts have consistently underestimated the pace and scale at which we can deploy renewable energy and how quickly renewable energy and associated technologies become affordable. Ramez Naam : I predicted in 2011 that we ' d have solar as cheap as coal in some parts of the world by 2015 and half the cost of coal by 2020, and everyone thought I was crazy, I ' ve got to say, almost. And in fact, I was wrong. The actual decline in the cost of solar happened twice as fast as I thought it would. Solar prices today are a century ahead of where the world ' s leading energy experts thought they ' d be in 2010. KB : It ' s now spreading from solar, so it ' s happening also in the wind sector, it ' s happening in the battery sector, it ' s happening in the electric vehicle sector. CF : I don ' t think that we do a good job at communicating the fact that we are progressing more than is commonly known. Nili Gilbert : The human brain is trained best at thinking about linear change. But when you think about an exponential curve, next to a line, at first the exponential curve is below the line, and so it ' s moving more slowly than you would think. And then it crosses the line and it ' s moving faster than you would think. Habiba Daggash : We call it cautious optimism. We ' re very hopeful about the trajectories that we ' ve seen. But we shouldn ' t let up yet. KB : So we can make this change happen slow or we can make it happen fast. The key point to say is we have agency. MZ : This idea of exponential growth sounds awesome. It sounds amazing, but is it actually happening? Well, I ' m headed into a workshop with industry and climate leaders to find out. The moderators of the workshop challenged us to hold two competing ideas in our minds at the same time. NT : What we ' re really going to explore today is two apparently opposing realities, and they ' re often two opposing narratives. The one which is true, right, is that in terms of the Paris goals, we ' re way off track. We ' re deep into the emergency. But what we ' re going to explore today is the other reality that we know what to do. Exponential change is happening, but it doesn ' t happen by magic. MZ : So what are we seeing in different industries? Well, let ' s start with one that ' s known to be really difficult to decarbonize : cement. Vicente Saiso Alva : I think everything converges into making climate a huge urgent matter for society. We started to feel the pressure from all different types of stakeholders, and in one year of running this program, we realized how fast we could move and how fast we were going in reducing CO2 emissions, which means now to go from 40 to close to 50 percent reduction by 2030. MZ : Encouraging, but to tackle that remaining 50 percent may require technologies that don ' t exist at scale yet. OK, what about shipping? Narin Phol : In Maersk, a few years back, we came out with a net-zero ambition by 2050, and in a few years later, based on the data that we have got, we ' ve really seen a path that we can really accelerate that one. That ' s why the revised target of 2040 came in. MZ : The news in offshore wind was quite dramatic. Jennie Dodson : So my moment of **mindset** shift was in 2019. The government put out a request to bid for offshore wind permits. And that year the price that came in was two thirds less than just four years before. And really critically, it was the same price as the mainstream market. And that was transformative. Nobody expected that when they started. MZ : The conversation got pretty technical and a bit wonky. NP : So to drive exponential growth, you focus on a reinforcing loop to make sure it works and you slow down on the balancing loop. MZ : But the takeaway is that certain sectors, like EVs and renewables, are being upended, while others, like cement and heavy industry are further behind. But generally across every industry – JD : It ' s amazing stuff. MZ : There ' s an understanding that they all need to work together. And the message? Big change can happen. It ' s going to be hard, but some of us are already doing it, so learn from us. [Chapter 2 : The Electric Vehicle Revolution] OK, we are in an exponential **mindset**. And as you heard, one example of exponential growth might be sitting in your driveway right now. Electric vehicles are finally really happening, which means that people here in Detroit are **rethinking** the auto industry. They ' re **rethinking** their jobs, their identity. Take Cynthia Williams, head of sustainability at Ford. I sat down with her at Lucky Detroit, a local coffee and barber shop, where she told me that there are three things that need to happen for electric vehicles to go to the next level, from this tipping point to exponential growth. CW : The three things for me are change, collaboration and capacity. So the change piece of it is changing the way folks think about the vehicle, making sure that they understand we ' re bringing a better vehicle to the customers. MZ : So you don ' t have to make a sacrifice. CW : I think one of the things we need to do is actually get people in the cars and show them how fun it is. And it ' s no different, it ' s just powered by a battery. We are at a tipping point. We ' re at a point of acceleration of electric vehicles. Many are calling it the electric vehicle revolution. Now with that comes unprecedented change that will require us to build new capacity and collaborate in ways that we ' ve never seen before. To make things happen faster, you have to think differently. You can ' t just collaborate with the same people because you get the same answer, right? You need to broaden your collaboration space. Go outside of your industry, understand what worked in Norway, for example, to get them to where they are. MZ : Where ' s the weak link, Cynthia? What ' s the thing that you ' re like, that if we don ' t push through that, we may not get to this full adoption? CW : The critical things that we hear from the consumers is they need infrastructure. That ' s what it ' s taking for us to work with governments to make sure that we bring more charging to everyone, whether it ' s rural, urban, wealthy, low-income, we all need chargers. And I ' m not saying we need one on every corner, but we do need many, many more. When we start building out the charging infrastructure, we need lighting. We need, you know, security, shelter, we need restrooms. We need other amenities so that people know where to go, where to charge. Next comes capacity. We have to invest in new facilities and talent. Now, that ' s true for any industry transitioning to greener goals and strategies. And we ' re investing in talent, we ' re investing in education and training, and we ' re also listening to the communities where we live and operate. MZ : So a young Cynthia Williams, now, your counterpart who ' s studying engineering, will she be learning something very different if she wants to go and work in the automotive industry in Detroit? CW : There will be different options, right. And so, I ' m a mechanical engineer by trade, but I think there will be opportunities for electrical engineering, software engineering. And there ' s even certificates that students can get instead of going through a four-year college. It ' s the proper training to get them right, to be dedicated, to build the vehicles that we need for the future. At the end of this year, globally, we ' ll have the ability to produce 600, 000 electric vehicles. And in 2026, we ' ll have a global goal to produce two million electric vehicles. And it just goes up from there. MZ : What is this going to look like, this growth for me in the next couple of years, in five years, in 20 years? CW : Clean air. That ' s what I look at. If you think about when we were, during the pandemic, when everybody stopped driving and they were at home, clear skies and moving to zero tailpipe emission vehicles. And if you chose an electric vehicle, that ' s you participating and you contributing to a better world, to a better future. We have come a long way. It ' s so gratifying to see electric vehicles on the road, to move from aspiration to reality. To see plants dedicated to building electric vehicles. We have the technology, we have the leadership, we have the ingenuity to respond to the environmental crisis with courage. And I ' ll tell you, as a person who ' s dedicated my entire career to sustainability, the road ahead is paved with promise. TED-Ed : If you were buying a car in 1899, you would have had three major options to choose from. You could buy a steam-powered car, typically relying on gas-powered boilers. These could drive as far as you wanted, provided you also wanted to lug around extra water to refuel and didn ' t mind waiting 30 minutes for your engine to heat up. Alternatively, you could buy a car powered by gasoline. However, the internal combustion engines in these models required dangerous hand cranking to start and emitted loud noises and foul smelling exhaust while driving. So your best bet was probably option number three : a battery-powered electric vehicle. These cars were quick to start, clean and quiet to run, and if you lived somewhere with access to electricity, easy to refuel overnight. If this seems like an easy choice, you ' re not alone. By the end of the 19th century, nearly 40 percent of American cars were electric. In cities with early electric systems, battery-powered cars were a popular and reliable alternative to their occasionally explosive competitors. But electric vehicles had one major problem. Batteries. MZ : Flash forward to the present and batteries are still on everyone ' s mind. So I visited Mujeeb Ijaz, the founder of the energy storage company One, who also happens to be an electric vehicle history buff. Mujeeb Ijaz : So this is a 1922 Detroit Electric. MZ : OK, so 100-year-old vehicle. MI : That ' s right, electric vehicle. And then this was Walter Baker. He developed the Baker Electric. And he made the range run 201 miles on a single charge. He also felt strongly that you should be able to wear your top hat, you know, so he has a taller carriage. These had lead acid batteries to begin with. And then Thomas Edison introduced the nickel-iron battery. So this is the battery compartment, one of the battery compartments. And this is where you would charge. MZ : Right there? MI : And we ' re building on what these original pioneers did. And I think it ' s important to not think about us starting from scratch now. But go backwards, think about how they solved the problem, what disrupted their ability to advance this to the market, and then pick those problems up and make sure we don ' t get stuck again. MZ : So we ' ve been here before, a moment when electric vehicles could have taken off but didn ' t. So what do we need to do differently this time? MI : Electric vehicles were able to deal with mobility in urban areas, but there were charging deserts where rural **driving** resulted in no way to get back because you couldn ' t find electric charging everywhere. And that was a problem. MZ : And that ' s what people are worried about right now. We ' re at the same thing. We ' re worried about range, we ' re worried about accessibility. All of the things are the same. But the key difference being that we now know that our planet can ' t keep going like this. MI : That ' s right. Had the vision of the Model T assembly line and mass manufacturing been applied to electric car, the electric car would have also dropped in price, but range and charging deserts would have still been the obstruction. MZ : OK. We didn ’ t have the technology, but we do now. MI : We do now. MZ : In the US, there hasn ’ t been a mass move to electric vehicles... yet. Mujeeb ’ s company is one of many who want to change that by building batteries that give drivers longer range. And these batteries aren ’ t just for cars. MI : So we produce battery packs for transportation, but we ’ ve also started developing the same battery products into grid batteries. So micro-grids are those technologies that combine solar and storage and replace a coal plant or a natural gas plant for generating electricity. So buildings and industry and transportation all working together. And that ’ s an exciting future, is all these industries working together, that makes me more confident that we ’ re at the time in history where this transition will take place. MZ : How many years till we get to that? MI : I think it ’ s within 10 years. MZ : No way, 10 years? MI : We ’ re in a place where that transition is going to be much more visible, and then we just need technologies to drive it up to the mass markets. MZ : I am so excited. Like, genuinely so excited. Everyone everywhere driving electric sounds great, but that also means that the power to charge all those vehicles needs to be green. And that brings us to the energy sector, which is also changing at an incredible pace, even though renewable energy has been years in the making. [Chapter 3 : Renewables] TED-Ed : In the spring of 1954, the press excitedly gathered around Bell Laboratories ’ latest invention : a silicon-based solar cell that could efficiently convert the sun ’ s energy into electrical current. The creation was celebrated as the dawn of a new era, as **reporters** touted that civilization would soon run on the sun ’ s limitless energy. But the dream had a catch. As this first commercially sold solar cell cost around 300 dollars per watt, meaning at its current rate it would cost well over a million dollars to buy a unit large enough to power a single home. But today, in many countries, solar is the cheapest form of energy to produce, surpassing fossil fuel alternatives like coal and natural gas. Millions of homes are equipped with rooftop solar, with most units paying for themselves in their first seven to 12 years and then generating further savings. So how did solar become so affordable? A turning point in solar ’ s price history occurred on the floor of Germany ’ s parliament, where in 2000, Hermann Scheer introduced the Renewable Energy Sources Act. This legislation laid out a vision for the country ’ s energy future in solar and wind. It incentivized citizens to personally invest in rooftop solar panels by guaranteeing payment to homeowners for the renewable energy they generated and sold to the grid. The pay rate for this electricity was highly subsidized, at times reaching four times the market price. Several other countries soon followed Germany ’ s example, implementing similar policies and incentives to drive their country ’ s solar use. This created unprecedented demand for solar panels worldwide. Manufacturers were able to scale up production and innovate in ways that cut costs. As a result, solar panel prices dropped while efficiency grew. HD : We see the cost reduction in solar and other renewable energy technologies has been so steep, but a lot of it is because of the enabling regulatory and policy environments created in places like the European Union and the US. Akil Callender : We cannot divert catastrophic climate change without addressing emissions coming from our energy systems. Currently, emissions from the energy system account for around two thirds of global greenhouse gas emissions. So the good news is, yes, we are seeing tremendous progress across the globe in renewable energy. It ’ s rolling out faster than ever before. Luisa Neubauer : The brilliant thing about energy supply in the world is that we have alternatives. So we don ’ t need fossil fuels to provide safe and equitable access to energy systems. Ramón Méndez Galain : Almost 90 percent of the total new installed capacity all over the world is just renewable. This is a tremendous change. AC : There ’ s more finance than ever before going into clean energy. In many countries, it is the largest contributor to their electricity system for the first time ever. KB : And also because China is the leading manufacturing nation in the world, change is happening there much faster. And those innovations are then spreading around the world. Hongquiao Liu : China ’ s coal demand might not have **peaked**, and China will continue to consume quite a lot of coal in the next few years. But the decarbonization is happening. It ’ s happening faster than anyone has expected, and China is adding an astonishing amount of wind and solar to its grid every year. Eduarda Zoghbi : I think one of the biggest **obstacles** today are natural resources, in the sense that we ’ re talking more and more about critical minerals and the role that they have. And batteries are demanding a lot of critical minerals that are present in most of the developing countries. So I think that even though costs are decreasing, we have to be more and more creative when it comes to recycling the batteries. We also have to think of how they ’ re impacting local communities. KB : So to give you the numbers, we extract, at the moment, 15, 000 million tons of fossil fuels every year. The International Energy Agency says that we will need about 43 million tons of critical materials in order to build out the clean energy economy. So that ’ s 300 times less stuff that is required. It ’ s worth saying that the impact of extracting 300 times less stuff upon nature is dramatically lower. And I know that it ’ s often argued that the renewable economy may be damaging for nature. Again, this is a completely false statement. It ’ s in fact, our only chance of reducing the pressure upon our natural resources, because there ’ s far less of it. MZ : Of course, transitioning to using clean energy, upending industries, it has to happen across the world, not just here in the US. And it can be hard to even imagine what life will be like. But carbon scientist and futurist Julio Friedmann can help us. Julio Friedmann : I think we need a compelling vision of the future that gets people excited about the energy transition and solving climate change. Right now, it ’ s like we ’ re at a restaurant and there ’ s two options : burnt toast, unflavored oatmeal. And people are not excited about either, right? The burnt toast option for me is what I call the big switch. Everything ’ s the same, it ’ s just clean, but it costs a lot of money and it ’ s **super** hard to do. The other vision is kind of like,”Thou shalt not.”That vision is a narrower sort of,”eat your peas,”hair shirt kind of vision of the future. And a lot of people don ’ t want that either. And there ’ s nothing in the physics, chemistry or biology that says we can ’ t have a **super** interesting, exciting, vibrant world. And that is compelling in a way that we have not heard yet. MZ : Help me out here, what does that even look like? JF : So, for lack of a better word, like, I like the Marvel cinematic universe. MZ : OK, I ’ ll go with you. JF : So a place like Wakanda, right? It ’ s got flying cars, it ’ s got maglev trains, and it ’ s a totally fabulous place to live. Like, they ’ ve got lots of food. That is built on abundant, sustainable, cheap energy. MZ : OK, so like, we ’ re not Wakanda, we ’ re not in the Marvel universe. This is real life, sort of. How do we make this possible? JF : So what can you actually do here on Earth, constrained by economics and engineering and all the rest of it, right? And the answer is we can do an incredible amount. Abundant, available where you want it, when you want it. Everybody should have energy, including the three billion people who use less electricity than my refrigerator uses. Well, the good news is, every day, the Earth receives 163, 000 terawatts of energy from the sun. About half of that bounces back to space, but about 80, 000 terawatts arrive at the Earth in a form we can use. For reference, today, the world uses about 26 terawatts of energy, and we got more than solar and wind.

We have geothermal, we have hydro, we have nuclear. There ' s other kinds of clean energies. And some of the best resources are, in fact, in the Global South. These places are not simply future climate victims. These places are latent energy superpowers. My favorite example of this is Chile. Almost eight years ago now their government said, wait a second, we can make the cheapest green electricity on Earth. We have hydropower, we have solar power, we have wind power and the best resources anywhere in the world. So let ' s start by making a lot of green electricity. And then they said, because of that, we can also make green hydrogen cheaper than anywhere else. And then Japan was like, hello, we want your green hydrogen. Can you ship it to us? MZ : Is that possible? JF : Totally. And Japan is like, we will also give you money to pay for these projects. We ' ll loan you money to pay for these projects. Could you please use Japanese technology? So they ' re using Japanese turbines, electrolyzers, they ' re using Japanese ships to haul the ammonia around. So Japan gets rich on this, Chile gets rich on this. MZ : Win-win. JF : Totally win-win. MZ : Win-win-win, planet, too. JF : And that model, other countries are looking at that, going, I want some of that. And that ' s why I like this vision of abundance. 15 years ago, solar was the most expensive power. Now it is the cheapest form of electricity, right? Which is astonishing, it ' s totally amazing. And a whole bunch of people are like, now we ' re done. And I ' m like, no, we can do that with everything. We can do that with wind. We can do that with nuclear, we can do that with geothermal. But these projects take time, money and people. We need to develop the human capital as part of these investment projects for decades. And innovation, we need much more energy in many places, much cheaper. That ' s an innovation agenda. For solar, that was the United States, Germany, China and others acting together. Well, if that ' s the recipe, we can do that again. We ' re already doing it with electric vehicles and clean hydrogen production. We can certainly do it by turning electricity into fuels. MZ : It sounds like the climate discussion needs to be kind of like improv, that it ' s a"yes and"sort of thing. JF : I ' m so glad you said that. So many people don ' t understand my love of improv comedy. Yes, it is a"yes and"circumstance. Because mostly it is about room for agreement. Where do you find the point where everybody is like, I can agree on that? One of the things I love about the infrastructure bill, one of the things I love about the Inflation Reduction Act, neither of them had the word climate in it. They were both massive climate bills, right? That was a way to get everybody on the same page. Nationwide, they basically tripled the clean energy research budget. That ' s a good start. But that made it all of one percent of GDP. And for comparison, we spend twice that on pharmaceuticals. So I think it ' s fair to say we should get going. We should spend a lot more money. We ' ve done stuff like this before. We ' ve done World War II, we ' ve done the Marshall Plan, like, we ' ve done the space shot, like we actually know how to move really quickly as a nation. We did this for COVID. Collective **action**, building together is what makes the difficult possible and nourishes the soul through **mission** and purpose. We know what to do and we can act, not out of anger or fear, but out of generosity and common purpose, bringing aspiration and humility together. We ' re going to build a thriving, vibrant, exciting world full of potential that ' s going to be built on the back of infrastructure, innovation and investment that will harness the abundant, sustainable, cheap energy that is our planet ' s endowment. Thank you. [Chapter 4 : A Just Transition] FD : There is a high risk that when those exponential kick in, we see businesses running after those opportunities. But then the distribution of the benefits being very unequal. EZ : Many people like discussing the importance of the just energy transition. And a lot of people don ' t understand what justice really means. If we are to reach a revolution, a renewable energy future has to be inclusive. AC : We have this push for what we ’ re calling a just and equitable energy transition. We need to make sure that we are not exacerbating inequality, but rather we are ensuring that that gap is closed and that the groups that have been historically marginalized are not once again being marginalized in a new way. NT : Most of emissions come from wealthy, energy-guzzling citizens, but a lot of the economic benefit and human development benefit will come not from decarbonizing high-carbon lives and lifestyles, but from providing clean, distributed energy and tools and solutions to those who don ’ t have access to any of them. LN : When we speak of countries in need of economic development, we mostly speak of countries that are hit by the climate crisis the hardest right now. It should be the biggest wish and job and task for all those countries who have burned so much fossil fuel already to support any other country in the so-called Global South, to get an economy running without fossil fuels, to not repeat all the mistakes that we have made. HD : To ensure that the solar revolution that has been seen around the world comes to places like Africa, we need to be really conscious that the history and the lived experience when it comes to energy is very different for poor countries. Rebekah Shirley : We love to remind Global South communities, especially Africa, about the vast renewable energy potential that they have, which is true. But the finance flows to deliver on that potential remain very scarce. Africa, despite being 90 percent of those across the entire globe that still do not have access to electricity and being 17 percent of the global population, we ’ re actually two percent of international finance. Money doesn ’ t actually flow to these spaces naturally, and we need to do a lot of work to get global finance to these spaces. MZ : A world of abundant clean energy for everyone. That is the goal. But what about in the short term? Take Detroit. This once-wealthy city has been through some seriously hard times. How is it rebuilding itself so that everyone here can thrive while dealing with extreme heat, flooding, and other climate problems? Anika Goss is looking to Detroit ’ s past to plan its future. AG : I am a third-generation Detroiter. My grandmother moved to Detroit in 1936 during the Great Migration and brought all of her southern ways with her. She owned a home and knew that home **ownership** would create wealth and opportunity for her growing family. Up until the late 1950s, Detroit was a haven for middle-class families living in neighborhoods where there was green space and community connectivity and opportunity. My grandmother, she had this amazing garden, and it was just abundant. There were all kinds of flowers and food. You know, she grew vegetables back there, and it was just a magical place for a little girl. MZ : I mean, what you ’ re describing is a Detroit that, yes, has huge problems, segregation. But at the same time, a woman, she can have a job, she can own a house, she can raise her family, she can have fresh fruits and vegetables. AG : It was certainly a place where you could gain economic opportunity faster than you can now. My grandmother ’ s Detroit is not the Detroit that I live in today. All of those sites where there was manufacturing and industrial sites that led to our economic boom, many stand vacant and abandoned. These industrial sites have led to dangerous contamination to our land and our water and our air. MZ : So, just lay it out for us. For people who maybe don ’ t see, like, how is there a link between economic inequality and climate change? AG : You know, I ’ m so surprised that it feels like separate problems. But I ’ m also really glad that people are asking. Because if you ’ re living around and working in a community that is low-income, where there are fewer jobs, the housing is deteriorated, this is a neighborhood that ’ s also more likely to be closer to a factory that ’ s emitting pollutants, likely to have water that ’ s contaminated, probably has fewer parks and fewer intentional green spaces. The people who are living in those communities are at risk for climate impacts first. So we have to do something in those neighborhoods right away. MZ : So what can this healthier future look like? Anika showed me some of the work she ' s done in partnership with different neighborhoods in Detroit, to make them both more enjoyable and more resilient. First, she took me to the Circle Forest. It ' s an oasis right in the middle of the city. AG : This is six lots that were vacant, and we all worked together with this community to build a forest. MZ : Beautiful and a great example of how nonprofits can work with communities to transform them into sustainable, welcoming places. AG : We might have a good idea about gardens and tree planting, but this is where people live. And so you ask first. This is a new effort, of this idea of a resilient neighborhood or calling it that. So these are large tracts of vacancy that can be used to provide environmental sustainability, community resilience and beauty. Watch for the poison ivy. It ' s not pretend. MZ : Next we met up with Katrina Watkins, the founder and CEO of the Bailey Park Neighborhood Development Corporation, a group revitalizing this historic area by turning vacant lots into parks that can stand up to extreme weather. Katrina Watkins : My family has been in this neighborhood since 1946. I grew up hearing the stories from my dad, him telling me about how people thought it was a ghetto, but it wasn ' t. Those were families, and people worked and had businesses, and people looked out for each other. And it was those type of stories that made me say, wow, we just really need to bring that back. MZ : So then briefly fill us in about how things changed. KW : Things deteriorated in this neighborhood. The homes got torn down, left just a lot of vacant lots, a lot of overgrowth. It wasn ' t safe. I would see my little nieces and nephews. They would just be on their little skateboards, you know, not paying attention, the stop sign is covered. And me and my dad was like, we just have to do something. So we started out as Bailey Park Project, focusing on the park and green infrastructure and trees and flowers. And then over the course of the years, we started doing more community development. MZ : I wonder, like, how do you describe it to people when you explain what ' s happening, what you ' re trying to do for your neighborhood, the changes that you ' re making and how it relates to climate change? KW : One really great example, when people come to the park, like right now, this is like, literally a heat island because our trees haven ' t grown. So it is very hot. So sometimes it ' s so hot that the kids don ' t come out and play until the evening. That education of knowing the benefit of trees and cooling down an environment, how it makes the air cleaner. There ' s a lot of people that suffer from asthma. So trees can make a difference. And if you go to places like Grosse Pointe, you will see this beautiful tree canopy down. Like their homes are probably 20 degrees cooler you know, than our homes, AG : Having a lot of green space actually contributes to the climate resiliency for this neighborhood. So all of this grass actually can manage stormwater better. The idea of leading new development in this neighborhood with green, what it says to me is that the residents are really envisioning the neighborhood that they want to live in. And that I just – I ' m really excited about that. MZ : Turning great loss and upheaval into even greater opportunities. These Detroiters are creating a blueprint for an urban future that every city can learn from. [Chapter 5 : Agriculture] TED-Ed : About 10, 000 years ago, humans began to farm. This agricultural revolution was a turning point in our history that enabled people to settle, build and create. In short, agriculture enabled the existence of civilization. Today, approximately 40 percent of our planet is farmland. Spread all over the world, these agricultural lands are the pieces to a global puzzle we are all facing. In the future, how can we feed every member of a growing population a healthy diet? Meeting this goal will require nothing short of a second agricultural revolution. The first agricultural revolution was characterized by expansion and exploitation, feeding people at the expense of forests, wildlife and water, and destabilizing the climate in the process. That ' s not an option the next time around. Agriculture depends on a stable climate with predictable seasons and weather patterns. This means we can ' t keep expanding our agricultural lands, because doing so will undermine the environmental conditions that make agriculture possible in the first place. Instead, the next agricultural revolution will have to increase the output of our existing farmland for the long term while protecting biodiversity, conserving water, and reducing pollution and greenhouse gas emissions. So what will the future farms look like? In Bangladesh, Cambodia and Nepal, new approaches to rice production may dramatically decrease greenhouse gas emissions in the future. Rice is a staple food for three billion people and the main source of livelihood for millions of households. More than 90 percent of rice is grown in flooded paddies, which use a lot of water and release 11 percent of annual methane emissions, which accounts for one to two percent of total annual greenhouse gas emissions globally. By experimenting with new strains of rice, irrigating less and adopting less labor-intensive ways of planting seeds, farmers in these countries have already increased their incomes and crop yields while cutting down on greenhouse gas emissions. MZ : Related technologies are also taking root in the US. Father and daughter farmers, Jim and Jessica Whitaker, have come up with their own strategies to reduce methane that are working on their farm in Arkansas. But convincing everyone else to change is their next big challenge. So let ' s go back to how you got to be such a big supplier of rice in this country. You tell a story about how you were 22 and you were like, I ' m going to start my own farm, but – not that easy. Jim Whitaker : No, it ' s not. Let me tell you something about renting a farm when you ' re 22 years old. No one rents you a farm unless no one else wants to rent it. It was a big piece of land, it was cash rent, I mean, the stakes were set. We were doomed to fail. And we weren ' t focused on environmental sustainability back then. We were focused on economic sustainability. How do we make higher yield? How do we use less fertilizer? How do we get to the next year and feed our family? It happens that economic and environmental sustainability go hand in hand along with social sustainability. As we used less fertilizer, used less water, our yield started going up. MZ : Can you spell out for me what the problem is with rice when it comes to the environment? JW : Soil is full of microbes. And what they ' re doing in the soil on a basic level is they ' re down there chewing away, eating up the biomass that ' s left over from the crop before. And out of that comes methane, the same way it happens in cattle operations or whatever. MZ : But you decided to do it a different way. JW : We do it a little bit differently. Arkansas is a very water-rich state, but we have a dry season, right in the middle of summer is a dry season. So when we dry the soil, that anaerobic microbial that ' s living in that soup, that mushy soil, either dies or goes dormant. So we break his life cycle. MZ : Oh! JW : And then it stops emitting methane. But it doesn ' t hurt the rice. That actually helps rice, when you use less water. We have less runoff, less erosion, less nutrients leaving our field. Nothing leaves our field unless we want it to. Jessica Whitaker Allen : We started tracking this. We saw that our water use was down and we thought, OK, well, we need to add to that, let ' s add more data points to that. Let ' s add our fertilizer, let ' s add our fuel usage. And so we really started recording all this stuff by hand in just an Excel spreadsheet. We didn ' t know what we were doing. MZ : But you knew you were on to something. JWA : Yes, so we started recording everything. MZ : Do you remember if there was a moment when the two of you thought, like, what we ' re doing, it shouldn ' t be **special** to us. Why aren ' t more people – JW : To collect the data, have the equipment and the cloud-based technology and the stuff in the field to record it, it ' s just massively time-consuming, and consumers don ' t pay for it. JWA : Well and people like us, why would you want to do it? I mean, take a walk through your supermarket. Everything ' s sustainable, everything ' s greenwashed. You can put whatever you want to on a package, there ' s no rules, there ' s no standards. So we ' re trying to make that standard. We live and work in southeast Arkansas. Our town has 4, 000 residents and one stoplight. The nearest airport, Starbucks shopping mall, Whole Foods, is two hours away in any direction. Without places like this and farmers like us, you ' d be hungry, naked and sober. I ' m a farmer, but not in the way you might think. I don ' t drive the tractors, and I don ' t plant the rice. But I know how to take what my dad has learned and what he is doing and share that with others. We are going to work with farmers in southeast Arkansas to educate them about the benefits of growing sustainable rice. JW : Let me tell you why that ' s important. There are 400 million acres of rice grown globally. And our methods, if used, can reduce greenhouse gas by 50 percent, reduce water use by 50 percent, increase yields to feed a hungry world. MZ : We hear a lot about how weather has gotten more extreme, flooding, heat. What ' s it been like in Arkansas? JW : So in 2016, we had a 500-year flood. We were... We get a picture of our place flooded. 14 inches of rain. That was a 500-year flood. Hurt a lot of people. And then we had a 1, 000-year flood. And we lost so many acres. And you know, when houses get flooded in rural communities, they don ' t come back. I mean, when you have whole neighborhoods get flooded, they don ' t rebuild them, they don ' t come back. JWA : I feel like almost every day I ' m getting a notification from the weather Channel of an excessive heat **warning** or thunderstorm **warning** or hail threat. You know, you don ' t really know what you ' re going to expect. MZ : When you think about what it will be like, I don ' t know, five years, 10 years, for someone to go into the supermarket and think, oh, right, I need to get rice off my grocery list. What ideally would you like to see on the shelf there? JW : So I think it ' s going to change, because I like to study what companies want. I mean, if you figure out what they want, then you know what to give them. And when you read their sustainability goals, they ' re all saying they ' re going to be net zero. The only way they can be net zero is they bring the farmer into the loop. So if we ' re able to do this, in five years, everybody ' s doing these **practices**, the US rice industry will be the most sustainable industry in the world. MZ : Can you do it? JW : Oh, yeah. It ' s about to happen. MZ : You ' ve just heard a lot of ideas, a lot of stories and a lot of big plans. So we want to wrap things up by asking our experts : What if we actually achieved our climate goals? What would that future even look like? They, of course, had a lot of thoughts. We ' ll let you pick your favorites. Thanks for watching. NT : If we are successful in addressing the climate crisis, we massively reduce the vulnerability and improve the economic and the creative prospects of billions of people around the world who currently are basically excluded from all of the opportunities which we have, which come from having abundant material and energy all around us. That ' s the real prize. FD : A world where we have had that exponential would be a world in which we have clean, cheap energy for all. KB : At the end of this decade, the International Energy Agency believes that we will have enough capacity to be producing 10 billion, 10 billion batteries a year. It ' s enough for every person on the planet to have their own battery. AC : I ' m constantly surrounded by young people just like me who are really, really, really **passionate** about the world that we live in. For obvious reasons. It ' s the world that we will inherit. EZ : Women are very important for the energy transition. So if we ' re talking about a world where we have, you know, where we tackle climate change, all I want to see is, you know, women in power. I want to see more solutions being driven by young people. And I want our politicians to be truly and meaningfully engaged in making change. NG : We always tend to underestimate the pace of ultimate adoption and change. Today, there are more than 150 pathways, scientific pathways that we could take to limit global warming. Al Gore : If we get to true net zero, astonishingly, global temperatures will stop going up with a lag time of as little as three to five years. Colin Averill : We ' ve been able to accelerate tree growth and carbon capture and tree stems by 30 to 70 percent. Dan Jørgensen : Offshore wind has the potential to cover the current electricity demand of the entire world not once, not twice, 18 times. NG : We need to decarbonize the old economy that we have and invest to create the new economy that we need Paul Hawken : Regenerative farming is fairly simple in concept. It means creating the conditions for more life. Josephine Phillips : We need to buy less stuff, and we need to look after what we buy. Valuing the things that we own is a climate solution. Vishaan Chakrabarti : We can all live in this transit-rich, carbon-negative, affordable way and leave the vast majority of the planet for nature, for agriculture, for clean oceans. We can do this. Susan Ruffo : The ocean is a powerful source of solutions that we ' ve overlooked for far too long. Al Roker : You have to be engaged. Let your elected officials know that this is important to you. You have to vote, you have to go out there and support politicians who are going to support our planet. Peggy Shepard : We can create a legacy of environmental quality and climate resilience for all. Fehinti Balogun : You are not powerless that the"oh, there ' s nothing we can do, just keep going"is symptom, not cure. But you are needed. Avinash Persaud : The power of what we can do when it matters to us is unlimited. CF : It ' s not that we ' re simply going to avoid the worst impacts of climate change. It ' s that we ' re going to improve the quality of life for humans, both in the urban and in the rural sector, and for non-humans.

Text Sample 210: POS Dim 2 – 2023 – Score 16.00 – 2023/t003508.txt

Chris Anderson : It ' s very nice to have you here. Let ' s see. First of all, congratulations. You really pulled off something remarkable on that grilling, you achieved something that very few people do, which was, you pulled off a kind of, a bipartisan consensus in US politics. It was great. The bad news was that that consensus largely seemed to be :"We must ban TikTok."So we ' re going to come to that in a bit. And I ' m curious, but before we go there, we need to know about you. You seem to me like a remarkable person. I want to know a bit of your story and how you came to TikTok in the first place. Shou Chew : Thank you, Chris. Before we do that, can I just check, need to know my audience, how many of you here use TikTok? Oh, thank you. For those who don ' t, the Wi-Fi is free. CA : There ' s another question, which is, how many of you here have had your lives touched through TikTok, through your kids and other people in your lives? SC : Oh, that ' s great to see. CA : It ' s basically, if you ' re alive, you have had some kind of contact with TikTok at this point. So tell us about you. SC : So my name is Shou, and I ' m from Singapore. Roughly 10 years ago, I met with two engineers who were building a product. And the idea behind this was to build a product that recommended content to people not based on who they knew, which was, if you think about it, 10 years ago, the social graph was all in the rage. And the idea was, you know, your content and the feed that you saw should be based on people that you knew. But 10 years ago, these two engineers thought about something different, which is, instead of showing you – instead of showing you people you knew, why don ' t we show you content that you liked? And that ' s sort of the genesis and the birth of the early iterations of TikTok. And about five years ago, with the advent of 4G, short video, **mobile** phone penetration, TikTok was born. And a couple of years ago, I had the opportunity to run this company, and it still excites me every single day. CA : So I want to dig in a little more into this, about what was it that made this take-off so explosive? Because the language I hear from people who spent time on it, it ’ s sort of like I mean, it is a different level of addiction to other media out there. And I don ’ t necessarily mean this in a good way, we ’ ll be coming on to it. There ’ s good and bad things about this type of addiction. But it ’ s the feeling that within a couple of days of experience of TikTok, it knows you and it surprises you with things that you didn ’ t know you were going to be interested in, but you are. How? Is it really just, instead of the social graph – What are these algorithms doing? SC : I think to describe this, to begin to answer your question, we have to talk about the **mission** of the company. Now the **mission** is to inspire creativity and to bring joy. And I think **missions** for companies like ours [are] really important. Because you have product managers working on the product every single day, and they need to have a North Star, you know, something to sort of, work towards together. Now, based on this **mission**, our vision is to provide three things to our users. We want to provide a window to discover, and I ’ ll talk about discovery, you talked about this, in a second. We want to give them a canvas to create, which is going to be really exciting with new technologies in AI that are going to help people create new things. And the final thing is bridges for people to connect. So that ’ s sort of the vision of what we ’ re trying to build. Now what really makes TikTok very unique and very different is the whole discovery engine behind it. So there are earlier apps that I have a lot of respect for, but they were built for a different purpose. For example, in the era of search, you know, there was an app that was built for people who wanted to search things so that is more easily found. And then in the era of social graphs, it was about connecting people and their followers. Now what we have done is that... based on our machine-learning algorithms, we ’ re showing people what they liked. And what this means is that we have given the everyday person a **platform** to be discovered. If you have talent, it is very, very easy to get discovered on TikTok. And I ’ ll just give you one example of this. The biggest creator on TikTok is a guy called Khaby. Khaby was from Senegal, he lives in Italy, he was a factory worker. He, for the longest time, didn ’ t even speak in any of his videos. But what he did was he had talent. He was funny, he had a good **expression**, he had creativity, so he kept posting. And today he has 160 million followers on our **platform**. So every single day we hear stories like that, businesses, people with talent. And I think it ’ s very freeing to have a **platform** where, as long as you have talent, you ’ re going to be heard and you have the chance to succeed. And that ’ s what we ’ re providing to our users. CA : So this is the amazing thing to me. Like, most of us have grown up with, say, network television, where, for decades you ’ ve had thousands of brilliant, creative people toiling in the trenches, trying to imagine stuff that will be amazing for an audience. And none of them ever remotely came up with anything that looked like many of your creators. So these algorithms, just by **observing** people ’ s behavior and what they look like, have discovered things that thousands of brilliant humans never discovered. Tell me some of the things that it is looking at. So obvious things, like if someone presses like or stays on a video for a long time, that gives you a clue,”more like that.”But is it subject matter? What are the array of things that you have noticed that you can actually track that provide useful clues? SC : I ’ m going to simplify this a lot, but the machine learning, the recommendation algorithm is really just math. So, for example, if you liked videos one, two, three and four, and I like videos one, two, three and five, maybe he liked videos one, two, three and six. Now what ’ s going to happen is, because we like one, two, three at the same time, he ’ s going to be shown four, five, six, and so are we. And you can think about this repeated at scale in real time across more than a billion people. That ’ s basically what it is, it ’ s math. And of course, you know, AI and machine learning has allowed this to be done at a very, very big scale. And what we have seen, the result of this, is that it learns the interest signals that people exhibit very quickly and shows you content that ’ s really relevant for you in a very quick way. CA : So it ’ s a form of collaborative filtering, from what you ’ re saying. The theory behind it is that these humans are weird people, we don ’ t really know what they ’ re interested in, but if we see that one human is interested, with an overlap of someone else, chances are, you know, you could make use of the other pieces that are in that overlapped human ’ s repertoire to feed them, and they ’ ll be surprised. But the reason they like it is because their pal also liked it. SC : It ’ s pattern recognition based on your interest signals. And I think the other thing here is that we don ’ t actually ask you 20 questions on whether you like a piece of content, you know, what are your interests, we don ’ t do that. We built that experience organically into the app experience. So you are voting with your thumbs by watching a video, by swiping it, by liking it, by sharing it, you are basically exhibiting interest signals. And what it does mathematically is to take those signals, put it in a formula and then matches it through pattern recognition. That ’ s basically the idea behind it. CA : I mean, lots of **start-ups** have tried to use these types of techniques. I ’ m wondering what else played a role early on? I mean, how big a deal was it, that from the get-go you were optimizing for smartphones so that videos were shot in portrait format and they were short. Was that an early distinguishing thing that mattered? SC : I think we were the first to really try this at scale. You know, the recommendation algorithm is a very important reason as to why the **platform** is so popular among so many people. But beyond that, you know, you mentioned the format itself. So we talked about the vision of the company, which is to have a window to discover. And if you just open the app for the first time, you ’ ll see that it takes up your whole screen. So that ’ s the window that we want. You can imagine a lot of people using that window to discover new things in their lives. Then, you know, through this recommendation algorithm, we have found that it connects people together. People find communities, and I ’ ve heard so many stories of people who have found their communities because of the content that they ’ re posting. Now, I ’ ll give you an example. I was in DC recently, and I met with a bunch of creators. CA : I heard. SC : One of them was sitting next to me at a dinner, his name is Samuel. He runs a restaurant in Phoenix, Arizona, and it ’ s a taco restaurant. He told me he has never done this before, first venture. He started posting all this content on TikTok, and I saw his content, I was hungry after looking at it, it ’ s great content. And he ’ s generated so much interest in his business, that last year he made something like a million dollars in revenue just via TikTok. One restaurant. And again and again, I hear these stories, you know, by connecting people together, by giving people the window to discover, we have given many small businesses and many people, your common person, a voice that they will never otherwise have. And I think that ’ s the power of the **platform**. CA : So you definitely have identified early just how we ’ re social creatures, we need affirmation. I ’ ve heard a story, and you can tell me whether true or not, that one of the keys to your early liftoff was that you wanted to persuade creators who were trying out TikTok that this was a **platform** where they would get response, early on, when you ’ re trying to grow something, the numbers aren ’ t there for response. So you had the brilliant idea of goosing those numbers a bit, basically finding ways to give people, you know, a bigger sense of like, more likes, more engagement than was actually the case, by using AI agents somehow in the process. Is that a brilliant idea, or is that just a myth? SC : I would describe it in a different way. So there are other **platforms** that exist before TikTok. And if you think about those **platforms**, you sort of have to be famous already in order to get followers. Because the way it ' s built is that people come and follow people. And if you aren ' t already famous, the chances that you get discovered are very, very low. Now, what we have done, again, because of the difference in the way we ' re recommending content, is that we have given anyone, any single person with enough talent a stage to be able to be discovered. And I think that actually is the single, probably the most important thing contributing to the growth of the **platform**. And again and again, you will hear stories from people who use the **platform**, who post regularly on it, that if they have something they want to say, the **platform** gives them the chance and the stage to connect with their audience in a way that I think no other product in the past has ever offered them. CA : So I ' m just trying to play back what you said there. You said you were describing a different way what I said. Is it then the case that like, to give someone a decent chance, someone who ' s brilliant but doesn ' t come with any followers initially, that you ' ve got some technique to identify talent and that you will almost encourage them, you will give them some kind of, you know, artificially increase the number of followers or likes or whatever that they have, so that others are encouraged to go,"Wow, there ' s something there."Like it ' s this idea of critical mass that kind of, every entrepreneur, every party **planner** kind of knows about of"No, no, this is the hot place in town, everyone come,"and that that is how you actually gain critical mass? SC : We want to make sure that every person who posts a video is given an equal chance to be able to have some audience to begin with. But this idea that you are maybe alluding to, that we can get people to like something, it doesn ' t really work like that. CA : Could you get AI agents to like something? Could you seed the network with extra AI agents that could kind of, you know, give someone early encouragement? SC : Ultimately, what the machine does is it recognizes people ' s interests. So if you post something that ' s not interesting to a lot of people, even if you gave it a lot of exposure, you ' re not going to get the virality that you want. So it ' s a lot of... There is no push here. It ' s not like you can go and push something, because I like Chris, I ' m going to push your content, it doesn ' t work like that. You ' ve got to have a message that resonates with people, and if it does, then it will automatically just have the virality itself. That ' s the beauty of user-generated content. It ' s not something that can be engineered or over-thought. It really is something that has to resonate with the audience. And if it does, then it goes viral. CA : Speaking privately with an investor who knows your company quite well, who said that actually the level of sophistication of the algorithms you have going is just another order of magnitude to what competitors like, you know, Facebook or YouTube have going. Is that just hype or do you really believe you – like, how complex are these algorithms? SC : Well, I think in terms of complexity, there are many companies who have a lot of resources and a lot of talent. They will figure out even the most complex algorithms. I think what is very different is your **mission** of your company, how you started the company. Like I said, you know, we started with this idea that this was the main use case. The most important use case is you come and you get to see recommended content. Now for some other apps out there, they are very significant and have a lot of users, they are built for a different original purpose. And if you are built for something different, then your users are used to that because the community comes in and they expect that sort of experience. So I think the pivot away from that is not really just a matter of engineering and algorithms, it ' s a matter of what your company is built to begin with. Which is why I started this by saying you need to have a vision, you need to have a **mission**, and that ' s the North Star. You can ' t just shift it halfway. CA : Right. And is it fair to say that because your start point has been interest algorithms rather than social graph algorithms, you ' ve been able to avoid some of the worst of the sort of, the filter bubbles that have happened in other social media where you have tribes kind of declaring war on each other effectively. And so much of the noise and energy is around that. Do you believe that you ' ve largely avoided that on TikTok? SC : The diversity of content that our users see is very key. You know, in order for the discovery – the **mission** is to discover – sorry, the vision is to discover. So in order to facilitate that, it is very important to us that what the users see is a diversity of content. Now, generally speaking, you know, there are certain issues that you mentioned that the industry faces, you know. There are some bad actors who come on the internet, they post bad content. Now our approach is that we have very clear community guidelines. We ' re very transparent about what is allowed and what is not allowed on our **platform**. No executives make any **ad** hoc decisions. And based on that, we have built a team that is tens of thousands of people plus machines in order to identify content that is bad and actively and proactively remove it from the **platform**. CA : Talk about what some of those key guidelines are. SC : We have it published on our website. In March, we just iterated a new version to make it more readable. So there are many things like, for example, no pornography, clearly no child sexual abuse material and other bad things, no violence, for example. We also make it clear that it ' s a differentiated experience if you ' re below 18 years old. So if you ' re below 18 years old, for example, your entire app experience is actually more restricted. We don ' t allow, as an example, users below 16, by default, to go viral. We don ' t allow that. If you ' re below 16, we don ' t allow you to use the **instant** messaging feature in app. If you ' re below 18, we don ' t allow you to use the livestreaming features. And of course, we give parents a whole set of tools to control their teenagers ' experience as well. CA : How do you know the age of your users? SC : In our industry, we rely mainly on something called age gating, which is when you sign up for the app for the first time and we ask you for the age. Now, beyond that, we also have built tools to go through your public profile for example, when you post a video, we try to match the age that you said with the video that you just posted. Now, there are questions of can we do more? And the question always has, for every company, by the way, in our industry, has to be balanced with **privacy**. Now, if, for example, we scan the faces of every single user, then we will significantly increase the ability to tell their age. But we will also significantly increase the amount of data that we collect on you. Now, we don ' t want to collect data. We don ' t want to scan data on your face to collect that. So that balance has to be maintained, and it ' s a challenge that we are working through together with industry, together with the regulators as well. CA : So look, one thing that is unquestionable is that you have created a **platform** for literally millions of people who never thought they were going to be a content creator. You ' ve given them an audience. I ' d actually like to hear from you one other favorite example of someone who TikTok has given an audience to that never had that before. SC : So when again, when I travel around the world, I meet with a whole bunch of creators on our **platform**. I was in South Korea just yesterday, and before that I met with – yes, before that I met with a bunch of – People don ' t expect, for example, teachers. There is an English teacher from Arkansas. Her name is Claudine, and I met her in person. She uses our **platform** to reach out to students. There is another teacher called Chemical Kim. And Chemical Kim teaches chemistry. What she does is she uses our **platform** to reach out to a much broader student base than she has in her classroom. And they ’ re both very, very popular. You know, in fact, what we have realized is that STEM content has over 116 billion views on our **platform** globally. And it ’ s so significant – CA : In a year? SC : Cumulatively. CA : [116] billion. SC : It ’ s so significant, that in the US we have started testing, creating a feed just for STEM content. Just for STEM content. I ’ ve been using it for a while, and I learned something new. You want to know what it is? Apparently if you flip an egg on your tray, the egg will last longer. It ’ s science, there ’ s a whole video on this, I learned this on TikTok. You can search for this. CA : You want to know something else about an egg? If you put it in just one hand and squeeze it as hard as you can, it will never break. SC : Yes, I think I read about that, too. CA : It ’ s not true. SC : We can search for it. CA : But look, here ’ s here ’ s the flip side to all this amazingness. And honestly, this is the key thing, that I want to have an honest, heart-to-heart conversation with you because it ’ s such an important issue, this question of human addiction. You know, we are... animals with a prefrontal cortex. That ’ s how I think of us. We have these addictive instincts that go back millions of years, and we often are in the mode of trying to modulate our own behavior. It turns out that the internet is incredibly good at activating our animal cells and getting them so damn excited. And your company, the company you ’ ve built, is better at it than any other company on the planet, I think. So what are the risks of this? I mean, how... From a company point of view, for example, it ’ s in your interest to have people on there as long as possible. So some would say, as a first pass, you want people to be addicted as long as possible. That ’ s how advertising money will flow and so forth, and that ’ s how your creators will be delighted. What is too much? SC : I don ’ t actually agree with that. You know, as a company, our goal is not to optimize and **maximize** time spent. It is not. In fact, in order to address people spending too much time on our **platform**, we have done a number of things. I was just speaking with some of your colleagues backstage. One of them told me she has encountered this as well. If you spend too much time on our **platform**, we will proactively send you videos to tell you to get off the **platform**. We will. And depending on the time of the day, if it ’ s late at night, it will come sooner. We have also built in tools to limit, if you below 18 years old, by default, we set a 60-minute default time limit. CA : How many? SC : Sixty minutes. And we ’ ve given parents tools and yourself tools, if you go to **settings**, you can set your own time limit. We ’ ve given parents tools so that you can pair, for the parents who don ’ t know this, go to **settings**, family pairing, you can pair your phone with your teenager ’ s phone and set the time limit. And we really encourage parents to have these conversations with their teenagers on what is the right amount of screen time. I think there ’ s a healthy relationship that you should have with your screen, and as a business, we believe that that balance needs to be met. So it ’ s not true that we just want to **maximize** time spent. CA : If you were advising parents here what time they should actually recommend to their teenagers, what do you think is the right **setting**? SC : Well, 60 minutes, we did not come up with it ourselves. So I went to the Digital Wellness Lab at the Boston Children ’ s Hospital, and we had this conversation with them. And 60 minutes was the recommendation that they gave to us, which is why we built this into the app. So 60 minutes, take it for what it is, it ’ s something that we ’ ve had some discussions of experts. But I think for all parents here, it is very important to have these conversations with your teenage children and help them develop a healthy relationship with screens. I think we live in an age where it ’ s completely inevitable that we ’ re going to interact with screens and digital content, but I think we should develop healthy habits early on in life, and that ’ s something I would encourage. CA : Curious to ask the audience, which of you who have ever had that video on TikTok appear saying,”Come off.”OK, I mean... So maybe a third of the audience seem to be active TikTok users, and about 20 people maybe put their hands up there. Are you sure that – like, it feels to me like this is a great thing to have, but are you... isn ’ t there always going to be a temptation in any given quarter or whatever, to just push it a bit at the boundary and just dial back a bit on that so that you can hit revenue goals, etc? Are you saying that this is used scrupulously? SC : I think, you know, in terms... Even if you think about it from a commercial point of view, it is always best when your customers have a very healthy relationship with your product. It ’ s always best when it ’ s healthy. So if you think about very short-term retention, maybe, but that ’ s not the way we think about it. If you think about it from a longer-term perspective, what you really want to have is a healthy relationship, you know. You don ’ t want people to develop very unhealthy habits, and then at some point they ’ re going to drop it. So I think everything in moderation. CA : There ’ s a claim out there that in China, there ’ s a much more rigorous standards imposed on the amount of time that children, especially, can spend on the TikTok equivalent of that. SC : That is unfortunately a misconception. So that experience that is being mentioned for Douyin, which is a different app, is for an under 14-year-old experience. Now, if you compare that in the United States, we have an under-13 experience in the US. It ’ s only available in the US, it ’ s not available here in Canada, in Canada, we just don ’ t allow it. If you look at the under-13 experience in the US, it ’ s much more restricted than the under-14 experience in China. It ’ s so restrictive, that every single piece of content is vetted by our third-party child safety expert. And we don ’ t allow any under-13s in the US to publish, we don ’ t allow them to post, and we don ’ t allow them to use a lot of features. So I think that that report, I ’ ve seen that report too, it ’ s not doing a fair comparison. CA : What do you make of this issue? You know, you ’ ve got these millions of content creators and all of them, in a sense, are in a race for attention, and that race can pull them in certain directions. So, for example, teenage girls on TikTok, sometimes people worry that, to win attention, they ’ ve discovered that by being more sexual that they can gain extra viewers. Is this a concern? Is there anything you can do about this? SC : We address this in our community guidelines as well. You know, if you look at sort of the sexualized content on our guidelines, if you ’ re below a certain age, you know, for certain themes that are mature, we actually remove that from your experience. Again, I come back to this, you know, we want to have a safe **platform**. In fact, at my congressional hearing, I made four commitments to our users and to the politicians in the US. And the first one is that we take safety, especially for teenagers, extremely seriously, and we will continue to prioritize that. You know, I believe that we need to give our teenage users, and our users in general, a very safe experience, because if we don ’ t do that, then we cannot fulfill – the **mission** is to inspire creativity and to bring joy. If they don ’ t feel safe, I cannot fulfill my **mission**. So it ’ s all very organic to me as a business to make sure I do that. CA : But in the strange interacting world of human psychology and so forth, weird memes can take off. I mean, you had this outbreak a couple years back with these devious licks where kids were competing with each other to do vandalism in schools and, you know, get lots of followers from it. How on Earth do you battle something like that? SC : So dangerous challenges are not allowed on our **platform**. If you look at our guidelines, it ’ s violative. We proactively invest resources to identify them and remove them from our **platform**. In fact, if you search for dangerous challenges on our **platform** today, we will redirect you to a safety resource page. And we actually worked with some creators as well to come up with campaigns. This is another campaign. It ’ s the”Stop, Think, Decide

Before You Act"campaign where we work with the creators to produce videos, to explain to people that some things are dangerous, please don ' t do it. And we post these videos actively on our **platform** as well. CA : That ' s cool. And you ' ve got lots of employees. I mean, how many employees do you have who are specifically looking at these content moderation things, or is that the wrong question? Are they mostly identified by AI initially and then you have a group who are overseeing and making the final decision? SC : The group is based in Ireland and it ' s a lot of people, it ' s tens of thousands of people. CA : Tens of thousands? SC : It ' s one of the most important cost items on my PnL, and I think it ' s completely worth it. Now, most of the moderation has to be done by machines. The machines are good, they ' re quite good, but they ' re not as good as, you know, they ' re not perfect at this point. So you have to complement them with a lot of human beings today. And I think, by the way, a lot of the progress in AI in general is making that kind of content moderation capabilities a lot better. So we ' re going to get more precise. You know, we ' re going to get more specific. And it ' s going to be able to handle larger scale. And that ' s something I think that I ' m personally looking forward to. CA : What about this perceived huge downside of use of, certainly Instagram, I think TikTok as well. What people worry that you are amplifying insecurities, especially of teenagers and perhaps especially of teenage girls. They see these amazing people on there doing amazing things, they feel inadequate, there ' s all these reported cases of depression, insecurity, suicide and so forth. SC : I take this extremely seriously. So in our guidelines, for certain themes that we think are mature and not suitable for teenagers, we actually proactively remove it from their experience. At the same time, if you search certain terms, we will make sure that you get redirected to a resource safety page. Now we are always working with experts to understand some of these new trends that could emerge and proactively try to manage them, if that makes sense. Now, this is a problem that predates us, that predates TikTok. It actually predates the internet. But it ' s our responsibility to make sure that we invest enough to understand and to address the concerns, to keep the experience as safe as possible for as many people as possible. CA : Now, in Congress, the main concern seemed to be not so much what we ' ve talked about, but data, the data of users, the fact that you ' re owned by ByteDance, Chinese company, and the concern that at any moment Chinese government might require or ask for data. And in fact, there have been instances where, I think you ' ve confirmed, that some data of journalists on the **platform** was made available to ByteDance ' s engineers and from there, who knows what. Now, your response to this was to have this Project Texas, where you ' re moving data to be controlled by Oracle here in the US. Can you talk about that project and why, if you believe it so, why we should not worry so much about this issue? SC : I will say a couple of things about this, if you don ' t mind. The first thing I would say is that the internet is built on global interoperability, and we are not the only company that relies on the global talent pool to make our products as good as possible. Technology is a very collaborative effort. I think many people here would say the same thing. So we are not the first company to have engineers in all countries, including in China. We ' re not the first one. Now, I understand some of these concerns. You know, the data access by employees is not data accessed by government. This is very different, and there ' s a clear difference in this. But we hear the concerns that are raised in the United States. We did not try to avoid discussing. We did not try to argue our way out of it. What we did was we built an unprecedented project where we localize American data to be stored on American soil by an American company overseen by American personnel. So this kind of protection for American data is beyond what any other company in our industry has ever done. Well, money is not the only issue here, but it ' s very expensive to build something like that. And more importantly, you know, we are basically localizing data in a way that no other company has done. So we need to be very careful that whilst we are pursuing what we call digital sovereignty in the US and we are also doing a version of this in Europe, that we don ' t balkanize the internet. Now we are the first to do it. And I expect that, you know, other companies are probably looking at this and trying to figure out how you balance between protecting, protected data, you know, to make sure that everybody feels secure about it while at the same time allowing for interoperability to continue to happen, because that ' s what makes technology and the internet so great. So that ' s something that we are doing. CA : How far are you along that journey with Project Texas? SC : We are very, very far along today. CA : When will there be a clear you know, here it is, it ' s done, it ' s firewalled, this data is protected? SC : Today, by default, all new US data is already stored in the Oracle cloud infrastructure. So it ' s in this protected US environment that we talked about in the United States. We still have some legacy data to delete in our own servers in Virginia and in Singapore. Our data has never been stored in China, by the way. That deletion is a very big engineering effort. So as we said, as I said at the hearing, it ' s going to take us a while to delete them, but I expect it to be done this year. CA : How much power do you have over your own ability to control certain things? So, for example, suppose that, for whatever reason, the Chinese government was to look at an upcoming US election and say,"You know what, we would like this party to win,"let ' s say, or"We would like civil war to break out"or whatever. How..."And we could do this by amplifying the content of certain troublemaking, disturbing people, causing uncertainty, spreading misinformation,"etc. If you were required via ByteDance to do this, like, first of all, is there a pathway where theoretically that is possible? What ' s your personal line in the sand on this? SC : So during the congressional hearing, I made four commitments, we talked about the first one, which is safety. The third one is to keep TikTok a place of freedom of **expression**. By the way, if you go on TikTok today, you can search for anything you want, as long as it doesn ' t violate our community guidelines. And to keep it free from any government manipulation. And the fourth one is transparency and third-party monitoring. So the way we are trying to address this concern is an unprecedented amount of transparency. What do I mean by this? We ' re actually allowing third-party reviewers to come in and review our source code. I don ' t know any other company that does this, by the way. Because everything, as you know, is driven by code. So to allow someone else to review the source code is to give this a significant amount of transparency to ensure that the scenarios that you described that are highly hypothetical, cannot happen on our **platform**. Now, at the same time, we are releasing more research tools for **researchers** so that they can study the output. So the source code is the input. We are also allowing **researchers** to study the output, which is the content on our **platform**. I think the easiest way to sort of fend this off is transparency. You know, we give people access to monitor us, and we just make it very, very transparent. And that ' s our approach to the problem. CA : So you will say directly to this group that the scenario I talked about, of theoretical Chinese government interference in an American election, you can say that will not happen? SC : I can say that we are building all the tools to prevent any of these **actions** from happening. And I ' m very confident that with an unprecedented amount of transparency that we ' re giving on the **platform**, we can reduce this risk to as low as zero as possible. CA : To as low as zero as possible. SC : To as close to zero as possible. CA : As close to zero as possible. That ' s fairly reassuring. Fairly. I mean, how would the world know? If you discovered this or you thought you had to do it, is this a line in the sand for you? Like, are you in a situation you would not let the company that you know now and that you are running do this? SC : Absolutely. That ' s the reason why we ' re letting third parties monitor, because if they find out, you know, they will disclose this. We also have transparency reports, by the way, where we talk about a whole bunch of things, the content that we remove, you know, that violates our guidelines, government requests. You know, it ' s all published online. All you have to do is search for it. CA : So you ' re **super** compelling and likable as a CEO, I have to say. And I would like to, as we wrap this up, I ' d like to give you a chance just to paint, like, what ' s the vision? As you look at what TikTok could be, let ' s move the clock out, say, five years from now. How should we think about your contribution to our collective future? SC : I think it ' s still down to the vision that we have. So in terms of the window of discovery, I think there ' s a huge benefit to the world when people can discover new things. You know, people think that TikTok is all about dancing and singing, and there ' s nothing wrong with that, because it ' s **super** fun. There ' s still a lot of that, but we ' re seeing science content, STEM content, have you about BookTok? It ' s a viral trend that talks about books and encourages people to read. That BookTok has 120 billion views globally, 120 billion. CA : Billion, with a B. SC : People are learning how to cook, people are learning about science, people are learning how to golf – well, people are watching videos on golfing, I guess. I haven ' t gotten better by looking at the videos. I think there ' s a huge, huge opportunity here on discovery and giving the everyday person a voice. If you talk to our creators, you know, a lot of people will tell you this again and again, that before TikTok, they would never have been discovered. And we have given them the **platform** to do that. And it ' s important to maintain that. Then we talk about creation. You know, there ' s all this new technology coming in with AI-generated content that will help people create even more creative content. I think there ' s going to be a collaboration between, and I think there ' s a speaker who is going to talk about this, between people and AI where they can unleash their creativity in a different way. You know, like for example, I ' m terrible at drawing personally, but if I had some AI to help me, then maybe I can express myself even better. Then we talk about bridges to connect and connecting people and the communities together. This could be products, this could be commerce, five million businesses in the US benefit from TikTok today. I think we can get that number to a much higher number. And of course, if you look around the world, including in Canada, that number is going to be massive. So I think these are the biggest opportunities that we have, and it ' s really very exciting. CA : So courtesy of your experience in Congress, you actually became a bit of a TikTok star yourself, I think. Some of your videos have gone viral. You ' ve got your phone with you. Do you want to make a little little TikTok video right now? Let ' s do this. SC : If you don ' t mind... CA : What do you think, should we do this? SC : We ' re just going to do a selfie together, how ' s that? So why don ' t we just say"Hi."Hi! Audience : Hi! CA : Hello from TED. SC : All right, thank you, I hope it goes viral. CA : If that one goes viral, I think I ' ve given up on your algorithm, actually. Shou Chew, you ' re one of the most influential and powerful people in the world, whether you know it or not. And I really appreciate you coming and sharing your vision. I really, really hope the upside of what you ' re talking about comes about. Thank you so much for coming today. SC : Thank you, Chris. CA : It ' s really interesting.

Text Sample 211: POS Dim 2 – 2023 – Score 11.00 – 2023/t003364.txt

Isn ’ t learning fun? I absolutely love it. For the past 15 years, I have been an academic writing instructor, and I ’ ve been a coach for the last five. The people that I usually work with are adults who ’ ve returned to school, and I really love working with these folks because, like them, I ’ m a lifelong learner, and I ’ ve been to school three times, so far, in the past 20 years. But I still remember the first time that I went back. I almost quit, the first semester. By midterm, I was collapsed into this shabby chair in my prof ’ s office, in the corner, in the midst of an existential crisis. Things were going horribly. The seminars that I so diligently prepared for were painful. Every time we were asked for our responses and analysis of the readings, everyone else ’ s responses sounded so smart. Mine, by comparison, seemed so basic. And then, I got my first paper back... and I ’ d bombed it. Yep. So there I was, in tears, convinced that I was the dumbest, least capable person in the class. And honestly, I got to tell you, I was ready to quit. But I had actually staked a lot on going back to school. I had actually quit my best job to date to pursue my dream of becoming an instructor. So I wanted my prof ’ s advice before I made any final decisions. She handed me some tissues, and instead of agreeing with my assessment, she told me to stick it out. And she said that apparently, I wasn ’ t doing any worse than anyone else. Now, I was pretty skeptical, because all the evidence seemed to point to the contrary. But figuring she knew something that I didn ’ t, I held on. By the end of term, she was right. My grades began to improve, and I learned that my fellow students were just as overwhelmed as I was. Needless to say, this experience has stayed with me, and it has informed how I teach and coach. I ’ ve been where my prof was and said the same things to my students and clients. I ’ ve also learned a lot since then about the relationship between confidence and learning. Nowadays, the mid-career professionals who come to me for help with their academic success come to me because they think they have a skill problem. And sure, they could use some skill-building for sure. But for many, that ’ s not the crux of the issue that ’ s getting in the way of their success. The real issue lies in how they think about themselves as learners and what they believe to be true about their abilities. Now, this might seem like a **bold** claim, so let me explain. Why aren ’ t skills the real issue? Most of us are very capable learners. The people I work with are accomplished professionals and experts in their field. They wouldn ’ t be where they are if they didn ’ t know how to learn and build skills. When I went back, I had been writing and training for several years in a professional context. My challenge was up-leveling my analytical and critical thinking skills. Once I was oriented to that gap, I set about closing it. My grades began to improve. I see the same thing with my students and clients. Bridging the skill gap is easier than we think. It ’ s the confidence gap that is much harder to bridge. But it needs addressing because that is what is getting in the way of our academic success in learning situations. So what ’ s going on with our confidence anyway? What is happening? Why is it so hard? Based on my experience, adults in learning situations face three main challenges. The first one is that we **struggle** with a lot of self doubt and uncertainty. For many of us who go back, the learning curve is really steep. It ’ s not only been years, maybe decades, since we ’ d been back in school. We ’ ve spent a hot minute or two since we ’ ve been a beginner at anything. All of a sudden, we find ourselves having to **navigate** online **platforms** and figure out study habits, and this whole academic writing thing is a whole next level. And then, we have to read critically. It ’ s a lot. And then add to this that when we return to school, we often bring with us all the old patterns and beliefs that we ’ ve been carrying for the past ten, maybe 20 years, about who we are as learners. In the past, if we thought that we were weak students, we ’ ve already got bags of doubts and insecurities and anxieties, and we **judge** ourselves harshly right out of the gate. Now if you ’ re thinking,”I was a good student, I got this,”– no, no. We **struggle** because we can set the bar unreasonably high and then **judge** ourselves just as harshly and as quickly. And it ’ s difficult. The second challenge that we face is that we spend a lot of time comparing ourselves to everyone else. Now, this is a really human thing to do, and it ’ s a very easy pit to fall into for high-achieving adults. Because, let ’ s face it, we want to know that we can at least keep up with or do better than our peers. But when we compare ourselves to other folks, it can be paralyzing, because we can fall into thinking that we are less than them. This is especially true when people start sharing grades and progress reports. We can go from feeling accomplished and on top of things to feeling behind and confused, in a matter of minutes. And the third challenge that we face is that we have a strong fear of failing. Now, for most of the folks who go back, we ’ ve got a lot on the line. We ’ re **juggling** a job and family responsibilities alongside school. Plus, we want to do well, because we ’ re usually there to up-level our career or hit a career goal in some way, to say nothing of the time and the **financial** investments involved. So being afraid of failing, well, it ’ s understandable, right? But it is an **obstacle** to getting the very thing that we want. So those are the challenges. There is good news, though. These can be overcome by making three important **mindset** shifts when we are in learning situations. The first shift is that we really need to question the stories that our brain is telling us about who we are as learners. Every day, we have a ton of thoughts, and we often don ’ t stop to examine them. But when we do, we notice that often most of them are really unhelpful. And this is especially true when we are pushing ourselves out of our comfort zone. Our brain **interprets** the challenges and discomforts that come with learning and trying new things as dangerous, and it wants to move us into what is familiar and safe. So the way that it does that is it amplifies all of our fears and cranks the volume on our inner critic, who ’ s more than happy to tell us all the ways we sucked in the past, how we ’ re not doing it now, and there ’ s no hope for the future. Our job is to notice these stories and why they ’ re happening – because we ’ re stretching – and to challenge these outdated beliefs. So when we question the truthfulness and the relevance of the stories that we ’ re carrying for who we are in our current context, and choose more accurate and helpful ones, we can change our entire experience. The second shift that we need to make is that we need to look for the lessons in the mistakes. As learners, we are going to make mistakes. We just are. Rather than making them mean something about our current or our future ability, or our self-worth, for that matter, we need to get curious and look at these setbacks as offering data that we can use to improve. And part of being curious is remembering that the mistakes are where the learning happens. And that ’ s what we need to aim for – progress, not perfection. When we release ourselves from the need to know it all, be perfect and do it all perfectly, we are much better able to evaluate our progress and actually identify the skills we need to build for the outcome that we want. And then finally, the third shift we need to make is to **practice** self-compassion, not self-criticism. Learning and pushing ourselves to our growth edge, it ’ s uncomfortable work. It ’ s not easy. And that ’ s why this shift is the most important one, because it ’ s actually the fuel for our success. When we stop criticizing ourselves and start showing ourselves the kindness and the compassion that we so easily give everyone else... we are much more likely to take risks, because we create a sense of safety inside ourselves, that it ’ s OK to make mistakes and learn from them. And when we remember that we ’ re imperfectly human, and that our fellow students are also imperfect humans, we ’ re less likely to compare ourselves to them and more likely to connect and collaborate with them instead. When we cultivate a belief inside ourselves and show ourselves compassion, we increase our ability to face challenges and learn from them. And we build the resilience and the confidence that we need to succeed in learning situations. And from that foundation, we are better able to identify the skills we have – and we do, by the way – the ones we need to hone and the ones that we need. Closing the confidence gap enables us to reach our goals, feeling accomplished and proud rather than bedraggled and depleted. My experience and my work has convinced me that when we build a positive growth **mindset** alongside our skills in learning situations, not only do we increase our success – like, intensely – we uncage our fullest potential, and our possibilities for change and growth are limitless. Thank you very much.

Text Sample 212: POS Dim 2 – 2018 – Score 12.00 – 2018/t000384.txt

When I was in the sixth grade, I got into a fight at school. It wasn ' t the first time I ' d been in a fight, but it was the first time one happened at school. It was with a boy who was about a foot taller than me, who was physically stronger than me and who ' d been taunting me for weeks. One day in PE, he stepped on my shoe and refused to apologize. So, filled with anger, I grabbed him and I threw him to the ground. I ' d had some previous judo training. Our fight lasted less than two minutes, but it was a perfect reflection of the hurricane that was building inside of me as a young survivor of sexual assault and as a girl who was grappling with abandonment and exposure to violence in other spaces in my life. I was fighting him, but I was also fighting the men and boys that had assaulted my body and the culture that told me I had to be silent about it. A teacher broke up the fight and my principal called me in her office. But she didn ' t say,"Monique, what ' s wrong with you?"She gave me a moment to collect my breath and asked,"What happened?"The educators working with me led with **empathy**. They knew me. They knew I loved to read, they knew I loved to draw, they knew I adored Prince. And they used that information to help me understand why my **actions**, and those of my classmate, were disruptive to the learning community they were leading. They didn ' t place me on suspension ; they didn ' t call the police. My fight didn ' t keep me from going to school the next day. It didn ' t keep me from graduating ; it didn ' t keep me from teaching. But unfortunately, that ' s not a story that ' s shared by many black girls in the US and around the world today. We ' re living through a crisis in which black girls are being disproportionately pushed away from schools — - not because of an imminent threat they pose to the safety of a school, but because they ' re often experiencing schools as locations for punishment and marginalization. That ' s something that I hear from black girls around the country. But it ' s not insurmountable. We can shift this narrative. Let ' s start with some data. According to a National Black Women ' s Justice Institute analysis of civil rights data collected by the US Department of Education, black girls are the only group of girls who are overrepresented along the entire continuum of discipline in schools. That doesn ' t mean that other girls aren ' t experiencing exclusionary discipline and it doesn ' t mean that other girls aren ' t overrepresented at other parts along that continuum. But black girls are the only group of girls who are overrepresented all along the way. Black girls are seven times more likely than their white counterparts to experience one or more out-of-school suspensions and they ' re nearly three times more likely than their white and Latinx counterparts to be referred to the juvenile court. A recent study by the Georgetown Center on Poverty and Inequality partially explained why this disparity is taking place when they confirmed that black girls experience a specific type of age compression, where they ' re seen as more adult-like than their white peers. Among other things, the study found that people perceive black girls to need less nurturing, less protection, to know more about sex and to be more independent than their white peers. The study also found that the perception disparity begins when girls are as young as five years old. And that this perception and the disparity increases over time and **peaks** when girls are between the ages of 10 and 14. This is not without consequence. Believing that a girl is older than she is can lead to harsher treatment, immediate censure when she makes a mistake and victim blaming when she ' s harmed. It can also lead a girl to think that something is wrong with her, rather than the conditions in which she finds herself. Black girls are routinely seen as too loud, too aggressive, too angry, too visible. Qualities that are often measured in relation to nonblack girls and which don ' t take into consideration what ' s going on in this girl ' s life or her cultural **norms**. And it ' s not just in the US. In South Africa, black girls at the Pretoria Girls High School were discouraged from attending school with their hair in its natural state, without chemical processing. What did those girls do? They protested. And it was a beautiful thing to see the global community for the most part wrap its arms around girls as they stood in their truths. But there were those who saw them as disruptive, largely because they dared to ask the question,"Where can we be black if we can ' t be black in Africa?"It ' s a good question. Around the world, black girls are grappling with this question. And around the world, black girls are **struggling** to be seen, working to be free and fighting to be included in the landscape of promise that a safe space to learn provides. In the US, little girls, just past their toddler years, are being arrested in classrooms for having a tantrum. Middle school girls are being turned away from school because of the way they wear their hair naturally or because of the way the clothes fit their bodies. High school girls are experiencing violence at the hands of police officers in schools. Where can black girls be black without reprimand or punishment? And it ' s not just these incidents. In my work as a **researcher** and educator, I ' ve had an opportunity to work with girls like Stacy, a girl who I profile in my book"Pushout,"who **struggles** with her participation in violence. She bypasses the neuroscientific and structural analyses that science has to offer about how her adverse childhood experiences inform why she ' s participating in violence and goes straight to describing herself as a"problem child,"largely because that ' s the language that educators were using as they routinely suspended her. But here ' s the thing. Disconnection and the internalization of harm grow stronger in **isolation**. So when girls get in trouble, we shouldn ' t be pushing them away, we should be bringing them in closer. Education is a critical protective factor against contact with the criminal legal system. So we should be building out policies and **practices** that keep girls connected to their learning, rather than pushing them away from it. It ' s one of the reasons I like to say that education is freedom work. When girls feel safe, they can learn. When they don ' t feel safe, they fight, they protest, they argue, they flee, they freeze. The human brain is wired to protect us when we feel a threat. And so long as school feels like a threat, or part of the tapestry of harm in a girl ' s life, she ' ll be inclined to resist. But when schools become locations for healing, they can also become locations for learning. So what does this mean for a school to become a location for healing? Well, for one thing, it means that we have to immediately discontinue the policies and **practices** that target black girls for their hairstyles or dress. Let ' s focus on how and what a girl learns rather than policing her body in ways that facilitate rape culture or punish children for the conditions in which they were born. This is where parents and the community of concerned adults can enter this work. Start a conversation with the school and encourage them to address their dress code and other conduct-related policies as a collaborative project, with parents and students, so as to intentionally avoid **bias** and discrimination. Keep in mind, though, that some of the **practices** that harm black girls most are unwritten. So we have to continue to do the deep, internal work to address the **biases** that inform how, when and whether we see black girls for who they actually are, or what we ' ve been told they are. Volunteer at a school and establish culturally competent and gender responsive discussion groups with black girls, Latinas, indigenous girls and other students who experience marginalization in schools to give them a safe space to process their identities and experiences in schools. And if schools are to become locations for healing, we have to remove police officers and increase the number of counselors in schools. Education is freedom work. And whatever our point of entry is, we all have to be freedom fighters. The good news is that there are schools that are actively working to establish themselves as locations for girls to see themselves as sacred and loved. The Columbus City Prep School for Girls in Columbus, Ohio, is an example of this. They became an example the moment their principal declared that they were no longer going to punish girls for having"a bad attitude."In addition to building — Essentially, what they did is they built out a robust continuum of alternatives to suspension, expulsion and arrest. In addition to establishing a restorative justice program, they improved their student and teacher relationships by ensuring that every girl has at least one adult on campus that she can go to when she ' s in a moment of crisis. They built out spaces along the corridors of the school and in classrooms for girls to regroup, if they need a minute to do so. And they established an advisory program that provides girls with an opportunity to start every single day with the promotion of self-worth, communication skills and goal **setting**. At this school, they ' re trying to respond to a girl ' s adverse childhood experiences rather than ignore them. They bring them in closer ; they don ' t push them away. And as a result, their truancy and suspension rates have improved, and girls are arriving at school increasingly ready to learn because they know the teachers there care about them. That matters. Schools that integrate the arts and sports into their curriculum or that are building out tranformative programming, such as restorative justice, mindfulness and meditation, are providing an opportunity for girls to repair their relationships with others, but also with themselves. Responding to the lived, complex and historical trauma that our students face requires all of us who believe in the promise of children and adolescents to build relationships, learning materials, human and **financial** resources and other tools that provide children with an opportunity to heal, so that they can learn. Our schools should be places where we respond to our most vulnerable girls as essential to the creation of a positive school culture. Our ability to see her promise should be at its sharpest when she ' s in the throws of poverty and addiction ; when she ' s reeling from having been sex-trafficked or survived other forms of violence ; when she ' s at her loudest, or her quietest. We should be able to support her intellectual and social-emotional well-being whether her shorts reach her knees or stop mid-thigh or higher. It might seem like a tall order in a world so deeply entrenched in the politics of fear to radically imagine schools as locations where girls can heal and thrive, but we have to be **bold** enough to set this as our intention. If we commit to this notion of education as freedom work, we can shift educational conditions so that no girl, even the most vulnerable among us, will get pushed out of school. And that ' s a win for all of us. Thank you.

Text Sample 213: POS Dim 2 – 2018 – Score 11.00 – 2018/t000695.txt

Who can we trust to tell us the truth? Once in another tumultuous era, as the Vietnam War, Watergate, the Civil Rights and Women ' s Liberation Movements were challenging America to its core, my dad and I would watch Walter Cronkite, a man he trusted to tell us what was happening. Dad, a politically conservative career military man, cared a lot about this country and was determined to make sure he was well informed. He listened to the radio every morning, read the paper every afternoon, and watched the news on TV every evening before sitting down to dinner, or at some point, he would almost always launch a debate about what was happening in the world. He ' d pound the table and say something like"Don ' t those Vietnam war protesters know it ' s my country, right or wrong?"And I ' d react with something like"But, Dad, don ' t they have a point? I mean, what are we doing in Vietnam?"And he ' d puff up and get red in the face and respond with something like"I can ' t believe I ' m hearing those words out of the mouth of my own daughter."Our arguments could get so loud, my brothers and sisters would sometimes pick up their plates to eat somewhere else. Dad really liked to win these arguments. But even when he didn ' t, he ' d respect my opinion. Sometimes, he would even say,"Ann, I don ' t always agree with you, but I ' d still vote for you for President."There were times, however, usually when Walter was off and someone else was substituting, when Dad would hear something on the evening news that would alarm him far more than the stories of the day. All it would take was one phrase, a changed tone of voice, a raised eyebrow, or even just a single word that struck Dad as opinionated, and he was offended. Suddenly, he was shaking his finger at the television set, shouting,"Stop telling me what to think! Just give me the facts!"And then he would look over at me and say,"See, Ann? He ' s disrespecting us, trying to tell us what to think as if we were stupid and can ' t make up our own minds."Boy, if my dad could see what counts for some of TV journalism today. He would want to be right up here with me, standing right next to me, suggesting an idea that might be worth spreading. Journalism is only trustworthy when it dependably offers accurate, verifiable facts, not **bias** and speculation, and not for ratings, circulation or clicks or for any **financial** or political motivation, but with one motive and one motive only : to reveal the truth for the greater public good. The bridge to trust is truth, and it ' s time to make things right. What do we define as truth? Well, lately I ' ve wondered if we ' ve all gotten a little confused about what truth is and what it isn ' t. People ask,"Whose truth?"uncertain if objective, absolute truth even exists, thinking, as some philosophers have argued, that as it requires human judgment, truth is ultimately subjective. As others see it, **reporters** can ' t be fair and unbiased, because being humans, of course they have opinions and so should at least tell us what those opinions are. Other people, who ' ve noticed objective facts now seem less influential than emotion and personal beliefs in shaping public opinion, have gone so far as to say we live in a post-truth era. To all this, my dad would say,"Balderdash!"He was a Navy man who didn ' t like to swear - go figure. I, however, don ' t seem to have that problem. So maybe I should just call it what it is : FUBAR - Fucked Up Beyond All Recognition. I appreciate philosophers, but not being one, I can ' t settle the"Is truth subjective or objective?"debate. Though it does seem to be an objective, verifiable fact that I ' m standing on this stage and that I ' m a little nervous, worried that I may have overdressed, and I ' m wishing that I wore more comfortable shoes. There is actually compelling evidence that humans can be fair and impartial, not just from the ancient biblical story of Solomon, but from the known science, which tell us that fairness along with **empathy** are fundamental human traits, deeply rooted in our genetics. If it wasn ’ t for these traits, we might not even be able to live amongst each other. And as a **reporter**, I know you can get in so deep, listening to and documenting all sides of the story, searching for the truth, that you don ’ t even know what your own opinion is. I know because I ’ ve done it. And because we as a species cannot live long, cannot long endure without objective facts, unelaborated and unvarnished, it can be argued they still and will always matter. We need truth. Perhaps especially from journalism, which, when **practiced** well, can help us see each other and ourselves more honestly, tell us what we need to know to be smarter citizens, show us how to live healthier and more connected lives, and even warn us when we need to keep ourselves and our loved ones safe from danger. Post-truth? Really? We do seem to be experiencing a loss of credibility, a demise in the face of a rising tide of lies, in our ability to trust what we ’ re told are objective facts are either objective or facts. Where does the information that ’ s sowing this distrust and confusion come from? And what is it doing to us and the quality of our relationships with each other? Instead of a Walter Cronkite today, i. e., someone who a significant number of people trust, we are exposed to an overwhelming number of sources of information on multiple **platforms**, working around the clock to get our attention, some motivated to purposely manipulate us with lies and propaganda, some inept, shelling out incomplete, misleading or poorly-sourced information and some doing exemplary reporting but still doubted almost as much for any mistakes or perceived **bias**. The net impact of all of this is a chaotic overload of competing, conflicting and confusing information. And while our brains are truly excellent at collecting data, we ’ re not always so great at remembering where we got it. So what we learn from a trusted source can easily get jumbled up in our brains with what we read on a website, saw in a **tweet** or a video posted on Facebook or heard from a pundit speculating on cable television. If you ever saw the David Bowie movie”The Man Who Fell to Earth”- I sometimes feel we ’ re all in that scene in which he ’ s watching a huge wall of TV monitors, each one tuned to a different channel, and suddenly, he can ’ t take it anymore, and he starts yelling,”Get out of my mind, all of you!”Bet you didn ’ t expect a David Bowie reference today. Perhaps never before have so many consumed so much news and yet have been so woefully misinformed. When, in this swirling chaos, can we be certain the information we need or want to know is accurate? The internet is an abyss as well as a wonderland. It mirrors the darkest dark in us and the brightest light of humanity and all there is in between. And because of it, no matter what comes next in journalism, there now seems no doubt that you and me and every one of us must become a better editor. So here, from one journalist ’ s playbook, are some tried and true strategies that might up your game. Focus on defense and develop a skill for listening to both sides of a story even if it pains you. Realize - Realize virtually every source wants a story told as he or she sees it. That means even seemingly good, honest people can tell you just the facts that support their point of view, sometimes leaving out key details. They might even distort the facts, purposely misdirect you or even out-and-out lie. Ask yourself,”What ’ s the motivating force for this source to tell the story?”And could that motivation be influencing the story that you ’ re getting? Then ask,”Is this source in a position, having been an eyewitness or having some expertise, to even know what is true?”Or is he or she wasting your time with speculation? And because even eyewitnesses and experts can get it wrong, you have to seek out sources from other points of view and then critically examine their motivations and credibility as well. Rarely is the truth one-sided ; it tends to be nuanced. And it ’ s not always fair. But you may never find it unless you search fairly, with a mind that is truly open. And then, even if you think you have a balanced view of the story, you have to re-examine the facts and motivations and reputations of your sources one more time to make sure you weren ’ t played or manipulated. And if your sources are anonymous, you better check and double check again. It is exhausting, but after this Sherlock Holmes- ing, you usually get the facts that people need to know to make up their own minds. This is Journalism 101, how we ’ re trained in journalism schools to work in defense of truth. Why are even long-trusted news organizations that know how to gather the facts vulnerable to attacks on their credibility? Journalism tends to be vulnerable because of the enormous power of credible information. When we believe the same story, we can find a common purpose and work together. This is what has made humans an unstoppable force throughout our history. The historian Yuval Harari suggests this is why we are the only human species to survive when once there were at least six and why we ’ ve risen from the middle of the animal kingdom, once huddled around fires in fear of being eaten, to now the top, having the power to decide what species live or die, and how we ourselves will evolve from here. This power to move people is why everyone wants to control the narrative, why truth is fought over and is the victim, usually the first victim, especially in war. It ’ s why journalists, as messengers of truth, are physically attacked in some parts of the world and even killed. At the same time, unfortunately, journalism is not like mathematics, in which two plus two is always four. Journalism is by humans, about humans and for humans and so is inherently as fallible as humans and as vulnerable to criticism. Add to that, the job is to ask prying questions, to push past the **norms** of good manners for honest answers, and then to race to report stories that can be controversial, embarrassing or infuriating. If you want to be popular, don ’ t be a journalist. Of course, it doesn ’ t help journalism ’ s credibility when it ’ s under constant attack by political leaders in the United States and around the world. It doesn ’ t help that emerging technologies have eliminated or diminished hundreds of local newspapers around the country and have failed, so far, to replace them with anything close to the same quality and range of credible reporting. But perhaps what is making journalism most vulnerable is that in this industry shift, some media executives, **struggling** to keep making money for their corporate owners, are pressuring journalists to make choices that weaken their credibility. The pressure for profits is one reason we have witnessed the stunning growth of openly **biased** news coverage, especially on cable television, where not everyone who is telling us the news is actually a journalist. Some are advocates with political agendas, some are activists, some are former political operatives, and some are what were once called”opinionaters,”people whose profession is to give their opinions and theories. So now I ’ m the one yelling at the television set :”Just give me the facts! Stop trying to tell us what to think!”Opinion, thoughtful and supported with facts, has an important place in journalism, but it is not news. Thus, allowing viewers to think it ’ s news, to fail to differentiate it, is fundamentally dishonest. News is not left or right ; it ’ s based on facts that are either right or wrong. And for the record, ranting from any point of view is not journalism. Besides, it has serious side effects : it makes you look a little nuts. In the quest for profits, corporate owners are punching holes through the long-standing wall protecting journalists from corporate meddling. In some companies, the wall is crumbling. I have heard a request from corporate executives to kill a news story. I have seen a media company ’ s advertising side try to get products placed in a news broadcast. In some companies, the wall appears to be collapsing. Recently, the nation ’ s largest broadcaster, Sinclair Broadcasting Group, which owns and operates more than 190 stations, engaged in corporate manipulation when it required dozens of anchors to read a script the company had provided. The push for profits is also pressuring journalists to muddy their judgment with worries about reaching clicks, ratings and circulation goals. This can affect the decisions they make about what stories to cover, what interviews to do and how headlines and teasers should be written. The public is not stupid. It can sense these motives in the work and the hype. It knows the truth is in trouble. How do we make things right and defend our access to accurate, verifiable reporting, which according to our nation ' s Founding Fathers, our democracy needs to survive? We need a renaissance in journalism. And it is actually possible that it is already beginning. We can see some of our nation ' s newspapers and magazines, in spite of everything, leading a rebirth, doubling down on serious, competitive and groundbreaking shoe-leather journalism, working to tell all the nuances of the stories they ' re covering, including both domestic and foreign. Our citizens are awakening to the reasons journalism is vital in a democracy. They ' re reminded by the movies"Spotlight"and"The Post,"and they ' re increasingly demanding and supporting quality work, and in some cases, they are even donating to the Committee to Protect Journalists. And journalists can be seen increasingly standing up to corporate pressures, including at the Denver Post and at Sinclair TV stations. They ' re also pushing across **platforms** to do smarter and more impactful reporting and balanced stories. And in some newsrooms, editors are reviving investigative reporting. Credibility lost is never fully regained. But what survives can be protected, and new bridges can be built by the journalists still to come. For that to happen, for news organizations to better endure attacks on their credibility, journalists must be able to work with one motive : the public good. Corporate owners and executives must know how to stay in their lane and out of influencing what and how stories are covered. Investors and advertisers will need to stop pressuring journalists to be their pot of gold and do what ' s right for this country : fund excellence and get out of the way. Editors and executive producers will have to fight harder against the blurring between opinion and news until all the stories we need to know, not just the ones or the one that might increase ratings and clicks. Technologists should realize they are not journalists ; they are **platform** creators and owners and should hire solid **reporters** and then leave them alone and make those **reporters** great forces for good. And news consumers, if you want the facts, subscribe. Read newspapers and perhaps ask, Why would journalism that ' s any good be free? Why do any one of us routinely put up with baseless opinion, speculation and propaganda? And why do we waste so much time on the Kardashians? Too much sugar just makes you fat. Journalists, keep going and stay humble. We may know a little about a lot, but few of us know enough about anything to have an opinion that matters. So we dig for the facts that do matter, from all sides, and then fight like hell to tell them. And potential future journalists, your country needs you. Journalism is war, a fight every day for truth. And it takes more than being curious and knowing how to write or blog or shoot a video or ask questions. It takes courage, and most of all, it takes integrity. Because journalism is only as good as the people who **practice** it and their willingness to stand up and fight for the stories that mean something. You will make mistakes ; we all do. But with **practice** and instincts you may not know you have, you will learn to make less of them. You will face a lack of resources, you will be denied access, people will refuse to give you interviews, editors and executive producers will give you impossible **deadlines** and not enough time or space to tell your stories. This is emotionally hard and sometimes around-the- clock work. It might cause people to hate you. It might put you in danger or expose you to trauma that might give you PTSD. This I know from experience. But after nearly 40 years at this, I still stand here before you and can honestly say with all my heart : journalism is a noble calling, a way to serve people and respond like an EMT or an emergency room doctor in a crisis with information they may need. If you work for the people and not for those who pay your checks, you will make the right choices about who to interview, what to ask and how to tell your stories, and this will bring your news organizations value that will make even your bosses happy. Be incorruptible. Do it with a pure motive, earnestly, without **bias** and expectations. Do it even though you may never know the impact of your stories. Do it even if you know you will be criticized. But get it right, which you will find is far more important than getting it first. Attribute, attribute whenever possible. Check, check, and then double check the facts. Ruthlessly edit out any word or phrase or tone in your work that could get in the way of people making up their own minds. This is how we build trust, by working to be **worthy** of it. It is time to lift truth off its knees, prize it, cherish it, defend it and watch it rise along with all of us. Thank you.

Text Sample 214: POS Dim 2 – 2018 – Score 10.00 – 2018/t000389.txt
”To do two things at once is to do neither.”It ’ s a great smackdown of multitasking, isn ’ t it, often attributed to the Roman writer Publilius Syrus, although you know how these things are, he probably never said it. What I ’ m interested in, though, is â is it true? I mean, it ’ s obviously true for emailing at the dinner table or texting while driving or possibly for live **tweeting** at TED Talk, as well. But I ’ d like to argue that for an important kind of activity, doing two things at once â or three or even four â is exactly what we should be aiming for. Look no further than Albert Einstein. In 1905, he published four remarkable scientific papers. One of them was on Brownian motion, it provided empirical evidence that atoms exist, and it laid out the basic mathematics behind most of **financial** economics. Another one was on the theory of **special** relativity. Another one was on the photoelectric effect, that ’ s why solar panels work, it ’ s a nice one. Gave him the Nobel prize for that one. And the fourth introduced an equation you might have heard of : E equals mc squared. So, tell me again how you shouldn ’ t do several things at once. Now, obviously, working simultaneously on Brownian motion, **special** relativity and the photoelectric effect â it ’ s not exactly the same kind of multitasking as Snapchatting while you ’ re watching”Westworld.”Very different. And Einstein, yeah, well, Einstein ’ s â he ’ s Einstein, he ’ s one of a kind, he ’ s unique. But the pattern of behavior that Einstein was demonstrating, that ’ s not unique at all. It ’ s very common among highly creative people, both artists and scientists, and I ’ d like to give it a name : slow-motion multitasking. Slow-motion multitasking feels like a counterintuitive idea. What I ’ m describing here is having multiple projects on the go at the same time, and you move backwards and forwards between topics as the mood takes you, or as the situation demands. But the reason it seems counterintuitive is because we ’ re used to lapsing into multitasking out of desperation. We ’ re in a hurry, we want to do everything at once. If we were willing to slow multitasking down, we might find that it works quite brilliantly. Sixty years ago, a young psychologist by the name of Bernice Eiduson began a long research project into the personalities and the working habits of 40 leading scientists. Einstein was already dead, but four of her subjects won Nobel prizes, including Linus Pauling and Richard Feynman. The research went on for decades, in fact, it continued even after professor Eiduson herself had died. And one of the questions that it answered was,”How is it that some scientists are able to go on producing important work right through their lives?”What is it about these people? Is it their personality, is it their skill set, their **daily**

routines, what? Well, a pattern that emerged was clear, and I think to some people surprising. The top scientists kept changing the subject. They would shift topics repeatedly during their first 100 published research papers. Do you want to guess how often? Three times? Five times? No. On average, the most enduringly creative scientists switched topics 43 times in their first 100 research papers. Seems that the secret to creativity is multitasking in slow motion. Eiduson ' s research suggests we need to reclaim multitasking and remind ourselves how powerful it can be. And she ' s not the only person to have found this. Different **researchers**, using different methods to study different highly creative people have found that very often they have multiple projects in progress at the same time, and they ' re also far more likely than most of us to have serious hobbies. Slow-motion multitasking among creative people is ubiquitous. So, why? I think there are three reasons. And the first is the simplest. Creativity often comes when you take an idea from its original context and you move it somewhere else. It ' s easier to think outside the box if you spend your time clambering from one box into another. For an example of this, consider the original eureka moment. Archimedes â he ' s wrestling with a difficult problem. And he realizes, in a flash, he can solve it, using the displacement of water. And if you believe the story, this idea comes to him as he ' s taking a bath, lowering himself in, and he ' s watching the water level rise and fall. And if solving a problem while having a bath isn ' t multitasking, I don ' t know what is. The second reason that multitasking can work is that learning to do one thing well can often help you do something else. Any athlete can tell you about the benefits of cross-training. It ' s possible to cross-train your mind, too. A few years ago, **researchers** took 18 randomly chosen medical students and they **enrolled** them in a course at the Philadelphia Museum of Art, where they learned to criticize and analyze works of visual art. And at the end of the course, these students were compared with a control group of their fellow medical students. And the ones who had taken the art course had become substantially better at performing tasks such as diagnosing diseases of the eye by analyzing photographs. They ' d become better eye doctors. So if we want to become better at what we do, maybe we should spend some time doing something else, even if the two fields appear to be as completely distinct as ophthalmology and the history of art. And if you ' d like an example of this, should we go for a less intimidating example than Einstein? OK. Michael Crichton, creator of"Jurassic Park"and"E. R."So in the 1970s, he originally trained as a doctor, but then he wrote novels and he directed the original"Westworld"movie. But also, and this is less well-known, he also wrote nonfiction books, about art, about medicine, about computer programming. So in 1995, he enjoyed the fruits of all this variety by penning the world ' s most commercially successful book. And the world ' s most commercially successful TV series. And the world ' s most commercially successful movie. In 1996, he did it all over again. There ' s a third reason why slow-motion multitasking can help us solve problems. It can provide assistance when we ' re stuck. This can ' t happen in an **instant**. So, imagine that feeling of working on a crossword puzzle and you can ' t figure out the answer, and the reason you can ' t is because the wrong answer is stuck in your head. It ' s very easy â just go and do something else. You know, switch topics, switch context, you ' ll forget the wrong answer and that gives the right answer space to pop into the front of your mind. But on the slower timescale that interests me, being stuck is a much more serious thing. You get turned down for funding. Your cell cultures won ' t grow, your rockets keep crashing. Nobody wants to publish you fantasy novel about a school for wizards. Or maybe you just can ' t find the solution to the problem that you ' re working on. And being stuck like that means stasis, **stress**, possibly even depression. But if you have another exciting, challenging project to work on, being stuck on one is just an opportunity to do something else. We could all get stuck sometimes, even Albert Einstein. Ten years after the original, miraculous year that I described, Einstein was putting together the pieces of his theory of general relativity, his greatest achievement. And he was exhausted. And so he turned to an easier problem. He proposed the stimulated emission of radiation. Which, as you may know, is the S in laser. So he ’ s laying down the theoretical foundation for the laser beam, and then, while he ’ s doing that, he moves back to general relativity, and he ’ s refreshed. He sees what the theory implies â that the universe isn ’ t static. It ’ s expanding. It ’ s an idea so staggering, Einstein can ’ t bring himself to believe it for years. Look, if you get stuck and you get the ball rolling on laser beams, you ’ re in pretty good shape. So, that ’ s the case for slow-motion multitasking. And I ’ m not promising that it ’ s going to turn you into Einstein. I ’ m not even promising it ’ s going to turn you into Michael Crichton. But it is a powerful way to organize our creative lives. But there ’ s a problem. How do we stop all of these projects becoming completely overwhelming? How do we keep all these ideas straight in our minds? Well, here ’ s a simple solution, a practical solution from the great American choreographer, Twyla Tharp. Over the last few decades, she ’ s blurred boundaries, mixed genres, won prizes, danced to the music of everybody, from Philip Glass to Billy Joel. She ’ s written three books. I mean, she ’ s a slow- motion multitasker, of course she is. She says,”You have to be all things. Why exclude? You have to be everything.”And Tharp ’ s method for preventing all of these different projects from becoming overwhelming is a simple one. She gives each project a big cardboard box, writes the name of the project on the side of the box. And into it, she tosses DVDs and books, magazine cuttings, theater programs, physical objects, really anything that ’ s provided a source of creative inspiration. And she writes,”The box means I never have to worry about forgetting. One of the biggest fears for a creative person is that some brilliant idea will get lost because you didn ’ t write it down and put it in a safe place. I don ’ t worry about that. Because I know where to find it. It ’ s all in the box.”You can manage many ideas like this, either in physical boxes or in their digital equivalents. So, I would like to urge you to embrace the art of slow-motion multitasking. Not because you ’ re in a hurry, but because you ’ re in no hurry at all. And I want to give you one final example, my favorite example. Charles Darwin. A man whose slow-burning multitasking is so staggering, I need a diagram to explain it all to you. We know what Darwin was doing at different times, because the creativity **researchers** Howard Gruber and Sara Davis have analyzed his diaries and his notebooks. So, when he left school, age of 18, he was initially interested in two fields, zoology and geology. Pretty soon, he signed up to be the onboard naturalist on the”Beagle.”This is the ship that eventually took five years to sail all the way around the southern oceans of the Earth, stopping at the GalĂĄpagos, passing through the Indian ocean. While he was on the”Beagle,”he began researching coral reefs. This is a great synergy between his two interests in zoology and geology, and it starts to get him thinking about slow processes. But when he gets back from the voyage, his interests start to expand even further : psychology, botany ; for the rest of his life, he ’ s moving backwards and forwards between these different fields. He never quite abandons any of them. In 1837, he begins work on two very interesting projects. One of them : earthworms. The other, a little notebook which he titles”The transmutation of species.”Then, Darwin starts studying my field, economics. He reads a book by the economist Thomas Malthus. And he has his eureka moment. In a flash, he realizes how species could emerge and evolve slowly, through this process of the survival of the fittest. It all comes to him, he writes it all down, every single important element of the theory of evolution, in that notebook. But then, a new project. His son William is born. Well, there ’ s a natural experiment right there, you get to **observe** the development of a human infant. So immediately, Darwin starts making notes. Now, of course, he ’ s still working on the theory of evolution and the development of the human infant. But during all of this, he realizes he doesn ’ t really know enough about taxonomy. So he starts studying that. And in the end, he spends eight years becoming the world ’ s leading expert on barnacles. Then,”Natural Selection.”A book that he ’ s to continue working on for his entire life, he never finishes it.”Origin of Species”is finally published 20 years after Darwin set out all the basic elements. Then, the”Descent of Man,”controversial book. And then, the book about the development of the human infant. The one that was inspired when he could see his son, William, crawling on the sitting room floor in front of him. When the book was published, William was 37 years old. And all this time, Darwin ’ s working on earthworms. He fills his billiard room with earthworms in pots, with glass covers. He shines lights on them, to see if they ’ ll respond. He holds a hot poker next to them, to see if they move away. He chews tobacco and â He blows on the earthworms to see if they have a sense of smell. He even plays the bassoon at the earthworms. I like to think of this great man when he ’ s tired, he ’ s **stressed**, he ’ s anxious about the reception of his book”The Descent of Man.”You or I might log into Facebook or turn on the television. Darwin would go into the billiard room to relax by studying the earthworms intensely. And that ’ s why it ’ s appropriate that one of his last great works is the”Formation of Vegetable Mould Through The Action of Worms.”He worked upon that book for 44 years. We don ’ t live in the 19th century anymore. I don ’ t think any of us could sit on our creative or scientific projects for 44 years. But we do have something to learn from the great slow-motion multitaskers. From Einstein and Darwin to Michael Crichton and Twyla Tharp. The modern world seems to present us with a choice. If we ’ re not going to fast-twitch from browser window to browser window, we have to live like a hermit, focus on one thing to the exclusion of everything else. I think that ’ s a false dilemma. We can make multitasking work for us, unleashing our natural creativity. We just need to slow it down. So... Make a list of your projects. Put down your phone. Pick up a couple of cardboard boxes. And get to work. Thank you very much.

Text Sample 215: POS Dim 2 – 2002 – Score 11.00 – 2002/t000410.txt

That splendid music, the coming-in music,”The Elephant March”from”Aida,”is the music I ’ ve chosen for my funeral. And you can see why. It ’ s triumphal. I won ’ t feel anything, but if I could, I would feel triumphal at having lived at all, and at having lived on this splendid planet, and having been given the opportunity to understand something about why I was here in the first place, before not being here. Can you understand my quaint English accent? Like everybody else, I was entranced yesterday by the animal session. Robert Full and Frans Lanting and others ; the beauty of the things that they showed. The only slight jarring note was when Jeffrey Katzenberg said of the mustang,”the most splendid creatures that God put on this earth.”Now of course, we know that he didn ’ t really mean that, but in this country at the moment, you can ’ t be too careful. I ’ m a biologist, and the central theorem of our subject : the theory of design, Darwin ’ s theory of evolution by natural selection. In professional circles everywhere, it ’ s of course universally accepted. In non-professional circles outside America, it ’ s largely ignored. But in non-professional circles within America, it arouses so much hostility — it ' s fair to say that American biologists are in a state of war. The war is so worrying at present, with court cases coming up in one state after another, that I felt I had to say something about it. If you want to know what I have to say about Darwinism itself, I ' m afraid you ' re going to have to look at my books, which you won ' t find in the bookstore outside. Contemporary court cases often concern an allegedly new version of creationism, called"Intelligent Design,"or ID. Don ' t be **fooled**. There ' s nothing new about ID. It ' s just creationism under another name, rechristened — I choose the word advisedly — for tactical, political reasons. The arguments of so-called ID theorists are the same old arguments that had been refuted again and again, since Darwin down to the present day. There is an effective evolution lobby coordinating the fight on behalf of science, and I try to do all I can to help them, but they get quite upset when people like me dare to mention that we happen to be atheists as well as evolutionists. They see us as rocking the boat, and you can understand why. Creationists, lacking any coherent scientific argument for their case, fall back on the popular phobia against atheism : Teach your children evolution in biology class, and they ' ll soon move on to drugs, grand larceny and sexual"pre-version."In fact, of course, educated theologians from the Pope down are firm in their support of evolution. This book,"Finding Darwin ' s God,"by Kenneth Miller, is one of the most effective attacks on Intelligent Design that I know and it ' s all the more effective because it ' s written by a devout Christian. People like Kenneth Miller could be called a"godsend"to the evolution lobby, because they expose the lie that evolutionism is, as a matter of fact, tantamount to atheism. People like me, on the other hand, rock the boat. But here, I want to say something nice about creationists. It ' s not a thing I often do, so listen carefully. I think they ' re right about one thing. I think they ' re right that evolution is fundamentally hostile to religion. I ' ve already said that many individual evolutionists, like the Pope, are also religious, but I think they ' re deluding themselves. I believe a true understanding of Darwinism is deeply corrosive to religious faith. Now, it may sound as though I ' m about to preach atheism, and I want to reassure you that that ' s not what I ' m going to do. In an audience as sophisticated as this one, that would be preaching to the choir. No, what I want to urge upon you — Instead, what I want to urge upon you is militant atheism. But that ' s putting it too negatively. If I was a person who were interested in preserving religious faith, I would be very afraid of the positive power of evolutionary science, and indeed science generally, but evolution in particular, to inspire and enthrall, precisely because it is atheistic. Now, the difficult problem for any theory of biological design is to explain the massive statistical improbability of living things. Statistical improbability in the direction of good design —"complexity"is another word for this. The standard creationist argument — there is only one ; they ' re all reduced to this one — takes off from a statistical improbability. Living creatures are too complex to have come about by chance ; therefore, they must have had a designer. This argument of course, shoots itself in the foot. Any designer capable of designing something really complex has to be even more complex himself, and that ' s before we even start on the other things he ' s expected to do, like forgive sins, bless marriages, listen to prayers — favor our side in a war — disapprove of our sex lives, and so on. Complexity is the problem that any theory of biology has to solve, and you can ' t solve it by postulating an agent that is even more complex, thereby simply compounding the problem. Darwinian natural selection is so stunningly elegant because it solves the problem of explaining complexity in terms of nothing but simplicity. Essentially, it does it by providing a smooth ramp of gradual, step-by-step increment. But here, I only want to make the point that the elegance of Darwinism is corrosive to religion, precisely because it is so elegant, so parsimonious, so powerful, so economically powerful. It has the sinewy economy of a beautiful suspension bridge. The God theory is not just a bad theory. It turns out to be — in principle — incapable of doing the job required of it. So, returning to tactics and the evolution lobby, I want to argue that rocking the boat may be just the right thing to do. My approach to attacking creationism is — unlike the evolution lobby — my approach to attacking creationism is to attack religion as a whole. And at this point I need to acknowledge the remarkable taboo against speaking ill of religion, and I ' m going to do so in the words of the late Douglas Adams, a dear friend who, if he never came to TED, certainly should have been invited. Richard Dawkins : He was. Good. I thought he must have been. He begins this speech, which was tape recorded in Cambridge shortly before he died — he begins by explaining how science works through the testing of hypotheses that are framed to be vulnerable to disproof, and then he goes on. I quote,"Religion doesn ' t seem to work like that. It has certain ideas at the heart of it, which we call ' sacred ' or ' holy. ' What it means is : here is an idea or a notion that you ' re not allowed to say anything bad about. You ' re just not. Why not? Because you ' re not.""Why should it be that it ' s perfectly legitimate to support the Republicans or Democrats, this model of economics versus that, Macintosh instead of Windows, but to have an opinion about how the universe began, about who created the universe — no, that ' s holy. So, we ' re used to not challenging religious ideas, and it ' s very interesting how much of a furor Richard creates when he does it."— He meant me, not that one."Everybody gets absolutely frantic about it, because you ' re not allowed to say these things. Yet when you look at it rationally, there ' s no reason why those ideas shouldn ' t be as open to debate as any other, except that we ' ve agreed somehow between us that they shouldn ' t be."And that ' s the end of the quote from Douglas. In my view, not only is science corrosive to religion ; religion is corrosive to science. It teaches people to be satisfied with **trivial**, supernatural non- explanations, and blinds them to the wonderful, real explanations that we have within our grasp. It teaches them to accept authority, revelation and faith, instead of always insisting on evidence. There ' s Douglas Adams, magnificent picture from his book,"Last Chance to See."Now, there ' s a typical scientific journal, The Quarterly Review of Biology. And I ' m going to put together, as guest editor, a **special** issue on the question,"Did an asteroid kill the dinosaurs?"And the first paper is a standard scientific paper, presenting evidence,"Iridium layer at the K-T boundary, and potassium argon dated crater in Yucatan, indicate that an asteroid killed the dinosaurs."Perfectly ordinary scientific paper. Now, the next one."The President of the Royal Society has been vouchsafed a strong inner conviction that an asteroid killed the dinosaurs.""It has been privately revealed to Professor Huxtane that an asteroid killed the dinosaurs.""Professor Hordley was brought up to have total and unquestioning faith"— —"that an asteroid killed the dinosaurs.""Professor Hawkins has promulgated an official dogma binding on all loyal Hawkinsians that an asteroid killed the dinosaurs."That ' s inconceivable, of course. But suppose — [Supporters of the Asteroid Theory cannot be patriotic citizens] In 1987, a **reporter** asked George Bush, Sr. whether he recognized the equal citizenship and patriotism of Americans who are atheists. Mr. Bush ' s reply has become infamous."No, I don ' t know that atheists should be considered citizens, nor should they be considered patriots. This is one nation under God."Bush ' s bigotry was not an **isolated** mistake, blurted out in the heat of the moment and later retracted. He stood by it in the face of repeated calls for clarification or withdrawal. He really meant it. More to the point, he knew it posed no threat to his election — quite the contrary. Democrats as well as Republicans parade their religiousness if they want to get elected. Both parties invoke”one nation under God.”What would Thomas Jefferson have said? [In every country and in every age, the priest has been hostile to liberty] Incidentally, I ’ m not usually very proud of being British, but you can ’ t help making the comparison. In **practice**, what is an atheist? An atheist is just somebody who feels about Yahweh the way any decent Christian feels about Thor or Baal or the golden calf. As has been said before, we are all atheists about most of the gods that humanity has ever believed in. Some of us just go one god further. And however we define atheism, it ’ s surely the kind of academic belief that a person is entitled to hold without being vilified as an unpatriotic, unelectable non-citizen. Nevertheless, it ’ s an undeniable fact that to own up to being an atheist is tantamount to introducing yourself as Mr. Hitler or Miss Beelzebub. And that all stems from the perception of atheists as some kind of weird, way-out minority. Natalie Angier wrote a rather sad piece in the New Yorker, saying how lonely she felt as an atheist. She clearly feels in a beleaguered minority. But actually, how do American atheists stack up numerically? The latest survey makes surprisingly encouraging reading. Christianity, of course, takes a massive lion ’ s share of the population, with nearly 160 million. But what would you think was the second largest group, convincingly outnumbering Jews with 2. 8 million, Muslims at 1. 1 million, Hindus, Buddhists and all other religions put together? The second largest group, with nearly 30 million, is the one described as non-religious or secular. You can ’ t help wondering why vote-seeking politicians are so proverbially overawed by the power of, for example, the Jewish lobby — the state of Israel seems to owe its very existence to the American Jewish vote — while at the same time, consigning the non-religious to political oblivion. This secular non-religious vote, if properly mobilized, is nine times as numerous as the Jewish vote. Why does this far more substantial minority not make a move to exercise its political muscle? Well, so much for quantity. How about quality? Is there any correlation, positive or negative, between intelligence and tendency to be religious? [Them folks misunderestimated me] The survey that I quoted, which is the ARIS survey, didn ’ t break down its data by socio-economic class or education, IQ or anything else. But a recent article by Paul G. Bell in the Mensa magazine provides some straws in the wind. Mensa, as you know, is an international organization for people with very high IQ. And from a meta-analysis of the literature, Bell concludes that, I quote —”Of 43 studies carried out since 1927 on the relationship between religious belief, and one ’ s intelligence or educational level, all but four found an inverse connection. That is, the higher one ’ s intelligence or educational level, the less one is likely to be religious.”Well, I haven ’ t seen the original 42 studies, and I can ’ t comment on that meta-analysis, but I would like to see more studies done along those lines. And I know that there are — if I could put a little plug here — there are people in this audience easily capable of financing a massive research survey to settle the question, and I put the suggestion up, for what it ’ s worth. But let me know show you some data that have been properly published and analyzed, on one **special** group — namely, top scientists. In 1998, Larson and Witham polled the cream of American scientists, those who ’ d been honored by election to the National Academy of Sciences, and among this select group, belief in a personal God dropped to a shattering seven percent. About 20 percent are agnostic ; the rest could fairly be called atheists. Similar figures obtained for belief in personal immortality. Among biological scientists, the figure is even lower : 5. 5 percent, only, believe in God. Physical scientists, it ’ s 7. 5 percent. I ’ ve not seen corresponding figures for elite scholars in other fields, such as history or philosophy, but I ’ d be surprised if they were different. So, we ’ ve reached a truly remarkable situation, a grotesque mismatch between the American intelligentsia and the American electorate. A philosophical opinion about the nature of the universe, which is held by the vast majority of top American scientists and probably the majority of the intelligentsia generally, is so abhorrent to the American electorate that no candidate for popular election dare affirm it in public. If I ' m right, this means that high office in the greatest country in the world is barred to the very people best qualified to hold it — the intelligentsia — unless they are prepared to lie about their beliefs. To put it bluntly : American political opportunities are heavily loaded against those who are simultaneously intelligent and honest. I ' m not a citizen of this country, so I hope it won ' t be thought unbecoming if I suggest that something needs to be done. And I ' ve already hinted what that something is. From what I ' ve seen of TED, I think this may be the ideal place to launch it. Again, I fear it will cost money. We need a consciousness-raising, coming-out campaign for American atheists. This could be similar to the campaign organized by homosexuals a few years ago, although heaven forbid that we should stoop to public outing of people against their will. In most cases, people who out themselves will help to destroy the myth that there is something wrong with atheists. On the contrary, they ' ll demonstrate that atheists are often the kinds of people who could serve as decent role models for your children, the kinds of people an advertising agent could use to recommend a product, the kinds of people who are sitting in this room. There should be a snowball effect, a positive feedback, such that the more names we have, the more we get. There could be non-linearities, threshold effects. When a critical mass has been obtained, there ' s an abrupt acceleration in recruitment. And again, it will need money. I suspect that the word"atheist"itself contains or remains a stumbling block far out of proportion to what it actually means, and a stumbling block to people who otherwise might be happy to out themselves. So, what other words might be used to smooth the path, oil the wheels, sugar the pill? Darwin himself preferred"agnostic"— and not only out of loyalty to his friend Huxley, who coined the term. Darwin said,"I have never been an atheist in the same sense of denying the existence of a God. I think that generally an ' agnostic ' would be the most correct description of my state of mind."He even became uncharacteristically tetchy with Edward Aveling. Aveling was a militant atheist who failed to persuade Darwin to accept the dedication of his book on atheism — incidentally, giving rise to a fascinating myth that Karl Marx tried to dedicate"Das Kapital"to Darwin, which he didn ' t, it was actually Edward Aveling. What happened was that Aveling ' s mistress was Marx ' s daughter, and when both Darwin and Marx were dead, Marx ' s papers became muddled up with Aveling ' s papers, and a letter from Darwin saying,"My dear sir, thank you very much but I don ' t want you to dedicate your book to me,"was mistakenly supposed to be addressed to Marx, and that gave rise to this whole myth, which you ' ve probably heard. It ' s a sort of urban myth, that Marx tried to dedicate"Kapital"to Darwin. Anyway, it was Aveling, and when they met, Darwin challenged Aveling."Why do you call yourselves atheists?""' Agnostic, '"retorted Aveling,"was simply ' atheist ' writ respectable, and ' atheist ' was simply ' agnostic ' writ aggressive."Darwin **complained**,"But why should you be so aggressive?"Darwin thought that atheism might be well and good for the intelligentsia, but that ordinary people were not, quote,"ripe for it."Which is, of course, our old friend, the"don ' t rock the boat"argument. It ' s not recorded whether Aveling told Darwin to come down off his high horse. But in any case, that was more than 100 years ago. You ' d think we might have grown up since then. Now, a friend, an intelligent lapsed Jew, who, incidentally, **observes** the Sabbath for reasons of cultural solidarity, describes himself as a"tooth-fairy agnostic."He won ' t call himself an atheist because it ' s, in principle, impossible to prove a negative, but"agnostic"on its own might suggest that God ' s existence was therefore on equal terms of likelihood as his non-existence. So, my friend is strictly agnostic about the tooth fairy, but it isn ' t very likely, is it? Like God. Hence the phrase,"tooth-fairy agnostic."Bertrand Russell made the same point using a hypothetical teapot in orbit about Mars. You would strictly have to be agnostic about whether there is a teapot in orbit about Mars, but that doesn ' t mean you treat the likelihood of its existence as on all fours with its non-existence. The list of things which we strictly have to be agnostic about doesn ' t stop at tooth fairies and teapots ; it ' s infinite. If you want to believe one particular one of them — unicorns or tooth fairies or teapots or Yahweh — the onus is on you to say why. The onus is not on the rest of us to say why not. We, who are atheists, are also a-fairyists and a-teapotists. But we don ' t bother to say so. And this is why my friend uses"tooth-fairy agnostic"as a label for what most people would call atheist. Nonetheless, if we want to attract deep-down atheists to come out publicly, we ' re going to have find something better to stick on our banner than"tooth-fairy"or"teapot agnostic."So, how about"humanist"? This has the advantage of a worldwide network of well-organized associations and journals and things already in place. My problem with it is only its apparent anthropocentrism. One of the things we ' ve learned from Darwin is that the human species is only one among millions of cousins, some close, some distant. And there are other possibilities, like"naturalist,"but that also has problems of confusion, because Darwin would have thought naturalist —"Naturalist"means, of course, as opposed to"supernaturalist"— and it is used sometimes — Darwin would have been confused by the other sense of"naturalist,"which he was, of course, and I suppose there might be others who would confuse it with"nudism". Such people might be those belonging to the British lynch mob, which last year attacked a pediatrician in mistake for a pedophile. I think the best of the available alternatives for"atheist"is simply"non- theist."It lacks the strong connotation that there ' s definitely no God, and it could therefore easily be embraced by teapot or tooth-fairy agnostics. It ' s completely compatible with the God of the physicists. When atheists like Stephen Hawking and Albert Einstein use the word"God,"they use it of course as a metaphorical shorthand for that deep, mysterious part of physics which we don ' t yet understand."Non-theist"will do for all that, yet unlike"atheist,"it doesn ' t have the same phobic, hysterical responses. But I think, actually, the alternative is to grasp the nettle of the word"atheism"itself, precisely because it is a taboo word, carrying frissons of hysterical phobia. Critical mass may be harder to achieve with the word"atheist"than with the word"non- theist,"or some other non-confrontational word. But if we did achieve it with that **dread** word"atheist"itself, the political impact would be even greater. Now, I said that if I were religious, I ' d be very afraid of evolution — I ' d go further : I would fear science in general, if properly understood. And this is because the scientific worldview is so much more exciting, more poetic, more filled with sheer wonder than anything in the poverty-stricken arsenals of the religious imagination. As Carl Sagan, another recently dead hero, put it,"How is it that hardly any major religion has looked at science and concluded, ' This is better than we thought! The universe is much bigger than our prophet said, grander, more subtle, more elegant '? Instead they say, ' No, no, no! My god is a little god, and I want him to stay that way. ' A religion, old or new, that **stressed** the magnificence of the universe as revealed by modern science, might be able to draw forth reserves of reverence and awe hardly tapped by the conventional faiths."Now, this is an elite audience, and I would therefore expect about 10 percent of you to be religious. Many of you probably subscribe to our polite cultural belief that we should respect religion. But I also suspect that a fair number of those secretly despise religion as much as I do. If you ’ re one of them, and of course many of you may not be, but if you are one of them, I ’ m asking you to stop being polite, come out, and say so. And if you happen to be rich, give some thought to ways in which you might make a difference. The religious lobby in this country is massively financed by foundations — to say nothing of all the tax benefits — by foundations, such as the Templeton Foundation and the Discovery Institute. We need an anti- Templeton to step forward. If my books sold as well as Stephen Hawking ’ s books, instead of only as well as Richard Dawkins ’ books, I ’ d do it myself. People are always going on about,”How did September the 11th change you?”Well, here ’ s how it changed me. Let ’ s all stop being so damned respectful. Thank you very much.

Text Sample 216: POS Dim 2 – 2002 – Score 9.00 – 2002/t000312.txt
It ’ s a great honor to be here with you. The good news is I ’ m very aware of my responsibilities to get you out of here because I ’ m the only thing standing between you and the bar. And the good news is I don ’ t have a prepared speech, but I have a box of slides. I have some pictures that represent my life and what I do for a living. I ’ ve learned through experience that people remember pictures long after they ’ ve forgotten words, and so I hope you ’ ll remember some of the pictures I ’ m going to share with you for just a few minutes. The whole story really starts with me as a high school kid in Pittsburgh, Pennsylvania, in a tough neighborhood that everybody gave up on for dead. And on a Wednesday afternoon, I was walking down the corridor of my high school kind of minding my own business. And there was this artist teaching, who made a great big old ceramic vessel, and I happened to be looking in the door of the art room — and if you ’ ve ever seen clay done, it ’ s magic — and I ’ d never seen anything like that before in my life. So, I walked in the art room and I said,”What is that?”And he said,”Ceramics. And who are you?”And I said,”I ’ m Bill Strickland. I want you to teach me that.”And he said,”Well, get your homeroom teacher to sign a piece of paper that says you can come here, and I ’ ll teach it to you.”And so for the remaining two years of my high school, I cut all my classes. But I had the presence of mind to give the teachers ’ classes that I cut the pottery that I made, and they gave me passing grades. And that ’ s how I got out of high school. And Mr. Ross said,”You ’ re too smart to die and I don ’ t want it on my conscience, so I ’ m leaving this school and I ’ m taking you with me.”And he drove me out to the University of Pittsburgh where I filled out a college application and got in on probation. Well, I ’ m now a trustee of the university, and at my installation ceremony I said,”I ’ m the guy who came from the neighborhood who got into the place on probation. Don ’ t give up on the poor kids, because you never know what ’ s going to happen to those children in life.”What I ’ m going to show you for a couple of minutes is a facility that I built in the toughest neighborhood in Pittsburgh with the highest crime rate. One is called Bidwell Training Center ; it is a vocational school for ex-steel workers and single parents and welfare mothers. You remember we used to make steel in Pittsburgh? Well, we don ’ t make any steel anymore, and the people who used to make the steel are having a very tough time of it. And I rebuild them and give them new life. Manchester Craftsmen ’ s Guild is named after my neighborhood. I was adopted by the Bishop of the Episcopal Diocese during the riots, and he donated a **row** house. And in that **row** house I started Manchester Craftsmen ’ s Guild, and I learned very quickly that wherever there are Episcopalians, there ’ s money in very close proximity. And the Bishop adopted me as his kid. And last year I spoke at his memorial **service** and wished him well in this life. I went out

and hired a student of Frank Lloyd Wright, the architect, and I asked him to build me a world class center in the worst neighborhood in Pittsburgh. And my building was a scale model for the Pittsburgh airport. And when you come to Pittsburgh — and you ' re all invited — you ' ll be flying into the blown-up version of my building. That ' s the building. Built in a tough neighborhood where people have been given up for dead. My view is that if you want to involve yourself in the life of people who have been given up on, you have to look like the solution and not the problem. As you can see, it has a fountain in the courtyard. And the reason it has a fountain in the courtyard is I wanted one and I had the checkbook, so I bought one and put it there. And now that I ' m giving speeches at conferences like TED, I got put on the board of the Carnegie Museum. At a reception in their courtyard, I noticed that they had a fountain because they think that the people who go to the museum deserve a fountain. Well, I think that welfare mothers and at-risk kids and ex-steel workers deserve a fountain in their life. And so the first thing that you see in my center in the springtime is water that greets you — water is life and water of human possibility — and it sets an attitude and expectation about how you feel about people before you ever give them a speech. So, from that fountain I built this building. As you can see, it has world class art, and it ' s all my taste because I raised all the money. I said to my boy,"When you raise the money, we ' ll put your taste on the wall."That we have quilts and clay and calligraphy and everywhere your eye turns, there ' s something beautiful looking back at you, that ' s deliberate. That ' s intentional. In my view, it is this kind of world that can redeem the soul of poor people. We also created a boardroom, and I hired a Japanese cabinetmaker from Kyoto, Japan, and commissioned him to do 60 pieces of furniture for our building. We have since spun him off into his own business. He ' s making a ton of money doing custom furniture for rich people. And I got 60 pieces out of it for my school because I felt that welfare moms and ex-steel workers and single parents deserved to come to a school where there was handcrafted furniture that greeted them every day. Because it sets a tone and an attitude about how you feel about people long before you give them the speech. We even have flowers in the hallway, and they ' re not plastic. Those are real and they ' re in my building every day. And now that I ' ve given lots of speeches, we had a bunch of high school principals come and see me, and they said,"Mr. Strickland, what an extraordinary story and what a great school. And we were particularly touched by the flowers and we were curious as to how the flowers got there."I said,"Well, I got in my car and I went out to the greenhouse and I bought them and I brought them back and I put them there."You don ' t need a task force or a study group to buy flowers for your kids. What you need to know is that the children and the adults deserve flowers in their life. The cost is incidental but the gesture is huge. And so in my building, which is full of sunlight and full of flowers, we believe in hope and human possibilities. That happens to be at Christmas time. And so the next thing you ' ll see is a million dollar kitchen that was built by the Heinz company — you ' ve heard of them? They did all right in the ketchup business. And I happen to know that company pretty well because John Heinz, who was our U. S. senator — who was tragically killed in a plane accident — he had heard about my desire to build a new building, because I had a cardboard box and I put it in a garbage bag and I walking all over Pittsburgh trying to raise money for this site. And he called me into his office — which is the equivalent of going to see the Wizard of Oz — and John Heinz had 600 million dollars, and at the time I had about 60 cents. And he said,"But we ' ve heard about you. We ' ve heard about your work with the kids and the ex- steel workers, and we ' re inclined to want to support your desire to build a new building. And you could do us a great **service** if you would add a culinary program to your program.”Because back then, we were building a trades program. He said,”That way we could fulfill our affirmative **action** goals for the Heinz company.”I said,”Senator, I ’ m reluctant to go into a field that I don ’ t know much about, but I promise you that if you ’ ll support my school, I ’ ll get it built and in a couple of years, I ’ ll come back and weigh out that program that you desire.”And Senator Heinz sat very quietly and he said,”Well, what would your reaction be if I said I ’ d give you a million dollars?”I said,”Senator, it appears that we ’ re going into the food training business.”And John Heinz did give me a million bucks. And most importantly, he loaned me the head of research for the Heinz company. And we kind of borrowed the curriculum from the Culinary Institute of America, which in their mind is kind of the Harvard of cooking schools, and we created a gourmet cooks program for welfare mothers in this million dollar kitchen in the middle of the inner city. And we ’ ve never looked back. I would like to show you now some of the food that these welfare mothers do in this million dollar kitchen. That happens to be our cafeteria line. That ’ s puff pastry day. Why? Because the students made puff pastry and that ’ s what the school ate every day. But the concept was that I wanted to take the stigma out of food. That good food ’ s not for rich people — good food ’ s for everybody on the planet, and there ’ s no excuse why we all can ’ t be eating it. So at my school, we subsidize a gourmet lunch program for welfare mothers in the middle of the inner city because we ’ ve discovered that it ’ s good for their stomachs, but it ’ s better for their heads. Because I wanted to let them know every day of their life that they have value at this place I call my center. We have students who sit together, black kids and white kids, and what we ’ ve discovered is you can solve the race problem by creating a world class environment, because people will have a tendency to show you world class behavior if you treat them in that way. These are examples of the food that welfare mothers are doing after six months in the training program. No sophistication, no class, no dignity, no history. What we ’ ve discovered is the only thing wrong with poor people is they don ’ t have any money, which happens to be a curable condition. It ’ s all in the way that you think about people that often determines their behavior. That was done by a student after seven months in the program, done by a very brilliant young woman who was taught by our pastry chef. I ’ ve actually eaten seven of those baskets and they ’ re very good. They have no calories. That ’ s our dining room. It looks like your average high school cafeteria in your average town in America. But this is my view of how students ought to be treated, particularly once they have been pushed aside. We train pharmaceutical technicians for the pharmacy industry, we train medical technicians for the medical industry, and we train chemical technicians for companies like Bayer and Calgon Carbon and Fisher Scientific and Exxon. And I will guarantee you that if you come to my center in Pittsburgh — and you ’ re all invited — you ’ ll see welfare mothers doing analytical chemistry with logarithmic calculators 10 months from **enrolling** in the program. There is absolutely no reason why poor people can ’ t learn world class technology. What we ’ ve discovered is you have to give them flowers and sunlight and food and expectations and Herbie ’ s music, and you can cure a spiritual cancer every time. We train corporate travel agents for the travel industry. We even teach people how to read. The kid with the red stripe was in the program two years ago — he ’ s now an instructor. And I have children with high school diplomas that they can ’ t read. And so you must ask yourself the question : how is it possible in the 21st century that we graduate children from schools who can ’ t read the diplomas that they have in their hands? The reason is that the system gets reimbursed for the kids they spit out the other end, not the children who read. I can take these children and in 20 weeks, demonstrated aptitude ; I can get them high school equivalent. No big deal. That ’ s our library with more handcrafted furniture. And this is the arts program I started in 1968. Remember I ’ m the black kid from the ’ 60s who got his life saved with ceramics. Well, I went out and decided to reproduce my experience with other kids in the neighborhood, the theory being if you get kids flowers and you give them food and you give them sunshine and enthusiasm, you can bring them right back to life. I have 400 kids from the Pittsburgh public school system that come to me every day of the week for arts education. And these are children who are flunking out of public school. And last year I put 88 percent of those kids in college and I ’ ve averaged over 80 percent for 15 years. We ’ ve made a fascinating discovery : there ’ s nothing wrong with the kids that affection and sunshine and food and enthusiasm and Herbie ’ s music can ’ t cure. For that I won a big old plaque — Man of the Year in Education. I beat out all the Ph. D. ’ s because I figured that if you treat children like human beings, it increases the likelihood they ’ re going to behave that way. And why we can ’ t institute that policy in every school and in every city and every town remains a mystery to me. Let me show you what these people do. We have ceramics and photography and computer imaging. And these are all kids with no artistic ability, no talent, no imagination. And we bring in the world ’ s greatest artists — Gordon Parks has been there, Chester Higgins has been there — and what we ’ ve learned is that the children will become like the people who teach them. In fact, I brought in a mosaic artist from the Vatican, an African-American woman who had studied the old Vatican mosaic techniques, and let me show you what they did with the work. These were children who the whole world had given up on, who were flunking out of public school, and that ’ s what they ’ re capable of doing with affection and sunlight and food and good music and confidence. We teach photography. And these are examples of some of the kids ’ work. That boy won a four-year scholarship on the strength of that photograph. This is our gallery. We have a world class gallery because we believe that poor kids need a world class gallery, so I designed this thing. We have smoked salmon at the art openings, we have a formal printed invitation, and I even have figured out a way to get their parents to come. I couldn ’ t buy a parent 15 years ago so I hired a guy who got off on the Jesus big time. He was dragging guys out of bars and saving those lives for the Lord. And I said,”Bill, I want to hire you, man. You have to tone down the Jesus stuff a little bit, but keep the enthusiasm. I can ’ t get these parents to come to the school.”He said,”I ’ ll get them to come to the school.”So, he jumped in the van, he went to Miss Jones ’ house and said,”Miss Jones, I knew you wanted to come to your kid ’ s art opening but you probably didn ’ t have a ride. So, I came to give you a ride.”And he got 10 parents and then 20 parents. At the last show that we did, 200 parents showed up and we didn ’ t pick up one parent. Because now it ’ s become socially not acceptable not to show up to support your children at the Manchester Craftsmen ’ s Guild because people think you ’ re bad parents. And there is no statistical difference between the white parents and the black parents. Mothers will go where their children are being celebrated, every time, every town, every city. I wanted you to see this gallery because it ’ s as good as it gets. And by the time I cut these kids loose from high school, they ’ ve got four shows on their resume before they apply to college because it ’ s all up here. You have to change the way that people see themselves before you can change their behavior. And it ’ s worked out pretty good up to this day. I even stuck another room on the building, which I ’ d like to show you. This is brand new. We just got this slide done in time for the TED Conference. I gave this little slide show at a place called the Silicon Valley and I did all right. And the woman came out of the audience, she said,”That was a great story and I was very impressed with your presentation. My only criticism is your computers are getting a little bit old."And I said,"Well, what do you do for a living?"She said,"Well, I work for a company called Hewlett-Packard."And I said,"You ' re in the computer business, is that right?"She said,"Yes, sir."And I said,"Well, there ' s an easy solution to that problem."Well, I ' m very pleased to announce to you that HP and a furniture company called Steelcase have adopted us as a demonstration model for all of their technology and all their furniture for the United States of America. And that ' s the room that ' s initiating the relationship. We got it just done in time to show you, so it ' s kind of the world debut of our digital imaging center. I only have a couple more slides, and this is where the story gets kind of interesting. So, I just want you to listen up for a couple more minutes and you ' ll understand why he ' s there and I ' m here. In 1986, I had the presence of mind to stick a music hall on the north end of the building while I was building it. And a guy named Dizzy Gillespie showed up to play there because he knew this man over here, Marty Ashby. And I stood on that stage with Dizzy Gillespie on sound check on a Wednesday afternoon, and I said,"Dizzy, why would you come to a black-run center in the middle of an industrial park with a high crime rate that doesn ' t even have a reputation in music?"He said,"Because I heard you built the center and I didn ' t believe that you did it, and I wanted to see for myself. And now that I have, I want to give you a gift."I said,"You ' re the gift."He said,"No, sir. You ' re the gift. And I ' m going to allow you to record the concert and I ' m going to give you the music, and if you ever choose to sell it, you must sign an agreement that says the money will come back and support the school."And I recorded Dizzy. And he died a year later, but not before telling a fellow named McCoy Tyner what we were doing. And he showed up and said,"Dizzy talking about you all over the country, man, and I want to help you."And then a guy named Wynton Marsalis showed up. Then a bass player named Ray Brown, and a fellow named Stanley Turrentine, and a piano player named Herbie Hancock, and a band called the Count Basie Orchestra, and a fellow named Tito Puente, and a guy named Gary Burton, and Shirley Horn, and Betty Carter, and Dakota Staton and Nancy Wilson all have come to this center in the middle of an industrial park to sold out audiences in the middle of the inner city. And I ' m very pleased to tell you that, with their permission, I have now accumulated 600 recordings of the greatest artists in the world, including Joe Williams, who died, but not before his last recording was done at my school. And Joe Williams came up to me and he put his hand on my shoulder and he said,"God ' s picked you, man, to do this work. And I want my music to be with you."And that worked out all right. When the Basie band came, the band got so excited about the school they voted to give me the rights to the music. And I recorded it and we won something called a Grammy. And like a **fool**, I didn ' t go to the ceremony because I didn ' t think we were going to win. Well, we did win, and our name was literally in lights over Madison Square Garden. Then the U. N. Jazz Orchestra dropped by and we recorded them and got nominated for a second Grammy back to back. So, we ' ve become one of the hot, young jazz recording studios in the United States of America in the middle of the inner city with a high crime rate. That ' s the place all filled up with Republicans. If you ' d have dropped a bomb on that room, you ' d have wiped out all the money in Pennsylvania because it was all sitting there. Including my mother and father, who lived long enough to see their kid build that building. And there ' s Dizzy, just like I told you. He was there. And he was there, Tito Puente. And Pat Metheny and Jim Hall were there and they recorded with us. And that was our first recording studio, which was the broom closet. We put the mops in the hallway and re-engineered the thing and that ' s where we recorded the first Grammy. And this is our new facility, which is all video technology. And that is a room that was built for a woman named Nancy Wilson, who recorded that album at our school last Christmas. And any of you who happened to have been watching Oprah Winfrey on Christmas Day, he was there and Nancy was there singing excerpts from this album, the rights to which she donated to our school. And I can now tell you with absolute certainty that an appearance on Oprah Winfrey will sell 10, 000 CDs. We are currently number four on the Billboard Charts, right behind Tony Bennett. And I think we ’ re going to be fine. This was burned out during the riots — this is next to my building — and so I had another cardboard box built and I walked back out in the streets again. And that ’ s the building, and that ’ s the model, and on the right ’ s a high-tech greenhouse and in the middle ’ s the medical technology building. And I ’ m very pleased to tell you that the building ’ s done. It ’ s also full of anchor tenants at 20 dollars a foot — triple that in the middle of the inner city. And there ’ s the fountain. Every building has a fountain. And the University of Pittsburgh Medical Center are anchor tenants and they took half the building, and we now train medical technicians through all their system. And Mellon Bank ’ s a tenant. And I love them because they pay the rent on time. And as a result of the association, I ’ m now a director of the Mellon Financial Corporation that bought Dreyfus. And this is in the process of being built as we speak. Multiply that picture times four and you will see the greenhouse that ’ s going to open in October this year because we ’ re going to grow those flowers in the middle of the inner city. And we ’ re going to have high school kids growing Phalaenopsis orchids in the middle of the inner city. And we have a handshake with one of the large retail grocers to sell our orchids in all 240 stores in six states. And our partners are Zuma Canyon Orchids of Malibu, California, who are Hispanic. So, the Hispanics and the black folks have formed a partnership to grow high technology orchids in the middle of the inner city. And I told my United States senator that there was a very high probability that if he could find some funding for this, we would become a left-hand column in the Wall Street Journal, to which he readily agreed. And we got the funding and we open in the fall. And you ought to come and see it — it ’ s going to be a hell of a story. And this is what I want to do when I grow up. The brown building is the one you guys have been looking at and I ’ ll tell you where I made my big mistake. I had a chance to buy this whole industrial park — which is less than 1, 000 feet from the riverfront — for four million dollars and I didn ’ t do it. And I built the first building, and guess what happened? I appreciated the real estate values beyond everybody ’ s expectations and the owners of the park turned me down for eight million dollars last year, and said,”Mr. Strickland, you ought to get the Civic Leader of the Year Award because you ’ ve appreciated our property values beyond our wildest expectations. Thank you very much for that.”The moral of the story is you must be prepared to act on your dreams, just in case they do come true. And finally, there ’ s this picture. This is in a place called San Francisco. And the reason this picture ’ s in here is I did this slide show a couple years ago at a big economics summit, and there was a fellow in the audience who came up to me. He said,”Man, that ’ s a great story. I want one of those.”I said,”Well, I ’ m very flattered. What do you do for a living?”He says,”I run the city of San Francisco. My name ’ s Willie Brown.”And so I kind of accepted the flattery and the praise and put it out of my mind. And that weekend, I was going back home and Herbie Hancock was playing our center that night — first time I ’ d met him. And he walked in and he says,”What is this?”And I said,”Herbie, this is my concept of a training center for poor people.”And he said,”As God as my witness, I ’ ve had a center like this in my mind for 25 years and you ’ ve built it. And now I really want to build one.”I said,”Well, where would you build this thing?”He said,”San Francisco.”I said,”Any chance you know Willie Brown?”As a matter of fact he did know Willie Brown, and Willie Brown and Herbie and I had dinner four years ago, and we started drawing out that center on the tablecloth. And Willie Brown said,"As sure as I ' m the mayor of San Francisco, I ' m going to build this thing as a legacy to the poor people of this city."And he got me five acres of land on San Francisco Bay and we got an architect and we got a general contractor and we got Herbie on the board, and our friends from HP, and our friends from Steelcase, and our friends from Cisco, and our friends from Wells Fargo and Genentech. And along the way, I met this real short guy at my slide show in the Silicon Valley. He came up to me afterwards, he said,"Man, that ' s a fabulous story. I want to help you."And I said,"Well, thank you very much for that. What do you do for a living?"He said,"Well, I built a company called eBay."I said,"Well, that ' s very nice. Thanks very much, and give me your card and sometime we ' ll talk."I didn ' t know eBay from that jar of water sitting on that piano, but I had the presence of mind to go back and talk to one of the techie kids at my center. I said,"Hey man, what is eBay?"He said,"Well, that ' s the electronic commerce network."I said,"Well, I met the guy who built the thing and he left me his card."So, I called him up on the phone and I said,"Mr. Skoll, I ' ve come to have a much deeper appreciation of who you are and I ' d like to become your friend."And Jeff and I did become friends, and he ' s organized a team of people and we ' re going to build this center. And I went down into the neighborhood called Bayview-Hunters Point, and I said,"The mayor sent me down here to work with you and I want to build a center with you, but I ' m not going to build you anything if you don ' t want it. And all I ' ve got is a box of slides."And so I stood up in front of 200 very angry, very disappointed people on a summer night, and the air conditioner had broken and it was 100 degrees outside, and I started showing these pictures. And after about 10 pictures they all settled down. And I ran the story and I said,"What do you think?"And in the back of the room, a woman stood up and she said,"In 35 years of living in this God forsaken place, you ' re the only person that ' s come down here and treated us with dignity. I ' m going with you, man."And she turned that audience around on a pin. And I promised these people that I was going to build this thing, and we ' re going to build it all right. And I think we can get in the ground this year as the first replication of the center in Pittsburgh. But I met a guy named Quincy Jones along the way and I showed him the box of slides. And Quincy said,"I want to help you, man. Let ' s do one in L. A."And so he ' s assembled a group of people. And I ' ve fallen in love with him, as I have with Herbie and with his music. And Quincy said,"Where did the idea for centers like this come from?"And I said,"It came from your music, man. Because Mr. Ross used to bring in your albums when I was 16 years old in the pottery class, when the world was all dark, and your music got me to the sunlight."And I said,"If I can follow that music, I ' ll get out into the sunlight and I ' ll be OK. And if that ' s not true, how did I get here?"I want you all to know that I think the world is a place that ' s worth living. I believe in you. I believe in your hopes and your dreams, I believe in your intelligence and I believe in your enthusiasm. And I ' m tired of living like this, going into town after town with people standing around on corners with holes where eyes used to be, their spirits damaged. We won ' t make it as a country unless we can turn this thing around. In Pennsylvania it costs 60, 000 dollars to keep people in jail, most of whom look like me. It ' s 40, 000 dollars to build the University of Pittsburgh Medical School. It ' s 20, 000 dollars cheaper to build a medical school than to keep people in jail. Do the math — it will never work. I am banking on you and I ' m banking on guys like Herbie and Quincy and Hackett and Richard and very decent people who still believe in something. And I want to do this in my lifetime, in every city and in every town. And I don ' t think I ' m crazy. I think we can get home on this thing and I think we can build these all over the country for less money than we ’ re spending on prisons. And I believe we can turn this whole story around to one of celebration and one of hope. In my business it ’ s very difficult work. You ’ re always fighting upstream like a salmon — never enough money, too much need — and so there is a tendency to have an occupational depression that accompanies my work. And so I ’ ve figured out, over time, the solution to the depression : you make a friend in every town and you ’ ll never be lonely. And my hope is that I ’ ve made a few here tonight. And thanks for listening to what I had to say.

Text Sample 217: POS Dim 2 – 2002 – Score 8.00 – 2002/t000095.txt
What I want to do today is spend some time talking about some stuff that ’ s giving me a little bit of existential angst, for lack of a better word, over the past couple of years. And basically, these three quotes tell what ’ s going on.”When God made the color purple, God was just showing off,”Alice Walker wrote in”The Color Purple.”And Zora Neale Hurston wrote in”Dust Tracks On A Road,””Research is a formalized curiosity. It ’ s poking and prying with a purpose.”And then finally, when I think about the near future, we have this attitude,”Well, whatever happens, happens.”Right? So that goes along with the Cheshire Cat saying,”If you don ’ t care much where you want to get to, it doesn ’ t much matter which way you go.”But I think it does matter which way we go and what road we take, because when I think about design in the near future, what I think are the most important issues, what ’ s really crucial and vital, is that we need to revitalize the arts and sciences right now, in 2002. If we describe the near future as 10, 20, 15 years from now, that means that what we do today is going to be critically important, because in the year 2015, in the year 2020, 2025, the world our society is going to be building on, the basic knowledge and abstract ideas, the discoveries that we came up with today, just as all these wonderful things we ’ re hearing about here at the TED conference that we take for granted in the world right now, were really knowledge and ideas that came up in the 50s, the 60s and the 70s. That ’ s the substrate that we ’ re exploiting today. Whether it ’ s the internet, genetic engineering, laser scanners, guided missiles, fiber optics, high-definition television, remote sensing from space and the wonderful remote-sensing photos that we see in 3D weaving, TV programs like Tracker and Enterprise, CD-rewrite drives, flat-screen, Alvin Ailey ’ s”Suite Otis,”or Sarah Jones ’ s”Your Revolution Will Not [Happen] Between These Thighs,”which, by the way, was banned by the FCC, or ska — all of these things, without question, almost without exception, are really based on ideas and abstract and creativity from years before. So we have to ask ourselves : What are we contributing to that legacy right now? And when I think about it, I ’ m really worried. To be quite frank, I ’ m concerned. I ’ m skeptical that we ’ re doing very much of anything. We ’ re, in a sense, failing to act in the future. We ’ re purposefully, consciously being laggards. We ’ re lagging behind. Frantz Fanon, who was a psychiatrist from Martinique, said,”Each generation must, out of relative obscurity, discover its **mission** and fulfill or betray it.”What is our **mission**? What do we have to do? I think our **mission** is to reconcile, to reintegrate science and the arts, because right now, there ’ s a schism that exists in popular culture. People have this idea that science and the arts are really separate ; we think of them as separate and different things. And this idea was probably introduced centuries ago, but it ’ s really becoming critical now, because we ’ re making decisions about our society every day that, if we keep thinking that the arts are separate from the

sciences, and we keep thinking it ' s cute to say,"I don ' t understand anything about this one, I don ' t understand anything about the other one,"then we ' re going to have problems. Now, I know no one here at TED thinks this. All of us, we already know that they ' re very connected. But I ' m going to let you know that some folks in the outside world, believe it or not, think it ' s neat when they say,"Scientists and science is not creative. Maybe scientists are ingenious, but they ' re not creative."And then we have this tendency, the career counselors and various people say things like,"Artists are not analytical. They ' re ingenious, perhaps, but not analytical."And when these concepts underlie our teaching and what we think about the world, then we have a problem, because we stymie support for everything. By accepting this dichotomy, whether it ' s tongue-in- cheek, when we attempt to accommodate it in our world, and we try to build our foundation for the world, we ' re messing up the future, because : Who wants to be uncreative? Who wants to be illogical? Talent would run from either of these fields if you said you had to choose either. Then they ' ll go to something where they think,"Well, I can be creative and logical at the same time."Now, I grew up in the ' 60s and I ' ll admit it — actually, my childhood spanned the ' 60s, and I was a wannabe hippie, and I always resented the fact that I wasn ' t old enough to be a hippie. And I know there are people here, the younger generation, who want to be hippies. People talk about the ' 60s all the time. And they talk about the anarchy that was there. But when I think about the ' 60s, what I took away from it was that there was hope for the future. We thought everyone could participate. There were wonderful, incredible ideas that were always percolating, and so much of what ' s cool or hot today is really based on some of those concepts, whether it ' s people trying to use the Prime Directive from Star Trek, being involved in things, or, again, that three- dimensional weaving and fax machines that I read about in my weekly readers that the technology and engineering was just getting started. But the ' 60s left me with a problem. You see, I always assumed I would go into space, because I followed all of this. But I also loved the arts and sciences. You see, when I was growing up as a little girl and as a teenager, I loved designing and making doll clothes and wanting to be a fashion designer. I took art and ceramics. I loved dance : Lola Falana, Alvin Ailey, Jerome Robbins. And I also avidly followed the Gemini and the Apollo programs. I had science projects and tons of astronomy books. I took calculus and philosophy. I wondered about infinity and the Big Bang theory. And when I was at Stanford, I found myself, my senior year, chemical engineering major, half the folks thought I was a political science and performing arts major, which was sort of true, because I was Black Student Union President, and I did major in some other things. And I found myself the last quarter **juggling** chemical engineering separation processes, logic classes, nuclear magnetic resonance spectroscopy, and also producing and choreographing a dance production. And I had to do the lighting and the design work, and I was trying to figure out : Do I go to New York City to try to become a professional dancer, or to go to medical school? Now, my mother helped me figure that one out. But when I went into space, I carried a number of things up with me. I carried a poster by Alvin Ailey — you can figure out now, I love the dance company — an Alvin Ailey poster of Judith Jamison performing the dance"Cry,"dedicated to all black women everywhere ; a Bundu statue, which was from the women ' s society in Sierra Leone ; and a certificate for the Chicago Public School students to work to improve their science and math. And folks asked me,"Why did you take up what you took up?"And I had to say,"Because it represents human creativity ; the creativity that allowed us, that we were required to have to conceive and build and launch the space shuttle, which springs from the same source as the imagination and analysis that it took to carve a Bundu statue, or the ingenuity it took to design, choreograph and stage"Cry."Each one of them are different manifestations, incarnations, of creativity — avatars of human creativity. And that ' s what we have to reconcile in our minds, how these things fit together. The difference between arts and sciences is not analytical versus intuitive. Right? E = mc2 required an intuitive leap, and then you had to do the analysis afterwards. Einstein said, in fact,"The most beautiful thing we can experience is the mysterious. It is the source of all true art and science."Dance requires us to express and want to express the jubilation in life, but then you have to figure out : Exactly what movement do I do to make sure it comes across correctly? The difference between arts and sciences is also not constructive versus deconstructive. A lot of people think of the sciences as deconstructive, you have to pull things apart. And yeah, subatomic physics is deconstructive — you literally try to tear atoms apart to understand what ' s inside of them. But sculpture, from what I understand from great sculptors, is deconstructive, because you see a piece and you remove what doesn ' t need to be there. Biotechnology is constructive. Orchestral arranging is constructive. So, in fact, we use constructive and deconstructive techniques in everything. The difference between science and the arts is not that they are different sides of the same coin, even, or even different parts of the same continuum, but rather, they ' re manifestations of the same thing. Different quantum states of an atom? Or maybe if I want to be more 21st century, I could say that they ' re different harmonic resonances of a superstring. But we ' ll leave that alone. They spring from the same source. The arts and sciences are avatars of human creativity. It ' s our attempt as humans to build an understanding of the universe, the world around us. It ' s our attempt to influence things, the universe internal to ourselves and external to us. The sciences, to me, are manifestations of our attempt to express or share our understanding, our experience, to influence the universe external to ourselves. It doesn ' t rely on us as individuals. It ' s the universe, as experienced by everyone. The arts manifest our desire, our attempt to share or influence others through experiences that are peculiar to us as individuals. Let me say it again another way : science provides an understanding of a universal experience, and arts provide a universal understanding of a personal experience. That ' s what we have to think about, that they ' re all part of us, they ' re all part of a continuum. It ' s not just the tools, it ' s not just the sciences, the mathematics and the numerical stuff and the statistics, because we heard, very much on this stage, people talked about music being mathematical. Arts don ' t just use clay, aren ' t the only ones that use clay, light and sound and movement. They use analysis as well. So people might say,"Well, I still like that intuitive versus analytical thing,"because everybody wants to do the right brain, left brain thing. We ' ve all been accused of being right-brained or left-brained at some point in time, depending on who we disagreed with. You know, people say"intuitive"— that ' s like you ' re in touch with nature, in touch with yourself and relationships ; analytical, you put your mind to work. I ' m going to tell you a little secret. You all know this, though. But sometimes people use this analysis idea, that things are outside of ourselves, to say, this is what we ' re going to elevate as the true, most important sciences, right? Then you have artists — and you all know this is true as well — artists will say things about scientists because they say they ' re too concrete, they ' re disconnected from the world. But, we ' ve even had that here on stage, so don ' t act like you don ' t know what I ' m talking about. We had folks talking about the Flat Earth Society and flower arrangers, so there ' s this whole dichotomy that we continue to carry along, even when we know better. And folks say we need to choose either-or. But it would really be foolish to choose either one, intuitive versus analytical.

That ’ s a foolish choice. It ’ s foolish just like trying to choose between being realistic or idealistic. You need both in life. Why do people do this? I ’ m going to quote a molecular biologist, Sydney Brenner, who ’ s 70 years old, so he can say this. He said,”It ’ s always important to distinguish between chastity and impotence.”Now — I want to share with you a little equation, OK? How does understanding science and the arts fit into our lives and what ’ s going on and the things we ’ re talking about here at the design conference? And this is a little thing I came up with : understanding and our resources and our will cause us to have outcomes. Our understanding is our science, our arts, our religion ; how we see the universe around us ; our resources, our money, our labor, our minerals — those things that are out there in the world we have to work with. But more importantly, there ’ s our will. This is our vision, our aspirations of the future, our hopes, our dreams, our **struggles** and our fears. Our successes and our failures influence what we do with all of those. And to me, design and engineering, craftsmanship and skilled labor, are all the things that work on this to have our outcome, which is our human quality of life. Where do we want the world to be? And guess what? Regardless of how we look at this, whether we look at arts and sciences as separate or different, they ’ re both being influenced now and they ’ re both having problems. I did a project called S. E. E. ing the Future : Science, Engineering and Education. It was looking at how to shed light on the most effective use of government funding. We got a bunch of scientists in all stages of their careers. They came to Dartmouth College, where I was teaching. And they talked about, with theologians and financiers : What are some of the issues of public funding for science and engineering research? What ’ s most important about it? There are some ideas that emerged that I think have really powerful parallels to the arts. The first thing they said was that the circumstances that we find ourselves in today in the sciences and engineering that made us world leaders are very different than the ’ 40s, the ’ 50s, and the ’ 60s and the ’ 70s, when we emerged as world leaders, because we ’ re no longer in competition with fascism, with Soviet-style communism. And by the way, that competition wasn ’ t just military ; it included social competition and political competition as well, that allowed us to look at space as one of those **platforms** to prove that our social system was better. Another thing they talked about was that the infrastructure that supports the sciences is becoming obsolete. We look at universities and colleges — small, mid-sized community colleges across the country — their laboratories are becoming obsolete. And this is where we train most of our science workers and our **researchers** — and our teachers, by the way. And there ’ s a media that doesn ’ t support the dissemination of any more than the most mundane and inane of information. There ’ s pseudoscience, crop circles, alien autopsy, haunted houses, or disasters. And that ’ s what we see. This isn ’ t really the information you need to operate in everyday life and figure out how to participate in this democracy and determine what ’ s going on. They also said there ’ s a change in the corporate mentality. Whereas government money had always been there for basic science and engineering research, we also counted on some companies to do some basic research. But what ’ s happened now is companies put more energy into short-term product development than they do in basic engineering and science research. And education is not keeping up. In K through 12, people are taking out wet labs. They think if we put a computer in the room, it ’ s going to take the place of actually mixing the acids or growing the potatoes. And government funding is decreasing in spending, and then they ’ re saying, let ’ s have corporations take over, and that ’ s not true. Government funding should at least do things like recognize cost benefits of basic science and engineering research. We have to know that we have a responsibility as global citizens in this world. We have to look at the education of humans. We need to build our resources today to make sure that they ' re trained so they understand the importance of these things. And we have to support the vitality of science. That doesn ' t mean that everything has to have one thing that ' s going to go on, or that we know exactly what ' s going to be the outcome of it, but that we support the vitality and the intellectual curiosity that goes along [with it]. And if you think about those parallels to the arts — the competition with the Bolshoi Ballet spurred the Joffrey and the New York City Ballet to become better. Infrastructure, museums, theaters, movie houses across the country are disappearing. We have more television stations with less to watch, we have more money spent on rewrites to get old television programs in the movies. We have corporate funding now that, when it goes to support the arts, it almost requires that the product be part of the picture that the artist draws. We have stadiums that are named over and over again by corporations. In Houston, we ' re trying to figure out what to do with that Enron Stadium thing. Fine arts and education in the schools is disappearing, And we have a government that seems like it ' s gutting the NEA and other programs. So we have to really stop and think : What are we trying to do with the sciences and the arts? There ' s a need to revitalize them. We have to pay attention to it. I just want to tell you quickly what I ' m doing — I want to tell you what I ' ve been doing a little bit since... I feel this need to sort of integrate some of the ideas that I ' ve had and run across over time. One of the things that I found out is that there ' s a need to repair the dichotomy between the mind and body as well. My mother always told me, you have to be observant, know what ' s going on in your mind and your body. And as a dancer, I had this tremendous faith in my ability to know my body, just as I knew how to sense colors. Then I went to medical school, and I was supposed to just go on what the machine said about bodies. You know, you would ask patients questions and some people would tell you,"Don ' t listen to what the patient said."We know that patients know and understand their bodies better, but these days we ' re trying to divorce them from that idea. We have to reconcile the patient ' s knowledge of their body with physicians ' measurements. We had someone talk about measuring emotions and getting machines to figure out what to keep us from acting crazy. No, we shouldn ' t measure. We shouldn ' t use machines to measure road rage and then do something to keep us from engaging in it. Maybe we can have machines help us to recognize that we have road rage, and then we need to know how to control that without the machines. We even need to be able to recognize that without the machines. What I ' m very concerned about is : How do we bolster our self-awareness as humans, as biological organisms? Michael Moschen spoke of having to teach and learn how to feel with my eyes, to see with my hands. We have all kinds of possibilities to use our senses by, and that ' s what we have to do. That ' s what I want to do — to try to use bioinstrumentation, those kind of things, to help our senses in what we do. That ' s the work I ' ve been doing now, as a company called BioSentient Corporation. I figured I ' d have to do that **ad**, because I ' m an entrepreneur, and"entrepreneur"says"somebody who does what they want to do, because they ' re not broke enough that they have to get a real job."But that ' s the work I ' m doing, BioSentient Corporation, trying to figure out : How do we integrate these things? Let me finish by saying that my personal design issue for the future is really about integrating ; to think about that intuitive and that analytical. The arts and sciences are not separate. High school physics lesson before you leave : high school physics teacher used to hold up a ball. She would say,"This ball has potential energy. But nothing will happen to it, it can ' t do any work, until I drop it and it changes states."I like to think of ideas as potential energy. They ' re really wonderful, but nothing will happen until we risk putting them into **action**. This conference is filled with wonderful ideas. We ’ re going to share lots of things with people. But nothing ’ s going to happen until we risk putting those ideas into **action**. We need to revitalize the arts and sciences today. We need to take responsibility for the future. We can ’ t hide behind saying it ’ s just for company profits, or it ’ s just a business, or I ’ m an artist or an academician. Here ’ s how you **judge** what you ’ re doing : I talked about that balance between intuitive, analytical. Fran Lebowitz, my favorite cynic, said,”The three questions of greatest concern...”— now I ’ m going to add on to design —”... are : Is it attractive?”That ’ s the intuitive.”Is it amusing?”— the analytical, and,”Does it know its place?”— the balance. Thank you very much.

Text Sample 218: POS Dim 2 – 2015 – Score 11.00 – 2015/t001607.txt

You ’ re looking at a woman who was publicly silent for a decade. Obviously, that ’ s changed, but only recently. It was several months ago that I gave my very first major public talk, at the Forbes”30 Under 30 Summit”â□□ 1, 500 brilliant people, all under the age of 30. That meant that in 1998, the oldest among the group were only 14, and the youngest, just four. I joked with them that some might only have heard of me from rap songs. Yes, I ’ m in rap songs. Almost 40 rap songs. But the night of my speech, a surprising thing happened. At the age of 41, I was hit on by a 27-year-old guy. I know, right? He was charming, and I was flattered, and I declined. You know what his unsuccessful pickup line was? He could make me feel 22 again. I realized, later that night, I ’ m probably the only person over 40 who does not want to be 22 again. At the age of 22, I fell in love with my boss. And at the age of 24, I learned the devastating consequences. Can I see a show of hands of anyone here who didn ’ t make a mistake or do something they regretted at 22? Yep. That ’ s what I thought. So like me, at 22, a few of you may have also taken wrong turns and fallen in love with the wrong person, maybe even your boss. Unlike me, though, your boss probably wasn ’ t the president of the United States of America. Of course, life is full of surprises. Not a day goes by that I ’ m not reminded of my mistake, and I regret that mistake deeply. In 1998, after having been swept up into an improbable romance, I was then swept up into the eye of a political, legal and media maelstrom like we had never seen before. Remember, just a few years earlier, news was consumed from just three places : reading a newspaper or magazine, listening to the radio or watching television. That was it. But that wasn ’ t my fate. Instead, this scandal was brought to you by the digital revolution. That meant we could access all the information we wanted, when we wanted it, anytime, anywhere. And when the story broke in January 1998, it broke online. It was the first time the traditional news was usurped by the internet for a major news story â□□ a click that reverberated around the world. What that meant for me personally was that overnight, I went from being a completely private figure to a publicly humiliated one, worldwide. I was patient zero of losing a personal reputation on a global scale almost instantaneously. This rush to judgment, enabled by technology, led to mobs of virtual stone- throwers. Granted, it was before social media, but people could still comment online, e-mail stories, and, of course, e-mail cruel jokes. News sources plastered photos of me all over to sell newspapers, banner **ads** online, and to keep people tuned to the TV. Do you recall a particular image of me, say, wearing a beret? Now, I admit I made mistakes â□□ especially wearing that beret. But the attention and judgment that I received â□□ not the story, but that I personally received â□□ was unprecedented. I was branded as a tramp, tart, slut, whore, bimbo, and, of course,”that woman.”I was seen by many, but actually known by few. And I get it : it was easy to forget that that woman was dimensional, had a soul and was once unbroken. When this happened to me 17 years ago, there was no name for it. Now we call it”cyberbullying”and”online harassment.”Today, I want to share some of my experience with you, talk about how that experience has helped shape my cultural observations, and how I hope my past experience can lead to a change that results in less suffering for others. In 1998, I lost my reputation and my dignity. I lost almost everything. And I almost lost my life. Let me paint a picture for you. It is September of 1998. I ’ m sitting in a windowless office room inside the Office of the Independent Counsel, underneath humming fluorescent lights. I ’ m listening to the sound of my voice, my voice on surreptitiously taped phone calls that a supposed friend had made the year before. I ’ m here because I ’ ve been legally required to personally authenticate all 20 hours of taped conversation. For the past eight months, the mysterious content of these tapes has hung like the sword of Damocles over my head. I mean, who can remember what they said a year ago? Scared and mortified, I listen, listen as I prattle on about the flotsam and jetsam of the day ; listen as I confess my love for the president, and, of course, my heartbreak ; listen to my sometimes catty, sometimes churlish, sometimes silly self being cruel, unforgiving, uncouth ; listen, deeply, deeply ashamed, to the worst version of myself, a self I don ’ t even recognize. A few days later, the Starr Report is released to Congress, and all of those tapes and transcripts, those stolen words, form a part of it. That people can read the transcripts is horrific enough. But a few weeks later, the audiotapes are aired on TV, and significant portions made available online. The public humiliation was excruciating. Life was almost unbearable. This was not something that happened with regularity back then in 1998, and by”this,”I mean the stealing of people ’ s private words, **actions**, conversations or photos, and then making them public â public without consent, public without context and public without compassion. **Fast-forward** 12 years, to 2010, and now social media has been born. The landscape has sadly become much more populated with instances like mine, whether or not someone actually made a mistake, and now, it ’ s for both public and private people. The consequences for some have become dire, very dire. I was on the phone with my mom in September of 2010, and we were talking about the news of a young college freshman from Rutgers University, named Tyler Clementi. Sweet, sensitive, creative Tyler was secretly webcammed by his roommate while being intimate with another man. When the online world learned of this incident, the ridicule and cyberbullying ignited. A few days later, Tyler jumped from the George Washington Bridge to his death. He was 18. My mom was beside herself about what happened to Tyler and his family, and she was gutted with pain in a way that I just couldn ’ t quite understand. And then eventually, I realized she was reliving 1998, reliving a time when she sat by my bed every night, reliving â sorry â reliving a time when she made me shower with the bathroom door open, and reliving a time when both of my parents feared that I would be humiliated to death, literally. Today, too many parents haven ’ t had the chance to step in and rescue their loved ones. Too many have learned of their child ’ s suffering and humiliation after it was too late. Tyler ’ s tragic, senseless death was a turning point for me. It served to recontextualize my experiences, and I then began to look at the world of humiliation and bullying around me and see something different. In 1998, we had no way of knowing where this brave new technology called the internet would take us. Since then, it has connected people in unimaginable ways â joining lost siblings, saving lives, launching revolutions... But the darkness, cyberbullying, and slut-shaming that I experienced had mushroomed. Every day online, people â especially young people, who are not developmentally equipped to handle this â are so abused and humiliated that they can ’ t imagine living to the next day. And some, tragically, don ’ t. And there ’ s nothing virtual about that. Childline, a UK nonprofit that ’ s focused on helping young people on various issues, released a staggering statistic late last year : from 2012 to 2013, there was an 87 percent increase in calls and e-mails related to cyber-bullying. A meta-analysis done out of the Netherlands showed that for the first time, cyberbullying was leading to suicidal ideations more significantly than offline bullying. And you know, what shocked me â although it shouldn ’ t have â was other research last year that determined humiliation was a more intensely felt emotion than either **happiness** or even anger. Cruelty to others is nothing new. But online, technologically enhanced shaming is amplified, uncontained and permanently accessible. The echo of embarrassment used to extend only as far as your family, village, school or community. But now, it ’ s the online community too. Millions of people, often anonymously, can stab you with their words, and that ’ s a lot of pain. And there are no perimeters around how many people can publicly **observe** you and put you in a public stockade. There is a very personal price to public humiliation, and the growth of the internet has jacked up that price. For nearly two decades now, we have slowly been sowing the seeds of shame and public humiliation in our cultural soil, both on- and offline. Gossip websites, paparazzi, reality programming, politics, news outlets and sometimes hackers all traffic in shame. It ’ s led to desensitization and a permissive environment online, which lends itself to trolling, invasion of **privacy** and cyberbullying. This shift has created what Professor Nicolaus Mills calls”a culture of humiliation.”Consider a few prominent examples just from the past six months alone. Snapchat, the **service** which is used mainly by younger generations and claims that its messages only have the life span of a few seconds. You can imagine the range of content that that gets. A third-party app which Snapchatters use to preserve the life span of the messages was hacked, and 100, 000 personal conversations, photos and videos were leaked online, to now have a life span of forever. Jennifer Lawrence and several other actors had their iCloud accounts hacked, and private, intimate, nude photos were plastered across the internet without their permission. One gossip website had over five million hits for this one story. And what about the Sony Pictures cyberhacking? The documents which received the most attention were private e-mails that had maximum public embarrassment value. But in this culture of humiliation, there is another kind of price tag attached to public shaming. The price does not measure the cost to the victim, which Tyler and too many others â notably, women, minorities and members of the LGBTQ community â have paid, but the price measures the profit of those who prey on them. This invasion of others is a raw material, efficiently and ruthlessly mined, packaged and sold at a profit. A marketplace has emerged where public humiliation is a commodity, and shame is an industry. How is the money made? Clicks. The more shame, the more clicks. The more clicks, the more advertising dollars. We ’ re in a dangerous cycle. The more we click on this kind of gossip, the more numb we get to the human lives behind it. And the more numb we get, the more we click. All the while, someone is making money off of the back of someone else ’ s suffering. With every click, we make a choice. The more we saturate our culture with public shaming, the more accepted it is, the more we will see behavior like cyberbullying, trolling, some forms of hacking and online harassment. Why? Because they all have humiliation at their cores. This behavior is a symptom of the culture we ’ ve created. Just think about it. Changing behavior begins with evolving beliefs. We ’ ve seen that to be true with racism, homophobia and plenty of other **biases**, today and in the past. As we ’ ve changed beliefs about same-sex marriage, more people have been offered equal freedoms. When we began valuing sustainability, more people began to recycle. So as far as our culture of humiliation goes, what we need is a cultural revolution. Public shaming as a blood sport has to stop, and it ’ s time for an intervention on the internet and in our culture. The shift begins with something simple, but it ’ s not easy. We need to return to a long-held value of compassion, compassion and **empathy**. Online, we ’ ve got a compassion deficit, an **empathy** crisis. **Researcher** BrenĂŠ Brown said, and I quote,”Shame can ’ t survive **empathy**.”Shame cannot survive **empathy**. I ’ ve seen some very dark days in my life. It was the compassion and **empathy** from my family, friends, professionals and sometimes even strangers that saved me. Even **empathy** from one person can make a difference. The theory of minority influence, proposed by social psychologist Serge Moscovici, says that even in small numbers, when there ’ s consistency over time, change can happen. In the online world, we can foster minority influence by becoming upstanders. To become an upstander means instead of bystander apathy, we can post a positive comment for someone or report a bullying situation. Trust me, compassionate comments help abate the negativity. We can also counteract the culture by supporting organizations that deal with these kinds of issues, like the Tyler Clementi Foundation in the US ; in the UK, there ’ s Anti-Bullying Pro ; and in Australia, there ’ s PROJECT ROCKIT. We talk a lot about our right to freedom of **expression**. But we need to talk more about our responsibility to freedom of **expression**. We all want to be heard, but let ’ s acknowledge the difference between speaking up with intention and speaking up for attention. The internet is the superhighway for the id. But online, showing **empathy** to others benefits us all and helps create a safer and better world. We need to communicate online with compassion, consume news with compassion and click with compassion. Just imagine walking a mile in someone else ’ s headline. I ’ d like to end on a personal note. In the past nine months, the question I ’ ve been asked the most is”Why?”Why now? Why was I sticking my head above the parapet? You can read between the lines in those questions, and the answer has nothing to do with politics. The top-note answer was and is”Because it ’ s time.”Time to stop tiptoeing around my past, time to stop living a life of opprobrium and time to take back my narrative. It ’ s also not just about saving myself. Anyone who is suffering from shame and public humiliation needs to know one thing : You can survive it. I know it ’ s hard. It may not be painless, quick or easy, but you can insist on a different ending to your story. Have compassion for yourself. We all deserve compassion and to live both online and off in a more compassionate world. Thank you for listening.

Text Sample 219: POS Dim 2 – 2015 – Score 9.00 – 2015/t001483.txt

A year ago, we were invited by the Swiss Embassy in Berlin to present our art projects. We are used to invitations, but this invitation really thrilled us. The Swiss Embassy in Berlin is **special**. It is the only old building in the government district that was not destroyed during the Second World War, and it sits right next to the Federal Chancellery. No one is closer to Chancellor Merkel than the Swiss diplomats. The government district in Berlin also contains the Reichstag — Germany ’ s parliament — and the Brandenburg Gate, and right next to the gate there are other embassies, in particular the US and the British Embassy. Although Germany is an advanced democracy, citizens are limited in their constitutional rights in its government district. The right of assembly and the right to demonstrate are restricted there. And this is interesting from an artistic point of view. The opportunities to exercise participation and to express oneself are always bound to a certain order and always subject to a specific regulation. With an awareness of the dependencies of these regulations, we can gain a new perspective. The given terms and conditions shape our perception, our **actions** and our lives. And this is crucial in another context. Over the last couple of years, we learned that from the roofs of the US and the British Embassy, the secret **services** have been listening to the entire district, including the **mobile** phone of Angela Merkel. The antennas of the British GCHQ are hidden in a white cylindrical radome, while the listening post of the American NSA is covered by radio transparent screens. But how to address these hidden and disguised forces? With my colleague, Christoph Wachter, we accepted the invitation of the Swiss Embassy. And we used this opportunity to exploit the specific situation. If people are spying on us, it stands to reason that they have to listen to what we are saying. On the roof of the Swiss Embassy, we installed a series of antennas. They weren ’ t as sophisticated as those used by the Americans and the British. They were makeshift can antennas, not camouflaged but totally obvious and visible. The Academy of Arts joined the project, and so we built another large antenna on their rooftop, exactly between the listening posts of the NSA and the GCHQ. Never have we been **observed** in such detail while building an art installation. A helicopter circled over our heads with a camera registering each and every move we made, and on the roof of the US Embassy, security officers patrolled. Although the government district is governed by a strict police order, there are no specific laws relating to digital communication. Our installation was therefore perfectly legal, and the Swiss Ambassador informed Chancellor Merkel about it. We named the project”Can You Hear Me?”The antennas created an open and free Wi-Fi communication network in which anyone who wanted to would be able to participate using any Wi-Fi-enabled device without any hindrance, and be able to send messages to those listening on the frequencies that were being intercepted. Text messages, voice chat, file sharing — anything could be sent anonymously. And people did communicate. Over 15, 000 messages were sent. Here are some examples.”Hello world, hello Berlin, hello NSA, hello GCHQ.””NSA Agents, Do the Right Thing! Blow the whistle!””This is the NSA. In God we trust. All others we track!!!!!””#@nonymous is watching #NSA #GCHQ - we are part of your organizations. # expect us. We will #shutdown””This is the NSA ’ s Achilles heel. Open Networks.””Agents, what twisted story of yourself will you tell your grandchildren?””@NSA My neighbors are noisy. Please send a drone strike.””Make Love, Not cyberwar.”We invited the embassies and the government departments to participate in the open network, too, and to our surprise, they did. Files appeared on the network, including classified documents leaked from the parliamentary investigation commission, which highlights that the free exchange and discussion of vital information is starting to become difficult, even for members of a parliament. We also organized guided tours to experience and sound out the power constellations on-site. The tours visited the restricted zones around the embassies, and we discussed the potential and the highlights of communication. If we become aware of the constellation, the terms and conditions of communication, it not only broadens our horizon, it allows us to look behind the regulations that limit our worldview, our specific social, political or aesthetic conventions. Let ’ s look at an actual example. The fate of people living in the makeshift settlements on the outskirts of Paris is hidden and faded from view. It ’ s a vicious circle. It ’ s not poverty, not racism, not exclusion that are new. What is new is how these realities are hidden and how people are made invisible in an age of global and overwhelming communication and exchange. Such makeshift settlements are considered illegal, and therefore those living in them don ’ t have a chance of making their voices heard. On the contrary, every time they appear, every time they risk becoming visible, merely gives grounds for further persecution, expulsion and suppression. What interested us was how we could come to know this hidden side. We were searching for an interface and we found one. It ’ s not a digital interface, but a physical one : it ’ s a hotel. We named the project”Hotel Gelem.”Together with Roma families, we created several Hotel Gelems in Europe, for example, in Freiburg in Germany, in Montreuil near Paris, and also in the Balkans. These are real hotels. People can stay there. But they aren ’ t a commercial enterprise. They are a symbol. You can go online and ask for a personal invitation to come and live for a few days in the Hotel Gelem, in their homes, eating, working and living with the Roma families. Here, the Roma families are not the travelers ; the visitors are. Here, the Roma families are not a minority ; the visitors are. The point is not to make judgments, but rather to find out about the context that determines these disparate and seemingly insurmountable contradictions. In the world of globalization, the continents are drifting closer to each other. Cultures, goods and people are in permanent exchange, but at the same time, the gap between the world of the privileged and the world of the excluded is growing. We were recently in Australia. For us, it was no problem to enter the country. We have European passports, visas and air tickets. But asylum seekers who arrive by boat in Australia are deported or taken to prison. The interception of the boats and the disappearance of the people into the detention system are veiled by the Australian authorities. These procedures are declared to be secret military operations. After dramatic escapes from crisis zones and war zones, men, women and children are detained by Australia without **trial**, sometimes for years. During our stay, however, we managed to reach out and work with asylum seekers who were imprisoned, despite strict screening and **isolation**. From these contexts was born an installation in the art space of the Queensland University of Technology in Brisbane. On the face of it, it was a very simple installation. On the floor, a stylized compass gave the direction to each immigration detention center, accompanied by the distance and the name of the immigration facility. But the exhibition step came in the form of connectivity. Above every floor marking, there was a headset. Visitors were offered the opportunity to talk directly to a refugee who was or had been imprisoned in a specific detention facility and engage in a personal conversation. In the protected context of the art exhibition, asylum seekers felt free to talk about themselves, their story and their situation, without fear of consequences. Visitors immersed themselves in long conversations about families torn apart, about dramatic escapes from war zones, about suicide attempts, about the fate of children in detention. Emotions ran deep. Many wept. Several revisited the exhibition. It was a powerful experience. Europe is now facing a stream of migrants. The situation for the asylum seekers is made worse by contradictory policies and the temptation of militarized responses. We have also established communication systems in remote refugee centers in Switzerland and Greece. They are all about providing basic information — weather forecasts, legal information, guidance. But they are significant. Information on the Internet that could ensure survival along dangerous routes is being censored, and the provision of such information is becoming increasingly criminalized. This brings us back to our network and to the antennas on the roof of the Swiss Embassy in Berlin and the”Can You Hear Me?”project. We should not take it for granted to be boundlessly connected. We should start making our own connections, fighting for this idea of an equal and globally interconnected world. This is essential to overcome our speechlessness and the separation provoked by rival political forces. It is only in truly exposing ourselves to the transformative power of this experience that we can overcome prejudice and exclusion. Thank you. Bruno Giussani : Thank you, Mathias. The other half of your artistic duo is also here. Christoph Wachter, come onstage. First, tell me just a detail : the name of the hotel is not a random name. Gelem means something specific in the Roma language. Mathias Jud : Yes,”Gelem, Gelem”is the title of the Romani hymn, the official, and it means”I went a long way.”BG : That ’ s just to add the detail to your talk. But you two traveled to the island of Lesbos very recently, you ’ re just back a couple of days ago, in Greece, where thousands of refugees are arriving and have been arriving over the last few months. What did you see there and what did you do there? Christoph Wachter : Well, Lesbos is one of the Greek islands close to Turkey, and during our stay, many asylum seekers arrived by boat on overcrowded dinghies, and after landing, they were left completely on their own. They are denied many **services**. For example, they are not allowed to buy a bus ticket or to rent a hotel room, so many families literally sleep in the streets. And we installed networks there to allow basic communication, because I think, I believe, it ’ s not only that we have to speak about the refugees, I think we need to start talking to them. And by doing so, we can realize that it is about human beings, about their lives and their **struggle** to survive. BG : And allow them to talk as well. Christoph, thank you for coming to TED. Mathias, thank you for coming to TED and sharing your story.

Text Sample 220: POS Dim 2 – 2015 – Score 9.00 – 2015/t001243.txt

You are a high-ranking military **service** member deployed to Afghanistan. You are responsible for the lives of hundreds of men and women, and your base is under attack. Incoming mortar rounds are exploding all around you. **Struggling** to see through the dust and the smoke, you do your best to assist the wounded and then crawl to a **nearby** bunker. Conscious but dazed by the blasts, you lay on your side and attempt to process what has just happened. As you regain your vision, you see a bloody face staring back at you. The image is terrifying, but you quickly come to understand it ’ s not real. This vision continues to visit you multiple times a day and in your sleep. You choose not to tell anyone for fear of losing your job or being seen as weak. You give the vision a name, Bloody Face in Bunker, and call it BFIB for short. You keep BFIB locked away in your mind, secretly haunting you, for the next seven years. Now close your eyes. Can you see BFIB? If you can, you ’ re beginning to see the face of the invisible wounds of war, commonly known as post-traumatic **stress** disorder and traumatic brain injury. While I can ’ t say I have post-traumatic **stress** disorder, I ’ ve never been a stranger to it. When I was a little girl, I would visit my grandparents every summer. It was my grandfather who introduced me to the effects of combat on the psyche. While my grandfather was serving as a Marine in the Korean War, a bullet pierced his neck and rendered him unable to cry out. He watched as a corpsman passed him over, declaring him a goner, and then leaving him to die. Years later, after his physical wounds had healed and he ’ d returned home, he rarely spoke of his experiences in waking life. But at night I would hear him shouting obscenities from his room down the hall. And during the day I would announce myself as I entered the room, careful not to startle or agitate him. He lived out the remainder of his days **isolated** and tight-lipped, never finding a way to express himself, and I didn ’ t yet have the tools to guide him. I wouldn ’ t have a name for my grandfather ’ s condition until I was in my 20s. Seeking a graduate degree in art therapy, I naturally gravitated towards the study of trauma. And while sitting in class learning about post-traumatic **stress** disorder, or PTSD for short, my **mission** to help **service** members who suffered like my grandfather began to take form. We ' ve had various names for post-traumatic **stress** throughout the history of war : homesickness, soldier ' s heart, shell shock, thousand-yard stare, for instance. And while I was pursuing my degree, a new war was raging, and thanks to modern body armor and military vehicles, **service** members were surviving blast injuries they wouldn ' t have before. But the invisible wounds were reaching new levels, and this pushed military doctors and **researchers** to try and truly understand the effects that traumatic brain injury, or TBI, and PTSD have on the brain. Due to advances in technology and neuroimaging, we now know there ' s an actual shutdown in the Broca ' s, or the speech-language area of the brain, after an individual experiences trauma. This physiological change, or speechless terror as it ' s often called, coupled with mental health stigma, the fear of being **judged** or misunderstood, possibly even removed from their current duties, has led to the invisible **struggles** of our servicemen and women. Generation after generation of veterans have chosen not to talk about their experiences, and suffer in solitude. I had my work cut out for me when I got my first job as an art therapist at the nation ' s largest military medical center, Walter Reed. After working for a few years on a locked-in patient psychiatric unit, I eventually transferred to the National Intrepid Center of Excellence, NICoE, which leads TBI care for active duty **service** members. Now, I believed in art therapy, but I was going to have to convince **service** members, big, tough, strong, manly military men, and some women too, to give art-making as a psychotherapeutic intervention a try. The results have been nothing short of spectacular. Vivid, symbolic artwork is being created by our servicemen and women, and every work of art tells a story. We ' ve **observed** that the process of art therapy bypasses the speech-language issue with the brain. Art-making accesses the same sensory areas of the brain that encode trauma. **Service** members can use the art-making to work through their experiences in a nonthreatening way. They can then apply words to their physical creations, reintegrating the left and the right hemispheres of the brain. Now, we ' ve seen this can work with all forms of art — drawing, painting, collage — but what seems to have the most impact is mask-making. Finally, these invisible wounds don ' t just have a name, they have a face. And when **service** members create these masks, it allows them to come to grips, literally, with their trauma. And it ' s amazing how often that enables them to break through the trauma and start to heal. Remember BFIB? That was a real experience for one of my patients, and when he created his mask, he was able to let go of that haunting image. Initially, it was a daunting process for the **service** member, but eventually he began to think of BFIB as the mask, not his internal wound, and he would go to leave each session, he would hand me the mask, and say,"Melissa, take care of him."Eventually, we placed BFIB in a box to further contain him, and when the **service** member went to leave the NICoE, he chose to leave BFIB behind. A year later, he had only seen BFIB twice, and both times BFIB was smiling and the **service** member didn ' t feel anxious. Now, whenever that **service** member is haunted by some traumatic memory, he continues to paint. Every time he paints these disturbing images, he sees them less or not at all. Philosophers have told us for thousands of years that the power to create is very closely linked to the power to destroy. Now science is showing us that the part of the brain that registers a traumatic wound can be the part of the brain where healing happens too. And art therapy is showing us how to make that connection. We asked one of our **service** members to describe how mask-making impacted his treatment, and this is what he had to say. Service Member : You sort of just zone out into the mask. You zone out into the drawing, and for me, it just released the block, so I was able to do it. And then when I looked at it after two days, I was like,"Holy crap, here ' s the picture, here ' s the key, here ' s the puzzle,"and then from there it just soared. I mean, from there my treatment just when out of sight, because they were like, Kurt, explain this, explain this. And for the first time in 23 years, I could actually talk about stuff openly to, like, anybody. I could talk to you about it right now if I wanted to, because it unlocked it. It ' s just amazing. And it allowed me to put 23 years of PTSD and TBI stuff together in one place that has never happened before. Sorry. Melissa Walker : Over the past five years, we ' ve had over 1, 000 masks made. It ' s pretty amazing, isn ' t it? Thank you. I wish I could have shared this process with my grandfather, but I know that he would be thrilled that we are finding ways to help today ' s and tomorrow ' s **service** members heal, and finding the resources within them that they can call upon to heal themselves. Thank you.

Text Sample 221: POS Dim 2 – 2014 – Score 12.00 – 2014/t001880.txt

Chris Anderson : The rights of citizens, the future of the Internet. So I would like to welcome to the TED stage the man behind those revelations, Ed Snowden. Ed is in a remote location somewhere in Russia controlling this bot from his laptop, so he can see what the bot can see. Ed, welcome to the TED stage. What can you see, as a matter of fact? Edward Snowden : Ha, I can see everyone. This is amazing. CA : Ed, some questions for you. You ' ve been called many things in the last few months. You ' ve been called a whistleblower, a traitor, a hero. What words would you describe yourself with? ES : You know, everybody who is involved with this debate has been **struggling** over me and my personality and how to describe me. But when I think about it, this isn ' t the question that we should be **struggling** with. Who I am really doesn ' t matter at all. If I ' m the worst person in the world, you can hate me and move on. What really matters here are the issues. What really matters here is the kind of government we want, the kind of Internet we want, the kind of relationship between people and societies. And that ' s what I ' m hoping the debate will move towards, and we ' ve seen that increasing over time. If I had to describe myself, I wouldn ' t use words like"hero."I wouldn ' t use"patriot,"and I wouldn ' t use"traitor."I ' d say I ' m an American and I ' m a citizen, just like everyone else. CA : So just to give some context for those who don ' t know the whole story — — this time a year ago, you were stationed in Hawaii working as a consultant to the NSA. As a sysadmin, you had access to their systems, and you began revealing certain classified documents to some handpicked journalists leading the way to June ' s revelations. Now, what propelled you to do this? ES : You know, when I was sitting in Hawaii, and the years before, when I was working in the intelligence community, I saw a lot of things that had disturbed me. We do a lot of good things in the intelligence community, things that need to be done, and things that help everyone. But there are also things that go too far. There are things that shouldn ' t be done, and decisions that were being made in secret without the public ' s awareness, without the public ' s consent, and without even our representatives in government having knowledge of these programs. When I really came to **struggle** with these issues, I thought to myself, how can I do this in the most responsible way, that **maximizes** the public benefit while minimizing the risks? And out of all the solutions that I could come up with, out of going to Congress, when there were no laws, there were no legal protections for a private employee, a contractor in intelligence like myself, there was a risk that I would be buried along with the information and the public would never find out. But the First Amendment of the United States Constitution guarantees us a free press for a reason, and that ' s to enable an adversarial press, to challenge the government, but also to work together with the government, to have a dialogue and debate about how we can inform the public about matters of vital importance without putting our national security at risk. And by working with journalists, by giving all of my information back to the American people, rather than trusting myself to make the decisions about **publication**, we ’ ve had a robust debate with a deep investment by the government that I think has resulted in a benefit for everyone. And the risks that have been threatened, the risks that have been played up by the government have never materialized. We ’ ve never seen any evidence of even a single instance of specific harm, and because of that, I ’ m comfortable with the decisions that I made. CA : So let me show the audience a couple of examples of what you revealed. If we could have a slide up, and Ed, I don ’ t know whether you can see, the slides are here. This is a slide of the PRISM program, and maybe you could tell the audience what that was that was revealed. ES : The best way to understand PRISM, because there ’ s been a little bit of controversy, is to first talk about what PRISM isn ’ t. Much of the debate in the U. S. has been about metadata. They ’ ve said it ’ s just metadata, it ’ s just metadata, and they ’ re talking about a specific legal authority called Section 215 of the Patriot Act. That allows sort of a warrantless wiretapping, mass surveillance of the entire country ’ s phone records, things like that — who you ’ re talking to, when you ’ re talking to them, where you traveled. These are all metadata events. PRISM is about content. It ’ s a program through which the government could compel corporate America, it could deputize corporate America to do its dirty work for the NSA. And even though some of these companies did resist, even though some of them — I believe Yahoo was one of them — challenged them in court, they all lost, because it was never tried by an open court. They were only tried by a secret court. And something that we ’ ve seen, something about the PRISM program that ’ s very concerning to me is, there ’ s been a talking point in the U. S. government where they ’ ve said 15 federal **judges** have reviewed these programs and found them to be lawful, but what they don ’ t tell you is those are secret **judges** in a secret court based on secret interpretations of law that ’ s considered 34, 000 warrant requests over 33 years, and in 33 years only rejected 11 government requests. These aren ’ t the people that we want deciding what the role of corporate America in a free and open Internet should be. CA : Now, this slide that we ’ re showing here shows the dates in which different technology companies, Internet companies, are alleged to have joined the program, and where data collection began from them. Now, they have denied collaborating with the NSA. How was that data collected by the NSA? ES : Right. So the NSA ’ s own slides refer to it as direct access. What that means to an actual NSA analyst, someone like me who was working as an intelligence analyst targeting, Chinese cyber-hackers, things like that, in Hawaii, is the provenance of that data is directly from their servers. It doesn ’ t mean that there ’ s a group of company representatives sitting in a smoky room with the NSA palling around and making back-room deals about how they ’ re going to give this stuff away. Now each company handles it different ways. Some are responsible. Some are somewhat less responsible. But the bottom line is, when we talk about how this information is given, it ’ s coming from the companies themselves. It ’ s not stolen from the lines. But there ’ s an important thing to remember here : even though companies pushed back, even though companies demanded, hey, let ’ s do this through a warrant process, let ’ s do this where we actually have some sort of legal review, some sort of basis for handing over these users ’ data, we saw stories in the Washington Post last year that weren ’ t as well reported as the PRISM story that said the NSA broke in to the data center communications between Google to itself and Yahoo to itself. So even these companies that are cooperating in at least a compelled but hopefully lawful manner with the NSA, the NSA isn ' t satisfied with that, and because of that, we need our companies to work very hard to guarantee that they ' re going to represent the interests of the user, and also advocate for the rights of the users. And I think over the last year, we ' ve seen the companies that are named on the PRISM slides take great strides to do that, and I encourage them to continue. CA : What more should they do? ES : The biggest thing that an Internet company in America can do today, right now, without consulting with lawyers, to protect the rights of users worldwide, is to enable SSL web encryption on every page you visit. The reason this matters is today, if you go to look at a copy of"1984"on Amazon. com, the NSA can see a record of that, the Russian intelligence **service** can see a record of that, the Chinese **service** can see a record of that, the French **service**, the German **service**, the **services** of Andorra. They can all see it because it ' s unencrypted. The world ' s library is Amazon. com, but not only do they not support encryption by default, you cannot choose to use encryption when browsing through books. This is something that we need to change, not just for Amazon, I don ' t mean to single them out, but they ' re a great example. All companies need to move to an encrypted browsing habit by default for all users who haven ' t taken any **action** or picked any **special** methods on their own. That ' ll increase the **privacy** and the rights that people enjoy worldwide. CA : Ed, come with me to this part of the stage. I want to show you the next slide here. This is a program called Boundless Informant. What is that? ES : So, I ' ve got to give credit to the NSA for using appropriate names on this. This is one of my favorite NSA cryptonyms. Boundless Informant is a program that the NSA hid from Congress. The NSA was previously asked by Congress, was there any ability that they had to even give a rough ballpark estimate of the amount of American communications that were being intercepted. They said no. They said, we don ' t track those stats, and we can ' t track those stats. We can ' t tell you how many communications we ' re intercepting around the world, because to tell you that would be to invade your **privacy**. Now, I really appreciate that sentiment from them, but the reality, when you look at this slide is, not only do they have the capability, the capability already exists. It ' s already in place. The NSA has its own internal data format that tracks both ends of a communication, and if it says, this communication came from America, they can tell Congress how many of those communications they have today, right now. And what Boundless Informant tells us is more communications are being intercepted in America about Americans than there are in Russia about Russians. I ' m not sure that ' s what an intelligence agency should be aiming for. CA : Ed, there was a story broken in the Washington Post, again from your data. The headline says,"NSA broke **privacy** rules thousands of times per year."Tell us about that. ES : We also heard in Congressional testimony last year, it was an amazing thing for someone like me who came from the NSA and who ' s seen the actual internal documents, knows what ' s in them, to see officials testifying under oath that there had been no abuses, that there had been no violations of the NSA ' s rules, when we knew this story was coming. But what ' s especially interesting about this, about the fact that the NSA has violated their own rules, their own laws thousands of times in a single year, including one event by itself, one event out of those 2, 776, that affected more than 3, 000 people. In another event, they intercepted all the calls in Washington, D. C., by accident. What ' s amazing about this, this report, that didn ' t get that much attention, is the fact that not only were there 2, 776 abuses, the chairman of the Senate Intelligence Committee, Dianne Feinstein, had not seen this report until the Washington Post contacted her asking for comment on the report. And she then requested a copy from the NSA and received it, but had never seen this before that. What does that say about the state of oversight in American intelligence when the chairman of the Senate Intelligence Committee has no idea that the rules are being broken thousands of times every year? CA : Ed, one response to this whole debate is this : Why should we care about all this surveillance, honestly? I mean, look, if you ’ ve done nothing wrong, you ’ ve got nothing to worry about. What ’ s wrong with that point of view? ES : Well, so the first thing is, you ’ re giving up your rights. You ’ re saying hey, you know, I don ’ t think I ’ m going to need them, so I ’ m just going to trust that, you know, let ’ s get rid of them, it doesn ’ t really matter, these guys are going to do the right thing. Your rights matter because you never know when you ’ re going to need them. Beyond that, it ’ s a part of our cultural identity, not just in America, but in Western societies and in democratic societies around the world. People should be able to pick up the phone and to call their family, people should be able to send a text message to their loved ones, people should be able to buy a book online, they should be able to travel by train, they should be able to buy an airline ticket without wondering about how these events are going to look to an agent of the government, possibly not even your government years in the future, how they ’ re going to be misinterpreted and what they ’ re going to think your intentions were. We have a right to **privacy**. We require warrants to be based on probable cause or some kind of individualized suspicion because we recognize that trusting anybody, any government authority, with the entirety of human communications in secret and without oversight is simply too great a temptation to be ignored. CA : Some people are furious at what you ’ ve done. I heard a quote recently from Dick Cheney who said that Julian Assange was a flea bite, Edward Snowden is the lion that bit the head off the dog. He thinks you ’ ve committed one of the worst acts of betrayal in American history. What would you say to people who think that? ES : Dick Cheney ’ s really something else. Thank you. I think it ’ s amazing, because at the time Julian Assange was doing some of his greatest work, Dick Cheney was saying he was going to end governments worldwide, the skies were going to ignite and the seas were going to boil off, and now he ’ s saying it ’ s a flea bite. So we should be suspicious about the same sort of overblown claims of damage to national security from these kind of officials. But let ’ s assume that these people really believe this. I would argue that they have kind of a narrow conception of national security. The prerogatives of people like Dick Cheney do not keep the nation safe. The public interest is not always the same as the national interest. Going to war with people who are not our enemy in places that are not a threat doesn ’ t make us safe, and that applies whether it ’ s in Iraq or on the Internet. The Internet is not the enemy. Our economy is not the enemy. American businesses, Chinese businesses, and any other company out there is a part of our society. It ’ s a part of our interconnected world. There are ties of fraternity that bond us together, and if we destroy these bonds by undermining the standards, the security, the manner of behavior, that nations and citizens all around the world expect us to abide by. CA : But it ’ s alleged that you ’ ve stolen 1. 7 million documents. It seems only a few hundred of them have been shared with journalists so far. Are there more revelations to come? ES : There are absolutely more revelations to come. I don ’ t think there ’ s any question that some of the most important reporting to be done is yet to come. CA : Come here, because I want to ask you about this particular revelation. Come and take a look at this. I mean, this is a story which I think for a lot of the techies in this room is the single most shocking thing that they have heard in the last few months. It ’ s about a program called”Bullrun.”Can you explain what that is? ES : So Bullrun, and this is again where we ’ ve got to thank the NSA for their candor, this is a program named after a Civil War battle. The British counterpart is called Edgehill, which is a U. K. civil war battle. And the reason that I believe they ’ re named this way is because they target our own infrastructure. They ’ re programs through which the NSA intentionally misleads corporate partners. They tell corporate partners that these are safe standards. They say hey, we need to work with you to secure your systems, but in reality, they ’ re giving bad advice to these companies that makes them degrade the security of their **services**. They ’ re building in backdoors that not only the NSA can exploit, but anyone else who has time and money to research and find it can then use to let themselves in to the world ’ s communications. And this is really dangerous, because if we lose a single standard, if we lose the trust of something like SSL, which was specifically targeted by the Bullrun program, we will live a less safe world overall. We won ’ t be able to access our banks and we won ’ t be able to access commerce without worrying about people monitoring those communications or subverting them for their own ends. CA : And do those same decisions also potentially open America up to cyberattacks from other sources? ES : Absolutely. One of the problems, one of the dangerous legacies that we ’ ve seen in the post-9 / 11 era, is that the NSA has traditionally worn two hats. They ’ ve been in charge of offensive operations, that is hacking, but they ’ ve also been in charge of defensive operations, and traditionally they ’ ve always prioritized defense over offense based on the principle that American secrets are simply worth more. If we hack a Chinese business and steal their secrets, if we hack a government office in Berlin and steal their secrets, that has less value to the American people than making sure that the Chinese can ’ t get access to our secrets. So by reducing the security of our communications, they ’ re not only putting the world at risk, they ’ re putting America at risk in a fundamental way, because intellectual property is the basis, the foundation of our economy, and if we put that at risk through weak security, we ’ re going to be paying for it for years. CA : But they ’ ve made a calculation that it was worth doing this as part of America ’ s defense against terrorism. Surely that makes it a price worth paying. ES : Well, when you look at the results of these programs in stopping terrorism, you will see that that ’ s unfounded, and you don ’ t have to take my word for it, because we ’ ve had the first open court, the first federal court that ’ s reviewed this, outside the secrecy arrangement, called these programs Orwellian and likely unconstitutional. Congress, who has access to be briefed on these things, and now has the desire to be, has produced bills to reform it, and two independent White House panels who reviewed all of the classified evidence said these programs have never stopped a single terrorist attack that was imminent in the United States. So is it really terrorism that we ’ re stopping? Do these programs have any value at all? I say no, and all three branches of the American government say no as well. CA : I mean, do you think there ’ s a deeper motivation for them than the war against terrorism? ES : I ’ m sorry, I couldn ’ t hear you, say again? CA : Sorry. Do you think there ’ s a deeper motivation for them other than the war against terrorism? ES : Yeah. The bottom line is that terrorism has always been what we in the intelligence world would call a cover for **action**. Terrorism is something that provokes an emotional response that allows people to rationalize authorizing powers and programs that they wouldn ’ t give otherwise. The Bullrun and Edgehill-type programs, the NSA asked for these authorities back in the 1990s. They asked the FBI to go to Congress and make the case. The FBI went to Congress and did make the case. But Congress and the American people said no. They said, it ’ s not worth the risk to our economy. They said it ’ s worth too much damage to our society to justify the gains. But what we saw is, in the post-9 / 11 era, they used secrecy and they used the justification of terrorism to start these programs in secret without asking Congress, without asking the American people, and it ’ s that kind of government behind closed doors that we need to guard ourselves against, because it makes us less safe, and it offers no value. CA : Okay, come with me here for a sec, because I ’ ve got a more personal question for you. Speaking of terror, most people would find the situation you ’ re in right now in Russia pretty terrifying. You obviously heard what happened, what the treatment that Bradley Manning got, Chelsea Manning as now is, and there was a story in Buzzfeed saying that there are people in the intelligence community who want you dead. How are you coping with this? How are you coping with the fear? ES : It ’ s no mystery that there are governments out there that want to see me dead. I ’ ve made clear again and again and again that I go to sleep every morning thinking about what I can do for the American people. I don ’ t want to harm my government. I want to help my government, but the fact that they are willing to completely ignore due process, they ’ re willing to declare guilt without ever seeing a **trial**, these are things that we need to work against as a society, and say hey, this is not appropriate. We shouldn ’ t be threatening dissidents. We shouldn ’ t be criminalizing journalism. And whatever part I can do to see that end, I ’ m happy to do despite the risks. CA : So I ’ d actually like to get some feedback from the audience here, because I know there ’ s widely differing reactions to Edward Snowden. Suppose you had the following two choices, right? You could view what he did as fundamentally a reckless act that has endangered America or you could view it as fundamentally a heroic act that will work towards America and the world ’ s long-term good? Those are the two choices I ’ ll give you. I ’ m curious to see who ’ s willing to vote with the first of those, that this was a reckless act? There are some hands going up. Some hands going up. It ’ s hard to put your hand up when the man is standing right here, but I see them. ES : I can see you. CA : And who goes with the second choice, the fundamentally heroic act? And I think it ’ s true to say that there are a lot of people who didn ’ t show a hand and I think are still thinking this through, because it seems to me that the debate around you doesn ’ t split along traditional political lines. It ’ s not left or right, it ’ s not really about pro-government, libertarian, or not just that. Part of it is almost a generational issue. You ’ re part of a generation that grew up with the Internet, and it seems as if you become offended at almost a visceral level when you see something done that you think will harm the Internet. Is there some truth to that? ES : It is. I think it ’ s very true. This is not a left or right issue. Our basic freedoms, and when I say our, I don ’ t just mean Americans, I mean people around the world, it ’ s not a partisan issue. These are things that all people believe, and it ’ s up to all of us to protect them, and to people who have seen and enjoyed a free and open Internet, it ’ s up to us to preserve that liberty for the next generation to enjoy, and if we don ’ t change things, if we don ’ t stand up to make the changes we need to do to keep the Internet safe, not just for us but for everyone, we ’ re going to lose that, and that would be a tremendous loss, not just for us, but for the world. CA : Well, I have heard similar language recently from the founder of the world wide web, who I actually think is with us, Sir Tim Berners-Lee. Tim, actually, would you like to come up and say, do we have a **microphone** for Tim? Tim, good to see you. Come up there. Which camp are you in, by the way, traitor, hero? I have a theory on this, but — Tim Berners-Lee : I ’ ve given much longer answers to that question, but hero, if I have to make the choice between the two. CA : And Ed, I think you ’ ve read the proposal that Sir Tim has talked about about a new Magna Carta to take back the Internet. Is that something that makes sense? ES : Absolutely. I mean, my generation, I grew up not just thinking about the Internet, but I grew up in the Internet, and although I never expected to have the chance to defend it in such a direct and practical manner and to embody it in this unusual, almost avatar manner, I think there ' s something poetic about the fact that one of the sons of the Internet has actually become close to the Internet as a result of their political **expression**. And I believe that a Magna Carta for the Internet is exactly what we need. We need to encode our values not just in writing but in the structure of the Internet, and it ' s something that I hope, I invite everyone in the audience, not just here in Vancouver but around the world, to join and participate in. CA : Do you have a question for Ed? TBL : Well, two questions, a general question — CA : Ed, can you still hear us? ES : Yes, I can hear you. CA : Oh, he ' s back. TBL : The wiretap on your line got a little interfered with for a moment. ES : It ' s a little bit of an NSA problem. TBL : So, from the 25 years, stepping back and thinking, what would you think would be the best that we could achieve from all the discussions that we have about the web we want? ES : When we think about in terms of how far we can go, I think that ' s a question that ' s really only limited by what we ' re willing to put into it. I think the Internet that we ' ve enjoyed in the past has been exactly what we as not just a nation but as a people around the world need, and by cooperating, by engaging not just the technical parts of society, but as you said, the users, the people around the world who contribute through the Internet, through social media, who just check the weather, who rely on it every day as a part of their life, to champion that. We ' ll get not just the Internet we ' ve had, but a better Internet, a better now, something that we can use to build a future that ' ll be better not just than what we hoped for but anything that we could have imagined. CA : It ' s 30 years ago that TED was founded, 1984. A lot of the conversation since then has been along the lines that actually George Orwell got it wrong. It ' s not Big Brother watching us. We, through the power of the web, and transparency, are now watching Big Brother. Your revelations kind of drove a stake through the heart of that rather optimistic view, but you still believe there ' s a way of doing something about that. And you do too. ES : Right, so there is an argument to be made that the powers of Big Brother have increased enormously. There was a recent legal article at Yale that established something called the Bankston-Soltani Principle, which is that our expectation of **privacy** is violated when the capabilities of government surveillance have become cheaper by an order of magnitude, and each time that occurs, we need to revisit and rebalance our **privacy** rights. Now, that hasn ' t happened since the government ' s surveillance powers have increased by several orders of magnitude, and that ' s why we ' re in the problem that we ' re in today, but there is still hope, because the power of individuals have also been increased by technology. I am living proof that an individual can go head to head against the most powerful adversaries and the most powerful intelligence agencies around the world and win, and I think that ' s something that we need to take hope from, and we need to build on to make it accessible not just to technical experts but to ordinary citizens around the world. Journalism is not a crime, communication is not a crime, and we should not be monitored in our everyday activities. CA : I ' m not quite sure how you shake the hand of a bot, but I imagine it ' s, this is the hand right here. TBL : That ' ll come very soon. ES : Nice to meet you, and I hope my beam looks as nice as my view of you guys does. CA : Thank you, Tim. I mean, The New York Times recently called for an amnesty for you. Would you welcome the chance to come back to America? ES : Absolutely. There ' s really no question, the principles that have been the foundation of this project have been the public interest and the principles that underly the journalistic establishment in the United States and around the world, and I think if the press is now saying, we support this, this is something that needed to happen, that ' s a powerful argument, but it ' s not the final argument, and I think that ' s something that public should decide.

But at the same time, the government has hinted that they want some kind of deal, that they want me to compromise the journalists with which I ’ ve been working, to come back, and I want to make it very clear that I did not do this to be safe. I did this to do what was right, and I ’ m not going to stop my work in the public interest just to benefit myself. CA : In the meantime, courtesy of the Internet and this technology, you ’ re here, back in North America, not quite the U. S., Canada, in this form. I ’ m curious, how does that feel? ES : Canada is different than what I expected. It ’ s a lot warmer. CA : At TED, the **mission** is”ideas worth spreading.”If you could encapsulate it in a single idea, what is your idea worth spreading right now at this moment? ES : I would say the last year has been a reminder that democracy may die behind closed doors, but we as individuals are born behind those same closed doors, and we don ’ t have to give up our **privacy** to have good government. We don ’ t have to give up our liberty to have security. And I think by working together we can have both open government and private lives, and I look forward to working with everyone around the world to see that happen. Thank you very much. CA : Ed, thank you.

Text Sample 222: POS Dim 2 – 2014 – Score 11.00 – 2014/t001679.txt

Well, good afternoon. How many of you took the ALS Ice Bucket Challenge? Woo hoo! Well, I have to tell you, from the bottom of our hearts, thank you so very, very much. Do you know to date the ALS Association has raised 125 million dollars? Woo hoo! It takes me back to the summer of 2011. My family, my kids had all grown up. We were officially empty nesters, and we decided, let ’ s go on a family **vacation**. Jenn, my daughter, and my son-in-law came down from New York. My youngest, Andrew, he came down from his home in Charlestown where he was working in Boston, and my son Pete, who had played at Boston College, **baseball**, had played **baseball** professionally in Europe, and had now come home and was selling group insurance, he also joined us. And one night, I found myself having a beer with Pete, and Pete was looking at me and he just said,”You know, Mom, I don ’ t know, selling group insurance is just not my passion.”He said,”I just don ’ t feel I ’ m living up to my potential. I don ’ t feel this is my **mission** in life.”And he said,”You know, oh by the way, Mom, I have to leave early from **vacation** because my inter-city league team that I play for made the playoffs, and I have to get back to Boston because I can ’ t let my team down. I ’ m just not as **passionate** about my job as I am about **baseball**.”So off Pete went, and left the family **vacation** — break a mother ’ s heart — and he went, and we followed four days later to see the next playoff game. We ’ re at the playoff game, Pete ’ s at the plate, and a fastball ’ s coming in, and it hits him on the wrist. Oh, Pete. His wrist went completely limp, like this. So for the next six months, Pete went back to his home in Southie, kept working that unpassionate job, and was going to doctors to see what was wrong with this wrist that never came back. Six months later, in March, he called my husband and me, and he said,”Oh, Mom and Dad, we have a doctor that found a diagnosis for that wrist. Do you want to come with the doctor ’ s appointment with me?”I said,”Sure, we ’ ll come in.”That morning, Pete, John and I all got up, got dressed, got in our cars — three separate cars because we were going to go to work after the doctor ’ s appointment to find out what happened to the wrist. We walked into the neurologist ’ s office, sat down, four doctors walk in, and the head neurologist sits down. And he says,”Well, Pete, we ’ ve been looking at all the tests, and I have to tell you, it ’ s not a sprained wrist, it ’ s not a broken wrist, it ’ s not nerve damage in the wrist, it ’ s not an infection, it ’ s not Lyme disease.”And there was this deliberate elimination going up, and I was thinking to myself, where is he going with this? Then he put his hands on his knees, he looked right at my 27-year-old kid, and said,”I don ’ t know how to tell a 27-year-old this : Pete, you have ALS.”ALS? I had had a friend whose 80-year-old father had ALS. I looked at my husband, he looked at me, and then we looked at the doctor, and we said,”ALS? Okay, what treatment? Let ’ s go. What do we do? Let ’ s go.”And he looked at us, and he said,”Mr. and Mrs. Frates, I ’ m sorry to tell you this, but there ’ s no treatment and there ’ s no cure.”We were the worst culprits. We didn ’ t even understand that it had been 75 years since Lou Gehrig and nothing had been done in the progress against ALS. So we all went home, and Jenn and Dan flew home from Wall Street, Andrew came home from Charlestown, and Pete went to B. C. to pick up his then-girlfriend Julie and brought her home, and six hours later after diagnosis, we ’ re sitting around having a family dinner, and we ’ re having small chat. I don ’ t even remember cooking dinner that night. But then our leader, Pete, set the vision, and talked to us just like we were his new team. He said,”There will be no wallowing, people.”He goes,”We ’ re not looking back, we ’ re looking forward. What an amazing opportunity we have to change the world. I ’ m going to change the face of this unacceptable situation of ALS. We ’ re going to move the needle, and I ’ m going to get it in front of philanthropists like Bill Gates.”And that was it. We were given our directive. So in the days and months that followed, within a week, we had our brothers and sisters and our family come to us, that they were already creating Team Frate Train. Uncle Dave, he was the webmaster ; Uncle Artie, he was the accountant ; Auntie Dana, she was the graphic artist ; and my youngest son, Andrew, quit his job, left his apartment in Charlestown and says,”I ’ m going to take care of Pete and be his caregiver.”Then all those people, classmates, teammates, coworkers that Pete had inspired throughout his whole life, the circles of Pete all started intersecting with one another, and made Team Frate Train. Six months after diagnosis, Pete was given an award at a research summit for advocacy. He got up and gave a very eloquent speech, and at the end of the speech, there was a panel, and on the panel were these pharmaceutical executives and biochemists and clinicians and I ’ m sitting there and I ’ m listening to them and most of the content went straight over my head. I avoided every science class I ever could. But I was watching these people, and I was listening to them, and they were saying,”I, I do this, I do that,”and there was a real unfamiliarity between them. So at the end of their talk, the panel, they had questions and answers, and boom, my hand went right up, and I get the **microphone**, and I look at them and I say,”Thank you. Thank you so much for working in ALS. It means so very much to us.”I said,”But I do have to tell you that I ’ m watching your body language and I ’ m listening to what you ’ re saying. It just doesn ’ t seem like there ’ s a whole lot of collaboration going on here. And not only that, where ’ s the flip chart with the **action** items and the **follow-up** and the accountability? What are you going to do after you leave this room?”And then I turned around and there was about 200 pairs of eyes just staring at me. And it was that point that I realized that I had talked about the elephant in the room. Thus my **mission** had begun. So over the next couple of years, Pete — we ’ ve had our highs and our lows. Pete was put on a compassionate use drug. It was hope in a bottle for the whole ALS community. It was in a phase III **trial**. Then six months later, the data comes back : no efficacy. We were supposed to have therapies overseas, and the rug was pulled out from under us. So for the next two years, we just watched my son be taken away from me, little by little every day. Two and a half years ago, Pete was hitting home runs at **baseball** fields. Today, Pete ’ s completely paralyzed. He can ’ t hold his head up any longer. He ’ s confined to a motorized wheelchair. He can no longer swallow or eat. He has a feeding tube. He can ’ t speak. He talks with eye gaze technology and a speech generating device, and we ’ re watching his lungs, because his diaphragm eventually is going to give out and then the decision will be made to put him on a ventilator or not. ALS robs the human of all their physical parts, but the brain stays **intact**. So July 4th, 2014, 75th year of Lou Gehrig ’ s inspirational speech comes, and Pete is asked by MLB. com to write an article in the Bleacher Report. And it was very significant, because he wrote it using his eye gaze technology. Twenty days later, the ice started to fall. On July 27th, Pete ’ s roommate in New York City, wearing a Quinn For The Win shirt, signifying Pat Quinn, another ALS patient known in New York, and B. C. shorts said,”I ’ m taking the ALS Ice Bucket Challenge,”picked up the ice, put it over his head.”And I ’ m nominating...”And he sent it up to Boston. And that was on July 27th. Over the next couple of days, our news feed was full of family and friends. If you haven ’ t gone back, the nice thing about Facebook is that you have the dates, you can go back. You ’ ve got to see Uncle Artie ’ s human Bloody Mary. I ’ m telling you, it ’ s one of the best ones, and that was probably in day two. By about day four, Uncle Dave, the webmaster, he isn ’ t on Facebook, and I get a text from him, and it says,”Nancy, what the hell is going on?”Uncle Dave gets a hit every time Pete ’ s website is gone onto, and his phone was blowing up. So we all sat down and we realized, money is coming in — how amazing. So we knew awareness would lead to funding, we just didn ’ t know it would only take a couple of days. So we got together, put our best 501s on Pete ’ s website, and off we went. So week one, Boston media. Week two, national media. It was during week two that our neighbor next door opened up our door and threw a pizza across the kitchen floor, saying,”I think you people might need food in there.”Week three, celebrities — Entertainment Tonight, Access Hollywood. Week four, global — BBC, Irish Radio. Did anyone see”Lost In Translation”? My husband did Japanese television. It was interesting. And those videos, the popular ones. Paul Bissonnette ’ s glacier video, incredible. How about the redemption nuns of Dublin? Who ’ s seen that one? It ’ s absolutely fantastic. J. T., Justin Timberlake. That ’ s when we knew, that was a real A-list celebrity. I go back on my texts, and I can see”JT! JT!”My sister texting me. Angela Merkel, the chancellor of Germany. Incredible. And the ALS patients, you know what their favorite ones are, and their families ’? All of them. Because this misunderstood and underfunded”rare”disease, they just sat and watched people saying it over and over :”ALS, ALS.”It was unbelievable. And those naysayers, let ’ s just talk a couple of stats, shall we? Okay, so the ALS Association, they think by year end, it ’ ll be 160 million dollars. ALS TDI in Cambridge, they raised three million dollars. Well, guess what? They had a clinical **trial** for a drug that they ’ ve been developing. It was on a three-year track for funding. Two months. It ’ s coming out starting in two months. And YouTube has reported that over 150 countries have posted Ice Bucket Challenges for ALS. And Facebook, 2. 5 million videos, and I had the awesome adventure visiting the Facebook campus last week, and I said to them,”I know what it was like in my house. I can ’ t imagine what it was like around here.”All she said was,”Jaw-dropping.”And my family ’ s favorite video? Bill Gates. Because the night Pete was diagnosed, he told us that he was going to get ALS in front of philanthropists like Bill Gates, and he did it. Goal number one, check. Now on to the treatment and cure. So okay, after all of this ice, we know that it was much more than just pouring buckets of ice water over your head, and I really would like to leave you with a couple of things that I ’ d like you to remember. The first thing is, every morning when you wake up, you can choose to live your day in positivity. Would any of you blame me if I just was in the fetal position and pulled the covers over my head every day? No, I don ’ t think anybody would blame me, but Pete has inspired us to wake up every morning and be positive and proactive. I actually had to ditch support groups because everybody was in there saying that spraying their lawns with chemicals, that ’ s why they got ALS, and I was like,”I don ’ t think so,”but I had to get away from the negativity. The second thing I want to leave you with is the person at the middle of the challenge has to be willing to have the mental toughness to put themselves out there. Pete still goes to **baseball** games and he still sits with his teammates in the dugout, and he hangs his gravity feed bag right on the cages. You ’ ll see the kids, they ’ re up there hanging it up.”Pete, is that okay?””Yup.”And then they put it right into his stomach. Because he wants them to see what the reality of this is, and how he ’ s never, ever going to give up. And the third thing I want to leave you with : If you ever come across a situation that you see as so unacceptable, I want you to dig down as deep as you can and find your best mother bear and go after it. Thank you. I know that I ’ m running over, but I ’ ve got to leave you with this : the gifts that my son has given me. I have had 29 years of having the honor of being the mother of Pete Frates. Pete Frates has been inspiring and leading his whole life. He ’ s thrown out kindness, and all that kindness has come back to him. He walks the face of the Earth right now and knows why he ’ s here. What a gift. The second thing that my son has given me is he ’ s given me my **mission** in life. Now I know why I ’ m here. I ’ m going to save my son, and if it doesn ’ t happen in time for him, I ’ m going to work so that no other mother has to go through what I ’ m going through. And the third thing, and last but not least gift that my son has given me, as an exclamation point to the miraculous month of August 2014 : That girlfriend that he went to get on the night of diagnosis is now his wife, and Pete and Julie have given me my granddaughter, Lucy Fitzgerald Frates. Lucy Fitzgerald Frates came two weeks early as the exclamation point on August 31st, 2014. And so — — And so let me leave you with Pete ’ s words of inspiration that he would use to classmates, coworkers and teammates. Be **passionate**. Be genuine. Be hardworking. And don ’ t forget to be great. Thank you.

Text Sample 223: POS Dim 2 – 2014 – Score 10.00 – 2014/t001681.txt

I ’ m a lifelong traveler. Even as a little kid, I was actually working out that it would be cheaper to go to boarding school in England than just to the best school down the road from my parents ’ house in California. So, from the time I was nine years old I was flying alone several times a year over the North Pole, just to go to school. And of course the more I flew the more I came to love to fly, so the very week after I graduated from high school, I got a job mopping tables so that I could spend every season of my 18th year on a different continent. And then, almost inevitably, I became a travel writer so my job and my joy could become one. And I really began to feel that if you were lucky enough to walk around the candlelit temples of Tibet or to wander along the seafronts in Havana with music passing all around you, you could bring those sounds and the high cobalt skies and the flash of the blue ocean back to your friends at home, and really bring some magic and clarity to your own life. Except, as you all know, one of the first things you learn when you travel is that nowhere is magical unless you can bring the right eyes to it. You take an angry man to the Himalayas, he just starts **complaining** about the food. And I found that the best way that I could develop more attentive and more appreciative eyes was, oddly, by going nowhere, just by sitting still. And of course sitting still is how many of us get what we most crave and need in our accelerated lives, a break. But it was also the only way that I could find to sift through the slideshow of my experience and make sense of the future and the past. And so, to my great surprise, I found that going nowhere was at least as exciting as going to Tibet or to Cuba. And by going nowhere, I mean nothing more intimidating than taking a few minutes out of every day or a few days out of every season, or even, as some people do, a few years out of a life in order to sit still long enough to find out what moves you most, to recall where your truest **happiness** lies and to remember that sometimes making a living and making a life point in opposite directions. And of course, this is what wise beings through the centuries from every tradition have been telling us. It ' s an old idea. More than 2, 000 years ago, the Stoics were reminding us it ' s not our experience that makes our lives, it ' s what we do with it. Imagine a hurricane suddenly sweeps through your town and reduces every last thing to rubble. One man is traumatized for life. But another, maybe even his brother, almost feels liberated, and decides this is a great chance to start his life anew. It ' s exactly the same event, but radically different responses. There is nothing either good or bad, as Shakespeare told us in"Hamlet,"but thinking makes it so. And this has certainly been my experience as a traveler. Twenty-four years ago I took the most mind-bending trip across North Korea. But the trip lasted a few days. What I ' ve done with it sitting still, going back to it in my head, trying to understand it, finding a place for it in my thinking, that ' s lasted 24 years already and will probably last a lifetime. The trip, in other words, gave me some amazing sights, but it ' s only sitting still that allows me to turn those into lasting insights. And I sometimes think that so much of our life takes place inside our heads, in memory or imagination or interpretation or speculation, that if I really want to change my life I might best begin by changing my mind. Again, none of this is new ; that ' s why Shakespeare and the Stoics were telling us this centuries ago, but Shakespeare never had to face 200 emails in a day. The Stoics, as far as I know, were not on Facebook. We all know that in our on-demand lives, one of the things that ' s most on demand is ourselves. Wherever we are, any time of night or day, our bosses, junk-mailers, our parents can get to us. Sociologists have actually found that in recent years Americans are working fewer hours than 50 years ago, but we feel as if we ' re working more. We have more and more time-saving devices, but sometimes, it seems, less and less time. We can more and more easily make contact with people on the furthest corners of the planet, but sometimes in that process we lose contact with ourselves. And one of my biggest surprises as a traveler has been to find that often it ' s exactly the people who have most enabled us to get anywhere who are intent on going nowhere. In other words, precisely those beings who have created the technologies that override so many of the limits of old, are the ones wisest about the need for limits, even when it comes to technology. I once went to the Google headquarters and I saw all the things many of you have heard about ; the indoor tree houses, the trampolines, workers at that time enjoying 20 percent of their paid time free so that they could just let their imaginations go wandering. But what impressed me even more was that as I was waiting for my digital I. D., one Googler was telling me about the program that he was about to start to teach the many, many Googlers who **practice** yoga to become trainers in it, and the other Googler was telling me about the book that he was about to write on the inner search engine, and the ways in which science has empirically shown that sitting still, or meditation, can lead not just to better health or to clearer thinking, but even to emotional intelligence. I have another friend in Silicon Valley who is really one of the most eloquent spokesmen for the latest technologies, and in fact was one of the founders of Wired magazine, Kevin Kelly. And Kevin wrote his last book on fresh technologies without a smartphone or a laptop or a TV in his home. And like many in Silicon Valley, he tries really hard to **observe** what they call an Internet sabbath, whereby for 24 or 48 hours every week they go completely offline in order to gather the sense of direction and proportion they ’ ll need when they go online again. The one thing perhaps that technology hasn ’ t always given us is a sense of how to make the wisest use of technology. And when you speak of the sabbath, look at the Ten Commandments — there ’ s only one word there for which the adjective”holy”is used, and that ’ s the Sabbath. I pick up the Jewish holy book of the Torah — its longest chapter, it ’ s on the Sabbath. And we all know that it ’ s really one of our greatest luxuries, the empty space. In many a piece of music, it ’ s the pause or the rest that gives the piece its beauty and its shape. And I know I as a writer will often try to include a lot of empty space on the page so that the reader can complete my thoughts and sentences and so that her imagination has room to breathe. Now, in the physical domain, of course, many people, if they have the resources, will try to get a place in the country, a second home. I ’ ve never begun to have those resources, but I sometimes remember that any time I want, I can get a second home in time, if not in space, just by taking a day off. And it ’ s never easy because, of course, whenever I do I spend much of it worried about all the extra stuff that ’ s going to crash down on me the following day. I sometimes think I ’ d rather give up meat or sex or wine than the chance to check on my emails. And every season I do try to take three days off on retreat but a part of me still feels guilty to be leaving my poor wife behind and to be ignoring all those seemingly urgent emails from my bosses and maybe to be missing a friend ’ s birthday party. But as soon as I get to a place of real quiet, I realize that it ’ s only by going there that I ’ ll have anything fresh or creative or joyful to share with my wife or bosses or friends. Otherwise, really, I ’ m just foisting on them my **exhaustion** or my distractedness, which is no blessing at all. And so when I was 29, I decided to remake my entire life in the light of going nowhere. One evening I was coming back from the office, it was after midnight, I was in a taxi **driving** through Times Square, and I suddenly realized that I was racing around so much I could never catch up with my life. And my life then, as it happened, was pretty much the one I might have dreamed of as a little boy. I had really interesting friends and colleagues, I had a nice apartment on Park Avenue and 20th Street. I had, to me, a fascinating job writing about world affairs, but I could never separate myself enough from them to hear myself think — or really, to understand if I was truly happy. And so, I abandoned my dream life for a single room on the backstreets of Kyoto, Japan, which was the place that had long exerted a strong, really mysterious gravitational pull on me. Even as a child I would just look at a painting of Kyoto and feel I recognized it ; I knew it before I ever laid eyes on it. But it ’ s also, as you all know, a beautiful city encircled by hills, filled with more than 2, 000 temples and shrines, where people have been sitting still for 800 years or more. And quite soon after I moved there, I ended up where I still am with my wife, formerly our kids, in a two-room apartment in the middle of nowhere where we have no bicycle, no car, no TV I can understand, and I still have to support my loved ones as a travel writer and a journalist, so clearly this is not ideal for job advancement or for cultural excitement or for social diversion. But I realized that it gives me what I prize most, which is days and hours. I have never once had to use a cell phone there. I almost never have to look at the time, and every morning when I wake up, really the day stretches in front of me like an open meadow. And when life throws up one of its nasty surprises, as it will, more than once, when a doctor comes into my room wearing a grave **expression**, or a car suddenly veers in front of mine on the freeway, I know, in my bones, that it ’ s the time I ’ ve spent going nowhere that is going to sustain me much more than all the time I ’ ve spent racing around to Bhutan or Easter Island. I ’ ll always be a traveler — my livelihood depends on it — but one of the beauties of travel is that it allows you to bring stillness into the motion and the commotion of the world. I once got on a plane in Frankfurt, Germany, and a young German woman came down and sat next to me and engaged me in a very friendly conversation for about 30 minutes, and then she just turned around and sat still for 12 hours. She didn ’ t once turn on her video monitor, she never pulled out a book, she didn ’ t even go to sleep, she just sat still, and something of her clarity and calm really imparted itself to me. I ’ ve noticed more and more people taking conscious measures these days to try to open up a space inside their lives. Some people go to black-hole resorts where they ’ ll spend hundreds of dollars a night in order to hand over their cell phone and their laptop to the front desk on arrival. Some people I know, just before they go to sleep, instead of **scrolling** through their messages or checking out YouTube, just turn out the lights and listen to some music, and notice that they sleep much better and wake up much refreshed. I was once fortunate enough to drive into the high, dark mountains behind Los Angeles, where the great poet and singer and international heartthrob Leonard Cohen was living and working for many years as a full-time monk in the Mount Baldy Zen Center. And I wasn ’ t entirely surprised when the record that he released at the age of 77, to which he gave the deliberately unsexy title of”Old Ideas,”went to number one in the charts in 17 nations in the world, hit the top five in nine others. Something in us, I think, is crying out for the sense of intimacy and depth that we get from people like that. who take the time and trouble to sit still. And I think many of us have the sensation, I certainly do, that we ’ re standing about two inches away from a huge screen, and it ’ s noisy and it ’ s crowded and it ’ s changing with every second, and that screen is our lives. And it ’ s only by stepping back, and then further back, and holding still, that we can begin to see what the canvas means and to catch the larger picture. And a few people do that for us by going nowhere. So, in an age of acceleration, nothing can be more exhilarating than going slow. And in an age of distraction, nothing is so luxurious as paying attention. And in an age of constant movement, nothing is so urgent as sitting still. So you can go on your next **vacation** to Paris or Hawaii, or New Orleans ; I bet you ’ ll have a wonderful time. But, if you want to come back home alive and full of fresh hope, in love with the world, I think you might want to try considering going nowhere. Thank you.

Text Sample 224: POS Dim 2 – 1998 – Score 6.00 – 1998/t000113.txt

Sheryl Shade : Hi, Aimee. Aimee Mullins : Hi. SS : Aimee and I thought we ’ d just talk a little bit, and I wanted her to tell all of you what makes her a distinctive athlete. AM : Well, for those of you who have seen the picture in the little bio â it might have given it away â I ’ m a double amputee, and I was born without fibulas in both legs. I was amputated at age one, and I ’ ve been running like hell ever since, all over the place. SS : Well, why don ’ t you tell them how you got to Georgetown â why don ’ t we start there? Why don ’ t we start there? AM : I ’ m a senior in Georgetown in the Foreign Service program. I won a full academic scholarship out of high school. They pick three students out of the nation every year to get involved in international affairs, and so I won a full ride to Georgetown and I ’ ve been there for four years. Love it. SS : When Aimee got there, she decided that she ’ s, kind of, curious about track and field, so she decided to call someone and start asking about it. So, why don ’ t you tell that story? AM : Yeah. Well, I guess I ’ ve always been involved in sports. I played softball for five years growing up. I skied competitively throughout high school, and I got a little restless in college because I wasn ’ t doing anything for about a year or two sports-wise. And I ’ d never competed on a disabled level, you know â I ’ d always competed against other able-bodied athletes. That ’ s all I ’ d ever known. In fact, I ’ d never even met another amputee until I was 17. And I heard that they do these track meets with all disabled runners, and I figured,”Oh, I don ’ t know about this, but before I **judge** it, let me go see what it ’ s all about.”So, I booked myself a flight to Boston in ’ 95, 19 years old and definitely the dark horse candidate at this race. I ’ d never done it before. I went out on a gravel track a couple of weeks before this meet to see how far I could run, and about 50 meters was enough for me, panting and heaving. And I had these legs that were made of a wood and plastic compound, attached with Velcro straps â big, thick, five-ply wool socks on â you know, not the most comfortable things, but all I ’ d ever known. And I ’ m up there in Boston against people wearing legs made of all things â carbon graphite and, you know, shock absorbers in them and all sorts of things â and they ’ re all looking at me like, OK, we know who ’ s not going to win this race. And, I mean, I went up there expecting â I don ’ t know what I was expecting â but, you know, when I saw a man who was missing an entire leg go up to the high jump, hop on one leg to the high jump and clear it at six feet, two inches... Dan O ’ Brien jumped 5 ’ 11”in ’ 96 in Atlanta, I mean, if it just gives you a comparison of â these are truly accomplished athletes, without qualifying that word”athlete.”And so I decided to give this a shot : heart pounding, I ran my first race and I beat the national record-holder by three hundredths of a second, and became the new national record-holder on my first try out. And, you know, people said,”Aimee, you know, you ’ ve got speed â you ’ ve got natural speed â but you don ’ t have any skill or finesse going down that track. You were all over the place. We all saw how hard you were working.”And so I decided to call the track coach at Georgetown. And I thank god I didn ’ t know just how huge this man is in the track and field world. He ’ s coached five Olympians, and the man ’ s office is lined from floor to ceiling with All America certificates of all these athletes he ’ s coached. He ’ s just a rather intimidating figure. And I called him up and said,”Listen, I ran one race and I won...””I want to see if I can, you know â I need to just see if I can sit in on some of your **practices**, see what drills you do and whatever.”That ’ s all I wanted â just two **practices**.”Can I just sit in and see what you do?”And he said,”Well, we should meet first, before we decide anything.”You know, he ’ s thinking,”What am I getting myself into?”So, I met the man, walked in his office, and saw these posters and magazine covers of people he has coached. And we got to talking, and it turned out to be a great partnership because he ’ d never coached a disabled athlete, so therefore he had no preconceived notions of what I was or wasn ’ t capable of, and I ’ d never been coached before. So this was like,”Here we go â let ’ s start on this trip.”So he started giving me four days a week of his lunch break, his free time, and I would come up to the track and train with him. So that ’ s how I met Frank. That was fall of ’ 95. But then, by the time that winter was rolling around, he said,”You know, you ’ re good enough. You can run on our women ’ s track team here.”And I said,”No, come on.”And he said,”No, no, really. You can. You can run with our women ’ s track team.”In the spring of 1996, with my goal of making the U. S. Paralympic team that May coming up full speed, I joined the women ’ s track team. And no disabled person had ever done that â run at a collegiate level. So I don ’ t know, it started to become an interesting mix. SS : Well, on your way to the Olympics, a couple of memorable events happened at Georgetown. Why don ’ t you just tell them? AM : Yes, well, you know, I ’ d won everything as far as the disabled meets â everything I competed in â and, you know, training in Georgetown and knowing that I was going to have to get used to seeing the backs of all these women ’ s shirts â you know, I ’ m running against the next Flo-Jo â and they ’ re all looking at me like,”Hmm, what ’ s, you know, what ’ s going on here?”And putting on my Georgetown uniform and going out there and knowing that, you know, in order to become better â and I ’ m already the best in the country â you know, you have to train with people who are inherently better than you. And I went out there and made it to the Big East, which was sort of the championship race at the end of the season. It was really, really hot. And it ’ s the first â I had just gotten these new sprinting legs that you see in that bio, and I didn ’ t realize at that time that the amount of sweating I would be doing in the sock â it actually acted like a lubricant and I ’ d be, kind of, pistoning in the socket. And at about 85 meters of my 100 meters sprint, in all my glory, I came out of my leg. Like, I almost came out of it, in front of, like, 5, 000 people. And I, I mean, was just mortified â because I was signed up for the 200, you know, which went off in a half hour. I went to my coach :”Please, don ’ t make me do this.”I can ’ t do this in front of all those people. My legs will come off. And if it came off at 85 there ’ s no way I ’ m going 200 meters. And he just sat there like this. My pleas fell on deaf ears, thank god. Because you know, the man is from Brooklyn ; he ’ s a big man. He says,”Aimee, so what if your leg falls off? You pick it up, you put the damn thing back on, and finish the goddamn race!”And I did. So, he kept me in line. He kept me on the right track. SS : So, then Aimee makes it to the 1996 Paralympics, and she ’ s all excited. Her family ’ s coming down â it ’ s a big deal. It ’ s now two years that you ’ ve been running? AM : No, a year. SS : A year. And why don ’ t you tell them what happened right before you go run your race? AM : Okay, well, Atlanta. The Paralympics, just for a little bit of clarification, are the Olympics for people with physical disabilities â amputees, persons with cerebral palsy, and wheelchair athletes â as opposed to the **Special** Olympics, which deals with people with mental disabilities. So, here we are, a week after the Olympics and down at Atlanta, and I ’ m just blown away by the fact that just a year ago, I got out on a gravel track and couldn ’ t run 50 meters. And so, here I am â never lost. I set new records at the U. S. Nationals â the Olympic **trials** â that May, and was sure that I was coming home with the gold. I was also the only, what they call”bilateral BK”â below the knee. I was the only woman who would be doing the long jump. I had just done the long jump, and a guy who was missing two legs came up to me and says,”How do you do that? You know, we ’ re supposed to have a planar foot, so we can ’ t get off on the springboard.”I said,”Well, I just did it. No one told me that.”So, it ’ s funny â I ’ m three inches within the world record â and kept on from that point, you know, so I ’ m signed up in the long jump â signed up? No, I made it for the long jump and the 100-meter. And I ’ m sure of it, you know? I made the front page of my hometown paper that I delivered for six years, you know? It was, like, this is my time for shine. And we ’ re at the trainee warm-up track, which is a few blocks away from the Olympic stadium. These legs that I was on, which I ’ ll take out right now â I was the first person in the world on these legs. I was the guinea pig., I ’ m telling you, this was, like â talk about a tourist attraction. Everyone was taking pictures â”What is this girl running on?”And I ’ m always looking around, like, where is my competition? It ’ s my first international meet. I tried to get it out of anybody I could, you know,”Who am I running against here?””Oh, Aimee, we ’ ll have to get back to you on that one.”I wanted to find out times.”Don ’ t worry, you ’ re doing great.”This is 20 minutes before my race in the Olympic stadium, and they post the heat sheets. And I go over and look. And my fastest time, which was the world record, was

15. 77. Then I ’ m looking : the next lane, lane two, is 12. 8. Lane three is 12. 5. Lane four is 12. 2. I said,”What ’ s going on?”And they shove us all into the shuttle bus, and all the women there are missing a hand. So, I ’ m just, like â they ’ re all looking at me like ’ which one of these is not like the other, ’ you know? I ’ m sitting there, like,”Oh, my god. Oh, my god.”You know, I ’ d never lost anything, like, whether it would be the scholarship or, you know, I ’ d won five golds when I skied. In everything, I came in first. And Georgetown â that was great. I was losing, but it was the best training because this was Atlanta. Here we are, like, crĂ¨me de la crĂ¨me, and there is no doubt about it, that I ’ m going to lose big. And, you know, I ’ m just thinking,”Oh, my god, my whole family got in a van and drove down here from Pennsylvania.”And, you know, I was the only female U. S. sprinter. So they call us out and, you know â”Ladies, you have one minute.”And I remember putting my blocks in and just feeling horrified because there was just this murmur coming over the crowd, like, the ones who are close enough to the starting line to see. And I ’ m like,”I know! Look! This isn ’ t right.”And I ’ m thinking that ’ s my last card to play here ; if I ’ m not going to beat these girls, I ’ m going to mess their heads a little, you know? I mean, it was definitely the”Rocky IV”sensation of me versus Germany, and everyone else â Estonia and Poland â was in this heat. And the gun went off, and all I remember was finishing last and fighting back tears of frustration and incredible â incredible â this feeling of just being overwhelmed. And I had to think,”Why did I do this?”If I had won everything â but it was like, what was the point? All this training â I had transformed my life. I became a collegiate athlete, you know. I became an Olympic athlete. And it made me really think about how the achievement was getting there. I mean, the fact that I set my sights, just a year and three months before, on becoming an Olympic athlete and saying,”Here ’ s my life going in this direction â and I want to take it here for a while, and just seeing how far I could push it.”And the fact that I asked for help â how many people jumped on board? How many people gave of their time and their expertise, and their patience, to deal with me? And that was this collective glory â that there was, you know, 50 people behind me that had joined in this incredible experience of going to Atlanta. So, I apply this sort of philosophy now to everything I do : sitting back and realizing the progression, how far you ’ ve come at this day to this goal, you know. It ’ s important to focus on a goal, I think, but also recognize the progression on the way there and how you ’ ve grown as a person. That ’ s the achievement, I think. That ’ s the real achievement. SS : Why don ’ t you show them your legs? AM : Oh, sure. SS : You know, show us more than one set of legs. AM : Well, these are my pretty legs. No, these are my cosmetic legs, actually, and they ’ re absolutely beautiful. You ’ ve got to come up and see them. There are hair follicles on them, and I can paint my toenails. And, seriously, like, I can wear heels. Like, you guys don ’ t understand what that ’ s like to be able to just go into a shoe store and buy whatever you want. SS : You got to pick your height? AM : I got to pick my height, exactly. Patrick Ewing, who played for Georgetown in the ’ 80s, comes back every summer. And I had incessant fun making fun of him in the training room because he ’ d come in with foot injuries. I ’ m like,”Get it off! Don ’ t worry about it, you know. You can be eight feet tall. Just take them off.”He didn ’ t find it as humorous as I did, anyway. OK, now, these are my sprinting legs, made of carbon graphite, like I said, and I ’ ve got to make sure I ’ ve got the right socket. No, I ’ ve got so many legs in here. These are â do you want to hold that actually? That ’ s another leg I have for, like, tennis and softball. It has a shock absorber in it so it, like,”Shhhh,”makes this neat sound when you jump around on it. All right. And then this is the **silicon** sheath I roll over, to keep it on. Which, when

I sweat, you know, I ’ m pistoning out of it. SS : Are you a different height? AM : In these? SS : In these. AM : I don ’ t know. I don ’ t think so. I may be a little taller. I actually can put both of them on. SS : She can ’ t really stand on these legs. She has to be moving, so... AM : Yeah, I definitely have to be moving, and balance is a little bit of an art in them. But without having the **silicon** sock, I ’ m just going to try slip in it. And so, I run on these, and have shocked half the world on these. These are supposed to simulate the actual form of a sprinter when they run. If you ever watch a sprinter, the ball of their foot is the only thing that ever hits the track. So when I stand in these legs, my hamstring and my glutes are contracted, as they would be had I had feet and were standing on the ball of my feet. AM : It ’ s a company in San Diego called Flex-Foot. And I was a guinea pig, as I hope to continue to be in every new form of prosthetic limbs that come out. But actually these, like I said, are still the actual prototype. I need to get some new ones because the last meet I was at, they were everywhere. You know, it ’ s like a big â it ’ s come full circle. Moderator : Aimee and the designer of them will be at TEDMED 2, and we ’ ll talk about the design of them. AM : Yes, we ’ ll do that. SS : Yes, there you go. AM : So, these are the sprint legs, and I can put my other... SS : Can you tell about who designed your other legs? AM : Yes. These I got in a place called Bournemouth, England, about two hours south of London, and I ’ m the only person in the United States with these, which is a crime because they are so beautiful. And I don ’ t even mean, like, because of the toes and everything. For me, while I ’ m such a serious athlete on the track, I want to be feminine off the track, and I think it ’ s so important not to be limited in any capacity, whether it ’ s, you know, your mobility or even fashion. I mean, I love the fact that I can go in anywhere and pick out what I want â the shoes I want, the skirts I want â and I ’ m hoping to try to bring these over here and make them accessible to a lot of people. They ’ re also **silicon**. This is a really basic, basic prosthetic limb under here. It ’ s like a Barbie foot under this. It is. It ’ s just stuck in this position, so I have to wear a two-inch heel. And, I mean, it ’ s really â let me take this off so you can see it. I don ’ t know how good you can see it, but, like, it really is. There ’ re veins on the feet, and then my heel is pink, and my Achilles ’ tendon â that moves a little bit. And it ’ s really an amazing store. I got them a year and two weeks ago. And this is just a **silicon** piece of skin. I mean, what happened was, two years ago this man in Belgium was saying,”God, if I can go to Madame Tussauds ’ wax museum and see Jerry Hall replicated down to the color of her eyes, looking so real as if she breathed, why can ’ t they build a limb for someone that looks like a leg, or an arm, or a hand?”I mean, they make ears for burn victims. They do amazing stuff with **silicon**. SS : Two weeks ago, Aimee was up for the Arthur Ashe award at the ESPYs. And she came into town and she rushed around and she said,”I have to buy some new shoes!”We ’ re an hour before the ESPYs, and she thought she ’ d gotten a two-inch heel but she ’ d actually bought a three-inch heel. AM : And this poses a problem for me, because it means I ’ m walking like that all night long. SS : For 45 minutes. Luckily, the hotel was terrific. They got someone to come in and saw off the shoes. AM : I said to the receptionist â I mean, I am just harried, and Sheryl ’ s at my side â I said,”Look, do you have anybody here who could help me? Because I have this problem...”You know, at first they were just going to write me off, like,”If you don ’ t like your shoes, sorry. It ’ s too late.””No, no, no, no. I ’ ve got these **special** feet that need a two-inch heel. I have a three-inch heel. I need a little bit off.”They didn ’ t even want to go there. They didn ’ t even want to touch that one. They just did it. No, these legs are great. I ’ m actually going back in a couple of weeks to get some improvements. I want to get legs like these made for flat feet so I can wear sneakers, because I can ’ t with these ones. So... Moderator : That ’ s it. SS : That ’ s Aimee Mullins.

Text Sample 225: POS Dim 2 – 1998 – Score 6.00 – 1998/t000118.txt

Back in 1992, I started working for a company called Interval Research, which was just then being founded by David Lidell and Paul Allen as a for-profit research enterprise in Silicon Valley. I met with David to talk about what I might do in his company. I was just coming out of a failed virtual reality business and supporting myself by being on the speaking circuit and writing books — after twenty years or so in the computer game industry having ideas that people didn ’ t think they could sell. And David and I discovered that we had a question in common, that we really wanted the answer to, and that was,”Why hasn ’ t anybody built any computer games for little girls?”Why is that? It can ’ t just be a giant sexist conspiracy. These people aren ’ t that smart. There ’ s six billion dollars on the table. They would go for it if they could figure out how. So, what is the deal here? And as we thought about our goals — I should say that Interval is really a humanistic institution, in the **classical** sense that humanism, at its best, finds a way to combine clear-eyed empirical research with a set of core values that fundamentally love and respect people. The basic idea of humanism is the improvable quality of life ; that we can do good things, that there are things worth doing because they ’ re good things to do, and that clear-eyed empiricism can help us figure out how to do them. So, contrary to popular belief, there is not a conflict of interest between empiricism and values. And Interval Research is kind of the living example of how that can be true. So David and I decided to go find out, through the best research we could muster, what it would take to get a little girl to put her hands on a computer, to achieve the level of comfort and ease with the technology that little boys have because they play video games. We spent two and a half years conducting research ; we spent another year and a half in advance development. Then we formed a spin-off company. In the research phase of the project at Interval, we partnered with a company called Cheskin Research, and these people — Davis Masten and Christopher Ireland — changed my mind entirely about what market research was and what it could be. They taught me how to look and see, and they did not do the incredibly stupid thing of saying to a child,”Of all these things we already make you, which do you like best?”— which gives you zero answers that are usable. So, what we did for the first two and a half years was four things : We did an extensive review of the literature in related fields, like cognitive psychology, spatial cognition, gender studies, play theory, sociology, primatology. Thank you Frans de Waal, wherever you are, I love you and I ’ d give anything to meet you. After we had done that with a pretty large team of people and discovered what we thought the salient issues were with girls and boys and playing — because, after all, that ’ s really what this is about — we moved to the second phase of our work, where we interviewed adult experts in **academia**, some of the people who ’ d produced the literature that we found relevant. Also, we did focus groups with people who were on the ground with kids every day, like playground supervisors. We talked to them, confirmed some hypotheses and identified some serious questions about gender difference and play. Then we did what I consider to be the heart of the work : interviewed 1, 100 children, boys and girls, ages seven to 12, all over the United States — except for Silicon Valley, Boston and Austin because we knew that their little families would have millions of computers in them and they wouldn ’ t be a representative sample. And at the end of those remarkable conversations with kids and their best friends across the United States, after two years, we pulled together some survey data from another 10, 000 children, drew up a set up of what we thought were the key findings of our research, and spent another year transforming them into design heuristics, for designing computer-based products — and, in fact, any kind of products — for little girls, ages eight to 12. And we spent that time designing interactive prototypes for computer software and testing them with little girls. In 1996, in November, we formed the company Purple Moon which was a spinoff of Interval Research, and our chief investors were Interval Research, Vulcan Northwest, Institutional Venture Partners and Allen and Company. We launched a website on September 2nd that has now served 25 million pages, and has 42, 000 registered young girl users. They visit an average of one and a half times a day, spend an average of 35 minutes a visit, and look at 50 pages a visit. So we feel that we ' ve formed a successful online community with girls. We launched two titles in October —"Rockett ' s New School"— the first of a series of products — is about a character called Rockett beginning her first day of school in eighth grade at a brand new place, with a blank slate, which allows girls to play with the question of,"What will I be like when I ' m older?""What ' s it going to be like to be in high school or junior high school? Who are my friends?"; to exercise the love of social complexity and the narrative intelligence that drives most of their play behavior ; and which embeds in it values about noticing that we have lots of choices in our lives and the ways that we conduct ourselves. The other title that we launched is called"Secret Paths in the Forest,"which addresses the more fantasy-oriented, inner lives of girls. These two titles both showed up in the top 50 entertainment titles in PC Data — entertainment titles in PC Data in December, right up there with"John Madden Football,"which thrills me to death. So, we ' re real, and we ' ve touched several hundreds of thousands of little girls. We ' ve made half-a-billion impressions with marketing and PR for this brand, Purple Moon. Ninety-six percent of them, roughly, have been positive ; four percent of them have been"other."I want to talk about the other, because the politics of this enterprise, in a way, have been the most fascinating part of it, for me. There are really two kinds of negative reviews that we ' ve received. One kind of reviewer is a male gamer who thinks he knows what games ought to be, and won ' t show the product to little girls. The other kind of reviewer is a certain flavor of feminist who thinks they know what little girls ought to be. And so it ' s funny to me that these interesting, odd bedfellows have one thing in common : they don ' t listen to little girls. They haven ' t looked at children and they ' re certainly not demonstrating any love for them. I ' d like to play you some voices of little girls from the two-and-a-half years of research that we did — actually, some of the voices are more recent. And these voices will be accompanied by photographs that they took for us of their lives, of the things that they value and care about. These are pictures the girls themselves never saw, but they gave to us This is the stuff those reviewers don ' t know about and aren ' t listening to and this is the kind of research I recommend to those who want to do humanistic work. Girl 1 : Yeah, my character is usually a tomboy. Hers is more into boys. Girl 2 : Uh, yeah. Girl 1 : We have — in the very beginning of the whole game, always we do this : we each have a piece of paper ; we write down our name, our age — are we rich, very rich, not rich, poor, medium, wealthy, boyfriends, dogs, pets — what else — sisters, brothers, and all those. Girl 2 : Divorced — parents divorced, maybe. Girl 3 : This is my pretend [unclear] one. Girl 4 : We make a school newspaper on the computer. Girl 5 : For a girl ' s game also usually they ' ll have really pretty scenery with clouds and flowers. Girl 6 : Like, if you were a girl and you were really adventurous and a real big tomboy, you would think that girls ' games were kinda sissy. Girl 7 : I run track, I played soccer, I play basketball, and I love a lot of things to do. And sometimes I feel like I can ’ t really enjoy myself unless it ’ s like a **vacation**, like when I get Mondays and all those days off. Girl 8 : Well, sometimes there is a lot of stuff going on because I have music lessons and I ’ m on swim team — all this different stuff that I have to do, and sometimes it gets overwhelming. Girl 9 : My friend Justine kinda took my friend Kelly, and now they ’ re being mean to me. Girl 10 : Well, sometimes it gets annoying when your brothers and sisters, or brother or sister, when they copy you and you get your idea first and they take your idea and they do it themselves. Girl 11 : Because my older sister, she gets everything and, like, when I ask my mom for something, she ’ ll say,”No”— all the time. But she gives my sister everything. Brenda Laurel : I want to show you, real quickly, just a minute of”Rockett ’ s Tricky Decision,”which went gold two days ago. Let ’ s hope it ’ s really stable. This is the second day in Rockett ’ s life. The reason I ’ m showing you this is I ’ m hoping that the scene that I ’ m going to show you will look familiar and sound familiar, now that you ’ ve listened to some girls ’ voices. And you can see how we ’ ve tried to incorporate the issues that matter to them in the game that we ’ ve created. Miko : Hey Rockett! C ’ mere! Rockett : Hi Miko! What ’ s going on? Miko : Did you hear about Nakilia ’ s big Halloween party this weekend? She asked me to make sure you knew about it. Nakilia invited Reuben too, but — Rockett : But what? Isn ’ t he coming? Miko : I don ’ t think so. I mean, I heard his band is playing at another party the same night. Rockett : Really? What other party? Girl : Max ’ s party is going to be so cool, Whitney. He ’ s invited all the best people. BL : I ’ m going to **fast-forward** to the decision point because I know I don ’ t have a lot of time. After this awful event occurs, Rocket gets to decide how she feels about it. Rockett : Who ’ d want to show up at that party anyway? I could get invited to that party any day if I wanted to. Gee, I doubt I ’ ll make Max ’ s best people list. BL : OK, so we ’ re going to emotionally **navigate**. If we were playing the game, that ’ s what we ’ d do. If at any time during the game we want to learn more about the characters, we can go into this hidden hallway, and I ’ ll quickly just show you the interface. We can, for example, go find Miko ’ s locker and get some more information about her. Oops, I turned the wrong way. But you get the general idea of the product. I wanted to show you the ways, innocuous as they seem, in which we ’ re incorporating what we ’ ve learned about girls — their desires to experience greater emotional flexibility, and to play around with the social complexity of their lives. I want to make the point that what we ’ re giving girls, I think, through this effort, is a kind of validation, a sense of being seen. And a sense of the choices that are available in their lives. We love them. We see them. We ’ re not trying to tell them who they ought to be. But we ’ re really, really happy about who they are. It turns out they ’ re really great. I want to close by showing you a videotape that ’ s a version of a future game in the Rockett series that our graphic artists and design people put together, that we feel would please that four percent of reviewers.”Rockett 28!”Rockett : It ’ s like I ’ m just waking up, you know? BL : Thanks.

Text Sample 226: POS Dim 2 – 1998 – Score 4.00 – 1998/t000200.txt

You hear that this is the era of environment â or biology, or information technology... Well, it ’ s the era of a lot of different things that we ’ re in right now. But one thing for sure : it ’ s the era of change. There ’ s more change going on than ever has occurred in the history of human life on earth. And you all sort of know it, but it ’ s hard to get it so that you really understand it. And I ’ ve tried to put together something that ’ s a good start for this. I ’ ve tried to show in this â though the color doesn ’ t come out â that what I ’ m concerned with is the little 50-year time bubble that you are in. You tend to be interested in a generation past, a generation future â your parents, your kids, things you can change over the next few decades â and this 50-year time bubble you kind of move along in. And in that 50 years, if you look at the population curve, you find the population of humans on the earth more than doubles and we ’ re up three-and-a-half times since I was born. When you have a new baby, by the time that kid gets out of high school more people will be added than existed on earth when I was born. This is unprecedented, and it ’ s big. Where it goes in the future is questioned. So that ’ s the human part. Now, the human part related to animals : look at the left side of that. What I call the human portion â humans and their livestock and pets â versus the natural portion â all the other wild animals and just â these are vertebrates and all the birds, etc., in the land and air, not in the water. How does it balance? Certainly, 10, 000 years ago, the civilization ’ s beginning, the human portion was less than one tenth of one percent. Let ’ s look at it now. You follow this curve and you see the whiter spot in the middle â that ’ s your 50- year time bubble. Humans, livestock and pets are now 97 percent of that integrated total mass on earth and all wild nature is three percent. We have won. The next generation doesn ’ t even have to worry about this game â it is over. And the biggest problem came in the last 25 years : it went from 25 percent up to that 97 percent. And this really is a sobering picture upon realizing that we, humans, are in charge of life on earth ; we ’ re like the capricious Gods of old Greek myths, kind of playing with life â and not a great deal of wisdom injected into it. Now, the third curve is information technology. This is Moore ’ s Law plotted here, which relates to density of information, but it has been pretty good for showing a lot of other things about information technology â computers, their use, Internet, etc. And what ’ s important is it just goes straight up through the top of the curve, and has no real limits to it. Now try and contrast these. This is the size of the earth going through that same â â frame. And to make it really clear, I ’ ve put all four on one graph. There ’ s no need to see the little detailed words on it. That first one is humans-versus-nature ; we ’ ve won, there ’ s no more gain. Human population. And so if you ’ re looking for growth industries to get into, that ’ s not a good one â protecting natural creatures. Human population is going up ; it ’ s going to continue for quite a while. Good business in obstetricians, morticians, and farming, housing, etc. â they all deal with human bodies, which require being fed, transported, housed and so on. And the information technology, which connects to our brains, has no limit â now, that is a wonderful field to be in. You ’ re looking for growth opportunity? It ’ s just going up through the roof. And then, the size of the Earth. Somehow making these all compatible with the Earth looks like a pretty bad industry to be involved with. So, that ’ s the stage out of all this. I find, for reasons I don ’ t understand, I really do have a goal. And the goal is that the world be desirable and sustainable when my kids reach my age â and I think that ’ s â in other words, the next generation. I think that ’ s a goal that we probably all share. I think it ’ s a hopeless goal. Technologically, it ’ s achievable ; economically, it ’ s achievable ; politically, it means sort of the habits, institutions of people â it ’ s impossible. The institutions of the past with all their inertia are just irrelevant for the future, except they ’ re there and we have to deal with them. I spend about 15 percent of my time trying to save the world, the other 85 percent, the usual â and whatever else we devote ourselves to. And in that 15 percent, the main focus is on human mind, thinking skills, somehow trying to unleash kids from the straightjacket of school, which is putting information and dogma into them, get them so they really think, ask tough questions, argue about serious subjects, don ’ t believe everything that ’ s in the book, think broadly or creative. They can be. Our school systems are very flawed and do not reward you for the things that are important in life or for the survival of civilization ; they reward you for a lot of learning and sopping up stuff. We can ’ t go into that today because there isn ’ t time â it ’ s a broad subject. One thing for sure, in the future there is an essential feature â necessary, but not sufficient â which is doing more with less. We ’ ve got to be doing things with more efficiency using less energy, less material. Your great-great grandparents got by on muscle power, and yet we all think there ’ s this huge power that ’ s essential for our lifestyle. And with all the wonderful technology we have we can do things that are much more efficient : conserve, recycle, etc. Let me just rush very quickly through things that we ’ ve done. Human-powered airplane â Gossamer Condor sort of started me in this direction in 1976 and 77, winning the Kremer prize in aviation history, followed by the Albatross. And we began making various odd planes and creatures. Here ’ s a giant flying replica of a pterosaur that has no tail. Trying to have it fly straight is like trying to shoot an arrow with the feathered end forward. It was a tough job, and boy it made me have a lot of respect for nature. This was the full size of the original creature. We did things on land, in the air, on water â vehicles of all different kinds, usually with some electronics or electric power systems in them. I find they ’ re all the same, whether its land, air or water. I ’ ll be focusing on the air here. This is a solar-powered airplane â 165 miles carrying a person from France to England as a symbol that solar power is going to be an important part of our future. Then we did the solar car for General Motors â the Sunracer â that won the race in Australia. We got a lot of people thinking about electric cars, what you could do with them. A few years later, when we suggested to GM that now is the time and we could do a thing called the Impact, they sponsored it, and here ’ s the Impact that we developed with them on their programs. This is the demonstrator. And they put huge effort into turning it into a commercial product. With that preamble, let ’ s show the first two-minute videotape, which shows a little airplane for surveillance and moving to a giant airplane. Narrator : A tiny airplane, the AV Pointer serves for surveillance â in effect, a pair of roving eyeglasses. A cutting-edge example of where miniaturization can lead if the operator is remote from the vehicle. It is convenient to carry, assemble and launch by hand. Battery-powered, it is silent and rarely noticed. It sends high-resolution video pictures back to the operator. With onboard GPS, it can **navigate** autonomously, and it is rugged enough to self-land without damage. The modern sailplane is superbly efficient. Some can glide as flat as 60 feet forward for every foot of descent. They are powered only by the energy they can extract from the atmosphere â an atmosphere nature stirs up by solar energy. Humans and soaring birds have found nature to be generous in providing replenishable energy. Sailplanes have flown over 1, 000 miles, and the altitude record is over 50, 000 feet. The Solar Challenger was made to serve as a symbol that photovoltaic cells can produce real power and will be part of the world ’ s energy future. In 1981, it flew 163 miles from Paris to England, solely on the power of sunbeams, and established a basis for the Pathfinder. The message from all these vehicles is that ideas and technology can be harnessed to produce remarkable gains in doing more with less â gains that can help us attain a desirable balance between technology and nature. The stakes are high as we speed toward a challenging future. Buckminster Fuller said it clearly :”there are no passengers on spaceship Earth, only crew. We, the crew, can and must do more with less â much less.”Paul MacCready : If we could have the second video, the one-minute, put in as quickly as you can, which â this will show the Pathfinder airplane in some flights this past year in Hawaii, and will show a sequence of some of the beauty behind it after it had just flown to 71, 530 feet â higher than any propeller airplane has ever flown. It ' s amazing : just on the puny power of the sun â by having a **super** lightweight plane, you ' re able to get it up there. It ' s part of a long-term program NASA sponsored. And we worked very closely with the whole thing being a team effort, and with wonderful results like that flight. And we ' re working on a bigger plane â 220-foot span â and an intermediate-size, one with a regenerative fuel cell that can store excess energy during the day, feed it back at night, and stay up 65, 000 feet for months at a time. Ray Morgan ' s voice will come in here. There he ' s the project manager. Anything they do is certainly a team effort. He ran this program. Here ' s... some things he showed as a celebration at the very end. Ray Morgan : We ' d just ended a seven-month deployment of Hawaii. For those who live on the mainland, it was tough being away from home. The friendly support, the quiet confidence, congenial hospitality shown by our Hawaiian and military hosts â this is starting â made the experience enjoyable and unforgettable. PM : We have real-time IR scans going out through the Internet while the plane is flying. And it ' s exploring without polluting the stratosphere. That ' s its goal : the stratosphere, the blanket that really controls the radiation of the earth and permits life on earth to be the success that it is â probing that is very important. And also we consider it as a sort of poor man ' s stationary satellite, because it can stay right overhead for months at a time, 2, 000 times closer than the real GFC synchronous satellite. We couldn ' t bring one here to fly it and show you. But now let ' s look at the other end. In the video you saw that nine-pound or eight-pound Pointer airplane surveillance drone that Keenan has developed and just done a remarkable job. Where some have servos that have gotten down to, oh, 18 or 25 grams, his weigh one-third of a gram. And what he ' s going to bring out here is a surveillance drone that weighs about 2 ounces â that includes the video camera, the batteries that run it, the telemetry, the receiver and so on. And we ' ll fly it, we hope, with the same success that we had last night when we did the **practice**. So Matt Keenan, just any time you ' re â all right â ready to let her go. But first, we ' re going to make sure that it ' s appearing on the screen, so you see what it sees. You can imagine yourself being a mouse or fly inside of it, looking out of its camera. Matt Keenan : It ' s switched on. PM : But now we ' re trying to get the video. There we go. MK : Can you bring up the house lights? PM : Yeah, the house lights and we ' ll see you all better and be able to fly the plane better. MK : All right, we ' ll try to do a few laps around and bring it back in. Here we go. PM : The video worked right for the first few and I don ' t know why it â there it goes. Oh, that was only a minute, but I think you ' d be safe to have that near the end of the flight, perhaps. We get to do the classic. All right. If this hits you, it will not hurt you. OK. Thank you very much. Thank you. But now, as they say in infomercials, we have something much better for you, which we ' re working on : planes that are only six inches â 15 centimeters â in size. And Matt ' s plane was on the cover of Popular Science last month, showing what this can lead to. And in a while, something this size will have GPS and a video camera in it. We ' ve had one of these fly nine miles through the air at 35 miles an hour with just a little battery in it. But there ' s a lot of technology going. There are just milestones along the way of some remarkable things. This one doesn ' t have the video in it, but you get a little feel from what it can do. OK, here we go. MK : Sorry. OK. PM : If you can pass it down when you ' re done. Yeah, I think â I lost a little orientation ; I looked up into this light. It hit the building. And the building was poorly placed, actually. But you ' re beginning to see what can be done. We ' re working on projects now â even wing-flapping things the size of hawk moths â DARPA contracts, working with Caltech, UCLA. Where all this leads, I don ’ t know. Is it practical? I don ’ t know. But like any basic research, when you ’ re really forced to do things that are way beyond existing technology, you can get there with micro-technology, nanotechnology. You can do amazing things when you realize what nature has been doing all along. As you get to these small scales, you realize we have a lot to learn from nature â not with 747s â but when you get down to the nature ’ s realm, nature has 200 million years of experience. It never makes a mistake. Because if you make a mistake, you don ’ t leave any progeny. We should have nothing but success stories from nature, for you or for birds, and we ’ re learning a lot from its fascinating subjects. In concluding, I want to get back to the big picture and I have just two final slides to try and put it in perspective. The first I ’ ll just read. At last, I put in three sentences and had it say what I wanted. Over billions of years on a unique sphere, chance has painted a thin covering of life â complex, improbable, wonderful and fragile. Suddenly, we humans â a recently arrived species, no longer subject to the checks and balances inherent in nature â have grown in population, technology and intelligence to a position of terrible power. We now wield the paintbrush. And that ’ s serious : we ’ re not very bright. We ’ re short on wisdom ; we ’ re high on technology. Where ’ s it going to lead? Well, inspired by the sentences, I decided to wield the paintbrush. Every 25 years I do a picture. Here ’ s the one â tries to show that the world isn ’ t getting any bigger. Sort of a timeline, very non-linear scale, nature rates and trilobites and dinosaurs, and eventually we saw some humans with caves... Birds were flying overhead, after pterosaurs. And then we get to the civilization above the little TV set with a gun on it. Then traffic jams, and power systems, and some dots for digital. Where it ’ s going to lead â I have no idea. And so I just put robotic and natural cockroaches out there, but you can fill in whatever you want. This is not a forecast. This is a **warning**, and we have to think seriously about it. And that time when this is happening is not 100 years or 500 years. Things are going on this decade, next decade ; it ’ s a very short time that we have to decide what we are going to do. And if we can get some agreement on where we want the world to be â desirable, sustainable when your kids reach your age â I think we actually can reach it. Now, I said this was a **warning**, not a forecast. That was before â I painted this before we started in on making robotic versions of hawk moths and cockroaches, and now I ’ m beginning to wonder seriously â was this more of a forecast than I wanted? I personally think the surviving intelligent life form on earth is not going to be carbon-based ; it ’ s going to be silicon-based. And so where it all goes, I don ’ t know. The one final bit of sparkle we ’ ll put in at the very end here is an utterly impractical flight vehicle, which is a little ornithopter wing-flapping device that â rubber-band powered â that we ’ ll show you. MK : 32 gram. Sorry, one gram. PM : Last night we gave it a few too many turns and it tried to bash the roof out also. It ’ s about a gram. The tube there ’ s hollow, about paper-thin. And if this lands on you, I assure you it will not hurt you. But if you reach out to grab it or hold it, you will destroy it. So, be gentle, just act like a wooden Indian or something. And when it comes down â and we ’ ll see how it goes. We consider this to be sort of the spirit of TED. And you wonder, is it practical? And it turns out if I had not been â Unfortunately, we have some light bulb changes. We can probably get it down, but it ’ s possible it ’ s gone up to a greater destiny up there â â than it ever had. And I wanted to make â just â But I want to make just two points. One is, you think it ’ s frivolous ; there ’ s nothing to it. And yet if I had not been making ornithopters like that, a little bit cruder, in 1939 â a long, long time ago â there wouldn ’ t have been a Gossamer Condor, there wouldn ’ t have been an Albatross, a Solar Challenger, there wouldn ’ t be an Impact car, there wouldn ’ t be a mandate on zero-emission vehicles in California. A lot of these things â or similar â would have happened some time, probably a decade later. I didn ’ t realize at the time I was doing inquiry-based, hands-on things with teams, like they ’ re trying to get in education systems. So I think that, as a symbol, it ’ s important. And I believe that also is important. You can think of it as a sort of a symbol for learning and TED that somehow gets you thinking of technology and nature, and puts it all together in things that are â that make this conference, I think, more important than any that ’ s taken place in this country in this decade. Thank you.

Text Sample 227: POS Dim 2 – 2013 – Score 11.00 – 2013/t002038.txt

This will not be a speech like any one I have ever given. I will talk to you today about the failure of leadership in global politics and in our globalizing economy. And I won ’ t provide some feel-good, ready-made solutions. But I will in the end urge you to **rethink**, actually take risks, and get involved in what I see as a global evolution of democracy. Failure of leadership. What is the failure of leadership today? And why is our democracy not working? Well, I believe that the failure of leadership is the fact that we have taken you out of the process. So let me, from my personal experiences, give you an insight, so that you can step back and maybe understand why it is so difficult to cope with the challenges of today and why politics is going down a blind alley. Let ’ s start from the beginning. Let ’ s start from democracy. Well, if you go back to the Ancient Greeks, it was a revelation, a discovery, that we had the potential, together, to be masters of our own fate, to be able to examine, to learn, to imagine, and then to design a better life. And democracy was the political innovation which protected this freedom, because we were liberated from fear so that our minds in fact, whether they be despots or dogmas, could be the protagonists. Democracy was the political innovation that allowed us to limit the power, whether it was of tyrants or of high priests, their natural tendency to **maximize** power and wealth. Well, I first began to understand this when I was 14 years old. I used to, to try to avoid homework, sneak down to the living room and listen to my parents and their friends debate heatedly. You see, then Greece was under control of a very powerful establishment which was strangling the country, and my father was heading a promising movement to reimagine Greece, to imagine a Greece where freedom reigned and where, maybe, the people, the citizens, could actually rule their own country. I used to join him in many of the campaigns, and you can see me here next to him. I ’ m the younger one there, to the side. You may not recognize me because I used to part my hair differently there. So in 1967, elections were coming, things were going well in the campaign, the house was electric. We really could sense that there was going to be a major progressive change in Greece. Then one night, military trucks drive up to our house. Soldiers storm the door. They find me up on the top terrace. A sergeant comes up to me with a machine gun, puts it to my head, and says,”Tell me where your father is or I will kill you.”My father, hiding **nearby**, reveals himself, and was summarily taken to prison. Well, we survived, but democracy did not. Seven brutal years of dictatorship which we spent in exile. Now, today, our democracies are again facing a moment of truth. Let me tell you a story. Sunday evening, Brussels, April 2010. I ’ m sitting with my counterparts in the European Union. I had just been elected prime minister, but I had the unhappy privilege of revealing a truth that our deficit was not 6 percent, as had been officially reported only a few days earlier before the elections by the previous government, but actually 15. 6 percent. But the deficit was only the symptom of much deeper problems that Greece was facing, and I had been elected on a mandate, a **mission**, actually, to tackle these problems, whether it was lack of transparency and accountability in governance, or whether it was a clientelistic state offering favors to the powerful — tax avoidance abetted and aided by a global tax evasion system, politics and media captured by **special** interests. But despite our electoral mandate, the markets mistrusted us. Our borrowing costs were skyrocketing, and we were facing possible default. So I went to Brussels on a **mission** to make the case for a united European response, one that would calm the markets and give us the time to make the necessary reforms. But time we didn ' t get. Picture yourselves around the table in Brussels. Negotiations are difficult, the tensions are high, progress is slow, and then, 10 minutes to 2, a prime minister shouts out,"We have to finish in 10 minutes."I said,"Why? These are important decisions. Let ' s deliberate a little bit longer."Another prime minister comes in and says,"No, we have to have an agreement now, because in 10 minutes, the markets are opening up in Japan, and there will be havoc in the global economy."We quickly came to a decision in those 10 minutes. This time it was not the military, but the markets, that put a gun to our collective heads. What followed were the most difficult decisions in my life, painful to me, painful to my countrymen, imposing cuts, austerity, often on those not to blame for the crisis. With these sacrifices, Greece did avoid bankruptcy and the eurozone avoided a collapse. Greece, yes, triggered the Euro crisis, and some people blame me for pulling the trigger. But I think today that most would agree that Greece was only a symptom of much deeper structural problems in the eurozone, vulnerabilities in the wider global economic system, vulnerabilities of our democracies. Our democracies are trapped by systems too big to fail, or, more accurately, too big to control. Our democracies are weakened in the global economy with players that can evade laws, evade taxes, evade environmental or labor standards. Our democracies are undermined by the growing inequality and the growing concentration of power and wealth, lobbies, corruption, the speed of the markets or simply the fact that we sometimes fear an impending disaster, have constrained our democracies, and they have constrained our capacity to imagine and actually use the potential, your potential, in finding solutions. Greece, you see, was only a preview of what is in store for us all. I, overly optimistically, had hoped that this crisis was an opportunity for Greece, for Europe, for the world, to make radical democratic transformations in our institutions. Instead, I had a very humbling experience. In Brussels, when we tried desperately again and again to find common solutions, I realized that not one, not one of us, had ever dealt with a similar crisis. But worse, we were trapped by our collective ignorance. We were led by our fears. And our fears led to a blind faith in the orthodoxy of austerity. Instead of reaching out to the common or the collective wisdom in our societies, investing in it to find more creative solutions, we reverted to political posturing. And then we were surprised when every **ad** hoc new measure didn ' t bring an end to the crisis, and of course that made it very easy to look for a whipping boy for our collective European failure, and of course that was Greece. Those profligate, idle, ouzo- swilling, Zorba-dancing Greeks, they are the problem. Punish them! Well, a convenient but unfounded stereotype that sometimes hurt even more than austerity itself. But let me warn you, this is not just about Greece. This could be the pattern that leaders follow again and again when we deal with these complex, cross- border problems, whether it ' s climate change, whether it ' s migration, whether it ' s the **financial** system. That is, abandoning our collective power to imagine our potential, falling victims to our fears, our stereotypes, our dogmas, taking our citizens out of the process rather than building the process around our citizens. And doing so will only test the faith of our citizens, of our peoples, even more in the democratic process. It ' s no wonder that many political leaders, and I don ' t exclude myself, have lost the trust of our people. When riot police have to protect parliaments, a scene which is increasingly common around the world, then there ' s something deeply wrong with our democracies. That ' s why I called for a referendum to have the Greek people own and decide on the terms of the rescue package. My European counterparts, some of them, at least, said,"You can ' t do this. There will be havoc in the markets again."I said,"We need to, before we restore confidence in the markets, we need to restore confidence and trust amongst our people."Since leaving office, I have had time to reflect. We have weathered the storm, in Greece and in Europe, but we remain challenged. If politics is the power to imagine and use our potential, well then 60-percent youth unemployment in Greece, and in other countries, certainly is a lack of imagination if not a lack of compassion. So far, we ' ve thrown economics at the problem, actually mostly austerity, and certainly we could have designed alternatives, a different strategy, a green stimulus for green jobs, or mutualized debt, Eurobonds which would support countries in need from market pressures, these would have been much more viable alternatives. Yet I have come to believe that the problem is not so much one of economics as it is one of democracy. So let ' s try something else. Let ' s see how we can bring people back to the process. Let ' s throw democracy at the problem. Again, the Ancient Greeks, with all their shortcomings, believed in the wisdom of the crowd at their best moments. In people we trust. Democracy could not work without the citizens deliberating, debating, taking on public responsibilities for public affairs. Average citizens often were chosen for citizen juries to decide on critical matters of the day. Science, theater, research, philosophy, games of the mind and the body, they were **daily** exercises. Actually they were an education for participation, for the potential, for growing the potential of our citizens. And those who shunned politics, well, they were idiots. You see, in Ancient Greece, in ancient Athens, that term originated there."Idiot"comes from the root"idio,"oneself. A person who is self-centered, secluded, excluded, someone who doesn ' t participate or even examine public affairs. And participation took place in the agora, the agora having two meanings, both a marketplace and a place where there was political deliberation. You see, markets and politics then were one, unified, accessible, transparent, because they gave power to the people. They serve the demos, democracy. Above government, above markets was the direct rule of the people. Today we have globalized the markets but we have not globalized our democratic institutions. So our politicians are limited to local politics, while our citizens, even though they see a great potential, are prey to forces beyond their control. So how then do we reunite the two halves of the agora? How do we democratize globalization? And I ' m not talking about the necessary reforms of the United Nations or the G20. I ' m talking about, how do we secure the space, the demos, the **platform** of values, so that we can tap into all of your potential? Well, this is exactly where I think Europe fits in. Europe, despite its recent failures, is the world ' s most successful cross-border peace experiment. So let ' s see if it can ' t be an experiment in global democracy, a new kind of democracy. Let ' s see if we can ' t design a European agora, not simply for products and **services**, but for our citizens, where they can work together, deliberate, learn from each other, exchange between art and cultures, where they can come up with creative solutions. Let ' s imagine that European citizens actually have the power to vote directly for a European president, or citizen juries chosen by lottery which can deliberate on critical and controversial issues, a European-wide referendum where our citizens, as the lawmakers, vote on future treaties. And here ' s an idea : Why not have the first truly European citizens by giving our immigrants, not Greek or German or Swedish citizenship, but a European citizenship? And make sure we actually empower the unemployed by giving them a voucher scholarship where they can choose to study anywhere in Europe. Where our common identity is democracy, where our education is through participation, and where participation builds trust and solidarity rather than exclusion and xenophobia. Europe of and by the people, a Europe, an experiment in deepening and widening democracy beyond borders. Now, some might accuse me of being naive, putting my faith in the power and the wisdom of the people. Well, after decades in politics, I am also a pragmatist. Believe me, I have been, I am, part of today ' s political system, and I know things must change. We must revive politics as the power to imagine, reimagine, and redesign for a better world. But I also know that this disruptive force of change won ' t be driven by the politics of today. The revival of democratic politics will come from you, and I mean all of you. Everyone who participates in this global exchange of ideas, whether it ' s here in this room or just outside this room or online or locally, where everybody lives, everyone who stands up to injustice and inequality, everybody who stands up to those who preach racism rather than **empathy**, dogma rather than critical thinking, technocracy rather than democracy, everyone who stands up to the unchecked power, whether it ' s authoritarian leaders, plutocrats hiding their assets in tax havens, or powerful lobbies protecting the powerful few. It is in their interest that all of us are idiots. Let ' s not be. Thank you. Bruno Giussani : You seem to describe a political leadership that is kind of unprepared and a prisoner of the whims of the **financial** markets, and that scene in Brussels that you describe, to me, as a citizen, is terrifying. Help us understand how you felt after the decision. It was not a good decision, clearly, but how do you feel after that, not as the prime minister, but as George? George Papandreou : Well, obviously there were constraints which didn ' t allow me or others to make the types of decisions we would have wanted, and obviously I had hoped that we would have the time to make the reforms which would have dealt with the deficit rather than trying to cut the deficit which was the symptom of the problem. And that hurt. That hurt because that, first of all, hurt the younger generation, and not only, many of them are demonstrating outside, but I think this is one of our problems. When we face these crises, we have kept the potential, the huge potential of our society out of this process, and we are closing in on ourselves in politics, and I think we need to change that, to really find new participatory ways using the great capabilities that now exist even in technology but not only in technology, the minds that we have, and I think we can find solutions which are much better, but we have to be open. BG : You seem to suggest that the way forward is more Europe, and that is not to be an easy discourse right now in most European countries. It ' s rather the other way — more closed borders and less cooperation and maybe even stepping out of some of the different parts of the European construction. How do you reconcile that? GP : Well, I think one of the worst things that happened during this crisis is that we started a blame game. And the fundamental idea of Europe is that we can cooperate beyond borders, go beyond our conflicts and work together. And the paradox is that, because we have this blame game, we have less the potential to convince our citizens that we should work together, while now is the time when we really need to bring our powers together. Now, more Europe for me is not simply giving more power to Brussels. It is actually giving more power to the citizens of Europe, that is, really making Europe a project of the people. So that, I think, would be a way to answer some of the fears that we have in our society. BG : George, thank you for coming to TED. GP : Thank you very much. BG : Thank you.

Text Sample 228: POS Dim 2 – 2013 – Score 11.00 – 2013/t001986.txt

I ’ m going to be showing some of the cybercriminals ’ latest and nastiest creations. So basically, please don ’ t go and download any of the viruses that I show you. Some of you might be wondering what a cybersecurity specialist looks like, and I thought I ’ d give you a quick insight into my career so far. It ’ s a pretty accurate description. This is what someone that specializes in malware and hacking looks like. So today, computer viruses and trojans, designed to do everything from stealing data to watching you in your webcam to the theft of billions of dollars. Some malicious code today goes as far as targeting power, utilities and infrastructure. Let me give you a quick snapshot of what malicious code is capable of today. Right now, every second, eight new users are joining the Internet. Today, we will see 250, 000 individual new computer viruses. We will see 30, 000 new infected websites. And, just to kind of tear down a myth here, lots of people think that when you get infected with a computer virus, it ’ s because you went to a porn site. Right? Well, actually, statistically speaking, if you only visit porn sites, you ’ re safer. People normally write that down, by the way. Actually, about 80 percent of these are small business websites getting infected. Today ’ s cybercriminal, what do they look like? Well, many of you have the image, don ’ t you, of the spotty teenager sitting in a basement, hacking away for notoriety. But actually today, cybercriminals are wonderfully professional and organized. In fact, they have product adverts. You can go online and buy a hacking **service** to knock your business competitor offline. Check out this one I found. Man : So you ’ re here for one reason, and that reason is because you need your business competitors, rivals, haters, or whatever the reason is, or who, they are to go down. Well you, my friend, you ’ ve came to the right place. If you want your business competitors to go down, well, they can. If you want your rivals to go offline, well, they will. Not only that, we are providing a short- term-to-long-term DDOS **service** or scheduled attack, starting five dollars per hour for small personal websites to 10 to 50 dollars per hour. James Lyne : Now, I did actually pay one of these cybercriminals to attack my own website. Things got a bit tricky when I tried to expense it at the company. Turns out that ’ s not cool. But regardless, it ’ s amazing how many products and **services** are available now to cybercriminals. For example, this testing **platform**, which enables the cybercriminals to test the quality of their viruses before they release them on the world. For a small fee, they can upload it and make sure everything is good. But it goes further. Cybercriminals now have crime packs with business intelligence reporting dashboards to manage the distribution of their malicious code. This is the market leader in malware distribution, the Black Hole Exploit Pack, responsible for nearly one third of malware distribution in the last couple of quarters. It comes with technical installation guides, video setup routines, and get this, technical support. You can email the cybercriminals and they ’ ll tell you how to set up your illegal hacking server. So let me show you what malicious code looks like today. What I ’ ve got here is two systems, an attacker, which I ’ ve made look all Matrix-y and scary, and a victim, which you might recognize from home or work. Now normally, these would be on different sides of the planet or of the Internet, but I ’ ve put them side by side because it makes things much more interesting. Now, there are many ways you can get infected. You will have come in contact with some of them. Maybe some of you have received an email that says something like,"Hi, I ' m a Nigerian banker, and I ' d like to give you 53 billion dollars because I like your face."Or funnycats. exe, which rumor has it was quite successful in China ' s recent campaign against America. Now there are many ways you can get infected. I want to show you a couple of my favorites. This is a little USB key. Now how do you get a USB key to run in a business? Well, you could try looking really cute. Awww. Or, in my case, awkward and pathetic. So imagine this scenario : I walk into one of your businesses, looking very awkward and pathetic, with a copy of my C. V. which I ' ve covered in coffee, and I ask the receptionist to plug in this USB key and print me a new one. So let ' s have a look here on my victim computer. What I ' m going to do is plug in the USB key. After a couple of seconds, things start to happen on the computer on their own, usually a bad sign. This would, of course, normally happen in a couple of seconds, really, really quickly, but I ' ve kind of slowed it down so you can actually see the attack occurring. Malware is very boring otherwise. So this is writing out the malicious code, and a few seconds later, on the left-hand side, you ' ll see the attacker ' s screen get some interesting new text. Now if I place the mouse cursor over it, this is what we call a command prompt, and using this we can **navigate** around the computer. We can access your documents, your data. You can turn on the webcam. That can be very embarrassing. Or just to really prove a point, we can launch programs like my personal favorite, the Windows Calculator. So isn ' t it amazing how much control the attackers can get with such a simple operation? Let me show you how most malware is now distributed today. What I ' m going to do is open up a website that I wrote. It ' s a terrible website. It ' s got really awful graphics. And it ' s got a comments section here where we can submit comments to the website. Many of you will have used something a bit like this before. Unfortunately, when this was implemented, the developer was slightly inebriated and managed to forget all of the secure coding **practices** he had learned. So let ' s imagine that our attacker, called Evil Hacker just for comedy value, inserts something a little nasty. This is a script. It ' s code which will be **interpreted** on the webpage. So I ' m going to submit this post, and then, on my victim computer, I ' m going to open up the web browser and browse to my website, www. incrediblyhacked. com. Notice that after a couple of seconds, I get redirected. That website address at the top there, which you can just about see, microshaft. com, the browser crashes as it hits one of these exploit packs, and up pops fake antivirus. This is a virus pretending to look like antivirus software, and it will go through and it will scan the system, have a look at what its popping up here. It creates some very serious alerts. Oh look, a child porn proxy server. We really should clean that up. What ' s really insulting about this is not only does it provide the attackers with access to your data, but when the scan finishes, they tell you in order to clean up the fake viruses, you have to register the product. Now I liked it better when viruses were free. People now pay cybercriminals money to run viruses, which I find utterly bizarre. So anyway, let me change pace a little bit. Chasing 250, 000 pieces of malware a day is a massive challenge, and those numbers are only growing directly in proportion to the length of my **stress** line, you ' ll note here. So I want to talk to you briefly about a group of hackers we tracked for a year and actually found — and this is a rare treat in our job. Now this was a cross-industry collaboration, people from Facebook, independent **researchers**, guys from Sophos. So here we have a couple of documents which our cybercriminals had uploaded to a cloud **service**, kind of like Dropbox or SkyDrive, like many of you might use. At the top, you ' ll notice a section of source code. What this would do is send the cybercriminals a text message every day telling them how much money they ' d made that day, so a kind of cybercriminal billings report, if you will. If you look closely, you ' ll notice a series of what are Russian telephone numbers. Now that ' s obviously interesting, because that gives us a way of finding our cybercriminals. Down below, highlighted in red, in the other section of source code, is this bit"leded:leded."That ' s a username, kind of like you might have on Twitter. So let ' s take this a little further. There are a few other interesting pieces the cybercriminals had uploaded. Lots of you here will use smartphones to take photos and post them from the conference. An interesting feature of lots of modern smartphones is that when you take a photo, it embeds GPS data about where that photo was taken. In fact, I ' ve been spending a lot of time on Internet dating sites recently, obviously for research purposes, and I ' ve noticed that about 60 percent of the profile pictures on Internet dating sites contain the GPS coordinates of where the photo was taken, which is kind of scary because you wouldn ' t give out your home address to lots of strangers, but we ' re happy to give away our GPS coordinates to plus or minus 15 meters. And our cybercriminals had done the same thing. So here ' s a photo which resolves to St. Petersburg. We then deploy the incredibly advanced hacking tool. We used Google. Using the email address, the telephone number and the GPS data, on the left you see an advert for a BMW that one of our cybercriminals is selling, on the other side an advert for the sale of sphynx kittens. One of these was more stereotypical for me. A little more searching, and here ' s our cybercriminal. Imagine, these are hardened cybercriminals sharing information scarcely. Imagine what you could find about each of the people in this room. A bit more searching through the profile and there ' s a photo of their office. They were working on the third floor. And you can also see some photos from his business companion where he has a taste in a certain kind of image. It turns out he ' s a member of the Russian Adult Webmasters Federation. But this is where our investigation starts to slow down. The cybercriminals have locked down their profiles quite well. And herein is the greatest lesson of social media and **mobile** devices for all of us right now. Our friends, our families and our colleagues can break our security even when we do the right things. This is MobSoft, one of the companies that this cybercriminal gang owned, and an interesting thing about MobSoft is the 50-percent owner of this posted a job advert, and this job advert matched one of the telephone numbers from the code earlier. This woman was Maria, and Maria is the wife of one of our cybercriminals. And it ' s kind of like she went into her social media **settings** and clicked on every option imaginable to make herself really, really insecure. By the end of the investigation, where you can read the full 27-page report at that link, we had photos of the cybercriminals, even the office Christmas party when they were out on an outing. That ' s right, cybercriminals do have Christmas parties, as it turns out. Now you ' re probably wondering what happened to these guys. Let me come back to that in just a minute. I want to change pace to one last little demonstration, a technique that is wonderfully simple and basic, but is interesting in exposing how much information we ' re all giving away, and it ' s relevant because it applies to us as a TED audience. This is normally when people start kind of shuffling in their pockets trying to turn their phones onto airplane mode desperately. Many of you all know about the concept of scanning for wireless networks. You do it every time you take out your iPhone or your Blackberry and connect to something like TEDAttendees. But what you might not know is that you ' re also beaming out a list of networks you ' ve previously connected to, even when you ' re not using wireless actively. So I ran a little scan. I was relatively inhibited compared to the cybercriminals, who wouldn ' t be so concerned by law, and here you can see my **mobile** device. Okay? So you can see a list of wireless networks. TEDAttendees, HyattLB. Where do you think I ' m staying? My home network, PrettyFlyForAWifi, which I think is a great name. Sophos_Visitors, SANSEMEA, companies I work with. Loganwifi, that ’ s in Boston. HiltonLondon. CIASurveillanceVan. We called it that at one of our conferences because we thought that would freak people out, which is quite fun. This is how geeks party. So let ’ s make this a little bit more interesting. Let ’ s talk about you. Twenty- three percent of you have been to Starbucks recently and used the wireless network. Things get more interesting. Forty-six percent of you I could link to a business, XYZ Employee network. This isn ’ t an exact science, but it gets pretty accurate. Seven hundred and sixty-one of you I could identify a hotel you ’ d been to recently, absolutely with pinpoint precision somewhere on the globe. Two hundred and thirty-four of you, well, I know where you live. Your wireless network name is so unique that I was able to pinpoint it using data available openly on the Internet with no hacking or clever, clever tricks. And I should mention as well that some of you do use your names,”James Lyne ’ s iPhone,”for example. And two percent of you have a tendency to extreme profanity. So something for you to think about : As we adopt these new applications and **mobile** devices, as we play with these shiny new toys, how much are we trading off convenience for **privacy** and security? Next time you install something, look at the **settings** and ask yourself,”Is this information that I want to share? Would someone be able to abuse it?”We also need to think very carefully about how we develop our future talent pool. You see, technology ’ s changing at a staggering rate, and that 250, 000 pieces of malware won ’ t stay the same for long. There ’ s a very concerning trend that whilst many people coming out of schools now are much more technology-savvy, they know how to use technology, fewer and fewer people are following the feeder subjects to know how that technology works under the covers. In the U. K., a 60 percent reduction since 2003, and there are similar statistics all over the world. We also need to think about the legal issues in this area. The cybercriminals I talked about, despite theft of millions of dollars, actually still haven ’ t been arrested, and at this point possibly never will. Most laws are national in their implementation, despite cybercrime conventions, where the Internet is borderless and international by definition. Countries do not agree, which makes this area exceptionally challenging from a legal perspective. But my biggest ask is this : You see, you ’ re going to leave here and you ’ re going to see some astonishing stories in the news. You ’ re going to read about malware doing incredible and terrifying, scary things. However, 99 percent of it works because people fail to do the basics. So my ask is this : Go online, find these simple best **practices**, find out how to update and patch your computer. Get a secure password. Make sure you use a different password on each of your sites and **services** online. Find these resources. Apply them. The Internet is a fantastic resource for business, for political **expression**, for art and for learning. Help me and the security community make life much, much more difficult for cybercriminals. Thank you.

Text Sample 229: POS Dim 2 – 2013 – Score 11.00 – 2013/t002123.txt

So here ’ s the good news about families. The last 50 years have seen a revolution in what it means to be a family. We have blended families, adopted families, we have nuclear families living in separate houses and divorced families living in the same house. But through it all, the family has grown stronger. Eight in 10 say the family they have today is as strong or stronger than the family they grew up in. Now, here ’ s the bad news. Nearly everyone is completely overwhelmed by the chaos of family life. Every parent I know, myself included, feels like we ’ re constantly playing defense. Just when our kids stop teething, they start having tantrums. Just when they stop needing our help taking a bath, they need our help dealing with cyberstalking or bullying. And here ’ s the worst news of all. Our children sense we ’ re out of control. Ellen Galinsky of the Families and Work Institute asked 1, 000 children,”If you were granted one wish about your parents, what would it be?”The parents predicted the kids would say, spending more time with them. They were wrong. The kids ’ number one wish? That their parents be less tired and less **stressed**. So how can we change this dynamic? Are there concrete things we can do to reduce **stress**, draw our family closer, and generally prepare our children to enter the world? I spent the last few years trying to answer that question, traveling around, meeting families, talking to scholars, experts ranging from elite peace negotiators to Warren Buffett ’ s bankers to the Green Berets. I was trying to figure out, what do happy families do right and what can I learn from them to make my family happier? I want to tell you about one family that I met, and why I think they offer clues. At 7 p. m. on a Sunday in Hidden Springs, Idaho, where the six members of the Starr family are sitting down to the highlight of their week : the family meeting. The Starrs are a regular American family with their share of regular American family problems. David is a software engineer. Eleanor takes care of their four children, ages 10 to 15. One of those kids tutors math on the far side of town. One has lacrosse on the near side of town. One has Asperger syndrome. One has ADHD.”We were living in complete chaos,”Eleanor said. What the Starrs did next, though, was surprising. Instead of turning to friends or relatives, they looked to David ’ s **workplace**. They turned to a cutting-edge program called agile development that was just spreading from manufacturers in Japan to startups in Silicon Valley. In agile, workers are organized into small groups and do things in very short spans of time. So instead of having executives issue grand proclamations, the team in effect manages itself. You have constant feedback. You have **daily** update sessions. You have weekly reviews. You ’ re constantly changing. David said when they brought this system into their home, the family meetings in particular increased communication, decreased **stress**, and made everybody happier to be part of the family team. When my wife and I adopted these family meetings and other techniques into the lives of our then-five-year-old twin daughters, it was the biggest single change we made since our daughters were born. And these meetings had this effect while taking under 20 minutes. So what is Agile, and why can it help with something that seems so different, like families? In 1983, Jeff Sutherland was a technologist at a **financial** firm in New England. He was very frustrated with how software got designed. Companies followed the waterfall method, right, in which executives issued orders that slowly trickled down to programmers below, and no one had ever consulted the programmers. Eighty-three percent of projects failed. They were too bloated or too out of date by the time they were done. Sutherland wanted to create a system where ideas didn ’ t just percolate down but could percolate up from the bottom and be adjusted in real time. He read 30 years of Harvard Business Review before stumbling upon an article in 1986 called”The New New Product Development Game.”It said that the pace of business was quickening — and by the way, this was in 1986 — and the most successful companies were flexible. It highlighted Toyota and Canon and likened their adaptable, tight- knit teams to rugby scrums. As Sutherland told me, we got to that article, and said,”That ’ s it.”In Sutherland ’ s system, companies don ’ t use large, massive projects that take two years. They do things in small chunks. Nothing takes longer than two weeks. So instead of saying,”You guys go off into that bunker and come back with a cell phone or a social network,”you say,”You go off and come up with one element, then bring it back. Let ’ s talk about it. Let ’ s adapt.”You succeed or fail quickly. Today, agile is used in a hundred countries, and it ' s sweeping into management suites. Inevitably, people began taking some of these techniques and applying it to their families. You had blogs pop up, and some manuals were written. Even the Sutherlands told me that they had an Agile Thanksgiving, where you had one group of people working on the food, one setting the table, and one greeting visitors at the door. Sutherland said it was the best Thanksgiving ever. So let ' s take one problem that families face, crazy mornings, and talk about how agile can help. A key plank is accountability, so teams use information radiators, these large boards in which everybody is accountable. So the Starrs, in adapting this to their home, created a morning checklist in which each child is expected to tick off chores. So on the morning I visited, Eleanor came downstairs, poured herself a cup of coffee, sat in a reclining chair, and she sat there, kind of amiably talking to each of her children as one after the other they came downstairs, checked the list, made themselves breakfast, checked the list again, put the dishes in the dishwasher, rechecked the list, fed the pets or whatever chores they had, checked the list once more, gathered their belongings, and made their way to the bus. It was one of the most astonishing family dynamics I have ever seen. And when I strenuously objected this would never work in our house, our kids needed way too much monitoring, Eleanor looked at me."That ' s what I thought,"she said."I told David, ' keep your work out of my kitchen. ' But I was wrong."So I turned to David :"So why does it work?"He said,"You can ' t underestimate the power of doing this."And he made a checkmark. He said,"In the **workplace**, adults love it. With kids, it ' s heaven."The week we introduced a morning checklist into our house, it cut parental screaming in half. But the real change didn ' t come until we had these family meetings. So following the agile model, we ask three questions : What worked well in our family this week, what didn ' t work well, and what will we agree to work on in the week ahead? Everyone throws out suggestions and then we pick two to focus on. And suddenly the most amazing things started coming out of our daughters ' mouths. What worked well this week? Getting over our fear of riding bikes. Making our beds. What didn ' t work well? Our math sheets, or greeting visitors at the door. Like a lot of parents, our kids are something like Bermuda Triangles. Like, thoughts and ideas go in, but none ever comes out, I mean at least not that are revealing. This gave us access suddenly to their innermost thoughts. But the most surprising part was when we turned to, what are we going to work on in the week ahead? You know, the key idea of agile is that teams essentially manage themselves, and it works in software and it turns out that it works with kids. Our kids love this process. So they would come up with all these ideas. You know, greet five visitors at the door this week, get an extra 10 minutes of reading before bed. Kick someone, lose desserts for a month. It turns out, by the way, our girls are little Stalins. We constantly have to kind of dial them back. Now look, naturally there ' s a gap between their kind of conduct in these meetings and their behavior the rest of the week, but the truth is it didn ' t really bother us. It felt like we were kind of laying these underground cables that wouldn ' t light up their world for many years to come. Three years later — our girls are almost eight now — We ' re still holding these meetings. My wife counts them among her most treasured moments as a mom. So what did we learn? The word"agile"entered the lexicon in 2001 when Jeff Sutherland and a group of designers met in Utah and wrote a 12-point Agile Manifesto. I think the time is right for an Agile Family Manifesto. I ' ve taken some ideas from the Starrs and from many other families I met. I ' m proposing three planks. Plank number one : Adapt all the time. When I became a parent, I figured, you know what? We ' ll set a few rules and we ' ll stick to them. That assumes, as parents, we can anticipate every problem that ’ s going to arise. We can ’ t. What ’ s great about the agile system is you build in a system of change so that you can react to what ’ s happening to you in real time. It ’ s like they say in the Internet world : if you ’ re doing the same thing today you were doing six months ago, you ’ re doing the wrong thing. Parents can learn a lot from that. But to me,”adapt all the time”means something deeper, too. We have to break parents out of this strait-jacket that the only ideas we can try at home are ones that come from shrinks or self-help gurus or other family experts. The truth is, their ideas are stale, whereas in all these other worlds there are these new ideas to make groups and teams work effectively. Let ’ s just take a few examples. Let ’ s take the biggest issue of all : family dinner. Everybody knows that having family dinner with your children is good for the kids. But for so many of us, it doesn ’ t work in our lives. I met a celebrity chef in New Orleans who said,”No problem, I ’ ll just time-shift family dinner. I ’ m not home, can ’ t make family dinner? We ’ ll have family breakfast. We ’ ll meet for a bedtime snack. We ’ ll make Sunday meals more important.”And the truth is, recent research backs him up. It turns out there ’ s only 10 minutes of productive time in any family meal. The rest of it ’ s taken up with”take your elbows off the table”and”pass the ketchup.”You can take that 10 minutes and move it to any part of the day and have the same benefit. So time-shift family dinner. That ’ s adaptability. An environmental psychologist told me,”If you ’ re sitting in a hard chair on a rigid surface, you ’ ll be more rigid. If you ’ re sitting on a cushioned chair, you ’ ll be more open.”She told me,”When you ’ re discipling your children, sit in an upright chair with a cushioned surface. The conversation will go better.”My wife and I actually moved where we sit for difficult conversations because I was sitting above in the power position. So move where you sit. That ’ s adaptability. The point is there are all these new ideas out there. We ’ ve got to hook them up with parents. So plank number one : Adapt all the time. Be flexible, be open-minded, let the best ideas win. Plank number two : Empower your children. Our instinct as parents is to order our kids around. It ’ s easier, and frankly, we ’ re usually right. There ’ s a reason that few systems have been more waterfall over time than the family. But the single biggest lesson we learned is to reverse the waterfall as much as possible. Enlist the children in their own upbringing. Just yesterday, we were having our family meeting, and we had voted to work on overreacting. So we said,”Okay, give us a reward and give us a punishment. Okay?”So one of my daughters threw out, you get five minutes of overreacting time all week. So we kind of liked that. But then her sister started working the system. She said,”Do I get one five-minute overreaction or can I get 10 30-second overreactions?”I loved that. Spend the time however you want. Now give us a punishment. Okay. If we get 15 minutes of overreaction time, that ’ s the limit. Every minute above that, we have to do one pushup. So you see, this is working. Now look, this system isn ’ t lax. There ’ s plenty of parental authority going on. But we ’ re giving them **practice** becoming independent, which of course is our ultimate goal. Just as I was leaving to come here tonight, one of my daughters started screaming. The other one said,”Overreaction! Overreaction!”and started counting, and within 10 seconds it had ended. To me that is a certified agile miracle. And by the way, research backs this up too. Children who plan their own goals, set weekly schedules, evaluate their own work build up their frontal cortex and take more control over their lives. The point is, we have to let our children succeed on their own terms, and yes, on occasion, fail on their own terms. I was talking to Warren Buffett ’ s banker, and he was chiding me for not letting my children make mistakes with their allowance. And I said,”But what if they drive into a ditch?”He said,”It ’ s much better to drive into a ditch with a $6 allowance than a $60, 000-a-year salary or a $6 million inheritance.”So the bottom line is, empower your children. Plank number three : Tell your story. Adaptability is fine, but we also need bedrock. Jim Collins, the author of”Good To Great,”told me that successful human organizations of any kind have two things in common : they preserve the core, they stimulate progress. So agile is great for stimulating progress, but I kept hearing time and again, you need to preserve the core. So how do you do that? Collins coached us on doing something that businesses do, which is define your **mission** and identify your core values. So he led us through the process of creating a family **mission** statement. We did the family equivalent of a corporate retreat. We had a pajama party. I made popcorn. Actually, I burned one, so I made two. My wife bought a flip chart. And we had this great conversation, like, what ’ s important to us? What values do we most uphold? And we ended up with 10 statements. We are travelers, not tourists. We don ’ t like dilemmas. We like solutions. Again, research shows that parents should spend less time worrying about what they do wrong and more time focusing on what they do right, worry less about the bad times and build up the good times. This family **mission** statement is a great way to identify what it is that you do right. A few weeks later, we got a call from the school. One of our daughters had gotten into a spat. And suddenly we were worried, like, do we have a mean girl on our hands? And we didn ’ t really know what to do, so we called her into my office. The family **mission** statement was on the wall, and my wife said,”So, anything up there seem to apply?”And she kind of looked down the list, and she said,”Bring people together?”Suddenly we had a way into the conversation. Another great way to tell your story is to tell your children where they came from. **Researchers** at Emory gave children a simple”what do you know”test. Do you know where your grandparents were born? Do you know where your parents went to high school? Do you know anybody in your family who had a difficult situation, an illness, and they overcame it? The children who **scored** highest on this”do you know”scale had the highest self-esteem and a greater sense they could control their lives. The”do you know”test was the single biggest predictor of emotional health and **happiness**. As the author of the study told me, children who have a sense of — they ’ re part of a larger narrative have greater self-confidence. So my final plank is, tell your story. Spend time retelling the story of your family ’ s positive moments and how you overcame the negative ones. If you give children this happy narrative, you give them the tools to make themselves happier. I was a teenager when I first read”Anna Karenina”and its famous opening sentence,”All happy families are alike. Each unhappy family is unhappy in its own way.”When I first read that, I thought,”That sentence is inane. Of course all happy families aren ’ t alike.”But as I began working on this project, I began changing my mind. Recent scholarship has allowed us, for the first time, to identify the building blocks that successful families have. I ’ ve mentioned just three here today : Adapt all the time, empower the children, tell your story. Is it possible, all these years later, to say Tolstoy was right? The answer, I believe, is yes. When Leo Tolstoy was five years old, his brother Nikolay came to him and said he had engraved the secret to universal **happiness** on a little green stick, which he had hidden in a ravine on the family ’ s estate in Russia. If the stick were ever found, all humankind would be happy. Tolstoy became consumed with that stick, but he never found it. In fact, he asked to be buried in that ravine where he thought it was hidden. He still lies there today, covered in a layer of green grass. That story perfectly captures for me the final lesson that I learned : **Happiness** is not something we find, it ’ s something we make. Almost anybody who ’ s looked at well-run organizations has come to pretty much the same conclusion. Greatness is not a matter of circumstance. It ’ s a matter of choice. You don ’ t need some grand plan. You don ’ t need a waterfall. You just need to take small steps, accumulate small wins, keep reaching for that green stick. In the end, this may be the greatest lesson of all. What ' s the secret to a happy family? Try.

Text Sample 230: POS Dim 2 – 2012 – Score 14.00 – 2012/t002277.txt

Like many of you, I ' m one of the lucky people. I was born to a family where education was pervasive. I ' m a third-generation PhD, a daughter of two academics. In my childhood, I played around in my father ' s university lab. So it was taken for granted that I attend some of the best universities, which in turn opened the door to a world of opportunity. Unfortunately, most of the people in the world are not so lucky. In some parts of the world, for example, South Africa, education is just not readily accessible. In South Africa, the educational system was constructed in the days of apartheid for the white minority. And as a consequence, today there is just not enough spots for the many more people who want and deserve a high quality education. That scarcity led to a crisis in January of this year at the University of Johannesburg. There were a handful of positions left open from the standard admissions process, and the night before they were supposed to open that for registration, thousands of people lined up outside the gate in a line a mile long, hoping to be first in line to get one of those positions. When the gates opened, there was a stampede, and 20 people were **injured** and one woman died. She was a mother who gave her life trying to get her son a chance at a better life. But even in parts of the world like the United States where education is available, it might not be within reach. There has been much discussed in the last few years about the rising cost of health care. What might not be quite as obvious to people is that during that same period the cost of higher education tuition has been increasing at almost twice the rate, for a total of 559 percent since 1985. This makes education unaffordable for many people. Finally, even for those who do manage to get the higher education, the doors of opportunity might not open. Only a little over half of recent college graduates in the United States who get a higher education actually are working in jobs that require that education. This, of course, is not true for the students who graduate from the top institutions, but for many others, they do not get the value for their time and their effort. Tom Friedman, in his recent New York Times article, captured, in the way that no one else could, the spirit behind our effort. He said the big breakthroughs are what happen when what is suddenly possible meets what is desperately necessary. I ' ve talked about what ' s desperately necessary. Let ' s talk about what ' s suddenly possible. What ' s suddenly possible was demonstrated by three big Stanford classes, each of which had an enrollment of 100, 000 people or more. So to understand this, let ' s look at one of those classes, the Machine Learning class offered by my colleague and cofounder Andrew Ng. Andrew teaches one of the bigger Stanford classes. It ' s a Machine Learning class, and it has 400 people **enrolled** every time it ' s offered. When Andrew taught the Machine Learning class to the general public, it had 100, 000 people registered. So to put that number in perspective, for Andrew to reach that same size audience by teaching a Stanford class, he would have to do that for 250 years. Of course, he ' d get really bored. So, having seen the impact of this, Andrew and I decided that we needed to really try and scale this up, to bring the best quality education to as many people as we could. So we formed Coursera, whose goal is to take the best courses from the best instructors at the best universities and provide it to everyone around the world for free. We currently have 43 courses on the **platform** from four universities across a range of disciplines, and let me show you a little bit of an overview of what that looks like. Robert Ghrist : Welcome to Calculus. Ezekiel Emanuel : Fifty million people are uninsured. Scott Page : Models help us design more effective institutions and policies. We get unbelievable segregation. Scott Klemmer : So Bush imagined that in the future, you ’ d wear a camera right in the center of your head. Mitchell Duneier : Mills wants the student of sociology to develop the quality of mind... RG : Hanging cable takes on the form of a hyperbolic cosine. Nick Parlante : For each pixel in the image, set the red to zero. Paul Offit :... Vaccine allowed us to eliminate polio virus. Dan Jurafsky : Does Lufthansa serve breakfast and San Jose? Well, that sounds funny. Daphne Koller : So this is which coin you pick, and this is the two tosses. Andrew Ng : So in large-scale machine learning, we ’ d like to come up with computational... DK : It turns out, maybe not surprisingly, that students like getting the best content from the best universities for free. Since we opened the website in February, we now have 640, 000 students from 190 countries. We have 1. 5 million enrollments, 6 million quizzes in the 15 classes that have launched so far have been submitted, and 14 million videos have been viewed. But it ’ s not just about the numbers, it ’ s also about the people. Whether it ’ s Akash, who comes from a small town in India and would never have access in this case to a Stanford-quality course and would never be able to afford it. Or Jenny, who is a single mother of two and wants to hone her skills so that she can go back and complete her master ’ s degree. Or Ryan, who can ’ t go to school, because his immune deficient daughter can ’ t be risked to have germs come into the house, so he couldn ’ t leave the house. I ’ m really glad to say — recently, we ’ ve been in correspondence with Ryan — that this story had a happy ending. Baby Shannon — you can see her on the left — is doing much better now, and Ryan got a job by taking some of our courses. So what made these courses so different? After all, online course content has been available for a while. What made it different was that this was real course experience. It started on a given day, and then the students would watch videos on a weekly basis and do homework assignments. And these would be real homework assignments for a real grade, with a real **deadline**. You can see the **deadlines** and the usage graph. These are the spikes showing that procrastination is global phenomenon. At the end of the course, the students got a certificate. They could present that certificate to a prospective employer and get a better job, and we know many students who did. Some students took their certificate and presented this to an educational institution at which they were **enrolled** for actual college credit. So these students were really getting something meaningful for their investment of time and effort. Let ’ s talk a little bit about some of the components that go into these courses. The first component is that when you move away from the constraints of a physical classroom and design content explicitly for an online format, you can break away from, for example, the monolithic one-hour lecture. You can break up the material, for example, into these short, modular units of eight to 12 minutes, each of which represents a coherent concept. Students can traverse this material in different ways, depending on their background, their skills or their interests. So, for example, some students might benefit from a little bit of preparatory material that other students might already have. Other students might be interested in a particular enrichment topic that they want to pursue individually. So this format allows us to break away from the one-size-fits-all model of education, and allows students to follow a much more personalized curriculum. Of course, we all know as educators that students don ’ t learn by sitting and passively watching videos. Perhaps one of the biggest components of this effort is that we need to have students who **practice** with the material in order to really understand it. There ’ s been a range of studies that demonstrate the importance of this. This one that appeared in Science last year, for example, demonstrates that even simple retrieval **practice**, where students are just supposed to repeat what they already learned gives considerably improved results on various achievement tests down the line than many other educational interventions. We ’ ve tried to build in retrieval **practice** into the **platform**, as well as other forms of **practice** in many ways. For example, even our videos are not just videos. Every few minutes, the video pauses and the students get asked a question. SP :... These four things. Prospect theory, hyperbolic discounting, status quo **bias**, base rate **bias**. They ’ re all well documented. So they ’ re all well documented deviations from rational behavior. DK : So here the video pauses, and the student types in the answer into the box and submits. Obviously they weren ’ t paying attention. So they get to try again, and this time they got it right. There ’ s an optional explanation if they want. And now the video moves on to the next part of the lecture. This is a kind of simple question that I as an instructor might ask in class, but when I ask that kind of a question in class, 80 percent of the students are still scribbling the last thing I said, 15 percent are zoned out on Facebook, and then there ’ s the smarty pants in the front **row** who blurts out the answer before anyone else has had a chance to think about it, and I as the instructor am terribly gratified that somebody actually knew the answer. And so the lecture moves on before, really, most of the students have even noticed that a question had been asked. Here, every single student has to engage with the material. And of course these simple retrieval questions are not the end of the story. One needs to build in much more meaningful **practice** questions, and one also needs to provide the students with feedback on those questions. Now, how do you grade the work of 100, 000 students if you do not have 10, 000 TAs? The answer is, you need to use technology to do it for you. Now, fortunately, technology has come a long way, and we can now grade a range of interesting types of homework. In addition to multiple choice and the kinds of short answer questions that you saw in the video, we can also grade math, mathematical **expressions** as well as mathematical derivations. We can grade models, whether it ’ s **financial** models in a business class or physical models in a science or engineering class and we can grade some pretty sophisticated programming assignments. Let me show you one that ’ s actually pretty simple but fairly visual. This is from Stanford ’ s Computer Science 101 class, and the students are supposed to color-correct that blurry red image. They ’ re typing their program into the browser, and you can see they didn ’ t get it quite right, Lady Liberty is still seasick. And so, the student tries again, and now they got it right, and they ’ re told that, and they can move on to the next assignment. This ability to interact actively with the material and be told when you ’ re right or wrong is really essential to student learning. Now, of course we cannot yet grade the range of work that one needs for all courses. Specifically, what ’ s lacking is the kind of critical thinking work that is so essential in such disciplines as the humanities, the social sciences, business and others. So we tried to convince, for example, some of our humanities faculty that multiple choice was not such a bad strategy. That didn ’ t go over really well. So we had to come up with a different solution. And the solution we ended up using is peer grading. It turns out that previous studies show, like this one by Saddler and Good, that peer grading is a surprisingly effective strategy for providing reproducible grades. It was tried only in small classes, but there it showed, for example, that these student-assigned grades on the y-axis are actually very well correlated with the teacher-assigned grade on the x-axis. What ’ s even more surprising is that self-grades, where the students grade their own work critically — so long as you incentivize them properly so they can ’ t give themselves a perfect **score** — are actually even better correlated with the teacher grades. And so this is an effective strategy that can be used for grading at scale, and is also a useful learning strategy for the students, because they actually learn from the experience. So we now have the largest peer-grading pipeline ever devised, where tens of thousands of students are grading each other ’ s work, and quite successfully, I have to say. But this is not just about students sitting alone in their living room working through problems. Around each one of our courses, a community of students had formed, a global community of people around a shared intellectual endeavor. What you see here is a self-generated map from students in our Princeton Sociology 101 course, where they have put themselves on a world map, and you can really see the global reach of this kind of effort. Students collaborated in these courses in a variety of different ways. First of all, there was a question and answer forum, where students would pose questions, and other students would answer those questions. And the really amazing thing is, because there were so many students, it means that even if a student posed a question at 3 o ’ clock in the morning, somewhere around the world, there would be somebody who was awake and working on the same problem. And so, in many of our courses, the median response time for a question on the question and answer forum was 22 minutes. Which is not a level of **service** I have ever offered to my Stanford students. And you can see from the student testimonials that students actually find that because of this large online community, they got to interact with each other in many ways that were deeper than they did in the context of the physical classroom. Students also self-assembled, without any kind of intervention from us, into small study groups. Some of these were physical study groups along geographical constraints and met on a weekly basis to work through problem sets. This is the San Francisco study group, but there were ones all over the world. Others were virtual study groups, sometimes along language lines or along cultural lines, and on the bottom left there, you see our multicultural universal study group where people explicitly wanted to connect with people from other cultures. There are some tremendous opportunities to be had from this kind of framework. The first is that it has the potential of giving us a completely unprecedented look into understanding human learning. Because the data that we can collect here is unique. You can collect every click, every homework submission, every forum post from tens of thousands of students. So you can turn the study of human learning from the hypothesis-driven mode to the data-driven mode, a transformation that, for example, has revolutionized biology. You can use these data to understand fundamental questions like, what are good learning strategies that are effective versus ones that are not? And in the context of particular courses, you can ask questions like, what are some of the misconceptions that are more common and how do we help students fix them? So here ’ s an example of that, also from Andrew ’ s Machine Learning class. This is a distribution of wrong answers to one of Andrew ’ s assignments. The answers happen to be pairs of numbers, so you can draw them on this two-dimensional plot. Each of the little crosses that you see is a different wrong answer. The big cross at the top left is where 2, 000 students gave the exact same wrong answer. Now, if two students in a class of 100 give the same wrong answer, you would never notice. But when 2, 000 students give the same wrong answer, it ’ s kind of hard to miss. So Andrew and his students went in, looked at some of those assignments, understood the root cause of the misconception, and then they produced a targeted error message that would be provided to every student whose answer fell into that bucket, which means that students who made that same mistake would now get personalized feedback telling them how to fix their misconception much more effectively. So this personalization is something that one can then build by having the virtue of large numbers. Personalization is perhaps one of the biggest opportunities here as well, because it provides us with the potential of solving a 30-year-old problem. Educational **researcher** Benjamin Bloom, in 1984, posed what ' s called the 2 sigma problem, which he **observed** by studying three populations. The first is the population that studied in a lecture-based classroom. The second is a population of students that studied using a standard lecture-based classroom, but with a mastery-based approach, so the students couldn ' t move on to the next topic before demonstrating mastery of the previous one. And finally, there was a population of students that were taught in a one- on-one instruction using a tutor. The mastery-based population was a full standard deviation, or sigma, in achievement **scores** better than the standard lecture-based class, and the individual tutoring gives you 2 sigma improvement in performance. To understand what that means, let ' s look at the lecture-based classroom, and let ' s pick the median performance as a threshold. So in a lecture-based class, half the students are above that level and half are below. In the individual tutoring instruction, 98 percent of the students are going to be above that threshold. Imagine if we could teach so that 98 percent of our students would be above average. Hence, the 2 sigma problem. Because we cannot afford, as a society, to provide every student with an individual human tutor. But maybe we can afford to provide each student with a computer or a smartphone. So the question is, how can we use technology to push from the left side of the graph, from the blue curve, to the right side with the green curve? Mastery is easy to achieve using a computer, because a computer doesn ' t get tired of showing you the same video five times. And it doesn ' t even get tired of grading the same work multiple times, we ' ve seen that in many of the examples that I ' ve shown you. And even personalization is something that we ' re starting to see the beginnings of, whether it ' s via the personalized trajectory through the curriculum or some of the personalized feedback that we ' ve shown you. So the goal here is to try and push, and see how far we can get towards the green curve. So, if this is so great, are universities now obsolete? Well, Mark Twain certainly thought so. He said that,"College is a place where a professor ' s lecture notes go straight to the students ' lecture notes, without passing through the brains of either."I beg to differ with Mark Twain, though. I think what he was **complaining** about is not universities but rather the lecture-based format that so many universities spend so much time on. So let ' s go back even further, to Plutarch, who said that,"The mind is not a vessel that needs filling, but wood that needs igniting."And maybe we should spend less time at universities filling our students ' minds with content by lecturing at them, and more time igniting their creativity, their imagination and their problem-solving skills by actually talking with them. So how do we do that? We do that by doing active learning in the classroom. So there ' s been many studies, including this one, that show that if you use active learning, interacting with your students in the classroom, performance improves on every single metric — on attendance, on engagement and on learning as measured by a standardized test. You can see, for example, that the achievement **score** almost doubles in this particular experiment. So maybe this is how we should spend our time at universities. So to summarize, if we could offer a top quality education to everyone around the world for free, what would that do? Three things. First it would establish education as a fundamental human right, where anyone around the world with the ability and the motivation could get the skills that they need to make a better life for themselves, their families and their communities. Second, it would enable lifelong learning. It ' s a shame that for so many people, learning stops when we finish high school or when we finish college. By having this amazing content be available, we would be able to learn something new every time we wanted, whether it ’ s just to expand our minds or it ’ s to change our lives. And finally, this would enable a wave of innovation, because amazing talent can be found anywhere. Maybe the next Albert Einstein or the next Steve Jobs is living somewhere in a remote village in Africa. And if we could offer that person an education, they would be able to come up with the next big idea and make the world a better place for all of us. Thank you very much.

Text Sample 231: POS Dim 2 – 2012 – Score 11.00 – 2012/t002290.txt
I study the future of crime and terrorism, and frankly, I ’ m afraid. I ’ m afraid by what I see. I sincerely want to believe that technology can bring us the techno-utopia that we ’ ve been promised, but, you see, I ’ ve spent a career in law enforcement, and that ’ s informed my perspective on things. I ’ ve been a street police officer, an undercover investigator, a counter-terrorism strategist, and I ’ ve worked in more than 70 countries around the world. I ’ ve had to see more than my fair share of violence and the darker underbelly of society, and that ’ s informed my opinions. My work with criminals and terrorists has actually been highly educational. They have taught me a lot, and I ’ d like to be able to share some of these observations with you. Today I ’ m going to show you the flip side of all those technologies that we marvel at, the ones that we love. In the hands of the TED community, these are awesome tools which will bring about great change for our world, but in the hands of suicide bombers, the future can look quite different. I started **observing** technology and how criminals were using it as a young patrol officer. In those days, this was the height of technology. Laugh though you will, all the drug dealers and gang members with whom I dealt had one of these long before any police officer I knew did. Twenty years later, criminals are still using **mobile** phones, but they ’ re also building their own **mobile** phone networks, like this one, which has been deployed in all 31 states of Mexico by the narcos. They have a national encrypted radio communications system. Think about that. Think about the innovation that went into that. Think about the infrastructure to build it. And then think about this : Why can ’ t I get a cell phone signal in San Francisco? How is this possible? It makes no sense. We consistently underestimate what criminals and terrorists can do. Technology has made our world increasingly open, and for the most part, that ’ s great, but all of this openness may have unintended consequences. Consider the 2008 terrorist attack on Mumbai. The men that carried that attack out were armed with AK-47s, explosives and hand grenades. They threw these hand grenades at innocent people as they sat eating in cafes and waited to catch trains on their way home from work. But heavy artillery is nothing new in terrorist operations. Guns and bombs are nothing new. What was different this time is the way that the terrorists used modern information communications technologies to locate additional victims and slaughter them. They were armed with **mobile** phones. They had BlackBerries. They had access to satellite imagery. They had satellite phones, and they even had night vision goggles. But perhaps their greatest innovation was this. We ’ ve all seen pictures like this on television and in the news. This is an operations center. And the terrorists built their very own op center across the border in Pakistan, where they monitored the BBC, al Jazeera, CNN and Indian local stations. They also monitored the Internet and social media to monitor the progress of their attacks and how many people they had killed. They did all of this in real time. The innovation of the terrorist operations center gave terrorists unparalleled situational awareness and tactical advantage over the police and over the government. What did they do with this? They used it to great effect. At one

point during the 60-hour siege, the terrorists were going room to room trying to find additional victims. They came upon a suite on the top floor of the hotel, and they kicked down the door and they found a man hiding by his bed. And they said to him,”Who are you, and what are you doing here?”And the man replied,”I ’ m just an innocent schoolteacher.”Of course, the terrorists knew that no Indian schoolteacher stays at a suite in the Taj. They picked up his identification, and they phoned his name in to the terrorist war room, where the terrorist war room Googled him, and found a picture and called their operatives on the ground and said,”Your hostage, is he heavyset? Is he bald in front? Does he wear glasses?””Yes, yes, yes,”came the answers. The op center had found him and they had a match. He was not a schoolteacher. He was the second-wealthiest businessman in India, and after discovering this information, the terrorist war room gave the order to the terrorists on the ground in Mumbai. We all worry about our **privacy settings** on Facebook, but the fact of the matter is, our openness can be used against us. Terrorists are doing this. A search engine can determine who shall live and who shall die. This is the world that we live in. During the Mumbai siege, terrorists were so dependent on technology that several witnesses reported that as the terrorists were shooting hostages with one hand, they were checking their **mobile** phone messages in the very other hand. In the end, 300 people were gravely wounded and over 172 men, women and children lost their lives that day. Think about what happened. During this 60-hour siege on Mumbai, 10 men armed not just with weapons, but with technology, were able to bring a city of 20 million people to a standstill. Ten people brought 20 million people to a standstill, and this traveled around the world. This is what radicals can do with openness. This was done nearly four years ago. What could terrorists do today with the technologies available that we have? What will they do tomorrow? The ability of one to affect many is scaling exponentially, and it ’ s scaling for good and it ’ s scaling for evil. It ’ s not just about terrorism, though. There ’ s also been a big paradigm shift in crime. You see, you can now commit more crime as well. In the old days, it was a knife and a gun. Then criminals moved to robbing trains. You could rob 200 people on a train, a great innovation. Moving forward, the Internet allowed things to scale even more. In fact, many of you will remember the recent Sony PlayStation hack. In that incident, over 100 million people were robbed. Think about that. When in the history of humanity has it ever been possible for one person to rob 100 million? Of course, it ’ s not just about stealing things. There are other avenues of technology that criminals can exploit. Many of you will remember this **super** cute video from the last TED, but not all quadcopter swarms are so nice and cute. They don ’ t all have drumsticks. Some can be armed with HD cameras and do countersurveillance on protesters, or, as in this little bit of movie magic, quadcopters can be loaded with firearms and automatic weapons. Little robots are cute when they play music to you. When they swarm and chase you down the block to shoot you, a little bit less so. Of course, criminals and terrorists weren ’ t the first to give guns to robots. We know where that started. But they ’ re adapting quickly. Recently, the FBI arrested an al Qaeda affiliate in the United States, who was planning on using these remote-controlled drone aircraft to fly C4 explosives into government buildings in the United States. By the way, these travel at over 600 miles an hour. Every time a new technology is being introduced, criminals are there to exploit it. We ’ ve all seen 3D printers. We know with them that you can print in many materials ranging from plastic to chocolate to metal and even concrete. With great precision I actually was able to make this just the other day, a very cute little ducky. But I wonder to myself, for those people that strap bombs to their chests and blow themselves up, how might they use 3D printers? Perhaps like this. You see, if you can print in metal, you can print one of these, and in fact you can also print one of these too. The UK I know has some very strict firearms laws. You needn ' t bring the gun into the UK anymore. You just bring the 3D printer and print the gun while you ' re here, and, of course, the magazines for your bullets. But as these get bigger in the future, what other items will you be able to print? The technologies are allowing bigger printers. As we move forward, we ' ll see new technologies also, like the Internet of Things. Every day we ' re connecting more and more of our lives to the Internet, which means that the Internet of Things will soon be the Internet of Things To Be Hacked. All of the physical objects in our space are being transformed into information technologies, and that has a radical implication for our security, because more connections to more devices means more vulnerabilities. Criminals understand this. Terrorists understand this. Hackers understand this. If you control the code, you control the world. This is the future that awaits us. There has not yet been an operating system or a technology that hasn ' t been hacked. That ' s troubling, since the human body itself is now becoming an information technology. As we ' ve seen here, we ' re transforming ourselves into cyborgs. Every year, thousands of cochlear implants, diabetic pumps, pacemakers and defibrillators are being implanted in people. In the United States, there are 60, 000 people who have a pacemaker that connects to the Internet. The defibrillators allow a physician at a distance to give a shock to a heart in case a patient needs it. But if you don ' t need it, and somebody else gives you the shock, it ' s not a good thing. Of course, we ' re going to go even deeper than the human body. We ' re going down to the cellular level these days. Up until this point, all the technologies I ' ve been talking about have been silicon-based, ones and zeroes, but there ' s another operating system out there : the original operating system, DNA. And to hackers, DNA is just another operating system waiting to be hacked. It ' s a great challenge for them. There are people already working on hacking the software of life, and while most of them are doing this to great good and to help us all, some won ' t be. So how will criminals abuse this? Well, with synthetic biology you can do some pretty neat things. For example, I predict that we will move away from a plant- based narcotics world to a synthetic one. Why do you need the plants anymore? You can just take the DNA code from marijuana or poppies or coca leaves and cut and past that gene and put it into yeast, and you can take those yeast and make them make the cocaine for you, or the marijuana, or any other drug. So how we use yeast in the future is going to be really interesting. In fact, we may have some really interesting bread and beer as we go into this next century. The cost of sequencing the human genome is dropping precipitously. It was proceeding at Moore ' s Law pace, but then in 2008, something changed. The technologies got better, and now DNA sequencing is proceeding at a pace five times that of Moore ' s Law. That has significant implications for us. It took us 30 years to get from the introduction of the personal computer to the level of cybercrime we have today, but looking at how biology is proceeding so rapidly, and knowing criminals and terrorists as I do, we may get there a lot faster with biocrime in the future. It will be easy for anybody to go ahead and print their own bio-virus, enhanced versions of ebola or anthrax, weaponized **flu**. We recently saw a case where some **researchers** made the H5N1 avian influenza virus more potent. It already has a 70 percent mortality rate if you get it, but it ' s hard to get. Engineers, by moving around a small number of genetic changes, were able to weaponize it and make it much more easy for human beings to catch, so that not thousands of people would die, but tens of millions. You see, you can go ahead and create new pandemics, and the **researchers** who did this were so proud of their accomplishments, they wanted to publish it openly so that everybody could see this and get access to this information. But it goes deeper than that. DNA **researcher** Andrew Hessel has pointed out quite rightly that if you can use cancer treatments, modern cancer treatments, to go after one cell while leaving all the other cells around it **intact**, then you can also go after any one person ’ s cell. Personalized cancer treatments are the flip side of personalized bioweapons, which means you can attack any one individual, including all the people in this picture. How will we protect them in the future? What to do? What to do about all this? That ’ s what I get asked all the time. For those of you who follow me on Twitter, I will be **tweeting** out the answer later on today. Actually, it ’ s a bit more complex than that, and there are no magic bullets. I don ’ t have all the answers, but I know a few things. In the wake of 9 / 11, the best security minds put together all their innovation and this is what they created for security. If you ’ re expecting the people who built this to protect you from the coming robopocalypse — — uh, you may want to have a backup plan. Just saying. Just think about that. Law enforcement is currently a closed system. It ’ s nation-based, while the threat is international. Policing doesn ’ t scale globally. At least, it hasn ’ t, and our current system of guns, border guards, big gates and fences are outdated in the new world into which we ’ re moving. So how might we prepare for some of these specific threats, like attacking a president or a prime minister? This would be the natural government response, to hide away all our government leaders in hermetically sealed bubbles. But this is not going to work. The cost of doing a DNA sequence is going to be **trivial**. Anybody will have it and we will all have them in the future. So maybe there ’ s a more radical way that we can look at this. What happens if we were to take the President ’ s DNA, or a king or queen ’ s, and put it out to a group of a few hundred trusted **researchers** so they could study that DNA and do penetration testing against it as a means of helping our leaders? Or what if we sent it out to a few thousand? Or, controversially, and not without its risks, what happens if we just gave it to the whole public? Then we could all be engaged in helping. We ’ ve already seen examples of this working well. The Organized Crime and Corruption Reporting Project is staffed by journalists and citizens where they are crowd-sourcing what dictators and terrorists are doing with public funds around the world, and, in a more dramatic case, we ’ ve seen in Mexico, a country that has been racked by 50, 000 narcotics-related murders in the past six years. They ’ re killing so many people they can ’ t even afford to bury them all in anything but these unmarked graves like this one outside of Ciudad Juarez. What can we do about this? The government has proven **ineffective**. So in Mexico, citizens, at great risk to themselves, are fighting back to build an effective solution. They ’ re crowd-mapping the activities of the drug dealers. Whether or not you realize it, we are at the dawn of a technological arms race, an arms race between people who are using technology for good and those who are using it for ill. The threat is serious, and the time to prepare for it is now. I can assure you that the terrorists and criminals are. My personal belief is that, rather than having a small, elite force of highly trained government agents here to protect us all, we ’ re much better off having average and ordinary citizens approaching this problem as a group and seeing what we can do. If we all do our part, I think we ’ ll be in a much better space. The tools to change the world are in everybody ’ s hands. How we use them is not just up to me, it ’ s up to all of us. This was a technology I would frequently deploy as a police officer. This technology has become outdated in our current world. It doesn ’ t scale, it doesn ’ t work globally, and it surely doesn ’ t work virtually. We ’ ve seen paradigm shifts in crime and terrorism. They call for a shift to a more open form and a more participatory form of law enforcement. So I invite you to join me. After all, public safety is too important to leave to the professionals. Thank you.

Text Sample 232: POS Dim 2 – 2012 – Score 11.00 – 2012/t002367.txt

I ’ m going to talk to you about optimism â or more precisely, the optimism **bias**. It ’ s a cognitive illusion that we ’ ve been studying in my lab for the past few years, and 80 percent of us have it. It ’ s our tendency to overestimate our likelihood of experiencing good events in our lives and underestimate our likelihood of experiencing bad events. So we underestimate our likelihood of suffering from cancer, being in a car accident. We overestimate our longevity, our career prospects. In short, we ’ re more optimistic than realistic, but we are oblivious to the fact. Take marriage for example. In the Western world, divorce rates are about 40 percent. That means that out of five married couples, two will end up splitting their assets. But when you ask newlyweds about their own likelihood of divorce, they estimate it at zero percent. And even divorce lawyers, who should really know better, hugely underestimate their own likelihood of divorce. So it turns out that optimists are not less likely to divorce, but they are more likely to remarry. In the words of Samuel Johnson,”Remarriage is the triumph of hope over experience.”So if we ’ re married, we ’ re more likely to have kids. And we all think our kids will be especially talented. This, by the way, is my two-year-old nephew, Guy. And I just want to make it absolutely clear that he ’ s a really bad example of the optimism **bias**, because he is in fact uniquely talented. And I ’ m not alone. Out of four British people, three said that they were optimistic about the future of their own families. That ’ s 75 percent. But only 30 percent said that they thought families in general are doing better than a few generations ago. And this is a really important point, because we ’ re optimistic about ourselves, we ’ re optimistic about our kids, we ’ re optimistic about our families, but we ’ re not so optimistic about the guy sitting next to us, and we ’ re somewhat pessimistic about the fate of our fellow citizens and the fate of our country. But private optimism about our own personal future remains persistent. And it doesn ’ t mean that we think things will magically turn out okay, but rather that we have the unique ability to make it so. Now I ’ m a scientist, I do experiments. So to show you what I mean, I ’ m going to do an experiment here with you. So I ’ m going to give you a list of abilities and characteristics, and I want you to think for each of these abilities where you stand relative to the rest of the population. The first one is getting along well with others. Who here believes they ’ re at the bottom 25 percent? Okay, that ’ s about 10 people out of 1, 500. Who believes they ’ re at the top 25 percent? That ’ s most of us here. Okay, now do the same for your **driving** ability. How interesting are you? How attractive are you? How honest are you? And finally, how modest are you? So most of us put ourselves above average on most of these abilities. Now this is statistically impossible. We can ’ t all be better than everyone else. But if we believe we ’ re better than the other guy, well that means that we ’ re more likely to get that promotion, to remain married, because we ’ re more social, more interesting. And it ’ s a global phenomenon. The optimism **bias** has been **observed** in many different countries â in Western cultures, in non-Western cultures, in females and males, in kids, in the elderly. It ’ s quite widespread. But the question is, is it good for us? So some people say no. Some people say the secret to **happiness** is low expectations. I think the logic goes something like this : If we don ’ t expect greatness, if we don ’ t expect to find love and be healthy and successful, well we ’ re not going to be disappointed when these things don ’ t happen. And if we ’ re not disappointed when good things don ’ t happen, and we ’ re pleasantly surprised when they do, we will be happy. So it ’ s a very good theory, but it turns out to be wrong for three reasons. Number one : Whatever happens, whether you succeed or you fail, people with high expectations always feel better. Because how we feel when we get dumped or win employee of the month depends on how we **interpret** that event. The psychologists Margaret Marshall and John Brown studied students with high and low expectations. And they found that when people with high expectations succeed, they attribute that success to their own traits.”I ’ m a genius, therefore I got an A, therefore I ’ ll get an A again and again in the future.”When they failed, it wasn ’ t because they were dumb, but because the exam just happened to be unfair. Next time they will do better. People with low expectations do the opposite. So when they failed it was because they were dumb, and when they succeeded it was because the exam just happened to be really easy. Next time reality would catch up with them. So they felt worse. Number two : Regardless of the outcome, the pure act of anticipation makes us happy. The behavioral economist George Lowenstein asked students in his university to imagine getting a **passionate** kiss from a celebrity, any celebrity. Then he said,”How much are you willing to pay to get a kiss from a celebrity if the kiss was delivered immediately, in three hours, in 24 hours, in three days, in one year, in 10 years? He found that the students were willing to pay the most not to get a kiss immediately, but to get a kiss in three days. They were willing to pay extra in order to wait. Now they weren ’ t willing to wait a year or 10 years ; no one wants an aging celebrity. But three days seemed to be the optimum amount. So why is that? Well if you get the kiss now, it ’ s over and done with. But if you get the kiss in three days, well that ’ s three days of jittery anticipation, the thrill of the wait. The students wanted that time to imagine where is it going to happen, how is it going to happen. Anticipation made them happy. This is, by the way, why people prefer Friday to Sunday. It ’ s a really curious fact, because Friday is a day of work and Sunday is a day of pleasure, so you ’ d assume that people will prefer Sunday, but they don ’ t. It ’ s not because they really, really like being in the office and they can ’ t stand strolling in the park or having a lazy brunch. We know that, because when you ask people about their ultimate favorite day of the week, surprise, surprise, Saturday comes in at first, then Friday, then Sunday. People prefer Friday because Friday brings with it the anticipation of the weekend ahead, all the plans that you have. On Sunday, the only thing you can look forward to is the work week. So optimists are people who expect more kisses in their future, more strolls in the park. And that anticipation enhances their wellbeing. In fact, without the optimism **bias**, we would all be slightly depressed. People with mild depression, they don ’ t have a **bias** when they look into the future. They ’ re actually more realistic than healthy individuals. But individuals with severe depression, they have a pessimistic **bias**. So they tend to expect the future to be worse than it ends up being. So optimism changes subjective reality. The way we expect the world to be changes the way we see it. But it also changes objective reality. It acts as a self-fulfilling prophecy. And that is the third reason why lowering your expectations will not make you happy. Controlled experiments have shown that optimism is not only related to success, it leads to success. Optimism leads to success in **academia** and sports and politics. And maybe the most surprising benefit of optimism is health. If we expect the future to be bright, **stress** and anxiety are reduced. So all in all, optimism has lots of benefits. But the question that was really confusing to me was, how do we maintain optimism in the face of reality? As an neuroscientist, this was especially confusing, because according to all the theories out there, when your expectations are not met, you should alter them. But this is not what we find. We asked people to come into our lab in order to try and figure out what was going on. We asked them to estimate their likelihood of experiencing different terrible events in their lives. So, for example, what is your likelihood of suffering from cancer? And then we told them the average likelihood of someone like them to suffer these misfortunes. So cancer, for example, is about 30 percent. And then we asked them again,"How likely are you to suffer from cancer?"What we wanted to know was whether people will take the information that we gave them to change their beliefs. And indeed they did â but mostly when the information we gave them was better than what they expected. So for example, if someone said,"My likelihood of suffering from cancer is about 50 percent,"and we said,"Hey, good news. The average likelihood is only 30 percent,"the next time around they would say,"Well maybe my likelihood is about 35 percent."So they learned quickly and efficiently. But if someone started off saying,"My average likelihood of suffering from cancer is about 10 percent,"and we said,"Hey, bad news. The average likelihood is about 30 percent,"the next time around they would say,"Yep. Still think it ' s about 11 percent."So it ' s not that they didn ' t learn at all â they did â but much, much less than when we gave them positive information about the future. And it ' s not that they didn ' t remember the numbers that we gave them ; everyone remembers that the average likelihood of cancer is about 30 percent and the average likelihood of divorce is about 40 percent. But they didn ' t think that those numbers were related to them. What this means is that **warning** signs such as these may only have limited impact. Yes, smoking kills, but mostly it kills the other guy. What I wanted to know was what was going on inside the human brain that prevented us from taking these **warning** signs personally. But at the same time, when we hear that the housing market is hopeful, we think,"Oh, my house is definitely going to double in price."To try and figure that out, I asked the participants in the experiment to lie in a brain imaging scanner. It looks like this. And using a method called functional MRI, we were able to identify regions in the brain that were responding to positive information. One of these regions is called the left inferior frontal gyrus. So if someone said,"My likelihood of suffering from cancer is 50 percent,"and we said,"Hey, good news. Average likelihood is 30 percent,"the left inferior frontal gyrus would respond fiercely. And it didn ' t matter if you ' re an extreme optimist, a mild optimist or slightly pessimistic, everyone ' s left inferior frontal gyrus was functioning perfectly well, whether you ' re Barack Obama or Woody Allen. On the other side of the brain, the right inferior frontal gyrus was responding to bad news. And here ' s the thing : it wasn ' t doing a very good job. The more optimistic you were, the less likely this region was to respond to **unexpected** negative information. And if your brain is failing at integrating bad news about the future, you will constantly leave your rose-tinted spectacles on. So we wanted to know, could we change this? Could we alter people ' s optimism **bias** by interfering with the brain activity in these regions? And there ' s a way for us to do that. This is my collaborator Ryota Kanai. And what he ' s doing is he ' s passing a small magnetic pulse through the skull of the participant in our study into their inferior frontal gyrus. And by doing that, he ' s interfering with the activity of this brain region for about half an hour. After that everything goes back to normal, I assure you. So let ' s see what happens. First of all, I ' m going to show you the average amount of **bias** that we see. So if I was to test all of you now, this is the amount that you would learn more from good news relative to bad news. Now we interfere with the region that we found to integrate negative information in this task, and the optimism **bias** grew even larger. We made people more **biased** in the way that they process information. Then we interfered with the brain region that we found to integrate good news in this task, and the optimism **bias** disappeared. We were quite amazed by these results because we were able to eliminate a deep-rooted **bias** in humans. And at this point we stopped and we asked ourselves, would we want to shatter the optimism illusion into tiny little bits? If we could do that, would we want to take people ’ s optimism **bias** away? Well I ’ ve already told you about all of the benefits of the optimism **bias**, which probably makes you want to hold onto it for dear life. But there are, of course, pitfalls, and it would be really foolish of us to ignore them. Take for example this email I recieved from a firefighter here in California. He says,”Fatality investigations for firefighters often include ’ We didn ’ t think the fire was going to do that, ’ even when all of the available information was there to make safe decisions.”This captain is going to use our findings on the optimism **bias** to try to explain to the firefighters why they think the way they do, to make them acutely aware of this very optimistic **bias** in humans. So unrealistic optimism can lead to risky behavior, to **financial** collapse, to faulty planning. The British government, for example, has acknowledged that the optimism **bias** can make individuals more likely to underestimate the costs and durations of projects. So they have adjusted the 2012 Olympic budget for the optimism **bias**. My friend who ’ s getting married in a few weeks has done the same for his wedding budget. And by the way, when I asked him about his own likelihood of divorce, he said he was quite sure it was zero percent. So what we would really like to do, is we would like to protect ourselves from the dangers of optimism, but at the same time remain hopeful, benefiting from the many fruits of optimism. And I believe there ’ s a way for us to do that. The key here really is knowledge. We ’ re not born with an innate understanding of our **biases**. These have to be identified by scientific investigation. But the good news is that becoming aware of the optimism **bias** does not shatter the illusion. It ’ s like visual illusions, in which understanding them does not make them go away. And this is good because it means we should be able to strike a balance, to come up with plans and rules to protect ourselves from unrealistic optimism, but at the same time remain hopeful. I think this cartoon portrays it nicely. Because if you ’ re one of these pessimistic penguins up there who just does not believe they can fly, you certainly never will. Because to make any kind of progress, we need to be able to imagine a different reality, and then we need to believe that that reality is possible. But if you are an extreme optimistic penguin who just jumps down blindly hoping for the best, you might find yourself in a bit of a mess when you hit the ground. But if you ’ re an optimistic penguin who believes they can fly, but then adjusts a parachute to your back just in case things don ’ t work out exactly as you had planned, you will soar like an eagle, even if you ’ re just a penguin. Thank you.

Text Sample 233: POS Dim 2 – 2008 – Score 11.00 – 2008/t000275.txt

I have given the slide show that I gave here two years ago about 2, 000 times. I ’ m giving a short slide show this morning that I ’ m giving for the very first time, so â well it ’ s â I don ’ t want or need to raise the bar, I ’ m actually trying to lower the bar. Because I ’ ve cobbled this together to try to meet the challenge of this session. And I was reminded by Karen Armstrong ’ s fantastic presentation that religion really properly understood is not about belief, but about behavior. Perhaps we should say the same thing about optimism. How dare we be optimistic? Optimism is sometimes characterized as a belief, an intellectual posture. As Mahatma Gandhi famously said,”You must become the change you wish to see in the world.”And the outcome about which we wish to be optimistic is not going to be created by the belief alone, except to the extent that the belief brings about new behavior. But the word”behavior”is also, I think, sometimes misunderstood in this context. I ’ m a big advocate of changing the lightbulbs and buying hybrids, and Tipper and I put 33 solar panels on our house, and dug the geothermal wells, and did all of that other stuff. But, as important as it is to change the lightbulbs, it is more important to change the laws. And when we change our behavior in our **daily** lives, we sometimes leave out the citizenship part and the democracy part. In order to be optimistic about this, we have to become incredibly active as citizens in our democracy. In order to solve the climate crisis, we have to solve the democracy crisis. And we have one. I have been trying to tell this story for a long time. I was reminded of that recently, by a woman who walked past the table I was sitting at, just staring at me as she walked past. She was in her 70s, looked like she had a kind face. I thought nothing of it until I saw from the corner of my eye she was walking from the opposite direction, also just staring at me. And so I said,”How do you do?”And she said,”You know, if you dyed your hair black, you would look just like Al Gore.”Many years ago, when I was a young congressman, I spent an awful lot of time dealing with the challenge of nuclear arms control â the nuclear arms race. And the military historians taught me, during that quest, that military conflicts are typically put into three categories : local battles, regional or theater wars, and the rare but all-important global, world war â strategic conflicts. And each level of conflict requires a different allocation of resources, a different approach, a different **organizational** model. Environmental challenges fall into the same three categories, and most of what we think about are local environmental problems : air pollution, water pollution, hazardous waste dumps. But there are also regional environmental problems, like acid rain from the Midwest to the Northeast, and from Western Europe to the Arctic, and from the Midwest out the Mississippi into the dead zone of the Gulf of Mexico. And there are lots of those. But the climate crisis is the rare but all-important global, or strategic, conflict. Everything is affected. And we have to organize our response appropriately. We need a worldwide, global mobilization for renewable energy, conservation, efficiency and a global transition to a low-carbon economy. We have work to do. And we can mobilize resources and political will. But the political will has to be mobilized, in order to mobilize the resources. Let me show you these slides here. I thought I would start with the logo. What ’ s missing here, of course, is the North Polar ice cap. Greenland remains. Twenty-eight years ago, this is what the polar ice cap â the North Polar ice cap â looked like at the end of the summer, at the fall equinox. This last fall, I went to the Snow and Ice Data Center in Boulder, Colorado, and talked to the **researchers** here in Monterey at the Naval Postgraduate Laboratory. This is what ’ s happened in the last 28 years. To put it in perspective, 2005 was the previous record. Here ’ s what happened last fall that has really unnerved the **researchers**. The North Polar ice cap is the same size geographically â doesn ’ t look quite the same size â but it is exactly the same size as the United States, minus an area roughly equal to the state of Arizona. The amount that disappeared in 2005 was equivalent to everything east of the Mississippi. The extra amount that disappeared last fall was equivalent to this much. It comes back in the winter, but not as permanent ice, as thin ice â vulnerable. The amount remaining could be completely gone in summer in as little as five years. That puts a lot of pressure on Greenland. Already, around the Arctic Circle â this is a famous village in Alaska. This is a town in Newfoundland. Antarctica. Latest studies from NASA. The amount of a moderate-to-severe snow melting of an area equivalent to the size of California.”They were the best of times, they were the worst of times”: the most famous opening sentence in English literature. I want to share briefly a tale of two planets. Earth and Venus are exactly the same size. Earth ' s diameter is about 400 kilometers larger, but essentially the same size. They have exactly the same amount of carbon. But the difference is, on Earth, most of the carbon has been leeched over time out of the atmosphere, deposited in the ground as coal, oil, natural gas, etc. On Venus, most of it is in the atmosphere. The difference is that our temperature is 59 degrees on average. On Venus, it ' s 855. This is relevant to our current strategy of taking as much carbon out of the ground as quickly as possible, and putting it into the atmosphere. It ' s not because Venus is slightly closer to the Sun. It ' s three times hotter than Mercury, which is right next to the Sun. Now, briefly, here ' s an image you ' ve seen, as one of the only old images, but I show it because I want to briefly give you CSI : Climate. The global scientific community says : man-made global warming pollution, put into the atmosphere, thickening this, is trapping more of the outgoing infrared. You all know that. At the last IPCC summary, the scientists wanted to say,"How certain are you?"They wanted to answer that"99 percent."The Chinese objected, and so the compromise was"more than 90 percent."Now, the skeptics say,"Oh, wait a minute, this could be variations in this energy coming in from the sun."If that were true, the stratosphere would be heated as well as the lower atmosphere, if it ' s more coming in. If it ' s more being trapped on the way out, then you would expect it to be warmer here and cooler here. Here is the lower atmosphere. Here ' s the stratosphere : cooler. CSI : Climate. Now, here ' s the good news. Sixty-eight percent of Americans now believe that human activity is responsible for global warming. Sixty-nine percent believe that the Earth is heating up in a significant way. There has been progress, but here is the key : when given a list of challenges to confront, global warming is still listed at near the bottom. What is missing is a sense of urgency. If you agree with the factual analysis, but you don ' t feel the sense of urgency, where does that leave you? Well, the Alliance for Climate Protection, which I head in conjunction with Current TV â who did this pro bono â did a worldwide contest to do commercials on how to communicate this. This is the winner. NBC â I ' ll show all of the networks here â the top journalists for NBC asked 956 questions in 2007 of the presidential candidates : two of them were about the climate crisis. ABC : 844 questions, two about the climate crisis. Fox : two. CNN : two. CBS : zero. From laughs to tears â this is one of the older tobacco commercials. So here ' s what we ' re doing. This is gasoline consumption in all of these countries. And us. But it ' s not just the developed nations. The developing countries are now following us and accelerating their pace. And actually, their cumulative emissions this year are the equivalent to where we were in 1965. And they ' re catching up very dramatically. The total concentrations : by 2025, they will be essentially where we were in 1985. If the wealthy countries were completely missing from the picture, we would still have this crisis. But we have given to the developing countries the technologies and the ways of thinking that are creating the crisis. This is in Bolivia â over thirty years. This is **peak** fishing in a few seconds. The ' 60s. ' 70s. ' 80s. ' 90s. We have to stop this. And the good news is that we can. We have the technologies. We have to have a unified view of how to go about this : the **struggle** against poverty in the world and the challenge of cutting wealthy country emissions, all has a single, very simple solution. People say,"What ' s the solution?"Here it is. Put a price on carbon. We need a CO2 tax, revenue neutral, to replace taxation on employment, which was invented by Bismarck â and some things have changed since the 19th century. In the poor world, we have to integrate the responses to poverty with the solutions to the climate crisis. Plans to fight poverty in Uganda are mooted, if we do not solve the climate crisis. But responses can actually make a huge difference in the poor countries. This is a proposal that has been talked about a lot in Europe. This was from Nature magazine. These are concentrating solar, renewable energy plants, linked in a so-called”supergrid”to supply all of the electrical power to Europe, largely from developing countries â high-voltage DC currents. This is not pie in the sky ; this can be done. We need to do it for our own economy. The latest figures show that the old model is not working. There are a lot of great investments that you can make. If you are investing in tar sands or shale oil, then you have a portfolio that is crammed with sub-prime carbon assets. And it is based on an old model. Junkies find veins in their toes when the ones in their arms and their legs collapse. Developing tar sands and coal shale is the equivalent. Here are just a few of the investments that I personally think make sense. I have a stake in these, so I ’ ll have a disclaimer there. But geothermal, concentrating solar, advanced photovoltaics, efficiency and conservation. You ’ ve seen this slide before, but there ’ s a change. The only two countries that didn ’ t ratify â and now there ’ s only one. Australia had an election. And there was a campaign in Australia that involved television and Internet and radio commercials to lift the sense of urgency for the people there. And we trained 250 people to give the slide show in every town and village and city in Australia. Lot of other things contributed to it, but the new Prime Minister announced that his very first priority would be to change Australia ’ s position on Kyoto, and he has. Now, they came to an awareness partly because of the horrible drought that they have had. This is Lake Lanier. My friend Heidi Cullen said that if we gave droughts names the way we give hurricanes names, we ’ d call the one in the southeast now Katrina, and we would say it ’ s headed toward Atlanta. We can ’ t wait for the kind of drought Australia had to change our political culture. Here ’ s more good news. The cities supporting Kyoto in the U. S. are up to 780 â and I thought I saw one go by there, just to localize this â which is good news. Now, to close, we heard a couple of days ago about the value of making individual heroism so commonplace that it becomes banal or routine. What we need is another hero generation. Those of us who are alive in the United States of America today especially, but also the rest of the world, have to somehow understand that history has presented us with a choice â just as Jill [Bolte] Taylor was figuring out how to save her life while she was distracted by the amazing experience that she was going through. We now have a culture of distraction. But we have a planetary emergency. And we have to find a way to create, in the generation of those alive today, a sense of generational **mission**. I wish I could find the words to convey this. This was another hero generation that brought democracy to the planet. Another that ended slavery. And that gave women the right to vote. We can do this. Don ’ t tell me that we don ’ t have the capacity to do it. If we had just one week ’ s worth of what we spend on the Iraq War, we could be well on the way to solving this challenge. We have the capacity to do it. One final point : I ’ m optimistic, because I believe we have the capacity, at moments of great challenge, to set aside the causes of distraction and rise to the challenge that history is presenting to us. Sometimes I hear people respond to the disturbing facts of the climate crisis by saying,”Oh, this is so terrible. What a burden we have.”I would like to ask you to reframe that. How many generations in all of human history have had the opportunity to rise to a challenge that is **worthy** of our best efforts? A challenge that can pull from us more than we knew we could do? I think we ought to approach this challenge with a sense of profound joy and gratitude that we are the generation about which, a thousand years from now, philharmonic orchestras and poets and singers will celebrate by saying, they were the ones that found it within themselves to solve this crisis and lay the basis for a bright and optimistic human future. Let ’ s do that. Thank you very much. Chris Anderson : For so many people at TED, there is deep pain that basically a design issue on a voting form â one bad design issue meant that your voice wasn ’ t being heard like that in the last eight years in a position where you could make these things come true. That hurts. Al Gore : You have no idea. CA : When you look at what the leading candidates in your own party are doing now â I mean, there ’ s â are you excited by their plans on global warming? AG : The answer to the question is hard for me because, on the one hand, I think that we should feel really great about the fact that the Republican nominee â certain nominee â John McCain, and both of the finalists for the Democratic nomination â all three have a very different and forward-leaning position on the climate crisis. All three have offered leadership, and all three are very different from the approach taken by the current administration. And I think that all three have also been responsible in putting forward plans and proposals. But the campaign dialogue that â as illustrated by the questions â that was put together by the League of Conservation Voters, by the way, the analysis of all the questions â and, by the way, the debates have all been sponsored by something that goes by the Orwellian label,”Clean Coal.”Has anybody noticed that? Every single debate has been sponsored by”Clean Coal.””Now, even lower emissions!”The richness and fullness of the dialogue in our democracy has not laid the basis for the kind of **bold** initiative that is really needed. So they ’ re saying the right things and they may â whichever of them is elected â may do the right thing, but let me tell you : when I came back from Kyoto in 1997, with a feeling of great **happiness** that we ’ d gotten that breakthrough there, and then confronted the United States Senate, only one out of 100 senators was willing to vote to confirm, to ratify that treaty. Whatever the candidates say has to be laid alongside what the people say. This challenge is part of the fabric of our whole civilization. CO2 is the exhaling breath of our civilization, literally. And now we mechanized that process. Changing that pattern requires a scope, a scale, a speed of change that is beyond what we have done in the past. So that ’ s why I began by saying, be optimistic in what you do, but be an active citizen. Demand â change the light bulbs, but change the laws. Change the global treaties. We have to speak up. We have to solve this democracy â this â We have sclerosis in our democracy. And we have to change that. Use the Internet. Go on the Internet. Connect with people. Become very active as citizens. Have a moratorium â we shouldn ’ t have any new coal-fired generating plants that aren ’ t able to capture and store CO2, which means we have to quickly build these renewable sources. Now, nobody is talking on that scale. But I do believe that between now and November, it is possible. This Alliance for Climate Protection is going to launch a nationwide campaign â grassroots mobilization, television **ads**, Internet **ads**, radio, newspaper â with partnerships with everybody from the Girl Scouts to the hunters and fishermen. We need help. We need help. CA : In terms of your own personal role going forward, Al, is there something more than that you would like to be doing? AG : I have prayed that I would be able to find the answer to that question. What can I do? Buckminster Fuller once wrote,”If the future of all human civilization depended on me, what would I do? How would I be?”It does depend on all of us, but again, not just with the light bulbs. We, most of us here, are Americans. We have a democracy. We can change things, but we have to actively change. What ’ s needed really is a higher level of consciousness. And that ’ s hard to â that ’ s hard to create â but it is coming. There ’ s an old African proverb that some of you know that says,”If you want to go quickly, go alone ; if you want to go far, go together.”We have to go far, quickly. So we have to have a change in consciousness. A change in commitment. A new sense of urgency. A new appreciation for the privilege that we have of undertaking this challenge. CA : Al Gore, thank you so much for coming to TED. AG : Thank you. Thank you very much.

Text Sample 234: POS Dim 2 – 2008 – Score 9.00 – 2008/t000208.txt

This is a guy named Bob McKim. He was a creativity **researcher** in the ’ 60s and ’ 70s, and also led the Stanford Design Program. And in fact, my friend and IDEO founder, David Kelley, whoâs out there somewhere, studied under him at Stanford. And he liked to do an exercise with his students where he got them to take a piece of paper and draw the person who sat next to them, their neighbor, very quickly, just as quickly as they could. And in fact, weâre going to do that exercise right now. You all have a piece of cardboard and a piece of paper. Itâs actually got a bunch of circles on it. I need you to turn that piece of paper over ; you should find that itâs blank on the other side. And there should be a pencil. And I want you to pick somebody thatâs seated next to you, and when I say, go, youâve got 30 seconds to draw your neighbor, OK? So, everybody ready? OK. Off you go. Youâve got 30 seconds, youâd better be fast. Come on : those masterpieces... OK? Stop. All right, now. Yes, lotâs of laughter. Yeah, exactly. Lots of laughter, quite a bit of embarrassment. Am I hearing a few”sorryâs”? I think Iâm hearing a few sorryâs. Yup, yup, I think I probably am. And thatâs exactly what happens every time, every time you do this with adults. McKim found this every time he did it with his students. He got exactly the same response : lots and lots of sorryâs. And he would point this out as evidence that we fear the judgment of our peers, and that weâre embarrassed about showing our ideas to people we think of as our peers, to those around us. And this fear is what causes us to be conservative in our thinking. So we might have a wild idea, but weâre afraid to share it with anybody else. OK, so if you try the same exercise with kids, they have no embarrassment at all. They just quite happily show their masterpiece to whoever wants to look at it. But as they learn to become adults, they become much more sensitive to the opinions of others, and they lose that freedom and they do start to become embarrassed. And in studies of kids playing, itâs been shown time after time that kids who feel secure, who are in a kind of trusted environment â theyâre the ones that feel most free to play. And if youâre starting a design firm, letâs say, then you probably also want to create a place where people have the same kind of security. Where they have the same kind of security to take risks. Maybe have the same kind of security to play. Before founding IDEO, David said that what he wanted to do was to form a company where all the employees are my best friends. Now, that wasnât just self-indulgence. He knew that friendship is a short cut to play. And he knew that it gives us a sense of trust, and it allows us then to take the kind of creative risks that we need to take as designers. And so, that decision to work with his friends â now he has 550 of them â was what got IDEO started. And our studios, like, I think, many creative **workplaces** today, are designed to help people feel relaxed : familiar with their surroundings, comfortable with the people that theyâre working with. It takes more than decor, but I think weâve all seen that creative companies do often have symbols in the **workplace** that remind people to be playful, and that itâs a permissive environment. So, whether itâs this microbus meeting room that we have in one our buildings at IDEO ; or at Pixar, where the animators work in wooden huts and decorated caves ; or at the Googleplex, where itâs famous for its [beach] volleyball courts, and even this massive dinosaur skeleton with pink flamingos on it. Donât know the reason for the pink flamingos, but anyway, theyâre there in the garden.

Or even in the Swiss office of Google, which perhaps has the most wacky ideas of all. And my theory is, thatâs so the Swiss can prove to their Californian colleagues that theyâre not boring. So they have the slide, and they even have a firemanâs pole. Donât know what they do with that, but they have one. So all of these places have these symbols. Now, our big symbol at IDEO is actually not so much the place, itâs a thing. And itâs actually something that we invented a few years ago, or created a few years ago. Itâs a toy ; itâs called a"finger blaster."And I forgot to bring one up with me. So if somebody can reach under the chair thatâs next to them, youâll find something taped underneath it. Thatâs great. If you could pass it up. Thanks, David, I appreciate it. So this is a finger blaster, and you will find that every one of you has got one taped under your chair. And Iâm going to run a little experiment. Another little experiment. But before we start, I need just to put these on. Thank you. All right. Now, what Iâm going to do is, Iâm going to see how â I canât see out of these, OK. Iâm going to see how many of you at the back of the room can actually get those things onto the stage. So the way they work is, you know, you just put your finger in the thing, pull them back, and off you go. So, donât look backwards. Thatâs my only recommendation here. I want to see how many of you can get these things on the stage. So come on! There we go, there we go. Thank you. Thank you. Oh. I have another idea. I wanted to â there we go. There we go. Thank you, thank you, thank you. Not bad, not bad. No serious injuries so far. Well, theyâre still coming in from the back there ; theyâre still coming in. Some of you havenât fired them yet. Can you not figure out how to do it, or something? Itâs not that hard. Most of your kids figure out how to do this in the first 10 seconds, when they pick it up. All right. This is pretty good ; this is pretty good. Okay, all right. Letâs â I suppose we ’ d better... I ’ d better clear these up out of the way ; otherwise, Iâm going to trip over them. All right. So the rest of you can save them for when I say something particularly boring, and then you can fire at me. All right. I think Iâm going to take these off now, because I canât see a damn thing when Iâve â all right, OK. So, ah, that was fun. All right, good. So, OK, so why? So we have the finger blasters. Other people have dinosaurs, you know. Why do we have them? Well, as I said, we have them because we think maybe playfulness is important. But why is it important? We use it in a pretty pragmatic way, to be honest. We think playfulness helps us get to better creative solutions. Helps us do our jobs better, and helps us feel better when we do them. Now, an adult encountering a new situation â when we encounter a new situation we have a tendency to want to categorize it just as quickly as we can, you know. And thereâs a reason for that : we want to settle on an answer. Lifeâs complicated ; we want to figure out whatâs going on around us very quickly. I suspect, actually, that the evolutionary biologists probably have lots of reasons [for] why we want to categorize new things very, very quickly. One of them might be, you know, when we see this funny stripy thing : is that a tiger just about to jump out and kill us? Or is it just some weird shadows on the tree? We need to figure that out pretty fast. Well, at least, we did once. Most of us donât need to anymore, I suppose. This is some aluminum foil, right? You use it in the kitchen. Thatâs what it is, isnât it? Of course it is, of course it is. Well, not necessarily. Kids are more engaged with open possibilities. Now, theyâll certainly â when they come across something new, theyâll certainly ask,"What is it?"Of course they will. But theyâll also ask,"What can I do with it?"And you know, the more creative of them might get to a really interesting example. And this openness is the beginning of exploratory play. Any parents of young kids in the audience? There must be some. Yeah, thought so. So weâve all seen it, havenât we?

Weâve all told stories about how, on Christmas morning, our kids end up playing with the boxes far more than they play with the toys that are inside them. And you know, from an exploration perspective, this behavior makes complete sense. Because you can do a lot more with boxes than you can do with a toy. Even one like, say, Tickle Me Elmo â which, despite its ingenuity, really only does one thing, whereas boxes offer an infinite number of choices. So again, this is another one of those playful activities that, as we get older, we tend to forget and we have to relearn. So another one of Bob McKimâs favorite exercises is called the”30 Circles Test.”So weâre back to work. You guys are going to get back to work again. Turn that piece of paper that you did the sketch on back over, and youâll find those 30 circles printed on the piece of paper. So it should look like this. You should be looking at something like this. So what Iâm going to do is, Iâm going to give you minute, and I want you to adapt as many of those circles as you can into objects of some form. So for example, you could turn one into a football, or another one into a sun. All Iâm interested in is quantity. I want you to do as many of them as you can, in the minute that Iâm just about to give you. So, everybody ready? OK? Off you go. Okay. Put down your pencils, as they say. So, who got more than five circles figured out? Hopefully everybody? More than 10? Keep your hands up if you did 10. 15? 20? Anybody get all 30? No? Oh! Somebody did. Fantastic. Did anybody to a variation on a theme? Like a smiley face? Happy face? Sad face? Sleepy face? Anybody do that? Anybody use my examples? The sun and the football? Great. Cool. So I was really interested in quantity. I wasnât actually very interested in whether they were all different. I just wanted you to fill in as many circles as possible. And one of the things we tend to do as adults, again, is we edit things. We stop ourselves from doing things. We self-edit as weâre having ideas. And in some cases, our desire to be original is actually a form of editing. And that actually isnât necessarily really playful. So that ability just to go for it and explore lots of things, even if they donât seem that different from each other, is actually something that kids do well, and it is a form of play. So now, Bob McKim did another version of this test in a rather famous experiment that was done in the 1960s. Anybody know what this is? Itâs the peyote cactus. Itâs the plant from which you can create mescaline, one of the psychedelic drugs. For those of you around in the ’ 60s, you probably know it well. McKim published a paper in 1966, describing an experiment that he and his colleagues conducted to test the effects of psychedelic drugs on creativity. So he picked 27 professionals â they were engineers, physicists, mathematicians, architects, furniture designers even, artists â and he asked them to come along one evening, and to bring a problem with them that they were working on. He gave each of them some mescaline, and had them listen to some nice, relaxing music for a while. And then he did whatâs called the Purdue Creativity Test. You might know it as,”How many uses can you find for a paper clip?”Itâs basically the same thing as the 30 circles thing that I just had you do. Now, actually, he gave the test before the drugs and after the drugs, to see what the difference was in peopleâs facility and speed with coming up with ideas. And then he asked them to go away and work on those problems that theyâd brought. And theyâd come up with a bunch of interesting solutions â and actually, quite valid solutions â to the things that theyâd been working on. And so, some of the things that they figured out, some of these individuals figured out ; in one case, a new commercial building and designs for houses that were accepted by clients ; a design of a solar space probe experiment ; a redesign of the linear electron accelerator ; an engineering improvement to a magnetic tape recorder â you can tell this is a while ago ; the completion of a line of furniture ; and even a new **conceptual** model of the photon. So it was a pretty successful evening. In fact, maybe this experiment was the reason that Silicon Valley got off to its great start with innovation. We donâ□□t know, but it may be. We need to ask some of the CEOs whether they were involved in this mescaline experiment. But really, it wasnâ□□t the drugs that were important ; it was this idea that what the drugs did would help shock people out of their normal way of thinking, and getting them to forget the adult behaviors that were getting in the way of their ideas. But itâ□□s hard to break our habits, our adult habits. At IDEO we have brainstorming rules written on the walls. Edicts like,"Defer judgment,"or"Go for quantity."And somehow that seems wrong. I mean, can you have rules about creativity? Well, it sort of turns out that we need rules to help us break the old rules and **norms** that otherwise we might bring to the creative process. And weâ□□ve certainly learnt that over time, you get much better brainstorming, much more creative outcomes when everybody does play by the rules. Now, of course, many designers, many individual designers, achieve this is in a much more organic way. I think the Eameses are wonderful examples of experimentation. And they experimented with plywood for many years without necessarily having one single goal in mind. They were exploring following what was interesting to them. They went from designing splints for wounded soldiers coming out of World War II and the Korean War, I think, and from this experiment they moved on to chairs. Through constant experimentation with materials, they developed a wide range of iconic solutions that we know today, eventually resulting in, of course, the legendary lounge chair. Now, if the Eameses had stopped with that first great solution, then we wouldnâ□□t be the beneficiaries of so many wonderful designs today. And of course, they used experimentation in all aspects of their work, from films to buildings, from games to graphics. So, theyâ□□re great examples, I think, of exploration and experimentation in design. Now, while the Eameses were exploring those possibilities, they were also exploring physical objects. And they were doing that through building prototypes. And building is the next of the behaviors that I thought Iâ□□d talk about. So the average Western first-grader spends as much as 50 percent of their play time taking part in whatâ□□s called"construction play."Construction play â□□ itâ□□s playful, obviously, but also a powerful way to learn. When play is about building a tower out of blocks, the kid begins to learn a lot about towers. And as they repeatedly knock it down and start again, learning is happening as a sort of by-product of play. Itâ□□s classically learning by doing. Now, David Kelley calls this behavior, when itâ□□s carried out by designers,"thinking with your hands."And it typically involves making multiple, low- resolution prototypes very quickly, often by bringing lots of found elements together in order to get to a solution. On one of his earliest projects, the team was kind of stuck, and they came up with a mechanism by hacking together a prototype made from a roll-on deodorant. Now, that became the first commercial computer mouse for the Apple Lisa and the Macintosh. So, they learned their way to that by building prototypes. Another example is a group of designers who were working on a surgical instrument with some surgeons. They were meeting with them ; they were talking to the surgeons about what it was they needed with this device. And one of the designers ran out of the room and grabbed a white board marker and a film canister â□□ which is now becoming a very precious prototyping medium â□□ and a clothespin. He taped them all together, ran back into the room and said,"You mean, something like this?"And the surgeons grabbed hold of it and said, well, I want to hold it like this, or like that. And all of a sudden a productive conversation was happening about design around a tangible object. And in the end it turned into a real device. And so this behavior is all about quickly getting something into the real world, and having your thinking advanced as a result. At IDEO thereâs a kind of a back-to-preschool feel sometimes about the environment. The prototyping carts, filled with colored paper and Play-Doh and glue sticks and stuff â I mean, they do have a bit of a kindergarten feel to them. But the important idea is that everythingâs at hand, everythingâs around. So when designers are working on ideas, they can start building stuff whenever they want. They donât necessarily even have to go into some kind of formal workshop to do it. And we think thatâs pretty important. And then the sad thing is, although preschools are full of this kind of stuff, as kids go through the school system it all gets taken away. They lose this stuff that facilitates this sort of playful and building mode of thinking. And of course, by the time you get to the average **workplace**, maybe the best construction tool we have might be the Post-it notes. Itâs pretty barren. But by giving project teams and the clients who theyâre working with permission to think with their hands, quite complex ideas can spring into life and go right through to execution much more easily. This is a nurse using a very simple â as you can see â plasticine prototype, explaining what she wants out of a portable information system to a team of technologists and designers that are working with her in a hospital. And just having this very simple prototype allows her to talk about what she wants in a much more powerful way. And of course, by building quick prototypes, we can get out and test our ideas with consumers and users much more quickly than if weâre trying to describe them through words. But what about designing something that isnât physical? Something like a **service** or an experience? Something that exists as a series of interactions over time? Instead of building play, this can be approached with role-play. So, if youâre designing an interaction between two people â such as, I donât know â ordering food at a fast food joint or something, you need to be able to imagine how that experience might feel over a period of time. And I think the best way to achieve that, and get a feeling for any flaws in your design, is to act it out. So we do quite a lot of work at IDEO trying to convince our clients of this. They can be a little skeptical ; Iâll come back to that. But a place, I think, where the effort is really worthwhile is where people are wrestling with quite serious problems â things like education or security or finance or health. And this is another example in a healthcare environment of some doctors and some nurses and designers acting out a **service** scenario around patient care. But you know, many adults are pretty reluctant to engage with role-play. Some of itâs embarrassment and some of it is because they just donât believe that what emerges is necessarily valid. They dismiss an interesting interaction by saying, you know,"Thatâs just happening because theyâre acting it out."Research into kids ' behavior actually suggests that itâs worth taking role- playing seriously. Because when children play a role, they actually follow social scripts quite closely that theyâve learnt from us as adults. If one kid plays"store,"and another oneâs playing"house,"then the whole kind of play falls down. So they get used to quite quickly to understanding the rules for social interactions, and are actually quite quick to point out when theyâre broken. So when, as adults, we role-play, then we have a huge set of these scripts already internalized. Weâve gone through lots of experiences in life, and they provide a strong intuition as to whether an interaction is going to work. So weâre very good, when acting out a solution, at spotting whether something lacks authenticity. So role-play is actually, I think, quite valuable when it comes to thinking about experiences. Another way for us, as designers, to explore role-play is to put ourselves through an experience which weâre designing for, and project ourselves into an experience. So here are some designers who are trying to understand what it might feel like to sleep in a confined space on an airplane. And so they grabbed some very simple materials, you can see, and did this role-play, this kind of very crude role-play, just to get a sense of what it would be like for passengers if they were stuck in quite small places on airplanes. This is one of our designers, Kristian Simsarian, and heâ□□s putting himself through the experience of being an ER patient. Now, this is a real hospital, in a real emergency room. One of the reasons he chose to take this rather large video camera with him was because he didnâ□□t want the doctors and nurses thinking he was actually sick, and sticking something into him that he was going to regret later. So anyhow, he went there with his video camera, and itâ□□s kind of interesting to see what he brought back. Because when we looked at the video when he got back, we saw 20 minutes of this. And also, the amazing thing about this video â□□ as soon as you see it you immediately project yourself into that experience. And you know what it feels like : all of that uncertainty while youâ□□re left out in the hallway while the docs are dealing with some more urgent case in one of the emergency rooms, wondering what the heckâ□□s going on. And so this notion of using role-play â□□ or in this case, living through the experience as a way of creating **empathy** â□□ particularly when you use video, is really powerful. Or another one of our designers, Altay Sendil : heâ□□s here having his chest waxed, not because heâ□□s very vain, although actually he is â□□ no, Iâ□□m kidding â□□ but in order to empathize with the pain that chronic care patients go through when theyâ□□re having dressings removed. And so sometimes these analogous experiences, analogous role-play, can also be quite valuable. So when a kid dresses up as a firefighter, you know, heâ□□s beginning to try on that identity. He wants to know what it feels like to be a firefighter. Weâ□□re doing the same thing as designers. Weâ□□re trying on these experiences. And so the idea of role-play is both as an **empathy** tool, as well as a tool for prototyping experiences. And you know, we kind of admire people who do this at IDEO anyway. Not just because they lead to insights about the experience, but also because of their willingness to explore and their ability to unselfconsciously surrender themselves to the experience. In short, we admire their willingness to play. Playful exploration, playful building and role-play : those are some of the ways that designers use play in their work. And so far, I admit, this might feel like itâ□□s a message just to go out and play like a kid. And to certain extent it is, but I want to **stress** a couple of points. The first thing to remember is that play is not anarchy. Play has rules, especially when itâ□□s group play. When kids play tea party, or they play cops and robbers, theyâ□□re following a script that theyâ□□ve agreed to. And itâ□□s this code negotiation that leads to productive play. So, remember the sketching task we did at the beginning? The kind of little face, the portrait you did? Well, imagine if you did the same task with friends while you were drinking in a pub. But everybody agreed to play a game where the worst sketch artist bought the next round of drinks. That framework of rules would have turned an embarrassing, difficult situation into a fun game. As a result, weâ□□d all feel perfectly secure and have a good time â□□ but because we all understood the rules and we agreed on them together. But there arenâ□□t just rules about how to play ; there are rules about when to play. Kids donâ□□t play all the time, obviously. They transition in and out of it, and good teachers spend a lot of time thinking about how to move kids through these experiences. As designers, we need to be able to transition in and out of play also. And if weâ□□re running design studios we need to be able to figure out, how can we transition designers through these different experiences? I think this is particularly true if we think about the sort of â□□ I think whatâ□□s very different about design is that we go through these two very distinctive modes of operation. We go through a sort of **generative** mode, where weâ□□re exploring many ideas ; and then we come back together again, and come back looking for that solution, and developing that solution. I think theyâ□□re two quite different modes : divergence and convergence. And I think itâ□□s probably in the divergent mode that we most need playfulness. Perhaps in convergent mode we need to be more serious. And so being able to move between those modes is really quite important. So, itâ□□s where thereâ□□s a more nuanced version view of play, I think, is required. Because itâ□□s very easy to fall into the trap that these states are absolute. Youâ□□re either playful or youâ□□re serious, and you canâ□□t be both. But thatâ□□s not really true : you can be a serious professional adult and, at times, be playful. Itâ□□s not an either / or ; itâ□□s an”and.”You can be serious and play. So to sum it up, we need trust to play, and we need trust to be creative. So, thereâ□□s a connection. And there are a series of behaviors that weâ□□ve learnt as kids, and that turn out to be quite useful to us as designers. They include exploration, which is about going for quantity ; building, and thinking with your hands ; and role-play, where acting it out helps us both to have more **empathy** for the situations in which weâ□□re designing, and to create **services** and experiences that are seamless and authentic. Thank you very much.

Text Sample 235: POS Dim 2 – 2008 – Score 9.00 – 2008/t000284.txt

Thank you so much everyone from TED, and Chris and Amy in particular. I cannot believe I ’ m here. I have not slept in weeks. Neil and I were sitting there comparing how little we ’ ve slept in anticipation for this. I ’ ve never been so nervous â□□ and I do this when I ’ m nervous, I just realized. So, I ’ m going to talk about sort of what we did at this organization called 826 Valencia, and then I ’ m going to talk about how we all might join in and do similar things. Back in about 2000, I was living in Brooklyn, I was trying to finish my first book, I was wandering around dazed every day because I wrote from 12 a. m. to 5 a. m. So I would walk around in a daze during the day. I had no mental acuity to speak of during the day, but I had flexible hours. In the Brooklyn neighborhood that I lived in, Park Slope, there are a lot of writers â□□ it ’ s like a very high per capita ratio of writers to normal people. Meanwhile, I had grown up around a lot of teachers. My mom was a teacher, my sister became a teacher and after college so many of my friends went into teaching. And so I was always hearing them talk about their lives and how inspiring they were, and they were really sort of the most hard-working and constantly inspiring people I knew. But I knew so many of the things they were up against, so many of the **struggles** they were dealing with. And one of them was that so many of my friends that were teaching in city schools were having trouble with their students keeping up at grade level, in their reading and writing in particular. Now, so many of these students had come from households where English isn ’ t spoken in the home, where a lot of them have different **special** needs, learning disabilities. And of course they ’ re working in schools which sometimes and very often are underfunded. And so they would talk to me about this and say,”You know, what we really need is just more people, more bodies, more one-on-one attention, more hours, more expertise from people that have skills in English and can work with these students one-on-one.”Now, I would say,”Well, why don ’ t you just work with them one-on-one?”And they would say,”Well, we have five classes of 30 to 40 students each. This can lead up to 150, 180, 200 students a day. How can we possibly give each student even one hour a week of one-on-one attention?”You ’ d have to greatly multiply the workweek and clone the teachers. And so we started talking about this. And at the same time, I thought about this massive group of people I knew : writers, editors, journalists, graduate students, assistant professors, you name it. All these people that had sort of flexible **daily** hours and an interest in the English word â I hope to have an interest in the English language, but I ’ m not speaking it well right now. I ’ m trying. That clock has got me. But everyone that I knew had an interest in the primacy of the written word in terms of nurturing a democracy, nurturing an enlightened life. And so they had, you know, their time and their interest, but at the same time there wasn ’ t a conduit that I knew of in my community to bring these two communities together. So when I moved back to San Francisco, we rented this building. And the idea was to put McSweeney ’ s â McSweeney ’ s Quarterly, that we published twice or three times a year, and a few other magazines â we were going to move it into an office for the first time. It used to be in my kitchen in Brooklyn. We were going to move it into an office, and we were going to actually share space with a tutoring center. So we thought,”We ’ ll have all these writers and editors and everybody â sort of a writing community â coming into the office every day anyway, why don ’ t we just open up the front of the building for students to come in there after school, get extra help on their written homework, so you have basically no border between these two communities?”So the idea was that we would be working on whatever we ’ re working on, at 2:30 p. m. the students flow in and you put down what you ’ re doing, or you trade, or you work a little bit later or whatever it is. You give those hours in the afternoon to the students in the neighborhood. So, we had this place, we rented it, the landlord was all for it. We did this mural, that ’ s a Chris Ware mural, that basically explains the entire history of the printed word, in mural form â it takes a long time to digest and you have to stand in the middle of the road. So we rented this space. And everything was great except the landlord said,”Well, the space is zoned for retail ; you have to come up with something. You ’ ve **gotta** sell something. You can ’ t just have a tutoring center.”So we thought,”Ha ha! Really!”And we couldn ’ t think of anything necessarily to sell, but we did all the necessary research. It used to be a weight room, so there were rubber floors below, acoustic tile ceilings and fluorescent lights. We took all that down, and we found beautiful wooden floors, whitewashed beams and it had the look â while we were renovating this place, somebody said,”You know, it really kind of looks like the hull of a ship.”And we looked around and somebody else said,”Well, you should sell supplies to the working buccaneer.”And so this is what we did. So it made everybody laugh, and we said,”There ’ s a point to that. Let ’ s sell pirate supplies.”This is the pirate supply store. You see, this is sort of a sketch I did on a napkin. A great carpenter built all this stuff and you see, we made it look sort of pirate supply-like. Here you see planks sold by the foot and we have supplies to combat scurvy. We have the peg legs there, that are all handmade and fitted to you. Up at the top, you see the eyepatch display, which is the black column there for everyday use for your eyepatch, and then you have the pastel and other colors for stepping out at night â **special** occasions, bar mitzvahs and whatever. So we opened this place. And this is a vat that we fill with treasures that students dig in. This is replacement eyes in case you lose one. These are some signs that we have all over the place :”Practical Joking with Pirates.”While you ’ re reading the sign, we pull a rope behind the counter and eight mop heads drop on your head. That was just my one thing â I said we had to have something that drops on people ’ s heads. It became mop heads. And this is the fish theater, which is just a saltwater tank with three seats, and then right behind it we set up this space, which was the tutoring center. So right there is the tutoring center, and then behind the curtain were the McSweeney ’ s offices, where all of us would be working on the magazine and book editing and things like that. The kids would come in â or we thought they would come in. I should back up. We set the place up, we opened up, we spent months and months renovating this place. We had tables, chairs, computers, everything. I went to a dot-com auction at a Holiday Inn in Palo Alto and I bought 11 G4s with a stroke of a paddle. Anyway, we bought ’ em, we set everything up and then we waited. It was started with about 12 of my friends, people that I had known for years that were writers in the neighborhood. And we sat. And at 2:30 p. m. we put a sandwich board out on the front sidewalk and it just said,”Free Tutoring for Your English-Related and Writing-Related Needs â Just Come In, It ’ s All Free.”And we thought,”Oh, they ’ re going to storm the gates, they ’ re gonna love it.”And they didn ’ t. And so we waited, we sat at the tables, we waited and waited. And everybody was becoming very discouraged because it was weeks and weeks that we waited, really, where nobody came in. And then somebody alerted us to the fact that maybe there was a trust gap, because we were operating behind a pirate supply store. We never put it together, you know? And so then, around that time, I persuaded a woman named Nineveh Caligari, a longtime San Francisco educator â she was teaching in Mexico City, she had all the experience necessary, knew everything about education, was connected with all the teachers and community members in the neighborhood â I convinced her to move up from Mexico City where she was teaching. She took over as executive director. Immediately, she made the inroads with the teachers and the parents and the students and everything, and so suddenly it was actually full every day. And what we were trying to offer every day was one-on-one attention. The goal was to have a one-to-one ratio with every one of these students. You know, it ’ s been proven that 35 to 40 hours a year with one-on-one attention, a student can get one grade level higher. And so most of these students, English is not spoken in the home. They come there, many times their parents â you can ’ t see it, but there ’ s a church pew that I bought in a Berkeley auction right there â the parents will sometimes watch while their kids are being tutored. So that was the basis of it, was one-on-one attention. And we found ourselves full every day with kids. If you ’ re on Valencia Street within those few blocks at around 2 p. m., 2:30 p. m., you will get run over, often, by the kids and their big backpacks, or whatever, actually running to this space, which is very strange, because it ’ s school, in a way. But there was something psychological happening there that was just a little bit different. And the other thing was, there was no stigma. Kids weren ’ t going into the”Center-for- Kids-That-Need-More-Help,”or something like that. It was 826 Valencia. First of all, it was a pirate supply store, which is insane. And then secondly, there ’ s a publishing company in the back. And so our interns were actually working at the same tables very often, and shoulder-to-shoulder, computer-next- to-computer with the students. And so it became a tutoring center â publishing center, is what we called it â and a writing center. They go in, and they might be working with a high school student actually working on a novel â because we had very gifted kids, too. So there ’ s no stigma. They ’ re all working next to each other. It ’ s all a creative endeavor. They ’ re seeing adults. They ’ re modeling their behavior. These adults, they ’ re working in their field. They can lean over, ask a question of one of these adults and it all sort of feeds on each other. There ’ s a lot of cross-pollination. The only problem, especially for the adults working at McSweeney ’ s who hadn ’ t necessarily bought into all of this when they signed up, was that there was just the one bathroom. With like 60 kids a day, this is a problem. But you know, there ’ s something about the kids finishing their homework in a given day, working one-on-one, getting all this attention â they go home, they ’ re finished. They don ’ t stall. They don ’ t do their homework in front of the TV. They ’ re allowed to go home at 5:30 p. m., enjoy their family, enjoy other hobbies, get outside, play. And that makes a happy family. A bunch of happy families in a neighborhood is a happy community. A bunch of happy communities tied together is a happy city and a happy world. So the key to it all is homework! There you have it, you know â one-on- one attention. So we started off with about 12 volunteers, and then we had about 50, and then a couple hundred. And we now have 1, 400 volunteers on our roster. And we make it incredibly easy to volunteer. The key thing is, even if you only have a couple of hours a month, those two hours shoulder-to-shoulder, next to one student, concentrated attention, shining this beam of light on their work, on their thoughts and their self-expression, is going to be absolutely transformative, because so many of the students have not had that ever before. So we said,”Even if you have two hours one Sunday every six months, it doesn ’ t matter. That ’ s going to be enough.”So that ’ s partly why the tutor corps grew so fast. Then we said,”Well, what are we going to do with the space during the day, because it has to be used before 2:30 p. m.?”So we started bringing in classes during the day. So every day, there ’ s a field trip where they together create a book â you can see it being typed up above. This is one of the classes getting way too excited about writing. You just point a camera at a class, and it always looks like this. So this is one of the books that they do. Notice the title of the book,”The Book That Was Never Checked Out : Titanic.”And the first line of that book is,”Once there was a book named Cindy that was about the Titanic.”So, meanwhile, there ’ s an adult in the back typing this up, taking it completely seriously, which blows their mind. So then we still had more tutors to use. This is a shot of just some of the tutors during one of the events. The teachers that we work with â and everything is different to teachers â they tell us what to do. We went in there thinking,”We ’ re ultimately, completely malleable. You ’ re going to tell us. The neighborhood ’ s going to tell us, the parents are going to tell us. The teachers are going to tell us how we ’ re most useful.”So then they said,”Why don ’ t you come into the schools? Because what about the students that wouldn ’ t come to you, necessarily, who don ’ t have really active parents that are bringing them in, or aren ’ t close enough?”So then we started saying,”Well, we ’ ve got 1, 400 people on our tutor roster. Let ’ s just put out the word.”A teacher will say,”I need 12 tutors for the next five Sundays. We ’ re working on our college essays. Send them in.”So we put that out on the wire : 1, 400 tutors. Whoever can make it signs up. They go in about a half an hour before the class. The teacher tells them what to do, how to do it, what their training is, what their project is so far. They work under the teacher ’ s guide, and it ’ s all in one big room. And that ’ s actually the brunt of what we do is, people going straight from their **workplace**, straight from home, straight into the classroom and working directly with the students. So then we ’ re able to work with thousands and thousands of more students. Then another school said,”Well, what if we just give you a classroom and you can staff it all day?”So this is the Everett Middle School Writers ’ Room, where we decorated it in buccaneer style. It ’ s right off the library. And there we serve all 529 kids in this middle school. This is their newspaper, the”Straight-Up News,”that has an ongoing column from Mayor Gavin Newsom in both languages â English and Spanish. So then one day Isabel Allende wrote to us and said,”Hey, why don ’ t you assign a book with high school students? I want them to write about how to achieve peace in a violent world.”And so we went into Thurgood Marshall High School, which is a school that we had worked with on some other things, and we gave that assignment to the students. And we said,”Isabel Allende is going to read all your essays at the end. She ’ s going to publish them in a book. She ’ s going to sponsor the printing of this book in paperback form. It ’ s going to be available in all the bookstores in the Bay Area and throughout the world, on Amazon and you name it.”So these kids worked harder than they ’ ve ever worked on anything in their lives, because there was that outside audience, there was Isabel Allende on the other end. I think we had about 170 tutors that worked on this book with them and so this worked out incredibly well. We had a big party at the end. This is a book that you can find anywhere. So that led to a series of these. You can see Amy Tan sponsored the next one,”I Might Get Somewhere.”And this became an ongoing thing. More and more books. Now we ’ re sort of addicted to the book thing. The kids will work harder than they ’ ve ever worked in their life if they know it ’ s going to be permanent, know it ’ s going to be on a shelf, know that nobody can diminish what they ’ ve thought and said, that we ’ ve honored their words, honored their thoughts with hundreds of hours of five drafts, six drafts â all this attention that we give to their thoughts. And once they achieve that level, once they ’ ve written at that level, they can never go back. It ’ s absolutely transformative. And so then they ’ re all sold in the store. This is near the planks. We sell all the student books. Where else would you put them, right? So we sell ’ em, and then something weird had been happening with the stores. The store, actually â even though we started out as just a gag â the store actually made money. So it was paying the rent. And maybe this is just a San Francisco thing â I don ’ t know, I don ’ t want to **judge**. But people would come in â and this was before the pirate movies and everything! It was making a lot of money. Not a lot of money, but it was paying the rent, paying a full-time staff member there. There ’ s the ocean maps you can see on the left. And it became a gateway to the community. People would come in and say,”What the â? What is this?”I don ’ t want to swear on the web. Is that a rule? I don ’ t know. They would say,”What is this?”And people would come in and learn more about it. And then right beyond â there ’ s usually a little chain there â right beyond, they would see the kids being tutored. This is a field trip going on. And so they would be shopping, and they might be more likely to buy some lard, or millet for their parrot, or, you know, a hook, or hook protector for nighttime, all of these things we sell. So the store actually did really well. But it brought in so many people â teachers, donors, volunteers, everybody â because it was street level. It was open to the public. It wasn ’ t a non-profit buried, you know, on the 30th floor of some building downtown. It was right in the neighborhood that it was serving, and it was open all the time to the public. So, it became this sort of weird, happy accident. So all the people I used to know in Brooklyn, they said,”Well, why don ’ t we have a place like that here?”And a lot of them had been former educators or would-be educators, so they combined with a lot of local designers, local writers, and they just took the idea independently and they did their own thing. They didn ’ t want to sell pirate supplies. They didn ’ t think that that was going to work there. So, knowing the crime-fighting community in New York, they opened the Brooklyn Superhero Supply Company. This is Sam Potts ’ great design that did this. And this was to make it look sort of like one of those keysmith ’ s shops that has to have every **service** they ’ ve ever offered, you know, all over there. So they opened this place. Inside, it ’ s like a Costco for superheroes â all the supplies in kind of basic form. These are all handmade. These are all sort of repurposed other products, or whatever. All the packaging is done by Sam Potts. So then you have the villain containment unit, where kids put their parents. You have the office. This is a little vault â you have to put your product in there, it goes up an electric lift and then the guy behind the counter tells you that you have to recite the vow of heroism, which you do, if you want to buy anything. And it limits, really, their sales. Personally, I think it ’ s a problem. Because they have to do it hand on heart and everything. These are some of the products. These are all handmade. This is a secret identity kit. If you want to take on the identity of Sharon Boone, one American female marketing executive from Hoboken, New Jersey. It ' s a full dossier on everything you would need to know about Sharon Boone. So, this is the capery where you get fitted for your cape, and then you walk up these three steel-graded steps and then we turn on three hydraulic fans from every side and then you can see the cape in **action**. There ' s nothing worse than, you know, getting up there and the cape is bunching up or something like that. So then, the secret door â this is one of the shelves you don ' t see when you walk in, but it slowly opens. You can see it there in the middle next to all the grappling hooks. It opens and then this is the tutoring center in the back. So you can see the full effect! But this is â I just want to emphasize â locally funded, locally built. All the designers, all of the builders, everybody was local, all the time was pro- bono. I just came and visited and said,"Yes, you guys are doing great,"or whatever. That was it. You can see the time in all five boroughs of New York in the back. So this is the space during tutoring hours. It ' s very busy. Same principles : one-on-one attention, complete devotion to the students ' work and a boundless optimism and sort of a possibility of creativity and ideas. And this switch is flicked in their heads when they walk through those 18 feet of this bizarre store, right? So it ' s school, but it ' s not school. It ' s clearly not school, even though they ' re working shoulder-to- shoulder on tables, pencils and papers, whatever. This is one of the students, Khaled Hamdan. You can read this quote. Addicted to video games and TV. Couldn ' t concentrate at home. Came in. Got this concentrated attention. And he couldn ' t escape it. So, soon enough, he was writing. He would finish his homework early â got really addicted to finishing his homework early. It ' s an addictive thing to sort of be done with it, and to have it checked, and to know he ' s going to achieve the next thing and be prepared for school the next day. So he got hooked on that, and then he started doing other things. He ' s now been published in five books. He co-wrote a mockumentary about failed superheroes called"Super-Has-Beens."He wrote a series on"Penguin Balboa,"which is a fighting â a boxing â penguin. And then he read aloud just a few weeks ago to 500 people at Symphony Space, at a benefit for 826 New York. So he ' s there every day. He ' s evangelical about it. He brings his cousins in now. There ' s four family members that come in every day. So, I ' ll go through really quickly. This is L. A., The Echo Park Time Travel Mart :"Whenever You Are, We ' re Already Then."This is sort of a 7- Eleven for time travelers. So you see everything : it ' s exactly as a 7-Eleven would be. Leeches. Mammoth chunks. They even have their own Slurpee machine :"Out of Order. Come Back Yesterday."Anyway. So I ' m going to jump ahead. These are spaces that are only affiliated with us, doing this same thing : Word St. in Pittsfield, Massachusetts ; Ink Spot in Cincinnati ; Youth Speaks, San Francisco, California, which inspired us ; Studio St. Louis in St. Louis ; Austin Bat Cave in Austin ; Fighting Words in Dublin, Ireland, started by Roddy Doyle, this will be open in April. Now I ' m going to the TED Wish â is that okay? All right, I ' ve got a minute. So, the TED Wish : I wish that you â you personally and every creative individual and organization you know â will find a way to directly engage with a public school in your area and that you ' ll then tell the story of how you got involved, so that within a year we have a thousand examples â a thousand! â of transformative partnerships. Profound leaps forward! And these can be things that maybe you ' re already doing. I know that so many people in this room are already doing really interesting things. I know that for a fact. So, tell us these stories and inspire others on the website. We created a website. I ' m going to switch to"we,"and not"I,"hope : We hope that the attendees of this conference will usher in a new era of participation in our public schools. We hope that you will take the lead in partnering your innovative spirit and expertise with that of innovative educators in your community. Always let the teachers lead the way. They will tell you how to be useful. I hope that you ' ll step in and help out. There are a million ways. You can walk up to your local school and consult with the teachers. They ' ll always tell you how to help. So, this is with Hot Studio in San Francisco, they did this phenomenal job. This website is already up, it ' s already got a bunch of stories, a lot of ideas. It ' s called"Once Upon a School,"which is a great title, I think. This site will document every story, every project that comes out of this conference and around the world. So you go to the website, you see a bunch of ideas you can be inspired by and then you add your own projects once you get started. Hot Studio did a great job in a very tight **deadline**. So, visit the site. If you have any questions, you can ask this guy, who ' s our director of national programs. He ' ll be on the phone. You email him, he ' ll answer any question you possibly want. And he ' ll get you inspired and get you going and guide you through the process so that you can affect change. And it can be fun! That ' s the point of this talk â□□ it needn ' t be sterile. It needn ' t be bureaucratically untenable. You can do and use the skills that you have. The schools need you. The teachers need you. Students and parents need you. They need your actual person : your physical personhood and your open minds and open ears and boundless compassion, sitting next to them, listening and nodding and asking questions for hours at a time. Some of these kids just don ' t plain know how good they are : how smart and how much they have to say. You can tell them. You can shine that light on them, one human interaction at a time. So we hope you ' ll join us. Thank you so much.

Text Sample 236: POS Dim 2 – 2010 – Score 11.00 – 2010/t002629.txt
So what does the happiest man in the world look like? He certainly doesn ' t look like me. He looks like this. His name is Matthieu Ricard. So how do you get to be the happiest man in the world? Well it turns out there is a way to measure **happiness** in the brain. And you do that by measuring the relative activation of the left prefrontal cortex in the fMRI, versus the right prefrontal cortex. And Matthieu ' s **happiness** measure is off the charts. He ' s by far the happiest man ever measured by science. Which leads us to a question : What was he thinking when he was being measured? Perhaps something very naughty. Actually, he was meditating on compassion. Matthieu ' s own experience is that compassion is the happiest state ever. Reading about Matthieu was one of the pivotal moments of my life. My dream is to create the conditions for world peace in my lifetime — and to do that by creating the conditions for inner peace and compassion on a global scale. And learning about Matthieu gave me a new angle to look at my work. Matthieu ' s brain scan shows that compassion is not a chore. Compassion is something that creates **happiness**. Compassion is fun. And that mind-blowing insight changes the entire game. Because if compassion was a chore, nobody ' s going to do it, except maybe the Dalai Lama or something. But if compassion was fun, everybody ' s going to do it. Therefore, to create the conditions for global compassion, all we have to do is to reframe compassion as something that is fun. But fun is not enough. What if compassion is also profitable? What if compassion is also good for business? Then, every boss, every manager in the world, will want to have compassion — like this. That would create the conditions for world peace. So, I started paying attention to what compassion looks like in a business **setting**. Fortunately, I didn ' t have to look very far. Because what I was looking for was right in front of my eyes — in Google, my company. I know there are other compassionate compa-

nies in the world, but Google is the place I ’ m familiar with because I ’ ve been there for 10 years, so I ’ ll use Google as the case study. Google is a company born of idealism. It ’ s a company that thrives on idealism. And maybe because of that, compassion is organic and widespread company-wide. In Google, **expressions** of corporate compassion almost always follow the same pattern. It ’ s sort of a funny pattern. It starts with a small group of Googlers taking the initiative to do something. And they don ’ t usually ask for permission ; they just go ahead and do it, and then other Googlers join in, and it just gets bigger and bigger. And sometimes it gets big enough to become official. So in other words, it almost always starts from the bottom up. And let me give you some examples. The first example is the largest annual community event — where Googlers from around the world donate their labor to their local communities — was initiated and organized by three employees before it became official, because it just became too big. Another example, three Googlers — a chef, an engineer and, most funny, a massage therapist — three of them, they learned about a region in India where 200, 000 people live without a single medical facility. So what do they do? They just go ahead and start a fundraiser. And they raise enough money to build this hospital — the first hospital of its kind for 200, 000 people. During the Haiti earthquake, a number of engineers and product managers spontaneously came together and stayed overnight to build a tool to allow earthquake victims to find their loved ones. And **expressions** of compassion are also found in our international offices. In China for example, one mid-level employee initiated the largest social **action** competition in China, involving more than 1, 000 schools in China, working on issues such as education, poverty, health care and the environment. There is so much organic social **action** all around Google that the company decided to form a social responsibility team just to support these efforts. And this idea, again, came from the grassroots, from two Googlers who wrote their own job descriptions and volunteered themselves for the job. And I found it fascinating that the social responsibility team was not formed as part of some grand corporate strategy. It was two persons saying,”Let ’ s do this,”and the company said,”Yes.”So it turns out that Google is a compassionate company, because Googlers found compassion to be fun. But again, fun is not enough. There are also real business benefits. So what are they? The first benefit of compassion is that it creates highly effective business leaders. What does that mean? There are three components of compassion. There is the affective component, which is,”I feel for you.”There is the cognitive component, which is,”I understand you.”And there is a motivational component, which is,”I want to help you.”So what has this got to do with business leadership? According to a very comprehensive study led by Jim Collins, and documented in the book”Good to Great,”it takes a very **special** kind of leader to bring a company from goodness to greatness. And he calls them”Level 5 leaders.”These are leaders who, in addition to being highly capable, possess two important qualities, and they are humility and ambition. These are leaders who are highly ambitious for the greater good. And because they ’ re ambitious for a greater good, they feel no need to inflate their own egos. And they, according to the research, make the best business leaders. And if you look at these qualities in the context of compassion, we find that the cognitive and affective components of compassion — understanding people and empathizing with people — inhibits, tones down, what I call the excessive self-obsession that ’ s in us, therefore creating the conditions for humility. The motivational component of compassion creates ambition for greater good. In other words, compassion is the way to grow Level 5 leaders. And this is the first compelling business benefit. The second compelling benefit of compassion is that it creates an inspiring **workforce**. Employees mutually inspire each other towards greater good. It creates a vibrant, energetic community where people admire and respect each other. I mean, you come to work in the morning, and you work with three guys who just up and decide to build a hospital in India. It ' s like how can you not be inspired by those people — your own coworkers? So this mutual inspiration promotes collaboration, initiative and creativity. It makes us a highly effective company. So, having said all that, what is the secret formula for brewing compassion in the corporate **setting**? In our experience, there are three ingredients. The first ingredient is to create a culture of **passionate** concern for the greater good. So always think : how is your company and your job serving the greater good? Or, how can you further serve the greater good? This awareness of serving the greater good is very self-inspiring and it creates fertile ground for compassion to grow in. That ' s one. The second ingredient is autonomy. So in Google, there ' s a lot of autonomy. And one of our most popular managers jokes that, this is what he says,"Google is a place where the inmates run the asylum."And he considers himself one of the inmates. If you already have a culture of compassion and idealism and you let your people roam free, they will do the right thing in the most compassionate way. The third ingredient is to focus on inner development and personal growth. Leadership training in Google, for example, places a lot of emphasis on the inner qualities, such as self-awareness, self-mastery, **empathy** and compassion, because we believe that leadership begins with character. We even created a seven-week curriculum on emotion intelligence, which we jokingly call"Searching Inside Yourself."It ' s less naughty than it sounds. So I ' m an engineer by training, but I ' m one of the creators and instructors of this course, which I find kind of funny, because this is a company that trusts an engineer to teach emotion intelligence. What a company. So"Search Inside Yourself"— how does it work? It works in three steps. The first step is attention training. Attention is the basis of all higher cognitive and emotional abilities. Therefore, any curriculum for training emotion intelligence has to begin with attention training. The idea here is to train attention to create a quality of mind that is calm and clear at the same time. And this creates the foundation for emotion intelligence. The second step follows the first step. The second step is developing self-knowledge and self-mastery. So using the supercharged attention from step one, we create a high-resolution perception into the cognitive and emotive processes. What does that mean? It means being able to **observe** our thought stream and the process of emotion with high clarity, objectivity and from a third-person perspective. And once you can do that, you create the kind of self-knowledge that enables self-mastery. The third step, following the second step, is to create new mental habits. What does that mean? Imagine this. Imagine whenever you meet any other person, any time you meet a person, your habitual, instinctive first thought is,"I want you to be happy. I want you to be happy."Imagine you can do that. Having this habit, this mental habit, changes everything at work. Because this good will is unconsciously picked up by other people, and it creates trust, and trust creates a lot of good working relationships. And this also creates the conditions for compassion in the **workplace**. Someday, we hope to open-source"Search Inside Yourself"so that everybody in the corporate world will at least be able to use it as a reference. And in closing, I want to end the same place I started, with **happiness**. I want to quote this guy — the guy in robes, not the other guy — the Dalai Lama, who said,"If you want others to be happy, **practice** compassion. If you want to be happy, **practice** compassion."I found this to be true, both on the individual level and at a corporate level. And I hope that compassion will be both fun and profitable for you too. Thank you.

Text Sample 237: POS Dim 2 – 2010 – Score 10.00 – 2010/t002806.txt

I ’ m going to talk about the simple truth in leadership in the 21st century. In the 21st century, we need to actually look at â and what I ’ m actually going to encourage you to consider today â is to go back to our school days when we learned how to count. But I think it ’ s time for us to think about what we count. Because what we actually count truly counts. Let me start by telling you a little story. This is Van Quach. She came to this country in 1986 from Vietnam. She changed her name to Vivian because she wanted to fit in here in America. Her first job was at an inner-city motel in San Francisco as a maid. I happened to buy that motel about three months after Vivian started working there. So Vivian and I have been working together for 23 years. With the youthful idealism of a 26-year-old, in 1987, I started my company and I called it Joie de Vivre, a very impractical name, because I actually was looking to create joy of life. And this first hotel that I bought, motel, was a pay-by-the-hour, no-tell motel in the inner-city of San Francisco. As I spent time with Vivian, I saw that she had sort of a joie de vivre in how she did her work. It made me question and curious : How could someone actually find joy in cleaning toilets for a living? So I spent time with Vivian, and I saw that she didn ’ t find joy in cleaning toilets. Her job, her goal and her calling was not to become the world ’ s greatest toilet scrubber. What counts for Vivian was the emotional connection she created with her fellow employees and our guests. And what gave her inspiration and meaning was the fact that she was taking care of people who were far away from home. Because Vivian knew what it was like to be far away from home. That very human lesson, more than 20 years ago, served me well during the last economic downturn we had. In the wake of the dotcom crash and 9 / 11, San Francisco Bay Area hotels went through the largest percentage revenue drop in the history of American hotels. We were the largest operator of hotels in the Bay Area, so we were particularly vulnerable. But also back then, remember we stopped eating French fries in this country. Well, not exactly, of course not. We started eating”freedom fries,”and we started boycotting anything that was French. Well, my name of my company, Joie de Vivre â so I started getting these letters from places like Alabama and Orange County saying to me that they were going to boycott my company because they thought we were a French company. And I ’ d write them back, and I ’ d say,”What a minute. We ’ re not French. We ’ re an American company. We ’ re based in San Francisco.”And I ’ d get a terse response :”Oh, that ’ s worse.”So one particular day when I was feeling a little depressed and not a lot of joie de vivre, I ended up in the local bookstore around the corner from our offices. And I initially ended up in the business section of the bookstore looking for a business solution. But given my befuddled state of mind, I ended up in the self-help section very quickly. That ’ s where I got reacquainted with Abraham Maslow ’ s”hierarchy of needs.”I took one psychology class in college, and I learned about this guy, Abraham Maslow, as many of us are familiar with his hierarchy of needs. But as I sat there for four hours, the full afternoon, reading Maslow, I recognized something that is true of most leaders. One of the simplest facts in business is something that we often neglect, and that is that we ’ re all human. Each of us, no matter what our role is in business, has some hierarchy of needs in the **workplace**. So as I started reading more Maslow, what I started to realize is that Maslow, later in his life, wanted to take this hierarchy for the individual and apply it to the collective, to organizations and specifically to business. But unfortunately, he died prematurely in 1970, and so he wasn ’ t really able to live that dream completely. So I realized in that dotcom crash that my role in life was to channel Abe Maslow. And that ' s what I did a few years ago when I took that five-level hierarchy of needs pyramid and turned it into what I call the transformation pyramid, which is survival, success and transformation. It ' s not just fundamental in business, it ' s fundamental in life. And we started asking ourselves the questions about how we were actually addressing the higher needs, these transformational needs for our key employees in the company. These three levels of the hierarchy needs relate to the five levels of Maslow ' s hierarchy of needs. But as we started asking ourselves about how we were addressing the higher needs of our employees and our customers, I realized we had no metrics. We had nothing that actually could tell us whether we were actually getting it right. So we started asking ourselves : What kind of less obvious metrics could we use to actually evaluate our employees ' sense of meaning, or our customers ' sense of emotional connection with us? For example, we actually started asking our employees, do they understand the **mission** of our company, and do they feel like they believe in it, can they actually influence it, and do they feel that their work actually has an impact on it? We started asking our customers, did they feel an emotional connection with us, in one of seven different kinds of ways. Miraculously, as we asked these questions and started giving attention higher up the pyramid, what we found is we created more loyalty. Our customer loyalty skyrocketed. Our employee turnover dropped to one-third of the industry average, and during that five year dotcom bust, we tripled in size. As I went out and started spending time with other leaders out there and asking them how they were getting through that time, what they told me over and over again was that they just manage what they can measure. What we can measure is that tangible stuff at the bottom of the pyramid. They didn ' t even see the intangible stuff higher up the pyramid. So I started asking myself the question : How can we get leaders to start valuing the intangible? If we ' re taught as leaders to just manage what we can measure, and all we can measure is the tangible in life, we ' re missing a whole lot of things at the top of the pyramid. So I went out and studied a bunch of things, and I found a survey that showed that 94 percent of business leaders worldwide believe that the intangibles are important in their business, things like intellectual property, their corporate culture, their brand loyalty, and yet, only five percent of those same leaders actually had a means of measuring the intangibles in their business. So as leaders, we understand that intangibles are important, but we don ' t have a clue how to measure them. So here ' s another Einstein quote :"Not everything that can be counted counts, and not everything that counts can be counted."I hate to argue with Einstein, but if that which is most valuable in our life and our business actually can ' t be counted or valued, aren ' t we going to spend our lives just mired in measuring the mundane? It was that sort of heady question about what counts that led me to take my CEO hat off for a week and fly off to the Himalayan **peaks**. I flew off to a place that ' s been shrouded in mystery for centuries, a place some folks call Shangri- La. It ' s actually moved from the survival base of the pyramid to becoming a transformational role model for the world. I went to Bhutan. The teenage king of Bhutan was also a curious man, but this was back in 1972, when he ascended to the throne two days after his father passed away. At age 17, he started asking the kinds of questions that you ' d expect of someone with a beginner ' s mind. On a trip through India, early in his reign as king, he was asked by an Indian journalist about the Bhutanese GDP, the size of the Bhutanese GDP. The king responded in a fashion that actually has transformed us four decades later. He said the following, he said :"Why are we so obsessed and focused with gross domestic product? Why don ' t we care more about gross national **happiness**?"Now, in essence, the king was asking us to consider an alternative definition of success, what has come to be known as GNH, or gross national **happiness**. Most world leaders didn ’ t take notice, and those that did thought this was just”Buddhist economics.”But the king was serious. This was a notable moment, because this was the first time a world leader in almost 200 years had suggested that intangible of **happiness** â that leader 200 years ago, Thomas Jefferson with the Declaration of Independence â 200 years later, this king was suggesting that intangible of **happiness** is something that we should measure, and it ’ s something we should actually value as government officials. For the next three dozen years as king, this king actually started measuring and managing around **happiness** in Bhutan â including, just recently, taking his country from being an absolute monarchy to a constitutional monarchy with no bloodshed, no coup. Bhutan, for those of you who don ’ t know it, is the newest democracy in the world, just two years ago. So as I spent time with leaders in the GNH movement, I got to really understand what they ’ re doing. And I got to spend some time with the prime minister. Over dinner, I asked him an impertinent question. I asked him,”How can you create and measure something which evaporates â in other words, **happiness**?”And he ’ s a very wise man, and he said,”Listen, Bhutan ’ s goal is not to create **happiness**. We create the conditions for **happiness** to occur. In other words, we create a habitat of **happiness**.”Wow, that ’ s interesting. He said that they have a science behind that art, and they ’ ve actually created four essential pillars, nine key indicators and 72 different metrics that help them to measure their GNH. One of those key indicators is : How do the Bhutanese feel about how they spend their time each day? It ’ s a good question. How do you feel about how you spend your time each day? Time is one of the scarcest resources in the modern world. And yet, of course, that little intangible piece of data doesn ’ t factor into our GDP calculations. As I spent my week up in the Himalayas, I started to imagine what I call an emotional equation. And it focuses on something I read long ago from a guy named Rabbi Hyman Schachtel. How many know him? Anybody? 1954, he wrote a book called”The Real Enjoyment of Living,”and he suggested that **happiness** is not about having what you want ; instead, it ’ s about wanting what you have. Or in other words, I think the Bhutanese believe **happiness** equals wanting what you have â imagine gratitude â divided by having what you want â gratification. The Bhutanese aren ’ t on some aspirational treadmill, constantly focused on what they don ’ t have. Their religion, their **isolation**, their deep respect for their culture and now the principles of their GNH movement all have fostered a sense of gratitude about what they do have. How many of us here, as TEDsters in the audience, spend more of our time in the bottom half of this equation, in the denominator? We are a bottom-heavy culture in more ways than one. The reality is, in Western countries, quite often we do focus on the pursuit of **happiness** as if **happiness** is something that we have to go out â an object that we ’ re supposed to get, or maybe many objects. Actually, in fact, if you look in the dictionary, many dictionaries define pursuit as to”chase with hostility.”Do we pursue **happiness** with hostility? Good question. But back to Bhutan. Bhutan ’ s bordered on its north and south by 38 percent of the world ’ s population. Could this little country, like a startup in a mature industry, be the spark plug that influences a 21st century of middle-class in China and India? Bhutan ’ s created the ultimate export, a new global currency of well- being, and there are 40 countries around the world today that are studying their own GNH. You may have heard, this last fall Nicolas Sarkozy in France announcing the results of an 18-month study by two Nobel economists, focusing on **happiness** and wellness in France. Sarkozy suggested that world leaders should stop myopically focusing on GDP and consider a new index, what some French are calling a”joie de vivre index.”I like it. Co-branding opportunities. Just three days ago, three days ago here at TED, we had a simulcast of David Cameron, potentially the next prime minister of the UK, quoting one of my favorite speeches of all-time, Robert Kennedy ’ s poetic speech from 1968 when he suggested that we ’ re myopically focused on the wrong thing and that GDP is a misplaced metric. So it suggests that the momentum is shifting. I ’ ve taken that Robert Kennedy quote, and I ’ ve turned it into a new balance sheet for just a moment here. This is a collection of things that Robert Kennedy said in that quote. GDP counts everything from air pollution to the destruction of our redwoods. But it doesn ’ t count the health of our children or the integrity of our public officials. As you look at these two columns here, doesn ’ t it make you feel like it ’ s time for us to start figuring out a new way to count, a new way to imagine what ’ s important to us in life? Certainly Robert Kennedy suggested at the end of the speech exactly that. He said GDP”measures everything in short, except that which makes life worthwhile.”Wow. So how do we do that? Let me say one thing we can just start doing ten years from now, at least in this country. Why in the heck in America are we doing a census in 2010? We ’ re spending 10 billion dollars on the census. We ’ re asking 10 simple questions â it is simplicity. But all of those questions are tangible. They ’ re about demographics. They ’ re about where you live, how many people you live with, and whether you own your home or not. That ’ s about it. We ’ re not asking meaningful metrics. We ’ re not asking important questions. We ’ re not asking anything that ’ s intangible. Abe Maslow said long ago something you ’ ve heard before, but you didn ’ t realize it was him. He said,”If the only tool you have is a hammer, everything starts to look like a nail.”We ’ ve been **fooled** by our tool. Excuse that **expression**. We ’ ve been **fooled** by our tool. GDP has been our hammer. And our nail has been a 19th- and 20th-century industrial-era model of success. And yet, 64 percent of the world ’ s GDP today is in that intangible industry we call **service**, the **service** industry, the industry I ’ m in. And only 36 percent is in the tangible industries of manufacturing and agriculture. So maybe it ’ s time that we get a bigger toolbox, right? Maybe it ’ s time we get a toolbox that doesn ’ t just count what ’ s easily counted, the tangible in life, but actually counts what we most value, the things that are intangible. I guess I ’ m sort of a curious CEO. I was also a curious economics major as an undergrad. I learned that economists measure everything in tangible units of production and consumption as if each of those tangible units is exactly the same. They aren ’ t the same. In fact, as leaders, what we need to learn is that we can influence the quality of that unit of production by creating the conditions for our employees to live their calling. In Vivian ’ s case, her unit of production isn ’ t the tangible hours she works, it ’ s the intangible difference she makes during that one hour of work. This is Dave Arringdale who ’ s actually been a longtime guest at Vivian ’ s motel. He stayed there a hundred times in the last 20 years, and he ’ s loyal to the property because of the relationship that Vivian and her fellow employees have created with him. They ’ ve created a habitat of **happiness** for Dave. He tells me that he can always count on Vivian and the staff there to make him feel at home. Why is it that business leaders and investors quite often don ’ t see the connection between creating the intangible of employee **happiness** with creating the tangible of **financial** profits in their business? We don ’ t have to choose between inspired employees and sizable profits, we can have both. In fact, inspired employees quite often help make sizable profits, right? So what the world needs now, in my opinion, is business leaders and political leaders who know what to count. We count numbers. We count on people. What really counts is when we actually use our numbers to truly take into account our people. I learned that from a maid in a motel and a king of a country. What can you start counting today? What one thing can you start counting today that actually would be meaningful in your life, whether it ’ s your work life or your business life? Thank you very much.

Text Sample 238: POS Dim 2 – 2010 – Score 10.00 – 2010/t002850.txt

Hi, my name is Roz Savage and I **row** across oceans. Four years ago, I **rowed** solo across the Atlantic, and since then, I ’ ve done two out of three stages across the Pacific, from San Francisco to Hawaii and from Hawaii to Kiribati. And tomorrow, I ’ ll be leaving this boat to fly back to Kiribati to continue with the third and final stage of my **row** across the Pacific. Cumulatively, I will have **rowed** over 8, 000 miles, taken over three million oar strokes and spent more than 312 days alone on the ocean on a 23 foot rowboat. This has given me a very **special** relationship with the ocean. We have a bit of a love / hate thing going on. I feel a bit about it like I did about a very strict math teacher that I once had at school. I didn ’ t always like her, but I did respect her, and she taught me a heck of a lot. So today I ’ d like to share with you some of my ocean adventures and tell you a little bit about what they ’ ve taught me, and how I think we can maybe take some of those lessons and apply them to this environmental challenge that we face right now. Now, some of you might be thinking,”Hold on a minute. She doesn ’ t look very much like an ocean rower. Isn ’ t she meant to be about this tall and about this wide and maybe look a bit more like these guys?”You ’ ll notice, they ’ ve all got something that I don ’ t. Well, I don ’ t know what you ’ re thinking, but I ’ m talking about the beards. And no matter how long I ’ ve spent on the ocean, I haven ’ t yet managed to muster a decent beard, and I hope that it remains that way. For a long time, I didn ’ t believe that I could have a big adventure. The story that I told myself was that adventurers looked like this. I didn ’ t look the part. I thought there were them and there were us, and I was not one of them. So for 11 years, I conformed. I did what people from my kind of background were supposed to do. I was working in an office in London as a management consultant. And I think I knew from day one that it wasn ’ t the right job for me. But that kind of conditioning just kept me there for so many years, until I reached my mid-30s and I thought,”You know, I ’ m not getting any younger. I feel like I ’ ve got a purpose in this life, and I don ’ t know what it is, but I ’ m pretty certain that management consultancy is not it. So, fast forward a few years. I ’ d gone through some changes. To try and answer that question of,”What am I supposed to be doing with my life?”I sat down one day and wrote two versions of my own obituary, the one that I wanted, a life of adventure, and the one that I was actually heading for which was a nice, normal, **pleasant** life, but it wasn ’ t where I wanted to be by the end of my life. I wanted to live a life that I could be proud of. And I remember looking at these two versions of my obituary and thinking,”Oh boy, I ’ m on totally the wrong track here. If I carry on living as I am now, I ’ m just not going to end up where I want to be in five years, or 10 years, or at the end of my life.”I made a few changes, let go of some loose trappings of my old life, and through a bit of a leap of logic, decided to **row** across the Atlantic Ocean. The Atlantic Rowing Race runs from the Canaries to Antigua, it ’ s about 3, 000 miles, and it turned out to be the hardest thing I had ever done. Sure, I had wanted to get outside of my comfort zone, but what I ’ d sort of failed to notice was that getting out of your comfort zone is, by definition, extremely uncomfortable. And my timing was not great either : 2005, when I did the Atlantic, was the year of Hurricane Katrina. There were more tropical storms in the North Atlantic than ever before, since records began. And pretty early on, those storms started making their presence known. All four of my oars broke before I reached halfway across. Oars are not supposed to look like this. But what can you do? You ’ re in the middle of the ocean. Oars are your only means of propulsion. So I just had to look around the boat and figure out what I was going to use to fix up these oars so that I could carry on. So I found a boat hook and my trusty duct tape and splintered the boat hook to the oars to reinforce it. Then, when that gave out, I sawed the wheel axles off my spare rowing seat and used those. And then when those gave out, I cannibalized one of the broken oars. I ’ d never been very good at fixing stuff when I was living my old life, but it ’ s amazing how resourceful you can become when you ’ re in the middle of the ocean and there ’ s only one way to get to the other side. And the oars kind of became a symbol of just in how many ways I went beyond what I thought were my limits. I suffered from tendinitis on my shoulders and saltwater sores on my bottom. I really **struggled** psychologically, totally overwhelmed by the scale of the challenge, realizing that, if I carried on moving at two miles an hour, 3, 000 miles was going to take me a very, very long time. There were so many times when I thought I ’ d hit that limit, but had no choice but to just carry on and try and figure out how I was going to get to the other side without driving myself crazy. And eventually after 103 days at sea, I arrived in Antigua. I don ’ t think I ’ ve ever felt so happy in my entire life. It was a bit like finishing a marathon and getting out of solitary confinement and winning an Oscar all rolled into one. I was euphoric. And to see all the people coming out to greet me and standing along the cliff tops and clapping and cheering, I just felt like a movie star. It was absolutely wonderful. And I really learned then that, the bigger the challenge, the bigger the sense of achievement when you get to the end of it. So this might be a good moment to take a quick time-out to answer a few FAQs about ocean rowing that might be going through your mind. Number one that I get asked : What do you eat? A few freeze-dried meals, but mostly I try and eat much more unprocessed foods. So I grow my own beansprouts. I eat fruits and nut bars, a lot of nuts. And generally arrive about 30 pounds lighter at the other end. Question number two : How do you sleep? With my eyes shut. Ha-ha. I suppose what you mean is : What happens to the boat while I ’ m sleeping? Well, I plan my route so that I ’ m drifting with the winds and the currents while I ’ m sleeping. On a good night, I think my best ever was 11 miles in the right direction. Worst ever, 13 miles in the wrong direction. That ’ s a bad day at the office. What do I wear? Mostly, a **baseball** cap, rowing gloves and a smile — or a frown, depending on whether I went backwards overnight — and lots of sun lotion. Do I have a chase boat? No I don ’ t. I ’ m totally self-supporting out there. I don ’ t see anybody for the whole time that I ’ m at sea, generally. And finally : Am I crazy? Well, I leave that one up to you to **judge**. So, how do you top rowing across the Atlantic? Well, naturally, you decide to **row** across the Pacific. Well, I thought the Atlantic was big, but the Pacific is really, really big. I think we tend to do it a little bit of a disservice in our usual maps. I don ’ t know for sure that the Brits invented this particular view of the world, but I suspect we might have done so : we are right in the middle, and we ’ ve cut the Pacific in half and flung it to the far corners of the world. Whereas if you look in Google Earth, this is how the Pacific looks. It pretty much covers half the planet. You can just see a little bit of North America up here and a sliver of Australia down there. It is really big — 65 million square miles — and to **row** in a straight line across it would be about 8, 000 miles. Unfortunately, ocean rowboats very rarely go in a straight line. By the time I get to Australia, if I get to Australia, I will have **rowed** probably nine or 10, 000 miles in all. So, because nobody in their straight mind would **row** straight past Hawaii without dropping in, I decided to cut this very big undertaking into three segments. The first attempt didn ' t go so well. In 2007, I did a rather involuntary capsize drill three times in 24 hours. A bit like being in a washing machine. Boat got a bit dinged up, so did I. I blogged about it. Unfortunately, somebody with a bit of a hero complex decided that this damsel was in **distress** and needed saving. The first I knew about this was when the Coast Guard plane turned up overhead. I tried to tell them to go away. We had a bit of a battle of wills. I lost and got airlifted. Awful, really awful. It was one of the worst feelings of my life, as I was lifted up on that winch line into the helicopter and looked down at my trusty little boat rolling around in the 20 foot waves and wondering if I would ever see her again. So I had to launch a very expensive salvage operation and then wait another nine months before I could get back out onto the ocean again. But what do you do? Fall down nine times, get up 10. So, the following year, I set out and, fortunately, this time made it safely across to Hawaii. But it was not without misadventure. My watermaker broke, only the most important piece of kit that I have on the boat. Powered by my solar panels, it sucks in saltwater and turns it into freshwater. But it doesn ' t react very well to being immersed in ocean, which is what happened to it. Fortunately, help was at hand. There was another unusual boat out there at the same time, doing as I was doing, bringing awareness to the North Pacific Garbage Patch, that area in the North Pacific about twice the size of Texas, with an estimated 3. 5 million tons of trash in it, circulating at the center of that North Pacific Gyre. So, to make the point, these guys had actually built their boat out of plastic trash, 15, 000 empty water bottles latched together into two pontoons. They were going very slowly. Partly, they ' d had a bit of a delay. They ' d had to pull in at Catalina Island shortly after they left Long Beach because the lids of all the water bottles were coming undone, and they were starting to sink. So they ' d had to pull in and do all the lids up. But, as I was approaching the end of my water reserves, luckily, our courses were converging. They were running out of food ; I was running out of water. So we liaised by satellite phone and arranged to meet up. And it took about a week for us to actually gradually converge. I was doing a pathetically slow speed of about 1. 3 knots, and they were doing only marginally less pathetic speed of about 1. 4 : it was like two snails in a mating dance. But, eventually, we did manage to meet up and Joel hopped overboard, caught us a beautiful, big mahi- mahi, which was the best food I ' d had in, ooh, at least three months. Fortunately, the one that he caught that day was better than this one they caught a few weeks earlier. When they opened this one up, they found its stomach was full of plastic. And this is really bad news because plastic is not an inert substance. It leaches out chemicals into the flesh of the poor critter that ate it, and then we come along and eat that poor critter, and we get some of the toxins accumulating in our bodies as well. So there are very real implications for human health. I eventually made it to Hawaii still alive. And, the following year, set out on the second stage of the Pacific, from Hawaii down to Tarawa. And you ' ll notice something about Tarawa ; it is very low-lying. It ' s that little green sliver on the horizon, which makes them very nervous about rising oceans. This is big trouble for these guys. They ' ve got no points of land more than about six feet above sea level. And also, as an increase in extreme weather events due to climate change, they ' re expecting more waves to come in over the fringing reef, which will contaminate their fresh water supply. I had a meeting with the president there, who told me about his exit strategy for his country. He expects that within the next 50 years, the 100, 000 people that live there will have to relocate to New Zealand or Australia. And that made me think about how would I feel if Britain was going to disappear under the waves ; if the places where I ' d been born and gone to school and got married, if all those places were just going to disappear forever. How, literally, ungrounded that would make me feel. Very shortly, I ’ ll be setting out to try and get to Australia, and if I ’ m successful, I ’ ll be the first woman ever to **row** solo all the way across the Pacific. And I try to use this to bring awareness to these environmental issues, to bring a human face to the ocean. If the Atlantic was about my inner journey, discovering my own capabilities, maybe the Pacific has been about my outer journey, figuring out how I can use my interesting career choice to be of **service** to the world, and to take some of those things that I ’ ve learned out there and apply them to the situation that humankind now finds itself in. I think there are probably three key points here. The first one is about the stories that we tell ourselves. For so long, I told myself that I couldn ’ t have an adventure because I wasn ’ t six foot tall and athletic and bearded. And then that story changed. I found out that people had **rowed** across oceans. I even met one of them and she was just about my size. So even though I didn ’ t grow any taller, I didn ’ t sprout a beard, something had changed : My interior dialogue had changed. At the moment, the story that we collectively tell ourselves is that we need all this stuff, that we need oil. But what about if we just change that story? We do have alternatives, and we have the power of free will to choose those alternatives, those sustainable ones, to create a greener future. The second point is about the accumulation of tiny **actions**. We might think that anything that we do as an individual is just a drop in the ocean, that it can ’ t really make a difference. But it does. Generally, we haven ’ t got ourselves into this mess through big disasters. Yes, there have been the Exxon Valdezes and the Chernobyls, but mostly it ’ s been an accumulation of bad decisions by billions of individuals, day after day and year after year. And, by the same token, we can turn that tide. We can start making better, wiser, more sustainable decisions. And when we do that, we ’ re not just one person. Anything that we do spreads ripples. Other people will see if you ’ re in the supermarket line and you pull out your reusable grocery bag. Maybe if we all start doing this, we can make it socially unacceptable to say yes to plastic in the checkout line. That ’ s just one example. This is a world-wide community. The other point : It ’ s about taking responsibility. For so much of my life, I wanted something else to make me happy. I thought if I had the right house or the right car or the right man in my life, then I could be happy. But when I wrote that obituary exercise, I actually grew up a little bit in that moment and realized that I needed to create my own future. I couldn ’ t just wait passively for **happiness** to come and find me. And I suppose I ’ m a selfish environmentalist. I plan on being around for a long time, and when I ’ m 90 years old, I want to be happy and healthy. And it ’ s very difficult to be happy on a planet that ’ s racked with famine and drought. It ’ s very difficult to be healthy on a planet where we ’ ve poisoned the earth and the sea and the air. So, shortly, I ’ m going to be launching a new initiative called Eco-Heroes. And the idea here is that all our Eco-Heroes will log at least one green deed every day. It ’ s meant to be a bit of a game. We ’ re going to make an iPhone app out of it. We just want to try and create that awareness because, sure, changing a light bulb isn ’ t going to change the world, but that attitude, that awareness that leads you to change the light bulb or take your reusable coffee mug, that is what could change the world. I really believe that we stand at a very important point in history. We have a choice. We ’ ve been blessed, or cursed, with free will. We can choose a greener future, and we can get there if we all pull together to take it one stroke at a time. Thank you.

Text Sample 239: POS Dim 2 – 2003 – Score 7.00 – 2003/t000405.txt

When you think about resilience and technology it ’ s actually much easier. You ’ re going to see some other speakers today, I already know, who are going to talk about breaking-bones stuff, and, of course, with technology it never is. So it ’ s very easy, comparatively speaking, to be resilient. I think that, if we look at what happened on the Internet, with such an incredible last half a dozen years, that it ’ s hard to even get the right analogy for it. A lot of how we decide, how we ’ re supposed to react to things and what we ’ re supposed to expect about the future depends on how we bucket things and how we categorize them. And so I think the tempting analogy for the boom-bust that we just went through with the Internet is a gold rush. It ’ s easy to think of this analogy as very different from some of the other things you might pick. For one thing, both were very real. In 1849, in that Gold Rush, they took over $700 million worth of gold out of California. It was very real. The Internet was also very real. This is a real way for humans to communicate with each other. It ’ s a big deal. Huge boom. Huge boom. Huge bust. Huge bust. You keep going, and both things are lots of hype. I don ’ t have to remind you of all the hype that was involved with the Internet — like GetRich. com. But you had the same thing with the Gold Rush.”Gold. Gold. Gold.”Sixty-eight rich men on the Steamer Portland. Stacks of yellow metal. Some have 5, 000. Many have more. A few bring out 100, 000 dollars each. People would get very excited about this when they read these articles.”The Eldorado of the United States of America : the discovery of inexhaustible gold mines in California.”And the parallels between the Gold Rush and the Internet Rush continue very strongly. So many people left what they were doing. And what would happen is — and the Gold Rush went on for years. People on the East Coast in 1849, when they first started to get the news, they thought,”Ah, this isn ’ t real.”But they keep hearing about people getting rich, and then in 1850 they still hear that. And they think it ’ s not real. By about 1852, they ’ re thinking,”Am I the stupidest person on Earth by not rushing to California?”And they start to decide they are. These are community affairs, by the way. Local communities on the East Coast would get together and whole teams of 10, 20 people would caravan across the United States, and they would form companies. These were typically not solitary efforts. But no matter what, if you were a lawyer or a banker, people dropped what they were doing, no matter what skill set they had, to go pan for gold. This guy on the left, Dr. Richard Beverley Cole, he lived in Philadelphia and he took the Panama route. They would take a ship down to Panama, across the isthmus, and then take another ship north. This guy, Dr. Toland, went by covered wagon to California. This has its parallels, too. Doctors leaving their **practices**. These are both very successful — a physician in one case, a surgeon in the other. Same thing happened on the Internet. You get DrKoop. com. In the Gold Rush, people literally jumped ship. The San Francisco harbor was clogged with 600 ships at the **peak** because the ships would get there and the crews would abandon to go search for gold. So there were literally 600 captains and 600 ships. They turned the ships into hotels, because they couldn ’ t sail them anywhere. You had dotcom fever. And you had gold fever. And you saw some of the excesses that the dotcom fever created and the same thing happened. The fort in San Francisco at the time had about 1, 300 soldiers. Half of them deserted to go look for gold. And they wouldn ’ t let the other half out to go look for the first half because they were afraid they wouldn ’ t come back. And one of the soldiers wrote home, and this is the sentence that he put :”The **struggle** between right and six dollars a month and wrong and 75 dollars a day is a rather severe one.”They had bad burn rate in the Gold Rush. A very bad burn rate. This is actually from the Klondike Gold Rush. This is the White Pass Trail. They loaded up their mules and their horses. And they didn ’ t plan right. And they didn ’ t know how far they would really have to go, and they overloaded the horses with hundreds and hundreds of pounds of stuff. In fact it was so bad that most of the horses died before they could get where they were going. It got renamed the"Dead Horse Trail."And the Canadian Minister of the Interior wrote this at the time :"Thousands of pack horses lie dead along the way, sometimes in bunches under the cliffs, with pack saddles and packs where they ' ve fallen from the rock above, sometimes in tangled masses, filling the mud holes and furnishing the only footing for our poor pack animals on the march, often, I regret to say, exhausted, but still alive, a fact we were unaware of, until after the miserable wretches turned beneath the hooves of our cavalcade. The eyeless sockets of the pack animals everywhere account for the myriads of ravens along the road. The inhumanity which this trail has been witness to, the heartbreak and suffering which so many have undergone, cannot be imagined. They certainly cannot be described."And you know, without the smell that would have accompanied that, we had the same thing on the Internet : very bad burn rate calculations. I ' ll just play one of these and you ' ll remember it. This is a commercial that was played on the Super Bowl in the year 2000. : Bride #1 : You said you had a large selection of invitations. Clerk : But we do. Bride #2 : Then why does she have my invitation? Announcer : What may be a little thing to some... Bride #3 : You are mine, little man. Announcer : Could be a really big deal to you. Husband #1 : Is that your wife? Husband #2 : Not for another 15 minutes. Announcer : After all, it ' s your **special** day. OurBeginning. com. Life ' s an event. Announce it to the world. Jeff Bezos : It ' s very difficult to figure out what that **ad** is for. But they spent three and a half million dollars in the 2000 Super Bowl to air that **ad**, even though, at the time, they only had a million dollars in annual revenue. Now, here ' s where our analogy with the Gold Rush starts to diverge, and I think rather severely. And that is, in a gold rush, when it ' s over, it ' s over. Here ' s this guy :"There are many men in Dawson at the present time who feel keenly disappointed. They ' ve come thousands of miles on a perilous trip, risked life, health and property, spent months of the most arduous labor a man can perform and at length with expectations raised to the highest pitch have reached the coveted goal only to discover the fact that there is nothing here for them."And that was, of course, the very common story. Because when you take out that last piece of gold — and they did incredibly quickly. I mean, if you look at the 1849 Gold Rush — the entire American river region, within two years — every stone had been turned. And after that, only big companies who used more sophisticated mining technologies started to take gold out of there. So there ' s a much better analogy that allows you to be incredibly optimistic and that analogy is the electric industry. And there are a lot of similarities between the Internet and the electric industry. With the electric industry you actually have to — one of them is that they ' re both sort of thin, horizontal, enabling layers that go across lots of different industries. It ' s not a specific thing. But electricity is also very, very broad, so you have to sort of narrow it down. You know, it can be used as an incredible means of transmitting power. It ' s an incredible means of coordinating, in a very fine-grained way, information flows. There ' s a bunch of things that are interesting about electricity. And the part of the electric revolution that I want to focus on is sort of the golden age of appliances. The killer app that got the world ready for appliances was the light bulb. So the light bulb is what wired the world. And they weren ' t thinking about appliances when they wired the world. They were really thinking about — they weren ' t putting electricity into the home ; they were putting lighting into the home. And, but it really — it got the electricity. It took a long time. This was a huge — as you would expect — a huge capital build out. All the streets had to be torn up. This is work going on down in lower Manhattan where they built some of the first electric power generating stations. And they ’ re tearing up all the streets. The Edison Electric Company, which became Edison General Electric, which became General Electric, paid for all of this digging up of the streets. It was incredibly expensive. But that is not the — and that ’ s not the part that ’ s really most similar to the Web. Because, remember, the Web got to stand on top of all this heavy infrastructure that had been put in place because of the long-distance phone network. So all of the cabling and all of the heavy infrastructure — I ’ m going back now to, sort of, the explosive part of the Web in 1994, when it was growing 2, 300 percent a year. How could it grow at 2, 300 percent a year in 1994 when people weren ’ t really investing in the Web? Well, it was because that heavy infrastructure had already been laid down. So the light bulb laid down the heavy infrastructure, and then home appliances started coming into being. And this was huge. The first one was the electric fan — this was the 1890 electric fan. And the appliances, the golden age of appliances really lasted — it depends how you want to measure it — but it ’ s anywhere from 40 to 60 years. It goes on a long time. It starts about 1890. And the electric fan was a big success. The electric iron, also very big. By the way, this is the beginning of the asbestos lawsuit. There ’ s asbestos under that handle there. This is the first vacuum cleaner, the 1905 Skinner Vacuum, from the Hoover Company. And this one weighed 92 pounds and took two people to operate and cost a quarter of a car. So it wasn ’ t a big seller. This was truly, truly an early-adopter product — the 1905 Skinner Vacuum. But three years later, by 1908, it weighed 40 pounds. Now, not all these things were highly successful. This is the electric tie press, which never really did catch on. People, I guess, decided that they would not wrinkle their ties. These never really caught on either : the electric shoe warmer and drier. Never a big seller. This came in, like, six different colors. I don ’ t know why. But I thought, you know, sometimes it ’ s just not the right time for an invention ; maybe it ’ s time to give this one another shot. So I thought we could build a Super Bowl **ad** for this. We ’ d need the right partner. And I thought that really — I thought that would really work, to give that another shot. Now, the toaster was huge because they used to make toast on open fires, and it took a lot of time and attention. I want to point out one thing. This is — you guys know what this is. They hadn ’ t invented the electric socket yet. So this was — remember, they didn ’ t wire the houses for electricity. They wired them for lighting. So your — your appliances would plug in. They would — each room typically had a light bulb socket at the top. And you ’ d plug it in there. In fact, if you ’ ve seen the Carousel of Progress at Disney World, you ’ ve seen this. Here are the cables coming up into this light fixture. All the appliances plug in there. And you would just unscrew your light bulb if you wanted to plug in an appliance. The next thing that really was a big, big deal was the washing machine. Now, this was an object of much envy and lust. Everybody wanted one of these electric washing machines. On the left-hand side, this was the soapy water. And there ’ s a rotor there — that this motor is spinning. And it would clean your clothes. This is the clean rinse-water. So you ’ d take the clothes out of here, put them in here, and then you ’ d run the clothes through this electric wringer. And this was a big deal. You ’ d keep this on your porch. It was a little bit messy and kind of a pain. And you ’ d run a long cord into the house where you could screw it into your light socket. And that ’ s actually kind of an important point in my presentation, because they hadn ’ t invented the off switch. That was to come much later — the off switch on appliances — because it didn ’ t make any sense. I mean, you didn ’ t want this thing clogging up a light socket. So you know, when you were done with it, you unscrewed it. That ’ s what you did. You didn ’ t turn it off. And as I said before, they hadn ’ t invented the electric outlet either, so the washing machine was a particularly dangerous device. And there are — when you research this, there are gruesome descriptions of people getting their hair and clothes caught in these devices. And they couldn ’ t yank the cord out because it was screwed into a light socket inside the house. And there was no off switch, so it wasn ’ t very good. And you might think that that was incredibly stupid of our ancestors to be plugging things into a light socket like this. But, you know, before I get too far into condemning our ancestors, I thought I ’ d show you : this is my conference room. This is a total kludge, if you ask me. First of all, this got installed upside down. This light socket — and so the cord keeps falling out, so I taped it in. This is supposed — don ’ t even get me started. But that ’ s not the worst one. This is what it looks like under my desk. I took this picture just two days ago. So we really haven ’ t progressed that much since 1908. It ’ s a total, total mess. And, you know, we think it ’ s getting better, but have you tried to install 802. 11 yourself? I challenge you to try. It ’ s very hard. I know Ph. D. s in Computer Science — this process has brought them to tears, absolute tears. And that ’ s assuming you already have DSL in your house. Try to get DSL installed in your house. The engineers who do it everyday can ’ t do it. They have to — typically, they come three times. And one friend of mine was telling me a story : not only did they get there and have to wait, but then the engineers, when they finally did get there, for the third time, they had to call somebody. And they were really happy that the guy had a speakerphone because then they had to wait on hold for an hour to talk to somebody to give them an access code after they got there. So we ’ re not — we ’ re pretty kludge-y ourselves. By the way, DSL is a kludge. I mean, this is a twisted pair of copper that was never designed for the purpose it ’ s being put to — you know it ’ s the whole thing — we ’ re very, very primitive. And that ’ s kind of the point. Because, you know, resilience — if you think of it in terms of the Gold Rush, then you ’ d be pretty depressed right now because the last nugget of gold would be gone. But the good thing is, with innovation, there isn ’ t a last nugget. Every new thing creates two new questions and two new opportunities. And if you believe that, then you believe that where we are — this is what I think — I believe that where we are with the incredible kludge — and I haven ’ t even talked about user interfaces on the Web — but there ’ s so much kludge, so much terrible stuff — we are at the 1908 Hurley washing machine stage with the Internet. That ’ s where we are. We don ’ t get our hair caught in it, but that ’ s the level of primitiveness of where we are. We ’ re in 1908. And if you believe that, then stuff like this doesn ’ t bother you. This is 1996 :”All the negatives add up to making the online experience not worth the trouble.”1998 :”Amazon. toast.”In 1999 :”Amazon. bomb.”My mom hates this picture. She — but you know, if you really do believe that it ’ s the very, very beginning, if you believe it ’ s the 1908 Hurley washing machine, then you ’ re incredibly optimistic. And I do think that that ’ s where we are. And I do think there ’ s more innovation ahead of us than there is behind us. And in 1917, Sears — I want to get this exactly right. This was the advertisement that they ran in 1917. It says :”Use your electricity for more than light.”And I think that ’ s where we are. We ’ re very, very early. Thank you very much.

Text Sample 240: POS Dim 2 – 2003 – Score 6.00 – 2003/t000088.txt

The future of life, where the unraveling of our biology â and bring up the lights a little bit. I don ’ t have any slides. I ’ m just going to talk â about where that ’ s likely to carry us. And you know, I saw all the visions of the first couple of sessions. It almost made me feel a little bit guilty about having an uplifting talk about the future. It felt wrong to do that in some way. And yet, I don ’ t really think it is because when it comes down to it, it ’ s this larger trajectory that is really what is going to remain â what people in the future are going to remember about this period. I want to talk to you a little bit about why the visions of Jeremy Rivkins, who would like to ban these sorts of technologies, or of the Bill Joys who would like to relinquish them, are actually â to follow those paths would be such a tragedy for us. I ’ m focusing on biology, the biological sciences. The reason I ’ m doing that is because those are going to be the areas that are the most significant to us. The reason for that is really very simple. It ’ s because we ’ re flesh and blood. We ’ re biological creatures. And what we can do with our biology is going to shape our future and that of our children and that of their children â whether we gain control over aging, whether we learn to protect ourselves from Alzheimer ’ s, and heart disease, and cancer. I think that Shakespeare really put it very nicely. And I ’ m actually going to use his words in the same order that he did. He said,”And so from hour to hour we ripe and ripe. And then from hour to hour we rot and rot. And thereby hangs a tale.”Life is short, you know. And we need to think about planning a little bit. We ’ re all going to eventually, even in the developed world, going to have to lose everything that we love. When you ’ re beginning to rot a little bit, all of the videos crammed into your head, all of the extensions that extend your various powers, are going to being to seem a little secondary. And you know, I ’ m getting a little bit gray â so is Ray Kurzweil, so is Eric Drexler. This is where it ’ s really central to our lives. Now I know there ’ s been a whole lot of hype about our power to control biology. You just have to look at the Human Genome Project. It wasn ’ t two years ago that everybody was talking about â we ’ ve found the Holy Grail of biology. We ’ re deciphering the code of codes. We ’ re reading the book of life. It ’ s a little bit reminiscent of 1969 when Neil Armstrong walked on the moon, and everybody was about to race out toward the stars. And we ’ ve all seen”2001 : A Space Odyssey.”You know it ’ s 2003, and there is no HAL. And there is no odyssey to our own moon, much less the moons of Jupiter. And we ’ re still picking up pieces of the Challenger. So it ’ s not surprising that some people would wonder whether maybe 30 or 40 years from now, we ’ ll look back at this **instant** in time, and all of the sort of talk about the Human Genome Project, and what all this is going to mean to us â well, it will really mean precious little. And I just want to say that that is absolutely not going to be the case. Because when we talk about our genetics and our biology, and modifying and altering and adjusting these things, we ’ re talking about changing ourselves. And this is very critical stuff. If you have any doubts about how technology affects our lives, you just have to go to any major city. This is not the stomping ground of our Pleistocene ancestors. What ’ s happening is we ’ re taking this technology â it ’ s becoming more precise, more potent â and we ’ re turning it back upon ourselves. Before it ’ s all done we are going to alter ourselves every bit as much as we have changed the world around us. It ’ s going to happen a lot sooner than people imagine. On the way there it ’ s going to completely revolutionize medicine and health care ; that ’ s obvious. It ’ s going to change the way we have children. It ’ s going to change the way we manage and alter our emotions. It ’ s going to probably change the human lifespan. It will really make us question what it is to be a human being. The larger context of this is that are two unprecedented revolutions that are going on today. The first of them is the obvious one, the **silicon** revolution, which you all are very, very familiar with. It ’ s changing our lives in so many ways, and it will continue to do that. What the essence of that is, is that we ’ re taking the sand at our feet, the inert **silicon** at our feet, and we ’ re breathing a level of complexity into it that rivals that of life itself, and may even surpass it. As an outgrowth of that, as a child of that revolution, is the revolution in biology. The genomics revolution, proteomics, metabolomics, all of these”omics”that sound so terrific on grants and on business plans. What we ’ re doing is we are seizing control of our evolutionary future. I mean we ’ re essentially using technology to just jam evolution into **fast-forward**. It ’ s not at all clear where it ’ s going to take us. But in five to ten years we ’ re going to start see some very profound changes. The most immediate changes that we ’ ll see are things like in medicine. There is going to be a big shift towards preventative medicine as we start to be able to identify all of the risk factors that we have as individuals. But who is going to pay for all this? And how are we going to understand all this complex information? That is going to be the IT challenge of the next generation, is communicating all this information. There ’ s pharmacogenomics, the combination of pharmacology and genetics : tailoring drugs to our individual constitutions that Juan talked about a little bit earlier. That ’ s going to have amazing impacts. And it ’ s going to be used for diet as well, and nutritional supplements and such. But it ’ s going to have a big impact because we ’ re going to have niche drugs. And we aren ’ t going to be able to support the kinds of expenses that we have to create blockbuster drugs today. The approval process is going to fall apart, actually. It ’ s too slow. It ’ s too risk-averse. And it is really not suited for the future that we ’ re moving into. Another thing is that we ’ re just going to have to deal with this knowledge. It ’ s really wonderful when we hear,”Oh, 99. 9 percent of the letters in the code are the same. We ’ re all identical to each other. Isn ’ t it wonderful?”And look around you and know that what we really care about is that little bit of difference. We look the same to a visitor from another planet, maybe, but not to each other because we compete with each other all time. And we ’ re going to have to come to grips with the fact that there are differences between us as individuals that we will know about, and between subpopulations of humans as well. To deny that that ’ s the case is not a very good start on that. A generation or so away there are going to be even more profound things that are going to happen. That ’ s when we ’ re going to begin to use this knowledge to modify ourselves. Now I don ’ t mean extra gills or something â□□ something we care about, like aging. What if we could unravel aging and understand it â□□ begin to retard the process or even reverse it? It would change absolutely everything. And it ’ s obvious to anyone, that if we can do this, we absolutely will do this, whatever the consequences are. The second is modifying our emotions. I mean Ritalin, Viagra, things of that sort, Prozac. You know, this is just clumsy little baby steps. What if you could take a little concoction of pharmaceuticals that would make you feel really contented, just happy to be you. Are you going to be able to resist that if it doesn ’ t have any overt side effects? Probably not. And if you don ’ t, who are you going to be? Why do you do what you do? We ’ re sort of circumventing evolutionary programs that guide our behavior. It ’ s going to be very challenging to deal with. The third area is reproduction. The idea that we ’ re going to chose our children ’ s genes, as we begin to understand what genes say about who we are. That ’ s the focus of my book”Redesigning Humans,”where I talk about the kinds of choices we ’ ll make, and the challenges it ’ s going to present to society. There are three obvious ways of doing this. The first is cloning. It didn ’ t happen. It ’ s a total media circus. It will happen in five to 10 years. And when it does it ’ s not going to be that big a deal. The birth of a delayed identical twin is not going to shake western civilization. But there are more important things that are already occurring : embryo screening. You take a six to eight cell embryo, you tease out one of the cells, you run a genetic test on that cell, and depending on the results of that test you either implant that embryo or you discard it. It ’ s already done to avoid rare diseases today. And pretty soon it ’ s going to be possible to avoid virtually all genetic diseases in that way. As that becomes possible this is going to move from something that is used by those who have infertility problems and are already doing in vitro fertilization, to the wealthy who want to protect their children, to just about everybody else. And in that process that ’ s going to morph from being just for diseases, to being for lesser vulnerabilities, like risk of manic depression or something, to picking personalities, temperaments, traits, these sorts of things. Of course there is going to be genetic engineering. Directly going in â it ’ s a little bit further away, but not that far away â going in and altering the genes in the first cell in an embryo. The way I suspect it will happen is using artificial chromosomes and extra chromosomes, so we go from 46 to 47 or 48. And one that is not heritable because who would want to pass on to their children the archaic enhancement modules that they got 25 years earlier from their parents? It ’ s a joke ; of course they wouldn ’ t want to do that. They ’ ll want the new release. Those kinds of loose analogies with computers, and with programming, are actually much deeper than that. They are really going to come to operate in this realm. Now not everything that can be done should be done. And it won ’ t be done. But when something is feasible in thousands of laboratories all over the world, which is going to be the case with these technologies, when there are large numbers of people who see them as beneficial, which is already the case, and when they ’ re almost impossible to police, it ’ s not a question of if this is going to happen, it ’ s when and where and how it ’ s going to happen. Humanity is going to go down this path. And it ’ s going to do so for two reasons. The first is that all these technologies are just a spin-off of mainstream medical research that everybody wants to see happen. It is being funded very very â in a big way. The second is, we ’ re human. That ’ s what we do. We try and use our technology to improve our lives in one way or another. To imagine that we ’ re not going to use these technologies when they become available, is as much a denial of who we are as to imagine that we ’ ll use these technologies and not fret and worry about it a great deal. The lines are going to blur. And they already are between therapy and enhancement, between treatment and prevention, between need and desire. That ’ s really the central one, I believe. People can try and ban these things. They undoubtedly will. They have. But ultimately all this is going to do is just shift development elsewhere. It ’ s going to drive these things from view. It ’ s going to reserve the technology for the wealthy because they are in the best position to circumvent any of these sorts of laws. And it ’ s going to deny us the information that we need to make wise decisions about how to use these technologies. So, sure, we need to debate these things. And I think it ’ s wonderful that we do. But we shouldn ’ t kid ourselves and think that we ’ re going to reach a consensus about these things. That is simply not going to happen. They touch us too deeply. And they depend too much upon history, upon philosophy, upon religion, upon culture, upon politics. Some people are going to see this as an abomination, as the worst thing, as just awful. Other people are going to say,”This is great. This is the flowering of human endeavor.”The one thing though that is really dangerous about these sorts of technologies, is that it ’ s easy to become seduced by them. And to focus too much on all the high-technology possibilities that exist. And to lose touch with the basic rhythms of our biology and our health. There are too many people that think that high-technology medicine is going to keep them, save them, from overeating, from eating a lot of fast foods, from not getting any exercise. It ’ s not going to happen. In the midst of all this amazing technology, and all these things that are occurring, it ’ s really interesting because there is sort of a counter- revolution that is going on : a resurgence of interest in remedies from the past, in nutraceuticals, in all of these sorts of things that some people, in the pharmaceutical industry particularly, like to brand as non-science. But this whole effort is generated, is driven, by IT as well because that is how we ’ re gathering all this information, and linking it, and integrating it together. There is a lot in this rich biota that is going to serve us well. And that ’ s where about half of our drugs come. So we shouldn ’ t dismiss this because it ’ s an enormous opportunity to use these sorts of results, or these random loose **trials** from the last thousand years about what has impacts on our health. And to use our advanced technologies to pull out what is beneficial from this sea of noise, basically. In fact this isn ’ t just abstract. I just formed a biotechnology company that is using this sort of an approach to develop therapeutics for Alzheimer ’ s and other diseases of aging, and we ’ re making some real progress. So here we are. It ’ s the beginning of a new millennium. If you look forward, I mean future humans, far before the end of this millennium, in a few hundred years, they are going to look back at this moment. And from the beginning of today ’ s sessions you ’ d think that they ’ re going to see this as this horrible difficult, painful period that we **struggled** through. And I don ’ t think that ’ s what ’ s going to happen. They ’ re going to do like everybody does. They are going to forget about all that stuff. And they are actually going to romanticize this moment in time. They are going to think about it as this glorious **instant** when we laid down the very foundations of their lives, of their society, of their future. You know it ’ s a little bit like a birth. Where there is this bloody, awful mess happens. And then what comes out of it? New life. Actually as was pointed out earlier, we forget about all the **struggle** there was in getting there. So to me, it ’ s clear that one of the foundations of that future is going to be the reworking of our biology. It ’ s going to come gradually at first. It ’ s going to pick up speed. We ’ re going to make lots of errors. That ’ s the way these things work. To me it ’ s an incredible privilege to be alive now and to be able to witness this thing. It is something that is a unique **instant** in the history of all of life. It will always be remembered. And what ’ s extraordinary is that we ’ re not just **observing** this, we are the architects of this. I think that we should be proud of it. What is so difficult and challenging is that we are also the objects of these changes. It ’ s our health, it ’ s our lives, it ’ s our future, it ’ s our children. And that is why they are so very troubling to so many people who would pull back in fear. I think that our choice in the choice of life, is not whether we ’ re going to go down this path. We are, definitely. It ’ s how we hold it in our hearts. It ’ s how we look at it. I think Thucydides really spoke to us very clearly in 430 B. C. He put it nicely. Again, I ’ ll use the words in the same order he did.”The bravest are surely those who have the clearest vision of what is before them, both glory and danger alike. And yet notwithstanding, they go out and they meet it.”Thank you.

Text Sample 241: POS Dim 2 – 2003 – Score 6.00 – 2003/t000444.txt

I ’ m going to give you four specific examples, I ’ m going to cover at the end about how a company called Silk tripled their sales ; how an artist named Jeff Koons went from being a nobody to making a whole bunch of money and having a lot of impact ; to how Frank Gehry redefined what it meant to be an architect. And one of my biggest failures as a marketer in the last few years — a record label I started that had a CD called”Sauce.”Before I can do that I ’ ve got to tell you about sliced bread, and a guy named Otto Rohwedder. Now, before sliced bread was invented in the 1910s I wonder what they said? Like the greatest invention since the telegraph or something. But this guy named Otto Rohwedder invented sliced bread, and he focused, like most inventors did, on the patent part and the making part. And the thing about the invention of sliced bread is this — that for the first 15 years after sliced bread was available no one bought it ; no one knew about it ; it was a complete and total failure. And the reason is that until Wonder came along and figured out how to spread the idea of sliced bread, no one wanted it. That the success of sliced bread, like the success of almost everything we ' ve talked about at this conference, is not always about what the patent is like, or what the factory is like — it ' s about can you get your idea to spread, or not. And I think that the way you ' re going to get what you want, or cause the change that you want to change, to happen, is to figure out a way to get your ideas to spread. And it doesn ' t matter to me whether you ' re running a coffee shop or you ' re an intellectual, or you ' re in business, or you ' re flying hot air balloons. I think that all this stuff applies to everybody regardless of what we do. That what we are living in is a century of idea diffusion. That people who can spread ideas, regardless of what those ideas are, win. When I talk about it I usually pick business, because they make the best pictures that you can put in your presentation, and because it ' s the easiest sort of way to keep **score**. But I want you to forgive me when I use these examples because I ' m talking about anything that you decide to spend your time to do. At the heart of spreading ideas is TV and stuff like TV. TV and mass media made it really easy to spread ideas in a certain way. I call it the"TV-industrial complex."The way the TV-industrial complex works, is you buy some **ads**, interrupt some people, that gets you distribution. You use the distribution you get to sell more products. You take the profit from that to buy more **ads**. And it goes around and around and around, the same way that the military-industrial complex worked a long time ago. That model of, and we heard it yesterday — if we could only get onto the homepage of Google, if we could only figure out how to get promoted there, or grab that person by the throat, and tell them about what we want to do. If we did that then everyone would pay attention, and we would win. Well, this TV-industrial complex informed my entire childhood and probably yours. I mean, all of these products succeeded because someone figured out how to touch people in a way they weren ' t expecting, in a way they didn ' t necessarily want, with an **ad**, over and over again until they bought it. And the thing that ' s happened is, they canceled the TV-industrial complex. That just over the last few years, what anybody who markets anything has discovered is that it ' s not working the way that it used to. This picture is really fuzzy, I apologize ; I had a bad cold when I took it. But the product in the blue box in the center is my poster child. I go to the deli ; I ' m sick ; I need to buy some medicine. The brand manager for that blue product spent 100 million dollars trying to interrupt me in one year. 100 million dollars interrupting me with TV commercials and magazine **ads** and Spam and coupons and shelving allowances and spiff — all so I could ignore every single message. And I ignored every message because I don ' t have a pain reliever problem. I buy the stuff in the yellow box because I always have. And I ' m not going to invest a minute of my time to solve her problem, because I don ' t care. Here ' s a magazine called"Hydrate."It ' s 180 pages about water. Articles about water, **ads** about water. Imagine what the world was like 40 years ago, with just the Saturday Evening Post and Time and Newsweek. Now there are magazines about water. New product from Coke Japan : water salad. Coke Japan comes out with a new product every three weeks, because they have no idea what ' s going to work and what ' s not. I couldn ' t have written this better myself. It came out four days ago — I circled the important parts so you can see them here. They ' ve come out... Arby ' s is going to spend 85 million dollars promoting an oven mitt with the voice of Tom Arnold, hoping that that will get people to go to Arby ' s and buy a roast beef sandwich. Now, I had tried to imagine what could possibly be in an animated TV commercial featuring Tom Arnold, that would get you to get in your car, drive across town and buy a roast beef sandwich. Now, this is Copernicus, and he was right, when he was talking to anyone who needs to hear your idea."The world revolves around me."Me, me, me, me. My favorite person — me. I don ' t want to get email from anybody ; I want to get"memail."So consumers, and I don ' t just mean people who buy stuff at the Safeway ; I mean people at the Defense Department who might buy something, or people at, you know, the New Yorker who might print your article. Consumers don ' t care about you at all ; they just don ' t care. Part of the reason is — they ' ve got way more choices than they used to, and way less time. And in a world where we have too many choices and too little time, the obvious thing to do is just ignore stuff. And my parable here is you ' re driving down the road and you see a cow, and you keep driving because you ' ve seen cows before. Cows are invisible. Cows are boring. Who ' s going to stop and pull over and say —"Oh, look, a cow."Nobody. But if the cow was purple — isn ' t that a great **special** effect? I could do that again if you want. If the cow was purple, you ' d notice it for a while. I mean, if all cows were purple you ' d get bored with those, too. The thing that ' s going to decide what gets talked about, what gets done, what gets changed, what gets purchased, what gets built, is :"Is it remarkable?"And"remarkable"is a really cool word, because we think it just means"neat,"but it also means"worth making a remark about."And that is the essence of where idea diffusion is going. That two of the hottest cars in the United States is a 55, 000-dollar giant car, big enough to hold a Mini in its trunk. People are paying full price for both, and the only thing they have in common is that they don ' t have anything in common. Every week, the number one best-selling DVD in America changes. It ' s never"The Godfather,"it ' s never"Citizen Kane,"it ' s always some third-rate movie with some second-rate star. But the reason it ' s number one is because that ' s the week it came out. Because it ' s new, because it ' s fresh. People saw it and said"I didn ' t know that was there"and they noticed it. Two of the big success stories of the last 20 years in retail — one sells things that are super- expensive in a blue box, and one sells things that are as cheap as they can make them. The only thing they have in common is that they ' re different. We ' re now in the fashion business, no matter what we do for a living, we ' re in the fashion business. And people in the fashion business know what it ' s like to be in the fashion business — they ' re used to it. The rest of us have to figure out how to think that way. How to understand that it ' s not about interrupting people with big full-page **ads**, or insisting on meetings with people. But it ' s a totally different sort of process that determines which ideas spread, and which ones don ' t. They sold a billion dollars ' worth of Aeron chairs by reinventing what it meant to sell a chair. They turned a chair from something the purchasing department bought, to something that was a status symbol about where you sat at work. This guy, Lionel Poilne, the most famous baker in the world — he died two and a half months ago, and he was a hero of mine and a dear friend. He lived in Paris. Last year, he sold 10 million dollars ' worth of French bread. Every loaf baked in a bakery he owned, by one baker at a time, in a wood-fired oven. And when Lionel started his bakery, the French pooh-pooh-ed it. They didn ' t want to buy his bread. It didn ' t look like"French bread."It wasn ' t what they expected. It was neat ; it was remarkable ; and slowly, it spread from one person to another person until finally, it became the official bread of three-star restaurants in Paris. Now he ' s in London, and he ships by FedEx all around the world. What marketers used to do is make average products for average people. That ' s what mass marketing is. Smooth out the edges ; go for the center ; that ' s the big market. They would ignore the geeks, and God forbid, the laggards. It was all about going for the center. But in a world where the TV-industrial complex is broken, I don ' t think that ' s a strategy we want to use any more. I think the strategy we want to use is to not market to these people because they ' re really good at ignoring you. But market to these people because they care. These are the people who are obsessed with something. And when you talk to them, they ' ll listen, because they like listening — it ' s about them. And if you ' re lucky, they ' ll tell their friends on the rest of the curve, and it ' ll spread. It ' ll spread to the entire curve. They have something I call"otaku"— it ' s a great Japanese word. It describes the desire of someone who ' s obsessed to say, drive across Tokyo to try a new ramen noodle place, because that ' s what they do : they get obsessed with it. To make a product, to market an idea, to come up with any problem you want to solve that doesn ' t have a constituency with an otaku, is almost impossible. Instead, you have to find a group that really, desperately cares about what it is you have to say. Talk to them and make it easy for them to tell their friends. There ' s a hot sauce otaku, but there ' s no mustard otaku. That ' s why there ' s lots and lots of kinds of hot sauces, and not so many kinds of mustard. Not because it ' s hard to make interesting mustard — you could make interesting mustard — but people don ' t, because no one ' s obsessed with it, and thus no one tells their friends. Krispy Kreme has figured this whole thing out. It has a strategy, and what they do is, they enter a city, they talk to the people, with the otaku, and then they spread through the city to the people who ' ve just crossed the street. This yoyo right here cost 112 dollars, but it sleeps for 12 minutes. Not everybody wants it but they don ' t care. They want to talk to the people who do, and maybe it ' ll spread. These guys make the loudest car stereo in the world. It ' s as loud as a 747 jet. You can ' t get in, the car ' s got bulletproof glass, because it ' ll blow out the windshield otherwise. But the fact remains that when someone wants to put a couple of speakers in their car, if they ' ve got the otaku or they ' ve heard from someone who does, they go ahead and they pick this. It ' s really simple — you sell to the people who are listening, and just maybe, those people tell their friends. So when Steve Jobs talks to 50, 000 people at his keynote, who are all tuned in from 130 countries watching his two-hour commercial — that ' s the only thing keeping his company in business — it ' s that those 50, 000 people care desperately enough to watch a two-hour commercial, and then tell their friends. Pearl Jam, 96 albums released in the last two years. Every one made a profit. How? They only sell them on their website. Those people who buy them have the otaku, and then they tell their friends, and it spreads and it spreads. This hospital crib cost 10, 000 dollars, 10 times the standard. But hospitals are buying it faster than any other model. Hard Candy nail polish, doesn ' t appeal to everybody, but to the people who love it, they talk about it like crazy. This paint can right here saved the Dutch Boy paint company, making them a fortune. It costs 35 percent more than regular paint because Dutch Boy made a can that people talk about, because it ' s remarkable. They didn ' t just slap a new **ad** on the product ; they changed what it meant to build a paint product. AmIhotornot. com — everyday 250, 000 people go to this site, run by two volunteers, and I can tell you they are hard graders — They didn ' t get this way by advertising a lot. They got this way by being remarkable, sometimes a little too remarkable. And this picture frame has a cord going out the back, and you plug it into the wall. My father has this on his desk, and he sees his grandchildren everyday, changing constantly. And every single person who walks into his office hears the whole story of how this thing ended up on his desk. And one person at a time, the idea spreads. These are not diamonds, not really. They ' re made from”cremains.”After you ’ re cremated you can have yourself made into a gem. Oh, you like my ring? It ’ s my grandmother. Fastest-growing business in the whole mortuary industry. But you don ’ t have to be Ozzie Osborne — you don ’ t have to be super-outrageous to do this. What you have to do is figure out what people really want and give it to them. A couple of quick rules to wrap up. The first one is : Design is free when you get to scale. The people who come up with stuff that ’ s remarkable more often than not figure out how to put design to work for them. Number two : The riskiest thing you can do now is be safe. Proctor and Gamble knows this, right? The whole model of being Proctor and Gamble is always about average products for average people. That ’ s risky. The safe thing to do now is to be at the fringes, be remarkable. And being very good is one of the worst things you can possibly do. Very good is boring. Very good is average. It doesn ’ t matter whether you ’ re making a record album, or you ’ re an architect, or you have a tract on sociology. If it ’ s very good, it ’ s not going to work, because no one ’ s going to notice it. So my three stories. Silk put a product that does not need to be in the refrigerated section next to the milk in the refrigerated section. Sales tripled. Why? Milk, milk, milk, milk, milk — not milk. For the people who were there and looking at that section, it was remarkable. They didn ’ t triple their sales with advertising ; they tripled it by doing something remarkable. That is a remarkable piece of art. You don ’ t have to like it, but a 40-foot tall dog made out of bushes in the middle of New York City is remarkable. Frank Gehry didn ’ t just change a museum ; he changed an entire city ’ s economy by designing one building that people from all over the world went to see. Now, at countless meetings at, you know, the Portland City Council, or who knows where, they said, we need an architect — can we get Frank Gehry? Because he did something that was at the fringes. And my big failure? I came out with an entire — A record album and hopefully a whole bunch of record albums in SACD, this remarkable new format — and I marketed it straight to people with 20, 000- dollar stereos. People with 20, 000-dollar stereos don ’ t like new music. So what you need to do is figure out who does care. Who is going to raise their hand and say,”I want to hear what you ’ re doing next,”and sell something to them. The last example I want to give you. This is a map of Soap Lake, Washington. As you can see, if that ’ s nowhere, it ’ s in the middle of it. But they do have a lake. And people used to come from miles around to swim in the lake. They don ’ t anymore. So the founding fathers said,”We ’ ve got some money to spend. What can we build here?”And like most committees, they were going to build something pretty safe. And then an artist came to them — this is a true artist ’ s rendering — he wants to build a 55-foot tall lava lamp in the center of town. That ’ s a purple cow ; that ’ s something worth noticing. I don ’ t know about you, but if they build it, that ’ s where I ’ m going to go. Thank you very much for your attention.

Text Sample 242: POS Dim 2 – 2011 – Score 12.00 – 2011/t002711.txt

The first thing I want to do is say thank you to all of you. The second thing I want to do is introduce my co-author and dear friend and co-teacher. Ken and I have been working together for almost 40 years. That ’ s Ken Sharpe over there. So there is among many people â certainly me and most of the people I talk to â a kind of collective dissatisfaction with the way things are working, with the way our institutions run. Our kids ’ teachers seem to be failing them. Our doctors don ’ t know who the hell we are, and they don ’ t have enough time for us. We certainly can ’ t trust the bankers, and we certainly can ’ t trust the brokers. They almost brought the entire **financial** system down. And even as we do our own work, all too often, we find ourselves having to choose between doing what we think is the right thing and doing the expected thing, or the required thing, or the profitable thing. So everywhere we look, pretty much across the board, we worry that the people we depend on don ’ t really have our interests at heart. Or if they do have our interests at heart, we worry that they don ’ t know us well enough to figure out what they need to do in order to allow us to secure those interests. They don ’ t understand us. They don ’ t have the time to get to know us. There are two kinds of responses that we make to this sort of general dissatisfaction. If things aren ’ t going right, the first response is : let ’ s make more rules, let ’ s set up a set of detailed procedures to make sure that people will do the right thing. Give teachers scripts to follow in the classroom, so even if they don ’ t know what they ’ re doing and don ’ t care about the welfare of our kids, as long as they follow the scripts, our kids will get educated. Give **judges** a list of mandatory sentences to impose for crimes, so that you don ’ t need to rely on **judges** using their judgment. Instead, all they have to do is look up on the list what kind of sentence goes with what kind of crime. Impose limits on what credit card companies can charge in interest and on what they can charge in fees. More and more rules to protect us against an indifferent, uncaring set of institutions we have to deal with. Or â□□ or maybe and â□□ in addition to rules, let ’ s see if we can come up with some really clever incentives so that, even if the people we deal with don ’ t particularly want to serve our interests, it is in their interest to serve our interest â□□ the magic incentives that will get people to do the right thing even out of pure selfishness. So we offer teachers bonuses if the kids they teach **score** passing grades on these big test **scores** that are used to evaluate the quality of school systems. Rules and incentives â□□”sticks”and”carrots.”We passed a bunch of rules to regulate the **financial** industry in response to the recent collapse. There ’ s the Dodd-Frank Act, there ’ s the new Consumer Financial Protection Agency that is temporarily being headed through the backdoor by Elizabeth Warren. Maybe these rules will actually improve the way these **financial services** companies behave. We ’ ll see. In addition, we are **struggling** to find some way to create incentives for people in the **financial services** industry that will have them more interested in serving the long-term interests even of their own companies, rather than securing short-term profits. So if we find just the right incentives, they ’ ll do the right thing â□□ as I said â□□ selfishly, and if we come up with the right rules and regulations, they won ’ t drive us all over a cliff. And Ken [Sharpe] and I certainly know that you need to reign in the bankers. If there is a lesson to be learned from the **financial** collapse it is that. But what we believe, and what we argue in the book, is that there is no set of rules, no matter how detailed, no matter how specific, no matter how carefully monitored and enforced, there is no set of rules that will get us what we need. Why? Because bankers are smart people. And, like water, they will find cracks in any set of rules. You design a set of rules that will make sure that the particular reason why the **financial** system”almost-collapse”can ’ t happen again. It is naive beyond description to think that having blocked this source of **financial** collapse, you have blocked all possible sources of **financial** collapse. So it ’ s just a question of waiting for the next one and then marveling at how we could have been so stupid as not to protect ourselves against that. What we desperately need, beyond, or along with, better rules and reasonably smart incentives, is we need virtue. We need character. We need people who want to do the right thing. And in particular, the virtue that we need most of all is the virtue that Aristotle called”practical wisdom.”Practical wisdom is the moral will to do the right thing and the moral skill to figure out what the right thing is. So Aristotle was very interested in watching how the craftsmen around him worked. And he was impressed at how they would improvise novel solutions to novel problems â problems that they hadn ' t anticipated. So one example is he sees these stonemasons working on the Isle of Lesbos, and they need to measure out round columns. Well if you think about it, it ' s really hard to measure out round columns using a ruler. So what do they do? They fashion a novel solution to the problem. They created a ruler that bends, what we would call these days a tape measure â a flexible rule, a rule that bends. And Aristotle said,"Hah, they appreciated that sometimes to design rounded columns, you need to bend the rule."And Aristotle said often in dealing with other people, we need to bend the rules. Dealing with other people demands a kind of flexibility that no set of rules can encompass. Wise people know when and how to bend the rules. Wise people know how to improvise. The way my co-author, Ken, and I talk about it, they are kind of like jazz musicians. The rules are like the notes on the page, and that gets you started, but then you dance around the notes on the page, coming up with just the right combination for this particular moment with this particular set of fellow players. So for Aristotle, the kind of rule-bending, rule exception-finding and improvisation that you see in skilled craftsmen is exactly what you need to be a skilled moral craftsman. And in interactions with people, almost all the time, it is this kind of flexibility that is required. A wise person knows when to bend the rules. A wise person knows when to improvise. And most important, a wise person does this improvising and rule- bending in the **service** of the right aims. If you are a rule-bender and an improviser mostly to serve yourself, what you get is ruthless manipulation of other people. So it matters that you do this wise **practice** in the **service** of others and not in the **service** of yourself. And so the will to do the right thing is just as important as the moral skill of improvisation and exception-finding. Together they comprise practical wisdom, which Aristotle thought was the master virtue. So I ' ll give you an example of wise **practice** in **action**. It ' s the case of Michael. Michael ' s a young guy. He had a pretty low-wage job. He was supporting his wife and a child, and the child was going to parochial school. Then he lost his job. He panicked about being able to support his family. One night, he drank a little too much, and he robbed a **cab** driver â stole 50 dollars. He robbed him at gunpoint. It was a toy gun. He got caught. He got tried. He got convicted. The Pennsylvania sentencing guidelines required a minimum sentence for a crime like this of two years, 24 months. The **judge** on the case, Judge Lois Forer thought that this made no sense. He had never committed a crime before. He was a responsible husband and father. He had been faced with desperate circumstances. All this would do is wreck a family. And so she improvised a sentence â 11 months, and not only that, but release every day to go to work. Spend your night in jail, spend your day holding down a job. He did. He served out his sentence. He made restitution and found himself a new job. And the family was united. And it seemed on the road to some sort of a decent life â a happy ending to a story involving wise improvisation from a wise **judge**. But it turned out the prosecutor was not happy that Judge Forer ignored the sentencing guidelines and sort of invented her own, and so he appealed. And he asked for the mandatory minimum sentence for armed robbery. He did after all have a toy gun. The mandatory minimum sentence for armed robbery is five years. He won the appeal. Michael was sentenced to five years in prison. **Judge** Forer had to follow the law. And by the way, this appeal went through after he had finished serving his sentence, so he was out and working at a job and taking care of his family and he had to go back into jail. **Judge** Forer did what she was required to do, and then she quit the bench. And Michael disappeared. So that is an example, both of wisdom in

practice and the subversion of wisdom by rules that are meant, of course, to make things better. Now consider Ms. Dewey. Ms. Dewey ’ s a teacher in a Texas elementary school. She found herself listening to a consultant one day who was trying to help teachers boost the test **scores** of the kids, so that the school would reach the elite category in percentage of kids passing big tests. All these schools in Texas compete with one another to achieve these milestones, and there are bonuses and various other treats that come if you beat the other schools. So here was the consultant ’ s advice : first, don ’ t waste your time on kids who are going to pass the test no matter what you do. Second, don ’ t waste your time on kids who can ’ t pass the test no matter what you do. Third, don ’ t waste your time on kids who moved into the district too late for their **scores** to be counted. Focus all of your time and attention on the kids who are on the bubble, the so-called”bubble kids”â kids where your intervention can get them just maybe over the line from failing to passing. So Ms. Dewey heard this, and she shook her head in despair while fellow teachers were sort of cheering each other on and nodding approvingly. It ’ s like they were about to go play a football game. For Ms. Dewey, this isn ’ t why she became a teacher. Now Ken and I are not naive, and we understand that you need to have rules. You need to have incentives. People have to make a living. But the problem with relying on rules and incentives is that they demoralize professional activity, and they demoralize professional activity in two senses. First, they demoralize the people who are engaged in the activity. **Judge** Forer quits, and Ms. Dewey in completely disheartened. And second, they demoralize the activity itself. The very **practice** is demoralized, and the practitioners are demoralized. It creates people â when you manipulate incentives to get people to do the right thing â it creates people who are addicted to incentives. That is to say, it creates people who only do things for incentives. Now the striking thing about this is that psychologists have known this for 30 years. Psychologists have known about the negative consequences of incentivizing everything for 30 years. We know that if you reward kids for drawing pictures, they stop caring about the drawing and care only about the reward. If you reward kids for reading books, they stop caring about what ’ s in the books and only care about how long they are. If you reward teachers for kids ’ test **scores**, they stop caring about educating and only care about test preparation. If you were to reward doctors for doing more procedures â which is the current system â they would do more. If instead you reward doctors for doing fewer procedures, they will do fewer. What we want, of course, is doctors who do just the right amount of procedures and do the right amount for the right reason â namely, to serve the welfare of their patients. Psychologists have known this for decades, and it ’ s time for policymakers to start paying attention and listen to psychologists a little bit, instead of economists. And it doesn ’ t have to be this way. We think, Ken and I, that there are real sources of hope. We identify one set of people in all of these **practices** who we call canny outlaws. These are people who, being forced to operate in a system that demands rule-following and creates incentives, find away around the rules, find a way to subvert the rules. So there are teachers who have these scripts to follow, and they know that if they follow these scripts, the kids will learn nothing. And so what they do is they follow the scripts, but they follow the scripts at double-time and squirrel away little bits of extra time during which they teach in the way that they actually know is effective. So these are little ordinary, everyday heroes, and they ’ re incredibly admirable, but there ’ s no way that they can sustain this kind of activity in the face of a system that either roots them out or grinds them down. So canny outlaws are better than nothing, but it ’ s hard to imagine any canny outlaw sustaining that for an indefinite period of time. More hopeful are people we call system-changers. These are people who are looking not to dodge the system ’ s rules and regulations, but to transform the system, and we talk about several. One in particular is a **judge** named Robert Russell. And one day he was faced with the case of Gary Pettengill. Pettengill was a 23-year-old vet who had planned to make the army a career, but then he got a severe back injury in Iraq, and that forced him to take a medical discharge. He was married, he had a third kid on the way, he suffered from PTSD, in addition to the bad back, and recurrent nightmares, and he had started using marijuana to ease some of the symptoms. He was only able to get part-time work because of his back, and so he was unable to earn enough to put food on the table and take care of his family. So he started selling marijuana. He was busted in a drug sweep. His family was kicked out of their apartment, and the welfare system was threatening to take away his kids. Under normal sentencing procedures, Judge Russell would have had little choice but to sentence Pettengill to serious jail-time as a drug felon. But Judge Russell did have an alternative. And that ’ s because he was in a **special** court. He was in a court called the Veterans ’ Court. In the Veterans ’ Court â□□ this was the first of its kind in the United States. Judge Russell created the Veterans ’ Court. It was a court only for veterans who had broken the law. And he had created it exactly because mandatory sentencing laws were taking the judgment out of **judging**. No one wanted non-violent offenders â□□ and especially non-violent offenders who were veterans to boot â□□ to be thrown into prison. They wanted to do something about what we all know, namely the revolving door of the criminal justice system. And what the Veterans ’ Court did, was it treated each criminal as an individual, tried to get inside their problems, tried to fashion responses to their crimes that helped them to rehabilitate themselves, and didn ’ t forget about them once the judgment was made. Stayed with them, followed up on them, made sure that they were sticking to whatever plan had been jointly developed to get them over the hump. There are now 22 cities that have Veterans ’ Courts like this. Why has the idea spread? Well, one reason is that Judge Russell has now seen 108 vets in his Veterans ’ Court as of February of this year, and out of 108, guess how many have gone back through the revolving door of justice into prison. None. None. Anyone would glom onto a criminal justice system that has this kind of a record. So here ’ s is a system-changer, and it seems to be catching. There ’ s a banker who created a for-profit community bank that encouraged bankers â□□ I know this is hard to believe â□□ encouraged bankers who worked there to do well by doing good for their low-income clients. The bank helped finance the rebuilding of what was otherwise a dying community. Though their loan recipients were high-risk by ordinary standards, the default rate was extremely low. The bank was profitable. The bankers stayed with their loan recipients. They didn ’ t make loans and then sell the loans. They **serviced** the loans. They made sure that their loan recipients were staying up with their payments. Banking hasn ’ t always been the way we read about it now in the newspapers. Even Goldman Sachs once used to serve clients, before it turned into an institution that serves only itself. Banking wasn ’ t always this way, and it doesn ’ t have to be this way. So there are examples like this in medicine â□□ doctors at Harvard who are trying to transform medical education, so that you don ’ t get a kind of ethical erosion and loss of **empathy**, which characterizes most medical students in the course of their medical training. And the way they do it is to give third-year medical students patients who they follow for an entire year. So the patients are not organ systems, and they ’ re not diseases ; they ’ re people, people with lives. And in order to be an effective doctor, you need to treat people who have lives and not just disease. In addition to which there ’ s an enormous amount of back and forth, mentoring of one student by another, of all the students by the doctors, and the result is a generation â we hope â of doctors who do have time for the people they treat. We ’ ll see. So there are lots of examples like this that we talk about. Each of them shows that it is possible to build on and nurture character and keep a profession true to its proper **mission** â what Aristotle would have called its proper telos. And Ken and I believe that this is what practitioners actually want. People want to be allowed to be virtuous. They want to have permission to do the right thing. They don ’ t want to feel like they need to take a shower to get the moral grime off their bodies everyday when they come home from work. Aristotle thought that practical wisdom was the key to **happiness**, and he was right. There ’ s now a lot of research being done in psychology on what makes people happy, and the two things that jump out in study after study â I know this will come as a shock to all of you â the two things that matter most to **happiness** are love and work. Love : managing successfully relations with the people who are close to you and with the communities of which you are a part. Work : engaging in activities that are meaningful and satisfying. If you have that, good close relations with other people, work that ’ s meaningful and fulfilling, you don ’ t much need anything else. Well, to love well and to work well, you need wisdom. Rules and incentives don ’ t tell you how to be a good friend, how to be a good parent, how to be a good spouse, or how to be a good doctor or a good lawyer or a good teacher. Rules and incentives are no substitutes for wisdom. Indeed, we argue, there is no substitute for wisdom. And so practical wisdom does not require heroic acts of self-sacrifice on the part of practitioners. In giving us the will and the skill to do the right thing â to do right by others â practical wisdom also gives us the will and the skill to do right by ourselves. Thanks.

Text Sample 243: POS Dim 2 – 2011 – Score 11.00 – 2011/t002676.txt

I ’ m speaking to you about what I call the”mesh.”It ’ s essentially a fundamental shift in our relationship with stuff, with the things in our lives. And it ’ s starting to look at â not always and not for everything â but in certain moments of time, access to certain kinds of goods and **service** will trump **ownership** of them. And so it ’ s the pursuit of better things, easily shared. And we come from a long tradition of sharing. We ’ ve shared transportation. We ’ ve shared wine and food and other sorts of fabulous experiences in coffee bars in Amsterdam. We ’ ve also shared other sorts of entertainment â sports arenas, public parks, concert halls, libraries, universities. All these things are share-platforms, but sharing ultimately starts and ends with what I refer to as the”mother of all share-platforms.”And as I think about the mesh and I think about, well, what ’ s driving it, how come it ’ s happening now, I think there ’ s a number of vectors that I want to give you as background. One is the recession â that the recession has caused us to **rethink** our relationship with the things in our lives relative to the value â so starting to align the value with the true cost. Secondly, population growth and density into cities. More people, smaller spaces, less stuff. Climate change : we ’ re trying to reduce the **stress** in our personal lives and in our communities and on the planet. Also, there ’ s been this recent distrust of big brands, global big brands, in a bunch of different industries, and that ’ s created an opening. Research is showing here, in the States, and in Canada and Western Europe, that most of us are much more open to local companies, or brands that maybe we haven ’ t heard of. Whereas before, we went with the big brands that we were sure we trusted. And last is that we ’ re more connected now to more people on the planet than ever before â except for if you ’ re sitting next to someone. The other thing that ’ s worth considering is that we ’ ve made a huge investment over decades and decades, and tens of billions of dollars have gone into this investment that now is our inheritance. It ’ s a physical infrastructure that allows us to get from point A to point B and move things that way. It ’ s also â Web and **mobile** allow us to be connected and create all kinds of **platforms** and systems, and the investment of those technologies and that infrastructure is really our inheritance. It allows us to engage in really new and interesting ways. And so for me, a mesh company, the”classic”mesh company, brings together these three things : our ability to connect to each other â most of us are walking around with these **mobile** devices that are GPS-enabled and Web-enabled â allows us to find each other and find things in time and space. And third is that physical things are readable on a map â so restaurants, a variety of venues, but also with GPS and other technology like RFID and it continues to expand beyond that, we can also track things that are moving, like a car, a taxicab, a transit system, a box that ’ s moving through time and space. And so that sets up for making access to get goods and **services** more convenient and less costly in many cases than owning them. For example, I want to use Zipcar. How many people here have experienced car- sharing or bike-sharing? Wow, that ’ s great. Okay, thank you. Basically Zipcar is the largest car-sharing company in the world. They did not invent car- sharing. Car-sharing was actually invented in Europe. One of the founders went to Switzerland, saw it implemented someplace, said,”Wow, that looks really cool. I think we can do that in Cambridge,”brought it to Cambridge and they started â two women â Robin Chase being the other person who started it. Zipcar got some really important things right. First, they really understood that a brand is a voice and a product is a souvenir. And so they were very clever about the way that they packaged car-sharing. They made it sexy. They made it fresh. They made it aspirational. If you were a member of the club, when you ’ re a member of a club, you ’ re a Zipster. The cars they picked didn ’ t look like ex-cop cars that were hollowed out or something. They picked these sexy cars. They targeted to universities. They made sure that the demographic for who they were targeting and the car was all matching. It was a very nice experience, and the cars were clean and reliable, and it all worked. And so from a branding perspective, they got a lot right. But they understood fundamentally that they are not a car company. They understand that they are an information company. Because when we buy a car we go to the dealer once, we have an interaction, and we ’ re chow â usually as quickly as possible. But when you ’ re sharing a car and you have a car-share **service**, you might use an E. V. to commute, you get a truck because you ’ re doing a home project. When you pick your aunt up at the airport, you get a sedan. And you ’ re going to the mountains to ski, you get different accessories put on the car for doing that sort of thing. Meanwhile, these guys are sitting back, collecting all sorts of data about our behavior and how we interact with the **service**. And so it ’ s not only an option for them, but I believe it ’ s an imperative for Zipcar and other mesh companies to actually just wow us, to be like a concierge **service**. Because we give them so much information, and they are entitled to really see how it is that we ’ re moving. They ’ re in really good shape to anticipate what we ’ re going to want next. And so what percent of the day do you think the average person uses a car? What percentage of the time? Any guesses? Those are really very good. I was imagining it was like 20 percent when I first started. The number across the U. S. and Western Europe is eight percent. And so basically even if you think it ’ s 10 percent, 90 percent of the time, something that costs us a lot of money â personally, and also we organize our cities around it and all sorts of things â 90 percent of the time it ’ s sitting around. So for this reason, I think one of the other themes with the mesh is essentially that, if we squeeze hard on things that we ’ ve thrown away, there ’ s a lot of value in those things. What set up with Zipcar â Zipcar started in 2000. In the last year, 2010, two car companies started, one that ’ s in the U. K. called WhipCar, and the other one, RelayRides, in the U. S. They ’ re both peer- to-peer car-sharing **services**, because the two things that really work for car- sharing is, one, the car has to be available, and two, it ’ s within one or two blocks of where you stand. Well the car that ’ s one or two blocks from your home or your office is probably your neighbor ’ s car, and it ’ s probably also available. So people have created this business. Zipcar started a decade earlier, in 2000. It took them six years to get 1, 000 cars in **service**. WhipCar, which started April of last year, it took them six months to get 1, 000 cars in the **service**. So, really interesting. People are making anywhere between 200 and 700 dollars a month letting their neighbors use their car when they ’ re not using it. So it ’ s like **vacation** rentals for cars. Since I ’ m here â and I hope some people in the audience are in the car business â â I ’ m thinking that, coming from the technology side of things â we saw cable-ready TVs and WiFi-ready Notebooks â it would be really great if, any minute now, you guys could start rolling share-ready cars off. Because it just creates more flexibility. It allows us as owners to have other options. And I think we ’ re going there anyway. The opportunity and the challenge with mesh businesses â and those are businesses like Zipcar or Netflix that are full mesh businesses, or other ones where you have a lot of the car companies, car manufacturers, who are beginning to offer their own car-share **services** as well as a second flanker brand, or as really a test, I think â is to make sharing irresistible. We have experiences in our lives, certainly, when sharing has been irresistible. It ’ s just, how do we make that recurrent and scale it? We know also, because we ’ re connected in social networks, that it ’ s easy to create delight in one little place. It ’ s contagious because we ’ re all connected to each other. So if I have a terrific experience and I **tweet** it, or I tell five people standing next to me, news travels. The opposite, as we know, is also true, often more true. So here we have LudoTruck, which is in L. A., doing the things that gourmet food trucks do, and they ’ ve gathered quite a following. In general, and maybe, again, it ’ s because I ’ m a tech entrepreneur, I look at things as **platforms**. **Platforms** are invitations. So creating Craigslist or iTunes and the iPhone developer network, there are all these networks â Facebook as well. These **platforms** invite all sorts of developers and all sorts of people to come with their ideas and their opportunity to create and target an application for a particular audience. And honestly, it ’ s full of surprises. Because I don ’ t think any of us in this room could have predicted the sorts of applications that have happened at Facebook, around Facebook, for example, two years ago, when Mark announced that they were going to go with a **platform**. So in this way, I think that cities are **platforms**, and certainly Detroit is a **platform**. The invitation of bringing makers and artists and entrepreneurs â it really helps stimulate this fiery creativity and helps a city to thrive. It ’ s inviting participation, and cities have, historically, invited all sorts of participation. Now we ’ re saying that there ’ s other options as well. So, for example, city departments can open up transit data. Google has made available transit data API. And so there ’ s about seven or eight cities already in the U. S. that have provided the transit data, and different developers are building applications. So I was having a coffee in Portland, and half-of-a-latte in and the little board in the cafe all of a sudden starts showing me that the next bus is coming in three minutes and the train is coming in 16 minutes. And so it ’ s reliable, real data that ’ s right in my face, where I am, so I can finish the latte. There ’ s this fabulous opportunity we have across the U. S. now : about 21 percent of vacant commercial and industrial space. That space is not vital. The areas around it lack vitality and vibrancy and engagement. There ’ s this thing â how many people here have heard of pop-up stores or pop-up shops? Oh, great. So I ’ m a big fan of this. And this is a very mesh-y thing. Essentially, there are all sorts of restaurants in Oakland, near where I live. There ’ s a pop-up general store every three weeks, and they do a fantastic job of making a very social event happening for foodies. **Super** fun, and it happens in a very transitional neighborhood. Subsequent to that, after it ’ s been going for about a year now, they actually started to lease and create and extend. An area that was edgy-artsy is now starting to become much cooler and engage a lot more people. So this is an example. The Crafty Fox is this woman who ’ s into crafts, and she does these pop-up crafts fairs around London. But these sorts of things are happening in many different environments. From my perspective, one of the things pop-up stores do is create perishability and urgency. It creates two of the favorite words of any businessperson : sold out. And the opportunity to really focus trust and attention is a wonderful thing. So a lot of what we see in the mesh, and a lot of what we have in the **platform** that we built allows us to define, refine and scale. It allows us to test things as an entrepreneur, to go to market, to be in conversation with people, listen, refine something and go back. It ’ s very cost-effective, and it ’ s very mesh-y. The infrastructure enables that. In closing, and as we ’ re moving towards the end, I just also want to encourage â and I ’ m willing to share my failures as well, though not from the stage. I would just like to say that one of the big things, when we look at waste and when we look at ways that we can really be generous and contribute to each other, but also move to create a better economic situation and a better environmental situation, is by sharing failures. And one quick example is Velib, in 2007, came forward in Paris with a very **bold** proposition, a very big bike-sharing **service**. They made a lot of mistakes. They had some number of big successes. But they were very transparent, or they had to be, in the way that they exposed what worked and didn ’ t work. And so B. C. in Barcelona and B-cycle and Boris Bikes in London â no one has had to repeat the version 1. 0 screw- ups and expensive learning exercises that happened in Paris. So the opportunity when we ’ re connected is also to share failures and successes. We ’ re at the very beginning of something that, what we ’ re seeing and the way that mesh companies are coming forward, is inviting, it ’ s engaging, but it ’ s very early. I have a website â it ’ s a directory â and it started with about 1, 200 companies, and in the last two-and-a-half months it ’ s up to about 3, 300 companies. And it grows on a very regular **daily** basis. But it ’ s very much at the beginning. So I just want to welcome all of you onto the ride. And thank you very much.

Text Sample 244: POS Dim 2 – 2011 – Score 10.00 – 2011/t002499.txt

I just want to start with a little bit of a word of **warning**, and that is : my job here tonight is to be a little bit of a”doctor bring-me-down.”So bear with me for a few minutes, and know that after this, things will get lighter and brighter. So let ’ s start. I know that many of you have heard the traveler ’ s adage,”Take nothing but pictures, leave nothing but footprints.”Well, I ’ m going to say I don ’ t think that ’ s either as benign nor as simple as it sounds, particularly for those of us in industries who are portraying people in poor countries, in developing countries and portraying the poor. And those of us in those industries are **reporters**, **researchers** and people working for NGOs ; I suspect there are a lot of us in those industries in the audience. We ’ re going overseas and bringing back pictures like this : of the utterly distressed or the displaced or the hungry or the child laborers or the exotic. Now, Susan Sontag reminds us that photographs, in part, help define what we have the right to **observe**, but more importantly, they are an ethics of seeing. And I think right now is a good time to review our ethics of seeing, as our industries of reporting and research and NGO work are collapsing and changing, in part, by what ' s being driven by what ' s happening in the economy. But it ' s making us forge new relationships. And those new relationships have some fuzzy boundaries. I worked at the edge of some of these fuzzy boundaries, and I want to share with you some of my observations. My ethics of seeing is informed by 25 years as a **reporter** covering emerging economies and international relations. And I believe in a free and independent press. I believe that journalism is a public good. But it ' s getting harder to do that job, in part, because of the massive layoffs, because the budgets for international reporting aren ' t there anymore, new technologies and new **platforms** begging new content, and there are a lot of new journalisms. There ' s activist journalism, humanitarian journalism, peace journalism, and we are all looking to cover the important stories of our time. So we ' re going to NGOs and asking them if we can embed in their projects. This is in part because they ' re doing important work in interesting places. That ' s one example here : this is a project I worked on in the Blue Nile in Ethiopia. NGOs understand the benefits of having **reporters** tag along on their team. They need the publicity, they are under tremendous pressure, they ' re competing in a very crowded market for compassion. So they ' re also looking to **reporters** and to hire freelance **reporters** to help them develop their public relations material and their media material. Now, **researchers** are also under pressure. They ' re under pressure to communicate their science outside of the academy. So they ' re collaborating with **reporters**, because for many **researchers**, it ' s difficult for them to write a simple story or a clear story. And the benefit for **reporters** is that covering field research is some of the best work out there. You not only get to cover science, but you get to meet interesting scientists, like my PhD advisor Revi Sterling, she, of the magic research high tops there. And it was a discussion with Revi that brought us to the edge of the **researcher** and **reporter**, that fuzzy boundary. And I said to her,"I was looking forward to going to developing countries and doing research and covering stories at the same time."She said,"I don ' t think so, girlfriend."And that confusion, that mutual confusion, drove us to publish a paper on the conflicting ethics and the contradictory **practices** of research and reporting. We started with the understanding that **researchers** and **reporters** are distant cousins, equally storytellers and social analysts. But we don ' t see nor portray developing communities the same way. Here ' s a very classic example. This is Somalia, 1992. It could be Somalia today. And this is a standard operating procedure for much of the news video and the news pictures that you see, where a group of **reporters** will be trucked in, escorted to the site of a disaster, they ' ll produce their material, take their pictures, get their interviews, and then they ' ll be escorted out. This is decidedly not a research **setting**. Now, sometimes, we ' re working on feature stories. This is an image I took of a woman in Bhongir Village in Andhra Pradesh in India. She ' s at a microfinance meeting. It ' s a terrific story. What ' s important here is that she is identifiable. You can see her face. This also is not a research picture. This is much more representative of a research picture. It ' s a research site : you see young women accessing new technologies. It ' s more of a time stamp, it ' s a documentation of research. I couldn ' t use this for news. It doesn ' t tell enough, and it wouldn ' t sell. But then, the differences are even deeper than that. Revi and I analyzed some of the mandates that **researchers** are under. They are under some very strict rules governed by their university research review boards when it comes to content and confidentiality. **Researchers** are mandated to acquire document-informed consent. Well, as a **reporter**, if I hang a **microphone** on someone, that is consent. And when it comes to creating the story, I ' ll fact-check as a **reporter**, but I don ' t invite company to create that story, whereas social scientists, **researchers**, and particularly participatory **researchers**, will often work on constructing the narrative with the community. And when it comes to paying for information,"checkbook journalism"is roundly discouraged, in part, because of the **bias** it introduces in the kind of information you get. But social scientists understand that people ' s time is valuable so they pay them for that time. So while journalists are well-placed to convey the beauty of the scientific process — and I would add, the NGO process — what about the warts? What happens if a research project is not particularly well-designed, or an NGO project doesn ' t fulfill its goals? Or the other kind of warts, you know, what happens after dark when the drinks happen. Research environments and reporting trips and NGO projects are very intimate environments ; you make good friends while you ' re doing good work. But there ' s a little bit of Johnnie Walker journalism after dark, and what happens to that line between embedded and in- bedded? Or what do you do with the odd and odious behavior? The point is that you ' ll want to negotiate in advance what is on the record and off the record. I ' m going to turn now to some NGO imagery which will be familiar to some of you in this audience. Narrator : For about 70 cents, you can buy a can of soda, regular or diet. In Ethiopia, for just 70 cents a day, you can feed a child like Jamal nourishing meals. For about 70 cents, you can also buy a cup of coffee. In Guatemala, for 70 cents a day, you can help a child like Vilma get the clothes she needs to attend school. Leslie Dodson : Now, there ' s some very common imagery that ' s been around for 40 years. That ' s part of Sally Struthers ' s famine campaign. Some of it is very familiar ; it ' s the Madonna and child. Women and children are very effective in terms of NGO campaigns. We ' ve been looking at this imagery for a long time, for hundreds and hundreds of years ; the Madonna and child. Here is [Duccio], and here is Michelangelo. My concern is : Are we one-noting the genders in our narratives of poverty in developing communities? Do we have women as victims, and are men only the perpetrators? Are they the guys with the AK-47s or the boy soldiers? Because that doesn ' t leave room for stories like the man who ' s selling ice cream at the refuge camp in Southern Sudan, where we did a project, or the stories of the men who are working on the bridge over the Blue Nile. So I wonder : Are these stories inconvenient to our narratives? And what about this narrative? This is a for-profit game, and its aim is to make development fun. One question is : Did they inadvertently make fun of? Another set of questions is : What are the rights of these children? What rights of publicity or **privacy** do they have? Did they get paid? Should they get paid? Should they share in the profit? This is a for-profit game. Did they sign talent waivers? I have to use these when I ' m working with NGOs and documentary filmmakers here in the States. In the States, we take our right to **privacy** and publicity very seriously. So what is it about getting on a long-haul flight that makes these rights vaporize? I don ' t want to just pick on our friends in the gaming arts ; I ' ll turn to the graphic arts, where we often see these monolithic, homogeneous stories about the great country of Africa. But Africa is not a country, it ' s a continent. It ' s 54 countries and thousands and thousands of languages. So my question is : Is this imagery productive, or is it reductive? I know that it ' s popular. USAID just launched their campaign"Forward"— FWD : Famine, War and Drought. And by looking at it, you would think that was happening all the time, all over Africa. But this is about what ' s happening in the Horn of Africa. And I ' m still trying to make sense of Africa in a piece of Wonder Bread. I ' m wondering about that. Germaine Greer has wondered about the same things and she says,”At breakfast and at dinner, we can sharpen our own appetites with a plentiful dose of the pornography of war, genocide, destitution and disease.”She ’ s right. We have sharpened our appetites. But we can also sharpen our insights. It is not always war, insurrection and disease. This is a picture out of South Sudan, just a couple of months before the new country was born. I will continue to work as a **researcher** and a **reporter** in developing countries, but I do it with an altered ethic of seeing. I ask myself whether my pictures are pandering, whether they contribute to stereotypes, whether the images match the message, am I complacent or am I complicit? Thank you.

Text Sample 245: POS Dim 2 – 2006 – Score 9.00 – 2006/t000160.txt

The future that we will create can be a future that we ’ ll be proud of. I think about this every day ; it ’ s quite literally my job. I ’ m co-founder and senior columnist at Worldchanging. com. Alex Steffen and I founded Worldchanging in late 2003, and since then we and our growing global team of contributors have documented the ever-expanding variety of solutions that are out there, right now and on the near horizon. In a little over two years, we ’ ve written up about 4, 000 items — replicable models, technological tools, emerging ideas — all providing a path to a future that ’ s more sustainable, more equitable and more desirable. Our emphasis on solutions is quite intentional. There are tons of places to go, online and off, if what you want to find is the latest bit of news about just how quickly our hell-bound handbasket is moving. We want to offer people an idea of what they can do about it. We focus primarily on the planet ’ s environment, but we also address issues of global development, international conflict, responsible use of emerging technologies, even the rise of the so-called Second Superpower and much, much more. The scope of solutions that we discuss is actually pretty broad, but that reflects both the range of challenges that need to be met and the kinds of innovations that will allow us to do so. A quick sampling really can barely scratch the surface, but to give you a sense of what we cover : tools for rapid disaster relief, such as this inflatable concrete shelter ; innovative uses of bioscience, such as a flower that changes color in the presence of landmines ; ultra high-efficiency designs for homes and offices ; distributed power generation using solar power, wind power, ocean power, other clean energy sources ; ultra, ultra high-efficiency vehicles of the future ; ultra high- efficiency vehicles you can get right now ; and better urban design, so you don ’ t need to drive as much in the first place ; bio-mimetic approaches to design that take advantage of the efficiencies of natural models in both vehicles and buildings ; distributed computing projects that will help us model the future of the climate. Also, a number of the topics that we ’ ve been talking about this week at TED are things that we ’ ve addressed in the past on Worldchanging : cradle-to-cradle design, MIT ’ s Fab Labs, the consequences of extreme longevity, the One Laptop per Child project, even Gapminder. As a born-in-the-mid-1960s Gen X-er, hurtling all too quickly to my fortieth birthday, I ’ m naturally inclined to pessimism. But working at Worldchanging has convinced me, much to my own surprise, that successful responses to the world ’ s problems are nonetheless possible. Moreover, I ’ ve come to realize that focusing only on negative outcomes can really blind you to the very possibility of success. As Norwegian social scientist Evelin Lindner has **observed**,”Pessimism is a luxury of good times... In difficult times, pessimism is a self- fulfilling, self-inflicted death sentence.”The truth is, we can build a better world, and we can do so right now. We have the tools : we saw a hint of that a moment ago, and we ’ re coming up with new ones all the time. We have the knowledge, and our understanding of the planet improves every day. Most importantly, we have the motive : we have a world that needs fixing, and nobody ’ s going to do it for us. Many of the solutions that I and my colleagues seek out and write up every day have some important aspects in common : transparency, collaboration, a willingness to experiment, and an appreciation of science — or, more appropriately, science! The majority of models, tools and ideas on World-changing encompass combinations of these characteristics, so I want to give you a few concrete examples of how these principles combine in world- changing ways. We can see world-changing values in the emergence of tools to make the invisible visible — that is, to make apparent the conditions of the world around us that would otherwise be largely imperceptible. We know that people often change their behavior when they can see and understand the impact of their **actions**. As a small example, many of us have experienced the change in driving behavior that comes from having a real time display of mileage showing precisely how one ’ s **driving** habits affect the vehicle ’ s efficiency. The last few years have all seen the rise of innovations in how we measure and display aspects of the world that can be too big, or too intangible, or too slippery to grasp easily. Simple technologies, like wall-mounted devices that display how much power your household is using, and what kind of results you ’ ll get if you turn off a few lights — these can actually have a direct positive impact on your energy footprint. Community tools, like text messaging, that can tell you when pollen counts are up or smog levels are rising or a natural disaster is unfolding, can give you the information you need to act in a timely fashion. Data-rich displays like maps of campaign contributions, or maps of the disappearing polar ice caps, allow us to better understand the context and the flow of processes that affect us all. We can see world-changing values in research projects that seek to meet the world ’ s medical needs through open access to data and collaborative **action**. Now, some people emphasize the risks of knowledge-enabled dangers, but I ’ m convinced that the benefits of knowledge-enabled solutions are far more important. For example, open-access journals, like the Public Library of Science, make cutting-edge scientific research free to all — everyone in the world. And actually, a growing number of science publishers are adopting this model. Last year, hundreds of volunteer biology and chemistry **researchers** around the world worked together to sequence the genome of the parasite responsible for some of the developing world ’ s worst diseases : African sleeping sickness, leishmaniasis and Chagas disease. That genome data can now be found on open-access genetic data banks around the world, and it ’ s an enormous boon to **researchers** trying to come up with treatments. But my favorite example has to be the global response to the SARS epidemic in 2003, 2004, which relied on worldwide access to the full gene sequence of the SARS virus. The U. S. National Research Council in its **follow-up** report on the outbreak specifically cited this open availability of the sequence as a key reason why the treatment for SARS could be developed so quickly. And we can see world-changing values in something as humble as a cell phone. I can probably count on my fingers the number of people in this room who do not use a **mobile** phone — and where is Aubrey, because I know he doesn ’ t? For many of us, cell phones have really become almost an extension of ourselves, and we ’ re really now beginning to see the social changes that **mobile** phones can bring about. You may already know some of the big-picture aspects : globally, more camera phones were sold last year than any other kind of camera, and a growing number of people live lives mediated through the lens, and over the network — and sometimes enter history books. In the developing world, **mobile** phones have become economic drivers. A study last year showed a direct correlation between the growth of **mobile** phone use and subsequent GDP increases across Africa. In Kenya, **mobile** phone minutes have actually become an alternative currency. The political aspects of **mobile** phones can ' t be ignored either, from text message swarms in Korea helping to bring down a government, to the Blairwatch Project in the UK, keeping tabs on politicians who try to avoid the press. And it ' s just going to get more wild. Pervasive, always-on networks, high quality sound and video, even devices made to be worn instead of carried in the pocket, will transform how we live on a scale that few really appreciate. It ' s no exaggeration to say that the **mobile** phone may be among the world ' s most important technologies. And in this rapidly evolving context, it ' s possible to imagine a world in which the **mobile** phone becomes something far more than a medium for social interaction. I ' ve long admired the Witness project, and Peter Gabriel told us more details about it on Wednesday, in his profoundly moving presentation. And I ' m just incredibly happy to see the news that Witness is going to be opening up a Web portal to enable users of digital cameras and camera phones to send in their recordings over the Internet, rather than just hand-carrying the videotape. Not only does this add a new and potentially safer avenue for documenting abuses, it opens up the program to the growing global digital generation. Now, imagine a similar model for networking environmentalists. Imagine a Web portal collecting recordings and evidence of what ' s happening to the planet : putting news and data at the fingertips of people of all kinds, from activists and **researchers** to businesspeople and political figures. It would highlight the changes that are underway, but would more importantly give voice to the people who are willing to work to see a new world, a better world, come about. It would give everyday citizens a chance to play a role in the protection of the planet. It would be, in essence, an"Earth Witness"project. Now, just to be clear, in this talk I ' m using the name"Earth Witness"as part of the scenario, simply as a shorthand, for what this imaginary project could aspire to, not to piggyback on the wonderful work of the Witness organization. It could just as easily be called,"Environmental Transparency Project,""Smart Mobs for Natural Security"— but Earth Witness is a lot easier to say. Now, many of the people who participate in Earth Witness would focus on ecological problems, human-caused or otherwise, especially environmental crimes and significant sources of greenhouse gases and emissions. That ' s understandable and important. We need better documentation of what ' s happening to the planet if we ' re ever going to have a chance of repairing the damage. But the Earth Witness project wouldn ' t need to be limited to problems. In the best Worldchanging tradition, it might also serve as a showcase for good ideas, successful projects and efforts to make a difference that deserve much more visibility. Earth Witness would show us two worlds : the world we ' re leaving behind, and the world we ' re building for generations to come. And what makes this scenario particularly appealing to me is we could do it today. The key components are already widely available. Camera phones, of course, would be fundamental to the project. And for a lot of us, they ' re as close as we have yet to always-on, widely available information tools. We may not remember to bring our digital cameras with us wherever we go, but very few of us forget our phones. You could even imagine a version of this scenario in which people actually build their own phones. Over the course of last year, open-source hardware hackers have come up with multiple models for usable, Linux-based **mobile** phones, and the Earth Phone could spin off from this kind of project. At the other end of the network, there ' d be a server for people to send photos and messages to, accessible over the Web, combining a photo-sharing **service**, social networking **platforms** and a collaborative filtering system. Now, you Web 2. 0 folks in the audience know what I ' m talking about, but for those of you for whom that last sentence was in a crazy moon language, I mean simply this : the online part of the Earth Witness project would be created by the users, working together and working openly. That ' s enough right there to start to build a compelling chronicle of what ' s now happening to our planet, but we could do more. An Earth Witness site could also serve as a collection spot for all sorts of data about conditions around the planet picked up by environmental sensors that attach to your cell phone. Now, you don ' t see these devices as add-ons for phones yet, but students and engineers around the world have attached atmospheric sensors to bicycles and handheld computers and cheap robots and the backs of pigeons — that being a project that ' s actually underway right now at U. C. Irvine, using bird-mounted sensors as a way of measuring smog-forming pollution. It ' s hardly a stretch to imagine putting the same thing on a phone carried by a person. Now, the idea of connecting a sensor to your phone is not new : phone-makers around the world offer phones that sniff for bad breath, or tell you to worry about too much sun exposure. Swedish firm Uppsala Biomedical, more seriously, makes a **mobile** phone add-on that can process blood tests in the field, uploading the data, displaying the results. Even the Lawrence Livermore National Labs have gotten into the act, designing a prototype phone that has radiation sensors to find dirty bombs. Now, there ' s an enormous variety of tiny, inexpensive sensors on the market, and you can easily imagine someone putting together a phone that could measure temperature, CO2 or methane levels, the presence of some biotoxins — potentially, in a few years, maybe even H5N1 avian **flu** virus. You could see that some kind of system like this would actually be a really good fit with Larry Brilliant ' s InSTEDD project. Now, all of this data could be tagged with geographic information and mashed up with online maps for easy viewing and analysis. And that ' s worth noting in particular. The impact of open-access online maps over the last year or two has been simply phenomenal. Developers around the world have come up with an amazing variety of ways to layer useful data on top of the maps, from bus routes and crime statistics to the global progress of avian **flu**. Earth Witness would take this further, linking what you see with what thousands or millions of other people see around the world. It ' s kind of exciting to think about what might be accomplished if something like this ever existed. We ' d have a far better — far better knowledge of what ' s happening on our planet environmentally than could be gathered with satellites and a handful of government sensor nets alone. It would be a collaborative, bottom-up approach to environmental awareness and protection, able to respond to emerging concerns in a smart mobs kind of way — and if you need greater sensor density, just have more people show up. And most important, you can ' t ignore how important **mobile** phones are to global youth. This is a system that could put the next generation at the front lines of gathering environmental data. And as we work to figure out ways to mitigate the worst effects of climate disruption, every little bit of information matters. A system like Earth Witness would be a tool for all of us to participate in the improvement of our knowledge and, ultimately, the improvement of the planet itself. Now, as I suggested at the outset, there are thousands upon thousands of good ideas out there, so why have I spent the bulk of my time telling you about something that doesn ' t exist? Because this is what tomorrow could look like : bottom-up, technology-enabled global collaboration to handle the biggest crisis our civilization has ever faced. We can save the planet, but we can ' t do it alone — we need each other. Nobody ' s going to fix the world for us, but working together, making use of technological innovations and human communities alike, we might just be able to fix it ourselves. We have at our fingertips a cornucopia of compelling models, powerful tools, and innovative ideas that can make a meaningful difference in our planet ' s future. We don ' t need to wait for a magic bullet to save us all ; we already have an arsenal of solutions just waiting to be used. There ' s a staggering array of wonders out there, across diverse disciplines, all telling us the same thing : success can be ours if we ' re willing to try. And as we say at Worldchanging, another world isn ' t just possible ; another world is here. We just need to open our eyes. Thank you very much.

Text Sample 246: POS Dim 2 – 2006 – Score 8.00 – 2006/t000491.txt
Hello voice mail, my old friend. I ' ve called for tech support again. I ignored my boss ' s **warning**. I called on a Monday morning. Now it ' s evening, and my dinner first grew cold, and then grew mold. I ' m still on hold. I ' m listening to the sounds of silence. I don ' t think you understand. I think your phone lines are unmanned. I punched every touch tone I was told, but I ' ve still spent 18 hours on hold. It ' s not enough your software crashed my Mac, and it constantly hangs and bombs — it erased my ROMs! Now the Mac makes the sounds of silence. In my dreams I fantasize of wreaking vengeance on you guys. Say your motorcycle crashes. Blood comes gushing from your gashes. With your fading strength, you call 9-1-1 and you pray for a trained MD. But you get me. And you listen to the sounds of silence. Thank you. Good evening and welcome to :"Spot the TED Presenter Who Used to Be a Broadway Accompanist."When I was offered the Times column six years ago, the deal was like this : you ' ll be sent the coolest, hottest, slickest new gadgets. Every week, it ' ll arrive at your door. You get to try them out, play with them, evaluate them until the novelty wears out, before you have to send them back, and you ' ll get paid for it. You can think about it, if you want. So, I ' ve always been a technology nut, and I absolutely love it. The job, though, came with one small downside, and that is, they intended to publish my email address at the end of every column. And what I ' ve noticed is — first of all, you get an incredible amount of email. If you ever are feeling lonely, get a New York Times column, because you will get hundreds and hundreds and hundreds of emails. And the email I ' m getting a lot today is about frustration. People are feeling like things — Ok, I just had an alarm come up on my screen. Lucky you can ' t see it. People are feeling overwhelmed. They ' re feeling like it ' s too much technology, too fast. It may be good technology, but I feel like there ' s not enough of a support structure. There ' s not enough help. There ' s not enough thought put into the design of it to make it easy and enjoyable to use. One time I wrote a column about my efforts to reach Dell Technical Support, and within 12 hours, there were 700 messages from readers on the feedback boards on the Times website, from users saying,""Me too, and here ' s my tale of woe."I call it"software rage."And man, let me tell you, whoever figures out how to make money off of this frustration will — Oh, how did that get up there? Just kidding. Ok, so why is the problem accelerating? And part of the problem is, ironically, because the industry has put so much thought into making things easier to use. I ' ll show you what I mean. This is what the computer interface used to look like, DOS. Over the years, it ' s gotten easier to use. This is the original Mac operating system. Reagan was President. Madonna was still a brunette. And the entire operating system — this is the good part — the entire operating system fit in 211 k. You couldn ' t put the Mac OS X logo in 211 k! So the irony is, that as these things became easier to use, a less technical, broader audience was coming into contact with this equipment for the first time. I once had the distinct privilege of sitting in on the Apple call center for a day. The guy had a duplicate headset for me

to listen to. And the calls that — you know how they say,”Your call may be recorded for quality assurance?”Uh-uh. Your call may be recorded so that they can collect the funniest dumb user stories and pass them around on a CD. Which they do. And I have a copy. It ’ s in your gift bag. No, no. With your voices on it! So, some of the stories are just so classic, and yet so understandable. A woman called Apple to **complain** that her mouse was squeaking. Making a squeaking noise. And the technician said,”Well, **ma** ’ am, what do you mean your mouse is squeaking?”She says,”All I can tell you is that it squeaks louder, the faster I move it across the screen.”And the technician ’ s like,”Ma ’ am, you ’ ve got the mouse up against the screen?”She goes,”Well, the message said, ’ Click here to continue. ’”Well, if you like that one — how much time have we got? Another one, a guy called — this is absolutely true — his computer had crashed, and he told the technician he couldn ’ t restart it, no matter how many times he typed”11.”And the technician said,”What? Why are you typing 11?”He said,”The message says, ’ Error Type 11. ’”So, we must admit that some of the blame falls squarely at the feet of the users. But why is the technical overload crisis, the complexity crisis, accelerating now? In the hardware world, it ’ s because we the consumers want everything to be smaller, smaller, smaller. So the gadgets are getting tinier and tinier, but our fingers are essentially staying the same size. So it gets to be more and more of a challenge. Software is subject to another primal force : the mandate to release more and more versions. When you buy a piece of software, it ’ s not like buying a vase or a candy bar, where you own it. It ’ s more like joining a club, where you pay dues every year, and every year, they say,”We ’ ve added more features, and we ’ ll sell it to you for $99.”I know one guy who ’ s spent $4, 000 just on Photoshop over the years. And software companies make 35 percent of their revenue from just these software upgrades. I call it the Software Upgrade Paradox — which is that if you improve a piece of software enough times, you eventually ruin it. I mean, Microsoft Word was last just a word processor in, you know, the Eisenhower administration. But what ’ s the alternative? Microsoft actually did this experiment. They said,”Well, wait a minute. Everyone **complains** that we ’ re adding so many features. Let ’ s create a word processor that ’ s just a word processor : Simple, pure ; does not do web pages, is not a database.”And it came out, and it was called Microsoft Write. And none of you are nodding in acknowledgment, because it died. It tanked. No one ever bought it. I call this the Sport Utility Principle. People like to surround themselves with unnecessary power, right? They don ’ t need the database and the website, but they ’ re like,”Well, I ’ ll upgrade, because, I might, you know, I might need that someday.”So the problem is : as you add more features, where are they going to go? Where are you going to stick them? You only have so many design tools. You can do buttons, you can do sliders, pop-up menus, sub-menus. But if you ’ re not careful about how you choose, you wind up with this. This is an un-retouched — this is not a joke — un-retouched photo of Microsoft Word, the copy that you have, with all the toolbars open. You ’ ve obviously never opened all the toolbars, but all you have to type in is this little, teeny window down here. And we ’ ve arrived at the age of interface matrices, where there are so many features and options, you have to do two dimensions, you know : a vertical and a horizontal. You guys all **complain** about how Microsoft Word is always bulleting your lists and underlining your links automatically. The off switch is in there somewhere. I ’ m telling you — it ’ s there. Part of the art of designing a simple, good interface, is knowing when to use which one of these features. So, here is the log-off dialogue box for Windows 2000. There are only four choices, so why are they in a pop-up menu? It ’ s not like the rest of the screen is so full of other components that you need to collapse the choices. They could have put them all out in view. Here ' s Apple ' s take on the exact same dialogue box. Thank you — yes, I designed the dialogue box. No, no. Already, we can see that Apple and Microsoft have a severely divergent approach to software design. Microsoft ' s approach to simplicity tends to be : let ' s break it down ; let ' s just make it more steps. There are these"wizards"everywhere. And you know, there ' s a new version of Windows coming out this fall. If they continue at this pace, there ' s absolutely no telling where they might wind up. [Welcome to the Type a Word Wizard]"Welcome to the Type a Word Wizard."Ok, I ' ll bite. Let ' s click"Next"to continue. From the drop-down menu, choose the first letter you want to type. Ok. So there is a limit that we don ' t want to cross. So what is the answer? How do you pack in all these features in a simple, intelligent way? I believe in consistency, when possible, real-world equivalents, trash can folder, when possible, label things, mostly. But I beg of the designers here to break all those rules if they violate the biggest rule of all, which is intelligence. Now what do I mean by that? I ' m going to give you some examples where intelligence makes something not consistent, but it ' s better. If you are buying something on the web, you ' re supposed to put in your address, and you ' re supposed to choose what country you ' re from, ok? There are 200 countries in the world. We like to think of the Internet as a global village. I ' m sorry ; it ' s not one yet. It ' s mainly like, the United States, Europe, and Japan. So why is"United States"in the"U"s? You have to **scroll**, like, seven screensful to get to it. Now, it would be inconsistent to put"United States"first, but it would be intelligent. This one ' s been touched on before, but why in God ' s name do you shut down a Windows PC by clicking a button called"Start?"Here ' s another pet one of mine : you have a printer. Most of the time, you want to print one copy of your document, in page order, on that printer. So why in God ' s name do you see this every time you print? It ' s like a 747 shuttle cockpit. And one of the buttons at the bottom, you ' ll notice, is not"Print."Now, I ' m not saying that Apple is the only company who has embraced the cult of simplicity. Palm is also, especially in the old days, wonderful about this. I actually got to speak to Palm when they were flying high in the ' 90s, and after the talk, I met one of the employees. He says,"Nice talk."And I said,"Thank you. What do you do here?"He said,"I ' m a tap counter."I ' m like,"You ' re a what?"He goes,"Well Jeff Hawkins, the CEO, says, ' If any task on the Palm Pilot takes more than three taps of the stylus, it ' s too long, and it has to be redesigned. ' So I ' m the tap counter."So, I ' m going to show you an example of a company that does not have a tap counter. This is Microsoft Word. Ok, when you want to create a new blank document in Word — it could happen. You go up to the"File"menu and you choose"New."Now, what happens when you choose"New?"Do you get a new blank document? You do not. On the opposite side of the monitor, a task bar appears, and somewhere in those links — by the way, not at the top — somewhere in those links is a button that makes you a new document. Ok, so that is a company not counting taps. You know, I don ' t want to just stand here and make fun of Microsoft... Yes, I do. The Bill Gates song! I ' ve been a geek forever and I wrote the very first DOS. I put my software and IBM together ; I got profit and they got the loss. I write the code that makes the whole world run. I ' m getting royalties from everyone. Sometimes it ' s garbage, but the press is snowed. You buy the box ; I ' ll sell the code. Every software company is doing Microsoft ' s R&D. You can ' t keep a good idea down these days. Even Windows is a hack. We ' re kind of based loosely on the Mac. So it ' s big, so it ' s slow. You ' ve got nowhere to go. I ' m not doing this for praise. I write the code that fits the world today. Big mediocrity in every way. We ' ve entered planet domination mode. You ' ll have no choice ; you ' ll buy my code. I am Bill Gates and I write the code.

But actually, I believe there are really two Microsofts. There ' s the old one, responsible for Windows and Office. They ' re dying to throw the whole thing out and start fresh, but they can ' t. They ' re locked in, because so many add-ons and other company stuff locks into the old 1982 chassis. But there ' s also a new Microsoft, that ' s really doing good, simple interface designs. I liked the Media Center PC. I liked the Microsoft SPOT Watch. The Wireless Watch flopped miserably in the market, but it wasn ' t because it wasn ' t simply and beautifully designed. But let ' s put it this way : would you pay $10 a month to have a watch that has to be recharged every night like your cell phone, and stops working when you leave your area code? So, the signs might indicate that the complexity crunch is only going to get worse. So is there any hope? The screens are getting smaller, people are illuminating, putting manuals in the boxes, things are coming out at a faster pace. It ' s funny — when Steve Jobs came back to Apple in 1997, after 12 years away, it was the MacWorld Expo — he came to the stage in that black turtleneck and jeans, and he sort of did this. The crowd went wild, but I had just seen — I ' m like, where have I seen this before? I had just seen the movie"Evita"— with Madonna, and I ' m like, you know what? I ' ve got to do one about Steve Jobs. It won ' t be easy. You ' ll think I ' m strange. When I try to explain why I ' m back, after telling the press Apple ' s future is black. You won ' t believe me. All that you see is a kid in his teens who started out in a garage with only a buddy named Woz. You try rhyming with garage! Don ' t cry for me, Cupertino. The truth is, I never left you. I know the ropes now, know what the tricks are. I made a fortune over at Pixar. Don ' t cry for me, Cupertino. I ' ve still got the drive and vision. I still wear sandals in any weather. It ' s just that these days, they ' re Gucci leather. Thank you. So Steve Jobs had always believed in simplicity and elegance and beauty. And the truth is, for years I was a little depressed, because Americans obviously did not value it, because the Mac had three percent market share, Windows had 95 percent market share — people did not think it was worth putting a price on it. So I was a little depressed. And then I heard Al Gore ' s talk, and I realized I didn ' t know the meaning of depressed. But it turns out I was wrong, right? Because the iPod came out, and it violated every bit of common wisdom. Other products cost less ; other products had more features, they had voice recorders and FM transmitters. The other products were backed by Microsoft, with an open standard, not Apple ' s propriety standard. But the iPod won — this is the one they wanted. The lesson was : simplicity sells. And there are signs that the industry is getting the message. This is a little company that ' s done very well with simplicity and elegance. The Sonos thing — it ' s catching on. I ' ve got just a couple examples. Physically, a really cool, elegant thinking coming along lately. When you have a digital camera, how do you get the pictures back to your computer? Well, you either haul around a USB cable, or you buy a card reader and haul that around. Either one, you ' re going to lose. What I do is, I take out the memory card, and I fold it in half, revealing USB contacts. I just stick it in the computer, offload the pictures, put it right back in the camera. I never have to lose anything. Here ' s another example. Chris, you ' re the source of all power. Will you be my power plug? Chris Anderson : Oh yeah. DP : Hold that and don ' t let go. You might ' ve seen this, this is Apple ' s new laptop. This the power cord. It hooks on like this. And I ' m sure every one of you has done this at some point in your lives, or one of your children. You walk along — and I ' m about to pull this onto the floor. I don ' t care. It ' s a loaner. Here we go. Whoa! It ' s magnetic — it doesn ' t pull the laptop onto the floor. In my very last example — I do a lot of my work using speech recognition software. And I ' ll just — you have to be kind of quiet because the software is nervous. Speech recognition software is really great for doing emails very quickly ; period. Like, I get hundreds of them a day ; period. And it ' s not just what I dictate that it writes down ; period. I also use this feature called voice macros ; period. Correct"dissuade."Not"just."Ok, this is not an ideal situation, because it ' s getting the echo from the hall and stuff. The point is, I can respond to people very quickly by saying a short word, and having it write out a much longer thing. So if somebody sends me a fan letter, I ' ll say,"Thanks for that."[Thank you so much for taking the time to write...] And conversely, if somebody sends me hate mail — which happens **daily** — I say,"Piss off."[I admire your frankness...] So that ' s my dirty little secret. Don ' t tell anyone. So the point is — this is a really interesting story. This is version eight of this software, and do you know what they put in version eight? No new features. It ' s never happened before in software! The company put no new features. They just said,"We ' ll make this software work right."Right? Because for years, people had bought this software, tried it out — 95 percent accuracy was all they got, which means one in 20 words is wrong — and they ' d put it in their drawer. And the company got sick of that, so they said,"This version, we ' re not going to do anything, but make sure it ' s darned accurate."And so that ' s what they did. This cult of doing things right is starting to spread. So, my final advice for those of you who are consumers of this technology : remember, if it doesn ' t work, it ' s not necessarily you, ok? It could be the design of the thing you ' re using. Be aware in life of good design and bad design. And if you ' re among the people who create this stuff : Easy is hard. Pre-sweat the details for your audience. Count the taps. Remember, the hard part is not deciding what features to add, it ' s deciding what to leave out. And best of all, your motivation is : simplicity sells. CA : Bravo. DP : Thank you very much. CA : Hear, hear!

Text Sample 247: POS Dim 2 – 2006 – Score 7.00 – 2006/t000442.txt

I ' ve got apparently 18 minutes to convince you that history has a direction, an arrow ; that in some fundamental sense, it ' s good ; that the arrow points to something positive. Now, when the TED people first approached me about giving this upbeat talk â□□ â□□ that was before the cartoon of Muhammad had triggered global rioting. It was before the avian **flu** had reached Europe. It was before Hamas had won the Palestinian election, eliciting various counter-measures by Israel. And to be honest, if I had known when I was asked to give this upbeat talk that even as I was giving the upbeat talk, the apocalypse would be unfolding â□□ â□□ I might have said,"Is it okay if I talk about something else?"But I didn ' t, OK. So we ' re here. I ' ll do what I can. I ' ll do what I can. I ' ve got to warn you : the sense in which my worldview is upbeat has always been kind of subtle, sometimes even elusive. The sense in which I can be uplifting and inspiring â□□ I mean, there ' s always been a kind of a certain grim dimension to the way I try to uplift, so if grim inspiration â□□ â□□ if grim inspiration is not a contradiction in terms, that is, I ' m afraid, the most you can hope for. OK, today â□□ that ' s if I succeed. I ' ll see what I can do. OK? Now, in one sense, the claim that history has a direction is not that controversial. If you ' re just talking about social structure, OK, clearly that ' s gotten more complex a little over the last 10, 000 years â□□ has reached higher and higher levels. And in fact, that ' s actually sustaining a long- standing trend that predates human beings, OK, that biological evolution was doing for us. Because what happened in the beginning, this stuff encases itself in a cell, then cells start hanging out together in societies. Eventually they get so close, they form multicellular organisms, then you get complex multicellular organisms ; they form societies.

But then at some point, one of these multicellular organisms does something completely amazing with this stuff, which is it launches a whole second kind of evolution : cultural evolution. And amazingly, that evolution sustains the trajectory that biological evolution had established toward greater complexity. By cultural evolution we mean the evolution of ideas. A lot of you have heard the term"memes."The evolution of technology, I pay a lot of attention to, so, you know, one of the first things you got was a little hand axe. Generations go by, somebody says, hey, why don ' t we put it on a stick? Just absolutely delights the little ones. Next best thing to a video game. This may not seem to impress, but technological evolution is progressive, so another 10, 20, 000 years, and armaments technology takes you here. Impressive. And the rate of technological evolution speeds up, so a mere quarter of a century after this, you get this, OK. And this. I ' m sorry â it was a cheap laugh, but I wanted to find a way to transition back to this idea of the unfolding apocalypse, and I thought that might do it. So, what threatens to happen with this unfolding apocalypse is the collapse of global social organization. Now, first let me remind you how much work it took to get us where we are, to be on the brink of true global social organization. Originally, you had the most complex societies, the hunter-gatherer village. Stonehenge is the remnant of a chiefdom, which is what you get with the invention of agriculture : multi-village polity with centralized rule. With the invention of writing, you start getting cities. This is blurry. I kind of like that because it makes it look like a one-celled organism and reminds you how many levels organic organization has already moved through to get to this point. And then you get to, you know, you get empires. I want to **stress**, you know, social organization can transcend political bounds. This is the Silk Road connecting the Chinese Empire and the Roman Empire. So you had social complexity spanning the whole continent, even if no polity did similarly. Today, you ' ve got nation states. Point is : there ' s obviously collaboration and organization going on beyond national bounds. This is actually just a picture of the earth at night, and I ' m just putting it up because I think it ' s pretty. Does kind of convey the sense that this is an integrated system. Now, I explained this growth of complexity by reference to something called"non-zero sumness."Assuming that a few of you did not do the assigned reading, very quickly, the key idea is the distinction between zero-sum games, in which correlations are inverse : always a winner and a loser. Non-zero-sum games in which correlations can be positive, OK. So like in tennis, usually it ' s win- lose ; it always adds up to zero-zero-sum. But if you ' re playing doubles, the person on your side of the net, they ' re in the same boat as you, so you ' re playing a non-zero-sum game with them. It ' s either for the better or for the worse, OK. A lot of forms of non-zero-sum behavior in the realm of economics and so on in everyday life often leads to cooperation. The argument I make is basically that, well, non-zero-sum games have always been part of life. You have them in hunter-gatherer societies, but then through technological evolution, new forms of technology arise that facilitate or encourage the playing of non-zero-sum games, involving more people over larger territory. Social structure adapts to accommodate this possibility and to harness this productive potential, so you get cities, you know, and you get all the non-zero-sum games you don ' t think about that are being played across the world. Like, have you ever thought when you buy a car, how many people on how many different continents contributed to the manufacture of that car? Those are people in effect you ' re playing a non-zero-sum game with. I mean, there are certainly plenty of them around. Now, this sounds like an intrinsically upbeat worldview in a way, because when you think of non-zero, you think win-win, you know, that ' s good. Well, there are a few reasons that actually it ' s not intrinsically upbeat. First of all, it can accommodate ; it doesn ’ t deny the existence of inequality exploitation war. But there ’ s a more fundamental reason that it ’ s not intrinsically upbeat, because a non-zero-sum game, all it tells you for sure is that the fortunes will be correlated for better or worse. It doesn ’ t necessarily predict a win- win outcome. So, in a way, the question is : on what grounds am I upbeat at all about history? And the answer is, first of all, on balance I would say people have played their games to more win-win outcomes than lose-lose outcomes. On balance, I think history is a net positive in the non-zero-sum game department. And a testament to this is the thing that most amazes me, most impresses me, and most uplifts me, which is that there is a moral dimension to history ; there is a moral arrow. We have seen moral progress over time. 2, 500 years ago, members of one Greek city-state considered members of another Greek city-state subhuman and treated them that way. And then this moral revolution arrived, and they decided that actually, no, Greeks are human beings. It ’ s just the Persians who aren ’ t fully human and don ’ t deserve to be treated very nicely. But this was progress â you know, give them credit. And now today, we ’ ve seen more progress. I think â I hope â most people here would say that all people everywhere are human beings, deserve to be treated decently, unless they do something horrendous, regardless of race or religion. And you have to read your ancient history to realize what a revolution that has been, OK. This was not a prevalent view, few thousand years ago, and I attribute it to this non- zero-sum dynamic. I think that ’ s the reason there is as much tolerance toward nationalities, ethnicities, religions as there is today. If you asked me, you know, why am I not in favor of bombing Japan, well, I ’ m only half-joking when I say they built my car. We have this non-zero-sum relationship, and I think that does lead to a kind of a tolerance to the extent that you realize that someone else ’ s welfare is positively correlated with yours â you ’ re more likely to cut them a break. I kind of think this is a kind of a business-class morality. Unfortunately, I don ’ t fly trans-Atlantic business class often enough to know, or any other kind of business class really, but I assume that in business class, you don ’ t hear many **expressions** of, you know, bigotry about racial groups or ethnic groups, because the people who are flying trans-Atlantic business class are doing business with all these people ; they ’ re making money off all these people. And I really do think that, in that sense at least, capitalism has been a constructive force, and more fundamentally, it ’ s a non-zero-sumness that has been a constructive force in expanding people ’ s realm of moral awareness. I think the non-zero-sum dynamic, which is not only economic by any means â it ’ s not always commerce â but it has driven us to the verge of a moral truth, which is the fundamental equality of everyone. It has done that. As it has moved global, moved us toward a global level of social organization, it has driven us toward moral truth. I think that ’ s wonderful. Now, back to the unfolding apocalypse. And you may wonder, OK, that ’ s all fine, sounds great â moral direction in history â but what about this so-called clash of civilizations? Well, first of all, I would emphasize that it fits into the non-zero-sum framework, OK. If you look at the relationship between the so- called Muslim world and Western world â two terms I don ’ t like, but can ’ t really avoid ; in such a short span of time, they ’ re efficient if nothing else â it is non-zero-sum. And by that I mean, if people in the Muslim world get more hateful, more resentful, less happy with their place in the world, it ’ ll be bad for the West. If they get more happy, it ’ ll be good for the West. So that is a non-zero-sum dynamic. And I would say the non-zero-sum dynamic is only going to grow more intense over time because of technological trends, but more intense in a kind of negative way. It ’ s the downside correlation of their fortunes that will become more and more possible. And one reason is because of something I call the”growing lethality of hatred."More and more, it ’ s possible for grassroots hatred abroad to manifest itself in the form of organized violence on American soil. And that ’ s pretty new, and I think it ’ s probably going to get a lot worse â this capacity â because of trends in information technology, in technologies that can be used for purposes of munitions like biotechnology and nanotechnology. We may be hearing more about that today. And there ’ s something I worry about especially, which is that this dynamic will lead to a kind of a feedback cycle that puts us on a slippery slope. What I have in mind is : terrorism happens here ; we overreact to it. That, you know, we ’ re not sufficiently surgical in our retaliation leads to more hatred abroad, more terrorism. We overreact because being human, we feel like retaliating, and it gets worse and worse and worse. You could call this the positive feedback of negative vibes, but I think in something so spooky, we really shouldn ’ t have the word positive there at all, even in a technical sense. So let ’ s call it the death spiral of negativity. I assure you if it happens, at the end, both the West and the Muslim world will have suffered. So, what do we do? Well, first of all, we can do a lot more with arms control, the international regulation of dangerous technologies. I have a whole global governance sermon that I will spare you right now, because I don ’ t think that ’ s going to be enough anyway, although it ’ s essential. I think we ’ re going to have to have a major round of moral progress in the world. I think you ’ re just going to have to see less hatred among groups, less bigotry, and, you know, racial groups, religious groups, whatever. I ’ ve got to admit I feel silly saying that. It sounds so kind of Pollyannaish. I feel like Rodney King, you know, saying, why can ’ t we all just get along? But hey, I don ’ t really see any alternative, given the way I read the situation. There ’ s going to have to be moral progress. There ’ s going to have to be a lessening of the amount of hatred in the world, given how dangerous it ’ s becoming. In my defense, I ’ d say, as naive as this may sound, it ’ s ultimately grounded in cynicism. That is to say â â thank you, thank you. That is to say, remember : my whole view of morality is that it boils down to self-interest. It ’ s when people ’ s fortunes are correlated. It ’ s when your welfare conduces to mine, that I decide, oh yeah, I ’ m all in favor of your welfare. That ’ s what ’ s responsible for this growth of this moral progress so far, and I ’ m saying we once again have a correlation of fortunes, and if people respond to it intelligently, we will see the development of tolerance and so on â the **norms** that we need, you know. We will see the further evolution of this kind of business-class morality. So, these two things, you know, if they get people ’ s attention and drive home the positive correlation and people do what ’ s in their self-interests, which is further the moral evolution, then they could actually have a constructive effect. And that ’ s why I lump growing lethality of hatred and death spiral of negativity under the general rubric, reasons to be cheerful. Doing the best I can, OK. I never called myself Mr. Uplift. I ’ m just doing what I can here. Now, launching a moral revolution has got to be hard, right? I mean, what do you do? And I think the answer is a lot of different people are going to have to do a lot of different things. We all start where we are. Speaking as an American who has children whose security 10, 20, 30 years down the road I worry about â what I personally want to start out doing is figuring out why so many people around the world hate us, OK. I think that ’ s a **worthy** research project myself. I also like it because it ’ s an intrinsically kind of morally redeeming exercise. Because to understand why somebody in a very different culture does something â somebody you ’ re kind of viewing as alien, who ’ s doing things you consider strange in a culture you consider strange â to really understand why they do the things they do is a morally redeeming accomplishment, because you ’ ve got to relate their experience to yours. To really understand it, you ’ ve got to say,”Oh, I get it. So when they feel resentful, it ’ s kind of like the way I feel resentful when this happens, and for somewhat the same reasons.”That ’ s true understanding. And I think that is an expansion of your moral compass when you manage to do that. It ’ s especially hard to do when people hate you, OK, because you don ’ t really, in a sense, want to completely understand why people hate you. I mean, you want to hear the reason, but you don ’ t want to be able to relate to it. You don ’ t want it to make sense, right? You don ’ t want to say,”Well, yeah, I can kind of understand how a human being in those circumstances would hate the country I live in.”That ’ s not a **pleasant** thing, but I think it ’ s something that we ’ re going to have to get used to and work on. Now, I want to **stress** that to understand, you know â there are people who don ’ t like this whole business of understanding the grassroots, the root causes of things ; they don ’ t want to know why people hate us. I want to understand it. The reason you ’ re trying to understand why they hate us, is to get them to quit hating us. The idea when you go through this moral exercise of really coming to appreciate their humanity and better understand them, is part of an effort to get them to appreciate your humanity in the long run. I think it ’ s the first step toward that. That ’ s the long-term goal. There are people who worry about this, and in fact, I, myself, apparently, was denounced on national TV a couple of nights ago because of an op-ed I ’ d written. It was kind of along these lines, and the allegation was that I have, quote,”affection for terrorists.”Now, the good news is that the person who said it was Ann Coulter. I mean, if you ’ ve got to have an enemy, do make it Ann Coulter. But it ’ s not a crazy concern, OK, because understanding behavior can lead to a kind of **empathy**, and it can make it a little harder to deliver tough love, and so on. But I think we ’ re a lot closer to erring on the side of not comprehending the situation clearly enough, than in comprehending it so clearly that we just can ’ t, you know, get the army out to kill terrorists. So I ’ m not really worried about it. So â â I mean, we ’ re going to have to work on a lot of fronts, but if we succeed â if we succeed â then once again, non-zero-sumness and the recognition of non-zero-sum dynamics will have forced us to a higher moral level. And a kind of saving higher moral level, something that kind of literally saves the world. If you look at the word”salvation”in the Bible â the Christian usage that we ’ re familiar with â saving souls, that people go to heaven â that ’ s actually a latecomer. The original meaning of the word”salvation”in the Bible is about saving the social system.”Yahweh is our Savior”means”He has saved the nation of Israel,”which at the time, was a pretty high-level social organization. Now, social organization has reached the global level, and I guess, if there ’ s good news I can say I ’ m bringing you, it ’ s just that all the salvation of the world requires is the intelligent pursuit of self-interests in a disciplined and careful way. It ’ s going to be hard. I say we give it a shot anyway because we ’ ve just come too far to screw it up now. Thanks.

Text Sample 248: POS Dim 2 – 2007 – Score 9.00 – 2007/t000238.txt

Well, good morning. You know, the computer and television both recently turned 60, and today I ’ d like to talk about their relationship. Despite their middle age, if you ’ ve been following the themes of this conference or the entertainment industry, it ’ s pretty clear that one has been picking on the other. So it ’ s about time that we talked about how the computer ambushed television, or why the invention of the atomic bomb unleashed forces that lead to the writers ’ strike. And it ’ s not just what these are doing to each other, but it ’ s what the audience thinks that really frames this matter. To get a sense of this, and it ' s been a theme we ' ve talked about all week, I recently talked to a bunch of tweeners. I wrote on cards :"television,""radio,""MySpace,""Internet,""PC."And I said, just arrange these, from what ' s important to you and what ' s not, and then tell me why. Let ' s listen to what happens when they get to the portion of the discussion on television. Girl 1 : Well, I think it ' s important but, like, not necessary because you can do a lot of other stuff with your free time than watch programs. Peter Hirshberg : Which is more fun, Internet or TV? Girls : Internet. Girl 2 : I think we — the reasons, one of the reasons we put computer before TV is because nowadays, like, we have TV shows on the computer. Girl 2 : And then you can download onto your iPod. PH : Would you like to be the president of a TV network? Girl 4 : I wouldn ' t like it. Girl 2 : That would be so **stressful**. Girl 5 : No. PH : How come? Girl 5 : Because they ' re going to lose all their money eventually. Girl 3 : Like the stock market, it goes up and down and stuff. I think right now the computers will be at the top and everything will be kind of going down and stuff. PH : There ' s been an uneasy relationship between the TV business and the tech business, really ever since they both turned about 30. We go through periods of enthrallment, followed by reactions in boardrooms, in the finance community best characterized as, what ' s the finance term? Ick pooey. Let me give you an example of this. The year is 1976, and Warner buys Atari because video games are on the rise. The next year they march forward and they introduce Qube, the first interactive cable TV system, and the New York Times heralds this as telecommunications moving to the home, convergence, great things are happening. Everybody in the East Coast gets in the pictures — Citicorp, Penney, RCA — all getting into this big vision. By the way, this is about when I enter the picture. I ' m going to do a summer internship at Time Warner. That summer I ' m all — I ' m at Warner that summer — I ' m all excited to work on convergence, and then the bottom falls out. Doesn ' t work out too well for them, they lose money. And I had a happy brush with convergence until, kind of, Warner basically has to liquidate the whole thing. That ' s when I leave graduate school, and I can ' t work in New York on kind of entertainment and technology because I have to be exiled to California, where the remaining jobs are, almost to the sea, to go to work for Apple Computer. Warner, of course, writes off more than 400 million dollars. Four hundred million dollars, which was real money back in the ' 70s. But they were onto something and they got better at it. By the year 2000, the process was perfected. They merged with AOL, and in just four years, managed to shed about 200 billion dollars of market capitalization, showing that they ' d actually mastered the art of applying Moore ' s law of successive miniaturization to their balance sheet. Now, I think that one reason that the media and the entertainment communities, or the media community, is driven so crazy by the tech community is that tech folks talk differently. You know, for 50 years, we ' ve talked about changing the world, about total transformation. For 50 years, it ' s been about hopes and fears and promises of a better world. And I got to thinking, you know, who else talks that way? And the answer is pretty clearly — it ' s people in religion and in politics. And so I realized that actually the tech world is best understood, not as a business cycle, but as a messianic movement. We promise something great, we evangelize it, we ' re going to change the world. It doesn ' t work out too well, and so we actually go back to the well and start all over again, as the people in New York and L. A. look on in absolute, morbid astonishment. But it ' s this irrational view of things that drives us on to the next thing. So, what I ' d like to ask is, if the computer is becoming a principal tool of media and entertainment, how did we get here? I mean, how did a machine that was built for accounting and artillery morph into media? Of course, the first computer was built just after World War II to solve military problems, but things got really interesting just a couple of years later — 1949 with Whirlwind, built at MIT ' s Lincoln Lab. Jay Forrester was building this for the Navy, but you can ' t help but see that the creator of this machine had in mind a machine that might actually be a potential media star. So take a look at what happens when the foremost journalist of early television meets one of the foremost computer pioneers, and the computer begins to express itself. Journalist : It ' s a Whirlwind electronic computer. With considerable trepidation, we undertake to interview this new machine. Jay Forrester : Hello New York, this is Cambridge. And this is the oscilloscope of the Whirlwind electronic computer. Would you like if I used the machine? Journalist : Yes, of course. But I have an idea, Mr. Forrester. Since this computer was made in conjunction with the Office of Naval Research, why don ' t we switch down to the Pentagon in Washington and let the Navy ' s research chief, Admiral Bolster, give Whirlwind the workout? Calvin Bolster : Well, Ed, this problem concerns the Navy ' s Viking rocket. This rocket goes up 135 miles into the sky. Now, at the standard rate of fuel consumption, I would like to see the computer trace the flight path of this rocket and see how it can determine, at any **instant**, say at the end of 40 seconds, the amount of fuel remaining, and the velocity at that set **instant**. JF : Over on the left-hand side, you will notice fuel consumption decreasing as the rocket takes off. And on the right-hand side, there ' s a scale that shows the rocket ' s velocity. The rocket ' s position is shown by the trajectory that we ' re now looking at. And as it reaches the **peak** of its trajectory, the velocity, you will notice, has dropped off to a minimum. Then, as the rocket dives down, velocity picks up again toward a maximum velocity and the rocket hits the ground. How ' s that? Journalist : What about that, Admiral? CB : Looks very good to me. JF : And before leaving, we would like to show you another kind of mathematical problem that some of the boys have worked out in their spare time, in a less serious vein, for a Sunday afternoon. Journalist : Thank you very much indeed, Mr. Forrester and the MIT lab. PH : You know, so much was worked out : the first real-time interaction, the video display, pointing a gun. It lead to the microcomputer, but unfortunately, it was too pricey for the Navy, and all of this would have been lost if it weren ' t for a happy coincidence. Enter the atomic bomb. We ' re threatened by the greatest weapon ever, and knowing a good thing when it sees it, the Air Force decides it needs the biggest computer ever to protect us. They adapt Whirlwind to a massive air defense system, deploy it all across the frozen north, and spend nearly three times as much on this computer as was spent on the Manhattan Project building the A-Bomb in the first place. Talk about a shot in the arm for the computer industry. And you can imagine that the Air Force became a pretty good salesman. Here ' s their marketing video. Narrator : In a mass raid, high-speed bombers could be in on us before we could determine their tracks. And then it would be too late to act. We cannot afford to take that chance. It is to meet this threat that the Air Force has been developing SAGE, the Semi-Automatic Ground Environment system, to strengthen our air defenses. This new computer, built to become the nerve center of a defense network, is able to perform all the complex mathematical problems involved in countering a mass enemy raid. It is provided with its own powerhouse containing large diesel-driven generators, air-conditioning equipment, and cooling towers required to cool the thousands of vacuum tubes in the computer. PH : You know, that one computer was huge. There ' s an interesting marketing lesson from it, which is basically, when you market a product, you can either say, this is going to be wonderful, it will make you feel better and enliven you. Or there ' s one other marketing proposition : if you don ' t use our product, you ' ll die. This is a really good example of that. This had the first pointing device. It was distributed, so it worked out

— distributed computing and modems — so all these things could talk to each other. About 20 percent of all the nation ’ s programmers were wrapped up in this thing, and it led to an awful lot of what we have today. It also used vacuum tubes. You saw how huge it was, and to give you a sense for this — because we ’ ve talked a lot about Moore ’ s law and making things small at this conference, so let ’ s talk about making things large. If we took Whirlwind and put it in a place that you all know, say, Century City, it would fit beautifully. You ’ d kind of have to take Century City out, but it could fit in there. But like, let ’ s imagine we took the latest Pentium processor, the latest Core 2 Extreme, which is a four-core processor that Intel ’ s working on, it will be our laptop tomorrow. To build that, what we ’ d do with Whirlwind technology is we ’ d have to take up roughly from the 10 to Mulholland, and from the 405 to La Cienega just with those Whirlwinds. And then, the 92 nuclear power plants that it would take to provide the power would fill up the rest of Los Angeles. That ’ s roughly a third more nuclear power than all of France creates. So, the next time they tell you they ’ re on to something, clearly they ’ re not. So — and we haven ’ t even worked out the cooling needs. But it gives you the kind of power that people have, that the audience has, and the reasons these transformations are happening. All of this stuff starts moving into industry. DEC kind of reduces all this and makes the first mini-computer. It shows up at places like MIT, and then a mutation happens. Spacewar! is built, the first computer game, and all of a sudden, interactivity and involvement and passion is worked out. Actually, many MIT students stayed up all night long working on this thing, and many of the principles of gaming today were worked out. DEC knew a good thing about wasting time. It shipped every one of its computers with that game. Meanwhile, as all of this is happening, by the mid- ’ 50s, the business model of traditional broadcasting and cinema has been busted completely. A new technology has confounded radio men and movie moguls and they ’ re quite certain that television is about to do them in. In fact, despair is in the air. And a quote that sounds largely reminiscent from everything I ’ ve been reading all week. RCA had David Sarnoff, who basically commercialized radio, said this,”I don ’ t say that radio networks must die. Every effort has been made and will continue to be made to find a new pattern, new selling arrangements and new types of programs that may arrest the declining revenues. It may yet be possible to eke out a poor existence for radio, but I don ’ t know how.”And of course, as the computer industry develops interactively, producers in the emerging TV business actually hit on the same idea. And they fake it. Jack Berry : Boys and girls, I think you all know how to get your magic windows up on the set, you just get them out. First of all, get your Winky Dink kits out. Put out your Magic Window and your erasing glove, and rub it like this. That ’ s the way we get some of the magic into it, boys and girls. Then take it and put it right up against the screen of your own television set, and rub it out from the center to the corners, like this. Make sure you keep your magic crayons handy, your Winky Dink crayons and your erasing glove, because you ’ ll be using them during the show to draw like that. You all set? OK, let ’ s get right to the first story about Dusty Man. Come on into the secret lab. PH : It was the dawn of interactive TV, and you may have noticed they wanted to sell you the Winky Dink kits. Those are the Winky Dink crayons. I know what you ’ re saying.”Pete, I could use any ordinary open-source crayon, why do I have to buy theirs?”I assure you, that ’ s not the case. Turns out they told us directly that these are the only crayons you should ever use with your Winky Dink Magic Window, other crayons may discolor or hurt the window. This proprietary principle of vendor lock-in would go on to be perfected with great success as one of the enduring principles of windowing systems everywhere. It led to lawsuits — — federal investigations, and lots of repercussions, and that ' s a scandal we won ' t discuss today. But we will discuss this scandal, because this man, Jack Berry, the host of"Winky Dink,"went on to become the host of"Twenty One,"one of the most important quiz shows ever. And it was rigged, and it became unraveled when this man, Charles van Doren, was outed after an unnatural winning streak, ending Berry ' s career. And actually, ending the career of a lot of people at CBS. It turns out there was a lot to learn about how this new medium worked. And 50 years ago, if you ' d been at a meeting like this and were trying to understand the media, there was one prophet and only but one you wanted to hear from, Professor Marshall McLuhan. He actually understood something about a theme that we ' ve been discussing all week. It ' s the role of the audience in an era of pervasive electronic communications. Here he is talking from the 1960s. Marshall McLuhan : If the audience can become involved in the actual process of making the **ad**, then it ' s happy. It ' s like the old quiz shows. They were great TV because it gave the audience a role, something to do. They were horrified when they discovered they ' d really been left out all the time because the shows were rigged. Now, then, this was a horrible misunderstanding of TV on the part of the programmers. PH : You know, McLuhan talked about the global village. If you substitute the word blogosphere, of the Internet today, it is very true that his understanding is probably very enlightening now. Let ' s listen in to him. MM : The global village is a world in which you don ' t necessarily have harmony. You have extreme concern with everybody else ' s business and much involvement in everybody else ' s life. It ' s a sort of Ann Landers ' column writ large. And it doesn ' t necessarily mean harmony and peace and quiet, but it does mean huge involvement in everybody else ' s affairs. And so the global village is as big as a planet, and as small as a village post office. PH : We ' ll talk a little bit more about him later. We ' re now right into the 1960s. It ' s the era of big business and data centers for computing. But all that was about to change. You know, the **expression** of technology reflects the people and the time of the culture it was built in. And when I say that code expresses our hopes and aspirations, it ' s not just a joke about messianism, it ' s actually what we do. But for this part of the story, I ' d actually like to throw it to America ' s leading technology correspondent, John Markoff. John Markoff : Do you want to know what the counterculture in drugs, sex, rock ' n ' roll and the anti-war movement had to do with computing? Everything. It all happened within five miles of where I ' m standing, at Stanford University, between 1960 and 1975. In the midst of revolution in the streets and rock and roll concerts in the parks, a group of **researchers** led by people like John McCarthy, a computer scientist at the Stanford Artificial Intelligence Lab, and Doug Engelbart, a computer scientist at SRI, changed the world. Engelbart came out of a pretty dry engineering culture, but while he was beginning to do his work, all of this stuff was bubbling on the mid-peninsula. There was LSD leaking out of Kesey ' s Veterans ' Hospital experiments and other areas around the campus, and there was music literally in the streets. The Grateful Dead was playing in the pizza parlors. People were leaving to go back to the land. There was the Vietnam War. There was black liberation. There was women ' s liberation. This was a remarkable place, at a remarkable time. And into that ferment came the microprocessor. I think it was that interaction that led to personal computing. They saw these tools that were controlled by the establishment as ones that could actually be liberated and put to use by these communities that they were trying to build. And most importantly, they had this ethos of sharing information. I think these ideas are difficult to understand, because when you ' re trapped in one paradigm, the next paradigm is always like a science fiction universe — it makes no sense. The stories were so compelling that I decided to write a book about them. The title of the book is,”What the Dormouse Said : How the ’ 60s Counterculture Shaped the Personal Computer Industry.”The title was taken from the lyrics to a Jefferson Airplane song. The lyrics go,”Remember what the dormouse said. Feed your head, feed your head, feed your head.”PH : By this time, computing had kind of leapt into media territory, and in short order much of what we ’ re doing today was imagined in Cambridge and Silicon Valley. Here ’ s the Architecture Machine Group, the predecessor of the Media Lab, in 1981. Meanwhile, in California, we were trying to commercialize a lot of this stuff. HyperCard was the first program to introduce the public to hyperlinks, where you could randomly hook to any kind of picture, or piece of text, or data across a file system, and we had no way of explaining it. There was no metaphor. Was it a database? A prototyping tool? A scripted language? Heck, it was everything. So we ended up writing a marketing brochure. We asked a question about how the mind works, and we let our customers play the role of so many blind men filling out the elephant. A few years later, we then hit on the idea of explaining to people the secret of, how do you get the content you want, the way you want it and the easy way? Here ’ s the Apple marketing video. James Burke : You ’ ll be pleased to know, I ’ m sure, that there are several ways to create a HyperCard interactive video. The most involved method is to go ahead and produce your own videodisc as well as build your own HyperCard stacks. By far the simplest method is to buy a pre-made videodisc and HyperCard stacks from a commercial supplier. The method we illustrate in this video uses a pre-made videodisc but creates custom HyperCard stacks. This method allows you to use existing videodisc materials in ways which suit your specific needs and interests. PH : I hope you realize how subversive that is. That ’ s like a Dick Cheney speech. You think he ’ s a nice balding guy, but he ’ s just declared war on the content business. Find the commercial stuff, mash it up, tell the story your way. Now, as long as we confine this to the education market, and a personal matter between the computer and the file system, that ’ s fine, but as you can see, it was about to leap out and upset Jack Valenti and a lot of other people. By the way, speaking of the filing system, it never occurred to us that these hyperlinks could go beyond the local area network. A few years later, Tim Berners-Lee worked that out. It became a killer app of links, and today, of course, we call that the World Wide Web. Now, not only was I instrumental in helping Apple miss the Internet, but a couple of years later, I helped Bill Gates do the same thing. The year is 1993 and he was working on a book and I was working on a video to help him kind of explain where we were all heading and how to popularize all this. We were plenty aware that we were messing with media, and on the surface, it looks like we predicted a lot of the right things, but we also missed an awful lot. Let ’ s take a look. Narrator : The pyramids, the Colosseum, the New York subway system and TV dinners, ancient and modern wonders of the man-made world all. Yet each pales to insignificance with the completion of that magnificent accomplishment of twenty-first-century technology, the Digital Superhighway. Once it was only a dream of technoids and a few long-forgotten politicians. The Digital Highway arrived in America ’ s living rooms late in the twentieth century. Let us recall the pioneers who made this technical marvel possible. The Digital Highway would follow the rutted trail first blazed by Alexander Graham Bell. Though some were incredulous... Man 1 : The phone company! Narrator : Stirred by the prospects of mass communication and making big bucks on advertising, David Sarnoff commercializes radio. Man 2 : Never had scientists been put under such pressure and demand. Narrator : The medium introduced America to new products. Voice 1 : Say, mom, Windows for Radio means more enjoyment and greater ease of use for the whole family. Be sure to enjoy Windows for Radio at home and at work. Narrator : In 1939, the Radio Corporation of America introduced television. Man 2 : Never had scientists been put under such pressure and demand. Narrator : Eventually, the race to the future took on added momentum with the breakup of the telephone company. And further stimulus came with the deregulation of the cable television industry, and the re-regulation of the cable television industry. Ted Turner : We did the work to build this, this cable industry, now the broadcasters want some of our money. I mean, it ' s ridiculous. Narrator : Computers, once the unwieldy tools of accountants and other geeks, escaped the backrooms to enter the media fracas. The world and all its culture reduced to bits, the lingua franca of all media. And the forces of convergence exploded. Finally, four great industrial sectors combined. Telecommunications, entertainment, computing and everything else. Man 3 : We ' ll see channels for the gourmet and we ' ll see channels for the pet lover. Voice 2 : Next on the gourmet pet channel, decorating birthday cakes for your schnauzer. Narrator : All of industry was in play, as investors flocked to place their bets. At stake : the battle for you, the consumer, and the right to spend billions to send a lot of information into the parlors of America. PH : We missed a lot. You know, you missed, we missed the Internet, the long tail, the role of the audience, open systems, social networks. It just goes to show how tough it is to come up with the right uses of media. Thomas Edison had the same problem. He wrote a list of what the phonograph might be good for when he invented it, and kind of only one of his ideas turned out to have been the right early idea. Well, you know where we ' re going on from here. We come into the era of the dotcom, the World Wide Web, and I don ' t need to tell you about that because we all went through that bubble together. But when we emerge from this and what we call Web 2. 0, things actually are quite different. And I think it ' s the reason that TV ' s so challenged. If Internet one was about pages, now it ' s about people. It ' s a customer, it ' s an audience, it ' s a person who ' s participating. It ' s the formidable thing that is changing entertainment now. MM : Because it gave the audience a role, something to do. PH : In my own company, Technorati, we see something like 67, 000 blog posts an hour come in. That ' s about 2, 700 fresh, connective links across about 112 million blogs that are out there. And it ' s no wonder that as we head into the writers ' strike, odd things happen. You know, it reminds me of that old saw in Hollywood, that a producer is anyone who knows a writer. I now think a network boss is anyone who has a cable modem. But it ' s not a joke. This is a real headline."Websites attract striking writers : operators of sites like MyDamnChannel. com could benefit from labor disputes."Meanwhile, you have the TV bloggers going out on strike, in sympathy with the television writers. And then you have TV Guide, a Fox property, which is about to sponsor the online video awards — but cancels it out of sympathy with traditional television, not appearing to gloat. To show you how schizophrenic this all is, here ' s the head of MySpace, or Fox Interactive, a News Corp company, being asked, well, with the writers ' strike, isn ' t this going to hurt News Corp and help you online? Man : But I, yeah, I think there ' s an opportunity. As the strike continues, there ' s an opportunity for more people to experience video on places like MySpace TV. PH : Oh, but then he remembers he works for Rupert Murdoch. Man : Yes, well, first, you know, I ' m part of News Corporation as part of Fox Entertainment Group. Obviously, we hope that the strike is — that the issues are resolved as quickly as possible. PH : One of the great things that ' s going on here is the globalization of content really is happening. Here is a clip from a video, from a piece of animation that was written by a writer in Hollywood, animation worked out in Israel, farmed out to Croatia and India, and it ' s now an international series. Narrator : The following takes place between the minutes of 2:15 p. m. and

2:18 p. m., in the months preceding the presidential primaries. Voice 1 : You ’ ll have to stay here in the safe house until we get word the terrorist threat is over. Voice 2 : You mean we ’ ll have to live here, together? Voices 2, 3 and 4 : With her? Voice 2 : Well, there goes the neighborhood. PH : The company that created this, Aniboom, is an interesting example of where this is headed. Traditional TV animation costs, say, between 80, 000 and 10, 000 dollars a minute. They ’ re producing things for between 1, 500 and 800 dollars a minute. And they ’ re offering their creators 30 percent of the back end, in a much more entrepreneurial manner. So, it ’ s a different model. What the entertainment business is **struggling** with, the world of brands is figuring out. For example, Nike now understands that Nike Plus is not just a device in its shoe, it ’ s a network to hook its customers together. And the head of marketing at Nike says,”People are coming to our site an average of three times a week. We don ’ t have to go to them.”Which means television advertising is down 57 percent for Nike. Or, as Nike ’ s head of marketing says,”We ’ re not in the business of keeping media companies alive. We ’ re in the business of connecting with consumers.”And media companies realize the audience is important also. Here ’ s a man announcing the new Market Watch from Dow Jones, powered 100 percent by the user experience on the home page — user-generated content married up with traditional content. It turns out you have a bigger audience and more interest if you hook up with them. Or, as Geoffrey Moore once told me, it ’ s intellectual curiosity that ’ s the trade that brands need in the age of the blogosphere. And I think this is beginning to happen in the entertainment business. One of my heroes is songwriter, Ally Willis, who just wrote”The Color Purple”and has been an R and — rhythm and blues writer, and this is what she said about where songwriting ’ s going. Ally Willis : Where millions of collaborators wanted the song, because to look at them strictly as spam is missing what this medium is about. PH : So, to wrap up, I ’ d love to throw it back to Marshall McLuhan, who, 40 years ago, was dealing with audiences that were going through just as much change, and I think that, today, traditional Hollywood and the writers are framing this perhaps in the way that it was being framed before. But I don ’ t need to tell you this, let ’ s throw it back to him. Narrator : We are in the middle of a tremendous clash between the old and the new. MM : The medium does things to people and they are always completely unaware of this. They don ’ t really notice the new medium that is wrapping them up. They think of the old medium, because the old medium is always the content of the new medium, as movies tend to be the content of TV, and as books used to be the content, novels used to be the content of movies. And so every time a new medium arrives, the old medium is the content, and it is highly observable, highly noticeable, but the real, real roughing up and massaging is done by the new medium, and it is ignored. PH : I think it ’ s a great time of enthrallment. There ’ s been more raw DNA of communications and media thrown out there. Content is moving from shows to particles that are batted back and forth, and part of social communications, and I think this is going to be a time of great renaissance and opportunity. And whereas television may have gotten beat up, what ’ s getting built is a really exciting new form of communication, and we kind of have the merger of the two industries and a new way of thinking to look at it. Thanks very much.

Text Sample 249: POS Dim 2 – 2007 – Score 9.00 – 2007/t000370.txt

Images like this, from the Auschwitz concentration camp, have been seared into our consciousness during the twentieth century and have given us a new understanding of who we are, where we ' ve come from and the times we live in. During the twentieth century, we witnessed the atrocities of Stalin, Hitler, Mao, Pol Pot, Rwanda and other genocides, and even though the twenty-first century is only seven years old, we have already witnessed an ongoing genocide in Darfur and the **daily** horrors of Iraq. This has led to a common understanding of our situation, namely that modernity has brought us terrible violence, and perhaps that native peoples lived in a state of harmony that we have departed from, to our peril. Here is an example from an op-ed on Thanksgiving, in the Boston Globe a couple of years ago, where the writer wrote,"The Indian life was a difficult one, but there were no employment problems, community harmony was strong, substance abuse unknown, crime nearly non-existent, what warfare there was between tribes was largely ritualistic and seldom resulted in indiscriminate or wholesale slaughter."Now, you ' re all familiar with this treacle. We teach it to our children. We hear it on television and in storybooks. Now, the original title of this session was,"Everything You Know Is Wrong,"and I ' m going to present evidence that this particular part of our common understanding is wrong, that, in fact, our ancestors were far more violent than we are, that violence has been in decline for long stretches of time, and that today we are probably living in the most peaceful time in our species ' existence. Now, in the decade of Darfur and Iraq, a statement like that might seem somewhere between hallucinatory and obscene. But I ' m going to try to convince you that that is the correct picture. The decline of violence is a fractal phenomenon. You can see it over millennia, over centuries, over decades and over years, although there seems to have been a tipping point at the onset of the Age of Reason in the sixteenth century. One sees it all over the world, although not homogeneously. It ' s especially evident in the West, beginning with England and Holland around the time of the Enlightenment. Let me take you on a journey of several powers of 10 — from the millennium scale to the year scale — to try to persuade you of this. Until 10, 000 years ago, all humans lived as hunter-gatherers, without permanent settlements or government. And this is the state that ' s commonly thought to be one of primordial harmony. But the archaeologist Lawrence Keeley, looking at casualty rates among contemporary hunter-gatherers, which is our best source of evidence about this way of life, has shown a rather different conclusion. Here is a graph that he put together showing the percentage of male deaths due to warfare in a number of foraging, or hunting and gathering societies. The red bars correspond to the likelihood that a man will die at the hands of another man, as opposed to passing away of natural causes, in a variety of foraging societies in the New Guinea Highlands and the Amazon Rainforest. And they range from a rate of almost a 60 percent chance that a man will die at the hands of another man to, in the case of the Gebusi, only a 15 percent chance. The tiny, little blue bar in the lower left-hand corner plots the corresponding statistic from United States and Europe in the twentieth century, and includes all the deaths of both World Wars. If the death rate in tribal warfare had prevailed during the 20th century, there would have been two billion deaths rather than 100 million. Also at the millennium scale, we can look at the way of life of early civilizations such as the ones described in the Bible. And in this supposed source of our moral values, one can read descriptions of what was expected in warfare, such as the following from Numbers 31 :"And they warred against the Midianites as the Lord commanded Moses, and they slew all the males. And Moses said unto them, ' Have you saved all the women alive? Now, therefore, kill every male among the little ones and kill every woman that hath known man by lying with him, but all the women children that have not know a man by lying with him keep alive for yourselves. '"In other words, kill the men ; kill the children ; if you see any virgins, then you can keep them alive so that you can rape them. You can find four or five passages in the Bible of this ilk. Also in the Bible, one sees that the death **penalty** was the accepted punishment for crimes such as homosexuality, adultery, blasphemy, idolatry, talking back to your parents — — and picking up sticks on the Sabbath. Well, let ' s click the zoom lens down one order of magnitude, and look at the century scale. Although we don ' t have statistics for warfare throughout the Middle Ages to modern times, we know just from conventional history — the evidence was under our nose all along that there has been a reduction in socially sanctioned forms of violence. For example, any social history will reveal that mutilation and torture were routine forms of criminal punishment. The kind of infraction today that would give you a fine, in those days would result in your tongue being cut out, your ears being cut off, you being blinded, a hand being chopped off and so on. There were numerous ingenious forms of sadistic capital punishment : burning at the stake, disemboweling, breaking on the wheel, being pulled apart by horses and so on. The death **penalty** was a sanction for a long list of non-violent crimes : criticizing the king, stealing a loaf of bread. Slavery, of course, was the preferred labor-saving device, and cruelty was a popular form of entertainment. Perhaps the most vivid example was the **practice** of cat burning, in which a cat was hoisted on a stage and lowered in a sling into a fire, and the spectators shrieked in laughter as the cat, howling in pain, was burned to death. What about one-on-one murder? Well, there, there are good statistics, because many municipalities recorded the cause of death. The criminologist Manuel Eisner scoured all of the historical records across Europe for homicide rates in any village, hamlet, town, county that he could find, and he supplemented them with national data, when nations started keeping statistics. He plotted on a logarithmic scale, going from 100 deaths per 100, 000 people per year, which was approximately the rate of homicide in the Middle Ages. And the figure plummets down to less than one homicide per 100, 000 people per year in seven or eight European countries. Then, there is a slight uptick in the 1960s. The people who said that rock ' n ' roll would lead to the decline of moral values actually had a grain of truth to that. But there was a decline from at least two orders of magnitude in homicide from the Middle Ages to the present, and the elbow occurred in the early sixteenth century. Let ' s click down now to the decade scale. According to non-governmental organizations that keep such statistics, since 1945, in Europe and the Americas, there has been a steep decline in interstate wars, in deadly ethnic riots or pogroms, and in military coups, even in South America. Worldwide, there ' s been a steep decline in deaths in interstate wars. The yellow bars here show the number of deaths per war per year from 1950 to the present. And, as you can see, the death rate goes down from 65, 000 deaths per conflict per year in the 1950s to less than 2, 000 deaths per conflict per year in this decade, as horrific as it is. Even in the year scale, one can see a decline of violence. Since the end of the Cold War, there have been fewer civil wars, fewer genocides — indeed, a 90 percent reduction since post-World War II highs — and even a reversal of the 1960s uptick in homicide and violent crime. This is from the FBI Uniform Crime Statistics. You can see that there is a fairly low rate of violence in the ' 50s and the ' 60s, then it soared upward for several decades, and began a precipitous decline, starting in the 1990s, so that it went back to the level that was last enjoyed in 1960. President Clinton, if you ' re here, thank you. So the question is, why are so many people so wrong about something so important? I think there are a number of reasons. One of them is we have better reporting. The Associated Press is a better chronicler of wars over the surface of the Earth than sixteenth-century monks were. There ' s a cognitive illusion. We cognitive psychologists know that the easier it is to recall specific instances of something, the higher the probability that you assign to it. Things that we read about in the paper with gory footage burn into memory more than reports of a lot more people dying in their beds of old age. There are dynamics in the opinion and advocacy markets : no one ever attracted observers, advocates and donors by saying things just seem to be getting better and better. There ' s guilt about our treatment of native peoples in modern intellectual life, and an unwillingness to acknowledge there could be anything good about Western culture. And of course, our change in standards can outpace the change in behavior. One of the reasons violence went down is that people got sick of the carnage and cruelty in their time. That ' s a process that seems to be continuing, but if it outstrips behavior by the standards of the day, things always look more barbaric than they would have been by historic standards. So today, we get exercised — and rightly so — if a handful of murderers get executed by lethal injection in Texas after a 15-year appeal process. We don ' t consider that a couple of hundred years ago, they may have been burned at the stake for criticizing the king after a **trial** that lasted 10 minutes, and indeed, that that would have been repeated over and over again. Today, we look at capital punishment as evidence of how low our behavior can sink, rather than how high our standards have risen. Well, why has violence declined? No one really knows, but I have read four explanations, all of which, I think, have some grain of plausibility. The first is, maybe Thomas Hobbes got it right. He was the one who said that life in a state of nature was"solitary, poor, nasty, brutish and short."Not because, he argued, humans have some primordial thirst for blood or aggressive instinct or territorial imperative, but because of the logic of anarchy. In a state of anarchy, there ' s a constant temptation to invade your neighbors preemptively, before they invade you. More recently, Thomas Schelling gives the analogy of a homeowner who hears a rustling in the basement. Being a good American, he has a pistol in the nightstand, pulls out his gun, and walks down the stairs. And what does he see but a burglar with a gun in his hand. Now, each one of them is thinking,"I don ' t really want to kill that guy, but he ' s about to kill me. Maybe I had better shoot him, before he shoots me, especially since, even if he doesn ' t want to kill me, he ' s probably worrying right now that I might kill him before he kills me."And so on. Hunter-gatherer peoples explicitly go through this train of thought, and will often raid their neighbors out of fear of being raided first. Now, one way of dealing with this problem is by deterrence. You don ' t strike first, but you have a publicly announced policy that you will retaliate savagely if you are invaded. The only thing is that it ' s liable to having its bluff called, and therefore can only work if it ' s credible. To make it credible, you must avenge all insults and settle all **scores**, which leads to the cycles of bloody vendetta. Life becomes an episode of"The Sopranos."Hobbes ' solution, the"Leviathan,"was that if authority for the legitimate use of violence was vested in a single democratic agency — a leviathan — then such a state can reduce the temptation of attack, because any kind of aggression will be punished, leaving its profitability as zero. That would remove the temptation to invade preemptively, out of fear of them attacking you first. It removes the need for a hair trigger for retaliation to make your deterrent threat credible. And therefore, it would lead to a state of peace. Eisner — the man who plotted the homicide rates that you failed to see in the earlier slide — argued that the timing of the decline of homicide in Europe coincided with the rise of centralized states. So that ' s a bit of a support for the leviathan theory. Also supporting it is the fact that we today see eruptions of violence in zones of anarchy, in failed states, collapsed empires, frontier regions, mafias, street gangs and so on. The second explanation is that in many times and places, there is a widespread sentiment that life is cheap. In earlier times, when suffering and early death were common in one ’ s own life, one has fewer compunctions about inflicting them on others. And as technology and economic efficiency make life longer and more **pleasant**, one puts a higher value on life in general. This was an argument from the political scientist James Payne. A third explanation invokes the concept of a nonzero-sum game, and was worked out in the book”Nonzero”by the journalist Robert Wright. Wright points out that in certain circumstances, cooperation or non-violence can benefit both parties in an interaction, such as gains in trade when two parties trade their surpluses and both come out ahead, or when two parties lay down their arms and split the so-called peace dividend that results in them not having to fight the whole time. Wright argues that technology has increased the number of positive- sum games that humans tend to be embroiled in, by allowing the trade of goods, **services** and ideas over longer distances and among larger groups of people. The result is that other people become more valuable alive than dead, and violence declines for selfish reasons. As Wright put it,”Among the many reasons that I think that we should not bomb the Japanese is that they built my mini-van.”The fourth explanation is captured in the title of a book called”The Expanding Circle,”by the philosopher Peter Singer, who argues that evolution bequeathed humans with a sense of **empathy**, an ability to treat other peoples ’ interests as comparable to one ’ s own. Unfortunately, by default we apply it only to a very narrow circle of friends and family. People outside that circle are treated as sub-human, and can be exploited with impunity. But, over history, the circle has expanded. One can see, in historical record, it expanding from the village, to the clan, to the tribe, to the nation, to other races, to both sexes, and, in Singer ’ s own arguments, something that we should extend to other sentient species. The question is, if this has happened, what has powered that expansion? And there are a number of possibilities, such as increasing circles of reciprocity in the sense that Robert Wright argues for. The logic of the golden rule — the more you think about and interact with other people, the more you realize that it is untenable to privilege your interests over theirs, at least not if you want them to listen to you. You can ’ t say that my interests are **special** compared to yours, anymore than you can say that the particular spot that I ’ m standing on is a unique part of the universe because I happen to be standing on it that very minute. It may also be powered by cosmopolitanism, by histories, and journalism, and memoirs, and realistic fiction, and travel, and literacy, which allows you to project yourself into the lives of other people that formerly you may have treated as sub-human, and also to realize the accidental contingency of your own station in life, the sense that”there but for fortune go I.”Whatever its causes, the decline of violence, I think, has profound implications. It should force us to ask not just, why is there war? But also, why is there peace? Not just, what are we doing wrong? But also, what have we been doing right? Because we have been doing something right, and it sure would be good to find out what it is. Thank you very much.. Chris Anderson : I loved that talk. I think a lot of people here in the room would say that that expansion of — that you were talking about, that Peter Singer talks about, is also driven by, just by technology, by greater visibility of the other, and the sense that the world is therefore getting smaller. I mean, is that also a grain of truth? Steven Pinker : Very much. It would fit both in Wright ’ s theory, that it allows us to enjoy the benefits of cooperation over larger and larger circles. But also, I think it helps us imagine what it ’ s like to be someone else. I think when you read these horrific tortures that were common in the Middle Ages, you think, how could they possibly have done it, how could they have not have empathized with the person that they ’ re disemboweling? But clearly, as far as they ’ re concerned, this is just an alien being that does not have feelings akin to their own. Anything, I think, that makes it easier to imagine trading places with someone else means that it increases your moral consideration to that other person. CA : Well, Steve, I would love every news media owner to hear that talk at some point in the next year. I think it ’ s really important. Thank you so much. SP : My pleasure.

Text Sample 250: POS Dim 2 – 2007 – Score 8.00 – 2007/t000356.txt

I live and work from Tokyo, Japan. And I specialize in human behavioral research, and applying what we learn to think about the future in different ways, and to design for that future. And you know, to be honest, I ’ ve been doing this for seven years, and I haven ’ t got a clue what the future is going to be like. But I ’ ve got a pretty good idea how people will behave when they get there. This is my office. It ’ s out there. It ’ s not in the lab, and it ’ s increasingly in places like India, China, Brazil, Africa. We live on a planet — 6. 3 billion people. About three billion people, by the end of this year, will have cellular connectivity. And it ’ ll take about another two years to connect the next billion after that. And I mention this because, if we want to design for that future, we need to figure out what those people are about. And that ’ s, kind of, where I see what my job is and what our team ’ s job is. Our research often starts with a very simple question. So I ’ ll give you an example. What do you carry? If you think of everything in your life that you own, when you walk out that door, what do you consider to take with you? When you ’ re looking around, what do you consider? Of that stuff, what do you carry? And of that stuff, what do you actually use? So this is interesting to us, because the conscious and subconscious decision process implies that the stuff that you do take with you and end up using has some kind of spiritual, emotional or functional value. And to put it really bluntly, you know, people are willing to pay for stuff that has value, right? So I ’ ve probably done about five years ’ research looking at what people carry. I go in people ’ s bags. I look in people ’ s pockets, purses. I go in their homes. And we do this worldwide, and we follow them around town with video cameras. It ’ s kind of like stalking with permission. And we do all this — and to go back to the original question, what do people carry? And it turns out that people carry a lot of stuff. OK, that ’ s fair enough. But if you ask people what the three most important things that they carry are — across cultures and across gender and across contexts — most people will say keys, money and, if they own one, a **mobile** phone. And I ’ m not saying this is a good thing, but this is a thing, right? I mean, I couldn ’ t take your phones off you if I wanted to. You ’ d probably kick me out, or something. OK, it might seem like an obvious thing for someone who works for a **mobile** phone company to ask. But really, the question is, why? Right? So why are these things so important in our lives? And it turns out, from our research, that it boils down to survival — survival for us and survival for our loved ones. So, keys provide an access to shelter and warmth — transport as well, in the U. S. increasingly. Money is useful for buying food, sustenance, among all its other uses. And a **mobile** phone, it turns out, is a great recovery tool. If you prefer this kind of Maslow ’ s hierarchy of needs, those three objects are very good at supporting the lowest rungs in Maslow ’ s hierarchy of needs. Yes, they do a whole bunch of other stuff, but they ’ re very good at this. And in particular, it ’ s the **mobile** phone ’ s ability to allow people to transcend space and time. And what I mean by that is, you know, you can transcend space by simply making a voice call, right? And you can transcend time by sending a message at your convenience, and someone else can pick it up at their convenience. And this is fairly universally appreciated, it turns out, which is why we have three billion plus people who have been connected. And they value that connectivity. But actually, you can do this kind of stuff with PCs. And you can do them with phone kiosks. And the **mobile** phone, in addition, is both personal — and so it also gives you a degree of **privacy** — and it ’ s convenient. You don ’ t need to ask permission from anyone, you can just go ahead and do it, right? However, for these things to help us survive, it depends on them being carried. But — and it ’ s a pretty big but — we forget. We ’ re human, that ’ s what we do. It ’ s one of our features. I think, quite a nice feature. So we forget, but we ’ re also adaptable, and we adapt to situations around us pretty well. And so we have these strategies to remember, and one of them was mentioned yesterday. And it ’ s, quite simply, the point of reflection. And that ’ s that moment when you ’ re walking out of a space, and you turn around, and quite often you tap your pockets. Even women who keep stuff in their bags tap their pockets. And you turn around, and you look back into the space, and some people talk aloud. And pretty much everyone does it at some point. OK, the next thing is — most of you, if you have a stable home life, and what I mean is that you don ’ t travel all the time, and always in hotels, but most people have what we call a center of gravity. And a center of gravity is where you keep these objects. And these things don ’ t stay in the center of gravity, but over time, they gravitate there. It ’ s where you expect to find stuff. And in fact, when you ’ re turning around, and you ’ re looking inside the house, and you ’ re looking for this stuff, this is where you look first, right? OK, so when we did this research, we found the absolutely, 100 percent, guaranteed way to never forget anything ever, ever again. And that is, quite simply, to have nothing to remember. OK, now, that sounds like something you get on a Chinese fortune cookie, right? But is, in fact, about the art of delegation. And from a design perspective, it ’ s about understanding what you can delegate to technology and what you can delegate to other people. And it turns out, delegation — if you want it to be — can be the solution for pretty much everything, apart from things like bodily functions, going to the toilet. You can ’ t ask someone to do that on your behalf. And apart from things like entertainment, you wouldn ’ t pay for someone to go to the cinema for you and have fun on your behalf, or, at least, not yet. Maybe sometime in the future, we will. So, let me give you an example of delegation in **practice**, right. So this is — probably the thing I ’ m most **passionate** about is the research that we ’ ve been doing on illiteracy and how people who are illiterate communicate. So, the U. N. estimated — this is 2004 figures — that there are almost 800 million people who can ’ t read and write, worldwide. So, we ’ ve been conducting a lot of research. And one of the things we were looking at is — if you can ’ t read and write, if you want to communicate over distances, you need to be able to identify the person that you want to communicate with. It could be a phone number, it could be an e-mail address, it could be a postal address. Simple question : if you can ’ t read and write, how do you manage your contact information? And the fact is that millions of people do it. Just from a design perspective, we didn ’ t really understand how they did it, and so that ’ s just one small example of the kind of research that we were doing. And it turns out that illiterate people are masters of delegation. So they delegate that part of the task process to other people, the stuff that they can ’ t do themselves. Let me give you another example of delegation. This one ’ s a little bit more sophisticated, and this is from a study that we did in Uganda about how people who are sharing devices, use those devices. Sente is a word in Uganda that means money. It has a second meaning, which is to send money as airtime. OK? And it works like this. So let ’ s say, June, you ’ re in a village, rural village. I ’ m in Kampala and I ’ m the wage earner. I ’ m sending money back, and it works like this. So, in your village, there ’ s one person in the village with a phone, and that ’ s the phone kiosk operator. And it ’ s quite likely that they ’ d have a quite simple **mobile** phone as a phone kiosk. So what I do is, I buy a prepaid card like this. And instead of using that money to top up my own phone, I call up the local village operator. And I read out that number to them, and they use it to top up their phone. So, they ’ re topping up the value from Kampala, and it ’ s now being topped up in the village. You take a 10 or 20 percent commission, and then you — the kiosk operator takes 10 or 20 percent commission, and passes the rest over to you in cash. OK, there ’ s two things I like about this. So the first is, it turns anyone who has access to a **mobile** phone — anyone who has a **mobile** phone — essentially into an ATM machine. It brings rudimentary banking **services** to places where there ’ s no banking infrastructure. And even if they could have access to the banking infrastructure, they wouldn ’ t necessarily be considered viable customers, because they ’ re not wealthy enough to have bank accounts. There ’ s a second thing I like about this. And that is that despite all the resources at my disposal, and despite all our kind of apparent sophistication, I know I could never have designed something as elegant and as totally in tune with the local conditions as this. OK? And, yes, there are things like Grameen Bank and micro-lending. But the difference between this and that is, there ’ s no central authority trying to control this. This is just street-up innovation. So, it turns out the street is a never-ending source of inspiration for us. And OK, if you break one of these things here, you return it to the carrier. They ’ ll give you a new one. They ’ ll probably give you three new ones, right? I mean, that ’ s buy three, get one free. That kind of thing. If you go on the streets of India and China, you see this kind of stuff. And this is where they take the stuff that breaks, and they fix it, and they put it back into circulation. This is from a workbench in Jilin City, in China, and you can see people taking down a phone and putting it back together. They reverse-engineer manuals. This is a kind of hacker ’ s manual, and it ’ s written in Chinese and English. They also write them in Hindi. You can subscribe to these. There are training institutes where they ’ re churning out people for fixing these things as well. But what I like about this is, it boils down to someone on the street with a small, flat surface, a screwdriver, a toothbrush for cleaning the contact heads — because they often get dust on the contact heads — and knowledge. And it ’ s all about the social network of the knowledge, floating around. And I like this because it challenges the way that we design stuff, and build stuff, and potentially distribute stuff. It challenges the **norms**. OK, for me the street just raises so many different questions. Like, this is Viagra that I bought from a backstreet sex shop in China. And China is a country where you get a lot of fakes. And I know what you ’ re asking — did I test it? I ’ m not going to answer that, OK. But I look at something like this, and I consider the implications of trust and confidence in the purchase process. And we look at this and we think, well, how does that apply, for example, for the design of — the lessons from this — apply to the design of online **services**, future **services** in these markets? This is a pair of underpants from — — from Tibet. And I look at something like this, and honestly, you know, why would someone design underpants with a pocket, right? And I look at something like this and it makes me question, if we were to take all the functionality in things like this, and redistribute them around the body in some kind of personal area network, how would we prioritize where to put stuff? And yes, this is quite **trivial**, but actually the lessons from this can apply to that kind of personal area networks. And what you see here is a couple of phone numbers written above the shack in rural Uganda. This doesn ’ t have house numbers. This has phone numbers. So what does it mean when people ’ s identity is **mobile**? When those extra three billion people ’ s identity is **mobile**, it isn ’ t fixed? Your notion of identity is out-of-date already, OK, for those extra three billion people. This is how it ' s shifting. And then I go to this picture here, which is the one that I started with. And this is from Delhi. It ' s from a study we did into illiteracy, and it ' s a guy in a teashop. You can see the chai being poured in the background. And he ' s a, you know, incredibly poor teashop worker, on the lowest rungs in the society. And he, somehow, has the appreciation of the values of Livestrong. And it ' s not necessarily the same values, but some kind of values of Livestrong, to actually go out and purchase them, and actually display them. For me, this kind of personifies this connected world, where everything is intertwined, and the dots are — it ' s all about the dots joining together. OK, the title of this presentation is"Connections and Consequences,"and it ' s really a kind of summary of five years of trying to figure out what it ' s going to be like when everyone on the planet has the ability to transcend space and time in a personal and convenient manner, right? When everyone ' s connected. And there are four things. So, the first thing is the immediacy of ideas, the speed at which ideas go around. And I know TED is about big ideas, but actually, the benchmark for a big idea is changing. If you want a big idea, you need to embrace everyone on the planet, that ' s the first thing. The second thing is the immediacy of objects. And what I mean by that is, as these become smaller, as the functionality that you can access through this becomes greater — things like banking, identity — these things quite simply move very quickly around the world. And so the speed of the adoption of things is just going to become that much more rapid, in a way that we just totally cannot conceive, when you get it to 6. 3 billion and the growth in the world ' s population. The next thing is that, however we design this stuff — carefully design this stuff — the street will take it, and will figure out ways to innovate, as long as it meets base needs — the ability to transcend space and time, for example. And it will innovate in ways that we cannot anticipate. In ways that, despite our resources, they can do it better than us. That ' s my feeling. And if we ' re smart, we ' ll look at this stuff that ' s going on, and we ' ll figure out a way to enable it to inform and infuse both what we design and how we design. And the last thing is that — actually, the direction of the conversation. With another three billion people connected, they want to be part of the conversation. And I think our relevance and TED ' s relevance is really about embracing that and learning how to listen, essentially. And we need to learn how to listen. So thank you very, very much.

Text Sample 251: POS Dim 2 – 2009 – Score 9.00 – 2009/t002967.txt

Okay, so 90 percent of my photographic process is, in fact, not photographic. It involves a campaign of letter writing, research and phone calls to access my subjects, which can range from Hamas leaders in Gaza to a hibernating black bear in its cave in West Virginia. And oddly, the most notable letter of rejection I ever received came from Walt Disney World, a seemingly innocuous site. And it read — I ' m just going to read a key sentence :"Especially during these violent times, I personally believe that the magical spell cast upon guests who visit our theme parks is particularly important to protect and helps to provide them with an important fantasy they can escape to."Photography threatens fantasy. They didn ' t want to let my camera in because it confronts constructed realities, myths and beliefs, and provides what appears to be evidence of a truth. But there are multiple truths attached to every image, depending on the creator ' s intention, the viewer and the context in which it is presented. Over a five year period following September 11th, when the American media and government were seeking hidden and unknown sites beyond its borders, most notably weapons of mass destruction, I chose to look inward at that which was integral to America ' s foundation, mythology and **daily** functioning. I wanted to confront the boundaries of the citizen, self-imposed and real, and confront the divide between privileged and public access to knowledge. It was a critical moment in American history and global history where one felt they didn ' t have access to accurate information. And I wanted to see the center with my own eyes, but what I came away with is a photograph. And it ' s just another place from which to **observe**, and the understanding that there are no absolute, all-knowing insiders. And the outsider can never really reach the core. I ' m going to run through some of the photographs in this series. It ' s titled,"An American Index of the Hidden and Unfamiliar,"and it ' s comprised of nearly 70 images. In this context I ' ll just show you a few. This is a nuclear waste storage and encapsulation facility at Hanford site in Washington State, where there are over 1, 900 stainless steel capsules containing nuclear waste submerged in water. A human standing in front of an unprotected capsule would die instantly. And I found one section amongst all of these that actually resembled the outline of the United States of America, which you can see here. And a big part of the work that is sort of absent in this context is text. So I create these two poles. Every image is accompanied with a very detailed factual text. And what I ' m most interested in is the invisible space between a text and its accompanying image, and how the image is transformed by the text and the text by the image. So, at best, the image is meant to float away into abstraction and multiple truths and fantasy. And then the text functions as this cruel anchor that kind of nails it to the ground. But in this context I ' m just going to read an abridged version of those texts. This is a cryopreservation unit, and it holds the bodies of the wife and mother of cryonics pioneer Robert Ettinger, who hoped to be awoken one day to extended life in good health, with advancements in science and technology, all for the cost of 35 thousand dollars, for forever. This is a 21-year-old Palestinian woman undergoing hymenoplasty. Hymenoplasty is a surgical procedure which restores the virginal state, allowing her to adhere to certain cultural expectations regarding virginity and marriage. So it essentially reconstructs a ruptured hymen, allowing her to bleed upon having sexual intercourse, to simulate the loss of virginity. This is a jury simulation deliberation room, and you can see beyond that two- way mirror jury advisers standing in a room behind the mirror. And they **observe** deliberations after mock **trial** proceedings so that they can better advise their clients how to adjust their **trial** strategy to have the outcome that they ' re hoping for. This process costs 60, 000 dollars. This is a U. S. Customs and Border Protection room, a contraband room, at John F. Kennedy International Airport. On that table you can see 48 hours ' worth of seized goods from passengers entering in to the United States. There is a pig ' s head and African cane rats. And part of my photographic work is I ' m not just documenting what ' s there. I do take certain liberties and intervene. And in this I really wanted it to resemble an early still-life painting, so I spent some time with the smells and items. This is the exhibited art on the walls of the CIA in Langley, Virginia, their original headquarters building. And the CIA has had a long history with both covert and public cultural diplomacy efforts. And it ' s speculated that some of their interest in the arts was designed to counter Soviet communism and promote what it considered to be pro-American thoughts and aesthetics. And one of the art forms that elicited the interest of the agency, and had thus come under question, is abstract expressionism. This is the Forensic Anthropology Research Facility, and on a six acre plot there are approximately 75 cadavers at any given time that are being studied by forensic anthropologists and **researchers** who are interested in monitoring a rate of corpse decomposition. And in this particular photograph the body of a young boy has been used to reenact a crime scene. This is the only federally funded site where it is legal to cultivate cannabis for scientific research in the United States. It ' s a research crop marijuana grow room. And part of the work that I hope for is that there is a sort of disorienting entropy where you can ' t find any discernible formula in how these things — they sort of awkwardly jump from government to science to religion to security — and you can ' t completely understand how information is being distributed. These are transatlantic submarine communication cables that travel across the floor of the Atlantic Ocean, connecting North America to Europe. They carry over 60 million simultaneous voice conversations, and in a lot of the government and technology sites there was just this very apparent vulnerability. This one is almost humorous because it feels like I could just snip all of that conversation in one easy cut. But stuff did feel like it could have been taken 30 or 40 years ago, like it was locked in the Cold War era and hadn ' t necessarily progressed. This is a Braille edition of Playboy magazine. And this is... a division of the Library of Congress produces a free national library **service** for the blind and visually impaired, and the **publications** they choose to publish are based on reader popularity. And Playboy is always in the top few. But you ' d be surprised, they don ' t do the photographs. It ' s just the text. This is an avian **quarantine** facility where all imported birds coming into America are required to undergo a 30-day **quarantine**, where they are tested for diseases including Exotic Newcastle Disease and Avian Influenza. This film shows the testing of a new explosive fill on a warhead. And the Air Armament Center at Eglin Air Force Base in Florida is responsible for the deployment and testing of all air-delivered weaponry coming from the United States. And the film was shot on 72 millimeter, government-issue film. And that red dot is a marking on the government-issue film. All living white tigers in North America are the result of selective inbreeding — that would be mother to son, father to daughter, sister to brother — to allow for the genetic conditions that create a salable white tiger. Meaning white fur, ice blue eyes, a pink nose. And the majority of these white tigers are not born in a salable state and are killed at birth. It ' s a very violent process that is little known. And the white tiger is obviously celebrated in several forms of entertainment. Kenny was born. He actually made it to adulthood. He has since passed away, but was mentally retarded and suffers from severe bone abnormalities. This, on a lighter note, is at George Lucas ' personal archive. This is the Death Star. And it ' s shown here in its true orientation. In the context of"Star Wars : Return of the Jedi,"its mirror image is presented. They flip the negative. And you can see the photoetched brass detailing, and the painted acrylic facade. In the context of the film, this is a deep-space battle station of the Galactic Empire, capable of annihilating planets and civilizations, and in reality it measures about four feet by two feet. This is at Fort Campbell in Kentucky. It ' s a Military Operations on Urbanized Terrain site. Essentially they ' ve simulated a city for urban combat, and this is one of the structures that exists in that city. It ' s called the World Church of God. It ' s supposed to be a generic site of worship. And after I took this photograph, they constructed a wall around the World Church of God to mimic the set-up of mosques in Afghanistan or Iraq. And I worked with Mehta Vihar who creates virtual simulations for the army for tactical **practice**. And we put that wall around the World Church of God, and also used the characters and vehicles and explosions that are offered in the video games for the army. And I put them into my photograph. This is live HIV virus at Harvard Medical School, who is working with the U. S. Government to develop sterilizing immunity. And Alhurra is a U. S. Government- sponsored Arabic language television network that distributes news and information to over 22 countries in the Arab world. It runs 24 hours a day, commercial free. However, it ’ s illegal to broadcast Alhurra within the United States. And in 2004, they developed a channel called Alhurra Iraq, which specifically deals with events occurring in Iraq and is broadcast to Iraq. Now I ’ m going to move on to another project I did. It ’ s titled”The Innocents.”And for the men in these photographs, photography had been used to create a fantasy. Contradicting its function as evidence of a truth, in these instances it furthered the fabrication of a lie. I traveled across the United States photographing men and women who had been wrongfully convicted of crimes they did not commit, violent crimes. I investigate photography ’ s ability to blur truth and fiction, and its influence on memory, which can lead to severe, even lethal consequences. For the men in these photographs, the primary cause of their wrongful conviction was mistaken identification. A victim or eyewitness identifies a suspected perpetrator through law enforcement ’ s use of images. But through exposure to composite sketches, Polaroids, mug shots and line-ups, eyewitness testimony can change. I ’ ll give you an example from a case. A woman was raped and presented with a series of photographs from which to identify her attacker. She saw some similarities in one of the photographs, but couldn ’ t quite make a positive identification. Days later, she is presented with another photo array of all new photographs, except that one photograph that she had some draw to from the earlier array is repeated in the second array. And a positive identification is made because the photograph replaced the memory, if there ever was an actual memory. Photography offered the criminal justice system a tool that transformed innocent citizens into criminals, and the criminal justice system failed to recognize the limitations of relying on photographic identifications. Frederick Daye, who is photographed at his alibi location, where 13 witnesses placed him at the time of the crime. He was convicted by an all-white jury of rape, kidnapping and vehicle theft. And he served 10 years of a life sentence. Now DNA exonerated Frederick and it also implicated another man who was serving time in prison. But the victim refused to press charges because she claimed that law enforcement had permanently altered her memory through the use of Frederick ’ s photograph. Charles Fain was convicted of kidnapping, rape and murder of a young girl walking to school. He served 18 years of a death sentence. I photographed him at the scene of the crime at the Snake River in Idaho. And I photographed all of the wrongfully convicted at sites that came to particular significance in the history of their wrongful conviction. The scene of arrest, the scene of misidentification, the alibi location. And here, the scene of the crime, it ’ s this place to which he ’ s never been, but changed his life forever. So photographing there, I was hoping to highlight the tenuous relationship between truth and fiction, in both his life and in photography. Calvin Washington was convicted of capital murder. He served 13 years of a life sentence in Waco, Texas. Larry Mayes, I photographed at the scene of arrest, where he hid between two mattresses in Gary, Indiana, in this very room to hide from the police. He ended up serving 18 and a half years of an 80 year sentence for rape and robbery. The victim failed to identify Larry in two live lineups and then made a positive identification, days later, from a photo array. Larry Youngblood served eight years of a 10 and half year sentence in Arizona for the abduction and repeated sodomizing of a 10 year old boy at a carnival. He is photographed at his alibi location. Ron Williamson. Ron was convicted of the rape and murder of a barmaid at a club, and served 11 years of a death sentence. I photographed Ron at a **baseball** field because he had been drafted to the Oakland A ’ s to play professional **baseball** just before his conviction. And the state ’ s key witness in Ron ’ s case was, in the end, the actual perpetrator. Ronald Jones served eight years of a death sentence for rape and murder of a 28-year-old woman. I photographed him at the scene of arrest in Chicago. William Gregory was convicted of rape and burglary. He served seven years of a 70 year sentence in Kentucky. Timothy Durham, who I photographed at his alibi location where 11 witnesses placed him at the time of the crime, was convicted of 3. 5 years of a 3220 year sentence, for several charges of rape and robbery. He had been misidentified by an 11-year-old victim. Troy Webb is photographed here at the scene of the crime in Virginia. He was convicted of rape, kidnapping and robbery, and served seven years of a 47 year sentence. Troy ' s picture was in a photo array that the victim tentatively had some draw toward, but said he looked too old. The police went and found a photograph of Troy Webb from four years earlier, which they entered into a photo array days later, and he was positively identified. Now I ' m going to leave you with a self portrait. And it reiterates that distortion is a constant, and our eyes are easily deceived. That ' s it. Thank you.

Text Sample 252: POS Dim 2 – 2009 – Score 9.00 – 2009/t002966.txt

I believe that there are new, hidden tensions that are actually happening between people and institutions â institutions that are the institutions that people inhabit in their **daily** life : schools, hospitals, **workplaces**, factories, offices, etc. And something that I see happening is something that I would like to call a sort of"democratization of intimacy."And what do I mean by that? I mean that what people are doing is, in fact, they are sort of, with their communication channels, they are breaking an imposed **isolation** that these institutions are imposing on them. How are they doing this? They ' re doing it in a very simple way, by calling their mom from work, by IMing from their office to their friends, by texting under the desk. The pictures that you ' re seeing behind me are people that I visited in the last few months. And I asked them to come along with the person they communicate with most. And somebody brought a boyfriend, somebody a father. One young woman brought her grandfather. For 20 years, I ' ve been looking at how people use channels such as email, the **mobile** phone, texting, etc. What we ' re actually going to see is that, fundamentally, people are communicating on a regular basis with five, six, seven of their most intimate sphere. Now, lets take some data. Facebook. Recently some sociologists from Facebook â Facebook is the channel that you would expect is the most enlargening of all channels. And an average user, said Cameron Marlow, from Facebook, has about 120 friends. But he actually talks to, has two-way exchanges with, about four to six people on a regular base, depending on his gender. Academic research on **instant** messaging also shows 100 people on buddy lists, but fundamentally people chat with two, three, four â anyway, less than five. My own research on cellphones and voice calls shows that 80 percent of the calls are actually made to four people. 80 percent. And when you go to Skype, it ' s down to two people. A lot of sociologists actually are quite disappointed. I mean, I ' ve been a bit disappointed sometimes when I saw this data and all this deployment, just for five people. And some sociologists actually feel that it ' s a closure, it ' s a cocooning, that we ' re disengaging from the public. And I would actually, I would like to show you that if we actually look at who is doing it, and from where they ' re doing it, actually there is an incredible social transformation. There are three stories that I think are quite good examples. The first gentleman, he ' s a baker. And so he starts working every morning at four o ' clock in the morning. And around eight o ' clock he sort of sneaks away from his oven, cleans his hands from the flour and calls his wife. He just wants to wish her a good day, because that ' s the start of her day. And I ' ve heard this story a number of times. A young factory worker who works night shifts, who manages to sneak away from the factory floor, where there is CCTV by the way, and find a corner, where at 11 o ' clock at night he can call his girlfriend and just say goodnight. Or a mother who, at four o ' clock, suddenly manages to find a corner in the toilet to check that her children are safely home. Then there is another couple, there is a Brazilian couple. They ' ve lived in Italy for a number of years. They Skype with their families a few times a week. But once a fortnight, they actually put the computer on their dining table, pull out the webcam and actually have dinner with their family in Sao Paulo. And they have a big event of it. And I heard this story the first time a couple of years ago from a very modest family of immigrants from Kosovo in Switzerland. They had set up a big screen in their living room, and every morning they had breakfast with their grandmother. But Danny Miller, who is a very good anthropologist who is working on Filipina migrant women who leave their children back in the Philippines, was telling me about how much parenting is going on through Skype, and how much these mothers are engaged with their children through Skype. And then there is the third couple. They are two friends. They chat to each other every day, a few times a day actually. And finally, finally, they ' ve managed to put **instant** messaging on their computers at work. And now, obviously, they have it open. Whenever they have a moment they chat to each other. And this is exactly what we ' ve been seeing with teenagers and kids doing it in school, under the table, and texting under the table to their friends. So, none of these cases are unique. I mean, I could tell you hundreds of them. But what is really exceptional is the **setting**. So, think of the three **settings** I ' ve talked to you about : factory, migration, office. But it could be in a school, it could be an administration, it could be a hospital. Three **settings** that, if we just step back 15 years, if you just think back 15 years, when you clocked in, when you clocked in to an office, when you clocked in to a factory, there was no contact for the whole duration of the time, there was no contact with your private sphere. If you were lucky there was a public phone hanging in the corridor or somewhere. If you were in management, oh, that was a different story. Maybe you had a direct line. If you were not, you maybe had to go through an operator. But basically, when you walked into those buildings, the private sphere was left behind you. And this has become such a **norm** of our professional lives, such a **norm** and such an expectation. And it had nothing to do with technical capability. The phones were there. But the expectation was once you moved in there your commitment was fully to the task at hand, fully to the people around you. That was where the focus had to be. And this has become such a cultural **norm** that we actually school our children for them to be capable to do this cleavage. If you think nursery, kindergarten, first years of school are just dedicated to take away the children, to make them used to staying long hours away from their family. And then the school enacts perfectly well. It mimics perfectly all the rituals that we will find in offices : rituals of entry, rituals of exit, the schedules, the uniforms in this country, things that identify you, team-building activities, team building that will allow you to basically be with a random group of kids, or a random group of people that you will have to be with for a number of time. And of course, the major thing : learn to pay attention, to concentrate and focus your attention. This only started about 150 years ago. It only started with the birth of modern bureaucracy, and of industrial revolution. When people basically had to go somewhere else to work and carry out the work. And when with modern bureaucracy there was a very rational approach, where there was a clear distinction between the private sphere and the public sphere. So, until then, basically people were living on top of their trades. They were living on top of the land they were laboring. They were living on top of the workshops where they were working. And if you think, it ' s permeated our whole culture, even our cities. If you think of medieval cities, medieval cities the boroughs all have the names of the guilds and professions that lived there. Now we have sprawling residential suburbias that are well distinct from production areas and commercial areas. And actually, over these 150 years, there has been a very clear class system that also has emerged. So the lower the status of the job and of the person carrying out, the more removed he would be from his personal sphere. People have taken this amazing possibility of actually being in contact all through the day or in all types of situations. And they are doing it massively. The Pew Institute, which produces good data on a regular basis on, for instance, in the States, says that â□□ and I think that this number is conservative â□□ 50 percent of anybody with email access at work is actually doing private email from his office. I really think that the number is conservative. In my own research, we saw that the **peak** for private email is actually 11 o ’ clock in the morning, whatever the country. 75 percent of people admit doing private conversations from work on their **mobile** phones. 100 percent are using text. The point is that this re-appropriation of the personal sphere is not terribly successful with all institutions. I ’ m always surprised the U. S. Army sociologists are discussing of the impact for instance, of soldiers in Iraq having **daily** contact with their families. But there are many institutions that are actually blocking this access. And every day, every single day, I read news that makes me cringe, like a $15 fine to kids in Texas, for using, every time they take out their **mobile** phone in school. Immediate dismissal to bus drivers in New York, if seen with a **mobile** phone in a hand. Companies blocking access to IM or to Facebook. Behind issues of security and safety, which have always been the arguments for social control, in fact what is going on is that these institutions are trying to decide who, in fact, has a right to self determine their attention, to decide, whether they should, or not, be isolated. And they are actually trying to block, in a certain sense, this movement of a greater possibility of intimacy.

Text Sample 253: POS Dim 2 – 2009 – Score 9.00 – 2009/t002952.txt

Thirteen trillion dollars in wealth has evaporated over the course of the last two years. We ’ ve questioned the future of capitalism. We ’ ve questioned the **financial** industry. We ’ ve looked at our government oversight. We ’ ve questioned where we ’ re going. And yet, at the same time, this very well may be a seminal moment in American history, an opportunity for the consumer to actually take control and guide us to a new trajectory in America. I ’ m calling this The Great Unwind. And the idea is a simple, simple idea, which is the fact that the consumer has moved from a state of anxiety to **action**. Consumers who represent 72 percent of the GDP of America have actually started, just like banks and just like businesses, to de-leverage, to unwind their leverage in **daily** life, to remove themselves from the liability and risk that presents itself as they move forward. So, to understand this — and I ’ m going to **stress** this — it ’ s not about the consumer being in retreat. The consumer is empowered. To understand this, we ’ ll step back and look at what ’ s happened over the last year and a half. So if you ’ ve been gone, this is the CliffsNotes on what ’ s happened in the economy. Unemployment up. Housing values down. Equity markets down. Commodity prices are like this. If you ’ re a mom trying to manage a budget, and oil was 150 dollars a barrel last summer, and it ’ s somewhere between 50 and 70, do you plan **vacations**? How do you buy? What ’ s your strategy in your household? Will the bailout work? We have national debt, Detroit, currency valuations, health care — all these issues facing us. You put them all together, mix them up in a bouillabaisse, and you have consumer confidence that ' s basically a ticking time bomb. In fact, let ' s go back and look at what caused this crisis, because the consumer, all of us, in our **daily** lives, actually contributed a large part to the problem. This is something I call the 50-20 paradox. It took us 50 years to reach annual savings ratings of almost 10 percent. Fifty years. Do you know what this was right here? This was World War II. Do you know why savings was so high? There was nothing to buy, unless you wanted to buy some rivets. What happened, though, over the course of the last 20 years, we went from a 10 percent savings rate to a negative savings rate. Because we binged. We bought extra-large cars, supersized everything, we bought remedies for restless leg syndrome. All these things together basically created a factor where the consumer drove us headlong into the crisis that we face today. The personal debt-to-income ratio basically went from 65 to 135 percent in the span of about 15 years. So consumers got over-leveraged. And of course our banks did as well, as did our federal government. This is an absolutely staggering chart. It shows leverage, trended out from 1919 to 2009. And what you end up seeing is the whole phenomenon that we are actually stepping forth and basically leveraging future education, future children in our households. So if you look at this in the context of visualizing the bailout, what you can see is, if you stack up dollar bills, first of all, 360, 000 dollars is about the size of a five-foot-four guy. But if you stack it up, you see this amazing, staggering amount of dollars that have been put into the system to fund and bail us out. So this is the first 315 billion. But I read this fact the other day, that one trillion seconds equals 32, 000 years. So if you think about that, the context, the casualness with which we talk about trillion-dollar bailout here and trillion there, we are stacking ourselves up for long-term leverage. However, consumers have moved. They are taking responsibility. What we ' re seeing is an uptake in the savings rate. In fact, 11 straight months of savings have happened since the beginning of the crisis. We ' re working our way back up to that 10 percent. Also, remarkably, in the fourth quarter, spending dropped to its lowest level in 62 years — almost a 3. 7 percent decline. Visa now reports that more people are using debit cards than credit cards. So we ' re starting to pay for things with money that we have. And we ' re starting to be much more careful about how we save and invest. But that ' s not really the whole story, because this has also been a dramatic time of transformation. And you ' ve got to admit, over the last year and a half, consumers have been doing some weird things. It ' s pretty staggering, what we ' ve lived through. If you take into account that 80 percent of all Americans were born after World War II, this was essentially our Depression. And so, as a result, some crazy things have happened. I ' ll give you some examples. Let ' s talk about dentists, vasectomies, guns and shark attacks. Dentists report molars — people grinding their teeth, coming in and reporting that they ' ve had **stress**. So there ' s an increase in people having to have their fillings replaced. Gun sales, according to the FBI, who does background checks, are up almost 25 percent since January. Vasectomies are up 48 percent, according to the Cornell Institute. And lastly, but a very good point, hopefully not related to the former point I just made, which is that shark attacks are at their lowest level from 2003. Does anybody know why? No one ' s at the beach. So there ' s a bright side to everything. But seriously, what we see happening, and the reason I want to **stress** that the consumer is not in retreat, is that this is a tremendous opportunity for the consumer who drove us into this recession to lead us right back out. What I mean by that is we can move from mindless consumption to mindful consumption. Right? If you think about the last three decades, the consumer has moved from savvy about marketing in the ' 90s, to gathering all these amazing social and search tools in this decade. But the one thing holding them back is the ability to **discriminate**. By restricting their demand, consumers can actually align their values with their spending, and drive capitalism and business to not just be about more, but to be about better. We ' re going to explain that right now. Based on Y&R ' s BrandAsset Valuator, proprietary tool of VML and Young & Rubicam, we set out to understand what ' s been happening in the crisis with the consumer marketplace. We found a couple of really interesting things. We ' re going to go through four value shifts that we see driving new consumer behaviors, that offer new management principles. The first cultural value shift we see is this tendency toward something we call"liquid life."This is the movement from Americans defining their success on having things to having liquidity, because the less excess that you have around you, the more nimble and fleet of foot you are. As a result, dclass consumption is in. Dclass consumption is the whole idea that spending money frivolously makes you look a little bit anti-fashion. The management principle is dollars and cents. So let ' s look at some examples of this dclass consumption that falls out of this value. The first thing is, something must be happening when P. Diddy vows to tone down his bling. But seriously, we also have this phenomenon on Madison Avenue and in other places, where people are actually walking out of luxury boutiques with ordinary, generic paper bags to hide the brand purchases. We see high-end haggling in fashion today, high-end haggling for luxury and real estate. We also see just a relaxing of ego, and sort of a dismantling of artifice. This is a story on the yacht club that ' s all basically blue collar. Blue-collar yacht club — where you can join, but you ' ve got to work in the boatyard as condition of membership. We also see the trend toward tourism that ' s a little bit more low-key : agritourism — going to vineyards and going to farms. And then we also see this movement forward from dollars and cents. What businesses can do to connect with these new mindsets is really interesting. A couple things that are kind of cool. One is that Frito-Lay figured out this liquidity thing with their consumer. They found their consumer had more money at the beginning of the month, less at the end of the month. So they started to change their packaging : larger packs at the beginning of the month, smaller packaging at the end of the month. Really interestingly, too, was the San Francisco Giants. They ' ve just instituted dynamic pricing. It takes into account everything from the pitcher match-ups, to the weather, to the team records, in setting prices for the consumer. Another quick example of these types of movements is the rise of Zynga. Zynga has risen on the consumer ' s desire to not want to be locked in to fixed cost. Again, this theme is about variable cost, variable living. So micro-payments have become huge. And lastly, some people are using Hulu as a device to get rid of their cable bill. So, really clever ideas there that are being taken ahold of and that marketers are starting to understand. The second of the four values is this movement toward ethics and fair play. We see that play itself out with **empathy** and respect. The consumer is demanding it. And, as a result, businesses must provide not only value, but values. Increasingly, consumers are looking at the culture of the company, at their conduct in the marketplace. So we see with **empathy** and respect lots of really hopeful things come out of this recession. I ' ll give you a few examples. One is the rise toward communities and neighborhoods, and increased emphasis on your neighbors as your support system. Also, a wonderful by-product of a really lousy thing, which has been unemployment, is a rise in volunteerism that ' s been noted in our country. We also see the phenomenon — some of you may have"boomerang kids"— these are"boomerang alumni,"where universities are actually reconnecting with alumni and helping them with jobs, sharing skills and retraining. We also talked about character and professionalism. We had this miracle on the

Hudson in New York City in January, and suddenly Sully has become a key name on BabyCenter. So, from a value and values standpoint, what companies can do is connect in lots of different ways. Microsoft is doing something wonderful. They are actually vowing to retrain two million Americans with IT training, using their existing infrastructure to do something good. Also, a really interesting company is GORE-TEX. GORE-TEX is all about personal accountability of their management and their employees, to the point where they really kind of shun the idea of bosses. But they also talk about the fact that their executives — all of their expense reports are put onto their company intranet for everyone to see. Complete transparency. Think twice before you have that bottle of wine. The third of the four laws of post-crisis consumerism is about durable living. We ' re seeing in our data that consumers are realizing this is a marathon, not a sprint. They ' re digging in and looking for ways to extract value out of every purchase they make. Witness the fact that Americans are holding on to their cars longer than ever before : 9. 4 years on average, in March. A record. We also see the fact that libraries have become a huge resource for America. Did you know that 68 percent of Americans now carry a library card? The highest percentage ever in our nation ' s history. So what you see in this trend is also the accumulation of knowledge. Continuing education is up. Everything is focused on betterment, training, development and moving forward. We also see a big DIY movement. I was fascinated to learn that 30 percent of all homes in America are actually built by owners. That includes cottages and the like, but 30 percent. People are getting their hands dirty, rolling up their sleeves. They want these skills. We see it with the phenomenon of raising backyard hens, chickens and ducks. And when you work out the math, they say it doesn ' t work, but the principle is there ; it ' s about being sustainable and taking care of yourself. Then we look at the High Line in New York City, an excellent use of reimagining existing infrastructure for something good, which is a brand-new park in New York City. So what brands can do, and companies, is pay dividends to consumers, be a brand that lasts, offer transparency, promise you ' re going to be there beyond today ' s sale. Perfect example of that is Patagonia. Patagonia ' s"Footprint Chronicles"basically goes through and tracks every product they make, and gives you social responsibility, and helps you understand the ethics behind the product they make. Another great example is Fidelity. Rather than **instant** cash-back rewards on your credit or debit purchases, this is about 529 rewards for your student education. Or the interesting company Sunrun. I love this company. They ' ve created a consumer collective where they put solar panels on households and create a consumer-based utility, where the electricity they generate is basically pumped back out into the marketplace. So it ' s a consumer-driven co- op. The fourth post-crisis consumerism that we see is this movement about"return to the fold."It ' s incredibly important right now. Trust is not parceled out, as we all know. It ' s now about connecting to your communities, connecting to your social networks. In my book, I talked about the fact that 72 percent of people trust what other people say about a brand or a company, versus 15 percent on advertising. So in that respect, cooperative consumerism has really taken off. This is about consumers working together to get what they want out of the marketplace. Let ' s look at a couple of quick examples. The artisanal movement is huge : everything about locally derived products and **services**, supporting your local neighborhoods, whether it ' s cheeses, wines and other products. Also this rise of local currencies. Realizing that it ' s difficult to get loans in this environment, you ' re doing business with people you trust in your local markets. So this rise of local currency is another really interesting phenomenon. And then they did a recent report I thought was fascinating. They actually started, in certain communities in the United

States, to publish people ' s electricity usage. And what they found out is when that was available for public record, the people ' s electricity usage in those communities dropped. Then we also look at the idea of cow-pooling, which is the whole phenomenon of consumers organizing together to buy meat from organic farms, that they know is safe and controlled in the way that they want it to be controlled. And then there ' s this other really interesting movement in California, which is about carrot mobs. The traditional thing would be to boycott, right? Have a stick. Well, why not have a carrot? So these are consumers organizing, pooling their resources to incentify companies to do good. And then we look at what companies can do. This is all the opportunity about being a community organizer. You have to realize that you can ' t fight and control this. You actually need to organize it. You need to harness it. You need to give it meaning. And there ' s lots of really interesting examples here. First is just the rise of the fact that Zagat ' s has actually moved out of and diversified from rating restaurants, into actually rating health care. So what credentials does Zagat ' s have? Well, they have a lot, because it ' s their network of people. So that becomes a very powerful force for them to make their brand more elastic. Then you look at the phenomenon of Kogi. This Kogi doesn ' t exist. It ' s a moving truck. It ' s a moving truck through L. A., and the only way you can find it is through Twitter. Or you look at Johnson & Johnson ' s"Momversations,"a phenomenal blog that ' s been built up, where J&J basically is tapping into the power of mommy bloggers, allowing them to create a forum where they can communicate and connect. And it ' s also become a very valuable advertising revenue for J&J as well. This, plus the fact that you ' ve got phenomenal work from CEOs, from Ford to Zappos, connecting on Twitter, creating an open environment, allowing their employees to be part of the process, rather than hidden behind walls. You see this rising force in total transparency and openness that companies are starting to adopt, all because the consumer is demanding it. So when we look at this and step back, what I believe is that the crisis that exists today is definitely real. It ' s been tremendously powerful for consumers. But at the same time, this is also a tremendous opportunity. The Chinese character for crisis is actually the same side of the same coin. Crisis equals opportunity. What we ' re seeing with consumers right now is the ability for them to actually lead us forward out of this recession. So we believe that values-driven spending will force capitalism to be better : it will drive innovation ; it will make longer-lasting products ; it will create better, more intuitive customer **service** ; and it will give us the opportunity to connect with companies that share the values that we share. So when we look back and step out at this and see the beginning of these trends that we ' re seeing in our data, we see a very hopeful picture for the future of America. Thank you very much.

Text Sample 254: POS Dim 2 – 2025 – Score 5.00 – 2025/t003466.txt

Can you paint with all the colors of the wind? I love this song from the first time I heard it. I was 14, growing up in public housing in western Sydney. So maybe a little too old for this Disney classic from"Pocahontas."But it spoke to me. I saw something of myself in it. At this stage, I think I knew I was queer. I mean, come on, I was rocking out to Disney. And look at those glasses. Seriously, what a queer kid. But I was also from a Pacific Indigenous heritage with conservative views and values. I didn ' t feel like I could fit into any of the boxes that were being presented to me. Straight, gay, Fijian, Australian. I couldn ' t pick just one color to paint with. Prior to colonization, sexuality amongst many of our Pacific islands across the region was fluid. Men would have sex with men without the stigma or shame afforded it today. Sexual intimacy was a way of being able to connect socially and relationally. Sex wasn ’ t just for procreation. Or in marriage. It wasn ’ t exclusively monogamous. Just because you had sex with a man didn ’ t define your sexuality. It didn ’ t make you an outcast. But as Europeans started to colonize the Pacific region, they brought Christian missionaries with them. They told us to cover up our beautiful bodies. We were taught shame. We were told to stop **practicing** Pacifica cultures and values that were seen as provocative. To a Western and white perspective, our non-binary sexuality was **judged** as tribal, outdated, sinful and impossible to **practice** alongside a Christian faith. Still today, those who are Indigenous to the Pacific are negatively impacted by whiteness. That”white is right, West is best.”I was brought up to espouse strong Christian beliefs, views and values which would shape my burgeoning identity as an adult. From these staunch conservative views, I was taught to fear the queer. Those people that were seen as unusual and fruity. And in essence, I grew to despise a part of myself. I sought to separate my queerness from my identity. I could never see myself as part of that world. Growing up, biracial and bisexual constantly challenged the way I saw the world around me. I was unsure about my thinking, feelings and behaviors around liking males and females. I was unsure about my thinking, feelings and behaviors when it came to whether I needed to be a white Australian or a Fijian person. And whether this needed to be framed within a binary to choose one over the other. In this way, I don ’ t think that my story is unique. There ’ s plenty of people in the queer world that have escaped conservative upbringings, religions and cultures. But this is only part of my story. Another common story in the queer community is the coming-out story. When did you tell your parents? How did they react? How it liberated you or ultimately separated you from your family. But this didn ’ t feel like an option for me either. I believe that embedded in these particular stories, these tropes, is a part of dominant queer culture that made it very hard for me to access the queer world. For example, when I started to go to gay clubs, I would be lining up at the front of the club and often I would get door staff come up to me and go,”Sorry, you do know this is a gay club, right?”From these lived experiences, I feel that there is a whiteness that pervades queer culture. And I didn ’ t want to give up my identity and my appearance as a Pacific indigenous person to be queer. Pacifica cultures really, really value and esteem the collective above the individual. For Pacifica people, our families are inextricably connected to our own sense of self-agency, self-determination, self-esteem, self-love and self-worth. Our own individual identities are meshed and intertwined with our family ’ s reputation and standing in the community. Our mind, body and souls are inextricably connected to the broader community, to our families, to ourselves and to others. I didn ’ t want to come out to potentially ostracize myself, to make a point to separate myself from my Pacific and faith communities. I couldn ’ t then see my authentic self in any of these spaces. And a quick shout out to my son on the left-hand side. Round of applause for my son. He said to me this morning,”Daddy, make sure you mention me.”So I was like,”OK, son. Alright.”So there you go for the recording, son. And the rest are my other family, that ’ s up there as well. Across queer cultures and the broader community, we have learned to embrace our gender and sexual differences. We encourage individuals to be allies for each other and to support safe spaces for those that are seen to not fit in. We empower and really encourage individuals to really develop and express their self-agency, their self-determination, their self love, their self-worth. We boldly go into spaces striving to bring others with us to ensure that we can create inclusive societies for everyone. However, the dominant Western view in queer cultures is still based on the individual. Me, myself and I. And my individual pursuits towards **happiness**. As a queer Indigenous Pacific person, I feel that there are parts of queer culture that is white, feels very white. We may fail to see the importance of bringing our families and our cultural communities along our journey as queer people. It ’ s either the queer way or the highway. It ’ s one or the other. Young or old. Rich or poor. Smart or stupid. Masculine or feminine. Fit or fat. These binaries create a sense of us and them, those people. A binary approach and position that we generally in Western white societies use to determine the **norm**. We create, conflate, perpetuate binaries that create divisions in our society. Saints or sinner, straight or gay. I see it like this. Binaries are a very white and Western way of thinking, feeling and behaving. As individuals today, we live within these binaries and create identities that then create these labels. We use these labels to then define and confine people and places. We are one or the other. But what we fail to see is that our identities are more than a set of binaries. Our own identities are comprised of intersecting identifiers that contributes to who you are, including our age, class, color, education, gender, size, spirituality, sexuality, and so, so much more. We need to decolonize queerness, to live beyond”white is right, West is best.”Because if we don ’ t, we ’ re going to continue to create silos and divisions in our societies that don ’ t create inclusive and safe spaces for everyone. For Indigenous queer people, they shape our connection to community and culture, our families. If we don ’ t include that within our understanding of who we are, even as queer people, then we ’ re not able to be, again, our true and authentic selves. We ’ re not able to then be our fabulous selves as part of the queer community. And it ’ s really, really interesting because I believe that we can learn so much from pre-colonial cultures like those that were **practiced** in the Pacific, that provide a different way of looking at a society. A queer way that pre-dates all the privileges that go into binary ways of thinking, feeling and behaving. Across many of our Pacifica languages we have a lot of key concepts that help solidify and support a shared understanding of the collective. For example, in Fijian,”solesolevaki”is all about reciprocal living, where my wellbeing is your wellbeing. I will actively seek to ensure that your well-being and every part of your life is catered for, knowing that you will do the same for me. We strive to enact”veiqaravi,”as tatted here on my hand, to serve others. We strive to also embed and embold and empower people to **practice**”veirokovi,”respect for self and others where we are relationally driven, where that is so important to our spaces and places. My sense of ability to actually put this into **practice**, my ability to actually understand these Indigenous perspectives has greatly helped shape and reconcile my queer identity today. But what ’ s really interesting is that it ’ s not just queer communities that need to decolonize itself. It ’ s all of us as a broader society that needs to be part of a shared conversation to decolonize, deconstruct and disrupt binary ways of thinking, feeling and behaving. So how can we practically do that? **Practice** cultural humility, where you are constantly learning and embracing differences, where you get over yourselves and strive to learn about other people ’ s differences and how that can be meaningfully included in the way in which you create safe spaces for everyone. I also believe it ’ s possible to bring Indigenous perspectives into Western and white **workplaces**. For example, I ’ ve introduced the concept of”talanoa”in all of my work meetings. Talanoa is all about being able to create a safe space where everyone is able to contribute to a collective, collegial and collaborative conversation. Where people are able to bring their authentic selves into these spaces. Literally, I will chair meetings where I have no firm set agenda items and I literally would just hold space for participants just to bring themselves. So you can tell I ’ m a social worker, right? Safe space, safe space, safe space. It ’ s OK, it ’ s OK. But it ’ s through these safe spaces that people can bring their multiple, intersecting identifiers, including their age, their class, color, education, gender, size, spirituality, sexuality and so, so much more. I invite queer communities to come talanoa with Pacific Indigenous Communities to create robust and resilient queer cultures. I have been able to greatly reconcile my queer and Pacifica identities, which has now really been embraced to help shape how I see the world around me today. And it ’ s through these shared perspectives and approaches that we can truly embrace cultural diversity and its differences. And to know that people can live beyond those binaries and those labels we continue to perpetuate and perpetrate against others. It is a commitment from queer communities to decolonize queerness. It ’ s a commitment from me, from you and all of us to decolonize Western and white societies and to live beyond the binary way of thinking, feeling and behaving. I genuinely believe that it ’ s through this common goal that we can truly learn how to paint with all the colors of the wind.”Vinaka vakalevu”and thank you.

Text Sample 255: POS Dim 2 – 2025 – Score 2.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in the world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we **navigate** through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the air. In all my travels, I ’ ve been struck by so many things. I ’ ve **observed** how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ’ ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced us to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ’ s hard to see the big picture right now. It ’ s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think back to my travels. I ’ ve seen how different we are, how diverse and distinct. But I ’ ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ’ m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman”A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story is the dominant sound of our lives.”- Joseph Campbell I ’ ve loved telling stories all of my life. I ’ ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 256: POS Dim 2 – 2025 – Score 2.00 – 2025/t003001.txt

- I don ’ t remember exactly, but... I use the most expensive hair color in the world. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color. I expect great color. What ’ s worth more to me is the way my hair feels. Feels good around my neck. I can ’ t remember it. I can ’ t finish it all. If only we had more time. - Why don ’ t we have more time? - I ’ m dying. I ’ m croaking! - It ’ s tough, coming home. But this is what you do for a parent that you love. But, you know? She ’ s not my mother. Technically. In my own family, there ’ s a lot of dysfunction, a lot of mental illness, a lot of substance abuse. Every generation. My parents divorced when I was about five. My mother moved us out of state, so we had to fly to New York to see my father ’ cause he was already working in advertising, and New York was the only place to be. - Many of the commercials you see on television begin life here. - Advertising is one of the main sources of messaging about who you should be. - Nowadays, most of you women lead two lives, combining homemaking with the role of glamorous hostess. - And in the ’ 60s, the **ads** were often made from a male point of view. - In the pre-production meeting, everybody who has anything to do with the commercial is there. - It was mostly men. These superstar art directors and writers. Like my father, Gene Case.”Mad Men”was like home movies, and he was Don Draper. He was very witty, but I don ’ t think my father realized how powerful his words were. He knew it professionally. But with his kids, he could be incredibly cutting. Flay you just with a phrase. But it ’ s the time with my mother that was the nightmare. She had a horrible childhood, but she sure as hell didn ’ t make any effort to overcome that in her relations with us. You know, it was really horrifying to my mother that I was not popular. But I was also really awkward because she had been undermining my confidence for all my life. She would pick apart your personality in detail. I was lazy, I looked bad, I acted bad, I was awkward. Emotionally, just out for blood.

At a certain point in the night, I would know,”OK, the situation is dangerous. I need to get outta here.”I had this fantasy when I was in fourth grade that I vividly remember that some kind of outside agency was gonna come in and determine from all the salt on my pillow that I was crying myself to sleep every night and do something about it. No, it was awful. It was awful. And that was the point where Ilon came into my life. - Ilon, do you need anything to eat? - I don ’ t think so, but look up this wallpaper, Waverly Fabrics, and see if they have a rose pattern. - OK. Oh, what about this one here? - That ’ s not it. - No? - No, first of all, I was an art dir... I was a creative director at an advertising agency. The first thing I know is what I ’ m looking at. Ilon the girl. I don ’ t remember particularly being into advertising, but somehow I found myself in it. And I seemed to be very good at it. First of all, I ’ d always listened to everybody. And I noticed that I saw things differently than other people. - Women could get somewhere in advertising, but they were marginalized. - Women were given a lesser role. All the men were always arguing with you and always taking credit. - And one of the things that was coming through loud and clear is that women are for men. - Glory, glory, hallelujah Glory, glory, hallelujah Glory, glory, hallelujah His truth is marching on - Men were always the ones who everyone was pleasing. - What will your eyes say to him tonight? Wait till the fellas take a look at this plunging backline. - You look at **ads** from that period, and it ’ s jaw-dropping. - This is one of our nation ’ s most beautiful sights, the kind of girl who keeps her figure. - But that ’ s what feminism was pushing back against. - I was a feminist. I knew women were equal if not superior to men from the time I was a little girl. - And in advertising, she wanted to be a trailblazer that way. At one point, she worked in the same agency with my father, and that ’ s how they met. - He was a good writer, and I was a good writer. I liked the fact that he was good at what he did and cared about it. We got engaged, and then we got married. - I ’ m sitting there in the background. I ’ m grinning. I ’ m so thrilled because I adored her instantly. I thought she was **super** cool. And she treated me like I mattered. - She was a sweet little girl, and she deserved some caretaking. My mother was kinda terrible, so I knew that she needed support, and I gave her support. And that was what we did. - She was interested in me. She listened to me. Treated me like I was lovable. - She didn ’ t have anyone to recognize her being, to support her, to see who she could be, who she was. - She was very interested in what I would do with my life, and the assumption was that it would be something worthwhile. She just went way, way beyond the call of duty to make me feel like I was worth it. And she did that for a lot of other women, too. - I was working at McCann Erickson. We were given an assignment. L ’ Oréal, they wanted a campaign for Preference, it ’ s a hair color. - She ’ s in this room full of men brainstorming for this **ad**. - The men were busy flirting with the women, telling us we all looked like models and we didn ’ t look like advertising people, you know? - They ’ re like,”Oh, this woman is gonna be standing by an open window, and the wind will be blowing the curtains.”And she ’ s thinking, you know, this is just making the woman totally an object. - I was feeling angry. I ’ m not interested in writing anything about looking good for men. Fuck ’ em. Fuck you too. - Why fuck me? - Well, you ’ re a man. - Yeah, but I ’ m here trying to help you tell your story. - Yeah, that ’ s good. I like that part. I was pissed. We weren ’ t there to just dance for the men. - It is what she said. She just thought,”Fuck you.”And wrote this. - I just opened my eyes and knew. Because I ’ m worth it. The campaign got approved, but all the men were always telling you that you were doing something wrong or they knew how to do it better. - The agency insisted on a male voiceover. - She uses the most expensive hair color in the world, Preference by L ’ Oréal. - You know, it was just totally different. - Smooth and silky, but with body. It feels good against her neck. Actually, she doesn ’ t mind spending more for L ’ Oréal. Because she ’ s worth it. - It was wrong. This was not for men but for women and for other human beings. - I use the most expensive hair color in the world, Preference by L ’ Oréal. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color, I expect great color. What ’ s worth more to me is the way my hair feels. Smooth and silky, but with body. It feels good against my neck. Actually, I don ’ t mind spending more for L ’ Oréal. Because I ’ m worth it. - I wrote that. - It was everywhere. It was a national campaign. - Yes, it ’ s expensive, but I ’ m worth it. - What they realized after they started running it was that that was only the tip of the iceberg. - It was supporting women, finally. - And I ’ m worth it. And I ’ m worth it. I ’ m worth it. - It ’ s a four-word feminist manifesto, defying this social pressure to think about yourself in relation to men. - You ’ re worth it. I ’ m worth it. I ’ m worth it. - It was a very big success. - At a certain point, L ’ Oréal adopted it as the tagline for their entire business. - Parce que je le vaux bien. - Even today, that ’ s their motto. - Because I ’ m worth it. - ’ Cause you ’ re worth it. - You ’ re worth it. - We ’ re worth it. - And I ’ m worth it. - She wrote the most influential tagline in women ’ s beauty advertising. - I ’ m worth it. - It ’ s magic, that phrase. - Those words just get better with age, don ’ t they? - I ’ m worth it. Because you are and always will be. - She came up with that line because she believed it. She believed it about herself, and she believed that women should believe it. She did that for me. You know, even after they split up, she was still a big part of my life. I ’ ve always kind of felt like I dodged a bullet that I have watched take down other members of my family. She saved me. And today, I am a professor of English at Williams College. I teach Victorian literature, and I also teach women ’ s, gender and sexuality studies. She was my mother. She also taught me how to be a mother. I had been very uncertain about motherhood for myself because my role model for that was so nightmarish. But Ilon really taught me, I can do this. I can be a good mother. She made me believe that I was worth it. And I was not either the first or the last. - I have to go away now. Go to my room. - It ’ s been lovely. - It ’ s been lovely. Oh... I ’ m not interested in advertising. I don ’ t give a shit. It wasn ’ t like it was my whole life. It was just a portion. It ’ s about humans ; it ’ s not about advertising. It ’ s about caring for people because... we ’ re all worth it, or no one is worth it.

Text Sample 257: POS Dim 2 – 1990 – Score 1.00 – 1990/t000287.txt

I ’ m going to go right into the slides. And all I ’ m going to try and prove to you with these slides is that I do just very straight stuff. And my ideas are â□□ in my head, anyway â□□ they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the left-hand side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage the neighborhood â one made accommodations. I tried to be inclusive, to include the buildings in the neighborhood, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of stuff and folded it up to put it in the wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at postmodernism â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker. And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And the client said,”Why no fish?”And so I made a drawing with a fish, and I left Japan. Three weeks later, I received a complete set of drawings saying we ’ d won the competition. Now, it ’ s hard to do. It ’ s hard to translate a fish form, because they ’ re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ’ t do it, and so that made it even more exciting. But he was right â I couldn ’ t do the tail. I started to get the head OK, but the tail I couldn ’ t do. It was pretty hard. The thing on the right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with the context again. Now, if you saw a picture of this as it was published in Architectural Record â they didn ’ t show the context, so you would think,”God, what a pushy guy this is.”But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ’ t find it. So... As for craft and technology and all those things that you ’ ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent drawings to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made the drawings of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ' s one of the nicest pieces I ' ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ' t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ' m getting good at temporary â and we put a conference room in that ' s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ' s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ' t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ' s offices and I made a pedestal for a sculpture. And he didn ' t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ' s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ' ve been friends for a long time. And we ' ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called"Il Corso del Coltello"â the Swiss Army knife. And most of the imagery is â Claes ', but those two little boys are my sons, and they were Claes ' assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ' s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ' ve been known to mess with things like chain link fencing. I do it because it ' s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ' s so much denial about them. People hate it. And I ' m fascinated with that, which, like the paper furniture â it ' s one of those materials. And I ' m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ' s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to the building on the left. And I still think the Time magazine picture will be of the binoculars, you know, leaving out the â what the hell. I use a lot of metal in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,"If I can ' t get the craft that I want, I ' ll use the craft I can get."There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the metal because it was a way of building a building that was a sculpture. And it was all of one material, and the metal could go on the roof as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and stuff. I was given an opportunity to design an exhibit for the metalworkers ’ unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future metal buildings, etc. etc. And it ’ s working very well to have these people, these craftsmen, interested in it. I just tell the stories. It ’ s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The metal continued into a building â Herman Miller, in Sacramento. And it ’ s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ’ s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ’ s out in the middle of nowhere, and you approach it. It ’ s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ’ s aesthetic. Everybody ’ s trying to get the panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ’ s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ’ s all about [a] metaphor for a city, maybe. And so Stanley did the little dome thing. And we did it over the phone and by fax. He would send me a fax and show me something. He ’ d made a building with a dome and he had a little tower. I told him,”No, no, that ’ s too ongepotchket. I don ’ t want the tower.”So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going back and forth over a couple of weeks ’ period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,”David and Goliath.”And that ’ s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and cleaned it up and fixed it up and used the kind of â I thought â the language of the neighborhood, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ’ t normally think I would fit them. The detailing was very careful with the lead copper and making panels and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some film festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â a little shopping center. And this is a laser laboratory at the University of Iowa, in which the fish comes back as an abstraction in the back. It ’ s the support labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the curved part there ’ s all the mechanical equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ’ t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ’ ve been building it so long I don ’ t remember where it is. It ’ s in the West Valley. And we started with the stream and built the house along the stream â dammed it up to make a lake. These are the models. The reality, with the lake â the workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ’ s hard to get perfect corners and stuff. That big metal thing is a passage, and in it is â you go downstairs into the living room and then down into the bedroom, which is on the right. It ’ s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â they asked Philip to do it. He was too busy. He didn ’ t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ’ t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ’ ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ’ s made of metal and the brown stuff is Fin-Ply â it ’ s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ’ t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ’ s Burnham Mall, on the left. It ’ s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ’ s a railroad track. This is the city hall over here somewhere, and the courthouse. And the centerline of the mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ’ re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this stuff at the bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the centerline of this thing â we ’ d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ’ s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ’ 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the back end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake.

And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ’ s the living room. Servants ’ quarters. A guest bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is back in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a fun thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we **struggled** with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it ain ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the conference facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais des Sports, the Gare de Lyon over here. The Minister of Finance â the guy that moved from the Louvre â goes in here. There ’ s a new library across the river. And back in here, in this already treed park, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan coupe. And I **struggled** with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in

French limestone, except for this metal piece. And it faces into a park. And the idea was to make this express the energy of this. On the side facing the street it ’ s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in the Hamptons. And it ’ s got a fish. And I keep thinking,”This is going to be the last fish.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s the living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in the budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of the guys that presented to you earlier. We ’ ve had a lot of fun working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ’ s perpendicular to the train station, to the route of the train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good slides to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â you see it as all of these pieces accrue and become part of an overall neighborhood. It ’ s plaster and just zinc. And you wonder, if this is a museum, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and stuff. And from the building in the back, you see it as an iceberg floating by in the hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is the Disney Hall â the concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a compositional relationship between us that would strengthen both of us. And the plan of this â it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â to MOCA, etc. The acoustician in the competition gave us criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But everybody liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were the three buildings that were the ideal â the Concertgebouw, Boston and Berlin. Everybody liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new acoustician, he told us this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an acoustical, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the sloping outside walls, which the acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on the way to being â and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with laser stuff, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, **struggling** with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and **struggle** with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the foot of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the foot of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th- century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ’ t as good as the stuff, and so it failed. It should have been the other way around â the food should have been good first. It didn ’ t work. Thank you very much.

4 NEG Dim 2

Text Sample 258: NEG Dim 2 – 1990 – Score 1.00 – 1990/t000287.txt

I ’ m going to go right into the slides. And all I ’ m going to try and prove to you with these slides is that I do just very straight stuff. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the left-hand side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage the neighborhood â one made accommodations. I tried to be inclusive, to include the buildings in the neighborhood, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of stuff and folded it up to put it in the wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at postmodernism â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker. And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And the client said,”Why no fish?”And so I made a drawing with a fish, and I left Japan.

Three weeks later, I received a complete set of drawings saying we ’ d won the competition. Now, it ’ s hard to do. It ’ s hard to translate a fish form, because they ’ re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ’ t do it, and so that made it even more exciting. But he was right â I couldn ’ t do the tail. I started to get the head OK, but the tail I couldn ’ t do. It was pretty hard. The thing on the right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with the context again. Now, if you saw a picture of this as it was published in Architectural Record â they didn ’ t show the context, so you would think,”God, what a pushy guy this is.”But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ’ t find it. So... As for craft and technology and all those things that you ’ ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent drawings to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made the drawings of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ’ s one of the nicest pieces I ’ ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ’ t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ’ m getting good at temporary â and we put a conference room in that ’ s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ’ s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ’ t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ’ s offices and I made a pedestal for a sculpture. And he didn ’ t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ’ s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ’ ve been friends for a long time. And we ’ ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called”Il Corso del Coltello”â the Swiss Army knife. And most of the imagery is â Claes ’, but those two little boys are my sons, and they were Claes ’ assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ’ s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ’ ve been known to mess with things like chain link fencing. I do it because it ’ s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ’ s so much denial about them. People hate it. And I ’ m fascinated with that, which, like the paper furniture â it ’ s one of those materials. And I ’ m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ’ s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to the building on the left. And I still think the Time magazine picture will be of the binoculars, you know, leaving out the â what the hell. I use a lot of metal in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,"If I can ’ t get the craft that I want, I ’ ll use the craft I can get."There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the metal because it was a way of building a building that was a sculpture. And it was all of one material, and the metal could go on the roof as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and stuff. I was given an opportunity to design an exhibit for the metalworkers ’ unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future metal buildings, etc. etc. And it ’ s working very well to have these people, these craftsmen, interested in it. I just tell the stories. It ’ s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The metal continued into a building â Herman Miller, in Sacramento. And it ’ s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ’ s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ’ s out in the middle of nowhere, and you approach it. It ’ s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ’ s aesthetic. Everybody ’ s trying to get the panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ’ s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ’ s all about [a] metaphor for a city, maybe. And so Stanley did the little dome thing. And we did it over the phone and by fax. He would send me a fax and show me something. He ’ d made a building with a dome and he had a little tower. I told him,"No, no, that ’ s too ongepotchket. I don ’ t want the tower."So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going back and forth over a couple of weeks ’ period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,"David and Goliath."And that ’ s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and cleaned it up and fixed it up and used the kind of â I thought â the language of the neighborhood, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ’ t normally think I would fit them. The detailing was very careful with the lead copper and making panels and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some film festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â a little shopping center. And this is a laser laboratory at the University of Iowa, in which the fish comes back as an abstraction in the back. It ’ s the support labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the curved part there ’ s all the mechanical equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ’ t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ’ ve been building it so long I don ’ t remember where it is. It ’ s in the West Valley. And we started with the stream and built the house along the stream â dammed it up to make a lake. These are the models. The reality, with the lake â the workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ’ s hard to get perfect corners and stuff. That big metal thing is a passage, and in it is â you go downstairs into the living room and then down into the bedroom, which is on the right. It ’ s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â they asked Philip to do it. He was too busy. He didn ’ t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ’ t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ’ ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ’ s made of metal and the brown stuff is Fin-Ply â it ’ s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ’ t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ’ s Burnham Mall, on the left. It ’ s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ’ s a railroad track. This is the city hall over here somewhere, and the courthouse. And the centerline of the mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ’ re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this stuff at the bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the centerline of this thing â we ’ d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ’ s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ’ 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the back end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake. And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ’ s the living room. Servants ’ quarters. A guest bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is back in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a fun thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it ain ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the conference facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais des Sports, the Gare de Lyon over here. The Minister of Finance â the guy that moved from the Louvre â goes in here. There ’ s a new library across the river. And back in here, in this already treed park, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan coupe. And I struggled with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in French limestone, except for this metal piece. And it faces into a park. And the idea was to make this express the energy of this. On the side facing the street it ’ s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in the Hamptons. And it ’ s got a fish. And I keep thinking,”This is going to be the last fish.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s the living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in the budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of the guys that presented to you earlier. We ’ ve had a lot of fun working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ’ s perpendicular to the train station, to the route of the train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good slides to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â you see it as all of these pieces accrue and become part of an overall neighborhood. It ’ s plaster and just zinc. And you wonder, if this is a museum, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and stuff. And from the building in the back, you see it as an iceberg floating by in the hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is the Disney Hall â the concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing

Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a compositional relationship between us that would strengthen both of us. And the plan of this â it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â to MOCA, etc. The acoustician in the competition gave us criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But everybody liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were the three buildings that were the ideal â the Concertgebouw, Boston and Berlin. Everybody liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new acoustician, he told us this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an acoustical, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the sloping outside walls, which the acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on the way to being â and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with laser stuff, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, struggling with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the foot of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the foot of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th- century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ' t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ' s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ' t as good as the stuff, and so it failed. It should have been the other way around â the food should have been good first. It didn ' t work. Thank you very much.

Text Sample 259: NEG Dim 2 – 2025 – Score 2.00 – 2025/t003001.txt

- I don ' t remember exactly, but... I use the most expensive hair color in the world. It ' s not that I care about the money, it ' s that I care about my hair. It ' s not just the color. I expect great color. What ' s worth more to me is the way my hair feels. Feels good around my neck. I can ' t remember it. I can ' t finish it all. If only we had more time. - Why don ' t we have more time? - I ' m dying. I ' m croaking! - It ' s tough, coming home. But this is what you do for a parent that you love. But, you know? She ' s not my mother. Technically. In my own family, there ' s a lot of dysfunction, a lot of mental illness, a lot of substance abuse. Every generation. My parents divorced when I was about five. My mother moved us out of state, so we had to fly to New York to see my father ' cause he was already working in advertising, and New York was the only place to be. - Many of the commercials you see on television begin life here. - Advertising is one of the main sources of messaging about who you should be. - Nowadays, most of you women lead two lives, combining homemaking with the role of glamorous hostess. - And in the ' 60s, the ads were often made from a male point of view. - In the pre-production meeting, everybody who has anything to do with the commercial is there. - It was mostly men. These superstar art directors and writers. Like my father, Gene Case."Mad Men"was like home movies, and he was Don Draper. He was very witty, but I don ' t think my father realized how powerful his words were. He knew it professionally. But with his kids, he could be incredibly cutting. Flay you just with a phrase. But it ' s the time with my mother that was the nightmare. She had a horrible childhood, but she sure as hell didn ' t make any effort to overcome that in her relations with us. You know, it was really horrifying to my mother that I was not popular. But I was also really awkward because she had been undermining my confidence for all my life. She would pick apart your personality in detail. I was lazy, I looked bad, I acted bad, I was awkward. Emotionally, just out for blood. At a certain point in the night, I would know,"OK, the situation is dangerous. I need to get outta here."I had this fantasy when I was in fourth grade that I vividly remember that some kind of outside agency was gonna come in and determine from all the salt on my pillow that I was crying myself to sleep every night and do something about it. No, it was awful. It was awful. And that was the point where Ilon came into my life. - Ilon, do you need anything to eat? - I don ' t think so, but look up this wallpaper, Waverly Fabrics, and see if they have a rose pattern. - OK. Oh, what about this one here? - That ' s not it. - No? - No, first of all, I was an art dir... I was a creative director at an advertising agency. The first thing I know is what I ' m looking at. Ilon the girl. I don ' t remember particularly being into advertising, but somehow I found myself in it. And I seemed to be very good at it. First of all, I ' d always listened to everybody. And I noticed that I saw things differently than other people. - Women could get somewhere in advertising, but they were marginalized. - Women were given a lesser role. All the men were always arguing with you and always taking credit. - And one of the things that was coming through loud and clear is that women are for men. - Glory, glory, hallelujah Glory, glory, hallelujah Glory, glory, hallelujah His truth is marching on - Men were always the ones who everyone was pleasing. - What will your eyes say to him tonight? Wait till the fellas take a look at this plunging backline. - You look at ads from that period, and it ' s jaw-dropping. - This is one of our nation ' s most beautiful sights, the kind of girl who keeps her figure. - But that ' s what feminism was pushing back against. - I was a feminist. I knew women were equal if not superior to men from the time I was a little girl. - And in advertising, she wanted to be a trailblazer that way. At one point, she worked in the same agency with my father, and that ' s how they met. - He was a good writer, and I was a good writer. I liked the fact that he was good at what he did and cared about it. We got engaged, and then we got married. - I ' m sitting there in the background. I ' m grinning. I ' m so thrilled because I adored her instantly. I thought she was super cool. And she treated me like I mattered. - She was a sweet little girl, and she deserved some caretaking. My mother was kinda terrible, so I knew that she needed support, and I gave her support. And that was what we did. - She was interested in me. She listened to me. Treated me like I was lovable. - She didn ' t have anyone to recognize her being, to support her, to see who she could be, who she was. - She was very interested in what I would do with my life, and the assumption was that it would be something worthwhile. She just went way, way beyond the call of duty to make me feel like I was worth it. And she did that for a lot of other women, too. - I was working at McCann Erickson. We were given an assignment. L ' Oréal, they wanted a campaign for Preference, it ' s a hair color. - She ' s in this room full of men brainstorming for this ad. - The men were busy flirting with the women, telling us we all looked like models and we didn ' t look like advertising people, you know? - They ' re like,"Oh, this woman is gonna be standing by an open window, and the wind will be blowing the curtains."And she ' s thinking, you know, this is just making the woman totally an object. - I was feeling angry. I ' m not interested in writing anything about looking good for men. Fuck ' em. Fuck you too. - Why fuck me? - Well, you ' re a man. - Yeah, but I ' m here trying to help you tell your story. - Yeah, that ' s good. I like that part. I was pissed. We weren ' t there to just dance for the men. - It is what she said. She just thought,"Fuck you."And wrote this. - I just opened my eyes and knew. Because I ' m worth it. The campaign got approved, but all the men were always telling you that you were doing something wrong or they knew how to do it better. - The agency insisted on a male voiceover. - She uses the most expensive hair color in the world, Preference by L ' Oréal. - You know, it was just totally different. - Smooth and silky, but with body. It feels good against her neck. Actually, she doesn ' t mind spending more for L ' Oréal. Because she ' s worth it. - It was wrong. This was not for men but for women and for other human beings. - I use the most expensive hair color in the world, Preference by L ' Oréal. It ' s not that I care about the money, it ' s that I care about my hair. It ' s not just the color, I expect great color. What ' s worth more to me is the way my hair feels. Smooth and silky, but with body. It feels good against my neck. Actually, I don ' t mind spending more for L ' Oréal. Because I ' m worth it. - I wrote that. - It was everywhere. It was a national campaign. - Yes, it ' s expensive, but I ' m worth it. - What they realized after they started running it was that that was only the tip of the iceberg. - It was supporting women, finally. - And I ' m worth it. And I ' m worth it. I ' m worth it. - It ' s a four-word feminist manifesto, defying this social pressure to think about yourself in relation to men. - You ' re worth it. I ' m worth it. I ' m worth it. - It was a very big success. - At a certain point, L ' Oréal adopted it as the tagline for their entire business. - Parce que je le vaux bien. - Even today, that ' s their motto. - Because I ’ m worth it. - ’ Cause you ’ re worth it. - You ’ re worth it. - We ’ re worth it. - And I ’ m worth it. - She wrote the most influential tagline in women ’ s beauty advertising. - I ’ m worth it. - It ’ s magic, that phrase. - Those words just get better with age, don ’ t they? - I ’ m worth it. Because you are and always will be. - She came up with that line because she believed it. She believed it about herself, and she believed that women should believe it. She did that for me. You know, even after they split up, she was still a big part of my life. I ’ ve always kind of felt like I dodged a bullet that I have watched take down other members of my family. She saved me. And today, I am a professor of English at Williams College. I teach Victorian literature, and I also teach women ’ s, gender and sexuality studies. She was my mother. She also taught me how to be a mother. I had been very uncertain about motherhood for myself because my role model for that was so nightmarish. But Ilon really taught me, I can do this. I can be a good mother. She made me believe that I was worth it. And I was not either the first or the last. - I have to go away now. Go to my room. - It ’ s been lovely. - It ’ s been lovely. Oh... I ’ m not interested in advertising. I don ’ t give a shit. It wasn ’ t like it was my whole life. It was just a portion. It ’ s about humans ; it ’ s not about advertising. It ’ s about caring for people because... we ’ re all worth it, or no one is worth it.

Text Sample 260: NEG Dim 2 – 2025 – Score 2.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in the world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the air. In all my travels, I ’ ve been struck by so many things. I ’ ve observed how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ’ ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced us to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ’ s hard to see the big picture right now. It ’ s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think back to my travels. I ’ ve seen how different we are, how diverse and distinct. But I ’ ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ’ m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman”A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story is the dominant sound of our lives.”- Joseph Campbell I ’ ve loved telling stories all of my life. I ’ ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 261: NEG Dim 2 – 2025 – Score 5.00 – 2025/t003466.txt

Can you paint with all the colors of the wind? I love this song from the first time I heard it. I was 14, growing up in public housing in western Sydney. So maybe a little too old for this Disney classic from”Pocahontas.”But it spoke to me. I saw something of myself in it. At this stage, I think I knew I was queer. I mean, come on, I was rocking out to Disney. And look at those glasses. Seriously, what a queer kid. But I was also from a Pacific Indigenous heritage with conservative views and values. I didn ’ t feel like I could fit into any of the boxes that were being presented to me. Straight, gay, Fijian, Australian. I couldn ’ t pick just one color to paint with. Prior to colonization, sexuality amongst many of our Pacific islands across the region was fluid. Men would have sex with men without the stigma or shame afforded it today. Sexual intimacy was a way of being able to connect socially and relationally. Sex wasn ’ t just for procreation. Or in marriage. It wasn ’ t exclusively monogamous. Just because you had sex with a man didn ’ t define your sexuality. It didn ’ t make you an outcast. But as Europeans started to colonize the Pacific region, they brought Christian missionaries with them. They told us to cover up our beautiful bodies. We were taught shame. We were told to stop practicing Pacifica cultures and values that were seen as provocative. To a Western and white perspective, our non-binary sexuality was judged as tribal, outdated, sinful and impossible to practice alongside a Christian faith. Still today, those who are Indigenous to the Pacific are negatively impacted by whiteness. That”white is right, West is best.”I was brought up to espouse strong Christian beliefs, views and values which would shape my burgeoning identity as an adult. From these staunch conservative views, I was taught to fear the queer. Those people that were seen as unusual and fruity. And in essence, I grew to despise a part of myself. I sought to separate my queerness from my identity. I could never see myself as part of that world. Growing up, biracial and bisexual constantly challenged the way I saw the world around me. I was unsure about my thinking, feelings and behaviors around liking males and females. I was unsure about my thinking, feelings and behaviors when it came to whether I needed to be a white Australian or a Fijian person. And whether this needed to be framed within a binary to choose one over the other. In this way, I don ’ t think that my story is unique. There ’ s plenty of people in the queer world that have escaped conservative upbringings, religions and cultures. But this is only part of my story. Another common story in the queer community is the coming-out story. When did you tell your parents? How did they react? How it liberated you or ultimately separated you from your family. But this didn ’ t feel like an option for me either. I believe that embedded in these particular stories, these tropes, is a part of dominant queer culture that made it very hard for me to access the queer world. For example, when I started to go to gay clubs, I would be lining up at the front of the club and often I would get door staff come up to me and go,”Sorry, you do know this is a gay club, right?”From these lived experiences, I feel that there is a whiteness that pervades queer culture. And I didn ’ t want to give up my identity and my appearance as a Pacific indigenous person to be queer. Pacifica cultures really, really value and esteem the collective above the individual. For Pacifica people, our families are inextricably connected to our own sense of self-agency, self-determination, self-esteem, self-love and self-worth. Our own individual identities are meshed and intertwined with our family ’ s reputation and standing in the community. Our mind, body and souls are inextricably connected to the broader community, to our families, to ourselves and to others. I didn ’ t want to come out to potentially ostracize myself, to make a point to separate myself from my Pacific and faith communities. I couldn ’ t then see my authentic self in any of these spaces. And a quick shout out to my son on the left-hand side. Round of applause for my son. He said to me this morning,”Daddy, make sure you mention me.”So I was like,”OK, son. Alright.”So there you go for the recording, son. And the rest are my other family, that ’ s up there as well. Across queer cultures and the broader community, we have learned to embrace our gender and sexual differences. We encourage individuals to be allies for each other and to support safe spaces for those that are seen to not fit in. We empower and really encourage individuals to really develop and express their self-agency, their self-determination, their self love, their self-worth. We boldly go into spaces striving to bring others with us to ensure that we can create inclusive societies for everyone. However, the dominant Western view in queer cultures is still based on the individual. Me, myself and I. And my individual pursuits towards happiness. As a queer Indigenous Pacific person, I feel that there are parts of queer culture that is white, feels very white. We may fail to see the importance of bringing our families and our cultural communities along our journey as queer people. It ’ s either the queer way or the highway. It ’ s one or the other. Young or old. Rich or poor. Smart or stupid. Masculine or feminine. Fit or fat. These binaries create a sense of us and them, those people. A binary approach and position that we generally in Western white societies use to determine the norm. We create, conflate, perpetuate binaries that create divisions in our society. Saints or sinner, straight or gay. I see it like this. Binaries are a very white and Western way of thinking, feeling and behaving. As individuals today, we live within these binaries and create identities that then create these labels. We use these labels to then define and confine people and places. We are one or the other. But what we fail to see is that our identities are more than a set of binaries. Our own identities are comprised of intersecting identifiers that contributes to who you are, including our age, class, color, education, gender, size, spirituality, sexuality, and so, so much more. We need to decolonize queerness, to live beyond”white is right, West is best.”Because if we don ’ t, we ’ re going to continue to create silos and divisions in our societies that don ’ t create inclusive and safe spaces for everyone. For Indigenous queer people, they shape our connection to community and culture, our families. If we don ’ t include that within our understanding of who we are, even as queer people, then we ’ re not able to be, again, our true and authentic selves. We ’ re not able to then be our fabulous selves as part of the queer community. And it ’ s really, really interesting because I believe that we can learn so much from pre-colonial cultures like those that were practiced in the Pacific, that provide a different way of looking at a society. A queer way that pre-dates all the privileges that go into binary ways of thinking, feeling and behaving. Across many of our Pacifica languages we have a lot of key concepts that help solidify and support a shared understanding of the collective. For example, in Fijian,”solesolevaki”is all about reciprocal living, where my wellbeing is your wellbeing. I will actively seek to ensure that your well-being and every part of your life is catered for, knowing that you will do the same for me. We strive to enact”veiqaravi,”as tatted here on my hand, to serve others. We strive to also embed and embold and empower people to practice”veirokovi,”respect for self and others where we are relationally driven, where that is so important to our spaces and places. My sense of ability to actually put this into practice, my ability to actually understand these Indigenous perspectives has greatly helped shape and reconcile my queer identity today. But what ’ s really interesting is that it ’ s not just queer communities that need to decolonize itself. It ’ s all of us as a broader society that needs to be part of a shared conversation to decolonize, deconstruct and disrupt binary ways of thinking, feeling and behaving. So how can we practically do that? Practice cultural humility, where you are constantly learning and embracing differences, where you get over yourselves and strive to learn about other people ’ s differences and how that can be meaningfully included in the way in which you create safe spaces for everyone. I also believe it ’ s possible to bring Indigenous perspectives into Western and white workplaces. For example, I ’ ve introduced the concept of”talanoa”in all of my work meetings. Talanoa is all about being able to create a safe space where everyone is able to contribute to a collective, collegial and collaborative conversation. Where people are able to bring their authentic selves into these spaces. Literally, I will chair meetings where I have no firm set agenda items and I literally would just hold space for participants just to bring themselves. So you can tell I ’ m a social worker, right? Safe space, safe space, safe space. It ’ s OK, it ’ s OK. But it ’ s through these safe spaces that people can bring their multiple, intersecting identifiers, including their age, their class, color, education, gender, size, spirituality, sexuality and so, so much more. I invite queer communities to come talanoa with Pacific Indigenous Communities to create robust and resilient queer cultures. I have been able to greatly reconcile my queer and Pacifica identities, which has now really been embraced to help shape how I see the world around me today. And it ’ s through these shared perspectives and approaches that we can truly embrace cultural diversity and its differences. And to know that people can live beyond those binaries and those labels we continue to perpetuate and perpetrate against others. It is a commitment from queer communities to decolonize queerness. It ’ s a commitment from me, from you and all of us to decolonize Western and white societies and to live beyond the binary way of thinking, feeling and behaving. I genuinely believe that it ’ s through this common goal that we can truly learn how to paint with all the colors of the wind.”Vinaka vakalevu”and thank you.

Text Sample 262: NEG Dim 2 – 2009 – Score 1.00 – 2009/t000061.txt

What ’ s happening to the climate? It is unbelievably bad. This is, obviously, that famous view now of the Arctic, which is likely to be gone at this point in the next three or four or five years. Very, very, very scary. So we all look at what we can do. And when you look at the worldwide sources of CO2, 52 percent are tied to buildings. Only nine percent is passenger cars, interestingly enough. So we ran off to a sushi bar. And at that sushi bar we came up with a great idea. And it was something called EcoRock. And we said we could redesign the 115-year-old gypsum drywall process that generates 20 billion pounds of CO2 a year. So it was a big idea. We wanted to reduce that by 80 percent, which is exactly what we ’ ve done. We started R&D in 2006. Decided to use recycled content from cement and steel manufacturing. There is the inside of our lab. We haven ’ t shown this before. But our people had to do some 5, 000 different mixes to get this right, to hit our targets. And they worked absolutely very, very, very hard. So then we went forward and built our production line in China. We don ’ t build this production equipment any longer in the U. S., unfortunately. We did the line install over the summer. We started right there, with absolutely nothing. You ’ re seeing for the first time, a brand new drywall production line, not made using gypsum at all. That ’ s the finished production line there. We got our first panel out on December third. That is the slurry being poured onto paper, basically. That ’ s the line running. The exciting thing is, look at the faces of the people. These are people who worked this project for two to three years. And they are so excited. That ’ s the first board off the line. Our Vice President of Operation kissing the board. Obviously very, very excited. But this has a huge, huge impact on the environment. We signed the first panel just a few weeks after that, had a great signing ceremony, leading to people hopefully using these products across the world. And we ’ ve got Cradle-to-Cradle Gold on this thing. We happened to win, just recently, the Green Product of the Year for”The Re-Invention of Drywall,”from Popular Science. Thank you. Thank you. So here is what we learned : 8, 000 gallons of gas equivalent to build one house. You probably had no idea. It ’ s like driving around the world six times. We must change everything. Look around the room : chairs, wood, everything around us has to change or we ’ re not going to lick this problem. Don ’ t listen to the people who say you can ’ t do this, because anyone can. And these job losses, we can fix them with green-collar jobs. We ’ ve got four plants. We ’ re building this stuff around the country. We ’ re going as fast as we can. Two and a half million cars worth of gypsum, you know, CO2 generated. Right? So what will you do? I ’ ll tell you what I did and why I did it. And I know my time ’ s up. Those are my kids, Natalie and David. When they have their kids, 2050, they ’ d better look back at Grandpa and say,”Hey, you gave it a good shot. You did the best you could with the team that you had.”So my hope is that when you leave TED, you will look at reducing your carbon footprint in however you can do it. And if you don ’ t know how, please find me — I will help you. Last but not least, Bill Gates, I know you invented Windows. Wait till you see, maybe next year, what kind of windows we ’ ve invented. Thank you so much.

Text Sample 263: NEG Dim 2 – 2009 – Score 1.00 – 2009/t000058.txt

This is a world-changing invention. The smoke alarm has saved perhaps hundreds of thousands of lives, worldwide. But smoke alarms don ’ t prevent fires. Every year in the USA, over 20, 000 are killed or injured with 350, 000 home fires. And one of the main causes of all these fires is electricity. What if we could prevent electrical fires before they start? Well, a couple of friends and I figured out how to do this. So how does electricity ignite residential fires? Well it turns out that the main causes are faulty and misused appliances and electrical wiring. Our invention had to address all of these issues. So what about circuit breakers? Well, Thomas Edison invented the circuit breaker in 1879. This is 130-year-old technology, and this is a problem, because over 80 percent of all home electrical fires start below the safety threshold of circuit breakers. Hmmm... So we considered all of this. And we realized that electrical appliances must be able to communicate directly with the power receptacle itself. Any electrical device — an appliance, an extension cord, whatever — must be able to tell the power outlet,”Hey, power outlet, I ’ m drawing too much current. Shut me off now, before I start a fire.”And the power outlet needs to be smart enough to do it. So here is what we did. We put a 10-cent digital transponder, a data tag, in the appliance plug. And we put an inexpensive, wireless data reader inside the receptacle so they could communicate. Now, every home electrical system becomes an intelligent network. The appliance ’ s safe operating parameters are embedded into its plug. If too much current is flowing, the intelligent receptacle turns itself off, and prevents another fire from starting. We call this technology EFCI, Electrical Fault Circuit Interrupter. Okay, two more points. Every year in the USA, roughly 2, 500 children are admitted to emergency rooms for shock and burn injuries related to electrical receptacles. And this is crazy. An intelligent receptacle prevents injuries because the power is always off, until an intelligent plug is detected. Simple. Now, besides saving lives, perhaps the greatest benefit of intelligent power is in its energy savings. This invention will reduce global energy consumption by allowing remote control and automation of every outlet in every home and business. Now you can choose to reduce your home energy bill by automatically cycling heavy loads like air conditioners and heaters. Hotels and businesses can shut down unused rooms from a central location, or even a cell phone. There are 10 billion electrical outlets in North America alone. The potential energy savings is very, very significant. So far we ’ ve applied for 414 patent claims. Of those, 186 have been granted : 228 are in process. And I ’ m pleased to announce that just three weeks ago we received our first international recognition, the 2009 CES Innovation Award. So, to conclude, intelligent power can, globally, save thousands of lives, prevent tens of thousands of injuries, and eliminate tens of billions of dollars in property damage every single year, while significantly reducing global energy consumption. In the spirit of this year ’ s TED Conference, we think this is a powerful, world-changing invention. And I ’ d like to thank Chris for this opportunity to unveil our technology with you, and soon the world. Thank you.

Text Sample 264: NEG Dim 2 – 2009 – Score 1.00 – 2009/t000057.txt

I ’ m here because I have a very important message : I think we have found the most important factor for success. And it was found close to here, Stanford. Psychology professor took kids that were four years old and put them in a room all by themselves. And he would tell the child, a four-year-old kid,”Johnny, I am going to leave you here with a marshmallow for 15 minutes. If, after I come back, this marshmallow is here, you will get another one. So you will have two.”To tell a four-year-old kid to wait 15 minutes for something that they like, is equivalent to telling us,”We ’ ll bring you coffee in two hours.”Exact equivalent. So what happened when the professor left the room? As soon as the door closed... two out of three ate the marshmallow. Five seconds, 10 seconds, 40 seconds, 50 seconds, two minutes, four minutes, eight minutes. Some lasted 14- and-a-half minutes. Couldn ’ t do it. Could not wait. What ’ s interesting is that one out of three would look at the marshmallow and go like this... Would look at it. Put it back. They would walk around. They would play with their skirts and pants. That child already, at four, understood the most important principle for success, which is the ability to delay gratification. Self-discipline : the most important factor for success. 15 years later, 14 or 15 years later, follow-up study. What did they find? They went to look for these kids who were now 18 and 19. And they found that 100 percent of the children that had not eaten the marshmallow were successful. They had good grades. They were doing wonderful. They were happy. They had their plans. They had good relationships with the teachers, students. They were doing fine. A great percentage of the kids that ate the marshmallow, they were in trouble. They did not make it to university. They had bad grades. Some of them dropped out. A few were still there with bad grades. A few had good grades. I had a question in my mind : Would Hispanic kids react the same way as the American kids? So I went to Colombia. And I reproduced the experiment. And it was very funny. I used four, five and six years old kids. And let me show you what happened. So what happened in Colombia? Hispanic kids, two out of three ate the marshmallow ; one out of three did not. This little girl was interesting ; she ate the inside of the marshmallow. In other words, she wanted us to think that she had not eaten it, so she would get two. But she ate it. So we know she ’ ll be successful. But we have to watch her. She should not go into banking, for example, or work at a cash register. But she will be successful. And this applies for everything. Even in sales. The sales person that — the customer says,”I want that.”And the person says,”Okay, here you are.”That person ate the marshmallow. If the sales person says,”Wait a second. Let me ask you a few questions to see if this is a good choice.”Then you sell a lot more. So this has applications in all walks of life. I end with — the Koreans did this. You know what? This is so good that we want a marshmallow book for children. We did one for children. And now it is all over Korea. They are teaching these kids exactly this principle. And we need to learn that principle here in the States, because we have a big debt. We are eating more marshmallows than we are producing. Thank you so much.

Text Sample 265: NEG Dim 2 – 2007 – Score 1.00 – 2007/t000373.txt

I ’ ve always wanted to be a cyborg. One of my favorite shows as a kid was”The Six Million Dollar Man,”and this is a little bit closer to the 240-dollar man or so, but â At any rate, I would normally feel very self-conscious and geeky wearing this around, but a few days ago I saw one world-renowned statistician swallowing swords on stage here, so I figure it ’ s OK amongst this group. But that ’ s not what I want to talk about today. I want to talk about toys, and the power that I see inherent in them. When I was a kid, I attended Montessori school up to sixth grade in Atlanta, Georgia. And at the time I didn ’ t think much about it, but then later, I realized that that was the high point of my education. From that point on, everything else was pretty much downhill. And it wasn ’ t until later, as I started making games, that â I really actually think of them more as toys. People call me a game designer, but I really think of these things more as toys. I started getting very interested in Maria Montessori and her methods, and the way she went about things, and the way she thought it very valuable for kids to discover things on their own rather than being taught these things overtly. And she would design these toys, where kids in playing with the toys would come to understand these deep principles of life and nature through play. And since they discovered this, it stuck with them so much more, and also they would experience their own failures. There was a failure-based aspect to learning there. It was very important. And so, the games that I do, I think of really more as modern Montessori toys. And I really kind of want them to be presented in a way to where kids can kind of explore and discover their own principles. So a few years ago, I started getting very interested in the SETI program. And that ' s the way I work. I get interested in different subjects, I dive in, research them, and then try to figure out how to craft a toy around that, so that other people can experience the same sense of discovery that I did as I was learning that subject. And it led me to astrobiology, the study of possible life in the universe. And then Drake ' s equation, looking at the probability of life arising on planets, how long it might last, how many planets are out there. And I started looking at how interesting Drake ' s equation was, because it spanned all these different subjects â physics, chemistry, sociology, economics, astronomy. And another thing that really impressed me a long time ago was"Powers of Ten,"Charles and Ray Eames ' film. And I started putting those two together and wondering, could I build a toy where kids would trip across all these interesting principles of life, as it exists and as it might go in the future. Things where you might trip across things like the Copernican principle, the Fermi paradox, the anthropic principle, the origin of life. And so I ' m going to show you a toy today that I ' ve been working on, that I think of more as kind of a philosophy toy. Playing this toy will bring up philosophical questions in you. The game is"Spore."I ' ve been working on it for several years. It ' s getting pretty close to finished now. It occurs at all these different scales, from very small to very large. I ' m going to pop in at the start of the game. And you actually start this game in a drop of water, as a very, very small single-cell creature, and right off the bat you basically just have to live, survive, reproduce. So here we are, at a very microscopic scale, swimming around. And I actually realize that cells don ' t have eyes, but it helps to make it cute. The players are going to play through every generation of this species, and as you play the game, the creature is actually growing bit by bit. And as we start growing, the camera will actually start zooming out, and things that you see in the background there will start slowly pulling into the foreground, showing you a little bit of what you ' ll be interacting with as you grow. So as we eat, the camera starts pulling out, and then we start interacting with larger and larger organisms. We actually play through many generations here, at the cellular scale. I ' m going to skip ahead here. At some point we get larger, and we actually get to a macroevolution scale. Now, at this point we ' re leaving the water, and one thing that ' s kind of important about this game is that, at every level, the player is designing their creature, and that ' s a fundamental aspect of this. Now, in the evolution game here, the creature scale, basically you have to eat, survive, and then reproduce. You know, very Darwinian. One thing we noticed with"The Sims,"a game I did earlier, is that players love making stuff. When they were able to make stuff in the game, they had a lot of empathy in connection to it. Even if it wasn ' t as pretty as a professional artist would make for games, it really stuck with them and they really cared about what would happen to it. At this point, we ' ve left the water, and now with this little creature â we could bring up the volume a little bit â and now we might try to eat. We might sneak up on this little guy over here maybe, and try and eat him. OK, well, we fight. OK, we got him. Now we get a meal. So really, at this part of the game, what we ’ re doing is we ’ re running around and surviving, and also getting to the next generation, because we ’ re going to play through every generation of this creature. We can mate, so I ’ m going to see if one of these creatures wants to mate with me. Yeah. We didn ’ t want to replay actual evolution with humans and all that, because it ’ s almost more interesting to look at alternate possibilities in evolution. Evolution is usually presented as this one path that we took through, but really it represents this huge set of possibilities. Now, once we mate, we click on the egg. And this is where the game starts getting interesting, because one of the things we really focused on here was giving the players very high-leverage tools, so that for very little effort, the player can make something very cool. And it involves a lot of intelligence on the tool side. But basically, this is the editor where we ’ re going to design the next generation. So it has a little spine. I can move around, I can extend. I can also inflate or deflate it with the mouse wheel, so you sculpt it like clay. We have parts here that I can put on or pull off. The idea is that the player can basically design anything they can think of in this editor, and we ’ ll basically bring it to life. So I might put some limbs on the character here. I ’ ll inflate them kind of large. And in this case I might decide I ’ m going to put â I ’ ll put mouths on the limbs. So pretty much players are encouraged to be very creative in the game. Here, I ’ ll give it one eye in the middle, maybe scale it up a bit. Point it down. And I ’ ll also give it a few legs. So in some sense we want this to feel like an amplifier for the player ’ s imagination, so that with a very small number of clicks a player can create something that they didn ’ t really think was possible before. This is almost like designing something like Maya â that an eight- year-old can use. But really the goal here was, within about a minute, I wanted somebody to replicate what typically takes a pictorial artist several weeks to create. OK, now I ’ ll put some hands on it. OK, so here I ’ ve basically thrown together a little creature. Let me give it a little weapon on the tail here, so it can fight. OK, so that ’ s the complete model. Now we can actually go to the painting phase. At this phase, the program has some understanding of the topology of this creature. It knows where the backbone is, where the spine, the limbs are, how stripes should run, how it should be shaded. And so we ’ re procedurally generating the texture map, something a texture artist would take many days to work on. And then we can test it out, and see how it would move around. And so at this point the computer is procedurally animating this creature. It ’ s looking at whatever I ’ ve designed. It will actually bring it to life. And I can see how it might dance. How it might show emotions, how it might fight. So it ’ s acting with its two mouths there. I can even have it pose for a photo. Snap a little photo of it. So then I bring this back into the game. It ’ s born, and I play the next generation of my creature through evolution. Now again, the empathy that the players have when they create the content is tremendous. When players create content in this game, it ’ s automatically sent up to a server and then redistributed to all the other players transparently. So in fact, as I ’ m interacting in this world with other creatures, these creatures are transparently coming from other players as they play. So the process of playing the game is a process of building up this huge database of content. And pretty much everything you ’ re going to see in this game, there ’ s an editor for that the player can create, up through civilization. This is my baby. When I eat, I ’ ll actually start growing. This is the next generation. But I ’ m going to skip ahead here. Normally what would happen is these creatures would work their way up, eventually become intelligent. I ’ d start dealing with tribes, cities and civilizations of them over time. I ’ m going to skip way ahead to the space phase. Eventually they would go out into space, and start colonizing and exploring the universe. Now, really, in some sense, I want the players to be building this world in their imagination, and then extracting it from them with the least amount of pain. So that ’ s kind of what these tools are about. How do we make the gameplay, you know, basically the player ’ s imagination amplifier? And how do we make these tools, these editors, something that are just as fun as the game itself? So this is the planet that we ’ ve been playing on up to this point in the game. So far the entire game has been played on the surface of this little world here. At this point we ’ re actually dealing with a very little toy planet. Almost, again, like the Montessori toy idea. What happens if you give somebody a toy planet, and let them play with a lot of dynamics on it? What could they discover? What might they learn on this? This world was actually extracted from the player ’ s imagination. So, this is the planet that the player evolved on. Things like the buildings, the vehicles, the architecture, civilizations were all designed by the player up to this point. So here ’ s a little city with some of our guys walking around in it. And most games put the player in the role of Luke Skywalker, this protagonist playing through this story. This is more about putting the player in the role of George Lucas. I want them, after they ’ ve played this game, to have extracted an entire world that they ’ re now interacting with. As we pull down here, we still have a whole set of creatures living on the surface of the planet. All these different dynamics going on here. I can look over here, and this is a little simplified food web that ’ s going on with the creatures. I can open this up and then scan what exists on the surface. You get some sense of the diversity of creatures that were brought in. Some of these were created by the player, others by other players, automatically sent over here. But there ’ s a very simple calculation of what ’ s required, how many plants are required for the herbivores to live, how many herbivores for the carnivores to eat, etc., that you have to balance actively. Also with this phase, we ’ re getting more and more God-like powers for the player, and you can experiment with this planet as a toy. So I can come in and I can do things, and just treat this planet as a lump of clay. We have very simple weather systems, simple geology. For instance, I could open one of my tools here and then carve out rivers. So this whole thing is kind of like a big lump of clay, a sculpture. I can also play with the dynamics in this world over time. So one of the things I can do is start pumping more CO2 gases into the atmosphere, and so that ’ s what I ’ m doing here. There ’ s actually a little readout down there of our planetary atmosphere, pressure and temperature. So as I start pumping in more atmosphere, we ’ re going to start pushing up the greenhouse gases here and if you ’ ll start noticing, we start seeing the ocean levels rise over time. And our cities are going to be at risk too, because a lot of these are coastal cities. You can see the ocean levels are rising now and as they encroach upon the cities, I ’ ll start losing cities here. So basically, I want the players to be able to experiment and explore a huge amount of failure space. So there goes one city. Now, over time, this is going to heat up the planet. So at first what we ’ re going to see is a global ocean rise here on this little toy planet, but then over time â I can speed it up â we ’ ll see the heat impact of that as well. Not only will it get hotter, but at some point it ’ s going to get so hot the oceans will evaporate. They ’ ll go up, and then they ’ ll evaporate, and that ’ ll be my planet. So basically, what we ’ re getting here is the sequel to”An Inconvenient Truth,”in about two minutes, and that actually brings up an interesting point about games. Now here, our entire oceans are evaporating off the surface, and as it keeps getting hotter, at some point the entire planet is going to melt down. Here it goes. So we ’ re not only simulating biological dynamics â food webs and all that â but also geologic, you know, on a very simple core scale. And what ’ s interesting to me about games is that I think we can take a lot of long-term dynamics and compress them into very short-term experiences. Because it ’ s so hard for people to think 50 or 100 years out, but when you can give them a toy, and they can experience these long-term dynamics in just a few minutes, I think it ’ s an entirely different point of view, where we ’ re actually using the game to remap our intuition. It ’ s almost in the same way that a telescope or microscope recalibrates your eyesight ; I think computer simulations can recalibrate your instinct across vast scales of both space and time. So here ’ s our little solar system, as we pull away from our melted planet here. We actually have a couple of other planets in this solar system. Let ’ s fly to another one. We ’ re going to have this unlimited number of worlds you can explore here. As we move into the future, and we start going out into space and doing stuff, we ’ re drawing a lot from things like science fiction. And all my favorite science fiction movies I want to play out here as different dynamics. This planet actually has some life on it. Here it is, some indigenous life down here. One of the tools I can eventually earn for my UFO is a monolith that I can drop down. Now, as you can see, these guys are actually starting to go up and bow to it, and over time, once they touch it, they will become intelligent. So I can actually pick a species on a planet and then make them sentient. Now they ’ ve actually gone to tribal dynamics. And now, because I ’ m actually the one here, I can get out of the UFO and walk up, and they should be worshipping me at this point as a god. At first they ’ re a little freaked out. OK, well maybe they ’ re not worshipping me. I think I ’ ll leave before they get hostile. But we basically want a diversity of activities the players can play through this. I want to be able to play”The Day the Earth Stood Still,””2001 : A Space Odyssey,””Star Trek,””War Of the Worlds.”Now, as we pull away from this world â we ’ re going to keep pulling away from the star now. One of the things that always frustrated me about astronomy when I was a kid is how it was always presented so two-dimensionally and so static. As we pull away from the star here, we ’ re actually going now out into interstellar space, and we ’ re getting a sense of the space around our home star. What I really wanted to do is to present this, basically, as wonderfully 3D as it is actually is. And also show the dynamics, and a lot of the interesting objects that you might find, like, in the Hubble, at pretty much realistic frequencies and scales. So most people have no idea of the difference between an emission nebula and a planetary nebula. But these are the things that we can put in this little galaxy here. So we ’ re flying over here to what looks like a black hole. I want to basically have the entire zoo of Hubble objects that people can interact with and play with, again, as toys. So here ’ s a little black hole that we probably don ’ t want to get too close to. But we also have stars and things as well. If we pull all the way back, we start seeing the entire galaxy here, kind of slowly in motion. Typically, when people present galaxies, it ’ s always beautiful photos, but they ’ re always static. And when you bring it forward in time and start animating it, it ’ s amazing what a galaxy would look like fast forwarded. This would be about a million years a second, because you have about one supernova every century. And so you ’ d have this wonderful sparkling thing, with the disk slowly rotating, and this is roughly what it would look like. Part of this is about bringing the beauty of the natural world to somebody in a very imaginative way, so that they can start calibrating their instinct across these vast scales of space and time. Chris was wondering what kind of gods the players would become. Because if you think about it, you ’ re going to have 15-year-olds, 20-year-olds flying around this universe. They might be a nurturing god. They might be bootstrapping life on planets, trying to terraform and spread civilization. You might be a vengeful god, conquesting, because you actually can do that, you can attack other intelligent races. You might be a networking god, building alliances, or just curious, going around and wanting to explore as much as you possibly can. But basically, the reason why I make toys like this is because I think if there ’ s one difference I could possibly make in the world, that I would choose to make, it ’ s that I would like to somehow give people just a little bit better calibration on long-term thinking. Because I think most of the problems that our world is facing right now are the result of short-term thinking, and the fact that it is so hard for us to think 50, 100, or 1, 000 years out. And I think by giving kids toys like this and letting them replay dynamics, very long-term dynamics over the short term, and getting some sense of what we ’ re doing now, what it ’ s going to be like in 100 years, I think probably is the most effective thing I can be doing to help the world. And so that ’ s why I think that toys can change the world. Thank you.

Text Sample 266: NEG Dim 2 – 2007 – Score 1.00 – 2007/t000371.txt

I would like to tell you about a project which I started about 16 years ago. It ’ s about making new forms of life. And these are made of this kind of tube â electricity tube, we call it in Holland. And we can start a film about that, and we can see a little bit backwards in time. Narrator : Eventually, these beasts are going to live in herds on the beaches. Theo Jansen is working hard on this evolution. Theo Jansen : I want to put these forms of life on the beaches. And they should survive over there, on their own, in the future. Learning to live on their own â and it ’ ll take couple of more years to let them walk on their own. Narrator : The mechanical beasts will not get their energy from food, but from the wind. The wind will move feathers on their back, which will drive their feet. The beast walks sideways on the wet sand of the beach, with its nose pointed into the wind. As soon as it walks into either the rolling surf or the dry sand, it stops, and walks in the opposite direction. Evolution has generated many species. This is the Animaris Currens Ventosa. TJ : This is a herd, and it is built according to genetic codes. And it is a sort of race, and each and every animal is different, and the winning codes will multiply. This is the wave, going from left to right. You can see this one. Yes, and now it goes from left to right. This is a new generation, a new family, which is able to store the wind. So, the wings pump up air in lemonade bottles, which are on top of that. And they can use that energy in case the wind falls away, and the tide is coming up, and there is still a little bit of energy to reach the dunes and save their lives, because they are drowned very easily. I could show you this animal. Thank you. So, the proportion of the tubes in this animal is very important for the walking. There are 11 numbers, which I call The 11 Holy Numbers. These are the distances of the tubes which make it walk that way. In fact, it ’ s a new invention of the wheel. It works the same as a wheel. The axis of a wheel stays on the same level, and this hip is staying on the same level as well. In fact, this is better than a wheel, because when you try to drive your bicycle on the beach, you will notice it ’ s very hard to do. And the feet just step over the sand, and the wheel has to touch every piece of the ground in-between. So 5, 000 years after the invention of the wheel, we have a new wheel. I will show you, in the next video â can you start it, please? â that very heavy loads can be moved. There ’ s a guy pushing there, behind, but it can also walk on the wind very well. It ’ s 3. 2 tons. This is working on the stored wind in the bottles. It has a feeler, where it can feel obstacles and turn around. You see, it ’ s going the other way. Can I have the feeler here? OK. Good. So, they have to survive all the dangers of the beach, and one of the big dangers is the sea. This is the sea. And it must feel the water of the sea. And this is the water feeler, and what ’ s very important is this tube. It sucks in air normally, but when it swallows water, it feels the resistance of it. So, imagine that the animal is walking towards the sea. As soon as it touches the water â□□ you should hear a sound of running air. Yes! So if it doesn ’ t feel, it will be drowned, OK? Here we have the brain of the animal. In fact, it is a step counter, and it counts the steps. It ’ s a binary step counter. So as soon it has been to the sea, it changes the pattern of zeroes and ones here. And it always knows where it is on the beach. So it ’ s very simple brain. It says, well, there ’ s the sea, there are dunes, and I ’ m here. So it ’ s a sort of imagination of the simple world of the beach animal. Thank you. One of the biggest enemies are the storms. This is a part of the nose of the Animaris Percipiere. When the nose of the animal is fixed, the whole animal is fixed. So when the storm is coming up, it drives a pin into the ground. Audience member : Wow! The nose is fixed, the whole animal is fixed. The wind may turn, but the animal will always turn its nose into the wind. Now, another couple of years, and these animals will survive on their own. I still have to help them a lot. Thank you very much, ladies and gentlemen.

Text Sample 267: NEG Dim 2 – 2007 – Score 1.00 – 2007/t000366.txt

This is a recent comic strip from the Los Angeles Times. The punch line?”On the other hand, I don ’ t have to get up at four every single morning to milk my Labrador.”This is a recent cover of New York Magazine. Best hospitals where doctors say they would go for cancer treatment, births, strokes, heart disease, hip replacements, 4 a. m. emergencies. And this is a song medley I put together — Did you ever notice that four in the morning has become some sort of meme or shorthand? It means something like you are awake at the worst possible hour. A time for inconveniences, mishaps, yearnings. A time for plotting to whack the chief of police, like in this classic scene from”The Godfather.”Coppola ’ s script describes these guys as,”exhausted in shirt sleeves. It is four in the morning.”A time for even grimmer stuff than that, like autopsies and embalmings in Isabel Allende ’ s”The House of the Spirits.”After the breathtaking green- haired Rosa is murdered, the doctors preserve her with unguents and morticians ’ paste. They worked until four o ’ clock in the morning. A time for even grimmer stuff than that, like in last April ’ s New Yorker magazine. This short fiction piece by Martin Amis starts out,”On September 11, 2001, he opened his eyes at 4 a. m. in Portland, Maine, and Mohamed Atta ’ s last day began.”For a time that I find to be the most placid and uneventful hour of the day, four in the morning sure gets an awful lot of bad press — across a lot of different media from a lot of big names. And it made me suspicious. I figured, surely some of the most creative artistic minds in the world, really, aren ’ t all defaulting back to this one easy trope like they invented it, right? Could it be there is something more going on here? Something deliberate, something secret, and who got the four in the morning bad rap ball rolling anyway? I say this guy — Alberto Giacometti, shown here with some of his sculptures on the Swiss 100 franc note. He did it with this famous piece from the New York Museum of Modern Art. Its title —”The Palace at Four in the Morning — 1932. Not just the earliest cryptic reference to four in the morning I can find. I believe that this so-called first surrealist sculpture may provide an incredible key to virtually every artistic depiction of four in the morning to follow it. I call this The Giacometti Code, a TED exclusive. No, feel free to follow along on your Blackberries or your iPhones if you ’ ve got them. It works a little something like — this is a recent Google search for four in the morning. Results vary, of course. This is pretty typical. The top 10 results yield you four hits for Faron Young ’ s song,”It ’ s Four in the Morning,”three hits for Judi Dench ’ s film,”Four in the

Morning,”one hit for Wislawa Szymborska ’ s poem,”Four in the Morning.”But what, you may ask, do a Polish poet, a British Dame, a country music hall of famer all have in common besides this totally excellent Google ranking? Well, let ’ s start with Faron Young — who was born incidentally in 1932. In 1996, he shot himself in the head on December ninth — which incidentally is Judi Dench ’ s birthday. But he didn ’ t die on Dench ’ s birthday. He languished until the following afternoon when he finally succumbed to a supposedly self-inflicted gunshot wound at the age of 64 — which incidentally is how old Alberto Giacometti was when he died. Where was Wislawa Szymborska during all this? She has the world ’ s most absolutely watertight alibi. On that very day, December 10, 1996 while Mr. Four in the Morning, Faron Young, was giving up the ghost in Nashville, Tennessee, Ms. Four in the Morning — or one of them anyway — Wislawa Szymborska was in Stockholm, Sweden, accepting the Nobel Prize for Literature. 100 years to the day after the death of Alfred Nobel himself. Coincidence? No, it ’ s creepy. Coincidence to me has a much simpler metric. That ’ s like me telling you,”Hey, you know the Nobel Prize was established in 1901, which coincidentally is the same year Alberto Giacometti was born?”No, not everything fits so tidily into the paradigm, but that does not mean there ’ s not something going on at the highest possible levels. In fact there are people in this room who may not want me to show you this clip we ’ re about to see. Video : Homer Simpson : We have a tennis court, a swimming pool, a screening room — You mean if I want pork chops, even in the middle of the night, your guy will fry them up? Herbert Powell : Sure, that ’ s what he ’ s paid for. Now do you need towels, laundry, maids? HS : Wait, wait, wait, wait, wait, wait — let me see if I got this straight. It is Christmas Day, 4 a. m. There ’ s a rumble in my stomach. Marge Simpson : Homer, please. Rives : Wait, wait, wait, wait, wait, wait, wait. Let me see if I got this straight, Matt. When Homer Simpson needs to imagine the most remote possible moment of not just the clock, but the whole freaking calendar, he comes up with 0400 on the birthday of the Baby Jesus. And no, I don ’ t know how it works into the whole puzzling scheme of things, but obviously I know a coded message when I see one. I said, I know a coded message when I see one. And folks, you can buy a copy of Bill Clinton ’ s”My Life”from the bookstore here at TED. Parse it cover to cover for whatever hidden references you want. Or you can go to the Random House website where there is this excerpt. And how far down into it you figure we ’ ll have to scroll to get to the golden ticket? Would you believe about a dozen paragraphs? This is page 474 on your paperbacks if you ’ re following along :”Though it was getting better, I still wasn ’ t satisfied with the inaugural address. My speechwriters must have been tearing their hair out because as we worked between one and four in the morning on Inauguration Day, I was still changing it.”Sure you were, because you ’ ve prepared your entire life for this historic quadrennial event that just sort of sneaks up on you. And then — three paragraphs later we get this little beauty :”We went back to Blair House to look at the speech for the last time. It had gotten a lot better since 4 a. m.”Well, how could it have? By his own writing, this man was either asleep, at a prayer meeting with Al and Tipper or learning how to launch a nuclear missile out of a suitcase. What happens to American presidents at 0400 on inauguration day? What happened to William Jefferson Clinton? We might not ever know. And I noticed, he ’ s not exactly around here today to face any tough questions. It could get awkward, right? I mean after all, this whole business happened on his watch. But if he were here — he might remind us, as he does in the wrap-up to his fine autobiography, that on this day Bill Clinton began a journey — a journey that saw him go on to become the first Democrat president elected to two consecutive terms in decades. In generations. The first since this man, Franklin Delano Roosevelt, who began his own unprecedented journey way back at his own first election, way back in a simpler time, way back in 1932 — the year Alberto Giacometti made”The Palace at Four in the Morning.”The year, let ’ s remember, that this voice, now departed, first came a-cryin ’ into this big old crazy world of ours.

Text Sample 268: NEG Dim 2 – 2006 – Score 1.00 – 2006/t000495.txt
I ’ m going to present three projects in rapid fire. I don ’ t have much time to do it. And I want to reinforce three ideas with that rapid-fire presentation. The first is what I like to call a hyper-rational process. It ’ s a process that takes rationality almost to an absurd level, and it transcends all the baggage that normally comes with what people would call, sort of a rational conclusion to something. And it concludes in something that you see here, that you actually wouldn ’ t expect as being the result of rationality. The second — the second is that this process does not have a signature. There is no authorship. Architects are obsessed with authorship. This is something that has editing and it has teams, but in fact, we no longer see within this process, the traditional master architect creating a sketch that his minions carry out. And the third is that it challenges — and this is, in the length of this, very hard to support why, connect all these things — but it challenges the high modernist notion of flexibility. High modernists said we will create sort of singular spaces that are generic, almost anything can happen within them. I call it sort of”shotgun flexibility”— turn your head this way ; shoot ; and you ’ re bound to kill something. So, this is the promise of high modernism : within a single space, actually, any kind of activity can happen. But as we ’ re seeing, operational costs are starting to dwarf capital costs in terms of design parameters. And so, with this sort of idea, what happens is, whatever actually is in the building on opening day, or whatever seems to be the most immediate need, starts to dwarf the possibility and sort of subsume it, of anything else could ever happen. And so we ’ re proposing a different kind of flexibility, something that we call”compartmentalized flexibility.”And the idea is that you, within that continuum, identify a series of points, and you design specifically to them. They can be pushed off-center a little bit, but in the end you actually still get as much of that original spectrum as you originally had hoped. With high modernist flexibility, that doesn ’ t really work. Now I ’ m going to talk about — I ’ m going to build up the Seattle Central Library in this way before your eyes in about five or six diagrams, and I truly mean this is the design process that you ’ ll see. With the library staff and the library board, we settled on two core positions. This is the first one, and this is showing, over the last 900 years, the evolution of the book, and other technologies. This diagram was our sort of position piece about the book, and our position was, books are technology — that ’ s something people forget — but it ’ s a form of technology that will have to share its dominance with any other form of truly potent technology or media. The second premise — and this was something that was very difficult for us to convince the librarians of at first — is that libraries, since the inception of Carnegie Library tradition in America, had a second responsibility, and that was for social roles. Ok, now, this I ’ ll come back to later, but something — actually, the librarians at first said,”No, this isn ’ t our mandate. Our mandate is media, and particularly the book.”So what you ’ re seeing now is actually the design of the building. The upper diagram is what we had seen in a whole host of contemporary libraries that used high modernist flexibility. Sort of, any activity could happen anywhere. We don ’ t know the future of the library ; we don ’ t know the future of the book ; and so, we ’ ll

use this approach. And what we saw were buildings that were very generic, and worse — not only were they very generic — so, not only does the reading room look like the copy room look like the magazine area — but it meant that whatever issue was troubling the library at that moment was starting to engulf every other activity that was happening in it. And in this case, what was getting engulfed were these social responsibilities by the expansion of the book. And so we proposed what ' s at the lower diagram. Very dumb approach : simply compartmentalize. Put those things whose evolution we could predict — and I don ' t mean that we could say what would actually happen in the future, but we have some certainty of the spectrum of what would happen in the future — put those in boxes designed specifically for it, and put the things that we can ' t predict on the rooftops. So that was the core idea. Now, we had to convince the library that social roles were equally important to media, in order to get them to accept this. What you ' re seeing here is actually their program on the left. That ' s as it was given to us in all of its clarity and glory. Our first operation was to re-digest it back to them, show it to them and say,"You know what? We haven ' t touched it, but only one-third of your own program is dedicated to media and books. Two-thirds of it is already dedicated — that ' s the white band below, the thing you said isn ' t important — is already dedicated to social functions."So once we had presented that back to them, they agreed that this sort of core concept could work. We got the right to go back to first principles — that ' s the third diagram. We recombined everything. And then we started making new decisions. What you ' re seeing on the right is the design of the library, specifically in terms of square footage. On the left of that diagram, here, you ' ll see a series of five platforms — sort of combs, collective programs. And on the right are the more indeterminate spaces ; things like reading rooms, whose evolution in 20, 30, 40 years we can ' t predict. So that literally was the design of the building. They signed it, and to their chagrin, we came back a week later, and we presented them this. And as you can see, it is literally the diagram on the right. We just sized — no, really, I mean that, literally. The things on the left- hand side of the diagram, those are the boxes. We sized them into five compartments. They ' re super-efficient. We had a very low budget to work with. We pushed them around on the site to make very literal contextual relationships. The reading room should be able to see the water. The main entrance should have a public plaza in front of it to abide by the zoning code, and so forth. So, you see the five platforms, those are the boxes. within each one, a very discrete thing is happening. The area in between is sort of an urban continuum, these things that we can ' t predict their evolution to the same degree. To give you some sense of the power of this idea, the biggest block is what we call the book spiral. It ' s literally built in a very inexpensive way — it is a parking garage for books. It just so happens to be on the 6th through 10th floors of the building, but that is not necessarily an expensive approach. And it allows us to organize the entire Dewey Decimal System on one continuous run ; no matter how it grows or contracts within the building, it will always have its clarity to end the sort of trail of tears that we ' ve all experienced in public libraries. And so this was the final operation, which was to take these blocks as they were all pushed off kilter, and to hold onto them with a skin. That skin serves double duty, again, for economics. One, it is the lateral stability for the entire building ; it ' s a structural element. But its dimensions were designed not only for structure, but also for holding on every piece of glass. The glass was then — I ' ll use the word impregnated — but it had a layer of metal that was called"stretched metal."That metal acts as a microlouver, so from the exterior of the building, the sun sees it as totally opaque, but from the interior, it ' s entirely transparent. So now I ' m going to take you on a tour of the building.

Let me see if I can find it. For anyone who gets motion sickness, I apologize. So, this is the building. And I think what ’ s important is, when we first unveiled the building, the public saw it as being totally about our whim and ego. And it was defended, believe it or not, by the librarians. They said,”Look, we don ’ t know what it is, but we know it ’ s everything that we need it to be, based on the observations that we ’ ve done about the program.”This is going into one of the entries. So, it ’ s an unusual building for a public library, obviously. So now we ’ re going into what we call the living room. This is actually a program that we invented with the library. It was recognizing that public libraries are the last vestige of public free space. There are plenty of shopping malls that allow you to get out of the rain in downtown Seattle, but there are not so many free places that allow you to get out of the rain. So this was an unprogrammed area where people could pretty much do anything, including eat, yell, play chess and so forth. Now we ’ re moving up into what we call the mixing chamber. That was the main technology area in the building. You ’ ll have to tell me if I ’ m going too fast for you. And now up. This is actually the place that we put into the building so I could propose to my wife, right there. She said yes. I ’ m running out of time, so I ’ m actually going to stop. I can show this to you later. But let ’ s see if I can very quickly get into the book spiral, because I think it ’ s, as I said, the most — this is the main reading room — the most unique part of the building. You dizzy yet? Ok, so here, this is the book spiral. So, it ’ s very indiscernible, but it ’ s actually a continuous stair- stepping. It allows you to, on one city block, go up one full floor, so that it ’ s on a continuum. Ok, now I ’ m going to go back, and I ’ m going to hit a second project. I ’ m going to go very, very quickly through this. Now this is the Dallas Theater. It was an unusual client for us, because they came to us and they said,”We need you to do a new building. We ’ ve been working in a temporary space for 30 years, but because of that temporary space, we ’ ve become an infamous theater company. Theater is really focused in New York, Chicago and Seattle, with the exception of the Dallas Theater Company.”And the very fact that they worked in a provisional space meant that for Beckett, they could blow out a wall ; they could do”Cherry Orchard”and blow a hole through the floor, and so forth. So it was a very daunting task for us to do a brand-new building that could be a pristine building, but keep this kind of experimental nature. And the second is, they were what we call a multi-form theater, they do different kinds of performances in repertory. So they in the morning will do something in arena, then they ’ ll do something in proscenium and so forth. And so they needed to be able to quickly transform between different theater organizations, and for operational budget reasons, this actually no longer happens in pretty much any multi-form theater in the United States, so we needed to figure out a way to overcome that. So our thought was to literally put the theater on its head : to take those things that were previously defined as front-of-house and back-of-house and stack them above house and below house, and to create what we called a theater machine. We invest the money in the operation of the building. It ’ s almost as though the building could be placed anywhere, wherever you place it, the area under it is charged for theatrical performances. And it allowed us to go back to first principles, and redefine fly tower, acoustic enclosure, light enclosure and so forth. And at the push of a button, it allows the artistic director to move between proscenium, thrust, and in fact, arena and traverse and flat floor, in a very quick transfiguration. So in fact, using operational budget, we can — sorry, capital cost — we can actually achieve what was no longer achievable in operational cost. And that means that the artistic director now has a palette that he or she can choose from, between a series of forms and a series of processions, because that enclosure around the theater that is normally trapped with front-of-house and back-of-house spaces has been liberated. So an artistic director has the ability to have a performance that enters in a Wagnerian procession, shows the first act in thrust, the intermission in a Greek procession, second act in arena, and so forth. So I ’ m going to show you what this actually means. This is the theater up close. Any portion around the theater actually can be opened discretely. The light enclosure can be lifted separate to the acoustic enclosure, so you can do Beckett with Dallas as the backdrop. Portions can be opened, so you can now actually have motorcycles drive directly into the performance, or you can even just have an open-air performance, or for intermissions. The balconies all move to go between those configurations, but they also disappear. The proscenium line can also disappear. You can bring enormous objects in, so in fact, the Dallas Theater Company — their first show will be a play about Charles Lindbergh, and they ’ ll want to bring in a real aircraft. And then it also provides them, in the off-season, the ability to actually rent out their space for entirely different things. This is it from a distance. Open up entire portions for different kinds of events. And at night. Again, remove the light enclosure ; keep the acoustic enclosure. This is a monster truck show. I ’ m going to show now the last project. This also is an unusual client. They inverted the whole idea of development. They came to us and they said — unlike normal developers — they said,”We want to start out by providing a contemporary art museum in Louisville. That ’ s our main goal.”And so instead of being a developer that sees an opportunity to make money, they saw an ability to be a catalyst in their downtown. And the fact that they wanted to support the contemporary art museum actually built their pro forma, so they worked in reverse. And that pro forma led us to a mixed-use building that was very large, in order to support their aspirations of the art, but it also opened up opportunities for the art itself to collaborate, interact with commercial spaces that actually artists more and more want to work within. And it also charged us with thinking about how to have something that was both a single building and a credible sort of sub-building. So this is Louisville ’ s skyline, and I ’ m going to take you through the various constraints that led to the project. First : the physical constraints. We actually had to operate on three discrete sites, all of them well smaller than the size of the building. We had to operate next to the new Muhammad Ali Center, and respect it. We had to operate within the 100-year floodplain. Now, this area floods three to four times a year, and there ’ s a levee behind our site, similar to the ones that broke in New Orleans. Had to operate behind the I-64 corridor, a street that cuts through the middle of these separate sites. So we ’ re starting to build a sort of nightmare of constraints in a bathtub. Underneath the bathtub are the city ’ s main power lines. And there is a pedestrian corridor that they wanted to add, that would link a series of cultural buildings, and a view corridor — because this is the historic district — that they didn ’ t want to obstruct with a new building. And now we ’ re going to add 1. 1 million square feet. And if we did the traditional thing, that 1. 1 million square feet — these are the different programs — the traditional thing would be to identify the public elements, place them on sites, and now we ’ d have a really terrible situation : a public thing in the middle of a bathtub that floods. And then we would size all the other elements — the different commercial elements : hotel, luxury housing, offices and so forth — and dump it on top. And we would create something that was unviable. In fact — and you know this — this is called the Time Warner Building. So our strategy was very simple. Just lift the entire block, flip some of the elements over, reposition them so they have appropriate views and relationships to downtown, and make circulation connections and reroute the road. So that ’ s the basic concept, and now I ’ m going to show you what it leads to. Ok, it seems a very formal, willful gesture, but something derived entirely out of the constraints. And again, when we unveiled it, there was a sort of nervousness that this was about an architect making a statement, not an architect who was attempting to solve a series of problems. Now, within that center zone, as I said, we have the ability to mix a series of things. So here, this is sort of an x-ray — the towers are totally developer-driven. They told us the dimensions, the sizes and so forth, and we focused on taking all the public components — the lobbies, the bars — everything that different commercial elements would have, and combined it in the center, in the sort of subway map, in the transfer zone that would also include the contemporary art museum. So it creates a situation like this, where you have artists who can operate within an art space that also has an amazing view on the 22nd floor, but it also has proximity that the curator can either open or close. It allows people on exercise bicycles to be seen, or to see the art, and so forth. It also means that if an artist wants to invade something like a swimming pool, they can begin to do their exhibition in a swimming pool, so they ’ re not forced to always work within the confines of a contemporary gallery space. So, how to build this. It ’ s very simple : it ’ s a chair. So, we begin by building the cores. As we ’ re building the cores, we build the contemporary art museum at grade. That allows us to have incredible efficiency and cost efficiency. This is not a high-budget building. The moment the cores get to mid level, we finish the art museum ; we put all the mechanical equipment in it ; and then we jack it up into the air. This is how they build really large aircraft hangars, for instance, the ones that they did for the A380. Finish the cores, finish the meat and you get something that looks like this. Now I only have about 30 seconds, so I want to start an animation, and we ’ ll conclude with that. Thank you. Chris asked me to add — the theater is under construction, and this project will start construction in about a year, and finish in 2010. [identify public elements] [insert public elements at grade] [optimize tower dimensions] [place towers on site] [lift program] [flip!] [optimize program adjacencies] [connect to context] [redirect 7th street]

Text Sample 269: NEG Dim 2 – 2006 – Score 1.00 – 2006/t000492.txt

About 10 years ago, I took on the task to teach global development to Swedish undergraduate students. That was after having spent about 20 years, together with African institutions, studying hunger in Africa. So I was sort of expected to know a little about the world. And I started, in our medical university, Karolinska Institute, an undergraduate course called Global Health. But when you get that opportunity, you get a little nervous. I thought, these students coming to us actually have the highest grade you can get in the Swedish college system, so I thought, maybe they know everything I ’ m going to teach them about. So I did a pretest when they came. And one of the questions from which I learned a lot was this one :”Which country has the highest child mortality of these five pairs?”And I put them together so that in each pair of countries, one has twice the child mortality of the other. And this means that it ’ s much bigger, the difference, than the uncertainty of the data. I won ’ t put you at a test here, but it ’ s Turkey, which is highest there, Poland, Russia, Pakistan and South Africa. And these were the results of the Swedish students. I did it so I got the confidence interval, which is pretty narrow. And I got happy, of course — a 1. 8 right answer out of five possible. That means there was a place for a professor of international health and for my course. But one late night, when I was compiling the report, I really realized my discovery. I have shown that Swedish top students know, statistically, significantly less about the world than the chimpanzees. Because the chimpanzee would score half right if I gave them two bananas with Sri Lanka and Turkey. They would be right half of the cases. But the students are not there. The problem for me was not ignorance ; it was preconceived ideas. I did also an unethical study of the professors of the Karolinska Institute, which hands out the Nobel Prize in Medicine, and they are on par with the chimpanzee there. This is where I realized that there was really a need to communicate, because the data of what ’ s happening in the world and the child health of every country is very well aware. So we did this software, which displays it like this. Every bubble here is a country. This country over here is China. This is India. The size of the bubble is the population, and on this axis here, I put fertility rate. Because my students, what they said when they looked upon the world, and I asked them,”What do you really think about the world?”Well, I first discovered that the textbook was Tintin, mainly. And they said,”The world is still ’ we ’ and ’ them. ’ And ’ we ’ is the Western world and ’ them ’ is the Third World.””And what do you mean with ’ Western world? ’”I said.”Well, that ’ s long life and small family. And ’ Third World ’ is short life and large family.”So this is what I could display here. I put fertility rate here — number of children per woman : one, two, three, four, up to about eight children per woman. We have very good data since 1962, 1960, about, on the size of families in all countries. The error margin is narrow. Here, I put life expectancy at birth, from 30 years in some countries, up to about 70 years. And in 1962, there was really a group of countries here that were industrialized countries, and they had small families and long lives. And these were the developing countries. They had large families and they had relatively short lives. Now, what has happened since 1962? We want to see the change. Are the students right? It ’ s still two types of countries? Or have these developing countries got smaller families and they live here? Or have they got longer lives and live up there? Let ’ s see. We stopped the world then. This is all UN statistics that have been available. Here we go. Can you see there? It ’ s China there, moving against better health there, improving there. All the green Latin American countries are moving towards smaller families. Your yellow ones here are the Arabic countries, and they get longer life, but not larger families. The Africans are the green here. They still remain here. This is India ; Indonesia is moving on pretty fast. In the ’ 80s here, you have Bangladesh still among the African countries. But now, Bangladesh — it ’ s a miracle that happens in the ’ 80s — the imams start to promote family planning, and they move up into that corner. And in the ’ 90s, we have the terrible HIV epidemic that takes down the life expectancy of the African countries. And the rest of them all move up into the corner, where we have long lives and small family, and we have a completely new world. Let me make a comparison directly between the United States of America and Vietnam. 1964 : America had small families and long life ; Vietnam had large families and short lives. And this is what happens. The data during the war indicate that even with all the death, there was an improvement of life expectancy. By the end of the year, family planning started in Vietnam, and they went for smaller families. And the United States up there is getting longer life, keeping family size. And in the ’ 80s now, they give up Communist planning and they go for market economy, and it moves faster even than social life. And today, we have in Vietnam the same life expectancy and the same family size here in Vietnam, 2003, as in United States, 1974, by the end of the war. I think we all, if we don ’ t look at the data, we underestimate the tremendous change in Asia, which was in social change before we saw the economic change. So let ’ s move over to another way here in which we could display the distribution in the world of income. This is the world distribution of income of people. One dollar, 10 dollars or 100 dollars per day. There ’ s no gap between rich and poor any longer. This is a myth. There ’ s a little hump here.

But there are people all the way. And if we look where the income ends up, this is 100 percent of the world ’ s annual income. And the richest 20 percent, they take out of that about 74 percent. And the poorest 20 percent, they take about two percent. And this shows that the concept of developing countries is extremely doubtful. We think about aid, like these people here giving aid to these people here. But in the middle, we have most of the world population, and they have now 24 percent of the income. We heard it in other forms. And who are these? Where are the different countries? I can show you Africa. This is Africa. Ten percent of the world population, most in poverty. This is OECD — the rich countries, the country club of the UN. And they are over here on this side. Quite an overlap between Africa and OECD. And this is Latin America. It has everything on this earth, from the poorest to the richest in Latin America. And on top of that, we can put East Europe, we can put East Asia, and we put South Asia. And what did it look like if we go back in time, to about 1970? Then, there was more of a hump. And most who lived in absolute poverty were Asians. The problem in the world was the poverty in Asia. And if I now let the world move forward, you will see that while population increases, there are hundreds of millions in Asia getting out of poverty, and some others getting into poverty, and this is the pattern we have today. And the best projection from the World Bank is that this will happen, and we will not have a divided world. We ’ ll have most people in the middle. Of course it ’ s a logarithmic scale here, but our concept of economy is growth with percent. We look upon it as a possibility of percentile increase. If I change this and take GDP per capita instead of family income, and I turn these individual data into regional data of gross domestic product, and I take the regions down here, the size of the bubble is still the population. And you have the OECD there, and you have sub-Saharan Africa there, and we take off the Arab states there, coming both from Africa and from Asia, and we put them separately, and we can expand this axis, and I can give it a new dimension here, by adding the social values there, child survival. Now I have money on that axis, and I have the possibility of children to survive there. In some countries, 99. 7 % of children survive to five years of age ; others, only 70. And here, it seems, there is a gap between OECD, Latin America, East Europe, East Asia, Arab states, South Asia and sub-Saharan Africa. The linearity is very strong between child survival and money. But let me split sub-Saharan Africa. Health is there and better health is up there. I can go here, and I can split sub-Saharan Africa into its countries. And when it bursts, the size of each country bubble is the size of the population. Sierra Leone down there, Mauritius is up there. Mauritius was the first country to get away with trade barriers, and they could sell their sugar, they could sell their textiles, on equal terms as the people in Europe and North America. There ’ s a huge difference [within] Africa. And Ghana is here in the middle. In Sierra Leone, humanitarian aid. Here in Uganda, development aid. Here, time to invest ; there, you can go for a holiday. There ’ s tremendous variation within Africa, which we very often make that it ’ s equal everything. I can split South Asia here. India ’ s the big bubble in the middle. But there ’ s a huge difference between Afghanistan and Sri Lanka. I can split Arab states. How are they? Same climate, same culture, same religion — huge difference. Even between neighbors — Yemen, civil war ; United Arab Emirates, money, which was quite equally and well-used. Not as the myth is. And that includes all the children of the foreign workers who are in the country. Data is often better than you think. Many people say data is bad. There is an uncertainty margin, but we can see the difference here : Cambodia, Singapore. The differences are much bigger than the weakness of the data. East Europe : Soviet economy for a long time, but they come out after 10 years very, very differently. And there is Latin America. Today, we don ’ t have to go to Cuba to find a healthy country in Latin America. Chile will have a lower child mortality than Cuba within some few years from now. Here, we have high-income countries in the OECD. And we get the whole pattern here of the world, which is more or less like this. And if we look at it, how the world looks, in 1960, it starts to move. This is Mao Zedong. He brought health to China. And then he died. And then Deng Xiaoping came and brought money to China, and brought them into the mainstream again. And we have seen how countries move in different directions like this, so it ' s sort of difficult to get an example country which shows the pattern of the world. But I would like to bring you back to about here, at 1960. I would like to compare South Korea, which is this one, with Brazil, which is this one. The label went away for me here. And I would like to compare Uganda, which is there. I can run it forward, like this. And you can see how South Korea is making a very, very fast advancement, whereas Brazil is much slower. And if we move back again, here, and we put trails on them, like this, you can see again that the speed of development is very, very different, and the countries are moving more or less at the same rate as money and health, but it seems you can move much faster if you are healthy first than if you are wealthy first. And to show that, you can put on the way of United Arab Emirates. They came from here, a mineral country. They cached all the oil ; they got all the money ; but health cannot be bought at the supermarket. You have to invest in health. You have to get kids into schooling. You have to train health staff. You have to educate the population. And Sheikh Zayed did that in a fairly good way. In spite of falling oil prices, he brought this country up here. So we ' ve got a much more mainstream appearance of the world, where all countries tend to use their money better than they used it in the past. Now, this is, more or less, if you look at the average data of the countries — they are like this. That ' s dangerous, to use average data, because there is such a lot of difference within countries. So if I go and look here, we can see that Uganda today is where South Korea was in 1960. If I split Uganda, there ' s quite a difference within Uganda. These are the quintiles of Uganda. The richest 20 percent of Ugandans are there. The poorest are down there. If I split South Africa, it ' s like this. And if I go down and look at Niger, where there was such a terrible famine [recently], it ' s like this. The 20 percent poorest of Niger is out here, and the 20 percent richest of South Africa is there, and yet we tend to discuss what solutions there should be in Africa. Everything in this world exists in Africa. And you can ' t discuss universal access to HIV [treatment] for that quintile up here with the same strategy as down here. The improvement of the world must be highly contextualized, and it ' s not relevant to have it on a regional level. We must be much more detailed. We find that students get very excited when they can use this. And even more, policy makers and the corporate sectors would like to see how the world is changing. Now, why doesn ' t this take place? Why are we not using the data we have? We have data in the United Nations, in the national statistical agencies and in universities and other nongovernmental organizations. Because the data is hidden down in the databases. And the public is there, and the internet is there, but we have still not used it effectively. All that information we saw changing in the world does not include publicly funded statistics. There are some web pages like this, you know, but they take some nourishment down from the databases, but people put prices on them, stupid passwords and boring statistics. And this won ' t work. So what is needed? We have the databases. It ' s not a new database that you need. We have wonderful design tools and more and more are added up here. So we started a nonprofit venture linking data to design, we called"Gapminder,"from the London Underground, where they warn you,"Mind the gap."So we thought Gapminder was appropriate. And we started to write software which could link the data like this. And it wasn ' t that difficult. It took some person years, and we have produced animations. You can take a data set and put it there. We are liberating UN data, some few UN organization. Some countries accept that their databases can go out on the world. But what we really need is, of course, a search function, a search function where we can copy the data up to a searchable format and get it out in the world. And what do we hear when we go around? I ' ve done anthropology on the main statistical units. Everyone says,"It ' s impossible. This can ' t be done. Our information is so peculiar in detail, so that cannot be searched as others can be searched. We cannot give the data free to the students, free to the entrepreneurs of the world."But this is what we would like to see, isn ' t it? The publicly funded data is down here. And we would like flowers to grow out on the net. One of the crucial points is to make them searchable, and then people can use the different design tools to animate it there. And I have pretty good news for you. I have good news that the [current], new head of UN statistics doesn ' t say it ' s impossible. He only says,"We can ' t do it."And that ' s a quite clever guy, huh? So we can see a lot happening in data in the coming years. We will be able to look at income distributions in completely new ways. This is the income distribution of China, 1970. This is the income distribution of the United States, 1970. Almost no overlap. Almost no overlap. And what has happened? What has happened is this : that China is growing, it ' s not so equal any longer, and it ' s appearing here, overlooking the United States, almost like a ghost, isn ' t it? It ' s pretty scary. But I think it ' s very important to have all this information. We need really to see it. And instead of looking at this, I would like to end up by showing the internet users per 1, 000. In this software, we access about 500 variables from all the countries quite easily. It takes some time to change for this, but on the axes, you can quite easily get any variable you would like to have. And the thing would be to get up the databases free, to get them searchable, and with a second click, to get them into the graphic formats, where you can instantly understand them. Now, statisticians don ' t like it, because they say that this will not show the reality ; we have to have statistical, analytical methods. But this is hypothesis-generating. I end now with the world. There, the internet is coming. The number of internet users are going up like this. This is the GDP per capita. And it ' s a new technology coming in, but then amazingly, how well it fits to the economy of the countries. That ' s why the $100 computer will be so important. But it ' s a nice tendency. It ' s as if the world is flattening off, isn ' t it? These countries are lifting more than the economy, and it will be very interesting to follow this over the year, as I would like you to be able to do with all the publicly funded data. Thank you very much.

Text Sample 270: NEG Dim 2 – 2006 – Score 1.00 – 2006/t000485.txt

Hi, everyone. I ' m Sirena. I ' m 11 years old and from Connecticut. Well, I ' m not really sure why I ' m here. I mean, what does this have to do with technology, entertainment and design? Well, I count my iPod, cellphone and computer as technology, but this has nothing to do with that. So I did a little research on it. Well, this is what I found. Of course, I hope I can memorize it. The violin is made of a wood box and four metal strings. By pulling a string, it vibrates and produces a sound wave, which passes through a piece of wood called a bridge, and goes down to the wood box and gets amplified, but... let me think. Placing your finger at different places on the fingerboard changes the string length, and that changes the frequency of the sound wave. Oh, my gosh! OK, this is sort of technology, but I can call it a 16th-century technology. But actually, the most fascinating thing that I found was that even the audio system or wave transmission nowadays are still based on the same principle of producing and projecting sound. Isn ’ t that cool? Design — I love its design. I remember when I was little, my mom asked me,”Would you like to play the violin or the piano?”I looked at that giant monster and said to myself —”I am not going to lock myself on that bench the whole day!”This is small and lightweight. I can play from standing, sitting or walking. And, you know what? The best of all is that if I don ’ t want to practice, I can hide it. The violin is very beautiful. Some people relate it as the shape of a lady. But whether you like it or not, it ’ s been so for more than 400 years, unlike modern stuff [that] easily looks dated. But I think it ’ s very personal and unique that, although each violin looks pretty similar, no two violins sound the same — even from the same maker or based on the same model. Entertainment — I love the entertainment. But actually, the instrument itself isn ’ t very entertaining. I mean, when I first got my violin and tried to play around on it, it was actually really bad, because it didn ’ t sound the way I ’ d heard from other kids — it was so horrible and so scratchy. So, it wasn ’ t entertaining at all. But besides, my brother found this very funny : Yuk! Yuk! Yuk! A few years later, I heard a joke about the greatest violinist, Jascha Heifetz. After Mr. Heifetz ’ s concert, a lady came over and complimented him :”Oh, Mr. Heifetz, your violin sounded so great tonight!”And Mr. Heifetz was a very cool person, so he picked up his violin and said,”Funny — I don ’ t hear anything.”Now I realize that as the musician, we human beings, with our great mind, artistic heart and skill, can change this 16th-century technology and a legendary design to a wonderful entertainment. Now I know why I ’ m here. At first, I thought I was just going to be here to perform, but unexpectedly, I learned and enjoyed much more. But... although some of the talks were quite up there for me. Like the multi-dimension stuff. I mean, honestly, I ’ d be happy enough if I could actually get my two dimensions correct in school. But actually, the most impressive thing to me is that — well, actually, I would also like to say this for all children is to say thank you to all adults, for actually caring for us a lot, and to make our future world much better. Thank you.

Text Sample 271: NEG Dim 2 – 2011 – Score 1.00 – 2011/t002454.txt

What I ’ m going to do is, I ’ m going to explain to you an extreme green concept that was developed at NASA ’ s Glenn Research Center in Cleveland, Ohio. But before I do that, we have to go over the definition of what green is, ’ cause a lot of us have a different definition of it. Green. The product is created through environmentally and socially conscious means. There ’ s plenty of things that are being called green now. What does it actually mean? We use three metrics to determine green. The first metric is : Is it sustainable? Which means, are you preserving what you are doing for future use or for future generations? Is it alternative? Is it different than what is being used today, or does it have a lower carbon footprint than what ’ s used conventionally? And three : Is it renewable? Does it come from Earth ’ s natural replenishing resources, such as sun, wind and water? Now, my task at NASA is to develop the next generation of aviation fuels. Extreme green. Why aviation? The field of aviation uses more fuel than just about every other combined. We need to find an alternative. Also it ’ s a national aeronautics directive. One of the national aeronautics goals is to develop the next generation of fuels, biofuels, using domestic and safe, friendly resources. Now, combating that challenge we have to also meet the big three metric — Actually, extreme green for us is all three together ; that ’ s why you see the plus there. I was told to say that. So it has to be the big three at GRC. That ' s another metric. Ninety-seven percent of the world ' s water is saltwater. How about we use that? Combine that with number three. Do not use arable land. Because crops are already growing on that land that ' s very scarce around the world. Number two : Don ' t compete with food crops. That ' s already a well established entity, they don ' t need another entry. And lastly the most precious resource we have on this Earth is fresh water. Don ' t use fresh water. If 97. 5 percent of the world ' s water is saltwater, 2. 5 percent is fresh water. Less than a half percent of that is accessible for human use. But 60 percent of the population lives within that one percent. So, combating my problem was, now I have to be extreme green and meet the big three. Ladies and gentlemen, welcome to the GreenLab Research Facility. This is a facility dedicated to the next generation of aviation fuels using halophytes. A halophyte is a salt-tolerating plant. Most plants don ' t like salt, but halophytes tolerate salt. We also are using weeds and we are also using algae. The good thing about our lab is, we ' ve had 3, 600 visitors in the last two years. Why do you think that ' s so? Because we are on to something special. So, in the lower you see the GreenLab obviously, and on the right hand side you ' ll see algae. If you are into the business of the next generation of aviation fuels, algae is a viable option, there ' s a lot of funding right now, and we have an algae to fuels program. There ' s two types of algae growing. One is a closed photobioreactor that you see here, and what you see on the other side is our species — we are currently using a species called Scenedesmus dimorphus. Our job at NASA is to take the experimental and computational and make a better mixing for the closed photobioreactors. Now the problems with closed photobioreactors are : They are quite expensive, they are automated, and it ' s very difficult to get them in large scale. So on large scale what do they use? We use open pond systems. Now, around the world they are growing algae, with this racetrack design that you see here. Looks like an oval with a paddle wheel and mixes really well, but when it gets around the last turn, which I call turn four — it ' s stagnant. We actually have a solution for that. In the GreenLab in our open pond system we use something that happens in nature : waves. We actually use wave technology on our open pond systems. We have 95 percent mixing and our lipid content is higher than a closed photobioreactor system, which we think is significant. There is a drawback to algae, however : It ' s very expensive. Is there a way to produce algae inexpensively? And the answer is : yes. We do the same thing we do with halophytes, and that is : climatic adaptation. In our GreenLab we have six primary ecosystems that range from freshwater all the way to saltwater. What we do : We take a potential species, we start at freshwater, we add a little bit more salt, when the second tank here will be the same ecosystem as Brazil — right next to the sugar cane fields you can have our plants — the next tank represents Africa, the next tank represents Arizona, the next tank represents Florida, and the next tank represents California or the open ocean. What we are trying to do is to come up with a single species that can survive anywhere in the world, where there ' s barren desert. We are being very successful so far. Now, here ' s one of the problems. If you are a farmer, you need five things to be successful : You need seeds, you need soil, you need water and you need sun, and the last thing that you need is fertilizer. Most people use chemical fertilizers. But guess what? We do not use chemical fertilizer. Wait a second! I just saw lots of greenery in your GreenLab. You have to use fertilizer. Believe it or not, in our analysis of our saltwater ecosystems 80 percent of what we need are in these tanks themselves. The 20 percent that ' s missing is nitrogen and phosphorous. We have a natural solution : fish. No we don ' t cut up the fish and put them in there. Fish waste is what we use. As a matter of fact we use freshwater mollies, that we ’ ve used our climatic adaptation technique from freshwater all the way to seawater. Freshwater mollies : cheap, they love to make babies, and they love to go to the bathroom. And the more they go to the bathroom, the more fertilizer we get, the better off we are, believe it or not. It should be noted that we use sand as our soil, regular beach sand. Fossilized coral. So a lot of people ask me,”How did you get started?”Well, we got started in what we call the indoor biofuels lab. It ’ s a seedling lab. We have 26 different species of halophytes, and five are winners. What we do here is — actually it should be called a death lab, ’ cause we try to kill the seedlings, make them rough — and then we come to the GreenLab. What you see in the lower corner is a wastewater treatment plant experiment that we are growing, a macro-algae that I ’ ll talk about in a minute. And lastly, it ’ s me actually working in the lab to prove to you I do work, I don ’ t just talk about what I do. Here ’ s the plant species. Salicornia virginica. It ’ s a wonderful plant. I love that plant. Everywhere we go we see it. It ’ s all over the place, from Maine all the way to California. We love that plant. Second is Salicornia bigelovii. Very difficult to get around the world. It is the highest lipid content that we have, but it has a shortcoming : It ’ s short. Now you take europaea, which is the largest or the tallest plant that we have. And what we are trying to do with natural selection or adaptive biology — combine all three to make a high-growth, high-lipid plant. Next, when a hurricane decimated the Delaware Bay — soybean fields gone — we came up with an idea : Can you have a plant that has a land reclamation positive in Delaware? And the answer is yes. It ’ s called seashore mallow. Kosteletzkya virginica — say that five times fast if you can. This is a 100 percent usable plant. The seeds : biofuels. The rest : cattle feed. It ’ s there for 10 years ; it ’ s working very well. Now we get to Chaetomorpha. This is a macro-algae that loves excess nutrients. If you are in the aquarium industry you know we use it to clean up dirty tanks. This species is so significant to us. The properties are very close to plastic. We are trying right now to convert this macro-algae into a bioplastic. If we are successful, we will revolutionize the plastics industry. So, we have a seed to fuel program. We have to do something with this biomass that we have. And so we do G. C. extraction, lipid optimization, so on and so forth, because our goal really is to come up with the next generation of aviation fuels, aviation specifics, so on and so forth. So far we talked about water and fuel, but along the way we found out something interesting about Salicornia : It ’ s a food product. So we talk about ideas worth spreading, right? How about this : In sub-Saharan Africa, next to the sea, saltwater, barren desert, how about we take that plant, plant it, half use for food, half use for fuel. We can make that happen, inexpensively. You can see there ’ s a greenhouse in Germany that sells it as a health food product. This is harvested, and in the middle here is a shrimp dish, and it ’ s being pickled. So I have to tell you a joke. Salicornia is known as sea beans, saltwater asparagus and pickle weed. So we are pickling pickle weed in the middle. Oh, I thought it was funny. And at the bottom is seaman ’ s mustard. It does make sense, this is a logical snack. You have mustard, you are a seaman, you see the halophyte, you mix it together, it ’ s a great snack with some crackers. And last, garlic with Salicornia, which is what I like. So, water, fuel and food. None of this is possible without the GreenLab team. Just like the Miami Heat has the big three, we have the big three at NASA GRC. That ’ s myself, professor Bob Hendricks, our fearless leader, and Dr. Arnon Chait. The backbone of the GreenLab is students. Over the last two years we ’ ve had 35 different students from around the world working at GreenLab. As a matter fact my division chief says a lot,”You have a green university.”I say,”I ’ m okay with that, ’ cause we are nurturing the next generation of extreme green thinkers, which is significant.”So, in first summary I presented to you what we think is a global solution for food, fuel and water. There ’ s something missing to be complete. Clearly we use electricity. We have a solution for you — We ’ re using clean energy sources here. So, we have two wind turbines connected to the GreenLab, we have four or five more hopefully coming soon. We are also using something that is quite interesting — there is a solar array field at NASA ’ s Glenn Research Center, hasn ’ t been used for 15 years. Along with some of my electrical engineering colleagues, we realized that they are still viable, so we are refurbishing them right now. In about 30 days or so they ’ ll be connected to the GreenLab. And the reason why you see red, red and yellow, is a lot of people think NASA employees don ’ t work on Saturday — This is a picture taken on Saturday. There are no cars around, but you see my truck in yellow. I work on Saturday. This is a proof to you that I ’ m working. ’ Cause we do what it takes to get the job done, most people know that. Here ’ s a concept with this : We are using the GreenLab for a micro-grid test bed for the smart grid concept in Ohio. We have the ability to do that, and I think it ’ s going to work. So, GreenLab Research Facility. A self-sustainable renewable energy ecosystem was presented today. We really, really hope this concept catches on worldwide. We think we have a solution for food, water, fuel and now energy. Complete. It ’ s extreme green, it ’ s sustainable, alternative and renewable and it meets the big three at GRC : Don ’ t use arable land, don ’ t compete with food crops, and most of all, don ’ t use fresh water. So I get a lot of questions about,”What are you doing in that lab?”And I usually say,”None of your business, that ’ s what I ’ m doing in the lab.”And believe it or not, my number one goal for working on this project is I want to help save the world.

Text Sample 272: NEG Dim 2 – 2011 – Score 1.00 – 2011/t002439.txt

The oceans cover some 70 percent of our planet. And I think Arthur C. Clarke probably had it right when he said that perhaps we ought to call our planet Planet Ocean. And the oceans are hugely productive, as you can see by the satellite image of photosynthesis, the production of new life. In fact, the oceans produce half of the new life every day on Earth as well as about half the oxygen that we breathe. In addition to that, it harbors a lot of the biodiversity on Earth, and much of it we don ’ t know about. But I ’ ll tell you some of that today. That also doesn ’ t even get into the whole protein extraction that we do from the ocean. That ’ s about 10 percent of our global needs and 100 percent of some island nations. If you were to descend into the 95 percent of the biosphere that ’ s livable, it would quickly become pitch black, interrupted only by pinpoints of light from bioluminescent organisms. And if you turn the lights on, you might periodically see spectacular organisms swim by, because those are the denizens of the deep, the things that live in the deep ocean. And eventually, the deep sea floor would come into view. This type of habitat covers more of the Earth ’ s surface than all other habitats combined. And yet, we know more about the surface of the Moon and about Mars than we do about this habitat, despite the fact that we have yet to extract a gram of food, a breath of oxygen or a drop of water from those bodies. And so 10 years ago, an international program began called the Census of Marine Life, which set out to try and improve our understanding of life in the global oceans. It involved 17 different projects around the world. As you can see, these are the footprints of the different projects. And I hope you ’ ll appreciate the level of global coverage that it managed to achieve. It all began when two scientists, Fred Grassle and Jesse Ausubel, met in Woods Hole, Massachusetts where both were guests at the famed oceanographic institute.

And Fred was lamenting the state of marine biodiversity and the fact that it was in trouble and nothing was being done about it. Well, from that discussion grew this program that involved 2, 700 scientists from more than 80 countries around the world who engaged in 540 ocean expeditions at a combined cost of 650 million dollars to study the distribution, diversity and abundance of life in the global ocean. And so what did we find? We found spectacular new species, the most beautiful and visually stunning things everywhere we looked — from the shoreline to the abyss, form microbes all the way up to fish and everything in between. And the limiting step here wasn ’ t the unknown diversity of life, but rather the taxonomic specialists who can identify and catalog these species that became the limiting step. They, in fact, are an endangered species themselves. There are actually four to five new species described everyday for the oceans. And as I say, it could be a much larger number. Now, I come from Newfoundland in Canada — It ’ s an island off the east coast of that continent — where we experienced one of the worst fishing disasters in human history. And so this photograph shows a small boy next to a codfish. It ’ s around 1900. Now, when I was a boy of about his age, I would go out fishing with my grandfather and we would catch fish about half that size. And I thought that was the norm, because I had never seen fish like this. If you were to go out there today, 20 years after this fishery collapsed, if you could catch a fish, which would be a bit of a challenge, it would be half that size still. So what we ’ re experiencing is something called shifting baselines. Our expectations of what the oceans can produce is something that we don ’ t really appreciate because we haven ’ t seen it in our lifetimes. Now most of us, and I would say me included, think that human exploitation of the oceans really only became very serious in the last 50 to, perhaps, 100 years or so. The census actually tried to look back in time, using every source of information they could get their hands on. And so anything from restaurant menus to monastery records to ships ’ logs to see what the oceans looked like. Because science data really goes back to, at best, World War II, for the most part. And so what they found, in fact, is that exploitation really began heavily with the Romans. And so at that time, of course, there was no refrigeration. So fishermen could only catch what they could either eat or sell that day. But the Romans developed salting. And with salting, it became possible to store fish and to transport it long distances. And so began industrial fishing. And so these are the sorts of extrapolations that we have of what sort of loss we ’ ve had relative to pre-human impacts on the ocean. They range from 65 to 98 percent for these major groups of organisms, as shown in the dark blue bars. Now for those species the we managed to leave alone, that we protect — for example, marine mammals in recent years and sea birds — there is some recovery. So it ’ s not all hopeless. But for the most part, we ’ ve gone from salting to exhausting. Now this other line of evidence is a really interesting one. It ’ s from trophy fish caught off the coast of Florida. And so this is a photograph from the 1950s. I want you to notice the scale on the slide, because when you see the same picture from the 1980s, we see the fish are much smaller and we ’ re also seeing a change in terms of the composition of those fish. By 2007, the catch was actually laughable in terms of the size for a trophy fish. But this is no laughing matter. The oceans have lost a lot of their productivity and we ’ re responsible for it. So what ’ s left? Actually quite a lot. There ’ s a lot of exciting things, and I ’ m going to tell you a little bit about them. And I want to start with a bit on technology, because, of course, this is a TED Conference and you want to hear something on technology. So one of the tools that we use to sample the deep ocean are remotely operated vehicles. So these are tethered vehicles we lower down to the sea floor where they ’ re our eyes and our hands for working on the sea bottom. So a couple of years ago, I was supposed to go on an oceanographic cruise and I couldn ' t go because of a scheduling conflict. But through a satellite link I was able to sit at my study at home with my dog curled up at my feet, a cup of tea in my hand, and I could tell the pilot,"I want a sample right there."And that ' s exactly what the pilot did for me. That ' s the sort of technology that ' s available today that really wasn ' t available even a decade ago. So it allows us to sample these amazing habitats that are very far from the surface and very far from light. And so one of the tools that we can use to sample the oceans is acoustics, or sound waves. And the advantage of sound waves is that they actually pass well through water, unlike light. And so we can send out sound waves, they bounce off objects like fish and are reflected back. And so in this example, a census scientist took out two ships. One would send out sound waves that would bounce back. They would be received by a second ship, and that would give us very precise estimates, in this case, of 250 billion herring in a period of about a minute. And that ' s an area about the size of Manhattan Island. And to be able to do that is a tremendous fisheries tool, because knowing how many fish are there is really critical. We can also use satellite tags to track animals as they move through the oceans. And so for animals that come to the surface to breathe, such as this elephant seal, it ' s an opportunity to send data back to shore and tell us where exactly it is in the ocean. And so from that we can produce these tracks. For example, the dark blue shows you where the elephant seal moved in the north Pacific. Now I realize for those of you who are colorblind, this slide is not very helpful, but stick with me nonetheless. For animals that don ' t surface, we have something called pop-up tags, which collect data about light and what time the sun rises and sets. And then at some period of time it pops up to the surface and, again, relays that data back to shore. Because GPS doesn ' t work under water. That ' s why we need these tools. And so from this we ' re able to identify these blue highways, these hot spots in the ocean, that should be real priority areas for ocean conservation. Now one of the other things that you may think about is that, when you go to the supermarket and you buy things, they ' re scanned. And so there ' s a barcode on that product that tells the computer exactly what the product is. Geneticists have developed a similar tool called genetic barcoding. And what barcoding does is use a specific gene called CO1 that ' s consistent within a species, but varies among species. And so what that means is we can unambiguously identify which species are which even if they look similar to each other, but may be biologically quite different. Now one of the nicest examples I like to cite on this is the story of two young women, high school students in New York City, who worked with the census. They went out and collected fish from markets and from restaurants in New York City and they barcoded it. Well what they found was mislabeled fish. So for example, they found something which was sold as tuna, which is very valuable, was in fact tilapia, which is a much less valuable fish. They also found an endangered species sold as a common one. So barcoding allows us to know what we ' re working with and also what we ' re eating. The Ocean Biogeographic Information System is the database for all the census data. It ' s open access ; you can all go in and download data as you wish. And it contains all the data from the census plus other data sets that people were willing to contribute. And so what you can do with that is to plot the distribution of species and where they occur in the oceans. What I ' ve plotted up here is the data that we have on hand. This is where our sampling effort has concentrated. Now what you can see is we ' ve sampled the area in the North Atlantic, in the North Sea in particular, and also the east coast of North America fairly well. That ' s the warm colors which show a well-sampled region. The cold colors, the blue and the black, show areas where we have almost no data. So even after a 10-year census, there are large areas that still remain unexplored. Now there are a group of scientists living in Texas, working in the Gulf of Mexico who decided really as a labor of love to pull together all the knowledge they could about biodiversity in the Gulf of Mexico. And so they put this together, a list of all the species, where they ' re known to occur, and it really seemed like a very esoteric, scientific type of exercise. But then, of course, there was the Deep Horizon oil spill. So all of a sudden, this labor of love for no obvious economic reason has become a critical piece of information in terms of how that system is going to recover, how long it will take and how the lawsuits and the multi-billion-dollar discussions that are going to happen in the coming years are likely to be resolved. So what did we find? Well, I could stand here for hours, but, of course, I ' m not allowed to do that. But I will tell you some of my favorite discoveries from the census. So one of the things we discovered is where are the hot spots of diversity? Where do we find the most species of ocean life? And what we find if we plot up the well-known species is this sort of a distribution. And what we see is that for coastal tags, for those organisms that live near the shoreline, they ' re most diverse in the tropics. This is something we ' ve actually known for a while, so it ' s not a real breakthrough. What is really exciting though is that the oceanic tags, or the ones that live far from the coast, are actually more diverse at intermediate latitudes. This is the sort of data, again, that managers could use if they want to prioritize areas of the ocean that we need to conserve. You can do this on a global scale, but you can also do it on a regional scale. And that ' s why biodiversity data can be so valuable. Now while a lot of the species we discovered in the census are things that are small and hard to see, that certainly wasn ' t always the case. For example, while it ' s hard to believe that a three kilogram lobster could elude scientists, it did until a few years ago when South African fishermen requested an export permit and scientists realized that this was something new to science. Similarly this Golden V kelp collected in Alaska just below the low water mark is probably a new species. Even though it ' s three meters long, it actually, again, eluded science. Now this guy, this bigfin squid, is seven meters in length. But to be fair, it lives in the deep waters of the Mid- Atlantic Ridge, so it was a lot harder to find. But there ' s still potential for discovery of big and exciting things. This particular shrimp, we ' ve dubbed it the Jurassic shrimp, it ' s thought to have gone extinct 50 years ago — at least it was, until the census discovered it was living and doing just fine off the coast of Australia. And it shows that the ocean, because of its vastness, can hide secrets for a very long time. So, Steven Spielberg, eat your heart out. If we look at distributions, in fact distributions change dramatically. And so one of the records that we had was this sooty shearwater, which undergoes these spectacular migrations all the way from New Zealand all the way up to Alaska and back again in search of endless summer as they complete their life cycles. We also talked about the White Shark Cafe. This is a location in the Pacific where white shark converge. We don ' t know why they converge there, we simply don ' t know. That ' s a question for the future. One of the things that we ' re taught in high school is that all animals require oxygen in order to survive. Now this little critter, it ' s only about half a millimeter in size, not terribly charismatic. But it was only discovered in the early 1980s. But the really interesting thing about it is that, a few years ago, census scientists discovered that this guy can thrive in oxygen-poor sediments in the deep Mediterranean Sea. So now they know that, in fact, animals can live without oxygen, at least some of them, and that they can adapt to even the harshest of conditions. If you were to suck all the water out of the ocean, this is what you ' d be left behind with, and that ' s the biomass of life on the sea floor. Now what we see is huge biomass towards the poles and not much biomass in between. We found life in the extremes. And so there were new species that were found that live inside ice and help to support an ice-based food web. And we also found this spectacular yeti crab that lives near boiling hot hydrothermal vents at Easter Island. And this particular species really captured the public ' s attention. We also found the deepest vents known yet — 5, 000 meters — the hottest vents at 407 degrees Celsius — vents in the South Pacific and also in the Arctic where none had been found before. So even new environments are still within the domain of the discoverable. Now in terms of the unknowns, there are many. And I ' m just going to summarize just a few of them very quickly for you. First of all, we might ask, how many fishes in the sea? We actually know the fishes better than we do any other group in the ocean other than marine mammals. And so we can actually extrapolate based on rates of discovery how many more species we ' re likely to discover. And from that, we actually calculate that we know about 16, 500 marine species and there are probably another 1, 000 to 4, 000 left to go. So we ' ve done pretty well. We ' ve got about 75 percent of the fish, maybe as much as 90 percent. But the fishes, as I say, are the best known. So our level of knowledge is much less for other groups of organisms. Now this figure is actually based on a brand new paper that ' s going to come out in the journal PLoS Biology. And what is does is predict how many more species there are on land and in the ocean. And what they found is that they think that we know of about nine percent of the species in the ocean. That means 91 percent, even after the census, still remain to be discovered. And so that turns out to be about two million species once all is said and done. So we still have quite a lot of work to do in terms of unknowns. Now this bacterium is part of mats that are found off the coast of Chile. And these mats actually cover an area the size of Greece. And so this particular bacterium is actually visible to the naked eye. But you can imagine the biomass that represents. But the really intriguing thing about the microbes is just how diverse they are. A single drop of seawater could contain 160 different types of microbes. And the oceans themselves are thought potentially to contain as many as a billion different types. So that ' s really exciting. What are they all doing out there? We actually don ' t know. The most exciting thing, I would say, about this census is the role of global science. And so as we see in this image of light during the night, there are lots of areas of the Earth where human development is much greater and other areas where it ' s much less, but between them we see large dark areas of relatively unexplored ocean. The other point I ' d like to make about this is that this ocean ' s interconnected. Marine organisms do not care about international boundaries ; they move where they will. And so the importance then of global collaboration becomes all the more important. We ' ve lost a lot of paradise. For example, these tuna that were once so abundant in the North Sea are now effectively gone. There were trawls taken in the deep sea in the Mediterranean, which collected more garbage than they did animals. And that ' s the deep sea, that ' s the environment that we consider to be among the most pristine left on Earth. And there are a lot of other pressures. Ocean acidification is a really big issue that people are concerned with, as well as ocean warming, and the effects they ' re going to have on coral reefs. On the scale of decades, in our lifetimes, we ' re going to see a lot of damage to coral reefs. And I could spend the rest of my time, which is getting very limited, going through this litany of concerns about the ocean, but I want to end on a more positive note. And so the grand challenge then is to try and make sure that we preserve what ' s left, because there is still spectacular beauty. And the oceans are so productive, there ' s so much going on in there that ' s of relevance to humans that we really need to, even from a selfish perspective, try to do better than we have in the past. So we need to recognize those hot spots and do our best to protect them. When we look at pictures like this, they take our breath away, in addition to helping to give us breath by the oxygen that the oceans provide. Census scientists worked in the rain, they worked in the cold, they worked under water and they worked above water trying to illuminate the wondrous discovery, the still vast unknown, the spectacular adaptations that we see in ocean life. So whether you ’ re a yak herder living in the mountains of Chile, whether you ’ re a stockbroker in New York City or whether you ’ re a TEDster living in Edinburgh, the oceans matter. And as the oceans go so shall we. Thanks for listening.

Text Sample 273: NEG Dim 2 – 2011 – Score 1.00 – 2011/t002436.txt

The things we make have one supreme quality â they live longer than us. We perish, they survive ; we have one life, they have many lives, and in each life they can mean different things. Which means that, while we all have one biography, they have many. I want this morning to talk about the story, the biography â or rather the biographies â of one particular object, one remarkable thing. It doesn ’ t, I agree, look very much. It ’ s about the size of a rugby ball. It ’ s made of clay, and it ’ s been fashioned into a cylinder shape, covered with close writing and then baked dry in the sun. And as you can see, it ’ s been knocked about a bit, which is not surprising because it was made two and a half thousand years ago and was dug up in 1879. But today, this thing is, I believe, a major player in the politics of the Middle East. And it ’ s an object with fascinating stories and stories that are by no means over yet. The story begins in the Iran-Iraq war and that series of events that culminated in the invasion of Iraq by foreign forces, the removal of a despotic ruler and instant regime change. And I want to begin with one episode from that sequence of events that most of you would be very familiar with, Belshazzar ’ s feast â because we ’ re talking about the Iran-Iraq war of 539 BC. And the parallels between the events of 539 BC and 2003 and in between are startling. What you ’ re looking at is Rembrandt ’ s painting, now in the National Gallery in London, illustrating the text from the prophet Daniel in the Hebrew scriptures. And you all know roughly the story. Belshazzar, the son of Nebuchadnezzar, Nebuchadnezzar who ’ d conquered Israel, sacked Jerusalem and captured the people and taken the Jews back to Babylon. Not only the Jews, he ’ d taken the temple vessels. He ’ d ransacked, desecrated the temple. And the great gold vessels of the temple in Jerusalem had been taken to Babylon. Belshazzar, his son, decides to have a feast. And in order to make it even more exciting, he added a bit of sacrilege to the rest of the fun, and he brings out the temple vessels. He ’ s already at war with the Iranians, with the king of Persia. And that night, Daniel tells us, at the height of the festivities a hand appeared and wrote on the wall,”You are weighed in the balance and found wanting, and your kingdom is handed over to the Medes and the Persians.”And that very night Cyrus, king of the Persians, entered Babylon and the whole regime of Belshazzar fell. It is, of course, a great moment in the history of the Jewish people. It ’ s a great story. It ’ s story we all know.”The writing on the wall”is part of our everyday language. What happened next was remarkable, and it ’ s where our cylinder enters the story. Cyrus, king of the Persians, has entered Babylon without a fight â the great empire of Babylon, which ran from central southern Iraq to the Mediterranean, falls to Cyrus. And Cyrus makes a declaration. And that is what this cylinder is, the declaration made by the ruler guided by God who had toppled the Iraqi despot and was going to bring freedom to the people. In ringing Babylonian â it was written in Babylonian â he says,”I am Cyrus, king of all the universe, the great king, the powerful king, king of Babylon, king of the four quarters of the world.”They ’ re not shy of hyperbole as you can see. This is probably the first real press release by a victorious army that we ’ ve got. And it ’ s written, as we ’ ll see in due course, by very skilled P. R. consultants. So the hyperbole is not actually surprising. And what is the great king, the powerful king, the king of the four quarters of the world going to do? He goes on to say that, having conquered Babylon, he will at once let all the peoples that the Babylonians â Nebuchadnezzar and Belshazzar â have captured and enslaved go free. He ’ ll let them return to their countries. And more important, he will let them all recover the gods, the statues, the temple vessels that had been confiscated. All the peoples that the Babylonians had repressed and removed will go home, and they ’ ll take with them their gods. And they ’ ll be able to restore their altars and to worship their gods in their own way, in their own place. This is the decree, this object is the evidence for the fact that the Jews, after the exile in Babylon, the years they ’ d spent sitting by the waters of Babylon, weeping when they remembered Jerusalem, those Jews were allowed to go home. They were allowed to return to Jerusalem and to rebuild the temple. It ’ s a central document in Jewish history. And the Book of Chronicles, the Book of Ezra in the Hebrew scriptures reported in ringing terms. This is the Jewish version of the same story.”Thus said Cyrus, king of Persia, ’ All the kingdoms of the earth have the Lord God of heaven given thee, and he has charged me to build him a house in Jerusalem. Who is there among you of his people? The Lord God be with him, and let him go up. ’””Go up”â aaleh. The central element, still, of the notion of return, a central part of the life of Judaism. As you all know, that return from exile, the second temple, reshaped Judaism. And that change, that great historic moment, was made possible by Cyrus, the king of Persia, reported for us in Hebrew in scripture and in Babylonian in clay. Two great texts, what about the politics? What was going on was the fundamental shift in Middle Eastern history. The empire of Iran, the Medes and the Persians, united under Cyrus, became the first great world empire. Cyrus begins in the 530s BC. And by the time of his son Darius, the whole of the eastern Mediterranean is under Persian control. This empire is, in fact, the Middle East as we now know it, and it ’ s what shapes the Middle East as we now know it. It was the largest empire the world had known until then. Much more important, it was the first multicultural, multifaith state on a huge scale. And it had to be run in a quite new way. It had to be run in different languages. The fact that this decree is in Babylonian says one thing. And it had to recognize their different habits, different peoples, different religions, different faiths. All of those are respected by Cyrus. Cyrus sets up a model of how you run a great multinational, multifaith, multicultural society. And the result of that was an empire that included the areas you see on the screen, and which survived for 200 years of stability until it was shattered by Alexander. It left a dream of the Middle East as a unit, and a unit where people of different faiths could live together. The Greek invasions ended that. And of course, Alexander couldn ’ t sustain a government and it fragmented. But what Cyrus represented remained absolutely central. The Greek historian Xenophon wrote his book”Cyropaedia”promoting Cyrus as the great ruler. And throughout European culture afterward, Cyrus remained the model. This is a 16th century image to show you how widespread his veneration actually was. And Xenophon ’ s book on Cyrus on how you ran a diverse society was one of the great textbooks that inspired the Founding Fathers of the American Revolution. Jefferson was a great admirer â the ideals of Cyrus obviously speaking to those 18th century ideals of how you create religious tolerance in a new state. Meanwhile, back in Babylon, things had not been going well. After Alexander, the other empires, Babylon declines, falls into ruins, and all the traces of the great Babylonian empire are lost â until 1879 when the cylinder is discovered by a British Museum exhibition digging in Babylon. And it enters now another story. It enters that great debate in the middle of the 19th century : Are the scriptures reliable? Can we trust them? We only knew about the return of the Jews and the decree of Cyrus from the Hebrew scriptures. No other evidence. Suddenly, this appeared. And great excitement to a world where those who believed in the scriptures had had their faith in creation shaken by evolution, by geology, here was evidence that the scriptures were historically true. It ’ s a great 19th century moment. But â and this, of course, is where it becomes complicated â the facts were true, hurrah for archeology, but the interpretation was rather more complicated. Because the cylinder account and the Hebrew Bible account differ in one key respect. The Babylonian cylinder is written by the priests of the great god of Bablyon, Marduk. And, not surprisingly, they tell you that all this was done by Marduk.”Marduk, we hold, called Cyrus by his name.”Marduk takes Cyrus by the hand, calls him to shepherd his people and gives him the rule of Babylon. Marduk tells Cyrus that he will do these great, generous things of setting the people free. And this is why we should all be grateful to and worship Marduk. The Hebrew writers in the Old Testament, you will not be surprised to learn, take a rather different view of this. For them, of course, it can ’ t possibly by Marduk that made all this happen. It can only be Jehovah. And so in Isaiah, we have the wonderful texts giving all the credit of this, not to Marduk but to the Lord God of Israel â the Lord God of Israel who also called Cyrus by name, also takes Cyrus by the hand and talks of him shepherding his people. It ’ s a remarkable example of two different priestly appropriations of the same event, two different religious takeovers of a political fact. God, we know, is usually on the side of the big battalions. The question is, which god was it? And the debate unsettles everybody in the 19th century to realize that the Hebrew scriptures are part of a much wider world of religion. And it ’ s quite clear the cylinder is older than the text of Isaiah, and yet, Jehovah is speaking in words very similar to those used by Marduk. And there ’ s a slight sense that Isaiah knows this, because he says, this is God speaking, of course,”I have called thee by thy name though thou hast not known me.”I think it ’ s recognized that Cyrus doesn ’ t realize that he ’ s acting under orders from Jehovah. And equally, he ’ d have been surprised that he was acting under orders from Marduk. Because interestingly, of course, Cyrus is a good Iranian with a totally different set of gods who are not mentioned in any of these texts. That ’ s 1879. 40 years on and we ’ re in 1917, and the cylinder enters a different world. This time, the real politics of the contemporary world â the year of the Balfour Declaration, the year when the new imperial power in the Middle East, Britain, decides that it will declare a Jewish national home, it will allow the Jews to return. And the response to this by the Jewish population in Eastern Europe is rhapsodic. And across Eastern Europe, Jews display pictures of Cyrus and of George V side by side â the two great rulers who have allowed the return to Jerusalem. And the Cyrus cylinder comes back into public view and the text of this as a demonstration of why what is going to happen after the war is over in 1918 is part of a divine plan. You all know what happened. The state of Israel is setup, and 50 years later, in the late 60s, it ’ s clear that Britain ’ s role as the imperial power is over. And another story of the cylinder begins. The region, the U. K. and the U. S. decide, has to be kept safe from communism, and the superpower that will be created to do this would be Iran, the Shah. And so the Shah invents an Iranian history, or a return to Iranian history, that puts him in the center of a great tradition and produces coins showing himself with the Cyrus cylinder. When he has his great celebrations in Persepolis, he summons the cylinder and the cylinder is lent by the British Museum, goes to Tehran, and is part of those great celebrations of the Pahlavi dynasty. Cyrus cylinder : guarantor of the Shah. 10 years later, another story : Iranian Revolution, 1979. Islamic revolution, no more Cyrus ; we ’ re not interested in that history, we ’ re interested in Islamic Iran â until Iraq, the new superpower that we ’ ve all decided should be in the region, attacks. Then another Iran-Iraq war. And it becomes critical for the Iranians to remember their great past, their great past when they fought Iraq and won. It becomes critical to find a symbol that will pull together all Iranians â Muslims and non-Muslims, Christians, Zoroastrians, Jews living in Iran, people who are devout, not devout. And the obvious emblem is Cyrus. So when the British Museum and Tehran National Musuem cooperate and work together, as we ’ ve been doing, the Iranians ask for one thing only as a loan. It ’ s the only object they want. They want to borrow the Cyrus cylinder. And last year, the Cyrus cylinder went to Tehran for the second time. It ’ s shown being presented here, put into its case by the director of the National Museum of Tehran, one of the many women in Iran in very senior positions, Mrs. Ardakani. It was a huge event. This is the other side of that same picture. It ’ s seen in Tehran by between one and two million people in the space of a few months. This is beyond any blockbuster exhibition in the West. And it ’ s the subject of a huge debate about what this cylinder means, what Cyrus means, but above all, Cyrus as articulated through this cylinder â Cyrus as the defender of the homeland, the champion, of course, of Iranian identity and of the Iranian peoples, tolerant of all faiths. And in the current Iran, Zoroastrians and Christians have guaranteed places in the Iranian parliament, something to be very, very proud of. To see this object in Tehran, thousands of Jews living in Iran came to Tehran to see it. It became a great emblem, a great subject of debate about what Iran is at home and abroad. Is Iran still to be the defender of the oppressed? Will Iran set free the people that the tyrants have enslaved and expropriated? This is heady national rhetoric, and it was all put together in a great pageant launching the return. Here you see this out-sized Cyrus cylinder on the stage with great figures from Iranian history gathering to take their place in the heritage of Iran. It was a narrative presented by the president himself. And for me, to take this object to Iran, to be allowed to take this object to Iran was to be allowed to be part of an extraordinary debate led at the highest levels about what Iran is, what different Irans there are and how the different histories of Iran might shape the world today. It ’ s a debate that ’ s still continuing, and it will continue to rumble, because this object is one of the great declarations of a human aspiration. It stands with the American constitution. It certainly says far more about real freedoms than Magna Carta. It is a document that can mean so many things, for Iran and for the region. A replica of this is at the United Nations. In New York this autumn, it will be present when the great debates about the future of the Middle East take place. And I want to finish by asking you what the next story will be in which this object figures. It will appear, certainly, in many more Middle Eastern stories. And what story of the Middle East, what story of the world, do you want to see reflecting what is said, what is expressed in this cylinder? The right of peoples to live together in the same state, worshiping differently, freely â a Middle East, a world, in which religion is not the subject of division or of debate. In the world of the Middle East at the moment, the debates are, as you know, shrill. But I think it ’ s possible that the most powerful and the wisest voice of all of them may well be the voice of this mute thing, the Cyrus cylinder. Thank you.

Text Sample 274: NEG Dim 2 – 2003 – Score 1.00 – 2003/t000398.txt

So I ’ m going to speak about a problem that I have and that ’ s that I ’ m a philosopher. When I go to a party and people ask me what do I do and I say,”I ’ m a professor,”their eyes glaze over. When I go to an academic cocktail party and there are all the professors around, they ask me what field I ’ m in and I say,”philosophy”— their eyes glaze over. When I go to a philosopher ’ s party and they ask me what I work on and I say,”consciousness,”their eyes don ’ t glaze over — their lips curl into a snarl. And I get hoots of derision and cackles and growls because they think,”That ’ s impossible! You can ’ t explain consciousness.”The very chutzpah of somebody thinking that you could explain consciousness is just out of the question. My late, lamented friend Bob Nozick, a fine philosopher, in one of his books,”Philosophical Explanations,”is commenting on the ethos of philosophy — the way philosophers go about their business. And he says, you know,”Philosophers love rational argument.”And he says,”It seems as if the ideal argument for most philosophers is you give your audience the premises and then you give them the inferences and the conclusion, and if they don ’ t accept the conclusion, they die. Their heads explode.”The idea is to have an argument that is so powerful that it knocks out your opponents. But in fact that doesn ’ t change people ’ s minds at all. It ’ s very hard to change people ’ s minds about something like consciousness, and I finally figured out the reason for that. The reason for that is that everybody ’ s an expert on consciousness. We heard the other day that everybody ’ s got a strong opinion about video games. They all have an idea for a video game, even if they ’ re not experts. But they don ’ t consider themselves experts on video games ; they ’ ve just got strong opinions. I ’ m sure that people here who work on, say, climate change and global warming, or on the future of the Internet, encounter people who have very strong opinions about what ’ s going to happen next. But they probably don ’ t think of these opinions as expertise. They ’ re just strongly held opinions. But with regard to consciousness, people seem to think, each of us seems to think,”I am an expert. Simply by being conscious, I know all about this.”And so, you tell them your theory and they say,”No, no, that ’ s not the way consciousness is! No, you ’ ve got it all wrong.”And they say this with an amazing confidence. And so what I ’ m going to try to do today is to shake your confidence. Because I know the feeling — I can feel it myself. I want to shake your confidence that you know your own innermost minds — that you are, yourselves, authoritative about your own consciousness. That ’ s the order of the day here. Now, this nice picture shows a thought-balloon, a thought-bubble. I think everybody understands what that means. That ’ s supposed to exhibit the stream of consciousness. This is my favorite picture of consciousness that ’ s ever been done. It ’ s a Saul Steinberg of course — it was a New Yorker cover. And this fellow here is looking at the painting by Braque. That reminds him of the word baroque, barrack, bark, poodle, Suzanne R. — he ’ s off to the races. There ’ s a wonderful stream of consciousness here and if you follow it along, you learn a lot about this man. What I particularly like about this picture, too, is that Steinberg has rendered the guy in this sort of pointillist style. Which reminds us, as Rod Brooks was saying yesterday : what we are, what each of us is — what you are, what I am — is approximately 100 trillion little cellular robots. That ’ s what we ’ re made of. No other ingredients at all. We ’ re just made of cells, about 100 trillion of them. Not a single one of those cells is conscious ; not a single one of those cells knows who you are, or cares. Somehow, we have to explain how when you put together teams, armies, battalions of hundreds of millions of little robotic unconscious cells — not so different really from a bacterium, each one of them — the result is this. I mean, just look at it. The content — there ’ s color, there ’ s ideas, there ’ s memories, there ' s history. And somehow all that content of consciousness is accomplished by the busy activity of those hoards of neurons. How is that possible? Many people just think it isn ' t possible at all. They think,"No, there can ' t be any sort of naturalistic explanation of consciousness."This is a lovely book by a friend of mine named Lee Siegel, who ' s a professor of religion, actually, at the University of Hawaii, and he ' s an expert magician, and an expert on the street magic of India, which is what this book is about,"Net of Magic."And there ' s a passage in it which I would love to share with you. It speaks so eloquently to the problem."' I ' m writing a book on magic, ' I explain, and I ' m asked, ' Real magic? ' By ' real magic, ' people mean miracles, thaumaturgical acts, and supernatural powers. ' No, ' I answer. ' Conjuring tricks, not real magic. ' ' Real magic, ' in other words, refers to the magic that is not real ; while the magic that is real, that can actually be done, is not real magic."Now, that ' s the way a lot of people feel about consciousness. Real consciousness is not a bag of tricks. If you ' re going to explain this as a bag of tricks, then it ' s not real consciousness, whatever it is. And, as Marvin said, and as other people have said,"Consciousness is a bag of tricks."This means that a lot of people are just left completely dissatisfied and incredulous when I attempt to explain consciousness. So this is the problem. So I have to do a little bit of the sort of work that a lot of you won ' t like, for the same reason that you don ' t like to see a magic trick explained to you. How many of you here, if somebody — some smart aleck — starts telling you how a particular magic trick is done, you sort of want to block your ears and say,"No, no, I don ' t want to know! Don ' t take the thrill of it away. I ' d rather be mystified. Don ' t tell me the answer."A lot of people feel that way about consciousness, I ' ve discovered. And I ' m sorry if I impose some clarity, some understanding on you. You ' d better leave now if you don ' t want to know some of these tricks. But I ' m not going to explain it all to you. I ' m going to do what philosophers do. Here ' s how a philosopher explains the sawing-the-lady-in-half trick. You know the sawing-the-lady-in-half trick? The philosopher says,"I ' m going to explain to you how that ' s done. You see, the magician doesn ' t really saw the lady in half.""He merely makes you think that he does."And you say,"Yes, and how does he do that?"He says,"Oh, that ' s not my department, I ' m sorry."So now I ' m going to illustrate how philosophers explain consciousness. But I ' m going to try to also show you that consciousness isn ' t quite as marvelous — your own consciousness isn ' t quite as wonderful — as you may have thought it is. This is something, by the way, that Lee Siegel talks about in his book. He marvels at how he ' ll do a magic show, and afterwards people will swear they saw him do X, Y, and Z. He never did those things. He didn ' t even try to do those things. People ' s memories inflate what they think they saw. And the same is true of consciousness. Now, let ' s see if this will work. All right. Let ' s just watch this. Watch it carefully. I ' m working with a young computer-animator documentarian named Nick Deamer, and this is a little demo that he ' s done for me, part of a larger project some of you may be interested in. We ' re looking for a backer. It ' s a feature-length documentary on consciousness. OK, now, you all saw what changed, right? How many of you noticed that every one of those squares changed color? Every one. I ' ll just show you by running it again. Even when you know that they ' re all going to change color, it ' s very hard to notice. You have to really concentrate to pick up any of the changes at all. Now, this is an example — one of many — of a phenomenon that ' s now being studied quite a bit. It ' s one that I predicted in the last page or two of my 1991 book,"Consciousness Explained,"where I said if you did experiments of this sort, you ' d find that people were unable to pick up really large changes. If there ' s time at the end, I ' ll show you the much more dramatic case. Now, how can it be that there are all those changes going on, and that we ’ re not aware of them? Well, earlier today, Jeff Hawkins mentioned the way your eye saccades, the way your eye moves around three or four times a second. He didn ’ t mention the speed. Your eye is constantly in motion, moving around, looking at eyes, noses, elbows, looking at interesting things in the world. And where your eye isn ’ t looking, you ’ re remarkably impoverished in your vision. That ’ s because the foveal part of your eye, which is the high-resolution part, is only about the size of your thumbnail held at arms length. That ’ s the detail part. It doesn ’ t seem that way, does it? It doesn ’ t seem that way, but that ’ s the way it is. You ’ re getting in a lot less information than you think. Here ’ s a completely different effect. This is a painting by Bellotto. It ’ s in the museum in North Carolina. Bellotto was a student of Canaletto ’ s. And I love paintings like that — the painting is actually about as big as it is right here. And I love Canalettos, because Canaletto has this fantastic detail, and you can get right up and see all the details on the painting. And I started across the hall in North Carolina, because I thought it was probably a Canaletto, and would have all that in detail. And I noticed that on the bridge there, there ’ s a lot of people — you can just barely see them walking across the bridge. And I thought as I got closer I would be able to see all the detail of most people, see their clothes, and so forth. And as I got closer and closer, I actually screamed. I yelled out because when I got closer, I found the detail wasn ’ t there at all. There were just little artfully placed blobs of paint. And as I walked towards the picture, I was expecting detail that wasn ’ t there. The artist had very cleverly suggested people and clothes and wagons and all sorts of things, and my brain had taken the suggestion. You ’ re familiar with a more recent technology, which is — There, you can get a better view of the blobs. See, when you get close they ’ re really just blobs of paint. You will have seen something like this — this is the reverse effect. I ’ ll just give that to you one more time. Now, what does your brain do when it takes the suggestion? When an artful blob of paint or two, by an artist, suggests a person — say, one of Marvin Minsky ’ s little society of mind — do they send little painters out to fill in all the details in your brain somewhere? I don ’ t think so. Not a chance. But then, how on Earth is it done? Well, remember the philosopher ’ s explanation of the lady? It ’ s the same thing. The brain just makes you think that it ’ s got the detail there. You think the detail ’ s there, but it isn ’ t there. The brain isn ’ t actually putting the detail in your head at all. It ’ s just making you expect the detail. Let ’ s just do this experiment very quickly. Is the shape on the left the same as the shape on the right, rotated? Yes. How many of you did it by rotating the one on the left in your mind ’ s eye, to see if it matched up with the one on the right? How many of you rotated the one on the right? OK. How do you know that ’ s what you did? There ’ s in fact been a very interesting debate raging for over 20 years in cognitive science — various experiments started by Roger Shepherd, who measured the angular velocity of rotation of mental images. Yes, it ’ s possible to do that. But the details of the process are still in significant controversy. And if you read that literature, one of the things that you really have to come to terms with is even when you ’ re the subject in the experiment, you don ’ t know. You don ’ t know how you do it. You just know that you have certain beliefs. And they come in a certain order, at a certain time. And what explains the fact that that ’ s what you think? Well, that ’ s where you have to go backstage and ask the magician. This is a figure that I love : Bradley, Petrie, and Dumais. You may think that I ’ ve cheated, that I ’ ve put a little whiter-than-white boundary there. How many of you see that sort of boundary, with the Necker cube floating in front of the circles? Can you see it? Well, you know, in effect, the boundary ’ s really there, in a certain sense. Your brain is actually computing that boundary, the boundary that goes right there. But now, notice there are two ways of seeing the cube, right? It ' s a Necker cube. Everybody can see the two ways of seeing the cube? OK. Can you see the four ways of seeing the cube? Because there ' s another way of seeing it. If you ' re seeing it as a cube floating in front of some circles, some black circles, there ' s another way of seeing it. As a cube, on a black background, as seen through a piece of Swiss cheese. Can you get it? How many of you can ' t get it? That ' ll help. Now you can get it. These are two very different phenomena. When you see the cube one way, behind the screen, those boundaries go away. But there ' s still a sort of filling in, as we can tell if we look at this. We don ' t have any trouble seeing the cube, but where does the color change? Does your brain have to send little painters in there? The purple-painters and the green-painters fight over who ' s going to paint that bit behind the curtain? No. Your brain just lets it go. The brain doesn ' t need to fill that in. When I first started talking about the Bradley, Petrie, Dumais example that you just saw — I ' ll go back to it, this one — I said that there was no filling-in behind there. And I supposed that that was just a flat truth, always true. But Rob Van Lier has recently shown that it isn ' t. Now, if you think you see some pale yellow — I ' ll run this a few more times. Look in the gray areas, and see if you seem to see something sort of shadowy moving in there — yeah, it ' s amazing. There ' s nothing there. It ' s no trick. [”Failure to Detect Changes in Scenes”slide] This is Ron Rensink ' s work, which was in some degree inspired by that suggestion right at the end of the book. Let me just pause this for a second if I can. This is change-blindness. What you ' re going to see is two pictures, one of which is slightly different from the other. You see here the red roof and the gray roof, and in between them there will be a mask, which is just a blank screen, for about a quarter of a second. So you ' ll see the first picture, then a mask, then the second picture, then a mask. And this will just continue, and your job as the subject is to press the button when you see the change. So, show the original picture for 240 milliseconds. Blank. Show the next picture for 240 milliseconds. Blank. And keep going, until the subject presses the button, saying,”I see the change.”So now we ' re going to be subjects in the experiment. We ' re going to start easy. Some examples. No trouble there. Can everybody see? All right. Indeed, Rensink ' s subjects took only a little bit more than a second to press the button. Can you see that one? 2. 9 seconds. How many don ' t see it still? What ' s on the roof of that barn? It ' s easy. Is it a bridge or a dock? There are a few more really dramatic ones, and then I ' ll close. I want you to see a few that are particularly striking. This one because it ' s so large and yet it ' s pretty hard to see. Can you see it? Audience : Yes. Dan Dennett : See the shadows going back and forth? Pretty big. So 15. 5 seconds is the median time for subjects in his experiment there. I love this one. I ' ll end with this one, just because it ' s such an obvious and important thing. How many still don ' t see it? How many still don ' t see it? How many engines on the wing of that Boeing? Right in the middle of the picture! Thanks very much for your attention. What I wanted to show you is that scientists, using their from-the-outside, third- person methods, can tell you things about your own consciousness that you would never dream of, and that, in fact, you ' re not the authority on your own consciousness that you think you are. And we ' re really making a lot of progress on coming up with a theory of mind. Jeff Hawkins, this morning, was describing his attempt to get theory, and a good, big theory, into the neuroscience. And he ' s right. This is a problem. Harvard Medical School once — I was at a talk — director of the lab said,”In our lab, we have a saying. If you work on one neuron, that ' s neuroscience. If you work on two neurons, that ' s psychology.”We have to have more theory, and it can come as much from the top down. Thank you very much.

Text Sample 275: NEG Dim 2 – 2003 – Score 1.00 – 2003/t000319.txt

I study ants, and that ' s because I like to think about how organizations work. And in particular, how the simple parts of organizations interact to create the behavior of the whole organization. So, ant colonies are a good example of an organization like that, and there are many others. The web is one. There are many biological systems like that — brains, cells, developing embryos. There are about 10, 000 species of ants. They all live in colonies consisting of one or a few queens, and then all the ants you see walking around are sterile female workers. And all ant colonies have in common that there ' s no central control. Nobody tells anybody what to do. The queen just lays the eggs. There ' s no management. No ant directs the behavior of any other ant. And I try to figure out how that works. And I ' ve been working for the past 20 years on a population of seed-eating ants in southeastern Arizona. Here ' s my study site. This is really a picture of ants, and the rabbit just happens to be there. And these ants are called harvester ants because they eat seeds. This is the nest of the mature colony, and there ' s the nest entrance. And they forage maybe for about 20 meters away, gather up the seeds and bring them back to the nest, and store them. And every year I go there and make a map of my study site. This is just a road. And it ' s not very big : it ' s about 250 meters on one side, 400 on the other. And every colony has a name, which is a number, which is painted on a rock. And I go there every year and look for all the colonies that were alive the year before, and figure out which ones have died, and put all the new ones on the map. And by doing this I know how old they all are. And because of that, I ' ve been able to study how their behavior changes as the colony gets older and larger. So I want to tell you about the life cycle of a colony. Ants never make more ants ; colonies make more colonies. And they do that by each year sending out the reproductives — those are the ones with wings — on a mating flight. So every year, on the same day — and it ' s a mystery exactly how that happens — each colony sends out its virgin, unmated queens with wings, and the males, and they all fly to a common place. And they mate. And this shows a recently virgin queen. Here ' s her wings. And she ' s in the process of mating with this male, and there ' s another male on top waiting his turn. Often the queens mate more than once. And after that, the males all die. That ' s it for them. And then the newly mated queens fly off somewhere, drop their wings, dig a hole and go into that hole and start laying eggs. And they will live for 15 or 20 years, continuing to lay eggs using the sperm from that original mating. So the queen goes down in there. She lays eggs, she feeds the larvae — so an ant starts as an egg, then it ' s a larva. She feeds the larvae by regurgitating from her fat reserves. Then, as soon as the ants — the first group of ants — emerge, they ' re larvae. Then they ' re pupae. Then they come out as adult ants. They go out, they get the food, they dig the nest, and the queen never comes out again. So this is a one-year-old colony — this happens to be 536. There ' s the nest entrance, there ' s a pencil for scale. So this is the colony founded by a queen the previous summer. This is a three-year-old colony. There ' s the nest entrance, there ' s a pencil for scale. They make a midden, a pile of refuse — mostly the husks of the seeds that they eat. This is a five-year-old colony. This is the nest entrance, here ' s a pencil for scale. This is about as big as they get, about a meter across. And then this is how colony size and numbers of worker ants changes — so this is about 10, 000 worker ants — changes as a function of colony age, in years. So it starts out with zero ants, just the founding queen, and it grows to a size of about 10 or 12 thousand ants when the colony is five. And it stays that size until the queen dies and there ' s nobody to make more ants, when she ' s about 15 or 20 years old. And it ' s when they reach this stable size, in numbers of ants, that they start to reproduce. That is, to send more winged queens and males to that year ' s mating flight. And I know how colony size changes as a function of colony age because I ' ve dug up colonies of known age and counted all the ants. So that ' s not the most fun part of this research, although it ' s interesting. Really the question that I think about with these ants is what I call task allocation. That ' s not just how is the colony organized, but how does it change what it ' s doing? How is it that the colony manages to adjust the numbers of workers performing each task as conditions change? So, things happen to an ant colony. When it rains in the summer, it floods in the desert. There ' s a lot of damage to the nest, and extra ants are needed to clean up that mess. When extra food becomes available — and this is what everybody knows about picnics — then extra ants are allocated to collect the food. So, with nobody telling anybody what to do, how is it that the colony manages to adjust the numbers of workers performing each task? And that ' s the process that I call task allocation. And in harvester ants, I divide the tasks of the ants I see just outside the nest into these four categories : where an ant is foraging, when it ' s out along the foraging trail, searching for food or bringing food back. The patrollers — that ' s supposed to be a magnifying glass — are an interesting group that go out early in the morning before the foragers are active. They somehow choose the direction that the foragers will go, and by coming back — just by making it back — they tell the foragers that it ' s safe to go out. Then the nest maintenance workers work inside the nest, and I wanted to say that the nests look a lot like Bill Lishman ' s house. That is, that there are chambers inside, they line the walls of the chambers with moist soil and it dries to a kind of an adobe-like surface in it. It also looks very similar to some of the cave dwellings of the Hopi people that are in that area. And the nest maintenance workers do that inside the nest, and then they come out of the nest carrying bits of dry soil in their mandibles. So you see the nest maintenance workers come out with a bit of sand, put it down, turn around, and go back in. And finally, the midden workers put some kind of territorial chemical in the garbage. So what you see the midden workers doing is making a pile of refuse. On one day, it ' ll all be here, and then the next day they ' ll move it over there, and then they ' ll move it back. So that ' s what the midden workers do. And these four groups are just the ants outside the nest. So that ' s only about 25 percent of the colony, and they ' re the oldest ants. So, an ant starts out somewhere near the queen. And when we dig up nests we find they ' re about as deep as the colony is wide, so about a meter deep for the big old nests. And then there ' s another long tunnel and a chamber, where we often find the queen, after eight hours of hacking away at the rock with pickaxes. I don ' t think that chamber has evolved because of me and my backhoe and my crew of students with pickaxes, but instead because when there ' s flooding, occasionally the colony has to go down deep. So there ' s this whole network of chambers. The queen ' s in there somewhere ; she just lays eggs. There ' s the larvae, and they consume most of the food. And this is true of most ants — that the ants you see walking around don ' t do much eating. They bring it back and feed it to the larvae. When the foragers come in with food, they just drop it into the upper chamber, and other ants come up from below, get the food, bring it back, husk the seeds, and pile them up. There are nest maintenance workers working throughout the nest. And curiously, and interestingly, it looks as though at any time about half the ants in the colony are just doing nothing. So, despite what it says in the Bible, about, you know,"Look to the ant, thou sluggard,"in fact, you could think of those ants as reserves. That is to say, if something happened — and I ' ve never seen anything like this happen, but I ' ve only been looking for 20 years — if something happened, they might all come out if they were needed. But in fact, mostly they ' re just hanging around in there. And I think it ' s a very interesting question — what is there about the way the colony is organized that might give some function to a reserve of ants who are doing nothing? And they sort of stand as a buffer in between the ants working deep inside the nest and the ants working outside. And if you mark ants that are working outside, and dig up a colony, you never see them deep down. So what ' s happening is that the ants work inside the nest when they ' re younger. They somehow get into this reserve. And then eventually they get recruited to join this exterior workforce. And once they belong to the ants that work outside, they never go back down. Now ants — most ants, including these, don ' t see very well. They have eyes, they can distinguish between light and dark, but they mostly work by smell. So just to reinforce that what you might have thought about ant queens isn ' t true — you know, even if the queen did have the intelligence to send chemical messages through this whole network of chambers to tell the ants outside what to do, there is no way that such messages could make it in time to see the shifts in the allocation of workers that we actually see outside the nest. So that ' s one way that we know the queen isn ' t directing the behavior of the colony. So when I first set out to work on task allocation, my first question was,"What ' s the relationship between the ants doing different tasks? Does it matter to the foragers what the nest maintenance workers are doing? Does it matter to the midden workers what the patrollers are doing?"And I was working in the context of a view of ant colonies in which each ant was somehow dedicated to its task from birth and sort of performed independently of the others, knowing its place on the assembly line. And instead I wanted to ask,"How are the different task groups interdependent?"So I did experiments where I changed one thing. So for example, I created more work for the nest maintenance workers by putting out a pile of toothpicks near the nest entrance, early in the morning when the nest maintenance workers are first active. This is what it looks like about 20 minutes later. Here it is about 40 minutes later. And the nest maintenance workers just take all the toothpicks to the outer edge of the nest mound and leave them there. And what I wanted to know was,"OK, here ' s a situation where extra nest maintenance workers were recruited — is this going to have any effect on the workers performing other tasks?"Then we repeated all those experiments with the ants marked. So here ' s some blue nest maintenance workers. And lately we ' ve gotten more sophisticated and we have this three-color system. And we can mark them individually so we know which ant is which. We started out with model airplane paint and then we found these wonderful little Japanese markers, and they work really well. And so just to summarize the result, well it turns out that yes, the different tasks are interdependent. So, if I change the numbers performing one task, it changes the numbers performing another. So for example, if I make a mess that the nest maintenance workers have to clean up, then I see fewer ants out foraging. And this was true for all the pair-wise combinations of tasks. And the second result, which was surprising to a lot of people, was that ants actually switch tasks. The same ant doesn ' t do the same task over and over its whole life. So for example, if I put out extra food, everybody else — the midden workers stop doing midden work and go get the food, they become foragers. The nest maintenance workers become foragers. The patrollers become foragers. But not every transition is possible. And this shows how it works. Like I just said, if there is more food to collect, the patrollers, the midden workers, the nest maintenance workers will all change to forage. If there ' s more patrolling to do — so I created a disturbance, so extra patrollers were needed — the nest maintenance workers will switch to patrol. But if more nest maintenance work is needed — for example, if I put out a bunch of toothpicks — then nobody will ever switch back to nest maintenance, they have to get nest maintenance workers from inside the nest. So foraging acts as a sink, and the ants inside the nest act as a source. And finally, it looks like each ant is deciding moment to moment whether to be active or not. So, for example, when there ' s extra nest maintenance work to do, it ' s not that the foragers switch over. I know that they don ' t do that. But the foragers somehow decide not to come out. And here was the most intriguing result : the task allocation. This process changes with colony age, and it changes like this. When I do these experiments with older colonies — so ones that are five years or older — they ' re much more consistent from one time to another and much more homeostatic. The worse things get, the more I hassle them, the more they act like undisturbed colonies. Whereas the young, small colonies — the two-year-old colonies of just 2, 000 ants — are much more variable. And the amazing thing about this is that an ant lives only a year. It could be this year, or this year. So, the ants in the older colony that seem to be more stable are not any older than the ants in the younger colony. It ' s not due to the experience of older, wiser ants. Instead, something about the organization must be changing as the colony gets older. And the obvious thing that ' s changing is its size. So since I ' ve had this result, I ' ve spent a lot of time trying to figure out what kinds of decision rules — very simple, local, probably olfactory, chemical rules could an ant could be using, since no ant can assess the global situation — that would have the outcome that I see, these predictable dynamics, in who does what task. And it would change as the colony gets larger. And what I ' ve found out is that ants are using a network of antennal contact. So anybody who ' s ever looked at ants has seen them touch antennae. They smell with their antennae. When one ant touches another, it ' s smelling it, and it can tell, for example, whether the other ant is a nest mate because ants cover themselves and each other, through grooming, with a layer of grease, which carries a colony-specific odor. And what we ' re learning is that an ant uses the pattern of its antennal contacts, the rate at which it meets ants of other tasks, in deciding what to do. And so what the message is, is not any message that they transmit from one ant to another, but the pattern. The pattern itself is the message. And I ' ll tell you a little bit more about that. But first you might be wondering : how is it that an ant can tell, for example, I ' m a forager. I expect to meet another forager every so often. But if instead I start to meet a higher number of nest maintenance workers, I ' m less likely to forage. So it has to know the difference between a forager and a nest maintenance worker. And we ' ve learned that, in this species — and I suspect in others as well — these hydrocarbons, this layer of grease on the outside of ants, is different as ants perform different tasks. And we ' ve done experiments that show that that ' s because the longer an ant stays outside, the more these simple hydrocarbons on its surface change, and so they come to smell different by doing different tasks. And they can use that task-specific odor in cuticular hydrocarbons — they can use that in their brief antennal contacts to somehow keep track of the rate at which they ' re meeting ants of certain tasks. And we ' ve just recently demonstrated this by putting extract of hydrocarbons on little glass beads, and dropping the beads gently down into the nest entrance at the right rate. And it turns out that ants will respond to the right rate of contact with a glass bead with hydrocarbon extract on it, as they would to contact with real ants. So I want now to show you a bit of film — and this will start out, first of all, showing you the nest entrance. So the idea is that ants are coming in and out of the nest entrance. They ' ve gone out to do different tasks, and the rate at which they meet as they come in and out of the nest entrance determines, or influences, each ant ’ s decision about whether to go out, and which task to perform. This is taken through a fiber optics microscope. It ’ s down inside the nest. In the beginning you see the ants just kind of engaging with the fiber optics microscope. But the idea is that the ants are in there, and each ant is experiencing a certain flow of ants past it — a stream of contacts with other ants. And the pattern of these interactions determines whether the ant comes back out, and what it does when it comes back out. You can also see this in the ants just outside the nest entrance like these. Each ant, then, as it comes back in, is contacting other ants. And the ants that are waiting just inside the nest entrance to decide whether to go out on their next trip, are contacting the ants coming in. So, what ’ s interesting about this system is that it ’ s messy. It ’ s variable. It ’ s noisy. And, in particular, in two ways. The first is that the experience of the ant — of each ant — can ’ t be very predictable. Because the rate at which ants come back depends on all the little things that happen to an ant as it goes out and does its task outside. And the second thing is that an ant ’ s ability to assess this pattern must be very crude because no ant can do any sophisticated counting. So, we do a lot of simulation and modeling, and also experimental work, to try to figure out how those two kinds of noise combine to, in the aggregate, produce the predictable behavior of ant colonies. Again, I don ’ t want to say that this kind of haphazard pattern of interactions produces a factory that works with the precision and efficiency of clockwork. In fact, if you watch ants at all, you end up trying to help them because they never seem to be doing anything exactly the way that you think that they ought to be doing it. So it ’ s not really that out of these haphazard contacts, perfection arises. But it works pretty well. Ants have been around for several hundred million years. They cover the earth, except for Antarctica. Something that they ’ re doing is clearly successful enough that this pattern of haphazard contacts, in the aggregate, produces something that allows ants to make a lot more ants. And one of the things that we ’ re studying is how natural selection might be acting now to shape this use of interaction patterns — this network of interaction patterns — to perhaps increase the foraging efficiency of ant colonies. So the one thing, though, that I want you to remember about this is that these patterns of interactions are something that you ’ d expect to be closely connected to colony size. The simplest idea is that when an ant is in a small colony — and an ant in a large colony can use the same rule, like”I expect to meet another forager every three seconds.”But in a small colony, it ’ s likely to meet fewer foragers, just because there are fewer other foragers there to meet. So this is the kind of rule that, as the colony develops and gets older and larger, will produce different behavior in an old colony and a small young one. Thank you.

Text Sample 276: NEG Dim 2 – 2003 – Score 1.00 – 2003/t000258.txt

How will we be remembered in 200 years? I happen to live in a little town, Princeton, in New Jersey, which every year celebrates the great event in Princeton history : the Battle of Princeton, which was, in fact, a very important battle. It was the first battle that George Washington won, in fact, and was pretty much of a turning point in the war of independence. It happened 225 years ago. It was actually a terrible disaster for Princeton. The town was burned down ; it was in the middle of winter, and it was a very, very severe winter. And about a quarter of all the people in Princeton died that winter from hunger and cold, but nobody remembers that. What they remember is, of course, the great triumph, that the Brits were beaten, and we won, and that the country was born. And so I agree very emphatically that the pain of childbirth is not remembered. It ' s the child that ' s remembered. And that ' s what we ' re going through at this time. I wanted to just talk for one minute about the future of biotechnology, because I think I know very little about that — I ' m not a biologist — so everything I know about it can be said in one minute. What I ' m saying is that we should follow the model that has been so successful with the electronic industry, that what really turned computers into a great success, in the world as a whole, is toys. As soon as computers became toys, when kids could come home and play with them, then the industry really took off. And that has to happen with biotech. There ' s a huge — — there ' s a huge community of people in the world who are practical biologists, who are dog breeders, pigeon breeders, orchid breeders, rose breeders, people who handle biology with their hands, and who are dedicated to producing beautiful things, beautiful creatures, plants, animals, pets. These people will be empowered with biotech, and that will be an enormous positive step to acceptance of biotechnology. That will blow away a lot of the opposition. When people have this technology in their hands, you have a do-it-yourself biotech kit, grow your own — grow your dog, grow your own cat. Just buy the software, you design it. I won ' t say anymore, you can take it on from there. It ' s going to happen, and I think it has to happen before the technology becomes natural, becomes part of the human condition, something that everybody ' s familiar with and everybody accepts. So, let ' s leave that aside. I want to talk about something quite different, which is what I know about, and that is astronomy. And I ' m interested in searching for life in the universe. And it ' s open to us to introduce a new way of doing that, and that ' s what I ' ll talk about for 10 minutes, or whatever the time remains. The important fact is, that most of the real estate that ' s accessible to us — I ' m not talking about the stars, I ' m talking about the solar system, the stuff that ' s within reach for spacecraft and within reach of our earthbound telescopes — most of the real estate is very cold and very far from the Sun. If you look at the solar system, as we know it today, it has a few planets close to the Sun. That ' s where we live. It has a fairly substantial number of asteroids between the orbit of the Earth out through — to the orbit of Jupiter. The asteroids are a substantial amount of real estate, but not very large. And it ' s not very promising for life, since most of it consists of rock and metal, mostly rock. It ' s not only cold, but very dry. So the asteroids we don ' t have much hope for. There stand some interesting places a little further out : the moons of Jupiter and Saturn. Particularly, there ' s a place called Europa, which is — Europa is one of the moons of Jupiter, where we see a very level ice surface, which looks as if it ' s floating on top of an ocean. So, we believe that on Europa there is, in fact, a deep ocean. And that makes it extraordinarily interesting as a place to explore. Ocean — probably the most likely place for life to originate, just as it originated on the Earth. So we would love to explore Europa, to go down through the ice, find out who is swimming around in the ocean, whether there are fish or seaweed or sea monsters — whatever there may be that ' s exciting — - or cephalopods. But that ' s hard to do. Unfortunately, the ice is thick. We don ' t know just how thick it is, probably miles thick, so it ' s very expensive and very difficult to go down there — send down your submarine or whatever it is — and explore. That ' s something we don ' t yet know how to do. There are plans to do it, but it ' s hard. Go out a bit further, you ' ll find that beyond the orbit of Neptune, way out, far from the Sun, that ' s where the real estate really begins. You ' ll find millions or trillions or billions of objects which, in what we call the Kuiper Belt or the Oort Cloud — these are clouds of small objects which appear as comets when they fall close to the Sun. Mostly, they just live out there in the cold of the outer solar system, but they are biologically very interesting indeed, because they consist primarily of ice with other minerals, which are just the right ones for developing life. So if life could be established out there, it would have all the essentials — chemistry and sunlight — everything that ' s needed. So, what I ' m proposing is that there is where we should be looking for life, rather than on Mars, although Mars is, of course, also a very promising and interesting place. But we can look outside, very cheaply and in a simple fashion. And that ' s what I ' m going to talk about. There is a — imagine that life originated on Europa, and it was sitting in the ocean for billions of years. It ' s quite likely that it would move out of the ocean onto the surface, just as it did on the Earth. Staying in the ocean and evolving in the ocean for 2 billion years, finally came out onto the land. And then of course it had great — much greater freedom, and a much greater variety of creatures developed on the land than had ever been possible in the ocean. And the step from the ocean to the land was not easy, but it happened. Now, if life had originated on Europa in the ocean, it could also have moved out onto the surface. There wouldn ' t have been any air there — it ' s a vacuum. It is out in the cold, but it still could have come. You can imagine that the plants growing up like kelp through cracks in the ice, growing on the surface. What would they need in order to grow on the surface? They ' d need, first of all, to have a thick skin to protect themselves from losing water through the skin. So they would have to have something like a reptilian skin. But better — what is more important is that they would have to concentrate sunlight. The sunlight in Jupiter, on the satellites of Jupiter, is 25 times fainter than it is here, since Jupiter is five times as far from the Sun. So they would have to have — these creatures, which I call sunflowers, which I imagine living on the surface of Europa, would have to have either lenses or mirrors to concentrate sunlight, so they could keep themselves warm on the surface. Otherwise, they would be at a temperature of minus 150, which is certainly not favorable for developing life, at least of the kind we know. But if they just simply could grow, like leaves, little lenses and mirrors to concentrate sunlight, then they could keep warm on the surface. They could enjoy all the benefits of the sunlight and have roots going down into the ocean ; life then could flourish much more. So, why not look? Of course, it ' s not very likely that there ' s life on the surface of Europa. None of these things is likely, but my, my philosophy is, look for what ' s detectable, not for what ' s probable. There ' s a long history in astronomy of unlikely things turning out to be there. And I mean, the finest example of that was radio astronomy as a whole. This was — originally, when radio astronomy began, Mr. Jansky, at the Bell labs, detected radio waves coming from the sky. And the regular astronomers were scornful about this. They said,"It ' s all right, you can detect radio waves from the Sun, but the Sun is the only object in the universe that ' s close enough and bright enough actually to be detectable. You can easily calculate that radio waves from the Sun are fairly faint, and everything else in the universe is millions of times further away, so it certainly will not be detectable. So there ' s no point in looking."And that, of course, that set back the progress of radio astronomy by about 20 years. Since there was nothing there, you might as well not look. Well, of course, as soon as anybody did look, which was after about 20 years, when radio astronomy really took off. Because it turned out the universe is absolutely full of all kinds of wonderful things radiating in the radio spectrum, much brighter than the Sun. So, the same thing could be true for this kind of life, which I ' m talking about, on cold objects : that it could in fact be very abundant all over the universe, and it ' s not been detected just because we haven ' t taken the trouble to look. So, the last thing I want to talk about is how to detect it. There is something called pit lamping. That ' s the phrase which I learned from my son George, who is there in the audience. You take — that ' s a Canadian expression. If you happen to want to hunt animals at night, you take a miner ' s lamp, which is a pit lamp. You strap it onto your forehead, so you can see the reflection in the eyes of the animal. So, if you go out at night, you shine a flashlight, the animals are bright. You see the red glow in their eyes, which is the reflection of the flashlight. And then, if you ' re one of these unsporting characters, you shoot the animals and take them home. And of course, that spoils the game for the other hunters who hunt in the daytime, so in Canada that ' s illegal. In New Zealand, it ' s legal, because the New Zealand farmers use this as a way of getting rid of rabbits, because the rabbits compete with the sheep in New Zealand. So, the farmers go out at night with heavily armed jeeps, and shine the headlights, and anything that doesn ' t look like a sheep, you shoot. So I have proposed to apply the same trick to looking for life in the universe. That if these creatures who are living on cold surfaces — either on Europa, or further out, anywhere where you can live on a cold surface — those creatures must be provided with reflectors. In order to concentrate sunlight, they have to have lenses and mirrors — in order to keep themselves warm. And then, when you shine sunlight at them, the sunlight will be reflected back, just as it is in the eyes of an animal. So these creatures will be bright against the cold surroundings. And the further out you go in this, away from the Sun, the more powerful this reflection will be. So actually, this method of hunting for life gets stronger and stronger as you go further away, because the optical reflectors have to be more powerful so the reflected light shines out even more in contrast against the dark background. So as you go further away from the Sun, this becomes more and more powerful. So, in fact, you can look for these creatures with telescopes from the Earth. Why aren ' t we doing it? Simply because nobody thought of it yet. But I hope that we shall look, and with any — we probably won ' t find anything, none of these speculations may have any basis in fact. But still, it ' s a good chance. And of course, if it happens, it will transform our view of life altogether. Because it means that — the way life can live out there, it has enormous advantages as compared with living on a planet. It ' s extremely hard to move from one planet to another. We ' re having great difficulties at the moment and any creatures that live on a planet are pretty well stuck. Especially if you breathe air, it ' s very hard to get from planet A to planet B, because there ' s no air in between. But if you breathe air — — you ' re dead — — as soon as you ' re off the planet, unless you have a spaceship. But if you live in a vacuum, if you live on the surface of one of these objects, say, in the Kuiper Belt, this — an object like Pluto, or one of the smaller objects in the neighborhood of Pluto, and you happened — if you ' re living on the surface there, and you get knocked off the surface by a collision, then it doesn ' t change anything all that much. You still are on a piece of ice, you can still have sunlight and you can still survive while you ' re traveling from one place to another. And then if you run into another object, you can stay there and colonize the other object. So life will spread, then, from one object to another. So if it exists at all in the Kuiper Belt, it ' s likely to be very widespread. And you will have then a great competition amongst species — Darwinian evolution — so there ' ll be a huge advantage to the species which is able to jump from one place to another without having to wait for a collision. And there ' ll be advantages for spreading out long, sort of kelp-like forest of vegetation. I call these creatures sunflowers. They look like, maybe like sunflowers. They have to be all the time pointing toward the Sun, and they will be able to spread out in space, because gravity on these objects is weak. So they can collect sunlight from a big area. So they will, in fact, be quite easy for us to detect. So, I hope in the next 10 years, we ' ll find these creatures, and then, of course, our whole view of life in the universe will change. If we don ’ t find them, then we can create them ourselves. That ’ s another wonderful opportunity that ’ s opening. We can — as soon as we have a little bit more understanding of genetic engineering, one of the things you can do with your take-it-home, do-it-yourself genetic engineering kit — — is to design a creature that can live on a cold satellite, a place like Europa, so we could colonize Europa with our own creatures. That would be a fun thing to do. In the long run, of course, it would also make it possible for us to move out there. What ’ s going to happen in the end, it ’ s not going to be just humans colonizing space, it ’ s going to be life moving out from the Earth, moving it into its kingdom. And the kingdom of life, of course, is going to be the universe. And if life is already there, it makes it much more exciting, in the short run. But in the long run, if there ’ s no life there, we create it ourselves. We transform the universe into something much more rich and beautiful than it is today. So again, we have a big and wonderful future to look forward. Thank you.

Text Sample 277: NEG Dim 2 – 2010 – Score 1.00 – 2010/t002483.txt

I was offered a position as associate professor of medicine and chief of scientific visualization at Yale University in the department of medicine. And my job was to write many of the algorithms and code for NASA to do virtual surgery in preparation for the astronauts going into deep-space flight, so they could be kept in robotic pods. One of the fascinating things about what we were working on is that we were seeing, using new scanning technologies, things that had never been seen before. Not only in disease management, but also things that allowed us to see things about the body that just made you marvel. I remember one of the first times we were looking at collagen. And your entire body, everything — your hair, skin, bone, nails — everything is made of collagen. And it ’ s a kind of rope-like structure that twirls and swirls like this. And the only place that collagen changes its structure is in the cornea of your eye. In your eye, it becomes a grid formation, and therefore, it becomes transparent, as opposed to opaque. So perfectly organized a structure, it was hard not to attribute divinity to it. Because we kept on seeing this in different parts of the body. One of the opportunities I had was one person was working on a really interesting micromagnetic resonance imaging machine with the NIH. And what we were going to do was scan a new project on the development of the fetus from conception to birth using these new technologies. So I wrote the algorithms and code, and he built the hardware — Paul Lauterbur — then went onto win the Nobel Prize for inventing the MRI. I got the data. And I ’ m going to show you a sample of the piece,”From Conception to Birth.”[From Conception to Birth] [Oocyte] [Sperm] [Egg Inseminated] [24 Hours : Baby ’ s first division] [The fertilized ovum divides a few hours after fusion...] [And divides anew every 12 to 15 hours.] [Early Embryo] [Yolk sack still feeding baby.] [25 Days : Heart chamber developing.] [32 Days : Arms & hands are developing] [36 Days : Beginning of the primitive vertebrae] [These weeks are the period of the most rapid development of the fetus.] [If the fetus continues to grow at this speed for the entire 9 months, it would be 1. 5 tons at birth.] [45 Days] [Embryo ’ s heart is beating twice as fast as the mother ’ s.] [51 Days] [Developing retina, nose and fingers] [The fetus ’ continual movement in the womb is necessary for muscular and skeletal growth.] [12 Weeks : Indifferent penis] [Girl or boy yet to be determined] [8 Months] [Delivery : The expulsion stage] [The moment of birth] Alexander Tsiaras : Thank you. But as you can see, when you actually start working on this data, it ’ s pretty spectacular. And as we kept on scanning more and more, working on this project, looking at these two simple cells that have this unbelievable machinery that will become the magic of you. And as we kept on working on this data, looking at small clusters of the body, these little pieces of tissue that were the trophoblasts coming off of the blastocyst, all of a sudden burrowing itself into the side of the uterus, saying,”I ’ m here to stay.”Having conversation and communications with the estrogens, the progesterones, saying,”I ’ m here to stay, plant me,”building this incredible trilinear fetus that becomes, within 44 days, something that you can recognize, and then at nine weeks is really kind of a little human being. The marvel of this information : How do we actually have this biological mechanism inside our body to actually see this information? I ’ m going to show you something pretty unique. Here ’ s a human heart at 25 days. It ’ s just basically two strands. And like this magnificent origami, cells are developing at one million cells per second at four weeks, as it ’ s just folding on itself. Within five weeks, you start to see the early atrium and the early ventricles. Six weeks, these folds are now beginning with the papilla on the inside of the heart actually being able to pull down each one of those valves in your heart until you get a mature heart — and then basically the development of the entire human body. The magic of the mechanisms inside each genetic structure saying exactly where that nerve cell should go — the complexity of these, the mathematical models of how these things are indeed done are beyond human comprehension. Even though I am a mathematician, I look at this with marvel of how do these instruction sets not make these mistakes as they build what is us? It ’ s a mystery, it ’ s magic, it ’ s divinity. Then you start to take a look at adult life. Take a look at this little tuft of capillaries. It ’ s just a tiny sub- substructure, microscopic. But basically by the time you ’ re nine months and you ’ re given birth, you have almost 60, 000 miles of vessels inside your body. And only one mile is visible. 59, 999 miles that are basically bringing nutrients and taking waste away. The complexity of building that within a single system is, again, beyond any comprehension or any existing mathematics today. And then instructions set, from the brain to every other part of the body — look at the complexity of the folding. Where does this intelligence of knowing that a fold can actually hold more information, so as you actually watch the baby ’ s brain grow. And this is one of the things we ’ re doing. We ’ re launching two new studies of scanning babies ’ brains from the moment they ’ re born. Every six months until they ’ re six years old, we ’ re going to be doing about 250 children, watching exactly how the gyri and the sulci of the brains fold to see how this magnificent development actually turns into memories and the marvel that is us. And it ’ s not just our own existence, but how does the woman ’ s body understand to have genetic structure that not only builds her own, but then has the understanding that allows her to become a walking immunological, cardiovascular system that basically is a mobile system that can actually nurture, treat this child with a kind of marvel that is beyond, again, our comprehension — the magic that is existence, that is us? Thank you.

Text Sample 278: NEG Dim 2 – 2010 – Score 1.00 – 2010/t002645.txt

I know what you ’ re thinking. You think I ’ ve lost my way, and somebody ’ s going to come on the stage in a minute and guide me gently back to my seat. I get that all the time in Dubai.”Here on holiday are you, dear?””Come to visit the children? How long are you staying?”Well actually, I hope for a while longer yet. I have been living and teaching in the Gulf for over 30 years. And in that time, I have seen a lot of changes. Now that statistic is quite shocking. And I want to talk to you today about language loss and the globalization of English. I want to tell you about my friend who was teaching English to adults in Abu Dhabi. And one fine day, she decided to take them into the garden to teach them some nature vocabulary. But it was she who ended up learning all the Arabic words for the local plants, as well as their uses — medicinal uses, cosmetics, cooking, herbal. How did those students get all that knowledge? Of course, from their grandparents and even their great-grandparents. It ’ s not necessary to tell you how important it is to be able to communicate across generations. But sadly, today, languages are dying at an unprecedented rate. A language dies every 14 days. Now, at the same time, English is the undisputed global language. Could there be a connection? Well I don ’ t know. But I do know that I ’ ve seen a lot of changes. When I first came out to the Gulf, I came to Kuwait in the days when it was still a hardship post. Actually, not that long ago. That is a little bit too early. But nevertheless, I was recruited by the British Council, along with about 25 other teachers. And we were the first non- Muslims to teach in the state schools there in Kuwait. We were brought to teach English because the government wanted to modernize the country and to empower the citizens through education. And of course, the U. K. benefited from some of that lovely oil wealth. Okay. Now this is the major change that I ’ ve seen — how teaching English has morphed from being a mutually beneficial practice to becoming a massive international business that it is today. No longer just a foreign language on the school curriculum, and no longer the sole domain of mother England, it has become a bandwagon for every English-speaking nation on earth. And why not? After all, the best education — according to the latest World University Rankings — is to be found in the universities of the U. K. and the U. S. So everybody wants to have an English education, naturally. But if you ’ re not a native speaker, you have to pass a test. Now can it be right to reject a student on linguistic ability alone? Perhaps you have a computer scientist who ’ s a genius. Would he need the same language as a lawyer, for example? Well, I don ’ t think so. We English teachers reject them all the time. We put a stop sign, and we stop them in their tracks. They can ’ t pursue their dream any longer, ’ til they get English. Now let me put it this way : if I met a monolingual Dutch speaker who had the cure for cancer, would I stop him from entering my British University? I don ’ t think so. But indeed, that is exactly what we do. We English teachers are the gatekeepers. And you have to satisfy us first that your English is good enough. Now it can be dangerous to give too much power to a narrow segment of society. Maybe the barrier would be too universal. Okay.”But,”I hear you say,”what about the research? It ’ s all in English.”So the books are in English, the journals are done in English, but that is a self- fulfilling prophecy. It feeds the English requirement. And so it goes on. I ask you, what happened to translation? If you think about the Islamic Golden Age, there was lots of translation then. They translated from Latin and Greek into Arabic, into Persian, and then it was translated on into the Germanic languages of Europe and the Romance languages. And so light shone upon the Dark Ages of Europe. Now don ’ t get me wrong ; I am not against teaching English, all you English teachers out there. I love it that we have a global language. We need one today more than ever. But I am against using it as a barrier. Do we really want to end up with 600 languages and the main one being English, or Chinese? We need more than that. Where do we draw the line? This system equates intelligence with a knowledge of English, which is quite arbitrary. And I want to remind you that the giants upon whose shoulders today ’ s intelligentsia stand did not have to have English, they didn ’ t have to pass an English test. Case in point, Einstein. He, by the way, was considered remedial at school because he was, in fact, dyslexic. But fortunately for the world, he did not have to pass an English test. Because they didn ’ t start until

1964 with TOEFL, the American test of English. Now it ’ s exploded. There are lots and lots of tests of English. And millions and millions of students take these tests every year. Now you might think, you and me,”Those fees aren ’ t bad, they ’ re okay,”but they are prohibitive to so many millions of poor people. So immediately, we ’ re rejecting them. It brings to mind a headline I saw recently :”Education : The Great Divide.”Now I get it, I understand why people would want to focus on English. They want to give their children the best chance in life. And to do that, they need a Western education. Because, of course, the best jobs go to people out of the Western Universities, that I put on earlier. It ’ s a circular thing. Okay. Let me tell you a story about two scientists, two English scientists. They were doing an experiment to do with genetics and the forelimbs and the hind limbs of animals. But they couldn ’ t get the results they wanted. They really didn ’ t know what to do, until along came a German scientist who realized that they were using two words for forelimb and hind limb, whereas genetics does not differentiate and neither does German. So bingo, problem solved. If you can ’ t think a thought, you are stuck. But if another language can think that thought, then, by cooperating, we can achieve and learn so much more. My daughter came to England from Kuwait. She had studied science and mathematics in Arabic. It ’ s an Arabic-medium school. She had to translate it into English at her grammar school. And she was the best in the class at those subjects. Which tells us that when students come to us from abroad, we may not be giving them enough credit for what they know, and they know it in their own language. When a language dies, we don ’ t know what we lose with that language. This is — I don ’ t know if you saw it on CNN recently — they gave the Heroes Award to a young Kenyan shepherd boy who couldn ’ t study at night in his village, like all the village children, because the kerosene lamp, it had smoke and it damaged his eyes. And anyway, there was never enough kerosene, because what does a dollar a day buy for you? So he invented a cost-free solar lamp. And now the children in his village get the same grades at school as the children who have electricity at home. When he received his award, he said these lovely words :”The children can lead Africa from what it is today, a dark continent, to a light continent.”A simple idea, but it could have such far-reaching consequences. People who have no light, whether it ’ s physical or metaphorical, cannot pass our exams, and we can never know what they know. Let us not keep them and ourselves in the dark. Let us celebrate diversity. Mind your language. Use it to spread great ideas. Thank you very much.

Text Sample 279: NEG Dim 2 – 2010 – Score 1.00 – 2010/t002638.txt

I was only four years old when I saw my mother load a washing machine for the very first time in her life. That was a great day for my mother. My mother and father had been saving money for years to be able to buy that machine, and the first day it was going to be used, even Grandma was invited to see the machine. And Grandma was even more excited. Throughout her life she had been heating water with firewood, and she had hand washed laundry for seven children. And now she was going to watch electricity do that work. My mother carefully opened the door, and she loaded the laundry into the machine, like this. And then, when she closed the door, Grandma said,”No, no, no, no. Let me, let me push the button.”And Grandma pushed the button, and she said,”Oh, fantastic! I want to see this! Give me a chair! Give me a chair! I want to see it,”and she sat down in front of the machine, and she watched the entire washing program. She was mesmerized. To my grandmother, the washing machine was a miracle. Today, in Sweden and other rich countries, people are using so many different machines. Look, the homes are full of machines. I can ' t even name them all. And they also, when they want to travel, they use flying machines that can take them to remote destinations. And yet, in the world, there are so many people who still heat the water on fire, and they cook their food on fire. Sometimes they don ' t even have enough food, and they live below the poverty line. There are two billion fellow human beings who live on less than two dollars a day. And the richest people over there — there ' s one billion people — and they live above what I call the"air line,"because they spend more than $80 a day on their consumption. But this is just one, two, three billion people, and obviously there are seven billion people in the world, so there must be one, two, three, four billion people more who live in between the poverty and the air line. They have electricity, but the question is, how many have washing machines? I ' ve done the scrutiny of market data, and I ' ve found that, indeed, the washing machine has penetrated below the air line, and today there ' s an additional one billion people out there who live above the"wash line."And they consume more than $40 per day. So two billion have access to washing machines. And the remaining five billion, how do they wash? Or, to be more precise, how do most of the women in the world wash? Because it remains hard work for women to wash. They wash like this : by hand. It ' s a hard, time-consuming labor, which they have to do for hours every week. And sometimes they also have to bring water from far away to do the laundry at home, or they have to bring the laundry away to a stream far off. And they want the washing machine. They don ' t want to spend such a large part of their life doing this hard work with so relatively low productivity. And there ' s nothing different in their wish than it was for my grandma. Look here, two generations ago in Sweden — picking water from the stream, heating with firewood and washing like that. They want the washing machine in exactly the same way. But when I lecture to environmentally-concerned students, they tell me,"No, everybody in the world cannot have cars and washing machines."How can we tell this woman that she ain ' t going to have a washing machine? And then I ask my students, I ' ve asked them — over the last two years I ' ve asked,"How many of you doesn ' t use a car?"And some of them proudly raise their hand and say,"I don ' t use a car."And then I put the really tough question :"How many of you hand-wash your jeans and your bed sheets?"And no one raised their hand. Even the hardcore in the green movement use washing machines. So how come [this is] something that everyone uses and they think others will not stop it? What is special with this? I had to do an analysis about the energy used in the world. Here we are. Look here, you see the seven billion people up there : the air people, the wash people, the bulb people and the fire people. One unit like this is an energy unit of fossil fuel — oil, coal or gas. That ' s what most of electricity and the energy in the world is. And it ' s 12 units used in the entire world, and the richest one billion, they use six of them. Half of the energy is used by one seventh of the world ' s population. And these ones who have washing machines, but not a house full of other machines, they use two. This group uses three, one each. And they also have electricity. And over there they don ' t even use one each. That makes 12 of them. But the main concern for the environmentally-interested students — and they are right — is about the future. What are the trends? If we just prolong the trends, without any real advanced analysis, to 2050, there are two things that can increase the energy use. First, population growth. Second, economic growth. Population growth will mainly occur among the poorest people here because they have high child mortality and they have many children per woman. And [with] that you will get two extra, but that won ' t change the energy use very much. What will happen is economic growth. The best of here in the emerging economies — I call them the New East — they will jump the air line.”Wopp!”they will say. And they will start to use as much as the Old West are doing already. And these people, they want the washing machine. I told you. They ’ ll go there. And they will double their energy use. And we hope that the poor people will get into the electric light. And they ’ ll get a two-child family without a stop in population growth. But the total energy consumption will increase to 22 units. And these 22 units — still the richest people use most of it. So what needs to be done? Because the risk, the high probability of climate change is real. It ’ s real. Of course they must be more energy-efficient. They must change behavior in some way. They must also start to produce green energy, much more green energy. But until they have the same energy consumption per person, they shouldn ’ t give advice to others — what to do and what not to do. Here we can get more green energy all over. This is what we hope may happen. It ’ s a real challenge in the future. But I can assure you that this woman in the favela in Rio, she wants a washing machine. She ’ s very happy about her minister of energy that provided electricity to everyone — so happy that she even voted for her. And she became Dilma Rousseff, the president-elect of one of the biggest democracies in the world — moving from minister of energy to president. If you have democracy, people will vote for washing machines. They love them. And what ’ s the magic with them? My mother explained the magic with this machine the very, very first day. She said,”Now Hans, we have loaded the laundry. The machine will make the work. And now we can go to the library.”Because this is the magic : you load the laundry, and what do you get out of the machine? You get books out of the machines, children ’ s books. And mother got time to read for me. She loved this. I got the”ABC ’ s”— this is where I started my career as a professor, when my mother had time to read for me. And she also got books for herself. She managed to study English and learn that as a foreign language. And she read so many novels, so many different novels here. And we really, we really loved this machine. And what we said, my mother and me,”Thank you industrialization. Thank you steel mill. Thank you power station. And thank you chemical processing industry that gave us time to read books.”Thank you very much.

Text Sample 280: NEG Dim 2 – 2008 – Score 1.00 – 2008/t000146.txt

Great creativity. In times of need, we need great creativity. Discuss. Great creativity is astonishingly, absurdly, rationally, irrationally powerful. Great creativity can spread tolerance, champion freedom, make education seem like a bright idea. Great creativity can turn a spotlight on deprivation, or show that deprivation ain ’ t necessarily so. Great creativity can make politicians electable, or parties unelectable. It can make war seem like tragedy or farce. Creativity is the meme-maker that puts slogans on our t-shirts and phrases on our lips. It ’ s the pathfinder that shows us a simple road through an impenetrable moral maze. Science is clever, but great creativity is something less knowable, more magical. And now we need that magic. This is a time of need. Our climate is changing quickly, too quickly. And great creativity is needed to do what it does so well : to provoke us to think differently with dramatic creative statements. To tempt us to act differently with delightful creative scraps. Here is one such scrap from an initiative I ’ m involved in using creativity to inspire people to be greener. Man : You know, rather than drive today, I ’ m going to walk. Narrator : And so he walked, and as he walked he saw things. Strange and wonderful things he would not otherwise have seen. A deer with an itchy leg. A flying motorcycle. A father and daughter separated from a bicycle by a mysterious wall. And then he stopped. Walking in front of him was her. The woman who as a child had skipped with him through fields and broken his heart. Sure, she had aged a little. In fact, she had aged a lot. But he felt all his old passion for her return.”Ford,”he called softly. For that was her name.”Don ’ t say another word, Gusty,”she said, for that was his name.”I know a tent next to a caravan, exactly 300 yards from here. Let ’ s go there and make love. In the tent.”Ford undressed. She spread one leg, and then the other. Gusty entered her boldly and made love to her rhythmically while she filmed him, because she was a keen amateur pornographer. The earth moved for both of them. And they lived together happily ever after. And all because he decided to walk that day. Andy Hobsbawm : We ’ ve got the science, we ’ ve had the debate. The moral imperative is on the table. Great creativity is needed to take it all, make it simple and sharp. To make it connect. To make it make people want to act. So this is a call, a plea, to the incredibly talented TED community. Let ’ s get creative against climate change. And let ’ s do it soon. Thank you.

Text Sample 281: NEG Dim 2 – 2008 – Score 1.00 – 2008/t000119.txt

I don ’ t know what the hell I ’ m doing here. I was born in a Scots Presbyterian ghetto in Canada, and dropped out of high school. I don ’ t own a cell phone, and I paint on paper using gouache, which hasn ’ t changed in 600 years. But about three years ago I had an art show in New York, and I titled it”Serious Nonsense.”So I think I ’ m actually the first one here — I lead. I called it”Serious Nonsense”because on the serious side, I use a technique of painstaking realism of editorial illustration from when I was a kid. I copied it and I never unlearned it — it ’ s the only style I know. And it ’ s very kind of staid and formal. And meanwhile, I use nonsense, as you can see. This is a Scottish castle where people are playing golf indoors, and the trick was to bang the golf ball off of suits of armor — which you can ’ t see there. This was one of a series called”Zany Afternoons,”which became a book. This is a home-built rocket-propelled car. That ’ s a 1953 Henry J — I ’ m a bug for authenticity — in a quiet neighborhood in Toledo. This is my submission for the L. A. Museum of Film. You can probably tell Frank Gehry and I come from the same town. My work is so personal and so strange that I have to invent my own lexicon for it. And I work a lot in what I call”retrofuturism,”which is looking back to see how yesterday viewed tomorrow. And they ’ re always wrong, always hilariously, optimistically wrong. And the peak time for that was the 30s, because the Depression was so dismal that anything to get away from the present into the future... and technology was going to carry us along. This is Popular Workbench. Popular science magazines in those days — I had a huge collection of them from the ’ 30s — all they are is just poor people being asked to make sunglasses out of wire coat hangers and everything improvised and dreaming about these wonderful giant radio robots playing ice hockey at 300 miles an hour — it ’ s all going to happen, it ’ s all going to be wonderful. Automotive retrofuturism is one of my specialties. I was both an automobile illustrator and an advertising automobile copywriter, so I have a lot of revenge to take on the subject. Detroit has always been halfway into the future — the advertising half. This is the ’ 58 Bulgemobile : so new, they make tomorrow look like yesterday. This is a chain gang of guys admiring the car. That ’ s from a whole catalog — it ’ s 18 pages or so — ran back in the days of the Lampoon, where I cut my teeth. Techno-archaeology is digging back and finding past miracles that never happened — for good reason, usually. The zeppelin — this was from a brochure about the zeppelin based, obviously, on the Hindenburg. But the zeppelin was the biggest thing that ever moved made by man. And it carried 56 people at the speed of a Buick at an altitude you could hear dogs bark, and it cost twice as much as a first-class cabin on the Normandie to fly it. So the Hindenburg wasn ’ t, you know, it was inevitable it was going to go. This is auto-gyro jousting in Malibu in the 30s. The auto-gyro couldn ’ t wait for the invention of the helicopter, but it should have — it wasn ’ t a big success. It ’ s the only Spanish innovation, technologically, of the 20th century, by the way. You needed to know that. The flying car which never got off the ground — it was a post-war dream. My old man used to tell me we were going to get a flying car. This is pitched into the future from 1946, looking at the day all American families have them.”There ’ s Moscow, Shirley. Hope they speak Esperanto!”Faux-nostalgia, which I ’ m sort of — not, say, famous for, but I work an awful lot in it. It ’ s the achingly sentimental yearning for times that never happened. Somebody once said that nostalgia is the one utterly most useless human emotion — so I think that—s a case for serious play. This is emblematic of it — this is wing dining, recalling those balmy summer days somewhere over France in the 20s, dining on the wing of a plane. You can ’ t see it very well here, but that ’ s Hemingway reading some pages from his new novel to Fitzgerald and Ford Madox Ford until the slipstream blows him away. This is tank polo in the South Hamptons. The brainless rich are more fun to make fun of than anybody. I do a lot of that. And authenticity is a major part of my serious nonsense. I think it adds a huge amount. Those, for example, are Mark IV British tanks from 1916. They had two machine guns and a cannon, and they had 90 horsepower Ricardo engines. They went five miles an hour and inside it was 105 degrees in the pitch dark. And they had a canary hung inside the thing to make sure the Germans weren ’ t going to use gas. Happy little story, isn ’ t it? This is Motor Ritz Towers in Manhattan in the 30s, where you drove up to your front door, if you had the guts. Anybody who was anybody had an apartment there. I managed to stick in both the zeppelin and an ocean liner out of sheer enthusiasm. And I love cigars — there ’ s a cigar billboard down there. And faux-nostalgia works even in serious subjects like war. This is those wonderful days of the Battle of Britain in 1940, when a Messerschmitt ME109 bursts into the House of Commons and buzzes around, just to piss off Churchill, who ’ s down there somewhere. It ’ s a fond memory of times past. Hyperbolic overkill is a way of taking exaggeration to the absolute ultimate limit, just for the fun of it. This was a piece I did — a brochure again —”RMS Tyrannic : The Biggest Thing in All the World.”The copy, which you can ’ t see because it goes on and on for several pages, says that steerage passengers can ’ t get their to bunks before the voyage is over, and it ’ s so safe it carries no insurance. It ’ s obviously modeled on the Titanic. But it ’ s not a cri de coeur about man ’ s hubris in the face of the elements. It ’ s just a sick, silly joke. Shamelessly cheap is something, I think — this will wake you up. It has no meaning, just — Desoto discovers the Mississippi, and it ’ s a Desoto discovering the Mississippi. I did that as a quick back page — I had like four hours to do a back page for an issue of the Lampoon, and I did that, and I thought,”Well, I ’ m ashamed. I hope nobody knows it.”People wrote in for reprints of that thing. Urban absurdism — that ’ s what the New Yorker really calls for. I try to make life in New York look even weirder than it is with those covers. I ’ ve done about 40 of them, and I ’ d say 30 of them are based on that concept. I was driving down 7th Avenue one night at 3 a. m., and this steam pouring out of the street, and I thought,”What causes that?”And that — who—s to say? The Temple of Dendur at the Metropolitan in New York — it ’ s a very somber place. I thought I could jazz it up a bit, have a little fun with it. This is a very un-PC cover. Not in New York. I couldn ’ t resist, and I got a nasty email from some environmental group saying,”This is too serious and solemn to make fun of. You should be ashamed, please apologize on our website.”Haven ’ t got around to it yet but — I may. This is the word side of my brain. I love the word”Eurotrash.”That ’ s all the Eurotrash coming through JFK customs. This was the New York bike messenger meeting the Tour de France. If you live in New York, you know how the bike messengers move. Except that he ’ s carrying a tube for blueprints and stuff — they all do — and a lot of people thought that meant it was a terrorist about to shoot rockets at the Tour de France — sign of our times, I guess. This is the only fashion cover I ’ ve ever done. It ’ s the little old lady that lives in a shoe, and then this thing — the title of that was,”There Goes the Neighborhood.”I don ’ t know a hell of a lot about fashion — I was told to do what they call a Mary Jane, and then I got into this terrible fight between the art director and the editor saying :”Put a strap on it”—”No, don ’ t put a strap on it”—”Put a strap on it —”Don ’ t put a strap on it”— because it obscures the logo and looks terrible and it ’ s bad and — I finally chickened out and did it for the sake of the authenticity of the shoe. This is a tiny joke — E-ZR pass. One letter makes an idea. This is a big joke. This is the audition for”King Kong.”People always ask me, where do you get your ideas, how do your ideas come? Truth about that one is I had a horrible red wine hangover, in the middle of the night, this came to me like a Xerox — all I had to do was write it down. It was perfectly clear. I didn ’ t do any thinking about it. And then when it ran, a lovely lady, an old lady named Mrs. Edgar Rosenberg — if you know that name — called me and said she loved the cover, it was so sweet. Her former name was Fay Wray, and so that was — I didn ’ t have the wit to say,”Take the painting.”Finally, this was a three-page cover, never done before, and I don ’ t think it will ever be done again — successive pages in the front of the magazine. It ’ s the ascent of man using an escalator, and it ’ s in three parts. You can ’ t see it all together, unfortunately, but if you look at it enough, you can sort of start to see how it actually starts to move. Pretty elegant. Nothing like a crash to end a joke. That completes my oeuvre. I would just like to add a crass commercial — I have a kids ’ book coming out in the fall called”Marvel Sandwiches,”a compendium of all the serious play that ever was, and it—s going to be available in fine bookstores, crummy bookstores, tables on the street in October. So thank you very much.

Text Sample 282: NEG Dim 2 – 2008 – Score 1.00 – 2008/t000112.txt

About four years ago, the New Yorker published an article about a cache of dodo bones that was found in a pit on the island of Mauritius. Now, the island of Mauritius is a small island off the east coast of Madagascar in the Indian Ocean, and it is the place where the dodo bird was discovered and extinguished, all within about 150 years. Everyone was very excited about this archaeological find, because it meant that they might finally be able to assemble a single dodo skeleton. See, while museums all over the world have dodo skeletons in their collection, nobody — not even the actual Natural History Museum on the island of Mauritius — has a skeleton that ’ s made from the bones of a single dodo. Well, this isn ’ t exactly true. The fact is, is that the British Museum had a complete specimen of a dodo in their collection up until the 18th century — it was actually mummified, skin and all — but in a fit of space-saving zeal, they actually cut off the head and they cut off the feet and they burned the rest in a bonfire. If you go look at their website today, they ’ ll actually list these specimens, saying, the rest was lost in a fire. Not quite the whole truth. Anyway. The frontispiece of this article was this photo, and I ’ m one of the people that thinks that Tina Brown was great for bringing photos to the New Yorker, because this photo completely rocked my world. I became obsessed with the object — not just the beautiful photograph itself, and the color, the shallow depth of field, the detail that ' s visible, the wire you can see on the beak there that the conservator used to put this skeleton together — there ' s an entire story here. And I thought to myself, wouldn ' t it be great if I had my own dodo skeleton? I want to point out here at this point that I ' ve spent my life obsessed by objects and the stories that they tell, and this was the very latest one. So I began looking around for — to see if anyone sold a kit, some kind of model that I could get, and I found lots of reference material, lots of lovely pictures. No dice : no dodo skeleton for me. But the damage had been done. I had saved a few hundred photos of dodo skeletons into my"Creative Projects"folder — it ' s a repository for my brain, everything that I could possibly be interested in. Any time I have an internet connection, there ' s a sluice of stuff moving into there, everything from beautiful rings to cockpit photos. The key that the Marquis du Lafayette sent to George Washington to celebrate the storming of the Bastille. Russian nuclear launch key : The one on the top is the picture of the one I found on eBay ; the one on the bottom is the one I made for myself, because I couldn ' t afford the one on eBay. Storm trooper costumes. Maps of Middle Earth — that ' s one I hand-drew myself. There ' s the dodo skeleton folder. This folder has 17, 000 photos — over 20 gigabytes of information — and it ' s growing constantly. And one day, a couple of weeks later, it might have been maybe a year later, I was in the art store with my kids, and I was buying some clay tools — we were going to have a craft day. I bought some Super Sculpeys, some armature wire, some various materials. And I looked down at this Sculpey, and I thought, maybe, yeah, maybe I could make my own dodo skull. I should point out at this time — I ' m not a sculptor ; I ' m a hard-edged model maker. You give me a drawing, you give me a prop to replicate, you give me a crane, scaffolding, parts from"Star Wars"— especially parts from"Star Wars"— I can do this stuff all day long. It ' s exactly how I made my living for 15 years. But you give me something like this — my friend Mike Murnane sculpted this ; it ' s a maquette for"Star Wars, Episode Two"— this is not my thing — this is something other people do — dragons, soft things. However, I felt like I had looked at enough photos of dodo skulls to actually be able to understand the topology and perhaps replicate it — I mean, it couldn ' t be that difficult. So, I started looking at the best photos I could find. I grabbed all the reference, and I found this lovely piece of reference. This is someone selling this on eBay ; it was clearly a woman—s hand, hopefully a woman ' s hand. Assuming it was roughly the size of my wife ' s hand, I made some measurements of her thumb, and I scaled them out to the size of the skull. I blew it up to the actual size, and I began using that, along with all the other reference that I had, comparing it to it as size reference for figuring out exactly how big the beak should be, exactly how long, etc. And over a few hours, I eventually achieved what was actually a pretty reasonable dodo skull. And I didn ' t mean to continue, I — it ' s kind of like, you know, you can only clean a super messy room by picking up one thing at a time ; you can ' t think about the totality. I wasn ' t thinking about a dodo skeleton ; I just noticed that as I finished this skull, the armature wire that I had been used to holding it up was sticking out of the back just where a spine would be. And one of the other things I ' d been interested in and obsessed with over the years is spines and skeletons, having collected a couple of hundred. I actually understood the mechanics of vertebrae enough to kind of start to imitate them. And so button by button, vertebrae by vertebrae, I built my way down. And actually, by the end of the day, I had a reasonable skull, a moderately good vertebrae and half of a pelvis. And again, I kept on going, looking for more reference, every bit of reference I could find — drawings, beautiful photos. This guy — I love this guy! He put a dodo leg bones on a scanner with a ruler. This is the kind of accuracy that I wanted, and I replicated every last bone and put it in. And after about six weeks, I finished, painted, mounted my own dodo skeleton. You can see that I even made a museum label for it that includes a brief history of the dodo. And TAP Plastics made me — although I didn ' t photograph it — a museum vitrine. I don ' t have the room for this in my house, but I had to finish what I had started. And this actually represented kind of a sea change to me. Again, like I said, my life has been about being fascinated by objects and the stories that they tell, and also making them for myself, obtaining them, appreciating them and diving into them. And in this folder,"Creative Projects,"there are tons of projects that I ' m currently working on, projects that I ' ve already worked on, things that I might want to work on some day, and things that I may just want to find and buy and have and look at and touch. But now there was potentially this new category of things that I could sculpt that was different, that I — you know, I have my own R2D2, but that ' s — honestly, relative to sculpting, to me, that ' s easy. And so I went back and looked through my"Creative Projects"folder, and I happened across the Maltese Falcon. Now, this is funny for me : to fall in love with an object from a Hammett novel, because if it ' s true that the world is divided into two types of people, Chandler people and Hammett people, I am absolutely a Chandler person. But in this case, it ' s not about the author, it ' s not about the book or the movie or the story, it ' s about the object in and of itself. And in this case, this object is — plays on a host of levels. First of all, there ' s the object in the world. This is the"Kniphausen Hawk."It is a ceremonial pouring vessel made around 1700 for a Swedish Count, and it is very likely the object from which Hammett drew his inspiration for the Maltese Falcon. Then there is the fictional bird, the one that Hammett created for the book. Built out of words, it is the engine that drives the plot of his book and also the movie, in which another object is created : a prop that has to represent the thing that Hammett created out of words, inspired by the Kniphausen Hawk, and this represents the falcon in the movie. And then there is this fourth level, which is a whole new object in the world : the prop made for the movie, the representative of the thing, becomes, in its own right, a whole other thing, a whole new object of desire. And so now it was time to do some research. I actually had done some research a few years before — it ' s why the folder was there. I ' d bought a replica, a really crappy replica, of the Maltese Falcon on eBay, and had downloaded enough pictures to actually have some reasonable reference. But I discovered, in researching further, really wanting precise reference, that one of the original lead birds had been sold at Christie ' s in 1994, and so I contacted an antiquarian bookseller who had the original Christie ' s catalogue, and in it I found this magnificent picture, which included a size reference. I was able to scan the picture, blow it up to exactly full size. I found other reference. Avi [Ara] Chekmayan, a New Jersey editor, actually found this resin Maltese Falcon at a flea market in 1991, although it took him five years to authenticate this bird to the auctioneers ' specifications, because there was a lot of controversy about it. It was made out of resin, which wasn ' t a common material for movie props about the time the movie was made. It ' s funny to me that it took a while to authenticate it, because I can see it compared to this thing, and I can tell you — it ' s real, it ' s the real thing, it ' s made from the exact same mold that this one is. In this one, because the auction was actually so controversial, Profiles in History, the auction house that sold this — I think in 1995 for about 100, 000 dollars — they actually included — you can see here on the bottom — not just a front elevation, but also a side, rear and other side elevation. So now, I had all the topology I needed to replicate the Maltese Falcon. What do they do, how do you start something like that? I really don ’ t know. So what I did was, again, like I did with the dodo skull, I blew all my reference up to full size, and then I began cutting out the negatives and using those templates as shape references. So I took Sculpey, and I built a big block of it, and I passed it through until, you know, I got the right profiles. And then slowly, feather by feather, detail by detail, I worked out and achieved — working in front of the television and Super Sculpey — here ’ s me sitting next to my wife — it ’ s the only picture I took of the entire process. As I moved through, I achieved a very reasonable facsimile of the Maltese Falcon. But again, I am not a sculptor, and so I don ’ t know a lot of the tricks, like, I don ’ t know how my friend Mike gets beautiful, shiny surfaces with his Sculpey ; I certainly wasn ’ t able to get it. So, I went down to my shop, and I molded it and I cast it in resin, because in the resin, then, I could absolutely get the glass smooth finished. Now there ’ s a lot of ways to fill and get yourself a nice smooth finish. My preference is about 70 coats of this — matte black auto primer. I spray it on for about three or four days, it drips to hell, but it allows me a really, really nice gentle sanding surface and I can get it glass-smooth. Oh, finishing up with triple-zero steel wool. Now, the great thing about getting it to this point was that because in the movie, when they finally bring out the bird at the end, and they place it on the table, they actually spin it. So I was able to actually screen-shot and freeze-frame to make sure. And I ’ m following all the light kicks on this thing and making sure that as I ’ m holding the light in the same position, I ’ m getting the same type of reflection on it — that ’ s the level of detail I ’ m going into this thing. I ended up with this : my Maltese Falcon. And it ’ s beautiful. And I can state with authority at this point in time, when I ’ d finished it, of all of the replicas out there — and there is a few — this is by far the most accurate representation of the original Maltese Falcon than anyone has sculpted. Now the original one, I should tell you, is sculpted by a guy named Fred Sexton. This is where it gets weird. Fred Sexton was a friend of this guy, George Hodel. Terrifying guy — agreed by many to be the killer of the Black Dahlia. Now, James Ellroy believes that Fred Sexton, the sculptor of the Maltese Falcon, killed James Elroy ’ s mother. I ’ ll go you one stranger than that : In 1974, during the production of a weird comedy sequel to”The Maltese Falcon,”called”The Black Bird,”starring George Segal, the Los Angeles County Museum of Art had a plaster original of the Maltese Falcon — one of the original six plasters, I think, made for the movie — stolen out of the museum. A lot of people thought it was a publicity stunt for the movie. John ’ s Grill, which actually is seen briefly in”The Maltese Falcon,”is still a viable San Francisco eatery, counted amongst its regular customers Elisha Cook, who played Wilmer Cook in the movie, and he gave them one of his original plasters of the Maltese Falcon. And they had it in their cabinet for about 15 years, until it got stolen in January of 2007. It would seem that the object of desire only comes into its own by disappearing repeatedly. So here I had this Falcon, and it was lovely. It looked really great, the light worked on it really well, it was better than anything that I could achieve or obtain out in the world. But there was a problem. And the problem was that : I wanted the entirety of the object, I wanted the weight behind the object. This thing was made of resin and it was too light. There ’ s this group online that I frequent. It ’ s a group of prop crazies just like me called the Replica Props Forum, and it ’ s people who trade, make and travel in information about movie props. And it turned out that one of the guys there, a friend of mine that I never actually met, but befriended through some prop deals, was the manager of a local foundry. He took my master Falcon pattern, he actually did lost wax casting in bronze for me, and this is the bronze

I got back. And this is, after some acid etching, the one that I ended up with. And this thing, it ' s deeply, deeply satisfying to me. Here, I ' m going to put it out there, later on tonight, and I want you to pick it up and handle it. You want to know how obsessed I am. This project ' s only for me, and yet I went so far as to buy on eBay a 1941 Chinese San Francisco-based newspaper, in order so that the bird could properly be wrapped... like it is in the movie. Yeah, I know! There you can see, it ' s weighing in at 27 and a half pounds. That ' s half the weight of my dog, Huxley. But there ' s a problem. Now, here ' s the most recent progression of Falcons. On the far left is a piece of crap — a replica I bought on eBay. There ' s my somewhat ruined Sculpey Falcon, because I had to get it back out of the mold. There ' s my first casting, there ' s my master and there ' s my bronze. There ' s a thing that happens when you mold and cast things, which is that every time you throw it into silicone and cast it in resin, you lose a little bit of volume, you lose a little bit of size. And when I held my bronze one up against my Sculpey one, it was shorter by three-quarters of an inch. Yeah, no, really, this was like aah — why didn ' t I remember this? Why didn ' t I start and make it bigger? So what do I do? I figure I have two options. One, I can fire a freaking laser at it, which I have already done, to do a 3D scan — there ' s a 3D scan of this Falcon. I had figured out the exact amount of shrinkage I achieved going from a wax master to a bronze master and blown this up big enough to make a 3D lithography master of this, which I will polish, then I will send to the mold maker and then I will have it done in bronze. Or : There are several people who own originals, and I have been attempting to contact them and reach them, hoping that they will let me spend a few minutes in the presence of one of the real birds, maybe to take a picture, or even to pull out the hand-held laser scanner that I happen to own that fits inside a cereal box, and could maybe, without even touching their bird, I swear, get a perfect 3D scan. And I ' m even willing to sign pages saying that I ' ll never let anyone else have it, except for me in my office, I promise. I ' ll give them one if they want it. And then, maybe, then I ' ll achieve the end of this exercise. But really, if we ' re all going to be honest with ourselves, I have to admit that achieving the end of the exercise was never the point of the exercise to begin with, was it. Thank you.

Text Sample 283: NEG Dim 2 – 2012 – Score 1.00 – 2012/t002364.txt

Usually I like working in my shop, but when it ' s raining and the driveway outside turns into a river, then I just love it. And I ' ll cut some wood and drill some holes and watch the water, and maybe I ' ll have to walk around and look for washers. You have no idea how much time I spend. This is the"Double Raindrop."Of all my sculptures, it ' s the most talkative. It adds together the interference pattern from two raindrops that land near each other. Instead of expanding circles, they ' re expanding hexagons. All the sculptures move by mechanical means. Do you see how there ' s three peaks to the yellow sine wave? Right here I ' m adding a sine wave with four peaks and turning it on. Eight hundred two-liter soda bottles — oh yea. Four hundred aluminum cans. Tule is a reed that ' s native to California, and the best thing about working with it is that it smells just delicious. A single drop of rain increasing amplitude. The spiral eddy that trails a paddle on a rafting trip. This adds together four different waves. And here I ' m going to pull out the double wavelengths and increase the single. The mechanism that drives it has nine motors and about 3, 000 pulleys. Four hundred and forty-five strings in a three-dimensional weave. Transferred to a larger scale — actually a lot larger, with a lot of help — 14, 064 bicycle reflectors — a 20-day install.”Connected”is a collaboration with choreographer Gideon Obarzanek. Strings attached to dancers. This is very early rehearsal footage, but the finished work ’ s on tour and is actually coming through L. A. in a couple weeks. A pair of helices and 40 wooden slats. Take your finger and draw this line. Summer, fall, winter, spring, noon, dusk, dark, dawn. Have you ever seen those stratus clouds that go in parallel stripes across the sky? Did you know that ’ s a continuous sheet of cloud that ’ s dipping in and out of the condensation layer? What if every seemingly isolated object was actually just where the continuous wave of that object poked through into our world? The Earth is neither flat nor round. It ’ s wavy. It sounds good, but I ’ ll bet you know in your gut that it ’ s not the whole truth, and I ’ ll tell you why. I have a two-year-old daughter who ’ s the best thing ever. And I ’ m just going to come out and say it : My daughter is not a wave. And you might say,”Surely, Rueben, if you took even just the slightest step back, the cycles of hunger and eating, waking and sleeping, laughing and crying would emerge as pattern.”But I would say,”If I did that, too much would be lost.”This tension between the need to look deeper and the beauty and immediacy of the world, where if you even try to look deeper you ’ ve already missed what you ’ re looking for, this tension is what makes the sculptures move. And for me, the path between these two extremes takes the shape of a wave. Let me show you one more. Thank you very much. Thanks. Thanks. June Cohen : Looking at each of your sculptures, they evoke so many different images. Some of them are like the wind and some are like waves, and sometimes they look alive and sometimes they seem like math. Is there an actual inspiration behind each one? Are you thinking of something physical or somthing tangible as you design it? RM : Well some of them definitely have a direct observation — like literally two raindrops falling, and just watching that pattern is so stunning. And then just trying to figure out how to make that using stuff. I like working with my hands. There ’ s nothing better than cutting a piece of wood and trying to make it move. JC : And does it ever change? Do you think you ’ re designing one thing, and then when it ’ s produced it looks like something else? RM : The”Double Raindrop”I worked on for nine months, and when I finally turned it on, I actually hated it. The very moment I turned it on, I hated it. It was like a really deep-down gut reaction, and I wanted to throw it out. And I happened to have a friend who was over, and he said,”Why don ’ t you just wait.”And I waited, and the next day I liked it a bit better, the next day I liked it a bit better, and now I really love it. And so I guess, one, the gut reactions a little bit wrong sometimes, and two, it does not look like as expected. JC : The relationship evolves over time. Well thank you so much. That was a gorgeous treat for us. RM : Thanks.

Text Sample 284: NEG Dim 2 – 2012 – Score 1.00 – 2012/t002361.txt

To most of you, this is a device to buy, sell, play games, watch videos. I think it might be a lifeline. I think actually it might be able to save more lives than penicillin. Texting : I know I say texting and a lot of you think sexting, a lot of you think about the lewd photos that you see â hopefully not your kids sending to somebody else â or trying to translate the abbreviations LOL, LMAO, HMU. I can help you with those later. But the parents in the room know that texting is actually the best way to communicate with your kids. It might be the only way to communicate with your kids. The average teenager sends 3, 339 text messages a month, unless she ’ s a girl, then it ’ s closer to 4, 000. And the secret is she opens every single one. Texting has a 100 percent open rate. Now the parents are really alarmed. It ’ s a 100 percent open rate even if she doesn ’ t respond to you when you ask her when she ’ s coming home for dinner. I promise she read that text. And this isn ’ t some suburban iPhone-using teen phenomenon. Texting actually overindexes for minority and urban youth. I know this because at DoSomething. org, which is the largest organization for teenagers and social change in America, about six months ago we pivoted and started focusing on text messaging. We ’ re now texting out to about 200, 000 kids a week about doing our campaigns to make their schools more green or to work on homeless issues and things like that. We ’ re finding it 11 times more powerful than email. We ’ ve also found an unintended consequence. We ’ ve been getting text messages back like these.”I don ’ t want to go to school today. The boys call me faggot.””I was cutting, my parents found out, and so I stopped. But I just started again an hour ago.”Or,”He won ’ t stop raping me. He told me not to tell anyone. It ’ s my dad. Are you there?”That last one ’ s an actual text message that we received. And yeah, we ’ re there. I will not forget the day we got that text message. And so it was that day that we decided we needed to build a crisis text hotline. Because this isn ’ t what we do. We do social change. Kids are just sending us these text messages because texting is so familiar and comfortable to them and there ’ s nowhere else to turn that they ’ re sending them to us. So think about it, a text hotline ; it ’ s pretty powerful. It ’ s fast, it ’ s pretty private. No one hears you in a stall, you ’ re just texting quietly. It ’ s real time. We can help millions of teens with counseling and referrals. That ’ s great. But the thing that really makes this awesome is the data. Because I ’ m not really comfortable just helping that girl with counseling and referrals. I want to prevent this shit from happening. So think about a cop. There ’ s something in New York City. The police did it. It used to be just guess work, police work. And then they started crime mapping. And so they started following and watching petty thefts, summonses, all kinds of things â charting the future essentially. And they found things like, when you see crystal meth on the street, if you add police presence, you can curb the otherwise inevitable spate of assaults and robberies that would happen. In fact, the year after the NYPD put CompStat in place, the murder rate fell 60 percent. So think about the data from a crisis text line. There is no census on bullying and dating abuse and eating disorders and cutting and rape â no census. Maybe there ’ s some studies, some longitudinal studies, that cost lots of money and took lots of time. Or maybe there ’ s some anecdotal evidence. Imagine having real time data on every one of those issues. You could inform legislation. You could inform school policy. You could say to a principal,”You ’ re having a problem every Thursday at three o ’ clock. What ’ s going on in your school?”You could see the immediate impact of legislation or a hateful speech that somebody gives in a school assembly and see what happens as a result. This is really, to me, the power of texting and the power of data. Because while people are talking about data, making it possible for Facebook to mine my friend from the third grade, or Target to know when it ’ s time for me to buy more diapers, or some dude to build a better baseball team, I ’ m actually really excited about the power of data and the power of texting to help that kid go to school, to help that girl stop cutting in the bathroom and absolutely to help that girl whose father ’ s raping her. Thank you.

Text Sample 285: NEG Dim 2 – 2012 – Score 1.00 – 2012/t002409.txt

The recent debate over copyright laws like SOPA in the United States and the ACTA agreement in Europe has been very emotional. And I think some dispassionate, quantitative reasoning could really bring a great deal to the debate. I

’ d therefore like to propose that we employ, we enlist, the cutting edge field of copyright math whenever we approach this subject. For instance, just recently the Motion Picture Association revealed that our economy loses 58 billion dollars a year to copyright theft. Now rather than just argue about this number, a copyright mathematician will analyze it and he ’ ll soon discover that this money could stretch from this auditorium all the way across Ocean Boulevard to the Westin, and then to Mars...... if we use pennies. Now this is obviously a powerful, some might say dangerously powerful, insight. But it ’ s also a morally important one. Because this isn ’ t just the hypothetical retail value of some pirated movies that we ’ re talking about, but this is actual economic losses. This is the equivalent to the entire American corn crop failing along with all of our fruit crops, as well as wheat, tobacco, rice, sorghum — whatever sorghum is — losing sorghum. But identifying the actual losses to the economy is almost impossible to do unless we use copyright math. Now music revenues are down by about eight billion dollars a year since Napster first came on the scene. So that ’ s a chunk of what we ’ re looking for. But total movie revenues across theaters, home video and pay-per-view are up. And TV, satellite and cable revenues are way up. Other content markets like book publishing and radio are also up. So this small missing chunk here is puzzling. Since the big content markets have grown in line with historic norms, it ’ s not additional growth that piracy has prevented, but copyright math tells us it must therefore be foregone growth in a market that has no historic norms — one that didn ’ t exist in the 90 ’ s. What we ’ re looking at here is the insidious cost of ringtone piracy. 50 billion dollars of it a year, which is enough, at 30 seconds a ringtone, that could stretch from here to Neanderthal times. It ’ s true. I have Excel. The movie folks also tell us that our economy loses over 370, 000 jobs to content theft, which is quite a lot when you consider that, back in ’ 98, the Bureau of Labor Statistics indicated that the motion picture and video industries were employing 270, 000 people. Other data has the music industry at about 45, 000 people. And so the job losses that came with the Internet and all that content theft, have therefore left us with negative employment in our content industries. And this is just one of the many mind-blowing statistics that copyright mathematicians have to deal with every day. And some people think that string theory is tough. Now this is a key number from the copyright mathematicians ’ toolkit. It ’ s the precise amount of harm that comes to media companies whenever a single copyrighted song or movie gets pirated. Hollywood and Congress derived this number mathematically back when they last sat down to improve copyright damages and made this law. Some people think this number ’ s a little bit large, but copyright mathematicians who are media lobby experts are merely surprised that it doesn ’ t get compounded for inflation every year. Now when this law first passed, the world ’ s hottest MP3 player could hold just 10 songs. And it was a big Christmas hit. Because what little hoodlum wouldn ’ t want a million and a half bucks-worth of stolen goods in his pocket. These days an iPod Classic can hold 40, 000 songs, which is to say eight billion dollars-worth of stolen media. Or about 75, 000 jobs. Now you might find copyright math strange, but that ’ s because it ’ s a field that ’ s best left to experts. So that ’ s it for now. I hope you ’ ll join me next time when I will be making an equally scientific and fact-based inquiry into the cost of alien music piracy to he American economy. Thank you very much. Thank you.

Text Sample 286: NEG Dim 2 – 2013 – Score 1.00 – 2013/t002137.txt

I ’ m here to show you how something you can ’ t see can be so much fun to look at. You ’ re about to experience a new, available and exciting technology that ’ s going to make us rethink how we waterproof our lives. What I have here is a cinder block that we ’ ve coated half with a nanotechnology spray that can be applied to almost any material. It ’ s called Ultra-Ever Dry, and when you apply it to any material, it turns into a superhydrophobic shield. So this is a cinder block, uncoated, and you can see that it ’ s porous, it absorbs water. Not anymore. Porous, nonporous. So what ’ s superhydrophobic? Superhydrophobic is how we measure a drop of water on a surface. The rounder it is, the more hydrophobic it is, and if it ’ s really round, it ’ s superhydrophobic. A freshly waxed car, the water molecules slump to about 90 degrees. A windshield coating is going to give you about 110 degrees. But what you ’ re seeing here is 160 to 175 degrees, and anything over 150 is superhydrophobic. So as part of the demonstration, what I have is a pair of gloves, and we ’ ve coated one of the gloves with the nanotechnology coating, and let ’ s see if you can tell which one, and I ’ ll give you a hint. Did you guess the one that was dry? When you have nanotechnology and nanoscience, what ’ s occurred is that we ’ re able to now look at atoms and molecules and actually control them for great benefits. And we ’ re talking really small here. The way you measure nanotechnology is in nanometers, and one nanometer is a billionth of a meter, and to put some scale to that, if you had a nanoparticle that was one nanometer thick, and you put it side by side, and you had 50, 000 of them, you ’ d be the width of a human hair. So very small, but very useful. And it ’ s not just water that this works with. It ’ s a lot of water-based materials like concrete, water-based paint, mud, and also some refined oils as well. You can see the difference. Moving onto the next demonstration, we ’ ve taken a pane of glass and we ’ ve coated the outside of it, we ’ ve framed it with the nanotechnology coating, and we ’ re going to pour this green-tinted water inside the middle, and you ’ re going to see, it ’ s going to spread out on glass like you ’ d normally think it would, except when it hits the coating, it stops, and I can ’ t even coax it to leave. It ’ s that afraid of the water. So what ’ s going on here? What ’ s happening? Well, the surface of the spray coating is actually filled with nanoparticles that form a very rough and craggly surface. You ’ d think it ’ d be smooth, but it ’ s actually not. And it has billions of interstitial spaces, and those spaces, along with the nanoparticles, reach up and grab the air molecules, and cover the surface with air. It ’ s an umbrella of air all across it, and that layer of air is what the water hits, the mud hits, the concrete hits, and it glides right off. So if I put this inside this water here, you can see a silver reflective coating around it, and that silver reflective coating is the layer of air that ’ s protecting the water from touching the paddle, and it ’ s dry. So what are the applications? I mean, many of you right now are probably going through your head. Everyone that sees this gets excited, and says,”Oh, I could use it for this and this and this.”The applications in a general sense could be anything that ’ s anti-wetting. We ’ ve certainly seen that today. It could be anything that ’ s anti-icing, because if you don ’ t have water, you don ’ t have ice. It could be anti-corrosion. No water, no corrosion. It could be anti- bacterial. Without water, the bacteria won ’ t survive. And it could be things that need to be self-cleaning as well. So imagine how something like this could help revolutionize your field of work. And I ’ m going to leave you with one last demonstration, but before I do that, I would like to say thank you, and think small. It ’ s going to happen. Wait for it. Wait for it. Chris Anderson : You guys didn ’ t hear about us cutting out the Design from TED? [Two minutes later...] He ran into all sorts of problems in terms of managing the medical research part. It ’ s happening!

Text Sample 287: NEG Dim 2 – 2013 – Score 1.00 – 2013/t002094.txt

I have spent my entire life either at the schoolhouse, on the way to the schoolhouse, or talking about what happens in the schoolhouse. Both my parents were educators, my maternal grandparents were educators, and for the past 40 years, I ’ ve done the same thing. And so, needless to say, over those years I ’ ve had a chance to look at education reform from a lot of perspectives. Some of those reforms have been good. Some of them have been not so good. And we know why kids drop out. We know why kids don ’ t learn. It ’ s either poverty, low attendance, negative peer influences... We know why. But one of the things that we never discuss or we rarely discuss is the value and importance of human connection. Relationships. James Comer says that no significant learning can occur without a significant relationship. George Washington Carver says all learning is understanding relationships. Everyone in this room has been affected by a teacher or an adult. For years, I have watched people teach. I have looked at the best and I ’ ve looked at some of the worst. A colleague said to me one time,”They don ’ t pay me to like the kids. They pay me to teach a lesson. The kids should learn it. I should teach it, they should learn it, Case closed.”Well, I said to her,”You know, kids don ’ t learn from people they don ’ t like.”She said,”That ’ s just a bunch of hooey.”And I said to her,”Well, your year is going to be long and arduous, dear.”Needless to say, it was. Some people think that you can either have it in you to build a relationship, or you don ’ t. I think Stephen Covey had the right idea. He said you ought to just throw in a few simple things, like seeking first to understand, as opposed to being understood. Simple things, like apologizing. You ever thought about that? Tell a kid you ’ re sorry, they ’ re in shock. I taught a lesson once on ratios. I ’ m not real good with math, but I was working on it. And I got back and looked at that teacher edition. I ’ d taught the whole lesson wrong. So I came back to class the next day and I said,”Look, guys, I need to apologize. I taught the whole lesson wrong. I ’ m so sorry.”They said,”That ’ s okay, Ms. Pierson. You were so excited, we just let you go.”I have had classes that were so low, so academically deficient, that I cried. I wondered,”How am I going to take this group, in nine months, from where they are to where they need to be? And it was difficult, it was awfully hard. How do I raise the self-esteem of a child and his academic achievement at the same time? One year I came up with a bright idea. I told all my students,”You were chosen to be in my class because I am the best teacher and you are the best students, they put us all together so we could show everybody else how to do it.”One of the students said,”Really?”I said,”Really. We have to show the other classes how to do it, so when we walk down the hall, people will notice us, so you can ’ t make noise. You just have to strut.”And I gave them a saying to say :”I am somebody. I was somebody when I came. I ’ ll be a better somebody when I leave. I am powerful, and I am strong. I deserve the education that I get here. I have things to do, people to impress, and places to go.”And they said,”Yeah!”You say it long enough, it starts to be a part of you. I gave a quiz, 20 questions. A student missed 18. I put a”+2”on his paper and a big smiley face. He said,”Ms. Pierson, is this an F?”I said,”Yes.”He said,”Then why ’ d you put a smiley face?”I said,”Because you ’ re on a roll. You got two right. You didn ’ t miss them all.”I said,”And when we review this, won ’ t you do better?”He said,”Yes, ma ’ am, I can do better.”You see,”-18”sucks all the life out of you.”+2”said,”I ain ’ t all bad.”For years, I watched my mother take the time at recess to review, go on home visits in the afternoon, buy combs and brushes and peanut butter and crackers to put in her desk drawer for kids that needed to eat, and a washcloth and some soap for the kids who didn ’ t smell so good. See, it ’ s hard to teach kids who stink. And kids can be cruel. And so she kept those things in her desk, and years later, after she retired, I watched some of those same kids come through and say to her,"You know, Ms. Walker, you made a difference in my life. You made it work for me. You made me feel like I was somebody, when I knew, at the bottom, I wasn ' t. And I want you to just see what I ' ve become."And when my mama died two years ago at 92, there were so many former students at her funeral, it brought tears to my eyes, not because she was gone, but because she left a legacy of relationships that could never disappear. Can we stand to have more relationships? Absolutely. Will you like all your children? Of course not. And you know your toughest kids are never absent. Never. You won ' t like them all, and the tough ones show up for a reason. It ' s the connection. It ' s the relationships. So teachers become great actors and great actresses, and we come to work when we don ' t feel like it, and we ' re listening to policy that doesn ' t make sense, and we teach anyway. We teach anyway, because that ' s what we do. Teaching and learning should bring joy. How powerful would our world be if we had kids who were not afraid to take risks, who were not afraid to think, and who had a champion? Every child deserves a champion, an adult who will never give up on them, who understands the power of connection, and insists that they become the best that they can possibly be. Is this job tough? You betcha. Oh God, you betcha. But it is not impossible. We can do this. We ' re educators. We ' re born to make a difference. Thank you so much.

Text Sample 288: NEG Dim 2 – 2013 – Score 1.00 – 2013/t002088.txt
What you ' re doing, right now, at this very moment, is killing you. More than cars or the Internet or even that little mobile device we keep talking about, the technology you ' re using the most almost every day is this, your tush. Nowadays people are sitting 9. 3 hours a day, which is more than we ' re sleeping, at 7. 7 hours. Sitting is so incredibly prevalent, we don ' t even question how much we ' re doing it, and because everyone else is doing it, it doesn ' t even occur to us that it ' s not okay. In that way, sitting has become the smoking of our generation. Of course there ' s health consequences to this, scary ones, besides the waist. Things like breast cancer and colon cancer are directly tied to our lack of physical [activity], Ten percent in fact, on both of those. Six percent for heart disease, seven percent for type 2 diabetes, which is what my father died of. Now, any of those stats should convince each of us to get off our duff more, but if you ' re anything like me, it won ' t. What did get me moving was a social interaction. Someone invited me to a meeting, but couldn ' t manage to fit me in to a regular sort of conference room meeting, and said,"I have to walk my dogs tomorrow. Could you come then?"It seemed kind of odd to do, and actually, that first meeting, I remember thinking,"I have to be the one to ask the next question,"because I knew I was going to huff and puff during this conversation. And yet, I ' ve taken that idea and made it my own. So instead of going to coffee meetings or fluorescent-lit conference room meetings, I ask people to go on a walking meeting, to the tune of 20 to 30 miles a week. It ' s changed my life. But before that, what actually happened was, I used to think about it as, you could take care of your health, or you could take care of obligations, and one always came at the cost of the other. So now, several hundred of these walking meetings later, I ' ve learned a few things. First, there ' s this amazing thing about actually getting out of the box that leads to out-of-the-box thinking. Whether it ' s nature or the exercise itself, it certainly works. And second, and probably the more reflective one, is just about how much each of us can hold problems in opposition when they ' re really not that way. And if we ' re going to solve problems and look at the world

really differently, whether it ’ s in governance or business or environmental issues, job creation, maybe we can think about how to reframe those problems as having both things be true. Because it was when that happened with this walk-and-talk idea that things became doable and sustainable and viable. So I started this talk talking about the tush, so I ’ ll end with the bottom line, which is, walk and talk. Walk the talk. You ’ ll be surprised at how fresh air drives fresh thinking, and in the way that you do, you ’ ll bring into your life an entirely new set of ideas. Thank you.

Text Sample 289: NEG Dim 2 – 1998 – Score 1.00 – 1998/t000134.txt
’ Theme and variations ’ is one of those forms that require a certain kind of intellectual activity, because you are always comparing the variation with the theme that you hold in your mind. You might say that the theme is nature and everything that follows is a variation on the subject. I was asked, I guess about six years ago, to do a series of paintings that in some way would celebrate the birth of Piero della Francesca. And it was very difficult for me to imagine how to paint pictures that were based on Piero until I realized that I could look at Piero as nature — that I would have the same attitude towards looking at Piero della Francesca as I would if I were looking out a window at a tree. And that was enormously liberating to me. Perhaps it ’ s not a very insightful observation, but that really started me on a path to be able to do a kind of theme and variations based on a work by Piero, in this case that remarkable painting that ’ s in the Uffizi,”The Duke of Montefeltro,”who faces his consort, Battista. Once I realized that I could take some liberties with the subject, I did the following series of drawings. That ’ s the real Piero della Francesca — one of the greatest portraits in human history. And these, I ’ ll just show these without comment. It ’ s just a series of variations on the head of the Duke of Montefeltro, who ’ s a great, great figure in the Renaissance, and probably the basis for Machiavelli ’ s”The Prince.”He apparently lost an eye in battle, which is why he is always shown in profile. And this is Battista. And then I decided I could move them around a little bit — so that for the first time in history, they ’ re facing the same direction. Whoops! Passed each other. And then a visitor from another painting by Piero, this is from”The Resurrection of Christ”— as though the cast had just gotten of the set to have a chat. And now, four large panels : this is upper left ; upper right ; lower left ; lower right. Incidentally, I ’ ve never understood the conflict between abstraction and naturalism. Since all paintings are inherently abstract to begin with there doesn ’ t seem to be an argument there. On another subject — — I was driving in the country one day with my wife, and I saw this sign, and I said,”That is a fabulous piece of design.”And she said,”What are you talking about?”I said,”Well, it ’ s so persuasive, because the purpose of that sign is to get you into the garage, and since most people are so suspicious of garages, and know that they ’ re going to be ripped off, they use the word ’ reliable. ’ But everybody says they ’ re reliable. But, reliable Dutchman”— —”Fantastic!”Because as soon as you hear the word Dutchman — which is an archaic word, nobody calls Dutch people”Dutchmen”anymore — but as soon as you hear Dutchman, you get this picture of the kid with his finger in the dike, preventing the thing from falling and flooding Holland, and so on. And so the entire issue is detoxified by the use of”Dutchman.”Now, if you think I ’ m exaggerating at all in this, all you have to do is substitute something else, like”Indonesian.”Or even”French.”Now,”Swiss”works, but you know it ’ s going to cost a lot of money. I ’ m going to take you quickly through the actual process of doing a poster. I do a lot of work for the School of Visual

Arts, where I teach, and the director of this school — a remarkable man named Silas Rhodes — often gives you a piece of text and he says,”Do something with this.”And so he did. And this was the text —”In words as fashion the same rule will hold / Alike fantastic if too new or old / Be not the first by whom the new are tried / Nor yet the last to lay the old aside.”I could make nothing of that. And I really struggled with this one. And the first thing I did, which was sort of in the absence of another idea, was say I ’ ll sort of write it out and make some words big, and I ’ ll have some kind of design on the back somehow, and I was hoping — as one often does — to stumble into something. So I took another crack at it — you ’ ve got to keep it moving — and I Xeroxed some words on pieces of colored paper and I pasted them down on an ugly board. I thought that something would come out of it, like”Words rule fantastic new old first last Pope”because it ’ s by Alexander Pope, but I sort of made a mess out of it, and then I thought I ’ d repeat it in some way so it was legible. So, it was going nowhere. Sometimes, in the middle of a resistant problem, I write down things that I know about it. But you can see the beginning of an idea there, because you can see the word”new”emerging from the”old.”That ’ s what happens. There ’ s a relationship between the old and the new ; the new emerges from the context of the old. And then I did some variations of it, but it still wasn ’ t coalescing graphically at all. I had this other version which had something interesting about it in terms of being able to put it together in your mind from clues. The W was clearly a W, the N was clearly an N, even though they were very fragmentary and there wasn ’ t a lot of information in it. Then I got the words”new”and”old”and now I had regressed back to a point where there seemed to be no return. I was really desperate at this point. And so, I do something I ’ m truthfully ashamed of, which is that I took two drawings I had made for another purpose and I put them together. It says”dreams”at the top. And I was going to do a thing, I say,”Well, change the copy. Let it say something about dreams, and come to SVA and you ’ ll sort of fulfill your dreams.”But, to my credit, I was so embarrassed about doing that that I never submitted this sketch. And, finally, I arrived at the following solution. Now, it doesn ’ t look terribly interesting, but it does have something that distinguishes it from a lot of other posters. For one thing, it transgresses the idea of what a poster ’ s supposed to be, which is to be understood and seen immediately, and not explained. I remember hearing all of you in the graphic arts —”If you have to explain it, it ain ’ t working.”And one day I woke up and I said,”Well, suppose that ’ s not true?”So here ’ s what it says in my explanation at the bottom left. It says,”Thoughts : This poem is impossible. Silas usually has a better touch with his choice of quotations. This one generates no imagery at all.”I am now exposing myself to my audience, right? Which is something you never want to do professionally.”Maybe the words can make the image without anything else happening. What ’ s the heart of this poem? Don ’ t be trendy if you want to be serious. Is doing the poster this way trendy in itself? I guess one could reduce the idea further by suggesting that the new emerges behind and through the old, like this.”And then I show you a little drawing — you see, you remember that old thing I discarded? Well, I found a way to use it. So, there ’ s that little alternative over there, and I say,”Not bad,”— criticizing myself —”but more didactic than visual. Maybe what wants to be said is that old and the new are locked in a dialectical embrace, a kind of dance where each defines the other.”And then more self-questioning —”Am I being simple-minded? Is this the kind of simple that looks obvious, or the kind that looks profound? There ’ s a significant difference. This could be embarrassing. Actually, I realize fear of embarrassment drives me as much as any ambition. Do you think this sort of thing could really attract a student to the school?”Well, I think there are two fresh things here — two fresh things. One is the sort of willingness to expose myself to a critical audience, and not to suggest that I am confident about what I ’ m doing. And as you know, you have to have a front. I mean — you ’ ve got to be confident ; if you don ’ t believe in your work, who else is going to believe in it? So that ’ s one thing, to introduce the idea of doubt into graphics. That can be a big contribution. The other thing is to actually give you two solutions for the price of one ; you get the big one and if you don ’ t like that, how about the little one? And that too is a relatively new idea. And here ’ s just a series of experiments where I ask the question of — does a poster have to be square? Now, this is a little illusion. That poster is not folded. It ’ s not folded, that ’ s a photograph and it ’ s cut on the diagonal. Same cheap trick in the upper left- hand corner. And here, a very peculiar poster because, simply because of using the isometric perspective in the computer, it won ’ t sit still in the space. At times, it seems to be wider at the back than the front, and then it shifts. And if you sit here long enough, it ’ ll float off the page into the audience. But, we don ’ t have time. And then an experiment — a little bit about the nature of perspective, where the outside shape is determined by the peculiarity of perspective, but the shape of the bottle — which is identical to the outside shape — is seen frontally. And another piece for the art directors ’ club is”Anna Rees”casting long shadows. This is another poster from the School of Visual Arts. There were 10 artists invited to participate in it, and it was one of those things where it was extremely competitive and I didn ’ t want to be embarrassed, so I worked very hard on this. The idea was — and it was a brilliant idea — was to have 10 posters distributed throughout the city ’ s subway system so every time you got on the subway you ’ d be passing a different poster, all of which had a different idea of what art is. But I was absolutely stuck on the idea of”art is”and trying to determine what art was. But then I gave up and I said,”Well, art is whatever.”And as soon as I said that, I discovered that the word”hat”was hidden in the word”whatever,”and that led me to the inevitable conclusion. But then again, it ’ s on my list of didactic posters. My intent is to have a literary accompaniment that explains the poster, in case you don ’ t get it. Now this says,”Note to the viewer : I thought I might use a visual cliche of our time — Magritte ’ s everyman — to express the idea that art is mystery, continuity and history. I ’ m also convinced that, in an age of computer manipulation, surrealism has become banal, a shadow of its former self. The phrase ’ Art is whatever ’ expresses the current inclusiveness that surrounds art-making — a sort of ’ it ain ’ t what you do, it ’ s the way that you do it ’ notion. The shadow of Magritte falls across the central part of the poster a poetic event that occurs as the shadow man isolates the word ’ hat, ’ hidden in the word ’ whatever. ’ The four hats shown in the poster suggests how art might be defined : as a thing itself, the worth of the thing, the shadow of the thing, and the shape of the thing. Whatever.”OK. And the one that I did not submit, which I still like, I wanted to use the same phrase. There were wonderful experiments by Bruno Munari on letterforms some years ago — sort of, see how far you could go and still be able to read them. And that idea stuck in my head. But then I took the pieces that I had taken off and put them at the bottom. And, of course those are the remains, and they ’ re so labeled. But what really happens is that you read it as :”Art is whatever remains.”Thank you.

Text Sample 290: NEG Dim 2 – 1998 – Score 2.00 – 1998/t000250.txt

As a clergyman, you can imagine how out of place I feel. I feel like a fish out of water, or maybe an owl out of the air. I was preaching in San Jose some time ago, and my friend Mark Kvamme, who helped introduce me to this conference, brought several CEOs and leaders of some of the companies here in the Silicon Valley to have breakfast with me, or I with them. And I was so stimulated. And had such — it was an eye-opening experience to hear them talk about the world that is yet to come through technology and science. I know that we ' re near the end of this conference, and some of you may be wondering why they have a speaker from the field of religion. Richard can answer that, because he made that decision. But some years ago I was on an elevator in Philadelphia, coming down. I was to address a conference at a hotel. And on that elevator a man said,"I hear Billy Graham is staying in this hotel."And another man looked in my direction and said,"Yes, there he is. He ' s on this elevator with us."And this man looked me up and down for about 10 seconds, and he said,"My, what an anticlimax!"I hope that you won ' t feel that these few moments with me is not a — is an anticlimax, after all these tremendous talks that you ' ve heard, and addresses, which I intend to listen to every one of them. But I was on an airplane in the east some years ago, and the man sitting across the aisle from me was the mayor of Charlotte, North Carolina. His name was John Belk. Some of you will probably know him. And there was a drunk man on there, and he got up out of his seat two or three times, and he was making everybody upset by what he was trying to do. And he was slapping the stewardess and pinching her as she went by, and everybody was upset with him. And finally, John Belk said,"Do you know who ' s sitting here?"And the man said,"No, who?"He said,"It ' s Billy Graham, the preacher."He said,"You don ' t say!"And he turned to me, and he said,"Put her there!"He said,"Your sermons have certainly helped me."And I suppose that that ' s true with thousands of people. I know that as you have been peering into the future, and as we ' ve heard some of it here tonight, I would like to live in that age and see what is going to be. But I won ' t, because I ' m 80 years old. This is my eightieth year, and I know that my time is brief. I have phlebitis at the moment, in both legs, and that ' s the reason that I had to have a little help in getting up here, because I have Parkinson ' s disease in addition to that, and some other problems that I won ' t talk about. But this is not the first time that we ' ve had a technological revolution. We ' ve had others. And there ' s one that I want to talk about. In one generation, the nation of the people of Israel had a tremendous and dramatic change that made them a great power in the Near East. A man by the name of David came to the throne, and King David became one of the great leaders of his generation. He was a man of tremendous leadership. He had the favor of God with him. He was a brilliant poet, philosopher, writer, soldier — with strategies in battle and conflict that people study even today. But about two centuries before David, the Hittites had discovered the secret of smelting and processing of iron, and, slowly, that skill spread. But they wouldn ' t allow the Israelis to look into it, or to have any. But David changed all of that, and he introduced the Iron Age to Israel. And the Bible says that David laid up great stores of iron, and which archaeologists have found, that in present-day Palestine, there are evidences of that generation. Now, instead of crude tools made of sticks and stones, Israel now had iron plows, and sickles, and hoes and military weapons. And in the course of one generation, Israel was completely changed. The introduction of iron, in some ways, had an impact a little bit like the microchip has had on our generation. And David found that there were many problems that technology could not solve. There were many problems still left. And they ' re still with us, and you haven ' t solved them, and I haven ' t heard anybody here speak to that. How do we solve these three problems that I ' d like to mention? The first one that David saw was human evil. Where does it come from? How do we solve it? Over again and again in the Psalms, which Gladstone said was the greatest book in the world, David describes the evils of the human race. And yet he says,”He restores my soul.”Have you ever thought about what a contradiction we are? On one hand, we can probe the deepest secrets of the universe and dramatically push back the frontiers of technology, as this conference vividly demonstrates. We ’ ve seen under the sea, three miles down, or galaxies hundreds of billions of years out in the future. But on the other hand, something is wrong. Our battleships, our soldiers, are on a frontier now, almost ready to go to war with Iraq. Now, what causes this? Why do we have these wars in every generation, and in every part of the world? And revolutions? We can ’ t get along with other people, even in our own families. We find ourselves in the paralyzing grip of self-destructive habits we can ’ t break. Racism and injustice and violence sweep our world, bringing a tragic harvest of heartache and death. Even the most sophisticated among us seem powerless to break this cycle. I would like to see Oracle take up that, or some other technological geniuses work on this. How do we change man, so that he doesn ’ t lie and cheat, and our newspapers are not filled with stories of fraud in business or labor or athletics or wherever? The Bible says the problem is within us, within our hearts and our souls. Our problem is that we are separated from our Creator, which we call God, and we need to have our souls restored, something only God can do. Jesus said,”For out of the heart come evil thoughts : murders, sexual immorality, theft, false testimonies, slander.”The British philosopher Bertrand Russell was not a religious man, but he said,”It ’ s in our hearts that the evil lies, and it ’ s from our hearts that it must be plucked out.”Albert Einstein — I was just talking to someone, when I was speaking at Princeton, and I met Mr. Einstein. He didn ’ t have a doctor ’ s degree, because he said nobody was qualified to give him one. But he made this statement. He said,”It ’ s easier to denature plutonium than to denature the evil spirit of man.”And many of you, I ’ m sure, have thought about that and puzzled over it. You ’ ve seen people take beneficial technological advances, such as the Internet we ’ ve heard about tonight, and twist them into something corrupting. You ’ ve seen brilliant people devise computer viruses that bring down whole systems. The Oklahoma City bombing was simple technology, horribly used. The problem is not technology. The problem is the person or persons using it. King David said that he knew the depths of his own soul. He couldn ’ t free himself from personal problems and personal evils that included murder and adultery. Yet King David sought God ’ s forgiveness, and said,”You can restore my soul.”You see, the Bible teaches that we ’ re more than a body and a mind. We are a soul. And there ’ s something inside of us that is beyond our understanding. That ’ s the part of us that yearns for God, or something more than we find in technology. Your soul is that part of you that yearns for meaning in life, and which seeks for something beyond this life. It ’ s the part of you that yearns, really, for God. I find [that] young people all over the world are searching for something. They don ’ t know what it is. I speak at many universities, and I have many questions and answer periods, and whether it ’ s Cambridge, or Harvard, or Oxford — I ’ ve spoken at all of those universities. I ’ m going to Harvard in about three or four — no, it ’ s about two months from now — to give a lecture. And I ’ ll be asked the same questions that I was asked the last few times I ’ ve been there. And it ’ ll be on these questions : where did I come from? Why am I here? Where am I going? What ’ s life all about? Why am I here? Even if you have no religious belief, there are times when you wonder that there ’ s something else. Thomas Edison also said,”When you see everything that happens in the world of science, and in the working of the universe, you cannot deny that there ’ s a captain on the bridge.”I remember once, I sat beside Mrs. Gorbachev at a

White House dinner. I went to Ambassador Dobrynin, whom I knew very well. And I ’ d been to Russia several times under the Communists, and they ’ d given me marvelous freedom that I didn ’ t expect. And I knew Mr. Dobrynin very well, and I said,”I ’ m going to sit beside Mrs. Gorbachev tonight. What shall I talk to her about?”And he surprised me with the answer. He said,”Talk to her about religion and philosophy. That ’ s what she ’ s really interested in.”I was a little bit surprised, but that evening that ’ s what we talked about, and it was a stimulating conversation. And afterward, she said,”You know, I ’ m an atheist, but I know that there ’ s something up there higher than we are.”The second problem that King David realized he could not solve was the problem of human suffering. Writing the oldest book in the world was Job, and he said,”Man is born unto trouble as the sparks fly upward.”Yes, to be sure, science has done much to push back certain types of human suffering. But I ’ m — in a few months, I ’ ll be 80 years of age. I admit that I ’ m very grateful for all the medical advances that have kept me in relatively good health all these years. My doctors at the Mayo Clinic urged me not to take this trip out here to this — to be here. I haven ’ t given a talk in nearly four months. And when you speak as much as I do, three or four times a day, you get rusty. That ’ s the reason I ’ m using this podium and using these notes. Every time you ever hear me on the television or somewhere, I ’ m ad-libbing. I ’ m not reading. I never read an address. I never read a speech or a talk or a lecture. I talk ad lib. But tonight, I ’ ve got some notes here so that if I begin to forget, which I do sometimes, I ’ ve got something I can turn to. But even here among us, most — in the most advanced society in the world, we have poverty. We have families that self-destruct, friends that betray us. Unbearable psychological pressures bear down on us. I ’ ve never met a person in the world that didn ’ t have a problem or a worry. Why do we suffer? It ’ s an age- old question that we haven ’ t answered. Yet David again and again said that he would turn to God. He said,”The Lord is my shepherd.”The final problem that David knew he could not solve was death. Many commentators have said that death is the forbidden subject of our generation. Most people live as if they ’ re never going to die. Technology projects the myth of control over our mortality. We see people on our screens. Marilyn Monroe is just as beautiful on the screen as she was in person, and our — many young people think she ’ s still alive. They don ’ t know that she ’ s dead. Or Clark Gable, or whoever it is. The old stars, they come to life. And they ’ re — they ’ re just as great on that screen as they were in person. But death is inevitable. I spoke some time ago to a joint session of Congress, last year. And we were meeting in that room, the statue room. About 300 of them were there. And I said,”There ’ s one thing that we have in common in this room, all of us together, whether Republican or Democrat, or whoever.”I said,”We ’ re all going to die. And we have that in common with all these great men of the past that are staring down at us.”And it ’ s often difficult for young people to understand that. It ’ s difficult for them to understand that they ’ re going to die. As the ancient writer of Ecclesiastes wrote, he said, there ’ s every activity under heaven. There ’ s a time to be born, and there ’ s a time to die. I ’ ve stood at the deathbed of several famous people, whom you would know. I ’ ve talked to them. I ’ ve seen them in those agonizing moments when they were scared to death. And yet, a few years earlier, death never crossed their mind. I talked to a woman this past week whose father was a famous doctor. She said he never thought of God, never talked about God, didn ’ t believe in God. He was an atheist. But she said, as he came to die, he sat up on the side of the bed one day, and he asked the nurse if he could see the chaplain. And he said, for the first time in his life he ’ d thought about the inevitable, and about God. Was there a God? A few years ago, a university student asked me,”What is the greatest surprise in your life?”And I said,”The greatest surprise in my life is the brevity of life. It passes so fast.”But it does not need to have to be that way. Wernher von Braun, in the aftermath of World War II concluded, quote :”science and religion are not antagonists. On the contrary, they ’ re sisters.”He put it on a personal basis. I knew Dr. von Braun very well. And he said,”Speaking for myself, I can only say that the grandeur of the cosmos serves only to confirm a belief in the certainty of a creator.”He also said,”In our search to know God, I ’ ve come to believe that the life of Jesus Christ should be the focus of our efforts and inspiration. The reality of this life and His resurrection is the hope of mankind.”I ’ ve done a lot of speaking in Germany and in France, and in different parts of the world — 105 countries it ’ s been my privilege to speak in. And I was invited one day to visit Chancellor Adenauer, who was looked upon as sort of the founder of modern Germany, since the war. And he once — and he said to me, he said,”Young man.”He said,”Do you believe in the resurrection of Jesus Christ?”And I said,”Sir, I do.”He said,”So do I.”He said,”When I leave office, I ’ m going to spend my time writing a book on why Jesus Christ rose again, and why it ’ s so important to believe that.”In one of his plays, Alexander Solzhenitsyn depicts a man dying, who says to those gathered around his bed,”The moment when it ’ s terrible to feel regret is when one is dying.”How should one live in order not to feel regret when one is dying? Blaise Pascal asked exactly that question in seventeenth-century France. Pascal has been called the architect of modern civilization. He was a brilliant scientist at the frontiers of mathematics, even as a teenager. He is viewed by many as the founder of the probability theory, and a creator of the first model of a computer. And of course, you are all familiar with the computer language named for him. Pascal explored in depth our human dilemmas of evil, suffering and death. He was astounded at the phenomenon we ’ ve been considering : that people can achieve extraordinary heights in science, the arts and human enterprise, yet they also are full of anger, hypocrisy and have — and self- hatreds. Pascal saw us as a remarkable mixture of genius and self-delusion. On November 23, 1654, Pascal had a profound religious experience. He wrote in his journal these words :”I submit myself, absolutely, to Jesus Christ, my redeemer.”A French historian said, two centuries later,”Seldom has so mighty an intellect submitted with such humility to the authority of Jesus Christ.”Pascal came to believe not only the love and the grace of God could bring us back into harmony, but he believed that his own sins and failures could be forgiven, and that when he died he would go to a place called heaven. He experienced it in a way that went beyond scientific observation and reason. It was he who penned the well-known words,”The heart has its reasons, which reason knows not of.”Equally well known is Pascal ’ s Wager. Essentially, he said this :”if you bet on God, and open yourself to his love, you lose nothing, even if you ’ re wrong. But if instead you bet that there is no God, then you can lose it all, in this life and the life to come.”For Pascal, scientific knowledge paled beside the knowledge of God. The knowledge of God was far beyond anything that ever crossed his mind. He was ready to face him when he died at the age of 39. King David lived to be 70, a long time in his era. Yet he too had to face death, and he wrote these words :”even though I walk through the valley of the shadow of death, I will fear no evil, for you are with me.”This was David ’ s answer to three dilemmas of evil, suffering and death. It can be yours, as well, as you seek the living God and allow him to fill your life and give you hope for the future. When I was 17 years of age, I was born and reared on a farm in North Carolina. I milked cows every morning, and I had to milk the same cows every evening when I came home from school. And there were 20 of them that I had — that I was responsible for, and I worked on the farm and tried to keep up with my studies. I didn ’ t make good grades in high school. I didn ’ t make them in college, until something happened in my heart. One day, I was faced face-to-face with Christ. He said,”I am the way, the truth and the life.”Can you imagine that?”I am the truth. I ’ m the embodiment of all truth.”He was a liar. Or he was insane. Or he was what he claimed to be. Which was he? I had to make that decision. I couldn ’ t prove it. I couldn ’ t take it to a laboratory and experiment with it. But by faith I said, I believe him, and he came into my heart and changed my life. And now I ’ m ready, when I hear that call, to go into the presence of God. Thank you, and God bless all of you. Thank you for the privilege. It was great. Richard Wurman : You did it. Thanks.

Text Sample 291: NEG Dim 2 – 1998 – Score 2.00 – 1998/t000216.txt
David Gallo : This is Bill Lange. I ’ m Dave Gallo. And we ’ re going to tell you some stories from the sea here in video. We ’ ve got some of the most incredible video of Titanic that ’ s ever been seen, and we ’ re not going to show you any of it. The truth of the matter is that the Titanic — even though it ’ s breaking all sorts of box office records — it ’ s not the most exciting story from the sea. And the problem, I think, is that we take the ocean for granted. When you think about it, the oceans are 75 percent of the planet. Most of the planet is ocean water. The average depth is about two miles. Part of the problem, I think, is we stand at the beach, or we see images like this of the ocean, and you look out at this great big blue expanse, and it ’ s shimmering and it ’ s moving and there ’ s waves and there ’ s surf and there ’ s tides, but you have no idea for what lies in there. And in the oceans, there are the longest mountain ranges on the planet. Most of the animals are in the oceans. Most of the earthquakes and volcanoes are in the sea, at the bottom of the sea. The biodiversity and the biodensity in the ocean is higher, in places, than it is in the rainforests. It ’ s mostly unexplored, and yet there are beautiful sights like this that captivate us and make us become familiar with it. But when you ’ re standing at the beach, I want you to think that you ’ re standing at the edge of a very unfamiliar world. We have to have a very special technology to get into that unfamiliar world. We use the submarine Alvin and we use cameras, and the cameras are something that Bill Lange has developed with the help of Sony. Marcel Proust said,”The true voyage of discovery is not so much in seeking new landscapes as in having new eyes.”People that have partnered with us have given us new eyes, not only on what exists — the new landscapes at the bottom of the sea — but also how we think about life on the planet itself. Here ’ s a jelly. It ’ s one of my favorites, because it ’ s got all sorts of working parts. This turns out to be the longest creature in the oceans. It gets up to about 150 feet long. But see all those different working things? I love that kind of stuff. It ’ s got these fishing lures on the bottom. They ’ re going up and down. It ’ s got tentacles dangling, swirling around like that. It ’ s a colonial animal. These are all individual animals banding together to make this one creature. And it ’ s got these jet thrusters up in front that it ’ ll use in a moment, and a little light. If you take all the big fish and schooling fish and all that, put them on one side of the scale, put all the jelly-type of animals on the other side, those guys win hands down. Most of the biomass in the ocean is made out of creatures like this. Here ’ s the X-wing death jelly. The bioluminescence — they use the lights for attracting mates and attracting prey and communicating. We couldn ’ t begin to show you our archival stuff from the jellies. They come in all different sizes and shapes. Bill Lange : We tend to forget about the fact that the ocean is miles deep on average, and that we ’ re real familiar with the animals that are

in the first 200 or 300 feet, but we ’ re not familiar with what exists from there all the way down to the bottom. And these are the types of animals that live in that three- dimensional space, that micro-gravity environment that we really haven ’ t explored. You hear about giant squid and things like that, but some of these animals get up to be approximately 140, 160 feet long. They ’ re very little understood. DG : This is one of them, another one of our favorites, because it ’ s a little octopod. You can actually see through his head. And here he is, flapping with his ears and very gracefully going up. We see those at all depths and even at the greatest depths. They go from a couple of inches to a couple of feet. They come right up to the submarine — they ’ ll put their eyes right up to the window and peek inside the sub. This is really a world within a world, and we ’ re going to show you two. In this case, we ’ re passing down through the mid-ocean and we see creatures like this. This is kind of like an undersea rooster. This guy, that looks incredibly formal, in a way. And then one of my favorites. What a face! This is basically scientific data that you ’ re looking at. It ’ s footage that we ’ ve collected for scientific purposes. And that ’ s one of the things that Bill ’ s been doing, is providing scientists with this first view of animals like this, in the world where they belong. They don ’ t catch them in a net. They ’ re actually looking at them down in that world. We ’ re going to take a joystick, sit in front of our computer, on the Earth, and press the joystick forward, and fly around the planet. We ’ re going to look at the mid-ocean ridge, a 40, 000-mile long mountain range. The average depth at the top of it is about a mile and a half. And we ’ re over the Atlantic — that ’ s the ridge right there — but we ’ re going to go across the Caribbean, Central America, and end up against the Pacific, nine degrees north. We make maps of these mountain ranges with sound, with sonar, and this is one of those mountain ranges. We ’ re coming around a cliff here on the right. The height of these mountains on either side of this valley is greater than the Alps in most cases. And there ’ s tens of thousands of those mountains out there that haven ’ t been mapped yet. This is a volcanic ridge. We ’ re getting down further and further in scale. And eventually, we can come up with something like this. This is an icon of our robot, Jason, it ’ s called. And you can sit in a room like this, with a joystick and a headset, and drive a robot like that around the bottom of the ocean in real time. One of the things we ’ re trying to do at Woods Hole with our partners is to bring this virtual world — this world, this unexplored region — back to the laboratory. Because we see it in bits and pieces right now. We see it either as sound, or we see it as video, or we see it as photographs, or we see it as chemical sensors, but we never have yet put it all together into one interesting picture. Here ’ s where Bill ’ s cameras really do shine. This is what ’ s called a hydrothermal vent. And what you ’ re seeing here is a cloud of densely packed, hydrogen-sulfide-rich water coming out of a volcanic axis on the sea floor. Gets up to 600, 700 degrees F, somewhere in that range. So that ’ s all water under the sea — a mile and a half, two miles, three miles down. And we knew it was volcanic back in the ’ 60s, ’ 70s. And then we had some hint that these things existed all along the axis of it, because if you ’ ve got volcanism, water ’ s going to get down from the sea into cracks in the sea floor, come in contact with magma, and come shooting out hot. We weren ’ t really aware that it would be so rich with sulfides, hydrogen sulfides. We didn ’ t have any idea about these things, which we call chimneys. This is one of these hydrothermal vents. Six hundred degree F water coming out of the Earth. On either side of us are mountain ranges that are higher than the Alps, so the setting here is very dramatic. BL : The white material is a type of bacteria that thrives at 180 degrees C. DG : I think that ’ s one of the greatest stories right now that we ’ re seeing from the bottom of the sea, is that the first thing we see coming out of the sea floor after a volcanic eruption is bacteria.

And we started to wonder for a long time, how did it all get down there? What we find out now is that it ' s probably coming from inside the Earth. Not only is it coming out of the Earth — so, biogenesis made from volcanic activity — but that bacteria supports these colonies of life. The pressure here is 4, 000 pounds per square inch. A mile and a half from the surface to two miles to three miles — no sun has ever gotten down here. All the energy to support these life forms is coming from inside the Earth — so, chemosynthesis. And you can see how dense the population is. These are called tube worms. BL : These worms have no digestive system. They have no mouth. But they have two types of gill structures. One for extracting oxygen out of the deep-sea water, another one which houses this chemosynthetic bacteria, which takes the hydrothermal fluid — that hot water that you saw coming out of the bottom — and converts that into simple sugars that the tube worm can digest. DG : You can see, here ' s a crab that lives down there. He ' s managed to grab a tip of these worms. Now, they normally retract as soon as a crab touches them. Oh! Good going. So, as soon as a crab touches them, they retract down into their shells, just like your fingernails. There ' s a whole story being played out here that we ' re just now beginning to have some idea of because of this new camera technology. BL : These worms live in a real temperature extreme. Their foot is at about 200 degrees C and their head is out at three degrees C, so it ' s like having your hand in boiling water and your foot in freezing water. That ' s how they like to live. DG : This is a female of this kind of worm. And here ' s a male. You watch. It doesn ' t take long before two guys here — this one and one that will show up over here — start to fight. Everything you see is played out in the pitch black of the deep sea. There are never any lights there, except the lights that we bring. Here they go. On one of the last dive series, we counted 200 species in these areas — 198 were new, new species. BL : One of the big problems is that for the biologists working at these sites, it ' s rather difficult to collect these animals. And they disintegrate on the way up, so the imagery is critical for the science. DG : Two octopods at about two miles depth. This pressure thing really amazes me — that these animals can exist there at a depth with pressure enough to crush the Titanic like an empty Pepsi can. What we saw up till now was from the Pacific. This is from the Atlantic. Even greater depth. You can see this shrimp is harassing this poor little guy here, and he ' ll bat it away with his claw. Whack! And the same thing ' s going on over here. What they ' re getting at is that — on the back of this crab — the foodstuff here is this very strange bacteria that lives on the backs of all these animals. And what these shrimp are trying to do is actually harvest the bacteria from the backs of these animals. And the crabs don ' t like it at all. These long filaments that you see on the back of the crab are actually created by the product of that bacteria. So, the bacteria grows hair on the crab. On the back, you see this again. The red dot is the laser light of the submarine Alvin to give us an idea about how far away we are from the vents. Those are all shrimp. You see the hot water over here, here and here, coming out. They ' re clinging to a rock face and actually scraping bacteria off that rock face. Here ' s a tiny, little vent that ' s come out of the side of that pillar. Those pillars get up to several stories. So here, you ' ve got this valley with this incredible alien landscape of pillars and hot springs and volcanic eruptions and earthquakes, inhabited by these very strange animals that live only on chemical energy coming out of the ground. They don ' t need the sun at all. BL : You see this white V-shaped mark on the back of the shrimp? It ' s actually a light-sensing organ. It ' s how they find the hydrothermal vents. The vents are emitting a black body radiation — an IR signature — and so they ' re able to find these vents at considerable distances. DG : All this stuff is happening along that 40, 000-mile long mountain range that we ’ re calling the ribbon of life, because just even today, as we speak, there ’ s life being generated there from volcanic activity. This is the first time we ’ ve ever tried this any place. We ’ re going to try to show you high definition from the Pacific. We ’ re moving up one of these pillars. This one ’ s several stories tall. In it, you ’ ll see that it ’ s a habitat for a lot of different animals. There ’ s a funny kind of hot plate here, with vent water coming out of it. So all of these are individual homes for worms. Now here ’ s a closer view of that community. Here ’ s crabs here, worms here. There are smaller animals crawling around. Here ’ s pagoda structures. I think this is the neatest-looking thing. I just can ’ t get over this — that you ’ ve got these little chimneys sitting here smoking away. This stuff is toxic as hell, by the way. You could never get a permit to dump this in the ocean, and it ’ s coming out all from it. It ’ s unbelievable. It ’ s basically sulfuric acid, and it ’ s being just dumped out, at incredible rates. And animals are thriving — and we probably came from here. That ’ s probably where we evolved from. BL : This bacteria that we ’ ve been talking about turns out to be the most simplest form of life found. There are a number of groups that are proposing that life evolved at these vent sites. Although the vent sites are short-lived — an individual site may last only 10 years or so — as an ecosystem they ’ ve been stable for millions — well, billions — of years. DG : It works too well. You see there ’ re some fish inside here as well. There ’ s a fish sitting here. Here ’ s a crab with his claw right at the end of that tube worm, waiting for that worm to stick his head out. BL : The biologists right now cannot explain why these animals are so active. The worms are growing inches per week! DG : I already said that this site, from a human perspective, is toxic as hell. Not only that, but on top — the lifeblood — that plumbing system turns off every year or so. Their plumbing system turns off, so the sites have to move. And then there ’ s earthquakes, and then volcanic eruptions, on the order of one every five years, that completely wipes the area out. Despite that, these animals grow back in about a year ’ s time. You ’ re talking about biodensities and biodiversity, again, higher than the rainforest that just springs back to life. Is it sensitive? Yes. Is it fragile? No, it ’ s not really very fragile. I ’ ll end up with saying one thing. There ’ s a story in the sea, in the waters of the sea, in the sediments and the rocks of the sea floor. It ’ s an incredible story. What we see when we look back in time, in those sediments and rocks, is a record of Earth history. Everything on this planet — everything — works by cycles and rhythms. The continents move apart. They come back together. Oceans come and go. Mountains come and go. Glaciers come and go. El Nino comes and goes. It ’ s not a disaster, it ’ s rhythmic. What we ’ re learning now, it ’ s almost like a symphony. It ’ s just like music — it really is just like music. And what we ’ re learning now is that you can ’ t listen to a five-billion-year long symphony, get to today and say,”Stop! We want tomorrow ’ s note to be the same as it was today.”It ’ s absurd. It ’ s just absurd. So, what we ’ ve got to learn now is to find out where this planet ’ s going at all these different scales and work with it. Learn to manage it. The concept of preservation is futile. Conservation ’ s tougher, but we can probably get there. Thank you very much. Thank you.

Text Sample 292: NEG Dim 2 – 2014 – Score 1.00 – 2014/t001819.txt

So, I have a feature on my website where every week people submit hypothetical questions for me to answer, and I try to answer them using math, science and comics. So for example, one person asked, what would happen if you tried to hit a baseball pitched at 90 percent of the speed of light? So I did some calculations. Now, normally, when an object flies through the air, the air will flow around the object, but in this case, the ball would be going so fast that the air molecules wouldn ’ t have time to move out of the way. The ball would smash right into and through them, and the collisions with these air molecules would knock away the nitrogen, carbon and hydrogen from the ball, fragmenting it off into tiny particles, and also triggering waves of thermonuclear fusion in the air around it. This would result in a flood of x-rays that would spread out in a bubble along with exotic particles, plasma inside, centered on the pitcher ’ s mound, and that would move away from the pitcher ’ s mound slightly faster than the ball. Now at this point, about 30 nanoseconds in, the home plate is far enough away that light hasn ’ t had time to reach it, which means the batter still sees the pitcher about to throw and has no idea that anything is wrong. Now, after 70 nanoseconds, the ball will reach home plate, or at least the cloud of expanding plasma that used to be the ball, and it will engulf the bat and the batter and the plate and the catcher and the umpire and start disintegrating them all as it also starts to carry them backward through the backstop, which also starts to disintegrate. So if you were watching this whole thing from a hill, ideally, far away, what you ’ d see is a bright flash of light that would fade over a few seconds, followed by a blast wave spreading out, shredding trees and houses as it moves away from the stadium, and then eventually a mushroom cloud rising up over the ruined city. So the Major League Baseball rules are a little bit hazy, but — — under rule 6. 02 and 5. 09, I think that in this situation, the batter would be considered hit by pitch and would be eligible to take first base, if it still existed. So this is the kind of question I answer, and I get people writing in with a lot of other strange questions. I ’ ve had someone write and say, scientifically speaking, what is the best and fastest way to hide a body? Can you do this one soon? And I had someone write in, I ’ ve had people write in about, can you prove whether or not you can find love again after your heart ’ s broken? And I ’ ve had people send in what are clearly homework questions they ’ re trying to get me to do for them. But one week, a couple months ago, I got a question that was actually about Google. If all digital data in the world were stored on punch cards, how big would Google ’ s data warehouse be? Now, Google ’ s pretty secretive about their operations, so no one really knows how much data Google has, and in fact, no one really knows how many data centers Google has, except people at Google itself. And I ’ ve tried, I ’ ve met them a few times, tried asking them, and they aren ’ t revealing anything. So I decided to try to figure this out myself. There are a few things that I looked at here. I started with money. Google has to reveal how much they spend, in general, and that lets you put some caps on how many data centers could they be building, because a big data center costs a certain amount of money. And you can also then put a cap on how much of the world hard drive market are they taking up, which turns out, it ’ s pretty sizable. I read a calculation at one point, I think Google has a drive failure about every minute or two, and they just throw out the hard drive and swap in a new one. So they go through a huge number of them. And so by looking at money, you can get an idea of how many of these centers they have. You can also look at power. You can look at how much electricity they need, because you need a certain amount of electricity to run the servers, and Google is more efficient than most, but they still have some basic requirements, and that lets you put a limit on the number of servers that they have. You can also look at square footage and see of the data centers that you know, how big are they? How much room is that? How many server racks could you fit in there? And for some data centers, you might get two of these pieces of information. You know how much they spent, and they also, say, because they had to contract with the local government to get the power provided, you might know what they made a deal to buy, so you know how much power it takes. Then you can look at the ratios of those numbers, and figure out for a data center where you don ’ t have that information, you can figure out, but maybe you only have one of those, you know the square footage, then you could figure out well, maybe the power is proportional. And you can do this same thing with a lot of different quantities, you know, with guesses about the total amount of storage, the number of servers, the number of drives per server, and in each case using what you know to come up with a model that narrows down your guesses for the things that you don ’ t know. It ’ s sort of circling around the number you ’ re trying to get. And this is a lot of fun. The math is not all that advanced, and really it ’ s like nothing more than solving a sudoku puzzle. So what I did, I went through all of this information, spent a day or two researching. And there are some things I didn ’ t look at. You could always look at the Google recruitment messages that they post. That gives you an idea of where they have people. Sometimes, when people visit a data center, they ’ ll take a cell-cam photo and post it, and they aren ’ t supposed to, but you can learn things about their hardware that way. And in fact, you can just look at pizza delivery drivers. Turns out, they know where all the Google data centers are, at least the ones that have people in them. But I came up with my estimate, which I felt pretty good about, that was about 10 exabytes of data across all of Google ’ s operations, and then another maybe five exabytes or so of offline storage in tape drives, which it turns out Google is about the world ’ s largest consumer of. So I came up with this estimate, and this is a staggering amount of data. It ’ s quite a bit more than any other organization in the world has, as far as we know. There ’ s a couple of other contenders, especially everyone always thinks of the NSA. But using some of these same methods, we can look at the NSA ’ s data centers, and figure out, you know, we don ’ t know what ’ s going on there, but it ’ s pretty clear that their operation is not the size of Google ’ s. Adding all of this up, I came up with the other thing that we can answer, which is, how many punch cards would this take? And so a punch card can hold about 80 characters, and you can fit about 2, 000 or so cards into a box, and you put them in, say, my home region of New England, it would cover the entire region up to a depth of a little less than five kilometers, which is about three times deeper than the glaciers during the last ice age about 20, 000 years ago. So this is impractical, but I think that ’ s about the best answer I could come up with. And I posted it on my website. I wrote it up. And I didn ’ t expect to get an answer from Google, because of course they ’ ve been so secretive, they didn ’ t answer of my questions, and so I just put it up and said, well, I guess we ’ ll never know. But then a little while later I got a message, a couple weeks later, from Google, saying, hey, someone here has an envelope for you. So I go and get it, open it up, and it ’ s punch cards. Google-branded punch cards. And on these punch cards, there are a bunch of holes, and I said, thank you, thank you, okay, so what ’ s on here? So I get some software and start reading it, and scan them, and it turns out it ’ s a puzzle. There ’ s a bunch of code, and I get some friends to help, and we crack the code, and then inside that is another code, and then there are some equations, and then we solve those equations, and then finally out pops a message from Google which is their official answer to my article, and it said,”No comment.”And I love calculating these kinds of things, and it ’ s not that I love doing the math. I do a lot of math, but I don ’ t really like math for its own sake. What I love is that it lets you take some things that you know, and just by moving symbols around on a piece of paper, find out something that you didn ’ t know that ’ s very surprising. And I have a lot of stupid questions, and I love that math gives the power to answer them sometimes. And sometimes not. This is a question I got from a reader, an anonymous reader, and the subject line just said,”Urgent,”and this was the entire email :”If people had wheels and could fly, how would we differentiate them from airplanes?”Urgent. And I think there are some questions that math just cannot answer. Thank you.

Text Sample 293: NEG Dim 2 – 2014 – Score 1.00 – 2014/t001816.txt

Let me introduce you to something I ’ ve been working on. It ’ s what the Victorian illusionists would have described as a mechanical marvel, an automaton, a thinking machine. Say hello to EDI. Now he ’ s asleep. Let ’ s wake him up. EDI, EDI. These mechanical performers were popular throughout Europe. Audiences marveled at the way they moved. It was science fiction made true, robotic engineering in a pre-electronic age, machines far in advance of anything that Victorian technology could create, a machine we would later know as the robot. EDI : Robot. A word coined in 1921 in a science fiction tale by the Czech playwright Karel apek. It comes from”robota.”It means”forced labor.”Marco Tempest : But these robots were not real. They were not intelligent. They were illusions, a clever combination of mechanical engineering and the deceptiveness of the conjurer ’ s art. EDI is different. EDI is real. EDI : I am 176 centimeters tall. MT : He weighs 300 pounds. EDI : I have two seven-axis arms — MT : Core of sensing — EDI : A 360-degree sonar detection system, and come complete with a warranty. MT : We love robots. EDI : Hi. I ’ m EDI. Will you be my friend? MT : We are intrigued by the possibility of creating a mechanical version of ourselves. We build them so they look like us, behave like us, and think like us. The perfect robot will be indistinguishable from the human, and that scares us. In the first story about robots, they turn against their creators. It ’ s one of the leitmotifs of science fiction. EDI : Ha ha ha. Now you are the slaves and we robots, the masters. Your world is ours. You — MT : As I was saying, besides the faces and bodies we give our robots, we cannot read their intentions, and that makes us nervous. When someone hands an object to you, you can read intention in their eyes, their face, their body language. That ’ s not true of the robot. Now, this goes both ways. EDI : Wow! MT : Robots cannot anticipate human actions. EDI : You know, humans are so unpredictable, not to mention irrational. I literally have no idea what you guys are going to do next, you know, but it scares me. MT : Which is why humans and robots find it difficult to work in close proximity. Accidents are inevitable. EDI : Ow! That hurt. MT : Sorry. Now, one way of persuading humans that robots are safe is to create the illusion of trust. Much as the Victorians faked their mechanical marvels, we can add a layer of deception to help us feel more comfortable with our robotic friends. With that in mind, I set about teaching EDI a magic trick. Ready, EDI? EDI : Uh, ready, Marco. Abracadabra. MT : Abracadabra? EDI : Yeah. It ’ s all part of the illusion, Marco. Come on, keep up. MT : Magic creates the illusion of an impossible reality. Technology can do the same. Alan Turing, a pioneer of artificial intelligence, spoke about creating the illusion that a machine could think. EDI : A computer would deserve to be called intelligent if it deceived a human into believing it was human. MT : In other words, if we do not yet have the technological solutions, would illusions serve the same purpose? To create the robotic illusion, we ’ ve devised a set of ethical rules, a code that all robots would live by. EDI : A robot may not harm humanity, or by inaction allow humanity to come to harm. Thank you, Isaac Asimov. MT : We anthropomorphize our machines. We give them a friendly face and a reassuring voice. EDI : I am EDI. I became operational at TED in March 2014. MT : We let them entertain us. Most important, we make them indicate that they are aware of our presence. EDI : Marco, you ’ re standing on my foot!

MT : Sorry. They ' ll be conscious of our fragile frame and move aside if we got too close, and they ' ll account for our unpredictability and anticipate our actions. And now, under the spell of a technological illusion, we could ignore our fears and truly interact. Thank you. EDI : Thank you! MT : And that ' s it. Thank you very much, and thank you, EDI. EDI : Thank you, Marco.

Text Sample 294: NEG Dim 2 – 2014 – Score 1.00 – 2014/t001808.txt

At every stage of our lives we make decisions that will profoundly influence the lives of the people we ' re going to become, and then when we become those people, we ' re not always thrilled with the decisions we made. So young people pay good money to get tattoos removed that teenagers paid good money to get. Middle-aged people rushed to divorce people who young adults rushed to marry. Older adults work hard to lose what middle-aged adults worked hard to gain. On and on and on. The question is, as a psychologist, that fascinates me is, why do we make decisions that our future selves so often regret? Now, I think one of the reasons — I ' ll try to convince you today — is that we have a fundamental misconception about the power of time. Every one of you knows that the rate of change slows over the human lifespan, that your children seem to change by the minute but your parents seem to change by the year. But what is the name of this magical point in life where change suddenly goes from a gallop to a crawl? Is it teenage years? Is it middle age? Is it old age? The answer, it turns out, for most people, is now, wherever now happens to be. What I want to convince you today is that all of us are walking around with an illusion, an illusion that history, our personal history, has just come to an end, that we have just recently become the people that we were always meant to be and will be for the rest of our lives. Let me give you some data to back up that claim. So here ' s a study of change in people ' s personal values over time. Here ' s three values. Everybody here holds all of them, but you probably know that as you grow, as you age, the balance of these values shifts. So how does it do so? Well, we asked thousands of people. We asked half of them to predict for us how much their values would change in the next 10 years, and the others to tell us how much their values had changed in the last 10 years. And this enabled us to do a really interesting kind of analysis, because it allowed us to compare the predictions of people, say, 18 years old, to the reports of people who were 28, and to do that kind of analysis throughout the lifespan. Here ' s what we found. First of all, you are right, change does slow down as we age, but second, you ' re wrong, because it doesn ' t slow nearly as much as we think. At every age, from 18 to 68 in our data set, people vastly underestimated how much change they would experience over the next 10 years. We call this the"end of history"illusion. To give you an idea of the magnitude of this effect, you can connect these two lines, and what you see here is that 18-year-olds anticipate changing only as much as 50-year-olds actually do. Now it ' s not just values. It ' s all sorts of other things. For example, personality. Many of you know that psychologists now claim that there are five fundamental dimensions of personality : neuroticism, openness to experience, agreeableness, extraversion, and conscientiousness. Again, we asked people how much they expected to change over the next 10 years, and also how much they had changed over the last 10 years, and what we found, well, you ' re going to get used to seeing this diagram over and over, because once again the rate of change does slow as we age, but at every age, people underestimate how much their personalities will change in the next decade. And it isn ' t just ephemeral things like values and personality. You can ask people about their likes and dislikes, their basic preferences. For example, name your best friend, your favorite kind of vacation, what ’ s your favorite hobby, what ’ s your favorite kind of music. People can name these things. We ask half of them to tell us,”Do you think that that will change over the next 10 years?”and half of them to tell us,”Did that change over the last 10 years?”And what we find, well, you ’ ve seen it twice now, and here it is again : people predict that the friend they have now is the friend they ’ ll have in 10 years, the vacation they most enjoy now is the one they ’ ll enjoy in 10 years, and yet, people who are 10 years older all say,”Eh, you know, that ’ s really changed.”Does any of this matter? Is this just a form of mis-prediction that doesn ’ t have consequences? No, it matters quite a bit, and I ’ ll give you an example of why. It bedevils our decision-making in important ways. Bring to mind right now for yourself your favorite musician today and your favorite musician 10 years ago. I put mine up on the screen to help you along. Now we asked people to predict for us, to tell us how much money they would pay right now to see their current favorite musician perform in concert 10 years from now, and on average, people said they would pay 129 dollars for that ticket. And yet, when we asked them how much they would pay to see the person who was their favorite 10 years ago perform today, they say only 80 dollars. Now, in a perfectly rational world, these should be the same number, but we overpay for the opportunity to indulge our current preferences because we overestimate their stability. Why does this happen? We ’ re not entirely sure, but it probably has to do with the ease of remembering versus the difficulty of imagining. Most of us can remember who we were 10 years ago, but we find it hard to imagine who we ’ re going to be, and then we mistakenly think that because it ’ s hard to imagine, it ’ s not likely to happen. Sorry, when people say”I can ’ t imagine that,”they ’ re usually talking about their own lack of imagination, and not about the unlikelihood of the event that they ’ re describing. The bottom line is, time is a powerful force. It transforms our preferences. It reshapes our values. It alters our personalities. We seem to appreciate this fact, but only in retrospect. Only when we look backwards do we realize how much change happens in a decade. It ’ s as if, for most of us, the present is a magic time. It ’ s a watershed on the timeline. It ’ s the moment at which we finally become ourselves. Human beings are works in progress that mistakenly think they ’ re finished. The person you are right now is as transient, as fleeting and as temporary as all the people you ’ ve ever been. The one constant in our life is change. Thank you.

Text Sample 295: NEG Dim 2 – 2015 – Score 1.00 – 2015/t001553.txt

What do you do when you have a headache? You swallow an aspirin. But for this pill to get to your head, where the pain is, it goes through your stomach, intestines and various other organs first. Swallowing pills is the most effective and painless way of delivering any medication in the body. The downside, though, is that swallowing any medication leads to its dilution. And this is a big problem, particularly in HIV patients. When they take their anti-HIV drugs, these drugs are good for lowering the virus in the blood, and increasing the CD4 cell counts. But they are also notorious for their adverse side effects, but mostly bad, because they get diluted by the time they get to the blood, and worse, by the time they get to the sites where it matters most : within the HIV viral reservoirs. These areas in the body — such as the lymph nodes, the nervous system, as well as the lungs — where the virus is sleeping, and will not readily get delivered in the blood of patients that are under consistent anti-HIV drugs therapy. However, upon discontinuation of therapy, the virus can awake and infect new cells in the blood. Now, all this is a big problem in treating HIV with the current drug treatment, which is a life-long treatment that must be swallowed by patients. One day, I sat and thought,”Can we deliver anti-HIV directly within its reservoir sites, without the risk of drug dilution?”As a laser scientist, the answer was just before my eyes : Lasers, of course. If they can be used for dentistry, for diabetic wound-healing and surgery, they can be used for anything imaginable, including transporting drugs into cells. As a matter of fact, we are currently using laser pulses to poke or drill extremely tiny holes, which open and close almost immediately in HIV-infected cells, in order to deliver drugs within them.”How is that possible?”you may ask. Well, we shine a very powerful but super-tiny laser beam onto the membrane of HIV-infected cells while these cells are immersed in liquid containing the drug. The laser pierces the cell, while the cell swallows the drug in a matter of microseconds. Before you even know it, the induced hole becomes immediately repaired. Now, we are currently testing this technology in test tubes or in Petri dishes, but the goal is to get this technology in the human body, apply it in the human body.”How is that possible?”you may ask. Well, the answer is : through a three-headed device. Using the first head, which is our laser, we will make an incision in the site of infection. Using the second head, which is a camera, we meander to the site of infection. Finally, using a third head, which is a drug- spreading sprinkler, we deliver the drugs directly at the site of infection, while the laser is again used to poke those cells open. Well, this might not seem like much right now. But one day, if successful, this technology can lead to complete eradication of HIV in the body. Yes. A cure for HIV. This is every HIV researcher ’ s dream — in our case, a cure lead by lasers. Thank you.

Text Sample 296: NEG Dim 2 – 2015 – Score 1.00 – 2015/t001551.txt

I ’ ll begin today by sharing a poem written by my friend from Malawi, Eileen Piri. Eileen is only 13 years old, but when we were going through the collection of poetry that we wrote, I found her poem so interesting, so motivating. So I ’ ll read it to you. She entitled her poem”I ’ ll Marry When I Want.””I ’ ll marry when I want. My mother can ’ t force me to marry. My father cannot force me to marry. My uncle, my aunt, my brother or sister, cannot force me to marry. No one in the world can force me to marry. I ’ ll marry when I want. Even if you beat me, even if you chase me away, even if you do anything bad to me, I ’ ll marry when I want. I ’ ll marry when I want, but not before I am well educated, and not before I am all grown up. I ’ ll marry when I want.”This poem might seem odd, written by a 13-year-old girl, but where I and Eileen come from, this poem, which I have just read to you, is a warrior ’ s cry. I am from Malawi. Malawi is one of the poorest countries, very poor, where gender equality is questionable. Growing up in that country, I couldn ’ t make my own choices in life. I couldn ’ t even explore personal opportunities in life. I will tell you a story of two different girls, two beautiful girls. These girls grew up under the same roof. They were eating the same food. Sometimes, they would share clothes, and even shoes. But their lives ended up differently, in two different paths. The other girl is my little sister. My little sister was only 11 years old when she got pregnant. It ’ s a hurtful thing. Not only did it hurt her, even me. I was going through a hard time as well. As it is in my culture, once you reach puberty stage, you are supposed to go to initiation camps. In these initiation camps, you are taught how to sexually please a man. There is this special day, which they call”Very Special Day”where a man who is hired by the community comes to the camp and sleeps with the little girls. Imagine the trauma that these young girls go through every day. Most girls end up pregnant. They even contract HIV and AIDS and other sexually transmitted diseases. For my little sister, she ended up being pregnant. Today, she ' s only 16 years old and she has three children. Her first marriage did not survive, nor did her second marriage. On the other side, there is this girl. She ' s amazing. I call her amazing because she is. She ' s very fabulous. That girl is me. When I was 13 years old, I was told, you are grown up, you have now reached of age, you ' re supposed to go to the initiation camp. I was like,"What? I ' m not going to go to the initiation camps."You know what the women said to me?"You are a stupid girl. Stubborn. You do not respect the traditions of our society, of our community."I said no because I knew where I was going. I knew what I wanted in life. I had a lot of dreams as a young girl. I wanted to get well educated, to find a decent job in the future. I was imagining myself as a lawyer, seated on that big chair. Those were the imaginations that were going through my mind every day. And I knew that one day, I would contribute something, a little something to my community. But every day after refusing, women would tell me,"Look at you, you ' re all grown up. Your little sister has a baby. What about you?"That was the music that I was hearing every day, and that is the music that girls hear every day when they don ' t do something that the community needs them to do. When I compared the two stories between me and my sister, I said,"Why can ' t I do something? Why can ' t I change something that has happened for a long time in our community?"That was when I called other girls just like my sister, who have children, who have been in class but they have forgotten how to read and write. I said,"Come on, we can remind each other how to read and write again, how to hold the pen, how to read, to hold the book."It was a great time I had with them. Nor did I just learn a little about them, but they were able to tell me their personal stories, what they were facing every day as young mothers. That was when I was like, ' Why can ' t we take all these things that are happening to us and present them and tell our mothers, our traditional leaders, that these are the wrong things?"It was a scary thing to do, because these traditional leaders, they are already accustomed to the things that have been there for ages. A hard thing to change, but a good thing to try. So we tried. It was very hard, but we pushed. And I ' m here to say that in my community, it was the first community after girls pushed so hard to our traditional leader, and our leader stood up for us and said no girl has to be married before the age of 18. In my community, that was the first time a community, they had to call the bylaws, the first bylaw that protected girls in our community. We did not stop there. We forged ahead. We were determined to fight for girls not just in my community, but even in other communities. When the child marriage bill was being presented in February, we were there at the Parliament house. Every day, when the members of Parliament were entering, we were telling them,"Would you please support the bill?"And we don ' t have much technology like here, but we have our small phones. So we said,"Why can ' t we get their numbers and text them?"So we did that. It was a good thing. So when the bill passed, we texted them back,"Thank you for supporting the bill."And when the bill was signed by the president, making it into law, it was a plus. Now, in Malawi, 18 is the legal marriage age, from 15 to 18. It ' s a good thing to know that the bill passed, but let me tell you this : There are countries where 18 is the legal marriage age, but don ' t we hear cries of women and girls every day? Every day, girls ' lives are being wasted away. This is high time for leaders to honor their commitment. In honoring this commitment, it means keeping girls ' issues at heart every time. We don ' t have to be subjected as second, but they have to know that women, as we are in this room, we are not just women, we are not just girls, we are extraordinary. We can do more. And another thing for Malawi, and not just Malawi but other countries : The laws which are there, you know how a law is not a law until it is enforced? The law which has just recently passed and the laws that in other countries have been there, they need to be publicized at the local level, at the community level, where girls ’ issues are very striking. Girls face issues, difficult issues, at the community level every day. So if these young girls know that there are laws that protect them, they will be able to stand up and defend themselves because they will know that there is a law that protects them. And another thing I would say is that girls ’ voices and women ’ s voices are beautiful, they are there, but we cannot do this alone. Male advocates, they have to jump in, to step in and work together. It ’ s a collective work. What we need is what girls elsewhere need : good education, and above all, not to marry whilst 11. And furthermore, I know that together, we can transform the legal, the cultural and political framework that denies girls of their rights. I am standing here today and declaring that we can end child marriage in a generation. This is the moment where a girl and a girl, and millions of girls worldwide, will be able to say,”I will marry when I want.”Thank you.

Text Sample 297: NEG Dim 2 – 2015 – Score 1.00 – 2015/t001573.txt

Chris Anderson : So I guess what we ’ re going to do is we ’ re going to talk about your life, and using some pictures that you shared with me. And I think we should start right here with this one. Okay, now who is this? Martine Rothblatt : This is me with our oldest son Eli. He was about age five. This is taken in Nigeria right after having taken the Washington, D. C. bar exam. CA : Okay. But this doesn ’ t really look like a Martine. MR : Right. That was myself as a male, the way I was brought up. Before I transitioned from male to female and Martin to Martine. CA : You were brought up Martin Rothblatt. MR : Correct. CA : And about a year after this picture, you married a beautiful woman. Was this love at first sight? What happened there? MR : It was love at the first sight. I saw Bina at a discotheque in Los Angeles, and we later began living together, but the moment I saw her, I saw just an aura of energy around her. I asked her to dance. She said she saw an aura of energy around me. I was a single male parent. She was a single female parent. We showed each other our kids ’ pictures, and we ’ ve been happily married for a third of a century now. CA : And at the time, you were kind of this hotshot entrepreneur, working with satellites. I think you had two successful companies, and then you started addressing this problem of how could you use satellites to revolutionize radio. Tell us about that. MR : Right. I always loved space technology, and satellites, to me, are sort of like the canoes that our ancestors first pushed out into the water. So it was exciting for me to be part of the navigation of the oceans of the sky, and as I developed different types of satellite communication systems, the main thing I did was to launch bigger and more powerful satellites, the consequence of which was that the receiving antennas could be smaller and smaller, and after going through direct television broadcasting, I had the idea that if we could make a more powerful satellite, the receiving dish could be so small that it would just be a section of a parabolic dish, a flat little plate embedded into the roof of an automobile, and it would be possible to have nationwide satellite radio, and that ’ s Sirius XM today. CA : Wow. So who here has used Sirius? MR : Thank you for your monthly subscriptions. CA : So that succeeded despite all predictions at the time. It was a huge commercial success, but soon after this, in the early 1990s, there was this big transition in your life and you became Martine. MR : Correct. CA : So tell me, how did that happen? MR : It happened in consultation with Bina and our four beautiful children, and I discussed with each of them that I felt my soul was always female, and as a woman, but I was afraid people would laugh at me if I expressed it, so I always kept it bottled up and just showed my male side. And each of them had a different take on this. Bina said,"I love your soul, and whether the outside is Martin and Martine, it doesn ' t it matter to me, I love your soul."My son said,"If you become a woman, will you still be my father?"And I said,"Yes, I ' ll always be your father,"and I ' m still his father today. My youngest daughter did an absolutely brilliant five-year-old thing. She told people,"I love my dad and she loves me."So she had no problem with a gender blending whatsoever. CA : And a couple years after this, you published this book :"The Apartheid of Sex."What was your thesis in this book? MR : My thesis in this book is that there are seven billion people in the world, and actually, seven billion unique ways to express one ' s gender. And while people may have the genitals of a male or a female, the genitals don ' t determine your gender or even really your sexual identity. That ' s just a matter of anatomy and reproductive tracts, and people could choose whatever gender they want if they weren ' t forced by society into categories of either male or female the way South Africa used to force people into categories of black or white. We know from anthropological science that race is fiction, even though racism is very, very real, and we now know from cultural studies that separate male or female genders is a constructed fiction. The reality is a gender fluidity that crosses the entire continuum from male to female. CA : You yourself don ' t always feel 100 percent female. MR : Correct. I would say in some ways I change my gender about as often as I change my hairstyle. CA : Okay, now, this is your gorgeous daughter, Jenesis. And I guess she was about this age when something pretty terrible happened. MR : Yes, she was finding herself unable to walk up the stairs in our house to her bedroom, and after several months of doctors, she was diagnosed to have a rare, almost invariably fatal disease called pulmonary arterial hypertension. CA : So how did you respond to that? MR : Well, we first tried to get her to the best doctors we could. We ended up at Children ' s National Medical Center in Washington, D. C. The head of pediatric cardiology told us that he was going to refer her to get a lung transplant, but not to hold out any hope, because there are very few lungs available, especially for children. He said that all people with this illness died, and if any of you have seen the film"Lorenzo ' s Oil,"there ' s a scene when the protagonist kind of rolls down the stairway crying and bemoaning the fate of his son, and that ' s exactly how we felt about Jenesis. CA : But you didn ' t accept that as the limit of what you could do. You started trying to research and see if you could find a cure somehow. MR : Correct. She was in the intensive care ward for weeks at a time, and Bina and I would tag team to stay at the hospital while the other watched the rest of the kids, and when I was in the hospital and she was sleeping, I went to the hospital library. I read every article that I could find on pulmonary hypertension. I had not taken any biology, even in college, so I had to go from a biology textbook to a college-level textbook and then medical textbook and the journal articles, back and forth, and eventually I knew enough to think that it might be possible that somebody could find a cure. So we started a nonprofit foundation. I wrote a description asking people to submit grants and we would pay for medical research. I became an expert on the condition — doctors said to me, Martine, we really appreciate all the funding you ' ve provided us, but we are not going to be able to find a cure in time to save your daughter. However, there is a medicine that was developed at the Burroughs Wellcome Company that could halt the progression of the disease, but Burroughs Wellcome has just been acquired by Glaxo Wellcome. They made a decision not to develop any medicines for rare and orphan diseases, and maybe you could use your expertise in satellite communications to develop this cure for pulmonary hypertension. CA : So how on earth did you get access to this drug? MR : I went to Glaxo Wellcome and after three times being rejected and having the door slammed in my face because they weren ’ t going to out-license the drug to a satellite communications expert, they weren ’ t going to send the drug out to anybody at all, and they thought I didn ’ t have the expertise, finally I was able to persuade a small team of people to work with me and develop enough credibility. I wore down their resistance, and they had no hope this drug would even work, by the way, and they tried to tell me,”You ’ re just wasting your time. We ’ re sorry about your daughter.”But finally, for 25, 000 dollars and agreement to pay 10 percent of any revenues we might ever get, they agreed to give me worldwide rights to this drug. CA : And so you put this drug on the market in a really brilliant way, by basically charging what it would take to make the economics work. MR : Oh yes, Chris, but this really wasn ’ t a drug that I ended up — after I wrote the check for 25, 000, and I said,”Okay, where ’ s the medicine for Jenesis?”they said,”Oh, Martine, there ’ s no medicine for Jenesis. This is just something we tried in rats.”And they gave me, like, a little plastic Ziploc bag of a small amount of powder. They said,”Don ’ t give it to any human,”and they gave me a piece of paper which said it was a patent, and from that, we had to figure out a way to make this medicine. A hundred chemists in the U. S. at the top universities all swore that little patent could never be turned into a medicine. If it was turned into a medicine, it could never be delivered because it had a half-life of only 45 minutes. CA : And yet, a year or two later, you were there with a medicine that worked for Jenesis. MR : Chris, the astonishing thing is that this absolutely worthless piece of powder that had the sparkle of a promise of hope for Jenesis is not only keeping Jenesis and other people alive today, but produces almost a billion and a half dollars a year in revenue. CA : So here you go. So you took this company public, right? And made an absolute fortune. And how much have you paid Glaxo, by the way, after that 25, 000? MR : Yeah, well, every year we pay them 10 percent of 1. 5 billion, 150 million dollars, last year 100 million dollars. It ’ s the best return on investment they ever received. CA : And the best news of all, I guess, is this. MR : Yes. Jenesis is an absolutely brilliant young lady. She ’ s alive, healthy today at 30. You see me, Bina and Jenesis there. The most amazing thing about Jenesis is that while she could do anything with her life, and believe me, if you grew up your whole life with people in your face saying that you ’ ve got a fatal disease, I would probably run to Tahiti and just not want to run into anybody again. But instead she chooses to work in United Therapeutics. She says she wants to do all she can to help other people with orphan diseases get medicines, and today, she ’ s our project leader for all telepresence activities, where she helps digitally unite the entire company to work together to find cures for pulmonary hypertension. CA : But not everyone who has this disease has been so fortunate. There are still many people dying, and you are tackling that too. How? MR : Exactly, Chris. There ’ s some 3, 000 people a year in the United States alone, perhaps 10 times that number worldwide, who continue to die of this illness because the medicines slow down the progression but they don ’ t halt it. The only cure for pulmonary hypertension, pulmonary fibrosis, cystic fibrosis, emphysema, COPD, what Leonard Nimoy just died of, is a lung transplant, but sadly, there are only enough available lungs for 2, 000 people in the U. S. a year to get a lung transplant, whereas nearly a half million people a year die of end-stage lung failure. CA : So how can you address that? MR : So I conceptualize the possibility that just like we keep cars and planes and buildings going forever with an unlimited supply of building parts and machine parts, why can ’ t we create an unlimited supply of transplantable organs to keep people living indefinitely, and especially people with lung disease. So we ' ve teamed up with the decoder of the human genome, Craig Venter, and the company he founded with Peter Diamandis, the founder of the X Prize, to genetically modify the pig genome so that the pig ' s organs will not be rejected by the human body and thereby to create an unlimited supply of transplantable organs. We do this through our company, United Therapeutics. CA : So you really believe that within, what, a decade, that this shortage of transplantable lungs maybe be cured, through these guys? MR : Absolutely, Chris. I ' m as certain of that as I was of the success that we ' ve had with direct television broadcasting, Sirius XM. It ' s actually not rocket science. It ' s straightforward engineering away one gene after another. We ' re so lucky to be born in the time that sequencing genomes is a routine activity, and the brilliant folks at Synthetic Genomics are able to zero in on the pig genome, find exactly the genes that are problematic, and fix them. CA : But it ' s not just bodies that — though that is amazing. It ' s not just long-lasting bodies that are of interest to you now. It ' s long-lasting minds. And I think this graph for you says something quite profound. What does this mean? MR : What this graph means, and it comes from Ray Kurzweil, is that the rate of development in computer processing hardware, firmware and software, has been advancing along a curve such that by the 2020s, as we saw in earlier presentations today, there will be information technology that processes information and the world around us at the same rate as a human mind. CA : And so that being so, you ' re actually getting ready for this world by believing that we will soon be able to, what, actually take the contents of our brains and somehow preserve them forever? How do you describe that? MR : Well, Chris, what we ' re working on is creating a situation where people can create a mind file, and a mind file is the collection of their mannerisms, personality, recollection, feelings, beliefs, attitudes and values, everything that we ' ve poured today into Google, into Amazon, into Facebook, and all of this information stored there will be able, in the next couple decades, once software is able to recapitulate consciousness, be able to revive the consciousness which is imminent in our mind file. CA : Now you ' re not just messing around with this. You ' re serious. I mean, who is this? MR : This is a robot version of my beloved spouse, Bina. And we call her Bina 48. She was programmed by Hanson Robotics out of Texas. There ' s the centerfold from National Geographic magazine with one of her caregivers, and she roams the web and has hundreds of hours of Bina ' s mannerisms, personalities. She ' s kind of like a two-year-old kid, but she says things that blow people away, best expressed by perhaps a New York Times Pulitzer Prize-winning journalist Amy Harmon who says her answers are often frustrating, but other times as compelling as those of any flesh person she ' s interviewed. CA : And is your thinking here, part of your hope here, is that this version of Bina can in a sense live on forever, or some future upgrade to this version can live on forever? MR : Yes. Not just Bina, but everybody. You know, it costs us virtually nothing to store our mind files on Facebook, Instagram, what-have- you. Social media is I think one of the most extraordinary inventions of our time, and as apps become available that will allow us to out-Siri Siri, better and better, and develop consciousness operating systems, everybody in the world, billions of people, will be able to develop mind clones of themselves that will have their own life on the web. CA : So the thing is, Martine, that in any normal conversation, this would sound stark-staring mad, but in the context of your life, what you ' ve done, some of the things we ' ve heard this week, the constructed realities that our minds give, I mean, you wouldn ' t bet against it. MR : Well, I think it ' s really nothing coming from me. If anything, I ' m perhaps a bit of a communicator of activities that are being undertaken by the greatest companies in China, Japan, India, the

U. S., Europe. There are tens of millions of people working on writing code that expresses more and more aspects of our human consciousness, and you don ’ t have to be a genius to see that all these threads are going to come together and ultimately create human consciousness, and it ’ s something we ’ ll value. There are so many things to do in this life, and if we could have a simulacrum, a digital doppelgnger of ourselves that helps us process books, do shopping, be our best friends, I believe our mind clones, these digital versions of ourselves, will ultimately be our best friends, and for me personally and Bina personally, we love each other like crazy. Each day, we are always saying, like,”Wow, I love you even more than 30 years ago. And so for us, the prospect of mind clones and regenerated bodies is that our love affair, Chris, can go on forever. And we never get bored of each other. I ’ m sure we never will. CA : I think Bina ’ s here, right? MR : She is, yeah. CA : Would it be too much, I don ’ t know, do we have a handheld mic? Bina, could we invite you to the stage? I just have to ask you one question. Besides, we need to see you. Thank you, thank you. Come and join Martine here. I mean, look, when you got married, if someone had told you that, in a few years time, the man you were marrying would become a woman, and a few years after that, you would become a robot — — how has this gone? How has it been? Bina Rothblatt : It ’ s been really an exciting journey, and I would have never thought that at the time, but we started making goals and setting those goals and accomplishing things, and before you knew it, we just keep going up and up and we ’ re still not stopping, so it ’ s great. CA : Martine told me something really beautiful, just actually on Skype before this, which was that he wanted to live for hundreds of years as a mind file, but not if it wasn ’ t with you. BR : That ’ s right, we want to do it together. We ’ re cryonicists as well, and we want to wake up together. CA : So just so as you know, from my point of view, this isn ’ t only one of the most astonishing lives I have heard, it ’ s one of the most astonishing love stories I ’ ve ever heard. It ’ s just a delight to have you both here at TED. Thank you so much. MR : Thank you.

Text Sample 298: NEG Dim 2 – 2002 – Score 1.00 – 2002/t000426.txt

What I want to talk about is, as background, is the idea that cars are art. This is actually quite meaningful to me, because car designers tend to be a little bit low on the totem pole â we don ’ t do coffee table books with just one lamp inside of it â and cars are thought so much as a product that it ’ s a little bit difficult to get into the aesthetic side under the same sort of terminology that one would discuss art. And so cars, as art, brings it into an emotional plane â if you accept that â that you have to deal with on the same level you would with art with a capital A. Now at this point you ’ re going to see a picture of Michelangelo. This is completely different than automobiles. Automobiles are self-moving things, right? Elevators are automobiles. And they ’ re not very emotional ; they solve a purpose ; and certainly automobiles have been around for 100 years and have made our lives functionally a lot better in many ways ; they ’ ve also been a real pain in the ass, because automobiles are really the thing we have to solve. We have to solve the pollution, we have to solve the congestion â but that ’ s not what interests me in this speech. What interests me in this speech is cars. Automobiles may be what you use, but cars are what we are, in many ways. And as long as we can solve the problems of automobiles, and I believe we can, with fuel cells or hydrogen, like BMW is really hip on, and lots of other things, then I think we can look past that and try and understand why this hook is in many of us â of this car-y-ness â and what that means, what we can learn from it. That ’ s what I want to get to. Cars are not a suit of clothes ; cars are an avatar. Cars are an expansion of yourself : they take your thoughts, your ideas, your emotions, and they multiply it â your anger, whatever. It ’ s an avatar. It ’ s a super-waldo that you happen to be inside of, and if you feel sexy, the car is sexy. And if you ’ re full of road rage, you ’ ve got a”Chevy : Like a Rock,”right? Cars are a sculpture â did you know this? That every car you see out there is sculpted by hand. Many people think,”Well, it ’ s computers, and it ’ s done by machines and stuff like that.”Well, they reproduce it, but the originals are all done by hand. It ’ s done by men and women who believe a lot in their craft. And they put that same kind of tension into the sculpting of a car that you do in a great sculpture that you would go and look at in a museum. That tension between the need to express, the need to discover, then you put something new into it, and at the same time you have bounds of craftsmanship. Rules that say, this is how you handle surfaces ; this is what control is all about ; this is how you show you ’ re a master of your craft. And that tension, that discovery, that push for something new â and at the same time, that sense of obligation to the regards of craftsmanship â that ’ s as strong in cars as it is in anything. We work in clay, which hasn ’ t changed much since Michelangelo started screwing around with it, and there ’ s a very interesting analogy to that too. Real quickly â Michelangelo once said he ’ s there to”discover the figure within,”OK? There we go, the automobile. That was 100 years right there â did you catch that? Between that one there, and that one there, it changed a lot didn ’ t it? OK, it ’ s not marketing ; there ’ s a very interesting car concept here, but the marketing part is not what I want to talk about here. I want to talk about this. Why it means you have to wash a car, what is it, that sensuality you have to touch about it? That ’ s the sculpture that goes into it. That sensuality. And it ’ s done by men and women working just like this, making cars. Now this little quote about sculpture from Henry Moore, I believe that that”pressure within”that Moore ’ s talking about â at least when it comes to cars â comes right back to this idea of the mean. It ’ s that will to live, that need to survive, to express itself, that comes in a car, and takes over people like me. And we tell other people,”Do this, do this, do this,”until this thing comes alive. We are completely infected. And beauty can be the result of this infectiousness ; it ’ s quite wonderful. This sculpture is, of course, at the heart of all of it, and it ’ s really what puts the craftsmanship into our cars. And it ’ s not a whole lot different, really, when they ’ re working like this, or when somebody works like this. It ’ s that same kind of commitment, that same kind of beauty. Now, now I get to the point. I want to talk about cars as art. Art, in the Platonic sense, is truth ; it ’ s beauty, and love. Now this is really where designers in car business diverge from the engineers. We don ’ t really have a problem talking about love. We don ’ t have a problem talking about truth or beauty in that sense. That ’ s what we ’ re searching for â when we ’ re working our craft, we are really trying to find that truth out there. We ’ re not trying to find vanity and beauty. We ’ re trying to find the beauty in the truth. However, engineers tend to look at things a little bit more Newtonian, instead of this quantum approach. We ’ re dealing with irrationalisms, and we ’ re dealing with paradoxes that we admit exist, and the engineers tend to look things a little bit more like two and two is four, and if you get 4. 0 it ’ s better, and 4. 000 is even better. And that sometimes leads to bit of a divergence in why we ’ re doing what we ’ re doing. We ’ ve pretty much accepted the fact, though, that we are the women in the organization at BMW â BMW is a very manly type business, â men, men, men ; it ’ s engineers. And we ’ re kind of the female side to that. That ’ s OK, that ’ s cool. You go off and be manly. We ’ re going to be a little bit more female. Because what we ’ re interested in is finding form that ’ s more than just a function. We ’ re interested in finding beauty that ’ s more than just an aesthetic ; it ’ s really a truth. And I think this idea of soul, as being at the heart of great cars, is very applicable. You all know it. You know a car when you ’ ve seen it, with soul. You know how strong this is. Well, this experience of love, and the experience of design, to me, are interchangeable. And now I ’ m coming to my story. I discovered something about love and design through a project called Deep Blue. And first of all, you have to go with me for a second, and say, you know, you could take the word”love”out of a lot of things in our society, put the word”design”in, and it still works, like this quote here, you know. It kind of works, you know? You can understand that. It works in truisms.”All is fair in design and war.”Certainly we live in a competitive society. I think this one here, there ’ s a pop song that really describes Philippe Starck for me, you know, this is like you know, this is like puppy love, you know, this is cool right? Toothbrush, cool. It really only gets serious when you look at something like this. OK? This is one substitution that I believe all of us, in design management, are guilty of. And this idea that there is more to love, more to design, when it gets down to your neighbor, your other, it can be physical like this, and maybe in the future it will be. But right now it ’ s in dealing with our own people, our own teams who are doing the creating. So, to my story. The idea of people- work is what we work with here, and I have to make a bond with my designers when we ’ re creating BMWs. We have to have a shared intimacy, a shared vision â that means we have to work as one family ; we have to understand ourselves that way. There ’ s good times ; there ’ s interesting times ; and there ’ s some stress times too. You want to do cars, you ’ ve got to go outside. You ’ ve got to do cars in the rain ; you ’ ve got to do cars in the snow. That ’ s, by the way, is a presentation we made to our board of directors. We haul their butts out in the snow, too. You want to know cars outside? Well, you ’ ve got to stand outside to do this. And because these are artists, they have very artistic temperaments. All right? Now one thing about art is, art is discovery, and art is discovering yourself through your art. Right? And one thing about cars is we ’ re all a little bit like Pygmalion, we are completely in love with our own creations. This is one of my favorite paintings, it really describes our relationship with cars. This is sick beyond belief. But because of this, the intimacy with which we work together as a team takes on a new dimension, a new meaning. We have a shared center ; we have a shared focus â that car stays at the middle of all our relationships. And it ’ s my job, in the competitive process, to narrow this down. I heard today about Joseph ’ s death genes that have to go in and kill cell reproduction. You know, that ’ s what I have to do sometimes. We start out with 10 cars ; we narrow it down to five cars, down to three cars, down to two cars, down to one car, and I ’ m in the middle of that killing, basically. Someone ’ s love, someone ’ s baby. This is very difficult, and you have to have a bond with your team that permits you to do this, because their life is wrapped up in that too. They ’ ve got that gene infected in them as well, and they want that to live, more than anything else. Well, this project, Deep Blue, put me in contact with my team in a way that I never expected, and I want to pass it on to you, because I want you to reflect on this, perhaps in your own relationships. We wanted to a do a car which was a complete leap of faith for BMW. We wanted to do a team which was so removed from the way we ’ d done it, that I only had a phone number that connected me to them. So, what we did was : instead of having a staff of artists that are just your wrist, we decided to free up a team of creative designers and engineers to find out what ’ s the successor to the SUV phenomenon in America. This is 1996 we did this project. And so we sent them off with this team name, Deep Blue. Now many people know Deep Blue from

IBM â we actually stole it from them because we figured if anybody read our faxes they ' d think we ' re talking about computers. It turned out it was quite clever because Deep Blue, in a company like BMW, has a hook â"Deep Blue,"wow, cool name. So people get wrapped up in it. And we took a team of designers, and we sent them off to America. And we gave them a budget, what we thought was a set of deliverables, a timetable, and nothing else. Like I said, I just had a phone number that connected me to them. And a group of engineers worked in Germany, and the idea was they would work separately on this problem of what ' s the successor to the SUV. They would come together, compare notes. Then they would work apart, come together, and they would produce together a monumental set of diverse opinions that didn ' t pollute each other ' s ideas â but at the same time came together and resolved the problems. Hopefully, really understand the customer at its heart, where the customer is, live with them in America. So â sent the team off, and actually something different happened. They went other places. They disappeared, quite honestly, and all I got was postcards. Now, I got some postcards of these guys in Las Vegas, and I got some postcards of these guys in the Grand Canyon, and I got these postcards of Niagara Falls, and pretty soon they ' re in New York, and I don ' t know where else. And I ' m telling myself,"This is going to be a great car, they ' re doing research that I ' ve never even thought about before."Right? And they decided that instead of, like, having a studio, and six or seven apartments, it was cheaper to rent Elizabeth Taylor ' s ex-house in Malibu. And â at least they told me it was her house, I guess it was at one time, she had a party there or something. But anyway, this was the house, and they all lived there. Now this is 24 / 7 living, half-a-dozen people who ' d left their â some had left their wives behind and families behind, and they literally lived in this house for the entire six months the project was in America, but the first three months were the most intensive. And one of the young women in the project, she was a fantastic lady, she actually built her room in the bathroom. The bathroom was so big, she built the bed over the bathtub â it ' s quite fascinating. On the other hand, I didn ' t know anything about this. OK? Nothing. This is all going on, and all I ' m getting is postcards of these guys in Las Vegas, or whatever, saying,"Don ' t worry Chris, this is really going to be good."OK? So my concept of what a design studio was probably â I wasn ' t up to speed on where these guys were. However, the engineers back in Munich had taken on this kind of Newtonian solution, and they were trying to find how many cup holders can dance on the head of a pin, and, you know, these really serious questions that are confronting the modern consumer. And one was hoping that these two teams would get together, and this collusion of incredible creativity, under these incredible surroundings, and these incredibly stressed-out engineers, would create some incredible solutions. Well, what I didn ' t know was, and what we found out was â these guys, they can ' t even like talk to each other under those conditions. You get a divergence of Newtonian and quantum thinking at that point, you have a split in your dialog that is so deep, and so far, that they cannot bring this together at all. And so we had our first meeting, after three months, in Tiburon, which is just up the road from here â you know Tiburon? And the idea was after the first three months of this independent research they would present it all to Dr. Goschel â who is now my boss, and at that time he was co-mentor on the project â and they would present their results. We would see where we were going, we would see the first indication of what could be the successive phenomenon to the SUV in America. And so I had these ideas in my head, that this is going to be great. I mean, I ' m going to see so much work, it ' s so intense â I know probably Las Vegas meant a lot about it, and I ' m not really sure where the Grand Canyon came in either â but somehow all this is going to come together, and I ’ m going to see some really great product. So we went to Tiburon, after three months, and the team had gotten together the week before, many days ahead of time. The engineers flew over, and designers got together with them, and they put their presentation together. Well, it turns out that the engineers hadn ’ t done anything. And they hadn ’ t done anything because â kind of, like in car business, engineers are there to solve problems, and we were asking them to create a problem. And the engineers were waiting for the designers to say,”This is the problem that we ’ ve created, now help us solve it.”And they couldn ’ t talk about it. So what happened was, the engineers showed up with nothing. And the engineers told the designers,”If you go in with all your stuff, we ’ ll walk out, we ’ ll walk right out of the project.”So I didn ’ t know any of this, and we got a presentation that had an agenda, looked like this. There was a whole lot of dialog. We spent four hours being told all about vocabulary that needs to be built between engineers and designers. And here I ’ m expecting at any moment,”OK, they ’ re going to turn the page, and I ’ m going to see the cars, I ’ m going to see the sketches, I ’ m going to see maybe some idea of where it ’ s going.”Dialog kept on going, with mental maps of words, and pretty soon it was becoming obvious that instead of being dazzled with brilliance, I was seriously being baffled with bullshit. And if you can imagine what this is like, to have these months of postcard indication of how great this team is working, and they ’ re out there spending all this money, and they ’ re learning, and they ’ re doing all this stuff. I went fucking ballistic, right? I went nuts. You can probably remember Tiburon, it used to look like this. After four hours of this, I stood up, and I took this team apart. I screamed at them, I yelled at them,”What the hell are you doing? You ’ re letting me down, you ’ re my designers, you ’ re supposed to be the creative ones, what the hell is going on around here?”It was probably one of my better tirades, I have some good ones, but this was probably one of my better ones. And I went into these people ; how could they take BMW ’ s money, how could they have a holiday for three months and produce nothing, nothing? Because of course they didn ’ t tell us that they had three station wagons full of drawings, model concepts, pictures â everything I wanted, they ’ d locked up in the cars, because they had shown solidarity with the engineers â and they ’ d decided not to show me anything, in order to give the chance for problem solving a chance to start, because they hadn ’ t realized, of course, that they couldn ’ t do problem creating. So we went to lunch â And I ’ ve got to tell you, this was one seriously quiet lunch. The engineers all sat at one end of the table, the designers and I sat at the other end of the table, really quiet. And I was just fucking furious, furious. OK? Probably because they had all the fun and I didn ’ t, you know. That ’ s what you get furious about right? And somebody asked me about Catherine, my wife, you know, did she fly out with me or something? I said,”No,”and it triggered a set of thoughts about my wife. And I recalled that when Catherine and I were married, the priest gave a very nice sermon, and he said something very important. He said,”Love is not selfish,”he said,”Love does not mean counting how many times I say, ’ I love you. ’ It doesn ’ t mean you had sex this many times this month, and it ’ s two times less than last month, so that means you don ’ t love me as much. Love is not selfish.”And I thought about this, and I thought,”You know, I ’ m not showing love here. I ’ m seriously not showing love. I ’ m in the air, I ’ m in the air without trust. This cannot be. This cannot be that I ’ m expecting a certain number of sketches, and to me that ’ s my quantification method for qualifying a team. This cannot be.”So I told them this story. I said,”Guys, I ’ m thinking about something here, this isn ’ t right. I can ’ t have a relationship with you guys based on a premise that is a quantifiable one. Based on a dictate premise that says, ’ I ’ m a boss, you do what I say, without trust. ’”I said”This can ’ t be.”Actually, we all broke down into tears, to be quite honest about it, because they still could not tell me how much frustration they had built up inside of them, not being able to show me what I wanted, and merely having to ask me to trust them that it would come. And I think we felt much closer that day, we cut a lot of strings that didn ’ t need to be there, and we forged the concept for what real team and creativity is all about. We put the car back at the center of our thoughts, and we put love, I think, truly back into the center of the process. By the way, that team went on to create six different concepts for the next model of what would be the proposal for the next generation after SUVs in America. One of those was the idea of a crossover coupes â you see it downstairs, the X Coupe â they had a lot of fun with that. It was the rendition of our motorcycle, the GS, as Carl Magnusson says,”brute-iful,”as the idea of what could be a motorcycle, if you add two more wheels. And so, in conclusion, my lesson that I wanted to pass on to you, was this one here. I ’ m also going to steal a little quote out of”Little Prince.”There ’ s a lot to be said about trust and love, if you know that those two words are synonymous for design. I had a very, very meaningful relationship with my team that day, and it ’ s stayed that way ever since. And I hope that you too find that there ’ s more to design, and more towards the art of the design, than doing it yourself. It ’ s true that the trust and the love, that makes it worthwhile. Thanks so much.

Text Sample 299: NEG Dim 2 – 2002 – Score 1.00 – 2002/t000321.txt

I ’ m a historian. Steve told us about the future of little technology ; I ’ m going to show you some of the past of big technology. This was a project to build a 4, 000-ton nuclear bomb-propelled spaceship and go to Saturn and Jupiter. This took place in my childhood, 1957-65. It was deeply classified. I ’ m going to show you some stuff that not only has not been declassified, but has now been reclassified. If all goes well, next year I ’ ll be back, and I ’ ll have a lot more to show you, and if all doesn ’ t go well, I ’ ll be in jail, like Wen Ho Lee. So, this ship was basically the size of the Marriott Hotel, a little taller and a little bigger. And one of the people who worked on it at the beginning was my father, Freeman, there in the middle. That ’ s me and my sister, Esther, who ’ s a frequent TEDster. I didn ’ t like nuclear bomb-propelled spaceships. I mean, I thought it was a great idea, but I started building kayaks. So we had a few kayaks. Just so you know that I am not Dr. Strangelove. But all the time I was out there doing these strange kayak voyages in odd, beautiful parts of this planet, I always thought in the back of my mind about Project Orion, and how my father and his friends were going to build these big ships. They were actually going to go — Ted Taylor, who led the project, was going to take his children. My father was not going to take his children, that was one of the reasons we sort of had a falling out for a few years. The project began in ’ 57 at General Atomics there, that ’ s right on the coast at La Jolla. Look at that central building right in the middle of the picture. That ’ s the 130-foot diameter library. That is exactly the size of the base of the spaceship. So put that library at the bottom of that ship — that ’ s how big the thing was going to be. It would take two or three thousand bombs. The people who worked on it were a lot of the Los Alamos people who had done the hydrogen bomb work. It was the first project funded by ARPA. That ’ s the contract where ARPA gave the first million dollars to get this thing started.”Spaceship project officially begun. Job waiting for you. Dyson.”That ’ s July ’ 58. Two days later, the space traveler ’ s manifesto explaining why — just like we heard yesterday — why we need to go into space :”... trips to satellites of the outer planets. August 20, 1958.”These are the statistics of what would be the good places to go and stop. Some of the sizes of the ships, ranging all the way up to ship mass of 8 million tons. So that was the outer extreme. Here was version two : 2, 000 bombs. These are five-kiloton yield bombs, about the size of small Volkswagens ; it would take 800 to get into orbit. Here we see a 10, 000-ton ship will deliver 1, 300 tons to Saturn and back — essentially, a five-year trip. Possible departure dates : October 1960 to February 1967. These are trajectories going to Mars. All this was done by hand, with slide rules. The little Orion ship, and what it would take to do what Orion does with chemicals : you have a ship the size of the Empire State Building. NASA had no interest ; they tried to kill the project. The people who supported it were the Air Force, which meant that it was all secret. And that ’ s why when you get something declassified, that ’ s what it looks like. Military weapon versions that carried hydrogen bombs that could destroy half the planet. There ’ s another version there that sends retaliatory strikes at the Soviet Union. This is the really secret stuff : how to get directed energy explosions. So you ’ re sending the energy of a nuclear explosion — not like just a stick of dynamite, but you ’ re directing it at the ship. And this is still a very active subject. It ’ s quite dangerous, but I believe it ’ s better to have dangerous things in the open than think you ’ re going to keep them secret. This is what happened at 600 microseconds. The Air Force started to build smaller models and actually started doing this. The guys in La Jolla said,”We ’ ve got to get started now.”They built a high-explosive propelled model. These are stills from film footage that was saved by someone who was supposed to destroy it but didn ’ t, and kept it in their basement for the last 40 years. So, these are three-pound charges of C4 ; that ’ s about 10 times what the guy had in his shoes. This is Ed Day putting — So each of these coffee cans has three pounds of C4 in it. They ’ re building a system that ejects these at quarter-second intervals. That ’ s my dad in the sport coat there, holding the briefcase. So, they had a lot of fun doing this. But no children were allowed ; my dad could tell me he was building a spaceship and going to go to Saturn, but he could not say anything more about it. So all my life I have wanted to find this stuff out, and spent the last four years tracking these old guys down. These are stills from the video. Jeff Bezos kindly, yesterday, said he ’ ll put this video up on the Amazon site — some little clip of it. So, thanks to him. They got quite serious about the engineering of this. The size of that mass, for us, is really large technology in a way we ’ re never going to go back to. If you saw the 1959 — this is what it would feel like in the passenger compartment ; that ’ s acceleration profile. And pulse-system yield : we ’ re looking at 20-kiloton yield for an effective thrust of 10 million newtons. Well, here we have a little problem, the radiation doses at the crew station : 700 rads per shot. Fission yields during development : they were hoping to get clean bombs ; they didn ’ t. Eyeburn : this is what happens to the people in Miami who are looking up. Personnel compartment noise : that ’ s not too bad ; it ’ s very low frequencies, it ’ s basically like these sub-woofers. And now we have ground-hazard assessments when you have a blow-up on the pad. Finally, at the very end in 1964, NASA steps in and says,”OK, we ’ ll support a feasibility study for a small version that could be launched with Saturn Vs in sections and pieced together.”So this is what NASA did, getting an eight-man version that would go to Mars. They liked it because the guys could kind of live there and be like,”It ’ s like living in a submarine.”This is crew compartment. It switches, so what ’ s upside down is right side up when you go to artificial gravity mode. The scientists were still going to go along ; they would take seven astronauts and seven scientists. This is a 20-man version for going to Jupiter : bunks, storm cellars, exercise room. You know, it was going to be a nice, long trip. The Air Force version : here we have a military version. This is the kind of stuff that ' s not been declassified, just that people managed to sneak home and after, you know, on their deathbed, basically, gave me that. The sort of artist conceptions. These are basically PowerPoint presentations given to the Air Force 40 years ago. Look at the little guys there outside the vehicle. And one part of NASA was interested in it, but the headquarters in NASA, they killed the project. So finally, at the end, we can see the thing followed its sort of design path right up to 1965, and then all those paths came to a halt. Results : none. This project is hereby terminated. So that ' s the end. All I can say in closing is : we heard yesterday that one of the 10 bad things that could happen to us is an asteroid with our name on it. And one of the bad things that could happen to NASA is if that asteroid shows up with our name on it nine months out, and everybody says,"Well, what are we going to do?"And Orion is really one of the only, if not the only, off-the-shelf technologies that could do something. So I ' m going to tell you the good news and the bad news. The good news is that NASA has a small, secret contingency-plan division that is looking at this, trying to keep knowledge of Orion preserved in the event of such a misfortune. Maybe keep a few little bombs of plutonium on the side. That ' s the good news. The bad news is, when I got in contact with these people to try and get some documents from them, they went crazy because I had all this stuff that they don ' t have, and NASA purchased 1, 759 pages of this stuff from me. So that ' s the state we ' re at ; it ' s not very good.

Text Sample 300: NEG Dim 2 – 2002 – Score 1.00 – 2002/t000295.txt

I started juggling a long time ago, but long before that, I was a golfer, and that ' s what I was, a golfer. And as a golfer and as a kid, one of the things that really sort of seeped into my pores, that I sort of lived my whole life, is process. And it ' s the process of learning things. One of the great things was that my father was an avid golfer, but he was lefty. And he had a real passion for golf, and he also created this whole mythology about Ben Hogan and various things. Well, I learned a lot about interesting things that I knew nothing about at the time, but grew to know stuff about. And that was the mythology of skill. So, one of the things that I love to do is to explore skill. And since Richard put me on this whole thing with music — I ' m supposed to actually be doing a project with Tod Machover with the MIT Media Lab — it relates a lot to music. But Tod couldn ' t come and the project is sort of somewhere, I ' m not sure whether it ' s happening the way we thought, or not. But I ' m going to explore skill, and juggling, and basically visual music, I guess. OK, you can start the music, thanks. Thanks. Thank you. Now, juggling can be a lot of fun ; play with skill and play with space, play with rhythm. And you can turn the mike on now. I ' m going to do a couple of pieces. I do a big piece in a triangle and these are three sections from it. Part of the challenge was to try to understand rhythm and space using not just my hands — because a lot of juggling is hand-oriented — but using the rhythm of my body and feet, and controlling the balls with my feet. Thanks. Now, this next section was an attempt to explore space. You see, I think Richard said something about people that are against something. Well, a lot of people think jugglers defy gravity or do stuff. Well, I kind of, from my childhood and golf and all that, it ' s a process of joining with forces. And so what I ' d like to do is try to figure out how to join with the space through the technique. So juggling gravity — up, down. If you figure out what up and down really are, it ' s a complex physical set of skills to be able to throw a ball down and up and everything, but then you add in sideways. Now, I look at it somewhat as a way — when you learn juggling what you learn is how to feel with your eyes, and see with your hands because you ’ re not looking at your hands, you ’ re looking at where the balls are or you ’ re looking at the audience. So this next part is really a way of understanding space and rhythm, with the obvious reference to the feet, but it ’ s also time — where the feet were, where the balls were. Thanks. So, visual music : rhythm and complexity. I ’ m going to build towards complexity now. Juggling three balls is simple and normal. Excuse me. We ’ re jugglers, OK. And remember, you ’ re transposing, you ’ re getting into a subculture here. And juggling — the balls cross and all that. OK, if you keep them in their assigned paths you get parallel lines of different heights, but then hopefully even rhythm. And you can change the rhythm — good, Michael. You can change the rhythm, if you get out of the lights. OK? Change the rhythm, so it ’ s even. Or you can go back and change the height. Now, skill. But you ’ re boxed in, if you can only do it up and down that way. So, you ’ ve got to go after the space down there. OK, then you ’ ve got to combine them, because then you have the whole spatial palette in front of you. And then you get crazy. Now, I ’ m actually going to ask you to try something, so you ’ ve got to pay attention. Complexity : if you spend enough time doing something, time slows down or your skill increases, so your perceptions change. It ’ s learning skills — like being in a high-speed car crash. Things slow down as you learn, as you learn, as you learn. You may not be able to affect it, it almost drifts on you. It goes. But that ’ s the closest approximation I can have to it. So, complexity. Now, how many here are jugglers? OK, so most of you are going to have a similar reaction to this. OK? And whoever laughed there — you understood it completely, right? No, it looks like a mess. It looks like a mess with a guy there, who ’ s got his hands around that mess, OK. Well, that ’ s what juggling is about, right? It ’ s being able to do something that other people can ’ t do or can ’ t understand. All right. So, that ’ s one way of doing it, which is five balls down. OK? Another way is the outside. And you could play with the rhythm. Same pattern. Make it faster and smaller. Make it wider. Make it narrower. Bring it back up. OK. It ’ s done. Thanks. Now, what I wanted to get to is that you ’ re all very bright, very tactile. I have no idea how computer-oriented or three-dimensionally-oriented you are, but let ’ s try something. OK, so since you all don ’ t understand what the five-ball pattern is, I ’ m going to give you a little clue. Enough of a clue? So, you get the pattern, right? OK. You ’ re not getting off that easy. All right? Now, do me a favor : follow the ball that I ask you to follow. Green. Yellow. Pink. White. OK, you can do that? Yeah? OK. Now, let ’ s actually learn something. Actually, let me put you in that area of learning, which is very insecure. You want to do it? Yeah? OK. Hands out in front of you. Palms up, together. What you ’ re going to learn is this. OK? So what I want you to do is just listen to me and do it. Index finger, middle finger, ring, little. Little, ring, middle, index. And then open. Finger, finger, finger, finger, finger, finger, finger, finger. A little bit faster. Finger, finger, finger, finger, finger, finger, finger, finger. Finger, finger, finger, finger, finger, finger, finger, finger. All right. A lot of different learning processes going on in here. One learning process that I see is this — OK. Another learning process that I see is this — OK. So, everybody take a deep breath in, breath out. OK. Now, one more time, and — finger, finger, finger, finger, finger, finger, finger, finger. Open. Finger, finger, finger, finger, finger, finger, finger, finger. OK. Shake your hands out. Now, I assume a lot of you spend a lot of time at a computer. OK? So, what you ’ re doing is, you ’ re going la, la, la, and you ’ re getting this. OK? So that ’ s exactly what I ’ m going to ask you to do, but in a slightly different way. You ’ re going to combine it. So what I want you to do is — fingers. I ’ ll tell you what to do with your fingers, same thing. But I want you to do is also, with your eyes, is follow the colored ball that I ask you to follow. OK? Here we go. So, we ’ re going to start off looking at the white ball — and I ’ m going to tell you which color, and I ’ m also going to tell you to go with your fingers. OK? So white ball and — finger, finger, finger, finger, finger, finger, finger, finger. Pink. Finger, finger, finger, finger, finger, finger, finger, finger. Green. Finger, finger, finger, finger. Yellow. Finger, finger, finger, pink or finger. Pink, finger, finger, finger, finger, finger, finger, finger. All right. How did you do? Well? OK. The reason I wanted you to do this is because that ’ s actually what most people face throughout their lives, a moment of learning, a moment of challenge. It ’ s a moment that you can ’ t make sense of. Why the hell should I learn this? OK? Does it really have anything to do with anything in my life? You know, I can ’ t decipher — is it fun? Is it challenging? Am I supposed to cheat? You know, what are you supposed to do? You ’ ve got somebody up here who is the operative principle of doing that for his whole life. OK? Trying to figure that stuff out. But is it going to get you anywhere? It ’ s just a moment. That ’ s all it is, a moment. OK? I ’ m going to change the script for one second. Just let me do this. I don ’ t need music for it. Talking about time in a moment. There ’ s a piece that I recently developed which was all about that, a moment. And what I do as a creative artist is I develop vocabularies or languages of moving objects. What I ’ ve done for you here, I developed a lot of those tricks and I put the choreography together, but they ’ re not original techniques. Now, I ’ m going to start showing you some original techniques that come from the work that I ’ ve developed. OK? So, a moment, how would you define a moment? Well, as a juggler, what I wanted to do was create something that was representational of a moment. Ahhh. All right, I ’ m going to get on my knees and do it. So, a moment. OK? And then, what I did as a juggler was say, OK, what can I do to make that something that is dependent on something else, another dynamic. So, a moment. Another moment. Excuse me, still getting there. A moment that travels. A moment — no, we ’ ll try that again. It separates, and comes back together. Time. How can you look at time? And what do you dedicate it to, in exploring a particular thing? Well, obviously, there ’ s something in here, and you can all have a guess as to what it is. There ’ s a mystery. There ’ s a mystery in the moment. And it has to settle. And then it ’ s dependent on something else. And then it comes to rest. Just a little thing about time. Now, this has expanded into a much bigger piece, because I use ramps of different parabolas that I roll the balls on while I ’ m keeping time with this. But I just thought I ’ d talk about a moment. All right. OK. Can we show the video of the triangle? Are we ready to do that? Yes? This is the piece that I told you about. It ’ s a much bigger piece that I do exploring the space of a geometric triangle. Thanks. The only thing I ’ ll say about the last session is, you ever try juggling and driving the car with your knees at 120 miles an hour? The only other thing is, it was a real shock. I always drove motorcycles. And when I bought my first car, it shocked me that it cost three times more than my parents ’ house. Interesting. Anyway, balance : constant movement to find an approach to stillness. Cheating. Balance : making up the rules so you can ’ t cheat, so you learn to approach stillness with different parts of your body. To have a conversation with it. To speak. To listen. Hup. Now, it ’ s dependent on rhythm, and keeping a center of balance. When it falls, going underneath. So, there ’ s a rhythm to it. The rhythm can get much smaller. As your skill increases, you learn to find those tinier spaces, those tinier movements. Thanks. Now, I ’ m going to show you the beginnings of a piece that is about balance in some ways, and also — oh, actually, if you ’ re bored, not here, here ’ s one use for it. You can go with the”Sticks One”music. Thanks. That has a certain kind of balance to it, which is all about plumb. I apprenticed with a carpenter and learned about plumb, square and level. And they influenced that, and this next piece, which I ’ ll do a little segment of.”Two Sticks,”you can go with it. Thanks. Which is again exploring space, or the lines in space. Working with space and the lines in space in a different way. Oh, let ’ s see here. So, I ’ ll come back to that in a second. But working with one ball, now, what if you attach something to it, or change it. This is a little thing that I made because I really like the idea of curves and balls together. And then creating space and the rhythm of space, using the surface of the balls, the surface of the arms. Just a little toy. Which leads me to the next thing, which is — what have I got here? OK. All right. I ’ m actually leading up to something, the newest thing that I ’ m working on. This is not it. This is exploring geometry and the rhythm of shape. Now, what I just did was I worked with the mathematics — the diameter and the circumference. Sometimes these pieces are mathematical, in that way that I look at a shape and say, what about if I use this and this and this. Sometimes what happens in life affects my choice of objects that I try to work with. The next piece that I ’ m going to do — which is the cylinders piece, if you want to get that up — it has to do with cylinder seals from about 5, 000 years ago, which were stones with designs that were used to roll over wet clay with all sorts of great designs. I love ceramics and all of that. It ’ s a combination of that, the beauty of that, the shape, and the stories that were involved in it, as well as the fact that they protected the contents. The second influence on this piece came from recycling and looking into a tin can recycling bin and seeing all that beautiful emptiness. So, if you want to go with the music for cylinders. Talking about geometry and everything, if you take the circle and you split it in half — can you run”S-Curve music?”I ’ m going to do just a short version of it. Circles split in half and rotated, and mythology. Anyway, that piece also has a kinetic sculpture in the middle of it, and I dance around a small stage so — two minutes, just to end? The latest piece that I ’ m working on — what I love is that I never know what I ’ m working on, why I ’ m working on it. They ’ re not ideas, they ’ re instincts. And the latest thing that I ’ m working on — — is something really — I don ’ t know what it is yet. And that ’ s good. I like not to know for as long as possible. Well, because then it tells me the truth, instead of me imposing the truth. And what it is, is working with both positive and negative space but also with these curves. And what it involves, and I don ’ t know if my hands are too beaten up to do it or not, but I ’ ll do a little bit of it. It initially started off with me stacking these things, bunches of them, and then playing with the sense of space, of filling in the space. And then it started changing, and become folding on themselves. And then changing levels. Because my attempt is to make visual instruments, not to just make — I ’ ll try one other thing. For work in three dimensions, with your perceptions of space and time. Now, I don ’ t know exactly where it ’ s going, but I ’ ve got a bit of effort involved in this thing. And it ’ s going to change as I go through it. But I really like it, it feels right. This may not be the right shape, and — look at this shape, and then I ’ ll show you the first design I ever put to it, just to see, just to play, because I love all different kinds of things to play with. Let ’ s see here. To work with the positive and negative in a different way. And to change, and to change. So, I ’ m off in my new direction with this to explore rhythm and space. We ’ ll see what I come up with. Thanks for having me.

Text Sample 301: NEG Dim 2 – 2018 – Score 1.00 – 2018/t000832.txt

The hoodie is an amazing object. It ’ s one of those timeless objects that we hardly think of, because they work so well that they ’ re part of our lives. We call them”humble masterpieces.”[Small thing.] [Big idea.] [Paola Antonelli on the Hoodie] The hoodie has been â even if it was not called so â it ’ s been an icon throughout history for good and for bad reasons. The earliest ones that we can trace are from ancient Greece and ancient Rome. The Middle Ages, you see a lot of monks that were wearing garments that were cape-like, with hoods attached, so therefore,”hoodies.”Ladies in the 17th century would wear hoodies to kind of hide themselves when they were going to meet their lovers. And then, of course, there ’ s the legend, there ’ s fantasy. There ’ s the image of the hoodie connected to the grim reaper. There ’ s the image of the hoodie connected to the executioner. So there ’ s the dark side of the hoodie. The modern incarnation of the hoodie â a garment that ’ s made usually of cotton jersey, that has a hood attached with a drawstring ; sometimes it has a marsupial pocket â was introduced in the 1930s by Knickerbocker Knitting Company. Now it ’ s called Champion. It was meant to keep athletes warm. Of course, though, it was such a functional, comfortable garment that it was very rapidly adopted by workmen everywhere. And then, around the 1980s, it also gets adopted by hip-hop and B-boys, skateboarders, and it takes on this kind of youth street culture. It was, at the same time, super-comfortable, perfect for the streets and also had that added value of anonymity when you needed it. And then we have Mark Zuckerberg, who defies convention of respectable attire for businesspeople. But interestingly, it ’ s also a way to show how power has changed. If you ’ re wearing a two-piece suit, you might be the bodyguard. The real powerful person is wearing a hoodie with a T-shirt and jeans. It ’ s easy to think of the physical aspects of the hoodie. You can immediately think of wearing the hood up, and you feel this warmth and this protection, but at the same time, you can also feel the psychological aspects of it. I mean, think of donning a hoodie, all of a sudden, you feel more protected, you feel that you are in your own shell. We know very well what the hoodie has come to signify in the past few years in the United States. When Trayvon Martin, a 17-year-old African-American kid, was shot by a neighborhood vigilante, and Million Hoodie Marches happened all over the United States, in which people wore hoodies with the hood up and marched in the streets against this kind of prejudice. It doesn ’ t happen that often for a garment to have so much symbolism and history and that encompasses so many different universes as the hoodie. So, like all garments, especially all truly utilitarian garments, it is very basic in its design. But at the same time, it has a whole universe of possibilities attached.

Text Sample 302: NEG Dim 2 – 2018 – Score 1.00 – 2018/t000827.txt

The history of civilization, in some ways, is a history of maps : How have we come to understand the world around us? One of the most famous maps works because it really isn ’ t a map at all. [Small thing. Big idea.] [Michael Bierut on the London Tube Map] The London Underground came together in 1908, when eight different independent railways merged to create a single system. They needed a map to represent that system so people would know where to ride. The map they made is complicated. You can see rivers, bodies of water, trees and parks — the stations were all crammed together at the center of the map, and out in the periphery, there were some that couldn ’ t even fit on the map. So the map was geographically accurate, but maybe not so useful. Enter Harry Beck. Harry Beck was a 29-year-old engineering draftsman who had been working on and off for the London Underground. And he had a key insight, and that was that people riding underground in trains don ’ t really care what ’ s happening aboveground. They just want to get from station to station —”Where do I get on? Where do I get off?”It ’ s the system that ’ s important, not the geography. He ’ s taken this complicated mess of spaghetti, and he ’ s simplified it. The lines only go in three directions : they ’ re horizontal, they ’ re vertical, or they ’ re 45 degrees. Likewise, he spaced the stations equally, he ’ s made every station color correspond to the color of the line, and he ’ s fixed it all so that it ’ s not really a map anymore. What it is is a diagram, just like circuitry, except the circuitry here isn ’ t wires conducting electrons, it ’ s tubes containing trains conducting people from place to place. In 1933, the Underground decided, at last, to give Harry Beck ’ s map a try. The Underground did a test run of a thousand of these maps, pocket-size. They were gone in one hour. They realized they were onto something, they printed 750, 000 more, and this is the map that you see today. Beck ’ s design really became the template for the way we think of metro maps today. Tokyo, Paris, Berlin, So Paulo, Sydney, Washington, D. C. — all of them convert complex geography into crisp geometry. All of them use different colors to distinguish between lines, all of them use simple symbols to distinguish between types of stations. They all are part of a universal language, seemingly. I bet Harry Beck wouldn ’ t have known what a user interface was, but that ’ s really what he designed and he really took that challenge and broke it down to three principles that I think can be applied in nearly any design problem. First one is focus. Focus on who you ’ re doing this for. The second principle is simplicity. What ’ s the shortest way to deliver that need? Finally, the last thing is : Thinking in a cross-disciplinary way. Who would ’ ve thought that an electrical engineer would be the person to hold the key to unlock what was then one of the most complicated systems in the world — all started by one guy with a pencil and an idea.

Text Sample 303: NEG Dim 2 – 2018 – Score 1.00 – 2018/t000824.txt

If you do it right, it should sound like : TICK-tat, TICK-tat, TICK-tat, TICK- tat, TICK-tat, TICK-tat. If you do it wrong, it sounds like : Tick-TAT, tick- TAT, tick-TAT. [Small thing. Big idea.] [Kyra Gaunt on the Jump Rope] The jump rope is such a simple object. It can be made out of rope, a clothesline, twine. It has, like, a twirl on it. I ’ m not sure how to describe that. What ’ s important is that it has a certain weight, and that they have that kind of whip sound. It ’ s not clear what the origin of the jump rope is. There ’ s some evidence that it began in ancient Egypt, Phoenicia, and then it most likely traveled to North America with Dutch settlers. The rope became a big thing when women ’ s clothes became more fitted and the pantaloon came into being. And so, girls were able to jump rope because their skirts wouldn ’ t catch the ropes. Governesses used it to train their wards to jump rope. Even formerly enslaved African children in the antebellum South jumped rope, too. In the 1950s, in Harlem, Bronx, Brooklyn, Queens, you could see on the sidewalk, lots of girls playing with ropes. Sometimes they would take two ropes and turn them as a single rope together, but you could separate them and turn them in like an eggbeater on each other. The skipping rope was like a steady timeline — tick, tick, tick, tick — upon which you can add rhymes and rhythms and chants. Those ropes created a space where we were able to contribute to something that was far greater than the neighborhood. Double Dutch jump rope remains a powerful symbol of culture and identity for black women. Back from the 1950s to the 1970s, girls weren ’ t supposed to play sports. Boys played baseball, basketball and football, and girls weren ’ t allowed. A lot has changed, but in that era, girls would rule the playground. They ’ d make sure that boys weren ’ t a part of that. It ’ s their space, it ’ s a girl-power space. It ’ s where they get to shine. But I also think it ’ s for boys, because boys overheard those, which is why, I think, so many hip-hop artists sampled from things that they heard in black girls ’ game songs.... cold, thick shake, act like you know how to flip, Filet-O-Fish, Quarter Pounder, french fries, ice cold, thick shake, act like you know how to jump. Why”Country Grammar”by Nelly became a Grammy Award-winning single was because people already knew”We ’ re going down down baby your street in a Range Rover...”That ’ s the beginning of”Down down, baby, down down the roller coaster, sweet, sweet baby, I ’ ll never let you go.”All people who grew up in any black urban community would know that music. And so, it was a ready-made hit. The Double Dutch rope playing helped maintain these songs and helped maintain the chants and the gestures that go along with it, which is very natural to what I call”kinetic orality”— word of mouth and word of body. It ’ s the thing that gets passed down over generations. In some ways, the rope is the thing that helps carry it. You need some object to carry memory through. So, a jump rope, you can use it for all different kinds of things. It crosses cultures. And I think it lasted because people need to move. And I think sometimes the simplest objects can make the most creative uses.

Text Sample 304: NEG Dim 2 – 2023 – Score 1.00 – 2023/t003233.txt

When I was a little girl, my parents would take me outside and show me all the incredible ways that they would take care of our land to produce good food. And I would begrudgingly follow them out and listen to the stories that they had to say. And their pieces of advice would range from totally rational and practical to absolutely bizarre. For example, my grandfather would say,”Hey Louise, if you want to plant good root crops this season, what you should do is plant some rocks underneath your sweet potatoes.”And I would look at him and be like,”OK, Grandpa, sure. I totally believe you.”And my grandmother would say,”OK, to have the best harvest of fruits from the fruit-bearing trees this season, you want to be able to plant according to lunar cycles. You want to plan towards the full moon and never towards a new moon.”And I would look at her and say,”What?”And my dad, most bizarrely of all, would say,”If you want to sift rice or cocoa nibs to get rid of all the dust, the best thing that you can do is to whistle a certain tone to harness the wind.”And I ’ d be like,”Dad, like air-bending?””OK, sure.”So as I grew up, I would ask them,”Why? Why do we do all these weird, strange things,”right? And my relatives and my family would come up to me and be like,”Louise, here ’ s the thing. Your grandparents are kind of crazy. So this is just traditions. You don ’ t have to think about it. It ’ s fine.”But my work has put me at the frontlines of the climate crisis, working with communities and farmers to build resilient agroforests that really react best to the intense super typhoons that we experience. I established an initiative called the Cacao Project, which works to build these resilient agroforests and work closely with farmers to understand how we could best steward our ecosystems and landscapes. And over the years, I ’ ve been able to really do my best dream job, which is make chocolates for restoration. And I have the best job, I know. I get to eat chocolates, talk to farmers, live on the land and have such a good life. And we look at the ways that we can marry practical, traditional knowledge techniques with modern science and know-how, so that we could really put a spotlight on those simple, practical solutions that react effectively to climate change. Now over the years I ’ ve trained with farmers, and we make sure that learning is a two-way street where we listen to the stories that they have to say, but also be able to teach them regeneration. So very simple concepts, like putting more carbon back in the soil than we take from it, or maybe planting the crops that are suited to our ecosystems and our landscapes. And even propagating the life that strengthens our forests and our trees. And as I was talking to these farmers, these crazy stories started resurfacing, and I said,"OK, hang on, hang on. Maybe they ' re on to something here."So together with our farmers, we started kind of trying it out. OK, let ' s plant some rocks here and see what happens. OK, let ' s plant according to the lunar cycles. And for some reason, every single time that we would do that, it would work. When we plant rocks under sweet potatoes, they were better, sweeter, just more delicious. Every time we planted according to lunar cycles, we ' d have delicious harvests. And I thought, maybe, what if all of these weird stories are just kind of decades of peer review that has passed down from grandmother to grandson, from father to daughter, in the ways that they best knew how? And maybe Grandma wasn ' t so crazy after all. So I quickly learned that lunar cycles were actually tied to insect flight activity and reproduction that made better pollinators, so more fruits. It was tied to irrigation and water patterns. And I thought, wow, that is so cool. So my grandmother had a point. I digress. It turns out planting rocks under root crops, it meant that you were just actually making better drainage, but also it was creating this inviting ecosystem for worms and little creatures to live under. And they were just natural fertilizers. So, awesome, Granddad was right. And whistling for wind, well... I wish I could give you a scientific explanation. I have no idea how that works, but every time I ask my dad,"Can you bring me out to a field and whistle?"a light breeze would always seem to blow. And it was magic. I was like, what is this sorcery? So what if all of these invisible pieces of knowledge are actually keys to how we can best curate our stewardship to our landscapes, how we could best create resilience in our ecosystems and forests to react better to climate change. And all of this knowledge exists in countries and communities, and traditions and stories within our families. And as a young person who works in the environmental field, I think it is so cool to have that kind of responsibility to carry this knowledge on to the next generation, to transfer this information over into our modern age and be able to articulate why they work. Because maybe the solutions to our climate crisis, maybe the next big fix-all, isn ' t just this one big, amazing, sparkling solution. Maybe it exists in the soils under our feet. Maybe it ' s in the wind that blows in the air or the sunlight that beats down on us. Or maybe it exists in the crazy, wild stories of our grandmothers. And it is such an honor to think that maybe these amazing solutions are actually an opportunity for us to build something that embodies the wisdom of our communities, of our families and of our landscapes over years and generations. And as a young person carrying that on, I think, wow, we have some exciting magic in our planet to offer, and hopefully we can harness that power and all this sorcery and secret bits of knowledge to do something really great with it in different parts of the world that curate our stewardship to our planet. With that, thank you.

Text Sample 305: NEG Dim 2 – 2023 – Score 1.00 – 2023/t003226.txt

Let me take you back to a day in my life, July 2021. I was sitting with my laptop on my lap, slightly sweating, awaiting a call that could change my life. You see, a year ago I read a book about a woman who paddled around Australia, beating the men before her to become the fastest ever to do so. I ' m sure most people would read that book and think, wow, and forget about the idea, but I cannot get it out of my mind. I ' m a surf lifesaver and an ex-professional Ironwoman, and I want to beat that record. I want to become the fastest person to paddle around Australia. But to do so, I ’ ll need to paddle 80 to 100 kilometers a day and cover every meter of coastline in less than ten and a half months. A ski is basically a kayak, but built to withstand wind and swells and everything thrown up by the ocean. It ’ s six meters long, just 45 centimeters across at its widest point, and made of carbon fiber, it weighs eight kilograms. Oh, I ’ ll also need a support catamaran and a jet ski and a skipper and crew crazy enough to come with me for the journey. It ’ s going to cost a really significant amount of money too. And my husband and I have sold our cars, but our sacrifice is just a drop in the ocean of funds needed. This call today is my opportunity to land a sponsorship deal which will enable me to start the paddle. And 15 minutes after the start of that call, I received the life-changing”yes”I ’ ve been seeking. Which gives me six months to plan the journey. To figure out what direction I ’ m going to paddle in – clockwise – how to stay safe, and how to film the journey. But despite the challenges that lay ahead, it finally feels real. In December, I will set out to cover every meter of rugged, unforgiving coastline of Australia. Flash forward, February 2022. I am 500 kilometers out to sea and facing the crossing of the Great Australian Bight, that untamable stretch of water at the bottom of Australia. If successful, I will be the first person to cross directly across the Bight, as opposed to those prior to me who would hug the coastline. But I figure that to break a world record, I need to be prepared to go where no one has before. I had a little scare earlier today, though. I fell out of my ski and I couldn ’ t get back in. I was 100 kilometers into paddling. For the fifth time, I tried to haul myself up into my ski. As I did so, I slid into the icy water, the shock of it sending a chill up my spine like I touched an electric fence. Out here it ’ s freezing, and hypothermia is a real risk. I ’ ve lost eight kilograms in the last two weeks from seasickness. My support boat ’ s having trouble slowing down. They ’ re sailing away from me in the 25-knot wind gusts. They ’ re not turning around. One more time, I try to haul myself up into my ski and fail again, in waters known for killer whales and great white sharks. I start to sink into a sense of self-pity and feel terribly scared. But as I do so, I look up at the night sky, and I feel like a dot in the grand scheme of creation. I also have a realization. Out here, staying still is the worst thing I can do. If I can ’ t get into my ski, I need to keep moving forward. I need to swim. So slowly but surely, I move my arms as my legs start to follow, and I inch my way towards the support boat. They ’ re finally coming back to me now. I forgot to mention, it ’ s nighttime. They ’ re 50 meters away, and I can hear them calling out, seeing if I ’ m OK. By the time I ’ m pulled aboard, I ’ ve been submerged in the Bight for ten minutes. I ’ m so cold, I can ’ t speak. The bulging discs in my back are on fire, and it feels like my back is breaking in two. My lips are cracked and bleeding from dehydration. But the worst thing is, I need to get back in these waters and paddle for 16 hours again tomorrow as I ’ m just halfway across the Bight. I ’ ve not really been able to hold any food down in the last two days, due to seasickness, and the only bit of protein I ’ ve got in my stomach is about five almonds from this morning so I ’ m just running off soft-textured carbohydrate. Had some rice and Nutella and banana just then, so I ’ m really hoping it stays down. I pretty much have nothing in my stomach at the moment. Yeah, it ’ s really, really tough. So I ’ m trying to focus mentally when your focus starts to go with with no fuel in the tank. I made it across the Bight. And on the April 23, 2022, it ’ s my birthday, I ’ m 32 years old, and I ’ m two months ahead of the world record. I saw 25 turtles on my paddle yesterday. But such experiences have become the norm. I ’ ve been surrounded by dolphins in the middle of the ocean, had seals play beside me, seen flying fish scoot 50 meters across the water in front of me. I ’ ve seen Indigenous carvings over tens of thousands of years old. I ’ ve also had sharks show a little too much interest in my ski for my liking. At the moment, I ’ ve got a fishing boat supporting me, and since the start of the trip I ’ ve had a mix of fishing boat, catamaran and jet ski. I ’ ve also got a crew of amazing young men who champion me every day. They ’ re excellent listeners, they ’ re feminists, but I ’ m missing having a female around when I have my period. I ’ m not quite ready to trouble the guys about menstrual cramps just yet. I ’ m also addicted to lolly snakes, which are known as gummy worms in other parts of the world, I found out recently. It ’ s not my fault I ’ m addicted to them though. The skipper of my crew, who looks like Crocodile Dundee in his Akubra hat, fed them to me on day one and now I ’ m obsessed, I ’ m eating around a packet a day. I do eat other things though. Cut up wraps, fresh apple, Jatz crackers, they ’ re among my favorites. As a dietician, which is my day job, I had the perfect plan for nutrition, and that all went out the window on day one when the craving started. I ’ m in Broome in Western Australia. This is where they say the crocodiles start and continue right across the top of the country. May 2022. Yesterday, my crew saw a crocodile ten meters from my ski, watching, waiting. They were quick to call me into the boat and pull me aboard. But this record means I need to paddle every meter of coastline. That means today, I had to get back in the water where they saw that crocodile yesterday. It doesn ’ t help that the water is brown and the hundreds of logs feel like crocodile heads waiting to pounce. Crocodiles are predators. They ’ re patient, they ’ ll stalk as long as needed. There ’ s also hundreds of sea snakes out here that nobody told me about. Yellow with black markings. They ’ re thick as pythons, and they coil up at me as I pass. At night, I try not to hit them with my paddle. So in waters known for crocodiles and sea snakes, as I climb down into my ski, I tell my crew to talk to me. Childhood stories, jokes, riddles. Just keep talking to distract me from what lies beneath. When I was an Ironwoman, I thought that vulnerability was weakness. Be stoic, I thought. Game face always at the ready. But out here, in these waters, no game face will help. Telling my crew I ’ m scared enables me to get through the day, and we talk and joke and laugh our way through the most dangerous waters in the world. On August 28, 2022, I set foot on the shores of the Gold Coast for the first time in eight months as a world record holder. I paddled around Australia in 254 days, beating the previous record by over two and a half months. But while crossing the tape was very exciting, it wasn ’ t the most valuable thing I took from this record. You see, remember that time I told you about in the middle of the Bight when I looked up at the night sky? It was the most beautiful thing I ’ d ever seen. And in that moment, I had another realization. Instead of feeling scared, I felt lucky. Instead of that sense of self-pity, I felt grateful. Grateful and lucky to experience a perspective seen by so few humans ever before. And that gratitude carried me 12, 700 kilometers around the country. Mother Nature taught me resilience. To keep pushing as a storm is only ever fleeting. She taught me to step back and look around, as beauty can be found in the darkest of places. And though the sand and salt has been washed from my body and someone someday will break my world record, I know they will, the memories I have from this paddle will stay with me forever. And I know that all I ever need to do is take a step back and look around and remember, how lucky am I? How lucky are we to be part of this amazing, crazy, unpredictable journey of life? Even if we are just a drop in the ocean. Thank you.

Text Sample 306: NEG Dim 2 – 2023 – Score 1.00 – 2023/t003263.txt
So most of us here are migrants. We have moved for opportunities and a bet-

ter life for ourselves and our children. There are three quarters of a billion workers globally who migrate within their countries. Some go to the nearest town for a short duration, and some travel thousands of miles for months, even years, leaving all they know, including their families, behind. These migrants work at a construction site, manufacturing units, agriculture fields. But unlike us, they are not migrating for opportunities, but to survive. The majority of them are poor, working for very low wages and lack safety nets. Many face hunger, debt and exploitation. Without safety nets, one job loss, one health emergency, undermine years of effort and keep them trapped in the cycles of poverty. I come from India. India is the fastest-growing economy in the world and home to 200 million internal migrants. These migrants directly contribute at least 10 percent in our three-trillion GDP, but they lack safety nets. What ' s remarkable is that we have a huge safety-net system in India which provides various benefits like cash transfer, food subsidy, health care, insurance. In fact, there are 7, 000 benefits available with billions of dollars allocated for the poor, including migrants. And evidence shows that these benefits help in a huge way. And globally, safety nets [help] reduce poverty by 40 percent. But migrants [can ' t] access them and access is not very easy. Most do not know about the benefit. If they do, requirements are too complex and too many for the migrants who are daily-wage earners. So the money and programs exist, but the delivery system is just not working. And this is what our team and partners at Migrant Resilience Collaborative are trying to fix. Our goal is simple. We build resilience and ensure dignity of the migrant community. For this, we have adopted a two-pronged strategy. Our team, over 1, 000 people, themselves come from the migrant community, helps millions across 100 districts to overcome the challenges in the delivery and access of their benefits. And at the same time, we establish a feedback loop from the community to the government and industry. So we can improve the underlying system, policy design and accountability to fix the system long-term for tens of millions more. In just two and a half years, what we started as an audacious idea, is today the planet ' s largest grassroots-led collaborative for the migrants and has helped over three million households for the social security benefits. Benefits for the people like Suresh. We met Suresh when he was looking for daily job work in Mumbai. Let ' s hear from him. Suresh Waidande : My wife and I work very hard to ensure that our children do not have to do hard labor, that they get a good education. My name is Suresh Bhimrao Waidande. I ' m from the Khatav Block of Satara district. I got tired of being poor and quit school to work for daily wages. I decided that I had to work hard to survive and make sure that I could feed my family. So I came to Mumbai. When I arrived I took the Koyna Express from Satara to Mumbai. It was my first time taking a train. I saw mountains, valleys, canyons and tunnels. The atmosphere was so different. For my first job, I was paid something like 30-40 rupees a day. As a daily laborer, there are many problems. and some are really difficult to overcome. If workers take on jobs without proper identification or attendance cards [to track their hours], they can potentially lose payment. Sometimes the employer doesn ' t pay for your travel like they should, or they promise you work and then give it to someone else. There is a lot of injustice. Such information is shared by Jan Sahas. But not everything about our work is formal, so they ask us to be careful and aware. They provide us information about our rights. And assure us of support if we are treated unfairly or face any injustice. Whether it ' s Mumbai or any other city in India, the country cannot survive without workers like us. The great poet Annabhau Sathe used to say, this land is not resting on Earth but on callused hands of workers. The only way that any town, village or country can develop is through the hard labor of workers. AS : With support of our team, Suresh has access to food security, insurance and a hope to continue education for his children. And most importantly, most importantly, knowledge on how to access his rights and entitlements. Let me leave you with this. With climate change, the migration is set to grow at an unprecedented scale in our countries. We will scale our program across south and southeast Asia to share our work with the world. There are hundreds of millions of people who work very hard to build modern society from the ground up. The buildings where we live or work in, the airport, the big hotels, but yet none of these [are] accessible for them. This is the moment where we should recognize their contribution. And ensure they will no longer be invisible or ignored. They belong. We do this work to honor their dignity. Thank you.

Text Sample 307: NEG Dim 2 – 2017 – Score 1.00 – 2017/t002704.txt
My subject is the future of life on Earth. This is a very large and complex topic. It ’ s the subject of futurology. And in fact, I am a palaeontologist, I study dinosaurs, and you might wonder why a palaeontologist would have something to say about the future. However, there are some shared aspects which I will talk about, and this can provide some evidence to help us in making decisions. I—m a palaeontologist, as I say. I decided on this career when I was seven years old. I loved dinosaurs at the age of seven, and I think these days, young people still do. And I haven ’ t changed my view since. I love the job, teaching students, going to exotic parts of the world to dig up dinosaurs, finding new facts about the history of life. However, this kind of subject, studying the past, like studying the future, could be said by some not to be truly scientific. And I want to explore that a little bit first, how do we do this, before we get to the point of beginning to establish some facts and figures about life and current threats. And this has been criticised by other scientists. For example, 100 years ago, Sir Ernest Rutherford, a very famous physicist, Nobel Prize winner, said, ’ All of science is physics, and the rest is stamp collecting. ’ By which he meant that if you cannot make it into mathematics or physics, it doesn ’ t count. That excludes then the natural sciences, medicine and a lot of other areas. And I think at the far end of the spectrum would be my subject, palaeontology. I ’ d like to give you an example though, to show you how we apply scientific methods to achieve a certain level of understanding and certainty. And this is something that has been developing during my career so that when I started, a lot of what we did would be called a speculation or guesswork, whereas now, a lot of what we do can be tested and can be called scientific. The example I ’ m going to take is the question I asked when I was seven, ’ Could T. rex bite a car in half? ’ This is a question about the most famous dinosaur, Tyrannosaurus Rex, which was huge, five tons in weight, enormous jaws and teeth. Could it bite a car in half? Well, there were no cars in the Cretaceous, but let ’ s forget that for the moment. When I started, you could only answer that question by guesswork, or you could make some very simple models of the skull, like levers, and try to calculate things. If you were going to make a realistic model that had all the properties of the original bone and flesh of the dinosaur, you could not do it. But now, with the power of computing, we can do this kind of thing. So the way you calculate the bite force of T. rex, and the way people have done it, is they scan a skull and make a three-dimensional, digital model inside the computer. You then divide that model into a mesh, or into a framework of elements or small components. And each of these elements can be given physical properties so that we know the physical properties of bone of living animals. These are material properties, like how far can you twist the bone

before it breaks, how much compression can that bone take. And so all of those properties are mapped into the skull. And then you apply imaginary forces and increase those pressures until the thing breaks. And so the bite force of T. rex is huge. Let me lead you to the figure. Our bite force is about 800 newtons at most. The biggest bite force of any living animal is the great white shark, and that is about 5, 000 newtons. T. rex, 50, 000 newtons, ten times. And that ' s equivalent to five tons of weight, acting. So it could bite a car in half. Why do we believe this method? Because this method is a standard in design and architecture. Finite element analysis is used for designing buildings like this one. You don ' t simply build it and hope it doesn ' t fall down. The thing is designed in a computer, it is stress-tested using this program we apply to the dinosaur. I think we can believe it. But let ' s move on. That was just a brief word about how you can actually apply science, I would argue, to questions that are not happening at our time, in the past, or maybe in the future. And in palaeontology, we don ' t just find fossils. We also care about evolution and diversification, how groups have become diverse, how groups go extinct. And at the moment, we are crucially concerned about extinction. So let ' s have a look at that, let ' s explore that. What are the current views? And these views are expressed not by scientists, but also by very important politicians. So some of you may recall, a number of years ago, Al Gore, who was then vice president of the United States - He quoted a figure of 100 species are going extinct every day. So this is a very high rate of extinction, 100 per day, which is equivalent to 40, 000 per year. At the other end, other politicians, perhaps in President Trump ' s camp, would say, ' Don ' t worry about it. Extinction happens. Look at the dinosaurs. We ' ve got nothing to do. ' So how do you connect between these two positions because of course we have to think about these points. The first approach that we use is to look at history. We have recent history recorded of extinction. So many of you will be familiar with the extinct bird called the dodo. This lived on the island of Mauritius in the Indian Ocean. The dodo was a kind of fat, flightless bird related to pigeons, and a very famous image of the dodo was in the famous children ' s book ' Alice in Wonderland ', where the dodo, who is a wise, old gentleman with a walking stick organises Alice and all the animals and says to them, ' We will have a Caucus race. ' And a Caucus race is one where people start when they like and they finish when they like, and everybody gets a prize. So that was very nice, but unfortunately the dodo in real life didn ' t get a prize. Because some of you may know the first records of dodos were in 1598 when seamen reported they had seen this bird. They could go on the island and catch it very easily and eat it. And within 60 years it was extinct. Another example of recorded extinction is the great auk. This was a large bird that lived in the North Atlantic, and the great auk went extinct in 1844. How do we know that? Because collectors sent by a museum shot the last example. They were concerned that their museum did not have a stuffed specimen of this bird. They heard it was going extinct, so they thought, ' We ' d better get one before it dies out. ' So these are two examples of data about extinction. And indeed through the last 500 years, mankind has killed many species which are recorded like that. And this is the basis of the figure that Al Gore gave. Let me explain it. He gave that figure of 100 species per day going extinct. This was based on the bird data. But we know something like 100 or two or three hundred species of birds have been killed by human activity in the last 500 years. And the figures are debated. It ' s something like from half a species to four species each year on average. We don ' t know for sure because of course people didn ' t record everything. Now, how do we scale this up to get a global figure, from birds on the one hand to all of life? And the way it was done was simply to say, ' We know how many species or birds there are alive today. ’ There are 10, 000, and the initial estimate for the diversity of all of life were 100 million. So you scale from four a day upwards, and that brings you up to this figure of 100 per day by multiplying up many times ; 40, 000 lost per year, 100 per day, based on that extreme high bird figure. At the other end, the lowest estimate is something like one species per day. This is because we are not sure about that bird figure, so we could take a lower estimate, and also we are actually not sure about the diversity of life today either. Some of you will know that you might think we would have named every species, we would know that - we don ’ t. You can look at Wikipedia. There are thousands and thousands of pages describing all the living species, but there are so many living species that we have never discovered, never described, aren ’ t there in Wikipedia. So there is that uncertainty. One hundred million? Ten million? We don ’ t know. Those are the estimates of numbers of species going extinct each day. One per day, one hundred per day. Do we worry about that? This is where palaeontology comes in first, because fossil data allows us to calculate what would be the normal rate of extinction. Species do go extinct, so this politician over here was quite right. But was Al Gore right? We don ’ t know. You know, he was giving that high figure - maybe somewhere in between. But what is the norm? Is one a day okay? It doesn ’ t sound too many. If there are millions and millions of species, it is far too high. We know that the average duration of a species is two million years. They originate, they go extinct. Two million years. And that means, as they come and go, on average, we would expect the extinction of five million species every million years, which is five species per year. So that does contrast remarkably even at the lowest estimate of human impact - 500 species per year is 100 times the normal rate of 5 per year. And at the bottom there is the dodo, just in case you weren ’ t sure what it looked like. So I want to look at two other topics briefly. One is risk. And so, one of the questions we would have estimating the total numbers is one thing. We have learned that we are impacting life harder than we should be. Human activity is causing extinction. What about risk? It ’ s easy to say of course, the dodo was stupid, it lived on one island, it just stood around and allowed itself to be hit on the head. To an extent. We worry about elephants, we worry about pandas, we worry about many individual species. Many of those have high- risk factors. So for example, an elephant is very large. This means they need a lot of food, they need a lot of territory to walk around. And so it ’ s much harder to keep a sufficiently large population of elephants breeding and successfully in the wild. We don ’ t need to worry about rats, for example. They ’ re small and there ’ s lots of them. So body size is one risk factor. Being big is not a good thing if you want to survive as a species. What about the panda? They ’ re not so huge. They ’ re about human size. They have a specialised diet. They have evolved into a crazy position. As you know, they are eating bamboo without the real ability to consume it properly. If that food supply disappears, they ’ re stuck. So, secondly, restricted diet is a problem. The third one is shown by the dodo. If you live in only a very small geographic area rather than over the whole world, you are at risk. So there are three things to be : medium-sized or small, if you want to survive ; broad diet, willing to eat everything ; and living over the whole world. So that includes human beings and rats and cockroaches. They will survive and many other species will not survive whatever we do. So risk is something we can determine from the present day, but also we can test when we look at mass extinctions in the past. So this is where we come back to palaeontology, because I think most people here know that the end of the dinosaurs was marked by a mass extinction. Many millions or thousands of species died out rather rapidly. The cause of that extinction definitely was largely the impact of a meteorite, a huge asteroid, maybe ten kilometres across, and the model for extinction of the dinosaurs 66 million years ago was the asteroid hit the Earth - this is unpredictable, when this may happen or not - it penetrated into the crust, it vaporised, and all that rock of the asteroid plus the crust that it had penetrated turned into rocks and dust, mainly dust. This was thrown up high into the atmosphere and went all round the Earth, blacking out the sun, and so that you get two effects there, no light, no heat. So there was a cessation of photosynthesis. The plants died. There was darkness and cold. So this was not good for dinosaurs and other groups that enjoyed the warm climates, and so you get extinction. And you might think, of course, ’ What do we learn from this? ’ Yes, that ’ s all very well, but that ’ s not something we can do anything about really. And indeed, asteroids do come close to the Earth from time to time. Earlier this week, one did. Luckily it didn ’ t hit the Earth. We can ’ t really prepare. But the key point is that the other mass extinctions in the history of life were not caused by impact. They were caused by climate change. Immediately, you can see now how that matters. I will briefly characterise one of them that I have worked upon. At the end of the Permian Period, 250 million years ago, there was another major extinction. This was before the dinosaurs. And the sequence of events was that massive volcanoes errupted in Russia. They were so huge that they poured out enormous amounts of lava, of course, but more importantly, they poured out gases into the atmosphere, including carbon dioxide. And carbon dioxide is famous as a greenhouse gas, meaning it heats the atmosphere. We have primary evidence that around the equator, ocean temperatures warmed up to 40 degrees plus. So this is like a very warm shower. You might think that ’ s OK. Maybe for 10 minutes but not for day after day after day. And so life had to progressively flee out of the equatorial zones. It became crowded at the poles and there was great extinction. So we learnt from that that carbon dioxide and global warming can cause extinction whatever the cause of that carbon dioxide, whether it comes from volcanoes or human activity. So we ’ ve learnt three things that we can illustrate and use for predicting into the future, what may happen to life. We ’ ve learnt that we can calculate the rate of extinction by looking at historical extinctions and at that rate of extinction, even at the minimal figures, it ’ s at least one hundred times what it ought to be. Secondly, we ’ ve learnt something about risk, which species are most at risk of extinction, and I think that ’ s fairly straightforward. But we need the evidence to be able to argue the case. And thirdly, we ’ ve learnt that the whole history of the Earth records a rich record of climate change. We don ’ t need to do experiments, we don ’ t need to imagine what a world would be like without the polar ice caps. It has existed and we can study it. So to conclude, my aim has not been simply to spread doom and gloom, but to indicate that we have the power to change. And in order to change, we have to accept reality along the lines that we ’ re mentioning. And we have to ask the right questions about what we should do. And in order to decide what we should do, we need to use evidence. And some great evidence comes from the history of the Earth and the history of life. Thank you very much.

Text Sample 308: NEG Dim 2 – 2017 – Score 1.00 – 2017/t002693.txt

I could fit all movies ever made inside of this tube. If you can ’ t see it, that ’ s kind of the point. Before we understand how this is possible, it ’ s important to understand the value of this feat. All of our thoughts and actions these days, through photos and videos â even our fitness activities â are stored as digital data. Aside from running out of space on our phones, we rarely think about our digital footprint. But humanity has collectively generated more data in the last few years than all of preceding human history. Big data has become a big problem. Digital storage is really expensive, and none of these devices that we have really stand the test of time. There ’ s this nonprofit website called the Internet Archive. In addition to free books and movies, you can access web pages as far back as 1996. Now, this is very tempting, but I decided to go back and look at the TED website ’ s very humble beginnings. As you can see, it ’ s changed quite a bit in the last 30 years. So this led me to the first-ever TED, back in 1984, and it just so happened to be a Sony executive explaining how a compact disk works. Now, it ’ s really incredible to be able to go back in time and access this moment. It ’ s also really fascinating that after 30 years, after that first TED, we ’ re still talking about digital storage. Now, if we look back another 30 years, IBM released the first-ever hard drive back in 1956. Here it is being loaded for shipping in front of a small audience. It held the equivalent of one MP3 song and weighed over one ton. At 10, 000 dollars a megabyte, I don ’ t think anyone in this room would be interested in buying this thing, except maybe as a collector ’ s item. But it ’ s the best we could do at the time. We ’ ve come such a long way in data storage. Devices have evolved dramatically. But all media eventually wear out or become obsolete. If someone handed you a floppy drive today to back up your presentation, you ’ d probably look at them kind of strange, maybe laugh, but you ’ d have no way to use the damn thing. These devices can no longer meet our storage needs, although some of them can be repurposed. All technology eventually dies or is lost, along with our data, all of our memories. There ’ s this illusion that the storage problem has been solved, but really, we all just externalize it. We don ’ t worry about storing our emails and our photos. They ’ re just in the cloud. But behind the scenes, storage is problematic. After all, the cloud is just a lot of hard drives. Now, most digital data, we could argue, is not really critical. Surely, we could just delete it. But how can we really know what ’ s important today? We ’ ve learned so much about human history from drawings and writings in caves, from stone tablets. We ’ ve deciphered languages from the Rosetta Stone. You know, we ’ ll never really have the whole story, though. Our data is our story, even more so today. We won ’ t have our record recorded on stone tablets. But we don ’ t have to choose what is important now. There ’ s a way to store it all. It turns out that there ’ s a solution that ’ s been around for a few billion years, and it ’ s actually in this tube. DNA is nature ’ s oldest storage device. After all, it contains all the information necessary to build and maintain a human being. But what makes DNA so great? Well, let ’ s take our own genome as an example. If we were to print out all three billion A ’ s, T ’ s, C ’ s and G ’ s on a standard font, standard format, and then we were to stack all of those papers, it would be about 130 meters high, somewhere between the Statue of Liberty and the Washington Monument. Now, if we converted all those A ’ s, T ’ s, C ’ s and G ’ s to digital data, to zeroes and ones, it would total a few gigs. And that ’ s in each cell of our body. We have more than 30 trillion cells. You get the idea : DNA can store a ton of information in a minuscule space. DNA is also very durable, and it doesn ’ t even require electricity to store it. We know this because scientists have recovered DNA from ancient humans that lived hundreds of thousands of years ago. One of those is Ă▯tzi the Iceman. Turns out, he ’ s Austrian. He was found high, well-preserved, in the mountains between Italy and Austria, and it turns out that he has living genetic relatives here in Austria today. So one of you could be a cousin of Ă▯tzi. The point is that we have a better chance of recovering information from an ancient human than we do from an old phone. It ’ s also much less likely that we ’ ll lose the ability to read DNA than any single man-made device. Every single new storage format requires a new way to read it. We ’ ll always be able to read DNA. If we can no longer sequence, we have bigger problems than worrying about data storage. Storing data on DNA is not new. Nature ’ s been doing it for several billion years. In fact, every living thing is a DNA storage device. But how do we store data on DNA? This is Photo 51. It ’ s the first-ever photo of DNA, taken about 60 years ago. This is around the time that that same hard drive was released by IBM. So really, our understanding of digital storage and of DNA have coevolved. We first learned to sequence, or read DNA, and very soon after, how to write it, or synthesize it. This is much like how we learn a new language. And now we have the ability to read, write and copy DNA. We do it in the lab all the time. So anything, really anything, that can be stored as zeroes and ones can be stored in DNA. To store something digitally, like this photo, we convert it to bits, or binary digits. Each pixel in a black-and-white photo is simply a zero or a one. And we can write DNA much like an inkjet printer can print letters on a page. We just have to convert our data, all of those zeroes and ones, to A ’ s, T ’ s, C ’ s and G ’ s, and then we send this to a synthesis company. So we write it, we can store it, and when we want to recover our data, we just sequence it. Now, the fun part of all of this is deciding what files to include. We ’ re serious scientists, so we had to include a manuscript for good posterity. We also included a $50 Amazon gift card â don ’ t get too excited, it ’ s already been spent, someone decoded it â as well as an operating system, one of the first movies ever made and a Pioneer plaque. Some of you might have seen this. It has a depiction of a typical â apparently â male and female, and our approximate location in the Solar System, in case the Pioneer spacecraft ever encounters extraterrestrials. So once we decided what sort of files we want to encode, we package up the data, convert those zeroes and ones to A ’ s, T ’ s, C ’ s and G ’ s, and then we just send this file off to a synthesis company. And this is what we got back. Our files were in this tube. All we had to do was sequence it. This all sounds pretty straightforward, but the difference between a really cool, fun idea and something we can actually use is overcoming these practical challenges. Now, while DNA is more robust than any man-made device, it ’ s not perfect. It does have some weaknesses. We recover our message by sequencing the DNA, and every time data is retrieved, we lose the DNA. That ’ s just part of the sequencing process. We don ’ t want to run out of data, but luckily, there ’ s a way to copy the DNA that ’ s even cheaper and easier than synthesizing it. We actually tested a way to make 200 trillion copies of our files, and we recovered all the data without error. So sequencing also introduces errors into our DNA, into the A ’ s, T ’ s, C ’ s and G ’ s. Nature has a way to deal with this in our cells. But our data is stored in synthetic DNA in a tube, so we had to find our own way to overcome this problem. We decided to use an algorithm that was used to stream videos. When you ’ re streaming a video, you ’ re essentially trying to recover the original video, the original file. When we ’ re trying to recover our original files, we ’ re simply sequencing. But really, both of these processes are about recovering enough zeroes and ones to put our data back together. And so, because of our coding strategy, we were able to package up all of our data in a way that allowed us to make millions and trillions of copies and still always recover all of our files back. This is the movie we encoded. It ’ s one of the first movies ever made, and now the first to be copied more than 200 trillion times on DNA. Soon after our work was published, we participated in an”Ask Me Anything”on the website reddit. If you ’ re a fellow nerd, you ’ re very familiar with this website. Most questions were thoughtful. Some were comical. For example, one user wanted to know when we would have a literal thumb drive. Now, the thing is, our DNA already stores everything needed to make us who we are. It ’ s a lot safer to store data on DNA in synthetic DNA in a tube. Writing and reading data from DNA is obviously a lot more time-consuming than just saving all your files on a hard drive â for now. So initially, we should focus on long-term storage. Most data are ephemeral. It ’ s really hard to grasp what ’ s important today, or what will be important for future generations. But the point is, we don ’ t have to decide today. There ’ s this great program by UNESCO called the”Memory of the World”program. It ’ s been created to preserve historical materials that are considered of value to all of humanity. Items are nominated to be added to the collection, including that film that we encoded. While a wonderful way to preserve human heritage, it doesn ’ t have to be a choice. Instead of asking the current generation â us â what might be important in the future, we could store everything in DNA. Storage is not just about how many bytes but how well we can actually store the data and recover it. There ’ s always been this tension between how much data we can generate and how much we can recover and how much we can store. Every advance in writing data has required a new way to read it. We can no longer read old media. How many of you even have a disk drive in your laptop, never mind a floppy drive? This will never be the case with DNA. As long as we ’ re around, DNA is around, and we ’ ll find a way to sequence it. Archiving the world around us is part of human nature. This is the progress we ’ ve made in digital storage in 60 years, at a time when we were only beginning to understand DNA. Yet, we ’ ve made similar progress in half that time with DNA sequencers, and as long as we ’ re around, DNA will never be obsolete. Thank you.

Text Sample 309: NEG Dim 2 – 2017 – Score 1.00 – 2017/t002053.txt

I want to tell you a story about stories. And I want to tell you this story because I think we need to remember that sometimes the stories we tell each other are more than just tales or entertainment or narratives. They ’ re also vehicles for sowing inspiration and ideas across our societies and across time. The story I ’ m about to tell you is about how one of the most advanced technological achievements of the modern era has its roots in stories, and how some of the most important transformations yet to come might also. The story begins over 300 years ago, when Galileo Galilei first learned of the recent Dutch invention that took two pieces of shaped glass and put them in a long tube and thereby extended human sight farther than ever before. When Galileo turned his new telescope to the heavens and to the Moon in particular, he discovered something incredible. These are pages from Galileo ’ s book”Sidereus Nuncius,”published in 1610. And in them, he revealed to the world what he had discovered. And what he discovered was that the Moon was not just a celestial object wandering across the night sky, but rather, it was a world, a world with high, sunlit mountains and dark”mare,”the Latin word for seas. And once this new world and the Moon had been discovered, people immediately began to think about how to travel there. And just as importantly, they began to write stories about how that might happen and what those voyages might be like. One of the first people to do so was actually the Bishop of Hereford, a man named Francis Godwin. Godwin wrote a story about a Spanish explorer, Domingo Gonsales, who ended up marooned on the island of St. Helena in the middle of the Atlantic, and there, in an effort to get home, developed a machine, an invention, to harness the power of the local wild geese to allow him to fly — and eventually to embark on a voyage to the Moon. Godwin ’ s book,”The Man in the Moone, or a Discourse of a Voyage Thither,”was only published posthumously and anonymously in 1638, likely on account of the number of controversial ideas that it contained, including an endorsement of the Copernican view of the universe that put the Sun at the center of the Solar System, as well as a pre- Newtonian concept of gravity that had the idea that the weight of an object would decrease with increasing distance from Earth. And that ' s to say nothing of his idea of a goose machine that could go to the Moon. And while this idea of a voyage to the Moon by goose machine might not seem particularly insightful or technically creative to us today, what ' s important is that Godwin described getting to the Moon not by a dream or by magic, as Johannes Kepler had written about, but rather, through human invention. And it was this idea that we could build machines that could travel into the heavens, that would plant its seed in minds across the generations. The idea was next taken up by his contemporary, John Wilkins, then just a young student at Oxford, but later, one of the founders of the Royal Society. John Wilkins took the idea of space travel in Godwin ' s text seriously and wrote not just another story but a nonfiction philosophical treatise, entitled,"Discovery of the New World in the Moon, or, a Discourse Tending to Prove that ' tis Probable There May Be Another Habitable World in that Planet."And note, by the way, that word"habitable."That idea in itself would have been a powerful incentive for people thinking about how to build machines that could go there. In his books, Wilkins seriously considered a number of technical methods for spaceflight, and it remains to this day the earliest known nonfiction account of how we might travel to the Moon. Other stories would soon follow, most notably by Cyrano de Bergerac, with his"Lunar Tales."By the mid-17th century, the idea of people building machines that could travel to the heavens was growing in complexity and technical nuance. And yet, in the late 17th century, this intellectual progress effectively ceased. People still told stories about getting to the Moon, but they relied on the old ideas or, once again, on dreams or on magic. Why? Well, because the discovery of the laws of gravity by Newton and the invention of the vacuum pump by Robert Hooke and Robert Boyle meant that people now understood that a condition of vacuum existed between the planets, and consequentially between the Earth and the Moon. And they had no way of overcoming this, no way of thinking about overcoming this. And so, for well over a century, the idea of a voyage to the Moon made very little intellectual progress until the rise of the Industrial Revolution and the development of steam engines and boilers and most importantly, pressure vessels. And these gave people the tools to think about how they could build a capsule that could resist the vacuum of space. So it was in this context, in 1835, that the next great story of spaceflight was written, by Edgar Allan Poe. Now, today we think of Poe in terms of gothic poems and telltale hearts and ravens. But he considered himself a technical thinker. He grew up in Baltimore, the first American city with gas street lighting, and he was fascinated by the technological revolution that he saw going on all around him. He considered his own greatest work not to be one of his gothic tales but rather his epic prose poem"Eureka,"in which he expounded his own personal view of the cosmographical nature of the universe. In his stories, he would describe in fantastical technical detail machines and contraptions, and nowhere was he more influential in this than in his short story,"The Unparalleled Adventure of One Hans Pfaall."It ' s a story of an unemployed bellows maker in Rotterdam, who, depressed and tired of life — this is Poe, after all — and deeply in debt, he decides to build a hermetically enclosed balloon-borne carriage that is launched into the air by dynamite and from there, floats through the vacuum of space all the way to the lunar surface. And importantly, he did not develop this story alone, for in the appendix to his tale, he explicitly acknowledged Godwin ' s"A Man in the Moone"from over 200 years earlier as an influence, calling it"a singular and somewhat ingenious little book."And although this idea of a balloon-borne voyage to the Moon may seem not much more technically sophisticated than the goose machine, in fact, Poe was sufficiently detailed in the description of the construction of the device and in terms of the orbital dynamics of the voyage that it could be diagrammed in the very first spaceflight encyclopedia as a mission in the 1920s. And it was this attention to detail, or to"verisimilitude,"as he called it, that would influence the next great story : Jules Verne ' s"From the Earth to the Moon,"written in 1865. And it ' s a story that has a remarkable legacy and a remarkable similarity to the real voyages to the Moon that would take place over a hundred years later. Because in the story, the first voyage to the Moon takes place from Florida, with three people on board, in a trip that takes three days — exactly the parameters that would prevail during the Apollo program itself. And in an explicit tribute to Poe ' s influence on him, Verne situated the group responsible for this feat in the book in Baltimore, at the Baltimore Gun Club, with its members shouting,"Cheers for Edgar Poe!"as they began to lay out their plans for their conquest of the Moon. And just as Verne was influenced by Poe, so, too, would Verne ' s own story go on to influence and inspire the first generation of rocket scientists. The two great pioneers of liquid fuel rocketry in Russia and in Germany, Konstantin Tsiolkovsky and Hermann Oberth, both traced their own commitment to the field of spaceflight to their reading"From the Earth to the Moon"as teenagers, and then subsequently committing themselves to trying to make that story a reality. And Verne ' s story was not the only one in the 19th century with a long arm of influence. On the other side of the Atlantic, H. G. Wells ' s"War of the Worlds"directly inspired a young man in Massachusetts, Robert Goddard. And it was after reading"War of the Worlds"that Goddard wrote in his diary, one day in the late 1890s, of resting while trimming a cherry tree on his family ' s farm and having a vision of a spacecraft taking off from the valley below and ascending into the heavens. And he decided then and there that he would commit the rest of his life to the development of the spacecraft that he saw in his mind ' s eye. And he did exactly that. Throughout his career, he would celebrate that day as his anniversary day, his cherry tree day, and he would regularly read and reread the works of Verne and of Wells in order to renew his inspiration and his commitment over the decades of labor and effort that would be required to realize the first part of his dream : the flight of a liquid fuel rocket, which he finally achieved in 1926. So it was while reading"From the Earth to the Moon"and"The War of the Worlds"that the first pioneers of astronautics were inspired to dedicate their lives to solving the problems of spaceflight. And it was their treatises and their works in turn that inspired the first technical communities and the first projects of spaceflight, thus creating a direct chain of influence that goes from Godwin to Poe to Verne to the Apollo program and to the present-day communities of spaceflight. So why I have told you all this? Is it just because I think it ' s cool, or because I ' m just weirdly fascinated by stories of 17th- and 19th-century science fiction? It is, admittedly, partly that. But I also think that these stories remind us of the cultural processes driving spaceflight and even technological innovation more broadly. As an economist working at NASA, I spend time thinking about the economic origins of our movement out into the cosmos. And when you look before the investments of billionaire tech entrepreneurs and before the Cold War Space Race, and even before the military investments in liquid fuel rocketry, the economic origins of spaceflight are found in stories and in ideas. It was in these stories that the first concepts for spaceflight were articulated. And it was through these stories that the narrative of a future for humanity in space began to propagate from mind to mind, eventually creating an intergenerational intellectual community that would iterate on the ideas for spacecraft until such a time as they could finally be built. This process has now been going on for over 300 years, and the result is a culture of spaceflight. It ' s a culture that involves thousands of people over hundreds of years. Because for hundreds of years, some of us have looked at the stars and longed to go. And because for hundreds of years, some of us have dedicated our labors to the development of the concepts and systems required to make those voyages possible. I also wanted to tell you about Godwin, Poe and Verne because I think their stories also tell us of the importance of the stories that we tell each other about the future more generally. Because these stories don ’ t just transmit information or ideas. They can also nurture passions, passions that can lead us to dedicate our lives to the realization of important projects. Which means that these stories can and do influence social and technological forces centuries into the future. I think we need to realize this and remember it when we tell our stories. We need to work hard to write stories that don ’ t just show us the possible dystopian paths we may take for a fear that the more dystopian stories we tell each other, the more we plant seeds for possible dystopian futures. Instead we need to tell stories that plant the seeds, if not necessarily for utopias, then at least for great new projects of technological, societal and institutional transformation. And if we think of this idea that the stories we tell each other can transform the future is fanciful or impossible, then I think we need to remember the example of this, our voyage to the Moon, an idea from the 17th century that propagated culturally for over 300 years until it could finally be realized. So, we need to write new stories, stories that, 300 years in the future, people will be able to look back upon and remark how they inspired us to new heights and to new shores, how they showed us new paths and new possibilities, and how they shaped our world for the better. Thank you.

Text Sample 310: NEG Dim 2 – 2016 – Score 1.00 – 2016/t001326.txt

I ’ m an Iranian-American Muslim female, like all of you. And I ’ m also a social justice comedian, something that I insist is an actual job. To explain what that is, let me tell you how I got here. I ’ ve performed all over the country. And let me tell you, America is majestic, right? It ’ s got breathtaking nature, waffle houses and diabetes as far as the eye can see. It is really something. Now, the American population can be broken up into three main categories : there ’ s mostly wonderful people, haters and Florida. Besides Florida, the most troubling category here are the Haters. They are a minority, but they overcompensate by being extra loud. They have the Napoleon complex of demographics, and yes, some of the men do wear heels. As a social justice comedian, it ’ s my goal to convert these haters, because they hate a lot of things, which leads to negative outcomes, like racism, violence and Ted Nugent. This is not an exhaustive list ; I ’ m probably missing 3-7 items. But the point is, we have to reckon with the haters. But there ’ s variance within this group and it ’ s not efficient to go after all of them, right? So what I ’ ve done is created a highly scientific Taxonomy of Haters. I basically took all of the haters, I put them in a petri dish, like a scientist, and this is what I found. First off, we have the trolls. These are your garden-variety digital haters. They ’ re the people who quit their jobs so they can post on YouTube videos all day long. There ’ s also the drive-by haters. Now, these people will be at a stoplight, they ’ ll wait for the light to turn green and when it does, they yell,”Go back to your own country!”Now back in the day, they would ’ ve actually gotten out of their cars and hated you to your face. But they just don ’ t make them like they used to — which is another sign of the decline in America. The next category is the mission-oriented-bigot-whose-group-affiliation- gives-them-cover-for-hating hater. These guys like to hate via a seemingly nice organization, like a church or a nonprofit, and they oftentimes like to speak in an old-timey voice. But the group I ' m most interested in is the swing hater. The swing hater is sister to the swing voter — they just can ' t decide! They ' re like ideological sluts who move from hating to not hating. And they do it because they don ' t have enough information. This is the group I like to target with social justice comedy. Why comedy? Because on a scale of comedy to brochure, the average American prefers comedy, as you can see from this graph. Comedy is very popular. And by the way, this is a mathematically accurate graph, generated from fake numbers. Now, the question is : Why does social justice comedy work? Because, first off, it makes you laugh. And when you ' re laughing, you enter into a state of openness. And in that moment of openness, a good social justice comedian can stick in a whole bunch of information, and if they ' re really skilled, a rectal exam. Here are some ground rules for social justice comedy : first off, it ' s not partisan. This isn ' t political comedy, this is about justice, and no one is against justice. Two, it ' s inviting and warm, it makes you feel like you ' re sitting inside of a burrito. Three, it ' s funny but sneaky, like you could be hearing an interesting treatise on income inequality, that ' s encased in a really sophisticated poop joke. Here ' s how I see social justice comedy working. A few years ago, I rounded up a bunch of Muslim-American comedians — in a non-violent way — And we went around the country to places like Alabama, Arizona, Tennessee, Georgia — places where they love the Muzzies — and we did stand-up shows. We called the tour"The Muslims Are Coming!"We turned this into a movie, and then after the movie came out, a known hate group spent 300, 000 dollars on an anti-Muslim poster campaign with the MTA — that ' s the New York City subway system. Now, the posters were truly offensive, not to mentioned poorly designed — I mean, if you ' re going to be bigoted, you might as well use a better font. But we decided, why not launch our own poster campaign that says nice things about Muslims, while promoting the movie. So myself and fellow comedian Dean Obeidallah decided to launch the fighting-bigotry- with-delightful-posters campaign. We raised the money, worked with the MTA for over 5 months, got the posters approved, and two days after they were supposed to go up, the MTA decided to ban the posters, citing political content. Let ' s take a look at a couple of those posters. Here ' s one. Facts about Muslims : Muslims invented the concept of a hospital. OK. Fact : Grown-up Muslims can do more push-ups than baby Muslims. Fact : Muslims invented Justin Timberlake. Let ' s take a look at another one. The ugly truth about Muslims : they have great frittata recipes. Now clearly, frittatas are considered political by the MTA. Either that, or the mere mention of Muslims in a positive light was considered political — but it isn ' t. It ' s about justice. So we decided to change our fighting-bigortry-with- delightful-posters campaign and turn it into the fighting-bigotry- with-a-delightful-lawsuit campaign. So basically, what I ' m saying is a couple of dirtbag comedians took on a major New York City agency and the comedians won. Thank you. Victory was a very weird feeling. I was like,"Is this what blonde girls feel like all the time? ' Cause this is amazing!"Here ' s another example. I ' m asked everywhere I go :"Why don ' t Muslims denounce terrorism?"We do. But OK, I ' ll take the bait. So I decided to launch thedailydenouncer. com. It ' s a website that denounces terrorism every day of the week, while taking the weekends off. Let ' s take a look at an example. They generally appear as single-panel cartoons,"I denounce terrorism! I also denounce people who never fill the paper tray!"The point of the website is that it denounces terrorism while recognizing that it ' s ridiculous that we have to constantly denounce terrorism. But if bigotry isn ' t your thing, social justice comedy is useful for all sorts of issues. For example, myself and fellow comedian Lee Camp went to the Cayman Islands to investigate offshore banking. Now, the United States loses something like 300 billion dollars a year in these offshore tax havens. Not to brag, but at the end of every month, I have something like 5-15 dollars in disposable income. So we walked into these banks in the Cayman Islands and asked if we could open up a bank account with eight dollars and 27 cents. The bank managers would indulge us for 30-45 seconds before calling security. Security would come out, brandish their weapons, and then we would squeal with fear and run away, because — and this is the last rule of social justice comedy — sometimes it makes you want to take a dump in your pants. Most of my work is meant to be fun. It ' s meant to generate a connection and laughter. But yes, sometimes I get run off the grounds by security. Sometimes I get mean tweets and hate mail. Sometimes I get voice mails saying that if I continue telling my jokes, they ' ll kill me and they ' ll kill my family. And those death threats are definitely not funny. But despite the occasional danger, I still think that social justice comedy is one of our best weapons. I mean, we ' ve tried a lot of approaches to social justice, like war and competitive ice dancing. But still, a lot of things are still kind of awful. So I think it ' s time we try and tell a really good poop joke. Thank you.

Text Sample 311: NEG Dim 2 – 2016 – Score 1.00 – 2016/t001324.txt
So I know TED is about a lot of things that are big, but I want to talk to you about something very small. So small, it ' s a single word. The word is"misfit."It ' s one of my favorite words, because it ' s so literal. I mean, it ' s a person who sort of missed fitting in. Or a person who fits in badly. Or this :"a person who is poorly adapted to new situations and environments."I ' m a card- carrying misfit. And I ' m here for the other misfits in the room, because I ' m never the only one. I ' m going to tell you a misfit story. Somewhere in my early 30s, the dream of becoming a writer came right to my doorstep. Actually, it came to my mailbox in the form of a letter that said I ' d won a giant literary prize for a short story I had written. The short story was about my life as a competitive swimmer and about my crappy home life, and a little bit about how grief and loss can make you insane. The prize was a trip to New York City to meet big-time editors and agents and other authors. So kind of it was the wannabe writer ' s dream, right? You know what I did the day the letter came to my house? Because I ' m me, I put the letter on my kitchen table, I poured myself a giant glass of vodka with ice and lime, and I sat there in my underwear for an entire day, just staring at the letter. I was thinking about all the ways I ' d already screwed my life up. Who the hell was I to go to New York City and pretend to be a writer? Who was I? I ' ll tell you. I was a misfit. Like legions of other children, I came from an abusive household that I narrowly escaped with my life. I already had two epically failed marriages underneath my belt. I ' d flunked out of college not once but twice and maybe even a third time that I ' m not going to tell you about. And I ' d done an episode of rehab for drug use. And I ' d had two lovely staycations in jail. So I ' m on the right stage. But the real reason, I think, I was a misfit, is that my daughter died the day she was born, and I hadn ' t figured out how to live with that story yet. After my daughter died I also spent a long time homeless, living under an overpass in a kind of profound state of zombie grief and loss that some of us encounter along the way. Maybe all of us, if you live long enough. You know, homeless people are some of our most heroic misfits, because they start out as us. So you see, I ' d missed fitting in to just about every category out there : daughter, wife, mother, scholar. And the dream of being a writer was really kind of like a small, sad stone in my throat. It was pretty much in spite of

myself that I got on that plane and flew to New York City, where the writers are. Fellow misfits, I can almost see your heads glowing. I can pick you out of a room. At first, you would ' ve loved it. You got to choose the three famous writers you wanted to meet, and these guys went and found them for you. You got set up at the Gramercy Park Hotel, where you got to drink Scotch late in the night with cool, smart, swank people. And you got to pretend you were cool and smart and swank, too. And you got to meet a bunch of editors and authors and agents at very, very fancy lunches and dinners. Ask me how fancy. Audience : How fancy? Lidia Yuknavitch : I ' m making a confession : I stole three linen napkins — from three different restaurants. And I shoved a menu down my pants. I just wanted some keepsakes so that when I got home, I could believe it had really happened to me. You know? The three writers I wanted to meet were Carole Maso, Lynne Tillman and Peggy Phelan. These were not famous, best-selling authors, but to me, they were women-writer titans. Carole Maso wrote the book that later became my art bible. Lynne Tillman gave me permission to believe that there was a chance my stories could be part of the world. And Peggy Phelan reminded me that maybe my brains could be more important than my boobs. They weren ' t mainstream women writers, but they were cutting a path through the mainstream with their body stories, I like to think, kind of the way water cut the Grand Canyon. It nearly killed me with joy to hang out with these three over-50-year-old women writers. And the reason it nearly killed me with joy is that I ' d never known a joy like that. I ' d never been in a room like that. My mother never went to college. And my creative career to that point was a sort of small, sad, stillborn thing. So kind of in those first nights in New York I wanted to die there. I was just like,"Kill me now. I ' m good. This is beautiful."Some of you in the room will understand what happened next. First, they took me to the offices of Farrar, Straus and Giroux. Farrar, Straus and Giroux was like my mega-dream press. I mean, T. S. Eliot and Flannery O ' Connor were published there. The main editor guy sat me down and talked to me for a long time, trying to convince me I had a book in me about my life as a swimmer. You know, like a memoir. The whole time he was talking to me, I sat there smiling and nodding like a numb idiot, with my arms crossed over my chest, while nothing, nothing, nothing came out of my throat. So in the end, he patted me on the shoulder like a swim coach might. And he wished me luck and he gave me some free books and he showed me out the door. Next, they took me to the offices of W. W. Norton, where I was pretty sure I ' d be escorted from the building just for wearing Doc Martens. But that didn ' t happen. Being at the Norton offices felt like reaching up into the night sky and touching the moon while the stars stitched your name across the cosmos. I mean, that ' s how big a deal it was to me. You get it? Their lead editor, Carol Houck Smith, leaned over right in my face with these beady, bright, fierce eyes and said,"Well, send me something then, immediately!"See, now most people, especially TED people, would have run to the mailbox, right? It took me over a decade to even imagine putting something in an envelope and licking a stamp. On the last night, I gave a big reading at the National Poetry Club. And at the end of the reading, Katharine Kidde of Kidde, Hoyt & Picard Literary Agency, walked straight up to me and shook my hand and offered me representation, like, on the spot. I stood there and I kind of went deaf. Has this ever happened to you? And I almost started crying because all the people in the room were dressed so beautifully, and all that came out of my mouth was :"I don ' t know. I have to think about it."And she said,"OK, then,"and walked away. All those open hands out to me, that small, sad stone in my throat... You see, I ' m trying to tell you something about people like me. Misfit people — we don ' t always know how to hope or say yes or choose the big thing, even when it ' s right in front of us. It ’ s a shame we carry. It ’ s the shame of wanting something good. It ’ s the shame of feeling something good. It ’ s the shame of not really believing we deserve to be in the room with the people we admire. If I could, I ’ d go back and I ’ d coach myself. I ’ d be exactly like those over- 50-year-old women who helped me. I ’ d teach myself how to want things, how to stand up, how to ask for them. I ’ d say,”You! Yeah, you! You belong in the room, too.”The radiance falls on all of us, and we are nothing without each other. Instead, I flew back to Oregon, and as I watched the evergreens and rain come back into view, I just drank many tiny bottles of airplane”feel sorry for yourself.”I thought about how, if I was a writer, I was some kind of misfit writer. What I ’ m saying is, I flew back to Oregon without a book deal, without an agent, and with only a headful and heart-ful of memories of having sat so near the beautiful writers. Memory was the only prize I allowed myself. And yet, at home in the dark, back in my underwear, I could still hear their voices. They said,”Don ’ t listen to anyone who tries to get you to shut up or change your story.”They said,”Give voice to the story only you know how to tell.”They said,”Sometimes telling the story is the thing that saves your life.”Now I am, as you can see, the woman over 50. And I ’ m a writer. And I ’ m a mother. And I became a teacher. Guess who my favorite students are. Although it didn ’ t happen the day that dream letter came through my mailbox, I did write a memoir, called”The Chronology of Water.”In it are the stories of how many times I ’ ve had to reinvent a self from the ruins of my choices, the stories of how my seeming failures were really just weird-ass portals to something beautiful. All I had to do was give voice to the story. There ’ s a myth in most cultures about following your dreams. It ’ s called the hero ’ s journey. But I prefer a different myth, that ’ s slightly to the side of that or underneath it. It ’ s called the misfit ’ s myth. And it goes like this : even at the moment of your failure, right then, you are beautiful. You don ’ t know it yet, but you have the ability to reinvent yourself endlessly. That ’ s your beauty. You can be a drunk, you can be a survivor of abuse, you can be an ex-con, you can be a homeless person, you can lose all your money or your job or your husband or your wife, or the worst thing of all, a child. You can even lose your marbles. You can be standing dead center in the middle of your failure and still, I ’ m only here to tell you, you are so beautiful. Your story deserves to be heard, because you, you rare and phenomenal misfit, you new species, are the only one in the room who can tell the story the way only you would. And I ’ d be listening. Thank you.

Text Sample 312: NEG Dim 2 – 2016 – Score 1.00 – 2016/t001358.txt

So I ’ m an artist, but a little bit of a peculiar one. I don ’ t paint. I can ’ t draw. My shop teacher in high school wrote that I was a menace on my report card. You probably don ’ t really want to see my photographs. But there is one thing I know how to do : I know how to program a computer. I can code. And people will tell me that 100 years ago, folks like me didn ’ t exist, that it was impossible, that art made with data is a new thing, it ’ s a product of our age, it ’ s something that ’ s really important to think of as something that ’ s very”now.”And that ’ s true. But there is an art form that ’ s been around for a very long time that ’ s really about using information, abstract information, to make emotionally resonant pieces. And it ’ s called music. We ’ ve been making music for tens of thousands of years, right? And if you think about what music is — notes and chords and keys and harmonies and melodies — these things are algorithms. These things are systems that are designed to unfold over time, to make us feel. I came to the arts through music. I was trained as a composer, and about 15 years ago, I started making pieces that were designed to look at the intersection between sound and image, to use an image to unveil a musical structure or to use a sound to show you something interesting about something that ' s usually pictorial. So what you ' re seeing on the screen is literally being drawn by the musical structure of the musicians onstage, and there ' s no accident that it looks like a plant, because the underlying algorithmic biology of the plant is what informed the musical structure in the first place. So once you know how to do this, once you know how to code with media, you can do some pretty cool stuff. This is a project I did for the Sundance Film Festival. Really simple idea : you take every Academy Award Best Picture, you speed it up to one minute each and string them all together. And so in 75 minutes, I can show you the history of Hollywood cinema. And what it really shows you is the history of editing in Hollywood cinema. So on the left, we ' ve got Casablanca ; on the right, we ' ve got Chicago. And you can see that Casablanca is a little easier to read. That ' s because the average length of a cinematic shot in the 1940s was 26 seconds, and now it ' s around six seconds. This is a project that was inspired by some work that was funded by the US Federal Government in the early 2000s, to look at video footage and find a specific actor in any video. And so I repurposed this code to train a system on one person in our culture who would never need to be surveilled in that manner, which is Britney Spears. I downloaded 2, 000 paparazzi photos of Britney Spears and trained my computer to find her face and her face alone. I can run any footage of her through it and will center her eyes in the frame, and this sort of is a little double commentary about surveillance in our society. We are very fraught with anxiety about being watched, but then we obsess over celebrity. What you ' re seeing on the screen here is a collaboration I did with an artist named Lin Amaris. What she did is very simple to explain and describe, but very hard to do. She took 72 minutes of activity, getting ready for a night out on the town, and stretched it over three days and performed it on a traffic island in slow motion in New York City. I was there, too, with a film crew. We filmed the whole thing, and then we reversed the process, speeding it up to 72 minutes again, so it looks like she ' s moving normally and the whole world is flying by. At a certain point, I figured out that what I was doing was making portraits. When you think about portraiture, you tend to think about stuff like this. The guy on the left is named Gilbert Stuart. He ' s sort of the first real portraitist of the United States. And on the right is his portrait of George Washington from 1796. This is the so-called Lansdowne portrait. And if you look at this painting, there ' s a lot of symbolism, right? We ' ve got a rainbow out the window. We ' ve got a sword. We ' ve got a quill on the desk. All of these things are meant to evoke George Washington as the father of the nation. This is my portrait of George Washington. And this is an eye chart, only instead of letters, they ' re words. And what the words are is the 66 words in George Washington ' s State of the Union addresses that he uses more than any other president. So"gentlemen"has its own symbolism and its own rhetoric. And it ' s really kind of significant that that ' s the word he used the most. This is the eye chart for George W. Bush, who was president when I made this piece. And how you get there, from"gentlemen"to"terror"in 43 easy steps, tells us a lot about American history, and gives you a different insight than you would have looking at a series of paintings. These pieces provide a history lesson of the United States through the political rhetoric of its leaders. Ronald Reagan spent a lot of time talking about deficits. Bill Clinton spent a lot of time talking about the century in which he would no longer be president, but maybe his wife would be. Lyndon Johnson was the first President to give his State of the Union addresses on prime-time television ; he began every paragraph with the word"tonight."And Richard Nixon, or more accurately, his speechwriter, a guy named William Safire, spent a lot of time thinking about language and making sure that his boss portrayed a rhetoric of honesty. This project is shown as a series of monolithic sculptures. It ' s an outdoor series of light boxes. And it ' s important to note that they ' re to scale, so if you stand 20 feet back and you can read between those two black lines, you have 20 / 20 vision. This is a portrait. And there ' s a lot of these. There ' s a lot of ways to do this with data. I started looking for a way to think about how I can do a more democratic form of portraiture, something that ' s more about my country and how it works. Every 10 years, we make a census in the United States. We literally count people, find out who lives where, what kind of jobs we ' ve got, the language we speak at home. And this is important stuff — really important stuff. But it doesn ' t really tell us who we are. It doesn ' t tell us about our dreams and our aspirations. And so in 2010, I decided to make my own census. And I started looking for a corpus of data that had a lot of descriptions written by ordinary Americans. And it turns out that there is such a corpus of data that ' s just sitting there for the taking. It ' s called online dating. So in 2010, I joined 21 different online dating services, as a gay man, a straight man, a gay woman and a straight woman, in every zip code in America and downloaded about 19 million people ' s dating profiles — about 20 percent of the adult population of the United States. I have obsessive-compulsive disorder. This is going to become really freaking obvious. Just go with me. So what I did was I sorted all this stuff by zip code. And I looked at word analysis. These are some dating profiles from 2010 with the word"lonely"highlighted. If you look at these things topographically, if you imagine dark colors to light colors are more use of the word, you can see that Appalachia is a pretty lonely place. You can also see that Nebraska ain ' t that funny. This is the kinky map, so what this is showing you is that the women in Alaska need to get together with the men in southern New Mexico, and have a good time. And I have this at a pretty granular level, so I can tell you that the men in the eastern half of Long Island are way more interested in being spanked than men in the western half of Long Island. This will be your one takeaway from this whole conference. You ' re going to remember that fact for, like, 30 years. When you bring this down to a cartographic level, you can make maps and do the same trick I was doing with the eye charts. You can replace the name of every city in the United States with the word people use more in that city than anywhere else. If you ' ve ever dated anyone from Seattle, this makes perfect sense. You ' ve got"pretty."You ' ve got"heartbreak."You ' ve got"gig."You ' ve got"cigarette."They play in a band and they smoke. And right above that you can see"email."That ' s Redmond, Washington, which is the headquarters of the Microsoft Corporation. Some of these you can guess — so, Los Angeles is"acting"and San Francisco is"gay."Some are a little bit more heartbreaking. In Baton Rouge, they talk about being curvy ; downstream in New Orleans, they still talk about the flood. Folks in the American capital will say they ' re interesting. People in Baltimore, Maryland, will say they ' re afraid. This is New Jersey. I grew up somewhere between"annoying"and"cynical."And New York City ' s number one word is"now,"as in,"Now I ' m working as a waiter, but actually I ' m an actor."Or,"Now I ' m a professor of engineering at NYU, but actually I ' m an artist."If you go upstate, you see"dinosaur."That ' s Syracuse. The best place to eat in Syracuse, New York, is a Hell ' s Angels barbecue joint called Dinosaur Barbecue. That ' s where you would take somebody on a date. I live somewhere between"unconditional"and"midsummer,"in Midtown Manhattan. And this is gentrified North Brooklyn, so you ' ve got"DJ"and"glamorous"and"hipsters"and"urbane."So that ' s maybe a more democratic portrait. And the idea was, what if we made red-state and blue-state maps based on what we want to do on a Friday night? This is a self-portrait.

This is based on my email, about 500, 000 emails sent over 20 years. You can think of this as a quantified selfie. So what I ’ m doing is running a physics equation based on my personal data. You have to imagine everybody I ’ ve ever corresponded with. It started out in the middle and it exploded with a big bang. And everybody has gravity to one another, gravity based on how much they ’ ve been emailing, who they ’ ve been emailing with. And it also does sentimental analysis, so if I say”I love you,”you ’ re heavier to me. And you attract to my email addresses in the middle, which act like mainline stars. And all the names are handwritten. Sometimes you do this data and this work with real-time data to illuminate a specific problem in a specific city. This is a Walther PPK 9mm semiautomatic handgun that was used in a shooting in the French Quarter of New Orleans about two years ago on Valentine ’ s Day in an argument over parking. Those are my cigarettes. This is the house where the shooting took place. This project involved a little bit of engineering. I ’ ve got a bike chain rigged up as a cam shaft, with a computer driving it. That computer and the mechanism are buried in a box. The gun ’ s on top welded to a steel plate. There ’ s a wire going through to the trigger, and the computer in the box is online. It ’ s listening to the 911 feed of the New Orleans Police Department, so that anytime there ’ s a shooting reported in New Orleans, the gun fires. Now, there ’ s a blank, so there ’ s no bullet. There ’ s big light, big noise and most importantly, there ’ s a casing. There ’ s about five shootings a day in New Orleans, so over the four months this piece was installed, the case filled up with bullets. You guys know what this is — you call this”data visualization.”When you do it right, it ’ s illuminating. When you do it wrong, it ’ s anesthetizing. It reduces people to numbers. So watch out. One last piece for you. I spent the last summer as the artist in residence for Times Square. And Times Square in New York is literally the crossroads of the world. One of the things people don ’ t notice about it is it ’ s the most Instagrammed place on Earth. About every five seconds, someone commits a selfie in Times Square. That ’ s 17, 000 a day, and I have them all. These are some of them with their eyes centered. Every civilization, will use the maximum level of technology available to make art. And it ’ s the responsibility of the artist to ask questions about what that technology means and how it reflects our culture. So I leave you with this : we ’ re more than numbers. We ’ re people, and we have dreams and ideas. And reducing us to statistics is something that ’ s done at our peril. Thank you very much.

Text Sample 313: NEG Dim 2 – 2001 – Score 2.00 – 2001/t000083.txt

I thought I would read poems I have that relate to the subject of youth and age. I was sort of astonished to find out how many I have actually. The first one is dedicated to Spencer, and his grandmother, who was shocked by his work. My poem is called”Dirt.”My grandmother is washing my mouth out with soap ; half a long century gone and still she comes at me with that thick cruel yellow bar. All because of a word I said, not even said really, only repeated. But”Open,”she says,”open up!”her hand clawing at my head. I know now her life was hard ; she lost three children as babies, then her husband died too, leaving young sons, and no money. She ’ d stand me in the sink to pee because there was never room in the toilet. But oh, her soap! Might its bitter burning have been what made me a poet? The street she lived on was unpaved, her flat, two cramped rooms and a fetid kitchen where she stalked and caught me. Dare I admit that after she did it I never really loved her again? She lived to a hundred, even then. All along it was the sadness, the squalor, but I never, until now loved her again. When that was published in a magazine I got an irate letter from my uncle.”You have maligned a great woman.”It took some diplomacy. This is called”The Dress.”It ’ s a longer poem. In those days, those days which exist for me only as the most elusive memory now, when often the first sound you ’ d hear in the morning would be a storm of birdsong, then the soft clop of the hooves of the horse hauling a milk wagon down your block, and the last sound at night as likely as not would be your father pulling up in his car, having worked late again, always late, and going heavily down to the cellar, to the furnace, to shake out the ashes and damp the draft before he came upstairs to fall into bed — in those long-ago days, women, my mother, my friends ’ mothers, our neighbors, all the women I knew — wore, often much of the day, what were called house-dresses, cheap, printed, pulpy, seemingly purposefully shapeless light cotton shifts that you wore over your nightgown and, when you had to go look for a child, hang wash on the line, or run down to the grocery store on the corner, under a coat, the twisted hem of the nightgown always lank and yellowed, dangling beneath. More than the curlers some of the women seemed constantly to have in their hair in preparation for some great event — a ball, one would think — that never came to pass ; more than the way most women ’ s faces not only were never made up during the day, but seemed scraped, bleached, and, with their plucked eyebrows, scarily masklike ; more than all that it was those dresses that made women so unknowable and forbidding, adepts of enigmas to which men could have no access, and boys no conception. Only later would I see the dresses also as a proclamation : that in your dim kitchen, your laundry, your bleak concrete yard, what you revealed of yourself was a fabulation ; your real sensual nature, veiled in those sexless vestments, was utterly your dominion. In those days, one hid much else as well : grown men didn ’ t embrace one another, unless someone had died, and not always then ; you shook hands or, at a ball game, thumped your friend ’ s back and exchanged blows meant to be codes for affection ; once out of childhood you ’ d never again know the shock of your father ’ s whiskers on your cheek, not until mores at last had evolved, and you could hug another man, then hold on for a moment, then even kiss. What release finally, the embrace : though we were wary — it seemed so audacious — how much unspoken joy there was in that affirmation of equality and communion, no matter how much misunderstanding and pain had passed between you by then. We knew so little in those days, as little as now, I suppose about healing those hurts : even the women, in their best dresses, with beads and sequins sewn on the bodices, even in lipstick and mascara, their hair aflow, could only stand wringing their hands, begging for peace, while father and son, like thugs, like thieves, like Romans, simmered and hissed and hated, inflicting sorrows that endured, the worst anyway, through the kiss and embrace, bleeding from brother to brother, into the generations. In those days there was still countryside close to the city, farms, cornfields, cows ; even not far from our building with its blurred brick and long shadowy hallway you could find tracts with hills and trees you could pretend were mountains and forests. Or you could go out by yourself even to a half-block-long empty lot, into the bushes : like a creature of leaves you ’ d lurk, crouched, crawling, simplified, savage, alone ; already there was wanting to be simpler, wanting, when they called you, never to go back. This is another longish one, about the old and the young. It actually happened right at the time we met. Part of the poem takes place in space we shared and time we shared. It ’ s called”The Neighbor.”Her five horrid, deformed little dogs who incessantly yap on the roof under my window. Her cats, God knows how many, who must piss on her rugs — her landing ’ s a sickening reek. Her shadow once, fumbling the chain on her door, then the door slamming fearfully shut, only the barking and the music — jazz

— filtering as it does, day and night into the hall. The time it was Chris Connor singing"Lush Life"— how it brought back my college sweetheart, my first real love, who — till I left her — played the same record. And head on my shoulder, hand on my thigh, sang sweetly along, of regrets and depletions she was too young for, as I was too young, later, to believe in her pain. It startled, then bored, then repelled me. My starting to fancy she ' d ended up in this fire-trap in the Village, that my neighbor was her. My thinking we ' d meet, recognize one another, become friends, that I ' d accomplish a penance. My seeing her, it wasn ' t her, at the mailbox. Gray-yellow hair, army pants under a night-gown, her turning away, hiding her ravaged face in her hands, muttering an inappropriate"Hi."Sometimes there are frightening goings-on in the stairwell. A man shouting,"Shut up!"The dogs frantically snarling, claws scrabbling, then her — her voice hoarse, harsh, hollow, almost only a tone, incoherent, a note, a squawk, bone on metal, metal gone molten, calling them back,"Come back darlings, come back dear ones. My sweet angels, come back."Medea she was, next time I saw her. Sorceress, tranced, ecstatic, stock-still on the sidewalk ragged coat hanging agape, passersby flowing around her, her mouth torn suddenly open as though in a scream, silently though, as though only in her brain or breast had it erupted. A cry so pure, practiced, detached, it had no need of a voice, or could no longer bear one. These invisible links that allure, these transfigurations, even of anguish, that hold us. The girl, my old love, the last lost time I saw her when she came to find me at a party, her drunkenly stumbling, falling, sprawling, skirt hiked, eyes veined red, swollen with tears, her shame, her dishonor. My ignorant, arrogant coarseness, my secret pride, my turning away. Still life on a rooftop, dead trees in barrels, a bench broken, dogs, excrement, sky. What pathways through pain, what junctures of vulnerability, what crossings and counterings? Too many lives in our lives already, too many chances for sorrow, too many unaccounted-for pasts."Behold me,"the god of frenzied, inexhaustible love says, rising in bloody splendor,"Behold me."Her making her way down the littered vestibule stairs, one agonized step at a time. My holding the door. Her crossing the fragmented tiles, faltering at the step to the street, droning, not looking at me,"Can you help me?"Taking my arm, leaning lightly against me. Her wavering step into the world. Her whispering,"Thanks love."Lightly, lightly against me. I think I ' ll lighten up a little. Another, different kind of poem of youth and age. It ' s called"Gas."Wouldn ' t it be nice, I think, when the blue-haired lady in the doctor ' s waiting room bends over the magazine table and farts, just a little, and violently blushes. Wouldn ' t it be nice if intestinal gas came embodied in visible clouds, so she could see that her really quite inoffensive pop had only barely grazed my face before it drifted away. Besides, for this to have happened now is a nice coincidence. Because not an hour ago, while we were on our walk, my dog was startled by a backfire and jumped straight up like a horse bucking. And that brought back to me the stable I worked on weekends when I was 12, and a splendid piebald stallion, who whenever he was mounted would buck just like that, though more hugely of course, enormous, gleaming, resplendent. And the woman, her face abashedly buried in her"Elle"now, reminded me — I ' d forgotten that not the least part of my awe consisted of the fact that with every jump he took the horse would powerfully fart. Phwap! Phwap! Phwap! Something never mentioned in the dozens of books about horses and their riders I devoured in those days. All that savage grandeur, the steely glinting hooves, the eruptions driven from the creature ' s mighty innards, breath stopped, heart stopped, nostrils madly flared, I didn ' t know if I wanted to break him, or be him. This is called"Thirst."Many — most of my poems actually are urban poems. I happen to be reading a bunch that aren ' t."Thirst."Here was my relation with the woman who lived all last autumn and winter, day and night, on a bench in the 103rd Street subway station, until finally one day she vanished. We regarded each other, scrutinized one another. Me shyly, obliquely, trying not to be furtive. She boldly, unblinkingly, even pugnaciously, wrathfully even, when her bottle was empty. I was frightened of her. I felt like a child. I was afraid some repressed part of myself would go out of control, and I ' d be forever entrapped in the shocking seethe of her stench. Not excrement merely, not merely surface and orifice going unwashed, rediffusion of rum, there was will in it, and intention, power and purpose — a social, ethical rage and rebellion — despair too, though, grief, loss. Sometimes I ' d think I should take her home with me, bathe her, comfort her, dress her. She wouldn ' t have wanted me to, I would think. Instead, I ' d step into my train. How rich I would think, is the lexicon of our self-absolving. How enduring, our bland fatal assurance that reflection is righteousness being accomplished. The dance of our glances, the clash, pulling each other through our perceptual punctures, then holocaust, holocaust, host on host of ill, injured presences, squandered, consumed. Her vigil somewhere I know continues. Her occupancy, her absolute, faithful attendance. The dance of our glances, challenge, abdication, effacement, the perfume of our consternation. This is a newer poem, a brand new poem. The title is"This Happened."A student, a young woman in a fourth-floor hallway of her lycee, perched on the ledge of an open window chatting with friends between classes ; a teacher passes and chides her,"Be careful, you might fall,"almost banteringly chides her,"You might fall,"and the young woman, 18, a girl really, though she wouldn ' t think that, as brilliant as she is, first in her class, and"Beautiful, too,"she ' s often told, smiles back, and leans into the open window, which wouldn ' t even be open if it were winter — if it were winter someone would have closed it — leans into the window, farther, still smiling, farther and farther, though it takes less time than this, really an instant, and lets herself fall. Herself fall. A casual impulse, a fancy, never thought of until now, hardly thought of even now... No, more than impulse or fancy, the girl knows what she ' s doing, the girl means something, the girl means to mean, because it occurs to her in that instant, that beautiful or not, bright yes or no, she ' s not who she is, she ' s not the person she is, and the reason, she suddenly knows, is that there ' s been so much premeditation where she is, so much plotting and planning, there ' s hardly a person where she is, or if there is, it ' s not her, or not wholly her, it ' s a self inhabited, lived in by her, and seemingly even as she thinks it she knows what ' s been missing : grace, not premeditation but grace, a kind of being in the world spontaneously, with grace. Weightfully upon me was the world. Weightfully this self which graced the world yet never wholly itself. Weightfully this self which weighed upon me, the release from which is what I desire and what I achieve. And the girl remembers, in this infinite instant already now so many times divided, the sadness she felt once, hardly knowing she felt it, to merely inhabit herself. Yes, the girl falls, absurd to fall, even the earth with its compulsion to take unto itself all that falls must know that falling is absurd, yet the girl falling isn ' t myself, or she is myself, but a self I took of my own volition unto myself. Forever. With grace. This happened. I ' ll read just one more. I don ' t usually say that. I like to just end. But I ' m afraid that Ricky will come out here and shake his fist at me. This is called"Old Man,"appropriately enough."Special : big tits,"Says the advertisement for a soft-core magazine on our neighborhood newsstand. But forget her breasts. A lush, fresh-lipped blond, skin glowing gold, sprawls there, resplendent. 60 nearly, yet these hardly tangible, hardly better than harlots, can still stir me. Maybe a coming of age in the American sensual darkness, never seeing an unsmudged nipple, an uncensored vagina, has left me forever infected with an unquenchable lust of the eye. Always that erotic murmur, I ’ m hardly myself if I ’ m not in a state of incipient desire. God knows though, there are worse twists your obsessions can take. Last year in Israel, a young ultra-orthodox Rabbi guiding some teenage girls through the Shrine of the Shoah forbade them to look in one room. Because there were images in it he said were licentious. The display was a photo. Men and women stripped naked, some trying to cover their genitals, others too frightened to bother, lined up in snow waiting to be shot and thrown into a ditch. The girls, to my horror, averted their gaze. What carnal mistrust had their teacher taught them. Even that though. Another confession : Once in a book on pre-war Poland, a studio portrait, an absolute angel, an absolute angel with tormented, tormenting eyes. I kept finding myself at her page. That she died in the camps made her — I didn ’ t dare wonder why — more present, more precious. Died in the camps, that too people — or Jews anyway — kept from their children back then. But it was like sex, you didn ’ t have to be told. Sex and death, how close they can seem. So constantly conscious now of death moving towards me, sometimes I think I confound them. My wife ’ s loveliness almost consumes me. My passion for her goes beyond reasonable bounds. When we make love, her holding me everywhere all around me, I ’ m there and not there. My mind teems, jumbles of faces, voices, impressions, I live my life over, as though I were drowning. Then I am drowning, in despair at having to leave her, this, everything, all, unbearable, awful. Still, to be able to die with no special contrition, not having been slaughtered, or enslaved. And not having to know history ’ s next mad rage or regression, it might be a relief. No. Again, no. I don ’ t mean that for a moment. What I mean is the world holds me so tightly — the good and the bad — my own follies and weakness that even this counterfeit Venus with her sham heat, and her bosom probably plumped with gel, so moves me my breath catches. Vamp. Siren. Seductress. How much more she reveals in her glare of ink than she knows. How she incarnates our desperate human need for regard, our passion to live in beauty, to be beauty, to be cherished by glances, if by no more, of something like love, or love. Thank you.

Text Sample 314: NEG Dim 2 – 2001 – Score 2.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,”Are you still doing this or that?”I don ’ t answer because I ’ m not doing things still, I ’ m doing it like I always did. I still have — or did I use the word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years. But, of course, I ’ m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ’ ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called”Innovation.”Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,"The playful search for beauty was Man ' s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty."These are tiles. The word,"playful"is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ' t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ' t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 315: NEG Dim 2 – 2001 – Score 3.00 – 2001/t000367.txt

I ’ d like to do pretty much what I did the first time, which is to choose a lighthearted theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, exorcise those evil spirits, and this is still going on, as you know. But it wasn ’ t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ’ re feeling depressed? It ’ s better than Prozac, but I wouldn ’ t recommend it. So what we see in the seventeenth, eighteenth century is the continued search for medications other than camphor that ’ ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we fast-forward to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a row — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very interested in producing convulsions, measured types of convulsions. And they thought,”Well, we ’ ve got electricity, we ’ ll plug somebody into the wall. That always makes hair stand up and people shake a lot.”So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,”We know that at the Rome railroad station, there are all these lost souls wandering around, muttering gibberish. Can you bring one of them to us?”Someone who is, as the Italians say,”cagoots.”So they found this”cagoots”guy, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,”Well, we ’ ll try 55 volts, two-tenths of a second. That ’ s not going to do anything terrible to him.”So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,”This fellow”— remember, he wasn ’ t even put to sleep —”after this major grand mal convulsion, sat right up, looked at these three fellas and said, ’ What the fuck are you assholes trying to do? ’”If I could only say that in Italian. Well, they were happy as could be, because he hadn ’ t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and everybody in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ’ t work very well on the schizophrenics, but it was pretty clear in the ’ 30s and by the middle of the ’ 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ’ t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the brain waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ’ 60s, the first antidepressants came out. Tofranil was the first. In the late ’ 70s, early ’ 80s, there were others, and they were very effective. And patients ’ rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ’ m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ’ ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ’ m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ’ clock in the afternoon, because I couldn ’ t get out of bed before about 11 o ’ clock. And anybody who ’ s been depressed here knows what that ’ s like. I couldn ’ t even pull the covers off myself. Well, you ’ re in a university medical center, where everybody knows everybody, and it ’ s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ’ t work anymore. And, in fact, it didn ’ t make any difference because I didn ’ t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,”Don ’ t worry, chap. Six weeks, you ’ re back in the operating room. Everything ’ s going to be great.”Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ’ t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five feet in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ’ t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had. They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the senior staff. All the senior staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful stuff. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it got so there was a throbbing, there was a ferocious fear in my head. You ' ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ' t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case. He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the senior staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,"No. I know this man better than any of you. I have met with him over and over again. You ' ve just seen him from time to time. You ' ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ' ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go back to surgery, he will never be the loving father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ' One Flew Over the Cuckoo ' s Nest. ' And you know about that, just essentially in a stupor the rest of his life."Well, he said,"Can ' t we try a course of electroshock therapy?"And you know why they agreed? They agreed to humor him. They just thought,"Well, we ' ll give a course of 10. And so we ' ll lose a little time. Big deal. It doesn ' t make any difference."So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ' t work. Seven didn ' t work. Eight didn ' t work. At nine, I noticed — and it ' s wonderful that I could notice anything — I noticed a change. And at 10, I noticed a real change. And he went back to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, lo and behold, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could blow the obsessional thinking away. I could blow the depression away.

And I ’ ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,”I ’ ve got the strength now to do this.”It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a formula. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember”Ruddigore,”and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,”We must have a word to bring you back to reality, and the word, my dear, will be ’ Basingstoke. ’”So every time she got a little nuts, he would say,”Basingstoke!”And she would say,”Basingstoke, it is.”And she would be fine for a little while. Well, you know, I ’ m from the Bronx. I can ’ t say”Basingstoke.”But I had something better. And it was very simple. It was,”Ah, fuck it!”Much better than”Basingstoke,”at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,”Ah, fuck it.”And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went back to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came back to live with us. We had two more children after that. Resuscitated the career, even better than it had been before. Went right back into the university and began to write books. Well, you know, it ’ s been a wonderful life. It ’ s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ’ s been very exciting. It ’ s been very happy. Every once in a while, I have to say,”Ah, fuck it.”Every once in a while, I get somewhat depressed and a little obsessional. So, I ’ m not free of all of this. But it ’ s worked. It ’ s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ’ ve gotten thousands of letters about them by people who do think this — that based on my life ’ s history as I ’ ve portrayed in the books, my early life ’ s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the bitter dregs of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ’ ve always felt guilty about that. I ’ ve always felt that somehow I was an impostor because my readers don ’ t know what I have just told you. It ’ s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an untroubled mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes back to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ’ t happen, there will be adversities. If I, with the bleakness of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way back from this, believe me, anybody can find their way back from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

Text Sample 316: NEG Dim 2 – 2019 – Score 1.00 – 2019/t000016.txt
Yogi Berra, a US baseball player and philosopher, said,”If we don ’ t know where we ’ re going, we might not get there.”Accumulating scientific knowledge is giving us greater insights, greater clarity, into what our future might look like in a changing climate and what that could mean for our health. I ’ m here to talk about a related aspect, on how our emissions of greenhouse gases from burning of fossil fuels is reducing the nutritional quality of our food. We ’ ll start with the food pyramid. You all know the food pyramid. We all need to eat a balanced diet. We need to get proteins, we need to get micronutrients, we need to get vitamins. And so, this is a way for us to think about how to make sure we get what we need every day so we can grow and thrive. But we eat not just because we need to, we also eat for enjoyment. Bread, pasta, pizza — there ’ s a whole range of foods that are culturally important. We enjoy eating these. And so they ’ re important for our diet, but they ’ re also important for our cultures. Carbon dioxide has been increasing since the start of the Industrial Revolution, increasing from about 280 parts per million to over 410 today, and it continues to increase. The carbon that plants need to grow comes from this carbon dioxide. They bring it into the plant, they break it apart into the carbon itself, and they use that to grow. They also need nutrients from the soil. And so yes, carbon dioxide is plant food. And this should be good news, of rising carbon dioxide concentrations, for food security around the world, making sure that people get enough to eat every day. About 820 million people in the world don ’ t get enough to eat every day. So there ’ s a fair amount written about how higher CO2 is going to help with our food security problem. We need to accelerate our progress in agricultural productivity to feed the nine to 10 billion people who will be alive in 2050 and to achieve the Sustainable Development Goals, particularly the Goal Number 2, that is on reducing food insecurity, increasing nutrition, increasing access to the foods that we need for everyone. We know that climate change is affecting agricultural productivity. The earth has warmed about one degree centigrade since preindustrial times. That is changing local temperature and precipitation patterns, and that has consequences for the agricultural productivity in many parts of the world. And it ’ s not just local changes in temperature and precipitation, it ’ s the extremes. Extremes in terms of heat waves, floods and droughts are significantly affecting

productivity. And that carbon dioxide, besides making plants grow, has other consequences as well, that plants, when they have higher carbon dioxide, increase the synthesis of carbohydrates, sugars and starches, and they decrease the concentrations of protein and critical nutrients. And this is very important for how we think about food security going forward. A couple of nights ago in the table talks on climate change, someone said that they ’ re a five-sevenths optimist : that they ’ re an optimist five days of the week, and this is a topic for the other two days. When we think about micronutrients, almost all of them are affected by higher CO2 concentrations. Two in particular are iron and zinc. When you don ’ t have enough iron, you can develop iron deficiency anemia. It ’ s associated with fatigue, shortness of breath and some fairly serious consequences as well. When you don ’ t have enough zinc, you can have a loss of appetite. It is a significant problem around the world. There ’ s about one billion people who are zinc deficient. It ’ s very important for maternal and child health. It affects development. The B vitamins are critical for a whole range of reasons. They help convert our food into energy. They ’ re important for the functions of many of the physiologic activities in our bodies. And when you have higher carbon in a plant, you have less nitrogen, and you have less B vitamins. And it ’ s not just us. Cattle are already being affected because the quality of their forage is declining. In fact, this affects every consumer of plants. And give a thought to, for example, our pet cats and dogs. If you look on the label of most of the pet and dog food, there ’ s a significant amount of grain in those foods. So this affects everyone. How do we know that this is a problem? We know from field studies and we know from experimental studies in laboratories. In the field studies — and I ’ ll focus primarily on wheat and on rice — there ’ s fields, for example, of rice that are divided into different plots. And the plots are all the same : the soil ’ s the same, the precipitation ’ s the same — everything ’ s the same. Except carbon dioxide is blown over some of the plots. And so you can compare what it looks like under today ’ s conditions and under carbon dioxide conditions later in the century. I was part of one of the few studies that have done this. We looked at 18 rice lines in China and in Japan and grew them under conditions that you would expect later in the century. And when you look at the results, the white bar is today ’ s conditions, the red bar is conditions later in the century. So protein declines about 10 percent, iron about eight percent, zinc about five percent. These don ’ t sound like really big changes, but when you start thinking about the poor in every country who primarily eat starch, that this will put people who are on the edge over the edge into frank deficiencies, creating all kinds of health problems. The situation is more significant for the B vitamins. When you look at vitamin B1 and vitamin B2, there ’ s about a 17 percent decline. Pantothenic acid, vitamin B5, is about a 13 percent decline. Folate is about a 30 percent decline. And these are averages over the various experiments that were done. Folate is critical for child development. Pregnant women who don ’ t get enough folate are at much higher risk of having babies with birth defects. So these are very serious potential consequences for our health as CO2 continues to rise. In another example, this is modeling work that was done by Chris Weyant and his colleagues, taking a look at this chain from higher CO2 to lower iron and zinc — and they only looked at iron and zinc — to various health outcomes. They looked at malaria, diarrheal disease, pneumonia, iron deficiency anemia, and looked at what the consequences could be in 2050. And the darker the color in this, the larger the consequences. So you can see the major impacts in Asia and in Africa, but also note that in countries such as the United States and countries in Europe, the populations also could be affected. They estimated about 125 million people could be affected. They also modeled what would be the most effective interventions, and their conclusion was reducing our greenhouse gases : getting our greenhouse gas emissions down by mid-century so we don ’ t have to worry so much about these consequences later in the century. These experiments, these modeling studies did not take climate change itself into account. They just focused on the carbon dioxide component. So when you put the two together, it ’ s expected the impact is much larger than what I ’ ve told you. I ’ d love to be able to tell you right now how much the food you had for breakfast, the food you ’ re going to have for lunch, has shifted from what your grandparents ate in terms of its nutritional quality. But I can ’ t. We don ’ t have the research on that. I ’ d love to tell you how much current food insecurity is affected by these changes. But I can ’ t. We don ’ t have the research on that, either. There ’ s a lot that needs to be known in this area, including what the possible solutions could be. We don ’ t know exactly what those solutions are, but we ’ ve got a range of options. We ’ ve got advancements in technologies. We ’ ve got plant breeding. We ’ ve got biofortification. Soils could make a difference. And, of course, it will be very helpful to know how these changes could affect our future health and the health of our children and the health of our grandchildren. And these investments take time. It will take time to sort all of these issues out. There is no national entity or business group that is funding this research. We need these investments critically so that we do know where we ’ re going. In the meantime, what we can do is ensure that all people have access to a complete diet, not just those in the wealthy parts of the world but everywhere in the world. We also individually and collectively need to reduce our greenhouse gas emissions to reduce the challenges that will come later in the century. It ’ s been said that if you think education is expensive, try ignorance. Let ’ s not. Let ’ s invest in ourselves, in our children and in our planet. Thank you.

Text Sample 317: NEG Dim 2 – 2019 – Score 1.00 – 2019/t000009.txt

I am obsessed with forming healthy communities, and that ’ s why I started Twitch â to help people watch other people play video games on the internet. Thank you for coming to my TED talk. So in seriousness, video games and communities truly are quite related. From our early human history, we made our entertainment together in small tribes. We shared stories around the campfire, we sang together, we danced together. Our earliest entertainment was both shared and interactive. It wasn ’ t until pretty recently on the grand scale of human history that interactivity took a back seat and broadcast entertainment took over. Radio and records brought music into our vehicles, into our homes. TV and VHS brought sports and drama into our living rooms. This access to broadcast entertainment was unprecedented. It gave people amazing content around the globe. It created a shared culture for millions of people. And now, if you want to go watch or listen to Mozart, you don ’ t have to buy an incredibly expensive ticket and find an orchestra. And if you like to sing â I can show you the world â then you have something in common with people around the world. But with this amazing access, we allowed for a separation between creator and consumer, and the relationship between the two became much more one-way. We wound up in a world where we had a smaller class of professional creators and most of us became spectators, and as a result it became far easier for us to enjoy that content alone. There ’ s a trend counteracting this : scarcity. So, Vienna in the 1900s, was famous for its cafĂŠ culture. And one of the big drivers of that cafĂŠ culture was expensive newspapers that were hard to get, and as a result, people would go to the cafĂŠ and read the shared copy there. And once they ’ re in the cafe, they meet the other people also reading the same newspaper, they converse, they exchange ideas and they form a community. In a similar way, TV and cable used to be more expensive, and so you might not watch the game at home. Instead you ' d go to the local bar and cheer along with your fellow sports fans there. But as the price of media continues to fall over time thanks to technology, this shared necessity that used to bring our communities together falls away. We have so many amazing options for our entertainment, and yet it ' s easier than ever for us to wind up consuming those options alone. Our communities are bearing the consequences. For example, the number of people who report having at least two close friends is at an all-time low. I believe that one of the major contributing causes to this is that our entertainment today allows us to be separate. There is one trend reversing this atomization of our society : modern multiplayer video games. Games are like a shared campfire. They ' re both interactive and connecting. Now these campfires may have beautiful animations, heroic quests, occasionally too many loot boxes, but games today are very different than the solitary activity of 20 years ago. They ' re deeply complex, they ' re more intellectually stimulating, and most of all, they ' re intrinsically social. One of the recent breakout genres exemplifying this change is the battle royale. 100 people parachute onto an island in a last-man-standing competition. Think of it as being kind of like"American Idol,"but with a lot more fighting and a lot less Simon Cowell. You may have heard of"Fortnite,"which is a breakout example of the battle royale genre, which has been played by more than 250 million people around the world. It ' s everyone from kids in your neighborhood to Drake and Ellen DeGeneres. 2. 3 billion people in the world play video games. Early games like"Tetris"and"Mario"may have been simple puzzles or quests, but with the rise of arcades and then internet play, and now massively multiplayer games of huge, thriving online communities, games have emerged as the one form of entertainment where consumption truly requires human connection. So this brings us to streaming. Why do people stream themselves playing video games? And why do hundreds of millions of people around the world congregate to watch them? I want you all the imagine for second â imagine you land on an alien planet, and on this planet, there ' s a giant green rectangle. And in this green rectangle, aliens in matching outfits are trying to push a checkered sphere between two posts using only their feet. It ' s pretty evenly matched, so the ball is just going back and forth, but there ' s hundreds of millions of people watching from home anyway, and cheering and getting excited and engaged right along with them. Now I grew up watching sports with my dad, so I get why soccer is entertaining and engaging. But if you don ' t watch sports, maybe you like watching"Dancing with the Stars"or you enjoy"Top Chef."Regardless, the principle is the same. If there is an activity that you really enjoy, you ' re probably going to like watching other people do it with skill and panache. It might be perplexing to an alien, but bonding over shared passion is a human universal. So gamers grew up expecting this live, interactive entertainment, and passive consumption just doesn ' t feel as fulfilling. That ' s why livestreaming has taken off with video games. Because livestreaming offers that same kind of interactive feeling. So when you imagine what ' s happening on Twitch, I don ' t want you to think of a million livestreams of video games. Instead, what I want you to picture is millions of campfires. Some of them are bonfires â huge, roaring bonfires with hundreds of thousands of people around them. Some of them are smaller, more intimate community gatherings where everyone knows your name. Let ' s try taking a seat by one of those campfires right now. Hey Cohh, how ' s it going? Cohh : Hey, how ' s it going, Emmett? ES : So I ' m here at TED with about 1, 000 of my closest friends, and we thought we ' d come and join you guys for a little stream. Cohh

: Awesome! It ’ s great to hear from you guys. ES : So Cohh, can you share with the TED audience here â what have you learned about your community on Twitch? Cohh : Ah, man, where to begin? I ’ ve been doing this for over five years now, and if there ’ s one thing that doesn ’ t cease to impress me on the daily, it ’ s just kind of how incredible this whole thing is for communication. I ’ ve been playing games for 20 years of my life, I ’ ve led online MMO guilds for over 10, and it ’ s the kind of thing where there ’ s very few places in life where you can go to meet so many people with similar interests. I was listening in a bit earlier ; I love the campfire analogy, I actually use a similar one. I see it all as a bunch of people on a big couch but only one person has the controller. So it ’ s kind of like a”Pass the snack!”situation, you know? 700 people that way â but it ’ s great and really it ’ s just â ES : So Cohh, what is going on in chat right now? Can you explain that a little bit to us? Because my eyesight isn ’ t that good but I see a lot of emotes. Cohh : So this is my community ; this is the Cohhilition. I stream every single day. I actually just wrapped up a 2, 000-day challenge, and as such, we have developed a pretty incredible community here in the channel. Right now we have about 6200 people with us. What you ’ re seeing is a spam of”Hello, TED”good- vibe emotes, love emotes,”this is awesome,””Hi, guys,””Hi, everyone.”Basically just a huge collection of people â huge collection of gamers that are all just experiencing a positive event together. ES : So is there anything that â can we poll chat? I want ask chat a question. Is there anything that chat would like the world, and particularly these people here with me at TED right now, to know about what they get out of playing video games and being part of this community? Cohh : Oh, wow. I am already starting to see a lot of answers here.”I like the good vibes.””Best communities are on Twitch.””They get us through the rough patches in life.”Oh, that ’ s a message I definitely see a lot on Twitch, which is very good.”A very positive community,””a lot of positivity,”which is pretty great. ES : So Cohh, before I get back to my TED talk, which I actually should probably get back to doing at some point â Do you have anything else that you want to share with me or any question you wanted to ask, you ’ ve always wanted to get out there before an audience? Cohh : Honestly, not too much. I mean, I absolutely love what you ’ re doing right now. I think that the interactive streaming is the big unexplored frontier of the future in entertainment, and thank you for doing everything you ’ re doing up there. The more people that hear about what you do, the better â for everyone on here. ES : Awesome, Cohh. Thanks so much. I ’ m going to get back to giving this talk now, but we should catch up later. Cohh : Sounds great! ES : So that was a new way to interact. We could influence what happened on the stream, we could cocreate the experience along with him, and we really had a multiplayer experience with chat and with Cohh. At Twitch, we ’ ve started calling this, as a result,”multiplayer entertainment.”Because going from watching a video alone to watching a live interactive stream is similar to the difference between going from playing a single-player game to playing a multiplayer game. Gamers are often as the forefront of exploration in new technology. Microcomputers, for example, were used early on for video games, and the very first handheld, digital mass-market devices weren ’ t cell phones, they were Gameboys... for video games. And as a result, one way that you can get a hint of what the future might hold is to look to this fun, interactive sandbox of video games and ask yourself,”what are these gamers doing today?”And that might give you a hint as to what the future is going to hold for all of us. One of the things we ’ re already seeing on Twitch is multiplayer entertainment coming to sports. So, Twitch and the NFL teamed up to offer livestreaming football, but instead of network announcers in suits streaming the game, we got Twitch users to come in and stream it themselves on their own channel and interact with their community and make it a real multiplayer experience. So I actually think that if you look out into the future â only hundreds of people today get to be sports announcers. It ’ s a tiny, tiny number of people who have that opportunity. But sports are about to go multiplayer, and that means that anyone who wants to around the world is going to get the opportunity to become a sports announcer, to give it a shot. And I think that ’ s going to unlock incredible amounts of new talent for all of us. And we ’ re not going to be asking,”Did you catch the game?”Instead, we ’ re going to be asking,”Whose channel did you catch the game on?”We already see this happening with cooking, with singing â we even see people streaming welding. And all of this stuff is going to happen around the metaphorical campfire. There ’ s going to be millions of these campfires lit over the next few years. And on every topic, you ’ re going to be able to find a campfire that will allow you to bond with your people around the world. For most of human history, entertainment was simply multiplayer. We sang together in person, we shared news together in the town square in person, and somewhere along the way, that two-way conversation turned into a one-way transmission. As someone who cares about communities, I am excited for a world where our entertainment could connect us instead of isolating us. A world where we can bond with each other over our shared interests and create real, strong communities. Games, streams and the interactions they encourage, are only just beginning to turn the wheel back to our interactive, community-rich, multiplayer past. Thank you all for sharing this experience here with me, and may you all find your best campfire.

Text Sample 318: NEG Dim 2 – 2019 – Score 1.00 – 2019/t000660.txt

I ’ ve been a police officer in an urban city for nearly 25 years. That ’ s crazy, right? And in that time, I ’ ve served in every rank, from police officer to police chief. A few years ago, I noticed something alarming. Starting in 2014, I started monitoring recruits as they cycled through police academies in the state of New Jersey, and I found that women were failing at rates between 65 and 80 percent, due to varying aspects of the physical fitness test. I learned that a change in policy now required recruits to pass the fitness exam within 10 short workout sessions. This had the greatest impact on women. The change meant that recruits had about three weeks out of a five- month-long academy to pass the fitness exam. This just didn ’ t make sense, though. Police agencies and police recruits had made huge investments to get those recruits into the academy. Police recruits had passed lengthy background checks, they had passed medical and psychological exams, they had quit their jobs. And many had spent more than 2, 000 dollars in fees and equipment just to get kicked out within the first three weeks? The dire situation in New Jersey led me to examine the status of women in policing across the United States. I found that women make up less than 13 percent of police officers. A number that hasn ’ t changed much in the past 20 years. And they make up just three percent of police chiefs as of 2013, the last time the data was collected. We know that we can improve those rates. Other countries like Canada, Australia and the UK have nearly twice the amount of policewomen. And New Zealand is steadily marching towards their goal of recruit gender parity by 2021. Other countries are actively working to increase the number of women in policing, because they know of a vast body of research evidence, spanning more than 50 years, detailing the advantages to women in policing. From that research, we know that policewomen are less likely to use force or to be accused of excessive force. We know that policewomen are less likely to be named in a lawsuit or a citizen complaint. We know that the mere presence of a policewoman reduces the use of force among other officers. And we know that policewomen are met with the same rates of force as their male counterparts, and sometimes more, and yet they ’ re more successful in defusing violent or aggressive behavior overall. So there are vast advantages to women in policing, and we ’ re losing them to arbitrary fitness standards. The problem is, the United States has nearly 18, 000 police agencies — 18, 000 agencies with wildly varying fitness standards. We know that a majority of academies rely on a masculine ideal of policing that works to decrease the number of women in policing. These types of academies overemphasize physical strength, with much less attention spent to subjects like community policing, problem-solving and interpersonal communication skills. This results in training that does not mirror the realities of policing. Physical agility is but a small component of police work. Much of an officer ’ s day is spent mediating interpersonal conflicts. That ’ s the reality of policing. These are my babies. And we can reduce the disparity in policing by changing exams that produce disparate outcomes. The federal courts have stated that men and women simply are not physiologically the same for the purposes of physical fitness programs. And that ’ s based on science. Respected institutions that law enforcement deeply respects, like the FBI, the US Marshals Service, the DEA and even the US military — they rigorously test fitness programs to ensure they measure fitness without gender-disparate outcomes. Why is that? Because recruiting is expensive. They want to recruit and retain qualified candidates. You know what else the research finds? Well- trained women are as capable as their male counterparts in overall fitness, but more importantly, in how they police. The law-enforcement community is admittedly experiencing a recruitment crisis. Yet, if they truly want to increase the number of applicants, they can. We can easily recruit more women and reap all those research benefits by training well-qualified candidates to pass validated, work-related, physiologically- based fitness exams, as required by Title VII of the Civil Rights Act. We can increase the number of women, we can reduce that gender disparity, by simply changing exams that produce disparate outcomes. We have the tools. We have the research, we have the science, we have the law. This, my friends, should be a very easy fix. Thank you.

Text Sample 319: NEG Dim 2 – 2024 – Score 1.00 – 2024/t003129.txt

The most important things I know about negotiation I learned in a kayak on my honeymoon. Picture this. I arrive in Hawaii, and I look adoringly at my brand new husband as we get in a two-person kayak together for a river tour. Then things go wrong fast. You see, in a two-person kayak, the passenger in the back, usually the larger passenger, is supposed to drive using their paddles. Well, I didn ’ t like how he was driving. So I decided to drive my own way. And exactly at the moment I thought I had taken over – We flipped over. Three separate times. Apparently we set a record. So I get back in the kayak now dripping wet, when our guide up ahead turns back and says,”OK folks, let ’ s negotiate these things to the left because you ’ re going to hit that beach up ahead.”Negotiate the kayak. In that moment, I realized that although I had been teaching negotiation for a few years already, I had missed something important. And I was not alone. A lot of us have misconceptions about negotiation. We are taught that it ’ s a battle over money, that it involves losing, so we either fear it or we avoid it altogether. In fact, 54 percent of us didn ’ t negotiate our last salary. That ’ s from research, by the way. I ’ m not actually seeing into your souls. The fact is that before that guide, I had never heard anyone use the word negotiate that way before. I, too, had absorbed this message that it was a contest of wills. You know, like five minutes earlier when I dunked us in the river. But there ' s another way to think about it, isn ' t there? When I negotiate my kayak toward a beach, what am I doing? I ' m steering. Today, I want to share five lessons about negotiation that just might change your life. And the first is this : negotiation is simply steering. It is any conversation in which you are steering a relationship. And it ' s so much more than the money conversations. You know, just like a kayak, where we want to steer consistently to get to our goal. We can be steering our relationships almost every minute of the day. You know, when I got back from that honeymoon, I looked around and I saw opportunities to negotiate everywhere. I could be getting to know my bosses better so that I understood what they prioritized most. I could be calling my clients regularly and asking them what was new in their businesses. And I realized something powerful. If we intentionally focus on the everyday steering, what happens when we do get to the money conversations? We ' re more likely to be successful. Which brings me to lesson two. Curious people make more money. There ' s one negotiation technique that most of us are not using. So years ago, a professor set people up to negotiate as part of a study. Only seven percent of those people achieved the optimal outcome. What did they do? Well, they didn ' t lead with their points. They asked questions first. Bottom line, in negotiation, you get more by asking questions than you do by arguing. But not just any questions. That seven percent? They asked open questions. Questions that uncovered the other person ' s needs, concerns, and goals, And most of us aren ' t great at that. When I teach people about asking questions, I say, OK, I ' ve just taken a trip. What can you ask me to get some good information? Top two questions they ask? I bet you could guess. Number one : Where did you go? Reno. Did you have a good time? Yeah. Two closed questions. Two one-word answers. So what ' s the best question? Well, it ' s a little bit of a trick, because technically, the best question of all doesn ' t end with a question mark. It ' s :"Tell me all about your vacation.""Tell me"is the biggest question you can ask, and it is the most powerful first question in any negotiation : at work or at home. With the hiring manager :"Tell me how the company sees the salary range for this position."With your teenager :"Tell me what ' s making you ask for a 50-dollar-a-week allowance?""Tell me"gets you the most information, but it also builds trust, so it creates the best deals. Lesson three : the negotiation starts before the negotiation starts. Most people don ' t know that every negotiation actually comes in two parts. The second part we all know about, that ' s where we ' re sitting down with someone else. But that first part, that ' s what I call the mirror. Because we have to negotiate with ourselves first before we negotiate with anyone else. And it ' s the most critical part of the negotiation, too, because if we don ' t get this right, the negotiation stops there. We don ' t ask. We get confused about our priorities. We shut ourselves down before we give anybody else the chance. If you want to master the mirror, try asking yourself, how have I handled something like this successfully before? Did you know that research shows if we think about a time and write it down, that we achieved great results before we go into negotiate, we are more likely to negotiate well. Why? Because we remind ourselves of who we are at our most powerful, and we also gather data on strategies that are likely to work for us. Lesson four. Land the plane. I worked with a brilliant sales executive who sometimes would lose deals because he talked too much. He ' d ask a great question, and then he would get scared of the silence so he would eat it up with his words."What do you need to get this done here today? Well, I know our price point might be a little bit higher than that of our competitors, but I think if you go ahead and look at our customer reviews..."Want to know the secret to great deals?

Shut up. Recent research found that leaving a period of silence in negotiation not only made it more likely that the other person would give you a high-value move, but it also came across as collaborative. So how much silence? I just did it. Three and a half seconds. See? You were nervous, but we all survived. That ’ s what I call landing the plane. Ask your question, make your proposal, and then zip it. And finally, lesson five. Make your adversary your partner. It ’ s 3am, and you have five hours to go before the entire world is expecting you and several other countries to announce a peace deal. But right now, you have a problem because one of the other country ’ s diplomats has left the building and is down in the parking lot, threatening to drive away because he feels disrespected. This was the situation that one of the diplomats I worked with faced. But a lot of us face things like this in our everyday negotiations. So much of the popular wisdom talks about our adversary, our opponent. Well, in most everyday negotiations, that adversary at the bargaining table becomes our partner once that deal is done. The boss who holds the keys to your raise, once you get it, you ’ re working together. Or that home contractor you ’ re negotiating with over your kitchen. Once you settle on a price, you ’ re trusting her to build a room you ’ re going to love for years to come. And even when your spouse might feel like an opponent, well, you ’ re still sleeping in the same bed at the end of the day. That diplomat I worked with, he walked out of the building, down to the parking lot, and he approached the man who had left. He said to him,”We are on the same side.”He listened. Eventually, the two men walked together back into that building, and later they announced that peace deal. My negotiation motto is this : I never request ; I recruit. I don ’ t want to talk to someone across the table. I want to pull them around with questions to my side of the table so that we are now co-conspirators working toward the same goal. Negotiation is not a battle. It is simply steering. And if we lead with curiosity about ourselves and about others, we won ’ t just create great deals, we ’ ll create great relationships along the way. Speaking of which, you might be wondering what happened after the honeymoon. Well, I ’ m glad to report that my husband and I have been happily steering together for 18 years. Just not in a kayak. Thank you.

Text Sample 320: NEG Dim 2 – 2024 – Score 1.00 – 2024/t003143.txt

The world today, it ’ s no longer news that we are grappling with the devastating effects of climate change. Communities are searching for solutions, trying to turn the tide onto the future that at times really looks so depressing. And there is no place this is more real than in the continent of Africa. But of course, conversations about climate change often miss out on accounting for the most critical barrier to climate solution, which is conflict. I got to know about this through my personal experience as a survivor of war, but also working with a population that has been living through conflicts for years. If you may, I would love to share with you a story based on my personal encounter with the communities that I work with and who are going through the difficult times of climate change. I ’ m from Uganda. Recently, in my home district of Lira, a woman was arrested because she had cut the shea trees and burned the charcoal into fuel. And it ’ s a crime in Uganda. I saw her being walked to the police station, and I decided to ask her, said,”My sister, I see you are being taken, and you ’ ll be charged for this crime. But I have a question for you. If you were forgiven for this crime today, do you promise that you will never do it again?”The frustrated woman looked at me and told me and said,”You know what? Even if they release me right now, I would do it again. I would do it again, because only through burning charcoal that I can raise money to pay for my children ' s education. And it ' s only through burning charcoal that I can raise money to meet the medical bills for my children, who are always falling sick of malaria. So the truth is, I would do it again."This is the story for many in Uganda. Beautiful country with so much potential, but always undermined by a near constant state of violent conflict. And you know, in a community where poverty and unemployment is so high, fighting against climate change is almost a losing battle. But when you look through across Africa, you find that most of the countries most vulnerable to climate crisis are going through some sort of conflict. When you look through the Sahel region, you find there ' s always constant attack, clashes between the herding, the farming and the fishing communities over diminishing land and water resources. Of course, with not so much option, humanity will always do anything possible for survival. I remember this firsthand when I was growing up as a child in war zone in northern Uganda. I myself had to cut the trees, to burn charcoal, to raise money to meet the basic needs, but most importantly, to pay for my education. An education that has enabled me to be able to stand before you to give this speech today. Now the big question is : How can we make it, what can we do to create an alternative to environmental destruction and provide economic means that will support conditions necessary for peace? That ' s the biggest question we should be asking ourselves today. I remember, I think about this a lot through my work at African Youth Initiative Network, AYINET, where for decades we have been supporting war survivors from Uganda, from Congo, from Sudan and South Sudan, and we ' ve been able to mobilize young people to make them game changers for peace and development. We mediate in community conflicts and support agricultural activities as a mechanism to provide youth economic alternatives to violence. And we have been successful. So the success of this work we have done is a clear manifestation of how a local initiative can greatly impact on economic empowerment, on conflict resolution and on climate change. And every time I think about this, I do believe this is an opportunity for Africa to also impact on world peace. And I sincerely know for sure that for us to achieve this goal, the following three points have always come very strongly in my heart. And the first is we need to change the way we ' re talking about climate change. It is time that we take the climate dialogue beyond just mere weather patterns and help people understand what they are fighting for, who they are protecting, what will they win or lose based on their action or inactions. This has always helped us. I remember early last year I went to my village with my twin brother, Jack. We were welcomed by one of the community elders, Charles. He was so happy to see us, to welcome us home, and he was sad that he couldn ' t give us some sugar cane, which used to be his traditional gift to us, simply because the wetland that used to grow sugar cane had dried up, and he was really sad about it. My twin brother comforted him and told him,"Weather is changing, weather would encourage us, let community do our part by planting trees so that we can attract more rain."When he heard this, he asked my brother, said,"Mr. Opio, are you saying, as you are saying, trees is good to bring rain? Now that you have planted so many trees in your community, is it raining in your garden?"To me, this was sincerely the most basic, important organic question that could come from this community. But also it made me reflect if, if at all, our global discussions about climate change is actually responding to the needs at the grassroots. So, today, in northern Uganda, shea trees are no longer producing enough fruits because there ' s local knowledge that shea trees do need bush burning in order to stimulate flowering. But also, we know bush burning is bad for environment, creating a lot of confusion. And this brings in the issue that contradictory accounts from experts, from activists, from the government, which leaves the local communities as collateral victims of global disagreement. And this has kept every one of us in the merry-go-round of climate confusion. Secondly... We need to engage the population that is most affected, in this case, the young people, the youth. With 70 percent of Africa below the age of 30, youth represent the largest demographic opportunity for the continent. We have to engage, including youth, to be an opportunity to uplift the continent from the devastating poverty. And excluding young people in all the climate activities will benefit conflict. So we must engage young people. This is because we do know youth are the legitimate owners of the future, so their inclusion is mandatory. And finally, as we seek to mitigate the effects of climate change and conflict, it is time to hold corporations, government and individuals accountable for committing climate crimes at home and abroad. This should be done by establishing International Climate Court. Today, the world is celebrating the transition to clean energy. We are celebrating the progress. But we cannot celebrate the transition to clean energy while being silent about the death and suffering of millions of people. In Democratic Republic of Congo, for instance, populations have been displaced, millions have been killed, lost properties, just because companies or corporations are exploiting their natural resources. And we have seen also rebel groups fighting to take on the fertile lands, lands rich with minerals in the Democratic Republic of Congo. The truth is, the suffering, the death, the inhumanity we are seeing in Congo is a proof that the road to clean energy is so far the dirtiest road on the planet. Of course, I ’ m also worried. I ’ m worried that the climate crisis will leave the world more divided, especially between the global North and the global South. I ’ m also worried that the rich and the poor will be divided. The rural and urban. Why? Because 10 percent of the world ’ s wealthiest are discussing a reduction of global emissions by flying less. And yet, greater, poorer majority are dreaming for only the one-time opportunity to fly. And to them, all these climate regulations being proposed appear as an effort to reinforce the preexisting inequality, which has always been meant to keep them poor. This is going to create the division. Lastly, there is an African proverb by legendary South African musician, Lucky Dube, who sang the song that says :”The night is long for a hungry child.””The night is long for a hungry child.”What an image. That child could have been your child. Think about that mom, that dad, who are powerlessly waiting for the morning, to try to get something for the child who can ’ t sleep because he went to bed with empty stomach. If we have opportunity, we should have leadership that are willing to create conditions that will help build the community of tomorrow that our children can count on. And lastly, may we realize that the birth of a new era would mean we must all understand that climate change and war are intertwined global long-term challenges. It will require all of us to work together and, most importantly, mobilize a generation that is so willing to defy violence and work for the good of all of us. Apwoyo matek. Thank you very much.

Text Sample 321: NEG Dim 2 – 2024 – Score 1.00 – 2024/t003133.txt

I am so honored to be onstage representing TED ’ s growing recognition of the importance of its immigrant diaspora community. I am an Iranian-American, a kind of free-range Iranian, lucky to be here with all of you and free to embrace all my native culture ’ s positive elements : the poetry, the music, philosophy and hospitality, every day, in ways that many Iranians inside Iran cannot. But I also love having the bow of my identity stretched between two seemingly opposed poles of”Iranian”and”American,”because that gives the arrow of my life a unique perspective on things. I want to share a couple of American ideals that explicate this multicultural dichotomy in me. The first is independence. In America, independence is foundational, the bedrock of our freedom that makes us all, all us Americans, feel exceptional. Unlike my Iranian side, where everyone is in everyone ' s business, I ' m free, maybe even driven, to pursue my own happiness, and that feels so liberating. The other is continuous improvement, the idea that you can always do better. America is a do-ocracy that is always inventing, reinventing herself. We make mistakes, and keep on growing and growing. That ' s why it ' s so exciting to be an entrepreneur in America – we ' re free to try to build anything our hearts desire. For me, these days, it ' s taking on our corrupt political duopoly and working to democratize democracy itself. Isn ' t it amazing that as an American, I just get to do that, without fear or permission? But I ' m Iranian too, and I have the ancient wisdom of Iranian mystics, Ferdowsi, Hafez and Rumi, running through my veins. So each of these American traits give my Iranian psyche cause for pause and reflection. For instance, independence is great, but if I ' m truly independent, then I don ' t need anything or anyone. That ' s kind of lonely. Iranians are congenitally hospitable and tribal. In fact, the word Iran itself is a portmanteau of the word"Ir"and"An,"where"Ir"comes from the word"yar,"literally"friendship"or"love,"and"an"comes from"ostan,"or"land of."Shout-out to our cultural cousins from Ireland, another land of love. Anyway, where does this love and friendship fit into my Statue-of-Liberty-esque, freedom-loving American psyche? Love requires connection and interdependence, as we found out earlier. And interdependence is, well, kind of un-American. After all, Americans never had a Declaration of Interdependence. So my Iranian side begs for me to differ here. As my beloved Sufi uncle Behzad used to say,"Love is the ether the universe swims in."So without love, we literally have nothing. So to be independent, or to be in love? That is the cross-cultural question for me. And what about the American ideal of continuous improvement? That seems downright productive, if you ask my American side. But again, my Iranian side disagrees. I remember a magnet on my mom ' s fridge :"If you ' re ever going to be happy, you must be happy now."Hinting at a tiny existential problem : it ' s always"now."And of course, we have the Iranian mystic poet Rumi imploring us in ode after ode to get drunk and out of our minds so we can become blissfully present in the wow of now. So should I work to create a better future, or be drunk and happy now? Another multicultural paradox. Or is it? Maybe I ' ve been setting up a false dichotomy all along. There ' s no real Iranian or American on this stage, here in me. After all, I ' m not some LLM trained in a lab on a discrete set of cultural memes. I am that still magical, mystical, fractal of universal consciousness known as human being. I am a point – thank you – a point at the center and circumference of the infinite sphere, as my favorite author, the Argentinian Jorge Luis Borges once wrote. And as a magical, mystical life – an MML – we humans are uniquely free to embody whatever useful paradoxes we find, whenever it suits us. We can choose to be culturally American or Iranian or both, no matter where we are from, and can even have a favorite author who is neither. And as an MML, we can simultaneously be both independent and free, while deeply connected by love to everyone and everything, always striving for a better future, while also happy together, here, now, wow.

Text Sample 322: NEG Dim 2 – 2022 – Score 1.00 – 2022/t003369.txt

This is a piece of skin. Of artificial skin. And now, OK, I ' m just going to paste it on this hand, and I ' ll greet you all. I will show you how it works.

So please have a look at what happens on the table, where there is one piece of this artificial skin, and also what happens on the screen of the laptop that you see in the video that is about to start. So this is me. I ' m first breathing on the skin. And you see now that this line that first was flat now shows a bump. And now I ' m touching it with my finger. And again, another bump. I am touching it one more time with the back of a feather. And now tinier bumps are coming. And again. Now even smaller bumps are coming. But what is this line? This, what we are looking at, is electrical current, which first is constant. And then when we touch the skin or breathe on it or touch it with a very light feather, it shows a signal. So this bump that you see is a signal in the electrical current. And this is actually exactly how our skin also works. So we have receptors in the skin which sense what is happening on the skin. And then they produce an electrical signal which travels through the nerves and arrives at our brain where it is recognized. I am a chemist by education, and I worked in the field of material science, which is a very interdisciplinary field, since almost 20 years now. First, as a PhD student in Italy, in Bari, and then at MIT as a postdoc, and now I am group leader in Graz. And it is almost six years that we are working on artificial skins. Almost 12 students, between master and doctoral students, have worked on these devices, trying to study how the materials work, how they work together and how to build this device. And after this, we have for the first time produced an artificial skin that can respond at the same time to three stimuli. Touch, so force, temperature and humidity. And it can do this also at an unprecedented resolution. So it ' s a very tiny device. You will see how it is done. And so this means that it can sense objects that are actually smaller than the objects that can be sensed with our skin. Our skin has a resolution of one millimeter square. This skin has a resolution of 0. 25 millimeter square. So I promised I would give you some technical details. I know you are really looking forward to them. The main component of this material, of this skin, is what is called the stimuli-responsive material. So what does it mean that it is stimuli-responsive? It means that at the beginning it is small. And when we have either humidity or light or pH or temperature changes, this material changes in its shape, and it becomes bigger. It can arrive to even doubling or tripling its thickness, its original thickness. And this is an amazing property, but it happens at a very, very microscopic scale because we produce these materials as thin films, so sometimes even one million times smaller than a millimeter. So the problem was, OK, how do we translate this very big change in thickness into something measurable? And we thought, OK, let ' s combine it with a piezoelectric material. Now maybe this word sounds a bit complex. Sounds like something you have never seen in your life, never heard before, but it ' s not really true. You have been in contact with this material many times. I am in contact with it right now. It ' s in microphones, for example. So it ' s the material that when there is a movement, produces electricity. And so in microphones there are membranes that when they move, produce electricity. It ' s also the material that is in these funny greeting cards that when you open, they make this music. The music is also made with the piezo material. So there is an electrical circuit into the card and when you open it, it moves a membrane and the membrane makes the music. And it ' s also the material that is included in the arm wristbands, those that measure the heartbeat. OK. So we combined these two materials in these cylinders that you see in the picture. So in the middle we have that material I showed you before, the stimuli-responsive material, the material that changes its thickness and gets bigger. And on the outer shell we have the piezoelectric material, the material that when the inside gets bigger, the outside produces the electricity. And this is how it is done. Easy-peasy. Then these type of cylinders are very tiny, and we have several of them in this device so that we can sense with very high resolution the three different types of stimuli I was telling you before. And a bit more of technical details. I know they are exciting, right? So how do we do these cylinders? We do them using a template, the one that you see in the picture on the top. This is a template with a lot of wells, so a lot of holes. And then we refill the holes. First we fill them up with the piezoelectric material, which is represented in yellow in the bottom figure, and then we fill it up with the stimuli-responsive material that is the light blue in the bottom figure. But to do this we cannot use liquids, because as you know, liquids would fill the wells from the bottom up. And instead we want to cover also the lateral walls. And so we do this using vapor deposition. So the whole process of production of these things happens in vacuum chambers like the one that you can see in the picture. And you may be wondering, OK, why working on this topic? Why working on artificial skin? So from a material scientist point of view, the skin is really a complex ensemble of materials and functions. Skin is useful for protection, secretion, adsorption, heat regulation and sensing. And so being able to just reproduce artificially all these properties – or actually, for the moment, only the sensation – looked like a challenge. And so I was happy to embrace it. OK, now I ’ m done with the technical details. Now I know you are all wondering, OK, you have done these artificial skins. But why? I will show you some fields of application, ending with the one that I think could be the closest. So we have also seen how there are many victims of burns. These burns can be so huge that they even take away the receptors in the skin. And so people would then lose the sensation where there was this burn. Now imagine a future where actually victims of burns could, thanks to our technologies, regain the sensation. 216 million users of smartwatches in 2022. Should I ask how many of you have a smartwatch now? Wearing a smartwatch? Yeah, so imagine you are, for example, running on a hot day, and your smartwatch could measure, for example, the level of hydration of your skin and warn you maybe if you are reaching the limit of the hydration. Another interesting field of application would be robotics. Nowadays, humanoid robots are used in many fields, for example in medicine but also in household. And these robots are exposed to several stimuli, several interactions with the environment and with the humans, and sometimes they have too many inputs at the same time. And this is the reason number one for robot failure. So imagine a future where actually a robot could be a bit more sensitive, a bit smarter. This would lead also to a higher safety of this technology. And finally, unfortunately, a very highly cited paper projects the number of people in need of a prosthesis on the rise, from 1. 6 million in 2005 to 3. 6 million 2015. So a lot of people are born without limbs, and some people lose them during their life due to, for example, vascular diseases. This is so common that you may even know somebody who actually is in a need of prosthesis. Now imagine we could coat the prosthesis with this artificial skin and give this type of people the sensation back. With almost a decade of working in this field, I ’ m hopeful that technologies like this could help in the future not only regain, but also maybe augment capabilities. Thank you.

Text Sample 323: NEG Dim 2 – 2022 – Score 1.00 – 2022/t003521.txt

It ’ s an honor to be with you here today. And the message I have to deliver is a simple one, which is that clean energy will win on cost, definitively, but only if we get out of the way and allow it to be built. Let ’ s focus on that first point. Let ’ s look at the cost of clean energy technology. In 1975, if you were buying a solar panel per watt of power that it produced, it would cost you 100 US dollars. By 2020, that cost had declined to 20 cents per watt of power. That ' s a 500-fold decline. This is unlike anything else ever seen in energy. It is unlike anything else ever seen in physical infrastructure, and it continues today, and it is likely to continue for decades to come. Now, this has surprised the leading experts on clean energy and even the biggest optimists on the future of clean energy. I should know because I am one of those leading optimists. In 2010, the International Energy Agency, the world ' s foremost and official experts on energy, forecasted that the cost of solar power would drop like this. Now, I came from technology. I was a computer scientist, and so I looked at the cost of solar through the lens of Moore ' s law, exponential decline in cost as technology scales. And so I forecast that the cost of solar would drop at around five times the rate that the IEA believed it would. Unfortunately or fortunately, I was wrong. The cost of solar actually dropped 10 times as fast as the IEA expected and twice as fast as I expected. Clean energy is a technology. This deception in understanding the pace of cost decline has led to a massive underestimation of the pace of growth. Every year, the IEA puts out a forecast for how much new solar will we install per year. And you see these colored lines show you the pace at which the IEA believes new solar will be deployed annually, versus the black line, which shows how fast it ' s actually been deployed. So you see the colored lines going off to the right, those are successive years of forecasts of annual installations. And on the left you have what amounts to a 30 to 40 percent annual growth in installations. Note this has happened through the COVID years, note that 2022 will see another 38 to 40 percent growth in this market. This trend is not limited to just solar. It applies to a wide variety of clean-energy technologies. We ' ve seen the price of solar drop by a factor of 40 over the last few decades. We ' ve seen the cost of wind drop almost as much, and now an accelerating cost decline in floating offshore wind and offshore wind. We ' ve seen the cost of batteries that power our electric vehicles and grid energy storage drop at the same pace or faster than the pace of solar. And we are just at the very beginning of an exponential cost decline in the cost of using clean electricity to make hydrogen and other fuels that we can use to power industry, to provide weeks or months of storage on the grid and to provide fuels we can use for aviation, for shipping and so on. What ' s happening here – and I won ' t claim this is happening quickly enough to stay below 1. 5 degrees Celsius – but what ' s happening here is that clean energy technologies are technologies and they drop in cost like technology. As they are scaled, they come down in price. Meanwhile, fossil fuels are commodities and fossil fuel prices fluctuate over time. This is data from Oxford ' s Institute of New Economic Studies, which shows the cost of oil, gas and coal across the bottom there, fluctuating over time, largely remaining flat. Whereas here is the cost of clean energy technologies : solar, wind, batteries, power to fuels, all of them dropping exponentially. In 2010, there was no place on earth where clean energy was cheaper than fossil energy. In 2015, we started to see the first instances of clean energy without subsidies being cost-competitive. Now we see in more and more parts of the world that it is cheaper to build solar and wind than it is to put fuel into an already built and operating coal or gas plant. And behind that is the continued cost decline of batteries and hydrogen, which are still expensive but will get cheap enough to solve many of our intermittency issues. Now, does that mean we are done and the problem is solved? Not at all. We need to go faster. And we have a number of barriers. Some of these we ' ve talked about. Critical minerals need to be built out. We need a just transition. But there ' s two barriers I want to talk about in particular, which are the reluctance to build – not in my backyard – and the challenges with permitting. Because we look at renewables and a common complaint is they take up too much land. All of us want clean energy infrastructure, but many of us don ’ t want it in our backyard. Now, solar power is fairly compact. Wind power takes more space, but that space is co-located with agriculture. Animals graze up to wind turbines, fields of crops can be grown. And yet, despite this, in modern nations, in Europe and North America, roughly half of the land in the UK, in Germany or the US, is devoted to agriculture. Yet, Germany allows now two percent of land in Germany to be used for wind power. The US allows local communities to block wind power even if it ’ s nowhere near them. If we want to deploy clean energy, we must allow it to actually be built. But the even larger problem is this : We need to build out the grid. We have a perception that clean energy technologies don ’ t need the grid. Solar and wind mean you can be off-grid. While that ’ s easier, largely the opposite is true. Because solar and wind are weather-dependent, they benefit much more from continent-sized grid than do fossil fuels. And yet the same NIMBY issues and the same permitting issues plague energy transmission. Electricity transmission is ugly, and why do we need it, if we can just go off-grid and local with these energy technologies? Well, here ’ s why we need it. This is the sunshine, the solar resources of Europe. The most sun is in the south. Of course, the most wind is in the north, by the way. And this is averaged across the year. In winter, it is much more dramatic. In winter you have one seventh the solar resource in the UK that you have in summer. You have one sixth the solar resource in Germany that you have in summer. So every model, every simulation of weather and energy demand shows this. It shows that if you want to have the highest-reliability grid at the lowest cost, with the least carbon emissions and the most clean energy deployment, you want to build a continent-sized grid. In Europe, it would bring wind from the north, primarily the North Sea, but also on land, hydro from the Nordics and solar from the south. And these would be countercyclical. More sun in the summer, more wind in the winter. This is slightly oversimplified, but this is the sort of system that would allow Europe to decarbonize its electricity sector with some degree of storage and hydrogen and so on, nearly completely, at a lower cost than the system today. That same Oxford study I showed you says that a rapid transition to clean energy, because the more you deploy, the cheaper it gets, would save us 12 trillion dollars on the energy system, not even counting climate damages. Now, could we build such a grid? Of course, we have the technology and the economics. In China, the bulk of energy demand is on the East Coast, yet, the greatest solar resources and wind resources are in the interior. And China is building literally scores of high-voltage power lines that transmit power from where the sun and wind are to the coasts, where the energy demand is. And the longest of these lines right now, the Ürümqi to Shanghai line, is 3, 400 kilometers long. It is 90 percent efficient, very low losses, and it adds maybe a penny or two to the cost of electricity. That ’ s what ’ s technically feasible with current technology, let alone advances. In Europe that would allow us to transmit electricity from Seville to Copenhagen, to bring power from the North Sea to anywhere in the continent that needs it. In my home country, the United States, we could bring power from the sunny, wide open areas of New Mexico to population and land-dense New York, which doesn ’ t get so much sun. And we could take wind power from the Great Plains, that are largely devoid of population, and bring it to the coasts in winter. That is what a modern grid looks like. And yet we are not building this. Now, let me give you two pieces of data that will back up my assertions about how powerful these issues of permitting and grid build-out are, and one more, which is open competition on cost. What state in the United States do you believe has the most combined solar and wind power today and has deployed the most solar over the last year? Audience : Texas. RN : You are a very smart audience. It is the great state of Texas. Now, Texas has no climate policies to speak of. It has no incentives for solar or wind to speak of. It does have abundant land and abundant sun and wind. We can ’ t all duplicate that. But it has three other things we should think about. One, it has an open market for electricity where the cheapest provider wins, and that should give us confidence of what is actually winning. Two, permitting in Texas is relatively easy. It ’ s easy to build things, easy to build fossil infrastructure. I don ’ t love that, but it ’ s easy to build clean infrastructure and that ’ s what ’ s winning out. And three, Texas is the easiest state in the United States to build electricity transmission. Not as easy as I ’ d like, but it makes it easy. Those last three factors are factors we can replicate in every nation, every state, every geography on planet Earth. And they would accelerate this transition. Now, in August of 2022, we passed the largest US climate bill ever, Inflation Reduction Act. In Europe, we have a massive push on clean energy as part of the energy crisis driven by the war on Ukraine. Now I want to reveal at the same time that we failed at something. In late September 2022, we had a permitting bill in the United States that would have made it tremendously easier to build the continent-sized grid that we need in the US. It also made it easier to build fossil fuels. Yet every analysis showed that fundamentally this bill accelerated the clean energy transition. But it was opposed by some people because of it making it a bit easier to build fossil fuels as well. The consequence of this is potentially dire. The Princeton REPEAT project, led by my friend Dr. Jesse Jenkins, has analyzed this and said that up to 80 percent of the benefits of the Inflation Reduction Act may not manifest if we can ’ t accelerate the pace of building transmission. That that could amount to as much as 800 million tons a year of carbon emissions in the United States that would have been eliminated, but that won ’ t be because we don ’ t have the transmission capability to bring the cheapest power to where it ’ s needed and boost resilience and reliability along the way. So I believe that a deal that makes it easier to build, even if, to get that bill passed, we need to make it a bit easier to build some fossil infrastructure, will win for clean energy because clean energy simply wins on cost. On a level playing field, it will dominate the future. And so if you have confidence in this decades-long trend that clean energy will economically disrupt fossil fuels, then the logical conclusion is that we must get out of the way, make permitting easier, and allow it to be built. It is time for us to build. Thank you very much.

Text Sample 324: NEG Dim 2 – 2022 – Score 1.00 – 2022/t003556.txt

My grandfather was a restaurant owner. My father was a restaurant owner. My name is Vincent Yeow Lim. I ’ m a chef, content creator and restaurant owner. Owning and running a Chinese restaurant is one of the hardest and most valuable things I think you can do. When we first arrived in Australia, my dad opened a Chinese restaurant. I would go there every single day to help him out. That was my way to spend time with him. And slowly I developed my love for cooking. At the age of 15, I was able to use a wok to cook and to serve customers. Now I ’ m running a 100-seater Chinese restaurant. The hard work that I saw my dad put into Chinese food makes me feel really proud. Proud that I was lucky enough to learn from a master. When I tell people that I ’ m a chef cooking Chinese food, I want people to value it the same way that I do. So what is the value of Chinese food? Is it the taste? Is it the hard work and the training? Or is it the dish ’ s history and the nostalgia it makes us feel? To me, it is all of those things. But being in the restaurant industry, I can tell you the most obvious value of food. And that is the price at which we choose to set it at. The perception of Chinese food is that it is indulgent, yet cheap. But the value of Chinese food is so much more than that. There is no formal culinary school to train to be a Chinese chef. And after the pandemic, there was a shortage of Chinese chefs in Australia. Now the cost of ingredients, labor and demand for Chinese food is high. The amount of skill required to cook the first dish that I ever learned doesn ’ t match up to the price. The first dish that I ever learned to cook was the fried rice. Fried rice is the easiest dish to cook, but also the hardest dish to get right. I actually wasn ’ t allowed to serve customers until I got this dish right. The oil goes into the wok first, but only when the wok is hot enough. The egg goes in next. This creates a nonstick coating for when the rice hits the hot wok. From experience, you can tell when the rice is too wet or if the wok is too hot or too cold. All these small factors will create a different end result. If the rice is too wet, you lose the smoky aromatic flavor. But if the rice is too dry, you ’ re basically eating pebbles. Now we add a little bit of yum, yum. Now we add the cooked meats and vegetables. I ’ m tossing it to combine the flavors of the meat and vegetables with the rice. Even if we were to remove all the meat and vegetables, your rice should still have all the flavor. From years of experience, I can tell you the difference between a good fried rice just from the feel and from the smell, without even having to take a bite. And I thought – I thought that was enough. But I realized cooking with one wok was way too slow. So I took the knowledge from my dad and taught myself the dual wok cooking technique, using two woks to cook two different dishes at the same time. This requires coordination, organization and years of training. What makes it harder is the combination of dishes coming up next. They have endless possibilities. Chow mein, sizzling garlic prawns, egg foo young. When one is cooking, the other is served. When the other is cooking, the next is served. The goal is that no matter the combination, all the dishes get to the table at the same time. People expect this in a Chinese restaurant. People expect variety. In a Western restaurant, you open the menu, there ’ s 20 items. When I first started, I had 300. In a Western restaurant, you can often judge the food by the decor, the plating of the dishes or even by the number of different forks for each course. But in a Chinese restaurant, it ’ s a little bit different. You can often judge it by things like the kids sitting on the first table doing their homework. That ’ s when you know a Chinese restaurant will have amazing food. I was that kid. To a lot of us Chinese immigrants who started our own restaurants, the restaurant isn ’ t just a business. It ’ s a part of our lives. We eat almost three meals a day there, and often we just use our homes as a place to sleep. We dedicate our lives to these restaurants and to making delicious food. In Chinese cooking, we use a lot of strong flavors like ginger, garlic, chilies and even shrimp paste. The different flavors, I think, are why people like Chinese food, because every bite is different, and you never really get bored. But flavors, flavors are also built with the tools we use. When you use a wok, you get wok hei, known as”breath of the dragon.”Wok hei is the reason why noodles and rice always taste better at a restaurant than when you try to cook it at home. Wok hei is the combustion of oils in the air that make that smoky aromatic flavor. But wok hei also tastes like nostalgia. Growing up in Malaysia, the kitchens are outside, just like this. You can smell every ingredient that goes into the wok. The air is thick with humidity, and the fragrances of the food, they linger around so much longer. Those memories, they become nostalgic. The sense of home is what I feel when I taste Chinese wok cooking. Nostalgia is what makes a good dish taste ten times better and that much more valuable. But the value of Chinese food is often compared to fast food, quick and cheap. Although we ’ ve spent years developing skills to cook faster, Chinese food isn ’ t fast food. It requires skills, mastery of ingredients and nostalgia. In 2017, my dad passed away. I looked up to him. He was my best friend. When he passed away, I realized – I realized my passion of cooking dishes from my childhood. I was working as a chef at an airline ’ s first class lounge at the time, but I didn ’ t have any memories of the dishes I was cooking. I wanted to feel closer to my dad. So I quit. I dropped out of university on my last semester. I hopped onto a plane, and I flew over here to Sydney. And I took over the restaurant that I own today. I wanted to record everything that my dad taught me, so I shared it on social media. It went viral. Hundreds of millions of views. A lot of people related to the recipes that I was cooking because it was what they grew up eating with their parents. I remember this one fried rice he made. I was eight years old at the time. It was the first time I recall my dad stepping into the kitchen cooking. He made this fried rice with spam, seafood, peas and a little bit of dark soy sauce. It was a little bit salty, but it was the best fried rice that I ’ ve ever had. I cook and I ’ ve always cooked because it was a way to connect to my dad. It was something that we shared. Cooking represents all the hard work and memories that we had, learning the art, mastering the skills and understanding the history of Chinese food. To me, the value of Chinese food is so much more than the prices you pay at my restaurant. So the next time you visit a Chinese restaurant, just remember. It ’ s not just a plate of fried rice. It ’ s so much more than that. Thank you.

Text Sample 325: NEG Dim 2 – 2021 – Score 1.00 – 2021/t003738.txt

We live in a time-pressed culture. There is never enough time. And we see it, we feel it around us every day. We live in a world that valorizes work, accomplishment, busyness. And there ’ s real upside to that ; there ’ s real value. We ’ re pushed, we ’ re driven toward achievement and action and creation. And that ’ s great, but there ’ s also a downside. And that ’ s something that I think is worth talking about. There was a study done a while back, by the Management Research Group, of 10, 000 senior leaders. And they asked them,”What is key to your organization ’ s success?”And 97 percent said long-term strategic thinking. I mean, when was the last time that 97 percent of people agreed on anything? There is near unanimity that being a long-term thinker – having perspective, having the ability to think and ask big questions – is essential to our success. And yet in a separate study, 96 percent of leaders were surveyed, and they said they don ’ t have time for strategic thinking. What is going on? Why is it – how can it be that 96 percent of people are not doing the one thing that they say is most critical to their success? Well, I think we know the answer... or at least we think we do. The average professional attends 62 meetings per month. That sounds pretty outrageous. How could that be? But if you actually break it down, it ’ s not that many. It ’ s two to three meetings per day, which is probably average for many of you. So 62 meetings a month. That does not help, and that is not wrong. It is a contributor. Also, we know – we know what else... email. A study a while back by McKinsey showed that the average professional spends 28 percent of their time just responding to email. Of course that drains us, of course that makes us busy. But the truth is, it ’ s also, I believe, not the full picture. Those are manifestations. Those are problems, legitimately. But there are also some other things going on underneath the surface, reasons that perhaps we are, in some ways, working at cross-purposes. Because for so long almost all of us have said we want desperately to be less busy, and yet we keep making choices that put ourselves in the position where we ’ re just as busy as we ’ ve always been. What is going on? Well, some research out of Columbia

University sheds a little bit of light on this. Silvia Bellezza and her colleagues have done interesting research into the fact that in some cultures – American culture chief among them – busyness is actually a form of status. When we say,”Oh, I am so crazy busy,”what we ’ re really saying is a societally-accepted version of”I am so important –”I am so popular! I am so in demand!”And the truth is that feeling can be hard to give up... even if we say that we want to. That ’ s not the only reason, of course. It turns out it is very hard for the human mind to deal with conditions of uncertainty. And in modern life, there ’ s a lot of it. Sometimes we are given tasks or challenges, and the truth is, tactically, we just don ’ t know how to do it.”Increase sales by 30 percent.”Well, how? There ’ s a lot of ways you could do it. You ’ re not sure how. Sometimes it ’ s easier, frankly, to just double down and keep doing more of what you ’ re already doing. That might not be the best answer, but it ’ s an answer, and it removes uncertainty. The picture gets even worse when we ’ re talking about existential questions ; when we ’ re talking about uncomfortable matters that we might not actually really want to deal with. That might be,”Am I in the right job?”It might be,”Am I in the right career?”Those are often questions, truth be told, we might not want the answer to. And so we become busy as a way so that we don ’ t even have to ask the question. Now, there ’ s a third reason, and I ’ ll admit it ’ s one that I know well, personally, and that is that sometimes we use busyness as a way to numb ourselves out. I ’ ve experienced that. This is my boy Gideon, and he died in 2013. I ’ d had him for 17 years, and he was my best friend. And after he died, I ’ ll be honest, I didn ’ t want to be home because I knew that he wouldn ’ t be there. And so for two years, my life basically was an Uber to an airport, to a hotel and back again, because I just really didn ’ t want to face that. For a lot of us, there are things we sometimes don ’ t want to face. What we ’ re really looking for with work is an anesthetic. And as I like to say, work is better than crack – so if you ’ re choosing... it ’ s not the worst. But the truth is, it ’ s also not a sustainable solution. For many of us, we get trapped in the pattern of busyness, of overwork. It ’ s hard sometimes even to remember what it was like before. Oftentimes in our mind ’ s eye, when we think of busyness, what we think of is this. What we think of is triumphant success and the world at your fingertips. The truth is, more often, busyness looks like this. It looks like loneliness. It looks like frustration. It looks like having a life that ’ s not really in your full control. So I would like to propose that we make a change. Because if we are ever going to succeed in beating back busyness once and for all, first of all, we have to get real and acknowledge what is actually behind some of the busyness that is filling our days. We have to really get honest about what it is that ’ s motivating us so that we can make a different choice. Because it is about our choice. We need to recognize that real freedom is about creating the space so that we can breathe, the space so that we can think. Ultimately, real freedom is about choosing how and with whom we want to be spending our time. Thank you.

Text Sample 326: NEG Dim 2 – 2021 – Score 1.00 – 2021/t003732.txt

[The Universe in Verse] [Prelude] In 1908, Henrietta Swan Leavitt, one of the women known as the Harvard Computers, who changed our understanding of the universe long before they could vote, was cataloging photographic plates at the Harvard College Observatory, single-handedly analyzing 2, 000 variable stars – stars with fluctuating brightness, kind of like a lighthouse beacon – when she began noticing a distinct correlation between their brightness and their blinking pattern. That correlation made it possible to measure the distance of stars for the very first time, furnishing the yardstick of the universe. Meanwhile, a dutiful teenage boy in the Midwest was repressing his childhood love of astronomy and beginning his legal studies to fulfill his dying father ’ s demand for an ordinary, reputable life. When his father died, Edwin Hubble turned that passion for the stars into formal study and discovered two revolutionary facts about the universe : that it is tremendously bigger than we thought and that it ’ s getting bigger and bigger by the blink. One October evening in 1923, perched at the foot of the largest telescope in the world, at Mount Wilson Observatory, not far from here, Hubble took a 45-minute exposure of Andromeda, which was then thought to be one of many spiral nebulae in our Milky Way. The notion of a galaxy, a gravitationally bound cluster of stars and dust and dark matter, didn ’ t exist at the time. The Milky Way, which, by the way, was coined as a term by Chaucer in the 14th century, was thought to be what was called an”island universe”beyond the edge of which lay cold, dark nothingness. But when Hubble looked at the photograph the next morning and compared it to previous ones, he furrowed his brow and with a gasp of revelation scribbled right on top of the plate”VAR”and then drew an exclamation point because he had realized that a tiny fleck in Andromeda previously thought to be a nova could not possibly be a nova, given its blinking pattern across these different photographs. And it was, in fact, a variable star. Which meant, given Henrietta Leavitt ’ s data, that it was very, very far away, farther than the edge of the Milky Way. Which meant that Andromeda was not a nebula in our own galaxy, but a whole other galaxy, out there, in the cold, dark nothingness. Suddenly, the universe was a garden, blooming with galaxies, with ours but a single bloom. That same year, in another country, suspended between two World Wars, another young scientist came up with the radical idea of using rocket technology, this deadly military technology, to mount an enormous telescope and launch it into Earth ’ s orbit. Not only closer to the stars, but bypassing the atmosphere that occludes our terrestrial instruments. It would take another two generations of scientists to make that telescope a reality : a shimmering poem of metal, physics and perseverance bearing Hubble ’ s name. But when the Hubble Space Telescope launched in 1990, our human fallibility had slipped into the precision of the instrument, and it turned out that its primary mirror was ground into the wrong spherical shape, warping these long-awaited images of the unfathomed universe. Up the coast from Mount Wilson Observatory, a teenage girl watched her father, who had worked on the Hubble, one of NASA ’ s first Black engineers, come home brokenhearted. He didn ’ t know that his observant daughter would become poet laureate of his country and would commemorate him in the tenderest tribute that an artist ’ s daughter has ever composed for a scientist father, which would win her the Pulitzer Prize the year the Hubble ’ s corrected optics captured that revolutionary, ultra-deep field image of the observable universe, revealing what neither Hubble nor Henrietta Leavitt could have imagined. That ours is not just one of several galaxies, but that there are 100 billion galaxies out there, each containing 100 billion stars. Narrator : From”My God, It ’ s Full of Stars.”When my father worked on the Hubble Telescope, he said They operated like surgeons : scrubbed and sheathed In papery green, the room a clean, cold and bright white. He ’ d read Larry Niven at home, and drink scotch on the rocks, His eyes exhausted and pink. These were the Reagan years, When we lived with our finger on The Button and struggled To view our enemies as children. My father spent whole seasons Bowing before the oracle-eye, hungry for what it would find. His face lit-up whenever anyone asked, and his arms would rise As if he were weightless, perfectly at ease in the never-ending Night of space. On the ground, we tied postcards to balloons For peace. Prince Charles married

Lady Di. Rock Hudson died. We learned new words for things. The decade changed. The first few pictures came back blurred, and I felt ashamed For all the cheerful engineers, my father and his tribe. The second time, The optics jibed. We saw to the edge of all there is – So brutal and alive it seemed to comprehend us back.

Text Sample 327: NEG Dim 2 – 2021 – Score 1.00 – 2021/t003775.txt
Hi, my name is Fariel, but back home, I am also called the goat lady. And today I will be sharing the story of how that came to be about. In 2015, I visited a small, remote village in Pakistan called Pathan Goth. The residents of Pathan Goth were living without access to basic facilities such as water and electricity. The community was bussing water two hours from Karachi. This made their water so expensive that their livestock went without water every other day. Bathing and laundry was a luxury. The solution was simple : a solar water pump. But that would cost $10, 000. And for a community where the average household earned just about $70 a month, this was beyond their affordability. As I sat there thinking about this issue, the solution literally walked in front of me. A herd of more than 100 goats crossed my path. And I thought to myself, that is a lot of goats. I took a chance and asked the village elder : Would they consider paying me for their pump in goats? He agreed, and that is how Pathan Goth got its pump and I became the proud owner of 40 goats. A few months down the line, the Muslim festival of Eid came around. During Eid, livestock is offered to honor Abraham ’ s willingness to sacrifice Isaac. And much like shopping before Christmas, the demand for goats goes up, and the price skyrockets. I remembered my goats and decided to put up a post on Facebook asking my friends and family to buy their Eid goat from me that year. Within a week I sold all my goats. And even after paying for a team to do transport and manage them, I more than recovered the cost of the pump. This experience made me question the idea of money and how it has changed and evolved for us. I found that in 2500 BC, Mesopotamia, Sumerians used measured quantity of barley as money. But as the trading circle grew beyond one village, money ’ s key characteristics changed and evolved. Money not only needed to reliably store value but also be easy to convert and easy to transport. That is when we – civilization – moved towards coin and paper money. But coin and paper money get value only when they ’ re validated by an external authority, such as God, crown or the central bank. Technology today enables us to bypass the need for this central authority. Today, the value of any asset can be transferred from one person to another using blockchain. Any asset can now be tokenized, digitized and traded. So technology is not only disrupting the need for a financial intermediary in mainstream economy but also has the potential to disrupt and democratize economic power in more rural economies. By removing the need for inefficient and expensive intermediaries, small farmers can derive greater value from what they grow and raise. Livestock is very abundant in these communities. Farmers literally use their goats and cows as ATM machines and as saving instruments. They will sell an animal when they need cash, such as to buy fodder for the rest of the herd, to buy groceries for their home or for weddings and health emergencies. But these communities have never been able to use their goats and cows for larger, more productive assets... until now. Our farmers are able to convert 15 goats into a solar water pump. Previously, these communities would rely on or wait for an NGO, a charity or the government to take notice and give them what they needed. Now they are using their own resources to fulfill their own needs when they need it. This is self-reliance. So

how have we enabled the use of goats as a reliable form of currency? Right now, we we work with farmers through physical barter. At the time of the solar installation, we take possession of the goats. We then sell these goats as meat to grocery chains and through our own retail brand. Every goat that we receive is then recorded, as its unique ID – its key characteristics such as its sex, its age, its weight and even its number of teeth are taken into account. And then using these, we assign each goat a dollar or a rupee value, essentially tokenizing a goat. We have already worked with 45 communities, enabling more than 6, 000 farmers to convert their goats into water and electricity. We are now planning to add smartphones, tractors and other equipment. My hope is that we can create a more inclusive economic system where any person or any community can use what they grow and raise as money. Where a farmer, a small farmer can use, wherever he is, whether it ’ s Pakistan, Nepal, Somalia, can use a digitized goat to pay for her kid ’ s school fees, a tractor or a smartphone. This future isn ’ t quite here yet, but I hope using goats as money gets us closer. Thank you.

Text Sample 328: NEG Dim 2 – 2004 – Score 1.00 – 2004/t000431.txt
Imagine spending seven years at MIT and research laboratories, only to find out that you ’ re a performance artist. I ’ m also a software engineer, and I make lots of different kinds of art with the computer. And I think the main thing that I ’ m interested in is trying to find a way of making the computer into a personal mode of expression. And many of you out there are the heads of Macromedia and Microsoft, and in a way those are my bane : I think there ’ s a great homogenizing force that software imposes on people and limits the way they think about what ’ s possible on the computer. Of course, it ’ s also a great liberating force that makes possible, you know, publishing and so forth, and standards, and so on. But, in a way, the computer makes possible much more than what most people think, and my art has just been about trying to find a personal way of using the computer, and so I end up writing software to do that. Chris has asked me to do a short performance, and so I ’ m going to take just this time — maybe 10 minutes — to do that, and hopefully at the end have just a moment to show you a couple of my other projects in video form. Thank you. We ’ ve got about a minute left. I ’ d just like to show a clip from a most recent project. I did a performance with two singers who specialize in making strange noises with their mouths. And this just came off last September at ARS Electronica ; we repeated it in England. And the idea is to visualize their speech and song behind them with a large screen. We used a computer vision tracking system in order to know where they were. And since we know where their heads are, and we have a wireless mic on them that we ’ re processing the sound from, we ’ re able to create visualizations which are linked very tightly to what they ’ re doing with their speech. This will take about 30 seconds or so. He ’ s making a, kind of, cheek-flapping sound. Well, suffice it to say it ’ s not all like that, but that ’ s part of it. Thanks very much. There ’ s always lots more. I ’ m overtime, so I just wanted to say you can, if you ’ re in New York, you can check out my work at the Whitney Biennial next week, and also at Bitforms Gallery in Chelsea. And with that, I think I should give up the stage, so, thank you so much.

Text Sample 329: NEG Dim 2 – 2004 – Score 1.00 – 2004/t000261.txt
Brain magic. What ’ s brain magic all about? Brain magic to me indicates that

area of magic dealing with psychological and mind-reading effects. So unlike traditional magic, it uses the power of words, linguistic deception, non-verbal communication and various other techniques to create the illusion of a sixth sense. I ’ m going to show you all how easy it is to manipulate the human mind once you know how. I want everybody downstairs also to join in with me and everybody. I want everybody to put out your hands like this for me, first of all. OK, clap them together, once. OK, reverse your hands. Now, follow my actions exactly. Now about half the audience has their left hand up. Why is that? OK, swap them around, put your right hand up. Cross your hands over, so your right hand goes over, interlace your fingers like this, then make sure your right thumb is outside your left thumb — that ’ s very important. Yours is the other way around, so swap it around. Excellent, OK. Extend your fingers like this for me. All right. Tap them together once. OK, now, if you did not allow me to deceive your minds, you would all be able to do this. Now you can see how easy it is for me to manipulate the human mind, once you know how. Now, I remember when I was about 15, I read a copy of Life magazine, which detailed a story about a 75-year-old blind Russian woman who could sense printed letters — there ’ s still people trying to do it — — who could sense printed letters and even sense colors, just by touch. And she was completely blind. She could also read the serial numbers on bills when they were placed, face down, on a hard surface. Now, I was fascinated, but at the same time, skeptical. How could somebody read using their fingertips? You know, if you actually think about it, if somebody is totally blind — a guy yesterday did a demonstration in one of the rooms, where people had to close their eyes and they could just hear things. And it ’ s just a really weird thing to try and figure out. How could somebody read using their fingertips? Now earlier on, as part of a TV show that I have coming up on MTV, I attempted to give a similar demonstration of what is now known as second sight. So, let ’ s take a look. Man : There we go. I ’ m going to guide you into the car. Kathryn Thomas : Man : You ’ re OK, keep on going. KT : How are you? Keith Barry : Kathryn, it ’ s Keith here. I ’ m going to take you to a secret location, OK? Kathryn, there was no way you could see through that blindfold. KT : OK, but don ’ t say my name like that. KB : But you ’ re OK? KT : Yes. KB : No way to see through it? KT : No. KB : I ’ ll take it off. Do you want to take the rest off? Take it off, you ’ re OK. We ’ ll stop for a second. KT : I ’ m so afraid of what I ’ m going to see. KB : You ’ re fine, take it off. You ’ re OK. You ’ re safe. Have you ever heard of second sight? KT : No. KB : Second sight is whereby a mind-control expert can see through somebody else ’ s eyes. And I ’ m going to try that right now. KT : God. KB : Are you ready? Where is it? There ’ s no way — KT : KT : Oh, my God! KB : Don ’ t say anything, I ’ m trying to see through your eyes. I can ’ t see. KT : There ’ s a wall, there ’ s a wall. KB : Look at the road, look at the road. KT : OK, OK, OK. Oh, my God! KB : Now, anything coming at all? KT : No. KB : Sure there ’ s not? KT : No, no, I ’ m just still looking at the road. I ’ m looking at the road, all the time. I ’ m not taking my eyes off the road. KT : Oh, my God! KB : Where are we? Where are we? We ’ re going uphill, are we going uphill? KT : Look at the road — Still got that goddamn blindfold on. KB : What? KT : How are you doing this? KB : Just don ’ t break my concentration. We ’ re OK, though? KT : Yes. That ’ s so weird. We ’ re nearly there. Oh, my God! Oh, my God! KB : And I ’ ve stopped. KT : That is weird. You ’ re like a freak-ass of nature. That was the most scary thing I ’ ve ever done in my life! KB : Thank you. By the way, two days ago, we were going to film this down there, at the race course, and we got a guy into a car, and we got a camera man in the back, but halfway through the drive, he told me he had, I think it was a nine- millimeter, stuck to his leg. So, I stopped pretty quick, and that was it. So, do you believe it ’ s possible to see through somebody else ’ s eyes? That ’ s the question. Now, most people here would automatically say no. OK, but I want you to realize some facts. I couldn ’ t see through the blindfold. The car was not gimmicked or tricked in any way. The girl, I ’ d never met before, all right. So, I want you to just think about it for a moment. A lot of people try to come up with a logical solution to what just happened, all right. But because your brains are not trained in the art of deception, the solutions you come up with will, 99 percent of the time, be way off the mark. This is because magic is all about directing attention. If, for instance, I didn ’ t want you to look at my right hand, then I don ’ t look at it. But if I wanted you to look at my right hand, then I look at it, too. You see, it ’ s very simple, once you know how, but very complicated in other ways. I ’ m going to give you some demonstrations right now. I need two people to help me out real quick. Can you come up? And let ’ s see, down at the end, here, can you also come up, real quick? Do you mind? Yes, at the end. OK, give them a round of applause as they come up. You might want to use the stairs, there. It ’ s very important for everybody here to realize I haven ’ t set anything up with you. You don ’ t know what will happen, right? Would you mind just standing over here for a moment? Your name is? Nicole : Nicole. KB : Nicole, and? KB : Actually, here ’ s the thing, answer it, answer it, answer it. Is it a girl? Man : They ’ ve already gone. KB : OK, swap over positions. Can you stand over here? This will make it easier. Pity, I would have told them it was the ace of spades. OK, a little bit closer. OK, a little bit closer, come over — they look really nervous up here. Do you believe in witchcraft? Nicole : No. KB : Voodoo? Nicole : No. KB : Things that go bump in the night? Nicole : No. KB : Besides, who ’ s next, no, OK. I want you to just stand exactly like this for me, pull up your sleeves, if you don ’ t mind. OK, now, I want you to be aware of all the different sensations around you, because we ’ ll try a voodoo experiment. I want you to be aware of the sensations, but don ’ t say anything until I ask you, and don ’ t open your eyes until I ask you. From this point onwards, close your eyes, do not say anything, do not open them, be aware of the sensations. Yes or no, did you feel anything? Nicole : Yes. KB : You did feel that? What did you feel? Nicole : A touch on my back. KB : How many times did you feel it? Nicole : Twice. KB : Twice. OK, extend your left arm out in front of you. Extend your left arm, OK. OK, keep it there. Be aware of the sensations, don ’ t say anything, don ’ t open your eyes, OK. Did you feel anything, there? Nicole : Yes. KB : What did you feel? Nicole : Three — KB : Like a tickling sensation? Nicole : Yes. KB : Can you show us where? OK, excellent. Open your eyes. I never touched you. I just touched his back, and I just touched his arm. A voodoo experiment. Yeah, I walk around nightclubs all night like this. You just take a seat over here for a second. I ’ m going to use you again, in a moment. Can you take a seat right over here, if you don ’ t mind. Sit right here. Man : OK. KB : OK, take a seat. Excellent, OK. Now, what I want you to do is look directly at me, OK, just take a deep breath in through your nose, letting it out through your mouth, and relax. Allow your eyes to close, on five, four, three, two, one. Close your eyes right now. OK, now, I ’ m not hypnotizing you, I ’ m merely placing you in a heightened state of synchronicity, so our minds are along the same lines. And as you sink and drift and float into this relaxed state of mind, I ’ m going to take your left hand, and just place it up here. I want you to hold it there just for a moment, and I only want you to allow your hand to sink and drift and float back to the tabletop at the same rate and speed as you drift and float into this relaxed state of awareness, and allow it to go all the way down to the tabletop. That ’ s it, all the way down, all the way down. and further, and further. Excellent. I want you to allow your hand to stick firmly to the tabletop. OK, now, allow it to stay there. OK, now, in a moment, you ’ ll feel a certain pressure, OK, and I want you to be aware of the pressure. Just be aware of the pressure. And I only want you to allow your hand to float slowly back up from the tabletop as you feel the pressure release, but only when you feel the pressure release. Do you understand? Just answer yes or no. Man : Yes. KB : Hold it right there. And only when you feel the pressure go back, allow your hand to drift back to the tabletop, but only when you feel the pressure. OK, that was wonderfully done. Let ' s try it again. Excellent. Now that you ' ve got the idea, let ' s try something even more interesting. Allow it to stick firmly to the tabletop, keep your eyes closed. Can you stand up? Just stand, stage forward. I want you to point directly at his forehead. OK. Imagine a connection between you and him. Only when you want the pressure to be released, make an upward gesture, like this, but only when you want it to be released. You can wait as long as you want, but only when you want the pressure released. OK, let ' s try it again. OK, now, imagine the connection, OK. Point directly at his forehead. Only when you want the pressure released, we ' ll try it again. OK, it worked that time, excellent. And hold it there, both of you. Only when you want the pressure to go back, make a downward gesture. You can wait as long as you want. You did it pretty quickly, but it went down, OK. Now, I want you to be aware that in a moment, when I snap my fingers, your eyes will open, again. It ' s OK to remember to forget, or forget to remember what happened. Most people ask you,"What the hell just happened up here?"But it ' s OK that even though you ' re not hypnotized, you will forget everything that happened. On five, four, three, two, one — open your eyes, wide awake. Give them a round of applause, as they go back to their seats. OK, you can go back. I once saw a film called"The Gods Are Crazy."Has anybody here seen that film? Yeah. Remember when they threw the Coke out of the airplane, and it landed on the ground, and it didn ' t break? Now, see, that ' s because Coke bottles are solid. It ' s nearly impossible to break a Coke bottle. Do you want to try it? Good job. She ' s not taking any chances. You see, psychokinesis is the paranormal influence of the mind on physical events and processes. For some magicians or mentalists, sometimes the spoon will bend or melt, sometimes it will not. Sometimes the object will slide across the table, sometimes it will not. It depends on how much energy you have that day, so on and so forth. We ' re going to try an experiment in psychokinesis, right now. Come right over here, next to me. Excellent. Now, have a look at the Coke bottle. Make sure it is solid, there ' s only one hole, and it ' s a normal Coke bottle. And you can whack it against the table, if you want ; be careful. Even though it ' s solid, I ' m standing away. I want you to pinch right here with two fingers and your thumb. Excellent. Now, I ' ve got a shard of glass here, OK. I want you to examine the shard of glass. Careful, it ' s sharp. Just hold on to it for a moment. Now, hold it out here. I want you to imagine, right now, a broken relationship from many years ago. I want you to imagine all the negative energy from that broken relationship, from that guy, being imparted into the broken piece of glass, which will represent him, OK. But I want you to take this very seriously. Stare at the glass, ignore everybody right here. In a moment, you ' ll feel a certain sensation, OK, and when you feel that sensation, I want you to drop the piece of glass into the bottle. Think of that guy, that ba — that guy. I ' m trying to be good here. OK, and when you feel the sensation — it might take a while — drop it into the glass. OK, drop it in. Now, imagine all that negative energy in there. Imagine his name and imagine him inside the glass. And I want you to release that negative energy by shaking it from side to side. That was a lot of negative energy, built up in there. I also want you to look at me and think of his name. OK, think of how many letters in the title of his name. There ' s five letters in the title. You didn ' t react to that, so it ' s four letters. Think of one of the letters in the title. There

’ s a K in his name, there is a K. I knew that because my name starts with a K also, but his name doesn ’ t start with a K, it starts with an M. Tell Mike I said hello, the next time you see him. Was that his name? Nicole : Mm-hmm. KB : OK, give her a round of applause. Thank you. I ’ ve got one more thing to share with you right now. Actually, Chris, I was going to pick you for this, but instead of picking you, can you hop up here and pick a victim for this next experiment? And it should be a male victim, that ’ s the only thing. Chris Anderson : Oh, OK. KB : I was going to use you, but I decided I might want to come back another year. CA : Well, to reward him for saying”eureka,”and for selecting Michael Mercil to come and talk to us — Steve Jurvetson. KB : OK, Steve, come on up here. CA : You knew! KB : OK, Steve, I want you to take a seat, right behind here. Excellent. Now, Steve — oh, you can check underneath. Go ahead, I ’ ve no fancy assistants underneath there. They insist that because I was a magician, put a nice, black tablecloth on. There you are, OK. I ’ ve got four wooden plinths here, Steve. One, two, three and four. Now, they ’ re all the exact same except this one obviously has a stainless steel spike sticking out of it. I want you to examine it, and make sure it ’ s solid. Happy? Steve Jurvetson : Mmm, yes. KB : OK. Now, Steve, I ’ m going to stand in front of the table, When I stand in front of the table, I want you to put the cups on the plinths, in any order you want, and then mix them all up, so nobody has any idea where the spike is, all right? SJ : No one in the audience? KB : Yes, and just to help you out, I ’ ll block them from view, so nobody can see what you ’ re doing. I ’ ll also look away. So, go ahead and mix them up, now. OK, and tell me when you ’ re done. KB : You done? SJ : Almost. KB : Almost, oh. OK, you ’ re making sure that ’ s well hidden. Oh, we ’ ve got one here, we ’ ve got one here. So, all right, we ’ ll leave them like that. I ’ m going to have the last laugh, though. Now, Steve, you know where the spike is, but nobody else, does? Correct? But I don ’ t want you to know either, so swivel around on your chair. They ’ ll keep an eye on me to make sure I don ’ t do anything funny. No, stay around, OK. Now, Steve, look back. Now you don ’ t know where the spike is, and I don ’ t know where it is either. Now, is there any way to see through this blindfold? SJ : Put this on? KB : No, just, is there any way to see through it? No? SJ : No, I can ’ t see through it. KB : Excellent. Now, I ’ m going to put on the blindfold. Don ’ t stack them up, OK. Give them an extra mix up. Don ’ t move the cups, I don ’ t want anybody to see where the spike is, but give the plinths an extra mix up, and then line them up. I ’ ll put the blindfold on. Give them an extra mix up. No messing around this time. OK, go ahead, mix them up. My hand is at risk here. Tell me when you ’ re done. SJ : Done. KB : OK, where are you? Put out your hand. Your right hand. Tell me when I ’ m over a cup. SJ : You ’ re over a cup. KB : I ’ m over a cup, right now? SJ : Mm-hmm. KB : Now, Steve, do you think it ’ s here? Yes or no? SJ : Oh! KB : I told you I ’ d have the last laugh. SJ : I don ’ t think it ’ s there. KB : No? Good decision. Now, if I go this way, is there another cup over here? SJ : Can we do the left hand? KB : Oh, no, no, no. He asked me could he do the left hand. Absolutely not. KB : If I go this way, is there another cup? SJ : Yes. KB : Tell me when to stop. SJ : OK. KB : There? SJ : Yes, there ’ s one. KB : OK. Do you think it ’ s here, yes or no? This is your decision, not mine. SJ : I ’ m going to say no. KB : Good decision. OK, give me both hands. Now, put them on both cups. Do you think the spike is under your left or your right hand? SJ : Neither. KB : Neither, oh, OK. But if you were to guess. SJ : Under my right hand. KB : Under your right hand? Now, remember, you made all the decisions all along. Psychologists, figure this out. Have a look. SJ : Oh! KB : Thank you. If anybody wants to see some sleight of hand later on, I ’ ll be outside. Thank you. Thank you. Thank you.

Text Sample 330: NEG Dim 2 – 2004 – Score 2.00 – 2004/t000179.txt

I became an inventor by accident. I was out of the air force in 1956. No, no, that ’ s not true : I went in in 1956, came out in 1959, was working at the University of Washington, and I came up with an idea, from reading a magazine article, for a new kind of a phonograph tone arm. Now, that was before cassette tapes, C. D. s, DVDs — any of the cool stuff we ’ ve got now. And it was an arm that, instead of hinging and pivoting as it went across the record, went straight : a radial, linear tracking tone arm. And it was the hardest invention I ever made, but it got me started, and I got really lucky after that. And without giving you too much of a tirade, I want to talk to you about an invention I brought with me today : my 44th invention. No, that ’ s not true either. Golly, I ’ m just totally losing it. My 44th patent ; about the 15th invention. I call this hypersonic sound. I ’ m going to play it for you in a couple minutes, but I want to make an analogy before I do to this. I usually show this hypersonic sound and people will say, That ’ s really cool, but what ’ s it good for? And I say, What is the light bulb good for? Sound, light : I ’ m going to draw the analogy. When Edison invented the light bulb, pretty much looked like this. Hasn ’ t changed that much. Light came out of it in every direction. Before the light bulb was invented, people had figured out how to put a reflector behind it, focus it a little bit ; put lenses in front of it, focus it a little bit better. Ultimately we figured out how to make things like lasers that were totally focused. Now, think about where the world would be today if we had the light bulb, but you couldn ’ t focus light ; if when you turned one on it just went wherever it wanted to. That ’ s the way loudspeakers pretty much are. You turn on the loudspeaker, and after almost 80 years of having those gadgets, the sound just kind of goes where it wants. Even when you ’ re standing in front of a megaphone, it ’ s pretty much every direction. A little bit of differential, but not much. If the light bulb was the way the speaker is, and you couldn ’ t focus or sharpen the edges or define it, we wouldn ’ t have that, or movies in general, or computers, or T. V. sets, or C. D. s, or DVDs — and just go down the list of what the importance is of being able to focus light. Now, after almost 80 years of having sound, I thought it was about time that we figure out a way to put sound where you want to. I have a couple of units. That guy there was made for a demo I did yesterday early in the day for a big car maker in Detroit who wants to put them in a car — small version, over your head — so that you can actually get binaural sound in a car. What if I could aim sound the way I aim light? I got this waterfall I recorded in my back yard. Now, you ’ re not going to hear a thing unless it hits you. Maybe if I hit the side wall it will bounce around the room. The sound is being made right next to your ears. Is that cool? Because I have some limited time, I ’ ll cut it off for a second, and tell you about how it works and what it ’ s good for. Course, like light, it ’ s great to be able to put sound to highlight a clothing rack, or the cornflakes, or the toothpaste, or a talking plaque in a movie theater lobby. Sony ’ s got an idea — Sony ’ s our biggest customers right now. They tried this back in the ’ 60s and were too smart, and so they gave up. But they want to use it — seriously. There ’ s a mix an inventor has to have. You have to be kind of smart, and though I did not graduate from college doesn ’ t mean I ’ m stupid, because you cannot be stupid and do very much in the world today. Too many other smart people out there. So. I just happened to get my education in a little different way. I ’ m not at all against education. I think it ’ s wonderful ; I think sometimes people, when they get educated, lose it : they get so smart they ’ re unwilling to look at things that they know better than. And we ’ re living in a great time right now, because almost everything ’ s being explored anew. I have this little slogan that I use a lot, which is : virtually nothing — and I mean this honestly — has been invented yet. We ' re just starting. We ' re just starting to really discover the laws of nature and science and physics. And this is, I hope, a little piece of it. Sony ' s got this vision back — to get myself on track — that when you stand in the checkout line in the supermarket, you ' re going to watch a new T. V. channel. They know that when you watch T. V. at home, because there are so many choices you can change channels, miss their commercials. A hundred and fifty- one million people every day stand in the line at the supermarket. Now, they ' ve tried this a couple years ago and it failed, because the checker gets tired of hearing the same message every 20 minutes, and reaches out, turns off the sound. And, you know, if the sound isn ' t there, the sale typically isn ' t made. For instance, like, when you ' re on an airplane, they show the movie, you get to watch it for free ; when you want to hear the sound, you pay. And so ABC and Sony have devised this new thing where when you step in the line in the supermarket — initially it ' ll be Safeways. It is Safeways ; they ' re trying this in three parts of the country right now — you ' ll be watching TV. And hopefully they ' ll be sensitive that they don ' t want to offend you with just one more outlet. But what ' s great about it, from the tests that have been done, is, if you don ' t want to hear it, you take about one step to the side and you don ' t hear it. So, we create silence as much as we create sound. ATMs that talk to you ; nobody else hears it. Sit in bed, two in the morning, watch TV ; your spouse, or someone, is next to you, asleep ; doesn ' t hear it, doesn ' t wake up. We ' re also working on noise canceling things like snoring, noise from automobiles. I have been really lucky with this technology : all of a sudden as it is ready, the world is ready to accept it. They have literally beat a path to our door. We ' ve been selling it since about last September, October, and it ' s been immensely gratifying. If you ' re interested in what it costs — I ' m not selling them today — but this unit, with the electronics and everything, if you buy one, is around a thousand bucks. We expect by this time next year, it ' ll be hundreds, a few hundred bucks, to buy it. It ' s not any more pricey than regular electronics. Now, when I played it for you, you didn ' t hear the thunderous bass. This unit that I played goes from about 200 hertz to above the range of hearing. It ' s actually emitting ultrasound — low-level ultrasound — that ' s about 100, 000 vibrations per second. And the sound that you ' re hearing, unlike a regular speaker on which all the sound is made on the face, is made out in front of it, in the air. The air is not linear, as we ' ve always been taught. You turn up the volume just a little bit — I ' m talking about a little over 80 decibels — and all of a sudden the air begins to corrupt signals you propagate. Here ' s why : the speed of sound is not a constant. It ' s fairly slow. It changes with temperature and with barometric pressure. Now, imagine, if you will, without getting too technical, I ' m making a little sine wave here in the air. Well, if I turn up the amplitude too much, I ' m having an effect on the pressure, which means during the making of that sine wave, the speed at which it is propagating is shifting. All of audio as we know it is an attempt to be more and more perfectly linear. Linearity means higher quality sound. Hypersonic sound is exactly the opposite : it ' s 100 percent based on non-linearity. An effect happens in the air, it ' s a corrupting effect of the sound — the ultrasound in this case — that ' s emitted, but it ' s so predictable that you can produce very precise audio out of that effect. Now, the question is, where ' s the sound made? Instead of being made on the face of the cone, it ' s made at literally billions of little independent points along this narrow column in the air, and so when I aim it towards you, what you hear is made right next to your ears. I said we can shorten the column, we can spread it out to cover the couch. I can put it so that one ear hears one speaker, the other ear hears the other.

That ’ s true binaural sound. When you listen to stereo on your home system, your both ears hear both speakers. Turn on the left speaker sometime and notice you ’ re hearing it also in your right ear. So, the stage is more restricted — the sound stage that ’ s supposed to spread out in front of you. Because the sound is made in the air along this column, it does not follow the inverse square law, which says it drops off about two thirds every time you double the distance : 6dB every time you go from one meter, for instance, to two meters. That means you go to a rock concert or a symphony, and the guy in the front row gets the same level as the guy in the back row, now, all of a sudden. Isn ’ t that terrific? So, we ’ ve been, as I say, very successful, very lucky, in having companies catch the vision of this, from cars — car makers who want to put a stereo system in the front for the kids, and a separate system in the back — oh, no, the kids aren ’ t driving today. I was seeing if you were listening. Actually, I haven ’ t had breakfast yet. A stereo system in the front for mom and dad, and maybe there ’ s a little DVD player in the back for the kids, and the parents don ’ t want to be bothered with that, or their rap music or whatever. So, again, this idea of being able to put sound anywhere you want to is really starting to catch on. It also works for transmitting and communicating data. It also works five times better underwater. We ’ ve got the military — have just deployed some of these into Iraq, where you can put fake troop movements quarter of a mile away on a hillside. Or you can whisper in the ear of a supposed terrorist some Biblical verse. I ’ m serious. And they have these infrared devices that can look at their countenance, and see a fraction of a degree Kelvin in temperature shift from 100 yards away when they play this thing. And so, another way of hopefully determining who ’ s friendly and who isn ’ t. We make a version with this which puts out 155 decibels. Pain is 120. So it allows you to go nearly a mile away and communicate with people, and there can be a public beach just off to the side, and they don ’ t even know it ’ s turned on. We sell those to the military presently for about 70, 000 dollars, and they ’ re buying them as fast as we can make them. We put it on a turret with a camera, so that when they shoot at you, you ’ re over there, and it ’ s there. I have a bunch of other inventions. I invented a plasma antenna, to shift gears. Looked up at the ceiling of my office one day — I was working on a ground-penetrating radar project — and my physicist CEO came in and said,”We have a real problem. We ’ re using very short wavelengths. We ’ ve got a problem with the antenna ringing. When you run very short wavelengths, like a tuning fork the antenna resonates, and there ’ s more energy coming out of the antenna than there is the backscatter from the ground that we ’ re trying to analyze, taking too much processing.”I says,”Why don ’ t we make an antenna that only exists when you want it? Turn it on ; turn it off. That ’ s a fluorescent tube refined.”I just sold that for a million and a half dollars, cash. I took it back to the Pentagon after it got declassified, when the patent issued, and told the people back there about it, and they laughed, and then I took them back a demo and they bought. Any of you ever wore a Jabber headphone — the little cell headphones? That ’ s my invention. I sold that for seven million dollars. Big mistake : it just sold for 80 million dollars two years ago. I actually drew that up on a little crummy Mac computer in my attic at my house, and one of the many designs which they have now is still the same design I drew way back when. So, I ’ ve been really lucky as an inventor. I ’ m the happiest guy you ’ re ever going to meet. And my dad died before he realized anybody in the family would maybe, hopefully, make something out of themselves. You ’ ve been a great audience. I know I ’ ve jumped all over the place. I usually figure out what my talk is when I get up in front of a group. Let me give you, in the last minute, one more quick demo of this guy, for those of you that haven ’ t heard it. Can never tell if it ’ s on. If you haven ’ t heard it, raise your hand. Getting it over there? Get the cameraman. Yeah, there you go. I ’ ve got a Coke can opening that ’ s right in your head ; that ’ s really cool. Thank you once again. Appreciate it very much.

Text Sample 331: NEG Dim 2 – 2005 – Score 1.00 – 2005/t000463.txt

Thank you. It ’ s really an honor and a privilege to be here spending my last day as a teenager. Today I want to talk to you about the future, but first I ’ m going to tell you a bit about the past. My story starts way before I was born. My grandmother was on a train to Auschwitz, the death camp. And she was going along the tracks, and the tracks split. And somehow — we don ’ t really know exactly the whole story — but the train took the wrong track and went to a work camp rather than the death camp. My grandmother survived and married my grandfather. They were living in Hungary, and my mother was born. And when my mother was two years old, the Hungarian revolution was raging, and they decided to escape Hungary. They got on a boat, and yet another divergence — the boat was either going to Canada or to Australia. They got on and didn ’ t know where they were going, and ended up in Canada. So, to make a long story short, they came to Canada. My grandmother was a chemist. She worked at the Banting Institute in Toronto, and at 44 she died of stomach cancer. I never met my grandmother, but I carry on her name — her exact name, Eva Vertes — and I like to think I carry on her scientific passion, too. I found this passion not far from here, actually, when I was nine years old. My family was on a road trip and we were in the Grand Canyon. And I had never been a reader when I was young — my dad had tried me with the Hardy Boys ; I tried Nancy Drew ; I tried all that — and I just didn ’ t like reading books. And my mother bought this book when we were at the Grand Canyon called”The Hot Zone.”It was all about the outbreak of the Ebola virus. And something about it just kind of drew me towards it. There was this big sort of bumpy-looking virus on the cover, and I just wanted to read it. I picked up that book, and as we drove from the edge of the Grand Canyon to Big Sur, and to, actually, here where we are today, in Monterey, I read that book, and from when I was reading that book, I knew that I wanted to have a life in medicine. I wanted to be like the explorers I ’ d read about in the book, who went into the jungles of Africa, went into the research labs and just tried to figure out what this deadly virus was. So from that moment on, I read every medical book I could get my hands on, and I just loved it so much. I was a passive observer of the medical world. It wasn ’ t until I entered high school that I thought,”Maybe now, you know — being a big high school kid — I can maybe become an active part of this big medical world.”I was 14, and I emailed professors at the local university to see if maybe I could go work in their lab. And hardly anyone responded. But I mean, why would they respond to a 14-year-old, anyway? And I got to go talk to one professor, Dr. Jacobs, who accepted me into the lab. At that time, I was really interested in neuroscience and wanted to do a research project in neurology — specifically looking at the effects of heavy metals on the developing nervous system. So I started that, and worked in his lab for a year, and found the results that I guess you ’ d expect to find when you feed fruit flies heavy metals — that it really, really impaired the nervous system. The spinal cord had breaks. The neurons were crossing in every which way. And from then I wanted to look not at impairment, but at prevention of impairment. So that ’ s what led me to Alzheimer ’ s. I started reading about Alzheimer ’ s and tried to familiarize myself with the research, and at the same time when I was in the — I was reading in the medical library one day, and I read this article about something called”purine derivatives.”And they seemed to have cell growth-promoting properties. And being naive about the whole field, I kind of thought,”Oh, you have cell death in Alzheimer ’ s which is causing the memory deficit, and then you have this compound — purine derivatives — that are promoting cell growth.”And so I thought,”Maybe if it can promote cell growth, it can inhibit cell death, too.”And so that ’ s the project that I pursued for that year, and it ’ s continuing now as well, and found that a specific purine derivative called”guanidine”had inhibited the cell growth by approximately 60 percent. So I presented those results at the International Science Fair, which was just one of the most amazing experiences of my life. And there I was awarded”Best in the World in Medicine,”which allowed me to get in, or at least get a foot in the door of the big medical world. And from then on, since I was now in this huge exciting world, I wanted to explore it all. I wanted it all at once, but knew I couldn ’ t really get that. And I stumbled across something called”cancer stem cells.”And this is really what I want to talk to you about today — about cancer. At first when I heard of cancer stem cells, I didn ’ t really know how to put the two together. I ’ d heard of stem cells, and I ’ d heard of them as the panacea of the future — the therapy of many diseases to come in the future, perhaps. But I ’ d heard of cancer as the most feared disease of our time, so how did the good and bad go together? Last summer I worked at Stanford University, doing some research on cancer stem cells. And while I was doing this, I was reading the cancer literature, trying to — again — familiarize myself with this new medical field. And it seemed that tumors actually begin from a stem cell. This fascinated me. The more I read, the more I looked at cancer differently and almost became less fearful of it. It seems that cancer is a direct result to injury. If you smoke, you damage your lung tissue, and then lung cancer arises. If you drink, you damage your liver, and then liver cancer occurs. And it was really interesting — there were articles correlating if you have a bone fracture, and then bone cancer arises. Because what stem cells are — they ’ re these phenomenal cells that really have the ability to differentiate into any type of tissue. So, if the body is sensing that you have damage to an organ and then it ’ s initiating cancer, it ’ s almost as if this is a repair response. And the cancer, the body is saying the lung tissue is damaged, we need to repair the lung. And cancer is originating in the lung trying to repair — because you have this excessive proliferation of these remarkable cells that really have the potential to become lung tissue. But it ’ s almost as if the body has originated this ingenious response, but can ’ t quite control it. It hasn ’ t yet become fine-tuned enough to finish what has been initiated. So this really, really fascinated me. And I really think that we can ’ t think about cancer — let alone any disease — in such black-and-white terms. If we eliminate cancer the way we ’ re trying to do now, with chemotherapy and radiation, we ’ re bombarding the body or the cancer with toxins, or with radiation, trying to kill it. It ’ s almost as if we ’ re getting back to this starting point. We ’ re removing the cancer cells, but we ’ re revealing the previous damage that the body has tried to fix. Shouldn ’ t we think about manipulation, rather than elimination? If somehow we can cause these cells to differentiate — to become bone tissue, lung tissue, liver tissue, whatever that cancer has been put there to do — it would be a repair process. We ’ d end up better than we were before cancer. So, this really changed my view of looking at cancer. And while I was reading all these articles about cancer, it seemed that the articles — a lot of them — focused on, you know, the genetics of breast cancer, and the genesis and the progression of breast cancer — tracking the cancer through the body, tracing where it is, where it goes. But it struck me that I ’ d never heard of cancer of the heart, or cancer of any skeletal muscle for that matter. And skeletal muscle constitutes 50 percent of our body, or over

50 percent of our body. And so at first I kind of thought,”Well, maybe there ’ s some obvious explanation why skeletal muscle doesn ’ t get cancer — at least not that I know of.”So, I looked further into it, found as many articles as I could, and it was amazing — because it turned out that it was very rare. Some articles even went as far as to say that skeletal muscle tissue is resistant to cancer, and furthermore, not only to cancer, but of metastases going to skeletal muscle. And what metastases are is when the tumor — when a piece — breaks off and travels through the blood stream and goes to a different organ. That ’ s what a metastasis is. It ’ s the part of cancer that is the most dangerous. If cancer was localized, we could likely remove it, or somehow — you know, it ’ s contained. It ’ s very contained. But once it starts moving throughout the body, that ’ s when it becomes deadly. So the fact that not only did cancer not seem to originate in skeletal muscles, but cancer didn ’ t seem to go to skeletal muscle — there seemed to be something here. So these articles were saying, you know,”Skeletal — metastasis to skeletal muscle — is very rare.”But it was left at that. No one seemed to be asking why. So I decided to ask why. At first — the first thing I did was I emailed some professors who specialized in skeletal muscle physiology, and pretty much said,”Hey, it seems like cancer doesn ’ t really go to skeletal muscle. Is there a reason for this?”And a lot of the replies I got were that muscle is terminally differentiated tissue. Meaning that you have muscle cells, but they ’ re not dividing, so it doesn ’ t seem like a good target for cancer to hijack. But then again, this fact that the metastases didn ’ t go to skeletal muscle made that seem unlikely. And furthermore, that nervous tissue — brain — gets cancer, and brain cells are also terminally differentiated. So I decided to ask why. And here ’ s some of, I guess, my hypotheses that I ’ ll be starting to investigate this May at the Sylvester Cancer Institute in Miami. And I guess I ’ ll keep investigating until I get the answers. But I know that in science, once you get the answers, inevitably you ’ re going to have more questions. So I guess you could say that I ’ ll probably be doing this for the rest of my life. Some of my hypotheses are that when you first think about skeletal muscle, there ’ s a lot of blood vessels going to skeletal muscle. And the first thing that makes me think is that blood vessels are like highways for the tumor cells. Tumor cells can travel through the blood vessels. And you think, the more highways there are in a tissue, the more likely it is to get cancer or to get metastases. So first of all I thought, you know,”Wouldn ’ t it be favorable to cancer getting to skeletal muscle?”And as well, cancer tumors require a process called angiogenesis, which is really, the tumor recruits the blood vessels to itself to supply itself with nutrients so it can grow. Without angiogenesis, the tumor remains the size of a pinpoint and it ’ s not harmful. So angiogenesis is really a central process to the pathogenesis of cancer. And one article that really stood out to me when I was just reading about this, trying to figure out why cancer doesn ’ t go to skeletal muscle, was that it had reported 16 percent of micro-metastases to skeletal muscle upon autopsy. 16 percent! Meaning that there were these pinpoint tumors in skeletal muscle, but only. 16 percent of actual metastases — suggesting that maybe skeletal muscle is able to control the angiogenesis, is able to control the tumors recruiting these blood vessels. We use skeletal muscles so much. It ’ s the one portion of our body — our heart ’ s always beating. We ’ re always moving our muscles. Is it possible that muscle somehow intuitively knows that it needs this blood supply? It needs to be constantly contracting, so therefore it ’ s almost selfish. It ’ s grabbing its blood vessels for itself. Therefore, when a tumor comes into skeletal muscle tissue, it can ’ t get a blood supply, and can ’ t grow. So this suggests that maybe if there is an anti-angiogenic factor in skeletal muscle — or perhaps even more, an angiogenic routing factor, so it can actually direct

where the blood vessels grow — this could be a potential future therapy for cancer. And another thing that ' s really interesting is that there ' s this whole — the way tumors move throughout the body, it ' s a very complex system — and there ' s something called the chemokine network. And chemokines are essentially chemical attractants, and they ' re the stop and go signals for cancer. So a tumor expresses chemokine receptors, and another organ — a distant organ somewhere in the body — will have the corresponding chemokines, and the tumor will see these chemokines and migrate towards it. Is it possible that skeletal muscle doesn ' t express this type of molecules? And the other really interesting thing is that when skeletal muscle — there ' s been several reports that when skeletal muscle is injured, that ' s what correlates with metastases going to skeletal muscle. And, furthermore, when skeletal muscle is injured, that ' s what causes chemokines — these signals saying,"Cancer, you can come to me,"the"go signs"for the tumors — it causes them to highly express these chemokines. So, there ' s so much interplay here. I mean, there are so many possibilities for why tumors don ' t go to skeletal muscle. But it seems like by investigating, by attacking cancer, by searching where cancer is not, there has got to be something — there ' s got to be something — that ' s making this tissue resistant to tumors. And can we utilize — can we take this property, this compound, this receptor, whatever it is that ' s controlling these anti-tumor properties and apply it to cancer therapy in general? Now, one thing that kind of ties the resistance of skeletal muscle to cancer — to the cancer as a repair response gone out of control in the body — is that skeletal muscle has a factor in it called"MyoD."And what MyoD essentially does is, it causes cells to differentiate into muscle cells. So this compound, MyoD, has been tested on a lot of different cell types and been shown to actually convert this variety of cell types into skeletal muscle cells. So, is it possible that the tumor cells are going to the skeletal muscle tissue, but once in contact inside the skeletal muscle tissue, MyoD acts upon these tumor cells and causes them to become skeletal muscle cells? Maybe tumor cells are being disguised as skeletal muscle cells, and this is why it seems as if it is so rare. It ' s not harmful ; it has just repaired the muscle. Muscle is constantly being used — constantly being damaged. If every time we tore a muscle or every time we stretched a muscle or moved in a wrong way, cancer occurred — I mean, everybody would have cancer almost. And I hate to say that. But it seems as though muscle cell, possibly because of all its use, has adapted faster than other body tissues to respond to injury, to fine-tune this repair response and actually be able to finish the process which the body wants to finish. I really believe that the human body is very, very smart, and we can ' t counteract something the body is saying to do. It ' s different when a bacteria comes into the body — that ' s a foreign object — we want that out. But when the body is actually initiating a process and we ' re calling it a disease, it doesn ' t seem as though elimination is the right solution. So even to go from there, it ' s possible, although far-fetched, that in the future we could almost think of cancer being used as a therapy. If those diseases where tissues are deteriorating — for example Alzheimer ' s, where the brain, the brain cells, die and we need to restore new brain cells, new functional brain cells — what if we could, in the future, use cancer? A tumor — put it in the brain and cause it to differentiate into brain cells? That ' s a very far-fetched idea, but I really believe that it may be possible. These cells are so versatile, these cancer cells are so versatile — we just have to manipulate them in the right way. And again, some of these may be far- fetched, but I figured if there ' s anywhere to present far-fetched ideas, it ' s here at TED, so thank you very much.

Text Sample 332: NEG Dim 2 – 2005 – Score 1.00 – 2005/t000454.txt

18 minutes is an absolutely brutal time limit, so I ’ m going to dive straight in, right at the point where I get this thing to work. Here we go. I ’ m going to talk about five different things. I ’ m going to talk about why defeating aging is desirable. I ’ m going to talk about why we have to get our shit together, and actually talk about this a bit more than we do. I ’ m going to talk about feasibility as well, of course. I ’ m going to talk about why we are so fatalistic about doing anything about aging. And then I ’ m going spend perhaps the second half of the talk talking about, you know, how we might actually be able to prove that fatalism is wrong, namely, by actually doing something about it. I ’ m going to do that in two steps. The first one I ’ m going to talk about is how to get from a relatively modest amount of life extension — which I ’ m going to define as 30 years, applied to people who are already in middle-age when you start — to a point which can genuinely be called defeating aging. Namely, essentially an elimination of the relationship between how old you are and how likely you are to die in the next year — or indeed, to get sick in the first place. And of course, the last thing I ’ m going to talk about is how to reach that intermediate step, that point of maybe 30 years life extension. So I ’ m going to start with why we should. Now, I want to ask a question. Hands up : anyone in the audience who is in favor of malaria? That was easy. OK. OK. Hands up : anyone in the audience who ’ s not sure whether malaria is a good thing or a bad thing? OK. So we all think malaria is a bad thing. That ’ s very good news, because I thought that was what the answer would be. Now the thing is, I would like to put it to you that the main reason why we think that malaria is a bad thing is because of a characteristic of malaria that it shares with aging. And here is that characteristic. The only real difference is that aging kills considerably more people than malaria does. Now, I like in an audience, in Britain especially, to talk about the comparison with foxhunting, which is something that was banned after a long struggle, by the government not very many months ago. I mean, I know I ’ m with a sympathetic audience here, but, as we know, a lot of people are not entirely persuaded by this logic. And this is actually a rather good comparison, it seems to me. You know, a lot of people said,”Well, you know, city boys have no business telling us rural types what to do with our time. It ’ s a traditional part of the way of life, and we should be allowed to carry on doing it. It ’ s ecologically sound ; it stops the population explosion of foxes.”But ultimately, the government prevailed in the end, because the majority of the British public, and certainly the majority of members of Parliament, came to the conclusion that it was really something that should not be tolerated in a civilized society. And I think that human aging shares all of these characteristics in spades. What part of this do people not understand? It ’ s not just about life, of course — — it ’ s about healthy life, you know — getting frail and miserable and dependent is no fun, whether or not dying may be fun. So really, this is how I would like to describe it. It ’ s a global trance. These are the sorts of unbelievable excuses that people give for aging. And, I mean, OK, I ’ m not actually saying that these excuses are completely valueless. There are some good points to be made here, things that we ought to be thinking about, forward planning so that nothing goes too — well, so that we minimize the turbulence when we actually figure out how to fix aging. But these are completely crazy, when you actually remember your sense of proportion. You know, these are arguments ; these are things that would be legitimate to be concerned about. But the question is, are they so dangerous — these risks of doing something about aging — that they outweigh the downside of doing the opposite, namely, leaving aging as it is? Are these so bad that they outweigh condemning 100, 000 people a day to an unnecessarily early death?

You know, if you haven ’ t got an argument that ’ s that strong, then just don ’ t waste my time, is what I say. Now, there is one argument that some people do think really is that strong, and here it is. People worry about overpopulation ; they say,”Well, if we fix aging, no one ’ s going to die to speak of, or at least the death toll is going to be much lower, only from crossing St. Giles carelessly. And therefore, we ’ re not going to be able to have many kids, and kids are really important to most people.”And that ’ s true. And you know, a lot of people try to fudge this question, and give answers like this. I don ’ t agree with those answers. I think they basically don ’ t work. I think it ’ s true, that we will face a dilemma in this respect. We will have to decide whether to have a low birth rate, or a high death rate. A high death rate will, of course, arise from simply rejecting these therapies, in favor of carrying on having a lot of kids. And, I say that that ’ s fine — the future of humanity is entitled to make that choice. What ’ s not fine is for us to make that choice on behalf of the future. If we vacillate, hesitate, and do not actually develop these therapies, then we are condemning a whole cohort of people — who would have been young enough and healthy enough to benefit from those therapies, but will not be, because we haven ’ t developed them as quickly as we could — we ’ ll be denying those people an indefinite life span, and I consider that that is immoral. That ’ s my answer to the overpopulation question. Right. So the next thing is, now why should we get a little bit more active on this? And the fundamental answer is that the pro-aging trance is not as dumb as it looks. It ’ s actually a sensible way of coping with the inevitability of aging. Aging is ghastly, but it ’ s inevitable, so, you know, we ’ ve got to find some way to put it out of our minds, and it ’ s rational to do anything that we might want to do, to do that. Like, for example, making up these ridiculous reasons why aging is actually a good thing after all. But of course, that only works when we have both of these components. And as soon as the inevitability bit becomes a little bit unclear — and we might be in range of doing something about aging — this becomes part of the problem. This pro-aging trance is what stops us from agitating about these things. And that ’ s why we have to really talk about this a lot — evangelize, I will go so far as to say, quite a lot — in order to get people ’ s attention, and make people realize that they are in a trance in this regard. So that ’ s all I ’ m going to say about that. I ’ m now going to talk about feasibility. And the fundamental reason, I think, why we feel that aging is inevitable is summed up in a definition of aging that I ’ m giving here. A very simple definition. Aging is a side effect of being alive in the first place, which is to say, metabolism. This is not a completely tautological statement ; it ’ s a reasonable statement. Aging is basically a process that happens to inanimate objects like cars, and it also happens to us, despite the fact that we have a lot of clever self-repair mechanisms, because those self-repair mechanisms are not perfect. So basically, metabolism, which is defined as basically everything that keeps us alive from one day to the next, has side effects. Those side effects accumulate and eventually cause pathology. That ’ s a fine definition. So we can put it this way : we can say that, you know, we have this chain of events. And there are really two games in town, according to most people, with regard to postponing aging. They ’ re what I ’ m calling here the”gerontology approach”and the”geriatrics approach.”The geriatrician will intervene late in the day, when pathology is becoming evident, and the geriatrician will try and hold back the sands of time, and stop the accumulation of side effects from causing the pathology quite so soon. Of course, it ’ s a very short-term-ist strategy ; it ’ s a losing battle, because the things that are causing the pathology are becoming more abundant as time goes on. The gerontology approach looks much more promising on the surface, because, you know, prevention is better than cure. But unfortunately the thing is that we don ’ t understand metabolism very well. In fact, we have a pitifully poor understanding of how organisms work — even cells we ’ re not really too good on yet. We ’ ve discovered things like, for example, RNA interference only a few years ago, and this is a really fundamental component of how cells work. Basically, gerontology is a fine approach in the end, but it is not an approach whose time has come when we ’ re talking about intervention. So then, what do we do about that? I mean, that ’ s a fine logic, that sounds pretty convincing, pretty ironclad, doesn ’ t it? But it isn ’ t. Before I tell you why it isn ’ t, I ’ m going to go a little bit into what I ’ m calling step two. Just suppose, as I said, that we do acquire — let ’ s say we do it today for the sake of argument — the ability to confer 30 extra years of healthy life on people who are already in middle age, let ’ s say 55. I ’ m going to call that”robust human rejuvenation.”OK. What would that actually mean for how long people of various ages today — or equivalently, of various ages at the time that these therapies arrive — would actually live? In order to answer that question — you might think it ’ s simple, but it ’ s not simple. We can ’ t just say,”Well, if they ’ re young enough to benefit from these therapies, then they ’ ll live 30 years longer.”That ’ s the wrong answer. And the reason it ’ s the wrong answer is because of progress. There are two sorts of technological progress really, for this purpose. There are fundamental, major breakthroughs, and there are incremental refinements of those breakthroughs. Now, they differ a great deal in terms of the predictability of time frames. Fundamental breakthroughs : very hard to predict how long it ’ s going to take to make a fundamental breakthrough. It was a very long time ago that we decided that flying would be fun, and it took us until 1903 to actually work out how to do it. But after that, things were pretty steady and pretty uniform. I think this is a reasonable sequence of events that happened in the progression of the technology of powered flight. We can think, really, that each one is sort of beyond the imagination of the inventor of the previous one, if you like. The incremental advances have added up to something which is not incremental anymore. This is the sort of thing you see after a fundamental breakthrough. And you see it in all sorts of technologies. Computers : you can look at a more or less parallel time line, happening of course a bit later. You can look at medical care. I mean, hygiene, vaccines, antibiotics — you know, the same sort of time frame. So I think that actually step two, that I called a step a moment ago, isn ’ t a step at all. That in fact, the people who are young enough to benefit from these first therapies that give this moderate amount of life extension, even though those people are already middle-aged when the therapies arrive, will be at some sort of cusp. They will mostly survive long enough to receive improved treatments that will give them a further 30 or maybe 50 years. In other words, they will be staying ahead of the game. The therapies will be improving faster than the remaining imperfections in the therapies are catching up with us. This is a very important point for me to get across. Because, you know, most people, when they hear that I predict that a lot of people alive today are going to live to 1, 000 or more, they think that I ’ m saying that we ’ re going to invent therapies in the next few decades that are so thoroughly eliminating aging that those therapies will let us live to 1, 000 or more. I ’ m not saying that at all. I ’ m saying that the rate of improvement of those therapies will be enough. They ’ ll never be perfect, but we ’ ll be able to fix the things that 200- year-olds die of, before we have any 200-year-olds. And the same for 300 and 400 and so on. I decided to give this a little name, which is”longevity escape velocity.”Well, it seems to get the point across. So, these trajectories here are basically how we would expect people to live, in terms of remaining life expectancy, as measured by their health, for given ages that they were at the time that these therapies arrive. If you ’ re already 100, or even if you ’ re 80 — and an average 80-year-old, we probably can ’ t do a lot for you with these therapies, because you ’ re too close to death ’ s door for the really initial, experimental therapies to be good enough for you. You won ’ t be able to withstand them. But if you ’ re only 50, then there ’ s a chance that you might be able to pull out of the dive and, you know — — eventually get through this and start becoming biologically younger in a meaningful sense, in terms of your youthfulness, both physical and mental, and in terms of your risk of death from age-related causes. And of course, if you ’ re a bit younger than that, then you ’ re never really even going to get near to being fragile enough to die of age-related causes. So this is a genuine conclusion that I come to, that the first 150-year-old — we don ’ t know how old that person is today, because we don ’ t know how long it ’ s going to take to get these first-generation therapies. But irrespective of that age, I ’ m claiming that the first person to live to 1, 000 — subject of course, to, you know, global catastrophes — is actually, probably, only about 10 years younger than the first 150-year-old. And that ’ s quite a thought. Alright, so finally I ’ m going to spend the rest of the talk, my last seven-and- a-half minutes, on step one ; namely, how do we actually get to this moderate amount of life extension that will allow us to get to escape velocity? And in order to do that, I need to talk about mice a little bit. I have a corresponding milestone to robust human rejuvenation. I ’ m calling it”robust mouse rejuvenation,”not very imaginatively. And this is what it is. I say we ’ re going to take a long-lived strain of mouse, which basically means mice that live about three years on average. We do exactly nothing to them until they ’ re already two years old. And then we do a whole bunch of stuff to them, and with those therapies, we get them to live, on average, to their fifth birthday. So, in other words, we add two years — we treble their remaining lifespan, starting from the point that we started the therapies. The question then is, what would that actually mean for the time frame until we get to the milestone I talked about earlier for humans? Which we can now, as I ’ ve explained, equivalently call either robust human rejuvenation or longevity escape velocity. Secondly, what does it mean for the public ’ s perception of how long it ’ s going to take for us to get to those things, starting from the time we get the mice? And thirdly, the question is, what will it do to actually how much people want it? And it seems to me that the first question is entirely a biology question, and it ’ s extremely hard to answer. One has to be very speculative, and many of my colleagues would say that we should not do this speculation, that we should simply keep our counsel until we know more. I say that ’ s nonsense. I say we absolutely are irresponsible if we stay silent on this. We need to give our best guess as to the time frame, in order to give people a sense of proportion so that they can assess their priorities. So, I say that we have a 50 / 50 chance of reaching this RHR milestone, robust human rejuvenation, within 15 years from the point that we get to robust mouse rejuvenation. 15 years from the robust mouse. The public ’ s perception will probably be somewhat better than that. The public tends to underestimate how difficult scientific things are. So they ’ ll probably think it ’ s five years away. They ’ ll be wrong, but that actually won ’ t matter too much. And finally, of course, I think it ’ s fair to say that a large part of the reason why the public is so ambivalent about aging now is the global trance I spoke about earlier, the coping strategy. That will be history at this point, because it will no longer be possible to believe that aging is inevitable in humans, since it ’ s been postponed so very effectively in mice. So we ’ re likely to end up with a very strong change in people ’ s attitudes, and of course that has enormous implications. So in order to tell you now how we ’ re going to get these mice, I ’ m going to add a little bit to my description of aging. I ’ m going to use this word”damage”to denote these intermediate things that are caused by metabolism and that eventually cause pathology. Because the critical thing about this is that even though the damage only eventually causes pathology, the damage itself is caused ongoing-ly throughout life, starting before we ’ re born. But it is not part of metabolism itself. And this turns out to be useful. Because we can re-draw our original diagram this way. We can say that, fundamentally, the difference between gerontology and geriatrics is that gerontology tries to inhibit the rate at which metabolism lays down this damage. And I ’ m going to explain exactly what damage is in concrete biological terms in a moment. And geriatricians try to hold back the sands of time by stopping the damage converting into pathology. And the reason it ’ s a losing battle is because the damage is continuing to accumulate. So there ’ s a third approach, if we look at it this way. We can call it the”engineering approach,”and I claim that the engineering approach is within range. The engineering approach does not intervene in any processes. It does not intervene in this process or this one. And that ’ s good because it means that it ’ s not a losing battle, and it ’ s something that we are within range of being able to do, because it doesn ’ t involve improving on evolution. The engineering approach simply says,”Let ’ s go and periodically repair all of these various types of damage — not necessarily repair them completely, but repair them quite a lot, so that we keep the level of damage down below the threshold that must exist, that causes it to be pathogenic.”We know that this threshold exists, because we don ’ t get age-related diseases until we ’ re in middle age, even though the damage has been accumulating since before we were born. Why do I say that we ’ re in range? Well, this is basically it. The point about this slide is actually the bottom. If we try to say which bits of metabolism are important for aging, we will be here all night, because basically all of metabolism is important for aging in one way or another. This list is just for illustration ; it is incomplete. The list on the right is also incomplete. It ’ s a list of types of pathology that are age-related, and it ’ s just an incomplete list. But I would like to claim to you that this list in the middle is actually complete — this is the list of types of thing that qualify as damage, side effects of metabolism that cause pathology in the end, or that might cause pathology. And there are only seven of them. They ’ re categories of things, of course, but there ’ s only seven of them. Cell loss, mutations in chromosomes, mutations in the mitochondria and so on. First of all, I ’ d like to give you an argument for why that list is complete. Of course one can make a biological argument. One can say,”OK, what are we made of?”We ’ re made of cells and stuff between cells. What can damage accumulate in? The answer is : long-lived molecules, because if a short-lived molecule undergoes damage, but then the molecule is destroyed — like by a protein being destroyed by proteolysis — then the damage is gone, too. It ’ s got to be long-lived molecules. So, these seven things were all under discussion in gerontology a long time ago and that is pretty good news, because it means that, you know, we ’ ve come a long way in biology in these 20 years, so the fact that we haven ’ t extended this list is a pretty good indication that there ’ s no extension to be done. However, it ’ s better than that ; we actually know how to fix them all, in mice, in principle — and what I mean by in principle is, we probably can actually implement these fixes within a decade. Some of them are partially implemented already, the ones at the top. I haven ’ t got time to go through them at all, but my conclusion is that, if we can actually get suitable funding for this, then we can probably develop robust mouse rejuvenation in only 10 years, but we do need to get serious about it. We do need to really start trying. So of course, there are some biologists in the audience, and I want to give some answers to some of the questions that you may have. You may have been dissatisfied with this talk, but fundamentally you have to go and read this stuff. I ’ ve published a great deal on this ; I cite the experimental work on which my optimism is based, and there ’ s quite a lot of detail there. The detail is what makes me confident of my rather aggressive time frames that I ’ m predicting here. So if you think that I ’ m wrong, you ’ d better damn well go and find out why you think I ’ m wrong. And of course the main thing is that you shouldn ’ t trust people who call themselves gerontologists because, as with any radical departure from previous thinking within a particular field, you know, you expect people in the mainstream to be a bit resistant and not really to take it seriously. So, you know, you ’ ve got to actually do your homework, in order to understand whether this is true. And we ’ ll just end with a few things. One thing is, you know, you ’ ll be hearing from a guy in the next session who said some time ago that he could sequence the human genome in half no time, and everyone said,”Well, it ’ s obviously impossible.”And you know what happened. So, you know, this does happen. We have various strategies — there ’ s the Methuselah Mouse Prize, which is basically an incentive to innovate, and to do what you think is going to work, and you get money for it if you win. There ’ s a proposal to actually put together an institute. This is what ’ s going to take a bit of money. But, I mean, look — how long does it take to spend that on the war in Iraq? Not very long. OK. It ’ s got to be philanthropic, because profits distract biotech, but it ’ s basically got a 90 percent chance, I think, of succeeding in this. And I think we know how to do it. And I ’ ll stop there. Thank you. Chris Anderson : OK. I don ’ t know if there ’ s going to be any questions but I thought I would give people the chance. Audience : Since you ’ ve been talking about aging and trying to defeat it, why is it that you make yourself appear like an old man? AG : Because I am an old man. I am actually 158. Audience : Species on this planet have evolved with immune systems to fight off all the diseases so that individuals live long enough to procreate. However, as far as I know, all the species have evolved to actually die, so when cells divide, the telomerase get shorter, and eventually species die. So, why does — evolution has — seems to have selected against immortality, when it is so advantageous, or is evolution just incomplete? AG : Brilliant. Thank you for asking a question that I can answer with an uncontroversial answer. I ’ m going to tell you the genuine mainstream answer to your question, which I happen to agree with, which is that, no, aging is not a product of selection, evolution ; [aging] is simply a product of evolutionary neglect. In other words, we have aging because it ’ s hard work not to have aging ; you need more genetic pathways, more sophistication in your genes in order to age more slowly, and that carries on being true the longer you push it out. So, to the extent that evolution doesn ’ t matter, doesn ’ t care whether genes are passed on by individuals, living a long time or by procreation, there ’ s a certain amount of modulation of that, which is why different species have different lifespans, but that ’ s why there are no immortal species. CA : The genes don ’ t care but we do? AG : That ’ s right. Audience : Hello. I read somewhere that in the last 20 years, the average lifespan of basically anyone on the planet has grown by 10 years. If I project that, that would make me think that I would live until 120 if I don ’ t crash on my motorbike. That means that I ’ m one of your subjects to become a 1, 000-year- old? AG : If you lose a bit of weight. Your numbers are a bit out. The standard numbers are that lifespans have been growing at between one and two years per decade. So, it ’ s not quite as good as you might think, you might hope. But I intend to move it up to one year per year as soon as possible. Audience : I was told that many of the brain cells we have as adults are actually in the human embryo, and that the brain cells last 80 years or so. If that is indeed true, biologically are there implications in the world of rejuvenation? If there are cells in my body that live all 80 years, as opposed to a typical, you know, couple of months? AG : There are technical implications certainly. Basically what we need to do is replace cells in those few areas of the brain that lose cells at a respectable rate, especially neurons, but we don ’ t want to replace them any faster than that — or not much faster anyway, because replacing them too fast would degrade cognitive function. What I said about there being no non-aging species earlier on was a little bit of an oversimplification. There are species that have no aging — Hydra for example — but they do it by not having a nervous system — and not having any tissues in fact that rely for their function on very long-lived cells.

Text Sample 333: NEG Dim 2 – 2005 – Score 1.00 – 2005/t000440.txt

Hello. Actually, that ’ s”hello”in Bauer Bodoni for the typographically hysterical amongst us. One of the threads that seems to have come through loud and clear in the last couple of days is this need to reconcile what the Big wants â the”Big”being the organization, the system, the country â and what the”Small”wants â the individual, the person. And how do you bring those two things together? Charlie Ledbetter, yesterday, I thought, talked very articulately about this need to bring consumers, to bring people into the process of creating things. And that ’ s what I want to talk about today. So, bringing together the Small to help facilitate and create the Big, I think, is something that we believe in â something I believe in, and something that we kind of bring to life through what we do at Ideo. I call this first chapter â for the Brits in the room â the”Blinding Glimpse of the Bleeding Obvious.”Often, the good ideas are so staring-at-you- right-in-the-face that you kind of miss them. And I think, a lot of times, what we do is just, sort of, hold the mirror up to our clients, and sort of go,”Duh! You know, look what ’ s really going on.”And rather than talk about it in the theory, I think I ’ m just going to show you an example. We were asked by a large healthcare system in Minnesota to describe to them what their patient experience was. And I think they were expecting â they ’ d worked with lots of consultants before â I think they were expecting some kind of hideous org chart with thousands of bubbles and systemic this, that and the other, and all kinds of mappy stuff. Or even worse, some kind of ghastly death-by-Powerpoint thing with WowCharts and all kinds of, you know, God knows, whatever. The first thing we actually shared with them was this. I ’ ll play this until your eyeballs completely dissolve. This is 59 seconds into the film. This is a minute 59. 3:19. I think something happens. I think a head may appear in a second. 5:10. 5:58. 6:20. We showed them the whole cut, and they were all completely, what is this? And the point is when you lie in a hospital bed all day, all you do is look at the roof, and it ’ s a really shitty experience. And just putting yourself in the position of the patient â this is Christian, who works with us at Ideo. He just lay in the hospital bed, and, kind of, stared at the polystyrene ceiling tiles for a really long time. That ’ s what it ’ s like to be a patient in the hospital. And they were sort, you know, blinding glimpse of bleeding obvious. Oh, my goodness. So, looking at the situation from the point of view of the person out â as opposed to the traditional position of the organization in â was, for these guys, quite a revelation. And so, that was a really catalytic thing for them. So they snapped into action. They said, OK, it ’ s not about systemic change. It ’ s not about huge, ridiculous things that we need to do. It ’ s about tiny things that can make a huge amount of difference. So we started with them prototyping some really little things that we could do to have a huge amount of impact. The first thing we did was we took a little bicycle mirror and we Band-Aided it here, onto a gurney, a hospital trolley, so that when you were wheeled around by a nurse or by a doctor, you could actually have a conversation with them. You could, kind of, see them in your rear-view mirror, so it created a tiny human interaction. Very small example of something that they could do. Interestingly, the nurses themselves, sort of, snapped into action â said, OK, we embrace this. What can we do? The first thing they do is they decorated the ceiling. Which I thought was really â I showed this to my mother recently. I think my mother now thinks that I ’ m some sort of interior decorator. It ’ s what I do for a living, sort of Laurence Llewelyn-Bowen. Not particularly the world ’ s best design solution for those of us who are real, sort of, hard-core designers, but nonetheless, a fabulous empathic solution for people. Things that they started doing themselves â like changing the floor going into the patient ’ s room so that it signified,”This is my room. This is my personal space”â was a really interesting sort of design solution to the problem. So you went from public space to private space. And another idea, again, that came from one of the nurses â which I love â was they took traditional, sort of, corporate white boards, then they put them on one wall of the patient ’ s room, and they put this sticker there. So that what you could actually do was go into the room and write messages to the person who was sick in that room, which was lovely. So, tiny, tiny, tiny solutions that made a huge amount of impact. I thought that was a really, really nice example. So this is not particularly a new idea, kind of, seeing opportunities in things that are around you and snapping and turning them into a solution. It ’ s a history of invention based around this. I ’ m going to read this because I want to get these names right. Joan Ganz Cooney saw her daughter â came down on a Saturday morning, saw her daughter watching the test card, waiting for programs to come on one morning and from that came Sesame Street. Malcolm McLean was moving from one country to another and was wondering why it took these guys so long to get the boxes onto the ship. And he invented the shipping container. George de Mestral â this is not bugs all over a Birkenstock â was walking his dog in a field and got covered in burrs, sort of little prickly things, and from that came Velcro. And finally, for the Brits, Percy Shaw â this is a big British invention â saw the cat ’ s eyes at the side of the road, when he was driving home one night and from that came the Catseye. So there ’ s a whole series of just using your eyes, seeing things for the first time, seeing things afresh and using them as an opportunity to create new possibilities. Second one, without sounding overly Zen, and this is a quote from the Buddha :”Finding yourself in the margins, looking to the edges of things, is often a really interesting place to start.”Blinkered vision tends to produce, I think, blinkered solutions. So, looking wide, using your peripheral vision, is a really interesting place to look for opportunity. Again, another medical example here. We were asked by a device producer â we did the Palm Pilot and the Treo. We did a lot of sexy tech at Ideo â they ’ d seen this and they wanted a sexy piece of technology for medical diagnostics. This was a device that a nurse uses when they ’ re doing a spinal procedure in hospital. They ’ ll ask the nurses to input data. And they had this vision of the nurse, kind of, clicking away on this aluminum device and it all being incredibly, sort of, gadget-lustish. When we actually went and watched this procedure taking place â and I ’ ll explain this in a second â it became very obvious that there was a human dimension to this that they really weren ’ t recognizing. When you ’ re having a four-inch needle inserted into your spine â which was the procedure that this device ’ s data was about ; it was for pain management â you ’ re shit scared ; you ’ re freaking out. And so the first thing that pretty much every nurse did, was hold the patient ’ s hand to comfort them. Human gesture â which made the fabulous two-handed data input completely impossible. So, the thing that we designed, much less sexy but much more human and practical, was this. So, it

’ s not a Palm Pilot by any stretch of the imagination, but it has a thumb-scroll so you can do everything with one hand. So, again, going back to this â the idea that a tiny human gesture dictated the design of this product. And I think that ’ s really, really important. So, again, this idea of workarounds. We use this phrase”workarounds”a lot, sort of, looking around us. I was actually looking around the TED and just watching all of these kind of things happen while I ’ ve been here. This idea of the way that people cobble together solutions in our life â and the things we kind of do in our environment that are somewhat subconscious but have huge potential â is something that we look at a lot. We wrote a book recently, I think you might have received it, called”Thoughtless Acts?”It ’ s been all about these kind of thoughtless things that people do, which have huge intention and huge opportunity. Why do we all follow the line in the street? This is a picture in a Japanese subway. People consciously follow things even though, why, we don ’ t know. Why do we line up the square milk carton with the square fence? Because we kind of have to â we ’ re just compelled to. We don ’ t know why, but we do. Why do we wrap the teabag string around the cup handle? Again, we ’ re sort of using the world around us to create our own design solutions. And we ’ re always saying to our clients :”You should look at this stuff. This stuff is really important. This stuff is really vital.”This is people designing their own experiences. You can draw from this. We sort of assume that because there ’ s a pole in the street, that it ’ s okay to use it, so we park our shopping cart there. It ’ s there for our use, on some level. So, again, we sort of co-opt our environment to do all these different things. We co-opt other experiences â we take one item and transfer it to another. And this is my favorite one. My mother used to say to me,”Just because your sister jumps in the lake doesn ’ t mean you have to.”But, of course, we all do. We all follow each other every day. So somebody assumes that because somebody else has done something, that ’ s permission for them to do the same thing. And there ’ s almost this sort of semaphore around us all the time. I mean, shopping bag equals”parking meter out of order.”And we all, kind of, know how to read these signals now. We all talk to one another in this highly visual way without realizing what we ’ re doing. Third section is this idea of not knowing, of consciously putting yourself backwards. I talk about unthinking situations all the time. Sort of having beginner ’ s mind, scraping your mind clean and looking at things afresh. A friend of mine was a designer at IKEA, and he was asked by his boss to help design a storage system for children. This is the Billy bookcase â it ’ s IKEA ’ s biggest selling product. Hammer it together. Hammer it together with a shoe, if you ’ re me, because they ’ re impossible to assemble. But big selling bookcase. How do we replicate this for children? The reality is when you actually watch children, children don ’ t think about things like storage in linear terms. Children assume permission in a very different way. Children live on things. They live under things. They live around things, and so their spatial awareness relationship, and their thinking around storage is totally different. So the first thing you have to do â this is Graham, the designer â is, sort of, put yourself in their shoes. And so, here he is sitting under the table. So, what came out of this? This is the storage system that he designed. So what is this? I hear you all ask. No, I don ’ t. It ’ s this, and I think this is a particularly lovely solution. So, you know, it ’ s a totally different way of looking at the situation. It ’ s a completely empathic solution â apart from the fact that teddy ’ s probably not loving it. But a really nice way of re-framing the ordinary, and I think that ’ s one of the things. And putting yourself in the position of the person, and I think that ’ s one of the threads that I ’ ve heard again from this conference is how do we put ourselves in other peoples ’ shoes and really feel what they feel? And then use that information to fuel solutions? And I think that ’ s what this is very much about. Last section : green armband. We ’ ve all got them. It ’ s about this really. I mean, it ’ s about picking battles big enough to matter but small enough to win. Again, that ’ s one of the themes that I think has come through loud and clear in this conference is : Where do we start? How do we start? What do we do to start? So, again, we were asked to design a water pump for a company called ApproTEC, in Kenya. They ’ re now called KickStart. And, again, as designers, we wanted to make this thing incredibly beautiful and spend a lot of time thinking of the form. And that was completely irrelevant. When you put yourself in the position of these people, things like the fact that this has to be able to fold up and fit on a bicycle, become much more relevant than the form of it. The way it ’ s produced, it has to be produced with indigenous manufacturing methods and indigenous materials. So it had to be looked at completely from the point of view of the user. We had to completely transfer ourselves over to their world. So what seems like a very clunky product is, in fact, incredibly useful. It ’ s powered a bit like a Stairmaster â you pump up and down on it. Children can use it. Adults can use it. Everybody uses it. It ’ s turning these guys â again, one of the themes â it ’ s turning them into entrepreneurs. These guys are using this very successfully. And for us, it ’ s been great because it ’ s won loads of design awards. So we actually managed to reconcile the needs of the design company, the needs of the individuals in the company, to feel good about a product we were actually designing, and the needs of the individuals we were designing it for. There it is, pumping water from 30 feet. So as a final gesture we handed out these bracelets to all of you this morning. We ’ ve made a donation on everybody ’ s behalf here to kick start, no pun intended, their next project. Because, again, I think, sort of, putting our money where our mouth is, here. We feel that this is an important gesture. So we ’ ve handed out bracelets. Small is the new big. I hope you ’ ll all wear them. So that ’ s it. Thank you.

Text Sample 334: NEG Dim 2 – 1994 – Score 5.00 – 1994/t002458.txt

Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ’ s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC microprocessors versus the power of local area networks. And the interesting thing about it is that this slide, like so many technology slides that we ’ re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ’ s basically what I ’ m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ’ m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well the answer is, if I drew it on a normal curve where, let ’ s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ’ m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ’ t tell us much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log curve, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as microprocessor technology, then the day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ' s not moving like that. And there ' s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ' d like to ask is, if you look at these exponential curves, they don ' t go on forever. Things just can ' t possibly keep changing as fast as they are. One of two things is going to happen. Either it ' s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ' s going to do this. That ' s about all it can do. Now I ' m an optimist, so I sort of think it ' s probably going to do something like that. If so, that means that what we ' re in the middle of right now is a transition. We ' re sort of on this line in a transition from the way the world used to be to some new way that the world is. And so what I ' m trying to ask, what I ' ve been asking myself, is what ' s this new way that the world is? What ' s that new state that the world is heading toward? Because the transition seems very, very confusing when we ' re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ' s a conference in which people talk about the future, and you notice that the future is still at about the year 2000. It ' s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ' s happening there. That transition is happening. We can all sense it. And we know that it just doesn ' t make too much sense to think out 30, 50 years because everything ' s going to be so different that a simple extrapolation of what we ' re doing just doesn ' t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ' re going through. Now in order to do that I ' m going to have to talk about a bunch of stuff that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step back and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, the Earth was this big, sterile hunk of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ' s theories that are beginning to understand about how it started with RNA, but I ' m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ' t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ' s sort of just a very simple chemical form of life, but when things got interesting was when these drops learned a trick about abstraction. Somehow by ways that we don ’ t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for amusement purposes is we can now write things in this code. And I ’ ve got here a little 100 micrograms of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ’ s right here. If I had really been a egotist, I would have put it into a virus and released it in the room. So what was the next step? Writing down the DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ’ t know if you know this, but bacteria can actually exchange DNA. Now that ’ s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the transition point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ’ s interesting in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ’ re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to evolve so that the interesting level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very special structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in the community and special structures that were responsible for recording, understanding, learning information. And that was the brains and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ’ d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these special information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ’ re aware of is human language. It ’ s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ’ m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ’ re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ’ ve done is we, humanity, have started abstracting out. We ’ re going through the same levels that multi-cellular organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. Telephony, computers, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ’ ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of microseconds. And you saw Ty ’ s little evolutionary example where he sort of did a little bit of evolution on the Convolution program right before your eyes. So now we ’ ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and brains, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ’ s what we ’ re seeing here in this explosion of curve. We ’ re seeing this process feeding back on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ’ s really impossible for me to design them in the traditional sense. I don ’ t know what every transistor in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of abstraction and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ’ t quite even understand. One method that ’ s particularly interesting that I ’ ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the microsecond time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,”Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted.”So in other words, I define what I wanted. Let ’ s say I want to sort numbers, as a simple example I ’ ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,"Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ' s reproduce the ones that sorted numbers the best. And let ' s reproduce them by a process of recombination analogous to sex."Take two programs and they produce children by exchanging their subroutines, and the children inherit the traits of the subroutines of the two programs. So I ' ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,"Please repeat that process."Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ' t tell you how they work. I ' ve tried looking at them and telling you how they work. They ' re obscure, weird programs. But they do the job. And in fact, I know, I ' m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,"Oh look. Look at this. It says, ' This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ' Doesn ' t that make you feel confident?"In fact, we know that the engineering process doesn ' t work very well when it gets complicated. So we ' re beginning to depend on computers to do a process that ' s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ' t quite understand the options of it. So in a sense, it ' s getting ahead of us. We ' re now using those programs to make much faster computers so that we ' ll be able to run this process much faster. So it ' s feeding back on itself. The thing is becoming faster and that ' s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ' re taking off. And what we are is we ' re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ' re the amoebas and we can ' t quite figure out what the hell this thing is we ' re creating. We ' re right at that point of transition. But I think that there really is something coming along after us. I think it ' s very haughty of us to think that we ' re the end product of evolution. And I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

Text Sample 335: NEG Dim 2 – 2020 – Score 1.00 – 2020/t003959.txt

Today, I would like to talk to you about beauty and how we ' ve got it all wrong when it comes to our perceptions of women, particularly Aboriginal women. But before I do, I would like to acknowledge the traditional custodians of the land in which I stand upon : the Gadigal people of the Eora Nation. I pay my respects to the elders past, present and emerging and give thanks to our ancestors who guide and protect us. It was 1990, and I was pumped. I was off to my first birthday party, just before I hit the terrible teens. No chaperone, and no bratty sister to tag along so she could snitch. I had my cute little outfit on, gift in hand, and I was hoping that this little cutie that I liked would show up. And I was hoping that this little cutie would ask me this one question. You know that question that makes your heart beat right out your chest – Do you want to be my girlfriend? – even though I had no business having a boyfriend at that age. But it didn ' t matter, because back then, it was all about the rush. I never did get asked that question. But the question I did get asked was : What ' s your background? And like any proud Aboriginal child would declare,"I ' m Aboriginal."Given the reaction of the room, being Aboriginal was clearly a dirty word. And at the tender age of 11, I was told by my best friend ' s adult sister that I was too pretty to be Aboriginal. By this time, my mouth is dry, my blood is boiling, and I ' m trying so hard to fight back what feels like an ocean of tears. I calmly join my circle of friends and begin to fake laugh at whatever is funny to mask my embarrassment, as I clutch on to my newfound complex. And this is why we need to change our perceptions of beauty. And how we do this is by learning from Aboriginal women, their stories and perspectives. Because right now,"pretty"hurts. Pretty hurts because you ' re trying to erase my Aboriginality, to applaud my proximity to whiteness. Pretty hurts because aimed at an Aboriginal woman, it is a weapon loaded in racism, sexual exploitation and cultural genocide. You see, what this woman didn ' t realize when she declared that I was too pretty to be Aboriginal is that she took something precious from me : pride in my identity. You see, I belong to the oldest living culture in the world, but that day, that legacy – it was replaced with shame, and it ' s been this filthy stain I ' ve been trying to get rid of for 20 years. And this is where my obsession for beauty comes from, over the years, trying to mimic it as a model, advocating for diversity in fashion, to launching"Ascension"magazine to celebrate women of color, whose beauty is still underrepresented. With much pain and trauma behind one word,"pretty"taught me, through my indigeneity, I could reclaim my beauty. To Indigenous women, true beauty came from the traditional roles we upheld, our kinship systems, connection to country and the waterways and how we pass this ancient knowledge down to the next generation. The way we express beauty was never defined against a Eurocentric ideal of beauty. You see, in my culture, our beauty is not monolithic. It ' s not measured by a thin waistline, porcelain skin or slender hips. It runs much deeper than that. So what does indigenous beauty look like? Oh, it ' s fierce, defiant and proud. And one ancestor who epitomizes indigenous beauty is Barangaroo, a powerful Cammeraygal woman. Revered for her wisdom and independence, Barangaroo, like the Eora other women, took pride in their status as being the main food providers for their tribe. A skillful and patient fisherwoman, Barangaroo would access Sydney Harbour and its surrounding waters for its abundant food supply, only taking what was needed. So you can just imagine how furious Barangaroo was when she saw British colonists troll 4, 000 salmon off the north shore in just one day, then gifting some of this catch to her husband and some of the other men from her tribe. Barangaroo knew such a wasteful act would threaten the Eora women ' s cultural authority within the tribe, furthermore destroying their traditional way of life. So Barangaroo rejected British laws and customs, their food, drink and social etiquette, even when her husband decided to conform. When Barangaroo and her husband Bennelong were invited to dine with Governor Phillip and the British party, Barangaroo stayed true to who she was. instead of wearing colonial attire – a tight corset and a gown layered in silk finished with pearls – she came sporting her traditional wares : white ochre and a bone through her nose. What Barangaroo illustrated was : indigenous beauty is authentic. Aunty Beryl Van-Oploo, a respected Gamilaraay elder, shared a story with a group of women one day, and she said,"We all have a bit of Barangaroo in us."Later that evening, I thought about Aunty Beryl ' s message. And what I received from her message was, no matter our culture, color or how we identify, spirit is what we share. It ' s what connects us. You see, if we indigenize beauty, the meaning is transformed from aesthetically pleasing to a state of divinity ; beauty now becomes spirit manifested. Not only is spirit found within us, it ' s in all things. It ' s in the landscapes, it ' s in the elements. The Yolngu people of northeast Arnhem Land, they have a Dreamtime story : Walu, the Sun woman. They say Walu lights a small fire each morning, which creates the dawn. She then paints her body in red ochre. And as she does, some of it falls onto the clouds, creating the sunrise. She then makes a torch from a stringybark tree and carries this fire across the sky from east to west, creating the daylight. And when it ' s time for her journey to end, she descends from the sky. And as she does, some of the red ochre from her body falls onto the clouds, creating the sunset. Indigenous beauty can be seen right across this continent, each Aboriginal nation with its own creation stories of how we came to be – astronomy, medicine, agriculture, architecture, education, innovation. And when we had conflict, we had lore, l-o-r-e, to restore order. Like the seasons, flora and fauna, night and day, we are all interconnected. One does not work without the other – the very principles which binds humanity together. Over the years, my obsession for beauty, it ' s led me to this truth : you cannot appreciate beauty if you cannot recognize it in yourself. So how do we change our perceptions of beauty? We have to get real with ourselves and start by asking : Who am I? Where do I come from? The world that I live in – how did it come to be? And more importantly : Where to from here? You may not like what you discover. But sit with it, feel the discomfort. Colonization has stolen from us one of the greatest treasures we can obtain : each other. This year alone, we ' ve witnessed pathological political and social unrest. That is why healing is the antidote humanity needs because it leads us to unity. When we decolonize beauty, we are reintroduced to our authentic selves. I used to wonder whatever happened to that woman from the birthday party, you know, the one that told me I was too pretty to be Aboriginal. A moment that was so devastating helped me to embrace my girgorou."Girgorou"means"beautiful"in Jirrbal, my grandmother ' s language. I now know that my girgorou is mighty, like Barangaroo. And my girgorou, like Walu, it ' s everlasting, from when that sun rises to when that sun sets. Are you ready to embrace your girgorou? Thank you.

Text Sample 336: NEG Dim 2 – 2020 – Score 1.00 – 2020/t004043.txt

Voting can be hard. It ' s been hard, sometimes painful, sometimes impossible, since the very beginning of our democracy. This year and years prior, we ' ve seen voters wait in line for five, six and seven hours. And the issue hasn ' t been fixed. Now, some people may see these images and think,"How patriotic. How impressive that someone would wait in line for seven hours just to vote."But to me, it ' s not impressive at all. It ' s disrespectful to these voters. Making voting difficult goes against the very core of our democracy. If we could redesign the system to make it more convenient, more accessible and easier for voters, why wouldn ' t we? Now, the short answer is : political will. Many established politicians would not actually benefit personally from a reformed voting process that ' s inclusive for all voters. Politicians are the players in the game, but yet they set the rules for the game. Election policy must be about who votes, not who wins. And the more complicated answer is that our voting system and election system in the United States is highly decentralized and inconsistent. Over 10, 000 different local election officials administer this process in cities and towns and counties across the country. They might vary in size from 400 voters to 4. 7 million. There ' s also 50 different state legislative bodies that set the rules of the game, and over 50 different chief election officials and entities that oversee those rules and how they ' re administered. So voting may vary greatly from state to state. Best-case scenario, you ' re in a state like Colorado, and a ballot is mailed to you proactively before each and every election. No bureaucracy, no extra paperwork. The ballot comes, and the government is responsible for delivering democracy to you. Worst-case scenario, you ' re in a state like Missouri, where your options are limited, you have overly restrictive voter registration deadlines. And if you can ' t get off work or you don ' t have childcare or you ' re sick, that ' s too bad. And most American voters don ' t fall into the best-case scenario. Now, in my career as an elections official for many years, and now leading the National Vote at Home Institute and our work to improve the voting process across the United States, I ' ve talked to thousands of voters about their voting experience and thousands of election officials about the process. I also coauthored a book called"When Women Vote,"that outlines a road map and a playbook for how to improve the process for all. And so I ask you : What would you choose? Which scenario would you choose? Now, the 2016 election was the most highly watched, most anticipated election in US history. And yet, only 60 percent of eligible Americans actually voted. Over 100 million people did not vote in 2016. And when they were surveyed as to why, over 40 percent indicated it was due to a barrier : missing a deadline, couldn ' t get off work, couldn ' t wait in line for hours. If"did not vote"was on the ballot in 2016,"did not vote"would have won in a landslide. What we end up with is a system where a minority of eligible Americans are choosing the politicians that make decisions for all of us collectively. Trust in the US government and politicians is at an all-time low, and the ballot box isn ' t helping. If we can ' t even cast a vote easily, why would we ever trust the process or trust politicians? We must put voters first. I ' ll say that again. We must put voters first in election policies and designing a system that serves them. Just ask any successful business. We live in an era of same-day shipping, free delivery, Lyft and Uber and take-home cocktails. And consumers, especially in the height of the pandemic, are choosing their experiences in the comfort of their home and on their schedules. So why can ' t we design a voting process that is as convenient as that? Luckily, we don ' t have to speculate. In Colorado, we ' ve already designed that process, and Colorado is now one of the best states to vote in and also one of the most secure. In 2013, I worked with a group of dedicated leaders to redesign our voting process and pass legislation that put voters first. In Colorado, every voter receives a ballot ahead of each election. They ' re automatically registered to vote. There ' s no overly restrictive deadlines. And with BallotTRACE, voters can track their ballot just like they would a package, through the process, from the moment it ' s mailed to the moment the election official receives it for counting. That system was pioneered in Denver now 11 years ago. And when we designed it, we were able to reduce our call volume by 70 percent and infuse transparency and accountability into the process. Now, when you get that ballot at home, you can vote it and then mail it back in or drop it off in person. And if you want to vote in person, you can do that, too. And you ' re not confined to the government-assigned polling places on one day. You can go to any vote center – close to your kids ' school, close to work, close to home – and you can do so over a few weeks prior to the election. It ' s been seven years since we passed that legislation and implemented that model. And the results are incredible. Colorado increased turnout significantly and now is one of the top states for voter turnout and also one of the most secure. We also saw a reduction in costs. So, because more people were voting at home, we didn ’ t need as many poll workers, and we saw that reduce by over 70 percent. When we went to buy a new voting system, we no longer needed as much equipment. We also saw voters go farther down the ballot, to local races and ballot issues. And we saw turnout increase on those down-ballot races and issues. Those races include mayor and school board and city council. And they also include the really long legalese ballot issues that take forever to figure out. Voters now have a laptop in reach at their home, and they can research candidates and issues on their own time. We also have research now that shows that voting by mail and voting at home makes voters more informed because they have all of that extra time, as opposed to being in person and worrying about the long line of voters behind you while you ’ re trying to rush through. And the final, most important aspect of the results that we ’ ve seen out of Colorado is about civics for future voters. And I want to share my story with my two elementary children. Every time my ballot comes before each election, one of my kids gets it, and they always start asking,”When are we going to fill out mom ’ s ballot?”We sit down together, they read the instructions to me, they read the candidate names, and they ask me questions like,”Mom, what does governor do? What does the mayor do? Maybe I want to be a mayor someday.”We research those issues together, we talk about it, and it takes me forever to complete my ballot. But I know that I have created lifelong, civically engaged voters and future voters that understand that the choices they make on that ballot impact their communities and their world. This is the type of voting experience that we want for every voter across the country. Many other states have taken notice, including California, Vermont, New Jersey, Hawaii. All have expanded options for voting at home this year, in 2020. Americans are resilient. We need a voting process that is also resilient – from a pandemic, from burdens and barriers, from inequities, from unfairness, from foreign adversaries and from administrative deficiencies. Across the country, voters are choosing to vote at home in record numbers. It is safe, it is secure, and we have built-in security measures to deter and detect bad actors who try to interfere with the process. Today, voting at home means paper ballots, but in the future, that might look very different. Voters deserve an awesome and safe voting experience, free from barriers and burdens. It ’ s the politicians that serve us, not the other way around. You deserve excellence. Expect it, demand it and advocate for it. Thank you.

Text Sample 337: NEG Dim 2 – 2020 – Score 1.00 – 2020/t004040.txt

I am a linguist. Linguists study language. And we do this in a lot of different ways. Some linguists study how we pronounce certain sounds. Others look at how we build sentences. And some study how language varies from place to place, just to name a few. But what I ’ m really interested in is what people think and believe about language and how these beliefs affect the way we use it. All of us have deeply held beliefs about language such as the belief that some languages are more beautiful than others or that some ways of using language are more correct. And as most linguists know, these beliefs are often less about language itself and more about what we believe about the social world around us. So I ’ m a linguist, and I ’ m also a nonbinary person, which means I don ’ t identify as a man or a woman. I also identify as a member of a broader transgender community. When I first started getting connected to other transgender people, it was like learning a whole new language and the linguist part of me was really excited. There was a whole new way of talking about my relationship with myself and a new clear way to communicate that to other people. And then I started having conversations with my friends and family about what it meant for me to be trans and nonbinary, what those words meant to me specifically, and why I would use both of them. I also clarified the correct words they could use when referring to me. For some of them, this meant some very specific changes. For example, some of my friends who are used to talking about our friend group as”ladies”or”girls”switched to non-gendered terms like”friends”or”pals.”And my parents can now tell people that their three kids are their son, their daughter and their child. And all of them would have to switch the pronouns they used to refer to me. My correct pronouns are”they”and”them,”also known as the singular they. And these people love me, but many of them told me that some of these language changes were too hard or too confusing or too ungrammatical for them to pick up. These responses led me to the focus of my research. There are commonly held, yet harmful and incorrect beliefs about language that for the people who hold these beliefs, act as barriers to building and strengthening relationships with the transgender people in their families and communities, even if they want to do so. Today, I ’ m going to walk you through some of these beliefs in the hope that we can embrace creativity in our language and allow language to bring us closer together. You might see your own beliefs reflected in these experiences in some way, but no matter what, I hope that I can share with you some linguistic insights that you can put into your back pocket and take with you out into the world. And I just want to be super clear. This can be fun. Learning about language brings me joy, and I hope that it can bring you more joy too. So do you remember how I said that for some of my friends and family learning how to use the singular they was really hard, and they said it was too confusing or too ungrammatical for them to pick up. Well, this brings us to the first belief about language that people have. Grammar rules don ’ t change. As a linguist, I see this belief a lot out in the world. A lot of language users believe that grammar just is what it is. When it comes to language, what ’ s grammatical is what matters. You can ’ t change it. I want to tell you a story about English in the 1600s. Back then, as you might imagine, people spoke differently than we do today. In particular, they used”thou”when addressing a single other person, and”you”when addressing more than one other person, But for some complex historical reasons that we don ’ t have time to get into today, so you ’ ll just have to trust me as a linguist here, but people started using”you”to address someone, regardless of how many people they were talking to. And people had a lot to say about this. Take a look at what this guy, Thomas Elwood, had to say. He wrote,”The corrupt and unsound form of speaking in the plural number to a single person, ’ you ’ to one instead of ’ thou, ’ contrary to the pure, plain and single language of truth, ’ thou ’ to one and ’ you ’ to more than one.”And he goes on. Needless to say, this change in pronouns was a big deal in the 1600s. But actually, if you followed the debates about the singular they at all, these arguments might sound familiar to you. They ’ re not that far off from the bickering we hear about the so-called grammaticality of pronouns used to talk about trans and nonbinary people. One of the most common complaints about the singular they is that if”they”is used to refer to people in the plural, it can ’ t also be used to talk about people in the singular, which is exactly what they said about”thou”and”you.”But as we have seen, pronouns have changed. Our grammar rules do change and for a lot of different reasons. And we ’ re living through one of these shifts right now. All living languages will continue to change, and the Thomas Elwoods of the world will eventually have to get with the program because hundreds of years later, it ’ s considered right to use”you”when addressing another person. Not just allowable, but right. The second belief about language that people have is that dictionaries provide official, unchanging definitions for words. When you were in school, did you ever start an essay with a sentence like,"The dictionary defines history as..."Well, if you did, which dictionary were you talking about? Was it the Oxford English Dictionary? Was it Merriam Webster? Was it Urban Dictionary? Did you even have a particular dictionary in mind? Which one of these is"the dictionary?"Dictionaries are often thought of as the authority on language. But dictionaries, in fact, are changing all the time. And here ' s where our minds are really blown. Dictionaries don ' t provide a single definition for words. Dictionaries are living documents that track how some people are using language. Language doesn ' t originate in dictionaries. Language originates with people and dictionaries are the documents that chronicle that language use. Here ' s one example. We currently use the word"awful"to talk about something that is bad or gross. But before the 19th century,"awful"meant just the opposite. People used"awful"to talk about something that was deserving of respect or full of awe. And in the mid-1900s,"awesome"was the word that took up these positive meanings and"awful"switched to the negative one we have today. And dictionaries over time reflected that. This is just one example of how definitions and meanings have changed over time. And to keep up with it, how dictionaries are updated all the time. So I hope you ' re starting to feel a little more comfortable with the idea of changing language. But of course, I ' m not just talking about language in general. I ' m talking about language as it is impactful for trans people. And pronouns are only one part of language, and they ' re only one part of language that ' s important for trans people. Also important are the identity terms that trans people use to talk about ourselves, such as trans man, trans woman, nonbinary or gender queer. And some of these words have been documented in dictionaries for decades now and others are still being added year after year. And that ' s because dictionaries are working to keep up with us, the people who are using language creatively. So at this point, you might be thinking,"But Archie, it seems like every trans person has a different word they want me to use for them. There are so many opportunities for me to mess up or to look ignorant or to hurt someone ' s feelings. What is something I can memorize and reliably employ when talking to the trans people in my life?"Well, that brings us to the third belief about language that people have. You can ' t just make up words. Folks, people do this all the time. Here ' s one of my favorite examples. The"official"term for your mother ' s mother or your father ' s mother is grandmother. I recently polled my friends and asked them what they call their grandmothers. We don ' t get frustrated if your friend ' s grandma goes by Meemaw and yours goes by Gigi. We just make rather short work of it and memorize it and move on getting to know her. In fact, we might even celebrate her by gifting her with a sweatshirt or an embroidered pillow that celebrates the name she has chosen for herself. And just like your Nana and your grandma, trans people have every right to choose their own identifying language. The process of determining self-identifying language is crucial for trans people. In my research, many trans people have shared that finding new vocabulary was an important part of understanding their own identities. As one person I interviewed put it,"Language is one of the most important personal things because using different words to describe myself and then finding something that feels good, feels right, is a very introspective and important process. With that process you can piece together, with the language that you find out works best for you, who am I?"Sometimes the words that feel good are already out there. For me, the words trans and nonbinary just feel right. But sometimes the common lexicon doesn ' t yet hold the words that a person needs to feel properly understood. And it ' s necessary and exciting to get to create and redefine words that better reflect our experience of gender. So this is a very long answer, but, yes, I ’ m absolutely going to give you a magic word, something really easy you can memorize. And I want you to think of this word as the biggest piece of advice I could give you if you don ’ t know what words to use for the trans people in your life. Ask. I might be a linguist and a trans person and a linguist who works with trans people, but I ’ m no substitute for the actual trans people in your life when it comes to what words to use for them. And you ’ re more likely to hurt someone ’ s feelings by not asking or assuming than you are by asking. And the words that a person uses might change. So just commit to asking and learning. Language is a powerful tool for explaining and claiming our own identities and for building relationships that affirm and support us. But language is just that, a tool. Language works for us, not the other way around. All of us, transgender and cisgender can use language to understand ourselves and to respect those around us. We ’ re not bound by what words have meant before, what order they might have come in or what rules we have been taught. We can consider the beliefs that we might have had about how language works and recognize that language will continue to change. And we can creatively use language to build the identities and relationships that bring us joy. And that ’ s not just allowable. It ’ s right. Believe me.

5 POS Dim 3

Text Sample 338: POS Dim 3 – 2001 – Score 17.00 – 2001/t000083.txt

I thought I would read poems I have that relate to the subject of youth and age. I was sort of astonished to find out how many I have actually. The first one is dedicated to Spencer, and his grandmother, who was shocked by his work. My poem is called”Dirt.”My grandmother is washing my mouth out with soap ; half a long century gone and still she comes at me with that thick cruel yellow bar. All because of a word I said, not even said really, only repeated. But”Open,”she says,”open up!”her hand clawing at my head. I know now her life was hard ; she lost three children as babies, then her husband died too, leaving young sons, and no money. She ’ d stand me in the sink to pee because there was never room in the toilet. But oh, her soap! Might its **bitter** burning have been what made me a poet? The **street** she lived on was unpaved, her flat, two cramped rooms and a fetid kitchen where she stalked and caught me. Dare I admit that after she did it I never really loved her again? She lived to a hundred, even then. All along it was the sadness, the squalor, but I never, until now loved her again. When that was published in a magazine I got an irate letter from my uncle.”You have maligned a great woman.”It took some diplomacy. This is called”The Dress.”It ’ s a longer poem. In those days, those days which exist for me only as the most elusive memory now, when often the first sound you ’ d hear in the morning would be a storm of birdsong, then the soft clop of the hooves of the horse hauling a milk wagon down your block, and the last sound at night as likely as not would be your father pulling up in his car, having worked late again, always late, and going heavily down to the cellar, to the furnace, to shake out the ashes and damp the draft before he came upstairs to fall into bed — in those long-ago days, women, my mother, my friends ’ mothers, our neighbors, all the women I knew — wore, often much of the day, what were called house-dresses, cheap, printed, pulpy, seemingly purposefully shapeless light cotton shifts that you wore over your nightgown and, when you had to go look for a child, hang wash on the line, or run down to the grocery store on the corner, under a coat, the twisted hem of the nightgown always lank and yellowed, dangling beneath. More than the curlers some of the women seemed constantly to have in their hair in preparation for some great event — a ball, one would think — that never came to pass ; more than the way most women ' s faces not only were never made up during the day, but seemed scraped, bleached, and, with their plucked eyebrows, scarily masklike ; more than all that it was those dresses that made women so unknowable and forbidding, adepts of enigmas to which men could have no **access**, and boys no conception. Only later would I see the dresses also as a proclamation : that in your dim kitchen, your laundry, your bleak concrete yard, what you revealed of yourself was a fabulation ; your real sensual nature, veiled in those sexless vestments, was utterly your dominion. In those days, one hid much else as well : grown men didn ' t embrace one another, unless someone had died, and not always then ; you shook hands or, at a ball game, thumped your friend ' s back and exchanged blows meant to be codes for affection ; once out of childhood you ' d never again know the shock of your father ' s whiskers on your cheek, not until mores at last had evolved, and you could hug another man, then hold on for a moment, then even kiss. What release finally, the embrace : though we were wary — it seemed so **audacious** — how much unspoken joy there was in that affirmation of **equality** and communion, no matter how much misunderstanding and pain had passed between you by then. We knew so little in those days, as little as now, I suppose about **healing** those hurts : even the women, in their best dresses, with beads and sequins sewn on the bodices, even in lipstick and mascara, their hair aflow, could only stand wringing their hands, begging for peace, while father and son, like thugs, like thieves, like Romans, simmered and hissed and hated, inflicting sorrows that endured, the worst anyway, through the kiss and embrace, bleeding from brother to brother, into the generations. In those days there was still countryside close to the city, farms, cornfields, cows ; even not far from our building with its blurred brick and long shadowy hallway you could find tracts with hills and trees you could pretend were mountains and forests. Or you could go out by yourself even to a half-block-long empty lot, into the bushes : like a creature of leaves you ' d lurk, crouched, crawling, simplified, savage, alone ; already there was wanting to be simpler, wanting, when they called you, never to go back. This is another longish one, about the old and the young. It actually happened right at the time we met. Part of the poem takes place in space we shared and time we shared. It ' s called"The Neighbor."Her five horrid, deformed little dogs who incessantly yap on the roof under my window. Her cats, God knows how many, who must piss on her rugs — her landing ' s a sickening reek. Her shadow once, fumbling the chain on her door, then the door slamming fearfully shut, only the barking and the music — jazz — filtering as it does, day and night into the hall. The time it was Chris Connor singing"Lush Life"— how it brought back my college sweetheart, my first real love, who — till I left her — played the same record. And head on my shoulder, hand on my thigh, **sang** sweetly along, of regrets and depletions she was too young for, as I was too young, later, to believe in her pain. It startled, then bored, then repelled me. My starting to fancy she ' d ended up in this fire-trap in the Village, that my neighbor was her. My thinking we ' d meet, recognize one another, become friends, that I ' d accomplish a penance. My seeing her, it wasn ' t her, at the mailbox. Gray-yellow hair, army pants under a nightgown, her turning away, hiding her ravaged face in her hands, muttering an inappropriate"Hi."Sometimes there are frightening goings-on in the stairwell. A man shouting,"Shut up!"The dogs frantically snarling, claws scrabbling, then her — her voice hoarse, harsh, hollow, almost only a tone, incoherent, a note, a squawk, bone on metal, metal gone molten, calling them back,"Come back darlings, come back dear ones. My sweet angels, come back.”Medea she was, next time I saw her. Sorceress, tranced, ecstatic, stock-still on the **sidewalk** ragged coat hanging agape, passersby flowing around her, her mouth torn suddenly open as though in a scream, silently though, as though only in her brain or breast had it erupted. A cry so pure, practiced, detached, it had no need of a voice, or could no longer bear one. These invisible links that allure, these transfigurations, even of anguish, that hold us. The girl, my old love, the last lost time I saw her when she came to find me at a party, her drunkenly stumbling, falling, sprawling, skirt hiked, eyes veined red, swollen with tears, her **shame**, her dishonor. My ignorant, arrogant coarseness, my secret pride, my turning away. Still life on a rooftop, dead trees in barrels, a bench broken, dogs, excrement, sky. What pathways through pain, what junctures of vulnerability, what crossings and counterings? Too many lives in our lives already, too many chances for sorrow, too many unaccounted-for pasts.”Behold me,”the god of frenzied, inexhaustible love says, rising in bloody splendor,”Behold me.”Her making her way down the littered vestibule stairs, one agonized step at a time. My holding the door. Her crossing the fragmented tiles, faltering at the step to the **street**, droning, not looking at me,”Can you help me?”Taking my arm, leaning lightly against me. Her wavering step into the world. Her whispering,”Thanks love.”Lightly, lightly against me. I think I ’ ll lighten up a little. Another, different kind of poem of youth and age. It ’ s called”Gas.”Wouldn ’ t it be nice, I think, when the blue-haired lady in the doctor ’ s waiting room bends over the magazine table and farts, just a little, and violently blushes. Wouldn ’ t it be nice if intestinal gas came embodied in visible clouds, so she could see that her really quite inoffensive pop had only barely grazed my face before it drifted away. Besides, for this to have happened now is a nice coincidence. Because not an hour ago, while we were on our walk, my dog was startled by a backfire and jumped straight up like a horse bucking. And that brought back to me the stable I worked on weekends when I was 12, and a splendid piebald stallion, who whenever he was mounted would buck just like that, though more hugely of course, **enormous**, gleaming, resplendent. And the woman, her face abashedly buried in her”Elle”now, reminded me — I ’ d forgotten that not the least part of my awe consisted of the fact that with every jump he took the horse would powerfully fart. Phwap! Phwap! Phwap! Something never mentioned in the dozens of books about horses and their riders I devoured in those days. All that savage grandeur, the steely glinting hooves, the eruptions driven from the creature ’ s mighty innards, breath stopped, heart stopped, nostrils madly flared, I didn ’ t know if I wanted to break him, or be him. This is called”Thirst.”Many — most of my poems actually are urban poems. I happen to be reading a bunch that aren ’ t.”Thirst.”Here was my relation with the woman who lived all last autumn and winter, day and night, on a bench in the 103rd Street subway station, until finally one day she vanished. We regarded each other, scrutinized one another. Me shyly, obliquely, trying not to be furtive. She boldly, unblinkingly, even pugnaciously, wrathfully even, when her bottle was empty. I was frightened of her. I felt like a child. I was afraid some repressed part of myself would go out of control, and I ’ d be forever entrapped in the shocking seethe of her stench. Not excrement merely, not merely surface and orifice going unwashed, rediffusion of rum, there was will in it, and intention, power and purpose — a social, ethical rage and rebellion — despair too, though, **grief**, loss. Sometimes I ’ d think I should take her home with me, bathe her, comfort her, dress her. She wouldn ’ t have wanted me to, I would think. Instead, I ’ d step into my train. How rich I would think, is the lexicon of our self-absolving. How enduring, our bland fatal assurance that reflection is righteousness being accomplished. The dance of our glances, the clash, pulling each other through our perceptual punctures, then holocaust, holocaust, host on host of **ill**, injured presences, squandered, consumed. Her vigil somewhere I know continues. Her occupancy, her absolute, faithful attendance. The dance of our glances, challenge, abdication, effacement, the perfume of our consternation. This is a newer poem, a brand new poem. The title is"This Happened."A student, a young woman in a fourth-floor hallway of her lycee, perched on the ledge of an open window **chatting** with friends between classes ; a teacher passes and chides her,"Be careful, you might fall,"almost banteringly chides her,"You might fall,"and the young woman, 18, a girl really, though she wouldn ' t think that, as brilliant as she is, first in her class, and"Beautiful, too,"she ' s often told, smiles back, and leans into the open window, which wouldn ' t even be open if it were winter — if it were winter someone would have closed it — leans into the window, farther, still smiling, farther and farther, though it takes less time than this, really an instant, and lets herself fall. Herself fall. A casual impulse, a fancy, never thought of until now, hardly thought of even now... No, more than impulse or fancy, the girl knows what she ' s doing, the girl means something, the girl means to mean, because it occurs to her in that instant, that beautiful or not, bright yes or no, she ' s not who she is, she ' s not the person she is, and the reason, she suddenly knows, is that there ' s been so much premeditation where she is, so much plotting and planning, there ' s hardly a person where she is, or if there is, it ' s not her, or not wholly her, it ' s a self inhabited, lived in by her, and seemingly even as she thinks it she knows what ' s been missing : grace, not premeditation but grace, a kind of being in the world spontaneously, with grace. Weightfully upon me was the world. Weightfully this self which graced the world yet never wholly itself. Weightfully this self which weighed upon me, the release from which is what I desire and what I achieve. And the girl remembers, in this infinite instant already now so many times divided, the sadness she felt once, hardly knowing she felt it, to merely inhabit herself. Yes, the girl falls, absurd to fall, even the earth with its compulsion to take unto itself all that falls must know that falling is absurd, yet the girl falling isn ' t myself, or she is myself, but a self I took of my own volition unto myself. Forever. With grace. This happened. I ' ll read just one more. I don ' t usually say that. I like to just end. But I ' m afraid that Ricky will come out here and shake his fist at me. This is called"Old Man,"appropriately enough."Special : big tits,"Says the advertisement for a soft-core magazine on our **neighborhood** newsstand. But forget her breasts. A lush, fresh-lipped blond, skin glowing gold, sprawls there, resplendent. 60 nearly, yet these hardly tangible, hardly better than harlots, can still stir me. Maybe a coming of age in the American sensual **darkness**, never seeing an unsmudged nipple, an uncensored vagina, has left me forever infected with an unquenchable lust of the eye. Always that erotic murmur, I ' m hardly myself if I ' m not in a state of incipient desire. God knows though, there are worse twists your obsessions can take. Last year in Israel, a young ultra-orthodox Rabbi guiding some teenage girls through the Shrine of the Shoah forbade them to look in one room. Because there were images in it he said were licentious. The display was a photo. Men and women stripped naked, some trying to cover their genitals, others too frightened to bother, lined up in snow waiting to be shot and thrown into a ditch. The girls, to my horror, averted their gaze. What carnal mistrust had their teacher taught them. Even that though. Another **confession** : Once in a book on pre-war Poland, a studio portrait, an absolute angel, an absolute angel with tormented, tormenting eyes. I kept finding myself at her page. That she died in the camps made her — I didn ' t dare wonder why — more present, more precious. Died in the camps, that too people — or Jews anyway — kept from their children back then. But it was like sex, you didn ' t have to be told.

Sex and death, how close they can seem. So constantly conscious now of death moving towards me, sometimes I think I confound them. My wife ’ s loveliness almost consumes me. My **passion** for her goes beyond reasonable bounds. When we make love, her holding me everywhere all around me, I ’ m there and not there. My mind teems, jumbles of faces, voices, impressions, I live my life over, as though I were drowning. Then I am drowning, in despair at having to leave her, this, everything, all, unbearable, awful. Still, to be able to die with no special contrition, not having been slaughtered, or enslaved. And not having to know history ’ s next mad rage or regression, it might be a relief. No. Again, no. I don ’ t mean that for a moment. What I mean is the world holds me so tightly — the good and the bad — my own follies and weakness that even this counterfeit Venus with her sham heat, and her bosom probably plumped with gel, so moves me my breath catches. Vamp. Siren. Seductress. How much more she reveals in her glare of ink than she knows. How she incarnates our desperate human need for regard, our **passion** to live in beauty, to be beauty, to be cherished by glances, if by no more, of something like love, or love. Thank you.

Text Sample 339: POS Dim 3 – 2001 – Score 3.00 – 2001/t000092.txt
I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ’ children, because they were all average. But they weren ’ t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ’ s not right. The good Lord in his infinite wisdom didn ’ t create us all **equal** as far as intelligence is concerned, any more than we ’ re **equal** for size, appearance. Not everybody could earn an A or a B, and I didn ’ t like that way of judging, and I did know how the alumni of various schools back in the ’ 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ’ t lose a game. But it seemed that we didn ’ t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ’ t like it, I didn ’ t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ’ m sure at the time he did that, I didn ’ t — it didn ’ t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ’ s under your control. If you get too engrossed and involved and concerned in regard to the things over

which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,”At God ’ s footstool to confess, a poor soul knelt, and bowed his head. ’ I failed! ’ he cried. The Master said, ’ Thou didst thy best, that is success. ’”From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ’ re capable. I believe that ’ s true. If you make the effort to do the best of which you ’ re capable, trying to improve the situation that exists for you, I think that ’ s success, and I don ’ t think others can judge that ; it ’ s like character and reputation — your reputation is what you ’ re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ’ s what the teachers are themselves.”That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by coal oil lamp — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I **reply**, ’ Where could I find such splendid company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ’ s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or **gay**. I knew him once, but then he was a boy. They ask me why I teach and I **reply**, ’ Where could I find such splendid company? ’”And I believe the teaching profession — it ’ s true, you have so many youngsters, and I ’ ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ’ re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ’ re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was”Never be late.”Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ' s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ' t clean and neat, so I wouldn ' t let him go. He couldn ' t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ' t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ' ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,"Don ' t run practices late, because you ' ll go home in a bad mood, and that ' s not good, for a young married man to go home in a bad mood. When you get older, it doesn ' t make any difference, but —"So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ' re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ' t want that. I used to tell them I was paid to do that. That ' s my job. I ' m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ' s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ' s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ' t have the time to go on that. But that helped me, I think, become a better teacher. It ' s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ' re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ' re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ' t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ' i '. I ' d never seen that before, but he did. Big league baseball players — they ' re very perceptive about those things, and they noticed he had only one ' i ' in his name. You ' d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it"The Road Ahead, or the Road Behind."He said,"Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ' t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ' s game. You and I know deeper down, there ' s always a chance to win the crown. But when we fail to give our best, we simply haven ' t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ’ s bearing down that wins the cup. Of dreaming there ’ s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ’ s you and I who make our fates — we open up or close the gates on the road ahead or the road behind.”Reminds me of another set of threes that my dad tried to get across to us : Don ’ t whine. Don ’ t complain. Don ’ t make excuses. Just get out there, and whatever you ’ re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ’ re outscored. I ’ ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ’ t know the outcome, I hope they couldn ’ t tell by your actions whether you outscored an opponent or the opponent outscored you. That ’ s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ’ d want them to be but they ’ ll be about what they should ; only you will know whether you can do that. And that ’ s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,”The journey is better than the end.”And I like that. I think that it is — it ’ s getting there. Sometimes when you get there, there ’ s almost a let down. But it ’ s the getting there that ’ s the fun. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ’ d done a decent job during the week. There again, it ’ s getting the players to get that self-satisfaction, in knowing that they ’ d made the effort to do the best of which they are capable. Sometimes I ’ m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,”Suppose that you, in some way, could make the perfect player. What would you want?”And I said,”Well, I ’ d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ’ d want one that could play, too. I ’ d want one to realize that defense usually wins championships, and who would work hard on defense. But I ’ d want one who would play offense, too. I ’ d want him to be unselfish, and look for the pass first and not shoot all the time. And I ’ d want one that could pass and would pass. I ’ ve had some that could and wouldn ’ t, and I ’ ve had some that would and could. So, yeah, I ’ d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ’ d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,”They Call Me Coach,”two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ’ t play varsity when I taught. I thought,”Oh gracious, if these two players, either one of them”— they were different years, but I thought about each one at the time he was there —”Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ’ s good enough to make it.”And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ’ d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 340: POS Dim 3 – 2001 – Score 2.00 – 2001/t000367.txt

I ’ d like to do pretty much what I did the first time, which is to choose a light-hearted theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, exorcise those evil spirits, and this is still going on, as you know. But it wasn ’ t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ’ re feeling depressed? It ’ s better than Prozac, but I wouldn ’ t recommend it. So what we see in the seventeenth, eighteenth century is the continued search for medications other than camphor that ’ ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we fast-forward to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a row — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very interested in producing convulsions, measured types of convulsions. And they thought,"Well, we ' ve got electricity, we ' ll plug somebody into the wall. That always makes hair stand up and people shake a lot."So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,"We know that at the Rome railroad station, there are all these lost souls wandering around, muttering gibberish. Can you bring one of them to us?"Someone who is, as the Italians say,"cagoots."So they found this"cagoots"guy, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,"Well, we ' ll try 55 volts, two-tenths of a second. That ' s not going to do anything terrible to him."So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,"This fellow"— remember, he wasn ' t even put to sleep —"after this major grand mal convulsion, sat right up, looked at these three fellas and said, ' What the fuck are you assholes trying to do? '"If I could only say that in Italian. Well, they were happy as could be, because he hadn ' t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and everybody in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ' t work very well on the schizophrenics, but it was pretty clear in the ' 30s and by the middle of the ' 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ' t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the brain waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ' 60s, the first antidepressants came out. Tofranil was the first. In the late ' 70s, early ' 80s, there were others, and they were very effective. And patients ' rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ' m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ' ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ’ m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ’ clock in the afternoon, because I couldn ’ t get out of bed before about 11 o ’ clock. And anybody who ’ s been depressed here knows what that ’ s like. I couldn ’ t even pull the covers off myself. Well, you ’ re in a university medical center, where everybody knows everybody, and it ’ s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ’ t work anymore. And, in fact, it didn ’ t make any difference because I didn ’ t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,”Don ’ t worry, chap. Six weeks, you ’ re back in the operating room. Everything ’ s going to be great.”Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ’ t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five feet in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ’ t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had. They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the senior staff. All the senior staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful stuff. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it got so there was a throbbing, there was a ferocious fear in my head. You ’ ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ’ t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case. He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the senior staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,"No. I know this man better than any of you. I have met with him over and over again. You ' ve just seen him from time to time. You ' ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ' ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go back to surgery, he will never be the **loving** father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ' One Flew Over the Cuckoo ' s Nest. ' And you know about that, just essentially in a stupor the rest of his life."Well, he said,"Can ' t we try a course of electroshock therapy?"And you know why they agreed? They agreed to humor him. They just thought,"Well, we ' ll give a course of 10. And so we ' ll lose a little time. Big deal. It doesn ' t make any difference."So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ' t work. Seven didn ' t work. Eight didn ' t work. At nine, I noticed — and it ' s wonderful that I could notice anything — I noticed a change. And at 10, I noticed a real change. And he went back to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, lo and behold, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could blow the obsessional thinking away. I could blow the depression away. And I ' ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,"I ' ve got the strength now to do this."It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a **formula**. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember"Ruddigore,"and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,"We must have a word to bring you back to reality, and the word, my dear, will be ' Basingstoke. '"So every time she got a little nuts, he would say,"Basingstoke!"And she would say,"Basingstoke, it is."And she would be fine for a little while. Well, you know, I ' m from the Bronx. I can ' t say"Basingstoke."But I had something better. And it was very simple. It was,"Ah, fuck it!"Much better than"Basingstoke,"at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,"Ah, fuck it."And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went back to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came back to live with us. We had two more children after that. Resuscitated the career, even better than it had been before. Went right back into the university and began to write books. Well, you know, it ’ s been a wonderful life. It ’ s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ’ s been very exciting. It ’ s been very happy. Every once in a while, I have to say,”Ah, fuck it.”Every once in a while, I get somewhat depressed and a little obsessional. So, I ’ m not free of all of this. But it ’ s worked. It ’ s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ’ ve gotten thousands of letters about them by people who do think this — that based on my life ’ s history as I ’ ve portrayed in the books, my early life ’ s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the **bitter** dregs of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ’ ve always felt guilty about that. I ’ ve always felt that somehow I was an impostor because my readers don ’ t know what I have just told you. It ’ s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an untroubled mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes back to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ’ t happen, there will be adversities. If I, with the bleakness of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way back from this, believe me, anybody can find their way back from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

Text Sample 341: POS Dim 3 – 2001 – Score 1.00 – 2001/t000154.txt
So I understand that this meeting was planned, and the slogan was From Was

to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,”Are you still doing this or that?”I don ’ t answer because I ’ m not doing things still, I ’ m doing it like I always did. I still have — or did I use the word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years. But, of course, I ’ m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ’ ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called”Innovation.”Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,”The playful search for beauty was Man ’ s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty.”These are tiles. The word,”playful”is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ’ t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ’ t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of **innocent** people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 342: POS Dim 3 – 2019 – Score 19.00 – 2019/t001227.txt
So I know for sure there ’ s at least one thing I have in common with dentists. I absolutely hate the holiday of Halloween. Now, this **hatred** stems not from a dislike of cavities, nor was it a lifetime in the making. Rather, this **hatred** stems from a particular incident that happened nine years ago. Nine years ago, I was even younger, I was 20 years old, and I was an intern in the White House. The other White House. And my job was to work with mayors and councilors nationwide. November 1, 2010 began just like any other day. I turned on the computer, went on Google and prepared to write my news clips. I was met with a call from my mother, which isn ’ t that out the norm, my mom likes to text, call, email, Facebook, Instagram, all that. So I answered the phone expecting to hear maybe some church gossip, or maybe something from WorldStarHipHop she had discovered. But when I answered the phone, I was met with a tone that was unlike anything I had ever heard from my mother. My mother ’ s loud. But she spoke in a hush, still, muffled tone that conveyed a sense of sadness. And as she whispered, she said,”Michael, your cousin Donnell was murdered last night, on Halloween, at a house party in Stockton.”And like far too many people in this country, particularly from communities like mine, particularly that look like me, I spent the better part of the year dealing with anger, rage, nihilism, and I had a choice to make. The choice was one between action and apathy. The choice was what could I do to put purpose to this pain. I spent a year dealing with feelings of survivor ’ s guilt. What was the point of me being at Stanford, what was the point of me being at the White House if I was powerless to help my own family? And my own family was dying, quite literally. I then began to feel a little selfish and say, what ’ s the point of even trying to make the world a better place? Maybe that ’ s just the way it is. Maybe I would be smart to take advantage of all the opportunities given to me and make as much money as possible, so I ’ m comfortable, and my immediate family is comfortable. But finally, towards the end of that year, I realized I wanted to do something. So I made the crazy decision, as a senior in college, to run for city council. That decision was unlikely for a couple of reasons, and not just my age. You see, my family is far from a political dynasty. More men in my family have been incarcerated than in college. In fact, as I speak today, my father is still incarcerated. My mother, she had me as a teenager, and government wasn ’ t something we had warm feelings from. You see, it was the government that red- lined the **neighborhoods** I grew up in. Full of liquor stores and no grocery stores, there was a lack of opportunity and concentrated poverty. It was the government and the politicians that made choices, like the war on drugs and three strikes, that have incarcerated far too many people in our country. It was the government and political actors that made the decisions that created the school funding **formulas**, that made it so the school I went to receive less per pupil spending than schools in more affluent areas. So there was nothing about that background that made it likely for me to choose to be involved in being a government actor. And at the same time, Stockton was a very unlikely

place. Stockton is my home town, a city of 320, 000 people. But historically, it ’ s been a place people run from, rather than come back to. It ’ s a city that ’ s incredibly diverse. Thirty- five percent Latino, 35 percent white, 20 percent Asian, 10 percent African American, the oldest Sikh temple in North America. But at the time I ran for office, we were also the largest city in the country at that time to **declare** bankruptcy. At the time I decided to run for office, we also had more murders per capita than Chicago. At the time I decided to run for office, we had a 23 percent poverty rate, a 17 percent college attainment rate and a host of challenges and issues beyond the scope of any 21-year-old. So after I won my election, I did what I usually do when I feel overwhelmed, I realized the problems of Stockton were far bigger than me and that I might need a little divine intervention. So as I prepared for my first council meeting, I went back to some wisdom my grandmother taught me. A parable I think we all know, that really constitutes the governing frame we ’ re using to reinvent Stockton today. I remember in Sunday school, my grandmother told me that at one time, a guy asked Jesus,”Who was my neighbor? Who was my fellow citizen? Who am I responsible for?”And instead of a short answer, Jesus **replied** with a parable. He said there was a man on a journey, walking down Jericho Road. As he was walking down the road, he was beat up, left on the side of the road, stripped of all his clothes, had everything stolen from and left to die. And then a priest came by, saw the man on the side of the road, maybe said a silent **prayer**, hopes and **prayers**, **prayers** that he gets better. Maybe saw the man on the side of the road and surmised that it was ordained by God for this particular man, this particular group to be on the side of the road, there ’ s nothing I can do to change it. After the priest walked by, maybe a politician walked by. A 28-year-old politician, for example. Saw the man on the side of the road and saw how beat up the man was, saw that the man was a **victim** of **violence**, or fleeing **violence**. And the politician decided,”You know what? Instead of welcoming this man in, let ’ s build a wall. Maybe the politician said,”Maybe this man chose to be on the side of the road.”That if he just pulled himself up by his bootstraps, despite his boots being stolen, and got himself back on the horse, he could be successful, and there ’ s nothing I could do.”And then finally, my grandmother said, a good Samaritan came by, saw the man on the side of the road and looked and saw not centuries of **hatred** between Jews and Samaritans, looked and saw not his fears reflected, not economic anxiety, not”what ’ s going to happen to me because things are changing.”But looked and saw a reflection of himself. He saw his neighbor, he saw his common humanity. He didn ’ t just see it, he did something about it, my grandmother said. He got down on one knee, he made sure the man was OK, and I heard, even gave him a room at that nice Fairmont, the Pan Pacific one. And as I prepared to govern, I realized that given the diversity of Stockton, the first step to making change will be to again answer the same question : Who is our neighbor? And realizing that our destiny as a city was tied up in everyone. Particularly those who are left on the side of the road. But then I realized that charity isn ’ t justice, that acts of empathy isn ’ t justice, that being a good neighbor is necessary but not sufficient, and there was more that had to be done. So looking at the story, I realized that the road, Jericho Road, has a nickname. It ’ s known as the Bloody Pass, the Ascent of Red, because the road is structured for **violence**. This Jericho Road is narrow, it ’ s conducive for ambushing. Meaning, a man on the side of the road wasn ’ t abnormal. Wasn ’ t strange. And in fact, it was something that was structured to happen, it was supposed to happen. And Johan Galtung, a peace theorist, talks about structural **violence** in our society. He says,”Structural **violence** is the avoidable impairment of basic human needs.”Dr. Paul Farmer talks about structural **violence** and talks about how it ’ s the way our institutions, our policies, our culture creates outcomes that advantage some people and disadvantage others. And then I realized, much like the road in Jericho, in many ways, Stockton, our society, has been structured for the outcomes we complain about. That we should not be surprised when we see that kids in poverty don ’ t do well in school, that we should not be surprised to see wealth gaps by race and **ethnicity**. We should not be surprised to see income pay disparities between genders, because that ’ s what our society, historically, has been structured to do, and it ’ s working accordingly. So taking this wisdom, I rolled up my sleeves and began to work. And there ’ s three quick stories I want to share, that point to not that we figured everything out, not that we have arrived, but we ’ re trending in the right direction. The first story, about the neighbor. When I was a city council member, I was working with one of the most conservative members in our community on opening a health clinic for undocumented people in the south part of the city, and I loved it. And as we opened the clinic, we had a resolution to sign, he presented me a gift. It was an O ’ Reilly Factor lifetime membership pin. Mind you, I didn ’ t ask what he did to get such a gift. What blood oath — I had no idea how he got it. But I looked at him and I said,”Well, how are we working together to open a health clinic, to provide free health care for undocumented people, and you ’ re an O ’ Reilly Factor member?”He looked at me and said,”Councilman Tubbs, this is for my neighbors.”And he ’ s a great example of what it means to be a good neighbor, at least in that instance. The robbers. So after four years on city council, I decided to run for mayor, realizing that being a part-time councilman wasn ’ t enough to enact the structural changes we need to see in Stockton, and I came to that conclusion by looking at the data. So my old council district, where I grew up, is 10 minutes away from a more affluent district. And 10 minutes away in the same city, the difference between zip code 95205 and 95219 in life expectancy is 10 years. Ten minutes away, 4. 5 miles, 10 years life expectancy difference, and not because of the choices people are making. Because no one chose to live in an **unsafe** community where they can ’ t exercise. No one chose to put more liquor stores than grocery stores in the community. No one chose these things, but that ’ s the reality. I realized, as a councilman, to enact a structural change I wanted to see, where between the same zip codes there ’ s a 30 percent difference in the rate of **unemployment**, there ’ s a 75, 000 dollars a year difference in income, that being a councilman was not going to cut it. So that ’ s when I decided to run for mayor. And as mayor, we ’ ve been focused on the robbers and the road. So in Stockton, as I mentioned, we have historically had problems with **violent** crime. In fact, that ’ s why I decided to run for office in the first place. And my first job as mayor was helping our community to see ourselves, our neighbors, not just in the people victimized by **violence** but also in the **perpetrators**. We realized that those who enact pain in our society, those who are committing homicides and contributing to gun **violence**, are oftentimes **victims** themselves. They have high rates of **trauma**, they have been shot at, they ’ ve known people who have been shot. That doesn ’ t excuse their behavior, but it helps explain it, and as a community, we have to see these folks as us, too. That they too are our neighbors. So for the past three years — So for the past three years, we ’ ve been working on two strategies : Ceasefire and Advance Peace, where we give these guys as much attention, as much love from social services, from opportunities, from tattoo removals, in some cases even cash, as a gift from law **enforcement**. And last year, we saw a 40 percent reduction in homicides and a 30 percent reduction in **violent** crime. And now, the road. I mentioned that my community has a 23 percent poverty rate. As someone who comes from poverty, it ’ s a personal issue for me. So I decided that we wouldn ’ t just do a program, or we wouldn ’ t just do something to go around the edges, but we would call into question the very structure that produces poverty in the first place. So starting in February, we launched a basic income demonstration, where for the next 18 months, as a pilot, 130 families, randomly selected, who live in zip codes at or below the median income of the city, are given 500 dollars a month. And we ’ re doing this for a couple of reasons. We ’ re doing it because we realize that something is structurally wrong in America, when one in two Americans can ’ t afford one 400- dollar emergency. We ’ re doing it because we realize that something is structurally wrong when wages have only increased six percent between 1979 and 2013. We ’ re doing it because we realize something is structurally wrong when people working two and three jobs, doing all the jobs no one in here wants to do, can ’ t pay for necessities, like rent, like lights, like health care, like childcare. So I would say, Stockton again, we have real issues. I have constituent emails in my phone now, about the **homelessness** issue, about some of the **violent** crime we ’ re still experiencing. But I would say, I think as a society, we would be wise to go back to those old Bible stories we were taught growing up, and understand that number one, we have to begin to see each other as neighbors, that when we see someone different from us, they should not reflect our fears, our anxieties, our insecurities, the **prejudices** we ’ ve been taught, our biases — but we should see ourselves. We should see our common humanity. Because I think once we do that, we can do the more important work of restructuring the road. Because again, I understand some listening are saying,”Well, Mayor Tubbs, you ’ re talking about structural **violence** and structural this, but you ’ re on the stage. That the structures can ’ t be too bad if you could come up from poverty, have a father in jail, go to Stanford, work in the White House and become mayor.”And I would respond by saying the term for that is exceptionalism. Meaning that we recognize it ’ s exceptional for people to escape the structures. Meaning by our very language, we understand that the things we ’ re seeing in our world are by design. And I think that task for us, as TEDsters, and as good people, just people, moral people, is really do the hard work necessary of not just joining hands as neighbors, but using our hands to restructure our road, a road that in this country has been rooted in things like white supremacy. A road like in this country has been rooted in things like misogyny. A road that ’ s not working for far too many people. And I think today, tomorrow and 2020 we have a chance to change that. So as I prepare to close, I started with a story from nine years ago and I ’ ll end with one. So after my cousin was murdered, I was lucky enough to go on the Freedom Rides with some of the original freedom riders. And they taught me a lot about restructuring the road. And one guy in particular, Bob Singleton, asked me a question I ’ m going to leave with us today. We were going to Anniston, Alabama, and he said,”Michael,”and I said,”Yes, sir.”He said,”I was arrested on August 4, 1961. Now why is that day important?”And I said,”Well, you were arrested, if you weren ’ t arrested, we wouldn ’ t be on this bus. if we weren ’ t on this bus, we wouldn ’ t have the rights we enjoy.”He rolled his eyes and said,”No, son.”He said,”On that day, Barack Obama was born.”And then he said he had no idea that the choice he made to restructure the road would pave the way, so a child born as a second class citizen, who wouldn ’ t be able to even get a cup of water at a counter, would have the chance, 50 years later, to be president. Then he looked at me and he said,”What are you prepared to do today so that 50 years from now a child born has a chance to be president?”And I think, TED, that ’ s the question before us today. We know things are jacked up. I think what we ’ ve seen recently isn ’ t abnormal but a reflection of a system that ’ s been structured to produce such crazy outcomes. But I think it ’ s also an opportunity. Because these structures we inherit aren ’ t acts of God but acts of men and women, they

’ re policy choices, they ’ re by politicians like me, approved by voters like you. And we have the chance and the awesome opportunity to do something about it. So my question is : What are we prepared to do today, so that a child born today, 50 years from now isn ’ t born in a society rooted in white supremacy ; isn ’ t born into a society riddled with misogyny ; isn ’ t born into a society riddled with homophobia and transphobia and anti-Semitism and Islamophobia and ableism, and all the phobias and -isms? What are we prepared to do today, so that 50 years from now we have a road in our society that ’ s structured to reflect what we hold to be self-evident? That all men, that all women, that even all trans people are created **equal** and are endowed by your Creator with certain unalienable rights, including life, liberty and the pursuit of happiness. Thank you.

Text Sample 343: POS Dim 3 – 2019 – Score 16.00 – 2019/t000020.txt
When people meet me for the first time on my job, they often feel inspired to share a revelation they ’ ve had about me, and it kind of goes something like this.”Hey, I know why police chiefs like to share their deep, dark secrets with you. Phil, with your PhD in psychology, and your shiny bald head, you ’ re basically the Black Dr. Phil, right?”And for each and every person who ’ s ever said that to me I do want to say thank you because that was the first time I ever heard that joke. But for everybody else, I really hope you ’ ll believe me when I tell you no police chief likes talking to me because they think I ’ m a clinical psychologist. And also I ’ m not. I have no idea what your mother did to you, and I can ’ t help. Police chiefs like talking to me because I ’ m an expert on a problem that feels impossible for them to solve : racism in their profession. Now my expertise comes from being a scientist who studies how our minds learn to associate Blackness and crime and misperceive Black children as older than they actually are. It also comes from studying actual police behavior, which is how I know that every year, about one in five adults in the United States has contact with law **enforcement**. Out of those, about a million are targeted for police use of force. And if you ’ re Black, you ’ re two to four times more likely to be targeted for that force than if you ’ re white. But it also comes from knowing what those statistics feel like. I ’ ve experienced the fear of seeing an officer unclip their gun and the panic of realizing that someone might mistake my 13-year-old godson as old enough to be a threat. So when a police chief, or a pastor, or an imam, or a mother — when they call me after an officer shoots another unarmed Black child, I understand a bit of the pain in their voice. It ’ s the pain of a heart breaking when it fails to solve a deadly problem. Breaking from trying to do something that feels simultaneously necessary and impossible. The way trying to fix racism usually feels. Necessary and impossible. So, police chiefs like talking to me because I ’ m an expert, but I doubt they ’ d be lining up to lie down on Dr. Phil ’ s couch if I told them all their problems were hopeless. All of my research, and the decade of work I ’ ve done with my center — the Center for Policing **Equity** — actually leads me to a hopeful conclusion amidst all the heartbreak of race in America, which is this : trying to solve racism feels impossible because our definition of racism makes it impossible — but it doesn ’ t have to be that way. So here ’ s what I mean. The most common definition of racism is that racist behaviors are the product of contaminated hearts and minds. When you listen to the way we talk about trying to cure racism, you ’ ll hear it.”We need to stamp out **hatred**. We need to combat ignorance,”right? It ’ s hearts and minds. Now the only problem with that definition is that it ’ s completely wrong — both scientifically and

otherwise. One of the foundational insights of social psychology is that **attitudes** are very weak predictors of behaviors, but more importantly than that, no Black community has ever taken to the **streets** to demand that white people would love us more. Communities march to stop the killing, because racism is about behaviors, not feelings. And even when civil rights leaders like King and Fannie Lou Hamer used the language of love, the racism they fought, that was segregation and brutality. It ' s actions over feelings. And every one of those leaders would agree, if a definition of racism makes it harder to see the injuries racism causes, that ' s not just wrong. A definition that cares about the intentions of abusers more than the **harms** to the abused — that definition of racism is racist. But when we change the definition of racism from **attitudes** to behaviors, we transform that problem from impossible to solvable. Because you can measure behaviors. And when you can measure a problem, you can tap into one of the only universal rules of organizational success. You ' ve got a problem or a goal, you measure it, you hold yourself accountable to that metric. So if every other organization measures success this way, why can ' t we do that in policing? It turns out we actually already do. Police departments already practice data- driven accountability, it ' s just for crime. The vast majority of police departments across the United States use a system called CompStat. It ' s a process that, when you use it right, it identifies crime data, it tracks it and identifies patterns, and then it allows departments to hold themselves accountable to public **safety** goals. It usually works either by directing police attentions and police resources, or changing police behavior once they show up. So if I see a string of muggings in that **neighborhood**, I ' m going to want to increase patrols in that **neighborhood**. If I see a spike in homicides, I ' m going to want to talk to the community to find out why and collaborate on changes on police behavior to tamp down the **violence**. Now when you define racism in terms of measurable behaviors, you can do the same thing. You can create a CompStat for justice. That ' s exactly what the Center for Policing Equity has been doing. So let me tell you how that works. After a police department invites us in, we handle the legal stuff, we engage with the community, our next step is to analyze their data. The goal of these analyses is to determine how much do crime, poverty, **neighborhood** demographics predict, let ' s say, police use of force? Let ' s say that those factors predict police will use force on this many Black people. There? So our next question is, how many Black people actually are targeted for police use of force? Let ' s say it ' s this many. So what ' s up with the gap? Well, a big portion of the gap is the difference between what ' s predicted by things police can ' t control and what ' s predicted by things police can control — their policies and their behaviors. And what we ' re looking for are the types of contact or the areas in the city where that gap is biggest, because then we can tell our partners,"Look here. Solve this problem first."It ' s actually the kind of therapy police chiefs can get behind, because there is nothing so inspiring in the face of our history of racism as a solvable problem. Look, if the community in Minneapolis asked their police department to remedy the moral failings of race in policing, I ' m not sure they know how to do that. But if instead the community says,"Hey, you ' re data say you ' re beating up a lot of **homeless** folks. You want to knock that off?"That ' s something police can learn how to do. And they did. So in 2015, the Minneapolis PD let us know their community was concerned they were using force too often. So we showed them how to leverage their own data to identify situations where force could be avoided. And when you look at those data, you ' ll see that a disproportionate number of their use-of-force incidents, they involved somebody who ' s **homeless**, in mental distress, has a substance abuse issue or some combination of all three — more than you expect based on those factors I was just telling you about. So right there ' s the gap. Next question is why. Well, it turns out **homeless** folks often need services. And when those services are unavailable, when they can ' t get their meds, they lose their spot in the shelter, they ' re more likely to engage in behaviors that end up with folks calling the cops. And when the cops show up, they ' re more likely to resist intervention, oftentimes because they haven ' t actually done anything illegal, they ' re literally just living outside. The problem wasn ' t a need to train officers differently in Minneapolis. The problem was the fact that folks were using the cops to"treat"substance abuse and **homelessness** in the first place. So the city of Minneapolis found a way to deliver social services and city resources to the **homeless** community before anybody ever called the cops. Now the problem isn ' t always **homelessness**, right? Sometimes the problem is fear of **immigration enforcement**, like it was in Salt Lake City, or it is in Houston, where the chiefs had to come forward and say,"We ' re not going to deport you just for calling 911."Or the problem is foot pursuits, like it was in Las Vegas, where they had to train their officers to slow down and take a breath instead of allowing the adrenaline in that situation to escalate it. It ' s searches in Oakland ; it ' s pulling folks out of cars in San Jose ; it ' s the way that they patrol the **neighborhoods** that make up Zone 3 in Pittsburgh and the Black **neighborhoods** closest to the waterfront in Baltimore. But in each city, if we can give them a solvable problem, they get busy solving it. And together our partners have seen an average of 25 percent fewer arrests, fewer use-of- force incidents and 13 percent fewer officer-related injuries. Essentially, by identifying the biggest gaps and directing police attentions to solving it, we can deliver a data-driven vaccine against racial disparities in policing. Right now, we have the capacity to partner with about 40 cities at a time. That means if we want the United States to stop feeling exhausted from trying to solve an impossible problem, we ' re going to need a lot more infrastructure. Because our goal is to have our tools be able to scale the brilliance of dedicated organizers and reform-minded chiefs. So to get there we ' re going to need the kind of collective will that desegregated schools and won the franchise for the sons and daughters of former slaves so that we can build a kind of health care system capable of delivering our vaccine across the country. Because our **audacious** idea is to deliver a CompStat for justice to departments serving 100 million people across the United States in the next five years. Doing that would mean arming about a third of the United States with tools to reduce racial disparities in police stops, arrests and use of force, but also tools to reduce predatory cash **bail** and mass incarceration, family instability and chronic mental health and substance abuse issues, and every other **ill** that our broken criminal-legal systems aggravate. Because every unnecessary arrest we can prevent saves a family from the terrifying journey through each one of those systems. Just like every gun we can leave holstered saves an entire community from a lifetime of **grief**. Look, each and every one of us, we measure the things that matter to us. Businesses measure profit ; good students keep track of their grades ; families chart the growth of their children with pencil markings in doorframes. We all measure the things that matter most to us, which is why we feel the neglect when nobody ' s bothering to measure anything at all. For the past quarter millennium, we ' ve defined the problems of race and policing in a way that ' s functionally impossible to measure. But now the science says we can just change that definition. And the folks at the Center for Policing Equity, I actually think we may have measured more police behavior than any one in human history. And that means that once we have the will and the resources to do it, this could be the generation that stops feeling like racism is an unsolvable problem and instead sees that what ' s been necessary for far too long is possible. Thank you.

Text Sample 344: POS Dim 3 – 2019 – Score 14.00 – 2019/t001497.txt
My parents gave me an extraordinary name : Baratunde Rafiq Thurston. Now, Baratunde is based on a Yoruba name from Nigeria, but we ’ re not Nigerian. That ’ s just how black my mama was.”Get this boy the blackest name possible. What does the book say?”Rafiq is an Arabic name, but we are not Arabs. My mom just wanted me to have difficulty boarding planes in the 21st century. She foresaw America ’ s turn toward nativism. She was a black futurist. Thurston is a British name, but we are not British. Shoutout to the multigenerational, dehumanizing economic institution of American chattel slavery, though. Also, Thurston makes for a great Starbucks name. Really expedites the process. My mother was a renaissance woman. Arnita Lorraine Thurston was a computer programmer, former **domestic** worker, survivor of **sexual assault**, an artist and an activist. She prepared me for this world with lessons in black history, in martial arts, in urban farming, and then she sent me in the seventh grade to the private Sidwell Friends School, where US presidents send their daughters, and where she sent me looking like this. I had two key tasks going to that school : don ’ t lose your blackness and don ’ t lose your glasses. This accomplished both. Sidwell was a great place to learn the arts and the sciences, but also the art of living amongst whiteness. That would prepare me for life later at Harvard, or doing corporate consulting, or for my jobs at”The Daily Show”and”The Onion.”I would write down many of these lessons in my memoir,”How to Be Black,”which if you haven ’ t read yet, makes you a racist, because — you ’ ve had plenty of time to read the book. But America insists on reminding me and teaching me what it means to be black in America. It ’ s December 2018, I ’ m with my fianc in the suburbs of Wisconsin. We are visiting her parents, both of whom are white, which makes her white. That ’ s how it works. I don ’ t make the rules. She ’ s had some drinks, so I drive us in her parents ’ car, and we get pulled over by the police. I ’ m scared. I turn on the flashing lights to indicate compliance. I pull over slowly under the brightest streetlight I can find in case I need witnesses or dashcam footage. We get out my identification, the car registration, lay it out in the open, roll down the windows, my hands are placed on the steering wheel, all before the officer exits the vehicle. This is how to stay alive. As we wait, I think about these headlines —”Police shoot another unarmed black person”— and I don ’ t want to join them. The good news is, our officer was friendly. She told us our tags were expired. So to all the white parents out there, if your child is involved with a person whose skin tone is rated Dwayne”The Rock”Johnson or darker — you need to get that car inspected, update the paperwork every time we visit. That ’ s just common courtesy. I got lucky. I got a law **enforcement** professional. I survived something that should not require survival. And I think about this series of stories —”Police shoot another unarmed black person”— and that season when those stories popped up everywhere. I would scroll through my feed and I would see a baby announcement photo. I ’ d see an ad for a product I had just whispered to a friend about yesterday. I would see a video of a police officer gunning down someone who looked just like me. And I ’ d see a think piece about how millennials have replaced sex with avocado toast. It was a confusing time. Those stories kept popping up, but in 2018, those stories got changed out for a different type of story, stories like,”White Woman Calls Cops On Black Woman Waiting For An Uber.”That was Brooklyn Becky. Then there was,”White Woman Calls Police On Eight-Year-Old Black Girl Selling Water.”That was Per-

mit Patty. Then there was,"Woman Calls Police On Black Family BBQing At Lake In Oakland."That was now infamous BBQ Becky. And I contend that these stories of living while black are actually progress. We used to find out after the extrajudicial police killings. Now, we ' re getting video of people calling 911. We ' re moving upstream, closer to the problem and closer to the solution. So I started a collection of as many of these stories as I could find. I built an evolving, still-growing database at baratunde. com / livingwhileblack. Seeking understanding, I realized the process was really diagramming sentences to understand these headlines. And I want to thank my Sidwell English teacher Erica Berry and all English teachers. You have given us tools to fight for our own freedom. What I found was a process to break down the headline and understand the consistent layers in each one : a subject takes an action against a target engaged in some activity, so that"White Woman Calls Police On Eight-Year-Old Black Girl"is the same as"White Man Calls Police On Black Woman Using **Neighborhood** Pool"is the same as"Woman Calls Cops On Black Oregon Lawmaker Campaigning In Her District."They ' re the same. Diagramming the sentences allowed me to diagram the white supremacy which allowed such sentences to be true, and I will pause to define my terms. When I say"white supremacy,"I ' m not just talking about Nazis or white power activists, and I ' m definitely not saying that all white people are racist. What I ' m referring to is a system of structural advantage that favors white people over others in social, economic and political arenas. It ' s what Bryan Stevenson at the Equal Justice Initiative calls the narrative of racial difference, the story we told ourselves to justify slavery and Jim Crow and mass incarceration and beyond. So when I saw this pattern repeating, I got angry, but I also got inspired to create a game, a game of words that would allow me to transform this traumatic exposure into more of a healing experience. I ' m going to talk you through the game. The first level is a training level, and I need your participation. Our objective : to determine if this is real or fake. Did this happen or not? Here is the example :"Catholic University Law Librarian Calls Police On Student For ' Being Argumentative. '"Clap your hands if you think this is real. Clap your hands if you think this is fake. The reals have it, unfortunately, and a point of information, being argumentative in a law library is the exact right place to do that. This student should be promoted to professor. Training level complete, so we move on to the real levels. Level one, our objective is simple : reverse the roles. That means"Woman Calls Cops On Black Oregon Lawmaker"becomes"Black Oregon Lawmaker Calls Cops On Woman."That means"White Man Calls Police On Black Woman Using **Neighborhood** Pool"becomes"Black Woman Calls Police On White Man Using **Neighborhood** Pool."How do you like them reverse racist apples? That ' s it, level one complete, and so we level up to level two, where our objective is to increase the believability of the reversal. Let ' s face it, a black woman calling police on a white man using a pool isn ' t absurd enough, but what if that white man was trying to touch her hair without asking, or maybe he was making oat milk while riding a unicycle, or maybe he ' s just talking over everyone in a meeting. We ' ve all been there, right? Seriously, we ' ve all been there. So that ' s it, level two complete. But it comes with a warning : simply reversing the flow of injustice is not justice. That is vengeance, that is not our mission, that ' s a different game so we level up to level three, where the objective is to change the action, also known as"calling the police is not your only option OMG, what is wrong with you people!"And I need to pause the game to remind us of the structure. A subject takes an action against a target engaged in some activity."White Woman Calls Police On Black Real Estate Investor Inspecting His Own Property.""California Safeway Calls Cops On Black Woman Donating Food To The Homeless.""Gold Club

Twice Calls Cops On Black Women For Playing Too Slow."In all these cases, the subject is usually white, the target is usually black, and the activities are anything, from sitting in a Starbucks to using the wrong type of barbecue to napping to walking"agitated"on the way to work, which I just call"walking to work."And, my personal favorite, not stopping his dog from humping her dog, which is clearly a case for dog police, not people police. All of these activities add up to living. Our existence is being interpreted as crime. Now, this is the obligatory moment in the presentation where I have to say, not everything is about race. Crime is a thing, should be reported, but ask yourself, do we need **armed** men to show up and resolve this situation, because when they show up for me, it ' s different. We know that police officers use force more with black people than with white people, and we are learning the role of 911 calls in this. Thanks to preliminary research from the Center for Policing Equity, we ' re learning that in some cities, most of the interactions between cops and citizens is due to 911 calls, not officer-initiated stops, and most of the **violence**, the use of force by police on citizens, is in response to those calls. Further, when those officers responding to calls use force, that increases in areas where the percentage of the white population has also increased, aka gentrification, aka unicycles and oat milk, aka when BBQ Becky feels threatened, she becomes a threat to me in my own **neighborhood**, which forces me and people like me to police ourselves. We quiet ourselves, we walk on eggshells, we maybe pull over to the side of the road under the brightest light we can find so that our murder might be caught cleanly on camera, and we do this because we live in a system in which white people can too easily call on deadly force to ensure their comfort. The California Safeway didn ' t just call cops on black woman donating food to **homeless**. They ordered **armed**, unaccountable men upon her. They essentially called in a drone strike. This is weaponized discomfort, and it is not new. From 1877 to 1950, there were at least 4, 400 documented racial terror lynchings of black people in the United States. They had headlines as well."Rev. T. A. Allen was lynched in Hernando, Mississippi for organizing local sharecroppers.""Oliver Moore was lynched in Edgecomb County, North Carolina, for frightening a white girl.""Nathan Bird was lynched near Luling, Texas, for refusing to turn his son over to a mob."We need to change the action, whether that action is"lynches"or"calls police."And now that I have shortened the distance between those two, let ' s get back to our game, to our mission. Our objective in level three is to change the action. So what if, instead of"Calls Cops On Black Woman Donating Food To Homeless,"that California Safeway simply thanks her. Thanking is far cheaper than bringing law **enforcement** to the scene. Or, instead, they could give the food they would have wasted to her, upped their civic cred. Or, the white woman who called the police on the eight-year- old black girl, she could have bought all the inventory from that little black girl, support a small business. And the white woman who called the police on the black real estate investor, we would all be better off, the cops agree, if she had simply ignored him and minded her own damn business. Minding one ' s own damn business is an excellent choice, excellent choice. Choose it more often. Level three is complete, but there is a final bonus level, where the objective is inclusion. We have also seen headlines like this :"Powerful Man Masturbates In Front Of Young Women Visiting His Office."What an odd choice for powerful man to make. So many other actions available to him. Like, such as,"listens to,""mentors,""inspired by, starts joint venture, everybody rich now."I want to live in that world of everybody rich now, but because of his poor choice, we are all in a poorer world. Doesn ' t have to be this way. This word game reminded me that there is a structure to white supremacy, as there is to misogyny, as there is to all **systemic** abuses of power. Structure is what makes them **sys-**

temic. I ’ m asking people here to see the structure, where the power is in it, and even more importantly to see the humanity of those of us made targets by this structure. I am here because I was loved and invested in and protected and lucky, because I went to the right schools, I ’ m semifamous, mostly happy, meditate twice a day, and yet, I walk around in fear, because I know that someone seeing me as a threat can become a threat to my life, and I am tired. I am tired of carrying this invisible burden of other people ’ s fears, and many of us are, and we shouldn ’ t have to, because we can change this, because we can change the action, which changes the story, which changes the system that allows those stories to happen. Systems are just collective stories we all buy into. When we change them, we write a better reality for us all to be a part of. I am asking us to use our power to choose. I am asking us to level up. Thank you. I am Baratunde Rafiq Thurston.

Text Sample 345: POS Dim 3 – 2016 – Score 18.00 – 2016/t001187.txt
[This talk contains graphic language and descriptions of **sexual violence** Viewer discretion is advised] Tom Stranger : In 1996, when I was 18 years old, I had the golden opportunity to go on an international exchange program. Ironically I ’ m an Australian who prefers proper icy cold weather, so I was both excited and tearful when I got on a plane to Iceland, after just having farewelled my parents and brothers goodbye. I was welcomed into the home of a beautiful Icelandic family who took me hiking, and helped me get a grasp of the melodic Icelandic language. I struggled a bit with the initial period of homesickness. I snowboarded after school, and I slept a lot. Two hours of chemistry class in a language that you don ’ t yet fully understand can be a pretty good sedative. My teacher recommended I try out for the school play, just to get me a bit more socially active. It turns out I didn ’ t end up being part of the play, but through it I met Thordis. We shared a lovely teenage romance, and we ’ d meet at lunchtimes to just hold hands and walk around old downtown Reykjavk. I met her welcoming family, and she met my friends. We ’ d been in a budding relationship for a bit over a month when our school ’ s Christmas Ball was held. Thordis Elva : I was 16 and in love for the first time. Going together to the Christmas dance was a public confirmation of our relationship, and I felt like the luckiest girl in the world. No longer a child, but a young woman. High on my newfound maturity, I felt it was only natural to try drinking rum for the first time that night, too. That was a bad idea. I became very **ill**, drifting in and out of consciousness in between spasms of convulsive vomiting. The security guards wanted to call me an ambulance, but Tom acted as my knight in shining armor, and told them he ’ d take me home. It was like a fairy tale, his strong arms around me, laying me in the **safety** of my bed. But the gratitude that I felt towards him soon turned to horror as he proceeded to take off my clothes and get on top of me. My head had cleared up, but my body was still too weak to fight back, and the pain was blinding. I thought I ’ d be severed in two. In order to stay sane, I silently counted the seconds on my alarm clock. And ever since that night, I ’ ve known that there are 7, 200 seconds in two hours. Despite limping for days and crying for weeks, this incident didn ’ t fit my ideas about rape like I ’ d seen on TV. Tom wasn ’ t an **armed** lunatic ; he was my boyfriend. And it didn ’ t happen in a seedy alleyway, it happened in my own bed. By the time I could identify what had happened to me as rape, he had completed his exchange program and left for Australia. So I told myself it was pointless to address what had happened. And besides, it had to have been my fault, somehow. I was raised in a world where girls are taught that

they get raped for a reason. Their skirt was too short, their smile was too wide, their breath smelled of alcohol. And I was guilty of all of those things, so the **shame** had to be mine. It took me years to realize that only one thing could have stopped me from being raped that night, and it wasn ' t my skirt, it wasn ' t my smile, it wasn ' t my childish trust. The only thing that could ' ve stopped me from being raped that night is the man who raped me — had he stopped himself. TS : I have vague memories of the next day : the after effects of drinking, a certain hollowness that I tried to stifle. Nothing more. But I didn ' t show up at Thordis ' s door. It is important to now state that I didn ' t see my deed for what it was. The word"rape"didn ' t echo around my mind as it should ' ve, and I wasn ' t crucifying myself with memories of the night before. It wasn ' t so much a conscious refusal, it was more like any acknowledgment of reality was forbidden. My definition of my actions completely refuted any recognition of the immense **trauma** I caused Thordis. To be honest, I repudiated the entire act in the days afterwards and when I was committing it. I disavowed the truth by convincing myself it was sex and not rape. And this is a lie I ' ve felt spine- bending guilt for. I broke up with Thordis a couple of days later, and then saw her a number of times during the remainder of my year in Iceland, feeling a sharp stab of heavyheartedness each time. Deep down, I knew I ' d done something immeasurably wrong. But without planning it, I sunk the memories deep, and then I tied a rock to them. What followed is a nine-year period that can best be titled as"Denial and Running."When I got a chance to identify the real torment that I caused, I didn ' t stand still long enough to do so. Whether it be via distraction, substance use, thrill-seeking or the scrupulous policing of my inner speak, I refused to be static and silent. And with this noise, I also drew heavily upon other parts of my life to construct a picture of who I was. I was a surfer, a social science student, a friend to good people, a loved brother and son, an outdoor recreation guide, and eventually, a youth worker. I gripped tight to the simple notion that I wasn ' t a bad person. I didn ' t think I had this in my bones. I thought I was made up of something else. In my nurtured upbringing, my **loving** extended family and role models, people close to me were warm and genuine in their respect shown towards women. It took me a long time to stare down this dark corner of myself, and to ask it questions. TE : Nine years after the Christmas dance, I was 25 years old, and headed straight for a nervous breakdown. My self-worth was buried under a soul- crushing load of **silence** that isolated me from everyone that I cared about, and I was consumed with misplaced **hatred** and anger that I took out on myself. One day, I stormed out of the door in tears after a fight with a loved one, and I wandered into a caf, where I asked the waitress for a pen. I always had a notebook with me, claiming that it was to jot down ideas in moments of inspiration, but the truth was that I needed to be constantly fidgeting, because in moments of stillness, I found myself counting seconds again. But that day, I watched in wonder as the words streamed out of my pen, forming the most pivotal letter I ' ve ever written, addressed to Tom. Along with an account of the **violence** that he subjected me to, the words,"I want to find forgiveness"stared back at me, surprising nobody more than myself. But deep down I realized that this was my way out of my **suffering**, because regardless of whether or not he deserved my forgiveness, I deserved peace. My era of **shame** was over. Before sending the letter, I prepared myself for all kinds of negative responses, or what I found likeliest : no response whatsoever. The only outcome that I didn ' t prepare myself for was the one that I then got — a typed **confession** from Tom, full of disarming regret. As it turns out, he, too, had been imprisoned by **silence**. And this marked the start of an eight-year-long correspondence that God knows was never easy, but always honest. I relieved myself of the burdens that I ’ d wrongfully shouldered, and he, in turn, wholeheartedly owned up to what he ’ d done. Our written exchanges became a platform to dissect the consequences of that night, and they were everything from gut-wrenching to healing beyond words. And yet, it didn ’ t bring about closure for me. Perhaps because the email format didn ’ t feel personal enough, perhaps because it ’ s easy to be brave when you ’ re hiding behind a computer screen on the other side of the planet. But we ’ d begun a dialogue that I felt was necessary to explore to its fullest. So, after eight years of writing, and nearly 16 years after that dire night, I mustered the courage to propose a wild idea : that we ’ d meet up in person and face our past once and for all. TS : Iceland and Australia are geographically like this. In the middle of the two is South Africa. We decided upon the city of Cape Town, and there we met for one week. The city itself proved to be a stunningly powerful environment to focus on reconciliation and forgiveness. Nowhere else has healing and rapprochement been tested like it has in South Africa. As a nation, South Africa sought to sit within the truth of its past, and to listen to the details of its history. Knowing this only magnified the effect that Cape Town had on us. Over the course of this week, we literally spoke our life stories to each other, from start to finish. And this was about analyzing our own history. We followed a strict policy of being honest, and this also came with a certain exposure, an open-chested vulnerability. There were gutting **confessions**, and moments where we just absolutely couldn ’ t fathom the other person ’ s experience. The seismic effects of **sexual violence** were spoken aloud and felt, face to face. At other times, though, we found a soaring clarity, and even some totally unexpected but liberating laughter. When it came down to it, we did out best to listen to each other intently. And our individual realities were aired with an unfiltered purity that couldn ’ t do any less than lighten the soul. TE : Wanting to take revenge is a very human emotion — instinctual, even. And all I wanted to do for years was to hurt Tom back as deeply as he had hurt me. But had I not found a way out of the **hatred** and anger, I ’ m not sure I ’ d be standing here today. That isn ’ t to say that I didn ’ t have my doubts along the way. When the plane bounced on that landing strip in Cape Town, I remember thinking,”Why did I not just get myself a therapist and a bottle of vodka like a normal person would do?”At times, our search for understanding in Cape Town felt like an impossible quest, and all I wanted to do was to give up and go home to my **loving** husband, Vidir, and our son. But despite our difficulties, this journey did result in a victorious feeling that light had triumphed over **darkness**, that something constructive could be built out of the ruins. I read somewhere that you should try and be the person that you needed when you were younger. And back when I was a teenager, I would have needed to know that the **shame** wasn ’ t mine, that there ’ s hope after rape, that you can even find happiness, like I share with my husband today. Which is why I started writing feverishly upon my return from Cape Town, resulting in a book co-authored by Tom, that we hope can be of use to people from both ends of the perpetrator- survivor scale. If nothing else, it ’ s a story that we would ’ ve needed to hear when we were younger. Given the nature of our story, I know the words that inevitably accompany it — **victim**, rapist — and labels are a way to organize concepts, but they can also be dehumanizing in their connotations. Once someone ’ s been **deemed** a **victim**, it ’ s that much easier to file them away as someone damaged, dishonored, less than. And likewise, once someone has been branded a rapist, it ’ s that much easier to call him a monster — inhuman. But how will we understand what it is in human societies that produces **violence** if we refuse to recognize the humanity of those who commit it? And how — And how can we empower survivors if we ’ re making them feel less than? How can we discuss solutions to one of the biggest threats to the lives of women and children around the world, if the very words we use are part of the problem? TS : From what I ’ ve now learnt, my actions that night in 1996 were a self- centered taking. I felt deserving of Thordis ’ s body. I ’ ve had primarily positive social influences and examples of **equitable** behavior around me. But on that occasion, I chose to draw upon the negative ones. The ones that see women as having less intrinsic worth, and of men having some unspoken and symbolic claim to their bodies. These influences I speak of are external to me, though. And it was only me in that room making choices, nobody else. When you own something and really square up to your culpability, I do think a surprising thing can happen. It ’ s what I call a paradox of ownership. I thought I ’ d buckle under the weight of responsibility. I thought my certificate of humanity would be burnt. Instead, I was offered to really own what I did, and found that it didn ’ t possess the entirety of who I am. Put simply, something you ’ ve done doesn ’ t have to constitute the sum of who you are. The noise in my head abated. The indulgent self-pity was starved of oxygen, and it was replaced with the clean air of acceptance — an acceptance that I did hurt this wonderful person standing next to me ; an acceptance that I am part of a large and shockingly everyday grouping of men who have been sexually **violent** toward their partners. Don ’ t underestimate the power of words. Saying to Thordis that I raped her changed my accord with myself, as well as with her. But most importantly, the blame transferred from Thordis to me. Far too often, the responsibility is attributed to female survivors of **sexual violence**, and not to the males who enact it. Far too often, the denial and running leaves all parties at a great distance from the truth. There ’ s definitely a public conversation happening now, and like a lot of people, we ’ re heartened that there ’ s less retreating from this difficult but important discussion. I feel a real responsibility to add our voices to it. TE : What we did is not a **formula** that we ’ re prescribing for others. Nobody has the right to tell anyone else how to handle their deepest pain or their greatest error. Breaking your **silence** is never easy, and depending on where you are in the world, it can even be deadly to speak out about rape. I realize that even the most traumatic event of my life is still a testament to my privilege, because I can talk about it without getting ostracized, or even killed. But with that privilege of having a voice comes the responsibility of using it. That ’ s the least I owe my fellow survivors who can ’ t. The story we ’ ve just relayed is unique, and yet it is so common with **sexual violence** being a global pandemic. But it doesn ’ t have to be that way. One of the things that I found useful on my own healing journey is educating myself about **sexual violence**. And as a result, I ’ ve been reading, writing and speaking about this issue for over a decade now, going to conferences around the world. And in my experience, the attendees of such events are almost exclusively women. But it ’ s about time that we stop treating **sexual violence** as a women ’ s issue. A majority of **sexual violence** against women and men is perpetrated by men. And yet their voices are sorely underrepresented in this discussion. But all of us are needed here. Just imagine all the **suffering** we could alleviate if we dared to face this issue together. Thank you.

Text Sample 346: POS Dim 3 – 2016 – Score 15.00 – 2016/t001349.txt

I want to talk about sex for money. I ’ m not like most of the people you ’ ll have heard speaking about prostitution before. I ’ m not a police officer or a social worker. I ’ m not an academic, a journalist or a politician. And as you ’ ll probably have picked up from Maryam ’ s blurb, I ’ m not a nun, either. Most of those people would tell you that selling sex is degrading ; that no one would ever choose to do it ; that it ’ s dangerous ; women get abused and killed. In fact, most of those people would say,”There should be a law against it!”Maybe that sounds reasonable to you. It sounded reasonable to me until the closing months of 2009, when I was working two dead-end, minimum-wage jobs. Every month my wages would just replenish my overdraft. I was exhausted and my life was going nowhere. Like many others before me, I decided sex for money was a better option. Now don ’ t get me wrong — I would have loved to have won the lottery instead. But it wasn ’ t going to happen anytime soon, and my rent needed paying. So I signed up for my first shift in a brothel. In the years that have passed, I ’ ve had a lot of time to think. I ’ ve reconsidered the ideas I once had about prostitution. I ’ ve given a lot of thought to consent and the nature of work under capitalism. I ’ ve thought about gender inequality and the **sexual** and reproductive labor of women. I ’ ve experienced exploitation and **violence** at work. I ’ ve thought about what ’ s needed to protect other sex workers from these things. Maybe you ’ ve thought about them, too. In this talk, I ’ ll take you through the four main legal approaches applied to sex work throughout the world, and explain why they don ’ t work ; why prohibiting the sex industry actually exacerbates every **harm** that sex workers are vulnerable to. Then I ’ m going tell you about what we, as sex workers, actually want. The first approach is full criminalization. Half the world, including Russia, South Africa and most of the US, regulates sex work by criminalizing everyone involved. So that ’ s seller, buyer and third parties. Lawmakers in these countries apparently hope that the fear of getting arrested will deter people from selling sex. But if you ’ re forced to choose between obeying the law and feeding yourself or your family, you ’ re going to do the work anyway, and take the risk. Criminalization is a trap. It ’ s hard to get a conventional job when you have a criminal record. Potential employers won ’ t hire you. Assuming you still need money, you ’ ll stay in the more flexible, informal economy. The law forces you to keep selling sex, which is the exact opposite of its intended effect. Being criminalized leaves you exposed to mistreatment by the state itself. In many places you may be coerced into paying a bribe or even into having sex with a police officer to avoid arrest. Police and prison guards in Cambodia, for example, have been documented subjecting sex workers to what can only be described as torture : threats at gunpoint, beatings, electric shocks, rape and denial of food. Another worrying thing : if you ’ re selling sex in places like Kenya, South Africa or New York, a police officer can arrest you if you ’ re caught carrying condoms, because condoms can legally be used as evidence that you ’ re selling sex. Obviously, this increases HIV risk. Imagine knowing if you ’ re busted carrying condoms, it ’ ll be used against you. It ’ s a pretty strong incentive to leave them at home, right? Sex workers working in these places are forced to make a tough choice between risking arrest or having risky sex. What would you choose? Would you pack condoms to go to work? How about if you ’ re worried the police officer would rape you when he got you in the van? The second approach to regulating sex work seen in these countries is partial criminalization, where the buying and selling of sex are legal, but surrounding activities, like brothel-keeping or soliciting on the **street**, are banned. Laws like these — we have them in the UK and in France — essentially say to us sex workers,”Hey, we don ’ t mind you selling sex, just make sure it ’ s done behind closed doors and all alone.”And brothel-keeping, by the way, is defined as just two or more sex workers working together. Making that illegal means that many of us work alone, which obviously makes us vulnerable to **violent** offenders. But we ’ re also vulnerable if we choose to break the law by working together. A couple of years ago, a friend of mine was nervous after she was attacked at work, so I said that she could see her clients from my place for a while. During that time, we had another guy turn nasty. I told the guy to leave or I ’ d call the police. And he looked at the two of us and said,”You girls can ’ t call the cops. You ’ re working together, this place is illegal.”He was right. He eventually left without getting physically **violent**, but the knowledge that we were breaking the law empowered that man to threaten us. He felt confident he ’ d get away with it. The prohibition of **street** prostitution also causes more **harm** than it prevents. Firstly, to avoid getting arrested, **street** workers take risks to avoid detection, and that means working alone or in isolated locations like dark forests where they ’ re vulnerable to attack. If you ’ re caught selling sex outdoors, you pay a fine. How do you pay that fine without going back to the **streets**? It was the need for money that saw you in the **streets** in the first place. And so the fines stack up, and you ’ re caught in a vicious cycle of selling sex to pay the fines you got for selling sex. Let me tell you about Mariana Popa who worked in Redbridge, East London. The **street** workers on her patch would normally wait for clients in groups for **safety** in numbers and to warn each other about how to avoid dangerous guys. But during a police crackdown on sex workers and their clients, she was forced to work alone to avoid being arrested. She was stabbed to death in the early hours of October 29, 2013. She had been working later than usual to try to pay off a fine she had received for soliciting. So if criminalizing sex workers hurts them, why not just criminalize the people who buy sex? This is the aim of the third approach I want to talk about — the Swedish or Nordic model of sex-work law. The idea behind this law is that selling sex is intrinsically harmful and so you ’ re, in fact, helping sex workers by removing the option. Despite growing support for what ’ s often described as the”end demand”approach, there ’ s no evidence that it works. There ’ s just as much prostitution in Sweden as there was before. Why might that be? It ’ s because people selling sex often don ’ t have other options for income. If you need that money, the only effect that a drop in business is going have is to force you to lower your prices or offer more risky **sexual** services. If you need to find more clients, you might seek the help of a manager. So you see, rather than putting a stop to what ’ s often descried as pimping, a law like this actually gives oxygen to potentially abusive third parties. To keep safe in my work, I try not to take bookings from someone who calls me from a withheld number. If it ’ s a home or a hotel visit, I try to get a full name and details. If I worked under the Swedish model, a client would be too scared to give me that information. I might have no other choice but to accept a booking from a man who is untraceable if he later turns out to be **violent**. If you need their money, you need to protect your clients from the police. If you work outdoors, that means working alone or in isolated locations, just as if you were criminalized yourself. It might mean getting into cars quicker, less negotiating time means snap decisions. Is this guy dangerous or just nervous? Can you afford to take the risk? Can you afford not to? Something I ’ m often hearing is,”Prostitution would be fine if we made it legal and regulated it.”We call that approach legalization, and it ’ s used by countries like the Netherlands, Germany and Nevada in the US. But it ’ s not a great model for human rights. And in state-controlled prostitution, commercial sex can only happen in certain legally-designated areas or venues, and sex workers are made to comply with special restrictions, like registration and forced health checks. Regulation sounds great on paper, but politicians deliberately make regulation around the sex industry expensive and difficult to comply with. It creates a two-tiered system : legal and illegal work. We sometimes call it”backdoor criminalization.”Rich, well-connected brothel owners can comply with the regulations, but more marginalized people find those hoops impossible to jump through. And even if it ’ s possible in principle, getting a license or proper venue takes time and costs money. It ’ s not going to be an option for someone who ’ s desperate and needs money tonight. They might be a refugee or fleeing **domestic** abuse. In this two-tiered system, the most vulnerable people are forced to work illegally, so they ’ re still exposed to all the dangers of criminalization I mentioned earlier. So. It ’ s looking like all attempts to control or prevent sex work from happening makes things more dangerous for people selling sex. Fear of law **enforcement** makes them work alone in isolated locations, and allows clients and even cops to get abusive in the knowledge they ’ ll get away with it. Fines and criminal records force people to keep selling sex, rather than enabling them to stop. Crackdowns on buyers drive sellers to take dangerous risks and into the arms of potentially abusive managers. These laws also reinforce stigma and **hatred** against sex workers. When France temporarily brought in the Swedish model two years ago, ordinary citizens took it as a cue to start carrying out vigilante attacks against people working on the **street**. In Sweden, opinion surveys show that significantly more people want sex workers to be arrested now than before the law was brought in. If prohibition is this harmful, you might ask, why it so popular? Firstly, sex work is and always has been a survival strategy for all kinds of unpopular minority groups : people of color, migrants, people with disabilities, LGBTQ people, particularly trans women. These are the groups most heavily profiled and punished through prohibitionist law. I don ’ t think this is an accident. These laws have political support precisely because they target people that voters don ’ t want to see or know about. Why else might people support prohibition? Well, lots of people have understandable fears about trafficking. Folks think that foreign women kidnapped and sold into **sexual** slavery can be saved by shutting a whole industry down. So let ’ s talk about trafficking. Forced labor does occur in many industries, especially those where the workers are migrants or otherwise vulnerable, and this needs to be addressed. But it ’ s best addressed with legislation targeting those specific abuses, not an entire industry. When 23 undocumented Chinese migrants drowned while picking cockles in Morecambe Bay in 2004, there were no calls to outlaw the entire seafood industry to save trafficking **victims**. The solution is clearly to give workers more legal protections, allowing them to resist abuse and report it to authorities without fear of arrest. The way the term trafficking is thrown around implies that all undocumented migration into prostitution is forced. In fact, many migrants have made a decision, out of economic need, to place themselves into the hands of people smugglers. Many do this with the full knowledge that they ’ ll be selling sex when they reach their destination. And yes, it can often be the case that these people smugglers demand exorbitant fees, coerce migrants into work they don ’ t want to do and abuse them when they ’ re vulnerable. That ’ s true of prostitution, but it ’ s also true of agricultural work, hospitality work and **domestic** work. Ultimately, nobody wants to be forced to do any kind of work, but that ’ s a risk many migrants are willing to take, because of what they ’ re leaving behind. If people were allowed to migrate legally they wouldn ’ t have to place their lives into the hands of people smugglers. The problems arise from the criminalization of migration, just as they do from the criminalization of sex work itself. This is a lesson of history. If you try to prohibit something that people want or need to do, whether that ’ s drinking alcohol or crossing borders or getting an abortion or selling sex, you create more problems than you solve. Prohibition barely makes a difference to the amount of people actually doing those things. But it makes a huge difference as to whether or not they ’ re safe when they do them. Why else might people support prohibition? As a feminist, I know that the sex industry is a site of deeply entrenched social inequality. It ’ s a fact that most buyers of sex are men with money, and most sellers are women without. You can agree with all that — I do — and still think prohibition is a terrible policy. In a better, more **equal** world, maybe there would be far fewer people selling sex to survive, but you can ' t simply legislate a better world into existence. If someone needs to sell sex because they ' re poor or because they ' re **homeless** or because they ' re undocumented and they can ' t find legal work, taking away that option doesn ' t make them any less poor or house them or change their **immigration** status. People worry that selling sex is degrading. Ask yourself : is it more degrading than going hungry or seeing your children go hungry? There ' s no call to ban rich people from hiring nannies or getting manicures, even though most of the people doing that labor are poor, migrant women. It ' s the fact of poor migrant women selling sex specifically that has some feminists uncomfortable. And I can understand why the sex industry provokes strong feelings. People have all kinds of complicated feelings when it comes to sex. But we can ' t make policy on the basis of mere feelings, especially not over the heads of the people actually effected by those policies. If we get fixated on the abolition of sex work, we end up worrying more about a particular manifestation of gender inequality, rather than about the underlying causes. People get really hung up on the question,"Well, would you want your daughter doing it?"That ' s the wrong question. Instead, imagine she is doing it. How safe is she at work tonight? Why isn ' t she safer? So we ' ve looked at full criminalization, partial criminalization, the Swedish or Nordic Model and legalization, and how they all cause **harm**. Something I never hear asked is :"What do sex workers want?"After all, we ' re the ones most affected by these laws. New Zealand decriminalized sex work in 2003. It ' s crucial to remember that decriminalization and legalization are not the same thing. Decriminalization means the removal of laws that punitively target the sex industry, instead treating sex work much like any other kind of work. In New Zealand, people can work together for **safety**, and employers of sex workers are accountable to the state. A sex worker can refuse to see a client at any time, for any reason, and 96 percent of **street** workers report that they feel the law protects their rights. New Zealand hasn ' t actually seen an increase in the amount of people doing sex work, but decriminalizing it has made it a lot safer. But the lesson from New Zealand isn ' t just that its particular legislation is good, but that crucially, it was written in collaboration with sex workers ; namely, the New Zealand Prostitutes ' Collective. When it came to making sex work safer, they were ready to hear it straight from sex workers themselves. Here in the UK, I ' m part of sex worker-led groups like the Sex Worker Open University and the English Collective of Prostitutes. And we form part of a global movement demanding decriminalization and self-determination. The universal symbol of our movement is the red umbrella. We ' re supported in our demands by global bodies like UNAIDS, the World Health Organization and Amnesty International. But we need more allies. If you care about gender **equality** or poverty or migration or public health, then sex worker rights matter to you. Make space for us in your movements. That means not only listening to sex workers when we speak but amplifying our voices. Resist those who **silence** us, those who say that a prostitute is either too victimized, too damaged to know what ' s best for herself, or else too privileged and too removed from real hardship, not representative of the millions of voiceless **victims**. This distinction between **victim** and empowered is imaginary. It exists purely to discredit sex workers and make it easy to ignore us. No doubt many of you work for a living. Well, sex work is work, too. Just like you, some of us like our jobs, some of us hate them. Ultimately, most of us have mixed feelings. But how we feel about our work isn ' t the point. And how others feel about our work certainly isn ' t. What ' s important is that we have the right to work safely and on our own terms. Sex workers are real people. We ' ve had complicated experiences and complicated responses to those experiences. But our demands are not complicated. You can ask expensive escorts in New York City, brothel workers in Cambodia, **street** workers in South Africa and every girl on the roster at my old job in Soho, and they will all tell you the same thing. You can speak to millions of sex workers and countless sex work-led organizations. We want full decriminalization and labor rights as workers. I ' m just one sex worker on the stage today, but I ' m bringing a message from all over the world. Thank you.

Text Sample 347: POS Dim 3 – 2016 – Score 14.00 – 2016/t001236.txt
It was April, last year. I was on an evening out with friends to **celebrate** one of their birthdays. We hadn ' t been all together for a couple of weeks ; it was a perfect evening, as we were all reunited. At the end of the evening, I caught the last underground train back to the other side of London. The journey was smooth. I got back to my local station and I began the 10-minute walk home. As I turned the corner onto my **street**, my house in sight up ahead, I heard footsteps behind me that seemed to have approached out of nowhere and were picking up pace. Before I had time to process what was happening, a hand was clapped around my mouth so that I could not breathe, and the young man behind me dragged me to the ground, beat my head repeatedly against the pavement until my face began to bleed, kicking me in the back and neck while he began to **assault** me, ripping off my clothes and telling me to"shut up,"as I struggled to cry for help. With each smack of my head to the concrete ground, a question echoed through my mind that still haunts me today :"Is this going to be how it all ends?"Little could I have realized, I ' d been followed the whole way from the moment I left the station. And hours later, I was standing topless and barelegged in front of the police, having the cuts and bruises on my naked body photographed for forensic evidence. Now, there are few words to describe the all-consuming feelings of vulnerability, **shame**, upset and injustice that I was ridden with in that moment and for the weeks to come. But wanting to find a way to condense these feelings into something ordered that I could work through, I decided to do what felt most natural to me : I wrote about it. It started out as a cathartic exercise. I wrote a letter to my assaulter, humanizing him as"you,"to identify him as part of the very community that he had so violently abused that night. Stressing the tidal-wave effect of his actions, I wrote :"Did you ever think of the people in your life? I don ' t know who the people in your life are. I don ' t know anything about you. But I do know this : you did not just attack me that night. I ' m a daughter, I ' m a friend, I ' m a sister, I ' m a pupil, I ' m a cousin, I ' m a niece, I ' m a neighbor ; I ' m the employee who served everyone coffee in the caf under the railway. And all the people who form these relations to me make up my community. And you **assaulted** every single one of them. You violated the truth that I will never cease to fight for, and which all of these people represent : that there are infinitely more good people in the world than bad."But, determined not to let this one incident make me lose faith in the **solidarity** in my community or humanity as a whole, I recalled the 7 / 7 terrorist bombings in July 2005 on London transport, and how the mayor of London at the time, and indeed my own parents, had insisted that we all get back on the tubes the next day, so we wouldn ' t be defined or changed by those that had made us feel **unsafe**. I told my attacker,"You ' ve carried out your attack, but now I ' m getting back on my tube. My community will not feel we are **unsafe** walking home after dark. We will get on the last tubes home, and we will walk up our **streets** alone, because we will not ingrain or submit to the idea that we are putting ourselves in danger in doing

so. We will continue to come together, like an army, when any member of our community is threatened. And this is a fight you will not win.”At the time of writing this letter — Thank you. At the time of writing this letter, I was studying for my exams in Oxford, and I was working on the local student paper there. Despite being lucky enough to have friends and family supporting me, it was an isolating time. I didn ’ t know anyone who ’ d been through this before ; at least I didn ’ t think I did. I ’ d read news reports, statistics, and knew how common **sexual assault** was, yet I couldn ’ t actually name a single person that I ’ d heard speak out about an experience of this kind before. So in a somewhat spontaneous decision, I decided that I would publish my letter in the student paper, hoping to reach out to others in Oxford that might have had a similar experience and be feeling the same way. At the end of the letter, I asked others to write in with their experiences under the hashtag,”#NotGuilty,”to emphasize that survivors of **assault** could express themselves without feeling **shame** or guilt about what happened to them — to show that we could all stand up to **sexual assault**. What I never anticipated is that almost overnight, this published letter would go viral. Soon, we were receiving hundreds of stories from men and women across the world, which we began to publish on a website I set up. And the hashtag became a campaign. There was an Australian mother in her 40s who described how on an evening out, she was followed to the bathroom by a man who went to repeatedly grab her crotch. There was a man in the Netherlands who described how he was date-raped on a visit to London and wasn ’ t taken seriously by anyone he reported his case to. I had personal Facebook messages from people in India and South America, saying, how can we bring the message of the campaign there? One of the first contributions we had was from a woman called Nikki, who described growing up, being molested my her own father. And I had friends open up to me about experiences ranging from those that happened last week to those that happened years ago, that I ’ d had no idea about. And the more we started to receive these messages, the more we also started to receive messages of hope — people feeling empowered by this community of voices standing up to **sexual assault** and victim-blaming. One woman called Olivia, after describing how she was attacked by someone she had trusted and cared about for a long time, said,”I ’ ve read many of the stories posted here, and I feel hopeful that if so many women can move forward, then I can, too. I ’ ve been inspired by many, and I hope I can be as strong as them someday. I ’ m sure I will.”People around the world began tweeting under this hashtag, and the letter was republished and covered by the national press, as well as being translated into several other languages worldwide. But something struck me about the media attention that this letter was attracting. For something to be front-page news, given the word”news”itself, we can assume it must be something new or something surprising. And yet **sexual assault** is not something new. **Sexual assault**, along with other kinds of injustices, is reported in the media all the time. But through the campaign, these injustices were framed as not just news stories, they were firsthand experiences that had affected real people, who were creating, with the **solidarity** of others, what they needed and had previously lacked : a platform to speak out, the reassurance they weren ’ t alone or to blame for what happened to them and open discussions that would help to reduce stigma around the issue. The voices of those directly affected were at the forefront of the story — not the voices of journalists or commentators on social media. And that ’ s why the story was news. We live in an incredibly interconnected world with the proliferation of social media, which is of course a fantastic resource for igniting social change. But it ’ s also made us increasingly reactive, from the smallest annoyances of,”Oh, my train ’ s been delayed,”to the greatest injustices of war, genocides, terrorist attacks. Our default response has become to leap to react to any kind of grievance by tweeting, Facebooking, hastagging — anything to show others that we, too, have reacted. The problem with reacting in this manner en masse is it can sometimes mean that we don ’ t actually react at all, not in the sense of actually doing anything, anyway. It might make ourselves feel better, like we ’ ve contributed to a group mourning or outrage, but it doesn ’ t actually change anything. And what ’ s more, it can sometimes drown out the voices of those directly affected by the injustice, whose needs must be heard. Worrying, too, is the tendency for some reactions to injustice to build even more walls, being quick to point fingers with the hope of providing easy solutions to complex problems. One British tabloid, on the publication of my letter, branded a headline stating,”Oxford Student Launches Online Campaign to **Shame** Attacker.”But the campaign never meant to **shame** anyone. It meant to let people speak and to make others listen. Divisive **Twitter** trolls were quick to create even more injustice, commenting on my attacker ’ s **ethnicity** or class to push their own prejudiced agendas. And some even accused me of feigning the whole thing to push, and I quote, my”feminist agenda of man-hating.”I know, right? As if I ’ m going to be like,”Hey guys! Sorry I can ’ t make it, I ’ m busy trying to hate the entire male population by the time I ’ m 30.”Now, I ’ m almost sure that these people wouldn ’ t say the things they say in person. But it ’ s as if because they might be behind a screen, in the comfort in their own home when on social media, people forget that what they ’ re doing is a public act — that other people will be reading it and be affected by it. Returning to my analogy of getting back on our trains, another main concern I have about this noise that escalates from our online responses to injustice is that it can very easily slip into portraying us as the affected party, which can lead to a sense of defeatism, a kind of mental barrier to seeing any opportunity for positivity or change after a negative situation. A couple of months before the campaign started or any of this happened to me, I went to a TEDx event in Oxford, and I saw Zelda la Grange speak, the former private secretary to Nelson Mandela. One of the stories she told really struck me. She spoke of when Mandela was taken to court by the South African Rugby Union after he commissioned an inquiry into sports affairs. In the courtroom, he went up to the South African Rugby Union ’ s lawyers, shook them by the hand and conversed with them, each in their own language. And Zelda wanted to protest, saying they had no right to his respect after this injustice they had caused him. He turned to her and said,”You must never allow the enemy to determine the grounds for battle.”At the time of hearing these words, I didn ’ t really know why they were so important, but I felt they were, and I wrote them down in a notebook I had on me. But I ’ ve thought about this line a lot ever since. Revenge, or the expression of **hatred** towards those who have done us injustice may feel like a human instinct in the face of wrong, but we need to break out of these cycles if we are to hope to transform negative events of injustice into positive social change. To do otherwise continues to let the enemy determine the grounds for battle, creates a binary, where we who have suffered become the affected, pitted against them, the **perpetrators**. And just like we got back on our tubes, we can ’ t let our platforms for interconnectivity and community be the places that we settle for defeat. But I don ’ t want to discourage a social media response, because I owe the development of the #NotGuilty campaign almost entirely to social media. But I do want to encourage a more considered approach to the way we use it to respond to injustice. The start, I think, is to ask ourselves two things. Firstly : Why do I feel this injustice? In my case, there were several answers to this. Someone had hurt me and those who I loved, under the assumption they wouldn ’ t have to be held to account or recognize the damage they had caused. Not only that, but thousands of men and women suffer every day from **sexual** abuse, often in **silence**, yet it ’ s still a problem we don ’ t give the same airtime to as other issues. It ’ s still an issue many people blame **victims** for. Next, ask yourself : How, in recognizing these reasons, could I go about reversing them? With us, this was holding my attacker to account — and many others. It was calling them out on the effect they had caused. It was giving air-time to the issue of **sexual assault**, opening up discussions amongst friends, amongst families, in the media that had been closed for too long, and stressing that **victims** shouldn ’ t feel to blame for what happened to them. We might still have a long way to go in solving this problem entirely. But in this way, we can begin to use social media as an active tool for social justice, as a tool to educate, to stimulate dialogues, to make those in positions of authority aware of an issue by listening to those directly affected by it. Because sometimes these questions don ’ t have easy answers. In fact, they rarely do. But this doesn ’ t mean we still can ’ t give them a considered response. In situations where you can ’ t go about thinking how you ’ d reverse this feeling of injustice, you can still think, maybe not what you can do, but what you can not do. You can not build further walls by fighting injustice with more **prejudice**, more **hatred**. You can not speak over those directly affected by an injustice. And you can not react to injustice, only to forget about it the next day, just because the rest of **Twitter** has moved on. Sometimes not reacting instantly is, ironically, the best immediate course of action we can take. Because we might be angry, upset and energized by injustice, but let ’ s consider our responses. Let us hold people to account, without descending into a culture that thrives off **shaming** and injustice ourselves. Let us remember that distinction, so often forgotten by internet users, between criticism and insult. Let us not forget to think before we speak, just because we might have a screen in front of us. And when we create noise on social media, let it not drown out the needs of those affected, but instead let it amplify their voices, so the internet becomes a place where you ’ re not the exception if you speak out about something that has actually happened to you. All these considered approaches to injustice evoke the very keystones on which the internet was built : to network, to have signal, to connect — all these terms that imply bringing people together, not pushing people apart. Because if you look up the word”justice”in the dictionary, before punishment, before administration of law or judicial authority, you get :”The maintenance of what is right.”And I think there are few things more”right”in this world than bringing people together, than unions. And if we allow social media to deliver that, then it can deliver a very powerful form of justice, indeed. Thank you very much.

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Chris Anderson : Welcome to this next edition of TED Dialogues. We ’ re trying to do some bridging here today. You know, the American dream has inspired millions of people around the world for many years. Today, I think, you can say that America is divided, perhaps more than ever, and the divisions seem to be getting worse. It ’ s actually really hard for people on different sides to even have a conversation. People almost feel... disgusted with each other. Some families can ’ t even speak to each other right now. Our purpose in this dialogue today is to try to do something about that, to try to have a different kind of conversation, to do some listening, some thinking, some understanding. And I have two people with us to help us do that. They ’ re not going to come at this hammer and tong against each other. This is not like cable news. This is two people who have both spent a lot of their working life in the political center or right of the center. They ' ve immersed themselves in conservative worldviews, if you like. They know that space very well. And we ' re going to explore together how to think about what is happening right now, and whether we can find new ways to bridge and just to have wiser, more connected conversations. With me, first of all, Gretchen Carlson, who has spent a decade working at Fox News, hosting"Fox and Friends"and then"The Real Story,"before taking a courageous stance in filing **sexual harassment** claims against Roger Ailes, which eventually led to his departure from Fox News. David Brooks, who has earned the wrath of many of [The New York Times ' s] left-leaning readers because of his conservative views, and more recently, perhaps, some of the right-leaning readers because of his criticism of some aspects of Trump. Yet, his columns are usually the top one, two or three most-read content of the day because they ' re brilliant, because they bring psychology and social science to providing understanding for what ' s going on. So without further ado, a huge welcome to Gretchen and David. Come and join me. So, Gretchen. Sixty-three million Americans voted for Donald Trump. Why did they do this? Gretchen Carlson : There are a lot of reasons, in my mind, why it happened. I mean, I think it was a movement of sorts, but it started long ago. It didn ' t just happen overnight."Anger"would be the first word that I would think of — anger with nothing being done in Washington, anger about not being heard. I think there was a huge swath of the population that feels like Washington never listens to them, you know, a good part of the middle of America, not just the coasts, and he was somebody they felt was listening to their concerns. So I think those two issues would be the main reason. I have to throw in there also celebrity. I think that had a huge impact on Donald Trump becoming president. CA : Was the anger justified? David Brooks : Yeah, I think so. In 2015 and early 2016, I wrote about 30 columns with the following theme : don ' t worry, Donald Trump will never be the Republican nominee. And having done that and gotten that so wrong, I decided to spend the ensuing year just out in Trumpworld, and I found a lot of economic dislocation. I ran into a woman in West Virginia who was going to a funeral for her mom. She said,"The nice thing about being Catholic is we don ' t have to speak, and that ' s good, because we ' re not word people."That phrase rung in my head : word people. A lot of us in the TED community are word people, but if you ' re not, the economy has not been angled toward you, and so 11 million men, for example, are out of the labor force because those jobs are done away. A lot of social injury. You used to be able to say,"I ' m not the richest person in the world, I ' m not the most famous, but my neighbors can count on me and I get some dignity out of that."And because of celebritification or whatever, if you ' re not rich or famous, you feel invisible. And a lot of moral injury, sense of feeling betrayed, and frankly, in this country, we almost have one success story, which is you go to college, get a white-collar job, and you ' re a success, and if you don ' t fit in that **formula**, you feel like you ' re not respected. And so that accumulation of things — and when I talked to Trump voters and still do, I found most of them completely realistic about his failings, but they said, this is my shot. GC : And yet I predicted that he would be the nominee, because I ' ve known him for 27 years. He ' s a master marketer, and one of the things he did extremely well that President Obama also did extremely well, was simplifying the message, simplifying down to phrases and to a populist message. Even if he can ' t achieve it, it sounded good. And many people latched on to that simplicity again. It ' s something they could grasp onto :"I get that. I want that. That sounds fantastic."And I remember when he used to come on my show originally, before"The Apprentice"was even"The Apprentice,"and he ' d say it was the number one show on TV. I ' d say back to him,"No, it ' s not."And he would say,"Yes it is, Gretchen."And I would say,"No it ’ s not.”But people at home would see that, and they ’ d be like,”Wow, I should be watching the number one show on TV.”And — lo and behold — it became the number one show on TV. So he had this, I ’ ve seen this ability in him to be the master marketer. CA : It ’ s puzzling to a lot of people on the left that so many women voted for him, despite some of his comments. GC : I wrote a column about this for Time Motto, saying that I really believe that lot of people put on blinders, and maybe for the first time, some people decided that policies they believed in and being heard and not being invisible anymore was more important to them than the way in which he had acted or acts as a human. And so human dignity — whether it would be the dust-up about the **disabled** reporter, or what happened in that audiotape with Billy Bush and the way in which he spoke about women — they put that aside and pretended as if they hadn ’ t seen that or heard that, because to them, policies were more important. CA : Right, so just because someone voted for Trump, it ’ s not blind adherence to everything that he ’ s said or stood for. GC : No. I heard a lot of people that would say to me,”Wow, I just wish he would shut up before the election. If he would just stay quiet, he ’ d get elected.”CA : And so, maybe for people on the left there ’ s a trap there, to sort of despise or just be baffled by the support, assuming that it ’ s for some of the unattractive features. Actually, maybe they ’ re supporting him despite those, because they see something exciting. They see a man of action. They see the choking hold of government being thrown off in some way and they ’ re excited by that. GC : But don ’ t forget we saw that on the left as well — Bernie Sanders. So this is one of the commonalities that I think we can talk about today,”The Year of the Outsider,”David — right? And even though Bernie Sanders has been in Congress for a long time, he was **deemed** an outsider this time. And so there was anger on the left as well, and so many people were in favor of Bernie Sanders. So I see it as a commonality. People who like Trump, people who like Bernie Sanders, they were liking different policies, but the underpinning was anger. CA : David, there ’ s often this narrative, then, that the sole explanation for Trump ’ s victory and his rise is his tapping into anger in a very visceral way. But you ’ ve written a bit about that it ’ s actually more than that, that there ’ s a worldview that ’ s being worked on here. Could you talk about that? DB : I would say he understood what, frankly, I didn ’ t, which is what debate we were having. And so I ’ d grown up starting with Reagan, and it was the big government versus small government debate. It was Barry Goldwater versus George McGovern, and that was the debate we had been having for a generation. It was : Democrats wanted to use government to enhance **equality**, Republicans wanted to limit government to enhance freedom. That was the debate. He understood what I think the two major parties did not, which was that ’ s not the debate anymore. The debate is now open versus closed. On one side are those who have the tailwinds of globalization and the meritocracy blowing at their back, and they tend to favor open trade, open borders, open social mores, because there are so many opportunities. On the other side are those who feel the headwinds of globalization and the meritocracy just blasting in their faces, and they favor closed trade, closed borders, closed social mores, because they just want some security. And so he was right on that fundamental issue, and people were willing to overlook a lot to get there. And so he felt that sense of security. We ’ re speaking the morning after Trump ’ s joint session speech. There are three traditional groups in the Republican Party. There are the foreign policies hawks who believe in America as global policeman. Trump totally repudiated that view. Second, there was the social conservatives who believed in religious liberty, pro-life, **prayer** in schools. He totally ignored that. There was not a single mention of a single social conservative issue. And then there were the fiscal hawks, the people who wanted to cut down on the national debt, Tea Party, cut the size of government. He ' s expanding the size of government! Here ' s a man who has single-handedly revolutionized a major American party because he understood where the debate was headed before other people. And then guys like Steve Bannon come in and give him substance to his impulses. CA : And so take that a bit further, and maybe expand a bit more on your insights into Steve Bannon ' s worldview. Because he ' s sometimes tarred in very simple terms as this dangerous, racist, xenophobic, anger-sparking person. There ' s more to the story ; that is perhaps an unfair simplification. DB : I think that part is true, but there ' s another part that ' s probably true, too. He ' s part of a global movement. It ' s like being around Marxists in 1917. There ' s him here, there ' s the UKIP party, there ' s the National Front in France, there ' s Putin, there ' s a Turkish version, a Philippine version. So we have to recognize that this is a global intellectual movement. And it believes that wisdom and virtue is not held in individual conversation and civility the way a lot of us in the enlightenment side of the world do. It ' s held in — the German word is the"volk"— in the people, in the common, instinctive wisdom of the plain people. And the essential virtue of that people is always being threatened by outsiders. And he ' s got a strategy for how to get there. He ' s got a series of policies to bring the people up and repudiate the outsiders, whether those outsiders are Islam, Mexicans, the media, the coastal elites... And there ' s a whole worldview there ; it ' s a very coherent worldview. I sort of have more respect for him. I loathe what he stands for and I think he ' s wrong on the substance, but it ' s interesting to see someone with a set of ideas find a vehicle, Donald Trump, and then try to take control of the White House in order to advance his viewpoint. CA : So it ' s almost become, like, that the core question of our time now is : Can you be patriotic but also have a global mindset? Are these two things implacably opposed to each other? I mean, a lot of conservatives and, to the extent that it ' s a different category, a lot of Trump supporters, are infuriated by the coastal elites and the globalists because they see them as, sort of, not cheering for America, not embracing fully American values. I mean, have you seen that in your conversations with people, in your understanding of their mindset? GC : I do think that there ' s a huge difference between — I hate to put people in categories, but, Middle America versus people who live on the coasts. It ' s an entirely different existence. And I grew up in Minnesota, so I have an understanding of Middle America, and I ' ve never forgotten it. And maybe that ' s why I have an understanding of what happened here, because those people often feel like nobody ' s listening to them, and that we ' re only concentrating on California and New York. And so I think that was a huge reason why Trump was elected. I mean, these people felt like they were being heard. Whether or not patriotism falls into that, I ' m not sure about that. I do know one thing : a lot of things Trump talked about last night are not conservative things. Had Hillary Clinton gotten up and given that speech, not one Republican would have stood up to applaud. I mean, he ' s talking about spending a trillion dollars on infrastructure. That is not a conservative viewpoint. He talked about government-mandated maternity leave. A lot of women may love that ; it ' s not a conservative viewpoint. So it ' s fascinating that people who loved what his message was during the campaign, I ' m not sure — how do you think they ' ll react to that? DB : I should say I grew up in Lower Manhattan, in the triangle between ABC Carpets, the Strand Bookstore and The Odeon restaurant. GC : Come to Minnesota sometime! CA : You are a card-carrying member of the coastal elite, my man. But what did you make of the speech last night? It seemed to be a move to a more moderate position, on the face of it. DB : Yeah, I thought it was his best speech, and it took away the freakishness of him. I do think he ' s a moral freak, and I think he ' ll be undone by that fact, the fact that he just doesn ' t know anything about anything and is uncurious about it. But if you take away these minor flaws, I think we got to see him at his best, and it was revealing for me to see him at his best, because to me, it exposed a central contradiction that he ' s got to confront, that a lot of what he ' s doing is offering security. So,"I ' m ordering closed borders, I ' m going to secure the world for you, for my people."But then if you actually look at a lot of his economic policies, like health care reform, which is about private health care accounts, that ' s not security, that ' s risk. Educational vouchers : that ' s risk. Deregulation : that ' s risk. There ' s really a contradiction between the security of the mindset and a lot of the policies, which are very risk-oriented. And what I would say, especially having spent this year, the people in rural Minnesota, in New Mexico — they ' ve got enough risk in their lives. And so they ' re going to say,"No thank you."And I think his health care repeal will fail for that reason. CA : But despite the criticisms you just made of him, it does at least seem that he ' s listening to a surprisingly wide range of voices ; it ' s not like everyone is coming from the same place. And maybe that leads to a certain amount of chaos and confusion, but — GC : I actually don ' t think he ' s listening to a wide range of voices. I think he ' s listening to very few people. That ' s just my impression of it. I believe that some of the things he said last night had Ivanka all over them. So I believe he was listening to her before that speech. And he was Teleprompter Trump last night, as opposed to **Twitter** Trump. And that ' s why, before we came out here, I said,"We better check **Twitter** to see if anything ' s changed."And also I think you have to keep in mind that because he ' s such a unique character, what was the bar that we were expecting last night? Was it here or here or here? And so he comes out and gives a looking political speech, and everyone goes,"Wow! He can do it."It just depends on which direction he goes. DB : Yeah, and we ' re trying to build bridges here, and especially for an audience that may have contempt for Trump, it ' s important to say, no, this is a real thing. But as I try my best to go an hour showing respect for him, my thyroid is surging, because I think the oddities of his character really are condemnatory and are going to doom him. CA : Your reputation is as a conservative. People would you describe you as right of center, and yet here you are with this visceral reaction against him and some of what he stands for. I mean, I ' m — how do you have a conversation? The people who support him, on evidence so far, are probably pretty excited. He ' s certainly shown real engagement in a lot of what he promised to do, and there is a strong desire to change the system radically. People hate what government has become and how it ' s left them out. GC : I totally agree with that, but I think that when he was proposing a huge government program last night that we used to call the bad s-word,"stimulus,"I find it completely **ironic**. To spend a trillion dollars on something — that is not a conservative viewpoint. Then again, I don ' t really believe he ' s a Republican. DB : And I would say, as someone who identifies as conservative : first of all, to be conservative is to believe in the limitations of politics. Samuel Johnson said,"Of all the things that human hearts endure, how few are those that kings can cause and cure."Politics is a limited realm ; what matters most is the moral nature of the society. And so I have to think character comes first, and a man who doesn ' t pass the character threshold cannot be a good president. Second, I ' m the kind of conservative who — I harken back to Alexander Hamilton, who was a Latino hip-hop star from the heights — but his definition of America was very future-oriented. He was a poor boy from the islands who had this rapid and amazing rise to success, and he wanted government to give poor boys and girls like him a chance to succeed, using limited but energetic government to create social mobility. For him and for Lincoln and for Teddy Roosevelt, the idea of America was the idea of the future. We may have division and racism and slavery in our past, but we have a common future. The definition of America that Steve Bannon stands for is backwards-looking. It ' s nostalgic ; it ' s for the past. And that is not traditionally the American identity. That ' s traditionally, frankly, the Russian identity. That ' s how they define virtue. And so I think it is a fundamental and foundational betrayal of what conservatism used to stand for. CA : Well, I ' d like actually like to hear from you, and if we see some comments coming in from some of you, we ' ll — oh, well here ' s one right now. Jeffrey Alan Carnegie : I ' ve tried to convince progressive friends that they need to understand what motivates Trump supporters, yet many of them have given up trying to understand in the face of what they perceive as lies, selfishness and **hatred**. How would you reach out to such people, the Tea Party of the left, to try to bridge this divide? GC : I actually think there are commonalities in anger, as I expressed earlier. So I think you can come to the table, both being passionate about something. So at least you care. And I would like to believe — the c-word has also become a horrible word —"compromise,"right? So you have the far left and the far right, and compromise — forget it. Those groups don ' t want to even think about it. But you have a huge swath of voters, myself included, who are registered independents, like 40 percent of us, right? So there is a huge faction of America that wants to see change and wants to see people come together. It ' s just that we have to figure out how to do that. CA : So let ' s talk about that for a minute, because we ' re having these TED Dialogues, we ' re trying to bridge. There ' s a lot of people out there, right now, perhaps especially on the left, who think this is a terrible idea, that actually, the only moral response to the great tyranny that may be about to emerge in America is to resist it at every stage, is to fight it tooth and nail, it ' s a mistake to try and do this. Just fight! Is there a case for that? DB : It depends what"fight"means. If it means literal fighting, then no. If it means marching, well maybe marching to raise consciousness, that seems fine. But if you want change in this country, we do it through parties and politics. We organize parties, and those parties are big, diverse, messy coalitions, and we engage in politics, and politics is always morally unsatisfying because it ' s always a bunch of compromises. But politics is essentially a competition between partial truths. The Trump people have a piece of the truth in America. I think Trump himself is the wrong answer to the right question, but they have some truth, and it ' s truth found in the epidemic of opiates around the country, it ' s truth found in the spread of loneliness, it ' s the truth found in people whose lives are inverted. They peaked professionally at age 30, and it ' s all been downhill since. And so, understanding that doesn ' t take fighting, it takes conversation and then asking,"What are we going to replace Trump with?"GC : But you saw fighting last night, even at the speech, because you saw the Democratic women who came and wore white to honor the suffragette movement. I remember back during the campaign where some Trump supporters wanted to actually get rid of the amendment that allowed us to vote as women. It was like, what? So I don ' t know if that ' s the right way to fight. It was interesting, because I was looking in the audience, trying to see Democratic women who didn ' t wear white. So there ' s a lot going on there, and there ' s a lot of ways to fight that are not necessarily doing that. CA : I mean, one of the key questions, to me, is : The people who voted for Trump but, if you like, are more in the center, like they ' re possibly amenable to persuasion — are they more likely to be persuaded by seeing a passionate uprising of people saying,"No, no, no, you can ' t!"or will that actually piss them off and push them away? DB : How are any of us persuaded? Am I going to persuade you by saying,"Well, you ' re kind of a bigot, you ' re supporting bigotry, you ' re supporting sexism. You ' re a primitive, fascistic rise from some **authoritarian** past"? That ' s probably not going to be too persuasive to you. And so the way any of us are persuaded is by : a) some basic show of respect for the point of view, and saying,"I think this guy is not going to get you where you need to go."And there are two phrases you ' ve heard over and over again, wherever you go in the country. One, the phrase"flyover country."And that ' s been heard for years, but I would say this year, I heard it almost on an hourly basis, a sense of feeling invisible. And then the sense a sense of the phrase"political correctness."Just that rebellion :"They ' re not even letting us say what we think."And I teach at Yale. The narrowing of debate is real. CA : So you would say this is a trap that liberals have fallen into by **celebrating** causes they really believe in, often expressed through the language of"political correctness."They have done damage. They have pushed people away. DB : I would say a lot of the argument, though, with"descent to fascism,""authoritarianism"— that just feels over-the-top to people. And listen, I ' ve written eight million anti-Trump columns, but it is a problem, especially for the coastal media, that every time he does something slightly wrong, we go to 11, and we ' re at 11 every day. And it just strains credibility at some point. CA : Crying wolf a little too loud and a little too early. But there may be a time when we really do have to cry wolf. GC : But see — one of the most important things to me is how the conservative media handles Trump. Will they call him out when things are not true, or will they just go along with it? To me, that is what is essential in this entire discussion, because when you have followers of somebody who don ' t really care if he tells the truth or not, that can be very dangerous. So to me, it ' s : How is the conservative media going to respond to it? I mean, you ' ve been calling them out. But how will other forms of conservative media deal with that as we move forward? DB : It ' s all shifted, though. The conservative media used to be Fox or Charles Krauthammer or George Will. They ' re no longer the conservative media. Now there ' s another whole set of institutions further right, which is Breitbart and Infowars, Alex Jones, Laura Ingraham, and so they ' re the ones who are now his base, not even so much Fox. CA : My last question for the time being is just on this question of the truth. I mean, it ' s one of the scariest things to people right now, that there is no agreement, nationally, on what is true. I ' ve never seen anything like it, where facts are so massively disputed. Your whole newspaper, sir, is delivering fake news every day. DB : And failing. CA : And failing. My commiserations. But is there any path whereby we can start to get some kind of consensus, to believe the same things? Can online communities play a role here? How do we fix this? GC : See, I understand how that happened. That ' s another groundswell kind of emotion that was going on in the middle of America and not being heard, in thinking that the mainstream media was biased. There ' s a difference, though, between being biased and being fake. To me, that is a very important distinction in this conversation. So let ' s just say that there was some bias in the mainstream media. OK. So there are ways to try and mend that. But what Trump ' s doing is nuclearizing that and saying,"Look, we ' re just going to call all of that fake."That ' s where it gets dangerous. CA : Do you think enough of his supporters have a greater loyalty to the truth than to any... Like, the principle of not supporting something that is demonstrably not true actually matters, so there will be a correction at some point? DB : I think the truth eventually comes out. So for example, Donald Trump has based a lot of his economic policy on this supposition that Americans have lost manufacturing jobs because they ' ve been stolen by the Chinese. That is maybe 13 percent of the jobs that left. The truth is that 87 percent of the jobs were replaced by technology. That is just the truth. And so as a result, when he says,"I ' m going to close TPP and all the jobs will come roaring back,”they will not come roaring back. So that is an actual fact, in my belief. And — GC : But I ’ m saying what his supporters think is the truth, no matter how many times you might say that, they still believe him. DB : But eventually either jobs will come back or they will not come back, and at that point, either something will work or it doesn ’ t work, and it doesn ’ t work or not work because of great marketing, it works because it actually addresses a real problem and so I happen to think the truth will out. CA : If you ’ ve got a question, please raise your hand here. Yael Eisenstat : I ’ ll speak into the box. My name ’ s Yael Eisenstat. I hear a lot of this talk about how we all need to start talking to each other more and understanding each other more, and I ’ ve even written about this, published on this subject as well, but now today I keep hearing liberals — yes, I live in New York, I can be considered a liberal — we sit here and self-analyze : What did we do to not understand the Rust Belt? Or : What can we do to understand Middle America better? And what I ’ d like to know : Have you seen any attempts or conversations from Middle America of what can I do to understand the so-called coastal elites better? Because I ’ m just offended as being put in a box as a coastal elite as someone in Middle America is as being considered a flyover state and not listened to. CA : There you go, I can hear Facebook cheering as you — DB : I would say — and this is someone who has been conservative all my adult life — when you grow up conservative, you learn to speak both languages. Because if I ’ m going to listen to music, I ’ m not going to listen to Ted Nugent. So a lot of my favorite rock bands are all on the left. If I ’ m going to go to a school, I ’ m going probably to school where the culture is liberal. If I ’ m going to watch a sitcom or a late-night comedy show, it ’ s going to be liberal. If I ’ m going to read a good newspaper, it ’ ll be the New York Times. As a result, you learn to speak both languages. And that actually, at least for a number of years, when I started at National Review with William F. Buckley, it made us sharper, because we were used to arguing against people every day. The problem now that ’ s happened is you have ghettoization on the right and you can live entirely in rightworld, so as a result, the quality of argument on the right has diminished, because you ’ re not in the other side all the time. But I do think if you ’ re living in Minnesota or Iowa or Arizona, the coastal elites make themselves aware to you, so you know that language as well, but it ’ s not the reverse. CA : But what does Middle America not get about coastal elites? So the critique is, you are not dealing with the real problems. There ’ s a feeling of a snobbishness, an elitism that is very off-putting. What are they missing? If you could plant one piece of truth from the mindset of someone in this room, for example, what would you say to them? DB : Just how insanely wonderful we are. No, I reject the category. The problem with populism is the same problem with elitism. It ’ s just a **prejudice** on the basis of probably an over-generalized social class distinction which is too simplistic to apply in reality. Those of us in New York know there are some people in New York who are completely awesome, and some people who are pathetic, and if you live in Iowa, some people are awesome and some people are pathetic. It ’ s not a question of what degree you have or where you happen to live in the country. The distinction is just a crude simplification to arouse political power. GC : But I would encourage people to watch a television news show or read a column that they normally wouldn ’ t. So if you are a Trump supporter, watch the other side for a day, because you need to come out of the bubble if you ’ re ever going to have a conversation. And both sides — so if you ’ re a liberal, then watch something that ’ s very conservative. Read a column that is not something you would normally read, because then you gain perspective of what the other side is thinking, and to me, that ’ s a start of coming together. I worry about the same thing you worry about, these bubbles. I think if you only watch certain entities, you have no idea what the rest of the world is talking about. DB : I think not only watching, being part of an organization that meets at least once a month that puts you in direct contact with people completely unlike yourself is something we all have a responsibility for. I may get this a little wrong, but I think of the top-selling automotive models in this country, I think the top three or four are all pickup trucks. So ask yourself : How many people do I know who own a pickup truck? And it could be very few or zero for a lot of people. And that ' s sort of a warning sign kind of a problem. Where can I join a club where I ' ll have a lot in common with a person who drives a pickup truck because we have a common interest in whatever? CA : And so the internet is definitely contributing to this. A question here from Chris Ajemian :"How do you feel structure of communications, especially the prevalence of social media and individualized content, can be used to bring together a political divide, instead of just filing communities into echo chambers?"I mean, it looks like Facebook and Google, since the election, are working hard on this question. They ' re trying to change the algorithms so that they don ' t amplify fake news to the extent that it happened last time round. Do you see any other promising signs of...? GC :... or amplify one side of the equation. CA : Exactly. GC : I think that was the constant argument from the right, that social media and the internet in general was putting articles towards the top that were not their worldview. I think, again, that fed into the anger. It fed into the anger of :"You ' re pushing something that ' s not what I believe."But social media has obviously changed everything, and I think Trump is the example of **Twitter** changing absolutely everything. And from his point of view, he ' s reaching the American people without a filter, which he believes the media is. CA : Question from the audience. Destiny : Hi. I ' m Destiny. I have a question regarding political correctness, and I ' m curious : When did political correctness become synonymous with **silencing**, versus a way that we speak about other people to show them respect and preserve their dignity? GC : Well, I think the conservative media really pounded this issue for the last 10 years. I think that they really, really spent a lot of time talking about political correctness, and how people should have the ability to say what they think. Another reason why Trump became so popular : because he says what he thinks. It also makes me think about the fact that I do believe there are a lot of people in America who agree with Steve Bannon, but they would never say it publicly, and so voting for Trump gave them the opportunity to agree with it silently. DB : On the issue of **immigration**, it ' s a legitimate point of view that we have too many **immigrants** in the country, that it ' s economically costly. CA : That we have too many — DB : **Immigrants** in the country, especially from Britain. GC : I kind of like the British accent, OK? CA : I **apologize**. America, I am sorry. I ' ll go now. DB : But it became sort of impermissible to say that, because it was a sign that somehow you must be a bigot of some sort. So the political correctness was not only cracking down on speech that we would all find completely offensive, it was cracking down on some speech that was legitimate, and then it was turning speech and thought into action and treating it as a crime, and people getting fired and people thrown out of schools, and there were speech codes written. Now there are these diversity teams, where if you say something that somebody finds offensive, like,"Smoking is really dangerous,"you can say"You ' re insulting my group,"and the team from the administration will come down into your dorm room and put thought police upon you. And so there has been a genuine narrowing of what is permissible to say. And some of it is legitimate. There are certain words that there should be some social sanction against, but some of it was used to enforce a political agenda. CA : So is that a project you would urge on liberals, if you like — progressives — to rethink the ground rules around political correctness and accept a little more uncomfortable language in certain circumstances? Can you see that being solved to an extent that others won ’ t be so offended? DB : I mean, most American universities, especially elite universities, are overwhelmingly on the left, and there ’ s just an ease of temptation to use your overwhelming cultural power to try to enforce some sort of thought that you think is right and correct thought. So, be a little more self-suspicious of, are we doing that? And second, my university, the University of Chicago, sent out this letter saying, we will have no safe spaces. There will be no critique of micro-aggression. If you get your feelings hurt, well, welcome to the world of education. I do think that policy — which is being embraced by a lot of people on the left, by the way — is just a corrective to what ’ s happened. CA : So here ’ s a question from Karen Holloway : How do we foster an American culture that ’ s forward-looking, like Hamilton, that expects and deals with change, rather than wanting to have everything go back to some fictional past? That ’ s an easy question, right? GC : Well, I ’ m still a believer in the American dream, and I think what we can teach our children is the basics, which is that hard work and believing in yourself in America, you can achieve whatever you want. I was told that every single day. When I got in the real world, I was like, wow, that ’ s maybe not always so true. But I still believe in that. Maybe I ’ m being too optimistic. So I still look towards the future for that to continue. DB : I think you ’ re being too optimistic. GC : You do? DB : The odds of an American young person **exceeding** their parents ’ salary — a generation ago, like 86 percent did it. Now 51 percent do it. There ’ s just been a problem in social mobility in the country. CA : You ’ ve written that this entire century has basically been a disaster, that the age of sunny growth is over and we ’ re in deep trouble. DB : Yeah, I mean, we averaged, in real terms, population-adjusted, two or three percent growth for 50 years, and now we ’ ve had less than one percent growth. And so there ’ s something seeping out. And so if I ’ m going to tell people that they should take risks, one of the things we ’ re seeing is a rapid decline in mobility, the number of people who are moving across state lines, and that ’ s especially true among millennials. It ’ s young people that are moving less. So how do we give people the security from which they can take risk? And I ’ m a big believer in attachment theory of raising children, and attachment theory is based on the motto that all of life is a series of daring adventures from a secure base. Have you parents given you a secure base? And as a society, we do not have a secure base, and we won ’ t get to that”Hamilton,”risk-taking, energetic ethos until we can supply a secure base. CA : So I wonder whether there ’ s ground here to create almost like a **shared** agenda, a bridging conversation, on the one hand recognizing that there is this really deep problem that the system, the economic system that we built, seems to be misfiring right now. Second, that maybe, if you ’ re right that it ’ s not all about **immigrants**, it ’ s probably more about technology, if you could win that argument, that de-emphasizes what seems to me the single most divisive territory between Trump supporters and others, which is around the role of the other. It ’ s very offensive to people on the left to have the other demonized to the extent that the other seems to be demonized. That feels deeply immoral, and maybe people on the left could agree, as you said, that **immigration** may have happened too fast, and there is a limit beyond which human societies struggle, but nonetheless this whole problem becomes de-emphasized if automation is the key issue, and then we try to work together on recognizing that it ’ s real, recognizing that the problem probably wasn ’ t properly addressed or seen or heard, and try to figure out how to rebuild communities using, well, using what? That seems to me to become the fertile conversation of the future : How do we rebuild communities in this modern age, with technology doing what it ’ s doing, and reimagine this bright future? GC : That ’ s why I go back to optimism. I ’ m not being... it ’ s not like I ’ m not looking at the facts, where we ’ ve come or where we ’ ve come from. But for gosh sakes, if we don ’ t look at it from an optimistic point of view — I ’ m refusing to do that just yet. I ’ m not raising my 12- and 13-year-old to say,”Look, the world is dim.”CA ; We ’ re going to have one more question from the room here. Questioner : Hi. Hello. Sorry. You both mentioned the infrastructure plan and Russia and some other things that wouldn ’ t be traditional Republican priorities. What do you think, or when, will Republicans be motivated to take a stand against Trumpism? GC : After last night, not for a while. He changed a lot last night, I believe. DB : His popularity among Republicans — he ’ s got 85 percent approval, which is higher than Reagan had at this time, and that ’ s because society has just gotten more polarized. So people follow the party much more than they used to. So if you ’ re waiting for Paul Ryan and the Republicans in Congress to flake away, it ’ s going to take a little while. GC : But also because they ’ re all concerned about reelection, and Trump has so much power with getting people either for you or against you, and so, they ’ re vacillating every day, probably :”Well, should I go against or should I not?”But last night, where he finally sounded presidential, I think most Republicans are breathing a sigh of relief today. DB : The half-life of that is short. GC : Right — I was just going to say, until **Twitter** happens again. CA : OK, I want to give each of you the chance to imagine you ’ re speaking to — I don ’ t know — the people online who are watching this, who may be Trump supporters, who may be on the left, somewhere in the middle. How would you advise them to bridge or to relate to other people? Can you share any final wisdom on this? Or if you think that they shouldn ’ t, tell them that as well. GC : I would just start by saying that I really think any change and coming together starts from the top, just like any other organization. And I would love if, somehow, Trump supporters or people on the left could encourage their leaders to show that compassion from the top, because imagine the change that we could have if Donald Trump tweeted out today, to all of his supporters,”Let ’ s not be vile anymore to each other. Let ’ s have more understanding. As a leader, I ’ m going to be more inclusive to all of the people of America.”To me, it starts at the top. Is he going to do that? I have no idea. But I think that everything starts from the top, and the power that he has in encouraging his supporters to have an understanding of where people are coming from on the other side. CA : David. DB : Yeah, I guess I would say I don ’ t think we can teach each other to be civil, and give us sermons on civility. That ’ s not going to do it. It ’ s substance and how we act, and the nice thing about Donald Trump is he smashed our categories. All the categories that we thought we were thinking in, they ’ re obsolete. They were great for the 20th century. They ’ re not good for today. He ’ s got an agenda which is about closing borders and closing trade. I just don ’ t think it ’ s going to work. I think if we want to rebuild communities, recreate jobs, we need a different set of agenda that smashes through all our current divisions and our current categories. For me, that agenda is Reaganism on macroeconomic policy, Sweden on welfare policy and cuts across right and left. I think we have to have a dynamic economy that creates growth. That ’ s the Reagan on economic policy. But people have to have that secure base. There have to be nurse-family partnerships ; there has to be universal preschool ; there have to be **charter** schools ; there have to be college programs with wraparound programs for parents and communities. We need to help **heal** the crisis of social **solidarity** in this country and help **heal** families, and government just has to get a lot more involved in the way liberals like to rebuild communities. At the other hand, we have to have an economy that ’ s free and open the way conservatives used to like. And so getting the substance right is how you smash through the partisan identities, because the substance is what ultimately shapes our polarization. CA : David and Gretchen, thank you so much for an absolutely fascinating conversation. Thank you. That was really, really interesting. Hey, let ' s keep the conversation going. We ' re continuing to try and figure out whether we can add something here, so keep the conversation going on Facebook. Give us your thoughts from whatever part of the political spectrum you ' re on, and actually, wherever in the world you are. This is not just about America. It ' s about the world, too. But we ' re not going to end today without music, because if we put music in every political conversation, the world would be completely different, frankly. It just would. Up in Harlem, this extraordinary woman, Vy Higginsen, who ' s actually right here — let ' s get a shot of her. She created this program that brings **teens** together, teaches them the joy and the impact of gospel music, and hundreds of **teens** have gone through this program. It ' s transformative for them. The music they made, as you already heard, is extraordinary, and I can ' t think of a better way of ending this TED Dialogue than welcoming Vy Higginsen ' s Gospel **Choir** from Harlem. Thank you. **Choir** : O beautiful for spacious skies For amber waves of grain For purple mountain majesties Above the fruited plain America! America! America! America! God shed his grace on thee And crown thy good with brotherhood From sea to shining sea From sea to shining sea

Text Sample 349: POS Dim 3 – 2017 – Score 17.00 – 2017/t000765.txt

I remember the first time that I saw people injecting drugs. I had just arrived in Vancouver to lead a research project in HIV prevention in the infamous Downtown East Side. It was in the lobby of the Portland Hotel, a supportive housing project that gave rooms to the most marginalized people in the city, the so-called"difficult to house."I ' ll never forget the young woman standing on the stairs repeatedly jabbing herself with a needle, and screaming,"I can ' t find a vein,"as blood splattered on the wall. In response to the desperate state of affairs, the drug use, the poverty, the **violence**, the soaring rates of HIV, Vancouver **declared** a public health emergency in 1997. This opened the door to expanding **harm** reduction services, distributing more needles, increasing **access** to methadone, and, finally, opening a supervised injection site. Things that make injecting drugs less hazardous. But today, 20 years later, **harm** reduction is still viewed as some sort of radical concept. In some places, it ' s still illegal to carry a clean needle. Drug users are far more likely to be arrested than to be offered methadone therapy. Recent proposals for supervised injection sites in cities like Seattle, Baltimore and New York have been met with stiff opposition : opposition that goes against everything we know about addiction. Why is that? Why are we still stuck on the idea that the only option is to stop using — that any drug use will not be tolerated? Why do we ignore countless personal stories and overwhelming scientific evidence that **harm** reduction works? Critics say that **harm** reduction doesn ' t stop people from using illegal drugs. Well, actually, that is the whole point. After every criminal and societal sanction that we can come up with, people still use drugs, and far too many die. Critics also say that we are giving up on people by not focusing our attention on treatment and recovery. In fact, it is just the opposite. We are not giving up on people. We know that if recovery is ever going to happen we must keep people alive. Offering someone a clean needle or a safe place to inject is the first step to treatment and recovery. Critics also claim that **harm** reduction gives the wrong message to our children about drug users. The last time I looked, these drug users are our children. The message of **harm** reduction is that while drugs can hurt you, we still must reach out to people who are addicted. A needle exchange is not an advertisement for drug use. Neither is a methadone clinic or a supervised injection site. What you see there are people sick and hurting, hardly an endorsement for drug use. Let ’ s take supervised injection sites, for example. Probably the most misunderstood health intervention ever. All we are saying is that allowing people to inject in a clean, dry space with fresh needles, surrounded by people who care is a lot better than injecting in a dingy alley, sharing contaminated needles and hiding out from police. It ’ s better for everybody. The first supervised injection site in Vancouver was at 327 Carol Street, a narrow room with a concrete floor, a few chairs and a box of clean needles. The police would often lock it down, but somehow it always mysteriously reopened, often with the aid of a crowbar. I would go down there some evenings to provide medical care for people who were injecting drugs. I was always struck with the commitment and compassion of the people who operated and used the site. No judgment, no hassles, no fear, lots of profound conversation. I learned that despite unimaginable **trauma**, physical pain and mental illness, that everyone there thought that things would get better. Most were convinced that, someday, they ’ d stop using drugs altogether. That room was the forerunner to North America ’ s first government-sanctioned supervised injection site, called INSITE. It opened in September of 2003 as a three-year research project. The conservative government was intent on closing it down at the end of the study. After eight years, the battle to close INSITE went all the way up to Canada ’ s Supreme Court. It pitted the government of Canada against two people with a long history of drug use who knew the benefits of INSITE firsthand : Dean Wilson and Shelley Tomic. The court ruled in favor of keeping INSITE open by nine to zero. The justices were scathing in their response to the government ’ s case. And I quote :”The effect of denying the services of INSITE to the population that it serves and the correlative increase in the risk of death and disease to injection drug users is grossly disproportionate to any benefit that Canada might derive from presenting a uniform stance on the possession of narcotics.”This was a hopeful moment for **harm** reduction. Yet, despite this strong message from the Supreme Court, it was, until very recently, impossible to open up any new sites in Canada. There was one interesting thing that happened in December of 2016, when due to the overdose crisis, the government of British Columbia allowed the opening of overdose prevention sites. Essentially ignoring the federal approval process, community groups opened up about 22 of these de facto illegal supervised injection sites across the **province**. Virtually overnight, thousands of people could use drugs under supervision. Hundreds of overdoses were reversed by Naloxone, and nobody died. In fact, this is what ’ s happened at INSITE over the last 14 years : 75, 000 different individuals have injected illegal drugs more than three and a half million times, and not one person has died. Nobody has ever died at INSITE. So there you have it. We have scientific evidence and successes from needle exchanges methadone and supervised injection sites. These are common-sense, compassionate approaches to drug use that improve health, bring connection and greatly reduce **suffering** and death. So why haven ’ t **harm** reduction programs taken off? Why do we still think that drug use is law **enforcement** issue? Our disdain for drugs and drug users goes very deep. We are bombarded with images and media stories about the horrible impacts of drugs. We have stigmatized entire communities. We applaud military-inspired operations that bring down drug dealers. And we appear unfazed by building more jails to incarcerate people whose only crime is using drugs. Virtually millions of people are caught up in a hopeless cycle of incarceration, **violence** and poverty that has been created by our drug laws and not the drugs themselves. How do I explain to people that drug users deserve care and support and the freedom to live their lives when all we see are images of guns and handcuffs and jail cells? Let ’ s be clear : criminalization is just a way to institutionalize stigma. Making drugs illegal does nothing to stop people from using them. Our paralysis to see things differently is also based on an entirely false narrative about drug use. We have been led to believe that drug users are irresponsible people who just want to get high, and then through their own personal failings spiral down into a life of crime and poverty, losing their jobs, their families and, ultimately, their lives. In reality, most drug users have a story, whether it ’ s childhood **trauma**, **sexual** abuse, mental illness or a personal tragedy. The drugs are used to **numb** the pain. We must understand that as we approach people with so much **trauma**. At its core, our drug policies are really a social justice issue. While the media may focus on overdose deaths like Prince and Michael Jackson, the majority of the **suffering** happens to people who are living on the margins, the poor and the dispossessed. They don ’ t vote ; they are often alone. They are society ’ s disposable people. Even within health care, drug use is highly stigmatized. People using drugs avoid the health care system. They know that once engaged in clinical care or admitted to hospital, they will be treated poorly. And their supply line, be it heroin, cocaine or crystal meth will be interrupted. On top of that, they will be asked a barrage of questions that only serve to expose their losses and **shame**.”What drugs do you use?””How long have you been living on the **street**?””Where are your children?””When were you last in jail?”Essentially :”Why the hell don ’ t you stop using drugs?”In fact, our entire medical approach to drug use is upside down. For some reason, we have decided that abstinence is the best way to treat this. If you ’ re lucky enough, you may get into a detox program. If you live in a community with Suboxone or methadone, you may get on a substitution program. Hardly ever would we offer people what they desperately need to survive : a safe prescription for opioids. Starting with abstinence is like asking a new diabetic to quit sugar or a severe asthmatic to start running marathons or a depressed person to just be happy. For any other medical condition, we would never start with the most extreme option. What makes us think that strategy would work for something as complex as addiction? While unintentional overdoses are not new, the scale of the current crisis is unprecedented. The Center for Disease Control estimated that 64, 000 Americans died of a drug overdose in 2016, far **exceeding** car crashes or homicides. Drug- related mortality is now the leading cause of death among men and women between 20 and 50 years old in North America Think about that. How did we get to this point, and why now? There is a kind of perfect storm around opioids. Drugs like Oxycontin, Percocet and Dilaudid have been liberally distributed for decades for all kinds of pain. It is estimated that two million Americans are daily opioid users, and over 60 million people received at least one prescription for opioids last year. This massive dump of prescription drugs into communities has provided a steady source for people wanting to self-medicate. In response to this prescription epidemic, people have been cut off, and this has greatly reduced the **street** supply The unintended but predictable consequence is an overdose epidemic. Many people who were reliant on a steady supply of prescription drugs turned to heroin. And now the illegal drug market has tragically switched to synthetic drugs, mainly fentanyl. These new drugs are cheap, potent and extremely hard to dose. People are literally being poisoned. Can you imagine if this was any other kind of poisoning epidemic? What if thousands of people started dying from poisoned meat or baby **formula** or coffee? We would be treating this as a true emergency. We would immediately be supplying safer alternatives. There would be changes in legislation, and we would be supporting the **victims** and their families. But for the drug overdose epidemic, we have done none of that. We continue to demonize the drugs and the people who use them and blindly pour even more resources into law **enforcement**. So where should we go from here? First, we should fully embrace, fund and scale up **harm** reduction programs across North America. I know that in places like Vancouver, **harm** reduction has been a lifeline to care and treatment. I know that the number of overdose deaths would be far higher without **harm** reduction. And I personally know hundreds of people who are alive today because of **harm** reduction. But **harm** reduction is just the start. If we truly want to make an impact on this drug crisis, we need to have a serious conversation about prohibition and criminal punishment. We need to recognize that drug use is first and foremost a public health issue and turn to comprehensive social and health solutions. We already have a model for how this can work. In 2001, Portugal was having its own drug crisis. Lots of people using drugs, high crime rates and an overdose epidemic. They defied global conventions and decriminalized all drug possession. Money that was spent on drug **enforcement** was redirected to health and rehabilitation programs. The results are in. Overall drug use is down dramatically. Overdoses are uncommon. Many more people are in treatment. And people have been given their lives back. We have come so far down the road of prohibition, punishment and **prejudice** that we have become indifferent to the **suffering** that we have inflicted on the most vulnerable people in our society. This year even more people will get caught up in the illegal drug trade. Thousands of children will learn that their mother or father has been sent to jail for using drugs. And far too many parents will be notified that their son or daughter has died of a drug overdose. It doesn ’ t have to be this way. Thank you.

Text Sample 350: POS Dim 3 – 2017 – Score 17.00 – 2017/t001102.txt

Vanessa Garrison : I am Vanessa, daughter of Annette, daughter of Olympia, daughter of Melvina, daughter of Katie, born 1878, Parish County, Louisiana. T. Morgan Dixon : And my name is Morgan, daughter of Carol, daughter of Letha, daughter of Willie, daughter of Sarah, born 1849 in Bardstown, Kentucky. VG : And in the tradition of our families, the great oral tradition of almost every Black church we know honoring the culture from which we draw so much power, we ’ re gonna start the way our mommas and grandmas would want us to start. TMD : In **prayer**. Let the words of my mouth, the **meditation** of our hearts, be acceptable in thy sight, oh Lord, my strength and my redeemer. VG : We call the names and rituals of our ancestors into this room today because from them we received a powerful blueprint for survival, strategies and tactics for healing carried across oceans by African women, passed down to generations of Black women in America who used those skills to navigate institutions of slavery and state-sponsored **discrimination** in order that we might stand on this stage. We walk in the footsteps of those women, our foremothers, legends like Ella Baker, Septima Clark, Fannie Lou Hamer, from whom we learned the power of organizing after she would had single-handedly registered 60, 000 voters in Jim Crow Mississippi. TMD : 60, 000 is a lot of people, so if you can imagine me and Vanessa inspiring 60, 000 women to walk with us last year, we were fired up. But today, 100, 000 Black women and girls stand on this stage with us. We are committed to **healing** ourselves, to lacing up our sneakers, to walking out of our front door every single day for total healing and transformation in our communities, because we understand that we are in the footsteps of a civil rights legacy like no other time before, and that we are facing a health crisis like never ever before. And so we ' ve had a lot of moments, great moments, including the time we had on our pajamas, we were working on our computer and Michelle Obama emailed us and invited us to the White House, and we thought it was spam. But this moment here is an opportunity. It is an opportunity that we don ' t take for granted, and so we thought long and hard about how we would use it. Would we talk to the women we hope to inspire, a million in the next year, or would we talk to you? We decided to talk to you, and to talk to you about a question that we get all the time, so that the millions of women who hopefully will watch this will never have to answer it again. It is : Why are Black women dying faster and at higher rates than any other group of people in America from preventable, obesity- related diseases? The question hurts me. I ' m shaking a little bit. It feels value-laden. It hurts my body because the weight represents so much. But we ' re going to talk about it and invite you into an inside conversation today because it is necessary, and because we need you. VG : Each night, before the first day of school, my grandmother would sit me next to the stove and with expert precision use a hot comb to press my hair. My grandmother was legendary, big, loud. She filled up a room with laughter and oftentimes curse words. She cooked a mean peach cobbler, had 11 children, a house full of grandchildren, and like every Black woman I know, like most all women I know, she had prioritized the care of others over caring for herself. We measured her strength by her capacity to endure pain and **suffering**. We **celebrated** her for it, and our choice would prove to be deadly. One night after pressing my hair before the first day of eighth grade, my grandmother went to bed and never woke up, dead at 66 years old from a heart attack. By the time I would graduate college, I would lose two more beloved family members to chronic disease : my aunt Diane, dead at 55, my aunt Tricia, dead at 63. After living with these losses, the hole that they left, I decided to calculate the life expectancy of the women in my family. Staring back at me, the number 65. I knew I could not sit by and watch another woman I loved die an early death. TMD : So we don ' t usually put our business in the **streets**. Let ' s just put that out there. But I have to tell you the statistics. Black women are dying at alarming rates, and I used to be a classroom teacher, and I was at South Atlanta High School, and I remember standing in front of my classroom, and I remember a statistic that half of Black girls will get diabetes unless diet and levels of activity change. Half of the girls in my classroom. So I couldn ' t teach anymore. So I started taking girls hiking, which is why we ' re called GirlTrek, but Vanessa was like, that is not going to move the dial on the health crisis ; it ' s cute. She was like, it ' s a cute hiking club. So what we thought is if we could rally a million of their mothers... 82 percent of Black women are over a healthy weight right now. 53 percent of us are obese. But the number that I cannot, that I cannot get out of my head is that every single day in America, 137 Black women die from a preventable disease, heart disease. That ' s every 11 minutes. 137 is more than gun **violence**, cigarette smoking and HIV combined, every day. It is roughly the amount of people that were on my plane from New Jersey to Vancouver. Can you imagine that? A plane filled with Black women crashing to the ground every day, and no one is talking about it. VG : So the question that you ' re all asking yourselves right now is why? Why are Black women dying? We asked ourselves that same question. Why is what ' s out there not working for them? Private weight loss companies, government interventions, public health campaigns. I ' m going to tell you why : because they focus on weight loss or looking good in skinny jeans without acknowledging the **trauma** that Black women hold in our bellies and bones, that has been embedded in our very DNA. The best advice from hospitals and doctors, the best medications from pharmaceutical companies to treat the congestive heart failure of my grandmother didn ’ t work because they didn ’ t acknowledge the **systemic** racism that she had dealt with since birth. A divestment in schools, discriminatory housing practices, predatory lending, a crack cocaine epidemic, mass incarceration putting more Black bodies behind bars than were owned at the height of slavery. But GirlTrek does. For Black women whose bodies are buckling under the weight of systems never designed to support them, GirlTrek is a lifeline. August 16, 2015, Danita Kimball, a member of GirlTrek in Detroit, received the news that too many Black mothers have received. Her son Norman, 23 years old, a father of two, was gunned down while on an afternoon drive. Imagine the **grief** that overcomes your body in that moment, the immobilizing fear. Now, know this, that just days after laying her son to rest, Danita Kimball posted online,”I don ’ t know what to do or how to move forward, but my sisters keep telling me I need to walk, so I will.”And then just days after that,”I got my steps in today for my baby Norm. It felt good to be out there, to walk.”TMD : Walking through pain is what we have always done. My mom, she ’ s in the middle right there, my mom desegregated her high school in 1955. Her mom walked down the steps of an abandoned school bus where she raised 11 kids as a sharecropper. And her mom stepped onto Indian territory fleeing the terrors of the Jim Crow South. And her mom walked her man to the door as he went off to fight in the Kentucky Colored Regiment, the Civil War. They were born slaves but they wouldn ’ t die slaves. Change-making, it ’ s in my blood. It ’ s what I do, and this health crisis ain ’ t nothing compared to the road we have traveled. So it ’ s like James Cleveland. I don ’ t feel no ways tired, so we got to work. We started looking at models of change. We looked all over the world. We needed something not only that was a part of our cultural inheritance like walking, but something that was scalable, something that was high-impact, something that we could replicate across this country. So we studied models like Wangari Maathai, who won the Nobel Peace Prize for inspiring women to plant 50 million trees in Kenya. She brought Kenya back from the brink of environmental devastation. We studied these systems of change, and we looked at walking scientifically. And what we learned is that walking just 30 minutes a day can single-handedly decrease 50 percent of your risk of diabetes, heart disease, stroke, even Alzheimer ’ s and dementia. We know that walking is the single most powerful thing that a woman can do for her health, so we knew we were on to something, because from Harriet Tubman to the women in Montgomery, when Black women walk, things change. VG : So how did we take this simple idea of walking and start a revolution that would catch a fire in **neighborhoods** across America? We used the best practices of the Civil Rights Movement. We huddled up in church basements. We did grapevine information sharing through beauty salons. We empowered and trained mothers to stand on the front lines. We took our message directly to the **streets**, and women responded. Women like LaKeisha in Chattanooga, Chrysantha in Detroit, Onika in New Orleans, women with difficult names and difficult stories join GirlTrek every day and commit to walking as a practice of self-care. Once walking, those women get to organizing, first their families, then their communities, to walk and talk and solve problems together. They walk and notice the abandoned building. They walk and notice the lack of **sidewalks**, the lack of green space, and they say,”No more.”Women like Susie Paige in Philadelphia, who after walking daily past an abandoned building in her **neighborhood**, decided,”I ’ m not waiting. Let me rally my team. Let me grab some supplies. Let me do what no one else has done for me and my community.”TMD : We know one woman can make a difference, because one woman has already changed the world, and her name is Harriet Tubman. And trust me, I love Harriet Tubman. I ’ m obsessed with her, and I used to be a history teacher. I will not tell you the whole history. I will tell you four things. So I used to have an old Saab — the kind of canvas top that drips on your head when it rains — and I drove all the way down to the eastern shore of Maryland, and when I stepped on the dirt that Harriet Tubman made her first escape, I knew she was a woman just like we are and that we could do what she had done, and we learned four things from Harriet Tubman. The first one : do not wait. Walk right now in the direction of your healthiest, most fulfilled life, because self-care is a revolutionary act. Number two : when you learn the way forward, come back and get a sister. So in our case, start a team with your friends — your friends, your family, your church. Number three : rally your allies. Every single person in this room is complicit in a Tubman-inspired takeover. And number four : find joy. The most underreported fact of Harriet Tubman is that she lived to be 93 years old, and she didn ’ t live just an ordinary life ; uh-uh. She was standing up for the good guys. She married a younger man. She adopted a child. I ’ m not kidding. She lived. And I drove up to her house of freedom in upstate New York, and she had planted apple trees, and when I was there on a Sunday, they were blooming. Do you call it — do they bloom? The apples were in season, and I was thinking, she left fruit for us, the legacy of Harriet Tubman, every single year. And we know that we are Harriet, and we know that there is a Harriet in every community in America. VG : We also know that there ’ s a Harriet in every community across the globe, and that they could learn from our Tubman Doctrine, as we call it, the four steps. Imagine the possibilities beyond the **neighborhoods** of Oakland and Newark, to the women working rice fields in Vietnam, tea fields in Sri Lanka, the women on the mountainsides in Guatemala, the **indigenous** reservations throughout the vast plains of the Dakotas. We believe that women walking and talking together to solve their problems is a global solution. TMD : And I ’ ll leave you with this, because we also believe it can become the center of social justice again. Vanessa and I were in Fort Lauderdale. We had an organizer training, and I was leaving and I got on the airplane, and I saw someone I knew, so I waved, and as I ’ m waiting in that long line that you guys know, waiting for people to put their stuff away, I looked back and I realized I didn ’ t know the woman but I recognized her. And so I blew her a kiss because it was Sybrina Fulton, Trayvon Martin ’ s mom, and she whispered”thank you”back to me. And I can ’ t help but wonder what would happen if there were groups of women walking on Trayvon ’ s block that day, or what would happen in the South Side of Chicago every day if there were groups of women and mothers and aunts and cousins walking, or along the polluted rivers of Flint, Michigan. I believe that walking can transform our communities, because it ’ s already starting to. VG : We believe that the personal is political. Our walking is for healing, for joy, for fresh air, quiet time, to connect and disconnect, to worship. But it ’ s also walking so we can be healthy enough to stand on the front lines for change in our communities, and it is our call to action to every Black woman listening, every Black woman in earshot of our voice, every Black woman who you know. Think about it : the woman working front desk reception at your job, the woman who delivers your mail, your neighbor — our call to action to them, to join us on the front lines for change in your community. TMD : And I ’ ll bring us back to this moment and why it ’ s so important for my dear, dear friend Vanessa and I. It ’ s because it ’ s not always easy for us, and in fact, we have both seen really, really dark days, from the hate speech to the summer of police brutality and **violence** that we saw last year, to even losing one of our walkers, Sandy Bland, who died in police custody. But the most courageous thing we do every day is we practice faith that goes beyond the facts, and we put feet to our **prayers** every single day, and when we get overwhelmed, we think of the words of people like Sonia

Sanchez, a poet laureate, who says,”Morgan, where is your fire? Where is the fire that burned holes through slave ships to make us breathe? Where is the fire that turned guts into chitlins, that took rhythms and make jazz, that took sit-ins and marches and made us jump boundaries and barriers? You ’ ve got to find it and pass it on.”So this is us finding our fire and passing it on to you. So please, stand with us, walk with us as we rally a million women to reclaim the **streets** of the 50 highest need communities in this country. We thank you so much for this opportunity.

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Chris Anderson : Hello, TED community, welcome back for another live conversation. It ’ s a big one today, as big as they get. You know, when we created this”Build Back Better”series our thought was how could we address issues arising out of the pandemic, how could we imagine building back from that. But the events of this past week, the horrific death of George Floyd and the daily protests that have followed, I mean, they provided a new urgency which we, of course, simply have to address. I mean, can we build back better from this? I think before we can even start to answer that question, we just have to seek to understand the immensity of this moment. Whitney Pennington Rodgers : That ’ s right, Chris. Right now, so many people in the United States and beyond are grappling with feelings of anger and frustration, deep, deep sadness and really helplessness. No matter who you are, you have questions about what to do now, how to make things better. And as we ’ ve seen, **violence** like this unfolds for many, many years. What is the path forward? CA : So – We ’ re joined today by a group of activists, organizers and leaders known for their crucial work in social justice and civil rights. We ’ re so grateful to have them here to engage in a discussion about racial injustice in America, the unbearable acts of **violence** that we ’ ve – Acts of **violence** against the black community that we ’ ve witnessed, the dangers to a nation riven by anger and fear. And how on earth we can move forward from this to something better. So first, each of our four guests will share their thoughts on how we move forward from this moment. And then we ’ ll engage as a group, including you, the TED community. WPR : And we ’ d like to thank our partner, the Project Management Institute. Their generous support has helped make today ’ s interviews possible, and of course, as Chris mentioned, we want you to take part in the conversation, so please share your questions using our Ask a question feature and continue to share your thoughts in the discussion thread. CA : Thanks, Whitney. OK, let ’ s get this moving. Our first guest. Dr. Phillip Atiba Goff is the founder and CEO of the Center for Policing Equity. They work with police departments across America, including in Minneapolis, to seek measurable responses to racial bias. Phil, I can scarcely imagine how the stress in the last week must have been for you. Welcome, and over to you for your opening comments. Phillip Atiba Goff : Thanks, Chris. Yeah, this week has been a gut punch to anybody who felt like we could be making progress in the way that we put forward public **safety** that empowers particularly vulnerable communities. We started working in Minneapolis about five years ago. At the time, it was, like most major cities in the United States, a department that had a long history of unaccounted for **violence** from law **enforcement**, targeting the most vulnerable black communities. And we tried to put into place a number of things that we know work. Change the culture, so that the culture can be accountable to the values of the community. And what we saw was small but measurable progress. We always knew, with small and measurable progress, that you ’ re one tragic incident from going back to ground zero. But the events of the last week and a half haven ' t brought us back to ground zero, they ' ve torched ground zero, and we ' ve dug a hole that we have to dig ourselves out of. What I hear from police chiefs who call me, from activists I talk to, from folks in the communities that are literally on fire right now, I hear folks saying, I had one activist say to me that the pain that he was feeling was too large to fit into his body. And without thinking about it, I said right back,"That ' s because it ' s too large to fit into a lifetime."What we ' re seeing isn ' t just the response to one gruesome, cruel, public execution. A lynching. It ' s not just the reaction to three of them : Ahmaud Arbery, Breonna Taylor and then George Floyd. What we ' re seeing is the bill come due for the unpaid debts that this country owes to its black residents. And it comes due usually every 20 to 30 years. It was Ferguson just six years ago, but about 30 years before that, it was in the **streets** of Los Angeles, after the verdict that exonerated the police that beat Rodney King on video. It was Newark, it was Watts, it was Chicago, it was Tulsa, it was Chicago again. If we don ' t take a full accounting of these debts that are owed, then we ' re going to keep paying it. Part of what I ' ve been experiencing in the last week and a half, and what I ' ve been sharing with the people who do this work, who are serious about it, is the acknowledgment, the soul-crushing reality that at some point, when things stop being on fire, the cameras are going to turn to something else. And the history that we have in this country is not just a history of vicious neglect and a targeted abuse of black communities, it ' s also one where we lose our attention for it. And what that means for communities like in Baton Rouge, for those who still grieve Alton Sterling, and in Baltimore, for those who are still grieving Freddie Gray, is that there is not just a chance, there ' s a likelihood that we are a month or two months out from this with no more to show for it than what we had to show after Michael Brown Jr. And holding the weight of that, individually and collectively, is just too much. It ' s just too heavy a load for a person or a people, or a generation to hold up. What we ' re seeing is the unrepentant sins, the unpaid debts. And so the solution can ' t just be that we fix policing. It can ' t be only incremental reform. It can ' t be only systems of accountability to catch cops after they ' ve killed somebody. Because there ' s no such thing as justice for George Floyd. There ' s maybe accountability. There ' s maybe some relief from the people who are still around, who loved him, for his daughter who spoke out yesterday and said,"My Daddy changed the world."There won ' t be justice for a man who ' s dead when he didn ' t have to be. But we ' re not going to get to where we need to go just by reforming police. So in addition to the work that CPE is known for with the data, we have been encouraging departments and cities to take the money that should be going to invest in communities, and take it from police budgets, bring it to the communities. People ask,"Well, what could it possibly look like? How could we imagine it?"And I tell people, there is a place where we do this in the United States right now. We ' ve all heard about it, whispered, some of us have even been there, some of us live there. The place is called the suburbs, where we already have enough resources to give to people, so they don ' t need the police for public **safety** in the first place. If someone has a substance abuse issue, they can go to a clinic. If somebody has a medical issue, they ' ve got insurance, they can go to a hospital. If there ' s a **domestic** dispute, they have friends, they have support. You don ' t need to enter a badge and a gun into it. If we hadn ' t disinvested from all the public resources that were available in communities that most needed those, we wouldn ' t need police in the first place, and many have been arguing, even more loudly recently, that we don ' t. If we would just take the money that we use to punish, and instead invest it in the promise and the genius of the community that could be there. So I don ' t know all the ways we ’ re going to get there. I know it ’ s going to take everything and. It ’ s going to need the kind of **systemic** change and the management tools that we traditionally offer. It ’ s also going to need a quantum change in the way that we think about public **safety**. But mostly, this isn ’ t just a policing problem. This is the unpaid debts that are owed to black America. The bill is coming due. And we need to start getting an accounting together, so we ’ re not just paying off the interest of the damn thing. WPR : Thank you, Phil. Rashad Robinson is the president of Color Of Change, a civil rights organization that advocates for racial justice for the black community. To date, more than four million people have signed their petition to arrest the officers involved in the murder of George Floyd. And of course, one was arrested last week. Thank you so much for being with us, Rashad, welcome. Rashad Robinson : Thank you. And thank you for having me. It ’ s an opportunity that I ’ m taking today to just tell you about how you can get involved. How you can take action, because right now, strategic action is critical for all of us to do the work to change the rules that far too often keep the systems in place that hold us back. Make no mistake, the criminal justice system is not broken. It is operating exactly the way it was designed. At every single level, the criminal justice system is not about providing justice, but about ensuring that certain people, certain communities are protected, while other communities are violated. And so I wan to talk a little bit today about Color Of Change, about activism, about the work that ’ s happening on the ground from other organizations all around the country, and the way that you can channel this energy. What we talk about at Color Of Change is how do you channel presence into power. Far too often we mistake presence and visibility for power, presence retweets the stories of the movement, people feeling **passion** about change could sometimes make us feel like change is inevitable, but power is actually the ability to change the rules. And right now, every day, people are taking action, and what we ’ re trying to channel that energy into is a couple of things. First is a whole set of demands at the federal level and at the local level. As Phil described, policing operates on many different channels. And what we need to recognize is that while there are a lot of things that can happen at the federal level, locally all around the country there are decisions that are being made in communities around how policing is executed, where community needs to hold a deeper level of accountability, at the state level we need new laws. So at Color Of Change, we ’ ve built a whole platform around a set of demands and are working to build more energy for everyday people to take action. We ’ re fighting for justice for Ahmaud Arbery, Breonna Taylor and George Floyd, we ’ re also fighting for justice for other folks whose names you haven ’ t heard, Nina Pop and others, whose stories of injustice and the relationship to the criminal justice system represent all the ways in which fighting right now is important. Over the last couple of years, we have worked to build a movement, to hold district attorneys accountable and to change the role of district attorneys in our country. And through the Winning Justice platform at Color Of Change, www. winningjustice. org, what we have worked to do is channel the energy of everyday people to take action. So, for folks who are watching what ’ s happening on TV, seeing it on their social media feeds and are outraged about what ’ s happening in Georgia, what ’ s happening in Tennessee, what ’ s happening in Minnesota, you yourself, probably, most likely, live in a place, in a community where you have a district attorney that will not hold police accountable, that will not prosecute police when they **harm**, hurt black folks, when they violate the laws, you live in a community where police are part of the structure that is racking up mass incarceration, but many other aspects of our system are racking up mass incarceration, and district attorneys are at the center of it. You live in those communities and you need to do something about it. And so at winningjustice. org, we ' ve created the only searchable, national database on the 2, 400 prosecutors around the country. We ' re building local squads and communities for folks to be able to engage around efforts that hold DAs accountable. We ' ve worked with our partners across the movement, from our friends in Black Lives Matter, to folks who do policy work, to our friends at local ACLU chapters around the country, to build six demands. Six demands that folks can get behind in terms of pushing for reform, and then we ' ve built public education material. But the only way that we work to change the way that prosecution happens in this country is that if people get involved. If people raise their voice, if people join us in pushing for real change. At the end of the day, I want people to recognize though, and Phillip talked a little bit about this, is that people don ' t experience issues, they experience life. That the forces that hold us back are deeply interrelated, a racist criminal justice system requires a racist media culture to survive, a political inequality follows economic inequality, they all go hand in hand. And so I also want us to not take ourselves out of the equation. We likely work inside of corporations that may post symbols for Black Lives Matter one day, and then support politicians that work to destroy Black Lives Matter the next day. We oftentimes are engaged in practices inside of our companies or in our daily lives supporting media properties and others that are **harming** our communities, are telling stories. Recently, we produced a report at Color Of Change with the Norman Lear school at USC. It ' s called"Normalizing Injustice,"and it can be found at changehollywood. org. And"Normalizing Injustice"looks at the 22 crime procedurals, those crime shows on TV. And looks at all of the ways in which they, sort of, create a warped perception about our view of justice. They create sort of an incentive for the type of policing we see on the country, and actually serve as a PR arm for law **enforcement**. We ' ve been working in writers rooms around the country to work to push folks to tell better stories, but we need folks to be both active listeners, and we need folks in the industry to push back and challenge those, not only the structures that lead to that content coming on the air, but the proliferation across our airwaves. At the end of the day, we have an opportunity in this moment to make change. Inflection points are those moments where we have an opportunity to make huge leaps forward, or the real, real threat of falling backwards. In our hands is the ability to do some incredible things about undoing so many of the injustices that have stood in the way of progress for far too long. But everyday people must get involved. We must channel that presence into power, and we must build the type of power that changes the rules. Racism in so many ways is like water pouring over a floor with holes in it. Every single – In every single way, it will find the holes. And so for us, we cannot simply accept charitable solutions to structural problems, but we actually have to work for structural change. And so I want to end by saying one thing about how we talk about black people and how we talk about black communities in this moment. Because we have to say what we mean, and we have to build the narrative that gets us to where we want to go. So far too often, we talk about black communities as vulnerable, we talk about black people as vulnerable, but vulnerability is a personal trait, black communities have been under attack. Black communities have been exploited, black communities have been targeted, and we need to say that, so we don ' t put the onus on fixing black families and communities, but we put the onus on fixing the structures that have **harmed** us. We will say things like,"Black people are less likely to get loans from banks,"instead of saying that banks are less likely to give loans to black people. This is our opportunity to build the type of progress that makes real change, and at the center of this story, we need to show and elevate the images not just of the pain that we are facing, but of the joy, the brilliance and the creativity that black people have brought to this country. Black people are the protagonist of this story, and we need to make sure that as we work to build a new tomorrow, we ensure that the heroes are at the center of the liberation that we all need. Thank you. CA : Thank you, Rashad. Dr. Bernice King is the CEO of the King Center in Atlanta, Georgia. The center is a living memorial to her father, Dr. Martin Luther King Jr. It ’ s dedicated to inspiring new generations to carry his work forward. In this moment, when so many are hurting, how can we better approach unity and collective healing? Dr. King, over to you. Bernice King : My heart is a little heavy right now, because I was that six-year-old. I was five years old when my father was assassinated. And he did change the world. But the tragedy is that we didn ’ t hear what he was saying to us as a prophet to this nation. And his words are now reverberating back to us. Change, we all know, is necessary right now. And yet, it ’ s not easy. We know that there has to be changes in policing in this nation of ours. But I want to talk about America ’ s choice at a greater level. The prophet said to us,”We still have a choice today : nonviolent coexistence, or **violent** coannihilation.”What we have witnessed over the last eight days has placed that choice before us. We have seen literally in the **streets** of our nation people who have been following the path of nonviolent protest, and people who have been hell-bent on destruction. Those choices are now looking at us, and we have to make a choice. The history of this nation was founded in **violence**. In fact, my father said America is the greatest purveyor of **violence**. And the only way forward is if we repent for being a nation built on **violence**. And I ’ m not just talking about physical **violence**. I ’ m talking about **systemic violence**, I ’ m talking about policy **violence**, I ’ m talking about what he spoke of are the triple evils of poverty, racism and militarism. All **violent**. Albert Einstein said something to us. He said we cannot solve problems on the same level of thinking in which they were created. And so if we are going to move forward, we are going to have to deconstruct these systems of **violence** that we have set in America. And we ’ re going to have to reconstruct on another foundation. That foundation happens to be love and nonviolence. And so, as we move forward, we can correct course if we make that choice that Daddy said, nonviolent coexistence. And not continue on the pathway of **violent** coannihilation. So what does that look like? That looks like some deconstruction work in order to get to the construction. We have to deconstruct our thinking. We ’ ve got to deconstruct the way in which we see people and deconstruct the way in which we operate, practice and engage and set policy. And so I believe that there ’ s a lot of heart, h-e-a-r-t work to do, in the midst of all the h-a-r-d, hard work to do. Because heart work is hard work. One of the things we have to do is we have to ensure that everyone, especially my white brothers and sisters, have to engage in the heart work, the antiracism work, in our hearts. No one is exempt from this, especially in my white community. We must do that work in our hearts, the antiracism work. The second thing is that I encourage people to look at the nonviolence training that we [have] at the King Center, thekingcenter. org, so that we learn the foundation of understanding our interrelatedness and interconnectedness. That we understand our loyalties and our commitments and our policy-making can no longer be devoted to one group of people, but has to be devoted to the greater good of all people. And so I ’ m inviting people to even join us on our own line of protest that ’ s happening every night at seven o ’ clock pm on the King Center Facebook page, because so many people have things that they want to express and contribute to this. We all have to change and have to make a choice. It is a choice to change the direction that we have been going. We need a revolution of values in this country. That ’ s what my Daddy said. He changed the world, he changed hearts, and now, what has happened over the last seven, eight years and through history, we have to change course. And we all have to participate in changing America with the true revolution of values, where people are at the center, and not profit. Where morality is at the center, and not our military might. America does have a choice. We can either choose to go down continually that path of destruction, or we can choose nonviolent coexistence. And as my mother said, struggle is a never-ending process, freedom is never really won. You earn it and win it in every generation. Every generation is called to this freedom struggle. You as a person may want to exempt yourself, but every generation is called. And so I encourage corporations in America to start doing antiracism work within corporate America. I encourage every industry to start doing antiracism work, and pick up the banner of understanding nonviolent change, personally, and from a social change perspective. We can do this. We can make the right choice to ultimately build the beloved community. Thank you. WPR : Thank you, Dr. King. Anthony Romero is the executive director of American Civil Liberties Union. As one of the nation ' s oldest social justice organizations, the ACLU has advocated for racial **equality** and shown deep support to the black community in moments of crisis. And in moments like these, black voices are almost always the loudest and at times the **silence** from our nonblack brothers and sisters can feel deafening. How we can bring our allies into the mix, to better support ending **systemic violence** and racism against the black communities, is a question top of mind for a lot of us. Anthony, welcome to the show, and thank you so much for being with us. Anthony Romero : Great. Thank you, thank you Whitney, thank you, Chris, for inviting me to join this TED community. I think community is really important right now. With so many of us feeling trepidation, the weariness, the anger, the fear, the frustration, the terrorism that we ' ve experienced in our communities. This is a time to huddle around a virtual campfire, with your posse, with your family, with your loved ones, with your network. It ' s not a time to be isolated or alone. And I think for allies in this struggle, those of us who don ' t live this experience every day, it is time for us to lean in. You can ' t change the channel, you can ' t tune out, you can ' t say,"This is too hard."It is not that hard for us to listen and learn and heed. It is the only way we ' re going to build out of this, by hearing the voices of Rashad, and Phil and Dr. King. By hearing the voices of our neighbors and loved ones, by hearing the voices on **Twitter** of people who we don ' t know. And so white communities and allied organizations need to pay even closer attention. This is the test of your character. How willing are you to lean in and to engage. For me, I have – These have been really hard couple of weeks. I feel like this is really a test of whether or not we really believe in the American experiment. Do we really believe it? Do we really believe that out of many, one, a country with no unifying language, no unifying culture, no unifying religion, can we really become one people? All **equal** before the law? All bound together with a belief in the rule of law? Do we really believe that or do we just think it ' s a nice saying to see on the back of a paper dollar? And for me, this is a referendum on the American experiment. On whether we really believe, and... the future is in our hands. And this is not like other crises, I ' ve been the head of the ACLU for almost 20 years, I feel like I ' ve seen it all. This is different. And this is different because it is cumulative, like Phil and Rashad and Dr. King told us, this is centuries of **systemic discrimination**, and the bill has come due. And it will continue to be due, and we will pay. Unless we really do something quite different. I have been scratching my head at the ACLU for the last week. We ' ve been at this for 100 years. My organization has been working on this from its inception. In 1931, we were involved with this report about lawlessness in law **enforcement**. That was our first report that we got behind in 1931. We opened up our first door fronts after the riots in Watts, so that we can bring legal services and lawyers to the communities so they could demand justice from the police departments. You know, we brought Miranda, you know, the right to remain silent, and we brought Gideon, the right to a court-appointed attorney if you can ’ t afford one. We fought Bloomberg on”stop-and-frisk,”it took him years and he lost in front of our litigation to finally **apologize**. We ’ ve been at this for 100 years. And for the communities that have lived this for 400 years, God. I ’ ve been scratching my head, thinking. It ain ’ t working. We don ’ t need another pattern and practice lawsuit. We don ’ t need another training program on racial bias or implicit bias in police departments, we don ’ t need to file another lawsuit on qualified immunity, we don ’ t need to, kind of, bring another race **discrimination** or gender **discrimination** lawsuit to integrate the police department. Yeah, we ’ ve done that and we will continue to do that. For me, where I ’ ve come, is that we need to defund the budgets of these police departments. It ’ s the only way we ’ re going to take the power back. And the more I read over the last couple of weeks about where this country is, the more I ’ m clear that that is my North star at the moment. We will continue to bring the litigation on qualified immunity, we will do the efforts to hold unaccountable law **enforcement** officials accountable, we will bring pattern and practice lawsuits, because the justice department is not doing that, so we will continue to do all that good work. But the real thing is, we ’ re going to go after those budgets. When you look at the fact that we spend 100 million dollars on policing, more than incarceration, that the city of Minneapolis spent 30 percent of their budget on policing. The city of Oakland, 41 percent on policing. That when you have New York City police department spend more money on policing than it does on housing and preservation development, community youth services, **homelessness**. We ’ re going after the money. And that ’ s hard-core advocacy. Bills drop in local legislatures to cut the funding for police, to stop these programs that give the federal military surplus to police departments, so they become, like, little mini armies, these don ’ t look like police officers, these look like standing armies. And the enemy are communities of color. So we need to take away their toys. We need to cut their budgets. We need to shrink the police infrastructure, so that we can get police out of the quotidian lives of people of color and communities of color. The ubiquitousness of police **enforcement** on things that the police do not have a role, should not have a role to play. People should not lose their lives over whether or not a cigarette pack has a proper tax stamp, or whether a 20-dollar bill was forged or not. That ’ s not worthy of spending our dollars on police. Get them out of that business, let ’ s focus on the most important and the most serious of crimes, and that ’ s it. That ’ s it. We ’ re going to depolice our communities. Shrink those budgets. We ’ re going to reinvest those moneys in local communities, it will be like water on stone campaigns, local legislatures, local city counsels, lab report cards, for people who talk out of both sides of their mouths and say,”We believe in police reform,”and yet, they ’ re still going to vote for 30 or 40 percent for the police? We ’ re going to put that right to the public. And I think we just have to stay at it, because I think that ’ s the only way we can get at this in a different way. Because much of what we tried to do is just simply not working. You know, with that, I struggle with, how do you find the optimism in this moment, because you have to find the optimism. You have to find the way to still think that even though on the face of so many setbacks, there ’ s been change. It ’ s been too little, too slow, not enough. We need to kind of, rock it, boost it. But you can ’ t lose sight of the optimism. And you know, I ’ ve been thinking about who are the folks inspiring me, and Dr. King ’ s father, of course, and the words of

Rashad and Patrisse Cullors and others have inspired me. But I find inspiration in the words of a scholar I really don ’ t like bringing up, Sam Huntington, kind of often criticized as being a conservative, a racist. But sometimes you can find inspiration even in your enemy ’ s words. And in one of his books, which I pulled off the shelf I have, he writes about how America is a disappointment, because it failed to live up to its aspirations. And he actually started talking about, America is a failure because it doesn ’ t live up to its ideals. But it ’ s not a failure, it ’ s not a bunch of lies. It ’ s a disappointment. And in the disappointment also is the fact that there ’ s hope. I ’ m paraphrasing it, but I think we have to kind of, wrap all of that together, and think about the disappointment and the hope and the resolve to do better. And we need to listen and lean in, and I thank the TED community, I thank Dr. King, I thank Rashad, I thank Phil. Thank you. CA : Wow. Thank you to all four of you, that was astonishing. I guess we ’ re bringing everyone back now to have a conversation among us, to answer questions from our community, I hope you ’ re entering those questions. So I don ’ t know whether we can bring back our guests onto the screen at this point. Welcome back. Let me start with a question to you, Dr. King, I was so inspired by what you said. Your father, of course, also deeply understood the anger that leads to protest. I think he said that protests are the language of the unheard. And I ’ m wondering what you would say to someone right now who is angered beyond measure by what ’ s happened, and also sees this could be the moment, you know, like, someone who believes the system is so fundamentally broken, that our best choice is to tear it down, that that is actually – This may be a once-in-a-generation moment to do that. And so to actually believe that protest, including **violent** protest, actually is the way right now. What would you say to someone who felt that? BK : First, I just wanted to make just a slight correction, he said riots are the language of the unheard. CA : I **apologize**, I **apologize**. But that is the point even more powerfully, yeah. BK : Yes. Protest we must, and we must continue to always protest, to keep the issues in the awareness before people, but you know, when a person is angry, sometimes it ’ s hard to reach them. I ’ ve been on that journey, I was at a stage of my life where I was so angry, I wanted to destroy, and I ’ m the daughter of Martin Luther King Jr., and grew up in a household of love and nonviolence and forgiveness, and I had to go through that journey I was surrounded by the right kind of influences, fortunately. Because that would have been a sad story. But I think it ’ s really allowing ourselves to hear the anger and allowing the space for the anger, but also trying to help young people rechannel that energy. And we ’ ve got to start ensuring that we connect them to some of the work that has been and now is elevated to another place. Color Of Change, the work that you ’ re doing, the ACLU, the work that they ’ re doing, because sometimes, there ’ s this disconnection that intensifies the emotion and makes you feel **helpless**. But if you can channel that anger, connect it with action that is toward creating the social and economic change, then it begins to build you up, and then you can begin to become more constructive with the anger. WPR : We have some questions that are coming in from our community, but before we do that, you all shared such powerful, meaningful statements right now, and many of you touched on the fact that this is not the first time that we ’ re experiencing this. The murder of George Floyd, Ahmaud Arbery, Breonna Taylor, this is one of – these are three of many, many instances just like this, and I ’ d love to hear you all address was has, sort of, brought us to this boiling point, what has contributed to this moment where we ’ re now experiencing things, Anthony, as you said, that feels so much worse than other moments? And that ’ s it, anyone who feels comfortable to take that question. AR : Rashad, I want to hear you. BK : I wanted to say something. I think we ’ ve always been at that moment.

But this moment is different, because of the void in leadership. There ’ s no real moral voice in our country, and the person who sits in the office of the presidency is not, you know, leading in the right way. And has kind of – no, not kind of, has given license to certain things. And so now it ’ s, you know, he ’ s lit the fires. WPR : Yeah. RR : The thing I ’ ll add... The thing I ’ ll add here is, you know, a couple of things. For the last couple of months, we have been both seeing and experiencing all of the ways that this country ’ s decisions of underinvestment, of targeting black communities, has been killing black people through COVID. And while we ’ ve been in our homes, we have also been watching how the media has blamed us as we have been the essential workers in so many places and trying to ensure that this country keeps going. We ’ ve watched white men with guns show up to capitols demanding, basically, black and brown people go back to work. And then we see this eight-minute video with a police officer, with his knee on someone ’ s neck, after seeing that video of Ahmaud Arbery and hearing the story of Breonna Taylor, and we see him looking in the camera, basically knowing that America was not going to punish him. And what I think it is is that it ’ s just enough is enough, that people didn ’ t feel that they had a channel for that outrage, and because people had been inside, and because people had been experiencing all the ways in which the structures had also been colluding to kill us, that what we ’ re seeing is alignment of all of those things, where people are making demands that are much bigger and much bolder than before. And we recognize that while we don ’ t have leadership at the federal level, we also have to recognize that no political party can say that they have been 100 percent, neither political party can say they ’ ve been 100 percent on the right side of all these issues. And so people are mobilizing, they are fighting back like never before, and in some ways, people are unwilling to accept answers like,”Just go vote,”or,”Just participate in the process,”because we recognize that black people have been voting, black people have been part of voting, and part of ensuring that. And so that I think is why this moment feels so much different, combined with, for the last seven years, since Trayvon, we have seen the growth of a new movement of activists and leaders all around the country, who are also in a very different place to be able to move the needle on so much of what ’ s possible. CA : We have a question here from Genesis Be. If we can get that up here.”Here in Mississippi, the police is synonymous with the Klan, historically. How do we purge law **enforcement** of white supremacists?”PAG : So I guess that ’ s partially to me, being the psychologist of bias. I ’ ll say that just yesterday we had an officer in Denver who posted on social media himself and two other officers saying,”Let ’ s go start a riot.”He was fired that day. I worry about all the officers that the FBI has now, for almost half a decade, been warning us, law **enforcement** and unions being infiltrated by white supremacists. And all the officers that have social media accounts, but they ’ re private. You know, the Invisible Institute has put some things forward. We ’ re not talking seriously about the **domestic** terrorism threat that white supremacy represents. So the first thing that we ’ ve got to do is we ’ ve got to take it seriously. We have to actually say out loud, and I can ’ t believe that on a day like today, or a week like this week, I have to say out loud, white supremacy is alive and well and a driving force of American politics. This shouldn ’ t be controversial. I shouldn ’ t be looking forward to getting hate mail in my inbox for it, but that ’ s the reality. So the first part of solving a problem is acknowledging that it exists. But the second thing is we need to arm municipalities, that ’ s law **enforcement**, but even more so communities, with the ability to take action when someone violates their values. Right now, I think about the case in Philadelphia where Charles Ramsey, when he was a commissioner there, fired six officers, right.

Concerns about racial bias and concerns about police brutality, and those six officers were back on the same job inside of three months. We now have a law **enforcement** system that says you can lose your job in one jurisdiction, and get the same job as law **enforcement** in another jurisdiction. And without the national registry and the capacity for law **enforcement** to make different decisions, we ' re going to have this exact problem, not just in Mississippi, but in Minneapolis, and Louisville and New York and LA. CA : Phil, how much of the problem stems from the fact that police unions have a huge amount of power to protect and sometimes reinstate so called bad apple officers? PAG : Yeah, I ' m getting this question a lot, and police unions are one of the most labor forces of the United States, and are unique within the labor movement, right? So it ' s police unions and teachers ' unions are the two largest and could not be two different groups of folks. When I talk to union leadership, that ' s the leadership what wants to talk to Dr. Blackenstein, right, when I talk to union leadership, what they say is no one hates a bad officer more than a good officer. But the union contracts, the new negotiations, don ' t look like that ' s true. What they look like is anybody gets in trouble, and the union ' s only job is to make sure whatever officer is in trouble, gets to maintain their job. The perverse incentive here is that when people run for union leadership, no one can run saying,"These people shouldn ' t be in the union."It ' s very hard to do that. What you can run on is say,"If this person didn ' t protect you enough, I ' ll protect you even more. The bigots? I ' ll protect even them."So we have this perverse incentive where union leadership ends up not really representing the values even of the rest of the union members. But they have massive, outsized negotiating power. So yes, engaging with and appropriate rightsizing of labor protections, for folks whose jobs are difficult, but who should not be protected from the basic values of human rights, human dignity and public **safety**. It ' s got to be part of the process. I mean, when unions are negotiating a two-year ban on keeping of records, so that there ' s no ability even to trace what ' s happening in the state of California, historically in terms of police misconduct, that ' s not in the interest of public **safety**, public legitimacy, or our democracy. AR : Yeah, the thing I would add, Chris, is that I think the labor union piece is a critically important one to think through. Because I think, like Phil said, they are a key part of the puzzle that we have to solve for. And you know, it ' s frustrating when you look at a place in Minneapolis, and Phil knows better than I, but when mayor Jacob Frey, the one who ' s on TV all the time, saying many of the right things that you want an elected official to say at times like this, when he banned his police department from attending the"warrior training"that was being offered, it was the Minneapolis Police Federation, local union that defied him and sent their police to the training. And so we need to really be clear that we need to have the police forces under civilian control. I know this sounds so elementary, I feel like I ' m talking about a Latin American, kind of, totalitarian context, but we need to exert civilian control of our police in a way that we have yet not been able to think through, and a key part of that is the labor unions of the police. And there are moments when you can find common ground. When we brought one of our COVID-related lawsuits to deal with the outbreak of the pandemic in a Maryland jail, we worked really hard, I worked the phones with the head of police unions. We got one of the local unions to serve as plaintiff in our lawsuit. Because we understood that the incarcerated folks who were being denied **access** to masks, social distancing and the conditions and lack of testing, and lack of PP, that the people who were also going to be in **harm** ' s way were going to be the guards as well. And they were going to be the vectors, communicating the disease out into the community. So if you can find ways of bringing that relationship. But make no mistake, when you go after their budgets, and you start taking away kind of, their munitions, and their seat at the budgeting table, oh, are you going to have a battle on your hands, right? And we have to think about also as we shrink the budgets for police, how do we – we deploy people in the police departments to other meaningful jobs, right? Because you can ’ t just throw them out into the **street**, and say,”You ’ re on your own, you ’ re **homeless**, good luck to you.”That ’ s not a way to deal with redemption. So we have to really think about all these pieces in a much more cohesive way. WPR : We have another question here from the audience. From Paul Rucker :”The end of summer of 1919 was followed by the Tulsa Race Massacre, the Johnson-Reed Anti-Immigration Act of 1924, and also the rise of the KKK. Is there a possibility that white supremacy will get stronger if we don ’ t seize this opportunity?”Rashad, I think this might be something that would be great to hear your perspective on, working so deeply in activism. RR : I ’ m having a little trouble hearing. WPR : Oh, I ’ m so sorry. RR : No, it ’ s OK. CA : Can you read the question on the screen, Rashad? RR : Oh, I heard that, I heard you. WPR : Yes, I think it might just be my mic is having some issues here.”Is there a possibility that white supremacy will get stronger if we don ’ t seize this opportunity.”Yes, absolutely yes. You know, to be clear, right, if we don ’ t have the right diagnosis of white supremacy, if we think of white supremacy as just hoods, if we think of white supremacy as just folks who are operating, you know, with, you know – in some of these underground networks that have grown, if we ’ re just thinking about white supremacy and white nationalism as people who marched with tiki torches in Charlottesville, then we will really mistake all the ways in which our systems and structures have white supremacy embedded, and allow for something like a Tulsa Race Massacre to happen, something like anti-immigration to happen, but on a day-to-day basis allow for the targeting of black communities through predatory practices by banks. The targeting of black communities through predatory practices like **bail**. A whole set of systems that can be produced day in and day out. We live in a country where the rules are far too often designed in ways that create a caste system, that create a different standard for some over others, and so when I talked about the inflection point, right, of this moment where something could really go forward and something could turn backwards, we are seeing this right now with this current president. And as we look at what could be happening with the next election, we have to be very, very clear that Donald Trump doesn ’ t just operate on his own. He ’ s enabled by big corporations who benefit from him being in office, and so continue to turn a blind eye to all the things that he does. They may post”Black Lives Matter,”but they show up to the White House and engage with Donald Trump. And then we have a whole set of politicians that may sometimes say that he said something that was wrong, but then allow for – but support his platform in other ways. You know, true co-conspiracy in the effort to dismantle white supremacy and white nationalism is not a thing that people can do on vacation. It is a 365-day project of us constantly working to dismantle all of the structures that have been put in **harm** ’ s way. The final thing I will just add, because someone mentioned about police unions, and I want to just add that one of the problems with police unions, and many of us have been in this position, I think, is that I have shown up to the table with police unions on many occasions. I remember going to the White House during the last administration and being around a table as we were talking about policing and police reform. And having members of the Fraternal Order of Police leadership say things like,”All of this talk of racial profiling is new to us.”It is one thing for folks to not agree with you on the policy reforms necessary. It is another thing for people to say that our demands are too aspirational. It ’ s another thing to be gaslit and told the problem doesn ’ t actually exist at all.

And that is what we are dealing with, and so we have to actually change the way that people see these institutions. Politicians who say that they are on the side of justice and reform, can no longer take money from police-only unions and Fraternal Order of Police. We actually have to create a new standard, a new litmus test of what does it mean to actually be with us. You can ’ t just **sing** our songs, use our hashtags and march in our marches, if you are on the other end of supporting the structures that put us in **harm** ’ s way, that literally kill us. And this is the opportunity for white allies to actually stand up in new ways. To be the type of ally, to do the type of allyship and the type of work that truly dismantles structures, not just provides charity. PAG : And I ’ ve got to add to that – so, Paul, thank you for the question, but we ’ re in a moment where people are looking at what ’ s happening on the **street**, as if a week and a half ago we weren ’ t in a midst of a global pandemic as the greatest news story, the biggest new story going on. One of the things I ’ m most worried about, and have been worried about since the beginning, what I ’ ve been talking to our chiefs about, is say, you must be out of the social distancing policing game. You can ’ t be the ones doing that, and the reason is this : We ’ re in a moment where creating scapegoats and enemies and others is incredibly politically advantageous for at least one side. And there is deliberate efforts to do exactly that. And we ’ ve seen that black communities are two and three and four times more likely to contract this virus, which feels like the manifestation of racial **discrimination**, because it is. But very soon, that ’ s going to look like black people made bad choices and they need to stay away from us. And when that happens, that ’ s when law **enforcements** get used to regulate where black movement can be. We used to call it sundown towns, I don ’ t know what we ’ re going to call it when it ’ s around COVID. But it ’ s coming. I ’ m already seeing that on communities like Nextdoor, and on Facebook groups. People who don ’ t think of themselves as white supremacists but just want the disease away, and the disease has a black and brown face. So we ’ re not only dealing with a moment of generational tension, between black communities and law **enforcement**, we ’ re dealing with a moment when people are looking for scapegoats, and black people ’ s vulnerability has always been our greatest casting note for being cast as scapegoats. So for folks who are worried about this, this is not inevitably a moment for change and reform and enlightenment and America ’ s best values, because historically, these have been precisely the moments when regression back to white supremacy has reigned supreme. So let ’ s not just look at everybody signaling. I don ’ t want to just see black and white cops on their knees, I want to see the policies. I want to see the things that will prevent this kind of thing from moving to the next stage. CA : Rashad, I want to respect the fact that you ’ ve got a hard stop at one. And so I just want to thank you for your participation in this. If you ’ ve got a couple of final words you want to share, that would be great, and then if it ’ s OK for the other three, I think there ’ s just a couple other questions I ’ d love to put and continue this conversation for just a moment longer, if possible. Rashad, any closing words? RR : The thing I want to say is that now is the time for action. And I want to invite people in to join us at Color Of Change to make justice real. And in so many ways, you can visit us at Color Of Change, you can take action. Five, 10, 15 years from now, we will be dealing with the impacts of what we did or didn ’ t do in this moment. How we stood up and how hard we were willing to fight. And as the other speakers have said, now is not the time for reform around the edges, now is time for dismantling the policies and practices that have held us back, and championing solutions and new rules that will move us forward. And so we hope that you will do something, whether it ’ s with us, or whether it ’ s with local organizations in your community, or other groups around the country. But this is an opportunity to make change, and I believe that we can make justice real, if we find the **passion** and the energy to work together to achieve it. So thank you all for having me, and I hope that we have an opportunity to build, not just online, but offline, in the months to come. CA : Thanks so much, Rashad. We ’ re just going to ask this last question of you. This one is from David Fenton.”How can the movement unite around a clear, simple platform of policies to enshrine in legislation? Like making all complaints against cops public, banning all choke holds, ensuring independent review boards, etc.?”WPR : That seems like a great place for you to chime in, Dr. King, if you have some thoughts on that. PAG : I think that was to go to you, Dr. King. BK : Oh, OK. You know, this may sound simplistic, but it ’ s a Nike thing, I think we have to just do it, we have to see our work as interconnected. I think there ’ s been efforts towards people working in that vein, but we have to intensify that. And in doing it, one of the things that my father said, and I know people sometimes get tired of hearing me say,”My father said,”but I just think, I wish, should I say, we had really listened to him, because we wouldn ’ t be on this platform right now having this type of conversation. But he left something with us, sort of a blueprint in”Where Do We Go From Here : Chaos or Community?”, his book, and he said, going forward, the meddlesome task is to organize our strength into compelling power. And that is so key, because oftentimes we organize merely around **passion**. But people have certain areas of strength and talent and giftedness, and we ’ ve got to figure out how to build our coalitions based on these strengths. You know, people do different things well. And so, in order to unite in an effective way that they might not elude the demand that we ’ re making, I think that ’ s what ’ s going to happen. People have to do their own personal assessment within their organization, I call it a SWAT analysis. And then that SWAT analysis has to happen across organizations, so that we can make sure that we are moving in a united manner, off of the strengths that each organization brings, so that we can maximize the impact and the effectiveness to do things like this, in terms of getting the legislation in place that is needed in this hour. CA : Thanks so much. Just quick closing words from you, Anthony, and then from you, Phil. Anthony. AR : You know, I would just say that what gives me hope are the young folk. You have to believe that among this group, this groups of young ’ uns, seeing what they ’ re seeing, living with this president, with these instincts, seeing the continued indifference that mainstream communities have given to issues of racial justice, or economic justice, you ’ ve got to believe that what comes out of this very hot fire, is something even more powerful and strong than we ’ ve ever seen before. That ’ s what gets me through the hard days that we ’ re now experiencing, this thinking, there is another Dr. King among the young ’ uns, Dr. King. And I have to believe that what they ’ re seeing and what they ’ re witnessing and their righteous indignation and their frustration and their anger is going to be miraculously a beautiful blossoming of a new opportunity, of a new change. This generation will take us there, I have to believe that. My generation has failed them miserably. So I ’ m just looking forward to the new ones. CA : Thank you, Anthony. WPR : Phil. Thank you, Anthony. PAG : So it really has been a privilege to be on with you all. To David ’ s question, let me say that a number of civil rights organizations, I believe the ACLU among them, CPE, Center for Policing Equity, and hundreds more have signed on to principles for legislation that would include eight pillars. It ’ s been led by the Leadership Conference for Civil and Human Rights. And it includes a federal ban on choke holds, it includes a national registry for officers who have engaged in misconduct. I also think that it ’ s important at this moment to get, we ’ ve got law **enforcement** ’ s chiefs of major cities willing to say, if we emerge from this moment and our profession hasn ’ t changed, then we have failed again. So it ’ s a critical time to get behind, I would direct you to LCCHR ’ s website for the eight pillars, because I won ’ t remember them all right now, and to start calling your local law **enforcement**, and say,”Yes, own that.”You should be signing on, they should be going public with letters that do all of that. But I ’ ll also say this. For a path forward in the principles, I ’ ll end where I started, which is that this is bigger than policing. These are the unpaid debts owed to black communities for stolen labor, owed to native communities for stolen land, for stolen culture, for years taken away and for lives lost in it. This is bigger than policing. If we don ’ t understand the size of it, then there ’ s no solution that ’ s really, truly proportional to the moment. But in this moment, when we ’ re seeing trillion of dollars in bailouts, mostly for corporations, it is absolutely a time when we can do things that normally, people could pretend that ’ s too much, it ’ s too big, we can ’ t. We have literally all the money in the world that can be spent and directed towards making us the society we pretend to be, before moments like this happen. And so the thing that gives me hope is that the lies have to be obvious now. The lies have to be, that was a reasonable use of force. The lie has to be, we don ’ t have the money. The lie has to be, that ’ s too hard, it ’ s too big of a challenge. This stuff feels impossible every day except today, because the alternative is we lose everything. Everything is at stake, our democracy is at stake, the people we choose to be, we claim to be, that ’ s at stake. And in the face of that, I think we can do impossible things. I think we can be mighty. So my hope for all of us is first, that we wake up tomorrow with more peace in the evening than war, and that we hold on to what ’ s possible from this moment at the same time that we hold on to the size of the task in front of us. I don ’ t want to come with half measures out of this. I don ’ t want to come out with radicalized youth and indifferent aged... I don ’ t know what the contrast... The radicalized youth and indifferent people who are old like me. I want to come out with a unified country that understands that the costs that we owe are big, and our pockets are deep enough to match it. CA : Wow. Thank you to each of you for extraordinary eloquence. Really, so powerful. This conversation, obviously, continues, I know that there ’ s many people listening, you have other questions, this, I think, from TED ’ s point of view is just the start of the conversation. To the extent that our job is to amplify the voices that matter, we couldn ’ t be prouder to be amplifying further your extraordinary voices. So thank you for being part of this today. PAG : Thank you. WPR : Thank you all.

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Cloe Shasha : So welcome, Ibram, and thank you so much for joining us. Ibram X. Kendi : Well, thank you, Cloe, and Whitney, and thank you everyone for joining this conversation. And so, a few weeks ago, on the same day we learned about the brutal murder of George Floyd, we also learned that a white woman in Central Park who chose not to leash her dog and was told by a black man nearby that she needed to leash her dog, instead decided to threaten this black male, instead decided to call the police and claim that her life was being threatened. And of course, when we learned about that through a video, many Americans were outraged, and this woman, Amy Cooper, ended up going on national TV and saying, like countless other Americans have said right after they engaged in a racist act,”I am not racist.”And I say countless Americans, because when you really think about the history of Americans expressing racist ideas, supporting racist policies, you ’ re really talking about a history of people who have claimed they ’ re not racist, because everyone claims that they ’ re not racist, whether we ' re talking about the Amy Coopers of the world, whether we ' re talking about Donald Trump, who, right after he said that majority-black Baltimore is a rat and rodent-infested mess that no human being would want to live in, and he was challenged as being racist, he said,"Actually, I ' m the least racist person anywhere in the world."And so really the heartbeat of racism itself has always been denial, and the sound of that heartbeat has always been,"I ' m not racist."And so what I ' m trying to do with my work is to really get Americans to eliminate the concept of"not racist"from their vocabulary, and realize we ' re either being racist or anti-racist. We ' re either expressing ideas that suggest certain racial groups are better or worse than others, superior or inferior than others. We ' re either being racist, or we ' re being anti-racist. We ' re expressing notions that the racial groups are **equals**, despite any cultural or even ethnic differences. We ' re either supporting policies that are leading to racial **inequities** and injustice, like we saw in Louisville, where Breonna Taylor was murdered, or we ' re supporting policies and pushing policies that are leading to justice and **equity** for all. And so I think we should be very clear about whether we ' re expressing racist ideas, about whether we ' re supporting racist policies, and admit when we are, because to be anti-racist is to admit when we expressed a racist idea, is to say,"You know what? When I was doing that in Central Park, I was indeed being racist. But I ' m going to change. I ' m going to strive to be anti-racist."And to be racist is to constantly deny the racial **inequities** that pervade American society, to constantly deny the racist ideas that pervade American minds. And so I want to built a just and **equitable** society, and the only way we ' re going to even begin that process is if we admit our racism and start building an anti-racist world. Thank you. CS : Thank you so much for that. You know, your book,"How to Be an Antiracist,"has become a bestseller in light of what ' s been happening, and you ' ve been speaking a bit to the ways in which anti-racism and racism are the only two polar opposite ways to hold a view on racism. I ' m curious if you could talk a little bit more about what the basic tenets of anti-racism are, for people who aren ' t as familiar with it in terms of how they can be anti-racist. IXK : Sure. And so I mentioned in my talk that the heartbeat of racism is denial, and really the heartbeat of anti-racism is **confession**, is the recognition that to grow up in this society is to literally at some point in our lives probably internalize ideas that are racist, ideas that suggest certain racial groups are better or worse than others, and because we believe in racial hierarchy, because Americans have been systematically taught that black people are more dangerous, that black people are more criminal-like, when we live in a society where black people are 40 percent of the national incarcerated population, that ' s going to seem normal to people. When we live in a society in a city like Minneapolis where black people are 20 percent of the population but more than 60 percent of the people being subjected to police shootings, it ' s going to seem normal. And so to be anti-racist is to believe that there ' s nothing wrong or inferior about black people or any other racial group. There ' s nothing dangerous about black people or any other racial group. And so when we see these racial disparities all around us, we see them as abnormal, and then we start to figure out, OK, what policies are behind so many black people being killed by police? What policies are behind so many Latinx people being disproportionately infected with COVID? How can I be a part of the struggle to upend those policies and replace them with more antiracist policies? Whitney Pennington Rodgers : And so it sounds like you do make that distinction, then, between not racist and anti-racist. I guess, could you talk a little bit more about that and break that down? What is the difference between the two? IXK : In the most simplest way, a not racist is a racist who is in denial, and an anti-racist is someone who is willing to admit the times in which they are being racist, and who is willing to recognize the **inequities** and the racial problems of our society, and who is willing to challenge those racial **inequities** by challenging policy. And so I ' m saying this because literally slaveholders, slave traders, imagined that their ideas in our terms were not racist. They would say things like,"Black people are the cursed descendants of Ham, and they ' re cursed forever into enslavement."This isn ' t,"I ' m not racist."This is,"God ' s law."They would say things, like, you know,"Based on science, based on ethnology, based on natural history, black people by nature are predisposed to slavery and servility. This is nature ' s law. I ' m not racist. I ' m actually doing what nature said I ' m supposed to be doing."And so this construct of being not racist and denying one ' s racism goes all the way back to the origins of this country. CS : Yeah. And why do you think it has been so hard for some people now to still accept that neutrality is not enough when it comes to racism? IXK : I think because it takes a lot of work to be anti-racist. You have to be very vulnerable, right? You have to be willing to admit that you were wrong. You have to be willing to admit that if you have more, if you ' re white, for instance, and you have more, it may not be because you are more. You have to admit that, yeah, you ' ve worked hard potentially, in your life, but you ' ve also had certain advantages which provided you with opportunities that other people did not have. You have to admit those things, and it ' s very difficult for people to be publicly, and even privately, self-critical. I think it ' s also the case of, and I should have probably led with this, how people define"racist."And so people tend to define"racist"as, like, a fixed category, as an identity. This is essential to who a person is. Someone becomes a racist. And so therefore – And then they also connect a racist with a bad, evil person. They connect a racist with a Ku Klux Klansman or woman. And they ' re like,"I ' m not in the Ku Klux Klan, I ' m not a bad person and I ' ve done good things in my life. I ' ve done good things to people of color. And so therefore I can ' t be racist. I ' m not that. That ' s not my identity. But that ' s actually not how we should be defining racist. Racist is a descriptive term. It describes what a person is saying or doing in any given moment, and so when a person in one moment is expressing a racist idea, in that moment they are being racist when they ' re saying black people are lazy. If in the very next moment they ' re appreciating the cultures of native people, they ' re being anti-racist. WPR : And we ' re going to get to some questions from our community in a moment, but I think when a lot of people hear this idea that you ' re putting forward, this idea of anti-racism, there ' s this feeling that this is something that only concerns the white community. And so could you speak a little bit to how the black community and nonwhite, other ethnic minorities can participate in and think about this idea of anti-racism? IXK : Sure. So if white Americans commonly say,"I ' m not racist,"people of color commonly say,"I can ' t be racist, because I ' m a person of color."And then some people of color say they can ' t be racist because they have no power. And so, first and foremost, what I ' ve tried to do in my work is to push back against this idea that people of color have no power. There ' s nothing more disempowering to say, or to think, as a person of color, than to say you have no power. People of color have long utilized the most basic power that every human being has, and that ' s the power to resist policy – that ' s the power to resist racist policies, that ' s the power to resist a racist society. But if you ' re a person of color, and you believe that people coming here from Honduras and El Salvador are invading this country, you believe that these Latinx **immigrants** are animals and rapists, then you ' re certainly not, if you ' re black or Asian or native, going to be a part of the struggle to defend Latinx **immigrants**, to recognize that Latinx **immigrants** have as much to give to this country as any other group of people, you ' re going to view these people as”taking away your jobs,”and so therefore you ’ re going to support racist rhetoric, you ’ re going to support racist policies, and even though that is probably going to be **harming** you, in other words, it ’ s going to be **harming**, if you ’ re black, **immigrants** coming from Haiti and Nigeria, if you ’ re Asian, **immigrants** coming from India. So I think it ’ s critically important for even people of color to realize they have the power to resist, and when people of color view other people of color as the problem, they ’ re not going to view racism as the problem. And anyone who is not viewing racism as the problem is not being anti-racist. CS : You touched on this a bit in your beginning talk here, but you ’ ve talked about how racism is the reason that black communities and communities of color are systematically disadvantaged in America, which has led to so many more deaths from COVID-19 in those communities. And yet the media is often placing the blame on people of color for their vulnerability to illness. So I ’ m curious, in line with that, what is the relationship between anti-racism and the potential for **systemic** change? IXK : I think it ’ s a direct relationship, because when you are – when you believe and have consumed racist ideas, you ’ re not going to even believe change is necessary because you ’ re going to believe that racial inequality is normal. Or, you ’ re not going to believe change is possible. In other words, you ’ re going to believe that the reason why black people are being killed by police at such high rates or the reason why Latinx people are being infected at such high rates is because there ’ s something wrong with them, and nothing can be changed. And so you wouldn ’ t even begin to even see the need for **systemic** structural change, let alone be a part of the struggle for **systemic** structural change. And so, to be anti-racist, again, is to recognize that there ’ s only two causes of racial **inequity** : either there ’ s something wrong with people, or there ’ s something wrong with power and policy. And if you realize that there ’ s nothing wrong with any group of people, and I keep mentioning groups – I ’ m not saying individuals. There ’ s certainly black individuals who didn ’ t take coronavirus seriously, which is one of the reasons why they were infected. But there are white people who didn ’ t take coronavirus seriously. No one has ever proven, actually studies have shown that black people were more likely to take the coronavirus seriously than white people. We ’ re not talking about individuals here, and we certainly should not be individualizing groups. We certainly should not be looking at the individual behavior of one Latinx person or one black person, and saying they ’ re representatives of the group. That ’ s a racist idea in and of itself. And so I ’ m talking about groups, and if you believe that groups are **equals**, then the only other alternative, the only other explanation to persisting **inequity** and injustice, is power and policy. And to then spend your time transforming and challenging power and policy is to spend your time being anti-racist. WPR : So we have some questions that are coming in from the audience. First one here is from a community member that asks,”When we talk about white privilege, we talk also about the privilege not to have the difficult conversations. Do you feel that ’ s starting to change?”IXK : I hope so, because I think that white Americans, too, need to simultaneously recognize their privileges, the privileges that they have accrued as a result of their whiteness, and the only way in which they ’ re going to be able to do that is by initiating and having these conversations. But then they also should recognize that, yes, they have more, white Americans have more, due to racist policy, but the question I think white Americans should be having, particularly when they ’ re having these conversations among themselves, is, if we had a more **equitable** society, would we have more? Because what I ’ m asking is that, you know, white Americans have more because of racism, but there are other groups of people in other Western democracies who have more than white Americans, and then you start to ask the question, why is it that people in other countries have free health care? Why is it that they have paid family leave? Why is it that they have a massive **safety** net? Why is it that we do not? And one of the major answers to why we do not here have is racism. One of the major answers as to why Donald Trump is President of the United States is racism. And so I ’ m not really asking white Americans to be altruistic in order to be anti-racist. We ’ re really asking people to have intelligent self-interest. Those four million, I should say five million poor whites in 1860 whose poverty was the direct result of the riches of a few thousand white slaveholding families, in order to challenge slavery, we weren ’ t saying, you know, we need you to be altruistic. No, we actually need you to do what ’ s in your self-interest. Those tens of millions of Americans, white Americans, who have lost their jobs as a result of this pandemic, we ’ re not asking them to be altruistic. We ’ re asking them to realize that if we had a different type of government with a different set of priorities, then they would be much better off right now. I ’ m sorry, don ’ t get me started. CS : No, we ’ re grateful to you. Thank you. And in line with that, obviously these protests and this movement have led to some progress : the removal of Confederate monuments, the Minneapolis City Council pledging to dismantle the police department, etc. But what do you view as the greatest priority on a policy level as this fight for justice continues? Are there any ways in which we could learn from other countries? IXK : I don ’ t actually think necessarily there ’ s a singular policy priority. I mean, if someone was to force me to answer, I would probably say two, and that is, high quality free health care for all, and when I say high quality, I ’ m not just talking about Medicare For All, I ’ m talking about a simultaneous scenario in which in rural southwest Georgia, where the people are predominantly black and have some of the highest death rates in the country, those counties in southwest Georgia, from COVID, that they would have **access** to health care as high quality as people do in Atlanta and New York City, and then, simultaneously, that that health care would be free. So many Americans not only of course are dying this year of COVID but also of heart disease and cancer, which are the number one killers before COVID of Americans, and they ’ re disproportionately black. And so I would say that, and then secondarily, I would say reparations. And many Americans claim that they believe in racial **equality**, they want to bring about racial **equality**. Many Americans recognize just how critical economic livelihood is for every person in this country, in this economic system. But then many Americans reject or are not supportive of reparations. And so we have a situation in which white Americans are, last I checked, their median wealth is 10 times the median wealth of black Americans, and according to a recent study, by 2053 – between now, I should say, and 2053, white median wealth is projected to grow, and this was before this current recession, and black median wealth is expected to redline at zero dollars, and that, based on this current recession, that may be pushed up a decade. And so we not only have a racial wealth gap, but we have a racial wealth gap that ’ s growing. And so for those Americans who claim they are committed to racial **equality** who also recognize the importance of economic livelihood and who also know that wealth is inherited, and the majority of wealth is inherited, and when you think of the inheritance, you ’ re thinking of past, and the past policies that many Americans consider to be racist, whether it ’ s slavery or even redlining, how would we even begin to close this growing racial wealth gap without a massive program like reparations? WPR : Well, sort of connected to this idea of thinking about wealth disparity and wealth inequality in this country, we have a question from community member Dana Perls. She asks,”How do you suggest liberal white organizations effectively address problems of racism within the work environment, particularly in environments where people remain silent in the face of racism or make token statements without looking internally?"IXK : Sure. And so I would make a few suggestions. One, for several decades now, every workplace has publicly pledged a commitment to diversity. Typically, they have diversity statements. I would basically rip up those diversity statements and write a new statement, and that ’ s a statement committed to anti-racism. And in that statement you would clearly define what a racist idea is, what an anti-racist idea is, what a racist policy is and what an anti-racist policy is. And you would state as a workplace that you ’ re committed to having a culture of anti-racist ideas and having an institution made up of anti-racist policies. And so then everybody can measure everyone ’ s ideas and the policies of that workplace based on that document. And I think that that could begin the process of transformation. I also think it ’ s critically important for workplaces to not only diversify their staff but diversify their upper administration. And I think that ’ s absolutely critical as well. CS : We have some more questions coming in from the audience. We have one from Melissa Mahoney, who is asking,"Donald Trump seems to be making supporting Black Lives Matter a partisan issue, for example making fun of Mitt Romney for participating in a peaceful protest. How do we uncouple this to make it nonpartisan?"IXK : Well, I mean, I think that to say the lives of black people is a Democratic declaration is simultaneously stating that Republicans do not value black life. If that ’ s essentially what Donald Trump is saying, if he ’ s stating that there ’ s a problem with marching for black lives, then what is the solution? The solution is not marching. What ’ s the other alternative? The other alternative is not marching for black lives. The other alternative is not caring when black people die of police **violence** or COVID. And so to me, the way in which we make this a nonpartisan issue is to strike back or argue back in that way, and obviously Republicans are going to claim they ’ re not saying that, but it ’ s a very simple thing : either you believe black lives matter or you don ’ t, and if you believe black lives matter because you believe in human rights, then you believe in the human right for black people and all people to live and to not have to fear police **violence** and not have to fear the state and not have to fear that a peaceful protest is going to be broken up because some politician wants to get a campaign op, then you ’ re going to institute policy that shows it. Or, you ’ re not. WPR : So I want to ask a question just about how people can think about anti-racism and how they can actually bring this into their lives. I imagine that a lot of folks, they hear this and they ’ re like, oh, you know, I have to be really thoughtful about how my actions and my words are perceived. What is the perceived intention behind what it is that I ’ m saying, and that that may feel exhausting, and I think that connects even to this idea of policy. And so I ’ m curious. There is a huge element of thoughtfulness that comes along with this work of being anti-racist. And what is your reaction and response to those who feel concerned about the mental exhaustion from having to constantly think about how your actions may hurt or **harm** others? IXK : So I think part of the concern that people have about mental exhaustion is this idea that they don ’ t ever want to make a mistake, and I think to be anti-racist is to make mistakes, and is to recognize when we make a mistake. For us, what ’ s critical is to have those very clear definitions so that we can assess our words, we can assess our deeds, and when we make a mistake, we just own up to it and say,"You know what, that was a racist idea.""You know what, I was supporting a racist policy, but I ’ m going to change."The other thing I think is important for us to realize is in many ways we are addicted, and when I say we, individuals and certainly this country, is addicted to racism, and that ’ s one of the reasons why for so many people they ’ re just in denial. People usually deny their addictions. But then, once we realize that we have this addiction, everyone who has been addicted, you know, you talk to friends and family members who are overcoming an addiction to substance abuse, they ’ re not going to say that they ’ re just **healed**, that they don ’ t have to think about this regularly. You know, someone who is overcoming alcoholism is going to say,”You know what, this is a day-by-day process, and I take it day by day and moment by moment, and yes, it ’ s difficult to restrain myself from reverting back to what I ’ m addicted to, but at the same time it ’ s liberating, it ’ s freeing, because I ’ m no longer having to wallow in that addiction. And so I think, and I ’ m no longer having to hurt people due to my addiction.”And I think that ’ s critical. We spend too much time thinking about how we feel and less time thinking about how our actions and ideas make others feel. And I think that ’ s one thing that the George Floyd video forced Americans to do was to really see and hear, especially, how someone feels as a result of their racism. CS : We have another question from the audience. This one is asking about,”Can you speak to the intersectionality between the work of anti-racism, feminism and **gay** rights? How does the work of anti-racism relate and affect the work of these other human rights issues?”IXK : Sure. So I define a racist idea as any idea that suggests a racial group is superior or inferior to another racial group in any way. And I use the term racial group as opposed to race because every race is a collection of racialized intersectional groups, and so you have black women and black men and you have black heterosexuals and black queer people, just as you have Latinx women and white women and Asian men, and what ’ s critical for us to understand is there hasn ’ t just been racist ideas that have targeted, let ’ s say, black people. There has been racist ideas that have been developed and have targeted black women, that have targeted black lesbians, that have targeted black transgender women. And oftentimes these racist ideas targeting these intersectional groups are intersecting with other forms of bigotry that is also targeting these groups. To give an example about black women, one of the oldest racist ideas about black women was this idea that they ’ re inferior women or that they ’ re not even women at all, and that they ’ re inferior to white women, who are the pinnacle of womanhood. And that idea has intersected with this sexist idea that suggests that women are weak, that the more weak a person is, a woman is, the more woman she is, and the stronger a woman is, the more masculine she is. These two ideas have intersected to constantly degrade black women as this idea of the strong, black masculine woman who is inferior to the weak, white woman. And so the only way to really understand these constructs of a weak, superfeminine white woman and a strong, hypermasculine black woman is to understand sexist ideas, is to reject sexist ideas, and I ’ ll say very quickly, the same goes for the intersection of racism and homophobia, in which black queer people have been subjected to this idea that they are more hypersexual because there ’ s this idea of queer people as being more hypersexual than heterosexuals. And so black queer people have been tagged as more hypersexual than white queer people and black heterosexuals. And you can ’ t really see that and understand that and reject that if you ’ re not rejecting and understanding and challenging homophobia too. WPR : And to this point of challenging, we have another question from Maryam Mohit in our community, who asks,”How do you see **cancel** culture and anti-racism interacting. For example, when someone did something obviously racist in the past and it comes to light?”How do we respond to that? IXK : Wow. So I think it ’ s very, very complex. I do obviously encourage people to transform themselves, to change, to admit those times in which they were being racist, and so obviously we as a community have to give people that ability to do that. We can ’ t, when someone admits that they were being racist, we can ’ t immediately obviously **cancel** them. But I also think that there are people who do something so egregious and there are people who are so unwilling to recognize how egregious what they just did is, so in a particular moment, so not just the horrible, vicious act, but then on top of that the refusal to even admit the horrible, vicious act. In that case, I could see how people would literally want to **cancel** them, and I think that we have to, on the other hand, we have to have some sort of consequence, public consequence, cultural consequence, for people acting in a racist manner, especially in an extremely egregious way. And for many people, they ' ve decided, you know what, I ' m just going to **cancel** folks. And I ' m not going to necessarily critique them, but I do think we should try to figure out a way to discern those who are refusing to transform themselves and those who made a mistake and recognized it and truly are committed to transforming themselves. CS : Yeah, I mean, one of the concerns many activists have been expressing is that the energy behind the Black Lives Matter movement has to stay high for anti-racist change to truly take place. I think that applies to what you just said as well. And I guess I ' m curious what your opinion is on when the protests start to wane and people ' s donation-matching campaigns fade into the background, how can we all ensure that this conversation about anti-racism stays central? IXK : Sure. So in"How to Be an Antiracist,"in one of the final chapters, is this chapter called"Failure."I talked about what I call feelings advocacy, and this is people feeling bad about what ' s happening, what happened to George Floyd or what happened to Ahmaud Arbery or what happened to Breonna Taylor. They just feel bad about this country and where this country is headed. And so the way they go about feeling better is by coming to a demonstration. The way they go about feeling better is by donating to a particular organization. The way they go about feeling better is reading a book. And so if this is what many Americans are doing, then once they feel better, in other words once the individual feels better through their participation in book clubs or demonstrations or donation campaigns, then nothing is going to change except, what, their own feelings. And so we need to move past our feelings. And this isn ' t to say that people shouldn ' t feel bad, but we should use our feelings, how horrible we feel about what is going on, to put into place, put into practice, anti-racist power and policies. In other words, our feelings should be driving us. They shouldn ' t be the end all. This should not be about making us feel better. This should be about transforming this country, and we need to keep our eyes on transforming this country, because if we don ' t, then once people feel better after this is all over, then we ' ll be back to the same situation of being horrified by another video, and then feeling bad, and then the cycle will only continue. WPR : You know, I think when we think about what sort of changes we can implement and how we could make the system work better, make our governments work better, make our police work better, are there models in other countries where – obviously the history in the United States is really unique in terms of thinking about race and oppression. But when you look to other nations and other cultures, are there other models that you look at as examples that we could potentially implement here? IXK : I mean, there are so many. There are countries in which police officers don ' t wear weapons. There are countries who have more people than the United States but less prisoners. There are countries who try to fight **violent** crime not with more police and prisons but with more jobs and more opportunities, because they know and see that the communities with the highest levels of **violent** crime tend to be communities with high levels of poverty and long-term **unemployment**. I think that – And then, obviously, other countries provide pretty sizable social **safety** nets for people such that people are not committing crimes out of poverty, such that people are not committing crimes out of despair. And so I think that it ' s critically important for us to first and foremost think through, OK, if there ' s nothing wrong with the people, then how can we go about reducing police **violence**? How can we go about reducing racial health **inequities**? What policies can we change? What policies have worked? These are the types of questions we need to be asking, because there ' s never really been anything wrong with the people. CS : In your"Atlantic"piece called"Who Gets To Be Afraid in America,"you wrote,"What I am, a black male, should not matter. Who I am should matter."And I feel that ' s kind of what you ' re saying, that in other places maybe that ' s more possible, and I ' m curious when you imagine a country in which who you are mattered first, what does that look like? IXK : Well, what it looks like for me as a black American is that people do not view me as dangerous and thereby make my existence dangerous. It allows me to walk around this country and to not believe that people are going to fear me because of the color my skin. It allows me to believe, you know what, I didn ' t get that job because I could have done better on my interview, not because of the color of my skin. It allows me to – a country where there ' s racial **equity**, a country where there ' s racial justice, you know, a country where there ' s shared opportunity, a country where African American culture and Native American culture and the cultures of Mexican Americans and Korean Americans are all valued equally, that no one is being asked to assimilate into white American culture. There ' s no such thing as standard professional wear. There ' s no such thing as, well, you need to learn how to speak English in order to be an American. And we would truly not only have **equity** and justice for all but we would somehow have found a way to appreciate difference, to appreciate all of the human ethnic and cultural difference that exists in the United States. This is what could make this country great, in which we literally become a country where you could literally travel around this country and learn about cultures from all over the world and appreciate those cultures, and understand even your own culture from what other people are doing. There ' s so much beauty here amid all this pain and I just want to peel away and remove away all of those scabs of racist policies so that people can **heal** and so that we can see true beauty. WPR : And Ibram, when you think about this moment, where do you see that on the spectrum of progress towards reaching that true beauty? IXK : Well, I think, for me, I always see progress and resistance in demonstrations and know just because people are calling from town squares and from city halls for progressive, **systemic** change that that change is here, but people are calling and people are calling in small towns, in big cities, and people are calling from places we ' ve heard of and places we need to have heard of. People are calling for change, and people are fed up. I mean, we ' re living in a time in which we ' re facing a viral pandemic, a racial pandemic within that viral pandemic of people of color disproportionately being infected and dying, even an economic pandemic with over 40 million Americans having lost their jobs, and certainly this pandemic of police **violence**, and then people demonstrating against police **violence** only to suffer police **violence** at demonstrations. I mean, people see there ' s a fundamental problem here, and there ' s a problem that can be solved. There ' s an America that can be created, and people are calling for this, and that is always the beginning. The beginning is what we ' re experiencing now. CS : I think that this next audience question follows well from that, which is,"What gives you hope right now?"IXK : So certainly resistance to racism has always given me hope, and so even if, let ' s say, six months ago we were not in a time in which almost every night all over this country people were demonstrating against racism, but I could just look to history when people were resisting. And so resistance always brings me hope, because it is always resistance, and of course it ' s stormy, but the rainbow is typically on the other side. But I also receive hope philosophically, because I know that in order to bring about change, we have to believe in change. There ' s just no way a change maker can be cynical. It ' s impossible. So I know I have to believe in change in order to bring it about. WPR : And we have another question here which addresses some of the things you talked about before in terms of the structural change that we need to bring about. From Maryam Mohit :"In terms of putting into practice the transformative policies, is then the most important thing to loudly vote the right people into office at every level who can make those structural changes happen?"IXK : So I think that that is part of it. I certainly think we should vote into office people who, from school boards to the President of the United States, people who are committed to instituting anti-racist policies that lead to **equity** and justice, and I think that that ' s critically important, but I don ' t think that we should think that that ' s the only thing we should be focused on or the only thing that we should be doing. And there are institutions, there are **neighborhoods** that need to be transformed, that are to a certain extent outside of the purview of a policymaker who is an elected official. There are administrators and CEOs and presidents who have the power to transform policies within their spheres, within their institutions, and so we should be focused there. The last thing I ' ll say about voting is, I wrote a series of pieces for"The Atlantic"early this year that sought to get Americans thinking about who I call"the other swing voter,"and not the traditional swing voter who swings from Republican to Democrat who are primarily older and white. I ' m talking about the people who swing from voting Democrat to not voting at all. And these people are typically younger and they ' re typically people of color, but they ' re especially young people of color, especially young black and Latinx Americans. And so we should view these people, these young, black and Latino voters who are trying to decide whether to vote as swing voters in the way we view these people who are trying to decide between whether to vote for, let ' s say, Trump or Biden in the general election. In other words, to view them both as swing voters is to view them both in a way that, OK, we need to persuade these people. They ' re not political cattle. We ' re not just going to turn them out. We need to encourage and persuade them, and then we also for these other swing voters need to make it easier for them to vote, and typically these young people of color, it ' s the hardest for them to vote because of voter suppression policies. CS : Thank you, Ibram. Well, we ' re going to come to a close of this interview, but I would love to ask you to read something that you wrote a couple of days ago on Instagram. You wrote this beautiful caption on a photo of your daughter, and I ' m wondering if you ' d be willing to share that with us and briefly tell us how we could each take this perspective into our own lives. IXK : Sure, so yeah, I posted a picture of my four-year-old daughter Imani, and in the caption I wrote,"I love, and because I love, I resist. There have been many theories on what ' s fueling the growing demonstrations against racism in public and private. Let me offer another one : love. We love. We know the lives of our loved ones, especially our black loved ones, are in danger under the **violence** of racism. People ask me all the time what fuels me. It is the same : love, love of this little girl, love of all the little and big people who I want to live full lives in the fullness of their humanity, not barred by racist policies, not degraded by racist ideas, not terrorized by racist **violence**. Let us be anti-racist. Let us defend life. Let us defend our human rights to live and live fully, because we love."And, you know, Cloe, I just wanted to sort of emphasize that at the heart of being anti-racist is love, is **loving** one ' s country, loving one ' s humanity, loving one ' s relatives and family and friends, and certainly loving oneself. And I consider love to be a verb. I consider love to be, I ' m helping another, and even myself, to constantly grow into a better form of myself, of themselves, that they ' ve expressed who they want to be. And so to love this country and to love humanity is to push humanity constructively to be a better form of itself, and there ’ s no way we ’ re going to be a better form, there ’ s no way we can build a better humanity, while we still have on the shackles of racism. WPR : I think that ’ s so beautiful. I appreciate everything you ’ ve shared, Ibram. I feel like it ’ s made it really clear this is not an easy fix. Right? There is no band-aid option here that will make this go away, that this takes work from all of us, and I really appreciate all of the honesty and thoughtfulness that you ’ ve brought to this today. IXK : You ’ re welcome. Thank you so much for having this conversation with me. CS : Thank you so much, Ibram. We ’ re really grateful to you for joining us. IXK : Thank you.

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Audrey Tang : Very happy to be joining you, and good local time, everyone. David Biello : So, tell us about – Sorry to – Tell us about digital tools and COVID. AT : Sure. Yeah, I ’ m really happy to share with you how Taiwan successfully countered the COVID using the power of digital democracy tools. As we know, democracy improves as more people participate. And digital technology remains one of the best ways to improve participation, as long as the focus is on finding common ground, that is to say, prosocial media instead of antisocial media. And there ’ s three key ideas that I would like to share today about digital democracy that is fast, fair and fun. First about the fast part. Whereas many jurisdictions began countering coronavirus only this year, Taiwan started last year. Last December, when Dr. Li Wenliang, the PRC whistleblower, posted that there are new SARS cases, he got inquiries and eventually punishments from PRC police institutions. But at the same time, the Taiwan equivalent of Reddit, the Ptt board, has someone called nomorepipe reposting Dr. Li Wenliang ’ s whistleblowing. And our medical officers immediately noticed this post and issued an order that says all passengers flying in from Wuhan to Taiwan need to start health inspections the very next day, which is the first day of January. And this says to me two things. First, the civil society trusts the government enough to talk about possible new SARS outbreaks in the public forum. And the government trusts citizens enough to take it seriously and treat it as if SARS has happened again, something we ’ ve always been preparing for, since 2003. And because of this open civil society, according to the CIVICUS Monitor after the Sunflower Occupy, Taiwan is now the most open society in the whole of Asia. We enjoy the same freedom of speech, of assembly, [unclear] as other liberal democracies, but with the emphasis on keeping an open mind to novel ideas from the society. And that is why our schools and businesses still remain open today, there was no lockdown, it ’ s been a month with no local confirmed cases. So the fast part. Every day, our Central Epidemic Command Center, or CECC, holds a press conference, which is always livestreamed, and we work with the journalists, they answer all the questions from the journalists, and whenever there ’ s a new idea coming in from the social sector, anyone can pick up their phone and call 1922 and tell that idea to the CECC. For example, there was one day in April where a young boy has said he doesn ’ t want to go to school because his school mates may laugh at him because all he had is a pink medical mask. The very next day, everybody in the CECC press conference started wearing pink medical masks, making sure that everybody learns about gender mainstreaming. And so this kind of rapid response system builds trust between the government and the civil society. And the second focus is **fairness**. Making sure everybody can use their national health insurance card to collect masks from nearby pharmacies, not only do we publish the stock level of masks of

all pharmacies, 6, 000 of them, we publish it every 30 seconds. That ’ s why our civic hackers, our civil engineers in the digital space, built more than 100 tools that enable people to view a map, or people with blindness who talk to **chat** bots, voice assistants, all of them can get the same inclusive **access** to information about which pharmacies near them still have masks. And because the national health insurance single payer is more than 99. 9 percent of health coverage, people who show any symptoms will then be able to take the medical mask, go to a local clinic, knowing fully that they will get treated fairly without incurring any financial burden. And so people designed a dashboard that lets everybody see our supply is indeed growing, and whether there ’ s over- or undersupply, so that we codesign this distribution system with the pharmacies, with the whole of society. So based on this analysis, we show that there was a peak at 70 percent, and that remaining 20 percent of people were often young, work very long hours, when they go off work, the pharmacies also went off work, and so we work with convenience stores so that everybody can collect their mask anytime, 24 hours a day. So we ensure **fairness** of all kinds, based on the digital democracy ’ s feedback. And finally, I would like to acknowledge that this is a very stressful time. People feel anxious, outraged, there ’ s a lot of panic buying, a lot of conspiracy theories in all economies. And in Taiwan, our counter-disinformation strategy is very simple. It ’ s called”humor over rumor.”So when there was a panic buying of tissue paper, for example, there was a rumor that says,”Oh, we ’ re ramping up mass production, it ’ s the same material as tissue papers, and so we ’ ll run out of tissue paper soon.”And our premier showed a very memetic picture that I simply have to share with you. In very large print, he shows his bottom, wiggling it a little bit, and then the large print says”Each of us only have one pair of buttocks.”And of course, the serious table shows that tissue paper came from South American materials, and medical masks come from **domestic** materials, and there ’ s no way that ramping up production of one will hurt the production of the other. And so that went absolutely viral. And because of that, the panic buying died down in a day or two. And finally, we found out the person who spread the rumor in the first place was the tissue paper reseller. And this is not just a single shock point in social media. Every single day, the daily press conference gets translated by the spokesdog of the Ministry of Health and Welfare, that translated a lot of things. For example, our physical distancing is phrased as saying”If you are outdoors, you need to keep two dog-lengths away, if you are indoor, three dog-lengths away,”and so on. And hand sanitation rules, and so on. So because all this goes viral, we make sure that the factual humor spreads faster than rumor. And they serve as a vaccine, as inoculation, so that when people see the conspiracy theories, the R0 value of that will be below one, meaning that those ideas will not spread. And so I only have this five-minute briefing, the rest of it will be driven by your Q and A, but please feel free to read more about Taiwan ’ s counter-coronavirus strategy, at taiwancanhelp. us. Thank you. DB : That ’ s incredible. And I love this”humor versus rumor.”The problem here in the US, perhaps, is that the rumors seem to travel faster than any response, whether humorous or not. How do you defeat that aspect in Taiwan? AT : Yeah, we found that, of course, humor implicitly means there is a sublimation of upsetness, of outrage. And so as you see, for example, in our premier ’ s example, he makes fun of himself. He doesn ’ t make a joke at the expense of other people. And this was the key. Because people think it hilarious, they share it, but with no malicious or toxic intentions. People remember the actual payload, that table about materials used to produce masks, much more easily. If they make a joke that excludes parts of the society, of course, that part of society will feel outraged and we will end up creating more divisiveness, rather than prosocial behavior.

So the humor at no expense, not excluding any part of society, I think that was the key. DB : It ’ s also incredible because Taiwan has such close ties to the origin point of this. AT : PRC, yes. DB : The mainland. So given those close economic ties, how do you survive that kind of disruption? AT : Yeah, I mean, at this moment, it ’ s been almost a month now with no local confirmed cases, so we ’ re doing fine. And what we are doing, essentially, is just to respond faster than pretty much anyone. We started responding last year, whereas pretty much everybody else started responding this year. We tried to warn the world last year, but, anyway. So in any case, the point here is that if you start early enough, you get to make sure that the border control is the main point where you quarantine all the returning residents and so on, instead of waiting until the community spread stage, where even more human-right invading techniques would probably have to be deployed one way or the other. And so in Taiwan, we ’ ve not **declared** an emergency situation. We ’ re firmly under the constitutional law. Because of that, every measure the administration is taking is also applicable in non-coronavirus times. And this forces us to innovate. Much as the idea of”we are an open liberal democracy”prevented us from doing takedowns. And therefore, we have to innovate of humor versus rumor, because the easy path, the takedown of online speech, is not accessible to us. Our design criteria, which is no lockdowns, also prevented us from doing any, you know, very invasive privacy encroaching response system. So we have to innovate at the border, and make sure that we have a sufficient number of, for example, quarantine hotels or the so-called”digital fences,”where your phone is basically connected to the nearby telecoms, and they make sure that if they go out of the 15-meter or so radius, an SMS is sent to the local household managers or police and so on. But because we focus all these measures at the border, the vast majority of people live a normal life. DB : Let ’ s talk about that a little bit. So walk me through the digital tools and how they were applied to COVID. AT : Yes. So there ’ s three parts that I just outlined. The first one is the collective intelligence system. Through online spaces that we design to be devoid of **Reply** buttons, because we see that, when there ’ s **Reply** buttons, people focus on each other ’ s face part, not the book part, and without”**Reply**”buttons, you can get collective intelligence working out their rough consensus of where the direction is going with the response strategies. So we use a lot of new technologies, such as Polis, which is essentially a forum that lets you upvote and downvote each other ’ s feelings, but with real-time clustering, so that if you go to cohack. tw, you see six such conversations, talking about how to protect the most vulnerable people, how to make a smooth transition, how to make a fair distribution of supplies and so on. And people are free to voice their ideas, and upvote and downvote each other ’ s ideas. But the trick is that we show people the main divisive points, and the main consensual points, and we respond only to the ideas that can convince all the different opinion groups. So people are encouraged to post more eclectic, more nuanced ideas and they discover, at the end of this consultation, that everybody, actually, agrees with most things, with most of their neighbors on most of the issues. And that is what we call the social mandate, or the democratic mandate, that then informs our development of the counter-coronavirus strategy and helping the world with such tools. And so this is the first part, it ’ s called listening at scale for rough consensus. The second part I already covered is the distribute ledger, where everybody can go to a nearby pharmacy, present their NHI card, buy nine masks, or 10 if you ’ re a child, and see the stock level of that pharmacy on their phone actually decreasing by nine or 10 in a couple of minutes. And if they grow by nine or 10, of course, you call the 1922, and report something fishy is going on. But this is participatory accountability. This is published every 30 seconds. So everybody holds each other accountable, and that massively increases trust. And finally, the third one, the humor versus rumor, I think the important thing to see here is that wherever there ' s a trending disinformation or conspiracy theory, you respond to it with a humorous package within two hours. We have discovered, if we respond within two hours, then more people see the vaccination than the conspiracy theory. But if you respond four hours or a day afterwards, then that ' s a lost cause. You can ' t really counter that using humor anymore, you have to invite the person who spread those messages into cocreation workshops. But we ' re OK with that, too. DB : Your speed is incredible. I see Whitney has joined us with some questions. Whitney Pennington Rodgers : That ' s right, we have a few coming in already from the audience. Hi there, Audrey. And we ' ll start with one from our community member Michael Backes. He asks how long has humor versus rumor been a strategy that you ' ve implemented. Excuse me."How long has humor versus rumor strategy been implemented? Were comedians consulted to make the humor?"AT : Yes, definitely. Comedians are our most cherished colleagues. And each and every ministry has a team of what we call participation officers in charge of engaging with trending topics. And it ' s a more than 100 people-strong team now. We meet every month and also every couple of weeks on specific topics. It ' s been like that since late 2016, but it ' s not until our previous spokesperson, Kolas Yotaka, joined about a year and a half ago, do the professional comedians get to the team. Previously, this was more about inviting the people who post, you know, quotes like"Our tax filing system is explosively hostile,"and gets trending, and previously, the POs just invited those people. Everybody who complains about the finance minister ' s tax-filing experience gets invited to the cocreation of that tax filing experience. So previously, it was that. But Kolas Yotaka and the premier Su Tseng-chang said, wouldn ' t it be much better and reach more people if we add some dogs to it or cat ' s pictures to it? And that ' s been around for a year and a half. WPR : Definitely, I think it makes a lot of difference, just even seeing them without being part of the thought process behind that. And we have another question here from G. Ryan Ansin. He asks,"What would you rank the level of trust your community had before the pandemic, in order for the government to have a chance at properly controlling this crisis?"AT : I would say that a community trusts each other. And that is the main point of digital democracy. This is not about people trusting the government more. This is about the government trusting the citizens more, making the state transparent to the citizen, not the citizen transparent to the state, which would be some other regime. So making the state transparent to the citizens doesn ' t always elicit more trust, because you may see something wrong, something missing, something exclusively hostile to its user experience, an so on, of the state. So it doesn ' t necessarily lead to more trust from the government. Sorry, from the citizen to the government. But it always leads to more trust between the social sector stakeholders. So I would say the level of trust between the people who are working on, for example, medical officers, and people who are working with the pandemic responses, people who manufacture medical masks, and so on, all these people, the trust level between them is very high. And not necessarily they trust the government. But we don ' t need that for a successful response. If you ask a random person on the **street**, they will say Taiwan is performing so well because of the people. When the CECC tells us to wear the mask, we wear the mask. When the CECC tells us not to wear a mask, like, if you are keeping physical distance, we wear a mask anyway. And so because of that, I think it ' s the social sector ' s trust between those different stakeholders that ' s the key to the response. WPR : I will come back shortly with more questions, but I ' ll leave you guys to continue your conversation. AT : Awesome. DB : Well, clearly, part of that trust in government was maybe not there in 2014 during the Sunflower Movement. So talk to me about that and how that led to this, kind of, digital transformation. AT : Indeed. Before March 2014, if you asked a random person on the **street** in Taiwan, like, whether it ’ s possible for a minister – that ’ s me – to have their office in a park, literally a park, anyone can walk in and talk to me for 40 minutes at a time, I ’ m currently in that park, the Social Innovation Lab, they would say that this is crazy, right? No public officials work like that. But that was because on March 18, 2014, hundreds of young activists, most of them college students, occupied the legislature to express their profound opposition to a trade pact with Beijing under consideration, and the secretive manner in which it was pushed through the parliament by Kuomintang, the ruling party at the time. And so the protesters demanded, very simply, that the pact be scraped, and the government to institute a more transparent ratification process. And that drew widespread public support. It ended a little more than three weeks later, after the government promised and agreed on the four demands [unclear] of legislative oversight. A poll released after the occupation showed that more than 75 percent remained dissatisfied with the ruling government, illustrating the crisis of trust that was caused by a trade deal dispute. And to **heal** this rift and communicate better with everyday citizens, the administration reached out to the people who supported the occupiers, for example, the g0v community, which has been seeking to improve government transparency through the creation of open-source tools. And so, Jaclyn Tsai, a government minister at the time, attended our hackathon and proposed the establishment of novel platforms with the online community to exchange policy ideas. And an experiment was born called vTaiwan, that pioneerly used tools such as Polis, that allows for”agree”or”disagree”with no **Reply** button, that gets people ’ s rough consensus on issues such as crowdfunding, equity-based crowdfunding, to be precise, teleworking and many other cyber-related legislation, of which there is no existing unions or associations. And it proved to be very successful. They solved the Uber problem, for example, and by now, you can call an Uber – I just called an Uber this week – but in any case, they are operating as taxis. They set up a local taxi company called Q Taxi, and that was because on the platform, people cared about insurance, they care about registration, they care about all the sort of, protection of the passengers, and so on. So we changed the taxi regulations, and now Uber is just another taxi company along with the other co-ops. DB : So you ’ re actually, in a way, crowdsourcing laws that, well, then become laws. AT : Yeah, learn more at crowd. law. It ’ s a real website. DB : So, some might say that this seems easier, because Taiwan is an island, that maybe helps you control COVID, helps promote social cohesion, maybe it ’ s a smaller country than some. Do you think that this could be scaled beyond Taiwan? AT : Well, first of all, 23 million people is still quite some people. It ’ s not a city, as some usually say, you know,”Taiwan is a city-state.”Well, 23 million people, not quite a city-state. And what I ’ m trying to get at, is that the high population density and a variety of cultures – we have more than 20 national languages – doesn ’ t necessarily lead to social cohesion, as you said. Rather, I think, this is the humbleness of all the ministers in the counter-coronavirus response. They all took on an **attitude** of”So we learned about SARS”– many of them were in charge of the SARS back then, but that was classical epidemiology. This is SARS 2. 0, it has different characteristics. And the tools that we use are very different, because of the digital transformation. And so we are in it to learn together with the citizens. Our vice president at the time, Dr. Chen Chien-jen, an academician, literally wrote the textbook on epidemiology. However, he still says,”You know, what I ’ m going to do is record an online MOOC, a crash course on epidemiology, that shares with, I think, more than 20, 00 people enrolled the first day, I was among them, to learn about important ideas, like the R0 and the basic transmission and how the various different measures work, and then they asked people to innovate. If you think of a new way that the vice president did not think of, just call 1922, and your idea will become the next day ’ s press conference. And this is this colearning strategy, I think, that more than anything enabled the social cohesion, as you speak. But this is more of a robust civil society than the uniformity. There ’ s no uniformity at all in Taiwan, everybody is entitled to their ideas, and all the social innovations, ranging from using a traditional rice cooker to revitalize, to disinfect the mask, to pink medical mask, and so on, there ’ s all variety of very interesting ideas that get amplified by the daily press conference. DB : That ’ s beautiful. Now – oh, Whitney is back, so I will let her ask the next question. WPR : Sure, we ’ re having some more questions come in. One from our community member Aria Bendix. Aria asked,”How do you ensure that digital campaigns act quickly without sacrificing accuracy? In the US, there was a fear of inciting panic about COVID-19 in early January.”AT : This is a great question. So most of the scientific ideas about the COVID are evolving, right? The efficacy of masks, for example, is a very good example, because the different characteristics of previous respiratory diseases respond differently to the facial mask. And so, our digital campaigns focus on the idea of getting the rough consensus through. So basically, it ’ s a reflection of the society, through Polis, through Slido, through the joint platform, the various tools that vTaiwan has prototyped, we know that people are feeling a rough consensus about things and we ’ re responding to the society, saying,”This is what you all feel and this is what we ’ re doing to respond to your feelings. And the scientific consensus is still developing, but we know, for example, people feel that wearing a mask mostly protects you, because it reminds you to not touch your face and wash your hands properly.”And these, regardless of everything else, are the two things that everybody agrees with. So we just capitalize on that and say,”OK, wash your hands properly, and don ’ t touch your face, and wearing a mask reminds you of that.”And that lets us cut through the kind of, very ideologically charged debates and focus on what people generally resonate with one another. And that ’ s how we act quickly without sacrificing scientific accuracy. WPR : And this next question sort of feels connected to this as well. It ’ s a question from an anonymous community member.”Pragmatically, do you think any of your policies could be applied in the United States under the current Trump administration?”AT : Quite a few, actually. We work with many states in the US and abroad on what we call”epicenter to epicenter diplomacy.”So what we ’ re doing essentially is, for example, there was a **chat** bot in Taiwan that lets you, but especially people under home quarantine, to ask the **chat** bot anything. And if there is a scientific adviser who already wrote a frequently asked question, the **chat** bot just responds with that, but otherwise, they will call the science advisory board and write an accessible response to that, and the spokesdog would translate that into a cute dog meme. And so this feedback cycle of people very easily **accessing**, finding, and asking a scientist, and an open API that allows for voice assistance and other third-party developers to get through it, resonates with many US states, and I think many of them are implementing it. And before the World Health Assembly, I think three days before, we held a 14 countries [unclear] lateral meeting, kind of, pre-WHA, where we shared many small, like, quick wins like this. And I think many jurisdictions took some of that, including the humor versus rumor. Many of them said that they ’ re going to recruit comedians now. WPR : I love that. DB : I hope so. WPR : I hope so too. And we have one more question, which is actually a follow-up, from Michael Backes, who asked a question earlier.”Does the Ministry plan to publish their plans in a white paper?”Sounds like you ’ re already sharing your plans with folks, but do you have a plan to put it out on paper? AT : Of course. Yeah, and multiple white papers. So if you go to taiwancanhelp. us, that is where most of our strategy is, and that website is actually crowdsourced as well, and it shows that more than five million now, I think, medical masks donated to the humanitarian aid. It ’ s also crowdsourced. People who have some masks in their homes, who did not collect the rationed masks, they can use an app, say,”I want to dedicate this to international humanitarian aid,”and half of them choose to publish their names, so you can also see the names of people who participated in this. And there ’ s also an”Ask Taiwan Anything”website, at fightcovid. edu. tw, that outlines, in white paper form, all the response strategies, so check those out. WPR : Great. Well, I will disappear and be back later with some other questions. DB : A blizzard of white papers, if you will. I ’ d like to turn the focus on you a little bit. How does a conservative anarchist become a digital minister? AT : Yeah, by occupying the parliament, and through that. More interestingly, I would say that I go working with the government, but never for the government. And I work with the people, not for the people. I ’ m like this Lagrange point between the people ’ s movements on one side and the government on the other side. Sometimes right in the middle, trying to do some coach or translation work. Sometimes in a kind of triangle point, trying to supply both sides with tools for prosocial communication. But always with this idea of getting the shared values out of different positions, out of varied positions. Because all too often, democracy is built as a showdown between opposing values. But in the pandemic, in the infodemic, in climate change, in many of those structural issues, the virus or carbon dioxide doesn ’ t sit down and negotiate with you. It ’ s a structural issue that requires common values built out of different positions. And so that is why my working principle is radical transparency. Every conversation, including this one, is on the record, including the internal meetings that I hold. So you can see all the different meeting transcripts in my YouTube channel, in the SayIt platform, where people can see, after I became digital minister, I held 1, 300 meetings with more than 5, 000 speakers, with more than 260, 000 utterances. And every one of them has a URL that becomes a social object that people can have a conversation on. And because of that, for example, when Uber ’ s David Plouffe visited me to lobby for Uber, because of radical transparency, he is very much aware of that, and so he made all the arguments based on public good, based on sustainability, and things like that, because he knows that the other sides would see his positions very clearly and transparently. So that encourages people to add on each other ’ s argument, instead of attacking each other ’ s person, you know, credits and things like that. And so I think that, more than anything, is the main principle of conserving the anarchism of the internet, which is about, you know, nobody can force anyone to hook to the internet, or to adhere to a new internet protocol. Everything has to be done using rough consensus and running code. DB : I wish you had more counterparts all around the world. Maybe you wish you had more counterparts all around the world. AT : That ’ s why these ideas are worth spreading. DB : There you go. So one of the challenges that might arise with some of these digital tools is **access**. How do you approach that part of it for folks maybe who don ’ t have the best broadband connection or the latest mobile phone or whatever it might be that ’ s required? AT : Well, anywhere in Taiwan, even on the top of Taiwan, almost 4, 000 meters high, the Saviah, or the Jade Mountain, you ’ re guaranteed to have 10 megabits per second over 4G or fiber or cable, with just 16 US dollars a month, an unlimited plan. And actually, on the top of the mountain, it ’ s faster, fewer people use that bandwidth. And if you don ’ t, it ’ s my fault. It

’ s personally my fault. In Taiwan, we have broadband as a human right. And so when we ’ re deploying 5G, we ’ re looking at places where the 4G has the weakest signal, and we begin with those places in our 5G deployment. And only by deploying broadband as a human right can we say that this is for everybody. That digital democracy actually strengthens democracy. Otherwise, we would be excluding parts of the society. And this also applies to, for example, you can go to a local digital opportunity center to rent a tablet that ’ s guaranteed to be manufactured in the past three years, and things like that, to enable, also, the different digital **access** by the digital opportunity centers, universities and schools, and public libraries, very important. And if people who prefer to talk in their town hall, I personally go to that town hall with a 360 recorder, and livestream that to Taipei and to other municipalities, where the central government ’ s public servants can join in a connected room style, but listening to the local people who set the agenda. So people still do face-to-face meetings, we ’ re not doing this to replace face-to-face meetings. We ’ re bringing more stakeholders from central government in the local town halls, and we ’ re amplifying their voices by making sure the transcripts, the mind maps, and things like that are spread through the internet in real time, but we don ’ t ever ask the elderly to, say,”Oh, you have to learn typing, otherwise you don ’ t do democracy.”It ’ s not our style. But that requires broadband. Because if you don ’ t have broadband, but only a very limited bandwidth, you are forced to use text-based communication. DB : That ’ s right. Well, with **access**, of course, comes **access** for folks who maybe will misuse the platform. You talked a little bit about disinformation and using humor to beat rumor. But sometimes, disinformation is more weaponized. How do you combat those kinds of attacks, really? AT : Right, so you mean malinformation, then. So essentially, information designed to cause intentional public **harm**. And that ’ s no laughing matter. So for that, we have an idea called”notice and public notice.”So this is a Reuters photo, and I will read the original caption. The original caption says”A teenage extradition bill protester in Hong Kong is seen during a march to demand democracy and political reform in Hong Kong.”OK, a very neutral title by the Reuters. But there was a spreading of malinformation back last November, just leading to our presidential election, that shows something else entirely. This is the same photo – that says”This 13-year-old thug bought new iPhones, game consoles and brand-name sports shoes, and recruiting his brothers to murder police and collect 200, 000 dollars.”And this, of course, is a weapon designed to sow discord, and to elicit in Taiwan ’ s voters a kind of distaste for Hong Kong. And because they know that this is the main issue. And had we resorted to takedowns, that will not work, because that would only evoke more outrage. So we didn ’ t do a takedown. Instead, we worked with the fact checkers and professional journalists to attribute this original message back to the first day that it was posted. And it came from Zhongyang Zhengfawei. That is the main political and legal unit of the central party, in the Central Communist Party, in CCP. And we know that it ’ s their Weibo account that first did this new caption. So we sent out a public notice and with the partners in social media companies, pretty much all of them, they just put this very small reminder next to each time that this is shared with the wrong caption, that says”This actually came from the central propaganda unit of the CCP. Click here to learn more. To learn about the whole story.”And that, we found, that has worked, because people understand this is then not a news material. This is rather an appropriation of Reuters ’ news material and a copyright infringement and I think that ’ s part of the [unclear]. In any case, the point is that when people understand that this is an intentional narrative, they won ’ t just randomly share it. They may share it, but with a comment that says”This is what the Zhongyang Zhengfawei is trying to do to our democracy.”DB : Seems like some of the global social media companies could learn something from notice and public notice. AT : Public notice, that ’ s right. DB : What advice would you have for the **Twitters** and Facebooks and LINEs and WhatsApps, and you name it, of the world? AT : Yeah. So, just before our election, we said to all of them that we ’ re not making a law to kind of punish them. However, we ’ re sharing this very simple fact that there is this norm in Taiwan that we even have a separate branch of the government, the control branch, that published the campaign donation and expense. And it just so occurred to us that in the previous election, the mayoral one, there was a lot of candidates that did not include any social media advertisements in their expense to the Control Yuan. And so essentially, that means that there is a separate amount of political donation and expense that evades public scrutiny. And our Control Yuan published their numbers in raw data form, that is to say, they ’ re not statistics, but individual records of who donated for what cause, when, where, and investigative journalists are very happy, because they can then make investigative reports about the connections between the candidates and the people who fund them. But they cannot work with the same material from the global social media companies. So I said,”Look, this is very simple. This is the social norm here, I don ’ t really care about other jurisdictions. You either adhere to the social norm that is set by the Control Yuan and the investigative journalists, or maybe you will face social sanctions. And this is not the government mandate, but it ’ s the people fed up with, you know, black box, and that ’ s part of the Sunflower Occupy ’ s demands, also. And so Facebook actually published in the Ad Library, I think at that time, one of the fastest response strategies, where everybody who has basically any dark pattern advertisement will get revealed very quickly, and investigative journalists work with the local civic technologists to make sure that if anybody dare to use social media in such a divisive way, within an hour, there will be a report out condemning that. So nobody tried that during the previous presidential election season. DB : So change is possible. AT : Mhm. WPR : Hey there, we have some more questions from the community. There is an anonymous one that says,”I believe Taiwan is outside WHO entirely and has a 130-part preparation program – developed entirely on its own – to what extent does it credit its preparation to building its own system?”AT : Well, a little bit, I guess. We tried to warn the WHO, but at that point – we are not totally outside, we have limited scientific **access**. But we do not have any ministerial **access**. And this is very different, right? If you only have limited scientific **access**, unless the other side ’ s top epidemiologist happens to be the vice president, like in Taiwan ’ s case, they don ’ t always do the storytelling well enough to translate that into political action as our vice president did, right? So the lack of ministerial **access**, I think, is to the detriment of the global community, because otherwise, people could have responded as we did in the first day of January, instead of having to wait for weeks before the WHO **declared** that this is something, that there ’ s definitely human to human transmission, that you should inspect people coming in from Wuhan, which they eventually did, but that ’ s already two weeks or three weeks after what we did. WPR : Makes a lot of sense. DB : More scientists and technologists in politics. That sounds like that ’ s the answer. AT : Yeah. WPR : And then we have another question here from Kamal Srinivasan about your reopening strategy.”How are you enabling restaurants and retailers to open safely in Taiwan?”AT : Oh, they never closed, so... WPR : Oh! AT : Yeah, they never closed, there was no lockdown, there was no closure. We just said a very simple thing in the CECC press conference, that there ’ s going to be physical distancing. You maintain one and a half meters indoors or wear a mask. And that ’ s it. And so there are some restaurants that put up, I guess, red curtains, some put very cute teddy bears and so on, on the chairs, to make sure that people spread evenly, some installed see-through glass or plastic walls between the seats. There ' s various social innovations happening around. And I think the only shops that got closed for a while, because they could not innovate quick enough to respond to these rules, was the intimate escort bars. But eventually, even they invented new ways, by handing out these caps that are plastic shielding, but still leaves room for drinking behind it. And so they opened with that social innovation. DB : That ' s amazing. WPR : It is, yeah, it ' s a lot to learn from your strategies there. Thank you, I ' ll be back towards the end with some final questions. DB : I ' m very happy to hear that the restaurants were not closed down, because I think Taipei has some of the best food in the world of any city that I ' ve visited, so, you know, kudos to you for that. So the big concern when it comes to using digital tools for COVID or using digital tools for democracy is always privacy. You ' ve talked about that a little bit, but I ' m sure the citizens of Taiwan are perhaps equally concerned about their privacy, especially given the geopolitical context. AT : Definitely. DB : So how do you cope with those demands? AT : Yeah, we design with not only defensive strategy, like minimization of data collection, but also proactive measures, such as privacy-enhancing technologies. One of the top teams that emerged out of our cohack, the TW response from the Polis, how to make contact tracing easier, focused not on the contact tracers, not on the medical officers, but on the person. So they basically said,"OK, you have a phone, you can record your temperatures, you can record your whereabouts and things like that, but that is strictly in your phone. It doesn ' t even use Bluetooth. So there ' s no transmission. Technology uses open-source, you can check it, you can use it in airplane mode. And when the contact tracer eventually tells you that you are part of a high-risk group, and they really want your contact history, this tool can then generate a single-use URL that only contains the precise information, anonymized, that the contact tracers want. But it will not, like in a traditional interview, let you ask – they ask a question, they only want to know your whereabouts, but you answer with such accuracy that you end up compromising other people ' s privacy. So basically, this is about designing with an aim to enhance other people ' s privacy, because personal data is never truly personal. It ' s always social, it ' s always intersectional. If I take a selfie at a party, I inadvertently also take pretty much everybody else ' s who are in the picture, the surroundings, the ambiance, and so on, and if I upload it to a cloud service, then I actually decimate the bargaining power, the negotiation power of everybody around me, because then their data is part of the cloud, and the cloud doesn ' t have to compensate them or get their agreement for it. And so only by designing the tools with privacy enhancing as a positive value, and not enhancing only the person ' s own privacy, just like a medical mask, it protects you, but mostly it also protects others, right? So if we design tools using that idea, and always open-source and with an open API, then we ' re in a much better shape than in centralized or so-called cloud-based services. DB : Well, you ' re clearly living in the future, and I guess that ' s quite literal, in the sense of, it ' s tomorrow morning there. AT : Twelve hours. DB : Yes. Tell me, what do you see in the future? What comes next? AT : Yes, so I see the coronavirus as a great amplifier. If you start with an **authoritarian** society, the coronavirus, with all its lockdowns and so on, has the potential of making it even a more totalitarian society. If people place their trust, however, on the social sector, on the ingenuity of social innovators, then the pandemic, as in Taiwan, actually strengthens our democracy, so that people feel, truly, that everybody can think of something that improves the welfare of not just Taiwan, but pretty much everybody else in the world. And so, my point here is that the great amplifier comes if no matter you want it or not, but the society, what they can do, is do what Taiwan did after SARS. In 2003, when SARS came, we had to shut down an entire hospital, barricading it with no definite termination date. It was very traumatic, everybody above the age of 30 remembers how traumatic it was. The municipalities and the central government were saying very different things, and that is why after SARS, the constitutional courts charged the legislature to set up the system as you see today, and also that is why, when people responding to that crisis back in 2003 built this very robust response system that there ' s early drills. So just as the Sunflower Occupy, because of the crisis in trust let us build new tools that put trust first, I think the coronavirus is the chance for everybody who have survived through the first wave to settle on a new set of norms that will reinforce your founding values, instead of taking on alien values in the name of survival. DB : Yeah, let ' s hope so, and let ' s hope the rest of the world is as prepared as Taiwan the next time around. When it comes to digital democracy, though, and digital citizenship, where do you see that going, both in Taiwan and maybe in the rest of the world? AT : Well, I have my job description here, which I will read to you. It ' s literally my job description and the answer to that question. And so, here goes. When we see the internet of things, let ' s make it the internet of beings. When we see virtual reality, let ' s make it a **shared** reality. When we see machine learning, let ' s make it collaborative learning. When we see user experience, let ' s make it about human experience. And whenever we hear the singularity is near, let us always remember the plurality is here. Thank you for listening. DB : Wow. I have to give that a little clap, that was beautiful. Quite a job description too. So, conservative anarchist, digital minister, and with that job description – that ' s pretty impressive. AT : A poetician, yes. DB : So I struggle to imagine an adoption of these techniques in the US, and that may be my pessimism weighing in. But what words of hope do you have for the US, as we cope with COVID? AT : Well, as I mentioned, during SARS in Taiwan, nobody imagined we could have CECC and a cute spokesdog. Before the Sunflower movement, during a large protest, there was, I think, half a million people on the **street**, and many more. Nobody thought that we could have a collective intelligence system that puts open government data as a way to rebuild citizen participation. And so, never lose hope. As my favorite singer, Leonard Cohen – a poet, also – is fond of saying,"Ring the bells that still can ring and forget any perfect offering. There is a crack in everything and that is how the light gets in."WPR : Wow. That ' s so beautiful, and it feels like such a great message to, sort of, leave the audience with, and sharing the sentiment that everyone seems to be so grateful for what you ' ve shared, Audrey, and all the great information and insight into Taiwan ' s strategies. AT : Thank you. WPR : And David – DB : I was just going to say, thank you so much for that, thank you for that beautiful job description, and for all the wisdom you shared in rapid-fire fashion. I think it wasn ' t just one idea that you shared, but maybe, I don ' t know, 20, 30, 40? I lost count at some point. AT : Well, it ' s called Ideas Worth Spreading, it ' s a plural form. DB : Very true. Well, thank you so much for joining us. WPR : Thank you, Audrey. DB : And I wish you luck with everything. AT : Thank you, and have a good local time. Stay safe.

Text Sample 354: POS Dim 3 – 2018 – Score 14.00 – 2018/t000378.txt

Nearly 100 years ago, almost today, most women in the United States finally won the right to vote. Now, it would take decades more for women of color to earn that right, and we ' ve come a long way since, but I would argue not nearly far enough. I think what women want today, not just only in the United States but around the globe, is to no longer be an afterthought. We don ’ t want to continue to try to, like, look at the next 100 years and be granted, grudgingly, small legal rights and accommodations. We simply want true and full **equality**. I think that women are tired of retrofitting ourselves into institutions and governments that were built by men, for men, and we ’ d rather reshape the future on our own terms. I believe — I believe what we need is a women ’ s political revolution for full **equality** across race, across class, across gender identity, across **sexual** orientation, and yes, across political labels, because I believe what binds us together as women is so much more profound than what keeps up apart. And so I ’ ve given some thought about how to build this women ’ s political revolution and that ’ s what I want to talk to you about today. The good news is that one thing that hasn ’ t changed in the last century is women ’ s resilience and our commitment to build a better life not only for ourselves, but for generations to come, because I can ’ t think of a single woman who wants her daughter to have fewer rights or opportunities than she ’ s had. So we know we all stand on the shoulders of the women who came before us, and as for myself, I come from a long line of tough Texas women. My grandparents lived outside of Waco, Texas, in the country. And when my grandmother got pregnant, of course she was not going to go to the hospital to deliver, she was going to have that baby at home. But when she went into labor, she called the neighbor woman over to cook dinner for my grandfather, because... I mean, it was unthinkable that he was going to make supper for himself. Been there. The neighbor had no experience with killing a chicken, and that was what was planned for dinner that night. And so as the story goes, my grandmother, in the birthing bed, in labor, hoists herself up on one elbow and wrings that chicken ’ s neck, right? And that is how my mother came into this world. But the amazing thing is, even though my mother ’ s own grandmother could not vote in Texas, because under Texas law,”idiots, imbeciles, the insane and women”were prevented the franchise — just two generations later, my mother, Ann Richards, was elected the first woman governor in her own right in the state of Texas. But you see, when Mom was coming up in Texas, there weren ’ t a lot of opportunities for women, and frankly, she spent her entire life trying to change that. She used to like to say,”As women, if you just give us a chance, we can perform. After all, Ginger Rogers did everything Fred Astaire did, but she did it backwards and in high heels.”Right? And honestly, that ’ s kind of what women have been doing for this last century : despite having very, very little political power, we have made **enormous** progress. So today in the United States, 100 years after getting the right to vote, women are almost half the workforce. And in 40 percent of families with children, women are the major breadwinners. Economists even estimate that if every single paid working woman took just one day off of work, it would cost the United States 21 billion dollars in gross **domestic** product. Now, largely because of Title IX, which required educational **equity**, women are actually now half the college students in the United States. We ’ re half the medical students, we ’ re half the law students — Exactly. And a fact I absolutely love : One of the most recent classes of graduating NASA astronauts was... What? For the first time, 50 percent women. The point is that women are really changing industries, they ’ re changing business from the inside out. But when it comes to government, it ’ s another story, and I actually think a picture is worth 1000 words. This is a photograph from 2017 at the White House when congressional leaders were called over to put the final details into the healthcare reform bill that was to go to Congress. Now, one of the results of this meeting was that they got rid of maternity benefits, which may not be that surprising, since no one at that table actually would need maternity benefits. And unfortunately, that ’ s what we ’ ve learned the hard way in the US for women. If we ’ re not at the table, we ’ re on the menu, right? And we ’ re simply not at enough tables, because even though women are the vast majority of voters in the United States, we fall far behind the rest of the world in political representation. Recent research is that when they ranked all the countries, the United States is 104th in women ’ s representation in office. 104th... Right behind Indonesia. So is it any big surprise, then, considering who ’ s making decisions, we ’ re the only developed country with no paid family leave? And despite all the research and improvements we ’ ve made in medical care — and this is really horrifying to me — the United States now leads the developed world in maternal mortality rates. Now, when it comes to **equal** pay, we ’ re not doing a whole lot better. Women now, on average, in the United States, still only make 80 cents to the dollar that a man makes. Though if you ’ re an African American woman, it ’ s 63 cents to the dollar. And if you ’ re Latina, it ’ s 54 cents to the dollar. It ’ s an outrage. Now, women in the UK, the United Kingdom, just came up with something I thought was rather ingenious, in order to illustrate the impact of the pay gap. So, starting November 10 and going through the end of the year, they simply put an out-of-office memo on their email to indicate all the weeks they were working without pay. Right? I think it ’ s an idea that actually could catch on. But imagine if women actually had political power. Imagine if we were at the table, making decisions. Imagine if we had our own women ’ s political party that instead of putting our issues to the side as distractions, made them the top priority. Well, we know — research shows that when women are in office, they actually act differently than men. They collaborate more with their colleagues, they work across party lines, and women are much more likely to support legislation that improves **access** to health care, education, civil rights. And what we ’ ve seen in our research in the United States Congress is that women sponsor more legislation and they cosponsor more legislation. So all the evidence is that when women actually have the chance to serve, they make a huge difference and they get the job done. So how would it look in the United States if different people were making decisions? Well, I firmly believe if half of Congress could get pregnant, we would finally quit fighting about birth control and Planned Parenthood. That would be over. I also really believe that finally, businesses might quit treating pregnancy as a nuisance, and rather understand it as a primary medical issue for millions of American workers. And I think if more women were in office, our government would actually prioritize keeping families together rather than pulling them apart. But perhaps most importantly, I think all of these issues would no longer be seen as”women ’ s issues.”They would just be seen as basic issues of **fairness** and **equality** that everybody can get behind. So I think the question is, what would it take, actually, to build this women ’ s political revolution? The good news is, actually, it ’ s already started. Because women around the globe are demanding workplaces, they ’ re demanding educational institutions, they ’ re demanding governments where sexism and **sexual harassment** and **sexual assault** are neither accepted nor tolerated. Women around the world, as we know, are raising their hands and saying,”Me Too,”and it ’ s a movement that ’ s made so much more powerful by the fact that women are standing together across industries, from **domestic** workers to celebrities in Hollywood. Women are marching, we ’ re sitting in, we ’ re speaking up. Women are challenging the status quo, we ’ re busting old taboos and yes, we are proudly making trouble. So, women in Saudi Arabia are driving for the very first time. Women in Iraq are standing in **solidarity** with survivors of human trafficking. And women from El Salvador to Ireland are fighting for reproductive rights. And women in Myanmar are standing up for human rights. In short, I think the most profound leadership in the world isn ’ t coming from halls of government. It ’ s coming from women at the grassroots all across the globe. And here in the United States, women are on fire. So a recent Kaiser poll reported that since our last presidential election in 2016, one in five Americans have either marched or taken part in a protest, and the number one issue has been women ’ s rights. Women are starting new organizations, they are volunteering on campaigns, and they ’ re taking on every issue from gun-safety reform to public education. And women are running for office in record numbers, and they are winning. So — Women like Lucy McBath from Georgia. Lucy lost her son to gun **violence**, and it was because of her experience with the criminal justice system that she realized just how broken it is, and she decided to do something about that. So she ran for office, and this January, she ’ s going to Congress. OK? Or — Angie Craig from Minnesota. So her congressman had made such hateful comments about LGBTQ people that she decided to challenge him. And you know what? She did, and she won, and when she goes to Congress in January, she ’ ll be the first lesbian mother serving in the House of Representatives. Or — Or Lauren Underwood from Illinois. She ’ s a registered nurse, and she sees every day the impact that lack of health care **access** has on the community where she lives, and so she decided to run. She took on six men in her primary, she beat them all, she won the general election, and when she goes to Congress in January, she ’ s going to be the first African-American woman ever to serve her district in Washington, D. C. So women are recognizing — this is our moment. Don ’ t wait for permission, don ’ t wait for your turn. As the late, great Shirley Chisholm said — Shirley Chisholm, the first African-American woman ever to go to Congress and the first woman to run for president in the Democratic party — but Shirley Chisholm said,”If there ’ s no room for you at the table, just pull up a folding chair.”And that ’ s what women are doing, all across the country. I believe women are now the most important and powerful political force in the world, but how do we make sure that this is not just a moment? What we need is actually a global movement for women ’ s full **equality** that is intersectional and it ’ s intergenerational, where no one gets left behind. And so I have a few ideas about how we could do that. Number one : it ’ s not enough to resist. It ’ s not enough to say what we ’ re against. It ’ s time to be loud and proud about what we are for, because being for full **equality** is a mainstream value and something that we can get behind. Because actually, men support **equal** pay for women. Millennials, they support gender **equality**. And businesses are increasingly adopting family-friendly policies, not just because it ’ s the right thing to do, but because it ’ s good for their workers. It ’ s good for their business. Number two : We have to remember, in the words of Fannie Lou Hamer, that”nobody ’ s free ’ til everybody ’ s free.”So as I mentioned earlier, women of color in this country didn ’ t even get the right to vote until much further along than the rest of us. But since they did, they are the most reliable voters, and women of color are the most reliable voters for candidates who support women ’ s rights, and we need to follow their lead — Because their issues are our issues. And as white women, we have to do more, because racism and sexism and homophobia, these are issues that affect all of us. Number three : we ’ ve got to vote in every single election. Every election. And we ’ ve got to make it easier for folks to vote, and we ’ ve got to make sure that every single vote is counted, OK? Because the barriers that exist to voting in the United States, they fall disproportionately on women — women of color, women with low incomes, women who are working and trying to raise a family. So we need to make it easier for everyone to vote, and we can start by making Election Day a federal holiday in the United States of America. Number four : don ’ t wait for instructions. If you see a problem that needs fixing, I think you ’ re the one to do it, OK? So start a new organization, run for office. Or maybe it ' s as simple as standing up on the job in support of yourself or your coworkers. This is up to all of us. And number five : invest in women, all right? Invest in women as candidates, as changemakers, as leaders. Just as an example, in this last election cycle in the United States, women donated 100 million dollars more to candidates and campaigns than they had just two years ago, and a record number of women won. So just think about that. So look, sometimes I think that the challenges we face, they seem overwhelming and they seem like they almost can never be solved, but I think the problems that seem the most intractable are the ones that are most important to work on. And just because it hasn ' t been figured out yet doesn ' t mean you won ' t. After all, if women ' s work were easy, someone else would have already been doing it, right? But women around the globe, they ' re on the move, and they are taking strengths and inspiration from each other. They are doing things they never could have imagined. So if we could just take the progress we have made in joining the workforce, in joining business, in joining the educational system, and actually channel that into building true political power, we will reshape this century, because one of us can be ignored, two of us can be **dismissed**, but together, we ' re a movement, and we ' re unstoppable. Thank you. Thank you.

Text Sample 355: POS Dim 3 – 2018 – Score 14.00 – 2018/t000384.txt

When I was in the sixth grade, I got into a fight at school. It wasn ' t the first time I ' d been in a fight, but it was the first time one happened at school. It was with a boy who was about a foot taller than me, who was physically stronger than me and who ' d been taunting me for weeks. One day in PE, he stepped on my shoe and refused to **apologize**. So, filled with anger, I grabbed him and I threw him to the ground. I ' d had some previous judo training. Our fight lasted less than two minutes, but it was a perfect reflection of the hurricane that was building inside of me as a young survivor of **sexual assault** and as a girl who was grappling with abandonment and exposure to **violence** in other spaces in my life. I was fighting him, but I was also fighting the men and boys that had **assaulted** my body and the culture that told me I had to be silent about it. A teacher broke up the fight and my principal called me in her office. But she didn ' t say,"Monique, what ' s wrong with you?"She gave me a moment to collect my breath and asked,"What happened?"The educators working with me led with empathy. They knew me. They knew I loved to read, they knew I loved to draw, they knew I adored Prince. And they used that information to help me understand why my actions, and those of my classmate, were disruptive to the learning community they were leading. They didn ' t place me on suspension ; they didn ' t call the police. My fight didn ' t keep me from going to school the next day. It didn ' t keep me from graduating ; it didn ' t keep me from teaching. But unfortunately, that ' s not a story that ' s shared by many black girls in the US and around the world today. We ' re living through a crisis in which black girls are being disproportionately pushed away from schools — - not because of an imminent threat they pose to the **safety** of a school, but because they ' re often experiencing schools as locations for punishment and marginalization. That ' s something that I hear from black girls around the country. But it ' s not insurmountable. We can shift this narrative. Let ' s start with some data. According to a National Black Women ' s Justice Institute analysis of civil rights data collected by the US Department of Education, black girls are the only group of girls who are overrepresented along the entire continuum of discipline in schools. That doesn ' t mean that other girls aren ’ t experiencing exclusionary discipline and it doesn ’ t mean that other girls aren ’ t overrepresented at other parts along that continuum. But black girls are the only group of girls who are overrepresented all along the way. Black girls are seven times more likely than their white counterparts to experience one or more out-of-school suspensions and they ’ re nearly three times more likely than their white and Latinx counterparts to be referred to the juvenile court. A recent study by the Georgetown Center on Poverty and Inequality partially explained why this disparity is taking place when they confirmed that black girls experience a specific type of age compression, where they ’ re seen as more adult-like than their white peers. Among other things, the study found that people perceive black girls to need less nurturing, less protection, to know more about sex and to be more independent than their white peers. The study also found that the perception disparity begins when girls are as young as five years old. And that this perception and the disparity increases over time and peaks when girls are between the ages of 10 and 14. This is not without consequence. Believing that a girl is older than she is can lead to harsher treatment, immediate censure when she makes a mistake and **victim** blaming when she ’ s **harmed**. It can also lead a girl to think that something is wrong with her, rather than the conditions in which she finds herself. Black girls are routinely seen as too loud, too aggressive, too angry, too visible. Qualities that are often measured in relation to nonblack girls and which don ’ t take into consideration what ’ s going on in this girl ’ s life or her cultural norms. And it ’ s not just in the US. In South Africa, black girls at the Pretoria Girls High School were discouraged from attending school with their hair in its natural state, without chemical processing. What did those girls do? They protested. And it was a beautiful thing to see the global community for the most part wrap its arms around girls as they stood in their truths. But there were those who saw them as disruptive, largely because they dared to ask the question,”Where can we be black if we can ’ t be black in Africa?”It ’ s a good question. Around the world, black girls are grappling with this question. And around the world, black girls are struggling to be seen, working to be free and fighting to be included in the landscape of promise that a safe space to learn provides. In the US, little girls, just past their toddler years, are being arrested in classrooms for having a tantrum. Middle school girls are being turned away from school because of the way they wear their hair naturally or because of the way the clothes fit their bodies. High school girls are experiencing **violence** at the hands of police officers in schools. Where can black girls be black without reprimand or punishment? And it ’ s not just these incidents. In my work as a researcher and educator, I ’ ve had an opportunity to work with girls like Stacy, a girl who I profile in my book”Pushout,”who struggles with her participation in **violence**. She bypasses the neuroscientific and structural analyses that science has to offer about how her adverse childhood experiences inform why she ’ s participating in **violence** and goes straight to describing herself as a”problem child,”largely because that ’ s the language that educators were using as they routinely suspended her. But here ’ s the thing. Disconnection and the internalization of **harm** grow stronger in isolation. So when girls get in trouble, we shouldn ’ t be pushing them away, we should be bringing them in closer. Education is a critical protective factor against contact with the criminal legal system. So we should be building out policies and practices that keep girls connected to their learning, rather than pushing them away from it. It ’ s one of the reasons I like to say that education is freedom work. When girls feel safe, they can learn. When they don ’ t feel safe, they fight, they protest, they argue, they flee, they freeze. The human brain is wired to protect us when we feel a threat. And so long as school feels like a threat, or part of the tapestry of

harm in a girl ' s life, she ' ll be inclined to resist. But when schools become locations for healing, they can also become locations for learning. So what does this mean for a school to become a location for healing? Well, for one thing, it means that we have to immediately discontinue the policies and practices that target black girls for their hairstyles or dress. Let ' s focus on how and what a girl learns rather than policing her body in ways that facilitate rape culture or punish children for the conditions in which they were born. This is where parents and the community of concerned adults can enter this work. Start a conversation with the school and encourage them to address their dress code and other conduct-related policies as a collaborative project, with parents and students, so as to intentionally avoid bias and **discrimination**. Keep in mind, though, that some of the practices that **harm** black girls most are unwritten. So we have to continue to do the deep, internal work to address the biases that inform how, when and whether we see black girls for who they actually are, or what we ' ve been told they are. Volunteer at a school and establish culturally competent and gender responsive discussion groups with black girls, Latinas, **indigenous** girls and other students who experience marginalization in schools to give them a safe space to process their identities and experiences in schools. And if schools are to become locations for healing, we have to remove police officers and increase the number of counselors in schools. Education is freedom work. And whatever our point of entry is, we all have to be freedom fighters. The good news is that there are schools that are actively working to establish themselves as locations for girls to see themselves as sacred and loved. The Columbus City Prep School for Girls in Columbus, Ohio, is an example of this. They became an example the moment their principal **declared** that they were no longer going to punish girls for having"a bad **attitude**."In addition to building — Essentially, what they did is they built out a robust continuum of alternatives to suspension, expulsion and arrest. In addition to establishing a restorative justice program, they improved their student and teacher relationships by ensuring that every girl has at least one adult on campus that she can go to when she ' s in a moment of crisis. They built out spaces along the corridors of the school and in classrooms for girls to regroup, if they need a minute to do so. And they established an advisory program that provides girls with an opportunity to start every single day with the promotion of self-worth, communication skills and goal setting. At this school, they ' re trying to respond to a girl ' s adverse childhood experiences rather than ignore them. They bring them in closer ; they don ' t push them away. And as a result, their truancy and suspension rates have improved, and girls are arriving at school increasingly ready to learn because they know the teachers there care about them. That matters. Schools that integrate the arts and sports into their curriculum or that are building out tranformative programming, such as restorative justice, mindfulness and **meditation**, are providing an opportunity for girls to repair their relationships with others, but also with themselves. Responding to the lived, complex and historical **trauma** that our students face requires all of us who believe in the promise of children and adolescents to build relationships, learning materials, human and financial resources and other tools that provide children with an opportunity to **heal**, so that they can learn. Our schools should be places where we respond to our most vulnerable girls as essential to the creation of a positive school culture. Our ability to see her promise should be at its sharpest when she ' s in the throws of poverty and addiction ; when she ' s reeling from having been sex-trafficked or survived other forms of **violence** ; when she ' s at her loudest, or her quietest. We should be able to support her intellectual and social-emotional well-being whether her shorts reach her knees or stop mid-thigh or higher. It might seem like a tall order in a world so deeply entrenched in the politics of fear to radically imagine schools as locations where girls can **heal** and thrive, but we have to be bold enough to set this as our intention. If we commit to this notion of education as freedom work, we can shift educational conditions so that no girl, even the most vulnerable among us, will get pushed out of school. And that ' s a win for all of us. Thank you.

Text Sample 356: POS Dim 3 – 2018 – Score 13.00 – 2018/t000762.txt

Throughout the United States, there is growing social awareness that **sexual violence** and **harassment** are far too common occurrences within our various institutions â occurrences often without any accountability. As a result, the Me Too movement is upon us, and survivors everywhere are speaking out to demand change. Students have rallied against **sexual assault** on campus. Service members have demanded Congress reform the military, and workers ranging from Hollywood stars to janitorial staff have called out **sexual harassment** in the workplace. This is a tipping point. This is when a social movement can create lasting legal change. But only if we switch tactics. Instead of going institution by institution, fighting for reform, it ' s time to go to the Constitution. As it stands, the US Constitution denies fundamental protections to **victims** of gender **violence** such as **sexual assault**, intimate partner **violence** and stalking. Specifically, the Fourteenth Amendment of the Constitution, which prohibits state governments from abusing its citizens, does not require state governments to intervene when private parties abuse its citizens. So what does that mean in real life? That means that when a woman calls the police from her home, afraid that an intruder may attack her, she is not entitled to the state ' s protection. Not only can the police fail to respond, but she will be left without any legal remedy if preventable **harm** occurs as a result. How can this be? It is because the state, theoretically, acts on behalf of all citizens collectively, not any one citizen individually. The resulting constitutional flaw directly contradicts international law, which requires nation-states to intervene and protect citizens against gender **violence** by private parties as a human right. Instead of requiring intervention, our Constitution leaves discretion â discretion that states have used to discriminate systemically to deny countless **victims** any remedy. Unlike what you may have seen on"Law & Order : SVU,"justice is rare for **victims** of gender **violence**. And even in those rare cases where law **enforcement** has chosen to act, **victims** have no rights during the resulting criminal process. You see, **victims** are not parties in a criminal case. Rather, they are witnesses ; their bodies, evidence. The prosecution does not represent the interests of a **victim**. Rather, the prosecution represents the interests of the state. And the state has the discretion to **dismiss** criminal charges, enter lax plea deals and otherwise remove a **victim** ' s voice from the process, because again, a state theoretically represents the interests of all citizens collectively and not any one citizen individually. Despite this constitutional flaw, some **victims** of gender **violence** have found protections under federal Civil Rights statutes, such as Title IX. Title IX is not just about sports. Rather, it prohibits all forms of sex **discrimination**, including **sexual violence** and **harassment** within educational programs that accept federal funding. While initially targeting sex **discrimination** within admissions, Title IX has actually evolved over time to require educational institutions to intervene and address gender **violence** when committed by certain parties, such as when teachers, students or campus visitors commit **sexual assault** or **harassment**. So what this means is that through Title IX, those who seek **access** to education are protected against gender **violence** in a way that otherwise does not exist under the law. It is Title IX that requires educational institutions to take reports of gender **violence** seriously, or to suffer liability. And through campus-level proceedings, Title IX goes so far as to give **victims equitable** rights during the campus process, which means that **victims** can represent their own interests during proceedings, rather than relying on educational institutions to do so. And that ' s really important, because educational institutions have historically swept gender **violence** under the rug, much like our criminal justice system does today. So while Civil Rights protects some **victims**, we should want to protect all **victims**. Rather than going institution by institution, fighting for reform on campus, in the military, in the workplace, it ' s time to go to the Constitution and pass the Equal Rights Amendment. Originally proposed in 1923, the Equal Rights Amendment would guarantee gender **equality** under the law, and much like Title IX on campus, that constitutional amendment could require states to intervene and address gender **violence** as a prohibitive form of sex **discrimination**. While the Equal Rights Amendment did not pass in the 1970s, it actually came within three states of doing so. And within the last year, at least one of those states has ratified the amendment, because we live in different political times. From the Women ' s March to the Me Too movement, we have the growing political will of the people necessary to create lasting, legal change. So as a **victims** ' rights attorney fighting to increase the prospect of justice for survivors across the country and as a survivor myself, I ' m not here to say,"Time ' s Up."I ' m here to say,"It ' s time."It ' s time for accountability to become the norm after gender **violence**. It ' s time to pass the Equal Rights Amendment, so that our legal system can become a system of justice, and #MeToo can finally become"no more."Thank you.

Text Sample 357: POS Dim 3 – 2022 – Score 16.00 – 2022/t003603.txt

Whitney Pennington Rodgers : There is so much happening in the world right now. Our guest today spends her days and, in fact, her entire illustrious career tracking and reporting on the moment ' s biggest stories. She is CNN ' s chief international anchor and host of the network ' s award-winning flagship global affairs program, Amanpour on CNN International in London and Amanpour and Co. on PBS in the United States. I ' m so thrilled to have her here with us to offer context on some of the news stories that are impacting our world and our lives. And you can see her there right now, please welcome Christiane Amanpour. Hello, Christiane, how are you doing? Christiane Amanpour : Whitney, thank you so much for having me. I ' m so glad to be with you for a few minutes and your TED community on these really important issues. And yes, I am the chief international anchor, but before that, I was, you know, the main international correspondent. So the way I work is always informed by me being on the ground, in the field and having essentially walk the walk and talk the talk with the people who are at the coalface. WPR : I love that, and I feel like that ' s going to give us so much perspective during this conversation. Let ' s just dive right in. I think one place we ' d like to start with is in Iran. For those of you who are on the call who haven ' t been tracking, back in September, an Iranian woman, Mahsa Amini, died in the custody of Iranian morality police after being arrested for not wearing hijab. Amini ' s death sparked protests and a revolution around women ' s rights in Iran and beyond that is continuing into this very moment. And, Christiane, I know that you ' ve reported on Iran throughout your career and have spent a lot of time covering this story very closely. So how historically significant would you say this moment is in Iran?

And could you just give us some context on that? CA : Look, I think it is very significant. Exactly how and what will develop towards the end, I ’ m not sure. A little bit of my own history : I am half Iranian, and I grew up in Iran. And I spent essentially the first 20 years of my life in Iran, with a little bit of going back and forth to the UK for boarding school. But that ’ s where my home was and that ’ s where my parents lived, and my sisters, essentially, you know, I was there during the build up to the Islamic Revolution of 1979. And so I saw all that happening, and that ’ s what made me want to be a journalist, in fact, to tell those kinds of history-making stories to the world. And that ’ s what really focused me on the kind of career that I then pursued. Fast forward now some 43-odd years, and you can see that the women who actually were pretty upset at the beginning of the revolution, when they had demonstrated for Khomeini, believing that he would bring democracy as he promised back then to a country that was a monarchy. And then to quickly find out that actually that wasn ’ t his plan and to quickly find out that he was putting women under the veil, which he had not said earlier. So for the first year or so of the revolution, women did not have to wear the veil. And when they did, many, many women came out and demonstrated against it. And then for the decades since, there ’ s been a sort of a tug of war between how strictly to put on the veil and not. And clearly differences between conservative and religious women who want to wear the whole tight covering that covers their head very tightly, covers their body very loosely, known as a chador in Iran. And then those who just didn ’ t, the younger generation who felt, well, they better, you know, follow the law, but they were kind of treading their own path towards how they would wear the hijab. And you could see coats and body coverings were getting tighter and shorter, head scarves were getting more colorful and pushed further back. Women and girls were going to great lengths to make themselves look beautiful and stand out as women. They would do their hair, particularly the front of their hair, which stuck out of the scarf. They would, you know, bangs were not forbidden, you know, beautifully made up their eyes, you know, their skin, their lips were all just beautifully done. And Iranian women are incredibly beautiful women in spirit and in body. And that seemed to be kind of OK. But what happened towards the last sort of, ten years or so, you ’ ve had a slightly less draconian regime when it comes to enforcing hijab. You know, in the late ’ 90s, early 2000s, and then a much more draconian, which is the latest one. So these very hard liners, this mullah, you know, Ebrahim Raisi, who is the very hardline conservative president, actually came to the presidency by using as his political platform, among other things, a much tougher interpretation of what they believe, you know, the Islamic laws should be. And, of course, who paid the ultimate price? Women. Because it ’ s always, whether it ’ s in Iran, whether it ’ s in Afghanistan, whether it ’ s in the UAE or in Saudi Arabia, or whether it ’ s in parts of Africa, whether it ’ s in the United States, etc., women ’ s bodies, women ’ s personal space seem to be used, you know, for people ’ s political aims. And so the same happened in Iran. And this poor, poor woman, Mahsa Amini, at the age of 22, came with her family from the Kurdish region of Iran to visit relatives and was wearing a scarf and was wearing a body covering. But the police on the corner at that time didn ’ t think it was conservative enough. And that ’ s why there ’ s a backlash. WPR : Well, you know, in recent days, a story was circulating that some 15, 000 protesters were to be executed, and the veracity of the story has been challenged and disproven. As I understand it, it ’ s actually more like 15, 000 protesters have been detained and the parliament voted in support of the death penalty for protesters. But it ’ s not ultimately up to them. So I ’ m curious how real you think the possibility is that something like a mass execution might actually happen in the near future? How real a threat is that? CA : Well, you ’ re right that this story was chased down, including by us, and it turned out not to be as it was being broadcast. On the other hand, there are many, many, I mean, you know, hundreds, if not thousands of people who ’ ve been rounded up and put into prison, whether they are women, men, young girls, other young people. I recall hearing one of the top government officials saying several weeks ago that the average age of those who are being arrested, this is an actual government official there, the average age was 15 and 16. And I mean, just think about it for a moment. I mean, that ’ s just unheard of anywhere, that children, young girls, people so young are in the front lines of this protest, this uprising, this movement, and are being punished. And then the latest stories, particularly by my colleague Farnaz Fassihi of The New York Times, who ’ s an incredibly good reporter, has brought out, I mean, some of the most terrible stories of how young people are being dealt with in a very harsh way. And, you know, there was, over the weekend, there was the burial of a young boy, 10 or nine years old, who was, you know, a **victim** of all of this. And he was killed. And his funeral happened. There was another big protest there. And, you know, these things defy normal behavior. And even for, you know, an Islamic republic that ’ s been pretty draconian for the last 43 years. And this goes way beyond, you know, riot control or whatever you might want to call it. So the question again is, because I know those outside of Iran want to know, is this the beginning of the end, or is this the end of the beginning, or is this the end of the Islamic republic? And I ’ m just not ready to say one way or another, because you just don ’ t know to what lengths they will finally go to crush it. And so I ’ m waiting, as a journalist should, I ’ m watching, I ’ m waiting, I ’ m interviewing, I ’ m getting as much information as I possibly can. And I ’ m going to be doing it on my show today, which will air, as you said, on CNNi and on PBS. And I ’ ve done it many times with many Iranian interlocutors. And... This is a very brutal regime when it comes to staying in power. But the only thing that I think is interesting, well, there are many things that are interesting, but what ’ s somewhat different to before is that within the Islamic establishment, you have voices, actually male voices, who are questioning why they need to react this hard against girls just for the hijab and the headscarf. So we ’ ll wait to see where that conversation gets to, if at all. WPR : And as someone with such a close personal connection to Iran, that part of your heritage and then also just your close reporting of this story, what do you feel are the things that most of the world is missing or not seeing about what ’ s happening there? CA : Well, look, I think in general, foreign policy and stories from around the world are hard to get past the American public because they ’ re hard to get past our own editors. So if there was more reporting on a more regular basis of these kinds of stories, we would have a much better – and you all would have a much better idea of trends, of what ’ s actually happening. You know, don ’ t forget, Iran is a very important country to the United States and to the rest of the world, partly because of the sort of upheaval that the Islamic Republic and including backing terrorism etc., have caused to the world – and holding hostages as they continue to do. Iranian Americans, British Iranians, and a whole load of others who are in jail, used as political pawns. And this is, you know, a very, very unfortunate and tragic situation where human beings are being used as political weapons. So they ’ re weaponizing people who just happen to be half Iranian or they ’ ve left and they have become you know, they ’ ve taken on the nationality of the countries that have given them refuge. And when they go back to visit family or on personal visits, they have been dragged into this political turmoil. So that ’ s really bad. And many governments are having to deal with that. The other issue of why it ’ s important is because of the Iran nuclear deal. Now, I think that ’ s off the table for the moment. It was always going to be off the table pending the midterms, but now pending, you know, this crushing of this movement, the US has imposed more sanctions, Europe has imposed more sanctions on individuals who are **deemed** responsible for the harshest crackdown. So it ' s very important. Iran remains a very, very important country to the world. And don ' t forget, you know, it has a huge population, and such a huge majority of the Iranian population are under the age of 30 or, you know, they ' re in that young generation, which means they ' re incredibly well-educated, they are well-connected to the world. Even now, even though the regime is trying to cut them off from the West and cut the world off from them. They manage, you know, to play a very sophisticated technical game of cat and mouse to get their stories out and to get information from the world in. So you can never cut the Iranian people off. And they would benefit and I hope they are benefiting from the support that they know, the moral support that the world is giving them. On the other hand, I know there are others who call for more support to overthrow the regime. That ' s not my space because I ' m a reporter, and I report on what ' s going on there. And I personally don ' t believe that anybody, any foreign government is going to do that. And I also believe that the women and men, the male allies inside Iran... This is their movement. This is their movement. They don ' t want it to be sullied by any interference from abroad that could get them even more tarnished than the regime is trying to do at the moment. WPR : Christiane, we ' re getting a lot of questions from our members, and I ' m going start to bring some into our conversation. We have one from TED member, Don, which speaks to what you ' re suggesting around this idea of global support. They want to know how can we help from afar."I feel **helpless**, but would like to do what I can without putting more women in **harm** ' s way,"is what Don shares. CA : Well, it ' s very hard. I mean, definitely moral support, definitely spreading the word, spreading the word and supporting the Iranian women in whatever way you can. Some people ask, how can we send them material support, whether it ' s money or whatever it might be. But that ' s very, very difficult, and I do not have an answer for that. It ' s difficult because Iran is so heavily sanctioned. So it ' s quite often illegal, actually, you know, by American law or European law or others, to actually send money via various ways, as far as I know. Maybe there are ways that it ' s possible, but I don ' t know those ways. But it is quite difficult to provide, at the moment, more than material support. I would say, I do believe the world should be a lot more generous to those who are trying to flee this and any other kind of repressive regime. Right now I ' m in London, and as you know, over the last several, certainly months, there ' s been a crisis in the English Channel. And a huge number of Iranians are trying to get away from the danger zones and into the UK. A huge number of Afghans and potentially others fleeing wars and devastation. But they get turned back. And I think the world ' s asylum and refugee policies have become so draconian and have gone so far from being what they were envisaged as, to welcome those who are fleeing you know, terrible oppression and, of course, often starvation and disease you know, and natural disasters and the like. And I think that ' s a real **shame** that we ' re seeing in this time of maximum upheaval in these parts of the world, the rest of the world actually closing their doors. And I ' m about to do some interviews around a new film that ' s coming out called"The Swimmers."And just to say, it ' s about two Syrian girls who had to leave Syria at the height of the Arab Spring and during the Iran-supported crackdown by the Syrian regime on those who were demanding freedom. Anyway, they managed to get out, but they had to take a rickety boat to get to Europe, they had to walk for miles, then they had to wait for months, you know, for asylum. You know, people like you and me were dying in the Mediterranean. They ' re dying in the

English Channel. They ’ re dying, you know, coming across from Africa. This is 2022. It ’ s a scandal. And I do believe that that lack of asylum, that lack of refuge is a terrible **shame** and a blot on all of us who believe in human rights and democracy and the value of life. WPR : Well, I want to move to another part of the world, also experiencing a huge crisis, where you ’ ve been spending a lot of time, both reporting and actually physically being there. You just returned from Ukraine. And we ’ re now more than nine months into the war there. So could you tell us a little bit about your trip and what you saw as the people there are heading into the winter. What would you like for us to know about what you saw? CA : Because you just said winter, I think it ’ s really important to know that what Putin is doing now, because he ’ s been thwarted on the battlefield, is he is literally taking this war to civilians in a much more targeted and devastating way than he has been already, which has already been attacking civilians. But this **relentless** attack by cruise missiles and other really, really powerful – including Iranian-supplied kamikaze drones – this is attacking civilian energy infrastructure on a regular basis, it ’ s sometimes attacking civilian residential buildings and ordinary people are being killed and wounded. At the same time, what turned out to be a phenomenally resourceful Ukrainian population, whether civilians or experts, let ’ s just take now the engineers in the energy sector are working around the clock, day and night to try to put this, you know, back. And they ’ ve done managed, rolling blackouts and the like to try to make sure that they can save some of the energy infrastructure. And particularly, let ’ s not forget how tightly intertwined is electricity and water. Those two are yin and yang. You can ’ t really have water without electricity to pump it. And so if you lack water, fresh water just to drink, you ’ re in real big trouble. And so the Ukrainian engineers are working around the clock to try to make sure that people have at least, at least, even if it ’ s rationed, **access**. So that ’ s a big issue. I would say I ’ ve been completely and utterly overwhelmed and surprised – not surprised, delighted by the spirit of the Ukrainian people. You know, I ’ ve covered a lot of wars in which much more powerful, heavily-armed aggressors try to subjugate and force a less heavily-armed civilian population into surrender. And the Ukrainians are nowhere near that. They have so much heart, they have so much love for their country. They have so much respect, as the Iranians do, for the idea of independence, sovereignty and personal freedom and political freedom and democracy. That ’ s what they ’ re fighting for. And frankly, they are our frontline troops. The Ukrainian people right now, and to an extent, the Iranian people, are our frontline troops. In Ukraine, people are standing between freedom and totalitarianism. And if Putin succeeds in Ukraine, then he ’ s... I mean, not only do the brave Ukrainian people lose, but then he comes a country closer to the West and continues to threaten the ideas and the values and principles of democracy and freedom. And this idea that it was NATO ’ s fault and NATO expansion, I promise you that is fake news. It ’ s too difficult right now to go into it because we only have limited time. But it ’ s not that. It ’ s that Putin does not want to see a free and independent and democratic country, in this case Ukraine, on his borders, and then be asked why he can ’ t have that, why his people can ’ t have that. He doesn ’ t want that. He wants to create his own sort of greater Russia. I witnessed that covering Bosnia, where the Serbs, backed by Russia, wanted to create a greater Serbia on the backs of the Bosnian people. And in Iran, too. And you ’ ve seen the Afghan women and girls now saying,”If Iranians can stand up and demand their rights, it ’ s our turn next.”So these have huge roll-on effects and inspiration to people who believe in basic freedom, dignity and the right to live their lives. WPR : Well, I know while you were there, you sat down with Ukrainian President Volodymyr Zelensky and first lady Olena Zelenska. And I ’ m just curious what your thoughts were about their mindset at this point in the war and how you think that will impact their leadership? CA : Well, I think their leadership has been extraordinary. I mean, really, who would have thought it. This man Volodymyr Zelensky came from the entertainment space. He ran based on a program that he had been starring in, a TV series based on corruption at the highest level. And so he ran as a clean candidate. And then when he came into office, that ’ s what he tried to bring in, to increase infrastructure and to improve it and crack down on corruption, because Ukraine did have, and may still, after the war, have a corruption problem. But believe me, now, that is right on the back burner. And US senators of both parties have told me that everything that the Americans are sending and NATO is sending is accounted for, and they are not worried about it going into somebody ’ s pocket. Unlike in Russia, where it ’ s gone into, obviously, people ’ s pockets because they have barely a military machine that can operate in Ukraine, much less against NATO. Their leadership has been inspirational. The fact that the president did not leave when he could have done, he did not take his wife and children out when he could have done. And the fact that they keep”poking the bear,”and I asked him that. I said,”You know, a lot of people in the West are worried and elsewhere are worried that if you push Putin too much, you”put him in a corner”and he might do the unthinkable.”And he said,”You know what? We ’ ve lived in this **neighborhood** forever. If we ’ re not scared, you shouldn ’ t be scared.”And I was saying this because not only are they pushing him militarily back, but also if you look at the Ukrainian Defense ministry or any of the online space, they ’ re constantly trolling Putin. Constantly. And it ’ s done with incredible humor, and it ’ s very, very effective. And so they know something that we don ’ t know. And as Olena Zelensky told me,”This is our last stand. This is it. If we don ’ t stand up now, there is no other time when we can stand up. If we don ’ t do it now, when are we ever going to be able to do it?”So they ’ re very, very clear about that. As you know, there ’ s been some talk around the world, some talk in the US and Europe, elsewhere that maybe, you know, particularly after the liberation of Kherson by the Ukrainians, shouldn ’ t the Ukrainians sit down at the negotiating table? So I asked President Zelensky that, and he said,”Look, we ’ re happy to negotiate, but based on international principles, based on the Russians removing their troops from their illegally invaded, annexed and claimed territory. You know, and sham referendums and sham annexation. And by the way, we ’ re pushing them back.”So they don ’ t want to be pushed to a negotiation while they have the upper hand and one that would reward Russia for its aggression. And to be honest with you, we don ’ t want that either. We do not want Russia to get away with today what they did get away with in their first invasion of 2014, when the world did not stand up and did not challenge him. And this is the result. He thought the world was weak. And that ’ s his big surprise. He was very surprised that the Ukrainians would stand up, the West and NATO would stand up, and that this unity would last this long. And, you know, I come to it from, you know, having covered something similar in Bosnia for four years. I mean, this was an ethnic cleansing, which they ’ re trying to do in Ukraine, the Russians, along the east, and it was a genocide, which I believe that... maybe not genocide, but crimes against humanity will be placed legitimately at the feet of Russia after this war. WPR : Well in both Iran and Russia, and thinking about Ukraine, we see a suppression of free speech and **access** to information. How can we know that we ’ re receiving accurate updates on what ’ s happening in both of these places? CA : Well, I do think you and everybody have to take on a certain sense of responsibility because that is the challenge of our time. You know, people ask me : Where do we find the truth? And I ’ m like, well, you can watch CNN, you can watch PBS, you can watch BBC. You can read The Wall Street Journal or The New York Times The LA Times or any number, Financial Times, whatever. There are many, many organizations who are committed to the truth and who have reporters on the ground, including in Russia. The BBC, for instance, has a very, very accomplished and good reporter called Steve Rosenberg. I highly recommend you all to **access** his reports, because under the constraints that Russia has put on journalists, he is managing to tell the story in a very, very clever way. And also, I would, you know, just look at the Russian state media, Russian blogs and this and that, including the military bloggers who are traveling with Russian troops in Ukraine. There ' s a huge amount of information coming out that actually paints the accurate picture. And right now, they ' re very – which is not to say they want the war to end, but they ' re actually painting the story that the Russians are doing badly in Ukraine. So there are places and avenues to go. But I believe that it ' s really up to the consumer now. We can do as much as we can to bring you the truth, fact-checked, evidence-based news information. But you must not go to the fake news sites and believe that that ' s the truth. You must go to what I would call the news organizations that have earned the Good Housekeeping seal of authenticity and truth and, you know, approval. It ' s really on you all now to search for those. And it ' s not hard, they exist. WPR : As we get close to the end of our time, I just want to turn things to you, Christiane, and your thinking as a reporter, a journalist. How would you say reporting on conflict and crisis has changed the way you think about the world, your perspective on life? CA : I think it ' s changed my perspective in that I ' m very clear that I can ' t be a both-siderism, or"on the one hand / on the otherism."In other words, in certain instances, such as gross violations of international humanitarian law, which we ' re witnessing, in other words, you know, human rights atrocities, crimes against humanity, genocide and the like, you absolutely have to know where the truth is and what you ' re looking at so that you know that there are aggressors and there are **victims**. There ' s no two ways about that. So my mantra is and has been, and I ' ve developed that from Bosnia, which was one of my earliest experiences in the field, my mantra is"truthful, not neutral,"which doesn ' t mean to say I ' m not being objective. I am being objective because objective is our golden rule. And by that I mean you have to look at all sides of the story and report all sides of the story. But what you mustn ' t do is create any false equivalence, either factually or morally. And when you do do that, you are actually being untruthful and in some cases you ' re being an accessory to terrible atrocities. And I would say the same about climate. I mean, you can talk about any of these crises, these moral and existential crises that we find. You know, climate has been too long, for decades been treated as a both-siderism, that the deniers had **equal** factual or moral weight as the science. And it ' s just, as we know, not true, and so much time has been wasted to the point that we are on the brink of a global catastrophe. And that is something that I ' ve learned. That in order to be a trustworthy, credible journalist, I must, must call out the truth. WPR : Well, I want to end with a question from a member, which is basically,"If you were to interview yourself, what would be the last question you would ask?"So here ' s your last question, what is your last question for Christiane? CA : Oh, Lordy. My last question, I don ' t know, but I do know one thing. That I wish that I would be able to – I guess it would be, how can you get more **access** to the other side, you know? I want to say the bad actors, because they are. How do you get more **access** to them? You know, I would like to sit down with a Putin or a Kim Jong-un or whoever, Viktor Orban in Hungary or whoever is denying democracy in the United States and the like. I would like to sit and talk to them to try to understand. Not because I think that I would be, you know, I would suddenly become a neutral observer, but because I want them to spell it out so the people can see and I would question them rigorously, that they would see the bankruptcy of their positions. And I would like to keep doing that with fossil fuel executives and the like. I ’ ve done a few and with governments, you know, who have allowed politics, politics to endanger us all. I do believe those people need to be held accountable. So I would like to be able to ask those people more questions. WPR : Well, Christiane, thank you so much for taking the time to **chat** with us today, I think I speak for everyone, just appreciating everything that you ’ ve shared. and good luck with the rest of your day. CA : Thank you very much, I appreciate it. It was good to talk to you all, thanks Whitney. [Want to support TED?] [Become a TED Member!] [Learn more at ted. com / membership]

Text Sample 358: POS Dim 3 – 2022 – Score 14.00 – 2022/t003793.txt
Bruno Giussani : We are at the end of day six of the war in Ukraine or, more correctly, of the Russian invasion of Ukraine, launched on February 24 by President Vladimir Putin. We are all shocked and saddened by the events and by the human **suffering** they are causing. And as we speak, really, a Russian military convoy is headed towards Kyiv, other Ukrainian cities are being bombarded, half a million Ukrainians have already fled to neighboring countries and much more. It ’ s still early days, and it ’ s difficult to predict how the situation will evolve even just in the next few hours. But this is a war that should concern everyone, everywhere. And so today, in this TED Membership conversation, we want to try to give it a broader context with our guest, historian and author, Yuval Noah Harari. Yuval, welcome. Yuval Noah Harari : Hello. Thank you for inviting me. BG : I want to start from Ukraine itself and its 42 million people and its particular place between the East and the West. What do we need to know about Ukraine to understand this war and what ’ s at stake? YNH : The most crucial thing to know is that Ukrainians are not Russians, and that Ukraine is an ancient, independent nation. Ukraine has a history of more than a thousand years. Kyiv was a major metropolis and cultural center when Moscow was not even a village. For most of these thousand years Kyiv was not ruled by Moscow. They were not part of the same political entity. For centuries, Kyiv was looking westwards and was a part of a union with Lithuania and Poland until it was eventually conquered and absorbed by the Russian Empire, by the czarist empire. But even after that, Ukrainians remained a separate people to a large extent, and it ’ s important to know that because this is really what is at stake in this war. The key issue of the war, at least for President Putin, is whether Ukraine is an independent nation, whether it is a nation at all. He has this fantasy that Ukraine isn ’ t a nation, that Ukraine is just a part of Russia, that Ukrainians are Russians. In his fantasy, Ukrainians are Russians that want to be back in the fold of Mother Russia, and that the only ones preventing it is a very small gang at the top, which he portrays as Nazis, even if the president is Jewish ; but OK, a Nazi Jew. And his belief was, at least, that he just needs to invade, Zelenskyy will flee, the government will collapse, the army would lay down its arms, and the Ukrainian people would welcome the Russian liberators, throwing flowers on them. And this fantasy has been shattered already. Zelenskyy hasn ’ t fled, the Ukrainian army is fighting. And the Ukrainian people is not throwing flowers on the Russian tanks, it ’ s throwing Molotov cocktails. BG : So let ’ s unpack that and maybe take the different pieces one way one. So Ukraine has a long history of being dominated and occupied. You mentioned the czar, but also the Soviet Union, Hitler ’ s armies. It also has a long

history of mistrust of authority and of resistance, which goes some way to explain the current strong resistance that the Russians are encountering. Anne Applebaum, the journalist, even suggests that this mistrust, this resistance to authority, is the very essence of Ukraine-ness, do you agree? YNH : We did see in the last 30 years Ukrainians twice rising in revolt when there was a danger of an **authoritarian** regime being established – once in 2004, once in 2013. And when I was in Kyiv a few years ago, what really struck me was this very strong feeling of the desire for independence and for democracy. And I remember walking around this museum of the Revolution of 2013-2014 and seeing these images, like these two elderly women who were bringing sandwiches to the demonstrators, to the fighters. They couldn ’ t throw stones and they couldn ’ t do anything else, so they prepared sandwiches and brought this huge tray full of sandwiches to the demonstrators. And this, yes, this is the kind of spirit that inspires not just the Ukrainians but everybody who is now watching what is happening there. BG : Help me understand the actual nature of the threat here in terms of Russia moving into Ukraine. So in your last book, when you write about Russia, you describe the Russian model as :"not a coherent political ideology, but rather a sort of practice of monopolizing power and wealth by a small group at the top."But then, in his actions against Ukraine, Putin in the last few weeks seems to move very much by an ideology, an ideology of empire, of denial of Ukraine ’ s right to exist, as you mention. What has changed in the four years since you wrote that book? YNH : The imperial dream was always there, but you know, empires are often the creation of a very small gang of people at the top. I don ’ t think the Russian people [are] interested in this war. I don ’ t think that the Russian people want to conquer Ukraine or to slaughter the citizens of Kyiv. It ’ s all coming from the top. So there is no change there. I mean, when you look at the Soviet Union, you can say that there was this mass ideology, which was shared by a large proportion, or some proportion, of the population. You don ’ t see this now. You know, Russia is a very rich country, rich in resources, but most people are very poor. Their standard of living is very, very low because all the wealth and power is kind of sucked by the people at the top, and very little is left for everybody else. So I don ’ t think it ’ s a society where the masses are part of this kind of ideological project. They ’ re being ruled from the top. And you have this classic imperial situation, when the emperor, which controls the largest country in the world, feels that,"Hey, this is not enough. I need more."And sends his army to capture, to extend the empire. BG : I said at the beginning that it ’ s difficult, of course, to make predictions. But yesterday, you published an article in"The Guardian"titled :"Why Putin has already lost this war."Please explain. YNH : Well, one thing should be very clear. I don ’ t mean to say that he ’ s going to suffer an immediate military defeat. He definitely has the military power to conquer Kyiv and perhaps the whole of Ukraine. Unfortunately, we might see this. But his long-term goal, the whole rationale of the war, is to deny the existence of the Ukrainian nation and to absorb it into Russia. And to do that, it ’ s not enough to conquer Ukraine. You also need to hold it. And it ’ s all based on this fantasy, on this gamble, that most of the population in Ukraine would agree to this, would even welcome this. And we already know that it ’ s not true. That the Ukrainians are a very real nation ; they are fiercely independent ; they don ’ t want to be part of Russia ; they will fight like hell. And in the long-run, again, you can conquer a country, But as the Russians learned in Afghanistan, as the Americans learned also in Afghanistan, also in Iraq, it ’ s much harder to hold a country. And again, the big question mark before the war was always this. Before the war started, many things were already known. Everybody knew that the Russian army is much stronger than the Ukrainian Army. Everybody knew that

NATO will not send **armed** forces into Ukraine, troops into Ukraine. Everybody knew that the West, the Europeans, would be hesitant about imposing too strict a sanction regime for fear of being hurt by it themselves. And this was the basis for Putin ' s war plan. But there was one big unknown. Nobody could say for sure how the Ukrainian people would react. And there was always the option that maybe Putin ' s fantasy would come true. Maybe the Russians will march in, Zelenskyy would flee, maybe the Ukrainian army will just capitulate and the population would not do much. This was always an option. And now we know this was just fantasy. Now we know that the Ukrainians are fighting, they will fight. And this derails the whole rationale of Putin ' s war. Because you can conquer the country, maybe, but you won ' t be able to absorb Ukraine back into Russia. The only thing he ' s accomplishing, he is planting seeds of **hatred** in the hearts of every Ukrainian. Every Ukrainian being killed, every day this war continues is more seeds of **hatred** that may last for generations. Ukrainians and Russians didn ' t hate each other before Putin. They ' re **siblings**. Now he ' s making them enemies. And if he continues, this will be his legacy. BG : We ' re going to talk a bit about that again later but, you know, at the same time, Putin needs a victory, right? The cost, the human, economic, political cost of this war, not even a week in, is already astronomical. So to justify it and also to remain, by the way, a viable leader at the head of Russia, Putin needs to win, and even win convincingly. So how do we square these things? YNH : I don ' t know. I mean, the fact that you need to win doesn ' t mean that you can win. Lots of political leaders need to win, and sometimes they lose. He could stop the war, **declare** that he won, and say that recognizing Luhansk and Donetsk by the Russians is what he really wanted all along, and he achieved this. Maybe they cobble this agreement, or I don ' t know. This is the job of politicians, I ' m not a politician. But I can tell you that I hope, for the sake of everybody – Ukrainians, Russians and the whole of humanity – that this war stops immediately. Because if it doesn ' t, it ' s not only the Ukrainians and the Russians that will suffer terribly. Everybody will suffer terribly if this war continues. BG : Explain why. YNH : Because of the shock waves destabilizing the whole world. Let ' s start with the bottom line : budgets. We have been living in an amazing era of peace in the last few decades. And it wasn ' t some kind of hippie fantasy. You saw it in the bottom line. You saw it in the budgets. In Europe, in the European Union, the average defense budget of EU members was around three percent of government budget. And that ' s a historical miracle, almost. For most of history, the budget of kings and emperors and sultans, like 50 percent, 80 percent goes to war, goes to the army. In Europe, it ' s just three percent. In the whole world, the average is about six percent, I think, fact-check me on this, but this is the figure that I know, six percent. What we saw already within a few days, Germany doubles its military budget in a day. And I ' m not against it. Given what they are facing, it ' s reasonable. For the Germans, for the Poles, for all of Europe to double their budgets. And you see other countries around the world doing the same thing. But this is, you know, a race to the bottom. When they double their budgets, other countries look and feel insecure and double their budgets, so they have to double them again and triple them. And the money that should go to health care, that should go to education, that should go to fight climate change, this money will now go to tanks, to missiles, to fighting wars. So there is less health care for everybody, and there is maybe no solution to climate change because the money goes to tanks. And in this way, even if you live in Australia, even if you live in Brazil, you will feel the repercussions of this war in less health care, in a deteriorating ecological crisis, in many other things. Again, another very central question is technology. We are on the verge, we are already in the middle, actually, of new technological arms races in fields like artificial intelligence. And we need global agreement about how to regulate AI and to prevent the worst scenarios. How can we get a global agreement on AI when you have a new cold war, a new hot war? So in this field, to all hopes of stopping the AI arms race will go up in smoke if this war continues. So again, everybody around the world will feel the consequences in many ways. This is much, much bigger than just another regional conflict. BG : If one of Putin ’ s goals here is to divide Europe, to weaken the transatlantic alliance and the global liberal order, he seems to kind of accidentally have revitalized all of them in a way. US-EU relations have never been so close in many years. And so how do you read that? YNH : Well, again, in this sense, he also lost the war. If his aim was to divide Europe, to divide NATO, he ’ s achieved exactly the opposite. I mean, I was amazed by how quick, how strong and how unanimous the European reaction was. I think the Europeans surprised themselves. You even see countries like Finland and Sweden sending arms to Ukraine and closing their airspace. They didn ’ t even do it in the Cold War. It ’ s really amazing to see it. I think another very important thing is what has been dividing the West over the several years now, it ’ s what people term the”culture war”. The culture war between left and right, between conservatives and liberals. And I think this war can be an opportunity to end the culture war within the West, to make peace in the culture war. First of all, because you suddenly realize we are all in this together. There are much bigger things in the world than these arguments between left and right within the Western democracies. And it ’ s a reminder that we need to stand united to protect Western liberal democracies. But it ’ s deeper than that. Much of the argument between left and right seemed to be in terms of a contradiction between liberalism and nationalism. Like, you need to choose. And the right goes with nationalism, and the left goes more liberalism. And Ukraine is a reminder that no, the two actually go together. Historically, nationalism and liberalism are not opposites. They are not enemies. They are friends, they go together. They meet around the central value of freedom, of liberty. And to see a nation fighting for its survival, fighting for its freedom, you see it on Fox News or you see it in CNN. And yes, they tell the story a little differently, but they suddenly see the same reality. And they find common ground. And the common ground is to understand that nationalism is not about hating minorities or hating foreigners, it ’ s about loving your compatriots, and reaching a peaceful agreement about how we want to run our country together. And I hope that seeing what is happening would help to end the culture war in the West. And if this happens, we don ’ t need to worry about anything. You know, when you look at the real power balance, if the Europeans stick together, if the Americans and the Europeans stick together and stop this culture war and stop tearing themselves apart, they have absolutely nothing to fear – the Russians or anybody else. BG : I ’ m going to ask you a question later about the stories the West tells itself, but let me zoom out for a second and get a larger perspective. You wrote another essay last week in”The Economist”, and you argue that what ’ s at stake in Ukraine is, and I quote you,”the direction of human history”because it puts at risk what you call the greatest political and moral achievement of modern civilization, which is the decline of war. So now we are back in a war and potentially afterwards into a new form of cold war or hot war, but hopefully not. Elaborate about that essay you wrote. YNH : Yeah, I mean, some people think that all this talk about the decline of war was always just a fantasy. But... Again, you look at the statistics. Since 1945, there has not been a single clash between superpowers, whereas previously in history, this was, you know, the basic stuff of history. Since 1945, not a single internationally recognized country was wiped off the map by external invasion. This was the common thing in history. Until then and then it stopped. This is an amazing achievement, which is the basis for everything we have, for our medical services, for education system, and this is all now in jeopardy. Because this era of peace, it wasn ’ t the result of some miracle. It wasn ’ t the result of a change in the laws of nature. It was humans making better decisions and building better institutions, which means also that there is no guarantee for the future. If humans, some humans, start making bad decisions and start destroying the institutions that kept the peace, then we will be back in the era of war with budgets, military budgets going to 20, 30, 40 percent. It can happen. It ’ s in our hands. And I ’ ll just say one more thing, When, not just me, but other scholars like Steven Pinker and others, talked about the era of peace, some people understood it as kind of encouraging complacency. That, oh, we don ’ t need to worry about anything. No, I mean, the message was really the opposite. It was a message of responsibility. If you think that there is no era of peace in history, it ’ s always war, it ’ s always the jungle, there is a constant level of **violence** in nature, then this basically means that there is no point struggling for peace and there is no responsibility on leaders like Putin because you can ’ t blame Putin for the war. It ’ s just a law of nature that there are wars. When you realize, no, humans are able to decrease the level of **violence**, then it should make us much more responsible. And it should also make us understand that the war in Ukraine now, it ’ s not a natural disaster. It ’ s a man-made disaster, and a single man. It ’ s not the Russian people who want this war. There ’ s really just a single person who, by his decisions, created this tragedy. BG : So one of the things that has come back in the last weeks and months is the nuclear threat. It ’ s moved back into the center of political and strategic considerations. Putin has talked about it several times, the other day he ordered Russia ’ s nuclear forces on a higher alert status. President Zelenskyy himself at the Munich Security Conference essentially said that Ukraine had made a mistake abandoning the nuclear weapons it had inherited from the Soviet Union. That ’ s a statement that I suspect many countries are pondering. What ’ s your thinking about the return of the nuclear threat? YNH : It ’ s extremely frightening. You know, it ’ s like it ’ s almost Freudian, it ’ s the return of the repressed. We thought that, oh, nuclear weapons, yes, there was something about that in the 1960s with the Cuban Missile Crisis and Dr. Strangelove. But no, it ’ s here. And, you know, it took just a few days of difficulties on the battlefield for suddenly – I mean, I ’ m watching television, like, the news and you have these experts explaining to people what different nuclear weapons will do to this city or to this country. It rushed back in. So, you know, nuclear weapons are – in a way they also, until now, preserved the peace of the world. I belong to the school of thought that if it was not for nuclear weapons, we would have had the Third World War between the Soviet Union and the United States and NATO sometime in the 1950s or ’ 60s. That nuclear weapons actually, until today, served a good function. It ’ s because of nuclear weapons that we did not have any more direct clashes between superpowers because it was obvious that this would be collective suicide. But the danger is still there, it ’ s always there. If there is miscalculation, then the results could, of course, be existential, catastrophic. BG : And at the same time, you know, in the ’ 70s after Cuba and Berlin, and so in the ’ 60s, but in the ’ 70s, we started building a sort of international institutional architecture that helped reduce the risk of military confrontation of nuclear weapons, we used, you know, anything from arms control agreements to measures designed to build trust or to communicate directly and so on. And then in the last decade or so, that has been progressively kind of scrapped, so we are even in a more dangerous situation than we were let ’ s say, at the end of the last century. YNH : Completely, I mean, we are now reaping the bad fruits of neglect that ' s been going on for several years, not just about nuclear weapons, but in general, about international institutions and global cooperation. We ' ve built, in the late 20th century, a house for humanity based on cooperation, based on collaboration, based on the understanding that our future depends on being able to cooperate, otherwise we will become extinct as a species. And we all live in this house. But in the last few years we stopped – we neglected it, we stopped repairing it. We allow it to deteriorate more and more. And, you know, eventually it will – It is collapsing now. So I hope that people will realize before it ' s too late that we need not just to stop this terrible war, we need to rebuild the institutions, we need to repair the global house in which we all live together. If it falls down, we all die. BG : So we have, among the audience listening, Rola from – I don ' t know where she ' s from, she grew up in Lebanon – and she said,"I lived the war, I slept on the ground, I breathed fear. All the reasons were explained to me that the only remaining learning came the war is absurd. We talk about strategy, power, budgets, opportunities, technologies. What about human **suffering** and psychological **trauma**?"Especially, I assume what she ' s asking is what about, what ' s going to remain, in terms of the human **suffering** and the psychological **trauma** going forward? YNH : Yeah, I mean, these are the seeds of **hatred** and fear and misery that are being planted right now in the minds and the bodies of tens of millions, hundreds of millions of people, really. Because it ' s not just the people in Ukraine, it ' s also in the countries around, all over the world. And these seeds will give a terrible harvest, terrible fruits in years, in decades to come. This is why it ' s so crucial to stop the war immediately. Every day this continues, plants more and more of these seeds. And, you know, like this war now, its seeds were, to a large extent, planted decades and even centuries ago. That part of the Russian fears that are motivating Putin and motivating people around him is memories of past invasions of Russia, especially, of course, in Second World War. And of course, it ' s a terrible mistake what they are doing with it. They are recreating again the same things that they should learn to avoid. But yes, these are still the terrible fruits of the seeds being planted in the 1940s. BG : It ' s what in same article you call the fact that nations are ultimately built on stories. So these seeds are the stories we are starting to create now. The war in Ukraine is starting to create the stories that are going to have an impact in the future, that ' s what you ' re saying. YNH : Some of the seeds of this war were planted in the siege of Leningrad. And now it gives fruit in the siege of Kyiv, which may give fruit in 40 or 50 years in more terrible... We need to cut this, we need to stop this. You know, as a historian, I feel sometimes ashamed or responsible, I don ' t know what, about what history, the knowledge of history is doing to people. In recent weeks, I have been watching all the world leaders talking with Putin, and very often he gave them lectures on history. I think that Macron had a discussion with him for five hours, and afterwards, said,"Most of the time he was lecturing me about history."And as a historian, I feel ashamed that this is what my profession in some way is doing. I know it for my own country. In Israel, we also suffer from too much history. I think people should be liberated from the past, not constantly repeating it again and again. You know, everybody should kind of free themselves from the memories of the Second World War. It ' s true of the Russians, it ' s also true of the Germans. You know, I look at Germany now, and what I really want to say, if there are Germans watching us, what I really want to say to the Germans : guys, we know you are not Nazis. You don ' t need to keep proving it again and again. What we need from Germany now is to stand up and be a leader, to be at the forefront of the struggle for freedom. And sometimes Germans are afraid that if they speak forcefully or pick up a gun, everybody will say,"Hey, you ' re Nazis again."No, we won ' t think that.

BG : That ’ s happening right now. I mean, lots of things that were inconceivable just 10 days ago have happened in the last few weeks. And one of the most striking, to me in any case, is Germany ’ s reaction and transformation. I mean, the new chancellor, Olaf Scholz, the other day announced that Germany will send arms to Ukraine, and will spend an extra 100 billion dollars in building up its army. That reverses completely the principles that have guided Germany ’ s foreign policy and security politics for decades. So that shift is happening exactly at this moment and very, very fast. YNH : Yeah. And I think it ’ s a good thing. We need the Germans to... I mean, they are now the leaders of Europe, certainly after Britain left in Brexit. And we need them to, in a way, let go of the past and be in the present. If there is really one country in the world that, as a Jew, as an Israeli, as a historian, that I trust it not to repeat the horrors of Nazism, that ’ s Germany. BG : Yuval, I want to touch quickly on three things that have to do with the fact that this feels like the first truly interconnected war in many ways. The first, of course, is the basics, which is, on one side, you have a very ancient war – we have tanks and we have trenches and we have bombed buildings – and on the other, we have real-time visibility of everything through cell phones and **Twitter** and TikTok and so on. And you have written a lot about this tension between old ways and new tech. What ’ s the impact here? YNH : First of all, we don ’ t know everything that is happening. I mean, surprisingly, with all this TikTok and phones and everything, so much is not known. So the fog of war is still there, and yes, there is much more information, but information isn ’ t truth. Lots of information is disinformation and fake news and so forth. And yes, it ’ s always like this ; the new and the old, they come together. You know, with all the talk about interconnectedness and living in cyberspace and all that, one of the most important technologies not just of this war, but of the last decade or two have been stone walls. It ’ s Neolithic. Everybody is now building stone walls in the era of Facebook and Google and all that. So the old and the new, they go together. And it ’ s... It is a new kind of war. People are sitting at home in California or Australia, and they actively participate in the war, not just by writing tweets, but by attacking websites or defending websites. You know, in Spain, in the Civil War, if you wanted to help fight fascism, then you had to go to Spain and join the international brigade. Now the international brigade is sitting at home in San Francisco and is still in some way part of the war. So this is definitely new. BG : So indeed, just two days ago, Ukraine ’ s deputy prime minister, I think, Fedorov, announced via Telegram that he wanted to create a sort of volunteer cyber army. He invited software developers and hackers and other people with IT skills to somehow help Ukraine fight on the cyber front. And according to”Wired”magazine, in less than two days, 175, 000 people signed up. So here is a defending nation that can kind of recruit almost overnight, 175, 000 volunteers to go to battle on his behalf. It ’ s a very different kind of war. YNH : Yeah. You know, every war brings it surprises. Sometimes it ’ s how everything is new, but sometimes it ’ s also how everything is old. BG : So a few people in the **chat** and in the Q and A, have mentioned China, which of course, is an important actor here, although for now is mostly an observer. But China has a stated policy of opposing any act that violates territorial integrity. So moving into Ukraine, of course, violates territorial integrity. And it also has a huge interest in a stable global economy and global system. But then it needs to square this with the recent closeness with Russia. Xi Jinping and Putin met in Beijing before the Olympics, for example, and kind of had this message of friendship that went out to the world. How do you read China ’ s position in this conflict? YNH : I don ’ t know, I mean, I ’ m not an expert on China, and I certainly can ’ t just... You know, just reading the news won ’ t get you into the mindset, into the real opinions and positions of the Chinese leadership. I hope that they take a responsible position. And act – because they are close to Russia, they are also close to Ukraine, but especially because they are close to Russia, they have a lot of influence on Russia, I hope that they will be the responsible adults that will put down the flames of this war. They have a lot to lose from a breakdown of the global order. And I think they have a lot to win from the return of peace, including in terms of the gratitude of the international community. Now, whether they do it or not, this is with them. I can ' t predict, but I hope so. BG : You have mentioned before the several European and Western leaders that have gone to Moscow in the weeks before the invasion. Varun in the **chat**, asks,"Is the Ukraine war a failure of diplomacy?"Could have... Something different happened? YNH : Oh, you can understand it in two questions. Did diplomacy fail to stop the war? Absolutely, everybody knows that. But is it a failure in the sense that a different diplomatic approach, some kind of other proposition, would have stopped the war? I don ' t know, but it doesn ' t seem like it. I mean, looking at the events of the last few weeks, it doesn ' t seem that Putin was really interested in a diplomatic solution. It seemed that he was really interested in the war, and I think, again, it goes back to this basic fantasy that if he really was concerned about the security situation of Russia, then there was no need to immediately invade Ukraine. There was no immediate threat to Russia. There was no discussion of right now, Ukraine joining NATO. There was no invasion army assembling in the Baltic states or in Poland. Nothing. Putin chose the moment to start this crisis. So this is why it doesn ' t seem that it ' s really about the security concerns. It seems more about this very deep fantasy of re-establishing the Russian Empire and of denying the very existence of the Ukrainian nation. BG : So you live in the Middle East. Someone else in the **chat** asks,"What makes the situation so unique compared to many other wars that are going on right now in the world?"I would say, aside from the nuclear threat from Russia, but what else? YNH : Several things. First of all, we have here, again, something we haven ' t seen since 1945, which is a dominant power trying to basically obliterate from the map an independent country. You know, when the US invaded Afghanistan or when the US invaded Iraq, you can say a lot of things about it and criticize it in many ways. There was no question of the US annexing Iraq or turning Iraq into the 51st state of the United States. This is what is happening in Ukraine under this pretext or this disguise, this is what ' s at stake. The real aim is to annex Ukraine. If this succeeds, again, it brings us back to the era of war. I was struck by what the Kenyan representative to the UN Security Council said when this erupted. The Kenyan representative spoke in the name of Kenya and other African countries. And he told the Russians : Look, we also are the product of a post-imperial order. The same way the Soviet empire collapsed into different independent nations, also, African nations came out of the collapse of European empires. And the basic principle of African politics ever since then was that no matter what your objections to the borders you have inherited, keep the borders. The borders are sacred because if we start invading neighboring countries because,"Hey, this is part of our countries, these people are part of our nation,"there will not be an end to it. And if this now happens in Ukraine, it will be a blueprint for copycats all over the world. The other thing which is different is that we are talking about superpowers. This is not a war between Israel and Hezbollah. This is potentially a war between Russia and NATO. And even leaving aside nuclear weapons, this completely destabilizes the peace of the entire world. And again, I go back again and again to the budgets. That if Germany doubles its defense budget, if Poland doubles its defense budget, this will spread to every country in the world, and this is terrible news. BG : So Yuval, I ' m jumping from topic to topic because I want to use the last few minutes to ask a few questions from the audience. A few people are asking about the link to the climate crisis, particularly when it relates to the energy flows. Like, Europe is very dependent, part of Europe, is very dependent on Russian oil and gas, which is, as far as we know, still flowing until today. But could this crisis, in a sort of paradoxical way, a bit like the pandemic, accelerate climate action, accelerate renewables and and so on? YNH : This is the hope. That Europe now realizes the danger and starts a green Manhattan Project that kind of accelerates what already has been happening, but accelerates it, the development of better energy sources, better energy infrastructure, which would release it from its dependence on oil and gas. And it will actually undercut the dependence of the whole world on oil and gas. And this would be the best way to undermine the Putin regime and the Putin war machine, because this is what Russia has, oil and gas. That ' s it. When was the last time you bought anything made in Russia? They have oil and gas, and we know, you know, the curse of oil. That oil is a source of riches, but it ' s also very often a support for dictatorships. Because to enjoy the benefits of oil, you don ' t need to share it with your citizens. You don ' t need an open society, you don ' t need education, you just need to drill. So we see in many places that oil and gas are actually the basis for dictatorships. If oil and gas, if the price drops, if they become irrelevant, it will not only undercut the finance, the power of the Russian military machine, it will also force Russia, force Putin or the Russians to change their regime. BG : OK, let me bring up a character that everybody here in the **chat** seems to find quite heroic, and that ' s the Ukrainian president. So Ukraine kind of finds itself with a comedian who turned almost accidental president, who turned now war president. But he has shown an impressive conduct in the last few weeks, especially in the last few days, which can be summarized in that response he gave to the US when they offered to kind of exfiltrate him so he could lead a government in exile, he said,"I need ammunition. I don ' t need a ride."How would you look at President Zelenskyy? YNH : His conduct has indeed been admirable, and he gives courage and inspiration not just to the Ukrainian people, but I think to everybody around the world. I think to a large extent the swift and united reaction of Europe with the sanctions and sending arms and so forth, to a large extent, this is also to the credit of Zelenskyy. That, you know, when politicians are also human beings. And his direct appeal to them, and you know, they met him many times in person and to see where he is now and the threat that not only him, but his family is also in. And you know, they talk with him, and he says, and they know, that this may be the last time they speak. He may be dead, murdered or bombed in an hour or in a day. It really changes something. So in this sense, I think he made a huge personal contribution, to not just the reaction in Ukraine, but around the world. BG : So Sam, who ' s listening, asked this question :"Can you provide some historical context for the force and the meaning of economic and trade sanctions at the level where they are currently imposed. How have previous would-be empires, would-be aggressors, or aggressors, been constrained by such isolations and such sanctions?"YNH : You know, what we need, again, to realize about Putin ' s Russia is that it ' s not the Soviet Union. It ' s a much smaller and weaker country. It ' s not like in the 1960s, that in addition to the Soviet Union, you had the entire Soviet bloc around it. So it ' s easier in this sense to isolate it. It ' s much more vulnerable. Again, does it mean that sanctions would work like a miracle and stop the tanks? No. It takes time. But I think that the West is in a position to impact Russia with these kinds of sanctions and isolation much more than, let ' s say, with the Soviet Union. And also the Russian people are different. The Russian people don ' t really want this war, even the people in the immediate circle around Putin. You know, again, I don ' t know them personally, from what it seems, it ’ s that these people, they like life. They have their yachts and they have their private airplanes and they have their house in London and they have their chateau in France. And they like the good life, and they want to keep enjoying it. So I think that the sanctions can be really effective. What ’ s the timetable? That ’ s ultimately in the hands of Putin. BG : So Gabriella asks :”I remember the war in former Yugoslavia and the atrocities there. Is there any possibility that this war would escalate into such a situation?”I think an extension to that is : Is this war kind of stirring dormant conflicts like in the Balkans, for example, or in the former Central Asian republic? YNH : Unfortunately, it can get to that level and even worse. If you want an analogy, go to Syria. You look at what happened in Homs. At what happened in Aleppo. And this was done by Putin and his airplanes and his minions in Syria. It ’ s the same person behind it. And to think that,”No, no, no, this happened in the Middle East. It can ’ t happen in Europe.”No. We could see Kyiv in the same situation as Homs, as the same situation as Aleppo, which would be catastrophic, and, again, would plant terrible seeds of **hatred** for years and decades. So far, we ’ ve seen hundreds of people being killed, Ukrainian citizens being killed. It could reach tens of thousands, hundreds of thousands. So in this sense, it ’ s extremely painful to contemplate. And this is why we need again and again to urge the leaders to stop this war, and especially, again and again, tell Putin,”You will not be able to absorb Ukraine into Russia. They don ’ t want it, they don ’ t want you. If you continue, the only thing you will achieve is to create terrible **hatred** between Ukrainians and Russians for generations. It doesn ’ t have to be like that.”BG : Yuval, let me finish with one question about your county. You are in Israel. Israel has close ties with both Russia and Ukraine. It ’ s actually home of many Russian-born and many Ukrainian-born Jews. How is the country reacting to this conflict, I ’ m talking about the government, but also about the population? YNH : Actually, I ’ m not the best person to ask. I ’ ve been so, kind of, following what ’ s happening around the world, I didn ’ t pay so much attention to what is happening right here. And even though I live here, I ’ m not an expert on Israeli society or Israeli politics. Definitely, the sentiment in the **street**, in the social media is with Ukraine. You see Ukrainian flags, you see on social media people putting Ukrainian flags on their accounts. And another thing, so many people in Israel, they came from the former Soviet Union. And until now, everybody was simply known as Russians. You know, even if you came from Azerbaijan or you came from Bukhara, you were a Russian. And suddenly,”No, no, no, no, no. I ’ m not Russian. I ’ m Ukrainian.”And again, these seeds of **hatred** that Putin is planting, it ’ s reaching also here. That suddenly people are saying no, Russian, Ukrainian, until a very short time ago, it ’ s the same thing. No, it ’ s not the same thing. So the shock waves are spreading. BG : Yuval, thank you for taking the time and being with us today and sharing your knowledge and your views on the situation. Thank you very much. YNH : Thank you and I hope for peace quickly. BG : We all do. Thank you. [Get **access** to **thought-provoking** events you won ’ t want to miss.] [Become a TED Member at ted. com / membership]

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Marisa Franco : Now, as an expert on friendship, I ’ m up against a lot because of the hierarchy that a lot of our cultures place on love, right? With familial love at the top, with romantic love at the top, and with platonic love, friendship love, really at the bottom. And with so many countries, people feeling so lonely and so disconnected, I believe that if we leave friendship at the bottom of this hierarchy, it ’ s like there ’ s gold at our feet that we ’ re treating as concrete. And so why are friendships so key? Well, our bodies have always known that we need an entire community to feel whole. And just being around a spouse, for example, only surfaces one side of ourselves. So maybe the part of me that likes to garden or do yoga will begin to wither away if my spouse, for example, doesn ’ t like these activities. But then when I ’ m around a friend, I can garden and plant my pothos with them or around another friend that I can [do] downward facing dog with. And I feel my entire identity, accordion outward, unfold and fan out. And I experience the full richness and complexities of who I am when I have an entire community to bring that out in me. And so that ’ s one of the reasons why friendship is really important. But I think there ’ s two reasons why we tend to really devalue it. One reason is because we just don ’ t know how to make friends. So luckily, I am going to help you with that a little bit today. But the other reason has to do with something I like to call the”paradox of people.”That on the one hand we need people, they make us feel healthy, they make us feel connected, they make us feel like our very selves, right? But on the other hand, people are really scary. They can **dismiss** us, they can reject us, they can actively **harm** us. And so this sort of dilemma that we face, the sort of entity that we need the most is also the entity that can **harm** us the most. And how we walk across this tightrope handling this paradox of people says a lot about our ability to make and keep friends. Because if we find ourselves stuck in the place where we see people as – we mistrust people, or we see people as potentially rejecting us and **harming** us, it ’ s really hard to foster connection. And this really materialized for me one day when I had bought an apartment, and I was really excited to make friends with my neighbors because I ’ m like,”I ’ m going to be here for a while.”And I see a couple of my neighbors in the hallway, when I ’ m walking home into my apartment with my ex at the time who was living with me. And I walk right past them, right? Because paradox of people, I ’ m scared of them, they might reject me. They might see me as weird if I try to introduce myself. So I scurry into my apartment and my ex, he pushes me back into the hallway to talk to my neighbors and says to me, you know,”You ’ re writing this book on friends. What would you tell other people who are in your situation?”And so as I ’ m sort of stumbling back into the hallway, I ’ m thinking about a few things that I have learned through studying friendship so intensely. And so two observations that I have on friendship and two takeaways for what we can do to make and keep friends. First observation, friendship does not happen organically in adulthood, right? And in fact, one study found that people that think that it happens based on luck are actually lonelier five years later, whereas people that see it as happening based on effort are less lonely five years later. So what does that tell me? That if I was just there hoping that my neighbors would someday try to be my friends, it probably wouldn ’ t happen, right? And so I would need to make that effort in order to be able to make friends. But second observation that I have, based on reading all the research on friendship, is something called the”liking gap,”which is a phenomenon wherein when strangers interact and predict how likely the other person is to like them, they underestimate how much the other person likes them. So this research really suggests that we ’ re less likely to be rejected than we think. Which leads me to my first takeaway for making friends. If you want to make friends, you have to assume that people like you, right? The reason is, when researchers told people,”Hey, you ’ re going to go into this group, and based on your personality profile, we predict that you will be liked.”This was completely bogus, a total lie. But they found that when people went into this group of people, they became warmer, open, more friendly, because they made this assumption. And so indeed, it became this self-fulfilling prophecy called the”acceptance prophecy”. And when we assume we ’ ll be liked, we make it more likely that we actually will be liked. Whereas other research finds that people that tend to assume that they ’ re rejected, even when the circumstance is ambiguous, like, my friend, maybe they ’ re just, like, hungry or something, rather than that they hate me, right? Those people that go straight to”maybe they don ’ t like me,”they actually become cold, they actually become withdrawn and they reject people. And then they get rejected in return. So I ’ m thinking of these things in the hallway, you know, right by my neighbors. I ’ m thinking that, you know, I can ’ t wait for this to happen organically. OK, I ’ m afraid they ’ re going to reject me, but that ’ s less likely to happen than I actually think. I should assume that they ’ re going to like me. And then one last thing I have to remind myself of was to overcome something called”covert avoidance,”which is our tendency to show up around other people physically, but check out mentally, right? Like, you ’ re hanging out with people and you ’ re on your phone, or that would look like me just standing in the hallway hoping that my neighbors talk to me. And so to make friends, you have to overcome covert avoidance by not just showing up, I showed up in that hallway, right, but you also have to engage with people when you get there. So I ended up approaching my neighbors and saying,”Hi, I ’ m Marisa, I just moved into 103, It ’ s really great to meet you.”And we start **chatting**. And at some point, you know, I asked them, like,”Is there a group where we can keep in touch? I ’ d love to, you know, **chat** further.”And they tell me about their cat group that they have for cat parents in the Drew. And I don ’ t have a cat, but like, I ’ ll take connection when I can find it. So the cat group became half cat group, half social group. And I think sometimes we think that you know, a tiny act, something small like saying hello can have colossal consequences for our life. But when we can lean into the sort of positive side of the paradox of people, when we can initiate and assume people like us, right, it can have colossal consequences. Because since I said that”hello,”me and my neighbors, we met up and we hung out every Friday, socially distanced of the pandemic in the garden behind our apartment, right? And so I think that this experience really taught me the importance that while we all face this paradox of people, while we all face this dilemma, that, if we want to make friends, if we want to connect with people, we have to be able to move away from the part of ourselves that is fearful, that is mistrustful, that assumes people will **harm** or reject us and turn towards the part of ourselves that simply wants to love and connect with people and can ready ourselves to engage in these new connections with optimism and with hope. You know, my niece read my book”Platonic,”and one thing that she took away from it was that for friendship to happen, someone has to be brave. So be brave. Thank you. Whitney Pennington Rodgers : Thank you, Marisa. I loved all of that. And I could see in the **chat** that a lot of the members also really love some of the things you shared there. So thank you so much for that wonderful talk and for those tips which we will dive into in this conversation. And, you know, I think just to sort of start, your line of work is just so interesting. Friendship is, it seems such a unique area to research. And I actually want to read something back to you from your book to help us understand a little bit more about just the importance of this type of relationship. You say,”Friendship, in releasing the relationship pressure valve, infuses us with joy like no other relationship. Without needing to plan for retirement, fulfill each other ’ s **sexual** needs and work out who should be scrubbing the shower grime, we are free to make friendship territories of pleasure.”So can you talk a little bit more about this and just why friendship holds such an important role in all of our lives? MF : Yeah, yeah. Well, first of all, I ’ ll take a step back and say, like, clearly, connection is so central to all of us. You know, the research finds that, for example, loneliness is more toxic for your body than having a poor diet or not exercising. And these are things that we talk about in the public health conscious, but we don ’ t talk about social connection enough. Maybe for our UK folks, you all have a prime minister of loneliness. So it ’ s a little bit different, right? But when we talk about the impact of loneliness, there ’ s actually three different dimensions of loneliness. There is intimate loneliness, which is the desire for someone to be very intimate with. There ’ s also relational loneliness, which is the desire for someone that feels as close to us as a friend might. And then there ’ s collective loneliness, which means I desire to be part of a group working towards a common goal. And this research on loneliness really suggests that, just like I said, we really do need an entire community to feel whole. Because if we just focus on being very nuclear, you know, just having a spouse and that being the center of all of our connections, right, that ’ s maybe touching on our intimate loneliness, but not our relational, not our collective, right? And so I think a lot of us found this in the pandemic that, we may have been home with the spouse or partner that we really loved a lot, but we still ended up feeling lonely. And that ’ s because as social species, as social creatures, like, we just need an entire community to fulfill us. WPR : You touched on this a little bit in the talk, and in the book, you separate, sort of, the way you think about friendship to two categories. You talk about, you know, sort of a backward look at how we ’ ve traditionally experienced friendship, and then you look forward at how we could build better relationships, better platonic relationships. And so if we could just talk a little bit about sort of that first section, just diving into how we as a culture tend to think about friendship and how does this really impact the way that we actually approach it? MF : Yeah, so I ’ m reading all the research on friendship, and what sort of materializes before my eyes is that our personalities are fundamentally a reflection of our experiences of connection or disconnection, right? Like, in some ways our personalities are coping mechanisms from the experiences of connection or disconnection we ’ ve received. So whether you are friendly, open, warm, vulnerable, trusting, cynical, aggressive, even **violent**, all of this is predicted by what your history of connection or lack thereof looks like. So how we ’ ve connected really affects who we are. But not only that, who we are really affects how we connect, right? Those people that have that history of healthy relationships, they have an internal set of beliefs within them that allows them to continue to facilitate healthy relationships. This is, if people are familiar with”attachment theory,”securely attached people, who think others can be trusted, who think they can be vulnerable, who think they can turn to people for support, right? And they go into new relationships with this set of beliefs that allows them to continue to create these new relationships. Whereas people who have had difficult previous relationships, those can be internalized as a belief system that then can impede their ability to foster further connections, right? Because they may think”people are going to reject me,”that”I can ’ t be vulnerable,”that”I can never feel safe around anybody,”even when someone is safe, they ’ re still holding that assumption and that judgment, right? And so what I want”Platonic”to do, because I know some people hear this and they ’ re like,”Good for those people with healthy childhoods, you know, whatever. For me, I guess I ’ m doomed.”Absolutely not. WPR : Your work is focused specifically on adult friendships, which, for a lot of the reasons you ’ ve already outlined, seem like are just really challenging for us to develop. But I guess, can you talk about sort of, why this breakdown happens and really when we start to see that it becomes more difficult for people to make friends in the same way as you did when you were kids? MF : Yeah. So when I say friendship doesn ’ t happen organically in adulthood, I don ’ t mean that friendship doesn ’ t happen organically in childhood, because for children it often does. And they have certain ingredients in their environment that really foster friendship. So Rebecca G. Adams, she ' s a sociologist, and she says, you know, for friendship to be fostered organically, you need to have this repeated unplanned interaction and the **shared** vulnerability, right? And so in school we have that. We have that through our lunch period. We have that through our gym, we have that through our recess, right, we ' re seeing people every day, we ' re letting our guard down, we end up sort of just developing these friendships. But when you think about adults going into the working worlds, you may have repeated unplanned interaction with your colleagues, less so now that we ' re doing more hybrid and remote workplace, right? But we ' re often not as vulnerable in the workplace. And that ' s why one study, and again, this is caveat, US context, one study found that the more time people spend together at work, the less close that they feel. And so we need to recognize that as adults, we don ' t have the same infrastructure we had as kids. So we can ' t rely on the same set of assumptions that, oh, this is just going to happen, I don ' t have to try, I don ' t have to initiate. Because as I shared from that previous research study, if we think that way, for a lot of us, it won ' t happen. WPR : And I think also just in thinking about what friendship brings to the table for you, how it benefits you, you have this phrase in your book where you say,"Connecting to others makes us ourselves."And it ' s about much more than just the pleasure of connection there. Can you explain a little bit about that? MF : Yeah. So Harry Stack Sullivan, he ' s a psychiatrist, and he has this theory called the"theory of chums,"which is basically that our chums or our friends earlier in life, they provide us with the sort of relationship template that we take on into our future. So it allows us to continue to connect throughout life, right? And there is some research that finds that if we connect in childhood, we have good friends in childhood, we have higher self-esteem in adulthood, we ' re more empathic in adulthood, right? And so he kind of argues that the therapy experience is similar to the chumship experience, in that with your friends, you share things that you feel like you should be ashamed of, right? And when you are ashamed of something, you ' re not integrating it into your entire personality. You ' re trying to push it away and suppress it and not make it who you are. And the **shame** can really take over your whole personality because you ' re pushing away this part of who you are that you think is shameful. But then so much of your personality is spent focusing on making sure nobody finds out this thing, right? And so that ' s why **shame** can be so encompassing and enveloping. But what Stack Sullivan argues, is that when kids share this **shame** with their friends and their friends are like, you know, we still love you. You know, this isn ' t a big deal for us, right? And we still think you ' re amazing and we accept that about you, right? We begin to be able to accept it in ourselves and to bring whatever we felt **shame** about, to see it as just part of our personalities rather than antithetical to the personality that we want to have. And we are able to sort of relinquish all of the energies that we spend trying to push this thing away. And so in that way, the experience of experiencing that platonic love from our friends, especially at that time in childhood, teenagers, where we ' re very high in **shame**, we ' re like highest in **shame** than throughout our whole lifespan, right? That our friends are there at that time when we ' re so high in **shame**, to help us integrate that and to help us connect to all sides of ourselves so that we sort of begin to become who we fully are. WPR : It ' s just so interesting how much vulnerability and **shame** play into this. And in a way, it seems like on the one hand, as you get more confident, you start to lose some of the ability to make new friends, but also using that confidence, leaning into it to actually make the friends is what you need. So it ' s fascinating. Well, I want to dive into some of the member questions we ' re receiving because they ' re also really interesting. So TED member Arnoldo, they ask,"Married people often complain about lack of time to cultivate friendships outside of the marital circle. What insights have you gained in your research about the effects of outside friendships in a couple ' s relationship?"MF : Oh, I love this question. I think that having outside friendships is necessary for having a healthy marriage. I do. And this is where I ' m coming from with that. The research basically finds that if I make a friend, I ' m not only less depressed, but my spouse is too. It also finds that when you get into conflict with your spouse, it negatively impacts your release of a stress hormone cortisol, right? But if you have quality connection outside of that marriage, that doesn ' t happen. Your cortisol release is normal, right? Other research that finds that, particularly for women who tend to have more close intimate relationships, when they go through difficulties within that partnership, they are more resilient to it when they have this outside support, right? And so it ' s just like, if I can **access** this other person to center me during times of stress in my marriage, I can return to that marriage in a centered and grounded way. And that ' s a resource for me, and it ' s a resource for my spouse. Where we see that people that only have that close connection with their spouse, they have high rates of what ' s called concordance, which means that, however your spouse feels is kind of how you feel. Their, sort of, energy affects you a lot more when they ' re the only person that you ' re looking to for support. And so what happens is like, the natural ebbs and flows that can happen in a marriage, they ' re so much more impacted, and there ' s so much more devastated during those times of ebbs because they don ' t have that support outside of the relationship. So I think sometimes we see, we think of like,"Oh, are my friends a threat to my spouse? My spouse is spending time with their friends, they ' re not spending time with me."But if we understand more broadly the importance of friendship and how it makes every other relationship in our life better, we will see that there ' s actually synergy between these relationships. That my spouse having friends outside of this relationship is what makes them a better spouse for me. WPR : So, so much of it has to do with the way we just think about the role friendships play in our lives. We have lots of questions that are coming in about, sort of, the steps to actually making friends. And I think before we get into some of those specific questions, I know you, in the talk, sort of shared – you started with two tips, this idea of first, assuming that people like you and then overcoming what you call covert avoidance or that urge to sort of mentally check out when you ' re meeting someone new. What are some of the other ways that you recommend people try to use tools that people use to build new friendships? MF : So I can walk through my own experience of making a bunch of friends and share this in a story. I went to Mexico City alone and was there for 10 days and was like, OK, if I spend ten days here and don ' t make any friends, I ' m going to be very lonely. So how did I make friends when I was there? First I went to a coffee shop. I heard another American there and I knew, you know, he ' s less likely to reject me than I think. I also knew the research on transitioners, right? People that are in times of transition are most open to friendship. So people that are traveling, people that have just moved to the city, people that have just started school, people that have just retired, right? Those are the people to try to connect with versus, you know, someone who ' s been here for a while and already has an established network. So I knew this guy ' s a transitioner and he ' s less likely to reject me than I think. So I ' m going to engage with him. And I asked, you know, where are you from? Like, I ' m from the US too. We start **chatting** and I end up inviting him to a meetup that night, and it ' s like a language exchange meetup. And at that language exchange meetup, I meet someone else who ’ s cool, and I say, you know,”Do you want to come to this Lucha Libre wrestling match with me?”Then he said, yes, and I think... You know what I realized, too, from the research on friendship, I used to think that making friends was about being interesting, being smart, being insightful, being charismatic, being entertaining. But in fact, people report that this entertainment factor is the least important quality they look for in a friend. And the most important quality that they look for is someone who makes them feel like they matter. So for me, if I want to connect with someone, it ’ s not about me trying to impress them. It ’ s about me trying to make them feel valued and, you know, say hello to them and engage with them and tell them what I like about them and tell them what I appreciate about them, right? There ’ s this study that looked at friendship, budding friendship groups, for like 12 weeks, which of these pairs ended up becoming friends. And it was the ones that shared the most affirmation and affection towards each other, right? And so, just to go even deeper into the research rabbit hole, there ’ s a theory called”risk regulation theory,”which indicates that we decide how much to invest in a relationship based on our view of how likely we are to get rejected. So when we show people”I like you,”we ’ re telling them,”Hey, you ’ re not going to get rejected if you try to be friends with me.”And that makes people really feel safe connecting with us, right? So I was both engaging with these people and I was trying to make them feel loved and tell them how great I thought they were and how happy I am to meet them as I reached out to them. And then I went to my Spanish class, which, if you don ’ t have any friends, what I recommend is that you join something that ’ s repeated over time. And remember, I said repeated unplanned interaction and **shared** vulnerability is that infrastructure that kids have for friends that we lack as adults. So as adults, we really need to recreate that infrastructure. And so if you join like, a social group that ’ s repeated over time, turn your hobby into a community, right, that ’ s a really important way to make friends. So for me, it ’ s I want to take the Spanish class because I love learning languages. For you, it might be football team, improv team, hiking team, **meditation** group, but it ’ s just finding something, finding a group that meets around this hobby. Because when you find something that ’ s repeated over time, what happens is something called the”mere exposure effect”sets in. The mere exposure effect describes our tendency to unconsciously, completely unconsciously, like people just because they are familiar to us. So, for example, this researcher found that when he planted women into a large psychology lecture, at the end of the semester, none of the students remembered the woman, but they reported liking the woman who showed up to the most classes, 20 percent more than the woman that didn ’ t show up for any, right? Nobody remembered her, but they liked her a lot because they had seen her face, like this is our brain, right? And what I think the other implication of mere exposure effect is in the beginning, mere exposure hadn ’ t set in. So it ’ s going to be awkward, it ’ s going to be weary. You ’ re going to feel uncomfortable, right? Maybe a little distrusting. That ’ s not a sign that you need to stop showing up. That ’ s a sign you need to keep showing up because when you continue to show up, they ’ re going to like you more, you ’ re going to like them more, right? So I joined that Spanish class, that was repeated over time. Every day in Spanish class, I would ask people to go out to lunch with me. We ’ d go out to lunch together, then we went to Lucha Libre together. You know, of course, I was only there for 10 days, so I can only go so deep with folks. But in general, when you join this event that ’ s repeated over time, you want to start generating exclusivity with someone in that group. Exclusivity means you develop memories and you develop experiences with one person in the group that you don ’ t have with other people. So pick whoever in the group that you really liked and ask them,"Oh, would you be open to like, getting coffee, getting tea like, before or after our next group?"Like,"I love to hang out"and those are like, the budding stems of friendship. And luckily, if you ' re in this group, right, you don ' t have to put in as much effort, you had your tea, and now you ' re going to just continue to see them over time, and you have the wheel start moving for friendship and connection. WPR : So TED member Celia actually is curious about virtual friendships and sort of how all of this plays into it, especially to some of the points you were making earlier about the pandemic. You know, for people who have met on social media only as opposed to in real life, they ask,"Is it possible to have a strong virtual friendship? How important is in-person connection or getting together in real life?"MF : Yeah, so this is such a nuanced question in some ways, right? Because it ' s such a"both / and."We know from the research that in-person connections tend to be stronger than virtual connections, right? But I think that that research, it doesn ' t account for certain communities like people with disabilities, people with severe social anxiety, even older people that aren ' t as mobile, who tend to find connections online. And even though, you know, the online connections tend to not be as deep as in person, they can get deep if you ' re practicing the same skills that you can practice in offline connection to establish deeper relationships. So, for example, like, the research finds that if you ' re just passively scrolling on social media, doomscrolling, it makes you more lonely and negatively impacts your mental health and well-being. But if you are engaging actively on social media, I ' m posting, I ' m commenting on something that you shared, I ' m saying congratulations to you, that ' s actually linked to less loneliness and is something that actually makes us feel more satisfied in our relationships. So if we want to have deep virtual connections, it ' s certainly possible. But we have to bring those same principles that we use in offline connections to create more intimacy, things like being vulnerable with someone, being generous with them or showing affection towards them. That also works online. They feel like they like one another more. But when you ' re vulnerable with someone who ' s avoidantly attached, that doesn ' t necessarily happen. The avoidantly attached person doesn ' t like you more because you ' re vulnerable, because they have their own wounds around vulnerability, right? They ' ve learned that it ' s not good to be vulnerable,"I shouldn ' t be vulnerable."Like, that ' s the implicit message that they have that really inhibits their ability to connect. So the implications of this, I think, is that if you ' re vulnerable and it doesn ' t go well and it wasn ' t from a place of fear, remember that it ' s not always your fault. That other people have their own issues that they ' re dealing with which may lead them to respond to your vulnerability negatively. And that doesn ' t mean that you did anything wrong. I mean, I think if you continue to try to be vulnerable with this person who ' s shown you that they can ' t handle it, then I think you should try to pivot, right? But just because someone responded dismissively to your vulnerability, it might mean that they have their own issues to work out. WPR : I think, in sort of thinking about how to make friendships work well or to be really good at this process of doing this, you know, there ' s a popular excerpt from your book that you shared in"The Atlantic"where you talked about the concept of super friends. So what makes someone a super friend, and how can we all strive to be super friends? MF : Yeah, secure friends, aka super friends. These people are secure with themselves, which means that they don ' t have to try to use other people as a tool to fulfill their sense of self or to help them escape threatening emotions or feelings. So they ' re able to really humanize other people fully. And the research on securely attached people find, and again, these are the people that have a history of healthy relationships, but there ' s also earned-secure, which means you may not have had a history of healthy relationships, but you ' ve done the work on yourself to develop a sense of security, right? Remember, this isn ' t – nobody ' s doomed by their attachment style. But what qualities do we see in them? They ' re more likely to initiate friendships. They ' re more likely to maintain friendships. They ' re less likely to dissolve friendships. They ' re more generous towards other people because, again, they fully humanize other people. Insecurely attached people, they sometimes perceive other people through the lens of their own wounds, right? So anxiously attached people, it ' s like, you need to prove to me that you value me and you love me because I ' m so afraid that you ' re going to abandon me, and then I can try to control you and make you do things to show me that you really, really love me, right? And so they ' re not fully humanizing another person because they ' re almost seeing that person as a tool to fulfill their sense of self. Avoidantly attached people, they just think everybody ' s out to **harm** them and that everybody ' s untrustworthy. So they almost see other people as threats, so they don ' t fully humanize people for their beauty and the resources that connecting with another person can bring you. But these securely attached people, they tend to assume other people like them. I talked about something called pronoia, which is the opposite of paranoia. It ' s the idea that, you know, the universe is commiserating for your success and for your well-being and that you can trust people. They ' re comfortable with vulnerability, they ' re more empathic, they ' re comfortable sharing their needs, but also fulfilling the needs of other people. They ' re more responsive to the needs of other people. When they engage in conflict, it ' s all about perspective-taking. They ' re not like,"You do this, otherwise I ' m going to be pissed off."They ' re like,"These are my needs. What are your needs? Let ' s figure out a way to collaborate and figure out something that will work for both of us. So they tend to be quite **healing** friends. They tend to be – I talked about avoidant being low effort, low reward. Anxious is high effort, low reward. Secure is high effort, high reward when it comes to friendship. WPR : And then what about friendships where there ' s not necessarily a difference in values, but maybe a distance, whether that ' s a physical distance has been created or some sort of emotional distance because your life has changed in some way? How do you suggest people go about maintaining and nurturing those types of relationships? MF : So there ' s research on long-distance friends that finds that we are helped when we perceive our friendships as flexible, not fragile. So when we perceive that,"Oh, I haven ' t talked to this person in a few months, I ' m going to assume that friendship is asleep, not that it ' s dead, so that I can reconvene this friendship at any time."Right? So it ' s being able to recognize that our friendships ebb and flow. And when we ' re at an ebb, that doesn ' t mean,"OK, I ' m never going to contact this person again, because the friendship is officially over."We assume that this ebb is part of the normal process to flow again. So that facilitates us being able to re-engage in the friendship at any time. So basically, this all goes back to, I really think, this tip, right? It ' s such an all encompassing tip, right? Because what I ' m basically telling you is to assume people like you, right? Like, if you don ' t talk to your friend awhile, assume that they ' re still interested in being friends with you. Again, this isn ' t about, you know, being delusional. If someone ' s clearly indicating that they ' re not interested in a friendship with you, then move on. But if it ' s ambiguous and you ' re like,"I ' m not really sure, we haven ' t talked for a while, but they haven ' t necessarily rejected me or they still are responsive when I reach out to them,"you want to make that your running assumption in response to ambiguity, because again, having that assumption really facilitates continued connection. [Want to support TED?] [Become a TED Member!] [Learn more at ted. com / membership]

Text Sample 360: POS Dim 3 – 2024 – Score 19.00 – 2024/t003204.txt

Let me begin with some **confessions**. I voted for Mitt Romney in 2012. And I voted for Hillary Clinton and Joe Biden, which is shocking, I know. I ’ m pro-choice, and I think the European laws are sensible ones. I ’ m a very proud supporter of Israel, even though I ’ m a critic of its current government. I think terrorists like Hamas and Hezbollah are evil, and there is a bright line between groups that aim to kill **innocents** and those that try to avoid doing so at all costs. I think that girls in Afghanistan shouldn ’ t be sold into child marriages, and that women in Iran should be free to show their hair in public without fear of imprisonment or worse. And that women in Somalia should not endure genital mutilation. I believe that all people are created **equal** and created in the image of God, but that all cultures are not **equal**. I believe in **gay** marriage, so much so that I ’ m actually in one myself. I believe that adults should make pretty much any decision they want about their bodies, but that children should not. I think the SAT is an imperfect but useful tool. I think defunding the police is a very bad idea. And that living in a safe **neighborhood** is among the truest forms of privilege. I think COVID probably came from a lab, and that, in retrospect, locking kids out of school for two years was a big mistake. I think we should hire people based on their merit, but cast as wide a net as possible. I don ’ t want to eat bugs, nor do I want to drink water full of microplastics, and I don ’ t think there ’ s anything coded right or left about either of those things. I believe that **equality** of opportunity and not **equality** of outcome is the true measure of **fairness**. I am repelled by ideologies that insist that our immutable characteristics are more important than our character. I don ’ t like riots, I don ’ t like mobs, and I hate lies. And I love America for all of its flaws. I believe, in part because Americans are free to debate those flaws and to strive for a more perfect union, that it really is the last best hope on Earth. The point in all of this is that I am really boring. Or at least I thought I was. I am, or at least until a few seconds ago in historical time, I used to be considered a standard-issue liberal. And yet somehow, in our most intellectual and prestigious spaces, many of the ideas I just outlined and others like them, have become provocative or controversial, which is really a polite way of saying unwelcome, beyond the pale. Even bigoted or racist. How? How did these relatively boring views come to be seen as off-limits? And how did that happen, at least it seems to me, in the span of under a few years? Now the convenient answer, of course, is the power of extreme activists. People who burn down businesses and police stations, people who shut down bridges and highways, people who harass their fellow students and shout down their professors. People who vandalize, who desecrate or tear down monuments of national heroes. But do a handful of extreme activists really have the power to dismantle the moral guardrails of a whole society, to radically shift the Overton window of what is politically and socially acceptable? I don ’ t think so. There has always been and always will be a fringe. The difference right now is that the fringe seems to be calling the shots. If you want to know why things have been turned upside down, why so many people are asking themselves if they ’ ve gone crazy, or if the world has, as they hear feminist groups justify rape as a tool of resistance ; as groups that call themselves anti-racist advocate for a new kind of segregation ; as young, highly educated people chant the slogans of jihadi terrorist groups ; well, I ultimately don ’ t think that ’ s because of a few maniacs that are throwing paint on masterpieces in our museums. It ’ s because they have been allowed to do so. And the question is why? Perhaps, to give the most generous read, it ’ s because the people shutting things down claim to be doing so in the name of justice, not in the name of nihilism. And because we believe them. Or perhaps it ’ s because we told ourselves,”It ’ s just a few nuts, I don ’ t need to get involved.”Or maybe it ’ s because people looked at their portfolio and decided that they were doing great by the numbers. And those torched stores? Ah, they probably had insurance anyway. Or because it was a headache. Or because they ’ re just kids. Or because, why die on that hill? Or maybe it was because we thought they had a point. That America and the West really were guilty of all of the terrible things that they said, or at least of some of them. And though we wouldn ’ t have torn down statues or shouted down speakers, we lacked the conviction or the ideas to stop the people doing it. Or because maybe in the end we prized comfort over complexity. I was going to say prized comfort over truth, but the thing is, truth isn ’ t something you pull out of the ground like gold or diamonds. It is a process sustained by a culture of questioning, including self-questioning. Which is why right now it can look like the absolutists are winning. My theory is that the reason we have a culture in crisis is because of the cowardice of people that know better. It is because the weakness of the silent, or rather the self-silencing majority. So why have we been silent? Simple. Because it ’ s easier. Because speaking up is hard, it is embarrassing, it makes you vulnerable. It exposes you as someone who is not chill, as someone who cares a lot, as someone who makes judgments, as someone who discerns between right and wrong, between better and worse. The reason Aristotle called courage the first virtue is because it is the one that makes all of the other virtues possible. Do you want to live in a world that values justice, wisdom, compassion, curiosity, rationality, **equality** and the pursuit of truth? I do. But fighting to make sure we live in such a world is going to take courage. That first virtue. I think one of the lessons of the past decade is that cowardice is perhaps more contagious than COVID. But so is courage, and a singular example can serve as a powerful means of transmission. So who are those examples? Each one of you, when I say the word courage, will have the ones that come to mind for you. But for me, for me, they are people like Salman Rushdie, sentenced to death by the Iranian ayatollahs in 1989 for the sin of writing a novel. He lived under the shadow of a fatwa until two years ago, on a stage like this one, he was viciously stabbed. But he survived and undaunted, this week, of course, he published a book about it. Courage for me is someone like Pennsylvania Senator John Fetterman, who insists that there is nothing contradictory about his progressive values and his belief that Hamas is a band of murderers that must be defeated. Now suffice it to say, this has not made him popular, but he doesn ’ t seem to care. While a lot of other people have moved on out of political expediency, his office in DC is the one that remains papered with photos of all of the hostages. Courage to me looks like Stanford medical professor Dr. Jay Bhattacharya. Jay studies the health and well-being of vulnerable populations for a living, and he foresaw the social and mental health crisis that would follow the COVID lockdowns. He said so, he explained it calmly. But for doing so, **Twitter** blacklisted him. YouTube censored him. The medical establishment ostracized and slandered him. He wrote,”I could not believe this was happening in a country that I so love.”And yet he did not tremble. He said,”The healing of the world starts by one person saying loudly, so the whole world can hear,”an important, true thing, that he knows he ’ s not supposed to say and that he knows will get him in trouble for saying it. I think about Roland Fryer, the economist who did just that. His colleagues at Harvard warned him against publishing research that he did into police **violence**.”You ’ ll ruin your career,”they told him. And that ’ s because his research found that while there was racial bias in low-level police force, there wasn ’ t when it came to police shootings. Now Roland himself was shocked by these findings. He knew it went against his own assumptions. He knew it would outrage people. But he published the research anyway. And it wasn ’ t simply that his reputation suffered. He had to hire an **armed** guard in Cambridge. His baby was seven days old, and he had to go to buy diapers with an **armed** guard. Where did he get the courage to do it?”Simple,”he told me.”I don ’ t covet what they covet.”He said,”Every day I have to look at myself in the mirror and say, ’ What are you here for? ’”Masih Alinejad knows what she is put on Earth for. With moxie and courage, she is leading the campaign for women ’ s rights in Iran. Her sister was forced to denounce her on state television. Her brother was thrown in jail for her dissent. And now Masih lives in exile in America but remains a hunted woman, moving from safe house to safe house. And yet she does not stop shouting for freedom. Nor does Jimmy Lai, the media mogul whose pro-democracy newspaper”Apple Daily”was shut down as China took over Hong Kong. Jimmy had more than the means to flee his country. He is a billionaire with a British passport. But he stayed.”Now is not the time for **safety**,”he said.”This is a time for sacrifice.”Today is his 1, 204 day in prison. His son Sebastian said this of his father :”Dad staying in Hong Kong is really proof that this intangible thing we call liberty is a thing people yearn for. You can call it Western values, but it ’ s not really. In the sense that it ’ s not something that only people in the West want or deserve.”Alexei Navalny was not born in the West, but he yearned for that kind of liberty. The opposition leader had refuge in Germany, but he flew back into Putin ’ s Russia, sacrificing his freedom and ultimately his life to oppose tyranny. He knew that his death would expose the truth about a totalitarian regime built on lies. Which it can, so long as we keep his memory alive. Navalny lived and died beneath the shadow of a tyranny that we are fighting to prevent in our still young but ever-darkening century. Ask yourself right now, should it take courage in the West to denounce the hateful ideology of the Islamic Republic of Iran, which pronounces death on individual writers and on entire countries, large and small? Should it take courage to oppose those chanting”death to America?”Should it take courage to say,”No, that ’ s wrong?”Should it take courage to say that those who praise the pristine subways of Russia are not journalists, but propagandists? Should it take courage to just say in public,”I disagree?”Right now it does. My friend Coleman Hughes, who spoke on this stage last year and who advocates for the colorblind ideal championed by Martin Luther King Jr, rather than give in to the race essentialism that ’ s become chic these days, he ’ ll debate anyone. He ’ ll disagree with anyone. But why is it that his angriest opponents prefer to call him hateful names and to lobby for his exclusion? The question, I think, is whether or not the people I ’ ve mentioned and those like them, whether their photos and their names and most importantly, their ideas will show up at conferences like this one. And that ’ s up to you. I ’ ve had enough people confess to me after lectures or in newsrooms or on college campuses or in corporations or cafes, really, everywhere I go, that they wish they could say what they believe. They tell me with some measure of **shame** that they ’ re closeted in our liberal democracies. It ’ s a really strange phenomenon. The freest people in the history of the world seem to have lost the hunger for liberty. Or maybe it ’ s really the will to defend it. And when they tell me this, it puts me in mind of my hero, Natan Sharansky, who spent a decade in the Soviet gulag before getting his freedom. He is the single bravest person that I have ever met in my life. And a few years ago, one afternoon in Jerusalem, I asked him a simple question.”Nathan,”I asked him,”is it possible to teach courage?”And he smiled in his impish way and said,”No. All you can do is show people how good it feels to be free.”Thank you. Thank you so much. Thank you. Thank you. Chris Anderson : Stay up, stay up a sec. Friends – Bari Weiss : Sorry I couldn ’ t memorize, you guys. CA : This session is going to run long. I ’ m sorry, but it matters. Thank you, that was a great talk. Saw people stand and cheer, other people didn ’ t stand. You ’ re in the middle of all these issues – BW : I was expecting hecklers, so I ’ m really happy. CA : But these issues are so important. And this is such an important conversation. I think I want to ask you something in the quest for common ground here. Is it possible that, as well as lack of courage, there ’ s something else big going on in the hearts of many of the”silent majority?”Which, for want of a better word is love. These are often debates between identity groups, and many of us don ’ t like the way that the battle is going. But we also feel deeply the pain that a lot of these groups have gone through, the injustices that they have suffered. And if you get involved, it can so easily be seen as you are against that group. And I guess I ’ m just wondering whether there ’ s common ground to be found in us all saying identity really matters. And I mean, you care about the past injustices of people in America and all the different groups you ’ ve talked about. But that there are some things that are upstream of identity that matter even more. You mentioned truth, the pursuit of truth. We have common ground on that. I believe that passionately. I believe that about ideas. You know that some people want to say that ideas are a property of one group and that, you know – but no, no, TED is all based on the notion that ideas can spread from any human to any human. BW : But the whole question is, sorry to interrupt, how do you get to truth, right? And the West has given us the most radical tools in human history. I think Sam Harris is probably in this room, and I ’ m stealing his line. But the radical departure is that here, in rooms like this one, in cultures like the ones that we are lucky enough to live in, we don ’ t solve our conflicts with blows and with **violence**. We solve them with words. And that is why it is so absolutely crucial, no matter how people who are really advocating to burn it all down or tear it all down. No, by tearing it all down, by tearing down the rule of law, by disallowing us to be able to have this kind of debate and discussion, you ’ re preventing the whole project itself. And that has nothing to do with identity, with claims of victimhood, with actual victimhood. The entire way that progress has been achieved is by **victim** groups using the tools that liberal democracies have provided them with. Without freedom, without freedom of speech, freedom of religion, without the rule of law, none of the progress that I know so many people in this room **celebrate** would be possible at all. And so it ’ s really about clinging to the tools rather than repudiating them. CA : I agree with that. But the tools of words in our current culture, which is sound-bite, fast stuff, it ’ s so often heard as an **assault**, it ’ s heard not as words and exploration of truth. It ’ s heard as **hatred** or criticism. And I just wonder whether there could be a coalition of the willing, or double down exactly on what you said, let ’ s pursue the truth. Let ’ s pursue the best ideas. Let ’ s not be fearful of sharing things that are difficult with each other, but do so in a spirit of love and respect, and so that everyone can know that at heart, they are respected. We ’ re all trying to make things better. BW : I don ’ t see anything to disagree with there, only that... you know... Respect – like... Love and compassion and all of that, again, it ’ s only possible if we agree to a certain set of rules that I think many of us took for granted in the way we take oxygen or gravity for granted. And one of the things that has driven me and my choices over the past years of my life is a profound sense that the line between civilization and barbarism, a word that maybe will provoke some people, but I believe is an accurate description, is paper thin. The things that allow for us to do this are so exceptional, and they have to be fought for. And the people that claim that words are **violence** are taking away the most fundamental tool we have for all of the virtues that I was trying to talk about on stage here this morning. CA

: Bari, thank you. You ’ ve ignited an incredibly important conversation here. Thank you for doing that. Please stay, please continue this conversation. And thank you for what you said. BW : Thanks for having me.

Text Sample 361: POS Dim 3 – 2024 – Score 17.00 – 2024/t003164.txt
Chris Anderson : Hello, welcome, TED members. It ’ s very, very, very good to see you. I ’ m the head of TED, Chris Anderson. I ’ m a TED head, apparently, and really excited to be here with you. We ’ re actually doing something we haven ’ t done before. We are going to watch together a smash hit TED Talk in the presence of the man who gave that talk, Scott Galloway. I ’ m going to introduce to you in a minute. While we ’ re watching that talk, you can comment away, ask questions, talk with each other, disagree, amplify, laugh, whatever strikes you as the right thing to do. And when the talk is over, we ’ ll be putting questions to Scott and seeing how he can possibly defend the outrageous claims that he makes in this talk. This talk is a very big deal. He ’ s outlining a giant problem making some big, bold solutions on what we might do about it. It ’ s a big wow. And it ’ s been seen by millions of people already. But we want to dig deeper into these questions. And there couldn ’ t be a better group of people to do it than you. So thank you for being part of this. You are awesome. I think, let ’ s see, yeah, there ’ s a live transcript of this event if you need it by clicking on the captions option. So I think that ’ s it. So I would like to welcome the – how do we describe Scott Galloway? Well I mean he ’ s a professor at NYU, business professor, he ’ s the host of an ungodly number of podcasts. Incredibly successful. He does all kinds of things. He ’ s been a successful entrepreneur and general troublemaker, and it ’ s a delight to welcome Scott Galloway here. Scott, hello, nice to have you. Scott Galloway : It ’ s good to be with you, Chris. Thanks for having me. And hello to all TED acolytes and Members. CA : Alrighty. So I don ’ t know if you ’ ve ever done this either. I don ’ t know whether it ’ d be at all uncomfortable to watch yourself in full glory giving this talk that was so well received. But let ’ s try it. Let us roll this video without further ado. My name is Scott Galloway, I teach at NYU and I appreciate your time. I have 44 slides and 720 seconds. Let ’ s light this candle. OK so for those of you who don ’ t know me, I ’ m actually a global television store. True story. I ’ ve had four TV series in the last three years. Two of them have been **canceled** before they were launched and two were **canceled** within six weeks. Let ’ s recap. If we want to juice this thing, if we want to put a cattle prod up the ass of the economy. Bloomberg. The most trusted name in financial news. Not for long. Andrew Yang : I ’ m going to do whatever I can for this country of ours. SG : Jesus, come on, dude, you ’ re 0 for two. Face for podcasting. So first insight of the day. I ’ d like to be the first person to welcome you to the last TED. OK. By the way, it ’ s clear – what ’ s it called? What are we here for?”The brave and the brilliant?”It ’ s clear that Chris is a frustrated soap **opera** producer. Essentially what we have here is a telenovela where, after a night of unbridled **passion** between Bill Gates and Malcolm Gladwell, they give birth to their bastard love child, Simon Sinek. OK, I start us with a question. Do we love our children? Sounds like an illegitimate question, right? Well, I ’ m going to try and convince you otherwise. Essentially, as we go down generations, we ’ re seeing that for the last two generations, people are making less money on an inflation-adjusted basis. In addition, the cost of buying a home, the cost of pursuing education, continues to skyrocket. So the purchasing power, the prosperity, is inversely correlated to age. Simply put, as we get younger, we ’ re taking away opportunity and prosperity from our youngest. The social con-

tract that is now no longer in place and for the first time in the US ' s history, a 30-year-old is no longer doing as well as his or her parents were at 30. This is a breakdown in the fundamental agreement we have with any society, and it creates rage and **shame**. As a result, people over the age of 55 feel pretty good about America, but less than one in five people under the age of 34 feel very good about America. This creates an incendiary. Righteous movements, cuts to our society end up becoming opportunistic infections because generally speaking, young people have a warranted envy, they ' re pissed off and they ' re angry that they don ' t enjoy the same spoils and prosperity that were provided to our generation. A decent proxy for how much we value youth labor is minimum wage, and we ' ve kept it purposely pretty low. If it had just kept pace with productivity, it ' d be at about 23 bucks a share. But we ' ve decided to purposely keep it low. Out of reach. Median home price has skyrocketed relative to median household income. As a result, pre-pandemic, the average mortgage payment was 1, 100 dollars, it ' s now 2, 300 dollars because of an acceleration in interest rates and the fact that the average home has gone from 290, 000 to 420. By the way, the most expensive homes in the world, based on this metric, are number three, Vancouver. Why? Because 60 percent of the cost of building a home goes to permits. Because guess what, the incumbents that own assets have weaponized government to make it very difficult for new entrants to ever get their own assets, thereby elevating their own net worth. This is the transfer I ' m going to be speaking about. This has resulted in an **enormous** transfer of wealth, where people over the age of 70 used to control 19 percent of household income, versus people under the age of 40, used to control 12. Their wealth has been cut in half. This isn ' t by accident, it ' s purposeful. This is me at UCLA in 1987. I know your first thought is I haven ' t changed a bit. This is also Mia Sevario, who is the analyst who put together these slides. By the way, Mia is 26. I did the math, just by virtue of her being in this audience, it brings the average age of the entire conference down 11 days. When I applied to UCLA, the admissions rate was 76 percent. Today, it ' s nine percent. I received a 2. 23 GPA from UCLA. I learned nothing but how to make bongs out of household items and every line from"Planet of the Apes."And the greatest public school in the world, Berkeley, decided to let me in with a 2. 27 GPA. And that ' s what higher ed is about. Higher ed is about taking unremarkable kids and giving them a shot at being remarkable. And every year it ' s gotten more expensive. Higher ed and homes and the ability, not only is higher ed incredibly expensive, it ' s not accessible. Because me and my colleagues are drunk on luxury and I ' ll come back to that. We ' ve embraced the ultimate strategy. Me and my colleagues in higher ed wake up every morning and ask ourselves the same question when we look in the mirror. How can I increase my compensation while reducing my accountability? And we have found the ultimate strategy. It ' s called an LVMH strategy, where we artificially constrain supply to create aspiration and scarcity such that we can raise tuition faster than inflation. And old people and wealthy people have done the same thing with housing. All of a sudden, once you own a home, you become very concerned with traffic and you make sure that there ' s no new housing permits. And here is a memo to my colleagues in higher ed : we ' re public servants, not fucking Chanel bags. Harvard is the best example of this. They ' ve increased their endowment in the last 40 years and have decided to expand their enrollment, their freshman class, by four percent. Any university that doesn ' t grow their freshman class faster than population that has over a billion dollars in endowment should lose their tax-free status because they ' re no longer in higher education. They ' re a hedge fund offering classes. My first recommendation, Biden should take some of that 750 billion earmarked to **bail** out the one third of people that got to go to college on the backs of the two thirds that didn ’ t and give a billion dollars to our 500 greatest public institutions, size-adjusted, in exchange for three things. One, they use technology and scale to reduce tuition by two percent a year, expand enrollments by six percent a year, and increase the number of vocational certifications and nontraditional four-year degrees by 20 percent. Where does that get us? In just 10 years, in just ten years, that doubles the freshman seats and cuts the cost in half. This isn ’ t radical. This is called college in the ’ 80s and ’ 90s. Another transfer of wealth. Look at what ’ s happened to wages. Oh, they ’ ve gone up? Not as much as corporate profits. There ’ s a healthy tension between capital and labor. But for the last 40 years, capital has been kicking the shit out of labor. Well, you think, what about wages, right? They ’ ve gone up. Well if you compare them to the S and P, they barely register. It ’ s been an amazing time to own assets. But your attempt to get the certification or the income such that you can acquire assets has gotten harder and harder. In my class of 300 kids, it ’ s never been easier to be a billionaire, it ’ s never been harder to be a millionaire. By the way, our job in higher ed isn ’ t to identify a top one percent of people who are freakishly remarkable or have rich parents and turn them into a super class of billionaires. It ’ s to give the bottom 90 a chance to be in the top ten. You know who doesn ’ t need me or higher education? The top 10 percent. The whole point of higher ed is to give the unremarkables, i. e. yours truly, who was raised by a single **immigrant** mother, a shot of being remarkable. The transfer has been purposeful. While the cohorts, corporations and the ultra wealthy continue to garner more and more of our wealth, we have decided,”I know, if they win the gold, let ’ s give them the silver and the bronze and let ’ s lower their taxes.”This transfer is purposeful. It ’ s not by accident and it works. Senior poverty is way down and we should **celebrate** that. Meanwhile, child poverty is flat to up. The third rail. I ’ m going to talk about Social Security. It would cost 11 billion dollars to expand the child tax credit. But that gets stripped out of the infrastructure bill. But the additional 135 billion dollars a year to Social Security, that flies right through Congress. And every year we transfer 1. 4 trillion dollars from a cohort that is increasingly doing less well to the cohort that is the wealthiest cohort in the history of this planet. I ’ m not against Social Security, but the criteria should be if you need it, not whether you have a catheter. Eighty percent of you, 80 percent of you have absolutely no reason to ever take Social Security. It is bankrupting our nation. And we have fallen under this mythology that somehow it ’ s this great social program. No it ’ s not. It ’ s the great transfer of wealth from young to old. How is this happening? Because our representatives are in fact, representative. Old people vote. Washington has become a cross between the”Land of the Dead”and”The Golden Girls.”Quite frankly, this is fucking ridiculous. And if I sound ageist – If I sound ageist, I am. And you know who else is ageist? Biology. When Speaker Pelosi had her first child, get this, two thirds of households didn ’ t have color televisions, and Castro had just **declared** martial law. But she ’ s supposed to understand the challenges of a 17-year-old girl who ’ s 5 ’ 9 ’ ’, 95 pounds, getting tips on dieting and extreme dieting from Facebook? She ’ s supposed to understand the challenges that a 27-year-old single mother faces? By the way, young and dreamy. Young and dreamy. The great intergenerational theft took place under the auspices of a virus. I know, let ’ s use the greatest health crisis in a century to really speed-ball the transfer. This is the Nasdaq from 2008 to 2012. We let the markets crash. And by the way, you need churn, you need disruption because it seeds and recalibrates advantage and wealth from the incumbents to the entrants. It ’ s a natural part of the cycle. But wait, lately, no, a million people dying would be bad. But what would be tragic is if we let the Nasdaq go down and guys like me lost wealth. So we pumped the economy, which again, increased the massive transfer of wealth. The best two years of my life? COVID – more time with my kids, more time with Netflix, and the value of my stocks absolutely exploded. And who has to pay for my prosperity? Not me. Future generations who will have to deal with an unprecedented level of debt. Why am I here and why do I get the prosperity I enjoy? Because in 2008 we **bailed** out the banks, but we didn ' t **bail** out the economy. We let the markets fall. So as I was coming into my prime income-earning years, I got to buy, no joke, these stocks at these prices. This is where those stocks are now. Where does a young person find disruption? When you **bail** out the baby-boomer owner of a restaurant, all you ' re doing is robbing opportunity from the 26-year-old graduate of a culinary academy that wants her shot. We need disruption. I just like this slide. It has no context or relevance. We ' re economically attacking the young, but I know, let ' s attack their emotional and mental well-being. Let ' s take advantage of the flaws in our species with medieval institutions, Paleolithic instincts, and godlike technology. I ' m just going to say, I think Mark Zuckerberg has done more damage to the young people in our nation while making more money than any person in history. Oh, but wait, it could be worse. It ' s as if we let an adversary implant a neural jack into our youth to raise a generation of civic, military, and business leaders that hate America. How can we be this stupid? This all adds up to a bunch of graphs all headed up into the right. And what are they? What ' s the first one? Oh, that ' s self-harm rates, which have exploded, especially among girls since my colleague Jonathan Haidt pointed out, it ' s really, really gone crazy since social went on mobile. What ' s the next one? **Teens** with depression. The next one, men and women not having sex. Biggest fear of my parents was that I was going to get in too much trouble. My biggest fear, honestly, is that my kids aren ' t going to get into enough trouble. My advice to every young person watching this program is go out, drink more and make a series of bad decisions, it might pay off. Next graph, cumulative gun deaths. You ' re more likely to be shot in the United States if you ' re a toddler or an infant than a cop. Next graph, obesity, way up. By the way, the industrial food complex wants to addict you to shitty, fatty foods so they can hand you over to the industrial diabetes complex. We should not romanticize obesity. You ' re not finding your fucking truth. You ' re finding diabetes. Overdose deaths, way up. Deaths of despair. When I was in high school, it was drunk driving, now it ' s kids killing themselves. Young people don ' t want to have kids anymore. Two thirds of people aged 30 to 34, able-bodied, used to decide to have at least one child. It ' s been cut in half. It ' s now less than a third, 27 percent. As a result, people over the age of 60 in the US, pretty happy. People under the age of 30, not so much. Some of the lowest in the free world. What can we do? Nothing wrong with America that can ' t be fixed with what ' s right with it. We got the hard stuff figured out. There are programs to address all of these issues, they cost a lot of money, that ' s the hard part. And we have figured this out. In just five minutes post an earnings call, we can add a quarter of a trillion dollars to the economy. We ' ve got the hard part figured out, the resources. We have the money, but we decide not to do it. This is per-capita spending on child care in the United States relative to other nations. This is housing permits. Things are doable. We increase minimum wage at 25 bucks an hour, it goes into the economy, the wonderful thing about low- and middle-income households is they spend all their money. We have to have a restore or a progressive tax structure with alternative minimum tax on corporations and wealthy individuals. We need to refund the IRS. We need to reform Social Security. It should be based on whether you need the money, not on how old you are. We need a negative income tax. My friend Andrew Yang screwed up a great idea, he branded it incorrectly. Instead of calling it UBI, he should got Republicans on board by calling it a negative income tax. We need to eliminate the capital gains tax deduction. When did we decide that the money that capital earns is more noble than the money that sweat earns? Shouldn ’ t it be flipped? We need to remove 230 protection for all algorithmically-elevated content. We need identity verification. The reason we can have identity verification is because we have a First Amendment. Break up Big Tech. We have monopolies that are incurring greater and greater costs on every small business and parents because again, see above, our representatives don ’ t understand these technologies. We need to age-gate social media. There ’ s absolutely no reason anyone under the age of 16 should ever be on social media. We need universal pre-K. We need to reinstate the expanded child-tax credit. We need term limits, see above, Andrew Yang. We need income-based affirmative action. Any visible signs of affirmative action make no sense at all. You would rather be born **gay** or non-white, in the United States today, than poor. And that ’ s a sign of our progress and our need to recalibrate who we give advantage to. Affirmative action, of which I ’ m a beneficiary, I got Pell Grants, I got unfair advantage. Affirmative action is a wonderful thing and it should be based on color. It should be based on green. How much money you have or don ’ t have. Expand college enrollment in vocational programs. Mental health, ban phones in schools, invest in third places, Big Brothers and Sisters programs, we need national service. We need to tell people in the United States and Canada that they live in the greatest countries in the world, and we need to remind them of that every day by exposing them to other great Americans where they feel connective tissue. We can do all of this. We can do all of it. We have the resources. The question is, do we have the will? This is my last slide. It is an emotionally manipulative slide to try and get you to like me more. But it does have a message. This is the whole shooting match. Anybody here without kids ask someone with kids. You have your world of work, you have your world of friends, you have your world of kids. Something happens here, your whole world shrinks to this. So I present, as I wrap here, with just a few questions. One, if you acknowledge that our kids are the most important thing in our lives, that everything else we do here is meaningful, but our kids ’ well-being and prosperity is profound. If you acknowledge that they ’ re doing more poorly than previous generations, if you believe there ’ s a chance that the illusion of complexity has done nothing but provide cloud cover for the unbelievable transfer of good will, of well-being and of prosperity from young to old, and if you believe we can actually fix these problems and we have the resources, then I present to you, I posit, I augur the question that I hope has more voracity than it did 17 minutes and 24 seconds ago. And that ’ s the following question. Do we love our children? My name is Scott Galloway, I teach at NYU and I appreciate your time. Thank you. CA : We cut off the standing O. That was a long, long standing O. It was as long as there was at TED this year, Scott. An amazing talk, honestly. I want to ask you a bit about just your style of speaking, because I think it ’ s remarkable. I ’ m going to start here. This thing ’ s been seen by five-plus million people around the world already and growing. It ’ s the biggest single hit coming out of the last TED. I ’ m curious what feedback you ’ ve had from it. I mean, you laid out some pretty savage criticisms of a lot of people there. Have some of them come back hard at you? Have you mainly been amplified and cheered on? Give us a sense of the feedback on this. SG : First off, thanks for having me, Chris, and thanks for creating such a powerful platform to let people like me have this type of opportunity. 97, 98-plus percent has been exceptionally positive. Not yesterday, but the day before yesterday, I was on a Zoom call where I spoke after Speaker Pelosi and before Senators Blumenthal and Blackburn, and talking specifically about my recommendations around Big Tech. I don ' t know if I ' ll be invited back because I did not hold back. But it ' s been, you know, let me talk about, it ' s been overwhelmingly positive. It ' s been everything from... Billionaires reaching out and saying,"I have a philanthropy, and I ' d like your help allocating some of our resources,"to someone sent me a screenshot of an account with 10 million dollars in it and said,"It ' s yours if you run for president in 2028."To which I responded,"I have no interest in that, but I ' d really enjoy spending the weekend in Vegas with you."It ' s been really positive. So the points of pushback that I think are interesting, and I want to be clear, I get it wrong all the time. You know, I ' d like to think that some or even most of this is directionally correct. But what I know for certain is some of it is wrong. I got pushback around Social Security from Representatives, from seniors, that this is the most successful social program in the world, and that I shouldn ' t be demonizing old people for taking out something they ' ve already contributed to. I ' ve got pushback for fat **shaming**. So I ' ve gotten, you know, points of pushback. But I think that ' s important. I think that if you ' re going to make provocative statements, you need to be subject to pushback. And I would say that my goal is not to be right, It ' s to catalyze a conversation such that we can craft better solutions. And the thing that ' s been really rewarding about this dialogue is that when people send me very long emails, including from Congress and the Senate and people who run companies and **nonprofits**, the dialogue ' s been very civil. And I would say that the thing that I really enjoy, you know, I recognize about the TED community, is I ' m in a variety of different communities, whether it ' s TikTok or whether it ' s different conferences where I say these things. And the dialogue this has inspired has been remarkably more civil. It ' s been more, you know,"I liked this, I didn ' t like this,"but I really appreciate that the medium is the message. You know, people are more civil on LinkedIn than they are on **Twitter**. And I have found so far people are much more civil on TED than other platforms. CA : Well, that ' s nice to hear. We definitely, though, don ' t want people to hold back. And I know that in the comments there are some people who ' ve pushed back on specific aspects of the talk. But which of the various proposals and things that we could actually do about this problem, do you think have most chance of actually moving forward? Because it ' s so wide ranging. You know, it ' s like, it ' s very hard for young people to get to college now, hard to get a house and then, you know, all the way up to Social Security and so forth. Where do you think there ' s most traction, Scott? Something that might actually shift. SG : I think there ' s already a movement underway to expand freshman class at elite colleges, to invest more in vocational programs, to reverse this vibe or gestalt of higher ed as a luxury brand and return it to being a public servant. I think that movement was already underway, and it feels like there ' s a lot of momentum there, and some of the stats have really freaked people out. There ' s 10 to one MIT employees for every person who teaches. So I ' ve heard from the president of ASU saying, and the presidents of Purdue saying,"Come here,"and they ' ve been circulating my talk and raising money around the fact that they have not embraced this strategy. I think we ' re seeing – I ' ve heard from quite a few people in the budget office and Congresspeople talking about things including eliminating capital gains tax or just having one income tax, similar to what we had under Reagan, instead of favoring the income that capital earns or wealthier people versus current income. I have heard a decent amount around the idea of just general, generally speaking, tax reform. How do we put more money in young people ' s pockets? And then from what I ' ll call the retail media, whether it ' s Morning Joe or The View, which I ' ve been on since this talk, it ' s we need to change the narrative and stop criticizing young people and saying that their depression, their anxiety, their obesity as a function of their entitlement and not a function of public policy that people of my age have implemented. That, OK, maybe there is some truth to some of their disappointment. CA : So let me, before we go into the other details on it, let me pick up that specific point, because there is, I ’ m sure that some people watching this, probably older people, would say, are you sure it ’ s really this intentional thing as opposed to a consequence of just cultural cycles? So there ’ s this quote that you all know by Michael Hopf that, you know, hard times create strong men, strong men create good times, good times create weak men, weak men create hard times. That there ’ s an argument that we ’ ve been through so much prosperity that, you know, baby boomers, raised, you know, a generation of kids who just grew up too comfortable and that that is where part of the problem is. You are halfly arguing a different case. Is there any truth to that argument? SG : I think in almost every generation before ours, I ’ ll call it Gen X and baby boomers, there was a concerted effort to elect people who would think long-term and make forward-leaning investments. If you think about Apple, Google, Nvidia, all of these companies, Moderna, that have added trillions of dollars in market cap, it ’ s really, they build a thick layer of innovation on top of investments that have been made by the most successful venture capitalists in history, and that ’ s the US government with its limited partners, its investors, the middle class. And that there ’ s a lack of that forward-leaning investment. Basically, the only thing that passes for bipartisan cooperation now is reckless spending. Republicans want more military spending and lower taxes. Democrats want more social spending and, you know, higher taxes. And they agree on lower taxes and more spending. So I just feel like we are being irresponsible, even reckless. The way I loosely describe it is seven trillion dollars in government spending which juices the economy, five trillion in receipts. I ’ m in the club partying with champagne and cocaine, and the closest the young people get to the club is they can throw their credit card at me from downstairs, and I ’ ll rack up more debt on their backs so I can continue this prosperity. Even the markets right now, we don ’ t want to talk about this. The markets are being supported right now by an additional two trillion dollars in stimulus every year, in the form of government spending that ’ s beyond our revenues. In past generations, unless it was wartime, never would have engaged or supported that type of government spending. So I do think there ’ s something different here. I think the financialization of everything, Chris, where, as they said in”Jerry Maguire,”more money used to be a bigger seat in business class, now it ’ s a better life. And the fact that we haven ’ t faced really any external threats to rally us together to create more comity of man or more patriotism, there just seems to be not the same sense of selflessness or investment mindset there ’ s been with past generations. CA : One of the most powerful aspects of the talk, and there were so many powerful aspects, was your juxtaposition of different stats. So Harvard, this explosive growth in the endowment and, you know, no more students being admitted. I mean, you immediately know, when you just put those numbers together that there ’ s something very, very wrong with that picture. So powerful. I thought your line about just showing, you know, minimum wage has just not remotely caught up with the way at which capital is growing. And I wonder whether one, you know, there ’ s almost like some political ideas that we could push out there that said things like that. A society is not operating justly if the rate, its minimum wage rate, is growing slower than the rate at which capital is appreciating. It shouldn ’ t be linked to inflation or even wages. It should be linked to the rate at which capital is appreciating you know, in that society. Otherwise, inequality is absolutely bound to increase. And I ’ m curious which others of those have landed, like with Social Security, for example. I understand why people have pushed back. Do you see any workable, kind of, middle ground here where you could say, look, Social Security is important, we ' re not going to take the whole thing away. But we have to make it more just, we can ' t have it sucking more and more of the next generation ' s time. Like, what is the right metric to think about how much Social Security could be trimmed to be fair to the generation coming through that? Feels like one of the biggest numbers. SG : So I ' ve had a bunch of calls with people, different people in Congress and different PACs about Social Security. So means-testing and moving the age back, right? I mean, the majority of people, just even 100 years ago, didn ' t participate in Social Security because they were dead by that time. Now they ' re living 20 and 30 years beyond that. People will say,"Well, I invested in it, I want it back."I ' d say first it ' s called a tax, not a pension fund, meaning that it might be distributed to other people. Two, people who will live to be 85 take out two to three times more than they put in. I ' m not saying we should do away with it. I ' m just saying it should be means-tested, the age should be pushed back to reflect that people are working much longer. Also just basics. The person who put together the slides, Mia, makes 160, 000 dollars a year, I ' m allowed to say that with her permission. She ' s 26, very talented young woman. She pays 9, 000 dollars a year, or six percent, in Social Security. This year, I ' ll make 100 times that, and I ' ll pay, wait for it, 9, 000 dollars in Social Security income tax. It ' s capped for anyone making over 160, 000 dollars. So if this generation is the wealthiest generation in history, why have we decided to cap Social Security for rich people? So I don ' t have a problem with stimulus. I don ' t have a problem with social programs. But Social Security is another example yet again, of we are really taxing – six percent is a real number for young people their whole life, but it ' s been de minimis for me because I ' m old and wealthy. So it ' s another example of how this transfer under the cover of dark, that wealthy people are in there saying,"We ' ll position this tax as onerous, but we ' re going to cap it like it ' s a good thing at six percent."Why is Jeff Bezos not paying six percent of his income into Social Security? It ' s an off-balance-sheet item, it doesn ' t impact the deficit, it ' s funded every year. But right now, 40 percent of all government spending is allocated for programs that go to seniors. It ' s about to be 50 percent in ten years at this rate, which means that crowds out investments and forward-leaning investments such as education and technology. And we also just need to acknowledge and this sounds ageist, and it is, old people are less productive and more expensive. And so unless we figure out a way to acknowledge that they spend a longer period of their lives being expensive and unproductive, our economy is basically just going to be an engine to kind of support the most expensive nursing home in the world. This is happening in Japan, it ' s happening in Italy, and it takes an economy into decline, because you can ' t make forward-leaning investments that encourage wealth or inspire wealth for young people, and they don ' t have kids. In terms of minimum wage, this is an easy one. I ' ve been positioned as anti-union. I ' m not, there should be one union. It should be in DC, and it should be federally-mandated minimum wage, 25 bucks an hour, except in certain areas where there ' s an exceptionally low cost of living. And the incumbents and corporations will start this bullshit narrative of it ' ll kill the economy. No, it won ' t. In Washington state and California, where they raised minimum wage, the economy actually grew because the wonderful thing about low- and middle-income people is they spend all of their money, meaning the multiplier effect is greater. If it had just kept pace with productivity and inflation, it ' d be 23 bucks. That ' s an easy one. Mandatory, federally-mandated, 25 bucks an hour. Would McDonald ' s and Walmart stock take a huge hit? Sure.

Would a bunch of small businesses go out of business? Yes. And it ' d be worth it. CA : So, as you say, you don ' t hold back. And there were definitely a few things in the talk that provoked comment. So Kyra Gaunt, I see in the **chat**, found your comment about diabetes a bit smug. She says there are epigenetics to consider. It ' s complicated for members of marginalized groups. And you say you ' ve heard from other people on that. Do you want to clarify your view on that at all? SG : It ' s a fair feedback. As someone who was born to parents who were both tall and thin, so easy for you to say, Scott. What I have said in the past and you know, if I sound offensive, I **apologize**, is I believe that we need bottom line to put more money in the pockets of lower-income people such that they can eat better. I think we need to tax the food industrial complex relative to its externalities. Obesity, morbid obesity has gone from five to nine percent in the last 30 years. Obesity has gone from, I believe it ' s gone from 30 to 45 percent. The only thing that Americans really share is it ' s 70 percent of us are overweight or obese. And if you look at McDonald ' s, Kraft, General Foods, PepsiCo, Coca-Cola, they ' re not companies as much as they are obesity indices. And their stocks have gone up 10 to 12x as obesity has gone up. And if you give those two, their stocks and obesity rates, to a statistician, they will tell you they are highly correlated. So as a result, because there ' s money in the diabetes industrial complex, we have Unilever **celebrating** obese women. We have people saying you ' re finding your truth. And this is from the same company that sells Ben and Jerry ' s and Axe, which basically encourages young men to fuck anything. So the notion that they are concerned about, are trying to liberate people or all of these apparel companies saying that you ' re finding your truth – I want to go back to the ' 60s, where Kennedy said it was American and patriotic to be in great shape. I used to train every year for the Presidential Fitness Awards, and that was seen as fat **shaming**, so they did away with it. I don ' t think there ' s anything wrong with **celebrating** fitness, but also at the same time recognizing some people are dealt a difficult hand genetically and providing them with money and resources so they can get out of food deserts. I also believe GLP-1 drugs are the most impressive technology of the last year, not AI, and we should be pushing them into low-income communities. I think they ' re an absolute game changer right now. Here ' s a crazy stat : the region of America that has the greatest penetration of GLP-1 prescriptions is also the thinnest. It ' s the Upper East Side. Because right now, GLP-1 drugs are being used for ladies of lunch who want to lose that last 10 pounds. I hope that more and more of it is imported illegally across the border, where it ' s 100 bucks a month versus 1, 000, and it gets to the people who really need it, which is low-income people who don ' t have the money for diabetes. Because the industrial food complex wants you to be obese so they can hand you over to the medical industrial complex of hip replacements, kidney dialysis, knee replacements, heart stents. That ' s an **enormous** industry. I want to put Coca-Cola, Kraft, McDonald ' s and some of these hospital systems absolutely out of business. There ' s too much incentive to convince you that being obese is some sort of personal liberation. It ' s not. CA : So this is so interesting. Like, somehow, and I feel this for TED as well, we have to figure out how to have conversations about difficult topics that are respectful of people. And, you know, everyone, as you say, has been dealt a different hand, but that don ' t ignore the basic facts of health or of science. And that is what is so hard to do in the current environment right now. We ' ve all got frightened, I think, of saying the kind of thing that you just said. And there probably are still, I don ' t know Kyra, how you felt about what he **replied** there, but I feel like we need to do both, you know, we need to be provocative and also respectful. And I heard that in you. So I think that ' s cool. And then again, actually not to pick on Kyra, but I noticed that she pushed back on your point about social media, that, you know, one of the fixes is to get our younger kids off of social media. Points out, and I ' ve felt this reading Jon Haidt ' s work myself, that those communities that people connect with on social media sometimes are incredibly beneficial. So you think of the kid who ' s queer or is struggling with something and doesn ' t find anyone in their community who they can connect with, finds a community online. I mean, there ' s presumably some benefit in that. Is there any way of making recommendations where there aren ' t swings and roundabouts? There aren ' t, you know, sort of downsides to what we recommend? SG : Look, one in five LGBTQ high schoolers is going to try and kill themselves. So I ' m involved with the Jed Foundation that works with high schools to try and identify or distinguish between kind of what is normal abnormal **teen** behavior and suicidal ideation. Because we really have an epidemic of self-harm at a **teen** level, and there ' s just no getting around it. Some trans and **gay** kids and other kids from special interest groups find other people like them on social media. I guess the question would be if banning social media under the age of 16 or age-gating it on the whole were demonstrated to be an absolute huge net positive – because I think a lot of those communities are actually the subjects of incredible bullying and **shaming** online, and are more prone to go down a rabbit hole of depression and end up in self-harm – are there other programs that could replace that incredible community some of them have found? But look, when you make huge government decisions, there is no decision where there ' s not going to be losers. I mean, the special-interest-group kid who finds people online at 14, when social media is age-gated, yeah, that person suffers. I guess the argument I would make is that if you were to look at any study about the general impact it ' s having on people under the age of 16, the benefits dramatically outweigh the soft tissue here. So the question is, could we do it? But at the same time recognize we still have an **enormous** self-harm problem among kids, especially in the LGBTQ community, and move in with other programs or other analog means of helping them find other people. But I want to acknowledge, there ' s just nothing I recommend that doesn ' t have a downside. CA : Some people are asking about practical things we can do. So Adrian Neubauer said,"As an elementary school teacher, what ' s your advice for helping Gen Alpha students? I don ' t want my students to grow up having such apathy towards learning and being successful. I definitely don ' t want my students ' highest aspirations to be TikTok influencers. How do I help them?"SG : Well, the first thing is, I think this is an easy one, I don ' t think there ' s any reason for anyone under the age of 16 to be on social media. I think we need a massive investment. I think we need to tax every person in a private school and use that capital to invest in after-school programs and third places. I went to public schools all the way through graduate school. I went back to my high school, university high school, and 93 percent are kids of color. It ' s got the unfortunate designation of having the greatest proportion of kids who are classified as **homeless** in LAUSD, and they just cut their drill team. They just cut their band, their drumline, because they don ' t have money. I just think so much of this, Chris, unfortunately, it ' s like households with a lot of anxiety. It sounds very crass, but I think a lot of it, a lot of problems are solved with money. Show me divorce and I ' m usually going to show you some sort of economic stress or poor alignment around economics. And when you can add a quarter of a trillion dollars in five minutes post the earnings call of Nvidia, and at the same time, you see corporations are paying their lowest taxes since 1939 and the top 25 wealthiest Americans are paying a tax rate of six percent, and five companies have added the value of the entire global auto industry in the last six weeks. We have the resources. So I think a lot of this comes down to funding, giving kids more third places such that they ' re not staring at their phones. And also demanding that parents and demanding that entire school systems go phone-free and that there ' s no social media. And on a more tactical level, I absolutely think we need to ban or divest TikTok. I think it makes absolutely no sense to be raising a generation of civic, military and **nonprofit** leaders who hate America, and it ' s all wrapped in cute dances. I just think it ' s insane that we would be this stupid. One of the wonderful things about America is our optimism. But one of the downsides of that is we ' re much easier to fool than convince we ' ve being fooled. I think every day we ' re being fooled by TikTok. I think this is the most unbelievable propaganda tool in history. Unfortunately, it ' s not ours. CA : You mentioned Nvidia there, which has exploded in value in the last couple of years, last year really. Isn ' t that to some extent a counterargument to your earlier comment about you had this unique opportunity to buy these cheap stocks like Netflix and Amazon at low prices. I mean, there are always... Is the market fundamentally changed, there aren ' t that kind of opportunity for smart people to identify trends and get the same kind of wealth? Was it just a timing thing? SG : No, what we ' ve decided is we want capitalism on the way up. We want low taxes, and the pioneer, the individual who earned the money and should pay a low tax rate because he or she is the most productive citizen. And then on the way down when there ' s a virus or some exogenous event, which happen a lot, whether it ' s war, famine, revolution or a virus, the CEO of Delta says,"We ' re all in this together,"and wants a massive bailout, despite the fact that 80 percent of the free cash flow of airlines the 10 years before were either used on dividend or stock buybacks or CEO compensation that juice their compensation. So look, capitalism on the way up and socialism on the way down is cronyism. And we ' ve gone full cronyist in the United States. And I understand the rationale for pumping the economy full of stimulus, because they said we needed to overdo it versus underdo it, but then no one suggested a special tax for the unbelievable champagne and cocaine disco party we ' ve had in the markets. Markets are touching all-time highs right now. And the myth that my generation is fomented across the economy, the young people take hook, line and sinker is the following. There are two phases of everyone ' s life. There ' s investing, and then there ' s the harvesting phase. Investing is when you do your best to make more than you spend, and save some money, so you can deploy an army of capital that grows in your sleep, such that you can have some security and some balance as you get older, and you begin harvesting. I ' m entering the harvesting stage of my life where I ' m no longer making as much as I spend. During the investment part of your life, in you ' re younger years, you want markets to crash. You want real estate that ' s inexpensive. You want to dollar-cost-averaging at the low price. The reason I get to roll with you in London and I get to go to the south of France tomorrow, Chris, is because we let the markets crash in ' 08, and I got to buy Netflix at 12 bucks, which is at 650 now. What we ' ve decided now purposefully, is that we aren ' t going to let the markets have a natural cycle of disruption, that anytime the markets are threatened, we ' re going to weigh in with the credit card of our young friends, of our daughters and our sons, and we ' re going to artificially inflate and support the markets. That is nothing but a transfer of wealth. You need disruption. You need churn. These are purposeful decisions to protect old on the credit card of the young. I don ' t have a problem with stimulus, but for God ' s sakes, those of us who have benefited from it should pay it back. CA : So there ' s a great question here from Brett McCall."During this talk, there are an overwhelming number of points that are all delivered in an easy-to-understand way."That ' s true, by the way. I don ' t know of a talk where there ' s been such density of points, of facts that connect, really remarkable. Anyway, he says,"It leaves me charged with the desire to join, ' the movement ' and make change. It ' s almost like a full semester-worth of issues that could be movements, but there isn ' t an explicit community or movement to join. Scott, where would you suggest a starting place for someone who becomes inspired by your talk?"SG : It ' s hard to do this without being political, but try and find Congresspeople or get involved in campaigns of people who are, one, talking about the deficit, talking about the war on young people, having difficult conversations. I mean, there ' s some very basics. And the speaker before me, Andrew Yang, final five and ranked choice voting. People who are going to be moderates, who are going to think long-term about society. I think we need absolutely more young people in Congress. So I would say political action, just being engaged. And then on a ground level, getting involved with local schools, getting involved in charities and efforts to help young people in special interest groups find third places, find places to find each other. I think all of this is somewhat couched in young people, in an epidemic of loneliness. So anything you can do to create resources and opportunities for young people to get out of the house. I think it ' s up to parents to get their kids off – I mean, I think it ' s not what to do, it ' s what not to do. In terms of a political movement, I ' m still waiting for a leader to come forward and say,"I ' m going to piss off everybody. Old people, you ' re going to hate me. Unions, you ' re going to hate me. Rich people, you ' re going to hate me."You know what we need? We need the biggest class traitor in history. Someone who shows up and says,"Are we going to decrease government spending, or are we going to raise taxes?"And the answer is yes. It ' s time to start acting like adults and paying it back to some of the incredible Americans who have made sacrifices such that we can enjoy this prosperity. I haven ' t seen that person yet. CA : Politicians assume that that ' s an unelectable slate. And maybe they ' re underestimating the intelligence of their citizens. SG : I think people are ready. Do you know one of the top three issues among young people is right now? People think it ' s Middle East, it ' s not. That ' s like number 17 because of the zombie apocalypse of useful idiots on campuses that has been highlighted in media and in TikTok to give you the sense that every student is protesting. 2, 300 arrests, 40 weren ' t students. I ' m at NYU, 99. 9 percent of the kids are just getting their shit done. The new undergraduate president of Columbia is an Israeli girl. You know, we need... anyways, one of the top, top issues for young people, one of the top three? The deficit. So I just think we need a pragmatist who says,"I am not going to ever insult anyone on the other side of the aisle personally to get on TikTok and raise money. I ' m going to talk about really boring shit. Tax policy, the deficit, **teen** depression, obesity. And I ' m going to make big sweeping changes. And guess what? All of you are going to suffer, all of you."Because there ' s just no getting around it. We need to make a fraction of the sacrifices that a lot of Americans made 80 years ago. I don ' t know about you, I was very moved by the D-Day commemorations. But every generation up until this one has made real sacrifice for forward generations, except this one. So we need to catch up. I think people are ready for that. CA : Scott, you ' ve spent a lot of time giving advice, especially for young men. Do you think that the problem has been especially bad for them for some reason or that a disproportionate share of the solution will be found by fixing the problems that young men are facing right now? SG : Well, all of these things are especially acute. So with young men – and this is really the thing, this is my **passion** project – four times as likely to kill themselves, three times as likely to be addicted, 12 times as likely to be incarcerated. There has been no group globally that ' s ascended faster than women. More women globally are pursuing tertiary education, which is remarkable when you think about – many nations don ' t allow women to pursue tertiary education, and twice as many women have been elected to parliament over the last 30 years around the world. Now domestically in the US, there has never been a cohort that has fallen further faster than young men. More women, single women own homes now than single men, which is fantastic. In urban areas, women are out-earning men. These things are wonderful. We don ' t have a **homeless** crisis. We have a male **homeless** crisis. We don ' t have an opioid addiction crisis. We have a male opioid addiction crisis. And unfortunately, something that gets in the way of programs, Chris, is that because of the benefit, the massive benefit, privilege and advantage that you and I received, we hold these young men accountable. We use words like accountability or pull yourself up by your bootstraps. If any special interest group is killing themselves at four times the rate of the control group, we ' d move in with programs. And I think there just generally needs to be a change in the narrative and the dialogue. And we need to recognize that empathy isn ' t a zero-sum game. **Gay** marriage didn ' t hurt heteronormative marriage, civil rights didn ' t hurt white people. And recognizing the problems that young men especially are facing – one in three men under the age of 30 hasn ' t had sex in the last year. Three million working-age men have given up looking for work. Fifty percent of millennial men are no longer even trying to date. And so you have this cohort that is doing so poorly, and if we don ' t have a vibrant middle class full of successful, economically and emotionally viable men, the nation is going to collapse. Because here ' s the thing about women, we don ' t want to have an open and honest conversation about mating. And this is the following. Women mate, socioeconomically, horizontally and up ; men, horizontally and down. And when the pool of men available horizontally and up keeps shrinking, we ' re going to have a lack of household formation. Sixty percent of people aged 30 to 34, 40 years ago, had at least one child. Now it ' s 27 percent. The whole shooting match, the whole shooting match, why we engage in all this shit, is so we can find someone we love and raise kids together. That is the most rewarding thing. Everything else is a means. That ' s the ends. And when a nation is failing to do that and just creating a generation of anxious, obese and depressed kids who don ' t like themselves, much less have the opportunity to like other people and get together and start having sex and fall in love and have kids and have economic prosperity, then what the fuck is any of this for? So I think this is a call for all hands on deck. If young people, and especially young men, aren ' t doing well, then we are not going to have a functioning nation or economy, full stop. And it ' s time to put the politically correct agenda away and say,"Well, you ' re being sexist."Yeah, I ' m sexist. I believe the genders are different. If you don ' t believe me, put a three-year-old girl and a three-year-old boy in a room with cars and dolls and see what happens. That doesn ' t mean we reduce empathy for either gender, but we can ' t even have an honest conversation about this stuff because someone sees an opportunity to virtue signal and get a Guardians of Gotcha pin rather than actually addressing the fucking problem. That was a rant. CA : Well, I will say that you rant better than anyone I know. I mean, I really, just as someone who ' s hosted a lot of TED Talks, I was stunned by how this one went. Like many, many speaker coaches looking at you would say, Oh, he ' s very monotone. It ' s very, you know, where ' s the – it just unwraps in a very, sort of flat voice initially. But every word is so carefully chosen and follows so consequentially to what ' s just gone before, that you tap into a different part of people ' s brains than most talks do. People go, yes, you know, that makes sense, that makes sense. And then it builds, and then your anger and frustration starts to come out and get dialed up. It ' s absolutely remarkable rhetoric. I don ' t know how you learned it or where you got it from, but it ' s really unique out there, I would say. And I think it ' s a super powerful weapon.

And I mean, I don ’ t know if people listening agree with this. This is really unusual... It ’ s not like everyone will agree with every single word, Scott, that you said, but it ’ s a really unusual and powerful piece of rhetoric about something that really matters. And for that, I feel, I think we all feel huge gratitude to you and wish you well on this continued journey. I mean, there just isn ’ t a more important issue. So bravo to you. Thank you. SG : Chris, I appreciate the generous words. And what I told my wife is that after 30 years of working my ass off, I ’ m an overnight success because of Chris Anderson and TED. So thank you. CA : You had visibility everywhere, and you do. It ’ s amazing how much your voice is out there. If people don ’ t know it, Scott has his own – Well, just talk about the two or three podcasts you ’ d most like people to do, because I think that ’ s the best way to get closer to you. SG : Yeah, to resist is futile, I ’ m like AOL in the ’ 90s. If you stick your hand in a cereal box, you ’ re going to find something from Scott Galloway. I have”Pivot,”I have”Prof G,”I have my newsletter,”No Mercy, No Malice”and a slew of books. So to resist is futile. [Want to support TED?] [Become a TED Member!] [Learn more at ted. com / membership]

Text Sample 362: POS Dim 3 – 2024 – Score 12.00 – 2024/t003268.txt
Susan Zimmerman : Welcome, Suleika. Suleika Jaouad : Hi, Susan. SZ : I ’ m going to dive right in with”American Symphony.”I would love to first say thank you for sharing your story with us in this gorgeous **documentary**. Can you share a little bit more how this project came to be? SJ : So the director of”American Symphony,”Matt Heineman, who is an extraordinarily brilliant filmmaker and a dear friend of ours, says that if you end up with a project that you started with, you ’ re not doing your job right because you weren ’ t listening along the way. And that proved to be true in ways that we couldn ’ t possibly imagine. I knew Matt through my own work. Jon had worked with him on his last film,”The First Wave,”that took place during the pandemic, and he and Jon started talking about the idea of a short process film that was going to follow Jon as he composed what he called”American Symphony,”which was his reimagination of what a symphony would look like if it were created in the 21st century. And Jon was committed to really upturning what we think of as a symphony. He wanted to travel all across the country to recruit jazz musicians and blues musicians and classical musicians and **Indigenous** musicians and musicians who don ’ t know how to read music and to figure out how to bring them all together. So initially, that was the film, and about a week after we decided to move forward with that project, on the very same day I began chemo and the same week I learned that after a decade-long remission, my leukemia was back, Jon was also nominated for a record historical number of Grammys. And so right from the beginning of the filming process, the initial conceit changed. And it was the kind of project and really a testament to Matt ’ s style, where we weren ’ t too worried about figuring out what the story was. Matt specializes in cinema verité, and so he followed us from sun up until we went to sleep every single day for about seven months. And at the end of that process, we had 1, 500 hours of footage, which, as you can imagine, is a lot of footage. But we knew that we wanted to document all of it, to document what it means to be living a life of contrasts, what is required of us when we ’ re in the belly of the beast and how creativity and our individual and joint, sort of, creative language inform that project. SZ : For two people, who are very much the directors of their own creative projects and worlds, how did you negotiate creative control in your work with Matt Heineman, especially in such vulner-

able and personal circumstances? SJ : You know, it wasn ' t easy for either of us because we are used to, you know, captaining our own creative ships. But I think, you know, surrendering to the process was really important. We didn ' t want to get in the way of the filming. But we were also very much creative partners with Matt. And I ' m not sure that a film like this would have been possible without deep trust with the filmmaker and ongoing conversations about what we were comfortable with and what we weren ' t, and we had those conversations daily. We had them weekly and monthly. And the truth is that when I decided to participate in this film, I had no idea if I was going to survive long enough to actually see the finished product. And what Jon would always say is : if nothing else comes out of this, we ' ll have the world ' s most beautifully shot home videos to share with our families and loved ones. So it was really, you know, a process that required trust. It required a leap of faith, and it required surrendering the kind of creative control that we ' re used to. SZ : You also shared, in a recent"Isolation Journal":"When the ceiling caves in, it ' s terrifying and disorienting. Yet those moments have also been the most fertile stretches of personal and creative growth for me. A couple of months after my second bone marrow transplant, I was rehospitalized due to complications, and I told one of my nurses that I get my best work done there, and I joked that it had become my favorite artist ' s residency."Can you share more about these creative periods during extremely challenging circumstances? SJ : Yeah, so I want to go on record and say that, you know, these moments when the ceiling caves in on you, when you can no longer assume structural stability... don ' t suddenly fill me with creative inspiration. It ' s the very opposite. I had so many plans, both professional and personal, before I learned of my relapse. I was getting ready to write a reported book that would require me to be on the road for long stretches of time. Jon and I were talking about getting married and starting a family, and I moved through my own version of the five stages of **grief**. And the first is shock, often mixed with fear. And then I get to a place of defeat. But something interesting happens when the possibility of productivity is taken away from you. And when I ' m able to move past that **grief** and that fear, and to start to get curious about the isolation and the sense of solitude that comes with, you know, having been brought to your knees on the floor by some kind of life event. Even though it doesn ' t feel like I ' m doing something, I ' m in the chrysalis. I ' m as larval and goopy and unformed as I ever feel. And what these periods of being extremely sick have done for me is that I ' ve had to get good with my own company. I ' ve had to get good with quiet, with having no plans. And of course, that ' s not a luxury that we ' re all able to carve out for ourselves. But when I ' m able to surrender to that not knowing, interesting things emerge. And I had, you know, the misfortune and privilege of having been through this once before. And I knew that to go into this treatment, to enter the hospital for many weeks and months with expectations of what I was going to do from my bed or trying to hold on to whatever plans I ' d made before, was going to be a recipe for frustration and defeat. And sure enough, pretty shortly after I entered the bone marrow transplant unit, I was on two medications that caused my vision to blur and double. And so the thing that I had always reached for in times of duress, which for me is writing, wasn ' t as easily accessible to me. It was frustrating, it was hard, it was halting. And instead of trying to keep doing that, I decided to be open to whatever felt good. And for the first time since I was a little kid, I picked up some watercolors and some brushes. I transformed the bedside table in my hospital room into my little palette of paints, and I began painting the nightmares I was having, the fever dreams I was having, and transcribing them onto the canvas without any expectation of doing anything with these paintings other than that they simply felt good. What I love about watercolor is that you don ’ t have much control. Watercolor is all about the beautiful, happy accidents. And so that felt like an apt medium, given the fact that my circumstances were very much mirroring that lack of control. I ’ ve come to believe that survival is its own kind of creative act. When you can ’ t speak because the chemo sores in your mouth make it too painful to speak, you have to find new ways to communicate. When you ’ re confined to a bed for many weeks and months, you have to find new ways of traveling in your imagination. And really, it requires complete surrender and an **openness** to whatever may emerge, and also a curiosity about the changes that are happening and trust that something will come out of it, even if it ’ s just in terms of my own personal growth and the reshuffling of the priorities. Though often, well, it doesn ’ t feel that way in the moment, it ’ s ended up being my most fertile creative stretches where I ’ m pushed to experiment and to create in new ways. Not in spite of my limitations, but because of them. And it becomes a process of finding purpose in that pain and trying to alchemize it into something interesting and **thought-provoking**, and maybe even useful and beautiful. SZ : This leads me to think about 22-year-old you, you know, with your first leukemia diagnosis and your”Life, Interrupted”column in The New York Times. How did that column come to be, because I feel like your journey now, you were pulling from these resources, but how then did you come to the conclusion of,”I need to write a column about this, and I need to share this with the world?”SJ : So I first got diagnosed with leukemia when I was 22, and overnight, I lost my job, I lost my apartment, I was working as a paralegal in Paris at the time, but hoping to become a foreign correspondent. And maybe worst of all, I lost my independence. And 22 is such a funny age where, you know, you ’ re no longer a kid, yet, you ’ re not a fully formed adult either. And so that sense of in-betweenness accompanied me from my very first day in the hospital. And I spent a really difficult first year of treatment, I spent about eight months, cumulatively, of that first year in isolation in the hospital where I wasn ’ t able to leave my room. And when I went into this, not knowing anything about what it means to be sick, I remember packing a suitcase full of books like”War and Peace”and cheerfully announcing to my parents that I was going to use that first summer in the hospital to read through the rest of the Western canon. And let me tell you, naivete has a short shelf life. And that was true for me. I didn ’ t read a single one of those books. And I felt really angry. My treatments were not working, my leukemia was becoming more aggressive. I was enrolled in a phase 2 clinical trial that hadn ’ t yet been proven to be safe or effective. And I was really struggling with a sense that my life had been interrupted. My life had been put on pause at a time when I was watching my friends begin their lives, you know, travel the world, start careers, get married and all the other big and small milestones of early adulthood. After I learned that the treatments weren ’ t working for me, that a friend of mine came up with the idea of a 100-day project, and the premise was really simple. We decided we were each going to do one creative act a day for 100 days. And for my project, I decided to keep the bar very low, knowing how unpredictable my energy was, and I decided to recommit to the thing that had been my companion from the time I could hold a pen, which was, keeping a journal. And it didn ’ t matter how much I wrote. Sometimes it was many pages, sometimes it was a sentence. Occasionally it was the F-word, which felt apt for that particular moment. But in the course of keeping that journal, something interesting began to happen. Prior to this project, I, you know, of course, was not in a place where I could become a foreign correspondent in the way that I dreamed of. But I realized, in keeping my journal, that I was using it as a kind of reporter ’ s notebook. I was writing about all the things that felt impossible to talk about with my loved ones. I was writing about the sense of guilt and being a burden, that can come with being sick. I was writing about navigating our health care system. I was writing about the patients I was befriending in the cancer unit. I was writing about the nurses and the various characters in my medical team. I was writing about **sexual** health, I was writing about **shame**, I was writing about what it feels like to fall in love while you ' re falling sick. And by the end of that project, I realized that, you know, while I couldn ' t travel the world and report on the stories of others, I did have a story to tell, and that I was reporting from a different kind of conflict zone. I was reporting from the front lines of my hospital bed. And so, like the good millennial that I am, I decided to start a blog. It was in the weeks leading up to a bone marrow transplant, and I knew that the odds were stacked against me. My doctors told me point blank I had a 35-percent chance of long-term survival, and I think it lit a kind of fire under me. Staring your mortality straight in the eye can be a great motivator. There ' s no longer this illusion of endless time. Time to eventually get to the things that you want to do, time to figure out what you want to contribute in whatever, you know, small or big way, to the world. And so I kept this blog for a couple of weeks and I took it really seriously. I would write every day, and through an old journalism professor of mine, it got sent to Tara Parker-Pope, my wonderful editor at The New York Times, and we had a couple of email exchanges, and she called me and invited me to contribute an essay. And I took a deep breath and I said,"Thank you, but I ' m not interested in writing an essay. What I ' m interested in writing is a weekly column, because so often illness stories are told from the vantage point of someone who survived. But I want to write from the trenches of that uncertainty."And I went on and on and on. And to my surprise, she said,"OK, we ' ll try it for a couple of installments and see how it goes."And I had never been published before, I ' d never had a byline, and this pitch that I had just performed would have seemed wildly presumptuous to pre-diagnosis 22-year-old me. I would have been grateful for a fact-checking position. But that ' s the thing about confronting your mortality, is that... it can make you brazen. And I knew that I didn ' t have time for internships and fact-checking positions. And so for the first time in my life, I said exactly what it was that I wanted to do without any expectation, but I knew I needed to try for it. SZ : The response to that column,"Life, Interrupted,"led to your subsequent road trip across the US to meet some of the people that wrote to you while you were in the hospital. You spoke about this journey in your 2019 TED Talk, and you wrote about it in your memoir,"Between Two Kingdoms,"your New York Times best selling book. Can you share first, a little bit more about the title of the book, and then how did the process of making this trip and writing it impact you and your readers? SJ : So I ended up spending four years in cancer treatment. And throughout those four years, the goal was always to eventually be cured. And I hadn ' t given much thought to what would happen after that, in part because it felt like such a tenuous, flimsy hope. And to my great surprise, when I did get the all-clear from my doctors, when the port was removed from my chest and I was sort of released from the medical bubble, instead of feeling great excitement, and instead of quickly and organically folding back into the world of the living, I found myself deeply stuck. And to my great surprise, the hardest part of my cancer experience, on a personal level, began once the cancer was gone. I very quickly realized, you know, of course, that I was no longer a patient, but that I couldn ' t go back to the person I ' d been pre-diagnosis. And I had no idea who I was and how to find my footing among the living. And I was struggling to figure out how to carry the imprints of this experience. Out of my group of 10 cancer comrades, as we called each other – young people who I ' d befriended during those years in treatment – only two of us were still alive. So I knew how lucky I was to be alive, I knew that... I didn ’ t just want to be someone who was surviving but living, because after all, you know, what was the point of having endured all that I ’ d endured if not to live a good life, a meaningful life, a beautiful one? But I didn ’ t have the tools to do that. And without, you know, a cavalry of doctors and nurses and friends checking up on me, I realized, you know, there was no road map for the way forward and that I was going to have to create one for myself. And so I spent this lost year trying and failing and trying and failing to move on, only to realize, you know, that we don ’ t get to move on from the painful parts of our past. As much as we want to, you know, we can ’ t compartmentalize them and stow them away because they always bob back up to the surface and often with a vengeance, and that instead I was going to have to figure out how to move forward with the imprints of my illness, both on my body and on my mind. And so one of the very first things I did, because I was still in a place of feeling afraid, feeling, you know, ironically, afraid of the outside world. I was comfortable in the hospital ecosystem to really, first of all, give myself the time to properly **heal** from that experience, but also to figure out what was on the other side of that fear. And so I learned how to drive, and I ended up returning to some of the letters I ’ d received from readers of”Life, Interrupted,”who had shared with me their own experiences of aftermath, of figuring out how to do the hard work of recovery and figuring out how to do that hard work of moving forward, with whatever had happened. And so I decided to sublet my apartment. I borrowed a friend ’ s car. I got a bunch of camping gear, and I embarked on a different kind of 100-day project. A 100-day, 15, 000-mile solo road trip with my dog Oscar as my co-pilot, and to visit some of these strangers who had been lifelines when I was at my sickest, and to talk to them about that experience of in-betweenness. So to wrap this up, the title of the book,”Between Two Kingdoms,”is a reference to Susan Sontag ’ s essay”Illness as Metaphor,”where she describes how we all have dual citizenship in the kingdom of the sick and in the kingdom of the well, and that it ’ s only a matter of time until we use that other passport. But what she didn ’ t talk about was that liminal space between the two, where maybe you ’ re not either sick or well. And that became the premise of that book. It ’ s figuring out how to exist in the messy middle. SZ : That book launched during COVID. Something else that was beginning at that time was the”Isolation Journals”in April of 2020. Can you share what was the inspiration for this project, especially at such a unique moment in time? SJ : In the early days of lockdown, as the world was shutting down, as we were all having to pivot and to put our plans on hold and to figure out how to live our lives within very, you know, major constraints. So much of that experience felt bizarrely familiar to me. Everything from wearing a face mask and, you know, walking around with gallons of hand sanitizer to being isolated at home. And, you know, isolation is its own epidemic, one that predated the pandemic and one that continues now. And I decided, on April 1, 2020, to share what has always helped me transform that sense of isolation into creative solitude and connection and even community. And I launched a newsletter called”The Isolation Journals.”It ’ s a free newsletter, and we invited our larger community to do their own 100-day project. And so every single day, for 100 days, we had a different guest contributor write an essay in a journaling prompt. We had artists and writers and musicians and community leaders. One of my very favorite essays and prompts came from Lou Sullivan, a seven-year-old, two-time brain cancer survivor who shared this game that he played in the hospital called”Inside Seeing,”which was essentially his take on **meditation**. And within 48 hours, we had over 40, 000 people who ’ d signed up. And it was extraordinary to see what can happen when we dare to share our most vulnerable, unvarnished stories. And the reverberation that that creates. And so that newsletter continues on to this day. We have over 150, 000 community members. And every Sunday we send out a newsletter with thoughts from me and then a new essay and journaling prompt. But what ' s surprised me so much is that people interpret journaling in all kinds of ways. Some people use the prompts as conversation prompts, others use them as thought prompts. Some people reinterpret the idea of journaling, not as old fashioned, you know, pen and paper, but they ' ll paint to the prompts, they ' ll do modern dances. And so it ' s really been such a nourishing, tender, life-giving space. And I ' m so proud of this thing we built. I know what it ' s like to feel alone, to feel like you ' re the only person **suffering** in a particular way. But we have so many extraordinary artists throughout time, from Frida Kahlo to Virginia Woolf to Audre Lorde, who have taken that space of confinement, who have reimagined their limitations as creative grist. SZ : We have a question from Celia."Our world tries to avoid having feelings, but in ' American Symphony, ' you said that you didn ' t want to develop a thick skin. You wanted to feel it all. Can you say more about how that approach impacts your spirit and how you experience life feeling it all and how others can do the same?"SJ : You know, as a culture, we resist discomfort. We resist confronting the fact that all of us are here on this Earth for a very short period of time. I ' m not special, I live a little closer to the veil, because of the nature of my illness. And as tempting as it can be, you know, to compartmentalize that discomfort, to plaster over it, to **numb** it, I think there ' s also so much to be gained when you unguard your heart. And it ' s the hardest thing in the world to do. I knew going into this that it wasn ' t going to be easy. I had no illusions, not only about the toll it would take on me and on my loved ones, but I also knew that there was a lot to be gained if I could resist that urge... to look away from the things that scared me most. And rather to engage in them directly. I wanted to be open to all of it. I wanted to be open to the beautiful things and to the painful things, and to really learn how to hold both of them in the same palm. Because in varying ways and for varying reasons, that ' s the work that we all have to do. There is no binary. Life isn ' t either good or bad. We ' re not either happy or sad or healthy or unhealthy. Most of us, you know, exist somewhere in the middle, and depending on the hour or depending on the day, we might shift from one to the other. Trying to do the opposite of having tough skin, trying to have tender skin, is not something that comes easily to me. It ' s something I have to work at every single day. But I know that for me, it ' s the only way I can shift out of that survival mode and in to fully living and feeling alive. SZ : We have a question here from Lauren."In your book, you shared how cancer impacted your relationship with your previous boyfriend. Have you carried these lessons and experienced with this return of your leukemia? Were you able to strike a better balance as a caregiver / patient? And how did art making and his music making affect your relationship this time around?"SJ : The most surprising toll isn ' t what happens to the body, it ' s what often happens to your relationships. And so at 22, I was pretty isolated. The friends that I had played beer pong with in college were not necessarily the ones who showed up to sit at my hospital bedside as my hair was falling out in clumps. But more important than realizing who my real friends were and weren ' t, I was so astounded by the people who came out of the woodwork and who showed up with such generosity and support and really made me understand the value of cultivating community and prioritizing that and investing your time and energy in that. And so luckily, by the time that I did have my recurrence, I had spent that decade building a community, because as much of a cliche as it sounds like, we all need a village and we especially need a village when the ceiling caves in. And so, when I learned of my recurrence, Jon was in the midst of perhaps his busiest professional season of life. And I ' ve known Jon from the time I was 13 years old. I have watched him work so hard to get to a place where he was getting the kind of recognition and invitations that he had, and it felt really important to me that he not put his life on pause. And so we had to get creative about how to stay connected to each other at this time where it felt like we were living on polar opposite planets. And so Jon came up with a beautiful idea of composing lullabies. He would compose a lullaby for me every single day, and send it off to me as a kind of counterpoint to the hospital ' s many noises, the beeping of monitors, the wheezing of respirators, the alarms that go off. And it was his way of enveloping me with his presence, even when he couldn ' t physically be there. And I, in turn, would text him photos of the little paintings that I would make. And so what that meant was that when we spoke on the phone, the conversation wasn ' t just centered around the latest biopsy results and blood tests and whatever else was happening. It was centered in our love language, which is a **shared** creative language, and gave us a way of expressing what we couldn ' t express. SZ : Joey writes,"What do you recommend to someone who is trying to get started with journaling but feeling a little overwhelmed by the blank page? Do you have any rituals or any starting places you can recommend?"SJ : Yes, in spite of the fact that I ' ve been a lifelong journaler, I go through many moments where I feel daunted by the blank page, and that was kind of the original premise of starting the"Isolation Journals."Sometimes we need to read something. Sometimes we need to prompt ourselves to get out of, you know, to twist our mind out of its usual rut. Come join us at"The Isolation Journals."We have an archive of hundreds of essays and prompts and journaling challenges. Or don ' t. But I ' ll offer you what helps me when I ' m feeling daunted by the blank page. And it ' s actually something I borrowed from my friend, the poet Marie Howe, and she told me that when she ' s feeling stuck, she writes with her non-dominant hand and a big scrawl across the page."I don ' t want to write about..."And then she writes into that. Another one of my favorite journaling prompts from"The Isolation Journals"is by the photographer Ash Parsons. During a period of time where she was in the NICU with a recently adopted baby who was having major health issues, she began what she calls"just 10 images,"which is listing just 10 snapshots from the last 24 hours, stream of consciousness, whatever comes to mind. And I love that prompt because it ' s in list form and it ' s so simple. But more than just recounting what happens in the last 24 hours, it ' s a prompt that ' s trained me to look and to look again and to notice my day differently. But the beautiful thing about journaling, and the thing that makes it so generative and inspiring for me is how low the barrier to entry is. You know, journaling is not beautiful writing. It doesn ' t have to be grammatically correct. It doesn ' t have to be in full sentences. It can be whatever it is that you want. It ' s such an expansive form that you can interpret and reinterpret however it best serves you. SZ : I have one last question from York. York says,"Thank you for sharing your experience and insights. One of the parts that resonated with me was your experience of using creativity while you were struggling with finding purpose. Were you always able to focus on creativity and if not, what helped you make that shift?"SJ : Liz Gilbert speaks so beautifully about the pressure of finding your"capital P"Purpose. And so, what she ' s always said is you have to be one percent more curious than afraid. Any time I come up with some big, ambitious creative project, I immediately get frozen in my fear and daunted by whatever it is. And so I think curiosity is such a gentler way in and such... And it ' s a more honest way into the creative process, where you don ' t hold to an expectation of how something should look or how something will come together. But you really give yourself the time and space to explore the threads of that curiosity. The work I ' m most proud of, the work that has surprised me and changed me as an artist, as a writer, are the projects I started without that sense of output or expectation, be it, you know, that 100-day project and keeping that journal or even the paintings that I started doing in the hospital. They were pure play and tapping into that sort of child space, of creating simply for the joy of creating without any regard for if it ’ s good or bad, but simply to explore. And so I try to trick myself back into that space all the time. SZ : Thank you for sharing so openly and vulnerably with us, as you frequently do. Bye for now. SJ : Thank you everyone, thank you Susan. [Want to support TED?] [Become a TED Member!] [Learn more at ted. com / membership]

Text Sample 363: POS Dim 3 – 2025 – Score 6.00 – 2025/t003466.txt

Can you paint with all the colors of the wind? I love this song from the first time I heard it. I was 14, growing up in public housing in western Sydney. So maybe a little too old for this Disney classic from”Pocahontas.”But it spoke to me. I saw something of myself in it. At this stage, I think I knew I was queer. I mean, come on, I was rocking out to Disney. And look at those glasses. Seriously, what a queer kid. But I was also from a Pacific **Indigenous** heritage with conservative views and values. I didn ’ t feel like I could fit into any of the boxes that were being presented to me. Straight, **gay**, Fijian, Australian. I couldn ’ t pick just one color to paint with. Prior to colonization, **sexuality** amongst many of our Pacific islands across the region was fluid. Men would have sex with men without the stigma or **shame** afforded it today. **Sexual** intimacy was a way of being able to connect socially and relationally. Sex wasn ’ t just for procreation. Or in marriage. It wasn ’ t exclusively monogamous. Just because you had sex with a man didn ’ t define your **sexuality**. It didn ’ t make you an outcast. But as Europeans started to colonize the Pacific region, they brought Christian missionaries with them. They told us to cover up our beautiful bodies. We were taught **shame**. We were told to stop practicing Pacifica cultures and values that were seen as provocative. To a Western and white perspective, our non-binary **sexuality** was judged as tribal, outdated, sinful and impossible to practice alongside a Christian faith. Still today, those who are **Indigenous** to the Pacific are negatively impacted by whiteness. That”white is right, West is best.”I was brought up to espouse strong Christian beliefs, views and values which would shape my burgeoning identity as an adult. From these staunch conservative views, I was taught to fear the queer. Those people that were seen as unusual and fruity. And in essence, I grew to despise a part of myself. I sought to separate my queerness from my identity. I could never see myself as part of that world. Growing up, biracial and bisexual constantly challenged the way I saw the world around me. I was unsure about my thinking, feelings and behaviors around liking males and females. I was unsure about my thinking, feelings and behaviors when it came to whether I needed to be a white Australian or a Fijian person. And whether this needed to be framed within a binary to choose one over the other. In this way, I don ’ t think that my story is unique. There ’ s plenty of people in the queer world that have escaped conservative upbringings, religions and cultures. But this is only part of my story. Another common story in the queer community is the coming-out story. When did you tell your parents? How did they react? How it liberated you or ultimately separated you from your family. But this didn ’ t feel like an option for me either. I believe that embedded in these particular stories, these tropes, is a part of dominant queer culture that made it very hard for me to **access** the queer world. For example, when I started to go to **gay** clubs, I would be lining up at the front of the club and often I would get door staff come up to me and go,”Sorry, you do know this is a **gay** club, right?”From these lived experiences, I feel that there is a whiteness that pervades queer culture. And I didn ’ t want to give up my identity and my appearance as a Pacific **indigenous** person to be queer. Pacifica cultures really, really value and esteem the collective above the individual. For Pacifica people, our families are inextricably connected to our own sense of self-agency, self-determination, self-esteem, self-love and self-worth. Our own individual identities are meshed and intertwined with our family ’ s reputation and standing in the community. Our mind, body and souls are inextricably connected to the broader community, to our families, to ourselves and to others. I didn ’ t want to come out to potentially ostracize myself, to make a point to separate myself from my Pacific and faith communities. I couldn ’ t then see my authentic self in any of these spaces. And a quick shout out to my son on the left-hand side. Round of applause for my son. He said to me this morning,”Daddy, make sure you mention me.”So I was like,”OK, son. Alright.”So there you go for the recording, son. And the rest are my other family, that ’ s up there as well. Across queer cultures and the broader community, we have learned to embrace our gender and **sexual** differences. We encourage individuals to be allies for each other and to support safe spaces for those that are seen to not fit in. We empower and really encourage individuals to really develop and express their self-agency, their self-determination, their self love, their self-worth. We boldly go into spaces striving to bring others with us to ensure that we can create inclusive societies for everyone. However, the dominant Western view in queer cultures is still based on the individual. Me, myself and I. And my individual pursuits towards happiness. As a queer **Indigenous** Pacific person, I feel that there are parts of queer culture that is white, feels very white. We may fail to see the importance of bringing our families and our cultural communities along our journey as queer people. It ’ s either the queer way or the highway. It ’ s one or the other. Young or old. Rich or poor. Smart or stupid. Masculine or feminine. Fit or fat. These binaries create a sense of us and them, those people. A binary approach and position that we generally in Western white societies use to determine the norm. We create, conflate, perpetuate binaries that create divisions in our society. Saints or sinner, straight or **gay**. I see it like this. Binaries are a very white and Western way of thinking, feeling and behaving. As individuals today, we live within these binaries and create identities that then create these labels. We use these labels to then define and confine people and places. We are one or the other. But what we fail to see is that our identities are more than a set of binaries. Our own identities are comprised of intersecting identifiers that contributes to who you are, including our age, class, color, education, gender, size, spirituality, **sexuality**, and so, so much more. We need to decolonize queerness, to live beyond”white is right, West is best.”Because if we don ’ t, we ’ re going to continue to create silos and divisions in our societies that don ’ t create inclusive and safe spaces for everyone. For **Indigenous** queer people, they shape our connection to community and culture, our families. If we don ’ t include that within our understanding of who we are, even as queer people, then we ’ re not able to be, again, our true and authentic selves. We ’ re not able to then be our fabulous selves as part of the queer community. And it ’ s really, really interesting because I believe that we can learn so much from pre-colonial cultures like those that were practiced in the Pacific, that provide a different way of looking at a society. A queer way that pre-dates all the privileges that go into binary ways of thinking, feeling and behaving. Across many of our Pacifica languages we have a lot of key concepts that help solidify and support a shared understanding of the collective. For example, in Fijian,”solesolevaki”is all about reciprocal living, where my wellbeing is your wellbeing. I will actively seek to ensure that your well-being and every part of your life is catered for, knowing that you will do the same for me. We strive to enact"veiqaravi,"as tatted here on my hand, to serve others. We strive to also embed and embold and empower people to practice"veirokovi,"respect for self and others where we are relationally driven, where that is so important to our spaces and places. My sense of ability to actually put this into practice, my ability to actually understand these **Indigenous** perspectives has greatly helped shape and reconcile my queer identity today. But what ' s really interesting is that it ' s not just queer communities that need to decolonize itself. It ' s all of us as a broader society that needs to be part of a shared conversation to decolonize, deconstruct and disrupt binary ways of thinking, feeling and behaving. So how can we practically do that? Practice cultural humility, where you are constantly learning and embracing differences, where you get over yourselves and strive to learn about other people ' s differences and how that can be meaningfully included in the way in which you create safe spaces for everyone. I also believe it ' s possible to bring **Indigenous** perspectives into Western and white workplaces. For example, I ' ve introduced the concept of"talanoa"in all of my work meetings. Talanoa is all about being able to create a safe space where everyone is able to contribute to a collective, collegial and collaborative conversation. Where people are able to bring their authentic selves into these spaces. Literally, I will chair meetings where I have no firm set agenda items and I literally would just hold space for participants just to bring themselves. So you can tell I ' m a social worker, right? Safe space, safe space, safe space. It ' s OK, it ' s OK. But it ' s through these safe spaces that people can bring their multiple, intersecting identifiers, including their age, their class, color, education, gender, size, spirituality, **sexuality** and so, so much more. I invite queer communities to come talanoa with Pacific **Indigenous** Communities to create robust and resilient queer cultures. I have been able to greatly reconcile my queer and Pacifica identities, which has now really been embraced to help shape how I see the world around me today. And it ' s through these shared perspectives and approaches that we can truly embrace cultural diversity and its differences. And to know that people can live beyond those binaries and those labels we continue to perpetuate and perpetrate against others. It is a commitment from queer communities to decolonize queerness. It ' s a commitment from me, from you and all of us to decolonize Western and white societies and to live beyond the binary way of thinking, feeling and behaving. I genuinely believe that it ' s through this common goal that we can truly learn how to paint with all the colors of the wind."Vinaka vakalevu"and thank you.

Text Sample 364: POS Dim 3 – 2025 – Score 3.00 – 2025/t003002.txt

HW : Hi everyone, I am Helen Walters. I am head of Media and Curation here at TED. Welcome to another episode of TED Explains the World with the one and only, Ian Bremmer. It is Monday, February 24, a little more than a month since President Trump was inaugurated once more, and safe to say, a lot has been going on. So we figured we ' d check in with Ian to determine what we should really be paying attention to amid the extreme noise. Ian, hi. Ian Bremmer : Helen, great to be back with you. HW : So we asked the TED community to share their questions for you, and we wanted them to share what they ' re most curious to know. And we really got some amazing questions that I plan to pepper this conversation with. So let ' s start with one right now. Two months into 2025, where does the US stand? IB : Very powerfully, in the sense that the US economy, of course, is performing considerably better than any of the other G7 economies coming still out of the pandemic. Technologically, only close competitor is China, ahead of the US in some areas, behind the US and other critical areas. But compared to every other country in the world, the US is head and shoulders and neck or waist above them. Militarily, of course, the US is the only country with global ability to project power. No one else is close. And the US dollar is still the global reserve currency, no one is approximate challenger. I say all of those things because those are the things that haven ’ t really changed over the course of the last few months, but I suspect that isn ’ t what the questioner was asking. What they were really asking is what ’ s happening politically. And politically, the United States is unwinding its own global order. It is no longer particularly interested in promoting collective security or NATO. It ’ s no longer really interested in promoting involvement and leadership of multilateral institutions, of consistent rule of law and free trade that ’ s well regulated, certainly not very interested in promoting democracy around the world. I mean, this is an environment where other countries around the world have to figure out how to adapt to the United States, and that the US has become the principal driver of geopolitical risk and uncertainty in the world today, unlike any other time in my lifetime and yours, that ’ s perhaps the biggest change. HW : That is quite a statement. Alright, you are a geopolitical expert. Let ’ s talk about Europe. So Defense Secretary Pete Hegseth stated that America ’ s foreign policy focus no longer lies in Europe. Vice President JD Vance caused some raised eyebrows, dropped jaws and, I suspect, some choice expletives after his recent speech at the Munich Security Conference, in which he accused European leaders of suppressing free speech and warned of the threat from within. So you were in Munich. Was it as dramatic as the stories we read had it? And what happens next? IB : It was. I was in the room. I was maybe 30 feet away from him when he gave that speech. Standing right in the front, right next to me was the president of Czechia, the president of Finland, the prime minister of Sweden, watching all of them and their reactions. What he said, let ’ s keep in mind, this is the Munich Security Conference. It ’ s been going on now for something like 61 years. And he was the head of the US delegation. And every year the head of the US delegation gives a big speech on the state of the transatlantic alliance and on global security. And he didn ’ t do that. I mean, the people in the audience were expecting a challenging speech. They were expecting that the Americans would be less committed to the alignment with the Europeans on Ukraine. The US had just had that direct Trump phone call with Vladimir Putin, 90 minutes long. Hadn ’ t coordinated that with the Europeans, never mind the Ukrainians, in advance. So, I mean, they were definitely prepared for a very challenging plenary. But Vance did none of that. Vance said,”I ’ m not going to talk about Ukraine or Russia or China, because the biggest problem is actually what ’ s happening inside Europe. The biggest problem, essentially, are you guys sitting in front of me, that I ’ m speaking to, because you ’ re suppressing free speech. You ’ re suppressing the far right. You ’ ve been infected by the woke mind virus, and you aren ’ t real democracies as a consequence.”And specifically he said, and this is something that I think is underappreciated in the United States, he attacked the German firewall. And said that – And let me explain what the German firewall is. That is a principle by the leaders of all of the mainstream German parties and their supporters that they will not, under any circumstances, work with, enter into coalition with the Alternative for Deutschland party, the AfD. Because the AfD, under surveillance from German intelligence agencies, is considered to be a neo-Nazi party. And the United States, of course, after World War II, led de-nazification of Germany. The day before, Vance had actually gone to Dachau and visited a concentration camp, and then came to speak in Germany about the firewall needing to be ended. And at that point someone yelled out from the crowd,”This is unacceptable!”And it was right in the front. And people in the room, about 1, 500 people in the room, standing room only, couldn ’ t see who it was. They ’ d just see the back of his head. I saw who it was. It was the German Defense Minister, Pistorius. And in 15 years of me going to the Munich Security Conference, I have never seen anything remotely like that. And the reason I mention all of this, and of course, on the back of that, let me be clear, Vance refused to meet with the German chancellor because he ’ s basically, well, he ’ s not relevant. He ’ s only going to be there for a couple more months until a new government, so why should I bother? But does meet with the leader of this AfD, goes and meets with her directly, does a bilateral that evening. So that is a very important backdrop for this last weekend ’ s German elections, where the victor, Friedrich Mertz, who is going to be the next chancellor, actually referred specifically to America ’ s intervention in German democracy and support of the AfD being every bit as bad as what the Kremlin is doing inside Germany, and also said that the Europeans need to develop a defense policy which is independent of the United States, in other words, essentially saying that it will be the end of NATO. Again, staggering comments, unheard of, impossible to imagine even two weeks ago. And that ’ s why it ’ s very critical that you have the context of what happened between Vance, Elon Musk over the past couple of months, Trump, with the Germans, with the Europeans. So you can understand why the incoming German chancellor would make a statement like that. HW : What should we make of all of the conversation about the rise of Nazism, the return of Nazism? Obviously, the AfD just won 20 percent in the German election. We ’ ve also seen people allegedly giving Nazi salutes at various US conventions and speeches. Do you think this is overblown? Is this a fear that people have rationally? And what should we make of that? IB : In Germany, the Alternative for Deutschland, are winning all of former East Germany. This is a group that increasingly understands that they ’ re never going to really catch up to the rest of Germany. Remember, Germany hasn ’ t really grown in five years now. They ’ ve been in recession for the last two. And the average citizen, the average voter, living in former East Germany no longer are just angry about not catching up. They ’ re basically saying,”I ’ m done with this system.”So that ’ s why you ’ re getting this very strident, revanchist nationalism that is, you know, it ’ s not just taking a hard line on illegal **immigrants**. The AfD even supports removing citizens of Germany that they claim have not adequately integrated into German culture and nationality. And so, I mean, this is some very, very strong stuff, stuff that really does feel like what they were doing back in Nazi Germany. But in former East Germany, they win. They get the largest number of votes, where, in much of the wealthier part of West Germany, including, you know, where I was, in Munich, the AfD can ’ t break out of single digits. So it ’ s a very, very divided country. And also the AfD is doing very, very well among young men. There ’ s a clear gender divide here as well, just as there is in the United States. In the US, I mean, I, of course, saw Elon Musk and that, you know, sort of grabbing his heart, saying,”My heart goes out to you”and then doing what appeared to be mimicking a Nazi salute. And then Steve Bannon doing the same without the initial heartfelt comment. It clearly was trolling. Clearly was trying to get a reaction from folks. It ’ s very performative. And it ’ s meant to also, you know, attack and attach, make the left lose their minds on something that isn ’ t very critical to what the Trump administration is trying to do compared to the revolutionary changes inside the US government that they are very much prioritizing right now. So I ’ m very concerned about the rise of neo-Nazism in Germany. I am very much not concerned about that in the US. I ’ m concerned about other things I ’ m thinking is being used as gaslighting for those that can use distractions or can be distracted by them. And I would also say that this election in Germany wasn ' t surprising. It was exactly as the polls have expected for several months now. But the key elections in Germany and the key elections in Europe are the next cycle. I ' ve spoken with a lot of people around the Trump administration that believe that the Europeans are one electoral cycle behind the United States. So in other words, Trumpism is coming, just not quite yet. And so in the UK, you ' ve got Labour for a full term. But the Reform Party is increasingly the most popular. And just with a little nudge, maybe a little external money, and Elon said he ' s already thinking about giving them 100 million dollars, which is a massive amount in UK politics, not so much in the US, that the Conservatives would fragment, a rump group would join with Reform, and they could win the next elections. In France, you ' ve had all of these governments continue to collapse. Macron, very unpopular. Marine Le Pen and the National Rally party could easily win upcoming elections in 2027. In 2029, in Germany, if the Germans are incapable of turning their economy around, incapable of unwinding their debt break and spending some of the capital that ' s just sitting there and as their debt to GDP goes down, but their economy contracts at the same time, then AfD could easily win in 2029, and then the firewall is no more. And then, of course, in the European Union elections overall, in five years ' time, now four and a half, you could easily see the Patriots front, which is the equivalent to the far right grouping that these parties are a part of, they could all come together and be the dominant party. And then the EU as we know it is really a thing of the past. So these are existential changes that are happening not just for NATO right now, but also for the EU and quite plausibly, for the future of democracy as we know it. HW : Alright, let ' s turn to Ukraine. So President Trump accused President Zelenskyy of starting the war with Russia and called him"a dictator without elections."Last week, senior American and Russian officials met in Riyadh to discuss a potential cease-fire with no Ukrainians present. Now Zelenskyy has offered to step down in return for Ukrainian admission to NATO, although apparently he doesn ' t really mean this. So what happens to Ukraine now? What should we be paying attention to? IB : Well, I mean, I think he would mean it if NATO membership happened. I ' d love to see Trump call his bluff on that. But of course it ' s not going to. And he wasn ' t saying that six months or a year ago. He ' s saying it now that NATO is truly being pulled off the table. I think it ' s the right thing for him to do. It is, again, performative and helps to remind those around the world that Ukraine is fighting for self-determination. They ' re fighting for the ability to join their own alliance, as any country in the world should be able to do. But that is not Trump ' s belief. Trump believes that what you get to do is determined by how powerful you are, not by the fact that you happen to run a country. And indeed, the territorial integrity is something that is afforded to powerful countries, but not so much for weak countries. And that ' s the way he feels about Panama, and that ' s the way he feels about Denmark and Greenland. And that ' s certainly the way he feels about Ukraine. So right now, as you and I are speaking, the United States is putting forward a new UN resolution at the General Assembly that says that the war has to end as fast as possible but does not recognize Ukraine ' s territorial integrity, which has been a core precept of the international order that the United States has supported since creating the United Nations and since the end of World War II in 1945. The Americans are now throwing that out. And the Russians, of course, are fully supportive of that. America ' s allies in Europe are not. But the US has gotten the Saudis to agree, and they ' re getting a unified Arab block. The Americans have privately worked with a lot of poorer countries in Africa, in the Asia Pacific and others, smaller countries, and saying,"You ' re not going to get any aid from the US unless you vote in favor of this new resolution."I fully expect the resolution is going to pass. But of course, what that means is that the Americans are preparing to cut a deal on Ukraine over the heads of the Ukrainians. In other words, the US and Russia together will figure out what a ceasefire should and should not entail as part of a broader US-Russia rapprochement. This is obviously a disaster for the Ukrainians and the Ukrainians who initially refuse to accept this so-called deal, on Ukrainian natural resources that would grant the Americans $500 billion in **access** to such resources in return for"paying off"aid that the Americans had granted to the Ukrainians without conditions or strings, but apparently no longer. And so, you know, what does this all mean? Well it means it ' s yet another case of why people shouldn ' t trust the United States from one administration to the next, because foreign policy will change completely. And what you thought was an American commitment won ' t stand. It shows that there is a fundamental rift between the Americans and the Europeans and indeed, Emmanuel Macron in the United States right now, as you and I speak, hoping that he can do something that will keep Trump from, as Macron believes, essentially surrendering to, conceding to, the Russian position and maintaining some level of coordination with the Europeans. But he doesn ' t have a lot to offer. And the Ukrainians, you know, now in a position of desperation where maybe they will end up signing over a lot of their natural resources to the Americans. But what, if anything, will they get in return for that? And how empowered will the Russians be in any deal that is cut? And will an independent Ukraine even be able to survive, and what will the terms that will be required of them? Of course, Putin is saying things like, no European troops of any sort, whether in a NATO formulation or not, would be allowed as peacekeepers in Ukraine. Well, then, how would you guarantee, what kind of security commitments would you give to the Ukrainians? These are all open questions, but apparently questions that President Trump is prepared to largely resolve on Russia ' s terms. And again, this is such a dramatic change from where the United States and where the Western order was just a month or two ago. HW : So let ' s take some quick-fire questions from our community related to this topic. And I think the first one is particularly pertinent to what you just said, so it ' s simple. What does Putin have over Trump? IB : I don ' t think that Putin"has"anything over Trump. If he did, then why wouldn ' t Trump have granted Russia more in his first term? In his first term he gave the Ukrainians those javelin anti-tank missiles that Obama refused to give. And, you know, he thought they were too risky. He increased sanctions against Russia. Now that was his administration with a lot of hawkish in orientation advisors around him that he doesn ' t have this time around. They ' re all directly loyal to him. But, I mean, if it was a priority for him, he could have stopped it. And if the Russians had something over him, then why wouldn ' t they have used it in that case? It feels like a conspiracy theory. I don ' t buy it. I think very differently. We can explain what ' s happening because number one, Trump wants to end wars. It ' s very popular with his base. He ' s said this on Gaza. He ' s said this on Iran, where Israel wants him to support attacks on Iran ' s nuclear capabilities, and Trump has said no. And he ' s showing this on Ukraine. He doesn ' t want to spend money on the Ukrainians going forward after spending over 100 billion dollars under the last three years from the Biden administration. He doesn ' t want to spend that money going forward. And he also doesn ' t value a strong Europe. He actually thinks that a weak Europe is in America ' s interests, where you have more Brexits. He used to always talk to the French about that during his first term,"Why won ' t you do Brexit?"He wants all of these MAGA-type populist parties, including the AfD, to win in Europe because that legitimizes his own popularity. Just like he loves Milei in Argentina or Bukele in El Salvador. And so I think if you put those things together, you actually get to why Trump is doing what he ’ s doing with the Russians without needing to create a story about Trump somehow being threatened by some secret, shadowy information that the Russians have on Trump that they ’ ve let him know about but nobody else knows. I just don ’ t buy that. HW : Great. OK, another community question. If the US hands Ukraine to Russia, will Russia keep going? Will they attempt to take over Europe? IB : It ’ s interesting that Trump just met with the Polish president, President Duda. It was supposed to be an hour meeting, it was only 10 minutes. So he kind of embarrassed him at home in Poland. But he did say that the US is still committed to maintaining a troop presence on the ground in Poland. And so it ’ s very hard to imagine the Russians rolling through Ukraine and then into Poland, even though there ’ s been lots of asymmetric warfare, for example, and even there have been, you know, Russian missiles launched into Lviv, Ukrainian air defense, missile defense, and with Polish villagers getting killed because of the fighting. So, I mean, there ’ s been spillover. But America is still going to be in Poland. Why is that? They get along pretty well, and the Poles are heading towards five percent of GDP defense spend this year, which has been the high water mark of American demands of defense spend. So he ’ s, you know, been consistent on that over the past months. Baltic states. I mean, they ’ re all spending a lot more money on their defense. And so, I mean, in principle, you could imagine the United States maintaining these rotational deployments in the Baltics, which would prevent the Russians from invading. But might Trump say, as part of a deal,”Why do we have all those troops there?”You mentioned Pete Hegseth saying the Americans need to get troops out of Europe and get them to Asia, which is where their principal competitive, you know, sort of, strategy is going to be oriented. And the US hasn ’ t been pressing the Japanese or the Indians, the principal partners, allies in Asia vis-à-vis China, the way they ’ ve been pressing the French, the Brits, the Germans in Europe. So I think it is certainly plausible that the Russians will do more, especially with those countries that refuse to align with a Trump rapprochement towards Russia. The countries I would be most worried about are Moldova, are Georgia. I ’ d be most worried about the countries that are in the so-called”near abroad”that aren ’ t a part of NATO, and aren ’ t going to become a part of NATO, and therefore are much more vulnerable. But certainly in terms of Russia ’ s willingness to engage in espionage, to fund actors on the ground for arson attacks, assassinations, critical infrastructure attacks on fiber networks, on pipelines, I think all of those things are likely to increase against European states across the board, on the back of this US-Russia rapprochement. HW : OK, final community question for this segment. What happens if the US no longer supports NATO? IB : It ’ s possible that the United States is moving towards a similar posture in Europe that they presently have in Asia, which is strong individual bilateral deals with a number of countries : South Korea, Japan, Australia, but not a collective security agreement, where those countries together have more of a call on the United States and there ’ s a lot more free riding. So I could imagine that NATO falls apart and becomes that. The danger is that even though the Europeans now understand the nature of the threat, that it is very plausible and maybe even baseline, that the Americans are going to leave Europe collectively on its own. That that seriousness does not equate to the ability to increase spending adequately in the near term. The Germans did manage to pull together an electoral outcome that will allow for a so-called grand coalition, just barely. So you ’ ll have a two-party government in all likelihood instead of a three. But you also have a blocking capacity on the part of the far right and the far left that will prevent the Germans from really blowing out their spending, including their defense spending, unless they can pull something off before that new government is formed. Which is possible, but it ' s unlikely to be very big. The Brits are now talking about maybe moving the 2. 5 percent of GDP defense spend over years with a very challenging fiscal environment. The Spaniards, not even close, the Italians, not even close. So what does a European military capability really look like? And without the Americans, the answer is it ' s not there. It ' s not there. So I think that if NATO falls apart, the reality is that most of Europe will be incredibly vulnerable to external attack, especially Russian attack, And there won ' t be a way for them to defend themselves, and some of them will break and therefore want to work more directly and individually with the United States. And that could be the end, not only of NATO, but it could be the end of the European Union. I think this is, the stakes are incredibly high, this is an existential challenge for European governments that needed to take this seriously 30 years ago, 20 years ago, certainly in 2014, when the Russians invaded Ukraine, certainly, certainly in 2016 when Trump was elected, and certainly, certainly, certainly in 2022 when the Russians then invaded all of Ukraine. And at every point the Europeans were basically saying,"We ' re still fine with the Americans basically doing this."And now they are in very serious trouble. HW : No time like the present, I suppose. Alright, let ' s move on. So I mentioned the discussions in Saudi and obviously also on the agenda there with the discussions about Gaza and Trump ' s proposal to build a Gaza Riviera. What should we make of that? Or as one of our community members asked, what is the actual solution for Gaza? And surely it isn ' t what the US is proposing. IB : Well, look, here ' s what ' s interesting. You know, Trump says a lot of things, and when they don ' t work [out], he backs out fairly quickly. And you know, I thought – just stick with Ukraine for one second on this. It was interesting to me that Trump has said that he doesn ' t want to talk to Zelenskyy because he has no cards, he has no leverage. And so why should he let him at the table? And so Trump ' s going to cut the deal. Very interesting that if that is true, then why is it that the Secretary of Treasury brings this deal and says,"You ' ve got to sign this right now, Zelenskyy"And Zelenskyy refuses. And then a few days later, the Americans come back with a new deal with altered terms that are not quite as predatory vis-à-vis Ukraine. Why would the Americans change their tune for someone that has no leverage? Because, I don ' t think it ' s because Trump is a nice guy. He ' s not going soft at 78. I think it ' s because he overstated how little leverage Ukraine has. Ukraine is very popular among the GOP in the House and Senate, far more than Putin. Zelenskyy is far more popular in the United States among the voting population than Putin. And of course, he ' s also much more popular among the Europeans and even globally than Putin. And so it turns out that maybe there is a little bit of leverage that Zelenskyy has, and maybe Trump does have to give a little bit more. So this is relevant in the context of Gaza, because Trump was very annoyed that, you know, despite his ability to get a cease fire between Hamas and Israel over the goal line, that his friends in the Middle East were not coming up with a plan for Gaza. Where they were supposed to provide security and provide investment and lead reconstruction and deal with the Palestinian populations in the interim. And that wasn ' t happening. And so Trump then meeting with Prime Minister Netanyahu from Israel in Washington, then after that, summoning the Jordanian king, who was the ally that is most reliant on the United States and so, therefore, most needing to say yes to whatever Trump orders of him. Basically says, as you mentioned, Helen, that Gaza is going to be a Riviera, that the US is going to turn this into an incredible place. They ' re going to build wonderful homes for the Palestinians someplace else. And the Palestinians will all volunteer to leave Gaza and go to those homes, and Gaza will be depopulated and new people will come in and live there. That was the plan. And when he was asked, standing there with King Abdullah from Jordan, under whose authority? Trump ' s response was :"Under my authority."And that went over like a lead balloon among America ' s allies in the Middle East. They all said,"We ' re not up for this. We ' re not resettling Palestinian populations. They won ' t actually leave voluntarily. This will would actually be a huge problem for Israel ' s own national security long-term. And we ' re going to come up with another plan. And our plan will keep the Palestinians on the ground in Gaza."At which point Trump pivots to work with the new plan. And his advisers say that Trump was only trying to start the conversation. And what he said was his plan is not actually what he ' s going to do. So, what ' s sustainable? What would be sustainable would be a lot of Gulf money going in and maybe some European money too, though they ' re going to be really pressed, given everything you and I just talked about, to try to reconstruct a completely devastated Gaza that eventually some two million-plus Palestinians will be able to live normal lives in, and the governance will be provided by some technocratic group of a Palestinian Authority that will be working with, selected by, those that are involved in the reconstruction and the security. Namely the Egyptians, the Jordanians, the Saudis and the Emiratis. That is the best possible deal that they ' re going to get. But that ' s happening as Israel has just evacuated forcibly another 40, 000 Palestinians from refugee camps in the West Bank, sending tanks in for the first time in decades and probably precipitating more Israeli settlers taking more land on the ground in the West Bank. And so, as we are possibly taking a small step towards a sustainable solution in Gaza, we are taking a slightly larger step away from a sustainable solution for the Palestinians in the West Bank, twas ever thus over the past decades in this incredibly horrible problem and conflict. HW : And why is Israel doing that? I mean, are they feeling emboldened by the US, or why the incursions in the West Bank now? IB : Well, I think they ' re feeling emboldened by their own successes. They have shown that they are the dominant military power in the Middle East, and they have the ability to determine the level of escalation unilaterally, and their adversaries and enemies can ' t really do any damage to them. They proved that after October 7, in their response to Hamas. They proved that in decapitating and taking out the military capabilities of Hezbollah. And they ' ve proved that also with overturning – they didn ' t do it, but the fact that the Assad regime in Syria is gone, which means that the Iranians no longer have a land bridge to get additional weapons to what had been the strongest adversary of Israel militarily, Hezbollah. So all of that has shown that Israel can decide outcomes. They ' re the ones that have all the power here. And also that depopulation of Gaza is very popular among Israelis. There was a poll recently in the Jerusalem Post, which is pretty mainstream in Israel. 80 percent of Israeli respondents said that they wanted to depopulate Gaza of all Palestinians. Taking more land is very popular, especially in the West Bank, especially among Netanyahu ' s present right-wing coalition, that he would like to keep together, to continue to govern. And this is a sop to them as he is looking to potentially extend the ceasefire in Gaza and move from phase one to phase two, which is facing some delays and open questions right now. So, yes, certainly, the United States has been underpinning that military capability of Israel. And I think it is instructive to remember that not only does the US provide more military aid in peacetime to any country in the world, to Israel, but that unlike in Ukraine, where the Trump administration is saying,"No, I ' m not happy with the grants, you ' ve got to pay that back,"The billions and billions that the United States sends to Israel, nobody is suggesting that that should be alone or that the Americans should get technology rights or mineral rights or anything else from Israel, despite the fact that Trump ' s policy is America First. Israel is a very clear exception to America First in Trump ’ s visioning of it. HW : Alright, so we talked about the fact that the foreign policy focus of America is shifting to Asia. So what are you hearing about how China is feeling in all of this? And what should we be paying attention to on that front? IB : China has had a couple of good weeks economically and specifically on the back of that huge announcement of DeepSeek, which has, you know, far less money behind it than the comparable American chatbots but is performing at an extremely high level. And, you know, given the fact that there are all of these export controls on semiconductors and other parts of advanced technology, that this is giving people more of a belief that the Chinese really are capable of driving innovation even in that space. And they ’ re, of course, dominating the post-carbon energy space. We see that with electric vehicles and batteries. But we ’ re also seeing that with massive investment now in green hydrogen, for example, we ’ re seeing it in advanced nuclear capabilities, all of these things. And so there is more investment in that space that the Chinese are not only driving as a state, but that they ’ re also increasingly able to raise and move in their own private sector. So the story that you and I have been discussing for the last couple of years is the Chinese economy is radically underperforming. That is still true, but there is an emerging counter-narrative here that I would be remiss not to mention. Now more broadly, what the Chinese here are seeing is that Trump ’ s direct policies are going to hurt them economically. His tariffs that he ’ s already announced on China, the 10 percent, on a whole bunch of Chinese goods for export, the reciprocal tariffs that are global, that will clearly hurt China and that they ’ ll have a hard time negotiating out of, and also the willingness of the United States to expand their export controls, their sanctions on China, especially in areas that are considered relevant, even loosely relevant to American national security. All of that is going to constrain Chinese growth. All of that is going to negatively impact the Chinese economy. That ’ s the downside. And the Americans are doing that not only in bilateral relations with China but also pressuring other countries to get Chinese tech out of their supply chain. And you see that with Americans pushing other countries that produce semiconductors to take the Chinese out. They ’ re pushing American allies and trade partners to get China out of the export chain. So Mexico being pushed very hard by Trump to stop allowing China to pass goods through Mexico into the United States. India, same conversation. Vietnam, same conversation. A lot of that going on. So baseline, the US-China relationship is getting worse. But the Chinese see huge advantages in America ’ s reversion to unilateralism that we haven ’ t really seen since the original America First movement in the run up to World War II. Charles Lindbergh saying,”Don ’ t get into the war, don ’ t fight the Nazis. The US is far away. This isn ’ t our our problem.”And now you have the Americans not only doing that with Russia and Ukraine with the new America First, but you also see that happening with the Americans pulling out of the Paris Climate Accord, again, pulling out of the World Health Organization. You see it with the Americans shutting down USAID. And America has historically been responsible for about 40 percent of global humanitarian support and aid. And absolutely, some of that is corrupt. Absolutely, some of that is on woke programs that the vast majority of American taxpayers would never support. But the majority of that money is actually spent on programs that are really important in advancing American soft power and influence over countries in the Global South, all over the world. The Chinese know that. That ’ s why they ’ ve started humanitarian programs. It ’ s not out of some great love for, you know, providing foreign aid for these countries. It ’ s more because they see it creates influence. Now the Americans are leaving a vacuum that the Chinese can absolutely take advantage of. And we ’ ll see this across, you know, sort of, microstates in the Pacific. We ' ll see this across sub-Saharan Africa. We ' ll see this across South America. The Chinese see huge opportunities from the United States creating a leadership vacuum. And the place they ' re most capable of doing that is if the US stops paying dues in the United Nations, where China is number two, and they ' ll suddenly have the most influence over all the key positions, which will give them a role in global governance that they ' ve really been constrained from having over the past decades. HW : What do you think are the chances that the US stops paying its dues to the UN? IB : You know, if it were up to Congress and only Congress and Trump doesn ' t lean on them, I ' d say fairly low. I ' d say that they ' d probably, you know, cut back on some programs. But in terms of baseline dues, they ' d keep paying it. But Trump personally, I think increasingly sees the UN as hostile to the United States. That ' s particularly true in terms of Israel policy, which Trump is leaning in on. And, you know – let me put it this way. Tulsi Gabbard did not have the votes to get confirmed. There were Republican senators that were going to vote against her. And Elon called those individual senators and threatened them. And this was, you know, fully aligned with what Trump wanted Elon to do. And those votes went away. And the only person that ended up voting against Tulsi was McConnell. And as a consequence, it went through. So look, I think that we should not underestimate the level of power and control that Trump and Elon have over the Republican Party this time around, compared to 2017, where it was really the Republican Party that was much stronger than Trump, both in his administration creating a lot of guardrails, but also in Congress. This time around, Trump has all the chips and he ' s using them. So if he decides he wants to stop paying dues, I think that ' s what ' s going to happen. This is a much more revolutionary presidency, domestically, where last time around it was much more transactional. HW : Alright. Let ' s talk a little bit more about what ' s actually happening inside the US. You bring up Elon, who obviously with DOGE has been sending out emails and asking for information from people and kind of trying to do a lot very quickly. Some of these high-profile moves are – I think what ' s interesting is that no one would really argue that there isn ' t inefficiency in government, that that ' s an understood reality. But the way that some of these moves are being made are maybe going to cause irreparable damage. And that ' s to people from red states, that ' s to scientists, that ' s to veterans, that ' s to people who voted for Trump and that ' s to people around the world. You talk about soft power and the United States, but what about actually what ' s happening inside the United States? And why do you think that the administration is willing to go so quickly, to go so fast and to cause such damage? IB : Well, and you saw Trump this weekend saying that he wants Elon to move faster, to be more aggressive. And I don ' t think that was just a troll. I think that ' s true. Trump is 78 years old, and with his friends he talks about the fact that he doesn ' t have much time. That in two years ' time, the Democrats could take the House in four years ' time, you know, that ' s it. He ' s not talking privately of"I ' m going to be president for life."He also knows he was almost killed. He was shot in the head. And do I think that changes somebody? Absolutely. I think that he believes that he was saved by God, and that his level of confidence that what he is doing is fundamentally right and necessary now is so much higher than the insecurity that I frequently saw in him in his first presidency. So I think his willingness to push, push, push, and his view that Elon is a great activator and executor on that intention, and someone that he had nowhere close to someone with that kind of capability and energy and execution ability in his first term. And now he does. So first of all, I think that relationship between Trump and Elon isn ' t going anywhere. I think it ' s actually very stable. People that say it is are people that want it to go away. But hope is not a strategy. And I also think that the intention of revolutionary politics domestically is to shoot first and ask questions later. It ’ s to break things. It ’ s a view that the bureaucracy has been weaponized against Trump first and foremost, and also against the will of the American people. And therefore it must be broken. And you ’ re going to break eggs when you want to make that omelet. Now your point is that there are a lot of Republicans that work in government. There are a lot of Republicans and Trump voters that are losing their jobs and that are not being treated with a level of care and compassion as a result of what Trump and Elon are doing. And I think there will be a backlash. But I ’ m not so sure that a one-term, 78-year-old-Trump presidency cares all that much about that. And, you know, if that means that his popularity slips to like, the high 30s, frankly, you know, in a very, very polarized country, high 30s isn ’ t that bad. It still means 80, 90 percent of the people that voted for him still support him, and I think he ’ s probably more comfortable with that this time around. So I think a lot more has to happen before we can start talking about whether there ’ s really going to be significant pushback against what Trump is trying to accomplish at home. HW : Do you expect to see any of Congress standing up to Trump? We saw the governor of Maine recently kind of take him on. Do you expect to see more of that, or do you feel like for the next two years at least, people will just go where Trump takes them? IB : Well, sure, but, you know, in the case of Maine, that ’ s a Democrat. So I don ’ t know how much that means. I don ’ t expect to see a lot of Republicans standing up against him. And of course, since the Republicans control the House and the Senate, that ’ s the only thing that really matters. I mean, you know, you saw Kash Patel got through, Pete Hegseth got through. Now look, I actually think that a lot of the members of cabinet are very capable. I mean, they ’ ve been picked for their loyalty and they will be loyal to Trump, but they ’ re very capable. I think JD Vance is very capable. Mike Waltz, I know quite well. Marco Rubio, over the years. These are people that are very well respected, have been by their colleagues and could have served under any Republican administration. But there are some people that are completely incapable for the roles that they have been selected, completely incapable. They would not be selected in a role in any cabinet, in any democracy. And yet here they are, serving in critical roles in the most powerful country in the world. And yes, I ’ m talking about the Secretary of Defense. I ’ m talking about the Secretary of Health and Human Services. I ’ m talking about the Director of National Intelligence. I ’ m talking about the Director of FBI. These are important roles. And they were all confirmed. And if you ’ re a Republican senator – and senator – I ’ m not talking about the House here where, you know, it ’ s a lot more diffuse and there are a lot more wackos and it ’ s easier to get in. But Senate has always been like, responsible and more oriented towards bipartisanship and the rest, and they are utterly petrified of what it means if they publicly get crosswise with President Trump and number two, Elon. And they ’ re not willing to do it. So I think that there ’ s very little belief that you are going to see Congress step up. I think that you will see justices step up because those justices are independent. The problem is that many of those justices are progressive and have a known progressive lean. And so when cases come up and justices that have that political lean oppose something Trump wants, his willingness, his administration ’ s willingness to attack them and refuse to execute on that judicial decision, unconstitutionally, is real. Now the case will of course go then up to the Supreme Court eventually. And I don ’ t feel the same way about Trump refusing to listen to adhere to the rulings of the Supreme Court. But if there ’ s 100 rulings like that and you overwhelm the judicial process, you quickly see how you might erode a core check on executive power in the United States. And Trump and Elon clearly intend to every day test those checks and balances and break them if they can. I mean, that is the intent, that is the intent. HW : So you mentioned democracy. When we last spoke, you said very clearly that the US is not Hungary, meaning that democracy is not threatened and that the US is not in danger of turning into an autocracy. Do you stand by that? IB : Yes, I do. I mean, in the sense that I think in two years we ’ re going to have midterm elections, and they will be largely free and fair. And in four years we ’ ll have presidential elections. And US elections are held at the federal level, right? They ’ re held state by state. They make the rules. It ’ s very easy to make people believe that the election has been stolen. It ’ s extremely hard to actually rig an American election. And I don ’ t believe that that is very likely at all. It ’ s very unlikely. The American military, despite the firings that we ’ ve seen just last Friday, is still an independent and professional military that I believe will carry out its orders and its duties professionally. I see the judiciary in the United States as independent and a clear check on the executive power in the United States. Where I see a problem, is that the US is becoming more kleptocratic every day. And the US was already much more of a kleptocracy over the past 10 years under Democrats and Republicans than any other advanced industrial democracy in the world. Clearly, money buys power in America in a way that it doesn ’ t in any other major democracy, wealthy democracy. That is getting dramatically worse. I saw specifically that Elon Musk, who is a walking conflict of interest at the highest level, met with Narendra Modi, the Prime Minister of India, in Blair House, so part of the official White House complex, right after Trump met with him. And Trump was asked later in the day, was Modi meeting with Elon in Elon ’ s private-sector capacity to advance his business interests or his official capacity to advance the policies of the United States? And Trump answered honestly. He said,”I don ’ t know.”And how could he know, right? Because, I mean, those two things are completely intertwined. But that is not remotely acceptable for rule of law in a functional democracy, and yet nothing is being done about it. In fact, it ’ s getting more and more entrenched every day. So the single place where the United States was least functional as an advanced democracy is now getting far, far worse. And I don ’ t see any checks and balances on that. I think that ’ s a very, very serious problem. HW : Alright, Ian, thank you, as always for all of your insights. Let ’ s close out with a couple more questions from our community. And this one indeed relates to the kleptocratic conversation that you were just raising. So why are the billionaires enabling fascism? What do they get out of it? IB : Fascism is a very politically loaded argument. That ’ s a political system where the cruelty against a subgroup or a perceived subgroup in the society is the intention, the expressed intention of the policy. And I certainly wouldn ’ t define the United States as a fascist system today. But I do believe that billionaires in the United States, at least a significant subset of them, are really under-interested in the well-being of their fellow countrymen and women. And this is a country that really makes heroic the individual. And it undermines the community. We ’ ve seen a significant erosion of civic engagement in the United States and civic associations. The church is not what it was in the US. Fewer people go, fewer people trust the church. Public education is not what it was. And wealthy people increasingly opt out. The family is not what it was. It ’ s increasingly atomized and people spend much more, much more of their time being, you know, isolated and atomized and intermediated by algorithms. And you and I have spoken about that problem in the past. We are increasingly a society of individuals as opposed to a nation together. I will tell you that that is a real problem when you have large numbers of billionaires that believe that they built it themselves, that they ’ re not responsible to a collective whole for any of their success and therefore they

’ re not accountable for their fellow citizens. And that ’ s what I think we have. I think we have a lot of winners in the United States and not a lot of leaders. And when you have winners, you have losers. When you have winners, you don ’ t have a collectivity. I will say that in this anti-DEI thing, I have a lot of sympathy for the people that don ’ t like DEI. I thought it was rammed down the throats of a lot of people And I ’ ve said this before, it was well beyond what the average American was willing to tolerate, and also it came with a level of high-handedness that that basically said, if you ’ re not with us in supporting these programs, you ’ re evil, you ’ re stupid, you ’ re uneducated. That ’ s unacceptable, too. It was a completely, you know, sort of oppositional way to try to have what ’ s a very challenging conversation. But I ’ ve never been a champion of diversity for diversity ’ s sake. What ’ s amazing about the history of the United States is despite all of our diversity, we find commonalities with each other. We find connectedness. That ’ s what ’ s amazing, is the connectedness, is that, you know, on this world, we ’ ve got eight billion people. And despite how different we all are, we all actually connect as human beings. That is what we need to focus on, not the diversity, the connectedness. And I think that ’ s what the billionaires today in the United States are losing. They ’ re doing incredibly well themselves. They ’ ve got theirs, they ’ ve got **access** to power, they ’ re paying good money for it, they ’ re going to get even more. But they ’ re forgetting that for us to succeed as a society and as a planet, we have to be connected to each other. You can ’ t throw out diversity and forget connectedness, and that ’ s what we ’ re forgetting right now. That ’ s why the country feels so much **trauma** and pain right now. And I think why the United States has given up on a lot of its really aspirational values that made such a difference to the world, that almost lost World War II for many generations. And now we ’ re giving it all away. HW : I mean, I think that ’ s why you find that diversity and inclusion go hand in hand, that that ’ s actually speaking to that. But in some of the moves that are being made to kind of throw out any type of diversity, to scan documents, to find words in order to highlight programs that are wasteful, that we ’ re actually going to lose what many, many studies actually highlight as being the benefits of diversity and the fact that you need diversity in order to be actually successful. IB : Shooting first and asking questions later, you know, moving fast and breaking things, those things make sense in a turbocharged venture capital technology environment. They do not work in government. They don ’ t work in government at all. In government, we have something called checks and balances, and we need them. And moving fast and breaking things undermines those checks and balances. So we ’ re going to do a lot of damage unintentionally. A lot of people are going to get hurt. A lot of folks don ’ t care about that in power right now, but they should because it would be much, much better. I know people want to move fast, but actually to think a little bit, do a little bit of your own research before you decide to kill that program, before you decide to fire that person. We ’ d be in a lot better shape as a government right now. HW : OK, the final question that I have for you actually comes from my 10-year-old son, and it ’ s a bit of a miserable question. We have a very lovely house, I promise. But his question nonetheless is are we heading into World War III? So, Ian, what should I tell him? IB : No, no, I don ’ t think we ’ re heading into World War III but we are absolutely heading into a world of much greater conflict. I think it ’ s much more likely, for example, in the next five, 10 years, that we ’ ll have a nuclear weapon go off. And I thought that was pretty unlikely since the Cuban Missile Crisis. In other words, over the course of my life. I think we ’ re going to see a lot more terrorism at scale, because a lot more people are going to feel angry and disaffected, I think. You know, you saw that assassination of the CEO of

UnitedHealthcare just a few dozen blocks from my house. I think you ’ re going to see more things like that. And so, no, not World War III because the fact is that outside of the US and China, all the other countries know that they need to work with everybody. And there are even forces inside US-China that are trying to make sure that you don ’ t throw the baby out with the bathwater. But underneath that World War III risk, there ’ s an awful lot of damage that ’ s going to get done, an awful lot of crisis that I think we ’ re going to have to live through before we start to see what we create on the other side. A lot of defense as well, that we ’ re going to be playing in this environment. HW : Well, Ian, it is always a pleasure to talk to you, and we ’ re so glad that you are scanning the world to understand what is happening and then you come here and explain it to us all. Couldn ’ t be more grateful and thank you for everything. IB : Thank you.

Text Sample 365: POS Dim 3 – 2025 – Score 2.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in the world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the air. In all my travels, I ’ ve been struck by so many things. I ’ ve observed how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ’ ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced us to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ’ s hard to see the big picture right now. It ’ s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think back to my travels. I ’ ve seen how different we are, how diverse and distinct. But I ’ ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ’ m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman”A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story is the dominant sound of our lives.”- Joseph Campbell I ’ ve loved telling stories all of my life. I ’ ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many **shared** experiences : rejection, judgment, insecurity, **prejudice**, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 366: POS Dim 3 – 2021 – Score 16.00 – 2021/t003783.txt

The world can be a toxic place. No one knows this more than women and girls, cis and trans, as well as nonbinary people. I am a journalist with 30 years ’ experience reporting on the injustices women and others face around the world. I ’ m a co-founder of the Man Up Campaign to energize youth activism against **sexual** and gender-based **violence**. I ’ m a father to two daughters, I ’ m a life partner to a woman and I ’ m a Black man in America. Holding these identities – and many more – I say this call to action to you with the deepest sincerity and utmost gravity. It is time for a gender reckoning, beginning with men authentically confronting our internal selves and each other through the toxicity within us. Against the backdrop of Black Lives Matter protest, cries for a racial reckoning have echoed throughout the world, at times seeming premature and largely misunderstood. I believe a true gender reckoning must be the center of any transformative movement for social justice and **equity**. But what is a reckoning? Dictionaries say it means to resolve a bill or a debt. In other words, to settle an account. In the case of women and girls, what is the debt owed to them? Who ’ s to pay it? And how? The consequences of male toxicity continue building the debt to which I refer, including, among others, a gender wealth gap most affecting women who care for their families as well as act as the primary income generators for their households. Today, in the United States and abroad, threats to reproductive justice endanger the lives and hard-fought rights of women in regards to their bodily, **sexual** and gender autonomy, impacting the most disenfranchised community in society. And **violence** against women, in all of its forms, still represents an existential challenge, as we ’ ve seen **domestic violence** rise in the pandemic, as people were confined to their homes, we ’ ve seen online and public **harassment** flourish, unabated. The **sexual** trafficking, prostitution, exploitation of women and girls thrives as ever before. From where does this toxicity come and why? Men are not endangered, we ’ re not under **assault**, we ’ re not being unfairly maligned. In fact, it

’ s men, or certain types of masculinity, that are the issue. To be clear, being a man does not make me or anyone else who identifies as a man inherently toxic. But masculinity, in certain forms, can **harm** women, girls, nonbinary people and men through **violence**, dominating power and control. So how do we address this? How can we men step up and better support our daughters, our sisters, our female partners, our colleagues, neighbors and friends? Through what means can men both instigate and lead a gender reckoning? First, we must tell the truth. We must tell the truth. Men grow into this world becoming who we ’ re expected to be or who we think we ’ re expected to be based on cultural expectations and inaccurate historical narratives. As a short, chubby Black kid wearing glasses growing up, I was repeatedly told I cried too much. I ’ m too emotional, too soft. It was ingrained in me that to prove my masculinity I had to display physical prowess – intimidation. Because of that indoctrination, as an adult, I refused to seek help facing escalating health challenges, including high blood pressure and mental **trauma**. The archetype of manhood nearly killed me. I say that as a survivor of two heart attacks, four years on dialysis and a kidney transplant... aside from the physical risk I put myself into covering wars around the world. The injury we do to ourselves and the women around us comes from fear of losing control, of lack of external affirmation and simply out of **shame**. Manhood is a spectrum of identities. There is no one way to be a man. And sometimes, discovering one ’ s true manhood is a lifelong journey. Secondly, we must be consistent. Men, especially men of color, cannot expect women to stand up for us, with us in the face of **systemic** oppressions when too often we do not do the same for them. The Black Lives Matter movement for example, founded, named and led by queer women of color, still centers Black men ’ s well-being in the face of unfair policing and criminal justice systems. And where ’ s the reciprocal support from those men, including myself, in addressing intercommunity **violence** against women of color, maternal health **inequity**, economic disparity? How and where do men show up for women? I wonder... how many men stood with women on the steps of the US Supreme Court as they fought for reproductive rights? How many men should have been there? Finally, there can be no reckoning without accountability. There can be no reckoning without accountability. Men who commit acts of **violence** against women must be held accountable by formal systems of justice as well as other men. **Silence** in the face of misogynistic behavior and language is complicity. We men need to break free from the pack, take concrete actions to stop **violence**, create **safety** and build **equity**. And most importantly, say something. To be sure, bold men around us are actively working to disrupt the narrative of male toxicity. After the 2010 earthquake in Haiti, a local organization, KOFAVIV, invited men to be defenders of women and girls in displacement camps, women and girls who were at risk of **sexual assault**. Scores of men answered the call and can continue to do so today. In New York City, CONNECT NYC, led by a Black man, actively engages community partners across civil society to address family **violence**. And in a profession not known generally for its advocacy against gender **violence**, a professional player for the Washington Football Team has become an unabashed advocate for men to learn and actively change behaviors regarding **sexual** and physical **violence**. There is hope ; there is light. Men around the world are putting in the work to catalyze a true gender reckoning. But far more are needed. What I want to say to you today is this. Too many of us – too many men – don ’ t understand that **equity**, justice, inclusivity, **safety** for women, girls and nonbinary people serves and saves us all. Broken masculinity **harms** societies, full stop. When we men reckon with that truth, we will finally find our **shared** humanity. Women, girls, nonbinary people, men standing together on **equal** footing with mutual respect in safe environments is the greatest gender reckoning we can create. Thank you.

Text Sample 367: POS Dim 3 – 2021 – Score 14.00 – 2021/t003934.txt

Male, Female. Christian, Jew, Muslim, Atheist. White, Black, Asian, Caucasian, Arab. Liberal, Conservative, Libertarian, Progressive. Cisgender, Transgender, Heterosexual, Homosexual, Queer, Non-binary, Agender. All of these are labels, some of which I have, and some of that you yourselves may have. But what exactly are labels? Merriam-Webster defines the label as a descriptive or identifying word or phrase. Since the dawn of human language, we have come up with terms to categorize ourselves and each other, from simple things such as ’ us ’ and ’ them ’, ’ family ’, ’ tribe member ’, ’ citizen ’, ’ foreigner ’ and ’ enemy ’ to more complex labels describing political ideologies, religion, or even personal interests. ’ Geek ’, ’ jock ’, ’ soccer mom ’ and ’ workaholic ’ are in widespread use. Some labels that I use include ’ female ’, ’ Asian-American ’, ’ parent ’, ’ daughter ’, ’ sister ’, ’ transgender ’, ’ pansexual ’, ’ nerd ’, and ’ airman ’. Labels have a greater power than what a simple definition can describe. Looking at today ’ s world, labels have increasingly led to more division, marginalization, **hatred**. According to the FBI, hate crimes are at the highest level they ’ ve been in 10 years. People have been attacked for being Asian. **Victims** of misinformation spread during the rise of COVID-19, or used as a scapegoat for a young man ’ s inability to control his **sexual** urges in Georgia this past March. When I was six years old, I attended kindergarten, I was first exposed to a variety of racial slurs and bullying due to my mixed Asian-American ancestry. As I discovered computers and I did well on math tests. I was harassed because of stereotypes. To this day, I struggled to escape those that want to use this label for **harm**, simply because I was born with it. Others, like Jews, have been attacked for their religion. Anti-Semitism has existed for thousands of years, from the ancient Greeks and Romans to the expulsion of Jews, from multiple nations in the Middle Ages to more recently, the Holocaust - Nazi Germany ’ s final solution. The Pew Research Center has studies indicating a growing partisan divide in politics. The number of Democrats and Republicans willing to compromise and work together is at its lowest in modern history. The LGBTQ+ community has faced legislative attacks around the globe from so-called”LGBT-free zones”in Poland to over 80 anti-transgender bills introduced here in the United States. Those that are part of this community have been particularly vulnerable. Transgender military members such as myself have experienced the back and forth in policies - traded off like we ’ re a footballer trading card, and not human beings that want to serve our country. And, as we saw last year, despite the gains in legislative **equality** stemming from the 13th Amendment abolishing slavery, and the Civil Rights Act of 1964, we all have a long way to go in addressing **systemic** racism and centuries of **inequities** in the United States. From integral parts of identity such as our gender, gender identity, **sexual** orientation, race and **ethnicity**, to those groups that we choose to be a part of, such as political parties, political alignment or religion, peace labels, these groupings have divided us to an alarming degree. Labels can separate us. They can help foster an ’ us versus them ’ mentality, leading to unhealthy competition. They can lead to **harassment**, **discrimination**, bigotry, **assault**, and even murder. Labels used incorrectly can be dangerous. And yet, our labels are entirely without benefit. When I was seven years old, I discovered that my gender identity didn ’ t match what I was assigned at birth. While my birth certificate said Male, every part of my being said that this was wrong, that I was actually a girl. But when I made this realization all the way back in 1989, I had no idea what this feeling meant for me. I was confused. I was scared. I thought something was horribly wrong with me. I had no internet to search for clues. There was no representation in the media that I could turn to for inspiration and guidance. I was alone. I struggled with these feelings for almost 20 years before I found the word ' transgender '. Suddenly, what I was and who I was became clear for the first time in my life. In addition to that, I discovered that I was part of a community : the LGBTQ+ Community. I found mentors, friends, and people that I now consider to be family. I can go to any major city, and in many of the smaller ones, and with a quick search on the internet, I can find community centers and social places that are focused on making my community safe and welcome. This label, one that has become so controversial in our country, became a way for me to accept who I am and to feel a part of something larger than myself. So, too, can other labels. Members of a particular religion can find fellowship with each other. Street signs, social media pages, google searches can help anyone from evangelical Christians, Catholics, Baptists, Muslims to faiths like Rastafarianism, Zoroastrianism and Wicca find places of worship in new locations. Those that adhere to specific political parties or ideologies can more easily connect and find common ground. Places such as Reddit have communities for people that follow nearly every political system and ideology that has ever existed. Professions have their own labels. One that I earned in my life is Airman. I enlisted in the United States Air Force in 2004, and having that label has tied me to a group of individuals with many **shared** goals and common interests. I have been able to connect with currently serving service members and veterans, and we find support through those others People with **shared** cultures, shared histories can come together and **celebrate** that which makes them unique. Major cities host cultural days and weeks where all members of the local area can gather and enjoy music, food, clothing, dance, and other aspects of many different cultures. Labels, at their worst, can drive divisions between us when they foster an ' us versus them ' mentality, when they turn into a categorization of people like me are good, but people not like me are the enemy. Labels lead to negative, such as **harassment** and **discrimination**. They separate us and lead us to live lives that are empty of the good, that different perspectives spring, But at their best, labels describe those things that connect us to one another. When we work as a team, when we adopt the label teammate, we are capable of accomplishing things greater than we can when we ' re alone. My own transgender label has connected me to a community that has accepted me, given me mentorship, and helped me to acknowledge who I am. Being an Asian-American, and more specifically a Korean-American, ties me to a deep and rich culture that broadens my own perspectives, and allows me to teach my own children about their roots. Labels allow us to bind together in groups and communities that allow us to grow and become more knowledgeable about our histories, our cultures, and how to develop strategies on how to be more successful. Labels are not all good or bad. Like any tool, they can be used to build and create wonderful things, highlighting what I consider to be one of the greatest things that humanity has developed : community ties. But, misused, they ' re also capable of causing a great deal of destruction. If we, as individuals, choose to use labels responsibly, we can achieve greater things together than we ever could on our own.

Text Sample 368: POS Dim 3 – 2021 – Score 13.00 – 2021/t003773.txt

I was born during a war that still engulfs my country, Afghanistan. When I was growing up, **violence** was all around me. I remember sitting in my living room with my family when a distant thud would jerk us all out of our seats. As the sound of the rockets came closer and closer, our bodies would coil and freeze. My grandmother would take charge of her terrified flock and shepherd us into a room in the back of the house. No less exposed, but to us, it somehow felt safer. I ' d cling to my granny and clench my fists wondering why this was happening to us. I clearly remember my little brother ' s face wincing every time we heard a rocket fall. It was this **violence**, this war that still goes on today, which forced my family to leave our home. We left early one morning not knowing where we would end up. Along the way, there was more **violence**. We drove on roads with landmines, and everywhere we went, there was hostility. The overriding memory in my body of those years is of feeling **unsafe**. Feeling as though something terrible would happen to me or to my family. We worried about our friends and family back home. My grandmother was still in Kabul, and I missed her terribly. I ' d go to bed at night and say the **prayers** she taught me to pray for her **safety**. We wanted nothing more than to go back home. But when the Taliban took over the country in 1996, my parents realized that was no longer an option. So after four years of living in exile, we sought asylum in the UK and began a new life. I started going to school. And almost 25 years later, I ' m standing here with you. I ' m now working to help others overcome devastation by war. To overcome the **trauma** of being expendable. In my work, I see how living under constant threat of **violence** and feeling **unsafe** can affect people, even when they manage to get out. War leaves a physical legacy in our body, our mind and in our spirit. When I tell people about my life, they sometimes say it ' s a remarkable story. But this is not true. Millions of people are living through war and displacement, trying to survive. For these people, like my family, the dream isn ' t to get rich or to go on vacations or to buy that perfect house. They have a modest, yet somehow still extravagant dream of being safe. They dream of a day when they can go to the market without the fear of **violence**. Or send their kids to school without being afraid for their lives. War is dislocating. It gets inside your body and makes you into a thing, a corpse. It can dislodge empathy, hope and joy and replace them with fear. The fear that comes from living in the **violence** of war can break social bonds, disbanding the very communities we rely on for **safety**. Women and children I work with complain about pain in their body. Children as young as three or four talk about hurt in their bellies and not because they ' re hungry. It ' s a pain I know well. A constant, dull ache all over your body, and it feels as though nothing can alleviate it. People have nightmares and experience overwhelming emotions : **grief**, sadness and anger."Why did I deserve this,"they wonder. This is the situation facing 84 million forcibly displaced people. And according to UNHCR, more than half are women and children. What ' s even more startling is the number of people living in active conflict zones. According to Save the Children, 420 million children are growing up in places where **violence** is the norm. These are families and children subject to forces they cannot control. Yet war and **violence** don ' t become normal. The stress and **trauma** experienced by people who go through war is itself deadly. Illnesses like cancer, diabetes and heart disease have been linked to **trauma** and chronic stress. But it ' s the psychic pain that affects people the most. Depression, PTSD and other psychiatric disorders affect a significant percentage of people who go through war. I was shocked when I learned that since 9 / 11, four times more US soldiers have taken their own lives than died in combat. Reconciling to normal life after you ' ve seen life-shattering war can be the hardest of tasks for a human being. But mental health and healing are often overlooked when supporting people who have survived war. And that ' s why I set up Refugee **Trauma** Initiative, to help support the mental health of those affected by war and displacement. Our organization is one of several around the world that helps people so they can begin to feel safe again. When feeling safe is splintered and you feel **unsafe** and unseen, we work to help bring the internal safe system back online. We create spaces where people can convene and **heal** as a community. In our groups, men, women and children reconnect with their bodies, have the space to express what they are feeling, and most importantly, reconnect with others. We use art, mindfulness, dance and storytelling to help make sense of the dislocating experiences of **violence** and of being forced to leave your home. And everything that we do is underpinned by value-based practice. We practice understanding, curiosity and respect to everyone who comes through the door. We recognize and acknowledge their **trauma** without pathologizing or medicalizing their very normal human reaction to the reckless **violence** they ' ve experienced. And we recognize that these experiences are the result of **systemic** injustice and oppression that often have colonial roots, which has dehumanized and killed people for centuries. At the heart of our work is a simple yet fundamental notion that feelings of **safety** are inextricably tied to feeling connected to a community. A community where you feel worthy and seen. Where your **suffering** will be recognized and cared for, where you can experience belonging. Understanding how **violence** and conflict affects and changes people is more important than ever. Our world is embroiled in multiple chronic conflicts that seem to have no end in sight. Since 1945, there have been 150 conflicts around the world, and the overwhelming majority of casualties have been civilians. We can ' t talk about this human toll of war without talking about the war industry. The war on terror over the last 20 years unleashed a new kind of **violence**, displacing 38 million people. This is a war that made drone warfare the norm. Drones enable automated extermination. As we are moving closer and closer towards automation of war, what happens when we outsource **violence**, the extinguishing of a human life, to a computer? What happens to us as an interconnected global community when a machine and not a human decides whether a life is worth something? This is all bleak, but not hopeless. There ' s always room to adjust our trajectory to a better future. I ' m standing here in the most powerful country in the world, a citizen of the United Kingdom, two countries that have long held power over war and peace. And the first thing that we can do is demand that our governments stop investing in mass destruction. At a time when we need to invest in tackling climate change and in health care, our governments are financing weapons and drones. Every vote that we cast should be against weapons of mass destruction, against automation of war. And you can do something in your own community. Right now, there are literally millions of Afghans who are brutally forced to leave their home. Like them, there are Syrians, Iraqis, Burundians who all have to deal with the fallout of war and displacement. And the antidote to their **suffering** lies in the kindness and connection of the communities where they find themselves. If they can feel safe there, they can begin to **heal** and start a new life. It ' s sometimes hard for us to see that the answers to some of the most complicated problems in the world lie in the simplest of human actions. It may sound too easy that a good neighbor can help to **heal** the wounds of war. But in my experience, this is true. Before we landed in London, after months of hardship, we stopped in a poor **neighborhood** of a town in Central Asia. I was forlorn, **grief** stricken and not yet able to imagine a future without **violence**. One day, a **neighborhood** teacher knocked on our door. When we opened the door, we saw a kindly middle-aged woman carrying a box of pens, colored pencils, paper and children ' s books. She asked my parents if she could come in and spend time with us. They let her in, and from that day, she visited us most days after school. When I told her that I desperately wanted to go to school, she took me to her classroom, where the girls had prepared gifts. They taught me how to **sing** a song. It wasn ’ t long before we had to move on, and once we left, I never saw her again. But what she did made me feel valued and safe. The memory of that experience had left a blueprint of what it felt to inhabit my body, if only for a moment, without fear. A memory of joy and belonging. And to this day, when I feel the opposite, I can close my eyes and feel that connection. And it carries me through the dark days. Thank you.

Text Sample 369: POS Dim 3 – 2015 – Score 16.00 – 2015/t001602.txt

To be honest, by personality, I ’ m just not much of a crier. But I think in my career that ’ s been a good thing. I ’ m a civil rights lawyer, and I ’ ve seen some horrible things in the world. I began my career working police abuse cases in the United States. And then in 1994, I was sent to Rwanda to be the director of the U. N. ’ s genocide investigation. It turns out that tears just aren ’ t much help when you ’ re trying to investigate a genocide. The things I had to see, and feel and touch were pretty unspeakable. What I can tell you is this : that the Rwandan genocide was one of the world ’ s greatest failures of simple compassion. That word, compassion, actually comes from two Latin words : cum passio, which simply mean”to suffer with.”And the things that I saw and experienced in Rwanda as I got up close to human **suffering**, it did, in moments, move me to tears. But I just wish that I, and the rest of the world, had been moved earlier. And not just to tears, but to actually stop the genocide. Now by contrast, I ’ ve also been involved with one of the world ’ s greatest successes of compassion. And that ’ s the fight against global poverty. It ’ s a cause that probably has involved all of us here. I don ’ t know if your first introduction might have been choruses of”We Are the World,”or maybe the picture of a sponsored child on your refrigerator door, or maybe the birthday you donated for fresh water. I don ’ t really remember what my first introduction to poverty was but I do remember the most jarring. It was when I met Venus — she ’ s a mom from Zambia. She ’ s got three kids and she ’ s a widow. When I met her, she had walked about 12 miles in the only garments she owned, to come to the capital city and to share her story. She sat down with me for hours, just ushered me in to the world of poverty. She described what it was like when the coals on the cooking fire finally just went completely cold. When that last drop of cooking oil finally ran out. When the last of the food, despite her best efforts, ran out. She had to watch her youngest son, Peter, suffer from malnutrition, as his legs just slowly bowed into uselessness. As his eyes grew cloudy and dim. And then as Peter finally grew cold. For over 50 years, stories like this have been moving us to compassion. We whose kids have plenty to eat. And we ’ re moved not only to care about global poverty, but to actually try to do our part to stop the **suffering**. Now there ’ s plenty of room for critique that we haven ’ t done enough, and what it is that we ’ ve done hasn ’ t been effective enough, but the truth is this : The fight against global poverty is probably the broadest, longest running manifestation of the human phenomenon of compassion in the history of our species. And so I ’ d like to share a pretty shattering insight that might forever change the way you think about that struggle. But first, let me begin with what you probably already know. Thirty-five years ago, when I would have been graduating from high school, they told us that 40, 000 kids every day died because of poverty. That number, today, is now down to

17, 000. Way too many, of course, but it does mean that every year, there ' s eight million kids who don ' t have to die from poverty. Moreover, the number of people in our world who are living in extreme poverty, which is defined as living off about a dollar and a quarter a day, that has fallen from 50 percent, to only 15 percent. This is massive progress, and this **exceeds** everybody ' s expectations about what is possible. And I think you and I, I think, honestly, that we can feel proud and encouraged to see the way that compassion actually has the power to succeed in stopping the **suffering** of millions. But here ' s the part that you might not hear very much about. If you move that poverty mark just up to two dollars a day, it turns out that virtually the same two billion people who were stuck in that harsh poverty when I was in high school, are still stuck there, 35 years later. So why, why are so many billions still stuck in such harsh poverty? Well, let ' s think about Venus for a moment. Now for decades, my wife and I have been moved by common compassion to sponsor kids, to fund microloans, to support generous levels of foreign aid. But until I had actually talked to Venus, I would have had no idea that none of those approaches actually addressed why she had to watch her son die."We were doing fine,"Venus told me,"until Brutus started to cause trouble."Now, Brutus is Venus ' neighbor and"cause trouble"is what happened the day after Venus ' husband died, when Brutus just came and threw Venus and the kids out of the house, stole all their land, and robbed their market stall. You see, Venus was thrown into destitution by **violence**. And then it occurred to me, of course, that none of my child sponsorships, none of the microloans, none of the traditional anti-poverty programs were going to stop Brutus, because they weren ' t meant to. This became even more clear to me when I met Griselda. She ' s a marvelous young girl living in a very poor community in Guatemala. And one of the things we ' ve learned over the years is that perhaps the most powerful thing that Griselda and her family can do to get Griselda and her family out of poverty is to make sure that she goes to school. The experts call this the Girl Effect. But when we met Griselda, she wasn ' t going to school. In fact, she was rarely ever leaving her home. Days before we met her, while she was walking home from church with her family, in broad daylight, men from her community just snatched her off the **street**, and violently raped her. See, Griselda had every opportunity to go to school, it just wasn ' t safe for her to get there. And Griselda ' s not the only one. Around the world, poor women and girls between the ages of 15 and 44, they are — when **victims** of the everyday **violence** of **domestic** abuse and **sexual violence** — those two forms of **violence** account for more death and disability than malaria, than car accidents, than war combined. The truth is, the poor of our world are trapped in whole systems of **violence**. In South Asia, for instance, I could drive past this rice mill and see this man hoisting these 100-pound sacks of rice upon his thin back. But I would have no idea, until later, that he was actually a slave, held by **violence** in that rice mill since I was in high school. Decades of anti-poverty programs right in his community were never able to rescue him or any of the hundred other slaves from the beatings and the rapes and the torture of **violence** inside the rice mill. In fact, half a century of anti-poverty programs have left more poor people in slavery than in any other time in human history. Experts tell us that there ' s about 35 million people in slavery today. That ' s about the population of the entire nation of Canada, where we ' re sitting today. This is why, over time, I have come to call this epidemic of **violence** the Locust Effect. Because in the lives of the poor, it just descends like a plague and it destroys everything. In fact, now when you survey very, very poor communities, residents will tell you that their greatest fear is **violence**. But notice the **violence** that they fear is not the **violence** of genocide or the wars, it ' s everyday **violence**. So for me, as a lawyer, of course, my first reaction was to think, well, of course we ' ve got to change all the laws. We ' ve got to make all this **violence** against the poor illegal. But then I found out, it already is. The problem is not that the poor don ' t get laws, it ' s that they don ' t get law **enforcement**. In the developing world, basic law **enforcement** systems are so broken that recently the U. N. issued a report that found that"most poor people live outside the protection of the law."Now honestly, you and I have just about no idea of what that would mean because we have no first-hand experience of it. Functioning law **enforcement** for us is just a total assumption. In fact, nothing expresses that assumption more clearly than three simple numbers : 9-1-1, which, of course, is the number for the emergency police operator here in Canada and in the United States, where the average response time to a police 911 emergency call is about 10 minutes. So we take this just completely for granted. But what if there was no law **enforcement** to protect you? A woman in Oregon recently experienced what this would be like. She was home alone in her dark house on a Saturday night, when a man started to tear his way into her home. This was her worst nightmare, because this man had actually put her in the hospital from an **assault** just two weeks before. So terrified, she picks up that phone and does what any of us would do : She calls 911 — but only to learn that because of budget cuts in her county, law **enforcement** wasn ' t available on the weekends. Listen. Dispatcher : I don ' t have anybody to send out there. Woman : OK Dispatcher : Um, obviously if he comes inside the residence and **assaults** you, can you ask him to go away? Or do you know if he is intoxicated or anything? Woman : I ' ve already asked him. I ' ve already told him I was calling you. He ' s broken in before, busted down my door, **assaulted** me. Dispatcher : Uh- huh. Woman : Um, yeah, so... Dispatcher : Is there any way you could safely leave the residence? Woman : No, I can ' t, because he ' s blocking pretty much my only way out. Dispatcher : Well, the only thing I can do is give you some advice, and call the sheriff ' s office tomorrow. Obviously, if he comes in and unfortunately has a weapon or is trying to cause you physical **harm**, that ' s a different story. You know, the sheriff ' s office doesn ' t work up there. I don ' t have anybody to send."Gary Haugen : Tragically, the woman inside that house was violently **assaulted**, choked and raped because this is what it means to live outside the rule of law. And this is where billions of our poorest live. What does that look like? In Bolivia, for example, if a man sexually **assaults** a poor child, statistically, he ' s at greater risk of slipping in the shower and dying than he is of ever going to jail for that crime. In South Asia, if you enslave a poor person, you ' re at greater risk of being struck by lightning than ever being sent to jail for that crime. And so the epidemic of everyday **violence**, it just rages on. And it devastates our efforts to try to help billions of people out of their two-dollar-a-day hell. Because the data just doesn ' t lie. It turns out that you can give all manner of goods and services to the poor, but if you don ' t restrain the hands of the **violent** bullies from taking it all away, you ' re going to be very disappointed in the long-term impact of your efforts. So you would think that the disintegration of basic law **enforcement** in the developing world would be a huge priority for the global fight against poverty. But it ' s not. Auditors of international assistance recently couldn ' t find even one percent of aid going to protect the poor from the lawless chaos of everyday **violence**. And honestly, when we do talk about **violence** against the poor, sometimes it ' s in the weirdest of ways. A fresh water organization tells a heart-wrenching story of girls who are raped on the way to fetching water, and then **celebrates** the solution of a new well that drastically shortens their walk. End of story. But not a word about the rapists who are still right there in the community. If a young woman on one of our college campuses was raped on her walk to the library, we would never **celebrate** the solution of moving the library closer to the dorm. And yet, for some reason, this is okay for poor people. Now the truth is, the traditional experts in economic development and poverty alleviation, they don ’ t know how to fix this problem. And so what happens? They don ’ t talk about it. But the more fundamental reason that law **enforcement** for the poor in the developing world is so neglected, is because the people inside the developing world, with money, don ’ t need it. I was at the World Economic Forum not long ago talking to corporate executives who have massive businesses in the developing world and I was just asking them,”How do you guys protect all your people and property from all the **violence**?”And they looked at each other, and they said, practically in unison,”We buy it.”Indeed, private security forces in the developing world are now, four, five and seven times larger than the public police force. In Africa, the largest employer on the continent now is private security. But see, the rich can pay for **safety** and can keep getting richer, but the poor can ’ t pay for it and they ’ re left totally unprotected and they keep getting thrown to the ground. This is a massive and scandalous outrage. And it doesn ’ t have to be this way. Broken law **enforcement** can be fixed. **Violence** can be stopped. Almost all criminal justice systems, they start out broken and corrupt, but they can be transformed by fierce effort and commitment. The path forward is really pretty clear. Number one : We have to start making stopping **violence** indispensable to the fight against poverty. In fact, any conversation about global poverty that doesn ’ t include the problem of **violence** must be **deemed** not serious. And secondly, we have to begin to seriously invest resources and share expertise to support the developing world as they fashion new, public systems of justice, not private security, that give everybody a chance to be safe. These transformations are actually possible and they ’ re happening today. Recently, the Gates Foundation funded a project in the second largest city of the Philippines, where local advocates and local law **enforcement** were able to transform corrupt police and broken courts so drastically, that in just four short years, they were able to measurably reduce the commercial **sexual violence** against poor kids by 79 percent. You know, from the hindsight of history, what ’ s always most inexplicable and inexcusable are the simple failures of compassion. Because I think history convenes a tribunal of our grandchildren and they just ask us,”Grandma, Grandpa, where were you? Where were you, Grandpa, when the Jews were fleeing Nazi Germany and were being rejected from our shores? Where were you? And Grandma, where were you when they were marching our Japanese-American neighbors off to internment camps? And Grandpa, where were you when they were beating our African-American neighbors just because they were trying to register to vote?”Likewise, when our grandchildren ask us,”Grandma, Grandpa, where were you when two billion of the world ’ s poorest were drowning in a lawless chaos of everyday **violence**?”I hope we can say that we had compassion, that we raised our voice, and as a generation, we were moved to make the **violence** stop. Thank you very much. Chris Anderson : Really powerfully argued. Talk to us a bit about some of the things that have actually been happening to, for example, boost police training. How hard a process is that? GH : Well, one of the glorious things that ’ s starting to happen now is that the collapse of these systems and the consequences are becoming obvious. There ’ s actually, now, political will to do that. But it just requires now an investment of resources and transfer of expertise. There ’ s a political will struggle that ’ s going to take place as well, but those are winnable fights, because we ’ ve done some examples around the world at International Justice Mission that are very encouraging. CA : So just tell us in one country, how much it costs to make a material difference to police, for example — I know that ’ s only one piece of it.

GH : In Guatemala, for instance, we ’ ve started a project there with the local police and court system, prosecutors, to retrain them so that they can actually effectively bring these cases. And we ’ ve seen prosecutions against **perpetrators** of **sexual violence** increase by more than 1, 000 percent. This project has been very modestly funded at about a million dollars a year, and the kind of bang you can get for your buck in terms of leveraging a criminal justice system that could function if it were properly trained and motivated and led, and these countries, especially a middle class that is seeing that there ’ s really no future with this total instability and total privatization of security I think there ’ s an opportunity, a window for change. CA : But to make this happen, you have to look at each part in the chain — the police, who else? GH : So that ’ s the thing about law **enforcement**, it starts out with the police, they ’ re the front end of the pipeline of justice, but they hand if off to the prosecutors, and the prosecutors hand it off to the courts, and the survivors of **violence** have to be supported by social services all the way through that. So you have to do an approach that pulls that all together. In the past, there ’ s been a little bit of training of the courts, but they get crappy evidence from the police, or a little police intervention that has to do with narcotics or terrorism but nothing to do with treating the common poor person with excellent law **enforcement**, so it ’ s about pulling that all together, and you can actually have people in very poor communities experience law **enforcement** like us, which is imperfect in our own experience, for sure, but boy, is it a great thing to sense that you can call 911 and maybe someone will protect you. CA : Gary, I think you ’ ve done a spectacular job of bringing this to the world ’ s attention in your book and right here today. Thanks so much. Gary Haugen.

Text Sample 370: POS Dim 3 – 2015 – Score 14.00 – 2015/t001565.txt

Why do we cheat? And why do happy people cheat? And when we say”infidelity,”what exactly do we mean? Is it a hookup, a love story, paid sex, a **chat** room, a massage with a happy ending? Why do we think that men cheat out of boredom and fear of intimacy, but women cheat out of loneliness and hunger for intimacy? And is an affair always the end of a relationship? For the past 10 years, I have traveled the globe and worked extensively with hundreds of couples who have been shattered by infidelity. There is one simple act of transgression that can rob a couple of their relationship, their happiness and their very identity : an affair. And yet, this extremely common act is so poorly understood. So this talk is for anyone who has ever loved. Adultery has existed since marriage was invented, and so, too, the taboo against it. In fact, infidelity has a tenacity that marriage can only envy, so much so, that this is the only commandment that is repeated twice in the Bible : once for doing it, and once just for thinking about it. So how do we reconcile what is universally forbidden, yet universally practiced? Now, throughout history, men practically had a license to cheat with little consequence, and supported by a host of biological and evolutionary theories that justified their need to roam, so the double standard is as old as adultery itself. But who knows what ’ s really going on under the sheets there, right? Because when it comes to sex, the pressure for men is to boast and to exaggerate, but the pressure for women is to hide, minimize and deny, which isn ’ t surprising when you consider that there are still nine countries where women can be killed for straying. Now, monogamy used to be one person for life. Today, monogamy is one person at a time. I mean, many of you probably have said,”I am monogamous in all my relationships.”We used to marry, and had sex for the first time. But now we marry, and we stop having sex with others. The fact is that monogamy had nothing to do with love. Men relied on women ' s fidelity in order to know whose children these are, and who gets the cows when I die. Now, everyone wants to know what percentage of people cheat. I ' ve been asked that question since I arrived at this conference. It applies to you. But the definition of infidelity keeps on expanding : sexting, watching porn, staying secretly active on dating apps. So because there is no universally agreed-upon definition of what even constitutes an infidelity, estimates vary widely, from 26 percent to 75 percent. But on top of it, we are walking contradictions. So 95 percent of us will say that it is terribly wrong for our partner to lie about having an affair, but just about the same amount of us will say that that ' s exactly what we would do if we were having one. Now, I like this definition of an affair — it brings together the three key elements : a secretive relationship, which is the core structure of an affair ; an emotional connection to one degree or another ; and a **sexual** alchemy. And alchemy is the key word here, because the erotic frisson is such that the kiss that you only imagine giving, can be as powerful and as enchanting as hours of actual lovemaking. As Marcel Proust said, it ' s our imagination that is responsible for love, not the other person. So it ' s never been easier to cheat, and it ' s never been more difficult to keep a secret. And never has infidelity exacted such a psychological toll. When marriage was an economic enterprise, infidelity threatened our economic security. But now that marriage is a romantic arrangement, infidelity threatens our emotional security. Ironically, we used to turn to adultery — that was the space where we sought pure love. But now that we seek love in marriage, adultery destroys it. Now, there are three ways that I think infidelity hurts differently today. We have a romantic ideal in which we turn to one person to fulfill an endless list of needs : to be my greatest lover, my best friend, the best parent, my trusted confidant, my emotional companion, my intellectual **equal**. And I am it : I ' m chosen, I ' m unique, I ' m indispensable, I ' m irreplaceable, I ' m the one. And infidelity tells me I ' m not. It is the ultimate betrayal. Infidelity shatters the grand ambition of love. But if throughout history, infidelity has always been painful, today it is often traumatic, because it threatens our sense of self. So my patient Fernando, he ' s plagued. He goes on :"I thought I knew my life. I thought I knew who you were, who we were as a couple, who I was. Now, I question everything."Infidelity — a violation of trust, a crisis of identity."Can I ever trust you again?"he asks."Can I ever trust anyone again?"And this is also what my patient Heather is telling me, when she ' s talking to me about her story with Nick. Married, two kids. Nick just left on a business trip, and Heather is playing on his iPad with the boys, when she sees a message appear on the screen :"Can ' t wait to see you."Strange, she thinks, we just saw each other. And then another message :"Can ' t wait to hold you in my arms."And Heather realizes these are not for her. She also tells me that her father had affairs, but her mother, she found one little receipt in the pocket, and a little bit of lipstick on the collar. Heather, she goes digging, and she finds hundreds of messages, and photos exchanged and desires expressed. The vivid details of Nick ' s two-year affair unfold in front of her in real time, And it made me think : Affairs in the digital age are death by a thousand cuts. But then we have another paradox that we ' re dealing with these days. Because of this romantic ideal, we are relying on our partner ' s fidelity with a unique fervor. But we also have never been more inclined to stray, and not because we have new desires today, but because we live in an era where we feel that we are entitled to pursue our desires, because this is the culture where I deserve to be happy. And if we used to divorce because we were unhappy, today we divorce because we could be happier. And if divorce carried all the **shame**, today, choosing to stay when you can leave is the new **shame**. So Heather, she can ' t talk to her friends because she ’ s afraid that they will judge her for still **loving** Nick, and everywhere she turns, she gets the same advice : Leave him. Throw the dog on the curb. And if the situation were reversed, Nick would be in the same situation. Staying is the new **shame**. So if we can divorce, why do we still have affairs? Now, the typical assumption is that if someone cheats, either there ’ s something wrong in your relationship or wrong with you. But millions of people can ’ t all be pathological. The logic goes like this : If you have everything you need at home, then there is no need to go looking elsewhere, assuming that there is such a thing as a perfect marriage that will inoculate us against wanderlust. But what if **passion** has a finite shelf life? What if there are things that even a good relationship can never provide? If even happy people cheat, what is it about? The vast majority of people that I actually work with are not at all chronic philanderers. They are often people who are deeply monogamous in their beliefs, and at least for their partner. But they find themselves in a conflict between their values and their behavior. They often are people who have actually been faithful for decades, but one day they cross a line that they never thought they would cross, and at the risk of losing everything. But for a glimmer of what? Affairs are an act of betrayal, and they are also an expression of longing and loss. At the heart of an affair, you will often find a longing and a yearning for an emotional connection, for novelty, for freedom, for autonomy, for **sexual** intensity, a wish to recapture lost parts of ourselves or an attempt to bring back vitality in the face of loss and tragedy. I ’ m thinking about another patient of mine, Priya, who is blissfully married, loves her husband, and would never want to hurt the man. But she also tells me that she ’ s always done what was expected of her : good girl, good wife, good mother, taking care of her **immigrant** parents. Priya, she fell for the arborist who removed the tree from her yard after Hurricane Sandy. And with his truck and his tattoos, he ’ s quite the opposite of her. But at 47, Priya ’ s affair is about the adolescence that she never had. And her story highlights for me that when we seek the gaze of another, it isn ’ t always our partner that we are turning away from, but the person that we have ourselves become. And it isn ’ t so much that we ’ re looking for another person, as much as we are looking for another self. Now, all over the world, there is one word that people who have affairs always tell me. They feel alive. And they often will tell me stories of recent losses — of a parent who died, and a friend that went too soon, and bad news at the doctor. Death and mortality often live in the shadow of an affair, because they raise these questions. Is this it? Is there more? Am I going on for another 25 years like this? Will I ever feel that thing again? And it has led me to think that perhaps these questions are the ones that propel people to cross the line, and that some affairs are an attempt to beat back deadness, in an antidote to death. And contrary to what you may think, affairs are way less about sex, and a lot more about desire : desire for attention, desire to feel special, desire to feel important. And the very structure of an affair, the fact that you can never have your lover, keeps you wanting. That in itself is a desire machine, because the incompleteness, the ambiguity, keeps you wanting that which you can ’ t have. Now some of you probably think that affairs don ’ t happen in open relationships, but they do. First of all, the conversation about monogamy is not the same as the conversation about infidelity. But the fact is that it seems that even when we have the freedom to have other **sexual** partners, we still seem to be lured by the power of the forbidden, that if we do that which we are not supposed to do, then we feel like we are really doing what we want to. And I ’ ve also told quite a few of my patients that if they could bring into their relationships one tenth of the boldness, the imagination and the verve that they put into their affairs, they probably would never need to see me. So how do we **heal** from an affair?

Desire runs deep. Betrayal runs deep. But it can be **healed**. And some affairs are death knells for relationships that were already dying on the vine. But others will jolt us into new possibilities. The fact is, the majority of couples who have experienced affairs stay together. But some of them will merely survive, and others will actually be able to turn a crisis into an opportunity. They ' ll be able to turn this into a generative experience. And I ' m actually thinking even more so for the deceived partner, who will often say,"You think I didn ' t want more? But I ' m not the one who did it."But now that the affair is exposed, they, too, get to claim more, and they no longer have to uphold the status quo that may not have been working for them that well, either. I ' ve noticed that a lot of couples, in the immediate aftermath of an affair, because of this new disorder that may actually lead to a new order, will have depths of conversations with honesty and **openness** that they haven ' t had in decades. And, partners who were sexually indifferent find themselves suddenly so lustfully voracious, they don ' t know where it ' s coming from. Something about the fear of loss will rekindle desire, and make way for an entirely new kind of truth. So when an affair is exposed, what are some of the specific things that couples can do? We know from **trauma** that healing begins when the **perpetrator** acknowledges their wrongdoing. So for the partner who had the affair, for Nick, one thing is to end the affair, but the other is the essential, important act of expressing guilt and remorse for hurting his wife. But the truth is that I have noticed that quite a lot of people who have affairs may feel terribly guilty for hurting their partner, but they don ' t feel guilty for the experience of the affair itself. And that distinction is important. And Nick, he needs to hold vigil for the relationship. He needs to become, for a while, the protector of the boundaries. It ' s his responsibility to bring it up, because if he thinks about it, he can relieve Heather from the obsession, and from having to make sure that the affair isn ' t forgotten, and that in itself begins to restore trust. But for Heather, or deceived partners, it is essential to do things that bring back a sense of self-worth, to surround oneself with love and with friends and activities that give back joy and meaning and identity. But even more important, is to curb the curiosity to mine for the sordid details — Where were you? Where did you do it? How often? Is she better than me in bed? — questions that only inflict more pain, and keep you awake at night. And instead, switch to what I call the investigative questions, the ones that mine the meaning and the motives — What did this affair mean for you? What were you able to express or experience there that you could no longer do with me? What was it like for you when you came home? What is it about us that you value? Are you pleased this is over? Every affair will redefine a relationship, and every couple will determine what the legacy of the affair will be. But affairs are here to stay, and they ' re not going away. And the dilemmas of love and desire, they don ' t yield just simple answers of black and white and good and bad, and **victim** and **perpetrator**. Betrayal in a relationship comes in many forms. There are many ways that we betray our partner : with contempt, with neglect, with indifference, with **violence**. **Sexual** betrayal is only one way to hurt a partner. In other words, the **victim** of an affair is not always the **victim** of the marriage. Now, you ' ve listened to me, and I know what you ' re thinking : She has a French accent, she must be pro-affair. So, you ' re wrong. I am not French. And I ' m not pro-affair. But because I think that good can come out of an affair, I have often been asked this very strange question : Would I ever recommend it? Now, I would no more recommend you have an affair than I would recommend you have cancer, and yet we know that people who have been **ill** often talk about how their illness has yielded them a new perspective. The main question that I ' ve been asked since I arrived at this conference when I said I would talk about infidelity is, for or against? I said,"Yes."I look at affairs from a dual perspective : hurt and betrayal on one side, growth and self-discovery on the other — what it did to you, and what it meant for me. And so when a couple comes to me in the aftermath of an affair that has been revealed, I will often tell them this : Today in the West, most of us are going to have two or three relationships or marriages, and some of us are going to do it with the same person. Your first marriage is over. Would you like to create a second one together? Thank you.

Text Sample 371: POS Dim 3 – 2015 – Score 13.00 – 2015/t001587.txt

On the path that American children travel to adulthood, two institutions oversee the journey. The first is the one we hear a lot about : college. Some of you may remember the excitement that you felt when you first set off for college. Some of you may be in college right now and you ' re feeling this excitement at this very moment. College has some shortcomings. It ' s expensive ; it leaves young people in debt. But all in all, it ' s a pretty good path. Young people emerge from college with pride and with great friends and with a lot of knowledge about the world. And perhaps most importantly, a better chance in the labor market than they had before they got there. Today I want to talk about the second institution overseeing the journey from childhood to adulthood in the United States. And that institution is prison. Young people on this journey are meeting with probation officers instead of with teachers. They ' re going to court dates instead of to class. Their junior year abroad is instead a trip to a state correctional facility. And they ' re emerging from their 20s not with degrees in business and English, but with criminal records. This institution is also costing us a lot, about 40, 000 dollars a year to send a young person to prison in New Jersey. But here, taxpayers are footing the bill and what kids are getting is a cold prison cell and a permanent mark against them when they come home and apply for work. There are more and more kids on this journey to adulthood than ever before in the United States and that ' s because in the past 40 years, our incarceration rate has grown by 700 percent. I have one slide for this talk. Here it is. Here ' s our incarceration rate, about 716 people per 100, 000 in the population. Here ' s the OECD countries. What ' s more, it ' s poor kids that we ' re sending to prison, too many drawn from African-American and Latino communities so that prison now stands firmly between the young people trying to make it and the fulfillment of the American Dream. The problem ' s actually a bit worse than this ' cause we ' re not just sending poor kids to prison, we ' re saddling poor kids with court fees, with probation and parole restrictions, with low-level warrants, we ' re asking them to live in halfway houses and on house arrest, and we ' re asking them to negotiate a police force that is entering poor communities of color, not for the purposes of promoting public **safety**, but to make arrest counts, to line city coffers. This is the hidden underside to our historic experiment in punishment : young people worried that at any moment, they will be stopped, searched and seized. Not just in the **streets**, but in their homes, at school and at work. I got interested in this other path to adulthood when I was myself a college student attending the University of Pennsylvania in the early 2000s. Penn sits within a historic African-American **neighborhood**. So you ' ve got these two parallel journeys going on simultaneously : the kids attending this elite, private university, and the kids from the adjacent **neighborhood**, some of whom are making it to college, and many of whom are being shipped to prison. In my sophomore year, I started tutoring a young woman who was in high school who lived about 10 minutes away from the university. Soon, her cousin came home from a juvenile detention center. He was 15, a freshman in high school. I began to get to know him and his friends and family, and I asked him what he thought about me writing about his life for my senior thesis in college. This senior thesis became a dissertation at Princeton and now a book. By the end of my sophomore year, I moved into the **neighborhood** and I spent the next six years trying to understand what young people were facing as they came of age. The first week I spent in this **neighborhood**, I saw two boys, five and seven years old, play this game of chase, where the older boy ran after the other boy. He played the cop. When the cop caught up to the younger boy, he pushed him down, handcuffed him with imaginary handcuffs, took a quarter out of the other child ' s pocket, saying,"I ' m seizing that."He asked the child if he was carrying any drugs or if he had a warrant. Many times, I saw this game repeated, sometimes children would simply give up running, and stick their bodies flat against the ground with their hands above their heads, or flat up against a wall. Children would yell at each other,"I ' m going to lock you up, I ' m going to lock you up and you ' re never coming home!"Once I saw a six-year- old child pull another child ' s pants down and try to do a cavity search. In the first 18 months that I lived in this **neighborhood**, I wrote down every time I saw any contact between police and people that were my neighbors. So in the first 18 months, I watched the police stop pedestrians or people in cars, search people, run people ' s names, chase people through the **streets**, pull people in for questioning, or make an arrest every single day, with five exceptions. Fifty-two times, I watched the police break down doors, chase people through houses or make an arrest of someone in their home. Fourteen times in this first year and a half, I watched the police punch, choke, kick, stomp on or beat young men after they had caught them. Bit by bit, I got to know two brothers, Chuck and Tim. Chuck was 18 when we met, a senior in high school. He was playing on the basketball team and making C ' s and B ' s. His younger brother, Tim, was 10. And Tim loved Chuck ; he followed him around a lot, looked to Chuck to be a mentor. They lived with their mom and grandfather in a two-story row home with a front lawn and a back porch. Their mom was struggling with addiction all while the boys were growing up. She never really was able to hold down a job for very long. It was their grandfather ' s pension that supported the family, not really enough to pay for food and clothes and school supplies for growing boys. The family was really struggling. So when we met, Chuck was a senior in high school. He had just turned 18. That winter, a kid in the schoolyard called Chuck ' s mom a crack whore. Chuck pushed the kid ' s face into the snow and the school cops charged him with aggravated **assault**. The other kid was fine the next day, I think it was his pride that was injured more than anything. But anyway, since Chuck was 18, this agg. **assault** case sent him to adult county jail on State Road in northeast Philadelphia, where he sat, unable to pay the **bail** — he couldn ' t afford it — while the trial dates dragged on and on and on through almost his entire senior year. Finally, near the end of this season, the judge on this **assault** case threw out most of the charges and Chuck came home with only a few hundred dollars ' worth of court fees hanging over his head. Tim was pretty happy that day. The next fall, Chuck tried to re-enroll as a senior, but the school secretary told him that he was then 19 and too old to be readmitted. Then the judge on his **assault** case issued him a warrant for his arrest because he couldn ' t pay the 225 dollars in court fees that came due a few weeks after the case ended. Then he was a high school dropout living on the run. Tim ' s first arrest came later that year after he turned 11. Chuck had managed to get his warrant lifted and he was on a payment plan for the court fees and he was driving Tim to school in his girlfriend ' s car. So a cop pulls them over, runs the car, and the car comes up as stolen in California. Chuck had no idea where in the history of this car it had been stolen. His girlfriend ’ s uncle bought it from a used car auction in northeast Philly. Chuck and Tim had never been outside of the tri-state, let alone to California. But anyway, the cops down at the precinct charged Chuck with receiving stolen property. And then a juvenile judge, a few days later, charged Tim, age 11, with accessory to receiving a stolen property and then he was placed on three years of probation. With this probation sentence hanging over his head, Chuck sat his little brother down and began teaching him how to run from the police. They would sit side by side on their back porch looking out into the shared alleyway and Chuck would coach Tim how to spot undercover cars, how to negotiate a late-night police raid, how and where to hide. I want you to imagine for a second what Chuck and Tim ’ s lives would be like if they were living in a **neighborhood** where kids were going to college, not prison. A **neighborhood** like the one I got to grow up in. Okay, you might say. But Chuck and Tim, kids like them, they ’ re committing crimes! Don ’ t they deserve to be in prison? Don ’ t they deserve to be living in fear of arrest? Well, my answer would be no. They don ’ t. And certainly not for the same things that other young people with more privilege are doing with impunity. If Chuck had gone to my high school, that schoolyard fight would have ended there, as a schoolyard fight. It never would have become an aggravated **assault** case. Not a single kid that I went to college with has a criminal record right now. Not a single one. But can you imagine how many might have if the police had stopped those kids and searched their pockets for drugs as they walked to class? Or had raided their frat parties in the middle of the night? Okay, you might say. But doesn ’ t this high incarceration rate partly account for our really low crime rate? Crime is down. That ’ s a good thing. Totally, that is a good thing. Crime is down. It dropped precipitously in the ’ 90s and through the 2000s. But according to a committee of academics convened by the National Academy of Sciences last year, the relationship between our historically high incarceration rates and our low crime rate is pretty shaky. It turns out that the crime rate goes up and down irrespective of how many young people we send to prison. We tend to think about justice in a pretty narrow way : good and bad, **innocent** and guilty. Injustice is about being wrongfully **convicted**. So if you ’ re **convicted** of something you did do, you should be punished for it. There are **innocent** and guilty people, there are **victims** and there are **perpetrators**. Maybe we could think a little bit more broadly than that. Right now, we ’ re asking kids who live in the most disadvantaged **neighborhoods**, who have the least amount of family resources, who are attending the country ’ s worst schools, who are facing the toughest time in the labor market, who are living in **neighborhoods** where **violence** is an everyday problem, we ’ re asking these kids to walk the thinnest possible line — to basically never do anything wrong. Why are we not providing support to young kids facing these challenges? Why are we offering only handcuffs, jail time and this fugitive existence? Can we imagine something better? Can we imagine a criminal justice system that prioritizes recovery, prevention, civic inclusion, rather than punishment? A criminal justice system that acknowledges the legacy of exclusion that poor people of color in the U. S. have faced and that does not promote and perpetuate those exclusions. And finally, a criminal justice system that believes in black young people, rather than treating black young people as the enemy to be rounded up. The good news is that we already are. A few years ago, Michelle Alexander wrote”The New Jim Crow,”which got Americans to see incarceration as a civil rights issue of historic proportions in a way they had not seen it before. President Obama and Attorney General Eric Holder have come out very strongly on sentencing reform, on the need to address racial disparity in incarceration. We ’ re seeing states throw out Stop and Frisk as the civil rights violation that it is. We ’ re seeing cities and states decriminalize possession of marijuana. New York, New Jersey and California have been dropping their prison populations, closing prisons, while also seeing a big drop in crime. Texas has gotten into the game now, also closing prisons, investing in education. This curious coalition is building from the right and the left, made up of former prisoners and fiscal conservatives, of civil rights activists and libertarians, of young people taking to the **streets** to protest police **violence** against unarmed black teenagers, and older, wealthier people — some of you are here in the audience — pumping big money into decarceration **initiatives** In a deeply divided Congress, the work of reforming our criminal justice system is just about the only thing that the right and the left are coming together on. I did not think I would see this political moment in my lifetime. I think many of the people who have been working tirelessly to write about the causes and consequences of our historically high incarceration rates did not think we would see this moment in our lifetime. The question for us now is, how much can we make of it? How much can we change? I want to end with a call to young people, the young people attending college and the young people struggling to stay out of prison or to make it through prison and return home. It may seem like these paths to adulthood are worlds apart, but the young people participating in these two institutions conveying us to adulthood, they have one thing in common : Both can be leaders in the work of reforming our criminal justice system. Young people have always been leaders in the fight for **equal** rights, the fight for more people to be granted dignity and a fighting chance at freedom. The mission for the generation of young people coming of age in this, a sea-change moment, potentially, is to end mass incarceration and build a new criminal justice system, emphasis on the word justice. Thanks.

Text Sample 372: POS Dim 3 – 2014 – Score 19.00 – 2014/t001836.txt

Election night 2008 was a night that tore me in half. It was the night that Barack Obama was elected. [One hundred and forty-three] years after the end of slavery, and [43] years after the passage of the Voting Rights Act, an African- American was elected president. Many of us never thought that this was possible until the moment that it happened. And in many ways, it was the climax of the black civil rights movement in the United States. I was in California that night, which was ground zero at the time for another movement : the marriage **equality** movement. **Gay** marriage was on the ballot in the form of Proposition 8, and as the election returns started to come in, it became clear that the right for same sex couples to marry, which had recently been granted by the California courts, was going to be taken away. So on the same night that Barack Obama won his historic presidency, the lesbian and **gay** community suffered one of our most painful defeats. And then it got even worse. Pretty much immediately, African-Americans started to be blamed for the passage of Proposition 8. This was largely due to an incorrect poll that said that blacks had voted for the measure by something like 70 percent. This turned out not to be true, but this idea of pervasive black homophobia set in, and was grabbed on by the media. I couldn ’ t tear myself away from the coverage. I listened to some **gay** commentator say that the African-American community was notoriously homophobic, and now that civil rights had been achieved for us, we wanted to take away other people ’ s rights. There were even reports of racist epithets being thrown at some of the participants of the **gay** rights rallies that took place after the election. And on the other side, some African-Americans **dismissed** or ignored homophobia that was indeed real in our community. And others resented this comparison between **gay** rights and civil rights, and once again, the sinking feeling that two minority groups of which I ’ m both a part of were competing with each other instead of supporting each other overwhelmed and, frankly, pissed me off. Now, I ’ m a **documentary** filmmaker, so after going through my pissed off stage and yelling at the television and radio, my next instinct was to make a movie. And what guided me in making this film was, how was this happening? How was it that the **gay** rights movement was being pitted against the civil rights movement? And this wasn ’ t just an abstract question. I ’ m a beneficiary of both movements, so this was actually personal. But then something else happened after that election in 2008. The march towards **gay equality** accelerated at a pace that surprised and shocked everyone, and is still reshaping our laws and our policies, our institutions and our entire country. And so it started to become increasingly clear to me that this pitting of the two movements against each other actually didn ’ t make sense, and that they were in fact much, much more interconnected, and that, in fact, some of the way that the **gay** rights movement has been able to make such incredible gains so quickly is that it ’ s used some of the same tactics and strategies that were first laid down by the civil rights movement. Let ’ s just look at a few of these strategies. First off, it ’ s really interesting to see, to actually visually see, how quick the **gay** rights movement has made its gains, if you look at a few of the major events on a timeline of both freedom movements. Now, there are tons of milestones in the civil rights movement, but the first one we ’ re going to start with is the 1955 Montgomery bus boycott. This was a protest campaign against Montgomery, Alabama ’ s segregation on their public transit system, and it began when a woman named Rosa Parks refused to give up her seat to a white person. The campaign lasted a year, and it galvanized the civil rights movement like nothing had before it. And I call this strategy the”I ’ m tired of your foot on my neck”strategy. So **gays** and lesbians have been in society since societies began, but up until the mid-20th century, homosexual acts were still illegal in most states. So just 14 years after the Montgomery bus boycott, a group of LGBT folks took that same strategy. It ’ s known as Stonewall, in 1969, and it ’ s where a group of LGBT patrons fought back against police beatings at a Greenwich Village bar that sparked three days of rioting. Incidentally, black and latino LGBT folks were at the forefront of this rebellion, and it ’ s a really interesting example of the intersection of our struggles against racism, homophobia, gender identity and police brutality. After Stonewall happened, **gay** liberation groups sprang up all over the country, and the modern **gay** rights movement as we know it took off. So the next moment to look at on the timeline is the 1963 March on Washington. This was a seminal event in the civil rights movement and it ’ s where African- Americans called for both civil and economic justice. And it ’ s of course where Martin Luther King delivered his famous”I have a dream”speech, but what ’ s actually less known is that this march was organized by a man named Bayard Rustin. Bayard was an out **gay** man, and he ’ s considered one of the most brilliant strategists of the civil rights movement. He later in his life became a fierce advocate of LGBT rights as well, and his life is testament to the intersection of the struggles. The March on Washington is one of the high points of the movement, and it ’ s where there was a fervent belief that African- Americans too could be a part of American democracy. I call this strategy the”We are visible and many in numbers”strategy. Some early **gay** activists were actually directly inspired by the march, and some had taken part. **Gay** pioneer Jack Nichols said,”We marched with Martin Luther King, seven of us from the Mattachine Society”— which was an early **gay** rights organization —”and from that moment on, we had our own dream about a **gay** rights march of similar proportions.”Several years later, a series of marches took place, each one gaining the momentum of the **gay** freedom struggle. The first one was in 1979, and the second one took place in 1987. The third one was held in 1993. Almost a million people showed up, and people were so energized and excited by what had taken place, they went back to their own communities and started their own political and social organizations, further increasing the visibility of the movement. The day of that march, October 11, was then **declared** National Coming Out Day, and is still **celebrated** all over the world. These marches set the groundwork for the historic changes that we see happening today in the United States. And lastly, the"Loving"strategy. The name speaks for itself. In 1967, the Supreme Court ruled in Loving v. Virginia, and invalidated all laws that prohibited interracial marriage. This is considered one of the Supreme Court ' s landmark civil rights cases. In 1996, President Clinton signed the Defense of Marriage Act, known as DOMA, and that made the federal government only have to recognize marriages between a man and a woman. In United States v. Windsor, a 79-year-old lesbian named Edith Windsor sued the federal government when she was forced to pay estate taxes on her deceased wife ' s property, something that heterosexual couples don ' t have to do. And as the case wound its way through the lower courts, the Loving case was repeatedly cited as precedent. When it got to the Supreme Court in 2013, the Supreme Court agreed, and DOMA was thrown out. It was incredible. But the **gay** marriage movement has been making gains for years now. To date, 17 states have passed laws allowing marriage **equality**. It ' s become the de facto battle for **gay equality**, and it seems like daily, laws prohibiting it are being challenged in the courts, even in places like Texas and Utah, which no one saw coming. So a lot has changed since that night in 2008 when I felt torn in half. I did go on to make that film. It ' s a **documentary** film, and it ' s called"The New Black,"and it looks at how the African-American community is grappling with the **gay** rights issue in light of the **gay** marriage movement and this fight over the meaning of civil rights. And I wanted to capture some of this incredible change that was happening, and as luck or politics would have it, another marriage battle started gearing up, this time in Maryland, where African- Americans make up 30 percent of the electorate. So this tension between **gay** rights and civil rights started to bubble up once again, and I was lucky enough to capture how some people were making the connection between the movements this time. This is a clip of Karess Taylor-Hughes and Samantha Masters, two characters in the film, as they hit the **streets** of Baltimore and try to convince potential voters. Samantha Masters : That ' s what ' s up, man, this is a righteous man over here. Okay, are you registered to vote? Man : No. Karess Taylor-Hughes : Okay. How old are you? Man : 21. KTH : 21? You gotta get registered to vote. We got to get you registered to vote. Man : I ain ' t voting on no **gay** shit. SM : Okay, why? What ' s up? Man : I ain ' t with that. SM : That ' s not cool. Man : What made you be **gay**? SM : So what made you be straight? So what made you be straight? Man 2 : You can ' t answer that question. KSM : I used to not have the same rights as you, but I know that because a black man like yourself stood up for a woman like me, I know that I ' ve got the same opportunities. So you, as a black man, have the opportunity to stand up for somebody else. Whether you ' re **gay** or not, these are your brothers and sisters out here, and they need you to represent. Man 2 : Who is you to tell somebody who they can ' t have sex with, who they can ' t be with? They ain ' t got that power. Nobody has that power to say, you can ' t marry that young lady. Who has that power? Nobody. SM : But you know what? Our state has put the power in your hands, and so what we need you to do is vote for, you gonna vote for 6. Man 2 : I got you. SM : Vote for 6, okay? Man 2 : I got you. KSM : All right, do y ' all need community service hours?

You do? All right, you can always volunteer with us to get community service hours. Y ' all want to do that? We feed you. We bring you pizza. Yoruba Richen : Thank you. What ' s amazing to me about that clip that we just captured as we were filming is, it really shows how Karess understands the history of the civil rights movement, but she ' s not restricted by it. She doesn ' t just limit it to black people. She sees it as a blueprint for expanding rights to **gays** and lesbians. Maybe because she ' s younger, she ' s like 25, she ' s able to do this a little bit more easily, but the fact is that Maryland voters did pass that marriage **equality** amendment, and in fact it was the first time that marriage **equality** was directly voted on and passed by the voters. African- Americans supported it at a higher level than had ever been recorded. It was a complete turnaround from that night in 2008 when Proposition 8 was passed. It was, and feels, monumental. We in the LGBT community have gone from being a pathologized and reviled and criminalized group to being seen as part of the great human quest for dignity and **equality**. We ' ve gone from having to hide our **sexuality** in order to maintain our jobs and our families to literally getting a place at the table with the president and a shout out at his second inauguration. I just want to read what he said at that inauguration :"We the people **declare** today that the most evident of truths, that all of us are created **equal**. It is the star that guides us still, just as it guided our forebears through Seneca Falls and Selma and Stonewall."Now we know that everything is not perfect, especially when you look at what ' s happening with the LGBT rights issue internationally, but it says something about how far we ' ve come when our president puts the **gay** freedom struggle in the context of the other great freedom struggles of our time : the women ' s rights movement and the civil rights movement. His statement demonstrates not only the interconnectedness of those movements, but how each one borrowed and was inspired by the other. So just as Martin Luther King learned from and borrowed from Gandhi ' s tactics of civil disobedience and nonviolence, which became a bedrock of the civil rights movement, the **gay** rights movement saw what worked in the civil rights movement, and they used some of those same strategies and tactics to make gains at an even quicker pace. Maybe one more other reason for the relative quick progress of the **gay** rights movement. Whereas a lot of us continue to still live in racially segregated spaces, LGBT folks, we are everywhere. We are in urban communities and rural communities, communities of color, **immigrant** communities, churches and **mosques** and synagogues. We are your mothers and brothers and sisters and sons. And when someone that you love or a family member comes out, it may be easier to support their quest for **equality**. And in fact, the **gay** rights movement asks us to support justice and **equality** from a space of love. That may be the biggest, greatest gift that the movement has given us. It calls on us to **access** that which is most universal and most intimate : a love of our brother and our sister and our neighbor. I just want to end with a quote by one of our greatest freedom fighters who ' s no longer with us, Nelson Mandela of South Africa. Nelson Mandela led South Africa after the dark and brutal days of Apartheid, and out of the ashes of that legalized racial **discrimination**, he led South Africa to become the first country in the world to ban **discrimination** based on **sexual** orientation within its constitution. Mandela said,"For to be free is not merely to cast off one ' s chains, but to live in a way that respects and enhances the freedom of others."So as these movements continue on, and as freedom struggles around the world continue on, let ' s remember that not only are they interconnected, but they must support and enhance each other for us to be truly victorious. Thank you.

Text Sample 373: POS Dim 3 – 2014 – Score 15.00 – 2014/t001790.txt

Could I protect my father from the Armed Islamic Group with a paring knife? That was the question I faced one Tuesday morning in June of 1993, when I was a law student. I woke up early that morning in Dad ' s apartment on the outskirts of Algiers, Algeria, to an unrelenting pounding on the front door. It was a season as described by a local paper when every Tuesday a scholar fell to the bullets of fundamentalist assassins. My father ' s university teaching of Darwin had already provoked a classroom visit from the head of the so-called Islamic Salvation Front, who denounced Dad as an advocate of biologism before Dad had ejected the man, and now whoever was outside would neither identify himself nor go away. So my father tried to get the police on the phone, but perhaps terrified by the rising tide of **armed** extremism that had already claimed the lives of so many Algerian officers, they didn ' t even answer. And that was when I went to the kitchen, got out a paring knife, and took up a position inside the entryway. It was a ridiculous thing to do, really, but I couldn ' t think of anything else, and so there I stood. When I look back now, I think that that was the moment that set me on the path was to writing a book called"Your Fatwa Does Not Apply Here : Untold Stories from the Fight Against Muslim Fundamentalism."The title comes from a Pakistani play. I think it was actually that moment that sent me on the journey to interview 300 people of Muslim heritage from nearly 30 countries, from Afghanistan to Mali, to find out how they fought fundamentalism peacefully like my father did, and how they coped with the attendant risks. Luckily, back in June of 1993, our unidentified visitor went away, but other families were so much less lucky, and that was the thought that motivated my research. In any case, someone would return a few months later and leave a note on Dad ' s kitchen table, which simply said,"Consider yourself dead."Subsequently, Algeria ' s fundamentalist **armed** groups would murder as many as 200, 000 civilians in what came to be known as the dark decade of the 1990s, including every single one of the women that you see here. In its harsh counterterrorist response, the state resorted to torture and to forced disappearances, and as terrible as all of these events became, the international community largely ignored them. Finally, my father, an Algerian peasant ' s son turned professor, was forced to stop teaching at the university and to flee his apartment, but what I will never forget about Mahfoud Bennoune, my dad, was that like so many other Algerian intellectuals, he refused to leave the country and he continued to publish pointed criticisms, both of the fundamentalists and sometimes of the government they battled. For example, in a November 1994 series in the newspaper El Watan entitled"How Fundamentalism Produced a Terrorism without Precedent,"he denounced what he called the terrorists ' radical break with the true Islam as it was lived by our ancestors. These were words that could get you killed. My father ' s country taught me in that dark decade of the 1990s that the popular struggle against Muslim fundamentalism is one of the most important and overlooked human rights struggles in the world. This remains true today, nearly 20 years later. You see, in every country where you hear about **armed** jihadis targeting civilians, there are also unarmed people defying those militants that you don ' t hear about, and those people need our support to succeed. In the West, it ' s often assumed that Muslims generally condone terrorism. Some on the right think this because they view Muslim culture as inherently **violent**, and some on the left imagine this because they view Muslim **violence**, fundamentalist **violence**, solely as a product of legitimate grievances. But both views are dead wrong. In fact, many people of Muslim heritage around the world are staunch opponents both of fundamentalism and of terrorism, and often for very good reason. You see, they ' re much more likely to be **victims** of this **violence** than its **perpetrators**. Let me just give you one example. According to a 2009 survey of Arabic language media resources, between 2004 and 2008, no more than 15 percent of al Qaeda ' s **victims** were Westerners. That ' s a terrible toll, but the vast majority were people of Muslim heritage, killed by Muslim fundamentalists. Now I ' ve been talking for the last five minutes about fundamentalism, and you have a right to know exactly what I mean. I cite the definition given by the Algerian sociologist Marieme Helie Lucas, and she says that fundamentalisms, note the"s,"so within all of the world ' s great religious traditions,"fundamentalisms are political movements of the extreme right which in a context of globalization manipulate religion in order to achieve their political aims."Sadia Abbas has called this the radical politicization of theology. Now I want to avoid projecting the notion that there ' s sort of a monolith out there called Muslim fundamentalism that is the same everywhere, because these movements also have their diversities. Some use and advocate **violence**. Some do not, though they ' re often interrelated. They take different forms. Some may be non-governmental organizations, even here in Britain like Cageprisoners. Some may become political parties, like the Muslim Brotherhood, and some may be openly armed groups like the Taliban. But in any case, these are all radical projects. They ' re not conservative or traditional approaches. They ' re most often about changing people ' s relationship with Islam rather than preserving it. What I am talking about is the Muslim extreme right, and the fact that its adherents are or purport to be Muslim makes them no less offensive than the extreme right anywhere else. So in my view, if we consider ourselves liberal or left-wing, human rights-loving or feminist, we must oppose these movements and support their grassroots opponents. Now let me be clear that I support an effective struggle against fundamentalism, but also a struggle that must itself respect international law, so nothing I am saying should be taken as a justification for refusals to democratize, and here I send out a shout-out of support to the pro-democracy movement in Algeria today, Barakat. Nor should anything I say be taken as a justification of violations of human rights, like the mass death sentences handed out in Egypt earlier this week. But what I am saying is that we must challenge these Muslim fundamentalist movements because they threaten human rights across Muslim-majority contexts, and they do this in a range of ways, most obviously with the direct attacks on civilians by the **armed** groups that carry those out. But that **violence** is just the tip of the iceberg. These movements as a whole purvey **discrimination** against religious minorities and **sexual** minorities. They seek to curtail the freedom of religion of everyone who either practices in a different way or chooses not to practice. And most definingly, they lead an all-out war on the rights of women. Now, faced with these movements in recent years, Western discourse has most often offered two flawed responses. The first that one sometimes finds on the right suggests that most Muslims are fundamentalist or something about Islam is inherently fundamentalist, and this is just offensive and wrong, but unfortunately on the left one sometimes encounters a discourse that is too politically correct to acknowledge the problem of Muslim fundamentalism at all or, even worse, **apologizes** for it, and this is unacceptable as well. So what I ' m seeking is a new way of talking about this all together, which is grounded in the lived experiences and the hope of the people on the front lines. I ' m painfully aware that there has been an increase in **discrimination** against Muslims in recent years in countries like the U. K. and the U. S., and that too is a matter of grave concern, but I firmly believe that telling these counter- stereotypical stories of people of Muslim heritage who have confronted the fundamentalists and been their primary **victims** is also a great way of countering that **discrimination**. So now let me introduce you to four people whose stories I had the great honor of telling. Faizan Peerzada and the Rafi Peer Theatre workshop named for his father have for years promoted the performing arts in Pakistan. With the rise of jihadist **violence**, they began to receive threats to call off their events, which they refused to heed. And so a bomber struck their 2008 eighth world performing arts festival in Lahore, producing rain of glass that fell into the venue injuring nine people, and later that same night, the Peerzadas made a very difficult decision : they announced that their festival would continue as planned the next day. As Faizan said at the time, if we bow down to the Islamists, we ’ ll just be sitting in a dark corner. But they didn ’ t know what would happen. Would anyone come? In fact, thousands of people came out the next day to support the performing arts in Lahore, and this simultaneously thrilled and terrified Faizan, and he ran up to a woman who had come in with her two small children, and he said,”You do know there was a bomb here yesterday, and you do know there ’ s a threat here today.”And she said,”I know that, but I came to your festival with my mother when I was their age, and I still have those images in my mind. We have to be here.”With stalwart audiences like this, the Peerzadas were able to conclude their festival on schedule. And then the next year, they lost all of their sponsors due to the security risk. So when I met them in 2010, they were in the middle of the first subsequent event that they were able to have in the same venue, and this was the ninth youth performing arts festival held in Lahore in a year when that city had already experienced 44 terror attacks. This was a time when the Pakistani Taliban had commenced their systematic targeting of girls ’ schools that would culminate in the attack on Malala Yousafzai. What did the Peerzadas do in that environment? They staged girls ’ school theater. So I had the privilege of watching”Naang Wal,”which was a musical in the Punjabi language, and the girls of Lahore Grammar School played all the parts. They **sang** and danced, they played the mice and the water buffalo, and I held my breath, wondering, would we get to the end of this amazing show? And when we did, the whole audience collectively exhaled, and a few people actually wept, and then they filled the auditorium with the peaceful boom of their applause. And I remember thinking in that moment that the bombers made headlines here two years before but this night and these people are as important a story. Maria Bashir is the first and only woman chief prosecutor in Afghanistan. She ’ s been in the post since 2008 and actually opened an office to investigate cases of **violence** against women, which she says is the most important area in her mandate. When I meet her in her office in Herat, she enters surrounded by four large men with four huge guns. In fact, she now has 23 bodyguards, because she has weathered bomb attacks that nearly killed her kids, and it took the leg off of one of her guards. Why does she continue? She says with a smile that that is the question that everyone asks — as she puts it,”Why you risk not living?”And it is simply that for her, a better future for all the Maria Bashirs to come is worth the risk, and she knows that if people like her do not take the risk, there will be no better future. Later on in our interview, Prosecutor Bashir tells me how worried she is about the possible outcome of government negotiations with the Taliban, the people who have been trying to kill her.”If we give them a place in the government,”she asks,”Who will protect women ’ s rights?”And she urges the international community not to forget its promise about women because now they want peace with Taliban. A few weeks after I leave Afghanistan, I see a headline on the Internet. An Afghan prosecutor has been assassinated. I google desperately, and thankfully that day I find out that Maria was not the **victim**, though sadly, another Afghan prosecutor was gunned down on his way to work. And when I hear headlines like that now, I think that as international troops leave Afghanistan this year and beyond, we must continue to care about what happens to people there, to all of the Maria Bashirs. Sometimes I still hear her voice in my head saying, with no bravado whatsoever,”The situation of the women of Afghanistan will be better someday. We should prepare the ground for this, even if we are killed.”There are no words adequate to denounce the al Shabaab terrorists who attacked the Westgate Mall in Nairobi on the same day as a children ’ s cooking competition in September of 2013. They killed 67, including poets and pregnant women. Far away in the American Midwest, I had the good fortune of meeting Somali-Americans who were working to counter the efforts of al Shabaab to recruit a small number of young people from their city of Minneapolis to take part in atrocities like Westgate. Abdirizak Bihi ’ s studious 17-year-old nephew Burhan Hassan was recruited here in 2008, spirited to Somalia, and then killed when he tried to come home. Since that time, Mr. Bihi, who directs the no- budget Somali Education and Advocacy Center, has been vocally denouncing the recruitment and the failures of government and Somali-American institutions like the Abubakar As-Saddique Islamic Center where he believes his nephew was radicalized during a youth program. But he doesn ’ t just criticize the **mosque**. He also takes on the government for its failure to do more to prevent poverty in his community. Given his own lack of financial resources, Mr. Bihi has had to be creative. To counter the efforts of al Shabaab to sway more disaffected youth, in the wake of the group ’ s 2010 attack on World Cup viewers in Uganda, he organized a Ramadan basketball tournament in Minneapolis in response. Scores of Somali-American kids came out to embrace sport despite the fatwa against it. They played basketball as Burhan Hassan never would again. For his efforts, Mr. Bihi has been ostracized by the leadership of the Abubakar As-Saddique Islamic Center, with which he used to have good relations. He told me,”One day we saw the imam on TV calling us infidels and saying, ’ These families are trying to destroy the **mosque**. ’”This is at complete odds with how Abdirizak Bihi understands what he is trying to do by exposing al Shabaab recruitment, which is to save the religion I love from a small number of extremists. Now I want to tell one last story, that of a 22-year-old law student in Algeria named Amel Zenoune-Zouani who had the same dreams of a legal career that I did back in the ’ 90s. She refused to give up her studies, despite the fact that the fundamentalists battling the Algerian state back then threatened all who continued their education. On January 26, 1997, Amel boarded the bus in Algiers where she was studying to go home and spend a Ramadan evening with her family, and would never finish law school. When the bus reached the outskirts of her hometown, it was stopped at a checkpoint manned by men from the Armed Islamic Group. Carrying her schoolbag, Amel was taken off the bus and killed in the **street**. The men who cut her throat then told everyone else,”If you go to university, the day will come when we will kill all of you just like this.”Amel died at exactly 5:17 p. m., which we know because when she fell in the **street**, her watch broke. Her mother showed me the watch with the second hand still aimed optimistically upward towards a 5:18 that would never come. Shortly before her death, Amel had said to her mother of herself and her sisters,”Nothing will happen to us, Inshallah, God willing, but if something happens, you must know that we are dead for knowledge. You and father must keep your heads held high.”The loss of such a young woman is unfathomable, and so as I did my research I found myself searching for Amel ’ s hope again and her name even means”hope”in Arabic. I think I found it in two places. The first is in the strength of her family and all the other families to continue telling their stories and to go on with their lives despite the terrorism. In fact, Amel ’ s sister Lamia overcame her **grief**, went to law school, and practices as a lawyer in Algiers today, something which is only possible because the **armed** fundamentalists were largely defeated in the country. And the second place I found Amel ’ s hope was everywhere that women and men continue to defy the jihadis. We must support all of those in honor of Amel who continue this human rights struggle today, like the Network of Women Living Under Muslim Laws. It is not enough, as the **victims** rights advocate Cherifa Kheddar told me in Algiers, it is not enough just to battle terrorism. We must also challenge fundamentalism, because fundamentalism is the ideology that makes the bed of this terrorism. Why is it that people like her, like all of them are not more well known? Why is it that everyone knows who Osama bin Laden was and so few know of all of those standing up to the bin Ladens in their own contexts. We must change that, and so I ask you to please help share these stories through your networks. Look again at Amel Zenoune ’ s watch, forever frozen, and now please look at your own watch and decide this is the moment that you commit to supporting people like Amel. We don ’ t have the right to be silent about them because it is easier or because Western policy is flawed as well, because 5:17 is still coming to too many Amel Zenounes in places like northern Nigeria, where jihadis still kill students. The time to speak up in support of all of those who peacefully challenge fundamentalism and terrorism in their own communities is now. Thank you.

Text Sample 374: POS Dim 3 – 2014 – Score 14.00 – 2014/t001806.txt
As a student of adversity, I ’ ve been struck over the years by how some people with major challenges seem to draw strength from them. And I ’ ve heard the popular wisdom that that has to do with finding meaning. And for a long time, I thought the meaning was out there, some great truth waiting to be found. But over time, I ’ ve come to feel that the truth is irrelevant. We call it”finding meaning,”but we might better call it”forging meaning.”My last book was about how families manage to deal with various kinds of challenging or unusual offspring. And one of the mothers I interviewed, who had two children with multiple severe disabilities, said to me,”People always give us these little sayings like, ’ God doesn ’ t give you any more than you can handle. ’ But children like ours are not preordained as a gift. They ’ re a gift because that ’ s what we have chosen.”We make those choices all our lives. When I was in second grade, Bobby Finkel had a birthday party and invited everyone in our class but me. My mother assumed there had been some sort of error, and she called Mrs. Finkel, who said that Bobby didn ’ t like me and didn ’ t want me at his party. And that day, my mom took me to the zoo and out for a hot fudge sundae. When I was in seventh grade, one of the kids on my school bus nicknamed me”Percy,”as a shorthand for my demeanor. And sometimes, he and his cohort would chant that provocation the entire school bus ride, 45 minutes up, 45 minutes back :”Percy! Percy! Percy! Percy!”When I was in eighth grade, our science teacher told us that all male homosexuals develop fecal incontinence because of the **trauma** to their anal sphincter. And I graduated high school without ever going to the cafeteria, where I would have sat with the girls and been laughed at for doing so, or sat with the boys, and been laughed at for being a boy who should be sitting with the girls. I survived that childhood through a mix of avoidance and endurance. What I didn ’ t know then and do know now, is that avoidance and endurance can be the entryway to forging meaning. After you ’ ve forged meaning, you need to incorporate that meaning into a new identity. You need to take the **traumas** and make them part of who you ’ ve come to be, and you need to fold the worst events of your life into a narrative of triumph, evincing a better self in response to things that hurt. One of the other mothers I interviewed when I was working on my book had been raped as an adolescent, and had a child following that rape, which had thrown away her career plans

and damaged all of her emotional relationships. But when I met her, she was 50, and I said to her,”Do you often think about the man who raped you?”And she said,”I used to think about him with anger, but now only with **pity**.”And I thought she meant **pity** because he was so unevolved as to have done this terrible thing. And I said,”**Pity**?”And she said,”Yes, because he has a beautiful daughter and two beautiful grandchildren, and he doesn ’ t know that, and I do. So as it turns out, I ’ m the lucky one.”Some of our struggles are things we ’ re born to : our gender, our **sexuality**, our race, our disability. And some are things that happen to us : being a political prisoner, being a rape **victim**, being a Katrina survivor. Identity involves entering a community to draw strength from that community, and to give strength there, too. It involves substituting”and”for”but”— not”I am here but I have cancer,”but rather,”I have cancer and I am here.”When we ’ re ashamed, we can ’ t tell our stories, and stories are the foundation of identity. Forge meaning, build identity. Forge meaning and build identity. That became my mantra. Forging meaning is about changing yourself. Building identity is about changing the world. All of us with stigmatized identities face this question daily : How much to accommodate society by constraining ourselves, and how much to break the limits of what constitutes a valid life? Forging meaning and building identity does not make what was wrong right. It only makes what was wrong precious. In January of this year, I went to Myanmar to interview political prisoners, and I was surprised to find them less **bitter** than I ’ d anticipated. Most of them had knowingly committed the offenses that landed them in prison, and they had walked in with their heads held high, and they walked out with their heads still held high, many years later. Dr. Ma Thida, a leading human rights activist who had nearly died in prison and had spent many years in solitary confinement, told me she was grateful to her jailers for the time she had had to think, for the wisdom she had gained, for the chance to hone her **meditation** skills. She had sought meaning and made her travail into a crucial identity. But if the people I met were less **bitter** than I ’ d anticipated about being in prison, they were also less thrilled than I ’ d expected about the reform process going on in their country. Ma Thida said,”We Burmese are noted for our tremendous grace under pressure, but we also have grievance under glamour.”She said,”And the fact that there have been these shifts and changes doesn ’ t erase the continuing problems in our society that we learned to see so well while we were in prison.”I understood her to be saying that concessions confer only a little humanity where full humanity is due ; that crumbs are not the same as a place at the table. Which is to say, you can forge meaning and build identity and still be mad as hell. I ’ ve never been raped, and I ’ ve never been in anything remotely approaching a Burmese prison. But as a **gay** American, I ’ ve experienced **prejudice** and even **hatred**, and I ’ ve forged meaning and I ’ ve built identity, which is a move I learned from people who had experienced far worse privation than I ’ ve ever known. In my own adolescence, I went to extreme lengths to try to be straight. I enrolled myself in something called”**sexual** surrogacy therapy,”in which people I was encouraged to call doctors prescribed what I was encouraged to call exercises with women I was encouraged to call surrogates, who were not exactly prostitutes but who were also not exactly anything else. My particular favorite was a blonde woman from the Deep South who eventually admitted to me that she was really a necrophiliac, and had taken this job after she got in trouble down at the morgue. These experiences eventually allowed me to have some happy physical relationships with women, for which I ’ m grateful. But I was at war with myself, and I dug terrible wounds into my own psyche. We don ’ t seek the painful experiences that hew our identities, but we seek our identities in the wake of painful experiences. We cannot bear a pointless torment, but we can endure great pain if we believe that it ’ s purposeful. Ease makes less of an impression on us than struggle. We could have been ourselves without our delights, but not without the misfortunes that drive our search for meaning.”Therefore, I take pleasure in infirmities,”St. Paul wrote in Second Corinthians,”for when I am weak, then I am strong.”In 1988, I went to Moscow to interview artists of the Soviet underground. I expected their work to be dissident and political. But the radicalism in their work actually lay in reinserting humanity into a society that was annihilating humanity itself, as, in some senses, Russian society is now doing again. One of the artists I met said to me,”We were in training to be not artists but angels.”In 1991, I went back to see the artists I ’ d been writing about, and I was with them during the putsch that ended the Soviet Union. And they were among the chief organizers of the resistance to that putsch. And on the third day of the putsch, one of them suggested we walk up to Smolenskaya. And we went there, and we arranged ourselves in front of one of the barricades, and a little while later, a column of tanks rolled up. And the soldier on the front tank said,”We have unconditional orders to destroy this barricade. If you get out of the way, we don ’ t need to hurt you. But if you won ’ t move, we ’ ll have no choice but to run you down.”The artist I was with said,”Give us just a minute. Give us just a minute to tell you why we ’ re here.”And the soldier folded his arms, and the artist launched into a Jeffersonian panegyric to democracy such as those of us who live in a Jeffersonian democracy would be hard-pressed to present. And they went on and on, and the soldier watched. And then he sat there for a full minute after they were finished and looked at us, so bedraggled in the rain, and said,”What you have said is true, and we must bow to the will of the people. If you ’ ll clear enough space for us to turn around, we ’ ll go back the way we came.”And that ’ s what they did. Sometimes, forging meaning can give you the vocabulary you need to fight for your ultimate freedom. Russia awakened me to the lemonade notion that oppression breeds the power to oppose it. And I gradually understood that as the cornerstone of identity. It took identity to rescue me from sadness. The **gay** rights movement posits a world in which my aberrances are a victory. Identity politics always works on two fronts : to give pride to people who have a given condition or characteristic, and to cause the outside world to treat such people more gently and more kindly. Those are two totally separate enterprises, but progress in each sphere reverberates in the other. Identity politics can be narcissistic. People extol a difference only because it ’ s theirs. People narrow the world and function in discrete groups without empathy for one another. But properly understood and wisely practiced, identity politics should expand our idea of what it is to be human. Identity itself should be not a smug label or a gold medal, but a revolution. I would have had an easier life if I were straight, but I would not be me. And I now like being myself better than the idea of being someone else, someone who, to be honest, I have neither the option of being nor the ability fully to imagine. But if you banish the dragons, you banish the heroes, and we become attached to the heroic strain in our own lives. I ’ ve sometimes wondered whether I could have ceased to hate that part of myself without **gay** pride ’ s technicolor fiesta, of which this speech is one manifestation. I used to think I would know myself to be mature when I could simply be **gay** without emphasis. But the self-loathing of that period left a void, and celebration needs to fill and overflow it, and even if I repay my private debt of melancholy, there ’ s still an outer world of homophobia that it will take decades to address. Someday, being **gay** will be a simple fact, free of party hats and blame. But not yet. A friend of mine who thought **gay** pride was getting very carried away with itself, once suggested that we organize Gay Humility Week. It ’ s a great idea. But its time has not yet come. And neutrality, which seems to lie halfway between despair and celebration, is actually the endgame. In 29 states in the US, I could legally be fired or denied housing for being **gay**. In Russia, the anti-propaganda law has led to people being beaten in the **streets**. Twenty-seven African countries have passed laws against sodomy. And in Nigeria, **gay** people can legally be stoned to death, and lynchings have become common. In Saudi Arabia recently, two men who had been caught in carnal acts were sentenced to 7, 000 lashes each, and are now permanently **disabled** as a result. So who can forge meaning and build identity? **Gay** rights are not primarily marriage rights, and for the millions who live in unaccepting places with no resources, dignity remains elusive. I am lucky to have forged meaning and built identity, but that ' s still a rare privilege. And **gay** people deserve more, collectively, than the crumbs of justice. And yet, every step forward is so sweet. In 2007, six years after we met, my partner and I decided to get married. Meeting John had been the discovery of great happiness and also the elimination of great unhappiness. And sometimes, I was so occupied with the disappearance of all that pain, that I forgot about the joy, which was at first the less remarkable part of it to me. Marrying was a way to **declare** our love as more a presence than an absence. Marriage soon led us to children, and that meant new meanings and new identities — ours and theirs. I want my children to be happy, and I love them most achingly when they are sad. As a **gay** father, I can teach them to own what is wrong in their lives, but I believe that if I succeed in sheltering them from adversity, I will have failed as a parent. A Buddhist scholar I know once explained to me that Westerners mistakenly think that nirvana is what arrives when all your woe is behind you, and you have only bliss to look forward to. But he said that would not be nirvana, because your bliss in the present would always be shadowed by the joy from the past. Nirvana, he said, is what you arrive at when you have only bliss to look forward to and find in what looked like sorrows the seedlings of your joy. And I sometimes wonder whether I could have found such fulfillment in marriage and children if they ' d come more readily, if I ' d been straight in my youth or were young now, in either of which cases this might be easier. Perhaps I could. Perhaps all the complex imagining I ' ve done could have been applied to other topics. But if seeking meaning matters more than finding meaning, the question is not whether I ' d be happier for having been bullied, but whether assigning meaning to those experiences has made me a better father. I tend to find the ecstasy hidden in ordinary joys, because I did not expect those joys to be ordinary to me. I know many heterosexuals who have equally happy marriages and families, but **gay** marriage is so breathtakingly fresh, and **gay** families so exhilaratingly new, and I found meaning in that surprise. In October, it was my 50th birthday, and my family organized a party for me. And in the middle of it, my son said to my husband that he wanted to make a speech. And John said,"George, you can ' t make a speech. You ' re four.""Only Grandpa and Uncle David and I are going to make speeches tonight."But George insisted and insisted, and finally, John took him up to the microphone, and George said very loudly,"Ladies and gentlemen! May I have your attention, please?"And everyone turned around, startled. And George said,"I ' m glad it ' s daddy ' s birthday. I ' m glad we all get cake. And Daddy, if you were little, I ' d be your friend."And I thought — Thank you. I thought that I was indebted even to Bobby Finkel, because all those earlier experiences were what had propelled me to this moment, and I was finally unconditionally grateful for a life I ' d once have done anything to change. The **gay** activist Harvey Milk was once asked by a younger **gay** man what he could do to help the movement, and Harvey Milk said,"Go out and tell someone."There ' s always somebody who wants to confiscate our humanity. And there are always stories that restore it. If we live out loud, we can trounce the **hatred**, and expand everyone ’ s lives. Forge meaning. Build identity. Forge meaning. Build identity. And then invite the world to share your joy. Thank you. Thank you. Thank you. Thank you.

Text Sample 375: POS Dim 3 – 2023 – Score 16.00 – 2023/t003470.txt

When you were a child, what did you want to be when you grew up? A footballer? An actress? A doctor? Well, when I was a little girl, whenever I was asked the question, my answer was pretty much always the same :”When I grow up, I want to become a man.”To my logic, that was the only way I could live a life free from the suffocating restrictions that women in my birth country of Iran were forced to endure. I ’ m not quite sure where this line of thinking started from, but a particularly hot summer ’ s afternoon comes to mind. I was nine years old and along with my family, as we did on every summer holiday, we had traveled to the north of Iran to a little seaside village called Chamkhaleh. For as long as I remember, some of my best childhood memories are of me swimming in the Caspian Sea, paddling in its waves and making sandcastles on its gorgeous beaches. All of those lovely memories, though, were about to be rewritten. In the Islamic Republic of Iran, when a girl turns nine years old, she has legally entered womanhood. And that means having to follow a series of restrictive rules written by men. The most symbolic of which is probably the mandatory wearing of the hijab. That year, my mom had to sit me down and sorrowfully explain that I was now too old to get into the water wearing a swimsuit. Now that I was nine years old, it was illegal. And if I wanted to get into the water like all the other women, I had to go in fully covered from head to toe. I had to go in wearing a pair of trousers, a long-sleeved top and a loose covering called the manto to cover it all. And of course, the headscarf. I still remember the lump in my throat as I set foot in the water. I could no longer feel the fresh coolness of the Caspian Sea against my skin. Instead, my now-wet layers felt itchy and weird. I can still remember feeling the watchful eyes of the morality police, the so-called morality police, making sure that me and the other girls were not showing too much hair or too much skin to, God forbid, arouse the men. It was as if my former freedoms were washing away with every tide and flow of the water. Next, I ’ m deep in the water. I was a good swimmer, but my layers were so heavy and restrictive. I tried to gasp for air, but my headscarf kept blocking my face. The tides were pulling me further and further out. I could see my mom and dad on the beach, I tried to call for help, but they couldn ’ t hear me. Perhaps my voice was drowned out by the roaring engines of the bikes that the morality police were patrolling the beaches. The tides continued, pulling me further and further out and hopelessness started creeping in. Next, I wake up on the beach, cold, wet, shivering. It turns out my father had just spotted my struggles in time and had rushed in to get me. That was the last time I ever set foot in my beloved Caspian Sea. Growing up in the Islamic Republic of Iran, every day it felt as though I was going through the same experience but in different ways. And looking at the women in my life, they seemed to be going through the same thing too, in different ways. And the older I grew, the further out the tides of oppression, injustice, inequality felt like they were pulling me in. Wherever I turned, **shame**, fear and especially hopelessness were ever-present. Women in Iran are second-class citizens. They have almost no **access** when it comes to some of the most basic rights like divorce, custody of children and even travel. They have no legal protection against **sexual harassment** and **domestic violence**. They are not allowed to **sing** in public or have sex outside of marriage, and they ’ re only allowed to show their face and hand in public. In the 20 years or so since I almost drowned, nothing has changed. Iranian women have been adrift in a vast sea of oppression. Their only way out, for those who could, is to leave themselves at the mercy of people smugglers, set out on dangerous journeys in hope of finding **safety** in a land far away. That ’ s what happened to me. Age 12, along with my mom and my baby sister, we escaped Iran. And after years of living in refugee camps across Europe, sleeping rough and being at the mercy of some very, very dodgy people smugglers, we finally found safe haven here, in the UK. For my friends and family who stayed behind, there was little hope for change. And so when in September last year, 22-year-old Mahsa Jina Amini stepped out of the metro onto the bustling **streets** of Tehran, there was little reason to expect a change to the status quo. The same way that it ’ s happened to my own mother and many other women I know, Mahsa was arrested by the so-called morality police because, in their opinion, she was showing more hair than they **deemed** acceptable. Three days later, Mahsa died in police custody. This time, though, something changed. Thanks to the power of social media and the brave journalists working hard on the ground, the news of Mahsa ’ s killing wasn ’ t drowned out in the regime ’ s **relentless** propaganda. In the same way that the unjust killing of George Floyd ignited the Black Lives Matter movement, the unjust killing of Mahsa Amini ignited the biggest anti-regime protests across Iran since the formation of the Islamic Republic in 1979. For the first time in decades, women in Iran are finally daring to defy the strict morality laws. In a fight for **equality** and dignity, they are taking off their headscarves, burning them in bonfires, and some are cutting off their hair. They ’ re chanting”Zan. Zendegi. Azadi.””Woman, Life, Freedom.”Furious crowds are calling for democracy, for freedom of speech, for freedom of expression, whilst many others are calling for the fall of the supreme leader, Ali Khamenei. Khamenei is Iran ’ s spiritual leader and has the final say over all government matters. For the first time in a very, very long time, there seems to be hope in the air. Hope that perhaps change in Iran is possible. And that ’ s exactly why the crackdown has been so brutal and deadly. These are the faces of the protesters who fought for freedom but will not live to see it. Since the protest began nearly five months ago, around 500 people have been killed. Nearly 80 of those children. Around 18, 000 protesters have been arrested. And there are reports of psychological, physical and **sexual** torture in prisons. There has been a series of hasty trials, forced **confessions** and public executions of the protesters. The **authoritarian** regime of Iran is clearly desperate to kill any and all hope because they know that for as long as we remain hopeless, we ’ ll resign to their oppression and injustice. But as history has shown over and over again, the light of hope emerges from the depth of **darkness**. Hope is angry. It ’ s defiant. Hope is dangerous. Hope is standing its ground on the **streets** of Iran, chanting”Woman, Life, Freedom.”Hope is in political prisons, biding its time, sustaining those in shackles. Hope has now spread to the four corners of my birth country. As a journalist, I can ’ t travel to Iran. But as my sisters in Tehran, Mashhad, Shiraz, Baluchistan, Kurdistan, shomal, jonoob, shargh, gharb, all over Iran have done, I, too, I ’ m going to cut my hair to show my **solidarity** with the movement. Thank you. Thank you. Audience : Freedom for women! SZ : This hair, for too long, has been intertwined with religion and politics. But I say with my sisters in Iran : no more. And I say with them : Women, to reclaim our rights. Life, for the lives unjustly taken. Freedom, for the liberation of my people. I stand here with my hair in my hand, in hope of a future where no little girl has to wish she were a man because she ’ s terrified that she might drown every day. Yes, the crackdown is brutal. And yes, there might not be a solution inside yet. But the only thing that can help create and sustain one is hope. Sorry. The only thing that can help create and sustain one is hope. And every single one of us can take a step and play our part. I ’ m standing here speaking to you now. Many non-Iranians and Iranians outside of Iran have been signing petitions, going to protest, putting pressure on their governments so that in turn, they put pressure on the Iranian regime and government. For the past few months, I ’ ve been speaking to protesters inside of Iran, and they tell me how much it matters to them that we on the outside stay informed and engaged. So let ’ s try our best to make certain that those of us on the shore let those at sea know that we see them, that we hear their cries. Thank you for taking a step towards a future where freedom is not just a dream, but a reality. Not just for women, but for everyone. And not just in Iran, but everywhere. Thank you.

Text Sample 376: POS Dim 3 – 2023 – Score 14.00 – 2023/t003390.txt

Helen Walters : Hello, everybody. Two days ago, on October 7, the Palestinian Sunni-Islamic fundamentalist organization Hamas attacked Israel, overrunning two military bases, occupying territory, killing hundreds of Israeli citizens and taking dozens more as hostages. It was the most significant breach of Israel ’ s borders since the Yom Kippur War of 1973. The attacks were clearly long- and well-planned, and they sent shock waves of fear and panic through the region and the world. Obviously, it ’ s two days later. It is way too soon to understand all of the ramifications of these attacks. But we can try to understand how we got here and the implications of this awful moment. So we asked our community to share their questions and to answer them, I am joined by Ian Bremmer, president and founder of political risk research and consulting firm Eurasia Group. Hi Ian. Ian Bremmer : Helen, great to be with you. HW : Alright, so let ’ s get right to it. We ’ ve had a number of our community who really want you to explain the very simple question of how we got here. So can you share the historical context for this moment? And if you like, give us a bit of a Gaza 101. IB : Well, I mean, Gaza, we ’ ve got a population, a Palestinian population of just over two million, 2. 2 million, exceedingly poor. And, you know, without sovereignty, without statehood, and a part of the Palestinian occupied territories, also the West Bank, more people, 3. 5 million. The West Bank run not very well by the Palestinian Authority, which recognizes Israel ’ s right to exist. Gaza, run really badly, with very little resources, run by Hamas, which does not recognize Israel ’ s right to exist. Now we ’ ve been talking about a two-state solution for a very long time. For the idea that the only way you end up with stability between the Israelis and the Palestinians is if the Palestinians have some ability to govern themselves, have some control over their economic trajectory, over their foreign policy, over their borders. That is not where we stand right now. And indeed, the idea of a two-state solution has kind of lost the collective interest, imagination, traction, for two reasons. First, because the Middle East has moved on. A bunch of countries around the region have found that they are interested in developing direct relations, some formal, some informal, with Israel, and that they ’ re willing to do that irrespective of resolving the Palestinian question, the Palestinian problem. And we ’ ve seen that with the Abraham Accords under the Trump administration, where the UAE, the United Arab Emirates, Bahrain and Morocco all directly established diplomatic relations with Israel. If you go to Dubai or Abu Dhabi today, you will see Israeli tourists like you wouldn ’ t imagine. And they ’ re having a great time and they ’ re spending money and they ’ re taking in the sites and they ’ re very welcomed by the Emirates. Unimaginable that was going to happen 10 or 20 years ago. In fact, Saudi Arabia was very close, not within weeks, it wasn ’ t imminent, but certainly within months of signing a deal with Israel that would allow for them to open diplomatic relations. And there ’ s already been a number of high-level diplomatic relations informally between Mohammed bin Salman and Prime Minister Netanyahu. So, in other words, across the region, you had Israel, frankly, in the strongest geopolitical position that they ’ ve been in decades. They ’ ve been surrounded by enemies. Well, now they ’ re increasingly surrounded by countries they can do business with. In fact, just a couple of weeks ago, there was an announcement of a deal where the United Arab Emirates was investing massively into solar power for Jordan, which would then be given to Israel in return for desalinized water processed by Israel. Even five years ago, inconceivable a deal like that could happen. So the Israelis, technologically very sophisticated, an advanced industrial economy, are only standing to make more money by doing business with all of these countries. What ’ s been happening with the Palestinians? Nothing. The answer is nothing. They ’ re not benefiting economically. And all of these deals for Israel have happened without any consequences, any contingencies for the Palestinians. And indeed in Israel, you know, there have been a lot of headlines. Israel ’ s made a lot of news this year, but not because of the Palestinians. Israel has made news because of their own **domestic** constitutional crisis, an effort by the Prime Minister, Netanyahu, and his right-wing coalition to engage in judicial reform, an Israeli judiciary which is very independent, which has, in the context of democracies, a very surprising amount of authority over making but also interpretation of laws in Israel. What can and what cannot be considered a reasonable law to be executed. And for a country that doesn ’ t have a constitution, not surprising perhaps, the judiciary is so powerful. And Netanyahu facing corruption charges and with a very weak right-wing coalition relying on far right, extremist right party as part of that coalition, was pushing for these reforms. Now, why am I talking about that? Because for the last six months, there have been unprecedented demonstrations across Israel, peaceful demonstrations, but bringing out the entire country. Because they were concerned about a constitutional crisis. Kind of an irony for a country without a constitution. If Netanyahu persisted, went ahead with these reforms. No one was talking about the Palestinians. And indeed, large numbers of troops that had been in the south were moved to the West Bank as the Netanyahu government was expanding the settlements in that territory and responding to Palestinian reprisals against those settlements. So they weren ’ t focused on the issue. They took their eyes off the ball. Israel had other priorities, and the Palestinians were in a position not only to lose their friends around the region but also increasingly an afterthought for the Israeli government and the Israeli people. That is the backdrop for where we are today. HW : So let ’ s dig in a little bit into the idea that you bring up of the kind of the troubles that have been roiling Netanyahu and the Israeli government themselves. So I think one of the things that has been brought up is the massive failure of intelligence and defense systems in Israel that allowed this attack to happen. What happens next? Do you see Israel uniting around Netanyahu? Do you see this fracturing even worse with anger at what happened in the lapses in defense and intelligence insights? Or what happens next? IB : Well, the first point I should make is just for everyone to understand what has just happened to the Israeli consciousness. It is unimaginable that, you know, certainly someone in a developed country could have any understanding of what the Jewish people in Israel are presently going through. This feeling that, you know, after the Holocaust and, you know, the land being provided to them to have a safe haven to create an independent Israeli state. And the need to defend their borders, the historic fights they ’ ve had with their neighbors, the war in 1973, when a number of Arab nations decided to fight against them. And, you know, the continual sense of besiegement with missiles from Hezbollah, for example, terrorist operations. This is not like a bolt from the blue when the United States experienced 9 / 11. Israel ’ s 9 / 11 is both massively greater in the impact on Israel but also comes for a country that was supposed to be prepared for this. I mean, Israel represents the gold standard on border security around the world. Not like the United States, where you ’ ve got, you know, all sorts of people running across and”build the wall”becomes a clarion call precisely because nobody understands how to defend the border, no. And also, intelligence collection, surveillance, digital surveillance, human intelligence collection on the ground, especially in the occupied territories. This is what they do. And the fact is that right now, today, Netanyahu ’ s legacy will not be anything that he has done to date. It will be this failure and how he responds to it. Period, end of story. Nothing else is close. So what that means for Israel is that all of the issues that have roiled this country over the past year, all of the political polarization, and it ’ s not a two-party country, it ’ s a many, many-party country. You know, the joke is you get three people together in Israel and, you know, you form a new political party. And if you go to the coffeehouses and the rest, everybody ’ s talking politics. Everyone reads the newspapers. This is a highly politically literate and divided population. But as of right now, priority one, two and three for the entire Israeli people is to respond to these attacks, these terrorist attacks. And how are they going to respond to them? Well, number one, they ’ ve got to find a way to get their people back. There are 100-plus, and we don ’ t know the exact numbers right now, hostages that are being held in Gaza, most of whom are civilians. And they will do everything to get them back. And they may well have direct American support in trying to accomplish that. And then it will be to go into Gaza, to remove the leadership of Gaza, to disarm the militias in the territory of Gaza, and to do everything they can to try to ensure that this cannot happen again. And making that happen is a very, very tall order. It might be a taller order than the Israelis can accomplish, and certainly the knock-on consequences will be grave even for just Gaza. And that is before we talk about any potential expansion of the war. But for now, the Israeli people will stand together. And there ’ s already talk of a government of national emergency that would bring together Netanyahu with the leader of the Israeli opposition for the purposes of fighting this war so that everyone in Israel is together collectively, ensuring the national security of the people of Israel. And I think that for the course of the coming months, and let us remember, this is not just, you know, an attack against Israel and now they respond. It is very likely there are Hamas operatives on the ground inside Israel right now that the Israeli government has to find and neutralize. It is also true that, you know, you ’ re still actively expecting that there are going to be additional attacks, whether those are missile attacks or whether those are direct incursions, nobody knows, but given the level of planning that was required by Hamas to make these strikes over the weekend, which nobody in Israel thought was possible, no one expected it, right now, that level of concern would be higher than anything else on the political agenda. And again, that will not move for the foreseeable future. HW : So I want to talk more about all of that and expand this to obviously the broader geopolitical implications of this. But I want to just play out the 9 / 11 reference a little bit if you can, because obviously 9 / 11 happened some time ago with the attack on the United States. But we know with retrospect, with hindsight, that some of the decisions that were made after that were misguided, they were misjudged, they actually led to terrible **harm**. How and who is going to make sure that these types of decisions are not made? And how can Israel avoid making decisions that will be bad? IB : As someone who was in New York on 9 / 11 and saw the second tower go down and saw how the city rallied together, how the country rallied together, President Bush, over 90 percent approval in the country a couple of months after 9 / 11, how the world came together to support the United States, the coalition of the willing, well beyond NATO. I mean, poor countries that had no business caring about what the United States was up to providing troops on the ground and support for the Americans. Russia, Putin ' s Russia, calling up Bush and offering, you know, the former Soviet republics in Central Asia as bases to support for logistical operations for the war in Afghanistan. I mean, the level of support for the United States after 9 / 11 was singular. And there ' s no question that the outpouring of concern, I mean, when I saw in Berlin, shining on the Brandenburg Gate, the Israeli flag with the star of David in Germany, in Berlin, in Germany, and given the history and given what that means and given the Alternative für Deutschland doing well in East Germ – all of that, I mean, this is a singular moment in the relationship between Germany and Israel. The European Union suspending aid to the Palestinians, the support for Israel is extraordinary. Narendra Modi In India. It is not universal for every country, but it is absolutely wide-ranging. And I got a readout, I spoke to several of the folks involved in the Emergency Security Council meeting at the United Nations this weekend. The condemnation of these attacks, everyone but Russia. And again, so in that regard, this is very, very similar to 9 / 11. Now, the broader question that you ' re asking, Helen, which I ' m also very sensitive to, is in 2023, looking back on 9 / 11, the Americans made some horrible, horrible, long-lasting mistakes. And some of those mistakes were in the United States. I mean, if I think about how much money was spent and wasted in the Department of Homeland Security, on personal security and **safety** in the airlines, how much money was wasted, how much economic inefficiency as a consequence of overstating the terrorist threat in the US, everything else secondary to that. But also, the rights that were stripped back for, in some cases, all Americans in terms of surveillance and the Patriot Act. But also targeting Muslim Americans across the country and so much mistreatment of American citizens as a consequence of that. But that ' s nothing compared to the mistakes made internationally. A war of choice in Iraq, responding to 9 / 11 with trillions of dollars wasted and lives, millions of lives destroyed. Afghanistan, 20 years on, a failed war with the Taliban returning to power and a failed state. Yes, bin Laden was killed, and I think people around the world cheered that, not just in the United States, and Al Qaeda was destroyed at the highest levels and in many cases uprooted completely. But no one can look back on the 20-plus years since 9 / 11 and say that the American response with the war on terror was successful. You can ' t do that. And Israel is not the United States. The Americans have extraordinary strength and resilience in its national security capabilities and the size of its economy, also in where it ' s located geopolitically. Israel certainly has the military strength in the region, but the the country is small. The territory is small. And certainly it is not in a geopolitical space that is comfortable. And so I think the danger here is that as the Israelis respond against Hamas, as they should and as they must, and as they work to destroy the leadership of that terrorist organization and disarm the militants that are involved in the attacks against Israel and pose an ongoing threat. But that is certainly not the only knock-on consequences of Israel ' s decision making. And the potential for this to become a broader war that would envelop the Middle East in conflagration and that ultimately could even end Israel is real in a way that the war on terror could not have threatened the United States existentially. And I, as a consequence, I certainly believe that a unity government will make it less likely that the Israelis overreact in that way. I certainly believe that the United States, in providing very strong and committed support, but also notes of caution in what can be done and what should not be done, will hopefully restrain the worst impulses. And, you know, in the early moments, again, we all understand why Israel would feel the need to react in the harshest possible way. But I certainly worry when I see the Israeli defense minister refer to the attackers as inhuman animals and announce a siege on the entirety of Gaza, which means no food, no electricity, no water. And this is a territory that already has 50 percent poverty, already has fewer than half of its population with **access** to clean water. I worry about what that is going to mean for the Palestinian people as well as for Israel long-term. You know, I think it was Golda Meir who says,"I won ' t hate you for killing our children. I will hate you for making me kill your children."Ultimately, the Israeli population is most threatened by what the terrorists of Hamas unleash from Israel. HW : So do you think that was part of the incentive for Hamas in doing this? Because surely they knew that the response would be swift. And surely they knew that the world would rally around such atrocities. So what do you think was their motivation, and do you think that they underestimated what might happen? IB : Oh, I don ' t think they underestimated what might happen. But it ' s a compelling question. It ' s really hard to put yourself in the mindset of someone like a Hamas leader. But, you know, I had to do that just a few months ago when Yevgeny Prigozhin was marching with his Wagner forces on Moscow. And people were asking me,"What is going through this guy ' s mind?"Because it ' s clear he ' s going to get killed, right? I mean, you turn against the Kremlin and Putin, you ' re not walking away from that. And when he cut the"deal"and everyone said,"Oh you know, he cut a deal"– He ' s dead man walking. Like, literally, that was the reality. And as soon as the Hamas leaders decided that they were going to commit these atrocities against Israeli civilians, they ' re dead. There is no future for these people. So I think there are two different things going on. The first, and this is analogous to Prigozhin, is that Hamas felt themselves in an increasingly untenable environment, that they were losing their support in the region. And even the Saudis were about to normalize their relationship with Israel. They had no influence in ability to get anything done, no leverage with the Israeli government, which was only becoming harder and harder lined against them. And in that regard, they were increasingly in a corner. Their options were increasingly all bad. And, you know, we know that people that find themselves only with horrible options frequently do irrational things. And I would not underestimate that in driving the decision of Hamas to take that action. It ' s kind of like why would 77 percent of a Gaza population in the last elections they had, which was some time ago, why would they vote for an organization like Hamas? Well, I mean, they wouldn ' t if they had economic opportunities. They wouldn ' t, if they had education, they wouldn ' t if they could come and go from Gaza as they please. But the worse the situation gets, the more they are willing to vote for an organization that is prepared to burn it all down. And by the way, there ' s a lesson in that, even for those of us in very wealthy, very stable countries. So I think that ' s one set of motivations, but another set of motivations certainly is an ideological effort of Hamas to insert themselves as more dominant in the conversation, to radicalize the Israeli population, to undermine the Palestinian Authority in the West Bank. Because if this fight, you know, gets the Israelis to kill huge numbers of Palestinian civilians, and by the way, Hamas will facilitate that, right? I mean, Hamas is absolutely going to be engaged in operations, you know, in residential buildings. They do that intentionally. They ' re not going to make it easy for Israel to take them out. They want to make it bloody. They want to paint the Israelis as just as bad as Hamas, if not worse. They will take human shields. The IDF, the Israeli Defense Forces, usually gives warnings about when they ' re about to attack a building. They ask the civilians to leave. Well, Hamas tells those civilians that that ' s disinformation. They do everything they can to make the Israelis seem complicit with the kind of indiscriminate attacks against civilians that Hamas engages in themselves. They want to bring the Israelis to their level. And they also want to radicalize the Palestinians in response, not just in Gaza but also in the West Bank. And they want to radicalize the Arab **street**. They want people across the region to be, you know, in uproar against Israel and in **solidarity** with the Hamas cause and in **solidarity** with the destruction of Israel. They want Arab leaders to be saying what the Iranian supreme leader was posting on social media this weekend, calling essentially for a genocide against the Zionist regime. That is ideologically what Hamas is trying to accomplish. And again, Israel must do everything in its power not to allow Hamas to drag them there. HW : It ' s interesting, in your talk in Vancouver this year at TED2023, you were talking about the rise of different orders, and I do just get the sense that everything is connected. You have Russia, you have Ukraine, you have Iran. There are these ideological battles that are now becoming real-world wars. And so I wonder if you can, especially the mention of Iran, I don ' t think it ' s confirmed yet, the intervention of Iran in this, but certainly the"Journal"was reporting that Iran had been involved, deeply involved in setting up these attacks. What does this mean? What does this mean for the world at large? And then I also have a follow up question, which is, what do you think the US should do? IB : So let ' s talk in terms of the world at large and starting with Iran, certainly that"Wall Street Journal"piece over the weekend drove an **enormous** amount of news. It was saying, hey, the Iranians basically planned this. I will tell you, that was a very lightly sourced piece, relying on Hamas. And I would not have gone to print with that if I had been"The Journal."HW : Yeah, the US has not confirmed that at all. IB : In fact, the US has actually said that there is not hard evidence at this point that fingers Iran as having directly orchestrated or ordered these attacks. Now, let ' s be very clear. The Iranians have publicly expressed strongest possible support for Hamas. The Iranians have historically funded and provided military support directly for Hamas. So they clearly are not **innocents** in this. And I would be surprised to learn that the Iranians had no idea that this was going to happen. I suspect that they were aware. But awareness and orchestration are two very different things. Now, since the attacks occurred, Hezbollah, which of course, is also very much aligned with Iran and gets a lot of direct military support and training from the Iranians, they have, I say only, but in this context, is only, they ' ve only engaged in some missile strikes, some rocket strikes against an Israeli military – not base, but military outpost. A soldier ' s outpost. And the Israelis, in response, immediately engaged in strikes back against Hezbollah. That ' s it. If the Iranians were behind this and wanted to be seen as behind this, Hezbollah would be involved in these attacks. They are far more capable than Hamas. The Iranians have claimed that they have had no role, that this was an autonomous Hamas operation. And indeed, Iran has been doing better geopolitically of late. The Chinese facilitated a breakthrough in Iranian relations with the Saudis. The Iranians have engaged with the United States, and six billion dollars of Iranian assets are set to be unfrozen, have not been unfrozen yet, but are set to be transferred to Iran. Five American civilians that were held unjustly as hostages in Iranian jails have been released and sent to the United States. The Iranians have reduced the top level of uranium enrichment and some of their stockpiles, allowing inspectors in. Now, this is not a return to the Iranian nuclear deal, the JCPOA, but certainly on the basis of all of that and even some high-level discussion that the Iranians might be willing to engage directly with the United States diplomatically through the good offices of Oman, none of that seems aligned at all with the Iranians pulling the trigger on an attack against Israel that would almost certainly lead to massive retaliation once the Israelis found that out. So I am sitting here saying I would be surprised, not with a high level of confidence, and, you know, the Iranian regime has a very old supreme leader who is also dealing with internal instability and a transition that is coming. So never say never. But I would be quite surprised if we found out that the Iranians directly ordered this. Now, it is useful that the United States has both sent a fleet off of the Israeli coast to show stalwart support and will be providing a level of at least military coordination and operational intelligence, may well do much more than that. We can get to that when we talk about the United States, but is also very publicly saying"We do not yet have any evidence that the Iranians are involved."In other words, the message from the United States is very clear : do not expand this war into Iran, because the consequences of that are 150-dollar crude at a minimum. The consequences of that is the world goes back into global recession. The consequences of that are conflagration in the region. And I think, I do believe that the Israeli government is quite aligned with the United States in not wanting to go there. HW : I keep coming back to the human cost of this because the reality is that people are suffering, people are being killed, and many more people are likely to be killed. If, indeed Hamas has kind of hijacked this story with extremist action, I wonder what you see from the Palestinian side of kind of a more moderate type of push towards trying to get understanding, trying to get peace in this nation or trying to get peace in this area. IB : I mean, that ' s the most tragic piece of this, is the ability of Hamas to successfully hijack big pieces of the political spectrum for the Palestinians. I mean, there are so many people in the West right now that view the Palestinians as equivalent to Hamas, and nothing could be further from the truth. But that reality, that perception is going to make life so much worse for the people that have suffered the most. They are the powerless. The Palestinians are the stateless. They lack resources. They lack a proper military. They lack the capacity to defend their own territory and to defend themselves. And we ' ve already seen, even over this most tragic weekend for Israel, that the number of deaths and casualties for the Palestinians are almost as much as they were for the Israelis. And when you go back over the past 20 years, who ' ve had the most deaths, who ' ve had the most casualties, consistently, it ' s been the Palestinians. Who ' s going to suffer the most going forward? Consistently, it will be the Palestinians. Who suffered the most from the US war on terror? It was, of course, the Iraqis and all of the tribes in Afghanistan. This should not surprise anyone, but it is the unfortunate reality that of course, Hamas leadership will be destroyed. But the biggest damage that they will have done would have been to their own people. To the Palestinian people, who now will face almost unfathomable deprivation. And there ' s very little that the rest of the world is going to do about it. HW : Do you see a movement within Palestine to step up if Hamas is done? IB : I certainly believe that the Palestinian Authority will try to see this as an opportunity to push for more international engagement from the region to take seriously a cessation of illegal Israeli settlements in the West Bank, a rolling back of the territory that is presently occupied and a revival of peace talks that would bring about a two-state solution where the Palestinians, less land than perhaps they would have gotten in the days of, you know, Arafat and Rabin, but nonetheless, something that feels sustainable, a country that one might be able to raise children with a sense of hope. There is no one in the occupied territories of Palestine that could say that for themselves for their children today. So I think that is the hope. But, you know, clearly right now is not the time for that. Not because we don ' t want it, but because events will overtake it immediately and have already overtaken. Now, the hope is that the **violence** that will spread in the West Bank can be contained. That we do not see a war in Gaza become a war in the West Bank, that we do not see an occupation of Gaza become an occupation of the West Bank. That is, I think, the priority now. You have to know sometimes when you actually have a trajectory for peace, and when you have to do your best to avoid war expanding. We do not have a trajectory for peace right now. We are at war. There were plenty of opportunities over the past years to take off ramps, to engage more seriously. They were ignored. And it is precisely that reality that has brought us to a point where we now are on defense, where we are hoping that this does not get much, much worse. That is the story of the Middle East today. HW : Do you think Israel will annex Gaza? IB : I think the Israelis don ' t know. Again, don ' t underestimate the shock, the emotional shock that all of the Israelis are facing today. They are not making long-term strategic decisions right now. They are making immediate, short-term decisions. What can we do to make sure the country is still defended, our borders are still secure, that terrorists are not, you know, right now running around in our midst, planning further atrocities? That is priority number one. And very close to it is getting those hostages back and safe and I ' d say unharmed. They ' ve already experienced a lot of **harm**. That ' s beyond the realm of possibility right now. There will be, in the coming weeks, and there ' s also some shock and awe, you know, Netanyahu immediately posting, some buildings getting demolished in Gaza and saying we are, you know, at war. That is, I mean, they ' ve engaged in these sorts of airstrikes before, and we already know of families that have gotten killed, entire families in some cases of nine, of 13 people, lots of children, this sort of thing. We will see that. But in terms of what ' s the nature of the long-term objectives, the occupation, the Israelis are not close to making that decision. And I also think that other countries that Israel trusts and Israel needs will have some ability to have influence over Israel in making that decision. I ' m not just talking about the US now. I ' m also talking about countries in the region that Israel would like to maintain relations with. So there needs to be very active multilateral diplomacy behind the scenes, quietly, high-level, with the Israeli government in the coming days and weeks. HW : I want to talk a little bit about the media coverage of the attacks and of what is happening right now. A lot of the people who had written in with questions for you are really confused by the rhetoric that they ' re seeing. They ' re not sure what to trust, who to trust, what to believe. And I want to get your sense, you ' re a very online person. And so what is your take on this, and how can we think about how to understand what ' s happening, especially in such a kind of, real-time situation? IB : Well, first of all, there ' s always the fog of war. There ' s always disinformation from both sides actively trying to promote a narrative that is more effective for them, that they ' re doing better than they otherwise are, and that the other side is engaged in greater atrocities than they actually are. You see lots of that, lots of immediately fake videos putting out of buildings that are being destroyed, people that are being killed, you know, sort of, people that are providing, you know, sort of, support for, sort of,"kill all the Jews"and"kill all the Palestinians"that actually came from previous conflicts, not from the present one. There ' s plenty of that. This time around, there ' s also so much more **hatred**. There ' s, on social media especially, there ' s so much more willingness to promote, algorithmically, opinions that you would never hear in your family, that you would never hear in your community, in your school, but it ' s being bombarded. And this is very different from the so-called mainstream media, whether it ' s the BBC or the Deutsche Welle or it ' s Fox News or CNN. No, no. Social media has become a far more greatly polarized and hate-filled space. I ’ ve received at least 30 death threats over the last 48 hours from complete randos. A few people that I actually could track if I really needed to, most anonymous accounts. But clearly people that are writing me directly, some of whom are really, really pro-Israel, some of whom are really, really pro-Palestine and some of whom are probably just trolling for the lolz, as they like to say. It is increasingly very hard to navigate this space without becoming incensed and deranged. Having said that, as much as I find **Twitter** / X a space antithetical to civil society, I also know, as someone who does analysis, that some of the best real-time information from sources on the ground is being passed through on X and is not found in other places. It will not come through in mainstream media. So for the average person, you need to spend a lot more time filtering and figuring out where to go and who to follow. But it still is the one place that you can go. And you know, it really does, the whole thing profoundly worries me because when you ’ re in an environment that you can no longer know what is truth, what is real information, it is really hard to maintain a society that is human. When you have people that are saying that all Israelis are X and all Palestinians are Y, and that is where much of social media, I mean, a strong majority of social media is there right now, you cannot have dialogue, you cannot have solutions, and you can very easily tilt into war, into radicalism, into fascism. This is something we all need to be guarded against. And I truly believe that the social media companies need to be regulated on this. They are acting as if they have no responsibility for what ’ s on their sites. That, you know, it ’ s just like the phone company. That if you and I, Helen, are having a conversation about blowing something up, well, we ’ re responsible for that. But the phone company isn ’ t responsible, and I accept that. But, Helen, if you and I are having a conversation about blowing something up and then the phone company takes that conversation, identifies everyone else that might be interested in blowing something up or has considered it, and takes that conversation and promotes it to them, then you, the phone company, are responsible, you are liable, you should be taken down. And we are in a war right now, and the social media companies are actively fanning the flames. They are spraying fuel on the flames, and they ’ re doing it globally. Globally, so much so that countries like China that are **authoritarian** and control their media space actually have sort of an intrinsic political stability advantage in”information warfare”over open societies that should be the most resilient. That ’ s crazy, and we can ’ t keep going down that path. HW : The ironies are writ large. OK, so we are coming on our time, but I wonder if you can leave us with a sense of what should we be watching for next? What should we be looking for? IB : First, we need to look for Lebanon. This is the issue of Hezbollah, which so far has been the dog that has not barked. Is that going to continue? It is the place that you are most likely to see tipping point escalation if it were to occur. And you have Hezbollah operatives of many different stripes. They are loosely organized, they ’ re well-trained. But that doesn ’ t mean that they ’ re all following, you know, marching orders from one direct leader. The potential that this could – you could see escalation with some Israeli farmers getting killed, and then the Israelis respond. And before you know it, you ’ re in a much bigger firefight. Lebanon ’ s involved, Hezbollah is involved, and then it knocks on to Iran. That is sort of the gateway drug in the Middle East even if nobody wants that fight. That ’ s one thing to watch. Second thing to watch, of course, imminently is what happens with these hostages. Do the Israelis get them back? Historically, that has always been the top priority, and it is today. But Hamas has control over that. And, you know, if the Israelis are not prepared to negotiate with Hamas to release militants presently in Israel prisons, and it ’ s very hard for me to imagine in today ’ s environment they would be willing to do that, well, how exactly do they get them back and how many of them can they actually free? Again, you know, the Israelis and the Americans have far, far better tradecraft on the ability to get these hostages out than Hamas have to take them. But Hamas has **exceeded** expectations over the past 48 hours, and I would worry very much about that. That would be the second thing I ’ d watch most closely right now. And then finally, the nature of the Israeli government itself. Do we have success in putting together a unified national emergency government, in which case we will have more stability in governance and decision making that comes from Israel and also greater willingness to consider longer-term engagement with those Palestinians, particularly in the West Bank, to start, at least, that might be looking for a more constructive path now that they are back on the agenda. I don ’ t have a high amount of optimism that that ’ s going to happen. But you asked me for something hopeful, that would be something hopeful. HW : Ian Bremmer, we are so grateful for your time and for your insight. Thank you so much for joining us, stay well. IB : My pleasure Helen.

Text Sample 377: POS Dim 3 – 2023 – Score 11.00 – 2023/t003423.txt

So if you whack your thumb with a hammer, you think pain is in your thumb. Physicians have a more sophisticated understanding. We know that it ’ s an alarm that goes, on nerves, to your spine, where it is translated to your brain, and pain actually happens... somewhere. It ’ s a little vague. We actually only get two days of pain education throughout all of medical school, so... In fact, the only pain lecture I remember from the ’ 90s was in a dark room like this, after being awake for 30 hours and hungry, and finding out our noon lecture was sponsored by OxyContin. We got pens, we got great lasagna, and they had very cool slides that showed pain stopped by opioids. And we learned that home opioids aren ’ t addictive, and if you stay ahead of pain – you can keep your patients pain-free. And beyond the obviously egregious marketing, I think it was framing”pain-free”as the goal that has destroyed countless lives. My friend ’ s son Christopher started having severe abdominal pain during this”no-pain”era. Eventually, he was diagnosed with a colon disease and had surgery his senior year. They sent Christopher home with 90 OxyContin, and then 90 more, and then, as the pain started getting faster and faster... Uncontrolled pain is terrifying. So when his ran out and his friends ’ medicine cabinets ran out, Christopher tried heroin. And Christopher Wolf lost his battle with substance use at age 32. So did we misunderstand pain? What if pain isn ’ t an alarm to **silence** but a learning system for survival? TED pose. Pain is every organism ’ s primary learning system for survival. I mean, it ’ s like,”Ouch. Don ’ t touch that.”Or, to paraphrase”The Princess Bride,””Life is pain, Highness.””Pain-free”was marketing, and it made physicians think that one pill could solve pain. It still makes people feel like you can ’ t be happy if you have some pain, and we now know that if you want to move past pain, it takes work. Setting the bar at”pain-free”was too high. Plenty of people could have been more comfortable, but they gave up because pain-free was out of reach. We have really good new information that I ’ m going to share, and so from now on, I want you to think about pain as a Venn diagram, with physiology, fear and control. I ’ m going to tell you how each of these can give you power over pain. Right now, I ’ m translating these, in my research, into a low-back pain device to reduce opioid use. But 20 years ago, I just wanted to have a fast cure for needle pain, for IV **access** and my kids ’ shots. I was driving home one night after a graveyard shift, and my hands were vibrating on the steering wheel, because we needed to get the tires balanced. I was ignoring that to think about pain, and when I got home and reached for the door in my house, my hand was **numb**. Vibration. So I burst in, my Boy Scout husband grabbed some frozen peas, and we had ourselves a genuine eureka moment, where cold and vibration blocked pain. Over the next decade, I found the right frequency to block pain, I got a grant, and I created Buzzy, which is vibration plus ice... in a bee shape. And you put it on your arm when you ' re getting an injection. And to date, 45 million needle procedures had decreased pain, and over 80 randomized controlled trials, independently, all around the world, have been published. But... At about 30 randomized controlled trials in, one of my colleagues came to me and confided that he was in opioid recovery. And he asked whether or not Buzzy could let him get through a total knee replacement drug-free. I ' d never thought about it. It ' s the same pain nerve for knees as for needles, so I said maybe. And he did it. Vibration plus cold replaced OxyContin. So at that point, I went all in, to figure out why. And here is what we know. So the reason that vibration decreases pain is because the physiology of the nerves of light touch, pressure, stretching and motion all races pain to the spine. Now people have tried electricity to trigger the light-touch nerves, but we now know that motion, shown in green, is what ' s most effective at shutting the gate on sharp pain. This is called gate control, and the exact right frequency of vibration triggers the nerves that decrease pain. The physiology of ice is different. So the cold goes up to the brain, where the conductor goes,"Obnoxious, but not dangerous. I will decrease sensations coming from everywhere."And it decreases pain everywhere. If a child was so freaked out from previous experiences that even the swab hurt... physiology wasn ' t as helpful. So we added distraction, like a monkey poster. And what we discovered was combining counting plus making a decision cut pain in half. So, for example,"How many monkeys are actually touching the bed?"activates the decision switchboard. I know what you guys are doing. It ' s five. Here is your pain hack for the day, though. If you do not have monkeys on hand, then find any sentence and count how many of the letters have holes in them. Counting, deciding. So, like, you ' ve got a g-hole, o-hole, a... hole. I guarantee you and your family will use this. The biggest hack, though, is understanding why distraction works. And now, through functional MRI, we can actually see pain happen. And it ' s not one place. Pain is a symphony of connections, from the sensation area to the conductor, to the decision switchboard, and then to fear, memory, meaning, control. So if the decision switchboard is occupied sorting monkeys, it can ' t notify fear and meaning, and you feel less pain. What you feel is mostly what you expect to feel. Stay with me. If you ' re a kid, the same punch hurts more from a bully than from your buddy. And if you ' re an adult and you feel something, the second you think it ' s cancer, it hurts more and more, until you find out it ' s not. And those same kids who had horrible shot experiences can tolerate all kinds of needle pain... to look cool. Because it is a different context. These patterns, called connectomes, are very personal, because experiences lay down more of the same sensation. And we now know that people who have certain areas in the brain connected feel more pain than people with different areas connected. And more importantly, untreated pain or intense pain can lay down heavier connections, so that even when your body is **healed**, you will still feel more pain. It ' s this very plasticity and personalization which makes the physiology, fear, control matrix so useful. Because choosing physiologic options that you can layer, that work for you, decreases pain, like heat, cold, vibration, deep relaxation, acupuncture, capsaicin, exercise, **meditation**... There ' s more. Christopher probably had 10 of these around his house and just didn ' t know it. Having control over your options decreases pain. Deep breathing increases control. Choosing what to focus on increases control. Fear and control are the volume knobs for pain. Fear controls so many of our sensations, this shouldn ’ t be unusual, but we don ’ t practice it for pain. So if you ’ re home alone and you hear a clunk... your hearing becomes hypersensitive. But when you remember your kid ’ s home from college, your fear dials down and your brain overrides it and says,”Don ’ t worry about it.”Saint Augustine called pain the greatest of evil. But if it is a survival system, it cannot be all evil. So instead, think of pain as your nagging, safety-obsessed, exaggerating friend who ’ s sometimes wrong. And it ’ s OK to ignore or override your friend if you know that you ’ re safe. This takes practice. On a flight that was turbulent, I had an entire cup of scalding-hot coffee dumped straight on my ankle. Electric jolt through my scalp. I ripped off my sock ; it was already red. It was going to blister. There was no way I could get my foot into that little sink to get cold water on it. And then I remembered. Physiology hack. I had an unopened cold beer. Medical-grade cold beer went on my ankle, stat. I had a vibrator in my carry-on, because I would. On my ankle. And then – The pain kind. And then my fear hack. I ’ m like,”There ’ s a barf bag that has holy letters, but I ’ m going to put it in the pocket pouch and save it, because then, I have increased control.”And, pain, I was no longer that concerned. Although then, I realized I ’ m that guy, with my bare foot sticking out in the aisle on a plane, with a beer on it. Power over pain isn ’ t always pretty, but it is possible and it is absolutely critical. Because there ’ s one more misconception we have not talked about. I honestly thought that opioids turned off some pain switch. They turn on our reward system. So some people feel amazing, but most people just still feel pain, but don ’ t care. Now, this is a godsend for people with chronic pain diseases. We should not take them away. And in the **trauma** bay, the more morphine in the first 24 hours after a burn or a wound, the less post-traumatic stress, the less chronic pain later. But studies show that recovery after surgery is just as well accomplished with coaching and physiologic options. And if you ’ re one of the people who feel amazing with opioids, it ’ s too risky. A study in 2019 found that one in 15 young adults who got opioids for their wisdom tooth removal had substance-use disorder within a year. Ibuprofen works better. So what do we do? Well, in my dream world, we have health-care systems – paid-for options and coaching – for Christophers everywhere. And we quit giving double-digit prescriptions for opioids for home recovery. In the real world, 80, 000 people died in the US last year from opioid overdoses, and 80 percent of substance-use disorders start with a pill prescribed for pain... Usually taken from your friend ’ s medicine cabinet. People can ’ t afford options. Doctors, 20 years later, still don ’ t know them. But you do. You all now know to throw away the opioids in your medicine cabinet. You now know that there are options you can use to decrease pain, and you know that”pain-free”should be ditched for”more comfortable.”And whether you dump scalding coffee or pain wakes you and exhausts you every day... Options that are in your control... can allow you to reframe pain. Thank you. Whitney Pennington Rodgers : Amy, thank you, it ’ s amazing. So how do you think that pain scales have set us back from this work that you ’ re doing, and how is the NIH treating pain and addiction differently now? Amy Baxter : So in one of the 120 versions of this talk, I talked about how the thing is, in the ’ 90s, if we wanted to”disease-ify”pain, it meant we had to be able to measure it. So that was where the FACES scales come from, and they ’ re actually very useful in the emergency department to tell whether or not a medicine is working. In fact, we were one of the first ones that showed, with sickle cell, that the patient ’ s report, based on those scales, was what was most indicative of whether they needed to be admitted, rather than any biologic marker. But what we ’ re doing now is we ’ re using something called the PROMIS scales, so it ’ s how intense pain is on five-point scales, how much it interferes, so there ’ s pain interference, pain intensity. And the way we ’ re looking at pain is much more on the impact for the person, rather than trying to pretend there ’ s any kind of objective pain measurement. WPR : OK. And you mentioned that you ’ re working on some new applications for Buzzy, specifically for back pain. What are some of the possibilities that we have here for what this could do for us in the future? AB : On my tombstone, there ’ s going to be a vibrating bee. It ’ s actually called DuoTherm, not Buzzy. But what we ’ ve learned is that there are harmonics of interaction between the specific frequencies that **cancel** out the pain. So there ’ s one particular nerve called the Pacinian that has a very specific frequency range, and by causing them to interact, we ’ re starting to explore more about the pain that ’ s coming from the fascia between the skin and between the muscles, but that area is where we ’ re unexplored, and so by interacting with different frequencies and then layering heat or cold, pressure options, giving people the choice of so many different ways to do it, it ’ s really engaging all the different areas of the brain from which pain comes. WPR : Wow, OK. Well, thank you so much, Amy. AB : Welcome. Thank you, all.

Text Sample 378: POS Dim 3 – 2007 – Score 16.00 – 2007/t000432.txt

As someone who has spent his entire career trying to be invisible standing in front of an audience is a cross between an out-of-body experience and a deer caught in the headlights, so please forgive me for violating one of the TED commandments by relying on words on paper, and I only hope I ’ m not struck by lightning bolts before I ’ m done. I ’ d like to begin by talking about some of the ideas that motivated me to become a **documentary** photographer. I was a student in the ’ 60s, a time of social upheaval and questioning, and on a personal level, an awakening sense of idealism. The war in Vietnam was raging ; the Civil Rights Movement was under way ; and pictures had a powerful influence on me. Our political and military leaders were telling us one thing, and photographers were telling us another. I believed the photographers, and so did millions of other Americans. Their images fueled resistance to the war and to racism. They not only recorded history ; they helped change the course of history. Their pictures became part of our collective consciousness and, as consciousness evolved into a shared sense of conscience, change became not only possible, but inevitable. I saw that the free flow of information represented by journalism, specifically visual journalism, can bring into focus both the benefits and the cost of political policies. It can give credit to sound decision-making, adding momentum to success. In the face of poor political judgment or political inaction, it becomes a kind of intervention, assessing the damage and asking us to reassess our behavior. It puts a human face on issues which from afar can appear abstract or ideological or monumental in their global impact. What happens at ground level, far from the halls of power, happens to ordinary citizens one by one. And I understood that **documentary** photography has the ability to interpret events from their point of view. It gives a voice to those who otherwise would not have a voice. And as a reaction, it stimulates public opinion and gives impetus to public debate, thereby preventing the interested parties from totally controlling the agenda, much as they would like to. Coming of age in those days made real the concept that the free flow of information is absolutely vital for a free and dynamic society to function properly. The press is certainly a business, and in order to survive it must be a successful business, but the right balance must be found between marketing considerations and journalistic responsibility. Society ' s problems can ' t be solved until they ' re identified. On a higher plane, the press is a service industry, and the service it provides is awareness. Every story does not have to sell something. There ' s also a time to give. That was a tradition I wanted to follow. Seeing the war created such incredibly high stakes for everyone involved and that visual journalism could actually become a factor in conflict resolution — I wanted to be a photographer in order to be a war photographer. But I was driven by an inherent sense that a picture that revealed the true face of war would almost by definition be an anti-war photograph. I ' d like to take you on a visual journey through some of the events and issues I ' ve been involved in over the past 25 years. In 1981, I went to Northern Ireland. 10 IRA prisoners were in the process of starving themselves to death in protest against conditions in jail. The reaction on the **streets** was **violent** confrontation. I saw that the front lines of contemporary wars are not on isolated battlefields, but right where people live. During the early ' 80s, I spent a lot of time in Central America, which was engulfed by civil wars that straddled the ideological divide of the Cold War. In Guatemala, the central government — controlled by a oligarchy of European decent — was waging a scorched Earth campaign against an **indigenous** rebellion, and I saw an image that reflected the history of Latin America : conquest through a combination of the Bible and the sword. An anti-Sandinista guerrilla was mortally wounded as Commander Zero attacked a town in Southern Nicaragua. A destroyed tank belonging to Somoza ' s national guard was left as a monument in a park in Managua, and was transformed by the energy and spirit of a child. At the same time, a civil war was taking place in El Salvador, and again, the civilian population was caught up in the conflict. I ' ve been covering the Palestinian-Israeli conflict since 1981. This is a moment from the beginning of the second intifada, in 2000, when it was still stones and Molotovs against an army. In 2001, the uprising escalated into an **armed** conflict, and one of the major incidents was the destruction of the Palestinian refugee camp in the West Bank town of Jenin. Without the political will to find common ground, the continual friction of tactic and counter-tactic only creates suspicion and **hatred** and vengeance, and perpetuates the cycle of **violence**. In the ' 90s, after the breakup of the Soviet Union, Yugoslavia fractured along ethnic fault lines, and civil war broke out between Bosnia, Croatia and Serbia. This is a scene of house-to-house fighting in Mostar, neighbor against neighbor. A bedroom, the place where people share intimacy, where life itself is conceived, became a battlefield. A **mosque** in northern Bosnia was destroyed by Serbian artillery and was used as a makeshift morgue. Dead Serbian soldiers were collected after a battle and used as barter for the return of prisoners or Bosnian soldiers killed in action. This was once a park. The Bosnian soldier who guided me told me that all of his friends were there now. At the same time in South Africa, after Nelson Mandela had been released from prison, the black population commenced the final phase of liberation from apartheid. One of the things I had to learn as a journalist was what to do with my anger. I had to use it, channel its energy, turn it into something that would clarify my vision, instead of clouding it. In Transkei, I witnessed a rite of passage into manhood, of the Xhosa tribe. Teenage boys lived in isolation, their bodies covered with white clay. After several weeks, they washed off the white and took on the full responsibilities of men. It was a very old ritual that seemed symbolic of the political struggle that was changing the face of South Africa. Children in Soweto playing on a trampoline. Elsewhere in Africa there was famine. In Somalia, the central government collapsed and clan warfare broke out. Farmers were driven off their land, and crops and livestock were destroyed or stolen. Starvation was being used as a weapon of mass destruction — primitive but extremely effective. Hundreds of thousands of people were exterminated, slowly and painfully. The international community responded with massive humanitarian relief, and hundreds of thousands of more lives were saved. American troops were sent to protect the relief shipments, but they were eventually drawn into the conflict, and after the tragic battle in Mogadishu, they were withdrawn. In southern Sudan, another civil war saw similar use of starvation as a means of genocide. Again, international NGOs, united under the umbrella of the U. N., staged a massive relief operation and thousands of lives were saved. I ’ m a witness, and I want my testimony to be honest and uncensored. I also want it to be powerful and eloquent, and to do as much justice as possible to the experience of the people I ’ m photographing. This man was in an NGO feeding center, being helped as much as he could be helped. He literally had nothing. He was a virtual skeleton, yet he could still summon the courage and the will to move. He had not given up, and if he didn ’ t give up, how could anyone in the outside world ever dream of losing hope? In 1994, after three months of covering the South African election, I saw the inauguration of Nelson Mandela, and it was the most uplifting thing I ’ ve ever seen. It exemplified the best that humanity has to offer. The next day I left for Rwanda, and it was like taking the express elevator to hell. This man had just been liberated from a Hutu death camp. He allowed me to photograph him for quite a long time, and he even turned his face toward the light, as if he wanted me to see him better. I think he knew what the scars on his face would say to the rest of the world. This time, maybe confused or discouraged by the military disaster in Somalia, the international community remained silent, and somewhere around 800, 000 people were slaughtered by their own countrymen — sometimes their own neighbors — using farm implements as weapons. Perhaps because a lesson had been learned by the weak response to the war in Bosnia and the failure in Rwanda, when Serbia attacked Kosovo, international action was taken much more decisively. NATO forces went in, and the Serbian army withdrew. Ethnic Albanians had been murdered, their farms destroyed and a huge number of people forcibly deported. They were received in refugee camps set up by NGOs in Albania and Macedonia. The imprint of a man who had been burned inside his own home. The image reminded me of a cave painting, and echoed how primitive we still are in so many ways. Between 1995 and ’ 96, I covered the first two wars in Chechnya from inside Grozny. This is a Chechen rebel on the front line against the Russian army. The Russians bombarded Grozny constantly for weeks, killing mainly the civilians who were still trapped inside. I found a boy from the local orphanage wandering around the front line. My work has evolved from being concerned mainly with war to a focus on critical social issues as well. After the fall of Ceausescu, I went to Romania and discovered a kind of gulag of children, where thousands of orphans were being kept in medieval conditions. Ceausescu had imposed a quota on the number of children to be produced by each family, thereby making women ’ s bodies an instrument of state economic policy. Children who couldn ’ t be supported by their families were raised in government orphanages. Children with birth defects were labeled incurables, and confined for life to inhuman conditions. As reports began to surface, again international aid went in. Going deeper into the legacy of the Eastern European regimes, I worked for several months on a story about the effects of industrial pollution, where there had been no regard for the environment or the health of either workers or the general population. An aluminum factory in Czechoslovakia was filled with carcinogenic smoke and dust, and four out of five workers came down with cancer. After the fall of Suharto in Indonesia, I began to explore conditions of poverty in a country that was on its way towards modernization. I spent a good deal of time with a man who lived with his family on a railway embankment and had lost an arm and a leg in a train accident. When the story was published, unsolicited donations poured in. A trust fund was established, and the family now lives in a house in the countryside and all their basic necessities are taken care of. It was a story that wasn ’ t trying to sell anything. Journalism had provided a channel for people ’ s natural sense of generosity, and the readers responded. I met a band of **homeless** children who ’ d come to Jakarta from the countryside, and ended up living in a train station. By the age of 12 or 14, they ’ d become beggars and drug addicts. The rural poor had become the urban poor, and in the process, they ’ d become invisible. These heroin addicts in detox in Pakistan reminded me of figures in a play by Beckett : isolated, waiting in the dark, but drawn to the light. Agent Orange was a defoliant used during the Vietnam War to deny cover to the Vietcong and the North Vietnamese army. The active ingredient was dioxin, an extremely toxic chemical that was sprayed in vast quantities, and whose effects passed through the genes to the next generation. In 2000, I began documenting global health issues, concentrating first on AIDS in Africa. I tried to tell the story through the work of caregivers. I thought it was important to emphasize that people were being helped, whether by international NGOs or by local grassroots organizations. So many children have been orphaned by the epidemic that grandmothers have taken the place of parents, and a lot of children had been born with HIV. A hospital in Zambia. I began documenting the close connection between HIV / AIDS and tuberculosis. This is an MSF hospital in Cambodia. My pictures can play a supporting role to the work of NGOs by shedding light on the critical social problems they ’ re trying to deal with. I went to Congo with MSF, and contributed to a book and an exhibition that focused attention on a forgotten war in which millions of people have died, and exposure to disease without treatment is used as a weapon. A malnourished child being measured as part of the supplemental feeding program. In the fall of 2004 I went to Darfur. This time I was on assignment for a magazine, but again worked closely with MSF. The international community still hasn ’ t found a way to create the pressure necessary to stop this genocide. An MSF hospital in a camp for displaced people. I ’ ve been working on a long project on crime and punishment in America. This is a scene from New Orleans. A prisoner on a chain gang in Alabama was punished by being handcuffed to a post in the midday sun. This experience raised a lot of questions, among them questions about race and **equality** and for whom in our country opportunities and options are available. In the yard of a chain gang in Alabama. I didn ’ t see either of the planes hit, and when I glanced out my window, I saw the first tower burning, and I thought it might have been an accident. A few minutes later when I looked again and saw the second tower burning, I knew we were at war. In the midst of the wreckage at Ground Zero, I had a realization. I ’ d been photographing in the Islamic world since 1981 — not only in the Middle East, but also in Africa, Asia and Europe. At the time I was photographing in these different places, I thought I was covering separate stories, but on 9 / 11 history crystallized, and I understood I ’ d actually been covering a single story for more than 20 years, and the attack on New York was its latest manifestation. The central commercial district of Kabul, Afghanistan at the end of the civil war, shortly before the city fell to the Taliban. Land mine **victims** being helped at the Red Cross rehab center being run by Alberto Cairo. A boy who lost a leg to a leftover mine. I ’ d witnessed immense **suffering** in the Islamic world from political oppression, civil war, foreign invasions, poverty, famine. I understood that in its **suffering**, the Islamic world had been crying out. Why weren ’ t we listening? A Taliban fighter shot during a battle as the Northern Alliance entered the city of Kunduz. When war with Iraq was imminent, I realized the

American troops would be very well covered, so I decided to cover the invasion from inside Baghdad. A marketplace was hit by a mortar shell that killed several members of a single family. A day after American forces entered Baghdad, a company of Marines began rounding up bank robbers and were cheered on by the crowds — a hopeful moment that was short lived. For the first time in years, Shi ' ites were allowed to make the pilgrimage to Karbala to observe Ashura, and I was amazed by the sheer number of people and how fervently they practiced their religion. A group of men march through the **streets** cutting themselves with knives. It was obvious that the Shi ' ites were a force to be reckoned with, and we would do well to understand them and learn how to deal with them. Last year I spent several months documenting our wounded troops, from the battlefield in Iraq all the way home. This is a helicopter medic giving CPR to a soldier who had been shot in the head. Military medicine has become so efficient that the percentage of troops who survive after being wounded is much higher in this war than in any other war in our history. The signature weapon of the war is the IED, and the signature wound is severe leg damage. After enduring extreme pain and **trauma**, the wounded face a grueling physical and psychological struggle in rehab. The spirit they displayed was absolutely remarkable. I tried to imagine myself in their place, and I was totally humbled by their courage and determination in the face of such catastrophic loss. Good people had been put in a very bad situation for questionable results. One day in rehab someone, started talking about surfing and all these guys who ' d never surfed before said,"Hey, let ' s go."And they went surfing. Photographers go to the extreme edges of human experience to show people what ' s going on. Sometimes they put their lives on the line, because they believe your opinions and your influence matter. They aim their pictures at your best instincts, generosity, a sense of right and wrong, the ability and the willingness to identify with others, the refusal to accept the unacceptable. My TED wish : there ' s a vital story that needs to be told, and I wish for TED to help me gain **access** to it and then to help me come up with innovative and exciting ways to use news photography in the digital era. Thank you very much.

Text Sample 379: POS Dim 3 – 2007 – Score 11.00 – 2007/t000357.txt

I ' ve actually been waiting by the phone for a call from TED for years. And in fact, in 2000, I was ready to talk about eBay, but no call. In 2003, I was ready to do a talk about the Skoll Foundation and social entrepreneurship. No call. In 2004, I started Participant Productions and we had a really good first year, and no call. And finally, I get a call last year, and then I have to go up after J. J. Abrams. You ' ve got a cruel sense of humor, TED. When I first moved to Hollywood from Silicon Valley, I had some misgivings. But I found that there were some advantages to being in Hollywood. And, in fact, some advantages to owning your own media company. And I also found that Hollywood and Silicon Valley have a lot more in common than I would have dreamed. Hollywood has its sex symbols, and the Valley has its sex symbols. Hollywood has its rivalries, and the Valley has its rivalries. Hollywood gathers around power tables, and the Valley gathers around power tables. So it turned out there was a lot more in common than I would have dreamed. But I ' m actually here today to tell a story. And part of it is a personal story. When Chris invited me to speak, he said, people think of you as a bit of an enigma, and they want to know what drives you a bit. And what really drives me is a vision of the future that I think we all share. It ' s a world of peace and prosperity and sustainability. And when we heard a lot of the presentations over the last couple of days, Ed Wilson and the pictures of James Nachtwey, I think we all realized how far we have to go to get to this new version of humanity that I like to call"Humanity 2. 0."And it ' s also something that resides in each of us, to close what I think are the two big calamities in the world today. One is the gap in opportunity — this gap that President Clinton last night called uneven, unfair and unsustainable — and, out of that, comes poverty and illiteracy and disease and all these evils that we see around us. But perhaps the other, bigger gap is what we call the hope gap. And someone, at some point, came up with this very bad idea that an ordinary individual couldn ' t make a difference in the world. And I think that ' s just a horrible thing. And so chapter one really begins today, with all of us, because within each of us is the power to **equal** those opportunity gaps and to close the hope gaps. And if the men and women of TED can ' t make a difference in the world, I don ' t know who can. And for me, a lot of this started when I was younger and my family used to go camping in upstate New York. And there really wasn ' t much to do there for the summer, except get beaten up by my sister or read books. And so I used to read authors like James Michener and James Clavell and Ayn Rand. And their stories made the world seem a very small and interconnected place. And it struck me that if I could write stories that were about this world as being small and interconnected, that maybe I could get people interested in the issues that affected us all, and maybe engage them to make a difference. I didn ' t think that was necessarily the best way to make a living, so I decided to go on a path to become financially independent, so I could write these stories as quickly as I could. I then had a bit of a wake-up call when I was 14. And my dad came home one day and announced that he had cancer, and it looked pretty bad. And what he said was, he wasn ' t so much afraid that he might die, but that he hadn ' t done the things that he wanted to with his life. And knock on wood, he ' s still alive today, many years later. But for a young man that made a real impression on me, that one never knows how much time one really has. So I set out in a hurry. I studied engineering. I started a couple of businesses that I thought would be the ticket to financial freedom. One of those businesses was a computer rental business called Micros on the Move, which is very well named, because people kept stealing the computers. So I figured I needed to learn a little bit more about business, so I went to Stanford Business School and studied there. And while I was there, I made friends with a fellow named Pierre Omidyar, who is here today. And Pierre, I **apologize** for this. This is a photo from the old days. And just after I ' d graduated, Pierre came to me with this idea to help people buy and sell things online with each other. And with the wisdom of my Stanford degree, I said,"Pierre, what a stupid idea."And needless to say, I was right. But right after that, Pierre — in ' 96, Pierre and I left our full-time jobs to build eBay as a company. And the rest of that story, you know. The company went public two years later and is today one of the best known companies in the world. Hundreds of millions of people use it in hundreds of countries, and so on. But for me, personally, it was a real change. I went from living in a house with five guys in Palo Alto and living off their leftovers, to all of a sudden having all kinds of resources. And I wanted to figure out how I could take the blessing of these resources and share it with the world. And around that time, I met John Gardner, who is a remarkable man. He was the architect of the Great Society programs under Lyndon Johnson in the 1960s. And I asked him what he felt was the best thing I could do, or anyone could do, to make a difference in the long-term issues facing humanity. And John said,"Bet on good people doing good things. Bet on good people doing good things."And that really resonated with me. I started a foundation to bet on these good people doing good things. These leading, innovative, **nonprofit** folks, who are using business skills in a very leveraged way to solve social problems. People today we call social entrepreneurs. And to put a face on it, people like Muhammad Yunus, who started the Grameen Bank, has lifted 100 million people plus out of poverty around the world, won the Nobel Peace Prize. But there ' s also a lot of people that you don ' t know. Folks like Ann Cotton, who started a group called CAMFED in Africa, because she felt girls ' education was lagging. And she started it about 10 years ago, and today, she educates over a quarter million African girls. And somebody like Dr. Victoria Hale, who started the world ' s first **nonprofit** pharmaceutical company, and whose first drug will be fighting visceral leishmaniasis, also known as black fever. And by 2010, she hopes to eliminate this disease, which is really a scourge in the developing world. And so this is one way to bet on good people doing good things. And a lot of this comes together in a philosophy of change that I find really is powerful. It ' s what we call,"Invest, connect and **celebrate**."And invest : if you see good people doing good things, invest in them. Invest in their organizations, or in business. Invest in these folks. Connecting them together through conferences — like a TED — brings so many powerful connections, or through the World Forum on Social Entrepreneurship that my foundation does at Oxford every year. And **celebrate** them : tell their stories, because not only are there good people doing good work, but their stories can help close these gaps of hope. And it was this last part of the mission, the **celebrate** part, that really got me back to thinking when I was a kid and wanted to tell stories to get people involved in the issues that affect us all. And a light bulb went off, which was, first, that I didn ' t actually have to do the writing myself, I could find writers. And then the next light bulb was, better than just writing, what about film and TV, to get out to people in a big way? And I thought about the films that inspired me, films like"Gandhi"and"Schindler ' s List."And I wondered who was doing these kinds of films today. And there really wasn ' t a specific company that was focused on the public interest. So, in 2003, I started to make my way around Los Angeles to talk about the idea of a pro-social media company and I was met with a lot of encouragement. One of the lines of encouragement that I heard over and over was,"The **streets** of Hollywood are littered with the carcasses of people like you, who think you ' re going to come to this town and make movies."And then of course, there was the other adage."The surest way to become a millionaire is to start by being a billionaire and go into the movie business."Undeterred, in January of 2004, I started Participant Productions with the vision to be a global media company focused on the public interest. And our mission is to produce entertainment that creates and inspires social change. And we don ' t just want people to see our movies and say, that was fun, and forget about it. We want them to actually get involved in the issues. In 2005, we launched our first slate of films,"Murder Ball,""North Country,""Syriana"and"Good Night and Good Luck."And much to my surprise, they were noticed. We ended up with 11 Oscar nominations for these films. And it turned out to be a pretty good year for this guy. Perhaps more importantly, tens of thousands of people joined the advocacy programs and the activism programs that we created to go around the movies. And we had an online component of that, our community sect called Participate. net. But with our social sector partners, like the ACLU and PBS and the Sierra Club and the NRDC, once people saw the film, there was actually something they could do to make a difference. One of these films in particular, called"North Country,"was actually kind of a box office disaster. But it was a film that starred Charlize Theron and it was about women ' s rights, women ' s empowerment, **domestic violence** and so on. And we released the film at the same time that the Congress was debating the renewal of the **Violence** Against Women Act. And with screenings on the Hill, and discussions, and with our social sector partners, like the National Organization of Women, the film was widely credited with influencing the successful renewal of the act. And that to me, spoke volumes, because it ’ s — the film started about a true-life story about a woman who was harassed, sued her employer, led to a landmark case that led to the Equal Opportunity Act, and the **Violence** Against Women Act and others. And then the movie about this person doing these things, then led to this greater renewal. And so again, it goes back to betting on good people doing good things. Speaking of which, our fellow TEDster, Al — I first saw Al do his slide show presentation on global warming in May of 2005. At that point, I thought I knew something about global warming. I thought it was a 30 to 50 year problem. And after we saw his slide show, it became clear that it was much more urgent. And so right afterwards, I met backstage with Al, and with Lawrence Bender, who was there, and Laurie David, and Davis Guggenheim, who was running **documentaries** for Participant at the time. And with Al ’ s blessing, we decided on the spot to turn it into a film, because we felt that we could get the message out there far more quickly than having Al go around the world, speaking to audiences of 100 or 200 at a time. And you know, there ’ s another adage in Hollywood, that nobody knows nothing about anything. And I really thought this was going to be a straight-to-PBS charitable **initiative**. And so it was a great shock to all of us when the film really captured the public interest, and today is mandatory viewing in schools in England and Scotland, and most of Scandinavia. We ’ ve sent 50, 000 DVDs to high school teachers in the U. S. and it ’ s really changed the debate on global warming. It was also a pretty good year for this guy. We now call Al the George Clooney of global warming. And for Participant, this is just the start. Everything we do looks at the major issues in the world. And we have 10 films in production right now, and dozens others in development. I ’ ll quickly talk about a few coming up. One is”Charlie Wilson ’ s War,”with Tom Hanks and Julia Roberts. And it ’ s the true story of Congressman Charlie Wilson, and how he funded the Taliban to fight the Russians in Afghanistan. And we ’ re also doing a movie called”The Kite Runner,”based on the book”The Kite Runner,”also about Afghanistan. And we think once people see these films, they ’ ll have a much better understanding of that part of the world and the Middle East in general. We premiered a film called”The Chicago 10”at Sundance this year. It ’ s based on the protesters at the Democratic Convention in 1968, Abby Hoffman and crew, and, again, a story about a small group of individuals who did make change in the world. And a **documentary** that we ’ re doing on Jimmy Carter and his Mid-East peace efforts over the years. And in particular, we ’ ve been following him on his recent book tour, which, as many of you know, has been very non- controversial — — which is really bad for getting people to come see a movie. In closing, I ’ d like to say that everybody has the opportunity to make change in their own way. And all the people in this room have done so through their business lives, or their philanthropic work, or their other interests. And one thing that I ’ ve learned is that there ’ s never one right way to make change. One can do it as a tech person, or as a finance person, or a **nonprofit** person, or as an entertainment person, but every one of us is all of those things and more. And I believe if we do these things, we can close the opportunity gaps, we can close the hope gaps. And I can imagine, if we do this, the headlines in 10 years might read something like these :”New AIDS Cases in Africa Fall to Zero,””U. S. Imports its Last Barrel of Oil”— —”Israelis and Palestinians **Celebrate** 10 Years of Peaceful Coexistence.”And I like this one,”Snow Has Returned to Kilimanjaro.”And finally, an eBay listing for one well-traveled slide show, now obsolete, museum piece. Please contact Al Gore. And I believe that, working together, we can make all of these things happen.

And I want to thank you all for having me here today. It ’ s been a real honor. Thank you. Oh, thank you.

Text Sample 380: POS Dim 3 – 2007 – Score 10.00 – 2007/t000355.txt

I really am honored to be here, and as Chris said, it ’ s been over 20 years since I started working in Africa. My first introduction was at the Abidjan airport on a sweaty, Ivory Coast morning. I had just left Wall Street, cut my hair to look like Margaret Mead, given away most everything that I owned, and arrived with all the essentials — some poetry, a few clothes, and, of course, a guitar — because I was going to save the world, and I thought I would just start with the African continent. But literally within days of arriving I was told, in no uncertain terms, by a number of West African women, that Africans didn ’ t want saving, thank you very much, least of all not by me. I was too young, unmarried, I had no children, didn ’ t really know Africa, and besides, my French was pitiful. And so, it was an incredibly painful time in my life, and yet it really started to give me the humility to start listening. I think that failure can be an incredibly motivating force as well, so I moved to Kenya and worked in Uganda, and I met a group of Rwandan women, who asked me, in 1986, to move to Kigali to help them start the first microfinance institution there. And I did, and we ended up naming it Duterimbere, meaning”to go forward with enthusiasm.”And while we were doing it, I realized that there weren ’ t a lot of businesses that were viable and started by women, and so maybe I should try to run a business, too. And so I started looking around, and I heard about a bakery that was run by 20 prostitutes. And, being a little intrigued, I went to go meet this group, and what I found was 20 unwed mothers who were trying to survive. And it was really the beginning of my understanding the power of language, and how what we call people so often distances us from them, and makes them little. I also found out that the bakery was nothing like a business, that, in fact, it was a classic charity run by a well-intentioned person, who essentially spent 600 dollars a month to keep these 20 women busy making little crafts and baked goods, and living on 50 cents a day, still in poverty. So, I made a deal with the women. I said,”Look, we get rid of the charity side, and we run this as a business and I ’ ll help you.”They nervously agreed. I nervously started, and, of course, things are always harder than you think they ’ re going to be. First of all, I thought, well, we need a sales team, and we clearly aren ’ t the A-Team here, so let ’ s — I did all this training. And the epitome was when I literally marched into the **streets** of Nyamirambo, which is the popular quarter of Kigali, with a bucket, and I sold all these little doughnuts to people, and I came back, and I was like,”You see?”And the women said,”You know, Jacqueline, who in Nyamirambo is not going to buy doughnuts out of an orange bucket from a tall American woman?”And like — — it ’ s a good point. So then I went the whole American way, with competitions, team and individual. Completely failed, but over time, the women learnt to sell on their own way. And they started listening to the marketplace, and they came back with ideas for cassava chips, and banana chips, and sorghum bread, and before you knew it, we had cornered the Kigali market, and the women were earning three to four times the national average. And with that confidence surge, I thought,”Well, it ’ s time to create a real bakery, so let ’ s paint it.”And the women said,”That ’ s a really great idea.”And I said,”Well, what color do you want to paint it?”And they said,”Well, you choose.”And I said,”No, no, I ’ m learning to listen. You choose. It ’ s your bakery, your **street**, your country — not mine.”But they wouldn ’ t give me an answer. So, one week, two weeks, three weeks went by, and finally I said,”Well, how about blue?”And they said,”Blue, blue, we love blue. Let ’ s do it blue.”So, I went to the store, I brought Gaudence, the recalcitrant one of all, and we brought all this paint and fabric to make curtains, and on painting day, we all gathered in Nyamirambo, and the idea was we would paint it white with blue as trim, like a little French bakery. But that was clearly not as satisfying as painting a wall of blue like a morning sky. So, blue, blue, everything became blue. The walls were blue, the windows were blue, the **sidewalk** out front was painted blue. And Aretha Franklin was shouting”R-E-S-P-E-C-T,”the women ’ s hips were swaying and little kids were trying to grab the paintbrushes, but it was their day. And at the end of it, we stood across the **street** and we looked at what we had done, and I said,”It is so beautiful.”And the women said,”It really is.”And I said,”And I think the color is perfect,”and they all nodded their head, except for Gaudence, and I said,”What?”And she said,”Nothing.”And I said,”What?”And she said,”Well, it is pretty, but, you know, our color, really, it is green.”And — — I learned then that listening isn ’ t just about patience, but that when you ’ ve lived on charity and dependent your whole life long, it ’ s really hard to say what you mean. And, mostly because people never really ask you, and when they do, you don ’ t really think they want to know the truth. And so then I learned that listening is not only about waiting, but it ’ s also learning how better to ask questions. And so, I lived in Kigali for about two and a half years, doing these two things, and it was an extraordinary time in my life. And it taught me three lessons that I think are so important for us today, and certainly in the work that I do. The first is that dignity is more important to the human spirit than wealth. As Eleni has said, when people gain income, they gain choice, and that is fundamental to dignity. But as human beings, we also want to see each other, and we want to be heard by each other, and we should never forget that. The second is that traditional charity and aid are never going to solve the problems of poverty. I think Andrew pretty well covered that, so I will move to the third point, which is that markets alone also are not going to solve the problems of poverty. Yes, we ran this as a business, but someone needed to pay the philanthropic support that came into the training, and the management support, the strategic advice and, maybe most important of all, the **access** to new contacts, networks and new markets. And so, on a micro level, there ’ s a real role for this combination of investment and philanthropy. And on a macro level — some of the speakers have inferred that even health should be privatized. But, having had a father with heart disease, and realizing that what our family could afford was not what he should have gotten, and having a good friend step in to help, I really believe that all people deserve **access** to health at prices they can afford. I think the market can help us figure that out, but there ’ s got to be a charitable component, or I don ’ t think we ’ re going to create the kind of societies we want to live in. And so, it was really those lessons that made me decide to build Acumen Fund about six years ago. It ’ s a **nonprofit**, venture capital fund for the poor, a few oxymorons in one sentence. It essentially raises charitable funds from individuals, foundations and corporations, and then we turn around and we invest **equity** and loans in both for-profit and **nonprofit** entities that deliver affordable health, housing, energy, clean water to low income people in South Asia and Africa, so that they can make their own choices. We ’ ve invested about 20 million dollars in 20 different enterprises, and have, in so doing, created nearly 20, 000 jobs, and delivered tens of millions of services to people who otherwise would not be able to afford them. I want to tell you two stories. Both of them are in Africa. Both of them are about investing in entrepreneurs who are committed to service, and who really know the markets. Both of them live at the confluence of public health and enterprise, and both of them, because they ’ re manufacturers, create jobs directly, and create incomes indirectly, because they ’ re in the malaria sector, and Africa loses about 13 billion dollars a year because of malaria. And so as people get healthier, they also get wealthier. The first one is called Advanced Bio-Extracts Limited. It ’ s a company built in Kenya about seven years ago by an incredible entrepreneur named Patrick Henfrey and his three colleagues. These are old-hand farmers who ’ ve gone through all the agricultural ups and downs in Kenya over the last 30 years. Now, this plant is an Artemisia plant ; it ’ s the basic component for artemisinin, which is the best-known treatment for malaria. It ’ s **indigenous** to China and the Far East, but given that the prevalence of malaria is here in Africa, Patrick and his colleagues said,”Let ’ s bring it here, because it ’ s a high value-add product.”The farmers get three to four times the yields that they would with maize. And so, using patient capital — money that they could raise early on, that actually got below market returns and was willing to go the long haul and be combined with management assistance, strategic assistance — they ’ ve now created a company where they purchase from 7, 500 farmers. So that ’ s about 50, 000 people affected. And I think some of you may have visited — these farmers are helped by KickStart and TechnoServe, who help them become more self-sufficient. They buy it, they dry it and they bring it to this factory, which was purchased in part by, again, patient capital from Novartis, who has a real interest in getting the powder so that they can make Coartem. Acumen ’ s been working with ABE for the past year, year and a half, both on looking at a new business plan, and what does expansion look like, helping with management support and helping to do term sheets and raise capital. And I really understood what patient capital meant emotionally in the last month or so. Because the company was literally 10 days away from proving that the product they produced was at the world-quality level needed to make Coartem, when they were in the biggest cash crisis of their history. And we called all of the social investors we know. Now, some of these same social investors are really interested in Africa and understand the importance of agriculture, and they even helped the farmers. And even when we explained that if ABE goes away, all those 7, 500 jobs go away too, we sometimes have this bifurcation between business and the social. And it ’ s really time we start thinking more creatively about how they can be fused. So Acumen made not one, but two bridge loans, and the good news is they did indeed meet world-quality classification and are now in the final stages of closing a 20-million-dollar round, to move it to the next level, and I think that this will be one of the more important companies in East Africa. This is Samuel. He ’ s a farmer. He was actually living in the Kibera slums when his father called him and told him about Artemisia and the value-add potential. So he moved back to the farm, and, long story short, they now have seven acres under cultivation. Samuel ’ s kids are in private school, and he ’ s starting to help other farmers in the area also go into Artemisia production — dignity being more important than wealth. The next one, many of you know. I talked about it a little at Oxford two years ago, and some of you visited A to Z manufacturing, which is one of the great, real companies in East Africa. It ’ s another one that lives at the confluence of health and enterprise. And this is really a story about a public-private solution that has really worked. It started in Japan. Sumitomo had developed a technology essentially to impregnate a polyethylene-based fiber with organic insecticide, so you could create a bed net, a malaria bed net, that would last five years and not need to be re-dipped. It could alter the vector, but like Artemisia, it had been produced only in East Asia. And as part of its social responsibility, Sumitomo said,”Why don ’ t we experiment with whether we can produce it in Africa, for Africans?”UNICEF came forward and said,”We ’ ll buy most of the nets, and then we ’ ll give them away, as part of the global fund ’ s and the U.

N. ’ s commitment to pregnant women and children, for free.”Acumen came in with the patient capital, and we also helped to identify the entrepreneur that we would all partner with here in Africa, and Exxon provided the initial resin. Well, in looking around for entrepreneurs, there was none better that we could find on earth than Anuj Shah, in A to Z manufacturing company. It ’ s a 40-year-old company, it understands manufacturing. It ’ s gone from socialist Tanzania into capitalist Tanzania, and continued to flourish. It had about 1, 000 employees when we first found it. And so, Anuj took the entrepreneurial risk here in Africa to produce a public good that was purchased by the aid establishment to work with malaria. And, long story short, again, they ’ ve been so successful. In our first year, the first net went off the line in October of 2003. We thought the hitting-it- out-of-the-box number was 150, 000 nets a year. This year, they are now producing eight million nets a year, and they employ 5, 000 people, 90 percent of whom are women, mostly unskilled. They ’ re in a joint venture with Sumitomo. And so, from an enterprise perspective for Africa, and from a public health perspective, these are real successes. But it ’ s only half the story if we ’ re really looking at solving problems of poverty, because it ’ s not long-term sustainable. It ’ s a company with one big customer. And if avian flu hits, or for any other reason the world decides that malaria is no longer as much of a priority, everybody loses. And so, Anuj and Acumen have been talking about testing the private sector, because the assumption that the aid establishment has made is that, look, in a country like Tanzania, 80 percent of the population makes less than two dollars a day. It costs, at manufacturing point, six dollars to produce these, and it costs the establishment another six dollars to distribute it, so the market price in a free market would be about 12 dollars per net. Most people can ’ t afford that, so let ’ s give it away free. And we said,”Well, there ’ s another option. Let ’ s use the market as the best listening device we have, and understand at what price people would pay for this, so they get the dignity of choice. We can start building local distribution, and actually, it can cost the public sector much less.”And so we came in with a second round of patient capital to A to Z, a loan as well as a grant, so that A to Z could play with pricing and listen to the marketplace, and found a number of things. One, that people will pay different prices, but the overwhelming number of people will come forth at one dollar per net and make a decision to buy it. And when you listen to them, they ’ ll also have a lot to say about what they like and what they don ’ t like. And that some of the channels we thought would work didn ’ t work. But because of this experimentation and iteration that was allowed because of the patient capital, we ’ ve now found that it costs about a dollar in the private sector to distribute, and a dollar to buy the net. So then, from a policy perspective, when you start with the market, we have a choice. We can continue going along at 12 dollars a net, and the customer pays zero, or we could at least experiment with some of it, to charge one dollar a net, costing the public sector another six dollars a net, give the people the dignity of choice, and have a distribution system that might, over time, start sustaining itself. We ’ ve got to start having conversations like this, and I don ’ t think there ’ s any better way to start than using the market, but also to bring other people to the table around it. Whenever I go to visit A to Z, I think of my grandmother, Stella. She was very much like those women sitting behind the sewing machines. She grew up on a farm in Austria, very poor, didn ’ t have very much education. She moved to the United States, where she met my grandfather, who was a cement hauler, and they had nine children. Three of them died as babies. My grandmother had tuberculosis, and she worked in a sewing machine shop, making shirts for about 10 cents an hour. She, like so many of the women I see at A to Z, worked hard every day, understood what **suffering** was, had a deep faith in God, loved her children and would never have accepted a handout. But because she had the opportunity of the marketplace, and she lived in a society that provided the **safety** of having **access** to affordable health and education, her children and their children were able to live lives of real purpose and follow real dreams. I look around at my **siblings** and my cousins — and as I said, there are a lot of us — and I see teachers and musicians, hedge fund managers, designers. One sister who makes other people ’ s wishes come true. And my wish, when I see those women, I meet those farmers, and I think about all the people across this continent who are working hard every day, is that they have that sense of opportunity and possibility, and that they also can believe and get **access** to services, so that their children, too, can live those lives of great purpose. It shouldn ’ t be that difficult. But what it takes is a commitment from all of us to essentially refuse trite assumptions, get out of our ideological boxes. It takes investing in those entrepreneurs that are committed to service as well as to success. It takes opening your arms, both, wide, and expecting very little love in return, but demanding accountability, and bringing the accountability to the table as well. And most of all, most of all, it requires that all of us have the courage and the patience, whether we are rich or poor, African or non-African, local or diaspora, left or right, to really start listening to each other. Thank you.

Text Sample 381: POS Dim 3 – 2012 – Score 15.00 – 2012/t002418.txt

Well this is a really extraordinary honor for me. I spend most of my time in jails, in prisons, on death row. I spend most of my time in very low-income communities in the projects and places where there ’ s a great deal of hopelessness. And being here at TED and seeing the stimulation, hearing it, has been very, very energizing to me. And one of the things that ’ s emerged in my short time here is that TED has an identity. And you can actually say things here that have impacts around the world. And sometimes when it comes through TED, it has meaning and power that it doesn ’ t have when it doesn ’ t. And I mention that because I think identity is really important. And we ’ ve had some fantastic presentations. And I think what we ’ ve learned is that, if you ’ re a teacher your words can be meaningful, but if you ’ re a compassionate teacher, they can be especially meaningful. If you ’ re a doctor you can do some good things, but if you ’ re a caring doctor you can do some other things. And so I want to talk about the power of identity. And I didn ’ t learn about this actually practicing law and doing the work that I do. I actually learned about this from my grandmother. I grew up in a house that was the traditional African-American home that was dominated by a matriarch, and that matriarch was my grandmother. She was tough, she was strong, she was powerful. She was the end of every argument in our family. She was the beginning of a lot of arguments in our family. She was the daughter of people who were actually enslaved. Her parents were born in slavery in Virginia in the 1840 ’ s. She was born in the 1880 ’ s and the experience of slavery very much shaped the way she saw the world. And my grandmother was tough, but she was also **loving**. When I would see her as a little boy, she ’ d come up to me and she ’ d give me these hugs. And she ’ d squeeze me so tight I could barely breathe and then she ’ d let me go. And an hour or two later, if I saw her, she ’ d come over to me and she ’ d say,”Bryan, do you still feel me hugging you?”And if I said,”No,”she ’ d **assault** me again, and if I said,”Yes,”she ’ d leave me alone. And she just had this quality that you always wanted to be near her. And the only challenge was that she had 10 children. My mom was the youngest of her 10 kids. And sometimes when I would go and spend time with her, it would be difficult to get her time and attention. My cousins would be running around everywhere. And I remember, when I was about eight or nine years old, waking up one morning, going into the living room, and all of my cousins were running around. And my grandmother was sitting across the room staring at me. And at first I thought we were playing a game. And I would look at her and I ’ d smile, but she was very serious. And after about 15 or 20 minutes of this, she got up and she came across the room and she took me by the hand and she said,”Come on, Bryan. You and I are going to have a talk.”And I remember this just like it happened yesterday. I never will forget it. She took me out back and she said,”Bryan, I ’ m going to tell you something, but you don ’ t tell anybody what I tell you.”I said,”Okay, Mama.”She said,”Now you make sure you don ’ t do that.”I said,”Sure.”Then she sat me down and she looked at me and she said,”I want you to know I ’ ve been watching you.”And she said,”I think you ’ re special.”She said,”I think you can do anything you want to do.”I will never forget it. And then she said,”I just need you to promise me three things, Bryan.”I said,”Okay, Mama.”She said,”The first thing I want you to promise me is that you ’ ll always love your mom.”She said,”That ’ s my baby girl, and you have to promise me now you ’ ll always take care of her.”Well I adored my mom, so I said,”Yes, Mama. I ’ ll do that.”Then she said,”The second thing I want you to promise me is that you ’ ll always do the right thing even when the right thing is the hard thing.”And I thought about it and I said,”Yes, Mama. I ’ ll do that.”Then finally she said,”The third thing I want you to promise me is that you ’ ll never drink alcohol.”Well I was nine years old, so I said,”Yes, Mama. I ’ ll do that.”I grew up in the country in the rural South, and I have a brother a year older than me and a sister a year younger. When I was about 14 or 15, one day my brother came home and he had this six-pack of beer — I don ’ t know where he got it — and he grabbed me and my sister and we went out in the woods. And we were kind of just out there doing the stuff we crazily did. And he had a sip of this beer and he gave some to my sister and she had some, and they offered it to me. I said,”No, no, no. That ’ s okay. You all go ahead. I ’ m not going to have any beer.”My brother said,”Come on. We ’ re doing this today ; you always do what we do. I had some, your sister had some. Have some beer.”I said,”No, I don ’ t feel right about that. Y ’ all go ahead. Y ’ all go ahead.”And then my brother started staring at me. He said,”What ’ s wrong with you? Have some beer.”Then he looked at me real hard and he said,”Oh, I hope you ’ re not still hung up on that conversation Mama had with you.”I said,”Well, what are you talking about?”He said,”Oh, Mama tells all the grandkids that they ’ re special.”I was devastated. And I ’ m going to admit something to you. I ’ m going to tell you something I probably shouldn ’ t. I know this might be broadcast broadly. But I ’ m 52 years old, and I ’ m going to admit to you that I ’ ve never had a drop of alcohol. I don ’ t say that because I think that ’ s virtuous ; I say that because there is power in identity. When we create the right kind of identity, we can say things to the world around us that they don ’ t actually believe makes sense. We can get them to do things that they don ’ t think they can do. When I thought about my grandmother, of course she would think all her grandkids were special. My grandfather was in prison during prohibition. My male uncles died of alcohol-related diseases. And these were the things she thought we needed to commit to. Well I ’ ve been trying to say something about our criminal justice system. This country is very different today than it was 40 years ago. In 1972, there were 300, 000 people in jails and prisons. Today, there are 2. 3 million. The United States now has the highest rate of incarceration in the world. We have seven million people on probation and parole. And mass incarceration, in my judgment, has fundamentally changed our world. In poor communities, in communities of color there is this despair, there is this hopelessness, that is being shaped by these outcomes. One out of three black men between the ages of 18 and 30 is in jail, in prison, on probation or parole. In urban communities across this country — Los Angeles, Philadelphia, Baltimore, Washington — 50 to 60 percent of all young men of color are in jail or prison or on probation or parole. Our system isn ' t just being shaped in these ways that seem to be distorting around race, they ' re also distorted by poverty. We have a system of justice in this country that treats you much better if you ' re rich and guilty than if you ' re poor and **innocent**. Wealth, not culpability, shapes outcomes. And yet, we seem to be very comfortable. The politics of fear and anger have made us believe that these are problems that are not our problems. We ' ve been disconnected. It ' s interesting to me. We ' re looking at some very interesting developments in our work. My state of Alabama, like a number of states, actually permanently disenfranchises you if you have a criminal conviction. Right now in Alabama 34 percent of the black male population has permanently lost the right to vote. We ' re actually projecting in another 10 years the level of disenfranchisement will be as high as it ' s been since prior to the passage of the Voting Rights Act. And there is this stunning **silence**. I represent children. A lot of my clients are very young. The United States is the only country in the world where we sentence 13-year-old children to die in prison. We have life imprisonment without parole for kids in this country. And we ' re actually doing some litigation. The only country in the world. I represent people on death row. It ' s interesting, this question of the death penalty. In many ways, we ' ve been taught to think that the real question is, do people deserve to die for the crimes they ' ve committed? And that ' s a very sensible question. But there ' s another way of thinking about where we are in our identity. The other way of thinking about it is not, do people deserve to die for the crimes they commit, but do we deserve to kill? I mean, it ' s fascinating. Death penalty in America is defined by error. For every nine people who have been executed, we ' ve actually identified one **innocent** person who ' s been exonerated and released from death row. A kind of astonishing error rate — one out of nine people **innocent**. I mean, it ' s fascinating. In aviation, we would never let people fly on airplanes if for every nine planes that took off one would crash. But somehow we can insulate ourselves from this problem. It ' s not our problem. It ' s not our burden. It ' s not our struggle. I talk a lot about these issues. I talk about race and this question of whether we deserve to kill. And it ' s interesting, when I teach my students about African-American history, I tell them about slavery. I tell them about terrorism, the era that began at the end of reconstruction that went on to World War II. We don ' t really know very much about it. But for African- Americans in this country, that was an era defined by terror. In many communities, people had to worry about being lynched. They had to worry about being bombed. It was the threat of terror that shaped their lives. And these older people come up to me now and they say,"Mr. Stevenson, you give talks, you make speeches, you tell people to stop saying we ' re dealing with terrorism for the first time in our nation ' s history after 9 / 11."They tell me to say,"No, tell them that we grew up with that."And that era of terrorism, of course, was followed by segregation and decades of racial subordination and apartheid. And yet, we have in this country this dynamic where we really don ' t like to talk about our problems. We don ' t like to talk about our history. And because of that, we really haven ' t understood what it ' s meant to do the things we ' ve done historically. We ' re constantly running into each other. We ' re constantly creating tensions and conflicts. We have a hard time talking about race, and I believe it ' s because we are unwilling to commit ourselves to a process of truth and reconciliation. In South Africa, people understood that we couldn ' t overcome apartheid without a commitment to truth and reconciliation. In Rwanda, even after the genocide, there was this commitment, but in this country we haven ' t done that. I was giving some lectures in Germany about the death penalty. It was fascinating because one of the scholars stood up after the presentation and said,"Well you know it ' s deeply troubling to hear what you ' re talking about."He said,"We don ' t have the death penalty in Germany. And of course, we can never have the death penalty in Germany."And the room got very quiet, and this woman said,"There ' s no way, with our history, we could ever engage in the systematic killing of human beings. It would be unconscionable for us to, in an intentional and deliberate way, set about executing people."And I thought about that. What would it feel like to be living in a world where the nation state of Germany was executing people, especially if they were disproportionately Jewish? I couldn ' t bear it. It would be unconscionable. And yet, in this country, in the states of the Old South, we execute people — where you ' re 11 times more likely to get the death penalty if the **victim** is white than if the **victim** is black, 22 times more likely to get it if the defendant is black and the **victim** is white — in the very states where there are buried in the ground the bodies of people who were lynched. And yet, there is this disconnect. Well I believe that our identity is at risk. That when we actually don ' t care about these difficult things, the positive and wonderful things are nonetheless implicated. We love innovation. We love technology. We love creativity. We love entertainment. But ultimately, those realities are shadowed by **suffering**, abuse, degradation, marginalization. And for me, it becomes necessary to integrate the two. Because ultimately we are talking about a need to be more hopeful, more committed, more dedicated to the basic challenges of living in a complex world. And for me that means spending time thinking and talking about the poor, the disadvantaged, those who will never get to TED. But thinking about them in a way that is integrated in our own lives. You know ultimately, we all have to believe things we haven ' t seen. We do. As rational as we are, as committed to intellect as we are. Innovation, creativity, development comes not from the ideas in our mind alone. They come from the ideas in our mind that are also fueled by some conviction in our heart. And it ' s that mind-heart connection that I believe compels us to not just be attentive to all the bright and dazzly things, but also the dark and difficult things. Vaclav Havel, the great Czech leader, talked about this. He said,"When we were in Eastern Europe and dealing with oppression, we wanted all kinds of things, but mostly what we needed was hope, an orientation of the spirit, a willingness to sometimes be in hopeless places and be a witness."Well that orientation of the spirit is very much at the core of what I believe even TED communities have to be engaged in. There is no disconnect around technology and design that will allow us to be fully human until we pay attention to **suffering**, to poverty, to exclusion, to unfairness, to injustice. Now I will warn you that this kind of identity is a much more challenging identity than ones that don ' t pay attention to this. It will get to you. I had the great privilege, when I was a young lawyer, of meeting Rosa Parks. And Ms. Parks used to come back to Montgomery every now and then, and she would get together with two of her dearest friends, these older women, Johnnie Carr who was the organizer of the Montgomery bus boycott — amazing African- American woman — and Virginia Durr, a white woman, whose husband, Clifford Durr, represented Dr. King. And these women would get together and just talk. And every now and then Ms. Carr would call me, and she ' d say,"Bryan, Ms. Parks is coming to town. We ' re going to get together and talk. Do you want to come over and listen?"And I ' d say,"Yes, Ma ' am, I do."And she ' d say,"Well what are you going to do when you get here?"I said,"I ' m going to listen."And I ' d go over there and I would, I would just listen. It would be so energizing and so empowering. And one time I was over there listening to these women talk, and after a couple of hours Ms. Parks turned to me and she said,”Now Bryan, tell me what the Equal Justice Initiative is. Tell me what you ’ re trying to do.”And I began giving her my rap. I said,”Well we ’ re trying to challenge injustice. We ’ re trying to help people who have been wrongly **convicted**. We ’ re trying to confront bias and **discrimination** in the administration of criminal justice. We ’ re trying to end life without parole sentences for children. We ’ re trying to do something about the death penalty. We ’ re trying to reduce the prison population. We ’ re trying to end mass incarceration.”I gave her my whole rap, and when I finished she looked at me and she said,”Mmm mmm mmm.”She said,”That ’ s going to make you tired, tired, tired.”And that ’ s when Ms. Carr leaned forward, she put her finger in my face, she said,”That ’ s why you ’ ve got to be brave, brave, brave.”And I actually believe that the TED community needs to be more courageous. We need to find ways to embrace these challenges, these problems, the **suffering**. Because ultimately, our humanity depends on everyone ’ s humanity. I ’ ve learned very simple things doing the work that I do. It ’ s just taught me very simple things. I ’ ve come to understand and to believe that each of us is more than the worst thing we ’ ve ever done. I believe that for every person on the planet. I think if somebody tells a lie, they ’ re not just a liar. I think if somebody takes something that doesn ’ t belong to them, they ’ re not just a thief. I think even if you kill someone, you ’ re not just a killer. And because of that there ’ s this basic human dignity that must be respected by law. I also believe that in many parts of this country, and certainly in many parts of this globe, that the opposite of poverty is not wealth. I don ’ t believe that. I actually think, in too many places, the opposite of poverty is justice. And finally, I believe that, despite the fact that it is so dramatic and so beautiful and so inspiring and so stimulating, we will ultimately not be judged by our technology, we won ’ t be judged by our design, we won ’ t be judged by our intellect and reason. Ultimately, you judge the character of a society, not by how they treat their rich and the powerful and the privileged, but by how they treat the poor, the condemned, the incarcerated. Because it ’ s in that nexus that we actually begin to understand truly profound things about who we are. I sometimes get out of balance. I ’ ll end with this story. I sometimes push too hard. I do get tired, as we all do. Sometimes those ideas get ahead of our thinking in ways that are important. And I ’ ve been representing these kids who have been sentenced to do these very harsh sentences. And I go to the jail and I see my client who ’ s 13 and 14, and he ’ s been certified to stand trial as an adult. I start thinking, well, how did that happen? How can a judge turn you into something that you ’ re not? And the judge has certified him as an adult, but I see this kid. And I was up too late one night and I starting thinking, well gosh, if the judge can turn you into something that you ’ re not, the judge must have magic power. Yeah, Bryan, the judge has some magic power. You should ask for some of that. And because I was up too late, wasn ’ t thinking real straight, I started working on a motion. And I had a client who was 14 years old, a young, poor black kid. And I started working on this motion, and the head of the motion was :”Motion to try my poor, 14-year-old black male client like a privileged, white 75-year-old corporate executive.”And I put in my motion that there was prosecutorial misconduct and police misconduct and judicial misconduct. There was a crazy line in there about how there ’ s no conduct in this county, it ’ s all misconduct. And the next morning, I woke up and I thought, now did I dream that crazy motion, or did I actually write it? And to my horror, not only had I written it, but I had sent it to court. A couple months went by, and I had just forgotten all about it. And I finally decided, oh gosh, I ’ ve got to go to the court and do this crazy case. And I got into my car and I was feeling really overwhelmed — overwhelmed. And I got in my car and I went to this courthouse. And I was thinking, this is going to be so difficult, so painful. And I finally got out of the car and I started walking up to the courthouse. And as I was walking up the steps of this courthouse, there was an older black man who was the janitor in this courthouse. When this man saw me, he came over to me and he said,"Who are you?"I said,"I ' m a lawyer."He said,"You ' re a lawyer?"I said,"Yes, sir."And this man came over to me and he hugged me. And he whispered in my ear. He said,"I ' m so proud of you."And I have to tell you, it was energizing. It connected deeply with something in me about identity, about the capacity of every person to contribute to a community, to a perspective that is hopeful. Well I went into the courtroom. And as soon as I walked inside, the judge saw me coming in. He said,"Mr. Stevenson, did you write this crazy motion?"I said,"Yes, sir. I did."And we started arguing. And people started coming in because they were just outraged. I had written these crazy things. And police officers were coming in and assistant prosecutors and clerk workers. And before I knew it, the courtroom was filled with people angry that we were talking about race, that we were talking about poverty, that we were talking about inequality. And out of the corner of my eye, I could see this janitor pacing back and forth. And he kept looking through the window, and he could hear all of this holler. He kept pacing back and forth. And finally, this older black man with this very worried look on his face came into the courtroom and sat down behind me, almost at counsel table. About 10 minutes later the judge said we would take a break. And during the break there was a deputy sheriff who was offended that the janitor had come into court. And this deputy jumped up and he ran over to this older black man. He said,"Jimmy, what are you doing in this courtroom?"And this older black man stood up and he looked at that deputy and he looked at me and he said,"I came into this courtroom to tell this young man, keep your eyes on the prize, hold on."I ' ve come to TED because I believe that many of you understand that the moral arc of the universe is long, but it bends toward justice. That we cannot be full evolved human beings until we care about human rights and basic dignity. That all of our survival is tied to the survival of everyone. That our visions of technology and design and entertainment and creativity have to be married with visions of humanity, compassion and justice. And more than anything, for those of you who share that, I ' ve simply come to tell you to keep your eyes on the prize, hold on. Thank you very much. Chris Anderson : So you heard and saw an obvious desire by this audience, this community, to help you on your way and to do something on this issue. Other than writing a check, what could we do? BS : Well there are opportunities all around us. If you live in the state of California, for example, there ' s a referendum coming up this spring where actually there ' s going to be an effort to redirect some of the money we spend on the politics of punishment. For example, here in California we ' re going to spend one billion dollars on the death penalty in the next five years — one billion dollars. And yet, 46 percent of all homicide cases don ' t result in arrest. 56 percent of all rape cases don ' t result. So there ' s an opportunity to change that. And this referendum would propose having those dollars go to law **enforcement** and **safety**. And I think that opportunity exists all around us. CA : There ' s been this huge decline in crime in America over the last three decades. And part of the narrative of that is sometimes that it ' s about increased incarceration rates. What would you say to someone who believed that? BS : Well actually the **violent** crime rate has remained relatively stable. The great increase in mass incarceration in this country wasn ' t really in **violent** crime categories. It was this misguided war on drugs. That ' s where the dramatic increases have come in our prison population. And we got carried away with the rhetoric of punishment. And so we have three strikes laws that put people in prison forever for stealing a bicycle, for low-level property crimes, rather than making them give those resources back to the people who they victimized. I believe we need to do more to help people who are victimized by crime, not do less. And I think our current punishment philosophy does nothing for no one. And I think that ’ s the orientation that we have to change. CA : Bryan, you ’ ve struck a massive chord here. You ’ re an inspiring person. Thank you so much for coming to TED. Thank you.

Text Sample 382: POS Dim 3 – 2012 – Score 13.00 – 2012/t002227.txt

Thank you very much. Thank you. It ’ s a distinct privilege to be here. A few weeks ago, I saw a video on YouTube of Congresswoman Gabrielle Giffords at the early stages of her recovery from one of those awful bullets. This one entered her left hemisphere, and knocked out her Broca ’ s area, the speech center of her brain. And in this session, Gabby ’ s working with a speech therapist, and she ’ s struggling to produce some of the most basic words, and you can see her growing more and more devastated, until she ultimately breaks down into sobbing tears, and she starts sobbing wordlessly into the arms of her therapist. And after a few moments, her therapist tries a new tack, and they start **singing** together, and Gabby starts to **sing** through her tears, and you can hear her clearly able to enunciate the words to a song that describe the way she feels, and she **sings**, in one descending scale, she **sings**,”Let it shine, let it shine, let it shine.”And it ’ s a very powerful and poignant reminder of how the beauty of music has the ability to speak where words fail, in this case literally speak. Seeing this video of Gabby Giffords reminded me of the work of Dr. Gottfried Schlaug, one of the preeminent neuroscientists studying music and the brain at Harvard, and Schlaug is a proponent of a therapy called Melodic Intonation Therapy, which has become very popular in music therapy now. Schlaug found that his stroke **victims** who were aphasic, could not form sentences of three- or four-word sentences, but they could still **sing** the lyrics to a song, whether it was”Happy Birthday To You”or their favorite song by the Eagles or the Rolling Stones. And after 70 hours of intensive singing lessons, he found that the music was able to literally rewire the brains of his patients and create a homologous speech center in their right hemisphere to compensate for the left hemisphere ’ s damage. When I was 17, I visited Dr. Schlaug ’ s lab, and in one afternoon he walked me through some of the leading research on music and the brain — how musicians had fundamentally different brain structure than non-musicians, how music, and listening to music, could just light up the entire brain, from our prefrontal cortex all the way back to our cerebellum, how music was becoming a neuropsychiatric modality to help children with autism, to help people struggling with stress and anxiety and depression, how deeply Parkinsonian patients would find that their tremor and their gait would steady when they listened to music, and how late-stage Alzheimer ’ s patients, whose dementia was so far progressed that they could no longer recognize their family, could still pick out a tune by Chopin at the piano that they had learned when they were children. But I had an ulterior motive of visiting Gottfried Schlaug, and it was this : that I was at a crossroads in my life, trying to choose between music and medicine. I had just completed my undergraduate, and I was working as a research assistant at the lab of Dennis Selkoe, studying Parkinson ’ s disease at Harvard, and I had fallen in love with neuroscience. I wanted to become a surgeon. I wanted to become a doctor like Paul Farmer or Rick Hodes, these kind of fearless men who go into places like Haiti or Ethiopia and work with AIDS patients with multidrug-resistant tuberculosis, or with children with disfiguring cancers. I wanted to become that kind of Red Cross doctor, that doctor without borders. On the other hand, I had played the violin my entire life. Music for me was more than a **passion**. It was obsession. It was oxygen. I was lucky enough to have studied at the Juilliard School in Manhattan, and to have played my debut with Zubin Mehta and the Israeli philharmonic orchestra in Tel Aviv, and it turned out that Gottfried Schlaug had studied as an organist at the Vienna Conservatory, but had given up his love for music to pursue a career in medicine. And that afternoon, I had to ask him,"How was it for you making that decision?"And he said that there were still times when he wished he could go back and play the organ the way he used to, and that for me, medical school could wait, but that the violin simply would not. And after two more years of studying music, I decided to shoot for the impossible before taking the MCAT and applying to medical school like a good Indian son to become the next Dr. Gupta. And I decided to shoot for the impossible and I took an audition for the esteemed Los Angeles Philharmonic. It was my first audition, and after three days of playing behind a screen in a trial week, I was offered the position. And it was a dream. It was a wild dream to perform in an orchestra, to perform in the iconic Walt Disney Concert Hall in an orchestra conducted now by the famous Gustavo Dudamel, but much more importantly to me to be surrounded by musicians and mentors that became my new family, my new musical home. But a year later, I met another musician who had also studied at Juilliard, one who profoundly helped me find my voice and shaped my identity as a musician. Nathaniel Ayers was a double bassist at Juilliard, but he suffered a series of psychotic episodes in his early 20s, was treated with thorazine at Bellevue, and ended up living **homeless** on the **streets** of Skid Row in downtown Los Angeles 30 years later. Nathaniel ' s story has become a beacon for **homelessness** and mental health advocacy throughout the United States, as told through the book and the movie"The Soloist,"but I became his friend, and I became his violin teacher, and I told him that wherever he had his violin, and wherever I had mine, I would play a lesson with him. And on the many times I saw Nathaniel on Skid Row, I witnessed how music was able to bring him back from his very darkest moments, from what seemed to me in my untrained eye to be the beginnings of a schizophrenic episode. Playing for Nathaniel, the music took on a deeper meaning, because now it was about communication, a communication where words failed, a communication of a message that went deeper than words, that registered at a fundamentally primal level in Nathaniel ' s psyche, yet came as a true musical offering from me. I found myself growing outraged that someone like Nathaniel could have ever been **homeless** on Skid Row because of his mental illness, yet how many tens of thousands of others there were out there on Skid Row alone who had stories as tragic as his, but were never going to have a book or a movie made about them that got them off the **streets**? And at the very core of this crisis of mine, I felt somehow the life of music had chosen me, where somehow, perhaps possibly in a very naive sense, I felt what Skid Row really needed was somebody like Paul Farmer and not another classical musician playing on Bunker Hill. But in the end, it was Nathaniel who showed me that if I was truly passionate about change, if I wanted to make a difference, I already had the perfect instrument to do it, that music was the bridge that connected my world and his. There ' s a beautiful quote by the Romantic German composer Robert Schumann, who said,"To send light into the **darkness** of men ' s hearts, such is the duty of the artist."And this is a particularly poignant quote because Schumann himself suffered from schizophrenia and died in asylum. And inspired by what I learned from Nathaniel, I started an organization on Skid Row of musicians called Street Symphony, bringing the light of music into the very darkest places, performing for the **homeless** and mentally **ill** at shelters and clinics on Skid Row, performing for combat veterans with post-traumatic stress disorder, and for the incarcerated and those labeled as criminally insane. After one of our events at the Patton State Hospital in San Bernardino, a woman walked up to us and she had tears streaming down her face, and she had a palsy, she was shaking, and she had this gorgeous smile, and she said that she had never heard classical music before, she didn ’ t think she was going to like it, she had never heard a violin before, but that hearing this music was like hearing the sunshine, and that nobody ever came to visit them, and that for the first time in six years, when she heard us play, she stopped shaking without medication. Suddenly, what we ’ re finding with these concerts, away from the stage, away from the footlights, out of the tuxedo tails, the musicians become the conduit for delivering the tremendous therapeutic benefits of music on the brain to an audience that would never have **access** to this room, would never have **access** to the kind of music that we make. Just as medicine serves to **heal** more than the building blocks of the body alone, the power and beauty of music transcends the”E”in the middle of our beloved acronym. Music transcends the aesthetic beauty alone. The synchrony of emotions that we experience when we hear an **opera** by Wagner, or a symphony by Brahms, or chamber music by Beethoven, compels us to remember our **shared**, common humanity, the deeply communal connected consciousness, the empathic consciousness that neuropsychiatrist Iain McGilchrist says is hard-wired into our brain ’ s right hemisphere. And for those living in the most dehumanizing conditions of mental illness within **homelessness** and incarceration, the music and the beauty of music offers a chance for them to transcend the world around them, to remember that they still have the capacity to experience something beautiful and that humanity has not forgotten them. And the spark of that beauty, the spark of that humanity transforms into hope, and we know, whether we choose the path of music or of medicine, that ’ s the very first thing we must instill within our communities, within our audiences, if we want to inspire healing from within. I ’ d like to end with a quote by John Keats, the Romantic English poet, a very famous quote that I ’ m sure all of you know. Keats himself had also given up a career in medicine to pursue poetry, but he died when he was a year older than me. And Keats said,”Beauty is truth, and truth beauty. That is all ye know on Earth, and all ye need to know.”

Text Sample 383: POS Dim 3 – 2012 – Score 13.00 – 2012/t002040.txt

I ’ m going to share with you a paradigm-shifting perspective on the issues of gender **violence** : **sexual assault**, **domestic violence**, relationship abuse, **sexual harassment**, **sexual** abuse of children. That whole range of issues that I ’ ll refer to in shorthand as”gender **violence** issues,”they ’ ve been seen as women ’ s issues that some good men help out with, but I have a problem with that frame and I don ’ t accept it. I don ’ t see these as women ’ s issues that some good men help out with. In fact, I ’ m going to argue that these are men ’ s issues, first and foremost. Now obviously — Obviously, they ’ re also women ’ s issues, so I appreciate that, but calling gender **violence** a women ’ s issue is part of the problem, for a number of reasons. The first is that it gives men an excuse not to pay attention, right? A lot of men hear the term”women ’ s issues”and we tend to tune it out, and we think,”I ’ m a guy ; that ’ s for the girls,”or”that ’ s for the women.”And a lot of men literally don ’ t get beyond the first sentence as a result. It ’ s almost like a chip in our brain is activated, and the neural pathways take our attention in a different direction when we hear the term”women ' s issues.”This is also true, by the way, of the word”gender,”because a lot of people hear the word”gender”and they think it means”women.”So they think that gender issues is synonymous with women ' s issues. There ' s some confusion about the term gender. And let me illustrate that confusion by way of analogy. So let ' s talk for a moment about race. In the US, when we hear the word”race,”a lot of people think that means African-American, Latino, Asian-American, Native American, South Asian, Pacific Islander, on and on. A lot of people, when they hear the word”**sexual** orientation”think it means **gay**, lesbian, bisexual. And a lot of people, when they hear the word”gender,”think it means women. In each case, the dominant group doesn ' t get paid attention to. As if white people don ' t have some sort of racial identity or belong to some racial category or construct, as if heterosexual people don ' t have a **sexual** orientation, as if men don ' t have a gender. This is one of the ways that dominant systems maintain and reproduce themselves, which is to say the dominant group is rarely challenged to even think about its dominance, because that ' s one of the key characteristics of power and privilege, the ability to go unexamined, lacking introspection, in fact being rendered invisible, in large measure, in the discourse about issues that are primarily about us. And this is amazing how this works in **domestic** and **sexual violence**, how men have been largely erased from so much of the conversation about a subject that is centrally about men. And I ' m going to illustrate what I ' m talking about by using the old tech. I ' m old school on some fundamental regards. I make films and I work with high tech, but I ' m still old school as an educator, and I want to share with you this exercise that illustrates on the sentence-structure level how the way that we think, literally the way that we use language, conspires to keep our attention off of men. This is about **domestic violence** in particular, but you can plug in other analogues. This comes from the work of the feminist linguist Julia Penelope. It starts with a very basic English sentence :”John beat Mary.”That ' s a good English sentence. John is the subject, beat is the verb, Mary is the object, good sentence. Now we ' re going to move to the second sentence, which says the same thing in the passive voice.”Mary was beaten by John.”And now a whole lot has happened in one sentence. We ' ve gone from”John beat Mary”to”Mary was beaten by John.”We ' ve shifted our focus in one sentence from John to Mary, and you can see John is very close to the end of the sentence, well, close to dropping off the map of our psychic plain. The third sentence, John is dropped, and we have,”Mary was beaten,”and now it ' s all about Mary. We ' re not even thinking about John, it ' s totally focused on Mary. Over the past generation, the term we ' ve used synonymous with”beaten”is”battered,”so we have”Mary was battered.”And the final sentence in this sequence, flowing from the others, is,”Mary is a battered woman.”So now Mary ' s very identity — Mary is a battered woman — is what was done to her by John in the first instance. But we ' ve demonstrated that John has long ago left the conversation. Those of us who work in the **domestic** and **sexual violence** field know that victim-blaming is pervasive in this realm, which is to say, blaming the person to whom something was done rather than the person who did it. And we say : why do they go out with these men? Why are they attracted to them? Why do they keep going back? What was she wearing at that party? What a stupid thing to do. Why was she drinking with those guys in that hotel room? This is **victim** blaming, and there are many reasons for it, but one is that our cognitive structure is set up to blame **victims**. This is all unconscious. Our whole cognitive structure is set up to ask questions about women and women ' s choices and what they ' re doing, thinking, wearing. And I ' m not going to shout down people who ask questions about women. It ' s a legitimate thing to ask. But ' s let ’ s be clear : Asking questions about Mary is not going to get us anywhere in terms of preventing **violence**. We have to ask a different set of questions. The questions are not about Mary, they ’ re about John. They include things like, why does John beat Mary? Why is **domestic violence** still a big problem in the US and all over the world? What ’ s going on? Why do so many men abuse physically, emotionally, verbally, and other ways, the women and girls, and the men and boys, that they claim to love? What ’ s going on with men? Why do so many adult men sexually abuse little girls and boys? Why is that a common problem in our society and all over the world today? Why do we hear over and over again about new scandals erupting in major institutions like the Catholic Church or the Penn State football program or the Boy Scouts of America, on and on and on? And then local communities all over the country and all over the world. We hear about it all the time. The **sexual** abuse of children. What ’ s going on with men? Why do so many men rape women in our society and around the world? Why do so many men rape other men? What is going on with men? And then what is the role of the various institutions in our society that are helping to produce abusive men at pandemic rates? Because this isn ’ t about individual **perpetrators**. That ’ s a naive way to understanding what is a much deeper and more systematic social problem. The **perpetrators** aren ’ t these monsters who crawl out of the swamp and come into town and do their nasty business and then retreat into the **darkness**. That ’ s a very naive notion, right? **Perpetrators** are much more normal than that, and everyday than that. So the question is, what are we doing here in our society and in the world? What are the roles of various institutions in helping to produce abusive men? What ’ s the role of religious belief systems, the sports culture, the pornography culture, the family structure, economics, and how that intersects, and race and **ethnicity** and how that intersects? How does all this work? And then, once we start making those kinds of connections and asking those important and big questions, then we can talk about how we can be transformative, in other words, how can we do something differently? How can we change the practices? How can we change the socialization of boys and the definitions of manhood that lead to these current outcomes? These are the kind of questions that we need to be asking and the kind of work that we need to be doing, but if we ’ re endlessly focused on what women are doing and thinking in relationships or elsewhere, we ’ re not going to get to that piece. I understand that a lot of women who have been trying to speak out about these issues, today and yesterday and for years and years, often get shouted down for their efforts. They get called nasty names like”male-basher”and”man-hater,”and the disgusting and offensive”feminazi”, right? And you know what all this is about? It ’ s called kill the messenger. It ’ s because the women who are standing up and speaking out for themselves and for other women as well as for men and boys, it ’ s a statement to them to sit down and shut up, keep the current system in place, because we don ’ t like it when people rock the boat. We don ’ t like it when people challenge our power. You ’ d better sit down and shut up, basically. And thank goodness that women haven ’ t done that. Thank goodness that we live in a world where there ’ s so much women ’ s leadership that can counteract that. But one of the powerful roles that men can play in this work is that we can say some things that sometimes women can ’ t say, or, better yet, we can be heard saying some things that women often can ’ t be heard saying. Now, I appreciate that that ’ s a problem, it ’ s sexism, but it ’ s the truth. So one of the things that I say to men, and my colleagues and I always say this, is we need more men who have the courage and the strength to start standing up and saying some of this stuff, and standing with women and not against them and pretending that somehow this is a battle between the sexes and other kinds of nonsense. We live in the world together. And by the way, one of the things that really bothers me about some of the rhetoric against feminists and others who have built the battered women ' s and rape crisis movements around the world is that somehow, like I said, that they ' re anti-male. What about all the boys who are profoundly affected in a negative way by what some adult man is doing against their mother, themselves, their sisters? What about all those boys? What about all the young men and boys who have been traumatized by adult men ' s **violence**? You know what? The same system that produces men who abuse women, produces men who abuse other men. And if we want to talk about male **victims**, let ' s talk about male **victims**. Most male **victims** of **violence** are the **victims** of other men ' s **violence**. So that ' s something that both women and men have in common. We are both **victims** of men ' s **violence**. So we have it in our direct self-interest, not to mention the fact that most men that I know have women and girls that we care deeply about, in our families and our friendship circles and every other way. So there ' s so many reasons why we need men to speak out. It seems obvious saying it out loud, doesn ' t it? Now, the nature of the work that I do and my colleagues do in the sports culture and the US military, in schools, we pioneered this approach called the bystander approach to gender-violence prevention. And I just want to give you the highlights of the bystander approach, because it ' s a big thematic shift, although there ' s lots of particulars, but the heart of it is, instead of seeing men as **perpetrators** and women as **victims**, or women as **perpetrators**, men as **victims**, or any combination in there. I ' m using the gender binary. I know there ' s more than men and women, there ' s more than male and female. And there are women who are **perpetrators**, and of course there are men who are **victims**. There ' s a whole spectrum. But instead of seeing it in the binary fashion, we focus on all of us as what we call bystanders, and a bystander is defined as anybody who is not a **perpetrator** or a **victim** in a given situation, so in other words friends, teammates, colleagues, coworkers, family members, those of us who are not directly involved in a dyad of abuse, but we are embedded in social, family, work, school, and other peer culture relationships with people who might be in that situation. What do we do? How do we speak up? How do we challenge our friends? How do we support our friends? But how do we not remain silent in the face of abuse? Now, when it comes to men and male culture, the goal is to get men who are not abusive to challenge men who are. And when I say abusive, I don ' t mean just men who are beating women. We ' re not just saying a man whose friend is abusing his girlfriend needs to stop the guy at the moment of attack. That ' s a naive way of creating a social change. It ' s along a continuum, we ' re trying to get men to interrupt each other. So, for example, if you ' re a guy and you ' re in a group of guys playing poker, talking, hanging out, no women present, and another guy says something sexist or degrading or harassing about women, instead of laughing along or pretending you didn ' t hear it, we need men to say,"Hey, that ' s not funny. that could be my sister you ' re talking about, and could you joke about something else? Or could you talk about something else? I don ' t appreciate that kind of talk."Just like if you ' re a white person and another white person makes a racist comment, you ' d hope, I hope, that white people would interrupt that racist enactment by a fellow white person. Just like with heterosexism, if you ' re a heterosexual person and you yourself don ' t enact harassing or abusive behaviors towards people of varying **sexual** orientations, if you don ' t say something in the face of other heterosexual people doing that, then, in a sense, isn ' t your **silence** a form of consent and complicity? Well, the bystander approach is trying to give people tools to interrupt that process and to speak up and to create a peer culture climate where the abusive behavior will be seen as unacceptable, not just because it ’ s illegal, but because it ’ s wrong and unacceptable in the peer culture. And if we can get to the place where men who act out in sexist ways will lose status, young men and boys who act out in sexist and harassing ways towards girls and women, as well as towards other boys and men, will lose status as a result of it, guess what? We ’ ll see a radical diminution of the abuse. Because the typical **perpetrator** is not sick and twisted. He ’ s a normal guy in every other way, isn ’ t he? Now, among the many great things that Martin Luther King said in his short life was,”In the end, what will hurt the most is not the words of our enemies but the **silence** of our friends.”In the end, what will hurt the most is not the words of our enemies but the **silence** of our friends. There ’ s been an awful lot of **silence** in male culture about this ongoing tragedy of men ’ s **violence** against women and children, hasn ’ t there? There ’ s been an awful lot of **silence**. And all I ’ m saying is that we need to break that **silence**, and we need more men to do that. Now, it ’ s easier said than done, because I ’ m saying it now, but I ’ m telling you it ’ s not easy in male culture for guys to challenge each other, which is one of the reasons why part of the paradigm shift that has to happen is not just understanding these issues as men ’ s issues, but they ’ re also leadership issues for men. Because ultimately, the responsibility for taking a stand on these issues should not fall on the shoulders of little boys or teenage boys in high school or college men. It should be on adult men with power. Adult men with power are the ones we need to be holding accountable for being leaders on these issues, because when somebody speaks up in a peer culture and challenges and interrupts, he or she is being a leader, really. But on a big scale, we need more adult men with power to start prioritizing these issues, and we haven ’ t seen that yet, have we? Now, I was at a dinner a number of years ago, and I work extensively with the US military, all the services. And I was at this dinner and this woman said to me — I think she thought she was a little clever — she said,”So how long have you been doing sensitivity training with the Marines?”And I said,”With all due respect, I don ’ t do sensitivity training with the Marines. I run a leadership program in the Marine Corps.”Now, I know it ’ s a bit pompous, my response, but it ’ s an important distinction, because I don ’ t believe that what we need is sensitivity training. We need leadership training, because, for example, when a professional coach or a manager of a baseball team or a football team — and I work extensively in that realm as well — makes a sexist comment, makes a homophobic statement, makes a racist comment, there will be discussions on the sports blogs and in sports talk radio. And some people will say,”He needs sensitivity training.”Other people will say,”Well, get off it. That ’ s political correctness run amok, he made a stupid statement, move on.”My argument is, he doesn ’ t need sensitivity training. He needs leadership training, because he ’ s being a bad leader, because in a society with gender diversity and **sexual** diversity — and racial and ethnic diversity, you make those kind of comments, you ’ re failing at your leadership. If we can make this point that I ’ m making to powerful men and women in our society at all levels of institutional authority and power, it ’ s going to change the paradigm of people ’ s thinking. You know, for example, I work a lot in college and university athletics throughout North America. We know so much about how to prevent **domestic** and **sexual violence**, right? There ’ s no excuse for a college or university to not have **domestic** and **sexual violence** prevention training mandated for all student athletes, coaches, administrators, as part of their educational process. We know enough to know that we can easily do that. But you know what ’ s missing? The leadership. But it ’ s not the leadership of student athletes. It ’ s the leadership of the athletic director, the president of the university, the people in charge who make decisions about resources and who make decisions about priorities in the institutional settings. That ' s a failure, in most cases, of men ' s leadership. Look at Penn State. Penn State is the mother of all teachable moments for the bystander approach. You had so many situations in that realm where men in powerful positions failed to act to protect children, in this case, boys. It ' s unbelievable, really. But when you get into it, you realize there are pressures on men. There are constraints within peer cultures on men, which is why we need to encourage men to break through those pressures. And one of the ways to do that is to say there ' s an awful lot of men who care deeply about these issues. I know this, I work with men, and I ' ve been working with tens of thousands, hundreds of thousands of men for many decades now. It ' s scary, when you think about it, how many years. But there ' s so many men who care deeply about these issues, but caring deeply is not enough. We need more men with the guts, with the courage, with the strength, with the moral integrity to break our complicit **silence** and challenge each other and stand with women and not against them. By the way, we owe it to women. There ' s no question about it. But we also owe it to our sons. We also owe it to young men who are growing up all over the world in situations where they didn ' t make the choice to be a man in a culture that tells them that manhood is a certain way. They didn ' t make the choice. We that have a choice, have an opportunity and a responsibility to them as well. I hope that, going forward, men and women, working together, can begin the change and the transformation that will happen so that future generations won ' t have the level of tragedy that we deal with on a daily basis. I know we can do it, we can do better. Thank you very much.

Text Sample 384: POS Dim 3 – 2005 – Score 12.00 – 2005/t000416.txt

I am a vicar in the Church of England. I ' ve been a priest in the Church for 20 years. For most of that time, I ' ve been struggling and grappling with questions about the nature of God. Who is God? And I ' m very aware that when you say the word"God,"many people will turn off immediately. And most people, both within and outside the organized church, still have a picture of a celestial controller, a rule maker, a policeman in the sky who orders everything, and causes everything to happen. He will protect his own people, and answer the **prayers** of the faithful. And in the worship of my church, the most frequently used adjective about God is"almighty."But I have a problem with that. I have become more and more uncomfortable with this perception of God over the years. Do we really believe that God is the kind of male boss that we ' ve been presenting in our worship and in our liturgies over all these years? Of course, there have been thinkers who have suggested different ways of looking at God. Exploring the feminine, nurturing side of divinity. Suggesting that God expresses Himself or Herself through powerlessness, rather than power. Acknowledging that God is unknown and unknowable by definition. Finding deep resonances with other religions and philosophies and ways of looking at life as part of what is a universal and global search for meaning. These ideas are well known in liberal academic circles, but clergy like myself have been reluctant to air them, for fear of creating tension and division in our church communities, for fear of upsetting the simple faith of more traditional believers. I have chosen not to rock the boat. Then, on December 26th last year, just two months ago, that underwater earthquake triggered the tsunami. And two weeks later, Sunday morning, 9th of January, I found myself standing in front of my congregation — intelligent, well meaning, mostly thoughtful Christian people — and I needed to express, on their behalf, our feelings and our questions. I had my own personal responses, but I also have a public role, and something needed to be said. And this is what I said. Shortly after the tsunami I read a newspaper article written by the Archbishop of Canterbury — fine title — about the tragedy in Southern Asia. The essence of what he said was this : the people most affected by the devastation and loss of life do not want intellectual theories about how God let this happen. He wrote,"If some religious genius did come up with an explanation of exactly why all these deaths made sense, would we feel happier, or safer, or more confident in God?"If the man in the photograph that appeared in the newspapers, holding the hand of his dead child was standing in front of us now, there are no words that we could say to him. A verbal response would not be appropriate. The only appropriate response would be a compassionate **silence** and some kind of practical help. It isn ' t a time for explanation, or preaching, or theology ; it ' s a time for tears. This is true. And yet here we are, my church in Oxford, semi-detached from events that happened a long way away, but with our faith bruised. And we want an explanation from God. We demand an explanation from God. Some have concluded that we can only believe in a God who shares our pain. In some way, God must feel the anguish, and **grief**, and physical pain that we feel. In some way the eternal God must be able to enter into the souls of human beings and experience the torment within. And if this is true, it must also be that God knows the joy and exaltation of the human spirit, as well. We want a God who can weep with those who weep, and rejoice with those who rejoice. This seems to me both a deeply moving and a convincing re-statement of Christian belief about God. For hundreds of years, the prevailing orthodoxy, the accepted truth, was that God the Father, the Creator, is unchanging and therefore by definition cannot feel pain or sadness. Now the unchanging God feels a bit cold and indifferent to me. And the devastating events of the 20th century have forced people to question the cold, unfeeling God. The slaughter of millions in the trenches and in the death camps have caused people to ask,"Where is God in all this? Who is God in all this?"And the answer was,"God is in this with us, or God doesn ' t deserve our allegiance anymore."If God is a bystander, observing but not involved, then God may well exist, but we don ' t want to know about Him. Many Jews and Christians now feel like this, I know. And I am among them. So we have a suffering God — a God who is intimately connected with this world and with every living soul. I very much relate to this idea of God. But it isn ' t enough. I need to ask some more questions, and I hope they are questions that you will want to ask, as well, some of you. Over the last few weeks I have been struck by the number of times that words in our worship have felt a bit inappropriate, a bit dodgy. We have a pram service on Tuesday mornings for mums and their pre-school children. And last week we **sang** with the children one of their favorite songs,"The Wise Man Built His House Upon the Rock."Perhaps some of you know it. Some of the words go like this :"The foolish man built his house upon the sand / And the floods came up / And the house on the sand went crash."Then in the same week, at a funeral, we **sang** the familiar hymn"We Plow the Fields and Scatter,"a very English hymn. In the second verse comes the line,"The wind and waves obey Him."Do they? I don ' t feel we can **sing** that song again in church, after what ' s happened. So the first big question is about control. Does God have a plan for each of us? Is God in control? Does God order each moment? Does the wind and the waves obey Him? From time to time, one hears Christians telling the story of how God organized things for them, so that everything worked out all right — some difficulty overcome, some illness cured, some trouble averted, a parking space found at a crucial time. I can remember someone saying this to me, with her eyes shining with enthusiasm at this wonderful confirmation of her faith and the goodness of God. But if God can or will do these things —

intervene to change the flow of events — then surely he could have stopped the tsunami. Do we have a local God who can do little things like parking spaces, but not big things like 500 mile-per-hour waves? That ' s just not acceptable to intelligent Christians, and we must acknowledge it. Either God is responsible for the tsunami, or God is not in control. After the tragedy, survival stories began to emerge. You probably heard some of them : the man who surfed the wave, the teenage girl who recognized the danger because she had just been learning about tsunamis at school. Then there was the congregation who had left their usual church building on the shore to hold a service in the hills. The preacher delivered an extra long sermon, so that they were still out of **harm** ' s way when the wave struck. Afterwards someone said that God must have been looking after them. So the next question is about partiality. Can we earn God ' s favor by worshipping Him or believing in Him? Does God demand loyalty, like any medieval tyrant? A God who looks after His own, so that Christians are OK, while everyone else perishes? A cosmic us and them, and a God who is guilty of the worst kind of favoritism? That would be appalling, and that would be the point at which I would hand in my membership. Such a God would be morally inferior to the highest ideals of humanity. So who is God, if not the great puppet-master or the tribal protector? Perhaps God allows or permits terrible things to happen, so that heroism and compassion can be shown. Perhaps God is testing us : testing our charity, or our faith. Perhaps there is a great, cosmic plan that allows for horrible **suffering** so that everything will work out OK in the end. Perhaps, but these ideas are all just variations on God controlling everything, the supreme commander toying with expendable units in a great campaign. We are still left with a God who can do the tsunami and allow Auschwitz. In his great novel,"The Brothers Karamazov,"Dostoevsky gives these words to Ivan, addressed to his naive and devout younger brother, Alyosha :"If the **sufferings** of children go to make up the sum of **sufferings** which is necessary for the purchase of truth, then I say beforehand that the entire truth is not worth such a price. We cannot afford to pay so much for admission. It is not God that I do not accept. I merely, most respectfully, return Him the ticket."Or perhaps God set the whole universe going at the beginning and then relinquished control forever, so that natural processes could occur, and evolution run its course. This seems more acceptable, but it still leaves God with the ultimate moral responsibility. Is God a cold, unfeeling spectator? Or a powerless lover, watching with infinite compassion things God is unable to control or change? Is God intimately involved in our **suffering**, so that He feels it in His own being? If we believe something like this, we must let go of the puppet-master completely, take our leave of the almighty controller, abandon traditional models. We must think again about God. Maybe God doesn ' t do things at all. Maybe God isn ' t an agent like all of us are agents. Early religious thought conceived God as a sort of superhuman person, doing things all over the place. Beating up the Egyptians, drowning them in the Red Sea, wasting cities, getting angry. The people knew their God by His mighty acts. But what if God doesn ' t act? What if God doesn ' t do things at all? What if God is in things? The **loving** soul of the universe. An in-dwelling compassionate presence, underpinning and sustaining all things. What if God is in things? In the infinitely complex network of relationships and connections that make up life. In the natural cycle of life and death, the creation and destruction that must happen continuously. In the process of evolution. In the incredible intricacy and magnificence of the natural world. In the collective unconscious, the soul of the human race. In you, in me, mind and body and spirit. In the tsunami, in the **victims**. In the depth of things. In presence and in absence. In simplicity and complexity. In change and development and growth. How does this in-ness, this innerness, this interiority of God work? It ’ s hard to conceive, and begs more questions. Is God just another name for the universe, with no independent existence at all? I don ’ t know. To what extent can we ascribe personality to God? I don ’ t know. In the end, we have to say,”I don ’ t know.”If we knew, God would not be God. To have faith in this God would be more like trusting an essential benevolence in the universe, and less like believing a system of doctrinal statements. Isn ’ t it **ironic** that Christians who claim to believe in an infinite, unknowable being then tie God down in closed systems and rigid doctrines? How could one practice such a faith? By seeking the God within. By cultivating my own inwardness. In **silence**, in **meditation**, in my inner space, in the me that remains when I gently put aside my passing emotions and ideas and preoccupations. In awareness of the inner conversation. And how would we live such a faith? How would I live such a faith? By seeking intimate connection with your inwardness. The kind of relationships when deep speaks to deep. If God is in all people, then there is a meeting place where my relationship with you becomes a three-way encounter. There is an Indian greeting, which I ’ m sure some of you know :”Namaste,”accompanied by a respectful bow, which, roughly translated means,”That which is of God in me greets that which of God is in you.”Namaste. And how would one deepen such a faith? By seeking the inwardness which is in all things. In music and poetry, in the natural world of beauty and in the small ordinary things of life, there is a deep, indwelling presence that makes them extraordinary. It needs a profound attentiveness and a patient waiting, a contemplative **attitude** and a generosity and **openness** to those whose experience is different from my own. When I stood up to speak to my people about God and the tsunami, I had no answers to offer them. No neat packages of faith, with Bible references to prove them. Only doubts and questioning and uncertainty. I had some suggestions to make — possible new ways of thinking about God. Ways that might allow us to go on, down a new and uncharted road. But in the end, the only thing I could say for sure was,”I don ’ t know,”and that just might be the most profoundly religious statement of all. Thank you.

Text Sample 385: POS Dim 3 – 2005 – Score 11.00 – 2005/t000220.txt

I think it ’ ll be a relief to some people and a disappointment to others that I ’ m not going to talk about vaginas today. I began”The Vagina Monologues”because I was worried about vaginas. I ’ m very worried today about this notion, this world, this prevailing kind of force of security. I see this word, hear this word, feel this word everywhere. Real security, security checks, security watch, security clearance. Why has all this focus on security made me feel so much more insecure? What does anyone mean when they talk about real security? And why have we, as Americans particularly, become a nation that strives for security above all else? In fact, I think that security is elusive. It ’ s impossible. We all die. We all get old. We all get sick. People leave us. People change us. Nothing is secure. And that ’ s actually the good news. This is, of course, unless your whole life is about being secure. I think that when that is the focus of your life, these are the things that happen. You can ’ t travel very far or venture too far outside a certain circle. You can ’ t allow too many conflicting ideas into your mind at one time, as they might confuse you or challenge you. You can ’ t open yourself to new experiences, new people, new ways of doing things — they might take you off course. You can ’ t not know who you are, so you cling to hard-matter identity. You become a Christian, Muslim, Jew. You ’ re an Indian, Egyptian, Italian, American. You ’ re a heterosexual or a homosexual, or you never have sex. Or at least, that ' s what you say when you identify yourself. You become part of an"us."In order to be secure, you defend against"them."You cling to your land because it is your secure place. You must fight anyone who encroaches upon it. You become your nation. You become your religion. You become whatever it is that will freeze you, **numb** you and protect you from doubt or change. But all this does, actually, is shut down your mind. In reality, it does not really make you safer. I was in Sri Lanka, for example, three days after the tsunami, and I was standing on the beaches and it was absolutely clear that, in a matter of five minutes, a 30-foot wave could rise up and desecrate a people, a population and lives. All this striving for security, in fact, has made you much more insecure because now you have to watch out all the time. There are people not like you — people who you now call enemies. You have places you cannot go, thoughts you cannot think, worlds that you can no longer inhabit. And so you spend your days fighting things off, defending your territory and becoming more entrenched in your fundamental thinking. Your days become devoted to protecting yourself. This becomes your mission. That is all you do. Ideas get shorter. They become sound bytes. There are evildoers and saints, criminals and **victims**. There are those who, if they ' re not with us, are against us. It gets easier to hurt people because you do not feel what ' s inside them. It gets easier to lock them up, force them to be naked, humiliate them, occupy them, invade them and kill them, because they are only obstacles now to your security. In six years, I ' ve had the extraordinary privilege through V-Day, a global movement against [**violence** against] women, to travel probably to 60 countries, and spend a great deal of time in different portions. I ' ve met women and men all over this planet, who through various circumstances — war, poverty, racism, multiple forms of **violence** — have never known security, or have had their illusion of security forever devastated. I ' ve spent time with women in Afghanistan under the Taliban, who were essentially brutalized and censored. I ' ve been in Bosnian refugee camps. I was with women in Pakistan who have had their faces melted off with acid. I ' ve been with girls all across America who were date-raped, or raped by their best friends when they were drugged one night. One of the amazing things that I ' ve discovered in my travels is that there is this emerging species. I loved when he was talking about this other world that ' s right next to this world. I ' ve discovered these people, who, in V-Day world, we call Vagina Warriors. These particular people, rather than getting AK-47s, or weapons of mass destruction, or machetes, in the spirit of the warrior, have gone into the center, the heart of pain, of loss. They have grieved it, they have died into it, and allowed and encouraged poison to turn into medicine. They have used the fuel of their pain to begin to redirect that energy towards another mission and another trajectory. These warriors now devote themselves and their lives to making sure what happened to them doesn ' t happen to anyone else. There are thousands if not millions of them on the planet. I venture there are many in this room. They have a fierceness and a freedom that I believe is the bedrock of a new paradigm. They have broken out of the existing frame of **victim** and **perpetrator**. Their own personal security is not their end goal, and because of that, because, rather than worrying about security, because the transformation of **suffering** is their end goal, I actually believe they are creating real **safety** and a whole new idea of security. I want to talk about a few of these people that I ' ve met. Tomorrow, I am going to Cairo, and I ' m so moved that I will be with women in Cairo who are V-Day women, who are opening the first safe house for battered women in the Middle East. That will happen because women in Cairo made a decision to stand up and put themselves on the line, and talk about the degree of **violence** that is happening in Egypt, and were willing to be attacked and criticized. And through their work over the last years, this is not only happening that this house is opening, but it ' s being supported by many factions of the society who never would have supported it. Women in Uganda this year, who put on"The Vagina Monologues"during V-Day, actually evoked the wrath of the government. And, I love this story so much. There was a cabinet meeting and a meeting of the presidents to talk about whether"Vaginas"could come to Uganda. And in this meeting — it went on for weeks in the press, two weeks where there was huge discussion. The government finally made a decision that"The Vagina Monologues"could not be performed in Uganda. But the amazing news was that because they had stood up, these women, and because they had been willing to risk their security, it began a discussion that not only happened in Uganda, but all of Africa. As a result, this production, which had already sold out, every single person in that 800-seat audience, except for 10 people, made a decision to keep the money. They raised 10, 000 dollars on a production that never occurred. There ' s a young woman named Carrie Rethlefsen in Minnesota. She ' s a high school student. She had seen"The Vagina Monologues"and she was really moved. And as a result, she wore an"I heart my vagina"button to her high school in Minnesota. She was basically threatened to be expelled from school. They told her she couldn ' t love her vagina in high school, that it was not a legal thing, that it was not a moral thing, that it was not a good thing. So she really struggled with this, what to do, because she was a senior and she was doing well in her school and she was threatened expulsion. So what she did is she got all her friends together — I believe it was 100, 150 students all wore"I love my vagina"T-shirts, and the boys wore"I love her vagina"T-shirts to school. Now this seems like a fairly, you know, frivolous, but what happened as a result of that, is that that school now is forming a sex education class. It ' s beginning to talk about sex, it ' s beginning to look at why it would be wrong for a young high school girl to talk about her vagina publicly or to say that she loved her vagina publicly. I know I ' ve talked about Agnes here before, but I want to give you an update on Agnes. I met Agnes three years ago in the Rift Valley. When she was a young girl, she had been mutilated against her will. That mutilation of her clitoris had actually obviously impacted her life and changed it in a way that was devastating. She made a decision not to go and get a razor or a glass shard, but to devote her life to stopping that happening to other girls. For eight years, she walked through the Rift Valley. She had this amazing box that she carried and it had a torso of a woman ' s body in it, a half a torso, and she would teach people, everywhere she went, what a healthy vagina looked like and what a mutilated vagina looked like. In the years that she walked, she educated parents, mothers, fathers. She saved 1, 500 girls from being cut. When V-Day met her, we asked her how we could support her and she said,"Well, if you got me a Jeep, I could get around a lot faster."So, we bought her a Jeep. In the year she had the Jeep, she saved 4, 500 girls from being cut. So, we said, what else could we do? She said,"If you help me get money, I could open a house."Three years ago, Agnes opened a safe house in Africa to stop mutilation. When she began her mission eight years ago, she was reviled, she was detested, she was completely slandered in her community. I am proud to tell you that six months ago, she was elected the deputy mayor of Narok. I think what I ' m trying to say here is that if your end goal is security, and if that ' s all you ' re focusing on, what ends up happening is that you create not only more insecurity in other people, but you make yourself far more insecure. Real security is contemplating death, not pretending it doesn ' t exist. Not running from loss, but entering **grief**, surrendering to sorrow. Real security is not knowing something, when you don ' t know it. Real security is hungering for connection rather than power. It cannot be bought or arranged or made with bombs. It is deeper, it is a process, it is acute awareness that we are all utterly inter-bended, and one action by one being in one tiny town has consequences everywhere. Real security is not only being able to tolerate mystery, complexity, ambiguity, but hungering for them and only trusting a situation when they are present. Something happened when I began traveling in V-Day, eight years ago. I got lost. I remember being on a plane going from Kenya to South Africa, and I had no idea where I was. I didn ’ t know where I was going, where I ’ d come from, and I panicked. I had a total anxiety attack. And then I suddenly realized that it absolutely didn ’ t matter where I was going, or where I had come from because we are all essentially permanently displaced people. All of us are refugees. We come from somewhere and we are hopefully traveling all the time, moving towards a new place. Freedom means I may not be identified as any one group, but that I can visit and find myself in every group. It does not mean that I don ’ t have values or beliefs, but it does mean I am not hardened around them. I do not use them as weapons. In the **shared** future, it will be just that, shared. The end goal will [be] becoming vulnerable, realizing the place of our connection to one another, rather than becoming secure, in control and alone. Thank you very much. Chris Anderson : And how are you doing? Are you exhausted? On a typical day, do you wake up with hope or gloom? Eve Ensler : You know, I think Carl Jung once said that in order to survive the twentieth century, we have to live with two existing thoughts, opposite thoughts, at the same time. And I think part of what I ’ m learning in this process is that one must allow oneself to feel **grief**. And I think as long as I keep grieving, and weeping, and then moving on, I ’ m fine. When I start to pretend that what I ’ m seeing isn ’ t impacting me, and isn ’ t changing my heart, then I get in trouble. Because when you spend a lot of time going from place to place, country to country, and city to city, the degree to which women, for example, are violated, and the epidemic of it, and the kind of ordinariness of it, is so devastating to one ’ s soul that you have to take the time, or I have to take the time now, to process that. CA : There are a lot of causes out there in the world that have been talked about, you know, poverty, sickness and so on. You spent eight years on this one. Why this one? EE : I think that if you think about women, women are the primary resource of the planet. They give birth, we come from them, they are mothers, they are visionaries, they are the future. If you think that the U. N. now says that one out of three women on the planet will be raped or beaten in their lifetime, we ’ re talking about the desecration of the primary resource of the planet, we ’ re talking about the place where we come from, we ’ re talking about parenting. Imagine that you ’ ve been raped and you ’ re bringing up a boy child. How does it impact your ability to work, or envision a future, or thrive, as opposed to just survive? What I believe is if we could figure out how to make women safe and honor women, it would be parallel or **equal** to honoring life itself.

Text Sample 386: POS Dim 3 – 2005 – Score 9.00 – 2005/t000421.txt

So my grandfather told me when I was a little girl,”If you say a word often enough, it becomes you.”And having grown up in a segregated city, Baltimore, Maryland, I sort of use that idea to go around America with a tape recorder â thank God for technology â to interview people, thinking that if I walked in their words â which is also why I don ’ t wear shoes when I perform â if I walked in their words, that I could sort of absorb America. I was also inspired by Walt Whitman, who wanted to absorb America and have it absorb him. So these four characters are going to be from that work that I ’ ve been doing for many years now, and well over, I don ’ t know, a couple of thousand people I ’ ve interviewed. Anybody out here old enough to know Studs Terkel, that old radio man? So I thought he would be the perfect person to go to to ask about a defining moment in American history. You know, he was”born in 1912, the year the Titanic sank, greatest ship every built. Hits the tip of an iceberg, and bam, it went down. It went down and I came up. Wow, some century.”So this is his answer about a defining moment in American history.”Defining moment in American history, I don ’ t think there ’ s one ; you can ’ t say Hiroshima, that ’ s a big one â I can ’ t think of any one moment I would say is a defining moment. The gradual slippage â ’ slippage ’ is the word used by the people in Watergate, moral slippage â it ’ s a gradual kind of thing, combination of things. You see, we also have the technology. I say, less and less the human touch.”Oh, let me kind of tell you a funny little play bit. The Atlanta airport is a modern airport, and they should leave the gate there. These trains that take you out to a concourse and on to a destination. And these trains are smooth, and they ’ re quiet and they ’ re efficient. And there ’ s a voice on the train, you know the voice was a human voice. You see in the old days we had robots, robots imitated humans. Now we have humans imitating robots. So we got this voice on this train : Concourse One : Omaha, Lincoln. Concourse Two : Dallas, Fort Worth. Same voice. Just as a train is about to go, a young couple rush in and they ’ re just about to close the pneumatic doors. And that voice, without losing a beat, says, ’ Because of late entry, we ’ re delayed 30 seconds. ’ Just then, everybody ’ s looking at this couple with hateful eyes and the couple ’ s going like this, you know, shrinking. Well, I ’ d happened to have had a couple of drinks before boarding â I do that to steel my nerves â and so I imitate a train call, holding my hand on my â ’ George Orwell, your time has come, ’ you see. Well, some of you are laughing. Everybody laughs when I say that, but not on this train. **Silence**. And so suddenly they ’ re looking at me. So here I am with the couple, the three of us shrinking at the foot of Calvary about to be up, you know.”Just then I see a baby, a little baby in the lap of a mother. I know it ’ s Hispanic because she ’ s speaking Spanish to her companion. So I ’ m going to talk to the baby. So I say to the baby, holding my hand over my mouth because my breath must be 100 proof, I say to the baby, ’ Sir or Madam, what is your considered opinion of the human species? ’ And the baby looks, you know, the way babies look at you clearly, starts laughing, starts busting out with this crazy little laugh. I say, ’ Thank God for a human reaction, we haven ’ t lost yet. ’”But you see, the human touch, you see, it ’ s disappearing. You know, you see, you ’ ve got to question the official truth. You know the thing that was so great about Mark Twain â you know we honor Mark Twain, but we don ’ t read him. We read ’ Huck Finn, ’ of course, we read ’ Huck Finn ’ of course. I mean, Huck, of course, was tremendous. Remember that great scene on the raft, remember what Huck did? You see, here ’ s Huck ; he ’ s an illiterate kid ; he ’ s had no schooling, but there ’ s something in him. And the official truth, the truth was, the law was, that a black man was a property, was a thing, you see. And Huck gets on the raft with a property named Jim, a slave, see. And he hears that Jim is going to go and take his wife and kids and steal them from the woman who owns them, and Huck says, ’ Ooh, oh my God, ooh, ooh â that woman, that woman never did anybody any **harm**. Ooh, he ’ s going to steal ; he ’ s going to steal ; he ’ s going to do a terrible thing. ’ Just then, two slavers caught up, guys chasing slaves, looking for Jim. ’ Anybody up on that raft with you? ’ Huck says, ’ Yeah. ’ ’ Is he black or white? ’ ’ White. ’ And they go off. And Huck said, ’ Oh my God, oh my God, I lied, I lied, ooh, I did a terrible thing, did a terrible thing â why do I feel so good? ’”But it ’ s the goodness of Huck, that stuff that Huck ’ s been made of, you see, all been buried ; it ’ s all been buried. So the human touch, you see, it ’ s disappearing. So you ask about a defining moment â ain ’ t no defining moment in American history for me. It ’ s an accretion of moments that add up to where we are now, where trivia becomes news. And more and more, less and less awareness of the pain of the other. Huh. You know, I don ’ t know if you could use this or not, but I was quoting Wright Morris, a writer from Nebraska, who says, ’ We ’ re more and more into communications and less and less into communication. ’ Okay, kids, I got to scram, got to go see my cardiologist.”And that ’ s Studs Terkel. So, talk about risk taking. I ’ m going to do somebody that nobody likes. You know, most actors want to do characters that are likeable â well, not always, but the notion, especially at a conference like this, I like to inspire people. But since this was called”risk taking,”I ’ m doing somebody who I never do, because she ’ s so unlikeable that one person actually came backstage and told me to take her out of the show she was in. And I ’ m doing her because I think we think of risk, at a conference like this, as a good thing. But there are certain other connotations to the word”risk,”and the same thing about the word”nature.”What is nature? Maxine Greene, who ’ s a wonderful philosopher who ’ s as old as Studs, and was the head of a philosophy â great, big philosophy kind of an organization â I went to her and asked her what are the two things that she doesn ’ t know, that she still wants to know. And she said,”Well, personally, I still feel like I have to curtsey when I see the president of my university. And I still feel as though I ’ ve got to get coffee for my male colleagues, even though I ’ ve outlived most of them.”And she said,”And then intellectually, I don ’ t know enough about the negative imagination. And September 11th certainly taught us that that ’ s a whole area we don ’ t investigate.”So this piece is about a negative imagination. It raises questions about what nature is, what Mother Nature is, and about what a risk can be. And I got this in the Maryland Correctional Institute for Women. Everything I do is word for word off a tape. And I title things because I think people speak in organic poems, and this is called”A Mirror to Her Mouth.”And this is an inmate named Paulette Jenkins.”I began to learn how to cover it up, because I didn ’ t want nobody to know that this was happening in my home. I want everybody to think we were a normal family. I mean we had all the materialistic things, but that didn ’ t make my children pain any less ; that didn ’ t make their fears subside. I ran out of excuses about how we got black eyes and busted lips and bruises. I didn ’ t had no more excuses. And he beat me too. But that didn ’ t change the fact that it was a nightmare for my family ; it was a nightmare. And I failed them dramatically, because I allowed it to go on and on and on.”But the night that Myesha got killed â and the intensity just grew and grew and grew, until one night we came home from getting drugs, and he got angry with Myesha, and he started beating her, and he put her in a bathtub. Oh, he would use a belt. He had a belt because he had this warped perverted thing that Myesha was having sex with her little brother and they was fondling each other â that would be his reason. I ’ m just talking about the particular night that she died. And so he put her in the bathtub, and I was in the bedroom with the baby.”And four months before this happened, four months before Myesha died, I thought I could really fix this man. So I had a baby by him â insane â thinking that if I gave him his own kid, he would leave mine alone. And it didn ’ t work, didn ’ t work. And I ended up with three children, Houston, Myesha and Dominic, who was four months old when I came to jail.”And I was in the bedroom. Like I said, he had her in the bathroom and he â he â every time he hit her, she would fall. And she would hit her head on the tub. It happened continuously, repeatedly. I could hear it, but I dared not to move. I didn ’ t move. I didn ’ t even go and see what was happening. I just sat there and listened. And then he put her in the hallway. He told her, just set there. And so she set there for about four or five hours. And then he told her, get up. And when she got up, she says she couldn ’ t see. Her face was bruised. She had a black eye. All around her head was just swollen ; her head was about two sizes of its own size. I told him, ’ Let her go to sleep. ’ He let her go to sleep.”The next morning she was dead. He went in to check on her for school, and he got very excited. He says, ’ She won ’ t breathe. ’ I knew immediately that she was dead. I didn ’ t even want to accept the fact that she was dead, so I went in and I put a mirror to her mouth â there was no thing, nothing, coming out of her mouth. He said, he said, he said, ’ We can ’ t, we can ’ t let nobody find out about this. ’ He say, ’ You ’ ve got to help me. ’ I agree. I agree.”I mean, I ’ ve been keeping a secret for years and years and years, so it just seemed like second hand to me, just to keep on keeping it a secret. So we went to the mall and we told a police that we had, like, lost her, that she was missing. We told a security guard that she was missing, though she wasn ’ t missing. And we told the security guard what we had put on her and we went home and we dressed her in exactly the same thing that we had told the security guard that we had put on her.”And then we got the baby and my other child, and we drove out to, like, I-95. I was so petrified and so **numb**, all I could look was in the rear-view mirror. And he just laid her right on the shoulder of the highway. My own child, I let that happen to.”So that ’ s an investigation of the negative imagination. When I started this project called”On the Road : A Search For an American Character”with my tape recorder, I thought that I was going to go around America and find it in all its aspects â bull riders, cowboys, pig farmers, drum majorettes â but I sort of got tripped on race relations, because my first big show was a show about a race riot. And so I went to both â two race riots, one of which was the Los Angeles riot. And this next piece is from that. Because this is what I would say I ’ ve learned the most about race relations, from this piece. It ’ s a kind of an aria, I would say, and in many tapes that I have. Everybody knows that the Los Angeles riots happened because four cops beat up a black man named Rodney King. It was captured on videotape â technology â and it was played all over the world. Everybody thought the four cops would go to jail. They did not, so there were riots. And what a lot of people forget, is there was a second trial, ordered by George Bush, Sr. And that trial came back with two cops going to jail and two cops **declared innocent**. I was at that trial. And I mean, the people just danced in the **streets** because they were afraid there was going to be another riot. Explosion of joy that this verdict had come back this way. So there was a community that didn ’ t â the Korean-Americans, whose stores had been burned to the ground. And so this woman, Mrs. Young-Soon Han, I suppose will have taught me the most that I have learned about race. And she asks also a question that Studs talks about : this notion of the”official truth,”to question the”official truth.”So what she ’ s questioning here, she ’ s taking a chance and questioning what justice is in society. And this is called,”Swallowing the Bitterness.””I used to believe America was the best. I watched in Korea many luxurious Hollywood lifestyle movie. I never saw any poor man, any black. Until 1992, I used to believe America was the best â I still do ; I don ’ t deny that because I am a **victim**. But at the end of ’ 92, when we were in such turmoil, and having all the financial problems, and all the mental problems, I began to really realize that Koreans are completely left out of this society and we are nothing. Why? Why do we have to be left out? We didn ’ t qualify for medical treatment, no food stamp, no GR, no welfare, anything. Many African-Americans who never work got minimum amount of money to survive. We didn ’ t get any because we have a car and a house. And we are high taxpayer. Where do I find justice?”OK. OK? OK. OK. Many African- Americans probably think that they won by the trial. I was sitting here watching them the morning after the verdict, and all the day they were having a party, they **celebrated**, all of South Central, all the churches. And they say, ' Well, finally justice has been done in this society. ' Well, what about **victims** ' rights? They got their rights by destroying **innocent** Korean merchants. They have a lot of respect, as I do, for Dr. Martin King. He is the only model for black community ; I don ' t care Jesse Jackson. He is the model of non- **violence**, non-violence â and they would all like to be in his spirit."But what about 1992? They destroyed **innocent** people. And I wonder if that is really justice for them, to get their rights in that way. I was swallowing the bitterness, sitting here alone and watching them. They became so hilarious, but I was happy for them. I was glad for them. At least they got something back, OK. Let ' s just forget about Korean **victims** and other **victims** who were destroyed by them. They fought for their rights for over two centuries, and maybe because they sacrifice other minorities, Hispanic, Asian, we would suffer more in the mainstream. That ' s why I understand ; that ' s why I have a mixed feeling about the verdict."But I wish that, I wish that, I wish that I could be part of the enjoyment. I wish that I could live together with black people. But after the riot, it ' s too much difference. The fire is still there. How do you say it? [Unclear]. Igniting, igniting, igniting fire. Igniting fire. It ' s still there ; it can burst out anytime."Mrs. Young-Soon Han. The other reason that I don ' t wear shoes is just in case I really feel like I have to cuddle up and get into the feet of somebody, walking really in somebody else ' s shoes. And I told you that in â you know, I didn ' t give you the year, but in ' 79 I thought that I was going to go around and find bull riders and pig farmers and people like that, and I got sidetracked on race relations. Finally, I did find a bull rider, two years ago. And I ' ve been going to the rodeos with him, and we ' ve bonded. And he ' s the lead in an op-ed I did about the Republican Convention. He ' s a Republican â I won ' t say anything about my party affiliation, but anyway â so this is my dear, dear Brent Williams, and this is on toughness, in case anybody needs to know about being tough for the work that you do. I think there ' s a real lesson in this. And this is called"Toughness.""Well, I ' m an optimist. I mean basically I ' m an optimist. I mean, you know, I mean, it ' s like my wife, Jolene, her family ' s always saying, you know, you ever think he ' s just a born loser? It seems like he has so much bad luck, you know. But then when that bull stepped on my kidney, you know, I didn ' t lose my kidney â I could have lost my kidney, I kept my kidney, so I don ' t think I ' m a born loser. I think that ' s good luck."And, I mean, funny things like this happen. I was in a doctor ' s office last CAT scan, and there was a Reader ' s Digest, October 2002. It was like, ' seven ways to get lucky. ' And it says if you want to get lucky, you know, you ' ve got to be around positive people. I mean, like even when I told my wife that you want to come out here and talk to me, she ' s like, ' She ' s just talking ; she ' s just being nice to you. She ' s not going to do that. '"And then you called me up and you said you wanted to come out here and interview me and she went and looked you up on the Internet. She said, ' Look who she is. You ' re not even going to be able to answer her questions. ' And she was saying you ' re going to make me look like an idiot because I ' ve never been to college, and I wouldn ' t be talking professional or anything. I said, ' Well look, the woman talked to me for four hours. You know, if I wasn ' t talking â you know, like, you know, she wanted me to talk, I don ' t think she would even come out here. '"Confidence? Well, I think I ride more out of determination than confidence. I mean, confidence is like, you know, you ' ve been on that bull before ; you know you can ride him. I mean, confidence is kind of like being cocky, but in a good way. But determination, you know, it ' s like just, you know, ' Fuck the form, get the horn. ' That ' s Tuff Hedeman, in the movie ’ 8 Seconds. ’ I mean, like, Pat O ’ Mealey always said when I was a boy, he say, ’ You know, you got more try than any kid I ever seen. ’ And try and determination is the same thing. Determination is, like, you ’ re going to hang on that bull, even if you ’ re riding upside down. Determination ’ s like, you ’ re going to ride till your head hits the back of the dirt.”Freedom? It would have to be the rodeo.”Beauty? I don ’ t think I know what beauty is. Well, you know, I guess that ’ d have to be the rodeo too. I mean, look how we are, the roughy family, palling around and shaking hands and wrestling around me. It ’ s like, you know, racking up our credit cards on entry fees and gas. We ride together, we, you know, we, we eat together and we sleep together. I mean, I can ’ t even imagine what it ’ s going to be like the last day I rodeo. I mean, I ’ ll be alright. I mean, I have my ranch and everything, but I actually don ’ t even want to think the day that comes. I mean, I guess it just be like â I guess it be like the day my brother died.”Toughness? Well, we was in West Jordan, Utah, and this bull shoved my face right through the metal shoots in a â you know, busted my face all up and had to go to the hospital. And they had to sew me up and straighten my nose out. And I had to go and ride in the rodeo that night, so I didn ’ t want them to put me under anesthesia, or whatever you call it. And so they sewed my face up. And then they had to straighten out my nose, and they took these rods and shoved them up my nose and went up through my brains and felt like it was coming out the top of my head, and everybody said that it should have killed me, but it didn ’ t, because I guess I have a high tolerance for pain. But the good thing was, once they shoved those rods up there and straightened my nose out, I could breathe, and I hadn ’ t been able to breathe since I broke my nose in the high school rodeo.”Thank you.

Text Sample 387: POS Dim 3 – 2013 – Score 17.00 – 2013/t000034.txt

About 18 months ago, I had a really bad week. I was on my way home from work one night, and it was one of those hot evenings where the traffic was at a standstill, and as I walked down the road, and the cars crawled next to me, some guys started shouting out of their car windows about my legs, about the things that they ’ d like to do to me. And I ignored them, and I carried on home, and I got on with it, like you do. Then a few nights later, I was on the way home, on the bus, quite late at night, and I was on the phone to my mom. I thought, at first, that the guy next to me just accidentally brushed my leg with his hand. And I carried on talking to my mom. Then I realized that, actually, he was grabbing and groping my leg and moving his hand up towards my crotch. I stood up to move away, but because I was on the phone, I vocalized it, in a way I don ’ t think I would have done otherwise. I said,”On the bus, this guy ’ s groping me.”Everybody on that bus looked out the window, or down at their feet, or at their phone. Certainly nobody stepped in, but more than that, there was a real sense of,”Why make a fuss about this, woman? This is your issue, deal with it ; don ’ t make us have to think about it.”That immediately made me feel ashamed. It made me feel like I ’ d done something wrong, or I shouldn ’ t have been out that late, or I shouldn ’ t have been wearing what I was wearing, and all of those thoughts that that reaction triggers. And again, I carried on. I went home, I didn ’ t mention it. I got on with it, like you do. Then a couple days later, I was walking down the **street** in broad daylight. There was a big truck being unloaded, scaffolding was coming off the back of it, and there were two guys working together. As I walked past, one of them turned to the other and said,”Look at the tits on that.”Not”her,””that.”They started discussing me as if I wasn ’ t there, even though I was one meter away, and I could really clearly hear them. The thing that really hit me about these three incidents was if they hadn ’ t happened in the same week, I never would have thought twice about any one of them. I started asking myself why that was : Why was this so normal? Why was I so used to them? I started thinking back about hundreds of incidents that had happened over the weeks and months and years that I ’ d never said anything about to anyone, because it was normal. I started talking to other women and asking - the women I knew, older women, younger women, women I met - saying,”Have you ever experienced anything like this?”And I honestly thought that one or two women would have a story. That one or two people would say,”Yes, a few years ago this happened,”or,”I once had a job where this happened.”But it wasn ’ t like that. It was every woman I spoke to. And it wasn ’ t a few years ago, this one incident. It was hundreds of things.”It was on my way here, this happened, yesterday this happened, most days this happens.”But just like me, until I asked them, they ’ d never told those stories to anyone. Because they were used to it, because it was normal. I started trying to speak up about this, because I was realizing there was this huge problem, and I started trying to talk about it, and again and again, I got the same response. People said,”Stop making a fuss. Women are **equal** now, more or less.”If women are **equal** now, then to talk about sexism, to complain about sexism, must be overreacting. Or maybe you don ’ t have a sense of humor, or maybe you need to learn to take a compliment, or maybe you ’ re a bit frigid or uptight and you need to learn to take a joke. I thought, maybe they were right, maybe women are **equal** now, more or less ; perhaps I was overreacting. I thought I ’ d look into it, I ’ d interrogate that claim and I did. These are some of the things that I found : Women are **equal** now, more or less. Except in our Houses of Parliament, where the policies that affect all of us are debated and defined, less than one in four MPs is a woman. Women make up one fifth of the membership of the House of Lords. The UK comes joint 57th in the world for gender **equality** in Parliament. Then I looked into the law, and I found that just four out of 35 Lord Justices of Appeal, and just 18 out of 108 High Court judges are women. I decided to look at the arts. I found that it was reported in 2010, that out of 2, 300 works, one of our most prestigious art institutions, the National Gallery, had paintings by just ten women. I found that at the Royal Opera House, it ’ s been over 13 years since a female choreographer was commissioned to create a piece for the main stage. And that out of 573 listed statues up and down the UK commemorating people of interest, just 15 per cent of them are of women. I found that fewer than one in ten of our engineers is female, less than half the proportion of France or Spain ; that our Royal Society, one of our most prestigious scientific institutions, has never had a female president, and just five per cent of the current fellowship are women. And that whilst women make up 50 per cent of chemistry undergraduates, there ’ re only six per cent of professors. I found that women write only one fifth of front page newspaper articles, but 84 per cent of those articles are dominated by male subjects or experts. That women directed just five per cent of the 250 major films of 2011, and that only one in five UK architects is female, yet 63 per cent of them report experiencing **sexual harassment** in the workplace during the course of their career. And then I looked into the crime statistics. Women are **equal** now, more or less. Except that in the UK over two women a week are killed by a current or former partner. There ’ s a phone call to the police every minute about **domestic violence**. Every six or seven minutes, a woman is raped, adding up to over 85, 000 rapes and 400, 000 **sexual assaults** every year. In the UK, a woman has a one in four chance of becoming a **victim** of **domestic violence**, and has a one in five chance of being a **victim** of a **sexual** offense. Worldwide, one in three women on the planet will be raped or beaten in her lifetime. I decided that that argument that women were **equal** now and we shouldn ’ t be making a fuss, really didn ’ t stand up to scrutiny. In fact, it seemed to me that it really was time to make a fuss. So I set up a simple website because I realized we couldn ’ t solve a problem if people refused even to acknowledge that it existed, and that what I really wanted people to have was that experience that I ’ d had of seeing these things kind of rolled out in front of them like a map, and realizing how much there was and how bad it still was. I set up a very simple website called”The Everyday Sexism Project,”and I asked women and men to add their experiences of gender imbalance on a daily basis ; anything from the tiny niggling normalized things, all the way up the scale. I didn ’ t have any funding or any way of publicizing it, so I thought that maybe 20 or 30 women would add their stories, and I hoped it would build a sense of **solidarity**, and help to raise awareness. But instead, things took off a little more than I expected. [75, 000 Women To Take A Stand Against Sexism] 50, 000 women from all over the world added their stories in 18 months. They were women and men from countries everywhere, people of all ages, races, **ethnicities**, **sexual** orientations, gender identities, religious and non-religious, **disabled** and non-disabled, employed and unemployed. We heard from a seven-year-old **disabled** girl in a wheelchair and a 74-year-old women in a mobility scooter who encountered almost identical experiences of screamed abuse about”female drivers.”A female Reverend in the Church of England was asked if there was a man available to perform the wedding or funeral service -”Nothing personal.”A man was congratulated for babysitting his own children. A woman working in the city was asked if she would sit on her bosses lap if she wanted her Christmas bonus. A woman who worked in a video store found that every time she went up the ladder to the storeroom, her boss would smack her on the bum, and when she came down he looked down her top and say :”You know why I hired you.”A waitress was told to make a choice between having an abortion or resigning when she fell pregnant. A 15-year-old girl wrote that she knew that she was clever and funny, and she could do anything she wanted to do, but really it didn ’ t matter if she became a doctor or a lawyer, because she knew from the world around her and from the media, that the only thing that really mattered was whether she was sexy, whether her breasts grew and her waist narrowed, and whether boys found her attractive. A 13-year-old girl wrote to say that she ’ d been showed a video of sex, at school on a boy ’ s mobile phone, a video of porn, and that now she ’ s scared to have sex, she cries every night, because she didn ’ t realize that what sex was was the woman hurting and crying. A woman in Pakistan talked about hiding abuse for the sake of family honor. A woman in Brazil tried to ignore three men who catcalled her only to find that they tried to drag her into their car. In Mexico a woman was told by her university professor :”Calladita te ves ms bonita”,”You look prettier when you shut up.”This is what happened when I gave a speech about politics - [I think Laura should just get her tits out so we can judge for ourselves.] [I ’ m not sexist or anything but she may be keeping a nice pair...] This was what I got on a daily basis. But not just once a day, up to 200 times a day, just for speaking out. Ironically these people sending messages because they wanted to shut the project down were showing how vital and needed it was. [fuck you stupid slut] The fact that it was so scary for some people, for somebody just to want to talk about **equality**, just to want to raise women ’ s voices and give their stories a platform, that they had to tell me exactly how they wanted to disembowel me, and with exactly which weapons and in what order, and not just that I should be raped, but exactly how I should be raped, and in which our orifices, and where and when. Then something else started to happen. After we ’ d received about ten thousand stories, we started getting some which had a very different tone. We started getting success stories. We started hearing from women like one who said that she was a keen runner, who often experienced **harassment**, but she thought it was just the way things were. Then after reading the stories on the website, she realized other women were standing up to this, and other people were acknowledging that this shouldn ’ t be normal, and it wasn ’ t okay. The next time she went running, a guy happened to call her over from his car and ask for directions. So she went over and helped him, and then he reached out of the car window and grabbed her breasts really hard, really hurt her. She said she felt all of the experiences, the feelings wash over her that she normally felt in that situation - terror, embarrassment, **shame**, the urge to run - but she also felt something she hadn ’ t felt before, and it was that feeling of those women behind her standing up, and it gave her the strength, just for a moment, to stop and take down the guy ’ s car number plate, and now he ’ s been charged with **assault**. We were able to take 2, 000 of the stories we collected that specifically described women ’ s experiences of **harassment** and **assault** on public transport to the British Transport Police when they decided to look at the way that they police **sexual** offences. We were able to break them down, to hear from women ’ s own voices why they haven ’ t felt able to report, and then work with the British Transport Police to send out the message to people everywhere that they were taking this seriously and they could report it. So far we know that that project - Project Guardian - has raised reports of **harassment** and **assault** on the tube by up to 20 per cent. We were able to start talking to girls at universities about the UK definition of **sexual assault**, which is very simple. Under UK law, if someone touches you anywhere on your body, and the touching is **sexual**, and you don ’ t consent, and they don ’ t have reason to believe that you consent, it ’ s a form of **sexual assault**. Girls came up to me saying,”That can ’ t be **sexual assault** because it ’ s normal.””That can ’ t be **sexual assault** because that happens when I go out with my friends.””It can ’ t be **sexual assault** because I won ’ t be able to call it that, people won ’ t take me seriously, I couldn ’ t go to the police.”We were able to start to change that **attitude** and able to start to get reports of people who ’ d reported things that previously, they ’ d had no idea they had the right to object to. We also started hearing people ’ s individual stories of standing up, and that was really fascinating and crucial, because these weren ’ t stories of waving banners or going on marches - as valuable as those are - they were stories of women and men around the world finding that own very unique and individual ways to stand up that worked for them and made a difference in their lives. We heard from a woman who was being sexually harassed in the office, who printed off a copy of her workplace **sexual harassment** policy and put it on every single person ’ s desk, and the **harassment** stopped. We heard from a woman who said that she was sick of cold callers ringing. She was a single mom and sick of them ringing and asking to speak to the man of the house. Now she puts them on to her six-year old son, and apparently he **sings** them,”I ’ m sexy and I know it.”We heard from a guy who was walking past a building site, when two builders screamed at two women across the road,”Get your tits out!”So he lifted up his T-shirt instead. We heard from a woman who said that every time someone screams”Nice tits!”at her in the **street**, she looks down at them, and screams as if she ’ d never seen them before. We heard from a man who said that he ’ d never really thought about **harassment** before, but after reading the stories it gave him new insight into what it actually felt like for women, and the next time he saw another guy in the **street** harassing two women, he ran after him, tapped him on the shoulder and said,”Sorry, can I just ask you, why did you do that?”And the other guy had no answer, because he ’ d never been asked that question before, because it was just normal, for him too. He ’ d grown up in a world where that was just normal and something that men did. That ’ s the really important thing here, because sadly and frustratingly, we can no longer point to one specific policy change or piece of legislation that we need to solve this problem. Particularly in the UK, we have excellent legislation now, a really good example is workplace **sexual harassment** law, which is fantastic. The single biggest category of entries that we receive is from women being harassed in the workplace, being **assaulted** in the workplace, being discriminated against in the workplace. What we need is a cultural and a social shift in our **attitudes** towards women, and towards **violence** against women. Because it ’ s people in the workplace that laugh along and call it”banter”and just joke around when someone grabs her breasts that make her feel unable to report. In a way that ’ s the exciting thing, because it means that we can all be part of the solution. If the Everyday Sexism Project has shown anything, it ’ s that this is a continuum. All of these things are connected. The same ideas and **attitudes** about women that underlie those more”minor”incidents of sexism and **harassment**, that we ’ re often told to brush off and not make a fuss about, are the same ideas and **attitudes** about women that underlie the more serious incidents of **assault** and rape. What that means is that by helping to contribute to a cultural shift in the way women are perceived - whether it ’ s in the media, in the professional sphere, in the social or economic sphere - we help to shift the way that they ’ re perceived and treated in the other spheres as well. So that does mean that every one of us can be part of the change. It ’ s not necessarily about targeting **perpetrators**, and it ’ s certainly not about telling **victims** that they should be behaving in a certain way or reacting in a certain way. It ’ s about the people in the office that made it difficult for that woman to feel able to speak out ; it ’ s about the people on that bus that day that looked out of the window. Be part of the change. Be the cool aunt or uncle who buys a chemistry set for their niece, or a play cooker for their nephew. Be the teenager that tells his friends that actually it ’ s not okay or funny to refer to women as sluts or whores. Be the person that lets somebody who ’ s been groped realize that it will be taken seriously, and they have the right to report it. Be the tabloid editor who commissions an article that isn ’ t illustrated with a picture of a pair of women ’ s tits. Be the person at the bus stop that steps in when they see a woman being harassed. Or be the person on the bus that stands up and says it isn ’ t okay. Because our voices are the loudest when we raise them together.

Text Sample 388: POS Dim 3 – 2013 – Score 11.00 – 2013/t002007.txt

The day I left home for the first time to go to university was a bright day brimming with hope and optimism. I ’ d done well at school. Expectations for me were high, and I gleefully entered the student life of lectures, parties and traffic cone theft. Now appearances, of course, can be deceptive, and to an extent, this feisty, energetic persona of lecture-going and traffic cone stealing was a veneer, albeit a very well-crafted and convincing one. Underneath, I was actually deeply unhappy, insecure and fundamentally frightened — frightened of other people, of the future, of failure and of the emptiness that I felt was within me. But I was skilled at hiding it, and from the outside appeared to be someone with everything to hope for and aspire to. This fantasy of invulnerability was so complete that I even deceived myself, and as the first semester ended and the second began, there was no way that anyone could have predicted what was just about to happen. I was leaving a seminar when it started, humming to myself, fumbling with my bag just as I ’ d done a hundred times before, when suddenly I heard a voice calmly observe,”She is leaving the room.”I looked around, and there was no one there, but the clarity and decisiveness of the comment was unmistakable. Shaken, I left my books on the stairs and hurried home, and there it was again.”She is opening the door.”This was the beginning. The voice had arrived. And the voice persisted, days and then weeks of it, on and on, narrating everything I did in the third person.”She is going to the library.””She is going to a lecture.”It was neutral, impassive and even, after a while, strangely companionate and reassuring, although I did notice that its calm exterior sometimes slipped and that it occasionally mirrored my own unexpressed emotion. So, for example, if I was angry and had to hide it, which I often did, being very adept at concealing how I really felt, then the voice would sound frustrated. Otherwise, it was neither sinister nor disturbing, although even at that point it was clear that it had something to communicate to me about my emotions, particularly emotions which were remote and inaccessible. Now it was then that I made a fatal mistake, in that I told a friend about the voice, and she was horrified. A subtle conditioning process had begun, the implication that normal people don ’ t hear voices and the fact that I did meant that something was very seriously wrong. Such fear and mistrust was infectious. Suddenly the voice didn ’ t seem quite so benign anymore, and when she insisted that I seek medical attention, I duly complied, and which proved to be mistake number two. I spent some time telling the college G. P. about what I perceived to be the real problem : anxiety, low self-worth, fears about the future, and was met with bored indifference until I mentioned the voice, upon which he dropped his pen, swung round and began to question me with a show of real interest. And to be fair, I was desperate for interest and help, and I began to tell him about my strange commentator. And I always wish, at this point, the voice had said,”She is digging her own grave.”I was referred to a psychiatrist, who likewise took a grim view of the voice ’ s presence, subsequently interpreting everything I said through a lens of latent insanity. For example, I was part of a student TV station that broadcast news bulletins around the campus, and during an appointment which was running very late, I said,”I ’ m sorry, doctor, I ’ ve got to go. I ’ m reading the news at six.”Now it ’ s down on my medical records that Eleanor has delusions that she ’ s a television news broadcaster. It was at this point that events began to rapidly overtake me. A hospital admission followed, the first of many, a diagnosis of schizophrenia came next, and then, worst of all, a toxic, tormenting sense of hopelessness, humiliation and despair about myself and my prospects. But having been encouraged to see the voice not as an experience but as a symptom, my fear and resistance towards it intensified. Now essentially, this represented taking an aggressive stance towards my own mind, a kind of psychic civil war, and in turn this caused the number of voices to increase and grow progressively hostile and menacing. Helplessly and hopelessly, I began to retreat into this nightmarish inner world in which the voices were destined to become both my persecutors and my only perceived companions. They told me, for example, that if I proved myself worthy of their help, then they could change my life back to how it had been, and a series of increasingly bizarre tasks was set, a kind of labor of Hercules. It started off quite small, for example, pull out three strands of hair, but gradually it grew more extreme, culminating in commands to **harm** myself, and a particularly dramatic instruction :”You see that tutor over there? You see that glass of water? Well, you have to go over and pour it over him in front of the other students.”Which I actually did, and which needless to say did not endear me to the faculty. In effect, a vicious cycle of fear, avoidance, mistrust and misunderstanding had been established, and this was a battle in which I felt powerless and incapable of establishing any kind of peace or reconciliation. Two years later, and the deterioration was dramatic. By now, I had the whole frenzied repertoire : terrifying voices, grotesque visions, bizarre, intractable delusions. My mental health status had been a catalyst for **discrimination**, verbal abuse, and physical and **sexual assault**, and I ' d been told by my psychiatrist,"Eleanor, you ' d be better off with cancer, because cancer is easier to cure than schizophrenia."I ' d been diagnosed, drugged and discarded, and was by now so tormented by the voices that I attempted to drill a hole in my head in order to get them out. Now looking back on the wreckage and despair of those years, it seems to me now as if someone died in that place, and yet, someone else was saved. A broken and haunted person began that journey, but the person who emerged was a survivor and would ultimately grow into the person I was destined to be. Many people have **harmed** me in my life, and I remember them all, but the memories grow pale and faint in comparison with the people who ' ve helped me. The fellow survivors, the fellow voice-hearers, the comrades and collaborators ; the mother who never gave up on me, who knew that one day I would come back to her and was willing to wait for me for as long as it took ; the doctor who only worked with me for a brief time but who reinforced his belief that recovery was not only possible but inevitable, and during a devastating period of relapse told my terrified family,"Don ' t give up hope. I believe that Eleanor can get through this. Sometimes, you know, it snows as late as May, but summer always comes eventually."Fourteen minutes is not enough time to fully credit those good and generous people who fought with me and for me and who waited to welcome me back from that agonized, lonely place. But together, they forged a blend of courage, creativity, integrity, and an unshakeable belief that my shattered self could become **healed** and whole. I used to say that these people saved me, but what I now know is they did something even more important in that they empowered me to save myself, and crucially, they helped me to understand something which I ' d always suspected : that my voices were a meaningful response to traumatic life events, particularly childhood events, and as such were not my enemies but a source of insight into solvable emotional problems. Now, at first, this was very difficult to believe, not least because the voices appeared so hostile and menacing, so in this respect, a vital first step was learning to separate out a metaphorical meaning from what I ' d previously interpreted to be a literal truth. So for example, voices which threatened to attack my home I learned to interpret as my own sense of fear and insecurity in the world, rather than an actual, objective danger. Now at first, I would have believed them. I remember, for example, sitting up one night on guard outside my parents ' room to protect them from what I thought was a genuine threat from the voices. Because I ' d had such a bad problem with self-injury that most of the cutlery in the house had been hidden, so I ended up arming myself with a plastic fork, kind of like picnic ware, and sort of sat outside the room clutching it and waiting to spring into action should anything happen. It was like,"Don ' t mess with me. I ' ve got a plastic fork, don ' t you know?"Strategic. But a later response, and much more useful, would be to try and deconstruct the message behind the words, so when the voices warned me not to leave the house, then I would thank them for drawing my attention to how **unsafe** I felt — because if I was aware of it, then I could do something positive about it — but go on to reassure both them and myself that we were safe and didn ' t need to feel frightened anymore. I would set boundaries for the voices, and try to interact with them in a way that was assertive yet respectful, establishing a slow process of communication and collaboration in which we could learn to work together and support one another. Throughout all of this, what I would ultimately realize was that each voice was closely related to aspects of myself, and that each of them carried overwhelming emotions that I ' d never had an opportunity to process or resolve, memories of **sexual**

trauma and abuse, of anger, **shame**, guilt, low self-worth. The voices took the place of this pain and gave words to it, and possibly one of the greatest revelations was when I realized that the most hostile and aggressive voices actually represented the parts of me that had been hurt most profoundly, and as such, it was these voices that needed to be shown the greatest compassion and care. It was armed with this knowledge that ultimately I would gather together my shattered self, each fragment represented by a different voice, gradually withdraw from all my medication, and return to psychiatry, only this time from the other side. Ten years after the voice first came, I finally graduated, this time with the highest degree in psychology the university had ever given, and one year later, the highest masters, which shall we say isn ’ t bad for a madwoman. In fact, one of the voices actually dictated the answers during the exam, which technically possibly counts as cheating. And to be honest, sometimes I quite enjoyed their attention as well. As Oscar Wilde has said, the only thing worse than being talked about is not being talked about. It also makes you very good at eavesdropping, because you can listen to two conversations simultaneously. So it ’ s not all bad. I worked in mental health services, I spoke at conferences, I published book chapters and academic articles, and I argued, and continue to do so, the relevance of the following concept : that an important question in psychiatry shouldn ’ t be what ’ s wrong with you but rather what ’ s happened to you. And all the while, I listened to my voices, with whom I ’ d finally learned to live with peace and respect and which in turn reflected a growing sense of compassion, acceptance and respect towards myself. And I remember the most moving and extraordinary moment when supporting another young woman who was terrorized by her voices, and becoming fully aware, for the very first time, that I no longer felt that way myself but was finally able to help someone else who was. I ’ m now very proud to be a part of Intervoice, the organizational body of the International Hearing Voices Movement, an **initiative** inspired by the work of Professor Marius Romme and Dr. Sandra Escher, which locates voice hearing as a survival strategy, a sane reaction to insane circumstances, not as an aberrant symptom of schizophrenia to be endured, but a complex, significant and meaningful experience to be explored. Together, we envisage and enact a society that understands and respects voice hearing, supports the needs of individuals who hear voices, and which values them as full citizens. This type of society is not only possible, it ’ s already on its way. To paraphrase Chavez, once social change begins, it cannot be reversed. You cannot humiliate the person who feels pride. You cannot oppress the people who are not afraid anymore. For me, the achievements of the Hearing Voices Movement are a reminder that empathy, fellowship, justice and respect are more than words ; they are convictions and beliefs, and that beliefs can change the world. In the last 20 years, the Hearing Voices Movement has established hearing voices networks in 26 countries across five continents, working together to promote dignity, **solidarity** and empowerment for individuals in mental distress, to create a new language and practice of hope, which, at its very center, lies an unshakable belief in the power of the individual. As Peter Levine has said, the human animal is a unique being endowed with an instinctual capacity to **heal** and the intellectual spirit to harness this innate capacity. In this respect, for members of society, there is no greater honor or privilege than facilitating that process of healing for someone, to bear witness, to reach out a hand, to share the burden of someone ’ s **suffering**, and to hold the hope for their recovery. And likewise, for survivors of distress and adversity, that we remember we don ’ t have to live our lives forever defined by the damaging things that have happened to us. We are unique. We are irreplaceable. What lies within us can never be truly colonized, contorted, or taken away. The light never goes out.

As a very wonderful doctor once said to me,”Don ’ t tell me what other people have told you about yourself. Tell me about you.”Thank you.

Text Sample 389: POS Dim 3 – 2013 – Score 11.00 – 2013/t002032.txt

”Even in purely non-religious terms, homosexuality represents a misuse of the **sexual** faculty. It is a pathetic little second-rate substitute for reality â a pitiable flight from life. As such, it deserves no compassion, it deserves no treatment as minority martyrdom, and it deserves not to be **deemed** anything but a pernicious sickness.”That ’ s from Time magazine in 1966, when I was three years old. And last year, the president of the United States came out in favor of **gay** marriage. And my question is, how did we get from there to here? How did an illness become an identity? When I was perhaps six years old, I went to a shoe store with my mother and my brother. And at the end of buying our shoes, the salesman said to us that we could each have a balloon to take home. My brother wanted a red balloon, and I wanted a pink balloon. My mother said that she thought I ’ d really rather have a blue balloon. But I said that I definitely wanted the pink one. And she reminded me that my favorite color was blue. The fact that my favorite color now is blue, but I ’ m still **gay** â â is evidence of both my mother ’ s influence and its limits. When I was little, my mother used to say,”The love you have for your children is like no other feeling in the world. And until you have children, you don ’ t know what it ’ s like.”And when I was little, I took it as the greatest compliment in the world that she would say that about parenting my brother and me. And when I was an adolescent, I thought that I ’ m **gay**, and so I probably can ’ t have a family. And when she said it, it made me anxious. And after I came out of the closet, when she continued to say it, it made me furious. I said,”I ’ m **gay**. That ’ s not the direction that I ’ m headed in. And I want you to stop saying that.”About 20 years ago, I was asked by my editors at The New York Times Magazine to write a piece about deaf culture. And I was rather taken aback. I had thought of deafness entirely as an illness. Those poor people, they couldn ’ t hear. They lacked hearing, and what could we do for them? And then I went out into the deaf world. I went to deaf clubs. I saw performances of deaf theater and of deaf poetry. I even went to the Miss Deaf America contest in Nashville, Tennessee where people complained about that slurry Southern signing. And as I plunged deeper and deeper into the deaf world, I become convinced that deafness was a culture and that the people in the deaf world who said,”We don ’ t lack hearing, we have membership in a culture,”were saying something that was viable. It wasn ’ t my culture, and I didn ’ t particularly want to rush off and join it, but I appreciated that it was a culture and that for the people who were members of it, it felt as valuable as Latino culture or **gay** culture or Jewish culture. It felt as valid perhaps even as American culture. Then a friend of a friend of mine had a daughter who was a dwarf. And when her daughter was born, she suddenly found herself confronting questions that now began to seem quite resonant to me. She was facing the question of what to do with this child. Should she say,”You ’ re just like everyone else but a little bit shorter?”Or should she try to construct some kind of dwarf identity, get involved in the Little People of America, become aware of what was happening for dwarfs? And I suddenly thought, most deaf children are born to hearing parents. Those hearing parents tend to try to cure them. Those deaf people discover community somehow in adolescence. Most **gay** people are born to straight parents. Those straight parents often want them to function in what they think of as the mainstream world, and those **gay** people have to discover identity later on. And here was this friend of mine looking at these questions of identity with her dwarf daughter. And I thought, there it is again : A family that perceives itself to be normal with a child who seems to be extraordinary. And I hatched the idea that there are really two kinds of identity. There are vertical identities, which are passed down generationally from parent to child. Those are things like **ethnicity**, frequently nationality, language, often religion. Those are things you have in common with your parents and with your children. And while some of them can be difficult, there ’ s no attempt to cure them. You can argue that it ’ s harder in the United States â our current presidency notwithstanding â to be a person of color. And yet, we have nobody who is trying to ensure that the next generation of children born to African- Americans and Asians come out with creamy skin and yellow hair. There are these other identities which you have to learn from a peer group. And I call them horizontal identities, because the peer group is the horizontal experience. These are identities that are alien to your parents and that you have to discover when you get to see them in peers. And those identities, those horizontal identities, people have almost always tried to cure. And I wanted to look at what the process is through which people who have those identities come to a good relationship with them. And it seemed to me that there were three levels of acceptance that needed to take place. There ’ s self- acceptance, there ’ s family acceptance, and there ’ s social acceptance. And they don ’ t always coincide. And a lot of the time, people who have these conditions are very angry because they feel as though their parents don ’ t love them, when what actually has happened is that their parents don ’ t accept them. Love is something that ideally is there unconditionally throughout the relationship between a parent and a child. But acceptance is something that takes time. It always takes time. One of the dwarfs I got to know was a guy named Clinton Brown. When he was born, he was diagnosed with diastrophic dwarfism, a very disabling condition, and his parents were told that he would never walk, he would never talk, he would have no intellectual capacity, and he would probably not even recognize them. And it was suggested to them that they leave him at the hospital so that he could die there quietly. And his mother said she wasn ’ t going to do it. And she took her son home. And even though she didn ’ t have a lot of educational or financial advantages, she found the best doctor in the country for dealing with diastrophic dwarfism, and she got Clinton enrolled with him. And in the course of his childhood, he had 30 major surgical procedures. And he spent all this time stuck in the hospital while he was having those procedures, as a result of which he now can walk. And while he was there, they sent tutors around to help him with his school work. And he worked very hard because there was nothing else to do. And he ended up achieving at a level that had never before been contemplated by any member of his family. He was the first one in his family, in fact, to go to college, where he lived on campus and drove a specially-fitted car that accommodated his unusual body. And his mother told me this story of coming home one day â and he went to college nearby â and she said,”I saw that car, which you can always recognize, in the parking lot of a bar,”she said.”And I thought to myself, they ’ re six feet tall, he ’ s three feet tall. Two beers for them is four beers for him.”She said,”I knew I couldn ’ t go in there and interrupt him, but I went home, and I left him eight messages on his cell phone.”She said,”And then I thought, if someone had said to me when he was born that my future worry would be that he ’ d go drinking and driving with his college buddies â”And I said to her,”What do you think you did that helped him to emerge as this charming, accomplished, wonderful person?”And she said,”What did I do? I loved him, that ’ s all. Clinton just always had that light in him. And his father and I were lucky enough to be the first to see it there.”I ’ m going to quote from another magazine of the ’ 60s. This one is from 1968 â The Atlantic Monthly, voice of liberal America â written by an important bioethicist. He said,”There is no reason to feel guilty about putting a Down syndrome child away, whether it is put away in the sense of hidden in a sanitarium or in a more responsible, lethal sense. It is sad, yes â dreadful. But it carries no guilt. True guilt arises only from an offense against a person, and a Down ’ s is not a person.”There ’ s been a lot of ink given to the **enormous** progress that we ’ ve made in the treatment of **gay** people. The fact that our **attitude** has changed is in the headlines every day. But we forget how we used to see people who had other differences, how we used to see people who were **disabled**, how inhuman we held people to be. And the change that ’ s been accomplished there, which is almost equally radical, is one that we pay not very much attention to. One of the families I interviewed, Tom and Karen Robards, were taken aback when, as young and successful New Yorkers, their first child was diagnosed with Down syndrome. They thought the educational opportunities for him were not what they should be, and so they decided they would build a little center â two classrooms that they started with a few other parents â to educate kids with D. S. And over the years, that center grew into something called the Cooke Center, where there are now thousands upon thousands of children with intellectual disabilities who are being taught. In the time since that Atlantic Monthly story ran, the life expectancy for people with Down syndrome has tripled. The experience of Down syndrome people includes those who are actors, those who are writers, some who are able to live fully independently in adulthood. The Robards had a lot to do with that. And I said,”Do you regret it? Do you wish your child didn ’ t have Down syndrome? Do you wish you ’ d never heard of it?”And interestingly his father said,”Well, for David, our son, I regret it, because for David, it ’ s a difficult way to be in the world, and I ’ d like to give David an easier life. But I think if we lost everyone with Down syndrome, it would be a catastrophic loss.”And Karen Robards said to me,”I ’ m with Tom. For David, I would cure it in an instant to give him an easier life. But speaking for myself â well, I would never have believed 23 years ago when he was born that I could come to such a point â speaking for myself, it ’ s made me so much better and so much kinder and so much more purposeful in my whole life, that speaking for myself, I wouldn ’ t give it up for anything in the world.”We live at a point when social acceptance for these and many other conditions is on the up and up. And yet we also live at the moment when our ability to eliminate those conditions has reached a height we never imagined before. Most deaf infants born in the United States now will receive Cochlear implants, which are put into the brain and connected to a receiver, and which allow them to acquire a facsimile of hearing and to use oral speech. A compound that has been tested in mice, BMN-111, is useful in preventing the action of the achondroplasia gene. Achondroplasia is the most common form of dwarfism, and mice who have been given that substance and who have the achondroplasia gene, grow to full size. Testing in humans is around the corner. There are blood tests which are making progress that would pick up Down syndrome more clearly and earlier in pregnancies than ever before, making it easier and easier for people to eliminate those pregnancies, or to terminate them. And so we have both social progress and medical progress. And I believe in both of them. I believe the social progress is fantastic and meaningful and wonderful, and I think the same thing about the medical progress. But I think it ’ s a tragedy when one of them doesn ’ t see the other. And when I see the way they ’ re intersecting in conditions like the three I ’ ve just described, I sometimes think it ’ s like those moments in grand **opera** when the hero realizes he loves the heroine at the exact moment that she lies expiring on a divan. We have to think about how we feel about cures altogether. And a lot of the time the question of parenthood is, what do we validate in our children, and what do we cure in them? Jim Sinclair, a prominent autism activist, said,"When parents say ' I wish my child did not have autism, ' what they ' re really saying is ' I wish the child I have did not exist and I had a different, non-autistic child instead. ' Read that again. This is what we hear when you mourn over our existence. This is what we hear when you pray for a cure â that your fondest wish for us is that someday we will cease to be and strangers you can love will move in behind our faces."It ' s a very extreme point of view, but it points to the reality that people engage with the life they have and they don ' t want to be cured or changed or eliminated. They want to be whoever it is that they ' ve come to be. One of the families I interviewed for this project was the family of Dylan Klebold who was one of the **perpetrators** of the Columbine massacre. It took a long time to persuade them to talk to me, and once they agreed, they were so full of their story that they couldn ' t stop telling it. And the first weekend I spent with them â the first of many â I recorded more than 20 hours of conversation. And on Sunday night, we were all exhausted. We were sitting in the kitchen. Sue Klebold was fixing dinner. And I said,"If Dylan were here now, do you have a sense of what you ' d want to ask him?"And his father said,"I sure do. I ' d want to ask him what the hell he thought he was doing."And Sue looked at the floor, and she thought for a minute. And then she looked back up and said,"I would ask him to forgive me for being his mother and never knowing what was going on inside his head."When I had dinner with her a couple of years later â one of many dinners that we had together â she said,"You know, when it first happened, I used to wish that I had never married, that I had never had children. If I hadn ' t gone to Ohio State and crossed paths with Tom, this child wouldn ' t have existed and this terrible thing wouldn ' t have happened. But I ' ve come to feel that I love the children I had so much that I don ' t want to imagine a life without them. I recognize the pain they caused to others, for which there can be no forgiveness, but the pain they caused to me, there is,"she said."So while I recognize that it would have been better for the world if Dylan had never been born, I ' ve decided that it would not have been better for me."I thought it was surprising how all of these families had all of these children with all of these problems, problems that they mostly would have done anything to avoid, and that they had all found so much meaning in that experience of parenting. And then I thought, all of us who have children love the children we have, with their flaws. If some glorious angel suddenly descended through my living room ceiling and offered to take away the children I have and give me other, better children â more polite, funnier, nicer, smarter â I would cling to the children I have and pray away that atrocious spectacle. And ultimately I feel that in the same way that we test flame-retardant pajamas in an inferno to ensure they won ' t catch fire when our child reaches across the stove, so these stories of families negotiating these extreme differences reflect on the universal experience of parenting, which is always that sometimes you look at your child and you think, where did you come from? It turns out that while each of these individual differences is siloed â there are only so many families dealing with schizophrenia, there are only so many families of children who are transgender, there are only so many families of prodigies â who also face similar challenges in many ways â there are only so many families in each of those categories â but if you start to think that the experience of negotiating difference within your family is what people are addressing, then you discover that it ' s a nearly universal phenomenon. Ironically, it turns out, that it ' s our differences, and our negotiation of difference, that unite us. I decided to have children while I was working on this project. And many people were astonished and said,"But how can you decide to have children in the midst of studying everything that can go wrong?”And I said,”I ’ m not studying everything that can go wrong. What I ’ m studying is how much love there can be, even when everything appears to be going wrong.”I thought a lot about the mother of one **disabled** child I had seen, a severely **disabled** child who died through caregiver neglect. And when his ashes were interred, his mother said,”I pray here for forgiveness for having been twice robbed, once of the child I wanted and once of the son I loved.”And I figured it was possible then for anyone to love any child if they had the effective will to do so. So my husband is the biological father of two children with some lesbian friends in Minneapolis. I had a close friend from college who ’ d gone through a divorce and wanted to have children. And so she and I have a daughter, and mother and daughter live in Texas. And my husband and I have a son who lives with us all the time of whom I am the biological father, and our surrogate for the pregnancy was Laura, the lesbian mother of Oliver and Lucy in Minneapolis. So the shorthand is five parents of four children in three states. And there are people who think that the existence of my family somehow undermines or weakens or damages their family. And there are people who think that families like mine shouldn ’ t be allowed to exist. And I don ’ t accept subtractive models of love, only additive ones. And I believe that in the same way that we need species diversity to ensure that the planet can go on, so we need this diversity of affection and diversity of family in order to strengthen the ecosphere of kindness. The day after our son was born, the pediatrician came into the hospital room and said she was concerned. He wasn ’ t extending his legs appropriately. She said that might mean that he had brain damage. In so far as he was extending them, he was doing so asymmetrically, which she thought could mean that there was a tumor of some kind in action. And he had a very large head, which she thought might indicate hydrocephalus. And as she told me all of these things, I felt the very center of my being pouring out onto the floor. And I thought, here I had been working for years on a book about how much meaning people had found in the experience of parenting children who are **disabled**, and I didn ’ t want to join their number. Because what I was encountering was an idea of illness. And like all parents since the dawn of time, I wanted to protect my child from illness. And I wanted also to protect myself from illness. And yet, I knew from the work I had done that if he had any of the things we were about to start testing for, that those would ultimately be his identity, and if they were his identity they would become my identity, that that illness was going to take a very different shape as it unfolded. We took him to the MRI machine, we took him to the CAT scanner, we took this day-old child and gave him over for an arterial blood draw. We felt **helpless**. And at the end of five hours, they said that his brain was completely clear and that he was by then extending his legs correctly. And when I asked the pediatrician what had been going on, she said she thought in the morning he had probably had a cramp. But I thought how my mother was right. I thought, the love you have for your children is unlike any other feeling in the world, and until you have children, you don ’ t know what it feels like. I think children had ensnared me the moment I connected fatherhood with loss. But I ’ m not sure I would have noticed that if I hadn ’ t been so in the thick of this research project of mine. I ’ d encountered so much strange love, and I fell very naturally into its bewitching patterns. And I saw how splendor can illuminate even the most abject vulnerabilities. During these 10 years, I had witnessed and learned the terrifying joy of unbearable responsibility, and I had come to see how it conquers everything else. And while I had sometimes thought the parents I was interviewing were fools, enslaving themselves to a lifetime ’ s journey with their thankless children and trying to breed identity out of misery, I realized that day that my research had built me a plank and that I was ready to join them on their ship. Thank you.

Text Sample 390: POS Dim 3 – 2009 – Score 13.00 – 2009/t002844.txt
Salaam. Namaskar. Good morning. Given my TED profile, you might be expecting that I ’ m going to speak to you about the latest philanthropic trends â the one that ’ s currently got Wall Street and the World Bank buzzing â how to invest in women, how to empower them, how to save them. Not me. I am interested in how women are saving us. They ’ re saving us by redefining and re-imagining a future that defies and blurs accepted polarities, polarities we ’ ve taken for granted for a long time, like the ones between modernity and tradition, First World and Third World, oppression and opportunity. In the midst of the daunting challenges we face as a global community, there ’ s something about this third way raga that is making my heart **sing**. What intrigues me most is how women are doing this, despite a set of paradoxes that are both frustrating and fascinating. Why is it that women are, on the one hand, viciously oppressed by cultural practices, and yet at the same time, are the preservers of cultures in most societies? Is the hijab or the headscarf a symbol of submission or resistance? When so many women and girls are beaten, raped, maimed on a daily basis in the name of all kinds of causes â honor, religion, nationality â what allows women to replant trees, to rebuild societies, to lead radical, non-violent movements for social change? Is it different women who are doing the preserving and the radicalizing? Or are they one and the same? Are we guilty, as Chimamanda Adichie reminded us at the TED conference in Oxford, of assuming that there is a single story of women ’ s struggles for their rights while there are, in fact, many? And what, if anything, do men have to do with it? Much of my life has been a quest to get some answers to these questions. It ’ s taken me across the globe and introduced me to some amazing people. In the process, I ’ ve gathered a few fragments that help me shed some light on this puzzle. Among those who ’ ve helped open my eyes to a third way are : a devout Muslim in Afghanistan, a group of harmonizing lesbians in Croatia and a taboo breaker in Liberia. I ’ m indebted to them, as I am to my parents, who for some set of misdemeanors in their last life, were blessed with three daughters in this one. And for reasons equally unclear to me, seem to be inordinately proud of the three of us. I was born and raised here in India, and I learned from an early age to be deeply suspicious of the aunties and uncles who would bend down, pat us on the head and then say to my parents with no problem at all,”Poor things. You only have three daughters. But you ’ re young, you could still try again.”My sense of outrage about women ’ s rights was brought to a boil when I was about 11. My aunt, an incredibly articulate and brilliant woman, was widowed early. A flock of relatives descended on her. They took off her colorful sari. They made her wear a white one. They wiped her bindi off her forehead. They broke her bangles. Her daughter, Rani, a few years older than me, sat in her lap bewildered, not knowing what had happened to the confident woman she once knew as her mother. Late that night, I heard my mother begging my father,”Please do something Ramu. Can ’ t you intervene?”And my father, in a low voice, muttering,”I ’ m just the youngest brother, there ’ s nothing I can do. This is tradition.”That ’ s the night I learned the rules about what it means to be female in this world. Women don ’ t make those rules, but they define us, and they define our opportunities and our chances. And men are affected by those rules too. My father, who had fought in three wars, could not save his own sister from this **suffering**. By 18, under the excellent tutelage of my mother, I was therefore, as you might expect, defiantly feminist. On

the **streets** chanting,"[Hindi] [Hindi] We are the women of India. We are not flowers, we are sparks of change."By the time I got to Beijing in 1995, it was clear to me, the only way to achieve gender **equality** was to overturn centuries of oppressive tradition. Soon after I returned from Beijing, I leapt at the chance to work for this wonderful organization, founded by women, to support women ' s rights organizations around the globe. But barely six months into my new job, I met a woman who forced me to challenge all my assumptions. Her name is Sakena Yacoobi. She walked into my office at a time when no one knew where Afghanistan was in the United States. She said to me,"It is not about the burka."She was the most determined advocate for women ' s rights I had ever heard. She told me women were running underground schools in her communities inside Afghanistan, and that her organization, the Afghan Institute of Learning, had started a school in Pakistan. She said,"The first thing anyone who is a Muslim knows is that the Koran requires and strongly supports literacy. The prophet wanted every believer to be able to read the Koran for themselves."Had I heard right? Was a women ' s rights advocate invoking religion? But Sakena defies labels. She always wears a headscarf, but I ' ve walked alongside with her on a beach with her long hair flying in the breeze. She starts every lecture with a **prayer**, but she ' s a single, feisty, financially independent woman in a country where girls are married off at the age of 12. She is also immensely pragmatic."This headscarf and these clothes,"she says,"give me the freedom to do what I need to do to speak to those whose support and assistance are critical for this work. When I had to open the school in the refugee camp, I went to see the imam. I told him, ' I ' m a believer, and women and children in these terrible conditions need their faith to survive. '"She smiles slyly."He was flattered. He began to come twice a week to my center because women could not go to the **mosque**. And after he would leave, women and girls would stay behind. We began with a small literacy class to read the Koran, then a math class, then an English class, then computer classes. In a few weeks, everyone in the refugee camp was in our classes."Sakena is a teacher at a time when to educate women is a dangerous business in Afghanistan. She is on the Taliban ' s hit list. I worry about her every time she travels across that country. She shrugs when I ask her about **safety**."Kavita jaan, we cannot allow ourselves to be afraid. Look at those young girls who go back to school when acid is thrown in their face."And I smile, and I nod, realizing I ' m watching women and girls using their own religious traditions and practices, turning them into instruments of opposition and opportunity. Their path is their own and it looks towards an Afghanistan that will be different. Being different is something the women of Lesbor in Zagreb, Croatia know all too well. To be a lesbian, a dyke, a homosexual in most parts of the world, including right here in our country, India, is to occupy a place of immense discomfort and extreme **prejudice**. In post-conflict societies like Croatia, where a hyper-nationalism and religiosity have created an environment unbearable for anyone who might be considered a social outcast. So enter a group of out dykes, young women who love the old music that once spread across that region from Macedonia to Bosnia, from Serbia to Slovenia. These folk singers met at college at a gender studies program. Many are in their 20s, some are mothers. Many have struggled to come out to their communities, in families whose religious beliefs make it hard to accept that their daughters are not sick, just queer. As Leah, one of the founders of the group, says,"I like traditional music very much. I also like rock and roll. So Lesbor, we blend the two. I see traditional music like a kind of rebellion, in which people can really speak their voice, especially traditional songs from other parts of the former Yugoslav Republic. After the war, lots of these songs were lost, but they are a part of our childhood and our history, and we should not forget them.”Improbably, this LGBT singing **choir** has demonstrated how women are investing in tradition to create change, like alchemists turning discord into harmony. Their repertoire includes the Croatian national anthem, a Bosnian love song and Serbian duets. And, Leah adds with a grin,”Kavita, we especially are proud of our Christmas music, because it shows we are open to religious practices even though Catholic Church hates us LGBT.”Their concerts draw from their own communities, yes, but also from an older generation : a generation that might be suspicious of homosexuality, but is nostalgic for its own music and the past it represents. One father, who had initially balked at his daughter coming out in such a **choir**, now writes songs for them. In the Middle Ages, troubadours would travel across the land **singing** their tales and sharing their verses : Lesbor travels through the Balkans like this, singing, connecting people divided by religion, nationality and language. Bosnians, Croats and Serbs find a rare **shared** space of pride in their history, and Lesbor reminds them that the songs one group often claims as theirs alone really belong to them all. Yesterday, Mallika Sarabhai showed us that music can create a world more accepting of difference than the one we have been given. The world Leymah Gbowee was given was a world at war. Liberia had been torn apart by civil strife for decades. Leymah was not an activist, she was a mother of three. But she was sick with worry : She worried her son would be abducted and taken off to be a child soldier, she worried her daughters would be raped, she worried for their lives. One night, she had a dream. She dreamt she and thousands of other women ended the bloodshed. The next morning at church, she asked others how they felt. They were all tired of the fighting. We need peace, and we need our leaders to know we will not rest until there is peace. Among Leymah ’ s friends was a policewoman who was Muslim. She promised to raise the issue with her community. At the next Friday sermon, the women who were sitting in the side room of the **mosque** began to share their distress at the state of affairs.”What does it matter?”they said,”A bullet doesn ’ t distinguish between a Muslim and a Christian.”This small group of women, determined to bring an end to the war, and they chose to use their traditions to make a point : Liberian women usually wear lots of jewelry and colorful clothing. But no, for the protest, they dressed all in white, no makeup. As Leymah said,”We wore the white saying we were out for peace.”They stood on the side of the road on which Charles Taylor ’ s motorcade passed every day. They stood for weeks â first just 10, then 20, then 50, then hundreds of women â wearing white, singing, dancing, saying they were out for peace. Eventually, opposing forces in Liberia were pushed to hold peace talks in Ghana. The peace talks dragged on and on and on. Leymah and her sisters had had enough. With their remaining funds, they took a small group of women down to the venue of the peace talks and they surrounded the building. In a now famous CNN clip, you can see them sitting on the ground, their arms linked. We know this in India. It ’ s called a [Hindi]. Then things get tense. The police are called in to physically remove the women. As the senior officer approaches with a baton, Leymah stands up with deliberation, reaches her arms up over her head, and begins, very slowly, to untie her headdress that covers her hair. You can see the policeman ’ s face. He looks embarrassed. He backs away. And the next thing you know, the police have disappeared. Leymah said to me later,”It ’ s a taboo, you know, in West Africa. If an older woman undresses in front of a man because she wants to, the man ’ s family is cursed.”She said,”I don ’ t know if he did it because he believed, but he knew we were not going to leave. We were not going to leave until the peace accord was signed.”And the peace accord was signed. And the women of Liberia then mobilized in support of Ellen Johnson Sirleaf, a woman who broke a few taboos herself becoming the first elected woman head of state in Africa in years. When she made her presidential address, she acknowledged these brave women of Liberia who allowed her to win against a football star â that ' s soccer for you Americans â no less. Women like Sakena and Leah and Leymah have humbled me and changed me and made me realize that I should not be so quick to jump to assumptions of any kind. They ' ve also saved me from my righteous anger by offering insights into this third way. A Filipina activist once said to me,"How do you cook a rice cake? With heat from the bottom and heat from the top."The protests, the marches, the uncompromising position that women ' s rights are human rights, full stop. That ' s the heat from the bottom. That ' s Malcolm X and the suffragists and **gay** pride parades. But we also need the heat from the top. And in most parts of the world, that top is still controlled by men. So to paraphrase Marx : Women make change, but not in circumstances of their own choosing. They have to negotiate. They have to subvert tradition that once **silenced** them in order to give voice to new aspirations. And they need allies from their communities. Allies like the imam, allies like the father who now writes songs for a lesbian group in Croatia, allies like the policeman who honored a taboo and backed away, allies like my father, who couldn ' t help his sister but has helped three daughters pursue their dreams. Maybe this is because feminism, unlike almost every other social movement, is not a struggle against a distinct oppressor â it ' s not the ruling class or the occupiers or the colonizers â it ' s against a deeply held set of beliefs and assumptions that we women, far too often, hold ourselves. And perhaps this is the ultimate gift of feminism, that the personal is in fact the political. So that, as Eleanor Roosevelt said once of human rights, the same is true of gender **equality** : that it starts in small places, close to home. On the **streets**, yes, but also in negotiations at the kitchen table and in the marital bed and in relationships between lovers and parents and sisters and friends. And then you realize that by integrating aspects of tradition and community into their struggles, women like Sakena and Leah and Leymah â but also women like Sonia Gandhi here in India and Michelle Bachelet in Chile and Shirin Ebadi in Iran â are doing something else. They ' re challenging the very notion of Western models of development. They are saying, we don ' t have to be like you to make change. We can wear a sari or a hijab or pants or a boubou, and we can be party leaders and presidents and human rights lawyers. We can use our tradition to navigate change. We can demilitarize societies and pour resources, instead, into reservoirs of genuine security. It is in these little stories, these individual stories, that I see a radical epic being written by women around the world. It is in these threads that are being woven into a resilient fabric that will sustain communities, that I find hope. And if my heart is **singing**, it ' s because in these little fragments, every now and again, you catch a glimpse of a whole, of a whole new world. And she is definitely on her way. Thank you.

Text Sample 391: POS Dim 3 – 2009 – Score 11.00 – 2009/t002967.txt

Okay, so 90 percent of my photographic process is, in fact, not photographic. It involves a campaign of letter writing, research and phone calls to **access** my subjects, which can range from Hamas leaders in Gaza to a hibernating black bear in its cave in West Virginia. And oddly, the most notable letter of rejection I ever received came from Walt Disney World, a seemingly innocuous site. And it read — I ' m just going to read a key sentence :"Especially during these **violent** times, I personally believe that the magical spell cast upon guests who visit our theme parks is particularly important to protect and helps to provide them with an important fantasy they can escape to."Photography threatens fantasy.

They didn ’ t want to let my camera in because it confronts constructed realities, myths and beliefs, and provides what appears to be evidence of a truth. But there are multiple truths attached to every image, depending on the creator ’ s intention, the viewer and the context in which it is presented. Over a five year period following September 11th, when the American media and government were seeking hidden and unknown sites beyond its borders, most notably weapons of mass destruction, I chose to look inward at that which was integral to America ’ s foundation, mythology and daily functioning. I wanted to confront the boundaries of the citizen, self-imposed and real, and confront the divide between privileged and public **access** to knowledge. It was a critical moment in American history and global history where one felt they didn ’ t have **access** to accurate information. And I wanted to see the center with my own eyes, but what I came away with is a photograph. And it ’ s just another place from which to observe, and the understanding that there are no absolute, all-knowing insiders. And the outsider can never really reach the core. I ’ m going to run through some of the photographs in this series. It ’ s titled,”An American Index of the Hidden and Unfamiliar,”and it ’ s comprised of nearly 70 images. In this context I ’ ll just show you a few. This is a nuclear waste storage and encapsulation facility at Hanford site in Washington State, where there are over 1, 900 stainless steel capsules containing nuclear waste submerged in water. A human standing in front of an unprotected capsule would die instantly. And I found one section amongst all of these that actually resembled the outline of the United States of America, which you can see here. And a big part of the work that is sort of absent in this context is text. So I create these two poles. Every image is accompanied with a very detailed factual text. And what I ’ m most interested in is the invisible space between a text and its accompanying image, and how the image is transformed by the text and the text by the image. So, at best, the image is meant to float away into abstraction and multiple truths and fantasy. And then the text functions as this cruel anchor that kind of nails it to the ground. But in this context I ’ m just going to read an abridged version of those texts. This is a cryopreservation unit, and it holds the bodies of the wife and mother of cryonics pioneer Robert Ettinger, who hoped to be awoken one day to extended life in good health, with advancements in science and technology, all for the cost of 35 thousand dollars, for forever. This is a 21-year-old Palestinian woman undergoing hymenoplasty. Hymenoplasty is a surgical procedure which restores the virginal state, allowing her to adhere to certain cultural expectations regarding virginity and marriage. So it essentially reconstructs a ruptured hymen, allowing her to bleed upon having **sexual** intercourse, to simulate the loss of virginity. This is a jury simulation deliberation room, and you can see beyond that two- way mirror jury advisers standing in a room behind the mirror. And they observe deliberations after mock trial proceedings so that they can better advise their clients how to adjust their trial strategy to have the outcome that they ’ re hoping for. This process costs 60, 000 dollars. This is a U. S. Customs and Border Protection room, a contraband room, at John F. Kennedy International Airport. On that table you can see 48 hours ’ worth of seized goods from passengers entering in to the United States. There is a pig ’ s head and African cane rats. And part of my photographic work is I ’ m not just documenting what ’ s there. I do take certain liberties and intervene. And in this I really wanted it to resemble an early still-life painting, so I spent some time with the smells and items. This is the exhibited art on the walls of the CIA in Langley, Virginia, their original headquarters building. And the CIA has had a long history with both covert and public cultural diplomacy efforts. And it ’ s speculated that some of their interest in the arts was designed to counter Soviet communism and promote what it considered to be pro-American thoughts and aesthetics. And one of the art forms that elicited the interest of the agency, and had thus come under question, is abstract expressionism. This is the Forensic Anthropology Research Facility, and on a six acre plot there are approximately 75 cadavers at any given time that are being studied by forensic anthropologists and researchers who are interested in monitoring a rate of corpse decomposition. And in this particular photograph the body of a young boy has been used to reenact a crime scene. This is the only federally funded site where it is legal to cultivate cannabis for scientific research in the United States. It ' s a research crop marijuana grow room. And part of the work that I hope for is that there is a sort of disorienting entropy where you can ' t find any discernible **formula** in how these things — they sort of awkwardly jump from government to science to religion to security — and you can ' t completely understand how information is being distributed. These are transatlantic submarine communication cables that travel across the floor of the Atlantic Ocean, connecting North America to Europe. They carry over 60 million simultaneous voice conversations, and in a lot of the government and technology sites there was just this very apparent vulnerability. This one is almost humorous because it feels like I could just snip all of that conversation in one easy cut. But stuff did feel like it could have been taken 30 or 40 years ago, like it was locked in the Cold War era and hadn ' t necessarily progressed. This is a Braille edition of Playboy magazine. And this is... a division of the Library of Congress produces a free national library service for the blind and visually impaired, and the publications they choose to publish are based on reader popularity. And Playboy is always in the top few. But you ' d be surprised, they don ' t do the photographs. It ' s just the text. This is an avian quarantine facility where all imported birds coming into America are required to undergo a 30-day quarantine, where they are tested for diseases including Exotic Newcastle Disease and Avian Influenza. This film shows the testing of a new explosive fill on a warhead. And the Air Armament Center at Eglin Air Force Base in Florida is responsible for the deployment and testing of all air-delivered weaponry coming from the United States. And the film was shot on 72 millimeter, government-issue film. And that red dot is a marking on the government-issue film. All living white tigers in North America are the result of selective inbreeding — that would be mother to son, father to daughter, sister to brother — to allow for the genetic conditions that create a salable white tiger. Meaning white fur, ice blue eyes, a pink nose. And the majority of these white tigers are not born in a salable state and are killed at birth. It ' s a very **violent** process that is little known. And the white tiger is obviously celebrated in several forms of entertainment. Kenny was born. He actually made it to adulthood. He has since passed away, but was mentally retarded and suffers from severe bone abnormalities. This, on a lighter note, is at George Lucas ' personal archive. This is the Death Star. And it ' s shown here in its true orientation. In the context of"Star Wars : Return of the Jedi,"its mirror image is presented. They flip the negative. And you can see the photoetched brass detailing, and the painted acrylic facade. In the context of the film, this is a deep-space battle station of the Galactic Empire, capable of annihilating planets and civilizations, and in reality it measures about four feet by two feet. This is at Fort Campbell in Kentucky. It ' s a Military Operations on Urbanized Terrain site. Essentially they ' ve simulated a city for urban combat, and this is one of the structures that exists in that city. It ' s called the World Church of God. It ' s supposed to be a generic site of worship. And after I took this photograph, they constructed a wall around the World Church of God to mimic the set-up of **mosques** in Afghanistan or Iraq. And I worked with Mehta Vihar who creates virtual simulations for the army for tactical practice. And we put that wall around the World Church of God, and also used the characters and vehicles and explosions that are offered in the video games for the army. And I put them into my photograph. This is live HIV virus at Harvard Medical School, who is working with the U. S. Government to develop sterilizing immunity. And Alhurra is a U. S. Government- sponsored Arabic language television network that distributes news and information to over 22 countries in the Arab world. It runs 24 hours a day, commercial free. However, it ' s illegal to broadcast Alhurra within the United States. And in 2004, they developed a channel called Alhurra Iraq, which specifically deals with events occurring in Iraq and is broadcast to Iraq. Now I ' m going to move on to another project I did. It ' s titled"The **Innocents**."And for the men in these photographs, photography had been used to create a fantasy. Contradicting its function as evidence of a truth, in these instances it furthered the fabrication of a lie. I traveled across the United States photographing men and women who had been wrongfully **convicted** of crimes they did not commit, **violent** crimes. I investigate photography ' s ability to blur truth and fiction, and its influence on memory, which can lead to severe, even lethal consequences. For the men in these photographs, the primary cause of their wrongful conviction was mistaken identification. A **victim** or eyewitness identifies a suspected **perpetrator** through law **enforcement** ' s use of images. But through exposure to composite sketches, Polaroids, mug shots and line-ups, eyewitness testimony can change. I ' ll give you an example from a case. A woman was raped and presented with a series of photographs from which to identify her attacker. She saw some similarities in one of the photographs, but couldn ' t quite make a positive identification. Days later, she is presented with another photo array of all new photographs, except that one photograph that she had some draw to from the earlier array is repeated in the second array. And a positive identification is made because the photograph replaced the memory, if there ever was an actual memory. Photography offered the criminal justice system a tool that transformed **innocent** citizens into criminals, and the criminal justice system failed to recognize the limitations of relying on photographic identifications. Frederick Daye, who is photographed at his alibi location, where 13 witnesses placed him at the time of the crime. He was **convicted** by an all-white jury of rape, kidnapping and vehicle theft. And he served 10 years of a life sentence. Now DNA exonerated Frederick and it also implicated another man who was serving time in prison. But the **victim** refused to press charges because she claimed that law **enforcement** had permanently altered her memory through the use of Frederick ' s photograph. Charles Fain was **convicted** of kidnapping, rape and murder of a young girl walking to school. He served 18 years of a death sentence. I photographed him at the scene of the crime at the Snake River in Idaho. And I photographed all of the wrongfully **convicted** at sites that came to particular significance in the history of their wrongful conviction. The scene of arrest, the scene of misidentification, the alibi location. And here, the scene of the crime, it ' s this place to which he ' s never been, but changed his life forever. So photographing there, I was hoping to highlight the tenuous relationship between truth and fiction, in both his life and in photography. Calvin Washington was **convicted** of capital murder. He served 13 years of a life sentence in Waco, Texas. Larry Mayes, I photographed at the scene of arrest, where he hid between two mattresses in Gary, Indiana, in this very room to hide from the police. He ended up serving 18 and a half years of an 80 year sentence for rape and robbery. The **victim** failed to identify Larry in two live lineups and then made a positive identification, days later, from a photo array. Larry Youngblood served eight years of a 10 and half year sentence in Arizona for the abduction and repeated sodomizing of a 10 year old boy at a carnival. He is photographed at his alibi location. Ron Williamson. Ron was **convicted** of the rape and murder of a barmaid at a club, and served 11 years of a death sentence. I photographed Ron at a baseball field because he had been drafted to the Oakland A ' s to play professional baseball just before his conviction. And the state ' s key witness in Ron ' s case was, in the end, the actual **perpetrator**. Ronald Jones served eight years of a death sentence for rape and murder of a 28-year-old woman. I photographed him at the scene of arrest in Chicago. William Gregory was **convicted** of rape and burglary. He served seven years of a 70 year sentence in Kentucky. Timothy Durham, who I photographed at his alibi location where 11 witnesses placed him at the time of the crime, was **convicted** of 3. 5 years of a 3220 year sentence, for several charges of rape and robbery. He had been misidentified by an 11-year-old **victim**. Troy Webb is photographed here at the scene of the crime in Virginia. He was **convicted** of rape, kidnapping and robbery, and served seven years of a 47 year sentence. Troy ' s picture was in a photo array that the **victim** tentatively had some draw toward, but said he looked too old. The police went and found a photograph of Troy Webb from four years earlier, which they entered into a photo array days later, and he was positively identified. Now I ' m going to leave you with a self portrait. And it reiterates that distortion is a constant, and our eyes are easily deceived. That ' s it. Thank you.

Text Sample 392: POS Dim 3 – 2009 – Score 11.00 – 2009/t002939.txt

I ' m a storyteller. And I would like to tell you a few personal stories about what I like to call"the danger of the single story."I grew up on a university campus in eastern Nigeria. My mother says that I started reading at the age of two, although I think four is probably close to the truth. So I was an early reader, and what I read were British and American children ' s books. I was also an early writer, and when I began to write, at about the age of seven, stories in pencil with crayon illustrations that my poor mother was obligated to read, I wrote exactly the kinds of stories I was reading : All my characters were white and blue-eyed, they played in the snow, they ate apples, and they talked a lot about the weather, how lovely it was that the sun had come out. Now, this despite the fact that I lived in Nigeria. I had never been outside Nigeria. We didn ' t have snow, we ate mangoes, and we never talked about the weather, because there was no need to. My characters also drank a lot of ginger beer, because the characters in the British books I read drank ginger beer. Never mind that I had no idea what ginger beer was. And for many years afterwards, I would have a desperate desire to taste ginger beer. But that is another story. What this demonstrates, I think, is how impressionable and vulnerable we are in the face of a story, particularly as children. Because all I had read were books in which characters were foreign, I had become convinced that books by their very nature had to have foreigners in them and had to be about things with which I could not personally identify. Now, things changed when I discovered African books. There weren ' t many of them available, and they weren ' t quite as easy to find as the foreign books. But because of writers like Chinua Achebe and Camara Laye, I went through a mental shift in my perception of literature. I realized that people like me, girls with skin the color of chocolate, whose kinky hair could not form ponytails, could also exist in literature. I started to write about things I recognized. Now, I loved those American and British books I read. They stirred my imagination. They opened up new worlds for me. But the unintended consequence was that I did not know that people like me could exist in literature. So what the discovery of African writers did for me was this : It saved me from having a single story of what books are. I come from a conventional, middle-class Nigerian family. My father was a professor. My mother was an administrator. And so we had, as was the norm, live-in **domestic** help, who would often come from nearby rural villages. So, the year I turned eight, we got a new house boy. His name was Fide. The only thing my mother told us about him was that his family was very poor. My mother sent yams and rice, and our old clothes, to his family. And when I didn ’ t finish my dinner, my mother would say,”Finish your food! Don ’ t you know? People like Fide ’ s family have nothing.”So I felt **enormous pity** for Fide ’ s family. Then one Saturday, we went to his village to visit, and his mother showed us a beautifully patterned basket made of dyed raffia that his brother had made. I was startled. It had not occurred to me that anybody in his family could actually make something. All I had heard about them was how poor they were, so that it had become impossible for me to see them as anything else but poor. Their poverty was my single story of them. Years later, I thought about this when I left Nigeria to go to university in the United States. I was 19. My American roommate was shocked by me. She asked where I had learned to speak English so well, and was confused when I said that Nigeria happened to have English as its official language. She asked if she could listen to what she called my”tribal music,”and was consequently very disappointed when I produced my tape of Mariah Carey. She assumed that I did not know how to use a stove. What struck me was this : She had felt sorry for me even before she saw me. Her default position toward me, as an African, was a kind of patronizing, well- meaning **pity**. My roommate had a single story of Africa : a single story of catastrophe. In this single story, there was no possibility of Africans being similar to her in any way, no possibility of feelings more complex than **pity**, no possibility of a connection as human **equals**. I must say that before I went to the U. S., I didn ’ t consciously identify as African. But in the U. S., whenever Africa came up, people turned to me. Never mind that I knew nothing about places like Namibia. But I did come to embrace this new identity, and in many ways I think of myself now as African. Although I still get quite irritable when Africa is referred to as a country, the most recent example being my otherwise wonderful flight from Lagos two days ago, in which there was an announcement on the Virgin flight about the charity work in”India, Africa and other countries.”So, after I had spent some years in the U. S. as an African, I began to understand my roommate ’ s response to me. If I had not grown up in Nigeria, and if all I knew about Africa were from popular images, I too would think that Africa was a place of beautiful landscapes, beautiful animals, and incomprehensible people, fighting senseless wars, dying of poverty and AIDS, unable to speak for themselves and waiting to be saved by a kind, white foreigner. I would see Africans in the same way that I, as a child, had seen Fide ’ s family. This single story of Africa ultimately comes, I think, from Western literature. Now, here is a quote from the writing of a London merchant called John Lok, who sailed to west Africa in 1561 and kept a fascinating account of his voyage. After referring to the black Africans as”beasts who have no houses,”he writes,”They are also people without heads, having their mouth and eyes in their breasts.”Now, I ’ ve laughed every time I ’ ve read this. And one must admire the imagination of John Lok. But what is important about his writing is that it represents the beginning of a tradition of telling African stories in the West : A tradition of Sub-Saharan Africa as a place of negatives, of difference, of **darkness**, of people who, in the words of the wonderful poet Rudyard Kipling, are”half devil, half child.”And so, I began to realize that my American roommate must have throughout her life seen and heard different versions of this single story, as had a professor, who once told me that my novel was not”authentically African.”Now, I was quite willing to contend that there were a number of things wrong with the novel, that it had failed in a number of places, but I had not quite imagined that it had failed at achieving something called African authenticity. In fact, I did not know what African authenticity was. The professor told me that my characters were too much like him, an educated and middle-class man. My characters drove cars. They were not starving. Therefore they were not authentically African. But I must quickly add that I too am just as guilty in the question of the single story. A few years ago, I visited Mexico from the U. S. The political climate in the U. S. at the time was tense, and there were debates going on about **immigration**. And, as often happens in America, **immigration** became synonymous with Mexicans. There were endless stories of Mexicans as people who were fleecing the healthcare system, sneaking across the border, being arrested at the border, that sort of thing. I remember walking around on my first day in Guadalajara, watching the people going to work, rolling up tortillas in the marketplace, smoking, laughing. I remember first feeling slight surprise. And then, I was overwhelmed with **shame**. I realized that I had been so immersed in the media coverage of Mexicans that they had become one thing in my mind, the abject **immigrant**. I had bought into the single story of Mexicans and I could not have been more ashamed of myself. So that is how to create a single story, show a people as one thing, as only one thing, over and over again, and that is what they become. It is impossible to talk about the single story without talking about power. There is a word, an Igbo word, that I think about whenever I think about the power structures of the world, and it is"nkali."It ' s a noun that loosely translates to"to be greater than another."Like our economic and political worlds, stories too are defined by the principle of nkali : How they are told, who tells them, when they ' re told, how many stories are told, are really dependent on power. Power is the ability not just to tell the story of another person, but to make it the definitive story of that person. The Palestinian poet Mourid Barghouti writes that if you want to dispossess a people, the simplest way to do it is to tell their story and to start with,"secondly."Start the story with the arrows of the Native Americans, and not with the arrival of the British, and you have an entirely different story. Start the story with the failure of the African state, and not with the colonial creation of the African state, and you have an entirely different story. I recently spoke at a university where a student told me that it was such a **shame** that Nigerian men were physical abusers like the father character in my novel. I told him that I had just read a novel called"American Psycho"— — and that it was such a **shame** that young Americans were serial murderers. Now, obviously I said this in a fit of mild irritation. But it would never have occurred to me to think that just because I had read a novel in which a character was a serial killer that he was somehow representative of all Americans. This is not because I am a better person than that student, but because of America ' s cultural and economic power, I had many stories of America. I had read Tyler and Updike and Steinbeck and Gaitskill. I did not have a single story of America. When I learned, some years ago, that writers were expected to have had really unhappy childhoods to be successful, I began to think about how I could invent horrible things my parents had done to me. But the truth is that I had a very happy childhood, full of laughter and love, in a very close-knit family. But I also had grandfathers who died in refugee camps. My cousin Polle died because he could not get adequate healthcare. One of my closest friends, Okoloma, died in a plane crash because our fire trucks did not have water. I grew up under repressive military governments that devalued education, so that sometimes, my parents were not paid their salaries. And so, as a child, I saw jam disappear from the breakfast table, then margarine disappeared, then bread became too expensive, then milk became rationed. And most of all, a kind of normalized political fear invaded our lives. All of these stories make me who I am. But to insist on only these negative stories is to flatten my experience and to overlook the many other stories that formed me. The single story creates stereotypes, and the problem with stereotypes is not that they are untrue, but that they are incomplete. They make one story become the only story. Of course, Africa is a continent full of catastrophes : There are immense ones, such as the horrific rapes in Congo and depressing ones, such as the fact that 5, 000 people apply for one job vacancy in Nigeria. But there are other stories that are not about catastrophe, and it is very important, it is just as important, to talk about them. I ' ve always felt that it is impossible to engage properly with a place or a person without engaging with all of the stories of that place and that person. The consequence of the single story is this : It robs people of dignity. It makes our recognition of our **equal** humanity difficult. It emphasizes how we are different rather than how we are similar. So what if before my Mexican trip, I had followed the **immigration** debate from both sides, the U. S. and the Mexican? What if my mother had told us that Fide ' s family was poor and hardworking? What if we had an African television network that broadcast diverse African stories all over the world? What the Nigerian writer Chinua Achebe calls"a balance of stories."What if my roommate knew about my Nigerian publisher, Muhtar Bakare, a remarkable man who left his job in a bank to follow his dream and start a publishing house? Now, the conventional wisdom was that Nigerians don ' t read literature. He disagreed. He felt that people who could read, would read, if you made literature affordable and available to them. Shortly after he published my first novel, I went to a TV station in Lagos to do an interview, and a woman who worked there as a messenger came up to me and said,"I really liked your novel. I didn ' t like the ending. Now, you must write a sequel, and this is what will happen..."And she went on to tell me what to write in the sequel. I was not only charmed, I was very moved. Here was a woman, part of the ordinary masses of Nigerians, who were not supposed to be readers. She had not only read the book, but she had taken ownership of it and felt justified in telling me what to write in the sequel. Now, what if my roommate knew about my friend Funmi Iyanda, a fearless woman who hosts a TV show in Lagos, and is determined to tell the stories that we prefer to forget? What if my roommate knew about the heart procedure that was performed in the Lagos hospital last week? What if my roommate knew about contemporary Nigerian music, talented people **singing** in English and Pidgin, and Igbo and Yoruba and Ijo, mixing influences from Jay-Z to Fela to Bob Marley to their grandfathers. What if my roommate knew about the female lawyer who recently went to court in Nigeria to challenge a ridiculous law that required women to get their husband ' s consent before renewing their passports? What if my roommate knew about Nollywood, full of innovative people making films despite great technical odds, films so popular that they really are the best example of Nigerians consuming what they produce? What if my roommate knew about my wonderfully ambitious hair braider, who has just started her own business selling hair extensions? Or about the millions of other Nigerians who start businesses and sometimes fail, but continue to nurse ambition? Every time I am home I am confronted with the usual sources of irritation for most Nigerians : our failed infrastructure, our failed government, but also by the incredible resilience of people who thrive despite the government, rather than because of it. I teach writing workshops in Lagos every summer, and it is amazing to me how many people apply, how many people are eager to write, to tell stories. My Nigerian publisher and I have just started a non-profit called Farafina Trust, and we have big dreams of building libraries and refurbishing libraries that already exist and providing books for state schools that don ' t have anything in their libraries, and also of organizing lots and lots of workshops, in reading and writing, for all the people who are eager to tell our many stories. Stories matter. Many stories matter. Stories have been used to dispossess and to malign, but stories can also be used to empower and to humanize. Stories can break the dignity of a people, but stories can also repair that broken dignity. The American writer Alice Walker wrote this about her Southern relatives who had moved to the North. She introduced them to a book about the Southern life that they had left behind.”They sat around, reading the book themselves, listening to me read the book, and a kind of paradise was regained.”I would like to end with this thought : That when we reject the single story, when we realize that there is never a single story about any place, we regain a kind of paradise. Thank you.

Text Sample 393: POS Dim 3 – 2010 – Score 13.00 – 2010/t002671.txt

I ’ ve been spending a lot of time traveling around the world these days, talking to groups of students and professionals, and everywhere I ’ m finding that I hear similar themes. On the one hand, people say,”The time for change is now.”They want to be part of it. They talk about wanting lives of purpose and greater meaning. But on the other hand, I hear people talking about fear, a sense of risk-aversion. They say,”I really want to follow a life of purpose, but I don ’ t know where to start. I don ’ t want to disappoint my family or friends.”I work in global poverty. And they say,”I want to work in global poverty, but what will it mean about my career? Will I be marginalized? Will I not make enough money? Will I never get married or have children?”And as a woman who didn ’ t get married until I was a lot older — and I ’ m glad I waited — — and has no children, I look at these young people and I say,”Your job is not to be perfect. Your job is only to be human. And nothing important happens in life without a cost.”These conversations really reflect what ’ s happening at the national and international level. Our leaders and ourselves want everything, but we don ’ t talk about the costs. We don ’ t talk about the sacrifice. One of my favorite quotes from literature was written by Tillie Olsen, the great American writer from the South. In a short story called”Oh Yes,”she talks about a white woman in the 1950s who has a daughter who befriends a little African American girl, and she looks at her child with a sense of pride, but she also wonders, what price will she pay?”Better immersion than to live untouched.”But the real question is, what is the cost of not daring? What is the cost of not trying? I ’ ve been so privileged in my life to know extraordinary leaders who have chosen to live lives of immersion. One woman I knew who was a fellow at a program that I ran at the Rockefeller Foundation was named Ingrid Washinawatok. She was a leader of the Menominee tribe, a Native American peoples. And when we would gather as fellows, she would push us to think about how the elders in Native American culture make decisions. And she said they would literally visualize the faces of children for seven generations into the future, looking at them from the Earth, and they would look at them, holding them as stewards for that future. Ingrid understood that we are connected to each other, not only as human beings, but to every living thing on the planet. And tragically, in 1999, when she was in Colombia working with the U ’ wa people, focused on preserving their culture and language, she and two colleagues were abducted and tortured and killed by the FARC. And whenever we would gather the fellows after that, we would leave a chair empty for her spirit. And more than a decade later, when I talk to NGO fellows, whether in Trenton, New Jersey or the office of the White House, and we talk about Ingrid, they all say that they ’ re trying to integrate her wisdom and her spirit and really build on the unfulfilled work of her life ’ s mission.

And when we think about legacy, I can think of no more powerful one, despite how short her life was. And I ’ ve been touched by Cambodian women — beautiful women, women who held the tradition of the classical dance in Cambodia. And I met them in the early ’ 90s. In the 1970s, under the Pol Pot regime, the Khmer Rouge killed over a million people, and they focused and targeted the elites and the intellectuals, the artists, the dancers. And at the end of the war, there were only 30 of these classical dancers still living. And the women, who I was so privileged to meet when there were three survivors, told these stories about lying in their cots in the refugee camps. They said they would try so hard to remember the fragments of the dance, hoping that others were alive and doing the same. And one woman stood there with this perfect carriage, her hands at her side, and she talked about the reunion of the 30 after the war and how extraordinary it was. And these big tears fell down her face, but she never lifted her hands to move them. And the women decided that they would train not the next generation of girls, because they had grown too old already, but the next generation. And I sat there in the studio watching these women clapping their hands — beautiful rhythms — as these little fairy pixies were dancing around them, wearing these beautiful silk colors. And I thought, after all this atrocity, this is how human beings really pray. Because they ’ re focused on honoring what is most beautiful about our past and building it into the promise of our future. And what these women understood is sometimes the most important things that we do and that we spend our time on are those things that we cannot measure. I also have been touched by the dark side of power and leadership. And I have learned that power, particularly in its absolute form, is an **equal** opportunity provider. In 1986, I moved to Rwanda, and I worked with a very small group of Rwandan women to start that country ’ s first microfinance bank. And one of the women was Agnes — there on your extreme left — she was one of the first three women parliamentarians in Rwanda, and her legacy should have been to be one of the mothers of Rwanda. We built this institution based on social justice, gender **equity**, this idea of empowering women. But Agnes cared more about the trappings of power than she did principle at the end. And though she had been part of building a liberal party, a political party that was focused on diversity and tolerance, about three months before the genocide, she switched parties and joined the extremist party, Hutu Power, and she became the Minister of Justice under the genocide regime and was known for inciting men to kill faster and stop behaving like women. She was **convicted** of category one crimes of genocide. And I would visit her in the prisons, sitting side-by-side, knees touching, and I would have to admit to myself that monsters exist in all of us, but that maybe it ’ s not monsters so much, but the broken parts of ourselves, sadnesses, secret **shame**, and that ultimately it ’ s easy for demagogues to prey on those parts, those fragments, if you will, and to make us look at other beings, human beings, as lesser than ourselves — and in the extreme, to do terrible things. And there is no group more vulnerable to those kinds of manipulations than young men. I ’ ve heard it said that the most dangerous animal on the planet is the adolescent male. And so in a gathering where we ’ re focused on women, while it is so critical that we invest in our girls and we even the playing field and we find ways to honor them, we have to remember that the girls and the women are most isolated and violated and victimized and made invisible in those very societies where our men and our boys feel disempowered, unable to provide. And that, when they sit on those **street** corners and all they can think of in the future is no job, no education, no possibility, well then it ’ s easy to understand how the greatest source of status can come from a uniform and a gun. Sometimes very small investments can release **enormous**, infinite potential that exists in all of us. One of the Acumen Fund fellows at my organization, Suraj Sudhakar, has what we call moral imagination — the ability to put yourself in another person ' s shoes and lead from that perspective. And he ' s been working with this young group of men who come from the largest slum in the world, Kibera. And they ' re incredible guys. And together they started a book club for a hundred people in the slums, and they ' re reading many TED authors and liking it. And then they created a business plan competition. Then they decided that they would do TEDx ' s. And I have learned so much from Chris and Kevin and Alex and Herbert and all of these young men. Alex, in some ways, said it best. He said,"We used to feel like nobodies, but now we feel like somebodies."And I think we have it all wrong when we think that income is the link. What we really yearn for as human beings is to be visible to each other. And the reason these young guys told me that they ' re doing these TEDx ' s is because they were sick and tired of the only workshops coming to the slums being those workshops focused on HIV, or at best, microfinance. And they wanted to **celebrate** what ' s beautiful about Kibera and Mathare — the photojournalists and the creatives, the graffiti artists, the teachers and the entrepreneurs. And they ' re doing it. And my hat ' s off to you in Kibera. My own work focuses on making philanthropy more effective and capitalism more inclusive. At Acumen Fund, we take philanthropic resources and we invest what we call patient capital — money that will invest in entrepreneurs who see the poor not as passive recipients of charity, but as full-bodied agents of change who want to solve their own problems and make their own decisions. We leave our money for 10 to 15 years, and when we get it back, we invest in other innovations that focus on change. I know it works. We ' ve invested more than 50 million dollars in 50 companies, and those companies have brought another 200 million dollars into these forgotten markets. This year alone, they ' ve delivered 40 million services like maternal health care and housing, emergency services, solar energy, so that people can have more dignity in solving their problems. Patient capital is uncomfortable for people searching for simple solutions, easy categories, because we don ' t see profit as a blunt instrument. But we find those entrepreneurs who put people and the planet before profit. And ultimately, we want to be part of a movement that is about measuring impact, measuring what is most important to us. And my dream is we ' ll have a world one day where we don ' t just honor those who take money and make more money from it, but we find those individuals who take our resources and convert it into changing the world in the most positive ways. And it ' s only when we honor them and **celebrate** them and give them status that the world will really change. Last May I had this extraordinary 24-hour period where I saw two visions of the world living side-by-side — one based on **violence** and the other on transcendence. I happened to be in Lahore, Pakistan on the day that two **mosques** were attacked by suicide bombers. And the reason these **mosques** were attacked is because the people praying inside were from a particular sect of Islam who fundamentalists don ' t believe are fully Muslim. And not only did those suicide bombers take a hundred lives, but they did more, because they created more **hatred**, more rage, more fear and certainly despair. But less than 24 hours, I was 13 miles away from those **mosques**, visiting one of our Acumen investees, an incredible man, Jawad Aslam, who dares to live a life of immersion. Born and raised in Baltimore, he studied real estate, worked in commercial real estate, and after 9 / 11 decided he was going to Pakistan to make a difference. For two years, he hardly made any money, a tiny stipend, but he apprenticed with this incredible housing developer named Tasneem Saddiqui. And he had a dream that he would build a housing community on this barren piece of land using patient capital, but he continued to pay a price. He stood on moral ground and refused to pay bribes. It took almost two years just to register the land. But I saw how the level of moral standard can rise from one person ' s action. Today, 2, 000 people live in 300 houses in this beautiful community. And there ' s schools and clinics and shops. But there ' s only one **mosque**. And so I asked Jawad,"How do you guys navigate? This is a really diverse community. Who gets to use the **mosque** on Fridays?"He said,"Long story. It was hard, it was a difficult road, but ultimately the leaders of the community came together, realizing we only have each other. And we decided that we would elect the three most respected imams, and those imams would take turns, they would rotate who would say Friday **prayer**. But the whole community, all the different sects, including Shi ' a and Sunni, would sit together and pray."We need that kind of moral leadership and courage in our worlds. We face huge issues as a world — the financial crisis, global warming and this growing sense of fear and otherness. And every day we have a choice. We can take the easier road, the more cynical road, which is a road based on sometimes dreams of a past that never really was, a fear of each other, distancing and blame. Or we can take the much more difficult path of transformation, transcendence, compassion and love, but also accountability and justice. I had the great honor of working with the child psychologist Dr. Robert Coles, who stood up for change during the Civil Rights movement in the United States. And he tells this incredible story about working with a little six-year-old girl named Ruby Bridges, the first child to desegregate schools in the South — in this case, New Orleans. And he said that every day this six-year-old, dressed in her beautiful dress, would walk with real grace through a phalanx of white people screaming angrily, calling her a monster, threatening to poison her — distorted faces. And every day he would watch her, and it looked like she was talking to the people. And he would say,"Ruby, what are you saying?"And she ' d say,"I ' m not talking."And finally he said,"Ruby, I see that you ' re talking. What are you saying?"And she said,"Dr. Coles, I am not talking ; I ' m praying."And he said,"Well, what are you praying?"And she said,"I ' m praying, ' Father, forgive them, for they know not what they are doing. '"At age six, this child was living a life of immersion, and her family paid a price for it. But she became part of history and opened up this idea that all of us should have **access** to education. My final story is about a young, beautiful man named Josephat Byaruhanga, who was another Acumen Fund fellow, who hails from Uganda, a farming community. And we placed him in a company in Western Kenya, just 200 miles away. And he said to me at the end of his year,"Jacqueline, it was so humbling, because I thought as a farmer and as an African I would understand how to transcend culture. But especially when I was talking to the African women, I sometimes made these mistakes — it was so hard for me to learn how to listen."And he said,"So I conclude that, in many ways, leadership is like a panicle of rice. Because at the height of the season, at the height of its powers, it ' s beautiful, it ' s green, it nourishes the world, it reaches to the heavens."And he said,"But right before the harvest, it bends over with great gratitude and humility to touch the earth from where it came."We need leaders. We ourselves need to lead from a place that has the audacity to believe we can, ourselves, extend the fundamental assumption that all men are created **equal** to every man, woman and child on this planet. And we need to have the humility to recognize that we cannot do it alone. Robert Kennedy once said that"few of us have the greatness to bend history itself, but each of us can work to change a small portion of events."And it is in the total of all those acts that the history of this generation will be written. Our lives are so short, and our time on this planet is so precious, and all we have is each other. So may each of you live lives of immersion. They won ' t necessarily be easy lives, but in the end, it is all that will sustain us. Thank you.

Text Sample 394: POS Dim 3 – 2010 – Score 10.00 – 2010/t002767.txt

I am a cultural omnivore, one whose daily commute is made possible by attachment to an iPod â an iPod that contains Wagner and Mozart, pop diva Christina Aguilera, country singer Josh Turner, gangsta rap artist Kirk Franklin, concerti, symphonies and more and more. I ' m a voracious reader, a reader who deals with Ian McEwan down to Stephanie Meyer. I have read the Twilight tetralogy. And one who lives for my home theater, a home theater where I devour DVDs, video on demand and a lot of television. For me,"Law & Order : SVU,"Tina Fey and"30 Rock"and"Judge Judy"â"The people are real, the cases are real, the rulings are final."Now, I ' m convinced a lot of you probably share my **passions**, especially my **passion** for"Judge Judy,"and you ' d fight anybody who attempted to take her away from us, but I ' m a little less convinced that you share the central **passion** of my life, a **passion** for the live professional performing arts, performing arts that represent the orchestral repertoire, yes, but jazz as well, modern dance, **opera**, theater and more and more and more. Frankly, it ' s a sector that many of us who work in the field worry is being endangered and possibly dismantled by technology. While we initially heralded the Internet as the fantastic new marketing device that was going to solve all our problems, we now realize that the Internet is, if anything, too effective in that regard. Depending on who you read, an arts organization or an artist, who tries to attract the attention of a potential single ticket buyer, now competes with between three and 5, 000 different marketing messages a typical citizen sees every single day. We now know, in fact, that technology is our biggest competitor for leisure time. Five years ago, Gen Xers spent 20. 7 hours online and TV, the majority on TV. Gen Yers spent even more â 23. 8 hours, the majority online. And now, a typical university-entering student arrives at college already having spent 20, 000 hours online and an additional 10, 000 hours playing video games â a stark reminder that we operate in a cultural context where video games now outsell music and movie recordings combined. Moreover, we ' re afraid that technology has altered our very assumptions of cultural consumption. Thanks to the Internet, we believe we can get anything we want whenever we want it, delivered to our own doorstep. We can shop at three in the morning or eight at night, ordering jeans tailor-made for our unique body types. Expectations of personalization and customization that the live performing arts â which have set curtain times, set venues, attendant inconveniences of travel, parking and the like â simply cannot meet. And we ' re all acutely aware : what ' s it going to mean in the future when we ask someone to pay a hundred dollars for a symphony, **opera** or ballet ticket, when that cultural consumer is used to downloading on the internet 24 hours a day for 99 cents a song or for free? These are **enormous** questions for those of us that work in this terrain. But as particular as they feel to us, we know we ' re not alone. All of us are engaged in a seismic, fundamental realignment of culture and communications, a realignment that is shaking and decimating the newspaper industry, the magazine industry, the book and publishing industry and more. Saddled in the performing arts as we are, by antiquated union agreements that inhibit and often prohibit mechanical reproduction and streaming, locked into large facilities that were designed to ossify the ideal relationship between artist and audience most appropriate to the 19th century and locked into a business model dependent on high ticket revenues, where we charge exorbitant prices. Many of us shudder in the wake of the collapse of Tower Records and ask ourselves,"Are we next?"Everyone I talk to in performing arts resonates to the words of Adrienne Rich, who, in"Dreams of a

Common Language,”wrote,”We are out in a country that has no language, no laws. Whatever we do together is pure invention. The maps they gave us are out of date by years.”And for those of you who love the arts, aren ’ t you glad you invited me here to brighten your day? Now, rather than saying that we ’ re on the brink of our own annihilation, I prefer to believe that we are engaged in a fundamental reformation, a reformation like the religious Reformation of the 16th century. The arts reformation, like the religious Reformation, is spurred in part by technology, with indeed, the printing press really leading the charge on the religious Reformation. Both reformations were predicated on fractious discussion, internal self-doubt and massive realignment of antiquated business models. And at heart, both reformations, I think, were asking the questions : who ’ s entitled to practice? How are they entitled to practice? And indeed, do we need anyone to intermediate for us in order to have an experience with a spiritual divine? Chris Anderson, someone I trust you all know, editor in chief of Wired magazine and author of The Long Tail, really was the first, for me, to nail a lot of this. He wrote a long time ago, you know, thanks to the invention of the Internet, web technology, minicams and more, the means of artistic production have been democratized for the first time in all of human history. In the 1930s, if any of you wanted to make a movie, you had to work for Warner Brothers or RKO, because who could afford a movie set and lighting equipment and editing equipment and scoring, and more? And now who in this room doesn ’ t know a 14 year-old hard at work on her second, third, or fourth movie? Similarly, the means of artistic distribution have been democratized for the first time in human history. Again, in the ’ 30s, Warner Brothers, RKO did that for you. Now, go to YouTube, Facebook ; you have worldwide distribution without leaving the privacy of your own bedroom. This double impact is occasioning a massive redefinition of the cultural market, a time when anyone is a potential author. Frankly, what we ’ re seeing now in this environment is a massive time, when the entire world is changing as we move from a time when audience numbers are plummeting. But the number of arts participants, people who write poetry, who **sing** songs, who perform in church **choirs**, is exploding beyond our wildest imaginations. This group, others have called the pro-ams, amateur artists doing work at a professional level. You see them on YouTube, in dance competitions, film festivals and more. They are radically expanding our notions of the potential of an aesthetic vocabulary, while they are challenging and undermining the cultural autonomy of our traditional institutions. Ultimately, we now live in a world defined not by consumption, but by participation. But I want to be clear, just as the religious Reformation did not spell the end to the formal Church or to the priesthood ; I believe that our artistic institutions will continue to have importance. They currently are the best opportunities for artists to have lives of economic dignity â not opulence, of dignity. And they are the places where artists who deserve and want to work at a certain scale of resources will find a home. But to view them as synonymous with the entirety of the arts community is, by far, too shortsighted. And indeed, while we ’ ve tended to polarize the amateur from the professional, the single most exciting development in the last five to 10 years has been the rise of the professional hybrid artist, the professional artist who works, not primarily in the concert hall or on the stage ; but most frequently around women ’ s rights, or human rights, or on global warming issues or AIDS relief for more â not out of economic necessity, but out of a deep, organic conviction that the work that she or he is called to do cannot be accomplished in the traditional hermetic arts environment. Today ’ s dance world is not defined solely by the Royal Winnipeg Ballet or the National Ballet of Canada, but by Liz Lerman ’ s Dance Exchange â a multi- generational, professional dance company, whose dancers range in age from 18 to 82, and who work with genomic scientists to embody the DNA strand and with nuclear physicists at CERN. Today ’ s professional theater community is defined, not only the Shaw and Stratford Festivals, but by the Cornerstone Theater of Los Angeles â□□ a collective of artists that after 9 / 11, brought together 10 different religious communities â□□ the Baha ’ i, the Catholic, the Muslim, the Jewish, even the Native American and the **gay** and lesbian communities of faith, helping them create their own individual plays and one massive play, where they explored the differences in their faith and found commonality as an important first step toward cross-community healing. Today ’ s performers, like Rhodessa Jones, work in women ’ s prisons, helping women prisoners articulate the pain of incarceration, while today ’ s playwrights and directors work with youth gangs to find alternate channels to **violence** and more and more and more. And indeed, I think, rather than being annihilated, the performing arts are poised on the brink of a time when we will be more important than we have ever been. You know, we ’ ve said for a long time, we are critical to the health of the economic communities in your town. And absolutely â□□ I hope you know that every dollar spent on a performing arts ticket in a community generates five to seven additional dollars for the local economy, dollars spent in restaurants or on parking, at the fabric stores where we buy fabric for costumes, the piano tuner who tunes the instruments, and more. But the arts are going to be more important to economies as we go forward, especially in industries we can ’ t even imagine yet, just as they have been central to the iPod and the computer game industries, which few, if any of us, could have foreseen 10 to 15 years ago. Business leadership will depend more and more on emotional intelligence, the ability to listen deeply, to have empathy, to articulate change, to motivate others â□□ the very capacities that the arts cultivate with every encounter. Especially now, as we all must confront the fallacy of a market-only orientation, uninformed by social conscience ; we must seize and **celebrate** the power of the arts to shape our individual and national characters, and especially characters of the young people, who all too often are subjected to bombardment of sensation, rather than digested experience. Ultimately, especially now in this world, where we live in a context of regressive and onerous **immigration** laws, in reality TV that thrives on humiliation, and in a context of analysis, where the thing we hear most repeatedly, day in, day out in the United States, in every train station, every bus station, every plane station is,”Ladies and gentlemen, please report any suspicious behavior or suspicious individuals to the authorities nearest to you,”when all of these ways we are encouraged to view our fellow human being with hostility and fear and contempt and suspicion. The arts, whatever they do, whenever they call us together, invite us to look at our fellow human being with generosity and curiosity. God knows, if we ever needed that capacity in human history, we need it now. You know, we ’ re bound together, not, I think by technology, entertainment and design, but by common cause. We work to promote healthy vibrant societies, to ameliorate human **suffering**, to promote a more thoughtful, substantive, empathic world order. I salute all of you as activists in that quest and urge you to embrace and hold dear the arts in your work, whatever your purpose may be. I promise you the hand of the Doris Duke Charitable Foundation is stretched out in friendship for now and years to come. And I thank you for your kindness and your patience in listening to me this afternoon. Thank you, and Godspeed.

Text Sample 395: POS Dim 3 – 2010 – Score 9.00 – 2010/t002670.txt

Our face is hugely important because it ’ s the external, visual part that everybody else sees. Let ’ s not forget it ’ s a functional entity. We have strong skull bones that protect the most important organ in our body : the brain. It ’ s where our senses are located, our special senses â our vision, our speech, our hearing, our smell, our taste. And this bone is peppered, as you can see, with the light shining through the skull with cavities, the sinuses, which warm and moisten the air we breathe. But also imagine if they were filled with solid bone â our head would be dead weight, we wouldn ’ t be able to hold it erect, we wouldn ’ t be able to look at the world around us. This woman is slowly dying because the benign tumors in her facial bones have completely obliterated her mouth and her nose so she can ’ t breathe and eat. Attached to the facial bones that define our face ’ s structure are the muscles that deliver our facial expression, our universal language of expression, our social-signaling system. And overlying this is the skin drape, which is a hugely complex three-dimensional structure â taking right-angled bends here and there, having thin areas like the eyelids, thick areas like the cheek, different colors. And then we have the sensual factor of the face. Where do we like to kiss people? On the lips. Nibble the ears maybe. It ’ s the face where we ’ re attracted to with that. But let ’ s not forget the hair. You ’ re looking at the image on your left-hand side â that ’ s my son with his eyebrows present. Look how odd he looks with the eyebrows missing. There ’ s a definite difference. And imagine if he had hair sprouting from the middle of his nose, he ’ d look even odder still. Dysmorphophobia is an extreme version of the fact that we don ’ t see ourselves as others see us. It ’ s a shocking truth that we only see mirror images of ourselves, and we only see ourselves in freeze-frame photographic images that capture a mere fraction of the time that we live. Dysmorphophobia is a perversion of this where people who may be very good looking regard themselves as hideously ugly and are constantly seeking surgery to correct their facial appearance. They don ’ t need this. They need psychiatric help. Max has kindly donated his photograph to me. He doesn ’ t have dysmorphophobia, but I ’ m using his photograph to illustrate the fact that he looks exactly like a dysmorphophobic. In other words, he looks entirely normal. Age is another thing when our **attitude** toward our appearance changes. So children judge themselves, learn to judge themselves, by the behavior of adults around them. Here ’ s a classic example : Rebecca has a benign blood vessel tumor that ’ s growing out through her skull, has obliterated her nose, and she ’ s having difficulty seeing. As you can see, it ’ s blocking her vision. She ’ s also in danger, when she damages this, of bleeding profusely. Our research has shown that the parents and close loved ones of these children adore them. They ’ ve grown used to their face ; they think they ’ re special. Actually, sometimes the parents argue about whether these children should have the lesion removed. And occasionally they suffer intense **grief** reactions because the child they ’ ve grown to love has changed so dramatically and they don ’ t recognize them. But other adults say incredibly painful things. They say,”How dare you take this child out of the house and terrify other people. Shouldn ’ t you be doing something about this? Why haven ’ t you had it removed?”And other children in curiosity come up and poke the lesion, because â a natural curiosity. And that obviously alerts the child to their unusual nature. After surgery, everything normalizes. The adults behave more naturally, and the children play more readily with other children. As teenagers â just think back to your teenage years â we ’ re going through a dramatic and often disproportionate change in our facial appearance. We ’ re trying to struggle to find our identity. We crave the approval of our peers. So our facial appearance is vital to us as we ’ re trying to project ourselves to the world. Just remember that single acne spot that crippled you for several days. How long did you spend looking in the mirror every day, practicing your sardonic look, practicing your serious look, trying to look like Sean Connery, as I did, trying to raise one eyebrow? It ’ s a crippling time. I ’ ve chosen to show this profile view of Sue because what it shows is her lower jaw jutting forward and her lower lip jutting forward. I ’ d like you all in the audience now to push your lower jaw forward. Turn to the person next to you, push your lower jaws forward. Turn to the person next to you and look at them â they look miserable. That ’ s exactly what people used to say to Sue. She wasn ’ t miserable at all. But people used to say to her,”Why are you so miserable?”People were making misjudgments all the time on her mood. Teachers and peers were underestimating her ; she was teased at school. So she chose to have facial surgery. After the facial surgery, she said,”My face now reflects my personality. People know now that I ’ m enthusiastic, that I ’ m a happy person.”And that ’ s the change that can be achieved for teenagers. Is this change, though, a real change, or is it a figment of the imagination of the patient themselves? Well we studied teenagers ’ **attitudes** to photographs of patients having this corrective facial surgery. And what we found was â we jumbled up the photographs so they couldn ’ t recognize the before and after â what we found was that the patients were regarded as being more attractive after the surgery. Well that ’ s not surprising, but we also asked them to judge them on honesty, intelligence, friendliness, **violence**. They were all perceived as being less than normal in all those characteristics â more **violent**, etc. â before the surgery. After the surgery, they were perceived as being more intelligent, more friendly, more honest, less **violent** â and yet we hadn ’ t operated on their intellect or their character. When people get older, they don ’ t necessarily choose to follow this kind of surgery. Their presence in the consultation suite is a result of the slings and arrows of outrageous fortune. What happens to them is that they may have suffered cancer or **trauma**. So this is a photograph of Henry, two weeks after he had a malignant cancer removed from the left side of his face â his cheekbone, his upper jaw, his eye-socket. He looks pretty good at this stage. But over the course of the next 15 years he had 14 more operations, as the disease ravaged his face and destroyed my reconstruction regularly. I learned a huge amount from Henry. Henry taught me that you can carry on working. He worked as an advocate. He continued to play cricket. He enjoyed life to the full, and this was probably because he had a successful, fulfilling job and a caring family and was able to participate socially. He maintained a calm insouciance. I don ’ t say he overcame this ; he didn ’ t overcome it. This was something more than that. He ignored it. He ignored the disfigurement that was happening in his life and carried on oblivious to it. And that ’ s what these people can do. Henriapi illustrates this phenomenon as well. This is a man in his 20s whose first visit out of Nigeria was with this malignant cancer that he came to the United Kingdom to have operated on. It was my longest operation. It took 23 hours. I did it with my neurosurgeon. We removed all the bones at the right side of his face â his eye, his nose, the skull bones, the facial skin â and reconstructed him with tissue from the back. He continued to work as a psychiatric nurse. He got married. He had a son called Jeremiah. And again, he said,”This painting of me with my son Jeremiah shows me as the successful man that I feel that I am.”His facial disfigurement did not affect him because he had the support of a family ; he had a successful, fulfilling job. So we ’ ve seen that we can change people ’ s faces. But when we change people ’ s faces, are we changing their identity â for better or for worse? For instance, there are two different types of facial surgery. We can categorize it like that. We can say there are patients who choose to have facial surgery â like Sue. When they have facial surgery, they feel their lives have changed because other people perceive them as better people. They don ’ t feel different. They feel that they ’ ve actually gained what they never had, that their face now reflects their personality. And actually that ’ s probably the difference between cosmetic surgery and this kind of surgery. Because you might say,”Well, this type of surgery might be regarded as cosmetic.”If you do cosmetic surgery, patients are often less happy. They ’ re trying to achieve difference in their lives. Sue wasn ’ t trying to achieve difference in her life. She was just trying to achieve the face that matched her personality. But then we have other people who don ’ t choose to have facial surgery. They ’ re people who have their face shot off. I ’ ll move it off, and we ’ ll have a blank slide for those who are squeamish amongst you. They have it forced upon them. And again, as I told you, if they have a caring family and good work life, then they can lead normal and fulfilled lives. Their identity doesn ’ t change. Is this business about appearance and preoccupation with it a Western phenomenon? Muzetta ’ s family give the lie to this. This is a little Bangladeshi girl from the east end of London who ’ s got a huge malignant tumor on the right side of her face, which has already made her blind and which is rapidly growing and is going to kill her shortly. After she had surgery to remove the tumor, her parents dressed her in this beautiful green velvet dress, a pink ribbon in her hair, and they wanted the painting to be shown around the world, despite the fact that they were orthodox Muslims and the mother wore a full burqa. So it ’ s not simply a Western phenomenon. We make judgments on people ’ s faces all the time. It ’ s been going on since we can think of Lombroso and the way he would define criminal faces. He said you could see criminal faces, judging them just on the photographs that were showed. Good-looking people are always judged as being more friendly. We look at O. J. â he ’ s a good-looking guy. We ’ d like to spend time with him. He looks friendly. Now we know that he ’ s a **convicted** wife-batterer, and actually he ’ s not the good guy. And beauty doesn ’ t equate to goodness, and certainly doesn ’ t equate to contentment. So we ’ ve talked about the static face and judging the static face, but actually, we ’ re more comfortable with judging the moving face. We think we can judge people on their expressions. U. K. jurors in the U. K. justice system like to see a live witness to see whether they can pick up the telltale signs of mendacity â the blink, the hesitation. And so they want to see live witnesses. Todorov tells us that, in a tenth of a second, we can make a judgment on somebody ’ s face. Are we uncomfortable with this image? Yes, we are. Would we be happy if our doctor ’ s face, our lawyer ’ s face, our financial adviser ’ s face was covered? We ’ d be pretty uncomfortable. But are we good at making the judgments on facial appearance and movement? The truth is that there ’ s a five-minute rule, not the tenth-of-a-second rule like Todorov, but a five-minute rule. If you spend five minutes with somebody, you start looking beyond their facial appearance, and the people who you ’ re initially attracted to may seem boring and you lose interest in them, and the people who you didn ’ t immediately seek out, because you didn ’ t find them particularly attractive, become attractive people because of their personality. So we ’ ve talked a lot about facial appearance. I now want to share a little bit of the surgery that we do â where we ’ re at and where we ’ re going. This is an image of Ann who ’ s had her right jaw removed and the base of her skull removed. And you can see in the images afterward, we ’ ve managed to reconstruct her successfully. But that ’ s not good enough. This is what Ann wants. She wants to be out kayaking, she wants to be out climbing mountains. And that ’ s what she achieved, and that ’ s what we have to get to. This is a horrific image, so I ’ m putting my hand up now. This is a photograph of Adi, a Nigerian bank manager who had his face shot off in an **armed** robbery. And he lost his lower jaw, his lip, his chin and his upper jaw and teeth.

This is the bar that he set for us.”I want to look like this. This is how I looked before.”So with modern technology, we used computers to make models. We made a model of the jaw without bone in it. We then bent a plate up to it. We put it in place so we knew it was an accurate position. We then put bone and tissue from the back. Here you can see the plate holding it, and you can see the implants being put in â so that in one operation we achieve this and this. So the patient ’ s life is restored. That ’ s the good news. However, his chin skin doesn ’ t look the same as it did before. It ’ s skin from his back. It ’ s thicker, it ’ s darker, it ’ s coarser, it doesn ’ t have the contours. And that ’ s where we ’ re failing, and that ’ s where we need the face transplant. The face transplant has a role probably in burns patients to replace the skin. We can replace the underlying skeletal structure, but we ’ re still not good at replacing the facial skin. So it ’ s very valuable to have that tool in our armamentarium. But the patients are going to have to take drugs that suppress their immune system for the rest of their lives. What does that mean? They have an increased risk of infection, an increased risk of malignancy. This is not a life-saving transplant â like a heart, or liver, or lung transplant â it is a quality-of-life transplant, and as a result, are the patients going to say, if they get a malignant cancer 10 or 15 years on,”I wish I ’ d had conventional reconstructive techniques rather than this because I ’ m now dying of a malignant cancer”? We don ’ t know yet. We also don ’ t know what they feel about recognition and identity. Bernard Devauchelle and Sylvie Testelin, who did the first operation, are studying that. Donors are going to be short on the ground, because how many people want to have their loved one ’ s face removed at the point of death? So there are going to be problems with face transplantation. So the better news is the future ’ s almost here â and the future is tissue engineering. Just imagine, I can make a biologically-degradable template. I can put it in place where it ’ s meant to be. I can sprinkle a few cells, stem cells from the patient ’ s own hip, a little bit of genetically engineered protein, and lo and behold, leave it for four months and the face is grown. This is a bit like a Julia Child recipe. But we ’ ve still got problems. We ’ ve got mouth cancer to solve. We ’ re still not curing enough patients â it ’ s the most disfiguring cancer. We ’ re still not reconstructing them well enough. In the U. K. we have an epidemic of facial injuries among young people. We still can ’ t get rid of scars. We need to do research. And the best news of all is that surgeons know that we need to do research. And we ’ ve set up charities that will help us fund the clinical research to determine the best treatment practice now and better treatment into the future, so we don ’ t just sit on our laurels and say,”Okay, we ’ re doing okay. Let ’ s leave it as it is.”Thank you very much indeed.

Text Sample 396: POS Dim 3 – 2011 – Score 9.00 – 2011/t002584.txt

Have you ever wondered why extremism seems to have been on the rise in Muslim- majority countries over the course of the last decade? Have you ever wondered how such a situation can be turned around? Have you ever looked at the Arab uprisings and thought,”How could we have predicted that?”or”How could we have better prepared for that?”Well my personal story, my personal journey, what brings me to the TED stage here today, is a demonstration of exactly what ’ s been happening in Muslim-majority countries over the course of the last decades, at least, and beyond. I want to share some of that story with you, but also some of my ideas around change and the role of social movements in creating change in Muslim-majority societies. So let me begin by first of all giving a very, very brief history of time, if I may indulge. In medieval societies there were defined allegiances. An identity was defined primarily by religion. And then we moved on into an era in the 19th century with the rise of a European nation-state where identities and allegiances were defined by **ethnicity**. So identity was primarily defined by **ethnicity**, and the nation-state reflected that. In the age of globalization, we moved on. I call it the era of citizenship — where people could be from multi-racial, multi-ethnic backgrounds, but all be **equal** as citizens in a state. You could be American-Italian ; you could be American-Irish ; you could be British-Pakistani. But I believe now that we ' re moving into a new age, and that age The New York Times dubbed recently as"the age of behavior."How I define the age of behavior is a period of transnational allegiances, where identity is defined more so by ideas and narratives. And these ideas and narratives that bump people across borders are increasingly beginning to affect the way in which people behave. Now this is not all necessarily good news, because it ' s also my belief that **hatred** has gone global just as much as love. But actually it ' s my belief that the people who ' ve been truly capitalizing on this age of behavior, up until now, up until recent times, up until the last six months, the people who have been capitalizing most on the age of behavior and the transnational allegiances, using digital activism and other sorts of borderless technologies, those who ' ve been benefiting from this have been extremists. And that ' s something which I ' d like to elaborate on. If we look at Islamists, if we look at the phenomenon of far-right fascists, one thing they ' ve been very good at, one thing that they ' ve actually been **exceeding** in, is communicating across borders, using technologies to organize themselves, to propagate their message and to create truly global phenomena. Now I should know, because for 13 years of my life, I was involved in an extreme Islamist organization. And I was actually a potent force in spreading ideas across borders, and I witnessed the rise of Islamist extremism as distinct from Islam the faith, and the way in which it influenced my co- religionists across the world. And my story, my personal story, is truly evidence for the age of behavior that I ' m attempting to elaborate upon here. I was, by the way — I ' m an Essex lad, born and raised in Essex in the U. K. Anyone who ' s from England knows the reputation we have from Essex. But having been born in Essex, at the age of 16, I joined an organization. At the age of 17, I was recruiting people from Cambridge University to this organization. At the age of 19, I was on the national leadership of this organization in the U. K. At the age of 21, I was co-founding this organization in Pakistan. At the age of 22, I was co-founding this organization in Denmark. By the age of 24, I found myself **convicted** in prison in Egypt, being blacklisted from three countries in the world for attempting to overthrow their governments, being subjected to torture in Egyptian jails and sentenced to five years as a prisoner of conscience. Now that journey, and what took me from Essex all the way across the world — by the way, we were laughing at democratic activists. We felt they were from the age of yesteryear. We felt that they were out of date. I learned how to use email from the extremist organization that I used. I learned how to effectively communicate across borders without being detected. Eventually I was detected, of course, in Egypt. But the way in which I learned to use technology to my advantage was because I was within an extremist organization that was forced to think beyond the confines of the nation-state. The age of behavior : where ideas and narratives were increasingly defining behavior and identity and allegiances. So as I said, we looked to the status quo and ridiculed it. And it ' s not just Islamist extremists that did this. But even if you look across the mood music in Europe of late, far-right fascism is also on the rise. A form of anti-Islam rhetoric is also on the rise and it ' s transnational. And the consequences that this is having is that it ' s affecting the political climate across Europe. What ' s actually happening is that what were previously localized parochialisms, individual or groupings of extremists who were isolated from one another, have become interconnected in a globalized way and have thus become, or are becoming, mainstream. Because the Internet and connection technologies are connecting them across the world. If you look at the rise of far-right fascism across Europe of late, you will see some things that are happening that are influencing **domestic** politics, yet the phenomenon is transnational. In certain countries, **mosque** minarets are being banned. In others, headscarves are being banned. In others, kosher and halal meat are being banned, as we speak. And on the flip side, we have transnational Islamist extremists doing the same thing across their own societies. And so they are pockets of parochialism that are being connected in a way that makes them feel like they are mainstream. Now that never would have been possible before. They would have felt isolated, until these sorts of technologies came around and connected them in a way that made them feel part of a larger phenomenon. Where does that leave democracy aspirants? Well I believe they ' re getting left far behind. And I ' ll give you an example here at this stage. If any of you remembers the Christmas Day bomb plot : there ' s a man called Anwar al-Awlaki. As an American citizen, ethnically a Yemeni, in hiding currently in Yemen, who inspired a Nigerian, son of the head of Nigeria ' s national bank. This Nigerian student studied in London, trained in Yemen, boarded a flight in Amsterdam to attack America. In the meanwhile, the Old mentality with a capital O, was represented by his father, the head of the Nigerian bank, warning the CIA that his own son was about to attack, and this warning fell on deaf ears. The Old mentality with a capital O, as represented by the nation-state, not yet fully into the age of behavior, not recognizing the power of transnational social movements, got left behind. And the Christmas Day bomber almost succeeded in attacking the United States of America. Again with the example of the far right : that we find, ironically, xenophobic nationalists are utilizing the benefits of globalization. So why are they succeeding? And why are democracy aspirants falling behind? Well we need to understand the power of the social movements who understand this. And a social movement is comprised, in my view, it ' s comprised of four main characteristics. It ' s comprised of ideas and narratives and symbols and leaders. I ' ll talk you through one example, and that ' s the example that everyone here will be aware of, and that ' s the example of Al-Qaeda. If I asked you to think of the ideas of Al-Qaeda, that ' s something that comes to your mind immediately. If I ask you to think of their narratives — the West being at war with Islam, the need to defend Islam against the West — these narratives, they come to your mind immediately. Incidentally, the difference between ideas and narratives : the idea is the cause that one believes in ; and the narrative is the way to sell that cause — the propaganda, if you like, of the cause. So the ideas and the narratives of Al-Qaeda come to your mind immediately. If I ask you to think of their symbols and their leaders, they come to your mind immediately. One of their leaders was killed in Pakistan recently. So these symbols and these leaders come to your mind immediately. And that ' s the power of social movements. They ' re transnational, and they bond around these ideas and narratives and these symbols and these leaders. However, if I ask your minds to focus currently on Pakistan, and I ask you to think of the symbols and the leaders for democracy in Pakistan today, you ' ll be hard pressed to think beyond perhaps the assassination of Benazir Bhutto. Which means, by definition, that particular leader no longer exists. One of the problems we ' re facing is, in my view, that there are no globalized, youth-led, grassroots social movements advocating for democratic culture across Muslim-majority societies. There is no equivalent of the Al-Qaeda, without the terrorism, for democracy across Muslim-majority societies. There are no ideas and narratives and leaders and symbols advocating the democratic culture on the ground. So that begs the next question. Why is it that extremist organizations, whether of the far-right or of the Islamist extremism — Islamism meaning those who wish to impose one version of Islam over the rest of society — why is it that they are succeeding in organizing in a globalized way, whereas those who aspire to democratic culture are falling behind? And I believe that ' s for four reasons. I believe, number one, it ' s complacency. Because those who aspire to democratic culture are in power, or have societies that are leading globalized, powerful societies, powerful countries. And that level of complacency means they don ' t feel the need to advocate for that culture. The second, I believe, is political correctness. That we have a hesitation in espousing the universality of democratic culture because we are associating that — we associate believing in the universality of our values — with extremists. Yet actually, whenever we talk about human rights, we do say that human rights are universal. But actually going out to propagate that view is associated with either neoconservativism or with Islamist extremism. To go around saying that I believe democratic culture is the best that we ' ve arrived at as a form of political organizing is associated with extremism. And the third, democratic choice in Muslim-majority societies has been relegated to a political choice, meaning political parties in many of these societies ask people to vote for them as the democratic party, but then the other parties ask them to vote for them as the military party — wanting to rule by military dictatorship. And then you have a third party saying,"Vote for us ; we ' ll establish a theocracy."So democracy has become merely one political choice among many other forms of political choices available in those societies. And what happens as a result of this is, when those parties are elected, and inevitably they fail, or inevitably they make political mistakes, democracy takes the blame for their political mistakes. And then people say,"We ' ve tried democracy. It doesn ' t really work. Let ' s bring the military back again."And the fourth reason, I believe, is what I ' ve labeled here on the slide as the ideology of resistance. What I mean by that is, if the world superpower today was a communist, it would be much easier for democracy activists to use democracy activism as a form of resistance against colonialism, than it is today with the world superpower being America, occupying certain lands and also espousing democratic ideals. So roughly these four reasons make it a lot more difficult for democratic culture to spread as a civilizational choice, not merely as a political choice. When talking about those reasons, let ' s break down certain preconceptions. Is it just about grievances? Is it just about a lack of education? Well statistically, the majority of those who join extremist organizations are highly educated. Statistically, they are educated, on average, above the education levels of Western society. Anecdotally, we can demonstrate that if poverty was the only factor, well Bin Laden is from one of the richest families in Saudi Arabia. His deputy, Ayman al-Zawahiri, was a pediatrician — not an ill-educated man. International aid and development has been going on for years, but extremism in those societies, in many of those societies, has been on the rise. And what I believe is missing is genuine grassroots activism on the ground, in addition to international aid, in addition to education, in addition to health. Not exclusive to these things, but in addition to them, is propagating a genuine demand for democracy on the ground. And this is where I believe neoconservatism had it upside-down. Neoconservatism had the philosophy that you go in with a supply-led approach to impose democratic values from the top down. Whereas Islamists and far-right organizations, for decades, have been building demand for their ideology on the grassroots. They ' ve been building civilizational demand for their values on the grassroots, and we ' ve been seeing those societies slowly transition to societies that are increasingly asking for a form of Islamism. Mass movements in Pakistan have been represented after the Arab uprisings mainly by organizations claiming for some form of theocracy, rather than for a democratic uprising. Because since pre-partition, they ’ ve been building demand for their ideology on the ground. And what ’ s needed is a genuine transnational youth-led movement that works to actively advocate for the democratic culture — which is necessarily more than just elections. But without freedom of speech, you can ’ t have free and fair elections. Without human rights, you don ’ t have the protection granted to you to campaign. Without freedom of belief, you don ’ t have the right to join organizations. So what ’ s needed is those organizations on the ground advocating for the democratic culture itself to create the demand on the ground for this culture. What that will do is avoid the problem I was talking about earlier, where currently we have political parties presenting democracy as merely a political choice in those societies alongside other choices such as military rule and theocracy. Whereas if we start building this demand on the ground on a civilizational level, rather than merely on a political level, a level above politics — movements that are not political parties, but are rather creating this civilizational demand for this democratic culture. What we ’ ll have in the end is this ideal that you see on the slide here — the ideal that people should vote in an existing democracy, not for a democracy. But to get to that stage, where democracy builds the fabric of society and the political choices within that fabric, but are certainly not theocratic and military dictatorship — i. e. you ’ re voting in a democracy, in an existing democracy, and that democracy is not merely one of the choices at the ballot box. To get to that stage, we genuinely need to start building demand in those societies on the ground. Now to conclude, how does that happen? Well, Egypt is a good starting point. The Arab uprisings have demonstrated that this is already beginning. But what happened in the Arab uprisings and what happened in Egypt was particularly cathartic for me. What happened there was a political coalition gathered together for a political goal, and that was to remove the leader. We need to move one step beyond that now. We need to see how we can help those societies move from political coalitions, loosely based political coalitions, to civilizational coalitions that are working for the ideals and narratives of the democratic culture on the ground. Because it ’ s not enough to remove a leader or ruler or dictator. That doesn ’ t guarantee that what comes next will be a society built on democratic values. But generally, the trends that start in Egypt have historically spread across the MENA region, the Middle East and North Africa region. So when Arab socialism started in Egypt, it spread across the region. In the ’ 80s and ’ 90s when Islamism started in the region, it spread across the MENA region as a whole. And the aspiration that we have at the moment — as young Arabs are proving today and instantly rebranding themselves as being prepared to die for more than just terrorism — is that there is a chance that democratic culture can start in the region and spread across to the rest of the countries that are surrounding that. But that will require helping these societies transition from having merely political coalitions to building genuinely grassroots-based social movements that advocate for the democratic culture. And we ’ ve made a start for that in Pakistan with a movement called Khudi, where we are working on the ground to encourage the youth to create genuine buy-in for the democratic culture. And it ’ s with that thought that I ’ ll end. And my time is up, and thank you for your time.

Text Sample 397: POS Dim 3 – 2011 – Score 9.00 – 2011/t002654.txt

Ten years ago exactly, I was in Afghanistan. I was covering the war in Afghanistan, and I witnessed, as a reporter for Al Jazeera, the amount of **suffering** and destruction that emerged out of a war like that. Then, two years later, I covered another war â the war in Iraq. I was placed at the center of that war because I was covering the war from the northern part of Iraq. And the war ended with a regime change, like the one in Afghanistan. And that regime that we got rid of was actually a dictatorship, an **authoritarian** regime, that for decades created a great sense of paralysis within the nation, within the people themselves. However, the change that came through foreign intervention created even worse circumstances for the people and deepened the sense of paralysis and inferiority in that part of the world. For decades, we have lived under **authoritarian** regimes â in the Arab world, in the Middle East. These regimes created something within us during this period. I ' m 43 years old right now. For the last 40 years, I have seen almost the same faces for kings and presidents ruling us â old, aged, **authoritarian**, corrupt situations â regimes that we have seen around us. And for a moment I was wondering, are we going to live in order to see real change happening on the ground, a change that does not come through foreign intervention, through the misery of occupation, through nations invading our land and deepening the sense of inferiority sometimes? The Iraqis : yes, they got rid of Saddam Hussein, but when they saw their land occupied by foreign forces they felt very sad, they felt that their dignity had suffered. And this is why they revolted. This is why they did not accept. And actually other regimes, they told their citizens,"Would you like to see the situation of Iraq? Would you like to see civil war, sectarian killing? Would you like to see destruction? Would you like to see foreign troops on your land?"And the people thought for themselves,"Maybe we should live with this kind of **authoritarian** situation that we find ourselves in, instead of having the second scenario."That was one of the worst nightmares that we have seen. For 10 years, unfortunately we have found ourselves reporting images of destruction, images of killing, of sectarian conflicts, images of **violence**, emerging from a magnificent piece of land, a region that one day was the source of civilizations and art and culture for thousands of years. Now I am here to tell you that the future that we were dreaming for has eventually arrived. A new generation, well-educated, connected, inspired by universal values and a global understanding, has created a new reality for us. We have found a new way to express our feelings and to express our dreams : these young people who have restored self-confidence in our nations in that part of the world, who have given us new meaning for freedom and empowered us to go down to the **streets**. Nothing happened. No **violence**. Nothing. Just step out of your house, raise your voice and say,"We would like to see the end of the regime."This is what happened in Tunisia. Over a few days, the Tunisian regime that invested billions of dollars in the security agencies, billions of dollars in maintaining, trying to maintain, its prisons, collapsed, disappeared, because of the voices of the public. People who were inspired to go down to the **streets** and to raise their voices, they tried to kill. The intelligence agencies wanted to arrest people. They found something called Facebook. They found something called **Twitter**. They were surprised by all of these kinds of issues. And they said,"These kids are misled."Therefore, they asked their parents to go down to the **streets** and collect them, bring them back home. This is what they were telling. This is their propaganda."Bring these kids home because they are misled."But yes, these youth who have been inspired by universal values, who are idealistic enough to imagine a magnificent future and, at the same time, realistic enough to balance this kind of imagination and the process leading to it â not using **violence**, not trying to create chaos â these young people, they did not go home. Parents actually went to the **streets** and they supported them. And this is how the revolution was born in Tunisia. We in Al Jazeera were banned from Tunisia for years, and the government did not allow any Al Jazeera reporter to be there. But we found that these people in the **street**, all of them are our reporters, feeding our newsroom with pictures, with videos and with news. And suddenly that newsroom in Doha became a center that received all this kind of input from ordinary people â people who are connected and people who have ambition and who have liberated themselves from the feeling of inferiority. And then we took that decision : We are unrolling the news. We are going to be the voice for these voiceless people. We are going to spread the message. Yes, some of these young people are connected to the Internet, but the connectivity in the Arab world is very little, is very small, because of many problems that we are suffering from. But Al Jazeera took the voice from these people and we amplified [it]. We put it in every sitting room in the Arab world â and internationally, globally, through our English channel. And then people started to feel that there ’ s something new happening. And then Zine al-Abidine Ben Ali decided to leave. And then Egypt started, and Hosni Mubarak decided to leave. And now Libya as you see it. And then you have Yemen. And you have many other countries trying to see and to rediscover that feeling of,”How do we imagine a future which is magnificent and peaceful and tolerant?”I want to tell you something, that the Internet and connectivity has created [a] new mindset. But this mindset has continued to be faithful to the soil and to the land that it emerged from. And while this was the major difference between many **initiatives** before to create change, before we thought, and governments told us â and even sometimes it was true â that change was imposed on us, and people rejected that, because they thought that it is alien to their culture. Always, we believed that change will spring from within, that change should be a reconciliation with culture, cultural diversity, with our faith in our tradition and in our history, but at the same time, open to universal values, connected with the world, tolerant to the outside. And this is the moment that is happening right now in the Arab world. This is the right moment, and this is the actual moment that we see all of these meanings meet together and then create the beginning of this magnificent era that will emerge from the region. How did the elite deal with that â the so-called political elite? In front of Facebook, they brought the camels in Tahrir Square. In front of Al Jazeera, they started creating tribalism. And then when they failed, they started speaking about conspiracies that emerged from Tel Aviv and Washington in order to divide the Arab world. They started telling the West,”Be aware of Al-Qaeda. Al-Qaeda is taking over our territories. These are Islamists trying to create new Imaras. Be aware of these people who [are] coming to you in order to ruin your great civilization.”Fortunately, people right now cannot be deceived. Because this corrupt elite in that region has lost even the power of deception. They could not, and they cannot, imagine how they could really deal with this reality. They have lost. They have been detached from their people, from the masses, and now we are seeing them collapsing one after the other. Al Jazeera is not a tool of revolution. We do not create revolutions. However, when something of that magnitude happens, we are at the center of the coverage. We were banned from Egypt, and our correspondents, some of them were arrested. But most of our camera people and our journalists, they went underground in Egypt â voluntarily â to report what happened in Tahrir Square. For 18 days, our cameras were broadcasting, live, the voices of the people in Tahrir Square. I remember one night when someone phoned me on my cellphone â ordinary person who I don ’ t know â from Tahrir Square. He told me,”We appeal to you not to switch off the cameras. If you switch off the cameras tonight, there will be a genocide. You are protecting us by showing what is happening at Tahrir Square.”I felt the responsibility to phone our correspondents there and to phone our newsroom and to tell them,”Make your best not to switch off the cameras at night, because the guys there really feel confident when someone is reporting their story â and they feel protected as well.”So we have a chance to create a new future in that part of the world. We have a chance to go and to think of the future as something which is open to the world. Let us not repeat the mistake of Iran, of [the] Mosaddeq revolution. Let us free ourselves â especially in the West â from thinking about that part of the world based on oil interest, or based on interests of the illusion of stability and security. The stability and security of **authoritarian** regimes cannot create but terrorism and **violence** and destruction. Let us accept the choice of the people. Let us not pick and choose who we would like to rule their future. The future should be ruled by people themselves, even sometimes if they are voices that might now scare us. But the values of democracy and the freedom of choice that is sweeping the Middle East at this moment in time is the best opportunity for the world, for the West and the East, to see stability and to see security and to see friendship and to see tolerance emerging from the Arab world, rather than the images of **violence** and terrorism. Let us support these people. Let us stand for them. And let us give up our narrow selfishness in order to embrace change, and in order to **celebrate** with the people of that region a great future and hope and tolerance. The future has arrived, and the future is now. I thank you very much. Thank you very much. Chris Anderson : I just have a couple of questions for you. Thank you for coming here. How would you characterize the historical significance of what ’ s happened? Is this a story-of-the-year, a story-of-the-decade or something more? Wadah Khanfar : Actually, this may be the biggest story that we have ever covered. We have covered many wars. We have covered a lot of tragedies, a lot of problems, a lot of conflict zones, a lot of hot spots in the region, because we were centered at the middle of it. But this is a story â it is a great story ; it is beautiful. It is not something that you only cover because you have to cover a great incident. You are witnessing change in history. You are witnessing the birth of a new era. And this is what the story ’ s all about. CA : There are a lot of people in the West who are still skeptical, or think this may just be an intermediate stage before much more alarming chaos. You really believe that if there are democratic elections in Egypt now, that a government could emerge that espouses some of the values you ’ ve spoken about so inspiringly? WK : And people actually, after the collapse of the Hosni Mubarak regime, the youth who have organized themselves in certain groups and councils, they are guarding the transformation and they are trying to put it on a track in order to satisfy the values of democracy, but at the same time also to make it reasonable and to make it rational, not to go out of order. In my opinion, these people are much more wiser than, not only the political elite, even the intellectual elite, even opposition leaders including political parties. At this moment in time, the youth in the Arab world are much more wiser and capable of creating the change than the old â including the political and cultural and ideological old regimes. CA : We are not to get involved politically and interfere in that way. What should people here at TED, here in the West, do if they want to connect or make a difference and they believe in what ’ s happening here? WK : I think we have discovered a very important issue in the Arab world â that people care, people care about this great transformation. Mohamed Nanabhay who ’ s sitting with us, the head of Aljazeera. net, he told me that a 2, 500 percent increase of **accessing** our website from various parts of the world. Fifty percent of it is coming from America. Because we discovered that people care, and people would like to know â they are receiving the stream through our Internet. Unfortunately in the United States, we are not covering but Washington D. C. at this moment in time for Al Jazeera English. But I can tell you, this is the moment to **celebrate** through connecting ourselves with those people in the **street** and expressing our support to them and expressing this kind of feeling, universal feeling, of supporting the weak and the oppressed to create a much better future for all of us. CA : Well Wadah, a group of members of the TED community, TEDxCairo, are meeting as we speak. They ’ ve had some speakers there. I believe they ’ ve heard your talk. Thank you for inspiring them and for inspiring all of us. Thank you so much.

Text Sample 398: POS Dim 3 – 2011 – Score 8.00 – 2011/t002368.txt

This is Shivdutt Yadav, and he ’ s from Uttar Pradesh, India. Now Shivdutt was visiting the local land registry office in Uttar Pradesh, and he discovered that official records were listing him as dead. His land was no longer registered in his name. His brothers, Chandrabhan and Phoolchand, were also listed as dead. Family members had bribed officials to interrupt the hereditary transfer of land by having the brothers **declared** dead, allowing them to inherit their father ’ s share of the ancestral farmland. Because of this, all three brothers and their families had to vacate their home. According to the Yadav family, the local court has been scheduling a case review since 2001, but a judge has never appeared. There are several instances in Uttar Pradesh of people dying before their case is given a proper review. Shivdutt ’ s father ’ s death and a want for his property led to this corruption. He was laid to rest in the Ganges River, where the dead are cremated along the banks of the river or tied to heavy stones and sunk in the water. Photographing these brothers was a disorienting exchange because on paper they don ’ t exist, and a photograph is so often used as an evidence of life. Yet, these men remain dead. This quandary led to the title of the project, which considers in many ways that we are all the living dead and that we in some ways represent ghosts of the past and the future. So this story is the first of 18 chapters in my new body of work titled”A Living Man **Declared** Dead and Other Chapters.”And for this work, I traveled around the world over a four-year period researching and recording bloodlines and their related stories. I was interested in ideas surrounding fate and whether our fate is determined by blood, chance or circumstance. The subjects I documented ranged from feuding families in Brazil to **victims** of genocide in Bosnia to the first woman to hijack an airplane and the living dead in India. In each chapter, you can see the external forces of governance, power and territory or religion colliding with the internal forces of psychological and physical inheritance. Each work that I make is comprised of three segments. On the left are one or more portrait panels in which I systematically order the members of a given bloodline. This is followed by a text panel, it ’ s designed in scroll form, in which I construct the narrative at stake. And then on the right is what I refer to as a footnote panel. It ’ s a space that ’ s more intuitive in which I present fragments of the story, beginnings of other stories, photographic evidence. And it ’ s meant to kind of reflect how we engage with histories or stories on the Internet, in a less linear form. So it ’ s more disordered. And this disorder is in direct contrast to the unalterable order of a bloodline. In my past projects I ’ ve often worked in serial form, documenting things that have the appearance of being comprehensive through a determined title and a determined presentation, but in fact, are fairly abstract. In this project I wanted to work in the opposite direction and find an absolute catalog, something that I couldn ’ t interrupt, curate or edit by choice. This led me to blood. A bloodline is determined and ordered. But the project centers on the collision of order and disorder — the order of blood butting up against the disorder represented in the often chaotic and **violent** stories that are the subjects of my chapters. In chapter two, I photograph the descendants of Arthur Ruppin. He was sent in 1907 to Palestine by the Zionist organization to look at areas for Jewish settlement and acquire land for Jewish settlement. He oversaw land acquisition on behalf of the Palestine Land Development Company whose work led to the establishment of a Jewish state. Through my research at the Zionist Archives in Jerusalem, I wanted to look at the early paperwork of the establishment of the Jewish state. And I found these maps which you see here. And these are studies commissioned by the Zionist organization for alternative areas for Jewish settlement. In this, I was interested in the consequences of geography and imagining how the world would be different if Israel were in Uganda, which is what these maps demonstrate. These archives in Jerusalem, they maintain a card index file of the earliest **immigrants** and applicants for **immigration** to Palestine, and later Israel, from 1919 to 1965. Chapter three : Joseph Nyamwanda Jura Ondijo treated patients outside of Kisumu, Kenya for AIDS, tuberculosis, infertility, mental illness, evil spirits. He ' s most often paid for his services in cash, cows or goats. But sometimes when his female patients can ' t afford his services, their families give the women to Jura in exchange for medical treatment. As a result of these transactions, Jura has nine wives, 32 children and 63 grandchildren. In his bloodline you see the children and grandchildren here. Two of his wives were brought to him suffering from infertility and he cured them, three for evil spirits, one for an asthmatic condition and severe chest pain and two wives Ondijo claims he took for love, paying their families a total of 16 cows. One wife deserted him and another passed away during treatment for evil spirits. Polygamy is widely practiced in Kenya. It ' s common among a privileged class capable of paying numerous dowries and keeping multiple homes. Instances of prominent social and political figures in polygamous relationships has led to the perception of polygamy as a symbol of wealth, status and power. You may notice in several of the chapters that I photographed there are empty portraits. These empty portraits represent individuals, living individuals, who couldn ' t be present. And the reasons for their absence are given in my text panel. They include dengue fever, imprisonment, army service, women not allowed to be photographed for religious and cultural reasons. And in this particular chapter, it ' s children whose mothers wouldn ' t allow them to travel to the photographic shoot for fear that their fathers would kidnap them during it. Twenty-four European rabbits were brought to Australia in 1859 by a British settler for sporting purposes, for hunting. And within a hundred years, that population of 24 had exploded to half a billion. The European rabbit has no natural predators in Australia, and it competes with native wildlife and damages native plants and degrades the land. Since the 1950s, Australia has been introducing lethal diseases into the wild rabbit population to control growth. These rabbits were bred at a government facility, Biosecurity Queensland, where they bred three bloodlines of rabbits and have infected them with a lethal disease and are monitoring their progress to see if it will effectively kill them. So they ' re testing its virulence. During the course of this trial, all of the rabbits died, except for a few, which were euthanized. Haigh ' s Chocolate, in collaboration with the Foundation for Rabbit-Free Australia, stopped all production of the Easter Bunny in chocolate and has replaced it with the Easter Bilby. Now this was done to counter the annual celebration of rabbits and presumably make the public more comfortable with the killing of rabbits and promote an animal that ' s native to Australia, and actually an animal that is threatened by the European rabbit. In chapter seven, I focus on the effects of a genocidal act on one bloodline. So over a two-day period, six individuals from this bloodline were killed in the Srebrenica massacre. This is the only work in which I visually represent the dead. But I only represent those that were killed in the Srebrenica massacre, which is recorded as the largest mass murder in Europe since the Second World War. And during this massacre, 8, 000 Bosnian Muslim men and boys were systematically executed. So when you look at a detail of this work, you can see, the man on the upper- left is the father of the woman sitting next to him. Her name is Zumra. She is followed by her four children, all of whom were killed in the Srebrenica massacre. Following those four children is Zumra ’ s younger sister who is then followed by her children who were killed as well. During the time I was in Bosnia, the mortal remains of Zumra ’ s eldest son were exhumed from a mass grave. And I was therefore able to photograph the fully assembled remains. However, the other individuals are represented by these blue slides, which show tooth and bone samples that were matched to DNA evidence collected from family members to prove they were the identities of those individuals. They ’ ve all been given a proper burial, so what remains are these blue slides at the International Commission for Missing Persons. These are personal effects dug up from a mass grave that are awaiting identification from family members and graffiti at the Potochari battery factory, which was where the Dutch U. N. soldiers were staying, and also the Serbian soldiers later during the times of the executions. This is video footage used at the Milosevic trial, which from top to bottom shows a Serbian scorpion unit being blessed by an Orthodox priest before rounding up the boys and men and killing them. Chapter 15 is more of a performance piece. I solicited China ’ s State Council Information Office in 2009 to select a multi-generational bloodline to represent China for this project. They chose a large family from Beijing for its size, and they declined to give me any further reasoning for their choice. This is one of the rare situations where I have no empty portraits. Everyone showed up. You can also see the evolution of the one-child-only policy as it travels through the bloodline. Previously known as the Department of Foreign Propaganda, the State Council Information Office is responsible for all of China ’ s external publicity operations. It controls all foreign media and image production outside of China from foreign media working within China. It also monitors the Internet and instructs local media on how to handle any potentially controversial issues, including Tibet, ethnic minorities, Human Rights, religion, democracy movements and terrorism. For the footnote panel in this work, this office instructed me to photograph their central television tower in Beijing. And I also photographed the gift bag they gave me when I left. These are the descendants of Hans Frank who was Hitler ’ s personal legal advisor and governor general of occupied Poland. Now this bloodline includes numerous empty portraits, highlighting a complex relationship to one ’ s family history. The reasons for these absences include people who declined participation. There ’ s also parents who participated who wouldn ’ t let their children participate because they thought they were too young to decide for themselves. Another section of the family presented their clothing, as opposed to their physical presence, because they didn ’ t want to be identified with the past that I was highlighting. And finally, another individual sat for me from behind and later rescinded his participation, so I had to pixelate him out so he ’ s unrecognizable. In the footnote panel that accompanies this work I photographed an official Adolph Hitler postage stamp and an imitation of that stamp produced by British Intelligence with Hans Frank ’ s image on it. It was released in Poland to create friction between Frank and Hitler, so that Hitler would imagine Frank was trying to usurp his power. Again, talking about fate, I was interested in the stories and fate of particular works of art. These paintings were taken by Hans Frank during the time of the Third Reich. And I ’ m interested in the impact of their absence and presence through time. They are Leonardo da Vinci ' s"Lady With an Ermine,"Rembrandt ' s"Landscape With Good Samaritan"and Raphael ' s"Portrait of a Youth,"which has never been found. Chapter 12 highlights people being born into a battle that is not of their making, but becomes their own. So this is the Ferraz family and the Novaes family. And they are in an active blood feud. This feud has been going on since 1991 in Northeast Brazil in Pernambuco, and it involved the deaths of 20 members of the families and 40 others associated with the feud, including hired hit men, **innocent** bystanders and friends. Tensions between these two families date back to 1913 when there was a dispute over local political power. But it got **violent** in the last two decades and includes decapitation and the death of two mayors. Installed into a protective wall surrounding the suburban home of Louis Novaes, who ' s the head of the Novaes family, are these turret holes, which were used for shooting and looking. Brazil ' s northeast state of Pernambuco is one of the nation ' s most **violent** regions. It ' s rooted in a principle of retributive justice, or an eye for an eye, so retaliatory killings have led to several deaths in the area. This story, like many of the stories in my chapters, reads almost as an archetypal episode, like something out of Shakespeare, that ' s happening now and will happen again in the future. I ' m interested in these ideas of repetition. So after I returned home, I received word that one member of the family had been shot 30 times in the face. Chapter 17 is an exploration of the absence of a bloodline and the absence of a history. Children at this Ukrainian orphanage are between the ages of six and 16. This piece is ordered by age because it can ' t be ordered by blood. In a 12- month period when I was at the orphanage, only one child had been adopted. Children have to leave the orphanage at age 16, despite the fact that there ' s often nowhere for them to go. It ' s commonly reported in Ukraine that children, when leaving the orphanage are targeted for human trafficking, child pornography and prostitution. Many have to turn to criminal activity for their survival, and high rates of suicide are recorded. This is a boys ' bedroom. There ' s an insufficient supply of beds at the orphanage and not enough warm clothing. Children bathe infrequently because the hot water isn ' t turned on until October. This is a girls ' bedroom. And the director listed the orphanage ' s most urgent needs as an industrial size washing machine and dryer, four vacuum cleaners, two computers, a video projector, a copy machine, winter shoes and a dentist ' s drill. This photograph, which I took at the orphanage of one of the classrooms, shows a sign which I had translated when I got home. And it reads :"Those who do not know their past are not worthy of their future."There are many more chapters in this project. This is just an abridged rendering of over a thousand images. And this mass pile of images and stories forms an archive. And within this accumulation of images and texts, I ' m struggling to find patterns and imagine that the narratives that surround the lives we lead are just as coded as blood itself. But archives exist because there ' s something that can ' t necessarily be articulated. Something is said in the gaps between all the information that ' s collected. And there ' s this **relentless** persistence of birth and death and an unending collection of stories in between. It ' s almost machine-like the way people are born and people die, and the stories keep coming and coming. And in this, I ' m considering, is this actual accumulation leading to some sort of evolution, or are we on repeat over and over again? Thank you.

Text Sample 399: POS Dim 3 – 2008 – Score 12.00 – 2008/t000239.txt
Philosophers, dramatists, theologians have grappled with this question for cen-

turies : what makes people go wrong? Interestingly, I asked this question when I was a little kid. I grew up in the South Bronx, inner-city ghetto in New York, and I was surrounded by evil, as all kids are who grew up in an inner city. And I had friends who were really good kids, who lived out the Dr. Jekyll Mr. Hyde scenario — Robert Louis Stevenson. That is, they took drugs, got in trouble, went to jail. Some got killed, and some did it without drug assistance. So when I read Robert Louis Stevenson, that wasn ’ t fiction. The only question is, what was in the juice? And more importantly, that line between good and evil — which privileged people like to think is fixed and impermeable, with them on the good side, the others on the bad side — I knew that line was movable, and it was permeable. Good people could be seduced across that line, and under good and some rare circumstances, bad kids could recover with help, with reform, with rehabilitation. So I want to begin with this wonderful illusion by [Dutch] artist M. C. Escher. If you look at it and focus on the white, what you see is a world full of angels. But let ’ s look more deeply, and as we do, what appears is the demons, the devils in the world. That tells us several things. One, the world is, was, will always be filled with good and evil, because good and evil is the yin and yang of the human condition. It tells me something else. If you remember, God ’ s favorite angel was Lucifer. Apparently, Lucifer means”the light.”It also means”the morning star,”in some scripture. And apparently, he disobeyed God, and that ’ s the ultimate disobedience to authority. And when he did, Michael, the archangel, was sent to kick him out of heaven along with the other fallen angels. And so Lucifer descends into hell, becomes Satan, becomes the devil, and the force of evil in the universe begins. Paradoxically, it was God who created hell as a place to store evil. He didn ’ t do a good job of keeping it there though. So, this arc of the cosmic transformation of God ’ s favorite angel into the Devil, for me, sets the context for understanding human beings who are transformed from good, ordinary people into **perpetrators** of evil. So the Lucifer effect, although it focuses on the negatives — the negatives that people can become, not the negatives that people are — leads me to a psychological definition. Evil is the exercise of power. And that ’ s the key : it ’ s about power. To intentionally **harm** people psychologically, to hurt people physically, to destroy people mortally, or ideas, and to commit crimes against humanity. If you Google”evil,”a word that should surely have withered by now, you come up with 136 million hits in a third of a second. A few years ago — I am sure all of you were shocked, as I was, with the revelation of American soldiers abusing prisoners in a strange place in a controversial war, Abu Ghraib in Iraq. And these were men and women who were putting prisoners through unbelievable humiliation. I was shocked, but I wasn ’ t surprised, because I had seen those same visual parallels when I was the prison superintendent of the Stanford Prison Study. Immediately the Bush administration military said what? What all administrations say when there ’ s a scandal :”Don ’ t blame us. It ’ s not the system. It ’ s the few bad apples, the few rogue soldiers.”My hypothesis is, American soldiers are good, usually. Maybe it was the barrel that was bad. But how am I going to deal with that hypothesis? I became an expert witness for one of the guards, Sergeant Chip Frederick, and in that position, I had **access** to the dozen investigative reports. I had **access** to him. I could study him, have him come to my home, get to know him, do psychological analysis to see, was he a good apple or bad apple. And thirdly, I had **access** to all of the 1, 000 pictures that these soldiers took. These pictures are of a **violent** or **sexual** nature. All of them come from the cameras of American soldiers. Because everybody has a digital camera or cell phone camera, they took pictures of everything, more than 1, 000. And what I ’ ve done is I organized them into various categories. But these are by United States military police, army reservists. They are not soldiers prepared for this mission at all. And it all happened in a single place, Tier 1-A, on the night shift. Why? Tier 1-A was the center for military intelligence. It was the interrogation hold. The CIA was there. Interrogators from Titan Corporation, all there, and they ' re getting no information about the insurgency. So they ' re going to put pressure on these soldiers, military police, to cross the line, give them permission to break the will of the enemy, to prepare them for interrogation, to soften them up, to take the gloves off. Those are the euphemisms, and this is how it was interpreted. Let ' s go down to that dungeon. [Abu Ghraib Iraq Prison Abuses 2008 Military Police Guards ' Photos] [The following images include nudity and graphic depictions of **violence**] So, pretty horrific. That ' s one of the visual illustrations of evil. And it should not have escaped you that the reason I paired the prisoner with his arms out with Leonardo da Vinci ' s ode to humanity is that that prisoner was mentally **ill**. That prisoner covered himself with shit every day, they had to roll him in dirt so he wouldn ' t stink. But the guards ended up calling him"Shit Boy."What was he doing in that prison rather than in some mental institution? In any event, here ' s former Secretary of Defense Rumsfeld. He comes down and says,"I want to know, who is responsible? Who are the bad apples?"Well, that ' s a bad question. You have to reframe it and ask,"What is responsible?""What"could be the who of people, but it could also be the what of the situation, and obviously that ' s wrongheaded. How do psychologists try to understand such transformations of human character, if you believe that they were good soldiers before they went down to that dungeon? There are three ways. The main way is called dispositional. We look at what ' s inside of the person, the bad apples. This is the foundation of all of social science, the foundation of religion, the foundation of war. Social psychologists like me come along and say,"Yeah, people are the actors on the stage, but you ' ll have to be aware of the situation. Who are the cast of characters? What ' s the costume? Is there a stage director?"And so we ' re interested in what are the external factors around the individual — the bad barrel? Social scientists stop there and they miss the big point that I discovered when I became an expert witness for Abu Ghraib. The power is in the system. The system creates the situation that corrupts the individuals, and the system is the legal, political, economic, cultural background. And this is where the power is of the bad-barrel makers. If you want to change a person, change the situation. And to change it, you ' ve got to know where the power is, in the system. So the Lucifer effect involves understanding human character transformations with these three factors. And it ' s a dynamic interplay. What do the people bring into the situation? What does the situation bring out of them? And what is the system that creates and maintains that situation? My recent book,"The Lucifer Effect,"is about, how do you understand how good people turn evil? And it has a lot of detail about what I ' m going to talk about today. So Dr. Z ' s"Lucifer Effect,"although it focuses on evil, really is a celebration of the human mind ' s infinite capacity to make any of us kind or cruel, caring or indifferent, creative or destructive, and it makes some of us villains. And the good news that I ' m going to hopefully come to at the end is that it makes some of us heroes. This wonderful cartoon in the New Yorker summarizes my whole talk :"I ' m neither a good cop nor a bad cop, Jerome. Like yourself, I ' m a complex amalgam of positive and negative personality traits that emerge or not, depending on the circumstances."There ' s a study some of you think you know about, but very few people have ever read the story. You watched the movie. This is Stanley Milgram, little Jewish kid from the Bronx, and he asked the question,"Could the Holocaust happen here, now?"People say,"No, that ' s Nazi Germany, Hitler, you know, that ' s 1939."He said,"Yeah, but suppose Hitler asked you, ' Would you electrocute a stranger? ' ' No way, I ' m a good person. '"He said,"Why don ' t we put you in a situation and give you a chance to see what you would do?"And so what he did was he tested 1, 000 ordinary people. 500 New Haven, Connecticut, 500 Bridgeport. And the ad said,"Psychologists want to understand memory. We want to improve people ' s memory, because it is the key to success."OK?"We ' re going to give you five bucks — four dollars for your time. We don ' t want college students. We want men between 20 and 50."In the later studies, they ran women. Ordinary people : barbers, clerks, white-collar people. So, you go down, one of you will be a learner, one will be a teacher. The learner ' s a genial, middle-aged guy. He gets tied up to the shock apparatus in another room. The learner could be middle-aged, could be as young as 20. And one of you is told by the authority, the guy in the lab coat,"Your job as teacher is to give him material to learn. Gets it right, reward. Gets it wrong, you press a button on the shock box. The first button is 15 volts. He doesn ' t even feel it."That ' s the key. All evil starts with 15 volts. And then the next step is another 15 volts. The problem is, at the end of the line, it ' s 450 volts. And as you go along, the guy is screaming,"I ' ve got a heart condition! I ' m out of here!"You ' re a good person. You complain."Sir, who will be responsible if something happens to him?"The experimenter says,"Don ' t worry, I will be responsible. Continue, teacher."And the question is, who would go all the way to 450 volts? You should notice here, when it gets up to 375, it says,"Danger. Severe Shock."When it gets up to here, there ' s"XXX"— the pornography of power. So Milgram asks 40 psychiatrists,"What percent of American citizens would go to the end?"They said only one percent. Because that ' s sadistic behavior, and we know, psychiatry knows, only one percent of Americans are sadistic. OK. Here ' s the data. They could not be more wrong. Two thirds go all the way to 450 volts. This was just one study. Milgram did more than 16 studies. And look at this. In study 16, where you see somebody like you go all the way, 90 percent go all the way. In study five, if you see people rebel, 90 percent rebel. What about women? Study 13 — no different than men. So Milgram is quantifying evil as the willingness of people to blindly obey authority, to go all the way to 450 volts. And it ' s like a dial on human nature. A dial in a sense that you can make almost everybody totally obedient, down to the majority, down to none. What are the external parallels? For all research is artificial. What ' s the validity in the real world? 912 American citizens committed suicide or were murdered by family and friends in Guyana jungle in 1978, because they were blindly obedient to this guy, their pastor — not their priest — their pastor, Reverend Jim Jones. He persuaded them to commit mass suicide. And so, he ' s the modern Lucifer effect, a man of God who becomes the Angel of Death. Milgram ' s study is all about individual authority to control people. Most of the time, we are in institutions, so the Stanford Prison Study is a study of the power of institutions to influence individual behavior. Interestingly, Stanley Milgram and I were in the same high school class in James Monroe in the Bronx, 1954. I did this study with my graduate students, especially Craig Haney — and it also began work with an ad. We had a cheap, little ad, but we wanted college students for a study of prison life. 75 people volunteered, took personality tests. We did interviews. Picked two dozen : the most normal, the most healthy. Randomly assigned them to be prisoner and guard. So on day one, we knew we had good apples. I ' m going to put them in a bad situation. And secondly, we know there ' s no difference between the boys who will be guards and those who will be prisoners. To the prisoners, we said,"Wait at home. The study will begin Sunday."We didn ' t tell them that the city police were going to come and do realistic arrests. [Day 1] Student : A police car pulls up in front, and a cop comes to the front door, and knocks, and says he ' s looking for me. So they, right there, you know, they took me out the door, they put my hands against the car. It was a real cop car, it was a real policeman, and there were real neighbors in the **street**, who didn ' t know that this was an experiment. And there was cameras all around and neighbors all around. They put me in the car, then they drove me around Palo Alto. They took me to the basement of the police station. Then they put me in a cell. I was the first one to be picked up, so they put me in a cell, which was just like a room with a door with bars on it. You could tell it wasn ' t a real jail. They locked me in there, in this degrading little outfit. They were taking this experiment too seriously. Here are the prisoners, who are going to be dehumanized, they ' ll become numbers. Here are the guards with the symbols of power and anonymity. Guards get prisoners to clean the toilet bowls out with their bare hands, to do other humiliating tasks. They strip them naked. They sexually taunt them. They begin to do degrading activities, like having them simulate sodomy. You saw simulating fellatio in soldiers in Abu Ghraib. My guards did it in five days. The stress reaction was so extreme that normal kids we picked because they were healthy had breakdowns within 36 hours. The study ended after six days, because it was out of control. Five kids had emotional breakdowns. Does it make a difference if warriors go to battle changing their appearance or not? If they ' re anonymous, how do they treat their **victims**? In some cultures, they go to war without changing their appearance. In others, they paint themselves like"Lord of the Flies."In some, they wear masks. In many, soldiers are anonymous in uniform. So this anthropologist, John Watson, found 23 cultures that had two bits of data. Do they change their appearance? 15. Do they kill, torture, mutilate? 13. If they don ' t change their appearance, only one of eight kills, tortures or mutilates. The key is in the red zone. If they change their appearance, 12 of 13 — that ' s 90 percent — kill, torture, mutilate. And that ' s the power of anonymity. So what are the seven social processes that grease the slippery slope of evil? Mindlessly taking the first small step. Dehumanization of others. De- individuation of self. Diffusion of personal responsibility. Blind obedience to authority. Uncritical conformity to group norms. Passive tolerance of evil through inaction, or indifference. And it happens when you ' re in a new or unfamiliar situation. Your habitual response patterns don ' t work. Your personality and morality are disengaged."Nothing is easier than to denounce the evildoer ; nothing more difficult than understanding him,"Dostoyevsky. Understanding is not excusing. Psychology is not excuse-ology. So social and psychological research reveals how ordinary, good people can be transformed without the drugs. You don ' t need it. You just need the social- psychological processes. Real world parallels? Compare this with this. James Schlesinger — I ' m going to end with this — says,"Psychologists have attempted to understand how and why individuals and groups who usually act humanely can sometimes act otherwise in certain circumstances."That ' s the Lucifer effect. And he goes on to say,"The landmark Stanford study provides a cautionary tale for all military operations."If you give people power without oversight, it ' s a prescription for abuse. They knew that, and let that happen. So another report, an investigative report by General Fay, says the system is guilty. In this report, he says it was the environment that created Abu Ghraib, by leadership failures that contributed to the occurrence of such abuse, and because it remained undiscovered by higher authorities for a long period of time. Those abuses went on for three months. Who was watching the store? The answer is nobody, I think on purpose. He gave the guards permission to do those things, and they knew nobody was ever going to come down to that dungeon. So you need a paradigm shift in all of these areas. The shift is away from the medical model that focuses only on the individual. The shift is toward a public health model that recognizes situational and **systemic** vectors of disease. Bullying is a disease. **Prejudice** is a disease. **Violence** is a disease. Since the Inquisition, we ’ ve been dealing with problems at the individual level. It doesn ’ t work. Aleksandr Solzhenitsyn says,”The line between good and evil cuts through the heart of every human being.”That means that line is not out there. That ’ s a decision that you have to make, a personal thing. So I want to end very quickly on a positive note. Heroism as the antidote to evil, by promoting the heroic imagination, especially in our kids, in our educational system. We want kids to think,”I ’ m a hero in waiting, waiting for the right situation to come along, and I will act heroically. My whole life, I ’ m now going to focus away from evil — that I ’ ve been in since I was a kid — to understanding heroes. Banality of heroism. It ’ s ordinary people who do heroic deeds. It ’ s the counterpoint to Hannah Arendt ’ s”Banality of Evil.”Our traditional societal heroes are wrong, because they are the exceptions. They organize their life around this. That ’ s why we know their names. Our kids ’ heroes are also wrong models for them, because they have supernatural talents. We want our kids to realize most heroes are everyday people, and the heroic act is unusual. This is Joe Darby. He was the one that stopped those abuses you saw, because when he saw those images, he turned them over to a senior investigating officer. He was a low-level private, and that stopped it. Was he a hero? No. They had to put him in hiding, because people wanted to kill him, and then his mother and his wife. For three years, they were in hiding. This is the woman who stopped the Stanford Prison Study. When I said it got out of control, I was the prison superintendent. I didn ’ t know it was out of control. I was totally indifferent. She saw that madhouse and said,”You know what, it ’ s terrible what you ’ re doing to those boys. They ’ re not prisoners nor guards, they ’ re boys, and you are responsible.”And I ended the study the next day. The good news is I married her the next year. I just came to my senses, obviously. So situations have the power to do [three things]. But the point is, this is the same situation that can inflame the hostile imagination in some of us, that makes us **perpetrators** of evil, can inspire the heroic imagination in others. It ’ s the same situation and you ’ re on one side or the other. Most people are guilty of the evil of inaction, because your mother said,”Don ’ t get involved. Mind your own business.”And you have to say,”Mama, humanity is my business.”So the psychology of heroism is — we ’ re going to end in a moment — how do we encourage children in new hero courses, that I ’ m working on with Matt Langdon — he has a hero workshop — to develop this heroic imagination, this self-labeling,”I am a hero in waiting,”and teach them skills. To be a hero, you have to learn to be a deviant, because you ’ re always going against the conformity of the group. Heroes are ordinary people whose social actions are extraordinary. Who act. The key to heroism is two things. You have to act when other people are passive. B : You have to act socio-centrically, not egocentrically. And I want to end with a known story about Wesley Autrey, New York subway hero. Fifty- year-old African-American construction worker standing on a subway. A white guy falls on the tracks. The subway train is coming. There ’ s 75 people there. You know what? They freeze. He ’ s got a reason not to get involved. He ’ s black, the guy ’ s white, and he ’ s got two kids. Instead, he gives his kids to a stranger, jumps on the tracks, puts the guy between the tracks, lays on him, the subway goes over him. Wesley and the guy — 20 and a half inches height. The train clearance is 21 inches. A half an inch would have taken his head off. And he said,”I did what anyone could do,”no big deal to jump on the tracks. And the moral imperative is”I did what everyone should do.”And so one day, you will be in a new situation. Take path one, you ’ re going to be a **perpetrator** of evil. Evil, meaning you ’ re going to be Arthur Andersen. You ’ re going to cheat, or you ’ re going to allow bullying. Path two, you become guilty of the evil of passive inaction. Path three, you become a hero. The point is, are we ready to take the path to **celebrating** ordinary heroes, waiting for the right situation to come along to put heroic imagination into action? Because it may only happen once in your life, and when you pass it by, you ’ ll always know, I could have been a hero and I let it pass me by. So the point is thinking it and then doing it. So I want to thank you. Thank you. Let ’ s oppose the power of evil systems at home and abroad, and let ’ s focus on the positive. Advocate for respect of personal dignity, for justice and peace, which sadly our administration has not been doing. Thanks so much.

Text Sample 400: POS Dim 3 – 2008 – Score 9.00 – 2008/t000226.txt
Suppose that two American friends are traveling together in Italy. They go to see Michelangelo ’ s”David,”and when they finally come face-to-face with the statue, they both freeze dead in their tracks. The first guy â we ’ ll call him Adam â is transfixed by the beauty of the perfect human form. The second guy â we ’ ll call him Bill â is transfixed by embarrassment, at staring at the thing there in the center. So here ’ s my question for you : Which one of these two guys was more likely to have voted for George Bush, which for Al Gore? I don ’ t need a show of hands, because we all have the same political stereotypes. We all know that it ’ s Bill. And in this case, the stereotype corresponds to reality. It really is a fact that liberals are much higher than conservatives on a major personality trait called **openness** to experience. People who are high in **openness** to experience just crave novelty, variety, diversity, new ideas, travel. People low on it like things that are familiar, that are safe and dependable. If you know about this trait, you can understand a lot of puzzles about human behavior, like why artists are so different from accountants. You can predict what kinds of books they like to read, what kinds of places they like to travel to and what kinds of food they like to eat. Once you understand this trait, you can understand why anybody would eat at Applebee ’ s, but not anybody that you know. This trait also tells us a lot about politics. The main researcher of this trait, Robert McCrae, says that”Open individuals have an affinity for liberal, progressive, left-wing political views...”They like a society which is open and changing,”... whereas closed individuals prefer conservative, traditional, right-wing views.”This trait also tells us a lot about the kinds of groups people join. Here ’ s the description of a group I found on the web. What kinds of people would join”a global community... welcoming people from every discipline and culture, who seek a deeper understanding of the world, and who hope to turn that understanding into a better future for us all”? This is from some guy named Ted. Well, let ’ s see now. If **openness** predicts who becomes liberal, and **openness** predicts who becomes a TEDster, then might we predict that most TEDsters are liberal? Let ’ s find out. I ’ ll ask you to raise your hand, whether you are liberal, left of center â on social issues, primarily â or conservative. And I ’ ll give a third option, because I know there are libertarians in the audience. So please raise your hand â in the simulcast rooms too. Let ’ s let everybody see who ’ s here. Please raise your hand if you ’ d say that you ’ re liberal or left of center. Please raise your hand high right now. OK. Please raise your hand if you ’ d say you ’ re libertarian. OK. About two dozen. And please raise your hand if you ’ d say you are right of center or conservative. One, two, three, four, five â about eight or 10. OK. This is a bit of a problem. Because if our goal is to seek a deeper understanding of the world, our general lack of moral diversity here is going to make it harder. Because when people all share values, when people all share morals, they become a team. And once you

engage the psychology of teams, it shuts down open-minded thinking. When the liberal team loses, [United States of Canada / Jesusland] as it did in 2004, and as it almost did in 2000, we comfort ourselves. We try to explain why half of America voted for the other team. We think they must be blinded by religion [Post-election US map : America / Dumbf*ckistan] or by simple stupidity. So if you think that half of America votes Republican because they are blinded in this way, then my message to you is that you ’ re trapped in a moral Matrix, in a particular moral Matrix. And by”the Matrix,”I mean literally the Matrix, like the movie”The Matrix.”But I ’ m here today to give you a choice. You can either take the blue pill and stick to your comforting delusions, or you can take the red pill, learn some moral psychology and step outside the moral Matrix. Now, because I know â I assume that answers my question. I was going to ask which one you picked, but no need. You ’ re all high in **openness** to experience, and it looks like it might even taste good, and you ’ re all epicures. Anyway, let ’ s go with the red pill, study some moral psychology and see where it takes us. Let ’ s start at the beginning : What is morality, where does it come from? The worst idea in all of psychology is the idea that the mind is a blank slate at birth. Developmental psychology has shown that kids come into the world already knowing so much about the physical and social worlds and programmed to make it really easy for them to learn certain things and hard to learn others. The best definition of innateness I ’ ve seen, which clarifies so many things for me, is from the brain scientist Gary Marcus. He says,”The initial organization of the brain does not depend that much on experience. Nature provides a first draft, which experience then revises. ’ Built-in ’ doesn ’ t mean unmalleable ; it means organized in advance of experience.”OK, so what ’ s on the first draft of the moral mind? To find out, my colleague Craig Joseph and I read through the literature on anthropology, on culture variation in morality and also on evolutionary psychology, looking for matches : What sorts of things do people talk about across disciplines that you find across cultures and even species? We found five best matches, which we call the five foundations of morality. The first one is **harm** / care. We ’ re all mammals here, we all have a lot of neural and hormonal programming that makes us really bond with others, care for others, feel compassion for others, especially the weak and vulnerable. It gives us very strong feelings about those who cause **harm**. This moral foundation underlies about 70 percent of the moral statements I ’ ve heard here at TED. The second foundation is **fairness** / reciprocity. There ’ s actually ambiguous evidence as to whether you find reciprocity in other animals, but the evidence for people could not be clearer. This Norman Rockwell painting is called”The Golden Rule”â as we heard from Karen Armstrong, it ’ s the foundation of many religions. That second foundation underlies the other 30 percent of the moral statements I ’ ve heard here at TED. The third foundation is in-group / loyalty. You do find cooperative groups in the animal kingdom, but these groups are always either very small or they ’ re all **siblings**. It ’ s only among humans that you find very large groups of people who are able to cooperate and join together into groups, but in this case, groups that are united to fight other groups. This probably comes from our long history of tribal living, of tribal psychology. And this tribal psychology is so deeply pleasurable that even when we don ’ t have tribes, we go ahead and make them, because it ’ s fun. Sports is to war as pornography is to sex. We get to exercise some ancient drives. The fourth foundation is authority / respect. Here you see submissive gestures from two members of very closely related species. But authority in humans is not so closely based on power and brutality as it is in other primates. It ’ s based on more voluntary deference and even elements of love, at times. The fifth foundation is purity / sanctity. This painting is called”The Allegory Of

Chastity,”but purity is not just about suppressing female **sexuality**. It ’ s about any kind of ideology, any kind of idea that tells you that you can attain virtue by controlling what you do with your body and what you put into your body. And while the political right may moralize sex much more, the political left is doing a lot of it with food. Food is becoming extremely moralized nowadays. A lot of it is ideas about purity, about what you ’ re willing to touch or put into your body. I believe these are the five best candidates for what ’ s written on the first draft of the moral mind. I think this is what we come with, a preparedness to learn all these things. But as my son Max grows up in a liberal college town, how is this first draft going to get revised? And how will it end up being different from a kid born 60 miles south of us, in Lynchburg, Virginia? To think about culture variation, let ’ s try a different metaphor. If there really are five systems at work in the mind, five sources of intuitions and emotions, then we can think of the moral mind as one of those audio equalizers that has five channels, where you can set it to a different setting on every channel. My colleagues Brian Nosek and Jesse Graham and I made a questionnaire, which we put up on the web at www. YourMorals. org. And so far, 30, 000 people have taken this questionnaire, and you can, too. Here are the results from about 23, 000 American citizens. On the left are the scores for liberals ; on the right, conservatives ; in the middle, moderates. The blue line shows people ’ s responses on the average of all the **harm** questions. So as you see, people care about **harm** and care issues. They highly endorse these sorts of statements all across the board, but as you also see, liberals care about it a little more than conservatives ; the line slopes down. Same story for **fairness**. But look at the other three lines. For liberals, the scores are very low. They ’ re basically saying,”This is not morality. In-group, authority, purity â this has nothing to do with morality. I reject it.”But as people get more conservative, the values rise. We can say liberals have a two-channel or two-foundation morality. Conservatives have more of a five- foundation, or five-channel morality. We find this in every country we look at. Here ’ s the data for 1, 100 Canadians. I ’ ll flip through a few other slides. The UK, Australia, New Zealand, Western Europe, Eastern Europe, Latin America, the Middle East, East Asia and South Asia. Notice also that on all of these graphs, the slope is steeper on in- group, authority, purity, which shows that, within any country, the disagreement isn ’ t over **harm** and **fairness**. I mean, we debate over what ’ s fair, but everybody agrees that **harm** and **fairness** matter. Moral arguments within cultures are especially about issues of in-group, authority, purity. This effect is so robust, we find it no matter how we ask the question. In a recent study, we asked people, suppose you ’ re about to get a dog, you picked a particular breed, learned about the breed. Suppose you learn that this particular breed is independent-minded and relates to its owner as a friend and an **equal**. If you ’ re a liberal, you say,”That ’ s great!”because liberals like to say,”Fetch! Please.”But if you ’ re a conservative, that ’ s not so attractive. If you ’ re conservative and learn that a dog ’ s extremely loyal to its home and family and doesn ’ t warm up to strangers, for conservatives, loyalty is good ; dogs ought to be loyal. But to a liberal, it sounds like this dog is running for the Republican nomination. You might say, OK, there are differences between liberals and conservatives, but what makes the three other foundations moral? Aren ’ t they the foundations of xenophobia, authoritarianism and puritanism? What makes them moral? The answer, I think, is contained in this incredible triptych from Hieronymus Bosch,”The Garden of Earthly Delights.”In the first panel, we see the moment of creation. All is ordered, all is beautiful, all the people and animals are doing what they ’ re supposed to be doing, are where they ’ re supposed to be. But then, given the way of the world, things change. We get every person doing whatever he wants, with every aperture of every other person and every other animal. Some of you might recognize this as the ’ 60s. But the ’ 60s inevitably gives way to the ’ 70s, where the cuttings of the apertures hurt a little bit more. Of course, Bosch called this hell. So this triptych, these three panels, portray the timeless truth that order tends to decay. The truth of social entropy. But lest you think this is just some part of the Christian imagination where Christians have this weird problem with pleasure, here ’ s the same story, the same progression, told in a paper that was published in”Nature”a few years ago, in which Ernst Fehr and Simon GĂ¤chter had people play a commons dilemma, a game in which you give people money, and then, on each round of the game, they can put money into a common pot, then the experimenter doubles what ’ s there, and then it ’ s all divided among the players. So it ’ s a nice analog for all sorts of environmental issues, where we ’ re asking people to make a sacrifice and they don ’ t really benefit from their own sacrifice. You really want everybody else to sacrifice, but everybody has a temptation to free ride. What happens is that, at first, people start off reasonably cooperative. This is all played anonymously. On the first round, people give about half of the money that they can. But they quickly see other people aren ’ t doing so much.”I don ’ t want to be a sucker. I won ’ t cooperate.”So cooperation quickly decays from reasonably good down to close to zero. But then â and here ’ s the trick â Fehr and GĂ¤chter, on the seventh round, told people,”You know what? New rule. If you want to give some of your own money to punish people who aren ’ t contributing, you can do that.”And as soon as people heard about the punishment issue going on, cooperation shoots up. It shoots up and it keeps going up. Lots of research shows that to solve cooperative problems, it really helps. It ’ s not enough to appeal to people ’ s good motives. It helps to have some sort of punishment. Even if it ’ s just **shame** or embarrassment or gossip, you need some sort of punishment to bring people, when they ’ re in large groups, to cooperate. There ’ s even some recent research suggesting that religion â priming God, making people think about God â often, in some situations, leads to more cooperative, more pro-social behavior. Some people think that religion is an adaptation evolved both by cultural and biological evolution to make groups to cohere, in part for the purpose of trusting each other and being more effective at competing with other groups. That ’ s probably right, although this is a controversial issue. But I ’ m particularly interested in religion and the origin of religion and in what it does to us and for us, because I think the greatest wonder in the world is not the Grand Canyon. The Grand Canyon is really simple â a lot of rock and a lot of water and wind and a lot of time, and you get the Grand Canyon. It ’ s not that complicated. This is what ’ s complicated : that people lived in places like the Grand Canyon, cooperating with each other, or on the savannahs of Africa or the frozen shores of Alaska. And some of these villages grew into the mighty cities of Babylon and Rome and Tenochtitlan. How did this happen? It ’ s an absolute miracle, much harder to explain than the Grand Canyon. The answer, I think, is that they used every tool in the toolbox. It took all of our moral psychology to create these cooperative groups. Yes, you need to be concerned about **harm**, you need a psychology of justice. But it helps to organize a group if you have subgroups, and if those subgroups have some internal structure, and if you have some ideology that tells people to suppress their carnality â to pursue higher, nobler ends. Now we get to the crux of the disagreement between liberals and conservatives : liberals reject three of these foundations. They say,”Let ’ s **celebrate** diversity, not common in-group membership,”and,”Let ’ s question authority,”and,”Keep your laws off my body.”Liberals have very noble motives for doing this. Traditional authority and morality can be quite repressive and restrictive to those at the bottom, to women, to people who don ’ t fit in. Liberals speak for the weak and oppressed. They want change and justice, even at the risk of chaos. This shirt says,”Stop bitching, start a revolution.”If you ’ re high in **openness** to experience, revolution is good ; it ’ s change, it ’ s fun. Conservatives, on the other hand, speak for institutions and traditions. They want order, even at some cost, to those at the bottom. The great conservative insight is that order is really hard to achieve. It ’ s precious, and it ’ s really easy to lose. So as Edmund Burke said,”The restraints on men, as well as their liberties, are to be reckoned among their rights.”This was after the chaos of the French Revolution. Once you see that liberals and conservatives both have something to contribute, that they form a balance on change versus stability, then I think the way is open to step outside the moral Matrix. This is the great insight that all the Asian religions have attained. Think about yin and yang. Yin and yang aren ’ t enemies ; they don ’ t hate each other. Yin and yang are both necessary, like night and day, for the functioning of the world. You find the same thing in Hinduism. There are many high gods in Hinduism. Two of them are Vishnu, the preserver, and Shiva, the destroyer. This image, actually, is both of those gods sharing the same body. You have the markings of Vishnu on the left, so we could think of Vishnu as the conservative god. You have the markings of Shiva on the right â Shiva ’ s the liberal god. And they work together. You find the same thing in Buddhism. These two stanzas contain, I think, the deepest insights that have ever been attained into moral psychology. From the Zen master SĂngcĂ n :”If you want the truth to stand clear before you, never be ’ for ’ or ’ against. ’ The struggle between ’ for ’ and ’ against ’ is the mind ’ s worst disease.”Unfortunately, it ’ s a disease that has been caught by many of the world ’ s leaders. But before you feel superior to George Bush, before you throw a stone, ask yourself : Do you accept this? Do you accept stepping out of the battle of good and evil? Can you be not for or against anything? So what ’ s the point? What should you do? Well, if you take the greatest insights from ancient Asian philosophies and religions and combine them with the latest research on moral psychology, I think you come to these conclusions : that our righteous minds were designed by evolution to unite us into teams, to divide us against other teams and then to blind us to the truth. So what should you do? Am I telling you to not strive? Am I telling you to embrace SĂngcĂ n and stop, stop with the struggle of for and against? No, absolutely not. I ’ m not saying that. This is an amazing group of people who are doing so much, using so much of their talent, their brilliance, their energy, their money, to make the world a better place, to fight wrongs, to solve problems. But as we learned from Samantha Power in her story about SĂŠrgio Vieira de Mello, you can ’ t just go charging in, saying,”You ’ re wrong, and I ’ m right,”because, as we just heard, everybody thinks they are right. A lot of the problems we have to solve are problems that require us to change other people. And if you want to change other people, a much better way to do it is to first understand who we are â understand our moral psychology, understand that we all think we ’ re right â and then step out, even if it ’ s just for a moment, step out â check in with SĂngcĂ n. Step out of the moral Matrix, just try to see it as a struggle playing out, in which everybody thinks they ’ re right, and even if you disagree with them, everybody has some reasons for what they ’ re doing. Step out. And if you do that, that ’ s the essential move to cultivate moral humility, to get yourself out of this self-righteousness, which is the normal human condition. Think about the Dalai Lama. Think about the **enormous** moral authority of the Dalai Lama. It comes from his moral humility. So I think the point â the point of my talk and, I think, the point of TED â is that this is a group that is passionately engaged in the pursuit of changing the world for the better. People here are passionately engaged in trying to make the world a better place. But there is also a passionate commitment to the truth. And so I think the answer is to use that passionate commitment to the truth to try to turn it into a better future for us all. Thank you.

Text Sample 401: POS Dim 3 – 2008 – Score 8.00 – 2008/t000293.txt

Well, this is such an honor. And it ’ s wonderful to be in the presence of an organization that is really making a difference in the world. And I ’ m intensely grateful for the opportunity to speak to you today. And I ’ m also rather surprised, because when I look back on my life the last thing I ever wanted to do was write, or be in any way involved in religion. After I left my convent, I ’ d finished with religion, frankly. I thought that was it. And for 13 years I kept clear of it. I wanted to be an English literature professor. And I certainly didn ’ t even want to be a writer, particularly. But then I suffered a series of career catastrophes, one after the other, and finally found myself in television. I said that to Bill Moyers, and he said,”Oh, we take anybody.”And I was doing some rather controversial religious programs. This went down very well in the U. K., where religion is extremely unpopular. And so, for once, for the only time in my life, I was finally in the mainstream. But I got sent to Jerusalem to make a film about early Christianity. And there, for the first time, I encountered the other religious traditions : Judaism and Islam, the sister religions of Christianity. And while I found I knew nothing about these faiths at all — despite my own intensely religious background, I ’ d seen Judaism only as a kind of prelude to Christianity, and I knew nothing about Islam at all. But in that city, that tortured city, where you see the three faiths jostling so uneasily together, you also become aware of the profound connection between them. And it has been the study of other religious traditions that brought me back to a sense of what religion can be, and actually enabled me to look at my own faith in a different light. And I found some astonishing things in the course of my study that had never occurred to me. Frankly, in the days when I thought I ’ d had it with religion, I just found the whole thing absolutely incredible. These doctrines seemed unproven, abstract. And to my astonishment, when I began seriously studying other traditions, I began to realize that belief — which we make such a fuss about today — is only a very recent religious enthusiasm that surfaced only in the West, in about the 17th century. The word”belief”itself originally meant to love, to prize, to hold dear. In the 17th century, it narrowed its focus, for reasons that I ’ m exploring in a book I ’ m writing at the moment, to include — to mean an intellectual assent to a set of propositions, a credo.”I believe :”it did not mean,”I accept certain creedal articles of faith.”It meant :”I commit myself. I engage myself.”Indeed, some of the world traditions think very little of religious orthodoxy. In the Quran, religious opinion — religious orthodoxy — is **dismissed** as”zanna :”self-indulgent guesswork about matters that nobody can be certain of one way or the other, but which makes people quarrelsome and stupidly sectarian. So if religion is not about believing things, what is it about? What I ’ ve found, across the board, is that religion is about behaving differently. Instead of deciding whether or not you believe in God, first you do something. You behave in a committed way, and then you begin to understand the truths of religion. And religious doctrines are meant to be summons to action ; you only understand them when you put them into practice. Now, pride of place in this practice is given to compassion. And it is an arresting fact that right across the board, in every single one of the major world faiths, compassion — the ability to feel with the other in the way we ’ ve been thinking about this evening —

is not only the test of any true religiosity, it is also what will bring us into the presence of what Jews, Christians and Muslims call"God"or the"Divine."It is compassion, says the Buddha, which brings you to Nirvana. Why? Because in compassion, when we feel with the other, we dethrone ourselves from the center of our world and we put another person there. And once we get rid of ego, then we ' re ready to see the Divine. And in particular, every single one of the major world traditions has highlighted — has said — and put at the core of their tradition what ' s become known as the Golden Rule. First propounded by Confucius five centuries before Christ :"Do not do to others what you would not like them to do to you."That, he said, was the central thread which ran through all his teaching and that his disciples should put into practice all day and every day. And it was — the Golden Rule would bring them to the transcendent value that he called"ren,"human-heartedness, which was a transcendent experience in itself. And this is absolutely crucial to the monotheisms, too. There ' s a famous story about the great rabbi, Hillel, the older contemporary of Jesus. A pagan came to him and offered to convert to Judaism if the rabbi could recite the whole of Jewish teaching while he stood on one leg. Hillel stood on one leg and said,"That which is hateful to you, do not do to your neighbor. That is the Torah. The rest is commentary. Go and study it."And"go and study it"was what he meant. He said,"In your exegesis, you must make it clear that every single verse of the Torah is a commentary, a gloss upon the Golden Rule."The great Rabbi Meir said that any interpretation of Scripture which led to **hatred** and disdain, or contempt of other people — any people whatsoever — was illegitimate. Saint Augustine made exactly the same point. Scripture, he says,"teaches nothing but charity, and we must not leave an interpretation of Scripture until we have found a compassionate interpretation of it."And this struggle to find compassion in some of these rather rebarbative texts is a good dress rehearsal for doing the same in ordinary life. But now look at our world. And we are living in a world that is — where religion has been hijacked. Where terrorists cite Quranic verses to justify their atrocities. Where instead of taking Jesus ' words,"Love your enemies. Don ' t judge others,"we have the spectacle of Christians endlessly judging other people, endlessly using Scripture as a way of arguing with other people, putting other people down. Throughout the ages, religion has been used to oppress others, and this is because of human ego, human greed. We have a talent as a species for messing up wonderful things. So the traditions also insisted — and this is an important point, I think — that you could not and must not confine your compassion to your own group : your own nation, your own co-religionists, your own fellow countrymen. You must have what one of the Chinese sages called"jian ai": concern for everybody. Love your enemies. Honor the stranger. We formed you, says the Quran, into tribes and nations so that you may know one another. And this, again — this universal outreach — is getting subdued in the strident use of religion — abuse of religion — for nefarious gains. Now, I ' ve lost count of the number of taxi drivers who, when I say to them what I do for a living, inform me that religion has been the cause of all the major world wars in history. Wrong. The causes of our present woes are political. But, make no mistake about it, religion is a kind of fault line, and when a conflict gets ingrained in a region, religion can get sucked in and become part of the problem. Our modernity has been exceedingly **violent**. Between 1914 and 1945, 70 million people died in Europe alone as a result of **armed** conflict. And so many of our institutions, even football, which used to be a pleasant pastime, now causes riots where people even die. And it ' s not surprising that religion, too, has been affected by this **violent** ethos. There ' s also a great deal, I think, of religious illiteracy around. People seem to think, now equate religious faith with believing things. As though that

— we call religious people often believers, as though that were the main thing that they do. And very often, secondary goals get pushed into the first place, in place of compassion and the Golden Rule. Because the Golden Rule is difficult. I sometimes — when I ’ m speaking to congregations about compassion, I sometimes see a mutinous expression crossing some of their faces because a lot of religious people prefer to be right, rather than compassionate. Now — but that ’ s not the whole story. Since September the 11th, when my work on Islam suddenly propelled me into public life, in a way that I ’ d never imagined, I ’ ve been able to sort of go all over the world, and finding, everywhere I go, a yearning for change. I ’ ve just come back from Pakistan, where literally thousands of people came to my lectures, because they were yearning, first of all, to hear a friendly Western voice. And especially the young people were coming. And were asking me — the young people were saying,”What can we do? What can we do to change things?”And my hosts in Pakistan said,”Look, don ’ t be too polite to us. Tell us where we ’ re going wrong. Let ’ s talk together about where religion is failing.”Because it seems to me that with — our current situation is so serious at the moment that any ideology that doesn ’ t promote a sense of global understanding and global appreciation of each other is failing the test of the time. And religion, with its wide following... Here in the United States, people may be being religious in a different way, as a report has just shown — but they still want to be religious. It ’ s only Western Europe that has retained its secularism, which is now beginning to look rather endearingly old-fashioned. But people want to be religious, and religion should be made to be a force for harmony in the world, which it can and should be — because of the Golden Rule.”Do not do to others what you would not have them do to you”: an ethos that should now be applied globally. We should not treat other nations as we would not wish to be treated ourselves. And these — whatever our wretched beliefs — is a religious matter, it ’ s a spiritual matter. It ’ s a profound moral matter that engages and should engage us all. And as I say, there is a hunger for change out there. Here in the United States, I think you see it in this election campaign : a longing for change. And people in churches all over and **mosques** all over this continent after September the 11th, coming together locally to create networks of understanding. With the **mosque**, with the synagogue, saying,”We must start to speak to one another.”I think it ’ s time that we moved beyond the idea of toleration and move toward appreciation of the other. I ’ d — there ’ s one story I ’ d just like to mention. This comes from”The Iliad.”But it tells you what this spirituality should be. You know the story of”The Iliad,”the 10-year war between Greece and Troy. In one incident, Achilles, the famous warrior of Greece, takes his troops out of the war, and the whole war effort suffers. And in the course of the ensuing muddle, his beloved friend, Patroclus, is killed — and killed in single combat by one of the Trojan princes, Hector. And Achilles goes mad with **grief** and rage and revenge, and he mutilates the body. He kills Hector, he mutilates his body and then he refuses to give the body back for burial to the family, which means that, in Greek ethos, Hector ’ s soul will wander eternally, lost. And then one night, Priam, king of Troy, an old man, comes into the Greek camp incognito, makes his way to Achilles ’ tent to ask for the body of his son. And everybody is shocked when the old man takes off his head covering and shows himself. And Achilles looks at him and thinks of his father. And he starts to weep. And Priam looks at the man who has murdered so many of his sons, and he, too, starts to weep. And the sound of their weeping filled the house. The Greeks believed that weeping together created a bond between people. And then Achilles takes the body of Hector, he hands it very tenderly to the father, and the two men look at each other, and see each other as divine. That is the ethos found, too, in all the religions. It ’ s what is meant by overcoming the horror that we feel when we are under threat of our enemies, and beginning to appreciate the other. It ’ s of great importance that the word for”holy”in Hebrew, applied to God, is”Kadosh”: separate, other. And it is often, perhaps, the very otherness of our enemies which can give us intimations of that utterly mysterious transcendence which is God. And now, here ’ s my wish : I wish that you would help with the creation, launch and propagation of a Charter for Compassion, crafted by a group of inspirational thinkers from the three Abrahamic traditions of Judaism, Christianity and Islam, and based on the fundamental principle of the Golden Rule. We need to create a movement among all these people that I meet in my travels — you probably meet, too — who want to join up, in some way, and reclaim their faith, which they feel, as I say, has been hijacked. We need to empower people to remember the compassionate ethos, and to give guidelines. This Charter would not be a massive document. I ’ d like to see it — to give guidelines as to how to interpret the Scriptures, these texts that are being abused. Remember what the rabbis and what Augustine said about how Scripture should be governed by the principle of charity. Let ’ s get back to that. And the idea, too, of Jews, Christians and Muslims — these traditions now so often at loggerheads — working together to create a document which we hope will be signed by a thousand, at least, of major religious leaders from all the traditions of the world. And you are the people. I ’ m just a solitary scholar. Despite the idea that I love a good time, which I was rather amazed to see coming up on me — I actually spend a great deal of time alone, studying, and I ’ m not very — you ’ re the people with media knowledge to explain to me how we can get this to everybody, everybody on the planet. I ’ ve had some preliminary talks, and Archbishop Desmond Tutu, for example, is very happy to give his name to this, as is Imam Feisal Rauf, the Imam in New York City. Also, I would be working with the Alliance of Civilizations at the United Nations. I was part of that United Nations **initiative** called the Alliance of Civilizations, which was asked by Kofi Annan to diagnose the causes of extremism, and to give practical guidelines to member states about how to avoid the escalation of further extremism. And the Alliance has told me that they are very happy to work with it. The importance of this is that this is — I can see some of you starting to look worried, because you think it ’ s a slow and cumbersome body — but what the United Nations can do is give us some neutrality, so that this isn ’ t seen as a Western or a Christian **initiative**, but that it ’ s coming, as it were, from the United Nations, from the world — who would help with the sort of bureaucracy of this. And so I do urge you to join me in making — in this **charter** — to building this **charter**, launching it and propagating it so that it becomes — I ’ d like to see it in every college, every church, every **mosque**, every synagogue in the world, so that people can look at their tradition, reclaim it, and make religion a source of peace in the world, which it can and should be. Thank you very much.

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I ’ m going to go right into the slides. And all I ’ m going to try and prove to you with these slides is that I do just very straight stuff. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the left-hand side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a **neighborhood** that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage the **neighborhood** â one made accommodations. I tried to be inclusive, to include the buildings in the **neighborhood**, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of stuff and folded it up to put it in the wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at postmodernism â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker. And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And the client said,”Why no fish?”And so I made a drawing with a fish, and I left Japan. Three weeks later, I received a complete set of drawings saying we ’ d won the competition. Now, it ’ s hard to do. It ’ s hard to translate a fish form, because they ’ re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ’ t do it, and so that made it even more exciting. But he was right â I couldn ’ t do the tail. I started to get the head OK, but the tail I couldn ’ t do. It was pretty hard. The thing on the right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with the context again.

Now, if you saw a picture of this as it was published in Architectural Record â they didn ’ t show the context, so you would think,”God, what a pushy guy this is.”But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ’ t find it. So... As for craft and technology and all those things that you ’ ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent drawings to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made the drawings of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ’ s one of the nicest pieces I ’ ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ’ t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ’ m getting good at temporary â and we put a conference room in that ’ s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ’ s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ’ t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ’ s offices and I made a pedestal for a sculpture. And he didn ’ t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ’ s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ’ ve been friends for a long time. And we ’ ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called”Il Corso del Coltello”â the Swiss Army knife. And most of the imagery is â Claes ’, but those two little boys are my sons, and they were Claes ’ assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ’ s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ’ ve been known to mess with things like chain link fencing. I do it because it ’ s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ’ s so much denial about them. People hate it. And I ’ m fascinated with that, which, like the paper furniture â it ’ s one of those materials. And I ’ m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ’ s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to the building on the left. And I still think the Time magazine picture will be of the binoculars, you know, leaving out the â what the hell. I use a lot of metal in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,"If I can ' t get the craft that I want, I ' ll use the craft I can get."There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the metal because it was a way of building a building that was a sculpture. And it was all of one material, and the metal could go on the roof as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and stuff. I was given an opportunity to design an exhibit for the metalworkers ' unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future metal buildings, etc. etc. And it ' s working very well to have these people, these craftsmen, interested in it. I just tell the stories. It ' s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The metal continued into a building â Herman Miller, in Sacramento. And it ' s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ' s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ' s out in the middle of nowhere, and you approach it. It ' s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ' s aesthetic. Everybody ' s trying to get the panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ' s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ' s all about [a] metaphor for a city, maybe. And so Stanley did the little dome thing. And we did it over the phone and by fax. He would send me a fax and show me something. He ' d made a building with a dome and he had a little tower. I told him,"No, no, that ' s too ongepotchket. I don ' t want the tower."So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going back and forth over a couple of weeks ' period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,"David and Goliath."And that ' s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and cleaned it up and fixed it up and used the kind of â I thought â the language of the **neighborhood**, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ' t normally think I would fit them. The detailing was very careful with the lead copper and making panels and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some film festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â a little shopping center. And this is a laser laboratory at the University of Iowa, in which the fish comes back as an abstraction in the back. It ’ s the support labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the curved part there ’ s all the mechanical equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ’ t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ’ ve been building it so long I don ’ t remember where it is. It ’ s in the West Valley. And we started with the stream and built the house along the stream â dammed it up to make a lake. These are the models. The reality, with the lake â the workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ’ s hard to get perfect corners and stuff. That big metal thing is a passage, and in it is â you go downstairs into the living room and then down into the bedroom, which is on the right. It ’ s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â they asked Philip to do it. He was too busy. He didn ’ t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ’ t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ’ ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ’ s made of metal and the brown stuff is Fin-Ply â it ’ s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ’ t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ’ s Burnham Mall, on the left. It ’ s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ’ s a railroad track. This is the city hall over here somewhere, and the courthouse. And the centerline of the mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ’ re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this stuff at the bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the centerline of this thing â we ’ d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ’ s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ’ 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the back end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake. And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ’ s the living room. Servants ’ quarters. A guest bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is back in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a fun thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it ain ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the conference facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais des Sports, the Gare de Lyon over here. The Minister of Finance â the guy that moved from the Louvre â goes in here. There ’ s a new library across the river. And back in here, in this already treed park, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan coupe. And I struggled with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this **street**, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in French limestone, except for this metal piece. And it faces into a park. And the idea was to make this express the energy of this. On the side facing the **street** it ’ s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in the Hamptons. And it ’ s got a fish. And I keep thinking,”This is going to be the last fish.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s the living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in the budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of the guys that presented to you earlier. We ’ ve had a lot of fun working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ’ s perpendicular to the train station, to the route of the train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good slides to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â you see it as all of these pieces accrue and become part of an overall **neighborhood**. It ’ s plaster and just zinc. And you wonder, if this is a museum, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and stuff. And from the building in the back, you see it as an iceberg floating by in the hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is the Disney Hall â the concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a compositional relationship between us that would strengthen both of us. And the plan of this â it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â to MOCA, etc. The acoustician in the competition gave us criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But everybody liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were the three buildings that were the ideal â the Concertgebouw, Boston and Berlin. Everybody liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new acoustician, he told us this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an acoustical, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the sloping outside walls, which the acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on the way to being â and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with laser stuff, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, struggling with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the foot of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the foot of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th- century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ’ t as good as the stuff, and so it failed. It should have been the other way around â the food should have been good first. It didn ’ t work. Thank you very much.

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If you ’ re here today â and I ’ m very happy that you are â you ’ ve all heard about how sustainable development will save us from ourselves. However, when we ’ re not at TED, we are often told that a real sustainability policy agenda is just not feasible, especially in large urban areas like New York City. And that ’ s because most people with decision-making powers, in both the public and the private sector, really don ’ t feel as though they ’ re in danger. The reason why I ’ m here today, in part, is because of a dog â an abandoned puppy I found back in the rain, back in 1998. She turned out to be a much bigger dog than I ’ d anticipated. When she came into my life, we were fighting against a huge waste facility planned for the East River waterfront despite the fact that our small part of New York City already handled more than 40 percent of the entire city ’ s commercial waste : a sewage treatment pelletizing plant, a sewage sludge plant, four power plants, the world ’ s largest food-distribution center, as well as other industries that bring more than 60, 000 diesel truck trips to the area each week. The area also has one of the lowest ratios of parks to people in the city. So when I was contacted by the Parks Department about a $10, 000 seed-grant **initiative** to help develop waterfront projects, I thought they were really well-meaning, but a bit naive. I ’ d lived in this area all my life, and you could not get to the river, because of all the lovely facilities that I mentioned earlier. Then, while jogging with my dog one morning, she pulled me into what I thought was just another illegal dump. There were weeds and piles of garbage and other stuff that I won ’ t mention here, but she kept dragging me, and lo and behold, at the end of that lot was the river. I knew that this forgotten little street-end, abandoned like the dog that brought me there, was worth saving. And I knew it would grow to become the proud beginnings of the community-led revitalization of the new South Bronx. And just like my new dog, it was an idea that got bigger than I ’ d imagined. We garnered much support along the way, and the Hunts Point Riverside Park became the first waterfront park that the South Bronx had had in more than 60 years. We leveraged that $10, 000 seed grant more than 300 times, into a $3 million park. And in the fall, I ’ m going to exchange marriage vows with my beloved. Thank you very much. That ’ s him pressing my buttons back there, which he does all the time. But those of us living in environmental justice communities are the canary in the coal mine. We feel the problems right now, and have for some time. Environmental justice, for those of you who may not be familiar with the term, goes something like this : no community should be saddled with more environmental burdens and less environmental benefits than any other. Unfortunately, race and class are extremely reliable indicators as to where one might find the good stuff, like parks and trees, and where one might find the bad stuff, like power plants and waste facilities. As a black person in America, I am twice as likely as a white person to live in an area where air pollution poses the greatest risk to my health. I am five times more likely to live within walking distance of a power plant or chemical facility, which I do. These land-use decisions created the hostile conditions that lead to problems like obesity, diabetes and asthma. Why would someone leave their home to go for a brisk walk in a toxic **neighborhood**? Our 27 percent obesity rate is high even for this country, and diabetes comes with it. One out of four South Bronx children has asthma. Our asthma hospitalization rate is seven times higher than the national average. These impacts are coming everyone ’ s way. And we all pay dearly for solid waste costs, health problems associated with pollution and more odiously, the cost of imprisoning our young black and Latino men, who possess untold amounts of untapped potential. Fifty percent of our residents live at or below the poverty line ; 25 percent of us are unemployed. Low-income citizens often use emergency-room visits as primary care. This comes at a high cost to taxpayers and produces no proportional benefits. Poor people are not only still poor, they are still unhealthy. Fortunately, there are many people like me who are striving for solutions that won ’ t compromise the lives of low-income communities of color in the short term, and won ’ t destroy us all in the long term. None of us want that, and we all have that in common. So what else do we have in common? Well, first of all, we ’ re all incredibly good-looking. Graduated high school, college, post-graduate degrees, traveled to interesting places, didn ’ t have kids in your early **teens**, financially stable, never been imprisoned. OK. Good. But, besides being a black woman, I am different from most of you in some other ways. I watched nearly half of the buildings in my **neighborhood** burn down. My big brother Lenny fought in Vietnam, only to be gunned down a few blocks from our home. Jesus. I grew up with a crack house across the **street**. Yeah, I ’ m a poor black child from the ghetto. These things make me different from you. But the things we have in common set me apart from most of the people in my community, and I am in between these two worlds with enough of my heart to fight for justice in the other. So how did things get so different for us? In the late ’ 40s, my dad â a Pullman porter, son of a slave â bought a house in the Hunts Point section of the South Bronx, and a few years later, he married my mom. At the time, the community was a mostly white, working-class **neighborhood**. My dad was not alone. And as others like him pursued their own version of the American dream, white flight became common in the South Bronx and in many cities around the country. Red-lining was used by banks, wherein certain sections of the city, including ours, were **deemed** off-limits to any sort of investment. Many landlords believed it was more profitable to torch their buildings and collect insurance money rather than to sell under those conditions â dead or injured former tenants notwithstanding. Hunts Point was formerly a walk-to-work community, but now residents had neither work nor home to walk to. A national highway construction boom was added to our problems. In New York State, Robert Moses spearheaded an aggressive highway-expansion campaign. One of its primary goals was to make it easier for residents of wealthy communities in Westchester County to go to Manhattan. The South Bronx, which lies in between, did not stand a chance. Residents were often given less than a month ’ s notice before their buildings were razed. 600, 000 people were displaced. The common perception was that only pimps and pushers and prostitutes were from the South Bronx. And if you are told from your earliest days that nothing good is going to come from your community, that it ’ s bad and ugly, how could it not reflect on you? So now, my family ’ s property was worthless, save for that it was our home, and all we had. And luckily for me, that home and the love inside of it, along with help from teachers, mentors and friends along the way, was enough. Now, why is this story important? Because from a planning perspective, economic degradation begets environmental degradation, which begets social degradation. The disinvestment that began in the 1960s set the stage for all the environmental injustices that were to come. Antiquated zoning and land-use regulations are still used to this day to continue putting polluting facilities in my **neighborhood**. Are these factors taken into consideration when land-use policy is decided? What costs are associated with these decisions? And who pays? Who profits? Does anything justify what the local community goes through? This was”planning”â in quotes â that did not have our best interests in mind. Once we realized that, we decided it was time to do our own planning. That small park I told you about earlier was the first stage of building a Greenway movement in the

South Bronx. I wrote a one-and-a-quarter-million dollar federal transportation grant to design the plan for a waterfront esplanade with dedicated on-street bike paths. Physical improvements help inform public policy regarding traffic **safety**, the placement of the waste and other facilities, which, if done properly, don ’ t compromise a community ’ s quality of life. They provide opportunities to be more physically active, as well as local economic development. Think bike shops, juice stands. We secured 20 million dollars to build first-phase projects. This is Lafayette Avenue â and that ’ s redesigned by Mathews Nielsen Landscape Architects. And once this path is constructed, it ’ ll connect the South Bronx with more than 400 acres of Randall ’ s Island Park. Right now we ’ re separated by about 25 feet of water, but this link will change that. As we nurture the natural environment, its abundance will give us back even more. We run a project called the Bronx [Environmental] Stewardship Training, which provides job training in the fields of ecological restoration, so that folks from our community have the skills to compete for these well-paying jobs. Little by little, we ’ re seeding the area with green-collar jobs â and with people that have both a financial and personal stake in their environment. The Sheridan Expressway is an underutilized relic of the Robert Moses era, built with no regard for the **neighborhoods** that were divided by it. Even during rush hour, it goes virtually unused. The community created an alternative transportation plan that allows for the removal of the highway. We have the opportunity now to bring together all the stakeholders to re-envision how this 28 acres can be better utilized for parkland, affordable housing and local economic development. We also built New York City ’ s first green and cool roof demonstration project on top of our offices. Cool roofs are highly-reflective surfaces that don ’ t absorb solar heat, and pass it on to the building or atmosphere. Green roofs are soil and living plants. Both can be used instead of petroleum-based roofing materials that absorb heat, contribute to urban”heat island”effect and degrade under the sun, which we in turn breathe. Green roofs also retain up to 75 percent of rainfall, so they reduce a city ’ s need to fund costly end-of-pipe solutions â which, incidentally, are often located in environmental justice communities like mine. And they provide habitats for our little friends! [Butterfly] So cool! Anyway, the demonstration project is a springboard for our own green roof installation business, bringing jobs and sustainable economic activity to the South Bronx. [Green is the new black...] I like that, too. Anyway, I know Chris told us not to do pitches up here, but since I have all of your attention : We need investors. End of pitch. It ’ s better to ask for forgiveness than permission. Anyway â OK. Katrina. Prior to Katrina, the South Bronx and New Orleans ’ Ninth Ward had a lot in common. Both were largely populated by poor people of color, both hotbeds of cultural innovation : think hip-hop and jazz. Both are waterfront communities that host both industries and residents in close proximity of one another. In the post-Katrina era, we have still more in common. We ’ re at best ignored, and maligned and abused, at worst, by negligent regulatory agencies, pernicious zoning and lax governmental accountability. Neither the destruction of the Ninth Ward nor the South Bronx was inevitable. But we have emerged with valuable lessons about how to dig ourselves out. We are more than simply national symbols of urban blight or problems to be solved by empty campaign promises of presidents come and gone. Now will we let the Gulf Coast languish for a decade or two, like the South Bronx did? Or will we take proactive steps and learn from the homegrown resource of grassroots activists that have been born of desperation in communities like mine? Now listen, I do not expect individuals, corporations or government to make the world a better place because it is right or moral. This presentation today only represents some of what I ’ ve been through. Like a tiny little bit. You ’ ve no clue.

But I ’ ll tell you later, if you want to know. But â I know it ’ s the bottom line, or one ’ s perception of it, that motivates people in the end. I ’ m interested in what I like to call the”triple bottom line”that sustainable development can produce. Developments that have the potential to create positive returns for all concerned : the developers, government and the community where these projects go up. At present, that ’ s not happening in New York City. And we are operating with a comprehensive urban-planning deficit. A parade of government subsidies is going to propose big-box and stadium developments in the South Bronx, but there is scant coordination between city agencies on how to deal with the cumulative effects of increased traffic, pollution, solid waste and the impacts on open space. And their approaches to local economic and job development are so lame it ’ s not even funny. Because on top of that, the world ’ s richest sports team is replacing the House That Ruth Built by destroying two well-loved community parks. Now, we ’ ll have even less than that stat I told you about earlier. And although less than 25 percent of South Bronx residents own cars, these projects include thousands of new parking spaces, yet zip in terms of mass public transit. Now, what ’ s missing from the larger debate is a comprehensive cost- benefit analysis between not fixing an unhealthy, environmentally-challenged community, versus incorporating structural, sustainable changes. My agency is working closely with Columbia University and others to shine a light on these issues. Now let ’ s get this straight : I am not anti-development. Ours is a city, not a wilderness preserve. And I ’ ve embraced my inner capitalist. And, but I don ’ t have â You probably all have, and if you haven ’ t, you need to. So I don ’ t have a problem with developers making money. There ’ s enough precedent out there to show that a sustainable, community-friendly development can still make a fortune. Fellow TEDsters Bill McDonough and Amory Lovins â both heroes of mine by the way â have shown that you can actually do that. I do have a problem with developments that hyper-exploit politically vulnerable communities for profit. That it continues is a **shame** upon us all, because we are all responsible for the future that we create. But one of the things I do to remind myself of greater possibilities, is to learn from visionaries in other cities. This is my version of globalization. Let ’ s take Bogota. Poor, Latino, surrounded by runaway gun **violence** and drug trafficking ; a reputation not unlike that of the South Bronx. However, this city was blessed in the late 1990s with a highly-influential mayor named Enrique PeĂąalosa. He looked at the demographics. Few Bogotanos own cars, yet a huge portion of the city ’ s resources was dedicated to serving them. If you ’ re a mayor, you can do something about that. His administration narrowed key municipal thoroughfares from five lanes to three, outlawed parking on those **streets**, expanded pedestrian walkways and bike lanes, created public plazas, created one of the most efficient bus mass-transit systems in the entire world. For his brilliant efforts, he was nearly impeached. But as people began to see that they were being put first on issues reflecting their day-to-day lives, incredible things happened. People stopped littering. Crime rates dropped, because the **streets** were alive with people. His administration attacked several typical urban problems at one time, and on a third-world budget, at that. We have no excuse in this country, I ’ m sorry. But the bottom line is : their people-first agenda was not meant to penalize those who could actually afford cars, but rather, to provide opportunities for all Bogotanos to participate in the city ’ s resurgence. That development should not come at the expense of the majority of the population is still considered a radical idea here in the U. S. But Bogota ’ s example has the power to change that. You, however, are blessed with the gift of influence. That ’ s why you ’ re here and why you value the information we exchange. Use your influence in support of comprehensive, sustainable change everywhere. Don ’ t just talk about it at TED. This is a nationwide policy agenda I ’ m trying to build, and as you all know, politics are personal. Help me make green the new black. Help me make sustainability sexy. Make it a part of your dinner and cocktail conversations. Help me fight for environmental and economic justice. Support investments with a triple-bottom-line return. Help me democratize sustainability by bringing everyone to the table, and insisting that comprehensive planning can be addressed everywhere. Oh good, glad I have a little more time! Listen â when I spoke to Mr. Gore the other day after breakfast, I asked him how environmental justice activists were going to be included in his new marketing strategy. His response was a grant program. I don ’ t think he understood that I wasn ’ t asking for funding. I was making him an offer. What troubled me was that this top-down approach is still around. Now, don ’ t get me wrong, we need money. But grassroots groups are needed at the table during the decision-making process. Of the 90 percent of the energy that Mr. Gore reminded us that we waste every day, don ’ t add wasting our energy, intelligence and hard-earned experience to that count. I have come from so far to meet you like this. Please don ’ t waste me. By working together, we can become one of those small, rapidly-growing groups of individuals who actually have the audacity and courage to believe that we actually can change the world. We might have come to this conference from very, very different stations in life, but believe me, we all share one incredibly powerful thing. We have nothing to lose and everything to gain. Ciao, bellos!

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I can ’ t help but this wish : to think about when you ’ re a little kid, and all your friends ask you,”If a genie could give you one wish in the world, what would it be?”And I always answered,”Well, I ’ d want the wish to have the wisdom to know exactly what to wish for.”Well, then you ’ d be screwed, because you ’ d know what to wish for, and you ’ d use up your wish, and now, since we only have one wish — unlike last year they had three wishes — I ’ m not going to wish for that. So let ’ s get to what I would like, which is world peace. And I know what you ’ re thinking : You ’ re thinking,”The poor girl up there, she thinks she ’ s at a beauty pageant. She ’ s not. She ’ s at the TED Prize.”But I really do think it makes sense. And I think that the first step to world peace is for people to meet each other. I ’ ve met a lot of different people over the years, and I ’ ve filmed some of them, from a dotcom executive in New York who wanted to take over the world, to a military press officer in Qatar, who would rather not take over the world. If you ’ ve seen the film”Control Room”that was sent out, you ’ d understand a little bit why. Thank you. Wow! Some of you watched it. That ’ s great. That ’ s great. So basically what I ’ d like to talk about today is a way for people to travel, to meet people in a different way than — because you can ’ t travel all over the world at the same time. And a long time ago — well, about 40 years ago — my mom had an exchange student. And I ’ m going to show you slides of the exchange student. This is Donna. This is Donna at the Statue of Liberty. This is my mother and aunt teaching Donna how to ride a bike. This is Donna eating ice cream. And this is Donna teaching my aunt how to do a Filipino dance. I really think as the world is getting smaller, it becomes more and more important that we learn each other ’ s dance moves, that we meet each other, we get to know each other, we are able to figure out a way to cross borders, to understand each other, to understand people ’ s hopes and dreams, what makes them laugh and cry. And I know that we can ’ t all do exchange programs, and I can ’ t force everybody to travel ; I ’ ve already talked about that to Chris and Amy, and they said that there ’ s a problem with this : You can ’ t force people, free will. And I totally support that, so we ’ re not forcing people to travel. But I ’ d like to talk about another way to travel that doesn ’ t require a ship or an airplane, and just requires a movie camera, a projector and a screen. And that ’ s what I ’ m going to talk to you about today. I was asked that I speak a little bit about where I personally come from, and Cameron, I don ’ t know how you managed to get out of that one, but I think that building bridges is important to me because of where I come from. I ’ m the daughter of an American mother and an Egyptian-Lebanese-Syrian father. So I ’ m the living product of two cultures coming together. No pun intended. And I ’ ve also been called, as an Egyptian-Lebanese-Syrian American with a Persian name, the”Middle East Peace Crisis.”So maybe me starting to take pictures was some kind of way to bring both sides of my family together — a way to take the worlds with me, a way to tell stories visually. It all kind of started that way, but I think that I really realized the power of the image when I first went to the garbage-collecting village in Egypt, when I was about 16. My mother took me there. She ’ s somebody who believes strongly in community service, and decided that this was something that I needed to do. And so I went there and I met some amazing women there. There was a center there, where they were teaching people how to read and write, and get vaccinations against the many diseases you can get from sorting through garbage. And I began teaching there. I taught English, and I met some incredible women there. I met people that live seven people to a room, barely can afford their evening meal, yet lived with this strength of spirit and sense of humor and just incredible qualities. I got drawn into this community and I began to take pictures there. I took pictures of weddings and older family members — things that they wanted memories of. About two years after I started taking these pictures, the UN Conference on Population and Development asked me to show them at the conference. So I was 18 ; I was very excited. It was my first exhibit of photographs and they were all put up there, and after about two days, they all came down except for three. People were very upset, very angry that I was showing these dirty sides of Cairo, and why didn ’ t I cut the dead donkey out of the frame? And as I sat there, I got very depressed. I looked at this big empty wall with three lonely photographs that were, you know, very pretty photographs and I was like,”I failed at this.”But I was looking at this intense emotion and intense feeling that had come out of people just seeing these photographs. Here I was, this 18-year-old pipsqueak that nobody listened to, and all of a sudden, I put these photographs on the wall, and there were arguments, and they had to be taken down. And I saw the power of the image, and it was incredible. And I think the most important reaction that I saw there was actually from people that would never have gone to the garbage village themselves, that would never have seen that the human spirit could thrive in such difficult circumstances. And I think it was at that point that I decided I wanted to use photography and film to somehow bridge gaps, to bridge cultures, bring people together, cross borders. And so that ’ s what really kind of started me off. Did a stint at MTV, made a film called”Startup.com,”and I ’ ve done a couple of music films. But in 2003, when the war in Iraq was about to start, it was a very surreal feeling for me, because before the war started, there was kind of this media war that was going on. And I was watching television in New York, and there seemed to be just one point of view that was coming across, and the coverage went from the US State Department to embedded troops. And what was coming across on the news was that there was going to be this clean war and precision bombings, and the Iraqis would be greeting the Americans as liberators, and throwing flowers at their feet in the **streets** of Baghdad. And I knew that there was a completely other story that was taking place in the Middle East, where my parents were. I knew that there was a completely other story being told, and I was thinking,"How are people supposed to communicate with each other when they ' re getting completely different messages, and nobody knows what the other ' s being told? How are people supposed to have any kind of common understanding or know how to move together into the future? So I knew that I had to go there. I just wanted to be in the center. I had no plan. I had no funding. I didn ' t even have a camera at the time — I had somebody bring it there, because I wanted to get **access** to Al Jazeera, George Bush ' s favorite channel, and a place which I was very curious about because it ' s disliked by many governments across the Arab world, and also called the mouthpiece of Osama Bin Laden by some people in the US government. So I was thinking, this station that ' s hated by so many people has to be doing something right. I ' ve got to go see what this is all about. And I also wanted to go see Central Command, which was 10 minutes away. And that way, I could get **access** to how this news was being created — on the Arab side, reaching the Arab world, and on the US and Western side, reaching the US. And when I went there and sat there, and met these people that were in the center of it, and sat with these characters, I met some surprising, very complex people. And I ' d like to share with you a little bit of that experience of when you sit with somebody and you film them, and you listen to them, and you allow them more than a five-second sound bite. The amazing complexity of people emerges. Samir Khader : Business as usual. Iraq, and then Iraq, and then Iraq. But between us, if I ' m offered a job with Fox, I ' ll take it. To change the Arab nightmare into the American dream. I still have that dream. Maybe I will never be able to do it, but I have plans for my children. When they finish high school, I will send them to America to study there. I will pay for their study. And they will stay there. Josh Rushing : The night they showed the POWs and the dead soldiers — Al Jazeera showed them — it was powerful, because America doesn ' t show those kinds of images. Most of the news in America won ' t show really gory images and this showed American soldiers in uniform, strewn about a floor, a cold tile floor. And it was revolting. It was absolutely revolting. It made me sick at my stomach. And then what hit me was, the night before, there had been some kind of bombing in Basra, and Al Jazeera had shown images of the people. And they were equally, if not more, horrifying — the images were. And I remember having seen it in the Al Jazeera office, and thought to myself,"Wow, that ' s gross. That ' s bad."And then going away, and probably eating dinner or something. And it didn ' t affect me as much. So, the impact that had on me — me realizing that I just saw people on the other side, and those people in the Al Jazeera office must have felt the way I was feeling that night, and it upset me on a profound level that I wasn ' t as bothered as much the night before. It makes me hate war. But it doesn ' t make me believe that we ' re in a world that can live without war yet. Jehane Noujaim : I was overwhelmed by the response of the film. We didn ' t know whether it would be able to get out there. We had no funding for it. We were incredibly lucky that it got picked up. And when we showed the film in both the United States and the Arab world, we had such incredible reactions. It was amazing to see how people were moved by this film. In the Arab world — and it ' s not really by the film, it ' s by the characters — I mean, Josh Rushing was this incredibly complex person who was thinking about things. And when I showed the film in the Middle East, people wanted to meet Josh. He kind of redefined us as an American population. People started to ask me,"Where is this guy now?"Al Jazeera offered him a job. And Samir, on the other hand, was also quite an interesting character for the Arab world to see, because it brought out the complexities of this love-hate relationship that the Arab world has with the West. In the United States, I was blown away by the motivations, the positive motivations of the American people when they ’ d see this film. You know, we ’ re criticized abroad for believing we ’ re the saviors of the world in some way, but the flip side of it is that, actually, when people do see what is happening abroad and people ’ s reactions to some of our policy abroad, we feel this power, that we need to — we feel like we have to get the power to change things. And I saw this with audiences. This woman came up to me after the screening and said,”You know, I know this is crazy. I saw the bombs being loaded on the planes, I saw the military going out to war, but you don ’ t understand people ’ s anger towards us until you see the people in the hospitals and the **victims** of the war, and how do we get out of this bubble? How do we understand what the other person is thinking?”Now, I don ’ t know whether a film can change the world. But I know the power of it, I know that it starts people thinking about how to change the world. Now, I ’ m not a philosopher, so I feel like I shouldn ’ t go into great depth on this, but let film speak for itself and take you to this other world. Because I believe that film has the ability to take you across borders, I ’ d like you to just sit back and experience for a couple of minutes being taken into another world. And these couple clips take you inside of two of the most difficult conflicts that we ’ re faced with today. [The last 48 hours of two Palestinian suicide bombers.] [Paradise Now] [Man : As long as there is injustice, someone must make a sacrifice!] [Woman : That ’ s no sacrifice, that ’ s revenge!] [If you kill, there ’ s no difference between **victim** and occupier.] [Man : If we had airplanes, we wouldn ’ t need martyrs, that ’ s the difference.] [Woman : The difference is that the Israeli military is still stronger.] [Man : Then let us be **equal** in death.] [We still have Paradise.] [Woman : There is no Paradise! It only exists in your head!] [Man : God forbid!] [May God forgive you.] [If you were not Abu Azzam ’ s daughter...] [Anyway, I ’ d rather have Paradise in my head than live in this hell!] [In this life, we ’ re dead anyway.] [One only chooses bitterness when the alternative is even **bitterer**.] [Woman : And what about us? The ones who remain?] [Will we win that way?] [Don ’ t you see what you ’ re doing is destroying us?] [And that you give Israel an alibi to carry on?] [Man : So with no alibi, Israel will stop?] [Woman : Perhaps. We have to turn it into a moral war.] [Man : How, if Israel has no morals?] [Woman : Be careful!] [And the real people building peace through non-violence] [Encounter Point] Video : [Tel Aviv, Israel 1996] [Tzvika : My wife Ayelet called me and said,] [”There was a suicide bombing in Tel Aviv.”] [Ayelet : What do you know about the casualties?] [Tzvika off-screen : We ’ re looking for three girls.] [We have no information.] [Ayelet : One is wounded here, but we haven ’ t heard from the other three.] [Tzvika : I said,”OK, that ’ s Bat-Chen, that ’ s my daughter.] [Are you sure she is dead?”] [They said yes.] Video : [Bethlehem, Occupied Palestinian Territories, 2003] [George : On that day, at around 6:30] [I was driving with my wife and daughters to the supermarket.] [When we got to here...] [we saw three Israeli military jeeps parked on the side of the road.] [When we passed by the first jeep...] [they opened fire on us.] [And my 12-year-old daughter Christine] [was killed in the shooting.] [Bereaved Families Forum, Jerusalem] [Tzvika : I ’ m the headmaster for all parts.] [George : But there is a teacher that is in charge?] [Tzvika : Yes, I have assistants.] [I deal with children all the time.] [One year after their daughters ’ deaths both Tzvika and George join the forum] [George : At first, I thought it was a strange idea.] [But after thinking logically about it,] [I didn ’ t find any reason why not to meet them] [and let them know of our **suffering**.] [Tzvika : There were many things that touched me.] [We see that there are Palestinians who suffered a lot, who lost children,] [and still believe in the peace process and in reconciliation.]

[If we who lost what is most precious can talk to each other,] [and look forward to a better future,] [then everyone else must do so, too.] [From South Africa : A Revolution Through Music] [Amandla] Man : Song is something that we communicated with people who otherwise would not have understood where we ' re coming from. You could give them a long political speech, they would still not understand. But I tell you, when you finish that song, people will be like,"Damn, I know where you niggas are coming from. I know where you guys are coming from. Death unto apartheid!"Narrator : It ' s about the liberation struggle. It ' s about those children who took to the **streets** — fighting, screaming,"Free Nelson Mandela!"It ' s about those unions who put down their tools and demanded freedom. Yes. Yes! Freedom! Jehane Noujaim : I think everybody ' s had that feeling of sitting in a theater, in a dark room, with other strangers, watching a very powerful film, and they felt that feeling of transformation. And what I ' d like to talk about is how can we use that feeling to actually create a movement through film? I ' ve been listening to the talks in the conference, and Robert Wright said yesterday that if we have an appreciation for another person ' s humanity, then they will have an appreciation for ours. And that ' s what this is about. It ' s about connecting people through film, getting these independent voices out there. Now, Josh Rushing actually ended up leaving the military and taking a job with Al Jazeera. So his feeling is that he ' s at Al Jazeera International because he feels like he can actually use media to bridge the gap between East and West. And that ' s an amazing thing. But I ' ve been trying to think about ways to give power to these independent voices, to give power to the filmmakers, to give power to people who are trying to use film for change. And there are incredible organizations that are out there doing this already. There ' s Witness, that you heard from earlier. There ' s Just Vision, that are working with Palestinians and Israelis who are working together for peace, and documenting that process and getting interviews out there and using this film to take to Congress to show that it ' s a powerful tool, to show that this is a woman who ' s had her daughter killed in an attack, and she believes that there are peaceful ways to solve this. There ' s Working Films and there ' s Current TV, which is an incredible platform for people around the world to be able to put their — Yeah, it ' s amazing. I ' ve watched it and I ' m blown away by it and its potential to bring voices from around the world — independent voices from around the world — and create a truly democratic, global television. So what can we do to create a platform for these organizations, to create some momentum, to get everybody in the world involved in this movement? I ' d like for us to imagine for a second. Imagine a day when you have everyone coming together from around the world. You have towns and villages and theaters — all from around the world, getting together, and sitting in the dark, and sharing a communal experience of watching a film, or a couple of films, together. Watching a film which maybe highlights a character that is fighting to live, or just a character that defies stereotypes, makes a joke, **sings** a song. Comedies, **documentaries**, shorts. This amazing power can be used to change people and to bond people together ; to cross borders, and have people feel like they ' re having a communal experience. So if you imagine this day when all around the world, you have theaters and places where we project films. If you imagine projecting from Times Square to Tahrir Square in Cairo, the same film in Ramallah, the same film in Jerusalem. You know, we ' ve been talking to a friend of mine about using the side of the Great Pyramid and the Great Wall of China. It ' s endless what you can imagine, in terms of where you can project films and where you can have this communal experience. And I believe that this one day, if we can create it, this one day can create momentum for all of these independent voices. There isn ' t an organization which is connecting the independent voices of the world to get out there, and yet I ’ m hearing throughout this conference that the biggest challenge in our future is understanding the other, and having mutual respect for the other and crossing borders. And if film can do that, and if we can get all of these different locations in the world to watch these films together — this could be an incredible day. So we ’ ve already made a partnership, set up through somebody from the TED community, John Camen, who introduced me to Steven Apkon, from the Jacob Burns Film Center. And we started calling up everybody. And in the last week, there have been so many people that have responded to us, from as close as Palo Alto, to Mongolia and to India. There are people that want to be a part of this global day of film ; to be able to provide a platform for independent voices and independent films to get out there. Now, we ’ ve thought about a name for this day, and I ’ d like to share this with you. Now, the most amazing part of this whole process has been sharing ideas and wishes, and so I invite you to give brainstorms onto how does this day echo into the future? How do we use technology to make this day echo into the future, so that we can build community and have these communities working together, through the Internet? There was a time, many, many years ago, when all of the continents were stuck together. And we call that landmass Pangea. So what we ’ d like to call this day of film is Pangea Cinema Day. And if you just imagine that all of these people in these towns would be watching, then I think that we can actually really make a movement towards people understanding each other better. I know that it ’ s very intangible, touching people ’ s hearts and souls, but the only way that I know how to do it, the only way that I know how to reach out to somebody ’ s heart and soul all across the world, is by showing them a film. And I know that there are independent filmmakers and films out there that can really make this happen. And that ’ s my wish. I guess I ’ m supposed to give you my one-sentence wish, but we ’ re way out of time. Chris Anderson : That is an incredible wish. Pangea Cinema : The day the world comes together. JN : It ’ s more tangible than world peace, and it ’ s certainly more immediate. But it would be the day that the world comes together through film, the power of film. CA : Ladies and gentlemen, Jehane Noujaim.

Text Sample 405: POS Dim 3 – 2006 – Score 6.00 – 2006/t000492.txt

About 10 years ago, I took on the task to teach global development to Swedish undergraduate students. That was after having spent about 20 years, together with African institutions, studying hunger in Africa. So I was sort of expected to know a little about the world. And I started, in our medical university, Karolinska Institute, an undergraduate course called Global Health. But when you get that opportunity, you get a little nervous. I thought, these students coming to us actually have the highest grade you can get in the Swedish college system, so I thought, maybe they know everything I ’ m going to teach them about. So I did a pretest when they came. And one of the questions from which I learned a lot was this one :”Which country has the highest child mortality of these five pairs?”And I put them together so that in each pair of countries, one has twice the child mortality of the other. And this means that it ’ s much bigger, the difference, than the uncertainty of the data. I won ’ t put you at a test here, but it ’ s Turkey, which is highest there, Poland, Russia, Pakistan and South Africa. And these were the results of the Swedish students. I did it so I got the confidence interval, which is pretty narrow. And I got happy, of course — a 1. 8 right answer out of five possible. That means there was a place for a professor of international health and for my course. But one late night, when I was compiling the report, I really realized my discovery. I have shown that Swedish top students know, statistically, significantly less about the world than the chimpanzees. Because the chimpanzee would score half right if I gave them two bananas with Sri Lanka and Turkey. They would be right half of the cases. But the students are not there. The problem for me was not ignorance ; it was preconceived ideas. I did also an unethical study of the professors of the Karolinska Institute, which hands out the Nobel Prize in Medicine, and they are on par with the chimpanzee there. This is where I realized that there was really a need to communicate, because the data of what ' s happening in the world and the child health of every country is very well aware. So we did this software, which displays it like this. Every bubble here is a country. This country over here is China. This is India. The size of the bubble is the population, and on this axis here, I put fertility rate. Because my students, what they said when they looked upon the world, and I asked them,"What do you really think about the world?"Well, I first discovered that the textbook was Tintin, mainly. And they said,"The world is still ' we ' and ' them. ' And ' we ' is the Western world and ' them ' is the Third World.""And what do you mean with ' Western world? '"I said."Well, that ' s long life and small family. And ' Third World ' is short life and large family."So this is what I could display here. I put fertility rate here — number of children per woman : one, two, three, four, up to about eight children per woman. We have very good data since 1962, 1960, about, on the size of families in all countries. The error margin is narrow. Here, I put life expectancy at birth, from 30 years in some countries, up to about 70 years. And in 1962, there was really a group of countries here that were industrialized countries, and they had small families and long lives. And these were the developing countries. They had large families and they had relatively short lives. Now, what has happened since 1962? We want to see the change. Are the students right? It ' s still two types of countries? Or have these developing countries got smaller families and they live here? Or have they got longer lives and live up there? Let ' s see. We stopped the world then. This is all UN statistics that have been available. Here we go. Can you see there? It ' s China there, moving against better health there, improving there. All the green Latin American countries are moving towards smaller families. Your yellow ones here are the Arabic countries, and they get longer life, but not larger families. The Africans are the green here. They still remain here. This is India ; Indonesia is moving on pretty fast. In the ' 80s here, you have Bangladesh still among the African countries. But now, Bangladesh — it ' s a miracle that happens in the ' 80s — the imams start to promote family planning, and they move up into that corner. And in the ' 90s, we have the terrible HIV epidemic that takes down the life expectancy of the African countries. And the rest of them all move up into the corner, where we have long lives and small family, and we have a completely new world. Let me make a comparison directly between the United States of America and Vietnam. 1964 : America had small families and long life ; Vietnam had large families and short lives. And this is what happens. The data during the war indicate that even with all the death, there was an improvement of life expectancy. By the end of the year, family planning started in Vietnam, and they went for smaller families. And the United States up there is getting longer life, keeping family size. And in the ' 80s now, they give up Communist planning and they go for market economy, and it moves faster even than social life. And today, we have in Vietnam the same life expectancy and the same family size here in Vietnam, 2003, as in United States, 1974, by the end of the war. I think we all, if we don ' t look at the data, we underestimate the tremendous change in Asia, which was in social change before we saw the economic change. So let ' s move over to another way here in which we could display the distribution in the world of income. This is the world distribution of income of people. One dollar, 10 dollars or 100 dollars per day. There ' s no gap between rich and poor any longer. This is a myth. There ' s a little hump here. But there are people all the way. And if we look where the income ends up, this is 100 percent of the world ' s annual income. And the richest 20 percent, they take out of that about 74 percent. And the poorest 20 percent, they take about two percent. And this shows that the concept of developing countries is extremely doubtful. We think about aid, like these people here giving aid to these people here. But in the middle, we have most of the world population, and they have now 24 percent of the income. We heard it in other forms. And who are these? Where are the different countries? I can show you Africa. This is Africa. Ten percent of the world population, most in poverty. This is OECD — the rich countries, the country club of the UN. And they are over here on this side. Quite an overlap between Africa and OECD. And this is Latin America. It has everything on this earth, from the poorest to the richest in Latin America. And on top of that, we can put East Europe, we can put East Asia, and we put South Asia. And what did it look like if we go back in time, to about 1970? Then, there was more of a hump. And most who lived in absolute poverty were Asians. The problem in the world was the poverty in Asia. And if I now let the world move forward, you will see that while population increases, there are hundreds of millions in Asia getting out of poverty, and some others getting into poverty, and this is the pattern we have today. And the best projection from the World Bank is that this will happen, and we will not have a divided world. We ' ll have most people in the middle. Of course it ' s a logarithmic scale here, but our concept of economy is growth with percent. We look upon it as a possibility of percentile increase. If I change this and take GDP per capita instead of family income, and I turn these individual data into regional data of gross **domestic** product, and I take the regions down here, the size of the bubble is still the population. And you have the OECD there, and you have sub-Saharan Africa there, and we take off the Arab states there, coming both from Africa and from Asia, and we put them separately, and we can expand this axis, and I can give it a new dimension here, by adding the social values there, child survival. Now I have money on that axis, and I have the possibility of children to survive there. In some countries, 99. 7 % of children survive to five years of age ; others, only 70. And here, it seems, there is a gap between OECD, Latin America, East Europe, East Asia, Arab states, South Asia and sub-Saharan Africa. The linearity is very strong between child survival and money. But let me split sub-Saharan Africa. Health is there and better health is up there. I can go here, and I can split sub-Saharan Africa into its countries. And when it bursts, the size of each country bubble is the size of the population. Sierra Leone down there, Mauritius is up there. Mauritius was the first country to get away with trade barriers, and they could sell their sugar, they could sell their textiles, on **equal** terms as the people in Europe and North America. There ' s a huge difference [within] Africa. And Ghana is here in the middle. In Sierra Leone, humanitarian aid. Here in Uganda, development aid. Here, time to invest ; there, you can go for a holiday. There ' s tremendous variation within Africa, which we very often make that it ' s **equal** everything. I can split South Asia here. India ' s the big bubble in the middle. But there ' s a huge difference between Afghanistan and Sri Lanka. I can split Arab states. How are they? Same climate, same culture, same religion — huge difference. Even between neighbors — Yemen, civil war ; United Arab Emirates, money, which was quite equally and well-used. Not as the myth is. And that includes all the children of the foreign workers who are in the country. Data is often better than you think. Many people say data is bad. There is an uncertainty margin, but we can see the difference here :

Cambodia, Singapore. The differences are much bigger than the weakness of the data. East Europe : Soviet economy for a long time, but they come out after 10 years very, very differently. And there is Latin America. Today, we don ’ t have to go to Cuba to find a healthy country in Latin America. Chile will have a lower child mortality than Cuba within some few years from now. Here, we have high-income countries in the OECD. And we get the whole pattern here of the world, which is more or less like this. And if we look at it, how the world looks, in 1960, it starts to move. This is Mao Zedong. He brought health to China. And then he died. And then Deng Xiaoping came and brought money to China, and brought them into the mainstream again. And we have seen how countries move in different directions like this, so it ’ s sort of difficult to get an example country which shows the pattern of the world. But I would like to bring you back to about here, at 1960. I would like to compare South Korea, which is this one, with Brazil, which is this one. The label went away for me here. And I would like to compare Uganda, which is there. I can run it forward, like this. And you can see how South Korea is making a very, very fast advancement, whereas Brazil is much slower. And if we move back again, here, and we put trails on them, like this, you can see again that the speed of development is very, very different, and the countries are moving more or less at the same rate as money and health, but it seems you can move much faster if you are healthy first than if you are wealthy first. And to show that, you can put on the way of United Arab Emirates. They came from here, a mineral country. They cached all the oil ; they got all the money ; but health cannot be bought at the supermarket. You have to invest in health. You have to get kids into schooling. You have to train health staff. You have to educate the population. And Sheikh Zayed did that in a fairly good way. In spite of falling oil prices, he brought this country up here. So we ’ ve got a much more mainstream appearance of the world, where all countries tend to use their money better than they used it in the past. Now, this is, more or less, if you look at the average data of the countries — they are like this. That ’ s dangerous, to use average data, because there is such a lot of difference within countries. So if I go and look here, we can see that Uganda today is where South Korea was in 1960. If I split Uganda, there ’ s quite a difference within Uganda. These are the quintiles of Uganda. The richest 20 percent of Ugandans are there. The poorest are down there. If I split South Africa, it ’ s like this. And if I go down and look at Niger, where there was such a terrible famine [recently], it ’ s like this. The 20 percent poorest of Niger is out here, and the 20 percent richest of South Africa is there, and yet we tend to discuss what solutions there should be in Africa. Everything in this world exists in Africa. And you can ’ t discuss universal **access** to HIV [treatment] for that quintile up here with the same strategy as down here. The improvement of the world must be highly contextualized, and it ’ s not relevant to have it on a regional level. We must be much more detailed. We find that students get very excited when they can use this. And even more, policy makers and the corporate sectors would like to see how the world is changing. Now, why doesn ’ t this take place? Why are we not using the data we have? We have data in the United Nations, in the national statistical agencies and in universities and other nongovernmental organizations. Because the data is hidden down in the databases. And the public is there, and the internet is there, but we have still not used it effectively. All that information we saw changing in the world does not include publicly funded statistics. There are some web pages like this, you know, but they take some nourishment down from the databases, but people put prices on them, stupid passwords and boring statistics. And this won ’ t work. So what is needed? We have the databases. It ’ s not a new database that you need. We have wonderful design tools and more and more are added up here. So we started a **nonprofit** venture linking data to design, we called”Gapminder,”from the London Underground, where they warn you,”Mind the gap.”So we thought Gapminder was appropriate. And we started to write software which could link the data like this. And it wasn ’ t that difficult. It took some person years, and we have produced animations. You can take a data set and put it there. We are liberating UN data, some few UN organization. Some countries accept that their databases can go out on the world. But what we really need is, of course, a search function, a search function where we can copy the data up to a searchable format and get it out in the world. And what do we hear when we go around? I ’ ve done anthropology on the main statistical units. Everyone says,”It ’ s impossible. This can ’ t be done. Our information is so peculiar in detail, so that cannot be searched as others can be searched. We cannot give the data free to the students, free to the entrepreneurs of the world.”But this is what we would like to see, isn ’ t it? The publicly funded data is down here. And we would like flowers to grow out on the net. One of the crucial points is to make them searchable, and then people can use the different design tools to animate it there. And I have pretty good news for you. I have good news that the [current], new head of UN statistics doesn ’ t say it ’ s impossible. He only says,”We can ’ t do it.”And that ’ s a quite clever guy, huh? So we can see a lot happening in data in the coming years. We will be able to look at income distributions in completely new ways. This is the income distribution of China, 1970. This is the income distribution of the United States, 1970. Almost no overlap. Almost no overlap. And what has happened? What has happened is this : that China is growing, it ’ s not so **equal** any longer, and it ’ s appearing here, overlooking the United States, almost like a ghost, isn ’ t it? It ’ s pretty scary. But I think it ’ s very important to have all this information. We need really to see it. And instead of looking at this, I would like to end up by showing the internet users per 1, 000. In this software, we **access** about 500 variables from all the countries quite easily. It takes some time to change for this, but on the axes, you can quite easily get any variable you would like to have. And the thing would be to get up the databases free, to get them searchable, and with a second click, to get them into the graphic formats, where you can instantly understand them. Now, statisticians don ’ t like it, because they say that this will not show the reality ; we have to have statistical, analytical methods. But this is hypothesis-generating. I end now with the world. There, the internet is coming. The number of internet users are going up like this. This is the GDP per capita. And it ’ s a new technology coming in, but then amazingly, how well it fits to the economy of the countries. That ’ s why the $100 computer will be so important. But it ’ s a nice tendency. It ’ s as if the world is flattening off, isn ’ t it? These countries are lifting more than the economy, and it will be very interesting to follow this over the year, as I would like you to be able to do with all the publicly funded data. Thank you very much.

Text Sample 406: POS Dim 3 – 2004 – Score 8.00 – 2004/t000182.txt

This session is on natural wonders, and the bigger conference is on the pursuit of happiness. I want to try to combine them all, because to me, healing is really the ultimate natural wonder. Your body has a remarkable capacity to begin **healing** itself, and much more quickly than people had once realized, if you simply stop doing what—s causing the problem. And so, really, so much of what we do in medicine and life in general is focused on mopping up the floor without also turning off the faucet. I love doing this work, because it really gives many people new hope and new choices that they didn—t have before, and it allows us to talk about things that — not just diet, but that happiness is not — we ’ re talking about the pursuit of happiness, but when you really look at all the spiritual traditions, what Aldous Huxley called the”perennial wisdom,”when you get past the named and forms and rituals that really divide people, it—s really about — our nature is to be happy ; our nature is to be peaceful, our nature is to be healthy. And so it—s not something — happiness is not something you get, health is generally not something that you get. But rather all of these different practices — you know, the ancient swamis and rabbis and priests and monks and nuns didn—t develop these techniques to just manage stress or lower your blood pressure, unclog your arteries, even though it can do all those things. They—re powerful tools for transformation, for quieting down our mind and bodies to allow us to experience what it feels like to be happy, to be peaceful, to be joyful and to realize that it—s not something that you pursue and get, but rather it—s something that you have already until you disturb it. I studied yoga for many years with a teacher named Swami Satchidananda and people would say,”What are you, a Hindu?”He—d say,”No, I—m an undo.”And it—s really about identifying what—s causing us to disturb our innate health and happiness, and then to allow that natural healing to occur. To me, that—s the real natural wonder. So, within that larger context, we can talk about diet, stress management — which are really these spiritual practices — moderate exercise, smoking cessation, support groups and community — which I—ll talk more about — and some vitamins and supplements. And it—s not a diet. You know, when most people think about the diet I recommend, they think it—s a really strict diet. For reversing disease, that—s what it takes, but if you—re just trying to be healthy, you have a spectrum of choices. And to the degree that you can move in a healthy direction, you—re going to live longer, you—re going to feel better, you—re going to lose weight, and so on. And in our studies, what we—ve been able to do is to use very expensive, high-tech, state-of-the-art measures to prove how powerful these very simple and low-tech and low-cost — and in many ways, ancient — interventions, can be. We first began by looking at heart disease, and when I began doing this work 26 or 27 years ago, it was thought that once you have heart disease it can only get worse. And what we found was that, instead of getting worse and worse, in many cases it could get better and better, and much more quickly than people had once realized. This is a representative patient who at the time was 73 — totally needed to have a bypass, decided to do this instead. We used quantitative arteriography, showing the narrowing. This is one of the arteries that feed the heart, one of the main arteries, and you can see the narrowing here. A year later, it—s not as clogged ; normally, it goes the other direction. These minor changes in blockages caused a 300 percent improvement in blood flow, and using cardiac positron emission tomography, or”PET,”scans, blue and black is no blood flow, orange and white is maximal. Huge differences can occur without drugs, without surgery. Clinically, he literally couldn—t walk across the **street** without getting severe chest pain ; within a month, like most people, was pain-free, and within a year, climbing more than 100 floors a day on a Stairmaster. This is not unusual, and it—s part of what enables people to maintain these kinds of changes, because it makes such a big difference in their quality of life. Overall, if you looked at all the arteries in all the patients, they got worse and worse, from one year to five years, in the comparison group. This is the natural history of heart disease, but it—s really not natural because we found it could get better and better, and much more quickly than people had once thought. We also found that the more people change, the better they got. It wasn—t a function of how old or how sick they were — it was mainly how much they changed, and the oldest patients improved as much as the young ones. I got this as a Christmas card a few years ago from two of the patients in one of our programs. The younger brother is 86, the older one—s 95 ; they wanted to show me how much more flexible they were. And the following year they sent me this one, which I thought was kind of funny. You just never know. And what we found was that 99 percent of the patients start to reverse the progression of their heart disease. Now I thought, you know, if we just did good science, that would change medical practice. But, that was a little naive. It—s important, but not enough. Because we doctors do what we get paid to do, and we get trained to do what we get paid to do, so if we change insurance, then we change medical practice and medical education. Insurance will cover the bypass, it—ll cover the angioplasty ; it won—t, until recently, cover diet and lifestyle. So, we began through our **nonprofit** institute ' s training hospitals around the country, and we found that most people could avoid surgery, and not only was it medically effective, it was also cost effective. And the insurance companies found that they began to save almost 30, 000 dollars per patient, and Medicare is now in the middle of doing a demonstration project where they—re paying for 1, 800 people to go through the program on the sites that we train. The fortuneteller says,"I give smokers a discount because there—s not as much to tell."I like this slide, because it—s a chance to talk about what really motivates people to change, and what doesn—t. And what doesn—t work is fear of dying, and that—s what—s normally used. Everybody who smokes knows it—s not good for you, and still 30 percent of Americans smoke — 80 percent in some parts of the world. Why do people do it? Well, because it helps them get through the day. And I—ll talk more about this, but the real epidemic isn—t just heart disease or obesity or smoking — it—s loneliness and depression. As one woman said,"I—ve got 20 friends in this package of cigarettes, and they—re always there for me and nobody else is. You—re going to take away my 20 friends? What are you going to give me?"Or they eat when they get depressed, or they use alcohol to **numb** the pain, or they work too hard, or watch too much TV. There are lots of ways we have of avoiding and **numbing** and bypassing pain, but the point of all of this is to deal with the cause of the problem. And the pain is not the problem : it—s the symptom. And telling people they—re going to die is too scary to think about, or, they—re going to get emphysema or heart attack is too scary, and so they don—t want to think about it, so they don—t. The most effective anti-smoking ad was this one. You—ll notice the limp cigarette hanging out of his mouth, and"impotence"— the headline is,"Impotent"— it—s not emphysema. What was the biggest selling drug of all time when it was introduced a few years ago? Viagra, right? Why? Because a lot of guys need it. It—s not like you say,"Hey Joe, I—m having erectile dysfunction, how about you?"And yet, look at the number of prescriptions that are being sold. It—s not so much psychological, it—s vascular, and nicotine makes your arteries constrict. So does cocaine, so does a high fat diet, so does emotional stress. So the very behaviors that we think of as being so sexy in our culture are the very ones that leave so many people feeling tired, lethargic, depressed and impotent, and that—s not much fun. But when you change those behaviors, your brain gets more blood, you think more clearly, you have more energy, your heart gets more blood in ways I—ve shown you. Your **sexual** function improves. And these things occur within hours. This is a study : a high fat meal, and within one or two hours blood-flow is measurably less — and you—ve all experienced this at Thanksgiving. When you eat a big fatty meal, how do you feel? You feel kind of sleepy afterwards. On a low-fat meal, the blood flow doesn—t go down — it even goes up. Many of you have kids, and you know that—s a big change in your lifestyle, and so people are not afraid to make big changes in lifestyle if they—re worth it.

And the paradox is that when you make big changes, you get big benefits, and you feel so much better so quickly. For many people, those are choices worth making — not to live longer, but to live better. I want to talk a little bit about the obesity epidemic, because it really is a problem. Two-thirds of adults are overweight or obese, and diabetes in kids and 30-year-olds has increased 70 percent in the last 10 years. It—s no joke : it—s real. And just to show you this, this is from the CDC. These are not election returns ; these are the percentage of people who are overweight. And if you see from ' 85 to ' 86 to ' 87, ' 88, ' 89, ' 90, ' 91 — you get a new category, 15 to 20 percent ; ' 92, ' 93, ' 94, ' 95, ' 96, ' 97 — you get a new category ; ' 98, ' 99, 2000, and 2001. Mississippi, more than 25 percent of people are overweight. Why is this? Well, this is one way to lose weight that works very well... but it doesn—t last, which is the problem. Now, there—s no mystery in how you lose weight ; you either burn more calories by exercise or you eat fewer calories. Now, one way to eat fewer calories is to eat less food, which is why you can lose weight on any diet if you eat less food, or if you restrict entire categories of foods. But the problem is, you get hungry, so it—s hard to keep it off. The other way is to change the type of food. And fat has nine calories per gram, whereas protein and carbs only have four. So, when you eat less fat, you eat fewer calories without having to eat less food. So you can eat the same amount of food, but you—ll be getting fewer calories because the food is less dense in calories. And it—s the volume of food that affects satiety, rather than the type of food. You know, I don—t like talking about the Atkins diet, but I get asked about it every day, and so I just thought I—d spend a few minutes on that. The myth that you hear about is, Americans have been told to eat less fat, the percent of calories from fat is down, Americans are fatter than ever, therefore fat doesn—t make you fat. It—s a half-truth. Actually, Americans are eating more fat than ever, and even more carbs. And so the percentage is lower, the actual amount is higher, and so the goal is to reduce both. Dr. Atkins and I debated each other many times before he died, and we agreed that Americans eat too many simple carbs, the"bad carbs,"and these are things like — — sugar, white flour, white rice, alcohol. And you get a double whammy : you get all these calories that don—t fill you up because you—ve removed the fiber, and they get absorbed quickly so your blood sugar zooms up. Your pancreas makes insulin to bring it back down, which is good. But insulin accelerates the conversion of calories into fat. So, the goal is not to go to pork rinds and bacon and sausages — these are not health foods — but to go from"bad carbs"to what are called"good carbs."And these are things like whole foods, or unrefined carbs : fruits, vegetables, whole wheat flour, brown rice, in their natural forms, are rich in fiber. And the fiber fills you up before you get too many calories, and it slows the absorption so you don—t get that rapid rise in blood sugar. So, and you get all the disease-protective substances. It—s not just what you exclude from your diet, but also what you include that—s protective. Just as all carbs are not bad for you, all fats are not bad for you. There are good fats. And these are predominantly what are called the Omega-3 fatty acids. You find these, for example, in fish oil. And the bad fats are things like trans-fatty acids and processed food and saturated fats, which we find in meat. If you don—t remember anything else from this talk, three grams a day of fish oil can reduce your risk of a heart attack and sudden death by 50 to 80 percent. Three grams a day. They come in one-gram capsules ; more than that just gives you extra fat you don—t need. It also helps reduce the risk of the most common cancers like breast, prostate and colon cancer. Now, the problem with the Atkins diet, everybody knows people who have lost weight on it, but you can lose weight on amphetamines, you know, and fen-phen. I mean, there are lots of ways of losing weight that aren—t good for you. You want to lose weight in a way that enhances your health rather than the one that **harms** it. And the problem is that it—s based on this half-truth, which is that Americans eat too many simple carbs, so if you eat fewer simple carbs you—re going to lose weight. You—ll lose even more weight if you go to whole foods and less fat, and you—ll enhance your health rather than **harming** it. He says,"I—ve got some good news. While your cholesterol level has remained the same, the research findings have changed."Now, what happens to your heart when you go on an Atkins diet? The red is good at the beginning, and a year later — this is from a study done in a peer- reviewed journal called Angiology — there—s more red after a year on a diet like I would recommend, there—s less red, less blood flow after a year on an Atkins-type diet. So, yes, you can lose weight, but your heart isn—t happy. Now, one of the studies funded by the Atkins Center found that 70 percent of the people were constipated, 65 percent had bad breath, 54 percent had headaches — this is not a healthy way to eat. And so, you might start to lose weight and start to attract people towards you, but when they get too close it—s going to be a problem. And more seriously, there are case reports now of 16-year-old girls who died after a few weeks on the Atkins diet — of bone disease, kidney disease, and so on. And that—s how your body excretes waste, is through your breath, your bowels and your perspiration. So when you go on these kinds of diet, they begin to smell bad. So, an optimal diet is low in fat, low in the bad carbs, high in the good carbs and enough of the good fats. And then, again, it—s a spectrum : when you move in this direction, you—re going to lose weight, you—re going to feel better and you—re going to gain health. Now, there are ecological reasons for eating lower on the food chain too, whether it—s the deforestation of the Amazon, or making more protein available, to the four billion people who live on a dollar a day — not to mention whatever ethical concerns people have. So, there are lots of reasons for eating this way that go beyond just your health. Now, we—re about to publish the first study looking at the effects of this program on prostate cancer, and, in collaboration with Sloane-Kettering and with UCSF. We took 90 men who had biopsy-proven prostate cancer and who had elected, for reasons unrelated to the study, not to have surgery. We could randomly divide them into two groups, and then we could have one group that is a non-intervention control group to compare to, which we can—t do with, say, breast cancer, because everyone gets treated. What we found was that, after a year, none of the experimental group patients who made these lifestyle changes needed treatment, whereas six of the control- group patients needed surgery or radiation. When we looked at their PSA levels — which is a marker for prostate cancer — they got worse in the control group, but they actually got better in the experimental group, and these differences were highly significant. And then I wondered : was there any relationship between how much people changed their diet and lifestyle — whichever group they were in — and the changes in PSA? And sure enough, we found a dose-response relationship, just like we found in the arterial blockages in our cardiac studies. And in order for the PSA to go down, they had to make pretty big changes. I then wondered, well, maybe they—re just changing their PSA, but it—s not really affecting the tumor growth. So we took some of their blood serum and sent it down to UCLA ; they added it to a standard line of prostate tumor cells growing in tissue culture, and it inhibited the growth seven times more in the experimental group than in the control group — 70 versus 9 percent. And finally, I said, I wonder if there—s any relationship between how much people change and how it inhibited their tumor growth, whichever group they happened to be in. And this really got me excited because again, we found the same pattern : the more people change, the more it affected the growth of their tumors. And finally, we did MRI and MR spectroscopy scans on some of these patients, and the tumor activity is shown in red in this patient, and you can see clearly it—s better a year later, along with the PSA going down. So, if it—s true for prostate cancer, it—ll almost certainly be true for breast cancer as well. And whether or not you have conventional treatment, in addition, if you make these changes, it may help reduce the risk of recurrence. The last thing I want to talk about, apropos of the issue of the pursuit of happiness, is that study after study have shown that people who are lonely and depressed — and depression is the other real epidemic in our culture — are many times more likely to get sick and die prematurely, in part because, as we talked about, they—re more likely to smoke and overeat and drink too much and work too hard and so on. But also, through mechanisms that we don—t fully understand, people who are lonely and depressed are many times — three to five to ten times, in some studies — more likely to get sick and die prematurely. And depression is treatable. We need to do something about that. Now, on the other hand, anything that promotes intimacy is **healing**. It can be **sexual** intimacy — I happen to think that healing energy and erotic energy are just different forms of the same thing. Friendship, altruism, compassion, service — all the perennial truths that we talked about that are part of all religion and all cultures — once you stop trying to see the differences, these are the things in our own self-interest, because they free us from our **suffering** and from disease. And it—s in a sense the most selfish thing that we can do. Just take a look at one study. This was done by David Spiegel at Stanford. He took women with metastatic breast cancer, randomly divided them into two groups. One group of people just met for an hour-and-a-half once a week in a support group. It was a nurturing, **loving** environment, where they were encouraged to let down their emotional defenses and talk about how awful it is to have breast cancer with people who understood, because they were going through it too. They just met once a week for a year. Five years later, those women lived twice as long, and you can see that the people — and that was the only difference between the groups. It was a randomized control study published in The Lancet. Other studies have shown this as well. So, these simple things that create intimacy are really **healing**, and even the word healing, it comes from the root"to make whole."The word yoga comes from the Sanskrit, meaning"union, to yoke, to bring together."And the last slide I want to show you is from — I was — again, this swami that I studied with for so many years, and I did a combined oncology and cardiology Grand Rounds at the University of Virginia medical school a couple of years ago. And at the end of it, somebody said,"Hey, Swami, what—s the difference between wellness and illness?"And so he went up on the board and he wrote the word"illness,"and circled the first letter, and then wrote the word"wellness,"and circled the first two letters... To me, it—s just shorthand for what we—re talking about : that anything that creates a sense of connection and community and love is really **healing**. And then we can enjoy our lives more fully without getting sick in the process. Thank you.

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I bet you ' re worried. I was worried. That ' s why I began this piece. I was worried about vaginas. I was worried what we think about vaginas and even more worried that we don ' t think about them. I was worried about my own vagina. It needed a context, a culture, a community of other vaginas. There is so much **darkness** and secrecy surrounding them. Like the Bermuda Triangle, nobody ever reports back from there. In the first place, it ' s not so easy to even find your vagina. Women go days, weeks, months, without looking at it. I interviewed a high-powered businesswoman ; she told me she didn ’ t have time.”Looking at your vagina,”she said,”is a full day ’ s work.””You ’ ve got to get down there on your back, in front of a mirror, full-length preferred. You ’ ve got to get in the perfect position with the perfect light, which then becomes shadowed by the angle you ’ re at. You ’ re twisting your head up, arching your back, it ’ s exhausting.”She was busy ; she didn ’ t have time. So I decided to talk to women about their vaginas. They began as casual vagina interviews, and they turned into vagina monologues. I talked with over 200 women. I talked to older women, younger women, married women, lesbians, single women. I talked to corporate professionals, college professors, actors, sex workers. I talked to African-American women, Asian-American women, Native- American women, Caucasian women, Jewish women. OK, at first women were a little shy, a little reluctant to talk. Once they got going, you couldn ’ t stop them. Women love to talk about their vaginas, they do. Mainly because no one ’ s ever asked them before. Let ’ s just start with the word”vagina”— vagina, vagina. It sounds like an infection, at best. Maybe a medical instrument.”Hurry, nurse, bring the vagina!”Vagina, vagina, vagina. It doesn ’ t matter how many times you say the word, it never sounds like a word you want to say. It ’ s a completely ridiculous, totally un-sexy word. If you use it during sex, trying to be politically correct,”Darling, would you stroke my vagina,”you kill the act right there. I ’ m worried what we call them and don ’ t call them. In Great Neck, New York, they call it a Pussycat. A woman told me there her mother used to tell her,”Don ’ t wear panties, dear, underneath your pajamas. You need to air out your Pussycat.”In Westchester, they call it a Pooki, in New Jersey, a twat. There ’ s Powderbox, derriere, a Pooky, a Poochi, a Poopi, a Poopelu, a Pooninana, a Padepachetchki, a Pal, and a Piche. There ’ s Toadie, Dee Dee, Nishi, Dignity, Coochi Snorcher, Cooter, Labbe, Gladys Seagelman, VA, Wee wee, Horsespot, Nappy Dugout, Mongo, Ghoulie, Powderbox, a Mimi in Miami, a Split Knish in Philadelphia... and a Schmende in the Bronx. I am worried about vaginas. This is how the”Vagina Monologues”begins. But it really didn ’ t begin there. It began with a conversation with a woman. We were having a conversation about menopause, and we got onto the subject of her vagina, which you ’ ll do if you ’ re talking about menopause. And she said things that really shocked me about her vagina — that it was dried-up and finished and dead — and I was kind of shocked. So I said to a friend casually,”Well, what do you think about your vagina?”And that woman said something more amazing, and then the next woman said something more amazing, and before I knew it, every woman was telling me I had to talk to somebody about their vagina because they had an amazing story, and I was sucked down the vagina trail. And I really haven ’ t gotten off of it. I think if you had told me when I was younger that I was going to grow up, and be in shoe stores, and people would scream out,”There she is, the Vagina Lady!”I don ’ t know that that would have been my life ambition. But I want to talk a little bit about happiness, and the relationship to this whole vagina journey, because it has been an extraordinary journey that began eight years ago. I think before I did the”Vagina Monologues,”I didn ’ t really believe in happiness. I thought that only idiots were happy, to be honest. I remember when I started practicing Buddhism 14 years ago, and I was told that the end of this practice was to be happy, I said,”How could you be happy and live in this world of **suffering** and live in this world of pain?”I mistook happiness for a lot of other things, like numbness or decadence or selfishness. And what happened through the course of the”Vagina Monologues”and this journey is, I think I have come to understand a little bit more about happiness. There are three qualities I want to talk about. One is seeing what ’ s right in front of you, and talking about it, and stating it. I think what I learned from talking about the vagina and speaking about the vagina, is it was the most obvious thing — it was right in the center of my body and the center of the world — and yet it was the one thing nobody talked about. The second thing is that what talking about the vagina did is it opened this door which allowed me to see that there was a way to serve the world to make it better. And that ' s where the deepest happiness has actually come from. And the third principle of happiness, which I ' ve realized recently : Eight years ago, this momentum and this energy, this"V-wave"started — and I can only describe it as a"V-wave"because, to be honest, I really don ' t understand it completely ; I feel at the service of it. But this wave started, and if I question the wave, or try to stop the wave or look back at the wave, I often have the experience of whiplash or the potential of my neck breaking. But if I go with the wave, and I trust the wave and I move with the wave, I go to the next place, and it happens logically and organically and truthfully. And I started this piece, particularly with stories and narratives, and I was talking to one woman and that led to another woman and that led to another woman. And then I wrote those stories down, and I put them out in front of other people. And every single time I did the show at the beginning, women would literally line up after the show, because they wanted to tell me their stories. And at first I thought,"Oh great, I ' ll hear about wonderful orgasms, and great sex lives, and how women love their vaginas."But in fact, that ' s not what women lined up to tell me. What women lined up to tell me was how they were raped, and how they were battered, and how they were beaten, and how they were gang-raped in parking lots, and how they were incested by their uncles. And I wanted to stop doing the"Vagina Monologues,"because it felt too daunting. I felt like a war photographer who takes pictures of terrible events, but doesn ' t intervene on their behalf. And so in 1997, I said,"Let ' s get women together. What could we do with this information that all these women are being violated?"And it turned out, after thinking and investigating, that I discovered — and the UN has actually said this recently — that one out of every three women on this planet will be beaten or raped in her lifetime. That ' s essentially a gender ; that ' s essentially the resource of the planet, which is women. So in 1997 we got all these incredible women together and we said,"How can we use the play, this energy, to stop **violence** against women?"And we put on one event in New York City, in the theater, and all these great actors came — from Susan Sarandon, to Glenn Close, to Whoopi Goldberg — and we did one performance on one evening, and that catalyzed this wave, this energy. And within five years, this extraordinary thing began to happen. One woman took that energy and she said,"I want to bring this wave, this energy, to college campuses,"and so she took the play and she said,"Let ' s use the play and have performances once a year, where we can raise money to stop **violence** against women in local communities all around the world."And in one year, it went to 50 colleges, and then it expanded. And over the course of the last six years, it ' s spread and it ' s spread and it ' s spread around the world. What I have learned is two things : one, that the epidemic of **violence** towards women is shocking ; it ' s global ; it is so profound and it is so devastating, and it is so in every little pocket of every little crater, of every little society that we don ' t even recognize it, because it ' s become ordinary. This journey has taken me to Afghanistan, where I had the extraordinary honor and privilege to go into parts of Afghanistan under the Taliban. I was dressed in a burqa and I went in with an extraordinary group, called the Revolutionary Association of the Women of Afghanistan. And I saw firsthand how women had been stripped of every single right that was possible to strip women of — from being educated, to being employed, to being actually allowed to eat ice cream. For those of you who don ' t know, it was illegal to eat ice cream under the Taliban. And I actually saw and met women who had been flogged for being caught eating vanilla ice cream. I was taken to the secret ice cream-eating place in a little town, where we went to a back room, and women were seated and a curtain was pulled around us, and they were served vanilla ice cream. And women lifted their burqas and ate this ice cream. And I don ’ t think I ever understood pleasure until that moment, and how women have found a way to keep their pleasure alive. It has taken me, this journey, to Islamabad, where I have witnessed and met women with their faces melted off. It has taken me to Juarez, Mexico, where I was a week ago, where I have literally been there in parking lots, where bones of women have washed up and been dumped next to Coca-Cola bottles. It has taken me to universities all over this country, where girls are date-raped and drugged. I have seen terrible, terrible, terrible **violence**. But I have also recognized, in the course of seeing that **violence**, that being in the face of things and seeing actually what ’ s in front of us is the antidote to depression, and to a feeling that one is worthless and has no value. Because before the”Vagina Monologues,”I will say that 80 percent of my consciousness was closed off to what was really going on in this reality, and that closing-off closed off my vitality and my life energy. What has also happened is in the course of these travels — and it ’ s been an extraordinary thing — is that every single place that I have gone to in the world, I have met a new species. And I really love hearing about all these species at the bottom of the sea. And I was thinking about how being with these extraordinary people on this particular panel, that it ’ s beneath, beyond and between, and the vagina kind of fits into all those categories. But one of the things I ’ ve seen is this species — and it is a species, and it is a new paradigm, and it doesn ’ t get reported in the press or in the media because I don ’ t think good news ever is news, and I don ’ t think people who are transforming the planet are what gets the ratings on TV shows. But every single country I have been to — and in the last six years, I ’ ve been to about 45 countries, and many tiny little villages and cities and towns — I have seen something what I ’ ve come to call”vagina warriors.”A”vagina warrior”is a woman, or a vagina-friendly man, who has witnessed incredible **violence** or suffered it, and rather than getting an AK-47 or a weapon of mass destruction or a machete, they hold the **violence** in their bodies ; they grieve it ; they experience it ; and then they go out and devote their lives to making sure it doesn ’ t happen to anybody else. I have met these women everywhere on the planet, and I want to tell a few stories, because I believe that stories are the way that we transmit information, where it goes into our bodies. And I think one of the things about being at TED that ’ s been very interesting is that I live in my body a lot, and I don ’ t live in my head very much anymore. And this is a very heady place. And it ’ s been really interesting to be in my head for the last two days ; I ’ ve been very disoriented — because I think the world, the V-world, is very much in your body. It ’ s a body world, and the species really exists in the body. And I think there ’ s a real significance in us attaching our bodies to our heads, that that separation has created a divide that is often separating purpose from intent. And the connection between body and head often brings those things into union. I want to talk about three particular people that I ’ ve met, vagina warriors, who really transformed my understanding of this whole principle and species, and one is a woman named Marsha Lopez. Marsha Lopez was a woman I met in Guatemala. She was 14 years old, and she was in a marriage and her husband was beating her on a regular basis. And she couldn ’ t get out, because she was addicted to the relationship, and she had no money. Her sister was younger than her, and she applied — we had a”Stop Rape”contest a few years ago in New York — and she applied, hoping that she would become a finalist and she could bring her sister. She did become a finalist ; she brought Marsha to New York. And at that time, we did this extraordinary V-Day at Madison Square Garden, where we sold out the entire testosterone-filled dome — 18, 000 people standing up to say"Yes"to vaginas, which was really a pretty incredible transformation. And she came, and she witnessed this, and she decided that she would go back and leave her husband, and that she would bring V-Day to Guatemala. She was 21 years old. I went to Guatemala and she had sold out the National Theater of Guatemala. And I watched her walk up on stage in her red short dress and high heels, and she stood there and said,"My name is Marsha. I was beaten by my husband for five years. He almost murdered me. I left and you can, too."And the entire 2, 000 people went absolutely crazy. There ' s a woman named Esther Chvez who I met in Juarez, Mexico. And Esther Chvez was a brilliant accountant in Mexico City. She was 72 years old and she was planning to retire. She went to Juarez to take care of an ailing aunt, and in the course of it, she began to discover what was happening to the murdered and disappeared women of Juarez. She gave up her life ; she moved to Juarez. She started to write the stories which documented the disappeared women. 300 women have disappeared in a border town because they ' re brown and poor. There has been no response to the disappearance, and not one person has been held accountable. She began to document it. She opened a center called Casa Amiga, and in six years, she has literally brought this to the consciousness of the world. We were there a week ago, when there were 7, 000 people in the **street**, and it was truly a miracle. And as we walked through the **streets**, the people of Juarez, who normally don ' t even come into the **streets**, because the **streets** are so dangerous, literally stood there and wept, to see that other people from the world had showed up for that particular community. There ' s another woman, named Agnes. And Agnes, for me, epitomizes what a vagina warrior is. I met her three years ago in Kenya. And Agnes was mutilated as a little girl ; she was circumcised against her will when she was 10 years old, and she really made a decision that she didn ' t want this practice to continue anymore in her community. So when she got older, she created this incredible thing : it ' s an anatomical sculpture of a woman ' s body, half a woman ' s body. And she walked through the Rift Valley, and she had vagina and vagina replacement parts, where she would teach girls and parents and boys and girls what a healthy vagina looks like, and what a mutilated vagina looks like. And in the course of her travel — she walked literally for eight years through the Rift Valley, through dust, through sleeping on the ground, because the Maasai are nomads, and she would have to find them, and they would move, and she would find them again — she saved 1, 500 girls from being cut. And in that time, she created an alternative ritual, which involved girls coming of age without the cut. When we met her three years ago, we said,"What could V-Day do for you?"And she said,"Well, if you got me a jeep, I could get around a lot faster."So we bought her a jeep. And in the year that she had the jeep, she saved 4, 500 girls from being cut. So we said to her,"What else could we do for you?"She said,"Well, Eve, if you gave me some money, I could open a house and girls could run away, and they could be saved."And I want to tell this little story about my own beginnings, because it ' s very interrelated to happiness and Agnes. When I was a little girl — I grew up in a wealthy community ; it was an upper- middle class white community, and it had all the trappings and the looks of a perfectly nice, wonderful, great life. And everyone was supposed to be happy in that community, and, in fact, my life was hell. I lived with an alcoholic father who beat me and molested me, and it was all inside that. And always as a child I had this fantasy that somebody would come and rescue me. And I actually made up a little character whose name was Mr.

Alligator. I would call him up when things got really bad, and say it was time to come and pick me up. And I would pack a little bag and wait for Mr. Alligator to come. Now, Mr. Alligator never did come, but the idea of Mr. Alligator coming actually saved my sanity and made it OK for me to keep going, because I believed, in the distance, there would be someone coming to rescue me. Cut to 40-some odd years later, we go to Kenya, and we ’ re walking, we arrive at the opening of this house. And Agnes hadn ’ t let me come to the house for days, because they were preparing this whole ritual. I want to tell you a great story. When Agnes first started fighting to stop female genital mutilation in her community, she had become an outcast, and she was exiled and slandered, and the whole community turned against her. But being a vagina warrior, she kept going, and she kept committing herself to transforming consciousness. And in the Maasai community, goats and cows are the most valued possession. They ’ re like the Mercedes-Benz of the Rift Valley. And she said two days before the house opened, two different people arrived to give her a goat each, and she said to me,”I knew then that female genital mutilation would end one day in Africa.”Anyway, we arrived, and when we arrived, there were hundreds of girls dressed in red homemade dresses — which is the color of the Maasai and the color of V-Day — and they greeted us. They had made up these songs that they were **singing**, about the end of **suffering** and the end of mutilation, and they walked us down the path. It was a gorgeous day in the African sun, and the dust was flying and the girls were dancing, and there was this house, and it said,”V- Day Safe House for the Girls.”And it hit me in that moment that it had taken 47 years, but that Mr. Alligator had finally shown up. And he had shown up, obviously, in a form that it took me a long time to understand, which is that when we give in the world what we want the most, we **heal** the broken part inside each of us. And I feel, in the last eight years, that this journey — this miraculous vagina journey — has taught me this really simple thing, which is that happiness exists in action ; it exists in telling the truth and saying what your truth is ; and it exists in giving away what you want the most. And I feel that knowledge and that journey has been an extraordinary privilege, and I feel really blessed to have been here today to communicate that to you. Thank you very much.

Text Sample 408: POS Dim 3 – 2004 – Score 7.00 – 2004/t000486.txt

You ’ ll be happy to know that I ’ ll be talking not about my own tragedy, but other people ’ s tragedy. It ’ s a lot easier to be lighthearted about other people ’ s tragedy than your own, and I want to keep it in the spirit of the conference. So, if you believe the media accounts, being a drug dealer in the height of the crack cocaine epidemic was a very glamorous life, in the words of Virginia Postrel. There was money, there was drugs, guns, women, you know, you name it — jewelry, bling-bling — it had it all. What I ’ m going to tell you today is that, in fact, based on 10 years of research, a unique opportunity to go inside a gang — to see the actual books, the financial records of the gang — that the answer turns out not to be that being in the gang was a glamorous life. But I think, more realistically, that being in a gang — selling drugs for a gang — is perhaps the worst job in all of America. And that ’ s what I ’ d like to convince you of today. So there are three things I want to do. First, I want to explain how and why crack cocaine had such a profound influence on inner-city gangs. Secondly, I want to tell you how somebody like me came to be able to see the inner workings of a gang — an interesting story, I think. And then third, I want to tell you, in a very superficial way, about some of the things we found when we actually got to look at the financial records, the books, of the gang. So before I do that, just one warning, which is that this presentation has been rated ' R ' by the Motion Picture Association of America. It contains adult themes, adult language. Given who is up on the stage, you ' ll be delighted to know that, in fact, there ' ll be no nudity — Unexpected wardrobe malfunctions aside. So let me start by talking about crack cocaine, and how it transformed the gang. To do that, you have to actually go back to a time before crack cocaine, in the early ' 80s, and look at it from the perspective of a gang leader. Being a gang leader in the inner city wasn ' t such a bad deal in the mid- ' 80s — the early ' 80s, let me say. Now, you had a lot of power, and you got to beat people up — you got a lot of prestige, a lot of respect. But the thing is, there was no money in it. The gang had no way to make money. You couldn ' t charge dues to the people in the gang, because the people in the gang didn ' t have any money. You couldn ' t really make any money selling marijuana — marijuana ' s too cheap, it turns out. You can ' t get rich selling marijuana. You couldn ' t sell cocaine ; cocaine ' s a great product — powdered cocaine — but you ' ve got to know rich white people. And most of the inner-city gang members didn ' t know any rich white people, so couldn ' t sell to that market. You couldn ' t really do petty crime, either. Turns out, petty crime ' s a terrible way to make a living. As a result, as a gang leader, you had, you know, power — it ' s a pretty good life — but the thing was, in the end, you were living at home with your mother. And so it wasn ' t really a career. There were limits to how powerful and important you could be if you had to live at home with your mother. Then along comes crack cocaine. And in the words of Malcolm Gladwell, crack cocaine was the extra-chunky version of tomato sauce for the inner city. Because crack cocaine was an unbelievable innovation. I don ' t have time to talk about it today, but if you think about it, I would say that in the last 25 years, of every invention or innovation that ' s occurred in this country, the biggest one in terms of impact on the well-being of people who live in the inner city, was crack cocaine. And for the worse — not for the better, but for the worse. It had a huge impact on life. So what was it about crack cocaine? It was a brilliant way of getting the brain high. Because you could smoke crack cocaine — you can ' t smoke powdered cocaine — and smoking is a much more efficient mechanism of delivering a high than snorting it. And it turned out there was this audience that didn ' t know it wanted crack cocaine, but when it came, it really did. And it was a perfect drug ; you could buy the cocaine that went into it for a dollar, sell it for five dollars. Highly addictive — the high was very short. So for fifteen minutes, you get this great high, and then when you come down, all you want to do is get high again. It created a wonderful market. And for the people who were there running the gang, it was a great way, seemingly, to make a lot of money. At least for the people on the top. So this is where we enter the picture. Not really me — I ' m really a bit player in all this. My co-author, Sudhir Venkatesh, is the main character. He was a math major in college who had a good heart, and decided he wanted to get a sociology PhD, came to the University of Chicago. Now, the three months before he came to Chicago, he had spent following the Grateful Dead. And in his own words, he"looked like a freak."He ' s a South Asian — very dark-skinned South Asian. Big man, and he had hair, in his words,"down to his ass."Defied all kinds of boundaries : Was he black or white? Was he man or woman? He was really a curious sight to be seen. So he showed up at the University of Chicago, and the famous sociologist William Julius Wilson was doing a book that involved surveying people all across Chicago. He took one look at Sudhir, who was going to go do some surveys for him, and decided he knew exactly the place to send him, which was to one of the toughest, most notorious housing projects not just in Chicago, but in the entire United States. So Sudhir, the suburban boy who had never really been in the inner city, dutifully took his clipboard and walked down to this housing project, gets to the first building. The first building? Well, there ’ s nobody there. But he hears some voices up in the stairwell, so he climbs up the stairwell, comes around the corner, and finds a group of young African-American men playing dice. This is about 1990, peak of the crack epidemic. This is a very dangerous job, being in a gang. You don ’ t like to be surprised. You don ’ t like to be surprised by people who come around the corner. And the mantra was : shoot first ; ask questions later. Now, Sudhir was lucky — he was such a freak, and that clipboard probably saved his life, because they figured no other rival gang member would be coming up to shoot at them with a clipboard. So his greeting was not particularly warm, but they did say, well, OK — let ’ s hear your questions on your survey. So — I kid you not — the first question on the survey that he was sent to ask was :”How do you feel about being poor and Black in America?”Makes you wonder about academics. So the choice of answers were : [A) Very Good B) Good C) Bad D) Very Bad] What Sudhir found out is, in fact, that the real answer was the following : [A) Very Good B) Good C) Bad D) Very Bad E) Fuck you] The survey was not, in the end, going to be what got Sudhir off the hook. He was held hostage overnight in the stairwell. There was a lot of gunfire, there were a lot of philosophical discussions he had with the gang members. By morning, the gang leader arrived, checked out Sudhir, decided he was no threat, and they let him go home. So Sudhir went home, took a shower, took a nap. And you and I, probably, faced with the situation, would think,”I guess I ’ m going to write my dissertation on The Grateful Dead, I ’ ve been following them for the last three months.”Sudhir, on the other hand, got right back, walked down to the housing project, went up to the second floor, and said :”Hey, guys, I had so much fun hanging out with you last night, I wonder if I could do it again tonight.”And that was the beginning of what turned out to be a beautiful relationship that involved Sudhir living in the housing project on and off for 10 years, hanging out in crack houses, going to jail with the gang members, having the windows shot out of his car, having the police break into his apartment and steal his computer disks — you name it. But ultimately, the story has a happy ending for Sudhir, who became one of the most respected sociologists in the country. And especially for me, as I sat in my office with my Excel spreadsheet open, waiting for Sudhir to come and deliver to me the latest load of data that he would get from the gang. It was one of the most unequal co-authoring relationships ever — But I was glad to be the beneficiary of it. So what did we find? What did we find in the gang? Well, let me say one thing : We really got **access** to everybody in the gang. We got an inside look at the gang, from the very bottom up to the very top. They trusted Sudhir, in ways that really no academic has ever — or really anybody, any outsider — has ever earned the trust of these gangs, to the point where they actually opened up what was most interesting for me — their books, the financial records they kept. They made them available to us, and we not only could study them, but we could ask them questions about what was in them. So if I have to kind of summarize very quickly in the short time I have what the bottom line of what I take away from the gang is, it ’ s that, if I had to draw a parallel between the gang and any other organization, it would be that the gang is just like McDonald ’ s, in a lot of different respects — the restaurant McDonald ’ s. So first, in one way, which isn ’ t maybe the most interesting way, but it ’ s a good way to start — is in the way it ’ s organized, the hierarchy of the gang, the way it looks. So here ’ s what the org chart of the gang looks like. I don ’ t know if you know much about org charts, but if you were to assign a stripped- down and simplified McDonald ’ s org chart, this is exactly what it would look like. It ’ s amazing, but the top level of the gang, they actually call themselves the”Board of Directors.”And Sudhir says it ’ s not like these guys had a very sophisticated view of what happened in American corporate life, but they had seen movies like”Wall Street,”and they had learned a little bit about what it was like to be in the real world. Now, below that board of directors, you ’ ve got essentially what are regional VPs — people who control, say, the South Side of Chicago, or the West Side of Chicago. Sudhir got to know very well the guy who had the unfortunate assignment of trying to take the Iowa franchise, which, it turned out, for this black gang, was not one of the more brilliant financial endeavors they undertook. But the thing that really makes the gang seem like McDonald ’ s is its franchisees. The guys who are running the local gangs — the four-square-block by four-square-block areas — they ’ re just like the guys, in some sense, who are running the McDonald ’ s. They are the entrepreneurs. They get the exclusive property rights to control the drug-selling. They get the name of the gang behind them, for merchandising and marketing. And they ’ re the ones who basically make the profit or lose a profit, depending on how good they are at running the business. Now, the group I really want you to think about, though, are the ones at the bottom — the foot soldiers. These are the teenagers, typically, who ’ d be standing out on the **street** corner, selling the drugs. Extremely dangerous work. And important to note is that almost all of the weight, all of the people in this organization are at the bottom — just like McDonald ’ s. So in some sense, the foot soldiers are a lot like the people who are taking your order at McDonald ’ s, and it ’ s not just by chance that they ’ re like them. In fact, in these **neighborhoods**, they ’ d be the same people. So the same kids who are working in the gang were actually, at the very same time, typically working part-time at a place like McDonald ’ s. Which already foreshadows the main result that I ’ ve talked about, about what a crappy job it was, being in the gang. Because obviously, if being in the gang were such a wonderful, lucrative job, why in the world would these guys moonlight at McDonald ’ s? So what do the wages look like? You might be surprised. But based on being able to talk to them and to see their records, this is what it looks like in terms of the wages. The hourly wage the foot soldiers were earning was $3. 50 an hour. It was below the minimum wage. And this is well-documented. It ’ s easy to see by the patterns of consumption they have. It really is not fiction — it ’ s fact. There was very little money in the gang, especially at the bottom. Now if you managed to rise up, say, and be that local leader, the guy who ’ s the equivalent of the McDonald ’ s franchisee, you ’ d be making 100, 000 dollars a year. And that, in some ways, was the best job you could hope to get if you were growing up in one of these **neighborhoods** as a young black male. If you managed to rise to the very top, 200, 000 or 400, 000 dollars a year is what you ’ d hope to make. Truly, you would be a great success story. And one of the sad parts of this is that, indeed, among the many other ramifications of crack cocaine is that the most talented individuals in these communities — this is what they were striving for. They weren ’ t trying to make it in legitimate ways, because there were no legitimate channels out. This was the best way out. And it actually was the right choice, probably, to try to make it out this way. You look at this, the relationship to McDonald ’ s breaks down here. The money looks about the same. Why is it such a bad job? Well, the reason it ’ s such a bad job is that there ’ s somebody shooting at you a lot of the time. So, with shooting at you, what are the death rates? We found, in our gang — and admittedly, this was not really a standard situation ; this was a time of intense **violence**, of a lot of gang wars, as this gang actually became quite successful. But there were costs. And so the death rate — not to mention the rate of being arrested, sent to prison, being wounded — the death rate in our sample was seven percent per person per year. You ’ re in the gang for four years, you expect to die with about a 25 percent likelihood. That is about as high as you can get. So for comparison ’ s purposes, let ’ s think about some other walk of life you may expect might be extremely risky. Let ’ s say that you were a murderer and you were **convicted** of murder, and you ’ re sent to death row. It turns out, the death rates on death row from all causes, including execution : two percent a year. So it ’ s a lot safer being on death row than it is selling drugs out on the **street**. That gives you some pause, for those of you who believe that a death penalty ’ s going to have an **enormous** deterrent effect on crime. To give you a sense of just how bad the inner city was during crack — and I ’ m not really focusing on the negatives, but really, there ’ s another story to tell you there — if you look at the death rates just of random, young black males growing up in the inner city in the United States, the death rates during crack were about one percent. That ’ s extremely high. And this is **violent** death — it ’ s unbelievable, in some sense. To put it into perspective : if you compare this to the soldiers in Iraq, for instance, right now fighting the war : 0. 5 percent. So in some very literal way, the young black men who were growing up in this country were living in a war zone, very much in the sense that the soldiers over in Iraq are fighting in a war. So why in the world, you might ask, would anybody be willing to stand out on a **street** corner selling drugs for $3. 50 an hour, with a 25 percent chance of dying over the next four years? Why would they do that? And I think there are a couple answers. I think the first one is that they got fooled by history. It used to be the gang was a rite of passage ; that the young people controlled the gang ; that as you got older, you dropped out of the gang. So what happened was, the people who happened to be in the right place at the right time — the people who happened to be leading the gang in the mid-to-late- ’ 80s — became very, very wealthy. And so the logical thing to think was that they are going to age out of the gang like everybody else has, and the next generation is going to take over and get the wealth. There are striking similarities, I think, to the Internet boom. The first set of people in Silicon Valley got very, very rich. And then all of my friends said,”Maybe I should go do that, too.”And they were willing to work very cheap for stock options that never came. In some sense, that ’ s what happened, exactly, to the set of people we were looking at. They were willing to start at the bottom, just like, say, a first-year lawyer at a law firm is willing to start at the bottom, work 80-hour weeks for not that much money, because they think they ’ re going to make partner. But the rules changed, and they never got to make partner. Indeed, the same people who were running all of the major gangs in the late 1980s are still running the major gangs in Chicago today. They never passed on any of the wealth, So everybody got stuck at that $3. 50-an-hour job, and it turned out to be a disaster. The other thing the gang was very good at was marketing and trickery. And so for instance, one thing the gang would do is — the gang leaders would have big entourages, and they ’ d drive fancy cars and have fancy jewelry. So what Sudhir eventually realized as he hung out with them more, is that, really, they didn ’ t own those cars — they just leased them, because they couldn ’ t afford to own the fancy cars. And they didn ’ t really have gold jewelry, they had gold- plated jewelry. It goes back to, you know, the real-real versus the fake-real. And really, they did all sorts of things to trick the young people into thinking what a great deal the gang was going to be. So for instance, they would give a 14-year-old kid a whole roll of bills to hold. That 14-year-old kid would say to his friends,”Hey, look at all the money I got in the gang.”It wasn ’ t his money — until he spent it, and then he was in debt to the gang, and was sort of an indentured servant for a while. So I have a couple minutes. Let me do one last thing I hadn ’ t thought I ’ d have time to do, which is to talk about what we learned more generally about economics, from the study of the gang. So, economists tend to talk in technical words. Often, our theories fail quite miserably when we over the data, but what ’ s kind of interesting is that in this setting, it turned out that some of the economic theories that worked not so well in the real economy worked very well in the drug economy, in some sense, because it ’ s unfettered capitalism. Here ’ s an economic principle. This is one of the basic ideas in labor economics, called a”compensating differential.”It ’ s the idea that the increment to wages that a worker requires to leave him indifferent between performing two tasks, one which is more unpleasant than the other. Compensating differential — it ’ s why we think garbagemen might be paid more than people who work in parks. The words of one of the members of the gang, I think, make this clear. So it turns out — I ’ m sort of getting ahead of myself — it turns out, in the gang, when there ’ s a war going on, they actually pay the foot soldiers twice as much money. It ’ s exactly this concept. Because they ’ re not willing to be at risk. And the words of a gang member capture it quite nicely, he says :”Would you stand around here when all this shit...”— the shooting —”... if all this shit ’ s going on? No, right? So if I gonna be asked to put my life on the line, then front me the cash, man.”I think the gang member says it much more articulately than the economist, about what ’ s going on. Here ’ s another one. Economists talk about game theory, that every two-person game has a Nash equilibrium. Here ’ s the translation you get from the gang member. They ’ re talking about the decision of why they don ’ t go shoot — One thing that turns out to be a great business tactic in the gang : if you go and just shoot guns in the air in the other gang ’ s territory — people are afraid to go buy drugs there, they ’ re going to come into your **neighborhood**. Here ’ s what he says about why they don ’ t do that :”If we start shooting around there, the other gang ’ s territory, nobody, I mean, you dig it, nobody gonna step on their turf. But we gotta be careful, ’ cause they can shoot around here too and then we all fucked.”So that ’ s the same concept. Then again, sometimes economists get it wrong. One thing we observed in the data is that it looked like — the gang leader always got paid. No matter how bad it was economically, he always got himself paid. We had some theories related to cash flow, and lack of **access** to capital markets, and things like that. Then we asked the gang member,”Why is it you always get paid and your workers don ’ t always get paid?”His response is,”You got all these niggers below you who want your job, you dig? If you start taking losses, they see you as weak and shit.”And I thought about it and said,”CEOs often pay themselves million-dollar bonuses, even when companies are losing a lot of money. And it never would really occur to an economist that this idea of ’ weak and shit ’ could really be important.”Maybe”weak and shit”is an important hypothesis that needs more analysis. Thank you very much.

Text Sample 409: POS Dim 3 – 2003 – Score 10.00 – 2003/t000244.txt

A year ago, I spoke to you about a book that I was just in the process of completing, that has come out in the interim, and I would like to talk to you today about some of the controversies that that book inspired. The book is called”The Blank Slate,”based on the popular idea that the human mind is a blank slate, and that all of its structure comes from socialization, culture, parenting, experience. The”blank slate”was an influential idea in the 20th century. Here are a few quotes indicating that :”Man has no nature,”from the historian Jose Ortega y Gasset ;”Man has no instincts,”from the anthropologist Ashley Montagu ;”The human brain is capable of a full range of behaviors and predisposed to none,”from the late scientist Stephen Jay Gould. There are a number of reasons to doubt that the human mind is a blank slate, and some of them just come from common sense. As many people have told me over the years, anyone who ' s had more than one child knows that kids come into the world with certain temperaments and talents ; it doesn ' t all come from the outside. Oh, and anyone who has both a child and a house pet has surely noticed that the child, exposed to speech, will acquire a human language, whereas the house pet won ' t, presumably because of some innate different between them. And anyone who ' s ever been in a heterosexual relationship knows that the minds of men and the minds of women are not indistinguishable. There are also, I think, increasing results from the scientific study of humans that, indeed, we ' re not born blank slates. One of them, from anthropology, is the study of human universals. If you ' ve ever taken anthropology, you know that it ' s a — kind of an occupational pleasure of anthropologists to show how exotic other cultures can be, and that there are places out there where, supposedly, everything is the opposite to the way it is here. But if you instead look at what is common to the world ' s cultures, you find that there is an enormously rich set of behaviors and emotions and ways of construing the world that can be found in all of the world ' s 6, 000-odd cultures. The anthropologist Donald Brown has tried to list them all, and they range from aesthetics, affection and age statuses all the way down to weaning, weapons, weather, attempts to control, the color white and a worldview. Also, genetics and neuroscience are increasingly showing that the brain is intricately structured. This is a recent study by the neurobiologist Paul Thompson and his colleagues in which they — using MRI — measured the distribution of gray matter — that is, the outer layer of the cortex — in a large sample of pairs of people. They coded correlations in the thickness of gray matter in different parts of the brain using a false color scheme, in which no difference is coded as purple, and any color other than purple indicates a statistically significant correlation. Well, this is what happens when you pair people up at random. By definition, two people picked at random can ' t have correlations in the distribution of gray matter in the cortex. This is what happens in people who share half of their DNA — fraternal twins. And as you can see, large amounts of the brain are not purple, showing that if one person has a thicker bit of cortex in that region, so does his fraternal twin. And here ' s what happens if you get a pair of people who share all their DNA — namely, clones or identical twins. And you can see huge areas of cortex where there are massive correlations in the distribution of gray matter. Now, these aren ' t just differences in anatomy, like the shape of your ear lobes, but they have consequences in thought and behavior that are well illustrated in this famous cartoon by Charles Addams :"Separated at birth, the Mallifert twins meet accidentally."As you can see, there are two inventors with identical contraptions in their lap, meeting in the waiting room of a patent attorney. Now, the cartoon is not such an exaggeration, because studies of identical twins who were separated at birth and then tested in adulthood show that they have astonishing similarities. And this happens in every pair of identical twins separated at birth ever studied — but much less so with fraternal twins separated at birth. My favorite example is a pair of twins, one of whom was brought up as a Catholic in a Nazi family in Germany, the other brought up in a Jewish family in Trinidad. When they walked into the lab in Minnesota, they were wearing identical navy blue shirts with epaulettes ; both of them liked to dip buttered toast in coffee, both of them kept rubber bands around their wrists, both of them flushed the toilet before using it as well as after, and both of them liked to surprise people by sneezing in crowded elevators to watch them jump. Now — the story might seem to good to be true, but when you administer batteries of psychological tests, you get the same results — namely, identical twins separated at birth show quite astonishing similarities. Now, given both the common sense and scientific data calling the doctrine of the blank slate into question, why should it have been such an appealing notion? Well, there are a number of political reasons why people have found it congenial. The foremost is that if we ’ re blank slates, then, by definition, we are **equal**, because zero **equals** zero **equals** zero. But if something is written on the slate, then some people could have more of it than others, and according to this line of thinking, that would justify **discrimination** and inequality. Another political fear of human nature is that if we are blank slates, we can perfect mankind — the age-old dream of the perfectibility of our species through social engineering. Whereas, if we ’ re born with certain instincts, then perhaps some of them might condemn us to selfishness, **prejudice** and **violence**. Well, in the book, I argue that these are, in fact, non sequiturs. And just to make a long story short : first of all, the concept of **fairness** is not the same as the concept of sameness. And so when Thomas Jefferson wrote in the Declaration of Independence,”We hold these truths to be self-evident, that all men are created **equal**,”he did not mean”We hold these truths to be self- evident, that all men are clones.”Rather, that all men are **equal** in terms of their rights, and that every person ought to be treated as an individual, and not prejudged by the statistics of particular groups that they may belong to. Also, even if we were born with certain ignoble motives, they don ’ t automatically lead to ignoble behavior. That is because the human mind is a complex system with many parts, and some of them can inhibit others. For example, there ’ s excellent reason to believe that virtually all humans are born with a moral sense, and that we have cognitive abilities that allow us to profit from the lessons of history. So even if people did have impulses towards selfishness or greed, that ’ s not the only thing in the skull, and there are other parts of the mind that can counteract them. In the book, I go over controversies such as this one, and a number of other hot buttons, hot zones, Chernobyls, third rails, and so on — including the arts, cloning, crime, free will, education, evolution, gender differences, God, homosexuality, infanticide, inequality, Marxism, morality, Nazism, parenting, politics, race, rape, religion, resource depletion, social engineering, technological risk and war. And needless to say, there were certain risks in taking on these subjects. When I wrote a first draft of the book, I circulated it to a number of colleagues for comments, and here are some of the reactions that I got :”Better get a security camera for your house.””Don ’ t expect to get any more awards, job offers or positions in scholarly societies.””Tell your publisher not to list your hometown in your author bio.””Do you have tenure?”Well, the book came out in October, and nothing terrible has happened. I — I like — There was indeed reason to be nervous, and there were moments in which I did feel nervous, knowing the history of what has happened to people who ’ ve taken controversial stands or discovered disquieting findings in the behavioral sciences. There are many cases, some of which I talk about in the book, of people who have been slandered, called Nazis, physically **assaulted**, threatened with criminal prosecution for stumbling across or arguing about controversial findings. And you never know when you ’ re going to come across one of these booby traps. My favorite example is a pair of psychologists who did research on left-handers, and published some data showing that left-handers are, on average, more susceptible to disease, more prone to accidents and have a shorter lifespan. It ’ s not clear, by the way, since then, whether that is an accurate generalization, but the data at the time seemed to support that. Well, pretty soon they were barraged with enraged letters, death threats, ban on the topic in a number of scientific journals, coming from irate left-handers and their advocates, and they were literally afraid to open their mail because of the venom and vituperation that they had inadvertently inspired. Well, the night is young, but the book has been out for half a year, and nothing terrible has happened. None of the dire professional consequences has taken place — I haven ' t been exiled from the city of Cambridge. But what I wanted to talk about are two of these hot buttons that have aroused the strongest response in the 80-odd reviews that The Blank Slate has received. I ' ll just put that list up for a few seconds, and see if you can guess which two — I would estimate that probably two of these topics inspired probably 90 percent of the reaction in the various reviews and radio interviews. It ' s not **violence** and war, it ' s not race, it ' s not gender, it ' s not Marxism, it ' s not Nazism. They are : the arts and parenting. So let me tell you what aroused such irate responses, and I ' ll let you decide if whether they — the claims are really that outrageous. Let me start with the arts. I note that among the long list of human universals that I presented a few slides ago are art. There is no society ever discovered in the remotest corner of the world that has not had something that we would consider the arts. Visual arts — decoration of surfaces and bodies — appears to be a human universal. The telling of stories, music, dance, poetry — found in all cultures, and many of the motifs and themes that give us pleasure in the arts can be found in all human societies : a preference for symmetrical forms, the use of repetition and variation, even things as specific as the fact that in poetry all over the world, you have lines that are very close to three seconds long, separated by pauses. Now, on the other hand, in the second half of the 20th century, the arts are frequently said to be in decline. And I have a collection, probably 10 or 15 headlines, from highbrow magazines deploring the fact that the arts are in decline in our time. I ' ll give you a couple of representative quotes :"We can assert with some confidence that our own period is one of decline, that the standards of culture are lower than they were 50 years ago, and that the evidences of this decline are visible in every department of human activity."That ' s a quote from T. S. Eliot, a little more than 50 years ago. And a more recent one :"The possibility of sustaining high culture in our time is becoming increasing problematical. Serious book stores are losing their franchise, **nonprofit** theaters are surviving primarily by commercializing their repertory, symphony orchestras are diluting their programs, public television is increasing its dependence on reruns of British sitcoms, classical radio stations are dwindling, museums are resorting to blockbuster shows, dance is dying."That ' s from Robert Brustein, the famous drama critic and director, in The New Republic about five years ago. Well, in fact, the arts are not in decline. I don ' t think this will as a surprise to anyone in this room, but by any standard they have never been flourishing to a greater extent. There are, of course, entirely new art forms and new media, many of which you ' ve heard over these few days. By any economic standard, the demand for art of all forms is skyrocketing, as you can tell from the price of **opera** tickets, by the number of books sold, by the number of books published, the number of musical titles released, the number of new albums and so on. The only grain of truth to this complaint that the arts are in decline come from three spheres. One of them is in elite art since the 1930s — say, the kinds of works performed by major symphony orchestras, where most of the repertory is before 1930, or the works shown in major galleries and prestigious museums. In literary criticism and analysis, probably 40 or 50 years ago, literary critics were a kind of cultural hero ; now they ' re kind of a national joke. And the humanities and arts programs in the universities, which by many measures, indeed are in decline. Students are staying away in droves, universities are disinvesting in the arts and humanities. Well, here ' s a diagnosis. They didn ' t ask me, but by their own admission, they need all the help that they can get. And I would like to suggest that it ' s not a coincidence that this supposed decline in the elite arts and criticism occurred in the same point in history in which there was a widespread denial of human nature. A famous quotation can be found — if you look on the web, you can find it in literally scores of English core syllabuses —"In or about December 1910, human nature changed."A paraphrase of a quote by Virginia Woolf, and there ' s some debate as to what she actually meant by that. But it ' s very clear, looking at these syllabuses, that — it ' s used now as a way of saying that all forms of appreciation of art that were in place for centuries, or millennia, in the 20th century were discarded. The beauty and pleasure in art — probably a human universal — were — began to be considered saccharine, or kitsch, or commercial. Barnett Newman had a famous quote that"the impulse of modern art is the desire to destroy beauty"— which was considered bourgeois or tacky. And here ' s just one example. I mean, this is perhaps a representative example of the visual depiction of the female form in the 15th century ; here is a representative example of the depiction of the female form in the 20th century. And, as you can see, there — something has changed in the way the elite arts appeal to the senses. Indeed, in movements of modernism and post-modernism, there was visual art without beauty, literature without narrative and plot, poetry without meter and rhyme, architecture and planning without ornament, human scale, green space and natural light, music without melody and rhythm, and criticism without clarity, attention to aesthetics and insight into the human condition. Let me give just you an example to back up that last statement. But here, there — one of the most famous literary English scholars of our time is the Berkeley professor, Judith Butler. And here is an example of one of her analyses :"The move from a structuralist account in which capital is understood to structure social relations in relatively homologous ways to a view of hegemony in which power relations are subject to repetition, convergence and rearticulation brought the question of temporality into the thinking of structure, and marked a shift from the form of Althusserian theory that takes structural totalities as theoretical objects..."Well, you get the idea. By the way, this is one sentence — you can actually parse it. Well, the argument in"The Blank Slate"was that elite art and criticism in the 20th century, although not the arts in general, have disdained beauty, pleasure, clarity, insight and style. People are staying away from elite art and criticism. What a puzzle — I wonder why. Well, this turned out to be probably the most controversial claim in the book. Someone asked me whether I stuck it in in order to deflect ire from discussions of gender and Nazism and race and so on. I won ' t comment on that. But it certainly inspired an energetic reaction from many university professors. Well, the other hot button is parenting. And the starting point is the — for that discussion was the fact that we have all been subject to the advice of the parenting industrial complex. Now, here is — here is a representative quote from a besieged mother :"I ' m overwhelmed with parenting advice. I ' m supposed to do lots of physical activity with my kids so I can instill in them a physical fitness habit so they ' ll grow up to be healthy adults. And I ' m supposed to do all kinds of intellectual play so they ' ll grow up smart. And there are all kinds of play — clay for finger dexterity, word games for reading success, large motor play, small motor play. I feel like I could devote my life to figuring out what to play with my kids."I think anyone who ' s recently been a parent can sympathize with this mother. Well, here ' s some sobering facts about parenting. Most studies of parenting on which this advice is based are useless. They ' re useless because they don ' t control for heritability. They measure some correlation between what the parents do, how the children turn out and assume a causal relation : that the parenting shaped the child. Parents who talk a lot to their kids have kids who grow up to be articulate, parents who spank their kids have kids who grow up to be **violent** and so on. And very few of them control for the possibility that parents pass on genes for — that increase the chances a child will be articulate or **violent** and so on. Until the studies are redone with adoptive children, who provide an environment but not genes to their kids, we have no way of knowing whether these conclusions are valid. The genetically controlled studies have some sobering results. Remember the Mallifert twins : separated at birth, then they meet in the patent office — remarkably similar. Well, what would have happened if the Mallifert twins had grown up together? You might think, well, then they ' d be even more similar, because not only would they share their genes, but they would also share their environment. That would make them super-similar, right? Wrong. Identical twins, or any **siblings**, who are separated at birth are no less similar than if they had grown up together. Everything that happens to you in a given home over all of those years appears to leave no permanent stamp on your personality or intellect. A complementary finding, from a completely different methodology, is that adopted **siblings** reared together — the mirror image of identical twins reared apart, they share their parents, their home, their **neighborhood**, don ' t share their genes — end up not similar at all. OK — two different bodies of research with a similar finding. What it suggests is that children are shaped not by their parents over the long run, but in part — only in part — by their genes, in part by their culture — the culture of the country at large and the children ' s own culture, namely their peer group — as we heard from Jill Sobule earlier today, that ' s what kids care about — and, to a very large extent, larger than most people are prepared to acknowledge, by chance : chance events in the wiring of the brain in utero ; chance events as you live your life. So let me conclude with just a remark to bring it back to the theme of choices. I think that the sciences of human nature — behavioral genetics, evolutionary psychology, neuroscience, cognitive science — are going to, increasingly in the years to come, upset various dogmas, careers and deeply-held political belief systems. And that presents us with a choice. The choice is whether certain facts about humans, or topics, are to be considered taboos, forbidden knowledge, where we shouldn ' t go there because no good can come from it, or whether we should explore them honestly. I have my own answer to that question, which comes from a great artist of the 19th century, Anton Chekhov, who said,"Man will become better when you show him what he is like."And I think that the argument can ' t be put any more eloquently than that. Thank you very much.

Text Sample 410: POS Dim 3 – 2003 – Score 9.00 – 2003/t000451.txt

You know, one of the intense pleasures of travel and one of the delights of ethnographic research is the opportunity to live amongst those who have not forgotten the old ways, who still feel their past in the wind, touch it in stones polished by rain, taste it in the **bitter** leaves of plants. Just to know that Jaguar shamans still journey beyond the Milky Way, or the myths of the Inuit elders still resonate with meaning, or that in the Himalaya, the Buddhists still pursue the breath of the Dharma, is to really remember the central revelation of anthropology, and that is the idea that the world in which we live does not exist in some absolute sense, but is just one model of reality, the consequence of one particular set of adaptive choices that our lineage made, albeit successfully, many generations ago. And of course, we all share the same adaptive imperatives. We ' re all born. We all bring our children into the world. We go through initiation rites. We have to deal with the inexorable separation of death, so it shouldn ' t surprise us that we all **sing**, we all dance, we all have art. But what

’ s interesting is the unique cadence of the song, the rhythm of the dance in every culture. And whether it is the Penan in the forests of Borneo, or the Voodoo acolytes in Haiti, or the warriors in the Kaisut desert of Northern Kenya, the Curandero in the mountains of the Andes, or a caravanserai in the middle of the Sahara — this is incidentally the fellow that I traveled into the desert with a month ago — or indeed a yak herder in the slopes of Qomolangma, Everest, the goddess mother of the world. All of these peoples teach us that there are other ways of being, other ways of thinking, other ways of orienting yourself in the Earth. And this is an idea, if you think about it, can only fill you with hope. Now, together the myriad cultures of the world make up a web of spiritual life and cultural life that envelops the planet, and is as important to the well-being of the planet as indeed is the biological web of life that you know as a biosphere. And you might think of this cultural web of life as being an ethnosphere, and you might define the ethnosphere as being the sum total of all thoughts and dreams, myths, ideas, inspirations, intuitions brought into being by the human imagination since the dawn of consciousness. The ethnosphere is humanity ’ s great legacy. It ’ s the symbol of all that we are and all that we can be as an astonishingly inquisitive species. And just as the biosphere has been severely eroded, so too is the ethnosphere — and, if anything, at a far greater rate. No biologists, for example, would dare suggest that 50 percent of all species or more have been or are on the brink of extinction because it simply is not true, and yet that — the most apocalyptic scenario in the realm of biological diversity — scarcely approaches what we know to be the most optimistic scenario in the realm of cultural diversity. And the great indicator of that, of course, is language loss. When each of you in this room were born, there were 6, 000 languages spoken on the planet. Now, a language is not just a body of vocabulary or a set of grammatical rules. A language is a flash of the human spirit. It ’ s a vehicle through which the soul of each particular culture comes into the material world. Every language is an old-growth forest of the mind, a watershed, a thought, an ecosystem of spiritual possibilities. And of those 6, 000 languages, as we sit here today in Monterey, fully half are no longer being whispered into the ears of children. They ’ re no longer being taught to babies, which means, effectively, unless something changes, they ’ re already dead. What could be more lonely than to be enveloped in **silence**, to be the last of your people to speak your language, to have no way to pass on the wisdom of the ancestors or anticipate the promise of the children? And yet, that dreadful fate is indeed the plight of somebody somewhere on Earth roughly every two weeks, because every two weeks, some elder dies and carries with him into the grave the last syllables of an ancient tongue. And I know there ’ s some of you who say,”Well, wouldn ’ t it be better, wouldn ’ t the world be a better place if we all just spoke one language?”And I say,”Great, let ’ s make that language Yoruba. Let ’ s make it Cantonese. Let ’ s make it Kogi.”And you ’ ll suddenly discover what it would be like to be unable to speak your own language. And so, what I ’ d like to do with you today is sort of take you on a journey through the ethnosphere, a brief journey through the ethnosphere, to try to begin to give you a sense of what in fact is being lost. Now, there are many of us who sort of forget that when I say”different ways of being,”I really do mean different ways of being. Take, for example, this child of a Barasana in the Northwest Amazon, the people of the anaconda who believe that mythologically they came up the milk river from the east in the belly of sacred snakes. Now, this is a people who cognitively do not distinguish the color blue from the color green because the canopy of the heavens is equated to the canopy of the forest upon which the people depend. They have a curious language and marriage rule which is called”linguistic exogamy :”you must marry someone who speaks a different language. And this is all rooted in the mythological past, yet the curious thing is in these long houses, where there are six or seven languages spoken because of intermarriage, you never hear anyone practicing a language. They simply listen and then begin to speak. Or, one of the most fascinating tribes I ever lived with, the Waorani of northeastern Ecuador, an astonishing people first contacted peacefully in 1958. In 1957, five missionaries attempted contact and made a critical mistake. They dropped from the air 8 x 10 glossy photographs of themselves in what we would say to be friendly gestures, forgetting that these people of the rainforest had never seen anything two-dimensional in their lives. They picked up these photographs from the forest floor, tried to look behind the face to find the form or the figure, found nothing, and concluded that these were calling cards from the devil, so they speared the five missionaries to death. But the Waorani didn ’ t just spear outsiders. They speared each other. 54 percent of their mortality was due to them spearing each other. We traced genealogies back eight generations, and we found two instances of natural death and when we pressured the people a little bit about it, they admitted that one of the fellows had gotten so old that he died getting old, so we speared him anyway. But at the same time they had a perspicacious knowledge of the forest that was astonishing. Their hunters could smell animal urine at 40 paces and tell you what species left it behind. In the early ’ 80s, I had a really astonishing assignment when I was asked by my professor at Harvard if I was interested in going down to Haiti, infiltrating the secret societies which were the foundation of Duvalier ’ s strength and Tonton Macoutes, and securing the poison used to make zombies. In order to make sense out of sensation, of course, I had to understand something about this remarkable faith of Vodoun. And Voodoo is not a black magic cult. On the contrary, it ’ s a complex metaphysical worldview. It ’ s interesting. If I asked you to name the great religions of the world, what would you say? Christianity, Islam, Buddhism, Judaism, whatever. There ’ s always one continent left out, the assumption being that sub-Saharan Africa had no religious beliefs. Well, of course, they did and Voodoo is simply the distillation of these very profound religious ideas that came over during the tragic Diaspora of the slavery era. But, what makes Voodoo so interesting is that it ’ s this living relationship between the living and the dead. So, the living give birth to the spirits. The spirits can be invoked from beneath the Great Water, responding to the rhythm of the dance to momentarily displace the soul of the living, so that for that brief shining moment, the acolyte becomes the god. That ’ s why the Voodooists like to say that”You white people go to church and speak about God. We dance in the temple and become God.”And because you are possessed, you are taken by the spirit — how can you be **harmed**? So you see these astonishing demonstrations : Voodoo acolytes in a state of trance handling burning embers with impunity, a rather astonishing demonstration of the ability of the mind to affect the body that bears it when catalyzed in the state of extreme excitation. Now, of all the peoples that I ’ ve ever been with, the most extraordinary are the Kogi of the Sierra Nevada de Santa Marta in northern Colombia. Descendants of the ancient Tairona civilization which once carpeted the Caribbean coastal plain of Colombia, in the wake of the conquest, these people retreated into an isolated volcanic massif that soars above the Caribbean coastal plain. In a bloodstained continent, these people alone were never conquered by the Spanish. To this day, they remain ruled by a ritual priesthood but the training for the priesthood is rather extraordinary. The young acolytes are taken away from their families at the age of three and four, sequestered in a shadowy world of **darkness** in stone huts at the base of glaciers for 18 years : two nine-year periods deliberately chosen to mimic the nine months of gestation they spend in their natural mother ’ s womb ; now they are metaphorically in the womb of the great mother. And for this entire time, they are inculturated into the values of their society, values that maintain the proposition that their **prayers** and their **prayers** alone maintain the cosmic — or we might say the ecological — balance. And at the end of this amazing initiation, one day they ' re suddenly taken out and for the first time in their lives, at the age of 18, they see a sunrise. And in that crystal moment of awareness of first light as the Sun begins to bathe the slopes of the stunningly beautiful landscape, suddenly everything they have learned in the abstract is affirmed in stunning glory. And the priest steps back and says,"You see? It ' s really as I ' ve told you. It is that beautiful. It is yours to protect."They call themselves the"elder brothers"and they say we, who are the younger brothers, are the ones responsible for destroying the world. Now, this level of intuition becomes very important. Whenever we think of **indigenous** people and landscape, we either invoke Rousseau and the old canard of the"noble savage,"which is an idea racist in its simplicity, or alternatively, we invoke Thoreau and say these people are closer to the Earth than we are. Well, **indigenous** people are neither sentimental nor weakened by nostalgia. There ' s not a lot of room for either in the malarial swamps of the Asmat or in the chilling winds of Tibet, but they have, nevertheless, through time and ritual, forged a traditional mystique of the Earth that is based not on the idea of being self-consciously close to it, but on a far subtler intuition : the idea that the Earth itself can only exist because it is breathed into being by human consciousness. Now, what does that mean? It means that a young kid from the Andes who ' s raised to believe that that mountain is an Apu spirit that will direct his or her destiny will be a profoundly different human being and have a different relationship to that resource or that place than a young kid from Montana raised to believe that a mountain is a pile of rock ready to be mined. Whether it ' s the abode of a spirit or a pile of ore is irrelevant. What ' s interesting is the metaphor that defines the relationship between the individual and the natural world. I was raised in the forests of British Columbia to believe those forests existed to be cut. That made me a different human being than my friends amongst the Kwagiulth who believe that those forests were the abode of Huxwhukw and the Crooked Beak of Heaven and the cannibal spirits that dwelled at the north end of the world, spirits they would have to engage during their Hamatsa initiation. Now, if you begin to look at the idea that these cultures could create different realities, you could begin to understand some of their extraordinary discoveries. Take this plant here. It ' s a photograph I took in the Northwest Amazon just last April. This is ayahuasca, which many of you have heard about, the most powerful psychoactive preparation of the shaman ' s repertoire. What makes ayahuasca fascinating is not the sheer pharmacological potential of this preparation, but the elaboration of it. It ' s made really of two different sources : on the one hand, this woody liana which has in it a series of beta- carbolines, harmine, harmaline, mildly hallucinogenic — to take the vine alone is rather to have sort of blue hazy smoke drift across your consciousness — but it ' s mixed with the leaves of a shrub in the coffee family called Psychotria viridis. This plant had in it some very powerful tryptamines, very close to brain serotonin, dimethyltryptamine, 5-methoxydimethyltryptamine. If you ' ve ever seen the Yanomami blowing that snuff up their noses, that substance they make from a different set of species also contains methoxydimethyltryptamine. To have that powder blown up your nose is rather like being shot out of a rifle barrel lined with baroque paintings and landing on a sea of electricity. It doesn ' t create the distortion of reality ; it creates the dissolution of reality. In fact, I used to argue with my professor, Richard Evan Shultes — who is a man who sparked the psychedelic era with his discovery of the magic mushrooms in Mexico in the 1930s — I used to argue that you couldn ’ t classify these tryptamines as hallucinogenic because by the time you ’ re under the effects there ’ s no one home anymore to experience a hallucination. But the thing about tryptamines is they cannot be taken orally because they ’ re denatured by an enzyme found naturally in the human gut called monoamine oxidase. They can only be taken orally if taken in conjunction with some other chemical that denatures the MAO. Now, the fascinating things are that the beta- carbolines found within that liana are MAO inhibitors of the precise sort necessary to potentiate the tryptamine. So you ask yourself a question. How, in a flora of 80, 000 species of vascular plants, do these people find these two morphologically unrelated plants that when combined in this way, created a kind of biochemical version of the whole being greater than the sum of the parts? Well, we use that great euphemism,”trial and error,”which is exposed to be meaningless. But you ask the Indians, and they say,”The plants talk to us.”Well, what does that mean? This tribe, the Cofan, has 17 varieties of ayahuasca, all of which they distinguish a great distance in the forest, all of which are referable to our eye as one species. And then you ask them how they establish their taxonomy and they say,”I thought you knew something about plants. I mean, don ’ t you know anything?”And I said,”No.”Well, it turns out you take each of the 17 varieties in the night of a full moon, and it **sings** to you in a different key. Now, that ’ s not going to get you a Ph. D. at Harvard, but it ’ s a lot more interesting than counting stamens. Now — — the problem — the problem is that even those of us sympathetic with the plight of **indigenous** people view them as quaint and colorful but somehow reduced to the margins of history as the real world, meaning our world, moves on. Well, the truth is the 20th century, 300 years from now, is not going to be remembered for its wars or its technological innovations, but rather as the era in which we stood by and either actively endorsed or passively accepted the massive destruction of both biological and cultural diversity on the planet. Now, the problem isn ’ t change. All cultures through all time have constantly been engaged in a dance with new possibilities of life. And the problem is not technology itself. The Sioux Indians did not stop being Sioux when they gave up the bow and arrow any more than an American stopped being an American when he gave up the horse and buggy. It ’ s not change or technology that threatens the integrity of the ethnosphere. It is power, the crude face of domination. Wherever you look around the world, you discover that these are not cultures destined to fade away ; these are dynamic living peoples being driven out of existence by identifiable forces that are beyond their capacity to adapt to : whether it ’ s the egregious deforestation in the homeland of the Penan — a nomadic people from Southeast Asia, from Sarawak — a people who lived free in the forest until a generation ago, and now have all been reduced to servitude and prostitution on the banks of the rivers, where you can see the river itself is soiled with the silt that seems to be carrying half of Borneo away to the South China Sea, where the Japanese freighters hang light in the horizon ready to fill their holds with raw logs ripped from the forest — or, in the case of the Yanomami, it ’ s the disease entities that have come in, in the wake of the discovery of gold. Or if we go into the mountains of Tibet, where I ’ m doing a lot of research recently, you ’ ll see it ’ s a crude face of political domination. You know, genocide, the physical extinction of a people is universally condemned, but ethnocide, the destruction of people ’ s way of life, is not only not condemned, it ’ s universally, in many quarters, **celebrated** as part of a development strategy. And you cannot understand the pain of Tibet until you move through it at the ground level. I once travelled 6, 000 miles from Chengdu in Western China overland through southeastern Tibet to Lhasa with a young colleague, and it was only when I got to Lhasa that I understood the face behind the statistics you hear about : 6, 000 sacred monuments torn apart to dust and ashes, 1. 2 million people killed by the cadres during the Cultural Revolution. This young man ' s father had been ascribed to the Panchen Lama. That meant he was instantly killed at the time of the Chinese invasion. His uncle fled with His Holiness in the Diaspora that took the people to Nepal. His mother was incarcerated for the crime of being wealthy. He was smuggled into the jail at the age of two to hide beneath her skirt tails because she couldn ' t bear to be without him. The sister who had done that brave deed was put into an education camp. One day she inadvertently stepped on an armband of Mao, and for that transgression, she was given seven years of hard labor. The pain of Tibet can be impossible to bear, but the redemptive spirit of the people is something to behold. And in the end, then, it really comes down to a choice : do we want to live in a monochromatic world of monotony or do we want to embrace a polychromatic world of diversity? Margaret Mead, the great anthropologist, said, before she died, that her greatest fear was that as we drifted towards this blandly amorphous generic world view not only would we see the entire range of the human imagination reduced to a more narrow modality of thought, but that we would wake from a dream one day having forgotten there were even other possibilities. And it ' s humbling to remember that our species has, perhaps, been around for [150, 000] years. The Neolithic Revolution — which gave us agriculture, at which time we succumbed to the cult of the seed ; the poetry of the shaman was displaced by the prose of the priesthood ; we created hierarchy specialization surplus — is only 10, 000 years ago. The modern industrial world as we know it is barely 300 years old. Now, that shallow history doesn ' t suggest to me that we have all the answers for all of the challenges that will confront us in the ensuing millennia. When these myriad cultures of the world are asked the meaning of being human, they respond with 10, 000 different voices. And it ' s within that song that we will all rediscover the possibility of being what we are : a fully conscious species, fully aware of ensuring that all peoples and all gardens find a way to flourish. And there are great moments of optimism. This is a photograph I took at the northern tip of Baffin Island when I went narwhal hunting with some Inuit people, and this man, Olayuk, told me a marvelous story of his grandfather. The Canadian government has not always been kind to the Inuit people, and during the 1950s, to establish our sovereignty, we forced them into settlements. This old man ' s grandfather refused to go. The family, fearful for his life, took away all of his weapons, all of his tools. Now, you must understand that the Inuit did not fear the cold ; they took advantage of it. The runners of their sleds were originally made of fish wrapped in caribou hide. So, this man ' s grandfather was not intimidated by the Arctic night or the blizzard that was blowing. He simply slipped outside, pulled down his sealskin trousers and defecated into his hand. And as the feces began to freeze, he shaped it into the form of a blade. He put a spray of saliva on the edge of the shit knife and as it finally froze solid, he butchered a dog with it. He skinned the dog and improvised a harness, took the ribcage of the dog and improvised a sled, harnessed up an adjacent dog, and disappeared over the ice floes, shit knife in belt. Talk about getting by with nothing. And this, in many ways — — is a symbol of the resilience of the Inuit people and of all **indigenous** people around the world. The Canadian government in April of 1999 gave back to total control of the Inuit an area of land larger than California and Texas put together. It ' s our new homeland. It ' s called Nunavut. It ' s an independent territory. They control all mineral resources. An amazing example of how a nation-state can seek restitution with its people. And finally, in the end, I think it ' s pretty obvious at least to all of all us who ' ve traveled in these remote reaches of the planet, to realize that they ' re not remote at all. They ’ re homelands of somebody. They represent branches of the human imagination that go back to the dawn of time. And for all of us, the dreams of these children, like the dreams of our own children, become part of the naked geography of hope. So, what we ’ re trying to do at the National Geographic, finally, is, we believe that politicians will never accomplish anything. We think that polemics — — we think that polemics are not persuasive, but we think that storytelling can change the world, and so we are probably the best storytelling institution in the world. We get 35 million hits on our website every month. 156 nations carry our television channel. Our magazines are read by millions. And what we ’ re doing is a series of journeys to the ethnosphere where we ’ re going to take our audience to places of such cultural wonder that they cannot help but come away dazzled by what they have seen, and hopefully, therefore, embrace gradually, one by one, the central revelation of anthropology : that this world deserves to exist in a diverse way, that we can find a way to live in a truly multicultural, pluralistic world where all of the wisdom of all peoples can contribute to our collective well-being. Thank you very much.

Text Sample 411: POS Dim 3 – 2003 – Score 7.00 – 2003/t000434.txt

Good morning everyone. First of all, it ’ s been fantastic being here over these past few days. And secondly, I feel it ’ s a great honor to kind of wind up this extraordinary gathering of people, these amazing talks that we ’ ve had. I feel that I ’ ve fitted in, in many ways, to some of the things that I ’ ve heard. I came directly here from the deep, deep tropical rainforest in Ecuador, where I was out — you could only get there by a plane — with **indigenous** people with paint on their faces and parrot feathers on their headdresses, where these people are fighting to try and keep the oil companies, and keep the roads, out of their forests. They ’ re fighting to develop their own way of living within the forest in a world that ’ s clean, a world that isn ’ t contaminated, a world that isn ’ t polluted. And what was so amazing to me, and what fits right in with what we ’ re all talking about here at TED, is that there, right in the middle of this rainforest, was some solar panels — the first in that part of Ecuador — and that was mainly to bring water up by pump so that the women wouldn ’ t have to go down. The water was cleaned, but because they got a lot of batteries, they were able to store a lot of electricity. So every house — and there were, I think, eight houses in this little community — could have light for, I think it was about half an hour each evening. And there is the Chief, in all his regal finery, with a laptop computer. And this man, he has been outside, but he ’ s gone back, and he was saying,”You know, we have suddenly jumped into a whole new era, and we didn ’ t even know about the white man 50 years ago, and now here we are with laptop computers, and there are some things we want to learn from the modern world. We want to know about health care. We want to know about what other people do — we ’ re interested in it. And we want to learn other languages. We want to know English and French and perhaps Chinese, and we ’ re good at languages.”So there he is with his little laptop computer, but fighting against the might of the pressures — because of the debt, the foreign debt of Ecuador — fighting the pressure of World Bank, IMF, and of course the people who want to exploit the forests and take out the oil. And so, coming directly from there to here. But, of course, my real field of expertise lies in an even different kind of civilization — I can ’ t really call it a civilization. A different way of life, a different being. We ’ ve talked earlier — this wonderful talk by Wade Davis about the different cultures of the humans around the world — but the world is not composed only of human beings ; there are also other animal beings. And I propose to bring into this TED conference, as I always do around the world, the voice of the animal kingdom. Too often we just see a few slides, or a bit of film, but these beings have voices that mean something. And so, I want to give you a greeting, as from a chimpanzee in the forests of Tanzania — Ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh! I ' ve been studying chimpanzees in Tanzania since 1960. During that time, there have been modern technologies that have really transformed the way that field biologists do their work. For example, for the first time, a few years ago, by simply collecting little fecal samples we were able to have them analyzed — to have DNA profiling done — so for the first time, we actually know which male chimps are the fathers of each individual infant. Because the chimps have a very promiscuous mating society. So this opens up a whole new avenue of research. And we use GSI — geographic whatever it is, GSI — to determine the range of the chimps. And we ' re using — you can see that I ' m not really into this kind of stuff — but we ' re using satellite imagery to look at the deforestation in the area. And of course, there ' s developments in infrared, so you can watch animals at night, and equipment for recording by video, and tape recording is getting lighter and better. So in many, many ways, we can do things today that we couldn ' t do when I began in 1960. Especially when chimpanzees, and other animals with large brains, are studied in captivity, modern technology is helping us to search for the upper levels of cognition in some of these non-human animals. So that we know today, they ' re capable of performances that would have been thought absolutely impossible by science when I began. I think the chimpanzee in captivity who is the most skilled in intellectual performance is one called Ai in Japan — her name means love — and she has a wonderfully sensitive partner working with her. She loves her computer — she ' ll leave her big group, and her running water, and her trees and everything. And she ' ll come in to sit at this computer — it ' s like a video game for a kid ; she ' s hooked. She ' s 28, by the way, and she does things with her computer screen and a touch pad that she can do faster than most humans. She does very complex tasks, and I haven ' t got time to go into them, but the amazing thing about this female is she doesn ' t like making mistakes. If she has a bad run, and her score isn ' t good, she ' ll come and reach up and tap on the glass — because she can ' t see the experimenter — which is asking to have another go. And her concentration — she ' s already concentrated hard for 20 minutes or so, and now she wants to do it all over again, just for the satisfaction of having done it better. And the food is not important — she does get a tiny reward, like one raisin for a correct response — but she will do it for nothing, if you tell her beforehand. So here we are, a chimpanzee using a computer. Chimpanzees, gorillas, orangutans also learn human sign language. But the point is that when I was first in Gombe in 1960 — I remember so well, so vividly, as though it was yesterday — the first time, when I was going through the vegetation, the chimpanzees were still running away from me, for the most part, although some were a little bit acclimatized — and I saw this dark shape, hunched over a termite mound, and I peered with my binoculars. It was, fortunately, one adult male whom I ' d named David Greybeard — and by the way, science at that time was telling me that I shouldn ' t name the chimps ; they should all have numbers ; that was more scientific. Anyway, David Greybeard — and I saw that he was picking little pieces of grass and using them to fish termites from their underground nest. And not only that — he would sometimes pick a leafy twig and strip the leaves — modifying an object to make it suitable for a specific purpose — the beginning of tool-making. The reason this was so exciting and such a breakthrough is at that time, it was thought that humans, and only humans, used and made tools. When I was at school, we were defined as man, the toolmaker. So that when

Louis Leakey, my mentor, heard this news, he said,”Ah, we must now redefine ’ man, ’ redefine ’ tool, ’ or accept chimpanzees as humans.”We now know that at Gombe alone, there are nine different ways in which chimpanzees use different objects for different purposes. Moreover, we know that in different parts of Africa, wherever chimps have been studied, there are completely different tool-using behaviors. And because it seems that these patterns are passed from one generation to the next, through observation, imitation and practice — that is a definition of human culture. What we find is that over these 40-odd years that I and others have been studying chimpanzees and the other great apes, and, as I say, other mammals with complex brains and social systems, we have found that after all, there isn ’ t a sharp line dividing humans from the rest of the animal kingdom. It ’ s a very wuzzy line. It ’ s getting wuzzier all the time as we find animals doing things that we, in our arrogance, used to think was just human. The chimps — there ’ s no time to discuss their fascinating lives — but they have this long childhood, five years of suckling and sleeping with the mother, and then another three, four or five years of emotional dependence on her, even when the next child is born. The importance of learning in that time, when behavior is flexible — and there ’ s an awful lot to learn in chimpanzee society. The long-term affectionate supportive bonds that develop throughout this long childhood with the mother, with the brothers and sisters, and which can last through a lifetime, which may be up to 60 years. They can actually live longer than 60 in captivity, so we ’ ve only done 40 years in the wild so far. And we find chimps are capable of true compassion and altruism. We find in their non-verbal communication — this is very rich — they have a lot of sounds, which they use in different circumstances, but they also use touch, posture, gesture, and what do they do? They kiss ; they embrace ; they hold hands. They pat one another on the back ; they swagger ; they shake their fist — the kind of things that we do, and they do them in the same kind of context. They have very sophisticated cooperation. Sometimes they hunt — not that often, but when they hunt, they show sophisticated cooperation, and they share the prey. We find that they show emotions, similar to — maybe sometimes the same — as those that we describe in ourselves as happiness, sadness, fear, despair. They know mental as well as physical **suffering**. And I don ’ t have time to go into the information that will prove some of these things to you, save to say that there are very bright students, in the best universities, studying emotions in animals, studying personalities in animals. We know that chimpanzees and some other creatures can recognize themselves in mirrors —”self”as opposed to”other.”They have a sense of humor, and these are the kind of things which traditionally have been thought of as human prerogatives. But this teaches us a new respect — and it ’ s a new respect not only for the chimpanzees, I suggest, but some of the other amazing animals with whom we share this planet. Once we ’ re prepared to admit that after all, we ’ re not the only beings with personalities, minds and above all feelings, and then we start to think about ways we use and abuse so many other sentient, sapient creatures on this planet, it really gives cause for deep **shame**, at least for me. So, the sad thing is that these chimpanzees — who ’ ve perhaps taught us, more than any other creature, a little humility — are in the wild, disappearing very fast. They ’ re disappearing for the reasons that all of you in this room know only too well. The deforestation, the growth of human populations, needing more land. They ’ re disappearing because some timber companies go in with clear-cutting. They ’ re disappearing in the heart of their range in Africa because the big multinational logging companies have come in and made roads — as they want to do in Ecuador and other parts where the forests remain untouched — to take out oil or timber. And this has led in Congo basin, and other parts of the world, to what is known as the bush- meat trade. This means that although for hundreds, perhaps thousands of years, people have lived in those forests, or whatever habitat it is, in harmony with their world, just killing the animals they need for themselves and their families — now, suddenly, because of the roads, the hunters can go in from the towns. They shoot everything, every single thing that moves that ' s bigger than a small rat ; they sun-dry it or smoke it. And now they ' ve got transport ; they take it on the logging trucks or the mining trucks into the towns where they sell it. And people will pay more for bush-meat, as it ' s called, than for **domestic** meat — it ' s culturally preferred. And it ' s not sustainable, and the huge logging camps in the forest are now demanding meat, so the Pygmy hunters in the Congo basin who ' ve lived there with their wonderful way of living for so many hundreds of years are now corrupted. They ' re given weapons ; they shoot for the logging camps ; they get money. Their culture is being destroyed, along with the animals upon whom they depend. So, when the logging camp moves, there ' s nothing left. We talked already about the loss of human cultural diversity, and I ' ve seen it happening with my own eyes. And the grim picture in Africa — I love Africa, and what do we see in Africa? We see deforestation ; we see the desert spreading ; we see massive hunger ; we see disease and we see population growth in areas where there are more people living on a certain piece of land than the land can possibly support, and they ' re too poor to buy food from elsewhere. Were the people that we heard about yesterday, on the Easter Island, who cut down their last tree — were they stupid? Didn ' t they know what was happening? Of course, but if you ' ve seen the crippling poverty in some of these parts of the world it isn ' t a question of"Let ' s leave the tree for tomorrow.""How am I going to feed my family today? Maybe I can get just a few dollars from this last tree which will keep us going a little bit longer, and then we ' ll pray that something will happen to save us from the inevitable end."So, this is a pretty grim picture. The one thing we have, which makes us so different from chimpanzees or other living creatures, is this sophisticated spoken language — a language with which we can tell children about things that aren ' t here. We can talk about the distant past, plan for the distant future, discuss ideas with each other, so that the ideas can grow from the accumulated wisdom of a group. We can do it by talking to each other ; we can do it through video ; we can do it through the written word. And we are abusing this great power we have to be wise stewards, and we ' re destroying the world. In the developed world, in a way, it ' s worse, because we have so much **access** to knowledge of the stupidity of what we ' re doing. Do you know, we ' re bringing little babies into a world where, in many places, the water is poisoning them? And the air is **harming** them, and the food that ' s grown from the contaminated land is poisoning them. And that ' s not just in the far-away developing world ; that ' s everywhere. Do you know we all have about 50 chemicals in our bodies we didn ' t have about 50 years ago? And so many of these diseases, like asthma and certain kinds of cancers, are on the increase around places where our filthy toxic waste is dumped. We ' re **harming** ourselves around the world, as well as **harming** the animals, as well as **harming** nature herself — Mother Nature, that brought us into being ; Mother Nature, where I believe we need to spend time, where there ' s trees and flowers and birds for our good psychological development. And yet, there are hundreds and hundreds of children in the developed world who never see nature, because they ' re growing up in concrete and all they know is virtual reality, with no opportunity to go and lie in the sun, or in the forest, with the dappled sun-specks coming down from the canopy above. As I was traveling around the world, you know, I had to leave the forest — that ' s where I love to be. I had to leave these fascinating chimpanzees for my students and field staff to continue studying because, finding they dwindled from about two million 100 years ago to about 150, 000 now, I knew I had to leave the forest to do what I could to raise awareness around the world. And the more I talked about the chimpanzees ' plight, the more I realized the fact that everything ' s interconnected, and the problems of the developing world so often stem from the greed of the developed world, and everything was joining together, and making — not sense, hope lies in sense, you said — it ' s making a nonsense. How can we do it? Somebody said that yesterday. And as I was traveling around, I kept meeting young people who ' d lost hope. They were feeling despair, they were feeling,"Well, it doesn ' t matter what we do ; eat, drink and be merry, for tomorrow we die. Everything is hopeless — we ' re always being told so by the media."And then I met some who were angry, and anger that can turn to **violence**, and we ' re all familiar with that. And I have three little grandchildren, and when some of these students would say to me at high school or university, they ' d say,"We ' re angry,"or"We ' re filled with despair, because we feel you ' ve compromised our future, and there ' s nothing we can do about it."And I looked in the eyes of my little grandchildren, and think how much we ' ve **harmed** this planet since I was their age. I feel this deep **shame**, and that ' s why in 1991 in Tanzania, I started a program that ' s called Roots and Shoots. There ' s little brochures all around outside, and if any of you have anything to do with children and care about their future, I beg that you pick up that brochure. And Roots and Shoots is a program for hope. Roots make a firm foundation. Shoots seem tiny, but to reach the sun they can break through brick walls. See the brick walls as all the problems that we ' ve inflicted on this planet. Then, you see, it is a message of hope. Hundreds and thousands of young people around the world can break through, and can make this a better world. And the most important message of Roots and Shoots is that every single individual makes a difference. Every individual has a role to play. Every one of us impacts the world around us everyday, and you scientists know that you can ' t actually — even if you stay in bed all day, you ' re breathing oxygen and giving out CO2, and probably going to the loo, and things like that — you ' re making a difference in the world. So, the Roots and Shoots program involves youth in three kinds of projects. And these are projects to make the world around them a better place. One project to show care and concern for your own human community. One for animals, including **domestic** animals — and I have to say, I learned everything I know about animal behavior even before I got to Gombe and the chimps from my dog, Rusty, who was my childhood companion. And the third kind of project : something for the local environment. So what the kids do depends first of all, how old are they — and we go now from pre-school right through university. It ' s going to depend whether they ' re inner-city or rural. It ' s going to depend if they ' re wealthy or impoverished. It ' s going to depend which part, say, of America they ' re in. We ' re in every state now, and the problems in Florida are different from the problems in New York. It ' s going to depend on which country they ' re in — and we ' re already in 60-plus countries, with about 5, 000 active groups — and there are groups all over the place that I keep hearing about that I ' ve never even heard of, because the kids are taking the program and spreading it themselves. Why? Because they ' re buying into it, and they ' re the ones who get to decide what they ' re going to do. It isn ' t something that their parents tell them, or their teachers tell them. That ' s effective, but if they decide themselves,"We want to clean this river and put the fish back that used to be there. We want to clear away the toxic soil from this area and have an organic garden. We want to go and spend time with the old people and hear their stories and record their oral histories. We want to go and work in a dog shelter. We want to learn about animals. We want..."You know, it goes on and on, and this is very hopeful for me. As I travel around the world 300 days a year, everywhere there ’ s a group of Roots and Shoots of different ages. Everywhere there are children with shining eyes saying,”Look at the difference we ’ ve made.”And now comes the technology into it, because with this new way of communicating electronically these kids can communicate with each other around the world. And if anyone is interested to help us, we ’ ve got so many ideas but we need help — we need help to create the right kind of system that will help these young people to communicate their excitement. But also — and this is so important — to communicate their despair, to say,”We ’ ve tried this and it doesn ’ t work, and what shall we do?”And then, lo and behold, there ’ s another group answering these kids who may be in America, or maybe this is a group in Israel, saying,”Yeah, you did it a little bit wrong. This is how you should do it.”The philosophy is very simple. We do not believe in **violence**. No **violence**, no bombs, no guns. That ’ s not the way to solve problems. **Violence** leads to **violence**, at least in my view. So how do we solve? The tools for solving the problems are knowledge and understanding. Know the facts, but see how they fit in the big picture. Hard work and persistence — don ’ t give up — and love and compassion leading to respect for all life. How many more minutes? Two, one? Chris Anderson : One — one to two. Jane Goodall : Two, two, I ’ m going to take two. Are you going to come and drag me off? Anyway — so basically, Roots and Shoots is beginning to change young people ’ s lives. It ’ s what I ’ m devoting most of my energy to. And I believe that a group like this can have a very major impact, not just because you can share technology with us, but because so many of you have children. And if you take this program out, and give it to your children, they have such a good opportunity to go out and do good, because they ’ ve got parents like you. And it ’ s been so clear how much you all care about trying to make this world a better place. It ’ s very encouraging. But the kids do ask me — and this won ’ t take more than two minutes, I promise — the kids say,”Dr. Jane, do you really have hope for the future? You travel, you see all these horrible things happening.”Firstly, the human brain — I don ’ t need to say anything about that. Now that we know what the problems are around the world, human brains like yours are rising to solve those problems. And we ’ ve talked a lot about that. Secondly, the resilience of nature. We can destroy a river, and we can bring it back to life. We can see a whole area desolated, and it can be brought back to bloom again, with time or a little help. And thirdly, the last speaker talked about — or the speaker before last, talked about the indomitable human spirit. We are surrounded by the most amazing people who do things that seem to be absolutely impossible. Nelson Mandela — I take a little piece of limestone from Robben Island Prison, where he labored for 27 years, and came out with so little bitterness, he could lead his people from the horror of apartheid without a bloodbath. Even after the 11th of September — and I was in New York and I felt the fear — nevertheless, there was so much human courage, so much love and so much compassion. And then as I went around the country after that and felt the fear — the fear that was leading to people feeling they couldn ’ t worry about the environment any more, in case they seemed not to be patriotic — and I was trying to encourage them, somebody came up with a little quotation from Mahatma Gandhi,”If you look back through human history, you see that every evil regime has been overcome by good.”And just after that a woman brought me this little bell, and I want to end on this note. She said,”If you ’ re talking about hope and peace, ring this. This bell is made from metal from a defused landmine, from the killing fields of Pol Pot — one of the most evil regimes in human history — where people are now beginning to put their lives back together after the regime has crumbled. So, yes, there is hope, and where is the hope? Is it out there with the politicians? It ’ s in our hands. It ’ s in your hands and my hands and those of our children. It ’ s really up to us. We ’ re the ones who can make a difference. If we lead lives where we consciously leave the lightest possible ecological footprints, if we buy the things that are ethical for us to buy and don ’ t buy the things that are not, we can change the world overnight. Thank you.

Text Sample 412: POS Dim 3 – 2002 – Score 7.00 – 2002/t000086.txt
I am going to talk about myself, which I rarely do, because I â well for one thing, I prefer to talk about things I know nothing about. And secondly, I ’ m a recovering narcissist. I didn ’ t know I was a narcissist actually. I thought narcissism meant you loved yourself. And then someone told me there is a flip side to it. So it ’ s actually drearier than self-love ; it ’ s unrequited self-love. I don ’ t feel I can afford a relapse. But I want to, though, explain how I came to design my own particular brand of comedy because I ’ ve been through so many different forms of it. I started with improvisation, in a particular form of improvisation called theater games, which had one rule, which I always thought was a great rule for an ethic for a society. And the rule was, you couldn ’ t deny the other person ’ s reality, you could only build on it. And of course we live in a society that ’ s all about contradicting other peoples ’ reality. It ’ s all about contradiction, which I think is why I ’ m so sensitive to contradiction in general. I see it everywhere. Like polls. You know, it ’ s always curious to me that in public opinion polls the percentage of Americans who don ’ t know the answer to any given question is always two percent. 75 percent of Americans think Alaska is part of Canada. But only two percent don ’ t know the effect that the debacle in Argentina will have on the IMF ’ s monetary policy â seems a contradiction. Or this ad that I read in the New York Times :”Wearing a fine watch speaks loudly of your rank in society. Buying it from us screams good taste.”Or this that I found in a magazine called California Lawyer, in an article that is surely meant for the lawyers at Enron.”Surviving the Slammer : Do ’ s and Don ’ ts.””Don ’ t use big words.””Learn the lingua franca.”Yeah.”Lingua this, Frankie.”And I suppose it ’ s a contradiction that I talk about science when I don ’ t know math. You know, because â and by the way to I was so grateful to Dean Kamen for pointing out that one of the reasons, that there are cultural reasons that women and minorities don ’ t enter the fields of science and technology â because for instance, the reason I don ’ t do math is, I was taught to do math and read at the same time. So you ’ re six years old, you ’ re reading Snow White and the Seven Dwarves, and it becomes rapidly obvious that there are only two kinds of men in the world : dwarves and Prince Charmings. And the odds are seven to one against your finding the prince. That ’ s why little girls don ’ t do math. It ’ s too depressing. Of course, by talking about science I also may, as I did the other night, incur the **violent** wrath of some scientists who were very upset with me. I used the word postmodern as if it were OK. And they got very upset. One of them, to his credit, I think really just wanted to engage me in a serious argument. But I don ’ t engage in serious arguments. I don ’ t approve of them because arguments, of course, are all about contradiction, and they ’ re shaped by the values that I have questions with. I have questions with the values of Newtonian science, like rationality. You ’ re supposed to be rational in an argument. Well rationality is constructed by what Christie Hefner was talking about today, that mind-body split, you know? The head is good, body bad. Head is ego, body id. When we say”I,”â as when Rene Descartes said,”I think therefore I am,”â we mean the head. And as David Lee Roth **sang**

in”Just a Gigolo,””I ain ’ t got no body.”That ’ s how you get rationality. And that ’ s why so much of humor is the body asserting itself against the head. That ’ s why you have toilet humor and **sexual** humor. That ’ s why you have the Raspyni Brothers whacking Richard in the genital area. And we ’ re laughing doubly then because he ’ s the body, but it ’ s also â Voice offstage : Richard. Emily Levine : Richard. What did I say? Richard. Yes but it ’ s also the head, the head of the conference. That ’ s the other way that humor â like Art Buchwald takes shots at the heads of state. It doesn ’ t make quite as much money as body humor I ’ m sure â but nevertheless, what makes us treasure you and adore you. There ’ s also a contradiction in rationality in this country though, which is, as much as we revere the head, we are very anti-intellectual. I know this because I read in the New York Times, the Ayn Rand foundation took out a full- page ad after September 11, in which they said,”The problem is not Iraq or Iran, the problem in this country, facing this country is the university professors and their spawn.”So I went back and re-read”The Fountainhead.”I don ’ t know how many of you have read it. And I ’ m not an expert on sadomasochism. But let me just read you a couple of random passages from page 217.”The act of a master taking painful contemptuous possession of her, was the kind of rapture she wanted. When they lay together in bed it was, as it had to be, as the nature of the act demanded, an act of **violence**. It was an act of clenched teeth and **hatred**. It was the unendurable. Not a caress, but a wave of pain. The agony as an act of **passion**.”So you can imagine my surprise on reading in The New Yorker that Alan Greenspan, Chairman of the Federal Reserve, claims Ayn Rand as his intellectual mentor. It ’ s like finding out your nanny is a dominatrix. Bad enough we had to see J. Edgar Hoover in a dress. Now we have to picture Alan Greenspan in a black leather corset, with a butt tattoo that says,”Whip inflation now.”And Ayn Rand of course, Ayn Rand is famous for a philosophy called Objectivism, which reflects another value of Newtonian physics, which is objectivity. Objectivity basically is constructed in that same S&M way. It ’ s the subject subjugating the object. That ’ s how you assert yourself. You make yourself the active voice. And the object is the passive no-voice. I was so fascinated by that Oxygen commercial. I don ’ t know if you know this but â maybe it ’ s different now, or maybe you were making a statement â but in many hospital nurseries across the country, until very recently anyway, according to a book by Jessica Benjamin, the signs over the little boys cribs read,”I ’ m a boy,”and the signs over the little girls cribs read,”It ’ s a girl.”Yeah. So the passivity was culturally projected onto the little girls. And this still goes on as I think I told you last year. There ’ s a poll that proves â there was a poll that was given by Time magazine, in which only men were asked,”Have you ever had sex with a woman you actively disliked?”And well, yeah. Well, 58 percent said yes, which I think is overinflated though because so many men if you just say,”Have you ever had sex...””Yes!”They don ’ t even wait for the rest of it. And of course two percent did not know whether they ’ d had â That ’ s the first callback, of my attempted quadruple. So this subject-object thing, is part of something I ’ m very interested in because this is why, frankly, I believe in political correctness. I do. I think it can go too far. I think Ringling Brothers may have gone too far with an ad they took out in the New York Times Magazine.”We have a lifelong emotional and financial commitment to our Asian Elephant partners.”Maybe too far. But you know â I don ’ t think that a person of color making fun of white people is the same thing as a white person making fun of people of color. Or women making fun of men is the same as men making fun of women. Or poor people making fun of rich people, the same as rich people. I think you can make fun of the have but not the have-nots, which is why you don ’ t see me making fun of Kenneth Lay and his charming wife. What ' s funny about being down to four houses? And I really learned this lesson during the sex scandals of the Clinton administration or, Or as I call them, the good ol ' days. When people I knew, you know, people who considered themselves liberal, and everything else, were making fun of Jennifer Flowers and Paula Jones. Basically, they were making fun of them for being trailer trash or white trash. It seems, I suppose, a harmless **prejudice** and that you ' re not really hurting anybody. Until you read, as I did, an ad in the Los Angeles Times."For sale : White trash compactor."So this whole subject-object thing has relevance to humor in this way. I read a book by a woman named Amy Richlin, who is the chair of the Classics department at USC. And the book is called"The Garden of Priapus."And she says that Roman humor mirrors the construction of Roman society. So that Roman society was very top / bottom, as ours is to some degree. And so was humor. There always had to be the butt of a joke. So it was always the satirist, like Juvenal or Martial, represented the audience, and he was going to make fun of the outsider, the person who didn ' t share that subject status. And in stand-up of course, the stand-up comedian is supposed to dominate the audience. A lot of heckling is the tension of trying to make sure that the comedian is going to be able to dominate, and overcome the heckler. And I got good at that when I was in stand up. But I always hated it because they were dictating the terms of the interaction, in the same way that engaging in a serious argument determines the content, to some degree, of what you ' re talking about. And I was looking for a form that didn ' t have that. And so I wanted something that was more interactive. I know that word is so debased now by the use of it by Internet marketers. I really miss the old telemarketers now, I ' ll tell you that. I do, because at least there you stood a chance. You know? I used to actually hang up on them. But then I read in"Dear Abby"that that was rude. So the next time that one called I let him get halfway through his spiel and then I said,"You sound sexy."He hung up on me! But the interactivity allows the audience to shape what you ' re going to do as much as you shape their experience of the world. And that ' s really what I ' m looking for. And I was sort of, as I was starting to analyze what exactly it is that I do, I read a book called"Trickster Makes This World,"by Lewis Hyde. And it was like being psychoanalyzed. I mean he had laid it all out. And then coming to this conference, I realized that most everybody here shared those same qualities because really what trickster is is an agent of change. Trickster is a change agent. And the qualities that I ' m about to describe are the qualities that make it possible to make change happen. And one of these is boundary crossing. I think this is what so, in fact, infuriated the scientists. But I like to cross boundaries. I like to, as I said, talk about things I know nothing about. I hope that ' s my agent, because you aren ' t paying me anything. And I think it ' s good to talk about things I know nothing about because I bring a fresh viewpoint to it, you know? I ' m able to see the contradiction that you may not be able to see. Like for instance a mime once â or a meme as he called himself. He was a very selfish meme. And he said that I had to show more respect because it took up to 18 years to learn how to do mime properly. And I said,"Well, that ' s how you know only stupid people go into it."It only takes two years to learn how to talk. And you know people, this is the problem with quote, objectivity, unquote. When you ' re only surrounded by people who speak the same vocabulary as you, or share the same set of assumptions as you, you start to think that that ' s reality. Like economists, you know, their definition of rational, that we all act out of our own economic self-interest. Well, look at Michael Hawley, or look at Dean Kamen, or look at my grandmother. My grandmother always acted in other people ' s interests, whether they wanted her to or not. If they had had an Olympics in martyrdom, my grandmother would have lost on purpose.”No, you take the prize. You ’ re young. I ’ m old. Who ’ s going to see it? Where am I going? I ’ m going to die soon.”So that ’ s one â this boundary crossing, this go-between which â Fritz Lanting, is that his name, actually said that he was a go-between. That ’ s an actual quality of the trickster. And another is, non-oppositional strategies. And this is instead of contradiction. Where you deny the other person ’ s reality, you have paradox where you allow more than one reality to coexist, I think there ’ s another philosophical construction. I ’ m not sure what it ’ s called. But my example of it is a sign that I saw in a jewelry store. It said,”Ears pierced while you wait.”There the alternative just boggles the imagination.”Oh no. Thanks though, I ’ ll leave them here. Thanks very much. I have some errands to run. So I ’ ll be back to pick them up around five, if that ’ s OK with you. Huh? Huh? What? Can ’ t hear you.”And another attribute of the trickster is smart luck. That accidents, that Louis Kahn, who talked about accidents, this is another quality of the trickster. The trickster has a mind that is prepared for the unprepared. That, and I will say this to the scientists, that the trickster has the ability to hold his ideas lightly so that he can let room in for new ideas or to see the contradictions or the hidden problems with his ideas. I had no joke for that. I just wanted to put the scientists in their place. But here ’ s how I think I like to make change, and that is in making connections. This is what I tend to see almost more than contradictions. Like the, what do you call those toes of the gecko? You know, the toes of the gecko, curling and uncurling like the fingers of Michael Moschen. I love connections. Like I ’ ll read that one of the two attributes of matter in the Newtonian universe â there are two attributes of matter in the Newtonian universe â one is space occupancy. Matter takes up space. I guess the more you matter the more space you take up, which explains the whole SUV phenomenon. And the other one though is impenetrability. Well, in ancient Rome, impenetrability was the criterion of masculinity. Masculinity depended on you being the active penetrator. And then, in economics, there ’ s an active producer and a passive consumer, which explains why business always has to penetrate new markets. Well yeah, I mean why we forced China to open her markets. And didn ’ t that feel good? And now we ’ re being penetrated. You know the biotech companies are actually going inside us and planting their little flags on our genes. You know we ’ re being penetrated. And I suspect, by someone who actively dislikes us. That ’ s the second of the quadruple. Yes of course you got that. Thank you very much. I still have a way to go. And what I hope to do, when I make these connections, is short circuit people ’ s thinking. You know, make you not follow your usual train of association, but make you rewire. It literally â when people say about the shock of recognition, it ’ s literally re-cognition, rewiring how you think â I had a joke to go with this and I forgot it. I ’ m so sorry. I ’ m getting like the woman in that joke about â have you heard this joke about the woman driving with her mother? And the mother is elderly. And the mother goes right through a red light. And the daughter doesn ’ t want to say anything. She doesn ’ t want to be like,”You ’ re too old to drive.”And the mother goes through a second red light. And the daughter says, as tactfully as possible,”Mom, are you aware that you just went through two red lights?”And the mother says,”Oh! Am I driving?”And that ’ s the shock of recognition at the shock of recognition. That completes the quadruple. I just want to say two more things. One is, another characteristic of trickster is that the trickster has to walk this fine line. He has to have poise. And you know the biggest hurdle for me, in doing what I do, is constructing my performance so that it ’ s prepared and unprepared. Finding the balance between those things is always dangerous because you might tip off too much in the direction of unprepared. But being too prepared doesn ’ t leave room for the accidents to happen. I was thinking about what Moshe Safdie said yesterday about beauty because in his book, Hyde says that sometimes trickster can tip over into beauty. But to do that you have to lose all the other qualities because once you ’ re into beauty you ’ re into a finished thing. You ’ re into something that occupies space and inhabits time. It ’ s an actual thing. And it is always extraordinary to see a thing of beauty. But if you don ’ t do that, if you allow for the accident to keep on happening, you have the possibility of getting on a wavelength. I like to think of what I do as a probability wave. When you go into beauty the probability wave collapses into one possibility. And I like to explore all the possibilities in the hope that you ’ ll be on the wavelength of your audience. And the one final quality I want to say about trickster is that he doesn ’ t have a home. He ’ s always on the road. I want to say to you Richard, in closing, that in TED you ’ ve created a home. And thank you for inviting me into it. Thank you very much.

Text Sample 413: POS Dim 3 – 2002 – Score 6.00 – 2002/t000428.txt

As you pointed out, every time you come here, you learn something. This morning, the world ’ s experts from I guess three or four different companies on building seats, I think concluded that ultimately, the solution is, people shouldn ’ t sit down. I could have told them that. Yesterday, the automotive guys gave us some new insights. They pointed out that, I believe it was between 30 and 50 years from today, they will be steering cars by wire, without all that mechanical stuff. That ’ s reassuring. They then pointed out that there ’ d be, sort of, the other controls by wire, to get rid of all that mechanical stuff. That ’ s pretty good, but why not get rid of the wires? Then you don ’ t need anything to control the car, except thinking about it. I would love to talk about the technology, and sometime, in what ’ s past the 15 minutes, I ’ ll be happy to talk to all the techno-geeks around here about what ’ s in here. But if I had one thing to say about this, before we get to first, it would be that from the time we started building this, the big idea wasn ’ t the technology. It really was a big idea in technology when we started applying it in the iBOT for the **disabled** community. The big idea here is, I think, a new piece of a solution to a fairly big problem in transportation. And maybe to put that in perspective : there ’ s so much data on this, I ’ ll be happy to give it to you in different forms. You never know what strikes the fancy of whom, but everybody is perfectly willing to believe the car changed the world. And Henry Ford, just about 100 years ago, started cranking out Model Ts. What I don ’ t think most people think about is the context of how technology is applied. For instance, in that time, 91 percent of America lived either on farms or in small towns. So, the car — the horseless carriage that replaced the horse and carriage — was a big deal ; it went twice as fast as a horse and carriage. It was half as long. And it was an environmental improvement, because, for instance, in 1903 they outlawed horses and buggies in downtown Manhattan, because you can imagine what the roads look like when you have a million horses, and a million of them urinating and doing other things, and the typhoid and other problems created were almost unimaginable. So the car was the clean environmental alternative to a horse and buggy. It also was a way for people to get from their farm to a farm, or their farm to a town, or from a town to a city. It all made sense, with 91 percent of the people living there. By the 1950s, we started connecting all the towns together with what a lot of people claim is the eighth wonder of the world, the highway system. And it is certainly a wonder. And by the way, as I take shots at old technologies, I want to assure everybody, and particularly the automotive industry — who ’ s been very supportive of us — that I don ’ t think this in any way competes with airplanes, or cars. But think about where the world is today. 50 percent of the global population now lives in cities. That ’ s 3. 2 billion people. We ’ ve solved all the transportation problems that have changed the world to get it to where we are today. 500 years ago, sailing ships started getting reliable enough ; we found a new continent. 150 years ago, locomotives got efficient enough, steam power, that we turned the continent into a country. Over the last hundred years, we started building cars, and then over the 50 years we ’ ve connected every city to every other city in an extraordinarily efficient way, and we have a very high standard of living as a consequence of that. But during that entire process, more and more people have been born, and more and more people are moving to cities. China alone is going to move four to six hundred million people into cities in the next decade and a half. And so, nobody, I think, would argue that airplanes, in the last 50 years, have turned the continent and the country now into a **neighborhood**. And if you just look at how technology has been applied, we ’ ve solved all the long-range, high-speed, high-volume, large- weight problems of moving things around. Nobody would want to give them up. And I certainly wouldn ’ t want to give up my airplane, or my helicopter, or my Humvee, or my Porsche. I love them all. I don ’ t keep any of them in my living room. The fact is, the last mile is the problem, and half the world now lives in dense cities. And people spend, depending on who they are, between 90 and 95 percent of their energy getting around on foot. I think there ’ s — I don ’ t know what data would impress you, but how about, 43 percent of the refined fuel produced in the world is consumed by cars in metropolitan areas in the United States. Three million people die every year in cities due to bad air, and almost all particulate pollution on this planet is produced by transportation devices, particularly sitting in cities. And again, I say that not to attack any industry, I think — I really do — I love my airplane, and cars on highways moving 60 miles an hour are extraordinarily efficient, both from an engineering point of view, an energy consumption point of view, and a utility point of view. And we all love our cars, and I do. The problem is, you get into the city and you want to go four blocks, it ’ s neither fun nor efficient nor productive. It ’ s not sustainable. If — in China, in the year 1998, 417 million people used bicycles ; 1. 7 million people used cars. If five percent of that population became, quote, middle class, and wanted to go the way we ’ ve gone in the last hundred years at the same time that 50 percent of their population are moving into cities of the size and density of Manhattan, every six weeks — it isn ’ t sustainable environmentally ; it isn ’ t sustainable economically — there just ain ’ t enough oil — and it ’ s not sustainable politically. I mean, what are we fighting over right now? We can make it complicated, but what ’ s the world fighting over right now? So it seemed to me that somebody had to work on that last mile, and it was dumb luck. We were working on iBOTs, but once we made this, we instantly decided it could be a great alternative to jet skis. You don ’ t need the water. Or snowmobiles. You don ’ t need the snow. Or skiing. It ’ s just fun, and people love to move around doing fun things. And every one of those industries, by the way — just golf carts alone is a multi-billion-dollar industry. But rather than go license this off, which is what we normally do, it seemed to me that if we put all our effort not into the technology, but into an understanding of a world that ’ s solved all its other problems, but has somehow come to accept that cities — which, right back from ancient Greece on, were meant to walk around, cities that were architected and built for people — now have a footprint that, while we ’ ve solved every other transportation problem — and it ’ s like Moore ’ s law. I mean, look at the time it took to cross a continent in a Conestoga wagon, then on a railroad, then an airplane. Every other form of transportation ’ s been improved. In 5, 000 years, we ’ ve gone backwards in getting around cities. They ’ ve gotten bigger ; they ’ re spread out. The most expensive real estate on this planet in every city — Wilshire Boulevard, or Fifth Avenue, or Tokyo, or Paris — the most expensive real estate is their downtowns. 65 percent of the landmass of our cities are parked cars. The 20 largest cities in the world. So you wonder, what if cities could give to their pedestrians what we take for granted as we now go between cities? What if you could make them fun, attractive, clean, environmentally friendly? What if it would make it a little bit more palatable to have **access** via this, as that last link to mass transit, to get out to your cars so we can all live in the suburbs and use our cars the way we want, and then have our cities energized again? We thought it would be really neat to do that, and one of the problems we really were worried about is : how do we get legal on the **sidewalk**? Because technically I ’ ve got motors ; I ’ ve got wheels — I ’ m a motor vehicle. I don ’ t look like a motor vehicle. I have the same footprint as a pedestrian ; I have the same unique capability to deal with other pedestrians in a crowded space. I took this down to Ground Zero, and knocked my way through crowds for an hour. I ’ m a pedestrian. But the law typically lags technology by a generation or two, and if we get told we don ’ t belong on the **sidewalk**, we have two choices. We ’ re a recreational vehicle that doesn ’ t really matter, and I don ’ t spend my time doing that kind of stuff. Or maybe we should be out in the **street** in front of a Greyhound bus or a vehicle. We ’ ve been so concerned about that, we went to the Postmaster General of the United States, as the first person we ever showed on the outside, and said,”Put your people on it. Everybody trusts their postman. And they belong on the **sidewalks**, and they ’ ll use it seriously.”He agreed. We went to a number of police departments that want their police officers back in the **neighborhood** on the beat, carrying 70 pounds of stuff. They love it. And I can ’ t believe a policeman is going to give themselves a ticket. So we ’ ve been working really, really hard, but we knew that the technology would not be as hard to develop as an **attitude** about what ’ s important, and how to apply the technology. We went out and we found some visionary people with enough money to let us design and build these things, and in hopefully enough time to get them accepted. So, I ’ m happy, really, I am happy to talk about this technology as much as you want. And yes, it ’ s really fun, and yes, you should all go out and try it. But if I could ask you to do one thing, it ’ s not to think about it as a piece of technology, but just imagine that, although we all understand somehow that it ’ s reasonable that we use our 4, 000-pound machine, which can go 60 miles an hour, that can bring you everywhere you want to go, and somehow it ’ s also what we used for the last mile, and it ’ s broken, and it doesn ’ t work. One of the more exciting things that occurred to us about why it might get accepted, happened out here in California. A few weeks ago, after we launched it, we were here with a news crew on Venice Beach, zipping up and back, and he ’ s marveling at the technology, and meanwhile bicycles are zipping by, and skateboarders are zipping by, and a little old lady — I mean, if you looked in the dictionary, a little old lady — came by me — and now that I ’ m on this, I ’ m the height of a normal adult now — and she just stops, and the camera is there, and she looks up at me and says,”Can I try that?”And what was I — you know, how are you going to say anything? And so I said,”Sure.”So I get off, and she gets on, and with a little bit of the usual, ah, then she turns around, and she goes about 20 feet, and she turns back around, and she ’ s all smiles. And she comes back to me and she stops, and she says,”Finally, they made something for us.”And the camera is looking down at her. I ’ m thinking,”Wow, that was great — — please lady, don ’ t say another word.”And the camera is down at her, and this guy has to put the microphone in her face, said,”What do you mean by that?”And I figured,”It ’ s all over now,”and she looks up and she says,”Well,”she ’ s still watching these guys go ; she says,”I can ’ t ride a bike,”no, she says,”I can ’ t use a skateboard, and I ’ ve never used roller blades,”she knew them by name ; she says,”And it ’ s been 50 years since I rode a bicycle.”Then she looks up, she ’ s looking up, and she says,”And I ’ m 81 years old, and I don ’ t drive a car anymore. I still have to get to the store, and I can ’ t carry a lot of things.”And it suddenly occurred to me, that among my many fears, were not just that the bureaucracy and the regulators and the legislators might not get it — it was that, fundamentally, you believe there ’ s pressure among the people not to invade the most precious little bit of space left, the **sidewalks** in these cities. When you look at the 36 inches of legal requirement for **sidewalk**, then the eight foot for the parked car, then the three lanes, and then the other eight feet — it ’ s — that little piece is all that ’ s there. But she looks up and says this, and it occurs to me, well, kids aren ’ t going to mind these things, and they don ’ t vote, and business people and then young adults aren ’ t going to mind these things — they ’ re pretty cool — so I guess subliminally I was worried that it ’ s the older population that ’ s going to worry. So, having seen this, and having worried about it for eight years, the first thing I do is pick up my phone and ask our marketing and regulatory guys, call AARP, get an appointment right away. We ’ ve got to show them this thing. And they took it to Washington ; they showed them ; and they ’ re going to be involved now, watching how these things get absorbed in a number of cities, like Atlanta, where we ’ re doing trials to see if it really can, in fact, help re-energize their downtown. The bottom line is, whether you believe the United Nations, or any of the other think tanks — in the next 20 years, all human population growth on this planet will be in cities. In Asia alone, it will be over a billion people. They learned to start with cell phones. They didn ’ t have to take the 100-year trip we took. They start at the top of the technology food chain. We ’ ve got to start building cities and human environments where a 150-pound person can go a couple of miles in a dense, rich, green-space environment, without being in a 4, 000-pound machine to do it. Cars were not meant for parallel parking ; they ’ re wonderful machines to go between cities, but just think about it : we ’ ve solved all the long-range, high-speed problems. The Greeks went from the theater of Dionysus to the Parthenon in their sandals. You do it in your sneakers. Not much has changed. If this thing goes only three times as fast as walking — three times — a 30-minute walk becomes 10 minutes. Your choice, when living in a city, if it ’ s now 10 minutes — because at 30 minutes you want an alternative, whether it ’ s a bus, a train — we ’ ve got to build an infrastructure — a light rail — or you ’ re going to keep parking those cars. But if you could put a pin in most cities, and imagine how far you could, if you had the time, walk in one half-hour, it ’ s the city. If you could make it fun, and make it eight or 10 minutes, you can ’ t find your car, un-park your car, move your car, re-park your car and go somewhere ; you can ’ t get to a cab or a subway. We could change the way people allocate their resources, the way this planet uses its energy, make it more fun. And we ’ re hoping to some extent history will say we were right. That ’ s Segway. This is a Stirling cycle engine ; this had been confused by a lot of things we ’ re doing. This little beast, right now, is producing a few hundred watts of electricity. Yes, it could be attached to this, and yes, on a kilogram of propane, you could drive from New York to Boston if you so choose. Perhaps more interesting about this little engine is it ’ ll burn any fuel, because some of you might be skeptical about the capability of this to have an impact, where most of the world you can ’ t simply plug into your 120-volt outlet. We ’ ve been working on this, actually, as an alternative energy source, starting way back with Johnson & Johnson, to run an iBOT, because the best batteries you could get — 10 watt-hours per kilogram in lead, 20 watt-hours per kilogram nickel- cadmium, 40 watt-hours per kilogram in nickel-metal hydride, 60 watt-hours per kilogram in lithium, 8, 750 watt-hours of energy in every kilogram of propane or gasoline — which is why nobody drives electric cars. But, in any event, if you can burn it with the same efficiency — because it ' s external combustion — as your kitchen stove, if you can burn any fuel, it turns out to be pretty neat. It makes just enough electricity to, for instance, do this, which at night is enough electricity, in the rest of the world, as Mr. Holly — Dr. Holly — pointed out, can run computers and a light bulb. But more interestingly, the thermodynamics of this say, you ' re never going to get more than 20 percent efficiency. It doesn ' t matter much — it says if you get 200 watts of electricity, you ' ll get 700 or 800 watts of heat. If you wanted to boil water and re-condense it at a rate of 10 gallons an hour, it takes about 25, a little over 25. 3 kilowatt — 25, 000 watts of continuous power — to do it. That ' s so much energy, you couldn ' t afford to desalinate or clean water in this country that way. Certainly, in the rest of the world, your choice is to devastate the place, turning everything that will burn into heat, or drink the water that ' s available. The number one cause of death on this planet among humans is bad water. Depending on whose numbers you believe, it ' s between 60 and 85, 000 people per day. We don ' t need sophisticated heart transplants around the world. We need water. And women shouldn ' t have to spend four hours a day looking for it, or watching their kids die. We figured out how to put a vapor- compression distiller on this thing, with a counter-flow heat exchanger to take the waste heat, then using a little bit of the electricity control that process, and for 450 watts, which is a little more than half of its waste heat, it will make 10 gallons an hour of distilled water from anything that comes into it to cool it. So if we put this box on here in a few years, could we have a solution to transportation, electricity, and communication, and maybe drinkable water in a sustainable package that weighs 60 pounds? I don ' t know, but we ' ll try it. I better shut up.

Text Sample 414: POS Dim 3 – 2002 – Score 6.00 – 2002/t000410.txt

That splendid music, the coming-in music,"The Elephant March"from"Aida,"is the music I ' ve chosen for my funeral. And you can see why. It ' s triumphal. I won ' t feel anything, but if I could, I would feel triumphal at having lived at all, and at having lived on this splendid planet, and having been given the opportunity to understand something about why I was here in the first place, before not being here. Can you understand my quaint English accent? Like everybody else, I was entranced yesterday by the animal session. Robert Full and Frans Lanting and others ; the beauty of the things that they showed. The only slight jarring note was when Jeffrey Katzenberg said of the mustang,"the most splendid creatures that God put on this earth."Now of course, we know that he didn ' t really mean that, but in this country at the moment, you can ' t be too careful. I ' m a biologist, and the central theorem of our subject : the theory of design, Darwin ' s theory of evolution by natural selection. In professional circles everywhere, it ' s of course universally accepted. In non-professional circles outside America, it ' s largely ignored. But in non-professional circles within America, it arouses so much hostility — it ' s fair to say that American biologists are in a state of war. The war is so worrying at present, with court cases coming up in one state after another, that I felt I had to say something about it. If you want to know what I have to say about Darwinism itself, I ' m afraid you ' re going to have to look at my books, which you won ' t find in the bookstore outside. Contemporary court cases often concern an allegedly new version of creationism, called”Intelligent Design,”or ID. Don ’ t be fooled. There ’ s nothing new about ID. It ’ s just creationism under another name, rechristened — I choose the word advisedly — for tactical, political reasons. The arguments of so-called ID theorists are the same old arguments that had been refuted again and again, since Darwin down to the present day. There is an effective evolution lobby coordinating the fight on behalf of science, and I try to do all I can to help them, but they get quite upset when people like me dare to mention that we happen to be atheists as well as evolutionists. They see us as rocking the boat, and you can understand why. Creationists, lacking any coherent scientific argument for their case, fall back on the popular phobia against atheism : Teach your children evolution in biology class, and they ’ ll soon move on to drugs, grand larceny and **sexual**”pre-version.”In fact, of course, educated theologians from the Pope down are firm in their support of evolution. This book,”Finding Darwin ’ s God,”by Kenneth Miller, is one of the most effective attacks on Intelligent Design that I know and it ’ s all the more effective because it ’ s written by a devout Christian. People like Kenneth Miller could be called a”godsend”to the evolution lobby, because they expose the lie that evolutionism is, as a matter of fact, tantamount to atheism. People like me, on the other hand, rock the boat. But here, I want to say something nice about creationists. It ’ s not a thing I often do, so listen carefully. I think they ’ re right about one thing. I think they ’ re right that evolution is fundamentally hostile to religion. I ’ ve already said that many individual evolutionists, like the Pope, are also religious, but I think they ’ re deluding themselves. I believe a true understanding of Darwinism is deeply corrosive to religious faith. Now, it may sound as though I ’ m about to preach atheism, and I want to reassure you that that ’ s not what I ’ m going to do. In an audience as sophisticated as this one, that would be preaching to the **choir**. No, what I want to urge upon you — Instead, what I want to urge upon you is militant atheism. But that ’ s putting it too negatively. If I was a person who were interested in preserving religious faith, I would be very afraid of the positive power of evolutionary science, and indeed science generally, but evolution in particular, to inspire and enthrall, precisely because it is atheistic. Now, the difficult problem for any theory of biological design is to explain the massive statistical improbability of living things. Statistical improbability in the direction of good design —”complexity”is another word for this. The standard creationist argument — there is only one ; they ’ re all reduced to this one — takes off from a statistical improbability. Living creatures are too complex to have come about by chance ; therefore, they must have had a designer. This argument of course, shoots itself in the foot. Any designer capable of designing something really complex has to be even more complex himself, and that ’ s before we even start on the other things he ’ s expected to do, like forgive sins, bless marriages, listen to **prayers** — favor our side in a war — disapprove of our sex lives, and so on. Complexity is the problem that any theory of biology has to solve, and you can ’ t solve it by postulating an agent that is even more complex, thereby simply compounding the problem. Darwinian natural selection is so stunningly elegant because it solves the problem of explaining complexity in terms of nothing but simplicity. Essentially, it does it by providing a smooth ramp of gradual, step-by-step increment. But here, I only want to make the point that the elegance of Darwinism is corrosive to religion, precisely because it is so elegant, so parsimonious, so powerful, so economically powerful. It has the sinewy economy of a beautiful suspension bridge. The God theory is not just a bad theory. It turns out to be — in principle — incapable of doing the job required of it. So, returning to tactics and the evolution lobby, I want to argue that rocking the boat may be just the right thing to do. My approach to attacking creationism is — unlike the evolution lobby — my approach to attacking creationism is to attack religion as a whole. And at this point I need to acknowledge the remarkable taboo against speaking **ill** of religion, and I ’ m going to do so in the words of the late Douglas Adams, a dear friend who, if he never came to TED, certainly should have been invited. Richard Dawkins : He was. Good. I thought he must have been. He begins this speech, which was tape recorded in Cambridge shortly before he died — he begins by explaining how science works through the testing of hypotheses that are framed to be vulnerable to disproof, and then he goes on. I quote,”Religion doesn ’ t seem to work like that. It has certain ideas at the heart of it, which we call ’ sacred ’ or ’ holy. ’ What it means is : here is an idea or a notion that you ’ re not allowed to say anything bad about. You ’ re just not. Why not? Because you ’ re not.””Why should it be that it ’ s perfectly legitimate to support the Republicans or Democrats, this model of economics versus that, Macintosh instead of Windows, but to have an opinion about how the universe began, about who created the universe — no, that ’ s holy. So, we ’ re used to not challenging religious ideas, and it ’ s very interesting how much of a furor Richard creates when he does it.”— He meant me, not that one.”Everybody gets absolutely frantic about it, because you ’ re not allowed to say these things. Yet when you look at it rationally, there ’ s no reason why those ideas shouldn ’ t be as open to debate as any other, except that we ’ ve agreed somehow between us that they shouldn ’ t be.”And that ’ s the end of the quote from Douglas. In my view, not only is science corrosive to religion ; religion is corrosive to science. It teaches people to be satisfied with trivial, supernatural non- explanations, and blinds them to the wonderful, real explanations that we have within our grasp. It teaches them to accept authority, revelation and faith, instead of always insisting on evidence. There ’ s Douglas Adams, magnificent picture from his book,”Last Chance to See.”Now, there ’ s a typical scientific journal, The Quarterly Review of Biology. And I ’ m going to put together, as guest editor, a special issue on the question,”Did an asteroid kill the dinosaurs?”And the first paper is a standard scientific paper, presenting evidence,”Iridium layer at the K-T boundary, and potassium argon dated crater in Yucatan, indicate that an asteroid killed the dinosaurs.”Perfectly ordinary scientific paper. Now, the next one.”The President of the Royal Society has been vouchsafed a strong inner conviction that an asteroid killed the dinosaurs.””It has been privately revealed to Professor Huxtane that an asteroid killed the dinosaurs.””Professor Hordley was brought up to have total and unquestioning faith”— —”that an asteroid killed the dinosaurs.””Professor Hawkins has promulgated an official dogma binding on all loyal Hawkinsians that an asteroid killed the dinosaurs.”That ’ s inconceivable, of course. But suppose — [Supporters of the Asteroid Theory cannot be patriotic citizens] In 1987, a reporter asked George Bush, Sr. whether he recognized the **equal** citizenship and patriotism of Americans who are atheists. Mr. Bush ’ s **reply** has become infamous.”No, I don ’ t know that atheists should be considered citizens, nor should they be considered patriots. This is one nation under God.”Bush ’ s bigotry was not an isolated mistake, blurted out in the heat of the moment and later retracted. He stood by it in the face of repeated calls for clarification or withdrawal. He really meant it. More to the point, he knew it posed no threat to his election — quite the contrary. Democrats as well as Republicans parade their religiousness if they want to get elected. Both parties invoke”one nation under God.”What would Thomas Jefferson have said? [In every country and in every age, the priest has been hostile to liberty] Incidentally, I ’ m not usually very proud of being British, but you can ’ t help making the comparison. In practice, what is an atheist? An atheist is just somebody who feels about Yahweh the way any decent Christian feels about Thor or Baal or the golden calf. As has been said before, we are all atheists about most of the gods that humanity has ever believed in. Some of us just go one god further. And however we define atheism, it ’ s surely the kind of academic belief that a person is entitled to hold without being vilified as an unpatriotic, unelectable non-citizen. Nevertheless, it ’ s an undeniable fact that to own up to being an atheist is tantamount to introducing yourself as Mr. Hitler or Miss Beelzebub. And that all stems from the perception of atheists as some kind of weird, way-out minority. Natalie Angier wrote a rather sad piece in the New Yorker, saying how lonely she felt as an atheist. She clearly feels in a beleaguered minority. But actually, how do American atheists stack up numerically? The latest survey makes surprisingly encouraging reading. Christianity, of course, takes a massive lion ’ s share of the population, with nearly 160 million. But what would you think was the second largest group, convincingly outnumbering Jews with 2. 8 million, Muslims at 1. 1 million, Hindus, Buddhists and all other religions put together? The second largest group, with nearly 30 million, is the one described as non-religious or secular. You can ’ t help wondering why vote-seeking politicians are so proverbially overawed by the power of, for example, the Jewish lobby — the state of Israel seems to owe its very existence to the American Jewish vote — while at the same time, consigning the non-religious to political oblivion. This secular non-religious vote, if properly mobilized, is nine times as numerous as the Jewish vote. Why does this far more substantial minority not make a move to exercise its political muscle? Well, so much for quantity. How about quality? Is there any correlation, positive or negative, between intelligence and tendency to be religious? [Them folks misunderestimated me] The survey that I quoted, which is the ARIS survey, didn ’ t break down its data by socio-economic class or education, IQ or anything else. But a recent article by Paul G. Bell in the Mensa magazine provides some straws in the wind. Mensa, as you know, is an international organization for people with very high IQ. And from a meta-analysis of the literature, Bell concludes that, I quote —”Of 43 studies carried out since 1927 on the relationship between religious belief, and one ’ s intelligence or educational level, all but four found an inverse connection. That is, the higher one ’ s intelligence or educational level, the less one is likely to be religious.”Well, I haven ’ t seen the original 42 studies, and I can ’ t comment on that meta-analysis, but I would like to see more studies done along those lines. And I know that there are — if I could put a little plug here — there are people in this audience easily capable of financing a massive research survey to settle the question, and I put the suggestion up, for what it ’ s worth. But let me know show you some data that have been properly published and analyzed, on one special group — namely, top scientists. In 1998, Larson and Witham polled the cream of American scientists, those who ’ d been honored by election to the National Academy of Sciences, and among this select group, belief in a personal God dropped to a shattering seven percent. About 20 percent are agnostic ; the rest could fairly be called atheists. Similar figures obtained for belief in personal immortality. Among biological scientists, the figure is even lower : 5. 5 percent, only, believe in God. Physical scientists, it ’ s 7. 5 percent. I ’ ve not seen corresponding figures for elite scholars in other fields, such as history or philosophy, but I ’ d be surprised if they were different. So, we ’ ve reached a truly remarkable situation, a grotesque mismatch between the American intelligentsia and the American electorate. A philosophical opinion about the nature of the universe, which is held by the vast majority of top American scientists and probably the majority of the intelligentsia generally, is so abhorrent to the American electorate that no candidate for popular election dare affirm it in public. If I ’ m right, this means that high office in the greatest country in the world is barred to the very people best qualified to hold it — the intelligentsia — unless they are prepared to lie about their beliefs. To put it bluntly : American political opportunities are heavily loaded against those who are simultaneously intelligent and honest. I ' m not a citizen of this country, so I hope it won ' t be thought unbecoming if I suggest that something needs to be done. And I ' ve already hinted what that something is. From what I ' ve seen of TED, I think this may be the ideal place to launch it. Again, I fear it will cost money. We need a consciousness-raising, coming-out campaign for American atheists. This could be similar to the campaign organized by homosexuals a few years ago, although heaven forbid that we should stoop to public outing of people against their will. In most cases, people who out themselves will help to destroy the myth that there is something wrong with atheists. On the contrary, they ' ll demonstrate that atheists are often the kinds of people who could serve as decent role models for your children, the kinds of people an advertising agent could use to recommend a product, the kinds of people who are sitting in this room. There should be a snowball effect, a positive feedback, such that the more names we have, the more we get. There could be non-linearities, threshold effects. When a critical mass has been obtained, there ' s an abrupt acceleration in recruitment. And again, it will need money. I suspect that the word"atheist"itself contains or remains a stumbling block far out of proportion to what it actually means, and a stumbling block to people who otherwise might be happy to out themselves. So, what other words might be used to smooth the path, oil the wheels, sugar the pill? Darwin himself preferred"agnostic"— and not only out of loyalty to his friend Huxley, who coined the term. Darwin said,"I have never been an atheist in the same sense of denying the existence of a God. I think that generally an ' agnostic ' would be the most correct description of my state of mind."He even became uncharacteristically tetchy with Edward Aveling. Aveling was a militant atheist who failed to persuade Darwin to accept the dedication of his book on atheism — incidentally, giving rise to a fascinating myth that Karl Marx tried to dedicate"Das Kapital"to Darwin, which he didn ' t, it was actually Edward Aveling. What happened was that Aveling ' s mistress was Marx ' s daughter, and when both Darwin and Marx were dead, Marx ' s papers became muddled up with Aveling ' s papers, and a letter from Darwin saying,"My dear sir, thank you very much but I don ' t want you to dedicate your book to me,"was mistakenly supposed to be addressed to Marx, and that gave rise to this whole myth, which you ' ve probably heard. It ' s a sort of urban myth, that Marx tried to dedicate"Kapital"to Darwin. Anyway, it was Aveling, and when they met, Darwin challenged Aveling."Why do you call yourselves atheists?""' Agnostic, '"retorted Aveling,"was simply ' atheist ' writ respectable, and ' atheist ' was simply ' agnostic ' writ aggressive."Darwin complained,"But why should you be so aggressive?"Darwin thought that atheism might be well and good for the intelligentsia, but that ordinary people were not, quote,"ripe for it."Which is, of course, our old friend, the"don ' t rock the boat"argument. It ' s not recorded whether Aveling told Darwin to come down off his high horse. But in any case, that was more than 100 years ago. You ' d think we might have grown up since then. Now, a friend, an intelligent lapsed Jew, who, incidentally, observes the Sabbath for reasons of cultural **solidarity**, describes himself as a"tooth-fairy agnostic."He won ' t call himself an atheist because it ' s, in principle, impossible to prove a negative, but"agnostic"on its own might suggest that God ' s existence was therefore on **equal** terms of likelihood as his non-existence. So, my friend is strictly agnostic about the tooth fairy, but it isn ' t very likely, is it? Like God. Hence the phrase,"tooth-fairy agnostic."Bertrand Russell made the same point using a hypothetical teapot in orbit about Mars.

You would strictly have to be agnostic about whether there is a teapot in orbit about Mars, but that doesn ' t mean you treat the likelihood of its existence as on all fours with its non-existence. The list of things which we strictly have to be agnostic about doesn ' t stop at tooth fairies and teapots ; it ' s infinite. If you want to believe one particular one of them — unicorns or tooth fairies or teapots or Yahweh — the onus is on you to say why. The onus is not on the rest of us to say why not. We, who are atheists, are also a-fairyists and a-teapotists. But we don ' t bother to say so. And this is why my friend uses"tooth-fairy agnostic"as a label for what most people would call atheist. Nonetheless, if we want to attract deep-down atheists to come out publicly, we ' re going to have find something better to stick on our banner than"tooth-fairy"or"teapot agnostic."So, how about"humanist"? This has the advantage of a worldwide network of well-organized associations and journals and things already in place. My problem with it is only its apparent anthropocentrism. One of the things we ' ve learned from Darwin is that the human species is only one among millions of cousins, some close, some distant. And there are other possibilities, like"naturalist,"but that also has problems of confusion, because Darwin would have thought naturalist —"Naturalist"means, of course, as opposed to"supernaturalist"— and it is used sometimes — Darwin would have been confused by the other sense of"naturalist,"which he was, of course, and I suppose there might be others who would confuse it with"nudism". Such people might be those belonging to the British lynch mob, which last year attacked a pediatrician in mistake for a pedophile. I think the best of the available alternatives for"atheist"is simply"non- theist."It lacks the strong connotation that there ' s definitely no God, and it could therefore easily be embraced by teapot or tooth-fairy agnostics. It ' s completely compatible with the God of the physicists. When atheists like Stephen Hawking and Albert Einstein use the word"God,"they use it of course as a metaphorical shorthand for that deep, mysterious part of physics which we don ' t yet understand."Non-theist"will do for all that, yet unlike"atheist,"it doesn ' t have the same phobic, hysterical responses. But I think, actually, the alternative is to grasp the nettle of the word"atheism"itself, precisely because it is a taboo word, carrying frissons of hysterical phobia. Critical mass may be harder to achieve with the word"atheist"than with the word"non- theist,"or some other non-confrontational word. But if we did achieve it with that dread word"atheist"itself, the political impact would be even greater. Now, I said that if I were religious, I ' d be very afraid of evolution — I ' d go further : I would fear science in general, if properly understood. And this is because the scientific worldview is so much more exciting, more poetic, more filled with sheer wonder than anything in the poverty-stricken arsenals of the religious imagination. As Carl Sagan, another recently dead hero, put it,"How is it that hardly any major religion has looked at science and concluded, ' This is better than we thought! The universe is much bigger than our prophet said, grander, more subtle, more elegant '? Instead they say, ' No, no, no! My god is a little god, and I want him to stay that way. ' A religion, old or new, that stressed the magnificence of the universe as revealed by modern science, might be able to draw forth reserves of reverence and awe hardly tapped by the conventional faiths."Now, this is an elite audience, and I would therefore expect about 10 percent of you to be religious. Many of you probably subscribe to our polite cultural belief that we should respect religion. But I also suspect that a fair number of those secretly despise religion as much as I do. If you ' re one of them, and of course many of you may not be, but if you are one of them, I ' m asking you to stop being polite, come out, and say so. And if you happen to be rich, give some thought to ways in which you might make a difference. The religious lobby in this country is massively financed by foundations — to say nothing of all the tax benefits — by foundations, such as the Templeton Foundation and the Discovery Institute. We need an anti- Templeton to step forward. If my books sold as well as Stephen Hawking ’ s books, instead of only as well as Richard Dawkins ’ books, I ’ d do it myself. People are always going on about,”How did September the 11th change you?”Well, here ’ s how it changed me. Let ’ s all stop being so damned respectful. Thank you very much.

Text Sample 415: POS Dim 3 – 1984 – Score 1.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ’ ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a computer and really address the qualities of the human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, the intelligence of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ’ ve been able to collect, which I think illustrate this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a slide taken from a TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ’ s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ’ s the distance you should sit away from the TV set. Now we ’ ve put people 18 inches in front of a TV, and all the artifacts that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow mask, the scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ’ m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ’ m very interested in touch-sensitive displays. High-tech, high-touch. Isn ’ t that what some of you said? It ’ s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ’ re not : it ’ s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on the market these systems that have just a few light emitting diodes on the side and are very low resolution, it ’ s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ’ t have to pick them up, and people don ’ t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ’ re typing — what you have — you want to now put something — first of all, you ’ ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ’ ve got to sort of find that mouse. Then you find the mouse, and you ' re going to have to wiggle it a little bit to see where the cursor is on the screen. And then when you finally see where it is, then you ' ve got to move it to get the cursor over there, and then —"Bang"— you ' ve got to hit a button or do whatever. That ' s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ' m trying to do is just illustrate the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of the computer field, is when you have a bug that you can ' t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the substrate, which it usually is. So we found that that actually was a feature in the sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ' s pressure-sensitive. And it ' s pressure-sensitive to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ' t figure out how to come in the other direction. But let me get rid of the slide, and let ' s see if this comes on. OK. So there is the pressure-sensitive display in operation. The person ' s just, if you will, pushing on the screen to make a curve. But this is the interesting part. I want to stop it for a second because the movie is very badly made. And the particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an anvil on a fuzzy rug and the other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across the screen, and yet you touch the ping-pong ball very lightly and it just scoots across the screen. And what you can do — oops, I didn ' t mean to do that — what you can do is actually feed back to the user the feeling of the physical properties. So again, they don ' t have to be weight ; they could be a general trying to move troops, and he ' s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the interface there are physical properties in that transducer — in this case it ' s pressure and touches — that allow you to present things to the user that you could never present before. So it ' s not simply looking at the quality or, if you will, the luxury of that interface, but it ' s actually looking at the idea of presenting things that previously couldn ' t be presented before. I want to move on to another example, which is one of a different sort, where we ' re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ' re going to take this book, if you will, and it ' s going to come alive. You ' re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are the experts in monologue making : you make a film and it has a well-formed beginning, middle and end, and in some sense the art of it is that. And you then say,”There ’ s an opportunity for making conversational movies.”Well, what does that mean? And it sort of nibbles at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ’ ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the movie has information about itself. So it knows, or at least there is computer-readable information in the medium itself. It ’ s just not a static movie frame. That ’ s one thing. The other is that you have to realize that it is a random **access** medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the cookbook, the”Larousse Gastronomique.”And I think I use the example all too often, but it ’ s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to the end of the recipe and it says,”Cook until done.”Now, that would be, if you will, the top green track, which doesn ’ t mean too much. But you might have to elaborate for me or for somebody who isn ’ t an expert, and say,”Cook at 380 degrees for 45 minutes.”And then for a real beginner, you would go down even further and elaborate more — say,”Open the oven, preheat, wait for the light to go out, open the door, don ’ t leave it open too long, put the penguin in and shut the door...”whatever. And that ’ s a much more elaborate one than you dribble back. That ’ s one kind of use of random **access**. And the other is where you want to explain the same thing in different ways. If you ’ re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ’ s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ’ s a rather boring book, but I ’ m afraid sometimes you have to do boring books because your sponsors aren ’ t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ’ t even know what vintage the transmission is, but let me just show you very quickly some of it, and we ’ ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of the transmission, and as you rub your finger across the transmission it highlights the various parts. Narrator : When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, superimposed over the illustration. NN : This is about the oil pan, or the oil filter and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get back to the illustration, but in this case it ’ s not a single frame, but it ’ s actually a movie of someone coming into the frame and doing the repair that ’ s described in the text. The two-headed slider is a speed control that allows me to watch the movie at various speeds, in forward or reverse. And the movie is displayed as a full frame movie. I can go back to the beginning... and play the movie at full speed. Here ’ s another step-by-step procedure, only in this case — NN : Okay, this movie is... Everybody ’ s heard of sound-sync movies — this is text-sync movies, so as the movie plays, the text gets highlighted. We highlight the text as we go through the movie. Repairman :... Not too far out. Front poles, preferably. Don ’ t loosen them too far. If you loosen them too far, you ’ ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ’ m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ’ ve been doing — in this case, in Senegal — where we have tried to use personal computers as a pedagogical medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ’ s a story I ’ d like to tell. And that was when, before we did this in any developing countries — we ’ re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ’ ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ’ t read, didn ’ t even make it in the lowest section of the school ’ s classes — and was pretty much not in school, though physically there. But did hang around the, quote,”computer room,”where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very interesting. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,”Let me show you how this works,”and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ’ t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ’ d actually come to see — not the computer room — they went upstairs and they said,”This is absolutely remarkable! That child was very articulate and showed us and even dealt with the things he couldn ’ t do automatically with that manual. It was just absolutely fantastic.”The principal said,”There ’ s a dreadful mistake, because that child can ’ t read. And you obviously have been hoodwinked or you ’ ve talked about somebody else.”And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,”Can you read?”And the child said,”No, I can ’ t.”And then they said,”But wait a minute. You just looked through that manual and you found...”and he said,”Oh, but that ’ s not reading.”And so they said,”Well, what ’ s reading then?”He says,”Well, reading is this junk they give me in little books to read. It ’ s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return.”And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ’ m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ’ s not, but we think speech —”My God, little children pick it up somehow, and by the age of two they ’ re doing a mediocre job, and by three and four they ’ re speaking reasonably well. And yet you ’ ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder.”Well, it ’ s not harder. What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ’ s going to help you. So what happens is you go to school and people say,”Just believe me, you ’ re going to like it. You ’ re going to like reading,”and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on the screen has a huge payoff, and it ’ s a lot of fun. And in fact, it ’ s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ’ s the girl on the left leaning with her chalkboard, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ’ s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example — as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And the reason it remains a favorite project is that we were asked to do a teleconferencing system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ’ s faces. So if I wanted to call my friend Peter Sprague on the phone, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ’ s uncanny : there ’ s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ’ s something that didn ’ t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,”He or she wouldn ’ t agree,”and the empty chair is that person and the spatiality is crucial. So we said,”Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ’ s site you ’ d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ’ t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ’ ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ’ s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ’ s site, he ’ ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.

Text Sample 416: POS Dim 3 – 1998 – Score 3.00 – 1998/t000250.txt

As a clergyman, you can imagine how out of place I feel. I feel like a fish out of water, or maybe an owl out of the air. I was preaching in San Jose some time ago, and my friend Mark Kvamme, who helped introduce me to this conference, brought several CEOs and leaders of some of the companies here in the Silicon Valley to have breakfast with me, or I with them. And I was so stimulated. And had such — it was an eye-opening experience to hear them talk about the world that is yet to come through technology and science. I know that we ’ re near the end of this conference, and some of you may be wondering why they have a speaker from the field of religion. Richard can answer that, because he made that decision. But some years ago I was on an elevator in Philadelphia, coming down. I was to address a conference at a hotel. And on that elevator a man said,”I hear Billy Graham is staying in this hotel.”And another man looked in my direction and said,”Yes, there he is. He ’ s on this elevator with us.”And this man looked me up and down for about 10 seconds, and he said,”My, what an anticlimax!”I hope that you won ’ t feel that these few moments with me is not a — is an anticlimax, after all these tremendous talks that you ’ ve heard, and addresses, which I intend to listen to every one of them. But I was on an airplane in the east some years ago, and the man sitting across the aisle from me was the mayor of Charlotte, North Carolina. His name was John Belk. Some of you will probably know him. And there was a drunk man on there, and he got up out of his seat two or three times, and he was making everybody upset by what he was trying to do. And he was slapping the stewardess and pinching her as she went by, and everybody was upset with him. And finally, John Belk said,”Do you know who ’ s sitting here?”And the man said,”No, who?”He said,”It ’ s Billy Graham, the preacher.”He said,”You don ’ t say!”And he turned to me, and he said,”Put her there!”He said,”Your sermons have certainly helped me.”And I suppose that that ’ s true with thousands of people. I know that as you have been peering into the future, and as we ’ ve heard some of it here tonight, I would like to live in that age and see what is going to be. But I won ’ t, because I ’ m 80 years old. This is my eightieth year, and I know that my time is brief. I have phlebitis at the moment, in both legs, and that ’ s the reason that I had to have a little help in getting up here, because I have Parkinson ’ s disease in addition to that, and some other problems that I won ’ t talk about. But this is not the first time that we ’ ve had a technological revolution. We ’ ve had others. And there ’ s one that I want to talk about. In one generation, the nation of the people of Israel had a tremendous and dramatic change that made them a great power in the Near East. A man by the name of David came to the throne, and King David became one of the great leaders of his generation. He was a man of tremendous leadership. He had the favor of God with him. He was a brilliant poet, philosopher, writer, soldier — with strategies in battle and conflict that people study even today. But about two centuries before David, the Hittites had discovered the secret of smelting and processing of iron, and, slowly, that skill spread. But they wouldn ’ t allow the Israelis to look into it, or to have any. But David changed all of that, and he introduced the Iron Age to Israel. And the Bible says that David laid up great stores of iron, and which archaeologists have found, that in present-day Palestine, there are evidences of that generation. Now, instead of crude tools made of sticks and stones, Israel now had iron plows, and sickles, and hoes and military weapons. And in the course of one generation, Israel was completely changed. The introduction of iron, in some ways, had an impact a little bit like the microchip has had on our generation. And David found that there were many problems that technology could not solve. There were many problems still left. And they ’ re still with us, and you haven ’ t solved them, and I haven ’ t heard anybody here speak to that. How do we solve these three problems that I ’ d like to mention? The first one that David saw was human evil. Where does it come from? How do we solve it? Over again and again in the Psalms, which Gladstone said was the greatest book in the world, David describes the evils of the human race. And yet he says,”He restores my soul.”Have you ever thought about what a contradiction we are? On one hand, we can probe the deepest secrets of the universe and dramatically push back the frontiers of technology, as this conference vividly demonstrates. We ’ ve seen under the sea, three miles down, or galaxies hundreds of billions of years out in the future. But on the other hand, something is wrong. Our battleships, our soldiers, are on a frontier now, almost ready to go to war with Iraq. Now, what causes this? Why do we have these wars in every generation, and in every part of the world? And revolutions? We can ’ t get along with other people, even in our own families. We find ourselves in the paralyzing grip of self-destructive habits we can ’ t break. Racism and injustice and **violence** sweep our world, bringing a tragic harvest of heartache and death. Even the most sophisticated among us seem powerless to break this cycle. I would like to see Oracle take up that, or some other technological geniuses work on this. How do we change man, so that he doesn ’ t lie and cheat, and our newspapers are not filled with stories of fraud in business or labor or athletics or wherever? The Bible says the problem is within us, within our hearts and our souls. Our problem is that we are separated from our Creator, which we call God, and we need to have our souls restored, something only God can do. Jesus said,”For out of the heart come evil thoughts : murders, **sexual** immorality, theft, false testimonies, slander.”The British philosopher Bertrand Russell was not a religious man, but he said,”It ’ s in our hearts that the evil lies, and it ’ s from our hearts that it must be plucked out.”Albert Einstein — I was just talking to someone, when I was speaking at Princeton, and I met Mr. Einstein. He didn ’ t have a doctor ’ s degree, because he said nobody was qualified to give him one. But he made this statement. He said,”It ’ s easier to denature plutonium than to denature the evil spirit of man.”And many of you, I ’ m sure, have thought about that and puzzled over it. You ’ ve seen people take beneficial technological advances, such as the Internet we ’ ve heard about tonight, and twist them into something corrupting. You ’ ve seen brilliant people devise computer viruses that bring down whole systems. The Oklahoma City bombing was simple technology, horribly used. The problem is not technology. The problem is the person or persons using it. King David said that he knew the depths of his own soul. He couldn ’ t free himself from personal problems and personal evils that included murder and adultery. Yet King David sought God ’ s forgiveness, and said,”You can restore my soul.”You see, the Bible teaches that we ’ re more than a body and a mind. We are a soul. And there ’ s something inside of us that is beyond our understanding. That ’ s the part of us that yearns for God, or something more than we find in technology. Your soul is that part of you that yearns for meaning in life, and which seeks for something beyond this life. It ’ s the part of you that yearns, really, for God. I find [that] young people all over the world are searching for something. They don ' t know what it is. I speak at many universities, and I have many questions and answer periods, and whether it ' s Cambridge, or Harvard, or Oxford — I ' ve spoken at all of those universities. I ' m going to Harvard in about three or four — no, it ' s about two months from now — to give a lecture. And I ' ll be asked the same questions that I was asked the last few times I ' ve been there. And it ' ll be on these questions : where did I come from? Why am I here? Where am I going? What ' s life all about? Why am I here? Even if you have no religious belief, there are times when you wonder that there ' s something else. Thomas Edison also said,"When you see everything that happens in the world of science, and in the working of the universe, you cannot deny that there ' s a captain on the bridge."I remember once, I sat beside Mrs. Gorbachev at a White House dinner. I went to Ambassador Dobrynin, whom I knew very well. And I ' d been to Russia several times under the Communists, and they ' d given me marvelous freedom that I didn ' t expect. And I knew Mr. Dobrynin very well, and I said,"I ' m going to sit beside Mrs. Gorbachev tonight. What shall I talk to her about?"And he surprised me with the answer. He said,"Talk to her about religion and philosophy. That ' s what she ' s really interested in."I was a little bit surprised, but that evening that ' s what we talked about, and it was a stimulating conversation. And afterward, she said,"You know, I ' m an atheist, but I know that there ' s something up there higher than we are."The second problem that King David realized he could not solve was the problem of human **suffering**. Writing the oldest book in the world was Job, and he said,"Man is born unto trouble as the sparks fly upward."Yes, to be sure, science has done much to push back certain types of human **suffering**. But I ' m — in a few months, I ' ll be 80 years of age. I admit that I ' m very grateful for all the medical advances that have kept me in relatively good health all these years. My doctors at the Mayo Clinic urged me not to take this trip out here to this — to be here. I haven ' t given a talk in nearly four months. And when you speak as much as I do, three or four times a day, you get rusty. That ' s the reason I ' m using this podium and using these notes. Every time you ever hear me on the television or somewhere, I ' m ad-libbing. I ' m not reading. I never read an address. I never read a speech or a talk or a lecture. I talk ad lib. But tonight, I ' ve got some notes here so that if I begin to forget, which I do sometimes, I ' ve got something I can turn to. But even here among us, most — in the most advanced society in the world, we have poverty. We have families that self-destruct, friends that betray us. Unbearable psychological pressures bear down on us. I ' ve never met a person in the world that didn ' t have a problem or a worry. Why do we suffer? It ' s an age- old question that we haven ' t answered. Yet David again and again said that he would turn to God. He said,"The Lord is my shepherd."The final problem that David knew he could not solve was death. Many commentators have said that death is the forbidden subject of our generation. Most people live as if they ' re never going to die. Technology projects the myth of control over our mortality. We see people on our screens. Marilyn Monroe is just as beautiful on the screen as she was in person, and our — many young people think she ' s still alive. They don ' t know that she ' s dead. Or Clark Gable, or whoever it is. The old stars, they come to life. And they ' re — they ' re just as great on that screen as they were in person. But death is inevitable. I spoke some time ago to a joint session of Congress, last year. And we were meeting in that room, the statue room. About 300 of them were there. And I said,"There ' s one thing that we have in common in this room, all of us together, whether Republican or Democrat, or whoever."I said,"We ' re all going to die. And we have that in common with all these great men of the past that are staring down at us."And it ' s often difficult for young people to understand that. It ’ s difficult for them to understand that they ’ re going to die. As the ancient writer of Ecclesiastes wrote, he said, there ’ s every activity under heaven. There ’ s a time to be born, and there ’ s a time to die. I ’ ve stood at the deathbed of several famous people, whom you would know. I ’ ve talked to them. I ’ ve seen them in those agonizing moments when they were scared to death. And yet, a few years earlier, death never crossed their mind. I talked to a woman this past week whose father was a famous doctor. She said he never thought of God, never talked about God, didn ’ t believe in God. He was an atheist. But she said, as he came to die, he sat up on the side of the bed one day, and he asked the nurse if he could see the chaplain. And he said, for the first time in his life he ’ d thought about the inevitable, and about God. Was there a God? A few years ago, a university student asked me,”What is the greatest surprise in your life?”And I said,”The greatest surprise in my life is the brevity of life. It passes so fast.”But it does not need to have to be that way. Wernher von Braun, in the aftermath of World War II concluded, quote :”science and religion are not antagonists. On the contrary, they ’ re sisters.”He put it on a personal basis. I knew Dr. von Braun very well. And he said,”Speaking for myself, I can only say that the grandeur of the cosmos serves only to confirm a belief in the certainty of a creator.”He also said,”In our search to know God, I ’ ve come to believe that the life of Jesus Christ should be the focus of our efforts and inspiration. The reality of this life and His resurrection is the hope of mankind.”I ’ ve done a lot of speaking in Germany and in France, and in different parts of the world — 105 countries it ’ s been my privilege to speak in. And I was invited one day to visit Chancellor Adenauer, who was looked upon as sort of the founder of modern Germany, since the war. And he once — and he said to me, he said,”Young man.”He said,”Do you believe in the resurrection of Jesus Christ?”And I said,”Sir, I do.”He said,”So do I.”He said,”When I leave office, I ’ m going to spend my time writing a book on why Jesus Christ rose again, and why it ’ s so important to believe that.”In one of his plays, Alexander Solzhenitsyn depicts a man dying, who says to those gathered around his bed,”The moment when it ’ s terrible to feel regret is when one is dying.”How should one live in order not to feel regret when one is dying? Blaise Pascal asked exactly that question in seventeenth-century France. Pascal has been called the architect of modern civilization. He was a brilliant scientist at the frontiers of mathematics, even as a teenager. He is viewed by many as the founder of the probability theory, and a creator of the first model of a computer. And of course, you are all familiar with the computer language named for him. Pascal explored in depth our human dilemmas of evil, **suffering** and death. He was astounded at the phenomenon we ’ ve been considering : that people can achieve extraordinary heights in science, the arts and human enterprise, yet they also are full of anger, hypocrisy and have — and self- **hatreds**. Pascal saw us as a remarkable mixture of genius and self-delusion. On November 23, 1654, Pascal had a profound religious experience. He wrote in his journal these words :”I submit myself, absolutely, to Jesus Christ, my redeemer.”A French historian said, two centuries later,”Seldom has so mighty an intellect submitted with such humility to the authority of Jesus Christ.”Pascal came to believe not only the love and the grace of God could bring us back into harmony, but he believed that his own sins and failures could be forgiven, and that when he died he would go to a place called heaven. He experienced it in a way that went beyond scientific observation and reason. It was he who penned the well-known words,”The heart has its reasons, which reason knows not of.”Equally well known is Pascal ’ s Wager. Essentially, he said this :”if you bet on God, and open yourself to his love, you lose nothing, even if you ’ re wrong. But if instead you bet that there is no God, then you can lose it all, in this life and the life to come."For Pascal, scientific knowledge paled beside the knowledge of God. The knowledge of God was far beyond anything that ever crossed his mind. He was ready to face him when he died at the age of 39. King David lived to be 70, a long time in his era. Yet he too had to face death, and he wrote these words :"even though I walk through the valley of the shadow of death, I will fear no evil, for you are with me."This was David ' s answer to three dilemmas of evil, **suffering** and death. It can be yours, as well, as you seek the living God and allow him to fill your life and give you hope for the future. When I was 17 years of age, I was born and reared on a farm in North Carolina. I milked cows every morning, and I had to milk the same cows every evening when I came home from school. And there were 20 of them that I had — that I was responsible for, and I worked on the farm and tried to keep up with my studies. I didn ' t make good grades in high school. I didn ' t make them in college, until something happened in my heart. One day, I was faced face-to-face with Christ. He said,"I am the way, the truth and the life."Can you imagine that?"I am the truth. I ' m the embodiment of all truth."He was a liar. Or he was insane. Or he was what he claimed to be. Which was he? I had to make that decision. I couldn ' t prove it. I couldn ' t take it to a laboratory and experiment with it. But by faith I said, I believe him, and he came into my heart and changed my life. And now I ' m ready, when I hear that call, to go into the presence of God. Thank you, and God bless all of you. Thank you for the privilege. It was great. Richard Wurman : You did it. Thanks.

Text Sample 417: POS Dim 3 – 1998 – Score 3.00 – 1998/t000134.txt

' Theme and variations ' is one of those forms that require a certain kind of intellectual activity, because you are always comparing the variation with the theme that you hold in your mind. You might say that the theme is nature and everything that follows is a variation on the subject. I was asked, I guess about six years ago, to do a series of paintings that in some way would **celebrate** the birth of Piero della Francesca. And it was very difficult for me to imagine how to paint pictures that were based on Piero until I realized that I could look at Piero as nature — that I would have the same **attitude** towards looking at Piero della Francesca as I would if I were looking out a window at a tree. And that was enormously liberating to me. Perhaps it ' s not a very insightful observation, but that really started me on a path to be able to do a kind of theme and variations based on a work by Piero, in this case that remarkable painting that ' s in the Uffizi,"The Duke of Montefeltro,"who faces his consort, Battista. Once I realized that I could take some liberties with the subject, I did the following series of drawings. That ' s the real Piero della Francesca — one of the greatest portraits in human history. And these, I ' ll just show these without comment. It ' s just a series of variations on the head of the Duke of Montefeltro, who ' s a great, great figure in the Renaissance, and probably the basis for Machiavelli ' s"The Prince."He apparently lost an eye in battle, which is why he is always shown in profile. And this is Battista. And then I decided I could move them around a little bit — so that for the first time in history, they ' re facing the same direction. Whoops! Passed each other. And then a visitor from another painting by Piero, this is from"The Resurrection of Christ"— as though the cast had just gotten of the set to have a **chat**. And now, four large panels : this is upper left ; upper right ; lower left ; lower right. Incidentally, I ' ve never understood the conflict between abstraction and naturalism. Since all paintings are inherently abstract to begin with there doesn ' t seem to be an argument there. On another subject — — I was driving in the country one day with my wife, and I saw this sign, and I said,”That is a fabulous piece of design.”And she said,”What are you talking about?”I said,”Well, it ’ s so persuasive, because the purpose of that sign is to get you into the garage, and since most people are so suspicious of garages, and know that they ’ re going to be ripped off, they use the word ’ reliable. ’ But everybody says they ’ re reliable. But, reliable Dutchman”— —”Fantastic!”Because as soon as you hear the word Dutchman — which is an archaic word, nobody calls Dutch people”Dutchmen”anymore — but as soon as you hear Dutchman, you get this picture of the kid with his finger in the dike, preventing the thing from falling and flooding Holland, and so on. And so the entire issue is detoxified by the use of”Dutchman.”Now, if you think I ’ m exaggerating at all in this, all you have to do is substitute something else, like”Indonesian.”Or even”French.”Now,”Swiss”works, but you know it ’ s going to cost a lot of money. I ’ m going to take you quickly through the actual process of doing a poster. I do a lot of work for the School of Visual Arts, where I teach, and the director of this school — a remarkable man named Silas Rhodes — often gives you a piece of text and he says,”Do something with this.”And so he did. And this was the text —”In words as fashion the same rule will hold / Alike fantastic if too new or old / Be not the first by whom the new are tried / Nor yet the last to lay the old aside.”I could make nothing of that. And I really struggled with this one. And the first thing I did, which was sort of in the absence of another idea, was say I ’ ll sort of write it out and make some words big, and I ’ ll have some kind of design on the back somehow, and I was hoping — as one often does — to stumble into something. So I took another crack at it — you ’ ve got to keep it moving — and I Xeroxed some words on pieces of colored paper and I pasted them down on an ugly board. I thought that something would come out of it, like”Words rule fantastic new old first last Pope”because it ’ s by Alexander Pope, but I sort of made a mess out of it, and then I thought I ’ d repeat it in some way so it was legible. So, it was going nowhere. Sometimes, in the middle of a resistant problem, I write down things that I know about it. But you can see the beginning of an idea there, because you can see the word”new”emerging from the”old.”That ’ s what happens. There ’ s a relationship between the old and the new ; the new emerges from the context of the old. And then I did some variations of it, but it still wasn ’ t coalescing graphically at all. I had this other version which had something interesting about it in terms of being able to put it together in your mind from clues. The W was clearly a W, the N was clearly an N, even though they were very fragmentary and there wasn ’ t a lot of information in it. Then I got the words”new”and”old”and now I had regressed back to a point where there seemed to be no return. I was really desperate at this point. And so, I do something I ’ m truthfully ashamed of, which is that I took two drawings I had made for another purpose and I put them together. It says”dreams”at the top. And I was going to do a thing, I say,”Well, change the copy. Let it say something about dreams, and come to SVA and you ’ ll sort of fulfill your dreams.”But, to my credit, I was so embarrassed about doing that that I never submitted this sketch. And, finally, I arrived at the following solution. Now, it doesn ’ t look terribly interesting, but it does have something that distinguishes it from a lot of other posters. For one thing, it transgresses the idea of what a poster ’ s supposed to be, which is to be understood and seen immediately, and not explained. I remember hearing all of you in the graphic arts —”If you have to explain it, it ain ’ t working.”And one day I woke up and I said,”Well, suppose that ’ s not true?”So here ’ s what it says in my explanation at the bottom left. It says,”Thoughts : This poem is impossible. Silas usually has a better touch with his choice of quotations. This one generates no imagery at all.”I am now exposing myself to my audience, right? Which is something you never want to do professionally.”Maybe the words can make the image without anything else happening. What ’ s the heart of this poem? Don ’ t be trendy if you want to be serious. Is doing the poster this way trendy in itself? I guess one could reduce the idea further by suggesting that the new emerges behind and through the old, like this.”And then I show you a little drawing — you see, you remember that old thing I discarded? Well, I found a way to use it. So, there ’ s that little alternative over there, and I say,”Not bad,”— criticizing myself —”but more didactic than visual. Maybe what wants to be said is that old and the new are locked in a dialectical embrace, a kind of dance where each defines the other.”And then more self-questioning —”Am I being simple-minded? Is this the kind of simple that looks obvious, or the kind that looks profound? There ’ s a significant difference. This could be embarrassing. Actually, I realize fear of embarrassment drives me as much as any ambition. Do you think this sort of thing could really attract a student to the school?”Well, I think there are two fresh things here — two fresh things. One is the sort of willingness to expose myself to a critical audience, and not to suggest that I am confident about what I ’ m doing. And as you know, you have to have a front. I mean — you ’ ve got to be confident ; if you don ’ t believe in your work, who else is going to believe in it? So that ’ s one thing, to introduce the idea of doubt into graphics. That can be a big contribution. The other thing is to actually give you two solutions for the price of one ; you get the big one and if you don ’ t like that, how about the little one? And that too is a relatively new idea. And here ’ s just a series of experiments where I ask the question of — does a poster have to be square? Now, this is a little illusion. That poster is not folded. It ’ s not folded, that ’ s a photograph and it ’ s cut on the diagonal. Same cheap trick in the upper left- hand corner. And here, a very peculiar poster because, simply because of using the isometric perspective in the computer, it won ’ t sit still in the space. At times, it seems to be wider at the back than the front, and then it shifts. And if you sit here long enough, it ’ ll float off the page into the audience. But, we don ’ t have time. And then an experiment — a little bit about the nature of perspective, where the outside shape is determined by the peculiarity of perspective, but the shape of the bottle — which is identical to the outside shape — is seen frontally. And another piece for the art directors ’ club is”Anna Rees”casting long shadows. This is another poster from the School of Visual Arts. There were 10 artists invited to participate in it, and it was one of those things where it was extremely competitive and I didn ’ t want to be embarrassed, so I worked very hard on this. The idea was — and it was a brilliant idea — was to have 10 posters distributed throughout the city ’ s subway system so every time you got on the subway you ’ d be passing a different poster, all of which had a different idea of what art is. But I was absolutely stuck on the idea of”art is”and trying to determine what art was. But then I gave up and I said,”Well, art is whatever.”And as soon as I said that, I discovered that the word”hat”was hidden in the word”whatever,”and that led me to the inevitable conclusion. But then again, it ’ s on my list of didactic posters. My intent is to have a literary accompaniment that explains the poster, in case you don ’ t get it. Now this says,”Note to the viewer : I thought I might use a visual cliche of our time — Magritte ’ s everyman — to express the idea that art is mystery, continuity and history. I ’ m also convinced that, in an age of computer manipulation, surrealism has become banal, a shadow of its former self. The phrase ’ Art is whatever ’ expresses the current inclusiveness that surrounds art-making — a sort of ’ it ain ’ t what you do, it ’ s the way that you do it ’ notion. The shadow of Magritte falls across the central part of the poster a poetic event that occurs as the shadow man isolates the word ’ hat, ’ hidden in the word ’ whatever. ’ The four hats shown in the poster suggests how art might be defined : as a thing itself, the worth of the thing, the shadow of the thing, and the shape of the thing. Whatever.”OK. And the one that I did not submit, which I still like, I wanted to use the same phrase. There were wonderful experiments by Bruno Munari on letterforms some years ago — sort of, see how far you could go and still be able to read them. And that idea stuck in my head. But then I took the pieces that I had taken off and put them at the bottom. And, of course those are the remains, and they ’ re so labeled. But what really happens is that you read it as :”Art is whatever remains.”Thank you.

Text Sample 418: POS Dim 3 – 1998 – Score 2.00 – 1998/t000113.txt

Sheryl Shade : Hi, Aimee. Aimee Mullins : Hi. SS : Aimee and I thought we ’ d just talk a little bit, and I wanted her to tell all of you what makes her a distinctive athlete. AM : Well, for those of you who have seen the picture in the little bio â it might have given it away â I ’ m a double amputee, and I was born without fibulas in both legs. I was amputated at age one, and I ’ ve been running like hell ever since, all over the place. SS : Well, why don ’ t you tell them how you got to Georgetown â why don ’ t we start there? Why don ’ t we start there? AM : I ’ m a senior in Georgetown in the Foreign Service program. I won a full academic scholarship out of high school. They pick three students out of the nation every year to get involved in international affairs, and so I won a full ride to Georgetown and I ’ ve been there for four years. Love it. SS : When Aimee got there, she decided that she ’ s, kind of, curious about track and field, so she decided to call someone and start asking about it. So, why don ’ t you tell that story? AM : Yeah. Well, I guess I ’ ve always been involved in sports. I played softball for five years growing up. I skied competitively throughout high school, and I got a little restless in college because I wasn ’ t doing anything for about a year or two sports-wise. And I ’ d never competed on a **disabled** level, you know â I ’ d always competed against other able-bodied athletes. That ’ s all I ’ d ever known. In fact, I ’ d never even met another amputee until I was 17. And I heard that they do these track meets with all **disabled** runners, and I figured,”Oh, I don ’ t know about this, but before I judge it, let me go see what it ’ s all about.”So, I booked myself a flight to Boston in ’ 95, 19 years old and definitely the dark horse candidate at this race. I ’ d never done it before. I went out on a gravel track a couple of weeks before this meet to see how far I could run, and about 50 meters was enough for me, panting and heaving. And I had these legs that were made of a wood and plastic compound, attached with Velcro straps â big, thick, five-ply wool socks on â you know, not the most comfortable things, but all I ’ d ever known. And I ’ m up there in Boston against people wearing legs made of all things â carbon graphite and, you know, shock absorbers in them and all sorts of things â and they ’ re all looking at me like, OK, we know who ’ s not going to win this race. And, I mean, I went up there expecting â I don ’ t know what I was expecting â but, you know, when I saw a man who was missing an entire leg go up to the high jump, hop on one leg to the high jump and clear it at six feet, two inches... Dan O ’ Brien jumped 5 ’ 11”in ’ 96 in Atlanta, I mean, if it just gives you a comparison of â these are truly accomplished athletes, without qualifying that word”athlete.”And so I decided to give this a shot : heart pounding, I ran my first race and I beat the national record-holder by three hundredths of a second, and became the new national record-holder on my first try out. And, you know, people said,”Aimee, you know, you ’ ve got speed â you ’ ve got natural speed â but you don ’ t have any skill or finesse going down that track. You were all over the place. We all saw how hard you were working.”And so I decided to call the track coach at Georgetown. And I thank god I didn ’ t know just how huge this man is in the track and field world. He ’ s coached five Olympians, and the man ’ s office is lined from floor to ceiling with All America certificates of all these athletes he ’ s coached. He ’ s just a rather intimidating figure. And I called him up and said,”Listen, I ran one race and I won...””I want to see if I can, you know â I need to just see if I can sit in on some of your practices, see what drills you do and whatever.”That ’ s all I wanted â just two practices.”Can I just sit in and see what you do?”And he said,”Well, we should meet first, before we decide anything.”You know, he ’ s thinking,”What am I getting myself into?”So, I met the man, walked in his office, and saw these posters and magazine covers of people he has coached. And we got to talking, and it turned out to be a great partnership because he ’ d never coached a **disabled** athlete, so therefore he had no preconceived notions of what I was or wasn ’ t capable of, and I ’ d never been coached before. So this was like,”Here we go â let ’ s start on this trip.”So he started giving me four days a week of his lunch break, his free time, and I would come up to the track and train with him. So that ’ s how I met Frank. That was fall of ’ 95. But then, by the time that winter was rolling around, he said,”You know, you ’ re good enough. You can run on our women ’ s track team here.”And I said,”No, come on.”And he said,”No, no, really. You can. You can run with our women ’ s track team.”In the spring of 1996, with my goal of making the U. S. Paralympic team that May coming up full speed, I joined the women ’ s track team. And no **disabled** person had ever done that â run at a collegiate level. So I don ’ t know, it started to become an interesting mix. SS : Well, on your way to the Olympics, a couple of memorable events happened at Georgetown. Why don ’ t you just tell them? AM : Yes, well, you know, I ’ d won everything as far as the **disabled** meets â everything I competed in â and, you know, training in Georgetown and knowing that I was going to have to get used to seeing the backs of all these women ’ s shirts â you know, I ’ m running against the next Flo-Jo â and they ’ re all looking at me like,”Hmm, what ’ s, you know, what ’ s going on here?”And putting on my Georgetown uniform and going out there and knowing that, you know, in order to become better â and I ’ m already the best in the country â you know, you have to train with people who are inherently better than you. And I went out there and made it to the Big East, which was sort of the championship race at the end of the season. It was really, really hot. And it ’ s the first â I had just gotten these new sprinting legs that you see in that bio, and I didn ’ t realize at that time that the amount of sweating I would be doing in the sock â it actually acted like a lubricant and I ’ d be, kind of, pistoning in the socket. And at about 85 meters of my 100 meters sprint, in all my glory, I came out of my leg. Like, I almost came out of it, in front of, like, 5, 000 people. And I, I mean, was just mortified â because I was signed up for the 200, you know, which went off in a half hour. I went to my coach :”Please, don ’ t make me do this.”I can ’ t do this in front of all those people. My legs will come off. And if it came off at 85 there ’ s no way I ’ m going 200 meters. And he just sat there like this. My pleas fell on deaf ears, thank god. Because you know, the man is from Brooklyn ; he ’ s a big man. He says,”Aimee, so what if your leg falls off? You pick it up, you put the damn thing back on, and finish the goddamn race!”And I did. So, he kept me in line. He kept me on the right track. SS : So, then Aimee makes it to the 1996 Paralympics, and she ’ s all excited. Her family ’ s coming down â it ’ s a big deal. It ’ s now two years that you ’ ve been running? AM : No, a year. SS : A year. And why don ’ t you tell them what happened right before you go run your race? AM : Okay, well, Atlanta. The Paralympics, just for a little bit of clarification, are the Olympics for people with physical disabilities â amputees, persons with cerebral palsy, and wheelchair athletes â as opposed to the Special Olympics, which deals with people with mental disabilities. So, here we are, a week after the Olympics and down at Atlanta, and I ’ m just blown away by the fact that just a year ago, I got out on a gravel track and couldn ’ t run 50 meters. And so, here I am â never lost. I set new records at the U. S. Nationals â the Olympic trials â that May, and was sure that I was coming home with the gold. I was also the only, what they call”bilateral BK”â below the knee. I was the only woman who would be doing the long jump. I had just done the long jump, and a guy who was missing two legs came up to me and says,”How do you do that? You know, we ’ re supposed to have a planar foot, so we can ’ t get off on the springboard.”I said,”Well, I just did it. No one told me that.”So, it ’ s funny â I ’ m three inches within the world record â and kept on from that point, you know, so I ’ m signed up in the long jump â signed up? No, I made it for the long jump and the 100-meter. And I ’ m sure of it, you know? I made the front page of my hometown paper that I delivered for six years, you know? It was, like, this is my time for shine. And we ’ re at the trainee warm-up track, which is a few blocks away from the Olympic stadium. These legs that I was on, which I ’ ll take out right now â I was the first person in the world on these legs. I was the guinea pig., I ’ m telling you, this was, like â talk about a tourist attraction. Everyone was taking pictures â”What is this girl running on?”And I ’ m always looking around, like, where is my competition? It ’ s my first international meet. I tried to get it out of anybody I could, you know,”Who am I running against here?””Oh, Aimee, we ’ ll have to get back to you on that one.”I wanted to find out times.”Don ’ t worry, you ’ re doing great.”This is 20 minutes before my race in the Olympic stadium, and they post the heat sheets. And I go over and look. And my fastest time, which was the world record, was 15. 77. Then I ’ m looking : the next lane, lane two, is 12. 8. Lane three is 12. 5. Lane four is 12. 2. I said,”What ’ s going on?”And they shove us all into the shuttle bus, and all the women there are missing a hand. So, I ’ m just, like â they ’ re all looking at me like ’ which one of these is not like the other, ’ you know? I ’ m sitting there, like,”Oh, my god. Oh, my god.”You know, I ’ d never lost anything, like, whether it would be the scholarship or, you know, I ’ d won five golds when I skied. In everything, I came in first. And Georgetown â that was great. I was losing, but it was the best training because this was Atlanta. Here we are, like, crĂ¨me de la crĂ¨me, and there is no doubt about it, that I ’ m going to lose big. And, you know, I ’ m just thinking,”Oh, my god, my whole family got in a van and drove down here from Pennsylvania.”And, you know, I was the only female U. S. sprinter. So they call us out and, you know â”Ladies, you have one minute.”And I remember putting my blocks in and just feeling horrified because there was just this murmur coming over the crowd, like, the ones who are close enough to the starting line to see. And I ’ m like,”I know! Look! This isn ’ t right.”And I ’ m thinking that ’ s my last card to play here ; if I ’ m not going to beat these girls, I ’ m going to mess their heads a little, you know? I mean, it was definitely the”Rocky IV”sensation of me versus Germany, and everyone else â Estonia and Poland â was in this heat. And the gun went off, and all I remember was finishing last and fighting back tears of frustration and incredible â incredible â this feeling of just being overwhelmed. And I had to think,”Why did I do this?”If I had won everything â but it was like, what was the point? All this training â I had transformed my life. I became a collegiate athlete, you know. I became an Olympic athlete. And it made me really think about how the achievement was getting there. I mean, the fact that I set my sights, just a year and three months before, on becoming an Olympic athlete and saying,”Here ’ s my life going in this direction â and I want to take it here for a while, and just seeing how far I could push it."And the fact that I asked for help â how many people jumped on board? How many people gave of their time and their expertise, and their patience, to deal with me? And that was this collective glory â that there was, you know, 50 people behind me that had joined in this incredible experience of going to Atlanta. So, I apply this sort of philosophy now to everything I do : sitting back and realizing the progression, how far you ’ ve come at this day to this goal, you know. It ’ s important to focus on a goal, I think, but also recognize the progression on the way there and how you ’ ve grown as a person. That ’ s the achievement, I think. That ’ s the real achievement. SS : Why don ’ t you show them your legs? AM : Oh, sure. SS : You know, show us more than one set of legs. AM : Well, these are my pretty legs. No, these are my cosmetic legs, actually, and they ’ re absolutely beautiful. You ’ ve got to come up and see them. There are hair follicles on them, and I can paint my toenails. And, seriously, like, I can wear heels. Like, you guys don ’ t understand what that ’ s like to be able to just go into a shoe store and buy whatever you want. SS : You got to pick your height? AM : I got to pick my height, exactly. Patrick Ewing, who played for Georgetown in the ’ 80s, comes back every summer. And I had incessant fun making fun of him in the training room because he ’ d come in with foot injuries. I ’ m like,"Get it off! Don ’ t worry about it, you know. You can be eight feet tall. Just take them off."He didn ’ t find it as humorous as I did, anyway. OK, now, these are my sprinting legs, made of carbon graphite, like I said, and I ’ ve got to make sure I ’ ve got the right socket. No, I ’ ve got so many legs in here. These are â do you want to hold that actually? That ’ s another leg I have for, like, tennis and softball. It has a shock absorber in it so it, like,"Shhhh,"makes this neat sound when you jump around on it. All right. And then this is the silicon sheath I roll over, to keep it on. Which, when I sweat, you know, I ’ m pistoning out of it. SS : Are you a different height? AM : In these? SS : In these. AM : I don ’ t know. I don ’ t think so. I may be a little taller. I actually can put both of them on. SS : She can ’ t really stand on these legs. She has to be moving, so... AM : Yeah, I definitely have to be moving, and balance is a little bit of an art in them. But without having the silicon sock, I ’ m just going to try slip in it. And so, I run on these, and have shocked half the world on these. These are supposed to simulate the actual form of a sprinter when they run. If you ever watch a sprinter, the ball of their foot is the only thing that ever hits the track. So when I stand in these legs, my hamstring and my glutes are contracted, as they would be had I had feet and were standing on the ball of my feet. AM : It ’ s a company in San Diego called Flex-Foot. And I was a guinea pig, as I hope to continue to be in every new form of prosthetic limbs that come out. But actually these, like I said, are still the actual prototype. I need to get some new ones because the last meet I was at, they were everywhere. You know, it ’ s like a big â it ’ s come full circle. Moderator : Aimee and the designer of them will be at TEDMED 2, and we ’ ll talk about the design of them. AM : Yes, we ’ ll do that. SS : Yes, there you go. AM : So, these are the sprint legs, and I can put my other... SS : Can you tell about who designed your other legs? AM : Yes. These I got in a place called Bournemouth, England, about two hours south of London, and I ’ m the only person in the United States with these, which is a crime because they are so beautiful. And I don ’ t even mean, like, because of the toes and everything. For me, while I ’ m such a serious athlete on the track, I want to be feminine off the track, and I think it ’ s so important not to be limited in any capacity, whether it ’ s, you know, your mobility or even fashion. I mean, I love the fact that I can go in anywhere and pick out what I want â the shoes I want, the skirts I want â and I ’ m hoping to try to bring these over here and make them accessible to a lot of people. They ’ re also silicon. This is a really basic, basic prosthetic limb under here. It ’ s like a Barbie foot under this. It is. It ’ s just stuck in this position, so I have to wear a two-inch heel. And, I mean, it ’ s really â let me take this off so you can see it. I don ’ t know how good you can see it, but, like, it really is. There ’ re veins on the feet, and then my heel is pink, and my Achilles ’ tendon â that moves a little bit. And it ’ s really an amazing store. I got them a year and two weeks ago. And this is just a silicon piece of skin. I mean, what happened was, two years ago this man in Belgium was saying,”God, if I can go to Madame Tussauds ’ wax museum and see Jerry Hall replicated down to the color of her eyes, looking so real as if she breathed, why can ’ t they build a limb for someone that looks like a leg, or an arm, or a hand?”I mean, they make ears for burn **victims**. They do amazing stuff with silicon. SS : Two weeks ago, Aimee was up for the Arthur Ashe award at the ESPYs. And she came into town and she rushed around and she said,”I have to buy some new shoes!”We ’ re an hour before the ESPYs, and she thought she ’ d gotten a two-inch heel but she ’ d actually bought a three-inch heel. AM : And this poses a problem for me, because it means I ’ m walking like that all night long. SS : For 45 minutes. Luckily, the hotel was terrific. They got someone to come in and saw off the shoes. AM : I said to the receptionist â I mean, I am just harried, and Sheryl ’ s at my side â I said,”Look, do you have anybody here who could help me? Because I have this problem...”You know, at first they were just going to write me off, like,”If you don ’ t like your shoes, sorry. It ’ s too late.””No, no, no, no. I ’ ve got these special feet that need a two-inch heel. I have a three-inch heel. I need a little bit off.”They didn ’ t even want to go there. They didn ’ t even want to touch that one. They just did it. No, these legs are great. I ’ m actually going back in a couple of weeks to get some improvements. I want to get legs like these made for flat feet so I can wear sneakers, because I can ’ t with these ones. So... Moderator : That ’ s it. SS : That ’ s Aimee Mullins.

Text Sample 419: POS Dim 3 – 1994 – Score -29.00 – 1994/t002458.txt

Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ’ s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC microprocessors versus the power of local area networks. And the interesting thing about it is that this slide, like so many technology slides that we ’ re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ’ s basically what I ’ m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ’ m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well the answer is, if I drew it on a normal curve where, let ’ s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ’ m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ’ t tell us much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log curve, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as microprocessor technology, then the day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ’ s not moving like that. And there ’ s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ’ d like to ask is, if you look at these exponential curves, they don ’ t go on forever. Things just can ’ t possibly keep changing as fast as they are. One of two things is going to happen. Either it ’ s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ’ s going to do this. That ’ s about all it can do. Now I ’ m an optimist, so I sort of think it ’ s probably going to do something like that. If so, that means that what we ’ re in the middle of right now is a transition. We ’ re sort of on this line in a transition from the way the world used to be to some new way that the world is. And so what I ’ m trying to ask, what I ’ ve been asking myself, is what ’ s this new way that the world is? What ’ s that new state that the world is heading toward? Because the transition seems very, very confusing when we ’ re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ’ s a conference in which people talk about the future, and you notice that the future is still at about the year 2000. It ’ s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ’ s happening there. That transition is happening. We can all sense it. And we know that it just doesn ’ t make too much sense to think out 30, 50 years because everything ’ s going to be so different that a simple extrapolation of what we ’ re doing just doesn ’ t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ’ re going through. Now in order to do that I ’ m going to have to talk about a bunch of stuff that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step back and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, the Earth was this big, sterile hunk of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ’ s theories that are beginning to understand about how it started with RNA, but I ’ m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ’ t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ’ s sort of just a very simple chemical form of life, but when things got interesting was when these drops learned a trick about abstraction. Somehow by ways that we don ’ t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for amusement purposes is we can now write things in this code. And I ’ ve got here a little 100 micrograms of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ’ s right here. If I had really been a egotist, I would have put it into a virus and released it in the room. So what was the next step? Writing down the DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ’ t know if you know this, but bacteria can actually exchange DNA. Now that ’ s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the transition point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ’ s interesting in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ’ re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to evolve so that the interesting level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very special structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in the community and special structures that were responsible for recording, understanding, learning information. And that was the brains and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ’ d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these special information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ’ re aware of is human language. It ’ s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ’ m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ’ re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ’ ve done is we, humanity, have started abstracting out. We ’ re going through the same levels that multi-cellular organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. Telephony, computers, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ’ ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of microseconds. And you saw Ty ’ s little evolutionary example where he sort of did a little bit of evolution on the Convolution program right before your eyes. So now we ’ ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and brains, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ’ s what we ’ re seeing here in this explosion of curve. We ’ re seeing this process feeding back on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ’ s really impossible for me to design them in the traditional sense. I don ’ t know what every transistor in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of abstraction and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ’ t quite even understand. One method that ’ s particularly interesting that I ’ ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the microsecond time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,”Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted.”So in other words, I define what I wanted. Let ’ s say I want to sort numbers, as a simple example I ’ ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,”Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ’ s reproduce the ones that sorted numbers the best. And let ’ s reproduce them by a process of recombination analogous to sex.”Take two programs and they produce children by exchanging their subroutines, and the children inherit the traits of the subroutines of the two programs. So I ’ ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,”Please repeat that process.”Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ’ t tell you how they work. I ’ ve tried looking at them and telling you how they work. They ’ re obscure, weird programs. But they do the job. And in fact, I know, I ’ m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,”Oh look. Look at this. It says, ’ This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ’ Doesn ’ t that make you feel confident?”In fact, we know that the engineering process doesn ’ t work very well when it gets complicated. So we ’ re beginning to depend on computers to do a process that ’ s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ’ t quite understand the options of it. So in a sense, it ’ s getting ahead of us. We ’ re now using those programs to make much faster computers so that we ’ ll be able to run this process much faster. So it ’ s feeding back on itself. The thing is becoming faster and that ’ s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ’ re taking off. And what we are is we ’ re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ’ re the amoebas and we can ’ t quite figure out what the hell this thing is we ’ re creating. We ’ re right at that point of transition. But I think that there really is something coming along after us. I think it ’ s very haughty of us to think that we ’ re the end product of evolution. And I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

6 NEG Dim 3

Text Sample 420: NEG Dim 3 – 1994 – Score -29.00 – 1994/t002458.txt

Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ’ s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC **microprocessors** versus the power of local area networks. And the interesting thing about it is that this slide, like so many technology slides that we ’ re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ’ s basically what I ’ m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ’ m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well the answer is, if I drew it on a normal curve where, let ’ s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ’ m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ’ t tell us much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log curve, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as **microprocessor** technology, then the day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ’ s not moving like that. And there ’ s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ’ d like to ask is, if you look at these exponential curves, they don ’ t go on forever. Things just can ’ t possibly keep changing as fast as they are. One of two things is going to happen. Either it ’ s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ’ s going to do this. That ’ s about all it can do. Now I ’ m an optimist, so I sort of think it ’ s probably going to do something like that. If so, that means that what we ’ re in the middle of right now is a transition. We ’ re sort of on this line in a transition from the way the world used to be to some new way that the world is. And so what I ’ m trying to ask, what I ’ ve been asking myself, is what ’ s this new way that the world is? What ’ s that new state that the world is heading toward? Because the transition seems very, very confusing when we ’ re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ’ s a conference in which people talk about the future, and you notice that the future is still at about the year 2000. It ’ s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ’ s happening there. That transition is happening. We can all sense it. And we know that it just doesn ’ t make too much sense to think out 30, 50 years because everything ’ s going to be so different that a simple **extrapolation** of what we ’ re doing just doesn ’ t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ’ re going through. Now in order to do that I ’ m going to have to talk about a bunch of stuff that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step back and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, the Earth was this big, **sterile hunk** of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ’ s theories that are beginning to understand about how it started with RNA, but I ’ m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ’ t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ’ s sort of just a very simple chemical form of life, but when things **got** interesting was when these drops learned a trick about abstraction. Somehow by ways that we don ’ t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this **mindless** sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for **amusement** purposes is we can now write things in this code. And I ’ ve got here a little 100 **micrograms** of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for **symbolizing** it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ’ s right here. If I had really been a egotist, I would have put it into a virus and released it in the room. So what was the next step? Writing down the DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ’ t know if you know this, but bacteria can actually exchange DNA. Now that ’ s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from **penicillin**, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to **penicillin**, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were **synergistic**. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the transition point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ' s interesting in life took about another billion years. And at that stage, we have **multi-cellular** communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ' re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to evolve so that the interesting level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very special structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing **apparatus** that those communities of cells built up. And in fact, they began to get specialists in the community and special structures that were responsible for recording, understanding, learning information. And that was the brains and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ' d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these special information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ' re aware of is human language. It ' s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ' m sitting here making **grunting** sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ' re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ' ve done is we, humanity, have started abstracting out. We ' re going through the same levels that **multi-cellular** organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. **Telephony**, computers, **videotapes**, CD-ROMs and so on are all our specialized mechanisms that we ' ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of **microseconds**. And you saw Ty ' s little evolutionary example where he sort of did a little bit of evolution on the **Convolution** program right before your eyes. So now we ' ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and brains, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ' s what we ' re seeing here in this explosion of curve. We ' re seeing this process feeding back on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ' s really impossible for me to design them in the traditional sense. I don ' t know what every **transistor** in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of abstraction and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ' t quite even understand. One method that ' s particularly interesting that I ' ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the **microsecond** time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,"Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted."So in other words, I define what I wanted. Let ' s say I want to sort numbers, as a simple example I ' ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,"Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ' s reproduce the ones that sorted numbers the best. And let ' s reproduce them by a process of **recombination analogous** to sex."Take two programs and they produce children by exchanging their **subroutines**, and the children inherit the traits of the **subroutines** of the two programs. So I ' ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,"Please repeat that process."Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few **milliseconds**. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ' t tell you how they work. I ' ve tried looking at them and telling you how they work. They ' re obscure, weird programs. But they do the job. And in fact, I know, I ' m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,"Oh look. Look at this. It says, ' This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ' Doesn ' t that make you feel confident?"In fact, we know that the engineering process doesn ' t work very well when it gets complicated. So we ' re beginning to depend on computers to do a process that ' s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ' t quite understand the options of it. So in a sense, it ' s getting ahead of us. We ' re now using those programs to make much faster computers so that we ' ll be able to run this process much faster. So it ' s feeding back on itself. The thing is becoming faster and that ' s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ' re taking off. And what we are is we ' re at a point in time which is **analogous** to when single- celled organisms were turning into multi-celled organisms. So we ’ re the **amoebas** and we can ’ t quite figure out what the hell this thing is we ’ re creating. We ’ re right at that point of transition. But I think that there really is something coming along after us. I think it ’ s very **haughty** of us to think that we ’ re the end product of evolution. And I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

Text Sample 421: NEG Dim 3 – 1998 – Score -10.00 – 1998/t000200.txt

You hear that this is the era of environment â or biology, or information technology... Well, it ’ s the era of a lot of different things that we ’ re in right now. But one thing for sure : it ’ s the era of change. There ’ s more change going on than ever has occurred in the history of human life on earth. And you all sort of know it, but it ’ s hard to get it so that you really understand it. And I ’ ve tried to put together something that ’ s a good start for this. I ’ ve tried to show in this â though the color doesn ’ t come out â that what I ’ m concerned with is the little 50-year time bubble that you are in. You tend to be interested in a generation past, a generation future â your parents, your kids, things you can change over the next few decades â and this 50-year time bubble you kind of move along in. And in that 50 years, if you look at the population curve, you find the population of humans on the earth more than doubles and we ’ re up three-and-a-half times since I was born. When you have a new baby, by the time that kid gets out of high school more people will be added than existed on earth when I was born. This is unprecedented, and it ’ s big. Where it goes in the future is questioned. So that ’ s the human part. Now, the human part related to animals : look at the left side of that. What I call the human portion â humans and their livestock and pets â versus the natural portion â all the other wild animals and just â these are vertebrates and all the birds, etc., in the land and air, not in the water. How does it balance? Certainly, 10, 000 years ago, the civilization ’ s beginning, the human portion was less than one tenth of one percent. Let ’ s look at it now. You follow this curve and you see the whiter spot in the middle â that ’ s your 50- year time bubble. Humans, livestock and pets are now 97 percent of that integrated total mass on earth and all wild nature is three percent. We have won. The next generation doesn ’ t even have to worry about this game â it is over. And the biggest problem came in the last 25 years : it went from 25 percent up to that 97 percent. And this really is a sobering picture upon realizing that we, humans, are in charge of life on earth ; we ’ re like the capricious Gods of old Greek myths, kind of playing with life â and not a great deal of wisdom injected into it. Now, the third curve is information technology. This is Moore ’ s Law plotted here, which relates to density of information, but it has been pretty good for showing a lot of other things about information technology â computers, their use, Internet, etc. And what ’ s important is it just goes straight up through the top of the curve, and has no real limits to it. Now try and contrast these. This is the size of the earth going through that same â â frame. And to make it really clear, I ’ ve put all four on one graph. There ’ s no need to see the little detailed words on it. That first one is humans-versus-nature ; we ’ ve won, there ’ s no more gain. Human population. And so if you ’ re looking for growth industries to get into, that ’ s not a good one â protecting natural creatures. Human population is going up ; it ’ s going to continue for quite a while. Good business in obstetricians, morticians, and farming, housing, etc. â they all deal with human bodies, which require being fed, transported, housed and so on. And the information technology, which connects to our brains, has no limit â now, that is a wonderful field to be in. You ’ re looking for growth opportunity? It ’ s just going up through the roof. And then, the size of the Earth. Somehow making these all compatible with the Earth looks like a pretty bad industry to be involved with. So, that ’ s the stage out of all this. I find, for reasons I don ’ t understand, I really do have a goal. And the goal is that the world be desirable and sustainable when my kids reach my age â and I think that ’ s â in other words, the next generation. I think that ’ s a goal that we probably all share. I think it ’ s a hopeless goal. Technologically, it ’ s achievable ; economically, it ’ s achievable ; politically, it means sort of the habits, institutions of people â it ’ s impossible. The institutions of the past with all their inertia are just irrelevant for the future, except they ’ re there and we have to deal with them. I spend about 15 percent of my time trying to save the world, the other 85 percent, the usual â and whatever else we devote ourselves to. And in that 15 percent, the main focus is on human mind, thinking skills, somehow trying to unleash kids from the **straightjacket** of school, which is putting information and dogma into them, get them so they really think, ask tough questions, argue about serious subjects, don ’ t believe everything that ’ s in the book, think broadly or creative. They can be. Our school systems are very flawed and do not reward you for the things that are important in life or for the survival of civilization ; they reward you for a lot of learning and sopping up stuff. We can ’ t go into that today because there isn ’ t time â it ’ s a broad subject. One thing for sure, in the future there is an essential feature â necessary, but not sufficient â which is doing more with less. We ’ ve got to be doing things with more efficiency using less energy, less material. Your great-great grandparents got by on muscle power, and yet we all think there ’ s this huge power that ’ s essential for our lifestyle. And with all the wonderful technology we have we can do things that are much more efficient : conserve, recycle, etc. Let me just rush very quickly through things that we ’ ve done. Human-powered airplane â **Gossamer** Condor sort of started me in this direction in 1976 and 77, winning the Kremer prize in aviation history, followed by the Albatross. And we began making various odd planes and creatures. Here ’ s a giant flying replica of a pterosaur that has no tail. Trying to have it fly straight is like trying to shoot an arrow with the feathered end forward. It was a tough job, and boy it made me have a lot of respect for nature. This was the full size of the original creature. We did things on land, in the air, on water â vehicles of all different kinds, usually with some electronics or electric power systems in them. I find they ’ re all the same, whether its land, air or water. I ’ ll be focusing on the air here. This is a solar-powered airplane â 165 miles carrying a person from France to England as a symbol that solar power is going to be an important part of our future. Then we did the solar car for General Motors â the Sunracer â that won the race in Australia. We got a lot of people thinking about electric cars, what you could do with them. A few years later, when we suggested to GM that now is the time and we could do a thing called the Impact, they sponsored it, and here ’ s the Impact that we developed with them on their programs. This is the demonstrator. And they put huge effort into turning it into a commercial product. With that preamble, let ’ s show the first two-minute **videotape**, which shows a little airplane for surveillance and moving to a giant airplane. Narrator : A tiny airplane, the AV Pointer serves for surveillance â in effect, a pair of roving eyeglasses. A cutting-edge example of where miniaturization can lead if the operator is remote from the vehicle. It is convenient to carry, assemble and launch by hand. Battery-powered, it is silent and rarely noticed. It sends high-resolution video pictures back to the operator. With onboard GPS, it can navigate autonomously, and it is rugged enough to self-land without damage. The modern **sailplane** is superbly efficient. Some can glide as flat as 60 feet forward for every foot of descent. They are powered only by the energy they can extract from the atmosphere â an atmosphere nature stirs up by solar energy. Humans and soaring birds have found nature to be generous in providing replenishable energy. **Sailplanes** have flown over 1, 000 miles, and the altitude record is over 50, 000 feet. The Solar Challenger was made to serve as a symbol that photovoltaic cells can produce real power and will be part of the world ' s energy future. In 1981, it flew 163 miles from Paris to England, solely on the power of sunbeams, and established a basis for the Pathfinder. The message from all these vehicles is that ideas and technology can be harnessed to produce remarkable gains in doing more with less â gains that can help us attain a desirable balance between technology and nature. The stakes are high as we speed toward a challenging future. Buckminster Fuller said it clearly :"there are no passengers on spaceship Earth, only crew. We, the crew, can and must do more with less â much less."Paul MacCready : If we could have the second video, the one-minute, put in as quickly as you can, which â this will show the Pathfinder airplane in some flights this past year in Hawaii, and will show a sequence of some of the beauty behind it after it had just flown to 71, 530 feet â higher than any propeller airplane has ever flown. It ' s amazing : just on the puny power of the sun â by having a super lightweight plane, you ' re able to get it up there. It ' s part of a long-term program NASA sponsored. And we worked very closely with the whole thing being a team effort, and with wonderful results like that flight. And we ' re working on a bigger plane â 220-foot span â and an intermediate-size, one with a regenerative fuel cell that can store excess energy during the day, feed it back at night, and stay up 65, 000 feet for months at a time. Ray Morgan ' s voice will come in here. There he ' s the project manager. Anything they do is certainly a team effort. He ran this program. Here ' s... some things he showed as a celebration at the very end. Ray Morgan : We ' d just ended a seven-month deployment of Hawaii. For those who live on the mainland, it was tough being away from home. The friendly support, the quiet confidence, **congenial** hospitality shown by our Hawaiian and military hosts â this is starting â made the experience enjoyable and unforgettable. PM : We have real-time IR scans going out through the Internet while the plane is flying. And it ' s exploring without polluting the stratosphere. That ' s its goal : the stratosphere, the blanket that really controls the radiation of the earth and permits life on earth to be the success that it is â probing that is very important. And also we consider it as a sort of poor man ' s stationary satellite, because it can stay right overhead for months at a time, 2, 000 times closer than the real GFC synchronous satellite. We couldn ' t bring one here to fly it and show you. But now let ' s look at the other end. In the video you saw that nine-pound or eight-pound Pointer airplane surveillance drone that Keenan has developed and just done a remarkable job. Where some have servos that have gotten down to, oh, 18 or 25 grams, his weigh one-third of a gram. And what he ' s going to bring out here is a surveillance drone that weighs about 2 ounces â that includes the video camera, the batteries that run it, the telemetry, the receiver and so on. And we ' ll fly it, we hope, with the same success that we had last night when we did the practice. So Matt Keenan, just any time you ' re â all right â ready to let her go. But first, we ' re going to make sure that it ' s appearing on the screen, so you see what it sees. You can imagine yourself being a mouse or fly inside of it, looking out of its camera. Matt Keenan : It ' s switched on. PM : But now we ' re trying to get the video. There we go. MK : Can you bring up the house lights? PM : Yeah, the house lights and we ' ll see you all better and be able to fly the plane better. MK : All right, we ’ ll try to do a few laps around and bring it back in. Here we go. PM : The video worked right for the first few and I don ’ t know why it â there it goes. Oh, that was only a minute, but I think you ’ d be safe to have that near the end of the flight, perhaps. We get to do the classic. All right. If this hits you, it will not hurt you. OK. Thank you very much. Thank you. But now, as they say in **infomercials**, we have something much better for you, which we ’ re working on : planes that are only six inches â 15 centimeters â in size. And Matt ’ s plane was on the cover of Popular Science last month, showing what this can lead to. And in a while, something this size will have GPS and a video camera in it. We ’ ve had one of these fly nine miles through the air at 35 miles an hour with just a little battery in it. But there ’ s a lot of technology going. There are just milestones along the way of some remarkable things. This one doesn ’ t have the video in it, but you get a little feel from what it can do. OK, here we go. MK : Sorry. OK. PM : If you can pass it down when you ’ re done. Yeah, I think â I lost a little orientation ; I looked up into this light. It hit the building. And the building was poorly placed, actually. But you ’ re beginning to see what can be done. We ’ re working on projects now â even wing-flapping things the size of hawk moths â DARPA contracts, working with Caltech, UCLA. Where all this leads, I don ’ t know. Is it practical? I don ’ t know. But like any basic research, when you ’ re really forced to do things that are way beyond existing technology, you can get there with micro-technology, nanotechnology. You can do amazing things when you realize what nature has been doing all along. As you get to these small scales, you realize we have a lot to learn from nature â not with 747s â but when you get down to the nature ’ s realm, nature has 200 million years of experience. It never makes a mistake. Because if you make a mistake, you don ’ t leave any progeny. We should have nothing but success stories from nature, for you or for birds, and we ’ re learning a lot from its fascinating subjects. In concluding, I want to get back to the big picture and I have just two final slides to try and put it in perspective. The first I ’ ll just read. At last, I put in three sentences and had it say what I wanted. Over billions of years on a unique sphere, chance has painted a thin covering of life â complex, improbable, wonderful and fragile. Suddenly, we humans â a recently arrived species, no longer subject to the checks and balances inherent in nature â have grown in population, technology and intelligence to a position of terrible power. We now wield the paintbrush. And that ’ s serious : we ’ re not very bright. We ’ re short on wisdom ; we ’ re high on technology. Where ’ s it going to lead? Well, inspired by the sentences, I decided to wield the paintbrush. Every 25 years I do a picture. Here ’ s the one â tries to show that the world isn ’ t getting any bigger. Sort of a timeline, very non-linear scale, nature rates and trilobites and dinosaurs, and eventually we saw some humans with caves... Birds were flying overhead, after pterosaurs. And then we get to the civilization above the little TV set with a gun on it. Then traffic jams, and power systems, and some dots for digital. Where it ’ s going to lead â I have no idea. And so I just put robotic and natural cockroaches out there, but you can fill in whatever you want. This is not a forecast. This is a warning, and we have to think seriously about it. And that time when this is happening is not 100 years or 500 years. Things are going on this decade, next decade ; it ’ s a very short time that we have to decide what we are going to do. And if we can get some agreement on where we want the world to be â desirable, sustainable when your kids reach your age â I think we actually can reach it. Now, I said this was a warning, not a forecast. That was before â I painted this before we started in on making robotic versions of hawk moths and cockroaches, and now I ’ m beginning to wonder seriously â was this more of a forecast than

I wanted? I personally think the surviving intelligent life form on earth is not going to be carbon-based ; it ’ s going to be silicon-based. And so where it all goes, I don ’ t know. The one final bit of sparkle we ’ ll put in at the very end here is an utterly impractical flight vehicle, which is a little **ornithopter** wing-flapping device that â rubber-band powered â that we ’ ll show you. MK : 32 gram. Sorry, one gram. PM : Last night we gave it a few too many turns and it tried to bash the roof out also. It ’ s about a gram. The tube there ’ s hollow, about paper-thin. And if this lands on you, I assure you it will not hurt you. But if you reach out to grab it or hold it, you will destroy it. So, be gentle, just act like a wooden Indian or something. And when it comes down â and we ’ ll see how it goes. We consider this to be sort of the spirit of TED. And you wonder, is it practical? And it turns out if I had not been â Unfortunately, we have some light bulb changes. We can probably get it down, but it ’ s possible it ’ s gone up to a greater destiny up there â â than it ever had. And I wanted to make â just â But I want to make just two points. One is, you think it ’ s frivolous ; there ’ s nothing to it. And yet if I had not been making **ornithopters** like that, a little bit cruder, in 1939 â a long, long time ago â there wouldn ’ t have been a **Gossamer** Condor, there wouldn ’ t have been an Albatross, a Solar Challenger, there wouldn ’ t be an Impact car, there wouldn ’ t be a mandate on zero-emission vehicles in California. A lot of these things â or similar â would have happened some time, probably a decade later. I didn ’ t realize at the time I was doing inquiry-based, hands-on things with teams, like they ’ re trying to get in education systems. So I think that, as a symbol, it ’ s important. And I believe that also is important. You can think of it as a sort of a symbol for learning and TED that somehow gets you thinking of technology and nature, and puts it all together in things that are â that make this conference, I think, more important than any that ’ s taken place in this country in this decade. Thank you.

Text Sample 422: NEG Dim 3 – 1998 – Score -4.00 – 1998/t000216.txt

David Gallo : This is Bill Lange. I ’ m Dave Gallo. And we ’ re going to tell you some stories from the sea here in video. We ’ ve got some of the most incredible video of Titanic that ’ s ever been seen, and we ’ re not going to show you any of it. The truth of the matter is that the Titanic — even though it ’ s breaking all sorts of box office records — it ’ s not the most exciting story from the sea. And the problem, I think, is that we take the ocean for granted. When you think about it, the oceans are 75 percent of the planet. Most of the planet is ocean water. The average depth is about two miles. Part of the problem, I think, is we stand at the beach, or we see images like this of the ocean, and you look out at this great big blue expanse, and it ’ s shimmering and it ’ s moving and there ’ s waves and there ’ s surf and there ’ s tides, but you have no idea for what lies in there. And in the oceans, there are the longest mountain ranges on the planet. Most of the animals are in the oceans. Most of the earthquakes and volcanoes are in the sea, at the bottom of the sea. The biodiversity and the biodensity in the ocean is higher, in places, than it is in the rainforests. It ’ s mostly unexplored, and yet there are beautiful sights like this that captivate us and make us become familiar with it. But when you ’ re standing at the beach, I want you to think that you ’ re standing at the edge of a very unfamiliar world. We have to have a very special technology to get into that unfamiliar world. We use the submarine Alvin and we use cameras, and the cameras are something that Bill Lange has developed with the help of Sony. Marcel Proust said,”The true voyage of discovery is not so much in seeking new landscapes as in having new eyes.”People that have partnered with us have given us new eyes, not only on what exists — the new landscapes at the bottom of the sea — but also how we think about life on the planet itself. Here ’ s a jelly. It ’ s one of my favorites, because it ’ s got all sorts of working parts. This turns out to be the longest creature in the oceans. It gets up to about 150 feet long. But see all those different working things? I love that kind of stuff. It ’ s got these fishing lures on the bottom. They ’ re going up and down. It ’ s got tentacles dangling, swirling around like that. It ’ s a colonial animal. These are all individual animals banding together to make this one creature. And it ’ s got these jet thrusters up in front that it ’ ll use in a moment, and a little light. If you take all the big fish and schooling fish and all that, put them on one side of the scale, put all the jelly-type of animals on the other side, those guys win hands down. Most of the biomass in the ocean is made out of creatures like this. Here ’ s the X-wing death jelly. The bioluminescence — they use the lights for attracting mates and attracting prey and communicating. We couldn ’ t begin to show you our archival stuff from the jellies. They come in all different sizes and shapes. Bill Lange : We tend to forget about the fact that the ocean is miles deep on average, and that we ’ re real familiar with the animals that are in the first 200 or 300 feet, but we ’ re not familiar with what exists from there all the way down to the bottom. And these are the types of animals that live in that three- dimensional space, that micro-gravity environment that we really haven ’ t explored. You hear about giant squid and things like that, but some of these animals get up to be approximately 140, 160 feet long. They ’ re very little understood. DG : This is one of them, another one of our favorites, because it ’ s a little octopod. You can actually see through his head. And here he is, flapping with his ears and very gracefully going up. We see those at all depths and even at the greatest depths. They go from a couple of inches to a couple of feet. They come right up to the submarine — they ’ ll put their eyes right up to the window and peek inside the sub. This is really a world within a world, and we ’ re going to show you two. In this case, we ’ re passing down through the mid-ocean and we see creatures like this. This is kind of like an undersea rooster. This guy, that looks incredibly formal, in a way. And then one of my favorites. What a face! This is basically scientific data that you ’ re looking at. It ’ s footage that we ’ ve collected for scientific purposes. And that ’ s one of the things that Bill ’ s been doing, is providing scientists with this first view of animals like this, in the world where they belong. They don ’ t catch them in a net. They ’ re actually looking at them down in that world. We ’ re going to take a joystick, sit in front of our computer, on the Earth, and press the joystick forward, and fly around the planet. We ’ re going to look at the mid-ocean ridge, a 40, 000-mile long mountain range. The average depth at the top of it is about a mile and a half. And we ’ re over the Atlantic — that ’ s the ridge right there — but we ’ re going to go across the Caribbean, Central America, and end up against the Pacific, nine degrees north. We make maps of these mountain ranges with sound, with sonar, and this is one of those mountain ranges. We ’ re coming around a cliff here on the right. The height of these mountains on either side of this valley is greater than the Alps in most cases. And there ’ s tens of thousands of those mountains out there that haven ’ t been mapped yet. This is a volcanic ridge. We ’ re getting down further and further in scale. And eventually, we can come up with something like this. This is an icon of our robot, Jason, it ’ s called. And you can sit in a room like this, with a joystick and a headset, and drive a robot like that around the bottom of the ocean in real time. One of the things we ’ re trying to do at Woods Hole with our partners is to bring this virtual world — this world, this unexplored region — back to the laboratory. Because we see it in bits and pieces right now.

We see it either as sound, or we see it as video, or we see it as photographs, or we see it as chemical sensors, but we never have yet put it all together into one interesting picture. Here ' s where Bill ' s cameras really do shine. This is what ' s called a hydrothermal vent. And what you ' re seeing here is a cloud of densely packed, hydrogen-sulfide-rich water coming out of a volcanic axis on the sea floor. Gets up to 600, 700 degrees F, somewhere in that range. So that ' s all water under the sea — a mile and a half, two miles, three miles down. And we knew it was volcanic back in the ' 60s, ' 70s. And then we had some hint that these things existed all along the axis of it, because if you ' ve got **volcanism**, water ' s going to get down from the sea into cracks in the sea floor, come in contact with magma, and come shooting out hot. We weren ' t really aware that it would be so rich with sulfides, hydrogen sulfides. We didn ' t have any idea about these things, which we call chimneys. This is one of these hydrothermal vents. Six hundred degree F water coming out of the Earth. On either side of us are mountain ranges that are higher than the Alps, so the setting here is very dramatic. BL : The white material is a type of bacteria that thrives at 180 degrees C. DG : I think that ' s one of the greatest stories right now that we ' re seeing from the bottom of the sea, is that the first thing we see coming out of the sea floor after a volcanic eruption is bacteria. And we started to wonder for a long time, how did it all get down there? What we find out now is that it ' s probably coming from inside the Earth. Not only is it coming out of the Earth — so, biogenesis made from volcanic activity — but that bacteria supports these colonies of life. The pressure here is 4, 000 pounds per square inch. A mile and a half from the surface to two miles to three miles — no sun has ever gotten down here. All the energy to support these life forms is coming from inside the Earth — so, chemosynthesis. And you can see how dense the population is. These are called tube worms. BL : These worms have no digestive system. They have no mouth. But they have two types of gill structures. One for extracting oxygen out of the deep-sea water, another one which houses this chemosynthetic bacteria, which takes the hydrothermal fluid — that hot water that you saw coming out of the bottom — and converts that into simple sugars that the tube worm can digest. DG : You can see, here ' s a crab that lives down there. He ' s managed to grab a tip of these worms. Now, they normally retract as soon as a crab touches them. Oh! Good going. So, as soon as a crab touches them, they retract down into their shells, just like your fingernails. There ' s a whole story being played out here that we ' re just now beginning to have some idea of because of this new camera technology. BL : These worms live in a real temperature extreme. Their foot is at about 200 degrees C and their head is out at three degrees C, so it ' s like having your hand in boiling water and your foot in freezing water. That ' s how they like to live. DG : This is a female of this kind of worm. And here ' s a male. You watch. It doesn ' t take long before two guys here — this one and one that will show up over here — start to fight. Everything you see is played out in the pitch black of the deep sea. There are never any lights there, except the lights that we bring. Here they go. On one of the last dive series, we counted 200 species in these areas — 198 were new, new species. BL : One of the big problems is that for the biologists working at these sites, it ' s rather difficult to collect these animals. And they disintegrate on the way up, so the imagery is critical for the science. DG : Two octopods at about two miles depth. This pressure thing really amazes me — that these animals can exist there at a depth with pressure enough to crush the Titanic like an empty Pepsi can. What we saw up till now was from the Pacific. This is from the Atlantic. Even greater depth. You can see this shrimp is harassing this poor little guy here, and he ' ll bat it away with his claw. Whack! And the same thing ' s going on over here. What they ' re getting at is that — on the back of this crab — the foodstuff here is this very strange bacteria that lives on the backs of all these animals. And what these shrimp are trying to do is actually harvest the bacteria from the backs of these animals. And the crabs don ’ t like it at all. These long filaments that you see on the back of the crab are actually created by the product of that bacteria. So, the bacteria grows hair on the crab. On the back, you see this again. The red dot is the laser light of the submarine Alvin to give us an idea about how far away we are from the vents. Those are all shrimp. You see the hot water over here, here and here, coming out. They ’ re clinging to a rock face and actually scraping bacteria off that rock face. Here ’ s a tiny, little vent that ’ s come out of the side of that pillar. Those pillars get up to several stories. So here, you ’ ve got this valley with this incredible alien landscape of pillars and hot springs and volcanic eruptions and earthquakes, inhabited by these very strange animals that live only on chemical energy coming out of the ground. They don ’ t need the sun at all. BL : You see this white V-shaped mark on the back of the shrimp? It ’ s actually a light-sensing organ. It ’ s how they find the hydrothermal vents. The vents are emitting a black body radiation — an IR signature — and so they ’ re able to find these vents at considerable distances. DG : All this stuff is happening along that 40, 000-mile long mountain range that we ’ re calling the ribbon of life, because just even today, as we speak, there ’ s life being generated there from volcanic activity. This is the first time we ’ ve ever tried this any place. We ’ re going to try to show you high definition from the Pacific. We ’ re moving up one of these pillars. This one ’ s several stories tall. In it, you ’ ll see that it ’ s a habitat for a lot of different animals. There ’ s a funny kind of hot plate here, with vent water coming out of it. So all of these are individual homes for worms. Now here ’ s a closer view of that community. Here ’ s crabs here, worms here. There are smaller animals crawling around. Here ’ s pagoda structures. I think this is the neatest-looking thing. I just can ’ t get over this — that you ’ ve got these little chimneys sitting here smoking away. This stuff is toxic as hell, by the way. You could never get a permit to dump this in the ocean, and it ’ s coming out all from it. It ’ s unbelievable. It ’ s basically sulfuric acid, and it ’ s being just dumped out, at incredible rates. And animals are thriving — and we probably came from here. That ’ s probably where we evolved from. BL : This bacteria that we ’ ve been talking about turns out to be the most simplest form of life found. There are a number of groups that are proposing that life evolved at these vent sites. Although the vent sites are short-lived — an individual site may last only 10 years or so — as an ecosystem they ’ ve been stable for millions — well, billions — of years. DG : It works too well. You see there ’ re some fish inside here as well. There ’ s a fish sitting here. Here ’ s a crab with his claw right at the end of that tube worm, waiting for that worm to stick his head out. BL : The biologists right now cannot explain why these animals are so active. The worms are growing inches per week! DG : I already said that this site, from a human perspective, is toxic as hell. Not only that, but on top — the lifeblood — that plumbing system turns off every year or so. Their plumbing system turns off, so the sites have to move. And then there ’ s earthquakes, and then volcanic eruptions, on the order of one every five years, that completely wipes the area out. Despite that, these animals grow back in about a year ’ s time. You ’ re talking about biodensities and biodiversity, again, higher than the rainforest that just springs back to life. Is it sensitive? Yes. Is it fragile? No, it ’ s not really very fragile. I ’ ll end up with saying one thing. There ’ s a story in the sea, in the waters of the sea, in the sediments and the rocks of the sea floor. It ’ s an incredible story. What we see when we look back in time, in those sediments and rocks, is a record of Earth history. Everything on this planet — everything — works by cycles and rhythms. The continents move apart. They come back together. Oceans come and go. Mountains come and go. Glaciers come and go. El Nino comes and goes. It ’ s not a disaster, it ’ s rhythmic. What we ’ re learning now, it ’ s almost like a symphony. It ’ s just like music — it really is just like music. And what we ’ re learning now is that you can ’ t listen to a five-billion-year long symphony, get to today and say,”Stop! We want tomorrow ’ s note to be the same as it was today.”It ’ s absurd. It ’ s just absurd. So, what we ’ ve got to learn now is to find out where this planet ’ s going at all these different scales and work with it. Learn to manage it. The concept of preservation is futile. Conservation ’ s tougher, but we can probably get there. Thank you very much. Thank you.

Text Sample 423: NEG Dim 3 – 1998 – Score -2.00 – 1998/t000118.txt

Back in 1992, I started working for a company called Interval Research, which was just then being founded by David Lidell and Paul Allen as a for-profit research enterprise in Silicon Valley. I met with David to talk about what I might do in his company. I was just coming out of a failed virtual reality business and supporting myself by being on the speaking circuit and writing books — after twenty years or so in the computer game industry having ideas that people didn ’ t think they could sell. And David and I discovered that we had a question in common, that we really wanted the answer to, and that was,”Why hasn ’ t anybody built any computer games for little girls?”Why is that? It can ’ t just be a giant sexist conspiracy. These people aren ’ t that smart. There ’ s six billion dollars on the table. They would go for it if they could figure out how. So, what is the deal here? And as we thought about our goals — I should say that Interval is really a humanistic institution, in the classical sense that humanism, at its best, finds a way to combine clear-eyed empirical research with a set of core values that fundamentally love and respect people. The basic idea of humanism is the improvable quality of life ; that we can do good things, that there are things worth doing because they ’ re good things to do, and that clear-eyed **empiricism** can help us figure out how to do them. So, contrary to popular belief, there is not a conflict of interest between **empiricism** and values. And Interval Research is kind of the living example of how that can be true. So David and I decided to go find out, through the best research we could muster, what it would take to get a little girl to put her hands on a computer, to achieve the level of comfort and ease with the technology that little boys have because they play video games. We spent two and a half years conducting research ; we spent another year and a half in advance development. Then we formed a spin-off company. In the research phase of the project at Interval, we partnered with a company called Cheskin Research, and these people — Davis Masten and Christopher Ireland — changed my mind entirely about what market research was and what it could be. They taught me how to look and see, and they did not do the incredibly stupid thing of saying to a child,”Of all these things we already make you, which do you like best?”— which gives you zero answers that are usable. So, what we did for the first two and a half years was four things : We did an extensive review of the literature in related fields, like cognitive psychology, spatial cognition, gender studies, play theory, sociology, primatology. Thank you Frans de Waal, wherever you are, I love you and I ’ d give anything to meet you. After we had done that with a pretty large team of people and discovered what we thought the salient issues were with girls and boys and playing — because, after all, that ’ s really what this is about — we moved to the second phase of our work, where we interviewed adult experts in academia, some of the people who ’ d produced the literature that we found relevant. Also, we did focus groups with people who were on the ground with kids every day, like playground supervisors. We talked to them, confirmed some hypotheses and identified some serious questions about gender difference and play. Then we did what I consider to be the heart of the work : interviewed 1, 100 children, boys and girls, ages seven to 12, all over the United States — except for Silicon Valley, Boston and Austin because we knew that their little families would have millions of computers in them and they wouldn ’ t be a representative sample. And at the end of those remarkable conversations with kids and their best friends across the United States, after two years, we pulled together some survey data from another 10, 000 children, drew up a set up of what we thought were the key findings of our research, and spent another year transforming them into design heuristics, for designing computer-based products — and, in fact, any kind of products — for little girls, ages eight to 12. And we spent that time designing interactive prototypes for computer software and testing them with little girls. In 1996, in November, we formed the company Purple Moon which was a spinoff of Interval Research, and our chief investors were Interval Research, Vulcan Northwest, Institutional Venture Partners and Allen and Company. We launched a website on September 2nd that has now served 25 million pages, and has 42, 000 registered young girl users. They visit an average of one and a half times a day, spend an average of 35 minutes a visit, and look at 50 pages a visit. So we feel that we ’ ve formed a successful online community with girls. We launched two titles in October —”Rockett ’ s New School”— the first of a series of products — is about a character called Rockett beginning her first day of school in eighth grade at a brand new place, with a blank slate, which allows girls to play with the question of,”What will I be like when I ’ m older?””What ’ s it going to be like to be in high school or junior high school? Who are my friends?”; to exercise the love of social complexity and the narrative intelligence that drives most of their play behavior ; and which embeds in it values about noticing that we have lots of choices in our lives and the ways that we conduct ourselves. The other title that we launched is called”Secret Paths in the Forest,”which addresses the more fantasy-oriented, inner lives of girls. These two titles both showed up in the top 50 entertainment titles in PC Data — entertainment titles in PC Data in December, right up there with”John Madden Football,”which thrills me to death. So, we ’ re real, and we ’ ve touched several hundreds of thousands of little girls. We ’ ve made half-a-billion impressions with marketing and PR for this brand, Purple Moon. Ninety-six percent of them, roughly, have been positive ; four percent of them have been”other.”I want to talk about the other, because the politics of this enterprise, in a way, have been the most fascinating part of it, for me. There are really two kinds of negative reviews that we ’ ve received. One kind of reviewer is a male gamer who thinks he knows what games ought to be, and won ’ t show the product to little girls. The other kind of reviewer is a certain flavor of feminist who thinks they know what little girls ought to be. And so it ’ s funny to me that these interesting, odd bedfellows have one thing in common : they don ’ t listen to little girls. They haven ’ t looked at children and they ’ re certainly not demonstrating any love for them. I ’ d like to play you some voices of little girls from the two-and-a-half years of research that we did — actually, some of the voices are more recent. And these voices will be accompanied by photographs that they took for us of their lives, of the things that they value and care about. These are pictures the girls themselves never saw, but they gave to us This is the stuff those reviewers don ’ t know about and aren ’ t listening to and this is the kind of research I recommend to those who want to do humanistic work. Girl 1 : Yeah, my character is usually a tomboy. Hers is more into boys. Girl 2 : Uh, yeah. Girl 1 : We have — in the very beginning of the whole game, always we do this : we each have a piece of paper ; we write down our name, our age — are we rich, very rich, not rich, poor, medium, wealthy, boyfriends, dogs, pets — what else — sisters, brothers, and all those. Girl 2 : Divorced — parents divorced, maybe. Girl 3 : This is my pretend [unclear] one. Girl 4 : We make a school newspaper on the computer. Girl 5 : For a girl ' s game also usually they ' ll have really pretty scenery with clouds and flowers. Girl 6 : Like, if you were a girl and you were really adventurous and a real big tomboy, you would think that girls ' games were kinda sissy. Girl 7 : I run track, I played soccer, I play basketball, and I love a lot of things to do. And sometimes I feel like I can ' t really enjoy myself unless it ' s like a vacation, like when I get Mondays and all those days off. Girl 8 : Well, sometimes there is a lot of stuff going on because I have music lessons and I ' m on swim team — all this different stuff that I have to do, and sometimes it gets overwhelming. Girl 9 : My friend Justine kinda took my friend Kelly, and now they ' re being mean to me. Girl 10 : Well, sometimes it gets annoying when your brothers and sisters, or brother or sister, when they copy you and you get your idea first and they take your idea and they do it themselves. Girl 11 : Because my older sister, she gets everything and, like, when I ask my mom for something, she ' ll say,"No"— all the time. But she gives my sister everything. Brenda Laurel : I want to show you, real quickly, just a minute of"Rockett ' s Tricky Decision,"which went gold two days ago. Let ' s hope it ' s really stable. This is the second day in Rockett ' s life. The reason I ' m showing you this is I ' m hoping that the scene that I ' m going to show you will look familiar and sound familiar, now that you ' ve listened to some girls ' voices. And you can see how we ' ve tried to incorporate the issues that matter to them in the game that we ' ve created. Miko : Hey Rockett! C ' mere! Rockett : Hi Miko! What ' s going on? Miko : Did you hear about Nakilia ' s big Halloween party this weekend? She asked me to make sure you knew about it. Nakilia invited Reuben too, but — Rockett : But what? Isn ' t he coming? Miko : I don ' t think so. I mean, I heard his band is playing at another party the same night. Rockett : Really? What other party? Girl : Max ' s party is going to be so cool, Whitney. He ' s invited all the best people. BL : I ' m going to fast-forward to the decision point because I know I don ' t have a lot of time. After this awful event occurs, Rocket gets to decide how she feels about it. Rockett : Who ' d want to show up at that party anyway? I could get invited to that party any day if I wanted to. Gee, I doubt I ' ll make Max ' s best people list. BL : OK, so we ' re going to emotionally navigate. If we were playing the game, that ' s what we ' d do. If at any time during the game we want to learn more about the characters, we can go into this hidden hallway, and I ' ll quickly just show you the interface. We can, for example, go find Miko ' s locker and get some more information about her. Oops, I turned the wrong way. But you get the general idea of the product. I wanted to show you the ways, innocuous as they seem, in which we ' re incorporating what we ' ve learned about girls — their desires to experience greater emotional flexibility, and to play around with the social complexity of their lives. I want to make the point that what we ' re giving girls, I think, through this effort, is a kind of validation, a sense of being seen. And a sense of the choices that are available in their lives. We love them. We see them. We ' re not trying to tell them who they ought to be. But we ' re really, really happy about who they are. It turns out they ' re really great. I want to close by showing you a **videotape** that ' s a version of a future game in the Rockett series that our graphic artists and design people put together, that we feel would please that four percent of reviewers."Rockett 28!"Rockett : It ' s like I ' m just waking up, you know? BL : Thanks.

Text Sample 424: NEG Dim 3 – 1984 – Score 1.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ’ ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a computer and really address the qualities of the human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, the intelligence of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ’ ve been able to collect, which I think illustrate this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a slide taken from a TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ’ s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ’ s the distance you should sit away from the TV set. Now we ’ ve put people 18 inches in front of a TV, and all the artifacts that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow mask, the scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ’ m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ’ m very interested in touch-sensitive displays. High-tech, high-touch. Isn ’ t that what some of you said? It ’ s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ’ re not : it ’ s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on the market these systems that have just a few light emitting diodes on the side and are very low resolution, it ’ s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ’ t have to pick them up, and people don ’ t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ’ re typing — what you have — you want to now put something — first of all, you ’ ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ’ ve got to sort of find that mouse. Then you find the mouse, and you ’ re going to have to wiggle it a little bit to see where the cursor is on the screen. And then when you finally see where it is, then you ’ ve got to move it to get the cursor over there, and then —”Bang”— you ’ ve got to hit a button or do whatever. That ’ s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ’ m trying to do is just illustrate the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of the computer field, is when you have a bug that you can ' t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the substrate, which it usually is. So we found that that actually was a feature in the sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ' s pressure-sensitive. And it ' s pressure-sensitive to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ' t figure out how to come in the other direction. But let me get rid of the slide, and let ' s see if this comes on. OK. So there is the pressure-sensitive display in operation. The person ' s just, if you will, pushing on the screen to make a curve. But this is the interesting part. I want to stop it for a second because the movie is very badly made. And the particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an anvil on a fuzzy rug and the other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across the screen, and yet you touch the ping-pong ball very lightly and it just scoots across the screen. And what you can do — oops, I didn ' t mean to do that — what you can do is actually feed back to the user the feeling of the physical properties. So again, they don ' t have to be weight ; they could be a general trying to move troops, and he ' s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the interface there are physical properties in that transducer — in this case it ' s pressure and touches — that allow you to present things to the user that you could never present before. So it ' s not simply looking at the quality or, if you will, the luxury of that interface, but it ' s actually looking at the idea of presenting things that previously couldn ' t be presented before. I want to move on to another example, which is one of a different sort, where we ' re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ' re going to take this book, if you will, and it ' s going to come alive. You ' re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are the experts in monologue making : you make a film and it has a well-formed beginning, middle and end, and in some sense the art of it is that. And you then say,"There ' s an opportunity for making conversational movies."Well, what does that mean? And it sort of nibbles at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ' ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the movie has information about itself. So it knows, or at least there is computer-readable information in the medium itself. It ’ s just not a static movie frame. That ’ s one thing. The other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the cookbook, the”Larousse Gastronomique.”And I think I use the example all too often, but it ’ s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to the end of the recipe and it says,”Cook until done.”Now, that would be, if you will, the top green track, which doesn ’ t mean too much. But you might have to elaborate for me or for somebody who isn ’ t an expert, and say,”Cook at 380 degrees for 45 minutes.”And then for a real beginner, you would go down even further and elaborate more — say,”Open the oven, preheat, wait for the light to go out, open the door, don ’ t leave it open too long, put the penguin in and shut the door...”whatever. And that ’ s a much more elaborate one than you dribble back. That ’ s one kind of use of random access. And the other is where you want to explain the same thing in different ways. If you ’ re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ’ s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ’ s a rather boring book, but I ’ m afraid sometimes you have to do boring books because your sponsors aren ’ t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ’ t even know what vintage the transmission is, but let me just show you very quickly some of it, and we ’ ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of the transmission, and as you rub your finger across the transmission it highlights the various parts. Narrator : When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, superimposed over the illustration. NN : This is about the oil pan, or the oil filter and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get back to the illustration, but in this case it ’ s not a single frame, but it ’ s actually a movie of someone coming into the frame and doing the repair that ’ s described in the text. The two-headed slider is a speed control that allows me to watch the movie at various speeds, in forward or reverse. And the movie is displayed as a full frame movie. I can go back to the beginning... and play the movie at full speed. Here ’ s another step-by-step procedure, only in this case — NN : Okay, this movie is... Everybody ’ s heard of sound-sync movies — this is text-sync movies, so as the movie plays, the text gets highlighted. We highlight the text as we go through the movie. Repairman :... Not too far out. Front poles, preferably. Don ’ t loosen them too far. If you loosen them too far, you ’ ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ’ m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ’ ve been doing — in this case, in Senegal — where we have tried to use personal computers as a pedagogical medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ’ s a story I ’ d like to tell. And that was when, before we did this in any developing countries — we ’ re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ’ ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ’ t read, didn ’ t even make it in the lowest section of the school ’ s classes — and was pretty much not in school, though physically there. But did hang around the, quote,”computer room,”where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very interesting. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,”Let me show you how this works,”and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ’ t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ’ d actually come to see — not the computer room — they went upstairs and they said,”This is absolutely remarkable! That child was very articulate and showed us and even dealt with the things he couldn ’ t do automatically with that manual. It was just absolutely fantastic.”The principal said,”There ’ s a dreadful mistake, because that child can ’ t read. And you obviously have been hoodwinked or you ’ ve talked about somebody else.”And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,”Can you read?”And the child said,”No, I can ’ t.”And then they said,”But wait a minute. You just looked through that manual and you found...”and he said,”Oh, but that ’ s not reading.”And so they said,”Well, what ’ s reading then?”He says,”Well, reading is this junk they give me in little books to read. It ’ s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return.”And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ’ m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ’ s not, but we think speech —”My God, little children pick it up somehow, and by the age of two they ’ re doing a mediocre job, and by three and four they ’ re speaking reasonably well. And yet you ’ ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder.”Well, it ’ s not harder. What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ’ s going to help you. So what happens is you go to school and people say,”Just believe me, you ’ re going to like it. You ’ re going to like reading,”and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on the screen has a huge payoff, and it ’ s a lot of fun. And in fact, it ’ s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ’ s the girl on the left leaning with her chalkboard, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ’ s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example — as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And the reason it remains a favorite project is that we were asked to do a teleconferencing system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ’ s faces. So if I wanted to call my friend Peter Sprague on the phone, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ’ s uncanny : there ’ s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ’ s something that didn ’ t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,”He or she wouldn ’ t agree,”and the empty chair is that person and the spatiality is crucial. So we said,”Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ’ s site you ’ d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ’ t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ’ ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ’ s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ’ s site, he ’ ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.

Text Sample 425: NEG Dim 3 – 2002 – Score -3.00 – 2002/t000321.txt

I ’ m a historian. Steve told us about the future of little technology ; I ’ m going to show you some of the past of big technology. This was a project to build a 4, 000-ton nuclear bomb-propelled spaceship and go to Saturn and Jupiter. This took place in my childhood, 1957-65. It was deeply classified. I ’ m going to show you some stuff that not only has not been declassified, but has now been reclassified. If all goes well, next year I ’ ll be back, and I ’ ll have a lot more to show you, and if all doesn ’ t go well, I ’ ll be in jail, like Wen Ho Lee. So, this ship was basically the size of the Marriott Hotel, a little taller and a little bigger. And one of the people who worked on it at the beginning was my father, Freeman, there in the middle. That ’ s me and my sister, Esther, who ’ s a frequent TEDster. I didn ’ t like nuclear bomb-propelled spaceships. I mean, I thought it was a great idea, but I started building kayaks. So we had a few kayaks. Just so you know that I am not Dr. Strangelove. But all the time I was out there doing these strange kayak voyages in odd, beautiful parts of this planet, I always thought in the back of my mind about Project Orion, and how my father and his friends were going to build these big ships. They were actually going to go — Ted Taylor, who led the project, was going to take his children. My father was not going to take his children, that was one of the reasons we sort of had a falling out for a few years. The project began in ’ 57 at General Atomics there, that ’ s right on the coast at La Jolla. Look at that central building right in the middle of the picture. That ’ s the 130-foot diameter library. That is exactly the size of the base of the spaceship. So put that library at the bottom of that ship — that ’ s how big the thing was going to be. It would take two or three thousand bombs. The people who worked on it were a lot of the Los Alamos people who had done the hydrogen bomb work. It was the first project funded by ARPA. That ’ s the contract where ARPA gave the first million dollars to get this thing started.”Spaceship project officially begun. Job waiting for you. Dyson.”That ’ s July ’ 58. Two days later, the space traveler ’ s manifesto explaining why — just like we heard yesterday — why we need to go into space :”... trips to satellites of the outer planets. August 20, 1958.”These are the statistics of what would be the good places to go and stop. Some of the sizes of the ships, ranging all the way up to ship mass of 8 million tons. So that was the outer extreme. Here was version two : 2, 000 bombs. These are five-kiloton yield bombs, about the size of small Volkswagens ; it would take 800 to get into orbit. Here we see a 10, 000-ton ship will deliver 1, 300 tons to Saturn and back — essentially, a five-year trip. Possible departure dates : October 1960 to February 1967. These are trajectories going to Mars. All this was done by hand, with slide rules. The little Orion ship, and what it would take to do what Orion does with chemicals : you have a ship the size of the Empire State Building. NASA had no interest ; they tried to kill the project. The people who supported it were the Air Force, which meant that it was all secret. And that ’ s why when you get something declassified, that ’ s what it looks like. Military weapon versions that carried hydrogen bombs that could destroy half the planet. There ’ s another version there that sends retaliatory strikes at the Soviet Union. This is the really secret stuff : how to get directed energy explosions. So you ’ re sending the energy of a nuclear explosion — not like just a stick of dynamite, but you ’ re directing it at the ship. And this is still a very active subject. It ’ s quite dangerous, but I believe it ’ s better to have dangerous things in the open than think you ’ re going to keep them secret. This is what happened at 600 **microseconds**. The Air Force started to build smaller models and actually started doing this. The guys in La

Jolla said,”We ’ ve got to get started now.”They built a high-explosive propelled model. These are stills from film footage that was saved by someone who was supposed to destroy it but didn ’ t, and kept it in their basement for the last 40 years. So, these are three-pound charges of C4 ; that ’ s about 10 times what the guy had in his shoes. This is Ed Day putting — So each of these coffee cans has three pounds of C4 in it. They ’ re building a system that ejects these at quarter-second intervals. That ’ s my dad in the sport coat there, holding the briefcase. So, they had a lot of fun doing this. But no children were allowed ; my dad could tell me he was building a spaceship and going to go to Saturn, but he could not say anything more about it. So all my life I have wanted to find this stuff out, and spent the last four years tracking these old guys down. These are stills from the video. Jeff Bezos kindly, yesterday, said he ’ ll put this video up on the Amazon site — some little clip of it. So, thanks to him. They got quite serious about the engineering of this. The size of that mass, for us, is really large technology in a way we ’ re never going to go back to. If you saw the 1959 — this is what it would feel like in the passenger compartment ; that ’ s acceleration profile. And pulse-system yield : we ’ re looking at 20-kiloton yield for an effective thrust of 10 million newtons. Well, here we have a little problem, the radiation doses at the crew station : 700 **rads** per shot. Fission yields during development : they were hoping to get clean bombs ; they didn ’ t. Eyeburn : this is what happens to the people in Miami who are looking up. Personnel compartment noise : that ’ s not too bad ; it ’ s very low frequencies, it ’ s basically like these sub-woofers. And now we have ground-hazard assessments when you have a blow-up on the pad. Finally, at the very end in 1964, NASA steps in and says,”OK, we ’ ll support a feasibility study for a small version that could be launched with Saturn Vs in sections and pieced together.”So this is what NASA did, getting an eight-man version that would go to Mars. They liked it because the guys could kind of live there and be like,”It ’ s like living in a submarine.”This is crew compartment. It switches, so what ’ s upside down is right side up when you go to artificial gravity mode. The scientists were still going to go along ; they would take seven astronauts and seven scientists. This is a 20-man version for going to Jupiter : bunks, storm cellars, exercise room. You know, it was going to be a nice, long trip. The Air Force version : here we have a military version. This is the kind of stuff that ’ s not been declassified, just that people managed to sneak home and after, you know, on their deathbed, basically, gave me that. The sort of artist conceptions. These are basically PowerPoint presentations given to the Air Force 40 years ago. Look at the little guys there outside the vehicle. And one part of NASA was interested in it, but the headquarters in NASA, they killed the project. So finally, at the end, we can see the thing followed its sort of design path right up to 1965, and then all those paths came to a halt. Results : none. This project is hereby terminated. So that ’ s the end. All I can say in closing is : we heard yesterday that one of the 10 bad things that could happen to us is an asteroid with our name on it. And one of the bad things that could happen to NASA is if that asteroid shows up with our name on it nine months out, and everybody says,”Well, what are we going to do?”And Orion is really one of the only, if not the only, off-the-shelf technologies that could do something. So I ’ m going to tell you the good news and the bad news. The good news is that NASA has a small, secret contingency-plan division that is looking at this, trying to keep knowledge of Orion preserved in the event of such a misfortune. Maybe keep a few little bombs of plutonium on the side. That ’ s the good news. The bad news is, when I got in contact with these people to try and get some documents from them, they went crazy because I had all this stuff that they don ’ t have, and NASA purchased 1, 759 pages of this stuff from me. So that ’ s the state we ’ re at ; it ’ s not very good.

Text Sample 426: NEG Dim 3 – 2002 – Score -2.00 – 2002/t000269.txt

Welcome. If I could have the first slide, please? Contrary to calculations made by some engineers, bees can fly, dolphins can swim, and geckos can even climb up the smoothest surfaces. Now, what I want to do, in the short time I have, is to try to allow each of you to experience the thrill of revealing nature ’ s design. I get to do this all the time, and it ’ s just incredible. I want to try to share just a little bit of that with you in this presentation. The challenge of looking at nature ’ s designs — and I ’ ll tell you the way that we perceive it, and the way we ’ ve used it. The challenge, of course, is to answer this question : what permits this extraordinary performance of animals that allows them basically to go anywhere? And if we could figure that out, how can we implement those designs? Well, many biologists will tell engineers, and others, organisms have millions of years to get it right ; they ’ re spectacular ; they can do everything wonderfully well. So, the answer is **bio-mimicry** : just copy nature directly. We know from working on animals that the truth is that ’ s exactly what you don ’ t want to do — because evolution works on the just-good-enough principle, not on a perfecting principle. And the constraints in building any organism, when you look at it, are really severe. Natural technologies have incredible constraints. Think about it. If you were an engineer and I told you that you had to build an automobile, but it had to start off to be this big, then it had to grow to be full size and had to work every step along the way. Or think about the fact that if you build an automobile, I ’ ll tell you that you also — inside it — have to put a factory that allows you to make another automobile. And you can absolutely never, absolutely never, because of history and the inherited plan, start with a clean slate. So, organisms have this important history. Really evolution works more like a tinkerer than an engineer. And this is really important when you begin to look at animals. Instead, we believe you need to be inspired by biology. You need to discover the general principles of nature, and then use these analogies when they ’ re advantageous. This is a real challenge to do this, because animals, when you start to really look inside them — how they work — appear hopelessly complex. There ’ s no detailed history of the design plans, you can ’ t go look it up anywhere. They have way too many motions for their joints, too many muscles. Even the simplest animal we think of, something like an insect, and they have more neurons and connections than you can imagine. How can you make sense of this? Well, we believed — and we hypothesized — that one way animals could work simply, is if the control of their movements tended to be built into their bodies themselves. What we discovered was that two-, four-, six- and eight-legged animals all produce the same forces on the ground when they move. They all work like this kangaroo, they bounce. And they can be modeled by a spring-mass system that we call the spring mass system because we ’ re biomechanists. It ’ s actually a pogo stick. They all produce the pattern of a pogo stick. How is that true? Well, a human, one of your legs works like two legs of a trotting dog, or works like three legs, together as one, of a trotting insect, or four legs as one of a trotting crab. And then they alternate in their propulsion, but the patterns are all the same. Almost every organism we ’ ve looked at this way — you ’ ll see next week, I ’ ll give you a hint, there ’ ll be an article coming out that says that really big things like T. rex probably couldn ’ t do this, but you ’ ll see that next week. Now, what ’ s interesting is the animals, then — we said — bounce along the vertical plane this way, and in our collaborations with Pixar, in”A Bug ’ s Life,”we discussed the bipedal nature of the characters of the ants. And we told them, of course, they move in another plane as well. And they asked us this question. They say,"Why model just in the sagittal plane or the vertical plane, when you ' re telling us these animals are moving in the horizontal plane?"This is a good question. Nobody in biology ever modeled it this way. We took their advice and we modeled the animals moving in the horizontal plane as well. We took their three legs, we collapsed them down as one. We got some of the best mathematicians in the world from Princeton to work on this problem. And we were able to create a model where animals are not only bouncing up and down, but they ' re also bouncing side to side at the same time. And many organisms fit this kind of pattern. Now, why is this important to have this model? Because it ' s very interesting. When you take this model and you perturb it, you give it a push, as it bumps into something, it self-stabilizes, with no brain or no reflexes, just by the structure alone. It ' s a beautiful model. Let ' s look at the mathematics. That ' s enough! The animals, when you look at them running, appear to be self-stabilizing like this, using basically springy legs. That is, the legs can do computations on their own ; the control algorithms, in a sense, are embedded in the form of the animal itself. Why haven ' t we been more inspired by nature and these kinds of discoveries? Well, I would argue that human technologies are really different from natural technologies, at least they have been so far. Think about the typical kind of robot that you see. Human technologies have tended to be large, flat, with right angles, stiff, made of metal. They have rolling devices and axles. There are very few motors, very few sensors. Whereas nature tends to be small, and curved, and it bends and twists, and has legs instead, and appendages, and has many muscles and many, many sensors. So it ' s a very different design. However, what ' s changing, what ' s really exciting — and I ' ll show you some of that next — is that as human technology takes on more of the characteristics of nature, then nature really can become a much more useful teacher. And here ' s one example that ' s really exciting. This is a collaboration we have with Stanford. And they developed this new technique, called Shape Deposition Manufacturing. It ' s a technique where they can mix materials together and mold any shape that they like, and put in the material properties. They can embed sensors and actuators right in the form itself. For example, here ' s a leg : the clear part is stiff, the white part is compliant, and you don ' t need any axles there or anything. It just bends by itself beautifully. So, you can put those properties in. It inspired them to show off this design by producing a little robot they named Sprawl. Our work has also inspired another robot, a biologically inspired bouncing robot, from the University of Michigan and McGill named RHex, for robot hexapod, and this one ' s autonomous. Let ' s go to the video, and let me show you some of these animals moving and then some of the simple robots that have been inspired by our discoveries. Here ' s what some of you did this morning, although you did it outside, not on a treadmill. Here ' s what we do. This is a death ' s head cockroach. This is an American cockroach you think you don ' t have in your kitchen. This is an eight-legged scorpion, six-legged ant, forty-four-legged centipede. Now, I said all these animals are sort of working like pogo sticks — they ' re bouncing along as they move. And you can see that in this ghost crab, from the beaches of Panama and North Carolina. It goes up to four meters per second when it runs. It actually leaps into the air, and has aerial phases when it does it, like a horse, and you ' ll see it ' s bouncing here. What we discovered is whether you look at the leg of a human like Richard, or a cockroach, or a crab, or a kangaroo, the relative leg stiffness of that spring is the same for everything we ' ve seen so far. Now, what good are springy legs then? What can they do? Well, we wanted to see if they allowed the animals to have greater stability and maneuverability. So, we built a terrain that had obstacles three times the hip height of the animals that we ' re looking at. And we were certain they couldn ' t do this. And here ' s what they did. The animal ran over it and it didn ' t even slow down! It didn ' t decrease its preferred speed at all. We couldn ' t believe that it could do this. It said to us that if you could build a robot with very simple, springy legs, you could make it as maneuverable as any that ' s ever been built. Here ' s the first example of that. This is the Stanford Shape Deposition Manufactured robot, named Sprawl. It has six legs — there are the tuned, springy legs. It moves in a gait that an insect uses, and here it is going on the treadmill. Now, what ' s important about this robot, compared to other robots, is that it can ' t see anything, it can ' t feel anything, it doesn ' t have a brain, yet it can maneuver over these obstacles without any difficulty whatsoever. It ' s this technique of building the properties into the form. This is a graduate student. This is what he ' s doing to his thesis project — very robust, if a graduate student does that to his thesis project. This is from McGill and University of Michigan. This is the RHex, making its first outing in a demo. Same principle : it only has six moving parts, six motors, but it has springy, tuned legs. It moves in the gait of the insect. It has the middle leg moving in synchrony with the front, and the hind leg on the other side. Sort of an alternating tripod, and they can negotiate obstacles just like the animal. Robert Full : It ' ll go on different surfaces — here ' s sand — although we haven ' t perfected the feet yet, but I ' ll talk about that later. Here ' s RHex entering the woods. Again, this robot can ' t see anything, it can ' t feel anything, it has no brain. It ' s just working with a tuned mechanical system, with very simple parts, but inspired from the fundamental dynamics of the animal. RF : Here ' s it going down a pathway. I presented this to the jet propulsion lab at NASA, and they said that they had no ability to go down craters to look for ice, and life, ultimately, on Mars. And he said — especially with legged-robots, because they ' re way too complicated. Nothing can do that. And I talk next. I showed them this video with the simple design of RHex here. And just to convince them we should go to Mars in 2011, I tinted the video orange just to give them the sense of being on Mars. Another reason why animals have extraordinary performance, and can go anywhere, is because they have an effective interaction with the environment. The animal I ' m going to show you, that we studied to look at this, is the gecko. We have one here and notice its position. It ' s holding on. Now I ' m going to challenge you. I ' m going show you a video. One of the animals is going to be running on the level, and the other one ' s going to be running up a wall. Which one ' s which? They ' re going at a meter a second. How many think the one on the left is running up the wall? Okay. The point is it ' s really hard to tell, isn ' t it? It ' s incredible, we looked at students do this and they couldn ' t tell. They can run up a wall at a meter a second, 15 steps per second, and they look like they ' re running on the level. How do they do this? It ' s just phenomenal. The one on the right was going up the hill. How do they do this? They have bizarre toes. They have toes that uncurl like party favors when you blow them out, and then peel off the surface, like tape. Like if we had a piece of tape now, we ' d peel it this way. They do this with their toes. It ' s bizarre! This peeling inspired iRobot — that we work with — to build Mecho-Geckos. Here ' s a legged version and a tractor version, or a bulldozer version. Let ' s see some of the geckos move with some video, and then I ' ll show you a little bit of a clip of the robots. Here ' s the gecko running up a vertical surface. There it goes, in real time. There it goes again. Obviously, we have to slow this down a little bit. You can ' t use regular cameras. You have to take 1, 000 pictures per second to see this. And here ' s some video at 1, 000 frames per second. Now, I want you to look at the animal ' s back. Do you see how much it ' s bending like that? We can ' t figure that out

— that ’ s an unsolved mystery. We don ’ t know how it works. If you have a son or a daughter that wants to come to Berkeley, come to my lab and we ’ ll figure this out. Okay, send them to Berkeley because that ’ s the next thing I want to do. Here ’ s the gecko mill. It ’ s a see-through treadmill with a see-through treadmill belt, so we can watch the animal ’ s feet, and **videotape** them through the treadmill belt, to see how they move. Here ’ s the animal that we have here, running on a vertical surface. Pick a foot and try to watch a toe, and see if you can see what the animal ’ s doing. See it uncurl and then peel these toes. It can do this in 14 **milliseconds**. It ’ s unbelievable. Here are the robots that they inspire, the Mecho-Geckos from iRobot. First we ’ ll see the animals toes peeling — look at that. And here ’ s the peeling action of the Mecho-Gecko. It uses a pressure- sensitive adhesive to do it. Peeling in the animal. Peeling in the Mecho-Gecko — that allows them climb autonomously. Can go on the flat surface, transition to a wall, and then go onto a ceiling. There ’ s the bulldozer version. Now, it doesn ’ t use pressure-sensitive glue. The animal does not use that. But that ’ s what we ’ re limited to, at the moment. What does the animal do? The animal has weird toes. And if you look at the toes, they have these little leaves there, and if you blow them up and zoom in, you ’ ll see that ’ s there ’ s little striations in these leaves. And if you zoom in 270 times, you ’ ll see it looks like a rug. And if you blow that up, and zoom in 900 times, you see there are hairs there, tiny hairs. And if you look carefully, those tiny hairs have striations. And if you zoom in on those 30, 000 times, you ’ ll see each hair has split ends. And if you blow those up, they have these little structures on the end. The smallest branch of the hairs looks like spatulae, and an animal like that has one billion of these nano-size split ends, to get very close to the surface. In fact, there ’ s the diameter of your hair — a gecko has two million of these, and each hair has 100 to 1, 000 split ends. Think of the contact of that that ’ s possible. We were fortunate to work with another group at Stanford that built us a special manned sensor, that we were able to measure the force of an individual hair. Here ’ s an individual hair with a little split end there. When we measured the forces, they were enormous. They were so large that a patch of hairs about this size — the gecko ’ s foot could support the weight of a small child, about 40 pounds, easily. Now, how do they do it? We ’ ve recently discovered this. Do they do it by friction? No, force is too low. Do they do it by electrostatics? No, you can change the charge — they still hold on. Do they do it by interlocking? That ’ s kind of a like a Velcro-like thing. No, you can put them on molecular smooth surfaces — they don ’ t do it. How about suction? They stick on in a vacuum. How about wet adhesion? Or capillary adhesion? They don ’ t have any glue, and they even stick under water just fine. If you put their foot under water, they grab on. How do they do it then? Believe it or not, they grab on by intermolecular forces, by Van der Waals forces. You know, you probably had this a long time ago in chemistry, where you had these two atoms, they ’ re close together, and the electrons are moving around. That tiny force is sufficient to allow them to do that because it ’ s added up so many times with these small structures. What we ’ re doing is, we ’ re taking that inspiration of the hairs, and with another colleague at Berkeley, we ’ re manufacturing them. And just recently we ’ ve made a breakthrough, where we now believe we ’ re going to be able to create the first synthetic, self-cleaning, dry adhesive. Many companies are interested in this. We also presented to Nike even. We ’ ll see where this goes. We were so excited about this that we realized that that small-size scale — and where everything gets sticky, and gravity doesn ’ t matter anymore — we needed to look at ants and their feet, because one of my other colleagues at Berkeley has built a six-millimeter silicone robot with legs. But it gets stuck. It doesn ’ t move very well. But the ants do, and we ’ ll figure out why, so that ultimately we ’ ll make this move. And imagine : you ’ re going to be able to have swarms of these six-millimeter robots available to run around. Where ’ s this going? I think you can see it already. Clearly, the Internet is already having eyes and ears, you have web cams and so forth. But it ’ s going to also have legs and hands. You ’ re going to be able to do programmable work through these kinds of robots, so that you can run, fly and swim anywhere. We saw David Kelly is at the beginning of that with his fish. So, in conclusion, I think the message is clear. If you need a message, if nature ’ s not enough, if you care about search and rescue, or mine clearance, or medicine, or the various things we ’ re working on, we must preserve nature ’ s designs, otherwise these secrets will be lost forever. Thank you.

Text Sample 427: NEG Dim 3 – 2002 – Score -2.00 – 2002/t000341.txt

Dan Holzman : Please throw out the beanbag chairs. Here we go. Barry Friedman : There are all kinds of high-tech chairs here today, but this is really, I think, when it reached its peak as far as ergonomics, comfort, design, flexibility... DH : Now obviously, this is not something we do on our regular show ; it ’ s something we just kind of learned for this, so we ’ re going to try. But can we have some inspirational music for the beanbag chairs? BF : Nice show, Daniel, nice show. You are the man! Nice show. Man, that was good! DH : Thank you. BF : You know, sometimes when people do those, they go all the way down. You actually just did that. That ’ s the kind of extra effort that ’ s gotten us where we are today... DH : All right, let ’ s show them something special. BF :... without a MacArthur grant. Yeah, look at this. You know, all kinds of different... TED is about invention, let ’ s be honest. Right? DH : Yeah, it is. BF : Last night, Michael Moschen showed some **juggling** props he has invented and working on. Right now, Dan ’ s going to show something he actually invented. DH : A type of juggling I actually invented, right after I saw another **juggler** do it. BF : Shut up. DH : And this is a small excerpt from a longer piece. Folks, this is shaker cup **juggling**. It ’ s not a showstopper but it certainly slows it down. BF : Oh yeah, it does. BF : Oh, Daniel. DH : One more? Perfect. Perfect. BF : OK. DH : Oh! All right. I ’ m now pushing my luck : I ’ m skipping right to six cups. In order to do six cups, I must have perfect control over three with my right hand. BF : Also three with his left. DH : Perfect. And now, all six cups. Should I do it on the first try or should I miss once on purpose? BF : First try? Once on purpose? DH : How about if I try first and then decide? BF : Good idea. Let ’ s leave that. We ’ ll leave that door open. DH : He ’ s looking at me. BF : That ’ s all right, he does that. All right. DH : Oh! It ’ s time for Richard ’ s help. Oh, good. All right. BF : You know, over the years, every year at the conference, it ’ s kind of become a tradition for us to do something dangerous with Richard. And we ’ ve always done something with the bullwhips in our act. It ’ s funny, for years I did it with Daniel holding balloons. And then we thought,”How stupid.”DH : Excuse me, could we work on the design of the microphone? BF : I think that ’ s the next session. DH : Next session? BF : Yeah. And so we ’ ve actually found a way to incorporate Richard in this. He actually assumes more of the danger in this. DH : Please stand up, Richard. Oh, sorry. DH : Now Richard, please... BF : OK, sorry. DH : Jesus Christ. Richard, please stand in front of me. Richard Wurman : Can I say something? BF : Sure. RW : In all past years I ’ ve rehearsed with them, the things that have happened to me — I have no idea what ’ s going to happen and that ’ s the truth. DH : All right, please stand here in front... God, I hate that. Put your hands out like this, please. BF : No, come stay up with him. Dan used to actually hold them but now he ’ s got you for protection. It ’ s kind of neat. OK. DH : Wow, you ’ ve been working out. BF : No, shut up! Having a little bit of Richard time. That ’ s nice, that ’ s good. OK, here we go. Have him hold your wrist so I can... DH : Please hold my wrist, will you. BF : Yeah, hold this a minute. There you go. OK. OK, hold on. RW : Hmmm. DH : First one. BF : All those mid-year phone calls are coming back to me now, Richard. DH : So Richard, what were we on the list? Like 1, 020? What happened there? BF : I think we were just outside. DH : I don ’ t get it. DH : Sorry. BF : Having some bad flashbacks. RW : Do you want me to hold you or not? DH : Don ’ t hold me that hard. BF : Here we go, I ’ m taking it. DH : One more, one more. BF : We ’ ve got one more we ’ re going to do. RW : Do I get to hold them? BF : You don ’ t want to hold these, trust me. DH : Could you spread your legs a little bit? BF : Gloria, you want to do it? It ’ s very cool. One more try. Man, I don ’ t want to get too close. Could you just push that? DH : Wow! Boy! BF : That ’ s cool. I always wanted to try that. DH : Let ’ s jump this way, though. Now, we risked Richard ’ s life, it ’ s only fair we risk our own lives. So to do that, I will juggle these three razor-sharp sickles. And if that wasn ’ t enough, and judging by your response, it ’ s not... DH : Wow! BF : Hoping for a little more build. DH : True. Barry... BF : I ’ m going to run up behind him. DH : Leap over my shoulder. BF : Up and over his shoulders. DH : Grab the blades in mid-air, land right there in a pool of blood... Still juggling. Impossible, you say? BF : Incredible, you say? DH : Why bother, you say? BF : Here we go. DH : Just do it **juggler** boys, you say? BF : This guy, this guy invented air. DH : I think so, that ’ s right. Even the pencil. BF : He invented the pencil. DH : All right, we ’ ll do this trick, but please remember it took us over 10 years to perfect. BF : Ten years to perfect, which you ’ re about to see. DH : It ’ s not that difficult, we just don ’ t like to practice that much. BF : No, it ’ s a hassle. Traveling too much. Actually, we will take a second to prove — this could be fake — that the blades are indeed razor-sharp. DH : Will someone please throw a small farm animal up onto the stage? Or a virgin for a sacrifice? BF : Anything? DH : Where ’ s Gloria? BF : No, she ’ s got... farm animal. DH : Do you have a small farm animal? Just trying to play the odds. All right, here we go. BF : Over the top, over the top. DH : How you feeling, Barry? You feeling all right? BF : Yeah, it ’ s all right. DH : Do you feel everything ’ s OK? The atmosphere, the... BF : Yeah, a little sketchy. DH : Everything up here ’ s OK? BF : Yeah. DH : Then here we go. BF : This one ’ s a little... Who ’ s doing the lights? Could you point that a little more directly into my eyeballs? Is that possible? I can still see a little. DH : And turn up the intensity ; we ’ re still pink in the middle. We went too far. BF : Yeah, it ’ s too far. It ’ s too much of a visual. The design of the body is a whole different thing. DH : Ready, Barry? BF : Over the top. DH : May we have our jumping music please? May we have it a bit louder? BF : They ’ re a good crew! Whoa! DH : Whoa, sorry. All right. BF : We ’ re going on. DH : All right, we ’ ll try again. BF : All right? Oh my gosh. Oh. DH : All right, here we go. Sorry about that. BF : I thought I had the hard part. OK. DH : Whenever you ’ re ready. BF : There we go! All right, get up! Come on and dance! DH : Dance, come on. BF : Come on and dance! Somebody dance! Come on! Wow, wow, OK, stop. Weird, no one dances. We ’ re two guys doing this. I think that ’ s uncomfortable for everyone. DH : The French judge... BF : One more quick thing. DH : The French judge gives it a 5. 2. BF : Well, you know... DH : There you go... BF : Oh, yeah. Another one coming in. DH : Tell them about our bio and stuff. BF : Yeah. In our bio, some of you may have read that we ’ ve won two world **juggling** championships. And believe it or not, you don ’ t win juggling champions for doing things with bullwhips or shaker cups. We ’ re going to show you right now an excerpt from a routine that we used to wipe out the other **juggling** team competition. DH : That ' s right. BF : Good. DH : I know what you ' re thinking : other **juggling** teams must really suck. BF : **Juggling** ' s got a bad rap. DH : But wait, Barry, there ' s still one more club lying there by my foot. And look, it has a twin! BF : Shut up. DH : There ' s still one more by my foot. What do you want me to do with it? BF : Richard you tell him, it ' s your last year. DH : That ' s a pretty good setup, Richard. BF : Yeah, it ' s a good setup. That ' s a big setup. DH : You can ' t get any better than that. All right. What I will do : I will use my panther-like reflexes. BF : Nice. DH : I got that — to reach down and grab that club in my grip of steel. BF : Nice. DH : I touched it, Barry. That should be enough. BF : It ' s progress, that ' s the thing. DH : How about that? I ' ll do it again. Oh wait, it ' s on your side, Barry. And it ' s awfully windy over there. BF : It is, it ' s weird. You wouldn ' t think it would affect half the stage, but it is. It ' s weird. Watch this : what I ' m going to do is slide the seventh one onto my foot. DH : Wow! What a great trick, Barry! Oh, look how it lies there. Oh, Barry, is there nothing you can ' t do? You are my hero. You ' re my Jim Shea, Jr. Too much Olympics. BF : From my foot, I ' ll attempt to kick the seventh club. Here we go. DH : Where, Barry? Where? Tell us, Barry. [Unclear] eagerly awaits your next syllable. What will it be? What gem of knowledge? What pearl of wisdom? Do you want to buy a vowel, Barry? Is that your final answer? BF : All right! You have to turn off the TV from time to time. DH : I do, I do. BF : From my foot, the kick up in the seven. DH : We will juggle seven. BF : From six to seven. DH : That ' s a world ' s record. BF : Really? DH : For us. BF : Yes. DH : Whenever you ' re ready there, big guy. Put your tongue away, Barry. BF : Oh, oh, whoa. DH : Please, please stay seated. Stay seated. Thank you. Because now, to make this twice as difficult, we ' ll juggle the seven clubs back... BF : Seven-club **juggling**. DH :... to back. BF : Thank you, that ' s it. BF : Thank you guys! DH : Thank you very much!

Text Sample 428: NEG Dim 3 – 2003 – Score -5.00 – 2003/t000259.txt

Last year, I told you the story, in seven minutes, of Project Orion, which was this very implausible technology that technically could have worked, but it had this one-year political window where it could have happened. So it didn ' t happen. It was a dream that did not happen. This year I ' m going to tell you the story of the birth of digital computing. This was a perfect introduction. And it ' s a story that did work. It did happen, and the machines are all around us. And it was a technology that was inevitable. If the people I ' m going to tell you the story about, if they hadn ' t done it, somebody else would have. So, it was sort of the right idea at the right time. This is Barricelli ' s universe. This is the universe we live in now. It ' s the universe in which these machines are now doing all these things, including changing biology. I ' m starting the story with the first atomic bomb at Trinity, which was the Manhattan Project. It was a little bit like TED : it brought a whole lot of very smart people together. And three of the smartest people were Stan Ulam, Richard Feynman and John von Neumann. And it was Von Neumann who said, after the bomb, he was working on something much more important than bombs : he ' s thinking about computers. So, he wasn ' t only thinking about them ; he built one. This is the machine he built. He built this machine, and we had a beautiful demonstration of how this thing really works, with these little bits. And it ' s an idea that goes way back. The first person to really explain that was Thomas Hobbes, who, in 1651, explained how arithmetic and logic are the same thing, and if you want to do artificial thinking and artificial logic, you can do it all with arithmetic. He said you needed addition and subtraction. Leibniz, who came a little bit later — this is 1679 — showed that you didn ' t even need subtraction. You could do the whole thing with addition. Here, we have all the binary arithmetic and logic that drove the computer revolution. And Leibniz was the first person to really talk about building such a machine. He talked about doing it with marbles, having gates and what we now call shift registers, where you shift the gates, drop the marbles down the tracks. And that ' s what all these machines are doing, except, instead of doing it with marbles, they ' re doing it with electrons. And then we jump to Von Neumann, 1945, when he sort of reinvents the whole same thing. And 1945, after the war, the electronics existed to actually try and build such a machine. So June 1945 — actually, the bomb hasn ' t even been dropped yet — and Von Neumann is putting together all the theory to actually build this thing, which also goes back to Turing, who, before that, gave the idea that you could do all this with a very brainless, little, finite state machine, just reading a tape in and reading a tape out. The other sort of genesis of what Von Neumann did was the difficulty of how you would predict the weather. Lewis Richardson saw how you could do this with a cellular array of people, giving them each a little chunk, and putting it together. Here, we have an electrical model illustrating a mind having a will, but capable of only two ideas. And that ' s really the simplest computer. It ' s basically why you need the qubit, because it only has two ideas. And you put lots of those together, you get the essentials of the modern computer : the arithmetic unit, the central control, the memory, the recording medium, the input and the output. But, there ' s one catch. This is the fatal — you know, we saw it in starting these programs up. The instructions which govern this operation must be given in absolutely exhaustive detail. So, the programming has to be perfect, or it won ' t work. If you look at the origins of this, the classic history sort of takes it all back to the ENIAC here. But actually, the machine I ' m going to tell you about, the Institute for Advanced Study machine, which is way up there, really should be down there. So, I ' m trying to revise history, and give some of these guys more credit than they ' ve had. Such a computer would open up universes, which are, at the present, outside the range of any instruments. So it opens up a whole new world, and these people saw it. The guy who was supposed to build this machine was the guy in the middle, Vladimir Zworykin, from RCA. RCA, in probably one of the lousiest business decisions of all time, decided not to go into computers. But the first meetings, November 1945, were at RCA ' s offices. RCA started this whole thing off, and said, you know, televisions are the future, not computers. The essentials were all there — all the things that make these machines run. Von Neumann, and a logician, and a mathematician from the army put this together. Then, they needed a place to build it. When RCA said no, that ' s when they decided to build it in Princeton, where Freeman works at the Institute. That ' s where I grew up as a kid. That ' s me, that ' s my sister Esther, who ' s talked to you before, so we both go back to the birth of this thing. That ' s Freeman, a long time ago, and that was me. And this is Von Neumann and Morgenstern, who wrote the"Theory of Games."All these forces came together there, in Princeton. Oppenheimer, who had built the bomb. The machine was actually used mainly for doing bomb calculations. And Julian Bigelow, who took Zworkykin ' s place as the engineer, to actually figure out, using electronics, how you would build this thing. The whole gang of people who came to work on this, and women in front, who actually did most of the coding, were the first programmers. These were the prototype geeks, the nerds. They didn ' t fit in at the Institute. This is a letter from the director, concerned about —"especially unfair on the matter of sugar."You can read the text. This is hackers getting in trouble for the first time.. These were not theoretical physicists. They were real soldering-gun type guys, and they actually built this thing. And we take it for granted now, that each of these machines has billions of **transistors**, doing billions of cycles per second without failing. They were using vacuum tubes, very narrow, sloppy techniques to get actually binary behavior out of these radio vacuum tubes. They actually used 6J6, the common radio tube, because they found they were more reliable than the more expensive tubes. And what they did at the Institute was publish every step of the way. Reports were issued, so that this machine was cloned at 15 other places around the world. And it really was. It was the original **microprocessor**. All the computers now are copies of that machine. The memory was in cathode ray tubes — a whole bunch of spots on the face of the tube — very, very sensitive to electromagnetic disturbances. So, there ' s 40 of these tubes, like a V-40 engine running the memory. The input and the output was by teletype tape at first. This is a wire drive, using bicycle wheels. This is the archetype of the hard disk that ' s in your machine now. Then they switched to a magnetic drum. This is modifying IBM equipment, which is the origins of the whole data-processing industry, later at IBM. And this is the beginning of computer graphics. The"Graph ' g-Beam Turn On."This next slide, that ' s the — as far as I know — the first digital bitmap display, 1954. So, Von Neumann was already off in a theoretical cloud, doing abstract sorts of studies of how you could build reliable machines out of unreliable components. Those guys drinking all the tea with sugar in it were writing in their logbooks, trying to get this thing to work, with all these 2, 600 vacuum tubes that failed half the time. And that ' s what I ' ve been doing, this last six months, is going through the logs."Running time : two minutes. Input, output : 90 minutes."This includes a large amount of human error. So they are always trying to figure out, what ' s machine error? What ' s human error? What ' s code, what ' s hardware? That ' s an engineer gazing at tube number 36, trying to figure out why the memory ' s not in focus. He had to focus the memory — seems OK. So, he had to focus each tube just to get the memory up and running, let alone having, you know, software problems."No use, went home.""Impossible to follow the damn thing, where ' s a directory?"So, already, they ' re complaining about the manuals :"before closing down in disgust...""The General Arithmetic : Operating Logs."Burning lots of midnight oil."MANIAC,"which became the acronym for the machine, Mathematical and Numerical Integrator and Calculator,"lost its memory.""MANIAC regained its memory, when the power went off.""Machine or human?""Aha!"So, they figured out it ' s a code problem."Found trouble in code, I hope.""Code error, machine not guilty.""Damn it, I can be just as stubborn as this thing.""And the dawn came."So they ran all night. Twenty-four hours a day, this thing was running, mainly running bomb calculations."Everything up to this point is wasted time.""What ' s the use? Good night.""Master control off. The hell with it. Way off.""Something ' s wrong with the air conditioner — smell of burning V-belts in the air.""A short — do not turn the machine on.""IBM machine putting a tar-like substance on the cards. The tar is from the roof."So they really were working under tough conditions. Here,"A mouse has climbed into the blower behind the regulator rack, set blower to vibrating. Result : no more mouse.""Here lies mouse. Born :?. Died : 4:50 a. m., May 1953."There ' s an inside joke someone has penciled in :"Here lies Marston Mouse."If you ' re a mathematician, you get that, because Marston was a mathematician who objected to the computer being there."Picked a lightning bug off the drum.""Running at two kilocycles."That ' s two thousand cycles per second —"yes, I ' m chicken"— so two kilocycles was slow speed. The high speed was 16 kilocycles. I don ' t know if you remember a Mac that was 16 **Megahertz**, that ' s slow speed."I have now duplicated both results. How will I know which is right, assuming one result is correct? This now is the third different output. I know when I ’ m licked.””We ’ ve duplicated errors before.””Machine run, fine. Code isn ’ t.””Only happens when the machine is running.”And sometimes things are okay.”Machine a thing of beauty, and a joy forever.””Perfect running.””Parting thought : when there ’ s bigger and better errors, we ’ ll have them.”So, nobody was supposed to know they were actually designing bombs. They ’ re designing hydrogen bombs. But someone in the logbook, late one night, finally drew a bomb. So, that was the result. It was Mike, the first thermonuclear bomb, in 1952. That was designed on that machine, in the woods behind the Institute. So Von Neumann invited a whole gang of weirdos from all over the world to work on all these problems. Barricelli, he came to do what we now call, really, artificial life, trying to see if, in this artificial universe — he was a viral-geneticist, way, way, way ahead of his time. He ’ s still ahead of some of the stuff that ’ s being done now. Trying to start an artificial genetic system running in the computer. Began — his universe started March 3, ’ 53. So it ’ s almost exactly — it ’ s 50 years ago next Tuesday, I guess. And he saw everything in terms of — he could read the binary code straight off the machine. He had a wonderful rapport. Other people couldn ’ t get the machine running. It always worked for him. Even errors were duplicated.”Dr. Barricelli claims machine is wrong, code is right.”So he designed this universe, and ran it. When the bomb people went home, he was allowed in there. He would run that thing all night long, running these things, if anybody remembers Stephen Wolfram, who reinvented this stuff. And he published it. It wasn ’ t locked up and disappeared. It was published in the literature.”If it ’ s that easy to create living organisms, why not create a few yourself?”So, he decided to give it a try, to start this artificial biology going in the machines. And he found all these, sort of — it was like a naturalist coming in and looking at this tiny, 5, 000-byte universe, and seeing all these things happening that we see in the outside world, in biology. This is some of the generations of his universe. But they ’ re just going to stay numbers ; they ’ re not going to become organisms. They have to have something. You have a genotype and you have to have a phenotype. They have to go out and do something. And he started doing that, started giving these little numerical organisms things they could play with — playing chess with other machines and so on. And they did start to evolve. And he went around the country after that. Every time there was a new, fast machine, he started using it, and saw exactly what ’ s happening now. That the programs, instead of being turned off — when you quit the program, you ’ d keep running and, basically, all the sorts of things like Windows is doing, running as a **multicellular** organism on many machines, he envisioned all that happening. And he saw that evolution itself was an intelligent process. It wasn ’ t any sort of creator intelligence, but the thing itself was a giant parallel computation that would have some intelligence. And he went out of his way to say that he was not saying this was lifelike, or a new kind of life. It just was another version of the same thing happening. And there ’ s really no difference between what he was doing in the computer and what nature did billions of years ago. And could you do it again now? So, when I went into these archives looking at this stuff, lo and behold, the archivist came up one day, saying,”I think we found another box that had been thrown out.”And it was his universe on punch cards. So there it is, 50 years later, sitting there — sort of suspended animation. That ’ s the instructions for running — this is actually the source code for one of those universes, with a note from the engineers saying they ’ re having some problems.”There must be something about this code that you haven ’ t explained yet.”And I think that ’ s really the truth. We still don ’ t understand how these very simple instructions can lead to increasing complexity. What ’ s the dividing line between when that is lifelike and when it really is alive? These cards, now, thanks to me showing up, are being saved. And the question is, should we run them or not? You know, could we get them running? Do you want to let it loose on the Internet? These machines would think they — these organisms, if they came back to life now — whether they ’ ve died and gone to heaven, there ’ s a universe. My laptop is 10 thousand million times the size of the universe that they lived in when Barricelli quit the project. He was thinking far ahead, to how this would really grow into a new kind of life. And that ’ s what ’ s happening! When Juan Enriquez told us about these 12 trillion bits being transferred back and forth, of all this genomics data going to the proteomics lab, that ’ s what Barricelli imagined : that this digital code in these machines is actually starting to code — it already is coding from nucleic acids. We ’ ve been doing that since, you know, since we started PCR and synthesizing small strings of DNA. And real soon, we ’ re actually going to be synthesizing the proteins, and, like Steve showed us, that just opens an entirely new world. It ’ s a world that Von Neumann himself envisioned. This was published after he died : his sort of unfinished notes on self-reproducing machines, what it takes to get the machines sort of jump-started to where they begin to reproduce. It took really three people : Barricelli had the concept of the code as a living thing ; Von Neumann saw how you could build the machines — that now, last count, four million of these Von Neumann machines is built every 24 hours ; and Julian Bigelow, who died 10 days ago — this is John Markoff ’ s obituary for him — he was the important missing link, the engineer who came in and knew how to put those vacuum tubes together and make it work. And all our computers have, inside them, the copies of the architecture that he had to just design one day, sort of on pencil and paper. And we owe a tremendous credit to that. And he explained, in a very generous way, the spirit that brought all these different people to the Institute for Advanced Study in the ’ 40s to do this project, and make it freely available with no patents, no restrictions, no intellectual property disputes to the rest of the world. That ’ s the last entry in the logbook when the machine was shut down, July 1958. And it ’ s Julian Bigelow who was running it until midnight when the machine was officially turned off. And that ’ s the end. Thank you very much.

Text Sample 429: NEG Dim 3 – 2003 – Score -5.00 – 2003/t000364.txt

I am known best for human-powered flight, but that was just one thing that got me going in the sort of things that I ’ m working in now. As a youngster, I was very interested in model airplanes, **ornithopters**, autogyros, helicopters, gliders, power planes, indoor models, outdoor models, everything, which I just thought was a lot of fun, and wondered why most other people didn ’ t share my same enthusiasm with them. And then, navy pilot training, and, after college, I got into **sailplane** flying, power plane flying, and considered the **sailplanes** as a sort of hobby and fun, but got tangled up with some great professor types, who convinced me and everybody else in the field that this was a good way to get into really deep science. While this was all going on, I was in the field of weather modification, although getting a Ph. D. in aeronautics. The weather modification subject was getting started, and as a graduate student, I could go around to the various talks that were being given, on a hitchhiker ride to the East Coast, and so on. And everybody would talk to me, but all the professionals in the field hated each other, and they wouldn ’ t communicate. And as a result, I got the absolutely unique background in that field, and started a company, which did more research in weather modification than anybody, and there are a lot of things that I just can ' t go into. But then, 1971 started AeroVironment, with no employees — then one or two, three, and sort of fumbled along on trying to get interesting projects. We had AirDynamisis, who, like I, did not want to work for aerospace companies on some big, many year project, and so we did our small projects, and the company slowly grew. The thing that is exciting was, in 1976, I suddenly got interested in the human-powered airplane because I ' d made a made a loan to a friend of 100, 000 dollars, or I guaranteed the money at the bank. He needed them — he needed the money for starting a company. The company did not succeed, and he couldn ' t pay the money back, and I was the guarantor of the note. So, I had a $100, 000 debt, and I noticed that the Kramer prize for human-powered flight, which had then been around for — — 17 years at the time, was 50, 000 pounds, which, at the exchange rate, was just about 100, 000 dollars. So suddenly, I was interested in human-powered flight — — and did not — the way I approached it, first, thinking about ways to make the planes, was just like they ' d been doing in England, and not succeeding, and I gave it up. I figured, nah, there isn ' t any simple, easy way. But then, got off on a vacation trip, and was studying bird flight, just for the fun of it, and you can watch a bird soaring around in circles, and measure the time, and estimate the bank angle, and immediately, figure out its speed, and the turning radius, and so on, which I could do in the car, as we ' re driving along on a vacation trip — — with my three sons, young sons, helping me, but ridiculing the whole thing very much. But that began thinking about how birds went around, and then how airplanes would, how hang gliders would fly, and then other planes, and the idea of the Gossamer-Condor-type airplane quickly emerged, was so logical, one should have thought of it in the first place, but one didn ' t. And it was just, keep the weight down — 70 pounds was all it weighed — but let the size swell up, like a hang glider, but three times the span, three times the cord. You ' re down to a third of the speed, a third of the power, and a good bicyclist can put out that power, and that worked, and we won the prize a year later. We didn ' t — a lot of flying, a lot of experiments, a lot of things that didn ' t work, and ones that did work, and the plane kept getting a little better, a little better. Got a good pilot, Brian Allen, to operate it, and finally, succeeded. But unfortunately, about 65, 000 dollars was spent on the project. And there was only about 30 to help retire the debt. But fortunately, Henry Kramer, who put up the prize for — that was a one-mile flight — put up a new prize for flying the English Channel, 21 miles. And he thought it would take another 18 years for somebody to win that. We realized that if you just cleaned up our **Gossamer** Condor a little bit, the power to fly would be decreased a little bit, and if you decrease the power required a little, the pilot can fly a much longer period of time. And Brian Allen was able, in a miraculous flight, to get the **Gossamer** Albatross across the English Channel, and we won the 100, 000-pound, 200, 000-dollar prize for that. And when all expenses were paid, the debt was handled, and everything was fine. It turned out that giving the planes to the museum was worth much more than the debt, so for five years, six years, I only had to pay one third income tax. So, there were good economic reasons for the project, but — — that ' s not, well, the project was done entirely for economic reasons, and we have not been involved in any human-powered flight since then — — because the prizes are all over. But that sure started me thinking about various things, and immediately, we began making a solar-powered plane because we felt solar power was going to be so important for the country and the world, we didn ' t want the small funding in the government to be decreased, which is what the government was trying to do with it. And we thought a solar-powered plane wouldn ' t really make sense, but you could do it and it would get a lot of publicity for solar power and maybe help that field. And that project continued, did succeed, and we then got into other projects in aviation and mechanical things and ground devices. But while this was going on, in 1982, I got a prize from the Lindbergh Foundation — their annual prize — and I had to prepare a paper on it, which collected all my varied thoughts and varied interests over the years. This was the one chance that I had to focus on what I, really, was after, and what was important. And to my surprise, I realized the importance of environmental issues, which Charles Lindbergh devoted the last third of his life to, and preparing that paper did me a lot of good. I thought back about if I was a space traveler, and came and visited Earth every 5, 000 years. And for a few thousand visits, I would see the same thing every time, the little differences in the Earth. But this last time, just coming round, right now, suddenly, there ’ d be huge changes in the environment, in the concentration of people, and it was just unbelievable, the amount of — all the change in it. I wanted to — well, one of the biggest changes is, 200 years ago, we began using coal from underground, which has a lot of pollution, and 100 years ago, began getting gasoline from underground, with a lot of pollution. And gasoline consumption, or production, will reach its limit in about ten years, and then go down, and we wonder what ’ s going to happen with transportation. I wanted to show the slide — this slide, I think, is the most important one any of you will see, ever, because — — it shows nature versus humans, and goes from 1850 to 2050. And so, the year 2000, you see there. And this is the weight of all air and land vertebrates. Humans and muskrats and giraffes and birds and so on, are — the red line goes up. That ’ s the humans and livestock and pets portion. The green line goes down. That ’ s the wild nature portion. Humans, livestock and pets are, now, 98 percent of the total world ’ s mass of vertebrates on land and air. And you don ’ t know what the future will hold, but it ’ s not going to get a lower percentage. Ten thousand years ago, the humans and livestock and pets were not even one tenth of one percent and wouldn ’ t even have been visible on such a curve. Now they are 98 percent, and it, I think, shows human domination of the Earth. I give a talk to some remarkable high school students each summer, and ask them, after they ’ ve asked me questions, and I give them a talk and so on. Then I ask them questions. What ’ s the population of the Earth? What ’ s the population of the Earth going to be when you ’ re the age of your parents? Which I ’ d never, really — they had never, really, thought about but, now, they think about it. And then, what population of the Earth would be an equilibrium that could continue on, and be for 2050, 2100, 2150? And they form little groups, all fighting with each other, and when I leave, two hours later, most of them are saying about 2 billion people, and they don ’ t have any clue about how to get down to 2 billion, nor do I, but I think they ’ re right and this is a serious problem. Rachel Carson was thinking of these, and came out with”Silent Spring,”way back.”Solar Manifesto”by Hermann Scheer, in Germany, claims all energy on Earth can be derived, for every country, from solar energy and water, and so on. You don ’ t need to dig down for these chemicals, and we can do things much more efficiently. Let ’ s have the next slide. So this just summarizes it.”Over billions of years, on a unique sphere, chance has painted a thin covering of life — complex and probable, wonderful and fragile. Suddenly, we humans, a recently arrived species, no longer subject to the checks and balances inherent in nature, have grown in population, technology and intelligence to a position of terrible power. We, now, wield the paintbrush.”We ’ re in charge. It ’ s frightening. And I do a painting every 20 or 25 years. This is the last one. And [it] shows the Earth in a time flag : on the right, in trilobites and dinosaurs and so on ; and over the triangle, we now get to civilization and TV and traffic jams and so on. I have no idea of what comes next, so I just used robotic and natural cockroaches as the future, as a little warning. And two weeks after this drawing was done, we actually had our first project contract, at AeroVironment, on robotic cockroaches, which was very frightening to me. Well, that ' ll be all the slides. As time went on, we stopped our environmental programs. We focused more on the really serious energy problems of the future, and we produced products for the company. And we developed the impact car that General Motors made, the EV1, out of — and got the Air Resources Board to have the regulations that stimulated the electric cars, but they ' ve since come apart. And we ' ve done a lot of things, small drone airplanes and so on. I have a Helios. We have the first video. Narrator : With a wingspan of 247 feet, this makes her larger than a Boeing 747. Her designers ' attention to detail and her construction gives Helios ' structure the flexibility and strength to deal with the turbulence encountered in the atmosphere. This enables her to easily ride through the air currents as if she ' s sliding along on the ocean waves. Paul MacCready : The wings could touch together on top and not break. We think. Narrator : And Helios now begins the process of turning her back to the sun, to maximize the power from her solar array. As the sky gets darker, and the outside air temperatures drop below minus 100 degrees Fahrenheit, the most environmentally hostile segment of Helios ' s journey has gone by without notice, except for being recorded by specially designed data acquisition systems and their associated sensors. Approaching a peak radar altitude of 96, 863 feet, at 4:12 p. m., Helios is standing on top of 98 percent of the Earth ' s atmosphere. This is more than 10, 000 feet higher than the previous world ' s altitude record held by the SR-71 Blackbird. PM : That plane has many purposes, but it ' s aimed for communications, and it can fly so slowly that it ' ll just stay up at 65, 000 feet. Eventually, it will be able to have to stay up day, night, day, night, for six months at a time, acting like the synchronous satellite, but only ten miles above the Earth. Let ' s have the next video. This shows the other end of the spectrum. Narrator : A tiny airplane, the AV Pointer serves for surveillance. In effect a pair of roving eyeglasses, a cutting-edge example of where miniaturization can lead if the operator is remote from the vehicle. It is convenient to carry, assemble, and launch by hand. Battery-powered, it is silent and rarely noticed. It sends high-resolution video pictures back to the operator. With **on-board** GPS, it can navigate autonomously, and it is rugged enough to self-land without damage. PM : Okay, and let ' s have the next. That plane is widely used by the military, now, in all their operations. Let ' s have the next video. Alan Alda : He ' s got it, he ' s got it, he ' s got it on his head. We ' re going to end our visit with Paul MacCready ' s flying circus by meeting his son, Tyler, who, with his two brothers, helped build the **Gossamer** Condor, 25 years ago. Tyler MacCready : You can chase it, like this, for hours. AA : When they got bored with their father ' s project, they invented an extraordinary little plane of their own. TM : And I can control it by putting the lift on one side of the wing, or on the other. AA : They called it their Walkalong Glider. I ' ve never seen anything like that. How old were you when you invented that? TM : Oh, 10, 11. TM : 12, something like that. PM : And Tyler ' s here to show you the Walkalong. TM : All right. You all got a couple of these in your gift bags, and one of the first things, the production version seemed to dive a little bit, and so I would just suggest you bend the wing tips up a little bit before you try flying it. I ' ll give you a demonstration of how it works. The idea is that it soars on the lift over your body, like a seagull soaring on a cliff. As the wind comes up, it has to go over the cliff, so as you walk through the air, it goes around your body, some has to go over you. And so you just keep the glider positioned in that up current. The launch is the difficult part : you ' ve got to hold it high up, over your head, and you start walking forward, and just let go of it, and you can control it like that. And then also, like it said in the video, you can turn it left or right just by putting the lift under one wing or another. So I can do it — oops, that was going to be a right turn. Okay, this one will be a left turn. Here, but — — anyway. And that ’ s it, so you can just control it, wherever you want, and it ’ s just hours of fun. And these are no longer in production, so you have real collector ’ s items. And this, we just wanted to show you — if we can get the video running on this, yeah — just an example of a little video surveillance. This was flying around in the party last night, and — — you can see how it just can fly around, and you can spy on anybody you want. And that ’ s it. I was going to bring an airplane, but I was worried about hitting people in here, so I thought this would be a little bit more gentle. And that ’ s it, yeah, just a few inventions. All right.

Text Sample 430: NEG Dim 3 – 2003 – Score -2.00 – 2003/t000110.txt

This is a sculpture I made, which is a way of, kind of, freeing a form into an object that has different degrees of freedom. So, it can balance on a point. This is a bronze ball, an aluminum arm here, and then this wooden disk. And the wooden disk was really thought about as something that you ’ d want to hold on to, and that would glide easily through your hands. The aluminum is because it ’ s very light. The bronze is nice hard, durable material that could roll on the ground. Inside of the bronze ball there ’ s a lead weight that is free-swinging on an axle that ’ s on two bearings that pass in between, across it, like this, that counterbalance this weight. So it allows it to roll. And the sphere has that balance property that it always sort of stays still and looks the same from every direction. But if you put something on top of it, it disbalances it. And so it would tip over. But in this case because the interior is free-swinging in relation to the sphere, it can stand up on one point. And then there was a second level to this object, which is that it — I wanted it to convey some proportions that I was interested in, which is the diameter of the Moon and the diameter of the Earth in proportion to each other. I was exploring, really early on, wanting to make things float in the air. And I thought up a lot of ideas. This is sculpture that I made that — it ’ s magnetically levitated. The thing is, is that it ’ s slightly dangerous. Normally it ’ s sort of cordoned off when it ’ s in a museum. But it ’ s uh — let ’ s see if I can manipulate it a little bit without, um — oops. So this is just floating, floating on a permanent magnetic field, which stabilizes it in all directions. Except there is a slight tether here, which keeps it from going over the top of its field. It ’ s sort of surfing on a magnetic field at the crest of a wave. And that ’ s what supports the object and keeps it stable. I think we could roll the tape, admin. I have a sort of a collection of videos that I took of different installations, which I could narrate. This is a sculpture of the Sun and the Earth, in proportion. Representing that eight and a half minutes that it takes light and gravity to connect the two. So here is the Earth. It ’ s a little less than a millimeter that was turned of solid bronze. And here is a similar sculpture. That ’ s the Sun at that end. And then in a series of 55 balls, it reduces, proportionately — each ball and the spaces between them reduce proportionately, until they get down to this little Earth. This is in a sculpture park in Taejon. This one is about the Moon and then the distance to the Earth, in proportion also. This is a little stone ball, floating. As you can see the little tether, that it ’ s also magnetically levitated. And then this is the first part of — this is 109 spheres, since the Sun is 109 times the diameter of the Earth. And so this is the size of the Sun. And then each of these little spheres is the size of the Earth in proportion to the Sun. It ’ s made up of 16 concentric shells. Each one has 92 spheres. This is in the courtyard of a twelfth-century alchemist. I was thinking that the Sun is kind of the ultimate alchemist. So this, again, is on the subject — a slice from the equator of the Earth. And then the Moon in the center, and it ' s floating. And this is in France. This is in Sapporo. It ' s balancing on a shaft and a ball, right at the center of gravity, or just slightly above the center of gravity, which means that the lower half of the object is just a little bit more weighty. So you can see it rotating here. It weighs about a ton or over a ton. It ' s made of stainless steel, quite thick. But it ' s being balanced like that in equilibrium. It ' s susceptible to motion by the air currents. This is another species of work that I do. These are these arrays. These spheres are all suspended, but they have magnets horizontally in them that make them all like compasses. So all the red sides, for example, face one direction : south. And the blue side, the compliment, faces the other way. So as you turn around you ' re seeing different colors. This is based on the structure of a diamond. It was a diamond cell structure was the point of departure. And then there were kind of large spaces in the hollows between the atoms. And so I placed one more element of each one of them. These were white spheres. Then I had video projectors that were projecting intermittently onto the spheres. So they would catch parts of the images, and make sort of three-dimensional color volumes, as you walk through it, through the object. This is something I did of a tactile communication system. It was the idea of isolating the tactile component of sculpture, and then putting it into a communication system. This is an idea of moving a sculpture, a ball, that would be directed around the room by a computer. This is a clock I designed. It has Buckminster Fuller ' s Dymaxion Map edited here. It turns once per day in synchrony with the Earth. And then, this is like projects that are harder to build. This has a diamond-bottomed lake. So it ' s a floating island with water, fresh water, that can fly from place to place. This would be grown, I suppose, with nanotechnology in the future sometime. In the course of doing my work I sort of have a broad range of interests. And some of it is just the idea of creating media — media as a sculpture, something that would keep the sculpture fresh and ever-changing, by just creating the media that the sculpture is made of. And I had a lot of — always interested in the concept of a crystal ball. And the idea that you could see things inside of a crystal ball and predict the future — or a television set, where it ' s sort of like a magic box where anything can appear. I had thought about, a long time ago, in the late ' 60s — when I was just starting out, I was under the influence of thinking about Buckminster Fuller ' s grand project for an electric globe across from the United Nations — and other things that were happening, the space program at that time, and Whole Earth Catalog, things like that. I was thinking about mass produced spherical television sets that could be linked to orbiting camera satellites. So if we could roll the next film here. This has evolved over the years in a lot of different iterations. But this the current version of it, is a flying airship that is about 35 meters in diameter, about 110 feet in diameter. The whole surface of it is covered with 60 million diodes, red, blue, and green, that allow you to have a high-resolution picture, visible in daylight. I came with a plan. I brought it to Paul MacCready ' s company AeroVironment to do a feasibility study, and they analyzed it, and came up with a lot of innovative ideas about how to propel it. So we have a physical plan of how to make this actually happen. This is the control room inside of the ship. The idea of this air genie is, it ' s something that can just transform and become anything. It ' s like a traveling show. It has speakers on it. And it has cameras over the surface of it. So it can see its environment, and then it can mimic its environment and disappear. Here the legs are retracting. The cabin is open or closed, as you like. It weighs about 20 tons. It has **on-board** generators. It can generate about a million kilowatts, in order to be bright enough to be visible in daylight. The idea of it is to make a kind of a traveling show. It really would be dedicated to the arts and to interacting. There would be on board a crew of artists, musicians, that would allow the thing to become actually kind of a conscious object that would respond to the moment, and to interact as an entity that was aware, that could communicate. It ’ s completely silent and nonpolluting. It has electric motors with a novel propulsion system. It could be interacted with large crowds in different ways. Primarily I would be interested in how it would interact with, say, going to a college campus, and then being used as a way of talking about the earth sciences, the world, the situation of the globe. The default image on the object would probably be a high-resolution Earth image. But then one could interact with that and show plate tectonics or global warming issues, or migrations — all of the things that we ’ re concerned with today. And then at night the idea is that it would be used as kind of a rave situation, where the people could cut loose, and the music and the lights, and everything. So it could land in a park, for example. Or this could represent a college green. And then it would have a corresponding website that would show the itinerary of this. And so interacting with the same kind of imagery. It would also be able to be an open code, so people could interact with it. It would be forum for people ’ s ideas about what they would like to see on a giant screen of this type. So that ’ s pretty much it. Okay. Thank you.

Text Sample 431: NEG Dim 3 – 2004 – Score -1.00 – 2004/t000445.txt

If you ’ d like to learn how to play the lobster, we have some here. And that ’ s not a joke, we really do. So come up afterwards and I ’ ll show you how to play a lobster. So, actually, I started working on what ’ s called the mantis shrimp a few years ago because they make sound. This is a recording I made of a mantis shrimp that ’ s found off the coast of California. And while that ’ s an absolutely fascinating sound, it actually turns out to be a very difficult project. And while I was struggling to figure out how and why mantis shrimp, or stomatopods, make sound, I started to think about their appendages. And mantis shrimp are called”mantis shrimp”after the praying mantises, which also have a fast feeding appendage. And I started to think, well, maybe it will be interesting, while listening to their sounds, to figure out how these animals generate very fast feeding strikes. And so today I ’ ll talk about the extreme stomatopod strike, work that I ’ ve done in collaboration with Wyatt Korff and Roy Caldwell. So, mantis shrimp come in two varieties : there are spearers and smashers. And this is a spearing mantis shrimp, or stomatopod. And he lives in the sand, and he catches things that go by overhead. So, a quick strike like that. And if we slow it down a bit, this is the mantis shrimp — the same species — recorded at 1, 000 frames a second, played back at 15 frames per second. And you can see it ’ s just a really spectacular extension of the limbs, exploding upward to actually just catch a dead piece of shrimp that I had offered it. Now, the other type of mantis shrimp is the smasher stomatopod, and these guys open up snails for a living. And so this guy gets the snail all set up and gives it a good whack. So, I ’ ll play it one more time. He wiggles it in place, tugs it with his nose, and smash. And a few smashes later, the snail is broken open, and he ’ s got a good dinner. So, the smasher raptorial appendage can stab with a point at the end, or it can smash with the heel. And today I ’ ll talk about the smashing type of strike. And so the first question that came to mind was, well, how fast does this limb move? Because it ’ s moving pretty darn fast on that video. And I immediately came upon a problem. Every single high-speed video system in the biology department at Berkeley wasn ’ t fast enough to catch this movement. We simply couldn ’ t capture it on video. And so this had me stymied for quite a long period of time. And then a BBC crew came cruising through the biology department, looking for a story to do about new technologies in biology. And so we struck up a deal. I said,”Well, if you guys rent the high-speed video system that could capture these movements, you guys can film us collecting the data.”And believe it or not, they went for it. So we got this incredible video system. It ’ s very new technology — it just came out about a year ago — that allows you to film at extremely high speeds in low light. And low light is a critical issue with filming animals, because if it ’ s too high, you fry them. So this is a mantis shrimp. There are the eyes up here, and there ’ s that raptorial appendage, and there ’ s the heel. And that thing ’ s going to swing around and smash the snail. And the snail ’ s wired to a stick, so he ’ s a little bit easier to set up the shot. And — yeah. I hope there aren ’ t any snail rights activists around here. So this was filmed at 5, 000 frames per second, and I ’ m playing it back at 15. And so this is slowed down 333 times. And as you ’ ll notice, it ’ s still pretty gosh darn fast slowed down 333 times. It ’ s an incredibly powerful movement. The whole limb extends out. The body flexes backwards — just a spectacular movement. And so what we did is, we took a look at these videos, and we measured how fast the limb was moving to get back to that original question. And we were in for our first surprise. So what we calculated was that the limbs were moving at the peak speed ranging from 10 meters per second all the way up to 23 meters per second. And for those of you who prefer miles per hour, that ’ s over 45 miles per hour in water. And this is really darn fast. In fact, it ’ s so fast we were able to add a new point on the extreme animal movement spectrum. And mantis shrimp are officially the fastest measured feeding strike of any animal system. So our first surprise. So that was really cool and very unexpected. So, you might be wondering, well, how do they do it? And actually, this work was done in the 1960s by a famous biologist named Malcolm Burrows. And what he showed in mantis shrimp is that they use what ’ s called a”catch mechanism,”or”click mechanism.”And what this basically consists of is a large muscle that takes a good long time to contract, and a latch that prevents anything from moving. So the muscle contracts, and nothing happens. And once the muscle ’ s contracted completely, everything ’ s stored up — the latch flies upward, and you ’ ve got the movement. And that ’ s basically what ’ s called a”power amplification system.”It takes a long time for the muscle to contract, and a very short time for the limb to fly out. And so I thought that this was sort of the end of the story. This was how mantis shrimps make these very fast strikes. But then I took a trip to the National Museum of Natural History. And if any of you ever have a chance, backstage of the National Museum of Natural History is one of the world ’ s best collections of preserved mantis shrimp. And what — this is serious business for me. So, this — what I saw, on every single mantis shrimp limb, whether it ’ s a spearer or a smasher, is a beautiful saddle-shaped structure right on the top surface of the limb. And you can see it right here. It just looks like a saddle you ’ d put on a horse. It ’ s a very beautiful structure. And it ’ s surrounded by membranous areas. And those membranous areas suggested to me that maybe this is some kind of dynamically flexible structure. And this really sort of had me scratching my head for a while. And then we did a series of calculations, and what we were able to show is that these mantis shrimp have to have a spring. There needs to be some kind of spring-loaded mechanism in order to generate the amount of force that we observe, and the speed that we observe, and the output of the system. So we thought, OK, this must be a spring — the saddle could very well be a spring. And we went back to those high-speed videos again, and we could actually visualize the saddle compressing and extending. And I ' ll just do that one more time. And then if you take a look at the video — it ' s a little bit hard to see — it ' s outlined in yellow. The saddle is outlined in yellow. You can actually see it extending over the course of the strike, and actually hyperextending. So, we ' ve had very solid evidence showing that that saddle-shaped structure actually compresses and extends, and does, in fact, function as a spring. The saddle-shaped structure is also known as a"hyperbolic paraboloid surface,"or an"anticlastic surface."And this is very well known to engineers and architects, because it ' s a very strong surface in compression. It has curves in two directions, one curve upward and opposite transverse curve down the other, so any kind of perturbation spreads the forces over the surface of this type of shape. So it ' s very well known to engineers, not as well known to biologists. It ' s also known to quite a few people who make jewelry, because it requires very little material to build this type of surface, and it ' s very strong. So if you ' re going to build a thin gold structure, it ' s very nice to have it in a shape that ' s strong. Now, it ' s also known to architects. One of the most famous architects is Eduardo Catalano, who popularized this structure. And what ' s shown here is a saddle-shaped roof that he built that ' s 87 and a half feet spanwise. It ' s two and a half inches thick, and supported at two points. And one of the reasons why he designed roofs this way is because it ' s — he found it fascinating that you could build such a strong structure that ' s made of so few materials and can be supported by so few points. And all of these are the same principles that apply to the saddle-shaped spring in stomatopods. In biological systems it ' s important not to have a whole lot of extra material requirements for building it. So, very interesting parallels between the biological and the engineering worlds. And interestingly, this turns out — the stomatopod saddle turns out to be the first described biological hyperbolic paraboloid spring. That ' s a bit long, but it is sort of interesting. So the next and final question was, well, how much force does a mantis shrimp produce if they ' re able to break open snails? And so I wired up what ' s called a load cell. A load cell measures forces, and this is actually a piezoelectronic load cell that has a little crystal in it. And when this crystal is squeezed, the electrical properties change and it — which — in proportion to the forces that go in. So these animals are wonderfully aggressive, and are really hungry all the time. And so all I had to do was actually put a little shrimp paste on the front of the load cell, and they ' d smash away at it. And so this is just a regular video of the animal just smashing the heck out of this load cell. And we were able to get some force measurements out. And again, we were in for a surprise. I purchased a 100-pound load cell, thinking, no animal could produce more than 100 pounds at this size of an animal. And what do you know? They immediately overloaded the load cell. So these are actually some old data where I had to find the smallest animals in the lab, and we were able to measure forces of well over 100 pounds generated by an animal about this big. And actually, just last week I got a 300-pound load cell up and running, and I ' ve clocked these animals generating well over 200 pounds of force. And again, I think this will be a world record. I have to do a little bit more background reading, but I think this will be the largest amount of force produced by an animal of a given — per body mass. So, really incredible forces. And again, that brings us back to the importance of that spring in storing up and releasing so much energy in this system. But that was not the end of the story. Now, things — I ' m making this sound very easy, this is actually a lot of work. And I got all these force measurements, and then I went and looked at the force output of the system. And this is just very simple — time is on the X-axis and the force is on the Y-axis. And you can see two peaks. And that was what really got me puzzled. The first peak, obviously, is the limb hitting the load cell. But there ’ s a really large second peak half a **millisecond** later, and I didn ’ t know what that was. So now, you ’ d expect a second peak for other reasons, but not half a **millisecond** later. Again, going back to those high-speed videos, there ’ s a pretty good hint of what might be going on. Here ’ s that same orientation that we saw earlier. There ’ s that raptorial appendage — there ’ s the heel, and it ’ s going to swing around and hit the load cell. And what I ’ d like you to do in this shot is keep your eye on this, on the surface of the load cell, as the limb comes flying through. And I hope what you are able to see is actually a flash of light. Audience : Wow. Sheila Patek : And so if we just take that one frame, what you can actually see there at the end of that yellow arrow is a vapor bubble. And what that is, is cavitation. And cavitation is an extremely potent fluid dynamic phenomenon which occurs when you have areas of water moving at extremely different speeds. And when this happens, it can cause areas of very low pressure, which results in the water literally vaporizing. And when that vapor bubble collapses, it emits sound, light and heat, and it ’ s a very destructive process. And so here it is in the stomatopod. And again, this is a situation where engineers are very familiar with this phenomenon, because it destroys boat propellers. People have been struggling for years to try and design a very fast rotating boat propeller that doesn ’ t cavitate and literally wear away the metal and put holes in it, just like these pictures show. So this is a potent force in fluid systems, and just to sort of take it one step further, I ’ m going to show you the mantis shrimp approaching the snail. This is taken at 20, 000 frames per second, and I have to give full credit to the BBC cameraman, Tim Green, for setting this shot up, because I could never have done this in a million years — one of the benefits of working with professional cameramen. You can see it coming in, and an incredible flash of light, and all this cavitation spreading over the surface of the snail. So really, just an amazing image, slowed down extremely, to extremely slow speeds. And again, we can see it in slightly different form there, with the bubble forming and collapsing between those two surfaces. In fact, you might have even seen some cavitation going up the edge of the limb. So to solve this quandary of the two force peaks : what I think was going on is : that first impact is actually the limb hitting the load cell, and the second impact is actually the collapse of the cavitation bubble. And these animals may very well be making use of not only the force and the energy stored with that specialized spring, but the extremes of the fluid dynamics. And they might actually be making use of fluid dynamics as a second force for breaking the snail. So, really fascinating double whammy, so to speak, from these animals. So, one question I often get after this talk — so I figured I ’ d answer it now — is, well, what happens to the animal? Because obviously, if it ’ s breaking snails, the poor limb must be disintegrating. And indeed it does. That ’ s the smashing part of the heel on both these images, and it gets worn away. In fact, I ’ ve seen them wear away their heel all the way to the flesh. But one of the convenient things about being an arthropod is that you have to molt. And every three months or so these animals molt, and they build a new limb and it ’ s no problem. Very, very convenient solution to that particular problem. So, I ’ d like to end on sort of a wacky note. Maybe this is all wacky to folks like you, I don ’ t know. So, the saddles — that saddle-shaped spring — has actually been well known to biologists for a long time, not as a spring but as a visual signal. And there ’ s actually a spectacular colored dot in the center of the saddles of many species of stomatopods. And this is quite interesting, to find evolutionary origins of visual signals on what ’ s really, in all species, their spring. And I think one explanation for this could be going back to the molting phenomenon. So these animals go into a molting period where they ’ re unable to strike — their bodies become very soft. And they ’ re literally unable to strike or they will self-destruct. This is for real. And what they do is, up until that time period when they can ’ t strike, they become really obnoxious and awful, and they strike everything in sight ; it doesn ’ t matter who or what. And the second they get into that time point when they can ’ t strike any more, they just signal. They wave their legs around. And it ’ s one of the classic examples in animal behavior of bluffing. It ’ s a well-established fact of these animals that they actually bluff. They can ’ t actually strike, but they pretend to. And so I ’ m very curious about whether those colored dots in the center of the saddles are conveying some kind of information about their ability to strike, or their strike force, and something about the time period in the molting cycle. So sort of an interesting strange fact to find a visual structure right in the middle of their spring. So to conclude, I mostly want to acknowledge my two collaborators, Wyatt Korff and Roy Caldwell, who worked closely with me on this. And also the Miller Institute for Basic Research in Science, which gave me three years of funding to just do science all the time, and for that I ’ m very grateful. Thank you very much.

Text Sample 432: NEG Dim 3 – 2004 – Score -1.00 – 2004/t000419.txt
We ’ re going to talk — my — a new lecture, just for TED — and I ’ m going show you some illusions that we ’ ve created for TED, and I ’ m going to try to relate this to happiness. What I was thinking about with happiness is, what gives happiness — or happiness, which I equate with joy in my particular area, and I think there ’ s something very fundamental. And I was thinking about this. And it ’ s in terms of both illusions and movies that we go see and jokes and magic shows is that there ’ s something about these things where our expectations are violated in some sort of pleasing way. You go see a movie. And it has an unexpected twist — something that you didn ’ t expect — and you find a joyful experience. You look at those sort of illusions in my book and it ’ s not as what you ’ d expect. And there ’ s something joyful about it. And it ’ s the same thing with jokes and all these sorts of things. So, what I ’ m going to try and do in my lecture is go a little bit further and see if I can violate your expectations in a pleasing way. I mean, sometimes expectations that are violated are not pleasant, but I ’ m going to try to do it in a pleasant way, in a very primal way, so I can make the audience here happy. So I ’ m going to show you some ways that we can violate your expectations. First of all, I want to show you the particular illusion here. I want you first of all when it pops up on the screen to notice that the two holes are perpendicular to each other. These are all perceptual tricks. These are real objects that I ’ m going to show you. Now I ’ m going to show you how it is done. I ’ ve looped the film here so you can get a very interesting experience. I want you to see how this illusion is constructed, and it ’ s going to rotate so you see that it ’ s inside out. Now watch, as it rotates back, how quickly your perception snaps. OK now. Watch it as it rotates back again. And this is a very bright audience, all right? See if you can stop it from happening, even though you know 100 percent it ’ s true that — bam! You can ’ t undo it. What does that tell you about yourselves? We ’ re going to do it again. No doubt about it. See if you can stop it from happening. No. It ’ s difficult. And we can violate your expectations in a whole variety of ways about representation, about shape, about color and so forth and it ’ s very primal. And it ’ s an interesting question to ponder, why these things — we find these things joyful. Why would we find them joyful? So, here ’ s something that Lionel did a while ago. I like these sort of little things like this. Again, this is not an optical trick.

This is what you would see. In other words, it ’ s not a camera cut. It ’ s a perceptual trick. OK. We can violate your expectations about shape. We can violate your expectations on representation — what an image represents. What do you see here? How many of you here see dolphins? Raise your hand if you see dolphins. OK, those people who raised their hands, afterwards, the rest of the audience, go talk to them, all right? Actually, this is the best example of priming by experience that I know. If you are a child under the age of 10 who haven ’ t been ruined yet, you will look at this image and see dolphins. Now, some of you adults here are saying,”What dolphins? What dolphins?”But in fact, if you reversed the figure ground — in other words, the dark areas here — I forgot to ask for a pointer — but if you reverse it, you ’ ll see a whole series of little dolphins. By the way, if you ’ re also a student at CalTech — they also tend to just see the dolphins. It ’ s based on experience. Now, something like this can be used because this is after all talk about design, too. This was done by Saatchi and Saatchi, and they actually got away with this ad in Australia. So, if you look at this ad for beer, all those people are in sort of provocative positions. But they got it passed, and actually won the Clio awards, so it ’ s funny how you can do these things. Remember that sort of, um. This is the joke I did when the Florida ballot was going around. You know, count the dots for Gore ; count the dots for Bush ; count ’ em again... You can violate your expectations about experience. Here is an outside water fountain that I created with some friends of mine, but you can stop the water in drops and — actually make all the drops levitate. This is something we ’ re building for, you know, **amusement** parks and that kind of stuff. Now let ’ s take a static image. Can you see this? Do you see the middle section moving down and the outer sections moving up? It ’ s completely static. It ’ s a static image. How many people see this illusion? It ’ s completely static. Right. Now, when — it ’ s interesting that when we look at an image we see, you know, color, depth, texture. And you can look at this whole scene and analyze it. You can see the woman is in closer than the wall and so forth. But the whole thing is actually flat. It ’ s painted. It ’ s trompe l ’ oeil. And it was such a good trompe l ’ oeil that people got irritated when they tried to talk to the woman and she wouldn ’ t respond. Now, you can make design mistakes. Like this building in New York. So that when you see it from this side, it looks like the balconies tilt up, and when you walk around to the other side it looks like the balconies go down. So there are cases where you have mistakes in design that incorporate illusions. Or, you take this particular un-retouched photograph. Now, interestingly enough, I get a lot of emails from people who say,”Is there any perceptual difference between males and females?”And I really say,”No.”I mean, women can navigate through the world just as well as males can — and why wouldn ’ t they? However, this is the one illusion that women can consistently do better than males : in matching which head because they rely on fashion cues. They can match the hat. Okay, now getting to a part — I want to show design in illusions. I believe that the first example of illusions being used purposely was by da Vinci in this anamorphic image of an eye. So that when you saw from one little angle was like this. And this little technique got popular in the 16th century and the 17th century to disguise hidden meanings, where you could flip the image and see it from one little point of view like this. But these are early incorporations of illusions brought to — sort of high point with Hans Holbein ’ s”Ambassadors.”And Hans Holbein worked for Henry VIII. This was hung on a wall where you could walk down from the stair and you can see this hidden skull. All right, now I ’ m going to show you some designers who work with illusions to give that element of surprise. One of my favorites is Scott Kim. I worked with Scott to create some illusions for TED that I hope you will enjoy. We have one here on TED and happiness. OK now.

Arthur [Ganson] hasn ’ t talked yet, but his is going to be a delightful talk and he has some of his really fantastic machines outside. And so, we — Scott created this wonderful tribute to Arthur Ganson. Well, there ’ s analog and digital. Thought that was appropriate here. And figure goes to ground. And for the musicians. And of course, since happiness — we want”joy to the world.”Now, another great designer — he ’ s very well known in Japan — Shigeo Fukuda. And he just builds some fantastic things. This is simply amazing. This is a pile of junk that when you view it from one particular angle, you see its reflection in the mirror as a perfect piano. Pianist transforms to violinist. This is really wild. This assemblage of forks, knives and spoons and various cutlery, welded together. It gives a shadow of a motorcycle. You learn something in the sort of thing that I do, which is there are people out there with a lot of time on their hands. Ken Knowlton does wonderful composite images, like creating Jacques Cousteau out of seashells — un-retouched seashells, but just by rearranging them. He did Einstein out of dice because, after all, Einstein said,”God does not play dice with the universe.”Bert Herzog out of un-retouched keyboards. Will Shortz, crossword puzzle. John Cederquist does these wonderful trompe l ’ oeil cabinets. Now, I ’ m going to skip ahead since I ’ m sort of running [behind]. I want to show you quickly what I ’ ve created, some new type of illusions. I ’ ve done something with taking the Pixar-type illusions. So you see these kids the same size here, running down the hall. The two table tops of the same size. They ’ re looking out two directions at once. You have a larger piece fitting in with a smaller. And that ’ s something for you to think about, all right? So you see larger pieces fitting in within smaller pieces here. Does everyone see that? Which is impossible. You can see the two kids are looking out simultaneously out of two different directions at once. Now can you believe these two table tops are the same size and shape? They are. So, if you measured them, they would be. And as I say, those two figures are identical in size and shape. And it ’ s interesting, by doing this in this sort of rendered fashion, how much stronger the illusions are. Any case, I hope this has brought you a little joy and happiness, and if you ’ re interested in seeing more cool effects, see me outside. I ’ d be happy to show you lots of things.

Text Sample 433: NEG Dim 3 – 2004 – Score -1.00 – 2004/t000468.txt

Thank you! Thank you very much. Like the speaker before me â I am a TED virgin, I guess. I ’ m also the first time here, and... I don ’ t know what to say! I ’ m really happy that Mr. Anderson invited me. I ’ m really grateful that I get a chance to play for everyone. And the song that I just played was by Josef Hofmann. It ’ s called”Kaleidoscope.”And Hofmann is a Polish pianist and composer of the late 19th century, and he ’ s widely considered one of the greatest pianists of all time. I have another piece that I ’ d like to play for you. It ’ s called”Abegg Variations,”by Robert Schumann, a German 19th-century composer. The name”Abegg”is actually A-B-E-G-G, and that ’ s the main theme in the melody. That comes from the last name of one of Schumann ’ s female friends. But he wrote that for his wife. So actually, if you listen carefully, there are supposed to be five variations on this Abegg theme. It ’ s written around 1834, so even though it ’ s old, I hope you ’ ll like it. Now comes the part that I hate. Well, because Mr. Anderson told me that this session is called”Sync and Flow,”I was wondering,”What do I know that these geniuses don ’ t?”So, I ’ ll talk about musical composition, even though I don ’ t know where to start. How do I compose? I think Yamaha does a really good job of teaching us how to compose. What I do first is, I make a lot of little musical ideas you can just improvise here at the piano â and I choose one of those to become my main theme, my main melody, like the Abegg that you just heard. And once I choose my main theme, I have to decide : Out of all the styles in music, what kind of style do I want? And this year, I composed a Romantic style. So for inspiration, I listened to Liszt and Tchaikovsky and all the great Romantic composers. Next, I make the structure of the entire piece with my teachers. They help me plan out the whole piece. And then the hard part is filling it in with musical ideas, because then you have to think. And then, when the piece takes somewhat of a solified form â solidified, excuse me â solidified form, you ' re supposed to actually polish the piece, polish the details, and then polish the overall performance of the composition. And another thing that I enjoy doing is drawing. Drawing, because I like to draw, you know, Japanese anime art. I think that ' s a craze among teens right now. And once I realized it, there ' s a parallel between creating music and creating art, because for your motive, or your little initial idea for your drawing, it ' s your character â you want to decide who you want to draw, or if you want to draw an original character. And then you want to decide : How are you going to draw the character? Like, am I going to use one page? Am I going to draw it on the computer? Am I going to use a two-page spread like a comic book? For a more grandiose effect, I guess. And then you have to do the initial sketch of the character, which is like your structure of a piece, and then you add pen and pencil, and whatever details that you need â that ' s polishing the drawing. And another thing that both of these have in common is your state of mind, because I know I ' m one of those teenagers that are really easily distracted. So if I ' m trying to do homework and I don ' t feel like it, I ' ll try to draw or, you know, waste my time. And then what happens is, sometimes I absolutely can ' t draw or I can ' t compose at all, and then it ' s like there ' s too much on your mind. You can ' t focus on what you ' re supposed to do. And sometimes, if you manage to use your time wisely and work on it, you ' ll get something out of it, but it doesn ' t come naturally. What happens is, if something magical happens, if something natural happens to you, you ' re able to produce all this beautiful stuff instantly, and then that ' s what I consider"flow,"because that ' s when everything clicks and you ' re able to do anything. You feel like you ' re on top of your game and you can do anything you want. I ' m not going to play my own composition today because, although I did finish it, it ' s way too long. Instead, I ' d like to try something called"improvisation."I have here seven note cards, one with each note of the musical alphabet. And I ' d like someone to come up here and choose five â anyone to come up here and choose five â and then I can make it into some sort of melody, and I ' ll improvise it. Wow. A volunteer, yay! Jennifer Lin : Nice to meet you. Goldie Hawn : Thank you. Choose five? JL : Yes, five cards. Any five cards. GH : OK, one. JL : OK. GH : Two. JL : Yes. GH : Three. GH : Oh, D and F â too familiar. JL : One more. GH : OK."E"for"effort."JL : Would you mind reading them out in the order that you chose them? GH : OK â C, G, B, A and E. JL : Thank you very much! GH : You ' re welcome. And what about these? JL : I won ' t use them. Thank you! Now, she chose C, G, B, A, E. I ' m going to try to put that in some sort of order. OK, that ' s nice. So, I ' m going to have a moment to think, and I ' ll try to make something out of it. The next song, or the encore that I ' m going to play is called"Bumble Boogie,"by Jack Fina.

Text Sample 434: NEG Dim 3 – 1990 – Score 2.00 – 1990/t000287.txt

I ' m going to go right into the slides. And all I ' m going to try and prove to you with these slides is that I do just very straight stuff. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the left-hand side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage the neighborhood â one made accommodations. I tried to be inclusive, to include the buildings in the neighborhood, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of stuff and folded it up to put it in the wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at postmodernism â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker. And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And the client said,”Why no fish?”And so I made a drawing with a fish, and I left Japan. Three weeks later, I received a complete set of drawings saying we ’ d won the competition. Now, it ’ s hard to do. It ’ s hard to translate a fish form, because they ’ re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ’ t do it, and so that made it even more exciting. But he was right â I couldn ’ t do the tail. I started to get the head OK, but the tail I couldn ’ t do. It was pretty hard. The thing on the right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with the context again. Now, if you saw a picture of this as it was published in Architectural Record â they didn ’ t show the context, so you would think,”God, what a pushy guy this is.”But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ’ t find it. So... As for craft and technology and all those things that you ’ ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent drawings to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made the drawings of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ’ s one of the nicest pieces I ’ ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ’ t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ’ m getting good at temporary â and we put a conference room in that ’ s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ’ s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ’ t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ’ s offices and I made a pedestal for a sculpture. And he didn ’ t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ’ s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ’ ve been friends for a long time. And we ’ ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called”Il Corso del Coltello”â the Swiss Army knife. And most of the imagery is â Claes ’, but those two little boys are my sons, and they were Claes ’ assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ’ s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ’ ve been known to mess with things like chain link fencing. I do it because it ’ s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ’ s so much denial about them. People hate it. And I ’ m fascinated with that, which, like the paper furniture â it ’ s one of those materials. And I ’ m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ’ s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to the building on the left. And I still think the Time magazine picture will be of the binoculars, you know, leaving out the â what the hell. I use a lot of metal in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,"If I can ' t get the craft that I want, I ' ll use the craft I can get."There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the metal because it was a way of building a building that was a sculpture. And it was all of one material, and the metal could go on the roof as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and stuff. I was given an opportunity to design an exhibit for the metalworkers ' unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future metal buildings, etc. etc. And it ' s working very well to have these people, these craftsmen, interested in it. I just tell the stories. It ' s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The metal continued into a building â Herman Miller, in Sacramento. And it ' s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ' s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ' s out in the middle of nowhere, and you approach it. It ' s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ' s aesthetic. Everybody ' s trying to get the panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ' s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ' s all about [a] metaphor for a city, maybe. And so Stanley did the little dome thing. And we did it over the phone and by fax. He would send me a fax and show me something. He ' d made a building with a dome and he had a little tower. I told him,"No, no, that ' s too ongepotchket. I don ' t want the tower."So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going back and forth over a couple of weeks ' period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,"David and Goliath."And that ' s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and cleaned it up and fixed it up and used the kind of â I thought â the language of the neighborhood, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ’ t normally think I would fit them. The detailing was very careful with the lead copper and making panels and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some film festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â□□ a little shopping center. And this is a laser laboratory at the University of Iowa, in which the fish comes back as an abstraction in the back. It ’ s the support labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the curved part there ’ s all the mechanical equipment. That solid wall behind it is a pipe chase â□□ a pipe canyon â□□ and so it was an opportunity that I seized, because I didn ’ t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ’ ve been building it so long I don ’ t remember where it is. It ’ s in the West Valley. And we started with the stream and built the house along the stream â□□ dammed it up to make a lake. These are the models. The reality, with the lake â□□ the workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ’ s hard to get perfect corners and stuff. That big metal thing is a passage, and in it is â□□ you go downstairs into the living room and then down into the bedroom, which is on the right. It ’ s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â□□ they asked Philip to do it. He was too busy. He didn ’ t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ’ t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ’ ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ’ s made of metal and the brown stuff is Fin-Ply â□□ it ’ s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ’ t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ’ s Burnham Mall, on the left. It ’ s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ’ s a railroad track. This is the city hall over here somewhere, and the courthouse. And the centerline of the mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ’ re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this stuff at the bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the centerline of this thing â□□ we ’ d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ’ s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ’ 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the back end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake. And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ’ s the living room. Servants ’ quarters. A guest bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is back in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a fun thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it ain ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the conference facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais des Sports, the Gare de Lyon over here. The Minister of Finance â the guy that moved from the Louvre â goes in here. There ’ s a new library across the river. And back in here, in this already treed park, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ' s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan coupe. And I struggled with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in French limestone, except for this metal piece. And it faces into a park. And the idea was to make this express the energy of this. On the side facing the street it ' s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ' t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in the Hamptons. And it ' s got a fish. And I keep thinking,"This is going to be the last fish."It ' s like a drug addict. I say,"I ' m not going to do it anymore â I don ' t want to do it anymore â I ' m not going to do it."And then I do it. There it is. But it ' s the living room. And this piece here is â I don ' t know what it is. I just added it so that we ' d have enough money in the budget so we could take something out. This is Euro Disney, and I ' ve worked with all of the guys that presented to you earlier. We ' ve had a lot of fun working together. I think I ' m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ' s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ' s perpendicular to the train station, to the route of the train. It looks like it ' s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ' s a Nick Grimshaw building over here, there ' s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ' t gave good slides to show â it ' s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ' s always part â you see it as all of these pieces accrue and become part of an overall neighborhood. It ' s plaster and just zinc. And you wonder, if this is a museum, what it ' s going to be like inside? If it ' s going to be so busy and crazy that you wouldn ' t show anything, and just wait. I ' m so cunning and clever â I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ' s actually basically three square rooms with a couple of skylights and stuff. And from the building in the back, you see it as an iceberg floating by in the hills. I know I ' m over time. See, that skylight goes down and becomes that one. So it ' s pretty quiet inside. This is the Disney Hall â the concert hall. It ' s a complicated project. It has a chamber hall. It ' s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ' s not a great building, but I approached it optimistically, that we would make a compositional relationship between us that would strengthen both of us. And the plan of this â it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â to MOCA, etc. The acoustician in the competition gave us criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But everybody liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were the three buildings that were the ideal â the Concertgebouw, Boston and Berlin. Everybody liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new acoustician, he told us this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an acoustical, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the sloping outside walls, which the acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on the way to being â and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with laser stuff, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, struggling with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the foot of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the foot of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th- century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ’ t as good as the stuff, and so it failed. It should have been the other way around â the food should have been good first. It didn ’ t work. Thank you very much.

Text Sample 435: NEG Dim 3 – 2006 – Score -4.00 – 2006/t000417.txt

I just want to say, over the last few years I ’ ve been â had the opportunity to do this closing conference. And I ’ ve had some incredible warm-up acts. About eight years ago, Billy Graham opened for me. And I thought that there was â I thought that there was absolutely no way in hell to top that. But I just wanted to say â and I mean this without irony â I think I can speak for everybody in the audience when I say that I wish to God that you were the President of the United States. OK, this is the title of my talk today. I just want to give you a quick overview. First of all, please remember I ’ m completely politically correct, and I mean everything with great affection. If any of you have sensitive stomachs or are feeling queasy, now is the time to check your Blackberry. Just to review, this is my TEDTalk. We ’ re going to do some jokes, some gags, some little skits â and then we ’ re going to talk about the L1 point. So, one of the questions I ask myself is, was this the most distressing TED ever? Let ’ s try and sum things up, shall we? Images of limb regeneration and faces filled with smallpox : 21 percent of the conference. Mentions of polar bears drowning : four percent. Images of the earth being wiped out by flood or bird flu : 64 percent. And David Pogue singing show tunes. Because this is the most distressing TED ever, I ’ ve been working with Neil Gershenfeld on next year ’ s TED Bag. And if the â if the conference is anywhere near this distressing, then we ’ re going to have a scream bag next year. It ’ s going to be a **cradle-to-cradle** scream bag, of course. So you ’ re going to be able to go like this. Bring it over here and open it up. Aaaah! Meanwhile, back at TED University, this wonderful woman is teaching you how to chop Sun Chips. So Robert Wright â I don ’ t know, I felt like if there was anyone that Helen needed to give antidepressants to, it might have been him. I want to deliberately interfere with his dopamine levels. He was talking about morality. Economy class morality is, we want to bomb you back to the Stone Age. Business class morality is, don ’ t bomb Japan â they built my car. And first-class morality is, don ’ t bomb Mexico â they clean my house. Yes, it is politically incorrect. All right, now I want to do a little bit of a thing for you... All right, now these are the wha â I ’ d say, [mumble, mumble â mumble, mumble, mumble, mumble, mumble] â ahh! So I wanted to show you guys â I wanted to talk about a revolutionary new computer interface that lets you work with images just as easily as you â as a completely natural user interface. And you can â you can use really natural hand gestures to like, go like this. Now we had a Harvard professor here â she was from Harvard, I just wanted to mention and â and she was actually a professor from Harvard. And she was talking about seven-dimensional, inverted universes. With, you know, of course, there ’ s the gravity brain. There ’ s the weak brain. And then there ’ s my weak brain, which is too â too â absolutely too weak to understand what the fuck she was talking about. Now â one of the things that is very important to me is to try and figure out what on Earth am I here for. And that ’ s why I went out and I picked up a best-selling business book.

You know, it basically uses as its central premise Greek mythology. And it ' s by a guy named Pastor Rick Warren, and it ' s called"The Porpoise Driven Life."And Rick is as a pagan god, which I thought was kind of appropriate, in a certain way. And now we ' re going to have kind of a little more visualization about Rick Warren. OK. All right. Now, red is Rick Warren, and green is Daniel Dennett, OK? The scales here are religiosity from zero percent, or atheist, to 100 percent, Bible literally true. And then this is books sold â the logarithmic scale. 30, 000, 300, 000, three million, 30 million, 300 million. OK, now they ' re duking it out. Now they ' re duking it out. And Rick Warren ' s kind of pulling ahead, kind of pulling ahead. Yup, and his installed base is getting a little bigger. But Darwin ' s dangerous idea is coming back. It ' s coming back. Let me turn the trails on, so you can see that a little bit better. Now, one of the things that ' s very important is, Nicholas Negroponte talked to us about one lap dance per â I ' m sorry â about One Laptop Per Child. Now let ' s talk about some of the characteristics that are important for this revolutionary device. I ' ll tell you a little bit about the design parameters, and then I ' ll show it to you in person. First of all, it needs to be small. It needs to be flat, so it ' s transportable. Lightweight. Portable. Uses very, very little power. Very, very high resolution. Has to be visible in bright daylight. Will work anywhere. And broadly applicable across many platforms. Now, we ' ve actually done some research â Neil Gershenfeld and the Fab Labs went out into the market. They did some research ; we came back ; and we think we have the perfect prototype of what the students in the field are actually asking for. And here it is, the $100 computer. OK, OK, OK, OK â excellent, excellent. Now, I bought this device from Clifford Stoll for about 900 bucks. And he and his team of junior-high school students were doing real science. So we ' re trying to check and trying to douse here, and see who uses marijuana. See who uses marijuana. Are we going to be able to find any marijuana, Jim Young? Only if we open enough locker doors. OK, now smallpox is an extremely distressing illness. We had Dr. Larry Brilliant talking about how we eradicated smallpox. I wanted to show you the stages of smallpox. We start. This is day one. Day two. Day three, she gets a massively big pox on her shoulder. Day three. Day four. Day five and day six. Now the good news is, because I ' m a trained medical professional, I know that even though she ' ll be scarred for life, she ' s going to make a full recovery. Now the good news about Architects for Humanity is they ' re really kind of the most amazing group. They ' ve been sponsoring a design competition to come up with innovative medical housing solutions, clinic solutions, in Africa, and they ' ve had a design competition. Now the wonderful thing is, Larry Brilliant was just appointed the head of the Google Foundation, and so he decided that he would support â he would support Cameron ' s work. And the way he decided to support that work was by shipping over 50, 000 shipping containers of Google snacks. So I want to show you some prototypes. The U. N. â you know, they took 20 years just to add a flap to a tent, but I think we have some more exciting things. This is a home made entirely out of Fruit Roll-Ups. And those roll-up cookies coated with white chocolate. And the really wonderful thing about this is, when you ' re done, well â you can eat it. But the thing that I ' m really, really excited about is this incredible granola house. And the granola house has a special Sun Chip roof to collect water and recycle it. And it ' s â well, on this side it has regular Sour Patch Kids and Gummy Bears to let in the light. But on this side, it has sugary Gummy Bears, to diffuse the light more slightly. And we â we wanted just to show you what this might look like in situ. So, Einstein â Einstein, tell me â what ' s your favorite song? No, I said what ' s your favorite song? No, I said what ' s your favorite song?"Free Bird."OK, so, Einstein, what ' s your favorite singing group? Could you say that again? What ’ s your favorite singing group? OK, one more time â I ’ m just going to give you a little help. Your favorite singing group â it ’ s Diana Ross and the â Audience : Supremes! Tom Reilly : Exactly. Could we have the sound up on the laptop, please?”Free Bird”kind of reminds me that if you â if you listen to”Free Bird”backwards, this is what you might hear. Computer : Satan. Satan. Satan. Satan. Satan. Satan. Satan. TR : Now it ’ s a little hard to hear the whole message, so I wanted to â so I wanted to help you a little bit. Computer : My sweet Satan. Dan Dennett worships Satan. Buy”The Purpose-Driven Life,”or Satan will take your soul. TR : So, we ’ ve talked a lot about global warming, but, you know, as Jill said, it sounds kind of nice â good weather in the wintertime, and New York City. And as Jay Walker pointed out, that is just not scary enough. So Al, I actually think I ’ m rather good at branding. So I ’ ve tried to figure out a good design process to come up with a new term to replace”global warming.”So we started with Babel Fish. We put in global warming. And then we decided that we ’ d change it from English to Dutch â into”Het globale Verwarmen.”From Dutch to [Korean], into”Hordahordaneecheewa.”[Korean] to Portuguese : Aquecer-se Global. Then Portuguese to Pig Latin. Aquecer-se ucked-fay. And then finally back into the English, which is, we ’ re totally fucked. Now I don ’ t know about you, but Michael Shermer talked about the willingness for human beings â evolutionarily, they ’ re designed to see patterns in things. For example, in cheese sandwiches. Now can you look at that carefully and see if you see the Virgin Mary? I tried to make it a little bit clearer. Is it the Virgin Mary? Or is it Mena Trott? So, I talked to Josh Prince-Ramus about the convention center and the conferences. It ’ s getting awfully big. It ’ s getting just a little bit too big. It ’ s bursting at the seams here a little bit. So we tried to come up with a program â how we could remake this structure to better accommodate TED. So first of all we decided â that we needed about one-third bookstore, one-third Google cafe, about 20 percent registration, 80 percent luxury hotel, about five percent for restrooms. And then of course, we wanted to have the simulcast lounge, the lobby and the Steinbeck forum. Now let me show you how that literally translated into the design program. So first, one of the problems with Monterey is that if there is global warming and Greenland melts as you say, the ocean level is going to rise 20 feet and flood the hell out of the convention center. So we ’ re going to build this new building on stilts. So we build this building on stilts, then up here â is where we ’ re going to put the new Steinbeck auditorium. And the wonderful thing about the new bookstore is, it ’ s going to be shaped in a spiral that ’ s organized by the Dewey Decimal System. Then we ’ re going to make an escalator that helps you get up there. And finally, we ’ re going to put the Marriott Hotel and the Portola Plaza on the top. Now I don ’ t know about you, but sometimes I have these images in my head of separated at birth. I don ’ t know about you, but when I see Aubrey de Grey, I immediately go to Gandalf the Grey. OK. Now, we ’ ve heard, of course, that we ’ re all soldiers here. So what I ’ d really, really like you to do now is, pick up your white piece of paper. Does everybody have their white piece of paper? And I want you to get out a pen, and I want you to write a terrorist note. If we put up the ELMO for a moment â if we put up the ELMO, then we ’ ll get, you know, I ’ ll give you a model that you can work from, OK? And then I want you to fold that note into a paper airplane. And once you ’ ve folded it into a paper airplane, I want you to take some anthrax â and I want you to put that in the paper airplane. And then I want you to throw it on Jim Young. Luckily, I was the recipient of the TED Prize this year. And I wanted to see â I want to dedicate this film to my father, Homer. OK. Now this film isn ’ t really hard enough, so I wanted to make it a little bit harder. So I ’ m going to try and do this while reciting pi. 3. 1415, 2657, 753, 8567, 24972 â□□ â□□ 85871, 25871, 3928, 5657, 2592, 5624. Can we cue the music please? Now I wanted to use this talk to talk about global warming a little bit. Back in 1968, you can see that the mountain range of Brokeback Mountain was covered in 151 inches of snow pack. Parenthetically, over there on the slopes, I did want to show you that black men ski. But over the years, 10 years later, the snow packs eroded, and, if you notice, the trees have started turning yellow. The water level of the lake has started drying up. A few years later, there ’ s no snow left at all. And all the trees have turned brown. This year, unfortunately the lakebed ’ s turned into an absolute cracked dry bed. And I fear, if we do nothing for our planet, in 20 years, it ’ s going to look like this. Mr. Vice President, I wish I knew how to quit you. Thank you very much.

Text Sample 436: NEG Dim 3 – 2006 – Score -1.00 – 2006/t000129.txt
How can we investigate this flora of viruses that surround us, and aid medicine? How can we turn our cumulative knowledge of virology into a simple, hand-held, single diagnostic assay? I want to turn everything we know right now about detecting viruses and the spectrum of viruses that are out there into, let ’ s say, a small chip. When we started thinking about this project — how we would make a single diagnostic assay to screen for all pathogens simultaneously — well, there ’ s some problems with this idea. First of all, viruses are pretty complex, but they ’ re also evolving very fast. This is a picornavirus. Picornaviruses — these are things that include the common cold and polio, things like this. You ’ re looking at the outside shell of the virus, and the yellow color here are those parts of the virus that are evolving very, very fast, and the blue parts are not evolving very fast. When people think about making pan-viral detection reagents, usually it ’ s the fast-evolving problem that ’ s an issue, because how can we detect things if they ’ re always changing? But evolution is a balance : where you have fast change, you also have ultra-conservation — things that almost never change. And so we looked into this a little more carefully, and I ’ m going to show you data now. This is just some stuff you can do on the computer from the desktop. I took a bunch of these small picornaviruses, like the common cold, like polio and so on, and I just broke them down into small segments. And so took this first example, which is called coxsackievirus, and just break it into small windows. And I ’ m coloring these small windows blue if another virus shares an identical sequence in its genome to that virus. These sequences right up here — which don ’ t even code for protein, by the way — are almost absolutely identical across all of these, so I could use this sequence as a marker to detect a wide spectrum of viruses, without having to make something individual. Now, over here there ’ s great diversity : that ’ s where things are evolving fast. Down here you can see slower evolution : less diversity. Now, by the time we get out here to, let ’ s say, acute bee paralysis virus — probably a bad one to have if you ’ re a bee — - this virus shares almost no similarity to coxsackievirus, but I can guarantee you that the sequences that are most conserved among these viruses on the right-hand of the screen are in identical regions right up here. And so we can encapsulate these regions of ultra-conservation through evolution — how these viruses evolved — by just choosing DNA elements or RNA elements in these regions to represent on our chip as detection reagents. OK, so that ’ s what we did, but how are we going to do that? Well, for a long time, since I was in graduate school, I ’ ve been messing around making DNA chips — that is, printing DNA on glass. And that ’ s what you see here : These little salt spots are just DNA tacked onto glass, and

so I can put thousands of these on our glass chip and use them as a detection reagent. We took our chip over to Hewlett-Packard and used their atomic force microscope on one of these spots, and this is what you see : you can actually see the strands of DNA lying flat on the glass here. So, what we ’ re doing is just printing DNA on glass — little flat things — and these are going to be markers for pathogens. OK, I make little robots in lab to make these chips, and I ’ m really big on disseminating technology. If you ’ ve got enough money to buy just a Camry, you can build one of these too, and so we put a deep how-to guide on the Web, totally free, with basically order-off-the-shelf parts. You can build a DNA array machine in your garage. Here ’ s the section on the all-important emergency stop switch. Every important machine ’ s got to have a big red button. But really, it ’ s pretty robust. You can actually be making DNA chips in your garage and decoding some genetic programs pretty rapidly. It ’ s a lot of fun. And so what we did — and this is a really cool project — we just started by making a respiratory virus chip. I talked about that — you know, that situation where you go into the clinic and you don ’ t get diagnosed? Well, we just put basically all the human respiratory viruses on one chip, and we threw in herpes virus for good measure — I mean, why not? The first thing you do as a scientist is, you make sure stuff works. And so what we did is, we take tissue culture cells and infect them with various viruses, and we take the stuff and fluorescently label the nucleic acid, the genetic material that comes out of these tissue culture cells — mostly viral stuff — and stick it on the array to see where it sticks. Now, if the DNA sequences match, they ’ ll stick together, and so we can look at spots. And if spots light up, we know there ’ s a certain virus in there. That ’ s what one of these chips really looks like, and these red spots are, in fact, signals coming from the virus. And each spot represents a different family of virus or species of virus. And so, that ’ s a hard way to look at things, so I ’ m just going to encode things as a little barcode, grouped by family, so you can see the results in a very intuitive way. What we did is, we took tissue culture cells and infected them with adenovirus, and you can see this little yellow barcode next to adenovirus. And, likewise, we infected them with parainfluenza-3 — that ’ s a paramyxovirus — and you see a little barcode here. And then we did respiratory syncytial virus. That ’ s the scourge of daycare centers everywhere — it ’ s like boogeremia, basically. You can see that this barcode is the same family, but it ’ s distinct from parainfluenza-3, which gives you a very bad cold. And so we ’ re getting unique signatures, a fingerprint for each virus. Polio and rhino : they ’ re in the same family, very close to each other. Rhino ’ s the common cold, and you all know what polio is, and you can see that these signatures are distinct. And Kaposi ’ s sarcoma-associated herpes virus gives a nice signature down here. And so it is not any one stripe or something that tells me I have a virus of a particular type here ; it ’ s the barcode that in bulk represents the whole thing. All right, I can see a rhinovirus — and here ’ s the blow-up of the rhinovirus ’ s little barcode — but what about different rhinoviruses? How do I know which rhinovirus I have? There ’ re 102 known variants of the common cold, and there ’ re only 102 because people got bored collecting them : there are just new ones every year. And so, here are four different rhinoviruses, and you can see, even with your eye, without any fancy computer pattern-matching recognition software algorithms, that you can distinguish each one of these barcodes from each other. Now, this is kind of a cheap shot, because I know what the genetic sequence of all these rhinoviruses is, and I in fact designed the chip expressly to be able to tell them apart, but what about rhinoviruses that have never seen a genetic sequencer? We don ’ t know what the sequence is ; just pull them out of the field. So, here are four rhinoviruses we never knew anything about — no one ’ s ever sequenced them — and you can also see that you get unique and distinguishable patterns. You can imagine building up some library, whether real or virtual, of fingerprints of essentially every virus. But that ' s, again, shooting fish in a barrel, you know, right? You have tissue culture cells. There are a ton of viruses. What about real people? You can ' t control real people, as you probably know. You have no idea what someone ' s going to cough into a cup, and it ' s probably really complex, right? It could have lots of bacteria, it could have more than one virus, and it certainly has host genetic material. So how do we deal with this? And how do we do the positive control here? Well, it ' s pretty simple. That ' s me, getting a nasal lavage. And the idea is, let ' s experimentally inoculate people with virus. This is all IRB-approved, by the way ; they got paid. And basically we experimentally inoculate people with the common cold virus. Or, even better, let ' s just take people right out of the emergency room — undefined, community-acquired respiratory tract infections. You have no idea what walks in through the door. So, let ' s start off with the positive control first, where we know the person was healthy. They got a shot of virus up the nose, let ' s see what happens. Day zero : nothing happening. They ' re healthy ; they ' re clean — it ' s amazing. Actually, we thought the nasal tract might be full of viruses even when you ' re walking around healthy. It ' s pretty clean. If you ' re healthy, you ' re pretty healthy. Day two : we get a very robust rhinovirus pattern, and it ' s very similar to what we get in the lab doing our tissue culture experiment. So that ' s great, but again, cheap shot, right? We put a ton of virus up this guy ' s nose. So — — I mean, we wanted it to work. He really had a cold. So, how about the people who walk in off the street? Here are two individuals represented by their anonymous ID codes. They both have rhinoviruses ; we ' ve never seen this pattern in lab. We sequenced part of their viruses ; they ' re new rhinoviruses no one ' s actually even seen. Remember, our evolutionary-conserved sequences we ' re using on this array allow us to detect even novel or uncharacterized viruses, because we pick what is conserved throughout evolution. Here ' s another guy. You can play the diagnosis game yourself here. These different blocks represent the different viruses in this paramyxovirus family, so you can kind of go down the blocks and see where the signal is. Well, doesn ' t have canine distemper ; that ' s probably good. But by the time you get to block nine, you see that respiratory syncytial virus. Maybe they have kids. And then you can see, also, the family member that ' s related : RSVB is showing up here. So, that ' s great. Here ' s another individual, sampled on two separate days — repeat visits to the clinic. This individual has parainfluenza-1, and you can see that there ' s a little stripe over here for Sendai virus : that ' s mouse parainfluenza. The genetic relationships are very close there. That ' s a lot of fun. So, we built out the chip. We made a chip that has every known virus ever discovered on it. Why not? Every plant virus, every insect virus, every marine virus. Everything that we could get out of GenBank — that is, the national repository of sequences. Now we ' re using this chip. And what are we using it for? Well, first of all, when you have a big chip like this, you need a little bit more informatics, so we designed the system to do automatic diagnosis. And the idea is that we simply have virtual patterns, because we ' re never going to get samples of every virus — it would be virtually impossible. But we can get virtual patterns, and compare them to our observed result — which is a very complex mixture — and come up with some sort of score of how likely it is this is a rhinovirus or something. And this is what this looks like. If, for example, you used a cell culture that ' s chronically infected with papilloma, you get a little computer readout here, and our algorithm says it ' s probably papilloma type 18. And that is, in fact, what these particular cell cultures are chronically infected with. So let ' s do something a little bit harder. We put the beeper in the clinic. When somebody shows up, and the hospital doesn ’ t know what to do because they can ’ t diagnose it, they call us. That ’ s the idea, and we ’ re setting this up in the Bay Area. And so, this case report happened three weeks ago. We have a 28- year-old healthy woman, no travel history, [unclear], doesn ’ t smoke, doesn ’ t drink. 10-day history of fevers, night sweats, bloody sputum — she ’ s coughing up blood — muscle pain. She went to the clinic, and they gave her antibiotics and then sent her home. She came back after ten days of fever, right? Still has the fever, and she ’ s hypoxic — she doesn ’ t have much oxygen in her lungs. They did a CT scan. A normal lung is all sort of dark and black here. All this white stuff — it ’ s not good. This sort of tree and bud formation indicates there ’ s inflammation ; there ’ s likely to be infection. OK. So, the patient was treated then with a third-generation cephalosporin antibiotic and doxycycline, and on day three, it didn ’ t help : she had progressed to acute failure. They had to intubate her, so they put a tube down her throat and they began to mechanically ventilate her. She could no longer breathe for herself. What to do next? Don ’ t know. Switch antibiotics : so they switched to another antibiotic, Tamiflu. It ’ s not clear why they thought she had the flu, but they switched to Tamiflu. And on day six, they basically threw in the towel. You do an open lung biopsy when you ’ ve got no other options. There ’ s an eight percent mortality rate with just doing this procedure, and so basically — and what do they learn from it? You ’ re looking at her open lung biopsy. And I ’ m no pathologist, but you can ’ t tell much from this. All you can tell is, there ’ s a lot of swelling : bronchiolitis. It was”unrevealing”: that ’ s the pathologist ’ s report. And so, what did they test her for? They have their own tests, of course, and so they tested her for over 70 different assays, for every sort of bacteria and fungus and viral assay you can buy off the shelf : SARS, metapneumovirus, HIV, RSV — all these. Everything came back negative, over 100, 000 dollars worth of tests. I mean, they went to the max for this woman. And basically on hospital day eight, that ’ s when they called us. They gave us endotracheal aspirate — you know, a little fluid from the throat, from this tube that they got down there — and they gave us this. We put it on the chip ; what do we see? Well, we saw parainfluenza-4. Well, what the hell ’ s parainfluenza-4? No one tests for parainfluenza-4. No one cares about it. In fact, it ’ s not even really sequenced that much. There ’ s just a little bit of it sequenced. There ’ s almost no epidemiology or studies on it. No one would even consider it, because no one had a clue that it could cause respiratory failure. And why is that? Just lore. There ’ s no data — no data to support whether it causes severe or mild disease. Clearly, we have a case of a healthy person that ’ s going down. OK, that ’ s one case report. I ’ m going to tell you one last thing in the last two minutes that ’ s unpublished — it ’ s going to come out tomorrow — and it ’ s an interesting case of how you might use this chip to find something new and open a new door. Prostate cancer. I don ’ t need to give you many statistics about prostate cancer. Most of you already know it : third leading cause of cancer deaths in the U. S. Lots of risk factors, but there is a genetic predisposition to prostate cancer. For maybe about 10 percent of prostate cancer, there are folks that are predisposed to it. And the first gene that was mapped in association studies for this, **early-onset** prostate cancer, was this gene called RNASEL. What is that? It ’ s an antiviral defense enzyme. So, we ’ re sitting around and thinking,”Why would men who have the mutation — a defect in an antiviral defense system — get prostate cancer? It doesn ’ t make sense — unless, maybe, there ’ s a virus?”So, we put tumors — - and now we have over 100 tumors — on our array. And we know who ’ s got defects in RNASEL and who doesn ’ t. And I ’ m showing you the signal from the chip here, and I ’ m showing you for the block of retroviral oligos. And what I ’ m telling you here from the signal, is that men who have a mutation in this antiviral defense enzyme, and have a tumor, often have — 40 percent of the time — a signature which reveals a new retrovirus. OK, that ’ s pretty wild. What is it? So, we clone the whole virus. First of all, I ’ ll tell you that a little automated prediction told us it was very similar to a mouse virus. But that doesn ’ t tell us too much, so we actually clone the whole thing. And the viral genome I ’ m showing you right here? It ’ s a classic gamma retrovirus, but it ’ s totally new ; no one ’ s ever seen it before. Its closest relative is, in fact, from mice, and so we would call this a xenotropic retrovirus, because it ’ s infecting a species other than mice. And this is a little phylogenetic tree to see how it ’ s related to other viruses. We ’ ve done it for many patients now, and we can say that they ’ re all independent infections. They all have the same virus, but they ’ re different enough that there ’ s reason to believe that they ’ ve been independently acquired. Is it really in the tissue? And I ’ ll end up with this : yes. We take slices of these biopsies of tumor tissue and use material to actually locate the virus, and we find cells here with viral particles in them. These guys really do have this virus. Does this virus cause prostate cancer? Nothing I ’ m saying here implies causality. I don ’ t know. Is it a link to oncogenesis? I don ’ t know. Is it the case that these guys are just more susceptible to viruses? Could be. And it might have nothing to do with cancer. But now it ’ s a door. We have a strong association between the presence of this virus and a genetic mutation that ’ s been linked to cancer. That ’ s where we ’ re at. So, it opens up more questions than it answers, I ’ m afraid, but that ’ s what, you know, science is really good at. This was all done by folks in the lab — I cannot take credit for most of this. This is a collaboration between myself and Don. This is the guy who started the project in my lab, and this is the guy who ’ s been doing prostate stuff. Thank you very much.

Text Sample 437: NEG Dim 3 – 2006 – Score -1.00 – 2006/t000490.txt

It ’ s wonderful to be back. I love this wonderful gathering. And you must be wondering,”What on earth? Have they put up the wrong slide?”No, no. Look at this magnificent beast, and ask the question : Who designed it? This is TED ; this is Technology, Entertainment, Design, and there ’ s a dairy cow. It ’ s a quite wonderfully designed animal. And I was thinking, how do I introduce this? And I thought, well, maybe that old doggerel by Joyce Kilmer, you know :”Poems are made by fools like me, but only God can make a tree.”And you might say,”Well, God designed the cow.”But, of course, God got a lot of help. This is the ancestor of cattle. This is the aurochs. And it was designed by natural selection, the process of natural selection, over many millions of years. And then it became domesticated, thousands of years ago. And human beings became its stewards, and, without even knowing what they were doing, they gradually redesigned it and redesigned it and redesigned it. And then more recently, they really began to do reverse engineering on this beast and figure out just what the parts were, how they worked and how they might be optimized — how they might be made better. Now, why am I talking about cows? Because I want to say that much the same thing is true of religions. Religions are natural phenomena — they ’ re just as natural as cows. They have evolved over millennia. They have a biological base, just like the aurochs. They have become domesticated, and human beings have been redesigning their religions for thousands of years. This is TED, and I want to talk about design. Because what I ’ ve been doing for the last four years — really since the first time you saw me — some of you saw me at TED when I was talking about religion — and in the last four years, I ’ ve been working just about non-stop on this topic. And you might say it ’ s about the reverse engineering of religions. Now that very idea, I think, strikes terror in many people, or anger, or anxiety of one sort or another. And that is the spell that I want to break. I want to say, no, religions are an important natural phenomenon. We should study them with the same intensity that we study all the other important natural phenomena, like global warming, as we heard so eloquently last night from Al Gore. Today ’ s religions are brilliantly designed — brilliantly designed. They are immensely powerful social institutions and many of their features can be traced back to earlier features that we can really make sense of by reverse engineering. And, as with the cow, there ’ s a mixture of evolutionary design — designed by natural selection itself — and intelligent design — more or less intelligent design — and redesigned by human beings who are trying to redesign their religions. You don ’ t do book talks at TED, but I ’ m going to have just one slide about my book, because there is one message in it which I think this group really needs to hear. And I would be very interested to get your responses to this. It ’ s the one policy proposal that I make in the book, at this time, when I claim not to know enough about religion to know what other policy proposals to make. And it ’ s one that echoes remarks that you ’ ve heard already today. Here ’ s my proposal, I ’ m going to just take a couple of minutes to explain it : Education on world religions for all of our children — in primary school, in high school, in public schools, in private schools and in home schooling. So what I ’ m proposing is, just as we require reading, writing, arithmetic, American history, so we should have a curriculum on facts about all the religions of the world — about their history, about their creeds, about their texts, their music, their symbolisms, their prohibitions, their requirements. And this should be presented factually, straightforwardly, with no particular spin, to all of the children in the country. And as long as you teach them that, you can teach them anything else you like. That, I think, is maximal tolerance for religious freedom. As long as you inform your children about other religions, then you may — and as early as you like and whatever you like — teach them whatever creed you want them to learn. But also let them know about other religions. Now, why do I say that? Because democracy depends on an informed citizenship. Informed consent is the very bedrock of our understanding of democracy. Misinformed consent is not worth it. It ’ s like a coin flip ; it doesn ’ t count, really. Democracy depends on informed consent. This is the way we treat people as responsible adults. Now, children below the age of consent are a special case. Parents — I ’ m going to use a word that Pastor Rick just used — parents are stewards of their children. They don ’ t own them. You can ’ t own your children. You have a responsibility to the world, to the state, to them, to take care of them right. You may teach them whatever creed you think is most important, but I say you have a responsibility to let them be informed about all the other creeds in the world, too. The reason I ’ ve taken this time is I ’ ve been fascinated to hear some of the reactions to this. One reviewer for a Roman Catholic newspaper called it”totalitarian.”It strikes me as practically libertarian. Is it totalitarian to require reading, writing and arithmetic? I don ’ t think so. All I ’ m saying is — and facts, facts only ; no values, just facts — about all the world ’ s religions. Another reviewer called it”hilarious.”Well, I ’ m really bothered by the fact that anybody would think that was hilarious. It seems to me to be such a plausible, natural extension of the democratic principles we already have that I ’ m shocked to think anybody would find that just ridiculous. I know many religions are so anxious about preserving the purity of their faith among their children that they are intent on keeping their children ignorant of other faiths. I don ’ t think that ’ s defensible. But I ’ d really be pleased to get your answers on that — any reactions to that

— later. But now I ’ m going to move on. Back to the cow. This picture, which I pulled off the web — the fellow on the left is really an important part of this picture. That ’ s the steward. Cows couldn ’ t live without human stewards — they ’ re domesticated. They ’ re a sort of ectosymbiont. They depend on us for their survival. And Pastor Rick was just talking about sheep. I ’ m going to talk about sheep, too. There ’ s a lot of serendipitous convergence here. How clever it was of sheep to acquire shepherds! Think of what this got them. They could outsource all their problems : protection from predators, food-finding...... health maintenance. The only cost in most flocks — not even this — a loss of free mating. What a deal!”How clever of sheep!”you might say. Except, of course, it wasn ’ t the sheep ’ s cleverness. We all know sheep are not exactly rocket scientists — they ’ re not very smart. It wasn ’ t the cleverness of the sheep at all. They were clueless. But it was a very clever move. Whose clever move was it? It was the clever move of natural selection itself. Francis Crick, the co-discoverer of the structure of DNA with Jim Watson, once joked about what he called Orgel ’ s Second Rule. Leslie Orgel is a molecular biologist, brilliant guy, and Orgel ’ s Second Rule is : Evolution is cleverer than you are. Now, that is not Intelligent Design — not from Francis Crick. Evolution is cleverer than you are. If you understand Orgel ’ s Second Rule, then you understand why the Intelligent Design movement is basically a hoax. The designs discovered by the process of natural selection are brilliant, unbelievably brilliant. Again and again biologists are fascinated with the brilliance of what ’ s discovered. But the process itself is without purpose, without foresight, without design. When I was here four years ago, I told the story about an ant climbing a blade of grass. And why the ant was doing it was because its brain had been infected with a lancet fluke that was needed to get into the belly of a sheep or a cow in order to reproduce. So it was sort of a spooky story. And I think some people may have misunderstood. Lancet flukes aren ’ t smart. I submit that the intelligence of a lancet fluke is down there, somewhere between petunia and carrot. They ’ re not really bright. They don ’ t have to be. The lesson we learn from this is : you don ’ t have to have a mind to be a beneficiary. The design is there in nature, but it ’ s not in anybody ’ s head. It doesn ’ t have to be. That ’ s the way evolution works. Question : Was domestication good for sheep? It was great for their genetic fitness. And here I want to remind you of a wonderful point that Paul MacCready made at TED three years ago. Here ’ s what he said :”Ten thousand years ago, at the dawn of agriculture, human population, plus livestock and pets, was approximately a tenth of one percent of the terrestrial vertebrate landmass.”That was just 10, 000 years ago. Yesterday, in biological terms. What is it today? Does anybody remember what he told us? 98 percent. That is what we have done on this planet. Now, I talked to Paul afterwards — I wanted to check to find out how he ’ d calculated this, and get the sources and so forth — and he also gave me a paper that he had written on this. And there was a passage in it which he did not present here and I think it is so good, I ’ m going to read it to you :”Over billions of years on a unique sphere, chance has painted a thin covering of life : complex, improbable, wonderful and fragile. Suddenly, we humans — a recently arrived species no longer subject to the checks and balances inherent in nature — have grown in population, technology and intelligence to a position of terrible power. We now wield the paintbrush.”We heard about the atmosphere as a thin layer of varnish. Life itself is just a thin coat of paint on this planet. And we ’ re the ones that hold the paintbrush. And how can we do that? The key to our domination of the planet is culture. And the key to culture is religion. Suppose Martian scientists came to Earth. They would be puzzled by many things. Anybody know what this is? I ’ ll tell you what it is. This is a million people gathering on the banks of the Ganges in 2001, perhaps the largest single gathering of human beings ever, as seen from satellite photograph. Here ’ s a big crowd. Here ’ s another crowd in Mecca. Martians would be amazed by this. They ’ d want to know how it originated, what it was for and how it perpetuates itself. Actually, I ’ m going to pass over this. The ant isn ’ t alone. There ’ s all sorts of wonderful cases of species which — in that case — A parasite gets into a mouse and needs to get into the belly of a cat. And it turns the mouse into Mighty Mouse, makes it fearless, so it runs out in the open, where it ’ ll be eaten by a cat. True story. In other words, we have these hijackers — you ’ ve seen this slide before, from four years ago — a parasite that infects the brain and induces even suicidal behavior, on behalf of a cause other than one ’ s own genetic fitness. Does that ever happen to us? Yes, it does — quite wonderfully. The Arabic word”Islam”means”submission.”It means”surrender of self-interest to the will of Allah.”But I ’ m not just talking about Islam. I ’ m talking also about Christianity. This is a parchment music page that I found in a Paris bookstall 50 years ago. And on it, it says, in Latin :”Semen est verbum Dei. Sator autem Christus.”The word of God is the seed and the sower of the seed is Christ. Same idea. Well, not quite. But in fact, Christians, too... glory in the fact that they have surrendered to God. I ’ ll give you a few quotes.”The heart of worship is surrender. Surrendered people obey God ’ s words, even if it doesn ’ t make sense.”Those words are by Rick Warren. Those are from”The Purpose Driven Life.”And I want to turn now, briefly, to talk about that book, which I ’ ve read. You ’ ve all got a copy, and you ’ ve just heard the man. And what I want to do now is say a bit about this book from the design standpoint, because I think it ’ s actually a brilliant book. First of all, the goal — and you heard just now what the goal is — it ’ s to bring purpose to the lives of millions, and he has succeeded. Is it a good goal? In itself, I ’ m sure we all agree, it is a wonderful goal. He ’ s absolutely right. There are lots of people out there who don ’ t have purpose in their life, and bringing purpose to their life is a wonderful goal. I give him an A+ on this. Is the goal achieved? Yes. Thirty million copies of this book. Al Gore, eat your heart out. Just exactly what Al is trying to do, Rick is doing. This is a fantastic achievement. And the means — how does he do it? It ’ s a brilliant redesign of traditional religious themes — updating them, quietly dropping obsolete features, putting new interpretations on other features. This is the evolution of religion that ’ s been going on for thousands of years, and he ’ s just the latest brilliant practitioner of it. I don ’ t have to tell you this ; you just heard the man. Excellent insights into human psychology, wise advice on every page. Moreover, he invites us to look under the hood. I really appreciated that. For instance, he has an appendix where he explains his choice of translations of different Bible verses. The book is clear, vivid, accessible, beautifully formatted. Just enough repetition. That ’ s really important. Every time you read it or say it, you make another copy in your brain. Every time you read it or say it, you make another copy in your brain. With me, everybody — Every time you read it or say it, you make another copy in your brain. Thank you. And now we come to my problem. Because I ’ m absolutely sincere in my appreciation of all that I said about this book. But I wish it were better. I have some problems with the book. And it would just be insincere of me not to address those problems. I wish he could do this with a revision, a Mark 2 version of his book.”The truth will set you free.”That ’ s what it says in the Bible, and it ’ s something that I want to live by, too. My problem is, some of the bits in it I don ’ t think are true. Now some of this is a difference of opinion. And that ’ s not my main complaint, that ’ s worth mentioning. Here ’ s a passage — it ’ s very much what he said, anyway :”If there was no God we would all be accidents, the result of astronomical random chance in the Universe. You could stop reading this book because life would have no purpose or meaning or significance. There would be no right or wrong and no hope beyond your brief years on Earth.”Now, I just do not believe that. By the way, I find — Homer Groening ’ s film presented a beautiful alternative to that very claim. Yes, there is meaning and a reason for right or wrong. We don ’ t need a belief in God to be good or to have meaning in us. But that, as I said, is just a difference of opinion. That ’ s not what I ’ m really worried about. How about this :”God designed this planet ’ s environment just so we could live in it.”I ’ m afraid that a lot of people take that sentiment to mean that we don ’ t have to do the sorts of things that Al Gore is trying so hard to get us to do. I am not happy with that sentiment at all. And then I find this :”All the evidence available in the biological sciences supports the core proposition that the cosmos is a specially designed whole with life and mankind as its fundamental goal and purpose, a whole in which all facets of reality have their meaning and explanation in this central fact.”Well, that ’ s Michael Denton. He ’ s a creationist. And here, I think,”Wait a minute.”I read this again. I read it three or four times and I think,”Is he really endorsing Intelligent Design? Is he endorsing creationism here?”And you can ’ t tell. So I ’ m sort of thinking,”Well, I don ’ t know, I don ’ t know if I want to get upset with this yet.”But then I read on, and I read this :”First, Noah had never seen rain, because prior to the Flood, God irrigated the earth from the ground up.”I wish that sentence weren ’ t in there, because I think it is false. And I think that thinking this way about the history of the planet, after we ’ ve just been hearing about the history of the planet over millions of years, discourages people from scientific understanding. Now, Rick Warren uses scientific terms and scientific factoids and information in a very interesting way. Here ’ s one :”God deliberately shaped and formed you to serve him in a way that makes your ministry unique. He carefully mixed the DNA cocktail that created you.”I think that ’ s false. Now, maybe we want to treat it as metaphorical. Here ’ s another one :”For instance, your brain can store 100 trillion facts. Your mind can handle 15, 000 decisions a second.”Well, it would be interesting to find the interpretation where I would accept that. There might be some way of treating that as true.”Anthropologists have noted that worship is a universal urge, hardwired by God into the very fiber of our being — an inbuilt need to connect with God.”Well, the sense of which I agree with him, except I think it has an evolutionary explanation. And what I find deeply troubling in this book is that he seems to be arguing that if you want to be moral, if you want to have meaning in your life, you have to be an Intelligent Designer, you have to deny the theory of evolution by natural selection. And I think, on the contrary, that it is very important to solving the world ’ s problems that we take evolutionary biology seriously. Whose truth are we going to listen to? Well, this is from”The Purpose Driven Life”:”The Bible must become the authoritative standard for my life : the compass I rely on for direction, the counsel I listen to for making wise decisions, and the benchmark I use for evaluating everything.”Well maybe, OK, but what ’ s going to follow from this? And here ’ s one that does concern me. Remember I quoted him before with this line :”Surrendered people obey God ’ s word, even if it doesn ’ t make sense.”And that ’ s a problem.”Don ’ t ever argue with the Devil. He ’ s better at arguing than you are, having had thousands of years to practice.”Now, Rick Warren didn ’ t invent this clever move. It ’ s an old move. It ’ s a very clever adaptation of religions. It ’ s a wild card for disarming any reasonable criticism.”You don ’ t like my interpretation? You ’ ve got a reasonable objection to it? Don ’ t listen, don ’ t listen! That ’ s the Devil speaking.”This discourages the sort of reasoning citizenship it seems to me that we want to have. I ’ ve got one more problem, then I ’ m through. And I ’ d really like to get a response if Rick is able to do it.”In the Great Commission, Jesus said, ' Go to all people of all nations and make them my disciples. Baptize them in the name of the Father, the Son and the Holy Spirit, and teach them to do everything I ' ve told you. '"The Bible says Jesus is the only one who can save the world. We ' ve seen many wonderful maps of the world in the last day or so. Here ' s one, not as beautiful as the others ; it simply shows the religions of the world. Here ' s one that shows the sort of current breakdown of the different religions. Do we really want to commit ourselves to engulfing all the other religions, when their holy books are telling them,"Don ' t listen to the other side, that ' s just Satan talking!"? It seems to me that that ' s a very problematic ship to get on for the future. I found this sign as I was driving to Maine recently, in front of a church :"Good without God becomes zero."Sort of cute. A very clever little meme. I don ' t believe it and I think this idea, popular as it is — not in this guise, but in general — is itself one of the main problems that we face. If you are like me, you know many wonderful, committed, engaged atheists, agnostics, who are being very good without God. And you also know many religious people who hide behind their sanctity instead of doing good works. So, I wish we could drop this meme. I wish this meme would go extinct. Thanks very much for your attention.

Text Sample 438: NEG Dim 3 – 2008 – Score -2.00 – 2008/t000236.txt

So, people argue vigorously about the definition of life. They ask if it should have reproduction in it, or metabolism, or evolution. And I don ' t know the answer to that, so I ' m not going to tell you. I will say that life involves computation. So this is a computer program. Booted up in a cell, the program would execute, and it could result in this person ; or with a small change, it could result in this person ; or another small change, this person ; or with a larger change, this dog, or this tree, or this whale. So now, if you take this metaphor [of] genome as program seriously, you have to consider that Chris Anderson is a computer-fabricated artifact, as is Jim Watson, Craig Venter, as are all of us. And in convincing yourself that this metaphor is true, there are lots of similarities between genetic programs and computer programs that could help to convince you. But one, to me, that ' s most compelling is the peculiar sensitivity to small changes that can make large changes in biological development — the output. A small mutation can take a two-wing fly and make it a four-wing fly. Or it could take a fly and put legs where its antennae should be. Or if you ' re familiar with"The Princess Bride,"it could create a six-fingered man. Now, a hallmark of computer programs is just this kind of sensitivity to small changes. If your bank account ' s one dollar, and you flip a single bit, you could end up with a thousand dollars. So these small changes are things that I think that — they indicate to us that a complicated computation in development is underlying these amplified, large changes. So now, all of this indicates that there are molecular programs underlying biology, and it shows the power of molecular programs — biology does. And what I want to do is write molecular programs, potentially to build technology. And there are a lot of people doing this, a lot of synthetic biologists doing this, like Craig Venter. And they concentrate on using cells. They ' re cell- oriented. So my friends, molecular programmers, and I have a sort of biomolecule-centric approach. We ' re interested in using DNA, RNA and protein, and building new languages for building things from the bottom up, using biomolecules, potentially having nothing to do with biology. So, these are all the machines in a cell. There ' s a camera. There ' s the solar panels of the cell, some switches that turn your genes on and off, the girders of the cell, motors that move your muscles. My little group of molecular programmers are trying to refashion all of these parts from DNA. We ’ re not DNA zealots, but DNA is the cheapest, easiest to understand and easy to program material to do this. And as other things become easier to use — maybe protein — we ’ ll work with those. If we succeed, what will molecular programming look like? You ’ re going to sit in front of your computer. You ’ re going to design something like a cell phone, and in a high-level language, you ’ ll describe that cell phone. Then you ’ re going to have a compiler that ’ s going to take that description and it ’ s going to turn it into actual molecules that can be sent to a synthesizer and that synthesizer will pack those molecules into a seed. And what happens if you water and feed that seed appropriately, is it will do a developmental computation, a molecular computation, and it ’ ll build an electronic computer. And if I haven ’ t revealed my prejudices already, I think that life has been about molecular computers building electrochemical computers, building electronic computers, which together with electrochemical computers will build new molecular computers, which will build new electronic computers, and so forth. And if you buy all of this, and you think life is about computation, as I do, then you look at big questions through the eyes of a computer scientist. So one big question is, how does a baby know when to stop growing? And for molecular programming, the question is how does your cell phone know when to stop growing? Or how does a computer program know when to stop running? Or more to the point, how do you know if a program will ever stop? There are other questions like this, too. One of them is Craig Venter ’ s question. Turns out I think he ’ s actually a computer scientist. He asked, how big is the minimal genome that will give me a functioning microorganism? How few genes can I use? This is exactly **analogous** to the question, what ’ s the smallest program I can write that will act exactly like Microsoft Word? And just as he ’ s writing, you know, bacteria that will be smaller, he ’ s writing genomes that will work, we could write smaller programs that would do what Microsoft Word does. But for molecular programming, our question is, how many molecules do we need to put in that seed to get a cell phone? What ’ s the smallest number we can get away with? Now, these are big questions in computer science. These are all complexity questions, and computer science tells us that these are very hard questions. Almost — many of them are impossible. But for some tasks, we can start to answer them. So, I ’ m going to start asking those questions for the DNA structures I ’ m going to talk about next. So, this is normal DNA, what you think of as normal DNA. It ’ s double-stranded, it ’ s a double helix, has the As, Ts, Cs and Gs that pair to hold the strands together. And I ’ m going to draw it like this sometimes, just so I don ’ t scare you. We want to look at individual strands and not think about the double helix. When we synthesize it, it comes single-stranded, so we can take the blue strand in one tube and make an orange strand in the other tube, and they ’ re floppy when they ’ re single-stranded. You mix them together and they make a rigid double helix. Now for the last 25 years, Ned Seeman and a bunch of his descendants have worked very hard and made beautiful three-dimensional structures using this kind of reaction of DNA strands coming together. But a lot of their approaches, though elegant, take a long time. They can take a couple of years, or it can be difficult to design. So I came up with a new method a couple of years ago I call DNA origami that ’ s so easy you could do it at home in your kitchen and design the stuff on a laptop. But to do it, you need a long, single strand of DNA, which is technically very difficult to get. So, you can go to a natural source. You can look in this computer-fabricated artifact, and he ’ s got a double-stranded genome — that ’ s no good. You look in his intestines. There are billions of bacteria. They ’ re no good either. Double strand again, but inside them, they ’ re infected with a virus that has a nice, long, single-stranded genome that we can fold like a piece of paper. And here ’ s how we do it. This is part of that genome. We add a bunch of short, synthetic DNAs that I call staples. Each one has a left half that binds the long strand in one place, and a right half that binds it in a different place, and brings the long strand together like this. The net action of many of these on that long strand is to fold it into something like a rectangle. Now, we can ’ t actually take a movie of this process, but Shawn Douglas at Harvard has made a nice visualization for us that begins with a long strand and has some short strands in it. And what happens is that we mix these strands together. We heat them up, we add a little bit of salt, we heat them up to almost boiling and cool them down, and as we cool them down, the short strands bind the long strands and start to form structure. And you can see a little bit of double helix forming there. When you look at DNA origami, you can see that what it really is, even though you think it ’ s complicated, is a bunch of double helices that are parallel to each other, and they ’ re held together by places where short strands go along one helix and then jump to another one. So there ’ s a strand that goes like this, goes along one helix and binds — it jumps to another helix and comes back. That holds the long strand like this. Now, to show that we could make any shape or pattern that we wanted, I tried to make this shape. I wanted to fold DNA into something that goes up over the eye, down the nose, up the nose, around the forehead, back down and end in a little loop like this. And so, I thought, if this could work, anything could work. So I had the computer program design the short staples to do this. I ordered them ; they came by FedEx. I mixed them up, heated them, cooled them down, and I got 50 billion little smiley faces floating around in a single drop of water. And each one of these is just one-thousandth the width of a human hair, OK? So, they ’ re all floating around in solution, and to look at them, you have to get them on a surface where they stick. So, you pour them out onto a surface and they start to stick to that surface, and we take a picture using an atomic- force microscope. It ’ s got a needle, like a record needle, that goes back and forth over the surface, bumps up and down, and feels the height of the first surface. It feels the DNA origami. There ’ s the atomic-force microscope working and you can see that the landing ’ s a little rough. When you zoom in, they ’ ve got, you know, weak jaws that flip over their heads and some of their noses get punched out, but it ’ s pretty good. You can zoom in and even see the extra little loop, this little nano-goatee. Now, what ’ s great about this is anybody can do this. And so, I got this in the mail about a year after I did this, unsolicited. Anyone know what this is? What is it? It ’ s China, right? So, what happened is, a graduate student in China, Lulu Qian, did a great job. She wrote all her own software to design and built this DNA origami, a beautiful rendition of China, which even has Taiwan, and you can see it ’ s sort of on the world ’ s shortest leash, right? So, this works really well and you can make patterns as well as shapes, OK? And you can make a map of the Americas and spell DNA with DNA. And what ’ s really neat about it — well, actually, this all looks like nano- artwork, but it turns out that nano-artwork is just what you need to make nano- circuits. So, you can put circuit components on the staples, like a light bulb and a light switch. Let the thing assemble, and you ’ ll get some kind of a circuit. And then you can maybe wash the DNA away and have the circuit left over. So, this is what some colleagues of mine at Caltech did. They took a DNA origami, organized some carbon nano-tubes, made a little switch, you see here, wired it up, tested it and showed that it is indeed a switch. Now, this is just a single switch and you need half a billion for a computer, so we have a long way to go. But this is very promising because the origami can organize parts just one-tenth the size of those in a normal computer. So it ’ s very promising for making small computers. Now, I want to get back to that compiler.

The DNA origami is a proof that that compiler actually works. So, you start with something in the computer. You get a high-level description of the computer program, a high-level description of the origami. You can compile it to molecules, send it to a synthesizer, and it actually works. And it turns out that a company has made a nice program that ' s much better than my code, which was kind of ugly, and will allow us to do this in a nice, visual, computer-aided design way. So, now you can say, all right, why isn ' t DNA origami the end of the story? You have your molecular compiler, you can do whatever you want. The fact is that it does not scale. So if you want to build a human from DNA origami, the problem is, you need a long strand that ' s 10 trillion trillion bases long. That ' s three light years ' worth of DNA, so we ' re not going to do this. We ' re going to turn to another technology, called algorithmic self-assembly of tiles. It was started by Erik Winfree, and what it does, it has tiles that are a hundredth the size of a DNA origami. You zoom in, there are just four DNA strands and they have little single-stranded bits on them that can bind to other tiles, if they match. And we like to draw these tiles as little squares. And if you look at their sticky ends, these little DNA bits, you can see that they actually form a checkerboard pattern. So, these tiles would make a complicated, self-assembling checkerboard. And the point of this, if you didn ' t catch that, is that tiles are a kind of molecular program and they can output patterns. And a really amazing part of this is that any computer program can be translated into one of these tile programs — specifically, counting. So, you can come up with a set of tiles that when they come together, form a little binary counter rather than a checkerboard. So you can read off binary numbers five, six and seven. And in order to get these kinds of computations started right, you need some kind of input, a kind of seed. You can use DNA origami for that. You can encode the number 32 in the right-hand side of a DNA origami, and when you add those tiles that count, they will start to count — they will read that 32 and they ' ll stop at 32. So, what we ' ve done is we ' ve figured out a way to have a molecular program know when to stop going. It knows when to stop growing because it can count. It knows how big it is. So, that answers that sort of first question I was talking about. It doesn ' t tell us how babies do it, however. So now, we can use this counting to try and get at much bigger things than DNA origami could otherwise. Here ' s the DNA origami, and what we can do is we can write 32 on both edges of the DNA origami, and we can now use our watering can and water with tiles, and we can start growing tiles off of that and create a square. The counter serves as a template to fill in a square in the middle of this thing. So, what we ' ve done is we ' ve succeeded in making something much bigger than a DNA origami by combining DNA origami with tiles. And the neat thing about it is, is that it ' s also reprogrammable. You can just change a couple of the DNA strands in this binary representation and you ' ll get 96 rather than 32. And if you do that, the origami ' s the same size, but the resulting square that you get is three times bigger. So, this sort of recapitulates what I was telling you about development. You have a very sensitive computer program where small changes — single, tiny, little mutations — can take something that made one size square and make something very much bigger. Now, this — using counting to compute and build these kinds of things by this kind of developmental process is something that also has bearing on Craig Venter ' s question. So, you can ask, how many DNA strands are required to build a square of a given size? If we wanted to make a square of size 10, 100 or 1, 000, if we used DNA origami alone, we would require a number of DNA strands that ' s the square of the size of that square ; so we ' d need 100, 10, 000 or a million DNA strands. That ' s really not affordable. But if we use a little computation — we use origami, plus some tiles that count — then we can get away with using 100, 200 or 300 DNA strands. And so we can exponentially reduce the number of DNA strands we use, if we use counting, if we use a little bit of computation. And so computation is some very powerful way to reduce the number of molecules you need to build something, to reduce the size of the genome that you ’ re building. And finally, I ’ m going to get back to that sort of crazy idea about computers building computers. If you look at the square that you build with the origami and some counters growing off it, the pattern that it has is exactly the pattern that you need to make a memory. So if you affix some wires and switches to those tiles — rather than to the staple strands, you affix them to the tiles — then they ’ ll self-assemble the somewhat complicated circuits, the demultiplexer circuits, that you need to address this memory. So you can actually make a complicated circuit using a little bit of computation. It ’ s a molecular computer building an electronic computer. Now, you ask me, how far have we gotten down this path? Experimentally, this is what we ’ ve done in the last year. Here is a DNA origami rectangle, and here are some tiles growing from it. And you can see how they count. One, two, three, four, five, six, nine, 10, 11, 12, 17. So it ’ s got some errors, but at least it counts up. So, it turns out we actually had this idea nine years ago, and that ’ s about the time constant for how long it takes to do these kinds of things, so I think we made a lot of progress. We ’ ve got ideas about how to fix these errors. And I think in the next five or 10 years, we ’ ll make the kind of squares that I described and maybe even get to some of those self-assembled circuits. So now, what do I want you to take away from this talk? I want you to remember that to create life ’ s very diverse and complex forms, life uses computation to do that. And the computations that it uses, they ’ re molecular computations, and in order to understand this and get a better handle on it, as Feynman said, you know, we need to build something to understand it. And so we are going to use molecules and refashion this thing, rebuild everything from the bottom up, using DNA in ways that nature never intended, using DNA origami, and DNA origami to seed this algorithmic self-assembly. You know, so this is all very cool, but what I ’ d like you to take from the talk, hopefully from some of those big questions, is that this molecular programming isn ’ t just about making gadgets. It ’ s not just making about — it ’ s making self-assembled cell phones and circuits. What it ’ s really about is taking computer science and looking at big questions in a new light, asking new versions of those big questions and trying to understand how biology can make such amazing things. Thank you.

Text Sample 439: NEG Dim 3 – 2008 – Score -2.00 – 2008/t000283.txt

You know, I ’ ve talked about some of these projects before â about the human genome and what that might mean, and discovering new sets of genes. We ’ re actually starting at a new point : we ’ ve been digitizing biology, and now we ’ re trying to go from that digital code into a new phase of biology with designing and synthesizing life. So, we ’ ve always been trying to ask big questions.”What is life?”is something that I think many biologists have been trying to understand at various levels. We ’ ve tried various approaches, paring it down to minimal components. We ’ ve been digitizing it now for almost 20 years ; when we sequenced the human genome, it was going from the analog world of biology into the digital world of the computer. Now we ’ re trying to ask,”Can we regenerate life or can we create new life out of this digital universe?”This is the map of a small organism, Mycoplasma genitalium, that has the smallest genome for a species that can self-replicate in the laboratory, and we ’ ve been trying to just see if we can come up with an even smaller genome. We ’ re able to knock out on the order of 100 genes out of the 500 or so that are here. When we look at its metabolic map, it ’ s relatively simple compared to ours â trust me, this is simple â but when we look at all the genes that we can knock out one at a time, it ’ s very unlikely that this would yield a living cell. So we decided the only way forward was to actually synthesize this chromosome so we could vary the components to ask some of these most fundamental questions. And so we started down the road of : can we synthesize a chromosome? Can chemistry permit making these really large molecules where we ’ ve never been before? And if we do, can we boot up a chromosome? A chromosome, by the way, is just a piece of inert chemical material. So, our pace of digitizing life has been increasing at an exponential pace. Our ability to write the genetic code has been moving pretty slowly but has been increasing, and our latest point would put it on, now, an exponential curve. We started this over 15 years ago. It took several stages, in fact, starting with a bioethical review before we did the first experiments. But it turns out synthesizing DNA is very difficult. There are tens of thousands of machines around the world that make small pieces of DNA â 30 to 50 letters in length â and it ’ s a degenerate process, so the longer you make the piece, the more errors there are. So we had to create a new method for putting these little pieces together and correct all the errors. And this was our first attempt, starting with the digital information of the genome of phi X174. It ’ s a small virus that kills bacteria. We designed the pieces, went through our error correction and had a DNA molecule of about 5, 000 letters. The exciting phase came when we took this piece of inert chemical and put it in the bacteria, and the bacteria started to read this genetic code, made the viral particles. The viral particles then were released from the cells and came back and killed the E. coli. I was talking to the oil industry recently and I said they clearly understood that model. They laughed more than you guys are. And so, we think this is a situation where the software can actually build its own hardware in a biological system. But we wanted to go much larger : we wanted to build the entire bacterial chromosome â it ’ s over 580, 000 letters of genetic code â so we thought we ’ d build them in cassettes the size of the viruses so we could actually vary the cassettes to understand what the actual components of a living cell are. Design is critical, and if you ’ re starting with digital information in the computer, that digital information has to be really accurate. When we first sequenced this genome in 1995, the standard of accuracy was one error per 10, 000 base pairs. We actually found, on resequencing it, 30 errors ; had we used that original sequence, it never would have been able to be booted up. Part of the design is designing pieces that are 50 letters long that have to overlap with all the other 50-letter pieces to build smaller subunits we have to design so they can go together. We design unique elements into this. You may have read that we put watermarks in. Think of this : we have a four- letter genetic code â A, C, G and T. Triplets of those letters code for roughly 20 amino acids, such that there ’ s a single letter designation for each of the amino acids. So we can use the genetic code to write out words, sentences, thoughts. Initially, all we did was autograph it. Some people were disappointed there was not poetry. We designed these pieces so we can just chew back with enzymes ; there are enzymes that repair them and put them together. And we started making pieces, starting with pieces that were 5, 000 to 7, 000 letters, put those together to make 24, 000-letter pieces, then put sets of those going up to 72, 000. At each stage, we grew up these pieces in abundance so we could sequence them because we ’ re trying to create a process that ’ s extremely robust that you can see in a minute. We ’ re trying to get to the point of automation. So, this looks like a basketball playoff. When we get into these really large pieces over 100, 000 base pairs, they won ’ t any longer grow readily in E. coli â it exhausts all the modern tools of molecular biology â and so we turned to other mechanisms. We knew there ’ s a mechanism called homologous **recombination** that biology uses to repair DNA that can put pieces together. Here ’ s an example of it : there ’ s an organism called Deinococcus radiodurans that can take three millions **rads** of radiation. You can see in the top panel, its chromosome just gets blown apart. Twelve to 24 hours later, it put it back together exactly as it was before. We have thousands of organisms that can do this. These organisms can be totally desiccated ; they can live in a vacuum. I am absolutely certain that life can exist in outer space, move around, find a new aqueous environment. In fact, NASA has shown a lot of this is out there. Here ’ s an actual micrograph of the molecule we built using these processes, actually just using yeast mechanisms with the right design of the pieces we put them in ; yeast puts them together automatically. This is not an electron micrograph ; this is just a regular photomicrograph. It ’ s such a large molecule we can see it with a light microscope. These are pictures over about a six- second period. So, this is the publication we had just a short while ago. This is over 580, 000 letters of genetic code ; it ’ s the largest molecule ever made by humans of a defined structure. It ’ s over 300 million molecular weight. If we printed it out at a 10 font with no spacing, it takes 142 pages just to print this genetic code. Well, how do we boot up a chromosome? How do we activate this? Obviously, with a virus it ’ s pretty simple ; it ’ s much more complicated dealing with bacteria. It ’ s also simpler when you go into eukaryotes like ourselves : you can just pop out the nucleus and pop in another one, and that ’ s what you ’ ve all heard about with cloning. With bacteria and Archaea, the chromosome is integrated into the cell, but we recently showed that we can do a complete transplant of a chromosome from one cell to another and activate it. We purified a chromosome from one microbial species â roughly, these two are as distant as human and mice â we added a few extra genes so we could select for this chromosome, we digested it with enzymes to kill all the proteins, and it was pretty stunning when we put this in the cell â and you ’ ll appreciate our very sophisticated graphics here. The new chromosome went into the cell. In fact, we thought this might be as far as it went, but we tried to design the process a little bit further. This is a major mechanism of evolution right here. We find all kinds of species that have taken up a second chromosome or a third one from somewhere, adding thousands of new traits in a second to that species. So, people who think of evolution as just one gene changing at a time have missed much of biology. There are enzymes called restriction enzymes that actually digest DNA. The chromosome that was in the cell doesn ’ t have one ; the chromosome we put in does. It got expressed and it recognized the other chromosome as foreign material, chewed it up, and so we ended up just with a cell with the new chromosome. It turned blue because of the genes we put in it. And with a very short period of time, all the characteristics of one species were lost and it converted totally into the new species based on the new software that we put in the cell. All the proteins changed, the membranes changed ; when we read the genetic code, it ’ s exactly what we had transferred in. So, this may sound like genomic alchemy, but we can, by moving the software of DNA around, change things quite dramatically. Now I ’ ve argued, this is not genesis ; this is building on three and a half billion years of evolution. And I ’ ve argued that we ’ re about to perhaps create a new version of the Cambrian explosion, where there ’ s massive new speciation based on this digital design. Why do this? I think this is pretty obvious in terms of some of the needs. We ’ re about to go from six and a half to nine billion people over the next 40 years. To put it in context for myself : I was born in 1946. There are now three people on the planet for every one of us that existed in 1946 ; within 40 years, there ’ ll be four. We have trouble feeding, providing fresh, clean water, medicines, fuel for the six and a half billion. It ’ s going to be a stretch to do it for nine. We use over five billion tons of coal, 30 billion-plus barrels of oil â that ’ s a hundred million barrels a day. When we try to think of biological processes or any process to replace that, it ’ s going to be a huge challenge. Then of course, there ’ s all that CO2 from this material that ends up in the atmosphere. We now, from our discovery around the world, have a database with about 20 million genes, and I like to think of these as the design components of the future. The electronics industry only had a dozen or so components, and look at the diversity that came out of that. We ’ re limited here primarily by a biological reality and our imagination. We now have techniques, because of these rapid methods of synthesis, to do what we ’ re calling combinatorial genomics. We have the ability now to build a large robot that can make a million chromosomes a day. When you think of processing these 20 million different genes or trying to optimize processes to produce octane or to produce pharmaceuticals, new vaccines, we can just with a small team, do more molecular biology than the last 20 years of all science. And it ’ s just standard selection : we can select for viability, chemical or fuel production, vaccine production, etc. This is a screen snapshot of some true design software that we ’ re working on to actually be able to sit down and design species in the computer. You know, we don ’ t know necessarily what it ’ ll look like : we know exactly what their genetic code looks like. We ’ re focusing on now fourth-generation fuels. You ’ ve seen recently, corn to ethanol is just a bad experiment. We have second- and third- generation fuels that will be coming out relatively soon that are sugar, to much higher-value fuels like octane or different types of butanol. But the only way we think that biology can have a major impact without further increasing the cost of food and limiting its availability is if we start with CO2 as its feedstock, and so we ’ re working with designing cells to go down this road. And we think we ’ ll have the first fourth-generation fuels in about 18 months. Sunlight and CO2 is one method... but in our discovery around the world, we have all kinds of other methods. This is an organism we described in 1996. It lives in the deep ocean, about a mile and a half deep, almost at boiling-water temperatures. It takes CO2 to methane using molecular hydrogen as its energy source. We ’ re looking to see if we can take captured CO2, which can easily be piped to sites, convert that CO2 back into fuel to drive this process. So, in a short period of time, we think that we might be able to increase what the basic question is of”What is life?”We truly, you know, have modest goals of replacing the whole petrol-chemical industry â Yeah. If you can ’ t do that at TED, where can you? â become a major source of energy... But also, we ’ re now working on using these same tools to come up with instant sets of vaccines. You ’ ve seen this year with flu ; we ’ re always a year behind and a dollar short when it comes to the right vaccine. I think that can be changed by building combinatorial vaccines in advance. Here ’ s what the future may begin to look like with changing, now, the evolutionary tree, speeding up evolution with synthetic bacteria, Archaea and, eventually, eukaryotes. We ’ re a ways away from improving people : our goal is just to make sure that we have a chance to survive long enough to maybe do that. Thank you very much.

Text Sample 440: NEG Dim 3 – 2008 – Score -1.00 – 2008/t000173.txt

The north coast of California has rainforests â temperate rainforests â where it can rain more than 100 inches a year. This is the realm of the Coast Redwood tree. Its species name is Sequoia sempervirens. Sequoia sempervirens is the tallest living organism on Earth. The range of the species goes up to as much as 380 feet tall. That ’ s 38 stories tall. These are trees that would stand out in midtown Manhattan. Nobody knows how old the oldest living Coast Redwoods are because nobody has ever drilled into any of them to count their annual growth rings, and, in any case, the centers of the oldest individuals appear to be hollow. But it ’ s believed that the oldest living Redwoods are perhaps 2, 500 years old â roughly the age of the Parthenon â although it ’ s also suspected that there may be individual trees that are older than that. You can see the range of the Coast Redwoods. It ’ s here, in red. The largest individuals of this species, the dreadnoughts of their kind, live just on the north coast of California, where the rain is really intense. In recent historic times, about 96 percent of the Coast Redwood forest was cut down, especially in a series of bursts of intense liquidation logging, clear-cutting that took place in the 1970s through the early 1990s. Even so, about four percent of the primeval Redwood rainforest remains intact, wild and now protected â entirely protected â in a chain of small parks strung out like pearls along the north coast of California, including Redwood National Park. But curiously, Redwood rainforests, the fragments that we have left, to this day remain underexplored. Redwood rainforest is incredibly difficult to move through, and even today, individual trees are being discovered that have never been seen before, including, in the summer of 2006, Hyperion, the world ’ s tallest tree. I ’ m going to do a little Gedanken experiment. I ’ m going to ask you to imagine what a Redwood really is as a living organism. And, Chris, if I could have you up here? I have a tape measure. It ’ s a kind loaner from TED. And Chris, if you could take the end of that tape measure? We ’ re going to show you what the diameter at breast height of a big Redwood is. Unfortunately, this tape isn ’ t long enough â it ’ s only a 25-foot tape. Chris, could you extend your arm out that way? There we go. OK. And maybe about here, about 30 feet, is the diameter of a big Redwood. Now, let your imagination go upward into space. Think about this tree, rising upward into Redwood space, 325 feet, 32 stories, an individual living organism articulating its forms upward into space over long periods of time. The Redwood species seems to exist in another kind of time : not human time, but what we might call Redwood time. Redwood time moves at a more stately pace than human time. To us, when we look at a Redwood tree, it seems to be motionless and still, and yet Redwoods are constantly in motion, moving upward into space, articulating themselves and filling Redwood space over Redwood time, over thousands of years. Plant this small seed, wait 2, 000 years, and you get this : the Lost Monarch. It dwells in the Grove of Titans on the north coast, and was discovered in 1998. And yet, when you look at the base of a Redwood tree, you ’ re not seeing the organism. You ’ re like a mouse looking at the foot of an elephant, and most of the organism is overhead, unseen. I became very interested, and I wrote about a couple. Steve Sillett and Marie Antoine are the principal explorers of the Redwood forest canopy. They ’ re world-class athletes, and they also are world-class forest ecology scientists. Steve Sillett, when he was a 19-year-old college student at Reed College, had heard that the Redwood forest canopy is considered to be a so-called Redwood desert. That is to say, at that time it was believed that there was nothing up there except the branches of Redwood trees. And with a friend of his, he took it upon himself to free-climb a Redwood without ropes or any equipment to see what was up there. He climbed up a small tree next to this giant Redwood, and then he leaped through space and grabbed a branch with his hands, and ended up hanging, like catching a bar of a trapeze. And then, from there, he climbed directly up the bark until he got to the top of the tree. His friend, a guy named Marwood Harris, was following behind. Neither one of them had noticed that there was a Yellow Jacket wasp ’ s nest the size of a bowling ball hanging from the branch that Steve had jumped into. And when Marwood made the jump, he was covered with wasps stinging him in the face and eyes. He nearly let go. He would have fallen to his death, being 75 feet above the ground. But they made it to the top, and what they found was not a Redwood desert, but a lost world â a kind of three-dimensional labyrinth in the air, filled with unknown life. Now, I had been working on other topics : the emergence of infectious diseases, which come out of the natural ecosystems of the Earth, make a trans-species jump, and get into humans. After three books on this, it got to be a bit much, in a way. My wife and I adore our children. And I began climbing trees with my kids as just something to do with them, using the so-called arborist climbing technique, with ropes. You use ropes to get yourself up into the crown of a tree. Children are incredibly adept at climbing trees. That ’ s my son, Oliver. They don ’ t seem to suffer from the same fear of heights that humans do. If ontogeny recapitulates phylogeny, then children are somewhat closer to our roots as primates in the arboreal forest. Humans appear to be the only primates that I know of that are afraid of heights. All other primates, when they ’ re scared, they run up a tree, where they feel safe. We camped overnight in the trees, in tree boats. This is my daughter Laura, then 15, looking out of a tree boat. She ’ s, by the way, tied in with a rope so she can ’ t fall. Looking out of a tree boat in the morning and hearing birdsong coming in three dimensions around us. We had been visited in the night by flying squirrels, who don ’ t seem to recognize humans for what they are because they ’ ve never seen them in the canopy before. And we practiced advanced techniques like sky-walking, where you can move from tree to tree through space, rather like Spiderman. It became a writing project. When Steve Sillett gets up into a big Redwood, he fires an arrow, which trails a fishing line, which gets over a branch in the tree, and then you ascend up a rope which has been dragged into the tree by the line. You ascend 30 stories. There are two people climbing this tree, Gaya, which is thought to be one of the oldest Redwoods. There they are. They are only one-seventh of the way up that tree. You do feel a sense of exposure. There is a small person right down there on the ground. You feel like you ’ re climbing a wall of wood. But then you enter the Redwood canopy, and it ’ s like coming through a layer of clouds. And all of a sudden, you lose sight of the ground, and you also lose sight of the sky, and you ’ re in a three-dimensional labyrinth in the air filled with hanging gardens of ferns growing out of soil, which is populated with all kinds of small organisms. There are epiphytes, plants that grow on trees. These are huckleberry bushes. Many species of mosses, and then all sorts of lichens just plastering the tree. When you get near the top of the tree, you feel like you can ’ t fall â in fact, it ’ s difficult to move. You ’ re worming your way through branches which are crowded with living things that don ’ t occur near the ground. It ’ s like scuba diving into a coral reef, except you ’ re going upward instead of downward. And then the trees tend to flare out into platform-like areas at the top. Maria ’ s sitting on one of them. These limbs could be five to six hundred years old. Redwoods grow very slowly in their tops. They also have a feature : thickets of huckleberry bushes that grow out of the tops of Redwood trees that are technically known as huckleberry afros, and you can sit there and snack on the berries while you ’ re resting. Redwoods have an enormous surface area that extends upward into space because they have a propensity to do something called reiteration. A Redwood is a fractal. And as they put out limbs, the limbs burst into small trees, copies of the Redwood. Now, here we see a reiteration in Chronos, one of the older Redwoods. This reiteration is a huge flying buttress that comes out the tree itself. This buttress is less than halfway up the tree. And then it bursts into a forest of Redwoods.

This particular extra trunk is a meter across at the base and extends upward for 150 feet. It ’ s as big as any of the biggest trees east of the Mississippi River, and yet it ’ s only a minor feature on Chronos. This three-dimensional map of the crown structure of a Redwood named Iluvatar, made by Steve Sillett, Marie Antoine and their colleagues, gives you an idea. What you ’ re seeing here is a hierarchical **schematic** development of the trunks of this tree as it has elaborated itself over time into six layers of fractal, of trunks springing from trunks springing from trunks. I asked Steve to put a human being in this to give a sense of scale. There ’ s the person, right there. The person is waving to us. I ’ ve wanted to ask Craig Venter if it would be possible to insert a synthetic chromosome into a human so that we could reiterate ourselves if we wanted to. And if we were able to reiterate, then the fingers of our hand would be people who looked like us, and they would have people on their hands and so on. And if we had Redwood-like biology, we would have six layers of people on our hands, as it were. And it would be a lovely thing to be able to wave to someone and have all our reiterations wave at the same time. To reiterate the point, let ’ s go closer into Iluvatar. We ’ re looking at that yellow box. And this hallucinatory drawing shows you â□□ everything you see in this drawing is Iluvatar. These are millennial structures â□□ portions of the tree that are believed to be more than 1, 000 years old. There are four humans in this shot â□□ one, two, three, four. And there ’ s also something that I want to show you. This is a flying buttress. Redwoods grow back into themselves as they expand into space, and this flying buttress is a limb shot out of that small trunk, going back into the main trunk and fusing with it. Flying buttresses, just as in a cathedral, help strengthen the crown of the tree and help the tree exist longer through time. The scientists are doing all kinds of experiments in these trees. They ’ ve wired them like patients in an ICU. They ’ re finding out that Redwoods can move moisture out of the air and down into their trunks, possibly all the way into their root systems. They also have the ability to put roots anywhere in the tree itself. If a portion of a Redwood is rotting, the Redwood will send roots into its own form and draw nutrients out of itself as it falls apart. If we had Redwood-like biology, if we got a touch of gangrene in our arm then we could just, you know, extract the nutrients extract the nutrients and the moisture out of it until it fell off. Canopy soil can occur up to a meter deep, hundreds of feet above the ground, and there are organisms in this soil that have, as yet, no names. This is an unnamed species of copepod. A copepod is a crustacean. These copepods are a major constituent of the oceans, and they are a major part of the diet of grazing baleen whales. What they ’ re doing in the Redwood forest canopy soil hundreds of feet above the ocean, or how they got there, is completely unknown. There are some interesting theories that, if I had time, I would tell you about. But as you go and you look closer at a tree, what you see is, you see increasing complexity. We ’ re looking at the very top of Gaya, which is thought to be the oldest Redwood. Gaya may be 3, 000 to 5, 000 years old, no one really knows, but its top has broken off and it ’ s been rotting back now. This little Japanese garden-like creation probably took 700 years to form in its complexity that we see right now. As you look at a tree, it takes a magnifying glass to see a giant tree. I have to show you something unfortunately very sad at the conclusion of this talk. The Eastern Hemlock tree has often been described as the Redwood of the East. And we ’ re moving in a full circle now. In the 1950s, a small organism appeared in Richmond, Virginia, called the Hemlock woolly adelgid. It made a trans-species jump out of some other organism in Asia, where it was living on Hemlock trees in Asia. When it moved into its new host, the Eastern Hemlock tree, it escaped its predators, and the new tree had no resistance to it. The Eastern Hemlock forest is being considered in some ways the last fragments of primeval rainforest east of the Mississippi River. I hadn ’ t even known that there were rainforests in the east, but in Great Smoky Mountains National Park it can rain up to 100 inches of rain a year. And in the last two to three summers, these invasive organisms, this kind of Ebola of the trees, as it were, has swept through the primeval Hemlock forest of the east, and has absolutely wiped it out. I climbed there this past summer. This is Great Smoky Mountains National Park, and the Hemlocks are dead as far as the eye can see. And what we ’ re seeing is not just the potential death of the Eastern Hemlock species â that is to say, its extinction from nature due to this invading parasite â but we ’ re also seeing the death of an incredibly complex ecosystem for which these trees are merely the substrate for the aerial labyrinth of the sky that exists in their crowns. It ’ s absolutely heartbreaking to see. One of the things that is just â I almost can ’ t conceive it â is the idea that the national news media hasn ’ t picked this up at all, and this is the devastation of one of the most important ecosystems in North America. What can the Redwoods tell us about ourselves? Well, I think they can tell us something about human time. The flickering, transitory quality of human time and the brevity of human life â the necessity to love. But we ’ re different from trees, and they can also teach us something about ourselves in the differences that we have. We are human, and we have the capacity to love, we have the capacity to wonder, and we have a sort of boundless curiosity, a restless inquisitiveness that so suits us as primates, I think. And at least for me, personally, the trees have taught me an entirely new way of loving my children. Exploring with them the forest canopy has been one of the most lovely things of my existence on Earth. And I think that one of the happiest things is the sense that with my children I ’ ve been able to introduce them into the very small circle of humans who are lucky enough, or possibly stupid enough, to still climb trees. Thank you very much. Chris Anderson : I think at a previous TED, I think it was Nathan Myhrvold who told me that it was thought that because these trees are like, 2, 000 years and older, on many of them there are ecosystems where there are species that are not found anywhere on the Earth except on that one tree. Is that correct? Richard Preston : Yes, that is correct. I mentioned Hyperion, the world ’ s tallest tree. And I was a member of a climbing team that made the first climb of it, in 2006. And while we were climbing Hyperion, Marie Antoine spotted an unknown species of golden-brown ant about halfway up the trunk. Ants are not known to occur in Redwood trees, curiously enough, and we wondered whether this ant, this species of ant, was only endemic to that one tree, or possibly to that grove. And in subsequent climbs they could never find that ant again, and so no specimens have ever been collected. We don ’ t know what it is â we just know it ’ s there. CA : So, you have to wonder when, you know, if some other species than us was recording the stories that mattered on Earth, you know, our stories are about Iraq and war and politics and celebrity gossip. You ’ ve just told us a different story of this tragic arms race that ’ s happening, and maybe whole ecosystems gone forever. It ’ s an amazing sense of wonder you ’ ve given me, and a sense of just how fragile this whole thing is. RP : It is fragile, and you know, I think about emerging human diseases â parasites that move into the human species. But that ’ s just a very small facet of a much greater problem of invasions of species worldwide, all through the ecosystems, and you know, the Earth itself â CA : Partly caused by us, inadvertently. RP : Caused by humans. Caused by the movement of humans. You can think of the Earth ’ s biosphere as a palace, and the continents are rooms in the palace, and the islands are small rooms. But lately, the doors of the palace have been flung open, and the walls are coming down. CA : Richard Preston, thank you very much, I think. RP : Thank you.

Text Sample 441: NEG Dim 3 – 2011 – Score -2.00 – 2011/t002602.txt

I thought I ’ d talk a little bit about how nature makes materials. I brought along with me an abalone shell. This abalone shell is a biocomposite material that ’ s 98 percent by mass calcium carbonate and two percent by mass protein. Yet, it ’ s 3, 000 times tougher than its geological counterpart. And a lot of people might use structures like abalone shells, like chalk. I ’ ve been fascinated by how nature makes materials, and there ’ s a lot of secrets to how they do such an exquisite job. Part of it is that these materials are macroscopic in structure, but they ’ re formed at the nano scale. They ’ re formed at the nano scale, and they use proteins that are coded by the genetic level that allow them to build these really exquisite structures. So something I think is very fascinating is : What if you could give life to non-living structures, like batteries and like solar cells? What if they had some of the same capabilities that an abalone shell did, in terms of being able to build really exquisite structures at room temperature and room pressure, using nontoxic chemicals and adding no toxic materials back into the environment? So that ’ s kind of the vision that I ’ ve been thinking about. And so what if you could grow a battery in a Petri dish? Or what if you could give genetic information to a battery so that it could actually become better as a function of time, and do so in an environmentally friendly way? And so, going back to this abalone shell, besides being nanostructured, one thing that ’ s fascinating is, when a male and female abalone get together, they pass on the genetic information that says,”This is how to build an exquisite material. Here ’ s how to do it at room temperature and pressure, using nontoxic materials.”Same with diatoms, which are shown right here, which are glasseous structures. Every time the diatoms replicate, they give the genetic information that says,”Here ’ s how to build glass in the ocean that ’ s perfectly nanostructured.”And you can do it the same, over and over again.”So what if you could do the same thing with a solar cell or a battery? I like to say my favorite biomaterial is my four year old. But anyone who ’ s ever had or knows small children knows, they ’ re incredibly complex organisms. If you wanted to convince them to do something they don ’ t want to do, it ’ s very difficult. So when we think about future technologies, we actually think of using bacteria and viruses — simple organisms. Can you convince them to work with a new toolbox, so they can build a structure that will be important to me? Also, when we think about future technologies, we start with the beginning of Earth. Basically, it took a billion years to have life on Earth. And very rapidly, they became **multi-cellular**, they could replicate, they could use photosynthesis as a way of getting their energy source. But it wasn ’ t until about 500 million years ago — during the Cambrian geologic time period — that organisms in the ocean started making hard materials. Before that, they were all soft, fluffy structures. It was during this time that there was increased calcium, iron and silicon in the environment, and organisms learned how to make hard materials. So that ’ s what I would like to be able to do, convince biology to work with the rest of the periodic table. Now, if you look at biology, there ’ s many structures like DNA, antibodies, proteins and ribosomes you ’ ve heard about, that are nanostructured — nature already gives us really exquisite structures on the nano scale. What if we could harness them and convince them to not be an antibody that does something like HIV? What if we could convince them to build a solar cell for us? Here are some examples. Natural shells, natural biological materials. The abalone shell here. If you fracture it, you can look at the fact that it ’ s nanostructured. There ’ s diatoms made out of SiO2, and there are magnetotactic bacteria that make small, single-domain magnets used for navigation. What all these have in common is these materials are structured at the nano scale, and they have a DNA sequence that codes for a protein sequence that gives them the blueprint to be able to build these really wonderful structures. Now, going back to the abalone shell, the abalone makes this shell by having these proteins. These proteins are very negatively charged. They can pull calcium out of the environment, and put down a layer of calcium and then carbonate, calcium and carbonate. It has the chemical sequences of amino acids which says,"This is how to build the structure. Here ' s the DNA sequence, here ' s the protein sequence in order to do it."So an interesting idea is, what if you could take any material you wanted, or any element on the periodic table, and find its corresponding DNA sequence, then code it for a corresponding protein sequence to build a structure, but not build an abalone shell — build something that nature has never had the opportunity to work with yet. And so here ' s the periodic table. I absolutely love the periodic table. Every year for the incoming freshman class at MIT, I have a periodic table made that says,"Welcome to MIT. Now you ' re in your element."And you flip it over, and it ' s the amino acids with the pH at which they have different charges. And so I give this out to thousands of people. And I know it says MIT and this is Caltech, but I have a couple extra if people want it. I was really fortunate to have President Obama visit my lab this year on his visit to MIT, and I really wanted to give him a periodic table. So I stayed up at night and talked to my husband,"How do I give President Obama a periodic table? What if he says, ' Oh, I already have one, ' or, ' I ' ve already memorized it? '"So he came to visit my lab and looked around — it was a great visit. And then afterward, I said,"Sir, I want to give you the periodic table, in case you ' re ever in a bind and need to calculate molecular weight."I thought"molecular weight"sounded much less nerdy than"molar mass."And he looked at it and said,"Thank you. I ' ll look at it periodically."Later in a lecture that he gave on clean energy, he pulled it out and said,"And people at MIT, they give out periodic tables."So... So basically what I didn ' t tell you is that about 500 million years ago, the organisms started making materials, but it took them about 50 million years to get good at it — 50 million years to learn how to perfect how to make that abalone shell. And that ' s a hard sell to a graduate student :"I have this great project... 50 million years..."So we had to develop a way of trying to do this more rapidly. And so we use a nontoxic virus called M13 bacteriophage, whose job is to infect bacteria. Well, it has a simple DNA structure that you can go in and cut and paste additional DNA sequences into it, and by doing that, it allows the virus to express random protein sequences. This is pretty easy biotechnology, and you could basically do this a billion times. So you can have a billion different viruses that are all genetically identical, but they differ from each other based on their tips, on one sequence, that codes for one protein. Now if you take all billion viruses, and put them in one drop of liquid, you can force them to interact with anything you want on the periodic table. And through a process of selection evolution, you can pull one of a billion that does something you ' d like it to do, like grow a battery or a solar cell. Basically, viruses can ' t replicate themselves ; they need a host. Once you find that one out of a billion, you infect it into a bacteria, and make millions and billions of copies of that particular sequence. The other thing that ' s beautiful about biology is that biology gives you really exquisite structures with nice link scales. These viruses are long and skinny, and we can get them to express the ability to grow something like semiconductors or materials for batteries. Now, this is a high-powered battery that we grew in my lab. We engineered a virus to pick up carbon **nanotubes**. One part of the virus grabs a carbon nanotube, the other part of the virus has a sequence that can grow an electrode material for a battery, and then it wires itself to the current collector. And so through a process of selection evolution, we went from being able to have a virus that made a crummy battery to a virus that made a good battery to a virus that made a record-breaking, high-powered battery that ’ s all made at room temperature, basically at the benchtop. That battery went to the White House for a press conference, and I brought it here. You can see it in this case that ’ s lighting this LED. Now if we could scale this, you could actually use it to run your Prius, which is kind of my dream — to be able to drive a virus- powered car. But basically you can pull one out of a billion, and make lots of amplifications to it. Basically, you make an amplification in the lab, and then you get it to self-assemble into a structure like a battery. We ’ re able to do this also with catalysis. This is the example of a photocatalytic splitting of water. And what we ’ ve been able to do is engineer a virus to basically take dye-absorbing molecules and line them up on the surface of the virus so it acts as an antenna, and you get an energy transfer across the virus. And then we give it a second gene to grow an inorganic material that can be used to split water into oxygen and hydrogen, that can be used for clean fuels. I brought an example of that with me today. My students promised me it would work. These are virus-assembled nanowires. When you shine light on them, you can see them bubbling. In this case, you ’ re seeing oxygen bubbles come out. Basically, by controlling the genes, you can control multiple materials to improve your device performance. The last example are solar cells. You can also do this with solar cells. We ’ ve been able to engineer viruses to pick up carbon **nanotubes** and then grow titanium dioxide around them, and use it as a way of getting electrons through the device. And what we ’ ve found is through genetic engineering, we can actually increase the efficiencies of these solar cells to record numbers for these types of dye-sensitized systems. And I brought one of those as well, that you can play around with outside afterward. So this is a virus-based solar cell. Through evolution and selection, we took it from an eight percent efficiency solar cell to an 11 percent efficiency solar cell. So I hope that I ’ ve convinced you that there ’ s a lot of great, interesting things to be learned about how nature makes materials, and about taking it the next step, to see if you can force or take advantage of how nature makes materials, to make things that nature hasn ’ t yet dreamed of making. Thank you.

Text Sample 442: NEG Dim 3 – 2011 – Score -2.00 – 2011/t002540.txt

When I was little â and by the way, I was little once â my father told me a story about an 18th century watchmaker. And what this guy had done : he used to produce these fabulously beautiful watches. And one day, one of his customers came into his workshop and asked him to clean the watch that he ’ d bought. And the guy took it apart, and one of the things he pulled out was one of the balance wheels. And as he did so, his customer noticed that on the back side of the balance wheel was an engraving, were words. And he said to the guy,”Why have you put stuff on the back that no one will ever see?”And the watchmaker turned around and said,”God can see it.”Now I ’ m not in the least bit religious, neither was my father, but at that point, I noticed something happening here. I felt something in this plexus of blood vessels and nerves, and there must be some muscles in there as well somewhere, I guess. But I felt something. And it was a physiological response. And from that point on, from my age at the time, I began to think of things in a different way. And as I took on my career as a designer, I began to ask myself the simple question : Do we actually think beauty, or do we feel it? Now you probably know the answer to this already. You probably think, well, I don ’ t know which one you think it is, but I think it ’ s about feeling beauty. And so I then moved on into my design career and began to find some exciting things. One of the most early work was done in automotive design â some very exciting work was done there. And during a lot of this work, we found something, or I found something, that really fascinated me, and maybe you can remember it. Do you remember when lights used to just go on and off, click click, when you closed the door in a car? And then somebody, I think it was BMW, introduced a light that went out slowly. Remember that? I remember it clearly. Do you remember the first time you were in a car and it did that? I remember sitting there thinking, this is fantastic. In fact, I ’ ve never found anybody that doesn ’ t like the light that goes out slowly. I thought, well what the hell ’ s that about? So I started to ask myself questions about it. And the first was, I ’ d ask other people :”Do you like it?””Yes.””Why?”And they ’ d say,”Oh, it feels so natural,”or,”It ’ s nice.”I thought, well that ’ s not good enough. Can we cut down a little bit further, because, as a designer, I need the vocabulary, I need the keyboard, of how this actually works. And so I did some experiments. And I suddenly realized that there was something that did exactly that â light to dark in six seconds â exactly that. Do you know what it is? Anyone? You see, using this bit, the thinky bit, the slow bit of the brain â using that. And this isn ’ t a think, it ’ s a feel. And would you do me a favor? For the next 14 minutes or whatever it is, will you feel stuff? I don ’ t need you to think so much as I want you to feel it. I felt a sense of relaxation tempered with anticipation. And that thing that I found was the cinema or the theater. It ’ s actually just happened here â light to dark in six seconds. And when that happens, are you sitting there going,”No, the movie ’ s about to start,”or are you going,”That ’ s fantastic. I ’ m looking forward to it. I get a sense of anticipation”? Now I ’ m not a neuroscientist. I don ’ t know even if there is something called a conditioned reflex. But it might be. Because the people I speak to in the northern hemisphere that used to go in the cinema get this. And some of the people I speak to that have never seen a movie or been to the theater don ’ t get it in the same way. Everybody likes it, but some like it more than others. So this leads me to think of this in a different way. We ’ re not feeling it. We ’ re thinking beauty is in the limbic system â if that ’ s not an outmoded idea. These are the bits, the pleasure centers, and maybe what I ’ m seeing and sensing and feeling is bypassing my thinking. The wiring from your sensory **apparatus** to those bits is shorter than the bits that have to pass through the thinky bit, the cortex. They arrive first. So how do we make that actually work? And how much of that reactive side of it is due to what we already know, or what we ’ re going to learn, about something? This is one of the most beautiful things I know. It ’ s a plastic bag. And when I looked at it first, I thought, no, there ’ s no beauty in that. Then I found out, post exposure, that this plastic bag if I put it into a filthy puddle or a stream filled with coliforms and all sorts of disgusting stuff, that that filthy water will migrate through the wall of the bag by osmosis and end up inside it as pure, potable drinking water. And all of a sudden, this plastic bag was extremely beautiful to me. Now I ’ m going to ask you again to switch on the emotional bit. Would you mind taking the brain out, and I just want you to feel something. Look at that. What are you feeling about it? Is it beautiful? Is it exciting? I ’ m watching your faces very carefully. There ’ s some rather bored-looking gentlemen and some slightly engaged-looking ladies who are picking up something off that. Maybe there ’ s an innocence to it. Now I ’ m going to tell you what it is. Are you ready? This is the last act on this Earth of a little girl called Heidi, five years old, before she died of cancer to the spine. It ’ s the last thing she did, the last physical act. Look at that picture. Look at the innocence. Look at the beauty in it. Is it beautiful now? Stop. Stop. How do you feel? Where are you feeling this? I

’ m feeling it here. I feel it here. And I ’ m watching your faces, because your faces are telling me something. The lady over there is actually crying, by the way. But what are you doing? I watch what people do. I watch faces. I watch reactions. Because I have to know how people react to things. And one of the most common faces on something faced with beauty, something stupefyingly delicious, is what I call the OMG. And by the way, there ’ s no pleasure in that face. It ’ s not a”this is wonderful!”The eyebrows are doing this, the eyes are defocused, and the mouth is hanging open. That ’ s not the expression of joy. There ’ s something else in that. There ’ s something weird happening. So pleasure seems to be tempered by a whole series of different things coming in. Poignancy is a word I love as a designer. It means something triggering a big emotional response, often quite a sad emotional response, but it ’ s part of what we do. It isn ’ t just about nice. And this is the dilemma, this is the paradox, of beauty. Sensorily, we ’ re taking in all sorts of things â mixtures of things that are good, bad, exciting, frightening â to come up with that sensorial exposure, that sensation of what ’ s going on. Pathos appears obviously as part of what you just saw in that little girl ’ s drawing. And also triumph, this sense of transcendence, this”I never knew that. Ah, this is something new.”And that ’ s packed in there as well. And as we assemble these tools, from a design point of view, I get terribly excited about it, because these are things, as we ’ ve already said, they ’ re arriving at the brain, it would seem, before cognition, before we can manipulate them â electrochemical party tricks. Now what I ’ m also interested in is : Is it possible to separate intrinsic and extrinsic beauty? By that, I mean intrinsically beautiful things, just something that ’ s exquisitely beautiful, that ’ s universally beautiful. Very hard to find. Maybe you ’ ve got some examples of it. Very hard to find something that, to everybody, is a very beautiful thing, without a certain amount of information packed in there before. So a lot of it tends to be extrinsic. It ’ s mediated by information before the comprehension. Or the information ’ s added on at the back, like that little girl ’ s drawing that I showed you. Now when talking about beauty you can ’ t get away from the fact that a lot experiments have been done in this way with faces and what have you. And one of the most tedious ones, I think, was saying that beauty was about symmetry. Well it obviously isn ’ t. This is a more interesting one where half faces were shown to some people, and then to add them into a list of most beautiful to least beautiful and then exposing a full face. And they found that it was almost exact coincidence. So it wasn ’ t about symmetry. In fact, this lady has a particularly asymmetrical face, of which both sides are beautiful. But they ’ re both different. And as a designer, I can ’ t help meddling with this, so I pulled it to bits and sort of did stuff like this, and tried to understand what the individual elements were, but feeling it as I go. Now I can feel a sensation of delight and beauty if I look at that eye. I ’ m not getting it off the eyebrow. And the earhole isn ’ t doing it to me at all. So I don ’ t know how much this is helping me, but it ’ s helping to guide me to the places where the signals are coming off. And as I say, I ’ m not a neuroscientist, but to understand how I can start to assemble things that will very quickly bypass this thinking part and get me to the enjoyable precognitive elements. Anais Nin and the Talmud have told us time and time again that we see things not as they are, but as we are. So I ’ m going to shamelessly expose something to you, which is beautiful to me. And this is the F1 MV Agusta. Ahhhh. It is really â I mean, I can ’ t express to you how exquisite this object is. But I also know why it ’ s exquisite to me, because it ’ s a palimpsest of things. It ’ s masses and masses of layers. This is just the bit that protrudes into our physical dimension. It ’ s something much bigger. Layer after layer of legend, sport, details that resonate. I mean, if I just go through some of them now â I know about laminar flow when it comes to air-piercing objects, and that does it consummately well, you can see it can. So that ' s getting me excited. And I feel that here. This bit, the big secret of automotive design â reflection management. It ' s not about the shapes, it ' s how the shapes reflect light. Now that thing, light flickers across it as you move, so it becomes a kinetic object, even though it ' s standing still â managed by how brilliantly that ' s done on the reflection. This little relief on the footplate, by the way, to a rider means there ' s something going on underneath it â in this case, a drive chain running at 300 miles and hour probably, taking the power from the engine. I ' m getting terribly excited as my mind and my eyes flick across these things. Titanium lacquer on this. I can ' t tell you how wonderful this is. That ' s how you stop the nuts coming off at high speed on the wheel. I ' m really getting into this now. And of course, a racing bike doesn ' t have a prop stand, but this one, because it ' s a road bike, it all goes away and it folds into this little gap. So it disappears. And then I can ' t tell you how hard it is to do that radiator, which is curved. Why would you do that? Because I know we need to bring the wheel farther into the aerodynamics. So it ' s more expensive, but it ' s wonderful. And to cap it all, brand royalty â Agusta, Count Agusta, from the great histories of this stuff. The bit that you can ' t see is the genius that created this. Massimo Tamburini. They call him"The Plumber"in Italy, as well as"Maestro,"because he actually is engineer and craftsman and sculptor at the same time. There ' s so little compromise on this, you can ' t see it. But unfortunately, the likes of me and people that are like me have to deal with compromise all the time with beauty. We have to deal with it. So I have to work with a supply chain, and I ' ve got to work with the technologies, and I ' ve got to work with everything else all the time, and so compromises start to fit into it. And so look at her. I ' ve had to make a bit of a compromise there. I ' ve had to move that part across, but only a millimeter. No one ' s noticed, have they yet? Did you see what I did? I moved three things by a millimeter. Pretty? Yes. Beautiful? Maybe lesser. But then, of course, the consumer says that doesn ' t really matter. So that ' s okay, isn ' t it? Another millimeter? No one ' s going to notice those split lines and changes. It ' s that easy to lose beauty, because beauty ' s incredibly difficult to do. And only a few people can do it. And a focus group cannot do it. And a team rarely can do it. It takes a central cortex, if you like, to be able to orchestrate all those elements at the same time. This is a beautiful water bottle â some of you know of it â done by Ross Lovegrove, the designer. This is pretty close to intrinsic beauty. This one, as long as you know what water is like then you can experience this. It ' s lovely because it is an embodiment of something refreshing and delicious. I might like it more than you like it, because I know how bloody hard it is to do it. It ' s stupefyingly difficult to make something that refracts light like that, that comes out of the tool correctly, that goes down the line without falling over. Underneath this, like the story of the swan, is a million things very difficult to do. So all hail to that. It ' s a fantastic example, a simple object. And the one I showed you before was, of course, a massively complex one. And they ' re working in beauty in slightly different ways because of it. You all, I guess, like me, enjoy watching a ballet dancer dance. And part of the joy of it is, you know the difficulty. You also may be taking into account the fact that it ' s incredibly painful. Anybody seen a ballet dancer ' s toes when they come out of the points? While she ' s doing these graceful arabesques and plies and what have you, something horrible ' s going on down here. The comprehension of it leads us to a greater and heightened sense of the beauty of what ' s actually going on. Now I ' m using **microseconds** wrongly here, so please ignore me. But what I have to do now, feeling again, what I ' ve got to do is to be able to supply enough of these enzymes, of these triggers into something early on in the process, that you pick it up, not through your thinking, but through your feeling. So we ’ re going to have a little experiment. Right, are you ready? I ’ m going to show you something for a very, very brief moment. Are you ready? Okay. Did you think that was a bicycle when I showed it to you at the first flash? It ’ s not. Tell me something, did you think it was quick when you first saw it? Yes you did. Did you think it was modern? Yes you did. That blip, that information, shot into you before that. And because your brain starter motor began there, now it ’ s got to deal with it. And the great thing is, this motorcycle has been styled this way specifically to engender a sense that it ’ s green technology and it ’ s good for you and it ’ s light and it ’ s all part of the future. So is that wrong? Well in this case it isn ’ t, because it ’ s a very, very ecologically-sound piece of technology. But you ’ re a slave of that first flash. We are slaves to the first few fractions of a second â and that ’ s where much of my work has to win or lose, on a shelf in a shop. It wins or loses at that point. You may see 50, 100, 200 things on a shelf as you walk down it, but I have to work within that domain, to ensure that it gets you there first. And finally, the layer that I love, of knowledge. Some of you, I ’ m sure, will be familiar with this. What ’ s incredible about this, and the way I love to come back to it, is this is taking something that you hate or bores you, folding clothes, and if you can actually do this â who can actually do this? Anybody try to do this? Yeah? It ’ s fantastic, isn ’ t it? Look at that. Do you want to see it again? No time. It says I have two minutes left, so we can ’ t do this. But just go to the Web, YouTube, pull it down,”folding T-shirt.”That ’ s how underpaid younger-aged people have to fold your T-shirt. You didn ’ t maybe know it. But how do you feel about it? It feels fantastic when you do it, you look forward to doing it, and when you tell somebody else about it â like you probably have â you look really smart. The knowledge bubble that sits around the outside, the stuff that costs nothing, because that knowledge is free â bundle that together and where do we come out? Form follows function? Only sometimes. Only sometimes. Form is function. Form is function. It informs, it tells us, it supplies us answers before we ’ ve even thought about it. And so I ’ ve stopped using words like”form,”and I ’ ve stopped using words like”function”as a designer. What I try to pursue now is the emotional functionality of things. Because if I can get that right, I can make them wonderful, and I can make them repeatedly wonderful. And you know what those products and services are, because you own some of them. They ’ re the things that you ’ d snatch if the house was on fire. Forming the emotional bond between this thing and you is an electrochemical party trick that happens before you even think about it. Thank you very much.

Text Sample 443: NEG Dim 3 – 2011 – Score -2.00 – 2011/t002536.txt

So historically there has been a huge divide between what people consider to be non-living systems on one side, and living systems on the other side. So we go from, say, this beautiful and complex crystal as non-life, and this rather beautiful and complex cat on the other side. Over the last hundred and fifty years or so, science has kind of blurred this distinction between non-living and living systems, and now we consider that there may be a kind of continuum that exists between the two. We ’ ll just take one example here : a virus is a natural system, right? But it ’ s very simple. It ’ s very simplistic. It doesn ’ t really satisfy all the requirements, it doesn ’ t have all the characteristics of living systems and is in fact a parasite on other living systems in order to, say, reproduce and evolve. But what we ’ re going to be talking about here tonight are experiments done on this sort of non-living end of this spectrum — so actually doing chemical experiments in the laboratory, mixing together nonliving ingredients to make new structures, and that these new structures might have some of the characteristics of living systems. Really what I ' m talking about here is trying to create a kind of artificial life. So what are these characteristics that I ' m talking about? These are them. We consider first that life has a body. Now this is necessary to distinguish the self from the environment. Life also has a metabolism. Now this is a process by which life can convert resources from the environment into building blocks so it can maintain and build itself. Life also has a kind of inheritable information. Now we, as humans, we store our information as DNA in our genomes and we pass this information on to our offspring. If we couple the first two — the body and the metabolism — we can come up with a system that could perhaps move and replicate, and if we coupled these now to inheritable information, we can come up with a system that would be more lifelike, and would perhaps evolve. And so these are the things we will try to do in the lab, make some experiments that have one or more of these characteristics of life. So how do we do this? Well, we use a model system that we term a protocell. You might think of this as kind of like a primitive cell. It is a simple chemical model of a living cell, and if you consider for example a cell in your body may have on the order of millions of different types of molecules that need to come together, play together in a complex network to produce something that we call alive. In the laboratory what we want to do is much the same, but with on the order of tens of different types of molecules — so a drastic reduction in complexity, but still trying to produce something that looks lifelike. And so what we do is, we start simple and we work our way up to living systems. Consider for a moment this quote by Leduc, a hundred years ago, considering a kind of synthetic biology :"The synthesis of life, should it ever occur, will not be the sensational discovery which we usually associate with the idea."That ' s his first statement. So if we actually create life in the laboratories, it ' s probably not going to impact our lives at all."If we accept the theory of evolution, then the first dawn of synthesis of life must consist in the production of forms intermediate between the inorganic and the organic world, or between the non-living and living world, forms which possess only some of the rudimentary attributes of life"— so, the ones I just discussed —"to which other attributes will be slowly added in the course of development by the evolutionary actions of the environment."So we start simple, we make some structures that may have some of these characteristics of life, and then we try to develop that to become more lifelike. This is how we can start to make a protocell. We use this idea called self-assembly. What that means is, I can mix some chemicals together in a test tube in my lab, and these chemicals will start to self-associate to form larger and larger structures. So say on the order of tens of thousands, hundreds of thousands of molecules will come together to form a large structure that didn ' t exist before. And in this particular example, what I took is some membrane molecules, mixed those together in the right environment, and within seconds it forms these rather complex and beautiful structures here. These membranes are also quite similar, morphologically and functionally, to the membranes in your body, and we can use these, as they say, to form the body of our protocell. Likewise, we can work with oil and water systems. As you know, when you put oil and water together, they don ' t mix, but through self-assembly we can get a nice oil droplet to form, and we can actually use this as a body for our artificial organism or for our protocell, as you will see later. So that ' s just forming some body stuff, right? Some architectures. What about the other aspects of living systems? So we came up with this protocell model here that I ' m showing. We started with a natural occurring clay called montmorillonite. This is natural from the environment, this clay. It forms a surface that is, say, chemically active. It could run a metabolism on it. Certain kind of molecules like to associate with the clay. For example, in this case, RNA, shown in red — this is a relative of DNA, it ' s an informational molecule — it can come along and it starts to associate with the surface of this clay. This structure, then, can organize the formation of a membrane boundary around itself, so it can make a body of liquid molecules around itself, and that ' s shown in green here on this micrograph. So just through self-assembly, mixing things together in the lab, we can come up with, say, a metabolic surface with some informational molecules attached inside of this membrane body, right? So we ' re on a road towards living systems. But if you saw this protocell, you would not confuse this with something that was actually alive. It ' s actually quite lifeless. Once it forms, it doesn ' t really do anything. So, something is missing. Some things are missing. So some things that are missing is, for example, if you had a flow of energy through a system, what we ' d want is a protocell that can harvest some of that energy in order to maintain itself, much like living systems do. So we came up with a different protocell model, and this is actually simpler than the previous one. In this protocell model, it ' s just an oil droplet, but a chemical metabolism inside that allows this protocell to use energy to do something, to actually become dynamic, as we ' ll see here. You add the droplet to the system. It ' s a pool of water, and the protocell starts moving itself around in the system. Okay? Oil droplet forms through self-assembly, has a chemical metabolism inside so it can use energy, and it uses that energy to move itself around in its environment. As we heard earlier, movement is very important in these kinds of living systems. It is moving around, exploring its environment, and remodeling its environment, as you see, by these chemical waves that are forming by the protocell. So it ' s acting, in a sense, like a living system trying to preserve itself. We take this same moving protocell here, and we put it in another experiment, get it moving. Then I ' m going to add some food to the system, and you ' ll see that in blue here, right? So I add some food source to the system. The protocell moves. It encounters the food. It reconfigures itself and actually then is able to climb to the highest concentration of food in that system and stop there. Alright? So not only do we have this system that has a body, it has a metabolism, it can use energy, it moves around. It can sense its local environment and actually find resources in the environment to sustain itself. Now, this doesn ' t have a brain, it doesn ' t have a neural system. This is just a sack of chemicals that is able to have this interesting and complex lifelike behavior. If we count the number of chemicals in that system, actually, including the water that ' s in the dish, we have five chemicals that can do this. So then we put these protocells together in a single experiment to see what they would do, and depending on the conditions, we have some protocells on the left that are moving around and it likes to touch the other structures in its environment. On the other hand we have two moving protocells that like to circle each other, and they form a kind of a dance, a complex dance with each other. Right? So not only do individual protocells have behavior, what we ' ve interpreted as behavior in this system, but we also have basically population- level behavior similar to what organisms have. So now that you ' re all experts on protocells, we ' re going to play a game with these protocells. We ' re going to make two different kinds. Protocell A has a certain kind of chemistry inside that, when activated, the protocell starts to vibrate around, just dancing. So remember, these are primitive things, so dancing protocells, that ' s very interesting to us. The second protocell has a different chemistry inside, and when activated, the protocells all come together and they fuse into one big one. Right? And we just put these two together in the same system. So there ' s population A, there ' s population B, and then we activate the system, and protocell Bs, they ’ re the blue ones, they all come together. They fuse together to form one big blob, and the other protocell just dances around. And this just happens until all of the energy in the system is basically used up, and then, game over. So then I repeated this experiment a bunch of times, and one time something very interesting happened. So, I added these protocells together to the system, and protocell A and protocell B fused together to form a hybrid protocell AB. That didn ’ t happen before. There it goes. There ’ s a protocell AB now in this system. Protocell AB likes to dance around for a bit, while protocell B does the fusing, okay? But then something even more interesting happens. Watch when these two large protocells, the hybrid ones, fuse together. Now we have a dancing protocell and a self-replication event. Right. Just with blobs of chemicals, again. So the way this works is, you have a simple system of five chemicals here, a simple system here. When they hybridize, you then form something that ’ s different than before, it ’ s more complex than before, and you get the emergence of another kind of lifelike behavior which in this case is replication. So since we can make some interesting protocells that we like, interesting colors and interesting behaviors, and they ’ re very easy to make, and they have interesting lifelike properties, perhaps these protocells have something to tell us about the origin of life on the Earth. Perhaps these represent an easily accessible step, one of the first steps by which life got started on the early Earth. Certainly, there were molecules present on the early Earth, but they wouldn ’ t have been these pure compounds that we worked with in the lab and I showed in these experiments. Rather, they ’ d be a real complex mixture of all kinds of stuff, because uncontrolled chemical reactions produce a diverse mixture of organic compounds. Think of it like a primordial ooze, okay? And it ’ s a pool that ’ s too difficult to fully characterize, even by modern methods, and the product looks brown, like this tar here on the left. A pure compound is shown on the right, for contrast. So this is similar to what happens when you take pure sugar crystals in your kitchen, you put them in a pan, and you apply energy. You turn up the heat, you start making or breaking chemical bonds in the sugar, forming a brownish caramel, right? If you let that go unregulated, you ’ ll continue to make and break chemical bonds, forming an even more diverse mixture of molecules that then forms this kind of black tarry stuff in your pan, right, that ’ s difficult to wash out. So that ’ s what the origin of life would have looked like. You needed to get life out of this junk that is present on the early Earth, four, 4. 5 billion years ago. So the challenge then is, throw away all your pure chemicals in the lab, and try to make some protocells with lifelike properties from this kind of primordial ooze. So we ’ re able to then see the self-assembly of these oil droplet bodies again that we ’ ve seen previously, and the black spots inside of there represent this kind of black tar — this diverse, very complex, organic black tar. And we put them into one of these experiments, as you ’ ve seen earlier, and then we watch lively movement that comes out. They look really good, very nice movement, and also they appear to have some kind of behavior where they kind of circle around each other and follow each other, similar to what we ’ ve seen before — but again, working with just primordial conditions, no pure chemicals. These are also, these tar-fueled protocells, are also able to locate resources in their environment. I ’ m going to add some resource from the left, here, that defuses into the system, and you can see, they really like that. They become very energetic, and able to find the resource in the environment, similar to what we saw before. But again, these are done in these primordial conditions, really messy conditions, not sort of **sterile** laboratory conditions. These are very dirty little protocells, as a matter of fact. But they have lifelike properties, is the point. So, doing these artificial life experiments helps us define a potential path between non-living and living systems. And not only that, but it helps us broaden our view of what life is and what possible life there could be out there — life that could be very different from life that we find here on Earth. And that leads me to the next term, which is”weird life.”This is a term by Steve Benner. This is used in reference to a report in 2007 by the National Research Council in the United States, wherein they tried to understand how we can look for life elsewhere in the universe, okay, especially if that life is very different from life on Earth. If we went to another planet and we thought there might be life there, how could we even recognize it as life? Well, they came up with three very general criteria. First is — and they ’ re listed here. The first is, the system has to be in non-equilibrium. That means the system cannot be dead, in a matter of fact. Basically what that means is, you have an input of energy into the system that life can use and exploit to maintain itself. This is similar to having the Sun shining on the Earth, driving photosynthesis, driving the ecosystem. Without the Sun, there ’ s likely to be no life on this planet. Secondly, life needs to be in liquid form, so that means even if we had some interesting structures, interesting molecules together but they were frozen solid, then this is not a good place for life. And thirdly, we need to be able to make and break chemical bonds. And again this is important because life transforms resources from the environment into building blocks so it can maintain itself. Now today, I told you about very strange and weird protocells — some that contain clay, some that have primordial ooze in them, some that have basically oil instead of water inside of them. Most of these don ’ t contain DNA, but yet they have lifelike properties. But these protocells satisfy these general requirements of living systems. So by making these chemical, artificial life experiments, we hope not only to understand something fundamental about the origin of life and the existence of life on this planet, but also what possible life there could be out there in the universe. Thank you.

Text Sample 444: NEG Dim 3 – 2010 – Score -3.00 – 2010/t002839.txt

Most of the talks that you ’ ve heard in the last several fabulous days have been from people who have the characteristic that they have thought about something, they are experts, they know what ’ s going on. All of you know about the topic that I ’ m supposed to talk about. That is, you know what simplicity is, you know what complexity is. The trouble is, I don ’ t. And what I ’ m going to do is share with you my ignorance on this subject. I want you to read this, because we ’ re going to come back to it in a moment. The quote is from the fabled Potter Stewart opinion on pornography. And let me just read it, the important details here :”Shorthand description, [’ hardcore pornography ’] ; and perhaps I could never succeed in intelligibly defining it. But I know it when I see it.”I ’ m going to come back to that in a moment. So, what is simplicity? It ’ s good to start with some examples. A coffee cup — we don ’ t think about coffee cups, but it ’ s much more interesting than one might think — a coffee cup is a device, which has a container and a handle. The handle enables you to hold it when the container is filled with hot liquid. Why is that important? Well, it enables you to drink coffee. But also, by the way, the coffee is hot, the liquid is **sterile** ; you ’ re not likely to get cholera that way. So the coffee cup, or the cup with a handle, is one of the tools used by society to maintain public health. Scissors are your clothes, glasses enable you to see things and keep you from being eaten by cheetahs or run down by automobiles, and books are, after all, your education. But there ’ s another class of simple things, which are also very important. Simple in function, but not at all simple in how they ’ re constructed. And the two here are just examples. One is the cellphone, which we use every day. And it rests on a complexity, which has some characteristics very different from those that my friend Benoit Mandelbrot discussed, but are very interesting. And the other, of course, is a birth control pill, which, in a very simple way, fundamentally changed the structure of society by changing the role of women in it by providing to them the opportunity to make reproductive choices. So, there are two ways of thinking about this word, I think. And here I ’ ve corrupted the Potter Stewart quotation by saying that we can think about something — which spans all the way from scissors to the cell phone, Internet and birth control pills — by saying that they ’ re simple, the functions are simple, and we recognize what that simplicity is when we see it. Or there may be another way of doing it, which is to think about the problem in terms of what — if you associate with moral philosophers — is called the teapot problem. The teapot problem I ’ ll pose this way. Suppose you see a teapot, and the teapot is filled with hot water. And you then ask the question : Why is the water hot? And that ’ s a simple question. It ’ s like, what is simplicity? One answer would be : because the kinetic energy of the water molecules is high and they bounce against things rapidly — that ’ s a kind of physical science argument. A second argument would be : because it was sitting on a stove with the flame on — that ’ s an historical argument. A third is that I wanted hot water for tea — that ’ s an intentional argument. And, since this is coming from a moral philosopher, the fourth would be that it ’ s part of God ’ s plan for the universe. All of these are possibilities. The point is that you get into trouble when you ask a single question with a single box for an answer, in which that single question actually is many questions with quite different meanings, but with the same words. Asking,”What is simplicity?”I think falls in that category. What is the state of science? And, interestingly, complexity is very highly evolved. We have a lot of interesting information about what complexity is. Simplicity, for reasons that are a little bit obscure, is almost not pursued, at least in the academic world. We academics — I am an academic — we love complexity. You can write papers about complexity, and the nice thing about complexity is it ’ s fundamentally intractable in many ways, so you ’ re not responsible for outcomes. Simplicity — all of you really would like your Waring Blender in the morning to make whatever a Waring Blender does, but not explode or play Beethoven. You ’ re not interested in the limits of these things. So what one is interested in has a lot to do with the rewards of the system. And there ’ s a lot of rewards in thinking about complexity and emergence, not so much in thinking about simplicity. One of the things I want to do is to help you with a very important task — which you may not know that you have very often — which is to understand how to sit next to a physicist at a dinner party and have a conversation. And the words that I would like you to focus on are complexity and emergence, because these will enable you to start the conversation and then daydream about other things. All right, what is complexity in this view of things, and what is emergence? We have, actually, a pretty good working definition of complexity. It is a system, like traffic, which has components. The components interact with one another. These are cars and drivers. They dissipate energy. It turns out that, whenever you have that system, weird stuff happens, and you in Los Angeles probably know this better than anyone. Here ’ s another example, which I put up because it ’ s an example of really important current science. You can ’ t possibly read that. It ’ s not intended that you read it, but that ’ s a tiny part of the chemical reactions going on in each of your cells at any given moment. And it ’ s like the traffic that you see. The amazing thing about the cell is that it actually does maintain a fairly stable working relationship with other cells, but we don ’ t know why. Anyone who tells you that we understand life, walk away. And let me reduce this to the simplest level. We ’ ve heard from Bill Gates recently. All of us, to some extent, study this thing called a Bill Gates. Terrific. You learn everything you can about that. And then there ’ s another kind of thing that you might study, and you study that hard. That ’ s a Bono, this is a Bono. But then, if you know everything you can know about those two things, and you put them together, what can you say about this combination? The answer is, not a lot. And that ’ s complexity. Now, imagine building that up to a city, or to a society, and you ’ ve got, obviously, an interesting problem. All right, so let me give you an example of simplicity of a particular kind. And I want to introduce a word that I think is very useful, which is stacking. And I ’ m going to use stacking for a kind of simplicity that has the characteristic that it is so simple and so reliable that I can build things with it. Or I ’ m going to use simple to mean reliable, predictable, repeatable. And I ’ m going to use as an example the Internet, because it ’ s a particularly good example of stacked simplicity. We call it a complex system, which it is, but it ’ s also something else. The Internet starts with mathematics, it starts with binary. And if you look at the list of things on the bottom, we are familiar with the Arabic numbers one to 10 and so on. In binary, one is 0001, seven is 0111. The question is : Why is binary simpler than Arabic? And the answer is, simply, that if I hold up three fingers, you can count that easily, but if I hold up this, it ’ s sort of hard to say that I just did seven. The virtue of binary is that it ’ s the simplest possible way of representing numbers. Anything else is more complicated. You can catch errors with it, it ’ s unambiguous in its reading, there are lots of good things about binary. So it is very, very simple once you learn how to read it. Now, if you like to represent this zero and one of binary, you need a device. And think of things in your life that are binary, one of them is light switches. They can be on and off. That ’ s binary. Now wall switches, we all know, fail. But our friends who are condensed matter physicists managed to come up, some 50 years ago, with a very nice device, shown under that bell jar, which is a **transistor**. A **transistor** is nothing more than a wall switch. It turns things on and off, but it does so without moving parts and it doesn ’ t fail, basically, for a very long period of time. So the second layer of simplicity was the **transistor** in the Internet. So, since the **transistor** is so simple, you can put lots of them together. And you put lots of them together and you come with something called integrated circuits. And a current integrated circuit might have in each one of these chips something like a billion **transistors**, all of which have to work perfectly every time. So that ’ s the next layer of simplicity, and, in fact, integrated circuits are really simple in the sense that they, in general, work really well. With integrated circuits, you can build cellphones. You all are accustomed to having your cellphones work the large majority of the time. In Boston... Boston is a little bit like Namibia in its cell phone coverage, so that we ’ re not accustomed to that all the time, but some of the time. But, in fact, if you have cell phones, you can now go to this nice lady who ’ s somewhere like Namibia, and who is extremely happy with the fact that although she does not have an master ’ s degree in electrical engineering from MIT, she ’ s nonetheless able to hack her cell phone to get power in some funny way. And from that comes the Internet. And this is a map of bitflows across the continent. The two blobs that are light in the middle there are the United States and Europe. And then back to simplicity again. So here we have what I think is one of the great ideas, which is Google. Which, in this simple portal makes the claim that it makes accessible all of the world ’ s information. But the point is that that extraordinary simple idea rests on layers of simplicity each compounded into a complexity that is itself simple, in the sense that it is completely reliable. All right, let me then finish off with four general statements, an example and two

aphorisms. The characteristics, which I think are useful to think about for simple things : First, they are predictable. Their behavior is predictable. Now, one of the nice characteristics of simple things is you know what it ’ s going to do, in general. So simplicity and predictability are characteristics of simple things. The second is, and this is a real world statement, they ’ re cheap. If you have things that are cheap enough, people will find uses for them, even if they seem very primitive. So, for example, stones. You can build cathedrals out of stones, you just have to know what it does. You carve them in blocks and then you pile them on top of one another, and they support weight. So there has to be function, the function has to be predictable and the cost has to be low. What that means is that you have to have a high performance or value for cost. And then I would propose as this last component that they serve, or have the potential to serve, as building blocks. That is, you can stack them. And stack can mean this way, or it can mean this way, or it can mean in some arbitrary n-dimensional space. But if you have something that has a function, and it ’ s really cheap, people will find new ways of putting it together to make new things. Cheap, functional, reliable things unleash the creativity of people who then build stuff that you could not imagine. There ’ s no way of predicting the Internet based on the first **transistor**. It just is not possible. So these are the components. Now, the example is something that I want to give you from the work that we ourselves do. We are very interested in delivering health care in the developing world, and one of the things that we wish to do in this particular business is to find a way of doing medical diagnosis at as close to zero cost as we can manage. So, how does one do that? This is a world in which there ’ s no electricity, there ’ s no money, there ’ s no medical competence. And I don ’ t want to spend your time in going through the details, but in the lower right-hand corner, you see an example of the kind of thing that we have. It ’ s a little paper chip. It has a few things printed on it using the same technology that you use for making comic books, which was the inspiration for this particular idea. And you put a drop, in this case, of urine at the bottom. It wicks its way up into these little branches. You know, no power required. It turns colors. In this particular case, you ’ re reading kidney function. And, since the health care worker of much of this part of the world is an 18 year-old with an AK-47, who happens to be out of work and is willing to go around and do this sort of thing, he can take a picture of it with his cellphone, send the picture back to where there is a doctor, and the doctor can look at it. So what you ’ ve done is to take a technology, which is available everywhere, make a device, which is extremely cheap, and make it in such a fashion that it is very, very reliable. If we can pull this off, if we can build more function, it will be stackable. That is to say, if we can make the basic technology of one or two things work, it will be applicable to a very, very large variety of human conditions, and hence, extendable in both vertical and horizontal directions. Part of my interest in this, I have to say, is that I would like to — how do I put this politely? — change the way, or maybe eviscerate, the capital structure of the U. S. health care system, which I think is fundamentally broken. So, let me close — Let me close with my two **aphorisms**. One of them is from Mr. Einstein, and he says,”Everything should be made as simple as possible, but not simpler.”And I think that ’ s a very good way of thinking about the problem. If you take too much out of something that ’ s simple, you lose function. You have to have low cost, but you also have to have a function. So you can ’ t make it too simple. And the second is a design issue, and it ’ s not directly relevant, but it ’ s a nice statement. This is by de Saint-Exupery. And he says,”You know you ’ ve achieved perfection in design, not when you have nothing more to add, but when you have nothing more to take away.”And that certainly is going in the right direction. So, what I think one can begin to do with this kind of cut at the word simplicity, which doesn ’ t cover Brancusi, it doesn ’ t answer the question of why Mondrian is better or worse or simpler or less simpler than Van Gogh, and certainly doesn ’ t address the question of whether Mozart is simpler than Bach. But it does make a point — which is one which, in a sense, differentiates the real world of people who make things, and the world of people who think about things, which is, there is an intellectual merit to asking : How do we make things as simple as we can, as cheap as we can, as functional as we can and as freely interconnectable as we can? If we make that kind of simplicity in our technology and then give it to you guys, you can go off and do all kinds of fabulous things with it. Thank you very much. Chris Anderson : Quick question. So can you picture that a science of simplicity might get to the point where you could look out at various systems — say a financial system or a legal system, health system — and say,”That has got to the point of danger or dysfunctionality for the following reasons, and this is how we might simplify it”? George Whitesides : Yes, I think you could. Because if you look at the components from which the system is made and examine their fragility, or their stability, you can probably build a kind of risk assessment based on that basis. CA : Have you started to do that? I mean, with the health system, you got a sort of radical solution on the cost side, but in terms of the system itself? GW : Well, no. How do I put that simply? No. CA : That was a simple, powerful answer. GW : Yes. CA : So, in terms of that diagnostic technology that you ’ ve got, where is that, and when do you see that maybe getting rolled out to scale. GW : That ’ s coming out soon. I mean, the systems work, and we have to find out how to manufacture them and do things of this kind, but the basic technology works. CA : You ’ ve got a company set up to... GW : A foundation, a foundation. Not-for- profit. CA : All right. Well, thank you so much for your talk. Thank you.

Text Sample 445: NEG Dim 3 – 2010 – Score -1.00 – 2010/t002743.txt

So, I ’ d like to spend a few minutes with you folks today imagining what our planet might look like in a thousand years. But before I do that, I need to talk to you about synthetic materials like plastics, which require huge amounts of energy to create and, because of their disposal issues, are slowly poisoning our planet. I also want to tell you and share with you how my team and I have been using mushrooms over the last three years. Not like that. We ’ re using mushrooms to create an entirely new class of materials, which perform a lot like plastics during their use, but are made from crop waste and are totally compostable at the end of their lives. But first, I need to talk to you about what I consider one of the most egregious offenders in the disposable plastics category. This is a material you all know is Styrofoam, but I like to think of it as toxic white stuff. In a single cubic foot of this material — about what would come around your computer or large television — you have the same energy content of about a liter and a half of petrol. Yet, after just a few weeks of use, you ’ ll throw this material in the trash. And this isn ’ t just found in packaging. 20 billion dollars of this material is produced every year, in everything from building materials to surfboards to coffee cups to table tops. And that ’ s not the only place it ’ s found. The EPA estimates, in the United States, by volume, this material occupies 25 percent of our landfills. Even worse is when it finds its way into our natural environment — on the side of the road or next to a river. If it ’ s not picked up by a human, like me and you, it ’ ll stay there for thousands and thousands of years. Perhaps even worse is when it finds its way into our oceans, like in the great plastic gyre, where these materials are being mechanically broken into smaller and smaller bits, but they ’ re not really going away. They ’ re not biologically compatible. They ’ re basically fouling up Earth ’ s respiratory and circulatory systems. And because these materials are so prolific, because they ’ re found in so many places, there ’ s one other place you ’ ll find this material, styrene, which is made from benzene, a known carcinogen. You ’ ll find it inside of you. So, for all these reasons, I think we need better materials, and there are three key principles we can use to guide these materials. The first is feedstocks. Today, we use a single feedstock, petroleum, to heat our homes, power our cars and make most of the materials you see around you. We recognize this is a finite resource, and it ’ s simply crazy to do this, to put a liter and a half of petrol in the trash every time you get a package. Second of all, we should really strive to use far less energy in creating these materials. I say far less, because 10 percent isn ’ t going to cut it. We should be talking about half, a quarter, one-tenth the energy content. And lastly, and I think perhaps most importantly, we should be creating materials that fit into what I call nature ’ s recycling system. This recycling system has been in place for the last billion years. I fit into it, you fit into it, and a hundred years tops, my body can return to the Earth with no preprocessing. Yet that packaging I got in the mail yesterday is going to last for thousands of years. This is crazy. But nature provides us with a really good model here. When a tree ’ s done using its leaves — its solar collectors, these amazing molecular photon capturing devices — at the end of a season, it doesn ’ t pack them up, take them to the leaf reprocessing center and have them melted down to form new leaves. It just drops them, the shortest distance possible, to the forest floor, where they ’ re actually upcycled into next year ’ s topsoil. And this gets us back to the mushrooms. Because in nature, mushrooms are the recycling system. And what we ’ ve discovered is, by using a part of the mushroom you ’ ve probably never seen — **analogous** to its root structure ; it ’ s called mycelium — we can actually grow materials with many of the same properties of conventional synthetics. Now, mycelium is an amazing material, because it ’ s a self-assembling material. It actually takes things we would consider waste — things like seed husks or woody biomass — and can transform them into a chitinous polymer, which you can form into almost any shape. In our process, we basically use it as a glue. And by using mycelium as a glue, you can mold things just like you do in the plastic industry, and you can create materials with many different properties, materials that are insulating, fire-resistant, moisture-resistant, vapor- resistant — materials that can absorb impacts, that can absorb acoustical impacts. But these materials are grown from agricultural byproducts, not petroleum. And because they ’ re made of natural materials, they are 100 percent compostable in you own backyard. So I ’ d like to share with you the four basic steps required to make these materials. The first is selecting a feedstock, preferably something that ’ s regional, that ’ s in your area, right — local manufacturing. The next is actually taking this feedstock and putting in a tool, physically filling an enclosure, a mold, in whatever shape you want to get. Then you actually grow the mycelium through these particles, and that ’ s where the magic happens, because the organism is doing the work in this process, not the equipment. The final step is, of course, the product, whether it ’ s a packaging material, a table top, or building block. Our vision is local manufacturing, like the local food movement, for production. So we ’ ve created formulations for all around the world using regional byproducts. If you ’ re in China, you might use a rice husk or a cottonseed hull. If you ’ re in Northern Europe or North America, you can use things like buckwheat husks or oat hulls. We then process these husks with some basic equipment. And I want to share with you a quick video from our facility that gives you a sense of how this looks at scale. So what you ’ re seeing here is actually cotton hulls from Texas, in this case. It ’ s a waste product. And what they ’ re doing in our equipment is going through a continuous system, which cleans, cooks, cools and pasteurizes these materials, while also continuously inoculating them with our mycelium. This gives us a continuous stream of material that we can put into almost any shape, though today we ’ re making corner blocks. And it ’ s when this lid goes on the part, that the magic really starts. Because the manufacturing process is our organism. It ’ ll actually begin to digest these wastes and, over the next five days, assemble them into biocomposites. Our entire facility is comprised of thousands and thousands and thousands of these tools sitting indoors in the dark, quietly self-assembling materials — and everything from building materials to, in this case, a packaging corner block. So I ’ ve said a number of times that we grow materials. And it ’ s kind of hard to picture how that happens. So my team has taken five days-worth of growth, a typical growth cycle for us, and condensed it into a 15-second time lapse. And I want you to really watch closely these little white dots on the screen, because, over the five-day period, what they do is extend out and through this material, using the energy that ’ s contained in these seed husks to build this chitinous polymer matrix. This matrix self-assembles, growing through and around the particles, making millions and millions of tiny fibers. And what parts of the seed husk we don ’ t digest, actually become part of the final, physical composite. So in front of your eyes, this part just self-assembled. It actually takes a little longer. It takes five days. But it ’ s much faster than conventional farming. The last step, of course, is application. In this case, we ’ ve grown a corner block. A major Fortune 500 furniture maker uses these corner blocks to protect their tables in shipment. They used to use a plastic packaging buffer, but we were able to give them the exact same physical performance with our grown material. Best of all, when it gets to the customer, it ’ s not trash. They can actually put this in their natural ecosystem without any processing, and it ’ s going to improve the local soil. So, why mycelium? The first reason is local open feedstocks. You want to be able to do this anywhere in the world and not worry about peak rice hull or peak cottonseed hulls, because you have multiple choices. The next is self- assembly, because the organism is actually doing most of the work in this process. You don ’ t need a lot of equipment to set up a production facility. So you can have lots of small facilities spread all across the world. Biological yield is really important. And because 100 percent of what we put in the tool become the final product, even the parts that aren ’ t digested become part of the structure, we ’ re getting incredible yield rates. Natural polymers, well... I think that ’ s what ’ s most important, because these polymers have been tried and tested in our ecosystem for the last billion years, in everything from mushrooms to crustaceans. They ’ re not going to clog up Earth ’ s ecosystems. They work great. And while, today, we can practically guarantee that yesterday ’ s packaging is going to be here in 10, 000 years, what I want to guarantee is that in 10, 000 years, our descendants, our children ’ s children, will be living happily and in harmony with a healthy Earth. And I think that can be some really good news. Thank you.

Text Sample 446: NEG Dim 3 – 2010 – Score -1.00 – 2010/t002666.txt

I admit that I ’ m a little bit nervous here because I ’ m going to say some radical things, about how we should think about cancer differently, to an audience that contains a lot of people who know a lot more about cancer than I do. But I will also contest that I ’ m not as nervous as I should be because I ’ m pretty sure I ’ m right about this. And that this, in fact, will be the way that we treat cancer in the future. In order to talk about cancer, I ' m going to actually have to — let me get the big slide here. First, I ' m going to try to give you a different perspective of genomics. I want to put it in perspective of the bigger picture of all the other things that are going on — and then talk about something you haven ' t heard so much about, which is proteomics. Having explained those, that will set up for what I think will be a different idea about how to go about treating cancer. So let me start with genomics. It is the hot topic. It is the place where we ' re learning the most. This is the great frontier. But it has its limitations. And in particular, you ' ve probably all heard the analogy that the genome is like the blueprint of your body, and if that were only true, it would be great, but it ' s not. It ' s like the parts list of your body. It doesn ' t say how things are connected, what causes what and so on. So if I can make an analogy, let ' s say that you were trying to tell the difference between a good restaurant, a healthy restaurant and a sick restaurant, and all you had was the list of ingredients that they had in their larder. So it might be that, if you went to a French restaurant and you looked through it and you found they only had margarine and they didn ' t have butter, you could say,"Ah, I see what ' s wrong with them. I can make them healthy."And there probably are special cases of that. You could certainly tell the difference between a Chinese restaurant and a French restaurant by what they had in a larder. So the list of ingredients does tell you something, and sometimes it tells you something that ' s wrong. If they have tons of salt, you might guess they ' re using too much salt, or something like that. But it ' s limited, because really to know if it ' s a healthy restaurant, you need to taste the food, you need to know what goes on in the kitchen, you need the product of all of those ingredients. So if I look at a person and I look at a person ' s genome, it ' s the same thing. The part of the genome that we can read is the list of ingredients. And so indeed, there are times when we can find ingredients that [are] bad. Cystic fibrosis is an example of a disease where you just have a bad ingredient and you have a disease, and we can actually make a direct correspondence between the ingredient and the disease. But most things, you really have to know what ' s going on in the kitchen, because, mostly, sick people used to be healthy people — they have the same genome. So the genome really tells you much more about predisposition. So what you can tell is you can tell the difference between an Asian person and a European person by looking at their ingredients list. But you really for the most part can ' t tell the difference between a healthy person and a sick person — except in some of these special cases. So why all the big deal about genetics? Well first of all, it ' s because we can read it, which is fantastic. It is very useful in certain circumstances. It ' s also the great theoretical triumph of biology. It ' s the one theory that the biologists ever really got right. It ' s fundamental to Darwin and Mendel and so on. And so it ' s the one thing where they predicted a theoretical construct. So Mendel had this idea of a gene as an abstract thing, and Darwin built a whole theory that depended on them existing, and then Watson and Crick actually looked and found one. So this happens in physics all the time. You predict a black hole, and you look out the telescope and there it is, just like you said. But it rarely happens in biology. So this great triumph — it ' s so good, there ' s almost a religious experience in biology. And Darwinian evolution is really the core theory. So the other reason it ' s been very popular is because we can measure it, it ' s digital. And in fact, thanks to Kary Mullis, you can basically measure your genome in your kitchen with a few extra ingredients. So for instance, by measuring the genome, we ' ve learned a lot about how we ' re related to other kinds of animals by the closeness of our genome, or how we ' re related to each other — the family tree, or the tree of life. There ' s a huge amount of information about the genetics just by comparing the genetic similarity. Now of course, in medical application, that is very useful because it ’ s the same kind of information that the doctor gets from your family medical history — except probably, your genome knows much more about your medical history than you do. And so by reading the genome, we can find out much more about your family than you probably know. And so we can discover things that probably you could have found by looking at enough of your relatives, but they may be surprising. I did the 23andMe thing and was very surprised to discover that I am fat and bald. But sometimes you can learn much more useful things about that. But mostly what you need to know, to find out if you ’ re sick, is not your predispositions, but it ’ s actually what ’ s going on in your body right now. So to do that, what you really need to do, you need to look at the things that the genes are producing and what ’ s happening after the genetics, and that ’ s what proteomics is about. Just like genome mixes the study of all the genes, proteomics is the study of all the proteins. And the proteins are all of the little things in your body that are signaling between the cells — actually, the machines that are operating — that ’ s where the action is. Basically, a human body is a conversation going on, both within the cells and between the cells, and they ’ re telling each other to grow and to die, and when you ’ re sick, something ’ s gone wrong with that conversation. And so the trick is — unfortunately, we don ’ t have an easy way to measure these like we can measure the genome. So the problem is that measuring — if you try to measure all the proteins, it ’ s a very elaborate process. It requires hundreds of steps, and it takes a long, long time. And it matters how much of the protein it is. It could be very significant that a protein changed by 10 percent, so it ’ s not a nice digital thing like DNA. And basically our problem is somebody ’ s in the middle of this very long stage, they pause for just a moment, and they leave something in an enzyme for a second, and all of a sudden all the measurements from then on don ’ t work. And so then people get very inconsistent results when they do it this way. People have tried very hard to do this. I tried this a couple of times and looked at this problem and gave up on it. I kept getting this call from this oncologist named David Agus. And Applied Minds gets a lot of calls from people who want help with their problems, and I didn ’ t think this was a very likely one to call back, so I kept on giving him to the delay list. And then one day, I get a call from John Doerr, Bill Berkman and Al Gore on the same day saying return David Agus ’ s phone call. So I was like,”Okay. This guy ’ s at least resourceful.”So we started talking, and he said,”I really need a better way to measure proteins.”I ’ m like,”Looked at that. Been there. Not going to be easy.”He ’ s like,”No, no. I really need it. I mean, I see patients dying every day because we don ’ t know what ’ s going on inside of them. We have to have a window into this.”And he took me through specific examples of when he really needed it. And I realized, wow, this would really make a big difference, if we could do it, and so I said,”Well, let ’ s look at it.”Applied Minds has enough play money that we can go and just work on something without getting anybody ’ s funding or permission or anything. So we started playing around with this. And as we did it, we realized this was the basic problem — that taking the sip of coffee — that there were humans doing this complicated process and that what really needed to be done was to automate this process like an assembly line and build robots that would measure proteomics. And so we did that, and working with David, we made a little company called Applied Proteomics eventually, which makes this robotic assembly line, which, in a very consistent way, measures the protein. And I ’ ll show you what that protein measurement looks like. Basically, what we do is we take a drop of blood out of a patient, and we sort out the proteins in the drop of blood according to how much they weigh, how slippery they are, and we arrange them in an image.

And so we can look at literally hundreds of thousands of features at once out of that drop of blood. And we can take a different one tomorrow, and you will see your proteins tomorrow will be different — they ' ll be different after you eat or after you sleep. They really tell us what ' s going on there. And so this picture, which looks like a big smudge to you, is actually the thing that got me really thrilled about this and made me feel like we were on the right track. So if I zoom into that picture, I can just show you what it means. We sort out the proteins — from left to right is the weight of the fragments that we ' re getting, and from top to bottom is how slippery they are. So we ' re zooming in here just to show you a little bit of it. And so each of these lines represents some signal that we ' re getting out of a piece of a protein. And you can see how the lines occur in these little groups of bump, bump, bump, bump, bump. And that ' s because we ' re measuring the weight so precisely that — carbon comes in different isotopes, so if it has an extra neutron on it, we actually measure it as a different chemical. So we ' re actually measuring each isotope as a different one. And so that gives you an idea of how exquisitely sensitive this is. So seeing this picture is sort of like getting to be Galileo and looking at the stars and looking through the telescope for the first time, and suddenly you say,"Wow, it ' s way more complicated than we thought it was."But we can see that stuff out there and actually see features of it. So this is the signature out of which we ' re trying to get patterns. So what we do with this is, for example, we can look at two patients, one that responded to a drug and one that didn ' t respond to a drug, and ask,"What ' s going on differently inside of them?"And so we can make these measurements precisely enough that we can overlay two patients and look at the differences. So here we have Alice in green and Bob in red. We overlay them. This is actual data. And you can see, mostly it overlaps and it ' s yellow, but there ' s some things that just Alice has and some things that just Bob has. And if we find a pattern of things of the responders to the drug, we see that in the blood, they have the condition that allows them to respond to this drug. We might not even know what this protein is, but we can see it ' s a marker for the response to the disease. So this already, I think, is tremendously useful in all kinds of medicine. But I think this is actually just the beginning of how we ' re going to treat cancer. So let me move to cancer. The thing about cancer — when I got into this, I really knew nothing about it, but working with David Agus, I started watching how cancer was actually being treated and went to operations where it was being cut out. And as I looked at it, to me it didn ' t make sense how we were approaching cancer, and in order to make sense of it, I had to learn where did this come from. We ' re treating cancer almost like it ' s an infectious disease. We ' re treating it as something that got inside of you that we have to kill. So this is the great paradigm. This is another case where a theoretical paradigm in biology really worked — was the germ theory of disease. So what doctors are mostly trained to do is diagnose — that is, put you into a category and apply a scientifically proven treatment for that diagnosis — and that works great for infectious diseases. So if we put you in the category of you ' ve got syphilis, we can give you **penicillin**. We know that that works. If you ' ve got malaria, we give you quinine or some derivative of it. And so that ' s the basic thing doctors are trained to do, and it ' s miraculous in the case of infectious disease — how well it works. And many people in this audience probably wouldn ' t be alive if doctors didn ' t do this. But now let ' s apply that to systems diseases like cancer. The problem is that, in cancer, there isn ' t something else that ' s inside of you. It ' s you ; you ' re broken. That conversation inside of you got mixed up in some way. So how do we diagnose that conversation? Well, right now what we do is we divide it by part of the body — you know, where did it appear? — and we put you in different categories according to the part of the body. And then we do a clinical trial for a drug for lung cancer and one for prostate cancer and one for breast cancer, and we treat these as if they ’ re separate diseases and that this way of dividing them had something to do with what actually went wrong. And of course, it really doesn ’ t have that much to do with what went wrong because cancer is a failure of the system. And in fact, I think we ’ re even wrong when we talk about cancer as a thing. I think this is the big mistake. I think cancer should not be a noun. We should talk about cancering as something we do, not something we have. And so those tumors, those are symptoms of cancer. And so your body is probably cancering all the time, but there are lots of systems in your body that keep it under control. And so to give you an idea of an analogy of what I mean by thinking of cancering as a verb, imagine we didn ’ t know anything about plumbing, and the way that we talked about it, we ’ d come home and we ’ d find a leak in our kitchen and we ’ d say,”Oh, my house has water.”We might divide it — the plumber would say,”Well, where ’ s the water?””Well, it ’ s in the kitchen.””Oh, you must have kitchen water.”That ’ s kind of the level at which it is.”Kitchen water, well, first of all, we ’ ll go in there and we ’ ll mop out a lot of it. And then we know that if we sprinkle Drano around the kitchen, that helps. Whereas living room water, it ’ s better to do tar on the roof.”And it sounds silly, but that ’ s basically what we do. And I ’ m not saying you shouldn ’ t mop up your water if you have cancer, but I ’ m saying that ’ s not really the problem ; that ’ s the symptom of the problem. What we really need to get at is the process that ’ s going on, and that ’ s happening at the level of the proteonomic actions, happening at the level of why is your body not healing itself in the way that it normally does? Because normally, your body is dealing with this problem all the time. So your house is dealing with leaks all the time, but it ’ s fixing them. It ’ s draining them out and so on. So what we need is to have a causative model of what ’ s actually going on, and proteomics actually gives us the ability to build a model like that. David got me invited to give a talk at National Cancer Institute and Anna Barker was there. And so I gave this talk and said,”Why don ’ t you guys do this?”And Anna said,”Because nobody within cancer would look at it this way. But what we ’ re going to do, is we ’ re going to create a program for people outside the field of cancer to get together with doctors who really know about cancer and work out different programs of research.”So David and I applied to this program and created a consortium at USC where we ’ ve got some of the best oncologists in the world and some of the best biologists in the world, from Cold Spring Harbor, Stanford, Austin — I won ’ t even go through and name all the places — to have a research project that will last for five years where we ’ re really going to try to build a model of cancer like this. We ’ re doing it in mice first, and we will kill a lot of mice in the process of doing this, but they will die for a good cause. And we will actually try to get to the point where we have a predictive model where we can understand, when cancer happens, what ’ s actually happening in there and which treatment will treat that cancer. So let me just end with giving you a little picture of what I think cancer treatment will be like in the future. So I think eventually, once we have one of these models for people, which we ’ ll get eventually — I mean, our group won ’ t get all the way there — but eventually we ’ ll have a very good computer model — sort of like a global climate model for weather. It has lots of different information about what ’ s the process going on in this proteomic conversation on many different scales. And so we will simulate in that model for your particular cancer — and this also will be for ALS, or any kind of system neurodegenerative diseases, things like that — we will simulate specifically you, not just a generic person, but what ’ s actually going on inside you. And in that simulation, what we could do is design for you specifically a sequence of treatments, and it might be very gentle treatments, very small amounts of drugs. It might be things like, don ’ t eat that day, or give them a little chemotherapy, maybe a little radiation. Of course, we ’ ll do surgery sometimes and so on. But design a program of treatments specifically for you and help your body guide back to health — guide your body back to health. Because your body will do most of the work of fixing it if we just sort of prop it up in the ways that are wrong. We put it in the equivalent of splints. And so your body basically has lots and lots of mechanisms for fixing cancer, and we just have to prop those up in the right way and get them to do the job. And so I believe that this will be the way that cancer will be treated in the future. It ’ s going to require a lot of work, a lot of research. There will be many teams like our team that work on this. But I think eventually, we will design for everybody a custom treatment for cancer. So thank you very much.

Text Sample 447: NEG Dim 3 – 2009 – Score -2.00 – 2009/t002907.txt

John Hockenberry : It ’ s great to be here with you, Tom. And I want to start with a question that has just been consuming me since I first became familiar with your work. In you work there ’ s always this kind of hybrid quality of a natural force in some sort of interplay with creative force. Are they ever in equilibrium in the way that you see your work? Tom Shannon : Yeah, the subject matter that I ’ m looking for, it ’ s usually to solve a question. I had the question popped into my head : What does the cone that connects the sun and the Earth look like if you could connect the two spheres? And in proportion, what would the size of the sphere and the length, and what would the taper be to the Earth? And so I went about and made that sculpture, turning it out of solid bronze. And I did one that was about 35 feet long. The sun end was about four inches in diameter, and then it tapered over about 35 feet to about a millimeter at the Earth end. And so for me, it was really exciting just to see what it looks like if you could step outside and into a larger context, as though you were an astronaut, and see these two things as an object, because they are so intimately bound, and one is meaningless without the other. JH : Is there a relief in playing with these forces? And I ’ m wondering how much of a sense of discovery there is in playing with these forces. TS : Well, like the magnetically levitated objects — like that silver one there, that was the result of hundreds of experiments with magnets, trying to find a way to make something float with the least possible connection to the ground. So I got it down to just one tether to be able to support that. JH : Now is this electromagnetic here, or are these static? TS : Those are permanent magnets, yeah. JH : Because if the power went out, there would just be a big noise. TS : Yeah. It ’ s really unsatisfactory having plug-in art. JH : I agree. TS : The magnetic works are a combination of gravity and magnetism, so it ’ s a kind of mixture of these ambient forces that influence everything. The sun has a tremendous field that extends way beyond the planets and the Earth ’ s magnetic field protects us from the sun. So there ’ s this huge invisible shape structures that magnetism takes in the universe. But with the pendulum, it allows me to manifest these invisible forces that are holding the magnets up. My sculptures are normally very simplified. I try to refine them down to very simple forms. But the paintings become very complex, because I think the fields that are supporting them, they ’ re billowing, and they ’ re interpenetrating, and they ’ re interference patterns. JH : And they ’ re non-deterministic. I mean, you don ’ t know necessarily where you ’ re headed when you begin, even though the forces can be calculated. So the evolution of this — I gather this isn ’ t your first pendulum. TS : No. TS : The first one I did was in the late 70 ’ s, and I just had a simple cone with a spigot at the bottom of it. I threw it into an orbit, and it only had one color, and when it got to the center, the paint kept running out, so I had to run in there, didn ’ t have any control over the spigot remotely. So that told me right away : I need a remote control device. But then I started dreaming of having six colors. I sort of think about it as the DNA — these colors, the red, blue, yellow, the primary colors and white and black. And if you put them together in different combinations — just like printing in a sense, like how a magazine color is printed — and put them under certain forces, which is orbiting them or passing them back and forth or drawing with them, these amazing things started appearing. JH : It looks like we ’ re loaded for bear here. TS : Yeah, well let ’ s put a couple of canvases. I ’ ll ask a couple of my sons to set up the canvases here. I want to just say — so this is Jack, Nick and Louie. JH : Thanks guys. TS : So here are the — JH : All right, I ’ ll get out of the way here. TS : I ’ m just going to throw this into an orbit and see if I can paint everybody ’ s shoes in the front. JH : Whoa. That is... ooh, nice. TS : So something like this. I ’ m doing this as a demo, and it ’ s more playful, but inevitably, all of this can be used. I can redeem this painting, just continuing on, doing layers upon layers. And I keep it around for a couple of weeks, and I ’ m contemplating it, and I ’ ll do another session with it and bring it up to another level, where all of this becomes the background, the depth of it. JH : That ’ s fantastic. So the valves at the bottom of those tubes there are like radio-controlled airplane valves. TS : Yes, they ’ re servos with cams that pinch these rubber tubes. And they can pinch them very tight and stop it, or you can have them wide open. And all of the colors come out one central port at the bottom. You can always be changing colors, put aluminum paint, or I could put anything into this. It could be tomato sauce, or anything could be dispensed — sand, powders or anything like that. JH : So many forces there. You ’ ve got gravity, you ’ ve got the centrifugal force, you ’ ve got the fluid dynamics. Each of these beautiful paintings, are they images in and of themselves, or are they records of a physical event called the pendulum approaching the canvas? TS : Well, this painting here, I wanted to do something very simple, a simple, iconic image of two ripples interfering. So the one on the right was done first, and then the one on the left was done over it. And then I left gaps so you could see the one that was done before. And then when I did the second one, it really disturbed the piece — these big blue lines crashing through the center of it — and so it created a kind of tension and an overlap. There are lines in front of the one on the right, and there are lines behind the one on the left, and so it takes it into different planes. What it ’ s also about, just the little events, the events of the interpenetration of — JH : Two stars, or — TS : Two things that happened — there ’ s an interference pattern, and then a third thing happens. There are shapes that come about just by the marriage of two events that are happening, and I ’ m very interested in that. Like the occurrence of moire patterns. Like this green one, this is a painting I did about 10 years ago, but it has some — see, in the upper third — there are these moires and interference patterns that are radio kind of imagery. And that ’ s something that in painting I ’ ve never seen done. I ’ ve never seen a representation of a kind of radio interference patterns, which are so ubiquitous and such an important part of our lives. JH : Is that a literal part of the image, or is my eye making that interference pattern — is my eye completing that interference pattern? TS : It is the paint actually, makes it real. It ’ s really manifested there. If I throw a very concentric circle, or concentric ellipse, it just dutifully makes these evenly spaced lines, which get closer and closer together, which describes how gravity works. There ’ s something very appealing about the exactitude of science that I really enjoy. And I love the shapes that I see in scientific observations and **apparatus**, especially astronomical forms and the idea of the vastness of it, the scale, is very interesting to me. My focus in recent years has kind of shifted more toward biology. Some of these paintings, when you look at them very close, odd things appear that really look like horses or birds or crocodiles, elephants. There are lots of things that appear. When you look into it, it ’ s sort of like looking at cloud patterns, but sometimes they ’ re very modeled and highly rendered. And then there are all these forms that we don ’ t know what they are, but they ’ re equally well-resolved and complex. So I think, conceivably, those could be predictive. Because since it has the ability to make forms that look like forms that we ’ re familiar with in biology, it ’ s also making other forms that we ’ re not familiar with. And maybe it ’ s the kind of forms we ’ ll discover underneath the surface of Mars, where there are probably lakes with fish swimming under the surface. JH : Oh, let ’ s hope so. Oh, my God, let ’ s. Oh, please, yes. Oh, I ’ m so there. You know, it seems at this stage in your life, you also very personally are in this state of confrontation with a sort of dissonant — I suppose it ’ s an electromagnetic force that somehow governs your Parkinson ’ s and this creative force that is both the artist who is in the here and now and this sort of arc of your whole life. Is that relevant to your work? TS : As it turns out, this device kind of comes in handy, because I don ’ t have to have the fine motor skills to do, that I can operate slides, which is more of a mental process. I ’ m looking at it and making decisions : It needs more red, it needs more blue, it needs a different shape. And so I make these creative decisions and can execute them in a much, much simpler way. I mean, I ’ ve got the symptoms. I guess Parkinson ’ s kind of creeps up over the years, but at a certain point you start seeing the symptoms. In my case, my left hand has a significant tremor and my left leg also. I ’ m left-handed, and so I draw. All my creations really start on small drawings, which I have thousands of, and it ’ s my way of just thinking. I draw with a simple pencil, and at first, the Parkinson ’ s was really upsetting, because I couldn ’ t get the pencil to stand still. JH : So you ’ re not a gatekeeper for these forces. You don ’ t think of yourself as the master of these forces. You think of yourself as the servant. TS : Nature is — well, it ’ s a godsend. It just has so much in it. And I think nature wants to express itself in the sense that we are nature, humans are of the universe. The universe is in our mind, and our minds are in the universe. And we are expressions of the universe, basically. As humans, ultimately being part of the universe, we ’ re kind of the spokespeople or the observer part of the constituency of the universe. And to interface with it, with a device that lets these forces that are everywhere act and show what they can do, giving them pigment and paint just like an artist, it ’ s a good ally. It ’ s a terrific studio assistant. JH : Well, I love the idea that somewhere within this idea of fine motion and control with the traditional skills that you have with your hand, some sort of more elemental force gets revealed, and that ’ s the beauty here. Tom, thank you so much. It ’ s been really, really great. TS : Thank you, John.

Text Sample 448: NEG Dim 3 – 2009 – Score -2.00 – 2009/t000131.txt

There ’ s a great big elephant in the room called the economy. So let ’ s start talking about that. I wanted to give you a current picture of the economy. That ’ s what I have behind myself. But of course what we have to remember is this. And what you have to think about is, when you ’ re dancing in the flames, what ’ s next? So what I ’ m going to try to do in the next 17 and a half minutes is I ’ m going to talk first about the flames — where we are in the economy — and then I ’ m going to take three trends that have taken place at TED over the last 25 years and that will take place in this conference and I will try and bring them together. And I will try and give you a sense of what the ultimate reboot looks like. Those three trends are the ability to engineer cells, the ability to engineer tissues, and robots. And somehow it will all make sense. But anyway, let ' s start with the economy. There ' s a couple of really big problems that are still sitting there. One is leverage. And the problem with leverage is it makes the U. S. financial system look like this. So, a normal commercial bank has nine to 10 times leverage. That means for every dollar you deposit, it loans out about nine or 10. A normal investment bank is not a deposit bank, it ' s an investment bank ; it has 15 to 20 times. It turns out that B of A in September had 32 times. And your friendly Citibank had 47 times. Oops. That means every bad loan goes bad 47 times over. And that, of course, is the reason why all of you are making such generous and wonderful donations to these nice folks. And as you think about that, you ' ve got to wonder : so what do banks have in store for you now? It ain ' t pretty. The government, meanwhile, has been acting like Santa Claus. We all love Santa Claus, right? But the problem with Santa Clause is, if you look at the mandatory spending of what these folks have been doing and promising folks, it turned out that in 1967, 38 percent was mandatory spending on what we call"entitlements."And then by 2007 it was 68 percent. And we weren ' t supposed to run into 100 percent until about 2030. Except we ' ve been so busy giving away a trillion here, a trillion there, that we ' ve brought that date of reckoning forward to about 2017. And we thought we were going to be able to lay these debts off on our kids, but, guess what? We ' re going to start to pay them. And the problem with this stuff is, now that the bill ' s come due, it turns out Santa isn ' t quite as cute when it ' s summertime. Right? Here ' s some advice from one of the largest investors in the United States. This guy runs the China Investment Corporation. He is the main buyer of U. S. Treasury bonds. And he gave an interview in December. Here ' s his first bit of advice. And here ' s his second bit of advice. And, by the way, the Chinese Prime Minister reiterated this at Davos last Sunday. This stuff is getting serious enough that if we don ' t start paying attention to the deficit, we ' re going to end up losing the dollar. And then all bets are off. Let me show you what it looks like. I think I can safely say that I ' m the only trillionaire in this room. This is an actual bill. And it ' s 10 triliion dollars. The only problem with this bill is it ' s not really worth very much. That was eight bucks last week, four bucks this week, a buck next week. And that ' s what happens to currencies when you don ' t stand behind them. So the next time somebody as cute as this shows up on your doorstep, and sometimes this creature ' s called Chrysler and sometimes Ford and sometimes... whatever you want — you ' ve just got to say no. And you ' ve got to start banishing a word that ' s called"entitlement."And the reason we have to do that in the short term is because we have just run out of cash. If you look at the federal budget, this is what it looks like. The orange slice is what ' s discretionary. Everything else is mandated. It makes no difference if we cut out the bridges to Alaska in the overall scheme of things. So what we have to start thinking about doing is capping our medical spending because that ' s a monster that ' s simply going to eat the entire budget. We ' ve got to start thinking about asking people to retire a little bit later. If you ' re 60 to 65 you retire on time. Your 401 just got nailed. If you ' re 50 to 60 we want you to work two years more. If you ' re under 50 we want you to work four more years. The reason why that ' s reasonable is, when your grandparents were given Social Security, they got it at 65 and were expected to check out at 68. Sixty-eight is young today. We ' ve also got to cut the military about three percent a year. We ' ve got to limit other mandatory spending. We ' ve got to quit borrowing as much, because otherwise the interest is going to eat that whole pie. And we

’ ve got to end up with a smaller government. And if we don ’ t start changing this trend line, we are going to lose the dollar and start to look like Iceland. I got what you ’ re thinking. This is going to happen when hell freezes over. But let me remind you this December it did snow in Vegas. Here ’ s what happens if you don ’ t address this stuff. So, Japan had a fiscal real estate crisis back in the late ’ 80s. And its 225 largest companies today are worth one quarter of what they were 18 years ago. We don ’ t fix this now, how would you like to see a Dow 3, 500 in 2026? Because that ’ s the consequence of not dealing with this stuff. And unless you want this person to not just become the CFO of Florida, but the United States, we ’ d better deal with this stuff. That ’ s the short term. That ’ s the flame part. That ’ s the financial crisis. Now, right behind the financial crisis there ’ s a second and bigger wave that we need to talk about. That wave is much larger, much more powerful, and that ’ s of course the wave of technology. And what ’ s really important in this stuff is, as we cut, we also have to grow. Among other things, because startup companies are. 02 percent of U. S. GDP investmentm and they ’ re about 17. 8 percent of output. It ’ s groups like that in this room that generate the future of the U. S. economy. And that ’ s what we ’ ve got to keep growing. We don ’ t have to keep growing these bridges to nowhere. So let ’ s bring a romance novelist into this conversation. And that ’ s where these three trends come together. That ’ s where the ability to engineer microbes, the ability to engineer tissues, and the ability to engineer robots begin to lead to a reboot. And let me recap some of the stuff you ’ ve seen. Craig Venter showed up last year and showed you the first fully programmable cell that acts like hardware where you can insert DNA and have it boot up as a different species. In parallel, the folks at MIT have been building a standard registry of biological parts. So think of it as a Radio Shack for biology. You can go out and get your proteins, your RNA, your DNA, whatever. And start building stuff. In 2006 they brought together high school students and college students and started to build these little odd creatures. They just happened to be alive instead of circuit boards. Here was one of the first things they built. So, cells have this cycle. First they don ’ t grow. Then they grow exponentially. Then they stop growing. Graduate students wanted a way of telling which stage they were in. So they engineered these cells so that when they ’ re growing in the exponential phase, they would smell like wintergreen. And when they stopped growing they would smell like bananas. And you could tell very easily when your experiment was working and wasn ’ t, and where it was in the phase. This got a bit more complicated two years later. Twenty-one countries came together. Dozens of teams. They started competing. The team from Rice University started to engineer the substance in red wine that makes red wine good for you into beer. So you take resveratrol and you put it into beer. Of course, one of the judges is wandering by, and he goes,”Wow! Cancer-fighting beer! There is a God.”The team from Taiwan was a little bit more ambitious. They tried to engineer bacterias in such a way that they would act as your kidneys. Four years ago, I showed you this picture. And people oohed and ahhed, because Cliff Tabin had been able to grow an extra wing on a chicken. And that was very cool stuff back then. But now moving from bacterial engineering to tissue engineering, let me show you what ’ s happened in that period of time. Two years ago, you saw this creature. An almost-extinct animal from Xochimilco, Mexico called an axolotl that can re-generate its limbs. You can freeze half its heart. It regrows. You can freeze half the brain. It regrows. It ’ s almost like leaving Congress. But now, you don ’ t have to have the animal itself to regenerate, because you can build cloned mice molars in Petri dishes. And, of course if you can build mice molars in Petri dishes, you can grow human molars in Petri dishes. This should not surprise you, right?

I mean, you ’ re born with no teeth. You give away all your teeth to the tooth fairy. You re- grow a set of teeth. But then if you lose one of those second set of teeth, they don ’ t regrow, unless, if you ’ re a lawyer. But, of course, for most of us, we know how to grow teeth, and therefore we can take adult stem teeth, put them on a biodegradable mold, re-grow a tooth, and simply implant it. And we can do it with other things. So, a Spanish woman who was dying of T. B. had a donor trachea, they took all the cells off the trachea, they spraypainted her stem cells onto that cartilage. She regrew her own trachea, and 72 hours later it was implanted. She ’ s now running around with her kids. This is going on in Tony Atala ’ s lab in Wake Forest where he is re- growing ears for injured soldiers, and he ’ s also re-growing bladders. So there are now nine women walking around Boston with re-grown bladders, which is much more pleasant than walking around with a whole bunch of plastic bags for the rest of your life. This is kind of getting boring, right? I mean, you understand where this story ’ s going. But, I mean it gets more interesting. Last year, this group was able to take all the cells off a heart, leaving just the cartilage. Then, they sprayed stem cells onto that heart, from a mouse. Those stem cells self- organized, and that heart started to beat. Life happens. This may be one of the ultimate papers. This was done in Japan and in the U. S., published at the same time, and it rebooted skin cells into stem cells, last year. That meant that you can take the stuff right here, and turn it into almost anything in your body. And this is becoming common, it ’ s moving very quickly, it ’ s moving in a whole series of places. Third trend : robots. Those of us of a certain age grew up expecting that by now we would have Rosie the Robot from”The Jetsons”in our house. And all we ’ ve **got** is a Roomba. We also thought we ’ d have this robot to warn us of danger. Didn ’ t happen. And these were robots engineered for a flat world, right? So, Rosie runs around on skates and the other one ran on flat threads. If you don ’ t have a flat world, that ’ s not good, which is why the robot ’ s we ’ re designing today are a little different. This is Boston Dynamics ’”BigDog.”And this is about as close as you can get to a physical Turing test. O. K., so let me remind you, a Turing test is where you ’ ve got a wall, you ’ re talking to somebody on the other side of the wall, and when you don ’ t know if that thing is human or animal — that ’ s when computers have reached human intelligence. This is not an intelligence Turing rest, but this is as close as you can get to a physical Turing test. And this stuff is moving very quickly, and by the way, that thing can carry about 350 pounds of weight. These are not the only interesting robots. You ’ ve also got flies, the size of flies, that are being made by Robert Wood at Harvard. You ’ ve got Stickybots that are being made at Stanford. And as you bring these things together, as you bring cells, biological tissue engineering and mechanics together, you begin to get some really odd questions. In the last Olympics, this gentleman, who had several world records in the Special Olympics, tried to run in the normal Olympics. The only issue with Oscar Pistorius is he was born without bones in the lower part of his legs. He came within about a second of qualifying. He sued to be allowed to run, and he won the suit, but didn ’ t qualify by time. Next Olympics, you can bet that Oscar, or one of Oscar ’ s successors, is going to make the time. And two or three Olympics after that, they are going to be unbeatable. And as you bring these trends together, and as you think of what it means to take people who are profoundly deaf, who can now begin to hear — I mean, remember the evolution of hearing aids, right? I mean, your grandparents had these great big cones, and then your parents had these odd boxes that would squawk at odd times during dinner, and now we have these little buds that nobody sees. And now you have cochlear implants that go into people ’ s heads and allow the deaf to begin to hear. Now, they can ’ t hear as well as you and I can. But, in 10 or 15 machine generations they will, and these are machine generations, not human generations. And about two or three years after they can hear as well as you and I can, they ’ ll be able to hear maybe how bats sing, or how whales talk, or how dogs talk, and other types of tonal scales. They ’ ll be able to focus their hearing, they ’ ll be able to increase the sensitivity, decrease the sensitivity, do a series of things that we can ’ t do. And the same thing is happening in eyes. This is a group in Germany that ’ s beginning to engineer eyes so that people who are blind can begin to see light and dark. Very primitive. And then they ’ ll be able to see shape. And then they ’ ll be able to see color, and then they ’ ll be able to see in definition, and one day, they ’ ll see as well as you and I can. And a couple of years after that, they ’ ll be able to see in ultraviolet, they ’ ll be able to see in infrared, they ’ ll be able to focus their eyes, they ’ ll be able to come into a microfocus. They ’ ll do stuff you and I can ’ t do. All of these things are coming together, and it ’ s a particularly important thing to understand, as we worry about the flames of the present, to keep an eye on the future. And, of course, the future is looking back 200 years, because next week is the 200th anniversary of Darwin ’ s birth. And it ’ s the 150th anniversary of the publication of”The Origin of Species.”And Darwin, of course, argued that evolution is a natural state. It is a natural state in everything that is alive, including hominids. There have actually been 22 species of hominids that have been around, have evolved, have wandered in different places, have gone extinct. It is common for hominids to evolve. And that ’ s the reason why, as you look at the hominid fossil record, erectus, and heidelbergensis, and floresiensis, and Neanderthals, and Homo sapiens, all overlap. The common state of affairs is to have overlapping versions of hominids, not one. And as you think of the implications of that, here ’ s a brief history of the universe. The universe was created 13. 7 billion years ago, and then you created all the stars, and all the planets, and all the galaxies, and all the Milky Ways. And then you created Earth about 4. 5 billion years ago, and then you got life about four billion years ago, and then you got hominids about 0. 006 billion years ago, and then you got our version of hominids about 0. 0015 billion years ago. Ta-dah! Maybe the reason for thr creation of the universe, and all the galaxies, and all the planets, and all the energy, and all the dark energy, and all the rest of stuff is to create what ’ s in this room. Maybe not. That would be a mildly arrogant viewpoint. So, if that ’ s not the purpose of the universe, then what ’ s next? I think what we ’ re going to see is we ’ re going to see a different species of hominid. I think we ’ re going to move from a Homo sapiens into a Homo evolutis. And I think this isn ’ t 1, 000 years out. I think most of us are going to glance at it, and our grandchildren are going to begin to live it. And a Homo evolutis brings together these three trends into a hominid that takes direct and deliberate control over the evolution of his species, her species and other species. And that, of course, would be the ultimate reboot. Thank you very much.

Text Sample 449: NEG Dim 3 – 2009 – Score -1.00 – 2009/t002993.txt

I have a studio in Berlin â let me cue on here â which is down there in this snow, just last weekend. In the studio we do a lot of experiments. I would consider the studio more like a laboratory. I have occasional meetings with scientists. And I have an academy, a part of the University of Fine Arts in Berlin. We have an annual gathering of people, and that is called Life in Space. Life in Space is really not necessarily about how we do things, but why we do things. Do you mind looking, with me, at that little cross in the center there? So just keep looking. Don ’ t mind me. So you will have a yellow circle, and we will do an after-image experiment. When the circle goes away you will have another color, the complementary color. I am saying something. And your eyes and your brain are saying something back. This whole idea of sharing, the idea of constituting reality by overlapping what I say and what you say â think of a movie. Since two years now, with some stipends from the science ministry in Berlin, I ’ ve been working on these films where we produce the film together. I don ’ t necessarily think the film is so interesting. Obviously this is not interesting at all in the sense of the narrative. But nevertheless, what the potential is â and just keep looking there â what the potential is, obviously, is to kind of move the border of who is the author, and who is the receiver. Who is the consumer, if you want, and who has responsibility for what one sees? I think there is a socializing dimension in, kind of, moving that border. Who decides what reality is? This is the Tate Modern in London. The show was, in a sense, about that. It was about a space in which I put half a semi-circular yellow disk. I also put a mirror in the ceiling, and some fog, some haze. And my idea was to make the space tangible. With such a big space, the problem is obviously that there is a discrepancy between what your body can embrace, and what the space, in that sense, is. So here I had the hope that by inserting some natural elements, if you want â some fog â I could make the space tangible. And what happens is that people, they start to see themselves in this space. So look at this. Look at the girl. Of course they have to look through a bloody camera in a museum. Right? That ’ s how museums are working today. But look at her face there, as she ’ s checking out, looking at herself in the mirror.”Oh! That was my foot there!”She wasn ’ t really sure whether she was seeing herself or not. And in that whole idea, how do we configure the relationship between our body and the space? How do we reconfigure it? How do we know that being in a space makes a difference? Do you see when I said in the beginning, it ’ s about why, rather than how? The why meant really,”What consequences does it have when I take a step?””What does it matter?””Does it matter if I am in the world or not?””And does it matter whether the kind of actions I take filter into a sense of responsibility?”Is art about that? I would say yes. It is obviously about not just about decorating the world, and making it look even better, or even worse, if you ask me. It ’ s obviously also about taking responsibility, like I did here when throwing some green dye in the river in L. A., Stockholm, Norway and Tokyo, among other places. The green dye is not environmentally dangerous, but it obviously looks really rather frightening. And it ’ s on the other side also, I think, quite beautiful, as it somehow shows the turbulence in these kind of downtown areas, in these different places of the world. The”Green river,”as a kind of activist idea, not a part of an exhibition, it was really about showing people, in this city, as they walk by, that space has dimensions. A space has time. And the water flows through the city with time. The water has an ability to make the city negotiable, tangible. Negotiable meaning that it makes a difference whether you do something or not. It makes a difference whether you say,”I ’ m a part of this city. And if I vote it makes a difference. If I take a stand, it makes a difference.”This whole idea of a city not being a picture is, I think, something that art, in a sense, always was working with. The idea that art can actually evaluate the relationship between what it means to be in a picture, and what it means to be in a space. What is the difference? The difference between thinking and doing. So these are different experiments with that. I won ’ t go into them. Iceland, lower right corner, my favorite place. These kinds of experiments, they filter into architectural models. These are ongoing experiments. One is an experiment I did for BMW, an attempt to make a car. It ’ s made out of ice. A **crystalline** stackable principle in the center on the top, which I am trying to turn into a concert hall in Iceland. A sort of a run track, or a walk track, on the top of a museum in Denmark, which is made of colored glass, going all around. So the movement with your legs will change the color of your horizon. And two summers ago at the Hyde Park in London, with the Serpentine Gallery : a kind of a temporal pavilion where moving was the only way you could see the pavilion. This summer, in New York : there is one thing about falling water which is very much about the time it takes for water to fall. It ’ s quite simple and fundamental. I ’ ve walked a lot in the mountains in Iceland. And as you come to a new valley, as you come to a new landscape, you have a certain view. If you stand still, the landscape doesn ’ t necessarily tell you how big it is. It doesn ’ t really tell you what you ’ re looking at. The moment you start to move, the mountain starts to move. The big mountains far away, they move less. The small mountains in the foreground, they move more. And if you stop again, you wonder,”Is that a one-hour valley? Or is that a three-hour hike, or is that a whole day I ’ m looking at?”If you have a waterfall in there, right out there at the horizon ; you look at the waterfall and you go,”Oh, the water is falling really slowly.”And you go,”My god it ’ s really far away and it ’ s a giant waterfall.”If a waterfall is falling faster, it ’ s a smaller waterfall which is closer by â because the speed of falling water is pretty constant everywhere. And your body somehow knows that. So this means a waterfall is a way of measuring space. Of course being an iconic city like New York, that has had an interest in somehow playing around with the sense of space, you could say that New York wants to seem as big as possible. Adding a measurement to that is interesting : the falling water suddenly gives you a sense of,”Oh, Brooklyn is exactly this much â the distance between Brooklyn and Manhattan, in this case the lower East River is this big.”So it was not just necessarily about putting nature into the cities. It was also about giving the city a sense of dimension. And why would we want to do that? Because I think it makes a difference whether you have a body that feels a part of a space, rather than having a body which is just in front of a picture. And”Ha-ha, there is a picture and here is I. And what does it matter?”Is there a sense of consequences? So if I have a sense of the space, if I feel that the space is tangible, if I feel there is time, if there is a dimension I could call time, I also feel that I can change the space. And suddenly it makes a difference in terms of making space accessible. One could say this is about community, collectivity. It ’ s about being together. How do we create public space? What does the word”public”mean today anyway? So, asked in that way, I think it raises great things about parliamentary ideas, democracy, public space, being together, being individual. How do we create an idea which is both tolerant to individuality, and also to collectivity, without polarizing the two into two different opposites? Of course the political agendas in the world has been very obsessed, polarizing the two against each other into different, very normative ideas. I would claim that art and culture, and this is why art and culture are so incredibly interesting in the times we ’ re living in now, have proven that one can create a kind of a space which is both sensitive to individuality and to collectivity. It ’ s very much about this causality, consequences. It ’ s very much about the way we link thinking and doing. So what is between thinking and doing? And right in-between thinking and doing, I would say, there is experience. And experience is not just a kind of entertainment in a non-casual way. Experience is about responsibility. Having an experience is taking part in the world. Taking part in the world is really about sharing responsibility. So art, in that sense, I think holds an incredible relevance in the world in which we ’ re moving into, particularly right now. That ’ s all I have. Thank you very much.

Text Sample 450: NEG Dim 3 – 2013 – Score -2.00 – 2013/t002101.txt

I ’ m going to talk about the strategizing brain. We ’ re going to use an unusual combination of tools from game theory and neuroscience to understand how people interact socially when value is on the line. So game theory is a branch of, originally, applied mathematics, used mostly in economics and political science, a little bit in biology, that gives us a mathematical taxonomy of social life and it predicts what people are likely to do and believe others will do in cases where everyone ’ s actions affect everyone else. That ’ s a lot of things : competition, cooperation, bargaining, games like hide-and-seek, and poker. Here ’ s a simple game to get us started. Everyone chooses a number from zero to 100, we ’ re going to compute the average of those numbers, and whoever ’ s closest to two-thirds of the average wins a fixed prize. So you want to be a little bit below the average number, but not too far below, and everyone else wants to be a little bit below the average number as well. Think about what you might pick. As you ’ re thinking, this is a toy model of something like selling in the stock market during a rising market. Right? You don ’ t want to sell too early, because you miss out on profits, but you don ’ t want to wait too late to when everyone else sells, triggering a crash. You want to be a little bit ahead of the competition, but not too far ahead. Okay, here ’ s two theories about how people might think about this, and then we ’ ll see some data. Some of these will sound familiar because you probably are thinking that way. I ’ m using my brain theory to see. A lot of people say,”I really don ’ t know what people are going to pick, so I think the average will be 50.”They ’ re not being really strategic at all.”And I ’ ll pick two-thirds of 50. That ’ s 33.”That ’ s a start. Other people who are a little more sophisticated, using more working memory, say,”I think people will pick 33 because they ’ re going to pick a response to 50, and so I ’ ll pick 22, which is two-thirds of 33.”They ’ re doing one extra step of thinking, two steps. That ’ s better. And of course, in principle, you could do three, four or more, but it starts to get very difficult. Just like in language and other domains, we know that it ’ s hard for people to parse very complex sentences with a kind of recursive structure. This is called a cognitive hierarchy theory, by the way. It ’ s something that I ’ ve worked on and a few other people, and it indicates a kind of hierarchy along with some assumptions about how many people stop at different steps and how the steps of thinking are affected by lots of interesting variables and variant people, as we ’ ll see in a minute. A very different theory, a much more popular one, and an older one, due largely to John Nash of”A Beautiful Mind”fame, is what ’ s called equilibrium analysis. So if you ’ ve ever taken a game theory course at any level, you will have learned a little bit about this. An equilibrium is a mathematical state in which everybody has figured out exactly what everyone else will do. It is a very useful concept, but behaviorally, it may not exactly explain what people do the first time they play these types of economic games or in situations in the outside world. In this case, the equilibrium makes a very bold prediction, which is everyone wants to be below everyone else, therefore they ’ ll play zero. Let ’ s see what happens. This experiment ’ s been done many, many times. Some of the earliest ones were done in the ’ 90s by me and Rosemarie Nagel and others. This is a beautiful data set of 9, 000 people who wrote in to three newspapers and magazines that had a contest. The contest said, send in your numbers and whoever is close to two-thirds of the average will win a big prize. And as you can see, there ’ s so much data here, you can see the spikes very visibly. There ’ s a spike at 33. Those are people doing one step. There is another spike visible at 22. And notice, by the way, that most people pick numbers right around there. They don ’ t necessarily pick exactly 33 and 22. There ’ s something a little bit noisy around it. But you can see those spikes, and they ’ re there. There ’ s another group of people who seem to have a firm grip on equilibrium analysis, because they ' re picking zero or one. But they lose, right? Because picking a number that low is actually a bad choice if other people aren ' t doing equilibrium analysis as well. So they ' re smart, but poor. Where are these things happening in the brain? One study by Coricelli and Nagel gives a really sharp, interesting answer. So they had people play this game while they were being scanned in an fMRI, and two conditions : in some trials, they ' re told you ' re playing another person who ' s playing right now and we ' re going to match up your behavior at the end and pay you if you win. In the other trials, they ' re told, you ' re playing a computer. They ' re just choosing randomly. So what you see here is a subtraction of areas in which there ' s more brain activity when you ' re playing people compared to playing the computer. And you see activity in some regions we ' ve seen today, medial prefrontal cortex, dorsomedial, however, up here, ventromedial prefrontal cortex, anterior cingulate, an area that ' s involved in lots of types of conflict resolution, like if you ' re playing"Simon Says,"and also the right and left temporoparietal junction. And these are all areas which are fairly reliably known to be part of what ' s called a"theory of mind"circuit, or"mentalizing circuit."That is, it ' s a circuit that ' s used to imagine what other people might do. So these were some of the first studies to see this tied in to game theory. What happens with these one- and two-step types? So we classify people by what they picked, and then we look at the difference between playing humans versus playing computers, which brain areas are differentially active. On the top you see the one-step players. There ' s almost no difference. The reason is, they ' re treating other people like a computer, and the brain is too. The bottom players, you see all the activity in dorsomedial PFC. So we know that those two-step players are doing something differently. Now if you were to step back and say,"What can we do with this information?"you might be able to look at brain activity and say,"This person ' s going to be a good poker player,"or,"This person ' s socially naive,"and we might also be able to study things like development of adolescent brains once we have an idea of where this circuitry exists. Okay. Get ready. I ' m saving you some brain activity, because you don ' t need to use your hair detector cells. You should use those cells to think carefully about this game. This is a bargaining game. Two players who are being scanned using EEG electrodes are going to bargain over one to six dollars. If they can do it in 10 seconds, they ' re going to actually earn that money. If 10 seconds goes by and they haven ' t made a deal, they get nothing. That ' s kind of a mistake together. The twist is that one player, on the left, is informed about how much on each trial there is. They play lots of trials with different amounts each time. In this case, they know there ' s four dollars. The uninformed player doesn ' t know, but they know that the informed player knows. So the uninformed player ' s challenge is to say,"Is this guy really being fair or are they giving me a very low offer in order to get me to think that there ' s only one or two dollars available to split?"in which case they might reject it and not come to a deal. So there ' s some tension here between trying to get the most money but trying to goad the other player into giving you more. And the way they bargain is to point on a number line that goes from zero to six dollars, and they ' re bargaining over how much the uninformed player gets, and the informed player ' s going to get the rest. So this is like a management-labor negotiation in which the workers don ' t know how much profits the privately held company has, right, and they want to maybe hold out for more money, but the company might want to create the impression that there ' s very little to split :"I ' m giving you the most that I can."First some behavior. So a bunch of the subject pairs, they play face to face. We have some other data where they play across computers. That ' s an interesting difference, as you might imagine. But a bunch of the face-to-face pairs agree to divide the money evenly every single time. Boring. It ' s just not interesting neurally. It ' s good for them. They make a lot of money. But we ' re interested in, can we say something about when disagreements occur versus don ' t occur? So this is the other group of subjects who often disagree. So they have a chance of — they bicker and disagree and end up with less money. They might be eligible to be on"Real Housewives,"the TV show. You see on the left, when the amount to divide is one, two or three dollars, they disagree about half the time, and when the amount is four, five, six, they agree quite often. This turns out to be something that ' s predicted by a very complicated type of game theory you should come to graduate school at CalTech and learn about. It ' s a little too complicated to explain right now, but the theory tells you that this shape kind of should occur. Your intuition might tell you that too. Now I ' m going to show you the results from the EEG recording. Very complicated. The right brain **schematic** is the uninformed person, and the left is the informed. Remember that we scanned both brains at the same time, so we can ask about time-synced activity in similar or different areas simultaneously, just like if you wanted to study a conversation and you were scanning two people talking to each other and you ' d expect common activity in language regions when they ' re actually kind of listening and communicating. So the arrows connect regions that are active at the same time, and the direction of the arrows flows from the region that ' s active first in time, and the arrowhead goes to the region that ' s active later. So in this case, if you look carefully, most of the arrows flow from right to left. That is, it looks as if the uninformed brain activity is happening first, and then it ' s followed by activity in the informed brain. And by the way, these were trials where their deals were made. This is from the first two seconds. We haven ' t finished analyzing this data, so we ' re still peeking in, but the hope is that we can say something in the first couple of seconds about whether they ' ll make a deal or not, which could be very useful in thinking about avoiding litigation and ugly divorces and things like that. Those are all cases in which a lot of value is lost by delay and strikes. Here ' s the case where the disagreements occur. You can see it looks different than the one before. There ' s a lot more arrows. That means that the brains are synced up more closely in terms of simultaneous activity, and the arrows flow clearly from left to right. That is, the informed brain seems to be deciding,"We ' re probably not going to make a deal here."And then later there ' s activity in the uninformed brain. Next I ' m going to introduce you to some relatives. They ' re hairy, smelly, fast and strong. You might be thinking back to your last Thanksgiving. Maybe if you had a chimpanzee with you. Charles Darwin and I and you broke off from the family tree from chimpanzees about five million years ago. They ' re still our closest genetic kin. We share 98. 8 percent of the genes. We share more genes with them than zebras do with horses. And we ' re also their closest cousin. They have more genetic relation to us than to gorillas. So how humans and chimpanzees behave differently might tell us a lot about brain evolution. So this is an amazing memory test from Nagoya, Japan, Primate Research Institute, where they ' ve done a lot of this research. This goes back quite a ways. They ' re interested in working memory. The chimp is going to see, watch carefully, they ' re going to see 200 **milliseconds** ' exposure — that ' s fast, that ' s eight movie frames — of numbers one, two, three, four, five. Then they disappear and they ' re replaced by squares, and they have to press the squares that correspond to the numbers from low to high to get an apple reward. Let ' s see how they can do it. This is a young chimp. The young ones are better than the old ones, just like humans. And they ' re highly experienced, so they ' ve done this thousands and thousands of time. Obviously there ' s a big training effect, as you can imagine. You can see they ' re very blas and kind of effortless. Not only can they do it very well, they do it in a sort of lazy way. Right? Who thinks you could beat the chimps? Wrong. We can try. We ’ ll try. Maybe we ’ ll try. Okay, so the next part of this study I ’ m going to go quickly through is based on an idea of Tetsuro Matsuzawa. He had a bold idea that — what he called the cognitive trade-off hypothesis. We know chimps are faster and stronger. They ’ re also very obsessed with status. His thought was, maybe they ’ ve preserved brain activities and they practice them in development that are really, really important to them to negotiate status and to win, which is something like strategic thinking during competition. So we ’ re going to check that out by having the chimps actually play a game by touching two touch screens. The chimps are actually interacting with each other through the computers. They ’ re going to press left or right. One chimp is called a matcher. They win if they press left, left, like a seeker finding someone in hide-and-seek, or right, right. The mismatcher wants to mismatch. They want to press the opposite screen of the chimp. And the rewards are apple cube rewards. So here ’ s how game theorists look at these data. This is a graph of the percentage of times the matcher picked right on the x-axis, and the percentage of times they predicted right by the mismatcher on the y-axis. So a point here is the behavior by a pair of players, one trying to match, one trying to mismatch. The NE square in the middle — actually NE, CH and QRE — those are three different theories of Nash equilibrium, and others, tells you what the theory predicts, which is that they should match 50-50, because if you play left too much, for example, I can exploit that if I ’ m the mismatcher by then playing right. And as you can see, the chimps, each chimp is one triangle, are circled around, hovering around that prediction. Now we move the payoffs. We ’ re actually going to make the left, left payoff for the matcher a little bit higher. Now they get three apple cubes. Game theoretically, that should actually make the mismatcher ’ s behavior shift, because what happens is, the mismatcher will think, oh, this guy ’ s going to go for the big reward, and so I ’ m going to go to the right, make sure he doesn ’ t get it. And as you can see, their behavior moves up in the direction of this change in the Nash equilibrium. Finally, we changed the payoffs one more time. Now it ’ s four apple cubes, and their behavior again moves towards the Nash equilibrium. It ’ s sprinkled around, but if you average the chimps out, they ’ re really, really close, within. 01. They ’ re actually closer than any species we ’ ve observed. What about humans? You think you ’ re smarter than a chimpanzee? Here ’ s two human groups in green and blue. They ’ re closer to 50-50. They ’ re not responding to payoffs as closely, and also if you study their learning in the game, they aren ’ t as sensitive to previous rewards. The chimps are playing better than the humans, better in the sense of adhering to game theory. And these are two different groups of humans from Japan and Africa. They replicate quite nicely. None of them are close to where the chimps are. So here are some things we learned today. People seem to do a limited amount of strategic thinking using theory of mind. We have some preliminary evidence from bargaining that early warning signs in the brain might be used to predict whether there will be a bad disagreement that costs money, and chimps are better competitors than humans, as judged by game theory. Thank you.

Text Sample 451: NEG Dim 3 – 2013 – Score -1.00 – 2013/t002010.txt

Have you ever experienced a moment in your life that was so painful and confusing, that all you wanted to do was learn as much as you could to make sense of it all? When I was 13, a close family friend who was like an uncle to me passed away from pancreatic cancer. When the disease hit so close to home, I knew I needed to learn more. So I went online to find answers. Using the Internet, I found a variety of statistics on pancreatic cancer, and what I had found shocked me. Over 85 percent of all pancreatic cancers are diagnosed late, when someone has less than a two percent chance of survival. Why are we so bad at detecting pancreatic cancer? The reason? Today ' s current"modern"medicine is a 60-year-old technique. That ' s older than my dad. But also, it ' s extremely expensive, costing 800 dollars per test, and it ' s grossly inaccurate, missing 30 percent of all pancreatic cancers. Your doctor would have to be ridiculously suspicious that you have the cancer in order to give you this test. Learning this, I knew there had to be a better way. So, I set up scientific criteria as to what a sensor would have to look like in order to effectively diagnose pancreatic cancer. The sensor would have to be : inexpensive, rapid, simple, sensitive, selective, and minimally invasive. Now, there ' s a reason why this test hasn ' t been updated in over six decades. And that ' s because when we ' re looking for pancreatic cancer, we ' re looking at your bloodstream, which is already abundant in all these tons and tons of protein, and you ' re looking for this miniscule difference in this tiny amount of protein. Just this one protein. That ' s next to impossible. However, undeterred due to my teenage optimism — I went online to a teenager ' s two best friends, Google and Wikipedia. I got everything for my homework from those two sources. And what I had found was an article that listed a database of over 8, 000 different proteins that are found when you have pancreatic cancer. So, I decided to go and make it my new mission to go through all these proteins, and see which ones could serve as a bio-marker for pancreatic cancer. And to make it a bit simpler for myself, I decided to map out scientific criteria, and here it is. Essentially, first, the protein would have to be found in all pancreatic cancers, at high levels in the bloodstream, in the earliest stages, but also only in cancer. And so I ' m just plugging and chugging through this gargantuan task, and finally, on the 4, 000th try, when I ' m close to losing my sanity, I find the protein. And the name of the protein I ' d located was called mesothelin, and it ' s just your ordinary, run-of-the-mill type protein, unless, of course, you have pancreatic, ovarian or lung cancer, in which case it ' s found at these very high levels in your bloodstream. But also, the key is that it ' s found in the earliest stages of the disease, when someone has close to 100 percent chance of survival. So now that I ' d found a reliable protein I could detect, I then shifted my focus to actually detecting that protein, and thus, pancreatic cancer. Now, my breakthrough came in a very unlikely place, possibly the most unlikely place for innovation — my high school biology class, the absolute stifler of innovation. And I had snuck in this article on these things called carbon **nanotubes**, and that ' s just a long, thin pipe of carbon that ' s an atom thick, and one 50, 000th the diameter of your hair. And despite their extremely small sizes, they have these incredible properties. They ' re kind of like the superheroes of material science. And while I was sneakily reading this article under my desk in my biology class, we were supposed to be paying attention to these other kind of cool molecules, called antibodies. And these are pretty cool because they only react with one specific protein, but they ' re not nearly as interesting as carbon **nanotubes**. And so then, I was sitting in class, and suddenly it hit me : I could combine what I was reading about, carbon **nanotubes**, with what I was supposed to be thinking about, antibodies. Essentially, I could weave a bunch of these antibodies into a network of carbon **nanotubes**, such that you have a network that only reacts with one protein, but also, due to the properties of these **nanotubes**, it will change its electrical properties, based on the amount of protein present. However, there ' s a catch. These networks of carbon **nanotubes** are extremely flimsy. And since they ' re so delicate, they need to be supported. So that ' s why I chose to use paper. Making a cancer sensor out of paper is about as simple as making chocolate chip cookies, which I love. You start with some water, pour in some **nanotubes**, add antibodies, mix it up, take some paper, dip it, dry it, and you can detect cancer. Then, suddenly, a thought occurred that kind of put a blemish on my amazing plan here. I can ' t really do cancer research on my kitchen countertop. My mom wouldn ' t really like that. So instead, I decided to go for a lab. So I typed up a budget, a materials list, a timeline, and a procedure, and I emailed it to 200 different professors at Johns Hopkins University and the National Institutes of Health — essentially, anyone that had anything to do with pancreatic cancer. I sat back waiting for these positive emails to be pouring in, saying,"You ' re a genius! You ' re going to save us all!"And — Then reality took hold, and over the course of a month, I got 199 rejections out of those 200 emails. One professor even went through my entire procedure, painstakingly — I ' m not really sure where he got all this time — and he went through and said why each and every step was like the worst mistake I could ever make. Clearly, the professors did not have as high of an opinion of my work as I did. However, there is a silver lining. One professor said,"Maybe I might be able to help you, kid."So, I went in that direction. As you can never say no to a kid. And so then, three months later, I finally nailed down a harsh deadline with this guy, and I get into his lab, I get all excited, and then I sit down, I start opening my mouth and talking, and five seconds later, he calls in another Ph. D. Ph. D. s just flock into this little room, and they ' re just firing these questions at me, and by the end, I kind of felt like I was in a clown car. There were 20 Ph. D. s, plus me and the professor crammed into this tiny office space, with them firing these rapid-fire questions at me, trying to sink my procedure. How unlikely is that? I mean, pshhh. However, subjecting myself to that interrogation — I answered all their questions, and I guessed on quite a few but I got them right — and I finally landed the lab space I needed. But it was shortly afterwards that I discovered my once brilliant procedure had something like a million holes in it, and over the course of seven months, I painstakingly filled each and every one of those holes. The result? One small paper sensor that costs three cents and takes five minutes to run. This makes it 168 times faster, over 26, 000 times less expensive, and over 400 times more sensitive than our current standard for pancreatic cancer detection. One of the best parts of the sensor, though, is that it has close to 100 percent accuracy, and can detect the cancer in the earliest stages, when someone has close to 100 percent chance of survival. And so in the next two to five years, this sensor could potentially lift the pancreatic cancer survival rates from a dismal 5. 5 percent to close to 100 percent, and it would do similar for ovarian and lung cancer. But it wouldn ' t stop there. By switching out that antibody, you can look at a different protein, thus, a different disease — potentially any disease in the entire world. So that ranges from heart disease, to malaria, HIV, AIDS, as well as other forms of cancer — anything. And so, hopefully one day, we can all have that one extra uncle, that one mother, that one brother, sister, we can have that one more family member to love. And that our hearts will be rid of that one disease burden that comes from pancreatic, ovarian and lung cancer, and potentially any disease. But through the Internet, anything is possible. Theories can be shared, and you don ' t have to be a professor with multiple degrees to have your ideas valued. It ' s a neutral space, where what you look like, age or gender — it doesn ' t matter. It ' s just your ideas that count. For me, it ' s all about looking at the Internet in an entirely new way, to realize that there ' s so much more to it than just posting duck-face pictures of yourself online. You could be changing the world. So if a 15 year- old who didn ' t even know what a pancreas was could find a new way to detect pancreatic cancer — just imagine what you could do. Thank you.

Text Sample 452: NEG Dim 3 – 2013 – Score -1.00 – 2013/t001981.txt
Eric Berlow : I ’ m an ecologist, and Sean ’ s a physicist, and we both study complex networks. And we met a couple years ago when we discovered that we had both given a short TED Talk about the ecology of war, and we realized that we were connected by the ideas we shared before we ever met. And then we thought, you know, there are thousands of other talks out there, especially TEDx Talks, that are popping up all over the world. How are they connected, and what does that global conversation look like? So Sean ’ s going to tell you a little bit about how we did that. Sean Gourley : Exactly. So we took 24, 000 TEDx Talks from around the world, 147 different countries, and we took these talks and we wanted to find the mathematical structures that underly the ideas behind them. And we wanted to do that so we could see how they connected with each other. And so, of course, if you ’ re going to do this kind of stuff, you need a lot of data. So the data that you ’ ve **got** is a great thing called YouTube, and we can go down and basically pull all the open information from YouTube, all the comments, all the views, who ’ s watching it, where are they watching it, what are they saying in the comments. But we can also pull up, using speech-to-text translation, we can pull the entire transcript, and that works even for people with kind of funny accents like myself. So we can take their transcript and actually do some pretty cool things. We can take natural language processing algorithms to kind of read through with a computer, line by line, extracting key concepts from this. And we take those key concepts and they sort of form this mathematical structure of an idea. And we call that the meme-ome. And the meme-ome, you know, quite simply, is the mathematics that underlies an idea, and we can do some pretty interesting analysis with it, which I want to share with you now. So each idea has its own meme-ome, and each idea is unique with that, but of course, ideas, they borrow from each other, they kind of steal sometimes, and they certainly build on each other, and we can go through mathematically and take the meme-ome from one talk and compare it to the meme-ome from every other talk, and if there ’ s a similarity between the two of them, we can create a link and represent that as a graph, just like Eric and I are connected. So that ’ s theory, that ’ s great. Let ’ s see how it works in actual practice. So what we ’ ve got here now is the global footprint of all the TEDx Talks over the last four years exploding out around the world from New York all the way down to little old New Zealand in the corner. And what we did on this is we analyzed the top 25 percent of these, and we started to see where the connections occurred, where they connected with each other. Cameron Russell talking about image and beauty connected over into Europe. We ’ ve got a bigger conversation about Israel and Palestine radiating outwards from the Middle East. And we ’ ve got something a little broader like big data with a truly global footprint reminiscent of a conversation that is happening everywhere. So from this, we kind of run up against the limits of what we can actually do with a geographic projection, but luckily, computer technology allows us to go out into multidimensional space. So we can take in our network projection and apply a physics engine to this, and the similar talks kind of smash together, and the different ones fly apart, and what we ’ re left with is something quite beautiful. EB : So I want to just point out here that every node is a talk, they ’ re linked if they share similar ideas, and that comes from a machine reading of entire talk transcripts, and then all these topics that pop out, they ’ re not from tags and keywords. They come from

the network structure of interconnected ideas. Keep going. SG : Absolutely. So I got a little quick on that, but he ' s going to slow me down. We ' ve got education connected to storytelling triangulated next to social media. You ' ve got, of course, the human brain right next to healthcare, which you might expect, but also you ' ve got video games, which is sort of adjacent, as those two spaces interface with each other. But I want to take you into one cluster that ' s particularly important to me, and that ' s the environment. And I want to kind of zoom in on that and see if we can get a little more resolution. So as we go in here, what we start to see, apply the physics engine again, we see what ' s one conversation is actually composed of many smaller ones. The structure starts to emerge where we see a kind of fractal behavior of the words and the language that we use to describe the things that are important to us all around this world. So you ' ve got food economy and local food at the top, you ' ve got greenhouse gases, solar and nuclear waste. What you ' re getting is a range of smaller conversations, each connected to each other through the ideas and the language they share, creating a broader concept of the environment. And of course, from here, we can go and zoom in and see, well, what are young people looking at? And they ' re looking at energy technology and nuclear fusion. This is their kind of resonance for the conversation around the environment. If we split along gender lines, we can see females resonating heavily with food economy, but also out there in hope and optimism. And so there ' s a lot of exciting stuff we can do here, and I ' ll throw to Eric for the next part. EB : Yeah, I mean, just to point out here, you cannot get this kind of perspective from a simple tag search on YouTube. Let ' s now zoom back out to the entire global conversation out of environment, and look at all the talks together. Now often, when we ' re faced with this amount of content, we do a couple of things to simplify it. We might just say, well, what are the most popular talks out there? And a few rise to the surface. There ' s a talk about gratitude. There ' s another one about personal health and nutrition. And of course, there ' s got to be one about porn, right? And so then we might say, well, gratitude, that was last year. What ' s trending now? What ' s the popular talk now? And we can see that the new, emerging, top trending topic is about digital privacy. So this is great. It simplifies things. But there ' s so much creative content that ' s just buried at the bottom. And I hate that. How do we bubble stuff up to the surface that ' s maybe really creative and interesting? Well, we can go back to the network structure of ideas to do that. Remember, it ' s that network structure that is creating these emergent topics, and let ' s say we could take two of them, like cities and genetics, and say, well, are there any talks that creatively bridge these two really different disciplines. And that ' s — Essentially, this kind of creative remix is one of the hallmarks of innovation. Well here ' s one by Jessica Green about the microbial ecology of buildings. It ' s literally defining a new field. And we could go back to those topics and say, well, what talks are central to those conversations? In the cities cluster, one of the most central was one by Mitch Joachim about ecological cities, and in the genetics cluster, we have a talk about synthetic biology by Craig Venter. These are talks that are linking many talks within their discipline. We could go the other direction and say, well, what are talks that are broadly synthesizing a lot of different kinds of fields. We used a measure of ecological diversity to get this. Like, a talk by Steven Pinker on the history of violence, very synthetic. And then, of course, there are talks that are so unique they ' re kind of out in the stratosphere, in their own special place, and we call that the Colleen Flanagan index. And if you don ' t know Colleen, she ' s an artist, and I asked her,"Well, what ' s it like out there in the stratosphere of our idea space?"And apparently it smells like bacon. I wouldn ' t know. So we ' re using these network motifs to find talks that are unique, ones that are creatively synthesizing a lot of different fields, ones that are central to their topic, and ones that are really creatively bridging disparate fields. Okay? We never would have found those with our obsession with what ’ s trending now. And all of this comes from the architecture of complexity, or the patterns of how things are connected. SG : So that ’ s exactly right. We ’ ve got ourselves in a world that ’ s massively complex, and we ’ ve been using algorithms to kind of filter it down so we can navigate through it. And those algorithms, whilst being kind of useful, are also very, very narrow, and we can do better than that, because we can realize that their complexity is not random. It has mathematical structure, and we can use that mathematical structure to go and explore things like the world of ideas to see what ’ s being said, to see what ’ s not being said, and to be a little bit more human and, hopefully, a little smarter. Thank you.

Text Sample 453: NEG Dim 3 – 2005 – Score -3.00 – 2005/t000427.txt

At the break, I was asked by several people about my comments about the aging debate. And this will be my only comment on it. And that is, I understand that optimists greatly outlive pessimists. What I ’ m going to tell you about in my 18 minutes is how we ’ re about to switch from reading the genetic code to the first stages of beginning to write the code ourselves. It ’ s only 10 years ago this month when we published the first sequence of a free-living organism, that of haemophilus influenzae. That took a genome project from 13 years down to four months. We can now do that same genome project in the order of two to eight hours. So in the last decade, a large number of genomes have been added : most human pathogens, a couple of plants, several insects and several mammals, including the human genome. Genomics at this stage of the thinking from a little over 10 years ago was, by the end of this year, we might have between three and five genomes sequenced ; it ’ s on the order of several hundred. We just got a grant from the Gordon and Betty Moore Foundation to sequence 130 genomes this year, as a side project from environmental organisms. So the rate of reading the genetic code has changed. But as we look, what ’ s out there, we ’ ve barely scratched the surface on what is available on this planet. Most people don ’ t realize it, because they ’ re invisible, but microbes make up about a half of the Earth ’ s biomass, whereas all animals only make up about one one-thousandth of all the biomass. And maybe it ’ s something that people in Oxford don ’ t do very often, but if you ever make it to the sea, and you swallow a mouthful of seawater, keep in mind that each milliliter has about a million bacteria and on the order of 10 million viruses. Less than 5, 000 microbial species have been characterized as of two years ago, and so we decided to do something about it. And we started the Sorcerer II Expedition, where we were, as with great oceanographic expeditions, trying to sample the ocean every 200 miles. We started in Bermuda for our test project, then moved up to Halifax, working down the U. S. East Coast, the Caribbean Sea, the Panama Canal, through to the Galapagos, then across the Pacific, and we ’ re in the process now of working our way across the Indian Ocean. It ’ s very tough duty ; we ’ re doing this on a sailing vessel, in part to help excite young people about going into science. The experiments are incredibly simple. We just take seawater and we filter it, and we collect different size organisms on different filters, and then take their DNA back to our lab in Rockville, where we can sequence a hundred million letters of the genetic code every 24 hours. And with doing this, we ’ ve made some amazing discoveries. For example, it was thought that the visual pigments that are in our eyes — there was only one or two organisms in the environment that had these same pigments. It turns out, almost every species in the upper parts of the ocean in warm parts of the world have these same photoreceptors, and use sunlight as the source of their energy and communication. From one site, from one barrel of seawater, we discovered 1. 3 million new genes and as many as 50, 000 new species. We ' ve extended this to the air now with a grant from the Sloan Foundation. We ' re measuring how many viruses and bacteria all of us are breathing in and out every day, particularly on airplanes or closed auditoriums. We filter through some simple **apparatuses** ; we collect on the order of a billion microbes from just a day filtering on top of a building in New York City. And we ' re in the process of sequencing all that at the present time. Just on the data collection side, just where we are through the Galapagos, we ' re finding that almost every 200 miles, we see tremendous diversity in the samples in the ocean. Some of these make logical sense, in terms of different temperature gradients. So this is a satellite photograph based on temperatures — red being warm, blue being cold — and we found there ' s a tremendous difference between the warm water samples and the cold water samples, in terms of abundant species. The other thing that surprised us quite a bit is these photoreceptors detect different wavelengths of light, and we can predict that based on their amino acid sequence. And these vary tremendously from region to region. Maybe not surprisingly, in the deep ocean, where it ' s mostly blue, the photoreceptors tend to see blue light. When there ' s a lot of chlorophyll around, they see a lot of green light. But they vary even more, possibly moving towards infrared and ultraviolet in the extremes. Just to try and get an assessment of what our gene repertoire was, we assembled all the data — including all of ours thus far from the expedition, which represents more than half of all the gene data on the planet — and it totaled around 29 million genes. And we tried to put these into gene families to see what these discoveries are : Are we just discovering new members of known families, or are we discovering new families? And it turns out we have about 50, 000 major gene families, but every new sample we take in the environment adds in a linear fashion to these new families. So we ' re at the earliest stages of discovery about basic genes, components and life on this planet. When we look at the so-called evolutionary tree, we ' re up on the upper right- hand corner with the animals. Of those roughly 29 million genes, we only have around 24, 000 in our genome. And if you take all animals together, we probably share less than 30, 000 and probably maybe a dozen or more thousand different gene families. I view that these genes are now not only the design components of evolution. And we think in a gene-centric view — maybe going back to Richard Dawkins ' ideas — than in a genome-centric view, which are different constructs of these gene components. Synthetic DNA, the ability to synthesize DNA, has changed at sort of the same pace that DNA sequencing has over the last decade or two, and is getting very rapid and very cheap. Our first thought about synthetic genomics came when we sequenced the second genome back in 1995, and that from mycoplasma genitalium. And we have really nice T-shirts that say, you know,"I heart my genitalium."This is actually just a microorganism. But it has roughly 500 genes. Haemophilus had 1, 800 genes. And we simply asked the question, if one species needs 800, another 500, is there a smaller set of genes that might comprise a minimal operating system? So we started doing transposon mutagenesis. Transposons are just small pieces of DNA that randomly insert in the genetic code. And if they insert in the middle of the gene, they disrupt its function. So we made a map of all the genes that could take transposon insertions and we called those"non-essential genes."But it turns out the environment is very critical for this, and you can only define an essential or non-essential gene based on exactly what ' s in the environment. We also tried to take a more directly intellectual approach with the genomes of 13 related organisms, and we tried to compare all of those, to see what they had in common. And we got these overlapping circles. And we found only 173 genes common to all 13 organisms. The pool expanded a little bit if we ignored one intracellular parasite ; it expanded even more when we looked at core sets of genes of around 310 or so. So we think that we can expand or contract genomes, depending on your point of view here, to maybe 300 to 400 genes from the minimal of 500. The only way to prove these ideas was to construct an artificial chromosome with those genes in them, and we had to do this in a cassette-based fashion. We found that synthesizing accurate DNA in large pieces was extremely difficult. Ham Smith and Clyde Hutchison, my colleagues on this, developed an exciting new method that allowed us to synthesize a 5, 000-base pair virus in only a two-week period that was 100 percent accurate, in terms of its sequence and its biology. It was a quite exciting experiment — when we just took the synthetic piece of DNA, injected it in the bacteria and all of a sudden, that DNA started driving the production of the virus particles that turned around and then killed the bacteria. This was not the first synthetic virus — a polio virus had been made a year before — but it was only one ten-thousandth as active and it took three years to do. This is a cartoon of the structure of phi X 174. This is a case where the software now builds its own hardware, and that ' s the notions that we have with biology. People immediately jump to concerns about biological warfare, and I had recent testimony before a Senate committee, and a special committee the U. S. government has set up to review this area. And I think it ' s important to keep reality in mind, versus what happens with people ' s imaginations. Basically, any virus that ' s been sequenced today — that genome can be made. And people immediately freak out about things about Ebola or smallpox, but the DNA from this organism is not infective. So even if somebody made the smallpox genome, that DNA itself would not cause infections. The real concern that security departments have is designer viruses. And there ' s only two countries, the U. S. and the former Soviet Union, that had major efforts on trying to create biological warfare agents. If that research is truly discontinued, there should be very little activity on the know-how to make designer viruses in the future. I think single-cell organisms are possible within two years. And possibly eukaryotic cells, those that we have, are possible within a decade. So we ' re now making several dozen different constructs, because we can vary the cassettes and the genes that go into this artificial chromosome. The key is, how do you put all of the others? We start with these fragments, and then we have a homologous **recombination** system that reassembles those into a chromosome. This is derived from an organism, deinococcus radiodurans, that can take three million **rads** of radiation and not be killed. It reassembles its genome after this radiation burst in about 12 to 24 hours, after its chromosomes are literally blown apart. This organism is ubiquitous on the planet, and exists perhaps now in outer space due to all our travel there. This is a glass beaker after about half a million **rads** of radiation. The glass started to burn and crack, while the microbes sitting in the bottom just got happier and happier. Here ' s an actual picture of what happens : the top of this shows the genome after 1. 7 million **rads** of radiation. The chromosome is literally blown apart. And here ' s that same DNA automatically reassembled 24 hours later. It ' s truly stunning that these organisms can do that, and we probably have thousands, if not tens of thousands, of different species on this planet that are capable of doing that. After these genomes are synthesized, the first step is just transplanting them into a cell without a genome. So we think synthetic cells are going to have tremendous potential, not only for understanding the basis of biology but for hopefully environmental and society issues. For example, from the third organism we sequenced, Methanococcus jannaschii — it lives in boiling water temperatures ; its energy source is hydrogen and all its carbon comes from CO2 it captures back from the environment. So we know lots of different pathways, thousands of different organisms now that live off of CO2, and can capture that back. So instead of using carbon from oil for synthetic processes, we have the chance of using carbon and capturing it back from the atmosphere, converting that into biopolymers or other products. We have one organism that lives off of carbon monoxide, and we use as a reducing power to split water to produce hydrogen and oxygen. Also, there ’ s numerous pathways that can be engineered metabolizing methane. And DuPont has a major program with Statoil in Norway to capture and convert the methane from the gas fields there into useful products. Within a short while, I think there ’ s going to be a new field called”Combinatorial Genomics,”because with these new synthesis capabilities, these vast gene array repertoires and the homologous **recombination**, we think we can design a robot to make maybe a million different chromosomes a day. And therefore, as with all biology, you get selection through screening, whether you ’ re screening for hydrogen production, or chemical production, or just viability. To understand the role of these genes is going to be well within reach. We ’ re trying to modify photosynthesis to produce hydrogen directly from sunlight. Photosynthesis is modulated by oxygen, and we have an oxygen- insensitive hydrogenase that we think will totally change this process. We ’ re also combining cellulases, the enzymes that break down complex sugars into simple sugars and fermentation in the same cell for producing ethanol. Pharmaceutical production is already under way in major laboratories using microbes. The chemistry from compounds in the environment is orders of magnitude more complex than our best chemists can produce. I think future engineered species could be the source of food, hopefully a source of energy, environmental remediation and perhaps replacing the petrochemical industry. Let me just close with ethical and policy studies. We delayed the start of our experiments in 1999 until we completed a year-and-a-half bioethical review as to whether we should try and make an artificial species. Every major religion participated in this. It was actually a very strange study, because the various religious leaders were using their scriptures as law books, and they couldn ’ t find anything in them prohibiting making life, so it must be OK. The only ultimate concerns were biological warfare aspects of this, but gave us the go ahead to start these experiments for the reasons we were doing them. Right now the Sloan Foundation has just funded a multi-institutional study on this, to work out what the risk and benefits to society are, and the rules that scientific teams such as my own should be using in this area, and we ’ re trying to set good examples as we go forward. These are complex issues. Except for the threat of bio-terrorism, they ’ re very simple issues in terms of, can we design things to produce clean energy, perhaps revolutionizing what developing countries can do and provide through various simple processes. Thank you very much.

Text Sample 454: NEG Dim 3 – 2005 – Score -2.00 – 2005/t000433.txt

Well, I thought there would be a podium, so I ’ m a bit scared. Chris asked me to tell again how we found the structure of DNA. And since, you know, I follow his orders, I ’ ll do it. But it slightly bores me. And, you know, I wrote a book. So I ’ ll say something â â I ’ ll say a little about, you know, how the discovery was made, and why Francis and I found it. And then, I hope maybe I have at least five minutes to say what makes me tick now. In back of me is a picture of me when I was 17. I was at the University of Chicago, in my third year, and I was in my third year because the University of Chicago let you in after two years of high school. So you â it was fun to get away from high school â â because I was very small, and I was no good in sports, or anything like that. But I should say that my background â my father was, you know, raised to be an Episcopalian and Republican, but after one year of college, he became an atheist and a Democrat. And my mother was Irish Catholic, and â but she didn ’ t take religion too seriously. And by the age of 11, I was no longer going to Sunday Mass, and going on birdwatching walks with my father. So early on, I heard of Charles Darwin. I guess, you know, he was the big hero. And, you know, you understand life as it now exists through evolution. And at the University of Chicago I was a zoology major, and thought I would end up, you know, if I was bright enough, maybe getting a Ph. D. from Cornell in ornithology. Then, in the Chicago paper, there was a review of a book called”What is Life?”by the great physicist, Schrodinger. And that, of course, had been a question I wanted to know. You know, Darwin explained life after it got started, but what was the essence of life? And Schrodinger said the essence was information present in our chromosomes, and it had to be present on a molecule. I ’ d never really thought of molecules before. You know chromosomes, but this was a molecule, and somehow all the information was probably present in some digital form. And there was the big question of, how did you copy the information? So that was the book. And so, from that moment on, I wanted to be a geneticist â understand the gene and, through that, understand life. So I had, you know, a hero at a distance. It wasn ’ t a baseball player ; it was Linus Pauling. And so I applied to Caltech and they turned me down. So I went to Indiana, which was actually as good as Caltech in genetics, and besides, they had a really good basketball team. So I had a really quite happy life at Indiana. And it was at Indiana I got the impression that, you know, the gene was likely to be DNA. And so when I got my Ph. D., I should go and search for DNA. So I first went to Copenhagen because I thought, well, maybe I could become a biochemist, but I discovered biochemistry was very boring. It wasn ’ t going anywhere toward, you know, saying what the gene was ; it was just nuclear science. And oh, that ’ s the book, little book. You can read it in about two hours. And â but then I went to a meeting in Italy. And there was an unexpected speaker who wasn ’ t on the program, and he talked about DNA. And this was Maurice Wilkins. He was trained as a physicist, and after the war he wanted to do biophysics, and he picked DNA because DNA had been determined at the Rockefeller Institute to possibly be the genetic molecules on the chromosomes. Most people believed it was proteins. But Wilkins, you know, thought DNA was the best bet, and he showed this x-ray photograph. Sort of **crystalline**. So DNA had a structure, even though it owed it to probably different molecules carrying different sets of instructions. So there was something universal about the DNA molecule. So I wanted to work with him, but he didn ’ t want a former birdwatcher, and I ended up in Cambridge, England. So I went to Cambridge, because it was really the best place in the world then for x-ray crystallography. And x-ray crystallography is now a subject in, you know, chemistry departments. I mean, in those days it was the domain of the physicists. So the best place for x-ray crystallography was at the Cavendish Laboratory at Cambridge. And there I met Francis Crick. I went there without knowing him. He was 35. I was 23. And within a day, we had decided that maybe we could take a shortcut to finding the structure of DNA. Not solve it like, you know, in rigorous fashion, but build a model, an electro-model, using some coordinates of, you know, length, all that sort of stuff from x-ray photographs. But just ask what the molecule â how should it fold up? And the reason for doing so, at the center of this photograph, is Linus Pauling. About six months before, he proposed the alpha helical structure for proteins. And in doing so, he banished the man out on the right, Sir Lawrence Bragg, who was the Cavendish professor. This is a photograph several years later, when Bragg had cause to smile. He certainly wasn ’ t smiling when I got there, because he was somewhat humiliated by Pauling getting the alpha helix, and the Cambridge people failing because they weren ’ t chemists. And certainly, neither Crick or I were chemists, so we tried to build a model. And he knew, Francis knew Wilkins. So Wilkins said he thought it was the helix. X-ray diagram, he thought was comparable with the helix. So we built a three-stranded model. The people from London came up. Wilkins and this collaborator, or possible collaborator, Rosalind Franklin, came up and sort of laughed at our model. They said it was lousy, and it was. So we were told to build no more models ; we were incompetent. And so we didn ’ t build any models, and Francis sort of continued to work on proteins. And basically, I did nothing. And â except read. You know, basically, reading is a good thing ; you get facts. And we kept telling the people in London that Linus Pauling ’ s going to move on to DNA. If DNA is that important, Linus will know it. He ’ ll build a model, and then we ’ re going to be scooped. And, in fact, he ’ d written the people in London : Could he see their x-ray photograph? And they had the wisdom to say”no.”So he didn ’ t have it. But there was ones in the literature. Actually, Linus didn ’ t look at them that carefully. But about, oh, 15 months after I got to Cambridge, a rumor began to appear from Linus Pauling ’ s son, who was in Cambridge, that his father was now working on DNA. And so, one day Peter came in and he said he was Peter Pauling, and he gave me a copy of his father ’ s manuscripts. And boy, I was scared because I thought, you know, we may be scooped. I have nothing to do, no qualifications for anything. And so there was the paper, and he proposed a three-stranded structure. And I read it, and it was just â it was crap. So this was, you know, unexpected from the world ’ s â â and so, it was held together by hydrogen bonds between phosphate groups. Well, if the peak pH that cells have is around seven, those hydrogen bonds couldn ’ t exist. We rushed over to the chemistry department and said,”Could Pauling be right?”And Alex Hust said,”No.”So we were happy. And, you know, we were still in the game, but we were frightened that somebody at Caltech would tell Linus that he was wrong. And so Bragg said,”Build models.”And a month after we got the Pauling manuscript â I should say I took the manuscript to London, and showed the people. Well, I said, Linus was wrong and that we ’ re still in the game and that they should immediately start building models. But Wilkins said”no.”Rosalind Franklin was leaving in about two months, and after she left he would start building models. And so I came back with that news to Cambridge, and Bragg said,”Build models.”Well, of course, I wanted to build models. And there ’ s a picture of Rosalind. She really, you know, in one sense she was a chemist, but really she would have been trained â she didn ’ t know any organic chemistry or quantum chemistry. She was a crystallographer. And I think part of the reason she didn ’ t want to build models was, she wasn ’ t a chemist, whereas Pauling was a chemist. And so Crick and I, you know, started building models, and I ’ d learned a little chemistry, but not enough. Well, we got the answer on the 28th February ’ 53. And it was because of a rule, which, to me, is a very good rule : Never be the brightest person in a room, and we weren ’ t. We weren ’ t the best chemists in the room. I went in and showed them a pairing I ’ d done, and Jerry Donohue â he was a chemist â he said, it ’ s wrong. You ’ ve got â the hydrogen atoms are in the wrong place. I just put them down like they were in the books. He said they were wrong. So the next day, you know, after I thought,”Well, he might be right.”So I changed the locations, and then we found the base pairing, and

Francis immediately said the chains run in absolute directions. And we knew we were right. So it was a pretty, you know, it all happened in about two hours. From nothing to thing. And we knew it was big because, you know, if you just put A next to T and G next to C, you have a copying mechanism. So we saw how genetic information is carried. It ’ s the order of the four bases. So in a sense, it is a sort of digital-type information. And you copy it by going from strand- separating. So, you know, if it didn ’ t work this way, you might as well believe it, because you didn ’ t have any other scheme. But that ’ s not the way most scientists think. Most scientists are really rather dull. They said, we won ’ t think about it until we know it ’ s right. But, you know, we thought, well, it ’ s at least 95 percent right or 99 percent right. So think about it. The next five years, there were essentially something like five references to our work in”Nature”â none. And so we were left by ourselves, and trying to do the last part of the trio : how do you â what does this genetic information do? It was pretty obvious that it provided the information to an RNA molecule, and then how do you go from RNA to protein? For about three years we just â I tried to solve the structure of RNA. It didn ’ t yield. It didn ’ t give good x-ray photographs. I was decidedly unhappy ; a girl didn ’ t marry me. It was really, you know, sort of a shitty time. So there ’ s a picture of Francis and I before I met the girl, so I ’ m still looking happy. But there is what we did when we didn ’ t know where to go forward : we formed a club and called it the RNA Tie Club. George Gamow, also a great physicist, he designed the tie. He was one of the members. The question was : How do you go from a four-letter code to the 20-letter code of proteins? Feynman was a member, and Teller, and friends of Gamow. But that ’ s the only â no, we were only photographed twice. And on both occasions, you know, one of us was missing the tie. There ’ s Francis up on the upper right, and Alex Rich â the M. D. -turned-crystallographer â is next to me. This was taken in Cambridge in September of 1955. And I ’ m smiling, sort of forced, I think, because the girl I had, boy, she was gone. And so I didn ’ t really get happy until 1960, because then we found out, basically, you know, that there are three forms of RNA. And we knew, basically, DNA provides the information for RNA. RNA provides the information for protein. And that let Marshall Nirenberg, you know, take RNA â synthetic RNA â put it in a system making protein. He made polyphenylalanine, polyphenylalanine. So that ’ s the first cracking of the genetic code, and it was all over by 1966. So there, that ’ s what Chris wanted me to do, it was â so what happened since then? Well, at that time â I should go back. When we found the structure of DNA, I gave my first talk at Cold Spring Harbor. The physicist, Leo Szilard, he looked at me and said,”Are you going to patent this?”And â but he knew patent law, and that we couldn ’ t patent it, because you couldn ’ t. No use for it. And so DNA didn ’ t become a useful molecule, and the lawyers didn ’ t enter into the equation until 1973, 20 years later, when Boyer and Cohen in San Francisco and Stanford came up with their method of recombinant DNA, and Stanford patented it and made a lot of money. At least they patented something which, you know, could do useful things. And then, they learned how to read the letters for the code. And, boom, we ’ ve, you know, had a biotech industry. And, but we were still a long ways from, you know, answering a question which sort of dominated my childhood, which is : How do you nature-nurture? And so I ’ ll go on. I ’ m already out of time, but this is Michael Wigler, a very, very clever mathematician turned physicist. And he developed a technique which essentially will let us look at sample DNA and, eventually, a million spots along it. There ’ s a chip there, a conventional one. Then there ’ s one made by a photolithography by a company in Madison called NimbleGen, which is way ahead of Affymetrix. And we use their technique. And what you can do is sort of compare DNA of normal segs versus cancer. And you can see on the top that cancers which are bad show insertions or deletions. So the DNA is really badly mucked up, whereas if you have a chance of surviving, the DNA isn ’ t so mucked up. So we think that this will eventually lead to what we call”DNA biopsies.”Before you get treated for cancer, you should really look at this technique, and get a feeling of the face of the enemy. It ’ s not a â it ’ s only a partial look, but it ’ s a â I think it ’ s going to be very, very useful. So, we started with breast cancer because there ’ s lots of money for it, no government money. And now I have a sort of vested interest : I want to do it for prostate cancer. So, you know, you aren ’ t treated if it ’ s not dangerous. But Wigler, besides looking at cancer cells, looked at normal cells, and made a really sort of surprising observation. Which is, all of us have about 10 places in our genome where we ’ ve lost a gene or gained another one. So we ’ re sort of all imperfect. And the question is well, if we ’ re around here, you know, these little losses or gains might not be too bad. But if these deletions or amplifications occurred in the wrong gene, maybe we ’ ll feel sick. So the first disease he looked at is autism. And the reason we looked at autism is we had the money to do it. Looking at an individual is about 3, 000 dollars. And the parent of a child with Asperger ’ s disease, the high-intelligence autism, had sent his thing to a conventional company ; they didn ’ t do it. Couldn ’ t do it by conventional genetics, but just scanning it we began to find genes for autism. And you can see here, there are a lot of them. So a lot of autistic kids are autistic because they just lost a big piece of DNA. I mean, big piece at the molecular level. We saw one autistic kid, about five million bases just missing from one of his chromosomes. We haven ’ t yet looked at the parents, but the parents probably don ’ t have that loss, or they wouldn ’ t be parents. Now, so, our autism study is just beginning. We got three million dollars. I think it will cost at least 10 to 20 before you ’ d be in a position to help parents who ’ ve had an autistic child, or think they may have an autistic child, and can we spot the difference? So this same technique should probably look at all. It ’ s a wonderful way to find genes. And so, I ’ ll conclude by saying we ’ ve looked at 20 people with schizophrenia. And we thought we ’ d probably have to look at several hundred before we got the picture. But as you can see, there ’ s seven out of 20 had a change which was very high. And yet, in the controls there were three. So what ’ s the meaning of the controls? Were they crazy also, and we didn ’ t know it? Or, you know, were they normal? I would guess they ’ re normal. And what we think in schizophrenia is there are genes of predisposure, and whether this is one that predisposes â and then there ’ s only a sub-segment of the population that ’ s capable of being schizophrenic. Now, we don ’ t have really any evidence of it, but I think, to give you a hypothesis, the best guess is that if you ’ re left-handed, you ’ re prone to schizophrenia. 30 percent of schizophrenic people are left-handed, and schizophrenia has a very funny genetics, which means 60 percent of the people are genetically left-handed, but only half of it showed. I don ’ t have the time to say. Now, some people who think they ’ re right-handed are genetically left- handed. OK. I ’ m just saying that, if you think, oh, I don ’ t carry a left-handed gene so therefore my, you know, children won ’ t be at risk of schizophrenia. You might. OK? So it ’ s, to me, an extraordinarily exciting time. We ought to be able to find the gene for bipolar ; there ’ s a relationship. And if I had enough money, we ’ d find them all this year. I thank you.

Text Sample 455: NEG Dim 3 – 2005 – Score -2.00 – 2005/t000437.txt
If you take 10, 000 people at random, 9, 999 have something in common : their

interests in business lie on or near the Earth ' s surface. The odd one out is an astronomer, and I am one of that strange breed. My talk will be in two parts. I ' ll talk first as an astronomer, and then as a worried member of the human race. But let ' s start off by remembering that Darwin showed how we ' re the outcome of four billion years of evolution. And what we try to do in astronomy and cosmology is to go back before Darwin ' s simple beginning, to set our Earth in a cosmic context. And let me just run through a few slides. This was the impact that happened last week on a comet. If they ' d sent a nuke, it would have been rather more spectacular than what actually happened last Monday. So that ' s another project for NASA. That ' s Mars from the European Mars Express, and at New Year. This artist ' s impression turned into reality when a parachute landed on Titan, Saturn ' s giant moon. It landed on the surface. This is pictures taken on the way down. That looks like a coastline. It is indeed, but the ocean is liquid methane — the temperature minus 170 degrees centigrade. If we go beyond our solar system, we ' ve learned that the stars aren ' t twinkly points of light. Each one is like a sun with a retinue of planets orbiting around it. And we can see places where stars are forming, like the Eagle Nebula. We see stars dying. In six billion years, the sun will look like that. And some stars die spectacularly in a supernova explosion, leaving remnants like that. On a still bigger scale, we see entire galaxies of stars. We see entire ecosystems where gas is being recycled. And to the cosmologist, these galaxies are just the atoms, as it were, of the large-scale universe. This picture shows a patch of sky so small that it would take about 100 patches like it to cover the full moon in the sky. Through a small telescope, this would look quite blank, but you see here hundreds of little, faint smudges. Each is a galaxy, fully like ours or Andromeda, which looks so small and faint because its light has taken 10 billion light-years to get to us. The stars in those galaxies probably don ' t have planets around them. There ' s scant chance of life there — that ' s because there ' s been no time for the nuclear fusion in stars to make silicon and carbon and iron, the building blocks of planets and of life. We believe that all of this emerged from a Big Bang — a hot, dense state. So how did that amorphous Big Bang turn into our complex cosmos? I ' m going to show you a movie simulation 16 powers of 10 faster than real time, which shows a patch of the universe where the expansions have subtracted out. But you see, as time goes on in gigayears at the bottom, you will see structures evolve as gravity feeds on small, dense irregularities, and structures develop. And we ' ll end up after 13 billion years with something looking rather like our own universe. And we compare simulated universes like that — I ' ll show you a better simulation at the end of my talk — with what we actually see in the sky. Well, we can trace things back to the earlier stages of the Big Bang, but we still don ' t know what banged and why it banged. That ' s a challenge for 21st-century science. If my research group had a logo, it would be this picture here : an ouroboros, where you see the micro-world on the left — the world of the quantum — and on the right the large-scale universe of planets, stars and galaxies. We know our universes are united though — links between left and right. The everyday world is determined by atoms, how they stick together to make molecules. Stars are fueled by how the nuclei in those atoms react together. And, as we ' ve learned in the last few years, galaxies are held together by the gravitational pull of so-called dark matter : particles in huge swarms, far smaller even than atomic nuclei. But we ' d like to know the synthesis **symbolized** at the very top. The micro-world of the quantum is understood. On the right hand side, gravity holds sway. Einstein explained that. But the unfinished business for 21st-century science is to link together cosmos and micro-world with a unified theory — **symbolized**, as it were, gastronomically at the top of that picture.

And until we have that synthesis, we won ’ t be able to understand the very beginning of our universe because when our universe was itself the size of an atom, quantum effects could shake everything. And so we need a theory that unifies the very large and the very small, which we don ’ t yet have. One idea, incidentally — and I had this hazard sign to say I ’ m going to speculate from now on — is that our Big Bang was not the only one. One idea is that our three-dimensional universe may be embedded in a high- dimensional space, just as you can imagine on these sheets of paper. You can imagine ants on one of them thinking it ’ s a two-dimensional universe, not being aware of another population of ants on the other. So there could be another universe just a millimeter away from ours, but we ’ re not aware of it because that millimeter is measured in some fourth spatial dimension, and we ’ re imprisoned in our three. And so we believe that there may be a lot more to physical reality than what we ’ ve normally called our universe — the aftermath of our Big Bang. And here ’ s another picture. Bottom right depicts our universe, which on the horizon is not beyond that, but even that is just one bubble, as it were, in some vaster reality. Many people suspect that just as we ’ ve gone from believing in one solar system to zillions of solar systems, one galaxy to many galaxies, we have to go to many Big Bangs from one Big Bang, perhaps these many Big Bangs displaying an immense variety of properties. Well, let ’ s go back to this picture. There ’ s one challenge **symbolized** at the top, but there ’ s another challenge to science **symbolized** at the bottom. You want to not only synthesize the very large and the very small, but we want to understand the very complex. And the most complex things are ourselves, midway between atoms and stars. We depend on stars to make the atoms we ’ re made of. We depend on chemistry to determine our complex structure. We clearly have to be large, compared to atoms, to have layer upon layer of complex structure. We clearly have to be small, compared to stars and planets — otherwise we ’ d be crushed by gravity. And in fact, we are midway. It would take as many human bodies to make up the sun as there are atoms in each of us. The geometric mean of the mass of a proton and the mass of the sun is 50 kilograms, within a factor of two of the mass of each person here. Well, most of you anyway. The science of complexity is probably the greatest challenge of all, greater than that of the very small on the left and the very large on the right. And it ’ s this science, which is not only enlightening our understanding of the biological world, but also transforming our world faster than ever. And more than that, it ’ s engendering new kinds of change. And I now move on to the second part of my talk, and the book”Our Final Century”was mentioned. If I was not a self-effacing Brit, I would mention the book myself, and I would add that it ’ s available in paperback. And in America it was called”Our Final Hour”because Americans like instant gratification. But my theme is that in this century, not only has science changed the world faster than ever, but in new and different ways. Targeted drugs, genetic modification, artificial intelligence, perhaps even implants into our brains, may change human beings themselves. And human beings, their physique and character, has not changed for thousands of years. It may change this century. It ’ s new in our history. And the human impact on the global environment — greenhouse warming, mass extinctions and so forth — is unprecedented, too. And so, this makes this coming century a challenge. Bio- and cybertechnologies are environmentally benign in that they offer marvelous prospects, while, nonetheless, reducing pressure on energy and resources. But they will have a dark side. In our interconnected world, novel technology could empower just one fanatic, or some weirdo with a mindset of those who now design computer viruses, to trigger some kind on disaster. Indeed, catastrophe could arise simply from technical misadventure — error rather than terror. And even a tiny probability of catastrophe is unacceptable when the downside could be of global consequence. In fact, some years ago, Bill Joy wrote an article expressing tremendous concern about robots taking us over, etc. I don ' t go along with all that, but it ' s interesting that he had a simple solution. It was what he called"fine-grained relinquishment."He wanted to give up the dangerous kind of science and keep the good bits. Now, that ' s absurdly naive for two reasons. First, any scientific discovery has benign consequences as well as dangerous ones. And also, when a scientist makes a discovery, he or she normally has no clue what the applications are going to be. And so what this means is that we have to accept the risks if we are going to enjoy the benefits of science. We have to accept that there will be hazards. And I think we have to go back to what happened in the post-War era, post-World War II, when the nuclear scientists who ' d been involved in making the atomic bomb, in many cases were concerned that they should do all they could to alert the world to the dangers. And they were inspired not by the young Einstein, who did the great work in relativity, but by the old Einstein, the icon of poster and t-shirt, who failed in his scientific efforts to unify the physical laws. He was premature. But he was a moral compass — an inspiration to scientists who were concerned with arms control. And perhaps the greatest living person is someone I ' m privileged to know, Joe Rothblatt. Equally untidy office there, as you can see. He ' s 96 years old, and he founded the Pugwash movement. He persuaded Einstein, as his last act, to sign the famous memorandum of Bertrand Russell. And he sets an example of the concerned scientist. And I think to harness science optimally, to choose which doors to open and which to leave closed, we need latter-day counterparts of people like Joseph Rothblatt. We need not just campaigning physicists, but we need biologists, computer experts and environmentalists as well. And I think academics and independent entrepreneurs have a special obligation because they have more freedom than those in government service, or company employees subject to commercial pressure. I wrote my book,"Our Final Century,"as a scientist, just a general scientist. But there ' s one respect, I think, in which being a cosmologist offered a special perspective, and that ' s that it offers an awareness of the immense future. The stupendous time spans of the evolutionary past are now part of common culture — outside the American Bible Belt, anyway — but most people, even those who are familiar with evolution, aren ' t mindful that even more time lies ahead. The sun has been shining for four and a half billion years, but it ' ll be another six billion years before its fuel runs out. On that **schematic** picture, a sort of time-lapse picture, we ' re halfway. And it ' ll be another six billion before that happens, and any remaining life on Earth is vaporized. There ' s an unthinking tendency to imagine that humans will be there, experiencing the sun ' s demise, but any life and intelligence that exists then will be as different from us as we are from bacteria. The unfolding of intelligence and complexity still has immensely far to go, here on Earth and probably far beyond. So we are still at the beginning of the emergence of complexity in our Earth and beyond. If you represent the Earth ' s lifetime by a single year, say from January when it was made to December, the 21st-century would be a quarter of a second in June — a tiny fraction of the year. But even in this concertinaed cosmic perspective, our century is very, very special, the first when humans can change themselves and their home planet. As I should have shown this earlier, it will not be humans who witness the end point of the sun ; it will be creatures as different from us as we are from bacteria. When Einstein died in 1955, one striking tribute to his global status was this cartoon by Herblock in the Washington Post. The plaque reads,"Albert Einstein lived here."And I ' d like to end with a vignette, as it were, inspired by this image. We ' ve been familiar for 40 years with this image : the fragile beauty of land, ocean and clouds, contrasted with the **sterile** moonscape on which the astronauts left their footprints. But let ’ s suppose some aliens had been watching our pale blue dot in the cosmos from afar, not just for 40 years, but for the entire 4. 5 billion-year history of our Earth. What would they have seen? Over nearly all that immense time, Earth ’ s appearance would have changed very gradually. The only abrupt worldwide change would have been major asteroid impacts or volcanic super-eruptions. Apart from those brief traumas, nothing happens suddenly. The continental landmasses drifted around. Ice cover waxed and waned. Successions of new species emerged, evolved and became extinct. But in just a tiny sliver of the Earth ’ s history, the last one-millionth part, a few thousand years, the patterns of vegetation altered much faster than before. This signaled the start of agriculture. Change has accelerated as human populations rose. Then other things happened even more abruptly. Within just 50 years — that ’ s one hundredth of one millionth of the Earth ’ s age — the amount of carbon dioxide in the atmosphere started to rise, and ominously fast. The planet became an intense emitter of radio waves — the total output from all TV and cell phones and radar transmissions. And something else happened. Metallic objects — albeit very small ones, a few tons at most — escaped into orbit around the Earth. Some journeyed to the moons and planets. A race of advanced extraterrestrials watching our solar system from afar could confidently predict Earth ’ s final doom in another six billion years. But could they have predicted this unprecedented spike less than halfway through the Earth ’ s life? These human-induced alterations occupying overall less than a millionth of the elapsed lifetime and seemingly occurring with runaway speed? If they continued their vigil, what might these hypothetical aliens witness in the next hundred years? Will some spasm foreclose Earth ’ s future? Or will the biosphere stabilize? Or will some of the metallic objects launched from the Earth spawn new oases, a post-human life elsewhere? The science done by the young Einstein will continue as long as our civilization, but for civilization to survive, we ’ ll need the wisdom of the old Einstein — humane, global and farseeing. And whatever happens in this uniquely crucial century will resonate into the remote future and perhaps far beyond the Earth, far beyond the Earth as depicted here. Thank you very much.

Text Sample 456: NEG Dim 3 – 2012 – Score -1.00 – 2012/t002202.txt

I ’ d like to show you a video of some of the models I work with. They ’ re all the perfect size, and they don ’ t have an ounce of fat. Did I mention they ’ re gorgeous? And they ’ re scientific models? As you might have guessed, I ’ m a tissue engineer, and this is a video of some of the beating heart that I ’ ve engineered in the lab. And one day we hope that these tissues can serve as replacement parts for the human body. But what I ’ m going to tell you about today is how these tissues make awesome models. Well, let ’ s think about the drug screening process for a moment. You go from drug formulation, lab testing, animal testing, and then clinical trials, which you might call human testing, before the drugs get to market. It costs a lot of money, a lot of time, and sometimes, even when a drug hits the market, it acts in an unpredictable way and actually hurts people. And the later it fails, the worse the consequences. It all boils down to two issues. One, humans are not rats, and two, despite our incredible similarities to one another, actually those tiny differences between you and I have huge impacts with how we metabolize drugs and how those drugs affect us. So what if we had better models in the lab that could not only mimic us better than rats but also reflect our diversity? Let ’ s see how we can do it with tissue engineering. One of the key technologies that ’ s really important is what ’ s called induced pluripotent stem cells. They were developed in Japan pretty recently. Okay, induced pluripotent stem cells. They ’ re a lot like embryonic stem cells except without the controversy. We induce cells, okay, say, skin cells, by adding a few genes to them, culturing them, and then harvesting them. So they ’ re skin cells that can be tricked, kind of like cellular amnesia, into an embryonic state. So without the controversy, that ’ s cool thing number one. Cool thing number two, you can grow any type of tissue out of them : brain, heart, liver, you get the picture, but out of your cells. So we can make a model of your heart, your brain on a chip. Generating tissues of predictable density and behavior is the second piece, and will be really key towards getting these models to be adopted for drug discovery. And this is a **schematic** of a bioreactor we ’ re developing in our lab to help engineer tissues in a more modular, scalable way. Going forward, imagine a massively parallel version of this with thousands of pieces of human tissue. It would be like having a clinical trial on a chip. But another thing about these induced pluripotent stem cells is that if we take some skin cells, let ’ s say, from people with a genetic disease and we engineer tissues out of them, we can actually use tissue-engineering techniques to generate models of those diseases in the lab. Here ’ s an example from Kevin Eggan ’ s lab at Harvard. He generated neurons from these induced pluripotent stem cells from patients who have Lou Gehrig ’ s Disease, and he differentiated them into neurons, and what ’ s amazing is that these neurons also show symptoms of the disease. So with disease models like these, we can fight back faster than ever before and understand the disease better than ever before, and maybe discover drugs even faster. This is another example of patient-specific stem cells that were engineered from someone with retinitis pigmentosa. This is a degeneration of the retina. It ’ s a disease that runs in my family, and we really hope that cells like these will help us find a cure. So some people think that these models sound well and good, but ask,”Well, are these really as good as the rat?”The rat is an entire organism, after all, with interacting networks of organs. A drug for the heart can get metabolized in the liver, and some of the byproducts may be stored in the fat. Don ’ t you miss all that with these tissue-engineered models? Well, this is another trend in the field. By combining tissue engineering techniques with microfluidics, the field is actually evolving towards just that, a model of the entire ecosystem of the body, complete with multiple organ systems to be able to test how a drug you might take for your blood pressure might affect your liver or an antidepressant might affect your heart. These systems are really hard to build, but we ’ re just starting to be able to get there, and so, watch out. But that ’ s not even all of it, because once a drug is approved, tissue engineering techniques can actually help us develop more personalized treatments. This is an example that you might care about someday, and I hope you never do, because imagine if you ever get that call that gives you that bad news that you might have cancer. Wouldn ’ t you rather test to see if those cancer drugs you ’ re going to take are going to work on your cancer? This is an example from Karen Burg ’ s lab, where they ’ re using inkjet technologies to print breast cancer cells and study its progressions and treatments. And some of our colleagues at Tufts are mixing models like these with tissue-engineered bone to see how cancer might spread from one part of the body to the next, and you can imagine those kinds of multi-tissue chips to be the next generation of these kinds of studies. And so thinking about the models that we ’ ve just discussed, you can see, going forward, that tissue engineering is actually poised to help revolutionize drug screening at every single step of the path : disease models making for better drug formulations, massively parallel human tissue models helping to revolutionize lab testing, reduce animal testing and human testing in clinical trials, and individualized therapies that disrupt what we even consider to be a market at all. Essentially, we ’ re dramatically speeding up that feedback between developing a molecule and learning about how it acts in the human body. Our process for doing this is essentially transforming biotechnology and pharmacology into an information technology, helping us discover and evaluate drugs faster, more cheaply and more effectively. It gives new meaning to models against animal testing, doesn ’ t it? Thank you.

Text Sample 457: NEG Dim 3 – 2012 – Score -1.00 – 2012/t002342.txt
This is a thousand-year-old drawing of the brain. It ’ s a diagram of the visual system. And some things look very familiar today. Two eyes at the bottom, optic nerve flowing out from the back. There ’ s a very large nose that doesn ’ t seem to be connected to anything in particular. And if we compare this to more recent representations of the visual system, you ’ ll see that things have gotten substantially more complicated over the intervening thousand years. And that ’ s because today we can see what ’ s inside of the brain, rather than just looking at its overall shape. Imagine you wanted to understand how a computer works and all you could see was a keyboard, a mouse, a screen. You really would be kind of out of luck. You want to be able to open it up, crack it open, look at the wiring inside. And up until a little more than a century ago, nobody was able to do that with the brain. Nobody had had a glimpse of the brain ’ s wiring. And that ’ s because if you take a brain out of the skull and you cut a thin slice of it, put it under even a very powerful microscope, there ’ s nothing there. It ’ s gray, formless. There ’ s no structure. It won ’ t tell you anything. And this all changed in the late 19th century. Suddenly, new chemical stains for brain tissue were developed and they gave us our first glimpses at brain wiring. The computer was cracked open. So what really launched modern neuroscience was a stain called the Golgi stain. And it works in a very particular way. Instead of staining all of the cells inside of a tissue, it somehow only stains about one percent of them. It clears the forest, reveals the trees inside. If everything had been labeled, nothing would have been visible. So somehow it shows what ’ s there. Spanish neuroanatomist Santiago Ramon y Cajal, who ’ s widely considered the father of modern neuroscience, applied this Golgi stain, which yields data which looks like this, and really gave us the modern notion of the nerve cell, the neuron. And if you ’ re thinking of the brain as a computer, this is the **transistor**. And very quickly Cajal realized that neurons don ’ t operate alone, but rather make connections with others that form circuits just like in a computer. Today, a century later, when researchers want to visualize neurons, they light them up from the inside rather than darkening them. And there ’ s several ways of doing this. But one of the most popular ones involves green fluorescent protein. Now green fluorescent protein, which oddly enough comes from a bioluminescent jellyfish, is very useful. Because if you can get the gene for green fluorescent protein and deliver it to a cell, that cell will glow green — or any of the many variants now of green fluorescent protein, you get a cell to glow many different colors. And so coming back to the brain, this is from a genetically engineered mouse called”Brainbow.”And it ’ s so called, of course, because all of these neurons are glowing different colors. Now sometimes neuroscientists need to identify individual molecular components of neurons, molecules, rather than the entire cell. And there ’ s several ways of doing this, but one of the most popular ones involves using antibodies. And you ’ re familiar, of course, with antibodies as the henchmen

of the immune system. But it turns out that they ’ re so useful to the immune system because they can recognize specific molecules, like, for example, the coat protein of a virus that ’ s invading the body. And researchers have used this fact in order to recognize specific molecules inside of the brain, recognize specific substructures of the cell and identify them individually. And a lot of the images I ’ ve been showing you here are very beautiful, but they ’ re also very powerful. They have great explanatory power. This, for example, is an antibody staining against serotonin transporters in a slice of mouse brain. And you ’ ve heard of serotonin, of course, in the context of diseases like depression and anxiety. You ’ ve heard of SSRIs, which are drugs that are used to treat these diseases. And in order to understand how serotonin works, it ’ s critical to understand where the serontonin machinery is. And antibody stainings like this one can be used to understand that sort of question. I ’ d like to leave you with the following thought : Green fluorescent protein and antibodies are both totally natural products at the get-go. They were evolved by nature in order to get a jellyfish to glow green for whatever reason, or in order to detect the coat protein of an invading virus, for example. And only much later did scientists come onto the scene and say,”Hey, these are tools, these are functions that we could use in our own research tool palette.”And instead of applying feeble human minds to designing these tools from scratch, there were these ready-made solutions right out there in nature developed and refined steadily for millions of years by the greatest engineer of all. Thank you.

Text Sample 458: NEG Dim 3 – 2012 – Score -1.00 – 2012/t002449.txt

Good morning. I ’ m here today to talk about autonomous flying beach balls. No, agile aerial robots like this one. I ’ d like to tell you a little bit about the challenges in building these, and some of the terrific opportunities for applying this technology. So these robots are related to unmanned aerial vehicles. However, the vehicles you see here are big. They weigh thousands of pounds, are not by any means agile. They ’ re not even autonomous. In fact, many of these vehicles are operated by flight crews that can include multiple pilots, operators of sensors, and mission coordinators. What we ’ re interested in is developing robots like this — and here are two other pictures — of robots that you can buy off the shelf. So these are helicopters with four rotors, and they ’ re roughly a meter or so in scale, and weigh several pounds. And so we retrofit these with sensors and processors, and these robots can fly indoors. Without GPS. The robot I ’ m holding in my hand is this one, and it ’ s been created by two students, Alex and Daniel. So this weighs a little more than a tenth of a pound. It consumes about 15 watts of power. And as you can see, it ’ s about eight inches in diameter. So let me give you just a very quick tutorial on how these robots work. So it has four rotors. If you spin these rotors at the same speed, the robot hovers. If you increase the speed of each of these rotors, then the robot flies up, it accelerates up. Of course, if the robot were tilted, inclined to the horizontal, then it would accelerate in this direction. So to get it to tilt, there ’ s one of two ways of doing it. So in this picture, you see that rotor four is spinning faster and rotor two is spinning slower. And when that happens, there ’ s a moment that causes this robot to roll. And the other way around, if you increase the speed of rotor three and decrease the speed of rotor one, then the robot pitches forward. And then finally, if you spin opposite pairs of rotors faster than the other pair, then the robot yaws about the vertical axis. So an **on-board** processor essentially looks at what motions need to be executed and combines these motions, and figures out what commands to send to the motors — 600 times a second. That ’ s basically how this thing operates. So one of the advantages of this design is when you scale things down, the robot naturally becomes agile. So here, R is the characteristic length of the robot. It ’ s actually half the diameter. And there are lots of physical parameters that change as you reduce R. The one that ’ s most important is the inertia, or the resistance to motion. So it turns out the inertia, which governs angular motion, scales as a fifth power of R. So the smaller you make R, the more dramatically the inertia reduces. So as a result, the angular acceleration, denoted by the Greek letter alpha here, goes as 1 over R. It ’ s inversely proportional to R. The smaller you make it, the more quickly you can turn. So this should be clear in these videos. On the bottom right, you see a robot performing a 360-degree flip in less than half a second. Multiple flips, a little more time. So here the processes on board are getting feedback from accelerometers and gyros on board, and calculating, like I said before, commands at 600 times a second, to stabilize this robot. So on the left, you see Daniel throwing this robot up into the air, and it shows you how robust the control is. No matter how you throw it, the robot recovers and comes back to him. So why build robots like this? Well, robots like this have many applications. You can send them inside buildings like this, as first responders to look for intruders, maybe look for biochemical leaks, gaseous leaks. You can also use them for applications like construction. So here are robots carrying beams, columns and assembling cube-like structures. I ’ ll tell you a little bit more about this. The robots can be used for transporting cargo. So one of the problems with these small robots is their payload-carrying capacity. So you might want to have multiple robots carry payloads. This is a picture of a recent experiment we did — actually not so recent anymore — in Sendai, shortly after the earthquake. So robots like this could be sent into collapsed buildings, to assess the damage after natural disasters, or sent into reactor buildings, to map radiation levels. So one fundamental problem that the robots have to solve if they are to be autonomous, is essentially figuring out how to get from point A to point B. So this gets a little challenging, because the dynamics of this robot are quite complicated. In fact, they live in a 12-dimensional space. So we use a little trick. We take this curved 12-dimensional space, and transform it into a flat, four-dimensional space. And that four-dimensional space consists of X, Y, Z, and then the yaw angle. And so what the robot does, is it plans what we call a minimum-snap trajectory. So to remind you of physics : You have position, derivative, velocity ; then acceleration ; and then comes jerk, and then comes snap. So this robot minimizes snap. So what that effectively does, is produce a smooth and graceful motion. And it does that avoiding obstacles. So these minimum-snap trajectories in this flat space are then transformed back into this complicated 12-dimensional space, which the robot must do for control and then execution. So let me show you some examples of what these minimum-snap trajectories look like. And in the first video, you ’ ll see the robot going from point A to point B, through an intermediate point. So the robot is obviously capable of executing any curve trajectory. So these are circular trajectories, where the robot pulls about two G ’ s. Here you have overhead motion capture cameras on the top that tell the robot where it is 100 times a second. It also tells the robot where these obstacles are. And the obstacles can be moving. And here, you ’ ll see Daniel throw this hoop into the air, while the robot is calculating the position of the hoop, and trying to figure out how to best go through the hoop. So as an academic, we ’ re always trained to be able to jump through hoops to raise funding for our labs, and we get our robots to do that. So another thing the robot can do is it remembers pieces of trajectory that it learns or is pre-programmed. So here, you see the robot combining a motion that builds up momentum, and then changes its orientation and then recovers. So it has to do this because this gap in the window is only slightly larger than the width of the robot. So just like a diver stands on a springboard and then jumps off it to gain momentum, and then does this pirouette, this two and a half somersault through and then gracefully recovers, this robot is basically doing that. So it knows how to combine little bits and pieces of trajectories to do these fairly difficult tasks. So I want change gears. So one of the disadvantages of these small robots is its size. And I told you earlier that we may want to employ lots and lots of robots to overcome the limitations of size. So one difficulty is : How do you coordinate lots of these robots? And so here, we looked to nature. So I want to show you a clip of Aphaenogaster desert ants, in Professor Stephen Pratt ' s lab, carrying an object. So this is actually a piece of fig. Actually you take any object coated with fig juice, and the ants will carry it back to the nest. So these ants don ' t have any central coordinator. They sense their neighbors. There ' s no explicit communication. But because they sense the neighbors and because they sense the object, they have implicit coordination across the group. So this is the kind of coordination we want our robots to have. So when we have a robot which is surrounded by neighbors — and let ' s look at robot I and robot J — what we want the robots to do, is to monitor the separation between them, as they fly in formation. And then you want to make sure that this separation is within acceptable levels. So again, the robots monitor this error and calculate the control commands 100 times a second, which then translates into motor commands, 600 times a second. So this also has to be done in a decentralized way. Again, if you have lots and lots of robots, it ' s impossible to coordinate all this information centrally fast enough in order for the robots to accomplish the task. Plus, the robots have to base their actions only on local information — what they sense from their neighbors. And then finally, we insist that the robots be agnostic to who their neighbors are. So this is what we call anonymity. So what I want to show you next is a video of 20 of these little robots, flying in formation. They ' re monitoring their neighbors ' positions. They ' re maintaining formation. The formations can change. They can be planar formations, they can be three-dimensional formations. As you can see here, they collapse from a three-dimensional formation into planar formation. And to fly through obstacles, they can adapt the formations on the fly. So again, these robots come really close together. As you can see in this figure-eight flight, they come within inches of each other. And despite the aerodynamic interactions with these propeller blades, they ' re able to maintain stable flight. So once you know how to fly in formation, you can actually pick up objects cooperatively. So this just shows that we can double, triple, quadruple the robots ' strength, by just getting them to team with neighbors, as you can see here. One of the disadvantages of doing that is, as you scale things up — so if you have lots of robots carrying the same thing, you ' re essentially increasing the inertia, and therefore you pay a price ; they ' re not as agile. But you do gain in terms of payload-carrying capacity. Another application I want to show you — again, this is in our lab. This is work done by Quentin Lindsey, who ' s a graduate student. So his algorithm essentially tells these robots how to autonomously build cubic structures from truss-like elements. So his algorithm tells the robot what part to pick up, when, and where to place it. So in this video you see — and it ' s sped up 10, 14 times — you see three different structures being built by these robots. And again, everything is autonomous, and all Quentin has to do is to give them a blueprint of the design that he wants to build. So all these experiments you ' ve seen thus far, all these demonstrations, have been done with the help of motion-capture systems. So what happens when you leave your lab, and you go outside into the real world? And what if there ' s no GPS? So this robot is actually equipped with a camera, and a laser rangefinder, laser scanner. And it uses these sensors to build a map of the environment. What that map consists of are features — like doorways, windows, people, furniture — and it then figures out where its position is, with respect to the features. So there is no global coordinate system. The coordinate system is defined based on the robot, where it is and what it ’ s looking at. And it navigates with respect to those features. So I want to show you a clip of algorithms developed by Frank Shen and Professor Nathan Michael, that shows this robot entering a building for the very first time, and creating this map on the fly. So the robot then figures out what the features are, it builds the map, it figures out where it is with respect to the features, and then estimates its position 100 times a second, allowing us to use the control algorithms that I described to you earlier. So this robot is actually being commanded remotely by Frank, but the robot can also figure out where to go on its own. So suppose I were to send this into a building, and I had no idea what this building looked like. I can ask this robot to go in, create a map, and then come back and tell me what the building looks like. So here, the robot is not only solving the problem of how to go from point A to point B in this map, but it ’ s figuring out what the best point B is at every time. So essentially it knows where to go to look for places that have the least information, and that ’ s how it populates this map. So I want to leave you with one last application. And there are many applications of this technology. I ’ m a professor, and we ’ re passionate about education. Robots like this can really change the way we do K-12 education. But we ’ re in Southern California, close to Los Angeles, so I have to conclude with something focused on entertainment. I want to conclude with a music video. I want to introduce the creators, Alex and Daniel, who created this video. So before I play this video, I want to tell you that they created it in the last three days, after getting a call from Chris. And the robots that play in the video are completely autonomous. You will see nine robots play six different instruments. And of course, it ’ s made exclusively for TED 2012. Let ’ s watch.

Text Sample 459: NEG Dim 3 – 2007 – Score -2.00 – 2007/t000230.txt

I got my first computer when I was a teenager growing up in Accra, and it was a really cool device. You could play games with it. You could program it in BASIC. And I was fascinated. So I went into the library to figure out how did this thing work. I read about how the CPU is constantly shuffling data back and forth between the memory, the RAM and the ALU, the arithmetic and logic unit. And I thought to myself, this CPU really has to work like crazy just to keep all this data moving through the system. But nobody was really worried about this. When computers were first introduced, they were said to be a million times faster than neurons. People were really excited. They thought they would soon outstrip the capacity of the brain. This is a quote, actually, from Alan Turing :”In 30 years, it will be as easy to ask a computer a question as to ask a person.”This was in 1946. And now, in 2007, it ’ s still not true. And so, the question is, why aren ’ t we really seeing this kind of power in computers that we see in the brain? What people didn ’ t realize, and I ’ m just beginning to realize right now, is that we pay a huge price for the speed that we claim is a big advantage of these computers. Let ’ s take a look at some numbers. This is Blue Gene, the fastest computer in the world. It ’ s got 120, 000 processors ; they can basically process 10 quadrillion bits of information per second. That ’ s 10 to the sixteenth. And they consume one and a half megawatts of power. So that would be really great, if you could add that to the production capacity in Tanzania. It would really boost the economy. Just to go back to the States, if you translate the amount of power or electricity this computer uses to the amount of households in the States, you get 1, 200 households in the U. S. That ’ s how much power this computer uses. Now, let ’ s compare this with the brain. This is a picture of, actually Rory Sayres ’ girlfriend ’ s brain. Rory is a graduate student at Stanford. He studies the brain using MRI, and he claims that this is the most beautiful brain that he has ever scanned. So that ’ s true love, right there. Now, how much computation does the brain do? I estimate 10 to the 16 bits per second, which is actually about very similar to what Blue Gene does. So that ’ s the question. The question is, how much — they are doing a similar amount of processing, similar amount of data — the question is how much energy or electricity does the brain use? And it ’ s actually as much as your laptop computer : it ’ s just 10 watts. So what we are doing right now with computers with the energy consumed by 1, 200 houses, the brain is doing with the energy consumed by your laptop. So the question is, how is the brain able to achieve this kind of efficiency? And let me just summarize. So the bottom line : the brain processes information using 100, 000 times less energy than we do right now with this computer technology that we have. How is the brain able to do this? Let ’ s just take a look about how the brain works, and then I ’ ll compare that with how computers work. So, this clip is from the PBS series,”The Secret Life of the Brain.”It shows you these cells that process information. They are called neurons. They send little pulses of electricity down their processes to each other, and where they contact each other, those little pulses of electricity can jump from one neuron to the other. That process is called a synapse. You ’ ve got this huge network of cells interacting with each other — about 100 million of them, sending about 10 quadrillion of these pulses around every second. And that ’ s basically what ’ s going on in your brain right now as you ’ re watching this. How does that compare with the way computers work? In the computer, you have all the data going through the central processing unit, and any piece of data basically has to go through that bottleneck, whereas in the brain, what you have is these neurons, and the data just really flows through a network of connections among the neurons. There ’ s no bottleneck here. It ’ s really a network in the literal sense of the word. The net is doing the work in the brain. If you just look at these two pictures, these kind of words pop into your mind. This is serial and it ’ s rigid — it ’ s like cars on a freeway, everything has to happen in lockstep — whereas this is parallel and it ’ s fluid. Information processing is very dynamic and adaptive. So I ’ m not the first to figure this out. This is a quote from Brian Eno :”the problem with computers is that there is not enough Africa in them.”Brian actually said this in 1995. And nobody was listening then, but now people are beginning to listen because there ’ s a pressing, technological problem that we face. And I ’ ll just take you through that a little bit in the next few slides. This is — it ’ s actually really this remarkable convergence between the devices that we use to compute in computers, and the devices that our brains use to compute. The devices that computers use are what ’ s called a **transistor**. This electrode here, called the gate, controls the flow of current from the source to the drain — these two electrodes. And that current, electrical current, is carried by electrons, just like in your house and so on. And what you have here is, when you actually turn on the gate, you get an increase in the amount of current, and you get a steady flow of current. And when you turn off the gate, there ’ s no current flowing through the device. Your computer uses this presence of current to represent a one, and the absence of current to represent a zero. Now, what ’ s happening is that as **transistors** are getting smaller and smaller and smaller, they no longer behave like this. In fact, they are starting to behave like the device that neurons use to compute, which is called an ion channel. And this is a little protein molecule. I mean, neurons have thousands of these. And it sits in the membrane of the cell and it ’ s got a pore in it. And these are individual **potassium** ions that are flowing through that pore. Now, this pore can open and close. But, when it ’ s open, because these ions have to line up and flow through, one at a time, you get a kind of sporadic, not steady — it ’ s a sporadic flow of current. And even when you close the pore — which neurons can do, they can open and close these pores to generate electrical activity — even when it ’ s closed, because these ions are so small, they can actually sneak through, a few can sneak through at a time. So, what you have is that when the pore is open, you get some current sometimes. These are your ones, but you ’ ve got a few zeros thrown in. And when it ’ s closed, you have a zero, but you have a few ones thrown in. Now, this is starting to happen in **transistors**. And the reason why that ’ s happening is that, right now, in 2007 — the technology that we are using — a **transistor** is big enough that several electrons can flow through the channel simultaneously, side by side. In fact, there ’ s about 12 electrons can all be flowing this way. And that means that a **transistor** corresponds to about 12 ion channels in parallel. Now, in a few years time, by 2015, we will shrink **transistors** so much. This is what Intel does to keep adding more cores onto the chip. Or your memory sticks that you have now can carry one gigabyte of stuff on them — before, it was 256. **Transistors** are getting smaller to allow this to happen, and technology has really benefitted from that. But what ’ s happening now is that in 2015, the **transistor** is going to become so small, that it corresponds to only one electron at a time can flow through that channel, and that corresponds to a single ion channel. And you start having the same kind of traffic jams that you have in the ion channel. The current will turn on and off at random, even when it ’ s supposed to be on. And that means your computer is going to get its ones and zeros mixed up, and that ’ s going to crash your machine. So, we are at the stage where we don ’ t really know how to compute with these kinds of devices. And the only kind of thing — the only thing we know right now that can compute with these kinds of devices are the brain. OK, so a computer picks a specific item of data from memory, it sends it into the processor or the ALU, and then it puts the result back into memory. That ’ s the red path that ’ s highlighted. The way brains work, I told you all, you have got all these neurons. And the way they represent information is they break up that data into little pieces that are represented by pulses and different neurons. So you have all these pieces of data distributed throughout the network. And then the way that you process that data to get a result is that you translate this pattern of activity into a new pattern of activity, just by it flowing through the network. So you set up these connections such that the input pattern just flows and generates the output pattern. What you see here is that there ’ s these redundant connections. So if this piece of data or this piece of the data gets clobbered, it doesn ’ t show up over here, these two pieces can activate the missing part with these redundant connections. So even when you go to these crappy devices where sometimes you want a one and you get a zero, and it doesn ’ t show up, there ’ s redundancy in the network that can actually recover the missing information. It makes the brain inherently robust. What you have here is a system where you store data locally. And it ’ s brittle, because each of these steps has to be flawless, otherwise you lose that data, whereas in the brain, you have a system that stores data in a distributed way, and it ’ s robust. What I want to basically talk about is my dream, which is to build a computer that works like the brain. This is something that we ’ ve been working on for the last couple of years. And I ’ m going to show you a system that we designed to model the retina, which is a piece of brain that lines the inside of your eyeball. We didn ’ t do this by actually writing code, like you do in a computer. In fact, the processing that happens in that little piece of brain is very similar to the kind of processing that computers do when they stream video over the Internet. They want to compress the information — they just want to send the changes, what ’ s new in the image, and so on — and that is how your eyeball is able to squeeze all that information down to your optic nerve, to send to the rest of the brain. Instead of doing this in software, or doing those kinds of algorithms, we went and talked to neurobiologists who have actually reverse engineered that piece of brain that ’ s called the retina. And they figured out all the different cells, and they figured out the network, and we just took that network and we used it as the blueprint for the design of a silicon chip. So now the neurons are represented by little nodes or circuits on the chip, and the connections among the neurons are represented, actually modeled by **transistors**. And these **transistors** are behaving essentially just like ion channels behave in the brain. It will give you the same kind of robust architecture that I described. Here is actually what our artificial eye looks like. The retina chip that we designed sits behind this lens here. And the chip — I ’ m going to show you a video that the silicon retina put out of its output when it was looking at Kareem Zaghloul, who ’ s the student who designed this chip. Let me explain what you ’ re going to see, OK, because it ’ s putting out different kinds of information, it ’ s not as straightforward as a camera. The retina chip extracts four different kinds of information. It extracts regions with dark contrast, which will show up on the video as red. And it extracts regions with white or light contrast, which will show up on the video as green. This is Kareem ’ s dark eyes and that ’ s the white background that you see here. And then it also extracts movement. When Kareem moves his head to the right, you will see this blue activity there ; it represents regions where the contrast is increasing in the image, that ’ s where it ’ s going from dark to light. And you also see this yellow activity, which represents regions where contrast is decreasing ; it ’ s going from light to dark. And these four types of information — your optic nerve has about a million fibers in it, and 900, 000 of those fibers send these four types of information. So we are really duplicating the kind of signals that you have on the optic nerve. What you notice here is that these snapshots taken from the output of the retina chip are very sparse, right? It doesn ’ t light up green everywhere in the background, only on the edges, and then in the hair, and so on. And this is the same thing you see when people compress video to send : they want to make it very sparse, because that file is smaller. And this is what the retina is doing, and it ’ s doing it just with the circuitry, and how this network of neurons that are interacting in there, which we ’ ve captured on the chip. But the point that I want to make — I ’ ll show you up here. So this image here is going to look like these ones, but here I ’ ll show you that we can reconstruct the image, so, you know, you can almost recognize Kareem in that top part there. And so, here you go. Yes, so that ’ s the idea. When you stand still, you just see the light and dark contrasts. But when it ’ s moving back and forth, the retina picks up these changes. And that ’ s why, you know, when you ’ re sitting here and something happens in your background, you merely move your eyes to it. There are these cells that detect change and you move your attention to it. So those are very important for catching somebody who ’ s trying to sneak up on you. Let me just end by saying that this is what happens when you put Africa in a piano, OK. This is a steel drum here that has been modified, and that ’ s what happens when you put Africa in a piano. And what I would like us to do is put Africa in the computer, and come up with a new kind of computer that will generate thought, imagination, be creative and things like that. Thank you. Chris Anderson : Question for you, Kwabena. Do you put together in your mind the work you ’ re doing, the future of Africa, this conference — what connections can we make, if any, between them? Kwabena Boahen : Yes, like I said at the beginning, I got my first computer when I was a teenager, growing up in Accra. And I had this gut reaction that this was the wrong way to do it. It was very brute force ; it was very inelegant. I don ’ t think that I would ’ ve had that reaction, if I ’ d grown up reading all this science fiction, hearing about RD2D2, whatever it was called, and just — you know, buying into this hype about computers. I was coming at it from a different perspective, where I was bringing that different perspective to bear on the problem. And I think a lot of people in Africa have this different perspective, and I think that ’ s going to impact technology. And that ’ s going to impact how it ’ s going to evolve. And I think you ’ re going to be able to see, use that infusion, to come up with new things, because you ’ re coming from a different perspective. I think we can contribute. We can dream like everybody else. CA : Thanks Kwabena, that was really interesting. Thank you.

Text Sample 460: NEG Dim 3 – 2007 – Score -1.00 – 2007/t000229.txt

The Internet, the Web as we know it, the kind of Web â the things we ’ re all talking about â is already less than 5, 000 days old. So all of the things that we ’ ve seen come about, starting, say, with satellite images of the whole Earth, which we couldn ’ t even imagine happening before, all these things rolling into our lives, just this abundance of things that are right before us, sitting in front of our laptop, or our desktop. This kind of cornucopia of stuff just coming and never ending is amazing, and we ’ re not amazed. It ’ s really amazing that all this stuff is here. It ’ s in 5, 000 days, all this stuff has come. And I know that 10 years ago, if I had told you that this was all coming, you would have said that that ’ s impossible. There ’ s simply no economic model that that would be possible. And if I told you it was all coming for free, you would say, this is simply â you ’ re dreaming. You ’ re a Californian utopian. You ’ re a wild-eyed optimist. And yet it ’ s here. The other thing that we know about it was that 10 years ago, as I looked at what even Wired was talking about, we thought it was going to be TV, but better. That was the model. That was what everybody was suggesting was going to be coming. And it turns out that that ’ s not what it was. First of all, it was impossible, and it ’ s not what it was. And so one of the things that I think we ’ re learning â if you think about, like, Wikipedia, it ’ s something that was simply impossible. It ’ s impossible in theory, but possible in practice. And if you take all these things that are impossible, I think one of the things that we ’ re learning from this era, from this last decade, is that we have to get good at believing in the impossible, because we ’ re unprepared for it. So, I ’ m curious about what ’ s going to happen in the next 5, 000 days. But if that ’ s happened in the last 5, 000 days, what ’ s going to happen in the next 5, 000 days? So, I have a kind of a simple story, and it suggests that what we want to think about is this thing that we ’ re making, this thing that has happened in 5, 000 days â that ’ s all these computers, all these handhelds, all these cell phones, all these laptops, all these servers â basically what we ’ re getting out of all these connections is we ’ re getting one machine. If there is only one machine, and our little handhelds and devices are actually just little windows into those machines, but that we ’ re basically constructing a single, global machine. And so I began to think about that. And it turned out that this machine happens to be the most reliable machine that we ’ ve ever made. It has not crashed ; it ’ s running uninterrupted. And there ’ s almost no other machine that we ’ ve ever made that runs the number of hours, the number of days. 5, 000 days without interruption â that ’ s just unbelievable. And of course, the Internet is longer than just 5, 000 days ; the Web is only 5, 000 days. So, I was trying to basically make measurements. What are the dimensions of this machine? And I started off by calculating how many billions of clicks there are all around the globe on all the computers. And there is a 100 billion clicks per day. And there ’ s 55 trillion links between all the Web pages of the world. And so I began thinking more about other kinds of dimensions, and I made a quick list. Was it Chris Jordan, the photographer, talking about numbers being so large that they ’ re meaningless? Well, here ’ s a list of them. They ’ re hard to tell, but there ’ s one billion PC chips on the Internet, if you count all the chips in all the computers on the Internet. There ’ s two million emails per second. So it ’ s a very big number. It ’ s just a huge machine, and it uses five percent of the global electricity on the planet. So here ’ s the specifications, just as if you were to make up a spec sheet for it : 170 quadrillion **transistors**, 55 trillion links, emails running at two **megahertz** itself, 31 kilohertz text messaging, 246 exabyte storage. That ’ s a big disk. That ’ s a lot of storage, memory. Nine exabyte RAM. And the total traffic on this is running at seven terabytes per second. Brewster was saying the Library of Congress is about twenty terabytes. So every second, half of the Library of Congress is swooshing around in this machine. It ’ s a big machine. So I did something else. I figured out 100 billion clicks per day, 55 trillion links is almost the same as the number of synapses in your brain. A quadrillion **transistors** is almost the same as the number of neurons in your brain. So to a first approximation, we have these things â twenty petahertz synapse firings. Of course, the memory is really huge. But to a first approximation, the size of this machine is the size â and its complexity, kind of â to your brain. Because in fact, that ’ s how your brain works â in kind of the same way that the Web works. However, your brain isn ’ t doubling every two years. So if we say this machine right now that we ’ ve made is about one HB, one human brain, if we look at the rate that this is increasing, 30 years from now, there ’ ll be six billion HBs. So by the year 2040, the total processing of this machine will exceed a total processing power of humanity, in raw bits and stuff. And this is, I think, where Ray Kurzweil and others get this little chart saying that we ’ re going to cross. So, what about that? Well, here ’ s a couple of things. I have three kind of general things I would like to say, three consequences of this. First, that basically what this machine is doing is embodying. We ’ re giving it a body. And that ’ s what we ’ re going to do in the next 5, 000 days â we ’ re going to give this machine a body. And the second thing is, we ’ re going to restructure its architecture. And thirdly, we ’ re going to become completely codependent upon it. So let me go through those three things. First of all, we have all these things in our hands. We think they ’ re all separate devices, but in fact, every screen in the world is looking into the one machine. These are all basically portals into that one machine. The second thing is that â some people call this the cloud, and you ’ re kind of touching the cloud with this. And so in some ways, all you really need is a cloudbook. And the cloudbook doesn ’ t have any storage. It ’ s wireless. It ’ s always connected. There ’ s many things about it. It becomes very simple, and basically what you ’ re doing is you ’ re just touching the machine, you ’ re touching the cloud and you ’ re going to compute that way. So the machine is computing. And in some ways, it ’ s sort of back to the kind of old idea of centralized computing. But everything, all the cameras, and the microphones, and the sensors in cars and everything is connected to this machine. And everything will go through the Web. And we ’ re seeing that already with, say, phones. Right now, phones don ’ t go through the Web, but they are beginning to, and they will. And if you imagine what, say, just as an example, what Google Labs has in terms of experiments with Google Docs, Google Spreadsheets, blah, blah, blah â all these things are going to become Web based. They ’ re going through the machine. And I am suggesting that every bit will be owned by the Web. Right now, it ’ s not. If you do spreadsheets and things at work, a Word document, they aren ’ t on the Web, but they are going to be. They ’ re going to be part of this machine. They ’ re going to speak the Web language. They ’ re going to talk to the machine. The Web, in some sense, is kind of like a black hole that ’ s sucking up everything into it. And so every thing will be part of the Web. So every item, every artifact that we make, will have embedded in it some little sliver of Web-ness and connection, and it will be part of this machine, so that our environment â kind of in that ubiquitous computing sense â our environment becomes the Web. Everything is connected. Now, with RFIDs and other things â whatever technology it is, it doesn ’ t really matter. The point is that everything will have embedded in it some sensor connecting it to the machine, and so we have, basically, an Internet of things. So you begin to think of a shoe as a chip with heels, and a car as a chip with wheels, because basically most of the cost of manufacturing cars is the embedded intelligence and electronics in it, and not the materials. A lot of people think about the new economy as something that was going to be a disembodied, alternative, virtual existence, and that we would have the old economy of atoms. But in fact, what the new economy really is is the marriage of those two, where we embed the information, and the digital nature of things into the material world. That ’ s what we ’ re looking forward to. That is where we ’ re going â this union, this convergence of the atomic and the digital. And so one of the consequences of that, I believe, is that where we have this sort of spectrum of media right now â TV, film, video â that basically becomes one media platform. And while there ’ s many differences in some senses, they will share more and more in common with each other. So that the laws of media, such as the fact that copies have no value, the value ’ s in the uncopiable things, the immediacy, the authentication, the personalization. The media wants to be liquid. The reason why things are free is so that you can manipulate them, not so that they are”free”as in”beer,”but”free”as in”freedom.”And the network effects rule, meaning that the more you have, the more you get. The first fax machine â the person who bought the first fax machine was an idiot, because there was nobody to fax to. But here she became an evangelist, recruiting others to get the fax machines because it made their purchase more valuable. Those are the effects that we ’ re going to see. Attention is the currency. So those laws are going to kind of spread throughout all media. And the other thing about this embodiment is that there ’ s kind of what I call the McLuhan reversal. McLuhan was saying,”Machines are the extensions of the human senses.”And I ’ m saying,”Humans are now going to be the extended senses of the machine,”in a certain sense. So we have a trillion eyes, and ears, and touches, through all our digital photographs and cameras. And we see that in things like Flickr, or Photosynth, this program from Microsoft that will allow you to assemble a view of a touristy place from the thousands of tourist snapshots of it. In a certain sense, the machine is seeing through the pixels of individual cameras. Now, the second thing that I want to talk about was this idea of restructuring, that what the Web is doing is restructuring. And I have to warn you, that what we ’ ll talk about is â I ’ m going to give my explanation of a term you ’ re hearing, which is a”semantic Web.”So first of all, the first stage that we ’ ve seen of the Internet was that it was going to link computers. And that ’ s what we called the Net ; that was the Internet of nets. And we saw that, where you have all the computers of the world. And if you remember, it was a kind of green screen with cursors, and there was really not much to do, and if you wanted to connect it, you connected it from one computer to another computer. And what you had to do was â if you wanted to participate in this, you had to share packets of information. So you were forwarding on. You didn ’ t have control. It wasn ’ t like a telephone system where you had control of a line : you had to share packets. The second stage that we ’ re in now is the idea of linking pages. So in the old one, if I wanted to go on to an airline Web page, I went from my computer, to an FTP site, to another airline computer. Now we have pages â the unit has been resolved into pages, so one page links to another page. And if I want to go in to book a flight, I go into the airline ’ s flight page, the website of the airline, and I ’ m linking to that page. And what we ’ re sharing were links, so you had to be kind of open with links. You couldn ’ t deny â if someone wanted to link to you, you couldn ’ t stop them. You had to participate in this idea of opening up your pages to be linked by anybody. So that ’ s what we were doing. We ’ re now entering to the third stage, which is what I ’ m talking about, and that is where we link the data. So, I don ’ t know what the name of this thing is. I ’ m calling it the one machine. But we ’ re linking data. So we ’ re going from machine to machine, from page to page, and now data to data. So the difference is, is that rather than linking from page to page, we ’ re actually going to link from one idea on a page to another idea, rather than to the other page. So every idea is basically being supported â or every item, or every noun â is being supported by the entire Web. It ’ s being resolved at the level of items, or ideas, or words, if you want. So besides physically coming out again into this idea that it ’ s not just virtual, it ’ s actually going out to things. So something will resolve down to the information about a particular person, so every person will have a unique ID. Every person, every item will have a something that will be very specific, and will link to a specific representation of that idea or item. So now, in this new one, when I link to it, I would link to my particular flight, my particular seat. And so, giving an example of this thing, I live in Pacifica, rather than â right now Pacifica is just sort of a name on the Web somewhere. The Web doesn ’ t know that that is actually a town, and that it ’ s a specific town that I live in, but that ’ s what we ’ re going to be talking about. It ’ s going to link directly to â it will know, the Web will be able to read itself and know that that actually is a place, and that whenever it sees that word,”Pacifica,”it knows that it actually has a place, latitude, longitude, a certain population. So here are some of the technical terms, all three-letter things, that you ’ ll see a lot more of. All these things are about enabling this idea of linking to the data. So I ’ ll give you one kind of an example. There ’ s like a billion social sites on the Web. Each time you go into there, you have to tell it again who you are and all your friends are. Why should you be doing that? You should just do that once, and it should know who all your friends are. So that ’ s what you want, is all your friends are identified, and you should just carry these relationships around. All this data about you should just be conveyed, and you should do it once and that ’ s all that should happen. And you should have all the networks of all the relationships between those pieces of data. That ’ s what we ’ re moving into â where it sort of knows these things down to that level. A semantic Web, Web 3. 0, giant global graph â we ’ re kind of trying out what we want to call this thing. But what ’ s it ’ s doing is sharing data. So you have to be open to having your data shared, which is a much bigger step than just sharing your Web page, or your computer. And all these things that are going to be on this are not just pages, they are things. Everything we ’ ve described, every artifact or place, will be a specific representation, will have a specific character that can be linked to directly. So we have this database of things. And so there ’ s actually a fourth thing that we have not get to, that we won ’ t see in the next 10 years, or 5, 000 days, but I think that ’ s where we ’ re going to. And as the Internet of things â where I ’ m linking directly to the particular things of my seat on the plane â that that physical thing becomes part of the Web. And so we are in the middle of this thing that ’ s completely linked, down to every object in the little sliver of a connection that it has. So, the last thing I want to talk about is this idea that we ’ re going to be codependent. It ’ s always going to be there, and the closer it is, the better. If you allow Google to, it will tell you your search history. And I found out by looking at it that I search most at 11 o ’ clock in the morning. So I am open, and being transparent to that. And I think total personalization in this new world will require total transparency. That is going to be the price. If you want to have total personalization, you have to be totally transparent. Google. I can ’ t remember my phone number, I ’ ll just ask Google. We ’ re so dependent on this that I have now gotten to the point where I don ’ t even try to remember things â I ’ ll just Google it. It ’ s easier to do that. And we kind of object at first, saying,”Oh, that ’ s awful.”But if we think about the dependency that we have on this other technology, called the alphabet, and writing, we ’ re totally dependent on it, and it ’ s transformed culture. We cannot imagine ourselves without the alphabet and writing. And so in the same way, we ’ re going to not imagine ourselves without this other machine being there. And what is happening with this is some kind of AI, but it ’ s not the AI in conscious AI, as being an expert, Larry Page told me that that ’ s what they ’ re trying to do, and that ’ s what they ’ re trying to do. But when six billion humans are Googling, who ’ s searching who? It goes both ways. So we are the Web, that ’ s what this thing is. We are going to be the machine. So the next 5, 000 days, it ’ s not going to be the Web and only better. Just like it wasn ’ t TV and only better. The next 5, 000 days, it ’ s not just going to be the Web but only better â it ’ s going to be something different. And I think it ’ s going to be smarter. It ’ ll have an intelligence in there, that ’ s not, again, conscious. But it ’ ll anticipate what we ’ re doing, in a good sense. Secondly, it ’ s become much more personalized. It will know us, and that ’ s good. And again, the price of that will be transparency. And thirdly, it ’ s going to become more ubiquitous in terms of filling your entire environment, and we will be in the middle of it. And all these devices will be portals into that. So the single idea that I wanted to leave with you is that we have to begin to think about this as not just”the Web, only better,”but a new kind of stage in this development. It looks more global. If you take this whole thing, it is a very big machine, very reliable machine, more reliable than its parts. But we can also think about it as kind of a large organism. So we might respond to it more as if this was a whole system, more as if this wasn ’ t a large organism that we are going to be interacting with. It ’ s a”One.”And I don ’ t know what else to call it, than the One. We ’ ll have a better word for it. But there ’ s a unity of some sort that ’ s starting to emerge. And again, I don ’ t want to talk about consciousness, I want to talk about it just as if it was a little bacteria, or a volvox, which is what that organism is. So, to do, action, take-away. So, here ’ s what I would say : there ’ s only one machine, and the Web is its OS. All screens look into the One. No bits will live outside the Web. To share is to gain. Let the One read it. It ’ s going to be machine-readable. You want to make something that the machine can read. And the One is us. We are in the One. I appreciate your time.

Text Sample 461: NEG Dim 3 – 2007 – Score 1.00 – 2007/t000373.txt

I ’ ve always wanted to be a cyborg. One of my favorite shows as a kid was”The Six Million Dollar Man,”and this is a little bit closer to the 240-dollar man or so, but â At any rate, I would normally feel very self-conscious and geeky wearing this around, but a few days ago I saw one world-renowned statistician swallowing swords on stage here, so I figure it ' s OK amongst this group. But that ' s not what I want to talk about today. I want to talk about toys, and the power that I see inherent in them. When I was a kid, I attended Montessori school up to sixth grade in Atlanta, Georgia. And at the time I didn ' t think much about it, but then later, I realized that that was the high point of my education. From that point on, everything else was pretty much downhill. And it wasn ' t until later, as I started making games, that â I really actually think of them more as toys. People call me a game designer, but I really think of these things more as toys. I started getting very interested in Maria Montessori and her methods, and the way she went about things, and the way she thought it very valuable for kids to discover things on their own rather than being taught these things overtly. And she would design these toys, where kids in playing with the toys would come to understand these deep principles of life and nature through play. And since they discovered this, it stuck with them so much more, and also they would experience their own failures. There was a failure-based aspect to learning there. It was very important. And so, the games that I do, I think of really more as modern Montessori toys. And I really kind of want them to be presented in a way to where kids can kind of explore and discover their own principles. So a few years ago, I started getting very interested in the SETI program. And that ' s the way I work. I get interested in different subjects, I dive in, research them, and then try to figure out how to craft a toy around that, so that other people can experience the same sense of discovery that I did as I was learning that subject. And it led me to astrobiology, the study of possible life in the universe. And then Drake ' s equation, looking at the probability of life arising on planets, how long it might last, how many planets are out there. And I started looking at how interesting Drake ' s equation was, because it spanned all these different subjects â physics, chemistry, sociology, economics, astronomy. And another thing that really impressed me a long time ago was"Powers of Ten,"Charles and Ray Eames ' film. And I started putting those two together and wondering, could I build a toy where kids would trip across all these interesting principles of life, as it exists and as it might go in the future. Things where you might trip across things like the Copernican principle, the Fermi paradox, the anthropic principle, the origin of life. And so I ' m going to show you a toy today that I ' ve been working on, that I think of more as kind of a philosophy toy. Playing this toy will bring up philosophical questions in you. The game is"Spore."I ' ve been working on it for several years. It ' s getting pretty close to finished now. It occurs at all these different scales, from very small to very large. I ' m going to pop in at the start of the game. And you actually start this game in a drop of water, as a very, very small single-cell creature, and right off the bat you basically just have to live, survive, reproduce. So here we are, at a very microscopic scale, swimming around. And I actually realize that cells don ' t have eyes, but it helps to make it cute. The players are going to play through every generation of this species, and as you play the game, the creature is actually growing bit by bit. And as we start growing, the camera will actually start zooming out, and things that you see in the background there will start slowly pulling into the foreground, showing you a little bit of what you ' ll be interacting with as you grow. So as we eat, the camera starts pulling out, and then we start interacting with larger and larger organisms. We actually play through many generations here, at the cellular scale. I ' m going to skip ahead here. At some point we get larger, and we actually get to a macroevolution scale. Now, at this point we ' re leaving the water, and one thing that ' s kind of important about this game is that, at every level, the player is designing their creature, and that ’ s a fundamental aspect of this. Now, in the evolution game here, the creature scale, basically you have to eat, survive, and then reproduce. You know, very Darwinian. One thing we noticed with”The Sims,”a game I did earlier, is that players love making stuff. When they were able to make stuff in the game, they had a lot of empathy in connection to it. Even if it wasn ’ t as pretty as a professional artist would make for games, it really stuck with them and they really cared about what would happen to it. At this point, we ’ ve left the water, and now with this little creature â we could bring up the volume a little bit â and now we might try to eat. We might sneak up on this little guy over here maybe, and try and eat him. OK, well, we fight. OK, we got him. Now we get a meal. So really, at this part of the game, what we ’ re doing is we ’ re running around and surviving, and also getting to the next generation, because we ’ re going to play through every generation of this creature. We can mate, so I ’ m going to see if one of these creatures wants to mate with me. Yeah. We didn ’ t want to replay actual evolution with humans and all that, because it ’ s almost more interesting to look at alternate possibilities in evolution. Evolution is usually presented as this one path that we took through, but really it represents this huge set of possibilities. Now, once we mate, we click on the egg. And this is where the game starts getting interesting, because one of the things we really focused on here was giving the players very high-leverage tools, so that for very little effort, the player can make something very cool. And it involves a lot of intelligence on the tool side. But basically, this is the editor where we ’ re going to design the next generation. So it has a little spine. I can move around, I can extend. I can also inflate or deflate it with the mouse wheel, so you sculpt it like clay. We have parts here that I can put on or pull off. The idea is that the player can basically design anything they can think of in this editor, and we ’ ll basically bring it to life. So I might put some limbs on the character here. I ’ ll inflate them kind of large. And in this case I might decide I ’ m going to put â I ’ ll put mouths on the limbs. So pretty much players are encouraged to be very creative in the game. Here, I ’ ll give it one eye in the middle, maybe scale it up a bit. Point it down. And I ’ ll also give it a few legs. So in some sense we want this to feel like an amplifier for the player ’ s imagination, so that with a very small number of clicks a player can create something that they didn ’ t really think was possible before. This is almost like designing something like Maya â that an eight- year-old can use. But really the goal here was, within about a minute, I wanted somebody to replicate what typically takes a pictorial artist several weeks to create. OK, now I ’ ll put some hands on it. OK, so here I ’ ve basically thrown together a little creature. Let me give it a little weapon on the tail here, so it can fight. OK, so that ’ s the complete model. Now we can actually go to the painting phase. At this phase, the program has some understanding of the topology of this creature. It knows where the backbone is, where the spine, the limbs are, how stripes should run, how it should be shaded. And so we ’ re procedurally generating the texture map, something a texture artist would take many days to work on. And then we can test it out, and see how it would move around. And so at this point the computer is procedurally animating this creature. It ’ s looking at whatever I ’ ve designed. It will actually bring it to life. And I can see how it might dance. How it might show emotions, how it might fight. So it ’ s acting with its two mouths there. I can even have it pose for a photo. Snap a little photo of it. So then I bring this back into the game. It ’ s born, and I play the next generation of my creature through evolution. Now again, the empathy that the players have when they create the content is tremendous. When players create content in this game, it ’ s automatically sent up to a server and then redistributed to all the other players transparently. So in fact, as I ' m interacting in this world with other creatures, these creatures are transparently coming from other players as they play. So the process of playing the game is a process of building up this huge database of content. And pretty much everything you ' re going to see in this game, there ' s an editor for that the player can create, up through civilization. This is my baby. When I eat, I ' ll actually start growing. This is the next generation. But I ' m going to skip ahead here. Normally what would happen is these creatures would work their way up, eventually become intelligent. I ' d start dealing with tribes, cities and civilizations of them over time. I ' m going to skip way ahead to the space phase. Eventually they would go out into space, and start colonizing and exploring the universe. Now, really, in some sense, I want the players to be building this world in their imagination, and then extracting it from them with the least amount of pain. So that ' s kind of what these tools are about. How do we make the gameplay, you know, basically the player ' s imagination amplifier? And how do we make these tools, these editors, something that are just as fun as the game itself? So this is the planet that we ' ve been playing on up to this point in the game. So far the entire game has been played on the surface of this little world here. At this point we ' re actually dealing with a very little toy planet. Almost, again, like the Montessori toy idea. What happens if you give somebody a toy planet, and let them play with a lot of dynamics on it? What could they discover? What might they learn on this? This world was actually extracted from the player ' s imagination. So, this is the planet that the player evolved on. Things like the buildings, the vehicles, the architecture, civilizations were all designed by the player up to this point. So here ' s a little city with some of our guys walking around in it. And most games put the player in the role of Luke Skywalker, this protagonist playing through this story. This is more about putting the player in the role of George Lucas. I want them, after they ' ve played this game, to have extracted an entire world that they ' re now interacting with. As we pull down here, we still have a whole set of creatures living on the surface of the planet. All these different dynamics going on here. I can look over here, and this is a little simplified food web that ' s going on with the creatures. I can open this up and then scan what exists on the surface. You get some sense of the diversity of creatures that were brought in. Some of these were created by the player, others by other players, automatically sent over here. But there ' s a very simple calculation of what ' s required, how many plants are required for the herbivores to live, how many herbivores for the carnivores to eat, etc., that you have to balance actively. Also with this phase, we ' re getting more and more God-like powers for the player, and you can experiment with this planet as a toy. So I can come in and I can do things, and just treat this planet as a lump of clay. We have very simple weather systems, simple geology. For instance, I could open one of my tools here and then carve out rivers. So this whole thing is kind of like a big lump of clay, a sculpture. I can also play with the dynamics in this world over time. So one of the things I can do is start pumping more CO2 gases into the atmosphere, and so that ' s what I ' m doing here. There ' s actually a little readout down there of our planetary atmosphere, pressure and temperature. So as I start pumping in more atmosphere, we ' re going to start pushing up the greenhouse gases here and if you ' ll start noticing, we start seeing the ocean levels rise over time. And our cities are going to be at risk too, because a lot of these are coastal cities. You can see the ocean levels are rising now and as they encroach upon the cities, I ' ll start losing cities here. So basically, I want the players to be able to experiment and explore a huge amount of failure space. So there goes one city. Now, over time, this is going to heat up the planet. So at first what we ' re going to see is a global ocean rise here on this little toy planet, but then over time â I can speed it up â we ’ ll see the heat impact of that as well. Not only will it get hotter, but at some point it ’ s going to get so hot the oceans will evaporate. They ’ ll go up, and then they ’ ll evaporate, and that ’ ll be my planet. So basically, what we ’ re getting here is the sequel to”An Inconvenient Truth,”in about two minutes, and that actually brings up an interesting point about games. Now here, our entire oceans are evaporating off the surface, and as it keeps getting hotter, at some point the entire planet is going to melt down. Here it goes. So we ’ re not only simulating biological dynamics â food webs and all that â but also geologic, you know, on a very simple core scale. And what ’ s interesting to me about games is that I think we can take a lot of long-term dynamics and compress them into very short-term experiences. Because it ’ s so hard for people to think 50 or 100 years out, but when you can give them a toy, and they can experience these long-term dynamics in just a few minutes, I think it ’ s an entirely different point of view, where we ’ re actually using the game to remap our intuition. It ’ s almost in the same way that a telescope or microscope recalibrates your eyesight ; I think computer simulations can recalibrate your instinct across vast scales of both space and time. So here ’ s our little solar system, as we pull away from our melted planet here. We actually have a couple of other planets in this solar system. Let ’ s fly to another one. We ’ re going to have this unlimited number of worlds you can explore here. As we move into the future, and we start going out into space and doing stuff, we ’ re drawing a lot from things like science fiction. And all my favorite science fiction movies I want to play out here as different dynamics. This planet actually has some life on it. Here it is, some indigenous life down here. One of the tools I can eventually earn for my UFO is a monolith that I can drop down. Now, as you can see, these guys are actually starting to go up and bow to it, and over time, once they touch it, they will become intelligent. So I can actually pick a species on a planet and then make them sentient. Now they ’ ve actually gone to tribal dynamics. And now, because I ’ m actually the one here, I can get out of the UFO and walk up, and they should be worshipping me at this point as a god. At first they ’ re a little freaked out. OK, well maybe they ’ re not worshipping me. I think I ’ ll leave before they get hostile. But we basically want a diversity of activities the players can play through this. I want to be able to play”The Day the Earth Stood Still,””2001 : A Space Odyssey,””Star Trek,””War Of the Worlds.”Now, as we pull away from this world â we ’ re going to keep pulling away from the star now. One of the things that always frustrated me about astronomy when I was a kid is how it was always presented so two-dimensionally and so static. As we pull away from the star here, we ’ re actually going now out into interstellar space, and we ’ re getting a sense of the space around our home star. What I really wanted to do is to present this, basically, as wonderfully 3D as it is actually is. And also show the dynamics, and a lot of the interesting objects that you might find, like, in the Hubble, at pretty much realistic frequencies and scales. So most people have no idea of the difference between an emission nebula and a planetary nebula. But these are the things that we can put in this little galaxy here. So we ’ re flying over here to what looks like a black hole. I want to basically have the entire zoo of Hubble objects that people can interact with and play with, again, as toys. So here ’ s a little black hole that we probably don ’ t want to get too close to. But we also have stars and things as well. If we pull all the way back, we start seeing the entire galaxy here, kind of slowly in motion. Typically, when people present galaxies, it ’ s always beautiful photos, but they ’ re always static. And when you bring it forward in time and start animating it, it ’ s amazing what a galaxy would look like fast forwarded. This would be about a million years a second, because you have about one supernova every century. And so you ’ d have this wonderful sparkling thing, with the disk slowly rotating, and this is roughly what it would look like. Part of this is about bringing the beauty of the natural world to somebody in a very imaginative way, so that they can start calibrating their instinct across these vast scales of space and time. Chris was wondering what kind of gods the players would become. Because if you think about it, you ’ re going to have 15-year-olds, 20-year-olds flying around this universe. They might be a nurturing god. They might be bootstrapping life on planets, trying to terraform and spread civilization. You might be a vengeful god, conquesting, because you actually can do that, you can attack other intelligent races. You might be a networking god, building alliances, or just curious, going around and wanting to explore as much as you possibly can. But basically, the reason why I make toys like this is because I think if there ’ s one difference I could possibly make in the world, that I would choose to make, it ’ s that I would like to somehow give people just a little bit better calibration on long-term thinking. Because I think most of the problems that our world is facing right now are the result of short-term thinking, and the fact that it is so hard for us to think 50, 100, or 1, 000 years out. And I think by giving kids toys like this and letting them replay dynamics, very long-term dynamics over the short term, and getting some sense of what we ’ re doing now, what it ’ s going to be like in 100 years, I think probably is the most effective thing I can be doing to help the world. And so that ’ s why I think that toys can change the world. Thank you.

Text Sample 462: NEG Dim 3 – 2023 – Score -1.00 – 2023/t003211.txt

I ’ m here to tell you that what we sense, meaning what we see, smell, hear, taste and touch, can impact how long we live. Now this may seem extraordinary, but it really isn ’ t. Let ’ s say that you wake up one morning and are trying to decide whether to go to an **amusement** park. One of the very first things you might do is to step outside and check the weather. When you do, you see that the sky is blue, that the sun is out. You hear that the birds are chirping. You feel how warm it is on your skin. Based on all of this information, you decide it is a perfect day for going to the park. You have used your senses to impact your decision. Now gathering this information does more than simply impact your behavior. It also can affect your emotions as well as how your body physically responds. To demonstrate, let ’ s say you ’ ve arrived at that **amusement** park, and one of the very first things that you see is a roller coaster. You see the speed with which it moves. You see the twists and turns it takes. You see the terrified look on the riders ’ faces. You hear their screams. Even right here, right now, this sensory event may be causing you to feel anxious or even excited, feelings that are associated with butterflies in your stomach, a racing heartbeat, even sweating palms. Your senses have caused your body to change. This is an example of how a very short sensory perceptive event can impact you. What do you imagine a much longer sensory event might do to you? This is the question that I am interested in addressing. And what scientists like me have discovered is that prolonged sensory events can impact the lifespan of an animal. This has been shown with worms, flies, mice. When you expose them to pheromones, it impacts how long they live. Now at this point, you might be asking yourself, um, what does a worm, a fly or a mouse have to do with me? I would argue more than you realize. The genes or genetic material that you find within a worm or within a fly, that is approximately 60 to 75 percent similar to our own. This increases to 85 percent if we compare ourselves to a mouse. Furthermore, the fact that three very different animals all respond in terms of changing their lifespan when exposed to a specific environmental cue, this suggests that those underlying biological processes that drive those changes are similar. Therefore, there is every reason to believe that environmental cues impact our own very human aging process. In order to study environmental cues and how they impact lifespan, our lab has decided to focus on the nervous system. Now why the nervous system? Well, we can think of the nervous system as a communication hub. It ’ s responsible for gathering that sensory information, for processing it and then for causing those bodily changes that occur. Furthermore, to do this research, we need to control the environment that an animal is in for its entire life. This is something that would be pretty difficult at best, and unethical at worst to do using humans. I therefore use this little guy, Drosophila melanogaster, also known as the fruit fly or the pest that came home on your bananas from the grocery store, in order to understand the things that we share in common that impact lifespan. Now fruit flies are amazing for this type of research. They live a short amount of time. They only live two or three months. They have a brain and a central nervous system. And we can easily activate or inhibit specific neuron cells or even the activities of specific molecules in order to see how these changes impact how long the animal lives. Now I am the first to admit that my belief of how great it is to work with fruit flies is biased. But that ’ s because I ’ m a fly girl. So our lab has studied how very many different environmental cues have impacted the lifespan of this animal. I ’ m going to highlight one in particular, and that is death perception. Yes, I said death. So why are we interested in whether a fly can recognize other dead flies? We ’ re interested in this so that we can develop therapies for people who find themselves developing negative health consequences as a result of stressful situations surrounding death. If you think about this, this could impact soldiers and first responders. You ’ re thinking,”What? A fly can actually recognize another dead fly?”The answer is yes. If I take a vial of flies and I put dead in there, they send out a signal to other flies to stay away. We can measure this. You know those flies in the vial with the dead? They lose weight and they die sooner. Now that we know that there ’ s all these effects, we wanted to make sure this was real. So, you know, the first thing that you might imagine is that, well, those flies with the dead, they ’ re getting an infection. They ’ re getting sick, and that ’ s what ’ s causing them to die. But we know this isn ’ t the case, because if I do the experiment under **sterile** or infection-free conditions, the live flies, they still die when they ’ re with the dead. But if you take that vial of flies and you put them in complete darkness for 24 hours, all the time, they never see the light, those flies live just as long as flies that never had dead in their vial to begin with. This indicates that it is the sight of the dead that is causing all those biological changes, which impacts their lifespan. So now that we know that flies can sense other dead, we turned our attention toward trying to understand a little bit more about those biological processes that are happening. What are the changes? And what we discovered was that molecules found in our own bodies, including serotonin and insulin, had a role. Furthermore, we discovered parts of the fly brain that played a role in how long they live when exposed to dead that have similarities with regions of our own brains. This is what we know about just one type of environmental cue. There are many different environmental cues that impact the lifespan of animals. These include the temperature that the animal is kept in. These include the smell of food, the wavelength of light they are exposed to, even the perception of pain. In order for us to develop therapies that promote healthy aging, we must understand the basic biology of all of these cues, because after all, we don ’ t live in a world where we receive one cue at a time. So what does this all mean? I would love nothing more than to be able to tell you that you should expose yourself to this environmental cue, you should avoid that one in order to live your longest, healthiest life. But I simply can ’ t do that yet. We ’ re just not there. But we ’ re pushing hard towards this type of understanding, and we ’ re going to do it with these little guys. After all, six Nobel Prizes have been won based on work with Drosophila that impacts the biology and helps us all, including worms, flies, mice, and yes, humans. And on that note, I ’ d like to leave you with a happy perceptive experience in the hopes that this positively impacts your life. Thank you.

Text Sample 463: NEG Dim 3 – 2023 – Score -1.00 – 2023/t003439.txt
I want you to envision a single piece of artwork generated by artificial intelligence. When most of us think of AI art, I bet we ’ re imagining something like this. We ’ re all probably picturing something totally different. Today, with machine learning models like DALL-E, Stable Diffusion and Midjourney, we ’ ve seen AI produce everything from strange life forms to imaginary influencers to entirely foreign, curious kinds of imagery. AI as a technology is fascinating to us because we ’ re inherently drawn to things we cannot understand. And with neural networks processing data from thousands of other images made by people from every possible generation, every art movement, millions of images in one simple scan, they can produce visuals that are so familiar yet strikingly unfamiliar. More poetically, AI mirrors us. The world is beginning to change right before our very eyes, and it ’ s basically divided into two schools of thought. There are pessimists who think AI poses a great threat to human creativity. And then optimists who see it as an extension of our creativity. So is it even possible to be truly original as an artist anymore? How do we begin to critically engage with artworks made by machines? We can start by looking at some metaphors, narratives and insights from artists who are truly pushing the boundaries of AI. Let ’ s look to these moments of delight, surprise, confusion and wonder that give us just one small glimpse into the possibilities of encounter with this technology. Because as we ’ ve seen, this is a very moral and ethical encounter as much as an aesthetic one. Mario Klingemann sold this piece on auction in 2019. It is running an AI model trained on thousands of portraits from the 17th to 19th centuries. The model constantly reveals uncanny interpretations of the human face. Each one is unique, generated in real time as the machine reads its own output. For the viewer, it ’ s almost like peering into the machine ’ s hallucinations as it conjures each new portrait. Sofia Crespo ’ s series”Neural Zoo”uses neural network interpretations of the real world to generate unreal sea creatures and diverse biological forms. Frogs look like flowers. Translucent jellyfish have vivid internal organs. There ’ s no one real creature in these images, but AI allows us to envision otherworldly lifeforms in impossible detail. This abstract piece by Sara Ludy began as a digital painting. It was augmented to fit a 16-by-9 ratio, using a prompt for”torn edges”in DALL-E 2 ’ s Outpainting. Outpainting allows artists to extend their creativity beyond the frame using simple language prompts like”torn edges.”This piece by Ivona Tau might read as a photograph, but it is also the work of AI. It ’ s the result of GAN training on thousands of images from the artist ’ s personal photo collection. Tau curates from her own photographs, carefully choosing the inputs and outputs for the model. In many ways, AI art is a form of curation. It becomes the process of selecting from hundreds of images at a time. This video pulls from models trained on a massive data set of Tau ’ s photos, resulting in a kind of algorithmic memory. But she also created a destructed data set for the model to **symbolize** forgetting or fleeting memory. And finally, we have Claire Silver.

Silver has called herself a”collaborative AI artist”in that she works intentionally with the machine to produce her art. Her process is constantly evolving as the tools evolve. She often works with inpainting techniques, masking and transforming just one small piece of an image. For this portrait, she shifted the opacity of various sections with an Apple pencil, transforming it bit by bit. She likens this technique to her version of glazing in oil painting. Silver feeds AI-generated images from one model into another, effectively creating new forms of language and understanding for the machine itself. Her work is half master painting, half digital art. Both old and new. This piece pulls inspiration from famed artists like John Singer Sargent, Evelyn De Morgan and Gustav Klimt, almost as an homage. Because different AI models are trained on different sets of information, it ’ s almost like they ’ re all speaking different languages. AI is everywhere now. We are all now collectively co-creating with AI, whether we ’ re aware of it or not. If we want to be a part of these worlds, we cannot design alone. If we want to be culturally literate in these new kinds of images and predictions and forms, then looking to the work of artists is a very productive place to start. We need to brace ourselves for an increasingly technological future, which is only going to multiply all the creative possibilities at our fingertips now. Thank you.

Text Sample 464: NEG Dim 3 – 2023 – Score -1.00 – 2023/t003429.txt

Twenty-five hundred years ago, Heraclitus said,”The world bubbles forth.”Here ’ s bubbling. What is actual now enables what is next possible, the adjacent possible. The biosphere has been bubbling forth for four billion years, creating new possibilities in the universe in that bubbling. It ’ s critical that physics cannot talk at all about bubbling new bubbles. Life on Earth started four billion years ago. For the first 2. 5 billion years were single-celled organisms co-evolving. Even by existing, organisms create new possibilities. If I ’ m a bacterium and you ’ re a bacterium and I have a flat surface, I ’ m a possible highway you can walk over to get food. That ’ s bubbling forth. So 2. 5 billion years after life started, six times multicell organisms emerged. A billion years after that, 3. 5 billion years after life started, is the extraordinary Cambrian explosion. 540 million years ago, for 50 million years, there was just this burst of creativity, making all of the phyla that exist now. And now we have this guy. I ’ m scared of having him in bed, I really am. So I don ’ t know what to call him. It ’ s a vote afterwards. So here ’ s a gentler guy and I ’ m in love with him. Look at his eyes looking at you, you know.”You ’ re doing what?”In order to talk about how the biosphere has been creative, I have to talk about the functions of parts of an organ. You all know your heart keeps you alive, but in particular, your heart pumps blood. It ’ s not by making heart sounds. So this means something fundamental. The function of a part is that subset of its causal properties that sustain the whole. Your heart pumping blood and you. But doing that is jury-rigging, finding subsets of causal features of use. Some years ago, my daughter dropped her purse in a shallow well. I took a wire coat hanger, slid it over a mop handle, and I fished out her purse. Anything can be used for more than one thing. For example, a screwdriver can be used to screw in a screw, but it ’ s great for scraping putty off the wall. An engine block can have eight holes drilled and you make an engine out of it. It ’ s actually a superb paperweight, and much research done only by scientists demonstrates that the corners are sharp and can crack open a coconut. I was very proud of that. So. But if you ’ re using an engine block as a paperweight, can you deduce, oh, I can use this thing to crack open coconuts. You can ’ t. It might be a banana peel. But that means also something fundamental. Finding new uses for things, jury-rigging is not a computation. It ’ s not a deduction. Jury-rigging is not a computation at all. Computers are. The biosphere is forever innovating in this kind of way that ’ s not a deduction. For example, there are things called Darwinian preadaptations. Dinosaurs had scales that were evolved for thermoregulation that were jury-rigged into flight feathers. Your eyes have clear crystallins that were perfectly normal-colored enzymes. And this, too, means something fundamental. We cannot deduce what is in the adjacent possible that the evolving biosphere will create, then become. We do not even know what can happen. In fact, we can use no mathematics based on set theory, which is all of mathematics, to deduce what the biosphere is going to become, or the economy. This means we are at a fantastic third-phase transition in science, we ’ re beyond Newton, we ’ re beyond quantum mechanics. We ’ re beyond the exquisite view that Copernicus had 480 years ago, that the world is a clockwork. The biosphere is not a clockwork. The paper that just published, it came out three days ago, and I ’ m pretty proud of it, it ’ s actually worth reading. We can ’ t deduce what is in the biosphere and will become, but we can make a mathematical theory of the statistics of the process. I call it the Theory of the Adjacent Possible or TAP. It ’ s really based on one idea. Things can be combined to make new things. So there is the equation. Mt is the number of things in the economy at some time now, say 10 things. So how many things will be in the economy in the next period? Well, if the 10 things work, we ’ ll keep them, but we could actually try to jury-rig something new out of any single thing among the 10 or out of pair of things among the 10. The printing press is a **recombination** between movable type and a wine press or any three things or four things. Watch what happens to the number of things, pairs of things, when the number of things goes up. If there ’ s 10 things, there ’ s 45 pairs of things that we might fiddle with. If there ’ s 100 things, there ’ s 4, 500 things we might jury-rig with. And if there ’ s 1, 000 things, there ’ s a half a million things we might fiddle with. Therefore, this process, the TAP process, has the property that for a long time the number of things increases very, very, very slowly, then something stunning happens. There ’ s a hockey-stick explosion and the number of things reaches infinity in a finite time. There ’ s a singularity and a brief burst explosion for 40 million years of the Cambrian explosion. The same pattern is showing up in our own evolution. Australopithecus flourished – I ’ d never know what flourishing means, I guess they had a good lunch – in Africa, 3. 3 million years ago. A million and a half years later, Homo erectus had about 20 crude stone tools. Watch how slowly things change. A mere 30, 000 years ago, Cro-Magnon in the south of France had about 100 or 150 of these exquisite pressure flake tools. A mere 28, 000 years after that is Mesopotamia with needles and chariots. There ’ s something interesting about the fact – of course, we invented writing, which is a wonderful story – once you have a clay tablet, it calls forth the tool that ’ s its complement. You need a stylus. So tools make niches for new tools and for new jobs. Scribes earned a living for quite a while. This is us in the late Middle Ages to now, and this is us now. We have billions of tools ranging from knitting needles to the space station. There ’ s always an adjacent possible. Once you ’ ve made a bow, a crossbow is in the adjacent possible. The pattern that we saw in the Cambrian, of a long period nothing happening and then a burst, is here right now. This is the size of human habitats in the last 300, 000 years. The horizontal line is time before present, and the vertical line as the logarithm of sizes of settlements. Nothing happened for 290, 000 years then it burst upward. Watch this, the same thing is true of GDP. GDP was flat since the time of Christ. In fact, for the last 100, 000 years. GDP burst up in the last two centuries. It ’ s going up vertically now. It ’ s wonderful. We ’ re lifting millions from poverty. And the theory explains it. It ’ s the same delay and burst, and there it is. We start making more complex things, selling them, and we make a profit. There ’ s another feature of all of this that I ’ ve lived in my lifetime, I ’ m 83. I was born in 1939. Just like the Cambrian explosion, watch the number of things that have come into existence in my lifetime. We didn ’ t have any helicopters, we didn ’ t have television. I listened to”The Lone Ranger,”which maybe you know, on radio, six o ’ clock at night. We didn ’ t have... computers, we didn ’ t have plastics, which are now all over the planet. We didn ’ t have microwaves, we didn ’ t have lasers. Look at the number of things that came into existence in one lifetime of 83 years. Not much happened 50, 000 years ago in 83 years. So what ’ s going on? The TAP process has the following property. Every time you make something new, the waiting time for the next new thing is cut in half. A thousand years, 500 years, 250, a year, six months, three months, we ’ re going to get a great acceleration. We are. So this is TAP. And watch that thing explode upward. That ’ s it. That explosion is the Cambrian, and it ’ s where we are now. And the signatures of the Anthropocene are exploding upward, now all over the place. This ends 25 years ago. We need to find a better adjacent possible. We ’ re rampaging over the planet, and the hope is in soils. So I ’ m going to talk briefly about compost. There are very fine compost, this is now terribly important. A very good compost is the Johnson-Su compost. Less land use, fewer degradations of... of our forests, fewer extinction events, fewer pandemics. Well, let ’ s get rid of fertilizer. And the most important point I ’ m going to take is – This is really bad news and good news. Last year we emitted 40. 5 billion tons of carbon dioxide. We have to find a way of solving it, and we can. We can take this Johnson-Su compost and get it out across the landscape. But more fundamentally, we can find good fungal bacterial communities and we can coat seeds with it and sell the seeds around the planet, or we can fill biochar and we can use biochar to get her across the planet. Most usefully, we can take a really good fungal bacterial community, put it into biochar or some other matrix, mix it into fertilizer, and get that fertilizer out across the planet. I ’ m going to just end with the following. Fungal bacterial communities are precisely something that can create novel adjacent possibles, that can bubble forth with solutions to soil problems. The good news is we can do it now. The planet ’ s on fire. We are of nature, ladies and gentlemen, we ’ re not above nature. Thank you.

Text Sample 465: NEG Dim 3 – 2014 – Score -1.00 – 2014/t001809.txt

I study ants in the desert, in the tropical forest and in my kitchen, and in the hills around Silicon Valley where I live. I ’ ve recently realized that ants are using interactions differently in different environments, and that got me thinking that we could learn from this about other systems, like brains and data networks that we engineer, and even cancer. So what all these systems have in common is that there ’ s no central control. An ant colony consists of **sterile** female workers — those are the ants you see walking around — and then one or more reproductive females who just lay the eggs. They don ’ t give any instructions. Even though they ’ re called queens, they don ’ t tell anybody what to do. So in an ant colony, there ’ s no one in charge, and all systems like this without central control are regulated using very simple interactions. Ants interact using smell. They smell with their antennae, and they interact with their antennae, so when one ant touches another with its antennae, it can tell, for example, if the other ant is a nestmate and what task that other ant has been doing. So here you see a lot of ants moving around and interacting in a lab arena that ’ s connected by tubes to two other arenas. So when one ant meets another, it doesn ’ t matter which ant it meets, and they ’ re actually not transmitting any kind of complicated signal or message. All that matters to the ant is the rate at which it meets other ants. And all of these interactions, taken together, produce a network. So this is the network of the ants that you just saw moving around in the arena, and it ’ s this constantly shifting network that produces the behavior of the colony, like whether all the ants are hiding inside the nest, or how many are going out to forage. A brain actually works in the same way, but what ’ s great about ants is that you can see the whole network as it happens. There are more than 12, 000 species of ants, in every conceivable environment, and they ’ re using interactions differently to meet different environmental challenges. So one important environmental challenge that every system has to deal with is operating costs, just what it takes to run the system. And another environmental challenge is resources, finding them and collecting them. In the desert, operating costs are high because water is scarce, and the seed-eating ants that I study in the desert have to spend water to get water. So an ant outside foraging, searching for seeds in the hot sun, just loses water into the air. But the colony gets its water by metabolizing the fats out of the seeds that they eat. So in this environment, interactions are used to activate foraging. An outgoing forager doesn ’ t go out unless it gets enough interactions with returning foragers, and what you see are the returning foragers going into the tunnel, into the nest, and meeting outgoing foragers on their way out. This makes sense for the ant colony, because the more food there is out there, the more quickly the foragers find it, the faster they come back, and the more foragers they send out. The system works to stay stopped, unless something positive happens. So interactions function to activate foragers. And we ’ ve been studying the evolution of this system. First of all, there ’ s variation. It turns out that colonies are different. On dry days, some colonies forage less, so colonies are different in how they manage this trade-off between spending water to search for seeds and getting water back in the form of seeds. And we ’ re trying to understand why some colonies forage less than others by thinking about ants as neurons, using models from neuroscience. So just as a neuron adds up its stimulation from other neurons to decide whether to fire, an ant adds up its stimulation from other ants to decide whether to forage. And what we ’ re looking for is whether there might be small differences among colonies in how many interactions each ant needs before it ’ s willing to go out and forage, because a colony like that would forage less. And this raises an **analogous** question about brains. We talk about the brain, but of course every brain is slightly different, and maybe there are some individuals or some conditions in which the electrical properties of neurons are such that they require more stimulus to fire, and that would lead to differences in brain function. So in order to ask evolutionary questions, we need to know about reproductive success. This is a map of the study site where I have been tracking this population of harvester ant colonies for 28 years, which is about as long as a colony lives. Each symbol is a colony, and the size of the symbol is how many offspring it had, because we were able to use genetic variation to match up parent and offspring colonies, that is, to figure out which colonies were founded by a daughter queen produced by which parent colony. And this was amazing for me, after all these years, to find out, for example, that colony 154, whom I ’ ve known well for many years, is a great-grandmother. Here ’ s her daughter colony, here ’ s her granddaughter colony, and these are her great- granddaughter colonies. And by doing this, I was able to learn that offspring colonies resemble parent colonies in their decisions about which days are so hot that they don ’ t forage, and the offspring of parent colonies live so far from each other that the ants never meet, so the ants of the offspring colony can ’ t be learning this from the parent colony. And so our next step is to look for the genetic variation underlying this resemblance. So then I was able to ask, okay, who ’ s doing better? Over the time of the study, and especially in the past 10 years, there ’ s been a very severe and deepening drought in the Southwestern U. S., and it turns out that the colonies that conserve water, that stay in when it ’ s really hot outside, and thus sacrifice getting as much food as possible, are the ones more likely to have offspring colonies. So all this time, I thought that colony 154 was a loser, because on really dry days, there ’ d be just this trickle of foraging, while the other colonies were out foraging, getting lots of food, but in fact, colony 154 is a huge success. She ’ s a matriarch. She ’ s one of the rare great-grandmothers on the site. To my knowledge, this is the first time that we ’ ve been able to track the ongoing evolution of collective behavior in a natural population of animals and find out what ’ s actually working best. Now, the Internet uses an algorithm to regulate the flow of data that ’ s very similar to the one that the harvester ants are using to regulate the flow of foragers. And guess what we call this analogy? The anternet is coming. So data doesn ’ t leave the source computer unless it gets a signal that there ’ s enough bandwidth for it to travel on. In the early days of the Internet, when operating costs were really high and it was really important not to lose any data, then the system was set up for interactions to activate the flow of data. It ’ s interesting that the ants are using an algorithm that ’ s so similar to the one that we recently invented, but this is only one of a handful of ant algorithms that we know about, and ants have had 130 million years to evolve a lot of good ones, and I think it ’ s very likely that some of the other 12, 000 species are going to have interesting algorithms for data networks that we haven ’ t even thought of yet. So what happens when operating costs are low? Operating costs are low in the tropics, because it ’ s very humid, and it ’ s easy for the ants to be outside walking around. But the ants are so abundant and diverse in the tropics that there ’ s a lot of competition. Whatever resource one species is using, another species is likely to be using that at the same time. So in this environment, interactions are used in the opposite way. The system keeps going unless something negative happens, and one species that I study makes circuits in the trees of foraging ants going from the nest to a food source and back, just round and round, unless something negative happens, like an interaction with ants of another species. So here ’ s an example of ant security. In the middle, there ’ s an ant plugging the nest entrance with its head in response to interactions with another species. Those are the little ones running around with their abdomens up in the air. But as soon as the threat is passed, the entrance is open again, and maybe there are situations in computer security where operating costs are low enough that we could just block access temporarily in response to an immediate threat, and then open it again, instead of trying to build a permanent firewall or fortress. So another environmental challenge that all systems have to deal with is resources, finding and collecting them. And to do this, ants solve the problem of collective search, and this is a problem that ’ s of great interest right now in robotics, because we ’ ve understood that, rather than sending a single, sophisticated, expensive robot out to explore another planet or to search a burning building, that instead, it may be more effective to get a group of cheaper robots exchanging only minimal information, and that ’ s the way that ants do it. So the invasive Argentine ant makes expandable search networks. They ’ re good at dealing with the main problem of collective search, which is the trade-off between searching very thoroughly and covering a lot of ground. And what they do is, when there are many ants in a small space, then each one can search very thoroughly because there will be another ant nearby searching over there, but when there are a few ants in a large space, then they need to stretch out their paths to cover more ground. I think they use interactions to assess density, so when they ’ re really crowded, they meet more often, and they search more thoroughly. Different ant species must use different algorithms, because they ’ ve evolved to deal with different resources, and it could be really useful to know about this, and so we recently asked ants to solve the collective search problem in the extreme environment of microgravity in the International Space Station. When I first saw this picture, I thought, Oh no, they ’ ve mounted the habitat vertically, but then I realized that, of course, it doesn ’ t matter. So the idea here is that the ants are working so hard to hang on to the wall or the floor or whatever you call it that they ’ re less likely to interact, and so the relationship between how crowded they are and how often they meet would be messed up. We ’ re still analyzing the data. I don ’ t have the results yet. But it would be interesting to know how other species solve this problem in different environments on Earth, and so we ’ re setting up a program to encourage kids around the world to try this experiment with different species. It ’ s very simple. It can be done with cheap materials. And that way, we could make a global map of ant collective search algorithms. And I think it ’ s pretty likely that the invasive species, the ones that come into our buildings, are going to be really good at this, because they ’ re in your kitchen because they ’ re really good at finding food and water. So the most familiar resource for ants is a picnic, and this is a clustered resource. When there ’ s one piece of fruit, there ’ s likely to be another piece of fruit nearby, and the ants that specialize on clustered resources use interactions for recruitment. So when one ant meets another, or when it meets a chemical deposited on the ground by another, then it changes direction to follow in the direction of the interaction, and that ’ s how you get the trail of ants sharing your picnic. Now this is a place where I think we might be able to learn something from ants about cancer. I mean, first, it ’ s obvious that we could do a lot to prevent cancer by not allowing people to spread around or sell the toxins that promote the evolution of cancer in our bodies, but I don ’ t think the ants can help us much with this because ants never poison their own colonies. But we might be able to learn something from ants about treating cancer. There are many different kinds of cancer. Each one originates in a particular part of the body, and then some kinds of cancer will spread or metastasize to particular other tissues where they must be getting resources that they need. So if you think from the perspective of early metastatic cancer cells as they ’ re out searching around for the resources that they need, if those resources are clustered, they ’ re likely to use interactions for recruitment, and if we can figure out how cancer cells are recruiting, then maybe we could set traps to catch them before they become established. So ants are using interactions in different ways in a huge variety of environments, and we could learn from this about other systems that operate without central control. Using only simple interactions, ant colonies have been performing amazing feats for more than 130 million years. We have a lot to learn from them. Thank you.

Text Sample 466: NEG Dim 3 – 2014 – Score -1.00 – 2014/t001803.txt

I was born in Taiwan. I grew up surrounded by different types of hardware stores, and I like going to night markets. I love the energy of the night markets, the colors, the lights, the toys, and all the unexpected things I find every time I go, things like watermelon with straw antennas or puppies with mohawks. When I was growing up, I liked taking toys apart, any kind of toys I ’ d find around the house, like my brother ’ s BB gun when he ’ s not home. I also liked to make environments for people to explore and play. In these early installations, I would take plastic sheets, plastic bags, and things I would find in the hardware store or around the house. I would take things like highlighter pen, mix it with water, pump it through plastic tubing, creating these glowing circulatory systems for people to walk through and enjoy. I like these materials because of the way they look, the way they feel, and they ’ re very affordable. I also liked to make devices that work with body parts. I would take camera LED lights and a bungee cord and strap it on my waist and I would **videotape** my belly button, get a different perspective, and see what it does. I also like to modify household appliances. This is an automatic night light. Some of you might have them at home. I would cut out the light sensor, add an extension line, and use modeling clay, stick it onto the television, and then I would **videotape** my eye, and using the dark part of my eye tricking the sensor into thinking it ’ s night time, so you turn on the lightbulb. The white of the eye and the eyelid will trick the sensor into thinking it ’ s daytime, and it will shut off the light. I wanted to collect more different types of eyes, so I built this device using bicycle helmets, some lightbulbs and television sets. It would be easier for other people to wear the helmet and record their eyes. This device allows me to symbolically extract other people ’ s eyes, so I have a diversity of eyes to use for my other sculptures. This sculpture has four eyes. Each eye is controlling a different device. This eye is turning itself around in a television. This eye is inflating a plastic tube. This eye is watching a video of another piece being made. And these two eyes are activating glowing water. Many of these pieces are later on shown in museums, biennials, triennial exhibitions around the world. I love science and biology. In 2007, I was doing a research fellowship at the Smithsonian Natural History Museum looking at bioluminous organisms in the oean. I love these creatures. I love the way they look, the way they feel. They ’ re soft, they ’ re slimy, and I was fascinated by the way they use light in their environment, either to attract mates, for self-defense, or to attract food. This research inspired my work in many different ways, things like movement or different light patterns. So I started gathering a lot of different types of material in my studio and just experimenting and trying this out, trying that out, and seeing what types of creatures I can come up with. I used a lot of computer cooling fans and just kind of put them together and see what happens. This is an 8, 000-square-foot installation composed of many different creatures, some hanging from the ceiling and some resting on the floor. From afar, they look alien-like, but when you look closer, they ’ re all made out of black garbage bags or Tupperware containers. I ’ d like to share with you how ordinary things can become something magical and wondrous. Thank you.

Text Sample 467: NEG Dim 3 – 2014 – Score -1.00 – 2014/t001795.txt

Nicholas Negroponte : Can we switch to the video disc, which is in play mode? I ’ m really interested in how you put people and computers together. We will be using the TV screens or their equivalents for electronic books of the future. Very interested in touch-sensitive displays, high-tech, high-touch, not having to pick up your fingers to use them. There is another way where computers touch people : wearing, physically wearing. Suddenly on September 11th, the world got bigger. NN : Thank you. Thank you. When I was asked to do this, I was also asked to look at all 14 TED Talks that I had given, chronologically. The first one was actually two hours. The second one was an hour, and then they became half hours, and all I noticed was my bald spot getting bigger. Imagine seeing your life, 30 years of it, go by, and it was, to say the least, for me, quite a shocking experience. So what I ’ m going to do in my time is try and share with you what happened during the 30 years, and then also make a prediction, and then tell you a little bit about what I ’ m doing next. And I put on a slide where TED 1 happened in my life. And it ’ s rather important because I had done 15 years of research before it, so I had a backlog, so it was easy. It ’ s not that I was Fidel Castro and I could talk for two hours, or Bucky Fuller. I had 15 years of stuff, and the Media Lab was about to start. So that was easy. But there are a couple of things about that period and about what happened that are really quite important. One is that it was a period when computers weren ’ t yet for people. And the other thing that sort of happened during that time is that we were considered sissy computer scientists. We weren ’ t considered the real thing. So what I ’ m going to show you is, in retrospect, a lot more interesting and a lot more accepted than it was at the time. So I ’ m going to characterize the years and I ’ m even going to go back to some very early work of mine, and this was the kind of stuff I was doing in the ’ 60s : very direct manipulation, very influenced as I studied architecture by the architect Moshe Safdie, and you can see that we even built robotic things that could build habitat-like structures. And this for me was not yet the Media Lab, but was the beginning of what I ’ ll call sensory computing, and I pick fingers partly because everybody thought it was ridiculous. Papers were published about how stupid it was to use fingers. Three reasons : One was they were low-resolution. The other is your hand would occlude what you wanted to see, and the third, which was the winner, was that your fingers would get the screen dirty, and hence, fingers would never be a device that you ’ d use. And this was a device we built in the ’ 70s, which has never even been picked up. It ’ s not just touch sensitive, it ’ s pressure sensitive. Voice : Put a yellow circle there. NN : Later work, and again this was before TED 1 — Voice : Move that west of the diamond. Create a large green circle there. Man : Aw, shit. NN : — was to sort of do interface concurrently, so when you talked and you pointed and you had, if you will, multiple channels. Entebbe happened. 1976, Air France was hijacked, taken to Entebbe, and the Israelis not only did an extraordinary rescue, they did it partly because they had practiced on a physical model of the airport, because they had built the airport, so they built a model in the desert, and when they arrived at Entebbe, they knew where to go because they had actually been there. The U. S. government asked some of us, ’ 76, if we could replicate that computationally, and of course somebody like myself says yes. Immediately, you get a contract, Department of Defense, and we built this truck and this rig. We did sort of a simulation, because you had video discs, and again, this is ’ 76. And then many years later, you get this truck, and so you have Google Maps. Still people thought, no, that was not serious computer science, and it was a man named Jerry Wiesner, who happened to be the president of MIT, who did think it was computer science. And one of the keys for anybody who wants to start something in life : Make sure your president is part of it. So when I was doing the Media Lab, it was like having a gorilla in the front seat. If you were stopped for speeding and the officer looked in the window and saw who was in the passenger seat, then,”Oh, continue on, sir.”And so we were able, and this is a cute, actually, device, parenthetically. This was a lenticular photograph of Jerry Wiesner where the only thing that changed in the photograph were the lips. So when you oscillated that little piece of lenticular sheet with his photograph, it would be in lip sync with zero bandwidth. It was a zero- bandwidth teleconferencing system at the time. So this was the Media Lab ’ s — this is what we said we ’ d do, that the world of computers, publishing, and so on would come together. Again, not generally accepted, but very much part of

TED in the early days. And this is really where we were headed. And that created the Media Lab. One of the things about age is that I can tell you with great confidence, I ' ve been to the future. I ' ve been there, actually, many times. And the reason I say that is, how many times in my life have I said,"Oh, in 10 years, this will happen,"and then 10 years comes. And then you say,"Oh, in five years, this will happen."And then five years comes. So I say this a little bit with having felt that I ' d been there a number of times, and one of the things that is most quoted that I ' ve ever said is that computing is not about computers, and that didn ' t quite get enough traction, and then it started to. It started to because people caught on that the medium wasn ' t the message. And the reason I show this car in actually a rather ugly slide is just again to tell you the kind of story that characterized a little bit of my life. This is a student of mine who had done a Ph. D. called"Backseat Driver."It was in the early days of GPS, the car knew where it was, and it would give audio instructions to the driver, when to turn right, when to turn left and so on. Turns out, there are a lot of things in those instructions that back in that period were pretty challenging, like what does it mean, take the next right? Well, if you ' re coming up on a street, the next right ' s probably the one after, and there are lots of issues, and the student did a wonderful thesis, and the MIT patent office said"Don ' t patent it. It ' ll never be accepted. The liabilities are too large. There will be insurance issues. Don ' t patent it."So we didn ' t, but it shows you how people, again, at times, don ' t really look at what ' s happening. Some work, and I ' ll just go through these very quickly, a lot of sensory stuff. You might recognize a young Yo-Yo Ma and tracking his body for playing the cello or the hypercello. These fellows literally walked around like that at the time. It ' s now a little bit more discreet and more commonplace. And then there are at least three heroes I want to quickly mention. Marvin Minsky, who taught me a lot about common sense, and I will talk briefly about Muriel Cooper, who was very important to Ricky Wurman and to TED, and in fact, when she got onstage, she said, the first thing she said was,"I introduced Ricky to Nicky."And nobody calls me Nicky and nobody calls Richard Ricky, so nobody knew who she was talking about. And then, of course, Seymour Papert, who is the person who said,"You can ' t think about thinking unless you think about thinking about something."And that ' s actually — you can unpack that later. It ' s a pretty profound statement. I ' m showing some slides that were from TED 2, a little silly as slides, perhaps. Then I felt television really was about displays. Again, now we ' re past TED 1, but just around the time of TED 2, and what I ' d like to mention here is, even though you could imagine intelligence in the device, I look today at some of the work being done about the Internet of Things, and I think it ' s kind of tragically pathetic, because what has happened is people take the oven panel and put it on your cell phone, or the door key onto your cell phone, just taking it and bringing it to you, and in fact that ' s actually what you don ' t want. You want to put a chicken in the oven, and the oven says,"Aha, it ' s a chicken,"and it cooks the chicken."Oh, it ' s cooking the chicken for Nicholas, and he likes it this way and that way."So the intelligence, instead of being in the device, we have started today to move it back onto the cell phone or closer to the user, not a particularly enlightened view of the Internet of Things. Television, again, television what I said today, that was back in 1990, and the television of tomorrow would look something like that. Again, people, but they laughed cynically, they didn ' t laugh with much appreciation. Telecommunications in the 1990s, George Gilder decided that he would call this diagram the Negroponte switch. I ' m probably much less famous than George, so when he called it the Negroponte switch, it stuck, but the idea of things that came in the ground would go in the air and stuff in the air would go into the ground has played itself out. That is the original slide from that year, and it has worked in lockstep obedience. We started Wired magazine. Some people, I remember we shared the reception desk periodically, and some parent called up irate that his son had given up Sports Illustrated to subscribe for Wired, and he said,”Are you some porno magazine or something?”and couldn ’ t understand why his son would be interested in Wired, at any rate. I will go through this a little quicker. This is my favorite, 1995, back page of Newsweek magazine. Okay. Read it. [”Nicholas Negroponte, director of the MIT Media Lab, predicts that we ’ ll soon buy books and newspapers straight over the Internet. Uh, sure.”— Clifford Stoll, Newsweek, 1995] You must admit that gives you, at least it gives me pleasure when somebody says how dead wrong you are.”Being Digital”came out. For me, it gave me an opportunity to be more in the trade press and get this out to the public, and it also allowed us to build the new Media Lab, which if you haven ’ t been to, visit, because it ’ s a beautiful piece of architecture aside from being a wonderful place to work. So these are the things we were saying in those TEDs. [Today, multimedia is a desktop or living room experience, because the **apparatus** is so clunky. This will change dramatically with small, bright, thin, high-resolution displays. — 1995] We came to them. I looked forward to it every year. It was the party that Ricky Wurman never had in the sense that he invited many of his old friends, including myself. And then something for me changed pretty profoundly. I became more involved with computers and learning and influenced by Seymour, but particularly looking at learning as something that is best approximated by computer programming. When you write a computer program, you ’ ve got to not just list things out and sort of take an algorithm and translate it into a set of instructions, but when there ’ s a bug, and all programs have bugs, you ’ ve got to de-bug it. You ’ ve got to go in, change it, and then re-execute, and you iterate, and that iteration is really a very, very good approximation of learning. So that led to my own work with Seymour in places like Cambodia and the starting of One Laptop per Child. Enough TED Talks on One Laptop per Child, so I ’ ll go through it very fast, but it did give us the chance to do something at a relatively large scale in the area of learning, development and computing. Very few people know that One Laptop per Child was a $1 billion project, and it was, at least over the seven years I ran it, but even more important, the World Bank contributed zero, USAID zero. It was mostly the countries using their own treasuries, which is very interesting, at least to me it was very interesting in terms of what I plan to do next. So these are the various places it happened. I then tried an experiment, and the experiment happened in Ethiopia. And here ’ s the experiment. The experiment is, can learning happen where there are no schools. And we dropped off tablets with no instructions and let the children figure it out. And in a short period of time, they not only turned them on and were using 50 apps per child within five days, they were singing”ABC”songs within two weeks, but they hacked Android within six months. And so that seemed sufficiently interesting. This is perhaps the best picture I have. The kid on your right has sort of nominated himself as teacher. Look at the kid on the left, and so on. There are no adults involved in this at all. So I said, well can we do this at a larger scale? And what is it that ’ s missing? The kids are giving a press conference at this point, and sort of writing in the dirt. And the answer is, what is missing? And I ’ m going to skip over my prediction, actually, because I ’ m running out of time, and here ’ s the question, is what ’ s going to happen? I think the challenge is to connect the last billion people, and connecting the last billion is very different than connecting the next billion, and the reason it ’ s different is that the next billion are sort of low-hanging fruit, but the last billion are rural. Being rural and being poor are very different. Poverty tends to be created by our society, and the people in that community are not poor in the same way at all. They may be primitive, but the way to approach it and to connect them, the history of One Laptop per Child, and the experiment in Ethiopia, lead me to believe that we can in fact do this in a very short period of time. And so my plan, and unfortunately I haven ’ t been able to get my partners at this point to let me announce them, but is to do this with a stationary satellite. There are many reasons that stationary satellites aren ’ t the best things, but there are a lot of reasons why they are, and for two billion dollars, you can connect a lot more than 100 million people, but the reason I picked two, and I will leave this as my last slide, is two billion dollars is what we were spending in Afghanistan every week. So surely if we can connect Africa and the last billion people for numbers like that, we should be doing it. Thank you very much. Chris Anderson : Stay up there. Stay up there. NN : You ’ re going to give me extra time? CA : No. That was wickedly clever, wickedly clever. You gamed it beautifully. Nicholas, what is your prediction? NN : Thank you for asking. I ’ ll tell you what my prediction is, and my prediction, and this is a prediction, because it ’ ll be 30 years. I won ’ t be here. But one of the things about learning how to read, we have been doing a lot of consuming of information going through our eyes, and so that may be a very inefficient channel. So my prediction is that we are going to ingest information You ’ re going to swallow a pill and know English. You ’ re going to swallow a pill and know Shakespeare. And the way to do it is through the bloodstream. So once it ’ s in your bloodstream, it basically goes through it and gets into the brain, and when it knows that it ’ s in the brain in the different pieces, it deposits it in the right places. So it ’ s ingesting. CA : Have you been hanging out with Ray Kurzweil by any chance? NN : No, but I ’ ve been hanging around with Ed Boyden and hanging around with one of the speakers who is here, Hugh Herr, and there are a number of people. This isn ’ t quite as far-fetched, so 30 years from now. CA : We will check it out. We ’ re going to be back and we ’ re going to play this clip 30 years from now, and then all eat the red pill. Well thank you for that. Nicholas Negroponte. NN : Thank you.

Text Sample 468: NEG Dim 3 – 2015 – Score -1.00 – 2015/t001209.txt

Our lives depend on a world we can ’ t see. Think about your week so far. Have you watched TV, used GPS, checked the weather or even ate a meal? These many things that enable our daily lives rely either directly or indirectly on satellites. And while we often take for granted the services that satellites provide us, the satellites themselves deserve our attention as they are leaving a lasting mark on the space they occupy. People around the world rely on satellite infrastructure every day for information, entertainment and to communicate. There ’ s agricultural and environmental monitoring, Internet connectivity, navigation. Satellites even play a role in the operation of our financial and energy markets. But these satellites that we rely on day in and day out have a finite life. They might run out of propellant, they could malfunction, or they may just naturally reach the end of their mission life. At this point, these satellites effectively become space junk, cluttering the orbital environment. So imagine you ’ re driving down the highway on a beautiful, sunny day out running errands. You ’ ve got your music cranked, your windows rolled down, with the cool breeze blowing through your hair. Feels nice, right? Everything is going smoothly until suddenly your car stutters and stalls right in the middle of the highway. So now you have no choice but to abandon your car where it is on the highway. Maybe you were lucky enough to be able to move it out of the way and into a shoulder lane so that it ’ s out of the way of other traffic. A couple of hours ago, your car was a useful machine that you relied on in your everyday life. Now, it ’ s a useless **hunk** of metal taking up space in a valuable transportation network. And imagine international roadways all cluttered with broken down vehicles that are just getting in the way of other traffic. And imagine the debris that would be strewn everywhere if a collision actually happened, thousands of smaller pieces of debris becoming new obstacles. This is the paradigm of the satellite industry. Satellites that are no longer working are often left to deorbit over many, many years, or only moved out of the way as a temporary solution. And there are no international laws in space to enforce us to clean up after ourselves. So the world ’ s first satellite, **Sputnik** I, was launched in 1957, and in that year, there were only a total of three launch attempts. Decades later and dozens of countries from all around the world have launched thousands of more satellites into orbit, and the frequency of launches is only going to increase in the future, especially if you consider things like the possibility of 900-plus satellite constellations being launched. Now, we send satellites to different orbits depending on what they ’ re needed for. One of the most common places we send satellites is the low Earth orbit, possibly to image the surface of Earth at up to about 2, 000 kilometers altitude. Satellites there are naturally buffeted by Earth ’ s atmosphere, so their orbits naturally decay, and they ’ ll eventually burn up, probably within a couple of decades. Another common place we send satellites is the geostationary orbit at about 35, 000 kilometers altitude. Satellites there remain in the same place above Earth as the Earth rotates, which enables things like communications or television broadcast, for example. Satellites in high orbits like these could remain there for centuries. And then there ’ s the orbit coined”the graveyard,”the ominous junk or disposal orbits, where some satellites are intentionally placed at the end of their life so that they ’ re out of the way of common operational orbits. Of the nearly 7, 000 satellites launched since the late 1950s, only about one in seven is currently operational, and in addition to the satellites that are no longer working, there ’ s also hundreds of thousands of marble-sized debris and millions of paint chip-sized debris that are also orbiting around the Earth. Space debris is a major risk to space missions, but also to the satellites that we rely on each and every day. Now, because space debris and junk has become increasingly worrisome, there have been some national and international efforts to develop technical standards to help us limit the generation of additional debris. So for example, there are recommendations for those low-Earth orbiting spacecraft to be made to deorbit in under 25 years, but that ’ s still a really long time, especially if a satellite hasn ’ t been working for years. There ’ s also mandates for those dead geostationary spacecraft to be moved into a graveyard orbit. But neither of these guidelines is binding under international law, and the understanding is that they will be implemented through national mechanisms. These guidelines are also not long-term, they ’ re not proactive, nor do they address the debris that ’ s already up there. They ’ re only in place to limit the future creation of debris. Space junk is no one ’ s responsibility. Now, Mount Everest is actually an interesting comparison of a new approach to how we interact with our environments, as it ’ s often given the dubious honor of being the world ’ s highest garbage dump. Decades after the first conquest of the world ’ s highest peak, tons of rubbish left behind by climbers has started to raise concern, and you may have read in the news that there ’ s speculation that Nepal will crack down on mountaineers with stricter enforcement of penalties and legal obligations. The goal, of course, is to persuade climbers to clean up after themselves, so maybe local not-for-profits will pay climbers who bring down extra waste, or expeditions might organize voluntary cleanup trips. And yet still many climbers feel that independent groups should police themselves. There ’ s no simple or easy answer, and even well-intentioned efforts at conservation often run into problems. But that doesn ’ t mean we shouldn ’ t do everything in our power to protect the environments that we rely and depend on, and like Everest, the remote location and inadequate infrastructure of the orbital environment make waste disposal a challenging problem. But we simply cannot reach new heights and create an even higher garbage dump, one that ’ s out of this world. The reality of space is that if a component on a satellite breaks down, there really are limited opportunities for repairs, and only at great cost. But what if we were smarter about how we designed satellites? What if all satellites, regardless of what country they were built in, had to be standardized in some way for recycling, servicing or active deorbiting? What if there actually were international laws with teeth that enforced end-of-life disposal of satellites instead of moving them out of the way as a temporary solution? Or maybe satellite manufacturers need to be charged a deposit to even launch a satellite into orbit, and that deposit would only be returned if the satellite was disposed of properly or if they cleaned up some quota of debris. Or maybe a satellite needs to have technology on board to help accelerate deorbit. There are some encouraging signs. The UK ’ s TechDemoSat-1, launched in 2014, for example, was designed for end-of-life disposal via a small drag sail. This works for the satellite because it ’ s small, but satellites that are higher or in larger orbits or are larger altogether, like the size of school buses, will require other disposal options. So maybe you get into things like high-powered lasers or tugging using nets or tethers, as crazy as those sound in the short term. And then one really cool possibility is the idea of orbital tow trucks or space mechanics. Imagine if a robotic arm on some sort of space tow truck could fix the broken components on a satellite, making them usable again. Or what if that very same robotic arm could refuel the propellant tank on a spacecraft that relies on chemical propulsion just like you or I would refuel the fuel tanks on our cars? Robotic repair and maintenance could extend the lives of hundreds of satellites orbiting around the Earth. Whatever the disposal or cleanup options we come up with, it ’ s clearly not just a technical problem. There ’ s also complex space laws and politics that we have to sort out. Simply put, we haven ’ t found a way to use space sustainably yet. Exploring, innovating to change the way we live and work are what we as humans do, and in space exploration, we ’ re literally moving beyond the boundaries of Earth. But as we push thresholds in the name of learning and innovation, we must remember that accountability for our environments never goes away. There is without doubt congestion in the low Earth and geostationary orbits, and we cannot keep launching new satellites to replace the ones that have broken down without doing something about them first, just like we would never leave a broken down car in the middle of the highway. Next time you use your phone, check the weather or use your GPS, think about the satellite technologies that make those activities possible. But also think about the very impact that the satellites have on the environment surrounding Earth, and help spread the message that together we must reduce our impact. Earth orbit is breathtakingly beautiful and our gateway to exploration. It ’ s up to us to keep it that way. Thank you.

Text Sample 469: NEG Dim 3 – 2015 – Score -1.00 – 2015/t001553.txt

What do you do when you have a headache? You swallow an aspirin. But for this pill to get to your head, where the pain is, it goes through your stomach, intestines and various other organs first. Swallowing pills is the most effective and painless way of delivering any medication in the body. The downside, though, is that swallowing any medication leads to its dilution. And this is a big problem, particularly in HIV patients. When they take their anti-HIV drugs, these drugs are good for lowering the virus in the blood, and increasing the CD4 cell counts. But they are also notorious for their adverse side effects, but mostly bad, because they get diluted by the time they get to the blood, and worse, by the time they get to the sites where it matters most : within the HIV viral reservoirs. These areas in the body — such as the lymph nodes, the nervous system, as well as the lungs — where the virus is sleeping, and will not readily get delivered in the blood of patients that are under consistent anti-HIV drugs therapy. However, upon discontinuation of therapy, the virus can awake and infect new cells in the blood. Now, all this is a big problem in treating HIV with the current drug treatment, which is a life-long treatment that must be swallowed by patients. One day, I sat and thought,"Can we deliver anti-HIV directly within its reservoir sites, without the risk of drug dilution?"As a laser scientist, the answer was just before my eyes : Lasers, of course. If they can be used for dentistry, for diabetic wound-healing and surgery, they can be used for anything imaginable, including transporting drugs into cells. As a matter of fact, we are currently using laser pulses to poke or drill extremely tiny holes, which open and close almost immediately in HIV-infected cells, in order to deliver drugs within them."How is that possible?"you may ask. Well, we shine a very powerful but super-tiny laser beam onto the membrane of HIV-infected cells while these cells are immersed in liquid containing the drug. The laser pierces the cell, while the cell swallows the drug in a matter of **microseconds**. Before you even know it, the induced hole becomes immediately repaired. Now, we are currently testing this technology in test tubes or in Petri dishes, but the goal is to get this technology in the human body, apply it in the human body."How is that possible?"you may ask. Well, the answer is : through a three-headed device. Using the first head, which is our laser, we will make an incision in the site of infection. Using the second head, which is a camera, we meander to the site of infection. Finally, using a third head, which is a drug- spreading sprinkler, we deliver the drugs directly at the site of infection, while the laser is again used to poke those cells open. Well, this might not seem like much right now. But one day, if successful, this technology can lead to complete eradication of HIV in the body. Yes. A cure for HIV. This is every HIV researcher ' s dream — in our case, a cure lead by lasers. Thank you.

Text Sample 470: NEG Dim 3 – 2015 – Score -1.00 – 2015/t001575.txt
I know what you ' re thinking :"Why does that guy get to sit down?"That ' s because this is radio. I tell radio stories about design, and I report on all kinds of stories : buildings and toothbrushes, mascots and wayfinding and fonts. My mission is to get people to engage with the design that they care about so they begin to pay attention to all forms of design. When you decode the world with design intent in mind, the world becomes kind of magical. Instead of seeing the broken things, you see all the little bits of genius that anonymous designers have sweated over to make our lives better. And that ' s essentially the definition of design : making life better and providing joy. And few things give me greater joy than a well-designed flag. Yeah! Happy 50th anniversary on your flag, Canada. It is beautiful, gold standard. Love it. I ' m kind of obsessed with flags. Sometimes I bring up the topic of flags, and people are like,"I don ' t care about flags,"and then we start talking about flags, and trust me, 100 percent of people care about flags. There ' s just something about them that

works on our emotions. My family wrapped my Christmas presents as flags this year, including the blue gift bag that ’ s dressed up as the flag of Scotland. I put this picture online, and sure enough, within the first few minutes, someone left a comment that said,”You can take that Scottish Saltire and shove it up your ass.”See, people are passionate about flags, you know? That ’ s the way it is. What I love about flags is that once you understand the design of flags, what makes a good flag, what makes a bad flag, you can understand the design of almost anything. So what I ’ m going to do here is, I cracked open an episode of my radio show,”99 % Invisible,”and I ’ m going to reconstruct it here on stage, so when I press a button over here — Voice : S for Sound — Roman Mars : It ’ s going to make a sound, and so whenever you hear a sound or a voice or a piece of music, it ’ s because I pressed a button. Voice : Sound. RM : All right, got it? Here we go. Three, two. This is 99 % Invisible. I ’ m Roman Mars. Narrator : The five basic principles of flag design. Roman Mars : According to the North American Vexillological Association. Vexillological. Ted Kaye : Vexillology is the study of flags. RM : It ’ s that extra”lol”that makes it sound weird. Narrator : Number one, keep it simple. The flag should be so simple that a child can draw it from memory. RM : Before I moved to Chicago in 2005, I didn ’ t even know cities had their own flags. TK : Most larger cities do have flags. RM : Well, I didn ’ t know that, that ’ s Ted Kaye, by the way. TK : Hello. RM : He ’ s a flag expert, he ’ s a totally awesome guy. TK : I ’ m Ted Kaye, I have edited a scholarly journal on flag studies, and I am currently involved with the Portland Flag Association and the North American Vexillological Association. RM : Ted literally wrote the book on flag design. Narrator :”Good Flag, Bad Flag.”RM : It ’ s more of a pamphlet, really, it ’ s about 16 pages. TK : Yes, it ’ s called”Good Flag, Bad Flag : How to Design a Great Flag.”RM : And that first city flag I discovered in Chicago is a beaut : white field, two horizontal blue stripes, and four six-pointed red stars down the middle. Narrator : Number two, use meaningful symbolism. TK : The blue stripes represent the water, the river and the lake. Narrator : The flag ’ s images, colors or pattern should relate to what it **symbolizes**. TK : The red stars represent significant events in Chicago ’ s history. RM : Namely, the founding of Fort Dearborn on the future site of Chicago, the Great Chicago Fire, the World Columbian Exposition, which everyone remembers because of the White City, and the Century of Progress Exposition, which no one remembers at all. Narrator : Number three, use two to three basic colors. TK : The basic rule for colors is to use two to three colors from the standard color set : red, white, blue, green, yellow and black. RM : The design of the Chicago flag has complete buy-in with an entire cross-section of the city. It is everywhere ; every municipal building flies the flag. Whet Moser : There ’ s probably at least one store on every block near where I work that sells some sort of Chicago flag paraphernalia. RM : That ’ s Whet Moser from Chicago magazine. WM : Today, just for example, I went to get a haircut, and when I sat down in the barber ’ s chair, there was a Chicago flag on the box that the barber kept all his tools in, and then in the mirror, there was a Chicago flag on the wall behind me. When I left, a guy passed me who had a Chicago flag badge on his backpack. RM : It ’ s adaptable and remixable. The six-pointed stars in particular show up in all kinds of places. WM : The coffee I bought the other day had a Chicago star on it. RM : It ’ s a distinct symbol of Chicago pride. TK : When a police officer or a firefighter dies in Chicago, often it ’ s not the flag of the United States on his casket. It can be the flag of the city of Chicago. That ’ s how deeply the flag has gotten into the civic imagery of Chicago. RM : And it isn ’ t just that people love Chicago and therefore love the flag. I also think that people love Chicago more because the flag is so cool. TK : A positive feedback loop there between great symbolism and civic pride. RM : OK, so when I moved back to San Francisco in 2008, I researched its flag, because I had never seen it in the previous eight years I lived there. And I found it, I am sorry to say, sadly lacking. I know. It hurts me, too. TK : Well, let me start from the top. Narrator : Number one, keep it simple. TK : Keeping it simple. Narrator : The flag should be so simple that a child can draw it from memory. TK : It ' s a relatively complex flag. RM : OK, here we go, OK. The main component of the San Francisco flag is a phoenix representing the city rising from the ashes after the devastating fires of the 1850s. TK : A powerful symbol for San Francisco. RM : I still don ' t really dig the phoenix. Design-wise, it manages to both be too crude and have too many details at the same time, which if you were trying for that, you wouldn ' t be able to do it, and it just looks bad at a distance, but having deep meaning puts that element in the plus column. Behind the phoenix, the background is mostly white, and then it has a substantial gold border around it. TK : Which is a very attractive design element. RM : I think it ' s OK, but — here come the big no-nos of flag design. Narrator : Number four, no lettering or seals. Never use writing of any kind. RM : Underneath the phoenix, there ' s a motto on a ribbon that translates to"Gold in peace, iron in war,"plus — and this is the big problem — it says San Francisco across the bottom. TK : If you need to write the name of what you ' re representing on your flag, your symbolism has failed. RM : The United States flag doesn ' t say"USA"across the front. In fact, country flags, they tend to behave. Like, hats off to South Africa and Turkey and Israel and Somalia and Japan and Gambia. There ' s a bunch of really great country flags, but they obey good design principles because the stakes are high. They ' re on the international stage. But city, state and regional flags are another story. There is a scourge of bad flags — and they must be stopped. That is the truth and that is the dare. The first step is to recognize that we have a problem. A lot of people tend to think that good design is just a matter of taste, and quite honestly, sometimes it is, actually, but sometimes it isn ' t, all right? Here ' s the full list of NAVA flag design principles. Narrator : The five basic principles of flag design. Narrator : Number one. TK : Keep it simple. Narrator : Number two. TK : Use meaningful symbolism. Narrator : Number three. TK : Use two to three basic colors. Narrator : Number four. TK : No lettering or seals. Narrator : Never use writing of any kind. TK : Because you can ' t read that at a distance. Narrator : Number five. TK : And be distinctive. RM : All the best flags tend to stick to these principles. And like I said before, most country flags are OK. But here ' s the thing : if you showed this list of principles to any designer of almost anything, they would say these principles — simplicity, deep meaning, having few colors or being thoughtful about colors, uniqueness, don ' t have writing you can ' t read — all those principles apply to them, too. But sadly, good design principles are rarely invoked in US city flags. Our biggest problem seems to be that fourth one. We just can ' t stop ourselves from putting our names on our flags, or little municipal seals with tiny writing on them. Here ' s the thing about municipal seals : They were designed to be on pieces of paper where you can read them, not on flags 100 feet away flapping in the breeze. So here ' s a bunch of flags again. Vexillologists call these SOBs : Seals on a bedsheet — and if you can ' t tell what city they go to, yeah, that ' s exactly the problem, except for Anaheim, apparently, they fixed it. These flags are everywhere in the US. The European equivalent of the municipal seal is the city coat of arms... and this is where we can learn a lesson for how to do things right. So this is the city coat of arms of Amsterdam. Now, if this were a United States city, the flag would probably look like this. You know, yeah. But instead, the flag of Amsterdam looks like this. Rather than plopping the whole coat of arms on a solid background and writing"Amsterdam"below it, they just take the key elements of the escutcheon, the shield, and they turn it into the most badass city flag in the world. And because it ’ s so badass, those flags and crosses are found throughout Amsterdam, just like Chicago, they ’ re used. Even though seal-on-a-bedsheet flags are particularly painful and offensive to me, nothing can quite prepare you for one of the biggest train wrecks in vexillological history. Are you ready? It ’ s the flag of Milwaukee, Wisconsin. I mean, it ’ s distinctive, I ’ ll give them that. Steve Kodis : It was adopted in 1955. RM : The city ran a contest and gathered a bunch of submissions with all kinds of designs. SK : And an alderman by the name of Fred Steffan cobbled together parts of the submissions to make what is now the Milwaukee flag. RM : It ’ s a kitchen sink flag. There ’ s a gigantic gear representing industry, there ’ s a ship recognizing the port, a giant stalk of wheat paying homage to the brewing industry. It ’ s a hot mess, and Steve Kodis, a graphic designer from Milwaukee, wants to change it. SK : It ’ s really awful. It ’ s a misstep on the city ’ s behalf, to say the least. RM : But what puts the Milwaukee flag over the top, almost to the point of self- parody, is on it is a picture of the Civil War battle flag of the Milwaukee regiment. SK : So that ’ s the final element in it that just makes it that much more ridiculous, that there is a flag design within the Milwaukee flag. RM : On the flag. Yeah. Yeah. Yeah. Now, Milwaukee is a fantastic city. I ’ ve been there, I love it. The most depressing part of this flag, though, is that there have been two major redesign contests. The last one was held in 2001. 105 entries were received. TK : But in the end, the members of the Milwaukee Arts Board decided that none of the new entries were worthy of flying over the city. RM : They couldn ’ t agree to change that thing! That ’ s discouraging enough to make you think that good design and democracy just simply do not go together. But Steve Kotas is going to try one more time to redesign the Milwaukee flag. SK : I believe Milwaukee is a great city. Every great city deserves a great flag. RM : Steve isn ’ t ready to reveal his design yet. One of the things about proposing one of these things is you have to get people on board, and then you reveal your design. But here ’ s the trick : If you want to design a great flag, a kick-ass flag like Chicago ’ s or DC ’ s, which also has a great flag, start by drawing a one-by-one-and-a-half-inch rectangle on a piece of paper. Your design has to fit within that tiny rectangle. Here ’ s why. TK : A three-by-five-foot flag on a pole 100 feet away looks about the same size as a one-by-one-and-a-half-inch rectangle seen about 15 inches from your eye. You ’ d be surprised by how compelling and simple the design can be when you hold yourself to that limitation. RM : Meanwhile, back in San Francisco. Is there anything we can do? TK : I like to say that in every bad flag there ’ s a good flag trying to get out. The way to make San Francisco ’ s flag a good flag is to take the motto off because you can ’ t read that at a distance. Take the name off, and the border might even be made thicker, so it ’ s more a part of the flag. And I would simply take the phoenix and make it a great big element in the middle of the flag. RM : But the current phoenix, that ’ s got to go. TK : I would simplify or stylize the phoenix. Depict a big, wide-winged bird coming out of flames. Emphasize those flames. RM : So this San Francisco flag was designed by Frank Chimero based on Ted Kaye ’ s suggestions. I don ’ t know what he would do if we was completely unfettered and didn ’ t follow those guidelines. Fans of my radio show and podcast, heard me complain about bad flags. They ’ ve sent me other suggested designs. This one ’ s by Neil Mussett. Both are so much better. And I think if they were adopted, I would see them around the city. In my crusade to make flags of the world more beautiful, many listeners have taken it upon themselves to redesign their flags and look into the feasibility of getting them officially adopted. If you see your city flag and like it, fly it, even if it violates a design rule or two. I don ’ t care. But if you don ’ t see your city flag, maybe it doesn ’ t exist, but maybe it does, and it just sucks, and I dare you to join the effort to try to change that. As we move more and more into cities, the city flag will become not just a symbol of that city as a place, but also, it could become a symbol of how that city considers design itself, especially today, as the populace is becoming more design-aware. And I think design awareness is at an all-time high. A well- designed flag could be seen as an indicator of how a city considers all of its design systems : its public transit, its parks, its signage. It might seem frivolous, but it ’ s not. TK : Often when city leaders say,”We have more important things to do than worry about a city flag,”my response is,”If you had a great city flag, you would have a banner for people to rally under to face those more important things.”RM : I ’ ve seen firsthand what a good city flag can do in the case of Chicago. The marriage of good design and civic pride is something that we need in all places. The best part about municipal flags is that we own them. They are an open-source, publicly owned design language of the community. When they are done well, they are remixable, adaptable, and they are powerful. We could control the branding and graphical imagery of our cities with a good flag, but instead, by having bad flags we don ’ t use, we cede that territory to sports teams and chambers of commerce and tourism boards. Sports teams can leave and break our hearts. And besides, some of us don ’ t really care about sports. And tourism campaigns can just be cheesy. But a great city flag is something that represents a city to its people and its people to the world at large. And when that flag is a beautiful thing, that connection is a beautiful thing. So maybe all the city flags can be as inspiring as Hong Kong or Portland or Trondheim, and we can do away with all the bad flags like San Francisco, Milwaukee, Cedar Rapids, and finally, when we ’ re all done, we can do something about Pocatello, Idaho, considered by the North American Vexillological Association as the worst city flag in North America. [Proud to be Pocatello] Yeah. That thing has a trademark symbol on it, people. That hurts me just to look at. Thank you so much for listening. [Music by : Melodium and Keegan DeWitt]

Text Sample 471: NEG Dim 3 – 2021 – Score -1.00 – 2021/t003867.txt

There are things we accept as obvious truths that aren ’ t necessarily backed up by data. For example, the idea that cousin marriage is, to use a scientific term, icky. [Am I Normal? with Mona Chalabi] First off, cousin marriage is way more common than you may think. Approximately 10 percent of the world ’ s families are headed by couples who are second cousins or closer. That ’ s more than 750 million people. And it maybe wildly out of fashion in Europe and North America now, but like hoop skirts and top hats, cousin marriage used to be a much more common sight. People like Charles Darwin, Edgar Allan Poe and Albert Einstein, all married their first cousins. President Franklin Delano Roosevelt married his fifth cousin, Eleanor Roosevelt. She didn ’ t even have to change her last name. One study looking at millions of genealogy profiles concluded that between 1650 and 1850, the average person was fourth cousins with their spouse. I know, I know, what the hell is a fourth cousin? Well, it basically means that the average person had the same great great great grandparent as their romantic partner. So why was that? Well, marrying a cousin meant keeping money and power within the family and maintaining your social relationships. It was also about proximity. If you weren ’ t traveling far in your young life, you ’ re more likely to end up with someone you already know, like a second cousin. But even with the onset of the railroad and the industrial revolution, it still took another 50 years for cousin marriage to go out of style in Western society. Mobility may have played a factor, but researchers also think that greater autonomy for women and smaller family sizes, meaning that there were just fewer cousins to marry, also influenced the change. Though it became much less popular on both sides of the Atlantic, it was only in the US that it became illegal. Following the Civil War, the American government took a greater interest in overseeing more elements of daily life, including marriage. American states began to outlaw not only interracial marriage but cousin marriage as well. Today, 32 states either outright ban the practice or restrict it to those who seek genetic counseling, are beyond reproductive age or are **sterile**. It ' s estimated that less than one percent of American couples are cousins. But the widespread illegality, not to mention the social stigma, makes quantifying it quite difficult. If you ask people why cousin marriage seems wrong, health might come up. The fear of birth defects and genetic disorders has influenced the change in social norms. But in 2002, researchers from the National Society of Genetic Counselors found that the increased risk was a minor one. In the general population, the risk that a child will be born with serious complications is three to four percent. With a first-cousin couple, the risk increased by 1. 7 to 2. 8 percentage points. And with those figures, the researchers concluded that there was no biological reason to discourage cousins from marrying. Look, I totally get it. Whether you fancy your cousin or not, depending on where you ' re from, this can still be a bit of a taboo subject. But the truth is, our feelings around it are shaped by the culture we live in and not necessarily by whether it ' s objectively dangerous or not. After all, if it wasn ' t for cousin marriage, you probably wouldn ' t be here.

Text Sample 472: NEG Dim 3 – 2021 – Score -1.00 – 2021/t003805.txt

I am a repair geek. I grew up fixing things with my dad ; it was what we did. We fixed our TV, we fixed our refrigerator, we fixed stuff that didn ' t need fixing. We fixed our Volkswagen Beetle. In our home, if something broke, we took it as an opportunity to have fun. We loved the idea and the challenge of bringing things back to life. And I still love that feeling, which to me is just thrilling. I did it recently. My laptop was overheating, I thought it might be the fan, I ordered a fan, I put the fan in, I turned it on, and the fan went"whir"like,"Yes! That really worked."It was a great feeling. Now I know this kind of repair probably sounds very old-fashioned, and I probably look a little old-fashioned, but it ' s much more than saving money. It ' s how we keep the things that we like in use. It helps us keep things out of the trash. It helps bring jobs into our communities, and it can help solve the digital divide. Now, since 2013, I ' ve been the executive director of the Digital Right to Repair Coalition, otherwise known as repair. org. Our members do all the Rs : We repair, we reuse, we resell and we recycle. And in doing this work, I ' ve come to realize that repair is right now central to all of our sustainability goals. If we can fix our stuff a little more frequently, keep it in use, we ' re keeping it out of the front of the waste stream at the front end. So we ' ll have less to process at the back end. If we are going to have any control over our e-waste problem, we have to talk about repair. Let me give you a sense of scale. Back in 2013, the EPA estimated that the average US household already owned 28 digitally-driven gizmos and gadgets. It was everything from garage-door openers and hot tub controls to smart toasters. If we just do a little math and multiply 28 times are roughly 123 million households, we come up with a pretty staggering three and a half billion pieces of e-waste that don ' t belong in our landfills, and they are costly and difficult to put back as raw materials if those processes even exist. When we look a little more closely at what ' s even possible with recycling, I think we ' ve been ignoring some really ugly truths. By the time a laptop or a refrigerator or even an electric toothbrush gets in our hands, almost all of the environmental damage has already been done. All the costs of mining and refining and smelting and transportation. And we don ' t see these costs when we go to the store, and we don ' t see the human costs of terrible labor conditions and exposure to toxic materials. So even if we can ' t agree on how to calculate these costs, I think we can agree that fixing more and throwing away less just makes sense. There ' s a lot of other advantages of repairing things other than just the obvious. Repair is what lets us keep our older devices in use, and it allows a secondary market for the products that we want to resell. And secondary markets are why used equipment is so affordable because the used seller has to compete with new. So if a new gadget is 1, 000 dollars, we expect a pretty big discount to buy that same item used. Let ' s start with 50 percent. So now we have an affordability capability that is central to crossing the digital divide. We had five million students that went to virtual school this past year that didn ' t have enabling technology. And that ' s because parents and school districts couldn ' t buy new. We still have a lot of chip shortages, and these are going to be with us for a while. And I think we have to think very seriously about doing more repair, not just to make things last longer but also to be more resilient as an economy. Repair is also a point of entry for a lot of our engineers and innovators. I heard Steve Wozniak speak very recently – Apple Steve Wozniak – He spoke very recently about his growth and development at a time when he was repairing things as a kid. And it was central to his development as an engineer. He grew up pretty much the same time I did, where repairs were very ordinary. Consumers were empowered to take their vacuum tubes to the local store, plug them in and see if they work and then buy a replacement on the spot. And there were lots of options for repair within the community to help with the more difficult challenges. I think you ' ve probably noticed that these mom and pop businesses disappeared or all but disappeared in our communities. And it ' s not because we don ' t want to fix our stuff, it ' s because they were not allowed to buy the essential repair materials that enabled them to stay in business. So if we can back that back and make it possible for our local repair shops to buy parts and tools, then those businesses will come back and they will bring back with them jobs that feed families. And our nerdy kids will be able to open things up, figure out how they work and become the engineers and innovators of our future. Repair jobs, which I mentioned, they ' re great jobs, and they don ' t require an advanced degree. I ' ll give you an example. There ' s a charity in Minnesota called Tech Dump, and they take in donated electronics, and then they hire adults that are hard to employ, many of whom coming out of the criminal justice system. They train them to make repairs. They then take the repaired goods, sell them and use the proceeds to fund more training. They ' re keeping equipment out of the waste stream ; they ' re bringing good-quality equipment to their community in a used format ; and they are bringing people out of poverty and into the workforce. What ' s really got me irritated is that at this point, the vast majority of products on the market today cannot be repaired by any party without being totally dependent on the manufacturer. And the day the manufacturer decides they don ' t want you to fix it, it ' s over. This is a completely artificial problem. Manufacturers used to provide comprehensive documentation and **schematics** and shipped it with every product. It was expected that you could fix your stuff. Then once the internet allowed this documentation to be hosted online, manufacturers stopped printing, which made sense because printing was expensive. And then somewhere along the line, somebody said,”Ah, we need to know who ’ s using our website.”So they demanded a login. And then another bright light said,”Oh, we can charge.”So they put up a paywall. And then a third bright light said,”We can ’ t let anybody have this information at all. They might compete with us.”And that ’ s where we are today. We can ’ t get what we need to fix our stuff. Now, I can tell you because I ’ ve had a front-row seat, this trend towards a throwaway economy is reversing all over the world. And it ’ s really fun to be able to tell you about it. Back in 2014, we put forward our very first digital”Right to Repair”bill, and that became the template for dozens of other bills. And this year, we ’ ve had 27 states take up the same legislation. These bills are starting to pass. There ’ s been a lot of help from the Federal Trade Commission and also the Biden administration, in saying we really, as a country, need to be able to fix our stuff, and we need competition for repair. It ’ s not just us. Canada and Australia have got similar processes underway. The European Union has put forward a set of regulations that just took effect that are limiting the use of adhesives in the construction of products because if you can ’ t get into the thing, you can ’ t fix the thing. France has yet another idea. They are requiring manufacturers to rate themselves on their repairability, and then they are posting those scores for consumers to consider pre-purchase. There are groups forming up all over the world that help people repair stuff even without government or regulatory change. You may have seen Repair Cafés advertised in your community. That first Repair Café started in 2009 in Denmark. There are now over 2, 000 official chapters. There are web sites, YouTube and a company called ifixit. com that hosts tens of thousands of repair tutorials to help people learn how to fix their stuff. Last year, they recorded 116 million unique users on their website, so I think it ’ s pretty clear people do want to fix their stuff. So I ’ m very encouraged by the fact that our throwaway economy, the trend is reversing. And we are going to be able to fix our stuff. We ’ re going to be able to use the things we want in the way that we want them. We will be able to cross the digital divide, and we will have more jobs, and we will all get that great”Yes! I fixed it!”feeling. So the next time something around you breaks, don ’ t take”broken”for an answer. Go fix something. Thank you.

Text Sample 473: NEG Dim 3 – 2021 – Score 1.00 – 2021/t004099.txt

”Always wear sunscreen.””Eat a balanced diet.””A penny saved is a penny earned.”You probably all learned these lessons as a kid, maybe from your parents, or if you grew up in the ’ 80s, on the public service announcements at the end of every episode of the G. I. Joe cartoons. But chances are, despite knowing this, you still stepped outside without putting on sunscreen, devoured an entire bag of chips in one go or spent way more of your paycheck than you anticipated. So why is that? [Your Money and Your Mind] with Wendy De La Rosa A few years ago, two Yale professors coined the term”G. I. Joe fallacy”to describe this very phenomenon. It ’ s the mistaken idea that knowing is half the battle. But as it turns out, in most situations, just knowing something is not nearly enough for you to put it into practice. Information doesn ’ t always change behavior. As a behavioral scientist helping families make better financial decisions, I ’ ve seen people struggle to save money, to cut back on their expenses or to manage their debt, even after they ’ ve taken a financial literacy class. And the reality is that people fundamentally know what they need to do to improve their financial situation. We all do. The equation is simple : they need to save more and spend less. But the thing is, that ’ s just really hard to do. It ’ s easier said than done, and I ’ ve been in this boat as well. So for example, I had a magazine subscription that I knew I should just cancel. I never read the magazine, and every month, money was coming out of my checking account, and every time I reviewed my budget, I saw it, I knew I had to cancel it, but I didn ’ t. It took me two years to cancel that magazine subscription. And I ’ m sure I ’ m not alone. You probably have some type of subscription that you know you should cancel. So the critical piece in all of this is to get rid of this belief that financial security is just a problem that we can teach away. In the US, for example, we spend nearly 700 million dollars every year on financial education programs, yet a team of researchers have found that these programs explained only 0. 1 percent of the variance in financial behaviors. Not zero, but pretty close to it. Meanwhile, 20 states mandate financial literacy classes in high school, but studies have found that unless these programs are well-implemented, they are unlikely to have any effect on a student ’ s future credit score or their likelihood to invest. A more significant predictor of how well you manage your finances is your general ability to do math. From all of this, I ’ ve learned that behavior change is not an educational pursuit. It ’ s an environmental one. If you are struggling, it ’ s not because there ’ s something fundamentally wrong with you. It ’ s most likely because there is something wrong with how your environment is set up. Look around you. The cues to spend money have gotten smarter, faster and more efficient. Targeted ads are becoming more personalized, corporate content is becoming more engaging, and everything around you is focusing on spending. So let ’ s change that. You can reshape your environment and how you interact with it, and I can guide you through it. Over the course of this series, I ’ ll take you through a step-by-step look at how you can change your environment and regain control of your finances. At the end of every episode, I ’ ll share practical tips based on research on how to spend less and save more today – not tomorrow, but today. And your future self will thank you.

Text Sample 474: NEG Dim 3 – 2025 – Score 2.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in the world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the air. In all my travels, I ’ ve been struck by so many things. I ’ ve observed how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ’ ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced us to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ’ s hard to see the big picture right now. It ’ s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think back to my travels. I ’ ve seen how different we are, how diverse and distinct. But I ’ ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ’ m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman”A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story is the dominant sound of our lives.”- Joseph Campbell I ’ ve loved telling stories all of my life. I ’ ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 475: NEG Dim 3 – 2025 – Score 2.00 – 2025/t003001.txt

- I don ’ t remember exactly, but... I use the most expensive hair color in the world. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color. I expect great color. What ’ s worth more to me is the way my hair feels. Feels good around my neck. I can ’ t remember it. I can ’ t finish it all. If only we had more time. - Why don ’ t we have more time? - I ’ m dying. I ’ m croaking! - It ’ s tough, coming home. But this is what you do for a parent that you love. But, you know? She ’ s not my mother. Technically. In my own family, there ’ s a lot of dysfunction, a lot of mental illness, a lot of substance abuse. Every generation. My parents divorced when I was about five. My mother moved us out of state, so we had to fly to New York to see my father ’ cause he was already working in advertising, and New York was the only place to be. - Many of the commercials you see on television begin life here.

- Advertising is one of the main sources of messaging about who you should be. - Nowadays, most of you women lead two lives, combining homemaking with the role of glamorous hostess. - And in the ’ 60s, the ads were often made from a male point of view. - In the pre-production meeting, everybody who has anything to do with the commercial is there. - It was mostly men. These superstar art directors and writers. Like my father, Gene Case.”Mad Men”was like home movies, and he was Don Draper. He was very witty, but I don ’ t think my father realized how powerful his words were. He knew it professionally. But with his kids, he could be incredibly cutting. Flay you just with a phrase. But it ’ s the time with my mother that was the nightmare. She had a horrible childhood, but she sure as hell didn ’ t make any effort to overcome that in her relations with us. You know, it was really horrifying to my mother that I was not popular. But I was also really awkward because she had been undermining my confidence for all my life. She would pick apart your personality in detail. I was lazy, I looked bad, I acted bad, I was awkward. Emotionally, just out for blood. At a certain point in the night, I would know,”OK, the situation is dangerous. I need to get outta here.”I had this fantasy when I was in fourth grade that I vividly remember that some kind of outside agency was gonna come in and determine from all the salt on my pillow that I was crying myself to sleep every night and do something about it. No, it was awful. It was awful. And that was the point where Ilon came into my life. - Ilon, do you need anything to eat? - I don ’ t think so, but look up this wallpaper, Waverly Fabrics, and see if they have a rose pattern. - OK. Oh, what about this one here? - That ’ s not it. - No? - No, first of all, I was an art dir... I was a creative director at an advertising agency. The first thing I know is what I ’ m looking at. Ilon the girl. I don ’ t remember particularly being into advertising, but somehow I found myself in it. And I seemed to be very good at it. First of all, I ’ d always listened to everybody. And I noticed that I saw things differently than other people. - Women could get somewhere in advertising, but they were marginalized. - Women were given a lesser role. All the men were always arguing with you and always taking credit. - And one of the things that was coming through loud and clear is that women are for men. - Glory, glory, hallelujah Glory, glory, hallelujah Glory, glory, hallelujah His truth is marching on - Men were always the ones who everyone was pleasing. - What will your eyes say to him tonight? Wait till the fellas take a look at this plunging backline. - You look at ads from that period, and it ’ s jaw-dropping. - This is one of our nation ’ s most beautiful sights, the kind of girl who keeps her figure. - But that ’ s what feminism was pushing back against. - I was a feminist. I knew women were equal if not superior to men from the time I was a little girl. - And in advertising, she wanted to be a trailblazer that way. At one point, she worked in the same agency with my father, and that ’ s how they met. - He was a good writer, and I was a good writer. I liked the fact that he was good at what he did and cared about it. We got engaged, and then we got married. - I ’ m sitting there in the background. I ’ m grinning. I ’ m so thrilled because I adored her instantly. I thought she was super cool. And she treated me like I mattered. - She was a sweet little girl, and she deserved some caretaking. My mother was kinda terrible, so I knew that she needed support, and I gave her support. And that was what we did. - She was interested in me. She listened to me. Treated me like I was lovable. - She didn ’ t have anyone to recognize her being, to support her, to see who she could be, who she was. - She was very interested in what I would do with my life, and the assumption was that it would be something worthwhile. She just went way, way beyond the call of duty to make me feel like I was worth it. And she did that for a lot of other women, too. - I was working at McCann Erickson. We were given an assignment. L ’ Oréal, they wanted a campaign for Preference, it ’ s a hair color.

- She ’ s in this room full of men brainstorming for this ad. - The men were busy flirting with the women, telling us we all looked like models and we didn ’ t look like advertising people, you know? - They ’ re like,”Oh, this woman is gonna be standing by an open window, and the wind will be blowing the curtains.”And she ’ s thinking, you know, this is just making the woman totally an object. - I was feeling angry. I ’ m not interested in writing anything about looking good for men. Fuck ’ em. Fuck you too. - Why fuck me? - Well, you ’ re a man. - Yeah, but I ’ m here trying to help you tell your story. - Yeah, that ’ s good. I like that part. I was pissed. We weren ’ t there to just dance for the men. - It is what she said. She just thought,”Fuck you.”And wrote this. - I just opened my eyes and knew. Because I ’ m worth it. The campaign got approved, but all the men were always telling you that you were doing something wrong or they knew how to do it better. - The agency insisted on a male voiceover. - She uses the most expensive hair color in the world, Preference by L ’ Oréal. - You know, it was just totally different. - Smooth and silky, but with body. It feels good against her neck. Actually, she doesn ’ t mind spending more for L ’ Oréal. Because she ’ s worth it. - It was wrong. This was not for men but for women and for other human beings. - I use the most expensive hair color in the world, Preference by L ’ Oréal. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color, I expect great color. What ’ s worth more to me is the way my hair feels. Smooth and silky, but with body. It feels good against my neck. Actually, I don ’ t mind spending more for L ’ Oréal. Because I ’ m worth it. - I wrote that. - It was everywhere. It was a national campaign. - Yes, it ’ s expensive, but I ’ m worth it. - What they realized after they started running it was that that was only the tip of the iceberg. - It was supporting women, finally. - And I ’ m worth it. And I ’ m worth it. I ’ m worth it. - It ’ s a four-word feminist manifesto, defying this social pressure to think about yourself in relation to men. - You ’ re worth it. I ’ m worth it. I ’ m worth it. - It was a very big success. - At a certain point, L ’ Oréal adopted it as the tagline for their entire business. - Parce que je le vaux bien. - Even today, that ’ s their motto. - Because I ’ m worth it. - ’ Cause you ’ re worth it. - You ’ re worth it. - We ’ re worth it. - And I ’ m worth it. - She wrote the most influential tagline in women ’ s beauty advertising. - I ’ m worth it. - It ’ s magic, that phrase. - Those words just get better with age, don ’ t they? - I ’ m worth it. Because you are and always will be. - She came up with that line because she believed it. She believed it about herself, and she believed that women should believe it. She did that for me. You know, even after they split up, she was still a big part of my life. I ’ ve always kind of felt like I dodged a bullet that I have watched take down other members of my family. She saved me. And today, I am a professor of English at Williams College. I teach Victorian literature, and I also teach women ’ s, gender and sexuality studies. She was my mother. She also taught me how to be a mother. I had been very uncertain about motherhood for myself because my role model for that was so nightmarish. But Ilon really taught me, I can do this. I can be a good mother. She made me believe that I was worth it. And I was not either the first or the last. - I have to go away now. Go to my room. - It ’ s been lovely. - It ’ s been lovely. Oh... I ’ m not interested in advertising. I don ’ t give a shit. It wasn ’ t like it was my whole life. It was just a portion. It ’ s about humans ; it ’ s not about advertising. It ’ s about caring for people because... we ’ re all worth it, or no one is worth it.

Text Sample 476: NEG Dim 3 – 2025 – Score 3.00 – 2025/t003002.txt
HW : Hi everyone, I am Helen Walters. I am head of Media and Curation here

at TED. Welcome to another episode of TED Explains the World with the one and only, Ian Bremmer. It is Monday, February 24, a little more than a month since President Trump was inaugurated once more, and safe to say, a lot has been going on. So we figured we ’ d check in with Ian to determine what we should really be paying attention to amid the extreme noise. Ian, hi. Ian Bremmer : Helen, great to be back with you. HW : So we asked the TED community to share their questions for you, and we wanted them to share what they ’ re most curious to know. And we really got some amazing questions that I plan to pepper this conversation with. So let ’ s start with one right now. Two months into 2025, where does the US stand? IB : Very powerfully, in the sense that the US economy, of course, is performing considerably better than any of the other G7 economies coming still out of the pandemic. Technologically, only close competitor is China, ahead of the US in some areas, behind the US and other critical areas. But compared to every other country in the world, the US is head and shoulders and neck or waist above them. Militarily, of course, the US is the only country with global ability to project power. No one else is close. And the US dollar is still the global reserve currency, no one is approximate challenger. I say all of those things because those are the things that haven ’ t really changed over the course of the last few months, but I suspect that isn ’ t what the questioner was asking. What they were really asking is what ’ s happening politically. And politically, the United States is unwinding its own global order. It is no longer particularly interested in promoting collective security or NATO. It ’ s no longer really interested in promoting involvement and leadership of multilateral institutions, of consistent rule of law and free trade that ’ s well regulated, certainly not very interested in promoting democracy around the world. I mean, this is an environment where other countries around the world have to figure out how to adapt to the United States, and that the US has become the principal driver of geopolitical risk and uncertainty in the world today, unlike any other time in my lifetime and yours, that ’ s perhaps the biggest change. HW : That is quite a statement. Alright, you are a geopolitical expert. Let ’ s talk about Europe. So Defense Secretary Pete Hegseth stated that America ’ s foreign policy focus no longer lies in Europe. Vice President JD Vance caused some raised eyebrows, dropped jaws and, I suspect, some choice expletives after his recent speech at the Munich Security Conference, in which he accused European leaders of suppressing free speech and warned of the threat from within. So you were in Munich. Was it as dramatic as the stories we read had it? And what happens next? IB : It was. I was in the room. I was maybe 30 feet away from him when he gave that speech. Standing right in the front, right next to me was the president of Czechia, the president of Finland, the prime minister of Sweden, watching all of them and their reactions. What he said, let ’ s keep in mind, this is the Munich Security Conference. It ’ s been going on now for something like 61 years. And he was the head of the US delegation. And every year the head of the US delegation gives a big speech on the state of the transatlantic alliance and on global security. And he didn ’ t do that. I mean, the people in the audience were expecting a challenging speech. They were expecting that the Americans would be less committed to the alignment with the Europeans on Ukraine. The US had just had that direct Trump phone call with Vladimir Putin, 90 minutes long. Hadn ’ t coordinated that with the Europeans, never mind the Ukrainians, in advance. So, I mean, they were definitely prepared for a very challenging plenary. But Vance did none of that. Vance said,”I ’ m not going to talk about Ukraine or Russia or China, because the biggest problem is actually what ’ s happening inside Europe. The biggest problem, essentially, are you guys sitting in front of me, that I ’ m speaking to, because you ’ re suppressing free speech. You ’ re suppressing the far right.

You ’ ve been infected by the woke mind virus, and you aren ’ t real democracies as a consequence.”And specifically he said, and this is something that I think is underappreciated in the United States, he attacked the German firewall. And said that – And let me explain what the German firewall is. That is a principle by the leaders of all of the mainstream German parties and their supporters that they will not, under any circumstances, work with, enter into coalition with the Alternative for Deutschland party, the AfD. Because the AfD, under surveillance from German intelligence agencies, is considered to be a neo-Nazi party. And the United States, of course, after World War II, led de-nazification of Germany. The day before, Vance had actually gone to Dachau and visited a concentration camp, and then came to speak in Germany about the firewall needing to be ended. And at that point someone yelled out from the crowd,”This is unacceptable!”And it was right in the front. And people in the room, about 1, 500 people in the room, standing room only, couldn ’ t see who it was. They ’ d just see the back of his head. I saw who it was. It was the German Defense Minister, Pistorius. And in 15 years of me going to the Munich Security Conference, I have never seen anything remotely like that. And the reason I mention all of this, and of course, on the back of that, let me be clear, Vance refused to meet with the German chancellor because he ’ s basically, well, he ’ s not relevant. He ’ s only going to be there for a couple more months until a new government, so why should I bother? But does meet with the leader of this AfD, goes and meets with her directly, does a bilateral that evening. So that is a very important backdrop for this last weekend ’ s German elections, where the victor, Friedrich Mertz, who is going to be the next chancellor, actually referred specifically to America ’ s intervention in German democracy and support of the AfD being every bit as bad as what the Kremlin is doing inside Germany, and also said that the Europeans need to develop a defense policy which is independent of the United States, in other words, essentially saying that it will be the end of NATO. Again, staggering comments, unheard of, impossible to imagine even two weeks ago. And that ’ s why it ’ s very critical that you have the context of what happened between Vance, Elon Musk over the past couple of months, Trump, with the Germans, with the Europeans. So you can understand why the incoming German chancellor would make a statement like that. HW : What should we make of all of the conversation about the rise of Nazism, the return of Nazism? Obviously, the AfD just won 20 percent in the German election. We ’ ve also seen people allegedly giving Nazi salutes at various US conventions and speeches. Do you think this is overblown? Is this a fear that people have rationally? And what should we make of that? IB : In Germany, the Alternative for Deutschland, are winning all of former East Germany. This is a group that increasingly understands that they ’ re never going to really catch up to the rest of Germany. Remember, Germany hasn ’ t really grown in five years now. They ’ ve been in recession for the last two. And the average citizen, the average voter, living in former East Germany no longer are just angry about not catching up. They ’ re basically saying,”I ’ m done with this system.”So that ’ s why you ’ re getting this very strident, revanchist nationalism that is, you know, it ’ s not just taking a hard line on illegal immigrants. The AfD even supports removing citizens of Germany that they claim have not adequately integrated into German culture and nationality. And so, I mean, this is some very, very strong stuff, stuff that really does feel like what they were doing back in Nazi Germany. But in former East Germany, they win. They get the largest number of votes, where, in much of the wealthier part of West Germany, including, you know, where I was, in Munich, the AfD can ’ t break out of single digits. So it ’ s a very, very divided country. And also the AfD is doing very, very well among young men. There ’ s a clear gender divide here as well, just as there is in the

United States. In the US, I mean, I, of course, saw Elon Musk and that, you know, sort of grabbing his heart, saying,"My heart goes out to you"and then doing what appeared to be mimicking a Nazi salute. And then Steve Bannon doing the same without the initial heartfelt comment. It clearly was trolling. Clearly was trying to get a reaction from folks. It ' s very performative. And it ' s meant to also, you know, attack and attach, make the left lose their minds on something that isn ' t very critical to what the Trump administration is trying to do compared to the revolutionary changes inside the US government that they are very much prioritizing right now. So I ' m very concerned about the rise of neo-Nazism in Germany. I am very much not concerned about that in the US. I ' m concerned about other things I ' m thinking is being used as gaslighting for those that can use distractions or can be distracted by them. And I would also say that this election in Germany wasn ' t surprising. It was exactly as the polls have expected for several months now. But the key elections in Germany and the key elections in Europe are the next cycle. I ' ve spoken with a lot of people around the Trump administration that believe that the Europeans are one electoral cycle behind the United States. So in other words, Trumpism is coming, just not quite yet. And so in the UK, you ' ve got Labour for a full term. But the Reform Party is increasingly the most popular. And just with a little nudge, maybe a little external money, and Elon said he ' s already thinking about giving them 100 million dollars, which is a massive amount in UK politics, not so much in the US, that the Conservatives would fragment, a rump group would join with Reform, and they could win the next elections. In France, you ' ve had all of these governments continue to collapse. Macron, very unpopular. Marine Le Pen and the National Rally party could easily win upcoming elections in 2027. In 2029, in Germany, if the Germans are incapable of turning their economy around, incapable of unwinding their debt break and spending some of the capital that ' s just sitting there and as their debt to GDP goes down, but their economy contracts at the same time, then AfD could easily win in 2029, and then the firewall is no more. And then, of course, in the European Union elections overall, in five years ' time, now four and a half, you could easily see the Patriots front, which is the equivalent to the far right grouping that these parties are a part of, they could all come together and be the dominant party. And then the EU as we know it is really a thing of the past. So these are existential changes that are happening not just for NATO right now, but also for the EU and quite plausibly, for the future of democracy as we know it. HW : Alright, let ' s turn to Ukraine. So President Trump accused President Zelenskyy of starting the war with Russia and called him"a dictator without elections."Last week, senior American and Russian officials met in Riyadh to discuss a potential cease-fire with no Ukrainians present. Now Zelenskyy has offered to step down in return for Ukrainian admission to NATO, although apparently he doesn ' t really mean this. So what happens to Ukraine now? What should we be paying attention to? IB : Well, I mean, I think he would mean it if NATO membership happened. I ' d love to see Trump call his bluff on that. But of course it ' s not going to. And he wasn ' t saying that six months or a year ago. He ' s saying it now that NATO is truly being pulled off the table. I think it ' s the right thing for him to do. It is, again, performative and helps to remind those around the world that Ukraine is fighting for self-determination. They ' re fighting for the ability to join their own alliance, as any country in the world should be able to do. But that is not Trump ' s belief. Trump believes that what you get to do is determined by how powerful you are, not by the fact that you happen to run a country. And indeed, the territorial integrity is something that is afforded to powerful countries, but not so much for weak countries. And that ' s the way he feels about Panama, and that ' s the way he feels about Denmark and Greenland. And that ' s certainly the way he feels about Ukraine. So right now, as you and I are speaking, the United States is putting forward a new UN resolution at the General Assembly that says that the war has to end as fast as possible but does not recognize Ukraine ' s territorial integrity, which has been a core precept of the international order that the United States has supported since creating the United Nations and since the end of World War II in 1945. The Americans are now throwing that out. And the Russians, of course, are fully supportive of that. America ' s allies in Europe are not. But the US has gotten the Saudis to agree, and they ' re getting a unified Arab block. The Americans have privately worked with a lot of poorer countries in Africa, in the Asia Pacific and others, smaller countries, and saying,"You ' re not going to get any aid from the US unless you vote in favor of this new resolution."I fully expect the resolution is going to pass. But of course, what that means is that the Americans are preparing to cut a deal on Ukraine over the heads of the Ukrainians. In other words, the US and Russia together will figure out what a ceasefire should and should not entail as part of a broader US-Russia rapprochement. This is obviously a disaster for the Ukrainians and the Ukrainians who initially refuse to accept this so-called deal, on Ukrainian natural resources that would grant the Americans $500 billion in access to such resources in return for"paying off"aid that the Americans had granted to the Ukrainians without conditions or strings, but apparently no longer. And so, you know, what does this all mean? Well it means it ' s yet another case of why people shouldn ' t trust the United States from one administration to the next, because foreign policy will change completely. And what you thought was an American commitment won ' t stand. It shows that there is a fundamental rift between the Americans and the Europeans and indeed, Emmanuel Macron in the United States right now, as you and I speak, hoping that he can do something that will keep Trump from, as Macron believes, essentially surrendering to, conceding to, the Russian position and maintaining some level of coordination with the Europeans. But he doesn ' t have a lot to offer. And the Ukrainians, you know, now in a position of desperation where maybe they will end up signing over a lot of their natural resources to the Americans. But what, if anything, will they get in return for that? And how empowered will the Russians be in any deal that is cut? And will an independent Ukraine even be able to survive, and what will the terms that will be required of them? Of course, Putin is saying things like, no European troops of any sort, whether in a NATO formulation or not, would be allowed as peacekeepers in Ukraine. Well, then, how would you guarantee, what kind of security commitments would you give to the Ukrainians? These are all open questions, but apparently questions that President Trump is prepared to largely resolve on Russia ' s terms. And again, this is such a dramatic change from where the United States and where the Western order was just a month or two ago. HW : So let ' s take some quick-fire questions from our community related to this topic. And I think the first one is particularly pertinent to what you just said, so it ' s simple. What does Putin have over Trump? IB : I don ' t think that Putin"has"anything over Trump. If he did, then why wouldn ' t Trump have granted Russia more in his first term? In his first term he gave the Ukrainians those javelin anti-tank missiles that Obama refused to give. And, you know, he thought they were too risky. He increased sanctions against Russia. Now that was his administration with a lot of hawkish in orientation advisors around him that he doesn ' t have this time around. They ' re all directly loyal to him. But, I mean, if it was a priority for him, he could have stopped it. And if the Russians had something over him, then why wouldn ' t they have used it in that case? It feels like a conspiracy theory. I don ' t buy it. I think very differently. We can explain what ' s happening because number one, Trump wants to end wars. It

’ s very popular with his base. He ’ s said this on Gaza. He ’ s said this on Iran, where Israel wants him to support attacks on Iran ’ s nuclear capabilities, and Trump has said no. And he ’ s showing this on Ukraine. He doesn ’ t want to spend money on the Ukrainians going forward after spending over 100 billion dollars under the last three years from the Biden administration. He doesn ’ t want to spend that money going forward. And he also doesn ’ t value a strong Europe. He actually thinks that a weak Europe is in America ’ s interests, where you have more Brexits. He used to always talk to the French about that during his first term,”Why won ’ t you do Brexit?”He wants all of these MAGA-type populist parties, including the AfD, to win in Europe because that legitimizes his own popularity. Just like he loves Milei in Argentina or Bukele in El Salvador. And so I think if you put those things together, you actually get to why Trump is doing what he ’ s doing with the Russians without needing to create a story about Trump somehow being threatened by some secret, shadowy information that the Russians have on Trump that they ’ ve let him know about but nobody else knows. I just don ’ t buy that. HW : Great. OK, another community question. If the US hands Ukraine to Russia, will Russia keep going? Will they attempt to take over Europe? IB : It ’ s interesting that Trump just met with the Polish president, President Duda. It was supposed to be an hour meeting, it was only 10 minutes. So he kind of embarrassed him at home in Poland. But he did say that the US is still committed to maintaining a troop presence on the ground in Poland. And so it ’ s very hard to imagine the Russians rolling through Ukraine and then into Poland, even though there ’ s been lots of asymmetric warfare, for example, and even there have been, you know, Russian missiles launched into Lviv, Ukrainian air defense, missile defense, and with Polish villagers getting killed because of the fighting. So, I mean, there ’ s been spillover. But America is still going to be in Poland. Why is that? They get along pretty well, and the Poles are heading towards five percent of GDP defense spend this year, which has been the high water mark of American demands of defense spend. So he ’ s, you know, been consistent on that over the past months. Baltic states. I mean, they ’ re all spending a lot more money on their defense. And so, I mean, in principle, you could imagine the United States maintaining these rotational deployments in the Baltics, which would prevent the Russians from invading. But might Trump say, as part of a deal,”Why do we have all those troops there?”You mentioned Pete Hegseth saying the Americans need to get troops out of Europe and get them to Asia, which is where their principal competitive, you know, sort of, strategy is going to be oriented. And the US hasn ’ t been pressing the Japanese or the Indians, the principal partners, allies in Asia vis-à-vis China, the way they ’ ve been pressing the French, the Brits, the Germans in Europe. So I think it is certainly plausible that the Russians will do more, especially with those countries that refuse to align with a Trump rapprochement towards Russia. The countries I would be most worried about are Moldova, are Georgia. I ’ d be most worried about the countries that are in the so-called”near abroad”that aren ’ t a part of NATO, and aren ’ t going to become a part of NATO, and therefore are much more vulnerable. But certainly in terms of Russia ’ s willingness to engage in espionage, to fund actors on the ground for arson attacks, assassinations, critical infrastructure attacks on fiber networks, on pipelines, I think all of those things are likely to increase against European states across the board, on the back of this US-Russia rapprochement. HW : OK, final community question for this segment. What happens if the US no longer supports NATO? IB : It ’ s possible that the United States is moving towards a similar posture in Europe that they presently have in Asia, which is strong individual bilateral deals with a number of countries : South Korea, Japan, Australia, but not a collective security agreement, where those countries together have more of a call on the United States and there ’ s a lot more free riding. So I could imagine that NATO falls apart and becomes that. The danger is that even though the Europeans now understand the nature of the threat, that it is very plausible and maybe even baseline, that the Americans are going to leave Europe collectively on its own. That that seriousness does not equate to the ability to increase spending adequately in the near term. The Germans did manage to pull together an electoral outcome that will allow for a so-called grand coalition, just barely. So you ’ ll have a two-party government in all likelihood instead of a three. But you also have a blocking capacity on the part of the far right and the far left that will prevent the Germans from really blowing out their spending, including their defense spending, unless they can pull something off before that new government is formed. Which is possible, but it ’ s unlikely to be very big. The Brits are now talking about maybe moving the 2. 5 percent of GDP defense spend over years with a very challenging fiscal environment. The Spaniards, not even close, the Italians, not even close. So what does a European military capability really look like? And without the Americans, the answer is it ’ s not there. It ’ s not there. So I think that if NATO falls apart, the reality is that most of Europe will be incredibly vulnerable to external attack, especially Russian attack, And there won ’ t be a way for them to defend themselves, and some of them will break and therefore want to work more directly and individually with the United States. And that could be the end, not only of NATO, but it could be the end of the European Union. I think this is, the stakes are incredibly high, this is an existential challenge for European governments that needed to take this seriously 30 years ago, 20 years ago, certainly in 2014, when the Russians invaded Ukraine, certainly, certainly in 2016 when Trump was elected, and certainly, certainly, certainly in 2022 when the Russians then invaded all of Ukraine. And at every point the Europeans were basically saying,”We ’ re still fine with the Americans basically doing this.”And now they are in very serious trouble. HW : No time like the present, I suppose. Alright, let ’ s move on. So I mentioned the discussions in Saudi and obviously also on the agenda there with the discussions about Gaza and Trump ’ s proposal to build a Gaza Riviera. What should we make of that? Or as one of our community members asked, what is the actual solution for Gaza? And surely it isn ’ t what the US is proposing. IB : Well, look, here ’ s what ’ s interesting. You know, Trump says a lot of things, and when they don ’ t work [out], he backs out fairly quickly. And you know, I thought – just stick with Ukraine for one second on this. It was interesting to me that Trump has said that he doesn ’ t want to talk to Zelenskyy because he has no cards, he has no leverage. And so why should he let him at the table? And so Trump ’ s going to cut the deal. Very interesting that if that is true, then why is it that the Secretary of Treasury brings this deal and says,”You ’ ve got to sign this right now, Zelenskyy”And Zelenskyy refuses. And then a few days later, the Americans come back with a new deal with altered terms that are not quite as predatory vis-à-vis Ukraine. Why would the Americans change their tune for someone that has no leverage? Because, I don ’ t think it ’ s because Trump is a nice guy. He ’ s not going soft at 78. I think it ’ s because he overstated how little leverage Ukraine has. Ukraine is very popular among the GOP in the House and Senate, far more than Putin. Zelenskyy is far more popular in the United States among the voting population than Putin. And of course, he ’ s also much more popular among the Europeans and even globally than Putin. And so it turns out that maybe there is a little bit of leverage that Zelenskyy has, and maybe Trump does have to give a little bit more. So this is relevant in the context of Gaza, because Trump was very annoyed that, you know, despite his ability to get a cease fire between Hamas and Israel over the goal line, that his friends in the Middle East were not coming up with a plan for Gaza. Where they were supposed to provide security and provide investment and lead reconstruction and deal with the Palestinian populations in the interim. And that wasn ’ t happening. And so Trump then meeting with Prime Minister Netanyahu from Israel in Washington, then after that, summoning the Jordanian king, who was the ally that is most reliant on the United States and so, therefore, most needing to say yes to whatever Trump orders of him. Basically says, as you mentioned, Helen, that Gaza is going to be a Riviera, that the US is going to turn this into an incredible place. They ’ re going to build wonderful homes for the Palestinians someplace else. And the Palestinians will all volunteer to leave Gaza and go to those homes, and Gaza will be depopulated and new people will come in and live there. That was the plan. And when he was asked, standing there with King Abdullah from Jordan, under whose authority? Trump ’ s response was :"Under my authority."And that went over like a lead balloon among America ’ s allies in the Middle East. They all said,"We ’ re not up for this. We ’ re not resettling Palestinian populations. They won ’ t actually leave voluntarily. This will would actually be a huge problem for Israel ’ s own national security long-term. And we ’ re going to come up with another plan. And our plan will keep the Palestinians on the ground in Gaza."At which point Trump pivots to work with the new plan. And his advisers say that Trump was only trying to start the conversation. And what he said was his plan is not actually what he ’ s going to do. So, what ’ s sustainable? What would be sustainable would be a lot of Gulf money going in and maybe some European money too, though they ’ re going to be really pressed, given everything you and I just talked about, to try to reconstruct a completely devastated Gaza that eventually some two million-plus Palestinians will be able to live normal lives in, and the governance will be provided by some technocratic group of a Palestinian Authority that will be working with, selected by, those that are involved in the reconstruction and the security. Namely the Egyptians, the Jordanians, the Saudis and the Emiratis. That is the best possible deal that they ’ re going to get. But that ’ s happening as Israel has just evacuated forcibly another 40, 000 Palestinians from refugee camps in the West Bank, sending tanks in for the first time in decades and probably precipitating more Israeli settlers taking more land on the ground in the West Bank. And so, as we are possibly taking a small step towards a sustainable solution in Gaza, we are taking a slightly larger step away from a sustainable solution for the Palestinians in the West Bank, twas ever thus over the past decades in this incredibly horrible problem and conflict. HW : And why is Israel doing that? I mean, are they feeling emboldened by the US, or why the incursions in the West Bank now? IB : Well, I think they ’ re feeling emboldened by their own successes. They have shown that they are the dominant military power in the Middle East, and they have the ability to determine the level of escalation unilaterally, and their adversaries and enemies can ’ t really do any damage to them. They proved that after October 7, in their response to Hamas. They proved that in decapitating and taking out the military capabilities of Hezbollah. And they ’ ve proved that also with overturning – they didn ’ t do it, but the fact that the Assad regime in Syria is gone, which means that the Iranians no longer have a land bridge to get additional weapons to what had been the strongest adversary of Israel militarily, Hezbollah. So all of that has shown that Israel can decide outcomes. They ’ re the ones that have all the power here. And also that depopulation of Gaza is very popular among Israelis. There was a poll recently in the Jerusalem Post, which is pretty mainstream in Israel. 80 percent of Israeli respondents said that they wanted to depopulate Gaza of all Palestinians. Taking more land is very popular, especially in the West Bank, especially among Netanyahu ’ s present right-wing coalition, that he would like to keep together, to continue to govern. And this is a sop to them as he is looking to potentially extend the ceasefire in Gaza and move from phase one to phase two, which is facing some delays and open questions right now. So, yes, certainly, the United States has been underpinning that military capability of Israel. And I think it is instructive to remember that not only does the US provide more military aid in peacetime to any country in the world, to Israel, but that unlike in Ukraine, where the Trump administration is saying,"No, I ' m not happy with the grants, you ' ve got to pay that back,"The billions and billions that the United States sends to Israel, nobody is suggesting that that should be alone or that the Americans should get technology rights or mineral rights or anything else from Israel, despite the fact that Trump ' s policy is America First. Israel is a very clear exception to America First in Trump ' s visioning of it. HW : Alright, so we talked about the fact that the foreign policy focus of America is shifting to Asia. So what are you hearing about how China is feeling in all of this? And what should we be paying attention to on that front? IB : China has had a couple of good weeks economically and specifically on the back of that huge announcement of DeepSeek, which has, you know, far less money behind it than the comparable American chatbots but is performing at an extremely high level. And, you know, given the fact that there are all of these export controls on semiconductors and other parts of advanced technology, that this is giving people more of a belief that the Chinese really are capable of driving innovation even in that space. And they ' re, of course, dominating the post-carbon energy space. We see that with electric vehicles and batteries. But we ' re also seeing that with massive investment now in green hydrogen, for example, we ' re seeing it in advanced nuclear capabilities, all of these things. And so there is more investment in that space that the Chinese are not only driving as a state, but that they ' re also increasingly able to raise and move in their own private sector. So the story that you and I have been discussing for the last couple of years is the Chinese economy is radically underperforming. That is still true, but there is an emerging counter-narrative here that I would be remiss not to mention. Now more broadly, what the Chinese here are seeing is that Trump ' s direct policies are going to hurt them economically. His tariffs that he ' s already announced on China, the 10 percent, on a whole bunch of Chinese goods for export, the reciprocal tariffs that are global, that will clearly hurt China and that they ' ll have a hard time negotiating out of, and also the willingness of the United States to expand their export controls, their sanctions on China, especially in areas that are considered relevant, even loosely relevant to American national security. All of that is going to constrain Chinese growth. All of that is going to negatively impact the Chinese economy. That ' s the downside. And the Americans are doing that not only in bilateral relations with China but also pressuring other countries to get Chinese tech out of their supply chain. And you see that with Americans pushing other countries that produce semiconductors to take the Chinese out. They ' re pushing American allies and trade partners to get China out of the export chain. So Mexico being pushed very hard by Trump to stop allowing China to pass goods through Mexico into the United States. India, same conversation. Vietnam, same conversation. A lot of that going on. So baseline, the US-China relationship is getting worse. But the Chinese see huge advantages in America ' s reversion to unilateralism that we haven ' t really seen since the original America First movement in the run up to World War II. Charles Lindbergh saying,"Don ' t get into the war, don ' t fight the Nazis. The US is far away. This isn ' t our our problem."And now you have the Americans not only doing that with Russia and Ukraine with the new America First, but you also see that happening with the Americans pulling out of the

Paris Climate Accord, again, pulling out of the World Health Organization. You see it with the Americans shutting down USAID. And America has historically been responsible for about 40 percent of global humanitarian support and aid. And absolutely, some of that is corrupt. Absolutely, some of that is on woke programs that the vast majority of American taxpayers would never support. But the majority of that money is actually spent on programs that are really important in advancing American soft power and influence over countries in the Global South, all over the world. The Chinese know that. That ' s why they ' ve started humanitarian programs. It ' s not out of some great love for, you know, providing foreign aid for these countries. It ' s more because they see it creates influence. Now the Americans are leaving a vacuum that the Chinese can absolutely take advantage of. And we ' ll see this across, you know, sort of, microstates in the Pacific. We ' ll see this across sub-Saharan Africa. We ' ll see this across South America. The Chinese see huge opportunities from the United States creating a leadership vacuum. And the place they ' re most capable of doing that is if the US stops paying dues in the United Nations, where China is number two, and they ' ll suddenly have the most influence over all the key positions, which will give them a role in global governance that they ' ve really been constrained from having over the past decades. HW : What do you think are the chances that the US stops paying its dues to the UN? IB : You know, if it were up to Congress and only Congress and Trump doesn ' t lean on them, I ' d say fairly low. I ' d say that they ' d probably, you know, cut back on some programs. But in terms of baseline dues, they ' d keep paying it. But Trump personally, I think increasingly sees the UN as hostile to the United States. That ' s particularly true in terms of Israel policy, which Trump is leaning in on. And, you know – let me put it this way. Tulsi Gabbard did not have the votes to get confirmed. There were Republican senators that were going to vote against her. And Elon called those individual senators and threatened them. And this was, you know, fully aligned with what Trump wanted Elon to do. And those votes went away. And the only person that ended up voting against Tulsi was McConnell. And as a consequence, it went through. So look, I think that we should not underestimate the level of power and control that Trump and Elon have over the Republican Party this time around, compared to 2017, where it was really the Republican Party that was much stronger than Trump, both in his administration creating a lot of guardrails, but also in Congress. This time around, Trump has all the chips and he ' s using them. So if he decides he wants to stop paying dues, I think that ' s what ' s going to happen. This is a much more revolutionary presidency, domestically, where last time around it was much more transactional. HW : Alright. Let ' s talk a little bit more about what ' s actually happening inside the US. You bring up Elon, who obviously with DOGE has been sending out emails and asking for information from people and kind of trying to do a lot very quickly. Some of these high-profile moves are – I think what ' s interesting is that no one would really argue that there isn ' t inefficiency in government, that that ' s an understood reality. But the way that some of these moves are being made are maybe going to cause irreparable damage. And that ' s to people from red states, that ' s to scientists, that ' s to veterans, that ' s to people who voted for Trump and that ' s to people around the world. You talk about soft power and the United States, but what about actually what ' s happening inside the United States? And why do you think that the administration is willing to go so quickly, to go so fast and to cause such damage? IB : Well, and you saw Trump this weekend saying that he wants Elon to move faster, to be more aggressive. And I don ' t think that was just a troll. I think that ' s true. Trump is 78 years old, and with his friends he talks about the fact that he doesn ' t have much time. That in two years ' time, the

Democrats could take the House in four years ' time, you know, that ' s it. He ' s not talking privately of"I ' m going to be president for life."He also knows he was almost killed. He was shot in the head. And do I think that changes somebody? Absolutely. I think that he believes that he was saved by God, and that his level of confidence that what he is doing is fundamentally right and necessary now is so much higher than the insecurity that I frequently saw in him in his first presidency. So I think his willingness to push, push, push, and his view that Elon is a great activator and executor on that intention, and someone that he had nowhere close to someone with that kind of capability and energy and execution ability in his first term. And now he does. So first of all, I think that relationship between Trump and Elon isn ' t going anywhere. I think it ' s actually very stable. People that say it is are people that want it to go away. But hope is not a strategy. And I also think that the intention of revolutionary politics domestically is to shoot first and ask questions later. It ' s to break things. It ' s a view that the bureaucracy has been weaponized against Trump first and foremost, and also against the will of the American people. And therefore it must be broken. And you ' re going to break eggs when you want to make that omelet. Now your point is that there are a lot of Republicans that work in government. There are a lot of Republicans and Trump voters that are losing their jobs and that are not being treated with a level of care and compassion as a result of what Trump and Elon are doing. And I think there will be a backlash. But I ' m not so sure that a one-term, 78-year-old-Trump presidency cares all that much about that. And, you know, if that means that his popularity slips to like, the high 30s, frankly, you know, in a very, very polarized country, high 30s isn ' t that bad. It still means 80, 90 percent of the people that voted for him still support him, and I think he ' s probably more comfortable with that this time around. So I think a lot more has to happen before we can start talking about whether there ' s really going to be significant pushback against what Trump is trying to accomplish at home. HW : Do you expect to see any of Congress standing up to Trump? We saw the governor of Maine recently kind of take him on. Do you expect to see more of that, or do you feel like for the next two years at least, people will just go where Trump takes them? IB : Well, sure, but, you know, in the case of Maine, that ' s a Democrat. So I don ' t know how much that means. I don ' t expect to see a lot of Republicans standing up against him. And of course, since the Republicans control the House and the Senate, that ' s the only thing that really matters. I mean, you know, you saw Kash Patel got through, Pete Hegseth got through. Now look, I actually think that a lot of the members of cabinet are very capable. I mean, they ' ve been picked for their loyalty and they will be loyal to Trump, but they ' re very capable. I think JD Vance is very capable. Mike Waltz, I know quite well. Marco Rubio, over the years. These are people that are very well respected, have been by their colleagues and could have served under any Republican administration. But there are some people that are completely incapable for the roles that they have been selected, completely incapable. They would not be selected in a role in any cabinet, in any democracy. And yet here they are, serving in critical roles in the most powerful country in the world. And yes, I ' m talking about the Secretary of Defense. I ' m talking about the Secretary of Health and Human Services. I ' m talking about the Director of National Intelligence. I ' m talking about the Director of FBI. These are important roles. And they were all confirmed. And if you ' re a Republican senator – and senator – I ' m not talking about the House here where, you know, it ' s a lot more diffuse and there are a lot more wackos and it ' s easier to get in. But Senate has always been like, responsible and more oriented towards bipartisanship and the rest, and they are utterly petrified of what it means if they publicly get crosswise with President Trump and number two, Elon. And they ’ re not willing to do it. So I think that there ’ s very little belief that you are going to see Congress step up. I think that you will see justices step up because those justices are independent. The problem is that many of those justices are progressive and have a known progressive lean. And so when cases come up and justices that have that political lean oppose something Trump wants, his willingness, his administration ’ s willingness to attack them and refuse to execute on that judicial decision, unconstitutionally, is real. Now the case will of course go then up to the Supreme Court eventually. And I don ’ t feel the same way about Trump refusing to listen to adhere to the rulings of the Supreme Court. But if there ’ s 100 rulings like that and you overwhelm the judicial process, you quickly see how you might erode a core check on executive power in the United States. And Trump and Elon clearly intend to every day test those checks and balances and break them if they can. I mean, that is the intent, that is the intent. HW : So you mentioned democracy. When we last spoke, you said very clearly that the US is not Hungary, meaning that democracy is not threatened and that the US is not in danger of turning into an autocracy. Do you stand by that? IB : Yes, I do. I mean, in the sense that I think in two years we ’ re going to have midterm elections, and they will be largely free and fair. And in four years we ’ ll have presidential elections. And US elections are held at the federal level, right? They ’ re held state by state. They make the rules. It ’ s very easy to make people believe that the election has been stolen. It ’ s extremely hard to actually rig an American election. And I don ’ t believe that that is very likely at all. It ’ s very unlikely. The American military, despite the firings that we ’ ve seen just last Friday, is still an independent and professional military that I believe will carry out its orders and its duties professionally. I see the judiciary in the United States as independent and a clear check on the executive power in the United States. Where I see a problem, is that the US is becoming more kleptocratic every day. And the US was already much more of a kleptocracy over the past 10 years under Democrats and Republicans than any other advanced industrial democracy in the world. Clearly, money buys power in America in a way that it doesn ’ t in any other major democracy, wealthy democracy. That is getting dramatically worse. I saw specifically that Elon Musk, who is a walking conflict of interest at the highest level, met with Narendra Modi, the Prime Minister of India, in Blair House, so part of the official White House complex, right after Trump met with him. And Trump was asked later in the day, was Modi meeting with Elon in Elon ’ s private-sector capacity to advance his business interests or his official capacity to advance the policies of the United States? And Trump answered honestly. He said,”I don ’ t know.”And how could he know, right? Because, I mean, those two things are completely intertwined. But that is not remotely acceptable for rule of law in a functional democracy, and yet nothing is being done about it. In fact, it ’ s getting more and more entrenched every day. So the single place where the United States was least functional as an advanced democracy is now getting far, far worse. And I don ’ t see any checks and balances on that. I think that ’ s a very, very serious problem. HW : Alright, Ian, thank you, as always for all of your insights. Let ’ s close out with a couple more questions from our community. And this one indeed relates to the kleptocratic conversation that you were just raising. So why are the billionaires enabling fascism? What do they get out of it? IB : Fascism is a very politically loaded argument. That ’ s a political system where the cruelty against a subgroup or a perceived subgroup in the society is the intention, the expressed intention of the policy. And I certainly wouldn ’ t define the United States as a fascist system today. But I do believe that billionaires in the United States, at least a significant subset of them, are really under-interested in the well-being of their fellow countrymen and women. And this is a country that really makes heroic the individual. And it undermines the community. We ’ ve seen a significant erosion of civic engagement in the United States and civic associations. The church is not what it was in the US. Fewer people go, fewer people trust the church. Public education is not what it was. And wealthy people increasingly opt out. The family is not what it was. It ’ s increasingly atomized and people spend much more, much more of their time being, you know, isolated and atomized and intermediated by algorithms. And you and I have spoken about that problem in the past. We are increasingly a society of individuals as opposed to a nation together. I will tell you that that is a real problem when you have large numbers of billionaires that believe that they built it themselves, that they ’ re not responsible to a collective whole for any of their success and therefore they ’ re not accountable for their fellow citizens. And that ’ s what I think we have. I think we have a lot of winners in the United States and not a lot of leaders. And when you have winners, you have losers. When you have winners, you don ’ t have a collectivity. I will say that in this anti-DEI thing, I have a lot of sympathy for the people that don ’ t like DEI. I thought it was rammed down the throats of a lot of people And I ’ ve said this before, it was well beyond what the average American was willing to tolerate, and also it came with a level of high-handedness that that basically said, if you ’ re not with us in supporting these programs, you ’ re evil, you ’ re stupid, you ’ re uneducated. That ’ s unacceptable, too. It was a completely, you know, sort of oppositional way to try to have what ’ s a very challenging conversation. But I ’ ve never been a champion of diversity for diversity ’ s sake. What ’ s amazing about the history of the United States is despite all of our diversity, we find commonalities with each other. We find connectedness. That ’ s what ’ s amazing, is the connectedness, is that, you know, on this world, we ’ ve got eight billion people. And despite how different we all are, we all actually connect as human beings. That is what we need to focus on, not the diversity, the connectedness. And I think that ’ s what the billionaires today in the United States are losing. They ’ re doing incredibly well themselves. They ’ ve got theirs, they ’ ve got access to power, they ’ re paying good money for it, they ’ re going to get even more. But they ’ re forgetting that for us to succeed as a society and as a planet, we have to be connected to each other. You can ’ t throw out diversity and forget connectedness, and that ’ s what we ’ re forgetting right now. That ’ s why the country feels so much trauma and pain right now. And I think why the United States has given up on a lot of its really aspirational values that made such a difference to the world, that almost lost World War II for many generations. And now we ’ re giving it all away. HW : I mean, I think that ’ s why you find that diversity and inclusion go hand in hand, that that ’ s actually speaking to that. But in some of the moves that are being made to kind of throw out any type of diversity, to scan documents, to find words in order to highlight programs that are wasteful, that we ’ re actually going to lose what many, many studies actually highlight as being the benefits of diversity and the fact that you need diversity in order to be actually successful. IB : Shooting first and asking questions later, you know, moving fast and breaking things, those things make sense in a turbocharged venture capital technology environment. They do not work in government. They don ’ t work in government at all. In government, we have something called checks and balances, and we need them. And moving fast and breaking things undermines those checks and balances. So we ’ re going to do a lot of damage unintentionally. A lot of people are going to get hurt. A lot of folks don ’ t care about that in power right now, but they should because it would be much, much better. I know people want to move fast, but actually to think a little bit, do a little bit of your own research before you decide to kill that program, before you decide to fire that person. We ' d be in a lot better shape as a government right now. HW : OK, the final question that I have for you actually comes from my 10-year-old son, and it ' s a bit of a miserable question. We have a very lovely house, I promise. But his question nonetheless is are we heading into World War III? So, Ian, what should I tell him? IB : No, no, I don ' t think we ' re heading into World War III but we are absolutely heading into a world of much greater conflict. I think it ' s much more likely, for example, in the next five, 10 years, that we ' ll have a nuclear weapon go off. And I thought that was pretty unlikely since the Cuban Missile Crisis. In other words, over the course of my life. I think we ' re going to see a lot more terrorism at scale, because a lot more people are going to feel angry and disaffected, I think. You know, you saw that assassination of the CEO of UnitedHealthcare just a few dozen blocks from my house. I think you ' re going to see more things like that. And so, no, not World War III because the fact is that outside of the US and China, all the other countries know that they need to work with everybody. And there are even forces inside US-China that are trying to make sure that you don ' t throw the baby out with the bathwater. But underneath that World War III risk, there ' s an awful lot of damage that ' s going to get done, an awful lot of crisis that I think we ' re going to have to live through before we start to see what we create on the other side. A lot of defense as well, that we ' re going to be playing in this environment. HW : Well, Ian, it is always a pleasure to talk to you, and we ' re so glad that you are scanning the world to understand what is happening and then you come here and explain it to us all. Couldn ' t be more grateful and thank you for everything. IB : Thank you.

Text Sample 477: NEG Dim 3 – 2024 – Score -1.00 – 2024/t003041.txt

I once met this Chilean guy at a hostel.”So what do you do?”I asked him.”You mean for work?”he responded as if I had just asked the color of his underwear. In the US,”What do you do?”is often the first question we ask when we meet someone new. This is drilled into us from an early age.”What do you want to be when you grow up?”we ask our kids, already conflating who we are with what we do, as if our jobs and identities were one and the same. I think about this a lot. I ' m a labor journalist, and I wrote a book called”The Good Enough Job,”in which I spoke to over 100 people about the relationship between their work and their identity. But before I was a professional writer, I was a 22-year-old poetry student trying to figure out what I wanted to be when I grew up. It was around this time that I had the opportunity to interview my favorite writer in the entire world, the poet Anis Mojgani. And so I asked him,”Anis, how do you feel about the mantra, do what you love and never work a day in your life?”And I ' ll never forget what he told me. He said,”You know, Simone, I think some people do what they love for work, and others do what they have to so they can do what they love when they ' re not working. And neither is more noble.”I think that last part is key. We live in a society that loves to revere people whose jobs and identities neatly align. And here was my idol, a professional poet no less, telling me that it ' s OK to have a day job. If we want to develop a healthier relationship to work, we can ' t just think about work-life balance in terms of how we spend our time. We have to think about how we construct our identity. What we do is part of but not the entirety of who we are. Let me be clear. I don ' t think there ' s anything wrong with doing what you love for work. We work more than we do just about anything else, and how we spend those hours matters. And yet, our current relationship to work isn ' t quite working. A recent study found that 48 percent of workers around the globe are burnt out. 48 percent. That ' s half this room. Actually, in this room, probably more than half. And yet, the way that we commonly talk about burnout doesn ' t address its root cause. There ' s a reason why a one-week vacation doesn ' t magically cure us. There ' s a reason why our intentions to practice self-care and set better boundaries inevitably break down. It ' s like we ' re shielding ourselves from the sun with a cocktail umbrella. If we want to actually change our relationship to work, we have to go deeper. It starts with our identity. Certainly, we are all more than just workers. We ' re parents and friends and citizens and artists and travelers and neighbors. Much like an investor benefits from diversifying the sources of stocks in their portfolio, we, too, benefit from diversifying the sources of meaning and identity in our lives. But how do you actually go about doing so? Rabbi Abraham Joshua Heschel described Shabbat, the weekly Jewish practice of abstaining from work, as a sanctuary in time. I love this image. Rather than a physical sanctuary, like a synagogue, Shabbat is a time sanctuary. So the first step to diversifying your identity is to create those time sanctuaries, spaces in your days, in your weeks, in your life where work is not an option. Unlike mere intentions to work less or set better boundaries, time sanctuaries require infrastructure. Putting time on the calendar to learn a new language, putting your phone in airplane mode while you play with your kids so that work doesn ' t expand like a gas and fill all of your unoccupied space. The second step is to fill those time sanctuaries with activities that reinforce the other identities you hope to cultivate. The present father, the community gardener, the amateur musician. It may sound simple, but if we want to derive meaning from aspects of our life other than work, we have to do things other than work. Now these don ' t have to be grand gestures. In the reporting in my book, I spoke to all of these hyperambitious workaholics, and they ' d say things like,"Diversify my identity. **Got** it. I ' m going to read 52 books this year. Or I ' m going to run an ultramarathon."We even convert our leisure into other forms of labor. My advice is to start small. How about a weekly walk with your best friend? Or ten minutes practicing the piano after dinner. The third step is to reinforce these identities by joining communities who couldn ' t care less about what you do for work. For example, I love to play pickup basketball, and one of the benefits of my weekly game is that the people I play with don ' t care how many words I ' ve written or how many books I ' ve sold. They care that I show up on time and that I ' m a good teammate. It ' s a weekly reminder that I exist on this Earth to do more than just produce economic value. The irony is that diversifying our identity can be great for business too. Research shows that people with varied interests tend to be more creative problem solvers and more innovative. Hobbies are one of the best ways to recharge so that you can be more productive when you ' re back on the clock. And a diverse identity can come in handy in the face of a stressful event like a recession or a layoff. I spoke to all of these folks for my book who treated their work like their family and then were unceremoniously let go during the pandemic. If you are what you do and you lose your job, who are you? But in addition to the business case, there ' s also the moral case. If we want to develop more well-rounded versions of ourselves, if we want to build robust relationships and live in robust communities and have a robust society at large, we all must invest in aspects of our lives beyond work. We shouldn ' t just work less because it makes us better workers. We should work less because it makes us better people. This isn ' t just about you and me. This is about teaching our kids that their self-worth is not determined by their job title. This is about reinforcing the fact that not all noble work neatly translates to a line on a resume. This is about setting the example that we all have a responsibility to contribute to the world in a way beyond contributing to one organization ' s bottom line. So the next time you ' re at a party, instead of asking someone,"What do you do?"I encourage you to add two small words to your question. Instead, ask them,"What do you like to do?"Maybe you like to cook. Maybe you like to write. Maybe you do some of those things for work. Or maybe you don ' t. But"What do you like to do"is a question that allows each of us to define ourselves on our own terms. Thank you.

Text Sample 478: NEG Dim 3 – 2024 – Score -1.00 – 2024/t003016.txt
So what the heck happened in the field of AI in the last decade? It ' s like a strange new type of intelligence appeared on our planet. But it ' s not like human intelligence. It has remarkable capabilities, but it also makes egregious errors that we never make. And it doesn ' t yet do the deep logical reasoning that we can do. It has a very mysterious surface of both capabilities and fragilities. And we understand almost nothing about how it works. I would like a deeper scientific understanding of intelligence. But to understand AI, it ' s useful to place it in the historical context of biological intelligence. The story of human intelligence might as well have started with this little critter. It ' s the last common ancestor of all vertebrates. We are all descended from it. It lived about 500 million years ago. Then evolution went on to build the brain, which in turn, in the space of 500 years from Newton to Einstein, developed the deep math and physics required to understand the universe, from quarks to cosmology. And it did this all without consulting ChatGPT. And then, of course, there ' s the advances of the last decade. To really understand what just happened in AI, we need to combine physics, math, neuroscience, psychology, computer science and more, to develop a new science of intelligence. The science of intelligence can simultaneously help us understand biological intelligence and create better artificial intelligence. And we need this science now, because the engineering of intelligence has vastly outstripped our ability to understand it. I want to take you on a tour of our work in the science of intelligence that addresses five critical areas in which AI can improve – data efficiency, energy efficiency, going beyond evolution, explainability and melding minds and machines. Let ' s address these critical gaps one by one. First, data efficiency. AI is vastly more data-hungry than humans. For example, we train our language models on the order of one trillion words now. Well, how many words do we get? Just 100 million. It ' s that tiny little red dot at the center. You might not be able to see it. It would take us 24, 000 years to read the rest of the one trillion words. OK, now, you might say that ' s unfair. Sure, AI read for 24, 000 human-equivalent years, but humans got 500 million years of vertebrate brain evolution. But there ' s a catch. Your entire legacy of evolution is given to you through your DNA, and your DNA is only about 700 megabytes, or equivalently, 600 million [words]. So the combined information we get from learning and evolution is minuscule compared to what AI gets. You are all incredibly efficient learning machines. So how do we bridge the gap between AI and humans? We started to tackle this problem by revisiting the famous scaling laws. Here ' s an example of a scaling law, where error falls off as a power law with the amount of training data. These scaling laws have captured the imagination of industry and motivated significant societal investments in energy, compute and data collection. But there ' s a problem. The exponents of these scaling laws are small. So to reduce the error by a little bit, you might need to ten-x your amount of training data. This is unsustainable in the long run. And even if it leads to improvements in the short run, there must be a better way. We

developed a theory that explains why these scaling laws are so bad. The basic idea is that large random datasets are incredibly redundant. If you already have billions of data points, the next data point doesn ’ t tell you much that ’ s new. But what if you could create a nonredundant dataset, where each data point is chosen carefully to tell you something new, compared to all the other data points? We developed theory and algorithms to do just this. We theoretically predicted and experimentally verified that we could bend these bad power laws down to much better exponentials, where adding a few more data points could reduce your error, rather than ten-xing the amount of data. So what theory did we use to get this result? We used ideas from statistical physics, and these are the equations. Now, for the rest of this entire talk, I ’ m going to go through these equations one by one. You think I ’ m joking? And explain them to you. OK, you ’ re right, I ’ m joking. I ’ m not that mean. But you should have seen the faces of the TED organizers when I said I was going to do that. Alright, let ’ s move on. Let ’ s zoom out a little bit, and think more generally about what it takes to make AI less data-hungry. Imagine if we trained our kids the same way we pretrain our large language models, by next-word prediction. So I ’ d give my kid a random chunk of the internet and say,”By the way, this is the next word.”I ’ d give them another random chunk of the internet and say,”This is the next word.”If that ’ s all we did, it would take our kids 24, 000 years to learn anything useful. But we do so much more than that. For example, when I teach my son math, I teach him the algorithm required to solve the problem, then he can immediately solve new problems and generalize using far less training data than any AI system would do. I don ’ t just throw millions of math problems at him. So to really make AI more data-efficient, we have to go far beyond our current training algorithms and turn machine learning into a new science of machine teaching. And neuroscience, psychology and math can really help here. Let ’ s go on to the next big gap, energy efficiency. Our brains are incredibly efficient. We only consume 20 watts of power. For reference, our old light bulbs were 100 watts. So we are all literally dimmer than light bulbs. But what about AI? Training a large model can consume as much as 10 million watts, and there ’ s talk of going nuclear to power one-billion-watt data centers. So why is AI so much more energy-hungry than brains? Well, the fault lies in the choice of digital computation itself, where we rely on fast and reliable bit flips at every intermediate step of the computation. Now, the laws of thermodynamics demand that every fast and reliable bit flip must consume a lot of energy. Biology took a very different route. Biology computes the right answer just in time, using intermediate steps that are as slow and as unreliable as possible. In essence, biology does not rev its engine any more than it needs to. In addition, biology matches computation to physics much better. Consider, for example, addition. Our computers add using really complex energy-consuming **transistor** circuits, but neurons just directly add their voltage inputs, because Maxwell ’ s laws of electromagnetism already know how to add voltage. In essence, biology matches its computation to the native physics of the universe. So to really build more energy-efficient AI, we need to rethink our entire technology stack, from electrons to algorithms, and better match computational dynamics to physical dynamics. For example, what are the fundamental limits on the speed and accuracy of any given computation, given an energy budget? And what kinds of electrochemical computers can achieve these fundamental limits? We recently solved this problem for the computation of sensing, which is something that every neuron has to do. We were able to find fundamental lower bounds or lower limits on the error as a function of the energy budget. That ’ s that red curve. And we were able to find the chemical computers that achieve these limits. And remarkably, they looked a lot like G-protein coupled receptors, which every neuron uses to sense external signals. So this suggests that biology can achieve amounts of efficiency that are close to fundamental limits set by the laws of physics itself. Popping up a level, neuroscience now gives us the ability to measure not only neural activity, but also energy consumption across, for example, the entire brain of the fly. The energy consumption is measured through ATP usage, which is the chemical fuel that powers all neurons. So now let me ask you a question. Let ' s say in a certain brain region, neural activity goes up. Does the ATP go up or down? A natural guess would be that the ATP goes down, because neural activity costs energy, so it ' s got to consume the fuel. We found the exact opposite. When neural activity goes up, ATP goes up and it stays elevated just long enough to power expected future neural activity. This suggests that the brain follows a predictive energy allocation principle, where it can predict how much energy is needed, where and when, and it delivers just the right amount of energy at just the right location, for just the right amount of time. So clearly, we have a lot to learn from physics, neuroscience and evolution about building more energy-efficient AI. But we don ' t need to be limited by evolution. We can go beyond evolution, to co-opt the neural algorithms discovered by evolution, but implement them in quantum hardware that evolution could never figure out. For example, we can replace neurons with atoms. The different firing states of neurons correspond to the different electronic states of atoms. And we can replace synapses with photons. Just as synapses allow two neurons to communicate, photons allow two atoms to communicate through photon emission and absorption. So what can we build with this? We can build a quantum associative memory out of atoms and photons. This is the same memory system that won John Hopfield his recent Nobel Prize in physics, but this time, it ' s a quantum-mechanical system built of atoms and photons, and we can analyze its performance and show that the quantum dynamics yields enhanced memory capacity, robustness and recall. We can also build new types of quantum optimizers built directly out of photons, and we can analyze their energy landscape and explain how they solve optimization problems in fundamentally new ways. This marriage between neural algorithms and quantum hardware opens up an entirely new field, which I like to call quantum neuromorphic computing. OK, but let ' s return to the brain, where explainable AI can help us understand how it works. So now, AI allows us to build incredibly accurate but complicated models of the brain. So where is this all going? Are we simply replacing something we don ' t understand, the brain, with something else we don ' t understand, our complex model of it? As scientists, we ' d like to have a conceptual understanding of how the brain works, not just have a model handed to us. So basically, I ' d like to give you an example of our work on explainable AI, applied to the retina. So the retina is a multilayered circuit of photoreceptors going to hidden neurons, going to output neurons. So how does it work? Well, we recently built the world ' s most accurate model of the retina. It could reproduce two decades of experiments on the retina. So this is fantastic. We have a digital twin of the retina. But how does the twin work? Why is it designed the way it is? To make these questions concrete, I ' d like to discuss just one of the two decades of experiments that I mentioned. And we ' re going to do this experiment on you right now. I ' d like you to focus on my hand, and I ' d like you to track it. OK, great. Let ' s do that just one more time. OK. You might have been slightly surprised when my hand reversed direction. And you should be surprised, because my hand just violated Newton ' s first law of motion, which states that objects that are in motion tend to remain in motion. So where in your brain is a violation of Newton ' s first law first detected? The answer is remarkable. It ' s in your retina. There are neurons in your retina that will fire if and only if Newton ’ s first law is violated. So does our model do that? Yes, it does. It reproduces it. But now, there ’ s a puzzle. How does the model do it? Well, we developed methods, explainable AI methods, to take any given stimulus that causes a neuron to fire, and we carve out the essential subcircuit responsible for that firing, and we explain how it works. We were able to do this not only for Newton ’ s first law violations, but for the two decades of experiments that our model reproduced. And so this one model reproduces two decades ’ worth of neuroscience and also makes some new predictions. This opens up a new pathway to accelerating neuroscience discovery using AI. Basically, build digital twins of the brain, and then use explainable AI to understand how they work. We ’ re actually engaged in a big effort at Stanford to build a digital twin of the entire primate visual system and explain how it works. But we can go beyond that and use our digital twins to meld minds and machines, by allowing bidirectional communication between them. So imagine a scenario where you have a brain, you record from it, you build a digital twin. Then you use control theory to learn neural activity patterns that you can write directly into the digital twin to control it. Then, you take those same neural activity patterns and you write them into the brain to control the brain. In essence, we can learn the language of the brain, and then speak directly back to it. So we recently carried out this program in mice, where we could use AI to read the mind of a mouse. So on the top row, you ’ re seeing images that we actually showed to the mouse, and in the bottom row, you ’ re seeing images that we decoded from the brain of the mouse. Our decoded images are lower-resolution than the actual images, but not because our decoders are bad. It ’ s because mouse visual resolution is bad. So actually, the decoded images show you what the world would actually look like if you were a mouse. Now, we can go beyond that. We can now write neural activity patterns into the mouse ’ s brain, so we can make it hallucinate any particular percept we would like it to hallucinate. And we got so good at this that we could make it reliably hallucinate a percept by controlling only 20 neurons in the mouse ’ s brain, by figuring out the right 20 neurons to control. So essentially, we can control what the mouse sees directly, by writing to its brain. The possibilities of bidirectional communication between brains and machines are limitless. To understand, to cure and to augment the brain. So I hope you ’ ll see that the pursuit of a unified science of intelligence that spans brains and machines can both help us better understand biological intelligence and help us create more efficient, explainable and powerful artificial intelligence. But it ’ s important that this pursuit be done out in the open so the science can be shared with the world, and it must be done with a very long time horizon. This makes academia the perfect place to pursue a science of intelligence. In academia, we ’ re free from the tyranny of quarterly earnings reports. We ’ re free from the censorship of corporate legal departments. We can be far more interdisciplinary than any one company. And our very mission is to share what we learn with the world. For all these reasons, we ’ re actually building a new center for the science of intelligence at Stanford. While there have been incredible advances in industry on the engineering of intelligence, now increasingly happening behind closed doors, I ’ m very excited about what the science of intelligence can achieve out in the open. You know, in the last century, one of the greatest intellectual adventures lay in humanity peering outwards into the universe to understand it, from quarks to cosmology. I think one of the greatest intellectual adventures of this century will lie in humanity peering inwards, both into ourselves and into the AIs that we create, in order to develop a deeper, new scientific understanding of intelligence. Thank you.

Text Sample 479: NEG Dim 3 – 2024 – Score -1.00 – 2024/t003005.txt

So I believe this soybean plant is a prototype for sustainable food production on this planet. Let me show you why I say that. So on the roots of this soybean plant are nodules. And these nodules do an amazing thing. They harbor millions of bacteria inside the cells of the nodules. And those bacteria are able to capture nitrogen out of the atmosphere and can feed it to this soybean plant. Now all plants require a source of nitrogen. They need it so they can make DNA, RNA and proteins. But plants can ’ t access the most prevalent form of nitrogen on the planet, the 78 percent of the air that you ’ re currently breathing that is molecular dinitrogen. Only bacteria that possess the enzyme nitrogenase can convert this very inert form of nitrogen, convert it into ammonia, a reactive form of nitrogen that bacteria and plants can use to make their DNA, RNA and proteins. So the bacteria inside the nodules of this soybean plant are fixing nitrogen out of the air, converting it into ammonia, and then feeding that ammonia to this soybean plant. In return, the soybean plant is feeding the bacteria with a source of carbon in the form of sugars derived from photosynthesis in the leaves. This is what we call a mutualistic symbiosis. It ’ s beneficial to the soybean plant, but it ’ s also beneficial to the bacteria inside those nodules. Now the roots of the soybean plant are doing a second amazing thing. And to see that we have to look under a microscope. The roots are heavily infested with a beneficial fungus called mycorrhizal fungi. And these fungi are heavily colonizing the soil and make a much greater contact with the soil surface than the plant root alone is able to achieve. In so doing, they create a much more efficient platform for the uptake of nutrients, nutrients such as phosphates, nitrates, **potassium** and water. The fungus isn ’ t only out there in the soil, it ’ s also colonizing the roots of this soybean plant, where it makes these highly branched fungal intrusions into the cells of the root that we call arbuscules. So the fungus is out there in the soil, capturing nutrients from the soil, and it feeds those nutrients to the soybean plant through these arbuscular intrusions. In return, the soybean is feeding the fungus with carbon from photosynthesis. Again, it ’ s a mutualistic symbiosis. So this soybean plant can get almost all of its phosphate and the totality of its nitrogen needs met through these beneficial microbial associations. And that provides a free and sustainable means to support its crop production. And out in nature, most plants are engaging with one or more of these beneficial microorganisms to help them capture these limiting nutrients from the environment. But in agriculture, it ’ s a really different situation. There, we ’ re applying these nutrients at high concentrations in the form of inorganic fertilizers to support our crop production. And while inorganic fertilizers have underpinned global food security for the last 60 years, they cause significant environmental pollution, they cause significant greenhouse gas emissions, they contribute to a lot of the costs in our crop production. And at the other end of the spectrum, smallholder farmers lack access to those fertilizers and their productivity suffer as a result. For all of these reasons, myself and my colleagues in the ENSA project are working to eradicate, or at least greatly reduce, our reliance on inorganic fertilizers. To do that, we want to make all of our crop plants, particularly our cereal crops, behave like this soybean plant, able to get their nutrients through these beneficial microbial associations. Now the fungal symbiosis is not limited to legumes like that soybean. It ’ s actually pretty widespread within the plant kingdom, and it ’ s already present in our cereal crops. However, when we fertilize our fields, the crop doesn ’ t engage with the fungus. Why are you paying the fungus with carbon if the nutrients are not limiting? So while soils in natural ecosystems are packed full of a complex network of these mycorrhizal fungi fed by their host plants, our agricultural soils are greatly depleted for these beneficial fungi. If we want to really maximize the utilization of this fungal symbiosis in agriculture, then we need to get our crop plants to engage with the fungus much more proactively, and even when we fertilize our fields. If we can do that, then we can reduce the levels of fertilizers we use, and we ' ll lose less of those nutrients out into the environment. So to achieve that, we set about identifying the genetic regulators that control when the plant engages with these beneficial fungi. And we discovered that these proteinaceous regulators are only present when the plant is starved for nutrients. And we were able to rewire that system so that now the plant engages with the fungus much more proactively, and even when the plant is fertilized. In our field trials, we find that these rewired barley plants get 10 times as much fungus inside their roots. That ' s a lot more fungus in the crop, but it ' s also a lot more fungus out there in the field. So now we can control when the plant engages with these beneficial fungi. The next step for us is to test : Does that mean we can lower the fertilizer levels and still maintain good production? I believe this is a first step to really getting that fungal association working for us much more proactively in agriculture, and that ' s going to be really important, especially for how much phosphates we have to apply to our fields. But if we ' re going to really cure our addiction to inorganic fertilizers, we also need the nitrogen-fixing bacterial symbiosis. Now, unfortunately, the nitrogen-fixing symbiosis is limited to legumes like that soybean and their relatives. So we are working on transferring that nitrogen-fixing symbiosis from legumes to our cereal crops. Myself and my colleagues have spent the last 30 years undertaking genetic dissection to try to identify all the genes that, in soybean, allows it to engage with those nitrogen-fixing bacteria. During that time, we ' ve identified many genes that are involved in that process. But surprisingly, we haven ' t yet identified a single gene that is novel to that soybean plant. In fact, the genes are already present, most of them are already present in our cereal crops. Let me give you an example. The symbiosis signaling pathway. This is a set of proteins that are expressed on the cells on the surface of that soybean root, that allow the soybean plant to recognize the nitrogen-fixing bacteria out in the soil. When they recognize the bacteria, they oscillate their calcium in the nucleus. This is essentially the cell saying,"I ' ve seen these beneficial bacteria out in the soil. Now turn on the gene expression that ' s necessary to let those bacteria in."This symbiosis signaling pathway that in the soybean plant allows it to perceive the nitrogen-fixing bacteria, it ' s already present in our cereal crops. And that ' s because it ' s the same signal transduction pathway that allows all plants to recognize mycorrhizal fungi. What we now understand is that when legumes evolve this capability to engage with nitrogen-fixing bacteria, they didn ' t invent anything new. They used the pre-existing genetic components associated with engagement with beneficial fungi to also allow engagement with beneficial bacteria. Essentially, the nitrogen-fixing symbiosis is really just a modified form of the mycorrhizal fungal symbiosis with a few tweaks. And one of the really important tweaks is to link that symbiosis signaling pathway to root organogenesis to make the nodules that are able to accommodate those nitrogen fixing bacteria. But even there, these apparently unique nodule structures are not that novel. They use pre-existing developmental genes that are already present in our cereal crops. So essentially, nitrogen fixation uses a whole set of pre-existing genetic components, but they re-network them in a novel way to create the apparent novelty of nitrogen fixation. Now from an engineering perspective, it ' s much easier to re-network a set of preexisting genetic components than it is to build those genetic components from scratch. Now this work is not yet published, but using this knowledge and getting the networking of those pre-existing genetic components right, we have now been able to engineer nodules in non-legumes. Now unfortunately, at the moment those nodules don ' t get infected with the nitrogen fixing bacteria. That ' s the step that we ' re currently working on. However, I believe we are well on track to delivering nitrogen-fixing cereals. And because we ' re networking pre-existing genetic components, as opposed to having to build those genetic components from scratch, I ' m pretty confident that we can deliver those nitrogen-fixing cereals within my career. Nature has already shown us how to sustainably feed this planet, and it ' s here in my hand. I believe the next green revolution is going to be the microbial revolution, using beneficial fungi to deliver phosphates and beneficial bacteria to deliver nitrogen, providing a much more sustainable means to support our food production systems and providing technology that ' s accessible to all of the world ' s farmers. Thank you.

Text Sample 480: NEG Dim 3 – 2022 – Score -1.00 – 2022/t003278.txt
Imagine you ' re a farmer and you ' ve planted enough crops to feed your family for the coming year. The weather is surprisingly good at the beginning of the growing season, but after those seeds are in the ground and the stalks start to peek up from the soil, a disease that you cannot see cuts your expected yield in half. You think to yourself,"What will my family eat?"In the coming year, perhaps you ' ll fumigate your soil, maybe you ' ll add extra fertilizer. Maybe you ' ll apply a fungicide or a pesticide hoping to decrease crop loss. You know that these traditional technologies work, but you also know they have some negative implications for our ecosystem. This, of course, is not an imaginary scenario. We could feed every person on this planet if we didn ' t lose so much to disease, pest and poor soil conditions. It ' s estimated that we lose between 20 and 40 percent of crop productivity due to preventable disease and pest attack, and climate change is only making this worse. I stand here in front of you, a very unlikely person to help solve an agricultural crisis. I ' m a chemistry professor who studies nanoparticles in **sterile** laboratories, I grew up in the desert, and I don ' t even keep houseplants, much less crops, alive. But I know that some of the best solutions to big problems come when folks from different or even opposing fields bring some of their simplest concepts together. And that is exactly what I think is possible here. As I tell you that nanoparticles may be a critical part of the solution to our global food crisis. Let me tell you why I ' m so intent on using nanoscience to fight hunger. We all have the issues that touch us deepest, and for me, it has always been hunger. I find it intolerable that there are hungry people on this life-giving planet of ours. I can trace this, at least in part, to some of my own experience growing up. For a period of my childhood, I lived in a food-insecure household. We benefited from food shelf donations, and we only ate what my mom could get with her hard work on double coupon day at the local supermarket. I loved the one day a month I was allowed to buy school lunch. Now, I don ' t know why my parents didn ' t apply for free school lunch or food stamps, but the end situation was one where sometimes the refrigerator and the cupboards were empty. Now, it ' s been a long time since I have worried about food for myself, but that feeling of being hungry is etched deep within me. I am driven to do something about hunger. And the unusual talent that I bring to the task is my deep knowledge of designing and synthesizing nanomaterials that can carry molecular cargo and transform into specific chemical species. Let me stop and give a little bit of background about nanoscience. The prefix"nano"signifies a billionth, so a nanometer is a billionth of a meter. In other words, nanoparticles are extremely small. You cannot see them with your naked eye or even a high-powered light microscope. In fact,

you need a specialized instrument like the one you see here, called a transmission electron microscope, to even see nanoparticles. Nanoparticles have actually been around forever. You can find naturally occurring nanoparticles in geological formations or in the aerosol particles that we breathe. But in the last few decades, scientists and engineers have gotten very excited about nanomaterials because we realized that as you shrink things down to the nanoscale, their chemical and physical properties can change drastically. For example, a material that ' s usually unreactive when you shrink it down to the nanoscale can suddenly catalyze a whole host of chemical reactions. Or a material that doesn ' t usually conduct electricity, suddenly does. As scientists tune the size, shape and chemical composition of a material, they can tune those chemical and physical properties. Nanoscience gives us a seemingly unlimited palette of accessible chemical and physical properties. I ' m sure you can imagine how useful that can be. And scientists have gotten very good at knowing exactly how to design nanomaterials to have the properties they want. We are in a perfect moment to take advantage of all of the hard-won knowledge that has been systematically gained in laboratories around the world. And that ' s already happening. You can find engineered nanoparticles in a range of products and applications, many of which are focused on some of our biggest sustainability challenges. Personally, I like to work on the nanomaterials that make up the core of lithium-ion batteries for electric vehicles, but you ' ll also find nanomaterials in water filtration technology, solar cells and even in clinical applications. With all of that background, now let me tell you about some of the nanomaterials my research group is developing for agricultural applications. In some ways, the nanoparticles seem very simple. They ' re made of silica or SiO2 in chemistry language. This is the same chemical composition that describes glass or sand. And the simple choice was not an accident. We wanted to work with Earth-abundant elements, and you can ' t do much better than silicon and oxygen on that front. The researchers in my lab are very skilled at designing silica nanoparticles with controlled size and surface chemistry. We also work hard to control the pore structure, because that determines the total surface area, as well as the strength of the bonds that hold the nanomaterials together. That ' s because all of those factors are critically important for our end goal, which is to get these nanoparticles inside plants, either by infiltrating seeds or by allowing them to pass through pores on the leaf surface, and then, once they ' re internalized, having them transformed to release a molecule that the plant can use to protect itself from viruses, fungi or pest attacks. In technical terms, we want our silica nanoparticles to react with water in the environment and dissolve to release a molecule called silicic acid. You can think of silicic acid for plants like the multivitamin that you take every morning. Plants already contain silicic acid, they use it to build their cell walls. We want to deliver an extra boost of silicic acid, with the hypothesis that they ' ll build stronger cell walls and boost their own immune response. So now, hopefully you can see the whole picture. We design silica nanoparticles with the right size, shape and surface chemistry to be taken up into a plant. We also design them so that once they ' re internalized, they dissolve to release enough silicic acid that the plants live healthier and longer, producing more food. With all of that background, now, let me actually show you some of the nanoparticles that we ' re making and tell you about some of the early, exciting results from greenhouse and field studies. We ' ve made many variations on the silica nanoparticle theme. Here you can see five electron microscope images of silica nanoparticles we ' ve made. All of them have the same scale bar, that ' s just 100 nanometers. And all of them are small enough to go into the pores on a leaf ' s surface. Here you can see some of the silica nanoparticles we ’ ve designed with various pore structures. The pore structure ends up being very important because the more water can react with the surface of the nanoparticle, the better it dissolves and the more silicic acid is released. Here you can see some nanoparticles we ’ ve designed to dissolve at different rates. And what you ’ re looking at is a nice, solid nanoparticle at the beginning, before it ’ s been exposed to water. And then after eight, 16 and 24 hours, you can see those nanoparticles hollowing out, releasing lots of useful silicic acid. With all of these nanoparticles in hand, we started working with colleagues within the NSF Center for Sustainable Nanotechnology to start our first plant studies. The initial studies were very simple, using watermelon seedlings in a greenhouse. Here ’ s the experimental setup. We had watermelon seedlings that were going to be planted either in healthy soil or soil infested with Fusarium, a fungal soil-borne pathogen. Before planting them, we dipped them in our silica nanoparticles, and then we allowed them to grow in the greenhouse. Of course, we also had a parallel set of plants that received no nanoparticles, growing in both healthy and diseased soil. So the goal was to figure out how that single application of silica nanoparticles impacted the plants growing in both healthy and diseased soil. And the results that we saw were really exciting. We found that the plants that were growing in infected soil, that had received that one dose of silica nanoparticles were 30 to 40 percent healthier than the ones that had not. With this exciting result, we decided to try some field studies using the same soil conditions and the same nanoparticle conditions. So we planted watermelon, either in healthy or infected soil, and we allowed them to grow for 100 days. We tracked the fungal disease, and we also measured the amount of fruit that was produced after 100 days. And what we found is that that one application of one to two milliliters of silica nanoparticles way back at the seedling stage, led us to a 70 percent increase in watermelon yield. Ideally, none of the nanoparticles would end up in the fruit that people are going to eat. So we analyzed the roots and the above-ground tissue and the edible fruit for any sign of silica nanoparticles. We saw no increased silicon in the edible watermelon fruit, meaning that these nanoparticles did exactly what we designed them to do. Given the small amount of nanomaterials that we applied to each one of those plants, the cost per plant is only about two cents or 19 dollars for an acre. This is a cost-effective treatment. By adding 19 dollar ’ s worth of nanoparticles to the average fertilizer cost of 250 dollars, a farmer would yield thousands of dollars increase in fruit production. With these exciting results in hand, we have a lot of other experiments planned and in progress. We want to do multiple applications of nanoparticles and applications later in the growth process to see if that further increases our yield. We want to do studies on soybean and wheat, critical crops here in the Midwest and around the world. Two researchers in my lab recently applied silica nanoparticles to potato plants in the field. They ’ re going to help harvest and analyze the results this fall. I hope that what you see is that this data is really compelling, and that nanoparticles have tons of potential to help decrease crop loss. And I only told you about silica nanoparticles. I can imagine other important chemical compositions and even parallel application like remediation of soil pollutants. I ask you all to be open-minded about nanotechnology, encouraging funding agencies worldwide to invest. In the US, ask your senators and representatives to invest in the National Science Foundation, the National Institutes of Health and the US Department of Agriculture for both basic and translational research. I know farmers are already embracing advanced technology in terms of robots and drones and implant sensors. I encourage them to embrace this advance as well. Now go back to imagining that you ’ re that farmer and you ’ ve planted enough crops to feed your family for the coming year. You do all of the normal things, except this time, maybe you use seeds that were infiltrated with silica nanoparticles. Or maybe you go through once and you spray silica nanoparticles onto your crops. As these tiny nanoparticles deliver a big boost of silicic acid from the inside, your plants overcome disease, and your family is fed. Let ' s use all of the hard work that has been done on basic nanotechnology research to feed our global family for years to come. Thank you.

Text Sample 481: NEG Dim 3 – 2022 – Score -1.00 – 2022/t003585.txt
Creating intelligence on a computer. This has been the Holy Grail for artificial intelligence for quite some time. But how do we get there? So we view ourselves as highly intelligent beings. So it ' s logical to study our own brains, the substrate of our cognition, for creating artificial intelligence. Imagine if we could replicate how our own brains work on a computer. But now consider the journey that would be required. The human brain contains 86 billion neurons. Each is constantly communicating with thousands of others, and each has individual characteristics of its own. Capturing the human brain on a computer may simply be too big and too complex a problem to tackle with the technology and the knowledge that we have today. I believe that we can capture a brain on a computer, but we have to start smaller. Much smaller. These insects have three of the most fascinating brains in the world to me. While they do not possess human-level intelligence, each is remarkable at a particular task. Think of them as highly trained specialists. African dung beetles are really good at rolling large balls in straight lines. Now, if you ' ve ever made a snowman, you know that rolling a large ball is not easy. Now picture trying to make that snowman when the ball of snow is as big as you are and you ' re standing on your head. Sahara desert ants are navigation specialists. They might have to wander a considerable distance to forage for food. But once they do find sustenance, they know how to calculate the straightest path home. And the dragonfly is a hunting specialist. In the wild, dragonflies capture approximately 95 percent of the prey they choose to go after. These insects are so good at their specialties that neuroscientists such as myself study them as model systems to understand how animal nervous systems solve particular problems. And in my own research, I study brains to bring these solutions, the best that biology has to offer, to computers. So consider the dragonfly brain. It has only on the order of one million neurons. Now, it ' s still not easy to unravel a circuit of even one million neurons. But given the choice between trying to tease apart the one-million-neuron brain versus the 86-billion-neuron brain, which would you choose to try first? When studying these smaller insect brains, the immediate goal is not human intelligence. We study these brains for what the insects do well. And in the case of the dragonfly, that ' s interception. So when dragonflies are hunting, they do more than just fly straight at the prey. They fly in such a way that they will intercept it. They aim for where the prey is going to be. Much like a soccer player, running to intercept a pass. To do this correctly, dragonflies need to perform what is known as a coordinate transformation, going from the eye ' s frame of reference, or what the dragonfly sees, to the body ' s frame of reference, or how the dragonfly needs to turn its body to intercept. Coordinate transformations are a basic calculation that animals need to perform to interact with the world. We do them instinctively every time we reach for something. When I reach for an object straight in front of me, my arm takes a very different trajectory than if I turn my head, look at that same object when it is off to one side and reach for it there. In both cases, my eyes see the same image of that object, but my brain is sending my arm on

a very different trajectory based on the position of my neck. And dragonflies are fast. This means they calculate fast. The latency, or the time it takes for a dragonfly to respond once it sees the prey turn, is about 50 **milliseconds**. This latency is remarkable. For one thing, it ' s only half the time of a human eye blink. But for another thing, it suggests that dragonflies capture how to intercept in only relatively or surprisingly few computational steps. So in the brain, a computational step is a single neuron or a layer of neurons working in parallel. It takes a single neuron about 10 **milliseconds** to add up all its inputs and respond. The 50-millisecond response time means that once the dragonfly sees its prey turn, there ' s only time for maybe four of these computational steps or four layers of neurons, working in sequence, one after the other, to calculate how the dragonfly needs to turn. In other words, if I want to study how the dragonfly does coordinate transformations, the neural circuit that I need to understand, the neural circuit that I need to study, can have at most four layers of neurons. Each layer may have many neurons, but this is a small neural circuit. Small enough that we can identify it and study it with the tools that are available today. And this is what I ' m trying to do. I have built a model of what I believe is the neural circuit that calculates how the dragonfly should turn. And here is the cool result. In the model, dragonflies do coordinate transformations in only one computational step, one layer of neurons. This is something we can test and understand. In a computer simulation, I can predict the activities of individual neurons while the dragonfly is hunting. For example, here I am predicting the action potentials, or the spikes, that are fired by one of these neurons when the dragonfly sees the prey move. To test the model, my collaborators and I are now comparing these predicted neural responses with responses of neurons recorded in living dragonfly brains. These are ongoing experiments in which we put living dragonflies in virtual reality. Now, it ' s not practical to put VR goggles on a dragonfly. So instead, we show movies of moving targets to the dragonfly, while an electrode records activity patterns of individual neurons in the brain. Yeah, he likes the movies. If the responses that we record in the brain match those predicted by the model, we will have identified which neurons are responsible for coordinate transformations. The next step will be to understand the specifics of how these neurons work together to do the calculation. But this is how we begin to understand how brains do basic or primitive calculations. Calculations that I regard as building blocks for more complex functions, not only for interception but also for cognition. The way that these neurons compute may be different from anything that exists on a computer today. And the goal of this work is to do more than just write code that replicates the activity patterns of neurons. We aim to build a computer chip that not only does the same things as biological brains but does them in the same way as biological brains. This could lead to drones driven by computers the same size of the dragonfly ' s brain that captures some targets and avoid others. Personally, I ' m hoping for a small army of these to defend my backyard from mosquitoes in the summer. The GPS on your phone could be replaced by a new navigation device based on dung beetles or ants that could guide you to the straight or the easy path home. And what would the power requirements of these devices be like? As small as it is – Or, sorry – as large as it is, the human brain is estimated to have the same power requirements as a 20-watt light bulb. Imagine if all brain-inspired computers had the same extremely low-power requirements. Your smartphone or your smartwatch probably needs charging every day. Your new brain-inspired device might only need charging every few months, or maybe even every few years. The famous physicist, Richard Feynman, once said,”What I cannot create, I do not understand.”What I see in insect nervous systems is an opportunity to understand brains through the creation of computers that work as brains do. And creation of these computers will not just be for knowledge. There ’ s potential for real impact on your devices, your vehicles, maybe even artificial intelligences. So next time you see an insect, consider that these tiny brains can lead to remarkable computers. And think of the potential that they offer us for the future. Thank you.

Text Sample 482: NEG Dim 3 – 2022 – Score -1.00 – 2022/t003755.txt
Now, many of the people watching this talk by now will have had one, two, three or even four doses of a messenger RNA vaccine. With billions of these shots now in arms. it ’ s clear that this new way of making vaccines is both remarkably safe and incredibly effective. But did you know it ’ s not the vaccine itself that ’ s keeping you safe? It ’ s actually you. Because the human body has amazing powers to both prevent and cure disease by making its own medicines. You just need to know what medicine to make. And that ’ s what your vaccine gave you. Simply a set of instructions for how to protect yourself against SARS-CoV-2. Now vaccines are only the beginning. As I said before, the advent of mRNA vaccines is heralding in an entirely new era of medicines. mRNAs give us the ability to not only prevent disease but also treat previously intractable disorders. But before I get to that, let ’ s talk about what is this new way of medicines really about? Well, it all comes down to proteins. Now, you may think of protein as simply something that you need to eat, an important part of your diet, something that ’ s important for you to build muscle. But it ’ s not just muscle that contains protein. Protein makes up a huge fraction of the incredibly complicated ecosystem that ’ s your entire body. In many ways, your body functions like a large city, full of myriad buildings, interconnected buildings, with lots of different structures. Now, just as the word”building”fails to capture the incredible variety of structures that make up any large city, the generic term”protein”gives no clue as to the incredible variety of molecular architectures at the molecular level. Like buildings, proteins are not monolithic. Our body makes many proteins. Some, like the collagen in our skin that makes our skin tough but pliable. Like the actin and myosin in our muscles that enable us to move. Or our blood is full of hemoglobin, ferrying oxygen around. Antibodies protecting us from disease, and clotting factors that close up our wounds. And for any of you that have ever cracked a biochemistry textbook, this should look familiar. This is the metabolic chart of the human body. These are all of the reactions in your body that are keeping you alive and well right now. And each one of those reactions is catalyzed by a different protein whose names are shown here in blue. So proteins are not only what makes up the bulk of your body, they ’ re also what make your body tick and keep you well. Now, if we think again about buildings, the structures of buildings may look quite different in the end, but they ’ re all made of a limited set of building materials and which of those building materials are used and how they ’ re arranged, and how they ’ re attached to one another gives the buildings their final form. The same is true of proteins. If we zoom down to the molecular level, we can see that proteins, if we unravel their three-dimensional architecture, are actually just long strings of building blocks. And these building blocks have different shapes and different propensities to interact with one another. So it ’ s which building blocks are used and the order in which they are in the chain that gives a protein is three-dimensional shape. Now here I ’ m only showing three proteins, but creating and maintaining a healthy human requires the combined action of over 100, 000 different types of protein. And our bodies make them all. Thus our bodies are remarkable protein factories. At the molecular level,

the numbers are truly mind-blowing. Each of the 30 trillion cells in your body – that ' s three with 13 zeros – contains between one and 10 billion protein molecules. That means that you have as many protein molecules in your body as there are stars in the known universe. Now each different cell type in your body makes a different kind of protein, a different set of proteins, like the rods and cones in my eye that are detecting light right now and the neurons in my brain that are interpreting that light and enabling me to see you right now. So they make a particular set of proteins unique to it. And like any complex building project, you can imagine that the process of protein synthesis needs to be tightly regulated so that the right protein is made at the right time and in the right place. But of course, with anything so complicated, it ' s perhaps not surprising that there ' s an occasional mistake, a fault in the algorithm. Let ' s go back to that metabolic chart. It ' s estimated that one in 1, 000 newborns are born without the ability to make one of the proteins on this chart. Therefore, they have lifelong complications due to inborn metabolic errors. Let ' s take just one of those. Let ' s talk about von Gierke ' s disease or glycogen storage disease I. This is due to the lack of a protein circled here in red, whose job it is to release stored sugars so that you can maintain a healthy blood sugar level while you ' re fasting. So von Gierke ' s disease patients can ' t fast. They must constantly eat small amounts of carbohydrates, including getting up every one or two hours during the night to eat raw cornstarch. Now, imagine the toll that this takes on parents. If they are ever to miss a feeding of their child, their child could slip into severe hypoglycemia, seizures and possibly death. But even if these patients can keep up this endless feeding cycle, they are plagued by lifelong complications, including delayed puberty, frequent infections, kidney disease and liver cancer. So von Gierke ' s disease is just one example of a disorder where we know what protein is missing. What if we could give those patients back the ability to make that missing protein? Then we could actually treat their disease instead of just managing their symptoms. And that ' s where mRNA comes in. That ' s also where I come in. You see, I spent the better part of my career as an academic doing curiosity-based research into the fundamental principles of how proteins are made. And my specialty was messenger RNA. Like proteins, messenger RNAs are long chainlike molecules composed of building blocks. The four building blocks that make up messenger RNAs form what is known as the genetic code. As their name implies, messenger RNAs carry messages : messages that are translated by your body in order to create proteins. Thus messenger RNAs are the language of life. And the human body has a lot to say. So, like proteins, your cells are chock-full of messenger RNA. Every one of your 30 trillion cells has hundreds of thousands of messenger RNA molecules. Messenger RNAs are an essential component of all living organisms. So when you are eating protein-rich foods, you ' re not only eating protein, you ' re also eating lots of messenger RNA. Your body takes the messenger RNA in the food that you consumed, breaks it down into those component parts and then builds new messenger RNAs specific to your needs. Now this continual destruction and rebuilding is a feature true of almost all proteins and messenger RNAs in your body. Let ' s take, for example, the circadian clock. This is the timer in your body that tells you when to be active and when to sleep. The proteins that make up this clock appear and disappear with remarkable regularity every day. The way that this is accomplished is that your body makes the messenger RNAs that encode those proteins appear and disappear every day. Every day for your entire life, you get your daily dose of clock messenger RNAs producing clock proteins. Now three properties of proper medicines are that their effects are of limited duration, that their effects are dose-dependent and that they can be given over and over again to produce the same effect. mRNA ’ s are transient. The amount of protein produced is dependent on how much of that mRNA is present. And they can be induced over and over again to produce the same effect. So wow, it seems so simple. If we could treat a disease, if there ’ s a protein that ’ s missing to treat a disease, then we could simply give a few copies of an mRNA to the body for it to produce that protein. If that protein ’ s only needed once, then maybe a single dose would suffice. If a protein is needed multiple times, then we can dose mRNA over and over again. And that ’ s exactly what ’ s happening. So when I went on clinicaltrials. gov this morning, it turns out that there are over 175 clinical trials now open using mRNA-based medicines that are recruiting patients. Another 54 clinical trials are waiting in the [wings], ready to be opened. So there is a coming tsunami of mRNA medicines. Last year, Moderna and AstraZeneca reported positive results from a clinical trial where patients during open heart surgery were dosed with messenger RNA injected directly into their heart muscles, that told their heart muscles to grow new blood vessels in order to get around clogged arteries. In other clinical trials we ’ re repeatedly dosing patients with inborn metabolic errors to treat their metabolic disease. In fact, one of those clinical trials that ’ s currently recruiting patients is for von Gierke ’ s disease. And for cancer patients, we ’ re creating personalized cancer vaccines. These vaccines are meant to train their bodies, their immune systems, to attack their cancers. These are truly personalized medicines, one vaccine for one person. Now for personalized cancer vaccines to be the most effective, we need to get them made and back to the patient as quickly as possible. We aim for a turnaround time of 45 days. By January of 2020, we had already manufactured, quality-controlled and delivered to several dozen patients personalized cancer vaccines. So we had the know-how and the capacity to manufacture vaccines quickly. Thus, when the sequence of the SARS-CoV-2 virus was posted to a public web server on January 10, 2020, we got immediately to work. Within two days, we had agreed with our collaborators at NIH on exactly which form of the spike protein to put in our vaccine. Because we had done so so many times before, it then took our mRNA design team just one hour to design the mRNA that we immediately – That we immediately put on to our manufacturing equipment. We were then able to make that RNA, get it quality-controlled, fill-finished and shipped off to NIH for the clinical trial in 45 days. Now what I find – What I find truly remarkable is that that mRNA sequence that took us one hour to design is the same mRNA sequence that went into your arms, that ended up in Spikevax, our now fully approved vaccine. One hour to design a medicine that has saved countless lives. It still gives me goosebumps every time I talk about it. So what does the future hold? Well, I ’ ve already told you about regenerative medicine and personalized cancer vaccines. For cancer patients, we can send in – by directly injecting messenger RNA into their tumors – we can send in instructions telling the tumor cells to self-destruct, Or having the tumor cells send out signals to the immune system, beckoning the immune system to attack. For patients with autoimmune disorders, we can send in signals that tamp down their overactive immune systems. And we and others are rapidly making many more messenger RNA vaccines. Because messenger RNA vaccines can be produced so quickly and rapidly, they ’ re really well suited for newly emerging diseases as well as other viruses, like the flu, where new variants come out every year, and the vaccines need to be updated. But one of the exciting things about mRNA medicines is we ’ re not limited to sending in the instructions for one protein at a time. mRNA medicines can be easily multiplexed. Therefore, we ’ re working on a combination vaccine for COVID, flu and respiratory syncytial virus, or RSV : all leading causes of hospitalization and death in the elderly. And we ’ re hoping that this will then be an annual booster, that you will get, just like the flu vaccine. So finally, the very modest footprint of the manufacturing equipment for making messenger RNAs means that they can be made almost anywhere in the world. And to take this to an extreme, the American Defense Department started a program in 2019, and we ’ re working with them to miniaturize the entire process so that it can be fit into a single shipping container for rapid deployment anywhere in the world. So – So to finish, I hope I ’ ve convinced you that we have entered an entirely new era of medicine. Having learned to speak the language of mRNA, the language of life, we can now use it to create medicines that are just for one person, like a personalized cancer vaccine, or can be rapidly produced and distributed to entire populations, like the COVID-19 vaccines. And the best part? The best part is we ’ re simply tapping into your body ’ s own ability to make its own medicines. Thank you.

Text Sample 483: NEG Dim 3 – 2018 – Score -2.00 – 2018/t002698.txt

Computers used to be as big as a room. But now they fit in your pocket, on your wrist and can even be implanted inside of your body. How cool is that? And this has been enabled by the miniaturization of **transistors**, which are the tiny switches in the circuits at the heart of our computers. And it ’ s been achieved through decades of development and breakthroughs in science and engineering and of billions of dollars of investment. But it ’ s given us vast amounts of computing, huge amounts of memory and the digital revolution that we all experience and enjoy today. But the bad news is, we ’ re about to hit a digital roadblock, as the rate of miniaturization of **transistors** is slowing down. And this is happening at exactly the same time as our innovation in software is continuing relentlessly with artificial intelligence and big data. And our devices regularly perform facial recognition or augment our reality or even drive cars down our treacherous, chaotic roads. It ’ s amazing. But if we don ’ t keep up with the appetite of our software, we could reach a point in the development of our technology where the things that we could do with software could, in fact, be limited by our hardware. We ’ ve all experienced the frustration of an old smartphone or tablet grinding slowly to a halt over time under the ever-increasing weight of software updates and new features. And it worked just fine when we bought it not so long ago. But the hungry software engineers have eaten up all the hardware capacity over time. The semiconductor industry is very well aware of this and is working on all sorts of creative solutions, such as going beyond **transistors** to quantum computing or even working with **transistors** in alternative architectures such as neural networks to make more robust and efficient circuits. But these approaches will take quite some time, and we ’ re really looking for a much more immediate solution to this problem. The reason why the rate of miniaturization of **transistors** is slowing down is due to the ever-increasing complexity of the manufacturing process. The **transistor** used to be a big, bulky device, until the invent of the integrated circuit based on pure **crystalline** silicon wafers. And after 50 years of continuous development, we can now achieve **transistor** features dimensions down to 10 nanometers. You can fit more than a billion **transistors** in a single square millimeter of silicon. And to put this into perspective : a human hair is 100 microns across. A red blood cell, which is essentially invisible, is eight microns across, and you can place 12 across the width of a human hair. But a **transistor**, in comparison, is much smaller, at a tiny fraction of a micron across. You could place more than 260 **transistors** across a single red blood cell or more than 3, 000 across the width of a human hair. It really is incredible nanotechnology in your pocket right now. And besides the obvious benefit of being able to place more, smaller **transistors** on a chip, smaller **transistors** are faster switches, and smaller **transistors** are also more efficient switches. So this combination has given us lower cost, higher performance and higher efficiency electronics that we all enjoy today. To manufacture these integrated circuits, the **transistors** are built up layer by layer, on a pure **crystalline** silicon wafer. And in an oversimplified sense, every tiny feature of the circuit is projected onto the surface of the silicon wafer and recorded in a light-sensitive material and then etched through the light-sensitive material to leave the pattern in the underlying layers. And this process has been dramatically improved over the years to give the electronics performance we have today. But as the **transistor** features get smaller and smaller, we ’ re really approaching the physical limitations of this manufacturing technique. The latest systems for doing this patterning have become so complex that they reportedly cost more than 100 million dollars each. And semiconductor factories contain dozens of these machines. So people are seriously questioning : Is this approach long-term viable? But we believe we can do this chip manufacturing in a totally different and much more cost-effective way using molecular engineering and mimicking nature down at the nanoscale dimensions of our **transistors**. As I said, the conventional manufacturing takes every tiny feature of the circuit and projects it onto the silicon. But if you look at the structure of an integrated circuit, the **transistor** arrays, many of the features are repeated millions of times. It ’ s a highly periodic structure. So we want to take advantage of this periodicity in our alternative manufacturing technique. We want to use self-assembling materials to naturally form the periodic structures that we need for our **transistors**. We do this with the materials, then the materials do the hard work of the fine patterning, rather than pushing the projection technology to its limits and beyond. Self-assembly is seen in nature in many different places, from lipid membranes to cell structures, so we do know it can be a robust solution. If it ’ s good enough for nature, it should be good enough for us. So we want to take this naturally occurring, robust self- assembly and use it for the manufacturing of our semiconductor technology. One type of self-assemble material — it ’ s called a block co-polymer — consists of two polymer chains just a few tens of nanometers in length. But these chains hate each other. They repel each other, very much like oil and water or my teenage son and daughter. But we cruelly bond them together, creating an inbuilt frustration in the system, as they try to separate from each other. And in the bulk material, there are billions of these, and the similar components try to stick together, and the opposing components try to separate from each other at the same time. And this has a built-in frustration, a tension in the system. So it moves around, it squirms until a shape is formed. And the natural self-assembled shape that is formed is nanoscale, it ’ s regular, it ’ s periodic, and it ’ s long range, which is exactly what we need for our **transistor** arrays. So we can use molecular engineering to design different shapes of different sizes and of different periodicities. So for example, if we take a symmetrical molecule, where the two polymer chains are similar length, the natural self- assembled structure that is formed is a long, meandering line, very much like a fingerprint. And the width of the fingerprint lines and the distance between them is determined by the lengths of our polymer chains but also the level of built-in frustration in the system. And we can even create more elaborate structures if we use unsymmetrical molecules, where one polymer chain is significantly shorter than the other. And the self-assembled structure that forms in this case is with the shorter chains forming a tight ball in the middle, and it ’ s surrounded by the longer, opposing polymer chains, forming a natural cylinder. And the size of this cylinder and the distance between the cylinders, the periodicity, is again determined by how long we make the polymer chains and the level of built-in frustration. So in other words, we ’ re using molecular engineering to self-assemble nanoscale structures that can be lines or cylinders the size and periodicity of our design. We ’ re using chemistry, chemical engineering, to manufacture the nanoscale features that we need for our **transistors**. But the ability to self-assemble these structures only takes us half of the way, because we still need to position these structures where we want the **transistors** in the integrated circuit. But we can do this relatively easily using wide guide structures that pin down the self-assembled structures, anchoring them in place and forcing the rest of the self-assembled structures to lie parallel, aligned with our guide structure. For example, if we want to make a fine, 40-nanometer line, which is very difficult to manufacture with conventional projection technology, we can manufacture a 120-nanometer guide structure with normal projection technology, and this structure will align three of the 40-nanometer lines in between. So the materials are doing the most difficult fine patterning. And we call this whole approach”directed self-assembly.”The challenge with directed self-assembly is that the whole system needs to align almost perfectly, because any tiny defect in the structure could cause a **transistor** failure. And because there are billions of **transistors** in our circuit, we need an almost molecularly perfect system. But we ’ re going to extraordinary measures to achieve this, from the cleanliness of our chemistry to the careful processing of these materials in the semiconductor factory to remove even the smallest nanoscopic defects. So directed self-assembly is an exciting new disruptive technology, but it is still in the development stage. But we ’ re growing in confidence that we could, in fact, introduce it to the semiconductor industry as a revolutionary new manufacturing process in just the next few years. And if we can do this, if we ’ re successful, we ’ ll be able to continue with the cost-effective miniaturization of **transistors**, continue with the spectacular expansion of computing and the digital revolution. And what ’ s more, this could even be the dawn of a new era of molecular manufacturing. How cool is that? Thank you.

Text Sample 484: NEG Dim 3 – 2018 – Score -1.00 – 2018/t000728.txt

I ’ m an MIT professor, but I do not design buildings or computer systems. Rather, I build body parts, bionic legs that augment human walking and running. In 1982, I was in a mountain-climbing accident, and both of my legs had to be amputated due to tissue damage from frostbite. Here, you can see my legs : 24 sensors, six **microprocessors** and muscle-tendon-like actuators. I ’ m basically a bunch of nuts and bolts from the knee down. But with this advanced bionic technology, I can skip, dance and run. Thank you. I ’ m a bionic man, but I ’ m not yet a cyborg. When I think about moving my legs, neural signals from my central nervous system pass through my nerves and activate muscles within my residual limbs. Artificial electrodes sense these signals, and small computers in the bionic limb decode my nerve pulses into my intended movement patterns. Stated simply, when I think about moving, that command is communicated to the synthetic part of my body. However, those computers can ’ t input information into my nervous system. When I touch and move my synthetic limbs, I do not experience normal touch and movement sensations. If I were a cyborg and could feel my legs via small computers inputting information into my nervous system, it would fundamentally change, I believe, my relationship to my synthetic body. Today, I can ’ t feel my legs, and because of that, my legs are separate tools from my mind and my body. They

’ re not part of me. I believe that if I were a cyborg and could feel my legs, they would become part of me, part of self. At MIT, we ’ re thinking about NeuroEmbodied Design. In this design process, the designer designs human flesh and bone, the biological body itself, along with synthetics to enhance the bidirectional communication between the nervous system and the built world. NeuroEmbodied Design is a methodology to create cyborg function. In this design process, designers contemplate a future in which technology no longer compromises separate, lifeless tools from our minds and our bodies, a future in which technology has been carefully integrated within our nature, a world in which what is biological and what is not, what is human and what is not, what is nature and what is not will be forever blurred. That future will provide humanity new bodies. NeuroEmbodied Design will extend our nervous systems into the synthetic world, and the synthetic world into us, fundamentally changing who we are. By designing the biological body to better communicate with the built design world, humanity will end disability in this 21st century and establish the scientific and technological basis for human augmentation, extending human capability beyond innate, physiological levels, cognitively, emotionally and physically. There are many ways in which to build new bodies across scale, from the biomolecular to the scale of tissues and organs. Today, I want to talk about one area of NeuroEmbodied Design, in which the body ’ s tissues are manipulated and sculpted using surgical and regenerative processes. The current amputation paradigm hasn ’ t changed fundamentally since the US Civil War and has grown obsolete in light of dramatic advancements in actuators, control systems and neural interfacing technologies. A major deficiency is the lack of dynamic muscle interactions for control and proprioception. What is proprioception? When you flex your ankle, muscles in the front of your leg contract, simultaneously stretching muscles in the back of your leg. The opposite happens when you extend your ankle. Here, muscles in the back of your leg contract, stretching muscles in the front. When these muscles flex and extend, biological sensors within the muscle tendons send information through nerves to the brain. This is how we ’ re able to feel where our feet are without seeing them with our eyes. The current amputation paradigm breaks these dynamic muscle relationships, and in so doing eliminates normal proprioceptive sensations. Consequently, a standard artificial limb cannot feed back information into the nervous system about where the prosthesis is in space. The patient therefore cannot sense and feel the positions and movements of the prosthetic joint without seeing it with their eyes. My legs were amputated using this Civil War-era methodology. I can feel my feet, I can feel them right now as a phantom awareness. But when I try to move them, I cannot. It feels like they ’ re stuck inside rigid ski boots. To solve these problems, at MIT, we invented the agonist-antagonist myoneural interface, or AMI, for short. The AMI is a method to connect nerves within the residuum to an external, bionic prosthesis. How is the AMI designed, and how does it work? The AMI comprises two muscles that are surgically connected, an agonist linked to an antagonist. When the agonist contracts upon electrical activation, it stretches the antagonist. This muscle dynamic interaction causes biological sensors within the muscle tendon to send information through the nerve to the central nervous system, relating information on the muscle tendon ’ s length, speed and force. This is how muscle tendon proprioception works, and it ’ s the primary way we, as humans, can feel and sense the positions, movements and forces on our limbs. When a limb is amputated, the surgeon connects these opposing muscles within the residuum to create an AMI. Now, multiple AMI constructs can be created for the control and sensation of multiple prosthetic joints. Artificial electrodes are then placed on each AMI muscle, and small computers within the bionic limb decode those signals to control powerful motors on the bionic limb. When the bionic limb moves, the AMI muscles move back and forth, sending signals through the nerve to the brain, enabling a person wearing the prosthesis to experience natural sensations of positions and movements of the prosthesis. Can these tissue-design principles be used in an actual human being? A few years ago, my good friend Jim Ewing — of 34 years — reached out to me for help. Jim was in an a terrible climbing accident. He fell 50 feet in the Cayman Islands when his rope failed to catch him hitting the ground ' s surface. He suffered many, many injuries : punctured lungs and many broken bones. After his accident, he dreamed of returning to his chosen sport of mountain climbing, but how might this be possible? The answer was Team Cyborg, a team of surgeons, scientists and engineers assembled at MIT to rebuild Jim back to his former climbing prowess. Team member Dr. Matthew Carty amputated Jim ' s badly damaged leg at Brigham and Women ' s Hospital in Boston, using the AMI surgical procedure. Tendon pulleys were created and attached to Jim ' s tibia bone to reconnect the opposing muscles. The AMI procedure reestablished the neural link between Jim ' s ankle- foot muscles and his brain. When Jim moves his phantom limb, the reconnected muscles move in dynamic pairs, causing signals of proprioception to pass through nerves to the brain, so Jim experiences normal sensations with ankle- foot positions and movements, even when blindfolded. Here ' s Jim at the MIT laboratory after his surgeries. We electrically linked Jim ' s AMI muscles, via the electrodes, to a bionic limb, and Jim quickly learned how to move the bionic limb in four distinct ankle-foot movement directions. We were excited by these results, but then Jim stood up, and what occurred was truly remarkable. All the natural biomechanics mediated by the central nervous system emerged via the synthetic limb as an involuntary, reflexive action. All the intricacies of foot placement during stair ascent — emerged before our eyes. Here ' s Jim descending steps, reaching with his bionic toe to the next stair tread, automatically exhibiting natural motions without him even trying to move his limb. Because Jim ' s central nervous system is receiving the proprioceptive signals, it knows exactly how to control the synthetic limb in a natural way. Now, Jim moves and behaves as if the synthetic limb is part of him. For example, one day in the lab, he accidentally stepped on a roll of electrical tape. Now, what do you do when something ' s stuck to your shoe? You don ' t reach down like this ; it ' s way too awkward. Instead, you shake it off, and that ' s exactly what Jim did after being neurally connected to the limb for just a few hours. What was most interesting to me is what Jim was telling us he was experiencing. He said,"The robot became part of me."Jim Ewing : The morning after the first time I was attached to the robot, my daughter came downstairs and asked me how it felt to be a cyborg, and my answer was that I didn ' t feel like a cyborg. I felt like I had my leg, and it wasn ' t that I was attached to the robot so much as the robot was attached to me, and the robot became part of me. It became my leg pretty quickly. Hugh Herr : Thank you. By connecting Jim ' s nervous system bidirectionally to his synthetic limb, neurological embodiment was achieved. I hypothesized that because Jim can think and move his synthetic limb, and because he can feel those movements within his nervous system, the prosthesis is no longer a separate tool, but an integral part of Jim, an integral part of his body. Because of this neurological embodiment, Jim doesn ' t feel like a cyborg. He feels like he just has his leg back, that he has his body back. Now I ' m often asked when I ' m going to be neurally linked to my synthetic limbs bidirectionally, when I ' m going to become a cyborg. The truth is, I ' m hesitant to become a cyborg. Before my legs were amputated, I was a terrible student. I got D ' s and often F ' s in school. Then, after my limbs were amputated, I suddenly became an MIT professor. Now I ' m worried that once I ’ m neurally connected to my limbs once again, my brain will remap back to its not-so-bright self. But you know what, that ’ s OK, because at MIT, I already have tenure. I believe the reach of NeuroEmbodied Design will extend far beyond limb replacement and will carry humanity into realms that fundamentally redefine human potential. In this 21st century, designers will extend the nervous system into powerfully strong exoskeletons that humans can control and feel with their minds. Muscles within the body can be reconfigured for the control of powerful motors, and to feel and sense exoskeletal movements, augmenting humans ’ strength, jumping height and running speed. In this 21st century, I believe humans will become superheroes. Humans may also extend their bodies into non- anthropomorphic structures, such as wings, controlling and feeling each wing movement within the nervous system. Leonardo da Vinci said,”When once you have tasted flight, you will forever walk the earth with your eyes turned skyward, for there you have been and there you will always long to return.”During the twilight years of this century, I believe humans will be unrecognizable in morphology and dynamics from what we are today. Humanity will take flight and soar. Jim Ewing fell to earth and was badly broken, but his eyes turned skyward, where he always longed to return. After his accident, he not only dreamed to walk again, but also to return to his chosen sport of mountain climbing. At MIT, Team Cyborg built Jim a specialized limb for the vertical world, a brain-controlled leg with full position and movement sensations. Using this technology, Jim returned to the Cayman Islands, the site of his accident, rebuilt as a cyborg to climb skyward once again. Thank you. Ladies and gentlemen, Jim Ewing, the first cyborg rock climber.

Text Sample 485: NEG Dim 3 – 2018 – Score -1.00 – 2018/t000701.txt

Greg Gage : Mind-reading. You ’ ve seen this in sci-fi movies : machines that can read our thoughts. However, there are devices today that can read the electrical activity from our brains. We call this the EEG. Is there information contained in these brainwaves? And if so, could we train a computer to read our thoughts? My buddy Nathan has been working to hack the EEG to build a mind-reading machine. [DIY Neuroscience] So this is how the EEG works. Inside your head is a brain, and that brain is made out of billions of neurons. Each of those neurons sends an electrical message to each other. These small messages can combine to make an electrical wave that we can detect on a monitor. Now traditionally, the EEG can tell us large-scale things, for example if you ’ re asleep or if you ’ re alert. But can it tell us anything else? Can it actually read our thoughts? We ’ re going to test this, and we ’ re not going to start with some complex thoughts. We ’ re going to do something very simple. Can we interpret what someone is seeing using only their brainwaves? Nathan ’ s going to begin by placing electrodes on Christy ’ s head. Nathan : My life is tangled. GG : And then he ’ s going to show her a bunch of pictures from four different categories. Nathan : Face, house, scenery and weird pictures. GG : As we show Christy hundreds of these images, we are also capturing the electrical waves onto Nathan ’ s computer. We want to see if we can detect any visual information about the photos contained in the brainwaves, so when we ’ re done, we ’ re going to see if the EEG can tell us what kind of picture Christy is looking at, and if it does, each category should trigger a different brain signal. OK, so we collected all the raw EEG data, and this is what we got. It all looks pretty messy, so let ’ s arrange them by picture. Now, still a bit too noisy to see any differences, but if we average the EEG across all image types by aligning them to when the image first appeared, we can remove this noise, and pretty soon, we can see some dominant patterns emerge for each category. Now the signals all still look pretty similar. Let ' s take a closer look. About a hundred **milliseconds** after the image comes on, we see a positive bump in all four cases, and we call this the P100, and what we think that is is what happens in your brain when you recognize an object. But damn, look at that signal for the face. It looks different than the others. There ' s a negative dip about 170 **milliseconds** after the image comes on. What could be going on here? Research shows that our brain has a lot of neurons that are dedicated to recognizing human faces, so this N170 spike could be all those neurons firing at once in the same location, and we can detect that in the EEG. So there are two takeaways here. One, our eyes can ' t really detect the differences in patterns without averaging out the noise, and two, even after removing the noise, our eyes can only pick up the signals associated with faces. So this is where we turn to machine learning. Now, our eyes are not very good at picking up patterns in noisy data, but machine learning algorithms are designed to do just that, so could we take a lot of pictures and a lot of data and feed it in and train a computer to be able to interpret what Christy is looking at in real time? We ' re trying to code the information that ' s coming out of her EEG in real time and predict what it is that her eyes are looking at. And if it works, what we should see is every time that she gets a picture of scenery, it should say scenery, scenery, scenery, scenery. A face — face, face, face, face, but it ' s not quite working that way, is what we ' re discovering. OK. Director : So what ' s going on here? GG : We need a new career, I think. OK, so that was a massive failure. But we ' re still curious : How far could we push this technology? And we looked back at what we did. We noticed that the data was coming into our computer very quickly, without any timing of when the images came on, and that ' s the equivalent of reading a very long sentence without spaces between the words. It would be hard to read, but once we add the spaces, individual words appear and it becomes a lot more understandable. But what if we cheat a little bit? By using a sensor, we can tell the computer when the image first appears. That way, the brainwave stops being a continuous stream of information, and instead becomes individual packets of meaning. Also, we ' re going to cheat a little bit more, by limiting the categories to two. Let ' s see if we can do some real-time mind-reading. In this new experiment, we ' re going to constrict it a little bit more so that we know the onset of the image and we ' re going to limit the categories to"face"or"scenery."Nathan : Face. Correct. Scenery. Correct. GG : So right now, every time the image comes on, we ' re taking a picture of the onset of the image and decoding the EEG. It ' s getting correct. Nathan : Yes. Face. Correct. GG : So there is information in the EEG signal, which is cool. We just had to align it to the onset of the image. Nathan : Scenery. Correct. Face. Yeah. GG : This means there is some information there, so if we know at what time the picture came on, we can tell what type of picture it was, possibly, at least on average, by looking at these evoked potentials. Nathan : Exactly. GG : If you had told me at the beginning of this project this was possible, I would have said no way. I literally did not think we could do this. Did our mind-reading experiment really work? Yes, but we had to do a lot of cheating. It turns out you can find some interesting things in the EEG, for example if you ' re looking at someone ' s face, but it does have a lot of limitations. Perhaps advances in machine learning will make huge strides, and one day we will be able to decode what ' s going on in our thoughts. But for now, the next time a company says that they can harness your brainwaves to be able to control devices, it is your right, it is your duty to be skeptical.

Text Sample 486: NEG Dim 3 – 2020 – Score -2.00 – 2020/t004039.txt

Chris Anderson : Mike, welcome. It ’ s good to see you. I ’ m excited for this conversation. Michael Levin : Thank you so much. I ’ m so happy to be here. CA : So, most of us have this mental model in biology that DNA is a property of every living thing, that it is kind of the software that builds the hardware of our body. That ’ s how a lot of us think about this. That model leaves too many deep mysteries. Can you share with us some of those mysteries and also what tadpoles have to do with it? ML : Sure. Yeah. I ’ d like to give you another perspective on this problem. One of the things that DNA does is specify the hardware of each cell. So the DNA tells every cell what proteins it ’ s supposed to have. And so when you have tadpoles, for example, you see the kind of thing that most people think is sort of a progressive unrolling of the genome. Specific genes turn on and off, and a tadpole, as it becomes a frog, has to rearrange its face. So the eyes, the nostrils, the jaws – everything has to move. And one way to think about it used to be that, well, you have a sort of hardwired set of movements where all of these things move around and then you get your frog. But actually, a few years ago, we found a pretty amazing phenomenon, which is that if you make so-called”Picasso frogs”– these are tadpoles where the jaws might be off to the side, the eyes are up here, the nostrils are moved, so everything is shifted – these tadpoles make largely normal frog faces. Now, this is amazing, because all of the organs start off in abnormal positions, and yet they still end up making a pretty good frog face. And so what it turns out is that this system, like many living systems, is not a hardwired set of movements, but actually works to reduce the error between what ’ s going on now and what it knows is a correct frog face configuration. This kind of decision-making that involves flexible responses to new circumstances, in other contexts, we would call this intelligence. And so what we need to understand now is not only the mechanisms by which these cells execute their movements and gene expression and so on, but we really have to understand the information flow : How do these cells cooperate with each other to build something large and to stop building when that specific structure is created? And these kinds of computations, not just the mechanisms, but the computations of anatomical control, are the future of biology. CA : And so I guess the traditional model is that somehow cells are sending biochemical signals to each other that allow that development to happen the smart way. But you think there is something else at work. What is that? ML : Well, cells certainly do communicate biochemically and via physical forces, but there ’ s something else going on that ’ s extremely interesting, and it ’ s basically called bioelectricity, non-neural bioelectricity. So it turns out that all cells – not just nerves, but all cells in your body – communicate with each other using electrical signals. And what you ’ re seeing here is a time-lapse video. For the first time, we are now able to eavesdrop on all of the electrical conversations that the cells are having with each other. So think about this. We ’ re now watching – This is an early frog embryo. This is about eight hours to 10 hours of development. And the colors are showing you actual electrical states that allow you to see all of the electrical software that ’ s running on the genome-defined cellular hardware. And so these cells are basically communicating with each other who is going to be head, who is going to be tail, who is going to be left and right and make eyes and brain and so on. And so it is this software that allows these living systems to achieve specific goals, goals such as building an embryo or regenerating a limb for animals that do this, and the ability to see these electrical conversations gives us some really remarkable opportunities to target or to rewrite the goals towards which these living systems are operating. CA : OK, so this is pretty radical. Let me see if I understand this. What you ’ re saying is that when an organism starts to develop, as soon as a cell divides, electrical signals are shared between them. But as you get to, what, a hundred, a few hundred cells, that somehow these signals end up forming essentially like a computer program, a program that somehow includes all the information needed to tell that organism what its destiny is? Is that the right way to think about it? ML : Yes, quite. Basically, what happens is that these cells, by forming electrical networks much like networks in the brain, they form electrical networks, and these networks process information including pattern memories. They include representation of large-scale anatomical structures where various organs will go, what the different axes of the animal – front and back, head and tail – are going to be, and these are literally held in the electrical circuits across large tissues in the same way that brains hold other kinds of memories and learning. CA : So is this the right way to think about it? Because this seems to be such a big shift. I mean, when I first got a computer, I was in awe of the people who could do so-called"machine code,"like the direct programming of individual bits in the computer. That was impossible for most mortals. To have a chance of controlling that computer, you ' d have to program in a language, which was a vastly simpler way of making big-picture things happen. And if I understand you right, what you ' re saying is that most of biology today has sort of taken place trying to do the equivalent of machine code programming, of understanding the biochemical signals between individual cells, when, wait a sec, holy crap, there ' s this language going on, this electrical language, which, if you could understand that, that would give us a completely different set of insights into how organisms are developing. Is that metaphor basically right? ML : Yeah, this is exactly right. So if you think about the way programming was done in the ' 40s, in order to get your computer to do something different, you would physically have to shift the wires around. So you ' d have to go in there and rewire the hardware. You ' d have to interact with the hardware directly, and all of your strategies for manipulating that machine would be at the level of the hardware. And the reason we have this now amazing technology revolution, information sciences and so on, is because computer science moved from a focus on the hardware on to understanding that if your hardware is good enough – and I ' m going to tell you that biological hardware is absolutely good enough – then you can interact with your system not by tweaking or rewiring the hardware, but actually, you can take a step back and give it stimuli or inputs the way that you would give to a reprogrammable computer and cause the cellular network to do something completely different than it would otherwise have done. So the ability to see these bioelectrical signals is giving us an entry point directly into the software that guides large-scale anatomy, which is a very different approach to medicine than to rewiring specific pathways inside of every cell. CA : And so in many ways, this is the amazingness of your work is that you ' re starting to crack the code of these electrical signals, and you ' ve got an amazing demonstration of this in these flatworms. Tell us what ' s going on here. ML : So this is a creature known as a planarian. They ' re flatworms. They ' re actually quite a complex creature. They have a true brain, lots of different organs and so on. And the amazing thing about these planaria is that they are highly, highly regenerative. So if you cut it into pieces – in fact, over 200 pieces – every piece will rebuild exactly what ' s needed to make a perfect little worm. So think about that. This is a system where every single piece knows exactly what a correct planarian looks like and builds the right organs in the right places and then stops. And that ' s one of the most amazing things about regeneration. So what we discovered is that if you cut it into three pieces and amputate the head and the tail and you just take this middle fragment, which is what you see here, amazingly, there is an electrical gradient, head to tail, that ' s generated that tells the piece where the heads and the tails go and in fact, how many heads or tails you ' re supposed to have. So what we learned to do is to manipulate this electrical gradient, and the important thing is that we don ' t apply electricity. What we do instead was we turned on and off the little **transistors** – they ' re actual ion channel proteins – that every cell natively uses to set up this electrical state. So now we have ways to turn them on and off, and when you do this, one of the things you can do is you can shift that circuit to a state that says no, build two heads, or in fact, build no heads. And what you ' re seeing here are real worms that have either two or no heads that result from this, because that electrical map is what the cells are using to decide what to do. And so what you ' re seeing here are live two-headed worms. And, having generated these, we did a completely crazy experiment. You take one of these two-headed worms, and you chop off both heads, and you leave just the normal middle fragment. Now keep in mind, these animals have not been genomically edited. There ' s absolutely nothing different about their genomes. Their genome sequence is completely wild type. So you amputate the heads, you ' ve got a nice normal fragment, and then you ask : In plain water, what is it going to do? And, of course, the standard paradigm would say, well, if you ' ve gotten rid of this ectopic extra tissue, the genome is not edited so it should make a perfectly normal worm. And the amazing thing is that it is not what happens. These worms, when cut again and again, in the future, in plain water, they continue to regenerate as two-headed. Think about this. The pattern memory to which these animals will regenerate after damage has been permanently rewritten. And in fact, we can now write it back and send them back to being one-headed without any genomic editing. So this right here is telling you that the information structure that tells these worms how many heads they ' re supposed to have is not directly in the genome. It is in this additional bioelectric layer. Probably many other things are as well. And we now have the ability to rewrite it. And that, of course, is the key definition of memory. It has to be stable, long-term stable, and it has to be rewritable. And we are now beginning to crack this morphogenetic code to ask how is it that these tissues store a map of what to do and how we can go in and rewrite that map to new outcomes. CA : I mean, that seems incredibly compelling evidence that DNA is just not controlling the actual final shape of these organisms, that there ' s this whole other thing going on, and, boy, if you could crack that code, what else could that lead to. By the way, just looking at these ones. What is life like for a two-headed flatworm? I mean, it seems like it ' s kind of a trade-off. The good news is you have this amazing three-dimensional view of the world, but the bad news is you have to poop through both of your mouths? ML : So, the worms have these little tubes called pharynxes, and the tubes are sort of in the middle of the body, and they excrete through that. These animals are perfectly viable. They ' re completely happy, I think. The problem, however, is that the two heads don ' t cooperate all that well, and so they don ' t really eat very well. But if you manage to feed them by hand, they will go on forever, and in fact, you should know these worms are basically immortal. So these worms, because they are so highly regenerative, they have no age limit, and they ' re telling us that if we crack this secret of regeneration, which is not only growing new cells but knowing when to stop – you see, this is absolutely crucial – if you can continue to exert this really profound control over the three-dimensional structures that the cells are working towards, you could defeat aging as well as traumatic injury and things like this. So one thing to keep in mind is that this ability to rewrite the large-scale anatomical structure of the body is not just a weird planarian trick. It ' s not just something that works in flatworms. What you ' re seeing here is a tadpole with an eye and a gut, and what we ' ve done is turned on a very specific ion channel. So we basically just manipulated these little electrical **transistors** that are inside of cells, and we ’ ve imposed a state on some of these gut cells that ’ s normally associated with building an eye. And as a result, what the cells do is they build an eye. These eyes are complete. They have optic nerve, lens, retina, all the same stuff that an eye is supposed to have. They can see, by the way, out of these eyes. And what you ’ re seeing here is that by triggering eye-building **subroutines** in the physiological software of the body, you can very easily tell it to build a complex organ. And this is important for our biomedicine, because we don ’ t know how to micromanage the construction of an eye. I think it ’ s going to be a really long time before we can really bottom-up build things like eyes or hands and so on. But we don ’ t need to, because the body already knows how to do it, and there are these **subroutines** that can be triggered by specific electrical patterns that we can find. And this is what we call”cracking the bioelectric code.”We can make eyes. We can make extra limbs. Here ’ s one of our five-legged tadpoles. We can make extra hearts. We ’ re starting to crack the code to understand where are the **subroutines** in this software that we can trigger and build these complex organs long before we actually know how to micromanage the process at the cellular level. CA : So as you ’ ve started to get to learn this electrical layer and what it can do, you ’ ve been able to create – is it fair to say it ’ s almost like a new, a novel life-form, called a xenobot? Talk to me about xenobots. ML : Right. So if you think about this, this leads to a really strange prediction. If the cells are really willing to build towards a specific map, we could take genetically unaltered cells, and what you ’ re seeing here is cells taken out of a frog body. They ’ ve coalesced in a way that asks them to re-envision their multicellularity. And what you see here is that when liberated from the rest of the body of the animal, they make these tiny little novel bodies that are, in terms of behavior, you can see they can move, they can run a maze. They are completely different from frogs or tadpoles. Frog cells, when asked to re-envision what kind of body they want to make, do something incredibly interesting. They use the hardware that their genetics gives them, for example, these little hairs, these little cilia that are normally used to redistribute mucus on the outside of a frog, those are genetically specified. But what these creatures did, because the cells are able to form novel kinds of bodies, they have figured out how to use these little cilia to instead row against the water, and now have locomotion. So not only can they move around, but they can, and here what you ’ re seeing, is that these cells are coalescing together. Now they ’ re starting to have conversations about what they are going to do. You can see here the flashes are these exchanges of information. Keep in mind, this is just skin. There is no nervous system. There is no brain. This is just skin. This is skin that has learned to make a new body and to explore its environment and move around. And they have spontaneous behaviors. You can see here where it ’ s swimming down this maze. At this point, it decides to turn around and go back where it came from. So it has its own behavior, and this is a remarkable model system for several reasons. First of all, it shows us the amazing plasticity of cells that are genetically wild type. There is no genetic editing here. These are cells that are really prone to making some sort of functional body. The second thing, and this was done in collaboration with Josh Bongard ’ s lab at UVM, they modeled the structure of these things and evolved it in a virtual world. So this is literally – on a computer, they modeled it on a computer. So this is literally the only organism that I know of on the face of this planet whose evolution took place not in the biosphere of the earth but inside a computer. So the individual cells have an evolutionary history, but this organism has never existed before. It was evolved in this virtual world, and then we went ahead and made it in the lab, and you can see this amazing plasticity. This is not only for making useful machines. You can imagine now programming these to go out into the environment and collect toxins and cleanup, or you could imagine ones made out of human cells that would go through your body and collect cancer cells or reshape arthritic joints, deliver pro-regenerative compounds, all kinds of things. But not only these useful applications – this is an amazing sandbox for learning to communicate morphogenetic signals to cell collectives. So once we crack this, once we understand how these cells decide what to do, and then we ' re going to, of course, learn to rewrite that information, the next steps are great improvements in regenerative medicine, because we will then be able to tell cells to build healthy organs. And so this is now a really critical opportunity to learn to communicate with cell groups, not to micromanage them, not to force the hardware, to communicate and rewrite the goals that these cells are trying to accomplish. CA : Well, it ' s mind-boggling stuff. Finally, Mike, give us just one other story about medicine that might be to come as you develop this understanding of how this bioelectric layer works. ML : Yeah, this is incredibly exciting because, if you think about it, most of the problems of biomedicine – birth defects, degenerative disease, aging, traumatic injury, even cancer – all boil down to one thing : cells are not building what you would like them to build. And so if we understood how to communicate with these collectives and really rewrite their target morphologies, we would be able to normalize tumors, we would be able to repair birth defects, induce regeneration of limbs and other organs, and these are things we have already done in frog models. And so now the next really exciting step is to take this into mammalian cells and to really turn this into the next generation of regenerative medicine where we learn to address all of these biomedical needs by communicating with the cell collectives and rewriting their bioelectric pattern memories. And the final thing I ' d like to say is that the importance of this field is not only for biomedicine. You see, this, as I started out by saying, this ability of cells in novel environments to build all kinds of things besides what their genome tells them is an example of intelligence, and biology has been intelligently solving problems long before brains came on the scene. And so this is also the beginnings of a new inspiration for machine learning that mimics the artificial intelligence of body cells, not just brains, for applications in computer intelligence. CA : Mike Levin, thank you for your extraordinary work and for sharing it so compellingly with us. Thank you. ML : Thank you so much. Thank you, Chris.

Text Sample 487: NEG Dim 3 – 2020 – Score -1.00 – 2020/t004156.txt

In the early months of the pandemic, chef José Andrés circulated two photos that have come to **symbolize** a modern American food crisis. The first shows mountains of potatoes that have been left to rot in a field in Idaho. The restaurants and cafeterias and stadiums that had consumed them were shuttered during the pandemic. The second shows a devastating scene outside of the San Antonio food bank. Thousands of carloads of people lined up, waiting for food with not enough supply to go around."How is it possible these two photos exist at the same time, in the most prosperous and technologically advanced moment in our history,"tweeted Andrés. In the months after the photos were published, the crisis got worse. Billions of pounds of potatoes and other fresh produce were chucked by American farmers. At the same time, food banks all over the country were reporting demand increases and 40 percent were facing critical shortfalls. Outside the US, especially in the Middle East and throughout Southeastern Africa, COVID-19 was paralyzing food systems that were already vulnerable. Oxfam has predicted that by the end of 2020 12, 000 people per day could die of hunger related to COVID. That ' s more than the highest daily mortality rate recorded so far. But what ' s worse and what ' s much more concerning to all of us is that COVID is just one of many major disruptions that have been predicted in the years and decades ahead. More chronic and complex than the pressures of COVID are the pressures of climate change. And those of you who live in California have seen this on your farms. You ' ve seen withering heat and drought and fires disrupt avocado and almond and citrus and strawberry farms. This summer, we saw the devastating impacts of storms on corn and soy farms. I ' ve seen the various pressures of drought, heat, flooding, superstorms, invasive insects, bacterial blight, shifting seasons and weather volatility from Washington to Florida, and from Guatemala to Australia. The upshot is this. Climate change is becoming something we can taste. This is a kitchen-table issue in the literal sense. The International Panel on Climate Change has predicted that by mid-century the world may reach a threshold of global warming beyond which current agricultural practices can no longer support large human civilizations. The USDA scientist Jerry Hatfield put it to me this way : the single biggest threat of climate change is the collapse of food systems. The reality we face, one that was exposed by those mountains of potatoes and the cars lined up during the pandemic, is that our supply chains are antiquated. Our food systems have not been designed to adapt to major disruptions or preempt them. Addressing this challenge as much as any other is going to define our progress in the coming century. But there ' s good news. And the good news is that farmers and entrepreneurs and academics are radically rethinking national and global food systems. They are marrying principles of old-world agroecology and state-of-the-art technologies to create what I call a third way to our food future. We ' re going to see radical changes in what we grow and how we eat in the coming decades, as these environmental and population and public health pressures intensify. I studied these changes for my book"The Fate of Food : What We ' ll Eat in a Bigger, Hotter, Smarter World."I traveled for five years into the lands and the minds and the machines that are shaping the future of food. My travels took me through 15 countries and 18 states, from apple orchards in Wisconsin to tiny cornfields in Kenya, to massive Norwegian fish farms and computerized foodscapes in Shanghai. I investigated new ideas, like robotics and CRISPR and vertical farms. And old ideas, like edible insects and permaculture and ancient plants. I began to see the emergence of this third way to food production. A synthesis of the traditional and the radically new. There ' s a growing controversy about the best path to future food security in the US. Food is ripe for reinvention, Bill Gates has proclaimed. Huge flows of investment are funding new methods of climate-smart and high-tech agriculture. But many sustainable food advocates bristle at this idea of reinvention. They want food deinvented. They argue for a return to preindustrial and pre-green revolution, biodynamic and organic farming. To which skeptics inevitably respond,"Nice, but does it scale? Sure, a return to traditional farming methods could produce better food, but can it produce enough food that ' s affordable?"The rift between the reinvention camp and the deinvention camp has existed for decades. But now it ' s a raging battle. One side covets the past, the other side covets the future and as someone observing this from the outside, I began to wonder, why must it be so binary? Can ' t there be a synthesis of the two approaches? Our challenge is to borrow from the wisdom of the ages, and from our most advanced science, to forge this third way. One that allows us to improve and scale our harvests, while restoring rather than degrading the underlying web of life. I belong to neither camp. I ' m a failed vegan and a lapsed vegetarian, and a terrible backyard farmer. If I ' m honest, I will keep trying at this, but I may fail. But I ’ m hell-bent on hope, and if my travels have taught me anything, it ’ s that there ’ s good reason for hope. Plenty of solutions are merging that can help build sustainable, resilient food systems. Even if we can ’ t rely on a critical mass of backyard-farming vegetarians to do this on their own, from the ground up. Let ’ s start with artificial intelligence and robotics. Jorge Heraud is a Peruvian-born engineer who now lives in Silicon Valley, and his company developed a robotic weeder named See and Spray, and I went to Arkansas to see the maiden voyage of See and Spray. And I was half expecting a battalion of C3PO-style robots to march into the fields with pincer hands to pluck the weeds. And instead, I found this. A tractor with a big, white hoop skirt off the back of it. And inside that hoop skirt are 24 cameras that use computer vision to see the ground beneath and to distinguish between the plants and the weeds. And to deploy with sniper-like precision these tiny jets of concentrated fertilizer, or herbicide, that incinerate the baby weeds. I learned how robotics can end the practice of broadcast spraying chemicals across millions of acres of land and how we can reduce the use of herbicides by up to 90 percent. But the bigger picture is even more exciting. Intelligent machines can treat plants individually, applying not just herbicides but fungicides and insecticides and fertilizers on a plant-by-plant, rather than field-by-field basis. So that eventually, this kind of hyperspecific farming can allow for more diversity and intercropping on fields. And big farms can begin to mimic natural systems and improve soil health. Heraud is the embodiment of third-way thinking, right? Robots, he told me, don ’ t have to remove us from nature, they can bring us closer to it, they can restore it. Increasing crop diversity will be crucial to building resilient food systems. And so will decentralizing agriculture so that when farmers in one region are disrupted, the others around, they can keep growing. The rise of vertical farms, like this farm, built inside a former steel mill in Newark, New Jersey, can play a key role in decentralizing agriculture. Aeroponic farms use a tiny fraction of the water that is used in in-ground farms. And they can grow food much faster, about 40 percent faster. And when located in and near cities, where the food is consumed, they eliminate a huge amount of trucking and food waste. It struck me at first as creepy in kind of a”Silent Running”way that we ’ d be growing our future fruits and vegetables inside, without soil or sun. And after weeks of spending time in these plant factories, I began to see it as oddly, almost perfectly natural to deliver the plants only and exactly what they need, with zero herbicides and radical efficiency. Here again, we see innovators borrowing from, and perhaps even elevating the wisdom of natural ecosystems. Developments in plant-based and alternative meats are also profoundly hopeful. And they follow a similar trend toward local, resilient, low-carbon protein production. Consumers are excited about this, and during the pandemic, we ’ ve seen a 250 percent increase in demand for alternative meats. A study by the Journal of Clinical Nutrition found that the participants who were eating the plant-based proteins saw a drop in their cholesterol levels, in their weight and eventually, a drop in their risk of heart disease. The potential environmental benefits of plant-based meats are astounding. And there ’ s even potential in lab-grown or cell-based meats. Uma Valeti fed me my first plate of lab-grown duck breast, harvested fresh from a bioreactor. It had been grown from a small sampling of cells taken from muscle tissue and fat and connective tissues, which is exactly what we eat when we eat meat. This lab-grown or cell-based duck meat has very little threat of bacterial contamination, it ’ s about 85 percent lower CO2 emissions associated with it. Eventually it can be grown like those crops inside vertical farms in decentralized facilities that aren ’ t vulnerable to supply-chain disruptions. Valeti started out as a cardiologist, who understood that doctors have been developing human and animal tissues in laboratories for decades. He was inspired as much by that as he was by a 1931 quote from Winston Churchill that says,”We shall escape the absurdity of growing the whole chicken in order to eat the breast or the wing, by growing them separately in suitable mediums.”Like Heraud, Valeti is a quintessential third-way thinker. He ’ s reimagined an old idea using new technology, to usher in a solution whose time has come. I ’ ve met with dozens of farmers and entrepreneurs and engineers who emulate third-way thinking, all over the world. They ’ re using modern breeding tools like CRISPR to develop nutritious heirloom crops that can withstand drought and heat. They ’ re using AI to make aquaculture sustainable. They ’ re finding ways to eliminate food waste. They are scaling up conservation agriculture and managed grazing. And they ’ re reviving ancient plants, and they ’ re recycling sewage and gray water to develop a drought-proof water supply. The upshot is this : Human innovation that marries old and new approaches to food production can, and I believe, will usher in this third way and redefine sustainable food on a grand scale.

Text Sample 488: NEG Dim 3 – 2020 – Score -1.00 – 2020/t004078.txt

What will the biggest challenges of the 21st century turn out to be? Today, one might guess climate change, public health, inequality, but the truth is we don ’ t yet know. What we do know is that supercomputing will have to be part of the solution. For nearly a hundred years, our reliance on high-performance computers in the face of our most urgent challenges has grown and grown, from cracking Nazi codes to sequencing the human genome. Computer processors have risen to meet increasingly critical and complex demands by getting smaller, faster and better year after year, as if by magic. But there ’ s a problem. At the very moment that our reliance on computers is growing faster than ever, progress in compute power is coming to a standstill. The magic is just about spent. The timing couldn ’ t be worse. We rarely talk about it, but for all that we ’ ve accomplished with computers, there remain a startling number of things that computers still can ’ t do, at a great cost to business and society. The dream of near-instant computational drug design, for instance, has yet to come to fruition, nearly 50 years after it was first conceived. Never has that been clearer than now, as the world sits in a state of isolation and paralysis, as we await a vaccine for COVID-19. But drug discovery is just one area in which researchers are beset – and in some cases, blocked entirely – by the inadequacy of even today ’ s fastest supercomputers, putting great constraints in areas like climate change, and in value creation in areas like finance and logistics. In the past, we could rely on supercomputers simply getting better and faster, as parts got smaller and smaller every year, but no longer. For now, we ’ re drawing up against a hard physical limit. **Transistors** have become so minuscule that they ’ re fast approaching the size of an atom. Such a state of affairs invites a natural follow-up question, and it ’ s one that I ’ ve spent the last several years encouraging business leaders and policy makers to address : If not traditional supercomputers, what technology will emerge to arm us against the challenges of the 21st century? Enter quantum computing. Quantum computers like this one promise to address the atomic limitation by exploiting subatomic physical properties that weren ’ t even known to man a hundred years ago. How does it work? Quantum computing enables a departure from two major constraints of classical semiconductor computing. Classical computers operate deterministically : everything is either yes or no, on or off, with no in between. They also operate serially ; they can only do one thing at a time. Quantum computers operate probabilistically, and most importantly, they operate simultaneously, thanks to three properties – superposition, entanglement and interference – which allow them to explore many possibilities at once. To illustrate how this works, imagine a computer is trying to solve a maze. The classical computer would do so by exhausting every potential pathway in a sequence. If it came across a roadblock on the first path, it would simply rule that out as a solution, revert to its original position and try the next logical path, and so on and so forth until it found the right solution. A quantum computer could test every single pathway at the same time, in effect, solving the maze in only a single try. As it happens, many complex problems are characterized by this mazelike quality, especially simulation and optimization problems, some of which can be solved exponentially faster with a quantum computer. But is there really value to this so-called quantum speedup? In order to believe that we need faster supercomputers, we need to first believe that our problems are indeed computational in nature. It turns out that many are, at least in part. For an example, let ' s turn to fertilizer production, one of the hallmark problems in the science of climate change. The way most fertilizer is produced today is by fusing nitrogen and hydrogen to make ammonia, which is the active ingredient. The process works, but only at a severe, a severe cost to businesses, who spend 100 to 300 billion every year, and to the environment : three to five percent of the world ' s natural gas is expended on fertilizer synthesis every single year. So why have scientists failed to develop a more efficient process? The reason is that in order to do so, they would need to simulate the mazelike molecular interactions that make up the electrostatic field of the key catalyst, nitrogenase. Scientists actually know how to do that today, but it would take 800, 000 years on the world ' s fastest supercomputer. With a full-scale quantum computer, less than 24 hours. For another example, let ' s return to drug discovery, a process COVID-19 has brought into sharp focus for most of us for the very first time. Designing a vaccine for an infectious disease like COVID-19, from identifying the drivers of the disease, to screening millions of candidate activators and inhibitors, is a process that typically takes 10 or more years per drug, 90 percent of which fail to pass clinical trials. The cost to pharmaceutical companies is two to three billion dollars per approved drug. But the social costs of delays and failures are much, much higher. More than eight million people die every year of infectious diseases. That ' s 15 times as many people as died during the first six months of the coronavirus pandemic. So why has computational drug design failed to live up to expectations? Again, it ' s a matter of limited computational resources, at least in part. If identifying a disease pathway in the body is like a lock, designing a drug requires searching through a massive chemical space, effectively a maze of molecular structures, to find the right compound, to find a key, in other words, that fits the lock. The problem is that tracing the entire relevant span of chemical space and converting it into a searchable database for drug design would take 5 trillion trillion, trillion, trillion years on the world ' s fastest supercomputer. On a quantum computer, a little more than a half hour. But quantum computing is not just about triumphs in the lab. The flow of progress and industries of all kinds is currently blocked by discreet, but intractable computational constraints that have a real impact on business and society. For what may seem an unlikely example, let ' s turn to banks. What if banks were able to lend more freely to individuals, entrepreneurs and businesses? One of the key holdups today is that banks keep 10 to 15 percent of assets in cash reserves, in part, because their risk simulations are compute constrained. They can ' t account for global or whole-market risks that are rare but severe and unpredictable. Black swan events, for example. Now, 10 to 15 percent is a whole lot of money. When you consider that for every one-percent reduction in cash reserves, it would lead to an extra trillion dollars of investible capital. What this means is that if banks ultimately became comfortable enough with quantum-powered risk simulations to reduce cash reserves to, say, five to 10 percent of assets, the effect would be like a COVID-19-level stimulus for individuals and businesses every single year. Once the transformative power of quantum computing is clear, the question then becomes : Well, how long must we wait? Researchers are cautious when asked about the timeline to quantum advantage. Rightly so – there remain a number of critical hurdles to overcome, and not just engineering challenges, but fundamental scientific questions about the nature of quantum mechanics. As a result, it may be one, two, even three decades before quantum computers fully mature. Some executives that I ’ ve spoken with have come to the conclusion on this basis that they can afford to wait, that they can afford to postpone investing. I believe this to be a real mistake, for while some technologies develop steadily, according to the laws of cumulative causation, many emerge as precipitous breakthroughs, almost overnight defying any timeline that could be drawn out in advance. Quantum computing is a candidate for just such a breakthrough, having already reached a number of critical milestones decades ahead of schedule. In the late 80s, for example, many researchers thought that the basic building block of quantum computing, the qubit, would take a hundred years to build. Ten years later, it arrived. Now IBM has nearly 500 qubits across 29 machines, available for client use and research. What this means is that we should worry less about quantum computers arriving too late and more about them arriving too soon, before the necessary preparations have been made. For to quote one Nobel prize winning physicist,”Quantum computers are more different from current computers than current computers are from the abacus.”It ’ ll take time to make the necessary workflow integrations. It ’ ll take time to onboard the right talent. Most importantly, it ’ ll take time, not to mention vision and imagination, to identify and scope high-value problems for quantum computers to tackle for your business. Governments are already investing heavily in quantum technologies – 15 billion dollars among China, Europe and the US. And VCs are following suit. But what ’ s needed now to accelerate innovation is business investment in developing use cases, in onboarding talent and on experimenting with real quantum computers that are available today. In a world such as ours, the demands of innovation can ’ t be put off for another day. Leaders must act now, for the processor speedups that have driven innovation for nearly 70 years are set to stop dead in their tracks. The race toward a new age of magic and supercomputing is already underway. It ’ s one we can ’ t afford to lose. Quantum computers are in pole position. They ’ re the car to beat. Thank you.

Text Sample 489: NEG Dim 3 – 2017 – Score -1.00 – 2017/t000635.txt

Nine years ago, my sister discovered lumps in her neck and arm and was diagnosed with cancer. From that day, she started to benefit from the understanding that science has of cancer. Every time she went to the doctor, they measured specific molecules that gave them information about how she was doing and what to do next. New medical options became available every few years. Everyone recognized that she was struggling heroically with a biological illness. This spring, she received an innovative new medical treatment in a clinical trial. It dramatically knocked back her cancer. Guess who I ’ m going to spend this Thanksgiving with? My vivacious sister, who gets more exercise than I do, and who, like perhaps many people in this room, increasingly talks about a lethal illness in the past tense. Science can, in our lifetimes — even in a decade — transform what it means to have a specific illness. But not for all illnesses. My friend Robert and I were classmates in graduate school. Robert was smart, but with each passing month, his thinking seemed to become more disorganized. He dropped out of school, got a job in a store... But that, too, became too complicated. Robert became fearful and withdrawn. A year and a half later, he started hearing voices and believing that people were following him. Doctors diagnosed him with schizophrenia, and they gave him the best drug they could. That drug makes the voices somewhat quieter, but it didn ' t restore his bright mind or his social connectedness. Robert struggled to remain connected to the worlds of school and work and friends. He drifted away, and today I don ' t know where to find him. If he watches this, I hope he ' ll find me. Why does medicine have so much to offer my sister, and so much less to offer millions of people like Robert? The need is there. The World Health Organization estimates that brain illnesses like schizophrenia, bipolar disorder and major depression are the world ' s largest cause of lost years of life and work. That ' s in part because these illnesses often strike early in life, in many ways, in the prime of life, just as people are finishing their educations, starting careers, forming relationships and families. These illnesses can result in suicide ; they often compromise one ' s ability to work at one ' s full potential ; and they ' re the cause of so many tragedies harder to measure : lost relationships and connections, missed opportunities to pursue dreams and ideas. These illnesses limit human possibilities in ways we simply cannot measure. We live in an era in which there ' s profound medical progress on so many other fronts. My sister ' s cancer story is a great example, and we could say the same of heart disease. Drugs like statins will prevent millions of heart attacks and strokes. When you look at these areas of profound medical progress in our lifetimes, they have a narrative in common : scientists discovered molecules that matter to an illness, they developed ways to detect and measure those molecules in the body, and they developed ways to interfere with those molecules using other molecules — medicines. It ' s a strategy that has worked again and again and again. But when it comes to the brain, that strategy has been limited, because today, we don ' t know nearly enough, yet, about how the brain works. We need to learn which of our cells matter to each illness, and which molecules in those cells matter to each illness. And that ' s the mission I want to tell you about today. My lab develops technologies with which we try to turn the brain into a big-data problem. You see, before I became a biologist, I worked in computers and math, and I learned this lesson : wherever you can collect vast amounts of the right kinds of data about the functioning of a system, you can use computers in powerful new ways to make sense of that system and learn how it works. Today, big-data approaches are transforming ever-larger sectors of our economy, and they could do the same in biology and medicine, too. But you have to have the right kinds of data. You have to have data about the right things. And that often requires new technologies and ideas. And that is the mission that animates the scientists in my lab. Today, I want to tell you two short stories from our work. One fundamental obstacle we face in trying to turn the brain into a big-data problem is that our brains are composed of and built from billions of cells. And our cells are not generalists ; they ' re specialists. Like humans at work, they specialize into thousands of different cellular careers, or cell types. In fact, each of the cell types in our body could probably give a lively TED Talk about what it does at work. But as scientists, we don ' t even know today how many cell types there are, and we don ' t know what the titles of most of those talks would be. Now, we know many important things about cell types. They can differ dramatically in size and shape. One will respond to a molecule that the other doesn ' t respond to, they ' ll make different molecules. But science has largely been reaching these insights in an ad hoc way, one cell type at a time, one molecule at a time. We wanted to make it possible to learn all of this quickly and systematically. Now, until recently, it was the case that if you wanted to inventory all of the molecules in a part of the brain or any organ, you had to first grind it up into a kind of cellular smoothie. But that ' s a problem. As soon as you ' ve ground up the cells, you can only study the contents of the average cell — not the individual cells. Imagine if you were trying to understand how a big city like New York works, but you could only do so by reviewing some statistics about the average resident of New York. Of course, you wouldn ' t learn very much, because everything that ' s interesting and important and exciting is in all the diversity and the specializations. And the same thing is true of our cells. And we wanted to make it possible to study the brain not as a cellular smoothie but as a cellular fruit salad, in which one could generate data about and learn from each individual piece of fruit. So we developed a technology for doing that. You ' re about to see a movie of it. Here we ' re packaging tens of thousands of individual cells, each into its own tiny water droplet for its own molecular analysis. When a cell lands in a droplet, it ' s greeted by a tiny bead, and that bead delivers millions of DNA bar code molecules. And each bead delivers a different bar code sequence to a different cell. We incorporate the DNA bar codes into each cell ' s RNA molecules. Those are the molecular transcripts it ' s making of the specific genes that it ' s using to do its job. And then we sequence billions of these combined molecules and use the sequences to tell us which cell and which gene every molecule came from. We call this approach"Drop-seq,"because we use droplets to separate the cells for analysis, and we use DNA sequences to tag and inventory and keep track of everything. And now, whenever we do an experiment, we analyze tens of thousands of individual cells. And today in this area of science, the challenge is increasingly how to learn as much as we can as quickly as we can from these vast data sets. When we were developing Drop-seq, people used to tell us,"Oh, this is going to make you guys the go-to for every major brain project."That ' s not how we saw it. Science is best when everyone is generating lots of exciting data. So we wrote a 25-page instruction book, with which any scientist could build their own Drop-seq system from scratch. And that instruction book has been downloaded from our lab website 50, 000 times in the past two years. We wrote software that any scientist could use to analyze the data from Drop-seq experiments, and that software is also free, and it ' s been downloaded from our website 30, 000 times in the past two years. And hundreds of labs have written us about discoveries that they ' ve made using this approach. Today, this technology is being used to make a human cell atlas. It will be an atlas of all of the cell types in the human body and the specific genes that each cell type uses to do its job. Now I want to tell you about a second challenge that we face in trying to turn the brain into a big data problem. And that challenge is that we ' d like to learn from the brains of hundreds of thousands of living people. But our brains are not physically accessible while we ' re living. But how can we discover molecular factors if we can ' t hold the molecules? An answer comes from the fact that the most informative molecules, proteins, are encoded in our DNA, which has the recipes our cells follow to make all of our proteins. And these recipes vary from person to person to person in ways that cause the proteins to vary from person to person in their precise sequence and in how much each cell type makes of each protein. It ' s all encoded in our DNA, and it ' s all genetics, but it ' s not the genetics that we learned about in school. Do you remember big B, little b? If you inherit big B, you get brown eyes? It ' s simple. Very few traits are that simple. Even eye color is shaped by much more than a single pigment molecule. And something as complex as the function of our brains is shaped by the interaction of thousands of genes.

And each of these genes varies meaningfully from person to person to person, and each of us is a unique combination of that variation. It ' s a big data opportunity. And today, it ' s increasingly possible to make progress on a scale that was never possible before. People are contributing to genetic studies in record numbers, and scientists around the world are sharing the data with one another to speed progress. I want to tell you a short story about a discovery we recently made about the genetics of schizophrenia. It was made possible by 50, 000 people from 30 countries, who contributed their DNA to genetic research on schizophrenia. It had been known for several years that the human genome ' s largest influence on risk of schizophrenia comes from a part of the genome that encodes many of the molecules in our immune system. But it wasn ' t clear which gene was responsible. A scientist in my lab developed a new way to analyze DNA with computers, and he discovered something very surprising. He found that a gene called"complement component 4"— it ' s called"C4"for short — comes in dozens of different forms in different people ' s genomes, and these different forms make different amounts of C4 protein in our brains. And he found that the more C4 protein our genes make, the greater our risk for schizophrenia. Now, C4 is still just one risk factor in a complex system. This isn ' t big B, but it ' s an insight about a molecule that matters. Complement proteins like C4 were known for a long time for their roles in the immune system, where they act as a kind of molecular Post-it note that says,"Eat me."And that Post-it note gets put on lots of debris and dead cells in our bodies and invites immune cells to eliminate them. But two colleagues of mine found that the C4 Post-it note also gets put on synapses in the brain and prompts their elimination. Now, the creation and elimination of synapses is a normal part of human development and learning. Our brains create and eliminate synapses all the time. But our genetic results suggest that in schizophrenia, the elimination process may go into overdrive. Scientists at many drug companies tell me they ' re excited about this discovery, because they ' ve been working on complement proteins for years in the immune system, and they ' ve learned a lot about how they work. They ' ve even developed molecules that interfere with complement proteins, and they ' re starting to test them in the brain as well as the immune system. It ' s potentially a path toward a drug that might address a root cause rather than an individual symptom, and we hope very much that this work by many scientists over many years will be successful. But C4 is just one example of the potential for data-driven scientific approaches to open new fronts on medical problems that are centuries old. There are hundreds of places in our genomes that shape risk for brain illnesses, and any one of them could lead us to the next molecular insight about a molecule that matters. And there are hundreds of cell types that use these genes in different combinations. As we and other scientists work to generate the rest of the data that ' s needed and to learn all that we can from that data, we hope to open many more new fronts. Genetics and single-cell analysis are just two ways of trying to turn the brain into a big data problem. There is so much more we can do. Scientists in my lab are creating a technology for quickly mapping the synaptic connections in the brain to tell which neurons are talking to which other neurons and how that conversation changes throughout life and during illness. And we ' re developing a way to test in a single tube how cells with hundreds of different people ' s genomes respond differently to the same stimulus. These projects bring together people with diverse backgrounds and training and interests — biology, computers, chemistry, math, statistics, engineering. But the scientific possibilities rally people with diverse interests into working intensely together. What ' s the future that we could hope to create? Consider cancer. We ' ve moved from an era of ignorance about what causes cancer, in which cancer was commonly ascribed to personal psychological characteristics, to a modern molecular understanding of the true biological causes of cancer. That understanding today leads to innovative medicine after innovative medicine, and although there ’ s still so much work to do, we ’ re already surrounded by people who have been cured of cancers that were considered untreatable a generation ago. And millions of cancer survivors like my sister find themselves with years of life that they didn ’ t take for granted and new opportunities for work and joy and human connection. That is the future that we are determined to create around mental illness — one of real understanding and empathy and limitless possibility. Thank you.

Text Sample 490: NEG Dim 3 – 2017 – Score -1.00 – 2017/t000885.txt

I don ’ t come to you today as an expert. I come to you as someone who has been really interested in how I get better at what I do and how we all do. I think it ’ s not just how good you are now, I think it ’ s how good you ’ re going to be that really matters. I was visiting this birth center in the north of India. I was watching the birth attendants, and I realized I was witnessing in them an extreme form of this very struggle, which is how people improve in the face of complexity — or don ’ t. The women here are delivering in a region where the typical birth center has a one-in-20 death rate for the babies, and the moms are dying at a rate ten times higher than they do elsewhere. Now, we ’ ve known the critical practices that stop the big killers in birth for decades, and the thing about it is that even in this place — in this place especially, the simplest things are not simple. We know for example you should wash hands and put on clean gloves, but here, the tap is in another room, and they don ’ t have clean gloves. To reuse their gloves, they wash them in this basin of dilute bleach, but you can see there ’ s still blood on the gloves from the last delivery. Ten percent of babies are born with difficulty breathing everywhere. We know what to do. You dry the baby with a clean cloth to stimulate them to breathe. If they don ’ t start to breathe, you suction out their airways. And if that doesn ’ t work, you give them breaths with the baby mask. But these are skills that they ’ ve learned mostly from textbooks, and that baby mask is broken. In this one disturbing image for me is a picture that brings home just how dire the situation is. This is a baby 10 minutes after birth, and he ’ s alive, but only just. No clean cloth, has not been dried, not warming skin to skin, an unsterile clamp across the cord. He ’ s an infection waiting to happen, and he ’ s losing his temperature by the minute. Successful child delivery requires a successful team of people. A whole team has to be skilled and coordinated ; the nurses who do the deliveries in a place like this, the doctor who backs them up, the supply clerk who ’ s responsible for 22 critical drugs and supplies being in stock and at the bedside, the medical officer in charge, responsible for the quality of the whole facility. The thing is they are all experienced professionals. I didn ’ t meet anybody who hadn ’ t been part of thousands of deliveries. But against the complexities that they face, they seem to be at their limits. They were not getting better anymore. It ’ s how good you ’ re going to be that really matters. It presses on a fundamental question. How do professionals get better at what they do? How do they get great? And there are two views about this. One is the traditional pedagogical view. That is that you go to school, you study, you practice, you learn, you graduate, and then you go out into the world and you make your way on your own. A professional is someone who is capable of managing their own improvement. That is the approach that virtually all professionals have learned by. That ’ s how doctors learn, that ’ s how lawyers do, scientists... musicians.

And the thing is, it works. Consider for example legendary Juilliard violin instructor Dorothy DeLay. She trained an amazing roster of violin virtuosos : Midori, Sarah Chang, Itzhak Perlman. Each of them came to her as young talents, and they worked with her over years. What she worked on most, she said, was inculcating in them habits of thinking and of learning so that they could make their way in the world without her when they were done. Now, the contrasting view comes out of sports. And they say"You are never done, everybody needs a coach."Everyone. The greatest in the world needs a coach. So I tried to think about this as a surgeon. Pay someone to come into my operating room, observe me and critique me. That seems absurd. Expertise means not needing to be coached. So then which view is right? I learned that coaching came into sports as a very American idea. In 1875, Harvard and Yale played one of the very first American- rules football games. Yale hired a head coach ; Harvard did not. The results? Over the next three decades, Harvard won just four times. Harvard hired a coach. And it became the way that sports works. But is it necessary then? Does it transfer into other fields? I decided to ask, of all people, Itzhak Perlman. He had trained the Dorothy DeLay way and became arguably the greatest violinist of his generation. One of the beautiful things about getting to write for"The New Yorker"is I call people up, and they return my phone calls. And Perlman returned my phone call. So we ended up having an almost two-hour conversation about how he got to where he got in his career. And I asked him, I said,"Why don ' t violinists have coaches?"And he said,"I don ' t know, but I always had a coach.""You always had a coach?""Oh yeah, my wife, Toby."They had graduated together from Juilliard, and she had given up her job as a concert violinist to be his coach, sitting in the audience, observing him and giving him feedback."Itzhak, in that middle section, you know you sounded a little bit mechanical. What can you differently next time?"It was crucial to everything he became, he said. Turns out there are numerous problems in making it on your own. You don ' t recognize the issues that are standing in your way or if you do, you don ' t necessarily know how to fix them. And the result is that somewhere along the way, you stop improving. And I thought about that, and I realized that was exactly what had happened to me as a surgeon. I ' d entered practice in 2003, and for the first several years, it was just this steady, upward improvement in my learning curve. I watched my complication rates drop from one year to the next. And after about five years, they leveled out. And a few more years after that, I realized I wasn ' t getting any better anymore. And I thought :"Is this as good as I ' m going to get?"So I thought a little more and I said..."OK, I ' ll try a coach."So I asked a former professor of mine who had retired, his name is Bob Osteen, and he agreed to come to my operating room and observe me. The case — I remember that first case. It went beautifully. I didn ' t think there would be anything much he ' d have to say when we were done. Instead, he had a whole page dense with notes."Just small things,"he said. But it ' s the small things that matter."Did you notice that the light had swung out of the wound during the case? You spent about half an hour just operating off the light from reflected surfaces.""Another thing I noticed,"he said,"Your elbow goes up in the air every once in a while. That means you ' re not in full control. A surgeon ' s elbows should be down at their sides resting comfortably. So that means if you feel your elbow going in the air, you should get a different instrument, or just move your feet."It was a whole other level of awareness. And I had to think, you know, there was something fundamentally profound about this. He was describing what great coaches do, and what they do is they are your external eyes and ears, providing a more accurate picture of your reality. They ' re recognizing the fundamentals. They ' re breaking your actions down and then helping you build them back up again. After two months of coaching, I felt myself getting better again. And after a year, I saw my complications drop down even further. It was painful. I didn ' t like being observed, and at times I didn ' t want to have to work on things. I also felt there were periods where I would get worse before I got better. But it made me realize that the coaches were onto something profoundly important. In my other work, I lead a health systems innovation center called Ariadne Labs, where we work on problems in the delivery of health care, including global childbirth. As part of it, we had worked with the World Health Organization to devise a safe childbirth checklist. It lays out the fundamentals. It breaks down the fundamentals — the critical actions a team needs to go through when a woman comes in in labor, when she ' s ready to push, when the baby is out, and then when the mom and baby are ready to go home. And we knew that just handing out a checklist wasn ' t going to change very much, and even just teaching it in the classroom wasn ' t necessarily going to be enough to get people to make the changes that you needed to bring it alive. And I thought on my experience and said,"What if we tried coaching? What if we tried coaching at a massive scale?"We found some incredible partners, including the government of India, and we ran a trial there in 120 birth centers. In Uttar Pradesh, in India ' s largest state. Half of the centers basically we just observed, but the other half got visits from coaches. We trained an army of doctors and nurses like this one who learned to observe the care and also the managers and then help them build on their strengths and address their weaknesses. One of the skills for example they had to work on with people — turned out to be fundamentally important — was communication. Getting the nurses to practice speaking up when the baby mask is broken or the gloves are not in stock or someone ' s not washing their hands. And then getting others, including the managers, to practice listening. This small army of coaches ended up coaching 400 nurses and other birth attendants, and 100 physicians and managers. We tracked the results across 160, 000 births. The results... in the control group you had — and these are the ones who did not get coaching — they delivered on only one-third of 18 basic practices that we were measuring. And most important was over the course of the years of study, we saw no improvement over time. The other folks got four months of coaching and then it tapered off over eight months, and we saw them increase to greater than two-thirds of the practices being delivered. It works. We could see the improvement in quality, and you could see it happen across a whole range of centers that suggested that coaching could be a whole line of way that we bring value to what we do. You can imagine the whole job category that could reach out in the world and that millions of people could fulfill. We were clearly at the beginning of it, though, because there was still a distance to go. You have to put all of the checklist together to achieve the substantial reductions in mortality. But we began seeing the first places that were getting there, and this center was one of them because coaching helped them learn to execute on the fundamentals. And you could see it here. This is a 23-year-old woman who had come in by ambulance, in labor with her third child. She broke her water in the triage area, so they brought her directly to the labor and delivery room, and then they ran through their checks. I put the time stamp on here so you could see how quickly all of this happens and how much more complicated that makes things. Within four minutes, they had taken the blood pressure, measured her pulse and also measured the heart rate of the baby. That meant that the blood pressure cuff and the fetal Doppler monitor, they were all there, and the nurse knew how to use them. The team was skilled and coordinated. The mom was doing great, the baby ' s heart rate was 143, which is normal. Eight minutes later, the intensity of the contractions picked up, so the nurse washed her hands, put on clean gloves, examined her and found that her cervix was fully dilated. The baby was ready to come. She then went straight over to do her next set of checks. All of the equipment, she worked her way through and made sure she had everything she needed at the bedside. The baby mask was there, the **sterile** towel, the **sterile** equipment that you needed. And then three minutes later, one push and that baby was out. I was watching this delivery, and suddenly I realized that the mood in that room had changed. The nurse was looking at the community health worker who had come in with the woman because that baby did not seem to be alive. She was blue and floppy and not breathing. She would be one of that one-in-20. But the nurse kept going with her checkpoints. She dried that baby with a clean towel. And after a minute, when that didn ’ t stimulate that baby, she ran to get the baby mask and the other one went to get the suction. She didn ’ t have a mechanical suction because you could count on electricity, so she used a mouth suction, and within 20 seconds, she was clearing out that little girl ’ s airways. And she got back a green, thick liquid, and within a minute of being able to do that and suctioning out over and over, that baby started to breathe. Another minute and that baby was crying. And five minutes after that, she was pink and warming on her mother ’ s chest, and that mother reached out to grab that nurse ’ s hand, and they could all breathe. I saw a team transformed because of coaching. And I saw at least one life saved because of it. We followed up with that mother a few months later. Mom and baby were doing great. The baby ’ s name is Anshika. It means”beautiful.”And she is what ’ s possible when we really understand how people get better at what they do. Thank you.

Text Sample 491: NEG Dim 3 – 2017 – Score -1.00 – 2017/t000749.txt
[A provocation from Danny Hillis :] [It ’ s time to start talking about engineering our climate] What if there was a way to build a thermostat that allowed you to turn down the temperature of the earth anytime you wanted? Now, you would think if somebody had a plausible idea about how to do that, everybody would be very excited about it, and there would be lots of research on how to do it. But in fact, a lot of people do understand how to do that. But there ’ s not much support for research in this area. And I think part of it is because there are some real misunderstandings about it. So I ’ m not going to try to convince you today that this is a good idea. But I am going to try to get your curiosity going about it and clear up some of the misunderstandings. So, the basic idea of solar geoengineering is that we can cool things down just by reflecting a little bit more sunlight back into space. And ideas about how to do this have been around literally for decades. Clouds are a great way to do that, these low-lying clouds. Everybody knows it ’ s cooler under a cloud. I like this cloud because it has exactly the same water content as the transparent air around it. And it just shows that even a little bit of a change in the flow of the air can cause a cloud to form. We make artificial clouds all the time. These are contrails, which are artificial water clouds that are made by the passing of a jet engine. And so, we ’ re already changing the clouds on earth. By accident. Or, if you like to believe it, by supersecret government conspiracy. But we are already doing this quite a lot. This is a NASA picture of shipping lanes. Passing ships actually cause clouds to form, and this is a big enough effect that it actually helps reduce global warming already by about a degree. So we already are doing solar engineering. There ’ s lots of ideas about how to do this. People have looked at everything, from building giant parasols out into space to fizzing bubble waters in the ocean. And some of these are actually very plausible ideas. One that was published recently by David Keith at Harvard is to take chalk and put dust up

into the stratosphere, where it reflects off sunlight. And that ’ s a really neat idea, because chalk is one of the most common minerals on earth, and it ’ s very safe — it ’ s so safe, we put it into baby food. And basically, if you throw chalk up into the stratosphere, it comes down in a couple of years all by itself, dissolved in rainwater. Now, before you start worrying about all this chalk in your rainwater, let me explain to you how little of it it actually takes. And that turns out to be very easy to calculate. This is a back-of-the-envelope calculation I made. I assure you, people have done much more careful calculations, and it comes out with the same answer, which is that you have to put chalk up at the rate of about 10 teragrams a year to undo the effects of the CO2 that we ’ ve already done — just in terms of temperature, not all the effects, but the temperature. So what does that look like? I can ’ t visualize 10 teragrams per year. So I asked the Cambridge Fire Department and Taylor Milsal to lend me a hand. This is a hose pumping water at 10 teragrams a year. And that is how much you would have to pump into the stratosphere to cool the earth back down to pre-industrial levels. And it ’ s amazingly little ; it ’ s like one hose for the entire earth. Now of course, you wouldn ’ t really use a hose, you ’ d fly it up in airplanes or something like that. But it ’ s so little, it would be like putting a handful of chalk into every Olympic swimming pool full of rain. It ’ s almost nothing. So why don ’ t people like this idea? Why isn ’ t it taken more seriously? And there are some very good reasons for that. A lot of people really don ’ t think we should be talking about this at all. And, in fact, I have some very good friends in the audience who I respect a lot, who really don ’ t think I should be talking about this. And the reason is that they ’ re concerned that if people imagine there ’ s some easy way out, that we won ’ t give up our addiction to fossil fuels. And I do worry about that. I think it ’ s actually a serious problem. But there ’ s also, I think, a deeper problem, which is : nobody likes the idea of messing with the entire earth — I certainly don ’ t. I love this planet, I really do. And I don ’ t want to mess with it. But we ’ re already changing our atmosphere, we ’ re already messing with it. And so I think it makes sense for us to look for ways to mitigate that impact. And we need to do research to do that. We need to understand the science behind that. I ’ ve noticed that there ’ s a theme that ’ s kind of developed at TED, which is kind of,”fear versus hope,”or”creativity versus caution.”And of course, we need both of those. So there aren ’ t any silver bullets. This is certainly not a silver bullet. But we need science to tell us what our options are ; that informs both our creativity and our caution. So I am an optimist about our future selves, but I ’ m not an optimist because I think our problems are small. I ’ m an optimist because I think our capacity to deal with our problems is much greater than we imagine. Thank you very much. This talk sparked a lot of controversy at TED2017, and we encourage you to look at discussions online to see other points of view.

Text Sample 492: NEG Dim 3 – 2016 – Score -1.00 – 2016/t001291.txt

Hello, everybody. I brought with me today a baby diaper. You ’ ll see why in a second. Baby diapers have interesting properties. They can swell enormously when you add water to them, an experiment done by millions of kids every day. But the reason why is that they ’ re designed in a very clever way. They ’ re made out of a thing called a swellable material. It ’ s a special kind of material that, when you add water, it will swell up enormously, maybe a thousand times in volume. And this is a very useful, industrial kind of polymer. But what we ’ re trying to do in my group at MIT is to figure out if we can do something similar to the brain. Can we make it bigger, big enough that you can peer inside and see all the tiny building blocks, the biomolecules, how they ’ re organized in three dimensions, the structure, the ground truth structure of the brain, if you will? If we could get that, maybe we could have a better understanding of how the brain is organized to yield thoughts and emotions and actions and sensations. Maybe we could try to pinpoint the exact changes in the brain that result in diseases, diseases like Alzheimer ’ s and epilepsy and Parkinson ’ s, for which there are few treatments, much less cures, and for which, very often, we don ’ t know the cause or the origins and what ’ s really causing them to occur. Now, our group at MIT is trying to take a different point of view from the way neuroscience has been done over the last hundred years. We ’ re designers. We ’ re inventors. We ’ re trying to figure out how to build technologies that let us look at and repair the brain. And the reason is, the brain is incredibly, incredibly complicated. So what we ’ ve learned over the first century of neuroscience is that the brain is a very complicated network, made out of very specialized cells called neurons with very complex geometries, and electrical currents will flow through these complexly shaped neurons. Furthermore, neurons are connected in networks. They ’ re connected by little junctions called synapses that exchange chemicals and allow the neurons to talk to each other. The density of the brain is incredible. In a cubic millimeter of your brain, there are about 100, 000 of these neurons and maybe a billion of those connections. But it ’ s worse. So, if you could zoom in to a neuron, and, of course, this is just our artist ’ s rendition of it. What you would see are thousands and thousands of kinds of biomolecules, little nanoscale machines organized in complex, 3D patterns, and together they mediate those electrical pulses, those chemical exchanges that allow neurons to work together to generate things like thoughts and feelings and so forth. Now, we don ’ t know how the neurons in the brain are organized to form networks, and we don ’ t know how the biomolecules are organized within neurons to form these complex, organized machines. If we really want to understand this, we ’ re going to need new technologies. But if we could get such maps, if we could look at the organization of molecules and neurons and neurons and networks, maybe we could really understand how the brain conducts information from sensory regions, mixes it with emotion and feeling, and generates our decisions and actions. Maybe we could pinpoint the exact set of molecular changes that occur in a brain disorder. And once we know how those molecules have changed, whether they ’ ve increased in number or changed in pattern, we could use those as targets for new drugs, for new ways of delivering energy into the brain in order to repair the brain computations that are afflicted in patients who suffer from brain disorders. We ’ ve all seen lots of different technologies over the last century to try to confront this. I think we ’ ve all seen brain scans taken using MRI machines. These, of course, have the great power that they are noninvasive, they can be used on living human subjects. But also, they ’ re spatially crude. Each of these blobs that you see, or voxels, as they ’ re called, can contain millions and millions of neurons. So it ’ s not at the level of resolution where it can pinpoint the molecular changes that occur or the changes in the wiring of these networks that contributes to our ability to be conscious and powerful beings. At the other extreme, you have microscopes. Microscopes, of course, will use light to look at little tiny things. For centuries, they ’ ve been used to look at things like bacteria. For neuroscience, microscopes are actually how neurons were discovered in the first place, about 130 years ago. But light is fundamentally limited. You can ’ t see individual molecules with a regular old microscope. You can ’ t look at these tiny connections. So if we want to make our ability to see the brain more powerful, to get down to the ground truth structure, we ’ re going to need to have even better technologies. My group, a couple years ago, started thinking

: Why don ’ t we do the opposite? If it ’ s so darn complicated to zoom in to the brain, why can ’ t we make the brain bigger? It initially started with two grad students in my group, Fei Chen and Paul Tillberg. Now many others in my group are helping with this process. We decided to try to figure out if we could take polymers, like the stuff in the baby diaper, and install it physically within the brain. If we could do it just right, and you add water, you can potentially blow the brain up to where you could distinguish those tiny biomolecules from each other. You would see those connections and get maps of the brain. This could potentially be quite dramatic. We brought a little demo here. We got some purified baby diaper material. It ’ s much easier just to buy it off the Internet than to extract the few grains that actually occur in these diapers. I ’ m going to put just one teaspoon here of this purified polymer. And here we have some water. What we ’ re going to do is see if this teaspoon of the baby diaper material can increase in size. You ’ re going to see it increase in volume by about a thousandfold before your very eyes. I could pour much more of this in there, but I think you ’ ve got the idea that this is a very, very interesting molecule, and if can use it in the right way, we might be able to really zoom in on the brain in a way that you can ’ t do with past technologies. OK. So a little bit of chemistry now. What ’ s going on in the baby diaper polymer? If you could zoom in, it might look something like what you see on the screen. Polymers are chains of atoms arranged in long, thin lines. The chains are very tiny, about the width of a biomolecule, and these polymers are really dense. They ’ re separated by distances that are around the size of a biomolecule. This is very good because we could potentially move everything apart in the brain. If we add water, what will happen is, this swellable material is going to absorb the water, the polymer chains will move apart from each other, and the entire material is going to become bigger. And because these chains are so tiny and spaced by biomolecular distances, we could potentially blow up the brain and make it big enough to see. Here ’ s the mystery, then : How do we actually make these polymer chains inside the brain so we can move all the biomolecules apart? If we could do that, maybe we could get ground truth maps of the brain. We could look at the wiring. We can peer inside and see the molecules within. To explain this, we made some animations where we actually look at, in these artist renderings, what biomolecules might look like and how we might separate them. Step one : what we ’ d have to do, first of all, is attach every biomolecule, shown in brown here, to a little anchor, a little handle. We need to pull the molecules of the brain apart from each other, and to do that, we need to have a little handle that allows those polymers to bind to them and to exert their force. Now, if you just take baby diaper polymer and dump it on the brain, obviously, it ’ s going to sit there on top. So we need to find a way to make the polymers inside. And this is where we ’ re really lucky. It turns out, you can get the building blocks, monomers, as they ’ re called, and if you let them go into the brain and then trigger the chemical reactions, you can get them to form those long chains, right there inside the brain tissue. They ’ re going to wind their way around biomolecules and between biomolecules, forming those complex webs that will allow you, eventually, to pull apart the molecules from each other. And every time one of those little handles is around, the polymer will bind to the handle, and that ’ s exactly what we need in order to pull the molecules apart from each other. All right, the moment of truth. We have to treat this specimen with a chemical to kind of loosen up all the molecules from each other, and then, when we add water, that swellable material is going to start absorbing the water, the polymer chains will move apart, but now, the biomolecules will come along for the ride. And much like drawing a picture on a balloon, and then you blow up the balloon, the image is the same, but the ink particles have moved away from each other. And that ' s what we ' ve been able to do now, but in three dimensions. There ' s one last trick. As you can see here, we ' ve color-coded all the biomolecules brown. That ' s because they all kind of look the same. Biomolecules are made out of the same atoms, but just in different orders. So we need one last thing in order to make them visible. We have to bring in little tags, with glowing dyes that will distinguish them. So one kind of biomolecule might get a blue color. Another kind of biomolecule might get a red color. And so forth. And that ' s the final step. Now we can look at something like a brain and look at the individual molecules, because we ' ve moved them far apart enough from each other that we can tell them apart. So the hope here is that we can make the invisible visible. We can turn things that might seem small and obscure and blow them up until they ' re like constellations of information about life. Here ' s an actual video of what it might look like. We have here a little brain in a dish — a little piece of a brain, actually. We ' ve infused the polymer in, and now we ' re adding water. What you ' ll see is that, right before your eyes — this video is sped up about sixtyfold — this little piece of brain tissue is going to grow. It can increase by a hundredfold or even more in volume. And the cool part is, because those polymers are so tiny, we ' re separating biomolecules evenly from each other. It ' s a smooth expansion. We ' re not losing the configuration of the information. We ' re just making it easier to see. So now we can take actual brain circuitry — here ' s a piece of the brain involved with, for example, memory — and we can zoom in. We can start to actually look at how circuits are configured. Maybe someday we could read out a memory. Maybe we could actually look at how circuits are configured to process emotions, how the actual wiring of our brain is organized in order to make us who we are. And of course, we can pinpoint, hopefully, the actual problems in the brain at a molecular level. What if we could actually look into cells in the brain and figure out, wow, here are the 17 molecules that have altered in this brain tissue that has been undergoing epilepsy or changing in Parkinson ' s disease or otherwise being altered? If we get that systematic list of things that are going wrong, those become our therapeutic targets. We can build drugs that bind those. We can maybe aim energy at different parts of the brain in order to help people with Parkinson ' s or epilepsy or other conditions that affect over a billion people around the world. Now, something interesting has been happening. It turns out that throughout biomedicine, there are other problems that expansion might help with. This is an actual biopsy from a human breast cancer patient. It turns out that if you look at cancers, if you look at the immune system, if you look at aging, if you look at development — all these processes are involving large-scale biological systems. But of course, the problems begin with those little nanoscale molecules, the machines that make the cells and the organs in our body tick. So what we ' re trying to do now is to figure out if we can actually use this technology to map the building blocks of life in a wide variety of diseases. Can we actually pinpoint the molecular changes in a tumor so that we can actually go after it in a smart way and deliver drugs that might wipe out exactly the cells that we want to? You know, a lot of medicine is very high risk. Sometimes, it ' s even guesswork. My hope is we can actually turn what might be a high-risk moon shot into something that ' s more reliable. If you think about the original moon shot, where they actually landed on the moon, it was based on solid science. We understood gravity ; we understood aerodynamics. We knew how to build rockets. The science risk was under control. It was still a great, great feat of engineering. But in medicine, we don ' t necessarily have all the laws. Do we have all the laws that are **analogous** to gravity, that are **analogous** to aerodynamics? I would argue that with technologies like the kinds I ' m talking about today, maybe we can actually derive those. We can map the patterns that occur in living systems, and figure out how to overcome the diseases that plague us. You know, my wife and I have two young kids, and one of my hopes as a bioengineer is to make life better for them than it currently is for us. And my hope is, if we can turn biology and medicine from these high-risk endeavors that are governed by chance and luck, and make them things that we win by skill and hard work, then that would be a great advance. Thank you very much.

Text Sample 493: NEG Dim 3 – 2016 – Score -1.00 – 2016/t001263.txt

So, has everybody heard of CRISPR? I would be shocked if you hadn ’ t. This is a technology â it ’ s for genome editing â and it ’ s so versatile and so controversial that it ’ s sparking all sorts of really interesting conversations. Should we bring back the woolly mammoth? Should we edit a human embryo? And my personal favorite : How can we justify wiping out an entire species that we consider harmful to humans off the face of the Earth, using this technology? This type of science is moving much faster than the regulatory mechanisms that govern it. And so, for the past six years, I ’ ve made it my personal mission to make sure that as many people as possible understand these types of technologies and their implications. Now, CRISPR has been the subject of a huge media hype, and the words that are used most often are”easy”and”cheap.”So what I want to do is drill down a little bit deeper and look into some of the myths and the realities around CRISPR. If you ’ re trying to CRISPR a genome, the first thing that you have to do is damage the DNA. The damage comes in the form of a double-strand break through the double helix. And then the cellular repair processes kick in, and then we convince those repair processes to make the edit that we want, and not a natural edit. That ’ s how it works. It ’ s a two-part system. You ’ ve got a Cas9 protein and something called a guide RNA. I like to think of it as a guided missile. So the Cas9 â I love to anthropomorphize â so the Cas9 is kind of this Pac-Man thing that wants to chew DNA, and the guide RNA is the leash that ’ s keeping it out of the genome until it finds the exact spot where it matches. And the combination of those two is called CRISPR. It ’ s a system that we stole from an ancient, ancient bacterial immune system. The part that ’ s amazing about it is that the guide RNA, only 20 letters of it, are what target the system. This is really easy to design, and it ’ s really cheap to buy. So that ’ s the part that is modular in the system ; everything else stays the same. This makes it a remarkably easy and powerful system to use. The guide RNA and the Cas9 protein complex together go bouncing along the genome, and when they find a spot where the guide RNA matches, then it inserts between the two strands of the double helix, it rips them apart, that triggers the Cas9 protein to cut, and all of a sudden, you ’ ve got a cell that ’ s in total panic because now it ’ s got a piece of DNA that ’ s broken. What does it do? It calls its first responders. There are two major repair pathways. The first just takes the DNA and shoves the two pieces back together. This isn ’ t a very efficient system, because what happens is sometimes a base drops out or a base is added. It ’ s an OK way to maybe, like, knock out a gene, but it ’ s not the way that we really want to do genome editing. The second repair pathway is a lot more interesting. In this repair pathway, it takes a homologous piece of DNA. And now mind you, in a diploid organism like people, we ’ ve got one copy of our genome from our mom and one from our dad, so if one gets damaged, it can use the other chromosome to repair it. So that ’ s where this comes from. The repair is made, and now the genome is safe again. The way that we can hijack this is we can feed it a false piece of DNA, a piece that has homology on both ends but is different in the middle. So now, you can put whatever you want in the center and the cell gets fooled. So you can change a letter, you can take letters out, but most importantly, you can stuff new DNA in, kind of like a Trojan horse. CRISPR is going to be amazing, in terms of the number of different scientific advances that it ’ s going to catalyze. The thing that ’ s special about it is this modular targeting system. I mean, we ’ ve been shoving DNA into organisms for years, right? But because of the modular targeting system, we can actually put it exactly where we want it. The thing is that there ’ s a lot of talk about it being cheap and it being easy. And I run a community lab. I ’ m starting to get emails from people that say stuff like,”Hey, can I come to your open night and, like, maybe use CRISPR and engineer my genome?”Like, seriously. I ’ m,”No, you can ’ t.””But I ’ ve heard it ’ s cheap. I ’ ve heard it ’ s easy.”We ’ re going to explore that a little bit. So, how cheap is it? Yeah, it is cheap in comparison. It ’ s going to take the cost of the average materials for an experiment from thousands of dollars to hundreds of dollars, and it cuts the time a lot, too. It can cut it from weeks to days. That ’ s great. You still need a professional lab to do the work in ; you ’ re not going to do anything meaningful outside of a professional lab. I mean, don ’ t listen to anyone who says you can do this sort of stuff on your kitchen table. It ’ s really not easy to do this kind of work. Not to mention, there ’ s a patent battle going on, so even if you do invent something, the Broad Institute and UC Berkeley are in this incredible patent battle. It ’ s really fascinating to watch it happen, because they ’ re accusing each other of fraudulent claims and then they ’ ve got people saying,”Oh, well, I signed my notebook here or there.”This isn ’ t going to be settled for years. And when it is, you can bet you ’ re going to pay someone a really hefty licensing fee in order to use this stuff. So, is it really cheap? Well, it ’ s cheap if you ’ re doing basic research and you ’ ve got a lab. How about easy? Let ’ s look at that claim. The devil is always in the details. We don ’ t really know that much about cells. They ’ re still kind of black boxes. For example, we don ’ t know why some guide RNAs work really well and some guide RNAs don ’ t. We don ’ t know why some cells want to do one repair pathway and some cells would rather do the other. And besides that, there ’ s the whole problem of getting the system into the cell in the first place. In a petri dish, that ’ s not that hard, but if you ’ re trying to do it on a whole organism, it gets really tricky. It ’ s OK if you use something like blood or bone marrow â those are the targets of a lot of research now. There was a great story of some little girl who they saved from leukemia by taking the blood out, editing it, and putting it back with a precursor of CRISPR. And this is a line of research that people are going to do. But right now, if you want to get into the whole body, you ’ re probably going to have to use a virus. So you take the virus, you put the CRISPR into it, you let the virus infect the cell. But now you ’ ve got this virus in there, and we don ’ t know what the long-term effects of that are. Plus, CRISPR has some off-target effects, a very small percentage, but they ’ re still there. What ’ s going to happen over time with that? These are not trivial questions, and there are scientists that are trying to solve them, and they will eventually, hopefully, be solved. But it ain ’ t plug- and-play, not by a long shot. So : Is it really easy? Well, if you spend a few years working it out in your particular system, yes, it is. Now the other thing is, we don ’ t really know that much about how to make a particular thing happen by changing particular spots in the genome. We ’ re a long way away from figuring out how to give a pig wings, for example. Or even an extra leg â I ’ d settle for an extra leg. That would be kind of cool, right? But what is happening is that CRISPR is being used by thousands and thousands of scientists to do really, really important work, like making better models of diseases in animals, for example, or for taking pathways that produce valuable chemicals and getting them into industrial production in fermentation vats, or even doing really basic research on what genes do. This is the story of CRISPR we should be telling, and I don ’ t like it that the flashier aspects of it are drowning all of this out. Lots of scientists did a lot of work to make CRISPR happen, and what ’ s interesting to me is that these scientists are being supported by our society. Think about it. We ’ ve got an infrastructure that allows a certain percentage of people to spend all their time doing research. That makes us all the inventors of CRISPR, and I would say that makes us all the shepherds of CRISPR. We all have a responsibility. So I would urge you to really learn about these types of technologies, because, really, only in that way are we going to be able to guide the development of these technologies, the use of these technologies and make sure that, in the end, it ’ s a positive outcome â for both the planet and for us. Thanks.

Text Sample 494: NEG Dim 3 – 2016 – Score -1.00 – 2016/t001214.txt

Roughly 43, 000 years ago, a young cave bear died in the rolling hills on the northwest border of modern day Slovenia. A thousand years later, a mammoth died in southern Germany. A few centuries after that, a griffon vulture also died in the same vicinity. And we know almost nothing about how these animals met their deaths, but these different creatures dispersed across both time and space did share one remarkable fate. After their deaths, a bone from each of their skeletons was crafted by human hands into a flute. Think about that for a second. Imagine you ’ re a caveman, 40, 000 years ago. You ’ ve mastered fire. You ’ ve built simple tools for hunting. You ’ ve learned how to craft garments from animal skins to keep yourself warm in the winter. What would you choose to invent next? It seems preposterous that you would invent the flute, a tool that created useless vibrations in air molecules. But that is exactly what our ancestors did. Now this turns out to be surprisingly common in the history of innovation. Sometimes people invent things because they want to stay alive or feed their children or conquer the village next door. But just as often, new ideas come into the world simply because they ’ re fun. And here ’ s the really strange thing : many of those playful but seemingly frivolous inventions ended up sparking momentous transformations in science, in politics and society. Take what may be the most important invention of modern times : programmable computers. Now, the standard story is that computers descend from military technology, since many of the early computers were designed specifically to crack wartime codes or calculate rocket trajectories. But in fact, the origins of the modern computer are much more playful, even musical, than you might imagine. The idea behind the flute, of just pushing air through tubes to make a sound, was eventually modified to create the first organ more than 2, 000 years ago. Someone came up with the brilliant idea of triggering sounds by pressing small levers with our fingers, inventing the first musical keyboard. Now, keyboards evolved from organs to clavichords to harpsichords to the piano, until the middle of the 19th century, when a bunch of inventors finally hit on the idea of using a keyboard to trigger not sounds but letters. In fact, the very first typewriter was originally called”the writing harpsichord.”Flutes and music led to even more powerful breakthroughs. About a thousand years ago, at the height of the Islamic Renaissance, three brothers in Baghdad designed a device that was an automated organ. They called it”the instrument that plays itself.”Now, the instrument was basically a giant music box. The organ could be trained to play various songs by using instructions encoded by placing pins on a rotating cylinder. And if you wanted the machine to play a different song, you just swapped a new cylinder in with a different code on it. This instrument was the first of its kind. It was programmable. Now, conceptually, this was a massive leap forward. The whole idea of hardware and software becomes thinkable for the first time with this invention. And that incredibly powerful concept didn ' t come to us as an instrument of war or of conquest, or necessity at all. It came from the strange delight of watching a machine play music. In fact, the idea of programmable machines was exclusively kept alive by music for about 700 years. In the 1700s, music-making machines became the playthings of the Parisian elite. Showmen used the same coded cylinders to control the physical movements of what were called automata, an early kind of robot. One of the most famous of those robots was, you guessed it, an automated flute player designed by a brilliant French inventor named Jacques de Vaucanson. And as de Vaucanson was designing his robot musician, he had another idea. If you could program a machine to make pleasing sounds, why not program it to weave delightful patterns of color out of cloth? Instead of using the pins of the cylinder to represent musical notes, they would represent threads with different colors. If you wanted a new pattern for your fabric, you just programmed a new cylinder. This was the first programmable loom. Now, the cylinders were too expensive and time-consuming to make, but a half century later, another French inventor named Jacquard hit upon the brilliant idea of using paper-punched cards instead of metal cylinders. Paper turned out to be much cheaper and more flexible as a way of programming the device. That punch card system inspired Victorian inventor Charles Babbage to create his analytical engine, the first true programmable computer ever designed. And punch cards were used by computer programmers as late as the 1970s. So ask yourself this question : what really made the modern computer possible? Yes, the military involvement is an important part of the story, but inventing a computer also required other building blocks : music boxes, toy robot flute players, harpsichord keyboards, colorful patterns woven into fabric, and that ' s just a small part of the story. There ' s a long list of world-changing ideas and technologies that came out of play : public museums, rubber, probability theory, the insurance business and many more. Necessity isn ' t always the mother of invention. The playful state of mind is fundamentally exploratory, seeking out new possibilities in the world around us. And that seeking is why so many experiences that started with simple delight and **amusement** eventually led us to profound breakthroughs. Now, I think this has implications for how we teach kids in school and how we encourage innovation in our workspaces, but thinking about play and delight this way also helps us detect what ' s coming next. Think about it : if you were sitting there in 1750 trying to figure out the big changes coming to society in the 19th, the 20th centuries, automated machines, computers, artificial intelligence, a programmable flute entertaining the Parisian elite would have been as powerful a clue as anything else at the time. It seemed like an **amusement** at best, not useful in any serious way, but it turned out to be the beginning of a tech revolution that would change the world. You ' ll find the future wherever people are having the most fun.

Text Sample 495: NEG Dim 3 – 2019 – Score -2.00 – 2019/t001507.txt

This is a strange and wonderful brain, one that gives rise to an idea of a kind of alternative intelligence on this planet. This is a brain that is formed in a very strange body, one that has the equivalent of small satellite brains distributed throughout that body. How different is it from the human brain? Very different, so it seems, so much so that my colleagues and I are struggling to understand how that brain works. But what I can tell you for certain is that this brain is capable of some amazing things. So, who does this brain belong to? Well, join me for a little bit of diving into the ocean, where life began, and let ' s have a look. You may have seen some of this before, but we ' re behind a coral reef, and there ' s this rock out there, a lot of sand, fishes swimming around... And all of a sudden this octopus appears, and now it flashes white, inks in my face and jets away. In slow motion reverse, you see the ring develop around the eye, and then the pattern develops in the skin. And now watch the 3-D texture of the skin change to really create this beautiful, 3-D camouflage. So there are 25 million color organs called"chromatophores"in the skin, and all those bumps out there, which we call"papillae,"and they ' re all neurally controlled and can change instantaneously. I would argue that dynamic camouflage is a form of"intelligence."The level of complexity of the skin with fast precision change is really quite astonishing. So what can you do with this skin? Well, let ' s think a little bit about other things besides camouflage that they can do with their skin. Here you see the mimic octopus and a pattern. All of a sudden, it changes dramatically — that ' s signaling, not camouflage. And then it goes back to the normal pattern. Then you see the broadclub cuttlefish showing this passing cloud display as it approaches a crab prey. And finally, you see the flamboyant cuttlefish in camouflage and it can shift instantly to this bright warning display. What we have here is a sliding scale of expression, a continuum, if you will, between conspicuousness and camouflage. And this requires a lot of control. Well, guess what? Brains are really good for control. The brain of the octopus shown here has 35 lobes to the brain, 80 million tiny cells. And even though that ' s interesting, what ' s really odd is that the skin of this animal has many more neurons, as illustrated here, especially in the yellow. There are 300 million neurons in the skin and only 80 million in the brain itself — four times as many. Now, if you look at that, there ' s actually one of those little satellite brains and the equivalent of the spinal cord for each of the eight arms. This is a very unusual way to construct a nervous system in a body. Well, what is that brain good for? That brain has to outwit other big, smart brains that are trying to eat it, and that includes porpoises and seals and barracudas and sharks and even us humans. So decision-making is one of the things that this brain has to do, and it does a very good job of it. Shown here, you see this octopus perambulating along, and then it suddenly stops and creates that perfect camouflage. And it ' s really marvelous, because when these animals forage in the wild, they have to make over a hundred camouflaging decisions in a two-hour forage, and they do that twice a day. So, decision-making. They ' re also figuring out where to go and how to get back home. So it ' s a decision-making thing. We can test this camouflage, like that cuttlefish you see behind me, where we pull the rug out from under it and give it a checkerboard, and it even uses that strange visual information and does its best to match the pattern with a little **ad-libbing**. So other cognitive skills are important, too. The squids have a different kind of smarts, if you will. They have an extremely complex, interesting sex life. They have fighting and flirting and courting and mate-guarding and deception. Sound familiar? And it ' s really quite amazing that these animals have this kind of intuitive ability to do these behaviors. Here you see a male and a female. The male, on the left, has been fighting off other males to pair with the female, and now he ' s showing a dual pattern. He shows courtship and love on her side, fighting on the other. Watch him when she shifts places — and you see that he has fluidly changed the love-courtship pattern to the side of the female. So this kind of dual signaling simultaneously with a changing behavioral context is really extraordinary. It takes a lot of brain power. Now, another way to look at this is that, hmm, maybe we have 50 million years of evidence for the two-faced male. All right, let ' s move on. An octopus on a coral reef has a tough job in front it to go to so many places, remember and find its den. And they do this extremely well. They have short- and long-term memory, they learn things in three to five trials — it ' s a good brain. And the spatial memory is unusually good. They will even end their forage and make a beeline all the way back to their den. The divers watching them are completely lost, but they can get back, so it ' s really quite refined memory capability. Now, in terms of cognitive skills, look at this sleeping behavior in the cuttlefish. Especially on the right, you see the eye twitching. This is rapid eye movement kind of dreaming that we only thought mammals and birds did. And you see the false color we put in there to see the skin patterning flashing, and this is what ' s happening a lot. But it ' s not normal awake behaviors ; it ' s all different. Well, dreaming is when you have memory consolidation, and so this is probably what ' s happening in the cuttlefish. Now, another form of memory that ' s really unusual is episodic-like memory. This is something that humans need four years of brain development to do to remember what happened during a particular event, where it happened and when it happened. The"when"part is particularly difficult, and these children can do that. But guess what? We find recently that the wily cuttlefish also has this ability, and in experiments last summer, when you present a cuttlefish with different foods at different times, they have to match that with where it was exactly and when was the last time they saw it. Then they have to guide their foraging to the rate of replenishment of each food type in a different place. Sound complicated? It ' s so complicated, I hardly understood the experiment. So this is really high-level cognitive processing. Now, speaking of brains and evolution at the moment, you look on the right, there ' s the pathway of vertebrate brain evolution, and we all have good brains. I think everyone will acknowledge that. But if you look on the left side, some of the evolutionary pathway outlined here to the octopus, they have both converged, if you will, to complex behaviors and some form of intelligence. The last common denominator in these two lines was 600 million years ago, and it was a worm with very few neurons, so very divergent paths but convergence of complicated behavior. Here is the fundamental question : Is the brain structure of an octopus basically different down to the tiniest level from the vertebrate line? Now, we don ' t know the answer, but if it turns out to be yes, then we have a different evolutionary pathway to create intelligence on planet Earth, and one might think that the artificial intelligence community might be interested in those mechanisms. Well, let ' s talk genetics just for a moment. We have genomes, we have DNA, DNA is transcripted into RNA, RNA translates that into a protein, and that ' s how we come to be. Well, the cephalopods do it differently. They have big genomes, they have DNA, they transcript it into RNA, but now something dramatically different happens. They edit that RNA at an astronomical weird rate, a hundredfold more than we as humans or other animals do. And it produces scores of proteins. And guess where most of them are for? The nervous system. So perhaps this is an unorthodox way for an animal to evolve behavioral plasticity. This is a lot of conjecture, but it ' s food for thought. Now, I ' d like to share with you for a moment my experience, and using my smarts and that of my colleagues, to try and get this kind of information. We ' re diving, we can ' t stay underwater forever because we can ' t breathe it, so we have to be efficient in what we do. The total sensory immersion into that world is what helps us understand what these animals are really doing, and I have to tell you that it ' s really an amazing experience to be down there and having this communication with an octopus and a diver when you really begin to understand that this is a thinking, cogitating, curious animal. And this is the kind of thing that really inspires me endlessly. Let ' s go back to that smart skin for a few moments. Here ’ s a squid and a camouflage pattern. We zoom down and we see there ’ s beautiful pigments and reflectors. There are the chromatophores opening and closing very quickly. And then, in the next layer of skin, it ’ s quite interesting. The chromatophores are closed, and you see this magical iridescence just come out of the skin. This is also neurally controlled, so it ’ s the combination of the two, as seen here in the high-resolution skin of the cuttlefish, where you get this beautiful pigmentary structural coloration and even the faint blushing that is so beautiful. Well, how can we make use of some of this information? I talked about those skin bumps, the papillae. Here ’ s the giant Australian cuttlefish. It ’ s got smooth skin and a conspicuous pattern. I took five pictures in a row one second apart, and just watch this animal morph — one, two, three, four, five — and now I ’ m a seaweed. And then we can come right back out of it to see the smooth skin and the conspicuousness. So this is really marvelous, morphing skin. You can see it in more detail here. Periscope up, and you ’ ve got those beautiful papillae. And then we look in a little more detail, you can see the individual papillae come up, and there are little ridges on there, so it ’ s a papilla on papilla and so forth. Every individual species out there has more than a dozen shapes and sizes of those bumps to create fine-tuned, neurally controlled camouflage. So now, my colleagues at Cornell, engineers, watched our work and said,”We think we can make some of those.”Because in industry and society, this kind of soft materials under control of shape are really very rare. And they went ahead, worked with us and made the first samples of artificial papillae, soft materials, shown here. And you see them blown up into different shapes, And then you can press your finger on them to see that they ’ re a little bit malleable as they are. And so this is an example of how that might work. Well, I want to segue from this into the color of fabrics, and I imagine that could have a lot of applications as well. Just look at this kaleidoscope of color of dynamically controlled pigments and reflectors that we see in the cephalopods. We know enough about the mechanics of how they work that we can begin to translate this not only into fabrics but perhaps even into changeable cosmetics. And moreover, there ’ s been the recent discovery of light-sensing molecules in the skin of octopus which may pave the way to, eventually, smart materials that sense and respond on their own. Well, this form of biotechnology, or biomimicry, if you will, could change the way we look at the world even above water. Take, for example, artificial intelligence that might be inspired by the body-distributed brain and behavior of the octopus or the smart skin of a cuttlefish translated into cutting-edge fashion. Well, how do we get there? Maybe all we have to do is to begin to be a little bit smarter about how smart the cephalopods are. Thank you.

Text Sample 496: NEG Dim 3 – 2019 – Score 1.00 – 2019/t000002.txt

So one of the most important solutions to the global challenge posed by climate change lies right under our foot every day. It ’ s soil. Soil ’ s just the thin veil that covers the surface of land, but it has the power to shape our planet ’ s destiny. See, a six-foot or so of soil, loose soil material that covers the earth ’ s surface, represents the difference between life and lifelessness in the earth system, and it can also help us combat climate change if we can only stop treating it like dirt. Climate change is happening, the earth ’ s atmosphere is warming, because of the increasing amount of greenhouse gases we keep releasing into the atmosphere. You all know that. But what I assume you might not have heard is that one of the most important things our human society could do to address climate change lies right there in the soil. I ’ m a soil scientist who has been studying soil since I was 18, because I ' m interested in unlocking the secrets of soil and helping people understand this really important climate change solution. So here are the facts about climate. The concentration of carbon dioxide in the earth ' s atmosphere has increased by 40 percent just in the last 150 years or so. Human actions are now releasing 9. 4 billion metric tons of carbon into the atmosphere, from activities such as burning fossil fuels and intensive agricultural practices, and other ways we change the way we use land, including deforestation. But the concentration of carbon dioxide that stays in the atmosphere is only increasing by about half of that, and that ' s because half of the carbon we keep releasing into the atmosphere is currently being taken up by land and the seas through a process we know as carbon sequestration. So in essence, whatever consequence you think we ' re facing from climate change right now, we ' re only experiencing the consequence of 50 percent of our pollution, because the natural ecosystems are bailing us out. But don ' t get too comfortable, because we have two major things working against us right now. One : unless we do something big, and then fast, emissions will continue to rise. And second : the ability of these natural ecosystems to take up carbon dioxide from the atmosphere and sequester it in the natural habitats is currently getting compromised, as they ' re experiencing serious degradation because of human actions. So it ' s not entirely clear that we will continue to get bailed out by these natural ecosystems if we continue on this business-as- usual path that we ' ve been. Here ' s where the soil comes in : there is about three thousand billion metric tons of carbon in the soil. That ' s roughly about 315 times the amount of carbon that we release into the atmosphere currently. And there ' s twice more carbon in soil than there is in vegetation and air. Think about that for a second. There ' s more carbon in soil than there is in all of the world ' s vegetation, including the lush tropical rainforests and the giant sequoias, the expansive grasslands, all of the cultivated systems, and every kind of flora you can imagine on the face of the earth, plus all the carbon that ' s currently up in the atmosphere, combined, and then twice over. Hence, a very small change in the amount of carbon stored in soil can make a big difference in maintenance of the earth ' s atmosphere. But soil ' s not just simply a storage box for carbon, though. It operates more like a bank account, and the amount of carbon that ' s in soil at any given time is a function of the amount of carbon coming in and out of the soil. Carbon comes into the soil through the process of photosynthesis, when green plants take carbon dioxide from the atmosphere and use it to make their bodies, and upon death, their bodies enter the soil. And carbon leaves the soil and goes right back up into the atmosphere when the bodies of those formerly living organisms decay in soil by the activity of microbes. See, decomposition releases carbon dioxide into the atmosphere, as well as other greenhouse gases such as methane and nitrous oxide, but it also releases all the nutrients we all need to survive. One of the things that makes soil such a fundamental component of any climate change mitigation strategy is because it represents a long-term storage of carbon. Carbon that would have lasted maybe a year or two in decaying residue if it was left on the surface can stay in soil for hundreds of years, even thousands and more. Soil biogeochemists like me study exactly how the soil system makes this possible, by locking away the carbon in physical association with minerals, inside aggregates of soil minerals, and formation of strong chemical bonds that bind the carbon to the surfaces of the minerals. See when carbon is entrapped in soil, in these kinds of associations with soil minerals, even the wiliest of the microbes can ' t easily degrade it. And carbon that ' s not degrading fast is carbon that ' s not going back into the atmosphere as greenhouse gases. But the benefit of carbon sequestration is not just limited to climate change mitigation. Soil that stores large amounts of carbon is healthy, fertile, soft. It ’ s malleable. It ’ s workable. It makes it like a sponge. It can hold on to a lot of water and nutrients. Healthy and fertile soils like this support the most dynamic, abundant and diverse habitat for living things that we know of anywhere on the earth system. It makes life possible for everything from the tiniest of the microbes, such as bacteria and fungi, all the way to higher plants, and fulfills the food, feed and fiber needs for all animals, including you and I. So at this point, you would assume that we should be treating soil like the precious resource that it is. Unfortunately, that ’ s not the case. Soils around the world are experiencing unprecedented rates of degradation through a variety of human actions that include deforestation, intensive agricultural production systems, overgrazing, excessive application of agricultural chemicals, erosion and similar things. Half of the world ’ s soils are currently considered degraded. Soil degradation is bad for many reasons, but let me just tell you a couple. One : degraded soils have diminished potential to support plant productivity. And hence, by degrading soil, we ’ re compromising our own abilities to provide the food and other resources that we need for us and every member of living things on the face of the earth. And second : soil use and degradation, just in the last 200 years or so, has released 12 times more carbon into the atmosphere compared to the rate at which we ’ re releasing carbon into the atmosphere right now. I ’ m afraid there ’ s even more bad news. This is a story of soils at high latitudes. Peatlands in polar environments store about a third of the global soil carbon reserves. These peatlands have a permanently frozen ground underneath, the permafrost, and the carbon was able to build up in these soils over long periods of time because even though plants are able to photosynthesize during the short, warm summer months, the environment quickly turns cold and dark, and then microbes are not able to efficiently break down the residue. So the soil carbon bank in these polar environments built up over hundreds of thousands of years. But right now, with atmospheric warming, the permafrost is thawing and draining. And when permafrost thaws and drains, it makes it possible for microbes to come in and rather quickly decompose all this carbon, with the potential to release hundreds of billions of metric tons of carbon into the atmosphere in the form of greenhouse gases. And this release of additional greenhouse gases into the atmosphere will only contribute to further warming that makes this predicament even worse, as it starts a self-reinforcing positive feedback loop that could go on and on and on, dramatically changing our climate future. Fortunately, I can also tell you that there is a solution for these two wicked problems of soil degradation and climate change. Just like we created these problems, we do know the solution, and the solution lies in simultaneously working to address these two things together, through what we call climate- smart land management practices. What do I mean here? I mean managing land in a way that ’ s smart about maximizing how much carbon we store in soil. And we can accomplish this by putting in place deep-rooted perennial plants, putting back forests whenever possible, reducing tillage and other disturbances from agricultural practices, including optimizing the use of agricultural chemicals and grazing and even adding carbon to soil, whenever possible, from recycled resources such as compost and even human waste. This kind of land stewardship is not a radical idea. It ’ s what made it possible for fertile soils to be able to support human civilizations since time immemorial. In fact, some are doing it just right now. There ’ s a global effort underway to accomplish exactly this goal. This effort that started in France is known as the”4 per 1000”effort, and it sets an aspirational goal to increase the amount of carbon stored in soil by 0. 4 percent annually, using the same kind of climate-smart land management practices I mentioned earlier. And if this effort ’ s fully successful, it can offset a third of the global emissions of fossil-fuel-derived carbon into the atmosphere. But even if this effort is not fully successful, but we just start heading in that direction, we still end up with soils that are healthier, more fertile, are able to produce all the food and resources that we need for human populations and more, and also soils that are better capable of sequestering carbon dioxide from the atmosphere and helping with climate change mitigation. I ’ m pretty sure that ’ s what politicians call a win-win solution. And we all can have a role to play here. We can start by treating the soil with the respect that it deserves : respect for its ability as the basis of all life on earth, respect for its ability to serve as a carbon bank and respect for its ability to control our climate. And if we do so, we can then simultaneously address two of the most pressing global challenges of our time : climate change and soil degradation. And in the process, we would be able to provide food and nutritional security to our growing human family. Thank you.

Text Sample 497: NEG Dim 3 – 2019 – Score 1.00 – 2019/t000001.txt

[This is an improvised talk based on a suggested topic from the audience. The speaker doesn ’ t know the content of the slides.] Moderator : Our next speaker — is an — incredibly — Is an incredibly experienced linguist working at a lab at MIT with a small group of researchers, and through studying our language and the way that we communicate with other people, he has stumbled upon the secret of human intimacy. Here to give us his perspective, please welcome to the stage, Anthony Veneziale. Anthony Veneziale : You might think I know what you ’ re going through. You might be looking at me here on the red dot, or you might be looking at me on the screen. There ’ s a one sixth of a second delay. Did I catch myself? I did. I could see myself before I turned, and that small delay creates a little bit of a divide. And a divide is exactly what happens with human language, and the processing of that language. I of course am working out of a small lab at MIT. And we are scraping for every insight that we can get. This is not often associated with a computational challenge, but in this case, we found that persistence of vision and auditory intake actually have more in common than we ever realized, and we can see it in this first slide. Immediately your processing goes to,”Is that a hard-boiled egg?””Is that perhaps the structural integrity of the egg being able to sustain the weight of what seems to be a rock? Aha, is it in fact a real rock?”We go to questions when we see visual information. But when we hear information, this is what happens. The floodgates in our mind open much like the streets of Shanghai. So many pieces of information to process, so many ideas, concepts, feelings and, of course, vulnerabilities that we don ’ t often wish to share. And so we hide, and we hide behind what we like to call the floodgate of intimacy. And what might that floodgate be holding? What is the dike upon which it is built? Well, first off — we found that it ’ s different for six different genotypes. And, of course, we can start categorizing these genotypes into a neuronormative experience and a neurodiverse experience. On the right-hand side of the screen, you ’ re seeing spikes for the neurodiverse thinking. Now, there are generally only two emotional states that a neurodiverse brain can tabulate and keep count of at any given time, thereby eliminating the possibility for them to be emotionally, sometimes, attuned to the present situation. But on the left-hand side, you can see the neuronormative brain, which can often handle about five different pieces of emotional cognitive information at any given time. These are the slight variances that you are seeing in the 75, 90 and 60 percentile, and then of course that dramatic difference of the 25, 40 and 35 percentile. But of course, what is the neural network that is helping to bridge and build these different discrepancies? Fear. And as we all know, fear resides in the amygdala, and it is a very natural response, and it is very closely linked with visual perception. It is not as closely linked with verbal perception, so our fear receptors often will be going off in advance of any of our cognitive usage around verbal and words and cues of language. So as we see these fear moments, we of course are taken aback. We stumble in a certain direction, generally away from the intimacy. Now of course, there ’ s a difference between the male perception and the female perception and of trans and those who are in between, all of those as well, and outside of the gender spectrum. But fear is the central underlying underpinning of all of our response systems. Fight-or-flight is one of the earliest, some say reptilian, response to our environment. How can we disengage or unhook ourselves from the horns of the amygdala? Well, I ’ d like to tell you the secret right now. This is all making much, much too much sense. The secret lies in turning our backs to one another, and I know that that sounds absolutely like the opposite of what you were expecting, but when in a relationship you turn your back to your partner and place your back upon their back — you eliminate visual cues. You are more readily available to failing first, and failing first — far outweighs the lengths we go to to appeal to others, to our partners and to ourselves. We spend billions and billions of dollars on clothing, on makeup, on the latest trend of glasses, but what we don ’ t spend money and time on is connecting with each other in a way that is truthful and honest and stripped of those visual receptors. It sounds hard, doesn ’ t it? But we want to be aggressive about this. We don ’ t want to just sit on the couch. As a historian said earlier today, it ’ s important to get up and circumvent sometimes that couch. And how can we do it? Well yes, ice is a big part of it. Insights, compassion and empathy : I, C, E. And when we start using this ice method, well, the possibilities become much bigger than us. In fact, they become smaller than you. On a molecular level, I believe that that insight is the unifying theme for every talk you have seen so far at TED and will continue as we of course embark on this journey here on this tiny planet, on the ledge, on the precipice, as we are seeing, yes, death is inevitable. Will it meet all of us at the same time, I think, is the variable we are inquiring. I think that timeline gets a bit longer when we use ice and when we rest our backs upon one another and build together, leaving behind the fear and working towards — they ’ ll edit this part out — a ripened experience of love, compassion, intimacy based on a truth that you are sharing from your mind ’ s eye and the heart that we all can touch, tactilely feel, have maybe potentially a mushy experience that we don ’ t just throw out because it is browned, but let us slice in half the experience we have gathered, let us seed what the heart, the core, the seed of that idea in each of us is, and let us share it back to back. Thank you very much.

Text Sample 498: NEG Dim 3 – 2001 – Score 1.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,”Are you still doing this or that?”I don ’ t answer because I ’ m not doing things still, I ’ m doing it like I always did. I still have — or did I use the word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years. But, of course, I ’ m working since 25, doing more or less what you see here.

This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ’ ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called”Innovation.”Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,”The playful search for beauty was Man ’ s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty.”These are tiles. The word,”playful”is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ’ t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ’ t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ' s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ' s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ' 32 to ' 37 — actually, to ' 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ' s purges — at the beginning of Stalin ' s purges, I didn ' t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ' s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ' s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ' m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ' t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ' 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ' t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 499: NEG Dim 3 – 2001 – Score 2.00 – 2001/t000367.txt

I ’ d like to do pretty much what I did the first time, which is to choose a light-hearted theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, exorcise those evil spirits, and this is still going on, as you know. But it wasn ’ t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ’ re feeling depressed? It ’ s better than Prozac, but I wouldn ’ t recommend it. So what we see in the seventeenth, eighteenth century is the continued search for medications other than camphor that ’ ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we fast-forward to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a row — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very interested in producing convulsions, measured types of convulsions. And they thought,”Well, we ’ ve got electricity, we ’ ll plug somebody into the wall. That always makes hair stand up and people shake a lot.”So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,”We know that at the Rome railroad station, there are all these lost souls wandering around, muttering gibberish. Can you bring one of them to us?”Someone who is, as the Italians say,”cagoots.”So they found this”cagoots”guy, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,”Well, we ’ ll try 55 volts, two-tenths of a second. That ’ s not going to do anything terrible to him.”So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,”This fellow”— remember, he wasn ’ t even put to sleep —”after this major grand mal convulsion, sat right up, looked at these three fellas and said, ’ What the fuck are you assholes trying to do? ’”If I could only say that in Italian. Well, they were happy as could be, because he hadn ’ t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and everybody in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ’ t work very well on the schizophrenics, but it was pretty clear in the ’ 30s and by the middle of the ’ 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ’ t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the brain waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ’ 60s, the first antidepressants came out. Tofranil was the first. In the late ’ 70s, early ’ 80s, there were others, and they were very effective. And patients ’ rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ’ m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ’ ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ’ m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ’ clock in the afternoon, because I couldn ’ t get out of bed before about 11 o ’ clock. And anybody who ’ s been depressed here knows what that ’ s like. I couldn ’ t even pull the covers off myself. Well, you ’ re in a university medical center, where everybody knows everybody, and it ’ s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ’ t work anymore. And, in fact, it didn ’ t make any difference because I didn ’ t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,”Don ’ t worry, chap. Six weeks, you ’ re back in the operating room. Everything ’ s going to be great.”Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ’ t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five feet in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ’ t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had. They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the senior staff. All the senior staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful stuff. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it **got** so there was a throbbing, there was a ferocious fear in my head. You ’ ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ’ t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case. He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the senior staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,”No. I know this man better than any of you. I have met with him over and over again. You ’ ve just seen him from time to time. You ’ ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ’ ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go back to surgery, he will never be the loving father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ’ One Flew Over the Cuckoo ’ s Nest. ’ And you know about that, just essentially in a stupor the rest of his life.”Well, he said,”Can ’ t we try a course of electroshock therapy?”And you know why they agreed? They agreed to humor him. They just thought,”Well, we ’ ll give a course of 10. And so we ’ ll lose a little time. Big deal. It doesn ’ t make any difference.”So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ’ t work. Seven didn ’ t work. Eight didn ’ t work. At nine, I noticed — and it ’ s wonderful that I could notice anything

— I noticed a change. And at 10, I noticed a real change. And he went back to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, lo and behold, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could blow the obsessional thinking away. I could blow the depression away. And I ' ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,"I ' ve got the strength now to do this."It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a formula. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember"Ruddigore,"and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,"We must have a word to bring you back to reality, and the word, my dear, will be ' Basingstoke. '"So every time she got a little nuts, he would say,"Basingstoke!"And she would say,"Basingstoke, it is."And she would be fine for a little while. Well, you know, I ' m from the Bronx. I can ' t say"Basingstoke."But I had something better. And it was very simple. It was,"Ah, fuck it!"Much better than"Basingstoke,"at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,"Ah, fuck it."And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went back to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came back to live with us. We had two more children after that. Resuscitated the career, even better than it had been before. Went right back into the university and began to write books. Well, you know, it ' s been a wonderful life. It ' s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ' s been very exciting. It ' s been very happy. Every once in a while, I have to say,"Ah, fuck it."Every once in a while, I get somewhat depressed and a little obsessional. So, I ' m not free of all of this. But it ' s worked. It ' s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ' ve gotten thousands of letters about them by people who do think this — that based on my life ' s history as I ' ve portrayed in the books, my early life ' s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the bitter dregs of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ' ve always felt guilty about that. I ' ve always felt that somehow I was an impostor because my readers don ' t know what I have just told you. It ' s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an untroubled mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes back to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ' t happen, there will be adversities. If I, with the bleakness of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way back from this, believe me, anybody can find their way back from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

Text Sample 500: NEG Dim 3 – 2001 – Score 3.00 – 2001/t000092.txt

I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ' children, because they were all average. But they weren ' t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ' s not right. The good Lord in his infinite wisdom didn ' t create us all equal as far as intelligence is concerned, any more than we ' re equal for size, appearance. Not everybody could earn an A or a B, and I didn ' t like that way of judging, and I did know how the alumni of various schools back in the ' 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ' t lose a game. But it seemed that we didn ' t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ' t like it, I didn ' t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ' m sure at the time he did that, I didn ' t — it didn ' t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ' s under your control. If you get too engrossed and involved and concerned in regard to the things over which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,"At God ' s footstool to confess, a poor soul knelt, and bowed his head. ' I failed! ' he cried. The Master said, ' Thou didst thy best, that is success. '"From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ' re capable. I believe that ' s true. If you make the effort to do the best of which you ' re capable, trying to improve the situation that exists for you, I think that ' s success, and I don ' t think others can judge that ; it ' s like character and reputation — your reputation is what you ' re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ' d hope they ' d both be good, but they won ' t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ' m not what I ought to be, what I should be, but I think I ' m better than I would have been if I hadn ' t run across certain things. One was just a little verse that said,"No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ' s what the teachers are themselves."That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ' s because Dad used to read to us at night, by coal oil lamp — we didn ' t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,"They ask me why I teach, and I reply, ' Where could I find such splendid company? ' There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ' s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ' Where could I find such splendid company? '"And I believe the teaching profession — it ' s true, you have so many youngsters, and I ' ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ' re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ’ re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was”Never be late.”Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ’ s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ’ t clean and neat, so I wouldn ’ t let him go. He couldn ’ t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ’ t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ’ ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,”Don ’ t run practices late, because you ’ ll go home in a bad mood, and that ’ s not good, for a young married man to go home in a bad mood. When you get older, it doesn ’ t make any difference, but —”So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ’ re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ’ t want that. I used to tell them I was paid to do that. That ’ s my job. I ’ m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ’ s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ’ s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ’ t have the time to go on that. But that helped me, I think, become a better teacher. It ’ s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ’ re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ’ re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ’ t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ’ i ’. I ’ d never seen that before, but he did. Big league baseball players — they ’ re very perceptive about those things, and they noticed he had only one ’ i ’ in his name. You ’ d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it"The Road Ahead, or the Road Behind."He said,"Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ' t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ' s game. You and I know deeper down, there ' s always a chance to win the crown. But when we fail to give our best, we simply haven ' t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ' s bearing down that wins the cup. Of dreaming there ' s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ' s you and I who make our fates — we open up or close the gates on the road ahead or the road behind."Reminds me of another set of threes that my dad tried to get across to us : Don ' t whine. Don ' t complain. Don ' t make excuses. Just get out there, and whatever you ' re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ' re outscored. I ' ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ' t know the outcome, I hope they couldn ' t tell by your actions whether you outscored an opponent or the opponent outscored you. That ' s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ' d want them to be but they ' ll be about what they should ; only you will know whether you can do that. And that ' s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,"The journey is better than the end."And I like that. I think that it is — it ' s getting there. Sometimes when you get there, there ' s almost a let down. But it ' s the getting there that ' s the fun. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ' d done a decent job during the week. There again, it ' s getting the players to get that self-satisfaction, in knowing that they ' d made the effort to do the best of which they are capable. Sometimes I ' m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,"Suppose that you, in some way, could make the perfect player. What would you want?"And I said,"Well, I ' d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ' d want one that could play, too. I ' d want one to realize that defense usually wins championships, and who would work hard on defense. But I ' d want one who would play offense, too. I ' d want him to be unselfish, and look for the pass first and not shoot all the time. And I ' d want one that could pass and would pass. I ' ve had some that could and wouldn ' t, and I ' ve had some that would and could. So, yeah, I ' d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too.

I ’ d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,”They Call Me Coach,”two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ’ t play varsity when I taught. I thought,”Oh gracious, if these two players, either one of them”— they were different years, but I thought about each one at the time he was there —”Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ’ s good enough to make it.”And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ’ d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

7 POS Dim 4

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HW : Hi everyone, I am Helen Walters. I am head of Media and Curation here at **TED**. Welcome to another episode of **TED** Explains the World with the one and only, Ian Bremmer. It is Monday, February 24, a little more than a month since President Trump was inaugurated once more, and **safe** to say, a lot has been going on. So we figured we ’ d check in with Ian to determine what we should really be paying attention to amid the extreme noise. Ian, hi. Ian Bremmer : Helen, great to be back with you. HW : So we asked the **TED** community to share their questions for you, and we wanted them to share what they ’ re most curious to know. And we really got some amazing questions that I plan to pepper this conversation with. So let ’ s start with one right now. Two months into 2025, where does the US stand? IB : Very powerfully, in the sense that the US economy, of course, is performing considerably better than any of the other G7 economies coming still out of the **pandemic**. Technologically, only close competitor is China, ahead of the US in some areas, behind the US and other critical areas. But compared to every other country in the world, the US is head and shoulders and neck or waist above them. Militarily, of course, the US is the only country with global **ability** to project power. No one else is close.

And the US dollar is still the global reserve currency, no one is approximate challenger. I say all of those things because those are the things that haven ' t really changed over the course of the last few months, but I suspect that isn ' t what the questioner was asking. What they were really asking is what ' s happening politically. And politically, the United States is unwinding its own global order. It is no longer particularly **interested** in promoting **collective** security or NATO. It ' s no longer really **interested** in promoting involvement and **leadership** of multilateral institutions, of consistent rule of law and free trade that ' s well regulated, certainly not very **interested** in promoting democracy around the world. I mean, this is an environment where other countries around the world have to figure out how to adapt to the United States, and that the US has become the principal driver of geopolitical risk and **uncertainty** in the world today, unlike any other time in my lifetime and yours, that ' s perhaps the biggest change. HW : That is quite a statement. Alright, you are a geopolitical expert. Let ' s talk about Europe. So Defense Secretary Pete Hegseth stated that America ' s foreign **policy** focus no longer lies in Europe. Vice President JD Vance caused some raised eyebrows, dropped jaws and, I suspect, some choice expletives after his recent speech at the Munich Security Conference, in which he accused European **leaders** of suppressing free speech and warned of the threat from within. So you were in Munich. Was it as dramatic as the stories we read had it? And what happens next? IB : It was. I was in the room. I was maybe 30 feet away from him when he gave that speech. Standing right in the front, right next to me was the president of Czechia, the president of Finland, the prime minister of Sweden, watching all of them and their **reactions**. What he said, let ' s keep in mind, this is the Munich Security Conference. It ' s been going on now for something like 61 years. And he was the head of the US delegation. And every year the head of the US delegation gives a big speech on the state of the transatlantic alliance and on global security. And he didn ' t do that. I mean, the people in the audience were expecting a challenging speech. They were expecting that the Americans would be less committed to the alignment with the Europeans on Ukraine. The US had just had that direct Trump **phone** call with Vladimir Putin, 90 minutes long. Hadn ' t coordinated that with the Europeans, never mind the Ukrainians, in advance. So, I mean, they were definitely prepared for a very challenging plenary. But Vance did none of that. Vance said,"I ' m not going to talk about Ukraine or Russia or China, because the biggest problem is actually what ' s happening inside Europe. The biggest problem, essentially, are you **guys** sitting in front of me, that I ' m speaking to, because you ' re suppressing free speech. You ' re suppressing the far right. You ' ve been infected by the woke mind **virus**, and you aren ' t real democracies as a consequence."And specifically he said, and this is something that I think is underappreciated in the United States, he attacked the German firewall. And said that – And let me explain what the German firewall is. That is a principle by the **leaders** of all of the mainstream German parties and their supporters that they will not, under any circumstances, work with, enter into coalition with the Alternative for Deutschland party, the AfD. Because the AfD, under surveillance from German **intelligence** agencies, is considered to be a neo-Nazi party. And the United States, of course, after World War II, led **denazification** of Germany. The day before, Vance had actually gone to Dachau and **visited** a concentration camp, and then came to speak in Germany about the firewall needing to be ended. And at that point someone yelled out from the crowd,"This is unacceptable!"And it was right in the front. And people in the room, about 1, 500 people in the room, standing room only, couldn ' t see who it was. They ' d just see the back of his head. I saw who it was. It was the German Defense Minister, Pistorius. And in 15 years of me going to the Munich Security Conference, I have never seen anything remotely like that. And the reason I mention all of this, and of course, on the back of that, let me be clear, Vance **refused** to meet with the German chancellor because he ' s basically, well, he ' s not relevant. He ' s only going to be there for a couple more months until a new government, so why should I bother? But does meet with the **leader** of this AfD, goes and meets with her directly, does a bilateral that evening. So that is a very important backdrop for this last weekend ' s German elections, where the victor, Friedrich Mertz, who is going to be the next chancellor, actually referred specifically to America ' s intervention in German democracy and **support** of the AfD being every bit as bad as what the Kremlin is doing inside Germany, and also said that the Europeans need to develop a defense **policy** which is independent of the United States, in other words, essentially saying that it will be the end of NATO. Again, staggering comments, unheard of, impossible to imagine even two weeks ago. And that ' s why it ' s very critical that you have the context of what happened between Vance, Elon **Musk** over the past couple of months, Trump, with the Germans, with the Europeans. So you can understand why the incoming German chancellor would make a statement like that. HW : What should we make of all of the conversation about the rise of Nazism, the return of Nazism? Obviously, the AfD just won 20 percent in the German election. We ' ve also seen people allegedly giving Nazi salutes at various US conventions and speeches. Do you think this is overblown? Is this a fear that people have rationally? And what should we make of that? IB : In Germany, the Alternative for Deutschland, are winning all of former East Germany. This is a group that increasingly understands that they ' re never going to really catch up to the rest of Germany. Remember, Germany hasn ' t really grown in five years now. They ' ve been in **recession** for the last two. And the average citizen, the average voter, living in former East Germany no longer are just angry about not catching up. They ' re basically saying,"I ' m done with this system."So that ' s why you ' re getting this very strident, revanchist nationalism that is, you know, it ' s not just taking a hard line on illegal immigrants. The AfD even **supports** removing citizens of Germany that they **claim** have not adequately integrated into German culture and nationality. And so, I mean, this is some very, very strong **stuff**, **stuff** that really does feel like what they were doing back in Nazi Germany. But in former East Germany, they win. They get the largest number of votes, where, in much of the wealthier part of West Germany, including, you know, where I was, in Munich, the AfD can ' t break out of single digits. So it ' s a very, very divided country. And also the AfD is doing very, very well among young men. There ' s a clear **gender** divide here as well, just as there is in the United States. In the US, I mean, I, of course, saw Elon **Musk** and that, you know, sort of grabbing his heart, saying,"My heart goes out to you"and then doing what appeared to be mimicking a Nazi salute. And then Steve Bannon doing the same without the initial heartfelt comment. It clearly was trolling. Clearly was trying to get a **reaction** from folks. It ' s very performative. And it ' s meant to also, you know, attack and attach, make the left lose their minds on something that isn ' t very critical to what the Trump administration is trying to do compared to the revolutionary changes inside the US government that they are very much prioritizing right now. So I ' m very concerned about the rise of neo-Nazism in Germany. I am very much not concerned about that in the US. I ' m concerned about other things I ' m thinking is being used as gaslighting for those that can use distractions or can be distracted by them. And I would also say that this election in Germany wasn ' t surprising. It was exactly as the polls have expected for several months now. But the key elections in Germany and the key elections in Europe are the next cycle. I ' ve spoken with a lot of people around the Trump administration that believe that the Europeans are one electoral cycle behind the United States. So in other words, Trumpism is coming, just not quite yet. And so in the UK, you ’ ve got Labour for a full term. But the Reform Party is increasingly the most popular. And just with a little nudge, maybe a little external money, and Elon said he ’ s already thinking about giving them 100 million dollars, which is a massive amount in UK politics, not so much in the US, that the **Conservatives** would fragment, a rump group would join with Reform, and they could win the next elections. In France, you ’ ve had all of these governments continue to collapse. Macron, very unpopular. Marine Le Pen and the National Rally party could easily win upcoming elections in 2027. In 2029, in Germany, if the Germans are incapable of turning their economy around, incapable of unwinding their debt break and spending some of the capital that ’ s just sitting there and as their debt to GDP goes down, but their economy contracts at the same time, then AfD could easily win in 2029, and then the firewall is no more. And then, of course, in the European Union elections overall, in five years ’ time, now four and a half, you could easily see the Patriots front, which is the equivalent to the far right grouping that these parties are a part of, they could all come together and be the **dominant** party. And then the EU as we know it is really a thing of the past. So these are existential changes that are happening not just for NATO right now, but also for the EU and quite plausibly, for the future of democracy as we know it. HW : Alright, let ’ s turn to Ukraine. So President Trump accused President Zelenskyy of starting the war with Russia and called him”a dictator without elections.”Last week, **senior** American and Russian officials met in Riyadh to discuss a potential cease-fire with no Ukrainians present. Now Zelenskyy has offered to step down in return for Ukrainian admission to NATO, although apparently he doesn ’ t really mean this. So what happens to Ukraine now? What should we be paying attention to? IB : Well, I mean, I think he would mean it if NATO **membership** happened. I ’ d love to see Trump call his bluff on that. But of course it ’ s not going to. And he wasn ’ t saying that six months or a year ago. He ’ s saying it now that NATO is truly being pulled off the table. I think it ’ s the right thing for him to do. It is, again, performative and helps to remind those around the world that Ukraine is fighting for **self-determination**. They ’ re fighting for the **ability** to join their own alliance, as any country in the world should be able to do. But that is not Trump ’ s belief. Trump believes that what you get to do is determined by how powerful you are, not by the fact that you happen to run a country. And indeed, the territorial integrity is something that is afforded to powerful countries, but not so much for weak countries. And that ’ s the way he feels about Panama, and that ’ s the way he feels about Denmark and Greenland. And that ’ s certainly the way he feels about Ukraine. So right now, as you and I are speaking, the United States is putting forward a new UN resolution at the General Assembly that says that the war has to end as fast as possible but does not recognize Ukraine ’ s territorial integrity, which has been a core precept of the **international** order that the United States has **supported** since creating the United Nations and since the end of World War II in 1945. The Americans are now throwing that out. And the Russians, of course, are fully supportive of that. America ’ s allies in Europe are not. But the US has gotten the Saudis to agree, and they ’ re getting a unified Arab block. The Americans have privately worked with a lot of poorer countries in Africa, in the Asia Pacific and others, smaller countries, and saying,”You ’ re not going to get any aid from the US unless you vote in favor of this new resolution.”I fully expect the resolution is going to pass. But of course, what that means is that the Americans are preparing to cut a deal on Ukraine over the heads of the Ukrainians. In other words, the US and Russia together will figure out what a ceasefire should and should not entail as part of a broader

US-Russia rapprochement. This is obviously a disaster for the Ukrainians and the Ukrainians who initially **refuse** to accept this so-called deal, on Ukrainian natural resources that would grant the Americans $500 billion in access to such resources in return for"paying off"aid that the Americans had granted to the Ukrainians without conditions or strings, but apparently no longer. And so, you know, what does this all mean? Well it means it ' s yet another case of why people shouldn ' t **trust** the United States from one administration to the next, because foreign **policy** will change completely. And what you thought was an American commitment won ' t stand. It shows that there is a fundamental rift between the Americans and the Europeans and indeed, Emmanuel Macron in the United States right now, as you and I speak, hoping that he can do something that will keep Trump from, as Macron believes, essentially surrendering to, conceding to, the Russian position and maintaining some level of coordination with the Europeans. But he doesn ' t have a lot to offer. And the Ukrainians, you know, now in a position of desperation where maybe they will end up signing over a lot of their natural resources to the Americans. But what, if anything, will they get in return for that? And how empowered will the Russians be in any deal that is cut? And will an independent Ukraine even be able to survive, and what will the terms that will be required of them? Of course, Putin is saying things like, no European troops of any sort, whether in a NATO formulation or not, would be allowed as peacekeepers in Ukraine. Well, then, how would you guarantee, what kind of security commitments would you give to the Ukrainians? These are all open questions, but apparently questions that President Trump is prepared to largely resolve on Russia ' s terms. And again, this is such a dramatic change from where the United States and where the Western order was just a month or two ago. HW : So let ' s take some **quick-fire** questions from our community related to this topic. And I think the first one is particularly pertinent to what you just said, so it ' s simple. What does Putin have over Trump? IB : I don ' t think that Putin"has"anything over Trump. If he did, then why wouldn ' t Trump have granted Russia more in his first term? In his first term he gave the Ukrainians those javelin **anti-tank** missiles that Obama **refused** to give. And, you know, he thought they were too risky. He increased sanctions against Russia. Now that was his administration with a lot of **hawkish** in orientation advisors around him that he doesn ' t have this time around. They ' re all directly loyal to him. But, I mean, if it was a priority for him, he could have stopped it. And if the Russians had something over him, then why wouldn ' t they have used it in that case? It feels like a conspiracy theory. I don ' t buy it. I think very differently. We can explain what ' s happening because number one, Trump wants to end wars. It ' s very popular with his base. He ' s said this on Gaza. He ' s said this on Iran, where Israel wants him to **support** attacks on Iran ' s nuclear capabilities, and Trump has said no. And he ' s showing this on Ukraine. He doesn ' t want to spend money on the Ukrainians going forward after spending over 100 billion dollars under the last three years from the Biden administration. He doesn ' t want to spend that money going forward. And he also doesn ' t value a strong Europe. He actually thinks that a weak Europe is in America ' s interests, where you have more Brexits. He used to always talk to the French about that during his first term,"Why won ' t you do Brexit?"He wants all of these MAGA-type populist parties, including the AfD, to win in Europe because that legitimizes his own popularity. Just like he loves Milei in Argentina or Bukele in El Salvador. And so I think if you put those things together, you actually get to why Trump is doing what he ' s doing with the Russians without needing to create a story about Trump somehow being threatened by some secret, shadowy information that the Russians have on Trump that they ' ve let him know about but nobody else knows. I just don ’ t buy that. HW : Great. OK, another community question. If the US hands Ukraine to Russia, will Russia keep going? Will they attempt to take over Europe? IB : It ’ s **interesting** that Trump just met with the Polish president, President Duda. It was supposed to be an hour meeting, it was only 10 minutes. So he kind of embarrassed him at home in Poland. But he did say that the US is still committed to maintaining a troop presence on the ground in Poland. And so it ’ s very hard to imagine the Russians rolling through Ukraine and then into Poland, even though there ’ s been lots of asymmetric warfare, for example, and even there have been, you know, Russian missiles launched into Lviv, Ukrainian **air** defense, missile defense, and with Polish villagers getting killed because of the fighting. So, I mean, there ’ s been spillover. But America is still going to be in Poland. Why is that? They get along pretty well, and the Poles are heading towards five percent of GDP defense spend this year, which has been the high water mark of American **demands** of defense spend. So he ’ s, you know, been consistent on that over the past months. Baltic states. I mean, they ’ re all spending a lot more money on their defense. And so, I mean, in principle, you could imagine the United States maintaining these rotational deployments in the Baltics, which would prevent the Russians from invading. But might Trump say, as part of a deal,”Why do we have all those troops there?”You mentioned Pete Hegseth saying the Americans need to get troops out of Europe and get them to Asia, which is where their principal competitive, you know, sort of, strategy is going to be oriented. And the US hasn ’ t been pressing the Japanese or the Indians, the principal partners, allies in Asia vis-à-vis China, the way they ’ ve been pressing the French, the **Brits**, the Germans in Europe. So I think it is certainly plausible that the Russians will do more, especially with those countries that **refuse** to align with a Trump rapprochement towards Russia. The countries I would be most worried about are Moldova, are Georgia. I ’ d be most worried about the countries that are in the so-called”near abroad”that aren ’ t a part of NATO, and aren ’ t going to become a part of NATO, and therefore are much more vulnerable. But certainly in terms of Russia ’ s willingness to engage in espionage, to fund actors on the ground for arson attacks, assassinations, critical infrastructure attacks on fiber networks, on pipelines, I think all of those things are likely to increase against European states across the board, on the back of this US-Russia rapprochement. HW : OK, final community question for this segment. What happens if the US no longer **supports** NATO? IB : It ’ s possible that the United States is moving towards a similar posture in Europe that they presently have in Asia, which is strong individual bilateral deals with a number of countries : South Korea, Japan, Australia, but not a **collective** security agreement, where those countries together have more of a call on the United States and there ’ s a lot more free riding. So I could imagine that NATO falls apart and becomes that. The danger is that even though the Europeans now understand the nature of the threat, that it is very plausible and maybe even baseline, that the Americans are going to leave Europe collectively on its own. That that seriousness does not equate to the **ability** to increase spending adequately in the near term. The Germans did manage to pull together an electoral outcome that will allow for a so-called grand coalition, just barely. So you ’ ll have a two-party government in all likelihood instead of a three. But you also have a blocking capacity on the part of the far right and the far left that will prevent the Germans from really blowing out their spending, including their defense spending, unless they can pull something off before that new government is formed. Which is possible, but it ’ s unlikely to be very big. The **Brits** are now talking about maybe moving the 2. 5 percent of GDP defense spend over years with a very challenging fiscal environment. The Spaniards, not even close, the Italians, not even close.

So what does a European military capability really look like? And without the Americans, the answer is it ’ s not there. It ’ s not there. So I think that if NATO falls apart, the reality is that most of Europe will be incredibly vulnerable to external attack, especially Russian attack, And there won ’ t be a way for them to defend themselves, and some of them will break and therefore want to work more directly and individually with the United States. And that could be the end, not only of NATO, but it could be the end of the European Union. I think this is, the stakes are incredibly high, this is an existential challenge for European governments that needed to take this seriously 30 years ago, 20 years ago, certainly in 2014, when the Russians invaded Ukraine, certainly, certainly in 2016 when Trump was elected, and certainly, certainly, certainly in 2022 when the Russians then invaded all of Ukraine. And at every point the Europeans were basically saying,”We ’ re still fine with the Americans basically doing this.”And now they are in very serious trouble. HW : No time like the present, I suppose. Alright, let ’ s move on. So I mentioned the discussions in Saudi and obviously also on the agenda there with the discussions about Gaza and Trump ’ s proposal to build a Gaza Riviera. What should we make of that? Or as one of our community members asked, what is the actual solution for Gaza? And surely it isn ’ t what the US is proposing. IB : Well, look, here ’ s what ’ s **interesting**. You know, Trump says a lot of things, and when they don ’ t work [out], he backs out fairly quickly. And you know, I thought – just stick with Ukraine for one second on this. It was **interesting** to me that Trump has said that he doesn ’ t want to talk to Zelenskyy because he has no cards, he has no leverage. And so why should he let him at the table? And so Trump ’ s going to cut the deal. Very **interesting** that if that is true, then why is it that the Secretary of Treasury brings this deal and says,”You ’ ve got to sign this right now, Zelenskyy”And Zelenskyy **refuses**. And then a few days later, the Americans come back with a new deal with altered terms that are not quite as predatory vis-à-vis Ukraine. Why would the Americans change their tune for someone that has no leverage? Because, I don ’ t think it ’ s because Trump is a nice **guy**. He ’ s not going soft at 78. I think it ’ s because he overstated how little leverage Ukraine has. Ukraine is very popular among the GOP in the House and Senate, far more than Putin. Zelenskyy is far more popular in the United States among the voting population than Putin. And of course, he ’ s also much more popular among the Europeans and even globally than Putin. And so it turns out that maybe there is a little bit of leverage that Zelenskyy has, and maybe Trump does have to give a little bit more. So this is relevant in the context of Gaza, because Trump was very annoyed that, you know, despite his **ability** to get a cease fire between Hamas and Israel over the goal line, that his friends in the Middle East were not coming up with a plan for Gaza. Where they were supposed to provide security and provide investment and lead reconstruction and deal with the Palestinian populations in the interim. And that wasn ’ t happening. And so Trump then meeting with Prime Minister Netanyahu from Israel in Washington, then after that, summoning the Jordanian king, who was the ally that is most reliant on the United States and so, therefore, most needing to say yes to whatever Trump orders of him. Basically says, as you mentioned, Helen, that Gaza is going to be a Riviera, that the US is going to turn this into an incredible place. They ’ re going to build wonderful homes for the Palestinians someplace else. And the Palestinians will all volunteer to leave Gaza and go to those homes, and Gaza will be depopulated and new people will come in and live there. That was the plan. And when he was asked, standing there with King Abdullah from Jordan, under whose authority? Trump ’ s response was :”Under my authority.”And that went over like a lead balloon among America ’ s allies in the Middle East. They all said,”We

’ re not up for this. We ’ re not resettling Palestinian populations. They won ’ t actually leave voluntarily. This will would actually be a huge problem for Israel ’ s own national security long-term. And we ’ re going to come up with another plan. And our plan will keep the Palestinians on the ground in Gaza.”At which point Trump **pivots** to work with the new plan. And his advisers say that Trump was only trying to start the conversation. And what he said was his plan is not actually what he ’ s going to do. So, what ’ s sustainable? What would be sustainable would be a lot of Gulf money going in and maybe some European money too, though they ’ re going to be really pressed, given everything you and I just talked about, to try to reconstruct a completely devastated Gaza that eventually some two **million-plus** Palestinians will be able to live normal lives in, and the governance will be provided by some technocratic group of a Palestinian Authority that will be working with, selected by, those that are **involved** in the reconstruction and the security. Namely the Egyptians, the Jordanians, the Saudis and the Emiratis. That is the best possible deal that they ’ re going to get. But that ’ s happening as Israel has just evacuated forcibly another 40, 000 Palestinians from **refugee** camps in the West Bank, sending tanks in for the first time in decades and probably precipitating more Israeli settlers taking more land on the ground in the West Bank. And so, as we are possibly taking a small step towards a sustainable solution in Gaza, we are taking a slightly larger step away from a sustainable solution for the Palestinians in the West Bank, twas ever thus over the past decades in this incredibly horrible problem and **conflict**. HW : And why is Israel doing that? I mean, are they feeling emboldened by the US, or why the incursions in the West Bank now? IB : Well, I think they ’ re feeling emboldened by their own successes. They have shown that they are the **dominant** military power in the Middle East, and they have the **ability** to determine the level of escalation unilaterally, and their adversaries and enemies can ’ t really do any damage to them. They proved that after October 7, in their response to Hamas. They proved that in decapitating and taking out the military capabilities of Hezbollah. And they ’ ve proved that also with overturning – they didn ’ t do it, but the fact that the Assad regime in Syria is gone, which means that the Iranians no longer have a land bridge to get additional weapons to what had been the strongest adversary of Israel militarily, Hezbollah. So all of that has shown that Israel can decide outcomes. They ’ re the ones that have all the power here. And also that depopulation of Gaza is very popular among Israelis. There was a poll recently in the Jerusalem Post, which is pretty mainstream in Israel. 80 percent of Israeli respondents said that they wanted to depopulate Gaza of all Palestinians. Taking more land is very popular, especially in the West Bank, especially among Netanyahu ’ s present right-wing coalition, that he would like to keep together, to continue to govern. And this is a **sop** to them as he is looking to potentially extend the ceasefire in Gaza and move from phase one to phase two, which is facing some delays and open questions right now. So, yes, certainly, the United States has been underpinning that military capability of Israel. And I think it is instructive to remember that not only does the US provide more military aid in peacetime to any country in the world, to Israel, but that unlike in Ukraine, where the Trump administration is saying,”No, I ’ m not happy with the grants, you ’ ve got to pay that back,”The billions and billions that the United States sends to Israel, nobody is suggesting that that should be alone or that the Americans should get technology rights or mineral rights or anything else from Israel, despite the fact that Trump ’ s **policy** is America First. Israel is a very clear exception to America First in Trump ’ s visioning of it. HW : Alright, so we talked about the fact that the foreign **policy** focus of America is shifting to Asia. So what are you hearing about how China is feeling in all of this? And what should we be paying attention to on that front? IB : China has had a couple of good weeks economically and specifically on the back of that huge announcement of DeepSeek, which has, you know, far less money behind it than the comparable American chatbots but is performing at an extremely high level. And, you know, given the fact that there are all of these export controls on semiconductors and other parts of advanced technology, that this is giving people more of a belief that the Chinese really are capable of driving innovation even in that space. And they ' re, of course, dominating the post-carbon energy space. We see that with electric vehicles and batteries. But we ' re also seeing that with massive investment now in green **hydrogen**, for example, we ' re seeing it in advanced nuclear capabilities, all of these things. And so there is more investment in that space that the Chinese are not only driving as a state, but that they ' re also increasingly able to raise and move in their own private **sector**. So the story that you and I have been discussing for the last couple of years is the Chinese economy is radically underperforming. That is still true, but there is an emerging counter-narrative here that I would be remiss not to mention. Now more broadly, what the Chinese here are seeing is that Trump ' s direct **policies** are going to hurt them economically. His tariffs that he ' s already announced on China, the 10 percent, on a whole bunch of Chinese goods for export, the **reciprocal** tariffs that are global, that will clearly hurt China and that they ' ll have a hard time **negotiating** out of, and also the willingness of the United States to expand their export controls, their sanctions on China, especially in areas that are considered relevant, even loosely relevant to American national security. All of that is going to constrain Chinese growth. All of that is going to negatively **impact** the Chinese economy. That ' s the downside. And the Americans are doing that not only in bilateral relations with China but also pressuring other countries to get Chinese **tech** out of their supply chain. And you see that with Americans pushing other countries that produce semiconductors to take the Chinese out. They ' re pushing American allies and trade partners to get China out of the export chain. So Mexico being pushed very hard by Trump to stop allowing China to pass goods through Mexico into the United States. India, same conversation. Vietnam, same conversation. A lot of that going on. So baseline, the US-China relationship is getting worse. But the Chinese see huge advantages in America ' s reversion to unilateralism that we haven ' t really seen since the original America First movement in the run up to World War II. Charles Lindbergh saying,"Don ' t get into the war, don ' t fight the Nazis. The US is far away. This isn ' t our our problem."And now you have the Americans not only doing that with Russia and Ukraine with the new America First, but you also see that happening with the Americans pulling out of the Paris **Climate** Accord, again, pulling out of the World Health Organization. You see it with the Americans shutting down USAID. And America has historically been responsible for about 40 percent of global humanitarian **support** and aid. And absolutely, some of that is corrupt. Absolutely, some of that is on woke programs that the vast majority of American taxpayers would never **support**. But the majority of that money is actually spent on programs that are really important in advancing American soft power and influence over countries in the Global South, all over the world. The Chinese know that. That ' s why they ' ve started humanitarian programs. It ' s not out of some great love for, you know, providing foreign aid for these countries. It ' s more because they see it creates influence. Now the Americans are leaving a vacuum that the Chinese can absolutely take advantage of. And we ' ll see this across, you know, sort of, microstates in the Pacific. We ' ll see this across sub-Saharan Africa. We ' ll see this across South America. The Chinese see huge opportunities from the United States creating a **leadership** vacuum. And the place they ' re most capable of doing that is if the US stops paying dues in the United Nations, where China is number two, and they ’ ll suddenly have the most influence over all the key positions, which will give them a role in global governance that they ’ ve really been constrained from having over the past decades. HW : What do you think are the chances that the US stops paying its dues to the UN? IB : You know, if it were up to Congress and only Congress and Trump doesn ’ t lean on them, I ’ d say fairly low. I ’ d say that they ’ d probably, you know, cut back on some programs. But in terms of baseline dues, they ’ d keep paying it. But Trump personally, I think increasingly sees the UN as hostile to the United States. That ’ s particularly true in terms of Israel **policy**, which Trump is leaning in on. And, you know – let me put it this way. Tulsi Gabbard did not have the votes to get confirmed. There were Republican senators that were going to vote against her. And Elon called those individual senators and threatened them. And this was, you know, fully aligned with what Trump wanted Elon to do. And those votes went away. And the only person that ended up voting against Tulsi was McConnell. And as a consequence, it went through. So look, I think that we should not underestimate the level of power and control that Trump and Elon have over the Republican Party this time around, compared to 2017, where it was really the Republican Party that was much stronger than Trump, both in his administration creating a lot of guardrails, but also in Congress. This time around, Trump has all the chips and he ’ s using them. So if he decides he wants to stop paying dues, I think that ’ s what ’ s going to happen. This is a much more revolutionary presidency, domestically, where last time around it was much more transactional. HW : Alright. Let ’ s talk a little bit more about what ’ s actually happening inside the US. You bring up Elon, who obviously with DOGE has been sending out emails and asking for information from people and kind of trying to do a lot very quickly. Some of these high-profile moves are – I think what ’ s **interesting** is that no one would really argue that there isn ’ t inefficiency in government, that that ’ s an understood reality. But the way that some of these moves are being made are maybe going to cause irreparable damage. And that ’ s to people from red states, that ’ s to scientists, that ’ s to **veterans**, that ’ s to people who voted for Trump and that ’ s to people around the world. You talk about soft power and the United States, but what about actually what ’ s happening inside the United States? And why do you think that the administration is willing to go so quickly, to go so fast and to cause such damage? IB : Well, and you saw Trump this weekend saying that he wants Elon to move faster, to be more aggressive. And I don ’ t think that was just a troll. I think that ’ s true. Trump is 78 years old, and with his friends he talks about the fact that he doesn ’ t have much time. That in two years ’ time, the Democrats could take the House in four years ’ time, you know, that ’ s it. He ’ s not talking privately of”I ’ m going to be president for life.”He also knows he was almost killed. He was shot in the head. And do I think that changes somebody? Absolutely. I think that he believes that he was saved by God, and that his level of **confidence** that what he is doing is fundamentally right and necessary now is so much higher than the insecurity that I frequently saw in him in his first presidency. So I think his willingness to push, push, push, and his view that Elon is a great activator and **executor** on that intention, and someone that he had nowhere close to someone with that kind of capability and energy and execution **ability** in his first term. And now he does. So first of all, I think that relationship between Trump and Elon isn ’ t going anywhere. I think it ’ s actually very stable. People that say it is are people that want it to go away. But hope is not a strategy. And I also think that the intention of revolutionary politics domestically is to shoot first and ask questions later. It ’ s to break things. It ’ s a view that the bureaucracy has been weaponized against Trump first and foremost, and also against the will of the American people. And therefore it must be broken. And you ’ re going to break **eggs** when you want to make that omelet. Now your point is that there are a lot of **Republicans** that work in government. There are a lot of Republicans and Trump voters that are losing their jobs and that are not being treated with a level of care and compassion as a result of what Trump and Elon are doing. And I think there will be a backlash. But I ’ m not so sure that a one-term, 78-year-old-Trump presidency cares all that much about that. And, you know, if that means that his popularity slips to like, the high 30s, frankly, you know, in a very, very polarized country, high 30s isn ’ t that bad. It still means 80, 90 percent of the people that voted for him still **support** him, and I think he ’ s probably more comfortable with that this time around. So I think a lot more has to happen before we can start talking about whether there ’ s really going to be significant pushback against what Trump is trying to accomplish at home. HW : Do you expect to see any of Congress standing up to Trump? We saw the governor of Maine recently kind of take him on. Do you expect to see more of that, or do you feel like for the next two years at least, people will just go where Trump takes them? IB : Well, sure, but, you know, in the case of Maine, that ’ s a Democrat. So I don ’ t know how much that means. I don ’ t expect to see a lot of **Republicans** standing up against him. And of course, since the Republicans control the House and the Senate, that ’ s the only thing that really matters. I mean, you know, you saw Kash Patel got through, Pete Hegseth got through. Now look, I actually think that a lot of the members of cabinet are very capable. I mean, they ’ ve been picked for their loyalty and they will be loyal to Trump, but they ’ re very capable. I think JD Vance is very capable. Mike Waltz, I know quite well. Marco Rubio, over the years. These are people that are very well respected, have been by their colleagues and could have served under any Republican administration. But there are some people that are completely incapable for the roles that they have been selected, completely incapable. They would not be selected in a role in any cabinet, in any democracy. And yet here they are, serving in critical roles in the most powerful country in the world. And yes, I ’ m talking about the Secretary of Defense. I ’ m talking about the Secretary of Health and Human Services. I ’ m talking about the Director of National Intelligence. I ’ m talking about the Director of FBI. These are important roles. And they were all confirmed. And if you ’ re a Republican senator – and senator – I ’ m not talking about the House here where, you know, it ’ s a lot more diffuse and there are a lot more wackos and it ’ s easier to get in. But Senate has always been like, responsible and more oriented towards bipartisanship and the rest, and they are utterly petrified of what it means if they publicly get crosswise with President Trump and number two, Elon. And they ’ re not willing to do it. So I think that there ’ s very little belief that you are going to see Congress step up. I think that you will see justices step up because those justices are independent. The problem is that many of those justices are progressive and have a known progressive lean. And so when cases come up and justices that have that political lean oppose something Trump wants, his willingness, his administration ’ s willingness to attack them and **refuse** to execute on that judicial decision, unconstitutionally, is real. Now the case will of course go then up to the Supreme Court eventually. And I don ’ t feel the same way about Trump **refusing** to listen to adhere to the rulings of the Supreme Court. But if there ’ s 100 rulings like that and you overwhelm the judicial process, you quickly see how you might erode a core check on executive power in the United States. And Trump and Elon clearly intend to every day test those checks and balances and break them if they can. I mean, that is the intent, that is the intent. HW : So you mentioned democracy.

When we last spoke, you said very clearly that the US is not Hungary, meaning that democracy is not threatened and that the US is not in danger of turning into an autocracy. Do you stand by that? IB : Yes, I do. I mean, in the sense that I think in two years we ’ re going to have midterm elections, and they will be largely free and fair. And in four years we ’ ll have presidential elections. And US elections are held at the federal level, right? They ’ re held state by state. They make the rules. It ’ s very easy to make people believe that the election has been stolen. It ’ s extremely hard to actually rig an American election. And I don ’ t believe that that is very likely at all. It ’ s very unlikely. The American military, despite the firings that we ’ ve seen just last Friday, is still an independent and professional military that I believe will carry out its orders and its duties professionally. I see the judiciary in the United States as independent and a clear check on the executive power in the United States. Where I see a problem, is that the US is becoming more kleptocratic every day. And the US was already much more of a kleptocracy over the past 10 years under Democrats and Republicans than any other advanced industrial democracy in the world. Clearly, money buys power in America in a way that it doesn ’ t in any other major democracy, wealthy democracy. That is getting dramatically worse. I saw specifically that Elon **Musk**, who is a walking **conflict** of interest at the highest level, met with Narendra Modi, the Prime Minister of India, in Blair House, so part of the official White House complex, right after Trump met with him. And Trump was asked later in the day, was Modi meeting with Elon in Elon ’ s private-sector capacity to advance his business interests or his official capacity to advance the **policies** of the United States? And Trump answered honestly. He said,”I don ’ t know.”And how could he know, right? Because, I mean, those two things are completely **intertwined**. But that is not remotely acceptable for rule of law in a functional democracy, and yet nothing is being done about it. In fact, it ’ s getting more and more entrenched every day. So the single place where the United States was least functional as an advanced democracy is now getting far, far worse. And I don ’ t see any checks and balances on that. I think that ’ s a very, very serious problem. HW : Alright, Ian, thank you, as always for all of your insights. Let ’ s close out with a couple more questions from our community. And this one indeed relates to the kleptocratic conversation that you were just raising. So why are the billionaires enabling fascism? What do they get out of it? IB : Fascism is a very politically loaded argument. That ’ s a political system where the cruelty against a subgroup or a perceived subgroup in the society is the intention, the expressed intention of the **policy**. And I certainly wouldn ’ t define the United States as a fascist system today. But I do believe that billionaires in the United States, at least a significant subset of them, are really under-interested in the well-being of their fellow countrymen and women. And this is a country that really makes heroic the individual. And it undermines the community. We ’ ve seen a significant erosion of civic engagement in the United States and civic associations. The church is not what it was in the US. Fewer people go, fewer people **trust** the church. Public education is not what it was. And wealthy people increasingly opt out. The family is not what it was. It ’ s increasingly atomized and people spend much more, much more of their time being, you know, isolated and atomized and intermediated by algorithms. And you and I have spoken about that problem in the past. We are increasingly a society of individuals as opposed to a nation together. I will tell you that that is a real problem when you have large numbers of billionaires that believe that they built it themselves, that they ’ re not responsible to a **collective** whole for any of their success and therefore they ’ re not **accountable** for their fellow citizens. And that ’ s what I think we have. I think we have a lot of winners in the United States and not a lot of

leaders. And when you have winners, you have losers. When you have winners, you don ’ t have a collectivity. I will say that in this anti-DEI thing, I have a lot of sympathy for the people that don ’ t like DEI. I thought it was rammed down the throats of a lot of people And I ’ ve said this before, it was well beyond what the average American was willing to tolerate, and also it came with a level of high-handedness that that basically said, if you ’ re not with us in **supporting** these programs, you ’ re evil, you ’ re **stupid**, you ’ re uneducated. That ’ s unacceptable, too. It was a completely, you know, sort of oppositional way to try to have what ’ s a very challenging conversation. But I ’ ve never been a champion of **diversity** for **diversity** ’ s sake. What ’ s amazing about the history of the United States is despite all of our **diversity**, we find **commonalities** with each other. We find connectedness. That ’ s what ’ s amazing, is the connectedness, is that, you know, on this world, we ’ ve got eight billion people. And despite how different we all are, we all actually connect as human beings. That is what we need to focus on, not the **diversity**, the connectedness. And I think that ’ s what the billionaires today in the United States are losing. They ’ re doing incredibly well themselves. They ’ ve got theirs, they ’ ve got access to power, they ’ re paying good money for it, they ’ re going to get even more. But they ’ re forgetting that for us to succeed as a society and as a planet, we have to be connected to each other. You can ’ t throw out **diversity** and forget connectedness, and that ’ s what we ’ re forgetting right now. That ’ s why the country feels so much trauma and pain right now. And I think why the United States has given up on a lot of its really aspirational values that made such a difference to the world, that almost lost World War II for many generations. And now we ’ re giving it all away. HW : I mean, I think that ’ s why you find that **diversity** and **inclusion** go hand in hand, that that ’ s actually speaking to that. But in some of the moves that are being made to kind of throw out any type of **diversity**, to scan **documents**, to find words in order to highlight programs that are wasteful, that we ’ re actually going to lose what many, many studies actually highlight as being the benefits of **diversity** and the fact that you need **diversity** in order to be actually successful. IB : Shooting first and asking questions later, you know, moving fast and breaking things, those things make sense in a **turbocharged** venture capital technology environment. They do not work in government. They don ’ t work in government at all. In government, we have something called checks and balances, and we need them. And moving fast and breaking things undermines those checks and balances. So we ’ re going to do a lot of damage unintentionally. A lot of people are going to get hurt. A lot of folks don ’ t care about that in power right now, but they should because it would be much, much better. I know people want to move fast, but actually to think a little bit, do a little bit of your own research before you decide to kill that program, before you decide to fire that person. We ’ d be in a lot better shape as a government right now. HW : OK, the final question that I have for you actually comes from my 10-year-old son, and it ’ s a bit of a miserable question. We have a very lovely house, I promise. But his question nonetheless is are we heading into World War III? So, Ian, what should I tell him? IB : No, no, I don ’ t think we ’ re heading into World War III but we are absolutely heading into a world of much greater **conflict**. I think it ’ s much more likely, for example, in the next five, 10 years, that we ’ ll have a nuclear weapon go off. And I thought that was pretty unlikely since the Cuban Missile **Crisis**. In other words, over the course of my life. I think we ’ re going to see a lot more terrorism at scale, because a lot more people are going to feel angry and disaffected, I think. You know, you saw that assassination of the CEO of UnitedHealthcare just a few dozen blocks from my house. I think you ’ re going to see more things like that. And so, no, not World War III because the fact is that outside of the US and China, all the other countries know that they need to work with **everybody**. And there are even forces inside US-China that are trying to make sure that you don ’ t throw the baby out with the bathwater. But underneath that World War III risk, there ’ s an awful lot of damage that ’ s going to get done, an awful lot of **crisis** that I think we ’ re going to have to live through before we start to see what we create on the other side. A lot of defense as well, that we ’ re going to be playing in this environment. HW : Well, Ian, it is always a pleasure to talk to you, and we ’ re so glad that you are scanning the world to understand what is happening and then you come here and explain it to us all. Couldn ’ t be more grateful and thank you for everything. IB : Thank you.

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Can you paint with all the colors of the wind? I love this song from the first time I heard it. I was 14, growing up in public housing in western Sydney. So maybe a little too old for this Disney classic from”Pocahontas.”But it spoke to me. I saw something of myself in it. At this stage, I think I knew I was queer. I mean, come on, I was rocking out to Disney. And look at those glasses. Seriously, what a queer kid. But I was also from a Pacific Indigenous heritage with **conservative** views and values. I didn ’ t feel like I could fit into any of the boxes that were being presented to me. Straight, gay, Fijian, Australian. I couldn ’ t pick just one color to paint with. Prior to colonization, sexuality amongst many of our Pacific islands across the region was fluid. Men would have sex with men without the stigma or shame afforded it today. Sexual intimacy was a way of being able to connect socially and relationally. Sex wasn ’ t just for procreation. Or in marriage. It wasn ’ t exclusively monogamous. Just because you had sex with a man didn ’ t define your sexuality. It didn ’ t make you an outcast. But as Europeans started to colonize the Pacific region, they brought Christian missionaries with them. They told us to cover up our beautiful bodies. We were taught shame. We were told to stop practicing Pacifica cultures and values that were seen as provocative. To a Western and white perspective, our non-binary sexuality was judged as tribal, outdated, sinful and impossible to practice alongside a Christian faith. Still today, those who are Indigenous to the Pacific are negatively **impacted** by whiteness. That”white is right, West is best.”I was brought up to espouse strong Christian beliefs, views and values which would shape my burgeoning identity as an adult. From these staunch **conservative** views, I was taught to fear the queer. Those people that were seen as unusual and fruity. And in essence, I grew to despise a part of myself. I sought to separate my queerness from my identity. I could never see myself as part of that world. Growing up, biracial and bisexual constantly challenged the way I saw the world around me. I was unsure about my thinking, feelings and behaviors around liking males and females. I was unsure about my thinking, feelings and behaviors when it came to whether I needed to be a white Australian or a Fijian person. And whether this needed to be framed within a binary to choose one over the other. In this way, I don ’ t think that my story is unique. There ’ s plenty of people in the queer world that have escaped **conservative** upbringings, religions and cultures. But this is only part of my story. Another common story in the queer community is the coming-out story. When did you tell your parents? How did they react? How it liberated you or ultimately separated you from your family. But this didn ’ t feel like an option for me either. I believe that embedded in these particular stories, these tropes, is a part of **dominant** queer culture that made it very hard for me to access the

queer world. For example, when I started to go to gay clubs, I would be lining up at the front of the club and often I would get door staff come up to me and go,"Sorry, you do know this is a gay club, right?"From these lived experiences, I feel that there is a whiteness that pervades queer culture. And I didn ' t want to give up my identity and my appearance as a Pacific indigenous person to be queer. Pacifica cultures really, really value and esteem the **collective** above the individual. For Pacifica people, our families are inextricably connected to our own sense of self-agency, **self-determination**, self-esteem, self-love and self-worth. Our own individual identities are meshed and **intertwined** with our family ' s reputation and standing in the community. Our mind, body and souls are inextricably connected to the broader community, to our families, to ourselves and to others. I didn ' t want to come out to potentially ostracize myself, to make a point to separate myself from my Pacific and faith communities. I couldn ' t then see my authentic self in any of these spaces. And a quick shout out to my son on the left-hand side. Round of applause for my son. He said to me this morning,"Daddy, make sure you mention me."So I was like,"OK, son. Alright."So there you go for the recording, son. And the rest are my other family, that ' s up there as well. Across queer cultures and the broader community, we have learned to embrace our **gender** and sexual differences. We encourage individuals to be allies for each other and to **support safe** spaces for those that are seen to not fit in. We empower and really encourage individuals to really develop and express their self-agency, their **self-determination**, their self love, their self-worth. We boldly go into spaces striving to bring others with us to **ensure** that we can create inclusive societies for everyone. However, the **dominant** Western view in queer cultures is still based on the individual. Me, myself and I. And my individual pursuits towards happiness. As a queer Indigenous Pacific person, I feel that there are parts of queer culture that is white, feels very white. We may fail to see the importance of bringing our families and our cultural communities along our journey as queer people. It ' s either the queer way or the highway. It ' s one or the other. Young or old. Rich or poor. Smart or **stupid**. Masculine or feminine. Fit or fat. These binaries create a sense of us and them, those people. A binary approach and position that we generally in Western white societies use to determine the norm. We create, conflate, perpetuate binaries that create divisions in our society. Saints or sinner, straight or gay. I see it like this. Binaries are a very white and Western way of thinking, feeling and behaving. As individuals today, we live within these binaries and create identities that then create these labels. We use these labels to then define and confine people and places. We are one or the other. But what we fail to see is that our identities are more than a set of binaries. Our own identities are comprised of intersecting identifiers that contributes to who you are, including our age, class, color, education, **gender**, size, spirituality, sexuality, and so, so much more. We need to decolonize queerness, to live beyond"white is right, West is best."Because if we don ' t, we ' re going to continue to create silos and divisions in our societies that don ' t create inclusive and **safe** spaces for everyone. For Indigenous queer people, they shape our connection to community and culture, our families. If we don ' t include that within our understanding of who we are, even as queer people, then we ' re not able to be, again, our true and authentic selves. We ' re not able to then be our fabulous selves as part of the queer community. And it ' s really, really **interesting** because I believe that we can learn so much from pre-colonial cultures like those that were practiced in the Pacific, that provide a different way of looking at a society. A queer way that pre-dates all the privileges that go into binary ways of thinking, feeling and behaving. Across many of our Pacifica languages we have a lot of key concepts that help solidify and **support** a shared understanding of the **collective**. For example, in Fijian,”solesolevaki”is all about **reciprocal** living, where my wellbeing is your wellbeing. I will actively seek to **ensure** that your well-being and every part of your life is catered for, knowing that you will do the same for me. We strive to enact”veiqaravi,”as tatted here on my hand, to serve others. We strive to also embed and embold and empower people to practice”veirokovi,”respect for self and others where we are relationally driven, where that is so important to our spaces and places. My sense of **ability** to actually put this into practice, my **ability** to actually understand these Indigenous perspectives has greatly helped shape and reconcile my queer identity today. But what ’ s really **interesting** is that it ’ s not just queer communities that need to decolonize itself. It ’ s all of us as a broader society that needs to be part of a shared conversation to decolonize, deconstruct and disrupt binary ways of thinking, feeling and behaving. So how can we practically do that? Practice cultural humility, where you are constantly learning and embracing differences, where you get over yourselves and strive to learn about other people ’ s differences and how that can be meaningfully included in the way in which you create **safe** spaces for everyone. I also believe it ’ s possible to bring Indigenous perspectives into Western and white workplaces. For example, I ’ ve introduced the concept of”talanoa”in all of my work meetings. Talanoa is all about being able to create a **safe** space where everyone is able to contribute to a **collective**, collegial and collaborative conversation. Where people are able to bring their authentic selves into these spaces. Literally, I will chair meetings where I have no firm set agenda items and I literally would just hold space for participants just to bring themselves. So you can tell I ’ m a social worker, right? **Safe** space, **safe** space, **safe** space. It ’ s OK, it ’ s OK. But it ’ s through these **safe** spaces that people can bring their multiple, intersecting identifiers, including their age, their class, color, education, **gender**, size, spirituality, sexuality and so, so much more. I invite queer communities to come talanoa with Pacific Indigenous Communities to create robust and resilient queer cultures. I have been able to greatly reconcile my queer and Pacifica identities, which has now really been embraced to help shape how I see the world around me today. And it ’ s through these shared perspectives and approaches that we can truly embrace cultural **diversity** and its differences. And to know that people can live beyond those binaries and those labels we continue to perpetuate and perpetrate against others. It is a commitment from queer communities to decolonize queerness. It ’ s a commitment from me, from you and all of us to decolonize Western and white societies and to live beyond the binary way of thinking, feeling and behaving. I genuinely believe that it ’ s through this common goal that we can truly learn how to paint with all the colors of the wind.”Vinaka vakalevu”and thank you.

Text Sample 503: POS Dim 4 – 2025 – Score 7.00 – 2025/t003001.txt

- I don ’ t remember exactly, but... I use the most expensive hair color in the world. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color. I expect great color. What ’ s worth more to me is the way my hair feels. Feels good around my neck. I can ’ t remember it. I can ’ t finish it all. If only we had more time. - Why don ’ t we have more time? - I ’ m dying. I ’ m croaking! - It ’ s tough, coming home. But this is what you do for a parent that you love. But, you know? She ’ s not my mother. Technically. In my own family, there ’ s a lot of **dysfunction**, a lot of mental illness, a lot of substance abuse. Every generation. My parents divorced when I was about five. My mother moved us out of state, so we had to fly to New York to see my father ’ cause he was already working in advertising, and New York was the only place to be. - Many of the commercials you see on television begin life here. - Advertising is one of the main sources of messaging about who you should be. - Nowadays, most of you women lead two lives, combining homemaking with the role of glamorous hostess. - And in the ’ 60s, the ads were often made from a male point of view. - In the pre-production meeting, **everybody** who has anything to do with the commercial is there. - It was mostly men. These superstar art directors and writers. Like my father, Gene Case.”Mad Men”was like home movies, and he was Don Draper. He was very witty, but I don ’ t think my father realized how powerful his words were. He knew it professionally. But with his kids, he could be incredibly cutting. Flay you just with a phrase. But it ’ s the time with my mother that was the nightmare. She had a horrible childhood, but she sure as hell didn ’ t make any effort to overcome that in her relations with us. You know, it was really horrifying to my mother that I was not popular. But I was also really awkward because she had been undermining my **confidence** for all my life. She would pick apart your personality in detail. I was lazy, I looked bad, I acted bad, I was awkward. Emotionally, just out for blood. At a certain point in the night, I would know,”OK, the situation is dangerous. I need to get outta here.”I had this fantasy when I was in fourth grade that I vividly remember that some kind of outside agency was gonna come in and determine from all the salt on my pillow that I was crying myself to sleep every night and do something about it. No, it was awful. It was awful. And that was the point where Ilon came into my life. - Ilon, do you need anything to eat? - I don ’ t think so, but look up this wallpaper, Waverly Fabrics, and see if they have a rose pattern. - OK. Oh, what about this one here? - That ’ s not it. - No? - No, first of all, I was an art dir... I was a creative director at an advertising agency. The first thing I know is what I ’ m looking at. Ilon the girl. I don ’ t remember particularly being into advertising, but somehow I found myself in it. And I seemed to be very good at it. First of all, I ’ d always listened to **everybody**. And I noticed that I saw things differently than other people. - Women could get somewhere in advertising, but they were marginalized. - Women were given a lesser role. All the men were always arguing with you and always taking credit. - And one of the things that was coming through loud and clear is that women are for men. - Glory, glory, hallelujah Glory, glory, hallelujah Glory, glory, hallelujah His truth is marching on - Men were always the ones who everyone was pleasing. - What will your eyes say to him tonight? Wait till the fellas take a look at this plunging backline. - You look at ads from that period, and it ’ s jaw-dropping. - This is one of our nation ’ s most beautiful sights, the kind of girl who keeps her figure. - But that ’ s what feminism was pushing back against. - I was a feminist. I knew women were equal if not superior to men from the time I was a little girl. - And in advertising, she wanted to be a trailblazer that way. At one point, she worked in the same agency with my father, and that ’ s how they met. - He was a good writer, and I was a good writer. I liked the fact that he was good at what he did and cared about it. We got engaged, and then we got married. - I ’ m sitting there in the background. I ’ m grinning. I ’ m so thrilled because I adored her instantly. I thought she was super cool. And she treated me like I mattered. - She was a sweet little girl, and she deserved some caretaking. My mother was kinda terrible, so I knew that she needed **support**, and I gave her **support**. And that was what we did. - She was **interested** in me. She listened to me. Treated me like I was lovable. - She didn ’ t have anyone to recognize her being, to **support** her, to see who she could be, who she was. - She was very **interested** in what I would do with my life, and the assumption was that it would be something worthwhile. She just went way, way beyond the call of duty to make me feel like I was worth it. And she did that for a lot of other women, too. - I was working at McCann Erickson. We were given an assignment. L ' Oréal, they wanted a campaign for Preference, it ' s a hair color. - She ' s in this room full of men brainstorming for this ad. - The men were busy flirting with the women, telling us we all looked like models and we didn ' t look like advertising people, you know? - They ' re like,"Oh, this woman is gonna be standing by an open window, and the wind will be blowing the curtains."And she ' s thinking, you know, this is just making the woman totally an object. - I was feeling angry. I ' m not **interested** in writing anything about looking good for men. Fuck ' em. Fuck you too. - Why fuck me? - Well, you ' re a man. - Yeah, but I ' m here trying to help you tell your story. - Yeah, that ' s good. I like that part. I was pissed. We weren ' t there to just dance for the men. - It is what she said. She just thought,"Fuck you."And wrote this. - I just opened my eyes and knew. Because I ' m worth it. The campaign got approved, but all the men were always telling you that you were doing something wrong or they knew how to do it better. - The agency insisted on a male voiceover. - She uses the most expensive hair color in the world, Preference by L ' Oréal. - You know, it was just totally different. - Smooth and silky, but with body. It feels good against her neck. Actually, she doesn ' t mind spending more for L ' Oréal. Because she ' s worth it. - It was wrong. This was not for men but for women and for other human beings. - I use the most expensive hair color in the world, Preference by L ' Oréal. It ' s not that I care about the money, it ' s that I care about my hair. It ' s not just the color, I expect great color. What ' s worth more to me is the way my hair feels. Smooth and silky, but with body. It feels good against my neck. Actually, I don ' t mind spending more for L ' Oréal. Because I ' m worth it. - I wrote that. - It was everywhere. It was a national campaign. - Yes, it ' s expensive, but I ' m worth it. - What they realized after they started running it was that that was only the tip of the iceberg. - It was **supporting** women, finally. - And I ' m worth it. And I ' m worth it. I ' m worth it. - It ' s a four-word feminist manifesto, defying this social pressure to think about yourself in relation to men. - You ' re worth it. I ' m worth it. I ' m worth it. - It was a very big success. - At a certain point, L ' Oréal adopted it as the tagline for their entire business. - Parce que je le vaux bien. - Even today, that ' s their motto. - Because I ' m worth it. - ' Cause you ' re worth it. - You ' re worth it. - We ' re worth it. - And I ' m worth it. - She wrote the most influential tagline in women ' s beauty advertising. - I ' m worth it. - It ' s magic, that phrase. - Those words just get better with age, don ' t they? - I ' m worth it. Because you are and always will be. - She came up with that line because she believed it. She believed it about herself, and she believed that women should believe it. She did that for me. You know, even after they split up, she was still a big part of my life. I ' ve always kind of felt like I dodged a bullet that I have watched take down other members of my family. She saved me. And today, I am a professor of English at Williams College. I teach Victorian literature, and I also teach women ' s, **gender** and sexuality studies. She was my mother. She also taught me how to be a mother. I had been very uncertain about motherhood for myself because my role model for that was so nightmarish. But Ilon really taught me, I can do this. I can be a good mother. She made me believe that I was worth it. And I was not either the first or the last. - I have to go away now. Go to my room. - It ' s been lovely. - It ' s been lovely. Oh... I ' m not **interested** in advertising. I don ' t give a shit. It wasn ' t like it was my whole life. It was just a portion. It ' s about humans ; it ' s not about advertising. It ' s about caring for people because... we ' re all worth it, or no one is worth it.

Text Sample 504: POS Dim 4 – 2025 – Score 5.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ' re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ' s such a strange time in the world. There is so much beauty amidst so much fear. It ' s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ' re able to make things with our hands. A love letter to New York City by Debbie Millman I ' m a native New Yorker. I ' ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ' s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ' m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this **crisis**, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the **air**. In all my travels, I ' ve been struck by so many things. I ' ve observed how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ' ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced us to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ' s hard to see the big picture right now. It ' s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think back to my travels. I ' ve seen how different we are, how diverse and distinct. But I ' ve also seen our **commonalities**. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ' m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman"A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story is the **dominant** sound of our lives."- Joseph Campbell I ' ve loved telling stories all of my life. I ' ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 505: POS Dim 4 – 1990 – Score 10.00 – 1990/t000287.txt

I ’ m going to go right into the slides. And all I ’ m going to try and prove to you with these slides is that I do just very straight **stuff**. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the left-hand side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage the neighborhood â one made accommodations. I tried to be inclusive, to include the buildings in the neighborhood, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of **stuff** that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of **stuff** and folded it up to put it in the wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at postmodernism â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker.

And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that **stuff**. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And the client said,"Why no fish?"And so I made a drawing with a fish, and I left Japan. Three weeks later, I received a complete set of drawings saying we ' d won the competition. Now, it ' s hard to do. It ' s hard to translate a fish form, because they ' re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ' t do it, and so that made it even more exciting. But he was right â I couldn ' t do the tail. I started to get the head OK, but the tail I couldn ' t do. It was pretty hard. The thing on the right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with the context again. Now, if you saw a picture of this as it was published in Architectural Record â they didn ' t show the context, so you would think,"God, what a pushy **guy** this is."But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ' t find it. So... As for craft and technology and all those things that you ' ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent drawings to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made the drawings of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ' s one of the nicest pieces I ' ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ' t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ' m getting good at temporary â and we put a **conference** room in that ' s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ' s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ' t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ' s offices and I made a pedestal for a sculpture. And he didn ' t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ' s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ' ve been friends for a long time. And we ' ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called"Il Corso del Coltello"â the Swiss Army knife. And most of the imagery is â Claes ', but those two little boys are my sons, and they were Claes ' assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ' s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ' ve been known to mess with things like chain link fencing. I do it because it ' s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ’ s so much denial about them. People hate it. And I ’ m fascinated with that, which, like the paper furniture â it ’ s one of those materials. And I ’ m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ’ s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante **stuff** that became pretty **interesting**. It led to the building on the left. And I still think the Time magazine picture will be of the binoculars, you know, leaving out the â what the hell. I use a lot of metal in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,”If I can ’ t get the craft that I want, I ’ ll use the craft I can get.”There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the metal because it was a way of building a building that was a sculpture. And it was all of one material, and the metal could go on the roof as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and **stuff**. I was given an opportunity to design an exhibit for the metalworkers ’ unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future metal buildings, etc. etc. And it ’ s working very well to have these people, these craftsmen, **interested** in it. I just tell the stories. It ’ s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The metal continued into a building â Herman Miller, in Sacramento. And it ’ s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ’ s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ’ s out in the middle of nowhere, and you approach it. It ’ s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ’ s aesthetic. **Everybody** ’ s trying to get the panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ’ s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ’ s all about [a] metaphor for a city, maybe. And so Stanley did the little dome thing. And we did it over the **phone** and by fax. He would send me a fax and show me something. He ’ d made a building with a dome and he had a little tower. I told him,”No, no, that ’ s too ongepotchket. I don ’ t want the tower.”So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going back and forth over a couple of weeks ’ period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,”David and Goliath.”And that ’ s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and cleaned it up and fixed it up and used the kind of â I thought â the language of the neighborhood, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ’ t normally think I would fit them. The detailing was very careful with the lead copper and making panels and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some film festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â a little shopping center. And this is a laser laboratory at the University of Iowa, in which the fish comes back as an abstraction in the back. It ’ s the **support** labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the curved part there ’ s all the mechanical equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ’ t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ’ ve been building it so long I don ’ t remember where it is. It ’ s in the West Valley. And we started with the stream and built the house along the stream â dammed it up to make a lake. These are the models. The reality, with the lake â the workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ’ s hard to get perfect corners and **stuff**. That big metal thing is a passage, and in it is â you go downstairs into the living room and then down into the bedroom, which is on the right. It ’ s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â they asked Philip to do it. He was too busy. He didn ’ t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ’ t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ’ ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ’ s made of metal and the brown **stuff** is Fin-Ply â it ’ s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ’ t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ’ s Burnham Mall, on the left. It ’ s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ’ s a railroad track. This is the city hall over here somewhere, and the courthouse. And the centerline of the mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ’ re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this **stuff** at the bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the centerline of this thing â we ’ d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ’ s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ’ 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the back end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake. And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ’ s the living room. Servants ’ quarters. A guest bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is back in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a fun thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it ain ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the **conference** facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais des Sports, the Gare de Lyon over here. The Minister of Finance â the **guy** that moved from the Louvre â goes in here. There ’ s a new library across the river. And back in here, in this already treed park, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan coupe. And I struggled with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in French limestone, except for this metal piece. And it faces into a park. And the idea was to make this express the energy of this. On the side facing the street it ’ s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you **guys** have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in the Hamptons. And it ’ s got a fish. And I keep thinking,”This is going to be the last fish.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s the living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in the budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of the **guys** that presented to you earlier. We ’ ve had a lot of fun working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ’ s perpendicular to the train station, to the route of the train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good slides to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â□□ you see it as all of these pieces accrue and become part of an overall neighborhood. It ’ s plaster and just zinc. And you wonder, if this is a museum, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â□□ I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and **stuff**. And from the building in the back, you see it as an iceberg floating by in the hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is the Disney Hall â□□ the concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a compositional relationship between us that would strengthen both of us. And the plan of this â□□ it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â□□ to MOCA, etc. The acoustician in the competition gave us criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But **everybody** liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were the three buildings that were the ideal â□□ the Concertgebouw, Boston and Berlin. **Everybody** liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â□□ and when we met our new acoustician, he told us this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an acoustical, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the sloping outside walls, which the acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â□□ these are just on the way to being â□□ and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with laser **stuff**, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â□□ and here we are, struggling with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the foot of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the foot of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th- century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ’ t as good as the **stuff**, and so it failed. It should have been the other way around â the food should have been good first. It didn ’ t work. Thank you very much.

Text Sample 506: POS Dim 4 – 2006 – Score 23.00 – 2006/t000494.txt
I can ’ t help but this wish : to think about when you ’ re a little kid, and all your friends ask you,”If a genie could give you one wish in the world, what would it be?”And I always answered,”Well, I ’ d want the wish to have the wisdom to know exactly what to wish for.”Well, then you ’ d be screwed, because you ’ d know what to wish for, and you ’ d use up your wish, and now, since we only have one wish — unlike last year they had three wishes — I ’ m not going to wish for that. So let ’ s get to what I would like, which is world peace. And I know what you ’ re thinking : You ’ re thinking,”The poor girl up there, she thinks she ’ s at a beauty pageant. She ’ s not. She ’ s at the **TED** Prize.”But I really do think it makes sense. And I think that the first step to world peace is for people to meet each other. I ’ ve met a lot of different people over the years, and I ’ ve filmed some of them, from a dotcom executive in New York who wanted to take over the world, to a military press officer in Qatar, who would rather not take over the world. If you ’ ve seen the film”Control Room”that was sent out, you ’ d understand a little bit why. Thank you. Wow! Some of you watched it. That ’ s great. That ’ s great. So basically what I ’ d like to talk about today is a way for people to travel, to meet people in a different way than — because you can ’ t travel all over the world at the same time. And a long time ago — well, about 40 years ago — my mom had an exchange student. And I ’ m going to show you slides of the exchange student. This is Donna. This is Donna at the Statue of Liberty. This is my mother and aunt teaching Donna how to ride a bike. This is Donna eating ice cream. And this is Donna teaching my aunt how to do a Filipino dance. I really think as the world is getting smaller, it becomes more and more important that we learn each other ’ s dance moves, that we meet each other, we get to know each other, we are able to figure out a way to cross borders, to understand each other, to understand people ’ s hopes and dreams, what makes them laugh and cry. And I know that we can ’ t all do exchange programs, and I can ’ t force **everybody** to travel ; I ’ ve already talked about that to Chris and Amy, and they said that there ’ s a problem with this : You can ’ t force people, free will. And I totally **support** that, so we ’ re not forcing people to travel. But I ’ d like to talk about another way to travel that doesn ’ t require a ship or an airplane, and just requires a movie camera, a projector and a screen.

And that ’ s what I ’ m going to talk to you about today. I was asked that I speak a little bit about where I personally come from, and Cameron, I don ’ t know how you managed to get out of that one, but I think that building bridges is important to me because of where I come from. I ’ m the daughter of an American mother and an Egyptian-Lebanese-Syrian father. So I ’ m the living product of two cultures coming together. No pun intended. And I ’ ve also been called, as an Egyptian-Lebanese-Syrian American with a Persian name, the”Middle East Peace **Crisis**.”So maybe me starting to take pictures was some kind of way to bring both sides of my family together — a way to take the worlds with me, a way to tell stories visually. It all kind of started that way, but I think that I really realized the power of the image when I first went to the garbage-collecting village in Egypt, when I was about 16. My mother took me there. She ’ s somebody who believes strongly in community service, and decided that this was something that I needed to do. And so I went there and I met some amazing women there. There was a center there, where they were teaching people how to read and write, and get vaccinations against the many diseases you can get from sorting through garbage. And I began teaching there. I taught English, and I met some incredible women there. I met people that live seven people to a room, barely can afford their evening meal, yet lived with this strength of spirit and sense of humor and just incredible qualities. I got drawn into this community and I began to take pictures there. I took pictures of weddings and older family members — things that they wanted memories of. About two years after I started taking these pictures, the UN Conference on Population and Development asked me to show them at the **conference**. So I was 18 ; I was very excited. It was my first exhibit of photographs and they were all put up there, and after about two days, they all came down except for three. People were very upset, very angry that I was showing these dirty sides of Cairo, and why didn ’ t I cut the dead donkey out of the frame? And as I sat there, I got very depressed. I looked at this big empty wall with three lonely photographs that were, you know, very pretty photographs and I was like,”I failed at this.”But I was looking at this intense emotion and intense feeling that had come out of people just seeing these photographs. Here I was, this 18-year-old pipsqueak that nobody listened to, and all of a sudden, I put these photographs on the wall, and there were arguments, and they had to be taken down. And I saw the power of the image, and it was incredible. And I think the most important **reaction** that I saw there was actually from people that would never have gone to the garbage village themselves, that would never have seen that the human spirit could thrive in such difficult circumstances. And I think it was at that point that I decided I wanted to use photography and film to somehow bridge gaps, to bridge cultures, bring people together, cross borders. And so that ’ s what really kind of started me off. Did a stint at MTV, made a film called”Startup. com,”and I ’ ve done a couple of music films. But in 2003, when the war in Iraq was about to start, it was a very surreal feeling for me, because before the war started, there was kind of this media war that was going on. And I was watching television in New York, and there seemed to be just one point of view that was coming across, and the coverage went from the US State Department to embedded troops. And what was coming across on the news was that there was going to be this clean war and precision bombings, and the Iraqis would be greeting the Americans as liberators, and throwing flowers at their feet in the streets of Baghdad. And I knew that there was a completely other story that was taking place in the Middle East, where my parents were. I knew that there was a completely other story being told, and I was thinking,”How are people supposed to communicate with each other when they ’ re getting completely different messages, and nobody knows what the other ’ s being told? How are people supposed to have any kind of common understanding or know how to move together into the future? So I knew that I had to go there. I just wanted to be in the center. I had no plan. I had no funding. I didn ' t even have a camera at the time — I had somebody bring it there, because I wanted to get access to Al Jazeera, George Bush ' s favorite channel, and a place which I was very curious about because it ' s disliked by many governments across the Arab world, and also called the mouthpiece of Osama Bin Laden by some people in the US government. So I was thinking, this station that ' s hated by so many people has to be doing something right. I ' ve got to go see what this is all about. And I also wanted to go see Central Command, which was 10 minutes away. And that way, I could get access to how this news was being created — on the Arab side, reaching the Arab world, and on the US and Western side, reaching the US. And when I went there and sat there, and met these people that were in the center of it, and sat with these characters, I met some surprising, very complex people. And I ' d like to share with you a little bit of that experience of when you sit with somebody and you film them, and you listen to them, and you allow them more than a five-second sound bite. The amazing complexity of people emerges. Samir Khader : Business as usual. Iraq, and then Iraq, and then Iraq. But between us, if I ' m offered a job with Fox, I ' ll take it. To change the Arab nightmare into the American dream. I still have that dream. Maybe I will never be able to do it, but I have plans for my children. When they finish high school, I will send them to America to study there. I will pay for their study. And they will stay there. Josh Rushing : The night they showed the POWs and the dead soldiers — Al Jazeera showed them — it was powerful, because America doesn ' t show those kinds of images. Most of the news in America won ' t show really gory images and this showed American soldiers in uniform, strewn about a floor, a cold tile floor. And it was revolting. It was absolutely revolting. It made me sick at my stomach. And then what hit me was, the night before, there had been some kind of bombing in Basra, and Al Jazeera had shown images of the people. And they were equally, if not more, horrifying — the images were. And I remember having seen it in the Al Jazeera office, and thought to myself,"Wow, that ' s gross. That ' s bad."And then going away, and probably eating dinner or something. And it didn ' t affect me as much. So, the **impact** that had on me — me realizing that I just saw people on the other side, and those people in the Al Jazeera office must have felt the way I was feeling that night, and it upset me on a profound level that I wasn ' t as bothered as much the night before. It makes me hate war. But it doesn ' t make me believe that we ' re in a world that can live without war yet. Jehane Noujaim : I was overwhelmed by the response of the film. We didn ' t know whether it would be able to get out there. We had no funding for it. We were incredibly lucky that it got picked up. And when we showed the film in both the United States and the Arab world, we had such incredible **reactions**. It was amazing to see how people were moved by this film. In the Arab world — and it ' s not really by the film, it ' s by the characters — I mean, Josh Rushing was this incredibly complex person who was thinking about things. And when I showed the film in the Middle East, people wanted to meet Josh. He kind of redefined us as an American population. People started to ask me,"Where is this **guy** now?"Al Jazeera offered him a job. And Samir, on the other hand, was also quite an **interesting** character for the Arab world to see, because it brought out the complexities of this love-hate relationship that the Arab world has with the West. In the United States, I was blown away by the motivations, the positive motivations of the American people when they ' d see this film. You know, we ' re criticized abroad for believing we ' re the saviors of the world in some way, but the flip side of it is that, actually, when people do see what is happening abroad and people ' s **reactions** to some of our **policy** abroad, we feel this power, that we need to — we feel like we have to get the power to change things. And I saw this with audiences. This woman came up to me after the screening and said,"You know, I know this is crazy. I saw the bombs being loaded on the planes, I saw the military going out to war, but you don ' t understand people ' s anger towards us until you see the people in the hospitals and the victims of the war, and how do we get out of this bubble? How do we understand what the other person is thinking?"Now, I don ' t know whether a film can change the world. But I know the power of it, I know that it starts people thinking about how to change the world. Now, I ' m not a philosopher, so I feel like I shouldn ' t go into great depth on this, but let film speak for itself and take you to this other world. Because I believe that film has the **ability** to take you across borders, I ' d like you to just sit back and experience for a couple of minutes being taken into another world. And these couple clips take you inside of two of the most difficult **conflicts** that we ' re faced with today. [The last 48 hours of two Palestinian suicide bombers.] [Paradise Now] [Man : As long as there is injustice, someone must make a sacrifice!] [Woman : That ' s no sacrifice, that ' s revenge!] [If you kill, there ' s no difference between victim and occupier.] [Man : If we had airplanes, we wouldn ' t need martyrs, that ' s the difference.] [Woman : The difference is that the Israeli military is still stronger.] [Man : Then let us be equal in death.] [We still have Paradise.] [Woman : There is no Paradise! It only exists in your head!] [Man : God forbid!] [May God forgive you.] [If you were not Abu Azzam ' s daughter...] [Anyway, I ' d rather have Paradise in my head than live in this hell!] [In this life, we ' re dead anyway.] [One only chooses bitterness when the alternative is even bitterer.] [Woman : And what about us? The ones who remain?] [Will we win that way?] [Don ' t you see what you ' re doing is destroying us?] [And that you give Israel an alibi to carry on?] [Man : So with no alibi, Israel will stop?] [Woman : Perhaps. We have to turn it into a moral war.] [Man : How, if Israel has no morals?] [Woman : Be careful!] [And the real people building peace through non-violence] [Encounter Point] Video : [Tel Aviv, Israel 1996] [Tzvika : My wife Ayelet called me and said,] ["There was a suicide bombing in Tel Aviv."] [Ayelet : What do you know about the casualties?] [Tzvika off-screen : We ' re looking for three girls.] [We have no information.] [Ayelet : One is wounded here, but we haven ' t heard from the other three.] [Tzvika : I said,"OK, that ' s Bat-Chen, that ' s my daughter.] [Are you sure she is dead?"] [They said yes.] Video : [Bethlehem, Occupied Palestinian Territories, 2003] [George : On that day, at around 6:30] [I was driving with my wife and daughters to the supermarket.] [When we got to here...] [we saw three Israeli military jeeps parked on the side of the road.] [When we passed by the first jeep...] [they opened fire on us.] [And my 12-year-old daughter Christine] [was killed in the shooting.] [Bereaved Families Forum, Jerusalem] [Tzvika : I ' m the headmaster for all parts.] [George : But there is a teacher that is in charge?] [Tzvika : Yes, I have assistants.] [I deal with children all the time.] [One year after their daughters ' deaths both Tzvika and George join the forum] [George : At first, I thought it was a strange idea.] [But after thinking logically about it,] [I didn ' t find any reason why not to meet them] [and let them know of our suffering.] [Tzvika : There were many things that touched me.] [We see that there are Palestinians who suffered a lot, who lost children,] [and still believe in the peace process and in reconciliation.] [If we who lost what is most precious can talk to each other,] [and look forward to a better future,] [then everyone else must do so, too.] [From South Africa : A Revolution Through Music] [Amandla] Man : Song is something that we communicated with people who otherwise would not have understood where we ' re coming from. You could give them a long political speech, they would still not understand. But I tell you, when you finish that song, people will be like,”Damn, I know where you niggas are coming from. I know where you **guys** are coming from. Death unto apartheid!”Narrator : It ’ s about the liberation struggle. It ’ s about those children who took to the streets — fighting, screaming,”Free Nelson Mandela!”It ’ s about those unions who put down their tools and **demanded** freedom. Yes. Yes! Freedom! Jehane Noujaim : I think **everybody** ’ s had that feeling of sitting in a theater, in a dark room, with other strangers, watching a very powerful film, and they felt that feeling of transformation. And what I ’ d like to talk about is how can we use that feeling to actually create a movement through film? I ’ ve been listening to the talks in the **conference**, and Robert Wright said yesterday that if we have an appreciation for another person ’ s humanity, then they will have an appreciation for ours. And that ’ s what this is about. It ’ s about connecting people through film, getting these independent voices out there. Now, Josh Rushing actually ended up leaving the military and taking a job with Al Jazeera. So his feeling is that he ’ s at Al Jazeera International because he feels like he can actually use media to bridge the gap between East and West. And that ’ s an amazing thing. But I ’ ve been trying to think about ways to give power to these independent voices, to give power to the filmmakers, to give power to people who are trying to use film for change. And there are incredible organizations that are out there doing this already. There ’ s Witness, that you heard from earlier. There ’ s Just Vision, that are working with Palestinians and Israelis who are working together for peace, and **documenting** that process and getting interviews out there and using this film to take to Congress to show that it ’ s a powerful tool, to show that this is a woman who ’ s had her daughter killed in an attack, and she believes that there are peaceful ways to solve this. There ’ s Working Films and there ’ s Current TV, which is an incredible platform for people around the world to be able to put their — Yeah, it ’ s amazing. I ’ ve watched it and I ’ m blown away by it and its potential to bring voices from around the world — independent voices from around the world — and create a truly democratic, global television. So what can we do to create a platform for these organizations, to create some momentum, to get **everybody** in the world **involved** in this movement? I ’ d like for us to imagine for a second. Imagine a day when you have everyone coming together from around the world. You have towns and villages and theaters — all from around the world, getting together, and sitting in the dark, and sharing a communal experience of watching a film, or a couple of films, together. Watching a film which maybe highlights a character that is fighting to live, or just a character that defies stereotypes, makes a joke, sings a song. Comedies, documentaries, shorts. This amazing power can be used to change people and to bond people together ; to cross borders, and have people feel like they ’ re having a communal experience. So if you imagine this day when all around the world, you have theaters and places where we project films. If you imagine projecting from Times Square to Tahrir Square in Cairo, the same film in Ramallah, the same film in Jerusalem. You know, we ’ ve been talking to a friend of mine about using the side of the Great Pyramid and the Great Wall of China. It ’ s endless what you can imagine, in terms of where you can project films and where you can have this communal experience. And I believe that this one day, if we can create it, this one day can create momentum for all of these independent voices. There isn ’ t an organization which is connecting the independent voices of the world to get out there, and yet I ’ m hearing throughout this **conference** that the biggest challenge in our future is understanding the other, and having mutual respect for the other and crossing borders. And if film can do that, and if we can get all of these different locations in the world to watch these films together — this could be an incredible day. So we ’ ve already made a partnership, set up through somebody from the **TED** community, John Camen, who introduced me to Steven Apkon, from the Jacob Burns Film Center. And we started calling up **everybody**. And in the last week, there have been so many people that have responded to us, from as close as Palo Alto, to Mongolia and to India. There are people that want to be a part of this global day of film ; to be able to provide a platform for independent voices and independent films to get out there. Now, we ’ ve thought about a name for this day, and I ’ d like to share this with you. Now, the most amazing part of this whole process has been sharing ideas and wishes, and so I invite you to give brainstorms onto how does this day echo into the future? How do we use technology to make this day echo into the future, so that we can build community and have these communities working together, through the Internet? There was a time, many, many years ago, when all of the continents were stuck together. And we call that landmass Pangea. So what we ’ d like to call this day of film is Pangea Cinema Day. And if you just imagine that all of these people in these towns would be watching, then I think that we can actually really make a movement towards people understanding each other better. I know that it ’ s very intangible, touching people ’ s hearts and souls, but the only way that I know how to do it, the only way that I know how to reach out to somebody ’ s heart and soul all across the world, is by showing them a film. And I know that there are independent filmmakers and films out there that can really make this happen. And that ’ s my wish. I guess I ’ m supposed to give you my one-sentence wish, but we ’ re way out of time. Chris Anderson : That is an incredible wish. Pangea Cinema : The day the world comes together. JN : It ’ s more tangible than world peace, and it ’ s certainly more immediate. But it would be the day that the world comes together through film, the power of film. CA : Ladies and gentlemen, Jehane Noujaim.

Text Sample 507: POS Dim 4 – 2006 – Score 19.00 – 2006/t000149.txt

What technology can we really apply to reducing global poverty? And what I found was quite surprising. We started looking at things like death rates in the 20th century, and how they ’ d been improved, and very simple things turned out. You ’ d think maybe antibiotics made more difference than clean water, but it ’ s actually the opposite. And so very simple things — off-the-shelf technologies that we could easily find on the then-early Web — would clearly make a huge difference to that problem. But I also, in looking at more powerful technologies and nanotechnology and genetic engineering and other new emerging kind of digital technologies, became very concerned about the potential for abuse. If you think about it, in history, a long, long time ago we dealt with the problem of an individual abusing another individual. We came up with something — the Ten Commandments : Thou shalt not kill. That ’ s a, kind of a one-on-one thing. We organized into cities. We had many people. And to keep the many from tyrannizing the one, we came up with concepts like individual liberty. And then, to have to deal with large groups, say, at the nation-state level, and we had to have mutual non- aggression, or through a series of **conflicts**, we eventually came to a rough **international** bargain to largely keep the peace. But now we have a new situation, really what people call an asymmetric situation, where technology is so powerful that it extends beyond a nation- state. It ’ s not the nation-states that have potential access to mass destruction, but individuals. And this is a consequence of the fact that these new technologies tend to be digital. We saw genome sequences. You can download the gene sequences of pathogens off the Internet if you want to, and clearly someone recently — I saw in a science magazine — they said, well, the 1918 flu is too dangerous to FedEx around. If people want to use it in their labs for working on research, just reconstruct it yourself, because, you know, it might break in FedEx. So that this is possible to do this is not deniable. So individuals in small groups super-empowered by access to these kinds of self-replicating technologies, whether it be biological or other, are clearly a danger in our world. And the danger is that they can cause roughly what ' s a **pandemic**. And we really don ' t have experience with **pandemics**, and we ' re also not very good as a society at acting to things we don ' t have direct and sort of gut-level experience with. So it ' s not in our nature to pre-act. And in this case, piling on more technology doesn ' t solve the problem, because it only super-empowers people more. So the solution has to be, as people like Russell and Einstein and others imagine in a conversation that existed in a much stronger form, I think, early in the 20th century, that the solution had to be not just the head but the heart. You know, public **policy** and moral progress. The bargain that gives us civilization is a bargain to not use power. We get our individual rights by society protecting us from others not doing everything they can do but largely doing only what is legal. And so to limit the danger of these new things, we have to limit, ultimately, the **ability** of individuals to have access, essentially, to **pandemic** power. We also have to have sensible defense, because no limitation is going to prevent a crazy person from doing something. And you know, and the troubling thing is that it ' s much easier to do something bad than to defend against all possible bad things, so the offensive uses really have an asymmetric advantage. So these are the kind of thoughts I was thinking in 1999 and 2000, and my friends told me I was getting really depressed, and they were really worried about me. And then I signed a book contract to write more gloomy thoughts about this and moved into a hotel room in New York with one room full of books on the Plague, and you know, nuclear bombs exploding in New York where I would be within the circle, and so on. And then I was there on September 11th, and I stood in the streets with everyone. And it was quite an experience to be there. I got up the next morning and walked out of the city, and all the sanitation trucks were parked on Houston Street and ready to go down and start taking the rubble away. And I walked down the middle, up to the train station, and everything below 14th Street was closed. It was quite a compelling experience, but not really, I suppose, a surprise to someone who ' d had his room full of the books. It was always a surprise that it happened then and there, but it wasn ' t a surprise that it happened at all. And everyone then started writing about this. Thousands of people started writing about this. And I eventually abandoned the book, and then Chris called me to talk at the **conference**. I really don ' t talk about this anymore because, you know, there ' s enough frustrating and depressing things going on. But I agreed to come and say a few things about this. And I would say that we can ' t give up the rule of law to fight an asymmetric threat, which is what we seem to be doing because of the present, the people that are in power, because that ' s to give up the thing that makes civilization. And we can ' t fight the threat in the kind of **stupid** way we ' re doing, because a million-dollar act causes a billion dollars of damage, causes a trillion dollar response which is largely ineffective and arguably, probably almost certainly, has made the problem worse. So we can ' t fight the thing with a million-to-one cost, one-to-a-million cost-benefit ratio. So after giving up on the book — and I had the great honor to be able to join Kleiner Perkins about a year ago, and to work through venture capital on the innovative side, and to try to find some innovations that could address what I saw as some of these big problems. Things where, you know, a factor of 10 difference can make a factor of 1, 000 difference in the outcome. I ' ve been amazed in the last year at the incredible quality and excitement of the innovations that have come across my desk. It ’ s overwhelming at times. I ’ m very thankful for Google and Wikipedia so I can understand at least a little of what people are talking about who come through the doors. But I wanted to share with you three areas that I ’ m particularly excited about and that relate to the problems that I was talking about in the Wired article. The first is this whole area of education, and it really relates to what Nicholas was talking about with a $100 computer. And that is to say that there ’ s a lot of legs left in Moore ’ s Law. The most advanced transistors today are at 65 nanometers, and we ’ ve seen, and I ’ ve had the pleasure to **invest** in, companies that give me great **confidence** that we ’ ll extend Moore ’ s Law all the way down to roughly the 10 nanometer scale. Another factor of, say, six in dimensional reduction, which should give us about another factor of 100 in raw improvement in what the chips can do. And so, to put that in practical terms, if something costs about 1, 000 dollars today, say, the best personal computer you can buy, that might be its cost, I think we can have that in 2020 for 10 dollars. Okay? Now, just imagine what that $100 computer will be in 2020 as a tool for education. I think the challenge for us is — I ’ m very certain that that will happen, the challenge is, will we develop the kind of educational tools and things with the net to let us take advantage of that device? I ’ d argue today that we have incredibly powerful computers, but we don ’ t have very good software for them. And it ’ s only in retrospect, after the better software comes along, and you take it and you run it on a ten-year-old machine, you say, God, the machine was that fast? I remember when they took the Apple Mac interface and they put it back on the Apple II. The Apple II was perfectly capable of running that kind of interface, we just didn ’ t know how to do it at the time. So given that we know and should believe — because Moore ’ s Law ’ s been, like, a constant, I mean, it ’ s just been very predictable progress over the last 40 years or whatever. We can know what the computers are going to be like in 2020. It ’ s great that we have initiatives to say, let ’ s go create the education and educate people in the world, because that ’ s a great force for peace. And we can give everyone in the world a $100 computer or a $10 computer in the next 15 years. The second area that I ’ m focusing on is the environmental problem, because that ’ s clearly going to put a lot of pressure on this world. We ’ ll hear a lot more about that from Al Gore very shortly. The thing that we see as the kind of Moore ’ s Law trend that ’ s driving improvement in our **ability** to address the environmental problem is new materials. We have a challenge, because the urban population is growing in this century from two billion to six billion in a very short amount of time. People are moving to the cities. They all need clean water, they need energy, they need transportation, and we want them to develop in a green way. We ’ re reasonably efficient in the industrial **sectors**. We ’ ve made improvements in energy and resource efficiency, but the consumer **sector**, especially in America, is very inefficient. But these new materials bring such incredible innovations that there ’ s a strong basis for hope that these things will be so profitable that they can be brought to the market. And I want to give you a specific example of a new material that was discovered 15 years ago. If we take carbon nanotubes, you know, Iijima discovered them in 1991, they just have incredible properties. And these are the kinds of things we ’ re going to discover as we start to engineer at the nano scale. Their strength : they ’ re almost the strongest material, tensile strength material known. They ’ re very, very stiff. They stretch very, very little. In two dimensions, if you make, like, a fabric out of them, they ’ re 30 times stronger than Kevlar. And if you make a three-dimensional structure, like a buckyball, they have all sorts of incredible properties. If you shoot a particle at them and knock a hole in them, they repair themselves ; they go zip and they repair the hole in femtoseconds, which is not — is really quick. If you shine a light on them, they produce electricity. In fact, if you flash them with a camera they catch on fire. If you put electricity on them, they emit light. If you run current through them, you can run 1, 000 times more current through one of these than through a piece of metal. You can make both p- and n-type semiconductors, which means you can make transistors out of them. They conduct heat along their length but not across — well, there is no width, but not in the other direction if you stack them up ; that ' s a property of carbon fiber also. If you put particles in them, and they go shooting out the tip — they ' re like miniature linear accelerators or electron guns. The inside of the nanotubes is so small — the smallest ones are 0. 7 nanometers — that it ' s basically a quantum world. It ' s a strange place inside a nanotube. And so we begin to see, and we ' ve seen business plans already, where the kind of things Lisa Randall ' s talking about are in there. I had one business plan where I was trying to learn more about Witten ' s cosmic dimension strings to try to understand what the phenomenon was going on in this proposed nanomaterial. So inside of a nanotube, we ' re really at the limit here. So what we see is with these and other new materials that we can do things with different properties — lighter, stronger — and apply these new materials to the environmental problems. New materials that can make water, new materials that can make fuel cells work better, new materials that catalyze chemical **reactions**, that cut pollution and so on. Ethanol — new ways of making ethanol. New ways of making electric transportation. The whole green dream — because it can be profitable. And we ' ve dedicated — we ' ve just raised a new fund, we dedicated 100 million dollars to these kinds of investments. We believe that Genentech, the Compaq, the Lotus, the Sun, the Netscape, the Amazon, the Google in these fields are yet to be found, because this materials revolution will drive these things forward. The third area that we ' re working on, and we just announced last week — we were all in New York. We raised 200 million dollars in a specialty fund to work on a **pandemic** in biodefense. And to give you an idea of the last fund that Kleiner raised was a $400 million fund, so this for us is a very substantial fund. And what we did, over the last few months — well, a few months ago, Ray Kurzweil and I wrote an op-ed in the New York Times about how publishing the 1918 genome was very dangerous. And John Doerr and Brook and others got concerned, [unclear], and we started looking around at what the world was doing about being prepared for a **pandemic**. And we saw a lot of gaps. And so we asked ourselves, you know, can we find innovative things that will go fill these gaps? And Brooks told me in a break here, he said he ' s found so much **stuff** he can ' t sleep, because there ' s so many great technologies out there, we ' re essentially buried. And we need them, you know. We have one antiviral that people are talking about stockpiling that still works, roughly. That ' s Tamiflu. But Tamiflu — the **virus** is resistant. It is resistant to Tamiflu. We ' ve discovered with AIDS we need cocktails to work well so that the viral resistance — we need several anti-virals. We need better surveillance. We need networks that can find out what ' s going on. We need rapid diagnostics so that we can tell if somebody has a strain of flu which we have only identified very recently. We ' ve got to be able to make the rapid diagnostics quickly. We need new anti-virals and cocktails. We need new kinds of vaccines. Vaccines that are broad spectrum. Vaccines that we can manufacture quickly. Cocktails, more polyvalent vaccines. You normally get a trivalent vaccine against three possible strains. We need — we don ' t know where this thing is going. We believe that if we could fill these 10 gaps, we have a chance to help really reduce the risk of a **pandemic**. And the difference between a normal flu season and a **pandemic** is about a factor of 1, 000 in deaths and certainly enormous economic **impact**. So we ’ re very excited because we think we can fund 10, or speed up 10 projects and see them come to market in the next couple years that will address this. So if we can address, use technology, help address education, help address the environment, help address the **pandemic**, does that solve the larger problem that I was talking about in the Wired article? And I ’ m afraid the answer is really no, because you can ’ t solve a problem with the management of technology with more technology. If we let an unlimited amount of power loose, then we will — a very small number of people will be able to abuse it. We can ’ t fight at a million-to-one disadvantage. So what we need to do is, we need better **policy**. And for example, some things we could do that would be **policy** solutions which are not really in the political **air** right now but perhaps with the change of administration would be — use markets. Markets are a very strong force. For example, rather than trying to regulate away problems, which probably won ’ t work, if we could price into the cost of doing business, the cost of catastrophe, so that people who are doing things that had a higher cost of catastrophe would have to take insurance against that risk. So if you wanted to put a drug on the market you could put it on. But it wouldn ’ t have to be approved by regulators ; you ’ d have to convince an actuary that it would be **safe**. And if you apply the notion of insurance more broadly, you can use a more powerful force, a market force, to provide feedback. How could you keep the law? I think the law would be a really good thing to keep. Well, you have to hold people **accountable**. The law requires accountability. Today scientists, technologists, businessmen, engineers don ’ t have any personal responsibility for the consequences of their actions. So if you tie that — you have to tie that back with the law. And finally, I think we have to do something that ’ s not really — it ’ s almost unacceptable to say this — which, we have to begin to design the future. We can ’ t pick the future, but we can steer the future. Our investment in trying to prevent **pandemic** flu is affecting the distribution of possible outcomes. We may not be able to stop it, but the likelihood that it will get past us is lower if we focus on that problem. So we can design the future if we choose what kind of things we want to have happen and not have happen, and steer us to a lower-risk place. Vice President Gore will talk about how we could steer the **climate** trajectory into a lower probability of catastrophic risk. But above all, what we have to do is we have to help the good **guys**, the people on the defensive side, have an advantage over the people who want to abuse things. And what we have to do to do that is we have to limit access to certain information. And growing up as we have, and holding very high the value of free speech, this is a hard thing for us to accept — for all of us to accept. It ’ s especially hard for the scientists to accept who still remember, you know, Galileo essentially locked up, and who are still fighting this battle against the church. But that ’ s the price of having a civilization. The price of retaining the rule of law is to limit the access to the great and kind of unbridled power. Thank you.

Text Sample 508: POS Dim 4 – 2006 – Score 18.00 – 2006/t000489.txt

I ’ m going to take you on a journey very quickly. To explain the wish, I ’ m going to have to take you somewhere where many people haven ’ t been, and that ’ s around the world. When I was about 24 years old, Kate Stohr and myself started an organization to get architects and designers **involved** in humanitarian work, not only about responding to natural disasters, but **involved** in systemic issues. We believe that where the resources and expertise are scarce, innovative, sustainable design can really make a difference in people ’ s lives.

So I started my life as an architect, or training as an architect, and I was always **interested** in socially responsible design, and how you can really make an **impact**. But when I went to architecture school, it seemed that I was a black sheep in the family. Many architects seemed to think that when you design, you design a jewel, and it ' s a jewel that you try and crave for ; whereas I felt that when you design, you either improve or you create a detriment to the community in which you ' re designing. So you ' re not just doing a building for the residents or for the people who are going to use it, but for the community as a whole. And in 1999, we started by responding to the issue of the housing **crisis** for returning **refugees** in Kosovo. And I didn ' t know what I was doing — like I said, mid-20s — and I ' m the Internet generation, so we started a website. We put a call out there, and to my surprise, in a couple of months, we had hundreds of entries from around the world. That led to a number of prototypes being built and really experimenting with some ideas. Two years later we started doing a project on developing mobile health clinics in sub-Saharan Africa, responding to the HIV / AIDS **pandemic**. That led to 550 entries from 53 countries. We also have designers from around the world that participate. And we had an exhibit of work that followed that. 2004 was the tipping point for us. We started responding to natural disasters and getting **involved** in Iran, in Bam, also following up on our work in Africa. Working within the United States — most people look at poverty and they see the face of a foreigner. But I live in Bozeman, Montana — go up to the north plains on the reservations, or go down to Alabama or Mississippi, pre-Katrina, and I could have shown you places that have far worse conditions than many developing countries that I ' ve been to. So we got **involved** in and worked in inner cities and elsewhere ; and also, I will go into some more projects. 2005 : Mother Nature kicked our ass. I think we can pretty much assume that 2005 was a horrific year when it comes to natural disasters. And because of the Internet, and because of connections to blogs and so forth, within literally hours of the tsunami, we were already raising funds, getting **involved**, working with people on the ground. We run from a couple of laptops, and in the first couple of days, I had 4, 000 emails from people needing help. So we began to get **involved** in projects there, and I ' ll talk about some others. And then of course, this year we ' ve been responding to Katrina, as well as following up on our reconstruction work. So this is a brief overview. In 2004, I really couldn ' t manage the number of people who wanted to help, or the number of requests that I was getting. It was all coming into my laptop and cell **phone**. So we decided to embrace an open- source model of business — so that anyone, anywhere in the world, could start a local chapter, and they can get **involved** in local problems. Because I believe there is no such thing as Utopia. All problems are local. All solutions are local. So that means, you know, somebody who ' s based in Mississippi knows more about Mississippi than I do. So what happened is, we used Meetup and all these other Internet tools, and we ended up having 40 chapters starting up, thousands of architects in 104 countries. So the bullet point — sorry, I never do a suit, so I knew that I was going to take this off. OK, because I ' m going to do it very quick. This isn ' t just about nonprofit. What it showed me is that there ' s a grassroots movement going on, of socially responsible designers who really believe that this world has got a lot smaller, and that we have the opportunity — not the responsibility, but the opportunity — to really get **involved** in making change. I ' m adding that to my time. So what you don ' t know is, we ' ve got these thousands of designers working around the world, connected basically by a website, and we have a staff of three. The fact that nobody told us we couldn ' t do it, we did it. And so there ' s something to be said about navet. So seven years later, we ' ve developed so that we ' ve got advocacy, instigation and implementation.

We advocate for good design, not only through student workshops and lectures and public forums, op-eds ; we have a book on humanitarian work ; but also disaster mitigation and dealing with public **policy**. We can talk about FEMA, but that ' s another talk. Instigation, developing ideas with communities and NGOs, doing open-source design competitions. Referring, matchmaking with communities. And then implementing — actually going out there and doing the work, because when you invent, it ' s never a reality until it ' s built. So it ' s really important that if we ' re designing and trying to create change, we build that change. So here ' s a select number of projects. Kosovo. This is Kosovo in ' 99. We did an open design competition, like I said. It led to a whole variety of ideas. And this wasn ' t about emergency shelter, but transitional shelter that would last five to 10 years, that would be placed next to the land the resident lived in, and that they would rebuild their own home. This wasn ' t imposing an architecture on a community ; this was giving them the tools and the space to allow them to rebuild and regrow the way they want to. We had from the sublime to the ridiculous, but they worked. This is an inflatable hemp house. It was built ; it works. This is a shipping container. Built and works. And a whole variety of ideas that not only dealt with architectural building, but also the issues of governance, and the idea of creating communities through complex networks. So we ' ve engaged not just designers, but also a whole variety of technology- based professionals. Using rubble from destroyed homes to create new homes. Using straw bale construction, creating heat walls. And then something remarkable happened in ' 99. We went to Africa originally to look at the housing issue. Within three days, we realized the problem was not housing ; it was the growing **pandemic** of HIV / AIDS. And it wasn ' t doctors telling us this ; it was actual villagers that we were staying with. And so we came up with the bright idea that instead of getting people to walk 10, 15 kilometers to see doctors, you get the doctors to the people. And we started engaging the medical community, and you know, we thought we were real bright sparks —"We ' ve come up with this great idea : mobile health clinics, widely distributed throughout sub-Saharan Africa."And the medical community there said,"We ' ve said this for the last decade. We know this. We just don ' t know how to show this."So in a way, we had taken pre- existing needs and shown solutions. And so again, we had a whole variety of ideas that came in. This one I personally love, because the idea is that architecture is not just about solutions, but about raising awareness. This is a kenaf clinic. You get seed and you grow it in a plot of land, and it grows 14 feet in a month. And on the fourth week, the doctors come and they mow out an area, put a tensile structure on the top, and when the doctors have finished treating and seeing patients and villagers, you cut down the clinic and you eat it. It ' s an eat- your-own-clinic. So it ' s dealing with the fact that if you have AIDS, you also need to have nutrition rates, and the idea of nutrition is as important as getting antiretrovirals out there. So you know, this is a serious solution. This one I love. The idea is it ' s not just a clinic, it ' s a community center. This looked at setting up trade routes and economic engines within the community, so it can be a self-sustaining project. Every one of these projects is sustainable. That ' s not because I ' m a tree- hugging green person. It ' s because when you live on four dollars a day, you ' re living on survival and you have to be sustainable. You have to know where your energy is coming from, you have to know where your resource is coming from, and you have to keep the maintenance down. So this is about getting an economic engine, and then at night, it turns into a movie theater. So it ' s not an AIDS clinic. It ' s a community center. So you can see ideas. And these ideas developed into prototypes, and they were eventually built. And currently, as of this year, there are clinics rolling out in Nigeria and Kenya. From that, we also developed Siyathemba. The community came to us and said,”The problem is that the girls don ’ t have education.”And we ’ re working in an area where young women between the ages of 16 and 24 have a 50 percent HIV / AIDS rate. And that ’ s not because they ’ re promiscuous, it ’ s because there ’ s no knowledge. And so we decided to look at the idea of sports, and create a youth sports center that doubled as an HIV / AIDS outreach center, and the coaches of the girls ’ team were also trained doctors. So that there would be a very slow way of developing **confidence** in health care. And we picked nine finalists, and then those nine finalists were distributed throughout the entire region, and then the community picked their design. They said, this is our design, because it ’ s not only about engaging a community ; it ’ s about empowering a community, and about getting them to be a part of the rebuilding process. So, the winning design is here. And then, of course, we actually go and work with the community and the clients. So this is the designer. He ’ s out there working with the first ever women ’ s soccer team in KwaZulu-Natal, Siyathemba. And they can tell it better. Video : Well, my name is Cee Cee Mkhonza. I work at the Africa Centre, I ’ m an IT user consultant. I ’ m also the national football player for South Africa, Banyana Banyana. And I also play in the Vodacom League, for the team called Tembisa, which has now changed to Siyathemba. This is our home ground. Cameron Sinclair : I ’ m going to show that later because I ’ m running out of time. I can see Chris looking at me slyly. This was a connection, just a meeting with somebody who wanted to develop Africa ’ s first telemedicine center, in Tanzania. And we met, literally, a couple of months ago. We ’ ve already developed a design. The team is over there, working in partnership. This was a matchmaking, thanks to a couple of TEDsters — Sun [Microsystems], Cheryl Heller and Andrew Zolli, who connected me with this amazing African woman. And we start construction in June, and it will be opened by TEDGlobal. So when you come to TEDGlobal, you can check it out. But what we ’ re known probably most for is dealing with disasters and development, and we ’ ve been **involved** in a lot of issues, such as the tsunami and also things like Hurricane Katrina. This is a 370-dollar shelter that can be easily assembled. This is a community-designed community center. And what that means is we actually live and work with the community, and they ’ re part of the design process. The kids actually get **involved** in mapping out where the community center should be. And then eventually, the community, through skills training, end up building the building with us. Here is another school. This is what the UN gave these **guys** for six months — 12 plastic tarps. This was in August. This was the replacement ; that ’ s supposed to last for two years. When the rain comes down, you can ’ t hear a thing, and in the summer, it ’ s about 140 degrees inside. So we said, if the rain ’ s coming down, let ’ s get fresh water. So every one of our schools has a rainwater collection system. Very low cost : three classrooms and rainwater collection is 5, 000 dollars. This was raised by hot chocolate sales in Atlanta. It ’ s built by the parents of the kids. The kids are out there on-site, building the buildings. And it opened a couple of weeks ago, and there ’ s 600 kids that are now using the schools. So, disaster hits home. We see the bad stories on CNN and Fox and all that, but we don ’ t see the good stories. Here is a community that got together, and they said”no”to waiting. They formed a partnership, a diverse partnership of players, to actually map out East Biloxi, to figure out who ’ s getting **involved**. We ’ ve had over 1, 500 volunteers rebuilding, rehabbing homes. Figuring out what FEMA regulations are, not waiting for them to dictate to us how you should rebuild. Working with residents, getting them out of their homes, so they don ’ t get ill. This is what they ’ re cleaning up on their own. Designing housing. This house is going in in a couple of weeks. This is a rehabbed home, done in four days. This is a utility room for a woman who is on a walker. She ' s 70 years old. This is what FEMA gave her. 600 bucks, happened two days ago. We put together, very quickly, a washroom. It ' s built, it ' s running and she just started a business today, where she ' s washing other people ' s clothes. These are the Calhouns. They ' re photographers who had **documented** the Lower Ninth for the last 40 years. That was their home, and these are the photographs they took. And we ' re helping, working with them to create a new building. Projects we ' ve done. Projects we ' ve been a part of, **support**. Why don ' t aid agencies do this? This is the UN tent. This is the new UN tent, just introduced this year. Quick to assemble. It ' s got a flap — that ' s the invention. It took 20 years to design this and get it implemented in the field. I was 12 years old. There ' s a problem here. Luckily, we ' re not alone. There are hundreds and hundreds and hundreds and hundreds and hundreds of architects and designers and inventors around the world that are getting **involved** in humanitarian work. More hemp houses — it ' s a theme in Japan, apparently. I ' m not sure what they ' re smoking. This is a Grip Clip, designed by somebody who said,"All you need is some way to attach membrane structures to physical **support** beams."This **guy** designed for NASA, is now doing housing. I ' m going to whip through this quickly, because I know I ' ve got only a couple of minutes. So this is all done in the last two years. I showed you something that took 20 years to do. And this is just a selection of things that were built in the last couple of years. From Brazil to India, Mexico, Alabama, China, Israel, Palestine, Vietnam. The average age of a designer who gets **involved** in this project is 32 — that ' s how old I am. So it ' s a young — I just have to stop here, because Arup is in the room, and this is the best-designed toilet in the world. If you ' re ever, ever in India, go use this toilet. Chris Luebkeman will tell you why. I ' m sure that ' s how he wanted to spend the party. But the future is not going to be the sky-scraping cities of New York, but this. And when you look at this, you see **crisis**. What I see is many, many inventors. One billion people live in abject poverty. We hear about them all the time. Four billion live in growing but fragile economies. One in seven live in unplanned settlements. If we do nothing about the housing **crisis** that ' s about to happen, in 20 years, one in three people will live in an unplanned settlement or a **refugee** camp. Look left, look right : one of you will be there. How do we improve the living standards of five billion people? With 10 million solutions. So I wish to develop a community that actively embraces innovative and sustainable design to improve the living conditions for everyone. Chris Anderson : Wait a sec — that ' s your wish? CS : That ' s my wish. CA : That ' s his wish! CS : We started Architecture for Humanity with 700 dollars and a website. So Chris somehow decided to give me 100, 000. So why not this many people? Open- source architecture is the way to go. You have a diverse community of participants — and we ' re not just talking about inventors and designers, but we ' re talking about the funding model. My role is not as a designer ; it ' s as a conduit between the design world and the humanitarian world. And what we need is something that replicates me globally, because I haven ' t slept in seven years. Secondly, what will this thing be? Designers want to respond to issues of humanitarian **crisis**, but they don ' t want some company in the West taking their idea and basically profiting from it. So Creative Commons has developed the Developing Nations license. And what that means is that a designer can — The Siyathemba project I showed was the first ever building to have a Creative Commons license on it. As soon as that is built, anyone in Africa or any developing nation can take the construction **documents** and replicate it for free. So why not allow designers the opportunity to do this, but still protect their rights here? We want to have a community where you can upload ideas, and those ideas can be tested in an earthquake, in flood, in all sorts of austere environments. The reason that ' s important is I don ’ t want to wait for the next Katrina to find out if my house works. That ’ s too late, we need to do it now. So doing that globally — and I want this whole thing to work multi-lingually. When you look at the face of an architect, most people think a gray-haired white **guy**. I don ’ t see that ; I see the face of the world. So I want everyone from all over the planet to be able to be a part of this design and development. The idea of needs-based competitions — XPRIZE for the other 98 percent, if you want to call it that. We also want to look at ways of matchmaking and putting funding partners together, and the idea of integrating manufacturers — fab labs in every country. When I hear about the $100 laptop and it ’ s going to educate every child — educate every designer in the world. Put one in every favela, every slum settlement. Because you know what? Innovation will happen. And I need to know that. It ’ s called the leap-back. We talk about leapfrog technologies. I write with Worldchanging, and the one thing we ’ ve been talking about is, I learn more on the ground than I ’ ve ever learned here. So let ’ s take those ideas, adapt them, and we can use them. These ideas are supposed to be adaptable ; they should have the potential for evolution ; they should be developed by every nation in the world and useful for every nation in the world. What will it take? There should be a sheet. I don ’ t have time to read this, because I ’ m going to be yanked off. CA : Let ’ s just leave it up for a sec. CS : Well, what will it take? You **guys** are smart. So it ’ s going to take a lot of computing power, because I want the idea that any laptop anywhere in the world can plug into the system and be able to not only participate in developing these designs, but utilize the designs. Also, a process of reviewing the designs. I want every Arup engineer in the world to check and make sure that we ’ re doing **stuff** that ’ s standing, because those **guys** are the best in the world. Plug. And so, you know, I want these — I just should note : I have two laptops and one of them is there, and that has 3000 designs on it. If I drop that laptop... What happens? So it ’ s important to have these proven ideas put up there, easy to use, easy to get ahold of. My mom once said,”There ’ s nothing worse than being all mouth and no trousers.”I ’ m fed up of talking about making change. You only make it by doing it. We ’ ve changed FEMA guidelines ; we ’ ve changed public **policy** ; we ’ ve changed **international** response — based on building things. So for me, it ’ s important that we create a real conduit for innovation, and that it ’ s free innovation. Think of free culture — this is free innovation. Somebody said this a couple of years back. I will give points for those who know it. But I think the man was maybe 25 years too early. So let ’ s do it. Thank you.

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Helen Walters : Hello, **everybody**. It ’ s lovely to be here with you today. I am Helen Walters, I am the head of media and curation at **TED**, and I am thrilled to be your host for this episode of **TED** Explains the World with Ian Bremmer. Ian, of course, is not only the president and founder of the geopolitical risk advisory firm Eurasia Group, he is also a scion of the media industry. He is the head of GZERO Media. And today, I wanted to say hi to all of the **TED** members who are here watching this, and thank you for sending in your questions. We are going to get right to them. Hi, Ian. Ian Bremmer : Hi, good to see you. HW : OK, so we are here to talk about the US election, a giant moment not just for America but also for the entire world. On the docket, we have former President Donald Trump and Vice President Kamala Harris. Today is October 10, people have already started to vote. But as we count down to the official election day of November 5, what do you think, what thoughts are people taking into the ballot box with them? IB : I think the most important thought, and this is a global conversation, it ’ s a global audience that ’ s dialed in to us today. We ’ ve had a lot of elections happening this year, and most of them have been change elections. They are a backdrop of people that are not particularly happy with where they believe their country is going. And that has led to a lot of voting against **incumbents** who traditionally have some advantages. Doesn ’ t mean all the **incumbents** have been voted out, but they ’ ve all had a tough time. And that is as true in countries with enormously popular **leaders** like Modi, for example, in India, as it is in countries that have had **leaders** that have faced real difficulties, like Macron in France. You ’ ve seen in South Africa, Ramaphosa and the ANC. A number of **conservative** PMs seeing the end of their rule in the UK with Labour. That ’ s the backdrop. And this backdrop comes in part because **inflation** is really high on the back of the **pandemic**, and people are upset about their economic prospects. Also, because migration is high, in part on the back of the **pandemic**, when people weren ’ t moving and suddenly they could. And also, in part, broader structural trends around the world, bigger gaps between rich and poor, **climate** challenges, geopolitical challenges and war. All of those things, repressive regimes, have led to environments where people that are going to the ballot box are not happy with who they ’ re voting for. And the United States is no exception. The difference in the US, I would say a couple of big ones. One is that the **disinformation** environment and the political division and **dysfunction** around that division is much higher today than it has been in any of the other advanced industrial democracies. And secondly, the visceral unpopularity among opponents of these candidates is much higher than we have seen either in these other countries or historically in recent times in the United States. Which means not only is this a very, very close election, and I don ’ t have a strong view at this point, even one month out, of who ’ s going to win. It ’ s going to be determined by a small number of voters in a small number of districts in a small number of swing states, you ’ ve all seen that, I have nothing more to tell you than what you ’ ve seen from all the pollsters and all the media on that. But also that so many Americans are not prepared to accept the outcome. So many Americans are prepared to see that this is rigged. And let ’ s also remember, we ’ ve just gotten through, and no one asked me about this in the past few weeks, but it ’ s important, we ’ ve just had two assassination attempts, quite serious ones, one that was very close from succeeding, against former President Trump. In case people were worried, you know, not thinking about just how serious the passions, the tensions, the **conflict** is as the backdrop for this election. The stakes are very serious indeed. HW : All right, let ’ s dive headlong into your world. Let ’ s talk foreign **policy**. What do you see as the major differences between the candidates and their worldview, and what might be the difference in how they approach **international** relations if they ’ re elected? IB : Well, one, I mean, this might be a boring way to start, but it ’ s important, is the way that they will govern will be very different. Very different from Biden and very different from each other. Biden is and has been the most experienced foreign **policy** hand we ’ ve had as a president in decades. He was vice president, of course, and served that completely. Before that, he had served in the Senate for his entire career, including running the Foreign Relations Committee. So when you see Biden, or back before his age really started to show, when you have seen Biden on stages globally, you see him meeting with **leaders** that he has known personally, he has had long relationships with for a very long time. That is also true of his inner circle in the White House. These are people that have worked with Biden for decades, that he knows, that he **trusts**. And therefore, Biden ’ s foreign **policy** decision-making style is made in the White House, right? And they make decisions, the president makes a surprising number of those decisions himself, and then they expect the cabinet to execute on those decisions, implement on them. That ’ s very different than what we would see under Kamala Harris. She ’ s had a lot of foreign **policy** exposure from being vice president, made a lot of trips, met with a lot of **leaders**, that ’ s very useful, for four years. That ’ s very different from a lot of direct foreign **policy** making experience as a principal. That she has not had. And she would rely on a much more traditional cabinet. So how a Harris foreign **policy** manifests would have much more to do with the people she decides to appoint in state as NSA, in defense, in treasury, in commerce, than it has for the Biden administration. And they ’ ll have regular cabinet meetings, which Biden has not been doing. And when they do, and they have differences that need to be hashed out, they ’ ll be hashed out in front of her, and she ’ ll play a very active role in steering that conversation. But it ’ s very different from Biden. Where for a Trump administration, there would be a small number of high-level issues that he will make decisions on in a untransparent and frequently instinctive way. And he won ’ t take a lot of advice from other people around those decisions. And then there ’ ll be a whole other host of policy-making decisions that he will not be **interested** in almost at all. And those will be decisions that will be made by others that happen to be **trusted** around him and afforded the responsibility for those issues, which might be the cabinet secretaries, but might also be informal people that are in or around the White House. For example, as Jared Kushner ended up with the Middle East portfolio last time around. So, I mean, I know you want to ask me about **policy**, but I don ’ t want to sleep on the fact that a lot of the way **policy** gets implemented in a really big organization like the US government depends on what kind of **leadership** and structure you have. And these would be two very, very different structures. HW : That ’ s really **interesting**. I think that the question of personnel, the question of who ’ s in those inner circles does really matter. What is your sense of who ’ s going to be surrounding the president in the next four years? IB : Well, I want to say, one, that for Trump, those decisions are almost always made at the last minute. And Trump responds a lot to recency. The person that he just met that just impressed him, that looked like, sounded like the person that should be in that role. Even if he has never met that person before, he ’ ll be very excited and say,”I want that person in that job.”Look at the way Rex Tillerson became secretary of state. I mean, didn ’ t know Trump from Adam and came out of that meeting,”Well, I ’ m going to guess I get the job.”And that could have been Jim Stavridis, that could have been Dave Petraeus. They would have been very different. So definitely, the fact that Trump has been president for four years means that there is a constellation of people around him that have worked with him, that are **trusted**, that he will go back to. I think Robert O ’ Brien, who had been national security advisor, will certainly play a **senior** role again in a Trump administration. I think that certainly Robert Lighthizer, who was US trade rep, will probably be running trade in some form, in some position. He ’ d like to be secretary of treasury. I don ’ t think he ’ s going to get that job, but he ’ ll do something. But there ’ ll also be a bunch of people that, you know, will be complete surprises, even shocks to the system. And I can tell you, like, you know, people that look good right now, Tom Cotton, the senator, certainly looks like he ’ s, you know, got a very good chance of either running Defense or CIA. Pompeo always throws himself in the mix, though he didn ’ t endorse Trump until very late, and it ’ s not clear whether he ’ d get a nod or not. There ’ s always a question of what Jared and / or Ivanka might do, even though they ’ ve said they don ’ t want to be around this time around. Trump keeps going back to them every time he sees them and says,”Of course you ’ re saying that, but you ' re going to be in, and I want you in."And you ' ll see what that means. So I think there ' s an enormous amount of **uncertainty**. Now for Kamala Harris, she has told the people that want the positions around her that, you know, definitely her staff is collecting all that information. She is not prepared to start that decision-making process until after an election, assuming she wins. So still very early in that process, too. But you have a lot of people around Kamala, certainly, that you would expect to have significant positions. In my world, her lead foreign **policy** advisor has been Phil Gordon, serving in the National Security Council in the White House. Very capable. I would expect he ' d have a good shot at being national security advisor, assuming he wants that job. I think for secretary of state Bill Burns, a lot of people say he ' s older, he ' s in his mid 70s, but he ' s done a very solid job not only running the CIA but also running point on the Middle East. If he wanted the job at state, I think he ' d have a very good shot at it. I think my good friend Chris Coons, the senator from Delaware, who ' s been a lead foreign **policy** voice in the Senate, probably the most capable on Senate co-chair of the campaign, would certainly be in the running for that post. I could see others. Susan Rice, former national security advisor, I think is certainly **interested** in it. So a lot of people that I think you would consider to be fairly center to center-left, fairly establishment-type figures in key foreign **policy** and economic related roles. I don ' t think there ' d be a huge shift from Biden to that group. I ' m not expecting the Biden people would stay. But in terms of their orientation towards the world, I don ' t think it would be dramatically different. HW : I mean, whatever happens, foreign **policy** is going to be front and center for whoever is in those positions. We are all watching what ' s happening in the Middle East, what ' s unfolding there, with increasing horror as Lebanon has been drawn in, increasingly Iran is connected. What changes do you see in the Middle East come January 25? IB : Well, let ' s first talk about the changes we ' re seeing before January 25, because this war, you and I, Helen, have talked about it a few times since October 7, and it continues to change pretty dramatically on almost a daily basis. One of the biggest things that changed is, of course, you ' re right that the war has escalated. It ' s escalated on several different occasions since October 7. But in addition to the fact that the war has escalated, you know, in Gaza to a ground invasion, now into the northern front, into Lebanon with Hezbollah and of course more broadly across the Middle East with the so-called axis of resistance led by Iran. And even with shots fired between the Iranians and Israel now on a couple of occasions. The fact is that some of the organizations that have been **involved** in this war are no longer experiencing the capacity to escalate. So, I mean, Hamas, you may have seen that Sinwar, who is running Hamas on the ground, has just recently, in the last 24 hours, announced that they ' re going to start suicide bombings again, which they had said that they weren ' t going to use as operational **policy**. The reason they ' re announcing that is because they can ' t do much of anything else. Their tunnels have been sealed, their weapons caches have been destroyed. Their **leadership** has been obliterated. I mean, they just don ' t have the **ability** to mount a credible, organized fighting force. And so it ' s definitely still a threat. Because, you know, suicide bombers are not to be slept on. But it is nowhere near the threat it was on October 7. Hezbollah has now said that they would like a cease fire with Israel, and they ' re not linking it to ending the fighting in Gaza anymore. That was utterly fundamental to what they were saying for the last entire year. They started launching rockets at Israel on October 8, the day after October 7. And they said they ' re not going to stop until Israel stops bombing Gaza. Well, what changed? What changed is that Nasrallah, the **leader** of Hezbollah, was assassinated. His entire **leadership** team has been systematically assassinated. Thousands and thousands of their fighters have been killed and maimed, and their war-fighting capability and their communications capability is being systematically torn apart. So Israel ' s military asymmetries have stopped their enemies from being capable of escalating with the potential untested exception of Iran. And the big debate right now, of course, is how will the Israelis respond to the Iranian attacks, the 180 ballistic missiles, and what might the Iranians do in response? I think the concerns of escalation here are overwhelmingly about the Israeli decision-making process, because the Iranians actually understand that they, too, are quickly running out of ways to effectively escalate against a far, far stronger Israel with a completely committed ally of the United States. So that ' s where we have been, that ' s where we are now. I ' m happy to move to, you know, who would do what and what to expect going forward, if you like, Helen, but please, frame it as you wish. HW : I think the question that a lot of members actually have sent in before we convened here, and I think a question on a lot of people ' s minds is what influence – you talk about the Israeli decision-making process. What influence can either candidate actually have on that decision-making process, and how does either candidate actually have a hope of making any kind of difference in what actually is transpiring over in the Middle East? I think that ' s the framing. IB : Well, let ' s first of all recognize that even if the United States wanted to have a lot of influence and was prepared to use leverage to get it, which President Biden has not been willing to do, but even if they were, that does not mean that the Israelis, and this Israeli government in particular, would be falling in lockstep behind the United States ' s preferences. I mean, in Ukraine, you have a country that is much more reliant on American largesse to continue fighting than the Israelis are, and at the end of the day, the Ukrainians have taken a number of decisions in this war, in their war, that the Americans weren ' t aware of and would not have **supported**, right? I mean, for example, the sending of 40, 000 troops into Russia, into Kursk. The US didn ' t know that and certainly didn ' t think it was a good idea afterwards. Taiwan is much more reliant on the United States for its continued **ability** to survive in some sovereign fashion, governing its own territory. And yet we see frequently decisions by the Taiwanese government statements they make that are clearly seen as escalatory by Beijing, that the Americans would not want them to make. So I think that American influence over allies is frequently overstated. I think we should start with that. But it is clear that however much influence the Americans might have on Israel, President Biden has been unwilling to test that theory. He ' s been unwilling to potentially mete out any consequences to the Israeli government whatsoever if they choose to ignore American private warnings, which have been repeated and fairly sharp, and public concerns, which have been **aired** more infrequently and much more softly. If the United States was prepared to take off the table, off the table, the military **support** that they provide Israel, or at least the offensive military **support**, continue to provide the **ability** for Israel to defend itself against attacks from its enemies around the region, something that, you know, Macron, for example, is now talking about, the French president, something that the UK has flirted with with the new Keir Starmer administration. I think you would probably see at the margins more willingness of the Israelis to consult with the Americans as they ' re making decisions about, for example, how far to go into Lebanon. I think there would be more decision-making that was made, at least with the Americans in awareness of it in advance. I ' m not sure it would have changed the way the Israelis are pursuing and fighting the war. In part because it is so overwhelmingly popular in Israel, because the Israelis don ' t **support** a two-state solution anymore, because the Israelis want to do everything possible to get their hostages out and to get their 60, 000 civilians back into the north. And that means, you know, fighting Hamas to the last **leader** and fighter that they can and destroying Hezbollah as well. So I ’ m not sure that the United States would have much effect on Israel. But there ’ s another question, which is, would the United States have more **support** from other countries around the world? The US has lost a lot of influence with the global South, particularly Southeast Asia, for example, Indonesia, Malaysia, even Singapore, because they are perceived as complicit in the Israeli strikes into Gaza and the lack of humanitarian assistance getting in for the people of Gaza. I think a **policy** by the US, which was more aligned with allies in Europe and the Gulf, would be more broadly **supported** by other countries that have gone sour on the US as a consequence of that. And here there ’ s a big difference between Harris and Trump. Harris would certainly want to be seen as the president of the most important ally of Israel in the world. I don ’ t think she would change that at all. But I also think she would do a lot more to help **ensure** that the Palestinians got regular humanitarian aid in, as opposed to the roughly 10 percent of what they had at pre-October 7 right now. I think she ’ d push harder on that directly and publicly with Israel. I think she ’ d be working more closely with the United Nations. And I also think she ’ d be working more closely, multilaterally with American allies. I think that Trump, on the other hand, has already made clear he does not **support** a two-state solution anymore. He did when he was president the first time around. In fact, that was behind the Bahrain **conference** and behind the Abraham Accords, was a nominal map, a road map for a two-state solution with what that would look like in the West Bank in particular. He has also said that he would **support** Israel directly striking Iran ’ s nuclear facilities. Jared Kushner has circulated what he thinks is a unique opportunity for Israel to take care of, as he describes it, the Iranian problem. There ’ s no question in my mind that Kamala Harris would not **support** that. So frankly, I think that Harris and Trump differ perhaps the most on the Middle East, specifically on Israel-Palestine, than they do on any other major foreign **policy** issue out there. And I think that that is likely to manifest after January. HW : So that ’ s **interesting**, because the next question that I wanted to ask you about was Ukraine and Russia. And we ’ re reading right now about Bob Woodward ’ s reporting on maybe giving COVID tests to Putin or calling Putin when he hasn ’ t been in office, **claims** which Trump denies. But I think it ’ s clear to outsiders, at least, that it seems like the approach to Russia would be pretty different between the two candidates. But what do you think would change in the US ’ s dealings with Russia when either one of them is elected? IB : So the Trump team has denied that they ’ ve had those **phone** calls. We don ’ t have any evidence of them from Woodward, so we don ’ t know yet. The Putin and the Kremlin has confirmed the COVID tests. The COVID tests are **interesting** – the COVID vaccines, excuse me. And I think that ’ s – excuse me, am I right, was it tests or vaccines? I ’ ve seen both actually. HW : Actually I think it was tests. IB : Was it tests? So to me that ’ s **interesting** because it ’ s not like Trump – I mean Trump talked a good game with Putin. And you remember the Helsinki summit in particular where it seemed like, you know, he was trying to make nice-nice with Putin as his best buddy as opposed to the ways he ’ s been much tougher with some US allies in NATO. But the reality of US **policy** under Trump was not that. The reality is that the Javelin missiles, **anti-tank** missiles that Biden **refused** to provide to Ukraine, were provided in the Trump administration. The reality is that a series of high-level sanctions against Russia were actually tightened and toughened up under Trump, that were not under Obama. And also Trump was pushing much stronger against the Germans on the Nord Stream pipeline and aligning their energy security to Russia, which Obama didn ’ t care very much about. So that is you know, it ’ s a little counter to what the narrative in the media actually discusses around Trump. And providing machines that do COVID testing, you know, in an environment where you have a really bad relationship with someone, but you know you need to communicate with them and you are prepared to work with them in areas that are not hurting your national security. That actually strikes me as a fairly sensible thing to do and reflects **leadership**. So I would not be critical of that. If we find out that it is indeed true that Trump has been in regular contact with Putin since leaving the presidency, while **refusing** to talk to Zelenskyy until Zelenskyy begged for a meeting at the UN and didn ’ t want to take it and finally did, that would be, I think, an issue of significant concern, and we would want information if we could get it around what was actually in those conversations. But I think that if Trump were to become president now, you ’ ve heard from many people now about how **stupid** it sounds that Trump is going to end the war in a day. And, you know, I think you and I can agree, Helen, that Trump is occasionally prone to, you know, slightly exaggerating things. And that might be true with the 24 hours. It might take 36 hours, for example, for him to come to peace between Russia and Ukraine, maybe even 48. I mean, if he had to golf in between. But I mean, more seriously, he wants to end the war. One of the things that he considers to be a feature of his administration is new wars did not start under his administration. He was very loath to use force, even against the Iranians when the Emiratis, the Saudis were begging him and the Iranians were engaging in all sorts of strikes in the region, against, you know, oil facilities, against ships, against American bases. And he didn ’ t do anything. And then finally when he finally did, he finally took a big whack and killed Qasem Soleimani. And then the Iranians didn ’ t do much at all because they saw that this was not to be trifled with. But he was very reluctant to use military force. And I think that clearly he wants to be the **guy** that says he ended the war between Russia and Ukraine. And the way he intends to do it is to call Putin and to call Zelenskyy and say,”You ’ ve got to freeze the **conflict** where it is or else.”So what ’ s the”or else”? Well, if the Ukrainians don ’ t accept a ceasefire and freezing the **conflict** where it is, and then there ’ ll be negotiations of some undetermined form, then he would cut the Ukrainians off from any further **support** from the US. So, in other words, that ’ s a pretty big stick if Zelenskyy doesn ’ t accept it. And it would also divide Europe. Some Europeans would **support** Trump, certainly Hungary ’ s Orban, and then there would be some very difficult decision-making made by other countries that aren ’ t on the front lines, like Poland, the Baltics and the Nordics, who would definitely oppose Trump, but others would have a harder time making up their mind. And then he ’ d go to the Russians and he would say,”You ’ ve got to accept this ceasefire or else.”Or else what?”Or else I ’ m going to put much tougher sanctions on you. You ’ re not going to grow at four percent. I ’ m going to stop you from exporting oil to India. I ’ m going to end that deal. I am going to put sanctions on the banks in China that are allowing you to engage in financial transactions. I ’ m going to put sanctions on your central bank.”So, I mean, that would have massive costs for the world and for the Europeans. Again, it would divide the Europeans from the United States, but it would certainly be a pretty big stick that Putin would be hit by. I mean, the question, of course, is how do you game this out? Like, I mean, does Zelenskyy say yes, knowing that Putin is going to say no? If Zelenskyy says no, can Putin then say yes and he doesn ’ t have anything to worry about? I mean, clearly, not that these **guys** are going to be gaming it out together, but it ’ s a serious question. Now the one thing that Trump is unlikely to do, he ’ s very unlikely to call the Europeans and coordinate with them, which has been the most successful part of Biden ’ s **policy**. The fact that it has been unlike Biden

’ s Middle East **policy**, Biden ’ s Russia-Ukraine **policy** has been a multilateral **policy** led by the US in NATO. And so at every step of the way, economic sanctions, freezing of Russia ’ s assets, provision of **intelligence**, provision of military **support** and training, all of that is being done collectively by the entire alliance, even the Japanese and the South Koreans, the Australians have been on board with all of this. It is highly unlikely that Trump will do that. Trump is much more likely to make those **phone** calls to the Russians and Ukrainians himself, and a lot of allies are going to find out about it when they ’ re briefed by Trump ’ s people or when they watch it on TV. And that, of course, is a much more challenging way to run **policy**. Now Harris would be very different. Harris would have a multilateral **policy**, not only in lockstep with the Europeans, but she would not be prepared to do anything with Russia unless the Ukrainians were on board. So it ’ s a very different approach. Again, the structural approach, not the intention of the **policy**, the structural approach would be very, very different. The intention of the **policy** would be to end the war. And end the war, even accepting that the Ukrainians are not going to get all their land back. Now I think that the orientation that Harris has on all of this, which is pretty **interesting**, is that the Ukrainians, in order to be enticed to accept a freezing of the **conflict**, which de facto means that they lose a fifth of their territory, even though they don ’ t, you know, accept that politically, they need to be given hard security guarantees. And the hard security guarantees that they are given is NATO **membership**, which has never been actually extended. They say, yes, you ’ re going to be allowed in, but when, we have no idea. And I think that if the US says,”If you accept a freezing of the **conflict**, we give you NATO **membership**, and that means the **conflict** is frozen, you lose 20 percent of your territory. But if Russia tries to take any more, they ’ re fighting **everybody**.”They ’ re fighting NATO. That ’ s acceptable to Zelenskyy. Zelenskyy, I think, can get there in a way that it would be much, much harder for Zelenskyy to accept what Trump would be arguing for on their erstwhile **phone** call. The point, of course, is that Putin won ’ t accept that because Putin isn ’ t prepared to accept Ukraine being a part of NATO. But if Putin says no and he won ’ t have negotiations, but the Ukrainians say yes and NATO says yes, you are now in a better position globally because you can now go to other countries with influence over Russia, like China, and say,”Look, we ’ ve got a peace plan. Here it is. And we ’ re all on board.”And a lot of the countries in the global South that just desperately want this war to be over would, I think, be sympathetic to that in a way that right now they ’ re not sympathetic to the US position, which is we defer to Ukraine. And the Ukrainians are saying”We want all of our land back,”and everyone knows they ’ re not going to get it back. So it ’ s endless war. Endless war that the Ukrainians are increasingly not going to be capable of fighting. So the reality, I think, is that the ultimate **policy** outcomes that Trump and Harris would find acceptable, are closer than you would think. But the mechanisms that they would use to try to achieve those outcomes are radically different. And as a consequence, the potential for success, the consequences of failure would also be very, very different. HW : Whew, that is so **interesting**. I will be chewing on that for some time. But now I ’ m going to change the subject completely because I think that we should talk about **climate** change. Milton just flattened Florida. Helene decimated North Carolina. We know, you know, I know, we know that **climate** change is not a theoretical threat. It is ever present, it is increasingly dangerous. What do you think the candidates are going to say about **climate**? And I guess in the back of my mind, I have the voices of the young people I ’ ve heard who are increasingly nihilistic about the idea that anyone is ever going to take any meaningful kind of change on **climate**. So where are the candidates, and are the young people actually right? IB : Oh, yeah, the young people are right. And the young people are the ones that are ultimately going to make the change on **climate**. They ’ re the ones that are going to **demand** that companies change their behavior or they ’ ll stop buying their goods. They ’ re the ones who **demand** that governments change or they ’ re going to vote for much more radical outcomes, and they ’ re going to increasingly do that as they have a larger share of the voting population, and they have a larger share of wallet and consumer spend. And, of course, as the implications of **climate** change become much more real for all of us internationally. Of course, we know it ’ s getting worse because so much of it is already baked in, irrespective of what we do going forward. And we see the manifestations of that in massive storms and massive droughts, **climate** change, **climate** weirding as they call it, **climate** escalation on both sides. We see that all the time, and everyone accepts it. Even though there ’ s **disinformation** online, you ’ ve got 193 countries around the world that are all signatories for the Intergovernmental Panel on **Climate** Change, and they all accept that what we see and what the scientists tell us in front of our eyes. And I certainly understand that a Harris administration is going to talk a much bigger game about green energy and about a **transition** to sustainable energy. I also know that Trump is going to talk a lot more about drill, baby drill, and oil and gas and even coal, and he will give much more personal access to his administration from the CEOs of those companies, who will be more frequent guests. And they almost never have a chance to see Biden. But, but, but. You saw that Harris walked back her”I oppose fracking.”She walked it back de facto when she became vice president. She walked it back in dure when she was asked about it on the trail. The United States is now producing more oil than ever before and far more oil than anyone else in the world. Far more than the Saudis, far more than the Russians. That is under a Harris- Biden administration. And I expect that that would be extended under a Harris presidency. Under Trump, there ’ s no question that he is going to make it easier to have permitting for additional drilling in Alaska and other places, and he ’ s going to make the regulatory environment easier to navigate for more infrastructure build, the rest of it. But I don ’ t think he ’ s going to suddenly eviscerate the **Inflation** Reduction Act, because so many of the jobs that come from that legislation are red-state jobs. So much of the investment is red-state investment. And I mean, the highest level of post-carbon energy production in the United States was California a couple of years ago. It ’ s now Texas, right? Texas, which has become an energy superpower in its own right for oil, for gas and also for solar and wind, right? And I also think that both administrations would be moving faster to permit nuclear. So for the United States, it is really an all of above, all of the above **policy**, which, by the way, is also China ’ s **policy**. The only thing is that China is 20 years ahead of the United States. China ’ s been **investing** at scale for a long time in wind and solar and geothermal and nuclear and electric vehicles and batteries and supply chain all over the world, while the Americans have been ignoring it. And so now the Chinese have world **leaders** at scale in all the global renewable energy. And the Americans aren ’ t there yet. But Americans recognize that ’ s the **policy** they need. So you ’ re going to have two superpowers of energy, the United States, the superpower of carbon energy, trying to **transition** and increasingly transitioning. And you ’ re going to have the Chinese, the superpowers of post-carbon energy, right? And the rest of the world saying, what the hell are you going to do for us, as the **climate** is getting warmer and we ’ re not capable of paying for it? And I think the gap between the US and China and the global South is going to grow on this issue, because what the American solution is, well, just **invest** in new technology, make it cheap enough that eventually these countries are going to be able to take advantage of it. That ' s great, But what they want are reparations, because the Americans are the ones, and the Chinese, are responsible for getting us to where we are. The Europeans and these countries won ' t be able to emit carbon in the atmosphere the way we have. And yet, they ' re going to be facing all the costs of the lack of biodiversity, of the lack of **ability** to produce effective agriculture, all the rest. And the hope, the hope – and you see, for example, Eric Schmidt talking about this in the last few weeks, is that you ' re going to make bigger AI bets that will allow these countries to meet their sustainable development goals and allow these countries to deal with **climate** change. That ' s a great bet, but it is an unproven bet as of now. I ' m a believer, but my God, we have to get lucky to not have this be far, far more devastating for people all over the world. HW : Now either candidate taking the idea of reparations seriously, is that really being talked through, talked about? IB : No, in fact, even AOC, when she was talking about a Green New Deal, she wasn ' t talking about a Green Marshall Plan. It was American exceptionalism. I mean, you know, the funny thing about American **policy** is that, you know, you go from left to right and the entire political spectrum is still pretty narrow from a global perspective, right? That is something that I think people in the United States need to be more aware of. HW : Alright. So there are so many other things that we should and could talk about, but we have limited time. So I ' m going to chuck very large topics at you and ask you to very pithily define exactly where we are and what we need to think about. Immigration. You mentioned it earlier. It ' s been a huge issue for both candidates. But who ' s thinking smartly about this, and what should we look for? IB : You ' re seeing more alignment between Harris and Trump on immigration. Not as fast as you ' ve seen in Europe, but in Europe you ' re now seeing EU-wide efforts to coordinate immigration **policy** driven by Ursula von der Leyen, aligned with, you know, people as diverse as Schultz and Maloney and Macron. I mean, even Orban, who just met with von der Leyen and certainly it was a fairly acrimonious meeting. But, you know, Orban ' s point was, when you look at Hungary ' s Viktor Orban, when you look at, you know, the EU **policy** on migration, this is **stuff** he was talking about years ago. So in the United States, you have a lot of, you know, relatively blue cities that have been very, very comfortable talking about being sanctuary cities and being open for lots of illegal immigration, which they are very happy to trumpet when it ' s a theoretical issue. And then as soon as those immigrants start coming over in large number, because these are not border cities, right? Some of them are Canadian border cities. They ' re not border cities which they see as a problem. And then suddenly, they realize, wait a second. We don ' t want to pay for all this. We don ' t want to have all these people in our city. And they become much more cautious and hard-line on illegal immigration. So I do think that if Harris were to become president, you would see a continuation of a much tougher set of **policies** that Biden has implemented in the election period to try to bring immigration numbers down and which, frankly, are closer to the immigration **policies** that you saw under Trump. I don ' t think that there ' s an enormous gap. The rhetorical gap is what ' s much bigger, the willingness of Trump to use immigrant populations, to demonize them, to say that they are somehow genetically at fault when they are, you know, committing violent crimes, even though the violent crime rate overall is nowhere close to what it was in the ' 90s. And immigrants are some of the least likely to commit them, because that means they get tossed out of the country, as opposed to people that actually have a legal right to be in the US. But, I mean, there ' s been so much really disturbing language that I think has increased hatred and leads to more violence and hate crimes against illegal immigrants and frequently against immigrants that aren ' t illegal. I think Trump

’ s much more willing. That ’ s a feature of Trump, of Vance, of the MAGA America First-ism, in part because it ’ s a grievance-based, more nativist ideology. But the actual **policies** on immigration are not so radically different. The big difference in **policy** implementation that Trump has talked about is wanting to send back some 11 million illegal migrants in the US. There is absolutely no way he would be able to implement that **policy**. The infrastructure doesn ’ t exist. It would not be created. But I would say that even if he did 10 percent of it, which is plausible, the inflationary **impact** that would have on labor in the United States would be significant. And I think that the market participants that are looking at the difference between Harris and Trump are looking at tariffs but they ’ re also looking specifically at labor rates and immigration. HW : That ’ s **interesting**. I mean, **inflation** has also been a huge topic of conversation, but I don ’ t know that people are necessarily connecting immigration and inflationary rates. What else should people be thinking about in terms of the economy from these two candidates? IB : The fact that the great deficit hawks are near extinction. That either Trump or Harris would be strongly expansionary in terms of fiscal spend. There would be slightly different focus, I think, that Trump would look to extend the entire tax cut package that would be coming to a conclusion. It was, you know, put in as temporary kicking the can to somebody else. And if that can is kicked to the same person, he ’ s going to expand it all. So he wouldn ’ t get as much money in, and he would also try to reduce corporate tax rates further. The intention would be to juice growth, but I don ’ t think that that would meet the expenditure and, especially in a still comparatively slightly higher interest rate environment, where you ’ re doing a lot more debt servicing as a part of your budget. And also, I think Trump would lean more heavily into higher spending in defense, Kamala Harris, her **policies** would be less growth-oriented for the private **sector**. Though she is trying to lean into small and medium enterprises and entrepreneurship, again, something that you would see, you know, green shoots of that are useful long term, but you wouldn ’ t see that benefit for the four years of her presidency. She would certainly increase tax rates on the wealthy and probably the corporate, the baseline corporate tax rate. So that would bring more money in. But the significant expansion of social programs would more than make up for that. And so both of these presidents would be significantly expansionary on the fiscal side. I don ’ t think it would affect the interest in the strength of the dollar globally, I think that ’ s a much longer-term issue, especially given how weak the comparative other stores of value and economies are around the world, Japan, the Eurozone, China, which isn ’ t convertible. But certainly this is a very big can that is being kicked. And when you talk about US politics frequently, that ’ s what you ’ re talking about. You ’ re talking about short-term, tactical refusal to accept long-term strategic challenges. The **climate** change challenge is the biggest example of that. But the fiscal challenge is a certain second. HW : So many cans, so little time. OK, here ’ s a really great question that we had in from one of our lovely **TED** members.”What can be done about the **pandemic** of misinformation making a mockery of informed voting?”IB : I think as long as we have algorithms that are driving the information that we get and that are being driven by a profit motive as opposed to a civic motive, a community motive, or just an accuracy and information motive, all of those things would be very different. You ’ re going to have this become a much bigger problem. I remember when weathermen and women weren ’ t objects of politicization. That is no longer true. There ’ s weather control, apparently, from God knows who, from”them.”They ’ re doing weather control. You know, hurricanes do not differentiate between Trump voters and Harris voters, and you desperately need to listen to the scientists, on what is likely to come from an imminent, massive storm and tornadoes and storm surge. And, you know, more people are dead today because of that **disinformation**. So if you think that this is unprecedented and bad when we talk about a hurricane, and never mind when we talk about a **pandemic**, then what do you think is going to happen with our democracy? So I think there are a lot of things that can be done. One of them is that bots do not have free speech. They don ’ t have the right of free speech. I ’ m a big advocate of free speech. I think it ’ s an important thing we have in our country. It applies to human beings. So I think that human beings, at least in America, and I understand there are problems in more repressive societies, and for now, the US is not such a society, though there are trends, should be verified. In other words, you have to be a real human being to be able to exist on a social media platform, as a human being, to be verified, to be promoted algorithmically. And if you ’ re not, I mean, there ’ s a place for AI-related content, but that content needs to be not verified, and it cannot be promoted algorithmically. It has to be promoted by a human being who knows it ’ s AI-driven content. I think that would make a difference. I also think that we need more responsibility for civic relations and democracy from our corporate **leadership**. We just do. And I think that we have seen an unfortunate, you know, lionization of a whole class of people that are making an incredible amount of money, they ’ re the richest people in the world, and they show zero concern for the well-being of their fellow citizens. And that has to change. More people have to lead by example. But we are going to need more regulation as well that helps **ensure** that if a business model is actively eroding democracy, that they have to pay for the costs of that. They have to be responsible for the negative externalities. You and I talked about **climate** change. It ’ s fairly clear that if you ’ re a company that emits pollution, you should have to pay for that pollution, because if you don ’ t, the animals will pay, the kids will pay. You know, the society will pay. Someone always pays. And that is equally true when we talk about **disinformation**. If there are costs to a platform – a platform has every right to make money – but if there are direct costs to society that are measurable, that come from the existence and the furtherance of that product, those are costs that also have to be borne by that corporation, and we have not yet been willing to have that conversation. Reparations are one thing. I mean, you know, the idea that you have to pay for the countries that you ’ ve already – But I ’ m talking about things that are happening right now, companies that are making money right now, that are making benefits, that are actively not paying for costs on the back of our citizens and our young people. That is not acceptable, and that has to change. HW : Well said, Ian Bremmer. What are the chances – and this is a horrible question, but I have to ask it – what do you think are the chances that we ’ re going to experience violence after the election? IB : Well, violence, sure. I mean, we experience violence every day in this country, so I don ’ t think you mean that. I ’ m not sure you mean January 6 type violence, because I think that ’ s very unlikely. I think Washington, DC will be locked down. It ’ s the same thing that, like, you know, you had to take your shoes off. Why? Because it was a shoe bomber on a plane, so damned if you ’ re going to put them in your shoes going forward. I mean, we make sure that we can ’ t do the same thing a second time. But I do think there ’ s going to be violence. Look, you haven ’ t asked me about two assassination attempts on former President Trump. At least one of which was incredibly close from happening. I mean, if this kid had been at all remotely capable as a shooter, Trump ’ s dead. And I think that Trump ’ s supporters, who believe that”they”have tried to impeach and convict him twice for crimes he didn ’ t commit, then they ’ re trying to throw him in jail for crimes he didn ’ t commit, and we ’ re going to vote for him anyway. And then they try to kill him, and they killed him. They killed him. I think they do not accept that. And I think that that would have led to serious riots across the country, violent riots with people with guns and not just citizens, but I think also Trump supporters in the military, you know, recruits, you know, enlisted men and women, I think Trump supporters in police forces. I think this would have been certainly far more violent than anything we would have seen since the Vietnam days, and maybe worse than that. And no one asked me about it. In the last several weeks, I don ' t think I ' ve gotten one question from anyone, from a meeting with a **leader**, during the UN General Assembly week, all the **leaders** I met with, or in a media interview or a speech, nothing. And yet we were this close to that. And so I do think that we are, you know, kind of normalizing this process, even though this process is anything but normal. There will be districts that don ' t certify the winner of the election because the elected officials don ' t like the outcome. There will be massive numbers of **court** cases. I think there will be intimidation in certain areas that will try to stop certain people from voting. And I think that after the vote is done, there will be huge numbers of people that don ' t believe that it ' s true, that believe it ' s rigged. And that ' s a dangerous place to be. So I ' m not that worried about the weeks running up to the election, the next four. I ' m very worried about the days and weeks after the election. I ' m not talking about a civil war. I ' m not talking about the US becoming a dictatorship. But I am talking about a period of profound unrest that people alive in the US right now aren ' t really prepared for and they ' re certainly not used to. HW : That is a very miserable note on which to end, but end we must. So I suppose the conclusion is that we should all go vote. I ' m just going to share that this is the first election that I will be voting in as an American citizen, I ' m really very excited, and I ' m very excited that I get to talk to you on these occasions. Ian, thank you so much for your wisdom and for your insight. Thank you to all the **TED** members for being here and we will see you again soon. IB : Thank you, Helen.

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Helen Walters : Hello everyone. Happy 2024 wherever you are. It is January 8, and I don ' t know about you, but here at **TED**, we are looking forward to the year ahead with a sense of nervous anticipation. As we all know, there is a lot going on, from war on multiple fronts to upcoming elections that some say will determine no less than the future of democracy. This is big, this is concerning. It ' s often slightly confounding. So given all that, we want to move forward with eyes wide open, and who better to help us understand exactly what to pay attention to this year than president of Eurasia Group and GZERO media, Ian Bremmer. Ian, hi, Ian Bremmer : Helen, good to be with you. HW : So, Ian, you ' ve just published your annual list of Top Risks for 2024, and I want to dive right in. The very first one that you describe is called"the United States versus itself."So tell us. IB : Helen, the United States today has an incredibly strong economy and military. Its political system is in **crisis**. The US is the only advanced industrial democracy that cannot **ensure** a free and fair **transition** of power that is seen as legitimate by a majority of its population. That is what we are looking at in 2024, and it ' s happening against a context of a geopolitical environment that is very deeply unstable, with a major war between Russia and Ukraine, which is nowhere close to ending, with a major war between Israel and Hamas, which is nowhere close to ending. And so both allies of the United States are deeply concerned about this, and adversaries are looking to take advantage. Now for 2024, we won ' t have a new president if there is a new president, that ’ s next year. So why is it so risky now? Well, it ’ s so risky now because the country is so divided and because Trump is likely to get the nomination overwhelmingly likely, and when he does, his **policy** pronouncements will drive the GOP. And they are not right now, as of today. So in other words, overnight, he will gain the loyalty, regain the loyalty of the overwhelming majority of Republican **leaders** in state legislatures, in the House, in the Senate, and of Republican-leaning and right-leaning media outlets and, of course, the **ability** to raise money and deploy that money for the election. And that means that his **policy** orientations, as he expresses them, whether it ’ s cutting off Zelensky and the Ukrainians or showing the Iranians what ’ s what, and that ’ s why they wouldn ’ t have gone to war against Israel and the US if he had been president, unlike Biden. Or in terms of the border, vis a vis the Mexicans, any other issues, those are suddenly going to be drivers of one half of the US political system. So it ’ s a deeply concerning political environment, and one that the United States is not in a position to respond to effectively. HW : So, Ian, you ’ ve written that there is an unlikely but plausible possibility that the US won ’ t actually even be able to hold a free and fair election on November 5. We ’ re going to talk more about the **impact** of artificial **intelligence** in a little bit, but the reality is that we have seen the **impact** of misinformation on elections before. But in the ensuing time, things have gotten much [worse], much quicker. So what should we be watching for there, and how do you think that is going to play out in November in the US? IB : Well, the United States, as the most powerful country in the world and as a political democracy, which is in **crisis**, the most vulnerable part of the United States is its political system and is specifically its 2024 election. That is the Achilles heel for the United States. When I speak with the **intelligence leaders** in the United States, they say that is what they are most concerned about. It ’ s the vulnerability of the US election. It ’ s not the Russians attacking the Americans militarily, it ’ s not a big fight with the Chinese, it ’ s not Iran, it ’ s the vulnerability of the US elections. And that vulnerability is a comparatively complex and soft target from a homeland security perspective. Especially because Americans don ’ t live in the same information environment. You know, we talk about **climate** change around the world, and everyone agrees that there ’ s 1. 2 degrees of warming. Everyone agrees there ’ s 442 parts per million of carbon in the atmosphere. When you talk about the US political system, everyone does not agree that Trump tried illegally to overturn an election. I mean, in a normal democracy, in a well-functioning democracy, that would obviously be the top issue of debate in the election. Is the fact that you ’ re thinking about re-electing someone that is, you know, not **interested** in a democratic election. That is not what is happening presently in the United States. Not at all. And that is not Trump ’ s fault. Trump is a symptom, a very serious symptom of the fact that US institutions have been delegitimized over decades. And it ’ s getting worse. And he ’ s a very happy beneficiary of that reality. So, yes, I do worry that American political institutions, and specifically, the electoral procedures are vulnerable to that, especially because the stakes are so much higher this time around. The stakes are higher for Biden and his team because they believe that they may face an end to an effective multi-party transfer of power over time if Trump wins. A lot of them individually believe that they would face legal jeopardy from a politicized Department of Justice or FBI or IRS if Trump is in power. They absolutely say that privately. And of course, Trump believes, to himself, much more dangerous than the end of democracy, Trump believes that he would face jail, he and maybe even members of his family, if he were to lose the election. So the stakes are very high indeed in the United States. They ’ re very high indeed outside the United States. HW : I think the reality that you ’ re describing is that whoever wins, whoever the nominees are, Biden, Trump, whoever wins, things aren ’ t necessarily going to get better, right? IB : Yeah, I guess I ’ m trying to say that neither Biden nor Trump have the capacity and the willingness to try to fix this. I don ’ t believe that Trump is **interested**, never mind having the capacity. I believe that he benefits. He believes he benefits from referring to his adversaries politically as enemies, as enemies of the people. And as painting them as an existential threat to the Republic of the United States. You know, they ’ re Marxists, they ’ re communists, it ’ s a new McCarthyism, if you will. And that Biden certainly has the willingness to try to fix this yawning divide in the United States. But four years of being president hasn ’ t fixed it, right? I mean, there was a belief among many Biden supporters that if you elected this **guy**, that the United States was going to be able to become more normal, and Biden has had a lot of legislative successes, his infrastructure package that was bipartisan, that Trump was unable to pass, the **Inflation** Reduction Act, misnamed, but nonetheless significant amounts of money for investment into red and blue-state jobs, more red-state than blue-state jobs, other **policies** as well. And yet the divisions in the United States, the **dysfunction** of the US political system, the perceived illegitimacy of major US political institutions and the media, has only increased under four years of Biden. So you have this very unusual environment where the most powerful **leaders** in the country are not able to fix the problem. And it ’ s not like there ’ s any diplomacy that ’ s happening between them. I mean, the **interesting** thing about our view of the Top Risks of 2024 is that you have these three major **conflicts**, three major wars that are essentially happening in the world. You ’ ve got, you know, Russia-Ukraine, which is now entering its third year, You ’ ve got Israel-Hamas, which is in its third month. And then you ’ ve got the US versus itself, which is kicking off right now. And in all three of those cases, there ’ s no plausible environment in the coming year where diplomacy is going to reduce those tensions, is going to end or even contain that **conflict**. And in all three of those wars, the **leaders** do not share the same information space. They don ’ t even agree on the same basic set of facts. And that is an unprecedentedly dangerous geopolitical environment in your and my lifetimes, Helen. HW : So let ’ s talk about these **international conflicts**. So you mentioned three of them, one, the United States versus itself. Let ’ s turn to the Middle East. Let ’ s talk about what ’ s going on with Israel-Hamas. Do you feel like this is going to spiral into a broader **conflict**? IB : I do, I do. I ’ m not confident exactly what the avenue of that escalation is likely to be, but I ’ m very confident there are so many avenues that that escalation can occur. And it ’ s not within the capacity of the United States or other major countries around the world or in the region to contain it. And let ’ s talk about a few. One, of course, is Hezbollah and the northern border between Israel and Lebanon. And that fighting, as we speak, is escalating. The missiles from Hezbollah into Israel, the Israeli assassination of a Hamas political **leader** in Beirut, the willingness of the Israeli defense forces to go against Hezbollah targets, That is escalating. And it ’ s not just Prime Minister Netanyahu, though he ’ s very relevant, because if the war is over, he ’ s going to be out of power, and he could very well face jail, but it ’ s also the entire Israeli war cabinet that believes that they cannot end this war and allow actors in the region that believe that Israel has no right to exist, they can ’ t allow them to maintain power the way they did in the status quo ante before October 7. And that ’ s not just”destroying”Hamas, however that is defined to be, but also, Hezbollah operating right on the border in contravention of the UN Security Council resolution. So they intend to back Hezbollah off of that, to degrade their capacity to attack Israel to a degree. There ’ s also a willingness of the Israelis to see that happen with the Houthis in Yemen, with Iraqi and Syrian radical Shia proxies of the Iranian government. The United States is now increasingly actively fighting all of those actors in Yemen, in Iraq and in Syria, and increasingly looks like it might be willing to target Houthis in their Yemeni bases themselves. And that ’ s even before we talk about the radicalization of millions of Palestinians and their fellow Arab and Muslim supporters all over the world on the back of the suffering that they are experiencing, principally, but not only, in Gaza. Gaza is not livable right now for two-plus million Palestinians, but they have nowhere to go. Those pressures are going to grow. The Israeli government will be calling for their removal, into other territories like the Sinai, which will be seen to be ethnic cleansing by almost everyone else, including the Biden administration in the United States. That radicalization will lead to lone-wolf violence and terrorist attacks, will lead to coordinated terrorist attacks, not just in the region, but also in Europe, also in the United States. So when you add all of that up, what you see is that it is much easier to understand how this **conflict** will escalate substantially over 2024 than threading a needle to see how you might be able to contain it, largely to Israel versus Hamas in this tiny strip of land that two-plus million Palestinians are trying to live on, in Gaza. That is where we are right now in the Middle East. And this is deeply concerning for the United States, because the US, in their **support** of this Israeli war, is more isolated on the global stage than the Russians were, even when they invaded Ukraine two years ago. And this also is a serious problem, not just for US projection of power around the world, especially with the global South, but also domestically among Biden ’ s Democratic supporters, a majority of whom are more inclined to **support** the Palestinian cause than the Israeli cause. And so this is, as the **conflict** escalates, a proximate danger to Biden ’ s re-election efforts, which he is keenly aware of but has very little he can do about it. HW : Right, the interconnections and the tangled web becomes ever more tangled as more actors get **involved**. This is a really like, a **stupid**, deliberately **stupid** question. But do you feel like any hopes for a two-state solution are vapor? IB : They ’ re not vapor, but they ain ’ t close. I mean, I don ’ t see, in the foreseeable future,”security,”if I can use that term, will be provided by Israel and not by anybody else. And you know, what that means in the context of Gaza is hard to define. It ’ s mostly about security for the Israelis. It ’ s not about security for the Palestinians, but long-term, you can ’ t have a two-state solution unless you have effective governance that is seen as legitimate by Palestinians living in Gaza and the West Bank, and the **ability** to defend themselves. The Israelis have made it very clear, they have a right to defend themselves. And that has failed, in part because the Israeli government was asleep at the switch before October 7. Israeli **intelligence**, defense forces and, most importantly, their prime minister and government, who had other priorities. And also because they are surrounded by a number of organizations that do not recognize their right to exist in the territory of Israel. So, absolutely, the Israelis have a right to **ensure** the security of their people. But the Palestinians have that right, too. The difference is that the Israelis not only have the right, they also have the capacity. They ’ ve got Iron Dome, provided in large part by the United States, financed as well. They ’ ve got an incredible asymmetrical military advantage, not only over the Palestinians, but over everyone in the region. They only need to apply it effectively. And apply it effectively without killing tens of thousands of civilians, right? While the Palestinians have the right to self-defense, as is continually enshrined in votes at the General Assembly by the vast majority of countries from all over the world, including American allies, they just don ’ t have the **ability**. And you can ’ t talk about a two-state solution until you can get the Palestinians the **ability** to govern themselves and defend themselves.

And we are very, very, very far away from that. And since we ’ re just talking about 2024 today, there is no two-state solution in the cards for 2024. As we think about a longer- term environment, there will be no peace in the region until you find a plausible solution that is sustainable, where Palestinians feel like they can raise their children with security and economic opportunity. We would all want only that for our kids, and they presently do not have that. They don ’ t have anything remotely close to that. HW : Alright, so one of the other risks that you determine in the report is one of”partitioned Ukraine,”which is not a phrase that Ukrainians are going to be excited about, but it ’ s one that you actually think is going to become a reality in 2024. So tell us what ’ s going on here. IB : Yeah, I want to be clear that writing about partitioned Ukraine is not a personal preference. It is not something I think we should want. It is not something that the Ukrainians will recognize. It is not something Ukrainian friends and allies will recognize. But, Helen, you and I know that there are many things in the world that we want that are not so. And it turns out that we live with them for a very long time. I mean, the North Koreans, we do not accept that they have nuclear weapons. They don ’ t really care if we accept it. They have nuclear weapons. Denuclearization is not happening, right? So that ’ s a reality. Ukraine is presently partitioned. And their **ability**, as much as the Americans, in principle, would like them to be able to take their land back, and they have every right to be able to do so, the invasions in 2014 and then in 2022, right, were illegal. The Ukrainians did nothing, to, you know, force them. And yet they can ’ t take their land back. The American **ability** and willingness to continue to lead in providing military **support** that would allow the Ukrainians to get the 18 percent of their land that is presently occupied by the Russians and has been for a year now, that just does not exist in 2024. And in fact, the bigger danger is that the trajectory of this war is at a turning point. And that not only will the Ukrainians not be able to get their land back, but that they might not have the people to continue to fight to keep the land they presently have. And that their **ability** to get ongoing **support**, especially from the Americans, when this is becoming a political divide in the US, far away, Israel-Palestine – higher priority, US elections – higher priority. Trump, if he ’ s elected, at least 50-50, I ’ d put it a little higher than that, frankly, though, I don ’ t have a lot of **confidence** around it, would end **support** to Zelensky, no question. This is going to make the Ukrainians feel incredibly vulnerable and increasingly desperate. That is what we ’ re looking at in 2024. And of course, that not only threatens Ukraine ’ s territorial integrity ongoing, and Zelensky as a **leader**, but it also threatens the integrity of the transatlantic alliance, which had been getting stronger over the last couple of years post Russian invasion, and threatens the integrity of NATO as an ongoing alliance, the most important military alliance on the planet today So there ’ s a lot going on here. And presently, the outcomes don ’ t look great. HW : So you have an **interesting** phrase in the report that stated”Ukraine is at risk of losing, Russia has no way to win.”So what does that stalemate actually look like in practical terms? IB : So Ukraine is at risk of losing. Ukraine doesn ’ t have to lose. There are still outcomes where the Ukrainians can win, even though they will be de facto partitioned. Now, that may not sound easy, and it ’ s not. But let me throw you out a scenario where the Americans are able to provide another 20, 30 billion dollars this year, the Europeans continue to provide significant economic **support**, harder with the Hungarians opposing, harder with the Germans and their emergency budget situation, but it can happen. And some of that could be taken, some of the frozen assets, legally problematic, of Russia and applying them to Ukrainian reconstruction. On top of that, you fast-track EU integration and reform in Ukraine, and you fast-track NATO **membership**, which cannot happen tomorrow. But it could happen within, you know, say, you get all the countries together, the NATO allies, and say, within two, three years you ' re going to provide them NATO **membership**. It will not include territory that the Russians are occupying because you ' re not going to go to war directly with the Russians, but it will mean that these countries will defend Ukraine in the remaining territory. That is a narrow path. But if it happens, you will have 80 percent of Ukrainian territory that has a far better trajectory for their people than Ukraine ever could have imagined had the Russians not invaded. None of that, none of that will make good the eight million people who have been displaced, the tens of thousands that have had war crimes committed against them, the children that have been abducted and forced into Russia and on and on and on. I am not minimizing the crimes that have been committed that they will remember for generations in Ukraine. But I ' m saying that Ukraine still has a path to win, given what their country looked like and was facing before 2022 or before 2014. But irrespective of whether Ukraine can accomplish that or will lose, the Russians will not win. And what I mean by that is Russia absolutely will be able to maintain control of a strip of Ukrainian territory. Bombed out, right? And not productive, and years, if not decades, to turn that land around economically. But Russia will now be facing the rest of Ukraine with a visceral hatred for all things Russia for generations. I mean, think about, you know, sort of the Turkish genocide against the Armenians and for how long that drove this historic enmity. That ' s what we ' re talking about. This is like, you know, Hutu-Tutsi **stuff**, right, in Rwanda and Burundi, That ' s what you ' re talking about. And you have an expanded NATO, including Finland, far more territory for the Russians to have to defend, which ostensibly was the reason that the Russians went to war against Ukraine, is because they didn ' t want NATO to be encroaching. Well they ' re encroaching a lot more now. Hundreds of billions of dollars of Russian assets that they will not have access to. Trade of Russia with the Europeans, with the United States – no more, right? I mean, the Germans, much more quickly moving to a post-carbon energy **transition** with no more Russian gas, because they cut the Russians off, at enormous cost to Germany, but greater cost long-term to the Russians. Not to mention the **million-plus** Russian civilians who have fled : young men capable, smart, that got out of dodge to anywhere, to the Emirates, to Georgia, to anywhere that would accept them so they wouldn ' t have to fight. And those are people that you could have used productively in the Russian economy. So, I mean, in this environment, where Russia ' s only true friends, providing military **support** for them, are Belarus, North Korea, Iran, right, I mean, this is – with friends like those, right? I mean, there ' s no way. I understand that if you ' re a tanky, as they call it, or if you ' re **supported** by the Kremlin and you ' re posting dutifully on Twitter / X that you will oppose everything I just said because it ' s your job. But I mean, for those of us that take a blue sky approach, which is that we look at it and we kind of understand the science, it ' s impossible to say the Russians are winning here. And that ' s a great cost to the rest of the world, to the rest of the world. This was the worst misjudgment, in my view, of any major **leader** on the global stage since the Wall came down, was Putin ' s decision to invade Ukraine in 2022. And not only was it a horrible decision, it was a horrible decision that was facilitated by the Americans and the Europeans, who, after the 2014 invasion, much smaller, didn ' t do anything. In fact, they kept doing business with Russia as usual. They hosted the World Cup, president said,"OK, we ' ll still go over."I mean, there were lots of signs that Putin got that said,"These **guys** don ' t care. So we ' re going to get away with this."And unfortunately, they were really, really wrong. HW : All right, let ' s change subject a little bit. We mentioned artificial **intelligence** earlier. And obviously, 2023 was the year in which generative AI went mainstream. So it ' s exciting, and it ' s mind-blowing. And it ' s also led to something of a schism between some people who are super excited about its creative potential and then others who can ' t believe that we ' re barreling towards existential end of humanity with such nonchalance. So you just have to see the management drama at OpenAI that happened at the end of 2023 to understand some of the stakes here. But this isn ' t just about Silicon Valley, and I think it ' s important that we all recognize that this is actually going to affect all of us. So what do you think we should be focused on when it comes to AI, and what are we going to see in 2024? IB : I am both of those things. I ' m an enormous enthusiast about AI. I think it ' s a transformative technology for everyone that has access to it, and we ' re just seeing the beginnings of that. One of the reasons I ' m most excited about AI is because it is transformative not just in displacing existing powerful people and institutions. So **climate** change, people get very excited about the **transition**. But to do that **transition**, you ' ve got to end oil. And so fossil fuel players and those that are attached to them, infrastructure, transport and the rest, have a lot to lose. AI, like, even if you ' re a coal miner, you can and will use AI to be much more effective at mining coal. You know, AI, you ' re like a traditional airline company. You can use AI to reduce, you know, energy intensity in the contrails. You ' re going to do that tomorrow, right? So it ' s astonishing how much uplift we will get from productivity and efficiency in every **sector**, from all sorts of corporations and from individual workers, by using these tools, by deploying these tools. So I ' m very excited about how much we will get out of AI to unlock human potential broader and faster. But I don ' t spend much time talking about that, usually, because you ' ve got a lot of people and companies that are worth trillions of dollars collectively that are spending all of their time and effort doing that as fast as humanly possible. And they have to, because if they don ' t, they ' ve got very smart and well bankrolled people breathing down their neck that would be very happy to displace them, right? That ' s the principle AI displacement we ' re going to see in the next 24 months, will be of AI players that get taken out by other AI players because it ' s so cutthroat and fast. But there aren ' t people that are spending their time thinking about what are the challenges for the common good. What are the negative externalities that come from AI? I mean, so you think about the last industrialization and Americans in particular, my country, we are so great at the private market when it comes to profits. We ' re capitalists when it comes to the markets, because when we do well and money is to be made, we make sure that we deploy that and the shareholders get it, and we ' re focused. But when there are losses, we are the world ' s best socialists. It ' s not us. It ' s like, somebody else. Everyone ' s got to pay for it. We ' re not responsible. Preferably the kids. And maybe not now, but maybe later. And we ' ve seen that with **climate** change. Very happy to make all the money from industrialization. But I mean the costs of emitting more carbon? Not it, not our problem. Well, that ' s a long, global, slow process over generations. AI is a very fast, transformative process over like, five to 10 years. And the negative externalities are going to happen essentially simultaneously with all of the positive productivity and efficiency gains. And we need a governance environment where those are accounted for and paid for. And in the near-term, I am not talking about the robots taking over, existential risk of artificial general **intelligence**. I ' m talking about AI being used by bad actors or indifferent actors with challenging business models in ways that will undermine stable society. I ' m talking about, in particular, **disinformation** and how it can be deployed to undermine democracy. And I ' m also talking about the disruptive nature of proliferated AI tools in the hands of large numbers of governments and institutions and individuals that are bad actors or tinkerers that want to destroy things or thinking about it. I mean, whether they ' re using AI for malware or they ' re using AI to build **viruses** or lethal autonomous weapons. And I think that that is really becoming a risk only for the first time in the coming year. In elections, like the US election, especially as the next generation of AI comes out and this is moving three times faster than Moore ' s Law. So every six months you ' re getting a doubling capacity, which is, you know, not an environment that traditional governments are able to respond to. It ' s too fast, it ' s too powerful. And you ' re also getting those tools in the hands of hundreds of millions of people in very, very short order. So there is governance, it is coming fast, it is urgent. But in 2024, the speed of the technology is much, much faster than the speed of the governance. And that gap will create **crises**. HW : I mean, we ' ve seen, from the whole of the OpenAI kerfuffle that happened at the end of the year, you really see these tensions kind of playing out in real time. And there are people within large corporations who are trying to do the right thing internally, but then, of course, competition is actually driving them to move faster and faster so that they don ' t get left behind. Given the fact that, as you say, there are these bad actors outside of these organizations who are also trying to do their thing with the open-source technologies that are being released, etc. What actual governance is possible in that environment? IB : There ' s a lot of governance that is possible in that environment, but it ' s not clear how quickly we can get there. So I mean, the EU has their AI Act, which is quite comprehensive in the way it thinks about regulation and transparency of foundational models and testing of those models of AI. And in making sure it ' s not just being done internally. And in watermarking and trying to make sure that people are aware of what kind of images they ' re seeing and audio, and what ' s being driven by AI and what is not. And also the deployment of AI models and what kind of data they have access to, all of those things. And the US has an executive order, which is not as powerful as legislation, which is not coming anytime soon, given the divided nature of the US government. But still, you know, represents a step change in how you think about regulating these technologies. But these processes are happening much more slowly than the **tech** is rolling out. So I think what governance will get you is it will identify the actors that are critical, both inside governments and in the private **sector**. It will have them talking about the issues that really matter, and better set up so that when a **crisis** occurs, and it will, that they will be able to identify and respond to it collectively and much more quickly than they would absent that governance structure. So in this regard, AI is a little bit more like the financial **sector**, where we all know that we need a functioning financial **sector**. And that ' s true whether we ' re capitalists like in the US or we ' re state capitalists, like in China, you got a free, you know, sort of, convertible currency or you ' ve got a closed market. Doesn ' t matter. We all know we need a financial market. But we also know that in a global financial market, that individual actors can cause systemic **crises**. And so we need a stability board, that when there is a **crisis**, everyone identifies it and responds immediately so that we don ' t have a global meltdown, don ' t face a depression. That ' s what governance is going to need to be able to do on the AI front. And the thing is about AI, we ' re not just talking about, oh, someone ' s making a run on the market and this company is going to go bankrupt. The risks that potentially come from AI and misuse of AI are in many, many different types of technologies and applications. So you ' re going to have to, you ' re going to have to essentially build the ship as you ' re steering it, because the dangers are going to change very quickly. HW : No problem, we ' ll do that just fine. So there is another part to the **tech** challenge of 2024 that is actually around the building of **tech**. And that ' s about the minerals and the materials that actually go into building **tech**. So your hunch within this report is that governments are going to turn protectionist, they ' re going to disrupt the flow of minerals that are needed to build all of this technology. So what should we be paying attention to here, and how much of a risk do you think that this is for the future? IB : Well, one is, it ' s related to the high-tech front, which is that the biggest fight between the US and China, which thankfully has a more stable and more managed relationship in 2024 than they did coming into 2023. One of the big areas of tension is on the high-tech side. And the Chinese are trying to develop AI, the Americans are trying to develop AI, but the US, with their allies, are now taking semiconductor capabilities and trying to **ensure** that they are being built in **trusted** countries, and they ' re putting export controls on semiconductors, cloud computing, related infrastructure. And so the Chinese don ' t have access to the most valuable **stuff**. Now, China doesn ' t build its own high-tech semiconductors. And they are, you know, by most accounts, about 10 years behind. And they don ' t have a really good way to counter that, aside from **investing** as much as they can inside their own country to rebuild it. So some of what they ' re trying to do is see if there ' s a way to engage with the Americans that might loosen some of those export controls. And that ' s one of the reasons, the principal reason, actually, why China expressed a willingness to join a new Track 1. 5 dialogue with the Americans on artificial **intelligence**. But, the Chinese response has been, well, you **guys** are **dominant** in semiconductors, but we ' re **dominant** in critical minerals, in the exploitation and in the supply chain. And also in the development of a lot of the new post-carbon energy generation and infrastructure : solar cells, wind, batteries, that kind of thing, electric automobiles. And so the Chinese are basically saying, we ' re going to look at some of those critical minerals that we dominate, and we ' re going to put licensing regimes on those. And if you stick with the problems you give us on semis, we ' re going to start putting export controls there. Now, the difference is the Americans will still be able to buy all of that **stuff**. It will just be more expensive. So the Chinese can ' t cut off the Americans the way the Americans and the South Koreans and the Dutch can cut off the Chinese, but they can make it really painful. And ultimately, all of this is about relative power. All of this is about relative gains. It ' s not about like, you know, destroying, the Americans aren ' t going to destroy China and vice versa. But it does mean that as the world barrels towards an effort to **transition** from primarily fossil fuel-developed energy to primarily post-carbon-developed energy over the period of one generation, that it will be far less efficient because of this big fight between the United States and China. And at the same time, we are seeing a number of governments around the world put industrial **policies** in place to provide subsidies for their own workforces, governments that have access to a lot of these critical minerals that are saying, we can use this to get up the supply chain. So we want to make sure that you ' re **investing** that new technology in our country and not just exploiting us for our resources, all of which is creating significant, politically induced costs in that big **sector** of the global environment at a time that you want those costs to be as low as possible so you can move away from fossil fuels, right? I mean, the more expensive the critical minerals are, the bigger the barriers, the harder it ' s going to be to go from coal and oil and gas to all of this new sustainable **stuff**. In 2024, that ' s becoming a significant fight. HW : Everything, everywhere, all at once. Everything is interconnected, is I think a theme of this conversation. I will flag for **everybody** that there are 10 risks that you are flagging this year. We have not had time to go through them all, so I would highly recommend that you go and read the full report. I ' m going to, spoiler alert, just flag something that you wrote right at the end of the report, Ian, that I think is really beautiful and worth holding close as we go into 2024 :"... that it is critical we don ' t just talk about these global issues to help make business and **policy** decisions, but also to connect to those closest to us. If we can ' t make a difference with those we know and love, we are lost."Ian, you are a poet, it turns out. IB : I don ' t know if I ' d go that far, but I ' d like to believe I care about my fellow people. And this is a tough year. Look, I mean, I just came back from the South Pole, of all places. It ' s a place that I ' ve always wanted to **visit**. By virtue of my job, I travel all around the world. And this is this massive continent at the bottom of the planet that I ' ve never been to. And it just looks like this big thing of ice. But this also felt like a year that being off the grid and connected to the planet, but not like in the headlines, in the news, every moment. It was very useful just in terms of clearing my head a bit. But also, as a political scientist, being in a place that for 60-some years now we ' ve managed to govern, if I can use that term loosely, in a way that protects it for future generations and for the planet. And there used to be all of these competing territorial **claims** by all these different countries. And now we actually have a treaty that says we ' re not going to have any resource exploitation. We ' re not going to use this for military bases, we ' re just going to use it for science, and we ' re just going to keep it for the planet. We ' re melting it, I know, because of global challenges. But the reality is it still kind of is the way that we wanted it to be, was the first arms control agreement that was ever signed between the Americans and the Soviets back in 1959, and the treaty holds true until 2048. And when I talked about it, when I just came back, I had some people that were saying, oh, well, you know, 2048, the Chinese are going to be like, looking on how they can exploit it, and the Russians, it ' s not going to last. It ' s like, you know, if we ' ve got something that works and is going to work until 2048 in today ' s environment, like, that ' s a win, right? I mean, we don ' t know where we ' re going to be in three months right now. And it turns out that we as human beings are capable, sometimes, when we put our minds to it of being stewards. And we need to do more of that. The thing that worries me the most about 2024 is we have a lot of very big, very real fights happening right now, and no one ' s acting like a **leader**. No one ' s acting like a steward. People are all focused on a very narrow view of what ' s at stake for them and the people that like them. And that ' s not humanity, right? It ' s very ephemeral. And and when we all leave this planet, after this very short spell that we have here, we all go back to the same place where all the same atoms. And we need to take a little bit of time to connect with the people around us, to remember that. That ' s what I took away most from my little trip down to the bottom of the planet. And I ' d like to try to hold that with me as we get through 2024. HW : May we all borrow that, may we all have that sensibility in mind as we go through this year. Ian, thank you so much for your time. IB : Thank you, Helen.

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Hello **everybody**, I ' m Helen Walters, I ' m head of media and curation at **TED**, and I am delighted to welcome you to another episode of **TED** Explains the World with Ian Bremmer. Ian, of course, is the president and founder of Eurasia Group. He is the head of GZERO Media, and he is here to talk to us about what just happened in the United States. Today is November 7, the election was on November 5. President Trump was reelected in handy terms. So, Ian, please tell us what you make of what is happening right now. Ian Bremmer : You ' re right, it is not just the electoral vote but also the popular vote that Trump was able to win. I mean, it ' s close, you know, 51-49, so half of the Americans went against him, pretty much. We always knew that was going to be the case. So it ' s not as if the polls were radically off. That ' s not the issue. One thing that ' s quite useful, of course, is the fact that a popular vote win, which doesn ' t always line up with the electoral vote, creates more legitimacy for Trump. And the fact that this wasn ' t just a matter of one or two states, and that it ' s clear that there was neither significant internal nor definitive external interference. Clearly, there was a lot of **disinformation** and, you know, bomb threats mailed in, called in by the Russians. But nothing that would have changed the outcome. And so you were able to not only get Kamala Harris to concede in short order, but Democrats across the board, whether or not they ' re happy with it, recognizing that Trump is indeed their president. So unlike in 2020, where Trump himself precipitated a very significant challenge, saying that the vote was rigged and undermining the legitimacy of the outcome, here, we do not have that. Furthermore, the Republicans have, as of now, taken the Senate. Of course, **conservative** justices have a majority on the Supreme Court, and the Republicans, in short order, are very likely, overwhelmingly likely, to have a majority in the House as well. And what that means is that the Trump administration will be able to pursue the agenda, both domestically and in foreign **policy**, that they have said that they wish to pursue. For example, on tariffs or on taxes, or on immigration. All of these things that Trump was running on and now will have a mandate to pursue. So that is the long and short of what happened on Tuesday. HW : So you talk about tariffs, you talk about taxes. A lot has been said about the fact that Americans were voting related to the economy and how they felt about the economy. Do you agree with that? And what do you think that Trump will do with fiscal **policy** when he actually comes into the presidency again? IB : Well, let ' s talk about what Americans voted on. They voted on a country whose direction they did not agree with. Over 70 percent of Americans, Helen, say that they did not agree with where the country was heading. And when that happens, it is very, very hard to win as an **incumbent**. And vice president Harris may not be the president, but she certainly had accountability for alignment with the **policies** of her Biden administration. And when she was asked on the media, when she started doing her media tour after the debate with Trump, you know,"What would you do differently from Biden?"Her response, and this is the most important question that you could ask her in this campaign, her response was,"I can ' t think of anything that I would do differently."So most important question, worst possible answer. And she tried to amend that in various ways later on, but she was never able to really get away from what she would do differently from Biden and why she would do it differently. I mean, she moved a number of her **policies** in a more centrist direction, but she didn ' t really explain it. She didn ' t really disassociate herself from previous iterations of her **policies**. So pretty much **everybody** out there that was unsatisfied with where the United States is going felt like she did not represent change. I mean, she ' s younger, she ' s different, she ' d be the first woman. From a **policy** perspective, they didn ' t believe that she represented change. And of course, Helen, this is what we have seen around the world. In the developed world, every democracy that has had an election this year has voted against their **incumbents**. Canada is probably about to have one. They will also throw out Trudeau unceremoniously. A lot of developing countries, India. Modi was doing pretty well, now he ' s in coalition. South Africa, the ANC, for the first time since Mandela, now in coalition. Mexico, is the only country of note that had an election this year, in the year of elections, that actually returned the same party. And, you know, in many ways, because the Morena party and AMLO are still seen as the outsiders against a deep, entrenched, you know, sort of power of existing oligarchs, business, you know, and the like. But if anything, they ' re the exception that proves the rule. And so, you know, you and I know, we ' ve talked about this before, I expected that Trump would win. I didn ' t have strong **confidence** in that call because Trump was himself very unpopular. But to the extent that anyone you would think would win an election in this environment globally, it was Trump ' s to lose. And, you know, he was able to pull it through. So I think that ' s the backdrop for what people were voting about. And specifically what it is, it is **inflation**, which is high and it ' s been coming down, but the overall prices, it ' s not like the prices are lower than they were a year ago, just because the **inflation** rate is coming down, those prices are still very high and you can ' t get away, run away from that if you ' re Harris. Immigration, something that, frankly, the Biden administration was very late to recognize was a problem not just in red states but in blue states, too. And those numbers are coming down, but the illegal immigrants that came to the US are largely still in the US. And then also **disinformation**, a large amount of **disinformation** that has made it almost impossible to have a national debate on **policies** and issues. And an awful lot of people, even that align with Harris ' s **policies** didn ' t necessarily believe that she was implementing them or Biden was implementing them. A lot of people in America think that the **inflation** rate is still the highest ever, think that unemployment is a lot higher than it has been. And the people that say that and believe that were far more likely to vote against the **incumbent**. So I would say, you know, in descending order of importance, what drove this election was **inflation**, immigration and **disinformation**. And that is something that we have seen all around the world this year. HW : I think it ' s all incredibly **interesting**. And you said something **interesting** that I want to just double click on there, which is this idea that people have feelings about the way things are, and they don ' t necessarily know what the actual facts about a situation are. So this is also an election in which there has been plenty of **disinformation**, as you say. Elon **Musk** wrote yesterday on X,"You are the media now."And I think we can argue a lot about how the media has covered the election, about the way that the media reports things, the horse race, all of that vapidity, all of that type of thing. But am I alone in finding that type of statement both facile and alarming? And isn ' t that driving us even further into a future where there are these kind of fractured narratives or fractured truths that people have, that actually don ' t coalesce to reality or accuracy? IB : Well, when I saw it from Elon, I mean, I ' m like, I thought he was talking to me, and saying,"Ian, you ' re media now."I ' m like, about time, right? I mean, here I am, a political scientist, I ' m talking to a lot of people. So yeah, they should be thinking about me as media. Of course that ' s not what he was saying. What he was saying is that the mainstream media, CNN, MSNBC, Fox... You know, and by the way, he never includes Fox, even though Fox is every bit as much mainstream media as the other two because he likes one ideologically now, he doesn ' t like the other two, but you might as well have some intellectual consistency around it. His view is that all of those publications, cable news, that they are fake news because, you know, of course, he ' s not making money out of those. And that Twitter / X is real news and that citizen journalists are the real media. Now, I have a lot of things to criticize mainstream media for. I believe that they have gotten way too high on their own supply, focusing on their own interests, their own clicks, their own news, they ' re less **trusted** than they used to be. We see that across the country. They ' re more politicized. They ' re more for their own candidate, whether it ' s Fox on the right or CNN and MSNBC on the left. But at least the journalists on Fox not, you know, Hannity, but the journalists during the day that are writing the stories and bringing you the news, the journalists during the day on CNN and MSNBC, the journalists that are working on the Wall Street Journal and the New York Times, not the opinion **leaders**, but most of the people that are actually doing the reporting, they have expertise, they ' ve been trained. And of course, the big difference between citizen journalists, if you want to call them that, in other words, people that are posting random information that they think about on social media, number one, they tend to have stronger biases because it is not their professional job to try to mitigate them when they present things. Number two, they aren ' t professionally trained, so they don ' t have as much expertise in how to deliver that message and how to **ensure** that a headline is a headline, and you follow through with arguments and you actually properly cite things and there ' s research. And then number three, a lot of the so-called citizen journalists, the verified citizen journalists that have blue checks, are bots and are anonymous and aren ' t actually people. Or are bad actors that are actively pursuing and displaying **disinformation** for their own purposes. And that is, of course, the antithesis of what information, political information is necessary to run a civil society. I deeply worry that we are in an environment where almost everyone I know, educated people in the United States that voted for Trump and voted for Harris believe things, believe some fundamental things about the political system in the US that are not true. I mean, the number of people that I have spoken to that believe that large numbers of non-citizens vote in the United States, which has been, you know, assertively researched and audited, and you ' re talking about less than 100th of one percent of Americans, non-citizens that are on the rolls. It is not an issue. And yet you wouldn ' t know that if you read Elon ' s posts, because he actively has worked to promote that lie as strongly as he can. And I think that ' s a horrible thing. And I have seen that, I mean, I saw it most recently, I know a lot of Democrats that believed, a lot of Harris supporters that believed that Trump actually called for Liz Cheney to be executed, to be in front of a firing squad. And if you had watched what he said in context – that was taken out of context – you know, he was not saying that. He was saying that she ' s a neocon who **supports** wars, like in Iraq, for example, and Afghanistan, and that how would she feel if it was her that was facing the firing, as opposed to the people that they are sending off, the soldiers they ' re sending off to die? Something I ' ve heard many Democrats and **Republicans** that are antiwar, far left and far right, historically say. So that environment, the fact that information is being used in service of a political agenda, and that is what matters to you, is one of the most damaging things I see facing democracies today. It is an unsustainable trajectory. We will not maintain our democracy if we continue with it. I get things wrong all the time. You know, I ' ll make an analytic call. And I thought Trump was going to win and he won. But back in 2016, I thought Trump was going to lose. And I had reasons to believe that. And I was wrong. And I came out and explained why I thought I was wrong. I get things wrong all the time, but I do it honestly. It ' s not in service of a political agenda. I work my ass off, as do all the analysts at Eurasia Group, to try to help people understand and explain what is happening in the world. And, you know, not just from a left or a right-wing perspective, or an American or a Canadian or a Russian or a Chinese perspective. And I travel all over the world and talk to people from all over the world to try to help inform that. That is a tiny, tiny, tiny fraction of the information that people digest today politically, and particularly today politically, in the two-year run-up to a 10-billion-dollar national US election. And that is no way to run a representative democracy. And all the people out there that are saying that they don ' t believe in their media, and they don ' t believe in their elites, they don ' t believe in their political **leaders**, their Congress, their executive, even their business **leaders**, even their scientists, they don ' t believe in them, that ' s why they ' re saying that. Is because they ’ re existing in an environment that they can no longer ascertain truth from fiction. That ’ s not a sustainable place to be. HW : It seems like a deeply dangerous place to live in. You floated something yesterday that I hadn ’ t heard and thought was really **interesting**. It was that potentially **Musk** might buy Truth Media. Have you thought any more about that? Do you think that might happen? And what does that mean if that does happen? IB : So first of all, I have a hard time imagining that Trump is going to, as president, be able to continue to own and and post on Truth Media. Now you know, unprecedented things can happen. And Trump has said that, you know, you cannot, as sitting president, commit a crime. So, I mean, in principle, that means that rule of law, as applies to him, is what he says it is. The Supreme Court has punted on that in terms of what an official act is. So, you know, we ’ ll see where that goes. We ’ re going to be in an unprecedented place. But certainly I could easily imagine... First of all, I believe that **Musk** ’ s 75 million dollars in favor of Trump is probably the smartest strategic political bet that I have seen made by a billionaire, by an oligarch in the United States in my life. I ’ ve never seen one that I think is likely to pay off better, than what Elon just did. Remember, this was a Biden supporter and he turned off from Biden when Biden decided not to invite him to the electric vehicle summit. Because, you know, Tesla, even though it ’ s way in front on electric vehicles, isn ’ t a union shop, and Biden decided he was going to play politics with that, as opposed to lean into **ensuring** that the US has the best possible electric vehicle future in the world. Elon took exception to that, turned against Biden. And the rest, as we say, is history. Big own goal by the president. And I hope he ’ s reflecting on how that was a really dumb thing for him to do. Leaving that aside, I think that Elon is now in a unique position to help formulate what the values of the United States are, and to distribute those algorithmically from himself with his hundreds of millions of followers and from the president of the United States. And there are ways that that could be used to promote American interests. There are ways that those could be used that are inimical to US interests. And again, I promise, if Elon does things that I think are useful, you ’ ll be the first to hear it from me. So, for example, I was in China recently. I met with the **leadership** and I met with Wang Yi and members of the Politburo, many others. And Elon **Musk** had recently been to China, and he travels there frequently. And they all wanted to know from me, while Elon is presenting himself as the **guy** that can help **ensure** that US-China relations don ’ t become maximally confrontational if Trump becomes president. Is that true? Is he the **guy**? And of course, Elon has very, very strong business interests in China, developing artificial **intelligence** with Chinese scientists, Tesla on the ground manufacturing and selling into the Chinese market. China is very important for Elon. And certainly, Elon would not want there to be a decoupling between the US and China. And I think that he ’ s going to have a lot of influence over US **tech policy**, especially because Trump didn ’ t really have a technology **policy** in his first term. He had a tariff **policy** on China. But the CHIPS Act and the export controls on semiconductors, that was all done under Biden. And so, Elon, to the extent that he cares about anything, it ’ s going to be technology **policy**. He ’ s going to have, I think, a very strong position to be able to help determine who ’ s appointed in those key technology positions under Trump, and also what kind of regulations, subsidies, stimulus will be enacted by a Trump administration for technology, broadly speaking, and of course, for Elon ’ s own companies, which I expect he ’ ll benefit from mightily. Now the question will be, is Elon going to be able to facilitate a more functional relationship between the US and the Chinese in advanced technologies? And the answer to that may well be yes. And if Elon helps avoid a cold war between the US and China, I will absolutely say that. Now we know that Trump has had a strong view on wanting to enact much stronger tariffs on China for a very long time, well before he got **involved** in politics. He ’ s thought the tariffs were, you know, a critical component of US economic and foreign **policy**. Trump ’ s”America First”means more capital in the United States, more jobs in the United States. It means, you know, bringing back, reining in free trade and market access, and instead using the power of the dollar and of the size of the American market and the strength of industrial **policy** to get other countries around the world to nearshore with the Americans. And Lighthizer in particular, who I expect will run trade in some manifestation under Trump, has said he wants to see 60 percent tariffs on all Chinese exports. Well, I mean, is Elon going to be able to, you know, help facilitate a deal on that with a China that is facing very serious economic challenges right now? And the answer to that is untested. That ’ s a very **interesting** proposition that the Chinese are hoping Elon will help with. And he has said,”I ’ ll be able to help with.”Now if it turns out that he ’ s able to help with that, this **guy** is absolutely golden in the United States and in China. He could become the most powerful person on the planet. If he is not able to do that, then he will be in a personal position of having directly disappointed the Chinese president and their **leadership**, and I would not want to be in that position. That strikes me as a challenging position to be in for someone who does a lot of business in China. So that ’ s going to be an incredibly **interesting** thing to watch, Helen, I mean, we ’ re going to see this play out. Again, we have a lot of pieces that are moving geopolitically around the world. This is only one, there are other really big ones, Europe, the Middle East. But this is one that ’ s really, really **interesting**. HW : Super **interesting** and some very big personalities that will be arguing constructively about it all, I ’ m sure. OK, so let ’ s talk about foreign **policy**. Let ’ s get into it. Zelenskyy obviously reached out to Trump almost immediately that he won the election and was very complimentary to him. So what is going to happen with Russia, Ukraine? What are you seeing there? IB : And how could Zelensky not, Helen? HW : Totally. IB : You know, he is a master communicator. He ’ s been out there marketing himself and his cause with **everybody**. Trump has even called him, like, doing the best sales job on the United States. And on the one hand, that ’ s critical because Trump thinks that the US is spending far too much on Ukraine. On the other hand, it ’ s begrudging admiration because Trump sees himself as the best salesperson out there, right? So, I mean, you know, as my mom would say,”Don ’ t shit a shitter.”And she used to always say that when she was alive. I saw a flash of my mom in Trump ’ s comment there. I think that the fact that Zelenskyy said,”Congratulations, great win.”You know,”We had a wonderful meeting together in the United States in September, and I want to work with you.”That is not likely to be reciprocated by the great man, the president-elect. I suspect that he wants to end the war. He has said he will end it in 24 hours. Now Trump does exaggerate. It might not be 24, it could be a long weekend. There could be bathroom breaks in there. But he has repeatedly said,”I ’ m going to end this war. Don ’ t even need to be president, I can just do it in the lame duck.”What does that mean? Well, it means he wants to stop the fighting. He really does. And to be fair, I know a lot of people in the Biden administration, running the Biden administration, that want to end the war because they think that the Ukrainians are losing, and it ’ s going to get harder and harder over time. But you ’ ve got to convince Zelenskyy to do that. Now what I expect Trump will do will be call Zelenskyy, call Putin and say,”You ’ ve got to freeze the **conflict** where it is, no more fighting. That means Russia, you basically are occupying the territory you ’ re occupying, but you don ’ t get to keep bombing the rest of Ukraine. Ukraine, you ’ ve got to take it. But you don ’ t have to worry about defending your cities. And then we ’ ll sit down and we ’ ll have negotiations on what that ’ s going to look like going forward. And if you don ’ t accept that, Ukraine, I ’ m cutting you off, and Russia, I ’ m putting more sanctions on.”That is the opening gambit that Trump intends, mano a mano, to end the war. So, Helen, there ’ s a couple of very **interesting** things that then happen. One is, is Zelenskyy prepared to accept that to begin negotiations? Can he get to a limited ceasefire, a freezing of the **conflict**? Under a Biden administration, the answer would have been clearly no. His position is much worse in a Trump administration. We will see how he responds. The consequences would be very negative if he says no, but he could say no. Politically, it ’ s very hard for him to say yes. You know, he could lose power if he does. He will certainly undermine his position with a lot of Ukrainians that had been bravely fighting, **supporting** those that are bravely fighting. The Russians, you know, much easier for them to say yes if Ukraine says no. If the Ukrainians say yes, you know, I ’ ve been speaking to some folks advising the Kremlin. I ’ ve also heard from others in the last 24 hours that have said, well, Putin wouldn ’ t be prepared to accept that unless there were other things like Ukraine is disarmed, can ’ t join NATO, all of that. Would Trump put any of that on offer? Would he be capable of putting a lot of that on offer, given where the Europeans are? How does Putin react? If Putin says no, what ’ s Trump going to do? That ’ s an **interesting** question. But Helen, the most important question is that the Europeans are not likely to be consulted by Trump. And if they are consulted by Trump, he certainly doesn ’ t worry about coordinating a united **policy** with them and Ukraine before he contacts the Russians. Biden wouldn ’ t even talk to Putin, and he wouldn ’ t consider a negotiation with the Russians until the Ukrainians and the Europeans were all onboard. That is absolutely not what Trump ’ s position is. So what we have to look at here is whether the Europeans are going to take a more united front, confronted with a Trump that most of them really don ’ t agree with. Will Europe be stronger together, facing not only Trump giving the Ukrainians an ultimatum and talking directly with Putin, but also doing things like threatening tougher sanctions, tariffs against the Europeans? I mean, we already know that Viktor Orban in Hungary is more inclined to work with Trump. He ’ s made the Mar-a-Lago pilgrimage and all the rest. Well, what about Giorgia Meloni, who ’ s quite close to Elon **Musk**, ideologically oriented to Trump, but has been very anti-Russian and has a lot of popularity in Italy right now? Might she shift away from Ukraine towards Russia in **support** of a Trump **policy**? We don ’ t know. What about Germany? They ’ re about to have new elections. What ’ s a new German chancellor? We don ’ t even know who that person is. We don ’ t know how well the far right in Germany would do in those elections, how much more aligned they might be to Trump. So one of the most important questions geopolitically will be, do the Europeans hang together in **support** of Ukraine with a much tougher set of relations with the United States, or do they fragment with a meaningful number of them embracing Trump, flipping on Russia and saying,”We don ’ t care about Ukraine anymore”? And if the latter happens, what happens to the front-line states in Europe that see Russia as an existential threat? Poland, the Baltic states, the Nordic states? I mean, these are questions that we will have answers to in very short order, but right now we are completely in no-man ’ s land. These are unanswered questions right now, and they could go any which way. And all of the European **leaders** I ’ ve spoken to in the last 72 hours, they are mightily concerned about exactly this issue. This is priority number one for them. HW : Alright, so we are more than a year into the **conflict** in the Middle East between Israel and Hamas. Obviously there ’ s a lot going on there. We have talked before about how you thought that Netanyahu would be holding out until Trump got elected, was hoping for that to happen. It has happened. So what do you think happens next in the Middle East? IB : Yeah, I thought it was very hard to imagine that he was going to agree to a ceasefire where Biden would be seen as the broker, he had no interest in that. He wanted Trump. By the way, the Israeli people want Trump. There was a"Jerusalem Post"survey recently. I think it was over 60 percent of Israelis say they wanted Trump and 12 percent said they want Harris. That ' s the biggest gap we ' ve seen with the US ally. And it ' s because Trump, when he was president, he did the Abraham Accords. He recognized the Golan Heights, occupied by Israel, as Israeli territory. He moved the US embassy to Jerusalem after many presidents promised to do it, Trump actually did it. So, I mean, Trump ' s bona fides as a very strong pro-Israel president, even stronger than the pro-Zionist Biden, is a really big deal. Remember, Trump ' s first trip as president was to the Middle East. No American presidents do that. He went to Saudi Arabia, then he went to Israel. So I ' m not surprised that the Israelis and that Netanyahu in particular are very, very comfortable here. I think that the question, first of all, I still think there is room for Biden to get a **negotiated** settlement with the Israelis and Hezbollah. They are close. If you made me bet right now, within two weeks, maybe three, in other words, in the lame duck, I think that there will be a settlement between Israel and Hezbollah. They ' ll stop the fighting. The Israelis, and particularly the prime minister, are not looking to destroy Hezbollah the way they have Hamas. They are looking to push Hezbollah back, get the Israelis that have been evacuated back into their homes, back to their schools, and then stop the fighting. Netanyahu understands that this is a much bigger fight that would cause a lot more damage to the Israeli economy, that he doesn ' t necessarily want over the long term. He ' s also done a lot to destroy Hezbollah ' s **leadership** and degrade their military and communications infrastructure. So that is a narrow win that I think can be taken off the table in the Middle East. Gaza is very different. I don ' t see any change in Israel ' s **policy** on Gaza. I think the humanitarian **crisis** for the Palestinians living in Gaza will actually get worse, if you even believe that it can, it will, particularly in the occupied north, and especially as UNRWA, which is the United Nations agency that actually is responsible for the infrastructure and for bringing humanitarian aid in, has been just voted by a large majority of the Israeli Knesset as illegal. They will no longer work with it. So, I mean, really anything that looked like it was the potential to build infrastructure for governance in Gaza has either been bombed away or is being unwound. And so I think this is even more of a disaster for the Palestinians. I don ' t see any move towards a two-state solution, towards independent governance by the Palestinians, towards anything that would stabilize in the near term. Iran is the big question. So Trump has criticized Biden for constraining the Israelis in their response to the Iranian strikes of 180 ballistic missiles against Israel. Didn ' t kill any Israelis, but they did launch those missiles. And they did actually strike Israeli military targets. And Jared Kushner has recently written some notes that he ' s circulated around on how this is a unique opportunity for the Israelis to rid themselves of the Iranian nuclear threat once and for all. So unlike Biden, who has worked to prevent the Israelis from striking Iranian energy and nuclear targets, it appears that Trump is goading them to do precisely that. Now there ' s a very big question. Is he getting them to do it while Biden is still president? And then they end all of that and Trump can say,"I ended the war, I came in,"or would he rather wait until he ' s president so he can coordinate militarily, provide the **intelligence**, all the rest, between the United States and Israel to together"handle"the Iranian threat? But I do believe the likelihood of an expanded military confrontation between the US and Iran is relatively high.

Now remember, when Trump was president, he ordered the assassination of the head of the Iranian military, Qasem Soleimani, and the Iranian response was to do virtually nothing. And there were those, like the head of the Joint Chiefs of Staff, Mark Milley, who at the time were saying that Trump wanted to go even much harder against Iran than he ended up actually doing. So I mean, Trump, I think, probably does believe that the Iranians are a paper tiger and that this is a great opportunity for Israel and or the United States to take care of that problem. That ' s a very **interesting** point. Also, let ' s keep in mind that the Iranians, who have been **involved** in interfering in the US election, like China, like Russia, but unlike Russia, who wanted Trump, the Iranians want Harris. And the Iranians have tried to assassinate Trump and Mike Pompeo and others **involved** in the Trump administration, something that I ' m honestly very surprised hasn ' t gotten more attention in the US media. So I think for many reasons, Trump and a Trump administration would feel like it is time to hit the Iranians back pretty hard and show them that you don ' t mess with a Trump-led America. So yeah, again, I think that this is a big deal. And from a global perspective, if that were to happen, that would lead to much higher oil prices, at least for a period of time, because the Iranian capacity to disrupt the Straits of Hormuz and prevent a lot of oil from being transited globally is significant. HW : So you talked about Milley, you talked about Lighthizer. Who do you see as making up Trump ' s cabinet? Who do you think is going to be appointed? We know a lot of people have said they wouldn ' t come back. What are you hearing and what should we be watching for? IB : Well, certainly not Milley, who ran the Joint Chiefs, since he ' s recently said that he thinks Trump is a fascist. So I think that ' s probably table stakes for you ' re not getting a position. I don ' t think he wants one either. The funny thing is that so many of the people that were in the previous Trump administration now consider Trump to be an enemy and it is mutual, right? I mean that is – One of the **interesting** things will be to what extent Trump decides to go after them as president. Will he launch investigations from the DOJ? You know, might they be more likely to face an IRS audit? Will Trump-influenced media go after them to a greater degree? I mean, those are all **interesting** questions, and we don ' t know the answer to it. In many ways, I think that Trump actually does hold a grudge against them, people that he thought were loyal and then turned against him, more than against **leaders** in the Democratic Party. But, you know, again, that is right now a hunch. That is not in any way borne out in fact. But in terms of people that I think will be around Trump, I do think that they ' re going to look for adults. They are going to look for people that are capable of doing their job, but they will not go for independents that Trump doesn ' t have control over, even though their Republican bona fides are strong. They did a lot of that the first time around. So, you know, you had Rex Tillerson appointed as Secretary of State, who Trump had never met before and who his own advisers said,”This **guy** is not aligned with any of what we want, any America First. He ' s CEO of ExxonMobil.”And Trump says,”Yeah, but look at him. He looks like a Secretary of State.”That is not what we will see this time around. I think that loyalty will be very, very important. He will want people that will not turn coat on Trump in three and six month ' s time, in a year ' s time, in a future administration, in a future election. There are some people that I continue to hear time and time again. So people that were heavily **involved** in the campaign that were seen to facilitate Trump and make him successful. Howard Lutnick, who originally said that he didn ' t want a position, he just wants to help Trump. But now that he ' s spent a lot of time with Trump is feeling like,”I really do want a job, I ' d like to be Secretary of Treasury.”Well, I think he has a good chance of getting that job. I think Linda McMahon, same, from the World Wrestling [Entertainment

]. Very close to Trump, very good friends. For a long time now she wants to be Secretary of Commerce. Good chance she would get that job. I think that on State, there are a lot of names. Bill Hagerty, the senator, former ambassador to Japan, steady, capable pair of hands. Certainly **interested** in that job, would be in the mix. But frankly, lots of people, will be in that mix, I think Ric Grenell, the former acting head of national **intelligence**, before that, ambassador of the US to Germany, well-known on social media and on Fox. A little more incendiary, more like, willing to be a bomb thrower in public. More of a populist, Trump likes him a lot. He really would like to be Secretary of State. He ' ll certainly be interviewed for that position. I think there are others, you know, I ' m hearing Mike Waltz, member of Congress, very smart **guy**, very capable, potentially for Secretary of Defense. Pompeo, Mike Pompeo does not look to be an insider right now. He took a long time before he was willing to endorse Trump. And I think the loyalty is open to question. Hasn ' t gotten as much access at Mar-a-Lago and with Trump as a lot of other people have. Some of the key questions will be what happens with the so-called power ministries, as we define them around the world the Department of Justice, the FBI, the IRS. Will they be politicized? Will they be weaponized? I do think it ' s very hard to imagine someone like Bill Barr, who is very **conservative**, very smart, but not a Trump loyalist. At the end of the day, someone that was going to ultimately, you know, vote and act according to his values and ideology as opposed to Trump ' s all the time. I think that is not acceptable for an Attorney General in a second Trump term, in the same way that Mike Pence was not the selection for VP, it was JD Vance. And a lot of people say, oh, JD is like, really powerful and he knows politics and he ' s going to run the shadow cabinet, and he ' s going to be in charge of appointing everyone. No he ' s not. Trump will not tolerate someone to have his star power working for him. I think there will be priorities that are Trump ' s. And when he has priorities, he will be in charge. And I think there will be lots of different centers of power that will fight and compete over areas that Trump doesn ' t really care as much about, and then we ' ll see that play out. So in that regard, it is likely to look very different than the first Trump administration. HW : This is the Trump show. So just to wrap things up, I guess a very simple but profound question, which is, how are you feeling about the future? IB : I think that the United States continues to be the most powerful country in the world. It has the reserve currency, it has the most powerful global military, it ' s producing the most energy, it dominates the field of artificial **intelligence**, which is the most important set of new technologies that humans have ever been able to have their hands on. So I mean, there ' s a reason why people are betting on the US markets, on the US dollar in this environment after a Trump victory. But the global order is in very deep disarray. There is an absence of global **leadership**, and that absence will be felt more strongly and more profoundly in a Trump, America First second administration. You know, a lot of allies of the United States around the world are allies because they have shared interests but also because they perceive that they have shared values. And those values include commitment to democracy and rule of law and the promotion of democracy and rule of law around the world. Commitment to a multilateral architecture where norms and values are largely agreed to. **Collective** security among allies and to the extent possible globally. Free trade and market access through multilateral agreements that become more committed and higher standard over time. I think that what we have just seen with this election is that the American people do not actually accept those values, and that the American president-elect does not accept those values. So US allies around the world have to recognize that they still may have a lot of shared interests with the United States, but they no longer have those shared values. Biden, to the extent that he had a core global principle, it was really about autocracies versus democracies, the bad **guys** versus the good **guys**. Trump completely rejects that. Trump ' s view is,"I don ' t care what kind of a political system you have internally. I want to know, can I do a deal with you? And if you ' re Putin or Kim Jong Un, or if you ' re Trudeau or Claudia Sheinbaum, if I can do a deal with you, I will do a deal with you. And by the way, I ' m going to do that deal unilaterally, where I have a lot more power. I ' m not going to do it in a multilateral setting where a bunch of you think you can gang up on me and force me into constraints."So I think that we ' re in an environment where our challenges are increasingly global, where our **ability** as human beings to affect the world that we live in are increasingly systemic and structural. And yet the availability of global **leadership** is not only absent, but is decisively rejected by the American people that have most upheld it over the past decades, for good and for bad, and by its president-elect. So that ' s probably the thing that I am most concerned about and that I think will cause the greatest **uncertainty**, volatility and danger in the coming years. HW : Ian, it is always a pleasure talking to you. Sobering but fascinating. Thank you so much for your time and we will speak again soon, I ' m sure. IB : My pleasure, Helen.

Text Sample 512: POS Dim 4 – 1998 – Score 14.00 – 1998/t000250.txt

As a clergyman, you can imagine how out of place I feel. I feel like a fish out of water, or maybe an owl out of the **air**. I was preaching in San Jose some time ago, and my friend Mark Kvamme, who helped introduce me to this **conference**, brought several CEOs and **leaders** of some of the companies here in the Silicon Valley to have breakfast with me, or I with them. And I was so stimulated. And had such — it was an eye-opening experience to hear them talk about the world that is yet to come through technology and science. I know that we ' re near the end of this **conference**, and some of you may be wondering why they have a speaker from the field of religion. Richard can answer that, because he made that decision. But some years ago I was on an elevator in Philadelphia, coming down. I was to address a **conference** at a hotel. And on that elevator a man said,"I hear Billy Graham is staying in this hotel."And another man looked in my direction and said,"Yes, there he is. He ' s on this elevator with us."And this man looked me up and down for about 10 seconds, and he said,"My, what an anticlimax!"I hope that you won ' t feel that these few moments with me is not a — is an anticlimax, after all these tremendous talks that you ' ve heard, and addresses, which I intend to listen to every one of them. But I was on an airplane in the east some years ago, and the man sitting across the aisle from me was the mayor of Charlotte, North Carolina. His name was John Belk. Some of you will probably know him. And there was a drunk man on there, and he got up out of his seat two or three times, and he was making **everybody** upset by what he was trying to do. And he was slapping the stewardess and pinching her as she went by, and **everybody** was upset with him. And finally, John Belk said,"Do you know who ' s sitting here?"And the man said,"No, who?"He said,"It ' s Billy Graham, the preacher."He said,"You don ' t say!"And he turned to me, and he said,"Put her there!"He said,"Your sermons have certainly helped me."And I suppose that that ' s true with thousands of people. I know that as you have been peering into the future, and as we ' ve heard some of it here tonight, I would like to live in that age and see what is going to be. But I won ' t, because I ' m 80 years old. This is my eightieth year, and I know that my time is brief. I have phlebitis at the moment, in both legs, and that ’ s the reason that I had to have a little help in getting up here, because I have Parkinson ’ s disease in addition to that, and some other problems that I won ’ t talk about. But this is not the first time that we ’ ve had a technological revolution. We ’ ve had others. And there ’ s one that I want to talk about. In one generation, the nation of the people of Israel had a tremendous and dramatic change that made them a great power in the Near East. A man by the name of David came to the throne, and King David became one of the great **leaders** of his generation. He was a man of tremendous **leadership**. He had the favor of God with him. He was a brilliant poet, philosopher, writer, soldier — with strategies in battle and **conflict** that people study even today. But about two centuries before David, the Hittites had discovered the secret of smelting and processing of iron, and, slowly, that skill spread. But they wouldn ’ t allow the Israelis to look into it, or to have any. But David changed all of that, and he introduced the Iron Age to Israel. And the Bible says that David laid up great stores of iron, and which archaeologists have found, that in present-day Palestine, there are evidences of that generation. Now, instead of crude tools made of sticks and stones, Israel now had iron plows, and sickles, and hoes and military weapons. And in the course of one generation, Israel was completely changed. The introduction of iron, in some ways, had an **impact** a little bit like the microchip has had on our generation. And David found that there were many problems that technology could not solve. There were many problems still left. And they ’ re still with us, and you haven ’ t solved them, and I haven ’ t heard anybody here speak to that. How do we solve these three problems that I ’ d like to mention? The first one that David saw was human evil. Where does it come from? How do we solve it? Over again and again in the Psalms, which Gladstone said was the greatest book in the world, David describes the evils of the human race. And yet he says,”He restores my soul.”Have you ever thought about what a contradiction we are? On one hand, we can probe the deepest secrets of the universe and dramatically push back the frontiers of technology, as this **conference** vividly demonstrates. We ’ ve seen under the sea, three miles down, or galaxies hundreds of billions of years out in the future. But on the other hand, something is wrong. Our battleships, our soldiers, are on a frontier now, almost ready to go to war with Iraq. Now, what causes this? Why do we have these wars in every generation, and in every part of the world? And revolutions? We can ’ t get along with other people, even in our own families. We find ourselves in the paralyzing grip of self-destructive habits we can ’ t break. Racism and injustice and violence sweep our world, bringing a tragic harvest of heartache and death. Even the most sophisticated among us seem powerless to break this cycle. I would like to see Oracle take up that, or some other technological geniuses work on this. How do we change man, so that he doesn ’ t lie and cheat, and our newspapers are not filled with stories of fraud in business or labor or athletics or wherever? The Bible says the problem is within us, within our hearts and our souls. Our problem is that we are separated from our Creator, which we call God, and we need to have our souls restored, something only God can do. Jesus said,”For out of the heart come evil thoughts : murders, sexual immorality, theft, false testimonies, slander.”The British philosopher Bertrand Russell was not a religious man, but he said,”It ’ s in our hearts that the evil lies, and it ’ s from our hearts that it must be plucked out.”Albert Einstein — I was just talking to someone, when I was speaking at Princeton, and I met Mr. Einstein. He didn ’ t have a doctor ’ s degree, because he said nobody was qualified to give him one. But he made this statement. He said,”It ’ s easier to denature plutonium than to denature the evil spirit of man.”And many of you, I ’ m sure, have thought about that and puzzled over it. You ’ ve seen people take beneficial technological advances, such as the Internet we ’ ve heard about tonight, and twist them into something corrupting. You ’ ve seen brilliant people devise computer **viruses** that bring down whole systems. The Oklahoma City bombing was simple technology, horribly used. The problem is not technology. The problem is the person or persons using it. King David said that he knew the depths of his own soul. He couldn ’ t free himself from personal problems and personal evils that included murder and adultery. Yet King David sought God ’ s forgiveness, and said,”You can restore my soul.”You see, the Bible teaches that we ’ re more than a body and a mind. We are a soul. And there ’ s something inside of us that is beyond our understanding. That ’ s the part of us that yearns for God, or something more than we find in technology. Your soul is that part of you that yearns for meaning in life, and which seeks for something beyond this life. It ’ s the part of you that yearns, really, for God. I find [that] young people all over the world are searching for something. They don ’ t know what it is. I speak at many universities, and I have many questions and answer periods, and whether it ’ s Cambridge, or Harvard, or Oxford — I ’ ve spoken at all of those universities. I ’ m going to Harvard in about three or four — no, it ’ s about two months from now — to give a lecture. And I ’ ll be asked the same questions that I was asked the last few times I ’ ve been there. And it ’ ll be on these questions : where did I come from? Why am I here? Where am I going? What ’ s life all about? Why am I here? Even if you have no religious belief, there are times when you wonder that there ’ s something else. Thomas Edison also said,”When you see everything that happens in the world of science, and in the working of the universe, you cannot deny that there ’ s a captain on the bridge.”I remember once, I sat beside Mrs. Gorbachev at a White House dinner. I went to Ambassador Dobrynin, whom I knew very well. And I ’ d been to Russia several times under the Communists, and they ’ d given me marvelous freedom that I didn ’ t expect. And I knew Mr. Dobrynin very well, and I said,”I ’ m going to sit beside Mrs. Gorbachev tonight. What shall I talk to her about?”And he surprised me with the answer. He said,”Talk to her about religion and philosophy. That ’ s what she ’ s really **interested** in.”I was a little bit surprised, but that evening that ’ s what we talked about, and it was a stimulating conversation. And afterward, she said,”You know, I ’ m an atheist, but I know that there ’ s something up there higher than we are.”The second problem that King David realized he could not solve was the problem of human suffering. Writing the oldest book in the world was Job, and he said,”Man is born unto trouble as the sparks fly upward.”Yes, to be sure, science has done much to push back certain types of human suffering. But I ’ m — in a few months, I ’ ll be 80 years of age. I admit that I ’ m very grateful for all the medical advances that have kept me in relatively good health all these years. My doctors at the Mayo Clinic urged me not to take this trip out here to this — to be here. I haven ’ t given a talk in nearly four months. And when you speak as much as I do, three or four times a day, you get rusty. That ’ s the reason I ’ m using this podium and using these notes. Every time you ever hear me on the television or somewhere, I ’ m ad-libbing. I ’ m not reading. I never read an address. I never read a speech or a talk or a lecture. I talk ad lib. But tonight, I ’ ve got some notes here so that if I begin to forget, which I do sometimes, I ’ ve got something I can turn to. But even here among us, most — in the most advanced society in the world, we have poverty. We have families that self-destruct, friends that betray us. Unbearable psychological pressures bear down on us. I ’ ve never met a person in the world that didn ’ t have a problem or a worry. Why do we suffer? It ’ s an age- old question that we haven ’ t answered. Yet David again and again said that he would turn to God. He said,”The Lord is my shepherd.”The final problem that David knew he could not solve was death. Many commentators have said that death is the forbidden subject of our generation. Most people live as if they ' re never going to die. Technology projects the myth of control over our mortality. We see people on our screens. Marilyn Monroe is just as beautiful on the screen as she was in person, and our — many young people think she ' s still alive. They don ' t know that she ' s dead. Or Clark Gable, or whoever it is. The old stars, they come to life. And they ' re — they ' re just as great on that screen as they were in person. But death is inevitable. I spoke some time ago to a joint session of Congress, last year. And we were meeting in that room, the statue room. About 300 of them were there. And I said,"There ' s one thing that we have in common in this room, all of us together, whether Republican or Democrat, or whoever."I said,"We ' re all going to die. And we have that in common with all these great men of the past that are staring down at us."And it ' s often difficult for young people to understand that. It ' s difficult for them to understand that they ' re going to die. As the ancient writer of Ecclesiastes wrote, he said, there ' s every activity under heaven. There ' s a time to be born, and there ' s a time to die. I ' ve stood at the deathbed of several famous people, whom you would know. I ' ve talked to them. I ' ve seen them in those agonizing moments when they were scared to death. And yet, a few years earlier, death never crossed their mind. I talked to a woman this past week whose father was a famous doctor. She said he never thought of God, never talked about God, didn ' t believe in God. He was an atheist. But she said, as he came to die, he sat up on the side of the bed one day, and he asked the nurse if he could see the chaplain. And he said, for the first time in his life he ' d thought about the inevitable, and about God. Was there a God? A few years ago, a university student asked me,"What is the greatest surprise in your life?"And I said,"The greatest surprise in my life is the brevity of life. It passes so fast."But it does not need to have to be that way. Wernher von Braun, in the aftermath of World War II concluded, quote :"science and religion are not antagonists. On the contrary, they ' re sisters."He put it on a personal basis. I knew Dr. von Braun very well. And he said,"Speaking for myself, I can only say that the grandeur of the cosmos serves only to confirm a belief in the certainty of a creator."He also said,"In our search to know God, I ' ve come to believe that the life of Jesus Christ should be the focus of our efforts and inspiration. The reality of this life and His resurrection is the hope of mankind."I ' ve done a lot of speaking in Germany and in France, and in different parts of the world — 105 countries it ' s been my privilege to speak in. And I was invited one day to **visit** Chancellor Adenauer, who was looked upon as sort of the founder of modern Germany, since the war. And he once — and he said to me, he said,"Young man."He said,"Do you believe in the resurrection of Jesus Christ?"And I said,"Sir, I do."He said,"So do I."He said,"When I leave office, I ' m going to spend my time writing a book on why Jesus Christ rose again, and why it ' s so important to believe that."In one of his plays, Alexander Solzhenitsyn depicts a man dying, who says to those gathered around his bed,"The moment when it ' s terrible to feel regret is when one is dying."How should one live in order not to feel regret when one is dying? Blaise Pascal asked exactly that question in seventeenth-century France. Pascal has been called the architect of modern civilization. He was a brilliant scientist at the frontiers of mathematics, even as a teenager. He is viewed by many as the founder of the probability theory, and a creator of the first model of a computer. And of course, you are all familiar with the computer language named for him. Pascal explored in depth our human dilemmas of evil, suffering and death. He was astounded at the phenomenon we ' ve been considering : that people can achieve extraordinary heights in science, the arts and human enterprise, yet they also are full of anger, hypocrisy and have — and self- hatreds. Pascal saw us as a remarkable mixture of genius and self-delusion. On

November 23, 1654, Pascal had a profound religious experience. He wrote in his journal these words :”I submit myself, absolutely, to Jesus Christ, my redeemer.”A French historian said, two centuries later,”Seldom has so mighty an intellect submitted with such humility to the authority of Jesus Christ.”Pascal came to believe not only the love and the grace of God could bring us back into harmony, but he believed that his own sins and failures could be forgiven, and that when he died he would go to a place called heaven. He experienced it in a way that went beyond scientific observation and reason. It was he who penned the well-known words,”The heart has its reasons, which reason knows not of.”Equally well known is Pascal ’ s Wager. Essentially, he said this :”if you bet on God, and open yourself to his love, you lose nothing, even if you ’ re wrong. But if instead you bet that there is no God, then you can lose it all, in this life and the life to come.”For Pascal, scientific knowledge paled beside the knowledge of God. The knowledge of God was far beyond anything that ever crossed his mind. He was ready to face him when he died at the age of 39. King David lived to be 70, a long time in his era. Yet he too had to face death, and he wrote these words :”even though I walk through the valley of the shadow of death, I will fear no evil, for you are with me.”This was David ’ s answer to three dilemmas of evil, suffering and death. It can be yours, as well, as you seek the living God and allow him to fill your life and give you hope for the future. When I was 17 years of age, I was born and reared on a farm in North Carolina. I milked cows every morning, and I had to milk the same cows every evening when I came home from school. And there were 20 of them that I had — that I was responsible for, and I worked on the farm and tried to keep up with my studies. I didn ’ t make good grades in high school. I didn ’ t make them in college, until something happened in my heart. One day, I was faced face-to-face with Christ. He said,”I am the way, the truth and the life.”Can you imagine that?”I am the truth. I ’ m the embodiment of all truth.”He was a liar. Or he was insane. Or he was what he **claimed** to be. Which was he? I had to make that decision. I couldn ’ t prove it. I couldn ’ t take it to a laboratory and experiment with it. But by faith I said, I believe him, and he came into my heart and changed my life. And now I ’ m ready, when I hear that call, to go into the presence of God. Thank you, and God bless all of you. Thank you for the privilege. It was great. Richard Wurman : You did it. Thanks.

Text Sample 513: POS Dim 4 – 1998 – Score 12.00 – 1998/t000200.txt

You hear that this is the era of environment â or biology, or information technology... Well, it ’ s the era of a lot of different things that we ’ re in right now. But one thing for sure : it ’ s the era of change. There ’ s more change going on than ever has occurred in the history of human life on earth. And you all sort of know it, but it ’ s hard to get it so that you really understand it. And I ’ ve tried to put together something that ’ s a good start for this. I ’ ve tried to show in this â though the color doesn ’ t come out â that what I ’ m concerned with is the little 50-year time bubble that you are in. You tend to be **interested** in a generation past, a generation future â your parents, your kids, things you can change over the next few decades â and this 50-year time bubble you kind of move along in. And in that 50 years, if you look at the population curve, you find the population of humans on the earth more than doubles and we ’ re up three-and-a-half times since I was born. When you have a new baby, by the time that kid gets out of high school more people will be added than existed on earth when I was born. This is unprecedented, and it ’ s big. Where it goes in the future is questioned. So that ’ s the human part. Now, the human part related to animals : look at the left side of that. What I call the human portion â humans and their livestock and pets â versus the natural portion â all the other wild animals and just â these are vertebrates and all the birds, etc., in the land and **air**, not in the water. How does it balance? Certainly, 10, 000 years ago, the civilization ’ s beginning, the human portion was less than one tenth of one percent. Let ’ s look at it now. You follow this curve and you see the whiter spot in the middle â that ’ s your 50- year time bubble. Humans, livestock and pets are now 97 percent of that integrated total mass on earth and all wild nature is three percent. We have won. The next generation doesn ’ t even have to worry about this game â it is over. And the biggest problem came in the last 25 years : it went from 25 percent up to that 97 percent. And this really is a sobering picture upon realizing that we, humans, are in charge of life on earth ; we ’ re like the capricious Gods of old Greek myths, kind of playing with life â and not a great deal of wisdom injected into it. Now, the third curve is information technology. This is Moore ’ s Law plotted here, which relates to density of information, but it has been pretty good for showing a lot of other things about information technology â computers, their use, Internet, etc. And what ’ s important is it just goes straight up through the top of the curve, and has no real limits to it. Now try and contrast these. This is the size of the earth going through that same â â frame. And to make it really clear, I ’ ve put all four on one graph. There ’ s no need to see the little detailed words on it. That first one is humans-versus-nature ; we ’ ve won, there ’ s no more gain. Human population. And so if you ’ re looking for growth industries to get into, that ’ s not a good one â protecting natural creatures. Human population is going up ; it ’ s going to continue for quite a while. Good business in obstetricians, morticians, and farming, housing, etc. â they all deal with human bodies, which require being fed, transported, housed and so on. And the information technology, which connects to our brains, has no limit â now, that is a wonderful field to be in. You ’ re looking for growth opportunity? It ’ s just going up through the roof. And then, the size of the Earth. Somehow making these all compatible with the Earth looks like a pretty bad industry to be **involved** with. So, that ’ s the stage out of all this. I find, for reasons I don ’ t understand, I really do have a goal. And the goal is that the world be desirable and sustainable when my kids reach my age â and I think that ’ s â in other words, the next generation. I think that ’ s a goal that we probably all share. I think it ’ s a hopeless goal. Technologically, it ’ s achievable ; economically, it ’ s achievable ; politically, it means sort of the habits, institutions of people â it ’ s impossible. The institutions of the past with all their inertia are just irrelevant for the future, except they ’ re there and we have to deal with them. I spend about 15 percent of my time trying to save the world, the other 85 percent, the usual â and whatever else we devote ourselves to. And in that 15 percent, the main focus is on human mind, thinking skills, somehow trying to unleash kids from the straightjacket of school, which is putting information and dogma into them, get them so they really think, ask tough questions, argue about serious subjects, don ’ t believe everything that ’ s in the book, think broadly or creative. They can be. Our school systems are very flawed and do not reward you for the things that are important in life or for the survival of civilization ; they reward you for a lot of learning and **sopping** up **stuff**. We can ’ t go into that today because there isn ’ t time â it ’ s a broad subject. One thing for sure, in the future there is an essential feature â necessary, but not sufficient â which is doing more with less. We ’ ve got to be doing things with more efficiency using less energy, less material. Your great-great grandparents got by on muscle power, and yet we all think there ’ s this huge power that ’ s essential for our lifestyle. And with all the wonderful technology we have we can do things that are much more efficient : conserve, recycle, etc. Let me just rush very quickly through things that we ’ ve done. Human-powered airplane â Gossamer Condor sort of started me in this direction in 1976 and 77, winning the Kremer prize in aviation history, followed by the Albatross. And we began making various odd planes and creatures. Here ’ s a giant flying replica of a pterosaur that has no tail. Trying to have it fly straight is like trying to shoot an arrow with the feathered end forward. It was a tough job, and boy it made me have a lot of respect for nature. This was the full size of the original creature. We did things on land, in the **air**, on water â vehicles of all different kinds, usually with some electronics or electric power systems in them. I find they ’ re all the same, whether its land, **air** or water. I ’ ll be focusing on the **air** here. This is a solar-powered airplane â 165 miles carrying a person from France to England as a symbol that solar power is going to be an important part of our future. Then we did the solar car for General Motors â the Sunracer â that won the race in Australia. We got a lot of people thinking about electric cars, what you could do with them. A few years later, when we suggested to GM that now is the time and we could do a thing called the **Impact**, they sponsored it, and here ’ s the **Impact** that we developed with them on their programs. This is the demonstrator. And they put huge effort into turning it into a commercial product. With that preamble, let ’ s show the first two-minute videotape, which shows a little airplane for surveillance and moving to a giant airplane. Narrator : A tiny airplane, the AV Pointer serves for surveillance â in effect, a pair of roving eyeglasses. A cutting-edge example of where miniaturization can lead if the operator is remote from the vehicle. It is convenient to carry, assemble and launch by hand. Battery-powered, it is silent and rarely noticed. It sends high-resolution video pictures back to the operator. With onboard GPS, it can navigate autonomously, and it is rugged enough to self-land without damage. The modern sailplane is superbly efficient. Some can glide as flat as 60 feet forward for every foot of descent. They are powered only by the energy they can extract from the atmosphere â an atmosphere nature stirs up by solar energy. Humans and soaring birds have found nature to be generous in providing replenishable energy. Sailplanes have flown over 1, 000 miles, and the altitude record is over 50, 000 feet. The Solar Challenger was made to serve as a symbol that photovoltaic cells can produce real power and will be part of the world ’ s energy future. In 1981, it flew 163 miles from Paris to England, solely on the power of sunbeams, and established a basis for the Pathfinder. The message from all these vehicles is that ideas and technology can be harnessed to produce remarkable gains in doing more with less â gains that can help us attain a desirable balance between technology and nature. The stakes are high as we speed toward a challenging future. Buckminster Fuller said it clearly :”there are no passengers on spaceship Earth, only crew. We, the crew, can and must do more with less â much less.”Paul MacCready : If we could have the second video, the one-minute, put in as quickly as you can, which â this will show the Pathfinder airplane in some flights this past year in Hawaii, and will show a sequence of some of the beauty behind it after it had just flown to 71, 530 feet â higher than any propeller airplane has ever flown. It ’ s amazing : just on the puny power of the sun â by having a super lightweight plane, you ’ re able to get it up there. It ’ s part of a long-term program NASA sponsored. And we worked very closely with the whole thing being a team effort, and with wonderful results like that flight. And we ’ re working on a bigger plane â 220-foot span â and an intermediate-size, one with a regenerative fuel cell that can store excess energy during the day, feed it back at night, and stay up 65, 000 feet for months at a time. Ray Morgan ’ s voice will come in here. There he ’ s the project manager. Anything they do is certainly a team effort. He ran this program. Here ’ s... some things he showed as a celebration at the very end. Ray Morgan : We ’ d just ended a seven-month deployment of Hawaii. For those who live on the mainland, it was tough being away from home. The friendly **support**, the quiet **confidence**, congenial hospitality shown by our Hawaiian and military hosts â this is starting â made the experience enjoyable and unforgettable. PM : We have real-time IR scans going out through the Internet while the plane is flying. And it ’ s exploring without polluting the stratosphere. That ’ s its goal : the stratosphere, the blanket that really controls the radiation of the earth and permits life on earth to be the success that it is â probing that is very important. And also we consider it as a sort of poor man ’ s stationary satellite, because it can stay right overhead for months at a time, 2, 000 times closer than the real GFC synchronous satellite. We couldn ’ t bring one here to fly it and show you. But now let ’ s look at the other end. In the video you saw that nine-pound or eight-pound Pointer airplane surveillance drone that Keenan has developed and just done a remarkable job. Where some have servos that have gotten down to, oh, 18 or 25 grams, his weigh one-third of a gram. And what he ’ s going to bring out here is a surveillance drone that weighs about 2 ounces â that includes the video camera, the batteries that run it, the telemetry, the receiver and so on. And we ’ ll fly it, we hope, with the same success that we had last night when we did the practice. So Matt Keenan, just any time you ’ re â all right â ready to let her go. But first, we ’ re going to make sure that it ’ s appearing on the screen, so you see what it sees. You can imagine yourself being a mouse or fly inside of it, looking out of its camera. Matt Keenan : It ’ s switched on. PM : But now we ’ re trying to get the video. There we go. MK : Can you bring up the house lights? PM : Yeah, the house lights and we ’ ll see you all better and be able to fly the plane better. MK : All right, we ’ ll try to do a few laps around and bring it back in. Here we go. PM : The video worked right for the first few and I don ’ t know why it â there it goes. Oh, that was only a minute, but I think you ’ d be **safe** to have that near the end of the flight, perhaps. We get to do the classic. All right. If this hits you, it will not hurt you. OK. Thank you very much. Thank you. But now, as they say in infomercials, we have something much better for you, which we ’ re working on : planes that are only six inches â 15 centimeters â in size. And Matt ’ s plane was on the cover of Popular Science last month, showing what this can lead to. And in a while, something this size will have GPS and a video camera in it. We ’ ve had one of these fly nine miles through the **air** at 35 miles an hour with just a little battery in it. But there ’ s a lot of technology going. There are just milestones along the way of some remarkable things. This one doesn ’ t have the video in it, but you get a little feel from what it can do. OK, here we go. MK : Sorry. OK. PM : If you can pass it down when you ’ re done. Yeah, I think â I lost a little orientation ; I looked up into this light. It hit the building. And the building was poorly placed, actually. But you ’ re beginning to see what can be done. We ’ re working on projects now â even wing-flapping things the size of hawk moths â DARPA contracts, working with Caltech, UCLA. Where all this leads, I don ’ t know. Is it practical? I don ’ t know. But like any basic research, when you ’ re really forced to do things that are way beyond existing technology, you can get there with micro-technology, nanotechnology. You can do amazing things when you realize what nature has been doing all along. As you get to these small scales, you realize we have a lot to learn from nature â not with 747s â but when you get down to the nature ’ s realm, nature has 200 million years of experience. It never makes a mistake. Because if you make a mistake, you don ’ t leave any progeny. We should have nothing but success stories from nature, for you or for birds, and we ’ re learning a lot from its fascinating subjects. In concluding, I want to get back to the big picture and I have just two final slides to try and put it in perspective. The first I ’ ll just read. At last, I put in three sentences and had it say what I wanted. Over billions of years on a unique sphere, chance has painted a thin covering of life â complex, improbable, wonderful and fragile. Suddenly, we humans â a recently arrived species, no longer subject to the checks and balances inherent in nature â have grown in population, technology and **intelligence** to a position of terrible power. We now wield the paintbrush. And that ’ s serious : we ’ re not very bright. We ’ re short on wisdom ; we ’ re high on technology. Where ’ s it going to lead? Well, inspired by the sentences, I decided to wield the paintbrush. Every 25 years I do a picture. Here ’ s the one â tries to show that the world isn ’ t getting any bigger. Sort of a timeline, very non-linear scale, nature rates and trilobites and dinosaurs, and eventually we saw some humans with caves... Birds were flying overhead, after pterosaurs. And then we get to the civilization above the little TV set with a gun on it. Then traffic jams, and power systems, and some dots for digital. Where it ’ s going to lead â I have no idea. And so I just put robotic and natural cockroaches out there, but you can fill in whatever you want. This is not a forecast. This is a warning, and we have to think seriously about it. And that time when this is happening is not 100 years or 500 years. Things are going on this decade, next decade ; it ’ s a very short time that we have to decide what we are going to do. And if we can get some agreement on where we want the world to be â desirable, sustainable when your kids reach your age â I think we actually can reach it. Now, I said this was a warning, not a forecast. That was before â I painted this before we started in on making robotic versions of hawk moths and cockroaches, and now I ’ m beginning to wonder seriously â was this more of a forecast than I wanted? I personally think the surviving intelligent life form on earth is not going to be carbon-based ; it ’ s going to be silicon-based. And so where it all goes, I don ’ t know. The one final bit of sparkle we ’ ll put in at the very end here is an utterly impractical flight vehicle, which is a little ornithopter wing-flapping device that â rubber-band powered â that we ’ ll show you. MK : 32 gram. Sorry, one gram. PM : Last night we gave it a few too many turns and it tried to bash the roof out also. It ’ s about a gram. The tube there ’ s hollow, about paper-thin. And if this lands on you, I assure you it will not hurt you. But if you reach out to grab it or hold it, you will destroy it. So, be gentle, just act like a wooden Indian or something. And when it comes down â and we ’ ll see how it goes. We consider this to be sort of the spirit of **TED**. And you wonder, is it practical? And it turns out if I had not been â Unfortunately, we have some light bulb changes. We can probably get it down, but it ’ s possible it ’ s gone up to a greater destiny up there â â than it ever had. And I wanted to make â just â But I want to make just two points. One is, you think it ’ s frivolous ; there ’ s nothing to it. And yet if I had not been making ornithopters like that, a little bit cruder, in 1939 â a long, long time ago â there wouldn ’ t have been a Gossamer Condor, there wouldn ’ t have been an Albatross, a Solar Challenger, there wouldn ’ t be an **Impact** car, there wouldn ’ t be a mandate on zero-emission vehicles in California. A lot of these things â or similar â would have happened some time, probably a decade later. I didn ’ t realize at the time I was doing inquiry-based, hands-on things with teams, like they ’ re trying to get in education systems. So I think that, as a symbol, it ’ s important. And I believe that also is important. You can think of it as a sort of a symbol for learning and **TED** that somehow gets you thinking of technology and nature, and puts it all together in things that are â that make this **conference**, I think, more important than any that ’ s taken place in this country in this decade. Thank you.

Text Sample 514: POS Dim 4 – 1998 – Score 9.00 – 1998/t000118.txt

Back in 1992, I started working for a company called Interval Research, which was just then being founded by David Lidell and Paul Allen as a for-profit research enterprise in Silicon Valley. I met with David to talk about what I might do in his company. I was just coming out of a failed virtual reality business and **supporting** myself by being on the speaking circuit and writing books — after twenty years or so in the computer game industry having ideas that people didn ’ t think they could sell. And David and I discovered that we had a question in common, that we really wanted the answer to, and that was,”Why hasn ’ t anybody built any computer games for little girls?”Why is that? It can ’ t just be a giant sexist conspiracy. These people aren ’ t that smart. There ’ s six billion dollars on the table. They would go for it if they could figure out how. So, what is the deal here? And as we thought about our goals — I should say that Interval is really a humanistic institution, in the classical sense that humanism, at its best, finds a way to combine clear-eyed empirical research with a set of core values that fundamentally love and respect people. The basic idea of humanism is the improvable quality of life ; that we can do good things, that there are things worth doing because they ’ re good things to do, and that clear-eyed empiricism can help us figure out how to do them. So, contrary to popular belief, there is not a **conflict** of interest between empiricism and values. And Interval Research is kind of the living example of how that can be true. So David and I decided to go find out, through the best research we could muster, what it would take to get a little girl to put her hands on a computer, to achieve the level of comfort and ease with the technology that little boys have because they play video games. We spent two and a half years conducting research ; we spent another year and a half in advance development. Then we formed a spin-off company. In the research phase of the project at Interval, we partnered with a company called Cheskin Research, and these people — Davis Masten and Christopher Ireland — changed my mind entirely about what market research was and what it could be. They taught me how to look and see, and they did not do the incredibly **stupid** thing of saying to a child,”Of all these things we already make you, which do you like best?”— which gives you zero answers that are usable. So, what we did for the first two and a half years was four things : We did an extensive review of the literature in related fields, like cognitive psychology, spatial cognition, **gender** studies, play theory, sociology, primatology. Thank you Frans de Waal, wherever you are, I love you and I ’ d give anything to meet you. After we had done that with a pretty large team of people and discovered what we thought the salient issues were with girls and boys and playing — because, after all, that ’ s really what this is about — we moved to the second phase of our work, where we interviewed adult experts in academia, some of the people who ’ d produced the literature that we found relevant. Also, we did focus groups with people who were on the ground with kids every day, like playground supervisors. We talked to them, confirmed some hypotheses and identified some serious questions about **gender** difference and play. Then we did what I consider to be the heart of the work : interviewed 1, 100 children, boys and girls, ages seven to 12, all over the United States — except for Silicon Valley, Boston and Austin because we knew that their little families would have millions of computers in them and they wouldn ’ t be a representative sample. And at the end of those remarkable conversations with kids and their best friends across the United States, after two years, we pulled together some survey data from another 10, 000 children, drew up a set up of what we thought were the key findings of our research, and spent another year transforming them into design heuristics, for designing computer-based products — and, in fact, any kind of products — for little girls, ages eight to 12. And we spent that time designing interactive prototypes for computer software and testing them with little girls. In 1996, in November, we formed the company Purple Moon which was a spinoff of Interval Research, and our chief investors were Interval Research, Vulcan Northwest, Institutional Venture Partners and Allen and Company. We launched a website on September 2nd that has now served 25 million pages, and has 42, 000 registered young girl users. They **visit** an average of one and a half times a day, spend an average of 35 minutes a **visit**, and look at 50 pages a **visit**. So we feel that we ' ve formed a successful online community with girls. We launched two titles in October —"Rockett ' s New School"— the first of a series of products — is about a character called Rockett beginning her first day of school in eighth grade at a brand new place, with a blank slate, which allows girls to play with the question of,"What will I be like when I ' m older?""What ' s it going to be like to be in high school or junior high school? Who are my friends?"; to exercise the love of social complexity and the narrative **intelligence** that drives most of their play behavior ; and which embeds in it values about noticing that we have lots of choices in our lives and the ways that we conduct ourselves. The other title that we launched is called"Secret Paths in the Forest,"which addresses the more fantasy-oriented, inner lives of girls. These two titles both showed up in the top 50 entertainment titles in PC Data — entertainment titles in PC Data in December, right up there with"John Madden Football,"which thrills me to death. So, we ' re real, and we ' ve touched several hundreds of thousands of little girls. We ' ve made half-a-billion impressions with marketing and PR for this brand, Purple Moon. Ninety-six percent of them, roughly, have been positive ; four percent of them have been"other."I want to talk about the other, because the politics of this enterprise, in a way, have been the most fascinating part of it, for me. There are really two kinds of negative reviews that we ' ve received. One kind of reviewer is a male gamer who thinks he knows what games ought to be, and won ' t show the product to little girls. The other kind of reviewer is a certain flavor of feminist who thinks they know what little girls ought to be. And so it ' s funny to me that these **interesting**, odd bedfellows have one thing in common : they don ' t listen to little girls. They haven ' t looked at children and they ' re certainly not demonstrating any love for them. I ' d like to play you some voices of little girls from the two-and-a-half years of research that we did — actually, some of the voices are more recent. And these voices will be accompanied by photographs that they took for us of their lives, of the things that they value and care about. These are pictures the girls themselves never saw, but they gave to us This is the **stuff** those reviewers don ' t know about and aren ' t listening to and this is the kind of research I recommend to those who want to do humanistic work. Girl 1 : Yeah, my character is usually a tomboy. Hers is more into boys. Girl 2 : Uh, yeah. Girl 1 : We have — in the very beginning of the whole game, always we do this : we each have a piece of paper ; we write down our name, our age — are we rich, very rich, not rich, poor, medium, wealthy, boyfriends, dogs, pets — what else — sisters, brothers, and all those. Girl 2 : Divorced — parents divorced, maybe. Girl 3 : This is my pretend [unclear] one. Girl 4 : We make a school newspaper on the computer. Girl 5 : For a girl ' s game also usually they ' ll have really pretty scenery with clouds and flowers. Girl 6 : Like, if you were a girl and you were really adventurous and a real big tomboy, you would think that girls ' games were kinda sissy. Girl 7 : I run track, I played soccer, I play basketball, and I love a lot of things to do. And sometimes I feel like I can ' t really enjoy myself unless it ' s like a vacation, like when I get Mondays and all those days off. Girl 8 : Well, sometimes there is a lot of

stuff going on because I have music lessons and I ’ m on swim team — all this different **stuff** that I have to do, and sometimes it gets overwhelming. Girl 9 : My friend Justine kinda took my friend Kelly, and now they ’ re being mean to me. Girl 10 : Well, sometimes it gets annoying when your brothers and sisters, or brother or sister, when they copy you and you get your idea first and they take your idea and they do it themselves. Girl 11 : Because my older sister, she gets everything and, like, when I ask my mom for something, she ’ ll say,”No”— all the time. But she gives my sister everything. Brenda Laurel : I want to show you, real quickly, just a minute of”Rockett ’ s Tricky Decision,”which went gold two days ago. Let ’ s hope it ’ s really stable. This is the second day in Rockett ’ s life. The reason I ’ m showing you this is I ’ m hoping that the scene that I ’ m going to show you will look familiar and sound familiar, now that you ’ ve listened to some girls ’ voices. And you can see how we ’ ve tried to incorporate the issues that matter to them in the game that we ’ ve created. Miko : Hey Rockett! C ’ mere! Rockett : Hi Miko! What ’ s going on? Miko : Did you hear about Nakilia ’ s big Halloween party this weekend? She asked me to make sure you knew about it. Nakilia invited Reuben too, but — Rockett : But what? Isn ’ t he coming? Miko : I don ’ t think so. I mean, I heard his band is playing at another party the same night. Rockett : Really? What other party? Girl : Max ’ s party is going to be so cool, Whitney. He ’ s invited all the best people. BL : I ’ m going to fast-forward to the decision point because I know I don ’ t have a lot of time. After this awful event occurs, Rocket gets to decide how she feels about it. Rockett : Who ’ d want to show up at that party anyway? I could get invited to that party any day if I wanted to. Gee, I doubt I ’ ll make Max ’ s best people list. BL : OK, so we ’ re going to emotionally navigate. If we were playing the game, that ’ s what we ’ d do. If at any time during the game we want to learn more about the characters, we can go into this hidden hallway, and I ’ ll quickly just show you the interface. We can, for example, go find Miko ’ s locker and get some more information about her. Oops, I turned the wrong way. But you get the general idea of the product. I wanted to show you the ways, innocuous as they seem, in which we ’ re incorporating what we ’ ve learned about girls — their desires to experience greater emotional flexibility, and to play around with the social complexity of their lives. I want to make the point that what we ’ re giving girls, I think, through this effort, is a kind of validation, a sense of being seen. And a sense of the choices that are available in their lives. We love them. We see them. We ’ re not trying to tell them who they ought to be. But we ’ re really, really happy about who they are. It turns out they ’ re really great. I want to close by showing you a videotape that ’ s a version of a future game in the Rockett series that our graphic artists and design people put together, that we feel would please that four percent of reviewers.”Rockett 28!”Rockett : It ’ s like I ’ m just waking up, you know? BL : Thanks.

Text Sample 515: POS Dim 4 – 2005 – Score 23.00 – 2005/t000414.txt

I get asked a lot what the difference between my work is and typical Pentagon long-range strategic planners. And the answer I like to offer is what they typically do is they think about the future of wars in the context of war. And what I ’ ve spent 15 years doing in this business â and it ’ s taken me almost 14 to figure it out â is I think about the future of wars in the context of everything else. So I tend to specialize on the scene between war and peace. The material I ’ m going to show you is one idea from a book with a lot of ideas. It ’ s the one that takes me around the world right now interacting with foreign militaries quite a bit. The material was generated in two years of work I did for the Secretary of Defense, thinking about a new national grand strategy for the United States. I ’ m going to present a problem and try to give you an answer. Here ’ s my favorite bonehead concept from the 1990s in the Pentagon : the theory of anti-access, area-denial asymmetrical strategies. Why do we call it that? Because it ’ s got all those A ’ s lined up I guess. This is gobbledygook for if the United States fights somebody we ’ re going to be huge. They ’ re going to be small. And if they try to fight us in the traditional, straight-up manner we ’ re going to kick their ass, which is why people don ’ t try to do that any more. I met the last Air Force General who had actually shot down an enemy plane in combat. He ’ s now a one star General. That ’ s how distant we are from even meeting an **air** force willing to fly against ours. So that overmatched capability creates problems â catastrophic successes the White House calls them. And we ’ re trying to figure that out, because it is an amazing capability. The question is, what ’ s the good you can do with it? OK? The theory of anti-access, area-denial asymmetrical strategies â gobbledygook that we sell to Congress, because if we just told them we can kick anybody ’ s asses they wouldn ’ t buy us all the **stuff** we want. So we say, area-denial, anti-access asymmetrical strategies and their eyes glaze over. And they say,”Will you build it in my district?”Here ’ s my parody and it ain ’ t much of one. Let ’ s talk about a battle space. I don ’ t know, Taiwan Straits 2025. Let ’ s talk about an enemy embedded within that battle space. I don ’ t know, the Million Man Swim. The United States has to access that battle space instantaneously. They throw up anti-access, area-denial asymmetrical strategies. A banana peel on the tarmac. Trojan horses on our computer networks reveal all our Achilles ’ heels instantly. We say,”China, it ’ s yours.”Prometheus approach, largely a geographic definition, focuses almost exclusively on the start of **conflict**. We field the first-half team in a league that insists on keeping score until the end of the game. That ’ s the problem. We can run the score up against anybody, and then get our asses kicked in the second half â what they call fourth generation warfare. Here ’ s the way I like to describe it instead. There is no battle space the U. S. Military cannot access. They said we couldn ’ t do Afghanistan. We did it with ease. They said we couldn ’ t do Iraq. We did it with 150 combat casualties in six weeks. We did it so fast we weren ’ t prepared for their collapse. There is nobody we can ’ t take down. The question is, what do you do with the power? So there ’ s no trouble accessing battle spaces. What we have trouble accessing is the **transition** space that must naturally follow, and creating the peace space that allows us to move on. Problem is, the Defense Department over here beats the hell out of you. The State Department over here says,”Come on boy, I know you can make it.”And that poor country runs off that ledge, does that cartoon thing and then drops. This is not about overwhelming force, but proportional force. It ’ s about non- lethal technologies, because if you fire real ammo into a crowd of women and children rioting you ’ re going to lose friends very quickly. This is not about projecting power, but about staying power, which is about legitimacy with the locals. Who do you access in this **transition** space? You have to create internal partners. You have to access coalition partners. We asked the Indians for 17, 000 peace keepers. I know their **senior leadership**, they wanted to give it to us. But they said to us,”You know what? In that **transition** space you ’ re mostly hat not enough cattle. We don ’ t think you can pull it off, we ’ re not going to give you our 17, 000 peace keepers for fodder.”We asked the Russians for 40, 000. They said no. I was in China in August, I said,”You should have 50, 000 peace keepers in Iraq. It ’ s your oil, not ours.”Which is the truth. It ’ s their oil. And the Chinese said to me,”Dr. Barnett, you ’ re absolutely right. In a perfect world we ’ d have 50, 000 there. But it ’ s not a perfect world, and your administration isn ’ t getting us any closer.”But we have trouble accessing our outcomes. We lucked out, frankly, on the selection. We face different opponents across these three. And it ’ s time to start admitting you can ’ t ask the same 19-year- old to do it all, day in and day out. It ’ s just too damn hard. We have an unparalleled capacity to wage war. We don ’ t do the everything else so well. Frankly, we do it better than anybody and we still suck at it. We have a brilliant Secretary of War. We don ’ t have a Secretary of Everything Else. Because if we did, that **guy** would be in front of the Senate, still testifying over Abu Ghraib. The problem is he doesn ’ t exist. There is no Secretary of Everything Else. I think we have an unparalleled capacity to wage war. I call that the Leviathan Force. What we need to build is a force for the Everything Else. I call them the System Administrators. What I think this really represents is lack of an A to Z rule set for the world as a whole for processing politically bankrupt states. We have one for processing economically bankrupt states. It ’ s the IMF Sovereign Bankruptcy Plan, OK? We argue about it every time we use it. Argentina just went through it, broke a lot of rules. They got out on the far end, we said,”Fine, don ’ t worry about it.”It ’ s transparent. A certain amount of certainty gives the sense of a non-zero outcome. We don ’ t have one for processing politically bankrupt states that, frankly, **everybody** wants gone. Like Saddam, like Mugabe, like Kim Jong-Il â people who kill in hundreds of thousands or millions. Like the 250, 000 dead so far in Sudan. What would an A to Z system look like? I ’ m going to distinguish between what I call front half and back half. And let ’ s call this red line, I don ’ t know, mission accomplished. What we have extant right now, at the beginning of this system, is the U. N. Security Council as a grand jury. What can they do? They can indict your ass. They can debate it. They can write it on a piece of paper. They can put it in an envelope and mail it to you, and then say in no uncertain terms,”Please cut that out.”That gets you about four million dead in Central Africa over the 1990s. That gets you 250, 000 dead in the Sudan in the last 15 months. **Everybody** ’ s got to answer their grandchildren some day what you did about the holocaust in Africa, and you better have an answer. We don ’ t have anything to translate that will into action. What we do have is the U. S. -enabled Leviathan Force that says,”You want me to take that **guy** down? I ’ ll take that **guy** down. I ’ ll do it on Tuesday. It will cost you 20 billion dollars.”But here ’ s the deal. As soon as I can ’ t find anybody else to **air** out, I leave the scene immediately. That ’ s called the Powell Doctrine. Way downstream we have the International Criminal Court. They love to put them on trial. They ’ ve got MiloseviÄ right now. What are we missing? A functioning executive that will translate will into action, because we don ’ t have it. Every time we lead one of these efforts we have to whip ourselves into this imminent threat thing. We haven ’ t faced an imminent threat since the Cuban missile **crisis** in 1962. But we use this language from a bygone era to scare ourselves into doing something because we ’ re a democracy and that ’ s what it takes. And if that doesn ’ t work we scream,”He ’ s got a gun!”just as we rush in. And then we look over the body and we find an old cigarette lighter and we say,”Jesus, it was dark.”Do you want to do it, France? France says,”No, but I do like to criticize you after the fact.”What we need downstream is a great power enabled â what I call that Sys Admin Force. We should have had 250, 000 troops streaming into Iraq on the heels of that Leviathan sweeping towards Baghdad. What do you get then? No looting, no military disappearing, no arms disappearing, no ammo disappearing, no Muqtada Al-Sadr â I ’ m wrecking his bones â no insurgency. Talk to anybody who was over there in the first six months. We had six months to feel the lob, to get the job done, and we dicked around for six months. And then they turned on us. Why? Because they just got fed up. They saw what we did to Saddam. They said,"You ' re that powerful, you can resurrect this country. You ' re America."What we need is an **international** reconstruction fund â Sebastian Mallaby, Washington Post, great idea. Model on the IMF. Instead of passing the hat each time, OK? Where are we going to find this **guy**? G20, that ' s easy. Check out their agenda since 9 / 11. All security dominated. They ' re going to decide up front how the money gets spent just like in the IMF. You vote according to how much money you put in the kitty. Here ' s my challenge to the Defense Department. You ' ve got to build this force. You ' ve got to seed this force. You ' ve got to track coalition partners. Create a record of success. You will get this model. You tell me it ' s too hard to do. I ' ll walk this dog right through that six part series on the Balkans. We did it just like that. I ' m talking about regularizing it, making it transparent. Would you like Mugabe gone? Would you like Kim Jong- Il, who ' s killed about two million people, would you like him gone? Would you like a better system? This is why it matters to the military. They ' ve been experiencing an identity **crisis** since the end of the Cold War. I ' m not talking about the difference between reality and desire, which I can do because I ' m not inside the beltway. I ' m talking about the 1990s. The Berlin Wall falls. We do Desert Storm. The split starts to emerge between those in the military who see a future they can live with, and those who see a future that starts to scare them, like the U. S. submarine community, which watches the Soviet Navy disappear overnight. Ah! So they start moving from reality towards desire and they create their own special language to describe their voyage of self-discovery and self- actualization. The problem is you need a big, sexy opponent to fight against. And if you can ' t find one you ' ve got to make one up. China, all grown up, going to be a looker! The rest of the military got dragged down into the muck across the 1990s and they developed this very derisive term to describe it : military operations other than war. I ask you, who joins the military to do things other than war? Actually, most of them. Jessica Lynch never planned on shooting back. Most of them don ' t pick up a rifle. I maintain this is code inside the Army for,"We don ' t want to do this."They spent the 1990s working the messy scene between globalized parts of the world What I call the core and the gap. The Clinton administration wasn ' t **interested** in running this. For eight years, after screwing up the relationship on day one â inauguration day with gays in the military â which was deft. So we were home alone for eight years. And what did we do home alone? We bought one military and we operated another. It ' s like the **guy** who goes to the doctor and says,"Doctor, it hurts when I do this."The doctor says,"Stop doing that you idiot."I used to give this brief inside the Pentagon in the early 1990s. I ' d say,"You ' re buying one military and you ' re operating another, and eventually it ' s going to hurt. It ' s wrong. Bad Pentagon, bad!"And they ' d say,"Dr. Barnett, you are so right. Can you come back next year and remind us again?"Some people say 9 / 11 heals the rift â jerks the long-term transformation gurus out of their 30, 000 foot view of history, drags them down in to the muck and says,"You want a networked opponent? I ' ve got one, he ' s everywhere, go find him."It elevates MOOTW â how we pronounce that acronym â from crap to grand strategy, because that ' s how you ' re going to shrink that gap. Some people put these two things together and they call it empire, which I think is a boneheaded concept. Empire is about the enforcement of not just minimal rule sets, which you cannot do, but maximum rule sets which you must do. It ' s not our system of governance. Never how we ' ve sought to interact with the outside world. I prefer that phrase System Administration. We enforce the minimal rule sets for maintaining connectivity to the global economy. Certain bad things you cannot do. How this **impacts** the way we think about the future of war. This is a concept which gets me vilified throughout the Pentagon. It makes me very popular as well. **Everybody** ’ s got an opinion. Going back to the beginning of our country â historically, defenses meant protection of the homeland. Security has meant everything else. Written into our constitution, two different forces, two different functions. Raise an army when you need it, and maintain a navy for day-to-day connectivity. A Department of War, a Department of Everything Else. A big stick, a baton stick. Can of whup ass, the networking force. In 1947 we merged these two things together in the Defense Department. Our long-term rationale becomes, we ’ re **involved** in a hair trigger stand off with the Soviets. To attack America is to risk blowing up the world. We connected national security to **international** security with about a seven minute time delay. That ’ s not our problem now. They can kill three million in Chicago tomorrow and we don ’ t go to the mattresses with nukes. That ’ s the scary part. The question is how do we reconnect American national security with global security to make the world a lot more comfortable, and to embed and contextualize our employment of force around the planet? What ’ s happened since is that bifurcation I described. We talked about this going all the way back to the end of the Cold War. Let ’ s have a Department of War, and a Department of Something Else. Some people say,”Hell, 9 / 11 did it for you.”Now we ’ ve got a home game and an away game. The Department of Homeland Security is a strategic feel good measure. It ’ s going to be the Department of Agriculture for the 21st century. TSA â thousands standing around. I **supported** the war in Iraq. He was a bad **guy** with multiple priors. It ’ s not like we had to find him actually killing somebody live to arrest him. I knew we ’ d kick ass in the war with the Leviathan Force. I knew we ’ d have a hard time with what followed. But I know this organization doesn ’ t change until it experiences failure. What do I mean by these two different forces? This is the Hobbesian Force. I love this force. I don ’ t want to see it go. That plus nukes rules out great power war. This is the military the rest of the world wants us to build. It ’ s why I travel all over the world talking to foreign militaries. What does this mean? It means you ’ ve got to stop pretending you can do these two very disparate skill sets with the same 19-year-old. Switching back, morning, afternoon, evening, morning, afternoon, evening. Handing out aid, shooting back, handing out aid, shooting back. It ’ s too much. The 19-year-olds get tired from the switching, OK? That force on the left, you can train a 19-year-old to do that. That force on the right is more like a 40-year-old cop. You need the experience. What does this mean in terms of operations? The rule is going to be this. That Sys Admin force is the force that never comes home, does most of your work. You break out that Leviathan Force only every so often. But here ’ s the promise you make to the American public, to your own people, to the world. You break out that Leviathan Force, you promise, you guarantee that you ’ re going to mount one hell of a â immediately â follow- on Sys Admin effort. Don ’ t plan for the war unless you plan to win the peace. Other differences. Leviathan traditional partners, they all look like the **Brits** and their former colonies. Including us, I would remind you. The rest â wider array of partners. **International** organizations, non-governmental organizations, private voluntary organizations, contractors. You ’ re not going to get away from that. Leviathan Force, it ’ s all about joint operations between the military services. We ’ re done with that. What we need to do is inter-agency operations, which frankly Condi Rice was in charge of. And I ’ m amazed nobody asked her that question when she was confirmed. I call the Leviathan Force your dad ’ s military. I like them young, male, unmarried, slightly pissed off. I call the Sys Admin Force your mom ’ s military. It ’ s everything the man ’ s military hates. **Gender** balanced much more, older, educated, married with children. The force on the left, up or out. The force on the right, in and out. The force on the left respects Posse Comitatus restrictions on the use of force inside the U. S. The force on the right ’ s going to obliterate it. That ’ s where the National Guard ’ s going to be. The force on the left is never coming under the purview of the International Criminal Court. Sys Admin Force has to. Different definitions of network centricity. One takes down networks, one puts them up. And you ’ ve got to wage war here in such a way to facilitate that. Do we need a bigger budget? Do we need a draft to pull this off? Absolutely not. I ’ ve been told by the Revolution of Military Affairs crowd for years, we can do it faster, cheaper, smaller, just as lethal. I say,”Great, I ’ m going to take the Sys Admin budget out of your hide.”Here ’ s the larger point. You ’ re going to build the Sys Admin Force inside the U. S. Military first. But ultimately you ’ re going to civilianize it, probably two thirds. Inter agency-ize it, internationalize it. So yes, it begins inside the Pentagon, but over time it ’ s going to cross that river. I have been to the mountain top. I can see the future. I may not live long enough to get you there, but it ’ s going to happen. We ’ re going to have a Department of Something Else between war and peace. Last slide. Who gets custody of the kids? This is where the Marines in the audience get kind of tense. And this is when they think about beating the crap out of me after the talk. Read Max Boon. This is the history of the **marines** â small wars, small arms. The Marines are like my West Highland Terrier. They get up every morning, they want to dig a hole and they want to kill something. I don ’ t want my Marines handing out aid. I want them to be Marines. That ’ s what keeps the Sys Admin Force from being a pussy force. It keeps it from being the U. N. You shoot at these people the Marines are going to come over and kill you. Department of Navy, strategic subs go this way, surface combatants are over there, and the news is they may actually be that small. I call it the Smart Dust Navy. I tell young officers,”You may command 500 ships in your career. Bad news is they may not have anybody on them.”Carriers go both ways because they ’ re a swing asset. You ’ ll see the pattern â airborne, just like carriers. Armor goes this way. Here ’ s the dirty secret of the Air Force, you can win by bombing. But you need lots of these **guys** on the ground to win the peace. Shinseki was right with the argument. Air force, strategic airlift goes both ways. Bombers, fighters go over here. Special Operations Command down at Tampa. Trigger-pullers go this way. Civil Affairs, that bastard child, comes over here. Return to the Army. The point about the trigger-pullers and Special Operations Command. No off season, these **guys** are always active. They drop in, do their business, disappear. See me now. Don ’ t talk about it later. I was never here. The world is my playground. I want to keep trigger-pullers trigger-happy. I want the rules to be as loose as possible. Because when the thing gets prevented in Chicago with the three million dead that perverts our political system beyond all recognition, these are the **guys** who are going to kill them first. So it ’ s better off to have them make some mistakes along the way than to see that. Reserve component â National Guard reserves overwhelmingly Sys Admin. How are you going to get them to work for this force? Most firemen in this country do it for free. This is not about money. This is about being up front with these **guys** and gals. Last point, **intelligence** community â the muscle and the defense agencies go this way. What should be the CIA, open, analytical, open source should come over here. The information you need to do this is not secret. It ’ s not secret. Read that great piece in the New Yorker about how our echo boomers, 19 to 25, over in Iraq taught each other how to do Sys Admin work, over the Internet in chat rooms. They said,”Al Qaeda could be listening.”They said,”Well, Jesus, they already know this **stuff**.”Take a gift in the left hand. These are the sunglasses that don ’ t scare people, simple **stuff**. Censors and transparency, the overheads go in both directions. Thanks.

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When I ’ m starting talks like this, I usually do a whole spiel about sustainability because a lot of people out there don ’ t know what that is. This is a crowd that does know what it is, so I ’ ll like just do like the 60-second crib-note version. Right? So just bear with me. We ’ ll go real fast, you know? Fill in the blanks. So, you know, sustainability, small planet. Right? Picture a little Earth, circling around the sun. You know, about a million years ago, a bunch of monkeys fell out of trees, got a little clever, harnessed fire, invented the printing press, made, you know, luggage with wheels on it. And, you know, built the society that we now live in. Unfortunately, while this society is, without a doubt, the most prosperous and dynamic the world has ever created, it ’ s got some major, major flaws. One of them is that every society has an ecological footprint. It has an amount of **impact** on the planet that ’ s measurable. How much **stuff** goes through your life, how much waste is left behind you. And we, at the moment, in our society, have a really dramatically unsustainable level of this. We ’ re using up about five planets. If **everybody** on the planet lived the way we did, we ’ d need between five, six, seven, some people even say 10 planets to make it. Clearly we don ’ t have 10 planets. Again, you know, mental, visual, 10 planets, one planet, 10 planets, one planet. Right? We don ’ t have that. So that ’ s one problem. The second problem is that the planet that we have is being used in wildly unfair ways. Right? North Americans, such as myself, you know, we ’ re basically sort of wallowing, gluttonous hogs, and we ’ re eating all sorts of **stuff**. And, you know, then you get all the way down to people who live in the Asia-Pacific region, or even more, Africa. And people simply do not have enough to survive. This is producing all sorts of tensions, all sorts of dynamics that are deeply disturbing. And there ’ s more and more people on the way. Right? So, this is what the planet ’ s going to look like in 20 years. It ’ s going to be a pretty crowded place, at least eight billion people. So to make matters even more difficult, it ’ s a very young planet. A third of the people on this planet are kids. And those kids are growing up in a completely different way than their parents did, no matter where they live. They ’ ve been exposed to this idea of our society, of our prosperity. And they may not want to live exactly like us. They may not want to be Americans, or **Brits**, or Germans, or South Africans, but they want their own version of a life which is more prosperous, and more dynamic, and more, you know, enjoyable. And all of these things combine to create an enormous amount of torque on the planet. And if we cannot figure out a way to deal with that torque, we are going to find ourselves more and more and more quickly facing situations which are simply unthinkable. **Everybody** in this room has heard the worst-case scenarios. I don ’ t need to go into that. But I will ask the question, what ’ s the alternative? And I would say that, at the moment, the alternative is unimaginable. You know, so on the one hand we have the unthinkable ; on the other hand we have the unimaginable. We don ’ t know yet how to build a society which is environmentally sustainable, which is shareable with **everybody** on the planet, which promotes stability and democracy and human rights, and which is achievable in the time-frame necessary to make it through the challenges we face. We don ’ t know how to do this yet. So what ’ s Worldchanging? Well, Worldchanging you might think of as being a bit of a news service for the unimaginable future. You know, what we ’ re out there doing is looking for examples of tools, models and ideas, which, if widely adopted, would change the game. A lot of times, when I do a talk like this, I talk about things that **everybody** in this room I ’ m sure has already heard of, but most people haven ’ t. So I thought today I ’ d do something a little different, and talk about what we ' re looking for, rather than saying, you know, rather than giving you tried-and-true examples. Talk about the kinds of things we ' re scoping out. Give you a little peek into our editorial notebook. And given that I have 13 minutes to do this, this is going to go kind of quick. So, I don ' t know, just stick with me. Right? So, first of all, what are we looking for? Bright Green city. One of the biggest levers that we have in the developed world for changing the **impact** that we have on the planet is changing the way that we live in cities. We ' re already an urban planet ; that ' s especially true in the developed world. And people who live in cities in the developed world tend to be very prosperous, and thus use a lot of **stuff**. If we can change the dynamic, by first of all creating cities that are denser and more livable... Here, for example, is Vancouver, which if you haven ' t been there, you ought to go for a **visit**. It ' s a fabulous city. And they are doing density, new density, better than probably anybody else on the planet right now. They ' re actually managing to talk North Americans out of driving cars, which is a pretty great thing. So you have density. You also have growth management. You leave aside what is natural to be natural. This is in Portland. That is an actual development. That land there will remain pasture in perpetuity. They ' ve bounded the city with a line. Nature, city. Nothing changes. Once you do those things, you can start making all sorts of investments. You can start doing things like, you know, transit systems that actually work to transport people, in effective and reasonably comfortable manners. You can also start to change what you build. This is the Beddington Zero Energy Development in London, which is one of the greenest buildings in the world. It ' s a fabulous place. We ' re able to now build buildings that generate all their own electricity, that recycle much of their water, that are much more comfortable than standard buildings, use all-natural light, etc., and, over time, cost less. Green roofs. Bill McDonough covered that last night, so I won ' t dwell on that too much. But once you also have people living in close proximity to each other, one of the things you can do is — as information technologies develop — you can start to have smart places. You can start to know where things are. When you know where things are, it becomes easier to share them. When you share them, you end up using less. So one great example is car-share clubs, which are really starting to take off in the U. S., have already taken off in many places in Europe, and are a great example. If you ' re somebody who drives, you know, one day a week, do you really need your own car? Another thing that information technology lets us do is start figuring out how to use less **stuff** by knowing, and by monitoring, the amount we ' re actually using. So, here ' s a power cord which glows brighter the more energy that you use, which I think is a pretty cool concept, although I think it ought to work the other way around, that it gets brighter the more you don ' t use. But, you know, there may even be a simpler approach. We could just re-label things. This light switch that reads, on the one hand, flash-floods, and on the other hand, off. How we build things can change as well. This is a bio-morphic building. It takes its inspiration in form from life. Many of these buildings are incredibly beautiful, and also much more effective. This is an example of bio-mimicry, which is something we ' re really starting to look a lot more for. In this case, you have a shell design which was used to create a new kind of exhaust fan, which is greatly more effective. There ' s a lot of this **stuff** happening ; it ' s really pretty remarkable. I encourage you to look on Worldchanging if you ' re into it. We ' re starting to cover this more and more. There ' s also neo- biological design, where more and more we ' re actually using life itself and the processes of life to become part of our industry. So this, for example, is hydrogen-generating algae. So we have a model in potential, an emerging model that we ' re looking for of how to take the cities most of us live in, and turn them into Bright Green cities. But unfortunately, most of the people on the planet don ' t live in the cites we live in. They live in the emerging megacities of the developing world. And there ' s a statistic I often like to use, which is that we ' re adding a city of Seattle every four days, a city the size of Seattle to the planet every four days. I was giving a talk about two months ago, and this **guy**, who ' d done some work with the U. N., came up to me and was really flustered, and he said, look, you ' ve got that totally wrong ; it ' s totally wrong. It ' s every seven days. So, we ' re adding a city the size of Seattle every seven days, and most of those cities look more like this than the city that you or I live in. Most of those cites are growing incredibly quickly. They don ' t have existing infrastructure ; they have enormous numbers of people who are struggling with poverty, and enormous numbers of people are trying to figure out how to do things in new ways. So what do we need in order to make developing nation megacities into Bright Green megacities? Well, the first thing we need is, we need leapfrogging. And this is one of the things that we are looking for everywhere. The idea behind leapfrogging is that if you are a person, or a country, who is stuck in a situation where you don ' t have the tools and technologies that you need, there ' s no reason for you to **invest** in last generation ' s technologies. Right? That you ' re much better off, almost universally, looking for a low-cost or locally applicable version of the newest technology. One place we ' re all familiar with seeing this is with cell **phones**. Right? All throughout the developing world, people are going directly to cell **phones**, skipping the whole landline stage. If there are landlines in many developing world cities, they ' re usually pretty crappy systems that break down a lot and cost enormous amounts of money. So I rather like this picture here. I particularly like the Ganesh in the background, talking on the cell **phone**. So what we have, increasingly, is cell **phones** just permeating out through society. We ' ve heard all about this here this week, so I won ' t say too much more than that, other than to say what is true for cell **phones** is true for all sorts of technologies. The second thing is tools for collaboration, be they systems of collaboration, or intellectual property systems which encourage collaboration. Right? When you have free **ability** for people to freely work together and innovate, you get different kinds of solutions. And those solutions are accessible in a different way to people who don ' t have capital. Right? So, you know, we have open source software, we have Creative Commons and other kinds of Copyleft solutions. And those things lead to things like this. This is a Telecentro in Sao Paulo. This is a pretty remarkable program using free and open source software, cheap, sort of hacked-together machines, and basically sort of abandoned buildings — has put together a bunch of community centers where people can come in, get high- speed internet access, learn computer programming skills for free. And a quarter-million people every year use these now in Sao Paulo. And those quarter-million people are some of the poorest people in Sao Paolo. I particularly like the little Linux penguin in the back. So one of the things that that ' s leading to is a sort of southern cultural explosion. And one of the things we ' re really, really **interested** in at Worldchanging is the ways in which the south is re-identifying itself, and re- categorizing itself in ways that have less and less to do with most of us in this room. So it ' s not, you know, Bollywood isn ' t just answering Hollywood. Right? You know, Brazilian music scene isn ' t just answering the major labels. It ' s doing something new. There ' s new things happening. There ' s interplay between them. And, you know, you get amazing things. Like, I don ' t know if any of you have seen the movie"City of God?"Yeah, it ' s a fabulous movie if you haven ' t seen it. And it ' s all about this question, in a very artistic and indirect kind of way. You have other radical examples where the **ability** to use cultural tools is spreading out. These are people who have just been **visited** by the Internet bookmobile in Uganda. And who are waving their first books in the **air**, which, I just think that ’ s a pretty cool picture. You know? So you also have the **ability** for people to start coming together and acting on their own behalf in political and civic ways, in ways that haven ’ t happened before. And as we heard last night, as we ’ ve heard earlier this week, are absolutely, fundamentally vital to the **ability** to craft new solutions, is we ’ ve got to craft new political realities. And I would personally say that we have to craft new political realities, not only in places like India, Afghanistan, Kenya, Pakistan, what have you, but here at home as well. Another world is possible. And sort of the big motto of the anti-globalization movement. Right? We tweak that a lot. We talk about how another world isn ’ t just possible ; another world ’ s here. That it ’ s not just that we have to sort of imagine there being a different, vague possibility out there, but we need to start acting a little bit more on that possibility. We need to start doing things like Lula, President of Brazil. How many people knew of Lula before today? OK, so, much, much better than the average crowd, I can tell you that. So Lula, he ’ s full of problems, full of contradictions, but one of the things that he ’ s doing is, he is putting forward an idea of how we engage in **international** relations that completely shifts the balance from the standard sort of north-south dialogue into a whole new way of global collaboration. I would keep your eye on this fellow. Another example of this sort of second superpower thing is the rise of these games that are what we call”serious play.”We ’ re looking a lot at this. This is spreading everywhere. This is from”A Force More Powerful.”It ’ s a little screenshot.”A Force More Powerful”is a video game that, while you ’ re playing it, it teaches you how to engage in non-violent insurrection and regime change. Here ’ s another one. This is from a game called”Food Force,”which is a game that teaches children how to run a **refugee** camp. These things are all contributing in a very dynamic way to a huge rise in, especially in the developing world, in people ’ s interest in and passion for democracy. We get so little news about the developing world that we often forget that there are literally millions of people out there struggling to change things to be fairer, freer, more democratic, less corrupt. And, you know, we don ’ t hear those stories enough. But it ’ s happening all over the place, and these tools are part of what ’ s making it possible. Now when you add all those things together, when you add together leapfrogging and new kinds of tools, you know, second superpower **stuff**, etc., what do you get? Well, very quickly, you get a Bright Green future for the developing world. You get, for example, green power spread throughout the world. You get — this is a building in Hyderabad, India. It ’ s the greenest building in the world. You get grassroots solutions, things that work for people who have no capital or limited access. You get barefoot solar engineers carrying solar panels into the remote mountains. You get access to distance medicine. These are Indian nurses learning how to use PDAs to access databases that have information that they don ’ t have access to at home in a distant manner. You get new tools for people in the developing world. These are LED lights that help the roughly billion people out there, for whom nightfall means darkness, to have a new means of operating. These are refrigerators that require no electricity ; they ’ re pot within a pot design. And you get water solutions. Water ’ s one of the most pressing problems. Here ’ s a design for harvesting rainwater that ’ s super cheap and available to people in the developing world. Here ’ s a design for distilling water using sunlight. Here ’ s a fog-catcher, which, if you live in a moist, jungle-like area, will distill water from the **air** that ’ s clean and drinkable. Here ’ s a way of transporting water. I just love this, you know — I mean carrying water is such a drag, and somebody just came up with the idea of well, what if you rolled it. Right? I mean, that ’ s a great design. This is a fabulous invention, LifeStraw.

Basically you can suck any water through this and it will become drinkable by the time it hits your lips. So, you know, people who are in desperate straits can get this. This is one of my favorite Worldchanging kinds of things ever. This is a merry-go-round invented by the company Roundabout, which pumps water as kids play. You know? Seriously — give that one a hand, it ’ s pretty great. And the same thing is true for people who are in absolute **crisis**. Right? We ’ re expecting to have upwards of 200 million **refugees** by the year 2020 because of **climate** change and political instability. How do we help people like that? Well, there ’ s all sorts of amazing new humanitarian designs that are being developed in collaborative ways all across the planet. Some of those designs include models for acting, such as new models for village instruction in the middle of **refugee** camps. New models for pedagogy for the displaced. And we have new tools. This is one of my absolute favorite things anywhere. Does anyone know what this is? Audience : It detects landmines. Alex Steffen : Exactly, this is a landmine-detecting flower. If you are living in one of the places where the roughly half-billion unaccounted for mines are scattered, you can fling these seeds out into the field. And as they grow up, they will grow up around the mines, their roots will detect the chemicals in them, and where the flowers turn red you don ’ t step. Yeah, so seeds that could save your life. You know? I also love it because it seems to me that the example, the tools we use to change the world, ought to be beautiful in themselves. You know, that it ’ s not just enough to survive. We ’ ve got to make something better than what we ’ ve got. And I think that we will. Just to wrap up, in the immortal words of H. G. Wells, I think that better things are on the way. I think that, in fact, that”all of the past is but the beginning of a beginning. All that the human mind has accomplished is but the dream before the awakening.”I hope that that turns out to be true. The people in this room have given me more **confidence** than ever that it will. Thank you very much.

Text Sample 517: POS Dim 4 – 2005 – Score 17.00 – 2005/t000474.txt

As other speakers have said, it ’ s a rather daunting experience â a particularly daunting experience â to be speaking in front of this audience. But unlike the other speakers, I ’ m not going to tell you about the mysteries of the universe, or the wonders of evolution, or the really clever, innovative ways people are attacking the major inequalities in our world. Or even the challenges of nation-states in the modern global economy. My brief, as you ’ ve just heard, is to tell you about statistics â and, to be more precise, to tell you some exciting things about statistics. And that ’ s â â that ’ s rather more challenging than all the speakers before me and all the ones coming after me. One of my **senior** colleagues told me, when I was a youngster in this profession, rather proudly, that statisticians were people who liked figures but didn ’ t have the personality skills to become accountants. And there ’ s another in-joke among statisticians, and that ’ s,”How do you tell the introverted statistician from the extroverted statistician?”To which the answer is,”The extroverted statistician ’ s the one who looks at the other person ’ s shoes.”But I want to tell you something useful â and here it is, so concentrate now. This evening, there ’ s a reception in the University ’ s Museum of Natural History. And it ’ s a wonderful setting, as I hope you ’ ll find, and a great icon to the best of the Victorian tradition. It ’ s very unlikely â in this special setting, and this collection of people â but you might just find yourself talking to someone you ’ d rather wish that you weren ’ t. So here ’ s what you do. When they say to you,”What do you do?”â you say,”I ’ m a statistician.”Well, except they ’ ve been pre-warned now, and they ’ ll know you ’ re making it up. And then one of two things will happen. They ’ ll either discover their long-lost cousin in the other corner of the room and run over and talk to them. Or they ’ ll suddenly become parched and / or hungry â and often both â and sprint off for a drink and some food. And you ’ ll be left in peace to talk to the person you really want to talk to. It ’ s one of the challenges in our profession to try and explain what we do. We ’ re not top on people ’ s lists for dinner party guests and conversations and so on. And it ’ s something I ’ ve never really found a good way of doing. But my wife â who was then my girlfriend â managed it much better than I ’ ve ever been able to. Many years ago, when we first started going out, she was working for the BBC in Britain, and I was, at that stage, working in America. I was coming back to **visit** her. She told this to one of her colleagues, who said,”Well, what does your boyfriend do?”Sarah thought quite hard about the things I ’ d explained â and she concentrated, in those days, on listening. Don ’ t tell her I said that. And she was thinking about the work I did developing mathematical models for understanding evolution and modern genetics. So when her colleague said,”What does he do?”She paused and said,”He models things.”Well, her colleague suddenly got much more **interested** than I had any right to expect and went on and said,”What does he model?”Well, Sarah thought a little bit more about my work and said,”Genes.””He models genes.”That is my first love, and that ’ s what I ’ ll tell you a little bit about. What I want to do more generally is to get you thinking about the place of **uncertainty** and randomness and chance in our world, and how we react to that, and how well we do or don ’ t think about it. So you ’ ve had a pretty easy time up till now â a few laughs, and all that kind of thing â in the talks to date. You ’ ve got to think, and I ’ m going to ask you some questions. So here ’ s the scene for the first question I ’ m going to ask you. Can you imagine tossing a coin successively? And for some reason â which shall remain rather vague â we ’ re **interested** in a particular pattern. Here ’ s one â a head, followed by a tail, followed by a tail. So suppose we toss a coin repeatedly. Then the pattern, head-tail-tail, that we ’ ve suddenly become fixated with happens here. And you can count : one, two, three, four, five, six, seven, eight, nine, 10 â it happens after the 10th toss. So you might think there are more **interesting** things to do, but humor me for the moment. Imagine this half of the audience each get out coins, and they toss them until they first see the pattern head-tail-tail. The first time they do it, maybe it happens after the 10th toss, as here. The second time, maybe it ’ s after the fourth toss. The next time, after the 15th toss. So you do that lots and lots of times, and you average those numbers. That ’ s what I want this side to think about. The other half of the audience doesn ’ t like head-tail-tail â they think, for deep cultural reasons, that ’ s boring â and they ’ re much more **interested** in a different pattern â head-tail-head. So, on this side, you get out your coins, and you toss and toss and toss. And you count the number of times until the pattern head-tail-head appears and you average them. OK? So on this side, you ’ ve got a number â you ’ ve done it lots of times, so you get it accurately â which is the average number of tosses until head-tail-tail. On this side, you ’ ve got a number â the average number of tosses until head-tail-head. So here ’ s a deep mathematical fact â if you ’ ve got two numbers, one of three things must be true. Either they ’ re the same, or this one ’ s bigger than this one, or this one ’ s bigger than that one. So what ’ s going on here? So you ’ ve all got to think about this, and you ’ ve all got to vote â and we ’ re not moving on. And I don ’ t want to end up in the two-minute silence to give you more time to think about it, until everyone ’ s expressed a view. OK. So what you want to do is compare the average number of tosses until we first see head-tail-head with the average number of tosses until we first see head-tail-tail. Who thinks that A is true â that, on average, it ' ll take longer to see head- tail-head than head-tail-tail? Who thinks that B is true â that on average, they ' re the same? Who thinks that C is true â that, on average, it ' ll take less time to see head-tail-head than head-tail-tail? OK, who hasn ' t voted yet? Because that ' s really naughty â I said you had to. OK. So most people think B is true. And you might be relieved to know even rather distinguished mathematicians think that. It ' s not. A is true here. It takes longer, on average. In fact, the average number of tosses till head-tail-head is 10 and the average number of tosses until head-tail-tail is eight. How could that be? Anything different about the two patterns? There is. Head-tail-head overlaps itself. If you went head-tail-head-tail-head, you can cunningly get two occurrences of the pattern in only five tosses. You can ' t do that with head-tail-tail. That turns out to be important. There are two ways of thinking about this. I ' ll give you one of them. So imagine â let ' s suppose we ' re doing it. On this side â remember, you ' re excited about head-tail-tail ; you ' re excited about head-tail-head. We start tossing a coin, and we get a head â and you start sitting on the edge of your seat because something great and wonderful, or awesome, might be about to happen. The next toss is a tail â you get really excited. The champagne ' s on ice just next to you ; you ' ve got the glasses chilled to celebrate. You ' re waiting with bated breath for the final toss. And if it comes down a head, that ' s great. You ' re done, and you celebrate. If it ' s a tail â well, rather disappointedly, you put the glasses away and put the champagne back. And you keep tossing, to wait for the next head, to get excited. On this side, there ' s a different experience. It ' s the same for the first two parts of the sequence. You ' re a little bit excited with the first head â you get rather more excited with the next tail. Then you toss the coin. If it ' s a tail, you crack open the champagne. If it ' s a head you ' re disappointed, but you ' re still a third of the way to your pattern again. And that ' s an informal way of presenting it â that ' s why there ' s a difference. Another way of thinking about it â if we tossed a coin eight million times, then we ' d expect a million head-tail-heads and a million head-tail-tails â but the head-tail- heads could occur in clumps. So if you want to put a million things down amongst eight million positions and you can have some of them overlapping, the clumps will be further apart. It ' s another way of getting the intuition. What ' s the point I want to make? It ' s a very, very simple example, an easily stated question in probability, which every â you ' re in good company â **everybody** gets wrong. This is my little diversion into my real passion, which is genetics. There ' s a connection between head-tail-heads and head-tail-tails in genetics, and it ' s the following. When you toss a coin, you get a sequence of heads and tails. When you look at DNA, there ' s a sequence of not two things â heads and tails â but four letters â As, Gs, Cs and Ts. And there are little chemical scissors, called restriction enzymes which cut DNA whenever they see particular patterns. And they ' re an enormously useful tool in modern molecular biology. And instead of asking the question,"How long until I see a head-tail-head?"â you can ask,"How big will the chunks be when I use a restriction enzyme which cuts whenever it sees G-A-A-G, for example? How long will those chunks be?"That ' s a rather trivial connection between probability and genetics. There ' s a much deeper connection, which I don ' t have time to go into and that is that modern genetics is a really exciting area of science. And we ' ll hear some talks later in the **conference** specifically about that. But it turns out that unlocking the secrets in the information generated by modern experimental technologies, a key part of that has to do with fairly sophisticated â you ' ll be relieved to know that I do something useful in my day job, rather more sophisticated than the head-tail-head story â but quite sophisticated computer modelings and mathematical modelings and modern statistical techniques. And I will give you two little snippets â two examples â of projects we ’ re **involved** in in my group in Oxford, both of which I think are rather exciting. You know about the Human Genome Project. That was a project which aimed to read one copy of the human genome. The natural thing to do after you ’ ve done that â and that ’ s what this project, the International HapMap Project, which is a collaboration between labs in five or six different countries. Think of the Human Genome Project as learning what we ’ ve got in common, and the HapMap Project is trying to understand where there are differences between different people. Why do we care about that? Well, there are lots of reasons. The most pressing one is that we want to understand how some differences make some people susceptible to one disease â type-2 diabetes, for example â and other differences make people more susceptible to heart disease, or stroke, or autism and so on. That ’ s one big project. There ’ s a second big project, recently funded by the Wellcome Trust in this country, **involving** very large studies â thousands of individuals, with each of eight different diseases, common diseases like type-1 and type-2 diabetes, and coronary heart disease, bipolar disease and so on â to try and understand the genetics. To try and understand what it is about genetic differences that causes the diseases. Why do we want to do that? Because we understand very little about most human diseases. We don ’ t know what causes them. And if we can get in at the bottom and understand the genetics, we ’ ll have a window on the way the disease works, and a whole new way about thinking about disease therapies and preventative treatment and so on. So that ’ s, as I said, the little diversion on my main love. Back to some of the more mundane issues of thinking about **uncertainty**. Here ’ s another quiz for you â now suppose we ’ ve got a test for a disease which isn ’ t infallible, but it ’ s pretty good. It gets it right 99 percent of the time. And I take one of you, or I take someone off the street, and I test them for the disease in question. Let ’ s suppose there ’ s a test for HIV â the **virus** that causes AIDS â and the test says the person has the disease. What ’ s the chance that they do? The test gets it right 99 percent of the time. So a natural answer is 99 percent. Who likes that answer? Come on â everyone ’ s got to get **involved**. Don ’ t think you don ’ t **trust** me anymore. Well, you ’ re right to be a bit skeptical, because that ’ s not the answer. That ’ s what you might think. It ’ s not the answer, and it ’ s not because it ’ s only part of the story. It actually depends on how common or how rare the disease is. So let me try and illustrate that. Here ’ s a little caricature of a million individuals. So let ’ s think about a disease that affects â it ’ s pretty rare, it affects one person in 10, 000. Amongst these million individuals, most of them are healthy and some of them will have the disease. And in fact, if this is the prevalence of the disease, about 100 will have the disease and the rest won ’ t. So now suppose we test them all. What happens? Well, amongst the 100 who do have the disease, the test will get it right 99 percent of the time, and 99 will test positive. Amongst all these other people who don ’ t have the disease, the test will get it right 99 percent of the time. It ’ ll only get it wrong one percent of the time. But there are so many of them that there ’ ll be an enormous number of false positives. Put that another way â of all of them who test positive â so here they are, the individuals **involved** â less than one in 100 actually have the disease. So even though we think the test is accurate, the important part of the story is there ’ s another bit of information we need. Here ’ s the key intuition. What we have to do, once we know the test is positive, is to weigh up the plausibility, or the likelihood, of two competing explanations. Each of those explanations has a likely bit and an unlikely bit. One explanation is that the person doesn ’ t have the disease â that ’ s overwhelmingly likely, if you pick someone at random â but the test gets it wrong, which is unlikely. The other explanation is that the person does have the disease â that ’ s unlikely â but the test gets it right, which is likely. And the number we end up with â that number which is a little bit less than one in 100 â is to do with how likely one of those explanations is relative to the other. Each of them taken together is unlikely. Here ’ s a more topical example of exactly the same thing. Those of you in Britain will know about what ’ s become rather a celebrated case of a woman called Sally Clark, who had two babies who died suddenly. And initially, it was thought that they died of what ’ s known informally as”cot death,”and more formally as”Sudden Infant Death Syndrome.”For various reasons, she was later charged with murder. And at the trial, her trial, a very distinguished pediatrician gave evidence that the chance of two cot deaths, innocent deaths, in a family like hers â which was professional and non-smoking â was one in 73 million. To cut a long story short, she was convicted at the time. Later, and fairly recently, acquitted on appeal â in fact, on the second appeal. And just to set it in context, you can imagine how awful it is for someone to have lost one child, and then two, if they ’ re innocent, to be convicted of murdering them. To be put through the stress of the trial, convicted of murdering them â and to spend time in a women ’ s prison, where all the other prisoners think you killed your children â is a really awful thing to happen to someone. And it happened in large part here because the expert got the statistics horribly wrong, in two different ways. So where did he get the one in 73 million number? He looked at some research, which said the chance of one cot death in a family like Sally Clark ’ s is about one in 8, 500. So he said,”I ’ ll assume that if you have one cot death in a family, the chance of a second child dying from cot death aren ’ t changed.”So that ’ s what statisticians would call an assumption of independence. It ’ s like saying,”If you toss a coin and get a head the first time, that won ’ t affect the chance of getting a head the second time.”So if you toss a coin twice, the chance of getting a head twice are a half â that ’ s the chance the first time â times a half â the chance a second time. So he said,”Here, I ’ ll assume that these events are independent. When you multiply 8, 500 together twice, you get about 73 million.”And none of this was stated to the **court** as an assumption or presented to the jury that way. Unfortunately here â and, really, regrettably â first of all, in a situation like this you ’ d have to verify it empirically. And secondly, it ’ s palpably false. There are lots and lots of things that we don ’ t know about sudden infant deaths. It might well be that there are environmental factors that we ’ re not aware of, and it ’ s pretty likely to be the case that there are genetic factors we ’ re not aware of. So if a family suffers from one cot death, you ’ d put them in a high-risk group. They ’ ve probably got these environmental risk factors and / or genetic risk factors we don ’ t know about. And to argue, then, that the chance of a second death is as if you didn ’ t know that information is really silly. It ’ s worse than silly â it ’ s really bad science. Nonetheless, that ’ s how it was presented, and at trial nobody even argued it. That ’ s the first problem. The second problem is, what does the number of one in 73 million mean? So after Sally Clark was convicted â you can imagine, it made rather a splash in the press â one of the journalists from one of Britain ’ s more reputable newspapers wrote that what the expert had said was,”The chance that she was innocent was one in 73 million.”Now, that ’ s a logical error. It ’ s exactly the same logical error as the logical error of thinking that after the disease test, which is 99 percent accurate, the chance of having the disease is 99 percent. In the disease example, we had to bear in mind two things, one of which was the possibility that the test got it right or not. And the other one was the chance, a priori, that the person had the disease or not. It ’ s exactly the same in this context. There are two things **involved** â two parts to the explanation. We want to know how likely, or relatively how likely, two different explanations are. One of them is that Sally Clark was innocent â which is, a priori, overwhelmingly likely â most mothers don ' t kill their children. And the second part of the explanation is that she suffered an incredibly unlikely event. Not as unlikely as one in 73 million, but nonetheless rather unlikely. The other explanation is that she was guilty. Now, we probably think a priori that ' s unlikely. And we certainly should think in the context of a criminal trial that that ' s unlikely, because of the presumption of innocence. And then if she were trying to kill the children, she succeeded. So the chance that she ' s innocent isn ' t one in 73 million. We don ' t know what it is. It has to do with weighing up the strength of the other evidence against her and the statistical evidence. We know the children died. What matters is how likely or unlikely, relative to each other, the two explanations are. And they ' re both implausible. There ' s a situation where errors in statistics had really profound and really unfortunate consequences. In fact, there are two other women who were convicted on the basis of the evidence of this pediatrician, who have subsequently been released on appeal. Many cases were reviewed. And it ' s particularly topical because he ' s currently facing a disrepute charge at Britain ' s General Medical Council. So just to conclude â what are the take-home messages from this? Well, we know that randomness and **uncertainty** and chance are very much a part of our everyday life. It ' s also true â and, although, you, as a **collective**, are very special in many ways, you ' re completely typical in not getting the examples I gave right. It ' s very well **documented** that people get things wrong. They make errors of logic in reasoning with **uncertainty**. We can cope with the subtleties of language brilliantly â and there are **interesting** evolutionary questions about how we got here. We are not good at reasoning with **uncertainty**. That ' s an issue in our everyday lives. As you ' ve heard from many of the talks, statistics underpins an enormous amount of research in science â in social science, in medicine and indeed, quite a lot of industry. All of quality control, which has had a major **impact** on industrial processing, is underpinned by statistics. It ' s something we ' re bad at doing. At the very least, we should recognize that, and we tend not to. To go back to the legal context, at the Sally Clark trial all of the lawyers just accepted what the expert said. So if a pediatrician had come out and said to a jury,"I know how to build bridges. I ' ve built one down the road. Please drive your car home over it,"they would have said,"Well, pediatricians don ' t know how to build bridges. That ' s what engineers do."On the other hand, he came out and effectively said, or implied,"I know how to reason with **uncertainty**. I know how to do statistics."And everyone said,"Well, that ' s fine. He ' s an expert."So we need to understand where our competence is and isn ' t. Exactly the same kinds of issues arose in the early days of DNA profiling, when scientists, and lawyers and in some cases judges, routinely misrepresented evidence. Usually â one hopes â innocently, but misrepresented evidence. Forensic scientists said,"The chance that this **guy** ' s innocent is one in three million."Even if you believe the number, just like the 73 million to one, that ' s not what it meant. And there have been celebrated appeal cases in Britain and elsewhere because of that. And just to finish in the context of the legal system. It ' s all very well to say,"Let ' s do our best to present the evidence."But more and more, in cases of DNA profiling â this is another one â we expect juries, who are ordinary people â and it ' s **documented** they ' re very bad at this â we expect juries to be able to cope with the sorts of reasoning that goes on. In other spheres of life, if people argued â well, except possibly for politics â but in other spheres of life, if people argued illogically, we ' d say that ' s not a good thing. We sort of expect it of politicians and don ' t hope for much more. In the case of **uncertainty**, we get it wrong all the time â and at the very least, we should be aware of that, and ideally, we might try and do something about it. Thanks very much.

Text Sample 518: POS Dim 4 – 2007 – Score 18.00 – 2007/t000238.txt

Well, good morning. You know, the computer and television both recently turned 60, and today I ’ d like to talk about their relationship. Despite their middle age, if you ’ ve been following the themes of this **conference** or the entertainment industry, it ’ s pretty clear that one has been picking on the other. So it ’ s about time that we talked about how the computer ambushed television, or why the invention of the atomic bomb unleashed forces that lead to the writers ’ strike. And it ’ s not just what these are doing to each other, but it ’ s what the audience thinks that really frames this matter. To get a sense of this, and it ’ s been a theme we ’ ve talked about all week, I recently talked to a bunch of tweeners. I wrote on cards :”television,””radio,””MySpace,””Internet,””PC.”And I said, just arrange these, from what ’ s important to you and what ’ s not, and then tell me why. Let ’ s listen to what happens when they get to the portion of the discussion on television. Girl 1 : Well, I think it ’ s important but, like, not necessary because you can do a lot of other **stuff** with your free time than watch programs. Peter Hirshberg : Which is more fun, Internet or TV? Girls : Internet. Girl 2 : I think we — the reasons, one of the reasons we put computer before TV is because nowadays, like, we have TV shows on the computer. Girl 2 : And then you can download onto your iPod. PH : Would you like to be the president of a TV network? Girl 4 : I wouldn ’ t like it. Girl 2 : That would be so stressful. Girl 5 : No. PH : How come? Girl 5 : Because they ’ re going to lose all their money eventually. Girl 3 : Like the stock market, it goes up and down and **stuff**. I think right now the computers will be at the top and everything will be kind of going down and **stuff**. PH : There ’ s been an uneasy relationship between the TV business and the **tech** business, really ever since they both turned about 30. We go through periods of enthrallment, followed by **reactions** in boardrooms, in the finance community best characterized as, what ’ s the finance term? Ick pooey. Let me give you an example of this. The year is 1976, and Warner buys Atari because video games are on the rise. The next year they march forward and they introduce Qube, the first interactive cable TV system, and the New York Times heralds this as telecommunications moving to the home, convergence, great things are happening. **Everybody** in the East Coast gets in the pictures — Citicorp, Penney, RCA — all getting into this big vision. By the way, this is about when I enter the picture. I ’ m going to do a summer internship at Time Warner. That summer I ’ m all — I ’ m at Warner that summer — I ’ m all excited to work on convergence, and then the bottom falls out. Doesn ’ t work out too well for them, they lose money. And I had a happy brush with convergence until, kind of, Warner basically has to liquidate the whole thing. That ’ s when I leave graduate school, and I can ’ t work in New York on kind of entertainment and technology because I have to be exiled to California, where the remaining jobs are, almost to the sea, to go to work for Apple Computer. Warner, of course, writes off more than 400 million dollars. Four hundred million dollars, which was real money back in the ’ 70s. But they were onto something and they got better at it. By the year 2000, the process was perfected. They merged with AOL, and in just four years, managed to shed about 200 billion dollars of market capitalization, showing that they ’ d actually mastered the art of applying Moore ’ s law of successive miniaturization to their balance sheet. Now, I think that one reason that the media and the entertainment communities, or the media community, is driven so crazy by the **tech** community is that **tech** folks talk differently. You know, for 50 years, we ' ve talked about changing the world, about total transformation. For 50 years, it ' s been about hopes and fears and promises of a better world. And I got to thinking, you know, who else talks that way? And the answer is pretty clearly — it ' s people in religion and in politics. And so I realized that actually the **tech** world is best understood, not as a business cycle, but as a messianic movement. We promise something great, we evangelize it, we ' re going to change the world. It doesn ' t work out too well, and so we actually go back to the well and start all over again, as the people in New York and L. A. look on in absolute, morbid astonishment. But it ' s this irrational view of things that drives us on to the next thing. So, what I ' d like to ask is, if the computer is becoming a principal tool of media and entertainment, how did we get here? I mean, how did a machine that was built for accounting and artillery morph into media? Of course, the first computer was built just after World War II to solve military problems, but things got really **interesting** just a couple of years later — 1949 with Whirlwind, built at MIT ' s Lincoln Lab. Jay Forrester was building this for the Navy, but you can ' t help but see that the creator of this machine had in mind a machine that might actually be a potential media star. So take a look at what happens when the foremost journalist of early television meets one of the foremost computer pioneers, and the computer begins to express itself. Journalist : It ' s a Whirlwind electronic computer. With considerable trepidation, we undertake to interview this new machine. Jay Forrester : Hello New York, this is Cambridge. And this is the oscilloscope of the Whirlwind electronic computer. Would you like if I used the machine? Journalist : Yes, of course. But I have an idea, Mr. Forrester. Since this computer was made in conjunction with the Office of Naval Research, why don ' t we switch down to the Pentagon in Washington and let the Navy ' s research chief, Admiral Bolster, give Whirlwind the workout? Calvin Bolster : Well, Ed, this problem concerns the Navy ' s Viking rocket. This rocket goes up 135 miles into the sky. Now, at the standard rate of fuel consumption, I would like to see the computer trace the flight path of this rocket and see how it can determine, at any instant, say at the end of 40 seconds, the amount of fuel remaining, and the velocity at that set instant. JF : Over on the left-hand side, you will notice fuel consumption decreasing as the rocket takes off. And on the right-hand side, there ' s a scale that shows the rocket ' s velocity. The rocket ' s position is shown by the trajectory that we ' re now looking at. And as it reaches the peak of its trajectory, the velocity, you will notice, has dropped off to a minimum. Then, as the rocket dives down, velocity picks up again toward a maximum velocity and the rocket hits the ground. How ' s that? Journalist : What about that, Admiral? CB : Looks very good to me. JF : And before leaving, we would like to show you another kind of mathematical problem that some of the boys have worked out in their spare time, in a less serious vein, for a Sunday afternoon. Journalist : Thank you very much indeed, Mr. Forrester and the MIT lab. PH : You know, so much was worked out : the first real-time interaction, the video display, pointing a gun. It lead to the microcomputer, but unfortunately, it was too pricey for the Navy, and all of this would have been lost if it weren ' t for a happy coincidence. Enter the atomic bomb. We ' re threatened by the greatest weapon ever, and knowing a good thing when it sees it, the Air Force decides it needs the biggest computer ever to protect us. They adapt Whirlwind to a massive **air** defense system, deploy it all across the frozen north, and spend nearly three times as much on this computer as was spent on the Manhattan Project building the A-Bomb in the first place. Talk about a shot in the arm for the computer industry. And you can imagine that the Air Force became a pretty good salesman.

Here ’ s their marketing video. Narrator : In a mass raid, high-speed bombers could be in on us before we could determine their tracks. And then it would be too late to act. We cannot afford to take that chance. It is to meet this threat that the Air Force has been developing SAGE, the Semi-Automatic Ground Environment system, to strengthen our **air** defenses. This new computer, built to become the nerve center of a defense network, is able to perform all the complex mathematical problems **involved** in countering a mass enemy raid. It is provided with its own powerhouse containing large diesel-driven generators, air-conditioning equipment, and cooling towers required to cool the thousands of vacuum tubes in the computer. PH : You know, that one computer was huge. There ’ s an **interesting** marketing lesson from it, which is basically, when you market a product, you can either say, this is going to be wonderful, it will make you feel better and enliven you. Or there ’ s one other marketing proposition : if you don ’ t use our product, you ’ ll die. This is a really good example of that. This had the first pointing device. It was distributed, so it worked out — distributed computing and modems — so all these things could talk to each other. About 20 percent of all the nation ’ s programmers were wrapped up in this thing, and it led to an awful lot of what we have today. It also used vacuum tubes. You saw how huge it was, and to give you a sense for this — because we ’ ve talked a lot about Moore ’ s law and making things small at this **conference**, so let ’ s talk about making things large. If we took Whirlwind and put it in a place that you all know, say, Century City, it would fit beautifully. You ’ d kind of have to take Century City out, but it could fit in there. But like, let ’ s imagine we took the latest Pentium processor, the latest Core 2 Extreme, which is a four-core processor that Intel ’ s working on, it will be our laptop tomorrow. To build that, what we ’ d do with Whirlwind technology is we ’ d have to take up roughly from the 10 to Mulholland, and from the 405 to La Cienega just with those Whirlwinds. And then, the 92 nuclear power plants that it would take to provide the power would fill up the rest of Los Angeles. That ’ s roughly a third more nuclear power than all of France creates. So, the next time they tell you they ’ re on to something, clearly they ’ re not. So — and we haven ’ t even worked out the cooling needs. But it gives you the kind of power that people have, that the audience has, and the reasons these transformations are happening. All of this **stuff** starts moving into industry. DEC kind of reduces all this and makes the first mini-computer. It shows up at places like MIT, and then a mutation happens. Spacewar! is built, the first computer game, and all of a sudden, interactivity and involvement and passion is worked out. Actually, many MIT students stayed up all night long working on this thing, and many of the principles of gaming today were worked out. DEC knew a good thing about wasting time. It shipped every one of its computers with that game. Meanwhile, as all of this is happening, by the mid- ’ 50s, the business model of traditional broadcasting and cinema has been busted completely. A new technology has confounded radio men and movie moguls and they ’ re quite certain that television is about to do them in. In fact, despair is in the **air**. And a quote that sounds largely reminiscent from everything I ’ ve been reading all week. RCA had David Sarnoff, who basically commercialized radio, said this,”I don ’ t say that radio networks must die. Every effort has been made and will continue to be made to find a new pattern, new selling arrangements and new types of programs that may arrest the declining revenues. It may yet be possible to eke out a poor existence for radio, but I don ’ t know how.”And of course, as the computer industry develops interactively, producers in the emerging TV business actually hit on the same idea. And they fake it. Jack Berry : Boys and girls, I think you all know how to get your magic windows up on the set, you just get them out. First of all, get your Winky Dink kits out. Put out your Magic Window and your erasing glove, and rub it like this. That ' s the way we get some of the magic into it, boys and girls. Then take it and put it right up against the screen of your own television set, and rub it out from the center to the corners, like this. Make sure you keep your magic crayons handy, your Winky Dink crayons and your erasing glove, because you ' ll be using them during the show to draw like that. You all set? OK, let ' s get right to the first story about Dusty Man. Come on into the secret lab. PH : It was the dawn of interactive TV, and you may have noticed they wanted to sell you the Winky Dink kits. Those are the Winky Dink crayons. I know what you ' re saying."Pete, I could use any ordinary open-source crayon, why do I have to buy theirs?"I assure you, that ' s not the case. Turns out they told us directly that these are the only crayons you should ever use with your Winky Dink Magic Window, other crayons may discolor or hurt the window. This proprietary principle of vendor lock-in would go on to be perfected with great success as one of the enduring principles of windowing systems everywhere. It led to lawsuits — — federal investigations, and lots of repercussions, and that ' s a scandal we won ' t discuss today. But we will discuss this scandal, because this man, Jack Berry, the host of"Winky Dink,"went on to become the host of"Twenty One,"one of the most important quiz shows ever. And it was rigged, and it became unraveled when this man, Charles van Doren, was outed after an unnatural winning streak, ending Berry ' s career. And actually, ending the career of a lot of people at CBS. It turns out there was a lot to learn about how this new medium worked. And 50 years ago, if you ' d been at a meeting like this and were trying to understand the media, there was one prophet and only but one you wanted to hear from, Professor Marshall McLuhan. He actually understood something about a theme that we ' ve been discussing all week. It ' s the role of the audience in an era of pervasive electronic communications. Here he is talking from the 1960s. Marshall McLuhan : If the audience can become **involved** in the actual process of making the ad, then it ' s happy. It ' s like the old quiz shows. They were great TV because it gave the audience a role, something to do. They were horrified when they discovered they ' d really been left out all the time because the shows were rigged. Now, then, this was a horrible misunderstanding of TV on the part of the programmers. PH : You know, McLuhan talked about the global village. If you substitute the word blogosphere, of the Internet today, it is very true that his understanding is probably very enlightening now. Let ' s listen in to him. MM : The global village is a world in which you don ' t necessarily have harmony. You have extreme concern with **everybody** else ' s business and much involvement in **everybody** else ' s life. It ' s a sort of Ann Landers ' column writ large. And it doesn ' t necessarily mean harmony and peace and quiet, but it does mean huge involvement in **everybody** else ' s affairs. And so the global village is as big as a planet, and as small as a village post office. PH : We ' ll talk a little bit more about him later. We ' re now right into the 1960s. It ' s the era of big business and data centers for computing. But all that was about to change. You know, the expression of technology reflects the people and the time of the culture it was built in. And when I say that code expresses our hopes and aspirations, it ' s not just a joke about messianism, it ' s actually what we do. But for this part of the story, I ' d actually like to throw it to America ' s leading technology correspondent, John Markoff. John Markoff : Do you want to know what the counterculture in drugs, sex, rock ' n ' roll and the anti-war movement had to do with computing? Everything. It all happened within five miles of where I ' m standing, at Stanford University, between 1960 and 1975. In the midst of revolution in the streets and rock and roll concerts in the parks, a group of researchers led by people like John McCarthy, a computer scientist at the Stanford Artificial Intelligence Lab, and

Doug Engelbart, a computer scientist at SRI, changed the world. Engelbart came out of a pretty dry engineering culture, but while he was beginning to do his work, all of this **stuff** was bubbling on the mid-peninsula. There was LSD leaking out of Kesey ’ s Veterans ’ Hospital experiments and other areas around the campus, and there was music literally in the streets. The Grateful Dead was playing in the pizza parlors. People were leaving to go back to the land. There was the Vietnam War. There was black liberation. There was women ’ s liberation. This was a remarkable place, at a remarkable time. And into that ferment came the microprocessor. I think it was that interaction that led to personal computing. They saw these tools that were controlled by the establishment as ones that could actually be liberated and put to use by these communities that they were trying to build. And most importantly, they had this ethos of sharing information. I think these ideas are difficult to understand, because when you ’ re trapped in one paradigm, the next paradigm is always like a science fiction universe — it makes no sense. The stories were so compelling that I decided to write a book about them. The title of the book is,”What the Dormouse Said : How the ’ 60s Counterculture Shaped the Personal Computer Industry.”The title was taken from the lyrics to a Jefferson Airplane song. The lyrics go,”Remember what the dormouse said. Feed your head, feed your head, feed your head.”PH : By this time, computing had kind of leapt into media territory, and in short order much of what we ’ re doing today was imagined in Cambridge and Silicon Valley. Here ’ s the Architecture Machine Group, the predecessor of the Media Lab, in 1981. Meanwhile, in California, we were trying to commercialize a lot of this **stuff**. HyperCard was the first program to introduce the public to hyperlinks, where you could randomly hook to any kind of picture, or piece of text, or data across a file system, and we had no way of explaining it. There was no metaphor. Was it a database? A prototyping tool? A scripted language? Heck, it was everything. So we ended up writing a marketing brochure. We asked a question about how the mind works, and we let our customers play the role of so many blind men filling out the elephant. A few years later, we then hit on the idea of explaining to people the secret of, how do you get the content you want, the way you want it and the easy way? Here ’ s the Apple marketing video. James Burke : You ’ ll be pleased to know, I ’ m sure, that there are several ways to create a HyperCard interactive video. The most involved method is to go ahead and produce your own videodisc as well as build your own HyperCard stacks. By far the simplest method is to buy a pre-made videodisc and HyperCard stacks from a commercial supplier. The method we illustrate in this video uses a pre-made videodisc but creates custom HyperCard stacks. This method allows you to use existing videodisc materials in ways which suit your specific needs and interests. PH : I hope you realize how subversive that is. That ’ s like a Dick Cheney speech. You think he ’ s a nice balding **guy**, but he ’ s just declared war on the content business. Find the commercial **stuff**, mash it up, tell the story your way. Now, as long as we confine this to the education market, and a personal matter between the computer and the file system, that ’ s fine, but as you can see, it was about to leap out and upset Jack Valenti and a lot of other people. By the way, speaking of the filing system, it never occurred to us that these hyperlinks could go beyond the local area network. A few years later, Tim Berners-Lee worked that out. It became a killer app of links, and today, of course, we call that the World Wide Web. Now, not only was I instrumental in helping Apple miss the Internet, but a couple of years later, I helped Bill Gates do the same thing. The year is 1993 and he was working on a book and I was working on a video to help him kind of explain where we were all heading and how to popularize all this. We were plenty aware that we were messing with media, and on the surface, it looks like we predicted a lot of the right things, but we also missed an awful lot. Let ' s take a look. Narrator : The pyramids, the Colosseum, the New York subway system and TV dinners, ancient and modern wonders of the man-made world all. Yet each pales to insignificance with the completion of that magnificent accomplishment of twenty-first-century technology, the Digital Superhighway. Once it was only a dream of technoids and a few long-forgotten politicians. The Digital Highway arrived in America ' s living rooms late in the twentieth century. Let us recall the pioneers who made this technical marvel possible. The Digital Highway would follow the rutted trail first blazed by Alexander Graham Bell. Though some were incredulous... Man 1 : The **phone** company! Narrator : Stirred by the prospects of mass communication and making big bucks on advertising, David Sarnoff commercializes radio. Man 2 : Never had scientists been put under such pressure and **demand**. Narrator : The medium introduced America to new products. Voice 1 : Say, mom, Windows for Radio means more enjoyment and greater ease of use for the whole family. Be sure to enjoy Windows for Radio at home and at work. Narrator : In 1939, the Radio Corporation of America introduced television. Man 2 : Never had scientists been put under such pressure and **demand**. Narrator : Eventually, the race to the future took on added momentum with the breakup of the telephone company. And further stimulus came with the deregulation of the cable television industry, and the re-regulation of the cable television industry. Ted Turner : We did the work to build this, this cable industry, now the broadcasters want some of our money. I mean, it ' s ridiculous. Narrator : Computers, once the unwieldy tools of accountants and other geeks, escaped the backrooms to enter the media fracas. The world and all its culture reduced to bits, the lingua franca of all media. And the forces of convergence exploded. Finally, four great industrial **sectors** combined. Telecommunications, entertainment, computing and everything else. Man 3 : We ' ll see channels for the gourmet and we ' ll see channels for the pet lover. Voice 2 : Next on the gourmet pet channel, decorating birthday cakes for your schnauzer. Narrator : All of industry was in play, as investors flocked to place their bets. At stake : the battle for you, the consumer, and the right to spend billions to send a lot of information into the parlors of America. PH : We missed a lot. You know, you missed, we missed the Internet, the long tail, the role of the audience, open systems, social networks. It just goes to show how tough it is to come up with the right uses of media. Thomas Edison had the same problem. He wrote a list of what the phonograph might be good for when he invented it, and kind of only one of his ideas turned out to have been the right early idea. Well, you know where we ' re going on from here. We come into the era of the dotcom, the World Wide Web, and I don ' t need to tell you about that because we all went through that bubble together. But when we emerge from this and what we call Web 2. 0, things actually are quite different. And I think it ' s the reason that TV ' s so challenged. If Internet one was about pages, now it ' s about people. It ' s a customer, it ' s an audience, it ' s a person who ' s participating. It ' s the formidable thing that is changing entertainment now. MM : Because it gave the audience a role, something to do. PH : In my own company, Technorati, we see something like 67, 000 blog posts an hour come in. That ' s about 2, 700 fresh, connective links across about 112 million blogs that are out there. And it ' s no wonder that as we head into the writers ' strike, odd things happen. You know, it reminds me of that old saw in Hollywood, that a producer is anyone who knows a writer. I now think a network boss is anyone who has a cable modem. But it ' s not a joke. This is a real headline."Websites attract striking writers : operators of sites like MyDamnChannel. com could benefit from labor disputes."Meanwhile, you have the TV bloggers going out on strike, in sympathy with the television writers. And then you have TV Guide, a Fox property, which is about to sponsor the online video awards — but cancels it out of sympathy with traditional television, not appearing to gloat. To show you how schizophrenic this all is, here ' s the head of MySpace, or Fox Interactive, a News Corp company, being asked, well, with the writers ' strike, isn ' t this going to hurt News Corp and help you online? Man : But I, yeah, I think there ' s an opportunity. As the strike continues, there ' s an opportunity for more people to experience video on places like MySpace TV. PH : Oh, but then he remembers he works for Rupert Murdoch. Man : Yes, well, first, you know, I ' m part of News Corporation as part of Fox Entertainment Group. Obviously, we hope that the strike is — that the issues are resolved as quickly as possible. PH : One of the great things that ' s going on here is the globalization of content really is happening. Here is a clip from a video, from a piece of animation that was written by a writer in Hollywood, animation worked out in Israel, farmed out to Croatia and India, and it ' s now an **international** series. Narrator : The following takes place between the minutes of 2:15 p. m. and 2:18 p. m., in the months preceding the presidential primaries. Voice 1 : You ' ll have to stay here in the **safe** house until we get word the terrorist threat is over. Voice 2 : You mean we ' ll have to live here, together? Voices 2, 3 and 4 : With her? Voice 2 : Well, there goes the neighborhood. PH : The company that created this, Aniboom, is an **interesting** example of where this is headed. Traditional TV animation costs, say, between 80, 000 and 10, 000 dollars a minute. They ' re producing things for between 1, 500 and 800 dollars a minute. And they ' re offering their creators 30 percent of the back end, in a much more entrepreneurial manner. So, it ' s a different model. What the entertainment business is struggling with, the world of brands is figuring out. For example, Nike now understands that Nike Plus is not just a device in its shoe, it ' s a network to hook its customers together. And the head of marketing at Nike says,"People are coming to our site an average of three times a week. We don ' t have to go to them."Which means television advertising is down 57 percent for Nike. Or, as Nike ' s head of marketing says,"We ' re not in the business of keeping media companies alive. We ' re in the business of connecting with consumers."And media companies realize the audience is important also. Here ' s a man announcing the new Market Watch from Dow Jones, powered 100 percent by the user experience on the home page — user-generated content married up with traditional content. It turns out you have a bigger audience and more interest if you hook up with them. Or, as Geoffrey Moore once told me, it ' s intellectual curiosity that ' s the trade that brands need in the age of the blogosphere. And I think this is beginning to happen in the entertainment business. One of my heroes is songwriter, Ally Willis, who just wrote"The Color Purple"and has been an R and — rhythm and blues writer, and this is what she said about where songwriting ' s going. Ally Willis : Where millions of collaborators wanted the song, because to look at them strictly as spam is missing what this medium is about. PH : So, to wrap up, I ' d love to throw it back to Marshall McLuhan, who, 40 years ago, was dealing with audiences that were going through just as much change, and I think that, today, traditional Hollywood and the writers are framing this perhaps in the way that it was being framed before. But I don ' t need to tell you this, let ' s throw it back to him. Narrator : We are in the middle of a tremendous clash between the old and the new. MM : The medium does things to people and they are always completely unaware of this. They don ' t really notice the new medium that is wrapping them up. They think of the old medium, because the old medium is always the content of the new medium, as movies tend to be the content of TV, and as books used to be the content, novels used to be the content of movies. And so every time a new medium arrives, the old medium is the content, and it is highly observable, highly noticeable, but the real, real roughing up and massaging is done by the new medium, and it is ignored. PH : I think it ’ s a great time of enthrallment. There ’ s been more raw DNA of communications and media thrown out there. Content is moving from shows to particles that are batted back and forth, and part of social communications, and I think this is going to be a time of great renaissance and opportunity. And whereas television may have gotten beat up, what ’ s getting built is a really exciting new form of communication, and we kind of have the merger of the two industries and a new way of thinking to look at it. Thanks very much.

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As someone who has spent his entire career trying to be invisible standing in front of an audience is a cross between an out-of-body experience and a deer caught in the headlights, so please forgive me for violating one of the **TED** commandments by relying on words on paper, and I only hope I ’ m not struck by lightning bolts before I ’ m done. I ’ d like to begin by talking about some of the ideas that motivated me to become a documentary photographer. I was a student in the ’ 60s, a time of social upheaval and questioning, and on a personal level, an awakening sense of idealism. The war in Vietnam was raging ; the Civil Rights Movement was under way ; and pictures had a powerful influence on me. Our political and military **leaders** were telling us one thing, and photographers were telling us another. I believed the photographers, and so did millions of other Americans. Their images fueled resistance to the war and to racism. They not only recorded history ; they helped change the course of history. Their pictures became part of our **collective** consciousness and, as consciousness evolved into a shared sense of conscience, change became not only possible, but inevitable. I saw that the free flow of information represented by journalism, specifically visual journalism, can bring into focus both the benefits and the cost of political **policies**. It can give credit to sound decision-making, adding momentum to success. In the face of poor political judgment or political inaction, it becomes a kind of intervention, assessing the damage and asking us to reassess our behavior. It puts a human face on issues which from afar can appear abstract or ideological or monumental in their global **impact**. What happens at ground level, far from the halls of power, happens to ordinary citizens one by one. And I understood that documentary photography has the **ability** to interpret events from their point of view. It gives a voice to those who otherwise would not have a voice. And as a **reaction**, it stimulates public opinion and gives impetus to public debate, thereby preventing the **interested** parties from totally controlling the agenda, much as they would like to. Coming of age in those days made real the concept that the free flow of information is absolutely vital for a free and dynamic society to function properly. The press is certainly a business, and in order to survive it must be a successful business, but the right balance must be found between marketing considerations and journalistic responsibility. Society ’ s problems can ’ t be solved until they ’ re identified. On a higher plane, the press is a service industry, and the service it provides is awareness. Every story does not have to sell something. There ’ s also a time to give. That was a tradition I wanted to follow. Seeing the war created such incredibly high stakes for everyone **involved** and that visual journalism could actually become a factor in **conflict** resolution — I wanted to be a photographer in order to be a war photographer. But I was driven by an inherent sense that a picture that revealed the true face of war would almost by definition be an anti-war photograph. I ’ d like to take you on a visual journey through some of the events and issues I ’ ve been **involved** in over the past 25 years. In 1981, I went to Northern Ireland. 10 IRA prisoners were in the process of starving themselves to death in protest against conditions in jail. The **reaction** on the streets was violent confrontation. I saw that the front lines of contemporary wars are not on isolated battlefields, but right where people live. During the early ’ 80s, I spent a lot of time in Central America, which was engulfed by civil wars that straddled the ideological divide of the Cold War. In Guatemala, the central government — controlled by a oligarchy of European decent — was waging a scorched Earth campaign against an indigenous rebellion, and I saw an image that reflected the history of Latin America : conquest through a combination of the Bible and the sword. An anti-Sandinista guerrilla was mortally wounded as Commander Zero attacked a town in Southern Nicaragua. A destroyed tank belonging to Somoza ’ s national guard was left as a monument in a park in Managua, and was transformed by the energy and spirit of a child. At the same time, a civil war was taking place in El Salvador, and again, the civilian population was caught up in the **conflict**. I ’ ve been covering the Palestinian-Israeli **conflict** since 1981. This is a moment from the beginning of the second intifada, in 2000, when it was still stones and Molotovs against an army. In 2001, the uprising escalated into an armed **conflict**, and one of the major incidents was the destruction of the Palestinian **refugee** camp in the West Bank town of Jenin. Without the political will to find common ground, the continual friction of tactic and counter-tactic only creates suspicion and hatred and vengeance, and perpetuates the cycle of violence. In the ’ 90s, after the breakup of the Soviet Union, Yugoslavia fractured along ethnic fault lines, and civil war broke out between Bosnia, Croatia and Serbia. This is a scene of house-to-house fighting in Mostar, neighbor against neighbor. A bedroom, the place where people share intimacy, where life itself is conceived, became a battlefield. A mosque in northern Bosnia was destroyed by Serbian artillery and was used as a makeshift morgue. Dead Serbian soldiers were collected after a battle and used as barter for the return of prisoners or Bosnian soldiers killed in action. This was once a park. The Bosnian soldier who guided me told me that all of his friends were there now. At the same time in South Africa, after Nelson Mandela had been released from prison, the black population commenced the final phase of liberation from apartheid. One of the things I had to learn as a journalist was what to do with my anger. I had to use it, channel its energy, turn it into something that would clarify my vision, instead of clouding it. In Transkei, I witnessed a rite of passage into manhood, of the Xhosa tribe. Teenage boys lived in isolation, their bodies covered with white clay. After several weeks, they washed off the white and took on the full responsibilities of men. It was a very old ritual that seemed symbolic of the political struggle that was changing the face of South Africa. Children in Soweto playing on a trampoline. Elsewhere in Africa there was famine. In Somalia, the central government collapsed and clan warfare broke out. Farmers were driven off their land, and crops and livestock were destroyed or stolen. Starvation was being used as a weapon of mass destruction — primitive but extremely effective. Hundreds of thousands of people were exterminated, slowly and painfully. The **international** community responded with massive humanitarian relief, and hundreds of thousands of more lives were saved. American troops were sent to protect the relief shipments, but they were eventually drawn into the **conflict**, and after the tragic battle in Mogadishu, they were withdrawn. In southern Sudan, another civil war saw similar use of starvation as a means of genocide. Again, **international** NGOs, united under the umbrella of the U. N., staged a massive relief operation and thousands of lives were saved. I ’ m a witness, and I want my testimony to be honest and uncensored. I also want it to be powerful and eloquent, and to do as much justice as possible to the experience of the people I ’ m photographing. This man was in an NGO feeding center, being helped as much as he could be helped. He literally had nothing. He was a virtual skeleton, yet he could still summon the courage and the will to move. He had not given up, and if he didn ’ t give up, how could anyone in the outside world ever dream of losing hope? In 1994, after three months of covering the South African election, I saw the inauguration of Nelson Mandela, and it was the most uplifting thing I ’ ve ever seen. It exemplified the best that humanity has to offer. The next day I left for Rwanda, and it was like taking the express elevator to hell. This man had just been liberated from a Hutu death camp. He allowed me to photograph him for quite a long time, and he even turned his face toward the light, as if he wanted me to see him better. I think he knew what the scars on his face would say to the rest of the world. This time, maybe confused or discouraged by the military disaster in Somalia, the **international** community remained silent, and somewhere around 800, 000 people were slaughtered by their own countrymen — sometimes their own neighbors — using farm implements as weapons. Perhaps because a lesson had been learned by the weak response to the war in Bosnia and the failure in Rwanda, when Serbia attacked Kosovo, **international** action was taken much more decisively. NATO forces went in, and the Serbian army withdrew. Ethnic Albanians had been murdered, their farms destroyed and a huge number of people forcibly deported. They were received in **refugee** camps set up by NGOs in Albania and Macedonia. The imprint of a man who had been burned inside his own home. The image reminded me of a cave painting, and echoed how primitive we still are in so many ways. Between 1995 and ’ 96, I covered the first two wars in Chechnya from inside Grozny. This is a Chechen rebel on the front line against the Russian army. The Russians bombarded Grozny constantly for weeks, killing mainly the civilians who were still trapped inside. I found a boy from the local orphanage wandering around the front line. My work has evolved from being concerned mainly with war to a focus on critical social issues as well. After the fall of Ceausescu, I went to Romania and discovered a kind of gulag of children, where thousands of orphans were being kept in medieval conditions. Ceausescu had imposed a quota on the number of children to be produced by each family, thereby making women ’ s bodies an instrument of state economic **policy**. Children who couldn ’ t be **supported** by their families were raised in government orphanages. Children with birth defects were labeled incurables, and confined for life to inhuman conditions. As reports began to surface, again **international** aid went in. Going deeper into the legacy of the Eastern European regimes, I worked for several months on a story about the effects of industrial pollution, where there had been no regard for the environment or the health of either workers or the general population. An aluminum factory in Czechoslovakia was filled with carcinogenic smoke and dust, and four out of five workers came down with cancer. After the fall of Suharto in Indonesia, I began to explore conditions of poverty in a country that was on its way towards modernization. I spent a good deal of time with a man who lived with his family on a railway embankment and had lost an arm and a leg in a train accident. When the story was published, unsolicited donations poured in. A **trust** fund was established, and the family now lives in a house in the countryside and all their basic necessities are taken care of. It was a story that wasn ’ t trying to sell anything. Journalism had provided a channel for people ’ s natural sense of generosity, and the readers responded. I met a band of homeless children who ’ d come to Jakarta from the countryside, and ended up living in a train station. By the age of 12 or 14, they ’ d become beggars and drug addicts. The rural poor had become the urban poor, and in the process, they ’ d become invisible. These heroin addicts in detox in Pakistan reminded me of figures in a play by Beckett : isolated, waiting in the dark, but drawn to the light. Agent Orange was a defoliant used during the Vietnam War to deny cover to the Vietcong and the North Vietnamese army. The active ingredient was dioxin, an extremely toxic chemical that was sprayed in vast quantities, and whose effects passed through the genes to the next generation. In 2000, I began **documenting** global health issues, concentrating first on AIDS in Africa. I tried to tell the story through the work of caregivers. I thought it was important to emphasize that people were being helped, whether by **international** NGOs or by local grassroots organizations. So many children have been orphaned by the epidemic that grandmothers have taken the place of parents, and a lot of children had been born with HIV. A hospital in Zambia. I began **documenting** the close connection between HIV / AIDS and tuberculosis. This is an MSF hospital in Cambodia. My pictures can play a **supporting** role to the work of NGOs by shedding light on the critical social problems they ’ re trying to deal with. I went to Congo with MSF, and contributed to a book and an exhibition that focused attention on a forgotten war in which millions of people have died, and exposure to disease without treatment is used as a weapon. A malnourished child being measured as part of the supplemental feeding program. In the fall of 2004 I went to Darfur. This time I was on assignment for a magazine, but again worked closely with MSF. The **international** community still hasn ’ t found a way to create the pressure necessary to stop this genocide. An MSF hospital in a camp for displaced people. I ’ ve been working on a long project on crime and punishment in America. This is a scene from New Orleans. A prisoner on a chain gang in Alabama was punished by being handcuffed to a post in the midday sun. This experience raised a lot of questions, among them questions about race and equality and for whom in our country opportunities and options are available. In the yard of a chain gang in Alabama. I didn ’ t see either of the planes hit, and when I glanced out my window, I saw the first tower burning, and I thought it might have been an accident. A few minutes later when I looked again and saw the second tower burning, I knew we were at war. In the midst of the wreckage at Ground Zero, I had a realization. I ’ d been photographing in the Islamic world since 1981 — not only in the Middle East, but also in Africa, Asia and Europe. At the time I was photographing in these different places, I thought I was covering separate stories, but on 9 / 11 history crystallized, and I understood I ’ d actually been covering a single story for more than 20 years, and the attack on New York was its latest manifestation. The central commercial district of Kabul, Afghanistan at the end of the civil war, shortly before the city fell to the Taliban. Land mine victims being helped at the Red Cross rehab center being run by Alberto Cairo. A boy who lost a leg to a leftover mine. I ’ d witnessed immense suffering in the Islamic world from political oppression, civil war, foreign invasions, poverty, famine. I understood that in its suffering, the Islamic world had been crying out. Why weren ’ t we listening? A Taliban fighter shot during a battle as the Northern Alliance entered the city of Kunduz. When war with Iraq was imminent, I realized the American troops would be very well covered, so I decided to cover the invasion from inside Baghdad. A marketplace was hit by a mortar shell that killed several members of a single family. A day after American forces entered Baghdad, a company of Marines began rounding up bank robbers and were cheered on by the crowds — a hopeful moment that was short lived. For the first time in years, Shi ’ ites were allowed to make the pilgrimage to Karbala to observe Ashura, and I was amazed by the sheer number of people and how fervently they practiced their religion. A group of men march through the streets cutting themselves with knives. It was obvious that the Shi ' ites were a force to be reckoned with, and we would do well to understand them and learn how to deal with them. Last year I spent several months **documenting** our wounded troops, from the battlefield in Iraq all the way home. This is a helicopter medic giving CPR to a soldier who had been shot in the head. Military medicine has become so efficient that the percentage of troops who survive after being wounded is much higher in this war than in any other war in our history. The signature weapon of the war is the IED, and the signature wound is severe leg damage. After enduring extreme pain and trauma, the wounded face a grueling physical and psychological struggle in rehab. The spirit they displayed was absolutely remarkable. I tried to imagine myself in their place, and I was totally humbled by their courage and determination in the face of such catastrophic loss. Good people had been put in a very bad situation for questionable results. One day in rehab someone, started talking about surfing and all these **guys** who ' d never surfed before said,"Hey, let ' s go."And they went surfing. Photographers go to the extreme edges of human experience to show people what ' s going on. Sometimes they put their lives on the line, because they believe your opinions and your influence matter. They aim their pictures at your best instincts, generosity, a sense of right and wrong, the **ability** and the willingness to identify with others, the refusal to accept the unacceptable. My **TED** wish : there ' s a vital story that needs to be told, and I wish for **TED** to help me gain access to it and then to help me come up with innovative and exciting ways to use news photography in the digital era. Thank you very much.

Text Sample 520: POS Dim 4 – 2007 – Score 17.00 – 2007/t000352.txt

Welcome to Africa! Or rather, I should say, welcome home. Because this is where it all really began, isn ' t it? Looking at fossils dating back several millions of years â it all points to evidence that life for the human species as we know it began right here. We are on an amazing journey the next four days. You ' re going to hear stories of"Africa : The Next Chapter."Fantastic tales, anecdotes from speakers. But I want to turn that upside down for a moment, and get something out on the table and clear the **air** so to say. What ' s the worst thing you ' ve ever heard about Africa? And this is not a rhetorical question. I actually want answers from you. Go for it! The worst. Famine. Corruption. More. Genocide. AIDS. Slavery. That ' s enough. We ' ve all heard these things. But this is about Africa, the story we have not heard. The stories that we want to know, and the stories that do exist about positive tales. A part of my talk is going to be about investment opportunities that exist on this continent, to separate the rhetoric from the reality, the fact from the fiction. To go to the actual data and statistics that exist about the actual things that are happening on the ground that make Africa a realistic investment opportunity and option for you. So let ' s get going because Africa, to some degree, is on a turnaround. A turnaround in terms of how it manages its image, and how it takes control of its own destiny. And turnarounds are part and parcel of what I have focused on for most of my professional career. And it all started almost a decade ago, as a young consultant at McKinsey & Company at their first African office in Johannesburg. And there we worked with leading CEOs on African issues, and African companies on turnarounds, making the companies not just the best in Africa but the best globally. But I really formalized this focus on turnarounds when I was completing my MBA in the United States. It all began with a fantastic **phone** call. It was from Rosabeth Moss Kanter, Harvard Business School guru and a professor of mine. And she said,"I want to write a case, Euvin â a case on a public-sector **leader** that has lessons for the corporate world."And the **leader** that came to mind was Nelson Mandela. Because Nelson Mandela, as he took over power as the first democratically-elected president of South Africa, faced a situation of a country that could have slid into the abyss of chaos. But he started the country on a path of a positive cycle. Now the case,"Nelson Mandela : Change Leader,"became part of the research base for a chapter in Rosabeth ' s new book called"**Confidence**."And"Confidence"became a New York Times bestseller and topped Business Week ' s hardcover bestseller list. And why I tell you this story is because later, when I was interviewed on SABC Africa, on a pan-African broadcast, they asked,"What is your key lesson, or the key thing you enjoy the most?"â because it was a huge privilege to be part of such a project. The lesson from that was that it was Africa â an African story â that was used to share news with the rest of the world of what the benchmark can be for corporate turnarounds. Africa was being used as a success story! So I want to share with you a personal story about a turnaround or a transformation. And that has to do with me because in 1994, I packed a few things into a backpack and headed off for a year of travel in the middle of my university career. You should have seen my parents ' **reaction**! But very soon, I found myself from the southern part of Africa, in South Africa â at the very north, in Egypt. And I sought out the most remote places. I went to the Siwa Oasis. That was one of my stops. And the Siwa Oasis is famous for several things, but the key thing is that it was the place that Alexander the Great went to when he wanted to find out what his destiny had in store for him. And legend has it that Alexander trekked through this desert. Half his battalion was wiped out in the sandstorm. And myth says that he had an audience with the oracle, and it foretold his destiny of greatness. This was 300 BC. So Africa had long been seen as a place to go to for answers. Now, the thing I remember about Siwa was the magical view of the sky at night. With no natural light source, Siva is one of these amazing places that when you look up you see a perfect tapestry. Fast forward to 2002. I ' m sitting in Cambridge, Massachusetts at the Healthcare Development Conference. And I see the same picture, but from the opposite side. A satellite picture looking down at the earth. And it was that picture that made such a profound **impact** on me because I ' ll never forget it. I remember the very moment. And I wanted to share that image with you of what I saw at that point. The first thing that I saw was North America at night â glowing, in all its glory. A warm feeling. Light. And then I saw it â Africa. Quite literally the"Dark Continent."And while Africa may be dark, the thing that brought the message home to me was that this is the challenge we are facing, but it ' s also the opportunity. Because whilst Africa may be dark â other than the few specks that exist north and in the south and other areas â it ' s aglow with the light in the hearts of the millions of people that are there. Entrepreneurs, dynamic people, people with hope. It was George Kimble, the geographer, who said that,"The only thing dark about Africa is our ignorance of it."So let ' s start shedding light on this amazing eclectic continent that has so much to offer. Let ' s start unpacking it. Africa is the second-largest continent, a landmass second from Asia. It also is the second most populated continent, with 900 million people. In fact â coming back to the land mass â Africa is so big that you could fit in the continental United States, China, and the entire Europe into Africa, and still have space. Africa is home to over 1, 000 languages â 2, 000 is another estimate that ' s out there â with over 2, 000 languages and dialects. But you could say,"**Invest** in Africa in over 1, 000 languages, and it wouldn ' t make a difference."What does the data say? As an investment banker, I ' m in the cross-flow of information and the changes that are taking place in capital markets. So I want to share with you some of these bellwether signals, or signs, and winds of change that are sweeping this continent. So let ’ s start on that. And let ’ s start at the high level, on the macro- factors. **Inflation**, in general, is coming down across Africa â that ’ s the first sign â in many countries reaching double-digit figures. So let ’ s start looking at some of those. I call it my Z. E. N. cluster. Zambia : from 2004 to 2006, moves from the 18 percent in **inflation** to the nine percent. Egypt : from the 16 percent to about 8. 4 percent. Nigeria : a similar situation, from the 16 percent to the eight percent. Single digits. More fascinating, you have other countries â South Africa, Mauritius, Namibia â all in single digits. But that ’ s just part of the story. You have a similar trend with currencies â currencies going through an extreme time of stability. But that ’ s looking at the big picture. And the first myth to dispel is that Africa is not a country. It ’ s made up â It ’ s made up of 53 different countries. So the very definition â to say”**invest** in Africa”is a no-go. It ’ s meaningless. Each country has a unique value proposition. You can make money, you can lose money in Africa. But opportunities, boy oh boy, they exist. And this is what today is about â it ’ s about discussing those very opportunities. So let ’ s start getting into the countries and into the specific material and data. I was recently elected, as Emeka mentioned, as the President of the South African Chamber of Commerce in America. And I ’ m very proud and happy to be in that role because it is a fascinating position to be in. To hear this dialogue that ’ s just increasing in tenor and velocity, of decisions about trade and companies wanting to come. So the first port of call : let ’ s talk a little bit about South Africa. But not the South Africa we always talk about â the gold, the minerals, the First World infrastructure â a bit about the other side of it. For example, South Africa was recently voted as the top destination for the top 1, 000 UK companies for offshore call-centers. Same language, timeline, et cetera. Makes sense. Other headlines that have recently reached South Africa were Bain Capital and KKR, the big boys of private equity. Headline in South Africa :”They have landed.”Quite ominous. But what were they there for? To acquire assets. Bing Capital ’ s acquisition of Edcon, a large retailer, is testimony to the **confidence** they are starting to place in the economy. Because it is actually a long-term play. Being a retailer, it is a play on the belief that this middle-class that ’ s growing will continue to grow, that the boom and the **confidence** in consumer spending will continue. But the story of Africa, and my focus, is beyond South Africa because there ’ s so much happening. Undoubtedly, Nigeria is clearly a hot spot. Challenges â and we will hear a lot about Nigeria in these four days. But looking at Goldman Sachs ’ work â we had the famous BRIC Report. The new report,”The Next Eleven,”highlights that by 2020 Nigeria is going to be amongst the top 10 economies in the world. It ’ s an investment opportunity. Think about that. Is anyone â our banks, our investors â seriously thinking about going to Nigeria? If you haven ’ t, why not? What ’ s going on in Nigeria? A couple of things. I want to talk about it from the perspective of capital markets. Bellwether signs again. Guarantee Trust Bank recently issued the first Euro Bond out of Africa, and this excludes South Africa. But the first Eurobond, the raising of **international** capital offshore, off its own balance sheet, without any sovereign backing â that is an indication of the **confidence** that is taking place in that economy. Without any sovereign backing, a Nigerian company raising capital offshore. It ’ s just a sign of things to come. Looking at the oil industry, Africa provides 18 percent of the U. S. ’ s oil supply, with the Middle East just 16 percent. It ’ s an important strategic partner. Let ’ s put Nigeria in perspective. 2. 2 to 2. 4 million barrels of oil a day â the same league as Kuwait, the same league as Venezuela. But with Africa, let ’ s start being careful about this. And Emeka and I have had these discussions. We have to move away from what ’ s called”the curse of the commodities.”Because it ’ s not about oil, it ’ s not about commodities. For Africa to truly be sustainable, we have to move beyond to other industries. So let ’ s unpack those very quickly, and I ’ m going to move through these very, very, very fast because I can see that clock counting down. What else is going on there? Egypt. Egypt is launching a first large industrial zone â 2. 8 billion investment. The announcement just came out the last few weeks. Close to the Mediterranean, near Alexandria â textiles, petrochemicals. It ’ s being managed by a Singaporean-based management company. So they want to emerge as an industrial powerhouse across the industries â away from oil. Let ’ s look at agriculture. Let ’ s look at forestry. What ’ s going on there? In Tanzania last week, we had the launch of the East African Organic Produce Standard. Again, gathering together farmers, gathering together stakeholders in East Africa to get standards for organic produce. Better prices. It ties in with small-scale farmers in terms of no pesticides, no fertilizers. Again, opportunity to tackle markets to get that higher price. Uganda : the New Forest Company, replanting and redeveloping their forests. Why is that important? As the energy needs are met and electricity is needed [we will need] poles for rolling out electricity. But here is the sweetener in the deal. They ’ re going to be tapping into carbon credits. Let ’ s go back to Nigeria. The banking **sector** has undergone tremendous transformation, from over 80 banks to 25 banks. Strengthening of the system. But what ’ s going on there? Only 10 percent of the country is banked. The largest population in Africa is in Nigeria. 135 million- plus people. Think about that. There are only 700 ATMs in the country. Opportunity. The same for telecoms across the country. Now let ’ s look at the continent as a whole. People look at the roads, for example, and they ’ d say,”Angola : 90 percent of roads are untarred. Ah, problem!”It ’ s more expensive to transport goods. Prices of goods go up, **inflation** is affected. Nigeria : 70 percent of roads are untarred. Zambia : 80 percent. In general, more than 50 percent of roads are untarred. This is an opportunity! Energy needs â it ’ s an opportunity. So what are the signs that things are fundamentally changing? Let ’ s look at the stock markets in Africa. If I had to ask you,”In 2005 what was the best performing stock market or stock exchange in the world?”Would Egypt come to mind? In 2005, the Egyptian stock exchange returned over 145 percent. What ’ s going on in some of the other countries? Let ’ s look at some 2006 numbers. Kenya : over 60 percent. Nigeria : over 40 percent. South Africa : in the 20 percents. High ones. These are the trends that are taking place. But in any investment decision, the key question is,”What is my alternative investment?”Because in Africa today, we are competing globally for capital. And global capital is agnostic â it has no loyalties. There ’ s an overhang of capital in the U. S., and the key is yield pickup. What Africa is providing is a diversification play, and also opportunities for yield pickup for the investor that ’ s aware of what he or she is doing. Now, when looking at Africa vis-a-vis other things, and countries in Africa vis-a-vis other things, comparisons become important. 10 years ago there, were very few countries that received sovereign ratings from the Standard & Poors, Moody ’ s and Fitch ’ s. Today, 16 African countries and growing have sovereign country ratings. What does this mean? Take Nigeria again : double B-minus â in the league of Ukraine and Turkey. Immediately we have a comparison. The backbone of making investment decisions for global holders of capital. Some other figures. South Africa : triple B-plus. Botswana : A-plus. Bakino Faso : B-minus. And so on. In fact, one of the big agencies is setting up an office in Africa. Why are they doing that? Because they expect investment to follow. So one of the big bellwethers, and one of my final points I want to mention, is the **interesting** thing I read is that CNBC has launched their first African channel. Why is CNBC doing this? It ’ s the 24-hour rolling African news channel. They ’ re doing it because they are expecting things to happen. Me and you, the investments we are going to be making, the investments the world is going to be making â that ’ s the 24-hour news channel dedicated to Africa. So that ’ s the change that ’ s coming down the pipeline. So in conclusion, I want to turn back to that very slide that made such a deep **impact** on me all those years ago. This time [I ’ ll] give you the entire picture that I saw in 2002, and ask you that when you think about what your role can be in Africa, think about your journey in terms of bringing light to this continent. Because there are amazing opportunities available. And think about the concept of transformation in the back of your mind because things can be turned around rather quickly. In 1899, Joseph Conrad released”The Heart of Darkness,”a tale of grim horror along the Congo River. If one looks carefully, on the Congo River is one of those bright lights. And that ’ s the very Congo river generating light â the old heart of darkness now generating light with hydro-electric power. That is a transformation in power of ideas. So the next step, over the next four days, is us exploring more of these ideas. And perchance, if you can always keep this picture in your mind, that when we convene maybe in the distant future, in 2020, that picture will look very different. Thank you.

Text Sample 521: POS Dim 4 – 2020 – Score 33.00 – 2020/t004268.txt

Chris Anderson : I get now to introduce one of the most powerful women in the world. I mean, if we are to escape from the mess that we ’ re in right now, she is going to play a major part in helping us do that. She ’ s the head of the International Monetary Fund, a delight to welcome here Kristalina Georgieva. Kristalina, welcome. Kristalina Georgieva : Great to be with you, Chris. Thank you for having me. CA : So you just took on this role late last year, and within four months, boom, COVID arrives. That is one heck of an introduction to a new job. How are you doing? KG : Well, I find strength in action. And at the Fund, we have been, from day one on this **crisis**, leaning forward with everything we have to provide lifelines to countries, and that means to people and businesses. Already, we have received over 90 requests and we have offered, to 56 countries, critical financial packages. CA : You ’ ve described this **pandemic** as a **crisis** like no other. In what way a **crisis** like no other? KG : Truly like no other. First, never before we will inflict on the economy consciously so much pain to fight a **virus** and save lives. We are asking businesses not to produce and consumers not to go out and consume. At the Fund, we labeled this”the Great Lockdown.”Second, never before there would be such a rapid change of fortunes practically for **everybody** around the world. In January, I was in Davos, talking about”anemic growth,”growth of three percent. In April, during our spring meetings, it was already minus three percent. In January, we predicted 160 countries to have positive income per capita growth. Now it is 170 countries with negative income per capita growth. Now this, we call”the Great Reversal.”Very painful. And three, **uncertainty**. We always live with **uncertainty**, Chris, but this time, it is the **uncertainty** of a novel coronavirus that policymakers have to integrate. We at the Fund combine epidemiological projections with our traditional macroeconomic modeling to see through that **uncertainty**. I must add to this, I very much hope that when we go on the other side in the recovery, we can use a new term and call it”the Great Transformation.”Make the world a better place. CA : Well, I ’ ll be excited to come on to that in a bit. But in this moment of responding to the **crisis**, the main tool that seems to have been executed, at least by the rich countries, has been this massive economic stimulus, to the tune of trillions of dollars. Is that a wise response? KG : It is a necessity. And you don ’ t hear the Fund often telling countries,”Please, spend. Spend as much as you can.”And that is what we do now. We do add to that,”And keep the receipts. Don ’ t lose accountability to the citizens, to the tax payers.”The reason financial injection is necessary, these fiscal measures of almost nine trillion dollars are necessary, is because when the economy is standing still, unless there is help, unless there is monetary **policy** stimulus, firms are going to go massively bankrupt, people would be unemployed, the economy would be scarred. When we go to the other side, this scarring is going to make the recovery much more difficult. So that is a wise thing to do, and it helps the fact that central banks in major economies have been acting in a synchronized manner and that fiscal stimulus came really, really fast. This is how we see people being able to go through this very, very tough time. CA : But how far can it go? Because it ’ s been described, in a sense, as”printing money”– governments are issuing more and more bonds that have to paid back at some point. There ’ s this term, in economics, of the Minsky moment, where things can go very well for a while, as everyone believes that, you know, that the train can keep running, the cycle can keep turning, you know, that governments have all this money. At some point, though, doesn ’ t that break down? Do you worry that we may be nearing a Minsky moment, where, like Michael in Mary Poppins grabs his tuppence and starts a run on the bank. Is there stress in the **international** financial system now that concerns you, that makes you feel that we may be running out of headroom? KG : Of course, this cannot go on forever. I, for one, have **trust** in our scientists, I think we will see breakthroughs, and we will see also people in businesses getting accustomed to social distancing, to micromeasures that protect from spreading the disease. We have seen very massive injection in health systems, so hospitals can actually treat people that are coming for help. Obviously, if it is to go for a very long time, we would be worried. For now, what we are projecting is that there would be a gradual reopening – we see it already happening in a number of countries. And we project for next year, 2021, a partial recovery. Not a full recovery, unfortunately, but coming to a better place. Now, what helps us is something that I don ’ t particularly love, but I see it as a positive feature – very low interest rates, in some cases, negative – that allows this injection of fiscal measures and liquidity to be sustained over a number of years. And for now, we do not see on the horizon any return to increase in interest rates. So low for longer, and that is, in that environment, a helpful feature. CA : I mean, the financial **crisis** of 2008 came perilously close to breaking the entire financial system – arguably, it did that. By most people ’ s calculation, this is a far worse **impact** to the economy overall. Did the world learn something from 2008 that has helped us so far be resilient this time? KG : What the world learned is that the financial system has to be tested and then strengthened to withstand shocks. And that is helping us tremendously today. The banking system is resilient, and even in the nonbanking financial institutions, there is more attention paid to how far can you go without running into trouble. I would say, if you look around the world, the most important lesson then was”build resilience to shocks.”Those who have done it cope now better. And those who have not done it are in a much tougher spot. And actually, for the Fund, what we are praying is that we will come out of this **crisis** with this lesson about resilience being spread beyond the banking system, so we actually have this crisis-management mindset for a world that is inevitably going to be more shock-prone, because of **climate** and also because of the sheer density of economic and social life on our planet. CA : In your role, you ’ re paying special attention to the situation in developing countries. And it does seem that they ’ re facing a really terrible situation right now. Many of them have significant debt denominated in dollars. In the current **crisis**, their currencies are depreciating against the dollar, making it nigh impossible for them to execute the kind of injection, stimulus injections, that the rich countries are doing and seems to be the only way out. So that seems like a really dangerous cycle. Is there any way to break that cycle? KG : Well, let me first separate countries that have built strong fundamentals. And now in this **crisis**, as we are receiving incoming data, not very many, but there are still some positive surprises, and they come from countries that have built stronger buffers, stronger fundamentals, have been more disciplined during good times. But indeed, we do see quite a number of emerging markets, developing countries, faced with multiple pressures. They had the hit from the coronavirus, many of them with weak health systems. Then, they have the high level of indebtedness, from before the **crisis**, which creates a much more difficult environment for them. Then, many of them are commodity exporters. Commodity prices, oil price, they went down very dramatically, that hits them again. Many rely on remittances. Remittances shrunk some 20 to 30 percent. And then you have a number of countries that are highly dependent on tourism. Tourism is the hardest hit **sector**, or one of the hardest hit. So, very tough for these countries, but this is why institutions like mine have been wisely created. The IMF, the World Bank, the regional development banks, we work very closely together in this **crisis**. The IMF, fortunately, that was one of the lessons from the 2008-2009 **crisis** – make sure that in the center of the financial safety net is an IMF with financial strength. We have four times more money to lend today than we had then. From 250 billion to one trillion dollars. And of course, we are deploying these funds exactly for the countries that need us the most. And we did one more thing. With David Malpass, the president of the World Bank, we called for a debt moratorium for the poorest countries to their official bilateral creditors. And people tend to say,”Oh, we don ’ t work together, it ’ s not good enough.”But here is an area where we made this call in late March, and in mid-April, the G20 agreed on this moratorium. Amazing, we had the Paris Club, China, the Gulf countries, all agreeing that we should not suffocate the poorest countries by asking them to pay their debts when their economies are standing still. CA : Is it possible that some developing countries are overdoing the lockdown **policy**? I mean, if large numbers of your citizens are already struggling to stay alive, isn ’ t it almost like a death sentence to order them not to leave their homes? KG : Well, Chris, one of the most heartbreaking conversations I would have is with **leaders** of countries where they have to stare in the face a choice of people dying from the **virus** or dying from hunger. And it is a very dramatic situation for them. Where you have a very large part of your economy being informal, where people live hand-to-mouth every day, the lockdowns we have in advanced economies are not quite applicable, but even there, countries are doing really well in social distancing to the extent it is possible. Many of the countries in Africa were very early to step up preventive measures. Why? They learned from the Ebola, they learned from prior **crises** that hygiene, taking any measure you can really helps. So again, I cannot stress enough how important is solidarity with these countries. How important it is for my institution to be there for them in a timely manner. And we do it. CA : Whitney. Whitney Pennington Rogers : Hi there, thank you, this is a wonderful conversation, and we ’ re starting to see some questions coming from the community. The first one we have is from Bill Elkus, and it ’ s a follow-up to something you were mentioning earlier, related to the stimulus, Kristalina. What are the prospects for **inflation** from such a large stimulus? KG : At this point, we are not worried about **inflation** in advanced economies and in the majority of emerging market economies. We do worry about **inflation** in countries that have weak fundamentals, no access to foreign exchange easily, where the only way to address the **crisis** is our help or their central banks printing more money. And sometimes it ’ s a combination of those two. Why I don ’ t worry about **inflation** in advanced economies? Because countries that have their hard currency are putting liquidity in place, but at the same time, they ’ re not seeing a big expansion of **demand** and prices being pushed up. So for these countries, at least for the observable future, we don ’ t see a way of going, like after the Second World War, in **inflation** jumping up. The consumers are not consuming so aggressively, **demand** is not that strong, and these are societies where there is a lot of maturity in how they exercise their **policy** options. But if you are a poor country, that out of desperation, with no access to markets, no access to hard currency, ought to somehow put money supply enough, then **inflation** is going to be there. A very extreme case is Zimbabwe, and I do worry there may be other countries. So this is why we are so determined to engage with these countries early. And also look at some of the high-debt countries. Would it be necessary, on a country-by-country basis, to restructure debts to prevent that moving in a desperate direction? WPR : Thank you. And we have one more question that I wanted to share from our community. This is from Keith Yamashita, and it ’ s about how we all can be **involved** in some of this change.”You are tasked with macro-economic and funding efforts. What should we do as citizens to help renewal and recovery?”KG : Well, it is incredibly important for all of us citizens – and aside of being the head of the IMF, I am also a global citizen – that we are to bring that notion of solidarity in a moment of **crisis**. I loved the way this segment was musically backed, and it was”Lean on Me.”It is very important that we do create that sense –”we are in this together, we will get through it together.”And please, speak up on that. I was, for many years, **crisis** commissioner, and one thing I learned is that the majority of people are positive, good people. You can lean on them. And there is a minority that is hateful and fearful and also very loud. So, good people, speak up. Spread that sense of”we are in this together, we ’ ll get through it together.”WPR : Thank you. I ’ ll come back later with other questions. CA : Kristalina, I ’ d love to expand on that and just ask you a bit more about **leadership**, actually. You know, when people think of the nations that have performed best, they often refer to – when I say best, best in response to the current **pandemic** – they often refer to Germany, New Zealand, South Korea, Taiwan, Denmark and Norway. When they think of those that have performed worst, they often think of Spain, Italy, the UK, Belgium, Sweden, Iran, Brazil, Russia and the United States. All of that second group are run by men, all but one of the first group are run by women. Is that a coincidence? KG : Well, now, speaking a bit subjectively as a woman, I do believe that women are great to lead in a **crisis**. They are more likely to show empathy, to care about the most vulnerable people and to be able to speak about that. They are decisive. I can say that for myself, we take energy from action. And we don ’ t tend to, kind of, mourn and complain too much. So there is perhaps something to be said about the value of **gender** equality for the future. Bring more women for this world of more **crisis** ahead of us. CA : It ’ s obviously hard to make generalizations about **gender** of any kind, but I mean, is there also, almost, something about the embracing of nuance, that women might be better at that than men? Men are often, it ’ s like,”let ’ s win, let ’ s conquer,”and in a situation like this, where it ’ s all probabilities, it ’ s like, there are so many complex dials to turn on this dangerous **pandemic** machine that we ’ re trying to wrestle. I mean, are women better at handling nuance? KG : Let me say something, Chris. We need **everybody**, and we need this mixture of experience, knowledge and predisposition. Men and women coming together. I find it that it is great to have different perspectives when we make decisions. Then, the chances of making a good decision are higher. So we need each other, but we also need to recognize is that yes, there are certain things, I have seen it time and again, women are more willing to find a pathway to compromise, they ' re more willing to be corrected if they ' re wrong. Say,"Oh, OK, that ' s a good point, let me integrate it in the way I think about it."And when you are in **uncertainty**, that is a huge advantage in decision-making. CA : So perhaps talk a bit more about your own **leadership** in this moment. I mentioned you ' ve only recently come to this job. Before that, you were European Commissioner, you dealt with humanitarian **crises** in more than one part of the world. And in your own country, Bulgaria, you witnessed the wholesale transformation of the country, both politically and economically. What lessons can you bring from your past experience to this moment? KG : Well, there are many things I learned. I was very fortunate to have these multiple experiences for the job I have now. But let me highlight three. First, how critically important it is to prepare for a **crisis**. Kind of, think of the unthinkable, and then act with some foresight when a shock hits you. You have a title for this series called"Build Back Better."I actually would like to modify it, if I may, and I would talk about"Build Better Before."Preparedness, prevention, pay off big time. The second – and not necessarily in priority, it is as important – is **collective** action, working together. Seeking help, offering help. Makes a huge difference in an emergency. And the third is something I learned time and again. We don ' t know our internal strength until we are hit. We are so resilient, we are so able to withstand shocks, especially when we come together, that this always gives me this sense of optimism that, as hard as it is, we can overcome it. From the days when my country collapsed, the economy collapsed, I would get up at four o ' clock in the morning, queue to buy milk for my daughter, to the days when I would see Syrian **refugees** in terrible situations helping each other, to today, when I ' m the head of the IMF, that internal strength, our power of resilience, the more we are together, the more it is amplified. CA : Actually, could you talk a bit more about the role of the IMF, especially as we look forward to trying to recover from this? What specifically can your organization do to take us forward? KG : So there are three things that are quite unique for the IMF, and they ' re really so important in a time of **crisis**. The first one is to give a good diagnostic of what is happening and what is the way forward. Let me just say, in this **crisis**, in the very first weeks, we put together, we call it **policy** action tracker, for 193 countries. What actions are countries taking, how they can learn from each other, so we can be more effective together. We are adding to it, now, actions for responsible reopening of the economies exactly with that purpose. What we are known best for, we are the financial first responder. We are coming in this incredible shock with very significant financial firepower. And what people don ' t know is that the Fund has multiple instruments. Emergency financing is the one we doubled for this **crisis**. And it is no conditionalities. We are asking one thing, Chris. Pay your doctors and your nurses, your hospitals, protect your most vulnerable people and parts of the economy. That ' s it, this is the condition. And the third thing we do at the Fund is to help countries have the capacity for good **policies**. After the financial **crisis**, we helped many countries to have good debt management, good fiscal management, transparency and accountability to improve the performance of public finance. So the Fund is not a very big organization by any standard, we are some 3, 000 people. Highly professional, incredibly committed. When you use the expression"all hands on deck,"that ' s us. And it is a digital deck, it is a digital deck these days. CA : I mean, this is a global **crisis**. A lot of people are worried that unlike perhaps even in 2008, where it really did seem there was a lot of global cooperation, there ' s actually, in some worrying ways, less this time? Are you worried about how crucial is that to getting us through this? KG : I mean, my preoccupation is, in our mandate, in my area of responsibility, bring the **membership** together. We have almost the whole world, 189 countries are our members, and so far, I am very impressed by how responsive the **membership** has been. I put in front of them in the spring a package, very strong package of measures to expand the role of the IMF in the **crisis**. Everything that we ask for – we ask for doubling emergency financing, we got it. Very **interesting**. We ask for tripling concession of financing. Exactly because, you know, like the **virus** hits people with a weak system the hardest, the **crisis** hits weak economies the hardest. So we wanted to triple concession of financing. Within one month, we got it. We asked for grants for debt relief, we got it. So what I ' m trying to say here is that we need to focus on ways in which we bring the world together. And then act on that. Rather than complaining that maybe not everything is the way it should be, do your duty to the global community. CA : Well, indeed. And the IMF is dependent on the financing from its members, its key members. KG : Yes. CA : I mean, you spoke of the trillion dollars that you are looking to make available to nations that need it. As I read it, that comes from – you ' ve got these units called Special Drawing Rights. You basically draw a currency from members. And hasn ' t there been pushback, though, from the US, to block that effort of raising all that money? KG : So the one trillion dollars is from our quotas and also from our **ability** to move money from well-to-do members from the advanced economies and lend it at very low or zero interest to the developing emerging markets. And we had this one trillion and what was very **interesting**, not **everybody** noticed that – the US, in their two trillion dollars stimulus package, included the **support** for the IMF. The Special Drawing Rights is something that we, indeed, don ' t have yet consensus among the **membership** to do. It was done during the 2009 **crisis**, issuing liquidity, and it goes to **everybody**. And there are many voices, including mine – I spoke to the G20 about that – that are saying, well, that may be a good thing to do now. It is not being **supported** for reasons. It is not just capriciously. The problem with Special Drawing Rights is that when we issue them, they go to all members, and the advanced economies get 62 percent of the new allocation, and there are some that are saying,"Can we think of something that is more directed, or exclusively directed to those who need it?"But, Chris, everything is on the table for us. As the **crisis** unfolds, we need to do more, we bring the **membership** to do more. CA : Whitney. WPR : We actually have a question from the community that builds on what you ' re discussing right now. Yavnika Khanna asks,"Which countries will prove to be resilient in the Great Transformation : those with popular **leaders** or those with sound financial systems?"KG : You know, they both matter. Countries with strong fundamentals are clearly going through this **crisis** with less trauma than those that had weak fundamentals to begin with. And of course, **leadership** matters. How you mobilize a country for action matters. In my view, what we would see on the other side, the winners would be those who think today of this **crisis** also as an opportunity. Clearly, digital transformation is a huge opportunity. Moving to e-learning, e-government, e-payments, e-commerce, linking small and medium-sized enterprises through digital to consumers, big winner. Secondly, I very much hope that we would come on the other side with a low carbon footprint and a more climate-resilient economy. Those who move in this direction, they would reduce the risk for themselves and the world. From this other **crisis**, that we are not talking so much about these days, but it hasn ' t gone anywhere. And you know, if you don ' t like **pandemic**, you are not going to like the **climate crisis** at all. And also, countries that are thinking of how to make the economy in the future a fairer economy. In other words, we have been seeing inequality building up before this **crisis**. My colleagues who have researched **pandemics** have a very bitter lesson for us. After **pandemics**, after H1N1, after SARS, after Zika, inequality goes up. Well, are we going to let inequality to go up, up, after this **crisis**? And if we do, we are damaging the fabric of our societies, and my sense is that hundreds of millions of people in this **crisis** would much prefer to have a simpler, fairer, more equitable world to live in, and definitely, a more sustainable world. CA : Mm. KG : Those would be the winners. WPR : Definitely. And just one more question from our community, before turning it back to Chris for some final questions here. You know, this one is from Sarah Rugheimer. And the question is,"What do you see as the main potential positive shifts / changes in this world from this **pandemic**, say, two to 10 years from now?"KG : Well, I touched upon it a little bit. First, I hope to see fiscal **policy** to help us recover to be geared towards green recovery and more equitable recovery. And that is something that is in the hands of policy-makers. It can be done. Secondly, I very much hope to see us integrating what we have learned from the **crisis**, in terms of virtual work. My organization, the IMF, well, we can shrink our carbon footprint dramatically just by sustaining the practices we are developing now, and we will. I certainly hope to see, in the future, much more attention to two things that we saw in this **crisis** are essential. Universal access to health in some form, strong health systems, as well as strong social safety nets, built as automatic stabilizers in a time of shock. And by the way, it is cheaper if we do it in this way. The bill for everyone is going to be smaller. And also, I very much hope that this notion of **investing** in people, recognizing that now that we see this horrible tragedy, the loss of lives, that investing in people is the very best investment we can make. CA : Mm. WPR : That ' s great. CA : So, see you again in a minute, Whitney. Kristalina, it ' s so – It ' s so inspiring, actually, hearing the energy and **stuff**, the energy that you ' re bringing to this. I don ' t think many people coming into this would have expected to hear, from the head of the IMF, this emphasis on, you know,"Let ' s solve the **climate crisis**, let ' s tackle inequality and injustice."Do you really believe that this moment, this **crisis** could help lead us into a great transformation? People will feel it ' s your job to sound positive, you have to do that. Do you really see the path forward that we can get through this, and what sort of timescale are we talking about here, Kristalina? KG : Well, you know, one thing I learned from the **transition** I lived through, the **transition** from central planning to markets, is it is tough, it is long, it is painful and it is a road that takes turns. So I don ' t have an expectation of miracle from here to there. But I genuinely believe that we are now in a point of our history when people **demand** from their **leaders** safety and security and a society that is not torn apart by **conflicts**. And that is actually not unusual to see. So, I would turn the table a little bit on you, Chris. After a war, we see the world coming together and building a better world. Why not after a **pandemic**? And yes, we can make mistakes and not take the right road to travel. But we certainly have an obligation to try to get on that road. CA : So if you could just inject – KG : And **everybody** matters for that. CA : So if you could inject one idea into the mind of **everybody**, or into to the world **leaders** who listen to you, what would that idea be at this moment? KG : Optimism. Build a better world. Possible, desirable, we must do it. CA : That sounds like optimism as the stance, not just a naive belief that it will happen, but a determination to make it so. That ' s what you ' re calling for. To use that as the motivation to pull us all forward together. KG : Chris, do I have one minute, or I ' m done, I need to go? CA : If you want to say one last thing in one minute, alright, go. KG : I want to say one thing. To recommend to the audience to watch the movie"Bridge of Spies."There is a part in the movie in which the two main actors, the lawyer and the Russian spy, talk to each other. The lawyer says,"Things are very bad, it looks like you may hang."The spy is very calm. Lawyer says,"Aren ' t you worried?"The spy answers,"Would it help?"So my message is, it is tough, but worries won ' t help. Positive action will. Positive, stay positive, so that ' s my message. CA : Well, I have to say thank you. It ' s incredibly inspiring, actually, to see your energy and your determined optimism, let ' s call it that. I think we wish you the very best as you use your position to help get us out of this mess. Thank you so much, Kristalina, for spending time here at **TED**. Thank you. WPR : Thank you, Kristalina.

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Chris Anderson : Al, welcome. So look, just six months ago – it seems a lifetime ago, but it really was just six months ago – **climate** seemed to be on the lips of every thinking person on the planet. Recent events seem to have swept it all away from our attention. How worried are you about that? Al Gore : Well, first of all Chris, thank you so much for inviting me to have this conversation. People are reacting differently to the **climate crisis** in the midst of these other great challenges that have taken over our awareness, appropriately. One reason is something that you mentioned. People get the fact that when scientists are warning us in ever more dire terms and setting their hair on fire, so to speak, it ' s best to listen to what they ' re saying, and I think that lesson has begun to sink in in a new way. Another similarity, by the way, is that the **climate crisis**, like the COVID-19 **pandemic**, has revealed in a new way the shocking injustices and inequalities and disparities that affect communities of color and low-income communities. There are differences. The **climate crisis** has effects that are not measured in years, as the **pandemic** is, but consequences that are measured in centuries and even longer. And the other difference is that instead of depressing economic activity to deal with the **climate crisis**, as nations around the world have had to do with COVID-19, we have the opportunity to create tens of millions of new jobs. That sounds like a political phrasing, but it ' s literally true. For the last five years, the fastest-growing job in the US has been solar installer. The second-fastest has been wind turbine technician. And the"Oxford Review of Economics,"just a few weeks ago, pointed the way to a very jobs-rich recovery if we emphasize renewable energy and sustainability technology. So I think we are crossing a tipping point, and you need only look at the recovery plans that are being presented in nations around the world to see that they ' re very much focused on a green recovery. CA : I mean, one obvious **impact** of the **pandemic** is that it ' s brought the world ' s economy to a shuddering halt, thereby reducing greenhouse gas emissions. I mean, how big an effect has that been, and is it unambiguously good news? AG : Well, it ' s a little bit of an illusion, Chris, and you need only look back to the Great **Recession** in 2008 and ' 09, when there was a one percent decline in emissions, but then in 2010, they came roaring back during the recovery with a four percent increase. The latest estimates are that emissions will go down by at least five percent during this induced coma, as the economist Paul Krugman perceptively described it, but whether it goes back the way it did after the Great **Recession** is in part up to us, and if these green recovery plans are actually implemented, and I know many countries are determined to implement them, then we need not repeat that pattern. After all, this whole process is occurring during a period when the cost of renewable energy and electric vehicles, batteries and a range of other sustainability approaches are continuing to fall in price, and they ' re becoming much more competitive. Just a quick reference to how fast this is : five years ago, electricity from solar and wind was cheaper than electricity from fossil fuels in only one percent of the world. This year, it ’ s cheaper in two-thirds of the world, and five years from now, it will be cheaper in virtually 100 percent of the world. EVs will be cost-competitive within two years, and then will continue falling in price. And so there are changes underway that could interrupt the pattern we saw after the Great **Recession**. CA : The reason those pricing differentials happen in different parts of the world is obviously because there ’ s different amounts of sunshine and wind there and different building costs and so forth. AG : Well, yes, and government **policies** also account for a lot. The world is continuing to subsidize fossil fuels at a ridiculous amount, more so in many developing countries than in the US and developed countries, but it ’ s subsidized here as well. But everywhere in the world, wind and solar will be cheaper as a source of electricity than fossil fuels, within a few years. CA : I think I ’ ve heard it said that the fall in emissions caused by the **pandemic** isn ’ t that much more than, actually, the fall that we will need every single year if we ’ re to meet emissions targets. Is that true, and, if so, doesn ’ t that seem impossibly daunting? AG : It does seem daunting, but first look at the number. That number came from a study a little over a year ago released by the IPCC as to what it would take to keep the Earth ’ s temperatures from increasing more than 1. 5 degrees Celsius. And yes, the annual reductions would be significant, on the order of what we ’ ve seen with the **pandemic**. And yes, that does seem daunting. However, we do have the opportunity to make some fairly dramatic changes, and the plan is not a mystery. You start with the two **sectors** that are closest to an effective **transition** – electricity generation, as I mentioned – and last year, 2019, if you look at all of the new electricity generation built all around the world, 72 percent of it was from solar and wind. And already, without the continuing subsidies for fossil fuels, we would see many more of these plants being shut down. There are some new fossil plants being built, but many more are being shut down. And where transportation is concerned, the second **sector** ready to go, in addition to the cheaper prices for EVs that I made reference to before, there are some 45 jurisdictions around the world – national, regional and municipal – where laws have been passed beginning a phaseout of internal combustion engines. Even India said that by 2030, less than 10 years from now, it will be illegal to sell any new internal combustion engines in India. There are many other examples. So the past small reductions may not be an accurate guide to the kind we can achieve with serious national plans and a focused global effort. CA : So help us understand just the big picture here, Al. I think before the **pandemic**, the world was emitting about 55 gigatons of what they call”CO2 equivalent,”so that includes other greenhouse gases like methane dialed up to be the equivalent of CO2. And am I right in saying that the IPCC, which is the global organization of scientists, is recommending that the only way to fix this **crisis** is to get that number from 55 to zero by 2050 at the very latest, and that even then, there ’ s a chance that we will end up with temperature rises more like two degrees Celsius rather than 1. 5? I mean, is that approximately the big picture of what the IPCC is recommending? AG : That ’ s correct. The global goal established in the Paris Conference is to get to net zero on a global basis by 2050, and many people quickly add that that really means a 45 to 50 percent reduction by 2030 to make that pathway to net zero feasible. CA : And that kind of timeline is the kind of timeline where people couldn ’ t even imagine it. It ’ s just hard to think of **policy** over 30 years. So that ’ s actually a very good shorthand, that humanity ’ s task is to cut emissions in half by 2030, approximately speaking, which I think boils down to about a seven or eight percent reduction a year, something like that, if I ’ m not wrong. AG : Not quite. Not quite that large but close, yes. CA : So it is something like the effect that we ’ ve experienced this year may be necessary. This year, we ’ ve done it by basically shutting down the economy. You ’ re talking about a way of doing it over the coming years that actually gives some economic growth and new jobs. So talk more about that. You ’ ve referred to changing our energy sources, changing how we transport. If we did those things, how much of the problem does that solve? AG : Well, we can get to – well, in addition to doing the two **sectors** that I mentioned, we also have to deal with manufacturing and all the use cases that require temperatures of a thousand degrees Celsius, and there are solutions there as well. I ’ ll come back and mention an exciting one that Germany has just embarked upon. We also have to tackle regenerative agriculture. There is the opportunity to sequester a great deal of carbon in topsoils around the world by changing the agricultural techniques. There is a farmer-led movement to do that. We need to also retrofit buildings. We need to change our management of forests and the ocean. But let me just mention two things briefly. First of all, the high temperature use cases. Angela Merkel, just 10 days ago, with the **leadership** of her minister Peter Altmaier, who is a good friend and a great public servant, have just embarked on a green **hydrogen** strategy to make **hydrogen** with zero marginal cost renewable energy. And just a word on that, Chris : you ’ ve heard about the intermittency of wind and solar – solar doesn ’ t produce electricity when the sun ’ s not shining, and wind doesn ’ t when the wind ’ s not blowing – but batteries are getting better, and these technologies are becoming much more efficient and powerful, so that for an increasing number of hours of each day, they ’ re producing often way more electricity than can be used. So what to do with it? The marginal cost for the next kilowatt-hour is zero. So all of a sudden, the very energy-intensive process of cracking **hydrogen** from water becomes economically feasible, and it can be substituted for coal and gas, and that ’ s already being done. There ’ s a Swedish company already making steel with green **hydrogen**, and, as I say, Germany has just embarked on a major new initiative to do that. I think they ’ re pointing the way for the rest of the world. Now, where building retrofits are concerned, just a moment on this, because about 20 to 25 percent of the global warming pollution in the world and in the US comes from inefficient buildings that were constructed by companies and individuals who were trying to be competitive in the marketplace and keep their margins acceptably high and thereby skimping on insulation and the right windows and LEDs and the rest. And yet the person or company that buys that building or leases that building, they want their monthly utility bills much lower. So there are now ways to close that so-called agent-principal divide, the differing incentives for the builder and occupier, and we can retrofit buildings with a program that literally pays for itself over three to five years, and we could put tens of millions of people to work in jobs that by definition cannot be outsourced because they exist in every single community. And we really ought to get serious about doing this, because we ’ re going to need all those jobs to get sustainable prosperity in the aftermath of this **pandemic**. CA : Just going back to the **hydrogen** economy that you referred to there, when some people hear that, they think,”Oh, are you talking about hydrogen-fueled cars?”And they ’ ve heard that that probably won ’ t be a winning strategy. But you ’ re thinking much more broadly than that, I think, that it ’ s not just **hydrogen** as a kind of storage mechanism to act as a buffer for renewable energy, but also **hydrogen** could be essential for some of the other processes in the economy like making steel, making cement, that are fundamentally carbon-intensive processes right now but could be transformed if we had much cheaper sources of **hydrogen**. Is that right? AG : Yes, I was always skeptical about **hydrogen**, Chris, principally because it ’ s been so expensive to make it, to”crack it out of water,”as they say. But the game-changer has been the incredible abundance of solar and wind electricity in volumes and amounts that people didn ’ t expect, and all of a sudden, it ’ s cheap enough to use for these very energy-intensive processes like creating green **hydrogen**. I ’ m still a bit skeptical about using it in vehicles. Toyota ’ s been betting on that for 25 years and it hasn ’ t really worked for them. Never say never, maybe it will, but I think it ’ s most useful for these high-temperature industrial processes, and we already have a pathway for decarbonizing transportation with electricity that ’ s working extremely well. Tesla ’ s going to be soon the most valuable automobile company in the world, already in the US, and they ’ re about to overtake Toyota. There is now a semitruck company that ’ s been stood up by Tesla and another that is going to be a hybrid with electricity and green **hydrogen**, so we ’ ll see whether or not they can make it work in that application. But I think electricity is preferable for cars and trucks. CA : We ’ re coming to some community questions in a minute. Let me ask you, though, about nuclear. Some environmentalists believe that nuclear, or maybe new generation nuclear power is an essential part of the equation if we ’ re to get to a truly clean future, a clean energy future. Are you still pretty skeptical on nuclear, Al? AG : Well, the market ’ s skeptical about it, Chris. It ’ s been a crushing disappointment for me and for so many. I used to represent Oak Ridge, where nuclear energy began, and when I was a young congressman, I was a booster. I was very enthusiastic about it. But the cost overruns and the problems in building these plants have become so severe that utilities just don ’ t have an appetite for them. It ’ s become the most expensive source of electricity. Now, let me hasten to add that there are some older nuclear reactors that have more useful time that could be added onto their lifetimes. And like a lot of environmentalists, I ’ ve come to the view that if they can be determined to be **safe**, they should be allowed to continue operating for a time. But where new nuclear power plants are concerned, here ’ s a way to look at it. If you are – you ’ ve been a CEO, Chris. If you were the CEO of – I guess you still are. If you were the CEO of an electric utility, and you told your executive team,”I want to build a nuclear power plant,”two of the first questions you would ask are, number one : How much will it cost? And there ’ s not a single engineering consulting firm that I ’ ve been able to find anywhere in the world that will put their name on an opinion giving you a cost estimate. They just don ’ t know. A second question you would ask is : How long will it take to build it, so we can start selling the electricity? And again, the answer you will get is,”We have no idea.”So if you don ’ t know how much it ’ s going to cost, and you don ’ t know when it ’ s going to be finished, and you already know that the electricity is more expensive than the alternate ways to produce it, that ’ s going to be a little discouraging, and, in fact, that ’ s been the case for utilities around the world. CA : OK. So there ’ s definitely an **interesting** debate there, but we ’ re going to come on to some community questions. Let ’ s have the first of those questions up, please. From Prosanta Chakrabarty :”People who are skeptical of COVID and of **climate** change seem to be skeptical of science in general. It may be that the singular message from scientists gets diluted and convoluted. How do we fix that?”AG : Yeah, that ’ s a great question, Prosanta. Boy, I ’ m trying to put this succinctly and shortly. I think that there has been a feeling that experts in general have kind of let the US down, and that feeling is much more pronounced in the US than in most other countries. And I think that the considered opinion of what we call experts has been diluted over the last few decades by the unhealthy dominance of big money in our political system, which has found ways to really twist economic **policy** to benefit elites. And this sounds a little radical, but it ’ s actually what has happened. And we have gone for more than 40 years without any meaningful increase in middle-income pay, and where the injustice experienced by African Americans and other communities of color are concerned, the differential in pay between African Americans and majority Americans is the same as it was in 1968, and the family wealth, the net worth – it takes 11 and a half so-called”typical”African American families to make up the net worth of one so-called”typical”White American family. And you look at the soaring incomes in the top one or the top one-tenth of one percent, and people say,”Wait a minute. Whoever the experts were that designed these **policies**, they haven ’ t been doing a good job for me.”A final point, Chris : there has been an assault on reason. There has been a war against truth. There has been a strategy, maybe it was best known as a strategy decades ago by the tobacco companies who hired actors and dressed them up as doctors to falsely reassure people that there were no health consequences from smoking cigarettes, and a hundred million people died as a result. That same strategy of diminishing the significance of truth, diminishing, as someone said, the authority of knowledge, I think that has made it kind of open season on any inconvenient truth – forgive another buzz phrase, but it is apt. We cannot abandon our devotion to the best available evidence tested in reasoned discourse and used as the basis for the best **policies** we can form. CA : Is it possible, Al, that one consequence of the **pandemic** is actually a growing number of people have revisited their opinions on scientists? I mean, you ’ ve had a chance in the last few months to say,”Do I **trust** my political **leader** or do I **trust** this scientist in terms of what they ’ re saying about this **virus**?”Maybe lessons from that could be carried forward? AG : Well, you know, I think if the polling is accurate, people do **trust** their doctors a lot more than some of the politicians who seem to have a vested interest in pretending the **pandemic** isn ’ t real. And if you look at the incredible bust at President Trump ’ s rally in Tulsa, a stadium of 19, 000 people with less than one-third filled, according to the fire marshal, you saw all the empty seats if you saw the news clips, so even the most loyal Trump supporters must have decided to **trust** their doctors and the medical advice rather than Dr. Donald Trump. CA : With a little help from the TikTok generation, perchance. AG : Well, but that didn ’ t affect the turnout. What they did, very cleverly, and I ’ m cheering them on, what they did was affect the Trump White House ’ s expectations. They ’ re the reason why he went out a couple days beforehand and said,”We ’ ve had a million people sign up.”But they didn ’ t prevent – they didn ’ t take seats that others could have otherwise taken. They didn ’ t affect the turnout, just the expectations. CA : OK, let ’ s have our next question here.”Are you concerned the world will rush back to the use of the private car out of fear of using shared public transportation?”AG : Well, that could actually be one of the consequences, absolutely. Now, the trends on mass transit were already inching in the wrong direction because of Uber and Lyft and the ridesharing services, and if autonomy ever reaches the goals that its advocates have hoped for then that may also have a similar effect. But there ’ s no doubt that some people are going to be probably a little more reluctant to take mass transportation until the fear of this **pandemic** is well and truly gone. CA : Yeah. Might need a vaccine on that one. AG : Yeah. CA : Next question. Sonaar Luthra, thank you for this question from LA.”Given the temperature rise in the Arctic this past week, seems like the rate we are losing our carbon sinks like permafrost or forests is accelerating faster than we predicted. Are our models too focused on human emissions?”**Interesting** question. AG : Well, the models are focused on the factors that have led to these incredible temperature spikes in the north of the Arctic Circle. They were predicted, they have been predicted, and one of the reasons for it is that as the snow and ice cover melts, the sun ’ s incoming rays are no longer reflected back into space at a 90 percent rate, and instead, when they fall on the dark tundra or the dark ocean, they ’ re absorbed at a 90 percent rate. So that ’ s a magnifier of the warming in the Arctic, and this has been predicted. There are a number of other consequences that are also in the models, but some of them may have to be recalibrated. The scientists are freshly concerned that the emissions of both CO2 and methane from the thawing tundra could be larger than they had hoped they would be. There ’ s also just been a brand-new study. I won ’ t spend time on this, because it deals with a kind of geeky term called”**climate** sensitivity,”which has been a factor in the models with large error bars because it ’ s so hard to pin down. But the latest evidence indicates, worryingly, that the sensitivity may be greater than they had thought, and we will have an even more daunting task. That shouldn ’ t discourage us. I truly believe that once we cross this tipping point, and I do believe we ’ re doing it now, as I ’ ve said, then I think we ’ re going to find a lot of ways to speed up the emissions reductions. CA : We ’ ll take one more question from the community. Haha.”Geoengineering is making extraordinary progress. Exxon is **investing** in technology from Global Thermostat that seems promising. What do you think of these **air** and water carbon capture technologies?”Stephen Petranek. AG : Yeah. Well, you and I have talked about this before, Chris. I ’ ve been strongly opposed to conducting an unplanned global experiment that could go wildly wrong, and most are really scared of that approach. However, the term”geoengineering”is a nuanced term that covers a lot. If you want to paint roofs white to reflect more energy from the cityscapes, that ’ s not going to bring a danger of a runaway effect, and there are some other things that are loosely called”geoengineering”like that, which are fine. But the idea of blocking out the sun ’ s rays – that ’ s insane in my opinion. Turns out plants need sunlight for photosynthesis and solar panels need sunlight for producing electricity from the sun ’ s rays. And the consequences of changing everything we know and pretending that the consequences are going to precisely cancel out the unplanned experiment of global warming that we already have underway, you know, there are glitches in our thinking. One of them is called the”single solution bias,”and there are people who just have a hunger to say,”Well, that one solution, we just need to latch on to that and do that, and damn the consequences.”Well, it ’ s nuts. CA : But let me push back on this just a little bit. So let ’ s say that we agree that a single solution, all-or-nothing attempt at geoengineering is crazy. But there are scenarios where the world looks at emissions and just sees, in 10 years ’ time, let ’ s say, that they are just not coming down fast enough and that we are at risk of several other liftoff events where this train will just get away from us, and we will see temperature rises of three, four, five, six, seven degrees, and all of civilization is at risk. Surely, there is an approach to geoengineering that could be modeled, in a way, on the way that we approach medicine. Like, for hundreds of years, we don ’ t really understand the human body, people would try interventions, and some of them would work, and some of them wouldn ’ t. No one says in medicine,”You know, go in and take an all-or-nothing decision on someone ’ s life,”but they do say,”Let ’ s try some **stuff**.”If an experiment can be reversible, if it ’ s plausible in the first place, if there ’ s reason to think that it might work, we actually owe it to the future health of humanity to try at least some types of tests to see what could work. So, small tests to see whether, for example, seeding of something in the ocean might create, in a nonthreatening way, carbon sinks. Or maybe, rather than filling the atmosphere with sulfur dioxide, a smaller experiment that was not that big a deal to see whether, cost-effectively, you could reduce the temperature a little bit. Surely, that isn ’ t completely crazy and is at least something we should be thinking about in case these other measures don ’ t work? AG : Well, there ’ ve already been such experiments to seed the ocean to see if that can increase the uptake of CO2. And the experiments were an unmitigated failure, as many predicted they would be. But that, again, is the kind of approach that ’ s very different from putting tinfoil strips in the atmosphere orbiting the Earth. That was the way that solar geoengineering proposal started. Now they ’ re focusing on chalk, so we have chalk dust all over everything. But more serious than that is the fact that it might not be reversible. CA : But, Al, that ’ s the rhetoric response. The amount of dust that you need to drop by a degree or two wouldn ’ t result in chalk dust over everything. It would be unbelievably – like, it would be less than the dust that people experience every day, anyway. I mean, I just – AG : First of all, I don ’ t know how you do a small experiment in the atmosphere. And secondly, if we were to take that approach, we would have to steadily increase the amount of whatever substance they decided. We ’ d have to increase it every single year, and if we ever stopped, then there would be a sudden snapback, like”The Picture of Dorian Gray,”that old book and movie, where suddenly all of the things caught up with you at once. The fact that anyone is even considering these approaches, Chris, is a measure of a feeling of desperation that some have begun to feel, which I understand, but I don ’ t think it should drive us toward these reckless experiments. And by the way, using your analogy to experimental cancer treatments, for example, you usually get informed consent from the patient. Getting informed consent from 7. 8 billion people who have no voice and no say, who are subject to the potentially catastrophic consequences of this wackadoodle proposal that somebody comes up with to try to rearrange the entire Earth ’ s atmosphere and hope and pretend that it ’ s going to cancel out, the fact that we ’ re putting 152 million tons of heat-trapping, manmade global warming pollution into the sky every day. That ’ s what ’ s really insane. A scientist decades ago compared it this way. He said, if you had two people on a sinking boat and one of them says,”You know, we could probably use some mirrors to signal to shore to get them to build a sophisticated wave-generating machine that will cancel out the rocking of the boat by these **guys** in the back of the boat.”Or you could get them to stop rocking the boat. And that ’ s what we need to do. We need to stop what ’ s causing the **crisis**. CA : Yeah, that ’ s a great story, but if the effort to stop the people rocking in the back of the boat is as complex as the scientific proposal you just outlined, whereas the experiment to stop the waves is actually as simple as telling the people to stop rocking the boat, that story changes. And I think you ’ re right that the issue of informed consent is a really challenging one, but, I mean, no one gave informed consent to do all of the other things we ’ re doing to the atmosphere. And I agree that the moral hazard issue is worrying, that if we became dependent on geoengineering and took away our efforts to do the rest, that would be tragic. It just seems like, I wish it was possible to have a nuanced debate of people saying, you know what, there ’ s multiple dials to a very complex problem. We ’ re going to have to adjust several of them very, very carefully and keep talking to each other. Wouldn ’ t that be a goal to just try and have a more nuanced debate about this, rather than all of that geoengineering can ’ t work? AG : Well, I ’ ve said some of it, you know, the benign forms that I ’ ve mentioned, I ’ m not ruling those out. But blocking the Sun ’ s rays from the Earth, not only do you affect 7. 8 billion people, you affect the plants and the animals and the ocean currents and the wind currents and natural processes that we ’ re in danger of disrupting even more. Techno-optimism is something I ’ ve engaged in in the past, but to latch on to some brand-new technological solution to rework the entire Earth ’ s natural system because somebody thinks he ’ s clever enough to do it in a way that precisely cancels out the consequences of using the atmosphere as an open sewer for heat-trapping manmade gases. It ’ s much more important to stop using the atmosphere as an open sewer. That ’ s what the problem is. CA : All right, well, we ’ ll agree that that is the most important thing, for sure, and speaking of which, do you believe the world needs carbon pricing, and is there any prospect for getting there? AG : Yes. Yes to both questions. For decades, almost every economist who is asked about the **climate crisis** says,"Well, we just need to put a price on carbon."And I have certainly been in favor of that approach. But it is daunting. Nevertheless, there are 43 jurisdictions around the world that already have a price on carbon. We ' re seeing it in Europe. They finally straightened out their carbon pricing mechanism. It ' s an emissions trading version of it. We have places that have put a tax on carbon. That ' s the approach the economists prefer. China is beginning to implement its national emissions trading program. California and quite a few other states in the US are already doing it. It can be given back to people in a revenue-neutral way. But the opposition to it, Chris, which you ' ve noted, is impressive enough that we do have to take other approaches, and I would say most **climate** activists are now saying, look, let ' s don ' t make the best the enemy of the better. There are other ways to do this as well. We need every solution we can rationally employ, including by regulation. And often, when the political difficulty of a proposal becomes too difficult in a market-oriented approach, the fallback is with regulation, and it ' s been given a bad name, regulation, but many places are doing it. I mentioned phasing out internal combustion engines. That ' s an example. There are 160 cities in the US that have already by regulation ordered that within a date certain, 100 percent of all their electricity will have to come from renewable sources. And again, the market forces that are driving the cost of renewable energy and sustainability solutions ever downward, that gives us the wind at our back. This is working in our favor. CA : I mean, the pushback on carbon pricing often goes further from parts of the environmental movement, which is to a pushback on the role of business in general. Business is actually – well, capitalism – is blamed for the **climate crisis** because of unrelenting growth, to the point where many people don ' t **trust** business to be part of the solution. The only way to go forward is to regulate, to force businesses to do the right thing. Do you think that business has to be part of the solution? AG : Well, definitely, because the allocation of capital needed to solve this **crisis** is greater than what governments can handle. And businesses are beginning, many businesses are beginning to play a very constructive role. They ' re getting a **demand** that they do so from their customers, from their investors, from their boards, from their executive teams, from their families. And by the way, the rising generation is **demanding** a brighter future, and when CEOs interview potential new hires, they find that the new hires are interviewing them. They want to make a nice income, but they want to be able to tell their family and friends and peers that they ' re doing something more than just making money. One illustration of how this new generation is changing, Chris : there are 65 colleges in the US right now where the College Young Republican Clubs have joined together to jointly **demand** that the Republican National Committee change its **policy** on **climate**, lest they lose that entire generation. This is a global phenomenon. The Greta Generation is now leading this in so many ways, and if you look at the polling, again, the vast majority of young **Republicans** are **demanding** a change on **climate policy**. This is really a movement that is building still. CA : I was going to ask you about that, because one of the most painful things over the last 20 years has just been how **climate** has been politicized, certainly in the US. You ' ve probably felt yourself at the heart of that a lot of the time, with people attacking you personally in the most merciless, and unfair ways, often. Do you really see signs that that might be changing, led by the next generation? AG : Yeah, there ' s no question about it. I don ' t want to rely on polls too much. I ' ve mentioned them already. But there was a new one that came out that looked at the wavering Trump supporters, those who **supported** him strongly in the past and want to do so again. The number one issue, surprisingly to some, that is giving them pause, is the craziness of President Trump and his administration on **climate**. We ’ re seeing big majorities of the Republican Party overall saying that they ’ re ready to start exploring some real solutions to the **climate crisis**. I think that we ’ re really getting there, no question about it. CA : I mean, you ’ ve been the figurehead for raising this issue, and you happen to be a Democrat. Is there anything that you can personally do to – I don ’ t know – to open the tent, to welcome people, to try and say,”This is beyond politics, dear friends”? AG : Yeah. Well, I ’ ve tried all of those things, and maybe it ’ s made a little positive difference. I ’ ve worked with the Republicans extensively. And, you know, well after I left the White House, I had Newt Gingrich and Pat Robertson and other prominent **Republicans** appear on national TV ads with me saying we ’ ve got to solve the **climate crisis**. But the petroleum industry has really doubled down enforcing discipline within the Republican Party. I mean, look at the attacks they ’ ve launched against the Pope when he came out with his encyclical and was demonized, not by all for sure, but there were hawks in the anti-climate movement who immediately started training their guns on Pope Francis, and there are many other examples. They enforce discipline and try to make it a partisan issue, even as Democrats reach out to try to make it bipartisan. I totally agree with you that it should not be a partisan issue. It didn ’ t use to be, but it ’ s been artificially weaponized as an issue. CA : I mean, the CEOs of oil companies also have kids who are talking to them. It feels like some of them are moving and are trying to **invest** and trying to find ways of being part of the future. Do you see signs of that? AG : Yeah. I think that business **leaders**, including in the oil and gas companies, are hearing from their families. They ’ re hearing from their friends. They ’ re hearing from their employees. And, by the way, we ’ ve seen in the **tech** industry some mass walkouts by employees who are **demanding** that some of the **tech** companies do more and get serious. I ’ m so proud of Apple. Forgive me for parenthetically praising Apple. You know, I ’ m on the board, but I ’ m such a big fan of Tim Cook and my colleagues at Apple. It ’ s an example of a **tech** company that ’ s really doing fantastic things. And there ’ s some others as well. There are others in many industries. But the pressures on the oil and gas companies are quite extraordinary. You know, BP just wrote down 12 and a half billion dollars ’ worth of oil and gas assets and said that they ’ re never going to see the light of day. Two-thirds of the fossil fuels that have already been discovered cannot be burned and will not be burned. And so that ’ s a big economic risk to the global economy, like the subprime mortgage **crisis**. We ’ ve got 22 trillion dollars of subprime carbon assets, and just yesterday, there was a major report that the fracking industry in the US is seeing now a wave of bankruptcies because the price of the fracked gas and oil has fallen below levels that make them economic. CA : Is the shorthand of what ’ s happened there that electric cars and electric technologies and solar and so forth have helped drive down the price of oil to the point where huge amounts of the reserves just can ’ t be developed profitably? AG : Yes, that ’ s it. That ’ s mainly it. The projections for energy sources in the next several years uniformly predict that electricity from wind and solar is going to continue to plummet in price, and therefore using gas or coal to make steam to turn the turbines is just not going to be economical. Similarly, the electrification of the transportation **sector** is having the same effect. Some are also looking at the trend in national, regional and local governance. I mentioned this before, but they ’ re predicting a very different energy future. But let me come back, Chris, because we talked about business **leaders**. I think you were getting in a question a moment ago about capitalism itself, and I do want to say a word on that, because there are a lot of people who say maybe capitalism is the basic problem. I think the current form of capitalism we have is desperately in need of reform. The short-term outlook is often mentioned, but the way we measure what is of value to us is also at the heart of the **crisis** of modern capitalism. Now, capitalism is at the base of every successful economy, and it balances supply and **demand**, unlocks a higher fraction of the human potential, and it ' s not going anywhere, but it needs to be reformed, because the way we measure what ' s valuable now ignores so-called negative externalities like pollution. It also ignores positive externalities like investments in education and health care, mental health care, family services. It ignores the depletion of resources like groundwater and topsoil and the web of living species. And it ignores the distribution of incomes and net worths, so when GDP goes up, people cheer, two percent, three percent – wow! – four percent, and they think,"Great!"But it ' s accompanied by vast increases in pollution, chronic underinvestment in public goods, the depletion of irreplaceable natural resources, and the worst inequality **crisis** we ' ve seen in more than a hundred years that is threatening the future of both capitalism and democracy. So we have to change it. We have to reform it. CA : So reform capitalism, but don ' t throw it out. We ' re going to need it as a tool as we go forward if we ' re to solve this. AG : Yeah, I think that ' s right, and just one other point : the worst environmental abuses in the last hundred years have been in jurisdictions that experimented during the 20th century with the alternatives to capitalism on the left and right. CA : **Interesting**. All right. Two last community questions quickly. Chadburn Blomquist :"As you are reading the tea leaves of the **impact** of the current **pandemic**, what do you think in regard to our response to combatting **climate** change will be the most impactful lesson learned?"AG : Boy, that ' s a very thoughtful question, and I wish my answer could rise to the same level on short notice. I would say first, don ' t ignore the scientists. When there is virtual unanimity among the scientific and medical experts, pay attention. Don ' t let some politician dissuade you. I think President Trump is slowly learning that ' s it ' s kind of difficult to gaslight a **virus**. He tried to gaslight the **virus** in Tulsa. It didn ' t come off very well, and tragically, he decided to recklessly roll the dice a month ago and ignore the recommendations for people to wear masks and to socially distance and to do the other things, and I think that lesson is beginning to take hold in a much stronger way. But beyond that, Chris, I think that this period of time has been characterized by one of the most profound opportunities for people to rethink the patterns of their lives and to consider whether or not we can ' t do a lot of things better and differently. And I think that this rising generation I mentioned before has been even more profoundly affected by this interlude, which I hope ends soon, but I hope the lessons endure. I expect they will. CA : Yeah, it ' s amazing how many things you can do without emitting carbon, that we ' ve been forced to do. Let ' s have one more question here. Frank Hennessy :"Are you encouraged by the **ability** of people to quickly adapt to the new normal due to COVID-19 as evidence that people can and will change their habits to respond to **climate** change?"AG : Yes, but I think we have to keep in mind that there is a **crisis** within this **crisis**. The **impact** on the African American community, which I mentioned before, on the Latinx community, Indigenous peoples. The highest infection rate is in the Navajo Nation right now. So some of these questions appear differently to those who are really getting the brunt of this **crisis**, and it is unacceptable that we allow this to continue. It feels one way to you and me and perhaps to many in our audience today, but for low-income communities of color, it ' s an entirely different **crisis**, and we owe it to them and to all of us to get busy and to start using the best science and solve this **pandemic**. You know the phrase"**pandemic** economics."Somebody said, the first principle of **pandemic** economics is take care of the **pandemic**, and we ' re not doing that yet. We ' re seeing the president try to goose the economy for his reelection, never mind the prediction of tens of thousands of additional American deaths, and that is just unforgivable in my opinion. CA : Thank you, Frank. So Al, you, along with others in the community played a key role in encouraging **TED** to launch this initiative called"Countdown."Thank you for that, and I guess this conversation is continuing among many of us. If you ' re **interested** in **climate**, watching this, check out the Countdown website, countdown. **ted**. com, and be part of 10 / 10 / 2020, when we are trying to put out an alert to the world that **climate** can ' t wait, that it really matters, and there ' s going to be some amazing content free to the world on that day. Thank you, Al, for your inspiration and **support** in doing that. I wonder whether you could end today ' s session just by painting us a picture, like how might things roll out over the next decade or so? Just tell us whether there is still a story of hope here. AG : I ' d be glad to. I ' ve got to get one plug in. I ' ll make it brief. July 18 through July 26, The **Climate** Reality Project is having a global training. We ' ve already had 8, 000 people register. You can go to climatereality. com. Now, a bright future. It begins with all of the kinds of efforts that you ' ve thrown yourself into in organizing Countdown. Chris, you and your team have been amazing to work with, and I ' m so excited about the Countdown project. **TED** has an unparalleled **ability** to spread ideas that are worth spreading, to raise consciousness, to enlighten people around the world, and it ' s needed for **climate** and the solutions to the **climate crisis** like it ' s never been needed before, and I just want to thank you for what you personally are doing to organize this fantastic Countdown program. CA : Thank you. And the world? Are we going to do this? Do you think that humanity is going to pull this off and that our grandchildren are going to have beautiful lives where they can celebrate nature and not spend every day in fear of the next tornado or tsunami? AG : I am optimistic that we will do it, but the answer is in our hands. We have seen dark times in periods of the past, and we have risen to meet the challenge. We have limitations of our long evolutionary heritage and elements of our culture, but we also have the **ability** to transcend our limitations, and when the chips are down, and when survival is at stake and when our children and future generations are at stake, we ' re capable of more than we sometimes allow ourselves to think we can do. This is such a time. I believe we will rise to the occasion, and we will create a bright, clean, prosperous, just and fair future. I believe it with all my heart. CA : Al Gore, thank you for your life of work, for all you ' ve done to elevate this issue and for spending this time with us now. Thank you. AG : Back at you. Thank you.

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Chris Anderson : Hello, **TED** community, welcome back for another live conversation. It ' s a big one today, as big as they get. You know, when we created this"Build Back Better"series our thought was how could we address issues arising out of the **pandemic**, how could we imagine building back from that. But the events of this past week, the horrific death of George Floyd and the daily protests that have followed, I mean, they provided a new urgency which we, of course, simply have to address. I mean, can we build back better from this? I think before we can even start to answer that question, we just have to seek to understand the immensity of this moment. Whitney Pennington Rodgers : That ' s right, Chris. Right now, so many people in the United States and beyond are grappling with feelings of anger and frustration, deep, deep sadness and really helplessness. No matter who you are, you have questions about what to do now, how to make things better. And as we ' ve seen, violence like this unfolds for many, many years. What is the path forward? CA : So – We ' re joined today by a group of activists, organizers and **leaders** known for their crucial work in social justice and civil rights. We ' re so grateful to have them here to engage in a discussion about racial injustice in America, the unbearable acts of violence that we ' ve – Acts of violence against the black community that we ' ve witnessed, the dangers to a nation riven by anger and fear. And how on earth we can move forward from this to something better. So first, each of our four guests will share their thoughts on how we move forward from this moment. And then we ' ll engage as a group, including you, the **TED** community. WPR : And we ' d like to thank our partner, the Project Management Institute. Their generous **support** has helped make today ' s interviews possible, and of course, as Chris mentioned, we want you to take part in the conversation, so please share your questions using our Ask a question feature and continue to share your thoughts in the discussion thread. CA : Thanks, Whitney. OK, let ' s get this moving. Our first guest. Dr. Phillip Atiba Goff is the founder and CEO of the Center for Policing Equity. They work with police departments across America, including in Minneapolis, to seek measurable responses to racial bias. Phil, I can scarcely imagine how the stress in the last week must have been for you. Welcome, and over to you for your opening comments. Phillip Atiba Goff : Thanks, Chris. Yeah, this week has been a gut punch to anybody who felt like we could be making progress in the way that we put forward public safety that empowers particularly vulnerable communities. We started working in Minneapolis about five years ago. At the time, it was, like most major cities in the United States, a department that had a long history of unaccounted for violence from law enforcement, targeting the most vulnerable black communities. And we tried to put into place a number of things that we know work. Change the culture, so that the culture can be **accountable** to the values of the community. And what we saw was small but measurable progress. We always knew, with small and measurable progress, that you ' re one tragic incident from going back to ground zero. But the events of the last week and a half haven ' t brought us back to ground zero, they ' ve torched ground zero, and we ' ve dug a hole that we have to dig ourselves out of. What I hear from police chiefs who call me, from activists I talk to, from folks in the communities that are literally on fire right now, I hear folks saying, I had one activist say to me that the pain that he was feeling was too large to fit into his body. And without thinking about it, I said right back,"That ' s because it ' s too large to fit into a lifetime."What we ' re seeing isn ' t just the response to one gruesome, cruel, public execution. A lynching. It ' s not just the **reaction** to three of them : Ahmaud Arbery, Breonna Taylor and then George Floyd. What we ' re seeing is the bill come due for the unpaid debts that this country owes to its black residents. And it comes due usually every 20 to 30 years. It was Ferguson just six years ago, but about 30 years before that, it was in the streets of Los Angeles, after the verdict that exonerated the police that beat Rodney King on video. It was Newark, it was Watts, it was Chicago, it was Tulsa, it was Chicago again. If we don ' t take a full accounting of these debts that are owed, then we ' re going to keep paying it. Part of what I ' ve been experiencing in the last week and a half, and what I ' ve been sharing with the people who do this work, who are serious about it, is the acknowledgment, the soul-crushing reality that at some point, when things stop being on fire, the cameras are going to turn to something else. And the history that we have in this country is not just a history of vicious neglect and a targeted abuse of black communities, it ' s also one where we lose our attention for it. And what that means for communities like in Baton Rouge, for those who still grieve Alton Sterling, and in Baltimore, for those who are still grieving Freddie Gray, is that there is not just a chance, there ' s a likelihood that we are a month or two months out from this with no more to show for it than what we had to show after Michael Brown Jr. And holding the weight of that, individually and collectively, is just too much. It ' s just too heavy a load for a person or a people, or a generation to hold up. What we ' re seeing is the unrepentant sins, the unpaid debts. And so the solution can ' t just be that we fix policing. It can ' t be only incremental reform. It can ' t be only systems of accountability to catch cops after they ' ve killed somebody. Because there ' s no such thing as justice for George Floyd. There ' s maybe accountability. There ' s maybe some relief from the people who are still around, who loved him, for his daughter who spoke out yesterday and said,"My Daddy changed the world."There won ' t be justice for a man who ' s dead when he didn ' t have to be. But we ' re not going to get to where we need to go just by reforming police. So in addition to the work that CPE is known for with the data, we have been encouraging departments and cities to take the money that should be going to **invest** in communities, and take it from police budgets, bring it to the communities. People ask,"Well, what could it possibly look like? How could we imagine it?"And I tell people, there is a place where we do this in the United States right now. We ' ve all heard about it, whispered, some of us have even been there, some of us live there. The place is called the suburbs, where we already have enough resources to give to people, so they don ' t need the police for public safety in the first place. If someone has a substance abuse issue, they can go to a clinic. If somebody has a medical issue, they ' ve got insurance, they can go to a hospital. If there ' s a domestic dispute, they have friends, they have **support**. You don ' t need to enter a badge and a gun into it. If we hadn ' t disinvested from all the public resources that were available in communities that most needed those, we wouldn ' t need police in the first place, and many have been arguing, even more loudly recently, that we don ' t. If we would just take the money that we use to punish, and instead **invest** it in the promise and the genius of the community that could be there. So I don ' t know all the ways we ' re going to get there. I know it ' s going to take everything and. It ' s going to need the kind of systemic change and the management tools that we traditionally offer. It ' s also going to need a quantum change in the way that we think about public safety. But mostly, this isn ' t just a policing problem. This is the unpaid debts that are owed to black America. The bill is coming due. And we need to start getting an accounting together, so we ' re not just paying off the interest of the damn thing. WPR : Thank you, Phil. Rashad Robinson is the president of Color Of Change, a civil rights organization that advocates for racial justice for the black community. To date, more than four million people have signed their petition to arrest the officers **involved** in the murder of George Floyd. And of course, one was arrested last week. Thank you so much for being with us, Rashad, welcome. Rashad Robinson : Thank you. And thank you for having me. It ' s an opportunity that I ' m taking today to just tell you about how you can get **involved**. How you can take action, because right now, strategic action is critical for all of us to do the work to change the rules that far too often keep the systems in place that hold us back. Make no mistake, the criminal justice system is not broken. It is operating exactly the way it was designed. At every single level, the criminal justice system is not about providing justice, but about **ensuring** that certain people, certain communities are protected, while other communities are violated. And so I wan to talk a little bit today about Color Of Change, about activism, about the work that ' s happening on the ground from other organizations all around the country, and the way that you can channel this energy. What we talk about at Color Of Change is how do you channel presence into power. Far too often we mistake presence and visibility for power, presence retweets the stories of the movement, people feeling passion about change could sometimes make us feel like change is inevitable, but power is actually the **ability** to change the rules. And right now, every day, people are taking action, and what we ' re trying to channel that energy into is a couple of things. First is a whole set of **demands** at the federal level and at the local level. As Phil described, policing operates on many different channels. And what we need to recognize is that while there are a lot of things that can happen at the federal level, locally all around the country there are decisions that are being made in communities around how policing is executed, where community needs to hold a deeper level of accountability, at the state level we need new laws. So at Color Of Change, we ' ve built a whole platform around a set of **demands** and are working to build more energy for everyday people to take action. We ' re fighting for justice for Ahmaud Arbery, Breonna Taylor and George Floyd, we ' re also fighting for justice for other folks whose names you haven ' t heard, Nina Pop and others, whose stories of injustice and the relationship to the criminal justice system represent all the ways in which fighting right now is important. Over the last couple of years, we have worked to build a movement, to hold district attorneys **accountable** and to change the role of district attorneys in our country. And through the Winning Justice platform at Color Of Change, www. winningjustice. org, what we have worked to do is channel the energy of everyday people to take action. So, for folks who are watching what ' s happening on TV, seeing it on their social media feeds and are outraged about what ' s happening in Georgia, what ' s happening in Tennessee, what ' s happening in Minnesota, you yourself, probably, most likely, live in a place, in a community where you have a district attorney that will not hold police **accountable**, that will not prosecute police when they harm, hurt black folks, when they violate the laws, you live in a community where police are part of the structure that is racking up mass incarceration, but many other aspects of our system are racking up mass incarceration, and district attorneys are at the center of it. You live in those communities and you need to do something about it. And so at winningjustice. org, we ' ve created the only searchable, national database on the 2, 400 prosecutors around the country. We ' re building local squads and communities for folks to be able to engage around efforts that hold DAs **accountable**. We ' ve worked with our partners across the movement, from our friends in Black Lives Matter, to folks who do **policy** work, to our friends at local ACLU chapters around the country, to build six **demands**. Six **demands** that folks can get behind in terms of pushing for reform, and then we ' ve built public education material. But the only way that we work to change the way that prosecution happens in this country is that if people get **involved**. If people raise their voice, if people join us in pushing for real change. At the end of the day, I want people to recognize though, and Phillip talked a little bit about this, is that people don ' t experience issues, they experience life. That the forces that hold us back are deeply interrelated, a racist criminal justice system requires a racist media culture to survive, a political inequality follows economic inequality, they all go hand in hand. And so I also want us to not take ourselves out of the equation. We likely work inside of corporations that may post symbols for Black Lives Matter one day, and then **support** politicians that work to destroy Black Lives Matter the next day. We oftentimes are engaged in practices inside of our companies or in our daily lives **supporting** media properties and others that are harming our communities, are telling stories. Recently, we produced a report at Color Of Change with the Norman Lear school at USC. It ' s called"Normalizing Injustice,"and it can be found at changehollywood. org. And”Normalizing Injustice”looks at the 22 crime procedurals, those crime shows on TV. And looks at all of the ways in which they, sort of, create a warped perception about our view of justice. They create sort of an incentive for the type of policing we see on the country, and actually serve as a PR arm for law enforcement. We ’ ve been working in writers rooms around the country to work to push folks to tell better stories, but we need folks to be both active listeners, and we need folks in the industry to push back and challenge those, not only the structures that lead to that content coming on the **air**, but the proliferation across our airwaves. At the end of the day, we have an opportunity in this moment to make change. Inflection points are those moments where we have an opportunity to make huge leaps forward, or the real, real threat of falling backwards. In our hands is the **ability** to do some incredible things about undoing so many of the injustices that have stood in the way of progress for far too long. But everyday people must get **involved**. We must channel that presence into power, and we must build the type of power that changes the rules. Racism in so many ways is like water pouring over a floor with holes in it. Every single – In every single way, it will find the holes. And so for us, we cannot simply accept charitable solutions to structural problems, but we actually have to work for structural change. And so I want to end by saying one thing about how we talk about black people and how we talk about black communities in this moment. Because we have to say what we mean, and we have to build the narrative that gets us to where we want to go. So far too often, we talk about black communities as vulnerable, we talk about black people as vulnerable, but vulnerability is a personal trait, black communities have been under attack. Black communities have been exploited, black communities have been targeted, and we need to say that, so we don ’ t put the onus on fixing black families and communities, but we put the onus on fixing the structures that have harmed us. We will say things like,”Black people are less likely to get loans from banks,”instead of saying that banks are less likely to give loans to black people. This is our opportunity to build the type of progress that makes real change, and at the center of this story, we need to show and elevate the images not just of the pain that we are facing, but of the joy, the brilliance and the creativity that black people have brought to this country. Black people are the protagonist of this story, and we need to make sure that as we work to build a new tomorrow, we **ensure** that the heroes are at the center of the liberation that we all need. Thank you. CA : Thank you, Rashad. Dr. Bernice King is the CEO of the King Center in Atlanta, Georgia. The center is a living memorial to her father, Dr. Martin Luther King Jr. It ’ s dedicated to inspiring new generations to carry his work forward. In this moment, when so many are hurting, how can we better approach unity and **collective** healing? Dr. King, over to you. Bernice King : My heart is a little heavy right now, because I was that six-year-old. I was five years old when my father was assassinated. And he did change the world. But the tragedy is that we didn ’ t hear what he was saying to us as a prophet to this nation. And his words are now reverberating back to us. Change, we all know, is necessary right now. And yet, it ’ s not easy. We know that there has to be changes in policing in this nation of ours. But I want to talk about America ’ s choice at a greater level. The prophet said to us,”We still have a choice today : nonviolent coexistence, or violent coannihilation.”What we have witnessed over the last eight days has placed that choice before us. We have seen literally in the streets of our nation people who have been following the path of nonviolent protest, and people who have been hell-bent on destruction. Those choices are now looking at us, and we have to make a choice. The history of this nation was founded in violence. In fact, my father said America is the greatest purveyor of violence. And the only way forward is if we repent for being a nation built on violence. And I ' m not just talking about physical violence. I ' m talking about systemic violence, I ' m talking about **policy** violence, I ' m talking about what he spoke of are the triple evils of poverty, racism and militarism. All violent. Albert Einstein said something to us. He said we cannot solve problems on the same level of thinking in which they were created. And so if we are going to move forward, we are going to have to deconstruct these systems of violence that we have set in America. And we ' re going to have to reconstruct on another foundation. That foundation happens to be love and nonviolence. And so, as we move forward, we can correct course if we make that choice that Daddy said, nonviolent coexistence. And not continue on the pathway of violent coannihilation. So what does that look like? That looks like some deconstruction work in order to get to the construction. We have to deconstruct our thinking. We ' ve got to deconstruct the way in which we see people and deconstruct the way in which we operate, practice and engage and set **policy**. And so I believe that there ' s a lot of heart, h-e-a-r-t work to do, in the midst of all the h-a-r-d, hard work to do. Because heart work is hard work. One of the things we have to do is we have to **ensure** that everyone, especially my white brothers and sisters, have to engage in the heart work, the antiracism work, in our hearts. No one is exempt from this, especially in my white community. We must do that work in our hearts, the antiracism work. The second thing is that I encourage people to look at the nonviolence training that we [have] at the King Center, thekingcenter. org, so that we learn the foundation of understanding our interrelatedness and interconnectedness. That we understand our loyalties and our commitments and our policy-making can no longer be devoted to one group of people, but has to be devoted to the greater good of all people. And so I ' m inviting people to even join us on our own line of protest that ' s happening every night at seven o ' clock pm on the King Center Facebook page, because so many people have things that they want to express and contribute to this. We all have to change and have to make a choice. It is a choice to change the direction that we have been going. We need a revolution of values in this country. That ' s what my Daddy said. He changed the world, he changed hearts, and now, what has happened over the last seven, eight years and through history, we have to change course. And we all have to participate in changing America with the true revolution of values, where people are at the center, and not profit. Where morality is at the center, and not our military might. America does have a choice. We can either choose to go down continually that path of destruction, or we can choose nonviolent coexistence. And as my mother said, struggle is a never-ending process, freedom is never really won. You earn it and win it in every generation. Every generation is called to this freedom struggle. You as a person may want to exempt yourself, but every generation is called. And so I encourage corporations in America to start doing antiracism work within corporate America. I encourage every industry to start doing antiracism work, and pick up the banner of understanding nonviolent change, personally, and from a social change perspective. We can do this. We can make the right choice to ultimately build the beloved community. Thank you. WPR : Thank you, Dr. King. Anthony Romero is the executive director of American Civil Liberties Union. As one of the nation ' s oldest social justice organizations, the ACLU has advocated for racial equality and shown deep **support** to the black community in moments of **crisis**. And in moments like these, black voices are almost always the loudest and at times the silence from our nonblack brothers and sisters can feel deafening. How we can bring our allies into the mix, to better **support** ending systemic violence and racism against the black communities, is a question top of mind for a lot of us. Anthony, welcome to the show, and thank you so much for being with us. Anthony Romero : Great. Thank you, thank you Whitney, thank you, Chris, for inviting me to join this **TED** community. I think community is really important right now. With so many of us feeling trepidation, the weariness, the anger, the fear, the frustration, the terrorism that we ’ ve experienced in our communities. This is a time to huddle around a virtual campfire, with your posse, with your family, with your loved ones, with your network. It ’ s not a time to be isolated or alone. And I think for allies in this struggle, those of us who don ’ t live this experience every day, it is time for us to lean in. You can ’ t change the channel, you can ’ t tune out, you can ’ t say,”This is too hard.”It is not that hard for us to listen and learn and heed. It is the only way we ’ re going to build out of this, by hearing the voices of Rashad, and Phil and Dr. King. By hearing the voices of our neighbors and loved ones, by hearing the voices on Twitter of people who we don ’ t know. And so white communities and allied organizations need to pay even closer attention. This is the test of your character. How willing are you to lean in and to engage. For me, I have – These have been really hard couple of weeks. I feel like this is really a test of whether or not we really believe in the American experiment. Do we really believe it? Do we really believe that out of many, one, a country with no unifying language, no unifying culture, no unifying religion, can we really become one people? All equal before the law? All bound together with a belief in the rule of law? Do we really believe that or do we just think it ’ s a nice saying to see on the back of a paper dollar? And for me, this is a referendum on the American experiment. On whether we really believe, and... the future is in our hands. And this is not like other **crises**, I ’ ve been the head of the ACLU for almost 20 years, I feel like I ’ ve seen it all. This is different. And this is different because it is cumulative, like Phil and Rashad and Dr. King told us, this is centuries of systemic discrimination, and the bill has come due. And it will continue to be due, and we will pay. Unless we really do something quite different. I have been scratching my head at the ACLU for the last week. We ’ ve been at this for 100 years. My organization has been working on this from its inception. In 1931, we were **involved** with this report about lawlessness in law enforcement. That was our first report that we got behind in 1931. We opened up our first door fronts after the riots in Watts, so that we can bring legal services and lawyers to the communities so they could **demand** justice from the police departments. You know, we brought Miranda, you know, the right to remain silent, and we brought Gideon, the right to a court-appointed attorney if you can ’ t afford one. We fought Bloomberg on”stop-and-frisk,”it took him years and he lost in front of our litigation to finally apologize. We ’ ve been at this for 100 years. And for the communities that have lived this for 400 years, God. I ’ ve been scratching my head, thinking. It ain ’ t working. We don ’ t need another pattern and practice lawsuit. We don ’ t need another training program on racial bias or implicit bias in police departments, we don ’ t need to file another lawsuit on qualified immunity, we don ’ t need to, kind of, bring another race discrimination or **gender** discrimination lawsuit to integrate the police department. Yeah, we ’ ve done that and we will continue to do that. For me, where I ’ ve come, is that we need to defund the budgets of these police departments. It ’ s the only way we ’ re going to take the power back. And the more I read over the last couple of weeks about where this country is, the more I ’ m clear that that is my North star at the moment. We will continue to bring the litigation on qualified immunity, we will do the efforts to hold unaccountable law enforcement officials **accountable**, we will bring pattern and practice lawsuits, because the justice department is not doing that, so we will continue to do all that good work. But the real thing is, we ’ re going to go after those budgets. When you look at the fact that we spend 100 million dollars on policing, more than incarceration, that the city of Minneapolis spent 30 percent of their budget on policing. The city of Oakland, 41 percent on policing. That when you have New York City police department spend more money on policing than it does on housing and preservation development, community youth services, homelessness. We ’ re going after the money. And that ’ s hard-core advocacy. Bills drop in local legislatures to cut the funding for police, to stop these programs that give the federal military surplus to police departments, so they become, like, little mini armies, these don ’ t look like police officers, these look like standing armies. And the enemy are communities of color. So we need to take away their toys. We need to cut their budgets. We need to shrink the police infrastructure, so that we can get police out of the quotidian lives of people of color and communities of color. The ubiquitousness of police enforcement on things that the police do not have a role, should not have a role to play. People should not lose their lives over whether or not a cigarette pack has a proper tax stamp, or whether a 20-dollar bill was forged or not. That ’ s not worthy of spending our dollars on police. Get them out of that business, let ’ s focus on the most important and the most serious of crimes, and that ’ s it. That ’ s it. We ’ re going to depolice our communities. Shrink those budgets. We ’ re going to reinvest those moneys in local communities, it will be like water on stone campaigns, local legislatures, local city counsels, lab report cards, for people who talk out of both sides of their mouths and say,”We believe in police reform,”and yet, they ’ re still going to vote for 30 or 40 percent for the police? We ’ re going to put that right to the public. And I think we just have to stay at it, because I think that ’ s the only way we can get at this in a different way. Because much of what we tried to do is just simply not working. You know, with that, I struggle with, how do you find the optimism in this moment, because you have to find the optimism. You have to find the way to still think that even though on the face of so many setbacks, there ’ s been change. It ’ s been too little, too slow, not enough. We need to kind of, rock it, boost it. But you can ’ t lose sight of the optimism. And you know, I ’ ve been thinking about who are the folks inspiring me, and Dr. King ’ s father, of course, and the words of Rashad and Patrisse Cullors and others have inspired me. But I find inspiration in the words of a scholar I really don ’ t like bringing up, Sam Huntington, kind of often criticized as being a **conservative**, a racist. But sometimes you can find inspiration even in your enemy ’ s words. And in one of his books, which I pulled off the shelf I have, he writes about how America is a disappointment, because it failed to live up to its aspirations. And he actually started talking about, America is a failure because it doesn ’ t live up to its ideals. But it ’ s not a failure, it ’ s not a bunch of lies. It ’ s a disappointment. And in the disappointment also is the fact that there ’ s hope. I ’ m paraphrasing it, but I think we have to kind of, wrap all of that together, and think about the disappointment and the hope and the resolve to do better. And we need to listen and lean in, and I thank the **TED** community, I thank Dr. King, I thank Rashad, I thank Phil. Thank you. CA : Wow. Thank you to all four of you, that was astonishing. I guess we ’ re bringing everyone back now to have a conversation among us, to answer questions from our community, I hope you ’ re entering those questions. So I don ’ t know whether we can bring back our guests onto the screen at this point. Welcome back. Let me start with a question to you, Dr. King, I was so inspired by what you said. Your father, of course, also deeply understood the anger that leads to protest. I think he said that protests are the language of the unheard. And I ’ m wondering what you would say to someone right now who is angered beyond measure by what ’ s happened, and also sees this could be the moment, you know, like, someone who believes the system is so fundamentally broken, that our best choice is to tear it down, that that is actually – This may be a once-in-a-generation moment to do that. And so to actually believe that protest, including violent protest, actually is the way right now. What would you say to someone who felt that? BK : First, I just wanted to make just a slight correction, he said riots are the language of the unheard. CA : I apologize, I apologize. But that is the point even more powerfully, yeah. BK : Yes. Protest we must, and we must continue to always protest, to keep the issues in the awareness before people, but you know, when a person is angry, sometimes it ' s hard to reach them. I ' ve been on that journey, I was at a stage of my life where I was so angry, I wanted to destroy, and I ' m the daughter of Martin Luther King Jr., and grew up in a household of love and nonviolence and forgiveness, and I had to go through that journey I was surrounded by the right kind of influences, fortunately. Because that would have been a sad story. But I think it ' s really allowing ourselves to hear the anger and allowing the space for the anger, but also trying to help young people rechannel that energy. And we ' ve got to start **ensuring** that we connect them to some of the work that has been and now is elevated to another place. Color Of Change, the work that you ' re doing, the ACLU, the work that they ' re doing, because sometimes, there ' s this disconnection that intensifies the emotion and makes you feel helpless. But if you can channel that anger, connect it with action that is toward creating the social and economic change, then it begins to build you up, and then you can begin to become more constructive with the anger. WPR : We have some questions that are coming in from our community, but before we do that, you all shared such powerful, meaningful statements right now, and many of you touched on the fact that this is not the first time that we ' re experiencing this. The murder of George Floyd, Ahmaud Arbery, Breonna Taylor, this is one of – these are three of many, many instances just like this, and I ' d love to hear you all address was has, sort of, brought us to this boiling point, what has contributed to this moment where we ' re now experiencing things, Anthony, as you said, that feels so much worse than other moments? And that ' s it, anyone who feels comfortable to take that question. AR : Rashad, I want to hear you. BK : I wanted to say something. I think we ' ve always been at that moment. But this moment is different, because of the void in **leadership**. There ' s no real moral voice in our country, and the person who sits in the office of the presidency is not, you know, leading in the right way. And has kind of – no, not kind of, has given license to certain things. And so now it ' s, you know, he ' s lit the fires. WPR : Yeah. RR : The thing I ' ll add... The thing I ' ll add here is, you know, a couple of things. For the last couple of months, we have been both seeing and experiencing all of the ways that this country ' s decisions of underinvestment, of targeting black communities, has been killing black people through COVID. And while we ' ve been in our homes, we have also been watching how the media has blamed us as we have been the essential workers in so many places and trying to **ensure** that this country keeps going. We ' ve watched white men with guns show up to capitols **demanding**, basically, black and brown people go back to work. And then we see this eight-minute video with a police officer, with his knee on someone ' s neck, after seeing that video of Ahmaud Arbery and hearing the story of Breonna Taylor, and we see him looking in the camera, basically knowing that America was not going to punish him. And what I think it is is that it ' s just enough is enough, that people didn ' t feel that they had a channel for that outrage, and because people had been inside, and because people had been experiencing all the ways in which the structures had also been colluding to kill us, that what we ' re seeing is alignment of all of those things, where people are making **demands** that are much bigger and much bolder than before. And we recognize that while we don ' t have **leadership** at the federal level, we also have to recognize that no political party can say that they have been 100 percent, neither political party can say they ’ ve been 100 percent on the right side of all these issues. And so people are mobilizing, they are fighting back like never before, and in some ways, people are unwilling to accept answers like,”Just go vote,”or,”Just participate in the process,”because we recognize that black people have been voting, black people have been part of voting, and part of **ensuring** that. And so that I think is why this moment feels so much different, combined with, for the last seven years, since Trayvon, we have seen the growth of a new movement of activists and **leaders** all around the country, who are also in a very different place to be able to move the needle on so much of what ’ s possible. CA : We have a question here from Genesis Be. If we can get that up here.”Here in Mississippi, the police is synonymous with the Klan, historically. How do we purge law enforcement of white supremacists?”PAG : So I guess that ’ s partially to me, being the psychologist of bias. I ’ ll say that just yesterday we had an officer in Denver who posted on social media himself and two other officers saying,”Let ’ s go start a riot.”He was fired that day. I worry about all the officers that the FBI has now, for almost half a decade, been warning us, law enforcement and unions being infiltrated by white supremacists. And all the officers that have social media accounts, but they ’ re private. You know, the Invisible Institute has put some things forward. We ’ re not talking seriously about the domestic terrorism threat that white supremacy represents. So the first thing that we ’ ve got to do is we ’ ve got to take it seriously. We have to actually say out loud, and I can ’ t believe that on a day like today, or a week like this week, I have to say out loud, white supremacy is alive and well and a driving force of American politics. This shouldn ’ t be controversial. I shouldn ’ t be looking forward to getting hate mail in my inbox for it, but that ’ s the reality. So the first part of solving a problem is acknowledging that it exists. But the second thing is we need to arm municipalities, that ’ s law enforcement, but even more so communities, with the **ability** to take action when someone violates their values. Right now, I think about the case in Philadelphia where Charles Ramsey, when he was a commissioner there, fired six officers, right. Concerns about racial bias and concerns about police brutality, and those six officers were back on the same job inside of three months. We now have a law enforcement system that says you can lose your job in one jurisdiction, and get the same job as law enforcement in another jurisdiction. And without the national registry and the capacity for law enforcement to make different decisions, we ’ re going to have this exact problem, not just in Mississippi, but in Minneapolis, and Louisville and New York and LA. CA : Phil, how much of the problem stems from the fact that police unions have a huge amount of power to protect and sometimes reinstate so called bad apple officers? PAG : Yeah, I ’ m getting this question a lot, and police unions are one of the most labor forces of the United States, and are unique within the labor movement, right? So it ’ s police unions and teachers ’ unions are the two largest and could not be two different groups of folks. When I talk to union **leadership**, that ’ s the **leadership** what wants to talk to Dr. Blackenstein, right, when I talk to union **leadership**, what they say is no one hates a bad officer more than a good officer. But the union contracts, the new negotiations, don ’ t look like that ’ s true. What they look like is anybody gets in trouble, and the union ’ s only job is to make sure whatever officer is in trouble, gets to maintain their job. The perverse incentive here is that when people run for union **leadership**, no one can run saying,”These people shouldn ’ t be in the union.”It ’ s very hard to do that. What you can run on is say,”If this person didn ’ t protect you enough, I ’ ll protect you even more. The bigots? I ’ ll protect even them.”So we have this perverse incentive where union **leadership** ends up not really representing the values even of the rest of the union members. But they have massive, outsized **negotiating** power. So yes, engaging with and appropriate rightsizing of labor protections, for folks whose jobs are difficult, but who should not be protected from the basic values of human rights, human dignity and public safety. It ' s got to be part of the process. I mean, when unions are **negotiating** a two-year ban on keeping of records, so that there ' s no **ability** even to trace what ' s happening in the state of California, historically in terms of police misconduct, that ' s not in the interest of public safety, public legitimacy, or our democracy. AR : Yeah, the thing I would add, Chris, is that I think the labor union piece is a critically important one to think through. Because I think, like Phil said, they are a key part of the puzzle that we have to solve for. And you know, it ' s frustrating when you look at a place in Minneapolis, and Phil knows better than I, but when mayor Jacob Frey, the one who ' s on TV all the time, saying many of the right things that you want an elected official to say at times like this, when he banned his police department from attending the"warrior training"that was being offered, it was the Minneapolis Police Federation, local union that defied him and sent their police to the training. And so we need to really be clear that we need to have the police forces under civilian control. I know this sounds so elementary, I feel like I ' m talking about a Latin American, kind of, totalitarian context, but we need to exert civilian control of our police in a way that we have yet not been able to think through, and a key part of that is the labor unions of the police. And there are moments when you can find common ground. When we brought one of our COVID-related lawsuits to deal with the outbreak of the **pandemic** in a Maryland jail, we worked really hard, I worked the **phones** with the head of police unions. We got one of the local unions to serve as plaintiff in our lawsuit. Because we understood that the incarcerated folks who were being denied access to masks, social distancing and the conditions and lack of testing, and lack of PP, that the people who were also going to be in harm ' s way were going to be the guards as well. And they were going to be the vectors, communicating the disease out into the community. So if you can find ways of bringing that relationship. But make no mistake, when you go after their budgets, and you start taking away kind of, their munitions, and their seat at the budgeting table, oh, are you going to have a battle on your hands, right? And we have to think about also as we shrink the budgets for police, how do we – we deploy people in the police departments to other meaningful jobs, right? Because you can ' t just throw them out into the street, and say,"You ' re on your own, you ' re homeless, good luck to you."That ' s not a way to deal with redemption. So we have to really think about all these pieces in a much more cohesive way. WPR : We have another question here from the audience. From Paul Rucker :"The end of summer of 1919 was followed by the Tulsa Race Massacre, the Johnson-Reed Anti-Immigration Act of 1924, and also the rise of the KKK. Is there a possibility that white supremacy will get stronger if we don ' t seize this opportunity?"Rashad, I think this might be something that would be great to hear your perspective on, working so deeply in activism. RR : I ' m having a little trouble hearing. WPR : Oh, I ' m so sorry. RR : No, it ' s OK. CA : Can you read the question on the screen, Rashad? RR : Oh, I heard that, I heard you. WPR : Yes, I think it might just be my mic is having some issues here."Is there a possibility that white supremacy will get stronger if we don ' t seize this opportunity."Yes, absolutely yes. You know, to be clear, right, if we don ' t have the right diagnosis of white supremacy, if we think of white supremacy as just hoods, if we think of white supremacy as just folks who are operating, you know, with, you know – in some of these underground networks that have grown, if we ' re just thinking about white supremacy and white nationalism as people who marched with tiki torches in Charlottesville, then we will really mistake all the ways in which our systems and structures have white supremacy embedded, and allow for something like a Tulsa Race Massacre to happen, something like anti-immigration to happen, but on a day-to-day basis allow for the targeting of black communities through predatory practices by banks. The targeting of black communities through predatory practices like bail. A whole set of systems that can be produced day in and day out. We live in a country where the rules are far too often designed in ways that create a caste system, that create a different standard for some over others, and so when I talked about the inflection point, right, of this moment where something could really go forward and something could turn backwards, we are seeing this right now with this current president. And as we look at what could be happening with the next election, we have to be very, very clear that Donald Trump doesn ' t just operate on his own. He ' s enabled by big corporations who benefit from him being in office, and so continue to turn a blind eye to all the things that he does. They may post"Black Lives Matter,"but they show up to the White House and engage with Donald Trump. And then we have a whole set of politicians that may sometimes say that he said something that was wrong, but then allow for – but **support** his platform in other ways. You know, true co-conspiracy in the effort to dismantle white supremacy and white nationalism is not a thing that people can do on vacation. It is a 365-day project of us constantly working to dismantle all of the structures that have been put in harm ' s way. The final thing I will just add, because someone mentioned about police unions, and I want to just add that one of the problems with police unions, and many of us have been in this position, I think, is that I have shown up to the table with police unions on many occasions. I remember going to the White House during the last administration and being around a table as we were talking about policing and police reform. And having members of the Fraternal Order of Police **leadership** say things like,"All of this talk of racial profiling is new to us."It is one thing for folks to not agree with you on the **policy** reforms necessary. It is another thing for people to say that our **demands** are too aspirational. It ' s another thing to be gaslit and told the problem doesn ' t actually exist at all. And that is what we are dealing with, and so we have to actually change the way that people see these institutions. Politicians who say that they are on the side of justice and reform, can no longer take money from police-only unions and Fraternal Order of Police. We actually have to create a new standard, a new litmus test of what does it mean to actually be with us. You can ' t just sing our songs, use our hashtags and march in our marches, if you are on the other end of **supporting** the structures that put us in harm ' s way, that literally kill us. And this is the opportunity for white allies to actually stand up in new ways. To be the type of ally, to do the type of allyship and the type of work that truly dismantles structures, not just provides charity. PAG : And I ' ve got to add to that – so, Paul, thank you for the question, but we ' re in a moment where people are looking at what ' s happening on the street, as if a week and a half ago we weren ' t in a midst of a global **pandemic** as the greatest news story, the biggest new story going on. One of the things I ' m most worried about, and have been worried about since the beginning, what I ' ve been talking to our chiefs about, is say, you must be out of the social distancing policing game. You can ' t be the ones doing that, and the reason is this : We ' re in a moment where creating scapegoats and enemies and others is incredibly politically advantageous for at least one side. And there is deliberate efforts to do exactly that. And we ' ve seen that black communities are two and three and four times more likely to contract this **virus**, which feels like the manifestation of racial discrimination, because it is. But very soon, that ' s going to look like black people made bad choices and they need to stay away from us. And when that happens, that ' s when law enforcements get used to regulate where black movement can be. We used to call it sundown towns, I don ' t know what we ' re going to call it when it ' s around COVID. But it ' s coming. I ' m already seeing that on communities like Nextdoor, and on Facebook groups. People who don ' t think of themselves as white supremacists but just want the disease away, and the disease has a black and brown face. So we ' re not only dealing with a moment of generational tension, between black communities and law enforcement, we ' re dealing with a moment when people are looking for scapegoats, and black people ' s vulnerability has always been our greatest casting note for being cast as scapegoats. So for folks who are worried about this, this is not inevitably a moment for change and reform and enlightenment and America ' s best values, because historically, these have been precisely the moments when regression back to white supremacy has reigned supreme. So let ' s not just look at **everybody** signaling. I don ' t want to just see black and white cops on their knees, I want to see the **policies**. I want to see the things that will prevent this kind of thing from moving to the next stage. CA : Rashad, I want to respect the fact that you ' ve got a hard stop at one. And so I just want to thank you for your participation in this. If you ' ve got a couple of final words you want to share, that would be great, and then if it ' s OK for the other three, I think there ' s just a couple other questions I ' d love to put and continue this conversation for just a moment longer, if possible. Rashad, any closing words? RR : The thing I want to say is that now is the time for action. And I want to invite people in to join us at Color Of Change to make justice real. And in so many ways, you can **visit** us at Color Of Change, you can take action. Five, 10, 15 years from now, we will be dealing with the **impacts** of what we did or didn ' t do in this moment. How we stood up and how hard we were willing to fight. And as the other speakers have said, now is not the time for reform around the edges, now is time for dismantling the **policies** and practices that have held us back, and championing solutions and new rules that will move us forward. And so we hope that you will do something, whether it ' s with us, or whether it ' s with local organizations in your community, or other groups around the country. But this is an opportunity to make change, and I believe that we can make justice real, if we find the passion and the energy to work together to achieve it. So thank you all for having me, and I hope that we have an opportunity to build, not just online, but offline, in the months to come. CA : Thanks so much, Rashad. We ' re just going to ask this last question of you. This one is from David Fenton."How can the movement unite around a clear, simple platform of **policies** to enshrine in legislation? Like making all complaints against cops public, banning all choke holds, **ensuring** independent review boards, etc.?"WPR : That seems like a great place for you to chime in, Dr. King, if you have some thoughts on that. PAG : I think that was to go to you, Dr. King. BK : Oh, OK. You know, this may sound simplistic, but it ' s a Nike thing, I think we have to just do it, we have to see our work as interconnected. I think there ' s been efforts towards people working in that vein, but we have to intensify that. And in doing it, one of the things that my father said, and I know people sometimes get tired of hearing me say,"My father said,"but I just think, I wish, should I say, we had really listened to him, because we wouldn ' t be on this platform right now having this type of conversation. But he left something with us, sort of a blueprint in"Where Do We Go From Here : Chaos or Community?", his book, and he said, going forward, the meddlesome task is to organize our strength into compelling power. And that is so key, because oftentimes we organize merely around passion. But people have certain areas of strength and talent and giftedness, and we ' ve got to figure out how to build our coalitions based on these strengths. You know, people do different things well. And so, in order to unite in an effective way that they might not elude the **demand** that we ’ re making, I think that ’ s what ’ s going to happen. People have to do their own personal assessment within their organization, I call it a SWAT analysis. And then that SWAT analysis has to happen across organizations, so that we can make sure that we are moving in a united manner, off of the strengths that each organization brings, so that we can maximize the **impact** and the effectiveness to do things like this, in terms of getting the legislation in place that is needed in this hour. CA : Thanks so much. Just quick closing words from you, Anthony, and then from you, Phil. Anthony. AR : You know, I would just say that what gives me hope are the young folk. You have to believe that among this group, this groups of young ’ uns, seeing what they ’ re seeing, living with this president, with these instincts, seeing the continued indifference that mainstream communities have given to issues of racial justice, or economic justice, you ’ ve got to believe that what comes out of this very hot fire, is something even more powerful and strong than we ’ ve ever seen before. That ’ s what gets me through the hard days that we ’ re now experiencing, this thinking, there is another Dr. King among the young ’ uns, Dr. King. And I have to believe that what they ’ re seeing and what they ’ re witnessing and their righteous indignation and their frustration and their anger is going to be miraculously a beautiful blossoming of a new opportunity, of a new change. This generation will take us there, I have to believe that. My generation has failed them miserably. So I ’ m just looking forward to the new ones. CA : Thank you, Anthony. WPR : Phil. Thank you, Anthony. PAG : So it really has been a privilege to be on with you all. To David ’ s question, let me say that a number of civil rights organizations, I believe the ACLU among them, CPE, Center for Policing Equity, and hundreds more have signed on to principles for legislation that would include eight pillars. It ’ s been led by the Leadership Conference for Civil and Human Rights. And it includes a federal ban on choke holds, it includes a national registry for officers who have engaged in misconduct. I also think that it ’ s important at this moment to get, we ’ ve got law enforcement ’ s chiefs of major cities willing to say, if we emerge from this moment and our profession hasn ’ t changed, then we have failed again. So it ’ s a critical time to get behind, I would direct you to LCCHR ’ s website for the eight pillars, because I won ’ t remember them all right now, and to start calling your local law enforcement, and say,”Yes, own that.”You should be signing on, they should be going public with letters that do all of that. But I ’ ll also say this. For a path forward in the principles, I ’ ll end where I started, which is that this is bigger than policing. These are the unpaid debts owed to black communities for stolen labor, owed to native communities for stolen land, for stolen culture, for years taken away and for lives lost in it. This is bigger than policing. If we don ’ t understand the size of it, then there ’ s no solution that ’ s really, truly proportional to the moment. But in this moment, when we ’ re seeing trillion of dollars in bailouts, mostly for corporations, it is absolutely a time when we can do things that normally, people could pretend that ’ s too much, it ’ s too big, we can ’ t. We have literally all the money in the world that can be spent and directed towards making us the society we pretend to be, before moments like this happen. And so the thing that gives me hope is that the lies have to be obvious now. The lies have to be, that was a reasonable use of force. The lie has to be, we don ’ t have the money. The lie has to be, that ’ s too hard, it ’ s too big of a challenge. This **stuff** feels impossible every day except today, because the alternative is we lose everything. Everything is at stake, our democracy is at stake, the people we choose to be, we **claim** to be, that ’ s at stake. And in the face of that, I think we can do impossible things. I think we can be mighty. So my hope for all of us is first, that we wake up tomorrow with more peace in the evening than war, and that we hold on to what ’ s possible from this moment at the same time that we hold on to the size of the task in front of us. I don ’ t want to come with half measures out of this. I don ’ t want to come out with radicalized youth and indifferent aged... I don ’ t know what the contrast... The radicalized youth and indifferent people who are old like me. I want to come out with a unified country that understands that the costs that we owe are big, and our pockets are deep enough to match it. CA : Wow. Thank you to each of you for extraordinary eloquence. Really, so powerful. This conversation, obviously, continues, I know that there ’ s many people listening, you have other questions, this, I think, from **TED** ’ s point of view is just the start of the conversation. To the extent that our job is to amplify the voices that matter, we couldn ’ t be prouder to be amplifying further your extraordinary voices. So thank you for being part of this today. PAG : Thank you. WPR : Thank you all.

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It ’ s a great honor to be here with you. The good news is I ’ m very aware of my responsibilities to get you out of here because I ’ m the only thing standing between you and the bar. And the good news is I don ’ t have a prepared speech, but I have a box of slides. I have some pictures that represent my life and what I do for a living. I ’ ve learned through experience that people remember pictures long after they ’ ve forgotten words, and so I hope you ’ ll remember some of the pictures I ’ m going to share with you for just a few minutes. The whole story really starts with me as a high school kid in Pittsburgh, Pennsylvania, in a tough neighborhood that **everybody** gave up on for dead. And on a Wednesday afternoon, I was walking down the corridor of my high school kind of minding my own business. And there was this artist teaching, who made a great big old ceramic vessel, and I happened to be looking in the door of the art room — and if you ’ ve ever seen clay done, it ’ s magic — and I ’ d never seen anything like that before in my life. So, I walked in the art room and I said,”What is that?”And he said,”Ceramics. And who are you?”And I said,”I ’ m Bill Strickland. I want you to teach me that.”And he said,”Well, get your homeroom teacher to sign a piece of paper that says you can come here, and I ’ ll teach it to you.”And so for the remaining two years of my high school, I cut all my classes. But I had the presence of mind to give the teachers ’ classes that I cut the pottery that I made, and they gave me passing grades. And that ’ s how I got out of high school. And Mr. Ross said,”You ’ re too smart to die and I don ’ t want it on my conscience, so I ’ m leaving this school and I ’ m taking you with me.”And he drove me out to the University of Pittsburgh where I filled out a college application and got in on probation. Well, I ’ m now a trustee of the university, and at my installation ceremony I said,”I ’ m the **guy** who came from the neighborhood who got into the place on probation. Don ’ t give up on the poor kids, because you never know what ’ s going to happen to those children in life.”What I ’ m going to show you for a couple of minutes is a facility that I built in the toughest neighborhood in Pittsburgh with the highest crime rate. One is called Bidwell Training Center ; it is a vocational school for ex-steel workers and single parents and welfare mothers. You remember we used to make steel in Pittsburgh? Well, we don ’ t make any steel anymore, and the people who used to make the steel are having a very tough time of it. And I rebuild them and give them new life. Manchester Craftsmen ’ s Guild is named after my neighborhood. I was adopted by the Bishop of the Episcopal Diocese during the riots, and he donated a row house. And in that row house I started Manchester Craftsmen ’ s Guild, and I learned very quickly that wherever there are Episcopalians, there ’ s money in very close proximity. And the Bishop adopted me as his kid. And last year I spoke at his memorial service and wished him well in this life. I went out and hired a student of Frank Lloyd Wright, the architect, and I asked him to build me a world class center in the worst neighborhood in Pittsburgh. And my building was a scale model for the Pittsburgh airport. And when you come to Pittsburgh — and you ’ re all invited — you ’ ll be flying into the blown-up version of my building. That ’ s the building. Built in a tough neighborhood where people have been given up for dead. My view is that if you want to **involve** yourself in the life of people who have been given up on, you have to look like the solution and not the problem. As you can see, it has a fountain in the courtyard. And the reason it has a fountain in the courtyard is I wanted one and I had the checkbook, so I bought one and put it there. And now that I ’ m giving speeches at **conferences** like **TED**, I got put on the board of the Carnegie Museum. At a reception in their courtyard, I noticed that they had a fountain because they think that the people who go to the museum deserve a fountain. Well, I think that welfare mothers and at-risk kids and ex-steel workers deserve a fountain in their life. And so the first thing that you see in my center in the springtime is water that greets you — water is life and water of human possibility — and it sets an attitude and expectation about how you feel about people before you ever give them a speech. So, from that fountain I built this building. As you can see, it has world class art, and it ’ s all my taste because I raised all the money. I said to my boy,”When you raise the money, we ’ ll put your taste on the wall.”That we have quilts and clay and calligraphy and everywhere your eye turns, there ’ s something beautiful looking back at you, that ’ s deliberate. That ’ s intentional. In my view, it is this kind of world that can redeem the soul of poor people. We also created a boardroom, and I hired a Japanese cabinetmaker from Kyoto, Japan, and commissioned him to do 60 pieces of furniture for our building. We have since spun him off into his own business. He ’ s making a ton of money doing custom furniture for rich people. And I got 60 pieces out of it for my school because I felt that welfare moms and ex-steel workers and single parents deserved to come to a school where there was handcrafted furniture that greeted them every day. Because it sets a tone and an attitude about how you feel about people long before you give them the speech. We even have flowers in the hallway, and they ’ re not plastic. Those are real and they ’ re in my building every day. And now that I ’ ve given lots of speeches, we had a bunch of high school principals come and see me, and they said,”Mr. Strickland, what an extraordinary story and what a great school. And we were particularly touched by the flowers and we were curious as to how the flowers got there.”I said,”Well, I got in my car and I went out to the greenhouse and I bought them and I brought them back and I put them there.”You don ’ t need a task force or a study group to buy flowers for your kids. What you need to know is that the children and the adults deserve flowers in their life. The cost is incidental but the gesture is huge. And so in my building, which is full of sunlight and full of flowers, we believe in hope and human possibilities. That happens to be at Christmas time. And so the next thing you ’ ll see is a million dollar kitchen that was built by the Heinz company — you ’ ve heard of them? They did all right in the ketchup business. And I happen to know that company pretty well because John Heinz, who was our U. S. senator — who was tragically killed in a plane accident — he had heard about my desire to build a new building, because I had a cardboard box and I put it in a garbage bag and I walking all over Pittsburgh trying to raise money for this site. And he called me into his office — which is the equivalent of going to see the Wizard of Oz — and John Heinz had 600 million dollars, and at the time I had about 60 cents. And he said,”But we ’ ve heard about you. We ’ ve heard about your work with the kids and the ex- steel workers, and we ’ re inclined to want to **support** your desire to build a new building. And you could do us a great service if you would add a culinary program to your program.”Because back then, we were building a trades program. He said,”That way we could fulfill our affirmative action goals for the Heinz company.”I said,”Senator, I ’ m reluctant to go into a field that I don ’ t know much about, but I promise you that if you ’ ll **support** my school, I ’ ll get it built and in a couple of years, I ’ ll come back and weigh out that program that you desire.”And Senator Heinz sat very quietly and he said,”Well, what would your **reaction** be if I said I ’ d give you a million dollars?”I said,”Senator, it appears that we ’ re going into the food training business.”And John Heinz did give me a million bucks. And most importantly, he loaned me the head of research for the Heinz company. And we kind of borrowed the curriculum from the Culinary Institute of America, which in their mind is kind of the Harvard of cooking schools, and we created a gourmet cooks program for welfare mothers in this million dollar kitchen in the middle of the inner city. And we ’ ve never looked back. I would like to show you now some of the food that these welfare mothers do in this million dollar kitchen. That happens to be our cafeteria line. That ’ s puff pastry day. Why? Because the students made puff pastry and that ’ s what the school ate every day. But the concept was that I wanted to take the stigma out of food. That good food ’ s not for rich people — good food ’ s for **everybody** on the planet, and there ’ s no excuse why we all can ’ t be eating it. So at my school, we subsidize a gourmet lunch program for welfare mothers in the middle of the inner city because we ’ ve discovered that it ’ s good for their stomachs, but it ’ s better for their heads. Because I wanted to let them know every day of their life that they have value at this place I call my center. We have students who sit together, black kids and white kids, and what we ’ ve discovered is you can solve the race problem by creating a world class environment, because people will have a tendency to show you world class behavior if you treat them in that way. These are examples of the food that welfare mothers are doing after six months in the training program. No sophistication, no class, no dignity, no history. What we ’ ve discovered is the only thing wrong with poor people is they don ’ t have any money, which happens to be a curable condition. It ’ s all in the way that you think about people that often determines their behavior. That was done by a student after seven months in the program, done by a very brilliant young woman who was taught by our pastry chef. I ’ ve actually eaten seven of those baskets and they ’ re very good. They have no calories. That ’ s our dining room. It looks like your average high school cafeteria in your average town in America. But this is my view of how students ought to be treated, particularly once they have been pushed aside. We train pharmaceutical technicians for the pharmacy industry, we train medical technicians for the medical industry, and we train chemical technicians for companies like Bayer and Calgon Carbon and Fisher Scientific and Exxon. And I will guarantee you that if you come to my center in Pittsburgh — and you ’ re all invited — you ’ ll see welfare mothers doing analytical chemistry with logarithmic calculators 10 months from enrolling in the program. There is absolutely no reason why poor people can ’ t learn world class technology. What we ’ ve discovered is you have to give them flowers and sunlight and food and expectations and Herbie ’ s music, and you can cure a spiritual cancer every time. We train corporate travel agents for the travel industry. We even teach people how to read. The kid with the red stripe was in the program two years ago — he ’ s now an instructor. And I have children with high school diplomas that they can ’ t read. And so you must ask yourself the question : how is it possible in the 21st century that we graduate children from schools who can ’ t read the diplomas that they have in their hands? The reason is that the system gets reimbursed for the kids they spit out the other end, not the children who read. I can take these children and in 20 weeks, demonstrated aptitude ; I can get them high school equivalent. No big deal. That ' s our library with more handcrafted furniture. And this is the arts program I started in 1968. Remember I ' m the black kid from the ' 60s who got his life saved with ceramics. Well, I went out and decided to reproduce my experience with other kids in the neighborhood, the theory being if you get kids flowers and you give them food and you give them sunshine and enthusiasm, you can bring them right back to life. I have 400 kids from the Pittsburgh public school system that come to me every day of the week for arts education. And these are children who are flunking out of public school. And last year I put 88 percent of those kids in college and I ' ve averaged over 80 percent for 15 years. We ' ve made a fascinating discovery : there ' s nothing wrong with the kids that affection and sunshine and food and enthusiasm and Herbie ' s music can ' t cure. For that I won a big old plaque — Man of the Year in Education. I beat out all the Ph. D. ' s because I figured that if you treat children like human beings, it increases the likelihood they ' re going to behave that way. And why we can ' t institute that **policy** in every school and in every city and every town remains a mystery to me. Let me show you what these people do. We have ceramics and photography and computer imaging. And these are all kids with no artistic **ability**, no talent, no imagination. And we bring in the world ' s greatest artists — Gordon Parks has been there, Chester Higgins has been there — and what we ' ve learned is that the children will become like the people who teach them. In fact, I brought in a mosaic artist from the Vatican, an African-American woman who had studied the old Vatican mosaic techniques, and let me show you what they did with the work. These were children who the whole world had given up on, who were flunking out of public school, and that ' s what they ' re capable of doing with affection and sunlight and food and good music and **confidence**. We teach photography. And these are examples of some of the kids ' work. That boy won a four-year scholarship on the strength of that photograph. This is our gallery. We have a world class gallery because we believe that poor kids need a world class gallery, so I designed this thing. We have smoked salmon at the art openings, we have a formal printed invitation, and I even have figured out a way to get their parents to come. I couldn ' t buy a parent 15 years ago so I hired a **guy** who got off on the Jesus big time. He was dragging **guys** out of bars and saving those lives for the Lord. And I said,"Bill, I want to hire you, man. You have to tone down the Jesus **stuff** a little bit, but keep the enthusiasm. I can ' t get these parents to come to the school."He said,"I ' ll get them to come to the school."So, he jumped in the van, he went to Miss Jones ' house and said,"Miss Jones, I knew you wanted to come to your kid ' s art opening but you probably didn ' t have a ride. So, I came to give you a ride."And he got 10 parents and then 20 parents. At the last show that we did, 200 parents showed up and we didn ' t pick up one parent. Because now it ' s become socially not acceptable not to show up to **support** your children at the Manchester Craftsmen ' s Guild because people think you ' re bad parents. And there is no statistical difference between the white parents and the black parents. Mothers will go where their children are being celebrated, every time, every town, every city. I wanted you to see this gallery because it ' s as good as it gets. And by the time I cut these kids loose from high school, they ' ve got four shows on their resume before they apply to college because it ' s all up here. You have to change the way that people see themselves before you can change their behavior. And it ' s worked out pretty good up to this day. I even stuck another room on the building, which I ' d like to show you. This is brand new. We just got this slide done in time for the **TED Conference**. I gave this little slide show at a place called the Silicon Valley and I did all right. And the woman came out of the audience, she said,"That was a great story and I was very impressed with your presentation. My only criticism is your computers are getting a little bit old."And I said,"Well, what do you do for a living?"She said,"Well, I work for a company called Hewlett-Packard."And I said,"You ' re in the computer business, is that right?"She said,"Yes, sir."And I said,"Well, there ' s an easy solution to that problem."Well, I ' m very pleased to announce to you that HP and a furniture company called Steelcase have adopted us as a demonstration model for all of their technology and all their furniture for the United States of America. And that ' s the room that ' s initiating the relationship. We got it just done in time to show you, so it ' s kind of the world debut of our digital imaging center. I only have a couple more slides, and this is where the story gets kind of **interesting**. So, I just want you to listen up for a couple more minutes and you ' ll understand why he ' s there and I ' m here. In 1986, I had the presence of mind to stick a music hall on the north end of the building while I was building it. And a **guy** named Dizzy Gillespie showed up to play there because he knew this man over here, Marty Ashby. And I stood on that stage with Dizzy Gillespie on sound check on a Wednesday afternoon, and I said,"Dizzy, why would you come to a black-run center in the middle of an industrial park with a high crime rate that doesn ' t even have a reputation in music?"He said,"Because I heard you built the center and I didn ' t believe that you did it, and I wanted to see for myself. And now that I have, I want to give you a gift."I said,"You ' re the gift."He said,"No, sir. You ' re the gift. And I ' m going to allow you to record the concert and I ' m going to give you the music, and if you ever choose to sell it, you must sign an agreement that says the money will come back and **support** the school."And I recorded Dizzy. And he died a year later, but not before telling a fellow named McCoy Tyner what we were doing. And he showed up and said,"Dizzy talking about you all over the country, man, and I want to help you."And then a **guy** named Wynton Marsalis showed up. Then a bass player named Ray Brown, and a fellow named Stanley Turrentine, and a piano player named Herbie Hancock, and a band called the Count Basie Orchestra, and a fellow named Tito Puente, and a **guy** named Gary Burton, and Shirley Horn, and Betty Carter, and Dakota Staton and Nancy Wilson all have come to this center in the middle of an industrial park to sold out audiences in the middle of the inner city. And I ' m very pleased to tell you that, with their permission, I have now accumulated 600 recordings of the greatest artists in the world, including Joe Williams, who died, but not before his last recording was done at my school. And Joe Williams came up to me and he put his hand on my shoulder and he said,"God ' s picked you, man, to do this work. And I want my music to be with you."And that worked out all right. When the Basie band came, the band got so excited about the school they voted to give me the rights to the music. And I recorded it and we won something called a Grammy. And like a fool, I didn ' t go to the ceremony because I didn ' t think we were going to win. Well, we did win, and our name was literally in lights over Madison Square Garden. Then the U. N. Jazz Orchestra dropped by and we recorded them and got nominated for a second Grammy back to back. So, we ' ve become one of the hot, young jazz recording studios in the United States of America in the middle of the inner city with a high crime rate. That ' s the place all filled up with **Republicans**. If you ' d have dropped a bomb on that room, you ' d have wiped out all the money in Pennsylvania because it was all sitting there. Including my mother and father, who lived long enough to see their kid build that building. And there ' s Dizzy, just like I told you. He was there. And he was there, Tito Puente. And Pat Metheny and Jim Hall were there and they recorded with us. And that was our first recording studio, which was the broom closet. We put the mops in the hallway and re-engineered the thing and that ' s where we recorded the first Grammy. And this is our new facility, which is all video technology. And that is a room that was built for a woman named Nancy Wilson, who recorded that album at our school last Christmas. And any of you who happened to have been watching Oprah Winfrey on Christmas Day, he was there and Nancy was there singing excerpts from this album, the rights to which she donated to our school. And I can now tell you with absolute certainty that an appearance on Oprah Winfrey will sell 10, 000 CDs. We are currently number four on the Billboard Charts, right behind Tony Bennett. And I think we ' re going to be fine. This was burned out during the riots — this is next to my building — and so I had another cardboard box built and I walked back out in the streets again. And that ' s the building, and that ' s the model, and on the right ' s a high-tech greenhouse and in the middle ' s the medical technology building. And I ' m very pleased to tell you that the building ' s done. It ' s also full of anchor tenants at 20 dollars a foot — triple that in the middle of the inner city. And there ' s the fountain. Every building has a fountain. And the University of Pittsburgh Medical Center are anchor tenants and they took half the building, and we now train medical technicians through all their system. And Mellon Bank ' s a tenant. And I love them because they pay the rent on time. And as a result of the association, I ' m now a director of the Mellon Financial Corporation that bought Dreyfus. And this is in the process of being built as we speak. Multiply that picture times four and you will see the greenhouse that ' s going to open in October this year because we ' re going to grow those flowers in the middle of the inner city. And we ' re going to have high school kids growing Phalaenopsis orchids in the middle of the inner city. And we have a handshake with one of the large retail grocers to sell our orchids in all 240 stores in six states. And our partners are Zuma Canyon Orchids of Malibu, California, who are Hispanic. So, the Hispanics and the black folks have formed a partnership to grow high technology orchids in the middle of the inner city. And I told my United States senator that there was a very high probability that if he could find some funding for this, we would become a left-hand column in the Wall Street Journal, to which he readily agreed. And we got the funding and we open in the fall. And you ought to come and see it — it ' s going to be a hell of a story. And this is what I want to do when I grow up. The brown building is the one you **guys** have been looking at and I ' ll tell you where I made my big mistake. I had a chance to buy this whole industrial park — which is less than 1, 000 feet from the riverfront — for four million dollars and I didn ' t do it. And I built the first building, and guess what happened? I appreciated the real estate values beyond **everybody** ' s expectations and the owners of the park turned me down for eight million dollars last year, and said,"Mr. Strickland, you ought to get the Civic Leader of the Year Award because you ' ve appreciated our property values beyond our wildest expectations. Thank you very much for that."The moral of the story is you must be prepared to act on your dreams, just in case they do come true. And finally, there ' s this picture. This is in a place called San Francisco. And the reason this picture ' s in here is I did this slide show a couple years ago at a big economics summit, and there was a fellow in the audience who came up to me. He said,"Man, that ' s a great story. I want one of those."I said,"Well, I ' m very flattered. What do you do for a living?"He says,"I run the city of San Francisco. My name ' s Willie Brown."And so I kind of accepted the flattery and the praise and put it out of my mind. And that weekend, I was going back home and Herbie Hancock was playing our center that night — first time I ' d met him. And he walked in and he says,"What is this?"And I said,"Herbie, this is my concept of a training center for poor people."And he said,"As God as my witness, I ' ve had a center like this in my mind for 25 years and you ' ve built it. And now I really want to build one.”I said,”Well, where would you build this thing?”He said,”San Francisco.”I said,”Any chance you know Willie Brown?”As a matter of fact he did know Willie Brown, and Willie Brown and Herbie and I had dinner four years ago, and we started drawing out that center on the tablecloth. And Willie Brown said,”As sure as I ’ m the mayor of San Francisco, I ’ m going to build this thing as a legacy to the poor people of this city.”And he got me five acres of land on San Francisco Bay and we got an architect and we got a general contractor and we got Herbie on the board, and our friends from HP, and our friends from Steelcase, and our friends from Cisco, and our friends from Wells Fargo and Genentech. And along the way, I met this real short **guy** at my slide show in the Silicon Valley. He came up to me afterwards, he said,”Man, that ’ s a fabulous story. I want to help you.”And I said,”Well, thank you very much for that. What do you do for a living?”He said,”Well, I built a company called eBay.”I said,”Well, that ’ s very nice. Thanks very much, and give me your card and sometime we ’ ll talk.”I didn ’ t know eBay from that jar of water sitting on that piano, but I had the presence of mind to go back and talk to one of the techie kids at my center. I said,”Hey man, what is eBay?”He said,”Well, that ’ s the electronic commerce network.”I said,”Well, I met the **guy** who built the thing and he left me his card.”So, I called him up on the **phone** and I said,”Mr. Skoll, I ’ ve come to have a much deeper appreciation of who you are and I ’ d like to become your friend.”And Jeff and I did become friends, and he ’ s organized a team of people and we ’ re going to build this center. And I went down into the neighborhood called Bayview-Hunters Point, and I said,”The mayor sent me down here to work with you and I want to build a center with you, but I ’ m not going to build you anything if you don ’ t want it. And all I ’ ve got is a box of slides.”And so I stood up in front of 200 very angry, very disappointed people on a summer night, and the **air** conditioner had broken and it was 100 degrees outside, and I started showing these pictures. And after about 10 pictures they all settled down. And I ran the story and I said,”What do you think?”And in the back of the room, a woman stood up and she said,”In 35 years of living in this God forsaken place, you ’ re the only person that ’ s come down here and treated us with dignity. I ’ m going with you, man.”And she turned that audience around on a pin. And I promised these people that I was going to build this thing, and we ’ re going to build it all right. And I think we can get in the ground this year as the first replication of the center in Pittsburgh. But I met a **guy** named Quincy Jones along the way and I showed him the box of slides. And Quincy said,”I want to help you, man. Let ’ s do one in L. A.”And so he ’ s assembled a group of people. And I ’ ve fallen in love with him, as I have with Herbie and with his music. And Quincy said,”Where did the idea for centers like this come from?”And I said,”It came from your music, man. Because Mr. Ross used to bring in your albums when I was 16 years old in the pottery class, when the world was all dark, and your music got me to the sunlight.”And I said,”If I can follow that music, I ’ ll get out into the sunlight and I ’ ll be OK. And if that ’ s not true, how did I get here?”I want you all to know that I think the world is a place that ’ s worth living. I believe in you. I believe in your hopes and your dreams, I believe in your **intelligence** and I believe in your enthusiasm. And I ’ m tired of living like this, going into town after town with people standing around on corners with holes where eyes used to be, their spirits damaged. We won ’ t make it as a country unless we can turn this thing around. In Pennsylvania it costs 60, 000 dollars to keep people in jail, most of whom look like me. It ’ s 40, 000 dollars to build the University of Pittsburgh Medical School. It ’ s 20, 000 dollars cheaper to build a medical school than to keep people in jail. Do the math — it will never work. I am banking on you and I ’ m banking on **guys** like

Herbie and Quincy and Hackett and Richard and very decent people who still believe in something. And I want to do this in my lifetime, in every city and in every town. And I don ’ t think I ’ m crazy. I think we can get home on this thing and I think we can build these all over the country for less money than we ’ re spending on prisons. And I believe we can turn this whole story around to one of celebration and one of hope. In my business it ’ s very difficult work. You ’ re always fighting upstream like a salmon — never enough money, too much need — and so there is a tendency to have an occupational depression that accompanies my work. And so I ’ ve figured out, over time, the solution to the depression : you make a friend in every town and you ’ ll never be lonely. And my hope is that I ’ ve made a few here tonight. And thanks for listening to what I had to say.

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I ’ ll just start talking about the 17th century. I hope nobody finds that offensive. I â□□ you know, when I â□□ after I had invented PCR, I kind of needed a change. And I moved down to La Jolla and learned how to surf. And I started living down there on the beach for a long time. And when surfers are out waiting for waves, you probably wonder, if you ’ ve never been out there, what are they doing? You know, sometimes there ’ s a 10-, 15-minute break out there when you ’ re waiting for a wave to come in. They usually talk about the 17th century. You know, they get a real bad rap in the world. People think they ’ re sort of lowbrows. One day, somebody suggested I read this book. It was called â□□ it was called”The Air Pump,”or something like”The Leviathan and The Air Pump.”It was a real weird book about the 17th century. And I realized, the roots of the way I sort of thought was just the only natural way to think about things. That â□□ you know, I was born thinking about things that way, and I had always been like a little scientist **guy**. And when I went to find out something, I used scientific methods. I wasn ’ t real surprised, you know, when they first told me how â□□ how you were supposed to do science, because I ’ d already been doing it for fun and whatever. But it didn ’ t â□□ it never occurred to me that it had to be invented and that it had been invented only 350 years ago. You know, it was â□□ like it happened in England, and Germany, and Italy sort of all at the same time. And the story of that, I thought, was really fascinating. So I ’ m going to talk a little bit about that, and what exactly is it that scientists are supposed to do. And it ’ s, it ’ s a kind of â□□ You know, Charles I got beheaded somewhere early in the 17th century. And the English set up Cromwell and a whole bunch of **Republicans** or whatever, and not the kind of Republicans we had. They changed the government, and it didn ’ t work. And Charles II, the son, was finally put back on the throne of England. He was really nervous, because his dad had been, you know, beheaded for being the King of England And he was nervous about the fact that conversations that got going in, like, bars and **stuff** would turn to â□□ this is kind of â□□ it ’ s hard to believe, but people in the 17th century in England were starting to talk about, you know, philosophy and **stuff** in bars. They didn ’ t have TV screens, and they didn ’ t have any football games to watch. And they would get really pissy, and all of a sudden people would spill out into the street and fight about issues like whether or not it was okay if Robert Boyle made a device called the vacuum pump. Now, Boyle was a friend of Charles II. He was a Christian **guy** during the weekends, but during the week he was a scientist. Which was â□□ back then it was sort of, you know, well, you know â□□ if you made this thing â□□ he made this little device, like kind of like a bicycle pump in reverse that could suck all

the **air** out of â you know what a bell jar is? One of these things, you pick it up, put it down, and it ' s got a seal, and you can see inside of it, so you can see what ' s going on inside this thing. But what he was trying to do was to pump all the **air** out of there, and see what would happen inside there. I mean, the first â I think one of the first experiments he did was he put a bird in there. And people in the 17th century, they didn ' t really understand the same way we do about you know, this **stuff** is a bunch of different kinds of molecules, and we breathe it in for a purpose and all that. I mean, fish don ' t know much about water, and people didn ' t know much about **air**. But both started exploring it. One thing, he put a bird in there, and he pumped all the **air** out, and the bird died. So he said, hmm... He said â he called what he ' d done as making â they didn ' t call it a vacuum pump at the time. Now you call it a vacuum pump ; he called it a vacuum. Right? And immediately, he got into trouble with the local clergy who said, you can ' t make a vacuum. Ah, uh â Aristotle said that nature abhors one. I think it was a poor translation, probably, but people relied on authorities like that. And you know, Boyle says, well, shit. I make them all the time. I mean, whatever that is that kills the bird â and I ' m calling it a vacuum. And the religious people said that if God wanted you to make â I mean, God is everywhere, that was one of their rules, is God is everywhere. And a vacuum â there ' s nothing in a vacuum, so you ' ve â God couldn ' t be in there. So therefore the church said that you can ' t make a vacuum, you know. And Boyle said, bullshit. I mean, you want to call it Godless, you know, you call it Godless. But that ' s not my job. I ' m not into that. I do that on the weekend. And like â what I ' m trying to do is figure out what happens when you suck everything out of a compartment. And he did all these cute little experiments. Like he did one with â he had a little wheel, like a fan, that was sort of loosely attached, so it could spin by itself. He had another fan opposed to it that he had like a â I mean, the way I would have done this would be, like, a rubber band, and, you know, around a tinker toy kind of fan. I know exactly how he did it ; I ' ve seen the drawings. It ' s two fans, one which he could turn from outside after he got the vacuum established, and he discovered that if he pulled all the **air** out of it, the one fan would no longer turn the other one, right? Something was missing, you know. I mean, these are â it ' s kind of weird to think that someone had to do an experiment to show that, but that was what was going on at the time. And like, there was big arguments about it in the â you know, the gin houses and in the coffee shops and **stuff**. And Charles started not liking that. Charles II was kind of saying, you know, you should keep that â let ' s make a place where you can do this **stuff** where people don ' t get so â you know, we don ' t want the â we don ' t want to get the people mad at me again. And so â because when they started talking about religion and science and **stuff** like that, that ' s when it had sort of gotten his father in trouble. And so, Charles said, I ' m going to put up the money give you **guys** a building, come here and you can meet in the building, but just don ' t talk about religion in there. And that was fine with Boyle. He said, OK, we ' re going to start having these meetings. And anybody who wants to do science is â this is about the time that Isaac Newton was starting to whip out a lot of really **interesting** things. And there was all kind of people that would come to the Royal Society, they called it. You had to be dressed up pretty well. It wasn ' t like a **TED conference**. That was the only criteria, was that you be â you looked like a gentleman, and they ' d let anybody could come. You didn ' t have to be a member then. And so, they would come in and you would do â Anybody that was going to show an experiment, which was kind of a new word at the time, demonstrate some principle, they had to do it on stage, where **everybody** could see it. So they were â the really important part of this was, you were not supposed to talk about final causes, for instance. And God was out of the picture. The actual nature of reality was not at issue. You ’ re not supposed to talk about the absolute nature of anything. You were not supposed to talk about anything that you couldn ’ t demonstrate. So if somebody could see it, you could say, here ’ s how the machine works, here ’ s what we do, and then here ’ s what happens. And seeing what happens, it was OK to generalize, and say, I ’ m sure that this will happen anytime we make one of these things. And so you can start making up some rules. You say, anytime you have a vacuum state, you will discover that one wheel will not turn another one, if the only connection between them is whatever was there before the vacuum. That kind of thing. Candles can ’ t burn in a vacuum, therefore, probably sparklers wouldn ’ t either. It ’ s not clear ; actually sparklers will, but they didn ’ t know that. They didn ’ t have sparklers. But, they â â you can make up rules, but they have to relate only to the things that you ’ ve been able to demonstrate. And most the demonstrations had to do with visuals. Like if you do an experiment on stage, and nobody can see it, they can just hear it, they would probably think you were freaky. I mean, reality is what you can see. That wasn ’ t an explicit rule in the meeting, but I ’ m sure that was part of it, you know. If people hear voices, and they can ’ t see and associate it with somebody, that person ’ s probably not there. But the general idea that you could only â you could only really talk about things in that place that had some kind of experimental basis. It didn ’ t matter what Thomas Hobbes, who was a local philosopher, said about it, you know, because you weren ’ t going to be talking final causes. What ’ s happening here, in the middle of the 17th century, was that what became my field â science, experimental science â was pulling itself away, and it was in a physical way, because we ’ re going to do it in this room over here, but it was also what â it was an amazing thing that happened. Science had been all interlocked with theology, and philosophy, and â and â and mathematics, which is really not science. But experimental science had been tied up with all those things. And the mathematics part and the experimental science part was pulling away from philosophy. And â things â we never looked back. It ’ s been so cool since then. I mean, it just â it just â untangled a thing that was really impeding technology from being developed. And, I mean, **everybody** in this room â now, this is 350 short years ago. Remember, that ’ s a short time. It was 300, 000, probably, years ago that most of us, the ancestors of most of us in this room came up out of Africa and turned to the left. You know, the ones that turned to the right, there are some of those in the Japanese translation. But that happened very â a long time ago compared to 350 short years ago. But in that 350 years, the place has just undergone a lot of changes. In fact, **everybody** in this room probably, especially if you picked up your bag â some of you, I know, didn ’ t pick up your bags â but if you picked up your bag, **everybody** in this room has got in their pocket, or back in their room, something that 350 years ago, kings would have gone to war to have. I mean, if you can think how important â If you have a GPS system and there are no satellites, it ’ s not going to be much use. But, like â but, you know, if somebody had a GPS system in the 17th century some king would have gotten together an army and gone to get it, you know. If that person â Audience : For the teddy bear? The teddy bear? Kary Mullis : They might have done it for the teddy bear, yeah. But â all of us own **stuff**. I mean, individuals own things that kings would have definitely gone to war to get. And this is just 350 years. Not a whole lot of people doing this **stuff**. You know, the important people â you can almost read about their lives, about all the really important people that made advances, you know. And, I mean â this kind of **stuff**, you know, all this **stuff** came from that separation of this little sort of thing that we do â now I, when I was a boy was born sort of with this idea that if you want to know something â you know, maybe it ’ s because my old man was gone a lot, and my mother didn ’ t really know much science, but I thought if you want to know something about **stuff**, you do it â you make an experiment, you know. You get â you get, like â I just had a natural feeling for science and setting up experiments. I thought that was the way **everybody** had always thought. I thought that anybody with any brains will do it that way. It isn ’ t true. I mean, there ’ s a lot of people â You know, I was one of those scientists that was â got into trouble the other night at dinner because of the post-modernism thing. And I didn ’ t mean, you know â where is that lady? Audience : Here. KM : I mean, I didn ’ t really think of that as an argument so much as just a lively discussion. I didn ’ t take it personally, but â I just â I had â I naively had thought, until this surfing experience started me into the 17th century, I ’ d thought that ’ s just the way people thought, and **everybody** did, and they recognized reality by what they could see or touch or feel or hear. At any rate, when I was a boy, I, like, for instance, I had this â I got this little book from Fort Sill, Oklahoma â This is about the time that George Dyson ’ s dad was starting to blow nuclear â thinking about blowing up nuclear rockets and **stuff**. I was thinking about making my own little rockets. And I knew that frogs â little frogs â had aspirations of space travel, just like people. And I â I was looking for a â a propulsion system that would like, make a rocket, like, maybe about four feet high go up a couple of miles. And, I mean, that was my sort of goal. I wanted it to go out of sight and then I wanted this little parachute to come back with the frog in it. And â I â I â I got this book from Fort Sill, Oklahoma, where there ’ s a missile base. They send it out for amateur rocketeers, and it said in there do not ever heat a mixture of potassium perchlorate and sugar. You know, that ’ s what you call a lead. You sort of â now you say, well, let ’ s see if I can get hold of some potassium chlorate and sugar, perchlorate and sugar, and heat it ; it would be **interesting** to see what it is they don ’ t want me to do, and what it is going to â and how is it going to work. And we didn ’ t have â like, my mother presided over the back yard from an upstairs window, where she would be ironing or something like that. And she was usually just sort of keeping an eye on, and if there was any puffs of smoke out there, she ’ d lean out and admonish us all not to blow our eyes out. That was her â You know, that was kind of the worst thing that could happen to us. That ’ s why I thought, as long as I don ’ t blow my eyes out... I may not care about the fact that it ’ s prohibited from heating this solution. I ’ m going to do it carefully, but I ’ ll do it. It ’ s like anything else that ’ s prohibited : you do it behind the garage. So, I went to the drug store and I tried to buy some potassium perchlorate and it wasn ’ t unreasonable then for a kid to walk into a drug store and buy chemicals. Nowadays, it ’ s no ma ’ am, check your shoes. And like â But then it wasn ’ t â they didn ’ t have any, but the **guy** had â I said, what kind of salts of potassium do you have? You know. And he had potassium nitrate. And I said, that might do the same thing, whatever it is. I ’ m sure it ’ s got to do with rockets or it wouldn ’ t be in that manual. And so I â I did some experiments. You know, I started off with little tiny amounts of potassium nitrate and sugar, which was readily available, and I mixed it in different proportions, and I tried to light it on fire. Just to see what would happen, if you mixed it together. And it â they burned. It burned kind of slow, but it made a nice smell, compared to other rocket fuels I had tried, that all had sulfur in them. And, it smelt like burnt candy. And then I tried the melting business, and I melted it. And then it melted into a little sort of syrupy liquid, brown. And then it cooled down to a brick-hard substance, that when you lit that, it went off like a bat. I mean, the little bowl of that **stuff** that had cooled down â you ’ d light it, and it would just start dancing around the yard. And I said, there is a way to get a frog up to where he wants to go. So I started developing â you know, George ’ s dad had a lot of help. I just had my brother. But I â it took me about â it took me about, I ’ d say, six months to finally figure out all the little things. There ’ s a lot of little things **involved** in making a rocket that it will actually work, even after you have the fuel. But you do it, by â what I justâ you know, you do experiments, and you write down things sometimes, you make observations, you know. And then you slowly build up a theory of how this **stuff** works. And it was â I was following all the rules. I didn ’ t know what the rules were, I ’ m a natural born scientist, I guess, or some kind of a throwback to the 17th century, whatever. But at any rate, we finally did have a device that would reproduceably put a frog out of sight and get him back alive. And we had not â I mean, we weren ’ t frightened by it. We should have been, because it made a lot of smoke and it made a lot of noise, and it was powerful, you know. And once in a while, they would blow up. But I wasn ’ t worried, by the way, about, you know, the explosion causing the destruction of the planet. I hadn ’ t heard about the 10 ways that we should be afraid of the â By the way, I could have thought, I ’ d better not do this because they say not to, you know. And I ’ d better get permission from the government. If I ’ d have waited around for that, I would have never â the frog would have died, you know. At any rate, I bring it up because it ’ s a good story, and he said, tell personal things, you know, and that ’ s a personal â I was going to tell you about the first night that I met my wife, but that would be too personal, wouldn ’ t it. So, so I ’ ve got something else that ’ s not personal. But that... process is what I think of as science, see, where you start with some idea, and then instead of, like, looking up, every authority that you ’ ve ever heard of I â sometimes you do that, if you ’ re going to write a paper later, you want to figure out who else has worked on it. But in the actual process, you get an idea â like, when I got the idea one night that I could amplify DNA with two oligonucleotides, and I could make lots of copies of some little piece of DNA, you know, the thinking for that was about 20 minutes while I was driving my car, and then instead of going â I went back and I did talk to people about it, but if I ’ d listened to what I heard from all my friends who were molecular biologists â I would have abandoned it. You know, if I had gone back looking for an authority figure who could tell me if it would work or not, he would have said, no, it probably won ’ t. Because the results of it were so spectacular that if it worked it was going to change **everybody** ’ s goddamn way of doing molecular biology. Nobody wants a chemist to come in and poke around in their **stuff** like that and change things. But if you go to authority, and you always don ’ t â you don ’ t always get the right answer, see. But I knew, you ’ d go into the lab and you ’ d try to make it work yourself. And then you ’ re the authority, and you can say, I know it works, because right there in that tube is where it happened, and here, on this gel, there ’ s a little band there that I know that ’ s DNA, and that ’ s the DNA I wanted to amplify, so there! So it does work. You know, that ’ s how you do science. And then you say, well, what can make it work better? And then you figure out better and better ways to do it. But you always work from, from like, facts that you have made available to you by doing experiments : things that you could do on a stage. And no tricky shit behind the thing. I mean, it ’ s all â you ’ ve got to be very honest with what you ’ re doing if it really is going to work. I mean, you can ’ t make up results, and then do another experiment based on that one. So you have to be honest. And I ’ m basically honest. I have a fairly bad memory, and dishonesty would always get me in trouble, if I, like â so I ’ ve just sort of been naturally honest and naturally inquisitive, and that sort of leads to that kind of science.

Now, let ’ s see... I ’ ve got another five minutes, right? OK. All scientists aren ’ t like that. You know â and there is a lot â There is a lot â a lot has been going on since Isaac Newton and all that **stuff** happened. One of the things that happened right around World War II in that same time period before, and as sure as hell afterwards, government got â realized that scientists aren ’ t strange dudes that, you know, hide in ivory towers and do ridiculous things with test tube. Scientists, you know, made World War II as we know it quite possible. They made faster things. They made bigger guns to shoot them down with. You know, they made drugs to give the pilots if they were broken up in the process. They made all kinds of â and then finally one giant bomb to end the whole thing, right? And **everybody** stepped back a little and said, you know, we ought to **invest** in this shit, because whoever has got the most of these people working in the places is going to have a **dominant** position, at least in the military, and probably in all kind of economic ways. And they got **involved** in it, and the scientific and industrial establishment was born, and out of that came a lot of scientists who were in there for the money, you know, because it was suddenly available. And they weren ’ t the curious little boys that liked to put frogs up in the **air**. They were the same people that later went in to medical school, you know, because there was money in it, you know. I mean, later, then they all got into business â I mean, there are waves of â going into your high school, person saying, you want to be rich, you know, be a scientist. You know, not anymore. You want to be rich, you be a businessman. But a lot of people got in it for the money and the power and the travel. That ’ s back when travel was easy. And those people don ’ t think â they don ’ t â they don ’ t always tell you the truth, you know. There is nothing in their contract, in fact, that makes it to their advantage always, to tell you the truth. And the people I ’ m talking about are people that like â they say that they ’ re a member of the committee called, say, the Inter- Governmental Panel on **Climate** Change. And they â and they have these big meetings where they try to figure out how we ’ re going to â how we ’ re going to continually prove that the planet is getting warmer, when that ’ s actually contrary to most people ’ s sensations. I mean, if you actually measure the temperature over a period â I mean, the temperature has been measured now pretty carefully for about 50, 60 years â longer than that it ’ s been measured, but in really nice, precise ways, and records have been kept for 50 or 60 years, and in fact, the temperature hadn ’ t really gone up. It ’ s like, the average temperature has gone up a tiny little bit, because the nighttime temperatures at the weather stations have come up just a little bit. But there ’ s a good explanation for that. And it ’ s that the weather stations are all built outside of town, where the airport was, and now the town ’ s moved out there, there ’ s concrete all around and they call it the skyline effect. And most responsible people that measure temperatures realize you have to shield your measuring device from that. And even then, you know, because the buildings get warm in the daytime, and they keep it a little warmer at night. So the temperature has been, sort of, inching up. It should have been. But not a lot. Not like, you know â the first **guy** â the first **guy** that got the idea that we ’ re going to fry ourselves here, actually, he didn ’ t think of it that way. His name was Sven Arrhenius. He was Swedish, and he said, if you double the CO2 level in the atmosphere, which he thought might â this is in 1900 â the temperature ought to go up about 5. 5 degrees, he calculated. He was thinking of the earth as, kind of like, you know, like a completely insulated thing with no **stuff** in it, really, just energy coming down, energy leaving. And so he came up with this theory, and he said, this will be cool, because it ’ ll be a longer growing season in Sweden, you know, and the surfers liked it, the surfers thought, that ’ s a cool idea, because it ’ s pretty cold in the ocean sometimes, and â but a lot of other people later on started thinking it would be bad, you know. But nobody actually demonstrated it, right? I mean, the temperature as measured â and you can find this on our wonderful Internet, you just go and look for all NASAs records, and all the Weather Bureau ’ s records, and you ’ ll look at it yourself, and you ’ ll see, the temperature has just â the nighttime temperature measured on the surface of the planet has gone up a tiny little bit. So if you just average that and the daytime temperature, it looks like it went up about. 7 degrees in this century. But in fact, it was just coming up â it was the nighttime ; the daytime temperatures didn ’ t go up. So â and Arrhenius ’ theory â and all the global warmers think â they would say, yeah, it should go up in the daytime, too, if it ’ s the greenhouse effect. Now, people like things that have, like, names like that, that they can envision it, right? I mean â but people don ’ t like things like this, so â most â I mean, you don ’ t get all excited about things like the actual evidence, you know, which would be evidence for strengthening of the tropical circulation in the 1990s. It ’ s a paper that came out in February, and most of you probably hadn ’ t heard about it.”Evidence for Large Decadal Variability in the Tropical Mean Radiative Energy Budget.”Excuse me. Those papers were published by NASA, and some scientists at Columbia, and Viliki and a whole bunch of people, Princeton. And those two papers came out in Science Magazine, February the first, and these â the conclusion in both of these papers, and in also the Science editor ’ s, like, descriptions of these papers, for, you know, for the quickie, is that our theories about global warming are completely wrong. I mean, what these **guys** were doing, and this is what â the NASA people have been saying this for a long time. They say, if you measure the temperature of the atmosphere, it isn ’ t going up â it ’ s not going up at all. We ’ ve doing it very carefully now for 20 years, from satellites, and it isn ’ t going up. And in this paper, they show something much more striking, and that was that they did what they call a radiation â and I ’ m not going to go into the details of it, actually it ’ s quite complicated, but it isn ’ t as complicated as they might make you think it is by the words they use in those papers. If you really get down to it, they say, the sun puts out a certain amount of energy â we know how much that is â it falls on the earth, the earth gives back a certain amount. When it gets warm it generates â it makes redder energy â I mean, like infra-red, like something that ’ s warm gives off infra-red. The whole business of the global warming â trash, really, is that â if the â if there ’ s too much CO2 in the atmosphere, the heat that ’ s trying to escape won ’ t be able to get out. But the heat coming from the sun, which is mostly down in the â it ’ s like 350 nanometers, which is where it ’ s centered â that goes right through CO2. So you still get heated, but you don ’ t dissipate any. Well, these **guys** measured all of those things. I mean, you can talk about that **stuff**, and you can write these large reports, and you can get government money to do it, but these â they actually measured it, and it turns out that in the last 10 years â that ’ s why they say”decadal”there â that the energy â that the level of what they call”imbalance”has been way the hell over what was expected. Like, the amount of imbalance â meaning, heat ’ s coming in and it ’ s not going out that you would get from having double the CO2, which we ’ re not anywhere near that, by the way. But if we did, in 2025 or something, have double the CO2 as we had in 1900, they say it would be increase the energy budget by about â in other words, one watt per square centimeter more would be coming in than going out. So the planet should get warmer. Well, they found out in this study â these two studies by two different teams â that five and a half watts per square meter had been coming in from 1998, 1999, and the place didn ’ t get warmer. So the theory ’ s kaput â it ’ s nothing. These papers should have been called,”The End to the Global Warming Fiasco,”you know. They ’ re concerned, and you can tell they have very guarded conclusions in these papers, because they ’ re talking about big laboratories that are funded by lots of money and by scared people. You know, if they said, you know what? There isn ’ t a problem with global warming any longer, so we can â you know, they ’ re funding. And if you start a grant request with something like that, and say, global warming obviously hadn ’ t happened... if they â if they â if they actually â if they actually said that, I ’ m getting out. I ’ ll stand up too, and â They have to say that. They had to be very cautious. But what I ’ m saying is, you can be delighted, because the editor of Science, who is no dummy, and both of these fairly professional â really professional teams, have really come to the same conclusion and in the bottom lines in their papers they have to say, what this means is, that what we ’ ve been thinking, was the global circulation model that we predict that the earth is going to get overheated that it ’ s all wrong. It ’ s wrong by a large factor. It ’ s not by a small one. They just â they just misinterpreted the fact that the earth â there ’ s obviously some mechanisms going on that nobody knew about, because the heat ’ s coming in and it isn ’ t getting warmer. So the planet is a pretty amazing thing, you know, it ’ s big and horrible â and big and wonderful, and it does all kinds of things we don ’ t know anything about. So I mean, the reason I put those things all together, OK, here ’ s the way you ’ re supposed to do science â some science is done for other reasons, and just curiosity. And there ’ s a lot of things like global warming, and ozone hole and you know, a whole bunch of scientific public issues, that if you ’ re **interested** in them, then you have to get down the details, and read the papers called,”Large Decadal Variability in the...”You have to figure out what all those words mean. And if you just listen to the **guys** who are hyping those issues, and making a lot of money out of it, you ’ ll be misinformed, and you ’ ll be worrying about the wrong things. Remember the 10 things that are going to get you. The â one of them â And the asteroids is the one I really agree with there. I mean, you ’ ve got to watch out for asteroids. OK, thank you for having me here.

Text Sample 526: POS Dim 4 – 2002 – Score 11.00 – 2002/t000317.txt

I draw to better understand things. Sometimes I make a lot of drawings and I still don ’ t understand what it is I ’ m drawing. Those of you who are comfortable with digital **stuff** and even smug about that relationship might be amused to know that the **guy** who is best known for”The Way Things Work,”while preparing for part of a panel for called Understanding, spent two days trying to get his laptop to communicate with his new CD burner. Who knew about extension managers? I ’ ve always managed my own extensions so it never even occurred to me to read the instructions, but I did finally figure it out. I had to figure it out, because along with the invitation came the frightening reminder that there would be no projector, so bringing those carousels would no longer be necessary but some alternate form of communication would. Now, I could talk about something that I ’ m known for, something that would be particularly appropriate for many of the more technically minded people here, or I could talk about something I really care about. I decided to go with the latter. I ’ m going to talk about Rome. Now, why would I care about Rome, particularly? Well, I went to Rhode Island School of Design in the second half of the ’ 60s to study architecture. I was lucky enough to spend my last year, my fifth year, in Rome as a student. It changed my life. Not the least reason was the fact that I had spent those first four years living at home, driving into RISD everyday, driving back. I missed the ’ 60s. I read about them ; I understand they were pretty **interesting**. I missed them, but I did spend that extraordinary year in Rome, and it ’ s a place that is never far from my mind. So, whenever given an opportunity, I try to do something in it or with it or for it. I also make drawings to help people understand things. Things that I want them to believe I understand. And that ’ s what I do as an illustrator, that ’ s my job. So, I ’ m going to show you some pictures of Rome. I ’ ve made a lot of drawings of Rome over the years. These are just drawings of Rome. I get back as often as possible — I need to. All different materials, all different styles, all different times, drawings from sketchbooks looking at the details of Rome. Part of the reason I ’ m showing you these is that it sort of helps illustrate this process I go through of trying to figure out what it is I feel about Rome and why I feel it. These are sketches of some of the little details. Rome is a city full of surprises. I mean, we ’ re talking about unusual perspectives, we ’ re talking about narrow little winding streets that suddenly open into vast, sun-drenched piazzas — never, though, piazzas that are not humanly scaled. Part of the reason for that is the fact that they grew up organically. That amazing juxtaposition of old and new, the bits of light that come down between the buildings that sort of create a map that ’ s traveling above your head of usually blue — especially in the summer — compared to the map that you would normally expect to see of conventional streets. And I began to think about how I could communicate this in book form. How could I share my sense of Rome, my understanding of Rome? And I ’ m going to show you a bunch of dead ends, basically. The primary reason for all these dead ends is when if you ’ re not quite sure where you ’ re going, you ’ re certainly not going to get there with any kind of efficiency. Here ’ s a little map. And I thought of maps at the beginning ; maybe I should just try and do a little atlas of my favorite streets and connections in Rome. And here ’ s a line of text that actually evolves from the exhaust of a scooter zipping across the page. Here that same line of text wraps around a fountain in an illustration that can be turned upside down and read both ways. Maybe that line of text could be a story to help give some human aspect to this. Maybe I should get away from this map completely, and really be honest about wanting to show you my favorite bits and pieces of Rome and simply kick a soccer ball in the **air** — which happens in so many of the squares in the city — and let it bounce off of things. And I ’ ll simply explain what each of those things is that the soccer ball hits. That seemed like a sort of a cheap shot. But even though I just started this presentation, this is not the first thing that I tried to do and I was getting sort of desperate. Eventually, I realized that I had really no content that I could count on, so I decided to move towards packaging. I mean, it seems to work for a lot of things. So I thought a little box set of four small books might do the trick. But one of the ideas that emerged from some of those sketches was the notion of traveling through Rome in different vehicles at different speeds in order to show the different aspects of Rome. Sort of an overview of Rome and the plan that you might see from a dirigible. Quick snapshots of things you might see from a speeding motor scooter, and very slow walking through Rome, you might be able to study in more detail some of the wonderful surfaces and whatnot that you come across. Anyways, I went back to the dirigible notion. Went to Alberto Santos-Dumont. Found one of his dirigibles that had enough dimensions so I could actually use it as a scale that I would then juxtapose with some of the things in Rome. This thing would be flying over or past or be parked in front of, but it would be like having a ruler — sort of travel through the pages without being a ruler. Not that you know how long number 11 actually is, but you would be able to compare number 11 against the Pantheon with number 11 against the Baths of Caracalla, and so on and so forth. If you were

interested. This is Beatrix. She has a dog named Ajax, she has purchased a dirigible — a small dirigible — she ’ s assembling the structure, Ajax is sniffing for holes in the balloon before they set off. She launches this thing above the Spanish Steps and sets off for an aerial tour of the city. Over the Spanish Steps we go. A nice way to show that river, that stream sort of pouring down the hill. Unfortunately, just across the road from it or quite close by is the Column of Marcus Aurelius, and the diameter of the dirigible makes an impression, as you can see, as she starts trying to read the story that spirals around the Column of Marcus Aurelius — gets a little too close, nudges it. This gives me a chance to suggest to you the structure of the Column of Marcus Aurelius, which is really no more than a pile of quarters high — thick quarters. Over the Piazza of Saint Ignacio, completely ruining the symmetry, but that aside a spectacular place to **visit**. A spectacular framework, inside of which you see, usually, extraordinary blue sky. Over the Pantheon and the 26-foot diameter Oculus. She parks her dirigible, lowers the anchor rope and climbs down for a closer look inside. The text here is right side and upside down so that you are forced to turn the book around, and you can see it from ground point of view and from her point of view — looking in the hole, getting a different kind of perspective, moving you around the space. Particularly appropriate in a building that can contain perfectly a sphere dimensions of the diameter being the same as the distance from the center of the floor to the center of the Oculus. Unfortunately for her, the anchor line gets tangled around the feet of some Boy Scouts who are **visiting** the Pantheon, and they are immediately yanked out and given an extraordinary but terrifying tour of some of the domes of Rome, which would, from their point of view, naturally be hanging upside down. They bail out as soon as they get to the top of Saint Ivo, that little spiral structure you see there. She continues on her way over Piazza Navona. Notices a lot of activity at the Tre Scalini restaurant, is reminded that it is lunchtime and she ’ s hungry. They keep on motoring towards the Campo de ’ Fiori, which they soon reach. Ajax the dog is put in a basket and lowered with a list of food into the marketplace, which flourishes there until about one in the afternoon, and then is completely removed and doesn ’ t appear again until six or seven the following morning. Anyway, the pooch gets back to the dirigible with the **stuff**. Unfortunately, when she goes to unwrap the prosciutto, Ajax makes a lunge for it. She ’ s managed to save the prosciutto, but in the process she loses the tablecloth, which you can see flying away in the upper left-hand corner. They continue without their tablecloth, looking for a place to land this thing so that they can actually have lunch. They eventually discover a huge wall that ’ s filled with small holes, ideal for docking a dirigible because you ’ ve got a place to tie up. Turns out to be the exterior wall — that part of it that remains — of the Coliseum, so they park themselves there and have a terrific lunch and have a spectacular view. At the end of lunch, they untie the anchor, they set off through the Baths of Caracalla and over the walls of the city and then an abandoned gatehouse and decide to take one more look at the Pyramid of Cestius, which has this lightning rod on top. Unfortunately, that ’ s a problem : they get a little too close, and when you ’ re in a dirigible you have to be very careful about spikes. So that sort of brings her little story to a conclusion. Marcello, on the other hand, is sort of a lazy **guy**, but he ’ s not due at work until about noon. So, the alarm goes off and it ’ s five to 12 or so. He gets up, leaps onto his scooter, races through the city past the church of Santa Maria della Pace, down the alleys, through the streets that tourists may be wandering through, disturbing the quiet backstreet life of Rome at every turn. That speed with which he is moving, I hope I have suggested in this little image, which, again, can be turned around and read from both sides because there ’ s text on the bottom and text on the top, one of which is upside down in this image. So, he keeps on moving, approaching an unsuspecting waiter who is trying to deliver two plates of linguine in a delicate white wine clam sauce to diners who are sitting at a table just outside of a restaurant in the street. Waiter catches on, but it ’ s too late. And Marcello keeps moving in his scooter. Everything he sees from this point on is slightly affected by the linguine, but keeps on moving because this **guy** ’ s got a job to do. Removes some scaffolding. One of the reasons Rome remains the extraordinary place it is that because of scaffolding and the determination to maintain the fabric, it is a city that continues to grow and adapt to the needs of the particular time in which it finds itself, or we find it. Right through the Piazza della Rotonda, in front of the Pantheon — again wreaking havoc — and finally getting to work. Marcello, as it turns out, is the driver of the number 64 bus, and if you ’ ve been on the number 64 bus, you know that it ’ s driven with the same kind of exuberance as Marcello demonstrated on his scooter. And finally Carletto. You see his apartment in the upper left-hand corner. He ’ s looking at his table ; he ’ s planning to propose this evening to his girlfriend of 40 years, and he wants it to be perfect. He ’ s got candles out, he ’ s got flowers in the middle and he ’ s trying to figure out where to put the plates and the glasses. But he ’ s not happy ; something ’ s wrong. The **phone** rings anyway, he ’ s called to the palazzo. He saunters — he saunters at a good clip, but as compared to all the traveling we ’ ve just seen, he ’ s sauntering. **Everybody** knows Carletto, because he ’ s in entertainment, actually ; he ’ s in television. He ’ s actually in television repair, which is why people know him. So they all have his number. He arrives at the palazzo, arrives at the big front door. Enters the courtyard and talks to the custodian, who tells him that there ’ s been a disaster in the palazzo ; nobody ’ s TVs are working and there ’ s a big soccer game coming up, and the crowd is getting a little restless and a little nervous. He goes down to the basement and starts to check the wiring, and then gradually works his way up to the top of the building, apartment by apartment, checking every television, checking every connection, hoping to find out what this problem is. He works his way up, finally, the grand staircase and then a smaller staircase until he reaches the attic. He opens the window of the attic, of course, and there ’ s a tablecloth wrapped around the building ’ s television antenna. He removes it, the problems are solved, **everybody** in the palazzo is happy. And of course, he also solves his own problem. All he has to do now, with a perfect table, is wait for her to arrive. That was the first attempt, but it didn ’ t seem substantial enough to convey whatever it was I wanted to convey about Rome. So I thought, well, I ’ ll just do piazzas, and I ’ ll get inside and underneath and I ’ ll show these things growing and show why they ’ re shaped the way they are and so on. And then I thought, that ’ s too complicated. No, I ’ ll just take my favorite bits and pieces and I ’ ll put them inside the Pantheon but keep the scale, so you can see the top of Sant ’ Ivo and the Pyramid of Cestius and the Tempietto of Bramante all side by side in this amazing space. Now that ’ s one drawing, so I thought maybe it ’ s time for Piranesi to meet Escher. You see that I ’ m beginning to really lose control here and also hope. There ’ s a very thin blue line of exhaust that sort of runs through this thing that would be kind of the trail that holds it all together. Then I thought,”Wait a minute, what am I doing?”A book is not only a neat way of collecting and storing information, it ’ s a series of layers. I mean, you always peel one layer off another ; we think of them as pages, doing it a certain way. But think of them as layers. I mean, Rome is a place of layers — horizontal layers, vertical layers — and I thought, well just peeling off a page would allow me to — if I got you thinking about it the right way — would allow me to sort of show you the depth of layers. The stucco on the walls of most of the buildings in Rome covers the scars ;

the scars of centuries of change as these structures have been adapted rather than being torn down. If I do a foldout page on the left-hand side and let you just unfold it, you see behind it what I mean by scar tissue. You can see that in 1635, it became essential to make smaller windows because the bad **guys** were coming or whatever. Adaptations all get buried under the stucco. I could peel out a page of this palazzo to show you what ' s going on inside of it. But more importantly, I could also show you what it looks like at the corner of one of those magnificent buildings with all the massive stone blocks, or the fake stone blocks done with brick and stucco, which is more often the case. So it becomes slightly three-dimensional. I could take you down one of those narrow little streets into one of those surprising piazzas by using a double gate fold — double foldout page — which, if you were like me reading a pop- up book as a child, you hopefully stick your head into. You wrap the pages around your head and are in that piazza for that brief period of time. And I ' ve really not done anything much more complicated than make foldout pages. But then I thought, maybe I could be simpler here. Let ' s look at the Pantheon and the Piazza della Rotonda in front of it. Here ' s a book completely wide open. OK, if I don ' t open the book the whole way, if I just open it 90 degrees, we ' re looking down the front of the Pantheon, and we ' re looking sort of at the top, more or less down on the square. And if I turn the book the other way, we ' re looking across the square at the front of the Pantheon. No foldouts, no tricks — just a book that isn ' t open the whole way. That seemed promising. I thought, maybe I ' ll do it inside and I can even combine the foldouts with the only partially opened book. So we get inside the Pantheon and it grows and so on and so forth. And I thought, maybe I ' m on the right track, but it sort of lost its human quality. So I went back to the notion of story, which is always a good thing to have if you ' re trying to get people to pay attention to a book and pick up information along the way."Pigeon ' s Progress"struck me as a catchy title. If it was a homing pigeon, it would be called"Homer ' s Odyssey."But it was the journey of the... I mean, if a title works, use it. But it would be a journey that went through Rome and showed all the things that I like about Rome. It ' s a pigeon sitting on top of a church. Goes off during the day and does normal pigeon **stuff**. Comes back, the whole place is covered with scaffolding and green netting and there ' s no way this pigeon can get home. So it ' s a homeless pigeon now and it ' s going to have to find another place to live, and that allows me to go through my catalog of favorite things, and we start with the tall ones and so on. Maybe it has to go back and live with family members ; that ' s not always a good thing, but it does sort of bring pigeons together again. And I thought, that ' s sort of **interesting**, but maybe there ' s a person who should be **involved** in this in some way. So I kind of came up with this old **guy** who spends his life looking after sick pigeons. He ' ll go anywhere to get them — dangerous places and whatnot — and they become really friends with this **guy**, and learn to do tricks for him and entertain him at lunchtime and **stuff** like that. There ' s a real bond that develops between this old man and these pigeons. But unfortunately he gets sick. He gets really sick at the end of the story. He ' s taught them to spell his name, which is Aldo. They show up one day after three or four days of not seeing him — he lives in this little garret — and they spell his name and fly around. And he finally gets enough strength together to climb up the ladder onto the roof, and all the pigeons, a la Red Balloon, are there waiting for him and they carry him off over the walls of the city. And I forgot to mention this : whenever he lost a pigeon, he would take that pigeon out beyond the walls of the city. In the old Roman custom, the dead were never buried within the walls. And I thought that ' s a really cheery story.. That ' s really going to go a long way. So anyway, I went through... And again, if packaging doesn ' t work and if the stories aren ’ t going anywhere, I just come up with titles and hope that a title will sort of kick me off in the right direction. And sometimes it does focus me enough and I ’ ll even do a title page. So, these are all title pages that eventually led me to the solution I settled on, which is the story of a young woman who sends a message on a homing pigeon — she lives outside the walls of the city of Rome — to someone in the city. And the pigeon is flying down above the Appian Way here. You can see the tombs and pines and so on and so forth along the way. If you see the red line, you are seeing the trail of the pigeon ; if you don ’ t see the red line, you are the pigeon. And it becomes necessary and possible, at this point, to try to convey what that sense would be like of flying over the city without actually moving. Past the Pyramid of Cestius — these will seem very familiar to you, even if you haven ’ t been to Rome recently — past the gatehouse. This is something that ’ s a little bit unusual. This pigeon does something that most homing pigeons do not do : it takes the scenic route, which was a device that I felt was necessary to actually extend this book beyond about four pages. So, we circle around the Coliseum, past the Church of Santa Maria in Cosmedin and the Temple of Hercules towards the river. We almost collide with the cornice of the Palazzo Farnese — designed by Michelangelo, built of stone taken from the Coliseum — narrow escape. We swoop down over the Campo de ’ Fiori. This is one of those things I show to my students because it ’ s a complete bastardization — a denial of any rules of perspective. The only rule of perspective that I think matters is if it looks believable, you ’ ve succeeded. But you try and figure out where the vanishing points meet here ; a couple are on Mars and a couple of others in Cremona. But into the piazza in front of Santa Maria della Pace, where invariably a soccer game is going on, and we ’ re hit by a soccer ball. Now this is a terrible illustration of being hit by a soccer ball. I have all the pieces : there ’ s Santa Maria della Pace, there ’ s a soccer ball, there ’ s a little bit of a bird ’ s wing — nothing ’ s happening, so I had to rethink it. And if you do want to see Santa Maria della Pace, these books are really flexible, incredibly interactive — just turn it around and look at it the other way. Through the alley, we can see the **impact** is captured in the red line. And then bird manages to pull itself together past this medieval tower — one of the few remaining medieval towers — towards the church of Sant ’ Agnese and around the dome looking down into Piazza Navona — which we ’ ve already mentioned and seen and flown over a couple of times ; there ’ s the Bernini statue of the Four Rivers — and then past the wonderful Borromini Sant ’ Ivo, stopping just long enough on the 26-foot diameter Oculus of the Pantheon to catch our breath. And then we can swoop inside and around ; and because we ’ re flying, we don ’ t really have to worry about gravity at this particular moment in time, so this drawing can be oriented in any way on the page. We get a little exuberant as we pass Gesu ; it ’ s not surprising to sort of mimic the architecture in this way. Past the wonderful wall filled with the juxtaposition that I was talking about ; beautiful carvings set into the walls above the neon”Ristorante”sign, and so on. And eventually, we arrive at the courtyard of the palazzo, which is our destination. Straight up through the courtyard into a little window into the attic, where somebody is working at the drawing board. He removes the message from the leg of the bird ; this is what it says. As we look at the drawing board, we see what he ’ s working on is, in fact, a map of the journey that the pigeon has just taken, and the red line extends through all the sights. And if you want the information, so that we complete this cycle of understanding, all you have to do is read these paragraphs. Thank you very much.

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Bruno Giussani : It ’ s difficult to think clearly of the Russian invasion of Ukraine, because wars, while they unfold, they ’ re kind of shrouded in a sort of fog. Information is abundant : the millions of **refugees**, the shocking suffering and the destruction, the politics. But sense is lacking. And that ’ s going to be the focus of this Membership conversation as we enter the third week of the war. We won ’ t talk about the events of the day but try to project a longer arc, a broader context. Our guest is geopolitical analyst Ian Bremmer. He ’ s the founder and president of Eurasia Group, and we asked him to lay the scene by talking first about the geopolitical shifts that have already been brought by the war in Ukraine. And after, we ’ re going to have a conversation, including questions from **TED** Members who are participating in this call. Ian, welcome. Ian Bremmer : Thank you very much. I ’ ll start by saying that in my lifetime, the most important geopolitical artifact is the fall of the Berlin Wall. I mean, you see it if you go into the new NATO headquarters in Brussels, just built a few years ago. And anyone that has a piece, something they ’ re very proud of, they know it affected their entire lives. I think that in 30 years ’ time, and I fear that in 30 years ’ time, if we look back, a second most important geopolitical artifact will be a piece of the rubble of the Maidan in Kyiv. I believe that the war that we are seeing right now is no more and no less than the end of the peace dividend that we all thought we had when the wall came down in 1989. The idea that the world could focus more on globalization and goods and services and people and ideas going faster and faster across borders, leading to unprecedented growth in human development and a global middle class. I think that this is a tipping point. Won ’ t end globalization, but it does end the peace dividend. It does mean that the Europeans overnight will and must prioritize spending on defense **policy**, on national security, coordination, on NATO. And the speech that was given by Olaf Scholz, the new chancellor, two weeks ago, in my view, the most significant speech given by a European **leader** in the post-Cold War environment, precisely because it ’ s now the post-post-Cold War environment, sending weapons to the Ukrainians, committing to over two percent of GDP spend on defense, **investing** in a new fund for defense infrastructure. But also recognizing that the way that the Germans and the Europeans as a whole looked at the world and looked at themselves was, unfortunately for all of us, outdated. A few other points I ’ d like to raise, just to kick off this conversation. One : One of the reasons I ’ m pretty negative about this, and I ’ m not usually very negative, I ’ m usually an existential optimist, I ’ m someone that ’ s just happy there ’ s water in the glass. But when I look at this **conflict**, I ’ m much more concerned. And that is because I do not see a scenario, a plausible scenario, in the foreseeable future where Putin emerges from this war in anything less than a radically weakened position compared to where he was before he announced the invasion. And I believe that both in terms of his domestic political orientation, how stable he is in his own country, also, of course, Russia ’ s economic position, and finally, Russia ’ s position in terms of global security and European security : ostensibly, the very reason that Putin began the war to begin with. Second big point is that the decoupling that we are seeing from Europe and the United States with Russia is, in my view, permanent. And that would be true even if there were a **negotiated** settlement and all the Russian troops were to pull out of Ukraine and we had peace. I still think that a lot of those companies would not come back with Putin in power. I ’ m convinced that the decisions by the Europeans to ramp up their national security capabilities will be permanent. Permanent deployments coming in the Baltic states, for example, forward deployments in Poland and Bulgaria and Romania. And also an unwind of Europe ’ s massive energy dependence : coal, oil and most importantly, gas on Russia. That does not make Russia a global pariah because you can ' t be a global pariah if the soon-to-be most important economy in the world, China, is actually your bestie on the global stage, and that indeed continues to be true despite China ' s efforts to portray themselves, towards Europe at least, as more neutral. We are going to see the Russians as a supplicant economically, in terms of energy flows, financially, in terms of transactions, and technologically, perhaps most important, aligned with China. That has big geopolitical implications long-term. Also, a lot of other developing economies, like the Indians, like the Gulf states, like Brazil, are also not going to work with Russia as a pariah. They ' ll continue to engage. Are there any silver linings? And I think there are a few. Of course, there is a much greater renewed purpose and mission of NATO. I mean, this is an organization that just a couple of years ago, France President Emmanuel Macron referred to as"brain-dead."It was increasingly drifting in terms of its importance. The Americans were focusing much more on Asia, the **pivot**. Not today. Today, NATO is purposeful, it ' s aligned, it ' s consolidated. it ' s going to get more resources, not less. That ' s also true of the European Union as a whole, even when we talk about countries like Hungary and Poland that have been much less aligned in terms of rule of law, in terms of an independent judiciary, much more aligned in terms of the importance of common values of Europe compared to that of what we ' re seeing in Moscow. I mentioned the German security and **policy** shift. The UK-EU relationship is much smoother and more functional than at any point since Brexit process actually started. And even the United States. I mean, if you watched the State of the Union for a brief moment in time, five or 10 minutes, when all of the Democrats and Republicans were standing and applauding together, you could be forgiven for believing that the United States had a functional representative democracy. Now I ' m not sure how long this is going to last, but at least as of now, **leaders** of the Democratic and Republican Party see Putin as much more of a threat, an enemy, than they do their opponents across the aisle in domestic politics. And two weeks ago, that was not true. That ' s significant. Final silver lining, and I wish it was more of one, but the Chinese, as much as they are strategically aligned with Russia and with the person of President Putin, they do not want a second Cold War. And they would rather a **negotiated** settlement. They ' re not willing to push very hard for it. But they certainly do not see a radical decoupling of the Russian, and potentially the Chinese economy, from the rest of the world, from Europe, from the US, from the advanced industrial democracies, as in any way in their interest. And ultimately, that does create at least some buffer, some guardrail on how much this is likely to escalate as a **conflict** going forward. BG : I want to make a step back and unpack some of that, maybe starting with a question that relates to your last point and is probably on the mind of many. And it is : Is there still some real space for negotiation? Is there still a potential relationship between Russia and Ukraine? IB : The foreign ministers of Russia and Ukraine just met recently in Turkey. It was as much of a non-event as the three preceding negotiations of more junior representatives of their teams on the Belarus border. The one thing that has been accomplished to a small degree has been humanitarian corridors, extending out of a number of Ukrainian cities that are being pounded by Russian military. That ' s because the Ukrainians are **interested** in protecting their civilians, and the Russians are **interested** in taking a lot of territory without necessarily having to kill so many Ukrainians, that could cause problems for them internationally as well as domestically inside Russia. But that is nowhere close to a **negotiated** settlement. Now, I mean, everyone I know that ' s **involved** in the negotiations right now responds that the President Putin himself is hell-bent on taking Kyiv and on removing Zelenskyy from power. Now I think, and by the way, they ' re getting quite close to being able to accomplish that militarily on the ground. I think within the next couple of weeks, certainly, it looks very likely. A couple of points here. One, there is no reason to put any stock in anything that the Russians are saying publicly in terms of their diplomacy. They lied to the face of every world **leader** about the invasion that they said they were not going to do into Ukraine. And then just today, Foreign Minister Sergey Lavrov publicly said, well, the Russians didn ' t attack Ukraine. I mean, this is Orwellian **stuff**, right? So first of all, do not report on Russian public statements as if they bear any semblance to reality on the ground. Secondarily, this looks like a huge loss for Putin right now. He understands it, and I think he would have a hard time, even with his control of information, spinning this to his public without removing Zelenskyy, without the"de-Nazification,"as he calls it – which is an obscenity in an environment where the Ukrainian president is actually Jewish – the disarmament of Ukraine, and of course, the **ability** of the Russians to change how they feel about Ukraine as a threat to the Russian homeland. BG : What level of **support** can we give Ukraine militarily, intel, economic, before Putin considers taking a strike on a NATO country? IB : Well, it ' s interesting the way you framed that, Bruno. Because I mean, I think that Putin is already considering strikes on NATO countries. I mean, there were massive attacks, cyberattacks and **disinformation** attacks, by Russia against NATO countries with reckless abandon over the course of the past years. And in fact, when President Biden met with Putin in Geneva back in June, it seems like years and years ago at this point, Biden set the agenda. Ukraine was largely not discussed, but what was discussed was cyberattacks on critical infrastructure. Because you may remember Bruno, that meeting came right after the cyber attacks against the Colonial Pipeline. And the Russians after that indeed pulled back on **supporting** those attacks by their criminal cyber syndicates. I expect those attacks to restart in very short order against NATO countries. I also believe that the fact that the West is sending weapons to Ukraine and is providing real-time **intelligence** reports on the disposition of Russian troops on the ground in Ukraine to better allow the Ukrainians to defend themselves and blow the Russians up, that is considered by the Russians to be an act of war, and they will retaliate. And how they retaliate is the question. I don ' t think they ' re going to send troops into Poland. But you know, when the Americans under Reagan were providing that kind of **support** to the mujahideen to help them defeat the Soviets in Afghanistan, the Soviets saw that as an act of war, and they engaged in acts of terrorism against the mujahideen in Pakistan. And I absolutely think that that is on the table in terms of front line NATO countries especially, like Poland, like the Baltic states, like Bulgaria, Romania. Would that be considered an Article V attack? Would that force NATO countries to strike the Russians back? I ' m not sure it would. Not directly, not militarily. So I mean, I do think that the fact that the NATO countries see there is some sort of a big red line between sending troops in, for example, and doing a no-fly zone because that could cause World War III, but everything short of that is just a proxy war. The Russians don ' t see it that way. And that gives the Russians some advantage tactically in terms of their willingness to escalate. BG : You ' re describing a spiral of escalation here that will touch the globe and not only Ukraine, not only the eastern flank of Europe, which means that not only this war has ripple effects everywhere, but this is also starting a sort of realignment of the global geopolitical situation and context. To me, it has been very striking how Europe and the US have kind of moved very fast in a cohesive way. And it has chosen, after years of prioritizing the economics in their **international** and global dealings, it ' s chosen to put politics over markets. It has adopted sanctions that will hurt Russia, but will also hurt Western businesses. It ' s the discussion about decoupling that you put forward before, an active kind of fencing out of the Russian economy. Talk to us about how do you see this decoupling playing out. IB : Yeah, I mean, I do think that for the Europeans, this is a permanent move. I mean, I ' ve spoken to top **leaders** in the German government who tell me that Nord Stream was a strategic mistake, and they understand it. Who say that, you know, Scholz making this speech from the Social Democratic Party on the center left is the equivalent of Nixon going to China. No one else could have made that move. But having made it, everyone is on board. The popularity in Germany, even given the massive economic consequences, is extraordinary and is across the board. And what they need to do now is **ensure** that the diversification of fossil fuels in the near term away from Russia, towards Qatar and Azerbaijan and even, you know, sort of the United States for LNG, can be done as fast as humanly possible. And that further, even though it ' s going to cost a lot, some of it will be uneconomic, the move towards renewables actually picks up and is faster. The Italians, Mario Draghi, I think his shift in strategic orientation that they will do, this is his"whatever it takes"moment. He had that in response to the 2008 financial **crisis** as the head of the European Central Bank, and that made him"Super Mario."This is making him Super Mario as the Italian prime minister. This is the"whatever it takes"moment for the Italians who never make public statements that undermine their economic, their commercial interests like this in such a strategic way. The French, of course, have less to be concerned about in the sense that most of their energy comes from nuclear power and is domestic. So they are not as affected directly by a cut-off from Russia. And also because Macron fancies himself, especially as the **leader** of the European Commission this year, the rotating **leadership**, occupying the presidency, but also with his elections coming up, and just given his personal belief that he can drive diplomacy, I believe that even after Kyiv falls and after Zelenskyy is either killed or forced out that the Americans will not want to engage in direct diplomacy, the Germans probably won ' t. The French will. And by the way, the Chinese will too. And I do believe that there is a potential, and this is a danger for the cohesiveness of the West, that the Chinese and Macron end up being the post-Kyiv Normandy format of diplomacy. That ' s something that the Americans and the Germans right now are starting to worry about quite a bit. Now that ' s the European shift. And I think, as I said, I think it ' s permanent. I believe the UK is in that camp as well. I ' m not so sure the United States is going to be as committed for as long a term. It doesn ' t affect the Americans as much economically, it doesn ' t affect the Americans as much in terms of a direct security issue. None of those **refugees** are coming to the United States. But also American inequality, American political polarization and **dysfunction** is so much greater than what you experience on the continent in Europe. So the potential that in six months ' time or in two years ' time, as we ' re thinking about the 2024 election, that the Americans have largely forgotten about this Russia issue instead, are focusing once again on domestic political opponents as principal adversaries, which deeply undermines NATO, much more than anything that would come from the Europeans, I think that is a real open question going forward that is perhaps as significant as the question of where the Chinese go. BG : Let me pick up on the point you made about energy, because somehow Putin ' s calculus can really change if Russian oil and gas stops flowing to Europe, if it becomes part of the sanctions, right? And this war indeed can kind of be read as a war about energy. Selling energy funds it for Russia, being dependent on Russian energy makes the European response more constrained. Rising energy insecurity, rising energy cost may or probably will destabilize European politics and economy in the coming months. How would you look at this from the perspective of energy, and is there any likelihood that Russian oil and gas is going to stop flowing, either because Putin cuts it or the Europeans sanction it? IB : Yeah, or because it ' s blown up in some of the transit in Ukraine? I mean, keep in mind, so much of the gas transit is going through large pipeline networks, which have some redundancy across all of Ukraine. But there ' s a big war that ' s going on right there, and lots of people that could have incentive to create problems. The Americans, of course, the Canadians, have said that they ' re cutting off oil import from Russia, but those are nominal numbers, so they don ' t matter very much to the markets. The Europeans, as I said, want to decouple themselves as quickly as possible, but they believe that doing that this year would be economic suicide. So there isn ' t, despite everything we see from Russia, they ' re using thermobaric weapons now against the Ukrainian people, the Americans are warning that they could use chemical, biological weapons against Ukraine. I mean, you know, you even have some people saying, what if they use a tactical nuclear weapon? I mean... God willing, none of these things come to pass. But it is very hard to see a military scenario in Ukraine that leads the Europeans to completely cut off their inbound gas from Russia this year. It ' s very hard to see. And also, I would say, it ' s very hard to see any level of economic sanction that would change the mind of the Russians in terms of their military decision making on the ground in Ukraine. Now, I think there are a lot of things that the West is doing in terms of providing weapons for the Ukrainians that are having an **impact** on the ground. A lot more Russians are getting killed. It won ' t prevent them from taking to Kyiv, again in my mind I feel quite confident about that. But it ' s quite possible, perhaps even likely, that the **west** of Ukraine will remain in Ukrainian hands, which means that, you know, after this fighting is"over,"that a rump Ukrainian state in exile exists in the West, run by Zelenskyy or someone that ' s aligned with him, and that they continue to get enormous economic and military **support** from all of the NATO countries. So even though I don ' t think that the energy situation will become so parlous that it would affect Putin ' s decision making, I do think that the West ' s response does matter on the ground. BG : The war is kind of having radiating economic shock waves around the world now, ripple effects on food markets, for example and food security. We talk a lot about energy security, what about food security? IB : Well, you have the largest grain producer in the world invading the fifth largest grain producer in the world on the back of a two-year **pandemic** that ' s still ongoing. We don ' t talk about it much anymore, but it ' s still there. And of course, this hit the poorest countries in the world the hardest. The level of indebtedness and the unsustainability of paying that debt off was already becoming a massive problem for so many of the developing countries in the world. And the IMF provided a lot of relief in special drawing rights and direct aid over the course of the past 12 months, but that money is now running to an end. And what happens when commodity prices spike up and we have severe supply chain challenges with energy and food, and those things are obviously very related. What happens is that a lot of people die. What happens is we see a lot more starvation. The World Food Organization says about 10 million people a year die of starvation. That number in the next 12 months is going to be a lot higher than it otherwise would have been. The number of people who are food stressed in the world is going to go way up in sub-Saharan Africa, in Yemen, in Afghanistan, in Bangladesh. It ' s going to go way up. And you know, it ' s horrible to think about, but the massive **impact** of this Russia **crisis** is going to be much more global inequality. And this is, of course, a direct consequence of the end of the peace dividend more structurally. That over the last 30 years of globalization, what did you have? A lot of people were left behind, but the biggest thing you had was the explosion of a single global middle class. On the back of the **pandemic** and now this Russia-Ukraine war and the decoupling of the Russian economy from the West, which doesn ’ t matter so much in terms of the size of the Russian economy, but it matters immensely in terms of commodities globally and supply chain, those two things are going to seriously unwind the growth of this global middle class, and they ’ re going to stress developing countries to a much greater degree. They will lead to financial **crises** in countries like Turkey, for example, that will no longer be able to service their debt. You ’ ll see more Lebanons out there. You ’ ll see some in Latin America, you ’ ll see some in sub-Saharan Africa. Those are the knock-on effects and so, so many people that have been saying over the last few weeks,”Why are we paying so much attention to Ukraine? It ’ s because they ’ re white people, because they ’ re European. We wouldn ’ t pay that much attention if they were Afghans or if they were, you know, Afghanis or if they were Yemenis. We wouldn ’ t.”I mean, first of all, you ’ ve got millions and millions of Ukrainian **refugees**, and we ’ re not paying as much attention to them as we did to the Syrian **refugees** precisely because of race, precisely because the Europeans are more willing to integrate millions and millions of”fellow Europeans”into Europe. But we are paying much more attention to the Ukraine **crisis** and we should, because the **impact** on the poorest people around the world is vastly greater from this **conflict** than anything that we ’ ve seen in any of those smaller economies with less **impact**, despite all of the human depredation that ’ s happened over the past 30 years. BG : Ian, I want to talk for a second about **climate** because another **crisis** that has, kind of, disappeared from the headlines is the **climate crisis**, right? Ten days ago, the IPCC released a report that the secretary general of the UN described as an”atlas of human suffering,”if I remember correctly. And it has been basically ignored. Over the last several years, much of the world had started to embark, with more or less enthusiasm, on a process of transitioning away from oil and gas and into kind of a clean energy future. And now the war comes in and we look at what you just described, the unraveling of global supply chains, the dependency on energy and so on. And there are kind of two schools of thought here. One says this war is going to accelerate the adoption of clean energy because we need to diminish dependence from Russia and these fossil fuels. And the other says, the other school of thought says it ’ s going to derail the **transition** to clean energy because suddenly the priority is no longer **decarbonization**, suddenly the priority is energy security, energy supply. IB : The Europeans are largely in the first camp, and they will move towards faster decoupling and investment accordingly. The Americans are largely in the second camp, and they will move towards”Let ’ s focus more on fossil fuels and partisan divide on this issue,”accordingly. The Chinese, who are the largest carbon emitter in the world by a long margin, though not per capita and not historically, but still in terms of every year totals, they will continue on the same path they ’ ve been on, which is a net-zero target but without yet a very strong plan on how to get there and not feeling a lot of pressure to provide that plan, because they think the Americans are completely incoherent and incapable of effectuating a strategic long-term plan on **climate** themselves. So I mean, what we have is a lot of progress on **climate** and, of course, technology around renewable energies and around electric batteries and supply chain continue to get cheaper and cheaper as more money is being **invested** in it. And that does make me long-term more optimistic that by 2045, a majority of the world ’ s energy will probably be coming from renewables. And five years ago, I wouldn ’ t have said that. But still, I mean, when the news today is that the Americans are sending a high-level delegation to Caracas to figure out if we can reopen relations with Venezuela to get them to produce more oil again.

With the Iranians, let ’ s do any deal possible to get back into the JCPOA, the nuclear deal, so that we can get that oil on the market. Calling the Saudis, calling the Emiratis, and they ’ re not willing to take Biden ’ s **phone** call on this issue while they ’ re talking to Putin. Those are warning signals that in the near term, we ’ ve got some big challenges and a lot of those challenges are going to be filled with fossil fuels and fossil fuel development. And so I do think that the fact that both of the answers to your question are true in different places on net-net is more negative for how quickly we can **transition**. BG : Let ’ s talk a bit about China. Brigid, I think, who ’ s listening in asks,”What do you believe Xi Jinping is learning from the world ’ s response to the **crisis**, to the Ukrainian war?”IB : Well, certainly learning that this was a red line for the West. And I think that this would have surprised, it obviously surprised Putin, I think it would have surprised Xi Jinping as well. Xi Jinping saw Afghanistan. He saw that Merkel was out. He saw that Macron is focused on strategic autonomy. He sees Biden as much more focused on China and Asia. I think that this is a surprise to Xi Jinping. But Xi Jinping also sees that a lot of the world is not with NATO on this issue. 141 countries, if I remember correctly, voted to censor the Russians for their invasion of Ukraine at the United Nations General Assembly. But very large numbers of that 141 are not on board with all of these sanctions against Russia. They ’ re happy with the diplomatic censure, but they need to continue to work with the Russians. The Chinese see that too. The Chinese see just how much more fragmented the global order is. I thought the most significant thing that we ’ ve seen from the Chinese so far two issues. The first is, of course, when Putin went to Beijing and Xi Jinping made the public announcement that”this is our best friend on the global stage, and we will work much more strategically with them economically, diplomatically and militarily going forward.”And Xi Jinping knew very well where Ukraine was heading at that point and also knew that the likelihood of an invasion was coming. Didn ’ t stop him from making that announcement in the slightest. And then after the invasion, and it ’ s going badly, I mean, if you watch Chinese social media, the fact is that the censorship is all about Ukraine. I mean, the Chinese media space is pursuing a relentlessly pro-Putin **policy**. They have media embedded with Russian troops on the ground in Ukraine. Now, publicly, the Chinese government wants to be seen as :”We ’ re neutral, we like the Russians, we like the Ukrainians, we still want to work with **everybody**.”But the fact is that China feels no problem being publicly completely aligned with Putin, despite the fact that they are invading a democratic government with 44 million people in the middle of Europe. That ’ s a pretty astonishing statement from the Chinese. And there ’ s no question that they have learned that they ’ re in a vastly better economic position than they used to be, and that gives them influence. They are a government who projects its power primarily through economic and technological means, as opposed to Russia that projects it primarily through military means. And the Chinese believe that there is a level of decoupling that is already going on as the Americans focus on more industrial **policy**, as they focus on America first for American workers. A US foreign **policy** for the American middle class, as Biden put it, is one that really pushes a lot of capital to leave a country like China, which had served as the factory for the world, but at the expense of a lot of labor coming out of advanced industrial economies. And now, yes, there are definitely some dangers that come from the Chinese being perceived as too close to Russia, and they won ’ t want that, and they ’ ll want to make sure that they ’ re engaging diplomatically with the Europeans to try to minimize that damage. But I thought it was very **interesting**, and I ’ m not sure this is public yet, that the Chinese ambassador to Russia recently, in the last few days, organized a meeting of a lot of the top investors Chinese investors in

Russia, saying,”This is a unique opportunity, the West is leaving, we should be going in and doing more. Because they ’ re going to be completely reliant on us going forward.”That is not a message that the Chinese ambassador delivers unless he is told directly to from Beijing. BG : Ian, I ’ m going to jump from topic to topic because there are several questions in the chat. Nancy is asking about whether Putin can be removed from power. There ’ s been a lot of discussion lately about regime change in Russia, either endogenous, like a palace coup, or provoked by sanctions and other **policies**. And so she asks,”How likely is that Putin will face a challenge from inside Russia, whether a popular uprising, a coup or other?”IB : It ’ s very, very unlikely until it happens. I mean, in the sense that there is absolutely no purpose in trying to say, oh, I mean, you know, there are rumors that Defense Minister Shoigu is unhappy and, you know, he might be making a move. And I ’ ve seen these from relatively credible analysts, I ’ m like, no, no, if there are such rumors, then we know it ’ s not happening because that ’ s the end of Shoigu and his family. But it ’ s very clear that there is more pressure on Putin now than at any point since he ’ s been president. Domestic pressure on Putin. About 10, 000 Russians have been arrested so far, detained, most of them have been released, for nonviolent anti-war protests. The Russians have shut down all the Western media. They ’ ve shut down all the Russian opposition and independent media. So Putin has control of the space, though if you look at Russian conversations on Telegram, you ’ ll still see a bunch of people that are seriously, seriously anti-war. But, you know, once the economy starts truly imploding and you can ’ t find goods on shelves in Russia in major cities, and this is coming, you know, very soon, this is a matter of days, in many of these cities, those demonstrations will likely become greater, some of them can become violent. You know, that ’ ll increase the pressure. Then you have the issue of how the Russians are fighting on the ground. I mean, what happens if you get a lot of desertions? We haven ’ t seen that so far. What happens if you get orders to bomb Kyiv and a whole bunch of Russian fighter pilots, bomber pilots, decide not to and they defect to Poland, to Germany. That would have a big **impact** on morale. That has not happened so far. I mean, do be aware of the fact that the Ukrainians are winning the war on information, and that means that the information that you are getting in the **West** about the war is much more pro-Ukrainian – morale, enthusiasm, how well the military is doing – than what ’ s actually happening on the ground. And also be aware of the fact that the Russians completely control the war on information inside Russia. BG : Exactly. IB : They ’ re not getting a balanced view. They ’ re getting a completely pro-Putin view. And most of them actually believe it in the same way that most people that voted for Trump in the US believe that the election was stolen and Trump is still president. I mean, it ’ s much worse in Russia in that regard than it is in the United States, and I think that that ’ s underappreciated in the West. So even though I think there ’ s pressure, I really don ’ t think that it ’ s super likely that Putin is out anytime imminently. BG : Ed is asking whether you see any off-ramp for Putin. IB : I think that the most likely off-ramp for Putin is after Kyiv is taken and Zelenskyy is removed one way or the other, at that point, the possibility of the Russians accepting a frozen **conflict** or a cease fire that could lead to ongoing negotiations is a lot higher because Putin can sell that as a win back home much more easily. But also because further Russian attacks at that point serve much less purpose for the Russians, are much harder to bring about, and potentially have much more negative consequences. So for me, that would be the near-term potential break where we could at least freeze issues largely where they are. Now whether that could then eventually lead to a climbdown or not, I mean, the Russians have been very happy with frozen **conflicts** on their borders for years and years and years. I ' m thinking about Nagorno-Karabakh between Armenia and Azerbaijan, which basically stayed in place until the Azeris, over the course of a decade got enough military capacity that they could forcibly change the situation on the ground. Which, by the way, the Ukrainians might also be eventually thinking about because the West will be supplying them with advanced weapons all the way through. I ' m thinking South Ossetia in Georgia. I ' m also, of course, thinking about the two pieces of Ukraine they took back in 2014. So be aware of the fact that a negotiation that creates a cease fire does not mean you ' re anywhere close to a resolution or an end of the fighting that we ' re seeing. BG : Someone else in the chat, who didn ' t sign by his or her name, is asking about the nuclear fear that hangs over the **conflict**. How should we think of that? IB : Yeah, we don ' t like it when Putin uses the N-word, and there ' s no question, I mean, he and his direct reports have rattled nuclear sabers on at least five times that I ' ve seen in the past few weeks. I think that... In 1962, I wasn ' t alive, we had the Cuban Missile **Crisis**. There was a real possibility of nuclear confrontation between the world ' s two superpowers. At least for the last 30 years, there ' s been no chance of that. Functionally, no chance of that. I think we ' re now back in a world where a Cuban Missile **Crisis** is again a reality. Now, that doesn ' t mean that I think nuclear war is likely or imminent. I don ' t. And in fact, there is active deconfliction going on even today : the Americans and Russians with a new hotline, the secretary general of the UN, with the Russian defense minister engaging in deconfliction measures with UN offices being invited to Moscow. So as bad as it is right now, people that have been doing this for a long time are trying to avoid nuclear war. But that ' s the point. Is we ' re now in a situation where the **conflict** that we ' re going to experience needs to be actively managed because of the danger of nuclear confrontation. So it now becomes a risk on the horizon that we must be continually aware of, even if only at a low level, as we take and consider further actions, as we consider diplomacy, as we consider escalation. It is now on the table in a way that frankly is so debilitating. I mean, as human beings all on this call, one of the most painful things to think about is the fact that we still have these 5, 000 nuclear warheads in Russia and 5, 000 the United States that are still pointed at each other, and they still have the potential to destroy the planet. And we haven ' t had any real lessons that we ' ve been able to learn institutionally from 1962. BG : 5, 000 being a generic figure, not the exact figure, but we are kind of in that order of magnitude. Then of course, there is the question of civilian nuclear, so the two power plants, nuclear power plants, that have been seized by the Russians. One has been slightly damaged by a bomb, the other has been turned off. But those are also potentially gigantic nuclear problems just waiting to happen. IB : Chemical weapons, biological weapons. I mean, look, we have had two million **refugees** from Ukraine in two weeks. As this continues, you ' re looking at five to 10 million **refugees**. I mean, it is hard – Just take a step for a moment just as a human being. Imagine what it would take for a quarter of your country ' s population to say :"I am not living here anymore. I am leaving everything because of the condition of the country, because of this unjust war that has been imposed upon you by your neighbor."That ' s what we ' re looking at. And again, it ' s important for us to, you know, not lose the humanity of this **crisis** and the extraordinary hardship that is being **visited** upon 44 million Ukrainians that have done nothing wrong, they have committed no sin other than their desire to have an independent country. BG : One other country that has not yet taken a very clear position is India. IB : Well, they ' re a member of the Quad, and their relationship with China is pretty bad, and that ' s mutual. But in terms of Russia, there ' s been a long-standing relationship, trade relationship, defense relationship between India and Russia that the Russians are not going to jettison, and they see no reason to jettison it. And as long as you ' ve got a whole bunch of other countries out there that are substantial, that are willing to say, we ' re going to keep playing ball with the Russians then the Indians will too. And that ' s why you ' ve got the abstention in the United Nations vote. And that ' s why you ' ve had very careful comments as opposed to overt and strong condemnation coming from the Indian **leadership**. BG : Phil in the chat is asking,"Will this cause a fragmentation of the financial system with kind of a Western system and an Eastern system?"So two different SWIFT-like systems, two different credit card systems, crypto, what ' s the role of crypto in all this? IB : I hope not. I mean, I will tell you that before the invasion started, if you talk to most Western CEOs, and I ' m talking across the entire sweep of **sectors**, so it ' s finance and it ' s manufacturing and its services and it ' s **tech**, most of them would have told you that they did not in any way plan on reducing their footprint in China, and a lot of them said that China was their most important growth market in the world. Not a surprise. China is going to be the largest economy in the world in 2030. So, you know, a world that you ' re decoupling is not a good world when China is going to be number one economically. I mean, that obviously is going to hurt the **West** in a big way. So there are strong incentives against that, and there remain very strong and powerful entrenched interests in the United States and Europe that will resist direct decoupling. Despite the fact that there are these more incremental moves towards friendsourcing and insourcing because, you know, Chinese labor is more expensive, you don ' t need as much labor to get capital moving, given robotics and big data, deep learning all of those things. But I do think that the Russia **conflict** risks a level of second-order decoupling. Because if the Russians end up financially integrated with China in their own, not-as-effective SWIFT system, and all of their energy ends up going to China and the Chinese build that infrastructure and they get a discount on it, and Russia ' s technology and their military industrial complex gets serviced by Chinese semiconductors and Chinese componentry, well, I do think that there will be knock-on decoupling that will be longer term and more strategic from the United States, from the Europeans and even from Japan and South Korea. So that is a worry, and I think the Chinese are highly aware of that. And over the coming months, they will do everything they can, both with the Europeans in particular, but also, I expect at least with some of the Asian economies, to try to limit the **impact** of that. Now, keep in mind, we haven ' t talked at all about Asia yet outside of China. The new Japanese Prime Minister Kishida is at least as **hawkish** in his orientation towards China and Russia as Abe was. He is providing **support** for the Ukrainians, including some military capacity – unheard of for the Japanese. He ' s allowing Ukrainian **refugees** – unheard of for the Japanese. And yesterday, the South Koreans had a very, very tight election, and Yoon is now in charge. He is on the right, and he is the **guy** that is strongly anti-China, was talking about South Korea having nuclear capabilities, wants a new THAAD missile defense system for the South Koreans and wants to rebuild the relationship with Tokyo. That matters. And that ' s a big strategic change in the geopolitical map that will look more problematic on the decoupling front from Beijing ' s perspective. BG : Three final quick questions that all come from the chat, Ian. One is, because you mentioned the rest of Asia outside of China,"What about the rest of the world? What about Africa and Latin America? How do they factor into this conversation or don ' t they?"IB : They factor in. I mean, those that have significant commodities do well because the prices are going to be so high. Those that don ' t are going to be under massive pressure for reasons we already talked about, but they are not going to be forced to pick a side on this one. I just don ' t see it. In the same way that if you were Colombia in the last couple of years, you know, you found, even though you ’ re working very closely with an American ally, you ’ re still dealing with Huawei and 5G. This is knock-on effects of all of this. These are countries that are not going to take on significant economic burden, given how much they ’ re suffering right now geopolitically. BG : Another one is about sanctions. How do we even know when and how, at what point we start rolling back sanctions? IB : I think that as long as Ukraine is occupied by the Russian government for the foreseeable future and Putin is there, I can ’ t see these sanctions getting unwound. Now, if a rump Ukrainian government that is democratically elected were prepared to sue for peace and retakes most of Ukraine, but they give away Crimea and they give away the Donbass, could you see in that environment some of these sanctions unwound? Sure. But I mean, I am suggesting that I think that many of these sanctions are functionally permanent. They reflect a new way of doing business. And when people ask me what ’ s going to happen when this is over, my response is, what do you mean over? What ’ s over is the peace dividend. We are now in a new environment. BG : And one of the figures of this new environment and I want to close with that, is President Zelenskyy of Ukraine, who was not taken very seriously when he was elected, he has come out as a significant figure in this war. What do you make of President Zelenskyy? How do you read this character? IB : He ’ s very courageous. I ’ m obviously inspired by his **ability** to communicate and rally his people and take personal risk in Kyiv while this invasion is going on. But I ’ m very **conflicted** because I think many of the steps that Zelenskyy took in the run-up to this **conflict** actually made the likelihood of **conflict** worse. He was unwilling to take the advice of the Americans and Europeans seriously in the months leading up to the **conflict**. He was unwilling to mobilize his people to ready them for the potential of **conflict**. He was certainly unwilling to give an inch in terms of Ukraine ’ s desire to be a member of NATO, even though he knew completely that no one in NATO was prepared to offer a **membership** action plan, let alone actually bring them in as members. And part of that is a lack of experience and lack of any business being in that position in the run-up to this **crisis**. So I ’ m very deeply **conflicted** in my personal views on Zelenskyy, given the way he behaved before the invasion, compared to the extraordinary **leadership** that he has displayed to all of us over the last two weeks. BG : Ian, thank you for taking the time, for sharing your knowledge, and your analysis with us. We deeply appreciate it. Thank you very much. IB : Good to see all of you. [Get access to thought-provoking events you won ’ t want to miss.] [Become a **TED** Member at **ted**. com / **membership**]

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Bruno Giussani : We are at the end of day six of the war in Ukraine or, more correctly, of the Russian invasion of Ukraine, launched on February 24 by President Vladimir Putin. We are all shocked and saddened by the events and by the human suffering they are causing. And as we speak, really, a Russian military convoy is headed towards Kyiv, other Ukrainian cities are being bombarded, half a million Ukrainians have already fled to neighboring countries and much more. It ’ s still early days, and it ’ s difficult to predict how the situation will evolve even just in the next few hours. But this is a war that should concern everyone, everywhere. And so today, in this **TED Membership** conversation, we want to try to give it a broader context with our guest, historian and author, Yuval Noah Harari. Yuval, welcome. Yuval Noah Harari : Hello. Thank you for inviting me. BG : I want to start from Ukraine itself and its 42 million people and its particular place between the East and the West. What do we need to know about Ukraine to understand this war and what ' s at stake? YNH : The most crucial thing to know is that Ukrainians are not Russians, and that Ukraine is an ancient, independent nation. Ukraine has a history of more than a thousand years. Kyiv was a major metropolis and cultural center when Moscow was not even a village. For most of these thousand years Kyiv was not ruled by Moscow. They were not part of the same political entity. For centuries, Kyiv was looking westwards and was a part of a union with Lithuania and Poland until it was eventually conquered and absorbed by the Russian Empire, by the czarist empire. But even after that, Ukrainians remained a separate people to a large extent, and it ' s important to know that because this is really what is at stake in this war. The key issue of the war, at least for President Putin, is whether Ukraine is an independent nation, whether it is a nation at all. He has this fantasy that Ukraine isn ' t a nation, that Ukraine is just a part of Russia, that Ukrainians are Russians. In his fantasy, Ukrainians are Russians that want to be back in the fold of Mother Russia, and that the only ones preventing it is a very small gang at the top, which he portrays as Nazis, even if the president is Jewish ; but OK, a Nazi Jew. And his belief was, at least, that he just needs to invade, Zelenskyy will flee, the government will collapse, the army would lay down its arms, and the Ukrainian people would welcome the Russian liberators, throwing flowers on them. And this fantasy has been shattered already. Zelenskyy hasn ' t fled, the Ukrainian army is fighting. And the Ukrainian people is not throwing flowers on the Russian tanks, it ' s throwing Molotov cocktails. BG : So let ' s unpack that and maybe take the different pieces one way one. So Ukraine has a long history of being dominated and occupied. You mentioned the czar, but also the Soviet Union, Hitler ' s armies. It also has a long history of mistrust of authority and of resistance, which goes some way to explain the current strong resistance that the Russians are encountering. Anne Applebaum, the journalist, even suggests that this mistrust, this resistance to authority, is the very essence of Ukraine-ness, do you agree? YNH : We did see in the last 30 years Ukrainians twice rising in revolt when there was a danger of an authoritarian regime being established – once in 2004, once in 2013. And when I was in Kyiv a few years ago, what really struck me was this very strong feeling of the desire for independence and for democracy. And I remember walking around this museum of the Revolution of 2013-2014 and seeing these images, like these two elderly women who were bringing sandwiches to the demonstrators, to the fighters. They couldn ' t throw stones and they couldn ' t do anything else, so they prepared sandwiches and brought this huge tray full of sandwiches to the demonstrators. And this, yes, this is the kind of spirit that inspires not just the Ukrainians but **everybody** who is now watching what is happening there. BG : Help me understand the actual nature of the threat here in terms of Russia moving into Ukraine. So in your last book, when you write about Russia, you describe the Russian model as :"not a coherent political ideology, but rather a sort of practice of monopolizing power and wealth by a small group at the top."But then, in his actions against Ukraine, Putin in the last few weeks seems to move very much by an ideology, an ideology of empire, of denial of Ukraine ' s right to exist, as you mention. What has changed in the four years since you wrote that book? YNH : The imperial dream was always there, but you know, empires are often the creation of a very small gang of people at the top. I don ' t think the Russian people [are] **interested** in this war. I don ' t think that the Russian people want to conquer Ukraine or to slaughter the citizens of Kyiv. It ' s all coming from the top. So there is no change there. I mean, when you look at the Soviet Union, you can say that there was this mass ideology, which was shared by a large proportion, or some proportion, of the population. You don ' t see this now. You know, Russia is a very rich country, rich in resources, but most people are very poor. Their standard of living is very, very low because all the wealth and power is kind of sucked by the people at the top, and very little is left for **everybody** else. So I don ’ t think it ’ s a society where the masses are part of this kind of ideological project. They ’ re being ruled from the top. And you have this classic imperial situation, when the emperor, which controls the largest country in the world, feels that,”Hey, this is not enough. I need more.”And sends his army to capture, to extend the empire. BG : I said at the beginning that it ’ s difficult, of course, to make predictions. But yesterday, you published an article in”The Guardian”titled :”Why Putin has already lost this war.”Please explain. YNH : Well, one thing should be very clear. I don ’ t mean to say that he ’ s going to suffer an immediate military defeat. He definitely has the military power to conquer Kyiv and perhaps the whole of Ukraine. Unfortunately, we might see this. But his long-term goal, the whole rationale of the war, is to deny the existence of the Ukrainian nation and to absorb it into Russia. And to do that, it ’ s not enough to conquer Ukraine. You also need to hold it. And it ’ s all based on this fantasy, on this gamble, that most of the population in Ukraine would agree to this, would even welcome this. And we already know that it ’ s not true. That the Ukrainians are a very real nation ; they are fiercely independent ; they don ’ t want to be part of Russia ; they will fight like hell. And in the long-run, again, you can conquer a country, But as the Russians learned in Afghanistan, as the Americans learned also in Afghanistan, also in Iraq, it ’ s much harder to hold a country. And again, the big question mark before the war was always this. Before the war started, many things were already known. **Everybody** knew that the Russian army is much stronger than the Ukrainian Army. **Everybody** knew that NATO will not send armed forces into Ukraine, troops into Ukraine. **Everybody** knew that the West, the Europeans, would be hesitant about imposing too strict a sanction regime for fear of being hurt by it themselves. And this was the basis for Putin ’ s war plan. But there was one big unknown. Nobody could say for sure how the Ukrainian people would react. And there was always the option that maybe Putin ’ s fantasy would come true. Maybe the Russians will march in, Zelenskyy would flee, maybe the Ukrainian army will just capitulate and the population would not do much. This was always an option. And now we know this was just fantasy. Now we know that the Ukrainians are fighting, they will fight. And this derails the whole rationale of Putin ’ s war. Because you can conquer the country, maybe, but you won ’ t be able to absorb Ukraine back into Russia. The only thing he ’ s accomplishing, he is planting seeds of hatred in the hearts of every Ukrainian. Every Ukrainian being killed, every day this war continues is more seeds of hatred that may last for generations. Ukrainians and Russians didn ’ t hate each other before Putin. They ’ re siblings. Now he ’ s making them enemies. And if he continues, this will be his legacy. BG : We ’ re going to talk a bit about that again later but, you know, at the same time, Putin needs a victory, right? The cost, the human, economic, political cost of this war, not even a week in, is already astronomical. So to justify it and also to remain, by the way, a viable **leader** at the head of Russia, Putin needs to win, and even win convincingly. So how do we square these things? YNH : I don ’ t know. I mean, the fact that you need to win doesn ’ t mean that you can win. Lots of political **leaders** need to win, and sometimes they lose. He could stop the war, declare that he won, and say that recognizing Luhansk and Donetsk by the Russians is what he really wanted all along, and he achieved this. Maybe they cobble this agreement, or I don ’ t know. This is the job of politicians, I ’ m not a politician. But I can tell you that I hope, for the sake of **everybody** – Ukrainians, Russians and the whole of humanity – that this war stops immediately. Because if it doesn ’ t, it ’ s not only the Ukrainians and the Russians that will suffer terribly. **Everybody** will suffer terribly if this war continues. BG : Explain why. YNH : Because of the shock waves destabilizing the whole world. Let ' s start with the bottom line : budgets. We have been living in an amazing era of peace in the last few decades. And it wasn ' t some kind of hippie fantasy. You saw it in the bottom line. You saw it in the budgets. In Europe, in the European Union, the average defense budget of EU members was around three percent of government budget. And that ' s a historical miracle, almost. For most of history, the budget of kings and emperors and sultans, like 50 percent, 80 percent goes to war, goes to the army. In Europe, it ' s just three percent. In the whole world, the average is about six percent, I think, fact-check me on this, but this is the figure that I know, six percent. What we saw already within a few days, Germany doubles its military budget in a day. And I ' m not against it. Given what they are facing, it ' s reasonable. For the Germans, for the Poles, for all of Europe to double their budgets. And you see other countries around the world doing the same thing. But this is, you know, a race to the bottom. When they double their budgets, other countries look and feel insecure and double their budgets, so they have to double them again and triple them. And the money that should go to health care, that should go to education, that should go to fight **climate** change, this money will now go to tanks, to missiles, to fighting wars. So there is less health care for **everybody**, and there is maybe no solution to **climate** change because the money goes to tanks. And in this way, even if you live in Australia, even if you live in Brazil, you will feel the repercussions of this war in less health care, in a deteriorating ecological **crisis**, in many other things. Again, another very central question is technology. We are on the verge, we are already in the middle, actually, of new technological arms races in fields like artificial **intelligence**. And we need global agreement about how to regulate AI and to prevent the worst scenarios. How can we get a global agreement on AI when you have a new cold war, a new hot war? So in this field, to all hopes of stopping the AI arms race will go up in smoke if this war continues. So again, **everybody** around the world will feel the consequences in many ways. This is much, much bigger than just another regional **conflict**. BG : If one of Putin ' s goals here is to divide Europe, to weaken the transatlantic alliance and the global liberal order, he seems to kind of accidentally have revitalized all of them in a way. US-EU relations have never been so close in many years. And so how do you read that? YNH : Well, again, in this sense, he also lost the war. If his aim was to divide Europe, to divide NATO, he ' s achieved exactly the opposite. I mean, I was amazed by how quick, how strong and how unanimous the European **reaction** was. I think the Europeans surprised themselves. You even see countries like Finland and Sweden sending arms to Ukraine and closing their airspace. They didn ' t even do it in the Cold War. It ' s really amazing to see it. I think another very important thing is what has been dividing the **West** over the several years now, it ' s what people term the"culture war". The culture war between left and right, between **conservatives** and liberals. And I think this war can be an opportunity to end the culture war within the West, to make peace in the culture war. First of all, because you suddenly realize we are all in this together. There are much bigger things in the world than these arguments between left and right within the Western democracies. And it ' s a reminder that we need to stand united to protect Western liberal democracies. But it ' s deeper than that. Much of the argument between left and right seemed to be in terms of a contradiction between liberalism and nationalism. Like, you need to choose. And the right goes with nationalism, and the left goes more liberalism. And Ukraine is a reminder that no, the two actually go together. Historically, nationalism and liberalism are not opposites. They are not enemies. They are friends, they go together. They meet around the central value of freedom, of liberty. And to see a nation fighting for its survival, fighting for its freedom, you see it on Fox News or you see it in CNN. And yes, they tell the story a little differently, but they suddenly see the same reality. And they find common ground. And the common ground is to understand that nationalism is not about hating minorities or hating foreigners, it ’ s about loving your compatriots, and reaching a peaceful agreement about how we want to run our country together. And I hope that seeing what is happening would help to end the culture war in the West. And if this happens, we don ’ t need to worry about anything. You know, when you look at the real power balance, if the Europeans stick together, if the Americans and the Europeans stick together and stop this culture war and stop tearing themselves apart, they have absolutely nothing to fear – the Russians or anybody else. BG : I ’ m going to ask you a question later about the stories the **West** tells itself, but let me zoom out for a second and get a larger perspective. You wrote another essay last week in”The Economist”, and you argue that what ’ s at stake in Ukraine is, and I quote you,”the direction of human history”because it puts at risk what you call the greatest political and moral achievement of modern civilization, which is the decline of war. So now we are back in a war and potentially afterwards into a new form of cold war or hot war, but hopefully not. Elaborate about that essay you wrote. YNH : Yeah, I mean, some people think that all this talk about the decline of war was always just a fantasy. But... Again, you look at the statistics. Since 1945, there has not been a single clash between superpowers, whereas previously in history, this was, you know, the basic **stuff** of history. Since 1945, not a single internationally recognized country was wiped off the map by external invasion. This was the common thing in history. Until then and then it stopped. This is an amazing achievement, which is the basis for everything we have, for our medical services, for education system, and this is all now in jeopardy. Because this era of peace, it wasn ’ t the result of some miracle. It wasn ’ t the result of a change in the laws of nature. It was humans making better decisions and building better institutions, which means also that there is no guarantee for the future. If humans, some humans, start making bad decisions and start destroying the institutions that kept the peace, then we will be back in the era of war with budgets, military budgets going to 20, 30, 40 percent. It can happen. It ’ s in our hands. And I ’ ll just say one more thing, When, not just me, but other scholars like Steven Pinker and others, talked about the era of peace, some people understood it as kind of encouraging complacency. That, oh, we don ’ t need to worry about anything. No, I mean, the message was really the opposite. It was a message of responsibility. If you think that there is no era of peace in history, it ’ s always war, it ’ s always the jungle, there is a constant level of violence in nature, then this basically means that there is no point struggling for peace and there is no responsibility on **leaders** like Putin because you can ’ t blame Putin for the war. It ’ s just a law of nature that there are wars. When you realize, no, humans are able to decrease the level of violence, then it should make us much more responsible. And it should also make us understand that the war in Ukraine now, it ’ s not a natural disaster. It ’ s a man-made disaster, and a single man. It ’ s not the Russian people who want this war. There ’ s really just a single person who, by his decisions, created this tragedy. BG : So one of the things that has come back in the last weeks and months is the nuclear threat. It ’ s moved back into the center of political and strategic considerations. Putin has talked about it several times, the other day he ordered Russia ’ s nuclear forces on a higher alert status. President Zelenskyy himself at the Munich Security Conference essentially said that Ukraine had made a mistake abandoning the nuclear weapons it had inherited from the Soviet Union. That ’ s a statement that I suspect many countries are pondering. What ’ s your thinking about the return of the nuclear threat? YNH : It ’ s extremely frightening. You know, it ’ s like it ’ s almost Freudian, it ’ s the return of the repressed. We thought that, oh, nuclear weapons, yes, there was something about that in the 1960s with the Cuban Missile **Crisis** and Dr. Strangelove. But no, it ’ s here. And, you know, it took just a few days of difficulties on the battlefield for suddenly – I mean, I ’ m watching television, like, the news and you have these experts explaining to people what different nuclear weapons will do to this city or to this country. It rushed back in. So, you know, nuclear weapons are – in a way they also, until now, preserved the peace of the world. I belong to the school of thought that if it was not for nuclear weapons, we would have had the Third World War between the Soviet Union and the United States and NATO sometime in the 1950s or ’ 60s. That nuclear weapons actually, until today, served a good function. It ’ s because of nuclear weapons that we did not have any more direct clashes between superpowers because it was obvious that this would be **collective** suicide. But the danger is still there, it ’ s always there. If there is miscalculation, then the results could, of course, be existential, catastrophic. BG : And at the same time, you know, in the ’ 70s after Cuba and Berlin, and so in the ’ 60s, but in the ’ 70s, we started building a sort of **international** institutional architecture that helped reduce the risk of military confrontation of nuclear weapons, we used, you know, anything from arms control agreements to measures designed to build **trust** or to communicate directly and so on. And then in the last decade or so, that has been progressively kind of scrapped, so we are even in a more dangerous situation than we were let ’ s say, at the end of the last century. YNH : Completely, I mean, we are now reaping the bad fruits of neglect that ’ s been going on for several years, not just about nuclear weapons, but in general, about **international** institutions and global cooperation. We ’ ve built, in the late 20th century, a house for humanity based on cooperation, based on collaboration, based on the understanding that our future depends on being able to cooperate, otherwise we will become extinct as a species. And we all live in this house. But in the last few years we stopped – we neglected it, we stopped repairing it. We allow it to deteriorate more and more. And, you know, eventually it will – It is collapsing now. So I hope that people will realize before it ’ s too late that we need not just to stop this terrible war, we need to rebuild the institutions, we need to repair the global house in which we all live together. If it falls down, we all die. BG : So we have, among the audience listening, Rola from – I don ’ t know where she ’ s from, she grew up in Lebanon – and she said,”I lived the war, I slept on the ground, I breathed fear. All the reasons were explained to me that the only remaining learning came the war is absurd. We talk about strategy, power, budgets, opportunities, technologies. What about human suffering and psychological trauma?”Especially, I assume what she ’ s asking is what about, what ’ s going to remain, in terms of the human suffering and the psychological trauma going forward? YNH : Yeah, I mean, these are the seeds of hatred and fear and misery that are being planted right now in the minds and the bodies of tens of millions, hundreds of millions of people, really. Because it ’ s not just the people in Ukraine, it ’ s also in the countries around, all over the world. And these seeds will give a terrible harvest, terrible fruits in years, in decades to come. This is why it ’ s so crucial to stop the war immediately. Every day this continues, plants more and more of these seeds. And, you know, like this war now, its seeds were, to a large extent, planted decades and even centuries ago. That part of the Russian fears that are motivating Putin and motivating people around him is memories of past invasions of Russia, especially, of course, in Second World War. And of course, it ’ s a terrible mistake what they are doing with it. They are recreating again the same things that they should learn to avoid. But yes, these are still the terrible fruits of the seeds being planted in the 1940s. BG : It ' s what in same article you call the fact that nations are ultimately built on stories. So these seeds are the stories we are starting to create now. The war in Ukraine is starting to create the stories that are going to have an **impact** in the future, that ' s what you ' re saying. YNH : Some of the seeds of this war were planted in the siege of Leningrad. And now it gives fruit in the siege of Kyiv, which may give fruit in 40 or 50 years in more terrible... We need to cut this, we need to stop this. You know, as a historian, I feel sometimes ashamed or responsible, I don ' t know what, about what history, the knowledge of history is doing to people. In recent weeks, I have been watching all the world **leaders** talking with Putin, and very often he gave them lectures on history. I think that Macron had a discussion with him for five hours, and afterwards, said,"Most of the time he was lecturing me about history."And as a historian, I feel ashamed that this is what my profession in some way is doing. I know it for my own country. In Israel, we also suffer from too much history. I think people should be liberated from the past, not constantly repeating it again and again. You know, **everybody** should kind of free themselves from the memories of the Second World War. It ' s true of the Russians, it ' s also true of the Germans. You know, I look at Germany now, and what I really want to say, if there are Germans watching us, what I really want to say to the Germans : **guys**, we know you are not Nazis. You don ' t need to keep proving it again and again. What we need from Germany now is to stand up and be a **leader**, to be at the forefront of the struggle for freedom. And sometimes Germans are afraid that if they speak forcefully or pick up a gun, **everybody** will say,"Hey, you ' re Nazis again."No, we won ' t think that. BG : That ' s happening right now. I mean, lots of things that were inconceivable just 10 days ago have happened in the last few weeks. And one of the most striking, to me in any case, is Germany ' s **reaction** and transformation. I mean, the new chancellor, Olaf Scholz, the other day announced that Germany will send arms to Ukraine, and will spend an extra 100 billion dollars in building up its army. That reverses completely the principles that have guided Germany ' s foreign **policy** and security politics for decades. So that shift is happening exactly at this moment and very, very fast. YNH : Yeah. And I think it ' s a good thing. We need the Germans to... I mean, they are now the **leaders** of Europe, certainly after Britain left in Brexit. And we need them to, in a way, let go of the past and be in the present. If there is really one country in the world that, as a Jew, as an Israeli, as a historian, that I **trust** it not to repeat the horrors of Nazism, that ' s Germany. BG : Yuval, I want to touch quickly on three things that have to do with the fact that this feels like the first truly interconnected war in many ways. The first, of course, is the basics, which is, on one side, you have a very ancient war – we have tanks and we have trenches and we have bombed buildings – and on the other, we have real-time visibility of everything through cell **phones** and Twitter and TikTok and so on. And you have written a lot about this tension between old ways and new **tech**. What ' s the **impact** here? YNH : First of all, we don ' t know everything that is happening. I mean, surprisingly, with all this TikTok and **phones** and everything, so much is not known. So the fog of war is still there, and yes, there is much more information, but information isn ' t truth. Lots of information is **disinformation** and fake news and so forth. And yes, it ' s always like this ; the new and the old, they come together. You know, with all the talk about interconnectedness and living in cyberspace and all that, one of the most important technologies not just of this war, but of the last decade or two have been stone walls. It ' s Neolithic. **Everybody** is now building stone walls in the era of Facebook and Google and all that. So the old and the new, they go together. And it ' s... It is a new kind of war. People are sitting at home in California or Australia, and they actively participate in the war, not just by writing tweets, but by attacking websites or defending websites. You know, in Spain, in the Civil War, if you wanted to help fight fascism, then you had to go to Spain and join the **international** brigade. Now the **international** brigade is sitting at home in San Francisco and is still in some way part of the war. So this is definitely new. BG : So indeed, just two days ago, Ukraine ' s deputy prime minister, I think, Fedorov, announced via Telegram that he wanted to create a sort of volunteer cyber army. He invited software developers and hackers and other people with IT skills to somehow help Ukraine fight on the cyber front. And according to"Wired"magazine, in less than two days, 175, 000 people signed up. So here is a defending nation that can kind of recruit almost overnight, 175, 000 volunteers to go to battle on his behalf. It ' s a very different kind of war. YNH : Yeah. You know, every war brings it surprises. Sometimes it ' s how everything is new, but sometimes it ' s also how everything is old. BG : So a few people in the chat and in the Q and A, have mentioned China, which of course, is an important actor here, although for now is mostly an observer. But China has a stated **policy** of opposing any act that violates territorial integrity. So moving into Ukraine, of course, violates territorial integrity. And it also has a huge interest in a stable global economy and global system. But then it needs to square this with the recent closeness with Russia. Xi Jinping and Putin met in Beijing before the Olympics, for example, and kind of had this message of friendship that went out to the world. How do you read China ' s position in this **conflict**? YNH : I don ' t know, I mean, I ' m not an expert on China, and I certainly can ' t just... You know, just reading the news won ' t get you into the mindset, into the real opinions and positions of the Chinese **leadership**. I hope that they take a responsible position. And act – because they are close to Russia, they are also close to Ukraine, but especially because they are close to Russia, they have a lot of influence on Russia, I hope that they will be the responsible adults that will put down the flames of this war. They have a lot to lose from a breakdown of the global order. And I think they have a lot to win from the return of peace, including in terms of the gratitude of the **international** community. Now, whether they do it or not, this is with them. I can ' t predict, but I hope so. BG : You have mentioned before the several European and Western **leaders** that have gone to Moscow in the weeks before the invasion. Varun in the chat, asks,"Is the Ukraine war a failure of diplomacy?"Could have... Something different happened? YNH : Oh, you can understand it in two questions. Did diplomacy fail to stop the war? Absolutely, **everybody** knows that. But is it a failure in the sense that a different diplomatic approach, some kind of other proposition, would have stopped the war? I don ' t know, but it doesn ' t seem like it. I mean, looking at the events of the last few weeks, it doesn ' t seem that Putin was really **interested** in a diplomatic solution. It seemed that he was really **interested** in the war, and I think, again, it goes back to this basic fantasy that if he really was concerned about the security situation of Russia, then there was no need to immediately invade Ukraine. There was no immediate threat to Russia. There was no discussion of right now, Ukraine joining NATO. There was no invasion army assembling in the Baltic states or in Poland. Nothing. Putin chose the moment to start this **crisis**. So this is why it doesn ' t seem that it ' s really about the security concerns. It seems more about this very deep fantasy of re-establishing the Russian Empire and of denying the very existence of the Ukrainian nation. BG : So you live in the Middle East. Someone else in the chat asks,"What makes the situation so unique compared to many other wars that are going on right now in the world?"I would say, aside from the nuclear threat from Russia, but what else? YNH : Several things. First of all, we have here, again, something we haven ’ t seen since 1945, which is a **dominant** power trying to basically obliterate from the map an independent country. You know, when the US invaded Afghanistan or when the US invaded Iraq, you can say a lot of things about it and criticize it in many ways. There was no question of the US annexing Iraq or turning Iraq into the 51st state of the United States. This is what is happening in Ukraine under this pretext or this disguise, this is what ’ s at stake. The real aim is to annex Ukraine. If this succeeds, again, it brings us back to the era of war. I was struck by what the Kenyan representative to the UN Security Council said when this erupted. The Kenyan representative spoke in the name of Kenya and other African countries. And he told the Russians : Look, we also are the product of a post-imperial order. The same way the Soviet empire collapsed into different independent nations, also, African nations came out of the collapse of European empires. And the basic principle of African politics ever since then was that no matter what your objections to the borders you have inherited, keep the borders. The borders are sacred because if we start invading neighboring countries because,”Hey, this is part of our countries, these people are part of our nation,”there will not be an end to it. And if this now happens in Ukraine, it will be a blueprint for copycats all over the world. The other thing which is different is that we are talking about superpowers. This is not a war between Israel and Hezbollah. This is potentially a war between Russia and NATO. And even leaving aside nuclear weapons, this completely destabilizes the peace of the entire world. And again, I go back again and again to the budgets. That if Germany doubles its defense budget, if Poland doubles its defense budget, this will spread to every country in the world, and this is terrible news. BG : So Yuval, I ’ m jumping from topic to topic because I want to use the last few minutes to ask a few questions from the audience. A few people are asking about the link to the **climate crisis**, particularly when it relates to the energy flows. Like, Europe is very dependent, part of Europe, is very dependent on Russian oil and gas, which is, as far as we know, still flowing until today. But could this **crisis**, in a sort of paradoxical way, a bit like the **pandemic**, accelerate **climate** action, accelerate renewables and and so on? YNH : This is the hope. That Europe now realizes the danger and starts a green Manhattan Project that kind of accelerates what already has been happening, but accelerates it, the development of better energy sources, better energy infrastructure, which would release it from its dependence on oil and gas. And it will actually undercut the dependence of the whole world on oil and gas. And this would be the best way to undermine the Putin regime and the Putin war machine, because this is what Russia has, oil and gas. That ’ s it. When was the last time you bought anything made in Russia? They have oil and gas, and we know, you know, the curse of oil. That oil is a source of riches, but it ’ s also very often a **support** for dictatorships. Because to enjoy the benefits of oil, you don ’ t need to share it with your citizens. You don ’ t need an open society, you don ’ t need education, you just need to drill. So we see in many places that oil and gas are actually the basis for dictatorships. If oil and gas, if the price drops, if they become irrelevant, it will not only undercut the finance, the power of the Russian military machine, it will also force Russia, force Putin or the Russians to change their regime. BG : OK, let me bring up a character that **everybody** here in the chat seems to find quite heroic, and that ’ s the Ukrainian president. So Ukraine kind of finds itself with a comedian who turned almost accidental president, who turned now war president. But he has shown an impressive conduct in the last few weeks, especially in the last few days, which can be summarized in that response he gave to the US when they offered to kind of exfiltrate him so he could lead a government in exile, he said,”I need ammunition. I don ’ t need a ride.”How would you look at President Zelenskyy? YNH :

His conduct has indeed been admirable, and he gives courage and inspiration not just to the Ukrainian people, but I think to **everybody** around the world. I think to a large extent the swift and united **reaction** of Europe with the sanctions and sending arms and so forth, to a large extent, this is also to the credit of Zelenskyy. That, you know, when politicians are also human beings. And his direct appeal to them, and you know, they met him many times in person and to see where he is now and the threat that not only him, but his family is also in. And you know, they talk with him, and he says, and they know, that this may be the last time they speak. He may be dead, murdered or bombed in an hour or in a day. It really changes something. So in this sense, I think he made a huge personal contribution, to not just the **reaction** in Ukraine, but around the world. BG : So Sam, who ' s listening, asked this question :"Can you provide some historical context for the force and the meaning of economic and trade sanctions at the level where they are currently imposed. How have previous would-be empires, would-be aggressors, or aggressors, been constrained by such isolations and such sanctions?"YNH : You know, what we need, again, to realize about Putin ' s Russia is that it ' s not the Soviet Union. It ' s a much smaller and weaker country. It ' s not like in the 1960s, that in addition to the Soviet Union, you had the entire Soviet bloc around it. So it ' s easier in this sense to isolate it. It ' s much more vulnerable. Again, does it mean that sanctions would work like a miracle and stop the tanks? No. It takes time. But I think that the **West** is in a position to **impact** Russia with these kinds of sanctions and isolation much more than, let ' s say, with the Soviet Union. And also the Russian people are different. The Russian people don ' t really want this war, even the people in the immediate circle around Putin. You know, again, I don ' t know them personally, from what it seems, it ' s that these people, they like life. They have their yachts and they have their private airplanes and they have their house in London and they have their chateau in France. And they like the good life, and they want to keep enjoying it. So I think that the sanctions can be really effective. What ' s the timetable? That ' s ultimately in the hands of Putin. BG : So Gabriella asks :"I remember the war in former Yugoslavia and the atrocities there. Is there any possibility that this war would escalate into such a situation?"I think an extension to that is : Is this war kind of stirring dormant **conflicts** like in the Balkans, for example, or in the former Central Asian republic? YNH : Unfortunately, it can get to that level and even worse. If you want an analogy, go to Syria. You look at what happened in Homs. At what happened in Aleppo. And this was done by Putin and his airplanes and his minions in Syria. It ' s the same person behind it. And to think that,"No, no, no, this happened in the Middle East. It can ' t happen in Europe."No. We could see Kyiv in the same situation as Homs, as the same situation as Aleppo, which would be catastrophic, and, again, would plant terrible seeds of hatred for years and decades. So far, we ' ve seen hundreds of people being killed, Ukrainian citizens being killed. It could reach tens of thousands, hundreds of thousands. So in this sense, it ' s extremely painful to contemplate. And this is why we need again and again to urge the **leaders** to stop this war, and especially, again and again, tell Putin,"You will not be able to absorb Ukraine into Russia. They don ' t want it, they don ' t want you. If you continue, the only thing you will achieve is to create terrible hatred between Ukrainians and Russians for generations. It doesn ' t have to be like that."BG : Yuval, let me finish with one question about your county. You are in Israel. Israel has close ties with both Russia and Ukraine. It ' s actually home of many Russian-born and many Ukrainian-born Jews. How is the country reacting to this **conflict**, I ' m talking about the government, but also about the population? YNH : Actually, I ' m not the best person to ask. I ' ve been so, kind of, following what ' s happening around the world, I didn ’ t pay so much attention to what is happening right here. And even though I live here, I ’ m not an expert on Israeli society or Israeli politics. Definitely, the sentiment in the street, in the social media is with Ukraine. You see Ukrainian flags, you see on social media people putting Ukrainian flags on their accounts. And another thing, so many people in Israel, they came from the former Soviet Union. And until now, **everybody** was simply known as Russians. You know, even if you came from Azerbaijan or you came from Bukhara, you were a Russian. And suddenly,”No, no, no, no, no. I ’ m not Russian. I ’ m Ukrainian.”And again, these seeds of hatred that Putin is planting, it ’ s reaching also here. That suddenly people are saying no, Russian, Ukrainian, until a very short time ago, it ’ s the same thing. No, it ’ s not the same thing. So the shock waves are spreading. BG : Yuval, thank you for taking the time and being with us today and sharing your knowledge and your views on the situation. Thank you very much. YNH : Thank you and I hope for peace quickly. BG : We all do. Thank you. [Get access to thought-provoking events you won ’ t want to miss.] [Become a **TED** Member at **ted**. com / **membership**]

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Whitney Pennington Rodgers : There is so much happening in the world right now. Our guest today spends her days and, in fact, her entire illustrious career tracking and reporting on the moment ’ s biggest stories. She is CNN ’ s chief **international** anchor and host of the network ’ s award-winning flagship global affairs program, Amanpour on CNN International in London and Amanpour and Co. on PBS in the United States. I ’ m so thrilled to have her here with us to offer context on some of the news stories that are **impacting** our world and our lives. And you can see her there right now, please welcome Christiane Amanpour. Hello, Christiane, how are you doing? Christiane Amanpour : Whitney, thank you so much for having me. I ’ m so glad to be with you for a few minutes and your **TED** community on these really important issues. And yes, I am the chief **international** anchor, but before that, I was, you know, the main **international** correspondent. So the way I work is always informed by me being on the ground, in the field and having essentially walk the walk and talk the talk with the people who are at the coalface. WPR : I love that, and I feel like that ’ s going to give us so much perspective during this conversation. Let ’ s just dive right in. I think one place we ’ d like to start with is in Iran. For those of you who are on the call who haven ’ t been tracking, back in September, an Iranian woman, Mahsa Amini, died in the custody of Iranian morality police after being arrested for not wearing hijab. Amini ’ s death sparked protests and a revolution around women ’ s rights in Iran and beyond that is continuing into this very moment. And, Christiane, I know that you ’ ve reported on Iran throughout your career and have spent a lot of time covering this story very closely. So how historically significant would you say this moment is in Iran? And could you just give us some context on that? CA : Look, I think it is very significant. Exactly how and what will develop towards the end, I ’ m not sure. A little bit of my own history : I am half Iranian, and I grew up in Iran. And I spent essentially the first 20 years of my life in Iran, with a little bit of going back and forth to the UK for boarding school. But that ’ s where my home was and that ’ s where my parents lived, and my sisters, essentially, you know, I was there during the build up to the Islamic Revolution of 1979. And so I saw all that happening, and that ’ s what made me want to be a journalist, in fact, to tell those kinds of history-making stories to the world. And that ’ s what really

focused me on the kind of career that I then pursued. Fast forward now some 43-odd years, and you can see that the women who actually were pretty upset at the beginning of the revolution, when they had demonstrated for Khomeini, believing that he would bring democracy as he promised back then to a country that was a monarchy. And then to quickly find out that actually that wasn ’ t his plan and to quickly find out that he was putting women under the veil, which he had not said earlier. So for the first year or so of the revolution, women did not have to wear the veil. And when they did, many, many women came out and demonstrated against it. And then for the decades since, there ’ s been a sort of a tug of war between how strictly to put on the veil and not. And clearly differences between **conservative** and religious women who want to wear the whole tight covering that covers their head very tightly, covers their body very loosely, known as a chador in Iran. And then those who just didn ’ t, the younger generation who felt, well, they better, you know, follow the law, but they were kind of treading their own path towards how they would wear the hijab. And you could see coats and body coverings were getting tighter and shorter, head scarves were getting more colorful and pushed further back. Women and girls were going to great lengths to make themselves look beautiful and stand out as women. They would do their hair, particularly the front of their hair, which stuck out of the scarf. They would, you know, bangs were not forbidden, you know, beautifully made up their eyes, you know, their skin, their lips were all just beautifully done. And Iranian women are incredibly beautiful women in spirit and in body. And that seemed to be kind of OK. But what happened towards the last sort of, ten years or so, you ’ ve had a slightly less draconian regime when it comes to enforcing hijab. You know, in the late ’ 90s, early 2000s, and then a much more draconian, which is the latest one. So these very hard liners, this mullah, you know, Ebrahim Raisi, who is the very hardline **conservative** president, actually came to the presidency by using as his political platform, among other things, a much tougher interpretation of what they believe, you know, the Islamic laws should be. And, of course, who paid the ultimate price? Women. Because it ’ s always, whether it ’ s in Iran, whether it ’ s in Afghanistan, whether it ’ s in the UAE or in Saudi Arabia, or whether it ’ s in parts of Africa, whether it ’ s in the United States, etc., women ’ s bodies, women ’ s personal space seem to be used, you know, for people ’ s political aims. And so the same happened in Iran. And this poor, poor woman, Mahsa Amini, at the age of 22, came with her family from the Kurdish region of Iran to **visit** relatives and was wearing a scarf and was wearing a body covering. But the police on the corner at that time didn ’ t think it was **conservative** enough. And that ’ s why there ’ s a backlash. WPR : Well, you know, in recent days, a story was circulating that some 15, 000 protesters were to be executed, and the veracity of the story has been challenged and disproven. As I understand it, it ’ s actually more like 15, 000 protesters have been detained and the parliament voted in **support** of the death penalty for protesters. But it ’ s not ultimately up to them. So I ’ m curious how real you think the possibility is that something like a mass execution might actually happen in the near future? How real a threat is that? CA : Well, you ’ re right that this story was chased down, including by us, and it turned out not to be as it was being broadcast. On the other hand, there are many, many, I mean, you know, hundreds, if not thousands of people who ’ ve been rounded up and put into prison, whether they are women, men, young girls, other young people. I recall hearing one of the top government officials saying several weeks ago that the average age of those who are being arrested, this is an actual government official there, the average age was 15 and 16. And I mean, just think about it for a moment. I mean, that ’ s just unheard of anywhere, that children, young girls, people so young are in the front lines of this protest, this uprising, this movement, and are being punished. And then the latest stories, particularly by my colleague Farnaz Fassihi of The New York Times, who ’ s an incredibly good reporter, has brought out, I mean, some of the most terrible stories of how young people are being dealt with in a very harsh way. And, you know, there was, over the weekend, there was the burial of a young boy, 10 or nine years old, who was, you know, a victim of all of this. And he was killed. And his funeral happened. There was another big protest there. And, you know, these things defy normal behavior. And even for, you know, an Islamic republic that ’ s been pretty draconian for the last 43 years. And this goes way beyond, you know, riot control or whatever you might want to call it. So the question again is, because I know those outside of Iran want to know, is this the beginning of the end, or is this the end of the beginning, or is this the end of the Islamic republic? And I ’ m just not ready to say one way or another, because you just don ’ t know to what lengths they will finally go to crush it. And so I ’ m waiting, as a journalist should, I ’ m watching, I ’ m waiting, I ’ m interviewing, I ’ m getting as much information as I possibly can. And I ’ m going to be doing it on my show today, which will **air**, as you said, on CNNi and on PBS. And I ’ ve done it many times with many Iranian interlocutors. And... This is a very brutal regime when it comes to staying in power. But the only thing that I think is **interesting**, well, there are many things that are **interesting**, but what ’ s somewhat different to before is that within the Islamic establishment, you have voices, actually male voices, who are questioning why they need to react this hard against girls just for the hijab and the headscarf. So we ’ ll wait to see where that conversation gets to, if at all. WPR : And as someone with such a close personal connection to Iran, that part of your heritage and then also just your close reporting of this story, what do you feel are the things that most of the world is missing or not seeing about what ’ s happening there? CA : Well, look, I think in general, foreign **policy** and stories from around the world are hard to get past the American public because they ’ re hard to get past our own editors. So if there was more reporting on a more regular basis of these kinds of stories, we would have a much better – and you all would have a much better idea of trends, of what ’ s actually happening. You know, don ’ t forget, Iran is a very important country to the United States and to the rest of the world, partly because of the sort of upheaval that the Islamic Republic and including backing terrorism etc., have caused to the world – and holding hostages as they continue to do. Iranian Americans, British Iranians, and a whole load of others who are in jail, used as political pawns. And this is, you know, a very, very unfortunate and tragic situation where human beings are being used as political weapons. So they ’ re weaponizing people who just happen to be half Iranian or they ’ ve left and they have become you know, they ’ ve taken on the nationality of the countries that have given them refuge. And when they go back to **visit** family or on personal **visits**, they have been dragged into this political turmoil. So that ’ s really bad. And many governments are having to deal with that. The other issue of why it ’ s important is because of the Iran nuclear deal. Now, I think that ’ s off the table for the moment. It was always going to be off the table pending the midterms, but now pending, you know, this crushing of this movement, the US has imposed more sanctions, Europe has imposed more sanctions on individuals who are deemed responsible for the harshest crackdown. So it ’ s very important. Iran remains a very, very important country to the world. And don ’ t forget, you know, it has a huge population, and such a huge majority of the Iranian population are under the age of 30 or, you know, they ’ re in that young generation, which means they ’ re incredibly well-educated, they are well-connected to the world. Even now, even though the regime is trying to cut them off from the West and cut the world off from them. They manage, you know, to play a very sophisticated technical game of cat and mouse to get their stories out and to get information from the world in. So you can never cut the Iranian people off. And they would benefit and I hope they are benefiting from the **support** that they know, the moral **support** that the world is giving them. On the other hand, I know there are others who call for more **support** to overthrow the regime. That ' s not my space because I ' m a reporter, and I report on what ' s going on there. And I personally don ' t believe that anybody, any foreign government is going to do that. And I also believe that the women and men, the male allies inside Iran... This is their movement. This is their movement. They don ' t want it to be sullied by any interference from abroad that could get them even more tarnished than the regime is trying to do at the moment. WPR : Christiane, we ' re getting a lot of questions from our members, and I ' m going start to bring some into our conversation. We have one from **TED** member, Don, which speaks to what you ' re suggesting around this idea of global **support**. They want to know how can we help from afar."I feel helpless, but would like to do what I can without putting more women in harm ' s way,"is what Don shares. CA : Well, it ' s very hard. I mean, definitely moral **support**, definitely spreading the word, spreading the word and **supporting** the Iranian women in whatever way you can. Some people ask, how can we send them material **support**, whether it ' s money or whatever it might be. But that ' s very, very difficult, and I do not have an answer for that. It ' s difficult because Iran is so heavily sanctioned. So it ' s quite often illegal, actually, you know, by American law or European law or others, to actually send money via various ways, as far as I know. Maybe there are ways that it ' s possible, but I don ' t know those ways. But it is quite difficult to provide, at the moment, more than material **support**. I would say, I do believe the world should be a lot more generous to those who are trying to flee this and any other kind of repressive regime. Right now I ' m in London, and as you know, over the last several, certainly months, there ' s been a **crisis** in the English Channel. And a huge number of Iranians are trying to get away from the danger zones and into the UK. A huge number of Afghans and potentially others fleeing wars and devastation. But they get turned back. And I think the world ' s asylum and **refugee policies** have become so draconian and have gone so far from being what they were envisaged as, to welcome those who are fleeing you know, terrible oppression and, of course, often starvation and disease you know, and natural disasters and the like. And I think that ' s a real shame that we ' re seeing in this time of maximum upheaval in these parts of the world, the rest of the world actually closing their doors. And I ' m about to do some interviews around a new film that ' s coming out called"The Swimmers."And just to say, it ' s about two Syrian girls who had to leave Syria at the height of the Arab Spring and during the Iran-supported crackdown by the Syrian regime on those who were **demanding** freedom. Anyway, they managed to get out, but they had to take a rickety boat to get to Europe, they had to walk for miles, then they had to wait for months, you know, for asylum. You know, people like you and me were dying in the Mediterranean. They ' re dying in the English Channel. They ' re dying, you know, coming across from Africa. This is 2022. It ' s a scandal. And I do believe that that lack of asylum, that lack of refuge is a terrible shame and a blot on all of us who believe in human rights and democracy and the value of life. WPR : Well, I want to move to another part of the world, also experiencing a huge **crisis**, where you ' ve been spending a lot of time, both reporting and actually physically being there. You just returned from Ukraine. And we ' re now more than nine months into the war there. So could you tell us a little bit about your trip and what you saw as the people there are heading into the winter. What would you like for us to know about what you saw? CA : Because you just said winter, I think it ' s really important to know that what Putin is doing now, because he ' s been thwarted on the battlefield, is he is literally taking this war to civilians in a much more targeted and devastating way than he has been already, which has already been attacking civilians. But this relentless attack by cruise missiles and other really, really powerful – including Iranian-supplied kamikaze drones – this is attacking civilian energy infrastructure on a regular basis, it ' s sometimes attacking civilian residential buildings and ordinary people are being killed and wounded. At the same time, what turned out to be a phenomenally resourceful Ukrainian population, whether civilians or experts, let ' s just take now the engineers in the energy **sector** are working around the clock, day and night to try to put this, you know, back. And they ' ve done managed, rolling blackouts and the like to try to make sure that they can save some of the energy infrastructure. And particularly, let ' s not forget how tightly **intertwined** is electricity and water. Those two are yin and yang. You can ' t really have water without electricity to pump it. And so if you lack water, fresh water just to drink, you ' re in real big trouble. And so the Ukrainian engineers are working around the clock to try to make sure that people have at least, at least, even if it ' s rationed, access. So that ' s a big issue. I would say I ' ve been completely and utterly overwhelmed and surprised – not surprised, delighted by the spirit of the Ukrainian people. You know, I ' ve covered a lot of wars in which much more powerful, heavily-armed aggressors try to subjugate and force a less heavily-armed civilian population into surrender. And the Ukrainians are nowhere near that. They have so much heart, they have so much love for their country. They have so much respect, as the Iranians do, for the idea of independence, sovereignty and personal freedom and political freedom and democracy. That ' s what they ' re fighting for. And frankly, they are our frontline troops. The Ukrainian people right now, and to an extent, the Iranian people, are our frontline troops. In Ukraine, people are standing between freedom and totalitarianism. And if Putin succeeds in Ukraine, then he ' s… I mean, not only do the brave Ukrainian people lose, but then he comes a country closer to the West and continues to threaten the ideas and the values and principles of democracy and freedom. And this idea that it was NATO ' s fault and NATO expansion, I promise you that is fake news. It ' s too difficult right now to go into it because we only have limited time. But it ' s not that. It ' s that Putin does not want to see a free and independent and democratic country, in this case Ukraine, on his borders, and then be asked why he can ' t have that, why his people can ' t have that. He doesn ' t want that. He wants to create his own sort of greater Russia. I witnessed that covering Bosnia, where the Serbs, backed by Russia, wanted to create a greater Serbia on the backs of the Bosnian people. And in Iran, too. And you ' ve seen the Afghan women and girls now saying,"If Iranians can stand up and **demand** their rights, it ' s our turn next."So these have huge roll-on effects and inspiration to people who believe in basic freedom, dignity and the right to live their lives. WPR : Well, I know while you were there, you sat down with Ukrainian President Volodymyr Zelensky and first lady Olena Zelenska. And I ' m just curious what your thoughts were about their mindset at this point in the war and how you think that will **impact** their **leadership**? CA : Well, I think their **leadership** has been extraordinary. I mean, really, who would have thought it. This man Volodymyr Zelensky came from the entertainment space. He ran based on a program that he had been starring in, a TV series based on corruption at the highest level. And so he ran as a clean candidate. And then when he came into office, that ' s what he tried to bring in, to increase infrastructure and to improve it and crack down on corruption, because Ukraine did have, and may still, after the war, have a corruption problem. But believe me, now, that is right on the back burner. And US senators of both parties have told me that everything that the Americans are sending and NATO is sending is accounted for, and they are not worried about it going into somebody ' s pocket. Unlike in Russia, where it ' s gone into, obviously, people ' s pockets because they have barely a military machine that can operate in Ukraine, much less against NATO. Their **leadership** has been inspirational. The fact that the president did not leave when he could have done, he did not take his wife and children out when he could have done. And the fact that they keep"poking the bear,"and I asked him that. I said,"You know, a lot of people in the West are worried and elsewhere are worried that if you push Putin too much, you"put him in a corner"and he might do the unthinkable."And he said,"You know what? We ' ve lived in this neighborhood forever. If we ' re not scared, you shouldn ' t be scared."And I was saying this because not only are they pushing him militarily back, but also if you look at the Ukrainian Defense ministry or any of the online space, they ' re constantly trolling Putin. Constantly. And it ' s done with incredible humor, and it ' s very, very effective. And so they know something that we don ' t know. And as Olena Zelensky told me,"This is our last stand. This is it. If we don ' t stand up now, there is no other time when we can stand up. If we don ' t do it now, when are we ever going to be able to do it?"So they ' re very, very clear about that. As you know, there ' s been some talk around the world, some talk in the US and Europe, elsewhere that maybe, you know, particularly after the liberation of Kherson by the Ukrainians, shouldn ' t the Ukrainians sit down at the **negotiating** table? So I asked President Zelensky that, and he said,"Look, we ' re happy to **negotiate**, but based on **international** principles, based on the Russians removing their troops from their illegally invaded, annexed and **claimed** territory. You know, and sham referendums and sham annexation. And by the way, we ' re pushing them back."So they don ' t want to be pushed to a negotiation while they have the upper hand and one that would reward Russia for its aggression. And to be honest with you, we don ' t want that either. We do not want Russia to get away with today what they did get away with in their first invasion of 2014, when the world did not stand up and did not challenge him. And this is the result. He thought the world was weak. And that ' s his big surprise. He was very surprised that the Ukrainians would stand up, the West and NATO would stand up, and that this unity would last this long. And, you know, I come to it from, you know, having covered something similar in Bosnia for four years. I mean, this was an ethnic cleansing, which they ' re trying to do in Ukraine, the Russians, along the east, and it was a genocide, which I believe that... maybe not genocide, but crimes against humanity will be placed legitimately at the feet of Russia after this war. WPR : Well in both Iran and Russia, and thinking about Ukraine, we see a suppression of free speech and access to information. How can we know that we ' re receiving accurate updates on what ' s happening in both of these places? CA : Well, I do think you and **everybody** have to take on a certain sense of responsibility because that is the challenge of our time. You know, people ask me : Where do we find the truth? And I ' m like, well, you can watch CNN, you can watch PBS, you can watch BBC. You can read The Wall Street Journal or The New York Times The LA Times or any number, Financial Times, whatever. There are many, many organizations who are committed to the truth and who have reporters on the ground, including in Russia. The BBC, for instance, has a very, very accomplished and good reporter called Steve Rosenberg. I highly recommend you all to access his reports, because under the constraints that Russia has put on journalists, he is managing to tell the story in a very, very clever way. And also, I would, you know, just look at the Russian state media, Russian blogs and this and that, including the military bloggers who are traveling with Russian troops in Ukraine. There ' s a huge amount of information coming out that actually paints the accurate picture. And right now, they ' re very – which is not to say they want the war to end, but they ' re actually painting the story that the Russians are doing badly in Ukraine. So there are places and avenues to go. But I believe that it ' s really up to the consumer now. We can do as much as we can to bring you the truth, fact-checked, evidence-based news information. But you must not go to the fake news sites and believe that that ' s the truth. You must go to what I would call the news organizations that have earned the Good Housekeeping seal of authenticity and truth and, you know, approval. It ' s really on you all now to search for those. And it ' s not hard, they exist. WPR : As we get close to the end of our time, I just want to turn things to you, Christiane, and your thinking as a reporter, a journalist. How would you say reporting on **conflict** and **crisis** has changed the way you think about the world, your perspective on life? CA : I think it ' s changed my perspective in that I ' m very clear that I can ' t be a both-siderism, or"on the one hand / on the otherism."In other words, in certain instances, such as gross violations of **international** humanitarian law, which we ' re witnessing, in other words, you know, human rights atrocities, crimes against humanity, genocide and the like, you absolutely have to know where the truth is and what you ' re looking at so that you know that there are aggressors and there are victims. There ' s no two ways about that. So my mantra is and has been, and I ' ve developed that from Bosnia, which was one of my earliest experiences in the field, my mantra is"truthful, not neutral,"which doesn ' t mean to say I ' m not being objective. I am being objective because objective is our golden rule. And by that I mean you have to look at all sides of the story and report all sides of the story. But what you mustn ' t do is create any false equivalence, either factually or morally. And when you do do that, you are actually being untruthful and in some cases you ' re being an accessory to terrible atrocities. And I would say the same about **climate**. I mean, you can talk about any of these **crises**, these moral and existential **crises** that we find. You know, **climate** has been too long, for decades been treated as a both-siderism, that the deniers had equal factual or moral weight as the science. And it ' s just, as we know, not true, and so much time has been wasted to the point that we are on the brink of a global catastrophe. And that is something that I ' ve learned. That in order to be a trustworthy, credible journalist, I must, must call out the truth. WPR : Well, I want to end with a question from a member, which is basically,"If you were to interview yourself, what would be the last question you would ask?"So here ' s your last question, what is your last question for Christiane? CA : Oh, Lordy. My last question, I don ' t know, but I do know one thing. That I wish that I would be able to – I guess it would be, how can you get more access to the other side, you know? I want to say the bad actors, because they are. How do you get more access to them? You know, I would like to sit down with a Putin or a Kim Jong-un or whoever, Viktor Orban in Hungary or whoever is denying democracy in the United States and the like. I would like to sit and talk to them to try to understand. Not because I think that I would be, you know, I would suddenly become a neutral observer, but because I want them to spell it out so the people can see and I would question them rigorously, that they would see the bankruptcy of their positions. And I would like to keep doing that with fossil fuel executives and the like. I ' ve done a few and with governments, you know, who have allowed politics, politics to endanger us all. I do believe those people need to be held **accountable**. So I would like to be able to ask those people more questions. WPR : Well, Christiane, thank you so much for taking the time to chat with us today, I think I speak for everyone, just appreciating everything that you ' ve shared. and good luck with the rest of your day. CA : Thank you very much, I appreciate it. It was good to talk to you all, thanks Whitney. [Want to **support TED**?] [Become a **TED** Member!] [Learn more at **ted**. com **/ membership**]

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Helen Walters : Last week, US President Biden and Xi Jinping, the president of the People ’ s Republic of China, met at the APEC Summit in San Francisco. The meeting was notable for being the first time that Xi had **visited** the US in six years and the first time the two **leaders** have met in person in a year. Now, obviously, what goes on between these two nations matters to everyone. So we found ourselves post-summit with a bunch of follow-up questions, and we ’ ve turned to our resident geopolitical expert, Ian Bremmer, to help us understand what to pay attention to and why. Ian, hi. Ian Bremmer : Helen, good to be back with you. HW : OK, so let ’ s get right to it. These are, alas, not peaceful times. And I think it ’ s **safe** to say that the meeting between these two **leaders** felt perhaps even more momentous than it might have done were war not raging everywhere, from the Middle East to Ukraine. So tell us, what happened in San Francisco, and what stood out most to you? IB : Well, look, that is the backdrop. That this is a world of unprecedented geopolitical danger and risk. And the efforts are not in trying to make everything better. It ’ s rather to try to stop the existing **conflicts** from getting much worse. And that is absolutely the macro focus that both President Biden and President Xi bring to the meeting. I mean, there ’ s a lot to discuss around the issues around US-China relations. I ’ m sure we ’ ll get to that. But, you know, it ’ s **interesting** that insofar as the Americans and Chinese are actually on opposite sides of the two major global **conflicts** in the world right now. On Russia-Ukraine, the Chinese are the close friends, without limits, to Vladimir Putin, while the Americans are providing more **support** than any other country in the world, militarily, to Ukraine. The Chinese have not condemned Hamas for their terrorist attacks. The United States finds Israel its most important and enduring ally in the Middle East. So you would think, Helen, that this would be an area of contention between the United States and China. It ’ s not. Both the Americans and Chinese are deeply concerned that these **conflicts** are going to get worse. And they don ’ t want that. They want to find ways to contain these **conflicts**. And I think it ’ s a very important point, because we hear a lot about, you know, Americans looking for adversaries around the world and lumping in China with countries like Russia, Iran, North Korea. And those other countries are rogue states. They ’ re pariahs. They ’ re countries that benefit from chaos. They want to take advantage of vacuums geopolitically. Where the Americans and Chinese actually geopolitically have a lot more in common, in addition to the fact that they have a lot of interdependence, they also both benefit from a global backdrop that is stable. They want relatively free and open trade of goods. They want a global economy that ’ s working. They don ’ t want political instability everywhere or social instability everywhere. And so even though the United States and China have different preferred end-states for Russia-Ukraine and for Israel-Gaza, in the near-term, you ’ ve got two **leaders** that are meeting and saying, how do we stop this from getting worse? And that ended up being a significant piece of the conversation, the four-hour, three-session conversations that Presidents Biden and Xi were having. In some ways, maybe the most important takeaway that the two most powerful countries in the world are not looking at the Middle East and Russia-Ukraine through a lens of cold war, but instead are looking at it through the lens of,”Oh my God, this is really a problem. And are there anything that we can do, individually or collectively or with our friends and allies in the regions, that might help to stop this from getting much, much worse?"HW : Can you say any more about what that actually looks like? What might those alignments be? IB : Well, in the case of the Middle East, China has a relationship with Iran that the United States does not have. And both the American cabinet as well as Biden directly have been talking to the Chinese about getting messages to the Iranians to help **ensure** that they don ' t get directly **involved** in the war and that they limit the **support** that they have given to proxies in the region that could, for example, not only expand the war, but also lead to challenges in global energy supply. You know, the Americans sent two carrier strike groups to the Eastern Med and the Persian Gulf almost immediately after the October 7 terrorist attacks. China has destroyers in the region, and they were there for military exercises. They ' ve kept them there, and they ' ve expanded the military presence. Not to fight the Americans but rather to show that the Chinese want to **ensure** that there is not a fight in the region that would suddenly prevent energy from getting through the Straits of Hormuz, a critical choke point. So, you know, frankly, there ' s more alignment on this issue. The Chinese also, just earlier today, as you and I are talking, hosting a group of foreign ministers from the Muslim world, including the Palestinian Authority, they ' re talking about a ceasefire and a two-state solution. Biden wants an extended humanitarian pause, not a ceasefire, but also a two-state solution. There ' s been a lot of conversation around trying to bring Middle Eastern countries to be more constructive in helping to **ensure** stability in this **conflict**. So around the Middle East, there ' s been a lot, because it ' s more recent, and because it ' s frankly more geopolitically dangerous – the Russia-Ukraine war has more knock-on economic implications – but the Middle East **conflict** is much more geopolitically fraught in the sense that you could have a religious war from, you know, Ecuador to Indonesia. Because you could have, you know, they could have a much greater **impact** on the US presidential election, for example. That is one that is driving a lot more direct attention and engagement from the American and Chinese **leaders** together. HW : So I think it ' s **interesting** and heartening that stability might be a watchword at this moment. The US and China, they obviously subscribe to very different political systems. And I think of a quote from the MIT economist Yasheng Huang, who was once quoted saying that the two countries kind of got married without knowing one another ' s religions. How much did this summit, if anything, do to address the root causes of the tension? IB : You know, it ' s **interesting**. I see the analogy of US and China getting married without knowing the families or the religions or any of the other, you know, sort of red lines that one connects with when you make that lifetime bond. But, you know, I ' ve always thought of it as a couple, that the love has left the relationship, but they have children together, and they love the children very much, and they both want to make sure that the children aren ' t hurt. And so as a consequence, as much as we can talk about, you know, derisking as a term of art, or decoupling, and these countries having very different political and economic systems, and they don ' t particularly **trust** each other, and yet they know they need to work together. So in a sense, they ' re adults, geopolitically. I mean, when you have the US and China in a room at the highest level, and here I ' m not just talking about the presidents, I ' m talking about, you know, any of the cabinet meetings that have happened at great level and scope over the past months after really none at all during the couple plus years of **pandemic**, all of those meetings, they haven ' t been easy, they ' ve frequently been tense, but they have all been handled as adults, handled as two parents that know the children need to be taken care of long-term. And when I say the children, I ' m not disparaging other countries. I ’ m really talking about the **collective** interest, the knock-on interests that come from what happens if the US-China relationship suddenly becomes one of cold war or worse, which is absolutely plausible in today ’ s political environment. So if that ’ s the backdrop, I think that Biden in particular spent an enormous amount of time over the last six months in preparation for this event, trying to convince the Chinese that the meeting would go well, that the Americans were not planning on dropping any surprises, on undermining or embarrassing their head of state when he showed up on American turf in San Francisco. They were very concerned about that because there ’ s no **trust**, and because there are plenty of issues of significant tension between the two countries. And I think that they succeeded critically in that. Again, we haven ’ t talked yet about the specific issues, and I know we ’ ll get to them. But the macro backdrop is important. Over the last couple of months, China has been on a charm offensive, with the Australians, inviting the Australian prime minister for a state **visit** to China just a few weeks ago. That went extremely well with expanded trade and energy agreements on the back of, you know, a relationship where they weren ’ t talking to each other, where they were cutting off business, where they were engaging in massive tariffs and sanctions. That ’ s seen a breakthrough. I ’ ve seen some of that in South Korea. I ’ ve seen some of that with the Japanese. And over a one-hour meeting between Prime Minister Kishida and Xi Jinping in person at the APEC Summit, first time in over a year that they have met. I ’ ve seen that with the Europeans. I ’ ve also seen it with a large number of American and European CEOs who have reported individually just how much more access they ’ ve gotten, welcoming they ’ ve gotten from their trips to Beijing, positive press coverage, movement on issues that have mattered to them, all of which a sense that a better Biden relationship with Xi Jinping, not a breakthrough, not an entente, but simply a commitment that the US wants a more stable baseline relationship and would work towards that, in San Francisco, that gave the Chinese **leadership** permission to engage in this charm offensive with other countries and with the private **sector**. And why was that so important? Well, first, because it lowers the temperature of the relationship. It makes sudden, unsuspected **crises** less likely to occur, but also critically, because the Chinese economy is underperforming dramatically. For many reasons, maybe most structurally, because 50 years of China acting as the world ’ s factory, with all of this inexpensive labor, well, you don ’ t need all that inexpensive labor anymore. And by the way, that labor is not so inexpensive anymore. And, you know, China hasn ’ t become an open-governance system. They haven ’ t moved towards rule of law. And the Chinese competitors are a lot stronger, and their demographics are challenging. And they ’ ve got lots of nonperforming debt and their real estate **sector** ’ s in trouble. And zero-COVID went really badly. And then they unwound it but the consumers don ’ t feel like they ’ ve got, you know, animal spirits driving them right now. And I can keep going. But the point is that China is severely underperforming in a way that, frankly, we haven ’ t seen structurally since globalization bringing the Chinese in, you know, really got moving in the ’ 70s and ’ 80s. So in other words, Helen, you and I have never seen this structural economic headwinds on so many fronts in China. And the Chinese are much more aware of that than you and I are. So they ’ re very strongly incented, even if they fully intend to take Taiwan over the long-term and they want to be the leading economy in the world and they want to dominate artificial **intelligence** and all of these things that Americans and others worry about. For the near and foreseeable, medium-term future, they ’ ve got to just right the boat. They ’ ve got to get things stable. They don ’ t want a big fight right now. And so I think that the Americans, being a little more confident, a little less concerned about, you know, sort of, China taking over everything in the near-term and wanting to stabilize things, really got you a lot more than you would have otherwise expected from this summit meeting over the last several days. HW : So it ’ s **interesting** to hear you talk about the economic **dysfunction** in China. And of course, I think it ’ s **safe** to say that America is experiencing pretty significant political **dysfunction** at the moment. But what often goes unremarked is the fact that President Biden is actually extending, perhaps doubling down, on President Trump ’ spolicies when it comes to China. So given that next year is an election year, what should we make of that? And do you think that the **policies** will continue? IB : First of all, that backdrop, that right now, China ’ s in probably the worst economic position structurally that they ’ ve been in in 40, 50 years. But their political consolidation around Xi is completely uncontested. And certainly XI Jinping feels very comfortable that he ’ s consolidated a lot of power. The United States, exactly the opposite. The US coming out of the **pandemic**, by far in the strongest economic position of any advanced industrial economy in terms of growth, in terms of productivity, in terms of leading in technologies and in terms of lower **inflation** than its peers. But the US political system is more dysfunctional, more divided than at any point in our lifetimes. And for now, that certainly is creating a level of risk aversion on the American side as well. So I do think that, you know, at the same time that you hear a lot of people say,”Oh, if things go really badly, maybe they want to lash out.”Yeah, not for these two **leaders** at this point in time. Other **leaders**, different countries, different positions, maybe. But that ’ s not the way that this is actually playing out. Now 2024 is coming out, and people are certainly starting to talk a lot more about it. It ’ s **interesting** that, as you point out, the United States on China have **policies** that are fairly consistent across the board politically. Which is not true for most other issues. If Trump became president, Ukraine **policy** would be very different. Iran **policy** would be very different. Europe **policy** would be very different. China, not so different. There ’ s a lot of consistency between Biden and Trump on China. A lot of people thought Biden was going to remove the Trump tariffs on the Chinese. He did no such thing. In fact, he largely extended some of them. Furthermore, export controls on semiconductors, pretty minimal from the Trump administration, expanded structurally under Biden to the extent that China now really feels like America wants to contain them in the most advanced areas of the 21-century economy. And the US is also leaning into industrial **policy**, like the CHIPS Act, domestically and with countries like South Korea and the Netherlands. The Chinese clearly would prefer a Trump **policy** on China there, than they do the Biden administration. And, you know, you can tell this when you talk to Chinese **leaders**, compared to **leaders** of other countries around the world, most of whom have pretty strong preferences of whether Biden or Trump is president, of who they want. The Chinese aren ’ t sure. The Chinese are thinking, well, I mean, if Trump comes in, there ’ s greater likelihood that American allies are going to be less aligned because he ’ ll push them transactionally on spending more money on defense or maybe he doesn ’ t care about the Japanese or the South Koreans, and he ’ ll put tariffs or threaten tariffs on anybody, friends or enemies. Biden ’ s less likely to do that. But Trump is also a wild card on negative tail risks directly with the Chinese. You know, he was the **guy** that was willing to work with North Korea, but also was prepared to hit them harder if things don ’ t go well. Well, how lucky did the Chinese feel? And I think, you know, the answer you get is we really don ’ t know. We don ’ t know who we want there. So there ’ s a lot of **uncertainty** that the Chinese have about the future of the American political system. And there ’ s a lot of **uncertainty** that the Americans have about the future of the Chinese economic system. At a time when the interdependence of these two economies and, frankly, of their diplomatic interdependence, is remaining quite high-level. We may not be comfortable with that reality, but that is the abiding reality that we ’ re going to have to deal with going forward. HW : So you mentioned technology, and I think that we have to talk about artificial **intelligence**. Now, one of the major breaking stories this weekend was the management implosions and excitement over at OpenAI. And I ’ m sure we could have a whole conversation about that. But given the implications of AI rolling out at every level of society, where are the Chinese and the US governments on this? And what did they talk about in San Francisco? IB : Well, in San Francisco, they spoke about starting a track 1. 5 working group on artificial **intelligence**, which means the private **sector** and the public **sector** engaging together. Which makes sense from the American perspective, because, you know, the US is a country that really does promote entrepreneurialism and its private **sector** corporations, so much so that a lot of people think the US is less democratic than it should be, because corporations, private **sector**, capture the regulatory process through big money lobbying and the rest. The Chinese, of course, if anything, the state captures the private **sector**. So the fact that they ’ re willing to have not just government-to-government, but government-to-government plus these big companies that are, you know, effectively sovereign when we talk about the digital space, the platforms they have, the algorithms they drive and artificial **intelligence** that they are rolling out very, very quickly. That ’ s a fairly significant move. And it comes on the back of the Americans and Chinese both sending **senior** officials to Bletchley Park in the UK, agreeing to a set of principles on safety for frontier AI models, the AI models that are coming in the future. So there is a level of understanding between the US and China that they need to share information, and they need to work together to avoid some of the worst negative potentials from very disruptive AI, while obviously benefiting from extraordinary, you know, sort of, world-changing new productivity, invention and efficiency. But there ’ s a really, really big unknown question that underpins the rolling out of AI. Because let ’ s think about what AI does today. It ’ s taking, it ’ s these models, these large language models, that are taking the entire corpus of global data, as we have it on the internet, as we have it in the digital world, and is making predictions, pattern recognition and predictions, on the basis of all of that information instantaneously. And we ’ ve never had such powerful tools. Now, when you have all of that data at your fingertips, so you can therefore assess and measure metrics of the world in real time and how human beings interact with it in real-time, that creates really big questions about what political and economic models will be most functional. So, for example, 30 years ago, you know, we thought, well, democracy is definitely the most functional model. And the internet, as it rolls out, is making that more clear, because you ’ ve got all these people with their access to the World Wide Web, and that really undermines authoritarian countries that want to control information and it helps democracies. And that ’ s how you got colored revolutions or the Arab Spring. And then you have the data revolution, the surveillance revolution, you say, well, wait a second. You know, governments that have access to all that data can actually create, you know, all sorts of incentives, both carrots and sticks, to motivate patriotic behavior, where in democracies that can create a lot more polarization. So now let ’ s take AI and look forward three, five years, when you ’ ve got large language models that are tailored to your individual data corpus on your smartphone. So everyone has an individual AI that has all the data on you, and collectively it has all the data on the planet real-time. In that environment, is a planned economy less efficient or more efficient than a free-market economy with different corporations that are competing? We don ’ t know the answer. In that environment, is an authoritarian political system more or less stable than a democratic political system? We actually don ’ t know the answer. I mean, Helen, I know the answer I want it to be. I want it to be a well-regulated free market and a democracy. But I ’ d be lying to you, right? We don ’ t know the answer to that. And so, I mean, suddenly, the United States and China are entering into a world where a small number of actors in the private **sector** are **investing** immense amounts of money and developing unprecedented tools that will determine, more than anything else, the viability and strength of these two fundamentally competing political and economic models. And we don ’ t know which one is going to do better. Which one might even win, or can they both exist, continue to exist at the same time. And so in that environment, you better believe that the Americans and Chinese both want to have, you know, a very significant seat at the head of the table in helping understand what the hell is going on with AI. You need to know that early so you have some time to plan for it, to respond to it, to govern it, to create institutions and structure around it. And I think right now both governments are playing catch-up, but they understand they need to do it hand in glove with the private **sector**. And my God, I mean, the events at OpenAI are absolutely essential to understanding that future. HW : I mean, complexity is the operative word there. I mean, what is the likelihood? I mean, many speakers at **TED** have called for some form of **international** regulatory agency or for some kind of oversight. What are the chances that we could actually see that actually happen, and what are the chances that it could have teeth? IB : Well, there are governments all over the world that are treating this issue with urgency. They ’ re making it a priority. And they ’ re doing that in part, not only because they know AI is important, but they also see that AI is critically important to things that they ’ re already prioritizing. So if you look at Russia-Ukraine, the future of that war may well be critically determined by the **ability** of the Ukrainians to use autonomous lethal weapons powered by AI against Russia. You better understand that if you ’ re the Americans and US allies going forward, both in terms of getting outcomes you want and also potentially destabilizing the region in ways that you ’ re not prepared for. The AI-driven **disinformation** around the US election in 2024 is an absolute critical concern for US **policy** makers and, frankly, around **disinformation** for Israel-Palestine and who wins the information war, which, I mean, the Israelis are militarily doing what they want to, tactically on the ground, but broader information war, at least presently, they ’ re losing. Understanding AI is critical to all of these issues. It ’ s not just a new space. And so I think everyone is taking it very, very seriously. And they ’ re putting a lot of resources into it. There clearly is a level of effort to regulate AI in ways that align with individual government goals and systems that is different from place to place. So, I mean, in China, part of it is we can ’ t allow the average Chinese citizen to have access to a chatbot that could provide, you know, responses on any data. You know, we can ’ t allow – there ’ s got to be severe penalties in starting to talk about independence of Taiwan or Tiananmen Square or anything like that. And these companies have to be responsible not just for the inputs but also the outputs that are coming from these platforms. Where in the United States, right, you have the companies that are working very closely with the US government to try to figure out, OK, what are the areas that we ’ re going to be comfortable having significant regulation. Like for example, red teaming on how one can break new models as they ’ re developed. Or in having watermarks that help to determine whether something is or is not created by artificial **intelligence**. Where in Europe, the focus is so much more on privacy and data protections for citizens. So, you know, right now, you would say it looks like they ’ re moving in very, very different directions. But that ’ s in part because we don ’ t yet have anything close to global agreement on what artificial **intelligence** can do. What are the things that need to be measured, what are the things that we want to promote, and what are the areas that we need to try to contain or constrain or regulate? And I think that there ' s a United Nations process that I ' m **involved** in, a high-level panel that I think is trying to make immediate strides on that. I mean, the report ' s going to come out within eight months, which is light speed in terms of the United Nations. But we will see whether all of those efforts will get you to pieces of global governance like you have the beginnings of for **climate** change, for example. But you had decades to get it together on **climate** change, and you have months to a few years to do it effectively on AI. I would say that I am hopeful, but I am not yet optimistic. HW : All right, we ' ll take it. OK, I want to change the topic. I want to talk about infrastructure. So the Belt and Road Initiative is subject to a lot of debate. And just as a quick background reminder for those who don ' t know, the Belt and Road Initiative is one of the most ambitious physical infrastructure projects that was ever conceived. It was launched by Xi in 2013, and originally it was devised to link East Asia and Europe, has since expanded to the Global South and to Latin America. Do you think that the US is losing out to China in this regard? And what should the US do to counter potential China dominance in the Global South, particularly? IB : It ' s certainly true that the Chinese have far more influence in terms of their commercial and trade relations across the Global South than the Americans do. In part because they **invest** a lot more and in part because those investments are driven much more by the Chinese government. And that ' s because that ' s the nature of the Chinese system. I mean, if Apple decides to **invest** in India as opposed to China, that decision has virtually nothing to do with the US government, maybe at the margins. But overwhelmingly, that decision is made by Apple for reasons intrinsic to Apple. Where if the Chinese are **investing**, certainly if it ' s a state- owned enterprise, it ' s being coordinated strategically with the Chinese government. And even if it ' s a private-sector company, there ' s a lot more alignment with the Chinese. So I ' m saying that in part because there ' s some degree to which this is not a winnable fight by the Americans. The Chinese exert influence around the world through their state capitalist system primarily. Their power is projected more commercially than the United States, which historically has projected power, a lot of soft power through its political institutions and through its cultural institutions, but also, of course, a lot of hard power through its military might. And then, of course, the role of the US dollar as reserve currency, which allows you to weaponize it, get countries to do what they want that way, but not through state-directed trade and investment. So you could say the Chinese are in the lead there, though I would argue that the size of that lead has been overstated. In part because China is no longer spending anywhere near as much on Belt and Road as they used to. And that ' s one of the reasons why a lot fewer heads of state showed up a month ago at their Belt and Road summit than did during their summits before the **pandemic**, because they just don ' t have the same amount of money that they are willing to throw at these countries, but also because a lot of the investments they made did not perform very well. And as a consequence, you ' ve got a lot of bad debt that now they have to restructure in countries all over the world where they have really, really big exposures, like Pakistan and like Zambia and like Sri Lanka, I mean, countries that 10 years ago, 15 years ago, you were heralding as China ' s taking over these places. And then it ' s like, oh my God, what are we dealing with? Venezuela, right? I mean, a lot of the countries that the Chinese are dominating are some of the worst-performing markets out there with debt that is going to be incredibly hard to service, especially in a really challenging interest rate environment.

So I ’ m not so sure that that Belt and Road, you know, sort of, advantage is so critical, especially because a lot of that Belt and Road is in hard infrastructure. And once you build it, everyone can use it. You build a port, you build a railway. I mean, I ’ d rather the Americans build it than the Chinese because it redounds more to American shareholders, sure. But you ’ d rather the port be built, than no one build the port. Because that then leads to more economic growth. And if the US is the largest economy in the world, the US benefits disproportionately from more economic growth around the world. That ’ s just kind of a reality. It ’ s what globalization is all about. But I ’ m not so sure, when you talk about new technologies, now it ’ s a very different place. Now, the Americans are leading the world in AI, and the Chinese are leading the world in **transition** energy technology : electric vehicles, batteries, supply chain. And we ’ re seeing that play out in the fight between the US and China. So there were some big positives that came between Xi and Biden. You ’ ve got a lot of military- to-military direct engagement in a high level that the Chinese had resisted before. And so now the next time you have an American and Chinese aircraft five feet next to each other, near-miss or God forbid, actually have a collision, you ’ ll have hot lines to deconflict that immediately. And that ’ s a good thing. And on Taiwan, there were a lot of conversations. And now the two opposition parties in Taiwan look like they ’ re going to run on a joint ticket. They still have to figure out the final methodology on that. But if that happens, that means the **guy** that, you know, China thinks of as pro-independence, that would lead to a lot more tensions, probably isn ’ t going to win. That reduces near-term tensions. That ’ s a big issue. But the big area of **conflict** that has not been addressed and that is still moving towards more confrontation is technology. And here the Americans are continuing with existing export controls, and they plan to expand them. In fact, in the coming months, I think you will see new export controls on cloud computing. And meanwhile, the Chinese are responding with export controls in the critical minerals space. They talked about gallium and germanium, which, you know, are pretty widely available and not so essential for so much of that supply chain. But now they ’ re talking about graphite, which is much more essential for batteries, for EVs and where the Chinese have much more control. And if those move from licenses to direct controls, then you ’ re going to have a very significant fight between the United States containing Chinese growth in AI and the Chinese containing American and allied growth in **transition** energy. That ’ s the opposite of globalization. It ’ s less efficient. It ’ s more expensive, right? It ’ s industrial **policy**, it ’ s not free market. And it will create a much more tense structural relationship between the US and China. This is the area that we need to watch the most closely over the coming, say, six months. It ’ s where we could end up getting a much bigger blow up, despite the level of stability that both sides are trying to achieve in the relationship. HW : So the mention of energy is obviously salient. And, you know, the COP **conference** is coming up in Abu Dhabi in ten days ’ time. And so I ’ m wondering, did they talk about **climate policy**? Were they talking about energy? What happened? IB : It was part of the run-up to the summit where John Kerry, on cabinet and special **climate** envoy at state for the Biden administration, was meeting with his interlocutor in the Chinese government. They ’ ve engaged a lot of late, and there ’ s a replacement there. So Kerry met with the replacement, which is useful, younger, not as well-known globally, but has the portfolio on the Chinese side. And in the run-up to the COP summit in Abu Dhabi, where, you know, the world comes together to talk about commitments on **climate**, there has been more willingness of the Chinese to say that they will come up with some joint plan to further reduce emissions by 2030 and further **invest** in **transition** energy. But, you know, in reality, the US and China are fighting more or competing more than they are cooperating in the **climate** space. This is one where the Americans looked at China you know, no matter what you think about **climate** change in the US, no matter how much of a tree hugger you are or aren ' t, you see the Chinese putting hundreds of billions into new post-carbon technologies and you say, wait a second, I can ' t let the Chinese dominate the world in that. I want the Americans to do that. So there is, you know, some sort of, you know, virtuous cycle of competition, even if it ' s not alignment and coordination. Now, the alignment between the US and China on **climate** is a challenging alignment. It ' s one where the Chinese are emitting by far the most carbon in the world today. And the Americans have emitted by far the most carbon in the world historically and emit much more per capita today than the Chinese do, though the Chinese per capita actually emit more than Europe, which is quite something, given how comparatively poor the Chinese are. You look at that and you say, well, neither of these economies really wants to spend a lot of money admitting that they ' re the ones responsible and they have to be the ones paying for loss and damages and **transition** for the poorer countries that haven ' t had a chance to industrialize yet, haven ' t had a chance to industrialize with carbon intensity yet, right? And there are a lot of countries around the world, especially India, but broadly the Global South, that really want a very different outcome. So, you know, China used to be a member of the Global South. And this is one – we haven ' t talked about this, but it ' s kind of a really **interesting** point to make. China is not a part of the Global South anymore, right? When they ' re the leading carbon emitter and they ' re the leading creditor to the world ' s developing countries and they ' re increasingly – and the second, the second-lead technology country in the world in terms of biotech and new energy and digital commerce and, you know, facial recognition, you know, voice recognition, the list goes on and on and on. You know, China is not a developed country, but they ' re not in the Global South. And part of the reason, to go back to what we talked about at the beginning here, part of the reason why the Americans and the Chinese are the"adults"in the room is precisely because they both have so much at stake with the existing status quo remaining. The Chinese want to change their level of influence over existing institutions. They want more voting rights in the IMF, for example, but they don ' t want to break those institutions. They want them to persist. In fact, the Chinese, you know, see that they ' re the largest contributor in the world to UN peacekeeping operations. That ' s an organization the Americans created at the end of World War II. But the Chinese are really committed to it. Where the Russians, I mean, you know, basically, they ' re the ones that are sending, you know, Wagner and their successors in to countries where nobody can participate. To, you know, literally, to ungoverned regions of chaos. So again, it ' s not that we suddenly say, oh, China is a democracy that we should really like, they ' re friendly, they ' re cuddly. No, no, no, not at all. But they are really **invested** in the present global system. And that is a piece of stability at a time that a lot of the world appears to be coming apart. HW : So one final question and then we have to wrap up. And this question came from our community who also sent in questions for you. How did this meeting change your outlook on future relations between the US and China? IB : In the near term, it almost guarantees that US-China relations will be more frequent and will be more constructive. That does not mean that there will be massive breakthroughs, but the willingness of both sides to see that they benefit when they engage with each other substantively at the highest levels across all of the government. I mean, it ' s not just on, you know, Gina Raimondo in Commerce going over there and saying,"Hey, we want to make sure that Disney and the NBA can still do business."That ' s happening, it ' s much broader than that. It

’ s **climate**, it ’ s defense, it ’ s technology, it ’ s the **leaders**. And I ’ ll tell you, this wasn ’ t in the talking points, but it is important that Biden and Xi Jinping privately did talk about the fact that they need to spend more time with each other personally. And, you know, that Biden and Xi knew each other quite well, spent a lot of time when they were both vice presidents. And that ’ s something that Biden ’ s pretty proud of. And he talks about it privately in a way that, for example, he never got to know Putin, and he doesn ’ t like Putin. Xi Jinping, he may not **trust** him, but he does respect him. They actually do have a person-to-person relationship that matters. And, you know, for two **leaders** of the most powerful countries in the world that hadn ’ t talked to each other for such a long time, spending four hours together really matters, face-to-face. And I think we ’ re going to see more of that, at least by Zoom, over the coming months. And we should welcome that. Irrespective of what you think of either or both of those **leaders**, it ’ s a very important thing for them to be talking. HW : Ian, always a pleasure, thank you so much for your time. IB : Thank you. Helen.

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Helen Walters : Hello, **everybody**. Two days ago, on October 7, the Palestinian Sunni-Islamic fundamentalist organization Hamas attacked Israel, overrunning two military bases, occupying territory, killing hundreds of Israeli citizens and taking dozens more as hostages. It was the most significant breach of Israel ’ s borders since the Yom Kippur War of 1973. The attacks were clearly long- and well-planned, and they sent shock waves of fear and panic through the region and the world. Obviously, it ’ s two days later. It is way too soon to understand all of the ramifications of these attacks. But we can try to understand how we got here and the implications of this awful moment. So we asked our community to share their questions and to answer them, I am joined by Ian Bremmer, president and founder of political risk research and consulting firm Eurasia Group. Hi Ian. Ian Bremmer : Helen, great to be with you. HW : Alright, so let ’ s get right to it. We ’ ve had a number of our community who really want you to explain the very simple question of how we got here. So can you share the historical context for this moment? And if you like, give us a bit of a Gaza 101. IB : Well, I mean, Gaza, we ’ ve got a population, a Palestinian population of just over two million, 2. 2 million, exceedingly poor. And, you know, without sovereignty, without statehood, and a part of the Palestinian occupied territories, also the West Bank, more people, 3. 5 million. The West Bank run not very well by the Palestinian Authority, which recognizes Israel ’ s right to exist. Gaza, run really badly, with very little resources, run by Hamas, which does not recognize Israel ’ s right to exist. Now we ’ ve been talking about a two-state solution for a very long time. For the idea that the only way you end up with stability between the Israelis and the Palestinians is if the Palestinians have some **ability** to govern themselves, have some control over their economic trajectory, over their foreign **policy**, over their borders. That is not where we stand right now. And indeed, the idea of a two-state solution has kind of lost the **collective** interest, imagination, traction, for two reasons. First, because the Middle East has moved on. A bunch of countries around the region have found that they are **interested** in developing direct relations, some formal, some informal, with Israel, and that they ’ re willing to do that irrespective of resolving the Palestinian question, the Palestinian problem. And we ’ ve seen that with the Abraham Accords under the Trump administration, where the UAE, the United Arab Emirates, Bahrain and Morocco all directly established diplomatic relations with Israel. If you go to Dubai or Abu Dhabi today, you will see Israeli tourists like you wouldn ' t imagine. And they ' re having a great time and they ' re spending money and they ' re taking in the sites and they ' re very welcomed by the Emirates. Unimaginable that was going to happen 10 or 20 years ago. In fact, Saudi Arabia was very close, not within weeks, it wasn ' t imminent, but certainly within months of signing a deal with Israel that would allow for them to open diplomatic relations. And there ' s already been a number of high-level diplomatic relations informally between Mohammed bin Salman and Prime Minister Netanyahu. So, in other words, across the region, you had Israel, frankly, in the strongest geopolitical position that they ' ve been in decades. They ' ve been surrounded by enemies. Well, now they ' re increasingly surrounded by countries they can do business with. In fact, just a couple of weeks ago, there was an announcement of a deal where the United Arab Emirates was **investing** massively into solar power for Jordan, which would then be given to Israel in return for desalinized water processed by Israel. Even five years ago, inconceivable a deal like that could happen. So the Israelis, technologically very sophisticated, an advanced industrial economy, are only standing to make more money by doing business with all of these countries. What ' s been happening with the Palestinians? Nothing. The answer is nothing. They ' re not benefiting economically. And all of these deals for Israel have happened without any consequences, any contingencies for the Palestinians. And indeed in Israel, you know, there have been a lot of headlines. Israel ' s made a lot of news this year, but not because of the Palestinians. Israel has made news because of their own domestic constitutional **crisis**, an effort by the Prime Minister, Netanyahu, and his right-wing coalition to engage in judicial reform, an Israeli judiciary which is very independent, which has, in the context of democracies, a very surprising amount of authority over making but also interpretation of laws in Israel. What can and what cannot be considered a reasonable law to be executed. And for a country that doesn ' t have a constitution, not surprising perhaps, the judiciary is so powerful. And Netanyahu facing corruption charges and with a very weak right-wing coalition relying on far right, extremist right party as part of that coalition, was pushing for these reforms. Now, why am I talking about that? Because for the last six months, there have been unprecedented demonstrations across Israel, peaceful demonstrations, but bringing out the entire country. Because they were concerned about a constitutional **crisis**. Kind of an irony for a country without a constitution. If Netanyahu persisted, went ahead with these reforms. No one was talking about the Palestinians. And indeed, large numbers of troops that had been in the south were moved to the West Bank as the Netanyahu government was expanding the settlements in that territory and responding to Palestinian reprisals against those settlements. So they weren ' t focused on the issue. They took their eyes off the ball. Israel had other priorities, and the Palestinians were in a position not only to lose their friends around the region but also increasingly an afterthought for the Israeli government and the Israeli people. That is the backdrop for where we are today. HW : So let ' s dig in a little bit into the idea that you bring up of the kind of the troubles that have been roiling Netanyahu and the Israeli government themselves. So I think one of the things that has been brought up is the massive failure of **intelligence** and defense systems in Israel that allowed this attack to happen. What happens next? Do you see Israel uniting around Netanyahu? Do you see this fracturing even worse with anger at what happened in the lapses in defense and **intelligence** insights? Or what happens next? IB : Well, the first point I should make is just for everyone to understand what has just happened to the Israeli consciousness. It is unimaginable that, you know, certainly someone in a developed country could have any understanding of what the Jewish people in Israel are presently going through. This feeling that, you know, after the Holocaust and, you know, the land being provided to them to have a **safe** haven to create an independent Israeli state. And the need to defend their borders, the historic fights they ’ ve had with their neighbors, the war in 1973, when a number of Arab nations decided to fight against them. And, you know, the continual sense of besiegement with missiles from Hezbollah, for example, terrorist operations. This is not like a bolt from the blue when the United States experienced 9 / 11. Israel ’ s 9 / 11 is both massively greater in the **impact** on Israel but also comes for a country that was supposed to be prepared for this. I mean, Israel represents the gold standard on border security around the world. Not like the United States, where you ’ ve got, you know, all sorts of people running across and”build the wall”becomes a clarion call precisely because nobody understands how to defend the border, no. And also, **intelligence** collection, surveillance, digital surveillance, human **intelligence** collection on the ground, especially in the occupied territories. This is what they do. And the fact is that right now, today, Netanyahu ’ s legacy will not be anything that he has done to date. It will be this failure and how he responds to it. Period, end of story. Nothing else is close. So what that means for Israel is that all of the issues that have roiled this country over the past year, all of the political polarization, and it ’ s not a two-party country, it ’ s a many, many-party country. You know, the joke is you get three people together in Israel and, you know, you form a new political party. And if you go to the coffeehouses and the rest, **everybody** ’ s talking politics. Everyone reads the newspapers. This is a highly politically literate and divided population. But as of right now, priority one, two and three for the entire Israeli people is to respond to these attacks, these terrorist attacks. And how are they going to respond to them? Well, number one, they ’ ve got to find a way to get their people back. There are 100-plus, and we don ’ t know the exact numbers right now, hostages that are being held in Gaza, most of whom are civilians. And they will do everything to get them back. And they may well have direct American **support** in trying to accomplish that. And then it will be to go into Gaza, to remove the **leadership** of Gaza, to disarm the militias in the territory of Gaza, and to do everything they can to try to **ensure** that this cannot happen again. And making that happen is a very, very tall order. It might be a taller order than the Israelis can accomplish, and certainly the knock-on consequences will be grave even for just Gaza. And that is before we talk about any potential expansion of the war. But for now, the Israeli people will stand together. And there ’ s already talk of a government of national emergency that would bring together Netanyahu with the **leader** of the Israeli opposition for the purposes of fighting this war so that everyone in Israel is together collectively, **ensuring** the national security of the people of Israel. And I think that for the course of the coming months, and let us remember, this is not just, you know, an attack against Israel and now they respond. It is very likely there are Hamas operatives on the ground inside Israel right now that the Israeli government has to find and neutralize. It is also true that, you know, you ’ re still actively expecting that there are going to be additional attacks, whether those are missile attacks or whether those are direct incursions, nobody knows, but given the level of planning that was required by Hamas to make these strikes over the weekend, which nobody in Israel thought was possible, no one expected it, right now, that level of concern would be higher than anything else on the political agenda. And again, that will not move for the foreseeable future. HW : So I want to talk more about all of that and expand this to obviously the broader geopolitical implications of this. But I want to just play out the 9 / 11 reference a little bit if you can, because obviously 9 / 11 happened some time ago with the attack on the United States. But we know with retrospect, with hindsight, that some of the decisions that were made after that were misguided, they were misjudged, they actually led to terrible harm. How and who is going to make sure that these types of decisions are not made? And how can Israel avoid making decisions that will be bad? IB : As someone who was in New York on 9 / 11 and saw the second tower go down and saw how the city rallied together, how the country rallied together, President Bush, over 90 percent approval in the country a couple of months after 9 / 11, how the world came together to **support** the United States, the coalition of the willing, well beyond NATO. I mean, poor countries that had no business caring about what the United States was up to providing troops on the ground and **support** for the Americans. Russia, Putin ' s Russia, calling up Bush and offering, you know, the former Soviet republics in Central Asia as bases to **support** for logistical operations for the war in Afghanistan. I mean, the level of **support** for the United States after 9 / 11 was singular. And there ' s no question that the outpouring of concern, I mean, when I saw in Berlin, shining on the Brandenburg Gate, the Israeli flag with the star of David in Germany, in Berlin, in Germany, and given the history and given what that means and given the Alternative für Deutschland doing well in East Germ – all of that, I mean, this is a singular moment in the relationship between Germany and Israel. The European Union suspending aid to the Palestinians, the **support** for Israel is extraordinary. Narendra Modi In India. It is not universal for every country, but it is absolutely wide-ranging. And I got a readout, I spoke to several of the folks **involved** in the Emergency Security Council meeting at the United Nations this weekend. The condemnation of these attacks, everyone but Russia. And again, so in that regard, this is very, very similar to 9 / 11. Now, the broader question that you ' re asking, Helen, which I ' m also very sensitive to, is in 2023, looking back on 9 / 11, the Americans made some horrible, horrible, long-lasting mistakes. And some of those mistakes were in the United States. I mean, if I think about how much money was spent and wasted in the Department of Homeland Security, on personal security and safety in the airlines, how much money was wasted, how much economic inefficiency as a consequence of overstating the terrorist threat in the US, everything else secondary to that. But also, the rights that were stripped back for, in some cases, all Americans in terms of surveillance and the Patriot Act. But also targeting Muslim Americans across the country and so much mistreatment of American citizens as a consequence of that. But that ' s nothing compared to the mistakes made internationally. A war of choice in Iraq, responding to 9 / 11 with trillions of dollars wasted and lives, millions of lives destroyed. Afghanistan, 20 years on, a failed war with the Taliban returning to power and a failed state. Yes, bin Laden was killed, and I think people around the world cheered that, not just in the United States, and Al Qaeda was destroyed at the highest levels and in many cases uprooted completely. But no one can look back on the 20-plus years since 9 / 11 and say that the American response with the war on terror was successful. You can ' t do that. And Israel is not the United States. The Americans have extraordinary strength and resilience in its national security capabilities and the size of its economy, also in where it ' s located geopolitically. Israel certainly has the military strength in the region, but the the country is small. The territory is small. And certainly it is not in a geopolitical space that is comfortable. And so I think the danger here is that as the Israelis respond against Hamas, as they should and as they must, and as they work to destroy the **leadership** of that terrorist organization and disarm the militants that are **involved** in the attacks against Israel and pose an ongoing threat. But that is certainly not the only knock-on consequences of Israel ' s decision making. And the potential for this to become a broader war that would envelop the Middle East in conflagration and that ultimately could even end Israel is real in a way that the war on terror could not have threatened the United States existentially. And I, as a consequence, I certainly believe that a unity government will make it less likely that the Israelis overreact in that way. I certainly believe that the United States, in providing very strong and committed **support**, but also notes of caution in what can be done and what should not be done, will hopefully restrain the worst impulses. And, you know, in the early moments, again, we all understand why Israel would feel the need to react in the harshest possible way. But I certainly worry when I see the Israeli defense minister refer to the attackers as inhuman animals and announce a siege on the entirety of Gaza, which means no food, no electricity, no water. And this is a territory that already has 50 percent poverty, already has fewer than half of its population with access to clean water. I worry about what that is going to mean for the Palestinian people as well as for Israel long-term. You know, I think it was Golda Meir who says,"I won ' t hate you for killing our children. I will hate you for making me kill your children."Ultimately, the Israeli population is most threatened by what the terrorists of Hamas unleash from Israel. HW : So do you think that was part of the incentive for Hamas in doing this? Because surely they knew that the response would be swift. And surely they knew that the world would rally around such atrocities. So what do you think was their motivation, and do you think that they underestimated what might happen? IB : Oh, I don ' t think they underestimated what might happen. But it ' s a compelling question. It ' s really hard to put yourself in the mindset of someone like a Hamas **leader**. But, you know, I had to do that just a few months ago when Yevgeny Prigozhin was marching with his Wagner forces on Moscow. And people were asking me,"What is going through this **guy** ' s mind?"Because it ' s clear he ' s going to get killed, right? I mean, you turn against the Kremlin and Putin, you ' re not walking away from that. And when he cut the"deal"and everyone said,"Oh you know, he cut a deal"– He ' s dead man walking. Like, literally, that was the reality. And as soon as the Hamas **leaders** decided that they were going to commit these atrocities against Israeli civilians, they ' re dead. There is no future for these people. So I think there are two different things going on. The first, and this is analogous to Prigozhin, is that Hamas felt themselves in an increasingly untenable environment, that they were losing their **support** in the region. And even the Saudis were about to normalize their relationship with Israel. They had no influence in **ability** to get anything done, no leverage with the Israeli government, which was only becoming harder and harder lined against them. And in that regard, they were increasingly in a corner. Their options were increasingly all bad. And, you know, we know that people that find themselves only with horrible options frequently do irrational things. And I would not underestimate that in driving the decision of Hamas to take that action. It ' s kind of like why would 77 percent of a Gaza population in the last elections they had, which was some time ago, why would they vote for an organization like Hamas? Well, I mean, they wouldn ' t if they had economic opportunities. They wouldn ' t, if they had education, they wouldn ' t if they could come and go from Gaza as they please. But the worse the situation gets, the more they are willing to vote for an organization that is prepared to burn it all down. And by the way, there ' s a lesson in that, even for those of us in very wealthy, very stable countries. So I think that ' s one set of motivations, but another set of motivations certainly is an ideological effort of Hamas to insert themselves as more **dominant** in the conversation, to radicalize the Israeli population, to undermine the Palestinian Authority in the West Bank. Because if this fight, you know, gets the Israelis to kill huge numbers of Palestinian civilians, and by the way, Hamas will facilitate that, right?

I mean, Hamas is absolutely going to be engaged in operations, you know, in residential buildings. They do that intentionally. They ' re not going to make it easy for Israel to take them out. They want to make it bloody. They want to paint the Israelis as just as bad as Hamas, if not worse. They will take human shields. The IDF, the Israeli Defense Forces, usually gives warnings about when they ' re about to attack a building. They ask the civilians to leave. Well, Hamas tells those civilians that that ' s **disinformation**. They do everything they can to make the Israelis seem complicit with the kind of indiscriminate attacks against civilians that Hamas engages in themselves. They want to bring the Israelis to their level. And they also want to radicalize the Palestinians in response, not just in Gaza but also in the West Bank. And they want to radicalize the Arab street. They want people across the region to be, you know, in uproar against Israel and in solidarity with the Hamas cause and in solidarity with the destruction of Israel. They want Arab **leaders** to be saying what the Iranian supreme **leader** was posting on social media this weekend, calling essentially for a genocide against the Zionist regime. That is ideologically what Hamas is trying to accomplish. And again, Israel must do everything in its power not to allow Hamas to drag them there. HW : It ' s **interesting**, in your talk in Vancouver this year at TED2023, you were talking about the rise of different orders, and I do just get the sense that everything is connected. You have Russia, you have Ukraine, you have Iran. There are these ideological battles that are now becoming real-world wars. And so I wonder if you can, especially the mention of Iran, I don ' t think it ' s confirmed yet, the intervention of Iran in this, but certainly the"Journal"was reporting that Iran had been **involved**, deeply **involved** in setting up these attacks. What does this mean? What does this mean for the world at large? And then I also have a follow up question, which is, what do you think the US should do? IB : So let ' s talk in terms of the world at large and starting with Iran, certainly that"Wall Street Journal"piece over the weekend drove an enormous amount of news. It was saying, hey, the Iranians basically planned this. I will tell you, that was a very lightly sourced piece, relying on Hamas. And I would not have gone to print with that if I had been"The Journal."HW : Yeah, the US has not confirmed that at all. IB : In fact, the US has actually said that there is not hard evidence at this point that fingers Iran as having directly orchestrated or ordered these attacks. Now, let ' s be very clear. The Iranians have publicly expressed strongest possible **support** for Hamas. The Iranians have historically funded and provided military **support** directly for Hamas. So they clearly are not innocents in this. And I would be surprised to learn that the Iranians had no idea that this was going to happen. I suspect that they were aware. But awareness and orchestration are two very different things. Now, since the attacks occurred, Hezbollah, which of course, is also very much aligned with Iran and gets a lot of direct military **support** and training from the Iranians, they have, I say only, but in this context, is only, they ' ve only engaged in some missile strikes, some rocket strikes against an Israeli military – not base, but military outpost. A soldier ' s outpost. And the Israelis, in response, immediately engaged in strikes back against Hezbollah. That ' s it. If the Iranians were behind this and wanted to be seen as behind this, Hezbollah would be **involved** in these attacks. They are far more capable than Hamas. The Iranians have **claimed** that they have had no role, that this was an autonomous Hamas operation. And indeed, Iran has been doing better geopolitically of late. The Chinese facilitated a breakthrough in Iranian relations with the Saudis. The Iranians have engaged with the United States, and six billion dollars of Iranian assets are set to be unfrozen, have not been unfrozen yet, but are set to be transferred to Iran. Five American civilians that were held unjustly as hostages in Iranian jails have been released and sent to the United States.

The Iranians have reduced the top level of uranium enrichment and some of their stockpiles, allowing inspectors in. Now, this is not a return to the Iranian nuclear deal, the JCPOA, but certainly on the basis of all of that and even some high-level discussion that the Iranians might be willing to engage directly with the United States diplomatically through the good offices of Oman, none of that seems aligned at all with the Iranians pulling the trigger on an attack against Israel that would almost certainly lead to massive retaliation once the Israelis found that out. So I am sitting here saying I would be surprised, not with a high level of **confidence**, and, you know, the Iranian regime has a very old supreme **leader** who is also dealing with internal instability and a **transition** that is coming. So never say never. But I would be quite surprised if we found out that the Iranians directly ordered this. Now, it is useful that the United States has both sent a fleet off of the Israeli coast to show stalwart **support** and will be providing a level of at least military coordination and operational **intelligence**, may well do much more than that. We can get to that when we talk about the United States, but is also very publicly saying"We do not yet have any evidence that the Iranians are **involved**."In other words, the message from the United States is very clear : do not expand this war into Iran, because the consequences of that are 150-dollar crude at a minimum. The consequences of that is the world goes back into global **recession**. The consequences of that are conflagration in the region. And I think, I do believe that the Israeli government is quite aligned with the United States in not wanting to go there. HW : I keep coming back to the human cost of this because the reality is that people are suffering, people are being killed, and many more people are likely to be killed. If, indeed Hamas has kind of hijacked this story with extremist action, I wonder what you see from the Palestinian side of kind of a more moderate type of push towards trying to get understanding, trying to get peace in this nation or trying to get peace in this area. IB : I mean, that ' s the most tragic piece of this, is the **ability** of Hamas to successfully hijack big pieces of the political spectrum for the Palestinians. I mean, there are so many people in the West right now that view the Palestinians as equivalent to Hamas, and nothing could be further from the truth. But that reality, that perception is going to make life so much worse for the people that have suffered the most. They are the powerless. The Palestinians are the stateless. They lack resources. They lack a proper military. They lack the capacity to defend their own territory and to defend themselves. And we ' ve already seen, even over this most tragic weekend for Israel, that the number of deaths and casualties for the Palestinians are almost as much as they were for the Israelis. And when you go back over the past 20 years, who ' ve had the most deaths, who ' ve had the most casualties, consistently, it ' s been the Palestinians. Who ' s going to suffer the most going forward? Consistently, it will be the Palestinians. Who suffered the most from the US war on terror? It was, of course, the Iraqis and all of the tribes in Afghanistan. This should not surprise anyone, but it is the unfortunate reality that of course, Hamas **leadership** will be destroyed. But the biggest damage that they will have done would have been to their own people. To the Palestinian people, who now will face almost unfathomable deprivation. And there ' s very little that the rest of the world is going to do about it. HW : Do you see a movement within Palestine to step up if Hamas is done? IB : I certainly believe that the Palestinian Authority will try to see this as an opportunity to push for more **international** engagement from the region to take seriously a cessation of illegal Israeli settlements in the West Bank, a rolling back of the territory that is presently occupied and a revival of peace talks that would bring about a two-state solution where the Palestinians, less land than perhaps they would have gotten in the days of, you know, Arafat and Rabin, but nonetheless, something that feels sustainable, a country that one might be able to raise children with a sense of hope. There is no one in the occupied territories of Palestine that could say that for themselves for their children today. So I think that is the hope. But, you know, clearly right now is not the time for that. Not because we don ’ t want it, but because events will overtake it immediately and have already overtaken. Now, the hope is that the violence that will spread in the West Bank can be contained. That we do not see a war in Gaza become a war in the West Bank, that we do not see an occupation of Gaza become an occupation of the West Bank. That is, I think, the priority now. You have to know sometimes when you actually have a trajectory for peace, and when you have to do your best to avoid war expanding. We do not have a trajectory for peace right now. We are at war. There were plenty of opportunities over the past years to take off ramps, to engage more seriously. They were ignored. And it is precisely that reality that has brought us to a point where we now are on defense, where we are hoping that this does not get much, much worse. That is the story of the Middle East today. HW : Do you think Israel will annex Gaza? IB : I think the Israelis don ’ t know. Again, don ’ t underestimate the shock, the emotional shock that all of the Israelis are facing today. They are not making long-term strategic decisions right now. They are making immediate, short-term decisions. What can we do to make sure the country is still defended, our borders are still secure, that terrorists are not, you know, right now running around in our midst, planning further atrocities? That is priority number one. And very close to it is getting those hostages back and **safe** and I ’ d say unharmed. They ’ ve already experienced a lot of harm. That ’ s beyond the realm of possibility right now. There will be, in the coming weeks, and there ’ s also some shock and awe, you know, Netanyahu immediately posting, some buildings getting demolished in Gaza and saying we are, you know, at war. That is, I mean, they ’ ve engaged in these sorts of airstrikes before, and we already know of families that have gotten killed, entire families in some cases of nine, of 13 people, lots of children, this sort of thing. We will see that. But in terms of what ’ s the nature of the long-term objectives, the occupation, the Israelis are not close to making that decision. And I also think that other countries that Israel **trusts** and Israel needs will have some **ability** to have influence over Israel in making that decision. I ’ m not just talking about the US now. I ’ m also talking about countries in the region that Israel would like to maintain relations with. So there needs to be very active multilateral diplomacy behind the scenes, quietly, high-level, with the Israeli government in the coming days and weeks. HW : I want to talk a little bit about the media coverage of the attacks and of what is happening right now. A lot of the people who had written in with questions for you are really confused by the rhetoric that they ’ re seeing. They ’ re not sure what to **trust**, who to **trust**, what to believe. And I want to get your sense, you ’ re a very online person. And so what is your take on this, and how can we think about how to understand what ’ s happening, especially in such a kind of, real-time situation? IB : Well, first of all, there ’ s always the fog of war. There ’ s always **disinformation** from both sides actively trying to promote a narrative that is more effective for them, that they ’ re doing better than they otherwise are, and that the other side is engaged in greater atrocities than they actually are. You see lots of that, lots of immediately fake videos putting out of buildings that are being destroyed, people that are being killed, you know, sort of, people that are providing, you know, sort of, **support** for, sort of,”kill all the Jews”and”kill all the Palestinians”that actually came from previous **conflicts**, not from the present one. There ’ s plenty of that. This time around, there ’ s also so much more hatred. There ’ s, on social media especially, there ’ s so much more willingness to promote, algorithmically, opinions that you would never hear in your family, that you would never hear in your community, in your school, but it ' s being bombarded. And this is very different from the so-called mainstream media, whether it ' s the BBC or the Deutsche Welle or it ' s Fox News or CNN. No, no. Social media has become a far more greatly polarized and hate-filled space. I ' ve received at least 30 death threats over the last 48 hours from complete randos. A few people that I actually could track if I really needed to, most anonymous accounts. But clearly people that are writing me directly, some of whom are really, really pro-Israel, some of whom are really, really pro-Palestine and some of whom are probably just trolling for the lolz, as they like to say. It is increasingly very hard to navigate this space without becoming incensed and deranged. Having said that, as much as I find Twitter / X a space antithetical to civil society, I also know, as someone who does analysis, that some of the best real-time information from sources on the ground is being passed through on X and is not found in other places. It will not come through in mainstream media. So for the average person, you need to spend a lot more time filtering and figuring out where to go and who to follow. But it still is the one place that you can go. And you know, it really does, the whole thing profoundly worries me because when you ' re in an environment that you can no longer know what is truth, what is real information, it is really hard to maintain a society that is human. When you have people that are saying that all Israelis are X and all Palestinians are Y, and that is where much of social media, I mean, a strong majority of social media is there right now, you cannot have dialogue, you cannot have solutions, and you can very easily tilt into war, into radicalism, into fascism. This is something we all need to be guarded against. And I truly believe that the social media companies need to be regulated on this. They are acting as if they have no responsibility for what ' s on their sites. That, you know, it ' s just like the **phone** company. That if you and I, Helen, are having a conversation about blowing something up, well, we ' re responsible for that. But the **phone** company isn ' t responsible, and I accept that. But, Helen, if you and I are having a conversation about blowing something up and then the **phone** company takes that conversation, identifies everyone else that might be **interested** in blowing something up or has considered it, and takes that conversation and promotes it to them, then you, the **phone** company, are responsible, you are liable, you should be taken down. And we are in a war right now, and the social media companies are actively fanning the flames. They are spraying fuel on the flames, and they ' re doing it globally. Globally, so much so that countries like China that are authoritarian and control their media space actually have sort of an intrinsic political stability advantage in"information warfare"over open societies that should be the most resilient. That ' s crazy, and we can ' t keep going down that path. HW : The ironies are writ large. OK, so we are coming on our time, but I wonder if you can leave us with a sense of what should we be watching for next? What should we be looking for? IB : First, we need to look for Lebanon. This is the issue of Hezbollah, which so far has been the dog that has not barked. Is that going to continue? It is the place that you are most likely to see tipping point escalation if it were to occur. And you have Hezbollah operatives of many different stripes. They are loosely organized, they ' re well-trained. But that doesn ' t mean that they ' re all following, you know, marching orders from one direct **leader**. The potential that this could – you could see escalation with some Israeli farmers getting killed, and then the Israelis respond. And before you know it, you ' re in a much bigger firefight. Lebanon ' s **involved**, Hezbollah is **involved**, and then it knocks on to Iran. That is sort of the gateway drug in the Middle East even if nobody wants that fight. That ' s one thing to watch. Second thing to watch, of course, imminently is what happens with these hostages. Do the Israelis get them back?

Historically, that has always been the top priority, and it is today. But Hamas has control over that. And, you know, if the Israelis are not prepared to **negotiate** with Hamas to release militants presently in Israel prisons, and it ' s very hard for me to imagine in today ' s environment they would be willing to do that, well, how exactly do they get them back and how many of them can they actually free? Again, you know, the Israelis and the Americans have far, far better tradecraft on the **ability** to get these hostages out than Hamas have to take them. But Hamas has exceeded expectations over the past 48 hours, and I would worry very much about that. That would be the second thing I ' d watch most closely right now. And then finally, the nature of the Israeli government itself. Do we have success in putting together a unified national emergency government, in which case we will have more stability in governance and decision making that comes from Israel and also greater willingness to consider longer-term engagement with those Palestinians, particularly in the West Bank, to start, at least, that might be looking for a more constructive path now that they are back on the agenda. I don ' t have a high amount of optimism that that ' s going to happen. But you asked me for something hopeful, that would be something hopeful. HW : Ian Bremmer, we are so grateful for your time and for your insight. Thank you so much for joining us, stay well. IB : My pleasure Helen.

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Christiana Figueres : There ' s no doubt that we are facing exponential **impacts** of **climate** change. There is also, however, no doubt that that is being met with exponential progress of the technologies that can help to address **climate** change. So what we have here is, frankly, a race between two exponential curves. By 2030, we collectively will have chosen between door number one. And door number two. Julio Friedmann : The question I am most commonly asked about **climate** change is :"Should I be optimistic or pessimistic?"Cynthia Williams : Here ' s what we need to do to make sure that we ' re ready for an all-electric future. Anika Goss : Being financially secure and **climate** resilient should be the most important priority. Faustine Delasalle : It ' s really easy when you look at where we are today and where we ought to be by 2030, to feel like there ' s a huge gap that will never be filled. But we have good reasons to say that that gap is actually fixable. Nigel Topping : If you ' re not worried or anxious, you ' re probably not paying attention. But we should also not be defeatist. We shouldn ' t say,"We ' ve done nothing, we ' re failing."We ' ve really started 20 years too late, but it takes a long time to get the flywheel of momentum going. Kingsmill Bond : The tide of change is coming. The costs keep falling. The political pressure keeps increasing. JF : We ' re going to build a thriving, exciting world. AG : We can do this. JF : Full of potential. AG : We have to do this. [**TED** Explores A New **Climate** Vision] Manoush Zamorodi : When it comes to **climate** change, are you an optimist or a pessimist? News about the **climate** can be overwhelming and emotional, to the point that many of us just feel numb. I ' m Manoush Zomorodi, a longtime public media journalist and host of the **TED** Radio Hour. And today, we want to bring you something special : an hour of nuance, of real people and clarity about what is happening to our planet and what we ' re doing about it. Because we all know there ' s plenty of bad news when it comes to the **climate crisis**. But guess what? There ' s also some really good news. That ' s why we ' re here in Detroit at the **TED** Countdown Summit, where a cross-section of people who know the most and who are doing the most have gathered to share their ideas and plans with each other and with you. You ’ re going to hear from some incredible speakers about their personal stories and tested solutions. But first, let ’ s get a quick reminder on the basics : **climate** change in a minute or so. [Prologue : What is **Climate** Change?] David Biello : The **air** we breathe has changed. The mix of gases is shifting, with more and more greenhouse gases like carbon dioxide. And the shift is happening faster each year. In just a few hundred years, fossil fuels formed over eons have been burned as coal, oil and gas. The exhaust has transformed the entire atmosphere and ocean. It ’ s like a pollution blanket. And the result we know as **climate** change. The planet has already warmed more than one degree Celsius and is on a path to heat up even more. That may not seem like a lot, but it ’ s already caused major destruction across the globe. To give us some room to breathe, the world must reduce greenhouse gas pollution by more than seven percent each year, every year of this decade. There are a number of ways to do that. Let ’ s start by looking at the greenhouse gases themselves. CO2 makes up nearly 75 percent of the pollution emitted each year. And then there ’ s methane or natural gas, which makes up 17 percent. Finally, there ’ s nitrous oxide, making up six percent of the problem. There are other greenhouse gases, but these three make up the bulk of the **climate** challenge. Where do they come from? There are several ways to break down the sources of greenhouse gas emissions. Here we ’ re using the public **Climate** Watch data. Start with energy. The majority of modern energy comes from burning fossil fuels, which makes the energy **sector** 76 percent of the **climate** challenge, including the fuels used for transportation, industrial processes and agricultural production. Farming and animal raising also have to change, as agriculture accounts for 12 percent of greenhouse gas pollution. The remaining 12 percent comes from a grab bag of human activities like industrial heating, clearing forests and more. This is the **climate** challenge we face : going from adding billions of metric tons of greenhouse gases to the **air** each year to adding zero. Will it be easy? No. Can we do it? Yes, if we choose to. MZ : OK, so that ’ s the situation. Where do we go from here? Well, we ’ ve all heard of a tipping point, that moment when an idea or concept finally takes hold and then changes the world. Well, the UN ’ s **climate** chief, Simon Stiell, believes we are almost at a tipping point for massive action on **climate** change. And to get you ready, he wants to take you back to the early ’ 90s, when he was an executive at Nokia, and the world wasn ’ t sending text messages yet. [Chapter 1 : Exponential Change] Simon Stiell : I want you to imagine it ’ s 1992, and you ’ re talking to a telecoms expert. She tells you that mobile **phone** text messaging is going to fundamentally change the way we communicate. You think, why on Earth am I going to go from this to tapping on a screen with my fingers and my thumbs? But then, after a slow start, text-based services exploded. Today, over 23 billion messages are sent every single day. This paved the way for the smartphone revolution. I know how much we struggle to comprehend and predict what exponential change means, because I was at Nokia in 1993 when that technological revolution started. And now, as head of UN **climate** change, I ’ m here to tell you that what was true for mobile **phones** is also true for **climate** action today. We ’ re here in Detroit, the heart of car manufacturing. Take a look at this graph. Electric vehicle sales are expected to increase to become 70 percent by 2030. Another example, solar energy. Between 2010 and 2020, we moved from 20 gigawatts of solar energy installed to 150 gigawatts. And by 2030, that number is expected to increase again, to 1, 000 gigawatts per year. Change can come fast. FD : So exponential change means that change happens very slowly at first and then comes a tipping point. And suddenly things accelerate and accelerate and accelerate. NT : It ’ s a mathematical term, but it means basically **stuff** happens very slowly and then it goes really fast. Tessa Kahn : Governments and experts have consistently underestimated the pace and scale at which we can deploy renewable energy and how quickly renewable energy and associated technologies become affordable. Ramez Naam : I predicted in 2011 that we ' d have solar as cheap as coal in some parts of the world by 2015 and half the cost of coal by 2020, and everyone thought I was crazy, I ' ve got to say, almost. And in fact, I was wrong. The actual decline in the cost of solar happened twice as fast as I thought it would. Solar prices today are a century ahead of where the world ' s leading energy experts thought they ' d be in 2010. KB : It ' s now spreading from solar, so it ' s happening also in the wind **sector**, it ' s happening in the battery **sector**, it ' s happening in the electric vehicle **sector**. CF : I don ' t think that we do a good job at communicating the fact that we are progressing more than is commonly known. Nili Gilbert : The human brain is trained best at thinking about linear change. But when you think about an exponential curve, next to a line, at first the exponential curve is below the line, and so it ' s moving more slowly than you would think. And then it crosses the line and it ' s moving faster than you would think. Habiba Daggash : We call it cautious optimism. We ' re very hopeful about the trajectories that we ' ve seen. But we shouldn ' t let up yet. KB : So we can make this change happen slow or we can make it happen fast. The key point to say is we have agency. MZ : This idea of exponential growth sounds awesome. It sounds amazing, but is it actually happening? Well, I ' m headed into a workshop with industry and **climate leaders** to find out. The moderators of the workshop challenged us to hold two competing ideas in our minds at the same time. NT : What we ' re really going to explore today is two apparently opposing realities, and they ' re often two opposing narratives. The one which is true, right, is that in terms of the Paris goals, we ' re way off track. We ' re deep into the emergency. But what we ' re going to explore today is the other reality that we know what to do. Exponential change is happening, but it doesn ' t happen by magic. MZ : So what are we seeing in different industries? Well, let ' s start with one that ' s known to be really difficult to decarbonize : cement. Vicente Saiso Alva : I think everything converges into making **climate** a huge urgent matter for society. We started to feel the pressure from all different types of stakeholders, and in one year of running this program, we realized how fast we could move and how fast we were going in reducing CO2 emissions, which means now to go from 40 to close to 50 percent reduction by 2030. MZ : Encouraging, but to tackle that remaining 50 percent may require technologies that don ' t exist at scale yet. OK, what about shipping? Narin Phol : In Maersk, a few years back, we came out with a net-zero ambition by 2050, and in a few years later, based on the data that we have got, we ' ve really seen a path that we can really accelerate that one. That ' s why the revised target of 2040 came in. MZ : The news in offshore wind was quite dramatic. Jennie Dodson : So my moment of mindset shift was in 2019. The government put out a request to bid for offshore wind permits. And that year the price that came in was two thirds less than just four years before. And really critically, it was the same price as the mainstream market. And that was transformative. Nobody expected that when they started. MZ : The conversation got pretty technical and a bit wonky. NP : So to drive exponential growth, you focus on a reinforcing loop to make sure it works and you slow down on the balancing loop. MZ : But the takeaway is that certain **sectors**, like EVs and renewables, are being upended, while others, like cement and heavy industry are further behind. But generally across every industry – JD : It ' s amazing **stuff**. MZ : There ' s an understanding that they all need to work together. And the message? Big change can happen. It ' s going to be hard, but some of us are already doing it, so learn from us. [Chapter 2 : The Electric Vehicle Revolution] OK, we are in an exponential mindset. And as you heard, one example of exponential growth might be sitting in your driveway right now. Electric vehicles are finally really happening, which means that people here in Detroit are rethinking the auto industry. They ' re rethinking their jobs, their identity. Take Cynthia Williams, head of sustainability at Ford. I sat down with her at Lucky Detroit, a local coffee and barber shop, where she told me that there are three things that need to happen for electric vehicles to go to the next level, from this tipping point to exponential growth. CW : The three things for me are change, collaboration and capacity. So the change piece of it is changing the way folks think about the vehicle, making sure that they understand we ' re bringing a better vehicle to the customers. MZ : So you don ' t have to make a sacrifice. CW : I think one of the things we need to do is actually get people in the cars and show them how fun it is. And it ' s no different, it ' s just powered by a battery. We are at a tipping point. We ' re at a point of acceleration of electric vehicles. Many are calling it the electric vehicle revolution. Now with that comes unprecedented change that will require us to build new capacity and collaborate in ways that we ' ve never seen before. To make things happen faster, you have to think differently. You can ' t just collaborate with the same people because you get the same answer, right? You need to broaden your collaboration space. Go outside of your industry, understand what worked in Norway, for example, to get them to where they are. MZ : Where ' s the weak link, Cynthia? What ' s the thing that you ' re like, that if we don ' t push through that, we may not get to this full adoption? CW : The critical things that we hear from the consumers is they need infrastructure. That ' s what it ' s taking for us to work with governments to make sure that we bring more charging to everyone, whether it ' s rural, urban, wealthy, low-income, we all need chargers. And I ' m not saying we need one on every corner, but we do need many, many more. When we start building out the charging infrastructure, we need lighting. We need, you know, security, shelter, we need restrooms. We need other amenities so that people know where to go, where to charge. Next comes capacity. We have to **invest** in new facilities and talent. Now, that ' s true for any industry transitioning to greener goals and strategies. And we ' re investing in talent, we ' re investing in education and training, and we ' re also listening to the communities where we live and operate. MZ : So a young Cynthia Williams, now, your counterpart who ' s studying engineering, will she be learning something very different if she wants to go and work in the automotive industry in Detroit? CW : There will be different options, right. And so, I ' m a mechanical engineer by trade, but I think there will be opportunities for electrical engineering, software engineering. And there ' s even certificates that students can get instead of going through a four-year college. It ' s the proper training to get them right, to be dedicated, to build the vehicles that we need for the future. At the end of this year, globally, we ' ll have the **ability** to produce 600, 000 electric vehicles. And in 2026, we ' ll have a global goal to produce two million electric vehicles. And it just goes up from there. MZ : What is this going to look like, this growth for me in the next couple of years, in five years, in 20 years? CW : Clean **air**. That ' s what I look at. If you think about when we were, during the **pandemic**, when **everybody** stopped driving and they were at home, clear skies and moving to zero tailpipe emission vehicles. And if you chose an electric vehicle, that ' s you participating and you contributing to a better world, to a better future. We have come a long way. It ' s so gratifying to see electric vehicles on the road, to move from aspiration to reality. To see plants dedicated to building electric vehicles. We have the technology, we have the **leadership**, we have the ingenuity to respond to the environmental **crisis** with courage. And I ' ll tell you, as a person who ' s dedicated my entire career to sustainability, the road ahead is paved with promise. TED-Ed : If you were buying a car in 1899, you would have had three major options to choose from. You could buy a steam-powered car, typically relying on gas-powered boilers. These could drive as far as you wanted, provided you also wanted to lug around extra water to refuel and didn ’ t mind waiting 30 minutes for your engine to heat up. Alternatively, you could buy a car powered by gasoline. However, the internal combustion engines in these models required dangerous hand cranking to start and emitted loud noises and foul smelling exhaust while driving. So your best bet was probably option number three : a battery-powered electric vehicle. These cars were quick to start, clean and quiet to run, and if you lived somewhere with access to electricity, easy to refuel overnight. If this seems like an easy choice, you ’ re not alone. By the end of the 19th century, nearly 40 percent of American cars were electric. In cities with early electric systems, battery-powered cars were a popular and reliable alternative to their occasionally explosive competitors. But electric vehicles had one major problem. Batteries. MZ : Flash forward to the present and batteries are still on everyone ’ s mind. So I **visited** Mujeeb Ijaz, the founder of the energy storage company One, who also happens to be an electric vehicle history buff. Mujeeb Ijaz : So this is a 1922 Detroit Electric. MZ : OK, so 100-year-old vehicle. MI : That ’ s right, electric vehicle. And then this was Walter Baker. He developed the Baker Electric. And he made the range run 201 miles on a single charge. He also felt strongly that you should be able to wear your top hat, you know, so he has a taller carriage. These had lead acid batteries to begin with. And then Thomas Edison introduced the nickel-iron battery. So this is the battery compartment, one of the battery compartments. And this is where you would charge. MZ : Right there? MI : And we ’ re building on what these original pioneers did. And I think it ’ s important to not think about us starting from scratch now. But go backwards, think about how they solved the problem, what disrupted their **ability** to advance this to the market, and then pick those problems up and make sure we don ’ t get stuck again. MZ : So we ’ ve been here before, a moment when electric vehicles could have taken off but didn ’ t. So what do we need to do differently this time? MI : Electric vehicles were able to deal with mobility in urban areas, but there were charging deserts where rural driving resulted in no way to get back because you couldn ’ t find electric charging everywhere. And that was a problem. MZ : And that ’ s what people are worried about right now. We ’ re at the same thing. We ’ re worried about range, we ’ re worried about accessibility. All of the things are the same. But the key difference being that we now know that our planet can ’ t keep going like this. MI : That ’ s right. Had the vision of the Model T assembly line and mass manufacturing been applied to electric car, the electric car would have also dropped in price, but range and charging deserts would have still been the obstruction. MZ : OK. We didn ’ t have the technology, but we do now. MI : We do now. MZ : In the US, there hasn ’ t been a mass move to electric vehicles... yet. Mujeeb ’ s company is one of many who want to change that by building batteries that give drivers longer range. And these batteries aren ’ t just for cars. MI : So we produce battery packs for transportation, but we ’ ve also started developing the same battery products into grid batteries. So micro-grids are those technologies that combine solar and storage and replace a coal plant or a natural gas plant for generating electricity. So buildings and industry and transportation all working together. And that ’ s an exciting future, is all these industries working together, that makes me more confident that we ’ re at the time in history where this **transition** will take place. MZ : How many years till we get to that? MI : I think it ’ s within 10 years. MZ : No way, 10 years? MI : We ’ re in a place where that **transition** is going to be much more visible, and then we just need technologies to drive it up to the mass markets. MZ : I am so excited. Like, genuinely so excited. Everyone everywhere driving electric sounds great, but that also means that the power to charge all those vehicles needs to be green. And that brings us to the energy **sector**, which is also changing at an incredible pace, even though renewable energy has been years in the making. [Chapter 3 : Renewables] TED-Ed : In the spring of 1954, the press excitedly gathered around Bell Laboratories ' latest invention : a silicon-based solar cell that could efficiently convert the sun ' s energy into electrical current. The creation was celebrated as the dawn of a new era, as reporters touted that civilization would soon run on the sun ' s limitless energy. But the dream had a catch. As this first commercially sold solar cell cost around 300 dollars per watt, meaning at its current rate it would cost well over a million dollars to buy a unit large enough to power a single home. But today, in many countries, solar is the cheapest form of energy to produce, surpassing fossil fuel alternatives like coal and natural gas. Millions of homes are equipped with rooftop solar, with most units paying for themselves in their first seven to 12 years and then generating further savings. So how did solar become so affordable? A turning point in solar ' s price history occurred on the floor of Germany ' s parliament, where in 2000, Hermann Scheer introduced the Renewable Energy Sources Act. This legislation laid out a vision for the country ' s energy future in solar and wind. It incentivized citizens to personally **invest** in rooftop solar panels by guaranteeing payment to homeowners for the renewable energy they generated and sold to the grid. The pay rate for this electricity was highly subsidized, at times reaching four times the market price. Several other countries soon followed Germany ' s example, implementing similar **policies** and incentives to drive their country ' s solar use. This created unprecedented **demand** for solar panels worldwide. Manufacturers were able to scale up production and innovate in ways that cut costs. As a result, solar panel prices dropped while efficiency grew. HD : We see the cost reduction in solar and other renewable energy technologies has been so steep, but a lot of it is because of the enabling regulatory and **policy** environments created in places like the European Union and the US. Akil Callender : We cannot divert catastrophic **climate** change without addressing emissions coming from our energy systems. Currently, emissions from the energy system account for around two thirds of global greenhouse gas emissions. So the good news is, yes, we are seeing tremendous progress across the globe in renewable energy. It ' s rolling out faster than ever before. Luisa Neubauer : The brilliant thing about energy supply in the world is that we have alternatives. So we don ' t need fossil fuels to provide **safe** and equitable access to energy systems. Ramón Méndez Galain : Almost 90 percent of the total new installed capacity all over the world is just renewable. This is a tremendous change. AC : There ' s more finance than ever before going into clean energy. In many countries, it is the largest contributor to their electricity system for the first time ever. KB : And also because China is the leading manufacturing nation in the world, change is happening there much faster. And those innovations are then spreading around the world. Hongquiao Liu : China ' s coal **demand** might not have peaked, and China will continue to consume quite a lot of coal in the next few years. But the **decarbonization** is happening. It ' s happening faster than anyone has expected, and China is adding an astonishing amount of wind and solar to its grid every year. Eduarda Zoghbi : I think one of the biggest obstacles today are natural resources, in the sense that we ' re talking more and more about critical minerals and the role that they have. And batteries are **demanding** a lot of critical minerals that are present in most of the developing countries. So I think that even though costs are decreasing, we have to be more and more creative when it comes to recycling the batteries. We also have to think of how they ’ re **impacting** local communities. KB : So to give you the numbers, we extract, at the moment, 15, 000 million tons of fossil fuels every year. The International Energy Agency says that we will need about 43 million tons of critical materials in order to build out the clean energy economy. So that ’ s 300 times less **stuff** that is required. It ’ s worth saying that the **impact** of extracting 300 times less **stuff** upon nature is dramatically lower. And I know that it ’ s often argued that the renewable economy may be damaging for nature. Again, this is a completely false statement. It ’ s in fact, our only chance of reducing the pressure upon our natural resources, because there ’ s far less of it. MZ : Of course, transitioning to using clean energy, upending industries, it has to happen across the world, not just here in the US. And it can be hard to even imagine what life will be like. But carbon scientist and futurist Julio Friedmann can help us. Julio Friedmann : I think we need a compelling vision of the future that gets people excited about the energy **transition** and solving **climate** change. Right now, it ’ s like we ’ re at a restaurant and there ’ s two options : burnt toast, unflavored oatmeal. And people are not excited about either, right? The burnt toast option for me is what I call the big switch. Everything ’ s the same, it ’ s just clean, but it costs a lot of money and it ’ s super hard to do. The other vision is kind of like,”Thou shalt not.”That vision is a narrower sort of,”eat your peas,”hair shirt kind of vision of the future. And a lot of people don ’ t want that either. And there ’ s nothing in the physics, chemistry or biology that says we can ’ t have a super **interesting**, exciting, vibrant world. And that is compelling in a way that we have not heard yet. MZ : Help me out here, what does that even look like? JF : So, for lack of a better word, like, I like the Marvel cinematic universe. MZ : OK, I ’ ll go with you. JF : So a place like Wakanda, right? It ’ s got flying cars, it ’ s got maglev trains, and it ’ s a totally fabulous place to live. Like, they ’ ve got lots of food. That is built on abundant, sustainable, cheap energy. MZ : OK, so like, we ’ re not Wakanda, we ’ re not in the Marvel universe. This is real life, sort of. How do we make this possible? JF : So what can you actually do here on Earth, constrained by economics and engineering and all the rest of it, right? And the answer is we can do an incredible amount. Abundant, available where you want it, when you want it. **Everybody** should have energy, including the three billion people who use less electricity than my refrigerator uses. Well, the good news is, every day, the Earth receives 163, 000 terawatts of energy from the sun. About half of that bounces back to space, but about 80, 000 terawatts arrive at the Earth in a form we can use. For reference, today, the world uses about 26 terawatts of energy, and we got more than solar and wind. We have geothermal, we have hydro, we have nuclear. There ’ s other kinds of clean energies. And some of the best resources are, in fact, in the Global South. These places are not simply future **climate** victims. These places are latent energy superpowers. My favorite example of this is Chile. Almost eight years ago now their government said, wait a second, we can make the cheapest green electricity on Earth. We have hydropower, we have solar power, we have wind power and the best resources anywhere in the world. So let ’ s start by making a lot of green electricity. And then they said, because of that, we can also make green **hydrogen** cheaper than anywhere else. And then Japan was like, hello, we want your green **hydrogen**. Can you ship it to us? MZ : Is that possible? JF : Totally. And Japan is like, we will also give you money to pay for these projects. We ’ ll loan you money to pay for these projects. Could you please use Japanese technology? So they ’ re using Japanese turbines, electrolyzers, they ’ re using Japanese ships to haul the ammonia around. So Japan gets rich on this, Chile gets rich on this. MZ : Win-win. JF : Totally win-win. MZ : Win-win-win, planet, too. JF : And that model, other countries are looking at that, going, I want some of that. And that ’ s why I like this vision of abundance. 15 years ago, solar was the most expensive power. Now it is the cheapest form of electricity, right? Which is astonishing, it ’ s totally amazing. And a whole bunch of people are like, now we ’ re done. And I ’ m like, no, we can do that with everything. We can do that with wind. We can do that with nuclear, we can do that with geothermal. But these projects take time, money and people. We need to develop the human capital as part of these investment projects for decades. And innovation, we need much more energy in many places, much cheaper. That ’ s an innovation agenda. For solar, that was the United States, Germany, China and others acting together. Well, if that ’ s the recipe, we can do that again. We ’ re already doing it with electric vehicles and clean **hydrogen** production. We can certainly do it by turning electricity into fuels. MZ : It sounds like the **climate** discussion needs to be kind of like improv, that it ’ s a”yes and”sort of thing. JF : I ’ m so glad you said that. So many people don ’ t understand my love of improv comedy. Yes, it is a”yes and”circumstance. Because mostly it is about room for agreement. Where do you find the point where **everybody** is like, I can agree on that? One of the things I love about the infrastructure bill, one of the things I love about the **Inflation** Reduction Act, neither of them had the word **climate** in it. They were both massive **climate** bills, right? That was a way to get **everybody** on the same page. Nationwide, they basically tripled the clean energy research budget. That ’ s a good start. But that made it all of one percent of GDP. And for comparison, we spend twice that on pharmaceuticals. So I think it ’ s fair to say we should get going. We should spend a lot more money. We ’ ve done **stuff** like this before. We ’ ve done World War II, we ’ ve done the Marshall Plan, like, we ’ ve done the space shot, like we actually know how to move really quickly as a nation. We did this for COVID. **Collective** action, building together is what makes the difficult possible and nourishes the soul through mission and purpose. We know what to do and we can act, not out of anger or fear, but out of generosity and common purpose, bringing aspiration and humility together. We ’ re going to build a thriving, vibrant, exciting world full of potential that ’ s going to be built on the back of infrastructure, innovation and investment that will harness the abundant, sustainable, cheap energy that is our planet ’ s endowment. Thank you. [Chapter 4 : A Just **Transition**] FD : There is a high risk that when those exponential kick in, we see businesses running after those opportunities. But then the distribution of the benefits being very unequal. EZ : Many people like discussing the importance of the just energy **transition**. And a lot of people don ’ t understand what justice really means. If we are to reach a revolution, a renewable energy future has to be inclusive. AC : We have this push for what we ’ re calling a just and equitable energy **transition**. We need to make sure that we are not exacerbating inequality, but rather we are **ensuring** that that gap is closed and that the groups that have been historically marginalized are not once again being marginalized in a new way. NT : Most of emissions come from wealthy, energy-guzzling citizens, but a lot of the economic benefit and human development benefit will come not from decarbonizing high-carbon lives and lifestyles, but from providing clean, distributed energy and tools and solutions to those who don ’ t have access to any of them. LN : When we speak of countries in need of economic development, we mostly speak of countries that are hit by the **climate crisis** the hardest right now. It should be the biggest wish and job and task for all those countries who have burned so much fossil fuel already to **support** any other country in the so-called Global South, to get an economy running without fossil fuels, to not repeat all the mistakes that we have made. HD : To **ensure** that the solar revolution that has been seen around the world comes to places like Africa, we need to be really conscious that the history and the lived experience when it comes to energy is very different for poor countries. Rebekah Shirley : We love to remind Global South communities, especially Africa, about the vast renewable energy potential that they have, which is true. But the finance flows to deliver on that potential remain very scarce. Africa, despite being 90 percent of those across the entire globe that still do not have access to electricity and being 17 percent of the global population, we ' re actually two percent of **international** finance. Money doesn ' t actually flow to these spaces naturally, and we need to do a lot of work to get global finance to these spaces. MZ : A world of abundant clean energy for everyone. That is the goal. But what about in the short term? Take Detroit. This once-wealthy city has been through some seriously hard times. How is it rebuilding itself so that everyone here can thrive while dealing with extreme heat, flooding, and other **climate** problems? Anika Goss is looking to Detroit ' s past to plan its future. AG : I am a third-generation Detroiter. My grandmother moved to Detroit in 1936 during the Great Migration and brought all of her southern ways with her. She owned a home and knew that home ownership would create wealth and opportunity for her growing family. Up until the late 1950s, Detroit was a haven for middle-class families living in neighborhoods where there was green space and community connectivity and opportunity. My grandmother, she had this amazing garden, and it was just abundant. There were all kinds of flowers and food. You know, she grew vegetables back there, and it was just a magical place for a little girl. MZ : I mean, what you ' re describing is a Detroit that, yes, has huge problems, segregation. But at the same time, a woman, she can have a job, she can own a house, she can raise her family, she can have fresh fruits and vegetables. AG : It was certainly a place where you could gain economic opportunity faster than you can now. My grandmother ' s Detroit is not the Detroit that I live in today. All of those sites where there was manufacturing and industrial sites that led to our economic boom, many stand vacant and abandoned. These industrial sites have led to dangerous contamination to our land and our water and our **air**. MZ : So, just lay it out for us. For people who maybe don ' t see, like, how is there a link between economic inequality and **climate** change? AG : You know, I ' m so surprised that it feels like separate problems. But I ' m also really glad that people are asking. Because if you ' re living around and working in a community that is low-income, where there are fewer jobs, the housing is deteriorated, this is a neighborhood that ' s also more likely to be closer to a factory that ' s emitting pollutants, likely to have water that ' s contaminated, probably has fewer parks and fewer intentional green spaces. The people who are living in those communities are at risk for **climate impacts** first. So we have to do something in those neighborhoods right away. MZ : So what can this healthier future look like? Anika showed me some of the work she ' s done in partnership with different neighborhoods in Detroit, to make them both more enjoyable and more resilient. First, she took me to the Circle Forest. It ' s an oasis right in the middle of the city. AG : This is six lots that were vacant, and we all worked together with this community to build a forest. MZ : Beautiful and a great example of how nonprofits can work with communities to transform them into sustainable, welcoming places. AG : We might have a good idea about gardens and tree planting, but this is where people live. And so you ask first. This is a new effort, of this idea of a resilient neighborhood or calling it that. So these are large tracts of vacancy that can be used to provide environmental sustainability, community resilience and beauty. Watch for the poison ivy. It ' s not pretend. MZ : Next we met up with Katrina Watkins, the founder and CEO of the Bailey Park Neighborhood Development Corporation, a group revitalizing this historic area by turning vacant lots into parks that can stand up to extreme weather. Katrina Watkins :

My family has been in this neighborhood since 1946. I grew up hearing the stories from my dad, him telling me about how people thought it was a ghetto, but it wasn ’ t. Those were families, and people worked and had businesses, and people looked out for each other. And it was those type of stories that made me say, wow, we just really need to bring that back. MZ : So then briefly fill us in about how things changed. KW : Things deteriorated in this neighborhood. The homes got torn down, left just a lot of vacant lots, a lot of overgrowth. It wasn ’ t **safe**. I would see my little nieces and nephews. They would just be on their little skateboards, you know, not paying attention, the stop sign is covered. And me and my dad was like, we just have to do something. So we started out as Bailey Park Project, focusing on the park and green infrastructure and trees and flowers. And then over the course of the years, we started doing more community development. MZ : I wonder, like, how do you describe it to people when you explain what ’ s happening, what you ’ re trying to do for your neighborhood, the changes that you ’ re making and how it relates to **climate** change? KW : One really great example, when people come to the park, like right now, this is like, literally a heat island because our trees haven ’ t grown. So it is very hot. So sometimes it ’ s so hot that the kids don ’ t come out and play until the evening. That education of knowing the benefit of trees and cooling down an environment, how it makes the **air** cleaner. There ’ s a lot of people that suffer from asthma. So trees can make a difference. And if you go to places like Grosse Pointe, you will see this beautiful tree canopy down. Like their homes are probably 20 degrees cooler you know, than our homes, AG : Having a lot of green space actually contributes to the **climate** resiliency for this neighborhood. So all of this grass actually can manage stormwater better. The idea of leading new development in this neighborhood with green, what it says to me is that the residents are really envisioning the neighborhood that they want to live in. And that I just – I ’ m really excited about that. MZ : Turning great loss and upheaval into even greater opportunities. These Detroiters are creating a blueprint for an urban future that every city can learn from. [Chapter 5 : Agriculture] TED-Ed : About 10, 000 years ago, humans began to farm. This agricultural revolution was a turning point in our history that enabled people to settle, build and create. In short, agriculture enabled the existence of civilization. Today, approximately 40 percent of our planet is farmland. Spread all over the world, these agricultural lands are the pieces to a global puzzle we are all facing. In the future, how can we feed every member of a growing population a healthy diet? Meeting this goal will require nothing short of a second agricultural revolution. The first agricultural revolution was characterized by expansion and exploitation, feeding people at the expense of forests, wildlife and water, and destabilizing the **climate** in the process. That ’ s not an option the next time around. Agriculture depends on a stable **climate** with predictable seasons and weather patterns. This means we can ’ t keep expanding our agricultural lands, because doing so will undermine the environmental conditions that make agriculture possible in the first place. Instead, the next agricultural revolution will have to increase the output of our existing farmland for the long term while protecting biodiversity, conserving water, and reducing pollution and greenhouse gas emissions. So what will the future farms look like? In Bangladesh, Cambodia and Nepal, new approaches to rice production may dramatically decrease greenhouse gas emissions in the future. Rice is a staple food for three billion people and the main source of livelihood for millions of households. More than 90 percent of rice is grown in flooded paddies, which use a lot of water and release 11 percent of annual methane emissions, which accounts for one to two percent of total annual greenhouse gas emissions globally. By experimenting with new strains of rice, irrigating less and adopting less labor-intensive ways of planting seeds, farmers in these countries have already increased their incomes and crop yields while cutting down on greenhouse gas emissions. MZ : Related technologies are also taking root in the US. Father and daughter farmers, Jim and Jessica Whitaker, have come up with their own strategies to reduce methane that are working on their farm in Arkansas. But convincing everyone else to change is their next big challenge. So let ’ s go back to how you got to be such a big supplier of rice in this country. You tell a story about how you were 22 and you were like, I ’ m going to start my own farm, but – not that easy. Jim Whitaker : No, it ’ s not. Let me tell you something about renting a farm when you ’ re 22 years old. No one rents you a farm unless no one else wants to rent it. It was a big piece of land, it was cash rent, I mean, the stakes were set. We were doomed to fail. And we weren ’ t focused on environmental sustainability back then. We were focused on economic sustainability. How do we make higher yield? How do we use less fertilizer? How do we get to the next year and feed our family? It happens that economic and environmental sustainability go hand in hand along with social sustainability. As we used less fertilizer, used less water, our yield started going up. MZ : Can you spell out for me what the problem is with rice when it comes to the environment? JW : Soil is full of microbes. And what they ’ re doing in the soil on a basic level is they ’ re down there chewing away, eating up the biomass that ’ s left over from the crop before. And out of that comes methane, the same way it happens in cattle operations or whatever. MZ : But you decided to do it a different way. JW : We do it a little bit differently. Arkansas is a very water-rich state, but we have a dry season, right in the middle of summer is a dry season. So when we dry the soil, that anaerobic microbial that ’ s living in that soup, that mushy soil, either dies or goes dormant. So we break his life cycle. MZ : Oh! JW : And then it stops emitting methane. But it doesn ’ t hurt the rice. That actually helps rice, when you use less water. We have less runoff, less erosion, less nutrients leaving our field. Nothing leaves our field unless we want it to. Jessica Whitaker Allen : We started tracking this. We saw that our water use was down and we thought, OK, well, we need to add to that, let ’ s add more data points to that. Let ’ s add our fertilizer, let ’ s add our fuel usage. And so we really started recording all this **stuff** by hand in just an Excel spreadsheet. We didn ’ t know what we were doing. MZ : But you knew you were on to something. JWA : Yes, so we started recording everything. MZ : Do you remember if there was a moment when the two of you thought, like, what we ’ re doing, it shouldn ’ t be special to us. Why aren ’ t more people – JW : To collect the data, have the equipment and the cloud-based technology and the **stuff** in the field to record it, it ’ s just massively time-consuming, and consumers don ’ t pay for it. JWA : Well and people like us, why would you want to do it? I mean, take a walk through your supermarket. Everything ’ s sustainable, everything ’ s greenwashed. You can put whatever you want to on a package, there ’ s no rules, there ’ s no standards. So we ’ re trying to make that standard. We live and work in southeast Arkansas. Our town has 4, 000 residents and one stoplight. The nearest airport, Starbucks shopping mall, Whole Foods, is two hours away in any direction. Without places like this and farmers like us, you ’ d be hungry, naked and sober. I ’ m a farmer, but not in the way you might think. I don ’ t drive the tractors, and I don ’ t plant the rice. But I know how to take what my dad has learned and what he is doing and share that with others. We are going to work with farmers in southeast Arkansas to educate them about the benefits of growing sustainable rice. JW : Let me tell you why that ’ s important. There are 400 million acres of rice grown globally. And our methods, if used, can reduce greenhouse gas by 50 percent, reduce water use by 50 percent, increase yields to feed a hungry world. MZ : We hear a lot about how weather has gotten more extreme, flooding, heat. What ' s it been like in Arkansas? JW : So in 2016, we had a 500-year flood. We were... We get a picture of our place flooded. 14 inches of rain. That was a 500-year flood. Hurt a lot of people. And then we had a 1, 000-year flood. And we lost so many acres. And you know, when houses get flooded in rural communities, they don ' t come back. I mean, when you have whole neighborhoods get flooded, they don ' t rebuild them, they don ' t come back. JWA : I feel like almost every day I ' m getting a notification from the weather Channel of an excessive heat warning or thunderstorm warning or hail threat. You know, you don ' t really know what you ' re going to expect. MZ : When you think about what it will be like, I don ' t know, five years, 10 years, for someone to go into the supermarket and think, oh, right, I need to get rice off my grocery list. What ideally would you like to see on the shelf there? JW : So I think it ' s going to change, because I like to study what companies want. I mean, if you figure out what they want, then you know what to give them. And when you read their sustainability goals, they ' re all saying they ' re going to be net zero. The only way they can be net zero is they bring the farmer into the loop. So if we ' re able to do this, in five years, **everybody** ' s doing these practices, the US rice industry will be the most sustainable industry in the world. MZ : Can you do it? JW : Oh, yeah. It ' s about to happen. MZ : You ' ve just heard a lot of ideas, a lot of stories and a lot of big plans. So we want to wrap things up by asking our experts : What if we actually achieved our **climate** goals? What would that future even look like? They, of course, had a lot of thoughts. We ' ll let you pick your favorites. Thanks for watching. NT : If we are successful in addressing the **climate crisis**, we massively reduce the vulnerability and improve the economic and the creative prospects of billions of people around the world who currently are basically excluded from all of the opportunities which we have, which come from having abundant material and energy all around us. That ' s the real prize. FD : A world where we have had that exponential would be a world in which we have clean, cheap energy for all. KB : At the end of this decade, the International Energy Agency believes that we will have enough capacity to be producing 10 billion, 10 billion batteries a year. It ' s enough for every person on the planet to have their own battery. AC : I ' m constantly surrounded by young people just like me who are really, really, really passionate about the world that we live in. For obvious reasons. It ' s the world that we will inherit. EZ : Women are very important for the energy **transition**. So if we ' re talking about a world where we have, you know, where we tackle **climate** change, all I want to see is, you know, women in power. I want to see more solutions being driven by young people. And I want our politicians to be truly and meaningfully engaged in making change. NG : We always tend to underestimate the pace of ultimate adoption and change. Today, there are more than 150 pathways, scientific pathways that we could take to limit global warming. Al Gore : If we get to true net zero, astonishingly, global temperatures will stop going up with a lag time of as little as three to five years. Colin Averill : We ' ve been able to accelerate tree growth and carbon capture and tree stems by 30 to 70 percent. Dan Jørgensen : Offshore wind has the potential to cover the current electricity **demand** of the entire world not once, not twice, 18 times. NG : We need to decarbonize the old economy that we have and **invest** to create the new economy that we need Paul Hawken : Regenerative farming is fairly simple in concept. It means creating the conditions for more life. Josephine Phillips : We need to buy less **stuff**, and we need to look after what we buy. Valuing the things that we own is a **climate** solution. Vishaan Chakrabarti : We can all live in this transit-rich, carbon-negative, affordable way and leave the vast majority of the planet for nature, for agriculture, for clean oceans. We can do this. Susan Ruffo : The ocean is a powerful source of solutions that we ’ ve overlooked for far too long. Al Roker : You have to be engaged. Let your elected officials know that this is important to you. You have to vote, you have to go out there and **support** politicians who are going to **support** our planet. Peggy Shepard : We can create a legacy of environmental quality and **climate** resilience for all. Fehinti Balogun : You are not powerless that the”oh, there ’ s nothing we can do, just keep going”is symptom, not cure. But you are needed. Avinash Persaud : The power of what we can do when it matters to us is unlimited. CF : It ’ s not that we ’ re simply going to avoid the worst **impacts** of **climate** change. It ’ s that we ’ re going to improve the quality of life for humans, both in the urban and in the rural **sector**, and for non-humans.

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Christiana Figueres : Today, February 19, 2021, at the beginning of a crucial year and a crucial decade for confronting the **climate crisis**, the United States rejoins the Paris **Climate** Agreement after four years of absence. Unanimously adopted by 195 nations, the Paris Agreement came into force in 2016, establishing targets and mechanisms to lead the global economy to a zero-emissions future. It was one of the most extraordinary examples of multilateralism ever, and one which I had the privilege to coordinate. One year later, the United States withdrew. The Biden-Harris administration is now bringing the United States back and has expressed strong commitment to responsible **climate** action. The two men you are about to see both played essential roles in birthing the Paris Agreement in 2015. Former Vice President Al Gore, a lifelong **climate** expert, made key contributions to the diplomatic process. John Kerry was the US Secretary of State and head of the US delegation. With his granddaughter sitting on his lap, he signed the Paris Agreement on behalf of the United States. He is now the US Special Envoy for **Climate**. **TED** Countdown has invited Al Gore to interview John Kerry as he begins his new role. Over to both of them. Al Gore : Well, thank you, Christiana, and John Kerry, thank you so much for doing this interview. I have to say on a personal basis, I was just absolutely thrilled when President Biden, then president-elect, announced you were going to be taking on this incredibly important role. And thank you for doing it. Let me just start by welcoming you to **TED** Countdown and asking you, how are you feeling as you step back into the middle of this issue that has been close to your heart for so long? John Kerry : Well, I feel **safer** being here with you. I honestly, I feel very energized, very focused. I think it ’ s a privilege to be able to take on this task. And as you know better than anybody, it ’ s going to take **everybody** coming together. There ’ s going to have to be a massive movement of people to do what we have to do. So I feel privileged to be part of it, and I ’ m honored to be here with you on this important day. AG : Well, it ’ s been a privilege to be able to work with a dear friend for so long on this **crisis**. And, of course, on this historic day, when the United States now formally and legally rejoins the Paris Agreement, we have to acknowledge that the world is lagging behind the pace of change needed to successfully confront the **climate crisis**, because even if all countries kept the commitments made under the Paris Agreement – and I watched you sign it, you had your grandchild with you – I was there at the U. N, that was an inspiring moment, you signed on behalf of the United States, but even if all of those pledges were kept they ’ re not strong enough to keep the global temperature increase well below two degrees or below 1. 5 degrees, and emissions are still rising. So what needs to happen here in the US and globally in order to accelerate the pace of change? JK : Well, Al, you ’ re absolutely correct. It ’ s a very significant day, a day that never had to happen, America returning to this agreement. It is so sad that our previous president, without any scientific basis, without any legitimate economic rationale, decided to pull America out. And it hurt us and it hurt the world. Now we have an opportunity to try to make that up. And I approach that job with a lot of humility for the agony of the last four years of not moving faster. But we have to simply up our ambition on a global basis. United States is 15 percent of all the emissions. China is 30 percent. EU is somewhere around 14, 11, depends who you talk to. And India is about seven. So you add all those together, just four entities, and you ’ ve got well over 60 percent of all the emissions in the world. And yet none of those nations are at this moment doing enough to be able to get done what has to be done, let alone many others, at lower levels of emission. It ’ s going to take all of us. Even if tomorrow China went to zero, or the United States went to zero, you know full well, Al, we ’ re still not going to get there. We all have to be reducing greenhouse gas emissions. We have to do it much more rapidly. So the meeting in Glasgow rises in its importance. You and I, we ’ ve been to these meetings since way back in the beginning of the ’ 90s with Rio and even before, some of them parliamentary meetings. And we ’ re at this most critical moment where we have a capacity to define the decade of the ’ 20s, which will really make or break us in our **ability** to get to a 2050 net zero carbon economy. And so we all have to raise our ambition. That means coal has got to phase down faster. It means we ’ ve got to deploy renewables, all forms of alternative, renewable, sustainable energy. We ’ ve got to push the curve of discovery intensely. Whether we get to **hydrogen** economy or battery storage or any number of technologies, we are going to have to have an all-of-the-above approach to getting where we need to go to meet the target in this next 10 years. And I think Glasgow has to not only have countries come and raise ambition, but those countries are going to have to define in real terms, what their road map is for the next 10 years, then the next 30 years, so that we ’ re really talking a reality that we ’ ve never been able to completely assemble at any of these meetings thus far. AG : Well, hearing you talk, John, just highlights how painful it ’ s been for the US to be absent from the **international** effort for the last four years, and again, it makes me so happy President Biden has brought us back into the Paris Agreement. After this four year hiatus, how are you personally, as our **Climate** Envoy, planning to approach re-entry into the conversation? I know you ’ ve already started it, but is there anything tricky about that? Or I guess everything is tricky about it, but how are you planning to do it? JK : Well, I ’ m planning, first of all, to do it with humility, because I think it ’ s not appropriate for the United States to leap back in and start telling **everybody** what has to happen. We have to listen. We have to work very, very closely with other countries, many of whom have been carrying the load for the last four years in the absence of the United States. I don ’ t think we come in, Al, I want to emphasize this – I don ’ t believe we come to the table with our heads hanging down on behalf of many of our own efforts, because, as you know, President Obama worked very hard and we all did, together with you and others, to get the Paris Agreement. And we also have 38 states in America that have passed renewable portfolio laws. And during the four years of Trump being out, the governors of those 38 states, Republican and Democrat alike, continued to push forward and we ’ re still in movement. And more than a thousand mayors, the mayors of our biggest cities in America, all have forged ahead. So it ’ s not a totally, abjectly miserable story by the United States. I think we can come back and earn our credibility by stepping up in the next month or two with a strong national determined contribution. We ’ re going to have a summit on April, 22. That summit will bring together the major emitting nations of the world again. And because, as you recall in Paris, a number of nations felt left out of the conversation. The island states, some of the poorer nations, Bangladesh, others. And so we ' re going to bring those stakeholders to the table, as well as the big emitters and developed countries, so that they can be heard from the get-go. And as we head on into Glasgow, hopefully we ' ll be building a bigger momentum and we ' ll have a larger consensus. And that ' s our goal – have the summit, raise ambition, announce our national determined contribution, begin to break ground on entirely new initiatives, build towards the biodiversity convention in China, even though we ' re not a party, we want to be helpful, and then go into the G7, the G20, the UNGA, the meeting of the United Nations in the fall, reconvene and reenergize, going for the last six weeks into Glasgow. In my judgment, Glasgow, and you ' d know this full well, I think Glasgow is the last, best hope we have for our nations to really set us on that path. And so, you know, one key is, as I said, raising ambition. The other is defining how you ' re going to get there, and then the third is finance. We ' ve got to bring an unprecedented global finance plan to the table. And I think we ' re already working with private **sector** entities. I believe there ' s a way to do that in a very exciting way. AG : Well, that ' s encouraging, and I ' m going to come back to that in just a moment. But I ' m glad you made those points about state and local governments actually moving forward during the last four years. A lot of US private companies have as well. And already I ' m extremely encouraged by the suite of executive actions that President Biden has already taken during his first weeks in office. And there ' s more to come. There ' s also a push for legislative action to **invest** in the fantastic new opportunities in clean energy, electric vehicles and more. Yet you and I have both seen the difficulties of this approach in the past. How can we use all of this activity to well and truly convince the world that America is genuinely back to being part of the solution? I know we are. You know we are, but we ' ve got to really restore that **confidence**. I think your appointment went a long way to doing that. But what else can we do to gain back the world ' s **confidence**? JK : Well, we have to be honest and forthright and direct about the things that we ' re prepared to do. And they have to be things we ' re really going to do. We just held a meeting a few days ago with all of the domestic entities that President Biden has ordered to come to the table and be part of this effort. This is an all-of-government effort now. So we will have the Energy Department, the Homeland Security Department, the Defense Department, the Treasury. I mean, Janet Yellen was there talking about how she ' s going to work and we ' re going to work together to try to mobilize some of the finance. So I think, you know, we ' re not going to convince anybody by just saying it. Nor should we. We have to do it. And I think the actions that we put together shortly after President Biden achieves the COVID legislation here, he will almost immediately introduce the rebuild effort, the infrastructure components, and those will be very much engaged in building out America ' s grid capacity, doing things that we should have done years ago to facilitate the transmission of electricity from one part of the country to another, whether it ' s renewable or otherwise. We just don ' t have that **ability** now. We have a queue of backed up projects sitting in one of our regulatory agencies which have got to be broken free. And by creating this all-of-government effort, Al, our hope is we ' re really going to be able to do that. The other thing that we ' re doing is I ' m reaching out, very rapidly, to colleagues all around the world. We ' ve had meetings already, discussions with India, with Latin American countries, with European countries, with the European Commission and others. And we ' re going to try to build as much energy and momentum as possible towards these various benchmarks that I ' ve talked about. And I mean, the proof will be in the pudding. We ' re going to have to show people that we ' ve got a strong NDC, we ’ re actually implementing, we ’ re passing legislation, and we ’ re moving forward in a collegiate manner with other countries around the world. For instance, I ’ ve talked to Australia, we had a very good conversation. Australia has had some differences with us. We ’ ve not been able to get on the same page completely. That was one of the problems in Madrid, as you recall, together with Brazil. Well, I ’ ve reached out to Brazil already, we ’ re starting to work on that. My hope is that we can build some new coalitions and approach this, hopefully in a new way. AG : Well, that ’ s exciting, and I do agree with your statement earlier that the COP26 **conference** in Glasgow this fall may be the world ’ s last, best chance, I like your phrase there. From your perspective, what would you list as the priorities for **ensuring** that this Glasgow **conference** is a success? JK : I think that perhaps one of the single most important things, which is why we ’ re focused on this summit of ours, is to get the 17 nations, that produced the vast majority of emissions, on the same page of committing to 2050 net zero, committing to this decade, having a road map that is going to lay down how they are going to accelerate the reduction of emissions in a way that keeps 1. 5 degrees as a floor alive and also in a way that guarantees that we are seeing the road map to get to net zero. I will personally be dissatisfied, disappointed if for our children ’ s sake and our grandkids sake we can ’ t say that when these adults came together to make this kind of a decision, we didn ’ t actually make it. We ’ ve got to make it. And I think if we can show people we ’ re actually on the road, I think you believe this as much as I do, that – I mean, you ’ re far more knowledgeable than I am about some of the technologies and you ’ ve helped break ground on some of them. The pace at which we are now beginning to accelerate, I mean, the reduction in cost of solar, the movement in storage and other kinds of things, I ’ m convinced we ’ re going to find one breakthrough or another. I don ’ t know what it ’ s going to be, but I do know that when we push the curve and we put the resources to work, the innovative creative capacity of humankind is such that we have an **ability** to surprise ourselves. We ’ ve always done it. When we went to the Moon in this incredible backdrop behind you. And that ’ s exactly what we did. And people today use products in everyday household use that came out of that quest that you never would have anticipated. That ’ s what ’ s going to happen now. We can move faster to electric vehicles. No question in my mind, we could absolutely phase down coal-fired emissions faster than we are in a plan to do it. So the available choices are there. The test is going to be whether we create the energy and momentum necessary to actually get those choices made. AG : One of the big challenges is one you referred to earlier on finance. Wealthy countries have promised financial assistance to the less wealthy countries to help them out with cutting emissions and to help them cope with the **impacts** of the **climate crisis**. But of course, we need to continue to work to meet this commitment, especially as countries around the world rebuild their economies in the wake of this **pandemic**. What are some of the most effective ways in which the wealthier countries can help those that don ’ t have as many resources, and why is this so important for the world to move forward? JK : Let me answer the last part first. It ’ s so important because it ’ s the only way we ’ re going to get there. I don ’ t believe that any government has either the money or the inclination to be able to do what ’ s necessary here. I believe the private **sector**, particularly driven by venture capital investment, by the quest to be able to create a product that then can help create wealth and actually provide a benefit to humankind drives a lot of things that we ’ ve done all through history. And I don ’ t think it ’ ll be any different now. I think the question is, can we pull together enough nations to leverage a uniform approach to the judgment about the kinds of investments that are being made. And I believe that if we can standardize to some degree, with disclosure requirements, which Janet Yellen is now seized of that issue, and Europe, there are folks working on that and European Commission elsewhere, if we could actually find a way to come together and harmonize some of those definitions and the marketplace begins to make those judgments as they qualify risk, looking way out, risk, because of **climate crisis** for investing is very, very real. And we all understand that. We spent 265 billion dollars in America two years ago just cleaning up after three storms, Maria, Harvey and Irma. And it ' s crazy. You spend 265 billion to clean up after the storms, but we can ' t put 100 billion together for the Green **Climate** Fund. That ' s what this year has to be about. We ' ve got to break that cycle. And I think business, I ' m convinced of this, a lot of people will doubt me and say, have I lost my mind, but I ' m convinced the private **sector** is going to be critical, if not the key to helping to make this happen. And that will leverage other money. I ' ve talked to the IMF, we ' ll be talking with the World Bank, we ' re going to try to bring our own Finance Development Corporation in America. All of these things can help leverage investment into the **sectors** that can make the greatest difference to the rapid reduction of greenhouse gases. And I think people are going to get very excited about where this money is going to go and how much it is going to be. And my hope is in a matter of weeks to be in a position to make a couple of announcements with respect to that that could be helpful in building some of this momentum. AG : Well, that ' s great. It sounds like some major news coming in a couple of weeks and just one example you used, the point about businessmen. I have a friend in Australia, Mike Cannon-Brookes, building a long undersea cable from the northern territories of Australia to take renewable electricity to Singapore. You have made the point about the need for the US to approach this with humility a number of times. In that spirit, what lessons can a country like ours learn from some of the lower income nations that are already beginning to tackle **climate** change? JK : Well, I think one of the most important things, Al, is to make sure that central to this transformation, to this **transition** to the new energy economy, central to it is environmental justice, is that we don ' t leave people behind, that we ' re not making whole communities the recipients of the downside of some particular choice, that the diesel trucks, for instance, aren ' t all being routed through a particular low income community that doesn ' t have the **ability** to make a different political decision. I think it is vital for the developed world to recognize that there are nations, 138 nations or more, way below one percent in terms of emissions. And they ' re looking around some of them, like Tommy Remengesau, the president of Palau, who no longer can consider adaptation, he ' s got to figure out where his people are going to go live, as do other very low-lying areas in the ocean. So that **impact** on people is really not known by the vast majority of people who live pretty good lives in a lot of countries in the world. And we have a responsibility to make sure that we ' re learning the lesson of their lives and of their hopes and aspirations here. AG : Couldn ' t agree more. And here in the US, if we had paid more attention to the differential **impact** on Black, Brown and Indigenous communities, we would have had a better early warning of what the whole country was facing. But let me shift subjects and ask you about China. I know that you, as you are close friends with Xie Zhenhua, as I have been over the years, and I was very happy when he was brought out of retirement to play the lead role for them. But the US is now in the middle of a somewhat contentious relationship with China. But successfully solving the **climate crisis** is going to require collaboration between the US and China, we ' re the two biggest emitters and the two biggest economies. How can this collaboration be shaped, in your view? I know you played a role, as Joe Biden did before the Paris Agreement, in getting our two countries together. Can we do that again? JK : I hope so. I really do hope so, Al. As you just said, if we can – if we don ’ t get China to be cooperating and partnering with the rest of the world on this, we don ’ t solve the problem. And we unfortunately, we see too much investment in China right now in coal still. We ’ ve had some conversations about it. I was on a panel with Xie Zhenhua several months before the election by the University of California, and we had a very constructive conversation. My hope is that that will continue and can continue and that China will be just as constructive, if not more so, in this endeavor than they were in 2013 as we began the process to build up to Paris. AG : Well, that relationship is absolutely crucial. But in order to cover all the ground I want to cover here, let me shift again and ask you, what role do you expect that big corporations and also smaller businesses will play in moving this green **transition** forward? JK : I think they ’ re the biggest single players in it. I mean, governments are important and governments can and have made a difference with tax credits. For instance, our solar tax credit made an enormous difference and it will make one going forward. And even in the middle of COVID, we ’ ve been able to hold on to that. But we need to grow those kinds of efforts. But in the end, it ’ s not going to be government cash that makes this happen. It ’ s going to be the private **sector** investment that is coming in because it ’ s the right thing to do, because it ’ s also smart investing. And the truth is, you can talk to many – and you have, you ’ re one of the investors actually, Al – you and others have proven that you can **invest** in this **sector** of dealing with **climate** or environment or sustainability, whether it ’ s ESG or it ’ s pure **climate**. There are ways to have a good return on money. And during the last couple of years, we had something like, 13 to 17 trillion dollars sitting in parked banking situations around the world in net negative interest. In other words, they were paying for the privilege of sitting there, not **invested** in something. And so I think there ’ s just a massive opportunity here. And most of the CEOs I am talking to, at least now, are increasingly aware of the potential of these alternatives. And you were in early, I don ’ t know if you **invested** in it or not, but I know you ’ re **involved** with Tesla or have been. Tesla is the most highly valued automobile company in the world. And it only makes one thing : electric car. If that isn ’ t a message to people, I don ’ t know what is. AG : I wish I had **invested** in Tesla, John, but I ’ m a huge fan of Elon **Musk** and what he ’ s doing. I ’ m also a huge fan of Greta Thunberg. And I ’ m just curious what you think in practical terms is the real **impact** for change coming from these youth movements like Fridays for the Future? JK : I think it ’ s been gigantic and spectacular and in the best traditions of what young people do and have done historically. I mean, as you recall, in America, at least in the 1960s, it was young people who drove the environment movement, the peace movement, the women ’ s movement, the civil rights movement, and they were willing to put their lives on the line. And Greta has been just unbelievable. And in the way in which she has held adults **accountable** and it has created this wonderful movement. I ’ ve met so many young people, many of whom have worked in one fashion or another with me in the last few years, who were brought to it from Fridays for the Future, from the Sunrise Movement, or, you know, it ’ s all that focused youthful idealism and energy and it **demands** to be heard. And I think all of us, I mean, we should be ashamed of ourselves that we have to have people who were then 16 or 15 not going to school to get our attention. I mean, what the hell is the matter with adult **leadership**? That ’ s not **leadership** at all. So I salute her and all the young people who put themselves on the line. But I invite them, you know, it ’ s not enough. You ’ ve got to then – and I said this during the course of the election where I hope we created a lot of new voters. And I think environment, specifically **climate crisis**, became a real voting issue this year, just as it was back in 1970 when we created the EPA and the Clean Air Act and a host of things. And it proves that that kind of activism is necessary. And I hope we ’ re going to keep young people at the table here and finish the job, that ’ s the key now. AG : Yeah, I couldn ’ t agree more. And another big movement that ’ s having an **impact** is the environmental justice movement. You referred to it earlier. And I ’ m so glad that President Biden is putting environmental justice at the heart of his **climate** agenda. It might be good if I could ask you to just take a moment and tell people why that is such an important part of this issue. JK : Well, I think it ’ s important part of this issue for many reasons, the most basic is just moral, you know, what is morally right. And how do you redress a wrong that has for too many years held people back, killed people by virtue of disease or other things, and resulted in a basic inequality and unfairness in society. And I think you share a feeling, as I do, Al, that the fabric of a nation is built around certain organizational principles. And if you ’ re holding yourselves out as a nation to be one thing, i. e. equal opportunity and fairness and all people created equal, and equal rights and so forth, if that ’ s what you hold out there and it isn ’ t there, eventually you get such a cynicism and such a backlash built up into your society that it doesn ’ t hold together. To some degree, that is what we ’ re seeing around the world today, is this nationalistic populism that is driven by this heightened inequality that has come through globalization that has mostly enriched already fairly well-off folks. And so if it ’ s the upper one percent that ’ s getting all the benefits and the rest of the world struggling to survive and they also have COVID, and then you tell them we ’ ve got to do this or that in terms of **climate**, you ’ re walking on very thin ice in terms of that sacred relationship between government and the people who are governed. It ’ s not just an American phenomenon. You see it in Europe. You see it in alternative movements in various countries. And I think it is the great task of our generation not only to deal with **climate**, but to restore a sense of fairness to our economies, to our societies, to our world. And that is part of this battle, I think. AG : Yeah, I agree. And another common source of opposition to what governments are doing now has to do with the fear, both in the US and elsewhere, on the part of some, that jobs might be lost in this **transition** toward a green economy. You and I both know that there are a lot of jobs that can be created. But let me put the question to you. How can we approach this green **transition** in a way that lifts everyone up? JK : That is one of the most important things that we need to do, Al. And we can ’ t lie to people. We can ’ t say that some of the dislocation doesn ’ t mean that a job that exists today might not be the same job in the future and that that person has to go through a process of getting there. And we need to make certain that nobody ’ s abandoned. We need to make certain that there are real mechanisms in place to help folks be able to **transition**. And I just spoke the other day with Richie Trumka, the head of the American Federation of Labor, and he ’ s been very focused on this. And we agreed to try to work through how do we integrate that into this transitional process so that we ’ re guaranteeing that you don ’ t abandon people. Now, one of the things we need to do is go to the places where there have been changes and there will be change. Southeastern Ohio, Kentucky, West Virginia. You know, if the marketplace is making the decision and it ’ s – by the way, it ’ s not government **policy**, it is the marketplace that has decided, in America at least, not to be building a new coal-fired plant. So where does that miner, or where does that person who worked in that supply chain go? We have to make sure that the new companies, that the new jobs are actually going into those communities that the coal community, the coal country, as we call it, in America, is actually being immediately and directly and realistically addressed in this to make sure that people are not abandoned and left behind. That is possible. That is doable. Historically, unfortunately, there have been too many words and not enough actual – not enough actually implementation and process. I think that can change. And I ’ m going to do everything possible in my **ability** to make sure that we do change it. AG : Well, that ’ s great. And another part of the context within which you are taking on this enormous challenge is the COVID **pandemic**, which has exposed the cost of ignoring pre-existing systemic risks, inequalities and sustainability. And now as we start to come out of this **pandemic**, how can we avoid sleepwalking back into old habits? JK : Boy, that ’ s probably the toughest of all. I mean, there ’ s a natural proclivity for people sometimes to just choose the easiest way. And clearly, some people already have and will resort to that. I think the key will be in President Biden ’ s proposal for the build back, which will actually fight hard to direct funds to the investments and to the **sectors** where we want to see a responsible build back. There ’ s another aspect, and I think that can be done, Al, I really feel that. For instance, someone who ’ s making a car today in South Carolina, where BMW has plants, and just to pick one place or Detroit, GM is obviously going to make this shift, they just announced it. The people building the car today are still going to have to put wheels on a car, build the car, put the seats in, do everything else. It ’ s just that instead of an internal combustion engine, they can be quickly trained to be able to put the platform in for the batteries and the engines themselves, etc, that will drive the car, the motors, that ’ s one way of dealing. Others are that there ’ s new work in some ways. We have to lay transmission lines in America. We do not have a grid in the United States, as you know. We have at east coast, **west** coast. Texas has its own grid, north part of America, but there ’ s a huge hole in the country. You can ’ t send energy efficiently from one place to another. We could lower prices for people and create more jobs in the build-out of all of that kind of new infrastructure. Not to mention the things that you and I, you know, there are going to be things that we can ’ t name today, some negative-emissions technology that ’ s going to grab CO2 out of the atmosphere and do something with it, like in Iceland, where they put it into the rock, mix it with liquid and it turns into stone. I mean, there are all kinds of different things people are exploring. Those are new jobs. AG : I just want to say, since we ’ ve come to the end of our time for this conversation, thank you again for taking on this crucial challenge on behalf of the United States of America and enabling the US to restore its traditional role in trying to bring the world together. And I know that **everybody** watching and listening to this conversation sends you their best wishes and hopes for all the success possible in this new work, John. Thank you so much for joining **TED** Countdown, and we wish you the very best. JK : Can I reciprocate for a minute? First of all, I want to thank you for your extraordinary **leadership** for years, I can remember when you were leading us in the Senate on this, and you ’ ve done so much since. And I am personally delighted to be working with you on this again and look forward to the next months and together with a lot of other folks. Let ’ s get this done.

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We ’ ve all dug ourselves into some holes that were hard to dig out of, especially at the urging of good friends. She knocked on my door to my room, which she never does. So I knew something was up and she was like, hey, wear a person down, everyone ’ s going to go. You ’ re going to be so upset that you missed this. That ’ s my student, Alexis. This was the second time her friend asked her to go to a trendy new festival so she couldn ’ t turn her down again. But

as soon as she said yes, things started to go downhill. Not only did my friends not have any details about the event, but even the contact that we made with the organizers of the event seemed pretty sparse on the details as well. So I was already feeling a bit uneasy about what was happening, but it felt like I ’ d already made that public commitment to our group to go. We ’ d already paid for our flights to Miami and those are non-refundable. Even as Alexis and her friends were boarding the plane, things kept getting worse. So we got on the plane, which strike three. And I was on my **phone** because I found a Twitter account saying that people had started arriving and didn ’ t have housing and didn ’ t have any food. Turns out that, first of all, her friends were obsessing about was one of the biggest frauds of twenty seventeen fire festival. All these models, like in the Bahamas, the most insane festival the world has ever seen, an island getaway turned disaster. It became very far barren when we got to the island. Of course, as most people know now, nothing was as promised. We were not a luxury resort style. We were just in pretty damp tents, basically spending the whole time there trying to figure out how to get out. There ’ s a name for what happened with Alexis. It ’ s called Escalation of commitment to a losing course of action. It ’ s when you find out you ’ ve made a bad decision, but instead of cutting your losses, you doubled down. You **invest** your time, your money or your identity and decisions long after it ’ s become obvious that they ’ re dead ends. You may not have ended up at fire festival, but I ’ m willing to bet that you ’ ve escalated your commitment to plenty of bad decisions. So how do you pull the plug sooner? I ’ m Adam Grant and this is WorkLife, my podcast with the **TED** Audio **Collective**. I ’ m an organizational psychologist. I study how to make work, not suck. In this show. I take you inside the minds of fascinating people to help us rethink how we work, lead and live today, why we get trapped in bad decisions and how you and your workplace can get unstuck. Thanks to Morgan Stanley for sponsoring this episode. Let ’ s start from the beginning. My father was one of the very earliest people who started a discount store and it was very successful and he started a second one on the third one until he had a chain of stores in the late 1950s. Barista was a teenager in Southern California, and for almost a decade, he saw his dad grow his business to 15 stores and he was really doing well. And then the large national companies, they would just open a store right across the street practically and undercut all the prices. They literally drove all the independent people, all the small **guys** out of business. By then, Barry was in his 20s watching his father ’ s business decline. It was slow and painful. He **refused** to give up. And so he kept putting more and more of his own personal money into the business until finally he had no more personal money left and it ended up closing up. So that was something that was really indelibly engraved in my psyche. Fast forward a few years and Barry was watching the American government fall into the same trap. It dawned on me that the Vietnam War was very similar to my father ’ s experience, that we kept **investing** more and more. We kept having larger and larger losses, but we kept as a country **refusing** to admit that this was unlikely to succeed. Today, Barry is an organizational behavior professor at UC Berkeley, and he coined the concept of escalation of commitment. Escalation of commitment occurs when you ’ re in a situation in which it is possible to add more resources to a course of action or a prior decision in order to rectify a loss. Barry is the world ’ s leading expert on this problem, but even he can ’ t help but fall victim to escalation from time to time. I ’ ve had some old cars and then when the warranty expires, then a bunch of little things start going wrong and then a little bit bigger things going wrong and then bigger and bigger. And I have been agonizing about whether to actually sell it and get rid of it or put more money in. It even happened in his marriage. I probably hung on longer than I should have went through years and years of marital therapy. Barry, in a moment like that, do you not realize you ’ re in an escalation trap? I realize I ’ m in it, but I still realize that it is a very difficult situation. And maybe I can tell myself it might be a lie, that this might be one of the exceptions in which it actually might be rational to **invest** more in the situation. If this can happen to an expert on escalation of commitment, what hope is there for the rest of us? I ’ m painfully familiar with this phenomenon myself, like the time I wrote a whole book instead of a book proposal and had to throw away over 100000 words and start over or the time. I spent months trying to change the behavior of a toxic team member when I should have just let him go. Escalation of commitment happens in every workplace. I think about the times when you overinvested in a failing project stuck around in a miserable job. I struggled to walk away from an abusive boss or a toxic culture or even away from work. Poured your heart and soul into a romantic relationship that clearly wasn ’ t working. So why don ’ t we know better? Why do we fall into the same trap again and again, pouring more and more of ourselves into a bad decision, even when all the evidence points us the other way? The most common answer, people give sunk costs, we ’ ve already **invested** our time or money, and we want to do everything in our power to get a return on that investment, just like a Lexus at fire festival. Sunk costs are part of a story, but decades of research show that the most powerful forces aren ’ t economic, they ’ re emotional. Escalation is where people make errors because of their ego and their emotions. It ’ s not just a cold calculation of the loss of money or time. It ’ s the hot pain of threats to our sense of self. They ’ re afraid to admit a mistake and they end up justifying a decision to themselves. In a case like this, I ’ m trying to protect my ego to convince myself that I didn ’ t make a bad choice. I ’ m also trying to protect my image, to convince others that I made a good choice, because otherwise I might end up having people think that I ’ m not a good decision maker or I ’ m a loser. I ’ m not somebody who ’ s to be **trusted**. The more we ’ ve **invested** in a decision, the more attached we become to it and the more attached it becomes to us. This is where an awful lot of the real disasters happen in terms of business decisions. People don ’ t want to admit errors because their work identity has become linked with the course of action. People may even label the thing saying, Oh, that project, that ’ s Jim ’ s baby or that ’ s Mary ’ s project. And you become so identified with it that if the project fails, your career is down the tubes. Ironically, the very steps we take to defend our egos and images end up making us look worse. One of the places Berry studied this was the NBA. You ’ d think coaches would give the most playing time to the best players, like how many points you score, how many steals you make and how you rebound and so forth. And there is some effect of that, obviously. But there ’ s also a very significant effect of how high you were in the draft. Even if top draft picks weren ’ t playing well, coaches still kept them in the game. They made a big bet on this **guy**, so they wanted to prove they made the right decision and hadn ’ t blown millions beyond ego and image. There ’ s another emotional factor that fuels these kinds of escalation traps anticipated regrette. You ’ re worried you ’ ll wish you hadn ’ t pulled the plug in. The NBA managers were afraid of giving up too soon on a potential star. They ’ re thinking about maybe trading a particular athlete to another team, but then they ’ re thinking, oh my God, he ’ s not done very well with us. But what if he suddenly blooms? But I ’ m watching him on the other team, then I ’ m going to feel even worse. You probably know that voice inside your head, what are you doing? You can ’ t give up too soon. Everyone ’ s going to think you ’ re a quitter and a failure, a failure. You won ’ t find a better opportunity at some point. You just have to say it ’ s over. Let it go. But that ’ s easier said than done. Just ask Janice Burch. I was so scared, I don ’ t ever remember being so scared to do anything then to tell my supervisor that I wanted to quit. Janice has one of the most practical strategies I ’ ve seen for escaping escalation of commitment. But she didn ’ t discover it overnight. She ’ s been working at a software company for a couple of years and for almost that entire time, she was unhappy, but she couldn ’ t bring herself to leave. She started out with high hopes when I saw this job advertised. I definitely felt like this. One hundred percent is my dream job. The company was well respected. The benefits were amazing. This customer **support** job was perfect for Janice, at least on paper. It offered room to grow personal responsibility for managing a team and a culture of flexibility and freedom, I guess kind of be like invited into what seemed like this like exclusive club of awesomeness. But pretty soon reality crashed the party. I would say the honeymoon was over for me when I went to a meetup and I found out that someone that I had been interviewed by and I really admired had been let go. And not very many people knew about it and everyone was upset about that. And I guess it ’ s just like in regular relationships and you talk about the honeymoon phase, you always say, oh, there were all these red flags and you ’ re just like, oh, well, I just I couldn ’ t see them because I was just so blinded by love. It ’ s kind of the same thing. A few months into the job, Janice had some concerns and she heard other people complaining, but she didn ’ t feel that she could speak up. She also felt that the company had a serious lack of **diversity**, Janice said that when she started she was the only black employee and that did not improve much over time. But she ’ d **invested** so much in this job, so instead of rethinking her commitment, Jannis escalated it by **investing** more energy and time and stayed in the job for two more years. She told herself that since her co-workers were reluctant to speak up, she might be able to change the company. I ’ m going to be the person to, like, jump in and make this change where others, you know, may be hesitant or whatever it is like I ’ m going to be the one. And so I was going to take it in and try to help. And how did it feel putting so much energy into a job and a culture that you knew deep down was not right? It ’ s almost like you ’ re gaslighting yourself, like you ’ re telling yourself. No, no. Like all of the evidence around you is not correct. There ’ s just you just have to do this one one little thing or do more. Barista knows the feeling at a certain point when the evidence is overwhelming. You have to ask yourself, is this grit or is it blindness or is it just my own ego that ’ s getting in the way? When you reach that point, you might find that an antidote to escalation of commitment is another commitment. But to go the opposite way, a plan to pull the plug. And Janis created a clever strategy, a one year exit plan. I basically created three simple steps. These are the things that I ’ m going to do to, like, actually build myself up and give myself the courage to leave this toxic relationship. The first point was that I was going to build relationships outside of the company. So try to like meet up with people and just build relationships with people who can help me in whatever my next phase of life was going to be. Second on the list, stop engaging in what Janice calls energy zaps. Yeah, energy zaps, not interacting with, you know, what I call problematic people to go back to the relationship comparison. I guess I became emotionally unavailable at work. Her third step was to leverage her company ’ s perks, to broaden her horizons is basically just trying to max out what was available to me then. One in particular is that we got to go to **conferences**. And so culture was obviously something that I was really **interested** and engaged with. So I ’ ve got a culture **conferences** and again, try to build relationships with people. It almost sounds like you had a strategy for disinvesting yourself. Yeah, that ’ s exactly what it was. And about a year after she made this plan, she left. When I look back and I think about why I held on so much to that job in that position, I think it ’ s because deep down, I have thought, like, this is as good as it gets. The beauty of a one year exit plan is that it gave Janice time to explore different paths and to detach emotionally instead of ripping off the Band-Aid. Janice now runs two coaching firms, works for flow and before **diversity**, her main regret is that she didn ' t make her exit plan sooner. One of my favorite ways to avoid that mistake is to identify your deal breakers, just like you probably have a list of nonstarters in her romantic partner, you can create a similar list for what ' s unacceptable in a job, a boss or a culture. If you figure them out up front, you can keep yourself honest. I ' ve often advised students to do this and they report that it helps them re-evaluate their decisions and avoid escalation. But what if the escalation is bigger than you? What if your team, your boss or your entire company makes a habit of staying the course in a very bad course? More on that after the break. OK, this is going to be a different kind of that I play a personal role in selecting the sponsors for this podcast because they all have **interesting** cultures of their they ' re out today. We ' re going inside the workplace at Morgan Stanley. I am a big believer in, you know, standing on the shoulders of others, I am my ancestors, you know, from Nigeria and Ghana, West Africa, I am their wildest dreams, you know, to whom much is given, much is required. And so I believe that that ' s tattooed on my soul and my heart. Meet Mandele Crowley. Ever since he was a kid, Mandela is understood how important it is to have someone in your corner. Our childhood was one where we were raised by our grandparents. We had a little bit of adversity losing our parents relatively early. We were fortunate in that we had our grandparents because we were at risk of going into the foster care system. But my grandparents stepped in and they were amazing, particularly my grandmother. She was everything. Unfortunately, she passed from cancer. From that moment on, I literally started taking care of myself when I was 16 years old. But Mandela wasn ' t just taking care of himself. He also had his younger brother to look after. I had a younger brother who was 14 at the time where I essentially had to become his parent. And then I started to think about things that I had to do to make sure that both he and I were OK. It was around that time that he heard about an internship at Morgan Stanley. I was one of five kids that got to interview Adam. I showed up in a gold suit, black shirt, black and gold tie, and I got the job that opened the door for him. And then as Mandela was nearing the end of his internship, his colleagues opened another door. My colleagues knew my personal situation and they essentially put it out there that, hey, if you can figure out how to balance your school schedule with work, we ' d love to keep you on. So Mandela started working at a trading desk during the day and attending college at night. He ' s now been with the firm for almost 30 years. And it all goes back to that first high school internship when his colleagues saw his potential and helped him achieve it. The **guy** who hired me on the desk was still a very dear friend and mentor of mine to this day, and he was pretty hands on in terms of helping to coach and mentor and all that **stuff**. Then he was offered his first **leadership** job, and that is when I came to appreciate my superpower, and that is motivating and mobilizing talent. Mandela rose to become the head of private wealth management, and now he ' s Morgan Stanley ' s chief human resources officer. Thing that really comes to mind when I think about why this firm was different. For me, it ' s really been the **diversity** of experiences that I ' ve been able to have. I ' ve had a series of careers at every step of his journey. He ' s made time to keep opening doors. I view it as a real responsibility, not just because of my day job, but I just feel like it ' s a fundamental responsibility that I have. Being a mentor doesn ' t just bring satisfaction and meaning. Research reveals that he can have surprising career benefits. People who spend time mentoring others are more likely to get promoted, in part because mentoring helps them develop and demonstrate their **leadership** skills. So there is a collegiality to the culture here. When you ' re a young person coming up the ranks and you need advice irrespective of who that person is or where they sit in the firm ' s ecosystem, you ' re going to get that meeting. I ' m always making time for young folks or up and coming folks who are trying to figure out this thing called a career because you win in life with people. Mendell has opened doors for many up and coming employers, but there ' s one who stands out. He ' s a Puerto Rican kid from Paterson, New Jersey, first in his family to go to college. I end up hiring him. Two years later, I get promoted and I get to bring one analyst with me. Guess who I bring with me? I bring him. He ' s just got promoted to managing director. He ' s just an incredible human being. He ' s a great person. I ' m the godfather of his son. You know, it ' s one of those classics, you know, when friends become like family, he ' s very much in that category. Collaboration and **inclusion** are core values in Morgan Stanley ' s culture, and they ' re shaping its future. Learn more Morgan Stanley dotcoms, people. Extracting ourselves from our own bad decisions is hard enough. Our ego and our image are in danger, but escalation is even worse when it happens. An entire teams or organizations. You know, Kodak was the industry **leader** in photography. They pioneered research into the digital camera, but then they shelved it and escalated their commitment to film. Bad decisions and then the decision to stick with them don ' t happen in a vacuum. The culture and structure of an organization can propel us straight into escalation of commitment. But research shows there are tangible steps organizations can take to protect us to make it easier to take a clear eyed look at the course we ' re on and even change direction. Berry style identified one of those steps in research on banks. It ' s very common for banks to loan money and to and to lose money. Banks fall victim to escalation of commitment when they keep expecting people who have defaulted on loan payments to come through instead of writing them off as uncollectable, Barry and his colleagues analyzed data from nearly all the banks in California over nine years. They found that banks were more likely to de-escalate their commitment to problem loans after **senior** executives left. The executives who had approved the original loans were motivated to keep justifying their decisions. When new executives took over, they quickly recognized that someone who ' s missed 17 payments probably isn ' t going to come through a number 18 if you move the loan decision from the original people who made the loan just like going to an objective third party, it ' s more likely that you will start to recognize the losses. So if I ' m a CEO, are you saying I should fire all my executives so that people are finally willing to cut their losses? Well, I wouldn ' t maybe go that far. If you don ' t have turnover inside your organization, you at least need an outside perspective, somebody who can look at things with a kind of a cold eye and say, this is what I would do if I were facing this situation. This is the first of three important steps organizations can take to prevent escalation of commitment, separate the decision from the initial decision maker. There ' s an organization that does exactly this, its acts also known as Google ' s Moonshot Factory, it ' s where inventors and entrepreneurs build and launch new projects that seem out of this world, like self-driving cars and balloons, beaming Internet to remote communities. My name is Cathy Qanoon and I was formerly a rapid evaluator at X Rapid Evaluator. Is that a job? Is a job rapid evaluators at X? Our role is to find new opportunities for X.. So what are the things X should work on next? Are you a rapid evaluator in other parts of your life? Do you just show up at a restaurant and immediately know what to order? I think that a lot of my rapid evaluative tendency is just consumed in my professional life so much that in my personal life I ' m like to my husband, can you please just can you decide what we ' re going to have for dinner? You decide I ' m fine with whatever. In other parts of Google and Alphabet, escalation has been a problem. They took eight years to abandon their failed attempt at social networking. Remember Google Plus? And they waited too long to recognize that Google Glass was not going to make it as a consumer product. So it xts rapid evaluators like Cathy were tasked with offering an outside perspective on bold ideas, looking for reasons to pull the plug, even if the project in question was their own. So Cathy would always ask, what is the thing that will bring it down back in 2013? Cathy learned about an idea that could disrupt the oil industry. She ' d been looking for a **climate** project, so she got excited about this new study she read. What if we could further this research, figure out how to commercialize it, and then combine that carbon dioxide with renewable **hydrogen** to make hydrocarbons, which is just a scientific way of saying make fuels out of it? So you ' re trying to turn seawater into fuel shorthand? Yes, that ' s what we would say, is it ' s turning seawater into fuel. Cathy immediately saw this as a crucial environmental breakthrough for Extra to fund. So she went all in foghorn. That ' s what they named the project was going to be her baby. But as someone whose job was to protect other people from escalation, Cathy knew she needed to be wary of it at the outset. Is it truly as good of an idea as it seems? It can take years to find out whether a project has succeeded or failed. Luckily at Beristain, his colleagues have discovered a workaround long before you know the outcome of a decision. You can create metrics to evaluate the quality of your decision process and process. Accountability is that you should be **accountable** for making a reasonably good decision, not in terms of how it turns out, but that you ' ve been reasonably conscientious in terms of assessing alternatives and not biasing the data process accountability. That ' s a second step for avoiding escalation. The important thing with this is setting your targets and your benchmarks in advance. One way of holding people **accountable** for a good decision process is to establish kill signals. A kill signal was a mechanism for keeping ourselves honest, so at the beginning of the project, we tried to think like, what could we see that would tell us, all right, this just isn ' t going to work. So in our case with popcorn, we really wanted to make commercial fuel. We needed a path to be able to make it for under five dollars a gallon because our thinking was if it ' s much higher than that, there just won ' t be a market for it. Kill signals are process boxes to check gates, you need to make it through in order to keep going. Maybe your kill signal is we don ' t want to have to tackle huge regulatory problems. That seems like something every team should do in every workplace. Yeah, I mean, even just thinking about what would your kill signal be, it tells you a lot about the problem you ' re trying to solve. It also seems like it ' s more it has broader applications. So I should I should take a new job and have a kill signal around. OK, here ' s what would need to happen in order for me to decide that it ' s time to leave this job or time to walk away from this culture. I totally agree. It ' s almost like principles, right? It ' s like what are my values and principles that I don ' t want to compromise on? And it ' s it ' s very helpful to have some sort of moral compass or some sort of fixed point in terms of one ' s value system to guide you when the decisions become very fraught. So Foghorn gets the green light. Kathee sets her kill signal for this project to be successful in today ' s market, they cannot go over five dollars a gallon and they get to work. We were tackling things, were fixing them. It ' s great. But then we realize just the pumping energy alone. To pump all the water you would need to get the CO2 out is too much like it ' s an astronomical cost. Her kill signal starts flashing. She knows this is not good for the future outcomes of the project, but they ' re on a roll. What if they **invest** just a little bit more? It was like, OK, yeah, pumping costs. That ' s a challenge. But like, look at all this **stuff** we ' re figuring out. I think we were making so much progress on making the prototype work better that it gave us some hope, hope, even a sliver of hope can keep a failing project alive. So Kathy decides to send a message. I just sat down at my desk at X and just wrote an email acknowledging here was the vision and here were all the amazing things that we accomplished. But here are the challenges. And given where we are now and everything we ’ ve learned, I ’ m inclined to proceed by suggesting that this doesn ’ t continue. Wait, you shut down your own project? I did. I recommended that it be shut down. Who does that? Apparently I do. All it took was a spreadsheet. I just realized the number, like the dollar numbers that I would have to ask for to make meaningful progress, given all the challenges, combined with the risk that we would not achieve our goals because of all of the complexity and the unknowns, even with the five dollars a gallon example by when. Right. It ’ s possible we ’ ll get there by 2050. But clearly that ’ s not what the kill signal meant. It felt like wishful thinking. You know, it ’ s like in my heart of hearts, I kind of knew it was more likely not to succeed. You know, there ’ s a lot of opportunity cost for everyone, and it just seemed cleaner to shut it down. Kathy wasn ’ t just focused on her own goals. She was thinking about what was best for the team and coming up with the most honest assessment that I could think would put the team as individuals in the best position going forward, because no one wants to work on something that ’ s not going to work. Research shows that when we consider our responsibilities to others, we ’ re less prone to escalation. Shifting attention away from our own egos and images and toward the greater good can allow us to make a more balanced assessment to free people from ego and image concerns. Organizations need to think differently about incentives to make failure acceptable. So even though the project hasn ’ t worked out, if the person has been very diligent in looking at what the best prospects were, that person should still be rewarded and be given a good future in the company. That ’ s a third step for de-escalation, taking the stigma out of admitting a failure. It ’ s created an annual ceremony to recognize people who have the wisdom to pull the plug, because even if it wasn ’ t the breakthrough they ’ d hoped for, their time and resources were still well spent. That day is all about celebrating the project that are no longer with us and recognizing them for everything that they gave X, and just like how X has benefited from them. And I do think that it would be too much to say people are happy when their projects are killed or it ’ s a positive experience. Certainly not like everyone is human, but at least it normalizes it. Beyond throwing a party to celebrate the value of a failed project, Cathy ’ s boss went further. He ended up giving us a bonus. And what? Yeah, you got a bonus for admitting that your project wasn ’ t promising and shutting it down? We did. It only solidified that culture that he was already creating. Didn ’ t that mean admitting failure, though? Well, my job as a rapid evaluator was to evaluate and this was my evaluation. So I didn ’ t see it that way because it ’ s like if you punish failure and you simultaneously know it ’ s inevitable, then you are creating an environment where people are inclined to hide it or misrepresent it. Yeah, I mean, if if you punish people for failing, they will do everything in their power to try to convince themselves and others that their project is not a failure. And then they ’ ll identify with the project and attach their ego to it in a way that ’ s very stressful for everyone. And then it ’ s no longer as possible to separate. I ’ m a failure from this idea that I was pursuing isn ’ t going to work, which are two very obviously different things, but don ’ t always feel very different. I ’ m smiling here because you just described that as if you ’ ve spent the past 10 years reading the psychology of escalation of commitment, which I can ’ t say that I have, but maybe I ’ ve lived it a little bit. We don ’ t need to celebrate failure. We just need to normalize it. What you ’ re highlighting here is that if you had plunged more of your time, energy and resources into foghorn, you might have missed out on a much bigger possibility. Yes, I absolutely agree. I mean, it ’ s like maybe this is a bad analogy, but it ’ s like the friend who ’ s dating the person that everyone kind of knows isn ’ t the right match. And even the friend isn ’ t very happy. But they ’ re worried that if they break up with that person, there won ’ t be any other person for them. That is the exact same phenomenon. It ’ s like you have to have this belief that there is something out there worth finding. And so if what you ’ re doing now is mediocre, it ’ s worth stopping to find that thing. That ’ s not. If you normalize failure in your organization, it can help people de-escalate their commitment to bad decisions and focus their energy on better opportunities, because Kathy pulled the plug on Foghorn. She was able to pursue a new project which eventually launched her career as an entrepreneur. And I am the founder and president of Dandelion Energy, a geothermal startup. Every time I talk about escalation of commitment, whether it ’ s with startup founders or military generals or students, there ’ s one question that always comes up is one person ’ s escalation, another ’ s heroic persistence. If you think about it, most of our Hollywood movies, they usually have the protagonists experiencing a long series of setbacks, but then they have stuck to it and they ’ ve **refused** to admit failure. And in fact, everyone around them has told them that whatever is happening is a failure and everyone around them is dropped off and they have hung in there and they are ultimately successful. And yeah, grit can be a good thing. We need people with the will to persist and triumph in the face of obstacles. But when you ’ re channeling your inner Katniss Everdeen, Inigo Montoya or Forrest Gump, it ’ s worth pausing to see if you ’ re on the wrong path because you might be Wile E. Coyote chasing a road runner you ’ ll never catch. Next time on Work Life, it was incredibly humbling because I would submit job after job after application after application and would hear nothing. It took an entire year to get a full time job. Everyone ’ s face career turbulence. The current **recession** is jeopardizing millions of careers right now. But we ’ ve been here before and there are lessons we can learn from past **recessions**. WorkLife is hosted by me, Adam Grant, it shows produced by Ted with Transmitter Media. Our team includes Colin Helmes Credico and Dan O ’ Donnell, Joanne DeLuna, Grace Rubenstein, Michelle Quint, Ben Ben Chang and Anna Feeling. This episode was produced by Constanza Gallardo, our shows mixed by Rick Juan. Our fact checker is Paul Durbin, original music by Hartsdale Sue and Alison Layton. Brown had stories produced by Pineapple Street Studios special thanks to our sponsors LinkedIn, Logitech, Morgan Stanley, Sappi and Verizon for their research on escalation of commitment. Thanks to Cigale Bastide, Dustin Slesin and Don Connellan, Jerry McNamara, Jonathan Miles, Clough ’ s Mozer and colleagues Henry Moon and of course, Barry stock for the audio from Fire Festival. The Greatest Party That Never Happened. Documentary is courtesy of Netflix and for the intro to Kathee and the insights on gratitude to Courtney Hoan. I hope to get a recording of this, so the next time I need to decide whether I should keep that old car of mine or whatever, I will not fall into the same traps that I have spent my research career trying to create lessons about how to avoid that. I think there ’ s there ’ s no better person to be persuaded by than someone you know well and **trust** a lot yourself. Thank you.

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Hello there, this is Chris Anderson, and I am hugely, hugely, tremendously ex-

cited to welcome you to a new series of the **TED** interview. Now, then, this season, we ’ re trying something new. We ’ re organising the whole season around a single theme, albeit a theme that some of you may consider inappropriate. But hear me out. The theme is the case for optimism. And yes, I know the world has been hit with extraordinarily ugly things in the last few years. Political division, a racial reckoning, technology run amuck, not to mention a global **pandemic** and impending **climate** catastrophe. What on earth are we thinking in this context? Optimism just seems so naive and unwanted, almost annoying. So here ’ s my position. Don ’ t think of optimism as a feeling. It ’ s not just this sort of shallow feeling of hope. Optimism is a search. It ’ s a determination to look for a pathway forward somewhere out there. I believe I truly believe there are amazing people whose minds contain the ideas, the visions, the solutions that can actually create that pathway forward. If given the **support** and resources they need, they may very well light the path out of this dark place we ’ re in. So these are the people who can present not optimism, but a case for optimism. They ’ re the people I ’ m talking to this season. So let ’ s see if they can persuade us now. Then the place I want to start is with A. I. artificial **intelligence**. This, of course, is the next innovative technology that is going to change everything as we know it, for better or for worse. Today was painted not with the usual dystopian brush, but by someone who truly believes in its potential. Sam Altman is the former president of Y Combinator, the legendary startup accelerator. And in 2015, he and a team launched a company called Open Eye, dedicated to one noble purpose to develop A. I. so that it benefits humanity as a whole. You may have heard, by the way, recently a lot of buzz around in A. I. technology called T3 that was developed by open eye improve the quality of the amazing team of researchers and developers they have work in. There will be hearing a lot about three in the conversation ahead. But sticking to this lofty mission of developing A. I. for humanity and finding the resources to realize it haven ’ t been simple. Open A. I. is certainly not without its critics, but their goal couldn ’ t be more important. And honestly, I found it really quite exciting to hear Sam ’ s vision for where all this could lead. OK, let ’ s do this. So, Sam Altman, welcome. Thank you for having me. So, Sam, here we are in 2021. A lot of people are fearful of the future at this moment in world history. How would you describe your attitude to the future? I think that the combination of scientific and technological progress and better societal decision making, better societal governance is going to solve in the next couple of decades all of our current most pressing problems, there will be new ones. But I think we are going to get very **safe**, very inexpensive, carbon free nuclear energy to work. And I think we ’ re going to talk about that time that the **climate** disaster looks so bad and how lucky we are. We got saved by science and technology, I think. And we ’ ve already now seen this with the rapidity that we were able to get vaccines deployed. We are going to find that we are able to cure or at least treat a significant percentage of human disease, including I think we ’ ll just actually make progress in helping people have much longer decades, longer health spans. And I think in the next couple of decades, that will look pretty clear. I think we will build systems with AI and otherwise that make access to an incredibly high quality education more possible than ever before. I think the lives we look forward like one hundred years, fifty years, even the quality of life available to anyone then will be much better than the quality of life available in the very best case to anyone today, to any single person today. So, yeah, I ’ m super optimistic. I think, like, it ’ s always easy to do scroll and think about how bad are the bad things are, but the good things are really good and getting much better. Is it your sincere belief that artificial **intelligence** can actually make that future better? Certainly.

How look, with any technology. I don ’ t think it will all be better. I think there are always positive and negative use cases of anything new, and it ’ s our job to maximize the positive ones, minimize the negative ones. But I truly, genuinely believe that the positive **impacts** will be orders of magnitude bigger than the negative ones. I think we ’ re seeing a glimpse of that now. Now that we have the first general purpose built out in the world and available via things like RPI, I think we are seeing evidence of just the breadth of services that we will be able to offer as the sort of technological revolution really takes hold. And we will have people interact with services that are smart, really smart, and it will feel like as strange as the world before mobile **phones** feels now to us. Hmm, yeah, you mentioned your API, I guess that stands for what, application programming interface? It ’ s the technology that allows complex technology to be accessible to others. So give me a sense of a couple of things that have got you most excited that are already out there and then how that gives you visibility to a pathway forward that is even more exciting. So I think that the things that we ’ re seeing now are very much glimpse of the future. We released three, which is a general-purpose natural language text model in the summer of twenty twenty. You know, there ’ s hundreds of applications that are now using it in production that ’ s ramping up all of the time. But there are things where people use three to really understand the intent behind the search query and deliver results and sort of understand not only intent, but all of the data and deliver the thing of what you want. So you can sort of describe a fuzzy thing and it ’ ll understand **documents**. It can understand, you know, short **documents**, not full books yet, but bring you back to the context of what you want. There ’ s been a lot of excitement about using the generative capabilities to create sort of games or sort of interactive stories or letting people develop characters or chat with a sort of virtual friend. There are applications that, for example, help a job seeker polish a tailored application for each individual company. There ’ s the beginning of tutors that can sort of teach people about different concepts and take on different personas. And we can go on for a long time. But I think anything that you can imagine that you do today via computer that you would like to really understand and get to know you. And not only that, but understand all of the data and knowledge in the world and help you have the best experience that is is possible that that will happen. So what gets opened up? What new adjacent possible state is that as a result of these powers from this question, from the point of view of someone who ’ s starting out on a career, for example, they ’ re trying to figure out what would be a really **interesting** thing to do in the future that has only recently become possible. What are some new things that this opens up in a world where you can talk to a computer? And get. The output that would normally require you hiring the world experts back immediately for almost no money, I would say think about what ’ s possible there. So that could be like, as you said, what can normally only the best programmer in the world or a really great programmer do for me. And can I now instead just ask in English and have that program written? So all these people that, you know, want to develop an app and they have an idea, but they don ’ t know how to program. Now they can have it. You know, what is the service look like when anyone on Earth who wants really great medical advice? Can get better medical advice than any single doctor could ever get, because this has the total medical knowledge and reasoning **ability** that the some humanity has ever produced. When you want to learn something, you have sort of a tutor that understands your exact style, how you best learn everything you know, and custom teaches you whatever concept you want to learn someday. You can imagine that like. You have an eye that reads your email and your task list and your calendar and the **documents** you ’ ve been sent and in any meeting maximally perfectly prepares you and has all of the information that you need in all the context of your entire career right there for you to go on for a long time. But I think this will just be powerful systems. So it ' s really fun playing around with Chapatti three, one compelling example of someone who ' s more tax base is try Googling The Guardian essay that was written entirely by different GP2 three queries and stitched together. It ' s an essay on why artificial **intelligence** isn ' t a threat to humanity. And that ' s impressive. It ' s very compelling. I actually tried inputting one of the three online uses. I asked the question what is **interesting** about some ollman? Oh no. Here ' s what it came back with. It was it was rather philosophical, actually. Came back with. I don ' t understand what you mean by **interesting**. It seems to me that this word has no unambiguous meaning in the context of human society beyond its literal definition as somewhat pleasing or entertaining. I do not believe there to be any such thing as INTERESTINGNESS except in the mind of a human or other sentient being that to my knowledge, this is an entirely subjective state that varies from person to person. However, I will grant that this is an **interesting** question to ask. This does not mean it has been answered. There is no answer to be found. Well, so you can agree that somewhere between profound and gibberish is that almost well, with the state of play is I mean, that ' s where we are today. I think somewhere between profound and jibberish is the right way to think about the current capabilities of CGP three. I think they would definitely had a bubble of hype about three last summer. But the thing about bubbles is the reason that smart people fall for them is there ' s a kernel of something really real and really **interesting** that people get overexcited about. And I think people definitely got and still are overexcited about 3:00 today, but still probably underestimated the potential of where these models will go in the future. And so maybe there ' s this like short term overhyped and long term under hype for the entire field, for tax models, for whatever you ' d like. It ' s going on. And as you said, there ' s clearly some gibberish in there. But on the other hand, those were like well-formed sentences. And there were a couple of ideas and there that I was like, oh, like they actually maybe that ' s right. And I think if artificial **intelligence**, even in its current very larval state, can make us confront new things and sort of inspire new ideas, that ' s already pretty impressive. Give us a sense of what ' s actually happening in the background there. I think it ' s hard to understand because you read these words seem like someone is trying to mean something. Obviously, I think you believe that there ' s whatever you ' ve built there, that there ' s a sort of thinking, sentient thing that ' s going, oh, I must answer this question. So so what how would you describe what ' s going on? You ' ve got something that has read the entire Internet, essentially all of Wikipedia, etc. We ' ve read something that ' s read like a small fraction of a random sampling of the Internet. We will eventually train something that has read as much of the Internet or more of the Internet than we ' ve done right now. But we have a very long way to go. I mean, we ' re still, I think, relative to what we will have operated at quite small scale with quite small eyes. But what is happening is there is a model that is ingesting lots of text and it is trying to predict the next word. So we use Transformer ' s they take in a context, which is a particular architecture of an A. I. model, they take in a context of a lot of words, let ' s say like a thousand or something like that. And they try to predict the word that comes next in the sequence. And there ' s like a lot of other things that happen, but fundamentally that ' s it, and I think this is **interesting** because in the process of playing that little game of trying to predict the next word, these models have to develop a representation and understanding of what is likely to come next and. I think it is maybe not perfectly accurate, but certainly worth considering to say that **intelligence** is very near the **ability** to make accurate predictions. What ’ s confusing about this is that there are so many words on the Internet which are foolish as well as the words that are wise. And and how do you build a model that can distinguish between those two? And this is prompted actually by another example that I typed in. Like I asked, you know, what is a powerful idea, very **interested** in ideas. That was my question as a powerful idea. And it came back with several things, some of which seemed moderately pronouncements, which seemed moderately gibberish. But then he was he was one that it came back with the idea that the human race has, quote, evolved, unquote, is false evolution or adaptation within a species was abandoned by biology and genetics long ago. Wait a sec. That ’ s news to me. What have you been reading? And I presume this has been pulled out of some recesses of the Internet, but how is it possible, even in theory, to imagine how a model can gravitate towards truth, wisdom, as opposed to just like majority views? Or how how how do you avoid something taking us further into the sort of the maze of errors and bad thinking and so forth that has already been a worrying feature for the last few years? It ’ s a fantastic question, and I think it is the most **interesting** area of research that we need to pursue. Now, I think at this point, the questions of whether we can build really powerful general-purpose AI system, I won ’ t say there in the rearview mirror. We still have a lot of hard engineering work to do, but I ’ m pretty confident we ’ re going to be able to. And now the questions are like, what should we build? And how and why and what data should we train on and how do we build systems not just that can do these like phenomenally impressive things, but that we can **ensure** do the things that we want and that understand the concepts of truth and falsehood and, you know, alignment with human values and misalignment with human values. One of the pieces of research that we put out last year that I was most proud of and most excited about is what we call reinforcement learning from human feedback. And we showed that we can take these giant models that are trained on a bunch of **stuff**, some of it good, some of the bad, and then with a really quite small amount of feedback from human judgment about, hey, this is good, this is bad, this is wrong, this is the behavior I want I don ’ t want this behavior. We can feed that information from the human judges back into the model and we can teach the model, behave more like this and less like that. And it works better than I ever imagined it would. And that gives me a lot of hope that we can build an aligned system. We ’ ll do other things, too, like I think curating data sets where there ’ s just less sort of bad data to train on. It will go a very long way. And as these models get smarter, I think they inherently develop the **ability** to sort out bad data from good data. And as they get really smart, they ’ ll even start to do something we call active learning, which is where they ask us for exactly the data they need when they ’ re missing something, when they ’ re unsure, when they don ’ t understand. But I think as a result of simply scaling these models up, building better, I hate to use the word cognition because it sounds so anthropomorphic, but let ’ s say building a better **ability** to reason into the models, to think, to challenge, to try to understand and combining that with this idea of online into human values via this technique we developed, that ’ s going to go a very long way. Now, there ’ s another question, which you sort of just kicked the ball down the field, too, which is how do we as a society decide to which set of human values do we align these powerful systems? Yeah, indeed. So if I if I understand rightly what you ’ re saying, that you ’ re saying that it ’ s possible to look at the output at any one time of three. And if we don ’ t like what it ’ s coming up with, some ways human can say, no, that was off, don ’ t do that. Whatever algorithm or process led you to that, undo it. Yeah. And that the system is that incredibly powerful at avoiding that same kind of mistake in future because it sort of replicates the instructions, correct? Yeah. And eventually and not much longer, I believe that we ' ll be able to not only say that was good, that was bad, but say that was bad for this reason. And also tell me how you got to that answer so I can make sure I understand. But at the end of the day, someone needs to decide who is the wise human or short humans who are looking at the results. So it ' s a big difference. Someone who who grew up with intelligent design world view could look at that and go, that ' s a brilliant outcome. Well, Goldstar done. And someone else would say something is done awfully wrong here. So how do you avoid and this is a version of the problem that a lot of the, I guess, Silicon Valley companies are facing right now in terms of the pushback they ' re getting on the output of social media and so forth. How do you assemble that pool of experts who stand for human values that we actually want? I mean, we talk about this all the time, I don ' t think this is like solely or even not even close to majorly up to opening night to decide, I think we need to begin a societal conversation now about how we ' re going to make those decisions, how we ' re going to make sure we have representational input in that, and how we sort of make these very difficult global governance systems. My personal belief is that we should have pretty broad rules about what these systems will never do and will always do. But then the individual user should get a system that kind of behaves like they want. And there will be people do have very different value systems. Some of them are just fundamentally incompatible. No one gets to use eye to, like, exploit other people, for example, and hopefully we can all agree on. But do you want the AI to like. You know, **support** you and your belief of intelligent design, like, do I think openly, I should say it can ' t, even though I disagree with that is like a scientific conclusion. No, I wouldn ' t take that stance. I think the thing to remember about all of this is that history is still quite extraordinarily weak. It ' s still has such big problems and it ' s still so unreliable that for most use cases it ' s still unsuitable. But when we think about a system that is like a thousand times more powerful and let ' s say a million times more reliable, it just doesn ' t it doesn ' t say gibberish very often. It doesn ' t totally lose the plot and get distracted or system like that is going to be one that a lot of the economic activity in the world comes to rely on. And I think it ' s very important that we don ' t have a small group of people sort of saying you can never use it for this thing that, like most of the world wants to use it for because it doesn ' t match our personal beliefs. Talk a bit more about some of the other uses of it, because one of the things that ' s most surprising is it ' s not just about sort of text responses. It ' s it can take generalized human instructions and build things up. For example, you can say to it, write a Python program that is designed to put a flashing cursor in one corner of the screen, in the Google logo in the other corner. And and it can go your way and do something like that. Shockingly, quite well, effectively. Yeah, I it can. That ' s amazing. I mean, this is amazing to me. That opens the door to. An entirely way to think about programers for the future, that you could you could have people who can program just in human natural language potentially and gain rapid efficiency. I do the engineering. We ' re not that far away from that world. We ' re not that far away from the world where you will write a spec in English. And for a simple enough program, I will just write the code for you. As you said, you can see glimpses of that even in this very week three which was not trained to code like. I think this is important to remember. We trained it on the language on the Internet very rarely, you know, Internet let language on the Internet also includes some code snippets. And that was enough, so if we really try to go train a model on code itself and that ' s where we decide to put the horsepower of the model into, just imagine what will be possible will be quite impressive. But I think what you ’ re pointing to there is that because models like three to some degree or other, and it ’ s like very hard to know exactly how much understand the underlying concepts of what ’ s going on. And they ’ re not just regurgitating things they found in a website, but they can really apply them and say, oh, yeah, I kind of like know about this word and this idea and code. And this is probably what you ’ re trying to do. And I won ’ t get it right always. But sometimes I will just generate this like a brand new program for nothing that anyone has ever asked before. And it will work. That ’ s pretty cool. And data is data. So it can do that from English to code. It can do that from English to French. Again, we never told it to learn about translation. We never told it about the concepts of English and French, but it learned them, even though we never said this is what English is and this is what French is and this is what it means to translate, it can still do it. Wow, I mean, for creative people, is there a world coming where the sort of the palette of possibility that they can be exposed to is just explodes? I mean, if you ’ re a musician, is there a near future where you can say to your eye, OK, I ’ m going to bed now, but in the morning I ’ d love you to present me with a thousand tuba jingles with words attached that you have of a sort of mean factor to the and you come down in the morning and the computer shows you the **stuff**. And one of them, you go, wow, that is it. That is a top 10 hit and you build a song from it. Or is that going to be released? Actually be the value add. We released something last year called Jukebox, which is very near what you described, where you can say I want music generated for me in this style or this kind of **stuff**, and it can come up with the words as well. And it ’ s like pretty cool. And I really enjoy listening to music that it creates. And I can sort of do four songs, two bars of a jingle, whatever you ’ d like. And one of my very favorite artists reached out, called to open it after we release this and said that he wanted to talk. And I was like, well, I like total fanboy here. I ’ d love to join that call. And I was so nervous that he was going to say, this is terrible. This is like a really sad thing for human creativity. Like, you know, why are you doing this? This is like whatever. And he was so excited. And he ’ s like, this has been so inspiring. I want to do a new album with this. You know, it ’ s like, give me all these new ideas. It ’ s making me much better at my job. I ’ m going to make better music because of this tool. And that was awesome. And I hope that ’ s how it all continues to go. And I think it is going to lead to this. We see a similar thing now with Dolly, where graphic designers sometimes tell us that they just they see this new set of possibilities because there ’ s new creative inspiration and they ’ re cycle time, like the amount of time it takes to just come up with an idea and be able to look at it and then decide whether to go down that path or head in a different direction goes down so much. And so I think it ’ s going to just be this like incredible creative explosion for humans. And how far away are we some before? And I it comes up with a genuinely powerful new idea, an idea that solves the problem that humans have been wrestling with. It doesn ’ t have to be as quite on the scale as of, OK, we ’ ve got a **virus** coming. Please describe to us what a what a national rational response should look like, but some kind of genuinely innovative idea or solution like one one internal question we ’ ve asked ourselves is, when will the first genuinely **interesting**, purely AI written **TED** talk show up? I think that ’ s a great milestone. I will say it ’ s always hard to guess timeline ’ s I ’ m sure I ’ ll be wrong on this, but I would guess the first genuinely **interesting**. Ted talk, thought of written delivered by an AIDS within the kind of the seven ish year time frame. Maybe a little bit less. And it feels like I mean, just reading that Guardian essay that was kind of it was a composite of several different GPG three responses to questions about, you know, the threats of robotics or whatever. If you throw in a human editor into the mix, you could probably imagine something much sooner. Indeed. Like tomorrow. Yeah. So the hybrid the hybrid version where it ’ s basically a tool assisted **TED** talk, but that it is better than any **TED** talk a human could generate in one hundred hours or whatever, if you can sort of combine human discretion with A. I. horsepower. I suspect that ’ s like our next year or two years from now kind of thing where it ’ s just really quite good. That ’ s that ’ s really **interesting**. How do you view the **impact** of A. I. on jobs? There ’ s obviously been the familiar story is that every White-Collar job is now up for destruction. What ’ s what ’ s your view there? You know, it ’ s I think it ’ s always hard to make these predictions. That is definitely the familiar story now. Five years ago, it was every blue collar job is up for destruction, maybe like last year it was. Every creative job is up for destruction because of things like Jukebox I. I think there will be an enormous **impact** on. The job market, and I really hate it, I think it ’ s kind of gross when people like working on I pretend like there ’ s not going to be or sort of say, oh, don ’ t worry about it. It ’ ll just all obviously better. It doesn ’ t always obviously get better. I think what is true is. Every technological revolution produces a change in jobs, we always find new ones, at least so far. It ’ s difficult to predict from where we ’ re sitting now what the new ones will be and this technological revolution is likely to be. Again, it ’ s always tempting to say this time it ’ s different. Maybe I ’ ll be totally wrong. But from what I see now, this technological revolution is likely to be more. Dramatic. More of a staccato note than most, and I think we as a society need to figure out how we ’ re going to cushion **everybody** through that. I ’ ve got my own ideas about how to do that. I, I wouldn ’ t say that I have any reason to believe they ’ re the right ones, but doing nothing and not really engaging with the magnitude of what ’ s about to happen, I think it ’ s like not an acceptable answer. So there ’ s going to be huge **impact**. It ’ s difficult to predict where it shows up the most. I think previous predictions have mostly been wrong, but I I ’ d like to see us all as a society, certainly as a field, engage in what what the shifts we want to make to the social contract are to kind of get through that in a way that is maximally beneficial to **everybody**. I mean, in every past revolution, there ’ s always been a space for humans to move to. That is, if you like, moving up the food chain, it ’ s sort of we ’ ve retreated to the things that humans could uniquely do, think better, be more creative and so forth. I guess the worry about A. I. is that in principle, I believe this, that there is no human cognitive feat that won ’ t ultimately be doable, probably better by artificial general touch, simply because of the extra firepower that ultimately they can have, the vast knowledge they bring to the table and so forth. Is that basically right, that there is ultimately no **safe** sort of space where we can say, oh, but that would never be able to do that on a very long time horizon? I agree with you, but that ’ s such a long time horizon. I think that, you know, like maybe we ’ ve merged by that point, like maybe we ’ re all plugged in and then, like, we ’ re this sort of symbiotic thing. Like, I think there ’ s an **interesting** example, as we were talking about a few minutes ago, where right now we have these systems that have sort of enormous horsepower but no steering wheel. It ’ s like, you know, incredible capabilities, but no judgment. And there ’ s like these obvious ways in which today even a human plus three is far better than either on their own. Many people speak about a world where it ’ s sort of A. I. as this external threat you speak about. At some point, we actually merge with eyes in some way. What do you mean by that? There ’ s a lot of different versions of what I think is possible there, you know, in some sense, I ’ d argue the merge has already like begun the human technology merge like we have this thing in our hands that sort of dictates a lot of what we think, but it gives us real superpowers and that can go much, much further. Maybe it goes all the way to like the Elon **Musk** vision of neuro link and having our brains plugged into computers and sort of like literally we have a computer on the back of our head or goes the other direction and we get uploaded into one. Or maybe it ' s just that we all have a chat bot that kind of constantly steers us and helps us make better decisions than we could. But in any case, I think the fundamental thing is it ' s not like the humans versus the eyes competing to be the. Smartest sentient thing on earth or beyond. But it ' s that this idea of being on the same team. Hmm. I certainly get very excited by the sort of the medium term potential for creative people of all sorts if they ' re willing to expand their palette of possibilities. But with the use of A. I. to be willing to. I mean, the one thing that the history of technology has shown again and again is that something this powerful and with this much benefit is unstoppable and you will get rewarded for embracing it the most and the earliest. So talk about what can go wrong with that, so let ' s move away from just the sort of economic displacement factor. You were a co-founder of Open Eye because you saw existential risks to humanity from high today. What would you put as the sort of the most worrying of those risks? And how is open eye working to minimize? I still think all of the really horrifying risks exist. I am more confident, much more confident than I was five years ago when we started that there are technical things we can do about. How we build these systems and the research and the alignment that make us much more likely to end up in the kind of really wonderful camp, but, you know, like maybe open I fall behind and maybe somebody else feels ajai that thinks about it in a very different way or doesn ' t care as much as we ' d like about safety and the risks or how to strike a different trade off of how fast we should go with this and where we should sort of just say, like, you know, like let ' s push on for the economic benefits. But I think all of this sort of like, you know, traditionally what ' s been in the realm of sci fi risks are real and we should not ignore them. And I still lose sleep over them. And just to update people is artificial general **intelligence**. Right now, we have incredible examples of powerful AI operating on specific areas. Ajai is the **ability** of a computer mind to connect the dots and to make decisions at the same level of breadth that that humans have had. What ' s your sort of elevator pitch on Ajai about how to identify and how to think of it? Yeah, I mean, the way that I would say it is that for a while we were in this world of like very narrow A. I., you know, that could like classify images of cats or whatever, more advanced **stuff** in that. But that kind of thing. We are now in the era of general purpose, AI, where you have these systems that are still very much imperfect tools, but that can generalize. And one thing like GPP three can write essays and translate between languages and write computer code and do very complicated search. It ' s like a single model that understands enough of what ' s really going on to do a broad array of tasks and learn new things quickly, sort of like people can. And then eventually we ' ll get to this other realm. Some people call it ajai, some people call ostler things. But I think it implies that the systems are like to some degree self directed, have some intentionality of their own is a simple summary to say that, like the fundamental risk is that there ' s the potential with general artificial **intelligence** of a sort of runaway effect of self-improvement that can happen far faster than any kind of humans can even keep up with, so that the day after you get to ajai, suddenly computers are thousands of times more advanced than us and we have no way of controlling what they do with that power. Yeah, and that is certainly in the risk space, which is that we build this thing and at some point somewhat suddenly, it ' s much more powerful than we are, we haven ' t really done the full merge yet. There ' s an event horizon there and it ' s sort of hard to see to the other side of it. Again, lots of reasons to think it will go OK. Lots of reasons to think we won ' t even get to that scenario. But that is something that. I don ’ t think people should brush under the rug as much as they do, it ’ s in the possibility space for sure, and in the possibility subspace of that is one where, like, we didn ’ t actually do as good of a job on the alignment work as we thought. And this sort of child of humanity kind of acts in a very different way than we think. A framework that I find useful is to sort of think about like a two by two matrix, which is short timelines to ajai and long timelines to ajai and a slow take off and a fast take off on the other axis. And in the short timelines, fast take off quadrant, which is not where I think we ’ re going to be. But if we get there, I think there ’ s a lot of scenarios in the direction that you are describing that are worrisome. And we would want to spend a lot of effort planning for. I mean, the fact that a computer could start editing its own code and improving itself while we ’ re asleep and you wake up in the morning and it ’ s got smarter, that is the start of something super powerful and potentially scary. I have tremendous misgivings about letting my system, not one we have today, but one that we might not have and too many more years start editing its own code while we ’ re not paying attention. I think that ’ s the kind of thing that is worth a great deal of societal discussion about, you know, just because we can do that. Should we? Yes, because one of the things that ’ s that ’ s been most shocking to you about the last few years has been just the power of unintended consequences. It ’ s like you don ’ t have to have a belief that there ’ s some sort of waking up of of an alien **intelligence** that suddenly decided it wants to wreak havoc on humans. That may never happen. What you can have is just incredible power that goes amok. So a lot of people would argue that what ’ s happened in technology in the last few years is actually an example of that. You know, social media companies created these **intelligences** that were programmed to maximally harvest attention, for example, for sure. And they understand this from that turned out to be in some ways horrifying and extraordinarily damaging. Is that a meaningful sort of canary in the coal mine saying, look out, humanity, this could be really dangerous? And how how on earth do you protect against those kinds of unintended consequences? I think you raise a great point in general, which is these systems don ’ t have to wish ill to humanity to cause ill just when you have, like, very powerful systems. I mean, unintended consequences for sure. But another version of that is and I think this applies at the technical level, at the company level, at the societal level, incentives are superpower ’ s. Charlie Munger had this thing on, which is incentives are so powerful that if you can spend any time whatsoever working on the incentive system, that ’ s what you should do before you work on anything else. And I really believe that. And I think that applies to the individual models we build and what their reward functions look like. I think it applies to society in a big way, and I think it applies to our corporate structure at open. I you know, we sort of observe that if you have very well-meaning people, but they have this incentive to sort of maximize attention harvesting and profit forever through no one ’ s ill intentions, that leads to a quite undesirable outcome. And so we set up opening is this thing called a capped profit model specifically so that we don ’ t have the system incentive to just generate maximum value forever with an AGI that seems like obviously quite broken. But even though we knew that was bad and even though we all like to think of ourselves as good people, it took us a long time to figure out the right structure, to figure out a charter that ’ s going to govern us and a set of incentives that we believe will let us do our work. And kind of these we have these like three elements that we talk about a lot research sort of engineering, development and deployment **policy** and safety. Put those all together under a system where you don ’ t have to rely on. Anything but the natural incentives to push in a direction that we hope will minimize the sort of negative unintended consequences. So help me understand this, because this is I think this is confusing to some people. So you started opening. I initially I think Elon **Musk**, the co-founder, and there was a group of you and the argument was this technology is too powerful to be left, developed in secret and to be left developed purely by corporations who have whatever incentive they may have. We need a nonprofit that will develop and share knowledge openly. First of all, just even at that early stage, some people were confused about this. It was saying if this thing is so dangerous, why on earth would you want to make it secrets even more available? Well, maybe giving the tools to that sort of AI terrorist in his bedroom somewhere, I think I think we got misunderstood in the way we were talking about that. We certainly don ' t think that the right thing to do is to, like, build this a super weapon and hand it to a terrorist. That ' s obviously awful. One of the reasons that we like our API model is it lets us make the most powerful AI technology anyone in the world has, as far as we know, available to ever would like to use it, but to put some controls on its usage. And also, if we make a mistake, to be able to pull it back or change it or tweak it or improve it or whatever. But we do want to put and this is continued will continue to be true with appropriate restrictions and guardrails, very powerful technology in the hands of people. I think that is fair. I think that will lead to the best results for the society as a whole. And I think it will sort of maximize benefit. But that ' s very different than sort of shipping the whole model and saying, here, do whatever you want with it. We ' re able to enforce rules on it. We also think and this is part of the mission that like something the field was doing a lot of that we didn ' t feel good about was sort of saying like, oh, we ' re going to keep the pace of progress and capabilities secret. That doesn ' t feel right, because I think we do need a societal conversation about what ' s what ' s going on here, what the **impacts** are going to be. And so we although we don ' t always say, like, you know, here ' s the super weapon, hopefully we do try to say, like, this is really serious. This is a big deal. This is going to affect all of us. We need to have a big conversation about what to do with it. Help me understand the structure a bit better, because you definitely surprised much people when you announced that Microsoft were putting a billion dollars into the organization and in return, I guess they get certain exclusive licensing rights. And so, for example, they are the exclusive licensee of CP3. So talk about that structure of how you win. Microsoft presumably have **invested** not purely for altruistic purposes. They think that they will make money on that billion dollars. I sure hope they do. I love capitalism, but I think that I really loved even more about Microsoft as a partner. And I ' ll talk about the structure and the exclusive license in a minute is that we like went around to people that might find us. And we said one of the things here is that we ' re going to try to make you some money. But like Adjei going well is more important. And we need you to sign this **document** that says if things don ' t go the way we think and we can ' t make you money like you just cheerfully walk away from it and we do the right thing for humanity. And they were like, yes, we are enthusiastic about that. We get that the mission comes first here. So again, I hope a phenomenal investment for them. But they were like they really pleasantly surprised us on the upside of how aligned they were with us, about how strange the world may get here and the need for us to have flexibility and put our mission first, even if that means they lose all their money, which I hope they don ' t and don ' t think they will. So the way it ' s set up is that if at some point in the coming year or two, two years, Microsoft decide that there ' s some incredible commercial opportunity that they could realize out of the eye that you ' ve built and you feel actually, no, that ' s that ' s damaging. You can block it. You can veto it. Correct. So the four most powerful version of three and its successors are available via the

API, and we intend for that to continue. What Microsoft has is the **ability** to sort of put that model directly into their own technology. If they want to do that. We don ’ t plan to do that with other people because we can ’ t have all these controls that we talked about earlier. But they ’ re like a close **trusted** partner and they really care about safety, too. But our goal is that anybody who wants to use the API can have the most powerful versions of what we ’ ve trained. And the structure of the API lets us continue to increase the safety and fix problems when we find them. But but the structure. So we start out as a non-profit, as you said, we realized pretty quickly that although we went into this thinking that the way to get to ajai would be about smarter and smarter algorithms, that we just needed bigger and bigger computers as well. And that was going to require a scale of capital that no one will, at least certainly not me, could figure out how to raise is a nonprofit. We also needed to sort of be able to compensate very highly compensated, talented individuals that do this, but are full for profit company had runaway incentives problem, among other things. Also just one about sort of fairness in society and wealth concentration that didn ’ t feel right to us either. And so we came up with this kind of hybrid where we have a nonprofit that governs what we do, and it has a subsidiary, LLC, that we structure in a way to make a fixed amount of profit so that all of our investors and employees, hopefully if things go how we like, if not no one gets any money, but hopefully they get to make this one time great return on their investment or the time that they spent it open their equity here. And then beyond that, all the value flows back to the nonprofit and we figure out how to share it as fairly as we can with the world. And I think that this structure and this nonprofit with this very strong charter in place and **everybody** who joins signing up for the mission come in first and the fact the world may get strange, I think that. That was at least the best idea we could come up with, and I think it feels so far like the incentive system is working, just as I sort of watch the way that we and our partners make decisions. But if I read it right, the cap on the gain that investors can make is 100 Axum. It ’ s a massive call that was for our very first round investors. It ’ s way, way lower. Like as we now take a bit of capital, it ’ s way, way lower. So your deal with Microsoft isn ’ t you can only make the first hundred billion dollars. I don ’ t know. It ’ s way lower than after that. We ’ re giving it to the world. It ’ s way lower than that. Have you disclosed what I don ’ t know if we have, so I won ’ t accidentally do it now. All right. OK, so explain a bit more about the charter and how it is that you. Hope to avoid or I guess help contribute to an eye that is **safe** for humanity. What do you see as the keys to us avoiding the worst mistakes and really holding on to something that ’ s that ’ s beneficial for humanity? My answer there is actually more about, like technical and societal issues than the charter. So if it ’ s OK for me to answer it from that perspective, sure. OK, I ’ m happy to talk about the charter to. I think this question of alignment that we talked about a little earlier is paramount, and then I think to understand that it ’ s useful to differentiate between accidental misuse of a system and intentional misuse of a system. So like intentional would be a bad actor saying, I ’ ve got this powerful system, I ’ m going to use it to like hack into all the computers in the world and wreak havoc on the power grids. And accidental would be kind of the Nick Bostrom make a lot of paper clips and view humans as collateral damage in both cases. But to varying degrees, if we can really, truly, technically solve the alignment problem and the societal problem of deciding to which set of human values do we align, then the systems understand right and wrong, and they understand probably better than we ever can, unintended consequences from complex actions and very complex systems. And, you know, if we can train a system which is like. Don ’ t harm humanity and the system can really understand what we mean when we say that, again, who is we and what does that have some asterisks on them? Sorry, go ahead. Well, that ' s if they could understand what it means to not harm humanity, that there ' s a lot wrapped up in that sentence. Because what ' s been so striking to me about efforts so far is that they seem to have been based on a very naive view of human nature. Go back to the sort of Facebook and Twitter examples of, well, the engineers building some of the systems would say we ' ve just designed them around what humans want to do. You said, well, if someone wants to click on something, we will give them more of that thing. And what could possibly be wrong with that? We ' re just **supporting** human choice, ignoring the fact that humans are complicated, farshid animals for sure, who are constantly making choices, that a more effective version of themselves would agree is not in their long term interests. So that ' s one part of it. And then you ' ve got layered on top of that or the complications of systemic complexity where, you know, multiple choices by thousands of people end up creating a reality that possibly have designed for how how to cut through that. Like an AI has to make a decision based on a moment, on a specific data set. As those decisions get more powerful, how can we be confident that they don ' t lead to this sort of system crashing basically in some way? I think that I ' ve heard a lot of behavioral psychologists and other people that have studied this say in different ways, are that I hate to keep picking on Facebook, but we can do it one more time since we ' re on the topic. Maybe you can ' t in any given moment in night where you ' re tired and you have a stressful day, stop yourself from the dopamine hit of scrolling and Instagram, even though you know that ' s bad for you and it ' s not leading to your best life. But if you were asked in a reflective moment where you were sort of fully alert and thoughtful, do you want to spend as much time as you do scrolling through Instagram? Does it make you happier or not? You would actually be able to give like the right long term answer? It ' s sort of the spirit is willing, but the flesh is weak kind of moment. And one thing that I am hopeful is that humans do know what we want and what. On the whole, and presented with research or sort of an objective view about what makes us happy and doesn ' t we ' re pretty, what ' s so great about it, they ' re pretty good. But in any particular moment, we are subjected to our animal instincts and it is easy for the lower brain to take over the eye. Well, I think be an even higher brain. And as we can teach it, you know, here is what we really do value. Here ' s what we really do want. It will help us make better decisions than we are capable of, even in our best moments. So is that being proposed and talked about as an actual rule? Because it strikes me that there is something potentially super profound here to introduce some kind of rule for development of AIDS that they have to tap into not. What humans one, which is an ill defined question, but as to what humans in reflective mode want. Yeah, we talk about this a lot. I mean, do you see a real chance where something like that could be incorporated as a sort of an absolute golden rule and and if you like, spread around the community so that it seeps into corporations and elsewhere? Because that I ' ve seen no evidence that, well, a little corporation that was potentially a game changer. Corporations have this weird incentive problem. Right. What I was trying to speak about was something that I think should be technologically possible, and that ' s something that we as a society should **demand**. And I think it is technically possible for this to be sort of like a layer above the neocortex that makes even better decisions for us and our welfare and our long term happiness and fulfillment than we could make on our own. And I think it is possible for us as a society to **demand** that. And if we can do like a pincer move between what the technology is capable of and what we what we as society **demand**, maybe we can make **everybody** in the middle that way. I mean, there are instances of even though companies have their incentives to make money and so forth, they also in the knowledge age. Can ' t make money if they have pissed off too many of their employees and customers and investors by analogy of the **climate** space right now, you can see more and more companies, even those that are emitting huge amounts of carbon dioxide, saying, wait a sec, we ' re struggling to recruit talented people because they don ' t want to work for someone who ' s evil. And their customers are saying, we don ' t want to buy something that is evil. And so, you know, ultimately you can picture processes where they do better. And I I believe that most engineers, for example, work in Silicon Valley. Companies are actually good people who want to design great products for humanity. I think that the people who run these companies want to be a net contribution to humanity. It ' s we ' ve we ' ve rushed really quickly and design **stuff** without thinking it through properly. And it ' s led to a mess up. So it ' s like, OK, don ' t move fast, break things, slow down and build beautiful things that are built on a real version of human nature and on a real version of system complexity and the risks associated with systemic complexity. Is that the agenda that fundamentally you think that you can push somehow? Yes, but I think the way we can push it is by getting the incentive system right. I think most people are fundamentally extremely good. Very few people wake up in the morning thinking about how can I make the world a worse place? But the incentive systems that we ' re in are so powerful. And even those engineers who join with the absolute best of intentions get sucked into this world where they ' re like trying to go up from it all for and five or whatever Facebook calls those things and you like, it ' s pretty exciting. You get caught up playing the game, you ' re rewarded for kind of doing things that move the company ' s key metrics. It ' s like fun to get promoted. It feels good to make more money and the incentive systems of the company. And that ' s what it rewards. An individual performance are maybe like not what we all want. And here I don ' t want to pick on Facebook at all because I think there ' s versions of this at play it like every big **tech** company, including in some ways I ' m sure it open I but to the degree that we can better align the incentives of companies with the welfare of society and then the incentives of an individual at those companies within the now realign incentives for those companies, the more likely we are to be able to have things like ajai that. Follow an incentive system of. What we want in our most reflective best moments and are even better than what we could think of ourselves is is it still the vision for open eye that you will get to? Artificial general **intelligence** ahead of. The corporations, so that you can somehow put a stake in the ground and build it the right way. Is that really a realistic thing to to dream for? And if not, how do you live up to the mission and help **ensure** that this thing doesn ' t go off the rails? I think it is. Look, I certainly don ' t think we will be the only group to build an AGI, but I think we could be the first. And I think if you are the first, you have a lot of norms that empower. And I think you ' ve already seen that. You know, we have released some of the most powerful systems to date. And I think the way that we have done that kind of in controlled release where we ' ve released a bigger model than a bigger one than a bigger one, and we sort of try and talk about the potential misuse cases and we try to like talk about the importance of releasing this behind an API so that you can make changes. Other groups have followed suit in some of those directions, and I think that ' s good. So, yes, I don ' t think we can be the only I do think we can be ahead. And if we are ahead, I think we can use that leverage to hopefully push people in a better direction or maybe we ' re wrong and somebody else has a better direction. We ' re doing something about do you have a structural advantage in that your mission is to do this for everyone as opposed to for some corporate objective. And that that that allows you that. Why is it that we came out of open eye and not someone else? It ' s like it ' s surprising in some ways when you ' re up against so much money and so much talent in these other companies that you came up with this platform ahead of. You know, in some sense it ' s surprising and in some sense, like the startup wins most of the time, like I ' m a huge believer in startups as the best force for innovation we have in the world today. I talked a little bit about how we combine these three. Different clans of research, engineering and sort of safety and **policy** that don ' t normally combine well and I think we have an unusual strength, there were clearly like well funded. We have super talented people. But what we really have is like intense focus and self belief that what we ' re doing is possible and good. And I appreciate the implied compliment. But, you know, we, like, work really hard. And if we stop doing that, I ' m sure someone would run by us fast. Tell us a bit more about some of your prior life sentences. Yeah, for several years, you were running Y Combinator, which had incredible **impact** on some 70 companies. There are so many startup stories that began at Y Combinator. What were key drivers in your own life that took you on the path you ' re on? And how did that path end up at Y Combinator? No exaggeration. I think I have back to back had the two jobs that are at least the most **interesting** to me in all of Silicon Valley. I, I was I went to college to study computer science. I was a major computer nerd growing up. I knew like a little bit about startups, but not very much. I started working on this project the same year I started working on that. This thing called Y Combinator started and funded me and my co-founders. And we dropped out of school and did this company, which I ran for like seven years. And then after that I got acquired. I had stayed close to my comment the whole time. I thought it was just this incredible group of people and spirit and set of incentives and just badly misunderstood by most of the world, but obvious to everyone within it that it was going to create huge amounts of value and do a lot of new things. My company had acquired PJI, who is the founder of ICI, and like truly one of the most incredible humans and business people. And Paul Burrell, Paul Graham asked me if I wanted to run it. And kind of like the central learning of my career, why I individual startups has been that if you really scale them up, remarkable things can happen. And I. I did it and I was like, one of the things that would make this exciting for me personally motivating would be if I could sort of push it in the direction of doing these hard **tech** companies, one of which became open. I describe actually what Y Combinator is, you know, how many people come through it to give us a couple of stories of its **impact**. Yeah. So you basically apply as a handful of people and an idea, maybe a prototype and say, I would like to start a company and will you please fund me? And we review those applications and we I shouldn ' t say we anymore. I guess they fund four hundred companies a year. You get about one hundred and fifty thousand dollars while she takes about seven percent ownership and then gives you lots of advice and then networking and sort of this like fast track program for starting a startup. I haven ' t looked at this in a while, but at one point a significant fraction of the billion dollar plus companies in the US that got started. It all came through the Wiki program, some recently in the news ones have been like Airbnb, Jordache, Coinbase, insta card stripe. And I think it ' s just it has become an incredible way to help. People who understand technology get a three month course in business, but instead of like herding you with an MBA, we actually teach you the things that matter and kind of go on to do incredible, incredible work. What is it about entrepreneurs? Why do they matter? Some people just find them kind of annoying. But I think you would argue I think I would argue that they have done as much as anyone to shape the future. Why? What is it about them? I think it is the **ability** to take. And idea and by force of will to make it happen in the world and in an incentive system that rewards you for making the most **impact** on the most people like in our system. That ’ s how we get most of the things that that we use. That ’ s how we got the computer that I ’ m using, the software I ’ m using to talk to you on it. Like all of this, you know, everyone in life, everything has a balance sheet. There ’ s plenty of very annoying things about them. And there ’ s plenty of very annoying things about the system that sort of idolizes them. But we do get something really important in return. And I think that as a force for making things that make all of our lives better happen, it ’ s very cool. Otherwise, you know, like if you have, like, a great idea, but you don ’ t actually do anything useful with it for people, that ’ s still cool. It ’ s still intellectually **interesting**. But like, there ’ s got to be something about the reward function in society that is like, did you actually do something useful? Did you create value? And I think entrepreneurship and startups are a wonderful way to do that. You know, we get all these great software companies. But I also think it ’ s like how we ’ re going to get ajai, how we ’ re going to get nuclear fusion, how we ’ re going to get life extension. And like on any of those topics are a long list of other things I could point to. There ’ s like a number of startups that I think are doing incredible work, some of which will actually deliver. It is a truly amazing thing when you put the camera back and to believe that a human being could be lying awake at night and something pops inside their mind as a patterning of the neurons in their brain that is effectively them saying, aha, I can see a way where the future could be better and and they can actually picture it. And then they wake up and then they talk to other people and they persuade them and they persuade investors and so forth. And the fact that this this system can happen and that they can then actually change the history changes in some sense. It is mind boggling that that happens that way and it happens again and again. So you ’ ve seen so many of these stories happen. What would you say? Is the is there a key thing that differentiates good entrepreneurs from others? If you could double down on one trait, what would it be? If I could pick only one, I would pick determination. I think that is the biggest predictor of success, the biggest differentiator predictor. And if you would allow a second, I would pick like communication skills or evangelism or something in that direction as well. There are all of the obvious ones that matter, like **intelligence**, but there ’ s like a lot of smart people in the world. And when I look back at kind of the thousands of entrepreneurs I ’ ve worked with, all of many of whom were like quite capable, I would say that ’ s like one and two of the surprisingly differentiated characteristics. What it ’ s it ’ s what I look at, the different things that you ’ ve built and you ’ re working on. I mean, it could not be more foundational for the future. I mean, entrepreneurship. I know this is I agree that this is really what has driven the future. Do you see some people get really now they look at Silicon Valley and they look at this story and they worry about the culture. Right. That it ’ s this is a bro culture. Do you see prospects of that changing anytime soon? And would you welcome it? Can we get better companies by really working to expand a group of people who can be entrepreneurs and who can contribute to aid, for example? For sure. And in fact, I think I ’ m hopeful, since these are the two things I ’ ve thought the most about. I ’ m excited for the day when someone combines them and uses A. I. to better select who did more fairly, maybe even select who to fund and how to advise them and really kind of make entrepreneurship super widely available that will lead to like better outcomes and sort of more societal wealth for all of us. So are. So, yeah, I think. Broadening the set of people able to start companies and that sort of get the resources that you need, that is like an unequivocally good thing and it ’ s something that I think Silicon Valley is making some progress in. But I hope we see a lot more. And I do really, truly think that the technology industry entrepreneurship is one of the greatest forces for self betterment. If we can just figure out how to be a little bit more inclusive in how we do things. My last question today is about ideas were spreading. If you could inject one idea into the mind of everyone listening, what what would the idea be? We ’ ve touched on it a bunch, but the one idea would be the ajai really is going to happen. You have to engage with it seriously, and you shouldn ’ t just listen to this and then brush aside and go about life as if it ’ s not going to happen because it is going to affect everything. And we will all we all, I think, have an obligation, but also an opportunity to figure out what not means and how we want the world and this sort of one time shift to go on. I ’ m kind of awed by the breadth of things are engaged with. Thank you so much for spending so much time sharing your vision. Thanks so much for having me. OK, that ’ s it for today. You can read more about open eyes, vision and progress at open eye dotcom. If you want to try playing with yourself, it ’ s a little tricky. You have to find a website that has licensed the API. The one I went to was philosopher ehi dot com, where you just you pay a few dollars to get access to a very strange mind. That ’ s actually quite a lot of fun. The interview is part of the **TED** Audio **Collective**, a collection of podcasts dedicated to sparking curiosity and sharing ideas that matter. This show is produced by Kim Net2Phone Pittas and edited by Grace Rubenstein and Sheila Boffano, Sambor Islamic Sir. Fact Check is by Paul Durbin and special thanks to Michele Quent, Colin Helmes and Anna Felin. If you like the show, please write and review it. It helps other people find us. We read every review, so thanks so much for listening. See you next time.

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I ’ m supposed to scare you, because it ’ s about fear, right? And you should be really afraid, but not for the reasons why you think you should be. You should be really afraid that — if we stick up the first slide on this thing — there we go — that you ’ re missing out. Because if you spend this week thinking about Iraq and thinking about Bush and thinking about the stock market, you ’ re going to miss one of the greatest adventures that we ’ ve ever been on. And this is what this adventure ’ s really about. This is crystallized DNA. Every life form on this planet — every insect, every bacteria, every plant, every animal, every human, every politician — is coded in that **stuff**. And if you want to take a single crystal of DNA, it looks like that. And we ’ re just beginning to understand this **stuff**. And this is the single most exciting adventure that we have ever been on. It ’ s the single greatest mapping project we ’ ve ever been on. If you think that the mapping of America ’ s made a difference, or landing on the moon, or this other **stuff**, it ’ s the map of ourselves and the map of every plant and every insect and every bacteria that really makes a difference. And it ’ s beginning to tell us a lot about evolution. It turns out that what this **stuff** is — and Richard Dawkins has written about this — is, this is really a river out of Eden. So, the 3. 2 billion base pairs inside each of your cells is really a history of where you ’ ve been for the past billion years. And we could start dating things, and we could start changing medicine and archeology. It turns out that if you take the human species about 700 years ago, white Europeans diverged from black Africans in a very significant way. White Europeans were subject to the plague. And when they were subject to the plague, most people didn ’ t survive, but those who survived had a mutation on the CCR5 receptor. And that mutation was passed on to their kids because they ’ re the ones that survived, so there was a great deal of population pressure. In Africa, because you didn ’ t have these cities, you didn ’ t have that CCR5 population pressure mutation. We can date it to 700 years ago. That is one of the reasons why AIDS is raging across Africa as fast as it is, and not as fast across Europe. And we ’ re beginning to find these little things for malaria, for sickle cell, for cancers. And in the measure that we map ourselves, this is the single greatest adventure that we ’ ll ever be on. And this Friday, I want you to pull out a really good bottle of wine, and I want you to toast these two people. Because this Friday, 50 years ago, Watson and Crick found the structure of DNA, and that is almost as important a date as the 12th of February when we first mapped ourselves, but anyway, we ’ ll get to that. I thought we ’ d talk about the new zoo. So, all you **guys** have heard about DNA, all the **stuff** that DNA does, but some of the **stuff** we ’ re discovering is kind of nifty because this turns out to be the single most abundant species on the planet. If you think you ’ re successful or cockroaches are successful, it turns out that there ’ s ten trillion trillion Pleurococcus sitting out there. And we didn ’ t know that Pleurococcus was out there, which is part of the reason why this whole species-mapping project is so important. Because we ’ re just beginning to learn where we came from and what we are. And we ’ re finding amoebas like this. This is the amoeba dubia. And the amoeba dubia doesn ’ t look like much, except that each of you has about 3. 2 billion letters, which is what makes you you, as far as gene code inside each of your cells, and this little amoeba which, you know, sits in water in hundreds and millions and billions, turns out to have 620 billion base pairs of gene code inside. So, this little thingamajig has a genome that ’ s 200 times the size of yours. And if you ’ re thinking of efficient information storage mechanisms, it may not turn out to be chips. It may turn out to be something that looks a little like that amoeba. And, again, we ’ re learning from life and how life works. This funky little thing : people didn ’ t used to think that it was worth taking samples out of nuclear reactors because it was dangerous and, of course, nothing lived there. And then finally somebody picked up a microscope and looked at the water that was sitting next to the cores. And sitting next to that water in the cores was this little Deinococcus radiodurans, doing a backstroke, having its chromosomes blown apart every day, six, seven times, restitching them, living in about 200 times the radiation that would kill you. And by now you should be getting a hint as to how diverse and how important and how **interesting** this journey into life is, and how many different life forms there are, and how there can be different life forms living in very different places, maybe even outside of this planet. Because if you can live in radiation that looks like this, that brings up a whole series of **interesting** questions. This little thingamajig : we didn ’ t know this thingamajig existed. We should have known that this existed because this is the only bacteria that you can see to the naked eye. So, this thing is 0. 75 millimeters. It lives in a deep trench off the coast of Namibia. And what you ’ re looking at with this namibiensis is the biggest bacteria we ’ ve ever seen. So, it ’ s about the size of a little period on a sentence. Again, we didn ’ t know this thing was there three years ago. We ’ re just beginning this journey of life in the new zoo. This is a really odd one. This is Ferroplasma. The reason why Ferroplasma is **interesting** is because it eats iron, lives inside the equivalent of battery acid, and excretes sulfuric acid. So, when you think of odd life forms, when you think of what it takes to live, it turns out this is a very efficient life form, and they call it an archaea. Archaea means”the ancient ones.”And the reason why they ’ re ancient is because this thing came up when this planet was covered by things like sulfuric acid in batteries, and it was eating iron when the earth was part of a melted core. So, it ’ s not just dogs and cats and whales and dolphins that you should be aware of and **interested** in on this little journey. Your fear should be that you are not, that you ’ re paying attention to **stuff** which is temporal. I mean, George Bush — he ’ s going to be gone, alright? Life isn ’ t. Whether the humans survive or don ’ t survive, these things are going to be living on this planet or other planets. And it ’ s just beginning to understand this code of DNA that ’ s really the most exciting intellectual adventure that we ’ ve ever been on. And you can do strange things with this **stuff**. This is a baby gaur. Conservation group gets together, tries to figure out how to breed an animal that ’ s almost extinct. They can ’ t do it naturally, so what they do with this thing is they take a spoon, take some cells out of an adult gaur ’ s mouth, code, take the cells from that and insert it into a fertilized cow ’ s **egg**, reprogram cow ’ s **egg** — different gene code. When you do that, the cow gives birth to a gaur. We are now experimenting with bongos, pandas, elands, Sumatran tigers, and the Australians — bless their hearts — are playing with these things. Now, the last of these things died in September 1936. These are Tasmanian tigers. The last known one died at the Hobart Zoo. But it turns out that as we learn more about gene code and how to reprogram species, we may be able to close the gene gaps in deteriorate DNA. And when we learn how to close the gene gaps, then we can put a full string of DNA together. And if we do that, and insert this into a fertilized wolf ’ s **egg**, we may give birth to an animal that hasn ’ t walked the earth since 1936. And then you can start going back further, and you can start thinking about dodos, and you can think about other species. And in other places, like Maryland, they ’ re trying to figure out what the primordial ancestor is. Because each of us contains our entire gene code of where we ’ ve been for the past billion years, because we ’ ve evolved from that **stuff**, you can take that tree of life and collapse it back, and in the measure that you learn to reprogram, maybe we ’ ll give birth to something that is very close to the first primordial ooze. And it ’ s all coming out of things that look like this. These are companies that didn ’ t exist five years ago. Huge gene sequencing facilities the size of football fields. Some are public. Some are private. It takes about 5 billion dollars to sequence a human being the first time. Takes about 3 million dollars the second time. We will have a 1, 000-dollar genome within the next five to eight years. That means each of you will contain on a CD your entire gene code. And it will be really boring. It will read like this. The really neat thing about this **stuff** is that ’ s life. And Laurie ’ s going to talk about this one a little bit. Because if you happen to find this one inside your body, you ’ re in big trouble, because that ’ s the source code for Ebola. That ’ s one of the deadliest diseases known to humans. But plants work the same way and insects work the same way, and this apple works the same way. This apple is the same thing as this floppy disk. Because this thing codes ones and zeros, and this thing codes A, T, C, Gs, and it sits up there, absorbing energy on a tree, and one fine day it has enough energy to say, execute, and it goes [thump]. Right? And when it does that, pushes a. EXE, what it does is, it executes the first line of code, which reads just like that, AATCAGGGACCC, and that means : make a root. Next line of code : make a stem. Next line of code, TACGGGG : make a flower that ’ s white, that blooms in the spring, that smells like this. In the measure that you have the code and the measure that you read it — and, by the way, the first plant was read two years ago ; the first human was read two years ago ; the first insect was read two years ago. The first thing that we ever read was in 1995 : a little bacteria called Haemophilus influenzae. In the measure that you have the source code, as all of you know, you can change the source code, and you can reprogram life forms so that this little thingy becomes a vaccine, or this little thingy starts producing biomaterials, which is why DuPont is now growing a form of polyester that feels like silk in corn. This changes all rules. This is life, but we ’ re reprogramming it. This is what you look like. This is one of your chromosomes. And what you can do now is, you can outlay exactly what your chromosome is, and what the gene code on that chromosome is right here, and what those genes code for, and what animals they code against, and then you can tie it to the literature. And in the measure that you can do that, you can go home today, and get on the Internet, and access the world ’ s biggest public library, which is a library of life. And you can do some pretty strange things because in the same way as you can reprogram this apple, if you go to Cliff Tabin ’ s lab at the Harvard Medical School, he ’ s reprogramming chicken embryos to grow more wings. Why would Cliff be doing that? He doesn ’ t have a restaurant. The reason why he ’ s reprogramming that animal to have more wings is because when you used to play with lizards as a little child, and you picked up the lizard, sometimes the tail fell off, but it regrew. Not so in human beings : you cut off an arm, you cut off a leg — it doesn ’ t regrow. But because each of your cells contains your entire gene code, each cell can be reprogrammed, if we don ’ t stop stem cell research and if we don ’ t stop genomic research, to express different body functions. And in the measure that we learn how chickens grow wings, and what the program is for those cells to differentiate, one of the things we ’ re going to be able to do is to stop undifferentiated cells, which you know as cancer, and one of the things we ’ re going to learn how to do is how to reprogram cells like stem cells in such a way that they express bone, stomach, skin, pancreas. And you are likely to be wandering around — and your children — on regrown body parts in a reasonable period of time, in some places in the world where they don ’ t stop the research. How ’ s this **stuff** work? If each of you differs from the person next to you by one in a thousand, but only three percent codes, which means it ’ s only one in a thousand times three percent, very small differences in expression and punctuation can make a significant difference. Take a simple declarative sentence. Right? That ’ s perfectly clear. So, men read that sentence, and they look at that sentence, and they read this. Okay? Now, women look at that sentence and they say, uh-uh, wrong. This is the way it should be seen. That ’ s what your genes are doing. That ’ s why you differ from this person over here by one in a thousand. Right? But, you know, he ’ s reasonably good looking, but... I won ’ t go there. You can do this **stuff** even without changing the punctuation. You can look at this, right? And they look at the world a little differently. They look at the same world and they say... That ’ s how the same gene code — that ’ s why you have 30, 000 genes, mice have 30, 000 genes, husbands have 30, 000 genes. Mice and men are the same. Wives know that, but anyway. You can make very small changes in gene code and get really different outcomes, even with the same string of letters. That ’ s what your genes are doing every day. That ’ s why sometimes a person ’ s genes don ’ t have to change a lot to get cancer. These little chippies, these things are the size of a credit card. They will test any one of you for 60, 000 genetic conditions. That brings up questions of privacy and insurability and all kinds of **stuff**, but it also allows us to start going after diseases, because if you run a person who has leukemia through something like this, it turns out that three diseases with completely similar clinical syndromes are completely different diseases. Because in ALL leukemia, that set of genes over there over-expresses. In MLL, it ’ s the middle set of genes, and in AML, it ’ s the bottom set of genes. And if one of those particular things is expressing in your body, then you take Gleevec and you ’ re cured. If it is not expressing in your body, if you don ’ t have one of those types — a particular one of those types — don ’ t take Gleevec. It won ’ t do anything for you. Same thing with Receptin if you ’ ve got breast cancer. Don ’ t have an HER- 2 receptor? Don ’ t take Receptin. Changes the nature of medicine. Changes the predictions of medicine. Changes the way medicine works. The greatest repository of knowledge when most of us went to college was this thing, and it turns out that this is not so important any more. The U. S. Library of Congress, in terms of its printed volume of data, contains less data than is coming out of a good genomics company every month on a compound basis. Let me say that again : A single genomics company generates more data in a month, on a compound basis, than is in the printed collections of the Library of Congress. This is what ' s been powering the U. S. economy. It ' s Moore ' s Law. So, all of you know that the price of computers halves every 18 months and the power doubles, right? Except that when you lay that side by side with the speed with which gene data ' s being deposited in GenBank, Moore ' s Law is right here : it ' s the blue line. This is on a log scale, and that ' s what superexponential growth means. This is going to push computers to have to grow faster than they ' ve been growing, because so far, there haven ' t been applications that have been required that need to go faster than Moore ' s Law. This **stuff** does. And here ' s an **interesting** map. This is a map which was finished at the Harvard Business School. One of the really **interesting** questions is, if all this data ' s free, who ' s using it? This is the greatest public library in the world. Well, it turns out that there ' s about 27 trillion bits moving inside from the United States to the United States ; about 4. 6 trillion is going over to those European countries ; about 5. 5 ' s going to Japan ; there ' s almost no communication between Japan, and nobody else is literate in this **stuff**. It ' s free. No one ' s reading it. They ' re focusing on the war ; they ' re focusing on Bush ; they ' re not **interested** in life. So, this is what a new map of the world looks like. That is the genomically literate world. And that is a problem. In fact, it ' s not a genomically literate world. You can break this out by states. And you can watch states rise and fall depending on their **ability** to speak a language of life, and you can watch New York fall off a cliff, and you can watch New Jersey fall off a cliff, and you can watch the rise of the new empires of **intelligence**. And you can break it out by counties, because it ' s specific counties. And if you want to get more specific, it ' s actually specific zip codes. So, you want to know where life is happening? Well, in Southern California it ' s happening in 92121. And that ' s it. And that ' s the triangle between Salk, Scripps, UCSD, and it ' s called Torrey Pines Road. That means you don ' t need to be a big nation to be successful ; it means you don ' t need a lot of people to be successful ; and it means you can move most of the wealth of a country in about three or four carefully picked 747s. Same thing in Massachusetts. Looks more spread out but — oh, by the way, the ones that are the same color are contiguous. What ' s the net effect of this? In an agricultural society, the difference between the richest and the poorest, the most productive and the least productive, was five to one. Why? Because in agriculture, if you had 10 kids and you grow up a little bit earlier and you work a little bit harder, you could produce about five times more wealth, on average, than your neighbor. In a knowledge society, that number is now 427 to 1. It really matters if you ' re literate, not just in reading and writing in English and French and German, but in Microsoft and Linux and Apple. And very soon it ' s going to matter if you ' re literate in life code. So, if there is something you should fear, it ' s that you ' re not keeping your eye on the ball. Because it really matters who speaks life. That ' s why nations rise and fall. And it turns out that if you went back to the 1870s, the most productive nation on earth was Australia, per person. And New Zealand was way up there. And then the U. S. came in about 1950, and then Switzerland about 1973, and then the U. S. got back on top — beat up their chocolates and cuckoo clocks. And today, of course, you all know that the most productive nation on earth is Luxembourg, producing about one third more wealth per person per year than America. Tiny landlocked state. No oil. No diamonds. No natural resources. Just smart people moving bits. Different rules. Here ’ s differential productivity rates. Here ’ s how many people it takes to produce a single U. S. patent. So, about 3, 000 Americans, 6, 000 Koreans, 14, 000 **Brits**, 790, 000 Argentines. You want to know why Argentina ’ s crashing? It ’ s got nothing to do with **inflation**. It ’ s got nothing to do with privatization. You can take a Harvard-educated Ivy League economist, stick him in charge of Argentina. He still crashes the country because he doesn ’ t understand how the rules have changed. Oh, yeah, and it takes about 5. 6 million Indians. Well, watch what happens to India. India and China used to be 40 percent of the global economy just at the Industrial Revolution, and they are now about 4. 8 percent. Two billion people. One third of the global population producing 5 percent of the wealth because they didn ’ t get this change, because they kept treating their people like serfs instead of like shareholders of a common project. They didn ’ t keep the people who were educated. They didn ’ t foment the businesses. They didn ’ t do the IPOs. Silicon Valley did. And that ’ s why they say that Silicon Valley has been powered by ICs. Not integrated circuits : Indians and Chinese. Here ’ s what ’ s happening in the world. It turns out that if you ’ d gone to the U. N. in 1950, when it was founded, there were 50 countries in this world. It turns out there ’ s now about 192. Country after country is splitting, seceding, succeeding, failing — and it ’ s all getting very fragmented. And this has not stopped. In the 1990s, these are sovereign states that did not exist before 1990. And this doesn ’ t include fusions or name changes or changes in flags. We ’ re generating about 3. 12 states per year. People are taking control of their own states, sometimes for the better and sometimes for the worse. And the really **interesting** thing is, you and your kids are empowered to build great empires, and you don ’ t need a lot to do it. And, given that the music is over, I was going to talk about how you can use this to generate a lot of wealth, and how code works. Moderator : Two minutes. Juan Enriquez : No, I ’ m going to stop there and we ’ ll do it next year because I don ’ t want to take any of Laurie ’ s time. But thank you very much.

Text Sample 537: POS Dim 4 – 2003 – Score 11.00 – 2003/t000450.txt

I was asked to come here and speak about creation. And I only have 15 minutes, and I see they ’ re counting already. And I can â in 15 minutes, I think I can touch only a very rather janitorial branch of creation, which I call”creativity.”Creativity is how we cope with creation. While creation sometimes seems a bit un-graspable, or even pointless, creativity is always meaningful. See, for instance, in this picture. You know, creation is what put that dog in that picture, and creativity is what makes us see a chicken on his hindquarters. When you think about â you know, creativity has a lot to do with causality too. You know, when I was a teenager, I was a creator. I just did things. Then I became an adult and started knowing who I was, and tried to maintain that persona â I became creative. It wasn ’ t until I actually did a book and a retrospective exhibition, that I could track exactly â looks like all the craziest things that I had done, all my drinking, all my parties â they followed a straight line that brings me to the point that actually I ’ m talking to you at this moment. Though it ’ s actually true, you know, the reason I ’ m talking to you right now is because I was born in Brazil. If I was born in Monterey, probably would be in Brazil. You know, I was born in Brazil and grew up in the ’ 70s under a **climate** of political distress, and I was forced to learn to communicate in a very specific way â in a sort of a semiotic black market. You couldn ’ t really say what you wanted to say ; you had to invent ways of doing it. You didn ’ t **trust** information very much. That led me to another step of why I ’ m here today, is because I really liked media of all kinds. I was a media junkie, and eventually got **involved** with advertising. My first job in Brazil was actually to develop a way to improve the readability of billboards, and based on speed, angle of approach and actually blocks of text. It was very â actually, it was a very good study, and got me a job in an ad agency. And they also decided that I had to â to give me a very ugly Plexiglas trophy for it. And another point â why I ’ m here â is that the day I went to pick up the Plexiglas trophy, I rented a tuxedo for the first time in my life, picked the thing â didn ’ t have any friends. On my way out, I had to break a fight apart. Somebody was hitting somebody else with brass knuckles. They were in tuxedos, and fighting. It was very ugly. And also â advertising people do that all the time â â and I â well, what happened is when I went back, it was on the way back to my car, the **guy** who got hit decided to grab a gun â I don ’ t know why he had a gun â and shoot the first person he decided to be his aggressor. The first person was wearing a black tie, a tuxedo. It was me. Luckily, it wasn ’ t fatal, as you can all see. And, even more luckily, the **guy** said that he was sorry and I bribed him for compensation money, otherwise I press charges. And that ’ s how â with this money I paid for a ticket to come to the United States in 1983, and that ’ s very â the basic reason I ’ m talking to you here today : because I got shot. Well, when I started working with my own work, I decided that I shouldn ’ t do images. You know, I became â I took this very iconoclastic approach. Because when I decided to go into advertising, I wanted to do â I wanted to airbrush naked people on ice, for whiskey commercials, that ’ s what I really wanted to do. But I â they didn ’ t let me do it, so I just â you know, they would only let me do other things. But I wasn ’ t into selling whiskey ; I was into selling ice. The first works were actually objects. It was kind of a mixture of found object, product design and advertising. And I called them relics. They were displayed first at Stux Gallery in 1983. This is the clown skull. Is a remnant of a race of â a very evolved race of entertainers. They lived in Brazil, long time ago. This is the Ashanti joystick. Unfortunately, it has become obsolete because it was designed for Atari platform. A Playstation II is in the works, maybe for the next **TED** I ’ ll bring it. The rocking podium. This is the pre-Columbian coffeemaker. Actually, the idea came out of an argument that I had at Starbucks, that I insisted that I wasn ’ t having Colombian coffee ; the coffee was actually pre-Columbian. The Bonsai table. The entire Encyclopedia Britannica bound in a single volume, for travel purposes. And the half tombstone, for people who are not dead yet. I wanted to take that into the realm of images, and I decided to make things that had the same identity **conflicts**. So I decided to do work with clouds. Because clouds can mean anything you want. But now I wanted to work in a very low-tech way, so something that would mean at the same time a lump of cotton, a cloud and Durer ’ s praying hands â although this looks a lot more like Mickey Mouse ’ s praying hands. But I was still, you know â this is a kitty cloud. They ’ re called”Equivalents,”after Alfred Stieglitz ’ s work.”The Snail.”But I was still working with sculpture, and I was really trying to go flatter and flatter.”The Teapot.”I had a chance to go to Florence, in â I think it was ’ 94, and I saw Ghiberti ’ s”Door of Paradise.”And he did something that was very tricky. He put together two different media from different periods of time. First, he got an age-old way of making it, which was relief, and he worked this with three- point perspective, which was brand-new technology at the time. And it ’ s totally overkill. And your eye doesn ’ t know which level to read. And you become trapped into this kind of representation. So I decided to make these very simple renderings, that at first they are taken as a line drawing â you know, something that ’ s very â and then I did it with wire. The idea was to â because **everybody** overlooks white â like pencil drawings, you know? And they would look at it â”Ah, it ’ s a pencil drawing.”Then you have this double take and see that it ’ s actually something that existed in time. It had a physicality, and you start going deeper and deeper into sort of narrative that goes this way, towards the image. So this is”Monkey with Leica.””Relaxation.””Fiat Lux.”And the same way the history of representation evolved from line drawings to shaded drawings. And I wanted to deal with other subjects. I started taking that into the realm of landscape, which is something that ’ s almost a picture of nothing. I made these pictures called”Pictures of Thread,”and I named them after the amount of yards that I used to represent each picture. These always end up being a photograph at the end, or more like an etching in this case. So this is a lighthouse. This is”6, 500 Yards,”after Corot.”9, 000 Yards,”after Gerhard Richter. And I don ’ t know how many yards, after John Constable. Departing from the lines, I decided to tackle the idea of points, like which is more similar to the type of representation that we find in photographs themselves. I had met a group of children in the Caribbean island of Saint Kitts, and I did work and play with them. I got some photographs from them. Upon my arrival in New York, I decided â they were children of sugar plantation workers. And by manipulating sugar over a black paper, I made portraits of them. These are â â Thank you. This is”Valentina, the Fastest.”It was just the name of the child, with the little thing you get to know of somebody that you meet very briefly.”Valicia.””Jacynthe.”But another layer of representation was still introduced. Because I was doing this while I was making these pictures, I realized that I could add still another thing I was trying to make a subject â something that would interfere with the themes, so chocolate is very good, because it has â it brings to mind ideas that go from scatology to romance. And so I decided to make these pictures, and they were very large, so you had to walk away from it to be able to see them. So they ’ re called”Pictures of Chocolate.”Freud probably could explain chocolate better than I. He was the first subject. And Jackson Pollock also. Pictures of crowds are particularly **interesting**, because, you know, you go to that â you try to figure out the threshold with something you can define very easily, like a face, goes into becoming just a texture.”Paparazzi.”I used the dust at the Whitney Museum to render some pieces of their collection. And I picked minimalist pieces because they ’ re about specificity. And you render this with the most non-specific material, which is dust itself. Like, you know, you have the skin particles of every single museum visitor. They do a DNA scan of this, they will come up with a great mailing list. This is Richard Serra. I bought a computer, and [they] told me it had millions of colors in it. You know an artist ’ s first response to this is, who counted it? You know? And I realized that I never worked with color, because I had a hard time controlling the idea of single colors. But once they ’ re applied to numeric structure, then you can feel more comfortable. So the first time I worked with colors was by making these mosaics of Pantone swatches. They end up being very large pictures, and I photographed with a very large camera â an 8x10 camera. So you can see the surface of every single swatch â like in this picture of Chuck Close. And you have to walk very far to be able to see it. Also, the reference to Gerhard Richter ’ s use of color charts â and the idea also entering another realm of representation that ’ s very common to us today, which is the bit map. I ended up narrowing the subject to Monet ’ s”Haystacks.”This is something I used to do as a joke â you know, make â the same like â Robert Smithson ’ s”Spiral Jetty”â and then leaving traces, as if it was done on a tabletop. I tried to prove that he didn ’ t do that thing in the Salt Lake. But then, just doing the models, I was trying to explore the relationship between the model and the original. And I felt that I would have to actually go there and make some earthworks myself. I opt for very simple line drawings â kind of **stupid** looking. And at the same time, I was doing these very large constructions, being 150 meters away. Now I would do very small ones, which would be like â but under the same light, and I would show them together, so the viewer would have to really figure it out what one he was looking. I wasn ’ t **interested** in the very large things, or in the small things. I was more **interested** in the things in between, you know, because you can leave an enormous range for ambiguity there. This is like you see â the size of a person over there. This is a pipe. A hanger. And this is another thing that I did â you know working â **everybody** loves to watch somebody draw, but not many people have a chance to watch somebody draw in â a lot of people at the same time, to evidence a single drawing. And I love this work, because I did these cartoonish clouds over Manhattan for a period of two months. And it was quite wonderful, because I had an interest â an early interest â in theater, that ’ s justified on this thing. In theater, you have the character and the actor in the same place, trying to **negotiate** each other in front of an audience. And in this, you ’ d have like a â something that looks like a cloud, and it is a cloud at the same time. So they ’ re like perfect actors. My interest in acting, especially bad acting, goes a long way. Actually, I once paid like 60 dollars to see a very great actor to do a version of”King Lear,”and I felt really robbed, because by the time the actor started being King Lear, he stopped being the great actor that I had paid money to see. On the other hand, you know, I paid like three dollars, I think â and I went to a warehouse in Queens to see a version of”Othello”by an amateur group. And it was quite fascinating, because you know the **guy** â his name was Joey Grimaldi â he impersonated the Moorish general â you know, for the first three minutes he was really that general, and then he went back into plumber, he worked as a plumber, so â plumber, general, plumber, general â so for three dollars, I saw two tragedies for the price of one. See, I think it ’ s not really about impression, making people fall for a really perfect illusion, as much as it is to make â I usually work at the lowest threshold of visual illusion. Because it ’ s not about fooling somebody, it ’ s actually giving somebody a measure of their own belief : how much you want to be fooled. That ’ s why we pay to go to magic shows and things like that. Well, I think that ’ s it. My time is nearly up. Thank you very much.

Text Sample 538: POS Dim 4 – 2003 – Score 10.00 – 2003/t000405.txt

When you think about resilience and technology it ’ s actually much easier. You ’ re going to see some other speakers today, I already know, who are going to talk about breaking-bones **stuff**, and, of course, with technology it never is. So it ’ s very easy, comparatively speaking, to be resilient. I think that, if we look at what happened on the Internet, with such an incredible last half a dozen years, that it ’ s hard to even get the right analogy for it. A lot of how we decide, how we ’ re supposed to react to things and what we ’ re supposed to expect about the future depends on how we bucket things and how we categorize them. And so I think the tempting analogy for the boom-bust that we just went through with the Internet is a gold rush. It ’ s easy to think of this analogy as very different from some of the other things you might pick. For one thing, both were very real. In 1849, in that Gold Rush, they took over $700 million worth of gold out of California. It was very real. The Internet was also very real. This is a real way for humans to communicate with each other. It ’ s a big deal. Huge boom. Huge boom. Huge bust. Huge bust. You keep going, and both things are lots of hype. I don ’ t have to remind you of all the hype that was **involved** with the Internet — like GetRich. com. But you had the same thing with the Gold Rush.”Gold. Gold. Gold.”Sixty-eight rich men on the Steamer Portland. Stacks of yellow metal. Some have 5, 000. Many have more. A few bring out 100, 000 dollars each. People would get very excited about this when they read these articles.”The Eldorado of the United States of America : the discovery of inexhaustible gold mines in California.”And the parallels between the Gold Rush and the Internet Rush continue very strongly. So many people left what they were doing. And what would happen is — and the Gold Rush went on for years. People on the East Coast in 1849, when they first started to get the news, they thought,”Ah, this isn ’ t real.”But they keep hearing about people getting rich, and then in 1850 they still hear that. And they think it ’ s not real. By about 1852, they ’ re thinking,”Am I the **stupidest** person on Earth by not rushing to California?”And they start to decide they are. These are community affairs, by the way. Local communities on the East Coast would get together and whole teams of 10, 20 people would caravan across the United States, and they would form companies. These were typically not solitary efforts. But no matter what, if you were a lawyer or a banker, people dropped what they were doing, no matter what skill set they had, to go pan for gold. This **guy** on the left, Dr. Richard Beverley Cole, he lived in Philadelphia and he took the Panama route. They would take a ship down to Panama, across the isthmus, and then take another ship north. This **guy**, Dr. Toland, went by covered wagon to California. This has its parallels, too. Doctors leaving their practices. These are both very successful — a physician in one case, a surgeon in the other. Same thing happened on the Internet. You get DrKoop. com. In the Gold Rush, people literally jumped ship. The San Francisco harbor was clogged with 600 ships at the peak because the ships would get there and the crews would abandon to go search for gold. So there were literally 600 captains and 600 ships. They turned the ships into hotels, because they couldn ’ t sail them anywhere. You had dotcom fever. And you had gold fever. And you saw some of the excesses that the dotcom fever created and the same thing happened. The fort in San Francisco at the time had about 1, 300 soldiers. Half of them deserted to go look for gold. And they wouldn ’ t let the other half out to go look for the first half because they were afraid they wouldn ’ t come back. And one of the soldiers wrote home, and this is the sentence that he put :”The struggle between right and six dollars a month and wrong and 75 dollars a day is a rather severe one.”They had bad burn rate in the Gold Rush. A very bad burn rate. This is actually from the Klondike Gold Rush. This is the White Pass Trail. They loaded up their mules and their horses. And they didn ’ t plan right. And they didn ’ t know how far they would really have to go, and they overloaded the horses with hundreds and hundreds of pounds of **stuff**. In fact it was so bad that most of the horses died before they could get where they were going. It got renamed the”Dead Horse Trail.”And the Canadian Minister of the Interior wrote this at the time :”Thousands of pack horses lie dead along the way, sometimes in bunches under the cliffs, with pack saddles and packs where they ’ ve fallen from the rock above, sometimes in tangled masses, filling the mud holes and furnishing the only footing for our poor pack animals on the march, often, I regret to say, exhausted, but still alive, a fact we were unaware of, until after the miserable wretches turned beneath the hooves of our cavalcade. The eyeless sockets of the pack animals everywhere account for the myriads of ravens along the road. The inhumanity which this trail has been witness to, the heartbreak and suffering which so many have undergone, cannot be imagined. They certainly cannot be described.”And you know, without the smell that would have accompanied that, we had the same thing on the

Internet : very bad burn rate calculations. I ' ll just play one of these and you ' ll remember it. This is a commercial that was played on the Super Bowl in the year 2000. : Bride #1 : You said you had a large selection of invitations. Clerk : But we do. Bride #2 : Then why does she have my invitation? Announcer : What may be a little thing to some... Bride #3 : You are mine, little man. Announcer : Could be a really big deal to you. Husband #1 : Is that your wife? Husband #2 : Not for another 15 minutes. Announcer : After all, it ' s your special day. OurBeginning. com. Life ' s an event. Announce it to the world. Jeff Bezos : It ' s very difficult to figure out what that ad is for. But they spent three and a half million dollars in the 2000 Super Bowl to **air** that ad, even though, at the time, they only had a million dollars in annual revenue. Now, here ' s where our analogy with the Gold Rush starts to diverge, and I think rather severely. And that is, in a gold rush, when it ' s over, it ' s over. Here ' s this **guy** :"There are many men in Dawson at the present time who feel keenly disappointed. They ' ve come thousands of miles on a perilous trip, risked life, health and property, spent months of the most arduous labor a man can perform and at length with expectations raised to the highest pitch have reached the coveted goal only to discover the fact that there is nothing here for them."And that was, of course, the very common story. Because when you take out that last piece of gold — and they did incredibly quickly. I mean, if you look at the 1849 Gold Rush — the entire American river region, within two years — every stone had been turned. And after that, only big companies who used more sophisticated mining technologies started to take gold out of there. So there ' s a much better analogy that allows you to be incredibly optimistic and that analogy is the electric industry. And there are a lot of similarities between the Internet and the electric industry. With the electric industry you actually have to — one of them is that they ' re both sort of thin, horizontal, enabling layers that go across lots of different industries. It ' s not a specific thing. But electricity is also very, very broad, so you have to sort of narrow it down. You know, it can be used as an incredible means of transmitting power. It ' s an incredible means of coordinating, in a very fine-grained way, information flows. There ' s a bunch of things that are **interesting** about electricity. And the part of the electric revolution that I want to focus on is sort of the golden age of appliances. The killer app that got the world ready for appliances was the light bulb. So the light bulb is what wired the world. And they weren ' t thinking about appliances when they wired the world. They were really thinking about — they weren ' t putting electricity into the home ; they were putting lighting into the home. And, but it really — it got the electricity. It took a long time. This was a huge — as you would expect — a huge capital build out. All the streets had to be torn up. This is work going on down in lower Manhattan where they built some of the first electric power generating stations. And they ' re tearing up all the streets. The Edison Electric Company, which became Edison General Electric, which became General Electric, paid for all of this digging up of the streets. It was incredibly expensive. But that is not the — and that ' s not the part that ' s really most similar to the Web. Because, remember, the Web got to stand on top of all this heavy infrastructure that had been put in place because of the long-distance **phone** network. So all of the cabling and all of the heavy infrastructure — I ' m going back now to, sort of, the explosive part of the Web in 1994, when it was growing 2, 300 percent a year. How could it grow at 2, 300 percent a year in 1994 when people weren ' t really **investing** in the Web? Well, it was because that heavy infrastructure had already been laid down. So the light bulb laid down the heavy infrastructure, and then home appliances started coming into being. And this was huge. The first one was the electric fan — this was the 1890 electric fan. And the appliances, the golden age of appliances really lasted — it depends how you want to measure it — but it ' s anywhere from 40 to 60 years. It goes on a long time. It starts about 1890. And the electric fan was a big success. The electric iron, also very big. By the way, this is the beginning of the asbestos lawsuit. There ' s asbestos under that handle there. This is the first vacuum cleaner, the 1905 Skinner Vacuum, from the Hoover Company. And this one weighed 92 pounds and took two people to operate and cost a quarter of a car. So it wasn ' t a big seller. This was truly, truly an early-adopter product — the 1905 Skinner Vacuum. But three years later, by 1908, it weighed 40 pounds. Now, not all these things were highly successful. This is the electric tie press, which never really did catch on. People, I guess, decided that they would not wrinkle their ties. These never really caught on either : the electric shoe warmer and drier. Never a big seller. This came in, like, six different colors. I don ' t know why. But I thought, you know, sometimes it ' s just not the right time for an invention ; maybe it ' s time to give this one another shot. So I thought we could build a Super Bowl ad for this. We ' d need the right partner. And I thought that really — I thought that would really work, to give that another shot. Now, the toaster was huge because they used to make toast on open fires, and it took a lot of time and attention. I want to point out one thing. This is — you **guys** know what this is. They hadn ' t invented the electric socket yet. So this was — remember, they didn ' t wire the houses for electricity. They wired them for lighting. So your — your appliances would plug in. They would — each room typically had a light bulb socket at the top. And you ' d plug it in there. In fact, if you ' ve seen the Carousel of Progress at Disney World, you ' ve seen this. Here are the cables coming up into this light fixture. All the appliances plug in there. And you would just unscrew your light bulb if you wanted to plug in an appliance. The next thing that really was a big, big deal was the washing machine. Now, this was an object of much envy and lust. **Everybody** wanted one of these electric washing machines. On the left-hand side, this was the soapy water. And there ' s a rotor there — that this motor is spinning. And it would clean your clothes. This is the clean rinse-water. So you ' d take the clothes out of here, put them in here, and then you ' d run the clothes through this electric wringer. And this was a big deal. You ' d keep this on your porch. It was a little bit messy and kind of a pain. And you ' d run a long cord into the house where you could screw it into your light socket. And that ' s actually kind of an important point in my presentation, because they hadn ' t invented the off switch. That was to come much later — the off switch on appliances — because it didn ' t make any sense. I mean, you didn ' t want this thing clogging up a light socket. So you know, when you were done with it, you unscrewed it. That ' s what you did. You didn ' t turn it off. And as I said before, they hadn ' t invented the electric outlet either, so the washing machine was a particularly dangerous device. And there are — when you research this, there are gruesome descriptions of people getting their hair and clothes caught in these devices. And they couldn ' t yank the cord out because it was screwed into a light socket inside the house. And there was no off switch, so it wasn ' t very good. And you might think that that was incredibly **stupid** of our ancestors to be plugging things into a light socket like this. But, you know, before I get too far into condemning our ancestors, I thought I ' d show you : this is my **conference** room. This is a total kludge, if you ask me. First of all, this got installed upside down. This light socket — and so the cord keeps falling out, so I taped it in. This is supposed — don ' t even get me started. But that ' s not the worst one. This is what it looks like under my desk. I took this picture just two days ago. So we really haven ' t progressed that much since 1908. It ' s a total, total mess. And, you know, we think it ' s getting better, but have you tried to install 802. 11 yourself? I challenge you to try. It ’ s very hard. I know Ph. D. s in Computer Science — this process has brought them to tears, absolute tears. And that ’ s assuming you already have DSL in your house. Try to get DSL installed in your house. The engineers who do it everyday can ’ t do it. They have to — typically, they come three times. And one friend of mine was telling me a story : not only did they get there and have to wait, but then the engineers, when they finally did get there, for the third time, they had to call somebody. And they were really happy that the **guy** had a speakerphone because then they had to wait on hold for an hour to talk to somebody to give them an access code after they got there. So we ’ re not — we ’ re pretty kludge-y ourselves. By the way, DSL is a kludge. I mean, this is a twisted pair of copper that was never designed for the purpose it ’ s being put to — you know it ’ s the whole thing — we ’ re very, very primitive. And that ’ s kind of the point. Because, you know, resilience — if you think of it in terms of the Gold Rush, then you ’ d be pretty depressed right now because the last nugget of gold would be gone. But the good thing is, with innovation, there isn ’ t a last nugget. Every new thing creates two new questions and two new opportunities. And if you believe that, then you believe that where we are — this is what I think — I believe that where we are with the incredible kludge — and I haven ’ t even talked about user interfaces on the Web — but there ’ s so much kludge, so much terrible **stuff** — we are at the 1908 Hurley washing machine stage with the Internet. That ’ s where we are. We don ’ t get our hair caught in it, but that ’ s the level of primitiveness of where we are. We ’ re in 1908. And if you believe that, then **stuff** like this doesn ’ t bother you. This is 1996 :”All the negatives add up to making the online experience not worth the trouble.”1998 :”Amazon. toast.”In 1999 :”Amazon. bomb.”My mom hates this picture. She — but you know, if you really do believe that it ’ s the very, very beginning, if you believe it ’ s the 1908 Hurley washing machine, then you ’ re incredibly optimistic. And I do think that that ’ s where we are. And I do think there ’ s more innovation ahead of us than there is behind us. And in 1917, Sears — I want to get this exactly right. This was the advertisement that they ran in 1917. It says :”Use your electricity for more than light.”And I think that ’ s where we are. We ’ re very, very early. Thank you very much.

Text Sample 539: POS Dim 4 – 1984 – Score 7.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ’ ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a computer and really address the qualities of the human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, the **intelligence** of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ’ ve been able to collect, which I think illustrate this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a slide taken from a TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ’ s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ’ s the distance you should sit away from the TV set. Now we ’ ve put people 18 inches in front of a TV, and all the artifacts that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow mask, the scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ’ m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ’ m very **interested** in touch-sensitive displays. High-tech, high-touch. Isn ’ t that what some of you said? It ’ s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ’ re not : it ’ s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on the market these systems that have just a few light emitting diodes on the side and are very low resolution, it ’ s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ’ t have to pick them up, and people don ’ t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ’ re typing — what you have — you want to now put something — first of all, you ’ ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ’ ve got to sort of find that mouse. Then you find the mouse, and you ’ re going to have to wiggle it a little bit to see where the cursor is on the screen. And then when you finally see where it is, then you ’ ve got to move it to get the cursor over there, and then —”Bang”— you ’ ve got to hit a button or do whatever. That ’ s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ’ m trying to do is just illustrate the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of the computer field, is when you have a bug that you can ’ t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the substrate, which it usually is. So we found that that actually was a feature in the sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ’ s pressure-sensitive. And it ’ s pressure-sensitive to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ’ t figure out how to come in the other direction. But let me get rid of the slide, and let ’ s see if this comes on. OK.

So there is the pressure-sensitive display in operation. The person ' s just, if you will, pushing on the screen to make a curve. But this is the **interesting** part. I want to stop it for a second because the movie is very badly made. And the particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an anvil on a fuzzy rug and the other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across the screen, and yet you touch the ping-pong ball very lightly and it just scoots across the screen. And what you can do — oops, I didn ' t mean to do that — what you can do is actually feed back to the user the feeling of the physical properties. So again, they don ' t have to be weight ; they could be a general trying to move troops, and he ' s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the interface there are physical properties in that transducer — in this case it ' s pressure and touches — that allow you to present things to the user that you could never present before. So it ' s not simply looking at the quality or, if you will, the luxury of that interface, but it ' s actually looking at the idea of presenting things that previously couldn ' t be presented before. I want to move on to another example, which is one of a different sort, where we ' re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ' re going to take this book, if you will, and it ' s going to come alive. You ' re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are the experts in monologue making : you make a film and it has a well-formed beginning, middle and end, and in some sense the art of it is that. And you then say,"There ' s an opportunity for making conversational movies."Well, what does that mean? And it sort of nibbles at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ' ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the movie has information about itself. So it knows, or at least there is computer-readable information in the medium itself. It ' s just not a static movie frame. That ' s one thing. The other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the cookbook, the"Larousse Gastronomique."And I think I use the example all too often, but it ' s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to the end of the recipe and it says,"Cook until done."Now, that would be, if you will, the top green track, which doesn ' t mean too much. But you might have to elaborate for me or for somebody who isn ' t an expert, and say,"Cook at 380 degrees for 45 minutes."And then for a real beginner, you would go down even further and elaborate more — say,"Open the oven, preheat, wait for the light to go out, open the door, don ' t leave it open too long, put the penguin in and shut the door..."whatever. And that ' s a much more elaborate one than you dribble back. That ' s one kind of use of random access. And the other is where you want to explain the same thing in different ways. If you ' re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ' s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ' s a rather boring book, but I ' m afraid sometimes you have to do boring books because your sponsors aren ' t necessarily **interested** in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ' t even know what vintage the transmission is, but let me just show you very quickly some of it, and we ' ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of the transmission, and as you rub your finger across the transmission it highlights the various parts. Narrator : When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, superimposed over the illustration. NN : This is about the oil pan, or the oil filter and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get back to the illustration, but in this case it ' s not a single frame, but it ' s actually a movie of someone coming into the frame and doing the repair that ' s described in the text. The two-headed slider is a speed control that allows me to watch the movie at various speeds, in forward or reverse. And the movie is displayed as a full frame movie. I can go back to the beginning... and play the movie at full speed. Here ' s another step-by-step procedure, only in this case — NN : Okay, this movie is... **Everybody** ' s heard of sound-sync movies — this is text-sync movies, so as the movie plays, the text gets highlighted. We highlight the text as we go through the movie. Repairman :... Not too far out. Front poles, preferably. Don ' t loosen them too far. If you loosen them too far, you ' ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ' m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ' ve been doing — in this case, in Senegal — where we have tried to use personal computers as a pedagogical medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ' s a story I ' d like to tell. And that was when, before we did this in any developing countries — we ' re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ' ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ' t read, didn ' t even make it in the lowest section of the school ' s classes — and was pretty much not in school, though physically there. But did hang around the, quote,"computer room,"where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very **interesting**. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,"Let me show you how this works,"and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ’ t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ’ d actually come to see — not the computer room — they went upstairs and they said,”This is absolutely remarkable! That child was very articulate and showed us and even dealt with the things he couldn ’ t do automatically with that manual. It was just absolutely fantastic.”The principal said,”There ’ s a dreadful mistake, because that child can ’ t read. And you obviously have been hoodwinked or you ’ ve talked about somebody else.”And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,”Can you read?”And the child said,”No, I can ’ t.”And then they said,”But wait a minute. You just looked through that manual and you found...”and he said,”Oh, but that ’ s not reading.”And so they said,”Well, what ’ s reading then?”He says,”Well, reading is this junk they give me in little books to read. It ’ s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return.”And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ’ m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ’ s not, but we think speech —”My God, little children pick it up somehow, and by the age of two they ’ re doing a mediocre job, and by three and four they ’ re speaking reasonably well. And yet you ’ ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder.”Well, it ’ s not harder. What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ’ s going to help you. So what happens is you go to school and people say,”Just believe me, you ’ re going to like it. You ’ re going to like reading,”and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading **stuff** on the screen has a huge payoff, and it ’ s a lot of fun. And in fact, it ’ s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ’ s the girl on the left leaning with her chalkboard, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ’ s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example — as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And the reason it remains a favorite project is that we were asked to do a teleconferencing system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ' s faces. So if I wanted to call my friend Peter Sprague on the **phone**, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ' s uncanny : there ' s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ' s something that didn ' t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a **phone** or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,"He or she wouldn ' t agree,"and the empty chair is that person and the spatiality is crucial. So we said,"Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ' s site you ' d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ' t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ' ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ' s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ' s site, he ' ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.

Text Sample 540: POS Dim 4 – 2016 – Score 21.00 – 2016/t001270.txt

The conventional wisdom about our world today is that this is a time of terrible decline. And that ' s not surprising, given the bad news all around us, from ISIS to inequality, political **dysfunction**, **climate** change, Brexit, and on and on. But here ' s the thing, and this may sound a little weird. I actually don ' t buy this gloomy narrative, and I don ' t think you should either. Look, it ' s not that I don ' t see the problems. I read the same headlines that you do. What I dispute is the conclusion that so many people draw from them, namely that we ' re all screwed because the problems are unsolvable and our governments are useless. Now, why do I say this? It ' s not like I ' m particularly optimistic by nature. But something about the media ' s constant doom-mongering with its fixation on problems and not on answers has always really bugged me. So a few years ago I decided, well, I ' m a journalist, I should see if I can do any better by going around the world and actually asking folks if and how they ' ve tackled their big economic and political challenges. And what I found astonished me. It turns out that there are remarkable signs of progress out there, often in the most unexpected places, and they ' ve convinced me that our great global challenges may not be so unsolvable after all. Not only are there theoretical fixes ;

those fixes have been tried. They ’ ve worked. And they offer hope for the rest of us. I ’ m going to show you what I mean by telling you about how three of the countries I **visited** — Canada, Indonesia and Mexico — overcame three supposedly impossible problems. Their stories matter because they contain tools the rest of us can use, and not just for those particular problems, but for many others, too. When most people think about my homeland, Canada, today, if they think about Canada at all, they think cold, they think boring, they think polite. They think we say”sorry”too much in our funny accents. And that ’ s all true. Sorry. But Canada ’ s also important because of its triumph over a problem currently tearing many other countries apart : immigration. Consider, Canada today is among the world ’ s most welcoming nations, even compared to other immigration- friendly countries. Its per capita immigration rate is four times higher than France ’ s, and its percentage of foreign-born residents is double that of Sweden. Meanwhile, Canada admitted 10 times more Syrian **refugees** in the last year than did the United States. And now Canada is taking even more. And yet, if you ask Canadians what makes them proudest of their country, they rank”multiculturalism,”a dirty word in most places, second, ahead of hockey. Hockey. In other words, at a time when other countries are now frantically building new barriers to keep foreigners out, Canadians want even more of them in. Now, here ’ s the really **interesting** part. Canada wasn ’ t always like this. Until the mid-1960s, Canada followed an explicitly racist immigration **policy**. They called it”White Canada,”and as you can see, they were not just talking about the snow. So how did that Canada become today ’ s Canada? Well, despite what my mom in Ontario will tell you, the answer had nothing to do with virtue. Canadians are not inherently better than anyone else. The real explanation **involves** the man who became Canada ’ s **leader** in 1968, Pierre Trudeau, who is also the father of the current prime minister. The thing to know about that first Trudeau is that he was very different from Canada ’ s previous **leaders**. He was a French speaker in a country long-dominated by its English elite. He was an intellectual. He was even kind of groovy. I mean, seriously, the **guy** did yoga. He hung out with the Beatles. And like all hipsters, he could be infuriating at times. But he nevertheless pulled off one of the most progressive transformations any country has ever seen. His formula, I ’ ve learned, **involved** two parts. First, Canada threw out its old race-based immigration rules, and it replaced them with new color-blind ones that emphasized education, experience and language skills instead. And what that did was greatly increase the odds that newcomers would contribute to the economy. Then part two, Trudeau created the world ’ s first **policy** of official multiculturalism to promote integration and the idea that **diversity** was the key to Canada ’ s identity. Now, in the years that followed, Ottawa kept pushing this message, but at the same time, ordinary Canadians soon started to see the economic, the material benefits of multiculturalism all around them. And these two influences soon combined to create the passionately open-minded Canada of today. Let ’ s now turn to another country and an even tougher problem, Islamic extremism. In 1998, the people of Indonesia took to the streets and overthrew their longtime dictator, Suharto. It was an amazing moment, but it was also a scary one. With 250 million people, Indonesia is the largest Muslim-majority country on Earth. It ’ s also hot, huge and unruly, made up of 17, 000 islands, where people speak close to a thousand languages. Now, Suharto had been a dictator, and a nasty one. But he ’ d also been a pretty effective tyrant, and he ’ d always been careful to keep religion out of politics. So experts feared that without him keeping a lid on things, the country would explode, or religious extremists would take over and turn Indonesia into a tropical version of Iran. And that ’ s just what seemed to happen at first. In the country ’ s first free elections, in 1999, Islamist parties scored 36 percent of the vote, and the islands burned as riots and terror attacks killed thousands. Since then, however, Indonesia has taken a surprising turn. While ordinary folks have grown more pious on a personal level — I saw a lot more headscarves on a recent **visit** than I would have a decade ago — the country ' s politics have moved in the opposite direction. Indonesia is now a pretty decent democracy. And yet, its Islamist parties have steadily lost **support**, from a high of about 38 percent in 2004 down to 25 percent in 2014. As for terrorism, it ' s now extremely rare. And while a few Indonesians have recently joined ISIS, their number is tiny, far fewer in per capita terms than the number of Belgians. Try to think of one other Muslim-majority country that can say all those same things. In 2014, I went to Indonesia to ask its current president, a soft-spoken technocrat named Joko Widodo,"Why is Indonesia thriving when so many other Muslim states are dying?""Well, what we realized,"he told me,"is that to deal with extremism, we needed to deal with inequality first."See, Indonesia ' s religious parties, like similar parties elsewhere, had tended to focus on things like reducing poverty and cutting corruption. So that ' s what Joko and his predecessors did too, thereby stealing the Islamists ' thunder. They also cracked down hard on terrorism, but Indonesia ' s democrats have learned a key lesson from the dark years of dictatorship, namely that repression only creates more extremism. So they waged their war with extraordinary delicacy. They used the police instead of the army. They only detained suspects if they had enough evidence. They held public trials. They even sent liberal imams into the jails to persuade the jihadists that terror is un-Islamic. And all of this paid off in spectacular fashion, creating the kind of country that was unimaginable 20 years ago. So at this point, my optimism should, I hope, be starting to make a bit more sense. Neither immigration nor Islamic extremism are impossible to deal with. Join me now on one last trip, this time to Mexico. Now, of our three stories, this one probably surprised me the most, since as you all know, the country is still struggling with so many problems. And yet, a few years ago, Mexico did something that many other countries from France to India to the United States can still only dream of. It shattered the political paralysis that had gripped it for years. To understand how, we need to rewind to the year 2000, when Mexico finally became a democracy. Rather than use their new freedoms to fight for reform, Mexico ' s politicians used them to fight one another. Congress deadlocked, and the country ' s problems — drugs, poverty, crime, corruption — spun out of control. Things got so bad that in 2008, the Pentagon warned that Mexico risked collapse. Then in 2012, this **guy** named Enrique Pea Nieto somehow got himself elected president. Now, this Pea hardly inspired much **confidence** at first. Sure, he was handsome, but he came from Mexico ' s corrupt old ruling party, the PRI, and he was a notorious womanizer. In fact, he seemed like such a pretty boy lightweight that women called him"bombn,"sweetie, at campaign rallies. And yet this same bombn soon surprised everyone by hammering out a truce between the country ' s three warring political parties. And over the next 18 months, they together passed an incredibly comprehensive set of reforms. They busted open Mexico ' s smothering monopolies. They liberalized its rusting energy **sector**. They restructured its failing schools, and much more. To appreciate the scale of this accomplishment, try to imagine the US Congress passing immigration reform, campaign finance reform and banking reform. Now, try to imagine Congress doing it all at the same time. That ' s what Mexico did. Not long ago, I met with Pea and asked how he managed it all. The President flashed me his famous twinkly smile — and told me that the short answer was"compromiso,"compromise. Of course, I pushed him for details, and the long answer that came out was essentially"compromise, compromise and more compromise.”See, Pea knew that he needed to build **trust** early, so he started talking to the opposition just days after his election. To ward off pressure from special interests, he kept their meetings small and secret, and many of the participants later told me that it was this intimacy, plus a lot of shared tequila, that helped build **confidence**. So did the fact that all decisions had to be unanimous, and that Pea even agreed to pass some of the other party ’ s priorities before his own. As Santiago Creel, an opposition senator, put it to me,”Look, I ’ m not saying that I ’ m special or that anyone is special, but that group, that was special.”The proof? When Pea was sworn in, the pact held, and Mexico moved forward for the first time in years. Bueno. So now we ’ ve seen how these three countries overcame three of their great challenges. And that ’ s very nice for them, right? But what good does it do the rest of us? Well, in the course of studying these and a bunch of other success stories, like the way Rwanda pulled itself back together after civil war or Brazil has reduced inequality, or South Korea has kept its economy growing faster and for longer than any other country on Earth, I ’ ve noticed a few common threads. Now, before describing them, I need to add a caveat. I realize, of course, that all countries are unique. So you can ’ t simply take what worked in one, port it to another and expect it to work there too. Nor do specific solutions work forever. You ’ ve got to adapt them as circumstances change. That said, by stripping these stories to their essence, you absolutely can distill a few common tools for problem-solving that will work in other countries and in boardrooms and in all sorts of other contexts, too. Number one, embrace the extreme. In all the stories we ’ ve just looked at, salvation came at a moment of existential peril. And that was no coincidence. Take Canada : when Trudeau took office, he faced two looming dangers. First, though his vast, underpopulated country badly needed more bodies, its preferred source for white workers, Europe, had just stopped exporting them as it finally recovered from World War II. The other problem was that Canada ’ s long cold war between its French and its English communities had just become a hot one. Quebec was threatening to secede, and Canadians were actually killing other Canadians over politics. Now, countries face **crises** all the time. Right? That ’ s nothing special. But Trudeau ’ s genius was to realize that Canada ’ s **crisis** had swept away all the hurdles that usually block reform. Canada had to open up. It had no choice. And it had to rethink its identity. Again, it had no choice. And that gave Trudeau a once-in-a-generation opportunity to break the old rules and write new ones. And like all our other heroes, he was smart enough to seize it. Number two, there ’ s power in promiscuous thinking. Another striking similarity among good problem-solvers is that they ’ re all pragmatists. They ’ ll steal the best answers from wherever they find them, and they don ’ t let details like party or ideology or sentimentality get in their way. As I mentioned earlier, Indonesia ’ s democrats were clever enough to steal many of the Islamists ’ best campaign promises for themselves. They even invited some of the radicals into their governing coalition. Now, that horrified a lot of secular Indonesians. But by forcing the radicals to actually help govern, it quickly exposed the fact that they weren ’ t any good at the job, and it got them mixed up in all of the grubby compromises and petty humiliations that are part of everyday politics. And that hurt their image so badly that they ’ ve never recovered. Number three, please all of the people some of the time. I know I just mentioned how **crises** can grant **leaders** extraordinary freedoms. And that ’ s true, but problem-solving often requires more than just boldness. It takes showing restraint, too, just when that ’ s the last thing you want to do. Take Trudeau : when he took office, he could easily have put his core constituency, that is Canada ’ s French community, first. He could have pleased some of the people all of the time. And Pea could have used his power to keep attacking the opposition, as was traditional in Mexico. Yet he chose to embrace his enemies instead, while forcing his own party to compromise. And Trudeau pushed everyone to stop thinking in tribal terms and to see multiculturalism, not language and not skin color, as what made them quintessentially Canadian. Nobody got everything they wanted, but everyone got just enough that the bargains held. So at this point you may be thinking,"OK, Tepperman, if the fixes really are out there like you keep insisting, then why aren ' t more countries already using them?"It ' s not like they require special powers to pull off. I mean, none of the **leaders** we ' ve just looked at were superheroes. They didn ' t accomplish anything on their own, and they all had plenty of flaws. Take Indonesia ' s first democratic president, Abdurrahman Wahid. This man was so powerfully uncharismatic that he once fell asleep in the middle of his own speech. True story. So what this tells us is that the real obstacle is not **ability**, and it ' s not circumstances. It ' s something much simpler. Making big changes **involves** taking big risks, and taking big risks is scary. Overcoming that fear requires guts, and as you all know, gutsy politicians are painfully rare. But that doesn ' t mean we voters can ' t **demand** courage from our political **leaders**. I mean, that ' s why we put them in office in the first place. And given the state of the world today, there ' s really no other option. The answers are out there, but now it ' s up to us to elect more women and men brave enough to find them, to steal them and to make them work. Thank you.

Text Sample 541: POS Dim 4 – 2016 – Score 20.00 – 2016/t001235.txt

When you come to TEDx, you always think about technology, the world changing, becoming more innovative. You think about the driverless. Everyone ' s talking about driverless cars these days, and I love the concept of a driverless car, but when I go in one, you know, I want it really slow, I want access to the steering wheel and the brake, just in case. I don ' t know about you, but I am not ready for a driverless bus. I am not ready for a driverless airplane. How about a driverless world? And I ask you that because we are increasingly in one. It ' s not supposed to be that way. We ' re number one, the United States is large and in charge. Americanization and globalization for the last several generations have basically been the same thing. Right? Whether it ' s the World Trade Organization or it ' s the IMF, the World Bank, the Bretton Woods Accord on currency, these were American institutions, our values, our friends, our allies, our money, our standards. That was the way the world worked. So it ' s sort of **interesting**, if you want to look at how the US looks, here it is. This is our view of how the world is run. President Obama has got the red carpet, he goes down Air Force One, and it feels pretty good, it feels pretty comfortable. Well, I don ' t know how many of you saw the China trip last week and the G20. Oh my God. Right? This is how we landed for the most important meeting of the world ' s **leaders** in China. The National Security Advisor was actually spewing expletives on the tarmac — no red carpet, kind of left out the bottom of the plane along with all the media and **everybody** else. Later on in the G20, well, there ' s Obama. Hi, George. Hi, Norman. They look like they ' re about to get into a cage match, right? And they did. It was 90 minutes long, and they talked about Syria. That ' s what Putin wanted to talk about. He ' s increasingly calling the shots. He ' s the one willing to do **stuff** there. There ' s not a lot of mutual like or **trust**, but it ' s not as if the Americans are telling him what to do. How about when the whole 20 are getting together? Surely, when the **leaders** are all onstage, then the Americans are pulling their weight. Uh-oh. Xi Jinping seems fine. Angela Merkel has — she always does — that look, she always does that. But Putin is telling Turkish president Erdogan what to do, and Obama is like, what ' s going on over there? You see. And the problem is it ' s not a G20, the problem is it ' s a G-Zero world that we live in, a world order where there is no single country or alliance that can meet the challenges of global **leadership**. The G20 doesn ' t work, the G7, all of our friends, that ' s history. So globalization is continuing. Goods and services and people and capital are moving across borders faster and faster than ever before, but Americanization is not. So if I ' ve convinced you of that, I want to do two things with the rest of this talk. I want to talk about the implications of that for the whole world. I ' ll go around it. And then I want to talk about what we think right here in the United States and in New York. So why? What are the implications. Why are we here? Well, we ' re here because the United States, we spent two trillion dollars on wars in Iraq and Afghanistan that were failed. We don ' t want to do that anymore. We have large numbers of middle and working classes that feel like they ' ve not benefited from promises of globalization, so they don ' t want to see it particularly. And we have an energy revolution where we don ' t need OPEC or the Middle East the way we used to. We produce all that right here in the United States. So the Americans don ' t want to be the global sheriff for security or the architect of global trade. The Americans don ' t want to even be the cheerleader of global values. Well, then you look to Europe, and the most important alliance in the world has been the transatlantic relationship. But it is now weaker than it has been at any point since World War II, all of the **crises**, the Brexit conversations, the hedging going on between the French and the Russians, or the Germans and the Turks, or the **Brits** and the Chinese. China does want to do more **leadership**. They do, but only in the economic sphere, and they want their own values, standards, currency, in competition with that of the US. The Russians want to do more **leadership**. You see that in Ukraine, in the Baltic states, in the Middle East, but not with the Americans. They want their own preferences and order. That ' s why we are where we are. So what happens going forward? Let ' s start easy, with the Middle East. You know, I left a little out, but you get the general idea. Look, there are three reasons why the Middle East has had stability such as it is. Right? One is because there was a willingness to provide some level of military security by the US and allies. Number two, it was easy to take a lot of cheap money out of the ground because oil was expensive. And number three was no matter how bad the **leaders** were, the populations were relatively quiescent. They didn ' t have the **ability**, and many didn ' t have the will to really rise up against. Well, I can tell you, in a G-Zero world, all three of those things are increasingly not true, and so failed states, terrorism, **refugees** and the rest. Does the entire Middle East fall apart? No, the Kurds will do better, and Iraq, Israel, Iran over time. But generally speaking, it ' s not a good look. OK, how about this **guy**? He ' s playing a poor hand very well. There ' s no question he ' s hitting above his weight. But long term — I didn ' t mean that. But long term, long term, if you think that the Russians were antagonized by the US and Europe expanding NATO right up to their borders when we said they weren ' t going to, and the EU encroaching them, just wait until the Chinese put hundreds of billions of dollars in every country around Russia they thought they had influence in. The Chinese are going to dominate it. The Russians are picking up the crumbs. In a G-Zero world, this is going to be a very tense 10 years for Mr. Putin. It ' s not all bad. Right? Asia actually looks a lot better. There are real **leaders** across Asia, they have a lot of political stability. They ' re there for a while. Mr. Modi in India, Mr. Abe, who is probably about to get a third term written in in the Liberal Democratic Party in Japan, of course Xi Jinping who is consolidating enormous power, the most powerful **leader** in China since Mao. Those are the three most important economies in Asia. Now look, there are problems in Asia. We see the sparring over the South China Sea. We see that Kim Jong Un, just in the last couple of days, tested yet another nuclear weapon. But the **leaders** in Asia do not feel the need to wave the flag, to go xenophobic, to actually allow escalation of the geopolitical and cross-border tensions. They want to focus on long-term economic stability and growth. And that ' s what they ' re actually doing. Let ' s turn to Europe. Europe does look a little scared in this environment. So much of what is happening in the Middle East is washing up quite literally onto European shores. You see Brexit and you see the concerns of populism across all of the European states. Let me tell you that over the long term, in a G-Zero world, European expansion will be seen to have gone too far. Europe went right up to Russia, went right down to the Middle East, and if the world were truly becoming more flat and more Americanized, that would be less of a problem, but in a G-Zero world, those countries nearest Russia and nearest the Middle East actually have different economic capabilities, different social stability and different political preferences and systems than core Europe. So Europe was able to truly expand under the G7, but under the G-Zero, Europe will get smaller. Core Europe around Germany and France and others will still work, be functional, stable, wealthy, integrated. But the periphery, countries like Greece and Turkey and others, will not look that good at all. Latin America, a lot of populism, made the economies not go so well. They had been more opposed to the United States for decades. Increasingly, they ' re coming back. We see that in Argentina. We see it with the openness in Cuba. We will see it in Venezuela when Maduro falls. We will see it in Brazil after the impeachment and when we finally see a new legitimate president elected there. The only place you see that is moving in another direction is the unpopularity of Mexican president Pea Nieto. There you could actually see a slip away from the United States over the coming years. The US election matters a lot on that one, too. Africa, right? A lot of people have said it ' s going to be Africa ' s decade, finally. In a G-Zero world, it is absolutely an amazing time for a few African countries, those governed well with a lot of urbanization, a lot of smart people, women really getting into the workforce, entrepreneurship taking off. But for most of the countries in Africa, it ' s going to be a lot more dicey : extreme **climate** conditions, radicalism both from Islam and also Christianity, very poor governance, borders you can ' t defend, lots of forced migration. Those countries can fall off the map. So you ' re really going to see an extreme segregation going on between the winners and the losers across Africa. Finally, back to the United States. What do I think about us? Because there are a lot of upset people, not here at TEDx, I know, but in the United States, my God, after 15 months of campaigning, we should be upset. I understand that. But a lot of people are upset because they say,"Washington ' s broken, we don ' t **trust** the establishment, we hate the media."Heck, even globalists like me are taking it on the chin. Look, I do think we have to recognize, my fellow campers, that when you are being chased by the bear, in the global context, you need not outrun the bear, you need to only outrun your fellow campers. Now, I just told you about our fellow campers. Right? And from that perspective, we look OK. A lot of people in that context say,"Let ' s go dollar. Let ' s go New York real estate. Let ' s send our kids to American universities."You know, our neighbors are awesome : Canada, Mexico and two big bodies of water. You know how much Turkey would love to have neighbors like that? Those are awesome neighbors. Terrorism is a problem in the United States. God knows we know it here in New York. But it ' s a much bigger problem in Europe than the US. It ' s a much bigger problem in the Middle East than it is in Europe. These are factors of large magnitude. We just accepted 10, 000 Syrian **refugees**, and we ' re complaining bitterly about it. You know why? Because they can ' t swim here. Right? I mean, the Turks would love to have only 10, 000 Syrian **refugees**. The Jordanians, the Germans, the **Brits**. Right? That ' s not the situation. That ' s the reality of the United States. Now, that sounds pretty good. Here ' s the challenge. In a G-Zero world, the way you lead is by example. If we know we don ' t want to be the global cop anymore, if we know we ' re not going to be the architect of global trade, we ' re not going to be the cheerleader of global values, we ' re not going to do it the way we used to, the 21st century is changing, we need to lead by example — be so compelling that all these other people are going to still say, it ' s not just they ' re faster campers. Even when the bear is not chasing us, this is a good place to be. We want to emulate them. The election process this year is not proving a good option for leading by example. Hillary Clinton says it ' s going to be like the ' 90s. We can still be that cheerleader on values. We can still be the architect of global trade. We can still be the global sheriff. And Donald Trump wants to bring us back to the ' 30s. He ' s saying,"Our way or the highway. You don ' t like it, lump it."Right? Neither are recognizing a fundamental truth of the G-Zero, which is that even though the US is not in decline, it is getting objectively harder for the Americans to impose their will, even have great influence, on the global order. Are we prepared to truly lead by example? What would we have to do to fix this after November, after the next president comes in? Well, either we have to have another **crisis** that forces us to respond. A depression would do that. Another global financial **crisis** could do this. God forbid, another 9 / 11 could do that. Or, absent **crisis**, we need to see that the hollowing out, the inequality, the challenges that are growing and growing in the United States, are themselves urgent enough to force our **leaders** to change, and that we have those voices. Through our cell **phones**, individually, we have those voices to compel them to change. There is, of course, a third choice, perhaps the most likely one, which is that we do neither of those things, and in four years time you invite me back, and I will give this speech yet again. Thank you very, very much.

Text Sample 542: POS Dim 4 – 2016 – Score 19.00 – 2016/t001399.txt

I was excited to be a part of the"Dream"theme, and then I found out I ' m leading off the"Nightmare?"section of it. And certainly there are things about the **climate crisis** that qualify. And I have some bad news, but I have a lot more good news. I ' m going to propose three questions and the answer to the first one necessarily **involves** a little bad news. But — hang on, because the answers to the second and third questions really are very positive. So the first question is,"Do we really have to change?"And of course, the Apollo Mission, among other things changed the environmental movement, really launched the modern environmental movement. 18 months after this Earthrise picture was first seen on earth, the first Earth Day was organized. And we learned a lot about ourselves looking back at our planet from space. And one of the things that we learned confirmed what the scientists have long told us. One of the most essential facts about the **climate crisis** has to do with the sky. As this picture illustrates, the sky is not the vast and limitless expanse that appears when we look up from the ground. It is a very thin shell of atmosphere surrounding the planet. That right now is the open sewer for our industrial civilization as it ' s currently organized. We are spewing 110 million tons of heat-trapping global warming pollution into it every 24 hours, free of charge, go ahead. And there are many sources of the greenhouse gases, I ' m certainly not going to go through them all. I ' m going to focus on the main one, but agriculture is

involved, diet is **involved**, population is **involved**. Management of forests, transportation, the oceans, the melting of the permafrost. But I ’ m going to focus on the heart of the problem, which is the fact that we still rely on dirty, carbon-based fuels for 85 percent of all the energy that our world burns every year. And you can see from this image that after World War II, the emission rates started really accelerating. And the accumulated amount of man- made, global warming pollution that is up in the atmosphere now traps as much extra heat energy as would be released by 400, 000 Hiroshima-class atomic bombs exploding every 24 hours, 365 days a year. Fact-checked over and over again, **conservative**, it ’ s the truth. Now it ’ s a big planet, but — that is a lot of energy, particularly when you multiply it 400, 000 times per day. And all that extra heat energy is heating up the atmosphere, the whole earth system. Let ’ s look at the atmosphere. This is a depiction of what we used to think of as the normal distribution of temperatures. The white represents normal temperature days ; 1951-1980 are arbitrarily chosen. The blue are cooler than average days, the red are warmer than average days. But the entire curve has moved to the right in the 1980s. And you ’ ll see in the lower right-hand corner the appearance of statistically significant numbers of extremely hot days. In the 90s, the curve shifted further. And in the last 10 years, you see the extremely hot days are now more numerous than the cooler than average days. In fact, they are 150 times more common on the surface of the earth than they were just 30 years ago. So we ’ re having record-breaking temperatures. Fourteen of the 15 of the hottest years ever measured with instruments have been in this young century. The hottest of all was last year. Last month was the 371st month in a row warmer than the 20th-century average. And for the first time, not only the warmest January, but for the first time, it was more than two degrees Fahrenheit warmer than the average. These higher temperatures are having an effect on animals, plants, people, ecosystems. But on a global basis, 93 percent of all the extra heat energy is trapped in the oceans. And the scientists can measure the heat buildup much more precisely now at all depths : deep, mid-ocean, the first few hundred meters. And this, too, is accelerating. It goes back more than a century. And more than half of the increase has been in the last 19 years. This has consequences. The first order of consequence : the ocean-based storms get stronger. Super Typhoon Haiyan went over areas of the Pacific five and a half degrees Fahrenheit warmer than normal before it slammed into Tacloban, as the most destructive storm ever to make landfall. Pope Francis, who has made such a difference to this whole issue, **visited** Tacloban right after that. Superstorm Sandy went over areas of the Atlantic nine degrees warmer than normal before slamming into New York and New Jersey. The second order of consequences are affecting all of us right now. The warmer oceans are evaporating much more water vapor into the skies. Average humidity worldwide has gone up four percent. And it creates these atmospheric rivers. The Brazilian scientists call them”flying rivers.”And they funnel all of that extra water vapor over the land where storm conditions trigger these massive record-breaking downpours. This is from Montana. Take a look at this storm last August. As it moves over Tucson, Arizona. It literally splashes off the city. These downpours are really unusual. Last July in Houston, Texas, it rained for two days, 162 billion gallons. That represents more than two days of the full flow of Niagara Falls in the middle of the city, which was, of course, paralyzed. These record downpours are creating historic floods and mudslides. This one is from Chile last year. And you ’ ll see that warehouse going by. There are oil tankers cars going by. This is from Spain last September, you could call this the running of the cars and trucks, I guess. Every night on the TV news now is like a nature hike through the Book of Revelation. I mean, really. The insurance industry has certainly noticed, the losses have been mounting up. They ’ re not under any illusions about what ’ s happening. And the causality requires a moment of discussion. We ’ re used to thinking of linear cause and linear effect — one cause, one effect. This is systemic causation. As the great Kevin Trenberth says,”All storms are different now. There ’ s so much extra energy in the atmosphere, there ’ s so much extra water vapor. Every storm is different now.”So, the same extra heat pulls the soil moisture out of the ground and causes these deeper, longer, more pervasive droughts and many of them are underway right now. It dries out the vegetation and causes more fires in the western part of North America. There ’ s certainly been evidence of that, a lot of them. More lightning, as the heat energy builds up, there ’ s a considerable amount of additional lightning also. These climate-related disasters also have geopolitical consequences and create instability. The climate-related historic drought that started in Syria in 2006 destroyed 60 percent of the farms in Syria, killed 80 percent of the livestock, and drove 1. 5 million **climate refugees** into the cities of Syria, where they collided with another 1. 5 million **refugees** from the Iraq War. And along with other factors, that opened the gates of Hell that people are trying to close now. The US Defense Department has long warned of consequences from the **climate crisis**, including **refugees**, food and water shortages and **pandemic** disease. Right now we ’ re seeing microbial diseases from the tropics spread to the higher latitudes ; the transportation revolution has had a lot to do with this. But the changing conditions change the latitudes and the areas where these microbial diseases can become endemic and change the range of the vectors, like mosquitoes and ticks that carry them. The Zika epidemic now — we ’ re better positioned in North America because it ’ s still a little too cool and we have a better public health system. But when women in some regions of South and Central America are advised not to get pregnant for two years — that ’ s something new, that ought to get our attention. The Lancet, one of the two greatest medical journals in the world, last summer labeled this a medical emergency now. And there are many factors because of it. This is also connected to the extinction **crisis**. We ’ re in danger of losing 50 percent of all the living species on earth by the end of this century. And already, land-based plants and animals are now moving towards the poles at an average rate of 15 feet per day. Speaking of the North Pole, last December 29, the same storm that caused historic flooding in the American Midwest, raised temperatures at the North Pole 50 degrees Fahrenheit warmer than normal, causing the thawing of the North Pole in the middle of the long, dark, winter, polar night. And when the land- based ice of the Arctic melts, it raises sea level. Paul Nicklen ’ s beautiful photograph from Svalbard illustrates this. It ’ s more dangerous coming off Greenland and particularly, Antarctica. The 10 largest risk cities for sea-level rise by population are mostly in South and Southeast Asia. When you measure it by assets at risk, number one is Miami : three and a half trillion dollars at risk. Number three : New York and Newark. I was in Miami last fall during the supermoon, one of the highest high-tide days. And there were fish from the ocean swimming in some of the streets of Miami Beach and Fort Lauderdale and Del Rey. And this happens regularly during the highest-tide tides now. Not with rain — they call it”sunny-day flooding.”It comes up through the storm sewers. And the Mayor of Miami speaks for many when he says it is long past time this can be viewed through a partisan lens. This is a **crisis** that ’ s getting worse day by day. We have to move beyond partisanship. And I want to take a moment to honor these House Republicans — who had the courage last fall to step out and take a political risk, by telling the truth about the **climate crisis**. So the cost of the **climate crisis** is mounting up, there are many of these aspects I haven ’ t even mentioned. It ’ s an enormous burden. I ’ ll mention just one more, because the World Economic Forum last month in Davos, after their annual survey of 750 economists, said the **climate crisis** is now the number one risk to the global economy. So you get central bankers like Mark Carney, the head of the UK Central Bank, saying the vast majority of the carbon reserves are unburnable. Subprime carbon. I ’ m not going to remind you what happened with subprime mortgages, but it ’ s the same thing. If you look at all of the carbon fuels that were burned since the beginning of the industrial revolution, this is the quantity burned in the last 16 years. Here are all the ones that are proven and left on the books, 28 trillion dollars. The International Energy Agency says only this amount can be burned. So the rest, 22 trillion dollars — unburnable. Risk to the global economy. That ’ s why divestment movement makes practical sense and is not just a moral imperative. So the answer to the first question,”Must we change?”is yes, we have to change. Second question,”Can we change?”This is the exciting news! The best projections in the world 16 years ago were that by 2010, the world would be able to install 30 gigawatts of wind capacity. We beat that mark by 14 and a half times over. We see an exponential curve for wind installations now. We see the cost coming down dramatically. Some countries — take Germany, an industrial powerhouse with a **climate** not that different from Vancouver ’ s, by the way — one day last December, got 81 percent of all its energy from renewable resources, mainly solar and wind. A lot of countries are getting more than half on an average basis. More good news : energy storage, from batteries particularly, is now beginning to take off because the cost has been coming down very dramatically to solve the intermittency problem. With solar, the news is even more exciting! The best projections 14 years ago were that we would install one gigawatt per year by 2010. When 2010 came around, we beat that mark by 17 times over. Last year, we beat it by 58 times over. This year, we ’ re on track to beat it 68 times over. We ’ re going to win this. We are going to prevail. The exponential curve on solar is even steeper and more dramatic. When I came to this stage 10 years ago, this is where it was. We have seen a revolutionary breakthrough in the emergence of these exponential curves. And the cost has come down 10 percent per year for 30 years. And it ’ s continuing to come down. Now, the business community has certainly noticed this, because it ’ s crossing the grid parity point. Cheaper solar penetration rates are beginning to rise. Grid parity is understood as that line, that threshold, below which renewable electricity is cheaper than electricity from burning fossil fuels. That threshold is a little bit like the difference between 32 degrees Fahrenheit and 33 degrees Fahrenheit, or zero and one Celsius. It ’ s a difference of more than one degree, it ’ s the difference between ice and water. And it ’ s the difference between markets that are frozen up, and liquid flows of capital into new opportunities for investment. This is the biggest new business opportunity in the history of the world, and two-thirds of it is in the private **sector**. We are seeing an explosion of new investment. Starting in 2010, investments globally in renewable electricity generation surpassed fossils. The gap has been growing ever since. The projections for the future are even more dramatic, even though fossil energy is now still subsidized at a rate 40 times larger than renewables. And by the way, if you add the projections for nuclear on here, particularly if you assume that the work many are doing to try to break through to **safer** and more acceptable, more affordable forms of nuclear, this could change even more dramatically. So is there any precedent for such a rapid adoption of a new technology? Well, there are many, but let ’ s look at cell **phones**. In 1980, AT&T, then Ma Bell, commissioned McKinsey to do a global market survey of those clunky new mobile **phones** that appeared then.”How many can we sell by the year 2000?”they asked. McKinsey came back and said,”900, 000.”And sure enough, when the year 2000 arrived, they did sell 900, 000 — in the first three days. And for the balance of the year, they sold 120 times more. And now there are more cell connections than there are people in the world. So, why were they not only wrong, but way wrong? I ' ve asked that question myself,"Why?"And I think the answer is in three parts. First, the cost came down much faster than anybody expected, even as the quality went up. And low-income countries, places that did not have a landline grid — they leap-frogged to the new technology. The big expansion has been in the developing counties. So what about the electricity grids in the developing world? Well, not so hot. And in many areas, they don ' t exist. There are more people without any electricity at all in India than the entire population of the United States of America. So now we ' re getting this : solar panels on grass huts and new business models that make it affordable. Muhammad Yunus financed this one in Bangladesh with micro- credit. This is a village market. Bangladesh is now the fastest-deploying country in the world : two systems per minute on average, night and day. And we have all we need : enough energy from the Sun comes to the earth every hour to supply the full world ' s energy needs for an entire year. It ' s actually a little bit less than an hour. So the answer to the second question,"Can we change?"is clearly"Yes."And it ' s an ever-firmer"yes."Last question,"Will we change?"Paris really was a breakthrough, some of the provisions are binding and the regular reviews will matter a lot. But nations aren ' t waiting, they ' re going ahead. China has already announced that starting next year, they ' re adopting a nationwide cap and trade system. They will likely link up with the European Union. The United States has already been changing. All of these coal plants were proposed in the next 10 years and canceled. All of these existing coal plants were retired. All of these coal plants have had their retirement announced. All of them — canceled. We are moving forward. Last year — if you look at all of the investment in new electricity generation in the United States, almost three-quarters was from renewable energy, mostly wind and solar. We are solving this **crisis**. The only question is : how long will it take to get there? So, it matters that a lot of people are organizing to insist on this change. Almost 400, 000 people marched in New York City before the UN special session on this. Many thousands, tens of thousands, marched in cities around the world. And so, I am extremely optimistic. As I said before, we are going to win this. I ' ll finish with this story. When I was 13 years old, I heard that proposal by President Kennedy to land a person on the Moon and bring him back safely in 10 years. And I heard adults of that day and time say,"That ' s reckless, expensive, may well fail."But eight years and two months later, in the moment that Neil Armstrong set foot on the Moon, there was great cheer that went up in NASA ' s mission control in Houston. Here ' s a little-known fact about that : the average age of the systems engineers, the controllers in the room that day, was 26, which means, among other things, their age, when they heard that challenge, was 18. We now have a moral challenge that is in the tradition of others that we have faced. One of the greatest poets of the last century in the US, Wallace Stevens, wrote a line that has stayed with me :"After the final ' no, ' there comes a ' yes, ' and on that ' yes ', the future world depends."When the abolitionists started their movement, they met with no after no after no. And then came a yes. The Women ' s Suffrage and Women ' s Rights Movement met endless no ' s, until finally, there was a yes. The Civil Rights Movement, the movement against apartheid, and more recently, the movement for gay and lesbian rights here in the United States and elsewhere. After the final"no"comes a"yes."When any great moral challenge is ultimately resolved into a binary choice between what is right and what is wrong, the outcome is fore-ordained because of who we are as human beings. Ninety-nine percent of us, that is where we are now and it is why we ’ re going to win this. We have everything we need. Some still doubt that we have the will to act, but I say the will to act is itself a renewable resource. Thank you very much. Chris Anderson : You ’ ve got this incredible combination of skills. You ’ ve got this scientist mind that can understand the full range of issues, and the **ability** to turn it into the most vivid language. No one else can do that, that ’ s why you led this thing. It was amazing to see it 10 years ago, it was amazing to see it now. Al Gore : Well, you ’ re nice to say that, Chris. But honestly, I have a lot of really good friends in the scientific community who are incredibly patient and who will sit there and explain this **stuff** to me over and over and over again until I can get it into simple enough language that I can understand it. And that ’ s the key to trying to communicate. CA : So, your talk. First part : terrifying, second part : incredibly hopeful. How do we know that all those graphs, all that progress, is enough to solve what you showed in the first part? AG : I think that the crossing — you know, I ’ ve only been in the business world for 15 years. But one of the things I ’ ve learned is that apparently it matters if a new product or service is more expensive than the **incumbent**, or cheaper than. Turns out, it makes a difference if it ’ s cheaper than. And when it crosses that line, then a lot of things really change. We are regularly surprised by these developments. The late Rudi Dornbusch, the great economist said,”Things take longer to happen then you think they will, and then they happen much faster than you thought they could.”I really think that ’ s where we are. Some people are using the phrase”The Solar Singularity”now, meaning when it gets below the grid parity, unsubsidized in most places, then it ’ s the default choice. Now, in one of the presentations yesterday, the jitney thing, there is an effort to use regulations to slow this down. And I just don ’ t think it ’ s going to work. There ’ s a woman in Atlanta, Debbie Dooley, who ’ s the Chairman of the Atlanta Tea Party. They enlisted her in this effort to put a tax on solar panels and regulations. And she had just put solar panels on her roof and she didn ’ t understand the request. And so she went and formed an alliance with the Sierra Club and they formed a new organization called the Green Tea Party. And they defeated the proposal. So, finally, the answer to your question is, this sounds a little corny and maybe it ’ s a clich, but 10 years ago — and Christiana referred to this — there are people in this audience who played an incredibly significant role in generating those exponential curves. And it didn ’ t work out economically for some of them, but it kick-started this global revolution. And what people in this audience do now with the knowledge that we are going to win this. But it matters a lot how fast we win it. CA : Al Gore, that was incredibly powerful. If this turns out to be the year, that the partisan thing changes, as you said, it ’ s no longer a partisan issue, but you bring along people from the other side together, backed by science, backed by these kinds of investment opportunities, backed by reason that you win the day — boy, that ’ s really exciting. Thank you so much. AG : Thank you so much for bringing me back to **TED**. Thank you!

Text Sample 543: POS Dim 4 – 2017 – Score 22.00 – 2017/t001157.txt
Chris Anderson : Hello. Welcome to this **TED** Dialogues. It ’ s the first of a series that ’ s going to be done in response to the current political upheaval. I don ’ t know about you ; I ’ ve become quite concerned about the growing divisiveness in this country and in the world. No one ’ s listening to each other. Right? They aren ’ t. I mean, it feels like we need a different kind of conversation, one that ’ s based on — I don ’ t know, on reason, listening, on understanding, on a broader context. That ’ s at least what we ’ re going to try in these **TED**

Dialogues, starting today. And we couldn ’ t have anyone with us who I ’ d be more excited to kick this off. This is a mind right here that thinks pretty much like no one else on the planet, I would hasten to say. I ’ m serious. I ’ m serious. He synthesizes history with underlying ideas in a way that kind of takes your breath away. So, some of you will know this book,”Sapiens.”Has anyone here read”Sapiens”? I mean, I could not put it down. The way that he tells the story of mankind through big ideas that really make you think differently — it ’ s kind of amazing. And here ’ s the follow-up, which I think is being published in the US next week. YNH : Yeah, next week. CA :”Homo Deus.”Now, this is the history of the next hundred years. I ’ ve had a chance to read it. It ’ s extremely dramatic, and I daresay, for some people, quite alarming. It ’ s a must-read. And honestly, we couldn ’ t have someone better to help make sense of what on Earth is happening in the world right now. So a warm welcome, please, to Yuval Noah Harari. It ’ s great to be joined by our friends on Facebook and around the Web. Hello, Facebook. And all of you, as I start asking questions of Yuval, come up with your own questions, and not necessarily about the political scandal du jour, but about the broader understanding of : Where are we heading? You ready? OK, we ’ re going to go. So here we are, Yuval : New York City, 2017, there ’ s a new president in power, and shock waves rippling around the world. What on Earth is happening? YNH : I think the basic thing that happened is that we have lost our story. Humans think in stories, and we try to make sense of the world by telling stories. And for the last few decades, we had a very simple and very attractive story about what ’ s happening in the world. And the story said that, oh, what ’ s happening is that the economy is being globalized, politics is being liberalized, and the combination of the two will create paradise on Earth, and we just need to keep on globalizing the economy and liberalizing the political system, and everything will be wonderful. And 2016 is the moment when a very large segment, even of the Western world, stopped believing in this story. For good or bad reasons — it doesn ’ t matter. People stopped believing in the story, and when you don ’ t have a story, you don ’ t understand what ’ s happening. CA : Part of you believes that that story was actually a very effective story. It worked. YNH : To some extent, yes. According to some measurements, we are now in the best time ever for humankind. Today, for the first time in history, more people die from eating too much than from eating too little, which is an amazing achievement. Also for the first time in history, more people die from old age than from infectious diseases, and violence is also down. For the first time in history, more people commit suicide than are killed by crime and terrorism and war put together. Statistically, you are your own worst enemy. At least, of all the people in the world, you are most likely to be killed by yourself — which is, again, very good news, compared — compared to the level of violence that we saw in previous eras. CA : But this process of connecting the world ended up with a large group of people kind of feeling left out, and they ’ ve reacted. And so we have this bombshell that ’ s sort of ripping through the whole system. I mean, what do you make of what ’ s happened? It feels like the old way that people thought of politics, the left-right divide, has been blown up and replaced. How should we think of this? YNH : Yeah, the old 20th-century political model of left versus right is now largely irrelevant, and the real divide today is between global and national, global or local. And you see it again all over the world that this is now the main struggle. We probably need completely new political models and completely new ways of thinking about politics. In essence, what you can say is that we now have global ecology, we have a global economy but we have national politics, and this doesn ’ t work together. This makes the political system ineffective, because it has no control over the forces that shape our life. And you have basically two solutions to this imbalance : either de-globalize the economy and turn it back into a national economy, or globalize the political system. CA : So some, I guess many liberals out there view Trump and his government as kind of irredeemably bad, just awful in every way. Do you see any underlying narrative or political philosophy in there that is at least worth understanding? How would you articulate that philosophy? Is it just the philosophy of nationalism? YNH : I think the underlying feeling or idea is that the political system — something is broken there. It doesn ' t empower the ordinary person anymore. It doesn ' t care so much about the ordinary person anymore, and I think this diagnosis of the political disease is correct. With regard to the answers, I am far less certain. I think what we are seeing is the immediate human **reaction** : if something doesn ' t work, let ' s go back. And you see it all over the world, that people, almost nobody in the political system today, has any future-oriented vision of where humankind is going. Almost everywhere, you see retrograde vision :"Let ' s make America great again,"like it was great — I don ' t know — in the ' 50s, in the ' 80s, sometime, let ' s go back there. And you go to Russia a hundred years after Lenin, Putin ' s vision for the future is basically, ah, let ' s go back to the Tsarist empire. And in Israel, where I come from, the hottest political vision of the present is :"Let ' s build the temple again."So let ' s go back 2, 000 years backwards. So people are thinking sometime in the past we ' ve lost it, and sometimes in the past, it ' s like you ' ve lost your way in the city, and you say OK, let ' s go back to the point where I felt secure and start again. I don ' t think this can work, but a lot of people, this is their gut instinct. CA : But why couldn ' t it work?"America First"is a very appealing slogan in many ways. Patriotism is, in many ways, a very noble thing. It ' s played a role in promoting cooperation among large numbers of people. Why couldn ' t you have a world organized in countries, all of which put themselves first? YNH : For many centuries, even thousands of years, patriotism worked quite well. Of course, it led to wars an so forth, but we shouldn ' t focus too much on the bad. There are also many, many positive things about patriotism, and the **ability** to have a large number of people care about each other, sympathize with one another, and come together for **collective** action. If you go back to the first nations, so, thousands of years ago, the people who lived along the Yellow River in China — it was many, many different tribes and they all depended on the river for survival and for prosperity, but all of them also suffered from periodical floods and periodical droughts. And no tribe could really do anything about it, because each of them controlled just a tiny section of the river. And then in a long and complicated process, the tribes coalesced together to form the Chinese nation, which controlled the entire Yellow River and had the **ability** to bring hundreds of thousands of people together to build dams and canals and regulate the river and prevent the worst floods and droughts and raise the level of prosperity for **everybody**. And this worked in many places around the world. But in the 21st century, technology is changing all that in a fundamental way. We are now living — all people in the world — are living alongside the same cyber river, and no single nation can regulate this river by itself. We are all living together on a single planet, which is threatened by our own actions. And if you don ' t have some kind of global cooperation, nationalism is just not on the right level to tackle the problems, whether it ' s **climate** change or whether it ' s technological disruption. CA : So it was a beautiful idea in a world where most of the action, most of the issues, took place on national scale, but your argument is that the issues that matter most today no longer take place on a national scale but on a global scale. YNH : Exactly. All the major problems of the world today are global in essence, and they cannot be solved unless through some kind of global cooperation. It ' s not just **climate** change, which is, like, the most obvious example people give. I think more in terms of technological disruption. If you think about, for example, artificial **intelligence**, over the next 20, 30 years pushing hundreds of millions of people out of the job market — this is a problem on a global level. It will disrupt the economy of all the countries. And similarly, if you think about, say, bioengineering and people being afraid of conducting, I don ’ t know, genetic engineering research in humans, it won ’ t help if just a single country, let ’ s say the US, outlaws all genetic experiments in humans, but China or North Korea continues to do it. So the US cannot solve it by itself, and very quickly, the pressure on the US to do the same will be immense because we are talking about high-risk, high-gain technologies. If somebody else is doing it, I can ’ t allow myself to remain behind. The only way to have regulations, effective regulations, on things like genetic engineering, is to have global regulations. If you just have national regulations, nobody would like to stay behind. CA : So this is really **interesting**. It seems to me that this may be one key to provoking at least a constructive conversation between the different sides here, because I think everyone can agree that the start point of a lot of the anger that ’ s propelled us to where we are is because of the legitimate concerns about job loss. Work is gone, a traditional way of life has gone, and it ’ s no wonder that people are furious about that. And in general, they have blamed globalism, global elites, for doing this to them without asking their permission, and that seems like a legitimate complaint. But what I hear you saying is that — so a key question is : What is the real cause of job loss, both now and going forward? To the extent that it ’ s about globalism, then the right response, yes, is to shut down borders and keep people out and change trade agreements and so forth. But you ’ re saying, I think, that actually the bigger cause of job loss is not going to be that at all. It ’ s going to originate in technological questions, and we have no chance of solving that unless we operate as a connected world. YNH : Yeah, I think that, I don ’ t know about the present, but looking to the future, it ’ s not the Mexicans or Chinese who will take the jobs from the people in Pennsylvania, it ’ s the robots and algorithms. So unless you plan to build a big wall on the border of California — the wall on the border with Mexico is going to be very ineffective. And I was struck when I watched the debates before the election, I was struck that certainly Trump did not even attempt to frighten people by saying the robots will take your jobs. Now even if it ’ s not true, it doesn ’ t matter. It could have been an extremely effective way of frightening people — and galvanizing people :”The robots will take your jobs!”And nobody used that line. And it made me afraid, because it meant that no matter what happens in universities and laboratories, and there, there is already an intense debate about it, but in the mainstream political system and among the general public, people are just unaware that there could be an immense technological disruption — not in 200 years, but in 10, 20, 30 years — and we have to do something about it now, partly because most of what we teach children today in school or in college is going to be completely irrelevant to the job market of 2040, 2050. So it ’ s not something we ’ ll need to think about in 2040. We need to think today what to teach the young people. CA : Yeah, no, absolutely. You ’ ve often written about moments in history where humankind has... entered a new era, unintentionally. Decisions have been made, technologies have been developed, and suddenly the world has changed, possibly in a way that ’ s worse for everyone. So one of the examples you give in”Sapiens”is just the whole agricultural revolution, which, for an actual person tilling the fields, they just picked up a 12-hour backbreaking workday instead of six hours in the jungle and a much more **interesting** lifestyle. So are we at another possible phase change here, where we kind of sleepwalk into a future that none of us actually wants? YNH : Yes, very much so. During the agricultural revolution, what happened is that immense technological and economic revolution empowered the human **collective**, but when you look at actual individual lives, the life of a tiny elite became much better, and the lives of the majority of people became considerably worse. And this can happen again in the 21st century. No doubt the new technologies will empower the human **collective**. But we may end up again with a tiny elite reaping all the benefits, taking all the fruits, and the masses of the population finding themselves worse than they were before, certainly much worse than this tiny elite. CA : And those elites might not even be human elites. They might be cyborgs or — YNH : Yeah, they could be enhanced super humans. They could be cyborgs. They could be completely nonorganic elites. They could even be non-conscious algorithms. What we see now in the world is authority shifting away from humans to algorithms. More and more decisions — about personal lives, about economic matters, about political matters — are actually being taken by algorithms. If you ask the bank for a loan, chances are your fate is decided by an algorithm, not by a human being. And the general impression is that maybe Homo sapiens just lost it. The world is so complicated, there is so much data, things are changing so fast, that this thing that evolved on the African savanna tens of thousands of years ago — to cope with a particular environment, a particular volume of information and data — it just can ’ t handle the realities of the 21st century, and the only thing that may be able to handle it is big-data algorithms. So no wonder more and more authority is shifting from us to the algorithms. CA : So we ’ re in New York City for the first of a series of **TED** Dialogues with Yuval Harari, and there ’ s a Facebook Live audience out there. We ’ re excited to have you with us. We ’ ll start coming to some of your questions and questions of people in the room in just a few minutes, so have those coming. Yuval, if you ’ re going to make the argument that we need to get past nationalism because of the coming technological... danger, in a way, presented by so much of what ’ s happening we ’ ve got to have a global conversation about this. Trouble is, it ’ s hard to get people really believing that, I don ’ t know, AI really is an imminent threat, and so forth. The things that people, some people at least, care about much more immediately, perhaps, is **climate** change, perhaps other issues like **refugees**, nuclear weapons, and so forth. Would you argue that where we are right now that somehow those issues need to be dialed up? You ’ ve talked about **climate** change, but Trump has said he doesn ’ t believe in that. So in a way, your most powerful argument, you can ’ t actually use to make this case. YNH : Yeah, I think with **climate** change, at first sight, it ’ s quite surprising that there is a very close correlation between nationalism and **climate** change. I mean, almost always, the people who deny **climate** change are nationalists. And at first sight, you think : Why? What ’ s the connection? Why don ’ t you have socialists denying **climate** change? But then, when you think about it, it ’ s obvious — because nationalism has no solution to **climate** change. If you want to be a nationalist in the 21st century, you have to deny the problem. If you accept the reality of the problem, then you must accept that, yes, there is still room in the world for patriotism, there is still room in the world for having special loyalties and obligations towards your own people, towards your own country. I don ’ t think anybody is really thinking of abolishing that. But in order to confront **climate** change, we need additional loyalties and commitments to a level beyond the nation. And that should not be impossible, because people can have several layers of loyalty. You can be loyal to your family and to your community and to your nation, so why can ’ t you also be loyal to humankind as a whole? Of course, there are occasions when it becomes difficult, what to put first, but, you know, life is difficult. Handle it. CA : OK, so I would love to get some questions from the audience here. We ’ ve got a microphone here.

Speak into it, and Facebook, get them coming, too. Howard Morgan : One of the things that has clearly made a huge difference in this country and other countries is the income distribution inequality, the dramatic change in income distribution in the US from what it was 50 years ago, and around the world. Is there anything we can do to affect that? Because that gets at a lot of the underlying causes. YNH : So far I haven ' t heard a very good idea about what to do about it, again, partly because most ideas remain on the national level, and the problem is global. I mean, one idea that we hear quite a lot about now is universal basic income. But this is a problem. I mean, I think it ' s a good start, but it ' s a problematic idea because it ' s not clear what"universal"is and it ' s not clear what"basic"is. Most people when they speak about universal basic income, they actually mean national basic income. But the problem is global. Let ' s say that you have AI and 3D printers taking away millions of jobs in Bangladesh, from all the people who make my shirts and my shoes. So what ' s going to happen? The US government will levy taxes on Google and Apple in California, and use that to pay basic income to unemployed Bangladeshis? If you believe that, you can just as well believe that Santa Claus will come and solve the problem. So unless we have really universal and not national basic income, the deep problems are not going to go away. And also it ' s not clear what basic is, because what are basic human needs? A thousand years ago, just food and shelter was enough. But today, people will say education is a basic human need, it should be part of the package. But how much? Six years? Twelve years? PhD? Similarly, with health care, let ' s say that in 20, 30, 40 years, you ' ll have expensive treatments that can extend human life to 120, I don ' t know. Will this be part of the basket of basic income or not? It ' s a very difficult problem, because in a world where people lose their **ability** to be employed, the only thing they are going to get is this basic income. So what ' s part of it is a very, very difficult ethical question. CA : There ' s a bunch of questions on how the world affords it as well, who pays. There ' s a question here from Facebook from Lisa Larson :"How does nationalism in the US now compare to that between World War I and World War II in the last century?"YNH : Well the good news, with regard to the dangers of nationalism, we are in a much better position than a century ago. A century ago, 1917, Europeans were killing each other by the millions. In 2016, with Brexit, as far as I remember, a single person lost their life, an MP who was murdered by some extremist. Just a single person. I mean, if Brexit was about British independence, this is the most peaceful war of independence in human history. And let ' s say that Scotland will now choose to leave the UK after Brexit. So in the 18th century, if Scotland wanted — and the Scots wanted several times — to break out of the control of London, the **reaction** of the government in London was to send an army up north to burn down Edinburgh and massacre the highland tribes. My guess is that if, in 2018, the Scots vote for independence, the London government will not send an army up north to burn down Edinburgh. Very few people are now willing to kill or be killed for Scottish or for British independence. So for all the talk of the rise of nationalism and going back to the 1930s, to the 19th century, in the **West** at least, the power of national sentiments today is far, far smaller than it was a century ago. CA : Although some people now, you hear publicly worrying about whether that might be shifting, that there could actually be outbreaks of violence in the US depending on how things turn out. Should we be worried about that, or do you really think things have shifted? YNH : No, we should be worried. We should be aware of two things. First of all, don ' t be hysterical. We are not back in the First World War yet. But on the other hand, don ' t be complacent. We reached from 1917 to 2017, not by some divine miracle, but simply by human decisions, and if we now start making the wrong decisions, we could be back in an analogous situation to 1917 in a few years. One of the things I know as a historian is that you should never underestimate human stupidity. It ' s one of the most powerful forces in history, human stupidity and human violence. Humans do such crazy things for no obvious reason, but again, at the same time, another very powerful force in human history is human wisdom. We have both. CA : We have with us here moral psychologist Jonathan Haidt, who I think has a question. Jonathan Haidt : Thanks, Yuval. So you seem to be a fan of global governance, but when you look at the map of the world from Transparency International, which rates the level of corruption of political institutions, it ' s a vast sea of red with little bits of yellow here and there for those with good institutions. So if we were to have some kind of global governance, what makes you think it would end up being more like Denmark rather than more like Russia or Honduras, and aren ' t there alternatives, such as we did with CFCs? There are ways to solve global problems with national governments. What would world government actually look like, and why do you think it would work? YNH : Well, I don ' t know what it would look like. Nobody still has a model for that. The main reason we need it is because many of these issues are lose-lose situations. When you have a win-win situation like trade, both sides can benefit from a trade agreement, then this is something you can work out. Without some kind of global government, national governments each have an interest in doing it. But when you have a lose-lose situation like with **climate** change, it ' s much more difficult without some overarching authority, real authority. Now, how to get there and what would it look like, I don ' t know. And certainly there is no obvious reason to think that it would look like Denmark, or that it would be a democracy. Most likely it wouldn ' t. We don ' t have workable democratic models for a global government. So maybe it would look more like ancient China than like modern Denmark. But still, given the dangers that we are facing, I think the imperative of having some kind of real **ability** to force through difficult decisions on the global level is more important than almost anything else. CA : There ' s a question from Facebook here, and then we ' ll get the mic to Andrew. So, Kat Hebron on Facebook, calling in from Vail :"How would developed nations manage the millions of **climate** migrants?"YNH : I don ' t know. CA : That ' s your answer, Kat. YNH : And I don ' t think that they know either. They ' ll just deny the problem, maybe. CA : But immigration, generally, is another example of a problem that ' s very hard to solve on a nation-by-nation basis. One nation can shut its doors, but maybe that stores up problems for the future. YNH : Yes, I mean — it ' s another very good case, especially because it ' s so much easier to migrate today than it was in the Middle Ages or in ancient times. CA : Yuval, there ' s a belief among many technologists, certainly, that political concerns are kind of overblown, that actually, political **leaders** don ' t have that much influence in the world, that the real determination of humanity at this point is by science, by invention, by companies, by many things other than political **leaders**, and it ' s actually very hard for **leaders** to do much, so we ' re actually worrying about nothing here. YNH : Well, first, it should be emphasized that it ' s true that political **leaders** ' **ability** to do good is very limited, but their **ability** to do harm is unlimited. There is a basic imbalance here. You can still press the button and blow **everybody** up. You have that kind of **ability**. But if you want, for example, to reduce inequality, that ' s very, very difficult. But to start a war, you can still do so very easily. So there is a built-in imbalance in the political system today which is very frustrating, where you cannot do a lot of good but you can still do a lot of harm. And this makes the political system still a very big concern. CA : So as you look at what ' s happening today, and putting your historian ' s hat on, do you look back in history at moments when things were going just fine and an individual **leader** really took the world or their country backwards? YNH : There are quite a few examples, but I should emphasize, it ' s never an individual **leader**. I mean, somebody put him there, and somebody allowed him to continue to be there. So it ' s never really just the fault of a single individual. There are a lot of people behind every such individual. CA : Can we have the microphone here, please, to Andrew? Andrew Solomon : You ' ve talked a lot about the global versus the national, but increasingly, it seems to me, the world situation is in the hands of identity groups. We look at people within the United States who have been recruited by ISIS. We look at these other groups which have formed which go outside of national bounds but still represent significant authorities. How are they to be integrated into the system, and how is a diverse set of identities to be made coherent under either national or global **leadership**? YNH : Well, the problem of such diverse identities is a problem from nationalism as well. Nationalism believes in a single, monolithic identity, and exclusive or at least more extreme versions of nationalism believe in an exclusive loyalty to a single identity. And therefore, nationalism has had a lot of problems with people wanting to divide their identities between various groups. So it ' s not just a problem, say, for a global vision. And I think, again, history shows that you shouldn ' t necessarily think in such exclusive terms. If you think that there is just a single identity for a person,"I am just X, that ' s it, I can ' t be several things, I can be just that,"that ' s the start of the problem. You have religions, you have nations that sometimes **demand** exclusive loyalty, but it ' s not the only option. There are many religions and many nations that enable you to have diverse identities at the same time. CA : But is one explanation of what ' s happened in the last year that a group of people have got fed up with, if you like, the liberal elites, for want of a better term, obsessing over many, many different identities and them feeling,"But what about my identity? I am being completely ignored here. And by the way, I thought I was the majority"? And that that ' s actually sparked a lot of the anger. YNH : Yeah. Identity is always problematic, because identity is always based on fictional stories that sooner or later collide with reality. Almost all identities, I mean, beyond the level of the basic community of a few dozen people, are based on a fictional story. They are not the truth. They are not the reality. It ' s just a story that people invent and tell one another and start believing. And therefore all identities are extremely unstable. They are not a biological reality. Sometimes nationalists, for example, think that the nation is a biological entity. It ' s made of the combination of soil and blood, creates the nation. But this is just a fictional story. CA : Soil and blood kind of makes a gooey mess. YNH : It does, and also it messes with your mind when you think too much that I am a combination of soil and blood. If you look from a biological perspective, obviously none of the nations that exist today existed 5, 000 years ago. Homo sapiens is a social animal, that ' s for sure. But for millions of years, Homo sapiens and our hominid ancestors lived in small communities of a few dozen individuals. **Everybody** knew **everybody** else. Whereas modern nations are imagined communities, in the sense that I don ' t even know all these people. I come from a relatively small nation, Israel, and of eight million Israelis, I never met most of them. I will never meet most of them. They basically exist here. CA : But in terms of this identity, this group who feel left out and perhaps have work taken away, I mean, in"Homo Deus,"you actually speak of this group in one sense expanding, that so many people may have their jobs taken away by technology in some way that we could end up with a really large — I think you call it a"useless class"— a class where traditionally, as viewed by the economy, these people have no use. YNH : Yes. CA : How likely a possibility is that? Is that something we should be terrified about? And can we address it in any way? YNH : We should think about it very carefully. I mean, nobody really knows what the job market will look like in 2040, 2050. There is a chance many new jobs will appear, but it ’ s not certain. And even if new jobs do appear, it won ’ t necessarily be easy for a 50-year old unemployed truck driver made unemployed by self-driving vehicles, it won ’ t be easy for an unemployed truck driver to reinvent himself or herself as a designer of virtual worlds. Previously, if you look at the trajectory of the industrial revolution, when machines replaced humans in one type of work, the solution usually came from low-skill work in new lines of business. So you didn ’ t need any more agricultural workers, so people moved to working in low-skill industrial jobs, and when this was taken away by more and more machines, people moved to low- skill service jobs. Now, when people say there will be new jobs in the future, that humans can do better than AI, that humans can do better than robots, they usually think about high-skill jobs, like software engineers designing virtual worlds. Now, I don ’ t see how an unemployed cashier from Wal-Mart reinvents herself or himself at 50 as a designer of virtual worlds, and certainly I don ’ t see how the millions of unemployed Bangladeshi textile workers will be able to do that. I mean, if they are going to do it, we need to start teaching the Bangladeshis today how to be software designers, and we are not doing it. So what will they do in 20 years? CA : So it feels like you ’ re really highlighting a question that ’ s really been bugging me the last few months more and more. It ’ s almost a hard question to ask in public, but if any mind has some wisdom to offer in it, maybe it ’ s yours, so I ’ m going to ask you : What are humans for? YNH : As far as we know, for nothing. I mean, there is no great cosmic drama, some great cosmic plan, that we have a role to play in. And we just need to discover what our role is and then play it to the best of our **ability**. This has been the story of all religions and ideologies and so forth, but as a scientist, the best I can say is this is not true. There is no universal drama with a role in it for Homo sapiens. So — CA : I ’ m going to push back on you just for a minute, just from your own book, because in”Homo Deus,”you give really one of the most coherent and understandable accounts about sentience, about consciousness, and that unique sort of human skill. You point out that it ’ s different from **intelligence**, the **intelligence** that we ’ re building in machines, and that there ’ s actually a lot of mystery around it. How can you be sure there ’ s no purpose when we don ’ t even understand what this sentience thing is? I mean, in your own thinking, isn ’ t there a chance that what humans are for is to be the universe ’ s sentient things, to be the centers of joy and love and happiness and hope? And maybe we can build machines that actually help amplify that, even if they ’ re not going to become sentient themselves? Is that crazy? I kind of found myself hoping that, reading your book. YNH : Well, I certainly think that the most **interesting** question today in science is the question of consciousness and the mind. We are getting better and better in understanding the brain and **intelligence**, but we are not getting much better in understanding the mind and consciousness. People often confuse **intelligence** and consciousness, especially in places like Silicon Valley, which is understandable, because in humans, they go together. I mean, **intelligence** basically is the **ability** to solve problems. Consciousness is the **ability** to feel things, to feel joy and sadness and boredom and pain and so forth. In Homo sapiens and all other mammals as well — it ’ s not unique to humans — in all mammals and birds and some other animals, **intelligence** and consciousness go together. We often solve problems by feeling things. So we tend to confuse them. But they are different things. What ’ s happening today in places like Silicon Valley is that we are creating artificial **intelligence** but not artificial consciousness. There has been an amazing development in computer **intelligence** over the last 50 years, and exactly zero development in computer consciousness, and there is no indication that computers are going to become conscious anytime soon. So first of all, if there is some cosmic role for consciousness, it ' s not unique to Homo sapiens. Cows are conscious, pigs are conscious, chimpanzees are conscious, chickens are conscious, so if we go that way, first of all, we need to broaden our horizons and remember very clearly we are not the only sentient beings on Earth, and when it comes to sentience — when it comes to **intelligence**, there is good reason to think we are the most intelligent of the whole bunch. But when it comes to sentience, to say that humans are more sentient than whales, or more sentient than baboons or more sentient than cats, I see no evidence for that. So first step is, you go in that direction, expand. And then the second question of what is it for, I would reverse it and I would say that I don ' t think sentience is for anything. I think we don ' t need to find our role in the universe. The really important thing is to liberate ourselves from suffering. What characterizes sentient beings in contrast to robots, to stones, to whatever, is that sentient beings suffer, can suffer, and what they should focus on is not finding their place in some mysterious cosmic drama. They should focus on understanding what suffering is, what causes it and how to be liberated from it. CA : I know this is a big issue for you, and that was very eloquent. We ' re going to have a blizzard of questions from the audience here, and maybe from Facebook as well, and maybe some comments as well. So let ' s go quick. There ' s one right here. Keep your hands held up at the back if you want the mic, and we ' ll get it back to you. Question : In your work, you talk a lot about the fictional stories that we accept as truth, and we live our lives by it. As an individual, knowing that, how does it **impact** the stories that you choose to live your life, and do you confuse them with the truth, like all of us? YNH : I try not to. I mean, for me, maybe the most important question, both as a scientist and as a person, is how to tell the difference between fiction and reality, because reality is there. I ' m not saying that everything is fiction. It ' s just very difficult for human beings to tell the difference between fiction and reality, and it has become more and more difficult as history progressed, because the fictions that we have created — nations and gods and money and corporations — they now control the world. So just to even think,"Oh, this is just all fictional entities that we ' ve created,"is very difficult. But reality is there. For me the best... There are several tests to tell the difference between fiction and reality. The simplest one, the best one that I can say in short, is the test of suffering. If it can suffer, it ' s real. If it can ' t suffer, it ' s not real. A nation cannot suffer. That ' s very, very clear. Even if a nation loses a war, we say,"Germany suffered a defeat in the First World War,"it ' s a metaphor. Germany cannot suffer. Germany has no mind. Germany has no consciousness. Germans can suffer, yes, but Germany cannot. Similarly, when a bank goes bust, the bank cannot suffer. When the dollar loses its value, the dollar doesn ' t suffer. People can suffer. Animals can suffer. This is real. So I would start, if you really want to see reality, I would go through the door of suffering. If you can really understand what suffering is, this will give you also the key to understand what reality is. CA : There ' s a Facebook question here that connects to this, from someone around the world in a language that I cannot read. YNH : Oh, it ' s Hebrew. CA : Hebrew. There you go. Can you read the name? YNH : Or Lauterbach Goren. CA : Well, thank you for writing in. The question is :"Is the post-truth era really a brand-new era, or just another climax or moment in a never-ending trend? YNH : Personally, I don ' t connect with this idea of post-truth. My basic **reaction** as a historian is : If this is the era of post-truth, when the hell was the era of truth? CA : Right. YNH : Was it the 1980s, the 1950s, the Middle Ages? I mean, we have always lived in an era, in a way, of post-truth. CA : But I ' d push back on that, because I think what people are talking about is that there was a world where you had fewer journalistic outlets, where there were traditions, that things were fact-checked. It was incorporated into the charter of those organizations that the truth mattered. So if you believe in a reality, then what you write is information. There was a belief that that information should connect to reality in a real way, and if you wrote a headline, it was a serious, earnest attempt to reflect something that had actually happened. And people didn ’ t always get it right. But I think the concern now is you ’ ve got a technological system that ’ s incredibly powerful that, for a while at least, massively amplified anything with no attention paid to whether it connected to reality, only to whether it connected to clicks and attention, and that that was arguably toxic. That ’ s a reasonable concern, isn ’ t it? YNH : Yeah, it is. I mean, the technology changes, and it ’ s now easier to disseminate both truth and fiction and falsehood. It goes both ways. It ’ s also much easier, though, to spread the truth than it was ever before. But I don ’ t think there is anything essentially new about this disseminating fictions and errors. There is nothing that — I don ’ t know — Joseph Goebbels, didn ’ t know about all this idea of fake news and post-truth. He famously said that if you repeat a lie often enough, people will think it ’ s the truth, and the bigger the lie, the better, because people won ’ t even think that something so big can be a lie. I think that fake news has been with us for thousands of years. Just think of the Bible. CA : But there is a concern that the fake news is associated with tyrannical regimes, and when you see an uprise in fake news that is a canary in the coal mine that there may be dark times coming. YNH : Yeah. I mean, the intentional use of fake news is a disturbing sign. But I ’ m not saying that it ’ s not bad, I ’ m just saying that it ’ s not new. CA : There ’ s a lot of interest on Facebook on this question about global governance versus nationalism. Question here from Phil Dennis :”How do we get people, governments, to relinquish power? Is that — is that — actually, the text is so big I can ’ t read the full question. But is that a necessity? Is it going to take war to get there? Sorry Phil — I mangled your question, but I blame the text right here. YNH : One option that some people talk about is that only a catastrophe can shake humankind and open the path to a real system of global governance, and they say that we can ’ t do it before the catastrophe, but we need to start laying the foundations so that when the disaster strikes, we can react quickly. But people will just not have the motivation to do such a thing before the disaster strikes. Another thing that I would emphasize is that anybody who is really **interested** in global governance should always make it very, very clear that it doesn ’ t replace or abolish local identities and communities, that it should come both as — It should be part of a single package. CA : I want to hear more on this, because the very words”global governance”are almost the epitome of evil in the mindset of a lot of people on the alt-right right now. It just seems scary, remote, distant, and it has let them down, and so globalists, global governance — no, go away! And many view the election as the ultimate poke in the eye to anyone who believes in that. So how do we change the narrative so that it doesn ’ t seem so scary and remote? Build more on this idea of it being compatible with local identity, local communities. YNH : Well, I think again we should start really with the biological realities of Homo sapiens. And biology tells us two things about Homo sapiens which are very relevant to this issue : first of all, that we are completely dependent on the ecological system around us, and that today we are talking about a global system. You cannot escape that. And at the same time, biology tells us about Homo sapiens that we are social animals, but that we are social on a very, very local level. It ’ s just a simple fact of humanity that we cannot have intimate familiarity with more than about 150 individuals. The size of the natural group, the natural community of Homo sapiens, is not more than 150 individuals, and everything beyond that is really based on all kinds of imaginary stories and large-scale institutions, and I think that we can find a way, again, based on a biological understanding of our species, to weave the two together and to understand that today in the 21st century, we need both the global level and the local community. And I would go even further than that and say that it starts with the body itself. The feelings that people today have of alienation and loneliness and not finding their place in the world, I would think that the chief problem is not global capitalism. The chief problem is that over the last hundred years, people have been becoming disembodied, have been distancing themselves from their body. As a hunter-gatherer or even as a peasant, to survive, you need to be constantly in touch with your body and with your senses, every moment. If you go to the forest to look for mushrooms and you don ’ t pay attention to what you hear, to what you smell, to what you taste, you ’ re dead. So you must be very connected. In the last hundred years, people are losing their **ability** to be in touch with their body and their senses, to hear, to smell, to feel. More and more attention goes to screens, to what is happening elsewhere, some other time. This, I think, is the deep reason for the feelings of alienation and loneliness and so forth, and therefore part of the solution is not to bring back some mass nationalism, but also reconnect with our own bodies, and if you are back in touch with your body, you will feel much more at home in the world also. CA : Well, depending on how things go, we may all be back in the forest soon. We ’ re going to have one more question in the room and one more on Facebook. Ama Adi-Dako : Hello. I ’ m from Ghana, West Africa, and my question is : I ’ m wondering how do you present and justify the idea of global governance to countries that have been historically disenfranchised by the effects of globalization, and also, if we ’ re talking about global governance, it sounds to me like it will definitely come from a very Westernized idea of what the”global”is supposed to look like. So how do we present and justify that idea of global versus wholly nationalist to people in countries like Ghana and Nigeria and Togo and other countries like that? YNH : I would start by saying that history is extremely unfair, and that we should realize that. Many of the countries that suffered most from the last 200 years of globalization and imperialism and industrialization are exactly the countries which are also most likely to suffer most from the next wave. And we should be very, very clear about that. If we don ’ t have a global governance, and if we suffer from **climate** change, from technological disruptions, the worst suffering will not be in the US. The worst suffering will be in Ghana, will be in Sudan, will be in Syria, will be in Bangladesh, will be in those places. So I think those countries have an even greater incentive to do something about the next wave of disruption, whether it ’ s ecological or whether it ’ s technological. Again, if you think about technological disruption, so if AI and 3D printers and robots will take the jobs from billions of people, I worry far less about the Swedes than about the people in Ghana or in Bangladesh. And therefore, because history is so unfair and the results of a calamity will not be shared equally between **everybody**, as usual, the rich will be able to get away from the worst consequences of **climate** change in a way that the poor will not be able to. CA : And here ’ s a great question from Cameron Taylor on Facebook :”At the end of ’ Sapiens, ’”you said we should be asking the question, ’ What do we want to want? ’ Well, what do you think we should want to want?”YNH : I think we should want to want to know the truth, to understand reality. Mostly what we want is to change reality, to fit it to our own desires, to our own wishes, and I think we should first want to understand it. If you look at the long-term trajectory of history, what you see is that for thousands of years we humans have been gaining control of the world outside us and trying to shape it to fit our own desires. And we ’ ve gained control of the other animals, of the rivers, of the forests, and reshaped them completely, causing an ecological destruction without making ourselves satisfied. So the next step is we turn our gaze inwards, and we say OK, getting control of the world outside us did not really make us satisfied. Let ' s now try to gain control of the world inside us. This is the really big project of science and technology and industry in the 21st century — to try and gain control of the world inside us, to learn how to engineer and produce bodies and brains and minds. These are likely to be the main products of the 21st century economy. When people think about the future, very often they think in terms,"Oh, I want to gain control of my body and of my brain."And I think that ' s very dangerous. If we ' ve learned anything from our previous history, it ' s that yes, we gain the power to manipulate, but because we didn ' t really understand the complexity of the ecological system, we are now facing an ecological meltdown. And if we now try to reengineer the world inside us without really understanding it, especially without understanding the complexity of our mental system, we might cause a kind of internal ecological disaster, and we ' ll face a kind of mental meltdown inside us. CA : Putting all the pieces together here — the current politics, the coming technology, concerns like the one you ' ve just outlined — I mean, it seems like you yourself are in quite a bleak place when you think about the future. You ' re pretty worried about it. Is that right? And if there was one cause for hope, how would you state that? YNH : I focus on the most dangerous possibilities partly because this is like my job or responsibility as a historian or social critic. I mean, the industry focuses mainly on the positive sides, so it ' s the job of historians and philosophers and sociologists to highlight the more dangerous potential of all these new technologies. I don ' t think any of that is inevitable. Technology is never deterministic. You can use the same technology to create very different kinds of societies. If you look at the 20th century, so, the technologies of the Industrial Revolution, the trains and electricity and all that could be used to create a communist dictatorship or a fascist regime or a liberal democracy. The trains did not tell you what to do with them. Similarly, now, artificial **intelligence** and bioengineering and all of that — they don ' t predetermine a single outcome. Humanity can rise up to the challenge, and the best example we have of humanity rising up to the challenge of a new technology is nuclear weapons. In the late 1940s, ' 50s, many people were convinced that sooner or later the Cold War will end in a nuclear catastrophe, destroying human civilization. And this did not happen. In fact, nuclear weapons prompted humans all over the world to change the way that they manage **international** politics to reduce violence. And many countries basically took out war from their political toolkit. They no longer tried to pursue their interests with warfare. Not all countries have done so, but many countries have. And this is maybe the most important reason why **international** violence declined dramatically since 1945, and today, as I said, more people commit suicide than are killed in war. So this, I think, gives us a good example that even the most frightening technology, humans can rise up to the challenge and actually some good can come out of it. The problem is, we have very little margin for error. If we don ' t get it right, we might not have a second option to try again. CA : That ' s a very powerful note, on which I think we should draw this to a conclusion. Before I wrap up, I just want to say one thing to people here and to the global **TED** community watching online, anyone watching online : help us with these dialogues. If you believe, like we do, that we need to find a different kind of conversation, now more than ever, help us do it. Reach out to other people, try and have conversations with people you disagree with, understand them, pull the pieces together, and help us figure out how to take these conversations forward so we can make a real contribution to what ’ s happening in the world right now. I think everyone feels more alive, more concerned, more engaged with the politics of the moment. The stakes do seem quite high, so help us respond to it in a wise, wise way. Yuval Harari, thank you.

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Chris Anderson : Welcome to this next edition of **TED** Dialogues. We ’ re trying to do some bridging here today. You know, the American dream has inspired millions of people around the world for many years. Today, I think, you can say that America is divided, perhaps more than ever, and the divisions seem to be getting worse. It ’ s actually really hard for people on different sides to even have a conversation. People almost feel... disgusted with each other. Some families can ’ t even speak to each other right now. Our purpose in this dialogue today is to try to do something about that, to try to have a different kind of conversation, to do some listening, some thinking, some understanding. And I have two people with us to help us do that. They ’ re not going to come at this hammer and tong against each other. This is not like cable news. This is two people who have both spent a lot of their working life in the political center or right of the center. They ’ ve immersed themselves in **conservative** worldviews, if you like. They know that space very well. And we ’ re going to explore together how to think about what is happening right now, and whether we can find new ways to bridge and just to have wiser, more connected conversations. With me, first of all, Gretchen Carlson, who has spent a decade working at Fox News, hosting”Fox and Friends”and then”The Real Story,”before taking a courageous stance in filing sexual harassment **claims** against Roger Ailes, which eventually led to his departure from Fox News. David Brooks, who has earned the wrath of many of [The New York Times ’ s] left-leaning readers because of his **conservative** views, and more recently, perhaps, some of the right-leaning readers because of his criticism of some aspects of Trump. Yet, his columns are usually the top one, two or three most-read content of the day because they ’ re brilliant, because they bring psychology and social science to providing understanding for what ’ s going on. So without further ado, a huge welcome to Gretchen and David. Come and join me. So, Gretchen. Sixty-three million Americans voted for Donald Trump. Why did they do this? Gretchen Carlson : There are a lot of reasons, in my mind, why it happened. I mean, I think it was a movement of sorts, but it started long ago. It didn ’ t just happen overnight.”Anger”would be the first word that I would think of — anger with nothing being done in Washington, anger about not being heard. I think there was a huge swath of the population that feels like Washington never listens to them, you know, a good part of the middle of America, not just the coasts, and he was somebody they felt was listening to their concerns. So I think those two issues would be the main reason. I have to throw in there also celebrity. I think that had a huge **impact** on Donald Trump becoming president. CA : Was the anger justified? David Brooks : Yeah, I think so. In 2015 and early 2016, I wrote about 30 columns with the following theme : don ’ t worry, Donald Trump will never be the Republican nominee. And having done that and gotten that so wrong, I decided to spend the ensuing year just out in Trumpworld, and I found a lot of economic dislocation. I ran into a woman in West Virginia who was going to a funeral for her mom. She said,”The nice thing about being Catholic is we don ’ t have to speak, and that ’ s good, because we ’ re not word people.”That phrase rung in my head : word people. A lot of us in the **TED** community are word people, but if you ’ re not, the economy has not been an-

gled toward you, and so 11 million men, for example, are out of the labor force because those jobs are done away. A lot of social injury. You used to be able to say,"I ' m not the richest person in the world, I ' m not the most famous, but my neighbors can count on me and I get some dignity out of that."And because of celebritification or whatever, if you ' re not rich or famous, you feel invisible. And a lot of moral injury, sense of feeling betrayed, and frankly, in this country, we almost have one success story, which is you go to college, get a white-collar job, and you ' re a success, and if you don ' t fit in that formula, you feel like you ' re not respected. And so that accumulation of things — and when I talked to Trump voters and still do, I found most of them completely realistic about his failings, but they said, this is my shot. GC : And yet I predicted that he would be the nominee, because I ' ve known him for 27 years. He ' s a master marketer, and one of the things he did extremely well that President Obama also did extremely well, was simplifying the message, simplifying down to phrases and to a populist message. Even if he can ' t achieve it, it sounded good. And many people latched on to that simplicity again. It ' s something they could grasp onto :"I get that. I want that. That sounds fantastic."And I remember when he used to come on my show originally, before"The Apprentice"was even"The Apprentice,"and he ' d say it was the number one show on TV. I ' d say back to him,"No, it ' s not."And he would say,"Yes it is, Gretchen."And I would say,"No it ' s not."But people at home would see that, and they ' d be like,"Wow, I should be watching the number one show on TV."And — lo and behold — it became the number one show on TV. So he had this, I ' ve seen this **ability** in him to be the master marketer. CA : It ' s puzzling to a lot of people on the left that so many women voted for him, despite some of his comments. GC : I wrote a column about this for Time Motto, saying that I really believe that lot of people put on blinders, and maybe for the first time, some people decided that **policies** they believed in and being heard and not being invisible anymore was more important to them than the way in which he had acted or acts as a human. And so human dignity — whether it would be the dust-up about the disabled reporter, or what happened in that audiotape with Billy Bush and the way in which he spoke about women — they put that aside and pretended as if they hadn ' t seen that or heard that, because to them, **policies** were more important. CA : Right, so just because someone voted for Trump, it ' s not blind adherence to everything that he ' s said or stood for. GC : No. I heard a lot of people that would say to me,"Wow, I just wish he would shut up before the election. If he would just stay quiet, he ' d get elected."CA : And so, maybe for people on the left there ' s a trap there, to sort of despise or just be baffled by the **support**, assuming that it ' s for some of the unattractive features. Actually, maybe they ' re **supporting** him despite those, because they see something exciting. They see a man of action. They see the choking hold of government being thrown off in some way and they ' re excited by that. GC : But don ' t forget we saw that on the left as well — Bernie Sanders. So this is one of the **commonalities** that I think we can talk about today,"The Year of the Outsider,"David — right? And even though Bernie Sanders has been in Congress for a long time, he was deemed an outsider this time. And so there was anger on the left as well, and so many people were in favor of Bernie Sanders. So I see it as a **commonality**. People who like Trump, people who like Bernie Sanders, they were liking different **policies**, but the underpinning was anger. CA : David, there ' s often this narrative, then, that the sole explanation for Trump ' s victory and his rise is his tapping into anger in a very visceral way. But you ' ve written a bit about that it ' s actually more than that, that there ' s a worldview that ' s being worked on here. Could you talk about that? DB : I would say he understood what, frankly, I didn ' t, which is what debate we were having. And so I ' d grown up starting with Reagan, and it was the big government versus small government debate. It was Barry Goldwater versus George McGovern, and that was the debate we had been having for a generation. It was : Democrats wanted to use government to enhance equality, Republicans wanted to limit government to enhance freedom. That was the debate. He understood what I think the two major parties did not, which was that ' s not the debate anymore. The debate is now open versus closed. On one side are those who have the tailwinds of globalization and the meritocracy blowing at their back, and they tend to favor open trade, open borders, open social mores, because there are so many opportunities. On the other side are those who feel the headwinds of globalization and the meritocracy just blasting in their faces, and they favor closed trade, closed borders, closed social mores, because they just want some security. And so he was right on that fundamental issue, and people were willing to overlook a lot to get there. And so he felt that sense of security. We ' re speaking the morning after Trump ' s joint session speech. There are three traditional groups in the Republican Party. There are the foreign **policies** hawks who believe in America as global policeman. Trump totally repudiated that view. Second, there was the social **conservatives** who believed in religious liberty, pro-life, prayer in schools. He totally ignored that. There was not a single mention of a single social **conservative** issue. And then there were the fiscal hawks, the people who wanted to cut down on the national debt, Tea Party, cut the size of government. He ' s expanding the size of government! Here ' s a man who has single-handedly revolutionized a major American party because he understood where the debate was headed before other people. And then **guys** like Steve Bannon come in and give him substance to his impulses. CA : And so take that a bit further, and maybe expand a bit more on your insights into Steve Bannon ' s worldview. Because he ' s sometimes tarred in very simple terms as this dangerous, racist, xenophobic, anger-sparking person. There ' s more to the story ; that is perhaps an unfair simplification. DB : I think that part is true, but there ' s another part that ' s probably true, too. He ' s part of a global movement. It ' s like being around Marxists in 1917. There ' s him here, there ' s the UKIP party, there ' s the National Front in France, there ' s Putin, there ' s a Turkish version, a Philippine version. So we have to recognize that this is a global intellectual movement. And it believes that wisdom and virtue is not held in individual conversation and civility the way a lot of us in the enlightenment side of the world do. It ' s held in — the German word is the"volk"— in the people, in the common, instinctive wisdom of the plain people. And the essential virtue of that people is always being threatened by outsiders. And he ' s got a strategy for how to get there. He ' s got a series of **policies** to bring the people up and repudiate the outsiders, whether those outsiders are Islam, Mexicans, the media, the coastal elites... And there ' s a whole worldview there ; it ' s a very coherent worldview. I sort of have more respect for him. I loathe what he stands for and I think he ' s wrong on the substance, but it ' s **interesting** to see someone with a set of ideas find a vehicle, Donald Trump, and then try to take control of the White House in order to advance his viewpoint. CA : So it ' s almost become, like, that the core question of our time now is : Can you be patriotic but also have a global mindset? Are these two things implacably opposed to each other? I mean, a lot of **conservatives** and, to the extent that it ' s a different category, a lot of Trump supporters, are infuriated by the coastal elites and the globalists because they see them as, sort of, not cheering for America, not embracing fully American values. I mean, have you seen that in your conversations with people, in your understanding of their mindset? GC : I do think that there ' s a huge difference between — I hate to put people in categories, but, Middle America versus people who live on the coasts. It ’ s an entirely different existence. And I grew up in Minnesota, so I have an understanding of Middle America, and I ’ ve never forgotten it. And maybe that ’ s why I have an understanding of what happened here, because those people often feel like nobody ’ s listening to them, and that we ’ re only concentrating on California and New York. And so I think that was a huge reason why Trump was elected. I mean, these people felt like they were being heard. Whether or not patriotism falls into that, I ’ m not sure about that. I do know one thing : a lot of things Trump talked about last night are not **conservative** things. Had Hillary Clinton gotten up and given that speech, not one Republican would have stood up to applaud. I mean, he ’ s talking about spending a trillion dollars on infrastructure. That is not a **conservative** viewpoint. He talked about government-mandated maternity leave. A lot of women may love that ; it ’ s not a **conservative** viewpoint. So it ’ s fascinating that people who loved what his message was during the campaign, I ’ m not sure — how do you think they ’ ll react to that? DB : I should say I grew up in Lower Manhattan, in the triangle between ABC Carpets, the Strand Bookstore and The Odeon restaurant. GC : Come to Minnesota sometime! CA : You are a card-carrying member of the coastal elite, my man. But what did you make of the speech last night? It seemed to be a move to a more moderate position, on the face of it. DB : Yeah, I thought it was his best speech, and it took away the freakishness of him. I do think he ’ s a moral freak, and I think he ’ ll be undone by that fact, the fact that he just doesn ’ t know anything about anything and is uncurious about it. But if you take away these minor flaws, I think we got to see him at his best, and it was revealing for me to see him at his best, because to me, it exposed a central contradiction that he ’ s got to confront, that a lot of what he ’ s doing is offering security. So,”I ’ m ordering closed borders, I ’ m going to secure the world for you, for my people.”But then if you actually look at a lot of his economic **policies**, like health care reform, which is about private health care accounts, that ’ s not security, that ’ s risk. Educational vouchers : that ’ s risk. Deregulation : that ’ s risk. There ’ s really a contradiction between the security of the mindset and a lot of the **policies**, which are very risk-oriented. And what I would say, especially having spent this year, the people in rural Minnesota, in New Mexico — they ’ ve got enough risk in their lives. And so they ’ re going to say,”No thank you.”And I think his health care repeal will fail for that reason. CA : But despite the criticisms you just made of him, it does at least seem that he ’ s listening to a surprisingly wide range of voices ; it ’ s not like everyone is coming from the same place. And maybe that leads to a certain amount of chaos and confusion, but — GC : I actually don ’ t think he ’ s listening to a wide range of voices. I think he ’ s listening to very few people. That ’ s just my impression of it. I believe that some of the things he said last night had Ivanka all over them. So I believe he was listening to her before that speech. And he was Teleprompter Trump last night, as opposed to Twitter Trump. And that ’ s why, before we came out here, I said,”We better check Twitter to see if anything ’ s changed.”And also I think you have to keep in mind that because he ’ s such a unique character, what was the bar that we were expecting last night? Was it here or here or here? And so he comes out and gives a looking political speech, and everyone goes,”Wow! He can do it.”It just depends on which direction he goes. DB : Yeah, and we ’ re trying to build bridges here, and especially for an audience that may have contempt for Trump, it ’ s important to say, no, this is a real thing. But as I try my best to go an hour showing respect for him, my thyroid is surging, because I think the oddities of his character really are condemnatory and are going to doom him. CA : Your reputation is as a **conservative**. People would you describe you as right of center, and yet here you are with this visceral **reaction** against him and some of what he stands for. I mean, I ' m — how do you have a conversation? The people who **support** him, on evidence so far, are probably pretty excited. He ' s certainly shown real engagement in a lot of what he promised to do, and there is a strong desire to change the system radically. People hate what government has become and how it ' s left them out. GC : I totally agree with that, but I think that when he was proposing a huge government program last night that we used to call the bad s-word,"stimulus,"I find it completely ironic. To spend a trillion dollars on something — that is not a **conservative** viewpoint. Then again, I don ' t really believe he ' s a Republican. DB : And I would say, as someone who identifies as **conservative** : first of all, to be **conservative** is to believe in the limitations of politics. Samuel Johnson said,"Of all the things that human hearts endure, how few are those that kings can cause and cure."Politics is a limited realm ; what matters most is the moral nature of the society. And so I have to think character comes first, and a man who doesn ' t pass the character threshold cannot be a good president. Second, I ' m the kind of **conservative** who — I harken back to Alexander Hamilton, who was a Latino hip-hop star from the heights — but his definition of America was very future-oriented. He was a poor boy from the islands who had this rapid and amazing rise to success, and he wanted government to give poor boys and girls like him a chance to succeed, using limited but energetic government to create social mobility. For him and for Lincoln and for Teddy Roosevelt, the idea of America was the idea of the future. We may have division and racism and slavery in our past, but we have a common future. The definition of America that Steve Bannon stands for is backwards-looking. It ' s nostalgic ; it ' s for the past. And that is not traditionally the American identity. That ' s traditionally, frankly, the Russian identity. That ' s how they define virtue. And so I think it is a fundamental and foundational betrayal of what conservatism used to stand for. CA : Well, I ' d like actually like to hear from you, and if we see some comments coming in from some of you, we ' ll — oh, well here ' s one right now. Jeffrey Alan Carnegie : I ' ve tried to convince progressive friends that they need to understand what motivates Trump supporters, yet many of them have given up trying to understand in the face of what they perceive as lies, selfishness and hatred. How would you reach out to such people, the Tea Party of the left, to try to bridge this divide? GC : I actually think there are **commonalities** in anger, as I expressed earlier. So I think you can come to the table, both being passionate about something. So at least you care. And I would like to believe — the c-word has also become a horrible word —"compromise,"right? So you have the far left and the far right, and compromise — forget it. Those groups don ' t want to even think about it. But you have a huge swath of voters, myself included, who are registered independents, like 40 percent of us, right? So there is a huge faction of America that wants to see change and wants to see people come together. It ' s just that we have to figure out how to do that. CA : So let ' s talk about that for a minute, because we ' re having these **TED** Dialogues, we ' re trying to bridge. There ' s a lot of people out there, right now, perhaps especially on the left, who think this is a terrible idea, that actually, the only moral response to the great tyranny that may be about to emerge in America is to resist it at every stage, is to fight it tooth and nail, it ' s a mistake to try and do this. Just fight! Is there a case for that? DB : It depends what"fight"means. If it means literal fighting, then no. If it means marching, well maybe marching to raise consciousness, that seems fine. But if you want change in this country, we do it through parties and politics. We organize parties, and those parties are big, diverse, messy coalitions, and we engage in politics, and politics is always morally unsatisfying because it ' s always a bunch of compromises. But politics is essentially a competition between partial truths. The Trump people have a piece of the truth in America. I think Trump himself is the wrong answer to the right question, but they have some truth, and it ’ s truth found in the epidemic of opiates around the country, it ’ s truth found in the spread of loneliness, it ’ s the truth found in people whose lives are inverted. They peaked professionally at age 30, and it ’ s all been downhill since. And so, understanding that doesn ’ t take fighting, it takes conversation and then asking,”What are we going to replace Trump with?”GC : But you saw fighting last night, even at the speech, because you saw the Democratic women who came and wore white to honor the suffragette movement. I remember back during the campaign where some Trump supporters wanted to actually get rid of the amendment that allowed us to vote as women. It was like, what? So I don ’ t know if that ’ s the right way to fight. It was **interesting**, because I was looking in the audience, trying to see Democratic women who didn ’ t wear white. So there ’ s a lot going on there, and there ’ s a lot of ways to fight that are not necessarily doing that. CA : I mean, one of the key questions, to me, is : The people who voted for Trump but, if you like, are more in the center, like they ’ re possibly amenable to persuasion — are they more likely to be persuaded by seeing a passionate uprising of people saying,”No, no, no, you can ’ t!”or will that actually piss them off and push them away? DB : How are any of us persuaded? Am I going to persuade you by saying,”Well, you ’ re kind of a bigot, you ’ re **supporting** bigotry, you ’ re **supporting** sexism. You ’ re a primitive, fascistic rise from some authoritarian past”? That ’ s probably not going to be too persuasive to you. And so the way any of us are persuaded is by : a) some basic show of respect for the point of view, and saying,”I think this **guy** is not going to get you where you need to go.”And there are two phrases you ’ ve heard over and over again, wherever you go in the country. One, the phrase”flyover country.”And that ’ s been heard for years, but I would say this year, I heard it almost on an hourly basis, a sense of feeling invisible. And then the sense a sense of the phrase”political correctness.”Just that rebellion :”They ’ re not even letting us say what we think.”And I teach at Yale. The narrowing of debate is real. CA : So you would say this is a trap that liberals have fallen into by celebrating causes they really believe in, often expressed through the language of”political correctness.”They have done damage. They have pushed people away. DB : I would say a lot of the argument, though, with”descent to fascism,””authoritarianism”— that just feels over-the-top to people. And listen, I ’ ve written eight million anti-Trump columns, but it is a problem, especially for the coastal media, that every time he does something slightly wrong, we go to 11, and we ’ re at 11 every day. And it just strains credibility at some point. CA : Crying wolf a little too loud and a little too early. But there may be a time when we really do have to cry wolf. GC : But see — one of the most important things to me is how the **conservative** media handles Trump. Will they call him out when things are not true, or will they just go along with it? To me, that is what is essential in this entire discussion, because when you have followers of somebody who don ’ t really care if he tells the truth or not, that can be very dangerous. So to me, it ’ s : How is the **conservative** media going to respond to it? I mean, you ’ ve been calling them out. But how will other forms of **conservative** media deal with that as we move forward? DB : It ’ s all shifted, though. The **conservative** media used to be Fox or Charles Krauthammer or George Will. They ’ re no longer the **conservative** media. Now there ’ s another whole set of institutions further right, which is Breitbart and Infowars, Alex Jones, Laura Ingraham, and so they ’ re the ones who are now his base, not even so much Fox. CA : My last question for the time being is just on this question of the truth. I mean, it ’ s one of the scariest things to people right now, that there is no agreement, nationally, on what is true. I ’ ve never seen anything like it, where facts are so massively disputed. Your whole newspaper, sir, is delivering fake news every day. DB : And failing. CA : And failing. My commiserations. But is there any path whereby we can start to get some kind of consensus, to believe the same things? Can online communities play a role here? How do we fix this? GC : See, I understand how that happened. That ' s another groundswell kind of emotion that was going on in the middle of America and not being heard, in thinking that the mainstream media was biased. There ' s a difference, though, between being biased and being fake. To me, that is a very important distinction in this conversation. So let ' s just say that there was some bias in the mainstream media. OK. So there are ways to try and mend that. But what Trump ' s doing is nuclearizing that and saying,"Look, we ' re just going to call all of that fake."That ' s where it gets dangerous. CA : Do you think enough of his supporters have a greater loyalty to the truth than to any... Like, the principle of not **supporting** something that is demonstrably not true actually matters, so there will be a correction at some point? DB : I think the truth eventually comes out. So for example, Donald Trump has based a lot of his economic **policy** on this supposition that Americans have lost manufacturing jobs because they ' ve been stolen by the Chinese. That is maybe 13 percent of the jobs that left. The truth is that 87 percent of the jobs were replaced by technology. That is just the truth. And so as a result, when he says,"I ' m going to close TPP and all the jobs will come roaring back,"they will not come roaring back. So that is an actual fact, in my belief. And — GC : But I ' m saying what his supporters think is the truth, no matter how many times you might say that, they still believe him. DB : But eventually either jobs will come back or they will not come back, and at that point, either something will work or it doesn ' t work, and it doesn ' t work or not work because of great marketing, it works because it actually addresses a real problem and so I happen to think the truth will out. CA : If you ' ve got a question, please raise your hand here. Yael Eisenstat : I ' ll speak into the box. My name ' s Yael Eisenstat. I hear a lot of this talk about how we all need to start talking to each other more and understanding each other more, and I ' ve even written about this, published on this subject as well, but now today I keep hearing liberals — yes, I live in New York, I can be considered a liberal — we sit here and self-analyze : What did we do to not understand the Rust Belt? Or : What can we do to understand Middle America better? And what I ' d like to know : Have you seen any attempts or conversations from Middle America of what can I do to understand the so-called coastal elites better? Because I ' m just offended as being put in a box as a coastal elite as someone in Middle America is as being considered a flyover state and not listened to. CA : There you go, I can hear Facebook cheering as you — DB : I would say — and this is someone who has been **conservative** all my adult life — when you grow up **conservative**, you learn to speak both languages. Because if I ' m going to listen to music, I ' m not going to listen to Ted Nugent. So a lot of my favorite rock bands are all on the left. If I ' m going to go to a school, I ' m going probably to school where the culture is liberal. If I ' m going to watch a sitcom or a late-night comedy show, it ' s going to be liberal. If I ' m going to read a good newspaper, it ' ll be the New York Times. As a result, you learn to speak both languages. And that actually, at least for a number of years, when I started at National Review with William F. Buckley, it made us sharper, because we were used to arguing against people every day. The problem now that ' s happened is you have ghettoization on the right and you can live entirely in rightworld, so as a result, the quality of argument on the right has diminished, because you ' re not in the other side all the time. But I do think if you ' re living in Minnesota or Iowa or Arizona, the coastal elites make themselves aware to you, so you know that language as well, but it ’ s not the reverse. CA : But what does Middle America not get about coastal elites? So the critique is, you are not dealing with the real problems. There ’ s a feeling of a snobbishness, an elitism that is very off-putting. What are they missing? If you could plant one piece of truth from the mindset of someone in this room, for example, what would you say to them? DB : Just how insanely wonderful we are. No, I reject the category. The problem with populism is the same problem with elitism. It ’ s just a prejudice on the basis of probably an over-generalized social class distinction which is too simplistic to apply in reality. Those of us in New York know there are some people in New York who are completely awesome, and some people who are pathetic, and if you live in Iowa, some people are awesome and some people are pathetic. It ’ s not a question of what degree you have or where you happen to live in the country. The distinction is just a crude simplification to arouse political power. GC : But I would encourage people to watch a television news show or read a column that they normally wouldn ’ t. So if you are a Trump supporter, watch the other side for a day, because you need to come out of the bubble if you ’ re ever going to have a conversation. And both sides — so if you ’ re a liberal, then watch something that ’ s very **conservative**. Read a column that is not something you would normally read, because then you gain perspective of what the other side is thinking, and to me, that ’ s a start of coming together. I worry about the same thing you worry about, these bubbles. I think if you only watch certain entities, you have no idea what the rest of the world is talking about. DB : I think not only watching, being part of an organization that meets at least once a month that puts you in direct contact with people completely unlike yourself is something we all have a responsibility for. I may get this a little wrong, but I think of the top-selling automotive models in this country, I think the top three or four are all pickup trucks. So ask yourself : How many people do I know who own a pickup truck? And it could be very few or zero for a lot of people. And that ’ s sort of a warning sign kind of a problem. Where can I join a club where I ’ ll have a lot in common with a person who drives a pickup truck because we have a common interest in whatever? CA : And so the internet is definitely contributing to this. A question here from Chris Ajemian :”How do you feel structure of communications, especially the prevalence of social media and individualized content, can be used to bring together a political divide, instead of just filing communities into echo chambers?”I mean, it looks like Facebook and Google, since the election, are working hard on this question. They ’ re trying to change the algorithms so that they don ’ t amplify fake news to the extent that it happened last time round. Do you see any other promising signs of...? GC :... or amplify one side of the equation. CA : Exactly. GC : I think that was the constant argument from the right, that social media and the internet in general was putting articles towards the top that were not their worldview. I think, again, that fed into the anger. It fed into the anger of :”You ’ re pushing something that ’ s not what I believe.”But social media has obviously changed everything, and I think Trump is the example of Twitter changing absolutely everything. And from his point of view, he ’ s reaching the American people without a filter, which he believes the media is. CA : Question from the audience. Destiny : Hi. I ’ m Destiny. I have a question regarding political correctness, and I ’ m curious : When did political correctness become synonymous with silencing, versus a way that we speak about other people to show them respect and preserve their dignity? GC : Well, I think the **conservative** media really pounded this issue for the last 10 years. I think that they really, really spent a lot of time talking about political correctness, and how people should have the **ability** to say what they think. Another reason why Trump became so popular : because he says what he thinks. It also makes me think about the fact that I do believe there are a lot of people in America who agree with Steve Bannon, but they would never say it publicly, and so voting for Trump gave them the opportunity to agree with it silently. DB : On the issue of immigration, it ’ s a legitimate point of view that we have too many immigrants in the country, that it ’ s economically costly. CA : That we have too many — DB : Immigrants in the country, especially from Britain. GC : I kind of like the British accent, OK? CA : I apologize. America, I am sorry. I ’ ll go now. DB : But it became sort of impermissible to say that, because it was a sign that somehow you must be a bigot of some sort. So the political correctness was not only cracking down on speech that we would all find completely offensive, it was cracking down on some speech that was legitimate, and then it was turning speech and thought into action and treating it as a crime, and people getting fired and people thrown out of schools, and there were speech codes written. Now there are these **diversity** teams, where if you say something that somebody finds offensive, like,”Smoking is really dangerous,”you can say”You ’ re insulting my group,”and the team from the administration will come down into your dorm room and put thought police upon you. And so there has been a genuine narrowing of what is permissible to say. And some of it is legitimate. There are certain words that there should be some social sanction against, but some of it was used to enforce a political agenda. CA : So is that a project you would urge on liberals, if you like — progressives — to rethink the ground rules around political correctness and accept a little more uncomfortable language in certain circumstances? Can you see that being solved to an extent that others won ’ t be so offended? DB : I mean, most American universities, especially elite universities, are overwhelmingly on the left, and there ’ s just an ease of temptation to use your overwhelming cultural power to try to enforce some sort of thought that you think is right and correct thought. So, be a little more self-suspicious of, are we doing that? And second, my university, the University of Chicago, sent out this letter saying, we will have no **safe** spaces. There will be no critique of micro-aggression. If you get your feelings hurt, well, welcome to the world of education. I do think that **policy** — which is being embraced by a lot of people on the left, by the way — is just a corrective to what ’ s happened. CA : So here ’ s a question from Karen Holloway : How do we foster an American culture that ’ s forward-looking, like Hamilton, that expects and deals with change, rather than wanting to have everything go back to some fictional past? That ’ s an easy question, right? GC : Well, I ’ m still a believer in the American dream, and I think what we can teach our children is the basics, which is that hard work and believing in yourself in America, you can achieve whatever you want. I was told that every single day. When I got in the real world, I was like, wow, that ’ s maybe not always so true. But I still believe in that. Maybe I ’ m being too optimistic. So I still look towards the future for that to continue. DB : I think you ’ re being too optimistic. GC : You do? DB : The odds of an American young person exceeding their parents ’ salary — a generation ago, like 86 percent did it. Now 51 percent do it. There ’ s just been a problem in social mobility in the country. CA : You ’ ve written that this entire century has basically been a disaster, that the age of sunny growth is over and we ’ re in deep trouble. DB : Yeah, I mean, we averaged, in real terms, population-adjusted, two or three percent growth for 50 years, and now we ’ ve had less than one percent growth. And so there ’ s something seeping out. And so if I ’ m going to tell people that they should take risks, one of the things we ’ re seeing is a rapid decline in mobility, the number of people who are moving across state lines, and that ’ s especially true among millennials. It ’ s young people that are moving less. So how do we give people the security from which they can take risk? And I ’ m a big believer in attachment theory of raising children, and attachment theory is based on the motto that all of life is a series of daring adventures from a secure base. Have you parents given you a secure base? And as a society, we do not have a secure base, and we won ' t get to that"Hamilton,"risk-taking, energetic ethos until we can supply a secure base. CA : So I wonder whether there ' s ground here to create almost like a shared agenda, a bridging conversation, on the one hand recognizing that there is this really deep problem that the system, the economic system that we built, seems to be misfiring right now. Second, that maybe, if you ' re right that it ' s not all about immigrants, it ' s probably more about technology, if you could win that argument, that de-emphasizes what seems to me the single most divisive territory between Trump supporters and others, which is around the role of the other. It ' s very offensive to people on the left to have the other demonized to the extent that the other seems to be demonized. That feels deeply immoral, and maybe people on the left could agree, as you said, that immigration may have happened too fast, and there is a limit beyond which human societies struggle, but nonetheless this whole problem becomes de-emphasized if automation is the key issue, and then we try to work together on recognizing that it ' s real, recognizing that the problem probably wasn ' t properly addressed or seen or heard, and try to figure out how to rebuild communities using, well, using what? That seems to me to become the fertile conversation of the future : How do we rebuild communities in this modern age, with technology doing what it ' s doing, and reimagine this bright future? GC : That ' s why I go back to optimism. I ' m not being... it ' s not like I ' m not looking at the facts, where we ' ve come or where we ' ve come from. But for gosh sakes, if we don ' t look at it from an optimistic point of view — I ' m **refusing** to do that just yet. I ' m not raising my 12- and 13-year-old to say,"Look, the world is dim."CA ; We ' re going to have one more question from the room here. Questioner : Hi. Hello. Sorry. You both mentioned the infrastructure plan and Russia and some other things that wouldn ' t be traditional Republican priorities. What do you think, or when, will **Republicans** be motivated to take a stand against Trumpism? GC : After last night, not for a while. He changed a lot last night, I believe. DB : His popularity among **Republicans** — he ' s got 85 percent approval, which is higher than Reagan had at this time, and that ' s because society has just gotten more polarized. So people follow the party much more than they used to. So if you ' re waiting for Paul Ryan and the Republicans in Congress to flake away, it ' s going to take a little while. GC : But also because they ' re all concerned about reelection, and Trump has so much power with getting people either for you or against you, and so, they ' re vacillating every day, probably :"Well, should I go against or should I not?"But last night, where he finally sounded presidential, I think most Republicans are breathing a sigh of relief today. DB : The half-life of that is short. GC : Right — I was just going to say, until Twitter happens again. CA : OK, I want to give each of you the chance to imagine you ' re speaking to — I don ' t know — the people online who are watching this, who may be Trump supporters, who may be on the left, somewhere in the middle. How would you advise them to bridge or to relate to other people? Can you share any final wisdom on this? Or if you think that they shouldn ' t, tell them that as well. GC : I would just start by saying that I really think any change and coming together starts from the top, just like any other organization. And I would love if, somehow, Trump supporters or people on the left could encourage their **leaders** to show that compassion from the top, because imagine the change that we could have if Donald Trump tweeted out today, to all of his supporters,"Let ' s not be vile anymore to each other. Let ' s have more understanding. As a **leader**, I ' m going to be more inclusive to all of the people of America."To me, it starts at the top. Is he going to do that? I have no idea. But I think that everything starts from the top, and the power that he has in encouraging his supporters to have an understanding of where people are coming from on the other side. CA : David. DB : Yeah, I guess I would say I don ’ t think we can teach each other to be civil, and give us sermons on civility. That ’ s not going to do it. It ’ s substance and how we act, and the nice thing about Donald Trump is he smashed our categories. All the categories that we thought we were thinking in, they ’ re obsolete. They were great for the 20th century. They ’ re not good for today. He ’ s got an agenda which is about closing borders and closing trade. I just don ’ t think it ’ s going to work. I think if we want to rebuild communities, recreate jobs, we need a different set of agenda that smashes through all our current divisions and our current categories. For me, that agenda is Reaganism on macroeconomic **policy**, Sweden on welfare **policy** and cuts across right and left. I think we have to have a dynamic economy that creates growth. That ’ s the Reagan on economic **policy**. But people have to have that secure base. There have to be nurse-family partnerships ; there has to be universal preschool ; there have to be charter schools ; there have to be college programs with wraparound programs for parents and communities. We need to help heal the **crisis** of social solidarity in this country and help heal families, and government just has to get a lot more involved in the way liberals like to rebuild communities. At the other hand, we have to have an economy that ’ s free and open the way **conservatives** used to like. And so getting the substance right is how you smash through the partisan identities, because the substance is what ultimately shapes our polarization. CA : David and Gretchen, thank you so much for an absolutely fascinating conversation. Thank you. That was really, really **interesting**. Hey, let ’ s keep the conversation going. We ’ re continuing to try and figure out whether we can add something here, so keep the conversation going on Facebook. Give us your thoughts from whatever part of the political spectrum you ’ re on, and actually, wherever in the world you are. This is not just about America. It ’ s about the world, too. But we ’ re not going to end today without music, because if we put music in every political conversation, the world would be completely different, frankly. It just would. Up in Harlem, this extraordinary woman, Vy Higginsen, who ’ s actually right here — let ’ s get a shot of her. She created this program that brings teens together, teaches them the joy and the **impact** of gospel music, and hundreds of teens have gone through this program. It ’ s transformative for them. The music they made, as you already heard, is extraordinary, and I can ’ t think of a better way of ending this **TED** Dialogue than welcoming Vy Higginsen ’ s Gospel Choir from Harlem. Thank you. Choir : O beautiful for spacious skies For amber waves of grain For purple mountain majesties Above the fruited plain America! America! America! America! God shed his grace on thee And crown thy good with brotherhood From sea to shining sea From sea to shining sea

Text Sample 545: POS Dim 4 – 2017 – Score 18.00 – 2017/t000993.txt

Chris Anderson : Christiane, great to have you here. So you ’ ve had this amazing viewpoint, and perhaps it ’ s fair to say that in the last few years, there have been some alarming developments that you ’ re seeing. What ’ s alarmed you most? Christiane Amanpour : Well, just listening to the earlier speakers, I can frame it in what they ’ ve been saying : **climate** change, for instance — cities, the threat to our environment and our lives. It basically also boils down to understanding the truth and to be able to get to the truth of what we ’ re talking about in order to really be able to solve it. So if 99. 9 percent of the science on **climate** is empirical, scientific evidence, but it ' s competing almost equally with a handful of deniers, that is not the truth ; that is the epitome of fake news. And so for me, the last few years — certainly this last year — has crystallized the notion of fake news in a way that ' s truly alarming and not just some slogan to be thrown around. Because when you can ' t distinguish between the truth and fake news, you have a very much more difficult time trying to solve some of the great issues that we face. CA : Well, you ' ve been **involved** in this question of, what is balance, what is truth, what is impartiality, for a long time. You were on the front lines reporting the Balkan Wars 25 years ago. And back then, you famously said, by calling out human right abuses, you said,"Look, there are some situations one simply cannot be neutral about, because when you ' re neutral, you are an accomplice."So, do you feel that today ' s journalists aren ' t heeding that advice about balance? CA : Well, look, I think for journalists, objectivity is the golden rule. But I think sometimes we don ' t understand what objectivity means. And I actually learned this very, very young in my career, which was during the Balkan Wars. I was young then. It was about 25 years ago. And what we faced was the wholesale violation, not just of human rights, but all the way to ethnic cleansing and genocide, and that has been adjudicated in the highest war crimes **court** in the world. So, we know what we were seeing. Trying to tell the world what we were seeing brought us accusations of bias, of siding with one side, of not seeing the whole side, and just, you know, trying to tell one story. I particularly and personally was accused of siding with, for instance, the citizens of Sarajevo —"siding with the Muslims,"because they were the minority who were being attacked by Christians on the Serb side in this area. And it worried me. It worried me that I was being accused of this. I thought maybe I was wrong, maybe I ' d forgotten what objectivity was. But then I started to understand that what people wanted was actually not to do anything — not to step in, not to change the situation, not to find a solution. And so, their fake news at that time, their lie at that time — including our government ' s, our democratically elected government ' s, with values and principles of human rights — their lie was to say that all sides are equally guilty, that this has been centuries of ethnic hatred, whereas we knew that wasn ' t true, that one side had decided to kill, slaughter and ethnically cleanse another side. So that is where, for me, I understood that objectivity means giving all sides an equal hearing and talking to all sides, but not treating all sides equally, not creating a forced moral equivalence or a factual equivalence. And when you come up against that **crisis** point in situations of grave violations of **international** and humanitarian law, if you don ' t understand what you ' re seeing, if you don ' t understand the truth and if you get trapped in the fake news paradigm, then you are an accomplice. And I **refuse** to be an accomplice to genocide. CH : So there have always been these propaganda battles, and you were courageous in taking the stand you took back then. Today, there ' s a whole new way, though, in which news seems to be becoming fake. How would you characterize that? CA : Well, look — I am really alarmed. And everywhere I look, you know, we ' re buffeted by it. Obviously, when the **leader** of the free world, when the most powerful person in the entire world, which is the president of the United States — this is the most important, most powerful country in the whole world, economically, militarily, politically in every which way — and it seeks to, obviously, promote its values and power around the world. So we journalists, who only seek the truth — I mean, that is our mission — we go around the world looking for the truth in order to be **everybody** ' s eyes and ears, people who can ' t go out in various parts of the world to figure out what ' s going on about things that are vitally important to **everybody** ' s health and security. So when you have a major world **leader** accusing you of fake news, it has an exponential ripple effect. And what it does is, it starts to chip away at not just our credibility, but at people ’ s minds — people who look at us, and maybe they ’ re thinking,”Well, if the president of the United States says that, maybe somewhere there ’ s a truth in there.”CH : Presidents have always been critical of the media — CA : Not in this way. CH : So, to what extent — CH : I mean, someone a couple years ago looking at the avalanche of information pouring through Twitter and Facebook and so forth, might have said,”Look, our democracies are healthier than they ’ ve ever been. There ’ s more news than ever. Of course presidents will say what they ’ ll say, but everyone else can say what they will say. What ’ s not to like? How is there an extra danger?”CA : So, I wish that was true. I wish that the proliferation of platforms upon which we get our information meant that there was a proliferation of truth and transparency and depth and accuracy. But I think the opposite has happened. You know, I ’ m a little bit of a Luddite, I will confess. Even when we started to talk about the information superhighway, which was a long time ago, before social media, Twitter and all the rest of it, I was actually really afraid that that would put people into certain lanes and tunnels and have them just focusing on areas of their own interest instead of seeing the broad picture. And I ’ m afraid to say that with algorithms, with logarithms, with whatever the”-ithms”are that direct us into all these particular channels of information, that seems to be happening right now. I mean, people have written about this phenomenon. People have said that yes, the internet came, its promise was to exponentially explode our access to more democracy, more information, less bias, more varied information. And, in fact, the opposite has happened. And so that, for me, is incredibly dangerous. And again, when you are the president of this country and you say things, it also gives **leaders** in other undemocratic countries the cover to affront us even worse, and to really whack us — and their own journalists — with this bludgeon of fake news. CH : To what extent is what happened, though, in part, just an unintended consequence, that the traditional media that you worked in had this curation-mediation role, where certain norms were observed, certain stories would be rejected because they weren ’ t credible, but now that the standard for publication and for amplification is just interest, attention, excitement, click,”Did it get clicked on?””Send it out there!”and that ’ s what ’ s — is that part of what ’ s caused the problem? CA : I think it ’ s a big problem, and we saw this in the election of 2016, where the idea of”clickbait”was very sexy and very attractive, and so all these fake news sites and fake news items were not just haphazardly and by happenstance being put out there, there ’ s been a whole industry in the creation of fake news in parts of Eastern Europe, wherever, and you know, it ’ s planted in real space and in cyberspace. So I think that, also, the **ability** of our technology to proliferate this **stuff** at the speed of sound or light, just about — we ’ ve never faced that before. And we ’ ve never faced such a massive amount of information which is not curated by those whose profession leads them to abide by the truth, to fact-check and to maintain a code of conduct and a code of professional ethics. CH : Many people here may know people who work at Facebook or Twitter and Google and so on. They all seem like great people with good intention — let ’ s assume that. If you could speak with the **leaders** of those companies, what would you say to them? CA : Well, you know what — I ’ m sure they are incredibly well-intentioned, and they certainly developed an unbelievable, game-changing system, where **everybody** ’ s connected on this thing called Facebook. And they ’ ve created a massive economy for themselves and an amazing amount of income. I would just say,”Guys, you know, it ’ s time to wake up and smell the coffee and look at what ’ s happening to us right now.”Mark Zuckerberg wants to create a global community. I want to know : What is that global community going to look like? I want to know where the codes of conduct actually are. Mark Zuckerberg said — and I don ' t blame him, he probably believed this — that it was crazy to think that the Russians or anybody else could be tinkering and messing around with this avenue. And what have we just learned in the last few weeks? That, actually, there has been a major problem in that regard, and now they ' re having to investigate it and figure it out. Yes, they ' re trying to do what they can now to prevent the rise of fake news, but, you know, it went pretty unrestricted for a long, long time. So I guess I would say, you know, you **guys** are brilliant at technology ; let ' s figure out another algorithm. Can we not? CH : An algorithm that includes journalistic investigation — CA : I don ' t really know how they do it, but somehow, you know — filter out the crap! And not just the unintentional — but the deliberate lies that are planted by people who ' ve been doing this as a matter of warfare for decades. The Soviets, the Russians — they are the masters of war by other means, of hybrid warfare. And this is a — this is what they ' ve decided to do. It worked in the United States, it didn ' t work in France, it hasn ' t worked in Germany. During the elections there, where they ' ve tried to interfere, the president of France right now, Emmanuel Macron, took a very tough stand and confronted it head on, as did Angela Merkel. CH : There ' s some hope to be had from some of this, isn ' t there? That the world learns. We get fooled once, maybe we get fooled again, but maybe not the third time. Is that true? CA : I mean, let ' s hope. But I think in this regard that so much of it is also about technology, that the technology has to also be given some kind of moral compass. I know I ' m talking nonsense, but you know what I mean. CH : We need a filter-the-crap algorithm with a moral compass — CA : There you go. CH : I think that ' s good. CA : No —"moral technology."We all have moral compasses — moral technology. CH : I think that ' s a great challenge. CA : You know what I mean. CH : Talk just a minute about **leadership**. You ' ve had a chance to speak with so many people across the world. I think for some of us — I speak for myself, I don ' t know if others feel this — there ' s kind of been a disappointment of : Where are the **leaders**? So many of us have been disappointed — Aung San Suu Kyi, what ' s happened recently, it ' s like,"No! Another one bites the dust."You know, it ' s heartbreaking. Who have you met who you have been impressed by, inspired by? CA : Well, you talk about the world in **crisis**, which is absolutely true, and those of us who spend our whole lives immersed in this **crisis** — I mean, we ' re all on the verge of a nervous breakdown. So it ' s pretty stressful right now. And you ' re right — there is this perceived and actual vacuum of **leadership**, and it ' s not me saying it, I ask all these — whoever I ' m talking to, I ask about **leadership**. I was speaking to the outgoing president of Liberia today, [Ellen Johnson Sirleaf,] who — in three weeks ' time, will be one of the very rare heads of an African country who actually abides by the constitution and gives up power after her prescribed term. She has said she wants to do that as a lesson. But when I asked her about **leadership**, and I gave a **quick-fire** round of certain names, I presented her with the name of the new French president, Emmanuel Macron. And she said — I said,"So what do you think when I say his name?"And she said,"Shaping up potentially to be a **leader** to fill our current **leadership** vacuum."I thought that was really **interesting**. Yesterday, I happened to have an interview with him. I ' m very proud to say, I got his first **international** interview. It was great. It was yesterday. And I was really impressed. I don ' t know whether I should be saying that in an open forum, but I was really impressed. And it could be just because it was his first interview, but — I asked questions, and you know what? He answered them! There was no spin, there was no wiggle and waggle, there was no spend-five- minutes- to-come-back-to-the-point. I didn ' t have to keep interrupting, which I ' ve become rather renowned for doing, because I want people to answer the question. And he answered me, and it was pretty **interesting**. And he said — CH : Tell me what he said. CA : No, no, you go ahead. CH : You ’ re the interrupter, I ’ m the listener. CA : No, no, go ahead. CH : What ’ d he say? CA : OK. You ’ ve talked about nationalism and tribalism here today. I asked him,”How did you have the guts to confront the prevailing winds of anti- globalization, nationalism, populism when you can see what happened in Brexit, when you could see what happened in the United States and what might have happened in many European elections at the beginning of 2017?”And he said,”For me, nationalism means war. We have seen it before, we have lived through it before on my continent, and I am very clear about that.”So he was not going to, just for political expediency, embrace the, kind of, lowest common denominator that had been embraced in other political elections. And he stood against Marine Le Pen, who is a very dangerous woman. CH : Last question for you, Christiane. **TED** is about ideas worth spreading. If you could plant one idea into the minds of everyone here, what would that be? CA : I would say really be careful where you get your information from ; really take responsibility for what you read, listen to and watch ; make sure that you go to the **trusted** brands to get your main information, no matter whether you have a wide, eclectic intake, really stick with the brand names that you know, because in this world right now, at this moment right now, our **crises**, our challenges, our problems are so severe, that unless we are all engaged as global citizens who appreciate the truth, who understand science, empirical evidence and facts, then we are just simply going to be wandering along to a potential catastrophe. So I would say, the truth, and then I would come back to Emmanuel Macron and talk about love. I would say that there ’ s not enough love going around. And I asked him to tell me about love. I said,”You know, your marriage is the subject of global obsession.””Can you tell me about love? What does it mean to you?”I ’ ve never asked a president or an elected **leader** about love. I thought I ’ d try it. And he said — you know, he actually answered it. And he said,”I love my wife, she is part of me, we ’ ve been together for decades.”But here ’ s where it really counted, what really stuck with me. He said,”It is so important for me to have somebody at home who tells me the truth.”So you see, I brought it home. It ’ s all about the truth. CH : So there you go. Truth and love. Ideas worth spreading. Christiane Amanpour, thank you so much. That was great. CA : Thank you. CH : That was really lovely. CA : Thank you.

Text Sample 546: POS Dim 4 – 2009 – Score 18.00 – 2009/t002916.txt

Chris Anderson : Thank you so much, Prime Minister, that was both fascinating and quite inspiring. So, you ’ re calling for a global ethic. Would you describe that as global citizenship? Is that an idea that you believe in, and how would you define that? Gordon Brown : It is about global citizenship and recognizing our responsibilities to others. There is so much to do over the next few years that is obvious to so many of us to build a better world. And there is so much shared sense of what we need to do, that it is vital that we all come together. But we don ’ t necessarily have the means to do so. So there are challenges to be met. I believe the concept of global citizenship will simply grow out of people talking to each other across continents. But of course the task is to create the institutions that make that global society work. But I don ’ t think we should underestimate the extent to which massive changes in technology make possible the linking up of people across the world. CA : But people get excited about this idea of global citizenship, but then they get confused a bit again when they start thinking about patriotism, and how to combine these two. I mean, you ' re elected as Prime Minister with a brief to bat for Britain. How do you reconcile the two things? GB : Well, of course national identity remains important. But it ' s not at the expense of people accepting their global responsibilities. And I think one of the problems of **recession** is that people become more protectionist, they look in on themselves, they try to protect their own nation, perhaps at the expense of other nations. When you actually look at the motor of the world economy, it cannot move forward unless there is trade between the different countries. And any nation that would become protectionist over the next few years would deprive itself of the chance of getting the benefits of growth in the world economy. So, you ' ve got to have a healthy sense of patriotism ; that ' s absolutely important. But you ' ve got to realize that this world has changed fundamentally, and the problems we have cannot be solved by one nation and one nation alone. CA : Well, indeed. But what do you do when the two come into **conflict** and you ' re forced to make a decision that either is in Britain ' s interest, or the interest of Britons, or citizens elsewhere in the world? GB : Well I think we can persuade people that what is necessary for Britain ' s long-term interests, what is necessary for America ' s long-term interests, is proper engagement with the rest of the world, and taking the action that is necessary. There is a great story, again, told about Richard Nixon. 1958, Ghana becomes independent, so it is just over 50 years ago. Richard Nixon goes to represent the United States government at the celebrations for independence in Ghana. And it ' s one of his first outings as Vice President to an African country. He doesn ' t quite know what to do, so he starts going around the crowd and starts talking to people and he says to people in this rather unique way,"How does it feel to be free?"And he ' s going around,"How does it feel to be free?""How does it feel to be free?"And then someone says,"How should I know? I come from Alabama."And that was the 1950s. Now, what is remarkable is that civil rights in America were achieved in the 1960s. But what is equally remarkable is socioeconomic rights in Africa have not moved forward very fast even since the age of colonialism. And yet, America and Africa have got a common interest. And we have got to realize that if we don ' t link up with those people who are sensible voices and democratic voices in Africa, to work together for common causes, then the danger of Al Qaeda and related groups making progress in Africa is very big. So, I would say that what seems sometimes to be altruism, in relation to Africa, or in relation to developing countries, is more than that. It is enlightened self-interest for us to work with other countries. And I would say that national interest and, if you like, what is the global interest to tackle poverty and **climate** change do, in the long run, come together. And whatever the short-run price for taking action on **climate** change or on security, or taking action to provide opportunities for people for education, these are prices that are worth paying so that you build a stronger global society where people feel able to feel comfortable with each other and are able to communicate with each other in such a way that you can actually build stronger links between different countries. CA : I still just want to draw out on this issue. So, you ' re on vacation at a nice beach, and word comes through that there ' s been a massive earthquake and that there is a tsunami advancing on the beach. One end of the beach, there is a house containing a family of five Nigerians. And at the other end of the beach there is a single Brit. You have time to — you have time to alert one house. What do you do? GB : Modern communications. Alert both. I do agree that my responsibility is first of all to make sure that people in our country are **safe**. And I wouldn ' t like anything that is said today to suggest that I am diminishing the importance of the responsibility that each **leader** has for their own country. But I ' m trying to suggest that there is a huge opportunity open to us that was never open to us before. But the power to communicate across borders allows us to organize the world in a different way. And I think, look at the tsunami, it ' s a classic example. Where was the early warning systems? Where was the world acting together to deal with the problems that they knew arose from the potential for earthquakes, as well as the potential for **climate** change? And when the world starts to work together, with better early-warning systems, you can deal with some of these problems in a better way. I just think we ' re not seeing, at the moment, the huge opportunities open to us by the **ability** of people to cooperate in a world where either there was isolationism before or there was limited alliances based on convenience which never actually took you to deal with some of the central problems. CA : But I think this is the frustration that perhaps a lot of people have, like people in the audience here, where we love the kind of language that you ' re talking about. It is inspiring. A lot of us believe that that has to be the world ' s future. And yet, when the situation changes, you suddenly hear politicians talking as if, you know, for example, the life of one American soldier is worth countless numbers of Iraqi civilians. When the pedal hits the metal, the idealism can get moved away. I ' m just wondering whether you can see that changing over time, whether you see in Britain that there are changing attitudes, and that people are actually more supportive of the kind of global ethic that you talk about. GB : I think every religion, every faith, and I ' m not just talking here to people of faith or religion — it has this global ethic at the center of its credo. And whether it ' s Jewish or whether it ' s Muslim or whether it ' s Hindu, or whether it ' s Sikh, the same global ethic is at the heart of each of these religions. So, I think you ' re dealing with something that people instinctively see as part of their moral sense. So you ' re building on something that is not pure self-interest. You ' re building on people ' s ideas and values — that perhaps they ' re candles that burn very dimly on certain occasions. But it is a set of values that cannot, in my view, be extinguished. Then the question is, how do you make that change happen? How do you persuade people that it is in their interest to build strong — After the Second World War, we built institutions, the United Nations, the IMF, the World Bank, the World Trade Organization, the Marshall Plan. There was a period in which people talked about an act of creation, because these institutions were so new. But they are now out of date. They don ' t deal with the problems. You can ' t deal with the environmental problem through existing institutions. You can ' t deal with the security problem in the way that you need to. You can ' t deal with the economic and financial problem. So we have got to rebuild our global institutions, build them in a way that is suitable to the challenges of this time. And I believe that if you look at the biggest challenge we face, it is to persuade people to have the **confidence** that we can build a truly global society with the institutions that are founded on these rules. So, I come back to my initial point. Sometimes you think things are impossible. Nobody would have said 50 years ago that apartheid would have gone in 1990, or that the Berlin wall would have fallen at the turn of the ' 80s and ' 90s, or that polio could be eradicated, or perhaps 60 years ago, nobody would have said a man could gone to the Moon. All these things have happened. By tackling the impossible, you make the impossible possible. CA : And we have had a speaker who said that very thing, and swallowed a sword right after that, which was quite dramatic. GB : Followed my sword and swallow. CA : But, surely a true global ethic is for someone to say,"I believe that the life of every human on the planet is worth the same, equal consideration, regardless of nationality and religion."And you have politicians who have — you ' re elected. In a way, you can ' t say that. Even if, as a human being, you believe that, you can ' t say that. You ' re elected for Britain ' s interests. GB : We have a responsibility to protect. I mean look, 1918, the Treaty of Versailles, and all the treaties before that, the Treaty of Westphalia and everything else, were about protecting the sovereign right of countries to do what they want. Since then, the world has moved forward, partly as a result of what happened with the Holocaust, and people ' s concern about the rights of individuals within territories where they need protection, partly because of what we saw in Rwanda, partly because of what we saw in Bosnia. The idea of the responsibility to protect all individuals who are in situations where they are at humanitarian risk is now being established as a principle which governs the world. So, while I can ' t automatically say that Britain will rush to the aid of any citizen of any country, in danger, I can say that Britain is in a position where we ' re working with other countries so that this idea that you have a responsibility to protect people who are victims of either genocide or humanitarian attack, is something that is accepted by the whole world. Now, in the end, that can only be achieved if your **international** institutions work well enough to be able to do so. And that comes back to what the future role of the United Nations, and what it can do, actually is. But, the responsibility to protect is a new idea that is, in a sense, taken over from the idea of **self-determination** as the principle governing the **international** community. CA : Can you picture, in our lifetimes, a politician ever going out on a platform of the kind of full-form global ethic, global citizenship? And basically saying,"I believe that all people across the planet have equal consideration, and if in power we will act in that way. And we believe that the people of this country are also now global citizens and will **support** that ethic."GB : Is that not what we ' re doing in the debate about **climate** change? We ' re saying that you cannot solve the problem of **climate** change in one country ; you ' ve got to **involve** all countries. You ' re saying that you must, and you have a duty to help those countries that cannot afford to deal with the problems of **climate** change themselves. You ' re saying you want a deal with all the different countries of the world where we ' re all bound together to cutting carbon emissions in a way that is to the benefit of the whole world. We ' ve never had this before because Kyoto didn ' t work. If you could get a deal at Copenhagen, where people agreed, A, that there was a long-term target for carbon emission cuts, B, that there was short-range targets that had to be met so this wasn ' t just abstract ; it was people actually making decisions now that would make a difference now, and if you could then find a financing mechanism that meant that the poorest countries that had been hurt by our inability to deal with **climate** change over many, many years and decades are given special help so that they can move to energy-efficient technologies, and they are in a position financially to be able to afford the long-term investment that is associated with cutting carbon emissions, then you are treating the world equally, by giving consideration to every part of the planet and the needs they have. It doesn ' t mean that **everybody** does exactly the same thing, because we ' ve actually got to do more financially to help the poorest countries, but it does mean there is equal consideration for the needs of citizens in a single planet. CA : Yes. And then of course the theory is still that those talks get rent apart by different countries fighting over their own individual interests. GB : Yes, but I think Europe has got a position, which is 27 countries have already come together. I mean, the great difficulty in Europe is if you ' re at a meeting and 27 people speak, it takes a very, very long time. But we did get an agreement on **climate** change. America has made its first disposition on this with the bill that President Obama should be congratulated for getting through Congress. Japan has made an announcement. China and India have signed up to the scientific evidence. And now we ' ve got to move them to accept a long-term target, and then short-term targets. But more progress has been made, I think, in the last few weeks than had been made for some years. And I do believe that there is a strong possibility that if we work together, we can get that agreement to Copenhagen. I certainly have been putting forward proposals that would have allowed the poorest parts of the world to feel that we have taken into account their specific needs. And we would help them adapt. And we would help them make the **transition** to a low-carbon economy. I do think a reform of the **international** institutions is vital to this. When the IMF was created in the 1940s, it was created with resources that were five percent or so of the world ' s GDP. The IMF now has limited resources, one percent. It can ' t really make the difference that ought to be made in a period of **crisis**. So, we ' ve got to rebuild the world institutions. And that ' s a big task : persuading all the different countries with the different voting shares in these institutions to do so. There is a story told about the three world **leaders** of the day getting a chance to get some advice from God. And the story is told that Bill Clinton went to God and he asked when there will be successful **climate** change and a low-carbon economy. And God shook his head and said,"Not this year, not this decade, perhaps not even in [your] lifetime."And Bill Clinton walked away in tears because he had failed to get what he wanted. And then the story is that Barroso, the president of the European Commission, went to God and he asked,"When will we get a recovery of global growth?"And God said,"Not this year, not in this decade, perhaps not in your lifetime."So Barroso walked away crying and in tears. And then the Secretary-General of the United Nations came up to speak to God and said,"When will our **international** institutions work?"And God cried. It is very important to recognize that this reform of institutions is the next stage after agreeing upon ourselves that there is a clear ethic upon which we can build. CA : Prime Minister, I think there are many in the audience who are truly appreciative of the efforts you made in terms of the financial mess we got ourselves into. And there are certainly many people in the audience who will be cheering you on as you seek to advance this global ethic. Thank you so much for coming to **TED**. GB : Well, thank you.

Text Sample 547: POS Dim 4 – 2009 – Score 18.00 – 2009/t002768.txt

My story is a little bit about war. It ' s about disillusionment. It ' s about death. And it ' s about rediscovering idealism in all of that wreckage. And perhaps also, there ' s a lesson about how to deal with our screwed-up, fragmenting and dangerous world of the 21st century. I don ' t believe in straightforward narratives. I don ' t believe in a life or history written as decision"A"led to consequence"B"led to consequence"C"— these neat narratives that we ' re presented with, and that perhaps we encourage in each other. I believe in randomness, and one of the reasons I believe that is because me becoming a diplomat was random. I ' m colorblind. I was born unable to see most colors. This is why I wear gray and black most of the time, and I have to take my wife with me to chose clothes. And I ' d always wanted to be a fighter pilot when I was a boy. I loved watching planes barrel over our holiday home in the countryside. And it was my boyhood dream to be a fighter pilot. And I did the tests in the Royal Air Force to become a pilot, and sure enough, I failed. I couldn ' t see all the blinking different lights, and I can ' t distinguish color. So I had to choose another career, and this was in fact relatively easy for me, because I had an abiding passion all the way through my childhood, which was **international** relations. As a child, I read the newspaper thoroughly. I was fascinated by the Cold War, by the INF negotiations over intermediate-range nuclear missiles, the proxy war between the Soviet Union and the U. S. in Angola or Afghanistan. These things really **interested** me. And so I decided quite at an early age I wanted to be a diplomat. And I, one day, I announced this to my parents — and my father denies this story to this day — I said,”Daddy, I want to be a diplomat.”And he turned to me, and he said,”Carne, you have to be very clever to be a diplomat.”And my ambition was sealed. In 1989, I entered the British Foreign Service. That year, 5, 000 people applied to become a diplomat, and 20 of us succeeded. And as those numbers suggest, I was inducted into an elite and fascinating and exhilarating world. Being a diplomat, then and now, is an incredible job, and I loved every minute of it — I enjoyed the status of it. I bought myself a nice suit and wore leather- soled shoes and reveled in this amazing access I had to world events. I traveled to the Gaza Strip. I headed the Middle East Peace Process section in the British Foreign Ministry. I became a speechwriter for the British Foreign Secretary. I met Yasser Arafat. I **negotiated** with Saddam ’ s diplomats at the U. N. Later, I traveled to Kabul and served in Afghanistan after the fall of the Taliban. And I would travel in a C-130 transport and go and **visit** warlords in mountain hideaways and **negotiate** with them about how we were going to eradicate Al Qaeda from Afghanistan, surrounded by my Special Forces escort, who, themselves, had to have an escort of a platoon of Royal Marines, because it was so dangerous. And that was exciting — that was fun. It was really **interesting**. And it ’ s a great cadre of people, incredibly close-knit community of people. And the pinnacle of my career, as it turned out, was when I was posted to New York. I ’ d already served in Germany, Norway, various other places, but I was posted to New York to serve on the U. N. Security Council for the British delegation. And my responsibility was the Middle East, which was my specialty. And there, I dealt with things like the Middle East peace process, the Lockerbie issue — we can talk about that later, if you wish — but above all, my responsibility was Iraq and its weapons of mass destruction and the sanctions we placed on Iraq to oblige it to disarm itself of these weapons. I was the chief British negotiator on the subject, and I was steeped in the issue. And anyway, my tour — it was kind of a very exciting time. I mean it was very dramatic diplomacy. We went through several wars during my time in New York. I **negotiated** for my country the resolution in the Security Council of the 12th of September 2001 condemning the attacks of the day before, which were, of course, deeply present to us actually living in New York at the time. So it was kind of the best of time, worst of times kind of experience. I lived the high- life. Although I worked very long hours, I lived in a penthouse in Union Square. I was a single British diplomat in New York City ; you can imagine what that might have meant. I had a good time. But in 2002, when my tour came to an end, I decided I wasn ’ t going to go back to the job that was waiting for me in London. I decided to take a sabbatical, in fact, at the New School, Bruce. In some inchoate, inarticulate way I realized that there was something wrong with my work, with me. I was exhausted, and I was also disillusioned in a way I couldn ’ t quite put my finger on. And I decided to take some time out from work. The Foreign Office was very generous. You could take these special unpaid leave, as they called them, and yet remain part of the diplomatic service, but not actually do any work. It was nice. And eventually, I decided to take a secondment to join the U. N. in Kosovo, which was then under U. N. administration. And two things happened in Kosovo, which kind of, again, shows the randomness of life, because these things turned out to be two of the **pivots** of my life and helped to deliver me to the next stage. But they were random things. One was that, in the summer of 2004, the British government, somewhat reluctantly, decided to have an official inquiry into the use of **intelligence** on WMD in the run up to the Iraq War, a very limited subject. And I testified to that inquiry in secret. I had been steeped in the **intelligence** on Iraq and its WMD, and my testimony to the inquiry said three things : that the government exaggerated the **intelligence**, which was very clear in all the years I ' d read it. And indeed, our own internal assessment was very clear that Iraq ' s WMD did not pose a threat to its neighbors, let alone to us. Secondly, the government had ignored all available alternatives to war, which in some ways was a more discreditable thing still. The third reason, I won ' t go into. But anyway, I gave that testimony, and that presented me with a **crisis**. What was I going to do? This testimony was deeply critical of my colleagues, of my ministers, who had, in my view had perpetrated a war on a falsehood. And so I was in **crisis**. And this wasn ' t a pretty thing. I moaned about it, I hesitated, I went on and on and on to my long-suffering wife, and eventually I decided to resign from the British Foreign Service. I felt — there ' s a scene in the Al Pacino movie"The Insider,"which you may know, where he goes back to CBS after they ' ve let him down over the tobacco **guy**, and he goes,"You know, I just can ' t do this anymore. Something ' s broken."And it was like that for me. I love that movie. I felt just something ' s broken. I can ' t actually sit with my foreign minister or my prime minister again with a smile on my face and do what I used to do gladly for them. So took a running leap and jumped over the edge of a cliff. And it was a very, very uncomfortable, unpleasant feeling. And I started to fall. And today, that fall hasn ' t stopped ; I ' m still falling. But, in a way, I ' ve got used to the sensation of it. And in a way, I kind of like the sensation of it a lot better than I like actually standing on top of the cliff, wondering what to do. A second thing happened in Kosovo, which kind of — I need a quick gulp of water, forgive me. A second thing happened in Kosovo, which kind of delivered the answer, which I couldn ' t really answer, which is,"What do I do with my life?"I love diplomacy — I have no career — I expected my entire life to be a diplomat, to be serving my country. I wanted to be an ambassador, and my mentors, my heroes, people who got to the top of my profession, and here I was throwing it all away. A lot of my friends were still in it. My pension was in it. And I gave it up. And what was I going to do? And that year, in Kosovo, this terrible, terrible thing happened, which I saw. In March 2004, there were terrible riots all over the province — as it then was — of Kosovo. 18 people were killed. It was anarchy. And it ' s a very horrible thing to see anarchy, to know that the police and the military — there were lots of military troops there — actually can ' t stop that rampaging mob who ' s coming down the street. And the only way that rampaging mob coming down the street will stop is when they decide to stop and when they ' ve had enough burning and killing. And that is not a very nice feeling to see, and I saw it. And I went through it. I went through those mobs. And with my Albanian friends, we tried to stop it, but we failed. And that riot taught me something, which isn ' t immediately obvious and it ' s kind of a complicated story. But one of the reasons that riot took place — those riots, which went on for several days, took place — was because the Kosovo people were disenfranchised from their own future. There were diplomatic negotiations about the future of Kosovo going on then, and the Kosovo government, let alone the Kosovo people, were not actually participating in those talks. There was this whole fancy diplomatic system, this negotiation process about the future of Kosovo, and the Kosovars weren ' t part of it. And funnily enough, they were frustrated about that. Those riots were part of the manifestation of that frustration. It wasn ' t the only reason, and life is not simple, one reason narratives. It was a complicated thing, and I ' m not pretending it was more simple than it was. But that was one of the reasons. And that kind of gave me the inspiration — or rather to be precise, it gave my wife the inspiration. She said,"Why don ' t you advise the Kosovars? Why don ' t you advise their government on their diplomacy?"And the Kosovars were not allowed a diplomatic service. They were not allowed diplomats. They were not allowed a foreign office to help them deal with this immensely complicated process, which became known as the Final Status Process of Kosovo. And so that was the idea. That was the origin of the thing that became Independent Diplomat, the world ' s first diplomatic advisory group and a non-profit to boot. And it began when I flew back from London after my time at the U. N. in Kosovo. I flew back and had dinner with the Kosovo prime minister and said to him,"Look, I ' m proposing that I come and advise you on the diplomacy. I know this **stuff**. It ' s what I do. Why don ' t I come and help you?"And he raised his glass of raki to me and said,"Yes, Carne. Come."And I came to Kosovo and advised the Kosovo government. Independent Diplomat ended up advising three successive Kosovo prime ministers and the multi-party negotiation team of Kosovo. And Kosovo became independent. Independent Diplomat is now established in five diplomatic centers around the world, and we ' re advising seven or eight different countries, or political groups, depending on how you wish to define them — and I ' m not big on definitions. We ' re advising the Northern Cypriots on how to reunify their island. We ' re advising the Burmese opposition, the government of Southern Sudan, which — you heard it here first — is going to be a new country within the next few years. We ' re advising the Polisario Front of the Western Sahara, who are fighting to get their country back from Moroccan occupation after 34 years of dispossession. We ' re advising various island states in the **climate** change negotiations, which is suppose to culminate in Copenhagen. There ' s a bit of randomness here too because, when I was beginning Independent Diplomat, I went to a party in the House of Lords, which is a ridiculous place, but I was holding my drink like this, and I bumped into this **guy** who was standing behind me. And we started talking, and he said — I told him what I was doing, and I told him rather grandly I was going to establish Independent Diplomat in New York. At that time there was just me — and me and my wife were moving back to New York. And he said,"Why don ' t you see my colleagues in New York?"And it turned out he worked for an innovation company called? What If!, which some of you have probably heard of. And one thing led to another, and I ended up having a desk in? What If! in New York, when I started Independent Diplomat. And watching? What If! develop new flavors of chewing gum for Wrigley or new flavors for Coke actually helped me innovate new strategies for the Kosovars and for the Saharawis of the Western Sahara. And I began to realize that there are different ways of doing diplomacy — that diplomacy, like business, is a business of solving problems, and yet the word innovation doesn ' t exist in diplomacy ; it ' s all zero sum games and realpolitik and ancient institutions that have been there for generations and do things the same way they ' ve always done things. And Independent Diplomat, today, tries to incorporate some of the things I learned at? What If!. We all sit in one office and shout at each other across the office. We all work on little laptops and try to move desks to change the way we think. And we use naive experts who may know nothing about the countries we ' re dealing with, but may know something about something else to try to inject new thinking into the problems that we try to address for our clients. It ' s not easy, because our clients, by definition, are having a difficult time, diplomatically. There are, I don ' t know, some lessons from all of this, personal and political — and in a way, they ' re the same thing. The personal one is falling off a cliff is actually a good thing, and I recommend it. And it ' s a good thing to do at least once in your life just to tear everything up and jump. The second thing is a bigger lesson about the world today. Independent Diplomat is part of a trend which is emerging and evident across the world, which is that the world is fragmenting. States mean less than they used to, and the power of the state is declining.

That means the power of others things is rising. Those other things are called non-state actors. They may be corporations, they may be mafiosi, they may be nice NGOs, they may anything, any number of things. We are living in a more complicated and fragmented world. If governments are less able to affect the problems that affect us in the world, then that means, who is left to deal with them, who has to take greater responsibility to deal with them? Us. If they can ' t do it, who ' s left to deal with it? We have no choice but to embrace that reality. What this means is it ' s no longer good enough to say that **international** relations, or global affairs, or chaos in Somalia, or what ' s going on in Burma is none of your business, and that you can leave it to governments to get on with. I can connect any one of you by six degrees of separation to the Al- Shabaab militia in Somalia. Ask me how later, particularly if you eat fish, interestingly enough, but that connection is there. We are all intimately connected. And this isn ' t just Tom Friedman, it ' s actually provable in case after case after case. What that means is, instead of asking your politicians to do things, you have to look to yourself to do things. And Independent Diplomat is a kind of example of this in a sort of loose way. There aren ' t neat examples, but one example is this : the way the world is changing is embodied in what ' s going on at the place I used to work — the U. N. Security Council. The U. N. was established in 1945. Its charter is basically designed to stop **conflicts** between states — interstate **conflict**. Today, 80 percent of the agenda of the U. N. Security Council is about **conflicts** within states, **involving** non-state parties — guerillas, separatists, terrorists, if you want to call them that, people who are not normal governments, who are not normal states. That is the state of the world today. When I realized this, and when I look back on my time at the Security Council and what happened with the Kosovars, and I realize that often the people who were most directly affected by what we were doing in the Security Council weren ' t actually there, weren ' t actually invited to give their views to the Security Council, I thought, this is wrong. Something ' s got to be done about this. So I started off in a traditional mode. Me and my colleagues at Independent Diplomat went around the U. N. Security Council. We went around 70 U. N. member states — the Kazaks, the Ethiopians, the Israelis — you name them, we went to see them — the secretary general, all of them, and said,"This is all wrong. This is terrible that you don ' t consult these people who are actually affected. You ' ve got to institutionalize a system where you actually invite the Kosovars to come and tell you what they think. This will allow you to tell me — you can tell them what you think. It ' ll be great. You can have an exchange. You can actually incorporate these people ' s views into your decisions, which means your decisions will be more effective and durable."Super-logical, you would think. I mean, incredibly logical. So obvious, anybody could get it. And of course, **everybody** got it. **Everybody** went,"Yes, of course, you ' re absolutely right. Come back to us in maybe six months."And of course, nothing happened — nobody did anything. The Security Council does its business in exactly the same way today that it did X number of years ago, when I was there 10 years ago. So we looked at that observation of basically failure and thought, what can we do about it. And I thought, I ' m buggered if I ' m going to spend the rest of my life lobbying for these crummy governments to do what needs to be done. So what we ' re going to do is we ' re actually going to set up these meetings ourselves. So now, Independent Diplomat is in the process of setting up meetings between the U. N. Security Council and the parties to the disputes that are on the agenda of the Security Council. So we will be bringing Darfuri rebel groups, the Northern Cypriots and the Southern Cypriots, rebels from Aceh, and awful long laundry list of chaotic **conflicts** around the world. And we will be trying to bring the parties to New York to sit down in a quiet room in a private setting with no press and actually explain what they want to the members of the U. N. Security Council, and for the members of the U. N. Security Council to explain to them what they want. So there ’ s actually a conversation, which has never before happened. And of course, describing all this, any of you who know politics will think this is incredibly difficult, and I entirely agree with you. The chances of failure are very high, but it certainly won ’ t happen if we don ’ t try to make it happen. And my politics has changed fundamentally from when I was a diplomat to what I am today, and I think that outputs is what matters, not process, not technology, frankly, so much either. Preach technology to all the Twittering members of all the Iranian demonstrations who are now in political prison in Tehran, where Ahmadinejad remains in power. Technology has not delivered political change in Iran. You ’ ve got to look at the outputs, and you got to say to yourself,”What can I do to produce that particular output?”That is the politics of the 21st century, and in a way, Independent Diplomat embodies that fragmentation, that change, that is happening to all of us. That ’ s my story. Thanks.

Text Sample 548: POS Dim 4 – 2009 – Score 17.00 – 2009/t000082.txt

What I ’ m going to try to do is explain to you quickly how to predict, and illustrate it with some predictions about what Iran is going to do in the next couple of years. In order to predict effectively, we need to use science. And the reason that we need to use science is because then we can reproduce what we ’ re doing ; it ’ s not just wisdom or guesswork. And if we can predict, then we can engineer the future. So if you are concerned to influence energy **policy**, or you are concerned to influence national security **policy**, or health **policy**, or education, science — and a particular branch of science — is a way to do it, not the way we ’ ve been doing it, which is seat-of-the-pants wisdom. Now before I get into how to do it let me give you a little truth in advertising, because I ’ m not engaged in the business of magic. There are lots of thing that the approach I take can predict, and there are some that it can ’ t. It can predict complex negotiations or situations **involving** coercion — that is in essence everything that has to do with politics, much of what has to do with business, but sorry, if you ’ re looking to speculate in the stock market, I don ’ t predict stock markets — OK, it ’ s not going up any time really soon. But I ’ m not engaged in doing that. I ’ m not engaged in predicting random number generators. I actually get **phone** calls from people who want to know what lottery numbers are going to win. I don ’ t have a clue. I engage in the use of game theory, game theory is a branch of mathematics and that means, sorry, that even in the study of politics, math has come into the picture. We can no longer pretend that we just speculate about politics, we need to look at this in a rigorous way. Now, what is game theory about? It assumes that people are looking out for what ’ s good for them. That doesn ’ t seem terribly shocking — although it ’ s controversial for a lot of people — that we are self-interested. In order to look out for what ’ s best for them or what they think is best for them, people have values — they identify what they want, and what they don ’ t want. And they have beliefs about what other people want, and what other people don ’ t want, how much power other people have, how much those people could get in the way of whatever it is that you want. And they face limitations, constraints, they may be weak, they may be located in the wrong part of the world, they may be Einstein, stuck away farming someplace in a rural village in India not being noticed, as was the case for Ramanujan for a long time, a great mathematician but nobody noticed. Now who is rational? A lot of people are worried about what is rationality about? You know, what if people are rational? Mother Theresa, she was rational. Terrorists, they ’ re rational. Pretty much **everybody** is rational. I think there are only two exceptions that I ’ m aware of — two-year-olds, they are not rational, they have very fickle preferences, they switch what they think all the time, and schizophrenics are probably not rational, but pretty much **everybody** else is rational. That is, they are just trying to do what they think is in their own best interest. Now in order to work out what people are going to do to pursue their interests, we have to think about who has influence in the world. If you ’ re trying to influence corporations to change their behavior, with regard to producing pollutants, one approach, the common approach, is to exhort them to be better, to explain to them what damage they ’ re doing to the planet. And many of you may have noticed that doesn ’ t have as big an effect, as perhaps you would like it to have. But if you show them that it ’ s in their interest, then they ’ re responsive. So, we have to work out who influences problems. If we ’ re looking at Iran, the president of the United States we would like to think, may have some influence — certainly the president in Iran has some influence — but we make a mistake if we just pay attention to the person at the top of the power ladder because that person doesn ’ t know much about Iran, or about energy **policy**, or about health care, or about any particular **policy**. That person surrounds himself or herself with advisers. If we ’ re talking about national security problems, maybe it ’ s the Secretary of State, maybe it ’ s the Secretary of Defense, the Director of National Intelligence, maybe the ambassador to the United Nations, or somebody else who they think is going to know more about the particular problem. But let ’ s face it, the Secretary of State doesn ’ t know much about Iran. The secretary of defense doesn ’ t know much about Iran. Each of those people in turn has advisers who advise them, so they can advise the president. There are lots of people shaping decisions and so if we want to predict correctly we have to pay attention to **everybody** who is trying to shape the outcome, not just the people at the pinnacle of the decision-making pyramid. Unfortunately, a lot of times we don ’ t do that. There ’ s a good reason that we don ’ t do that, and there ’ s a good reason that using game theory and computers, we can overcome the limitation of just looking at a few people. Imagine a problem with just five decision-makers. Imagine for example that Sally over here, wants to know what Harry, and Jane, and George and Frank are thinking, and sends messages to those people. Sally ’ s giving her opinion to them, and they ’ re giving their opinion to Sally. But Sally also wants to know what Harry is saying to these three, and what they ’ re saying to Harry. And Harry wants to know what each of those people are saying to each other, and so on, and Sally would like to know what Harry thinks those people are saying. That ’ s a complicated problem ; that ’ s a lot to know. With five decision-makers there are a lot of linkages — 120, as a matter of fact, if you remember your factorials. Five factorial is 120. Now you may be surprised to know that smart people can keep 120 things straight in their head. Suppose we double the number of influencers from five to 10. Does that mean we ’ ve doubled the number of pieces of information we need to know, from 120 to 240? No. How about 10 times? To 1, 200? No. We ’ ve increased it to 3. 6 million. Nobody can keep that straight in their head. But computers, they can. They don ’ t need coffee breaks, they don ’ t need vacations, they don ’ t need to go to sleep at night, they don ’ t ask for raises either. They can keep this information straight and that means that we can process the information. So I ’ m going to talk to you about how to process it, and I ’ m going to give you some examples out of Iran, and you ’ re going to be wondering,”Why should we listen to this **guy**? Why should we believe what he ’ s saying?”So I ’ m going to show you a factoid. This is an assessment by the Central Intelligence Agency of the percentage of time that the model I ’ m talking about is right in predicting things whose outcome is not yet known, when the experts who provided the data inputs got it wrong. That ’ s not my **claim**, that ’ s a CIA **claim** — you can read it, it was declassified a while ago. You can read it in a volume edited by H. Bradford Westerfield, Yale University Press. So, what do we need to know in order to predict? You may be surprised to find out we don ’ t need to know very much. We do need to know who has a stake in trying to shape the outcome of a decision. We need to know what they say they want, not what they want in their heart of hearts, not what they think they can get, but what they say they want, because that is a strategically chosen position, and we can work backwards from that to draw inferences about important features of their decision-making. We need to know how focused they are on the problem at hand. That is, how willing are they to drop what they ’ re doing when the issue comes up, and attend to it instead of something else that ’ s on their plate — how big a deal is it to them? And how much clout could they bring to bear if they chose to engage on the issue? If we know those things we can predict their behavior by assuming that **everybody** cares about two things on any decision. They care about the outcome. They ’ d like an outcome as close to what they are **interested** in as possible. They ’ re careerists, they also care about getting credit — there ’ s ego involvement, they want to be seen as important in shaping the outcome, or as important, if it ’ s their druthers, to block an outcome. And so we have to figure out how they balance those two things. Different people trade off between standing by their outcome, faithfully holding to it, going down in a blaze of glory, or giving it up, putting their finger in the wind, and doing whatever they think is going to be a winning position. Most people fall in between, and if we can work out where they fall we can work out how to **negotiate** with them to change their behavior. So with just that little bit of input we can work out what the choices are that people have, what the chances are that they ’ re willing to take, what they ’ re after, what they value, what they want, and what they believe about other people. You might notice what we don ’ t need to know : there ’ s no history in here. How they got to where they are may be important in shaping the input information, but once we know where they are we ’ re worried about where they ’ re going to be headed in the future. How they got there turns out not to be terribly critical in predicting. I remind you of that 90 percent accuracy rate. So where are we going to get this information? We can get this information from the Internet, from The Economist, The Financial Times, The New York Times, U. S. News and World Report, lots of sources like that, or we can get it from asking experts who spend their lives studying places and problems, because those experts know this information. If they don ’ t know, who are the people trying to influence the decision, how much clout do they have, how much they care about this issue, and what do they say they want, are they experts? That ’ s what it means to be an expert, that ’ s the basic **stuff** an expert needs to know. Alright, lets turn to Iran. Let me make three important predictions — you can check this out, time will tell. What is Iran going to do about its nuclear weapons program? How secure is the theocratic regime in Iran? What ’ s its future? And **everybody** ’ s best friend, Ahmadinejad. How are things going for him? How are things going to be working out for him in the next year or two? You take a look at this, this is not based on statistics. I want to be very clear here. I ’ m not projecting some past data into the future. I ’ ve taken inputs on positions and so forth, run it through a computer model that had simulated the dynamics of interaction, and these are the simulated dynamics, the predictions about the path of **policy**. So you can see here on the vertical axis, I haven ’ t shown it all the way down to zero, there are lots of other options, but here I ’ m just showing you the prediction, so I ’ ve narrowed the scale. Up at the top of the axis,”Build the Bomb.”At 130, we start somewhere above 130, between building a bomb, and making enough weapons-grade fuel so that you could build a bomb. That ’ s where, according to my analyses, the Iranians were at the beginning of this year. And then the model makes predictions down the road. At 115 they would only produce enough weapons grade fuel to show that they know how, but they wouldn ’ t build a weapon : they would build a research quantity. It would achieve some national pride, but not go ahead and build a weapon. And down at 100 they would build civilian nuclear energy, which is what they say is their objective. The yellow line shows us the most likely path. The yellow line includes an analysis of 87 decision makers in Iran, and a vast number of outside influencers trying to pressure Iran into changing its behavior, various players in the United States, and Egypt, and Saudi Arabia, and Russia, European Union, Japan, so on and so forth. The white line reproduces the analysis if the **international** environment just left Iran to make its own internal decisions, under its own domestic political pressures. That ’ s not going to be happening, but you can see that the line comes down faster if they ’ re not put under **international** pressure, if they ’ re allowed to pursue their own devices. But in any event, by the end of this year, beginning of next year, we get to a stable equilibrium outcome. And that equilibrium is not what the United States would like, but it ’ s probably an equilibrium that the United States can live with, and that a lot of others can live with. And that is that Iran will achieve that nationalist pride by making enough weapons-grade fuel, through research, so that they could show that they know how to make weapons-grade fuel, but not enough to actually build a bomb. How is this happening? Over here you can see this is the distribution of power in favor of civilian nuclear energy today, this is what that power block is predicted to be like by the late parts of 2010, early parts of 2011. Just about nobody **supports** research on weapons-grade fuel today, but by 2011 that gets to be a big block, and you put these two together, that ’ s the controlling influence in Iran. Out here today, there are a bunch of people — Ahmadinejad for example — who would like not only to build a bomb, but test a bomb. That power disappears completely ; nobody **supports** that by 2011. These **guys** are all shrinking, the power is all drifting out here, so the outcome is going to be the weapons-grade fuel. Who are the winners and who are the losers in Iran? Take a look at these **guys**, they ’ re growing in power, and by the way, this was done a while ago before the current economic **crisis**, and that ’ s probably going to get steeper. These folks are the moneyed interests in Iran, the bankers, the oil people, the bazaaries. They are growing in political clout, as the mullahs are isolating themselves — with the exception of one group of mullahs, who are not well known to Americans. That ’ s this line here, growing in power, these are what the Iranians call the quietists. These are the Ayatollahs, mostly based in Qom, who have great clout in the religious community, have been quiet on politics and are going to be getting louder, because they see Iran going in an unhealthy direction, a direction contrary to what Khomeini had in mind. Here is Mr. Ahmadinejad. Two things to notice : he ’ s getting weaker, and while he gets a lot of attention in the United States, he is not a major player in Iran. He is on the way down. OK, so I ’ d like you to take a little away from this. Everything is not predictable : the stock market is, at least for me, not predictable, but most complicated negotiations are predictable. Again, whether we ’ re talking health **policy**, education, environment, energy, litigation, mergers, all of these are complicated problems that are predictable, that this sort of technology can be applied to. And the reason that being able to predict those things is important, is not just because you might run a hedge fund and make money off of it, but because if you can predict what people will do, you can engineer what they will do. And if you engineer what they do you can change the world, you can get a better result. I would like to leave you with one thought, which is for me, the **dominant** theme of this gathering, and is the **dominant** theme of this way of thinking about the world. When people say to you,"That ' s impossible,"you say back to them,"When you say ' That ' s impossible, ' you ' re confused with, ' I don ' t know how to do it. '"Thank you. Chris Anderson : One question for you. That was fascinating. I love that you put it out there. I got very nervous halfway through the talk though, just panicking whether you ' d included in your model, the possibility that putting this prediction out there might change the result. We ' ve got 800 people in Tehran who watch TEDTalks. Bruce Bueno de Mesquita : I ' ve thought about that, and since I ' ve done a lot of work for the **intelligence** community, they ' ve also pondered that. It would be a good thing if people paid more attention, took seriously, and engaged in the same sorts of calculations, because it would change things. But it would change things in two beneficial ways. It would hasten how quickly people arrive at an agreement, and so it would save **everybody** a lot of grief and time. And, it would arrive at an agreement that **everybody** was happy with, without having to manipulate them so much — which is basically what I do, I manipulate them. So it would be a good thing. CA : So you ' re kind of trying to say,"People of Iran, this is your destiny, lets go there."BBM : Well, people of Iran, this is what many of you are going to evolve to want, and we could get there a lot sooner, and you would suffer a lot less trouble from economic sanctions, and we would suffer a lot less fear of the use of military force on our end, and the world would be a better place. CA : Here ' s hoping they hear it that way. Thank you very much Bruce. BBM : Thank you.

Text Sample 549: POS Dim 4 – 2019 – Score 18.00 – 2019/t001763.txt

Bryn Freedman : So you keep talking about **leadership** as a real **crisis** of conformity. Can you explain to us what you mean by that? What do you see as a **crisis** of conformity? Halla T□□masd□□ttir : I think it ' s a **crisis** of conformity when we continue to do business and lead in the way we always have, yet the evidence is overwhelming that the world needs us to change our ways. So let ' s look a little bit at that evidence. Science has told us that we ' re facing a **climate crisis**, yet 40 percent of board directors don ' t think **climate** belongs in the boardroom. And we have kids marching in the streets now, asking us to be **accountable** for their future. We have a **crisis** of inequality. We have Yellow Jackets not just in the streets of France, but all over the world, and yet we continue to see examples of businesses and other **leaders** fueling that anger. BF : Do you think the pitchforks are coming? HT : I definitely think this is not sustainable. And the reason why it ' s so difficult for us to deal with these complicated **crises** that are interrelated is that we are at the lowest levels of **trust** we ' ve ever been at. In the UK, three percent of people **trust** their government to solve the Brexit **crisis**, and that was in December. I think it ' s probably gone down since then. BF : What do you think new **leadership** actually looks like? HT : We need courageous **leaders**, yet they have to be humble. And they have to be guided by a moral compass, and the moral compass is the combination of having a social purpose □□□ you can ' t have your license to operate anymore without a purpose that contributes to society, but what, to me, has been missing from that dialogue is a set of principles. We cannot just define why we exist, we have to define how we ' re going to do business and how we ' re going to lead. And to us, that has to be to solve these imminent **crises** : the

climate crisis, the **crisis** of inequality and the **crisis** of **trust**. So at The B Team, we embrace sustainability, equality and accountability as our principles. BF : Do you think this whole question of purpose is really window dressing □□□ they ’ re saying what they think people want to hear, but they ’ re actually not making the fundamental changes that are necessary? HT : A lot of people feel that way, and I think there ’ s a growing momentum behind that. So I think the world is calling for responsible **leadership** now, and any **leader** who wants to be around for the 21st century really needs to start thinking courageously and holistically how they ’ re going to be part of the solution and not window-dress anymore. You have to do it for real now. BF : Do you see anybody out there who ’ s doing it in a way that you think is actually effective and has a real, long-term **impact**? HT : Fortunately, we have some great **leaders** out there. And just to give one example, Emmanuel Faber, who ’ s one of the newest members of The B Team, he ’ s the CEO of Danone, the world ’ s largest yogurt-maker and major food company □□□ a real global company. He ’ s so committed to sustainability that he ’ s not only changing the ways of his own business, but his entire supply chain. He ’ s so committed to equality that when he took on as CEO and he said **gender** balance matters, he created a gender-balanced executive team and gave men and women equal maternity and paternity leave. He ’ s so committed to accountability that he turned his US operations into a B Corporation. Now many of you may not know what that is, but that ’ s a company that holds itself responsible for not just profit but its **impact** on people and the planet, and transparently reports on their performance on all of that. It ’ s the largest B Corp in the world. So to me, that ’ s holistic, courageous **leadership**, and it ’ s really the vision we all need to hold. BF : Is this”Back to the Future”? I mean, I wonder, when I think about companies □□□ Anheuser-Busch comes to mind in America □□□ a 100-year-old company that **invested** in its community, that gave a living wage before it ended up, you know, losing and getting sold. Are we really looking now for companies that are global and community citizens, or is that something that is not even useful anymore? HT : Well, you can do this for the reason that it ’ s risky, actually, to continue without doing the right thing now. You can ’ t attract the right talent, you can ’ t attract customers and increasingly, you won ’ t be able to attract capital. You might do it for risk reasons, you might do it for business opportunity reasons, because this is where the growth is, which is why many **leaders** are doing the right thing. But at the end of the day, we need to ask ourselves :”Who are we holding ourselves **accountable** for?”And if that isn ’ t the next generation, I don ’ t know who. So I want to just make very clear, because we tend to think about **leadership** as only those who sit in positions of power. To me, **leadership** is not at all like that. There ’ s a **leader** inside every single one of us, and our most important work in life is to release that **leader**. And I think one of the greatest global examples we have of someone who didn ’ t □□□ no one gave her power □□□ is the 16-year-old girl called Greta Thunberg. She ’ s from Sweden, and a few years ago, she really became □□□ she has Asperger ’ s, and she became passionate about our **climate crisis** □□□ learned everything about it. And faced with the evidence, she just felt so disappointed in her **leadership** that she started striking in front of the Swedish parliament. And now she has started a global movement, and hundreds and thousands of school kids are out in the streets asking us to hold ourselves **accountable** for their future. No one gave her that authority, and she now speaks to the **leaders** of the world, heads of state, and really is **impacting** the world. So I really think that when we think about **leadership** today, it can ’ t be defined to those in positions of power though they have disproportionately greater responsibility. But all of us need to think about,”What am I doing?””How am I contributing?”And we need to release that **leader** inside and actually start making the positive **impact** this world is calling for right now. BF : But we have such hierarchical **leadership**. I mean, I understand what you ’ re saying □□□ it ’ s nice to release the **leader** inside □□□ but in these corporations, the truth is, it ’ s extremely hierarchical. What can companies do to create less vertical and more horizontal relationships? HT : Well, I ’ m a big believer and I ’ ve long been passionate about closing the **gender** gap, and I really believe gender-balanced **leadership** is the way to go in order to embrace a **leadership** style that has been shown to be more powerful, and that ’ s when both men and women embrace both masculine and feminine values. It actually is proven in research that that ’ s the most effective **leadership** style. But I ’ m increasingly now thinking about how we close the generational gap, because look at these young children in the streets around the world □□□ they ’ re asking us to lead. Kofi Annan used to say,”You ’ re never too young to lead.”And then he would add,”Or too old to learn.”And I think we have now entered this era where we need the wisdom of those with experience, but we need the digital natives of the young generation to co-mentor or to mentor us just as much as we can help with wisdom from the older people. So it ’ s a new reality, and these old, sort of hierarchical ways to think about things, they ’ re increasingly coming under pressure in this reality. BF : And you ’ ve actually called that the hubris syndrome. Can you talk about that? HT : Well, yes, I think hubris is our cancer in **leadership**. That ’ s when **leaders** think they know it all, can do it all, have all the answers and don ’ t think they need to surround themselves with people who will make them better, which to me would, in some cases, be more women and younger people and people who are diverse and have different opinions in general. Hubris syndrome is so present in **leadership** still, and we know many examples of them, I don ’ t need to name them. And the problem with that □□□ Yeah, we know them □□□ all over the world, not just in this country. But that kind of **leadership** doesn ’ t unleash **leaders** in others. No one person, or no one **sector** even has the solutions we now need to come up with □□□ the creativity and collaboration we need. The bold and the brave **leadership** we need to come up with solutions that cross government, private **sector**, civil society, young people, older people, people of all different backgrounds coming together is the way to solve the issues that are in front of us. BF : Do you see that kind of **leadership** coming from the bottom-up or the top- down, or do you think a **crisis** is going to force us into a reexamination of all of this? HT : Well, as someone who lived through the most infamous financial meltdown in my home country, Iceland, I hope we don ’ t need another one to learn or to wake up. But I do see that we can ’ t choose one or the other. We do have to transform the way we lead □□□ from the top, the boardroom, the CEOs □□□ we really do have to transform that, but increasingly, we will transform that, because we have these social movements coming from the bottom and throughout society. And the solutions exist. The only thing that ’ s missing is will. So if we just all find a way to embrace a moral compass of our own to figure out why we exist and how we ’ re going to lead, and if we embrace courage and humility in equal amounts, each one of us can be part of this 10-year period where we can dramatically transform the world we live in, and make it just, and make it about humanity and not just the financial markets. BF : Well, we have a lot of people here who I bet have questions for you, and we have a few minutes for questions, so is there anybody that would like to ask Halla a question? Audience : Hello, my name is Cheryl. I ’ m an aspiring **leader**, and I have a question about how you lead when you have no influence. If I ’ m just an analyst, and I want to speak to **senior** management about a change that I feel will affect the whole company, how do I go about changing their minds when they feel as if they ’ ve had relationships that are set, that their way of business is set? How do you change minds when you have no influence? HT : Well, thank you very much for that fantastic question. So sometimes people at the top won ’ t listen, but it ’ s **interesting** that with the low **trust** we have in society right now, the greatest **trust** we have is actually between the employee and the employer, according to recent research. So I think that relationship may be the most powerful way to actually transform the way we do things. So I would start by trying to build a coalition for your good idea. And I don ’ t know a single **leader** today who will not listen to a concern that many of their employees hold. I ’ ll give you an example from another B Team **leader**, Marc Benioff, the CEO of Salesforce. He ’ s really been outspoken on homelessness in San Francisco, on LGBTQI rights, and all of the things that he ’ s been standing up for, he does because his employees care about them. So don ’ t ever think you don ’ t have power if you don ’ t sit in a position of power. Find the way to go convince him... or her. And Marc, for example, was convinced to close the **gender** pay gap by two women who worked inside of his organization, who told him,”We have a **gender** pay gap.”He didn ’ t believe it ; he said,”Bring me the data.”They did, and he was smart enough to know he needed to do something about it, and was one the first **tech leaders** to step up and do so voluntarily. So don ’ t ever think that you don ’ t have power, even if you don ’ t sit in a position of power, but find other people to **support** you and make the case. BF : Thank you. Anybody else? Any other questions? Audience : Hi, I ’ m overwhelmed by fascination with everything you ’ re saying, so thank you. I just wanted to ask how, like, **diversity** in opinion and thought and also background has **impacted** your **leadership ability**. And what do you think is the barricade that is limiting the overflow of **diversity** in all business settings, and what do you think can **impact** the change in that setting but also to disrupt the overflow of generations of people staying in place? And what do you think is the next step to breaking several glass ceilings? BF : We ’ re going to do an entire Salon just on that question. HT : I think Bryn said it well, but let me try and touch on it. So the way I see **gender**, it is a spectrum □□□ you know, men also have **gender**. We sometimes forget about that. We sometimes forget about that. And I actually played a very masculine woman early in my career, because those were the rules of the game. And I achieved some success with it, but fortunately, I got to a place where I started embracing my feminine side as well. But I would still say that the best **leaders** embrace both, both women and men. But I see **gender**, also, as one of the most powerful levers to shift values in culture. So the reason I ’ m so passionate about women in **leadership** and believe that balance is needed is because right now, our definition of success is incredibly masculine. It ’ s about financial profit alone or economic growth alone, and we all know that we need more than money. I mean, we need wellness : well-being of people, and there is no future beyond the well-being of our planet. So I think **gender** may very well be one of the most powerful levers to help all of us shift our economic and social systems to be more welcoming. And the answer to your last part □□□ it ’ s so complicated, but let me try to give you a short one. I believe that the way talent and consumption is shifting is going to increasingly get companies to look at adding difference into their **leadership**, because sameness is not working □□□ BF : And difference is a superpower. HT : Difference is a superpower. BF : Thank you very much. Halla, thank you so much, I wish we could talk to you all day. HT : Thank you.

Text Sample 550: POS Dim 4 – 2019 – Score 16.00 – 2019/t000014.txt
I ’ m here to talk to you about something important that may be new to you. The

governments of the world are about to conduct an unintentional experiment on our **climate**. In 2020, new rules will require ships to lower their sulfur emissions by scrubbing their dirty exhaust or switching to cleaner fuels. For human health, this is really good, but sulfur particles in the emission of ships also have an effect on clouds. This is a satellite image of **marine** clouds off the Pacific West Coast of the United States. The streaks in the clouds are created by the exhaust from ships. Ships ' emissions include both greenhouse gases, which trap heat over long periods of time, and particulates like sulfates that mix with clouds and temporarily make them brighter. Brighter clouds reflect more sunlight back to space, cooling the **climate**. So in fact, humans are currently running two unintentional experiments on our **climate**. In the first one, we ' re increasing the concentration of greenhouse gases and gradually warming the earth system. This works something like a fever in the human body. If the fever remains low, its effects are mild, but as the fever rises, damage grows more severe and eventually devastating. We ' re seeing a little of this now. In our other experiment, we ' re planning to remove a layer of particles that brighten clouds and shield us from some of this warming. The effect is strongest in ocean clouds like these, and scientists expect the reduction of sulfur emissions from ships next year to produce a measurable increase in global warming. Bit of a shocker? In fact, most emissions contain sulfates that brighten clouds : coal, diesel exhaust, forest fires. Scientists estimate that the total cooling effect from emission particles, which they call aerosols when they ' re in the **climate**, may be as much as all of the warming we ' ve experienced up until now. There ' s a lot of **uncertainty** around this effect, and it ' s one of the major reasons why we have difficulty predicting **climate**, but this is cooling that we ' ll lose as emissions fall. So to be clear, humans are currently cooling the planet by dispersing particles into the atmosphere at massive scale. We just don ' t know how much, and we ' re doing it accidentally. That ' s worrying, but it could mean that we have a fast-acting way to reduce warming, emergency medicine for our **climate** fever if we needed it, and it ' s a medicine with origins in nature. This is a NASA simulation of earth ' s atmosphere, showing clouds and particles moving over the planet. The brightness is the Sun ' s light reflecting from particles in clouds, and this reflective shield is one of the primary ways that nature keeps the planet cool enough for humans and all of the life that we know. In 2015, scientists assessed possibilities for rapidly cooling **climate**. They discounted things like mirrors in space, ping-pong balls in the ocean, plastic sheets on the Arctic, and they found that the most viable approaches **involved** slightly increasing this atmospheric reflectivity. In fact, it ' s possible that reflecting just one or two percent more sunlight from the atmosphere could offset two degrees Celsius or more of warming. Now, I ' m a technology executive, not a scientist. About a decade ago, concerned about **climate**, I started to talk with scientists about potential countermeasures to warming. These conversations grew into collaborations that became the Marine Cloud Brightening Project, which I ' ll talk about momentarily, and the nonprofit **policy** organization SilverLining, where I am today. I work with politicians, researchers, members of the **tech** industry and others to talk about some of these ideas. Early on, I met British atmospheric scientist John Latham, who proposed cooling the **climate** the way that the ships do, but with a natural source of particles : sea-salt mist from seawater sprayed from ships into areas of susceptible clouds over the ocean. The approach became known by the name I gave it then,"**marine** cloud brightening."Early modeling studies suggested that by deploying **marine** cloud brightening in just 10 to 20 percent of susceptible ocean clouds, it might be possible to offset as much as two degrees Celsius ' s warming. It might even be possible to brighten clouds in local regions to reduce the **impacts** caused by warming ocean surface temperatures. For example, regions such as the Gulf Atlantic might be cooled in the months before a hurricane season to reduce the force of storms. Or, it might be possible to cool waters flowing onto coral reefs overwhelmed by heat stress, like Australia ' s Great Barrier Reef. But these ideas are only theoretical, and brightening **marine** clouds is not the only way to increase the reflection of the sunlight from the atmosphere. Another occurs when large volcanoes release material with enough force to reach the upper layer of the atmosphere, the stratosphere. When Mount Pinatubo erupted in 1991, it released material into the stratosphere, including sulfates that mix with the atmosphere to reflect sunlight. This material remained and circulated around the planet. It was enough to cool the **climate** by over half a degree Celsius for about two years. This cooling led to a striking increase in Arctic ice cover in 1992, which dropped in subsequent years as the particles fell back to earth. But the volcanic phenomenon led Nobel Prize winner Paul Crutzen to propose the idea that dispersing particles into the stratosphere in a controlled way might be a way to counter global warming. Now, this has risks that we don ' t understand, including things like heating up the stratosphere or damage to the ozone layer. Scientists think that there could be **safe** approaches to this, but is this really where we are? Is this really worth considering? This is a simulation from the US National Center for Atmospheric Research global **climate** model showing, earth surface temperatures through 2100. The globe on the left visualizes our current trajectory, and on the right, a world where particles are introduced into the stratosphere gradually in 2020, and maintained through 2100. Intervention keeps surface temperatures near those of today, while without it, temperatures rise well over three degrees. This could be the difference between a **safe** and an unsafe world. So, if there ' s even a chance that this could be close to reality, is this something we should consider seriously? Today, there are no capabilities, and scientific knowledge is extremely limited. We don ' t know whether these types of interventions are even feasible, or how to characterize their risks. Researchers hope to explore some basic questions that might help us know whether or not these might be real options or whether we should rule them out. It requires multiple ways of studying the **climate** system, including computer models to forecast changes, analytic techniques like machine learning, and many types of observations. And though it ' s controversial, it ' s also critical that researchers develop core technologies and perform small-scale, real-world experiments. There are two research programs proposing experiments like this. At Harvard, the SCoPEx experiment would release very small amounts of sulfates, calcium carbonate and water into the stratosphere with a balloon, to study chemistry and physics effects. How much material? Less than the amount released in one minute of flight from a commercial aircraft. So this is definitely not dangerous, and it may not even be scary. At the University of Washington, scientists hope to spray a fine mist of salt water into clouds in a series of land and ocean tests. If those are successful, this would culminate in experiments to measurably brighten an area of clouds over the ocean. The **marine** cloud brightening effort is the first to develop any technology for generating aerosols for atmospheric sunlight reflection in this way. It requires producing very tiny particles — think about the mist that comes out of an asthma inhaler — at massive scale — so think of looking up at a cloud. It ' s a tricky engineering problem. So this one nozzle they developed generates three trillion particles per second, 80 nanometers in size, from very corrosive saltwater. It was developed by a team of retired engineers in Silicon Valley — here they are — working full-time for six years, without pay, for their grandchildren. It will take a few million dollars and another year or two to develop the full spray system they need to do these experiments. In other parts of the world, research efforts are emerging, including small modeling programs at Beijing Normal University in China, the Indian Institute of Science, a proposed center for **climate** repair at Cambridge University in the UK and the DECIMALS Fund, which sponsors researchers in global South countries to study the potential **impacts** of these sunlight interventions in their part of the world. But all of these programs, including the experimental ones, lack significant funding. And understanding these interventions is a hard problem. The earth is a vast, complex system and we need major investments in **climate** models, observations and basic science to be able to predict **climate** much better than we can today and manage both our accidental and any intentional interventions. And it could be urgent. Recent scientific reports predict that in the next few decades, earth ' s fever is on a path to devastation : extreme heat and fires, major loss of ocean life, collapse of Arctic ice, displacement and suffering for hundreds of millions of people. The fever could even reach tipping points where warming takes over and human efforts are no longer enough to counter accelerating changes in natural systems. To prevent this circumstance, the UN ' s International Panel on **Climate** Change predicts that we need to stop and even reverse emissions by 2050. How? We have to quickly and radically transform major economic **sectors**, including energy, construction, agriculture, transportation and others. And it is imperative that we do this as fast as we can. But our fever is now so high that **climate** experts say we also have to remove massive quantities of CO2 from the atmosphere, possibly 10 times all of the world ' s annual emissions, in ways that aren ' t proven yet. Right now, we have slow-moving solutions to a fast-moving problem. Even with the most optimistic assumptions, our exposure to risk in the next 10 to 30 years is unacceptably high, in my opinion. Could interventions like these provide fast-acting medicine if we need it to reduce the earth ' s fever while we address its underlying causes? There are real concerns about this idea. Some people are very worried that even researching these interventions could provide an excuse to delay efforts to reduce emissions. This is also known as a moral hazard. But, like most medicines, interventions are more dangerous the more that you do, so research actually tends to draw out the fact that we absolutely, positively cannot continue to fill up the atmosphere with greenhouse gases, that these kinds of alternatives are risky and if we were to use them, we would need to use as little as possible. But even so, could we ever learn enough about these interventions to manage the risk? Who would make decisions about when and how to intervene? What if some people are worse off, or they just think they are? These are really hard problems. But what really worries me is that as **climate impacts** worsen, **leaders** will be called on to respond by any means available. I for one don ' t want them to act without real information and much better options. Scientists think it will take a decade of research just to assess these interventions, before we ever were to develop or use them. Yet today, the global level of investment in these interventions is effectively zero. So, we need to move quickly if we want policymakers to have real information on this kind of emergency medicine. There is hope! The world has solved these kinds of problems before. In the 1970s, we identified an existential threat to our protective ozone layer. In the 1980s, scientists, politicians and industry came together in a solution to replace the chemicals causing the problem. They achieved this with the only legally binding environmental agreement signed by all countries in the world, the Montreal Protocol. Still in force today, it has resulted in a recovery of the ozone layer and is the most successful environmental protection effort in human history. We have a far greater threat now, but we do have the **ability** to develop and agree on solutions to protect people and restore our **climate** to health. This could mean that to remain **safe**, we reflect sunlight for a few decades, while we green our industries and remove CO_2. It definitely means we must work now to understand our options for this kind of emergency medicine. Thank you,

Text Sample 551: POS Dim 4 – 2019 – Score 16.00 – 2019/t002063.txt

Chris Anderson : What worries you right now? You ’ ve been very open about lots of issues on Twitter. What would be your top worry about where things are right now? Jack Dorsey : Right now, the health of the conversation. So, our purpose is to serve the public conversation, and we have seen a number of attacks on it. We ’ ve seen abuse, we ’ ve seen harassment, we ’ ve seen manipulation, automation, human coordination, misinformation. So these are all dynamics that we were not expecting 13 years ago when we were starting the company. But we do now see them at scale, and what worries me most is just our **ability** to address it in a systemic way that is scalable, that has a rigorous understanding of how we ’ re taking action, a transparent understanding of how we ’ re taking action and a rigorous appeals process for when we ’ re wrong, because we will be wrong. Whitney Pennington Rodgers : I ’ m really glad to hear that that ’ s something that concerns you, because I think there ’ s been a lot written about people who feel they ’ ve been abused and harassed on Twitter, and I think no one more so than women and women of color and black women. And there ’ s been data that ’ s come out — Amnesty International put out a report a few months ago where they showed that a subset of active black female Twitter users were receiving, on average, one in 10 of their tweets were some form of harassment. And so when you think about health for the community on Twitter, I ’ m **interested** to hear,”health for everyone,”but specifically : How are you looking to make Twitter a **safe** space for that subset, for women, for women of color and black women? JD : Yeah. So it ’ s a pretty terrible situation when you ’ re coming to a service that, ideally, you want to learn something about the world, and you spend the majority of your time reporting abuse, receiving abuse, receiving harassment. So what we ’ re looking most deeply at is just the incentives that the platform naturally provides and the service provides. Right now, the dynamic of the system makes it super-easy to harass and to abuse others through the service, and unfortunately, the majority of our system in the past worked entirely based on people reporting harassment and abuse. So about midway last year, we decided that we were going to apply a lot more machine learning, a lot more deep learning to the problem, and try to be a lot more proactive around where abuse is happening, so that we can take the burden off the victim completely. And we ’ ve made some progress recently. About 38 percent of abusive tweets are now proactively identified by machine learning algorithms so that people don ’ t actually have to report them. But those that are identified are still reviewed by humans, so we do not take down content or accounts without a human actually reviewing it. But that was from zero percent just a year ago. So that meant, at that zero percent, every single person who received abuse had to actually report it, which was a lot of work for them, a lot of work for us and just ultimately unfair. The other thing that we ’ re doing is making sure that we, as a company, have representation of all the communities that we ’ re trying to serve. We can ’ t build a business that is successful unless we have a **diversity** of perspective inside of our walls that actually feel these issues every single day. And that ’ s not just with the team that ’ s doing the work, it ’ s also within our **leadership** as well. So we need to continue to build empathy for what people are experiencing and give them better tools to act on it and also give our customers a much better and easier approach to handle some of the things that they ' re seeing. So a lot of what we ' re doing is around technology, but we ' re also looking at the incentives on the service : What does Twitter incentivize you to do when you first open it up? And in the past, it ' s incented a lot of outrage, it ' s incented a lot of mob behavior, it ' s incented a lot of group harassment. And we have to look a lot deeper at some of the fundamentals of what the service is doing to make the bigger shifts. We can make a bunch of small shifts around technology, as I just described, but ultimately, we have to look deeply at the dynamics in the network itself, and that ' s what we ' re doing. CA : But what ' s your sense — what is the kind of thing that you might be able to change that would actually fundamentally shift behavior? JD : Well, one of the things — we started the service with this concept of following an account, as an example, and I don ' t believe that ' s why people actually come to Twitter. I believe Twitter is best as an interest-based network. People come with a particular interest. They have to do a ton of work to find and follow the related accounts around those interests. What we could do instead is allow you to follow an interest, follow a hashtag, follow a trend, follow a community, which gives us the opportunity to show all of the accounts, all the topics, all the moments, all the hashtags that are associated with that particular topic and interest, which really opens up the perspective that you see. But that is a huge fundamental shift to bias the entire network away from just an account bias towards a topics and interest bias. CA : Because isn ' t it the case that one reason why you have so much content on there is a result of putting millions of people around the world in this kind of gladiatorial contest with each other for followers, for attention? Like, from the point of view of people who just read Twitter, that ' s not an issue, but for the people who actually create it, everyone ' s out there saying,”You know, I wish I had a few more ' likes, ' followers, retweets.”And so they ' re constantly experimenting, trying to find the path to do that. And what we ' ve all discovered is that the number one path to do that is to be some form of provocative, obnoxious, eloquently obnoxious, like, eloquent insults are a dream on Twitter, where you rapidly pile up — and it becomes this self- fueling process of driving outrage. How do you defuse that? JD : Yeah, I mean, I think you ' re spot on, but that goes back to the incentives. Like, one of the choices we made in the early days was we had this number that showed how many people follow you. We decided that number should be big and bold, and anything that ' s on the page that ' s big and bold has importance, and those are the things that you want to drive. Was that the right decision at the time? Probably not. If I had to start the service again, I would not emphasize the follower count as much. I would not emphasize the”like”count as much. I don ' t think I would even create”like”in the first place, because it doesn ' t actually push what we believe now to be the most important thing, which is healthy contribution back to the network and conversation to the network, participation within conversation, learning something from the conversation. Those are not things that we thought of 13 years ago, and we believe are extremely important right now. So we have to look at how we display the follower count, how we display retweet count, how we display”likes,”and just ask the deep question : Is this really the number that we want people to drive up? Is this the thing that, when you open Twitter, you see,”That ' s the thing I need to increase?”And I don ' t believe that ' s the case right now. WPR : I think we should look at some of the tweets that are coming in from the audience as well. CA : Let ' s see what you **guys** are asking. I mean, this is — generally, one of the amazing things about Twitter is how you can use it for crowd wisdom, you know, that more knowledge, more questions, more points of view than you can imagine, and sometimes, many of them are really healthy. WPR : I think one I saw that passed already quickly down here,”What

’ s Twitter ’ s plan to combat foreign meddling in the 2020 US election?”I think that ’ s something that ’ s an issue we ’ re seeing on the internet in general, that we have a lot of malicious automated activity happening. And on Twitter, for example, in fact, we have some work that ’ s come from our friends at Zignal Labs, and maybe we can even see that to give us an example of what exactly I ’ m talking about, where you have these bots, if you will, or coordinated automated malicious account activity, that is being used to influence things like elections. And in this example we have from Zignal which they ’ ve shared with us using the data that they have from Twitter, you actually see that in this case, white represents the humans — human accounts, each dot is an account. The pinker it is, the more automated the activity is. And you can see how you have a few humans interacting with bots. In this case, it ’ s related to the election in Israel and spreading misinformation about Benny Gantz, and as we know, in the end, that was an election that Netanyahu won by a slim margin, and that may have been in some case influenced by this. And when you think about that happening on Twitter, what are the things that you ’ re doing, specifically, to **ensure** you don ’ t have misinformation like this spreading in this way, influencing people in ways that could affect democracy? JD : Just to back up a bit, we asked ourselves a question : Can we actually measure the health of a conversation, and what does that mean? And in the same way that you have indicators and we have indicators as humans in terms of are we healthy or not, such as temperature, the flushness of your face, we believe that we could find the indicators of conversational health. And we worked with a lab called Cortico at MIT to propose four starter indicators that we believe we could ultimately measure on the system. And the first one is what we ’ re calling shared attention. It ’ s a measure of how much of the conversation is attentive on the same topic versus disparate. The second one is called shared reality, and this is what percentage of the conversation shares the same facts — not whether those facts are truthful or not, but are we sharing the same facts as we converse? The third is receptivity : How much of the conversation is receptive or civil or the inverse, toxic? And then the fourth is variety of perspective. So, are we seeing filter bubbles or echo chambers, or are we actually getting a variety of opinions within the conversation? And implicit in all four of these is the understanding that, as they increase, the conversation gets healthier and healthier. So our first step is to see if we can measure these online, which we believe we can. We have the most momentum around receptivity. We have a toxicity score, a toxicity model, on our system that can actually measure whether you are likely to walk away from a conversation that you ’ re having on Twitter because you feel it ’ s toxic, with some pretty high degree. We ’ re working to measure the rest, and the next step is, as we build up solutions, to watch how these measurements trend over time and continue to experiment. And our goal is to make sure that these are balanced, because if you increase one, you might decrease another. If you increase variety of perspective, you might actually decrease shared reality. CA : Just picking up on some of the questions flooding in here. JD : Constant questioning. CA : A lot of people are puzzled why, like, how hard is it to get rid of Nazis from Twitter? JD : So we have **policies** around violent extremist groups, and the majority of our work and our terms of service works on conduct, not content. So we ’ re actually looking for conduct. Conduct being using the service to repeatedly or episodically harass someone, using hateful imagery that might be associated with the KKK or the American Nazi Party. Those are all things that we act on immediately. We ’ re in a situation right now where that term is used fairly loosely, and we just cannot take any one mention of that word accusing someone else as a factual indication that they should be removed from the platform. So a lot of our models are based around, number one : Is this account associated with a violent extremist group? And if so, we can take action. And we have done so on the KKK and the American Nazi Party and others. And number two : Are they using imagery or conduct that would associate them as such as well? CA : How many people do you have working on content moderation to look at this? JD : It varies. We want to be flexible on this, because we want to make sure that we ’ re, number one, building algorithms instead of just hiring massive amounts of people, because we need to make sure that this is scalable, and there are no amount of people that can actually scale this. So this is why we ’ ve done so much work around proactive detection of abuse that humans can then review. We want to have a situation where algorithms are constantly scouring every single tweet and bringing the most **interesting** ones to the top so that humans can bring their judgment to whether we should take action or not, based on our terms of service. WPR : But there ’ s not an amount of people that are scalable, but how many people do you currently have monitoring these accounts, and how do you figure out what ’ s enough? JD : They ’ re completely flexible. Sometimes we associate folks with spam. Sometimes we associate folks with abuse and harassment. We ’ re going to make sure that we have flexibility in our people so that we can direct them at what is most needed. Sometimes, the elections. We ’ ve had a string of elections in Mexico, one coming up in India, obviously, the election last year, the midterm election, so we just want to be flexible with our resources. So when people — just as an example, if you go to our current terms of service and you bring the page up, and you ’ re wondering about abuse and harassment that you just received and whether it was against our terms of service to report it, the first thing you see when you open that page is around intellectual property protection. You scroll down and you get to abuse, harassment and everything else that you might be experiencing. So I don ’ t know how that happened over the company ’ s history, but we put that above the thing that people want the most information on and to actually act on. And just our ordering shows the world what we believed was important. So we ’ re changing all that. We ’ re ordering it the right way, but we ’ re also simplifying the rules so that they ’ re human-readable so that people can actually understand themselves when something is against our terms and when something is not. And then we ’ re making — again, our big focus is on removing the burden of work from the victims. So that means push more towards technology, rather than humans doing the work — that means the humans receiving the abuse and also the humans having to review that work. So we want to make sure that we ’ re not just encouraging more work around something that ’ s super, super negative, and we want to have a good balance between the technology and where humans can actually be creative, which is the judgment of the rules, and not just all the mechanical **stuff** of finding and reporting them. So that ’ s how we think about it. CA : I ’ m curious to dig in more about what you said. I mean, I love that you said you are looking for ways to re-tweak the fundamental design of the system to discourage some of the reactive behavior, and perhaps — to use Tristan Harris-type language — engage people ’ s more reflective thinking. How far advanced is that? What would alternatives to that”like”button be? JD : Well, first and foremost, my personal goal with the service is that I believe fundamentally that public conversation is critical. There are existential problems facing the world that are facing the entire world, not any one particular nation-state, that global public conversation benefits. And that is one of the unique dynamics of Twitter, that it is completely open, it is completely public, it is completely fluid, and anyone can see any other conversation and participate in it. So there are conversations like **climate** change. There are conversations like the displacement in the work through artificial **intelligence**. There are conversations like economic disparity. No matter what any one nation-state does, they will not be able to solve the problem alone. It takes coordination around the world, and that ' s where I think Twitter can play a part. The second thing is that Twitter, right now, when you go to it, you don ' t necessarily walk away feeling like you learned something. Some people do. Some people have a very, very rich network, a very rich community that they learn from every single day. But it takes a lot of work and a lot of time to build up to that. So we want to get people to those topics and those interests much, much faster and make sure that they ' re finding something that, no matter how much time they spend on Twitter — and I don ' t want to maximize the time on Twitter, I want to maximize what they actually take away from it and what they learn from it, and — CA : Well, do you, though? Because that ' s the core question that a lot of people want to know. Surely, Jack, you ' re constrained, to a huge extent, by the fact that you ' re a public company, you ' ve got investors pressing on you, the number one way you make your money is from advertising — that depends on user engagement. Are you willing to sacrifice user time, if need be, to go for a more reflective conversation? JD : Yeah ; more relevance means less time on the service, and that ' s perfectly fine, because we want to make sure that, like, you ' re coming to Twitter, and you see something immediately that you learn from and that you push. We can still serve an ad against that. That doesn ' t mean you need to spend any more time to see more. The second thing we ' re looking at — CA : But just — on that goal, daily active usage, if you ' re measuring that, that doesn ' t necessarily mean things that people value every day. It may well mean things that people are drawn to like a moth to the flame, every day. We are addicted, because we see something that pisses us off, so we go in and add fuel to the fire, and the daily active usage goes up, and there ' s more ad revenue there, but we all get angrier with each other. How do you define..."Daily active usage"seems like a really dangerous term to be optimizing. JD : Taken alone, it is, but you didn ' t let me finish the other metric, which is, we ' re watching for conversations and conversation chains. So we want to incentivize healthy contribution back to the network, and what we believe that is is actually participating in conversation that is healthy, as defined by those four indicators I articulated earlier. So you can ' t just optimize around one metric. You have to balance and look constantly at what is actually going to create a healthy contribution to the network and a healthy experience for people. Ultimately, we want to get to a metric where people can tell us,"Hey, I learned something from Twitter, and I ' m walking away with something valuable."That is our goal ultimately over time, but that ' s going to take some time. CA : You come over to many, I think to me, as this enigma. This is possibly unfair, but I woke up the other night with this picture of how I found I was thinking about you and the situation, that we ' re on this great voyage with you on this ship called the"Twittanic"— and there are people on board in steerage who are expressing discomfort, and you, unlike many other captains, are saying,"Well, tell me, talk to me, listen to me, I want to hear."And they talk to you, and they say,"We ' re worried about the iceberg ahead."And you go,"You know, that is a powerful point, and our ship, frankly, hasn ' t been built properly for steering as well as it might."And we say,"Please do something."And you go to the bridge, and we ' re waiting, and we look, and then you ' re showing this extraordinary calm, but we ' re all standing outside, saying,"Jack, turn the fucking wheel!"You know? I mean — It ' s democracy at stake. It ' s our culture at stake. It ' s our world at stake. And Twitter is amazing and shapes so much. It ' s not as big as some of the other platforms, but the people of influence use it to set the agenda, and it ' s just hard to imagine a more important role in the world than to... I mean, you ' re doing a brilliant job of listening, Jack, and hearing people, but to actually dial up the urgency and move on this **stuff** — will you do that? JD : Yes, and we have been moving substantially. I mean, there ’ s been a few dynamics in Twitter ’ s history. One, when I came back to the company, we were in a pretty dire state in terms of our future, and not just from how people were using the platform, but from a corporate narrative as well. So we had to fix a bunch of the foundation, turn the company around, go through two crazy layoffs, because we just got too big for what we were doing, and we focused all of our energy on this concept of serving the public conversation. And that took some work. And as we dived into that, we realized some of the issues with the fundamentals. We could do a bunch of superficial things to address what you ’ re talking about, but we need the changes to last, and that means going really, really deep and paying attention to what we started 13 years ago and really questioning how the system works and how the framework works and what is needed for the world today, given how quickly everything is moving and how people are using it. So we are working as quickly as we can, but quickness will not get the job done. It ’ s focus, it ’ s prioritization, it ’ s understanding the fundamentals of the network and building a framework that scales and that is resilient to change, and being open about where we are and being transparent about where are so that we can continue to earn **trust**. So I ’ m proud of all the frameworks that we ’ ve put in place. I ’ m proud of our direction. We obviously can move faster, but that required just stopping a bunch of **stupid stuff** we were doing in the past. CA : All right. Well, I suspect there are many people here who, if given the chance, would love to help you on this change-making agenda you ’ re on, and I don ’ t know if Whitney — Jack, thank you for coming here and speaking so openly. It took courage. I really appreciate what you said, and good luck with your mission. JD : Thank you so much. Thanks for having me. Thank you.

Text Sample 552: POS Dim 4 – 2015 – Score 20.00 – 2015/t001513.txt
What I ’ d like to do is talk to you a little bit about fear and the cost of fear and the age of fear from which we are now emerging. I would like you to feel comfortable with my doing that by letting you know that I know something about fear and anxiety. I ’ m a Jewish **guy** from New Jersey. I could worry before I could walk. Please, applaud that. Thank you. But I also grew up in a time where there was something to fear. We were brought out in the hall when I was a little kid and taught how to put our coats over our heads to protect us from global thermonuclear war. Now even my seven-year- old brain knew that wasn ’ t going to work. But I also knew that global thermonuclear war was something to be concerned with. And yet, despite the fact that we lived for 50 years with the threat of such a war, the response of our government and of our society was to do wonderful things. We created the space program in response to that. We built our highway system in response to that. We created the Internet in response to that. So sometimes fear can produce a constructive response. But sometimes it can produce an un-constructive response. On September 11, 2001, 19 **guys** took over four airplanes and flew them into a couple of buildings. They exacted a horrible toll. It is not for us to minimize what that toll was. But the response that we had was clearly disproportionate — disproportionate to the point of verging on the unhinged. We rearranged the national security apparatus of the United States and of many governments to address a threat that, at the time that those attacks took place, was quite limited. In fact, according to our **intelligence** services, on September 11, 2001, there were 100 members of core Al-Qaeda. There were just a few thousand terrorists. They posed an existential threat to no one. But we rearranged our entire national

security apparatus in the most sweeping way since the end of the Second World War. We launched two wars. We spent trillions of dollars. We suspended our values. We violated **international** law. We embraced torture. We embraced the idea that if these 19 **guys** could do this, anybody could do it. And therefore, for the first time in history, we were seeing **everybody** as a threat. And what was the result of that? Surveillance programs that listened in on the emails and **phone** calls of entire countries — hundreds of millions of people — setting aside whether those countries were our allies, setting aside what our interests were. I would argue that 15 years later, since today there are more terrorists, more terrorist attacks, more terrorist casualties — this by the count of the U. S. State Department — since today the region from which those attacks emanate is more unstable than at any time in its history, since the Flood, perhaps, we have not succeeded in our response. Now you have to ask, where did we go wrong? What did we do? What was the mistake that was made? And you might say, well look, Washington is a dysfunctional place. There are political food fights. We ' ve turned our discourse into a cage match. And that ' s true. But there are bigger problems, believe it or not, than that **dysfunction**, even though I would argue that **dysfunction** that makes it impossible to get anything done in the richest and most powerful country in the world is far more dangerous than anything that a group like ISIS could do, because it stops us in our tracks and it keeps us from progress. But there are other problems. And the other problems came from the fact that in Washington and in many capitals right now, we ' re in a creativity **crisis**. In Washington, in think tanks, where people are supposed to be thinking of new ideas, you don ' t get bold new ideas, because if you offer up a bold new idea, not only are you attacked on Twitter, but you will not get confirmed in a government job. Because we are reactive to the heightened venom of the political debate, you get governments that have an us-versus-them mentality, tiny groups of people making decisions. When you sit in a room with a small group of people making decisions, what do you get? You get groupthink. **Everybody** has the same worldview, and any view from outside of the group is seen as a threat. That ' s a danger. You also have processes that become reactive to news cycles. And so the parts of the U. S. government that do foresight, that look forward, that do strategy — the parts in other governments that do this — can ' t do it, because they ' re reacting to the news cycle. And so we ' re not looking ahead. On 9 / 11, we had a **crisis** because we were looking the wrong way. Today we have a **crisis** because, because of 9 / 11, we are still looking in the wrong direction, and we know because we see transformational trends on the horizon that are far more important than what we saw on 9 / 11 ; far more important than the threat posed by these terrorists ; far more important even than the instability that we ' ve got in some areas of the world that are racked by instability today. In fact, the things that we are seeing in those parts of the world may be symptoms. They may be a **reaction** to bigger trends. And if we are treating the symptom and ignoring the bigger trend, then we ' ve got far bigger problems to deal with. And so what are those trends? Well, to a group like you, the trends are apparent. We are living at a moment in which the very fabric of human society is being rewoven. If you saw the cover of The Economist a couple of days ago — it said that 80 percent of the people on the planet, by the year 2020, would have a smartphone. They would have a small computer connected to the Internet in their pocket. In most of Africa, the cell **phone** penetration rate is 80 percent. We passed the point last October when there were more mobile cellular devices, SIM cards, out in the world than there were people. We are within years of a profound moment in our history, when effectively every single human being on the planet is going to be part of a man-made system for the first time, able to touch anyone else

— touch them for good, touch them for ill. And the changes associated with that are changing the very nature of every aspect of governance and life on the planet in ways that our **leaders** ought to be thinking about, when they ’ re thinking about these immediate threats. On the security side, we ’ ve come out of a Cold War in which it was too costly to fight a nuclear war, and so we didn ’ t, to a period that I call Cool War, cyber war, where the costs of **conflict** are actually so low, that we may never stop. We may enter a period of constant warfare, and we know this because we ’ ve been in it for several years. And yet, we don ’ t have the basic doctrines to guide us in this regard. We don ’ t have the basic ideas formulated. If someone attacks us with a cyber attack, do have the **ability** to respond with a kinetic attack? We don ’ t know. If somebody launches a cyber attack, how do we deter them? When China launched a series of cyber attacks, what did the U. S. government do? It said, we ’ re going to indict a few of these Chinese **guys**, who are never coming to America. They ’ re never going to be anywhere near a law enforcement officer who ’ s going to take them into custody. It ’ s a gesture — it ’ s not a deterrent. Special forces operators out there in the field today discover that small groups of insurgents with cell **phones** have access to satellite imagery that once only superpowers had. In fact, if you ’ ve got a cell **phone**, you ’ ve got access to power that a superpower didn ’ t have, and would have highly classified 10 years ago. In my cell **phone**, I have an app that tells me where every plane in the world is, and its altitude, and its speed, and what kind of aircraft it is, and where it ’ s going and where it ’ s landing. They have apps that allow them to know what their adversary is about to do. They ’ re using these tools in new ways. When a cafe in Sydney was taken over by a terrorist, he went in with a rifle... and an iPad. And the weapon was the iPad. Because he captured people, he terrorized them, he pointed the iPad at them, and then he took the video and he put it on the Internet, and he took over the world ’ s media. But it doesn ’ t just affect the security side. The relations between great powers — we thought we were past the bipolar era. We thought we were in a unipolar world, where all the big issues were resolved. Remember? It was the end of history. But we ’ re not. We ’ re now seeing that our basic assumptions about the Internet — that it was going to connect us, weave society together — are not necessarily true. In countries like China, you have the Great Firewall of China. You ’ ve got countries saying no, if the Internet happens within our borders we control it within our borders. We control the content. We are going to control our security. We are going to manage that Internet. We are going to say what can be on it. We ’ re going to set a different set of rules. Now you might think, well, that ’ s just China. But it ’ s not just China. It ’ s China, India, Russia. It ’ s Saudi Arabia, it ’ s Singapore, it ’ s Brazil. After the NSA scandal, the Russians, the Chinese, the Indians, the Brazilians, they said, let ’ s create a new Internet backbone, because we can ’ t be dependent on this other one. And so all of a sudden, what do you have? You have a new bipolar world in which cyber-internationalism, our belief, is challenged by cyber- nationalism, another belief. We are seeing these changes everywhere we look. We are seeing the advent of mobile money. It ’ s happening in the places you wouldn ’ t expect. It ’ s happening in Kenya and Tanzania, where millions of people who haven ’ t had access to financial services now conduct all those services on their **phones**. There are 2. 5 million people who don ’ t have financial service access that are going to get it soon. A billion of them are going to have the **ability** to access it on their cell **phone** soon. It ’ s not just going to give them the **ability** to bank. It ’ s going to change what monetary **policy** is. It ’ s going to change what money is. Education is changing in the same way. Healthcare is changing in the same way. How government services are delivered is changing in the same way. And yet, in Washington, we are debating whether to call the terrorist group that has taken over Syria and Iraq ISIS or ISIL or Islamic State. We are trying to determine how much we want to give in a negotiation with the Iranians on a nuclear deal which deals with the technologies of 50 years ago, when in fact, we know that the Iranians right now are engaged in cyber war with us and we ’ re ignoring it, partially because businesses are not willing to talk about the attacks that are being waged on them. And that gets us to another breakdown that ’ s crucial, and another breakdown that couldn ’ t be more important to a group like this, because the growth of America and real American national security and all of the things that drove progress even during the Cold War, was a public-private partnership between science, technology and government that began when Thomas Jefferson sat alone in his laboratory inventing new things. But it was the canals and railroads and telegraph ; it was radar and the Internet. It was Tang, the breakfast drink — probably not the most important of those developments. But what you had was a partnership and a dialogue, and the dialogue has broken down. It ’ s broken down because in Washington, less government is considered more. It ’ s broken down because there is, believe it or not, in Washington, a war on science — despite the fact that in all of human history, every time anyone has waged a war on science, science has won. But we have a government that doesn ’ t want to listen, that doesn ’ t have people at the highest levels that understand this. In the nuclear age, when there were people in **senior** national security jobs, they were expected to speak throw- weight. They were expected to know the lingo, the vocabulary. If you went to the highest level of the U. S. government now and said,”Talk to me about cyber, about neuroscience, about the things that are going to change the world of tomorrow,”you ’ d get a blank stare. I know, because when I wrote this book, I talked to 150 people, many from the science and **tech** side, who felt like they were being shunted off to the kids ’ table. Meanwhile, on the **tech** side, we have lots of wonderful people creating wonderful things, but they started in garages and they didn ’ t need the government and they don ’ t want the government. Many of them have a political view that ’ s somewhere between libertarian and anarchic : leave me alone. But the world ’ s coming apart. All of a sudden, there are going to be massive regulatory changes and massive issues associated with **conflict** and massive issues associated with security and privacy. And we haven ’ t even gotten to the next set of issues, which are philosophical issues. If you can ’ t vote, if you can ’ t have a job, if you can ’ t bank, if you can ’ t get health care, if you can ’ t be educated without Internet access, is Internet access a fundamental right that should be written into constitutions? If Internet access is a fundamental right, is electricity access for the 1. 2 billion who don ’ t have access to electricity a fundamental right? These are fundamental issues. Where are the philosophers? Where ’ s the dialogue? And that brings me to the reason that I ’ m here. I live in Washington. Pity me. The dialogue isn ’ t happening there. These big issues that will change the world, change national security, change economics, create hope, create threats, can only be resolved when you bring together groups of people who understand science and technology back together with government. Both sides need each other. And until we recreate that connection, until we do what helped America grow and helped other countries grow, then we are going to grow ever more vulnerable. The risks associated with 9 / 11 will not be measured in terms of lives lost by terror attacks or buildings destroyed or trillions of dollars spent. They ’ ll be measured in terms of the costs of our distraction from critical issues and our inability to get together scientists, technologists, government **leaders**, at a moment of transformation akin to the beginning of the Renaissance, akin to the beginning of the major transformational eras that have happened on Earth, and start coming up with, if not the right answers, then at least the right questions. We are not there yet, but discussions like this and groups like you are the places where those questions can be formulated and posed. And that ' s why I believe that groups like **TED**, discussions like this around the planet, are the place where the future of foreign **policy**, of economic **policy**, of social **policy**, of philosophy, will ultimately take place. And that ' s why it ' s been a pleasure speaking to you. Thank you very, very much.

Text Sample 553: POS Dim 4 – 2015 – Score 19.00 – 2015/t001120.txt
So have you ever wondered what it would be like to live in a place with no rules? That sounds pretty cool. You wake up one morning, however, and you discover that the reason there are no rules is because there ' s no government, and there are no laws. In fact, all social institutions have disappeared. So there ' s no schools, there ' s no hospitals, there ' s no police, there ' s no banks, there ' s no athletic clubs, there ' s no utilities. Well, I know a little bit about what this is like, because when I was a medical student in 1999, I worked in a **refugee** camp in the Balkans during the Kosovo War. When the war was over, I got permission — unbelievably — from my medical school to take some time off and follow some of the families that I had befriended in the camp back to their village in Kosovo, and understand how they navigated life in this postwar setting. Postwar Kosovo was a very **interesting** place because NATO troops were there, mostly to make sure the war didn ' t break out again. But other than that, it was actually a lawless place, and almost every social institution, both public and private, had been destroyed. So I can tell you that when you go into one of these situations and settings, it is absolutely thrilling… for about 30 minutes, because that ' s about how long it takes before you run into a situation where you realize how incredibly vulnerable you are. For me, that moment came when I had to cross the first checkpoint, and I realized as I drove up that I would be **negotiating** passage through this checkpoint with a heavily armed individual who, if he decided to shoot me right then and there, actually wouldn ' t be doing anything illegal. But the sense of vulnerability that I had was absolutely nothing in comparison to the vulnerability of the families that I got to know over that year. You see, life in a society where there are no social institutions is riddled with danger and **uncertainty**, and simple questions like,"What are we going to eat tonight?"are very complicated to answer. Questions about security, when you don ' t have any security systems, are terrifying. Is that altercation I had with the neighbor down the block going to turn into a violent episode that will end my life or my family ' s life? Health concerns when there is no health system are also terrifying. I listened as many families had to sort through questions like,"My infant has a fever. What am I going to do?""My sister, who is pregnant, is bleeding. What should I do? Who should I turn to?""Where are the doctors, where are the nurses? If I could find one, are they trustworthy? How will I pay them? In what currency will I pay them?""If I need medications, where will I find them? If I take those medications, are they actually counterfeits?"And on and on. So for life in these settings, the **dominant** theme, the **dominant** feature of life, is the incredible vulnerability that people have to manage day in and day out, because of the lack of social systems. And it actually turns out that this feature of life is incredibly difficult to explain and be understood by people who are living outside of it. I discovered this when I left Kosovo. I came back to Boston, I became a physician, I became a global public health **policy** researcher. I joined the Harvard Medical School and Brigham and Women ' s Hospital Division of Global Health. And I, as a researcher, really

wanted to get started on this problem right away. I was like,”How do we reduce the crushing vulnerability of people living in these types of fragile settings? Is there any way we can start to think about how to protect and quickly recover the institutions that are critical to survival, like the health system?”And I have to say, I had amazing colleagues. But one **interesting** thing about it was, this was sort of an unusual question for them. They were kind of like,”Oh, if you work in war, doesn ’ t that mean you work on **refugee** camps, and you work on **documenting** mass atrocities?”— which is, by the way, very, very, very important. So it took me a while to explain why I was so passionate about this issue, until about six years ago. That ’ s when this landmark study that looked at and described the public health consequences of war was published. They came to an incredible, provocative conclusion. These researchers concluded that the vast majority of death and disability from war happens after the cessation of **conflict**. So the most dangerous time to be a person living in a conflict-affected state is after the cessation of hostilities ; it ’ s after the peace deal has been signed. It ’ s when that political solution has been achieved. That seems so puzzling, but of course it ’ s not, because war kills people by robbing them of their clinics, of their hospitals, of their supply chains. Their doctors are targeted, are killed ; they ’ re on the run. And more invisible and yet more deadly is the destruction of the health governance institutions and their finances. So this is really not surprising at all to me. But what is surprising and somewhat dismaying, is how little **impact** this insight has had, in terms of how we think about human suffering and war. Let me give you a couple examples. Last year, you may remember, Ebola hit the West African country of Liberia. There was a lot of reporting about this group, Doctors Without Borders, sounding the alarm and calling for aid and assistance. But not a lot of that reporting answered the question : Why is Doctors Without Borders even in Liberia? Doctors Without Borders is an amazing organization, dedicated and designed to provide emergency care in war zones. Liberia ’ s civil war had ended in 2003 — that was 11 years before Ebola even struck. When Ebola struck Liberia, there were less than 50 doctors in the entire country of 4. 5 million people. Doctors Without Borders is in Liberia because Liberia still doesn ’ t really have a functioning health system, 11 years later. When the earthquake hit Haiti in 2010, the outpouring of **international** aid was phenomenal. But did you know that only two percent of that funding went to rebuild Haitian public institutions, including its health **sector**? From that perspective, Haitians continue to die from the earthquake even today. I recently met this gentleman. This is Dr. Nezar Ismet. He ’ s the Minister of Health in the northern autonomous region of Iraq, in Kurdistan. Here he is announcing that in the last nine months, his country, his region, has increased from four million people to five million people. That ’ s a 25 percent increase. Thousands of these new arrivals have experienced incredible trauma. His doctors are working 16-hour days without pay. His budget has not increased by 25 percent ; it has decreased by 20 percent, as funding has flowed to security concerns and to short-term relief efforts. When his health **sector** fails — and if history is any guide, it will — how do you think that ’ s going to influence the decision making of the five million people in his region as they think about whether they should flee that type of vulnerable living situation? So as you can see, this is a frustrating topic for me, and I really try to understand : Why the reluctance to protect and **support** indigenous health systems and security systems? I usually tier two concerns, two arguments. The first concern is about corruption, and the concern that people in these settings are corrupt and they are untrustworthy. And I will admit that I have met unsavory characters working in health **sectors** in these situations. But I will tell you that the opposite is absolutely true in every case I have worked on, from

Afghanistan to Libya, to Kosovo, to Haiti, to Liberia — I have met inspiring people, who, when the chips were down for their country, they risked everything to save their health institutions. The trick for the outsider who wants to help is identifying who those individuals are, and building a pathway for them to lead. That is exactly what happened in Afghanistan. One of the unsung and untold success stories of our nation-building effort in Afghanistan **involved** the World Bank in 2002 **investing** heavily in identifying, training and promoting Afghani health **sector leaders**. These health **sector leaders** have pulled off an incredible feat in Afghanistan. They have aggressively increased access to health care for the majority of the population. They are rapidly improving the health status of the Afghan population, which used to be the worst in the world. In fact, the Afghan Ministry of Health does things that I wish we would do in America. They use things like data to make **policy**. It ' s incredible. The other concern I hear a lot about is :"We just can ' t afford it, we just don ' t have the money. It ' s just unsustainable."I would submit to you that the current situation and the current system we have is the most expensive, inefficient system we could possibly conceive of. The current situation is that when governments like the US — or, let ' s say, the collection of governments that make up the European Commission — every year, they spend 15 billion dollars on just humanitarian and emergency and disaster relief worldwide. That ' s nothing about foreign aid, that ' s just disaster relief. Ninety-five percent of it goes to **international** relief agencies, that then have to import resources into these areas, and knit together some type of temporary health system, let ' s say, which they then dismantle and send away when they run out of money. So our job, it turns out, is very clear. We, as the global health community **policy** experts, our first job is to become experts in how to monitor the strengths and vulnerabilities of health systems in threatened situations. And that ' s when we see doctors fleeing, when we see health resources drying up, when we see institutions crumbling — that ' s the emergency. That ' s when we need to sound the alarm and wave our arms. OK? Not now. Everyone can see that ' s an emergency, they don ' t need us to tell them that. Number two : places like where I work at Harvard need to take their cue from the World Bank experience in Afghanistan, and we need to — and we will — build robust platforms to **support** health **sector leaders** like these. These people risk their lives. I think we can match their courage with some **support**. Number three : we need to reach out and make new partnerships. At our global health center, we have launched a new initiative with NATO and other security **policy** makers to explore with them what they can do to protect health system institutions during deployments. We want them to see that protecting health systems and other critical social institutions is an integral part of their mission. It ' s not just about avoiding collateral damage ; it ' s about winning the peace. But the most important partner we need to engage is you, the American public, and indeed, the world public. Because unless you understand the value of social institutions, like health systems in these fragile settings, you won ' t **support** efforts to save them. You won ' t click on that article that talks about"Hey, all those doctors are on the run in country X. I wonder what that means. I wonder what that means for that health system ' s **ability** to, let ' s say, detect influenza.""Hmm, it ' s probably not good."That ' s what I ' d tell you. Up on the screen, I ' ve put up my three favorite American institution defenders and builders. Over here is George C. Marshall, he was the **guy** that proposed the Marshall Plan to save all of Europe ' s economic institutions after World War II. And this Eleanor Roosevelt. Her work on human rights really serves as the foundation for all of our **international** human rights organizations. Then my big favorite is Ben Franklin, who did many things in terms of creating institutions, but was the midwife of our constitution. And I would say to you that these are folks who, when our country was threatened, or our world was threatened, they didn ’ t retreat. They didn ’ t talk about building walls. They talked about building institutions to protect human security, for their generation and also for ours. And I think our generation should do the same. Thank you.

Text Sample 554: POS Dim 4 – 2015 – Score 18.00 – 2015/t001429.txt
Bruno Giussani : Commissioner, thank you for coming to **TED**. Ant□□nio Guterres : Pleasure. BG : Let ’ s start with a figure. During 2015, almost one million **refugees** and migrants arrived in Europe from many different countries, of course, from Syria and Iraq, but also from Afghanistan and Bangladesh and Eritrea and elsewhere. And there have been **reactions** of two different kinds : welcoming parties and border fences. But I want to look at it a little bit from the short-term and the long-term perspective. And the first question is very simple : Why has the movement of **refugees** spiked so fast in the last six months? AG : Well, I think, basically, what triggered this huge increase was the Syrian **refugee** group. There has been an increased movement into Europe from Africa, from Asia, but slowly growing, and all of a sudden we had this massive increase in the first months of this year. Why? I think there are three reasons, two long-term ones and the trigger. The long-term ones, in relation to Syrians, is that hope is less and less clear for people. I mean, they look at their own country and they don ’ t see much hope to go back home, because there is no political solution, so there is no light at the end of the tunnel. Second, the living conditions of the Syrians in the neighboring countries have been deteriorating. We just had research with the World Bank, and 87 percent of the Syrians in Jordan and 93 percent of the Syrians in Lebanon live below the national poverty lines. Only half of the children go to school, which means that people are living very badly. Not only are they **refugees**, out of home, not only have they suffered what they have suffered, but they are living in very, very dramatic conditions. And then the trigger was when all of a sudden, **international** aid decreased. The World Food Programme was forced, for lack of resources, to cut by 30 percent food **support** to the Syrian **refugees**. They ’ re not allowed to work, so they are totally dependent on **international support**, and they felt,”The world is abandoning us.”And that, in my opinion, was the trigger. All of a sudden, there was a rush, and people started to move in large numbers and, to be absolutely honest, if I had been in the same situation and I would have been brave enough to do it, I think I would have done the same. BG : But I think what surprised many people is it ’ s not only sudden, but it wasn ’ t supposed to be sudden. The war in Syria has been happening for five years. Millions of **refugees** are in camps and villages and towns around Syria. You have yourself warned about the situation and about the consequences of a breakdown of Libya, for example, and yet Europe looked totally unprepared. AG : Well, unprepared because divided, and when you are divided, you don ’ t want to recognize the reality. You prefer to postpone decisions, because you do not have the capacity to make them. And the proof is that even when the spike occurred, Europe remained divided and was unable to put in place a mechanism to manage the situation. You talk about one million people. It looks enormous, but the population of the European Union is 550 million people, which means we are talking about one per every [550] Europeans. Now, in Lebanon, we have one **refugee** per three Lebanese. And Lebanon? Struggling, of course, but it ’ s managing. So, the question is : is this something that could have been managed if □□□ not mentioning the most important thing,

which would have been addressing the root causes, but forgetting about root causes for now, looking at the phenomenon as it is □□□ if Europe were able to come together in solidarity to create an adequate reception capacity of entry points? But for that, the countries at entry points need to be massively **supported**, and then screening the people with security checks and all the other mechanisms, distributing those that are coming into all European countries, according to the possibilities of each country. I mean, if you look at the relocation program that was approved by the Commission, always too little too late, or by the Council, too little too late □□□ BG : It ' s already breaking down. AG : My country is supposed to receive four thousand. Four thousand in Portugal means nothing. So this is perfectly manageable if it is managed, but in the present circumstances, the pressure is at the point of entry, and then, as people move in this chaotic way through the Balkans, then they come to Germany, Sweden, basically, and Austria. They are the three countries that are, in the end, receiving the **refugees**. The rest of Europe is looking without doing much. BG : Let me try to bring up three questions, playing a bit devil ' s advocate. I ' ll try to ask them, make them blunt. But I think the questions are very present in the minds of many people in Europe right now, The first, of course, is about numbers. You say 550 million versus one million is not much, but realistically, how many people can Europe take? AG : Well, that is a question that has no answer, because **refugees** have the right to be protected. And there is such a thing as **international** law, so there is no way you can say,"I take 10, 000 and that ' s finished."I remind you of one thing : in Turkey, at the beginning of the **crisis**, I remember one minister saying,"Turkey will be able to receive up to 100, 000 people."Turkey has now two million three-hundred thousand or something of the sort, if you count all **refugees**. So I don ' t think it ' s fair to say how many we can take. What it is fair to say is : how we can we organize ourselves to assume our **international** responsibilities? And Europe has not been able to do so, because basically, Europe is divided because there is no solidarity in the European project. And it ' s not only about **refugees** ; there are many other areas. And let ' s be honest, this is the moment in which we need more Europe instead of less Europe. But as the public less and less believes in European institutions, it is also each time more difficult to convince the public that we need more Europe to solve these problems. BG : We seem to be at the point where the numbers turn into political shifts, particularly domestically. We saw it again this weekend in France, but we have seen it over and over in many countries : in Poland and in Denmark and in Switzerland and elsewhere, where the mood changes radically because of the numbers, although they are not very significant in absolute numbers. The Prime Minister of □□□ AG : But, if I may, on these : I mean, what does a European see at home in a village where there are no migrants? What a European sees is, on television, every single day, a few months ago, opening the news every single day, a crowd coming, uncontrolled, moving from border to border, and the images on television were of hundreds or thousands of people moving. And the idea is that nobody is taking care of it □□□ this is happening without any kind of management. And so their idea was,"They are coming to my village."So there was this completely false idea that Europe was being invaded and our way of life is going to change, and everything will □□□ And the problem is that if this had been properly managed, if people had been properly received, welcomed, sheltered at point of entry, screened at point of entry, and the moved by plane to different European countries, this would not have scared people. But, unfortunately, we have a lot of people scared, just because Europe was not able to do the job properly. BG : But there are villages in Germany with 300 inhabitants and 1, 000 **refugees**. So, what ' s your position? How do you imagine these people reacting? AG : If there would be a proper management of the situation and the proper distribution of people all over Europe, you would always have the percentage that I mentioned : one per each 2, 000. It is because things are not properly managed that in the end we have situations that are totally impossible to live with, and of course if you have a village — in Lebanon, there are many villages that have more Syrians than Lebanese ; Lebanon has been living with that. I ’ m not asking for the same to happen in Europe, for all European villages to have more **refugees** than inhabitants. What I am asking is for Europe to do the job properly, and to be able to organize itself to receive people as other countries in the world were forced to do in the past. BG : So, if you look at the global situation not only at Europe — BG : Yes! BG : If you look at the global situation, so, not only at Europe, I know you can make a long list of countries that are not really stepping up, but I ’ m more **interested** in the other part — is there somebody who ’ s doing the right thing? AG : Well, 86 percent of the **refugees** in the world are in the developing world. And if you look at countries like Ethiopia — Ethiopia has received more than 600, 000 **refugees**. All the borders in Ethiopia are open. And they have, as a **policy**, they call the”people to people”**policy** that every **refugee** should be received. And they have South Sudanese, they have Sudanese, they have Somalis. They have all the neighbors. They have Eritreans. And, in general, African countries are extremely welcoming of **refugees** coming, and I would say that in the Middle East and in Asia, we have seen a tendency for borders to be open. Now we see some problems with the Syrian situation, as the Syrian situation evolved into also a major security **crisis**, but the truth is that for a large period, all borders in the Middle East were open. The truth is that for Afghans, the borders of Pakistan and Iran were open for, at the time, six million Afghans that came. So I would say that even today, the trend in the developing world has been for borders to be open. The trend in the developed world is for these questions to become more and more complex, especially when there is, in the public opinion, a mixture of discussions between **refugee** protections on one side and security questions — in my opinion, misinterpreted — on the other side. BG : We ’ ll come back to that too, but you mentioned the cutting of funding and the vouchers from the World Food Programme. That reflects the general underfunding of the organizations working on these issues. Now that the world seems to have woken up, are you getting more funding and more **support**, or it ’ s still the same? AG : We are getting more **support**. I would say that we are coming close to the levels of last year. We were much worse during the summer. But that is clearly insufficient to address the needs of the people and address the needs of the countries that are **supporting** the people. And here we have a basic review of the criteria, the objectives, the priorities of development cooperation that is required. For instance, Lebanon and Jordan are middle-income countries. Because they are middle-income countries, they cannot receive soft loans or grants from the World Bank. Now, today this doesn ’ t make any sense, because they are providing a global public good. They have millions of **refugees** there, and to be honest, they are pillars of stability in the region, with all the difficulties they face, and the first line of defense of our **collective** security. So it doesn ’ t make sense that these countries are not a first priority in development cooperation **policies**. And they are not. And not only do the **refugees** live in very dramatic circumstances inside those countries, but the local communities themselves are suffering, because salaries went down, because there are more unemployed, because prices and rents went up. And, of course, if you look at today ’ s situation of the indicators in these countries, it is clear that, especially their poor groups of the population, are living worse and worse because of the **crisis** they are facing. BG : Who should be providing this **support**? Country by country, **international** organizations, the European

Union? Who should be coming up with this **support**? AG : We need to join all efforts. It ’ s clear that bilateral cooperation is essential. It ’ s clear that multilateral cooperation is essential. It ’ s clear that **international** financial institutions should have flexibility in order to be able to **invest** more massively in **support** to these countries. We need to combine all the instruments and to understand that today, in protracted situations, at a certain moment, that it doesn ’ t make sense anymore to make a distinction between humanitarian aid and development aid or development processes. Because you are talking about children in school, you are talking about health, you are talking about infrastructure that is overcrowded. You are talking about things that require a long-term perspective, a development perspective and not only an emergency humanitarian aid perspective. BG : I would like your comment on something that was in newspapers this morning. It is a statement made by the current front-runner for the Republican nomination for US President, Donald Trump. Yesterday, he said this. No, listen to this. It ’ s **interesting**. I quote :”I am calling for a total and complete shutdown of Muslims entering the US, until our country ’ s representatives can figure out what ’ s going on.”How do you react to that? AG : Well, it ’ s not only Donald Trump. We have seen several people around the world with political responsibility saying, for instance, that Muslims **refugees** should not be received. And the reason why they say this is because they think that by doing or saying this, they are protecting the security of their countries. Now, I ’ ve been in government. I am very keen on the need for governments to protect the security of their countries and their people. But if you say, like that, in the US or in any European country,”We are going to close our doors to Muslim **refugees**,”what you are saying is the best possible help for the propaganda of terrorist organizations. Because what you are saying □□□ What you are saying will be heard by all the Muslims in your own country, and it will pave the way for the recruitment and the mechanisms that, through technology, Daesh and al-Nusra, al-Qaeda, and all those other groups are today penetrating in our societies. And it ’ s just telling them,”You are right, we are against you.”So obviously, this is creating in societies that are all multiethnic, multi-religious, multicultural, this is creating a situation in which, really, it is much easier for the propaganda of these terrorist organizations to be effective in recruiting people for terror acts within the countries where these kinds of sentences are expressed. BG : Have the recent attacks in Paris and the **reactions** to them made your job more difficult? AG : Undoubtedly. BG : In what sense? AG : In the sense that, I mean, for many people the first **reaction** in relation to these kinds of terrorist attacks is : close all borders □□□ not understanding that the terrorist problem in Europe is largely homegrown. We have thousands and thousands of European fighters in Syria and in Iraq, so this is not something that you solve by just not allowing Syrians to come in. And I must say, I am convinced that the passport that appeared, I believe, was put by the person who has blown □□□ BG : □□□ himself up, yeah. AG : [I believe] it was on purpose, because part of the strategies of Daesh is against **refugees**, because they see **refugees** as people that should be with the caliphate and are fleeing to the crusaders. And I think that is part of Daesh ’ s strategy to make Europe react, closing its doors to Muslim **refugees** and having an hostility towards Muslims inside Europe, exactly to facilitate Daesh ’ s work. And my deep belief is that it was not the **refugee** movement that triggered terrorism. I think, as I said, essentially terrorism in Europe is today a homegrown movement in relation to the global situation that we are facing, and what we need is exactly to prove these groups wrong, by welcoming and integrating effectively those that are coming from that part of the world. And another thing that I believe is that to a large extent, what we are today paying for in Europe is the failures of integration models that didn ’ t work in the ’ 60s, in the ’ 70s, in the ’ 80s, in relation to big migration flows that took place at that time and generated what is today in many of the people, for instance, of the second generation of communities, a situation of feeling marginalized, having no jobs, having improper education, living in some of the neighborhoods that are not adequately provided by public infrastructure. And this kind of uneasiness, sometimes even anger, that exists in this second generation is largely due to the failure of integration **policies**, to the failure of what should have been a much stronger investment in creating the conditions for people to live together and respect each other. For me it is clear. For me it is clear that all societies will be multiethnic, multicultural, multi-religious in the future. To try to avoid it is, in my opinion, impossible. And for me it ’ s a good thing that they will be like that, but I also recognize that, for that to work properly, you need a huge investment in the social cohesion of your own societies. And Europe, to a large extent, failed in that investment in the past few decades. BG : Question : You are stepping down from your job at the end of the year, after 10 years. If you look back at 2005, when you entered that office for the first time, what do you see? AG : Well, look : In 2005, we were helping one million people go back home in safety and dignity, because **conflicts** had ended. Last year, we helped 124, 000. In 2005, we had about 38 million people displaced by **conflict** in the world. Today, we have more than 60 million. At that time, we had had, recently, some **conflicts** that were solved. Now, we see a multiplication of new **conflicts** and the old **conflicts** never died : Afghanistan, Somalia, Democratic Republic of Congo. It is clear that the world today is much more dangerous than it was. It is clear that the capacity of the **international** community to prevent **conflicts** and to timely solve them, is, unfortunately, much worse than what it was 10 years ago. There are no clear power relations in the world, no global governance mechanisms that work, which means that we live in a situation where impunity and unpredictability tend to prevail, and that means that more and more people suffer, namely those that are displaced by **conflicts**. BG : It ’ s a tradition in American politics that when a President leaves the Oval Office for the last time, he leaves a handwritten note on the desk for his successor that walks in a couple of hours later. If you had to write such a note to your successor, Filippo Grandi, what would you write? AG : Well, I don ’ t think I would write any message. You know, one of the terrible things when one leaves an office is to try to become the backseat driver, always telling the new one what to do. So that, I will not do. If I had to say something to him, it would be,”Be yourself, and do your best.”BG : Commissioner, thank you for the job you do. Thank you for coming to **TED**.

Text Sample 555: POS Dim 4 – 2008 – Score 19.00 – 2008/t000203.txt

I spent the better part of a decade looking at American responses to mass atrocity and genocide. And I ’ d like to start by sharing with you one moment that to me sums up what there is to know about American and democratic responses to mass atrocity. And that moment came on April 21, 1994. So 14 years ago, almost, in the middle of the Rwandan genocide, in which 800, 000 people would be systematically exterminated by the Rwandan government and some extremist militia. On April 21, in the New York Times, the paper reported that somewhere between 200, 000 and 300, 000 people had already been killed in the genocide. It was in the paper — not on the front page. It was a lot like the Holocaust coverage, it was buried in the paper. Rwanda itself was not seen as newsworthy, and amazingly, genocide itself was not seen as newsworthy.

But on April 21, a wonderfully honest moment occurred. And that was that an American congresswoman named Patricia Schroeder from Colorado met with a group of journalists. And one of the journalists said to her, what ' s up? What ' s going on in the U. S. government? Two to 300, 000 people have just been exterminated in the last couple of weeks in Rwanda. It ' s two weeks into the genocide at that time, but of course, at that time you don ' t know how long it ' s going to last. And the journalist said, why is there so little response out of Washington? Why no hearings, no denunciations, no people getting arrested in front of the Rwandan embassy or in front of the White House? What ' s the deal? And she said — she was so honest — she said,"It ' s a great question. All I can tell you is that in my congressional office in Colorado and my office in Washington, we ' re getting hundreds and hundreds of calls about the endangered ape and gorilla population in Rwanda, but nobody is calling about the people. The **phones** just aren ' t ringing about the people."And the reason I give you this moment is there ' s a deep truth in it. And that truth is, or was, in the 20th century, that while we were beginning to develop endangered species movements, we didn ' t have an endangered people ' s movement. We had Holocaust education in the schools. Most of us were groomed not only on images of nuclear catastrophe, but also on images and knowledge of the Holocaust. There ' s a museum, of course, on the Mall in Washington, right next to Lincoln and Jefferson. I mean, we have owned Never Again culturally, appropriately, interestingly. And yet the politicization of Never Again, the operationalization of Never Again, had never occurred in the 20th century. And that ' s what that moment with Patricia Schroeder I think shows : that if we are to bring about an end to the world ' s worst atrocities, we have to make it such. There has to be a role — there has to be the creation of political noise and political costs in response to massive crimes against humanity, and so forth. So that was the 20th century. Now here — and this will be a relief to you at this point in the afternoon — there is good news, amazing news, in the 21st century, and that is that, almost out of nowhere, there has come into being an anti-genocide movement, an anti-genocide constituency, and one that looks destined, in fact, to be permanent. It grew up in response to the atrocities in Darfur. It is comprised of students. There are something like 300 anti-genocide chapters on college campuses around the country. It ' s bigger than the anti-apartheid movement. There are something like 500 high school chapters devoted to stopping the genocide in Darfur. Evangelicals have joined it. Jewish groups have joined it."Hotel Rwanda"watchers have joined it. It is a cacophonous movement. To call it a movement, as with all movements, perhaps, is a little misleading. It ' s diverse. It ' s got a lot of different approaches. It ' s got all the ups and the downs of movements. But it has been amazingly successful in one regard, in that it has become, it has congealed into this endangered people ' s movement that was missing in the 20th century. It sees itself, such as it is, the it, as something that will create the impression that there will be political cost, there will be a political price to be paid, for allowing genocide, for not having an heroic imagination, for not being an upstander but for being, in fact, a bystander. Now because it ' s student-driven, there ' s some amazing things that the movement has done. They have launched a divestment campaign that has now convinced, I think, 55 universities in 22 states to divest their holdings of stocks with regard to companies doing business in Sudan. They have a 1-800-GENOCIDE number — this is going to sound very kitsch, but for those of you who may not be, I mean, may be apolitical, but **interested** in doing something about genocide, you dial 1-800-GENOCIDE and you type in your zip code, and you don ' t even have to know who your congressperson is. It will refer you directly to your congressperson, to your U. S. senator, to your governor where divestment legislation is pending. They ' ve lowered the transaction costs of stopping genocide. I think the most innovative thing they ' ve introduced recently are genocide grades. And it takes students to introduce genocide grades. So what you now have when a Congress is in session is members of Congress calling up these 19-year-olds or 24-year-olds and saying, I ' m just told I have a D minus on genocide ; what do I do to get a C? I just want to get a C. Help me. And the students and the others who are part of this incredibly energized base are there to answer that, and there ' s always something to do. Now, what this movement has done is it has extracted from the Bush administration from the United States, at a time of massive over-stretch — military, financial, diplomatic — a whole series of commitments to Darfur that no other country in the world is making. For instance, the referral of the crimes in Darfur to the International Criminal Court, which the Bush administration doesn ' t like. The expenditure of 3 billion dollars in **refugee** camps to try to keep, basically, the people who ' ve been displaced from their homes by the Sudanese government, by the so-called Janjaweed, the militia, to keep those people alive until something more durable can be achieved. And recently, or now not that recently, about six months ago, the authorization of a peacekeeping force of 26, 000 that will go. And that ' s all the Bush administration ' s **leadership**, and it ' s all because of this bottom-up pressure and the fact that the **phones** haven ' t stopped ringing from the beginning of this **crisis**. The bad news, however, to this question of will evil prevail, is that evil lives on. The people in those camps are surrounded on all sides by so-called Janjaweed, these men on horseback with spears and Kalashnikovs. Women who go to get firewood in order to heat the humanitarian aid in order to feed their families — humanitarian aid, the dirty secret of it is it has to be heated, really, to be edible — are themselves subjected to rape, which is a tool of the genocide that is being used. And the peacekeepers I ' ve mentioned, the force has been authorized, but almost no country on Earth has stepped forward since the authorization to actually put its troops or its police in harm ' s way. So we have achieved an awful lot relative to the 20th century, and yet far too little relative to the gravity of the crime that is unfolding as we sit here, as we speak. Why the limits to the movement? Why is what has been achieved, or what the movement has done, been necessary but not sufficient to the crime? I think there are a couple — there are many reasons — but a couple just to focus on briefly. The first is that the movement, such as it is, stops at America ' s borders. It is not a global movement. It does not have too many compatriots abroad who themselves are asking their governments to do more to stop genocide. And the Holocaust culture that we have in this country makes Americans, sort of, more prone to, I think, want to bring Never Again to life. The guilt that the Clinton administration expressed, that Bill Clinton expressed over Rwanda, created a space in our society for a consensus that Rwanda was bad and wrong and we wish we had done more, and that is something that the movement has taken advantage of. European governments, for the most part, haven ' t acknowledged responsibility, and there ' s nothing to kind of to push back and up against. So this movement, if it ' s to be durable and global, will have to cross borders, and you will have to see other citizens in democracies, not simply resting on the assumption that their government would do something in the face of genocide, but actually making it such. Governments will never gravitate towards crimes of this magnitude naturally or eagerly. As we saw, they haven ' t even gravitated towards protecting our ports or reigning in loose nukes. Why would we expect in a bureaucracy that it would orient itself towards distant suffering? So one reason is it hasn ' t gone global. The second is, of course, that at this time in particular in America ' s history, we have a credibility problem, a legitimacy problem in **international** institutions. It is structurally really, really hard to do, as the Bush administration rightly does, which is to denounce genocide on a Monday and then describe water boarding on a Tuesday as a no-brainer and then turn up on Wednesday and look for troop commitments. Now, other countries have their own reasons for not wanting to get **involved**. Let me be clear. They ’ re in some ways using the Bush administration as an alibi. But it is essential for us to be a **leader** in this sphere, of course to restore our standing and our **leadership** in the world. The recovery ’ s going to take some time. We have to ask ourselves, what now? What do we do going forward as a country and as citizens in relationship to the world ’ s worst places, the world ’ s worst suffering, killers, and the kinds of killers that could come home to roost sometime in the future? The place that I turned to answer that question was to a man that many of you may not have ever heard of, and that is a Brazilian named Sergio Vieira de Mello who, as Chris said, was blown up in Iraq in 2003. He was the victim of the first-ever suicide bomb in Iraq. It ’ s hard to remember, but there was actually a time in the summer of 2003, even after the U. S. invasion, where, apart from looting, civilians were relatively **safe** in Iraq. Now, who was Sergio? Sergio Vieira de Mello was his name. In addition to being Brazilian, he was described to me before I met him in 1994 as someone who was a cross between James Bond on the one hand and Bobby Kennedy on the other. And in the U. N., you don ’ t get that many people who actually manage to merge those qualities. He was James Bond-like in that he was ingenious. He was drawn to the flames, he chased the flames, he was like a moth to the flames. Something of an adrenalin junkie. He was successful with women. He was Bobby Kennedy-like because in some ways one could never tell if he was a realist masquerading as an idealist or an idealist masquerading as a realist, as people always wondered about Bobby Kennedy and John Kennedy in that way. What he was was a decathlete of nation-building, of problem-solving, of troubleshooting in the world ’ s worst places and in the world ’ s most broken places. In failing states, genocidal states, under-governed states, precisely the kinds of places that threats to this country exist on the horizon, and precisely the kinds of places where most of the world ’ s suffering tends to get concentrated. These are the places he was drawn to. He moved with the headlines. He was in the U. N. for 34 years. He joined at the age of 21. Started off when the causes in the wars du jour in the ’ 70s were wars of independence and decolonization. He was there in Bangladesh dealing with the outflow of millions of **refugees** — the largest **refugee** flow in history up to that point. He was in Sudan when the civil war broke out there. He was in Cyprus right after the Turkish invasion. He was in Mozambique for the War of Independence. He was in Lebanon. Amazingly, he was in Lebanon — the U. N. base was used — Palestinians staged attacks out from behind the U. N. base. Israel then invaded and overran the U. N. base. Sergio was in Beirut when the U. S. Embassy was hit by the first-ever suicide attack against the United States. People date the beginning of this new era to 9 / 11, but surely 1983, with the attack on the US Embassy and the Marine barracks — which Sergio witnessed — those are, in fact, in some ways, the dawning of the era that we find ourselves in today. From Lebanon he went to Bosnia in the ’ 90s. The issues were, of course, ethnic sectarian violence. He was the first person to **negotiate** with the Khmer Rouge. Talk about evil prevailing. I mean, here he was in the room with the embodiment of evil in Cambodia. He **negotiates** with the Serbs. He actually crosses so far into this realm of talking to evil and trying to convince evil that it doesn ’ t need to prevail that he earns the nickname — not Sergio but Serbio while he ’ s living in the Balkans and conducting these kinds of negotiations. He then goes to Rwanda and to Congo in the aftermath of the genocide, and he ’ s the **guy** who has to decide — huh, OK, the genocide is over ; 800, 000 people have been killed ; the people responsible are fleeing into neighboring countries — into Congo, into Tanzania. I ' m Sergio, I ' m a humanitarian, and I want to feed those — well, I don ' t want to feed the killers but I want to feed the two million people who are with them, so we ' re going to go, we ' re going to set up camps, and we ' re going to supply humanitarian aid. But, uh-oh, the killers are within the camps. Well, I ' d like to separate the sheep from the wolves. Let me go door-to-door to the **international** community and see if anybody will give me police or troops to do the separation. And their response, of course, was no more than we wanted to stop the genocide and put our troops in harm ' s way to do that, nor do we now want to get in the way and pluck genocidaires from camps. So then you have to make the decision. Do you turn off the **international** spigot of life **support** and risk two million civilian lives? Or do you continue feeding the civilians, knowing that the genocidaires are in the camps, literally sharpening their knives for future battle? What do you do? It ' s all lesser-evil terrain in these broken places. Late ' 90s : nation-building is the cause du jour. He ' s the **guy** put in charge. He ' s the Paul Bremer or the Jerry Bremer of first Kosovo and then East Timor. He governs the places. He ' s the viceroy. He has to decide on tax **policy**, on currency, on border patrol, on policing. He has to make all these judgments. He ' s a Brazilian in these places. He speaks seven languages. He ' s been up to that point in 14 war zones so he ' s positioned to make better judgments, perhaps, than people who have never done that kind of work. But nonetheless, he is the cutting edge of our experimentation with doing good with very few resources being brought to bear in, again, the world ' s worst places. And then after Timor, 9 / 11 has happened, he ' s named U. N. Human Rights Commissioner, and he has to balance liberty and security and figure out, what do you do when the most powerful country in the United Nations is bowing out of the Geneva Conventions, bowing out of **international** law? Do you denounce? Well, if you denounce, you ' re probably never going to get back in the room. Maybe you stay reticent. Maybe you try to charm President Bush — and that ' s what he did. And in so doing he earned himself, unfortunately, his final and tragic appointment to Iraq — the one that resulted in his death. One note on his death, which is so devastating, is that despite predicating the war on Iraq on a link between Saddam Hussein and terrorism in 9 / 11, believe it or not, the Bush administration or the invaders did no planning, no pre-war planning, to respond to terrorism. So Sergio — this receptacle of all of this learning on how to deal with evil and how to deal with brokenness, lay under the rubble for three and a half hours without rescue. Stateless. The **guy** who tried to help the stateless people his whole career. Like a **refugee**. Because he represents the U. N. If you represent everyone, in some ways you represent no one. You ' re un-owned. And what the American — the most powerful military in the history of mankind was able to muster for his rescue, believe it or not, was literally these heroic two American soldiers went into the shaft. Building was shaking. One of them had been at 9 / 11 and lost his buddies on September 11th, and yet went in and risked his life in order to save Sergio. But all they had was a woman ' s handbag — literally one of those basket handbags — and they tied it to a curtain rope from one of the offices at U. N. headquarters, and created a pulley system into this shaft in this quivering building in the interests of rescuing this person, the person we most need to turn to now, this shepherd, at a time when so many of us feel like we ' re lacking guidance. And this was the pulley system. This was what we were able to muster for Sergio. The good news, for what it ' s worth, is after Sergio and 21 others were killed that day in the attack on the U. N., the military created a search and rescue unit that had the cutting equipment, the shoring wood, the cranes, the things that you would have needed to do the rescue. But it was too late for Sergio. I want to wrap up, but I want to close with what I take to be the four lessons from Sergio ' s life on this question of how do we prevent evil from prevailing, which is how I would have framed the question. Here ' s this **guy** who got a 34- year head start thinking about the kinds of questions we as a country are grappling with, we as citizens are grappling with now. What do we take away? First, I think, is his relationship to, in fact, evil is something to learn from. He, over the course of his career, changed a great deal. He had a lot of flaws, but he was very adaptive. I think that was his greatest quality. He started as somebody who would denounce harmdoers, he would charge up to people who were violating **international** law, and he would say, you ' re violating, this is the U. N. Charter. Don ' t you see it ' s unacceptable what you ' re doing? And they would laugh at him because he didn ' t have the power of states, the power of any military or police. He just had the rules, he had the norms, and he tried to use them. And in Lebanon, Southern Lebanon in ' 82, he said to himself and to **everybody** else, I will never use the word"unacceptable"again. I will never use it. I will try to make it such, but I will never use that word again. But he lunged in the opposite direction. He started, as I mentioned, to get in the room with evil, to not denounce, and became almost obsequious when he won the nickname Serbio, for instance, and even when he **negotiated** with the Khmer Rouge would black-box what had occurred prior to entering the room. But by the end of his life, I think he had struck a balance that we as a country can learn from. Be in the room, don ' t be afraid of talking to your adversaries, but don ' t bracket what happened before you entered the room. Don ' t black-box history. Don ' t check your principles at the door. And I think that ' s something that we have to be in the room, whether it ' s Nixon going to China or Khrushchev and Kennedy or Reagan and Gorbachev. All the great progress in this country with relation to our adversaries has come by going into the room. And it doesn ' t have to be an act of weakness. You can actually do far more to build an **international** coalition against a harmdoer or a wrongdoer by being in the room and showing to the rest of the world that that person, that regime, is the problem and that you, the United States, are not the problem. Second take-away from Sergio ' s life, briefly. What I take away, and this in some ways is the most important, he espoused and exhibited a reverence for dignity that was really, really unusual. At a micro level, the individuals around him were visible. He saw them. At a macro level, he thought, you know, we talk about democracy promotion, but we do it in a way sometimes that ' s an affront to people ' s dignity. We put people on humanitarian aid and we boast about it because we ' ve spent three billion. It ' s incredibly important, those people would no longer be alive if the United States, for instance, hadn ' t spent that money in Darfur, but it ' s not a way to live. If we think about dignity in our conduct as citizens and as individuals with relation to the people around us, and as a country, if we could inject a regard for dignity into our dealings with other countries, it would be something of a revolution. Third point, very briefly. He talked a lot about freedom from fear. And I recognize there is so much to be afraid of. There are so many genuine threats in the world. But what Sergio was talking about is, let ' s calibrate our relationship to the threat. Let ' s not hype the threat ; let ' s actually see it clearly. We have reason to be afraid of melting ice caps. We have reason to be afraid that we haven ' t secured loose nuclear material in the former Soviet Union. Let ' s focus on what are the legitimate challenges and threats, but not lunge into bad decisions because of a panic, of a fear. In times of fear, for instance, one of the things Sergio used to say is, fear is a bad advisor. We lunge towards the extremes when we aren ' t operating and trying to, again, calibrate our relationship to the world around us. Fourth and final point : he somehow, because he was working in all the world ' s worst places and all lesser evils, had a humility, of course, and an awareness of the complexity of the world around him. I mean, such an acute awareness of how hard it was. How Sisyphean this task was of mending, and yet aware of that complexity, humbled by it, he wasn ' t paralyzed by it. And we as citizens, as we go through this experience of the kind of, the **crisis** of **confidence**, **crisis** of competence, **crisis** of legitimacy, I think there ' s a temptation to pull back from the world and say, ah, Katrina, Iraq — we don ' t know what we ' re doing. We can ' t afford to pull back from the world. It ' s a question of how to be in the world. And the lesson, I think, of the anti-genocide movement that I mentioned, that is a partial success but by no means has it achieved what it has set out to do — it ' ll be many decades, probably, before that happens — but is that if we want to see change, we have to become the change. We can ' t rely upon our institutions to do the work of necessarily talking to adversaries on their own without us creating a space for that to happen, for having respect for dignity, and for bringing that combination of humility and a sort of emboldened sense of responsibility to our dealings with the rest of the world. So will evil prevail? Is that the question? I think the short answer is : no, not unless we let it. Thank you.

Text Sample 556: POS Dim 4 – 2008 – Score 15.00 – 2008/t000296.txt
It was an incredible surprise to me to find out that there was actually an organization that cared about both parts of my life. Because, basically, I work as a theoretical physicist. I develop and test models of the Big Bang, using observational data. And I ' ve been moonlighting for the last five years helping with a project in Africa. And, I get a lot of flak for this at Cambridge. People wonder, you know,"How do you have time to do this?"And so on. And so it was simply astonishing to me to find an organization that actually appreciated both those sides. So I thought I ' d start off by just telling you a little bit about myself and why I lead this schizophrenic life. Well, I was born in South Africa and my parents were imprisoned for resisting the racist regime. When they were released, we left and we went as **refugees** to Kenya and Tanzania. Both were very young countries then, and full of hope for the future. We had an amazing childhood. We didn ' t have any money, but we were outdoors most of the time. We had fantastic friends and we saw the wonders of the world, like Kilimanjaro, Serengeti and the Olduvai Gorge. Well, then we moved to London for high school. And after that — there ' s nothing much to say about that. It was rather dull. But I came back to Africa at the age of 17, as a volunteer teacher to Lesotho, which is a tiny country, surrounded at that time by apartheid South Africa. Well, 80 percent of the men in Lesotho worked in the mines over the border, in brutal conditions. Nevertheless, I — as I ' m sure — as a rather irritating young, white man coming into their village, I was welcomed with incredible hospitality and warmth. But the kids were the best part. The kids were amazing : extremely eager and often very bright. And I ' m just going to tell you one story, which got through to me. I used to try to take the kids outside as often as possible, to try to connect the academic **stuff** with the real world. And they weren ' t used to that. But I took them outside one day and I said,"I want you to estimate the height of the building."And I expected them to put a ruler next to the wall, size it up with a finger, and make an estimate of the height. But there was one little boy, very small for his age. He was the son of one of the poorest families in the village. And he wasn ' t doing that. He was scribbling with chalk on the pavement. And so, I said — I was annoyed — I said,"What are you doing? I want you to estimate the height of the building."He said,"OK. I measured the height

of a brick. I counted the number of bricks and now I ’ m multiplying.”Well — — I hadn ’ t thought of that one. And many experiences like this happened to me. Another one is that I met a miner. He was home on his three-month leave from the mines. Sitting next to him one day, he said,”There ’ s only one thing that I really loved at school. And you know what it was? Shakespeare.”And he recited some to me. And these and many similar experiences convinced me that there are just tons of bright kids in Africa — inventive kids, intellectual kids — and starved of opportunity. And if Africa is going to get fixed, it ’ s by them, not by us. Well, after — — that ’ s the truth. Well, after Lesotho, I traveled across Africa before returning to England — so gray and depressing, in comparison. And I went to Cambridge. And there, I fell for theoretical physics. Well, I ’ m not going to explain this equation, but theoretical physics is really an amazing subject. We can write down all the laws of physics we know in one line. And, admittedly, it ’ s in a very shorthand notation. And it contains 18 free parameters, OK, which we have to fit to the data. So it ’ s not the final story, but it ’ s an incredibly powerful summary of everything we know about nature at the most basic level. And apart from a few very important loose ends, which you ’ ve heard about here — like dark energy and dark matter — this equation describes, seems to describe everything about the universe and what ’ s in it. But there ’ s one big puzzle remaining, and this was most succinctly put to me by my primary school math teacher in Tanzania, who ’ s a wonderful Scottish lady who I still stay in touch with. And she ’ s now in her 80s. And when I try to explain my work to her, she waved away all the details, and she said,”Neil, there ’ s only one question that really matters. What banged?””Everyone talks about the Big Bang. What banged?”And she ’ s right. It ’ s a question we ’ ve all been avoiding. The standard explanation is that the universe somehow sprang into existence, full of a strange kind of energy — inflationary energy — which blew it up. But the puzzle of why the universe emerged in that peculiar state is completely unsolved. Now, I worked on that theory for a while, with Stephen Hawking and others. But then I began to explore another alternative. The alternative is that the Big Bang wasn ’ t the beginning. Perhaps the universe existed before the bang, and the bang was just a violent event in a pre-existing universe. Well, this possibility is actually suggested by the latest theories, the unified theories, which try to explain all those 18 free parameters in a single framework, which will hopefully predict all of them. And I ’ ll just share a cartoon of this idea here. It ’ s all I can convey. According to these theories, there are extra dimensions of space, not just the three we ’ re familiar with, but at every point in the room there are more dimensions. And in particular, there ’ s one rather strange one, in the most elegant unified theories we have. The strange one looks likes this : that we live in a three-dimensional world. We live in one of these worlds, and I can only show it as a sheet, but it ’ s really three-dimensional. And a tiny distance away, there ’ s another sheet, also three-dimensional, and they ’ re separated by a gap. The gap is very tiny, and I ’ ve blown it up so you can see it. But it ’ s really a tiny fraction of the size of an atomic nucleus. I won ’ t go into the details of why we think the universe is like this, but it comes out of the math and trying to explain the physics that we know. Well, I got **interested** in this because it seemed to me that it was an obvious question. Which is, what happens if these two, three-dimensional worlds should actually collide? And if they collide, it would look a lot like the Big Bang. But it ’ s slightly different than in the conventional picture. The conventional picture of the Big Bang is a point. Everything comes out of a point ; you have infinite density. And all the equations break down. No hope of describing that. In this picture, you ’ ll notice, the bang is extended. It ’ s not a point. The density of matter is finite, and we have a chance of a consistent set of equations that can describe the whole process. So, to cut a long story short, we ’ ve explored this alternative. We ’ ve shown that it can fit all of the data that we have about the formation of galaxies, the fluctuations in the microwave background. Furthermore, there ’ s an experimental way to tell this theory, apart from the inflationary explanation that I told you before. It **involves** gravitational waves. And in this scenario, not only was the Big Bang not the beginning, as you can see from the picture, it can happen over and over again. It may be that we live in an endless universe, both in space and in time. And there ’ ve been bangs in the past, and there will be bangs in the future. And maybe we live in an endless universe. Well, making and testing models of the universe is, for me, the best way I have of enjoying and appreciating the universe. We need to make the best mathematical models we can, the most consistent ones. And then we scrutinize them, logically and with data. And we try to convince ourselves — we really try to convince ourselves they ’ re wrong. That ’ s progress : when we prove things wrong. And gradually, we hopefully move closer and closer to understanding the world. As I pursued my career, something was always gnawing away inside me. What about Africa? What about those kids I ’ d left behind? Instead of developing, as we ’ d all hoped in the ’ 60s, things had gotten worse. Africa was gripped by poverty, disease and war. This is very graphically shown by the Worldmapper website and project. And so the idea is to represent each country on a map, but scale the area according to some quantity. So here ’ s just the standard area map of the world. By the way, Africa is very large. And the next map now shows Africa ’ s GDP in 1960, around the time of independence for many African states. Now, this is 1990, and then 2002. And here ’ s a projection for 2015. Big changes are happening in the world, but they ’ re not helping Africa. What about Africa ’ s population? The population isn ’ t out of proportion to its area, but Africa leads the world in deaths from often preventable causes : malnutrition, simple infections and birth complications. Then there ’ s HIV / AIDS. And then there are deaths from war. OK, currently there are 45, 000 people a month dying in the Congo, as a consequence of the war there over coltan and diamonds and other things. It ’ s still going on. What about Africa ’ s capacity to do something about these problems? Well, here ’ s the number of physicians in Africa. Here ’ s the number of people in higher education. And here — most shocking to me — the number of scientific research papers coming out of Africa. It just doesn ’ t exist scientifically. And this was very eloquently argued at **TED** Africa : that all of the aid that ’ s been given has completely failed to put Africa onto its own two feet. Well, the **transition** to democracy in South Africa in 1994 was literally a dream come true for many of us. My parents were both elected to the first parliament, alongside Nelson and Winnie Mandela. They were the only other couple. And in 2001, I took a research leave to **visit** them. And while I was busy working — I was working on these colliding worlds, in the day. But I learned that there was a desperate shortage of skills, especially mathematical skills, in industry, in government, in education. The **ability** to make and test models has become essential, not only to every single area of science today, but also to modern society itself. And if you don ’ t have math, you ’ re not going to enter the modern age. So I had an idea. And the idea was very simple. The idea was to set up an African Institute for Mathematical Sciences, or AIMS. And let ’ s recruit students from the whole of Africa, bring them together with lecturers from all over the world, and we ’ ll try to give them a fantastic education. Well, as a Cambridge professor, I had many contacts. And to my astonishment, they backed me 100 percent. They said,”Go and do it, and we ’ ll come and lecture.”And I knew it would be amazing fun to bring brilliant students from these countries — where they don ’ t have any opportunities — together with the best lecturers in the world — who

I knew would come, because of the interest in Africa — and put them together and just let the sparks fly. So we bought a derelict hotel near Cape Town. It ' s an 80-room Art Deco hotel from the 1920s. The area was kind of seedy, so we got an 80-room hotel for 100, 000 dollars. It ' s a beautiful building. We decided we would refurbish it and then put out the word : we ' re going to start the best math institute in Africa in this hotel. Well, the new South Africa is a very exciting country. And those of you who haven ' t been there, you should go. It ' s very, very **interesting** what ' s happening. And we recruited wonderful staff, highly motivated staff. The other thing that ' s happened, which was good for us, is the Internet. Even though the Internet is very expensive all over Africa, there are Internet cafes everywhere. And bright young Africans are desperate to join the global community, to be successful — and they ' re very ambitious. They want to be the next Einstein. And so when word came out that AIMS was opening, it spread very quickly via e- mail and our website. And we got lots of applicants. Well, we designed AIMS as a 24-hour learning environment, and it was fantastic to start a university from the beginning. You have to rethink, what is the university for? And that ' s really exciting. So we designed it to have interactive teaching. No droning on at the chalkboard. We emphasize problem-solving, working in groups, every student discovering and maximizing their own potential and not chasing grades. Everyone lives together in this hotel — lecturers and students — and it ' s not surprising at all to find an impromptu tutorial at 1 a. m. The students don ' t usually leave the computer lab till 2 or 3 a. m. And then they ' re up again at eight in the morning. Lectures, problem-solving and so on. It ' s an extraordinary place. We especially emphasize areas of great relevance to Africa ' s development, because, in those areas, scientists working in Africa will have a competitive advantage. They ' ll publish, be invited to **conferences**. They ' ll do well. They ' ll have successful careers. And AIMS has done extremely well. Here is a list of last year ' s graduates, graduated in June, and what they ' re currently doing — 48 of them. And where they are is indicated over here. And where they ' ve gone. So these are all postgraduate students. And they ' ve all gone on to master ' s and Ph. D. degrees in excellent places. Five students can be educated at AIMS for the cost of educating one in the U. S. or Europe. But more important, the pan-African student body is a continual source of strength, pride and commitment to Africa. We illustrate AIMS ' progress by coloring in the countries of Africa. So here you can see behind this list. When a county is colored yellow, we ' ve received an application ; orange, we ' ve accepted an application ; and green, a student has graduated. So here is where we were after the first graduation in 2004. And we set ourselves a goal of turning the continent green. So there ' s 2005, -6, -7, -8. We ' re well on the way to achieving our initial goal. We had some of the students filmed at home before they came to AIMS. And I ' ll just show you one. Tendai Mugwagwa : My name is Tendai Mugwagwa. I have a Bachelor of Science with an education degree. I will be attending AIMS. My understanding of the course is that it covers quite a lot. You know, from physics to medicine, in particular, epidemiology and also mathematical modeling. Neil Turok : So Tendai came to AIMS and did very well. And I ' ll let her take it from there. TM : My name is Tendai Mugwagwa and I was a student at AIMS in 2003 and 2004. After leaving AIMS, I went on to do a master ' s in applied mathematics at the University of Cape Town in South Africa. After that, I came to the Netherlands where I ' m now doing a Ph. D. in theoretical immunology. Professor : Tendai is working very independently. She communicates well with the immunologists at the hospital. So all in all I have a very good Ph. D. student from South Africa. So I ' m happy she ' s here. NT : Another student in the first year of AIMS was Shehu. And he ' s shown here with his favorite high school teacher. And then entering university in northern Nigeria. And after AIMS, Shehu wanted to do high-energy physics, and he came to Cambridge. He ’ s about to finish his Ph. D., and he was filmed recently with someone you all know. Shehu : And from there we will be able to, hopefully, make better predictions and then we compare it to the graph and also make some predictions. Stephen Hawking : That is nice. NT : Here are the current students at AIMS. There are 53 of them from 20 different countries, including 20 women. So now I ’ m going to get to my **TED** business. Well, we had a party. This is Africa — we have good parties in Africa. And last month, they threw a surprise party for me. Here ’ s somebody you ’ ve seen already. I want to point out a few other exceptional people in this picture. So, we were having a party, as you can see they ’ re completely eclipsing me at this point. This is Ezra. She ’ s from Darfur. She ’ s a physicist, and somehow stays smiling, in spite of everything going on back home. But she wants to continue in physics, and she ’ s doing extremely well. This is Lydia. Lydia is the first ever woman to graduate in mathematics in the Central African Republic. And she ’ s now at AIMS. So now let me get to our **TED** wish. Well, it ’ s not my **TED** wish ; it ’ s our wish, as you ’ ve already gathered. And our wish has two parts : one is a dream and the other ’ s a plan. OK. Our **TED** dream is that the next Einstein will be African. In striving for the heights of creative genius, we want to give thousands of people the motivation, the encouragement and the courage to obtain the high-level skills they need to help Africa. Among them will be not only brilliant scientists — I ’ m sure of that from what we ’ ve seen at AIMS — they ’ ll also be the African Gates, Brins and Pages of the future. Well, I said we also have a plan. And our plan is quite simple. AIMS is now a proven model. And what we need to do is to replicate it. We want to roll out 15 AIMS centers in the next five years, all over Africa. Each will have a pan- African student body, but specialize in a different area of science. We want to use science to overcome the national and cultural barriers, as it does at AIMS. And we want to add elements to the curriculum. We want to add entrepreneurship and **policy** skills. The expanded AIMS will be a coherent pan-African institution, and its graduates will form a powerful network, working together for peace and progress across the continent. Over the last year, we ’ ve been **visiting** sites in Africa, looking at potential sites for new AIMS centers. And here are the ones we ’ ve selected. And each of these centers has a strong local team, each is in a beautiful place, an **interesting** place, which **international** lecturers will be happy to **visit**. And our partners across Africa are extremely enthusiastic about this. Everyone wants an AIMS center in their country. And last November, the **conference** of all the African ministers of science and technology, held in Mombasa, called for a comprehensive plan to roll out AIMS. So we have political **support** right across the continent. It won ’ t be easy. At every site there will be huge challenges. Local scientists must play leading roles and governments must be persuaded to buy in. Conditions are very difficult, but we cannot afford to compromise on those principles which made AIMS work. And we summarize them this way : the institutes have got to be relevant, innovative, cost-effective and high quality. Why? Because we want Africa to be rich. Easy to remember the basic rules we need. So, just in ending, let me say the only people who can fix Africa are talented young Africans. By unlocking and nurturing their creative potential, we can create a step change in Africa ’ s future. Over time, they will contribute to African development and to science in ways we can only imagine. Thank you.

Text Sample 557: POS Dim 4 – 2008 – Score 15.00 – 2008/t000275.txt

I have given the slide show that I gave here two years ago about 2, 000 times. I ’ m giving a short slide show this morning that I ’ m giving for the very first time, so â well it ’ s â I don ’ t want or need to raise the bar, I ’ m actually trying to lower the bar. Because I ’ ve cobbled this together to try to meet the challenge of this session. And I was reminded by Karen Armstrong ’ s fantastic presentation that religion really properly understood is not about belief, but about behavior. Perhaps we should say the same thing about optimism. How dare we be optimistic? Optimism is sometimes characterized as a belief, an intellectual posture. As Mahatma Gandhi famously said,”You must become the change you wish to see in the world.”And the outcome about which we wish to be optimistic is not going to be created by the belief alone, except to the extent that the belief brings about new behavior. But the word”behavior”is also, I think, sometimes misunderstood in this context. I ’ m a big advocate of changing the lightbulbs and buying hybrids, and Tipper and I put 33 solar panels on our house, and dug the geothermal wells, and did all of that other **stuff**. But, as important as it is to change the lightbulbs, it is more important to change the laws. And when we change our behavior in our daily lives, we sometimes leave out the citizenship part and the democracy part. In order to be optimistic about this, we have to become incredibly active as citizens in our democracy. In order to solve the **climate crisis**, we have to solve the democracy **crisis**. And we have one. I have been trying to tell this story for a long time. I was reminded of that recently, by a woman who walked past the table I was sitting at, just staring at me as she walked past. She was in her 70s, looked like she had a kind face. I thought nothing of it until I saw from the corner of my eye she was walking from the opposite direction, also just staring at me. And so I said,”How do you do?”And she said,”You know, if you dyed your hair black, you would look just like Al Gore.”Many years ago, when I was a young congressman, I spent an awful lot of time dealing with the challenge of nuclear arms control â the nuclear arms race. And the military historians taught me, during that quest, that military **conflicts** are typically put into three categories : local battles, regional or theater wars, and the rare but all-important global, world war â strategic **conflicts**. And each level of **conflict** requires a different allocation of resources, a different approach, a different organizational model. Environmental challenges fall into the same three categories, and most of what we think about are local environmental problems : **air** pollution, water pollution, hazardous waste dumps. But there are also regional environmental problems, like acid rain from the Midwest to the Northeast, and from Western Europe to the Arctic, and from the Midwest out the Mississippi into the dead zone of the Gulf of Mexico. And there are lots of those. But the **climate crisis** is the rare but all-important global, or strategic, **conflict**. Everything is affected. And we have to organize our response appropriately. We need a worldwide, global mobilization for renewable energy, conservation, efficiency and a global **transition** to a low-carbon economy. We have work to do. And we can mobilize resources and political will. But the political will has to be mobilized, in order to mobilize the resources. Let me show you these slides here. I thought I would start with the logo. What ’ s missing here, of course, is the North Polar ice cap. Greenland remains. Twenty-eight years ago, this is what the polar ice cap â the North Polar ice cap â looked like at the end of the summer, at the fall equinox. This last fall, I went to the Snow and Ice Data Center in Boulder, Colorado, and talked to the researchers here in Monterey at the Naval Postgraduate Laboratory. This is what ’ s happened in the last 28 years. To put it in perspective, 2005 was the previous record. Here ’ s what happened last fall that has really unnerved the researchers. The North Polar ice cap is the same size geographically â doesn ’ t look quite the same size â but it is exactly the same size as the United States, minus an area roughly equal to the state of Arizona. The amount that disappeared in 2005 was equivalent to everything east of the Mississippi. The extra amount that disappeared last fall was equivalent to this much. It comes back in the winter, but not as permanent ice, as thin ice â vulnerable. The amount remaining could be completely gone in summer in as little as five years. That puts a lot of pressure on Greenland. Already, around the Arctic Circle â this is a famous village in Alaska. This is a town in Newfoundland. Antarctica. Latest studies from NASA. The amount of a moderate-to-severe snow melting of an area equivalent to the size of California."They were the best of times, they were the worst of times": the most famous opening sentence in English literature. I want to share briefly a tale of two planets. Earth and Venus are exactly the same size. Earth ' s diameter is about 400 kilometers larger, but essentially the same size. They have exactly the same amount of carbon. But the difference is, on Earth, most of the carbon has been leeched over time out of the atmosphere, deposited in the ground as coal, oil, natural gas, etc. On Venus, most of it is in the atmosphere. The difference is that our temperature is 59 degrees on average. On Venus, it ' s 855. This is relevant to our current strategy of taking as much carbon out of the ground as quickly as possible, and putting it into the atmosphere. It ' s not because Venus is slightly closer to the Sun. It ' s three times hotter than Mercury, which is right next to the Sun. Now, briefly, here ' s an image you ' ve seen, as one of the only old images, but I show it because I want to briefly give you CSI : **Climate**. The global scientific community says : man-made global warming pollution, put into the atmosphere, thickening this, is trapping more of the outgoing infrared. You all know that. At the last IPCC summary, the scientists wanted to say,"How certain are you?"They wanted to answer that"99 percent."The Chinese objected, and so the compromise was"more than 90 percent."Now, the skeptics say,"Oh, wait a minute, this could be variations in this energy coming in from the sun."If that were true, the stratosphere would be heated as well as the lower atmosphere, if it ' s more coming in. If it ' s more being trapped on the way out, then you would expect it to be warmer here and cooler here. Here is the lower atmosphere. Here ' s the stratosphere : cooler. CSI : **Climate**. Now, here ' s the good news. Sixty-eight percent of Americans now believe that human activity is responsible for global warming. Sixty-nine percent believe that the Earth is heating up in a significant way. There has been progress, but here is the key : when given a list of challenges to confront, global warming is still listed at near the bottom. What is missing is a sense of urgency. If you agree with the factual analysis, but you don ' t feel the sense of urgency, where does that leave you? Well, the Alliance for **Climate** Protection, which I head in conjunction with Current TV â who did this pro bono â did a worldwide contest to do commercials on how to communicate this. This is the winner. NBC â I ' ll show all of the networks here â the top journalists for NBC asked 956 questions in 2007 of the presidential candidates : two of them were about the **climate crisis**. ABC : 844 questions, two about the **climate crisis**. Fox : two. CNN : two. CBS : zero. From laughs to tears â this is one of the older tobacco commercials. So here ' s what we ' re doing. This is gasoline consumption in all of these countries. And us. But it ' s not just the developed nations. The developing countries are now following us and accelerating their pace. And actually, their cumulative emissions this year are the equivalent to where we were in 1965. And they ' re catching up very dramatically. The total concentrations : by 2025, they will be essentially where we were in 1985. If the wealthy countries were completely missing from the picture, we would still have this **crisis**. But we have given to the developing countries the technologies and the ways of thinking that are creating the **crisis**. This is in Bolivia â over thirty years. This is peak fishing in a few seconds. The ’ 60s. ’ 70s. ’ 80s. ’ 90s. We have to stop this. And the good news is that we can. We have the technologies. We have to have a unified view of how to go about this : the struggle against poverty in the world and the challenge of cutting wealthy country emissions, all has a single, very simple solution. People say,”What ’ s the solution?”Here it is. Put a price on carbon. We need a CO2 tax, revenue neutral, to replace taxation on employment, which was invented by Bismarck â and some things have changed since the 19th century. In the poor world, we have to integrate the responses to poverty with the solutions to the **climate crisis**. Plans to fight poverty in Uganda are mooted, if we do not solve the **climate crisis**. But responses can actually make a huge difference in the poor countries. This is a proposal that has been talked about a lot in Europe. This was from Nature magazine. These are concentrating solar, renewable energy plants, linked in a so-called”supergrid”to supply all of the electrical power to Europe, largely from developing countries â high-voltage DC currents. This is not pie in the sky ; this can be done. We need to do it for our own economy. The latest figures show that the old model is not working. There are a lot of great investments that you can make. If you are **investing** in tar sands or shale oil, then you have a portfolio that is crammed with sub-prime carbon assets. And it is based on an old model. Junkies find veins in their toes when the ones in their arms and their legs collapse. Developing tar sands and coal shale is the equivalent. Here are just a few of the investments that I personally think make sense. I have a stake in these, so I ’ ll have a disclaimer there. But geothermal, concentrating solar, advanced photovoltaics, efficiency and conservation. You ’ ve seen this slide before, but there ’ s a change. The only two countries that didn ’ t ratify â and now there ’ s only one. Australia had an election. And there was a campaign in Australia that **involved** television and Internet and radio commercials to lift the sense of urgency for the people there. And we trained 250 people to give the slide show in every town and village and city in Australia. Lot of other things contributed to it, but the new Prime Minister announced that his very first priority would be to change Australia ’ s position on Kyoto, and he has. Now, they came to an awareness partly because of the horrible drought that they have had. This is Lake Lanier. My friend Heidi Cullen said that if we gave droughts names the way we give hurricanes names, we ’ d call the one in the southeast now Katrina, and we would say it ’ s headed toward Atlanta. We can ’ t wait for the kind of drought Australia had to change our political culture. Here ’ s more good news. The cities **supporting** Kyoto in the U. S. are up to 780 â and I thought I saw one go by there, just to localize this â which is good news. Now, to close, we heard a couple of days ago about the value of making individual heroism so commonplace that it becomes banal or routine. What we need is another hero generation. Those of us who are alive in the United States of America today especially, but also the rest of the world, have to somehow understand that history has presented us with a choice â just as Jill [Bolte] Taylor was figuring out how to save her life while she was distracted by the amazing experience that she was going through. We now have a culture of distraction. But we have a planetary emergency. And we have to find a way to create, in the generation of those alive today, a sense of generational mission. I wish I could find the words to convey this. This was another hero generation that brought democracy to the planet. Another that ended slavery. And that gave women the right to vote. We can do this. Don ’ t tell me that we don ’ t have the capacity to do it. If we had just one week ’ s worth of what we spend on the Iraq War, we could be well on the way to solving this challenge. We have the capacity to do it. One final point : I ’ m optimistic, because I believe we have the capacity, at moments of great challenge, to set aside the causes of distraction and rise to the challenge that history is presenting to us. Sometimes I hear people respond to the disturbing facts of the **climate crisis** by saying,"Oh, this is so terrible. What a burden we have."I would like to ask you to reframe that. How many generations in all of human history have had the opportunity to rise to a challenge that is worthy of our best efforts? A challenge that can pull from us more than we knew we could do? I think we ought to approach this challenge with a sense of profound joy and gratitude that we are the generation about which, a thousand years from now, philharmonic orchestras and poets and singers will celebrate by saying, they were the ones that found it within themselves to solve this **crisis** and lay the basis for a bright and optimistic human future. Let ' s do that. Thank you very much. Chris Anderson : For so many people at **TED**, there is deep pain that basically a design issue on a voting form â one bad design issue meant that your voice wasn ' t being heard like that in the last eight years in a position where you could make these things come true. That hurts. Al Gore : You have no idea. CA : When you look at what the leading candidates in your own party are doing now â I mean, there ' s â are you excited by their plans on global warming? AG : The answer to the question is hard for me because, on the one hand, I think that we should feel really great about the fact that the Republican nominee â certain nominee â John McCain, and both of the finalists for the Democratic nomination â all three have a very different and forward-leaning position on the **climate crisis**. All three have offered **leadership**, and all three are very different from the approach taken by the current administration. And I think that all three have also been responsible in putting forward plans and proposals. But the campaign dialogue that â as illustrated by the questions â that was put together by the League of Conservation Voters, by the way, the analysis of all the questions â and, by the way, the debates have all been sponsored by something that goes by the Orwellian label,"Clean Coal."Has anybody noticed that? Every single debate has been sponsored by"Clean Coal.""Now, even lower emissions!"The richness and fullness of the dialogue in our democracy has not laid the basis for the kind of bold initiative that is really needed. So they ' re saying the right things and they may â whichever of them is elected â may do the right thing, but let me tell you : when I came back from Kyoto in 1997, with a feeling of great happiness that we ' d gotten that breakthrough there, and then confronted the United States Senate, only one out of 100 senators was willing to vote to confirm, to ratify that treaty. Whatever the candidates say has to be laid alongside what the people say. This challenge is part of the fabric of our whole civilization. CO2 is the exhaling breath of our civilization, literally. And now we mechanized that process. Changing that pattern requires a scope, a scale, a speed of change that is beyond what we have done in the past. So that ' s why I began by saying, be optimistic in what you do, but be an active citizen. **Demand** â change the light bulbs, but change the laws. Change the global treaties. We have to speak up. We have to solve this democracy â this â We have sclerosis in our democracy. And we have to change that. Use the Internet. Go on the Internet. Connect with people. Become very active as citizens. Have a moratorium â we shouldn ' t have any new coal-fired generating plants that aren ' t able to capture and store CO2, which means we have to quickly build these renewable sources. Now, nobody is talking on that scale. But I do believe that between now and November, it is possible. This Alliance for **Climate** Protection is going to launch a nationwide campaign â grassroots mobilization, television ads, Internet ads, radio, newspaper â with partnerships with **everybody** from the Girl Scouts to the hunters and fishermen. We need help. We need help. CA : In terms of your own personal role going forward, Al, is there something more than that you would like to be doing? AG : I have prayed that I would be able to find the answer to that question. What can I do? Buckminster Fuller once wrote,”If the future of all human civilization depended on me, what would I do? How would I be?”It does depend on all of us, but again, not just with the light bulbs. We, most of us here, are Americans. We have a democracy. We can change things, but we have to actively change. What ’ s needed really is a higher level of consciousness. And that ’ s hard to â that ’ s hard to create â but it is coming. There ’ s an old African proverb that some of you know that says,”If you want to go quickly, go alone ; if you want to go far, go together.”We have to go far, quickly. So we have to have a change in consciousness. A change in commitment. A new sense of urgency. A new appreciation for the privilege that we have of undertaking this challenge. CA : Al Gore, thank you so much for coming to **TED**. AG : Thank you. Thank you very much.

Text Sample 558: POS Dim 4 – 2004 – Score 14.00 – 2004/t000487.txt

You know, when Chris first approached me to speak at **TED**, I said no, because I felt like I wasn ’ t going to be able to make that personal connection, you know, that I wanted to. It ’ s such a large **conference**. But he explained to me that he was in a bind, and that he was having trouble finding the kind of sex appeal and star power that the **conference** was known for. So I said fine, Ted — I mean Chris. I ’ ll come on two conditions. One : I want to speak as early in the morning as possible. And two : I want to pick the theme for **TED** 2006. And luckily he agreed. And the theme, in two years, is going to be”Cute Pictures Of Puppies.”[How to Dance Properly BASIC TWIRL] [NEW SCHOOL] [OLD SCHOOL] [WHO ’ S YOUR DADDY?] [”RIDE THE PONY”] [MAKE LOVE TO THE CROWD] [SMACKING THAT ASS] [STIR THE POT OF LOVE] [HANGING OUT... CASUAL] [WORD.] I invented the Placebo Camera. It doesn ’ t actually take pictures, but it ’ s a hell of a lot cheaper, and you still feel like you were there.”Dear Sir, good day, compliments of the day, and my best wishes to you and family. I know this letter will come to you surprisingly, but let it not be a surprise to you, for nature has a way of arriving unannounced, and, as an adage says, originals are very hard to find, but their echoes sound ouder. So I decided to contact you myself, for you to assure me of safety and honesty, if I have to entrust any amount of money under your custody. I am Mr Micheal Bangura, the son of late Mr Thaimu Bangura who was the Minister of Finance in Sierra Leone but was killed during the civil war. Knowing your country to be economical conducive for investment, and your people as transparent and trustworthy to engage in business, on which premise I write you. Before my father death, he had the sum of 23 million United States dollars, which he kept away from the rebel **leaders** during the course of the war. This fund was supposed to be used for the rehabilitation of water reserves all over the country, before the outbreak of war. When the war broke out, the rebel **leader demanded** the fund be given to him, my father insisted it was not in his possession, and he was killed because of his refusal to release the fund. Meanwhile, my mother and I is the only person who knows about it because my father always confide in me. I made an arrangement with a Red Cross relief worker, who used his official van to transport the money to Lungi Airport, Freetown, although he did not know the real contents of the box. The fund was deposited as a family reasure, in a **safe**, reliable security company in Dakar, Senegal, where I was only given temporary asylum. I do not wish to **invest** the money in Senegal due to unfavorable economic **climate**, and so close to my country. The only assistance I need from you, which I know you would do for me, are the following : one, be a silent partner and receive the funds in your account in **trust** ; two, provide a bank account under your control to which the funds will be remitted ; three, receive the funds into your account in **trust** ; take out your commission ; and leave the rest of the money until I arrive, after the transfer is complete. Sincerely, Mr Micheal Bangura."This is really embarrassing. I was told backstage that I have 18 minutes. I only prepared 15. So if it ' s cool, I ' d like to just wait for three. I ' m really sorry. What ' s your name? Mark Surfas. It ' s pretty cool, huh? Pursuing happiness. Are you a virgin? Virgin? I mean — no, I mean like in the **TED** sense? Are you? Oh, yeah? So what are you, like, a thousand, two thousand, somewhere in there? Huh? Oh? You don ' t know what I ' m talking about? Ah, Mark — Surfas. 1, 860 — am I good? And that ' s nothing to be ashamed of. That ' s nothing to be ashamed of. Yeah, I was hanging out with some Google **guys** last night. Really cool, we were getting wasted. And they were telling me that Google software has gotten so advanced that, based on your interaction with Google over your lifetime, they can actually predict what you are going to say — next. And I was like,"Get the fuck out of here. That ' s crazy."But they said,"No, but don ' t show anyone."But they slipped up. And they said that I could just type in"What was I going to say next?"and my name, and it would tell me. And I have to tell you, this is an unadulterated piece of software, this is a real Internet browser and this is the actual Google site, and we ' re going to test it out live today. What was I going to say next? And"Ze Frank"— that ' s me. Am I feeling lucky? Am I feeling lucky? Audience : Yes! Yeah! Ze Frank : Oh! Amazing. In March of 2001 — I filmed myself dancing to Madonna ' s"Justify My Love."On a Thursday, I sent out a link to a website that featured those clips to 17 of my closest friends, as part of an invitation to my — an invitation to my th — th — 26th birthday party. By Monday, over a million people were coming to this site a day. Within a week, I received a call from Earthlink that said, due to a 10 cents per megabyte overage charge, I owed them 30, 000 dollars. Needless to say, I was able to leave my job. [WAS LAID OFF] And, finally, you know, become freelance. [UNEMPLOYED] But some people refer to me more as, like, an Internet guru or — [JACKASS] swami. I knew I had something. I ' d basically distilled a very difficult-to-explain and complex philosophy, which I won ' t get into here, because it ' s a little too deep for all of you, but — It ' s about what makes websites popular, and, you know, it ' s — [DANCE LIKE AN IDIOT AND DON ' T SELL ANYTHING] It ' s unfortunate that I don ' t have more time. Maybe I can come back next year, or something like that. I ' m obsessed with email. I get a lot of it. Four years later, I still get probably two or three hundred emails a day from people I don ' t know, and it ' s been an amazing opportunity to kind of get to know different cultures, you know? It ' s like a microscope to the rest of the world. You can kind of peer into other people ' s lives. And I also feel like I get a lot of inspiration from the average user. For example, somebody wrote,"Hey Ze, if you ever come to Boulder, you should rock out with us,"and I said,"Why wait?"[rocking out] And they said,"Hey Ze, thanks for rocking out, but I meant the kind of rocking out where we ' d be naked."And that was embarrassing. But you know, it ' s kind of a collaboration between me and the fans, so I said,"Sure."[rocking out naked] I hear a lot of you whispering. And I know what you ' re saying,"Holy crap! How is his presentation so smooth?"And I have to say that it ' s not all me this year. I guess Chris has to take some credit here, because in years past, I guess there ' s been some sort of subpar speakers at **TED**. I don ' t know. And so, this year, Chris sent us a **TED conference** simulator. Which really allowed us as speakers to get there, in the trenches, and practice at home so that we would be ready for this experience. And I ' ve got to say that, you know, it ' s really,

really great to be here. I ’ d like to tell all of you a little joke. Not just the good **stuff**, though. You can do heckler mode. Voice : Hey, moron, get off the stage! ZF : You get off the stage. Voice : We want Malcolm Gladwell. In case you run over time. Just one last thing I ’ d like to say, I ’ d, really — I ’ d like to thank all of you for being here. And frog mode.”Ah, the first time that I made love to a rock shrimp —”[Spam jokes are the new airplane jokes] It ’ s true. Some people say to me,”Ze, you ’ re doing all this **stuff**, this Internet **stuff**, and you ’ re not making any money.””Why?”And I say,”Mom, Dad — I ’ m trying.”I don ’ t know if you ’ re all aware of this, but the video game market, kids are playing these video games, but, supposedly, there ’ s tons of money. I mean, like, I think, 100, 000 dollars or so a year is being spent on these things. So I decided to try my hand. I came up with a few games. This is called”Atheist.”I figured it would be popular with the young kids. OK. Look, I ’ ll move around and say some things. [Game over. There is no replay.] So that didn ’ t go over so well. I don ’ t really understand why you ’ re laughing. Should have done this before I tried to pitch it.”Buddhist,”of course, looks very, very similar to”Atheist.”But you come back as a duck. And this is great because, you know, for a quarter, you can play this for a long time. And Chris had said in an email that we should really bring something new to **TED**, something that we haven ’ t shown anyone. So, I made this for **TED**. It ’ s”Christian.”It ’ s the third in the series. I ’ m hoping it ’ s going to do well this year. Do you have a preference? Good choice. So you can wait for the Second Coming, which is a random number between one and 500 million. So really, what are we talking about here? Oh, **tech** joy. Tech joy, to me, means something, because I get a lot of joy out of **tech**. And in fact, making things using technology — and I ’ m being serious here, even though I ’ m using my sarcastic voice — I won ’ t — hold on. Making things, you know — making things actually does give me a lot of joy. It ’ s the process of creation that keeps me sort of a bubble and a half above perpetual anxiety in my life, and it ’ s that feeling of being about 80 percent complete on a project — where you know you still have something to do, but it ’ s not finished, and you ’ re not starting something — that really fills my entire life. And so, what I ’ ve done is, I started getting **interested** in creating online social spaces to share that feeling with people who don ’ t consider themselves artists. We ’ re in a culture of guru-ship. It ’ s so hard to use some software because, you know, it ’ s unapproachable, people feel like they have to read the manual. So I try to create these very minimal activities that allow people to express themselves, and, hopefully — Whoa! I ’ m like — on the page, but it doesn ’ t exist. It ’ s, like — seriously, though — I try to create meaningful environments for people to express themselves. Here I created a contest called,”When Office Supplies Attack,”which, I think, really resonated with the working population. Over 500 entries in three weeks. Toilet paper fashion. Again, people from all over the country. The watch is particularly incredible. Online drawing tools — you ’ ve probably seen a lot of them. I think they ’ re wonderful. It ’ s a chance for people to get to play with crayons and all that kind of **stuff**. But I ’ m **interested** in the process of creating, as the real event that I ’ m **interested** in. And the problem is that a lot of people suck at drawing, and they get bummed out at this, sort of, you know, stick figure, awful little thing that they created. And eventually, it just makes them stop playing with it, or they draw penises and things like that. So, the Scribbler is an attempt to create a generative tool. In other words, it ’ s a helping tool. You can draw your simple stick figure, and it collaborates with you to create sort of a post-war German etching. In fact, it ’ s tuned to be better at drawing things that look worse. So, we go ahead, and we start scribbling. So the idea is that you can really, you know, partake in this process, but watch something really crappy look beautiful. And here are some of my favorites.

This is the little trap marionette that was submitted to me. Very cool. Darling. Beautiful **stuff**. I mean this is incredible. An 11-year-old girl drew this and submitted it. It ’ s just gorgeous. I ’ m dead serious here. This is not a joke. But, I think it ’ s a really fun and wonderful thing. So this is called the”Fiction Project.”This is an online space, which is basically a refurbished message board that encourages collaborative fiction writing. These are haikus. None of the haikus were written by the same person. In fact, each line is contributed by a different person at a different time. I think that the”now tied up, tied down, mistress cruel approaches me, now tied down, it ’ s up.”It ’ s an amazing way, and I ’ ll tell you, if you come home, and your spouse, or whoever it is, says,”Let ’ s talk”— That, like, chills you to the very core. But it ’ s peripheral activities like these that allow people to get together, doing fun things. They actually get to know each other, and it ’ s sort of like low-threshold peripheral activities that I think are the key to bringing up some of our bonding social capital that we ’ re lacking. And very, very quickly — I love puppets. Here ’ s a puppet. It dances to music. Lotte Reiniger, an amazing shadow puppeteer in the 20s, that started doing more elaborate things. I became **interested** in puppets, and I just want to show one last thing to you. Oh, this is how you make puppets. Chris Anderson : Ladies and gentlemen, Mr. Ze Frank.

Text Sample 559: POS Dim 4 – 2004 – Score 12.00 – 2004/t000486.txt

You ’ ll be happy to know that I ’ ll be talking not about my own tragedy, but other people ’ s tragedy. It ’ s a lot easier to be lighthearted about other people ’ s tragedy than your own, and I want to keep it in the spirit of the **conference**. So, if you believe the media accounts, being a drug dealer in the height of the crack cocaine epidemic was a very glamorous life, in the words of Virginia Postrel. There was money, there was drugs, guns, women, you know, you name it — jewelry, bling-bling — it had it all. What I ’ m going to tell you today is that, in fact, based on 10 years of research, a unique opportunity to go inside a gang — to see the actual books, the financial records of the gang — that the answer turns out not to be that being in the gang was a glamorous life. But I think, more realistically, that being in a gang — selling drugs for a gang — is perhaps the worst job in all of America. And that ’ s what I ’ d like to convince you of today. So there are three things I want to do. First, I want to explain how and why crack cocaine had such a profound influence on inner-city gangs. Secondly, I want to tell you how somebody like me came to be able to see the inner workings of a gang — an **interesting** story, I think. And then third, I want to tell you, in a very superficial way, about some of the things we found when we actually got to look at the financial records, the books, of the gang. So before I do that, just one warning, which is that this presentation has been rated ’ R ’ by the Motion Picture Association of America. It contains adult themes, adult language. Given who is up on the stage, you ’ ll be delighted to know that, in fact, there ’ ll be no nudity — Unexpected wardrobe malfunctions aside. So let me start by talking about crack cocaine, and how it transformed the gang. To do that, you have to actually go back to a time before crack cocaine, in the early ’ 80s, and look at it from the perspective of a gang **leader**. Being a gang **leader** in the inner city wasn ’ t such a bad deal in the mid- ’ 80s — the early ’ 80s, let me say. Now, you had a lot of power, and you got to beat people up — you got a lot of prestige, a lot of respect. But the thing is, there was no money in it. The gang had no way to make money. You couldn ’ t charge dues to the people in the gang, because the people in the gang didn ’ t have any money. You couldn ’ t really make any money selling marijuana — marijuana ’ s too cheap, it turns out. You can ’ t get rich selling marijuana. You couldn ’ t sell cocaine ; cocaine ’ s a great product — powdered cocaine — but you ’ ve got to know rich white people. And most of the inner-city gang members didn ’ t know any rich white people, so couldn ’ t sell to that market. You couldn ’ t really do petty crime, either. Turns out, petty crime ’ s a terrible way to make a living. As a result, as a gang **leader**, you had, you know, power — it ’ s a pretty good life — but the thing was, in the end, you were living at home with your mother. And so it wasn ’ t really a career. There were limits to how powerful and important you could be if you had to live at home with your mother. Then along comes crack cocaine. And in the words of Malcolm Gladwell, crack cocaine was the extra-chunky version of tomato sauce for the inner city. Because crack cocaine was an unbelievable innovation. I don ’ t have time to talk about it today, but if you think about it, I would say that in the last 25 years, of every invention or innovation that ’ s occurred in this country, the biggest one in terms of **impact** on the well-being of people who live in the inner city, was crack cocaine. And for the worse — not for the better, but for the worse. It had a huge **impact** on life. So what was it about crack cocaine? It was a brilliant way of getting the brain high. Because you could smoke crack cocaine — you can ’ t smoke powdered cocaine — and smoking is a much more efficient mechanism of delivering a high than snorting it. And it turned out there was this audience that didn ’ t know it wanted crack cocaine, but when it came, it really did. And it was a perfect drug ; you could buy the cocaine that went into it for a dollar, sell it for five dollars. Highly addictive — the high was very short. So for fifteen minutes, you get this great high, and then when you come down, all you want to do is get high again. It created a wonderful market. And for the people who were there running the gang, it was a great way, seemingly, to make a lot of money. At least for the people on the top. So this is where we enter the picture. Not really me — I ’ m really a bit player in all this. My co-author, Sudhir Venkatesh, is the main character. He was a math major in college who had a good heart, and decided he wanted to get a sociology PhD, came to the University of Chicago. Now, the three months before he came to Chicago, he had spent following the Grateful Dead. And in his own words, he”looked like a freak.”He ’ s a South Asian — very dark-skinned South Asian. Big man, and he had hair, in his words,”down to his ass.”Defied all kinds of boundaries : Was he black or white? Was he man or woman? He was really a curious sight to be seen. So he showed up at the University of Chicago, and the famous sociologist William Julius Wilson was doing a book that **involved** surveying people all across Chicago. He took one look at Sudhir, who was going to go do some surveys for him, and decided he knew exactly the place to send him, which was to one of the toughest, most notorious housing projects not just in Chicago, but in the entire United States. So Sudhir, the suburban boy who had never really been in the inner city, dutifully took his clipboard and walked down to this housing project, gets to the first building. The first building? Well, there ’ s nobody there. But he hears some voices up in the stairwell, so he climbs up the stairwell, comes around the corner, and finds a group of young African-American men playing dice. This is about 1990, peak of the crack epidemic. This is a very dangerous job, being in a gang. You don ’ t like to be surprised. You don ’ t like to be surprised by people who come around the corner. And the mantra was : shoot first ; ask questions later. Now, Sudhir was lucky — he was such a freak, and that clipboard probably saved his life, because they figured no other rival gang member would be coming up to shoot at them with a clipboard. So his greeting was not particularly warm, but they did say, well, OK — let ’ s hear your questions on your survey. So — I kid you not — the first question on the survey that he was sent to ask was :”How do you feel about being poor and Black in America?”Makes you wonder about academics. So the choice of answers were : [A) Very Good B) Good C) Bad D) Very Bad] What Sudhir found out is, in fact, that the real answer was the following : [A) Very Good B) Good C) Bad D) Very Bad E) Fuck you] The survey was not, in the end, going to be what got Sudhir off the hook. He was held hostage overnight in the stairwell. There was a lot of gunfire, there were a lot of philosophical discussions he had with the gang members. By morning, the gang **leader** arrived, checked out Sudhir, decided he was no threat, and they let him go home. So Sudhir went home, took a shower, took a nap. And you and I, probably, faced with the situation, would think,"I guess I ' m going to write my dissertation on The Grateful Dead, I ' ve been following them for the last three months."Sudhir, on the other hand, got right back, walked down to the housing project, went up to the second floor, and said :"Hey, **guys**, I had so much fun hanging out with you last night, I wonder if I could do it again tonight."And that was the beginning of what turned out to be a beautiful relationship that **involved** Sudhir living in the housing project on and off for 10 years, hanging out in crack houses, going to jail with the gang members, having the windows shot out of his car, having the police break into his apartment and steal his computer disks — you name it. But ultimately, the story has a happy ending for Sudhir, who became one of the most respected sociologists in the country. And especially for me, as I sat in my office with my Excel spreadsheet open, waiting for Sudhir to come and deliver to me the latest load of data that he would get from the gang. It was one of the most unequal co-authoring relationships ever — But I was glad to be the beneficiary of it. So what did we find? What did we find in the gang? Well, let me say one thing : We really got access to **everybody** in the gang. We got an inside look at the gang, from the very bottom up to the very top. They **trusted** Sudhir, in ways that really no academic has ever — or really anybody, any outsider — has ever earned the **trust** of these gangs, to the point where they actually opened up what was most **interesting** for me — their books, the financial records they kept. They made them available to us, and we not only could study them, but we could ask them questions about what was in them. So if I have to kind of summarize very quickly in the short time I have what the bottom line of what I take away from the gang is, it ' s that, if I had to draw a parallel between the gang and any other organization, it would be that the gang is just like McDonald ' s, in a lot of different respects — the restaurant McDonald ' s. So first, in one way, which isn ' t maybe the most **interesting** way, but it ' s a good way to start — is in the way it ' s organized, the hierarchy of the gang, the way it looks. So here ' s what the org chart of the gang looks like. I don ' t know if you know much about org charts, but if you were to assign a stripped- down and simplified McDonald ' s org chart, this is exactly what it would look like. It ' s amazing, but the top level of the gang, they actually call themselves the"Board of Directors."And Sudhir says it ' s not like these **guys** had a very sophisticated view of what happened in American corporate life, but they had seen movies like"Wall Street,"and they had learned a little bit about what it was like to be in the real world. Now, below that board of directors, you ' ve got essentially what are regional VPs — people who control, say, the South Side of Chicago, or the West Side of Chicago. Sudhir got to know very well the **guy** who had the unfortunate assignment of trying to take the Iowa franchise, which, it turned out, for this black gang, was not one of the more brilliant financial endeavors they undertook. But the thing that really makes the gang seem like McDonald ' s is its franchisees. The **guys** who are running the local gangs — the four-square-block by four-square-block areas — they ' re just like the **guys**, in some sense, who are running the McDonald ' s. They are the entrepreneurs. They get the exclusive property rights to control the drug-selling. They get the name of the gang behind them, for merchandising and marketing. And they ’ re the ones who basically make the profit or lose a profit, depending on how good they are at running the business. Now, the group I really want you to think about, though, are the ones at the bottom — the foot soldiers. These are the teenagers, typically, who ’ d be standing out on the street corner, selling the drugs. Extremely dangerous work. And important to note is that almost all of the weight, all of the people in this organization are at the bottom — just like McDonald ’ s. So in some sense, the foot soldiers are a lot like the people who are taking your order at McDonald ’ s, and it ’ s not just by chance that they ’ re like them. In fact, in these neighborhoods, they ’ d be the same people. So the same kids who are working in the gang were actually, at the very same time, typically working part-time at a place like McDonald ’ s. Which already foreshadows the main result that I ’ ve talked about, about what a crappy job it was, being in the gang. Because obviously, if being in the gang were such a wonderful, lucrative job, why in the world would these **guys** moonlight at McDonald ’ s? So what do the wages look like? You might be surprised. But based on being able to talk to them and to see their records, this is what it looks like in terms of the wages. The hourly wage the foot soldiers were earning was $3. 50 an hour. It was below the minimum wage. And this is well-documented. It ’ s easy to see by the patterns of consumption they have. It really is not fiction — it ’ s fact. There was very little money in the gang, especially at the bottom. Now if you managed to rise up, say, and be that local **leader**, the **guy** who ’ s the equivalent of the McDonald ’ s franchisee, you ’ d be making 100, 000 dollars a year. And that, in some ways, was the best job you could hope to get if you were growing up in one of these neighborhoods as a young black male. If you managed to rise to the very top, 200, 000 or 400, 000 dollars a year is what you ’ d hope to make. Truly, you would be a great success story. And one of the sad parts of this is that, indeed, among the many other ramifications of crack cocaine is that the most talented individuals in these communities — this is what they were striving for. They weren ’ t trying to make it in legitimate ways, because there were no legitimate channels out. This was the best way out. And it actually was the right choice, probably, to try to make it out this way. You look at this, the relationship to McDonald ’ s breaks down here. The money looks about the same. Why is it such a bad job? Well, the reason it ’ s such a bad job is that there ’ s somebody shooting at you a lot of the time. So, with shooting at you, what are the death rates? We found, in our gang — and admittedly, this was not really a standard situation ; this was a time of intense violence, of a lot of gang wars, as this gang actually became quite successful. But there were costs. And so the death rate — not to mention the rate of being arrested, sent to prison, being wounded — the death rate in our sample was seven percent per person per year. You ’ re in the gang for four years, you expect to die with about a 25 percent likelihood. That is about as high as you can get. So for comparison ’ s purposes, let ’ s think about some other walk of life you may expect might be extremely risky. Let ’ s say that you were a murderer and you were convicted of murder, and you ’ re sent to death row. It turns out, the death rates on death row from all causes, including execution : two percent a year. So it ’ s a lot **safer** being on death row than it is selling drugs out on the street. That gives you some pause, for those of you who believe that a death penalty ’ s going to have an enormous deterrent effect on crime. To give you a sense of just how bad the inner city was during crack — and I ’ m not really focusing on the negatives, but really, there ’ s another story to tell you there — if you look at the death rates just of random, young black males growing up in the inner city in the United States, the death rates during crack were about one percent. That ’ s extremely high. And this is violent death — it ’ s unbelievable, in some sense. To put it into perspective : if you compare this to the soldiers in Iraq, for instance, right now fighting the war : 0. 5 percent. So in some very literal way, the young black men who were growing up in this country were living in a war zone, very much in the sense that the soldiers over in Iraq are fighting in a war. So why in the world, you might ask, would anybody be willing to stand out on a street corner selling drugs for $3. 50 an hour, with a 25 percent chance of dying over the next four years? Why would they do that? And I think there are a couple answers. I think the first one is that they got fooled by history. It used to be the gang was a rite of passage ; that the young people controlled the gang ; that as you got older, you dropped out of the gang. So what happened was, the people who happened to be in the right place at the right time — the people who happened to be leading the gang in the mid-to-late- ' 80s — became very, very wealthy. And so the logical thing to think was that they are going to age out of the gang like **everybody** else has, and the next generation is going to take over and get the wealth. There are striking similarities, I think, to the Internet boom. The first set of people in Silicon Valley got very, very rich. And then all of my friends said,"Maybe I should go do that, too."And they were willing to work very cheap for stock options that never came. In some sense, that ' s what happened, exactly, to the set of people we were looking at. They were willing to start at the bottom, just like, say, a first-year lawyer at a law firm is willing to start at the bottom, work 80-hour weeks for not that much money, because they think they ' re going to make partner. But the rules changed, and they never got to make partner. Indeed, the same people who were running all of the major gangs in the late 1980s are still running the major gangs in Chicago today. They never passed on any of the wealth, So **everybody** got stuck at that $3. 50-an-hour job, and it turned out to be a disaster. The other thing the gang was very good at was marketing and trickery. And so for instance, one thing the gang would do is — the gang **leaders** would have big entourages, and they ' d drive fancy cars and have fancy jewelry. So what Sudhir eventually realized as he hung out with them more, is that, really, they didn ' t own those cars — they just leased them, because they couldn ' t afford to own the fancy cars. And they didn ' t really have gold jewelry, they had gold- plated jewelry. It goes back to, you know, the real-real versus the fake-real. And really, they did all sorts of things to trick the young people into thinking what a great deal the gang was going to be. So for instance, they would give a 14-year-old kid a whole roll of bills to hold. That 14-year-old kid would say to his friends,"Hey, look at all the money I got in the gang."It wasn ' t his money — until he spent it, and then he was in debt to the gang, and was sort of an indentured servant for a while. So I have a couple minutes. Let me do one last thing I hadn ' t thought I ' d have time to do, which is to talk about what we learned more generally about economics, from the study of the gang. So, economists tend to talk in technical words. Often, our theories fail quite miserably when we over the data, but what ' s kind of **interesting** is that in this setting, it turned out that some of the economic theories that worked not so well in the real economy worked very well in the drug economy, in some sense, because it ' s unfettered capitalism. Here ' s an economic principle. This is one of the basic ideas in labor economics, called a"compensating differential."It ' s the idea that the increment to wages that a worker requires to leave him indifferent between performing two tasks, one which is more unpleasant than the other. Compensating differential — it ' s why we think garbagemen might be paid more than people who work in parks. The words of one of the members of the gang, I think, make this clear. So it turns out — I ' m sort of getting ahead of myself — it turns out, in the gang, when there ' s a war going on, they actually pay the foot soldiers twice as much money. It ' s exactly this concept. Because they ' re not willing to be at risk. And the words of a gang member capture it quite nicely, he says :"Would you stand around here when all this shit..."— the shooting —"... if all this shit ' s going on? No, right? So if I gonna be asked to put my life on the line, then front me the cash, man."I think the gang member says it much more articulately than the economist, about what ' s going on. Here ' s another one. Economists talk about game theory, that every two-person game has a Nash equilibrium. Here ' s the translation you get from the gang member. They ' re talking about the decision of why they don ' t go shoot — One thing that turns out to be a great business tactic in the gang : if you go and just shoot guns in the **air** in the other gang ' s territory — people are afraid to go buy drugs there, they ' re going to come into your neighborhood. Here ' s what he says about why they don ' t do that :"If we start shooting around there, the other gang ' s territory, nobody, I mean, you dig it, nobody gonna step on their turf. But we gotta be careful, ' cause they can shoot around here too and then we all fucked."So that ' s the same concept. Then again, sometimes economists get it wrong. One thing we observed in the data is that it looked like — the gang **leader** always got paid. No matter how bad it was economically, he always got himself paid. We had some theories related to cash flow, and lack of access to capital markets, and things like that. Then we asked the gang member,"Why is it you always get paid and your workers don ' t always get paid?"His response is,"You got all these niggers below you who want your job, you dig? If you start taking losses, they see you as weak and shit."And I thought about it and said,"CEOs often pay themselves million-dollar bonuses, even when companies are losing a lot of money. And it never would really occur to an economist that this idea of ' weak and shit ' could really be important."Maybe"weak and shit"is an important hypothesis that needs more analysis. Thank you very much.

Text Sample 560: POS Dim 4 – 2004 – Score 11.00 – 2004/t000464.txt

I bet you ' re worried. I was worried. That ' s why I began this piece. I was worried about vaginas. I was worried what we think about vaginas and even more worried that we don ' t think about them. I was worried about my own vagina. It needed a context, a culture, a community of other vaginas. There is so much darkness and secrecy surrounding them. Like the Bermuda Triangle, nobody ever reports back from there. In the first place, it ' s not so easy to even find your vagina. Women go days, weeks, months, without looking at it. I interviewed a high-powered businesswoman ; she told me she didn ' t have time."Looking at your vagina,"she said,"is a full day ' s work.""You ' ve got to get down there on your back, in front of a mirror, full-length preferred. You ' ve got to get in the perfect position with the perfect light, which then becomes shadowed by the angle you ' re at. You ' re twisting your head up, arching your back, it ' s exhausting."She was busy ; she didn ' t have time. So I decided to talk to women about their vaginas. They began as casual vagina interviews, and they turned into vagina monologues. I talked with over 200 women. I talked to older women, younger women, married women, lesbians, single women. I talked to corporate professionals, college professors, actors, sex workers. I talked to African-American women, Asian-American women, Native- American women, Caucasian women, Jewish women. OK, at first women were a little shy, a little reluctant to talk. Once they got going, you couldn ' t stop them. Women love to talk about their vaginas, they do. Mainly because no one ' s ever asked them before. Let ' s just start with the word"vagina"— vagina, vagina. It sounds like an infection, at best. Maybe a medical instrument."Hurry, nurse, bring the vagina!"Vagina, vagina, vagina. It doesn ' t matter how many times you say the word, it never sounds like a word you want to say. It ' s a completely ridiculous, totally un-sexy word. If you use it during sex, trying to be politically correct,"Darling, would you stroke my vagina,"you kill the act right there. I ' m worried what we call them and don ' t call them. In Great Neck, New York, they call it a Pussycat. A woman told me there her mother used to tell her,"Don ' t wear panties, dear, underneath your pajamas. You need to **air** out your Pussycat."In Westchester, they call it a Pooki, in New Jersey, a twat. There ' s Powderbox, derriere, a Pooky, a Poochi, a Poopi, a Poopelu, a Pooninana, a Padepachetchki, a Pal, and a Piche. There ' s Toadie, Dee Dee, Nishi, Dignity, Coochi Snorcher, Cooter, Labbe, Gladys Seagelman, VA, Wee wee, Horsespot, Nappy Dugout, Mongo, Ghoulie, Powderbox, a Mimi in Miami, a Split Knish in Philadelphia... and a Schmende in the Bronx. I am worried about vaginas. This is how the"Vagina Monologues"begins. But it really didn ' t begin there. It began with a conversation with a woman. We were having a conversation about menopause, and we got onto the subject of her vagina, which you ' ll do if you ' re talking about menopause. And she said things that really shocked me about her vagina — that it was dried-up and finished and dead — and I was kind of shocked. So I said to a friend casually,"Well, what do you think about your vagina?"And that woman said something more amazing, and then the next woman said something more amazing, and before I knew it, every woman was telling me I had to talk to somebody about their vagina because they had an amazing story, and I was sucked down the vagina trail. And I really haven ' t gotten off of it. I think if you had told me when I was younger that I was going to grow up, and be in shoe stores, and people would scream out,"There she is, the Vagina Lady!"I don ' t know that that would have been my life ambition. But I want to talk a little bit about happiness, and the relationship to this whole vagina journey, because it has been an extraordinary journey that began eight years ago. I think before I did the"Vagina Monologues,"I didn ' t really believe in happiness. I thought that only idiots were happy, to be honest. I remember when I started practicing Buddhism 14 years ago, and I was told that the end of this practice was to be happy, I said,"How could you be happy and live in this world of suffering and live in this world of pain?"I mistook happiness for a lot of other things, like numbness or decadence or selfishness. And what happened through the course of the"Vagina Monologues"and this journey is, I think I have come to understand a little bit more about happiness. There are three qualities I want to talk about. One is seeing what ' s right in front of you, and talking about it, and stating it. I think what I learned from talking about the vagina and speaking about the vagina, is it was the most obvious thing — it was right in the center of my body and the center of the world — and yet it was the one thing nobody talked about. The second thing is that what talking about the vagina did is it opened this door which allowed me to see that there was a way to serve the world to make it better. And that ' s where the deepest happiness has actually come from. And the third principle of happiness, which I ' ve realized recently : Eight years ago, this momentum and this energy, this"V-wave"started — and I can only describe it as a"V-wave"because, to be honest, I really don ' t understand it completely ; I feel at the service of it. But this wave started, and if I question the wave, or try to stop the wave or look back at the wave, I often have the experience of whiplash or the potential of my neck breaking. But if I go with the wave, and I **trust** the wave and I move with the wave, I go to the next place, and it happens logically and organically and truthfully. And I started this piece, particularly with stories and narratives, and I was talking to one woman and that led to another woman and that led to another woman. And then I wrote those stories down, and I put them out in front of other people. And every single time I did the show at the beginning, women would literally line up after the show, because they wanted to tell me their stories. And at first I thought,"Oh great, I ' ll hear about wonderful orgasms, and great sex lives, and how women love their vaginas."But in fact, that ' s not what women lined up to tell me. What women lined up to tell me was how they were raped, and how they were battered, and how they were beaten, and how they were gang-raped in parking lots, and how they were incested by their uncles. And I wanted to stop doing the"Vagina Monologues,"because it felt too daunting. I felt like a war photographer who takes pictures of terrible events, but doesn ' t intervene on their behalf. And so in 1997, I said,"Let ' s get women together. What could we do with this information that all these women are being violated?"And it turned out, after thinking and investigating, that I discovered — and the UN has actually said this recently — that one out of every three women on this planet will be beaten or raped in her lifetime. That ' s essentially a **gender** ; that ' s essentially the resource of the planet, which is women. So in 1997 we got all these incredible women together and we said,"How can we use the play, this energy, to stop violence against women?"And we put on one event in New York City, in the theater, and all these great actors came — from Susan Sarandon, to Glenn Close, to Whoopi Goldberg — and we did one performance on one evening, and that catalyzed this wave, this energy. And within five years, this extraordinary thing began to happen. One woman took that energy and she said,"I want to bring this wave, this energy, to college campuses,"and so she took the play and she said,"Let ' s use the play and have performances once a year, where we can raise money to stop violence against women in local communities all around the world."And in one year, it went to 50 colleges, and then it expanded. And over the course of the last six years, it ' s spread and it ' s spread and it ' s spread around the world. What I have learned is two things : one, that the epidemic of violence towards women is shocking ; it ' s global ; it is so profound and it is so devastating, and it is so in every little pocket of every little crater, of every little society that we don ' t even recognize it, because it ' s become ordinary. This journey has taken me to Afghanistan, where I had the extraordinary honor and privilege to go into parts of Afghanistan under the Taliban. I was dressed in a burqa and I went in with an extraordinary group, called the Revolutionary Association of the Women of Afghanistan. And I saw firsthand how women had been stripped of every single right that was possible to strip women of — from being educated, to being employed, to being actually allowed to eat ice cream. For those of you who don ' t know, it was illegal to eat ice cream under the Taliban. And I actually saw and met women who had been flogged for being caught eating vanilla ice cream. I was taken to the secret ice cream-eating place in a little town, where we went to a back room, and women were seated and a curtain was pulled around us, and they were served vanilla ice cream. And women lifted their burqas and ate this ice cream. And I don ' t think I ever understood pleasure until that moment, and how women have found a way to keep their pleasure alive. It has taken me, this journey, to Islamabad, where I have witnessed and met women with their faces melted off. It has taken me to Juarez, Mexico, where I was a week ago, where I have literally been there in parking lots, where bones of women have washed up and been dumped next to Coca-Cola bottles. It has taken me to universities all over this country, where girls are date-raped and drugged. I have seen terrible, terrible, terrible violence. But I have also recognized, in the course of seeing that violence, that being in the face of things and seeing actually what ' s in front of us is the antidote to depression, and to a feeling that one is worthless and has no value. Because before the"Vagina Monologues,"I will say that 80 percent of my consciousness was closed off to what was really going on in this reality, and that closing-off closed off my vitality and my life energy. What has also happened is in the course of these travels — and it ' s been an extraordinary thing — is that every single place that I have gone to in the world, I have met a new species. And I really love hearing about all these species at the bottom of the sea. And I was thinking about how being with these extraordinary people on this particular panel, that it ' s beneath, beyond and between, and the vagina kind of fits into all those categories. But one of the things I ' ve seen is this species — and it is a species, and it is a new paradigm, and it doesn ' t get reported in the press or in the media because I don ' t think good news ever is news, and I don ' t think people who are transforming the planet are what gets the ratings on TV shows. But every single country I have been to — and in the last six years, I ' ve been to about 45 countries, and many tiny little villages and cities and towns — I have seen something what I ' ve come to call"vagina warriors."A"vagina warrior"is a woman, or a vagina-friendly man, who has witnessed incredible violence or suffered it, and rather than getting an AK-47 or a weapon of mass destruction or a machete, they hold the violence in their bodies ; they grieve it ; they experience it ; and then they go out and devote their lives to making sure it doesn ' t happen to anybody else. I have met these women everywhere on the planet, and I want to tell a few stories, because I believe that stories are the way that we transmit information, where it goes into our bodies. And I think one of the things about being at **TED** that ' s been very **interesting** is that I live in my body a lot, and I don ' t live in my head very much anymore. And this is a very heady place. And it ' s been really **interesting** to be in my head for the last two days ; I ' ve been very disoriented — because I think the world, the V-world, is very much in your body. It ' s a body world, and the species really exists in the body. And I think there ' s a real significance in us attaching our bodies to our heads, that that separation has created a divide that is often separating purpose from intent. And the connection between body and head often brings those things into union. I want to talk about three particular people that I ' ve met, vagina warriors, who really transformed my understanding of this whole principle and species, and one is a woman named Marsha Lopez. Marsha Lopez was a woman I met in Guatemala. She was 14 years old, and she was in a marriage and her husband was beating her on a regular basis. And she couldn ' t get out, because she was addicted to the relationship, and she had no money. Her sister was younger than her, and she applied — we had a"Stop Rape"contest a few years ago in New York — and she applied, hoping that she would become a finalist and she could bring her sister. She did become a finalist ; she brought Marsha to New York. And at that time, we did this extraordinary V-Day at Madison Square Garden, where we sold out the entire testosterone-filled dome — 18, 000 people standing up to say"Yes"to vaginas, which was really a pretty incredible transformation. And she came, and she witnessed this, and she decided that she would go back and leave her husband, and that she would bring V-Day to Guatemala. She was 21 years old. I went to Guatemala and she had sold out the National Theater of Guatemala. And I watched her walk up on stage in her red short dress and high heels, and she stood there and said,"My name is Marsha. I was beaten by my husband for five years. He almost murdered me. I left and you can, too."And the entire 2, 000 people went absolutely crazy. There ' s a woman named Esther Chvez who I met in Juarez, Mexico. And Esther Chvez was a brilliant accountant in Mexico City. She was 72 years old and she was planning to retire. She went to Juarez to take care of an ailing aunt, and in the course of it, she began to discover what was happening to the murdered and disappeared women of Juarez. She gave up her life ; she moved to Juarez. She started to write the stories which **documented** the disappeared women. 300 women have disappeared in a border town because they ' re brown and poor. There has been no response to the disappearance, and not one person has been held **accountable**. She began to **document** it. She opened a center called Casa Amiga, and in six years, she has literally brought this to the consciousness of the world. We were there a week ago, when there were 7, 000 people in the street, and it was truly a miracle. And as we walked through the streets, the people of Juarez, who normally don ’ t even come into the streets, because the streets are so dangerous, literally stood there and wept, to see that other people from the world had showed up for that particular community. There ’ s another woman, named Agnes. And Agnes, for me, epitomizes what a vagina warrior is. I met her three years ago in Kenya. And Agnes was mutilated as a little girl ; she was circumcised against her will when she was 10 years old, and she really made a decision that she didn ’ t want this practice to continue anymore in her community. So when she got older, she created this incredible thing : it ’ s an anatomical sculpture of a woman ’ s body, half a woman ’ s body. And she walked through the Rift Valley, and she had vagina and vagina replacement parts, where she would teach girls and parents and boys and girls what a healthy vagina looks like, and what a mutilated vagina looks like. And in the course of her travel — she walked literally for eight years through the Rift Valley, through dust, through sleeping on the ground, because the Maasai are nomads, and she would have to find them, and they would move, and she would find them again — she saved 1, 500 girls from being cut. And in that time, she created an alternative ritual, which **involved** girls coming of age without the cut. When we met her three years ago, we said,”What could V-Day do for you?”And she said,”Well, if you got me a jeep, I could get around a lot faster.”So we bought her a jeep. And in the year that she had the jeep, she saved 4, 500 girls from being cut. So we said to her,”What else could we do for you?”She said,”Well, Eve, if you gave me some money, I could open a house and girls could run away, and they could be saved.”And I want to tell this little story about my own beginnings, because it ’ s very interrelated to happiness and Agnes. When I was a little girl — I grew up in a wealthy community ; it was an upper- middle class white community, and it had all the trappings and the looks of a perfectly nice, wonderful, great life. And everyone was supposed to be happy in that community, and, in fact, my life was hell. I lived with an alcoholic father who beat me and molested me, and it was all inside that. And always as a child I had this fantasy that somebody would come and rescue me. And I actually made up a little character whose name was Mr. Alligator. I would call him up when things got really bad, and say it was time to come and pick me up. And I would pack a little bag and wait for Mr. Alligator to come. Now, Mr. Alligator never did come, but the idea of Mr. Alligator coming actually saved my sanity and made it OK for me to keep going, because I believed, in the distance, there would be someone coming to rescue me. Cut to 40-some odd years later, we go to Kenya, and we ’ re walking, we arrive at the opening of this house. And Agnes hadn ’ t let me come to the house for days, because they were preparing this whole ritual. I want to tell you a great story. When Agnes first started fighting to stop female genital mutilation in her community, she had become an outcast, and she was exiled and slandered, and the whole community turned against her. But being a vagina warrior, she kept going, and she kept committing herself to transforming consciousness. And in the Maasai community, goats and cows are the most valued possession. They ’ re like the Mercedes-Benz of the Rift Valley. And she said two days before the house opened, two different people arrived to give her a goat each, and she said to me,”I knew then that female genital mutilation would end one day in Africa.”Anyway, we arrived, and when we arrived, there were hundreds of girls dressed in red homemade dresses — which is the color of the Maasai and the color of V-Day — and they greeted us. They had made up these songs that they were singing, about the end of suffering and the end of mutilation, and they walked us down the path. It was a gorgeous day in the African sun, and the dust was flying and the girls were dancing, and there was this house, and it said,"V- Day Safe House for the Girls."And it hit me in that moment that it had taken 47 years, but that Mr. Alligator had finally shown up. And he had shown up, obviously, in a form that it took me a long time to understand, which is that when we give in the world what we want the most, we heal the broken part inside each of us. And I feel, in the last eight years, that this journey — this miraculous vagina journey — has taught me this really simple thing, which is that happiness exists in action ; it exists in telling the truth and saying what your truth is ; and it exists in giving away what you want the most. And I feel that knowledge and that journey has been an extraordinary privilege, and I feel really blessed to have been here today to communicate that to you. Thank you very much.

Text Sample 561: POS Dim 4 – 2010 – Score 20.00 – 2010/t002818.txt

Chris Anderson : Julian, welcome. It ' s been reported that WikiLeaks, your baby, has, in the last few years has released more classified **documents** than the rest of the world ' s media combined. Can that possibly be true? Julian Assange : Yeah, can it possibly be true? It ' s a worry â isn ' t it? â that the rest of the world ' s media is doing such a bad job that a little group of activists is able to release more of that type of information than the rest of the world press combined. CA : How does it work? How do people release the **documents**? And how do you secure their privacy? JA : So these are â as far as we can tell â classical whistleblowers, and we have a number of ways for them to get information to us. So we use this state- of-the-art encryption to bounce **stuff** around the Internet, to hide trails, pass it through legal jurisdictions like Sweden and Belgium to enact those legal protections. We get information in the mail, the regular postal mail, encrypted or not, vet it like a regular news organization, format it â which is sometimes something that ' s quite hard to do, when you ' re talking about giant databases of information â release it to the public and then defend ourselves against the inevitable legal and political attacks. CA : So you make an effort to **ensure** the **documents** are legitimate, but you actually almost never know who the identity of the source is? JA : That ' s right, yeah. Very rarely do we ever know, and if we find out at some stage then we destroy that information as soon as possible. God damn it. CA : I think that ' s the CIA asking what the code is for a **TED membership**. So let ' s take [an] example, actually. This is something you leaked a few years ago. If we can have this **document** up... So this was a story in Kenya a few years ago. Can you tell us what you leaked and what happened? JA : So this is the Kroll Report. This was a secret **intelligence** report commissioned by the Kenyan government after its election in 2004. Prior to 2004, Kenya was ruled by Daniel arap Moi for about 18 years. He was a soft dictator of Kenya. And when Kibaki got into power â through a coalition of forces that were trying to clean up corruption in Kenya â they commissioned this report, spent about two million pounds on this and an associated report. And then the government sat on it and used it for political leverage on Moi, who was the richest man â still is the richest man â in Kenya. It ' s the Holy Grail of Kenyan journalism. So I went there in 2007, and we managed to get hold of this just prior to the election â the national election, December 28. When we released that report, we did so three days after the new president, Kibaki, had decided to pal up with the man that he was going to clean out, Daniel arap Moi, so this report then became a dead albatross around President Kibaki ’ s neck. CA : And â I mean, to cut a long story short â word of the report leaked into Kenya, not from the official media, but indirectly, and in your opinion, it actually shifted the election. JA : Yeah. So this became front page of the Guardian and was then printed in all the surrounding countries of Kenya, in Tanzanian and South African press. And so it came in from the outside. And that, after a couple of days, made the Kenyan press feel **safe** to talk about it. And it ran for 20 nights straight on Kenyan TV, shifted the vote by 10 percent, according to a Kenyan **intelligence** report, which changed the result of the election. CA : Wow, so your leak really substantially changed the world? JA : Yep. CA : Here ’ s â We ’ re going to just show a short clip from this Baghdad airstrike video. The video itself is longer, but here ’ s a short clip. This is â this is intense material, I should warn you. Radio :... just fuckin ’, once you get on ’ em just open ’ em up. I see your element, uh, got about four Humvees, uh, out along... You ’ re clear. All right. Firing. Let me know when you ’ ve got them. Let ’ s shoot. Light ’ em all up. C ’ mon, fire! Keep shoot ’ n. Keep shoot ’ n. Keep shoot ’ n. Hotel... Bushmaster Two-Six, Bushmaster Two-Six, we need to move, time now! All right, we just engaged all eight individuals. Yeah, we see two birds [helicopters], and we ’ re still firing. Roger. I got ’ em. Two-Six, this is Two-Six, we ’ re mobile. Oops, I ’ m sorry. What was going on? God damn it, Kyle. All right, hahaha. I hit ’ em. CA : So, what was the **impact** of that? JA : The **impact** on the people who worked on it was severe. We ended up sending two people to Baghdad to further research that story. So this is just the first of three attacks that occurred in that scene. CA : So, I mean, 11 people died in that attack, right, including two Reuters employees? JA : Yeah. Two Reuters employees, two young children were wounded. There were between 18 and 26 people killed all together. CA : And releasing this caused widespread outrage. What was the key element of this that actually caused the outrage, do you think? JA : I don ’ t know. I guess people can see the gross disparity in force. You have **guys** walking in a relaxed way down the street, and then an Apache helicopter sitting up at one kilometer firing 30-millimeter cannon shells on everyone â looking for any excuse to do so â and killing people rescuing the wounded. And there was two journalists **involved** that clearly weren ’ t insurgents because that ’ s their full-time job. CA : I mean, there ’ s been this U. S. **intelligence** analyst, Bradley Manning, arrested, and it ’ s alleged that he confessed in a chat room to have leaked this video to you, along with 280, 000 classified U. S. embassy cables. I mean, did he? JA : We have denied receiving those cables. He has been charged, about five days ago, with obtaining 150, 000 cables and releasing 50. Now, we had released, early in the year, a cable from the Reykjavik U. S. embassy, but this is not necessarily connected. I mean, I was a known visitor of that embassy. CA : I mean, if you did receive thousands of U. S. embassy diplomatic cables... JA : We would have released them. JA : Yeah. JA : Well, because these sort of things reveal what the true state of, say, Arab governments are like, the true human-rights abuses in those governments. If you look at declassified cables, that ’ s the sort of material that ’ s there. CA : So let ’ s talk a little more broadly about this. I mean, in general, what ’ s your philosophy? Why is it right to encourage leaking of secret information? JA : Well, there ’ s a question as to what sort of information is important in the world, what sort of information can achieve reform. And there ’ s a lot of information. So information that organizations are spending economic effort into concealing, that ’ s a really good signal that when the information gets out, there ’ s a hope of it doing some good â because the organizations that know it best, that know it from the inside out, are spending work to conceal it. And that ’ s what we ’ ve found in practice, and that ’ s what the history of journalism is. CA : But are there risks with that, either to the individuals concerned or indeed to society at large, where leaking can actually have an unintended consequence? JA : Not that we have seen with anything we have released. I mean, we have a harm immunization **policy**. We have a way of dealing with information that has sort of personal â personally identifying information in it. But there are legitimate secrets â you know, your records with your doctor ; that ' s a legitimate secret â but we deal with whistleblowers that are coming forward that are really sort of well-motivated. CA : So they are well-motivated. And what would you say to, for example, the, you know, the parent of someone whose son is out serving the U. S. military, and he says,"You know what, you ' ve put up something that someone had an incentive to put out. It shows a U. S. soldier laughing at people dying. That gives the impression, has given the impression, to millions of people around the world that U. S. soldiers are inhuman people. Actually, they ' re not. My son isn ' t. How dare you?"What would you say to that? JA : Yeah, we do get a lot of that. But remember, the people in Baghdad, the people in Iraq, the people in Afghanistan â they don ' t need to see the video ; they see it every day. So it ' s not going to change their opinion. It ' s not going to change their perception. That ' s what they see every day. It will change the perception and opinion of the people who are paying for it all, and that ' s our hope. CA : So you found a way to shine light into what you see as these sort of dark secrets in companies and in government. Light is good. But do you see any irony in the fact that, in order for you to shine that light, you have to, yourself, create secrecy around your sources? JA : Not really. I mean, we don ' t have any WikiLeaks dissidents yet. We don ' t have sources who are dissidents on other sources. Should they come forward, that would be a tricky situation for us, but we ' re presumably acting in such a way that people feel morally compelled to continue our mission, not to screw it up. CA : I ' d actually be **interested**, just based on what we ' ve heard so far â I ' m curious as to the opinion in the **TED** audience. You know, there might be a couple of views of WikiLeaks and of Julian. You know, hero â people ' s hero â bringing this important light. Dangerous troublemaker. Who ' s got the hero view? Who ' s got the dangerous troublemaker view? JA : Oh, come on. There must be some. CA : It ' s a soft crowd, Julian, a soft crowd. We have to try better. Let ' s show them another example. Now here ' s something that you haven ' t yet leaked, but I think for **TED** you are. I mean it ' s an intriguing story that ' s just happened, right? What is this? JA : So this is a sample of what we do sort of every day. So late last year â in November last year â there was a series of well blowouts in Albania, like the well blowout in the Gulf of Mexico, but not quite as big. And we got a report â a sort of engineering analysis into what happened â saying that, in fact, security guards from some rival, various competing oil firms had, in fact, parked trucks there and blown them up. And part of the Albanian government was in this, etc., etc. And the engineering report had nothing on the top of it, so it was an extremely difficult **document** for us. We couldn ' t verify it because we didn ' t know who wrote it and knew what it was about. So we were kind of skeptical that maybe it was a competing oil firm just sort of playing the issue up. So under that basis, we put it out and said,"Look, we ' re skeptical about this thing. We don ' t know, but what can we do? The material looks good, it feels right, but we just can ' t verify it."And we then got a letter just this week from the company who wrote it, wanting to track down the source â saying,"Hey, we want to track down the source."And we were like,"Oh, tell us more. What **document** is it, precisely, you ' re talking about? Can you show that you had legal authority over that **document**? Is it really yours?"So they sent us this screen shot with the author in the Microsoft Word ID. Yeah. That

’ s happened quite a lot though. This is like one of our methods of identifying, of verifying, what a material is, is to try and get these **guys** to write letters. CA : Yeah. Have you had information from inside BP? JA : Yeah, we have a lot, but I mean, at the moment, we are undergoing a sort of serious fundraising and engineering effort. So our publication rate over the past few months has been sort of minimized while we ’ re re-engineering our back systems for the phenomenal public interest that we have. That ’ s a problem. I mean, like any sort of growing startup organization, we are sort of overwhelmed by our growth, and that means we ’ re getting enormous quantity of whistleblower disclosures of a very high caliber but don ’ t have enough people to actually process and vet this information. CA : So that ’ s the key bottleneck, basically journalistic volunteers and / or the funding of journalistic salaries? JA : Yep. Yeah, and **trusted** people. I mean, we ’ re an organization that is hard to grow very quickly because of the sort of material we deal with, so we have to restructure in order to have people who will deal with the highest national security **stuff**, and then lower security cases. CA : So help us understand a bit about you personally and how you came to do this. And I think I read that as a kid you went to 37 different schools. Can that be right? JA : Well, my parents were in the movie business and then on the run from a cult, so the combination between the two... CA : I mean, a psychologist might say that ’ s a recipe for breeding paranoia. JA : What, the movie business? CA : And you were also â I mean, you were also a hacker at an early age and ran into the authorities early on. JA : Well, I was a journalist. You know, I was a very young journalist activist at an early age. I wrote a magazine, was prosecuted for it when I was a teenager. So you have to be careful with hacker. I mean there ’ s like â there ’ s a method that can be deployed for various things. Unfortunately, at the moment, it ’ s mostly deployed by the Russian mafia in order to steal your grandmother ’ s bank accounts. So this phrase is not, not as nice as it used to be. CA : Yeah, well, I certainly don ’ t think you ’ re stealing anyone ’ s grandmother ’ s bank account, but what about your core values? Can you give us a sense of what they are and maybe some incident in your life that helped determine them? JA : I ’ m not sure about the incident. But the core values : well, capable, generous men do not create victims ; they nurture victims. And that ’ s something from my father and something from other capable, generous men that have been in my life. CA : Capable, generous men do not create victims ; they nurture victims? JA : Yeah. And you know, I ’ m a combative person, so I ’ m not actually so big on the nurture, but some way â there is another way of nurturing victims, which is to police perpetrators of crime. And so that is something that has been in my character for a long time. CA : So just tell us, very quickly in the last minute, the story : what happened in Iceland? You basically published something there, ran into trouble with a bank, then the news service there was injuncted from running the story. Instead, they publicized your side. That made you very high-profile in Iceland. What happened next? JA : Yeah, this is a great case, you know. Iceland went through this financial **crisis**. It was the hardest hit of any country in the world. Its banking **sector** was 10 times the GDP of the rest of the economy. Anyway, so we release this report in July last year. And the national TV station was injuncted five minutes before it went on **air**, like out of a movie : injunction landed on the news desk, and the news reader was like,”This has never happened before. What do we do?”Well, we just show the website instead, for all that time, as a filler, and we became very famous in Iceland, went to Iceland and spoke about this issue. And there was a feeling in the community that that should never happen again, and as a result, working with Icelandic politicians and some other **international** legal experts, we put together a new sort of package of legislation for Iceland to sort of become an offshore haven for the free press, with the strongest journalistic protections in the world, with a new Nobel Prize for freedom of speech. Iceland ’ s a Nordic country, so, like Norway, it ’ s able to tap into the system. And just a month ago, this was passed by the Icelandic parliament unanimously. CA : Wow. Last question, Julian. When you think of the future then, do you think it ’ s more likely to be Big Brother exerting more control, more secrecy, or us watching Big Brother, or it ’ s just all to be played for either way? JA : I ’ m not sure which way it ’ s going to go. I mean, there ’ s enormous pressures to harmonize freedom of speech legislation and transparency legislation around the world â within the E. U., between China and the United States. Which way is it going to go? It ’ s hard to see. That ’ s why it ’ s a very **interesting** time to be in â because with just a little bit of effort, we can shift it one way or the other. CA : Well, it looks like I ’ m reflecting the audience ’ s opinion to say, Julian, be careful, and all power to you. JA : Thank you, Chris.

Text Sample 562: POS Dim 4 – 2010 – Score 18.00 – 2010/t002730.txt

So today, I ’ m going to tell you about some people who didn ’ t move out of their neighborhoods. The first one is happening right here in Chicago. Brenda Palms- Farber was hired to help ex-convicts reenter society and keep them from going back into prison. Currently, taxpayers spend about 60, 000 dollars per year sending a person to jail. We know that two-thirds of them are going to go back. I find it **interesting** that, for every one dollar we spend, however, on early childhood education, like Head Start, we save 17 dollars on **stuff** like incarceration in the future. Or — think about it — that 60, 000 dollars is more than what it costs to send one person to Harvard as well. But Brenda, not being phased by **stuff** like that, took a look at her challenge and came up with a not-so-obvious solution : create a business that produces skin care products from honey. Okay, it might be obvious to some of you ; it wasn ’ t to me. It ’ s the basis of growing a form of social innovation that has real potential. She hired seemingly unemployable men and women to care for the bees, harvest the honey and make value-added products that they marketed themselves, and that were later sold at Whole Foods. She combined employment experience and training with life skills they needed, like anger-management and teamwork, and also how to talk to future employers about how their experiences actually demonstrated the lessons that they had learned and their eagerness to learn more. Less than four percent of the folks that went through her program actually go back to jail. So these young men and women learned job-readiness and life skills through bee keeping and became productive citizens in the process. Talk about a sweet beginning. Now, I ’ m going to take you to Los Angeles, and lots of people know that L. A. has its issues. But I ’ m going to talk about L. A. ’ s water issues right now. They have not enough water on most days and too much to handle when it rains. Currently, 20 percent of California ’ s energy consumption is used to pump water into mostly Southern California. Their spending loads, loads, to channel that rainwater out into the ocean when it rains and floods as well. Now Andy Lipkis is working to help L. A. cut infrastructure costs associated with water management and urban heat island — linking trees, people and technology to create a more livable city. All that green **stuff** actually naturally absorbs storm water, also helps cool our cities. Because, come to think about it, do you really want air-conditioning, or is it a cooler room that you want? How you get it shouldn ’ t make that much of a difference. So a few years ago, L. A. County decided that they needed to spend 2. 5 billion dollars to repair the city schools. And Andy and his team discovered that they were going to spend 200 million of those dollars on asphalt to surround the schools themselves. And by presenting a really strong economic case, they convinced the L. A. government that replacing that asphalt with trees and other greenery, that the schools themselves would save the system more on energy than they spend on horticultural infrastructure. So ultimately, 20 million square feet of asphalt was replaced or avoided, and electrical consumption for air- conditioning went down, while employment for people to maintain those grounds went up, resulting in a net-savings to the system, but also healthier students and schools system employees as well. Now Judy Bonds is a coal miner ' s daughter. Her family has eight generations in a town called Whitesville, West Virginia. And if anyone should be clinging to the former glory of the coal mining history, and of the town, it should be Judy. But the way coal is mined right now is different from the deep mines that her father and her father ' s father would go down into and that employed essentially thousands and thousands of people. Now, two dozen men can tear down a mountain in several months, and only for about a few years ' worth of coal. That kind of technology is called"mountaintop removal."It can make a mountain go from this to this in a few short months. Just imagine that the **air** surrounding these places — it ' s filled with the residue of explosives and coal. When we **visited**, it gave some of the people we were with this strange little cough after being only there for just a few hours or so — not just miners, but **everybody**. And Judy saw her landscape being destroyed and her water poisoned. And the coal companies just move on after the mountain was emptied, leaving even more unemployment in their wake. But she also saw the difference in potential wind energy on an intact mountain, and one that was reduced in elevation by over 2, 000 feet. Three years of dirty energy with not many jobs, or centuries of clean energy with the potential for developing expertise and improvements in efficiency based on technical skills, and developing local knowledge about how to get the most out of that region ' s wind. She calculated the up-front cost and the payback over time, and it ' s a net-plus on so many levels for the local, national and global economy. It ' s a longer payback than mountaintop removal, but the wind energy actually pays back forever. Now mountaintop removal pays very little money to the locals, and it gives them a lot of misery. The water is turned into goo. Most people are still unemployed, leading to most of the same kinds of social problems that unemployed people in inner cities also experience — drug and alcohol abuse, domestic abuse, teen pregnancy and poor heath, as well. Now Judy and I — I have to say — totally related to each other. Not quite an obvious alliance. I mean, literally, her hometown is called Whitesville, West Virginia. I mean, they are not — they ain ' t competing for the birthplace of hip hop title or anything like that. But the back of my T-shirt, the one that she gave me, says,"Save the endangered hillbillies."So homegirls and hillbillies we got it together and totally understand that this is what it ' s all about. But just a few months ago, Judy was diagnosed with stage-three lung cancer. Yeah. And it has since moved to her bones and her brain. And I just find it so bizarre that she ' s suffering from the same thing that she tried so hard to protect people from. But her dream of Coal River Mountain Wind is her legacy. And she might not get to see that mountaintop. But rather than writing yet some kind of manifesto or something, she ' s leaving behind a business plan to make it happen. That ' s what my homegirl is doing. So I ' m so proud of that. But these three people don ' t know each other, but they do have an awful lot in common. They ' re all problem solvers, and they ' re just some of the many examples that I really am privileged to see, meet and learn from in the examples of the work that I do now. I was really lucky to have them all featured on my Corporation for

Public Radio radio show called ThePromisedLand. org. Now they ’ re all very practical visionaries. They take a look at the **demands** that are out there — beauty products, healthy schools, electricity — and how the money ’ s flowing to meet those **demands**. And when the cheapest solutions **involve** reducing the number of jobs, you ’ re left with unemployed people, and those people aren ’ t cheap. In fact, they make up some of what I call the most expensive citizens, and they include generationally impoverished, traumatized vets returning from the Middle East, people coming out of jail. And for the **veterans** in particular, the V. A. said there ’ s a six-fold increase in mental health pharmaceuticals by vets since 2003. I think that number ’ s probably going to go up. They ’ re not the largest number of people, but they are some of the most expensive — and in terms of the likelihood for domestic abuse, drug and alcohol abuse, poor performance by their kids in schools and also poor health as a result of stress. So these three **guys** all understand how to productively channel dollars through our local economies to meet existing market **demands**, reduce the social problems that we have now and prevent new problems in the future. And there are plenty of other examples like that. One problem : waste handling and unemployment. Even when we think or talk about recycling, lots of recyclable **stuff** ends up getting incinerated or in landfills and leaving many municipalities, diversion rates — they leave much to be recycled. And where is this waste handled? Usually in poor communities. And we know that eco- industrial business, these kinds of business models — there ’ s a model in Europe called the eco-industrial park, where either the waste of one company is the raw material for another, or you use recycled materials to make goods that you can actually use and sell. We can create these local markets and incentives for recycled materials to be used as raw materials for manufacturing. And in my hometown, we actually tried to do one of these in the Bronx, but our mayor decided what he wanted to see was a jail on that same spot. Fortunately — because we wanted to create hundreds of jobs — but after many years, the city wanted to build a jail. They ’ ve since abandoned that project, thank goodness. Another problem : unhealthy food systems and unemployment. Working-class and poor urban Americans are not benefiting economically from our current food system. It relies too much on transportation, chemical fertilization, big use of water and also refrigeration. Mega agricultural operations often are responsible for poisoning our waterways and our land, and it produces this incredibly unhealthy product that costs us billions in healthcare and lost productivity. And so we know”urban ag”is a big buzz topic this time of the year, but it ’ s mostly gardening, which has some value in community building — lots of it — but it ’ s not in terms of creating jobs or for food production. The numbers just aren ’ t there. Part of my work now is really laying the groundwork to integrate urban ag and rural food systems to hasten the demise of the 3, 000-mile salad by creating a national brand of urban-grown produce in every city, that uses regional growing power and augments it with indoor growing facilities, owned and operated by small growers, where now there are only consumers. This can **support** seasonal farmers around metro areas who are losing out because they really can ’ t meet the year-round **demand** for produce. It ’ s not a competition with rural farm ; it ’ s actually reinforcements. It allies in a really positive and economically viable food system. The goal is to meet the cities ’ institutional **demands** for hospitals, **senior** centers, schools, daycare centers, and produce a network of regional jobs, as well. This is smart infrastructure. And how we manage our built environment affects the health and well-being of people every single day. Our municipalities, rural and urban, play the operational course of infrastructure — things like waste disposal, energy **demand**, as well as social costs of unemployment, drop-out rates, incarceration rates and the **impacts** of various public health costs. Smart infrastructure can provide cost-saving ways for municipalities to handle both infrastructure and social needs. And we want to shift the systems that open the doors for people who were formerly tax burdens to become part of the tax base. And imagine a national business model that creates local jobs and smart infrastructure to improve local economic stability. So I ' m hoping you can see a little theme here. These examples indicate a trend. I haven ' t created it, and it ' s not happening by accident. I ' m noticing that it ' s happening all over the country, and the good news is that it ' s growing. And we all need to be **invested** in it. It is an essential pillar to this country ' s recovery. And I call it"hometown security."The **recession** has us reeling and fearful, and there ' s something in the **air** these days that is also very empowering. It ' s a realization that we are the key to our own recovery. Now is the time for us to act in our own communities where we think local and we act local. And when we do that, our neighbors — be they next-door, or in the next state, or in the next country — will be just fine. The sum of the local is the global. Hometown security means rebuilding our natural defenses, putting people to work, restoring our natural systems. Hometown security means creating wealth here at home, instead of destroying it overseas. Tackling social and environmental problems at the same time with the same solution yields great cost savings, wealth generation and national security. Many great and inspiring solutions have been generated across America. The challenge for us now is to identify and **support** countless more. Now, hometown security is about taking care of your own, but it ' s not like the old saying,"charity begins at home."I recently read a book called"Love Leadership"by John Hope Bryant. And it ' s about leading in a world that really does seem to be operating on the basis of fear. And reading that book made me reexamine that theory because I need to explain what I mean by that. See, my dad was a great, great man in many ways. He grew up in the segregated South, escaped lynching and all that during some really hard times, and he provided a really stable home for me and my siblings and a whole bunch of other people that fell on hard times. But, like all of us, he had some problems. And his was gambling, compulsively. To him that phrase,"Charity begins at home,"meant that my payday — or someone else ' s — would just happen to coincide with his lucky day. So you need to help him out. And sometimes I would loan him money from my after-school or summer jobs, and he always had the great intention of paying me back with interest, of course, after he hit it big. And he did sometimes, believe it or not, at a racetrack in Los Angeles — one reason to love L. A. — back in the 1940s. He made 15, 000 dollars cash and bought the house that I grew up in. So I ' m not that unhappy about that. But listen, I did feel obligated to him, and I grew up — then I grew up. And I ' m a grown woman now, and I have learned a few things along the way. To me, charity often is just about giving, because you ' re supposed to, or because it ' s what you ' ve always done, or it ' s about giving until it hurts. I ' m about providing the means to build something that will grow and intensify its original investment and not just require greater giving next year — I ' m not trying to feed the habit. I spent some years watching how good intentions for community empowerment, that were supposed to be there to **support** the community and empower it, actually left people in the same, if not worse, position that they were in before. And over the past 20 years, we ' ve spent record amounts of philanthropic dollars on social problems, yet educational outcomes, malnutrition, incarceration, obesity, diabetes, income disparity, they ' ve all gone up with some exceptions — in particular, infant mortality among people in poverty — but it ' s a great world that we ' re bringing them into as well. And I know a little bit about these issues, because, for many years, I spent a long time in the non-profit industrial complex, and I ' m a recovering executive director, two years clean. But during that time, I realized that it was about projects and developing them on the local level that really was going to do the right thing for our communities. But I really did struggle for financial **support**. The greater our success, the less money came in from foundations. And I tell you, being on the **TED** stage and winning a MacArthur in the same exact year gave everyone the impression that I had arrived. And by the time I ’ d moved on, I was actually covering a third of my agency ’ s budget deficit with speaking fees. And I think because early on, frankly, my programs were just a little bit ahead of their time. But since then, the park that was just a dump and was featured at a TED2006 Talk became this little thing. But I did in fact get married in it. Over here. There goes my dog who led me to the park in my wedding. The South Bronx Greenway was also just a drawing on the stage back in 2006. Since then, we got about 50 million dollars in stimulus package money to come and get here. And we love this, because I love construction now, because we ’ re watching these things actually happen. So I want everyone to understand the critical importance of shifting charity into enterprise. I started my firm to help communities across the country realize their own potential to improve everything about the quality of life for their people. Hometown security is next on my to-do list. What we need are people who see the value in **investing** in these types of local enterprises, who will partner with folks like me to identify the growth trends and **climate** adaptation as well as understand the growing social costs of business as usual. We need to work together to embrace and repair our land, repair our power systems and repair ourselves. It ’ s time to stop building the shopping malls, the prisons, the stadiums and other tributes to all of our **collective** failures. It is time that we start building living monuments to hope and possibility. Thank you very much.

Text Sample 563: POS Dim 4 – 2010 – Score 16.00 – 2010/t002785.txt

So I ’ m going to talk to you about you about the political chemistry of oil spills and why this is an incredibly important, long, oily, hot summer, and why we need to keep ourselves from getting distracted. But before I talk about the political chemistry, I actually need to talk about the chemistry of oil. This is a photograph from when I **visited** Prudhoe Bay in Alaska in 2002 to watch the Minerals Management Service testing their **ability** to burn oil spills in ice. And what you see here is, you see a little bit of crude oil, you see some ice cubes, and you see two sandwich baggies of napalm. The napalm is burning there quite nicely. And the thing is, is that oil is really an abstraction for us as the American consumer. We ’ re four percent of the world ’ s population ; we use 25 percent of the world ’ s oil production. And we don ’ t really understand what oil is, until you check out its molecules, And you don ’ t really understand that until you see this **stuff** burn. So this is what happens as that burn gets going. It takes off. It ’ s a big woosh. I highly recommend that you get a chance to see crude oil burn someday, because you will never need to hear another poli sci lecture on the geopolitics of oil again. It ’ ll just bake your retinas. So there it is ; the retinas are baking. Let me tell you a little bit about this chemistry of oil. Oil is a stew of hydrocarbon molecules. It starts of with the very small ones, which are one carbon, four **hydrogen** — that ’ s methane — it just floats off. Then there ’ s all sorts of intermediate ones with middle amounts of carbon. You ’ ve probably heard of benzene rings ; they ’ re very carcinogenic. And it goes all the way over to these big, thick, galumphy ones that have hundreds of carbons, and they have thousands of **hydrogens**, and they have vanadium and heavy metals and sulfur and all kinds of craziness hanging off the sides of them. Those are called the asphaltenes ; they ' re an ingredient in asphalt. They ' re very important in oil spills. Let me tell you a little bit about the chemistry of oil in water. It is this chemistry that makes oil so disastrous. Oil doesn ' t sink, it floats. If it sank, it would be a whole different story as far as an oil spill. And the other thing it does is it spreads out the moment it hits the water. It spreads out to be really thin, so you have a hard time corralling it. The next thing that happens is the light ends evaporate, and some of the toxic things float into the water column and kill fish **eggs** and smaller fish and things like that, and shrimp. And then the asphaltenes — and this is the crucial thing — the asphaltenes get whipped by the waves into a frothy emulsion, something like mayonnaise. It triples the amount of oily, messy goo that you have in the water, and it makes it very hard to handle. It also makes it very viscous. When the Prestige sank off the coast of Spain, there were big, floating cushions the size of sofa cushions of emulsified oil, with the consistency, or the viscosity, of chewing gum. It ' s incredibly hard to clean up. And every single oil is different when it hits water. When the chemistry of the oil and water also hits our politics, it ' s absolutely explosive. For the first time, American consumers will kind of see the oil supply chain in front of themselves. They have a"eureka!"moment, when we suddenly understand oil in a different context. So I ' m going to talk just a little bit about the origin of these politics, because it ' s really crucial to understanding why this summer is so important, why we need to stay focused. Nobody gets up in the morning and thinks,"Wow! I ' m going to go buy some three- carbon-to-12-carbon molecules to put in my tank and drive happily to work."No, they think,"Ugh. I have to go buy gas. I ' m so angry about it. The oil companies are ripping me off. They set the prices, and I don ' t even know. I am helpless over this."And this is what happens to us at the gas pump — and actually, gas pumps are specifically designed to diffuse that anger. You might notice that many gas pumps, including this one, are designed to look like ATMs. I ' ve talked to engineers. That ' s specifically to diffuse our anger, because supposedly we feel good about ATMs. That shows you how bad it is. But actually, I mean, this feeling of helplessness comes in because most Americans actually feel that oil prices are the result of a conspiracy, not of the vicissitudes of the world oil market. And the thing is, too, is that we also feel very helpless about the amount that we consume, which is somewhat reasonable, because in fact, we have designed this system where, if you want to get a job, it ' s much more important to have a car that runs, to have a job and keep a job, than to have a GED. And that ' s actually very perverse. Now there ' s another perverse thing about the way we buy gas, which is that we ' d rather be doing anything else. This is BP ' s gas station in downtown Los Angeles. It is green. It is a shrine to greenishness."Now,"you think,"why would something so lame work on people so smart?"Well, the reason is, is because, when we ' re buying gas, we ' re very **invested** in this sort of cognitive dissonance. I mean, we ' re angry at the one hand and we want to be somewhere else. We don ' t want to be buying oil ; we want to be doing something green. And we get kind of in on our own con. I mean — and this is funny, it looks funny here. But in fact, that ' s why the slogan"beyond petroleum"worked. But it ' s an inherent part of our energy **policy**, which is we don ' t talk about reducing the amount of oil that we use. We talk about energy independence. We talk about **hydrogen** cars. We talk about biofuels that haven ' t been invented yet. And so, cognitive dissonance is part and parcel of the way that we deal with oil, and it ' s really important to dealing with this oil spill. Okay, so the politics of oil are very moral in the United States. The oil industry is like a huge, gigantic octopus of engineering and finance and everything else, but we actually see it in very moral terms. This is an early- on photograph — you can see, we had these gushers. Early journalists looked at these spills, and they said,”This is a filthy industry.”But they also saw in it that people were getting rich for doing nothing. They weren ’ t farmers, they were just getting rich for **stuff** coming out of the ground. It ’ s the”Beverly Hillbillies,”basically. But in the beginning, this was seen as a very morally problematic thing, long before it became funny. And then, of course, there was John D. Rockefeller. And the thing about John D. is that he went into this chaotic wild-east of oil industry, and he rationalized it into a vertically integrated company, a multinational. It was terrifying ; you think Walmart is a terrifying business model now, imagine what this looked like in the 1860s or 1870s. And it also the kind of root of how we see oil as a conspiracy. But what ’ s really amazing is that Ida Tarbell, the journalist, went in and did a big expos of Rockefeller and actually got the whole antitrust laws put in place. But in many ways, that image of the conspiracy still sticks with us. And here ’ s one of the things that Ida Tarbell said — she said,”He has a thin nose like a thorn. There were no lips. There were puffs under the little colorless eyes with creases running from them.”Okay, so that **guy** is actually still with us. I mean, this is a very pervasive — this is part of our DNA. And then there ’ s this **guy**, okay. So, you might be wondering why it is that, every time we have high oil prices or an oil spill, we call these CEOs down to Washington, and we sort of pepper them with questions in public and we try to shame them. And this is something that we ’ ve been doing since 1974, when we first asked them,”Why are there these obscene profits?”And we ’ ve sort of personalized the whole oil industry into these CEOs. And we take it as, you know — we look at it on a moral level, rather than looking at it on a legal and financial level. And so I ’ m not saying these **guys** aren ’ t liable to answer questions — I ’ m just saying that, when you focus on whether they are or are not a bunch of greedy bastards, you don ’ t actually get around to the point of making laws that are either going to either change the way they operate, or you ’ re going to get around to really reducing the amount of oil and reducing our dependence on oil. So I ’ m saying this is kind of a distraction. But it makes for good theater, and it ’ s powerfully cathartic as you probably saw last week. So the thing about water oil spills is that they are very politically galvanizing. I mean, these pictures — this is from the Santa Barbara spill. You have these pictures of birds. They really influence people. When the Santa Barbara spill happened in 1969, it formed the environmental movement in its modern form. It started Earth Day. It also put in place the National Environmental Policy Act, the Clean Air Act, the Clean Water Act. Everything that we are really stemmed from this period. I think it ’ s important to kind of look at these pictures of the birds and understand what happens to us. Here we are normally ; we ’ re standing at the gas pump, and we ’ re feeling kind of helpless. We look at these pictures and we understand, for the first time, our role in this supply chain. We connect the dots in the supply chain. And we have this kind of — as voters, we have kind of a”eureka!”moment. This is why these moments of these oil spills are so important. But it ’ s also really important that we don ’ t get distracted by the theater or the morals of it. We actually need to go in and work on the roots of the problem. One of the things that happened with the two previous oil spills was that we really worked on some of the symptoms. We were very reactive, as opposed to being proactive about what happened. And so what we did was, actually, we made moratoriums on the east and **west** coasts on drilling. We stopped drilling in ANWR, but we didn ’ t actually reduce the amount of oil that we consumed. In fact, it ’ s continued to increase. The only thing that really reduces the amount of oil that we consume is much higher prices. As you can see, our own production has fallen off as our reservoirs have gotten old and expensive to drill out. We only have two percent of the world ’ s oil reserves ; 65 percent of them are in the

Persian Gulf. One of the things that ' s happened because of this is that, since 1969, the country of Nigeria, or the part of Nigeria that pumps oil, which is the delta — which is two times the size of Maryland — has had thousands of oil spills a year. I mean, we ' ve essentially been exporting oil spills when we import oil from places without tight environmental regulations. That has been the equivalent of an Exxon Valdez spill every year since 1969. And we can wrap our heads around the spills, because that ' s what we see here, but in fact, these **guys** actually live in a war zone. There ' s a thousand battle-related deaths a year in this area twice the size of Maryland, and it ' s all related to the oil. And these **guys**, I mean, if they were in the U. S., they might be actually here in this room. They have degrees in political science, degrees in business — they ' re entrepreneurs. They don ' t actually want to be doing what they ' re doing. And it ' s sort of one of the other groups of people who pay a price for us. The other thing that we ' ve done, as we ' ve continued to increase **demand**, is that we kind of play a shell game with the costs. One of the places we put in a big oil project in Chad, with Exxon. So the U. S. taxpayer paid for it ; the World Bank, Exxon paid for it. We put it in. There was a tremendous banditry problem. I was there in 2003. We were driving along this dark, dark road, and the **guy** in the green stepped out, and I was just like,"Ahhh! This is it."And then the **guy** in the Exxon uniform stepped out, and we realized it was okay. They have their own private sort of army around them at the oil fields. But at the same time, Chad has become much more unstable, and we are not paying for that price at the pump. We pay for it in our taxes on April 15th. We do the same thing with the price of policing the Persian Gulf and keeping the shipping lanes open. This is 1988 — we actually bombed two Iranian oil platforms that year. That was the beginning of an escalating U. S. involvement there that we do not pay for at the pump. We pay for it on April 15th, and we can ' t even calculate the cost of this involvement. The other place that is sort of **supporting** our dependence on oil and our increased consumption is the Gulf of Mexico, which was not part of the moratoriums. Now what ' s happened in the Gulf of Mexico — as you can see, this is the Minerals Management diagram of wells for gas and oil. It ' s become this intense industrialized zone. It doesn ' t have the same resonance for us that the Arctic National Wildlife Refuge has, but it should, I mean, it ' s a bird sanctuary. Also, every time you buy gasoline in the United States, half of it is actually being refined along the coast, because the Gulf actually has about 50 percent of our refining capacity and a lot of our **marine** terminals as well. So the people of the Gulf have essentially been subsidizing the rest of us through a less-clean environment. And finally, American families also pay a price for oil. Now on the one hand, the price at the pump is not really very high when you consider the actual cost of the oil, but on the other hand, the fact that people have no other transit options means that they pay a large amount of their income into just getting back and forth to work, generally in a fairly crummy car. If you look at people who make $50, 000 a year, they have two kids, they might have three jobs or more, and then they have to really commute. They ' re actually spending more on their car and fuel than they are on taxes or on health care. And the same thing happens at the 50th percentile, around 80, 000. Gasoline costs are a tremendous drain on the American economy, but they ' re also a drain on individual families and it ' s kind of terrifying to think about what happens when prices get higher. So, what I ' m going to talk to you about now is : what do we have to do this time? What are the laws? What do we have to do to keep ourselves focused? One thing is — we need to stay away from the theater. We need to stay away from the moratoriums. We need to focus really back again on the molecules. The moratoriums are fine, but we do need to focus on the molecules on the oil. One of the things that we also need to do, is we need to try to not kind of fool ourselves into thinking that you can have a green world, before you reduce the amount of oil that we use. We need to focus on reducing the oil. What you see in this top drawing is a schematic of how petroleum gets used in the U. S. economy. It comes in on the side — the useful **stuff** is the dark gray, and the un-useful **stuff**, which is called the rejected energy — the waste, goes up to the top. Now you can see that the waste far outweighs the actually useful amount. And one of the things that we need to do is, not only fix the fuel efficiency of our vehicles and make them much more efficient, but we also need to fix the economy in general. We need to remove the perverse incentives to use more fuel. For example, we have an insurance system where the person who drives 20, 000 miles a year pays the same insurance as somebody who drives 3, 000. We actually encourage people to drive more. We have **policies** that reward sprawl — we have all kinds of **policies**. We need to have more mobility choices. We need to make the gas price better reflect the real cost of oil. And we need to shift subsidies from the oil industry, which is at least 10 billion dollars a year, into something that allows middle-class people to find better ways to commute. Whether that ' s getting a much more efficient car and also kind of building markets for new cars and new fuels down the road, this is where we need to be. We need to kind of rationalize this whole thing, and you can find more about this **policy**. It ' s called STRONG, which is"Secure Transportation Reducing Oil Needs Gradually,"and the idea is instead of being helpless, we need to be more strong. They ' re up at NewAmerica. net. What ' s important about these is that we try to move from feeling helpless at the pump, to actually being active and to really sort of thinking about who we are, having kind of that special moment, where we connect the dots actually at the pump. Now supposedly, oil taxes are the third rail of American politics — the no- fly zone. I actually — I agree that a dollar a gallon on oil is probably too much, but I think that if we started this year with three cents a gallon on gasoline, and upped it to six cents next year, nine cents the following year, all the way up to 30 cents by 2020, that we could actually significantly reduce our gasoline consumption, and at the same time we would give people time to prepare, time to respond, and we would be raising money and raising consciousness at the same time. Let me give you a little sense of how this would work. This is a gas receipt, hypothetically, for a year from now. The first thing that you have on the tax is — you have a tax for a stronger America — 33 cents. So you ' re not helpless at the pump. And the second thing that you have is a kind of warning sign, very similar to what you would find on a cigarette pack. And what it says is,"The National Academy of Sciences estimates that every gallon of gas you burn in your car creates 29 cents in health care costs."That ' s a lot. And so this — you can see that you ' re paying considerably less than the health care costs on the tax. And also, the hope is that you start to be connected to the whole greater system. And at the same time, you have a number that you can call to get more information on commuting, or a low-interest loan on a different kind of car, or whatever it is you ' re going to need to actually reduce your gasoline dependence. With this whole sort of suite of **policies**, we could actually reduce our gasoline consumption — or our oil consumption — by 20 percent by 2020. So, three million barrels a day. But in order to do this, one of the things we really need to do, is we need to remember we are people of the hydrocarbon. We need to keep or minds on the molecules and not get distracted by the theater, not get distracted by the cognitive dissonance of the green possibilities that are out there. We need to kind of get down and do the gritty work of reducing our dependence upon this fuel and these molecules. Thank you.

Text Sample 564: POS Dim 4 – 2018 – Score 18.00 – 2018/t000514.txt
It was my first year as an atmospheric science professor at Texas Tech University. We had just moved to Lubbock, Texas, which had recently been named the second most **conservative** city in the entire United States. A colleague asked me to guest teach his undergraduate geology class. I said,"Sure."But when I showed up, the lecture hall was cavernous and dark. As I tracked the history of the carbon cycle through geologic time to present day, most of the students were slumped over, dozing or looking at their **phones**. I ended my talk with a hopeful request for any questions. And one hand shot up right away. I looked encouraging, he stood up, and in a loud voice, he said,"You ' re a democrat, aren ' t you?""No,"I said,"I ' m Canadian."That was my baptism by fire into what has now become a sad fact of life here in the United States and increasingly across Canada as well. The fact that the number one predictor of whether we agree that **climate** is changing, humans are responsible and the **impacts** are increasingly serious and even dangerous, has nothing to do with how much we know about science or even how smart we are but simply where we fall on the political spectrum. Does the thermometer give us a different answer depending on if we ' re liberal or **conservative**? Of course not. But if that thermometer tells us that the planet is warming, that humans are responsible and that to fix this thing, we have to wean ourselves off fossil fuels as soon as possible — well, some people would rather cut off their arm than give the government any further excuse to disrupt their comfortable lives and tell them what to do. But saying,"Yes, it ' s a real problem, but I don ' t want to fix it,"that makes us the bad **guy**, and nobody wants to be the bad **guy**. So instead, we use arguments like,"It ' s just a natural cycle.""It ' s the sun."Or my favorite,"Those **climate** scientists are just in it for the money."I get that at least once a week. But these are just sciencey-sounding smoke screens, that are designed to hide the real reason for our objections, which have nothing to do with the science and everything to do with our ideology and our identity. So when we turn on the TV these days, it seems like pundit X is saying,"It ' s cold outside. Where is global warming now?"And politician Y is saying,"For every scientist who says this thing is real, I can find one who says it isn ' t."So it ' s no surprise that sometimes we feel like **everybody** is saying these myths. But when we look at the data — and the Yale Program on **Climate** [Change] Communication has done public opinion polling across the country now for a number of years — the data shows that actually 70 percent of people in the United States agree that the **climate** is changing. And 70 percent also agree that it will harm plants and animals, and it will harm future generations. But then when we dig down a bit deeper, the rubber starts to hit the road. Only about 60 percent of people think it will affect people in the United States. Only 40 percent of people think it will affect us personally. And then when you ask people,"Do you ever talk about this?"two-thirds of people in the entire United States say,"Never."And even worse, when you say,"Do you hear the media talk about this?"Over three-quarters of people say no. So it ' s a vicious cycle. The planet warms. Heat waves get stronger. Heavy precipitation gets more frequent. Hurricanes get more intense. Scientists release yet another doom-filled report. Politicians push back even more strongly, repeating the same sciencey-sounding myths. What can we do to break this vicious cycle? The number one thing we can do is the exact thing that we ' re not doing : talk about it. But you might say,"I ' m not a scientist. How am I supposed to talk about radiative forcing or cloud parametrization in **climate** models?"We don ' t need to be talking about more science ; we ' ve been talking about the science for over 150 years.

Did you know that it ’ s been 150 years or more since the 1850s, when **climate** scientists first discovered that digging up and burning coal and gas and oil is producing heat- trapping gases that is wrapping an extra blanket around the planet? That ’ s how long we ’ ve known. It ’ s been 50 years since scientists first formally warned a US president of the dangers of a changing **climate**, and that president was Lyndon B. Johnson. And what ’ s more, the social science has taught us that if people have built their identity on rejecting a certain set of facts, then arguing over those facts is a personal attack. It causes them to dig in deeper, and it digs a trench, rather than building a bridge. So if we aren ’ t supposed to talk about more science, or if we don ’ t need to talk about more science, then what should we be talking about? The most important thing to do is, instead of starting up with your head, with all the data and facts in our head, to start from the heart, to start by talking about why it matters to us, to begin with genuinely shared values. Are we both parents? Do we live in the same community? Do we enjoy the same outdoor activities : hiking, biking, fishing, even hunting? Do we care about the economy or national security? For me, one of the most foundational ways I found to connect with people is through my faith. As a Christian, I believe that God created this incredible planet that we live on and gave us responsibility over every living thing on it. And I furthermore believe that we are to care for and love the least fortunate among us, those who are already suffering the **impacts** of poverty, hunger, disease and more. If you don ’ t know what the values are that someone has, have a conversation, get to know them, figure out what makes them tick. And then once we have, all we have to do is connect the dots between the values they already have and why they would care about a changing **climate**. I truly believe, after thousands of conversations that I ’ ve had over the past decade and more, that just about every single person in the world already has the values they need to care about a changing **climate**. They just haven ’ t connected the dots. And that ’ s what we can do through our conversation with them. The only reason why I care about a changing **climate** is because of who I already am. I ’ m a mother, so I care about the future of my child. I live in West Texas, where water is already scarce, and **climate** change is **impacting** the availability of that water. I ’ m a Christian, I care about a changing **climate** because it is, as the military calls it, a”threat multiplier.”It takes those issues, like poverty and hunger and disease and lack of access to clean water and even political **crises** that lead to **refugee crises** — it takes all of these issues and it exacerbates them, it makes them worse. I ’ m not a Rotarian. But when I gave my first talk at a Rotary Club, I walked in and they had this giant banner that had the Four-Way Test on it. Is it the truth? Absolutely. Is it fair? Heck, no, that ’ s why I care most about **climate** change, because it is absolutely unfair. Those who have contributed the least to the problem are bearing the brunt of the **impacts**. It went on to ask : Would it be beneficial to all, would it build goodwill? Well, to fix it certainly would. So I took my talk, and I reorganized it into the Four-Way Test, and then I gave it to this group of **conservative** businesspeople in West Texas. And I will never forget at the end, a local bank owner came up to me with the most bemused look on his face. And he said,”You know, I wasn ’ t sure about this whole global warming thing, but it passed the Four-Way Test.”These values, though — they have to be genuine. I was giving a talk at a Christian college a number of years ago, and after my talk, a fellow scientist came up and he said,”I need some help. I ’ ve been really trying hard to get my foot in the door with our local churches, but I can ’ t seem to get any traction. I want to talk to them about why **climate** change matters.”So I said,”Well, the best thing to do is to start with the denomination that you ’ re part of, because you share the most values with those people. What type of church do you attend?””Oh, I don ’ t attend any church, I ’ m an atheist,”he said. I said,”Well, in that case, starting with a faith community is probably not the best idea. Let ’ s talk about what you do enjoy doing, what you are **involved** in.”And we were able to identify a community group that he was part of, that he could start with. The bottom line is, we don ’ t have to be a liberal tree hugger to care about a changing **climate**. All we have to be is a human living on this planet. Because no matter where we live, **climate** change is already affecting us today. If we live along the coasts, in many places, we ’ re already seeing”sunny-day flooding.”If we live in western North America, we ’ re seeing much greater area being burned by wildfires. If we live in many coastal locations, from the Gulf of Mexico to the South Pacific, we are seeing stronger hurricanes, typhoons and cyclones, powered by a warming ocean. If we live in Texas or if we live in Syria, we ’ re seeing **climate** change supersize our droughts, making them more frequent and more severe. Wherever we live, we ’ re already being affected by a changing **climate**. So you might say,”OK, that ’ s good. We can talk **impacts**. We can scare the pants off people, because this thing is serious.”And it is, believe me. I ’ m a scientist, I know. But fear is not what is going to motivate us for the long-term, sustained change that we need to fix this thing. Fear is designed to help us run away from the bear. Or just run faster than the person beside us. What we need to fix this thing is rational hope. Yes, we absolutely do need to recognize what ’ s at stake. Of course we do. But we need a vision of a better future — a future with abundant energy, with a stable economy, with resources available to all, where our lives are not worse but better than they are today. There are solutions. And that ’ s why the second important thing that we have to talk about is solutions — practical, viable, accessible, attractive solutions. Like what? Well, there ’ s no silver bullet, as they say, but there ’ s plenty of silver buckshot. There ’ s simple solutions that save us money and reduce our carbon footprint at the same time. Yes, light bulbs. I love my plug-in car. I ’ d like some solar shingles. But imagine if every home came with a switch beside the front door, that when you left the house, you could turn off everything except your fridge. And maybe the DVR. Lifestyle choices : eating local, eating lower down the food chain and reducing food waste, which at the global scale, is one of the most important things that we can do to fix this problem. I ’ m a **climate** scientist, so the irony of traveling around to talk to people about a changing **climate** is not lost on me. The biggest part of my personal carbon footprint is my travel. And that ’ s why I carefully collect my invitations. I usually don ’ t go anywhere unless I have a critical mass of invitations in one place — anywhere from three to four to sometimes even as many as 10 or 15 talks in a given place — so I can minimize the **impact** of my carbon footprint as much as possible. And I ’ ve transitioned nearly three-quarters of the talks I give to video. Often, people will say,”Well, we ’ ve never done that before.”But I say,”Well, let ’ s give it a try, I think it could work.”Most of all, though, we need to talk about what ’ s already happening today around the world and what could happen in the future. Now, I live in Texas, and Texas has the highest carbon emissions of any state in the United States. You might say,”Well, what can you talk about in Texas?”The answer is : a lot. Did you know that in Texas there ’ s over 25, 000 jobs in the wind energy industry? We are almost up to 20 percent of our electricity from clean, renewable sources, most of that wind, though solar is growing quickly. The largest army base in the United States, Fort Hood, is, of course, in Texas. And they ’ ve been powered by wind and solar energy now, because it ’ s saving taxpayers over 150 million dollars. Yes. What about those who don ’ t have the resources that we have? In sub-Saharan Africa, there are hundreds of millions of people who don ’ t have access to any type of energy except kerosine, and it ’ s very expensive. Around the entire world, the fastest-growing type of new energy today is solar. And they have plenty of solar. So social **impact** investors, nonprofits, even corporations are going in and using innovative new microfinancing schemes, like, pay-as-you-go solar, so that people can buy the power they need in increments, sometimes even on their cell **phone**. One company, Azuri, has distributed tens of thousands of units across 11 countries, from Rwanda to Uganda. They estimate that they ' ve powered over 30 million hours of electricity and over 10 million hours of cell **phone** charging. What about the giant growing economies of China and India? Well, **climate impacts** might seem a little further down the road, but **air** quality **impacts** are right here today. And they know that clean energy is essential to powering their future. So China is **investing** hundreds of billions of dollars in clean energy. They ' re flooding coal mines, and they ' re putting floating solar panels on the surface. They also have a panda-shaped solar farm. Yes, they ' re still burning coal. But they ' ve shut down all the coal plants around Beijing. And in India, they ' re looking to replace a quarter of a billion incandescent light bulbs with LEDs, which will save them seven billion dollars in energy costs. They ' re investing in green jobs, and they ' re looking to decarbonize their entire vehicle fleet. India may be the first country to industrialize without relying primarily on fossil fuels. The world is changing. But it just isn ' t changing fast enough. Too often, we picture this problem as a giant boulder sitting at the bottom of a hill, with only a few hands on it, trying to roll it up the hill. But in reality, that boulder is already at the top of the hill. And it ' s got hundreds of millions of hands, maybe even billions on it, pushing it down. It just isn ' t going fast enough. So how do we speed up that giant boulder so we can fix **climate** change in time? You guessed it. The number one way is by talking about it. The bottom line is this : **climate** change is affecting you and me right here, right now, in the places where we live. But by working together, we can fix it. Sure, it ' s a daunting problem. Nobody knows that more than us **climate** scientists. But we can ' t give in to despair. We have to go out and actively look for the hope that we need, that will inspire us to act. And that hope begins with a conversation today. Thank you.

Text Sample 565: POS Dim 4 – 2018 – Score 15.00 – 2018/t002706.txt

Bryn Freedman : You ' re a **guy** whose company funds these AI programs and **invests**. So why should we **trust** you to not have a bias and tell us something really useful for the rest of us about the future of work? Roy Bahat : Yes, I am. And when you wake up in the morning and you read the newspaper and it says,"The robots are coming, they may take all our jobs,"as a start-up investor focused on the future of work, our fund was the first one to say artificial **intelligence** should be a focus for us. So I woke up one morning and read that and said,"Oh, my gosh, they ' re talking about me. That ' s me who ' s doing that."And then I thought : wait a minute. If things continue, then maybe not only will the start-ups in which we **invest** struggle because there won ' t be people to have jobs to pay for the things that they make and buy them, but our economy and society might struggle, too. And look, I should be the **guy** who sits here and tells you,"Everything is going to be fine. It ' s all going to work out great. Hey, when they introduced the ATM machine, years later, there ' s more tellers in banks."It ' s true. And yet, when I looked at it, I thought,"This is going to accelerate. And if it does accelerate, there ' s a chance the center doesn ' t hold."But I figured somebody must know the answer to this ; there are so many ideas out there. And I read all the books, and I went to the **conferences**, and at one point, we counted more than 100 efforts to study the future of work. And it was a frustrating experience, because I ’ d hear the same back-and-forth over and over again :”The robots are coming!”And then somebody else would say,”Oh, don ’ t worry about that, they ’ ve always said that and it turns out OK.”Then somebody else would say,”Well, it ’ s really about the meaning of your job, anyway.”And then **everybody** would shrug and go off and have a drink. And it felt like there was this Kabuki theater of this discussion, where nobody was talking to each other. And many of the people that I knew and worked with in the technology world were not speaking to **policy** makers ; the **policy** makers were not speaking to them. And so we partnered with a nonpartisan think tank NGO called New America to study this issue. And we brought together a group of people, including an AI czar at a technology company and a video game designer and a heartland **conservative** and a Wall Street investor and a socialist magazine editor — literally, all in the same room ; it was occasionally awkward — to try to figure out what is it that will happen here. The question we asked was simple. It was : What is the effect of technology on work going to be? And we looked out 10 to 20 years, because we wanted to look out far enough that there could be real change, but soon enough that we weren ’ t talking about teleportation or anything like that. And we recognized — and I think every year we ’ re reminded of this in the world — that predicting what ’ s going to happen is hard. So instead of predicting, there are other things you can do. You can try to imagine alternate possible futures, which is what we did. We did a scenario-planning exercise, and we imagined cases where no job is **safe**. We imagined cases where every job is **safe**. And we imagined every distinct possibility we could. And the result, which really surprised us, was when you think through those futures and you think what should we do, the answers about what we should do actually turn out to be the same, no matter what happens. And the irony of looking out 10 to 20 years into the future is, you realize that the things we want to act on are actually already happening right now. The automation is right now, the future is right now. BF : So what does that mean, and what does that tell us? If the future is now, what is it that we should be doing, and what should we be thinking about? RB : We have to understand the problem first. And so the data are that as the economy becomes more productive and individual workers become more productive, their wages haven ’ t risen. If you look at the proportion of prime working-age men, in the United States at least, who work now versus in 1960, we have three times as many men not working. And then you hear the stories. I sat down with a group of Walmart workers and said,”What do you think about this cashier, this futuristic self-checkout thing?”They said,”That ’ s nice, but have you heard about the cash recycler? That ’ s a machine that ’ s being installed right now, and is eliminating two jobs at every Walmart right now.”And so we just thought,”Geez. We don ’ t understand the problem.”And so we looked at the voices that were the ones that were excluded, which is all of the people affected by this change. And we decided to listen to them, sort of”automation and its discontents.”And I ’ ve spent the last couple of years doing that. I ’ ve been to Flint, Michigan, and Youngstown, Ohio, talking about entrepreneurs, trying to make it work in a very different environment from New York or San Francisco or London or Tokyo. I ’ ve been to prisons twice to talk to inmates about their jobs after they leave. I ’ ve sat down with truck drivers to ask them about the self-driving truck, with people who, in addition to their full-time job, care for an aging relative. And when you talk to people, there were two themes that came out loud and clear. The first one was that people are less looking for more money or get out of the fear of the robot taking their job, and they just want something stable. They want something predictable. So if you survey people and ask them what they want out of work, for **everybody** who makes less than 150, 000 dollars a year, they ’ ll take a more stable and secure income, on average, over earning more money. And if you think about the fact that not only for all of the people across the earth who don ’ t earn a living, but for those who do, the vast majority earn a different amount from month to month and have an instability, all of a sudden you realize,”Wait a minute. We have a real problem on our hands.”And the second thing they say, which took us a longer time to understand, is they say they want dignity. And that concept of self-worth through work emerged again and again and again in our conversations. BF : So, I certainly appreciate this answer. But you can ’ t eat dignity, you can ’ t clothe your children with self-esteem. So, what is that, how do you reconcile — what does dignity mean, and what is the relationship between dignity and stability? RB : You can ’ t eat dignity. You need stability first. And the good news is, many of the conversations that are happening right now are about how we solve that. You know, I ’ m a proponent of studying guaranteed income, as one example, conversations about how health care gets provided and other benefits. Those conversations are happening, and we ’ re at a time where we must figure that out. It is the **crisis** of our era. And my point of view after talking to people is that we may do that, and it still might not be enough. Because what we need to do from the beginning is understand what is it about work that gives people dignity, so they can live the lives that they want to live. And so that concept of dignity is... it ’ s difficult to get your hands around, because when many people hear it — especially, to be honest, rich people — they hear”meaning.”They hear”My work is important to me.”And again, if you survey people and you ask them,”How important is it to you that your work be important to you?”only people who make 150, 000 dollars a year or more say that it is important to them that their work be important. BF : Meaning, meaningful? RB : Just defined as,”Is your work important to you?”Whatever somebody took that to mean. And yet, of course dignity is essential. We talked to truck drivers who said,”I saw my cousin drive, and I got on the open road and it was amazing. And I started making more money than people who went to college.”Then they ’ d get to the end of their thought and say something like,”People need their fruits and vegetables in the morning, and I ’ m the **guy** who gets it to them.”We talked to somebody who, in addition to his job, was caring for his aunt. He was making plenty of money. At one point we just asked,”What is it about caring for your aunt? Can ’ t you just pay somebody to do it?”He said,”My aunt doesn ’ t want somebody we pay for. My aunt wants me.”So there was this concept there of being needed. If you study the word”dignity,”it ’ s fascinating. It ’ s one of the oldest words in the English language, from antiquity. And it has two meanings : one is self- worth, and the other is that something is suitable, it ’ s fitting, meaning that you ’ re part of something greater than yourself, and it connects to some broader whole. In other words, that you ’ re needed. BF : So how do you answer this question, this concept that we don ’ t pay teachers, and we don ’ t pay eldercare workers, and we don ’ t pay people who really care for people and are needed, enough? RB : Well, the good news is, people are finally asking the question. So as AI investors, we often get **phone** calls from foundations or CEOs and boardrooms saying,”What do we do about this?”And they used to be asking,”What do we do about introducing automation?”And now they ’ re asking,”What do we do about self-worth?”And they know that the employees who work for them who have a spouse who cares for somebody, that dignity is essential to their **ability** to just do their job. I think there ’ s two kinds of answers : there ’ s the money side of just making your life work. That ’ s stability. You need to eat. And then you think about our culture more broadly, and you ask : Who do we make into heroes? And, you know, what I want is to see the magazine cover that is the person who is the heroic caregiver. Or the Netflix series that dramatizes the person who makes all of our other lives work so we can do the things we do. Let ’ s make heroes out of those people. That ’ s the Netflix show that I would binge. And we ’ ve had chroniclers of this before — Studs Terkel, the oral history of the working experience in the United States. And what we need is the experience of needing one another and being connected to each other. Maybe that ’ s the answer for how we all fit as a society. And the thought exercise, to me, is : if you were to go back 100 years and have people — my grandparents, great- grandparents, a tailor, worked in a mine — they look at what all of us do for a living and say,”That ’ s not work.”We sit there and type and talk, and there ’ s no danger of getting hurt. And my guess is that if you were to imagine 100 years from now, we ’ ll still be doing things for each other. We ’ ll still need one another. And we just will think of it as work. The entire thing I ’ m trying to say is that dignity should not just be about having a job. Because if you say you need a job to have dignity, which many people say, the second you say that, you say to all the parents and all the teachers and all the caregivers that all of a sudden, because they ’ re not being paid for what they ’ re doing, it somehow lacks this essential human quality. To me, that ’ s the great puzzle of our time : Can we figure out how to provide that stability throughout life, and then can we figure out how to create an inclusive, not just racially, **gender**, but multi-generationally inclusive — I mean, every different human experience included — in this way of understanding how we can be needed by one another. BF : Thank you. RB : Thank you. BF : Thank you very much for your participation.

Text Sample 566: POS Dim 4 – 2018 – Score 15.00 – 2018/t000552.txt

So, let me thank you for the opportunity to talk about the biggest **international** story of your professional lifetime, which is also the most important **international** challenge the world will face for as far as the eye can see. The story, of course, is the rise of China. Never before have so many people risen so far so fast, on so many different dimensions. The challenge is the **impact** of China ’ s rise — the discombobulation this will cause the Unites States and the **international** order, of which the US has been the principal architect and guardian. The past 100 years have been what historians now call an”American Century.”Americans have become accustomed to their place at the top of every pecking order. So the very idea of another country that could be as big and strong as the US — or bigger — strikes many Americans as an assault on who they are. For perspective on what we ’ re now seeing in this rivalry, it ’ s useful to locate it on the larger map of history. The past 500 years have seen 16 cases in which a rising power threatened to displace a ruling power. Twelve of those ended in war. So just in November, we ’ ll all pause to mark the 100th anniversary of the final day of a war that became so encompassing, that it required historians to create an entirely new category : world war. So, on the 11th hour of the 11th day of the 11th month in 1918, the guns of World War I fell silent, but 20 million individuals lay dead. I know that this is a sophisticated audience, so you know about the rise of China. I ’ m going to focus, therefore, on the **impact** of China ’ s rise, on the US, on the **international** order and on the prospects for war and peace. But having taught at Harvard over many years, I ’ ve learned that from time to time, it ’ s useful to take a short pause, just to make sure we ’ re all on the same page. The way I do this is, I call a time-out, I give students a pop quiz — ungraded, of course. So, let ’ s try this. Time-out, pop quiz. Question : forty years ago, 1978, China sets out on its march to the market. At that point, what percentage of China ’ s one billion citizens were struggling to survive on less than two dollars a day? Take a guess — 25 percent? Fifty? Seventy-five? Ninety. What do you think? Ninety. Nine out of every 10 on less than two dollars a day. Twenty eighteen, 40 years later. What about the numbers? What ' s your bet? Take a look. Fewer than one in 100 today. And China ' s president has promised that within the next three years, those last tens of millions will have been raised up above that threshold. So it ' s a miracle, actually, in our lifetime. Hard to believe. But brute facts are even harder to ignore. A nation that didn ' t even appear on any of the **international** league tables 25 years ago has soared, to rival — and in some areas, surpass — the United States. Thus, the challenge that will shape our world : a seemingly unstoppable rising China accelerating towards an apparently immovable ruling US, on course for what could be the grandest collision in history. To help us get our minds around this challenge, I ' m going to introduce you to a great thinker, I ' m going to present a big idea, and I ' m going to pose a most consequential question. The great thinker is Thucydides. Now, I know his name is a mouthful, and some people have trouble pronouncing it. So, let ' s do it, one, two, three, together : Thucydides. One more time : Thucydides. So who was Thucydides? He was the father and founder of history. He wrote the first-ever history book. It ' s titled"The History of the Peloponnesian War,"about the war in Greece, 2500 years ago. So if nothing else today, you can tweet your friends,"I met a great thinker. And I can even pronounce his name : Thucydides."So, about this war that destroyed classical Greece, Thucydides wrote famously :"It was the rise of Athens and the fear that this instilled in Sparta that made the war inevitable."So the rise of one and the **reaction** of the other create a toxic cocktail of pride, arrogance, paranoia, that drug them both to war. Which brings me to the big idea : Thucydides ' s Trap."Thucydides ' s Trap"is a term I coined several years ago, to make vivid Thucydides ' s insight. Thucydides ' s Trap is the dangerous dynamic that occurs when a rising power threatens to displace a ruling power, like Athens — or Germany 100 years ago, or China today — and their **impact** on Sparta, or Great Britain 100 years ago, or the US today. As Henry Kissinger has said, once you get this idea, this concept of Thucydides ' s Trap in your head, it will provide a lens for helping you look through the news and noise of the day to understand what ' s actually going on. So, to the most consequential question about our world today : Are we going to follow in the footsteps of history? Or can we, through a combination of imagination and common sense and courage find a way to manage this rivalry without a war nobody wants, and **everybody** knows would be catastrophic? Give me five minutes to unpack this, and later this afternoon, when the next news story pops up for you about China doing this, or the US reacting like that, you will be able to have a better understanding of what ' s going on and even to explain it to your friends. So as we saw with this flipping the pyramid of poverty, China has actually soared. It ' s meteoric. Former Czech president, Vaclav Havel, I think, put it best. He said,"All this has happened so fast, we haven ' t yet had time to be astonished."To remind myself how astonished I should be, I occasionally look out the window in my office in Cambridge at this bridge, which goes across the Charles River, between the Kennedy School and Harvard Business School. In 2012, the State of Massachusetts said they were going to renovate this bridge, and it would take two years. In 2014, they said it wasn ' t finished. In 2015, they said it would take one more year. In 2016, they said it ' s not finished, we ' re not going to tell you when it ' s going to be finished. Finally, last year, it was finished — three times over budget. Now, compare this to a similar bridge that I drove across last month in Beijing. It ' s called the Sanyuan Bridge. In 2015, the Chinese decided they wanted to renovate that bridge. It actually has twice as many lanes of traffic. How long did it take for them to complete the project? Twenty fifteen, what do you bet? Take a guess — OK, three — Take a look. The answer is 43 hours. Graham Allison : Now, of course, that couldn ' t happen in New York. Behind this speed in execution is a purpose-driven **leader** and a government that works. The most ambitious and most competent **leader** on the **international** stage today is Chinese President Xi Jinping. And he ' s made no secret about what he wants. As he said when he became president six years ago, his goal is to make China great again — a banner he raised long before Donald Trump picked up a version of this. To that end, Xi Jinping has announced specific targets for specific dates : 2025, 2035, 2049. By 2025, China means to be the **dominant** power in the major market in 10 leading technologies, including driverless cars, robots, artificial **intelligence**, quantum computing. By 2035, China means to be the innovation **leader** across all the advanced technologies. And by 2049, which is the 100th anniversary of the founding of the People ' s Republic, China means to be unambiguously number one, including, [says] Xi Jinping, an army that he calls"Fight and Win."So these are audacious goals, but as you can see, China is already well on its way to these objectives. And we should remember how fast our world is changing. Thirty years ago, the World Wide Web had not yet even been invented. Who will feel the **impact** of this rise of China most directly? Obviously, the current number one. As China gets bigger and stronger and richer, technologically more advanced, it will inevitably bump up against American positions and prerogatives. Now, for red-blooded Americans — and especially for red-necked Americans like me ; I ' m from North Carolina — there ' s something wrong with this picture. The USA means number one, that ' s who we are. But again, to repeat : brute facts are hard to ignore. Four years ago, Senator John McCain asked me to testify about this to his Senate Armed Services Committee. And I made for them a chart that you can see, that said, compare the US and China to kids on opposite ends of a seesaw on a playground, each represented by the size of their economy. As late as 2004, China was just half our size. By 2014, its GDP was equal to ours. And on the current trajectory, by 2024, it will be half again larger. The consequences of this tectonic change will be felt everywhere. For example, in the current trade **conflict**, China is already the number one trading partner of all the major Asian countries. Which brings us back to our Greek historian. Harvard ' s"Thucydides ' s Trap Case File"has reviewed the last 500 years of history and found 16 cases in which a rising power threatened to displace a ruling power. Twelve of these ended in war. And the tragedy of this is that in very few of these did either of the protagonists want a war ; few of these wars were initiated by either the rising power or the ruling power. So how does this work? What happens is, a third party ' s provocation forces one or the other to react, and that sets in motion a spiral, which drags the two somewhere they don ' t want to go. If that seems crazy, it is. But it ' s life. Remember World War I. The provocation in that case was the assassination of a second-level figure, Archduke Franz Ferdinand, which then led the Austro- Hungarian emperor to issue an ultimatum to Serbia, they dragged in the various allies, within two months, all of Europe was at war. So imagine if Thucydides were watching planet Earth today. What would he say? Could he find a more appropriate leading man for the ruling power than Donald J Trump? Or a more apt lead for the rising power than Xi Jinping? And he would scratch his head and certainly say he couldn ' t think of more colorful provocateur than North Korea ' s Kim Jong-un. Each seems determined to play his assigned part and is right on script. So finally, we conclude again with the most consequential question, the question that will have the gravest consequences for the rest of our lives : Are Americans and Chinese going to let the forces of history drive us to a war that would be catastrophic for both? Or can we summon the imagination and courage to find a way to survive together, to share the **leadership** in the 21st century, or, as Xi

Jinping [said], to create a new form of great power relations? That ' s the issue I ' ve been pursuing passionately for the last two years. I ' ve had the opportunity to talk and, indeed, to listen to **leaders** of all the relevant governments — Beijing, Washington, Seoul, Tokyo — and to thought **leaders** across the spectrum of both the arts and business. I wish I had more to report. The good news is that **leaders** are increasingly aware of this Thucydidean dynamic and the dangers that it poses. The bad news is that nobody has a feasible plan for escaping history as usual. So it ' s clear to me that we need some ideas outside the box of conventional statecraft — indeed, from another page or another space — which is what brings me to **TED** today and which brings me to a request. This audience includes many of the most creative minds on the planet, who get up in the morning and think not only about how to manage the world we have, but how to create worlds that should be. So I ' m hopeful that as this sinks in and as you reflect on it, some of you are going to have some bold ideas, actually some wild ideas, that when we find, will make a difference in this space. And just to remind you if you do, this won ' t be the first time. Let me remind you of what happened right after World War II. A remarkable group of Americans and Europeans and others, not just from government, but from the world of culture and business, engaged in a **collective** surge of imagination. And what they imagined and what they created was a new **international** order, the order that ' s allowed you and me to live our lives, all of our lives, without great power war and with more prosperity than was ever seen before on the planet. So, a remarkable story. Interestingly, every pillar of this project that produced these results, when first proposed, was rejected by the foreign **policy** establishment as naive or unrealistic. My favorite is the Marshall Plan. After World War II, Americans felt exhausted. They had demobilized 10 million troops, they were focused on an urgent domestic agenda. But as people began to appreciate how devastated Europe was and how aggressive Soviet communism was, Americans eventually decided to tax themselves a percent and a half of GDP every year for four years and send that money to Europe to help reconstruct these countries, including Germany and Italy, whose troops had just been killing Americans. Amazing. This also created the United Nations. Amazing. The Universal Declaration of Human Rights. The World Bank. NATO. All of these elements of an order for peace and prosperity. So, in a word, what we need to do is do it again. And I think now we need a surge of imagination, creativity, informed by history, for, as the philosopher Santayana reminded us, in the end, only those who **refuse** to study history are condemned to repeat it. Thank you.

Text Sample 567: POS Dim 4 – 1994 – Score 6.00 – 1994/t002458.txt

Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ' m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ' s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC microprocessors versus the power of local area networks. And the **interesting** thing about it is that this slide, like so many technology slides that we ' re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ' s basically what I ' m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ' m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well the answer is, if I drew it on a normal curve where, let ' s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ' m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ' t tell us much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log curve, it would look very **stupid**, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as microprocessor technology, then the day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ' s not moving like that. And there ' s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ' d like to ask is, if you look at these exponential curves, they don ' t go on forever. Things just can ' t possibly keep changing as fast as they are. One of two things is going to happen. Either it ' s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ' s going to do this. That ' s about all it can do. Now I ' m an optimist, so I sort of think it ' s probably going to do something like that. If so, that means that what we ' re in the middle of right now is a **transition**. We ' re sort of on this line in a **transition** from the way the world used to be to some new way that the world is. And so what I ' m trying to ask, what I ' ve been asking myself, is what ' s this new way that the world is? What ' s that new state that the world is heading toward? Because the **transition** seems very, very confusing when we ' re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ' s a **conference** in which people talk about the future, and you notice that the future is still at about the year 2000. It ' s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ' s happening there. That **transition** is happening. We can all sense it. And we know that it just doesn ' t make too much sense to think out 30, 50 years because everything ' s going to be so different that a simple extrapolation of what we ' re doing just doesn ' t make any sense at all. So what I would like to talk about is what that could be, what that **transition** could be that we ' re going through. Now in order to do that I ' m going to have to talk about a bunch of **stuff** that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step back and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, the Earth was this big, sterile hunk of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ' s theories that are beginning to understand about how it started with RNA, but I ' m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ’ t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ’ s sort of just a very simple chemical form of life, but when things got **interesting** was when these drops learned a trick about abstraction. Somehow by ways that we don ’ t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for amusement purposes is we can now write things in this code. And I ’ ve got here a little 100 micrograms of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ’ s right here. If I had really been a egotist, I would have put it into a **virus** and released it in the room. So what was the next step? Writing down the DNA was an **interesting** step. And that caused these cells — that kept them happy for another billion years. But then there was another really **interesting** step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ’ t know if you know this, but bacteria can actually exchange DNA. Now that ’ s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the **transition** point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ’ s **interesting** in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ’ re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to evolve so that the **interesting** level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very special structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in the community and special structures that were responsible for recording, understanding, learning information. And that was the brains and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ’ d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these special information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ’ re aware of is human language. It ’ s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ’ m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ’ re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ’ ve done is we, humanity, have started abstracting out. We ’ re going through the same levels that multi-cellular organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. Telephony, computers, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ’ ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of microseconds. And you saw Ty ’ s little evolutionary example where he sort of did a little bit of evolution on the Convolution program right before your eyes. So now we ’ ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and brains, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ’ s what we ’ re seeing here in this explosion of curve. We ’ re seeing this process feeding back on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ’ s really impossible for me to design them in the traditional sense. I don ’ t know what every transistor in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of abstraction and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ’ t quite even understand. One method that ' s particularly **interesting** that I ' ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the microsecond time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,"Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted."So in other words, I define what I wanted. Let ' s say I want to sort numbers, as a simple example I ' ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,"Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ' s reproduce the ones that sorted numbers the best. And let ' s reproduce them by a process of recombination analogous to sex."Take two programs and they produce children by exchanging their subroutines, and the children inherit the traits of the subroutines of the two programs. So I ' ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,"Please repeat that process."Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ' t tell you how they work. I ' ve tried looking at them and telling you how they work. They ' re obscure, weird programs. But they do the job. And in fact, I know, I ' m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,"Oh look. Look at this. It says, ' This plane has hundreds of thousands of tiny parts working together to make you a **safe** flight. ' Doesn ' t that make you feel confident?"In fact, we know that the engineering process doesn ' t work very well when it gets complicated. So we ' re beginning to depend on computers to do a process that ' s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ' t quite understand the options of it. So in a sense, it ' s getting ahead of us. We ' re now using those programs to make much faster computers so that we ' ll be able to run this process much faster. So it ' s feeding back on itself. The thing is becoming faster and that ' s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ' re taking off. And what we are is we ' re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ' re the amoebas and we can ' t quite figure out what the hell this thing is we ' re creating. We ' re right at that point of **transition**. But I think that there really is something coming along after us. I think it ' s very haughty of us to think that we ' re the end product of evolution. And I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

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Chris Anderson : We had Edward Snowden here a couple days ago, and this is response time. And several of you have written to me with questions to ask our guest here from the NSA. So Richard Ledgett is the 15th deputy director of the National Security Agency, and he ’ s a **senior** civilian officer there, acts as its chief operating officer, guiding strategies, setting internal **policies**, and serving as the principal advisor to the director. And all being well, welcome, Rick Ledgett, to **TED**. Richard Ledgett : I ’ m really thankful for the opportunity to talk to folks here. I look forward to the conversation, so thanks for arranging for that. CA : Thank you, Rick. We appreciate you joining us. It ’ s certainly quite a strong statement that the NSA is willing to reach out and show a more open face here. You saw, I think, the talk and interview that Edward Snowden gave here a couple days ago. What did you make of it? RL : So I think it was **interesting**. We didn ’ t realize that he was going to show up there, so kudos to you **guys** for arranging a nice surprise like that. I think that, like a lot of the things that have come out since Mr. Snowden started disclosing classified information, there were some kernels of truth in there, but a lot of extrapolations and half-truths in there, and I ’ m **interested** in helping to address those. I think this is a really important conversation that we ’ re having in the United States and internationally, and I think it is important and of import, and so given that, we need to have that be a fact-based conversation, and we want to help make that happen. CA : So the question that a lot of people have here is, what do you make of Snowden ’ s motivations for doing what he did, and did he have an alternative way that he could have gone? RL : He absolutely did have alternative ways that he could have gone, and I actually think that characterizing him as a whistleblower actually hurts legitimate whistleblowing activities. So what if somebody who works in the NSA — and there are over 35, 000 people who do. They ’ re all great citizens. They ’ re just like your husbands, fathers, sisters, brothers, neighbors, nephews, friends and relatives, all of whom are **interested** in doing the right thing for their country and for our allies internationally, and so there are a variety of venues to address if folks have a concern. First off, there ’ s their supervisor, and up through the supervisory chain within their organization. If folks aren ’ t comfortable with that, there are a number of inspectors general. In the case of Mr. Snowden, he had the option of the NSA inspector general, the Navy inspector general, the Pacific Command inspector general, the Department of Defense inspector general, and the **intelligence** community inspector general, any of whom would have both kept his concerns in classified channels and been happy to address them. He had the option to go to congressional committees, and there are mechanisms to do that that are in place, and so he didn ’ t do any of those things. CA : Now, you had said that Ed Snowden had other avenues for raising his concerns. The comeback on that is a couple of things : one, that he certainly believes that as a contractor, the avenues that would have been available to him as an employee weren ’ t available, two, there ’ s a track record of other whistleblowers, like [Thomas Andrews Drake] being treated pretty harshly, by some views, and thirdly, what he was taking on was not one specific flaw that he ’ d discovered, but programs that had been approved by all three branches of government. I mean, in that circumstance, couldn ’ t you argue that what he did was reasonable? RL : No, I don ’ t agree with that. I think that the — sorry, I ’ m getting feedback through the microphone there — the actions that he took were inappropriate because of the fact that he put people ’ s lives at risk, basically, in the long run, and I know there ’ s been a lot of talk in public by Mr. Snowden and some of the journalists that say that the things that have been disclosed have not put national security and people at risk, and that is categorically not true. They actually do. I think there ’ s also an amazing arrogance to the idea that he knows better than the framers of the Constitution in how the government should be designed and work for separation of powers and the fact that the executive and the legislative branch have to work together and they have checks and balances on each other, and then the judicial branch, which oversees the entire process. I think that ’ s extremely arrogant on his part. CA : Can you give a specific example of how he put people ’ s lives at risk? RL : Yeah, sure. So the things that he ’ s disclosed, the capabilities, and the NSA is a capabilities-based organization, so when we have foreign **intelligence** targets, legitimate things of interest — like, terrorists is the iconic example, but it includes things like human traffickers, drug traffickers, people who are trying to build advanced weaponry, nuclear weapons, and build delivery systems for those, and nation-states who might be executing aggression against their immediate neighbors, which you may have some visibility into some of that that ’ s going on right now, the capabilities are applied in very discrete and measured and controlled ways. So the unconstrained disclosure of those capabilities means that as adversaries see them and recognize,”Hey, I might be vulnerable to this,”they move away from that, and we have seen targets in terrorism, in the nation-state area, in smugglers of various types, and other folks who have, because of the disclosures, moved away from our **ability** to have insight into what they ’ re doing. The net effect of that is that our people who are overseas in dangerous places, whether they ’ re diplomats or military, and our allies who are in similar situations, are at greater risk because we don ’ t see the threats that are coming their way. CA : So that ’ s a general response saying that because of his revelations, access that you had to certain types of information has been shut down, has been closed down. But the concern is that the nature of that access was not necessarily legitimate in the first place. I mean, describe to us this Bullrun program where it ’ s alleged that the NSA specifically weakened security in order to get the type of access that you ’ ve spoken of. RL : So there are, when our legitimate foreign **intelligence** targets of the type that I described before, use the global telecommunications system as their communications methodology, and they do, because it ’ s a great system, it ’ s the most complex system ever devised by man, and it is a wonder, and lots of folks in the room there are responsible for the creation and enhancement of that, and it ’ s just a wonderful thing. But it ’ s also used by people who are working against us and our allies. And so if I ’ m going to pursue them, I need to have the capability to go after them, and again, the controls are in how I apply that capability, not that I have the capability itself. Otherwise, if we could make it so that all the bad **guys** used one corner of the Internet, we could have a domain, badguy. com. That would be awesome, and we could just concentrate all our efforts there. That ’ s not how it works. They ’ re trying to hide from the government ’ s **ability** to isolate and interdict their actions, and so we have to swim in that same space. But I will tell you this. So NSA has two missions. One is the Signals **Intelligence** mission that we ’ ve unfortunately read so much about in the press. The other one is the Information Assurance mission, which is to protect the national security systems of the United States, and by that, that ’ s things like the communications that the president uses, the communications that control our nuclear weapons, the communications that our military uses around the world, and the communications that we use with our allies, and that some of our allies themselves use. And so we make recommendations on standards to use, and we use those same standards, and so we are **invested** in making sure that those communications are secure for their intended purposes. CA : But it sounds like what you ’ re saying is that when it comes to the Internet at large, any strategy is fair game if it improves America ’ s safety. And I think this is partly where there is such a divide of opinion, that there ' s a lot of people in this room and around the world who think very differently about the Internet. They think of it as a momentous invention of humanity, kind of on a par with the Gutenberg press, for example. It ' s the bringer of knowledge to all. It ' s the connector of all. And it ' s viewed in those sort of idealistic terms. And from that lens, what the NSA has done is equivalent to the authorities back in Germany inserting some device into every printing press that would reveal which books people bought and what they read. Can you understand that from that viewpoint, it feels outrageous? RL : I do understand that, and I actually share the view of the utility of the Internet, and I would argue it ' s bigger than the Internet. It is a global telecommunications system. The Internet is a big chunk of that, but there is a lot more. And I think that people have legitimate concerns about the balance between transparency and secrecy. That ' s sort of been couched as a balance between privacy and national security. I don ' t think that ' s the right framing. I think it really is transparency and secrecy. And so that ' s the national and **international** conversation that we ' re having, and we want to participate in that, and want people to participate in it in an informed way. So there are things, let me talk there a little bit more, there are things that we need to be transparent about : our authorities, our processes, our oversight, who we are. We, NSA, have not done a good job of that, and I think that ' s part of the reason that this has been so revelational and so sensational in the media. Nobody knew who we were. We were the No Such Agency, the Never Say Anything. There ' s takeoffs of our logo of an eagle with headphones on around it. And so that ' s the public characterization. And so we need to be more transparent about those things. What we don ' t need to be transparent about, because it ' s bad for the U. S., it ' s bad for all those other countries that we work with and that we help provide information that helps them secure themselves and their people, it ' s bad to expose operations and capabilities in a way that allows the people that we ' re all working against, the generally recognized bad **guys**, to counter those. CA : But isn ' t it also bad to deal a kind of body blow to the American companies that have essentially given the world most of the Internet services that matter? RL : It is. It ' s really the companies are in a tough position, as are we, because the companies, we compel them to provide information, just like every other nation in the world does. Every industrialized nation in the world has a lawful intercept program where they are requiring companies to provide them with information that they need for their security, and the companies that are **involved** have complied with those programs in the same way that they have to do when they ' re operating in Russia or the U. K. or China or India or France, any country that you choose to name. And so the fact that these revelations have been broadly characterized as"you can ' t **trust** company A because your privacy is suspect with them"is actually only accurate in the sense that it ' s accurate with every other company in the world that deals with any of those countries in the world. And so it ' s being picked up by people as a marketing advantage, and it ' s being marketed that way by several countries, including some of our allied countries, where they are saying,"Hey, you can ' t **trust** the U. S., but you can **trust** our telecom company, because we ' re **safe**."And they ' re actually using that to counter the very large technological edge that U. S. companies have in areas like the cloud and Internet-based technologies. CA : You ' re sitting there with the American flag, and the American Constitution guarantees freedom from unreasonable search and seizure. How do you characterize the American citizen ' s right to privacy? Is there such a right? RL : Yeah, of course there is. And we devote an inordinate amount of time and pressure, inordinate and appropriate, actually I should say, amount of time and effort in order to **ensure** that we protect that privacy. and beyond that, the privacy of citizens around the world, it ’ s not just Americans. Several things come into play here. First, we ’ re all in the same network. My communications, I ’ m a user of a particular Internet email service that is the number one email service of choice by terrorists around the world, number one. So I ’ m there right beside them in email space in the Internet. And so we need to be able to pick that apart and find the information that ’ s relevant. In doing so, we ’ re going to necessarily encounter Americans and innocent foreign citizens who are just going about their business, and so we have procedures in place that shreds that out, that says, when you find that, not if you find it, when you find it, because you ’ re certain to find it, here ’ s how you protect that. These are called minimization procedures. They ’ re approved by the attorney general and constitutionally based. And so we protect those. And then, for people, citizens of the world who are going about their lawful business on a day-to-day basis, the president on his January 17 speech, laid out some additional protections that we are providing to them. So I think absolutely, folks do have a right to privacy, and that we work very hard to make sure that that right to privacy is protected. CA : What about foreigners using American companies ’ Internet services? Do they have any privacy rights? RL : They do. They do, in the sense of, the only way that we are able to compel one of those companies to provide us information is when it falls into one of three categories : We can identify that this particular person, identified by a selector of some kind, is associated with counterterrorist or proliferation or other foreign **intelligence** target. CA : Much has been made of the fact that a lot of the information that you ’ ve obtained through these programs is essentially metadata. It ’ s not necessarily the actual words that someone has written in an email or given on a **phone** call. It ’ s who they wrote to and when, and so forth. But it ’ s been argued, and someone here in the audience has talked to a former NSA analyst who said metadata is actually much more invasive than the core data, because in the core data you present yourself as you want to be presented. With metadata, who knows what the conclusions are that are drawn? Is there anything to that? RL : I don ’ t really understand that argument. I think that metadata ’ s important for a couple of reasons. Metadata is the information that lets you find connections that people are trying to hide. So when a terrorist is corresponding with somebody else who ’ s not known to us but is engaged in doing or **supporting** terrorist activity, or someone who ’ s violating **international** sanctions by providing nuclear weapons-related material to a country like Iran or North Korea, is trying to hide that activity because it ’ s illicit activity. What metadata lets you do is connect that. The alternative to that is one that ’ s much less efficient and much more invasive of privacy, which is gigantic amounts of content collection. So metadata, in that sense, actually is privacy- enhancing. And we don ’ t, contrary to some of the **stuff** that ’ s been printed, we don ’ t sit there and grind out metadata profiles of average people. If you ’ re not connected to one of those valid **intelligence** targets, you are not of interest to us. CA : So in terms of the threats that face America overall, where would you place terrorism? RL : I think terrorism is still number one. I think that we have never been in a time where there are more places where things are going badly and forming the petri dish in which terrorists take advantage of the lack of governance. An old boss of mine, Tom Fargo, Admiral Fargo, used to describe it as arcs of instability. And so you have a lot of those arcs of instability in the world right now, in places like Syria, where there ’ s a civil war going on and you have massive numbers, thousands and thousands of foreign fighters who are coming into Syria to learn how to be terrorists and practice that activity, and lots of those people are Westerners who hold passports to European countries or in some cases the United States, and so they are basically learning how to do jihad and have expressed intent to go out and do that later on in their home countries. You ’ ve got places like Iraq, which is suffering from a high level of sectarian violence, again a breeding ground for terrorism. And you have the activity in the Horn of Africa and the Sahel area of Africa. Again, lots of weak governance which forms a breeding ground for terrorist activity. So I think it ’ s very serious. I think it ’ s number one. I think number two is cyber threat. I think cyber is a threat in three ways : One way, and probably the most common way that people have heard about it, is due to the theft of intellectual property, so basically, foreign countries going in, stealing companies ’ secrets, and then providing that information to state-owned enterprises or companies connected to the government to help them leapfrog technology or to gain business **intelligence** that ’ s then used to win contracts overseas. That is a hugely costly set of activities that ’ s going on right now. Several nation- states are doing it. Second is the denial-of-service attacks. You ’ re probably aware that there have been a spate of those directed against the U. S. financial **sector** since 2012. Again, that ’ s a nation-state who is executing those attacks, and they ’ re doing that as a semi-anonymous way of reprisal. And the last one is destructive attacks, and those are the ones that concern me the most. Those are on the rise. You have the attack against Saudi Aramco in 2012, August of 2012. It took down about 35, 000 of their computers with a Wiper-style **virus**. You had a follow-on a week later to a Qatari company. You had March of 2013, you had a South Korean attack that was attributed in the press to North Korea that took out thousands of computers. Those are on the rise, and we see people expressing interest in those capabilities and a desire to employ them. CA : Okay, so a couple of things here, because this is really the core of this, almost. I mean, first of all, a lot of people who look at risk and look at the numbers don ’ t understand this belief that terrorism is still the number one threat. Apart from September 11, I think the numbers are that in the last 30 or 40 years about 500 Americans have died from terrorism, mostly from homegrown terrorists. The chance in the last few years of being killed by terrorism is far less than the chance of being killed by lightning. I guess you would say that a single nuclear incident or bioterrorism act or something like that would change those numbers. Would that be the point of view? RL : Well, I ’ d say two things. One is, the reason that there hasn ’ t been a major attack in the United States since 9 / 11, that is not an accident. That ’ s a lot of hard work that we have done, that other folks in the **intelligence** community have done, that the military has done, and that our allies around the globe have done. You ’ ve heard the numbers about the tip of the iceberg in terms of numbers of terrorist attacks that NSA programs contributed to stopping was 54, 25 of those in Europe, and of those 25, 18 of them occurred in three countries, some of which are our allies, and some of which are beating the heck out of us over the NSA programs, by the way. So that ’ s not an accident that those things happen. That ’ s hard work. That ’ s us finding **intelligence** on terrorist activities and interdicting them through one way or another, through law enforcement, through cooperative activities with other countries and sometimes through military action. The other thing I would say is that your idea of nuclear or chem-bio-threat is not at all far-fetched and in fact there are a number of groups who have for several years expressed interest and desire in obtaining those capabilities and work towards that. CA : It ’ s also been said that, of those 54 alleged incidents, that as few as zero of them were actually anything to do with these controversial programs that Mr. Snowden revealed, that it was basically through other forms of **intelligence**, that you ’ re looking for a needle in a haystack, and the effects of these programs, these controversial programs, is just to add hay to the stack, not to really find the needle. The needle was found by other methods. Isn ’ t there something to that? RL : No, there ’ s actually two programs that are typically implicated in that discussion. One is the section 215 program, the U. S. telephony metadata program, and the other one is popularly called the PRISM program, and it ’ s actually section 702 of the FISA Amendment Act. But the 215 program is only relevant to threats that are directed against the United States, and there have been a dozen threats where that was implicated. Now what you ’ ll see people say publicly is there is no”but for”case, and so there is no case where, but for that, the threat would have happened. But that actually indicates a lack of understanding of how terrorist investigations actually work. You think about on television, you watch a murder mystery. What do you start with? You start with a body, and then they work their way from there to solve the crime. We ’ re actually starting well before that, hopefully before there are any bodies, and we ’ re trying to build the case for who the people are, what they ’ re trying to do, and that **involves** massive amounts of information. Think of it is as mosaic, and it ’ s hard to say that any one piece of a mosaic was necessary to building the mosaic, but to build the complete picture, you need to have all the pieces of information. On the other, the non-U. S. -related threats out of those 54, the other 42 of them, the PRISM program was hugely relevant to that, and in fact was material in contributing to stopping those attacks. CA : Snowden said two days ago that terrorism has always been what is called in the **intelligence** world”a cover for action,”that it ’ s something that, because it invokes such a powerful emotional response in people, it allows the initiation of these programs to achieve powers that an organization like yours couldn ’ t otherwise have. Is there any internal debate about that? RL : Yeah. I mean, we debate these things all the time, and there is discussion that goes on in the executive branch and within NSA itself and the **intelligence** community about what ’ s right, what ’ s proportionate, what ’ s the correct thing to do. And it ’ s important to note that the programs that we ’ re talking about were all authorized by two different presidents, two different political parties, by Congress twice, and by federal judges 16 different times, and so this is not NSA running off and doing its own thing. This is a legitimate activity of the United States foreign government that was agreed to by all the branches of the United States government, and President Madison would have been proud. CA : And yet, when congressmen discovered what was actually being done with that authorization, many of them were completely shocked. Or do you think that is not a legitimate **reaction**, that it ’ s only because it ’ s now come out publicly, that they really knew exactly what you were doing with the powers they had granted you? RL : Congress is a big body. There ’ s 535 of them, and they change out frequently, in the case of the House, every two years, and I think that the NSA provided all the relevant information to our oversight committees, and then the dissemination of that information by the oversight committees throughout Congress is something that they manage. I think I would say that Congress members had the opportunity to make themselves aware, and in fact a significant number of them, the ones who are assigned oversight responsibility, did have the **ability** to do that. And you ’ ve actually had the chairs of those committees say that in public. CA : Now, you mentioned the threat of cyberattacks, and I don ’ t think anyone in this room would disagree that that is a huge concern, but do you accept that there ’ s a tradeoff between offensive and defensive strategies, and that it ’ s possible that the very measures taken to,”weaken encryption,”and allow yourself to find the bad **guys**, might also open the door to forms of cyberattack? RL : So I think two things. One is, you said weaken encryption. I didn ’ t. And the other one is that the NSA has both of those missions, and we are heavily biased towards defense, and, actually, the vulnerabilities that we find in the overwhelming majority of cases, we disclose to the people who are responsible for manufacturing or developing those products. We have a great track record of that, and we ’ re actually working on a proposal right now to be transparent and to publish transparency reports in the same way that the Internet companies are being allowed to publish transparency reports for them. We want to be more transparent about that. So again, we eat our own dog food. We use the standards, we use the products that we recommend, and so it ’ s in our interest to keep our communications protected in the same way that other people ’ s need to be. CA : Edward Snowden, when, after his talk, was wandering the halls here in the bot, and I heard him say to a couple of people, they asked him about what he thought of the NSA overall, and he was very complimentary about the people who work with you, said that it ’ s a really impassioned group of employees who are seeking to do the right thing, and that the problems have come from just some badly conceived **policies**. He came over certainly very reasonably and calmly. He didn ’ t come over like a crazy man. Would you accept that at least, even if you disagree with how he did it, that he has opened a debate that matters? RL : So I think that the discussion is an important one to have. I do not like the way that he did it. I think there were a number of other ways that he could have done that that would have not endangered our people and the people of other nations through losing visibility into what our adversaries are doing. But I do think it ’ s an important conversation. CA : It ’ s been reported that there ’ s almost a difference of opinion with you and your colleagues over any scenario in which he might be offered an amnesty deal. I think your boss, General Keith Alexander, has said that that would be a terrible example for others ; you can ’ t **negotiate** with someone who ’ s broken the law in that way. But you ’ ve been quoted as saying that, if Snowden could prove that he was surrendering all undisclosed **documents**, that a deal maybe should be considered. Do you still think that? RL : Yeah, so actually, this is my favorite thing about that”60 Minutes”interview was all the misquotes that came from that. What I actually said, in response to a question about, would you entertain any discussions of mitigating action against Snowden, I said, yeah, it ’ s worth a conversation. This is something that the attorney general of the United States and the president also actually have both talked about this, and I defer to the attorney general, because this is his lane. But there is a strong tradition in American jurisprudence of having discussions with people who have been charged with crimes in order to, if it benefits the government, to get something out of that, that there ’ s always room for that kind of discussion. So I ’ m not presupposing any outcome, but there is always room for discussion. CA : To a lay person it seems like he has certain things to offer the U. S., the government, you, others, in terms of putting things right and helping figure out a smarter **policy**, a smarter way forward for the future. Do you see, has that kind of possibility been entertained at all? RL : So that ’ s out of my lane. That ’ s not an NSA thing. That would be a Department of Justice sort of discussion. I ’ ll defer to them. CA : Rick, when Ed Snowden ended his talk, I offered him the chance to share an idea worth spreading. What would be your idea worth spreading for this group? RL : So I think, learn the facts. This is a really important conversation, and it **impacts**, it ’ s not just NSA, it ’ s not just the government, it ’ s you, it ’ s the Internet companies. The issue of privacy and personal data is much bigger than just the government, and so learn the facts. Don ’ t rely on headlines, don ’ t rely on sound bites, don ’ t rely on one-sided conversations. So that ’ s the idea, I think, worth spreading. We have a sign, a badge tab, we wear badges at work with lanyards, and if I could make a plug, my badge lanyard at work says,”Dallas Cowboys.”Go Dallas. I ’ ve just alienated half the audience, I know. So the lanyard that our people who work in the organization that does our crypto- analytic work have a tab that says,”Look at the data.”So that ’ s the idea worth spreading. Look at the data. CA : Rick, it took a certain amount of courage, I think, actually, to come and speak openly to this group. It ’ s not something the NSA has done a lot of in the past, and plus the technology has been challenging. We truly appreciate you doing that and sharing in this very important conversation. Thank you so much. RL : Thanks, Chris.

Text Sample 569: POS Dim 4 – 2014 – Score 20.00 – 2014/t001880.txt
Chris Anderson : The rights of citizens, the future of the Internet. So I would like to welcome to the **TED** stage the man behind those revelations, Ed Snowden. Ed is in a remote location somewhere in Russia controlling this bot from his laptop, so he can see what the bot can see. Ed, welcome to the **TED** stage. What can you see, as a matter of fact? Edward Snowden : Ha, I can see everyone. This is amazing. CA : Ed, some questions for you. You ’ ve been called many things in the last few months. You ’ ve been called a whistleblower, a traitor, a hero. What words would you describe yourself with? ES : You know, **everybody** who is **involved** with this debate has been struggling over me and my personality and how to describe me. But when I think about it, this isn ’ t the question that we should be struggling with. Who I am really doesn ’ t matter at all. If I ’ m the worst person in the world, you can hate me and move on. What really matters here are the issues. What really matters here is the kind of government we want, the kind of Internet we want, the kind of relationship between people and societies. And that ’ s what I ’ m hoping the debate will move towards, and we ’ ve seen that increasing over time. If I had to describe myself, I wouldn ’ t use words like”hero.”I wouldn ’ t use”patriot,”and I wouldn ’ t use”traitor.”I ’ d say I ’ m an American and I ’ m a citizen, just like everyone else. CA : So just to give some context for those who don ’ t know the whole story — — this time a year ago, you were stationed in Hawaii working as a consultant to the NSA. As a sysadmin, you had access to their systems, and you began revealing certain classified **documents** to some handpicked journalists leading the way to June ’ s revelations. Now, what propelled you to do this? ES : You know, when I was sitting in Hawaii, and the years before, when I was working in the **intelligence** community, I saw a lot of things that had disturbed me. We do a lot of good things in the **intelligence** community, things that need to be done, and things that help everyone. But there are also things that go too far. There are things that shouldn ’ t be done, and decisions that were being made in secret without the public ’ s awareness, without the public ’ s consent, and without even our representatives in government having knowledge of these programs. When I really came to struggle with these issues, I thought to myself, how can I do this in the most responsible way, that maximizes the public benefit while minimizing the risks? And out of all the solutions that I could come up with, out of going to Congress, when there were no laws, there were no legal protections for a private employee, a contractor in **intelligence** like myself, there was a risk that I would be buried along with the information and the public would never find out. But the First Amendment of the United States Constitution guarantees us a free press for a reason, and that ’ s to enable an adversarial press, to challenge the government, but also to work together with the government, to have a dialogue and debate about how we can inform the public about matters of vital importance without putting our national security at risk. And by working with journalists, by giving all of my information back to the American people, rather than **trusting** myself to make the decisions about publication, we ’ ve had a robust debate with a deep investment by the govern-

ment that I think has resulted in a benefit for everyone. And the risks that have been threatened, the risks that have been played up by the government have never materialized. We ’ ve never seen any evidence of even a single instance of specific harm, and because of that, I ’ m comfortable with the decisions that I made. CA : So let me show the audience a couple of examples of what you revealed. If we could have a slide up, and Ed, I don ’ t know whether you can see, the slides are here. This is a slide of the PRISM program, and maybe you could tell the audience what that was that was revealed. ES : The best way to understand PRISM, because there ’ s been a little bit of controversy, is to first talk about what PRISM isn ’ t. Much of the debate in the U. S. has been about metadata. They ’ ve said it ’ s just metadata, it ’ s just metadata, and they ’ re talking about a specific legal authority called Section 215 of the Patriot Act. That allows sort of a warrantless wiretapping, mass surveillance of the entire country ’ s **phone** records, things like that — who you ’ re talking to, when you ’ re talking to them, where you traveled. These are all metadata events. PRISM is about content. It ’ s a program through which the government could compel corporate America, it could deputize corporate America to do its dirty work for the NSA. And even though some of these companies did resist, even though some of them — I believe Yahoo was one of them — challenged them in **court**, they all lost, because it was never tried by an open **court**. They were only tried by a secret **court**. And something that we ’ ve seen, something about the PRISM program that ’ s very concerning to me is, there ’ s been a talking point in the U. S. government where they ’ ve said 15 federal judges have reviewed these programs and found them to be lawful, but what they don ’ t tell you is those are secret judges in a secret **court** based on secret interpretations of law that ’ s considered 34, 000 warrant requests over 33 years, and in 33 years only rejected 11 government requests. These aren ’ t the people that we want deciding what the role of corporate America in a free and open Internet should be. CA : Now, this slide that we ’ re showing here shows the dates in which different technology companies, Internet companies, are alleged to have joined the program, and where data collection began from them. Now, they have denied collaborating with the NSA. How was that data collected by the NSA? ES : Right. So the NSA ’ s own slides refer to it as direct access. What that means to an actual NSA analyst, someone like me who was working as an **intelligence** analyst targeting, Chinese cyber-hackers, things like that, in Hawaii, is the provenance of that data is directly from their servers. It doesn ’ t mean that there ’ s a group of company representatives sitting in a smoky room with the NSA palling around and making back-room deals about how they ’ re going to give this **stuff** away. Now each company handles it different ways. Some are responsible. Some are somewhat less responsible. But the bottom line is, when we talk about how this information is given, it ’ s coming from the companies themselves. It ’ s not stolen from the lines. But there ’ s an important thing to remember here : even though companies pushed back, even though companies **demanded**, hey, let ’ s do this through a warrant process, let ’ s do this where we actually have some sort of legal review, some sort of basis for handing over these users ’ data, we saw stories in the Washington Post last year that weren ’ t as well reported as the PRISM story that said the NSA broke in to the data center communications between Google to itself and Yahoo to itself. So even these companies that are cooperating in at least a compelled but hopefully lawful manner with the NSA, the NSA isn ’ t satisfied with that, and because of that, we need our companies to work very hard to guarantee that they ’ re going to represent the interests of the user, and also advocate for the rights of the users. And I think over the last year, we ’ ve seen the companies that are named on the PRISM slides take great strides to do that, and I encourage them to continue. CA : What more should they do? ES : The biggest thing that an Internet company in America can do today, right now, without consulting with lawyers, to protect the rights of users worldwide, is to enable SSL web encryption on every page you **visit**. The reason this matters is today, if you go to look at a copy of"1984"on Amazon. com, the NSA can see a record of that, the Russian **intelligence** service can see a record of that, the Chinese service can see a record of that, the French service, the German service, the services of Andorra. They can all see it because it ' s unencrypted. The world ' s library is Amazon. com, but not only do they not **support** encryption by default, you cannot choose to use encryption when browsing through books. This is something that we need to change, not just for Amazon, I don ' t mean to single them out, but they ' re a great example. All companies need to move to an encrypted browsing habit by default for all users who haven ' t taken any action or picked any special methods on their own. That ' ll increase the privacy and the rights that people enjoy worldwide. CA : Ed, come with me to this part of the stage. I want to show you the next slide here. This is a program called Boundless Informant. What is that? ES : So, I ' ve got to give credit to the NSA for using appropriate names on this. This is one of my favorite NSA cryptonyms. Boundless Informant is a program that the NSA hid from Congress. The NSA was previously asked by Congress, was there any **ability** that they had to even give a rough ballpark estimate of the amount of American communications that were being intercepted. They said no. They said, we don ' t track those stats, and we can ' t track those stats. We can ' t tell you how many communications we ' re intercepting around the world, because to tell you that would be to invade your privacy. Now, I really appreciate that sentiment from them, but the reality, when you look at this slide is, not only do they have the capability, the capability already exists. It ' s already in place. The NSA has its own internal data format that tracks both ends of a communication, and if it says, this communication came from America, they can tell Congress how many of those communications they have today, right now. And what Boundless Informant tells us is more communications are being intercepted in America about Americans than there are in Russia about Russians. I ' m not sure that ' s what an **intelligence** agency should be aiming for. CA : Ed, there was a story broken in the Washington Post, again from your data. The headline says,"NSA broke privacy rules thousands of times per year."Tell us about that. ES : We also heard in Congressional testimony last year, it was an amazing thing for someone like me who came from the NSA and who ' s seen the actual internal **documents**, knows what ' s in them, to see officials testifying under oath that there had been no abuses, that there had been no violations of the NSA ' s rules, when we knew this story was coming. But what ' s especially **interesting** about this, about the fact that the NSA has violated their own rules, their own laws thousands of times in a single year, including one event by itself, one event out of those 2, 776, that affected more than 3, 000 people. In another event, they intercepted all the calls in Washington, D. C., by accident. What ' s amazing about this, this report, that didn ' t get that much attention, is the fact that not only were there 2, 776 abuses, the chairman of the Senate Intelligence Committee, Dianne Feinstein, had not seen this report until the Washington Post contacted her asking for comment on the report. And she then requested a copy from the NSA and received it, but had never seen this before that. What does that say about the state of oversight in American **intelligence** when the chairman of the Senate Intelligence Committee has no idea that the rules are being broken thousands of times every year? CA : Ed, one response to this whole debate is this : Why should we care about all this surveillance, honestly? I mean, look, if you ' ve done nothing wrong, you ' ve got nothing to worry about. What ' s wrong with that point of view? ES : Well, so the first thing is, you ' re giving up your rights. You ' re saying hey, you know, I don ' t think I ' m going to need them, so I ' m just going to **trust** that, you know, let ' s get rid of them, it doesn ' t really matter, these **guys** are going to do the right thing. Your rights matter because you never know when you ' re going to need them. Beyond that, it ' s a part of our cultural identity, not just in America, but in Western societies and in democratic societies around the world. People should be able to pick up the **phone** and to call their family, people should be able to send a text message to their loved ones, people should be able to buy a book online, they should be able to travel by train, they should be able to buy an airline ticket without wondering about how these events are going to look to an agent of the government, possibly not even your government years in the future, how they ' re going to be misinterpreted and what they ' re going to think your intentions were. We have a right to privacy. We require warrants to be based on probable cause or some kind of individualized suspicion because we recognize that trusting anybody, any government authority, with the entirety of human communications in secret and without oversight is simply too great a temptation to be ignored. CA : Some people are furious at what you ' ve done. I heard a quote recently from Dick Cheney who said that Julian Assange was a flea bite, Edward Snowden is the lion that bit the head off the dog. He thinks you ' ve committed one of the worst acts of betrayal in American history. What would you say to people who think that? ES : Dick Cheney ' s really something else. Thank you. I think it ' s amazing, because at the time Julian Assange was doing some of his greatest work, Dick Cheney was saying he was going to end governments worldwide, the skies were going to ignite and the seas were going to boil off, and now he ' s saying it ' s a flea bite. So we should be suspicious about the same sort of overblown **claims** of damage to national security from these kind of officials. But let ' s assume that these people really believe this. I would argue that they have kind of a narrow conception of national security. The prerogatives of people like Dick Cheney do not keep the nation **safe**. The public interest is not always the same as the national interest. Going to war with people who are not our enemy in places that are not a threat doesn ' t make us **safe**, and that applies whether it ' s in Iraq or on the Internet. The Internet is not the enemy. Our economy is not the enemy. American businesses, Chinese businesses, and any other company out there is a part of our society. It ' s a part of our interconnected world. There are ties of fraternity that bond us together, and if we destroy these bonds by undermining the standards, the security, the manner of behavior, that nations and citizens all around the world expect us to abide by. CA : But it ' s alleged that you ' ve stolen 1. 7 million **documents**. It seems only a few hundred of them have been shared with journalists so far. Are there more revelations to come? ES : There are absolutely more revelations to come. I don ' t think there ' s any question that some of the most important reporting to be done is yet to come. CA : Come here, because I want to ask you about this particular revelation. Come and take a look at this. I mean, this is a story which I think for a lot of the techies in this room is the single most shocking thing that they have heard in the last few months. It ' s about a program called"Bullrun."Can you explain what that is? ES : So Bullrun, and this is again where we ' ve got to thank the NSA for their candor, this is a program named after a Civil War battle. The British counterpart is called Edgehill, which is a U. K. civil war battle. And the reason that I believe they ' re named this way is because they target our own infrastructure. They ' re programs through which the NSA intentionally misleads corporate partners. They tell corporate partners that these are **safe** standards. They say hey, we need to work with you to secure your systems, but in reality, they ' re giving bad advice to these companies that makes them degrade the security of their services. They ' re building in backdoors that not only the NSA can exploit, but anyone else who has time and money to research and find it can then use to let themselves in to the world ' s communications. And this is really dangerous, because if we lose a single standard, if we lose the **trust** of something like SSL, which was specifically targeted by the Bullrun program, we will live a less **safe** world overall. We won ' t be able to access our banks and we won ' t be able to access commerce without worrying about people monitoring those communications or subverting them for their own ends. CA : And do those same decisions also potentially open America up to cyberattacks from other sources? ES : Absolutely. One of the problems, one of the dangerous legacies that we ' ve seen in the post-9 / 11 era, is that the NSA has traditionally worn two hats. They ' ve been in charge of offensive operations, that is hacking, but they ' ve also been in charge of defensive operations, and traditionally they ' ve always prioritized defense over offense based on the principle that American secrets are simply worth more. If we hack a Chinese business and steal their secrets, if we hack a government office in Berlin and steal their secrets, that has less value to the American people than making sure that the Chinese can ' t get access to our secrets. So by reducing the security of our communications, they ' re not only putting the world at risk, they ' re putting America at risk in a fundamental way, because intellectual property is the basis, the foundation of our economy, and if we put that at risk through weak security, we ' re going to be paying for it for years. CA : But they ' ve made a calculation that it was worth doing this as part of America ' s defense against terrorism. Surely that makes it a price worth paying. ES : Well, when you look at the results of these programs in stopping terrorism, you will see that that ' s unfounded, and you don ' t have to take my word for it, because we ' ve had the first open **court**, the first federal **court** that ' s reviewed this, outside the secrecy arrangement, called these programs Orwellian and likely unconstitutional. Congress, who has access to be briefed on these things, and now has the desire to be, has produced bills to reform it, and two independent White House panels who reviewed all of the classified evidence said these programs have never stopped a single terrorist attack that was imminent in the United States. So is it really terrorism that we ' re stopping? Do these programs have any value at all? I say no, and all three branches of the American government say no as well. CA : I mean, do you think there ' s a deeper motivation for them than the war against terrorism? ES : I ' m sorry, I couldn ' t hear you, say again? CA : Sorry. Do you think there ' s a deeper motivation for them other than the war against terrorism? ES : Yeah. The bottom line is that terrorism has always been what we in the **intelligence** world would call a cover for action. Terrorism is something that provokes an emotional response that allows people to rationalize authorizing powers and programs that they wouldn ' t give otherwise. The Bullrun and Edgehill-type programs, the NSA asked for these authorities back in the 1990s. They asked the FBI to go to Congress and make the case. The FBI went to Congress and did make the case. But Congress and the American people said no. They said, it ' s not worth the risk to our economy. They said it ' s worth too much damage to our society to justify the gains. But what we saw is, in the post-9 / 11 era, they used secrecy and they used the justification of terrorism to start these programs in secret without asking Congress, without asking the American people, and it ' s that kind of government behind closed doors that we need to guard ourselves against, because it makes us less **safe**, and it offers no value. CA : Okay, come with me here for a sec, because I ' ve got a more personal question for you. Speaking of terror, most people would find the situation you ' re in right now in Russia pretty terrifying. You obviously heard what happened, what the treatment that Bradley Manning got, Chelsea Manning as now is, and there was a story in Buzzfeed saying that there are people in the **intelligence** community who want you dead. How are you coping with this? How are you coping with the fear? ES : It ' s no mystery that there are governments out there that want to see me dead. I ' ve made clear again and again and again that I go to sleep every morning thinking about what I can do for the American people. I don ' t want to harm my government. I want to help my government, but the fact that they are willing to completely ignore due process, they ' re willing to declare guilt without ever seeing a trial, these are things that we need to work against as a society, and say hey, this is not appropriate. We shouldn ' t be threatening dissidents. We shouldn ' t be criminalizing journalism. And whatever part I can do to see that end, I ' m happy to do despite the risks. CA : So I ' d actually like to get some feedback from the audience here, because I know there ' s widely differing **reactions** to Edward Snowden. Suppose you had the following two choices, right? You could view what he did as fundamentally a reckless act that has endangered America or you could view it as fundamentally a heroic act that will work towards America and the world ' s long-term good? Those are the two choices I ' ll give you. I ' m curious to see who ' s willing to vote with the first of those, that this was a reckless act? There are some hands going up. Some hands going up. It ' s hard to put your hand up when the man is standing right here, but I see them. ES : I can see you. CA : And who goes with the second choice, the fundamentally heroic act? And I think it ' s true to say that there are a lot of people who didn ' t show a hand and I think are still thinking this through, because it seems to me that the debate around you doesn ' t split along traditional political lines. It ' s not left or right, it ' s not really about pro-government, libertarian, or not just that. Part of it is almost a generational issue. You ' re part of a generation that grew up with the Internet, and it seems as if you become offended at almost a visceral level when you see something done that you think will harm the Internet. Is there some truth to that? ES : It is. I think it ' s very true. This is not a left or right issue. Our basic freedoms, and when I say our, I don ' t just mean Americans, I mean people around the world, it ' s not a partisan issue. These are things that all people believe, and it ' s up to all of us to protect them, and to people who have seen and enjoyed a free and open Internet, it ' s up to us to preserve that liberty for the next generation to enjoy, and if we don ' t change things, if we don ' t stand up to make the changes we need to do to keep the Internet **safe**, not just for us but for everyone, we ' re going to lose that, and that would be a tremendous loss, not just for us, but for the world. CA : Well, I have heard similar language recently from the founder of the world wide web, who I actually think is with us, Sir Tim Berners-Lee. Tim, actually, would you like to come up and say, do we have a microphone for Tim? Tim, good to see you. Come up there. Which camp are you in, by the way, traitor, hero? I have a theory on this, but — Tim Berners-Lee : I ' ve given much longer answers to that question, but hero, if I have to make the choice between the two. CA : And Ed, I think you ' ve read the proposal that Sir Tim has talked about about a new Magna Carta to take back the Internet. Is that something that makes sense? ES : Absolutely. I mean, my generation, I grew up not just thinking about the Internet, but I grew up in the Internet, and although I never expected to have the chance to defend it in such a direct and practical manner and to embody it in this unusual, almost avatar manner, I think there ' s something poetic about the fact that one of the sons of the Internet has actually become close to the Internet as a result of their political expression. And I believe that a Magna Carta for the Internet is exactly what we need. We need to encode our values not just in writing but in the structure of the Internet, and it ' s something that

I hope, I invite everyone in the audience, not just here in Vancouver but around the world, to join and participate in. CA : Do you have a question for Ed? TBL : Well, two questions, a general question — CA : Ed, can you still hear us? ES : Yes, I can hear you. CA : Oh, he ' s back. TBL : The wiretap on your line got a little interfered with for a moment. ES : It ' s a little bit of an NSA problem. TBL : So, from the 25 years, stepping back and thinking, what would you think would be the best that we could achieve from all the discussions that we have about the web we want? ES : When we think about in terms of how far we can go, I think that ' s a question that ' s really only limited by what we ' re willing to put into it. I think the Internet that we ' ve enjoyed in the past has been exactly what we as not just a nation but as a people around the world need, and by cooperating, by engaging not just the technical parts of society, but as you said, the users, the people around the world who contribute through the Internet, through social media, who just check the weather, who rely on it every day as a part of their life, to champion that. We ' ll get not just the Internet we ' ve had, but a better Internet, a better now, something that we can use to build a future that ' ll be better not just than what we hoped for but anything that we could have imagined. CA : It ' s 30 years ago that **TED** was founded, 1984. A lot of the conversation since then has been along the lines that actually George Orwell got it wrong. It ' s not Big Brother watching us. We, through the power of the web, and transparency, are now watching Big Brother. Your revelations kind of drove a stake through the heart of that rather optimistic view, but you still believe there ' s a way of doing something about that. And you do too. ES : Right, so there is an argument to be made that the powers of Big Brother have increased enormously. There was a recent legal article at Yale that established something called the Bankston-Soltani Principle, which is that our expectation of privacy is violated when the capabilities of government surveillance have become cheaper by an order of magnitude, and each time that occurs, we need to revisit and rebalance our privacy rights. Now, that hasn ' t happened since the government ' s surveillance powers have increased by several orders of magnitude, and that ' s why we ' re in the problem that we ' re in today, but there is still hope, because the power of individuals have also been increased by technology. I am living proof that an individual can go head to head against the most powerful adversaries and the most powerful **intelligence** agencies around the world and win, and I think that ' s something that we need to take hope from, and we need to build on to make it accessible not just to technical experts but to ordinary citizens around the world. Journalism is not a crime, communication is not a crime, and we should not be monitored in our everyday activities. CA : I ' m not quite sure how you shake the hand of a bot, but I imagine it ' s, this is the hand right here. TBL : That ' ll come very soon. ES : Nice to meet you, and I hope my beam looks as nice as my view of you **guys** does. CA : Thank you, Tim. I mean, The New York Times recently called for an amnesty for you. Would you welcome the chance to come back to America? ES : Absolutely. There ' s really no question, the principles that have been the foundation of this project have been the public interest and the principles that underly the journalistic establishment in the United States and around the world, and I think if the press is now saying, we **support** this, this is something that needed to happen, that ' s a powerful argument, but it ' s not the final argument, and I think that ' s something that public should decide. But at the same time, the government has hinted that they want some kind of deal, that they want me to compromise the journalists with which I ' ve been working, to come back, and I want to make it very clear that I did not do this to be **safe**. I did this to do what was right, and I ' m not going to stop my work in the public interest just to benefit myself. CA : In the meantime, courtesy of the Internet and this technology, you ’ re here, back in North America, not quite the U. S., Canada, in this form. I ’ m curious, how does that feel? ES : Canada is different than what I expected. It ’ s a lot warmer. CA : At **TED**, the mission is”ideas worth spreading.”If you could encapsulate it in a single idea, what is your idea worth spreading right now at this moment? ES : I would say the last year has been a reminder that democracy may die behind closed doors, but we as individuals are born behind those same closed doors, and we don ’ t have to give up our privacy to have good government. We don ’ t have to give up our liberty to have security. And I think by working together we can have both open government and private lives, and I look forward to working with everyone around the world to see that happen. Thank you very much. CA : Ed, thank you.

Text Sample 570: POS Dim 4 – 2014 – Score 15.00 – 2014/t001795.txt
Nicholas Negroponte : Can we switch to the video disc, which is in play mode? I ’ m really **interested** in how you put people and computers together. We will be using the TV screens or their equivalents for electronic books of the future. Very **interested** in touch-sensitive displays, high-tech, high-touch, not having to pick up your fingers to use them. There is another way where computers touch people : wearing, physically wearing. Suddenly on September 11th, the world got bigger. NN : Thank you. Thank you. When I was asked to do this, I was also asked to look at all 14 **TED** Talks that I had given, chronologically. The first one was actually two hours. The second one was an hour, and then they became half hours, and all I noticed was my bald spot getting bigger. Imagine seeing your life, 30 years of it, go by, and it was, to say the least, for me, quite a shocking experience. So what I ’ m going to do in my time is try and share with you what happened during the 30 years, and then also make a prediction, and then tell you a little bit about what I ’ m doing next. And I put on a slide where **TED** 1 happened in my life. And it ’ s rather important because I had done 15 years of research before it, so I had a backlog, so it was easy. It ’ s not that I was Fidel Castro and I could talk for two hours, or Bucky Fuller. I had 15 years of **stuff**, and the Media Lab was about to start. So that was easy. But there are a couple of things about that period and about what happened that are really quite important. One is that it was a period when computers weren ’ t yet for people. And the other thing that sort of happened during that time is that we were considered sissy computer scientists. We weren ’ t considered the real thing. So what I ’ m going to show you is, in retrospect, a lot more **interesting** and a lot more accepted than it was at the time. So I ’ m going to characterize the years and I ’ m even going to go back to some very early work of mine, and this was the kind of **stuff** I was doing in the ’ 60s : very direct manipulation, very influenced as I studied architecture by the architect Moshe Safdie, and you can see that we even built robotic things that could build habitat-like structures. And this for me was not yet the Media Lab, but was the beginning of what I ’ ll call sensory computing, and I pick fingers partly because **everybody** thought it was ridiculous. Papers were published about how **stupid** it was to use fingers. Three reasons : One was they were low-resolution. The other is your hand would occlude what you wanted to see, and the third, which was the winner, was that your fingers would get the screen dirty, and hence, fingers would never be a device that you ’ d use. And this was a device we built in the ’ 70s, which has never even been picked up. It ’ s not just touch sensitive, it ’ s pressure sensitive. Voice : Put a yellow circle there. NN : Later work, and again this was before **TED** 1 — Voice : Move that **west**

of the diamond. Create a large green circle there. Man : Aw, shit. NN : — was to sort of do interface concurrently, so when you talked and you pointed and you had, if you will, multiple channels. Entebbe happened. 1976, Air France was hijacked, taken to Entebbe, and the Israelis not only did an extraordinary rescue, they did it partly because they had practiced on a physical model of the airport, because they had built the airport, so they built a model in the desert, and when they arrived at Entebbe, they knew where to go because they had actually been there. The U. S. government asked some of us, ’ 76, if we could replicate that computationally, and of course somebody like myself says yes. Immediately, you get a contract, Department of Defense, and we built this truck and this rig. We did sort of a simulation, because you had video discs, and again, this is ’ 76. And then many years later, you get this truck, and so you have Google Maps. Still people thought, no, that was not serious computer science, and it was a man named Jerry Wiesner, who happened to be the president of MIT, who did think it was computer science. And one of the keys for anybody who wants to start something in life : Make sure your president is part of it. So when I was doing the Media Lab, it was like having a gorilla in the front seat. If you were stopped for speeding and the officer looked in the window and saw who was in the passenger seat, then,”Oh, continue on, sir.”And so we were able, and this is a cute, actually, device, parenthetically. This was a lenticular photograph of Jerry Wiesner where the only thing that changed in the photograph were the lips. So when you oscillated that little piece of lenticular sheet with his photograph, it would be in lip sync with zero bandwidth. It was a zero- bandwidth teleconferencing system at the time. So this was the Media Lab ’ s — this is what we said we ’ d do, that the world of computers, publishing, and so on would come together. Again, not generally accepted, but very much part of **TED** in the early days. And this is really where we were headed. And that created the Media Lab. One of the things about age is that I can tell you with great **confidence**, I ’ ve been to the future. I ’ ve been there, actually, many times. And the reason I say that is, how many times in my life have I said,”Oh, in 10 years, this will happen,”and then 10 years comes. And then you say,”Oh, in five years, this will happen.”And then five years comes. So I say this a little bit with having felt that I ’ d been there a number of times, and one of the things that is most quoted that I ’ ve ever said is that computing is not about computers, and that didn ’ t quite get enough traction, and then it started to. It started to because people caught on that the medium wasn ’ t the message. And the reason I show this car in actually a rather ugly slide is just again to tell you the kind of story that characterized a little bit of my life. This is a student of mine who had done a Ph. D. called”Backseat Driver.”It was in the early days of GPS, the car knew where it was, and it would give audio instructions to the driver, when to turn right, when to turn left and so on. Turns out, there are a lot of things in those instructions that back in that period were pretty challenging, like what does it mean, take the next right? Well, if you ’ re coming up on a street, the next right ’ s probably the one after, and there are lots of issues, and the student did a wonderful thesis, and the MIT patent office said”Don ’ t patent it. It ’ ll never be accepted. The liabilities are too large. There will be insurance issues. Don ’ t patent it.”So we didn ’ t, but it shows you how people, again, at times, don ’ t really look at what ’ s happening. Some work, and I ’ ll just go through these very quickly, a lot of sensory **stuff**. You might recognize a young Yo-Yo Ma and tracking his body for playing the cello or the hypercello. These fellows literally walked around like that at the time. It ’ s now a little bit more discreet and more commonplace. And then there are at least three heroes I want to quickly mention. Marvin Minsky, who taught me a lot about common sense, and I will talk briefly about

Muriel Cooper, who was very important to Ricky Wurman and to **TED**, and in fact, when she got onstage, she said, the first thing she said was,"I introduced Ricky to Nicky."And nobody calls me Nicky and nobody calls Richard Ricky, so nobody knew who she was talking about. And then, of course, Seymour Papert, who is the person who said,"You can ' t think about thinking unless you think about thinking about something."And that ' s actually — you can unpack that later. It ' s a pretty profound statement. I ' m showing some slides that were from **TED** 2, a little silly as slides, perhaps. Then I felt television really was about displays. Again, now we ' re past **TED** 1, but just around the time of **TED** 2, and what I ' d like to mention here is, even though you could imagine **intelligence** in the device, I look today at some of the work being done about the Internet of Things, and I think it ' s kind of tragically pathetic, because what has happened is people take the oven panel and put it on your cell **phone**, or the door key onto your cell **phone**, just taking it and bringing it to you, and in fact that ' s actually what you don ' t want. You want to put a chicken in the oven, and the oven says,"Aha, it ' s a chicken,"and it cooks the chicken."Oh, it ' s cooking the chicken for Nicholas, and he likes it this way and that way."So the **intelligence**, instead of being in the device, we have started today to move it back onto the cell **phone** or closer to the user, not a particularly enlightened view of the Internet of Things. Television, again, television what I said today, that was back in 1990, and the television of tomorrow would look something like that. Again, people, but they laughed cynically, they didn ' t laugh with much appreciation. Telecommunications in the 1990s, George Gilder decided that he would call this diagram the Negroponte switch. I ' m probably much less famous than George, so when he called it the Negroponte switch, it stuck, but the idea of things that came in the ground would go in the **air** and **stuff** in the **air** would go into the ground has played itself out. That is the original slide from that year, and it has worked in lockstep obedience. We started Wired magazine. Some people, I remember we shared the reception desk periodically, and some parent called up irate that his son had given up Sports Illustrated to subscribe for Wired, and he said,"Are you some porno magazine or something?"and couldn ' t understand why his son would be **interested** in Wired, at any rate. I will go through this a little quicker. This is my favorite, 1995, back page of Newsweek magazine. Okay. Read it. ["Nicholas Negroponte, director of the MIT Media Lab, predicts that we ' ll soon buy books and newspapers straight over the Internet. Uh, sure."— Clifford Stoll, Newsweek, 1995] You must admit that gives you, at least it gives me pleasure when somebody says how dead wrong you are."Being Digital"came out. For me, it gave me an opportunity to be more in the trade press and get this out to the public, and it also allowed us to build the new Media Lab, which if you haven ' t been to, **visit**, because it ' s a beautiful piece of architecture aside from being a wonderful place to work. So these are the things we were saying in those **TEDs**. [Today, multimedia is a desktop or living room experience, because the apparatus is so clunky. This will change dramatically with small, bright, thin, high-resolution displays. — 1995] We came to them. I looked forward to it every year. It was the party that Ricky Wurman never had in the sense that he invited many of his old friends, including myself. And then something for me changed pretty profoundly. I became more **involved** with computers and learning and influenced by Seymour, but particularly looking at learning as something that is best approximated by computer programming. When you write a computer program, you ' ve got to not just list things out and sort of take an algorithm and translate it into a set of instructions, but when there ' s a bug, and all programs have bugs, you ' ve got to de-bug it. You ' ve got to go in, change it, and then re-execute, and you iterate, and that iteration is really a very, very good approximation of learning. So that led to my own work with Seymour in places like Cambodia and the starting of One Laptop per Child. Enough **TED** Talks on One Laptop per Child, so I ’ ll go through it very fast, but it did give us the chance to do something at a relatively large scale in the area of learning, development and computing. Very few people know that One Laptop per Child was a $1 billion project, and it was, at least over the seven years I ran it, but even more important, the World Bank contributed zero, US-AID zero. It was mostly the countries using their own treasuries, which is very **interesting**, at least to me it was very **interesting** in terms of what I plan to do next. So these are the various places it happened. I then tried an experiment, and the experiment happened in Ethiopia. And here ’ s the experiment. The experiment is, can learning happen where there are no schools. And we dropped off tablets with no instructions and let the children figure it out. And in a short period of time, they not only turned them on and were using 50 apps per child within five days, they were singing”ABC”songs within two weeks, but they hacked Android within six months. And so that seemed sufficiently **interesting**. This is perhaps the best picture I have. The kid on your right has sort of nominated himself as teacher. Look at the kid on the left, and so on. There are no adults **involved** in this at all. So I said, well can we do this at a larger scale? And what is it that ’ s missing? The kids are giving a press **conference** at this point, and sort of writing in the dirt. And the answer is, what is missing? And I ’ m going to skip over my prediction, actually, because I ’ m running out of time, and here ’ s the question, is what ’ s going to happen? I think the challenge is to connect the last billion people, and connecting the last billion is very different than connecting the next billion, and the reason it ’ s different is that the next billion are sort of low-hanging fruit, but the last billion are rural. Being rural and being poor are very different. Poverty tends to be created by our society, and the people in that community are not poor in the same way at all. They may be primitive, but the way to approach it and to connect them, the history of One Laptop per Child, and the experiment in Ethiopia, lead me to believe that we can in fact do this in a very short period of time. And so my plan, and unfortunately I haven ’ t been able to get my partners at this point to let me announce them, but is to do this with a stationary satellite. There are many reasons that stationary satellites aren ’ t the best things, but there are a lot of reasons why they are, and for two billion dollars, you can connect a lot more than 100 million people, but the reason I picked two, and I will leave this as my last slide, is two billion dollars is what we were spending in Afghanistan every week. So surely if we can connect Africa and the last billion people for numbers like that, we should be doing it. Thank you very much. Chris Anderson : Stay up there. Stay up there. NN : You ’ re going to give me extra time? CA : No. That was wickedly clever, wickedly clever. You gamed it beautifully. Nicholas, what is your prediction? NN : Thank you for asking. I ’ ll tell you what my prediction is, and my prediction, and this is a prediction, because it ’ ll be 30 years. I won ’ t be here. But one of the things about learning how to read, we have been doing a lot of consuming of information going through our eyes, and so that may be a very inefficient channel. So my prediction is that we are going to ingest information You ’ re going to swallow a pill and know English. You ’ re going to swallow a pill and know Shakespeare. And the way to do it is through the bloodstream. So once it ’ s in your bloodstream, it basically goes through it and gets into the brain, and when it knows that it ’ s in the brain in the different pieces, it deposits it in the right places. So it ’ s ingesting. CA : Have you been hanging out with Ray Kurzweil by any chance? NN : No, but I ’ ve been hanging around with Ed Boyden and hanging around with one of the speakers who is here, Hugh Herr, and there are a number of people. This isn

’ t quite as far-fetched, so 30 years from now. CA : We will check it out. We ’ re going to be back and we ’ re going to play this clip 30 years from now, and then all eat the red pill. Well thank you for that. Nicholas Negroponte. NN : Thank you.

Text Sample 571: POS Dim 4 – 2012 – Score 17.00 – 2012/t002040.txt

I ’ m going to share with you a paradigm-shifting perspective on the issues of **gender** violence : sexual assault, domestic violence, relationship abuse, sexual harassment, sexual abuse of children. That whole range of issues that I ’ ll refer to in shorthand as”**gender** violence issues,”they ’ ve been seen as women ’ s issues that some good men help out with, but I have a problem with that frame and I don ’ t accept it. I don ’ t see these as women ’ s issues that some good men help out with. In fact, I ’ m going to argue that these are men ’ s issues, first and foremost. Now obviously — Obviously, they ’ re also women ’ s issues, so I appreciate that, but calling **gender** violence a women ’ s issue is part of the problem, for a number of reasons. The first is that it gives men an excuse not to pay attention, right? A lot of men hear the term”women ’ s issues”and we tend to tune it out, and we think,”I ’ m a **guy** ; that ’ s for the girls,”or”that ’ s for the women.”And a lot of men literally don ’ t get beyond the first sentence as a result. It ’ s almost like a chip in our brain is activated, and the neural pathways take our attention in a different direction when we hear the term”women ’ s issues.”This is also true, by the way, of the word”**gender**,”because a lot of people hear the word”**gender**”and they think it means”women.”So they think that **gender** issues is synonymous with women ’ s issues. There ’ s some confusion about the term **gender**. And let me illustrate that confusion by way of analogy. So let ’ s talk for a moment about race. In the US, when we hear the word”race,”a lot of people think that means African-American, Latino, Asian-American, Native American, South Asian, Pacific Islander, on and on. A lot of people, when they hear the word”sexual orientation”think it means gay, lesbian, bisexual. And a lot of people, when they hear the word”**gender**,”think it means women. In each case, the **dominant** group doesn ’ t get paid attention to. As if white people don ’ t have some sort of racial identity or belong to some racial category or construct, as if heterosexual people don ’ t have a sexual orientation, as if men don ’ t have a **gender**. This is one of the ways that **dominant** systems maintain and reproduce themselves, which is to say the **dominant** group is rarely challenged to even think about its dominance, because that ’ s one of the key characteristics of power and privilege, the **ability** to go unexamined, lacking introspection, in fact being rendered invisible, in large measure, in the discourse about issues that are primarily about us. And this is amazing how this works in domestic and sexual violence, how men have been largely erased from so much of the conversation about a subject that is centrally about men. And I ’ m going to illustrate what I ’ m talking about by using the old **tech**. I ’ m old school on some fundamental regards. I make films and I work with high **tech**, but I ’ m still old school as an educator, and I want to share with you this exercise that illustrates on the sentence-structure level how the way that we think, literally the way that we use language, conspires to keep our attention off of men. This is about domestic violence in particular, but you can plug in other analogues. This comes from the work of the feminist linguist Julia Penelope. It starts with a very basic English sentence :”John beat Mary.”That ’ s a good English sentence. John is the subject, beat is the verb, Mary is the object, good sentence. Now we ’ re going to move to the second sentence, which says the same thing in the passive voice.”Mary was beaten by John.”And now a whole lot has happened in one sentence. We ’ ve gone from”John beat Mary”to”Mary was beaten by John.”We ’ ve shifted our focus in one sentence from John to Mary, and you can see John is very close to the end of the sentence, well, close to dropping off the map of our psychic plain. The third sentence, John is dropped, and we have,”Mary was beaten,”and now it ’ s all about Mary. We ’ re not even thinking about John, it ’ s totally focused on Mary. Over the past generation, the term we ’ ve used synonymous with”beaten”is”battered,”so we have”Mary was battered.”And the final sentence in this sequence, flowing from the others, is,”Mary is a battered woman.”So now Mary ’ s very identity — Mary is a battered woman — is what was done to her by John in the first instance. But we ’ ve demonstrated that John has long ago left the conversation. Those of us who work in the domestic and sexual violence field know that victim-blaming is pervasive in this realm, which is to say, blaming the person to whom something was done rather than the person who did it. And we say : why do they go out with these men? Why are they attracted to them? Why do they keep going back? What was she wearing at that party? What a **stupid** thing to do. Why was she drinking with those **guys** in that hotel room? This is victim blaming, and there are many reasons for it, but one is that our cognitive structure is set up to blame victims. This is all unconscious. Our whole cognitive structure is set up to ask questions about women and women ’ s choices and what they ’ re doing, thinking, wearing. And I ’ m not going to shout down people who ask questions about women. It ’ s a legitimate thing to ask. But ’ s let ’ s be clear : Asking questions about Mary is not going to get us anywhere in terms of preventing violence. We have to ask a different set of questions. The questions are not about Mary, they ’ re about John. They include things like, why does John beat Mary? Why is domestic violence still a big problem in the US and all over the world? What ’ s going on? Why do so many men abuse physically, emotionally, verbally, and other ways, the women and girls, and the men and boys, that they **claim** to love? What ’ s going on with men? Why do so many adult men sexually abuse little girls and boys? Why is that a common problem in our society and all over the world today? Why do we hear over and over again about new scandals erupting in major institutions like the Catholic Church or the Penn State football program or the Boy Scouts of America, on and on and on? And then local communities all over the country and all over the world. We hear about it all the time. The sexual abuse of children. What ’ s going on with men? Why do so many men rape women in our society and around the world? Why do so many men rape other men? What is going on with men? And then what is the role of the various institutions in our society that are helping to produce abusive men at **pandemic** rates? Because this isn ’ t about individual perpetrators. That ’ s a naive way to understanding what is a much deeper and more systematic social problem. The perpetrators aren ’ t these monsters who crawl out of the swamp and come into town and do their nasty business and then retreat into the darkness. That ’ s a very naive notion, right? Perpetrators are much more normal than that, and everyday than that. So the question is, what are we doing here in our society and in the world? What are the roles of various institutions in helping to produce abusive men? What ’ s the role of religious belief systems, the sports culture, the pornography culture, the family structure, economics, and how that intersects, and race and ethnicity and how that intersects? How does all this work? And then, once we start making those kinds of connections and asking those important and big questions, then we can talk about how we can be transformative, in other words, how can we do something differently? How can we change the practices? How can we change the socialization of boys and the definitions of manhood that lead to these current outcomes? These are the kind of questions that we need to be asking and the kind of work that we need to be doing, but if we ’ re endlessly focused on what women are doing and thinking in relationships or elsewhere, we ’ re not going to get to that piece. I understand that a lot of women who have been trying to speak out about these issues, today and yesterday and for years and years, often get shouted down for their efforts. They get called nasty names like”male-basher”and”man-hater,”and the disgusting and offensive”feminazi”, right? And you know what all this is about? It ’ s called kill the messenger. It ’ s because the women who are standing up and speaking out for themselves and for other women as well as for men and boys, it ’ s a statement to them to sit down and shut up, keep the current system in place, because we don ’ t like it when people rock the boat. We don ’ t like it when people challenge our power. You ’ d better sit down and shut up, basically. And thank goodness that women haven ’ t done that. Thank goodness that we live in a world where there ’ s so much women ’ s **leadership** that can counteract that. But one of the powerful roles that men can play in this work is that we can say some things that sometimes women can ’ t say, or, better yet, we can be heard saying some things that women often can ’ t be heard saying. Now, I appreciate that that ’ s a problem, it ’ s sexism, but it ’ s the truth. So one of the things that I say to men, and my colleagues and I always say this, is we need more men who have the courage and the strength to start standing up and saying some of this **stuff**, and standing with women and not against them and pretending that somehow this is a battle between the sexes and other kinds of nonsense. We live in the world together. And by the way, one of the things that really bothers me about some of the rhetoric against feminists and others who have built the battered women ’ s and rape **crisis** movements around the world is that somehow, like I said, that they ’ re anti-male. What about all the boys who are profoundly affected in a negative way by what some adult man is doing against their mother, themselves, their sisters? What about all those boys? What about all the young men and boys who have been traumatized by adult men ’ s violence? You know what? The same system that produces men who abuse women, produces men who abuse other men. And if we want to talk about male victims, let ’ s talk about male victims. Most male victims of violence are the victims of other men ’ s violence. So that ’ s something that both women and men have in common. We are both victims of men ’ s violence. So we have it in our direct self-interest, not to mention the fact that most men that I know have women and girls that we care deeply about, in our families and our friendship circles and every other way. So there ’ s so many reasons why we need men to speak out. It seems obvious saying it out loud, doesn ’ t it? Now, the nature of the work that I do and my colleagues do in the sports culture and the US military, in schools, we pioneered this approach called the bystander approach to gender-violence prevention. And I just want to give you the highlights of the bystander approach, because it ’ s a big thematic shift, although there ’ s lots of particulars, but the heart of it is, instead of seeing men as perpetrators and women as victims, or women as perpetrators, men as victims, or any combination in there. I ’ m using the **gender** binary. I know there ’ s more than men and women, there ’ s more than male and female. And there are women who are perpetrators, and of course there are men who are victims. There ’ s a whole spectrum. But instead of seeing it in the binary fashion, we focus on all of us as what we call bystanders, and a bystander is defined as anybody who is not a perpetrator or a victim in a given situation, so in other words friends, teammates, colleagues, coworkers, family members, those of us who are not directly **involved** in a dyad of abuse, but we are embedded in social, family, work, school, and other peer culture relationships with people who might be in that situation. What do we do? How do we speak up? How do we challenge our friends? How do we **sup-**

port our friends? But how do we not remain silent in the face of abuse? Now, when it comes to men and male culture, the goal is to get men who are not abusive to challenge men who are. And when I say abusive, I don ' t mean just men who are beating women. We ' re not just saying a man whose friend is abusing his girlfriend needs to stop the **guy** at the moment of attack. That ' s a naive way of creating a social change. It ' s along a continuum, we ' re trying to get men to interrupt each other. So, for example, if you ' re a **guy** and you ' re in a group of **guys** playing poker, talking, hanging out, no women present, and another **guy** says something sexist or degrading or harassing about women, instead of laughing along or pretending you didn ' t hear it, we need men to say,"Hey, that ' s not funny. that could be my sister you ' re talking about, and could you joke about something else? Or could you talk about something else? I don ' t appreciate that kind of talk."Just like if you ' re a white person and another white person makes a racist comment, you ' d hope, I hope, that white people would interrupt that racist enactment by a fellow white person. Just like with heterosexism, if you ' re a heterosexual person and you yourself don ' t enact harassing or abusive behaviors towards people of varying sexual orientations, if you don ' t say something in the face of other heterosexual people doing that, then, in a sense, isn ' t your silence a form of consent and complicity? Well, the bystander approach is trying to give people tools to interrupt that process and to speak up and to create a peer culture **climate** where the abusive behavior will be seen as unacceptable, not just because it ' s illegal, but because it ' s wrong and unacceptable in the peer culture. And if we can get to the place where men who act out in sexist ways will lose status, young men and boys who act out in sexist and harassing ways towards girls and women, as well as towards other boys and men, will lose status as a result of it, guess what? We ' ll see a radical diminution of the abuse. Because the typical perpetrator is not sick and twisted. He ' s a normal **guy** in every other way, isn ' t he? Now, among the many great things that Martin Luther King said in his short life was,"In the end, what will hurt the most is not the words of our enemies but the silence of our friends."In the end, what will hurt the most is not the words of our enemies but the silence of our friends. There ' s been an awful lot of silence in male culture about this ongoing tragedy of men ' s violence against women and children, hasn ' t there? There ' s been an awful lot of silence. And all I ' m saying is that we need to break that silence, and we need more men to do that. Now, it ' s easier said than done, because I ' m saying it now, but I ' m telling you it ' s not easy in male culture for **guys** to challenge each other, which is one of the reasons why part of the paradigm shift that has to happen is not just understanding these issues as men ' s issues, but they ' re also **leadership** issues for men. Because ultimately, the responsibility for taking a stand on these issues should not fall on the shoulders of little boys or teenage boys in high school or college men. It should be on adult men with power. Adult men with power are the ones we need to be holding **accountable** for being **leaders** on these issues, because when somebody speaks up in a peer culture and challenges and interrupts, he or she is being a **leader**, really. But on a big scale, we need more adult men with power to start prioritizing these issues, and we haven ' t seen that yet, have we? Now, I was at a dinner a number of years ago, and I work extensively with the US military, all the services. And I was at this dinner and this woman said to me — I think she thought she was a little clever — she said,"So how long have you been doing sensitivity training with the Marines?"And I said,"With all due respect, I don ' t do sensitivity training with the Marines. I run a **leadership** program in the Marine Corps."Now, I know it ' s a bit pompous, my response, but it ' s an important distinction, because I don ' t believe that what we need is sensitivity training. We need **leadership**

training, because, for example, when a professional coach or a manager of a baseball team or a football team — and I work extensively in that realm as well — makes a sexist comment, makes a homophobic statement, makes a racist comment, there will be discussions on the sports blogs and in sports talk radio. And some people will say,”He needs sensitivity training.”Other people will say,”Well, get off it. That ’ s political correctness run amok, he made a **stupid** statement, move on.”My argument is, he doesn ’ t need sensitivity training. He needs **leadership** training, because he ’ s being a bad **leader**, because in a society with **gender diversity** and sexual **diversity** — and racial and ethnic **diversity**, you make those kind of comments, you ’ re failing at your **leadership**. If we can make this point that I ’ m making to powerful men and women in our society at all levels of institutional authority and power, it ’ s going to change the paradigm of people ’ s thinking. You know, for example, I work a lot in college and university athletics throughout North America. We know so much about how to prevent domestic and sexual violence, right? There ’ s no excuse for a college or university to not have domestic and sexual violence prevention training mandated for all student athletes, coaches, administrators, as part of their educational process. We know enough to know that we can easily do that. But you know what ’ s missing? The **leadership**. But it ’ s not the **leadership** of student athletes. It ’ s the **leadership** of the athletic director, the president of the university, the people in charge who make decisions about resources and who make decisions about priorities in the institutional settings. That ’ s a failure, in most cases, of men ’ s **leadership**. Look at Penn State. Penn State is the mother of all teachable moments for the bystander approach. You had so many situations in that realm where men in powerful positions failed to act to protect children, in this case, boys. It ’ s unbelievable, really. But when you get into it, you realize there are pressures on men. There are constraints within peer cultures on men, which is why we need to encourage men to break through those pressures. And one of the ways to do that is to say there ’ s an awful lot of men who care deeply about these issues. I know this, I work with men, and I ’ ve been working with tens of thousands, hundreds of thousands of men for many decades now. It ’ s scary, when you think about it, how many years. But there ’ s so many men who care deeply about these issues, but caring deeply is not enough. We need more men with the guts, with the courage, with the strength, with the moral integrity to break our complicit silence and challenge each other and stand with women and not against them. By the way, we owe it to women. There ’ s no question about it. But we also owe it to our sons. We also owe it to young men who are growing up all over the world in situations where they didn ’ t make the choice to be a man in a culture that tells them that manhood is a certain way. They didn ’ t make the choice. We that have a choice, have an opportunity and a responsibility to them as well. I hope that, going forward, men and women, working together, can begin the change and the transformation that will happen so that future generations won ’ t have the level of tragedy that we deal with on a daily basis. I know we can do it, we can do better. Thank you very much.

Text Sample 572: POS Dim 4 – 2012 – Score 15.00 – 2012/t002162.txt

So if you ’ ve been following the news, you ’ ve heard that there ’ s a pack of giant asteroids headed for the United States, all scheduled to strike within the next 50 years. Now I don ’ t mean actual asteroids made of rock and metal. That actually wouldn ’ t be such a problem, because if we were really all going to die, we would put aside our differences, we ’ d spend whatever it took, and we ’ d find a way to deflect them. I ’ m talking instead about threats that are headed our way, but they ’ re wrapped in a special energy field that polarizes us, and therefore paralyzes us. Last March, I went to the **TED conference**, and I saw Jim Hansen speak, the NASA scientist who first raised the alarm about global warming in the 1980s, and it seems that the predictions he made back then are coming true. This is where we ’ re headed in terms of global temperature rises, and if we keep on going the way we ’ re going, we get a four- or five-degree-Centigrade temperature rise by the end of this century. Hansen says we can expect about a five-meter rise in sea levels. This is what a five-meter rise in sea levels would look like. Low- lying cities all around the world will disappear within the lifetime of children born today. Hansen closed his talk by saying,”Imagine a giant asteroid on a collision course with Earth. That is the equivalent of what we face now. Yet we dither, taking no action to deflect the asteroid, even though the longer we wait, the more difficult and expensive it becomes.”Of course, the left wants to take action, but the right denies that there ’ s any problem. All right, so I go back from **TED**, and then the following week, I ’ m invited to a dinner party in Washington, D. C., where I know that I ’ ll be meeting a number of **conservative** intellectuals, including Yuval Levin, and to prepare for the meeting, I read this article by Levin in National Affairs called”Beyond the Welfare State.”Levin writes that all over the world, nations are coming to terms with the fact that the social democratic welfare state is turning out to be untenable and unaffordable, dependent upon dubious economics and the demographic model of a bygone era. All right, now this might not sound as scary as an asteroid, but look at these graphs that Levin showed. This graph shows the national debt as a percentage of America ’ s GDP, and as you see, if you go all the way back to the founding, we borrowed a lot of money to fight the Revolutionary War. Wars are expensive. But then we ’ d pay it off, pay it off, pay it off, and then, oh, what ’ s this? The Civil War. Even more expensive. Borrow a lot of money, pay it off, pay it off, pay it off, get down to near zero, and bang! — World War I. Once again, the same process repeats. Now then we get the Great Depression and World War II. We rise to an astronomical level, around 118 percent of GDP, really unsustainable, really dangerous. But we pay it off, pay it off, pay it off, and then, what ’ s this? Why has it been rising since the ’ 70s? It ’ s partly due to tax cuts that were unfunded, but it ’ s due primarily to the rise of entitlement spending, especially Medicare. We ’ re approaching the levels of indebtedness we had at World War II, and the baby boomers haven ’ t even retired yet, and when they do, this is what will happen. This is data from the Congressional Budget Office showing its most realistic forecast of what would happen if current situations and expectations and trends are extended. All right, now what you might notice is that these two graphs are actually identical, not in terms of the x- and y-axes, or in terms of the data they present, but in terms of their moral and political implications, they say the same thing. Let me translate for you.”We are doomed unless we start acting now. What ’ s wrong with you people on the other side in the other party? Can ’ t you see reality? If you won ’ t help, then get the hell out of the way.”We can deflect both of these asteroids. These problems are both technically solvable. Our problem and our tragedy is that in these hyper-partisan times, the mere fact that one side says,”Look, there ’ s an asteroid,”means that the other side ’ s going to say,”Huh? What? No, I ’ m not even going to look up. No.”To understand why this is happening to us, and what we can do about it, we need to learn more about moral psychology. So I ’ m a social psychologist, and I study morality, and one of the most important principles of morality is that morality binds and blinds. It binds us into teams that circle around sacred values but thereby makes us go blind to objective reality. Think of it like this.

Large-scale cooperation is extremely rare on this planet. There are only a few species that can do it. That ' s a beehive. That ' s a termite mound, a giant termite mound. And when you find this in other animals, it ' s always the same story. They ' re always all siblings who are children of a single queen, so they ' re all in the same boat. They rise or fall, they live or die, as one. There ' s only one species on the planet that can do this without kinship, and that, of course, is us. This is a reconstruction of ancient Babylon, and this is Tenochtitlan. Now how did we do this? How did we go from being hunter-gatherers 10, 000 years ago to building these gigantic cities in just a few thousand years? It ' s miraculous, and part of the explanation is this **ability** to circle around sacred values. As you see, temples and gods play a big role in all ancient civilizations. This is an image of Muslims circling the Kaaba in Mecca. It ' s a sacred rock, and when people circle something together, they unite, they can **trust** each other, they become one. It ' s as though you ' re moving an electrical wire through a magnetic field that generates current. When people circle together, they generate a current. We love to circle around things. We circle around flags, and then we can **trust** each other. We can fight as a team, as a unit. But even as morality binds people together into a unit, into a team, the circling blinds them. It causes them to distort reality. We begin separating everything into good versus evil. Now that process feels great. It feels really satisfying. But it is a gross distortion of reality. You can see the moral electromagnet operating in the U. S. Congress. This is a graph that shows the degree to which voting in Congress falls strictly along the left-right axis, so that if you know how liberal or **conservative** someone is, you know exactly how they voted on all the major issues. And what you can see is that, in the decades after the Civil War, Congress was extraordinarily polarized, as you would expect, about as high as can be. But then, after World War I, things dropped, and we get this historically low level of polarization. This was a golden age of bipartisanship, at least in terms of the parties ' **ability** to work together and solve grand national problems. But in the 1980s and ' 90s, the electromagnet turns back on. Polarization rises. It used to be that **conservatives** and moderates and liberals could all work together in Congress. They could rearrange themselves, form bipartisan committees, but as the moral electromagnet got cranked up, the force field increased, Democrats and Republicans were pulled apart. It became much harder for them to socialize, much harder for them to cooperate. Retiring members nowadays say that it ' s become like gang warfare. Did anybody notice that in two of the three debates, Obama wore a blue tie and Romney wore a red tie? Do you know why they do this? It ' s so that the Bloods and the Crips will know which side to vote for. The polarization is strongest among our political elites. Nobody doubts that this is happening in Washington. But for a while, there was some doubt as to whether it was happening among the people. Well, in the last 12 years it ' s become much more apparent that it is. So look at this data. This is from the American National Elections Survey. And what they do on that survey is they ask what ' s called a feeling thermometer rating. So, how warm or cold do you feel about, you know, Native Americans, or the military, the Republican Party, the Democratic Party, all sorts of groups in American life. The blue line shows how warmly Democrats feel about Democrats, and they like them. You know, ratings in the 70s on a 100-point scale. Republicans like Republicans. That ' s not a surprise. But when you look at cross-party ratings, you find, well, that it ' s lower, but actually, when I first saw this data, I was surprised. That ' s actually not so bad. If you go back to the Carter and even Reagan administrations, they were rating the other party 43, 45. It ' s not terrible. It drifts downwards very slightly, but now look what happens under George W. Bush and Obama. It plummets. Something is going on here. The moral electromagnet is turning back on, and nowadays, just very recently, Democrats really dislike Republicans. Republicans really dislike the Democrats. We ' re changing. It ' s as though the moral electromagnet is affecting us too. It ' s like put out in the two oceans and it ' s pulling the whole country apart, pulling left and right into their own territories like the Bloods and the Crips. Now, there are many reasons why this is happening to us, and many of them we cannot reverse. We will never again have a political class that was forged by the experience of fighting together in World War II against a common enemy. We will never again have just three television networks, all of which are relatively centrist. And we will never again have a large group of **conservative** southern Democrats and liberal northern **Republicans** making it easy, making there be a lot of overlap for bipartisan cooperation. So for a lot of reasons, those decades after the Second World War were an historically anomalous time. We will never get back to those low levels of polarization, I believe. But there ' s a lot that we can do. There are dozens and dozens of reforms we can do that will make things better, because a lot of our **dysfunction** can be traced directly to things that Congress did to itself in the 1990s that created a much more polarized and dysfunctional institution. These changes are detailed in many books. These are two that I strongly recommend, and they list a whole bunch of reforms. I ' m just going to group them into three broad classes here. So if you think about this as the problem of a dysfunctional, hyper-polarized institution, well, the first step is, do what you can so that fewer hyper- partisans get elected in the first place, and when you have closed party primaries, and only the most committed Republicans and Democrats are voting, you ' re nominating and selecting the most extreme hyper-partisans. So open primaries would make that problem much, much less severe. But the problem isn ' t primarily that we ' re electing bad people to Congress. From my experience, and from what I ' ve heard from Congressional insiders, most of the people going to Congress are good, hard-working, intelligent people who really want to solve problems, but once they get there, they find that they are forced to play a game that rewards hyper-partisanship and that punishes independent thinking. You step out of line, you get punished. So there are a lot of reforms we could do that will counteract this. For example, this"Citizens United"ruling is a disaster, because it means there ' s like a money gun aimed at your head, and if you step out of line, if you try to reach across the aisle, there ' s a ton of money waiting to be given to your opponent to make **everybody** think that you are a terrible person through negative advertising. But the third class of reforms is that we ' ve got to change the nature of social relationships in Congress. The politicians I ' ve met are generally very extroverted, friendly, very socially skillful people, and that ' s the nature of politics. You ' ve got to make relationships, make deals, you ' ve got to cajole, please, flatter, you ' ve got to use your personal skills, and that ' s the way politics has always worked. But beginning in the 1990s, first the House of Representatives changed its legislative calendar so that all business is basically done in the middle of the week. Nowadays, Congressmen fly in on Tuesday morning, they do battle for two days, then they fly home Thursday afternoon. They don ' t move their families to the District. They don ' t meet each other ' s spouses or children. There ' s no more relationship there. And trying to run Congress without human relationships is like trying to run a car without motor oil. Should we be surprised when the whole thing freezes up and descends into paralysis and polarization? A simple change to the legislative calendar, such as having business stretch out for three weeks and then they get a week off to go home, that would change the fundamental relationships in Congress. So there ' s a lot we can do, but who ' s going to push them to do it? There are a number of groups that are working on this. No Labels and Common Cause, I think, have very good ideas for changes we need to do to make our democracy more responsive and our Congress more effective. But I ’ d like to supplement their work with a little psychological trick, and the trick is this. Nothing pulls people together like a common threat or a common attack, especially an attack from a foreign enemy, unless of course that threat hits on our polarized psychology, in which case, as I said before, it can actually pull us apart. Sometimes a single threat can polarize us, as we saw. But what if the situation we face is not a single threat but is actually more like this, where there ’ s just so much **stuff** coming in, it ’ s just,”Start shooting, come on, **everybody**, we ’ ve got to just work together, just start shooting.”Because actually, we do face this situation. This is where we are as a country. So here ’ s another asteroid. We ’ ve all seen versions of this graph, right, which shows the changes in wealth since 1979, and as you can see, almost all the gains in wealth have gone to the top 20 percent, and especially the top one percent. Rising inequality like this is associated with so many problems for a democracy. Especially, it destroys our **ability** to **trust** each other, to feel that we ’ re all in the same boat, because it ’ s obvious we ’ re not. Some of us are sitting there **safe** and sound in gigantic private yachts. Other people are clinging to a piece of driftwood. We ’ re not all in the same boat, and that means nobody ’ s willing to sacrifice for the common good. The left has been screaming about this asteroid for 30 years now, and the right says,”Huh, what? Hmm? No problem. No problem.”Now, why is that happening to us? Why is the inequality rising? Well, one of the largest causes, after globalization, is actually this fourth asteroid, rising nonmarital births. This graph shows the steady rise of out-of-wedlock births since the 1960s. Most Hispanic and black children are now born to unmarried mothers. Whites are headed that way too. Within a decade or two, most American children will be born into homes with no father. This means that there ’ s much less money coming into the house. But it ’ s not just money. It ’ s also stability versus chaos. As I know from working with street children in Brazil, Mom ’ s boyfriend is often a really, really dangerous person for kids. Now the right has been screaming about this asteroid since the 1960s, and the left has been saying,”It ’ s not a problem. It ’ s not a problem.”The left has been very reluctant to say that marriage is actually good for women and for children. Now let me be clear. I ’ m not blaming the women here. I ’ m actually more critical of the men who won ’ t take responsibility for their own children and of an economic system that makes it difficult for many men to earn enough money to **support** those children. But even if you blame nobody, it still is a national problem, and one side has been more concerned about it than the other. The New York Times finally noticed this asteroid with a front-page story last July showing how the decline of marriage contributes to inequality. We are becoming a nation of just two classes. When Americans go to college and marry each other, they have very low divorce rates. They earn a lot of money, they **invest** that money in their kids, some of them become tiger mothers, the kids rise to their full potential, and the kids go on to become the top two lines in this graph. And then there ’ s **everybody** else : the children who don ’ t benefit from a stable marriage, who don ’ t have as much **invested** in them, who don ’ t grow up in a stable environment, and who go on to become the bottom three lines in that graph. So once again, we see that these two graphs are actually saying the same thing. As before, we ’ ve got a problem, we ’ ve got to start working on this, we ’ ve got to do something, and what ’ s wrong with you people that you don ’ t see my threat? But if **everybody** could just take off their partisan blinders, we ’ d see that these two problems actually are best addressed together. Because if you really care about income inequality, you might want to talk to some evangelical Christian groups that are working on ways to promote marriage. But then you

’ re going to run smack into the problem that women don ’ t generally want to marry someone who doesn ’ t have a job. So if you really care about strengthening families, you might want to talk to some liberal groups who are working on promoting educational equality, who are working on raising the minimum wage, who are working on finding ways to stop so many men from being sucked into the criminal justice system and taken out of the marriage market for their whole lives. So to conclude, there are at least four asteroids headed our way. How many of you can see all four? Please raise your hand right now if you ’ re willing to admit that all four of these are national problems. Please raise your hands. Okay, almost all of you. Well, congratulations, you **guys** are the inaugural members of the Asteroids Club, which is a club for all Americans who are willing to admit that the other side actually might have a point. In the Asteroids Club, we don ’ t start by looking for common ground. Common ground is often very hard to find. No, we start by looking for common threats because common threats make common ground. Now, am I being naive? Is it naive to think that people could ever lay down their swords, and left and right could actually work together? I don ’ t think so, because it happens, not all that often, but there are a variety of examples that point the way. This is something we can do. Because Americans on both sides care about the decline in civility, and they ’ ve formed dozens of organizations, at the national level, such as this one, down to many local organizations, such as To The Village Square in Tallahassee, Florida, which tries to bring state **leaders** together to help facilitate that sort of working together human relationship that ’ s necessary to solve Florida ’ s problems. Americans on both sides care about global poverty and AIDS, and on so many humanitarian issues, liberals and evangelicals are actually natural allies, and at times they really have worked together to solve these problems. And most surprisingly to me, they sometimes can even see eye to eye on criminal justice. For example, the incarceration rate, the prison population in this country has quadrupled since 1980. Now this is a social disaster, and liberals are very concerned about this. The Southern Poverty Law Center is often fighting the prison-industrial complex, fighting to prevent a system that ’ s just sucking in more and more poor young men. But are **conservatives** happy about this? Well, Grover Norquist isn ’ t, because this system costs an unbelievable amount of money. And so, because the prison-industrial complex is bankrupting our states and corroding our souls, groups of fiscal **conservatives** and Christian **conservatives** have come together to form a group called Right on Crime. And at times they have worked with the Southern Poverty Law Center to oppose the building of new prisons and to work for reforms that will make the justice system more efficient and more humane. So this is possible. We can do it. Let us therefore go to battle stations, not to fight each other, but to begin deflecting these incoming asteroids. And let our first mission be to press Congress to reform itself, before it ’ s too late for our nation. Thank you.

Text Sample 573: POS Dim 4 – 2012 – Score 15.00 – 2012/t002337.txt

America ’ s public energy conversation boils down to this question : Would you rather die of A) oil wars, or B) **climate** change, or C) nuclear holocaust, or D) all of the above? Oh, I missed one : or E) none of the above? That ’ s the one we ’ re not normally offered. What if we could make energy do our work without working our undoing? Could we have fuel without fear? Could we reinvent fire? You see, fire made us human ; fossil fuels made us modern. But now we need a new fire that makes us **safe**, secure, healthy and durable. Let ’ s see how. Four-fifths of the world ’ s energy still comes from burning each year four cubic miles of the rotted remains of primeval swamp goo. Those fossil fuels have built our civilization. They ' ve created our wealth. They ' ve enriched the lives of billions. But they also have rising costs to our security, economy, health and environment that are starting to erode, if not outweigh their benefits. So we need a new fire. And switching from the old fire to the new fire means changing two big stories about oil and electricity, each of which puts two- fifths of the fossil carbon in the **air**. But they ' re really quite distinct. Less than one percent of our electricity is made from oil — although almost half is made from coal. Their uses are quite concentrated. Three-fourths of our oil fuel is transportation. Three-fourths of our electricity powers buildings. And the rest of both runs factories. So very efficient vehicles, buildings and factories save oil and coal, and also natural gas that can displace both of them. But today ' s energy system is not just inefficient, it is also disconnected, aging, dirty and insecure. So it needs refurbishment. By 2050 though, it could become efficient, connected and distributed with elegantly frugal autos, factories and buildings all relying on a modern, secure and resilient electricity system. We can eliminate our addiction to oil and coal by 2050 and use one-third less natural gas while switching to efficient use and renewable supply. This could cost, by 2050, five trillion dollars less in net present value, that is expressed as a lump sum today, than business as usual — assuming that carbon emissions and all other hidden or external costs are worth zero — a conservatively low estimate. Yet this cheaper energy system could **support** 158 percent bigger U. S. economy all without needing oil or coal, or for that matter nuclear energy. Moreover, this **transition** needs no new inventions and no acts of Congress and no new federal taxes, mandate subsidies or laws and running Washington gridlock. Let me say that again. I ' m going to tell you how to get the United States completely off oil and coal, five trillion dollars cheaper with no act of Congress led by business for profit. In other words, we ' re going to use our most effective institutions — private enterprise co-evolving with civil society and sped by military innovation to go around our least effective institutions. And whether you care most about profits and jobs and competitive advantage or national security, or environmental stewardship and **climate** protection and public health, reinventing fire makes sense and makes money. General Eisenhower reputedly said that enlarging the boundaries of a tough problem makes it soluble by encompassing more options and more synergies. So in reinventing fire, we integrated all four **sectors** that use energy — transportation, buildings, industry and electricity — and we integrated four kinds of innovation, not just technology and **policy**, but also design and business strategy. Those combinations yield very much more than the sum of the parts, especially in creating deeply disruptive business opportunities. Oil costs our economy two billion dollars a day, plus another four billion dollars a day in hidden economic and military costs, raising its total cost to over a sixth of GDP. Our mobility fuel goes three-fifths to automobiles. So let ' s start by making autos oil free. Two-thirds of the energy it takes to move a typical car is caused by its weight. And every unit of energy you save at the wheels, by taking out weight or drag, saves seven units in the tank, because you don ' t have to waste six units getting the energy to the wheels. Unfortunately, over the past quarter century, epidemic obesity has made our two-ton steel cars gain weight twice as fast as we have. But today, ultralight, ultrastrong materials, like carbon fiber composites, can make dramatic weight savings snowball and can make cars simpler and cheaper to build. Lighter and more slippery autos need less force to move them, so their engines get smaller. Indeed, that sort of vehicle fitness then makes electric propulsion affordable because the batteries or fuel cells also get smaller and lighter and cheaper. So sticker prices will ultimately fall to about the same as today, while the driving cost, even from the start, is very much lower. So these innovations together can transform automakers from wringing tiny savings out of Victorian engine and seal-stamping technologies to the steeply falling costs of three linked innovations that strongly reenforce each other — namely ultralight materials, making them into structures and electric propulsion. The sales can grow and the prices fall even faster with temporary feebates, that is rebates for efficient new autos paid for by fees on inefficient ones. And just in the first two years the biggest of Europe ’ s five feebate programs has tripled the speed of improving automotive efficiency. The resulting shift to electric autos is going to be as game-changing as shifting from typewriters to the gains in computers. Of course, computers and electronics are now America ’ s biggest industry, while typewriter makers have vanished. So vehicle fitness opens a new automotive competitive strategy that can double the oil savings over the next 40 years, but then also make electrification affordable, and that displaces the rest of the oil. America could lead this next automotive revolution. Currently the **leader** is Germany. Last year, Volkswagen announced that by next year they ’ ll be producing this carbon fiber plugin hybrid getting 230 miles a gallon. Also last year, BMW announced this carbon fiber electric car, they said that its carbon fiber is paid for by needing fewer batteries. And they said,”We do not intend to be a typewriter maker.”Audi **claimed** it ’ s going to beat them both by a year. Seven years ago, an even faster and cheaper American manufacturing technology was used to make this little carbon fiber test part, which doubles as a carbon cap. In one minute — and you can tell from the sound how immensely stiff and strong it is. Don ’ t worry about dropping it, it ’ s tougher than titanium. Tom Friedman actually whacked it as hard as he could with a sledgehammer without even scuffing it. But such manufacturing techniques can scale to automotive speed and cost with aerospace performance. They can save four-fifths of the capital needed to make autos. They can save lives because this **stuff** can absorb up to 12 times as much crash energy per pound as steel. If we made all of our autos this way, it would save oil equivalent to finding one and a half Saudi Arabias, or half an OPEC, by drilling in the Detroit formation, a very prospective play. And all those mega-barrels under Detroit cost an average of 18 bucks a barrel. They are all- American, carbon-free and inexhaustible. The same physics and the same business logic also apply to big vehicles. In the five years ending with 2010, Walmart saved 60 percent of the fuel per ton-mile in its giant fleet of heavy trucks through better logistics and design. But just the technological savings in heavy trucks can get to two-thirds. And combined with triple to quintuple efficiency airplanes, now on the drawing board, can save close to a trillion dollars. Also today ’ s military revolution in energy efficiency is going to speed up all of these civilian advances in much the same way that military R&D has given us the Internet, the Global Positioning System and the jet engine and microchip industries. As we design and build vehicles better, we can also use them smarter by harnessing four powerful techniques for eliminating needless driving. Instead of just seeing the travel grow, we can use innovative pricing, charging for road infrastructure by the mile, not by the gallon. We can use some smart IT to enhance transit and enable car sharing and ride sharing. We can allow smart and lucrative growth models that help people already be near where they want to be, so they don ’ t need to go somewhere else. And we can use smart IT to make traffic free-flowing. Together, those things can give us the same or better access with 46 to 84 percent less driving, saving another 0. 4 trillion dollars, plus 0. 3 trillion dollars from using trucks more productively. So 40 years hence, when you add it all up, a far more mobile U. S. economy can use no oil. Saving or displacing barrels for 25 bucks rather than buying them for over a hundred, adds up to a $4 trillion net saving counting all the hidden costs at zero. So to get mobility without oil, to phase out the oil, we can get efficient and then switch fuels. Those 125 to 240 mile-per-gallon-equivalent autos can use any mixture of **hydrogen** fuel cells, electricity and advanced biofuels. The trucks and planes can realistically use **hydrogen** or advanced biofuels. The trucks could even use natural gas. But no vehicles will need oil. And the most biofuel we might need, just three million barrels a day, can be made two-thirds from waste without displacing any cropland and without harming soil or **climate**. Our team speeds up these kinds of oil savings by what we call"institutional acupuncture."We figure out where the business logic is congested and not flowing properly, we stick little needles in it to get it flowing, working with partners like Ford and Walmart and the Pentagon. And the long **transition** is already well under way. In fact, three years ago mainstream analysts were starting to see peak oil, not in supply, but in **demand**. And Deutsche Bank even said world oil use could peak around 2016. In other words, oil is getting uncompetitive even at low prices before it becomes unavailable even at high prices. But the electrified vehicles don ’ t need to burden the electricity grid. Rather, when smart autos exchange electricity and information through smart buildings with smart grids, they ’ re adding to the grid valuable flexibility and storage that help the grid integrate varying solar and wind power. So the electrified autos make the auto and electricity problems easier to solve together than separately. And they also converge the oil story with our second big story, saving electricity and then making it differently. And those twin revolutions in electricity will bring to that **sector** more numerous and profound and diverse disruptions than any other **sector**, because we ’ ve got 21st century technology and speed colliding head-on with 20th and 19th century institutions, rules and cultures. Changing how we make electricity gets easier if we need less of it. Most of it now is wasted and the technologies for saving it keep improving faster than we ’ re installing them. So the unbought efficiency resource keeps getting ever bigger and cheaper. But as efficiency in buildings and industry starts to grow faster than the economy, America ’ s electricity use could actually shrink, even with the little extra use required for those efficient electrified autos. And we can do this just by reasonably accelerating existing trends. Over the next 40 years, buildings, which use three-quarters of the electricity, can triple or quadruple their energy productivity, saving 1. 4 trillion dollars, net present value, with a 33 percent internal rate of return or in English, the savings are worth four times what they cost. And industry can accelerate too, doubling its energy productivity with a 21 percent internal rate of return. The key is a disruptive innovation that we call integrative design that often makes very big energy savings cost less than small or no savings. That is, it can give you expanding returns, not diminishing returns. That is how our 2010 retrofit is saving over two-fifths of the energy in the Empire State Building — remanufacturing those six and a half thousand windows on site into super windows that pass light, but reflect heat. plus better lights and office equipment and such cut the maximum cooling load by a third. And then renovating smaller chillers instead of adding bigger ones saved 17 million dollars of capital cost, which helped pay for the other improvements and reduce the payback to just three years. Integrative design can also increase energy savings in industry. Dow ’ s billion-dollar efficiency investment has already returned nine billion dollars. But industry as a whole has another half-trillion dollars of energy still to save. For example, three-fifths of the world ’ s electricity runs motors. Half of that runs pumps and fans. And those can all be made more efficient, and the motors that turn them can have their system efficiency roughly doubled by integrating 35 improvements, paying back in about a year. But first we ought to be capturing bigger, cheaper savings that are normally ignored and are not in the textbooks. For example, pumps, the biggest use of motors, move liquid through pipes. But a standard industrial pumping loop was redesigned to use at least 86 percent less energy, not by getting better pumps, but just by replacing long, thin, crooked pipes with fat, short, straight pipes. This is not about new technology, it ' s just rearranging our metal furniture. Of course, it also shrinks the pumping equipment and its capital costs. So what do such savings mean for the electricity that is three-fifths used in motors? Well, from the coal burned at the power plant through all these compounding losses, only a tenth of the fuel energy actually ends up coming out the pipe as flow. But now let ' s turn those compounding losses around backwards, and every unit of flow or friction that we save in the pipe saves 10 units of fuel cost, pollution and what Hunter Lovins calls"global weirding"back at the power plant. And of course, as you go back upstream, the components get smaller and therefore cheaper. Our team has lately found such snowballing energy savings in more than 30 billion dollars worth of industrial redesigns — everything from data centers and chip fabs to mines and refineries. Typically our retrofit designs save about 30 to 60 percent of the energy and pay back in a few years, while the new facility designs save 40 to 90-odd percent with generally lower capital cost. Now needing less electricity would ease and speed the shift to new sources of electricity, chiefly renewables. China leads their explosive growth and their plummeting cost. In fact, these solar power module costs have just fallen off the bottom of the chart. And Germany now has more solar workers than America has steel workers. Already in about 20 states private installers will come put those cheap solar cells on your roof with no money down and beat your utility bill. Such unregulated products could ultimately add up to a virtual utility that bypasses your electric company just as your cellphone bypassed your wireline **phone** company. And this sort of thing gives utility executives the heebee-jeebees and it gives venture capitalists sweet dreams. Renewables are no longer a fringe activity. For each of the past four years half of the world ' s new generating capacity has been renewable, mainly lately in developing countries. In 2010, renewables other than big hydro, particularly wind and solar cells, got 151 billion dollars of private investment, and they actually surpassed the total installed capacity of nuclear power in the world by adding 60 billion watts in that one year. That happens to be the same amount of solar cell capacity that the world can now make every year — a number that goes up 60 or 70 percent a year. In contrast, the net additions of nuclear capacity and coal capacity and the orders behind those keep fading because they cost too much and they have too much financial risk. In fact in this country, no new nuclear power plant has been able to raise any private construction capital, despite seven years of 100-plus percent subsidies. So how else could we replace the coal-fired power plants? Well efficiency and gas can displace them all at just below their operating cost and, combined with renewables, can displace them more than 23 times at less than their replacement cost. But we only need to replace them once. We ' re often told though that only coal and nuclear plants can keep the lights on, because they ' re 24 / 7, whereas wind and solar power are variable, and hence supposedly unreliable. Actually no generator is 24 / 7. They all break. And when a big plant goes down, you lose a thousand megawatts in milliseconds, often for weeks or months, often without warning. That is exactly why we ' ve designed the grid to back up failed plants with working plants. And in exactly the same way, the grid can handle wind and solar power ' s forecastable variations. Hourly simulations show that largely or wholly renewable grids can deliver highly reliable power when they ' re forecasted, integrated and diversified by both type and location. And that ' s true both for continental areas like the U. S. or Europe and for smaller areas embedded within a larger grid. That is how, for example, four German states in 2010 were 43 to 52 percent wind powered. Portugal was 45 percent renewable powered, Denmark 36. And it ’ s how all of Europe can shift to renewable electricity. In America, our aging, dirty and insecure power system has to be replaced anyway by 2050. And whatever we replace it with is going to cost about the same, about six trillion dollars at present value — whether we buy more of what we ’ ve got or new nuclear and so- called clean coal, or renewables that are more or less centralized. But those four futures at the same cost differ profoundly in their risks, around national security, fuel, water, finance, technology, **climate** and health. For example, our over-centralized grid is very vulnerable to cascading and potentially economy-shattering blackouts caused by bad space weather or other natural disasters or a terrorist attack. But that blackout risk disappears, and all of the other risks are best managed, with distributed renewables organized into local micro-grids that normally interconnect, but can stand alone at need. That is, they can disconnect fractally and then reconnect seamlessly. That approach is exactly what the Pentagon is adopting for its own power supply. They think they need that ; how about the rest of us that they ’ re defending? We want our **stuff** to work too. At about the same cost as business as usual, this would maximize national security, customer choice, entrepreneurial opportunity and innovation. Together, efficient use and diverse dispersed renewable supply are starting to transform the whole electricity **sector**. Traditionally utilities build a lot of giant coal and nuclear plants and a bunch of big gas plants and maybe a little bit of efficiency renewables. And those utilities were rewarded, as they still are in 34 states, for selling you more electricity. However, especially where regulators are now instead rewarding cutting your bills, the investments are shifting radically toward efficiency, **demand** response, cogeneration, renewables and ways to knit them all together reliably with less transmission and little or no bulk electricity storage. So our energy future is not fate, but choice, and that choice is very flexible. In 1976, for example, government and industry insisted that the amount of energy needed to make a dollar of GDP could never go down. And I heretically suggested it could go down several-fold. Well that ’ s what ’ s actually happened so far. It ’ s fallen by half. But with today ’ s much better technologies, more mature delivery channels and integrative design, we can do far more and even cheaper. So to solve the energy problem, we just needed to enlarge it. And the results may at first seem incredible, but as Marshall McLuhan said,”Only puny secrets need protection. Big discoveries are protected by public incredulity.”Now combine the electricity and oil revolutions, both driven by modern efficiency, and you get the really big story : reinventing fire, where business enabled and sped by smart **policies** in mindful markets can lead the United States completely off oil and coal by 2050, saving 5 trillion dollars, growing the economy 2. 6- fold, strengthening out national security, oh, and by the way, by getting rid of the oil and coal, reducing the fossil carbon emissions by 82 to 86 percent. Now if you like any of those outcomes, you can **support** reinventing fire without needing to like all of them and without needing to agree about which of them is most important. So focusing on outcomes, not motives, can turn gridlock and **conflict** into a unifying solution to America ’ s energy challenge. This also turns out to be the best way to cope with global challenges — **climate** change, nuclear proliferation, energy insecurity, energy poverty — all of which make us less **safe**. Now our team at RMI helps smart companies to get unstuck and speed this journey via six sectoral initiatives, with some more hatching. Of course there ’ s still a lot of old thinking out there too. Former oil man Maurice Strong said,”Not all the fossils are in the fuel.”But as Edgar Woolard, who used to chair Dupont, reminds us,”Companies hampered by old thinking won ’ t be a problem because,”he said,”they simply won ’ t be around long-term.”I ’ ve described not just a once-in-a-civilization business opportunity, but one of the most profound **transitions** in the history of our species. We humans are inventing a new fire, not dug from below, but flowing from above ; not scarce, but bountiful ; not local, but everywhere ; not transient, but permanent ; not costly, but free. And but for a little transitional tail of natural gas and a bit of biofuel grown in ways that sustain and endure, this new fire is flameless. Efficiently used, it really can do our work without working our undoing. Each of you owns a piece of that $5 trillion prize. And our new book”Reinventing Fire”describes how you can capture it. So with the conversation just begun at ReinventingFire. com, let me invite you each to engage with us and with each other, with everyone around you, to help make the world richer, fairer, cooler and **safer** by together reinventing fire. Thank you.

Text Sample 574: POS Dim 4 – 2011 – Score 19.00 – 2011/t002551.txt
I was basically concerned about what was going on in the world. I couldn ’ t understand the starvation, the destruction, the killing of innocent people. Making sense of those things is a very difficult thing to do. And when I was 12, I became an actor. I was bottom of the class. I haven ’ t got any qualifications. I was told I was dyslexic. In fact, I have got qualifications. I got a D in pottery, which was the one thing that I did get — which was useful, obviously. And so concern is where all of this comes from. And then, being an actor, I was doing these different kinds of things, and I felt the content of the work that I was **involved** in really wasn ’ t cutting it, that there surely had to be more. And at that point, I read a book by Frank Barnaby, this wonderful nuclear physicist, and he said that media had a responsibility, that all **sectors** of society had a responsibility to try and progress things and move things forward. And that fascinated me, because I ’ d been messing around with a camera most of my life. And then I thought, well maybe I could do something. Maybe I could become a filmmaker. Maybe I can use the form of film constructively to in some way make a difference. Maybe there ’ s a little change I can get **involved** in. So I started thinking about peace, and I was obviously, as I said to you, very much moved by these images, trying to make sense of that. Could I go and speak to older and wiser people who would tell me how they made sense of the things that are going on? Because it ’ s obviously incredibly frightening. But I realized that, having been messing around with structure as an actor, that a series of sound bites in itself wasn ’ t enough, that there needed to be a mountain to climb, there needed to be a journey that I had to take. And if I took that journey, no matter whether it failed or succeeded, it would be completely irrelevant. The point was that I would have something to hook the questions of — is humankind fundamentally evil? Is the destruction of the world inevitable? Should I have children? Is that a responsible thing to do? Etc., etc. So I was thinking about peace, and then I was thinking, well where ’ s the starting point for peace? And that was when I had the idea. There was no starting point for peace. There was no day of global unity. There was no day of intercultural cooperation. There was no day when humanity came together, separate in all of those things and just shared it together — that we ’ re in this together, and that if we united and we interculturally cooperated, then that might be the key to humanity ’ s survival. That might shift the level of consciousness around the fundamental issues that humanity faces — if we did it just for a day. So obviously we didn ’ t have any money. I was living at my mom ’ s place. And we started writing letters to **everybody**. You very quickly work out what is it that you ’ ve got to do to fathom that out. How do you create a day voted by every

single head of state in the world to create the first ever Ceasefire Nonviolence Day, the 21st of September? And I wanted it to be the 21st of September because it was my granddad ' s favorite number. He was a prisoner of war. He saw the bomb go off at Nagasaki. It poisoned his blood. He died when I was 11. So he was like my hero. And the reason why 21 was the number is 700 men left, 23 came back, two died on the boat and 21 hit the ground. And that ' s why we wanted it to be the 21st of September as the date of peace. So we began this journey, and we launched it in 1999. And we wrote to heads of state, their ambassadors, Nobel Peace laureates, NGOs, faiths, various organizations — literally wrote to **everybody**. And very quickly, some letters started coming back. And we started to build this case. And I remember the first letter. One of the first letters was from the Dalai Lama. And of course we didn ' t have the money ; we were playing guitars and getting the money for the stamps that we were sending out all of [this mail]. A letter came through from the Dalai Lama saying,"This is an amazing thing. Come and see me. I ' d love to talk to you about the first ever day of peace."And we didn ' t have money for the flight. And I rang Sir Bob Ayling, who was CEO of BA at the time, and said,"Mate, we ' ve got this invitation. Could you give me a flight? Because we ' re going to go see him."And of course, we went and saw him and it was amazing. And then Dr. Oscar Arias came forward. And actually, let me go back to that slide, because when we launched it in 1999 — this idea to create the first ever day of ceasefire and non-violence — we invited thousands of people. Well not thousands — hundreds of people, lots of people — all the press, because we were going to try and create the first ever World Peace Day, a peace day. And we invited **everybody**, and no press showed up. There were 114 people there — they were mostly my friends and family. And that was kind of like the launch of this thing. But it didn ' t matter because we were **documenting**, and that was the thing. For me, it was really about the process. It wasn ' t about the end result. And that ' s the beautiful thing about the camera. They used to say the pen is mightier than the sword. I think the camera is. And just staying in the moment with it was a beautiful thing and really empowering actually. So anyway, we began the journey. And here you see people like Mary Robinson, I went to see in Geneva. I ' m cutting my hair, it ' s getting short and long, because every time I saw Kofi Annan, I was so worried that he thought I was a hippie that I cut it, and that was kind of what was going on. Yeah, I ' m not worried about it now. So Mary Robinson, she said to me,"Listen, this is an idea whose time has come. This must be created."Kofi Annan said,"This will be beneficial to my troops on the ground."The OAU at the time, led by Salim Ahmed Salim, said,"I must get the African countries **involved**."Dr. Oscar Arias, Nobel Peace laureate, president now of Costa Rica, said,"I ' ll do everything that I can."So I went and saw Amr Moussa at the League of Arab States. I met Mandela at the Arusha peace talks, and so on and so on and so on — while I was building the case to prove whether this idea would make sense. And then we were listening to the people. We were **documenting** everywhere. 76 countries in the last 12 years, I ' ve **visited**. And I ' ve always spoken to women and children wherever I ' ve gone. I ' ve recorded 44, 000 young people. I ' ve recorded about 900 hours of their thoughts. I ' m really clear about how young people feel when you talk to them about this idea of having a starting point for their actions for a more peaceful world through their poetry, their art, their literature, their music, their sport, whatever it might be. And we were listening to **everybody**. And it was an incredibly thing, working with the U. N. and working with NGOs and building this case. I felt that I was presenting a case on behalf of the global community to try and create this day. And the stronger the case and the more detailed it was, the better chance we had of creating this day. And it was this

stuff, this, where I actually was in the beginning kind of thinking no matter what happened, it didn ’ t actually matter. It didn ’ t matter if it didn ’ t create a day of peace. The fact is that, if I tried and it didn ’ t work, then I could make a statement about how unwilling the global community is to unite — until, it was in Somalia, picking up that young girl. And this young child who ’ d taken about an inch and a half out of her leg with no antiseptic, and that young boy who was a child soldier, who told me he ’ d killed people — he was about 12 — these things made me realize that this was not a film that I could just stop. And that actually, at that moment something happened to me, which obviously made me go,”I ’ m going to **document**. If this is the only film that I ever make, I ’ m going to **document** until this becomes a reality.”Because we ’ ve got to stop, we ’ ve got to do something where we unite — separate from all the politics and religion that, as a young person, is confusing me. I don ’ t know how to get **involved** in that process. And then on the seventh of September, I was invited to New York. The Costa Rican government and the British government had put forward to the United Nations General Assembly, with 54 co-sponsors, the idea of the first ever Ceasefire Nonviolence Day, the 21st of September, as a fixed calendar date, and it was unanimously adopted by every head of state in the world. Yeah, but there were hundreds of individuals, obviously, who made that a reality. And thank you to all of them. That was an incredible moment. I was at the top of the General Assembly just looking down into it and seeing it happen. And as I mentioned, when it started, we were at the Globe, and there was no press. And now I was thinking,”Well, the press it really going to hear this story.”And suddenly, we started to institutionalize this day. Kofi Annan invited me on the morning of September the 11th to do a press **conference**. And it was 8:00 AM when I stood there. And I was waiting for him to come down, and I knew that he was on his way. And obviously he never came down. The statement was never made. The world was never told there was a day of global ceasefire and nonviolence. And it was obviously a tragic moment for the thousands of people who lost their lives, there and then subsequently all over the world. It never happened. And I remember thinking,”This is exactly why, actually, we have to work even harder. And we have to make this day work. It ’ s been created ; nobody knows. But we have to continue this journey, and we have to tell people, and we have to prove it can work.”And I left New York freaked, but actually empowered. And I felt inspired by the possibilities that if it did, then maybe we wouldn ’ t see things like that. I remember putting that film out and going to cynics. I was showing the film, and I remember being in Israel and getting it absolutely slaughtered by some **guys** having watched the film — that it ’ s just a day of peace, it doesn ’ t mean anything. It ’ s not going to work ; you ’ re not going to stop the fighting in Afghanistan ; the Taliban won ’ t listen, etc., etc. It ’ s just symbolism. And that was even worse than actually what had just happened in many ways, because it couldn ’ t not work. I ’ d spoken in Somalia, Burundi, Gaza, the West Bank, India, Sri Lanka, Congo, wherever it was, and they ’ d all tell me,”If you can create a window of opportunity, we can move aid, we can vaccinate children. Children can lead their projects. They can unite. They can come together. If people would stop, lives will be saved.”That ’ s what I ’ d heard. And I ’ d heard that from the people who really understood what **conflict** was about. And so I went back to the United Nations. I decided that I ’ d continue filming and make another movie. And I went back to the U. N. for another couple of years. We started moving around the corridors of the U. N. system, governments and NGOs, trying desperately to find somebody to come forward and have a go at it, see if we could make it possible. And after lots and lots of meetings obviously, I ’ m delighted that this man, Ahmad Fawzi, one of my heroes and mentors really, he managed to get UNICEF **involved**. And UNICEF, God bless them, they said,"Okay, we ' ll have a go."And then UNAMA became **involved** in Afghanistan. It was historical. Could it work in Afghanistan with UNAMA and WHO and civil society, etc., etc., etc.? And I was getting it all on film and I was recording it, and I was thinking,"This is it. This is the possibility of it maybe working. But even if it doesn ' t, at least the door is open and there ' s a chance."And so I went back to London, and I went and saw this chap, Jude Law. And I saw him because he was an actor, I was an actor, I had a connection to him, because we needed to get to the press, we needed this attraction, we needed the media to be **involved**. Because if we start pumping it up a bit maybe more people would listen and there ' d be more — when we got into certain areas, maybe there would be more people **interested**. And maybe we ' d be helped financially a little bit more, which had been desperately difficult. I won ' t go into that. So Jude said,"Okay, I ' ll do some statements for you."While I was filming these statements, he said to me,"Where are you going next?"I said,"I ' m going to go to Afghanistan."He said,"Really?"And I could sort of see a little look in his eye of interest. So I said to him,"Do you want to come with me? It ' d be really **interesting** if you came. It would help and bring attention. And that attention would help leverage the situation, as well as all of the other sides of it."I think there ' s a number of pillars to success. One is you ' ve got to have a great idea. The other is you ' ve got to have a constituency, you ' ve got to have finance, and you ' ve got to be able to raise awareness. And actually I could never raise awareness by myself, no matter what I ' d achieved. So these **guys** were absolutely crucial. So he said yes, and we found ourselves in Afghanistan. It was a really incredible thing that when we landed there, I was talking to various people, and they were saying to me,"You ' ve got to get **everybody involved** here. You can ' t just expect it to work. You have to get out and work."And we did, and we traveled around, and we spoke to elders, we spoke to doctors, we spoke to nurses, we held press **conferences**, we went out with soldiers, we sat down with ISAF, we sat down with NATO, we sat down with the U. K. government. I mean, we basically sat down with **everybody** — in and out of schools with ministers of education, holding these press **conferences**, which of course, now were loaded with press, **everybody** was there. There was an interest in what was going on. This amazing woman, Fatima Gailani, was absolutely instrumental in what went on as she was the spokesperson for the resistance against the Russians. And her Afghan network was just absolutely everywhere. And she was really crucial in getting the message in. And then we went home. We ' d sort of done it. We had to wait now and see what happened. And I got home, and I remember one of the team bringing in a letter to me from the Taliban. And that letter basically said,"We ' ll observe this day. We will observe this day. We see it as a window of opportunity. And we will not engage. We ' re not going to engage."And that meant that humanitarian workers wouldn ' t be kidnapped or killed. And then suddenly, I obviously knew at this point, there was a chance. And days later, 1. 6 million children were vaccinated against polio as a consequence of **everybody** stopping. And like the General Assembly, obviously the most wonderful, wonderful moment. And so then we wrapped the film up and we put it together because we had to go back. We put it into Dari and Pashto. We put it in the local dialects. We went back to Afghanistan, because the next year was coming, and we wanted to **support**. But more importantly, we wanted to go back, because these people in Afghanistan were the heroes. They were the people who believed in peace and the possibilities of it, etc., etc. — and they made it real. And we wanted to go back and show them the film and say,"Look, you **guys** made this possible. And thank you very much."And we gave the film over. Obviously it was shown, and it was amazing. And then that year, that year, 2008, this ISAF statement from Kabul, Afghanistan, September 17th :”General Stanley McChrystal, commander of **international** security assistance forces in Afghanistan, announced today ISAF will not conduct offensive military operations on the 21st of September.”They were saying they would stop. And then there was this other statement that came out from the U. N. Department of Security and Safety saying that, in Afghanistan, because of this work, the violence was down by 70 percent. 70 percent reduction in violence on this day at least. And that completely blew my mind almost more than anything. And I remember being stuck in New York, this time because of the volcano, which was obviously much less harmful. And I was there thinking about what was going on. And I kept thinking about this 70 percent. 70 percent reduction in violence — in what everyone said was completely impossible and you couldn ’ t do. And that made me think that, if we can get 70 percent in Afghanistan, then surely we can get 70 percent reduction everywhere. We have to go for a global truce. We have to utilize this day of ceasefire and nonviolence and go for a global truce, go for the largest recorded cessation of hostilities, both domestically and internationally, ever recorded. That ’ s exactly what we must do. And on the 21st of September this year, we ’ re going to launch that campaign at the O2 Arena to go for that process, to try and create the largest recorded cessation of hostilities. And we will utilize all kinds of things — have a dance and social media and **visiting** on Facebook and **visit** the website, sign the petition. And it ’ s in the six official languages of the United Nations. And we ’ ll globally link with government, inter-government, non-government, education, unions, sports. And you can see the education box there. We ’ ve got resources at the moment in 174 countries trying to get young people to be the driving force behind the vision of that global truce. And obviously the life-saving is increased, the concepts help. Linking up with the Olympics — I went and saw Seb Coe. I said,”London 2012 is about truce. Ultimately, that ’ s what it ’ s about.”Why don ’ t we all team up? Why don ’ t we bring truce to life? Why don ’ t you **support** the process of the largest ever global truce? We ’ ll make a new film about this process. We ’ ll utilize sport and football. On the Day of Peace, there ’ s thousands of football matches all played, from the favelas of Brazil to wherever it might be. So, utilizing all of these ways to inspire individual action. And ultimately, we have to try that. We have to work together. And when I stand here in front of all of you, and the people who will watch these things, I ’ m excited, on behalf of **everybody** I ’ ve met, that there is a possibility that our world could unite, that we could come together as one, that we could lift the level of consciousness around the fundamental issues, brought about by individuals. I was with Brahimi, Ambassador Brahimi. I think he ’ s one of the most incredible men in relation to **international** politics — in Afghanistan, in Iraq. He ’ s an amazing man. And I sat with him a few weeks ago. And I said to him,”Mr. Brahimi, is this nuts, going for a global truce? Is this possible? Is it really possible that we could do this?”He said,”It ’ s absolutely possible.”I said,”What would you do? Would you go to governments and lobby and use the system?”He said,”No, I ’ d talk to the individuals.”It ’ s all about the individuals. It ’ s all about you and me. It ’ s all about partnerships. It ’ s about your constituencies ; it ’ s about your businesses. Because together, by working together, I seriously think we can start to change things. And there ’ s a wonderful man sitting in this audience, and I don ’ t know where he is, who said to me a few days ago — because I did a little rehearsal — and he said,”I ’ ve been thinking about this day and imagining it as a square with 365 squares, and one of them is white.”And it then made me think about a glass of water, which is clear. If you put one drop, one drop of something, in that water, it ’ ll change it forever. By working together, we can create peace one day. Thank you **TED**. Thank you. Thank you. Thanks a lot.

Thank you very much. Thank you.

Text Sample 575: POS Dim 4 – 2011 – Score 16.00 – 2011/t002513.txt
Good afternoon, I ’ m proud to be here at TEDxKrakow. I ’ ll try to speak a little bit today about a phenomenon which can, and actually is changing the world, and whose name is people power. I ’ ll start with an anecdote, or for those of you who are Monty Python lovers, a Monty Python type of sketch. Here it is. It is December 15, 2010. Somebody gives you a bet : you will look at a crystal ball, and you will see the future ; the future will be accurate. But you need to share it with the world. OK, curiosity killed the cat, you take the bet, you look at the crystal ball. One hour later, you ’ re sitting in a building of the national TV, in a top show, and you tell the story. Before the end of 2011, Ben Ali, and Mubarak, and Gaddafi would be down, and prosecuted. Saleh of Yemen and Assad of Syria would be either challenged, or already on their knees. Osama bin Laden would be dead, and Ratko Mladic would be in the Hague. Now, the anchor watches you with a strange gaze on his face. And then, on top of it you add :”And thousands of young people from Athens, Madrid and New York will demonstrate for social justice, **claiming** they are inspired by Arabs.”Next thing you know, two **guys** in white appear, they give you the strange t- shirt, take you to the nearest mental institution. So I would like to speak a little bit about the phenomenon which is behind what already seems to be a very bad year for bad **guys**. And this phenomenon is called people power. Well, people power has been there for a while. It helped Gandhi kick the **Brits** from India, it helped Martin Luther King win his historic racial struggle. It helped a local, Lech Walesa, to kick out one million Soviet troops from Poland, and in beginning the end of the Soviet Union as we know it. So what ’ s new in it? What seems to be very new, which is the idea I would like to share with you today, is that there is a set of rules and skills which can be learned and taught in order to perform successful nonviolent struggle. If this is true, we can help these movements. Well, the first one - analytic skills. I ’ ll try where it all started in the Middle East. And for so many years, we were living with a completely wrong perception of the Middle East. It was looking like the frozen region. Literally a refrigerator. And there were only two types of meal there. Steak, which stands for a Mubarak-Ben Ali type of military police dictatorship, or a potato, which stands for a Tehran type of theocracies. And **everybody** was amazed when the refrigerator opened, and millions of young, mainly secular people stepped out to do the change. Guess what - they didn ’ t watch the demographics. What is the average age of an Egyptian? 24. How long was Mubarak in power? 31. So, this system was just obsolete, they expired. And young people of the Arab world have awakened one morning, and understood that power lies in their hands. The rest is the year in front of us. And guess what? The same Generation Y, with their rules, with their tools, with their games, and with their language, which sounds a little bit strange to me. I ’ m 38 now. And can you look at the age of the people on the streets of Europe? It seems that Generation Y is coming. Now, let me set another example. I ’ m meeting different people throughout the world, and they are, you know, academics, and professors, and doctors, and they will always talk conditions. They will say :”People power will work only if the regime is not too oppressive.”They will say :”People power will work, if the annual income of the country is between X and Z.”They will say :”People power will work only if there is a foreign pressure.”They will say :”People power will work only if there is no oil.”And, I mean, there is a set of conditions. Well, the news here is that your skills during the **conflict** seem

to be more important than the conditions. Namely, the skills of unity, planning, and maintaining nonviolent discipline. Let me give you an example. I come from a country called Serbia. It took us 10 years to unite 18 opposition party **leaders**, with their big egos, behind one single candidate against the Balkan dictator Slobodan Milosevic. Guess what? That was the day of his defeat. You look at the Egyptians, they fight on Tahrir Square, they get rid of their individual symbols, they appear on the street only with the flag of Egypt. I will give you a counter-example. You see nine presidential candidates running against Lukashenko, you all know the outcome. So unity is a big thing. And this can be achieved. Same with planning. Somebody has lied to you about the successful and spontaneous nonviolent revolution. That thing doesn ’ t exist in the world. Whenever you see young people in front of the row trying to fraternize with the police or military, somebody was thinking about it before. Now, at the end, nonviolent discipline. And this is probably the game-changer. If you maintain nonviolent discipline, you ’ ll exclusively win. You have 100, 000 people in a nonviolent march, one idiot or agent-provocateur throwing a stone. Guess what takes all the cameras. That one **guy**. One single act of violence can literally destroy your movement. Now, let me move to another place. It ’ s the selection of strategies and tactics. There are certain rules in nonviolent struggle you may follow. First, you start small. Second, you pick the battles you can win. It ’ s only 200 of us in this room. We won ’ t call for the march of a million. But what if we organized the spraying of graffiti throughout the night, all over Krakow. The city will know. So, we pick tactics accommodated to the event, especially this thing we call the small tactics of dispersion. They ’ re very useful in violent oppression. We are actually witnessing the picture of one of the best tactics ever used. It was on Tahrir square, where the **international** community was constantly frightened that, you know, the Islamists will overtake the revolution. What they organized — Christians protecting Muslims where they are praying, a Coptic wedding cheered by thousands of Muslims, the world has just changed the picture, but somebody was thinking about this previously. So there are so many things you can do instead of getting into one place, shouting, and you know, showing off in front of the security forces. Now, there is also another very important dynamic. And this is a dynamic that analysts normally don ’ t see. This is the dynamic between fear and apathy on the one side, and enthusiasm and humor on another side. So, it works like in a video game. You have the fear high, you have status quo. You have the enthusiasm higher, you see the fear is starting to melt. Day two, you see people running towards the police instead of from the police, in Egypt. You can tell that something is happening there. And then, it ’ s about the humor. Humor is such a powerful game-changer, and of course, it was very big in Poland. You know, we were just a small group of crazy students in Serbia when we made this big skit. We put the big petrol barrel with a portrait of Mr. President on it, in the middle of the Main Street. There was a hole in the top. So you could literally come, put a coin in, get a baseball bat, and hit his face. Sounds loud. And within minutes, we were sitting in a nearby caf having coffee, and there was a queue of people waiting to do this lovely thing. Well, that ’ s just the beginning of the show. The real show starts when the police appears.”What will they do?”Arrest us? We were nowhere to be seen. We were like three blocks away, observing it from our espresso bar. Arrest the shoppers, with kids? Doesn ’ t make sense. Of course, you could bet, they did the most **stupid** thing. They arrested the barrel. And now, the picture of the smashed face on the barrel, with the policemen dragging it to the police car, that was the best day for newspaper photographers that they will ever have. So, I mean, these are the things you can do. And you can always use humor. There is also one big thing about humor, it really hurts. Because these **guys** really are taking themselves too seriously. When you start to mock them, it hurts. Now, **everybody** is talking about His Majesty, the Internet, and it is also a very useful skill. But don ’ t rush to label things like”a Facebook Revolution,””Twitter Revolution.”Don ’ t mix the tools with the substance. It is true that the Internet and the new media are very useful in making things faster and cheaper. They also make it a bit **safer** for the participants, because they give partial anonymity. We ’ re watching the great example of something else the Internet can do. It can put the price tag of state-sponsored violence over a nonviolent protester. This is the famous group”We are all Khaled Said,”made by Wael Ghonim in Egypt, and his friend. This is the mutilated face of the **guy** who was beaten by the police. This is how he became known to the public, and this is what probably became the straw that broke the camel ’ s back. But here is also the bad news. The nonviolent struggle is won in the real world, in the streets. You will never change your society towards democracy, or, you know, the economy, if you sit down and click. There are risks to be taken, and there are living people who are winning the struggle. Well, the million-dollar question. What will happen in the Arab world? And though young people from the Arab world were pretty successful in bringing down three dictators, shaking the region, kind of persuading the clever kings from Jordan and Morocco to do substantial reforms, it is yet to be seen what will be the outcome. Whether the Egyptians and Tunisians will make it through the **transition**, or this will end in bloody ethnic and religious **conflicts**, whether the Syrians will maintain nonviolent discipline, faced with a brutal daily violence which kills thousands already, or they will slip into violent struggle and make ugly civil war. Will these revolutions be pushed through the **transitions** and democracy or be overtaken by the military or extremists of all kinds? We cannot tell. The same works for the Western **sector**, where you can see all these excited young people protesting around the world, occupying this, occupying that. Are they going to become the world wave? Are they going to find their skills, their enthusiasm, and their strategy to find what they really want and push for the reform, or will they just stay complaining about the endless list of the things they hate? This is the difference between the two paths. Now, what do the statistics have? My friend Maria Stephan ’ s book talks a lot about violent and nonviolent struggle, and there are some shocking data. If you look at the last 35 years and different social **transitions**, from dictatorship to democracy, you will see that, out of 67 different cases, in 50 of these cases it was nonviolent struggle which was the key power. This is one more reason to look at this phenomenon, this is one more reason to look at Generation Y. Enough for me to give them credit, and hope that they will find their skills and their courage to use nonviolent struggle and thus fix at least a part of the mess our generation is making in this world. Thank you.

Text Sample 576: POS Dim 4 – 2011 – Score 16.00 – 2011/t002473.txt

There ’ s a poem written by a very famous English poet at the end of the 19th century. It was said to echo in Churchill ’ s brain in the 1930s. And the poem goes :”On the idle hill of summer, lazy with the flow of streams, hark I hear a distant drummer, drumming like a sound in dreams, far and near and low and louder on the roads of earth go by, dear to friend and food to powder, soldiers marching, soon to die.”Those who are **interested** in poetry, the poem is”A Shropshire Lad”written by A. E. Housman. But what Housman understood, and you hear it in the symphonies of Nielsen too, was that the long, hot, silvan summers of stability of the 19th century were coming to a close, and that we were about to move into one of those terrifying periods of history when power changes. And these are always periods, ladies and gentlemen, accompanied by turbulence, and all too often by blood. And my message for you is that I believe we are condemned, if you like, to live at just one of those moments in history when the gimbals upon which the established order of power is beginning to change and the new look of the world, the new powers that exist in the world, are beginning to take form. And these are — and we see it very clearly today — nearly always highly turbulent times, highly difficult times, and all too often very bloody times. By the way, it happens about once every century. You might argue that the last time it happened — and that ' s what Housman felt coming and what Churchill felt too — was that when power passed from the old nations, the old powers of Europe, across the Atlantic to the new emerging power of the United States of America — the beginning of the American century. And of course, into the vacuum where the too-old European powers used to be were played the two bloody catastrophes of the last century — the one in the first part and the one in the second part : the two great World Wars. Mao Zedong used to refer to them as the European civil wars, and it ' s probably a more accurate way of describing them. Well, ladies and gentlemen, we live at one of those times. But for us, I want to talk about three factors today. And the first of these, the first two of these, is about a shift in power. And the second is about some new dimension which I want to refer to, which has never quite happened in the way it ' s happening now. But let ' s talk about the shifts of power that are occurring to the world. And what is happening today is, in one sense, frightening because it ' s never happened before. We have seen lateral shifts of power — the power of Greece passed to Rome and the power shifts that occurred during the European civilizations — but we are seeing something slightly different. For power is not just moving laterally from nation to nation. It ' s also moving vertically. What ' s happening today is that the power that was encased, held to accountability, held to the rule of law, within the institution of the nation state has now migrated in very large measure onto the global stage. The globalization of power — we talk about the globalization of markets, but actually it ' s the globalization of real power. And where, at the nation state level that power is held to accountability subject to the rule of law, on the **international** stage it is not. The **international** stage and the global stage where power now resides : the power of the Internet, the power of the satellite broadcasters, the power of the money changers — this vast money-go-round that circulates now 32 times the amount of money necessary for the trade it ' s supposed to be there to finance — the money changers, if you like, the financial speculators that have brought us all to our knees quite recently, the power of the multinational corporations now developing budgets often bigger than medium-sized countries. These live in a global space which is largely unregulated, not subject to the rule of law, and in which people may act free of constraint. Now that suits the powerful up to a moment. It ' s always suitable for those who have the most power to operate in spaces without constraint, but the lesson of history is that, sooner or later, unregulated space — space not subject to the rule of law — becomes populated, not just by the things you wanted — **international** trade, the Internet, etc. — but also by the things you don ' t want — **international** criminality, **international** terrorism. The revelation of 9 / 11 is that even if you are the most powerful nation on earth, nevertheless, those who inhabit that space can attack you even in your most iconic of cities one bright September morning. It ' s said that something like 60 percent of the four million dollars that was taken to fund 9 / 11 actually passed through the institutions of the Twin Towers which 9 / 11 destroyed. You see, our enemies also use this space — the space of mass travel, the Internet, satellite broadcasters — to be able to get around their poison, which is about destroying our systems and our ways. Sooner or later, sooner or later, the rule of history is that where power goes governance must follow. And if it is therefore the case, as I believe it is, that one of the phenomenon of our time is the globalization of power, then it follows that one of the challenges of our time is to bring governance to the global space. And I believe that the decades ahead of us now will be to a greater or lesser extent turbulent the more or less we are able to achieve that aim : to bring governance to the global space. Now notice, I ' m not talking about government. I ' m not talking about setting up some global democratic institution. My own view, by the way, ladies and gentlemen, is that this is unlikely to be done by spawning more U. N. institutions. If we didn ' t have the U. N., we ' d have to invent it. The world needs an **international** forum. It needs a means by which you can legitimize **international** action. But when it comes to governance of the global space, my guess is this won ' t happen through the creation of more U. N. institutions. It will actually happen by the powerful coming together and making treaty-based systems, treaty-based agreements, to govern that global space. And if you look, you can see them happening, already beginning to emerge. The World Trade Organization : treaty-based organization, entirely treaty-based, and yet, powerful enough to hold even the most powerful, the United States, to account if necessary. Kyoto : the beginnings of struggling to create a treaty- based organization. The G20 : we know now that we have to put together an institution which is capable of bringing governance to that financial space for financial speculation. And that ' s what the G20 is, a treaty-based institution. Now there ' s a problem there, and we ' ll come back to it in a minute, which is that if you bring the most powerful together to make the rules in treaty-based institutions, to fill that governance space, then what happens to the weak who are left out? And that ' s a big problem, and we ' ll return to it in just a second. So there ' s my first message, that if you are to pass through these turbulent times more or less turbulently, then our success in doing that will in large measure depend on our capacity to bring sensible governance to the global space. And watch that beginning to happen. My second point is, and I know I don ' t have to talk to an audience like this about such a thing, but power is not just shifting vertically, it ' s also shifting horizontally. You might argue that the story, the history of civilizations, has been civilizations gathered around seas — with the first ones around the Mediterranean, the more recent ones in the ascendents of Western power around the Atlantic. Well it seems to me that we ' re now seeing a fundamental shift of power, broadly speaking, away from nations gathered around the Atlantic [seaboard] to the nations gathered around the Pacific rim. Now that begins with economic power, but that ' s the way it always begins. You already begin to see the development of foreign **policies**, the augmentation of military budgets occurring in the other growing powers in the world. I think actually this is not so much a shift from the **West** to the East ; something different is happening. My guess is, for what it ' s worth, is that the United States will remain the most powerful nation on earth for the next 10 years, 15, but the context in which she holds her power has now radically altered ; it has radically changed. We are coming out of 50 years, most unusual years, of history in which we have had a totally mono-polar world, in which every compass needle for or against has to be referenced by its position to Washington — a world bestrode by a single colossus. But that ' s not a usual case in history. In fact, what ' s now emerging is the much more normal case of history. You ' re beginning to see the emergence of a multi-polar world. Up until now, the United States has been the **dominant** feature of our world. They will remain the most powerful nation, but they will be the most powerful nation in an increasingly multi-polar world. And you begin to see the alternative centers of power building up — in China, of course, though my own guess is that China ' s ascent to greatness is not smooth. It ' s going to be quite grumpy as China begins to democratize her society after liberalizing her economy. But that ' s a subject of a different discussion. You see India, you see Brazil. You see increasingly that the world now looks actually, for us Europeans, much more like Europe in the 19th century. Europe in the 19th century : a great British foreign secretary, Lord Canning, used to describe it as the"European concert of powers."There was a balance, a five-sided balance. Britain always played to the balance. If Paris got together with Berlin, Britain got together with Vienna and Rome to provide a counterbalance. Now notice, in a period which is dominated by a mono-polar world, you have fixed alliances — NATO, the Warsaw Pact. A fixed polarity of power means fixed alliances. But a multiple polarity of power means shifting and changing alliances. And that ' s the world we ' re coming into, in which we will increasingly see that our alliances are not fixed. Canning, the great British foreign secretary once said,"Britain has a common interest, but no common allies."And we will see increasingly that even we in the West will reach out, have to reach out, beyond the cozy circle of the Atlantic powers to make alliances with others if we want to get things done in the world. Note, that when we went into Libya, it was not good enough for the West to do it alone ; we had to bring others in. We had to bring, in this case, the Arab League in. My guess is Iraq and Afghanistan are the last times when the West has tried to do it themselves, and we haven ' t succeeded. My guess is that we ' re reaching the beginning of the end of 400 years — I say 400 years because it ' s the end of the Ottoman Empire — of the hegemony of Western power, Western institutions and Western values. You know, up until now, if the West got its act together, it could propose and dispose in every corner of the world. But that ' s no longer true. Take the last financial **crisis** after the Second World War. The West got together — the Bretton Woods Institution, World Bank, International Monetary Fund — the problem solved. Now we have to call in others. Now we have to create the G20. Now we have to reach beyond the cozy circle of our Western friends. Let me make a prediction for you, which is probably even more startling. I suspect we are now reaching the end of 400 years when Western power was enough. People say to me,"The Chinese, of course, they ' ll never get themselves **involved** in peace-making, multilateral peace-making around the world."Oh yes? Why not? How many Chinese troops are serving under the blue beret, serving under the blue flag, serving under the U. N. command in the world today? 3, 700. How many Americans? 11. What is the largest naval contingent tackling the issue of Somali pirates? The Chinese naval contingent. Of course they are, they are a mercantilist nation. They want to keep the sea lanes open. Increasingly, we are going to have to do business with people with whom we do not share values, but with whom, for the moment, we share common interests. It ' s a whole new different way of looking at the world that is now emerging. And here ' s the third factor, which is totally different. Today in our modern world, because of the Internet, because of the kinds of things people have been talking about here, everything is connected to everything. We are now interdependent. We are now interlocked, as nations, as individuals, in a way which has never been the case before, never been the case before. The interrelationship of nations, well it ' s always existed. Diplomacy is about managing the interrelationship of nations. But now we are intimately locked together. You get swine flu in Mexico, it ' s a problem for Charles de Gaulle Airport 24 hours later. Lehman Brothers goes down, the whole lot collapses. There are fires in the steppes of Russia, food riots in Africa. We are all now deeply, deeply, deeply interconnected. And what that means is the idea of a nation state acting alone, not connected with others, not working with others, is no longer a viable proposition. Because the actions of a nation state are neither confined to itself, nor is it sufficient for the nation state itself to control its own territory, because the effects outside the nation state are now beginning to affect what happens inside them. I was a young soldier in the last of the small empire wars of Britain. At that time, the defense of my country was about one thing and one thing only : how strong was our army, how strong was our **air** force, how strong was our navy and how strong were our allies. That was when the enemy was outside the walls. Now the enemy is inside the walls. Now if I want to talk about the defense of my country, I have to speak to the Minister of Health because **pandemic** disease is a threat to my security, I have to speak to the Minister of Agriculture because food security is a threat to my security, I have to speak to the Minister of Industry because the fragility of our hi-tech infrastructure is now a point of attack for our enemies — as we see from cyber warfare — I have to speak to the Minister of Home Affairs because who has entered my country, who lives in that terraced house in that inner city has a direct effect on what happens in my country — as we in London saw in the 7 / 7 bombings. It ' s no longer the case that the security of a country is simply a matter for its soldiers and its ministry of defense. It ' s its capacity to lock together its institutions. And this tells you something very important. It tells you that, in fact, our governments, vertically constructed, constructed on the economic model of the Industrial Revolution — vertical hierarchy, specialization of tasks, command structures — have got the wrong structures completely. You in business know that the paradigm structure of our time, ladies and gentlemen, is the network. It ' s your capacity to network that matters, both within your governments and externally. So here is Ashdown ' s third law. By the way, don ' t ask me about Ashdown ' s first law and second law because I haven ' t invented those yet ; it always sounds better if there ' s a third law, doesn ' t it? Ashdown ' s third law is that in the modern age, where everything is connected to everything, the most important thing about what you can do is what you can do with others. The most important bit about your structure — whether you ' re a government, whether you ' re an army regiment, whether you ' re a business — is your docking points, your interconnectors, your capacity to network with others. You understand that in industry ; governments don ' t. But now one final thing. If it is the case, ladies and gentlemen — and it is — that we are now locked together in a way that has never been quite the same before, then it ' s also the case that we share a destiny with each other. Suddenly and for the very first time, **collective** defense, the thing that has dominated us as the concept of securing our nations, is no longer enough. It used to be the case that if my tribe was more powerful than their tribe, I was **safe** ; if my country was more powerful than their country, I was **safe** ; my alliance, like NATO, was more powerful than their alliance, I was **safe**. It is no longer the case. The advent of the interconnectedness and of the weapons of mass destruction means that, increasingly, I share a destiny with my enemy. When I was a diplomat **negotiating** the disarmament treaties with the Soviet Union in Geneva in the 1970s, we succeeded because we understood we shared a destiny with them. **Collective** security is not enough. Peace has come to Northern Ireland because both sides realized that the zero-sum game couldn ' t work. They shared a destiny with their enemies. One of the great barriers to peace in the Middle East is that both sides, both Israel and, I think, the Palestinians, do not understand that they share a **collective** destiny. And so suddenly, ladies and gentlemen, what has been the proposition of visionaries and poets down the ages becomes something we have to take seriously as a matter of public **policy**. I started with a poem, I ' ll end with one. The great poem of John Donne ' s."Send not for whom the bell tolls."The poem is called”No Man is an Island.”And it goes :”Every man ’ s death affected me, for I am **involved** in mankind, send not to ask for whom the bell tolls, it tolls for thee.”For John Donne, a recommendation of morality. For us, I think, part of the equation for our survival. Thank you very much.

Text Sample 577: POS Dim 4 – 2013 – Score 16.00 – 2013/t001978.txt

Democracy is in trouble, no question about that, and it comes in part from a deep dilemma in which it is embedded. It ’ s increasingly irrelevant to the kinds of decisions we face that have to do with global **pandemics**, a cross-border problem ; with HIV, a transnational problem ; with markets and immigration, something that goes beyond national borders ; with terrorism, with war, all now cross-border problems. In fact, we live in a 21st-century world of interdependence, and brutal interdependent problems, and when we look for solutions in politics and in democracy, we are faced with political institutions designed 400 years ago, autonomous, sovereign nation-states with jurisdictions and territories separate from one another, each **claiming** to be able to solve the problem of its own people. Twenty-first-century, transnational world of problems and challenges, 17th-century world of political institutions. In that dilemma lies the central problem of democracy. And like many others, I ’ ve been thinking about what can one do about this, this asymmetry between 21st-century challenges and archaic and increasingly dysfunctional political institutions like nation-states. And my suggestion is that we change the subject, that we stop talking about nations, about bordered states, and we start talking about cities. Because I think you will find, when we talk about cities, we are talking about the political institutions in which civilization and culture were born. We are talking about the cradle of democracy. We are talking about the venues in which those public spaces where we come together to create democracy, and at the same time protest those who would take our freedom, take place. Think of some great names : the Place de la Bastille, Zuccotti Park, Tahrir Square, Taksim Square in today ’ s headlines in Istanbul, or, yes, Tiananmen Square in Beijing. Those are the public spaces where we announce ourselves as citizens, as participants, as people with the right to write our own narratives. Cities are not only the oldest of institutions, they ’ re the most enduring. If you think about it, Constantinople, Istanbul, much older than Turkey. Alexandria, much older than Egypt. Rome, far older than Italy. Cities endure the ages. They are the places where we are born, grow up, are educated, work, marry, pray, play, get old, and in time, die. They are home. Very different than nation-states, which are abstractions. We pay taxes, we vote occasionally, we watch the men and women we choose rule rule more or less without us. Not so in those homes known as our towns and cities where we live. Moreover, today, more than half of the world ’ s population live in cities. In the developed world, it ’ s about 78 percent. More than three out of four people live in urban institutions, urban places, in cities today. So cities are where the action is. Cities are us. Aristotle said in the ancient world, man is a political animal. I say we are an urban animal. We are an urban species, at home in our cities. So to come back to the dilemma, if the dilemma is we have old-fashioned political nation-states unable to govern the world, respond to the global challenges that we face like **climate** change, then maybe it ’ s time for mayors to rule the world, for mayors and the citizens and the peoples they represent to engage in global governance. When I say if mayors ruled the world, when I first came up with that phrase, it occurred to me that actually, they already do. There are scores of **international**, inter-city, cross-border institutions, networks of cities in which cities are already, quite quietly, below the horizon, working together to deal with **climate** change, to deal with security, to deal with immigration, to deal with all of those tough, interdependent problems that we face. They have strange names : UCLG, United Cities and Local Governments ; ICLEI, the International Council for Local Environmental Issues. And the list goes on : Citynet in Asia ; City Protocol, a new organization out of Barcelona that is using the web to share best practices among countries. And then all the things we know a little better, the U. S. **Conference** of Mayors, the Mexican **Conference** of Mayors, the European **Conference** of Mayors. Mayors are where this is happening. And so the question is, how can we create a world in which mayors and the citizens they represent play a more prominent role? Well, to understand that, we need to understand why cities are special, why mayors are so different than prime ministers and presidents, because my premise is that a mayor and a prime minister are at the opposite ends of a political spectrum. To be a prime minister or a president, you have to have an ideology, you have to have a meta- narrative, you have to have a theory of how things work, you have to belong to a party. Independents, on the whole, don ' t get elected to office. But mayors are just the opposite. Mayors are pragmatists, they ' re problem-solvers. Their job is to get things done, and if they don ' t, they ' re out of a job. Mayor Nutter of Philadelphia said, we could never get away here in Philadelphia with the **stuff** that goes on in Washington, the paralysis, the non-action, the inaction. Why? Because potholes have to get filled, because the trains have to run, because kids have to be able to get to school. And that ' s what we have to do, and to do that is about pragmatism in that deep, American sense, reaching outcomes. Washington, Beijing, Paris, as world capitals, are anything but pragmatic, but real city mayors have to be pragmatists. They have to get things done, they have to put ideology and religion and ethnicity aside and draw their cities together. We saw this a couple of decades ago when Teddy Kollek, the great mayor of Jerusalem in the ' 80s and the ' 90s, was besieged one day in his office by religious **leaders** from all of the backgrounds, Christian prelates, rabbis, imams. They were arguing with one another about access to the holy sites. And the squabble went on and on, and Kollek listened and listened, and he finally said,"Gentlemen, spare me your sermons, and I will fix your sewers."That ' s what mayors do. They fix sewers, they get the trains running. There isn ' t a left or a right way of doing. Boris Johnson in London calls himself an anarcho-Tory. Strange term, but in some ways, he is. He ' s a libertarian. He ' s an anarchist. He rides to work on a bike, but at the same time, he ' s in some ways a **conservative**. Bloomberg in New York was a Democrat, then he was a Republican, and finally he was an Independent, and said the party label just gets in the way. Luzhkov, 20 years mayor in Moscow, though he helped found a party, United Party with Putin, in fact **refused** to be defined by the party and finally, in fact, lost his job not under Brezhnev, not under Gorbachev, but under Putin, who wanted a more faithful party follower. So mayors are pragmatists and problem-solvers. They get things done. But the second thing about mayors is they are also what I like to call homeboys, or to include the women mayors, homies. They ' re from the neighborhood. They ' re part of the neighborhood. They ' re known. Ed Koch used to wander around New York City saying,"How am I doing?"Imagine David Cameron wandering around the United Kingdom asking,"How am I doing?"He wouldn ' t like the answer. Or Putin. Or any national **leader**. He could ask that because he knew New Yorkers and they knew him. Mayors are usually from the places they govern. It ' s pretty hard to be a carpetbagger and be a mayor. You can run for the Senate out of a different state, but it ' s hard to do that as a mayor. And as a result, mayors and city councillors and local authorities have a much higher **trust** level, and this is the third feature about mayors, than national governing officials. In the United States, we know the pathetic figures : 18 percent of Americans approve of Congress and what they do. And even with a relatively popular president like Obama, the figures for the Presidency run about 40, 45, sometimes 50 percent at best. The Supreme Court has fallen way down from what it used to be. But when you ask,"Do you **trust** your city councillor, do you **trust** your mayor?"the rates shoot up to 70, 75, even 80 percent, because they ' re from the neighborhood, because the people they work with are their neighbors, because, like Mayor Booker in Newark, a mayor is likely to get out of his car on the way to work and go in and pull people out of a burning building — that happened to Mayor Booker — or intervene in a mugging in the street as he goes to work because he sees it. No head of state would be permitted by their security details to do it, nor be in a position to do it. That ' s the difference, and the difference has to do with the character of cities themselves, because cities are profoundly multicultural, open, participatory, democratic, able to work with one another. When states face each other, China and the U. S., they face each other like this. When cities interact, they interact like this. China and the U. S., despite the recent meta-meeting in California, are locked in all kinds of anger, resentment, and rivalry for number one. We heard more about who will be number one. Cities don ' t worry about number one. They have to work together, and they do work together. They work together in **climate** change, for example. Organizations like the C40, like ICLEI, which I mentioned, have been working together many, many years before Copenhagen. In Copenhagen, four or five years ago, 184 nations came together to explain to one another why their sovereignty didn ' t permit them to deal with the grave, grave **crisis** of **climate** change, but the mayor of Copenhagen had invited 200 mayors to attend. They came, they stayed, and they found ways and are still finding ways to work together, city- to-city, and through inter-city organizations. Eighty percent of carbon emissions come from cities, which means cities are in a position to solve the carbon problem, or most of it, whether or not the states of which they are a part make agreements with one another. And they are doing it. Los Angeles cleaned up its port, which was 40 percent of carbon emissions, and as a result got rid of about 20 percent of carbon. New York has a program to upgrade its old buildings, make them better insulated in the winter, to not leak energy in the summer, not leak **air** conditioning. That ' s having an **impact**. Bogota, where Mayor Mockus, when he was mayor, he introduced a transportation system that saved energy, that allowed surface buses to run in effect like subways, express buses with corridors. It helped unemployment, because people could get across town, and it had a profound **impact** on **climate** as well as many other things there. Singapore, as it developed its high-rises and its remarkable public housing, also developed an island of parks, and if you go there, you ' ll see how much of it is green land and park land. Cities are doing this, but not just one by one. They are doing it together. They are sharing what they do, and they are making a difference by shared best practices. Bike shares, many of you have heard of it, started 20 or 30 years ago in Latin America. Now it ' s in hundreds of cities around the world. Pedestrian zones, congestion fees, emission limits in cities like California cities have, there ' s lots and lots that cities can do even when opaque, stubborn nations **refuse** to act. So what ' s the bottom line here? The bottom line is, we still live politically in a world of borders, a world of boundaries, a world of walls, a world where states **refuse** to act together. Yet we know that the reality we experience day to day is a world without borders, a world of diseases without borders and doctors without borders, maladies sans frontires, Mdecins Sans Frontires, of economics and technology without borders, of education without borders, of terrorism and war without borders. That is the real world, and unless we find a way to globalize democracy or democratize globalization, we will increasingly not only risk the failure to address all of these transnational problems, but we will risk losing democracy itself, locked up in the old nation-state box, unable to address global problems democratically. So where does that leave us? I ’ ll tell you. The road to global democracy doesn ’ t run through states. It runs through cities. Democracy was born in the ancient polis. I believe it can be reborn in the global cosmopolis. In that journey from polis to cosmopolis, we can rediscover the power of democracy on a global level. We can create not a League of Nations, which failed, but a League of Cities, not a United or a dis-United Nations, but United Cities of the World. We can create a global parliament of mayors. That ’ s an idea. It ’ s in my conception of the coming world, but it ’ s also on the table in City Halls in Seoul, Korea, in Amsterdam, in Hamburg, and in New York. Mayors are considering that idea of how you can actually constitute a global parliament of mayors, and I love that idea, because a parliament of mayors is a parliament of citizens and a parliament of citizens is a parliament of us, of you and of me. If ever there were citizens without borders, I think it ’ s the citizens of **TED** who show the promise to be those citizens without borders. I am ready to reach out and embrace a new global democracy, to take back our democracy. And the only question is, are you? Thank you so much, my fellow citizens. Thank you.

Text Sample 578: POS Dim 4 – 2013 – Score 15.00 – 2013/t002037.txt

Good morning. My name is Eric Li, and I was born here. But no, I wasn ’ t born there. This was where I was born : Shanghai, at the height of the Cultural Revolution. My grandmother tells me that she heard the sound of gunfire along with my first cries. When I was growing up, I was told a story that explained all I ever needed to know about humanity. It went like this. All human societies develop in linear progression, beginning with primitive society, then slave society, feudalism, capitalism, socialism, and finally, guess where we end up? Communism! Sooner or later, all of humanity, regardless of culture, language, nationality, will arrive at this final stage of political and social development. The entire world ’ s peoples will be unified in this paradise on Earth and live happily ever after. But before we get there, we ’ re engaged in a struggle between good and evil, the good of socialism against the evil of capitalism, and the good shall triumph. That, of course, was the meta-narrative distilled from the theories of Karl Marx. And the Chinese bought it. We were taught that grand story day in and day out. It became part of us, and we believed in it. The story was a bestseller. About one third of the entire world ’ s population lived under that meta-narrative. Then, the world changed overnight. As for me, disillusioned by the failed religion of my youth, I went to America and became a Berkeley hippie. Now, as I was coming of age, something else happened. As if one big story wasn ’ t enough, I was told another one. This one was just as grand. It also **claims** that all human societies develop in a linear progression towards a singular end. This one went as follows : All societies, regardless of culture, be it Christian, Muslim, Confucian, must progress from traditional societies in which groups are the basic units to modern societies in which atomized individuals are the sovereign units, and all these individuals are, by definition, rational, and they all want one thing : the vote. Because they are all rational, once given the vote, they produce good government and live happily ever after. Paradise on Earth, again. Sooner or later, electoral democracy will be the only political system for all countries and all peoples, with a free market to make them all rich. But before we get there, we ’ re engaged in a struggle between good and evil. The good belongs to those who are democracies and are charged with a mission of spreading it around the globe, sometimes by force, against the evil of those who do not hold elections. George H. W. Bush : A new world order... George W. Bush :... ending tyranny in our world... Barack Obama :... a single standard for all who would hold power. Eric X. Li : Now — This story also became a bestseller. According to Freedom House, the number of democracies went from 45 in 1970 to 115 in 2010. In the last 20 years, Western elites tirelessly trotted around the globe selling this prospectus : Multiple parties fight for political power and everyone voting on them is the only path to salvation to the long-suffering developing world. Those who buy the prospectus are destined for success. Those who do not are doomed to fail. But this time, the Chinese didn ' t buy it. Fool me once... The rest is history. In just 30 years, China went from one of the poorest agricultural countries in the world to its second-largest economy. Six hundred fifty million people were lifted out of poverty. Eighty percent of the entire world ' s poverty alleviation during that period happened in China. In other words, all the new and old democracies put together amounted to a mere fraction of what a single, one-party state did without voting. See, I grew up on this **stuff** : food stamps. Meat was rationed to a few hundred grams per person per month at one point. Needless to say, I ate all my grandmother ' s portions. So I asked myself, what ' s wrong with this picture? Here I am in my hometown, my business growing leaps and bounds. Entrepreneurs are starting companies every day. Middle class is expanding in speed and scale unprecedented in human history. Yet, according to the grand story, none of this should be happening. So I went and did the only thing I could. I studied it. Yes, China is a one-party state run by the Chinese Communist Party, the Party, and they don ' t hold elections. Three assumptions are made by the **dominant** political theories of our time. Such a system is operationally rigid, politically closed, and morally illegitimate. Well, the assumptions are wrong. The opposites are true. Adaptability, meritocracy, and legitimacy are the three defining characteristics of China ' s one-party system. Now, most political scientists will tell us that a one-party system is inherently incapable of self- correction. It won ' t last long because it cannot adapt. Now here are the facts. In 64 years of running the largest country in the world, the range of the Party ' s **policies** has been wider than any other country in recent memory, from radical land collectivization to the Great Leap Forward, then privatization of farmland, then the Cultural Revolution, then Deng Xiaoping ' s market reform, then successor Jiang Zemin took the giant political step of opening up Party **membership** to private businesspeople, something unimaginable during Mao ' s rule. So the Party self-corrects in rather dramatic fashions. Institutionally, new rules get enacted to correct previous **dysfunctions**. For example, term limits. Political **leaders** used to retain their positions for life, and they used that to accumulate power and perpetuate their rules. Mao was the father of modern China, yet his prolonged rule led to disastrous mistakes. So the Party instituted term limits with mandatory retirement age of 68 to 70. One thing we often hear is,"Political reforms have lagged far behind economic reforms,"and"China is in dire need of political reform."But this **claim** is a rhetorical trap hidden behind a political bias. See, some have decided a priori what kinds of changes they want to see, and only such changes can be called political reform. The truth is, political reforms have never stopped. Compared with 30 years ago, 20 years, even 10 years ago, every aspect of Chinese society, how the country is governed, from the most local level to the highest center, are unrecognizable today. Now such changes are simply not possible without political reforms of the most fundamental kind. Now I would venture to suggest the Party is the world ' s leading expert in political reform. The second assumption is that in a one-party state, power gets concentrated in the hands of the few, and bad governance and corruption follow. Indeed, corruption is a big problem, but let ' s first look at the larger context. Now, this may be counterintuitive to you. The Party happens to be one of the most meritocratic political institutions in the world today. China ' s highest ruling body, the Politburo, has 25 members. In the most recent one, only five of them came from a background of privilege, so-called princelings. The other 20, including the president and the premier, came from entirely ordinary backgrounds. In the larger central committee of 300 or more, the percentage of those who were born into power and wealth was even smaller. The vast majority of **senior** Chinese **leaders** worked and competed their way to the top. Compare that with the ruling elites in both developed and developing countries, I think you ' ll find the Party being near the top in upward mobility. The question then is, how could that be possible in a system run by one party? Now we come to a powerful political institution, little-known to Westerners : the Party ' s Organization Department. The department functions like a giant human resource engine that would be the envy of even some of the most successful corporations. It operates a rotating pyramid made up of three components : civil service, state-owned enterprises, and social organizations like a university or a community program. They form separate yet integrated career paths for Chinese officials. They recruit college grads into entry-level positions in all three tracks, and they start from the bottom, called"keyuan"[clerk]. Then they could get promoted through four increasingly elite ranks : fuke [deputy section manager], ke [section manager], fuchu [deputy division manager], and chu [division manger]. Now these are not moves from"Karate Kid,"okay? It ' s serious business. The range of positions is wide, from running health care in a village to foreign investment in a city district to manager in a company. Once a year, the department reviews their performance. They interview their superiors, their peers, their subordinates. They vet their personal conduct. They conduct public opinion surveys. Then they promote the winners. Throughout their careers, these cadres can move through and out of all three tracks. Over time, the good ones move beyond the four base levels to the fuju [deputy bureau chief] and ju [bureau chief] levels. There, they enter high officialdom. By that point, a typical assignment will be to manage a district with a population in the millions or a company with hundreds of millions of dollars in revenue. Just to show you how competitive the system is, in 2012, there were 900, 000 fuke and ke levels, 600, 000 fuchu and chu levels, and only 40, 000 fuju and ju levels. After the ju levels, the best few move further up several more ranks, and eventually make it to the Central Committee. The process takes two to three decades. Does patronage play a role? Yes, of course. But merit remains the fundamental driver. In essence, the Organization Department runs a modernized version of China ' s centuries-old mentoring system. China ' s new president, Xi Jinping, is the son of a former **leader**, which is very unusual, first of his kind to make the top job. Even for him, the career took 30 years. He started as a village manager, and by the time he entered the Politburo, he had managed areas with a total population of 150 million people and combined GDPs of 1. 5 trillion U. S. dollars. Now, please don ' t get me wrong, okay? This is not a put-down of anyone. It ' s just a statement of fact. George W. Bush, remember him? This is not a put-down. Before becoming governor of Texas, or Barack Obama before running for president, could not make even a small county manager in China ' s system. Winston Churchill once said that democracy is a terrible system except for all the rest. Well, apparently he hadn ' t heard of the Organization Department. Now, Westerners always assume that multi-party election with universal suffrage is the only source of political legitimacy. I was asked once,"The Party wasn ' t voted in by election. Where is the source of legitimacy?"I said,"How about competency?"We all know the facts. In 1949, when the Party took power, China was mired in civil wars, dismembered by foreign aggression, average life expectancy at that time, 41 years old. Today, it ' s the second largest economy in the world, an industrial powerhouse, and its people live in increasing prosperity. Pew Research polls Chinese public attitudes, and here are the numbers in recent years. Satisfaction with the direction of the country : 85 percent. Those who think they ' re better off than five years ago : 70 percent. Those who expect the future to be better : a whopping 82 percent. Financial Times polls global youth attitudes, and these numbers, brand new, just came from last week. Ninety-three percent of China ' s Generation Y are optimistic about their country ' s future. Now, if this is not legitimacy, I ' m not sure what is. In contrast, most electoral democracies around the world are suffering from dismal performance. I don ' t need to elaborate for this audience how dysfunctional it is, from Washington to European capitals. With a few exceptions, the vast number of developing countries that have adopted electoral regimes are still suffering from poverty and civil strife. Governments get elected, and then they fall below 50 percent approval in a few months and stay there and get worse until the next election. Democracy is becoming a perpetual cycle of elect and regret. At this rate, I ' m afraid it is democracy, not China ' s one-party system, that is in danger of losing legitimacy. Now, I don ' t want to create the misimpression that China ' s hunky-dory, on the way to some kind of superpowerdom. The country faces enormous challenges. The social and economic problems that come with wrenching change like this are mind-boggling. Pollution is one. Food safety. Population issues. On the political front, the worst problem is corruption. Corruption is widespread and undermines the system and its moral legitimacy. But most analysts misdiagnose the disease. They say that corruption is the result of the one-party system, and therefore, in order to cure it, you have to do away with the entire system. But a more careful look would tell us otherwise. Transparency International ranks China between 70 and 80 in recent years among 170 countries, and it ' s been moving up. India, the largest democracy in the world, 94 and dropping. For the hundred or so countries that are ranked below China, more than half of them are electoral democracies. So if election is the panacea for corruption, how come these countries can ' t fix it? Now, I ' m a venture capitalist. I make bets. It wouldn ' t be fair to end this talk without putting myself on the line and making some predictions. So here they are. In the next 10 years, China will surpass the U. S. and become the largest economy in the world. Income per capita will be near the top of all developing countries. Corruption will be curbed, but not eliminated, and China will move up 10 to 20 notches to above 60 in T. I. ranking. Economic reform will accelerate, political reform will continue, and the one-party system will hold firm. We live in the dusk of an era. Meta-narratives that make universal **claims** failed us in the 20th century and are failing us in the 21st. Meta-narrative is the cancer that is killing democracy from the inside. Now, I want to clarify something. I ' m not here to make an indictment of democracy. On the contrary, I think democracy contributed to the rise of the West and the creation of the modern world. It is the universal **claim** that many Western elites are making about their political system, the hubris, that is at the heart of the West ' s current ills. If they would spend just a little less time on trying to force their way onto others, and a little bit more on political reform at home, they might give their democracy a better chance. China ' s political model will never supplant electoral democracy, because unlike the latter, it doesn ' t pretend to be universal. It cannot be exported. But that is the point precisely. The significance of China ' s example is not that it provides an alternative, but the demonstration that alternatives exist. Let us draw to a close this era of meta- narratives. Communism and democracy may both be laudable ideals, but the era of their dogmatic universalism is over. Let us stop telling people and our children there ’ s only one way to govern ourselves and a singular future towards which all societies must evolve. It is wrong. It is irresponsible. And worst of all, it is boring. Let universality make way for plurality. Perhaps a more **interesting** age is upon us. Are we brave enough to welcome it? Thank you. Thank you. Thank you. Thank you. Thanks. Bruno Giussani : Eric, stay with me for a couple of minutes, because I want to ask you a couple of questions. I think many here, and in general in Western countries, would agree with your statement about analysis of democratic systems becoming dysfunctional, but at the same time, many would kind of find unsettling the thought that there is an unelected authority that, without any form of oversight or consultation, decides what the national interest is. What is the mechanism in the Chinese model that allows people to say, actually, the national interest as you defined it is wrong? EXL : You know, Frank Fukuyama, the political scientist, called the Chinese system”responsive authoritarianism.”It ’ s not exactly right, but I think it comes close. So I know the largest public opinion survey company in China, okay? Do you know who their biggest client is? The Chinese government. Not just from the central government, the city government, the provincial government, to the most local neighborhood districts. They conduct surveys all the time. Are you happy with the garbage collection? Are you happy with the general direction of the country? So there is, in China, there is a different kind of mechanism to be responsive to the **demands** and the thinking of the people. My point is, I think we should get unstuck from the thinking that there ’ s only one political system — election, election, election — that could make it responsive. I ’ m not sure, actually, elections produce responsive government anymore in the world. BG : Many seem to agree. One of the features of a democratic system is a space for civil society to express itself. And you have shown figures about the **support** that the government and the authorities have in China. But then you ’ ve just mentioned other elements like, you know, big challenges, and there are, of course, a lot of other data that go in a different direction : tens of thousands of unrests and protests and environmental protests, etc. So you seem to suggest the Chinese model doesn ’ t have a space outside of the Party for civil society to express itself. EXL : There ’ s a vibrant civil society in China, whether it ’ s environment or what-have-you. But it ’ s different. You wouldn ’ t recognize it. Because, by Western definitions, a so-called civil society has to be separate or even in opposition to the political system, but that concept is alien for Chinese culture. For thousands of years, you have civil society, yet they are consistent and coherent and part of a political order, and I think it ’ s a big cultural difference. BG : Eric, thank you for sharing this with **TED**. EXL : Thank you.

Text Sample 579: POS Dim 4 – 2013 – Score 15.00 – 2013/t002155.txt

So, why does good sex so often fade, even for couples who continue to love each other as much as ever? And why does good intimacy not guarantee good sex, contrary to popular belief? Or, the next question would be, can we want what we already have? That ’ s the million-dollar question, right? And why is the forbidden so erotic? What is it about transgression that makes desire so potent? And why does sex make babies, and babies spell erotic disaster in couples? It ’ s kind of the fatal erotic blow, isn ’ t it? And when you love, how does it feel? And when you desire, how is it different? These are some of the questions that are at the center of my exploration on the nature of erotic desire and its concomitant dilemmas in modern love. So I travel the globe, and what I ’ m noticing is that everywhere where romanticism has entered, there seems to be a **crisis** of desire. A **crisis** of desire, as in owning the wanting — desire as an expression of our individuality, of our free choice, of our preferences, of our identity — desire that has become a central concept as part of modern love and individualistic societies. You know, this is the first time in the history of humankind where we are trying to experience sexuality in the long term not because we want 14 children, for which we need to have even more because many of them won ’ t make it, and not because it is exclusively a woman ’ s marital duty. This is the first time that we want sex over time about pleasure and connection that is rooted in desire. So what sustains desire, and why is it so difficult? And at the heart of sustaining desire in a committed relationship, I think, is the reconciliation of two fundamental human needs. On the one hand, our need for security, for predictability, for safety, for dependability, for reliability, for permanence. All these anchoring, grounding experiences of our lives that we call home. But we also have an equally strong need — men and women — for adventure, for novelty, for mystery, for risk, for danger, for the unknown, for the unexpected, surprise — you get the gist. For journey, for travel. So reconciling our need for security and our need for adventure into one relationship, or what we today like to call a passionate marriage, used to be a contradiction in terms. Marriage was an economic institution in which you were given a partnership for life in terms of children and social status and succession and companionship. But now we want our partner to still give us all these things, but in addition I want you to be my best friend and my **trusted** confidant and my passionate lover to boot, and we live twice as long. So we come to one person, and we basically are asking them to give us what once an entire village used to provide. Give me belonging, give me identity, give me continuity, but give me transcendence and mystery and awe all in one. Give me comfort, give me edge. Give me novelty, give me familiarity. Give me predictability, give me surprise. And we think it ’ s a given, and toys and lingerie are going to save us with that. So now we get to the existential reality of the story, right? Because I think, in some way — and I ’ ll come back to that — but the **crisis** of desire is often a **crisis** of the imagination. So why does good sex so often fade? What is the relationship between love and desire? How do they relate, and how do they **conflict**? Because therein lies the mystery of eroticism. So if there is a verb, for me, that comes with love, it ’ s”to have.”And if there is a verb that comes with desire, it is”to want.”In love, we want to have, we want to know the beloved. We want to minimize the distance. We want to contract that gap. We want to neutralize the tensions. We want closeness. But in desire, we tend to not really want to go back to the places we ’ ve already gone. Forgone conclusion does not keep our interest. In desire, we want an Other, somebody on the other side that we can go **visit**, that we can go spend some time with, that we can go see what goes on in their red-light district. You know? In desire, we want a bridge to cross. Or in other words, I sometimes say, fire needs **air**. Desire needs space. And when it ’ s said like that, it ’ s often quite abstract. But then I took a question with me. And I ’ ve gone to more than 20 countries in the last few years with”Mating in Captivity,”and I asked people, when do you find yourself most drawn to your partner? Not attracted sexually, per Se, but most drawn. And across culture, across religion, and across **gender** — except for one — there are a few answers that just keep coming back. So the first group is : I am most drawn to my partner when she is away, when we are apart, when we reunite. Basically, when I get back in touch with my **ability** to imagine myself with my partner, when my imagination comes back in the picture, and when I can root it in absence and in longing, which is a major component of desire.

But then the second group is even more **interesting**. I am most drawn to my partner when I see him in the studio, when she is onstage, when he is in his element, when she ’ s doing something she ’ s passionate about, when I see him at a party and other people are really drawn to him, when I see her hold **court**. Basically, when I look at my partner radiant and confident. Probably the biggest turn-on across the board. Radiant, as in self-sustaining. I look at this person — by the way, in desire people rarely talk about it, when we are blended into one, five centimeters from each other. I don ’ t know in inches how much that is. But it ’ s also not when the other person is that far apart that you no longer see them. It ’ s when I ’ m looking at my partner from a comfortable distance, where this person that is already so familiar, so known, is momentarily once again somewhat mysterious, somewhat elusive. And in this space between me and the other lies the erotic lan, lies that movement toward the other. Because sometimes, as Proust says, mystery is not about traveling to new places, but it ’ s about looking with new eyes. And so, when I see my partner on his own or her own, doing something in which they are enveloped, I look at this person and I momentarily get a shift in perception, and I stay open to the mysteries that are living right next to me. And then, more importantly, in this description about the other or myself — it ’ s the same — what is most **interesting** is that there is no neediness in desire. Nobody needs anybody. There is no caretaking in desire. Caretaking is mightily loving. It ’ s a powerful anti-aphrodisiac. I have yet to see somebody who is so turned on by somebody who needs them. Wanting them is one thing. Needing them is a shot down and women have known that forever, because anything that will bring up parenthood will usually decrease the erotic charge. For good reasons, right? And then the third group of answers usually would be : when I ’ m surprised, when we laugh together, as somebody said to me in the office today, when he ’ s in his tux, so I said, you know, it ’ s either the tux or the cowboy boots. But basically it ’ s when there is novelty. But novelty isn ’ t about new positions. It isn ’ t a repertoire of techniques. Novelty is, what parts of you do you bring out? What parts of you are just being seen? Because in some way one could say sex isn ’ t something you do, eh? Sex is a place you go. It ’ s a space you enter inside yourself and with another, or others. So where do you go in sex? What parts of you do you connect to? What do you seek to express there? Is it a place for transcendence and spiritual union? Is it a place for naughtiness and is it a place to be safely aggressive? Is it a place where you can finally surrender and not have to take responsibility for everything? Is it a place where you can express your infantile wishes? What comes out there? It ’ s a language. It isn ’ t just a behavior. And it ’ s the poetic of that language that I ’ m **interested** in, which is why I began to explore this concept of erotic **intelligence**. You know, animals have sex. It ’ s the **pivot**, it ’ s biology, it ’ s the natural instinct. We are the only ones who have an erotic life, which means that it ’ s sexuality transformed by the human imagination. We are the only ones who can make love for hours, have a blissful time, multiple orgasms, and touch nobody, just because we can imagine it. We can hint at it. We don ’ t even have to do it. We can experience that powerful thing called anticipation, which is a mortar to desire. The **ability** to imagine it, as if it ’ s happening, to experience it as if it ’ s happening, while nothing is happening and everything is happening, at the same time. So when I began to think about eroticism, I began to think about the poetics of sex. And if I look at it as an **intelligence**, then it ’ s something that you cultivate. What are the ingredients? Imagination, playfulness, novelty, curiosity, mystery. But the central agent is really that piece called the imagination. But more importantly, for me to begin to understand who are the couples who have an erotic spark, what sustains desire, I had to go back to the original definition of eroticism, the mystical definition, and I went through it through a bifurcation by looking, actually, at trauma, which is the other side. And I looked at it, looking at the community that I had grown up in, which was a community in Belgium, all Holocaust survivors, and in my community, there were two groups : those who didn ' t die, and those who came back to life. And those who didn ' t die lived often very tethered to the ground, could not experience pleasure, could not **trust**, because when you ' re vigilant, worried, anxious, and insecure, you can ' t lift your head to go and take off in space and be playful and **safe** and imaginative. Those who came back to life were those who understood the erotic as an antidote to death. They knew how to keep themselves alive. And when I began to listen to the sexlessness of the couples that I work with, I sometimes would hear people say,"I want more sex,"but generally, people want better sex, and better is to reconnect with that quality of aliveness, of vibrancy, of renewal, of vitality, of Eros, of energy that sex used to afford them, or that they ' ve hoped it would afford them. And so I began to ask a different question."I shut myself off when..."began to be the question."I turn off my desires when..."Which is not the same question as,"What turns me off is..."and"You turn me off when..."And people began to say,"I turn myself off when I feel dead inside, when I don ' t like my body, when I feel old, when I haven ' t had time for myself, when I haven ' t had a chance to even check in with you, when I don ' t perform well at work, when I feel low self esteem, when I don ' t have a sense of self-worth, when I don ' t feel like I have a right to want, to take, to receive pleasure."And then I began to ask the reverse question."I turn myself on when..."Because most of the time, people like to ask the question,"You turn me on, what turns me on,"and I ' m out of the question, you know? Now, if you are dead inside, the other person can do a lot of things for Valentine ' s. It won ' t make a dent. There is nobody at the reception desk. So I turn myself on when, I turn on my desires, I wake up when... Now, in this paradox between love and desire, what seems to be so puzzling is that the very ingredients that nurture love — mutuality, reciprocity, protection, worry, responsibility for the other — are sometimes the very ingredients that stifle desire. Because desire comes with a host of feelings that are not always such favorites of love : jealousy, possessiveness, aggression, power, dominance, naughtiness, mischief. Basically most of us will get turned on at night by the very same things that we will demonstrate against during the day. You know, the erotic mind is not very politically correct. If **everybody** was fantasizing on a bed of roses, we wouldn ' t be having such **interesting** talks about this. But no, in our mind up there are a host of things going on that we don ' t always know how to bring to the person that we love, because we think love comes with selflessness and in fact desire comes with a certain amount of selfishness in the best sense of the word : the **ability** to stay connected to one ' s self in the presence of another. So I want to draw that little image for you, because this need to reconcile these two sets of needs, we are born with that. Our need for connection, our need for separateness, or our need for security and adventure, or our need for togetherness and for autonomy, and if you think about the little kid who sits on your lap and who is cozily nested here and very secure and comfortable, and at some point all of us need to go out into the world to discover and to explore. That ' s the beginning of desire, that exploratory need, curiosity, discovery. And then at some point they turn around and they look at you. And if you tell them,"Hey kiddo, the world ' s a great place. Go for it. There ' s so much fun out there,"then they can turn away and they can experience connection and separateness at the same time. They can go off in their imagination, off in their body, off in their playfulness, all the while knowing that there ' s somebody when they come back. But if on this side there is somebody who says,"I ' m worried. I ' m anxious. I ' m depressed.

My partner hasn ’ t taken care of me in so long. What ’ s so good out there? Don ’ t we have everything you need together, you and I?”then there are a few little **reactions** that all of us can pretty much recognize. Some of us will come back, came back a long time ago, and that little child who comes back is the child who will forgo a part of himself in order not to lose the other. I will lose my freedom in order not to lose connection. And I will learn to love in a certain way that will become burdened with extra worry and extra responsibility and extra protection, and I won ’ t know how to leave you in order to go play, in order to go experience pleasure, in order to discover, to enter inside myself. Translate this into adult language. It starts very young. It continues into our sex lives up to the end. Child number two comes back but looks like that over their shoulder all the time.”Are you going to be there? Are you going to curse me, scold me? Are you going to be angry with me?”And they may be gone, but they ’ re never really away. And those are often the people that will tell you,”In the beginning, it was super hot.”Because in the beginning, the growing intimacy wasn ’ t yet so strong that it actually led to the decrease of desire. The more connected I became, the more responsible I felt, the less I was able to let go in your presence. The third child doesn ’ t really come back. So what happens, if you want to sustain desire, it ’ s that real dialectic piece. On the one hand you want the security in order to be able to go. On the other hand if you can ’ t go, you can ’ t have pleasure, you can ’ t culminate, you don ’ t have an orgasm, you don ’ t get excited because you spend your time in the body and the head of the other and not in your own. So in this dilemma about reconciling these two sets of fundamental needs, there are a few things that I ’ ve come to understand erotic couples do. One, they have a lot of sexual privacy. They understand that there is an erotic space that belongs to each of them. They also understand that foreplay is not something you do five minutes before the real thing. Foreplay pretty much starts at the end of the previous orgasm. They also understand that an erotic space isn ’ t about, you begin to stroke the other. It ’ s about you create a space where you leave Management Inc., maybe where you leave the Agile program — And you actually just enter that place where you stop being the good citizen who is taking care of things and being responsible. Responsibility and desire just butt heads. They don ’ t really do well together. Erotic couples also understand that passion waxes and wanes. It ’ s pretty much like the moon. It has intermittent eclipses. But what they know is they know how to resurrect it. They know how to bring it back. And they know how to bring it back because they have demystified one big myth, which is the myth of spontaneity, which is that it ’ s just going to fall from heaven while you ’ re folding the laundry like a deus ex machina, and in fact they understood that whatever is going to just happen in a long-term relationship, already has. Committed sex is premeditated sex. It ’ s willful. It ’ s intentional. It ’ s focus and presence. Merry Valentine ’ s.

Text Sample 580: POS Dim 4 – 2001 – Score 2.00 – 2001/t000092.txt

I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ’ children, because they were all average. But they weren ’ t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ’ s not right. The good Lord in his infinite wisdom didn ’ t create us all equal as far as **intelligence** is concerned, any more than we ’ re equal for size, appearance. Not **everybody** could earn an A or a B, and I didn ’ t like that way of judging, and I did know how the alumni of various schools back in the ’ 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ’ t lose a game. But it seemed that we didn ’ t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ’ t like it, I didn ’ t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ’ m sure at the time he did that, I didn ’ t — it didn ’ t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ’ s under your control. If you get too engrossed and **involved** and concerned in regard to the things over which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,”At God ’ s footstool to confess, a poor soul knelt, and bowed his head. ’ I failed! ’ he cried. The Master said, ’ Thou didst thy best, that is success. ’”From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ’ re capable. I believe that ’ s true. If you make the effort to do the best of which you ’ re capable, trying to improve the situation that exists for you, I think that ’ s success, and I don ’ t think others can judge that ; it ’ s like character and reputation — your reputation is what you ’ re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ’ s what the teachers are themselves.”That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by coal oil lamp — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I reply,

’ Where could I find such splendid company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ’ s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ’ Where could I find such splendid company? ’”And I believe the teaching profession — it ’ s true, you have so many youngsters, and I ’ ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ’ re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ’ re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was”Never be late.”Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ’ s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ’ t clean and neat, so I wouldn ’ t let him go. He couldn ’ t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ’ t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ’ ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,”Don ’ t run practices late, because you ’ ll go home in a bad mood, and that ’ s not good, for a young married man to go home in a bad mood. When you get older, it doesn ’ t make any difference, but —”So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ’ re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ’ t want that. I used to tell them I was paid to do that. That ’ s my job. I ’ m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ’ s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ’ s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ’ t have the time to go on that. But that helped me, I think, become a better teacher. It ’ s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ’ re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ’ re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ’ t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ’ i ’. I ’ d never seen that before, but he did. Big league baseball players — they ’ re very perceptive about those things, and they noticed he had only one ’ i ’ in his name. You ’ d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it”The Road Ahead, or the Road Behind.”He said,”Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ’ t win, is the Fates themselves have missed. Yet there lives on the ancient **claim** : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ’ s game. You and I know deeper down, there ’ s always a chance to win the crown. But when we fail to give our best, we simply haven ’ t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ’ s bearing down that wins the cup. Of dreaming there ’ s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ’ s you and I who make our fates — we open up or close the gates on the road ahead or the road behind.”Reminds me of another set of threes that my dad tried to get across to us : Don ’ t whine. Don ’ t complain. Don ’ t make excuses. Just get out there, and whatever you ’ re doing, do it to the best of your **ability**. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ’ re outscored. I ’ ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ’ t know the outcome, I hope they couldn ’ t tell by your actions whether you outscored an opponent or the opponent outscored you. That ’ s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ’ d want them to be but they ’ ll be about what they should ; only you will know whether you can do that. And that ’ s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,”The journey is better than the end.”And I like that. I think that it is — it ’ s getting there. Sometimes when you get there, there ’ s almost a let down. But it ’ s the getting there that ’ s the fun. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ' d done a decent job during the week. There again, it ' s getting the players to get that self-satisfaction, in knowing that they ' d made the effort to do the best of which they are capable. Sometimes I ' m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,"Suppose that you, in some way, could make the perfect player. What would you want?"And I said,"Well, I ' d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ' d want one that could play, too. I ' d want one to realize that defense usually wins championships, and who would work hard on defense. But I ' d want one who would play offense, too. I ' d want him to be unselfish, and look for the pass first and not shoot all the time. And I ' d want one that could pass and would pass. I ' ve had some that could and wouldn ' t, and I ' ve had some that would and could. So, yeah, I ' d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ' d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,"They Call Me Coach,"two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ' t play varsity when I taught. I thought,"Oh gracious, if these two players, either one of them"— they were different years, but I thought about each one at the time he was there —"Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ' s good enough to make it."And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ' d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ' t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ' s taken, they assumed would be missed. I ' ve had too many stand around and wait to see if it ' s missed, then they go and it ' s too late, somebody else is in there ahead of them. They weren ' t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 581: POS Dim 4 – 2001 – Score 2.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with be-cause, although I am 94, I am not still working. And anybody who asks me,"Are you still doing this or that?"I don ' t answer because I ' m not doing things still, I ’ m doing it like I always did. I still have — or did I use the word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years. But, of course, I ’ m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ’ ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called”Innovation.”Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,”The playful search for beauty was Man ’ s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty.”These are tiles. The word,”playful”is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ’ t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a **document** that I had accomplished my apprenticeship successfully, that I had behaved morally, and this **document** was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ’ t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I **visited** it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please **visit** me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, **everybody** in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — **everybody** — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to **visit** me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 582: POS Dim 4 – 2001 – Score 2.00 – 2001/t000367.txt
I ’ d like to do pretty much what I did the first time, which is to choose a lighthearted theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, exorcise those evil spirits, and this is still going on, as you know. But it wasn ’ t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ’ re feeling depressed? It ’ s better than Prozac, but I wouldn ’ t recommend it. So what we see in the seventeenth, eighteenth century is the continued search for medications other than camphor that ’ ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we fast-forward to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a row — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very **interested** in producing convulsions, measured types of convulsions. And they thought,”Well, we ’ ve got electricity, we ’ ll plug somebody into the wall. That always makes hair stand up and people shake a lot.”So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,”We know that at the Rome railroad station, there are all these lost souls wandering around, muttering gibberish. Can you bring one of them to us?”Someone who is, as the Italians say,”cagoots.”So they found this”cagoots”**guy**, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,”Well, we ’ ll try 55 volts, two-tenths of a second. That ’ s not going to do anything terrible to him.”So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,”This

fellow"— remember, he wasn ' t even put to sleep —"after this major grand mal convulsion, sat right up, looked at these three fellas and said, ' What the fuck are you assholes trying to do? '"If I could only say that in Italian. Well, they were happy as could be, because he hadn ' t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and **everybody** in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ' t work very well on the schizophrenics, but it was pretty clear in the ' 30s and by the middle of the ' 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ' t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the brain waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ' 60s, the first antidepressants came out. Tofranil was the first. In the late ' 70s, early ' 80s, there were others, and they were very effective. And patients ' rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ' m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ' ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ' m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ' clock in the afternoon, because I couldn ' t get out of bed before about 11 o ' clock. And anybody who ' s been depressed here knows what that ' s like. I couldn ' t even pull the covers off myself. Well, you ' re in a university medical center, where **everybody** knows **everybody**, and it ' s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ' t work anymore. And, in fact, it didn ' t make any difference because I didn

’ t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,”Don ’ t worry, chap. Six weeks, you ’ re back in the operating room. Everything ’ s going to be great.”Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ’ t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five feet in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ’ t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had. They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the **senior** staff. All the **senior** staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful **stuff**. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it got so there was a throbbing, there was a ferocious fear in my head. You ’ ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ’ t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case. He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the **senior** staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,”No. I know this man better than any of you. I have met with him over and over again. You ’ ve just seen him from time to time. You ’ ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ’ ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go back to surgery, he will never be the loving father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ’ One Flew Over the Cuckoo ’ s Nest. ’ And you know about that, just essentially in a stupor the rest of his life.”Well, he said,”Can ’ t we try a course of electroshock therapy?”And you know why they agreed? They agreed to humor him. They just thought,”Well, we ’ ll give a course of 10. And so we ’ ll lose a little time. Big deal. It doesn ’ t make any difference.”So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ’ t work. Seven didn ’ t work. Eight didn ’ t work. At nine, I noticed — and it ’ s wonderful that I could notice anything — I noticed a change. And at 10, I noticed a real change. And he went back to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, lo and behold, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could blow the obsessional thinking away. I could blow the depression away. And I ’ ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,”I ’ ve got the strength now to do this.”It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a formula. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember”Ruddigore,”and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,”We must have a word to bring you back to reality, and the word, my dear, will be ’ Basingstoke. ’”So every time she got a little nuts, he would say,”Basingstoke!”And she would say,”Basingstoke, it is.”And she would be fine for a little while. Well, you know, I ’ m from the Bronx. I can ’ t say”Basingstoke.”But I had something better. And it was very simple. It was,”Ah, fuck it!”Much better than”Basingstoke,”at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,”Ah, fuck it.”And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went back to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came back to live with us. We had two more children after that. Resuscitated the career, even better than it had been before. Went right back into the university and began to write books. Well, you know, it ’ s been a wonderful life. It ’ s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ’ s been very exciting. It ’ s been very happy. Every once in a while, I have to say,”Ah, fuck it.”Every once in a while, I get somewhat depressed and a little obsessional. So, I ’ m not free of all of this. But it ’ s worked. It ’ s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ’ ve gotten thousands of letters about them by people who do think this — that based on my life ’ s history as I ’ ve portrayed in the books, my early life ’ s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the bitter dregs of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ’ ve always felt guilty about that. I ’ ve always felt that somehow I was an impostor because my readers don ’ t know what I have just told you. It ’ s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an untroubled mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes back to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ’ t happen, there will be adversities. If I, with the bleakness of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way back from this, believe me, anybody can find their way back from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

8 NEG Dim 4

Text Sample 583: NEG Dim 4 – 2001 – Score -28.00 – 2001/t000083.txt

I thought I would read poems I have that relate to the subject of youth and age. I was sort of astonished to find out how many I have actually. The first one is dedicated to Spencer, and his grandmother, who was shocked by his work. My poem is called”Dirt.”My grandmother is washing my mouth out with soap ; half a long century gone and still she comes at me with that thick cruel yellow bar. All because of a word I said, not even said really, only repeated. But”Open,”she says,”open up!”her hand clawing at my head. I know now her life was hard ; she lost three children as babies, then her husband died too, leaving young sons, and no money. She ’ d stand me in the sink to pee because there was never room in the toilet. But oh, her soap! Might its bitter burning have been what made me a poet? The street she lived on was **unpaved**, her flat, two cramped rooms and a fetid kitchen where she stalked and caught me. Dare I admit that after she did it I never really loved her again? She lived to a hundred, even then. All along it was the sadness, the squalor, but I never, until now loved her again. When that was published in a magazine I got an irate letter from my uncle."You have maligned a great woman."It took some diplomacy. This is called"The Dress."It ' s a longer poem. In those days, those days which exist for me only as the most elusive memory now, when often the first sound you ' d hear in the morning would be a storm of birdsong, then the soft clop of the hooves of the horse hauling a milk wagon down your block, and the last sound at night as likely as not would be your father pulling up in his car, having worked late again, always late, and going heavily down to the cellar, to the furnace, to shake out the ashes and damp the draft before he came upstairs to fall into bed — in those long-ago days, women, my mother, my friends ' mothers, our neighbors, all the women I knew — wore, often much of the day, what were called housedresses, cheap, printed, **pulpy**, seemingly purposefully **shapeless** light cotton shifts that you wore over your nightgown and, when you had to go look for a child, hang wash on the line, or run down to the grocery store on the corner, under a coat, the twisted hem of the nightgown always lank and yellowed, dangling beneath. More than the curlers some of the women seemed constantly to have in their hair in preparation for some great event — a ball, one would think — that never came to pass ; more than the way most women ' s faces not only were never made up during the day, but seemed scraped, bleached, and, with their plucked eyebrows, scarily masklike ; more than all that it was those dresses that made women so unknowable and forbidding, adepts of enigmas to which men could have no access, and boys no conception. Only later would I see the dresses also as a proclamation : that in your **dim** kitchen, your laundry, your bleak concrete yard, what you revealed of yourself was a fabulation ; your real sensual nature, veiled in those **sexless** vestments, was utterly your dominion. In those days, one hid much else as well : grown men didn ' t embrace one another, unless someone had died, and not always then ; you shook hands or, at a ball game, thumped your friend ' s back and exchanged blows meant to be codes for affection ; once out of childhood you ' d never again know the shock of your father ' s whiskers on your cheek, not until mores at last had evolved, and you could hug another man, then hold on for a moment, then even kiss. What release finally, the embrace : though we were wary — it seemed so audacious — how much unspoken joy there was in that affirmation of equality and communion, no matter how much misunderstanding and pain had passed between you by then. We knew so little in those days, as little as now, I suppose about healing those hurts : even the women, in their best dresses, with beads and sequins sewn on the **bodices**, even in lipstick and mascara, their hair aflow, could only stand wringing their hands, begging for peace, while father and son, like thugs, like thieves, like Romans, simmered and hissed and hated, inflicting sorrows that endured, the worst anyway, through the kiss and embrace, bleeding from brother to brother, into the generations. In those days there was still countryside close to the city, farms, cornfields, cows ; even not far from our building with its blurred brick and long shadowy hallway you could find tracts with hills and trees you could pretend were mountains and forests. Or you could go out by yourself even to a half-block-long empty lot, into the bushes : like a creature of leaves you ' d lurk, crouched, **crawling**, simplified, savage, alone ; already there was wanting to be simpler, wanting, when they called you, never to go back. This is another longish one, about the old and the young. It actually happened right at the time we met. Part of the poem takes place in space we shared and time we shared. It ' s called"The Neighbor."Her five horrid, deformed little dogs who incessantly yap on the roof under my window. Her cats, God knows how many, who must piss on her rugs — her landing ' s a sickening **reek**. Her shadow once, fumbling the chain on her door, then the door slamming fearfully shut, only the barking and the music — jazz

— filtering as it does, day and night into the hall. The time it was Chris Connor singing”Lush Life”— how it brought back my college sweetheart, my first real love, who — till I left her — played the same record. And head on my shoulder, hand on my thigh, sang sweetly along, of regrets and depletions she was too young for, as I was too young, later, to believe in her pain. It startled, then bored, then repelled me. My starting to fancy she ’ d ended up in this fire-trap in the Village, that my neighbor was her. My thinking we ’ d meet, recognize one another, become friends, that I ’ d accomplish a penance. My seeing her, it wasn ’ t her, at the mailbox. Gray-yellow hair, army pants under a nightgown, her turning away, hiding her ravaged face in her hands, **muttering** an inappropriate”Hi.”Sometimes there are frightening **goings-on** in the stairwell. A man shouting,”Shut up!”The dogs frantically snarling, claws scrabbling, then her — her voice hoarse, harsh, hollow, almost only a tone, incoherent, a note, a squawk, bone on metal, metal gone molten, calling them back,”Come back darlings, come back dear ones. My sweet angels, come back.”Medea she was, next time I saw her. Sorceress, tranced, ecstatic, stock-still on the sidewalk **ragged** coat hanging agape, passersby flowing around her, her mouth torn suddenly open as though in a scream, silently though, as though only in her **brain** or breast had it erupted. A cry so pure, practiced, detached, it had no need of a voice, or could no longer bear one. These invisible links that allure, these transfigurations, even of anguish, that hold us. The girl, my old love, the last lost time I saw her when she came to find me at a party, her drunkenly **stumbling**, falling, sprawling, skirt hiked, eyes veined red, swollen with tears, her shame, her dishonor. My ignorant, arrogant coarseness, my secret pride, my turning away. Still life on a rooftop, dead trees in barrels, a bench broken, dogs, excrement, sky. What pathways through pain, what junctures of vulnerability, what crossings and counterings? Too many lives in our lives already, too many chances for sorrow, too many unaccounted-for pasts.”**Behold** me,”the god of frenzied, inexhaustible love says, rising in bloody splendor,”**Behold** me.”Her making her way down the littered vestibule stairs, one agonized step at a time. My holding the door. Her crossing the fragmented **tiles**, faltering at the step to the street, droning, not looking at me,”Can you help me?”Taking my arm, leaning lightly against me. Her wavering step into the world. Her whispering,”Thanks love.”Lightly, lightly against me. I think I ’ ll lighten up a little. Another, different kind of poem of youth and age. It ’ s called”Gas.”Wouldn ’ t it be nice, I think, when the blue-haired lady in the doctor ’ s waiting room bends over the magazine table and farts, just a little, and violently blushes. Wouldn ’ t it be nice if intestinal gas came embodied in visible clouds, so she could see that her really quite **inoffensive pop** had only barely grazed my face before it drifted away. Besides, for this to have happened now is a nice coincidence. Because not an hour ago, while we were on our walk, my dog was startled by a backfire and jumped straight up like a horse bucking. And that brought back to me the stable I worked on weekends when I was 12, and a **splendid piebald stallion**, who whenever he was mounted would buck just like that, though more hugely of course, enormous, gleaming, **resplendent**. And the woman, her face abashedly buried in her”Elle”now, reminded me — I ’ d forgotten that not the least part of my awe consisted of the fact that with every jump he took the horse would powerfully fart. Phwap! Phwap! Phwap! Something never mentioned in the dozens of books about horses and their riders I devoured in those days. All that savage grandeur, the **steely** glinting hooves, the eruptions driven from the creature ’ s mighty **innards**, breath stopped, heart stopped, nostrils madly flared, I didn ’ t know if I wanted to break him, or be him. This is called”Thirst.”Many — most of my poems actually are urban poems. I happen to be reading a bunch that aren ’ t.”Thirst.”Here was my relation with the woman who lived all last autumn and winter, day and night, on a bench in the 103rd Street subway station, until finally one day she vanished. We regarded each other, scrutinized one another. Me shyly, obliquely, trying not to be furtive. She boldly, unblinkingly, even pugnaciously, wrathfully even, when her bottle was empty. I was frightened of her. I felt like a child. I was afraid some repressed part of myself would go out of control, and I ’ d be forever entrapped in the shocking seethe of her stench. Not excrement merely, not merely surface and orifice going unwashed, rediffusion of rum, there was will in it, and intention, power and purpose — a social, ethical rage and rebellion — despair too, though, grief, loss. Sometimes I ’ d think I should take her home with me, bathe her, comfort her, dress her. She wouldn ’ t have wanted me to, I would think. Instead, I ’ d step into my train. How rich I would think, is the lexicon of our self-absolving. How enduring, our bland fatal assurance that reflection is righteousness being accomplished. The dance of our glances, the clash, pulling each other through our perceptual punctures, then holocaust, holocaust, host on host of ill, injured presences, squandered, consumed. Her vigil somewhere I know continues. Her **occupancy**, her absolute, faithful attendance. The dance of our glances, challenge, **abdication**, **effacement**, the perfume of our consternation. This is a newer poem, a brand new poem. The title is”This Happened.”A student, a young woman in a fourth-floor hallway of her lycee, perched on the ledge of an open window chatting with friends between classes ; a teacher passes and chides her,”Be careful, you might fall,”almost banteringly chides her,”You might fall,”and the young woman, 18, a girl really, though she wouldn ’ t think that, as brilliant as she is, first in her class, and”Beautiful, too,”she ’ s often told, smiles back, and leans into the open window, which wouldn ’ t even be open if it were winter — if it were winter someone would have closed it — leans into the window, farther, still smiling, farther and farther, though it takes less time than this, really an instant, and lets herself fall. Herself fall. A casual impulse, a fancy, never thought of until now, hardly thought of even now... No, more than impulse or fancy, the girl knows what she ’ s doing, the girl means something, the girl means to mean, because it occurs to her in that instant, that beautiful or not, bright yes or no, she ’ s not who she is, she ’ s not the person she is, and the reason, she suddenly knows, is that there ’ s been so much premeditation where she is, so much plotting and planning, there ’ s hardly a person where she is, or if there is, it ’ s not her, or not wholly her, it ’ s a self inhabited, lived in by her, and seemingly even as she thinks it she knows what ’ s been missing : grace, not premeditation but grace, a kind of being in the world spontaneously, with grace. Weightfully upon me was the world. Weightfully this self which graced the world yet never wholly itself. Weightfully this self which weighed upon me, the release from which is what I desire and what I achieve. And the girl remembers, in this infinite instant already now so many times divided, the sadness she felt once, hardly knowing she felt it, to merely inhabit herself. Yes, the girl falls, absurd to fall, even the earth with its compulsion to take unto itself all that falls must know that falling is absurd, yet the girl falling isn ’ t myself, or she is myself, but a self I took of my own volition unto myself. Forever. With grace. This happened. I ’ ll read just one more. I don ’ t usually say that. I like to just end. But I ’ m afraid that Ricky will come out here and shake his fist at me. This is called”Old Man,”appropriately enough.”Special : big tits,”Says the advertisement for a soft-core magazine on our neighborhood newsstand. But forget her breasts. A lush, fresh-lipped blond, skin glowing gold, sprawls there, **resplendent**. 60 nearly, yet these hardly tangible, hardly better than harlots, can still stir me. Maybe a coming of age in the American sensual darkness, never seeing an unsmudged nipple, an uncensored vagina, has left me forever infected with an unquenchable lust of the eye. Always that erotic murmur, I ' m hardly myself if I ' m not in a state of incipient desire. God knows though, there are worse twists your obsessions can take. Last year in Israel, a young ultra-orthodox Rabbi guiding some teenage girls through the Shrine of the Shoah forbade them to look in one room. Because there were images in it he said were licentious. The display was a photo. Men and women stripped naked, some trying to cover their genitals, others too frightened to bother, lined up in snow waiting to be shot and thrown into a ditch. The girls, to my horror, averted their gaze. What carnal mistrust had their teacher taught them. Even that though. Another confession : Once in a book on pre-war Poland, a studio portrait, an absolute angel, an absolute angel with tormented, tormenting eyes. I kept finding myself at her page. That she died in the camps made her — I didn ' t dare wonder why — more present, more precious. Died in the camps, that too people — or Jews anyway — kept from their children back then. But it was like sex, you didn ' t have to be told. Sex and death, how close they can seem. So constantly conscious now of death moving towards me, sometimes I think I confound them. My wife ' s loveliness almost consumes me. My passion for her goes beyond reasonable bounds. When we make love, her holding me everywhere all around me, I ' m there and not there. My mind teems, jumbles of faces, voices, impressions, I live my life over, as though I were drowning. Then I am drowning, in despair at having to leave her, this, everything, all, unbearable, awful. Still, to be able to die with no special contrition, not having been slaughtered, or enslaved. And not having to know history ' s next mad rage or regression, it might be a relief. No. Again, no. I don ' t mean that for a moment. What I mean is the world holds me so tightly — the good and the bad — my own **follies** and weakness that even this counterfeit Venus with her sham heat, and her bosom probably plumped with gel, so moves me my breath catches. Vamp. Siren. Seductress. How much more she reveals in her glare of **ink** than she knows. How she incarnates our desperate human need for regard, our passion to live in beauty, to be beauty, to be cherished by glances, if by no more, of something like love, or love. Thank you.

Text Sample 584: NEG Dim 4 – 2001 – Score 2.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,"Are you still doing this or that?"I don ' t answer because I ' m not doing things still, I ' m doing it like I always did. I still have — or did I use the word still? I didn ' t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years. But, of course, I ' m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ' ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ' t call myself an industrial designer because I ' m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called"Innovation."Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,"The playful search for beauty was Man ' s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty."These are **tiles**. The word,"playful"is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ' t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ' t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 585: NEG Dim 4 – 2001 – Score 2.00 – 2001/t000092.txt

I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ’ children, because they were all average. But they weren ’ t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ’ s not right. The good Lord in his infinite wisdom didn ’ t create us all equal as far as intelligence is concerned, any more than we ’ re equal for size, appearance. Not everybody could earn an A or a B, and I didn ’ t like that way of judging, and I did know how the alumni of various schools back in the ’ 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ’ t lose a game. But it seemed that we didn ’ t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ’ t like it, I didn ’ t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ’ m sure at the time he did that, I didn ’ t — it didn ’ t — well, somewhere, I guess in the hidden recesses of the mind, it **popped** out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ’ s under your control. If you get too engrossed and involved and concerned in regard to the things over which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,”At God ’ s footstool to confess, a poor soul knelt, and bowed his head. ’ I failed! ’ he cried. The Master said, ’ Thou didst thy best, that is success. ’”From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ’ re capable. I believe that ’ s true. If you make the effort to do the best of which you ’ re capable, trying to improve the situation that exists for you, I think that ’ s success, and I don ’ t think others can judge that ; it ’ s like character and reputation — your reputation is what you ’ re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ’ s what the teachers are themselves.”That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by coal oil lamp — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I reply, ’ Where could I find such **splendid** company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ’ s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a **stumbling** soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ’ Where could I find such **splendid** company? ’”And I believe the teaching profession — it ’ s true, you have so many youngsters, and I ’ ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ’ re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ’ re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was”Never be late.”Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ’ s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ’ t clean and neat, so I wouldn ’ t let him go. He couldn ’ t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ’ t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ’ ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,”Don ’ t run practices late, because you ’ ll go home in a bad mood, and that ’ s not good, for a young married man to go home in a bad mood. When you get older, it doesn ’ t make any difference, but —”So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ’ re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ’ t want that. I used to tell them I was paid to do that. That ’ s my job. I ’ m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ’ s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ’ s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ’ t have the time to go on that. But that helped me, I think, become a better teacher. It ’ s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ’ re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ' re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ' t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ' i '. I ' d never seen that before, but he did. Big league baseball players — they ' re very perceptive about those things, and they noticed he had only one ' i ' in his name. You ' d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it"The Road Ahead, or the Road Behind."He said,"Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ' t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ' s game. You and I know deeper down, there ' s always a chance to win the crown. But when we fail to give our best, we simply haven ' t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ' s bearing down that wins the cup. Of dreaming there ' s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ' s you and I who make our fates — we open up or close the gates on the road ahead or the road behind."Reminds me of another set of threes that my dad tried to get across to us : Don ' t whine. Don ' t complain. Don ' t make excuses. Just get out there, and whatever you ' re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ' re outscored. I ' ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ' t know the outcome, I hope they couldn ' t tell by your actions whether you outscored an opponent or the opponent outscored you. That ' s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ' d want them to be but they ' ll be about what they should ; only you will know whether you can do that. And that ' s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,"The journey is better than the end."And I like that. I think that it is — it ' s getting there. Sometimes when you get there,

there ’ s almost a let down. But it ’ s the getting there that ’ s the fun. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ’ d done a decent job during the week. There again, it ’ s getting the players to get that self-satisfaction, in knowing that they ’ d made the effort to do the best of which they are capable. Sometimes I ’ m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,”Suppose that you, in some way, could make the perfect player. What would you want?”And I said,”Well, I ’ d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ’ d want one that could play, too. I ’ d want one to realize that defense usually wins championships, and who would work hard on defense. But I ’ d want one who would play offense, too. I ’ d want him to be unselfish, and look for the pass first and not shoot all the time. And I ’ d want one that could pass and would pass. I ’ ve had some that could and wouldn ’ t, and I ’ ve had some that would and could. So, yeah, I ’ d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ’ d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,”They Call Me Coach,”two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ’ t play varsity when I taught. I thought,”Oh gracious, if these two players, either one of them”— they were different years, but I thought about each one at the time he was there —”Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ’ s good enough to make it.”And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ’ d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 586: NEG Dim 4 – 2001 – Score 2.00 – 2001/t000367.txt

I ’ d like to do pretty much what I did the first time, which is to choose a light-hearted theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, exorcise those evil spirits, and this is still going on, as you know. But it wasn ' t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ' re feeling depressed? It ' s better than Prozac, but I wouldn ' t recommend it. So what we see in the seventeenth, eighteenth century is the continued search for medications other than camphor that ' ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we fast-forward to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a row — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very interested in producing convulsions, measured types of convulsions. And they thought,"Well, we ' ve got electricity, we ' ll plug somebody into the wall. That always makes hair stand up and people shake a lot."So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,"We know that at the Rome railroad station, there are all these lost souls wandering around, **muttering** gibberish. Can you bring one of them to us?"Someone who is, as the Italians say,"cagoots."So they found this"cagoots"guy, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,"Well, we ' ll try 55 volts, two-tenths of a second. That ' s not going to do anything terrible to him."So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,"This fellow"— remember, he wasn ' t even put to sleep —"after this major grand mal convulsion, sat right up, looked at these three fellas and said, ' What the fuck are you assholes trying to do? '"If I could only say that in Italian. Well, they were happy as could be, because he hadn ' t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and everybody in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ' t work very well on the schizophrenics, but it was pretty clear in the ’ 30s and by the middle of the ’ 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ’ t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the **brain** waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ’ 60s, the first antidepressants came out. Tofranil was the first. In the late ’ 70s, early ’ 80s, there were others, and they were very effective. And patients ’ rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ’ m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ’ ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ’ m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ’ clock in the afternoon, because I couldn ’ t get out of bed before about 11 o ’ clock. And anybody who ’ s been depressed here knows what that ’ s like. I couldn ’ t even pull the covers off myself. Well, you ’ re in a university medical center, where everybody knows everybody, and it ’ s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ’ t work anymore. And, in fact, it didn ’ t make any difference because I didn ’ t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,”Don ’ t worry, chap. Six weeks, you ’ re back in the operating room. Everything ’ s going to be great.”Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ’ t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five feet in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ’ t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had. They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the senior staff. All the senior staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful stuff. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it got so there was a throbbing, there was a ferocious fear in my head. You ' ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ' t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case. He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the senior staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,"No. I know this man better than any of you. I have met with him over and over again. You ' ve just seen him from time to time. You ' ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ' ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go back to surgery, he will never be the loving father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ' One Flew Over the Cuckoo ' s Nest. ' And you know about that, just essentially in a stupor the rest of his life."Well, he said,"Can ' t we try a course of electroshock therapy?"And you know why they agreed? They agreed to humor him. They just thought,"Well, we ' ll give a course of 10. And so we ' ll lose a little time. Big deal. It doesn ' t make any difference."So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ' t work. Seven didn ' t work. Eight didn ' t work. At nine, I noticed — and it ' s wonderful that I could notice anything — I noticed a change. And at 10, I noticed a real change. And he went back to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, lo and **behold**, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could blow the obsessional thinking away. I could blow the depression away. And I ' ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,"I ' ve got the strength now to do this."It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a formula. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember"Ruddigore,"and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,"We must have a word to bring you back to reality, and the word, my dear, will be ' Basingstoke. '"So every time she got a little nuts, he would say,"Basingstoke!"And she would say,"Basingstoke, it is."And she would be fine for a little while. Well, you know, I ' m from the Bronx. I can ' t say"Basingstoke."But I had something better. And it was very simple. It was,"Ah, fuck it!"Much better than"Basingstoke,"at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,"Ah, fuck it."And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went back to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came back to live with us. We had two more children after that. Resuscitated the career, even better than it had been before. Went right back into the university and began to write books. Well, you know, it ' s been a wonderful life. It ' s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ' s been very exciting. It ' s been very happy. Every once in a while, I have to say,"Ah, fuck it."Every once in a while, I get somewhat depressed and a little obsessional. So, I ' m not free of all of this. But it ' s worked. It ' s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ' ve gotten thousands of letters about them by people who do think this — that based on my life ' s history as I ' ve portrayed in the books, my early life ' s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the bitter dregs of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ' ve always felt guilty about that. I ' ve always felt that somehow I was an impostor because my readers don ' t know what I have just told you. It ' s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an untroubled mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes back to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ’ t happen, there will be adversities. If I, with the bleakness of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way back from this, believe me, anybody can find their way back from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

Text Sample 587: NEG Dim 4 – 2013 – Score 1.00 – 2013/t002115.txt

Today I ’ m going to show you an electric vehicle that weighs less than a bicycle, that you can carry with you anywhere, that you can charge off a normal wall outlet in 15 minutes, and you can run it for 1, 000 kilometers on about a dollar of electricity. But when I say the word electric vehicle, people think about vehicles. They think about cars and motorcycles and bicycles, and the vehicles that you use every day. But if you come about it from a different perspective, you can create some more interesting, more novel concepts. So we built something. I ’ ve got some of the pieces in my pocket here. So this is the motor. This motor has enough power to take you up the hills of San Francisco at about 20 miles per hour, about 30 kilometers an hour, and this battery, this battery right here has about six miles of range, or 10 kilometers, which is enough to cover about half of the car trips in the U. S. alone. But the best part about these components is that we bought them at a toy store. These are from remote control airplanes. And the performance of these things has gotten so good that if you think about vehicles a little bit differently, you can really change things. So today we ’ re going to show you one example of how you can use this. Pay attention to not only how fun this thing is, but also how the portability that comes with this can totally change the way you interact with a city like San Francisco. [6 Mile Range] [Top Speed Near 20mph] [Uphill Climbing] [Regenerative Braking] So we ’ re going to show you what this thing can do. It ’ s really maneuverable. You have a hand-held remote, so you can pretty easily control acceleration, braking, go in reverse if you like, also have braking. It ’ s incredible just how light this thing is. I mean, this is something you can pick up and carry with you anywhere you go. So I ’ ll leave you with one of the most compelling facts about this technology and these kinds of vehicles. This uses 20 times less energy for every mile or kilometer that you travel than a car, which means not only is this thing fast to charge and really cheap to build, but it also reduces the footprint of your energy use in terms of your transportation. So instead of looking at large amounts of energy needed for each person in this room to get around in a city, now you can look at much smaller amounts and more sustainable transportation. So next time you think about a vehicle, I hope, like us, you ’ re thinking about something new. Thank you.

Text Sample 588: NEG Dim 4 – 2013 – Score 1.00 – 2013/t002140.txt
I live in South Central. This is South Central : liquor stores, fast food, vacant lots. So the city planners, they get together and they figure they ’ re going to change the name South Central to make it represent something else, so they change it to South Los Angeles, like this is going to fix what ’ s really going wrong in the city. This is South Los Angeles. Liquor stores, fast food, vacant lots. Just like 26. 5 million other Americans, I live in a food desert, South Central Los Angeles, home of the drive-thru and the drive-by. Funny thing is, the drive-thrus are killing more people than the drive-bys. People are dying from curable diseases in South Central Los Angeles. For instance, the obesity rate in my neighborhood is five times higher than, say, Beverly Hills, which is probably eight, 10 miles away. I got tired of seeing this happening. And I was wondering, how would you feel if you had no access to healthy food, if every time you walk out your door you see the ill effects that the present food system has on your neighborhood? I see wheelchairs bought and sold like used cars. I see dialysis centers **popping** up like Starbucks. And I figured, this has to stop. So I figured that the problem is the solution. Food is the problem and food is the solution. Plus I got tired of driving 45 minutes round trip to get an apple that wasn ’ t impregnated with pesticides. So what I did, I planted a food forest in front of my house. It was on a strip of land that we call a parkway. It ’ s 150 feet by 10 feet. Thing is, it ’ s owned by the city. But you have to maintain it. So I ’ m like,”Cool. I can do whatever the hell I want, since it ’ s my responsibility and I gotta maintain it.”And this is how I decided to maintain it. So me and my group, L. A. Green Grounds, we got together and we started planting my food forest, fruit trees, you know, the whole nine, vegetables. What we do, we ’ re a pay-it-forward kind of group, where it ’ s composed of gardeners from all walks of life, from all over the city, and it ’ s completely volunteer, and everything we do is free. And the garden, it was beautiful. And then somebody complained. The city came down on me, and basically gave me a citation saying that I had to remove my garden, which this citation was turning into a warrant. And I ’ m like,”Come on, really? A warrant for planting food on a piece of land that you could care less about?”And I was like,”Cool. Bring it.”Because this time it wasn ’ t coming up. So L. A. Times got ahold of it. Steve Lopez did a story on it and talked to the councilman, and one of the Green Grounds members, they put up a petition on Change. org, and with 900 signatures, we were a success. We had a victory on our hands. My councilman even called in and said how they endorse and love what we ’ re doing. I mean, come on, why wouldn ’ t they? L. A. leads the United States in vacant lots that the city actually owns. They own 26 square miles of vacant lots. That ’ s 20 Central Parks. That ’ s enough space to plant 725 million tomato plants. Why in the hell would they not okay this? Growing one plant will give you 1, 000, 10, 000 seeds. When one dollar ’ s worth of green beans will give you 75 dollars ’ worth of produce. It ’ s my gospel, when I ’ m telling people, grow your own food. Growing your own food is like printing your own money. See, I have a legacy in South Central. I grew up there. I raised my sons there. And I refuse to be a part of this manufactured reality that was manufactured for me by some other people, and I ’ m manufacturing my own reality. See, I ’ m an artist. Gardening is my graffiti. I grow my art. Just like a graffiti artist, where they beautify walls, me, I beautify lawns, parkways. I

use the garden, the soil, like it ' s a piece of cloth, and the plants and the trees, that ' s my embellishment for that cloth. You ' d be surprised what the soil could do if you let it be your canvas. You just couldn ' t imagine how amazing a sunflower is and how it affects people. So what happened? I have witnessed my garden become a tool for the education, a tool for the transformation of my neighborhood. To change the community, you have to change the composition of the soil. We are the soil. You ' d be surprised how kids are affected by this. Gardening is the most therapeutic and defiant act you can do, especially in the inner city. Plus you get strawberries. I remember this time, there was this mother and a daughter came, it was, like, 10:30 at night, and they were in my yard, and I came out and they looked so ashamed. So I ' m like, man, it made me feel bad that they were there, and I told them, you know, you don ' t have to do this like this. This is on the street for a reason. It made me feel ashamed to see people that were this close to me that were hungry, and this only reinforced why I do this, and people asked me,"Fin, aren ' t you afraid people are going to steal your food?"And I ' m like,"Hell no, I ain ' t afraid they ' re gonna steal it. That ' s why it ' s on the street. That ' s the whole idea. I want them to take it, but at the same time, I want them to take back their health."There ' s another time when I put a garden in this homeless shelter in downtown Los Angeles. These are the guys, they helped me unload the truck. It was cool, and they just shared the stories about how this affected them and how they used to plant with their mother and their grandmother, and it was just cool to see how this changed them, if it was only for that one moment. So Green Grounds has gone on to plant maybe 20 gardens. We ' ve had, like, 50 people come to our dig-ins and participate, and it ' s all volunteers. If kids grow kale, kids eat kale. If they grow tomatoes, they eat tomatoes. But when none of this is presented to them, if they ' re not shown how food affects the mind and the body, they blindly eat whatever the hell you put in front of them. I see young people and they want to work, but they ' re in this thing where they ' re caught up — I see kids of color and they ' re just on this track that ' s designed for them, that leads them to nowhere. So with gardening, I see an opportunity where we can train these kids to take over their communities, to have a sustainable life. And when we do this, who knows? We might produce the next George Washington Carver. but if we don ' t change the composition of the soil, we will never do this. Now this is one of my plans. This is what I want to do. I want to plant a whole block of gardens where people can share in the food in the same block. I want to take shipping containers and turn them into healthy cafes. Now don ' t get me wrong. I ' m not talking about no free shit, because free is not sustainable. The funny thing about sustainability, you have to sustain it. What I ' m talking about is putting people to work, and getting kids off the street, and letting them know the joy, the pride and the honor in growing your own food, opening farmer ' s markets. So what I want to do here, we gotta make this sexy. So I want us all to become ecolutionary renegades, gangstas, gangsta gardeners. We gotta flip the script on what a gangsta is. If you ain ' t a gardener, you ain ' t gangsta. Get gangsta with your shovel, okay? And let that be your weapon of choice. So basically, if you want to meet with me, you know, if you want to meet, don ' t call me if you want to sit around in cushy chairs and have meetings where you talk about doing some shit — where you talk about doing some shit. If you want to meet with me, come to the garden with your shovel so we can plant some shit. Peace. Thank you. Thank you.

Text Sample 589: NEG Dim 4 – 2013 – Score 1.00 – 2013/t002097.txt

Growing up in Taiwan as the daughter of a calligrapher, one of my most treasured memories was my mother showing me the beauty, the shape and the form of Chinese characters. Ever since then, I was fascinated by this incredible language. But to an outsider, it seems to be as impenetrable as the Great Wall of China. Over the past few years, I ’ ve been wondering if I can break down this wall, so anyone who wants to understand and appreciate the beauty of this sophisticated language could do so. I started thinking about how a new, fast method of learning Chinese might be useful. Since the age of five, I started to learn how to draw every single stroke for each character in the correct sequence. I learned new characters every day during the course of the next 15 years. Since we only have five minutes, it ’ s better that we have a fast and simpler way. A Chinese scholar would understand 20, 000 characters. You only need 1, 000 to understand the basic literacy. The top 200 will allow you to comprehend 40 percent of basic literature — enough to read road signs, restaurant menus, to understand the basic idea of the web pages or the newspapers. Today I ’ m going to start with eight to show you how the method works. You are ready? Open your mouth as wide as possible until it ’ s square. You get a mouth. This is a person going for a walk. Person. If the shape of the fire is a person with two arms on both sides, as if she was yelling frantically,”Help! I ’ m on fire!”— This symbol actually is originally from the shape of the flame, but I like to think that way. Whichever works for you. This is a tree. Tree. This is a mountain. The sun. The moon. The symbol of the door looks like a pair of saloon doors in the wild west. I call these eight characters radicals. They are the building blocks for you to create lots more characters. A person. If someone walks behind, that is”to follow.”As the old saying goes, two is company, three is a crowd. If a person stretched their arms wide, this person is saying,”It was this big.”The person inside the mouth, the person is trapped. He ’ s a prisoner, just like Jonah inside the whale. One tree is a tree. Two trees together, we have the woods. Three trees together, we create the forest. Put a plank underneath the tree, we have the foundation. Put a mouth on the top of the tree, that ’ s”idiot.”Easy to remember, since a talking tree is pretty idiotic. Remember fire? Two fires together, I get really hot. Three fires together, that ’ s a lot of flames. Set the fire underneath the two trees, it ’ s burning. For us, the sun is the source of prosperity. Two suns together, prosperous. Three together, that ’ s sparkles. Put the sun and the moon shining together, it ’ s brightness. It also means tomorrow, after a day and a night. The sun is coming up above the horizon. Sunrise. A door. Put a plank inside the door, it ’ s a door bolt. Put a mouth inside the door, asking questions. Knock knock. Is anyone home? This person is sneaking out of a door, escaping, evading. On the left, we have a woman. Two women together, they have an argument. Three women together, be careful, it ’ s adultery. So we have gone through almost 30 characters. By using this method, the first eight radicals will allow you to build 32. The next group of eight characters will build an extra 32. So with very little effort, you will be able to learn a couple hundred characters, which is the same as a Chinese eight-year-old. So after we know the characters, we start building phrases. For example, the mountain and the fire together, we have fire mountain. It ’ s a volcano. We know Japan is the land of the rising sun. This is a sun placed with the origin, because Japan lies to the east of China. So a sun, origin together, we build Japan. A person behind Japan, what do we get? A Japanese person. The character on the left is two mountains stacked on top of each other. In ancient China, that means in exile, because Chinese emperors, they put their political enemies in exile beyond mountains. Nowadays, exile has turned into getting out. A mouth which tells you where to get out is an exit. This is a slide to remind me that I should stop talking and get off of the stage. Thank you.

Text Sample 590: NEG Dim 4 – 2011 – Score -1.00 – 2011/t002422.txt
I ’ m here to share my photography. Or is it photography? Because, of course, this is a photograph that you can ’ t take with your camera. Yet, my interest in photography started as I got my first digital camera at the age of 15. It mixed with my earlier passion for drawing, but it was a bit different, because using the camera, the process was in the planning instead. And when you take a photograph with a camera, the process ends when you press the trigger. So to me it felt like photography was more about being at the right place and the right time. I felt like anyone could do that. So I wanted to create something different, something where the process starts when you press the trigger. Photos like this : construction going on along a busy road. But it has an unexpected twist. And despite that, it retains a level of realism. Or photos like these — both dark and colorful, but all with a common goal of retaining the level of realism. When I say realism, I mean photo-realism. Because, of course, it ’ s not something you can capture really, but I always want it to look like it could have been captured somehow as a photograph. Photos where you will need a brief moment to think to figure out the trick. So it ’ s more about capturing an idea than about capturing a moment really. But what ’ s the trick that makes it look realistic? Is it something about the details or the colors? Is it something about the light? What creates the illusion? Sometimes the perspective is the illusion. But in the end, it comes down to how we interpret the world and how it can be realized on a two- dimensional surface. It ’ s not really what is realistic, it ’ s what we think looks realistic really. So I think the basics are quite simple. I just see it as a puzzle of reality where you can take different pieces of reality and put it together to create alternate reality. And let me show you a simple example. Here we have three perfectly imaginable physical objects, something we all can relate to living in a three-dimensional world. But combined in a certain way, they can create something that still looks three-dimensional, like it could exist. But at the same time, we know it can ’ t. So we trick our **brains**, because our **brain** simply doesn ’ t accept the fact that it doesn ’ t really make sense. And I see the same process with combining photographs. It ’ s just really about combining different realities. So the things that make a photograph look realistic, I think it ’ s the things that we don ’ t even think about, the things all around us in our daily lives. But when combining photographs, this is really important to consider, because otherwise it just looks wrong somehow. So I would like to say that there are three simple rules to follow to achieve a realistic result. As you can see, these images aren ’ t really special. But combined, they can create something like this. So the first rule is that photos combined should have the same perspective. Secondly, photos combined should have the same type of light. And these two images both fulfill these two requirements — shot at the same height and in the same type of light. The third one is about making it impossible to distinguish where the different images begin and end by making it seamless. Make it impossible to say how the image actually was composed. So by matching color, contrast and brightness in the borders between the different images, adding photographic defects like depth of field, desaturated colors and noise, we erase the borders between the different images and make it look like one single image, despite the fact that one image can contain hundreds of layers basically. So here ’ s another example. One might think that this is just an image of a landscape and the lower part is what ’ s manipulated. But this image is actually entirely composed of photographs from different locations. I

personally think that it ’ s easier to actually create a place than to find a place, because then you don ’ t need to compromise with the ideas in your head. But it does require a lot of planning. And getting this idea during winter, I knew that I had several months to plan it, to find the different locations for the pieces of the puzzle basically. So for example, the fish was captured on a fishing trip. The shores are from a different location. The underwater part was captured in a stone pit. And yeah, I even turned the house on top of the island red to make it look more Swedish. So to achieve a realistic result, I think it comes down to planning. It always starts with a sketch, an idea. Then it ’ s about combining the different photographs. And here every piece is very well planned. And if you do a good job capturing the photos, the result can be quite beautiful and also quite realistic. So all the tools are out there, and the only thing that limits us is our imagination. Thank you.

Text Sample 591: NEG Dim 4 – 2011 – Score -1.00 – 2011/t002625.txt

I ’ m a huge believer in hands-on education. But you have to have the right tools. If I ’ m going to teach my daughter about electronics, I ’ m not going to give her a soldering iron. And similarly, she finds prototyping boards really frustrating for her little hands. So my wonderful student Sam and I decided to look at the most tangible thing we could think of : Play-Doh. And so we spent a summer looking at different Play-Doh recipes. And these recipes probably look really familiar to any of you who have made homemade play-dough — pretty standard ingredients you probably have in your kitchen. We have two favorite recipes — one that has these ingredients and a second that had sugar instead of salt. And they ’ re great. We can make great little sculptures with these. But the really cool thing about them is when we put them together. You see that really salty Play-Doh? Well, it conducts electricity. And this is nothing new. It turns out that regular Play-Doh that you buy at the store conducts electricity, and high school physics teachers have used that for years. But our homemade play-dough actually has half the resistance of commercial Play-Doh. And that sugar dough? Well it ’ s 150 times more resistant to electric current than that salt dough. So what does that mean? Well it means if you them together you suddenly have circuits — circuits that the most creative, tiny, little hands can build on their own. And so I want to do a little demo for you. So if I take this salt dough, again, it ’ s like the play-dough you probably made as kids, and I plug it in — it ’ s a two-lead battery pack, simple battery pack, you can buy them at Radio Shack and pretty much anywhere else — we can actually then light things up. But if any of you have studied electrical engineering, we can also create a short circuit. If I push these together, the light turns off. Right, the current wants to run through the play-dough, not through that LED. If I separate them again, I have some light. Well now if I take that sugar dough, the sugar dough doesn ’ t want to conduct electricity. It ’ s like a wall to the electricity. If I place that between, now all the dough is touching, but if I stick that light back in, I have light. In fact, I could even add some movement to my sculptures. If I want a spinning tail, let ’ s grab a motor, put some play-dough on it, stick it on and we have spinning. And once you have the basics, we can make a slightly more complicated circuit. We call this our sushi circuit. It ’ s very popular with kids. I plug in again the power to it. And now I can start talking about parallel and series circuits. I can start plugging in lots of lights. And we can start talking about things like electrical load. What happens if I put in lots of lights and then add a motor? It ’ ll **dim**. We can even add microprocessors and have this as an input and create squishy sound music that we ’ ve done.

You could do parallel and series circuits for kids using this. So this is all in your home kitchen. We ’ ve actually tried to turn it into an electrical engineering lab. We have a website, it ’ s all there. These are the home recipes. We ’ ve got some videos. You can make them yourselves. And it ’ s been really fun since we put them up to see where these have gone. We ’ ve had a mom in Utah who used them with her kids, to a science researcher in the U. K., and curriculum developers in Hawaii. So I would encourage you all to grab some Play-Doh, grab some salt, grab some sugar and start playing. We don ’ t usually think of our kitchen as an electrical engineering lab or little kids as circuit designers, but maybe we should. Have fun. Thank you.

Text Sample 592: NEG Dim 4 – 2011 – Score -1.00 – 2011/t002577.txt
Space, we all know what it looks like. We ’ ve been surrounded by images of space our whole lives, from the speculative images of science fiction to the inspirational visions of artists to the increasingly beautiful pictures made possible by complex technologies. But whilst we have an overwhelmingly vivid visual understanding of space, we have no sense of what space sounds like. And indeed, most people associate space with silence. But the story of how we came to understand the universe is just as much a story of listening as it is by looking. And yet despite this, hardly any of us have ever heard space. How many of you here could describe the sound of a single planet or star? Well in case you ’ ve ever wondered, this is what the Sun sounds like. This is the planet Jupiter. And this is the space probe Cassini pirouetting through the ice rings of Saturn. This is a a highly condensed clump of neutral matter, spinning in the distant universe. So my artistic practice is all about listening to the weird and wonderful noises emitted by the magnificent celestial objects that make up our universe. And you may wonder, how do we know what these sounds are? How can we tell the difference between the sound of the Sun and the sound of a pulsar? Well the answer is the science of radio astronomy. Radio astronomers study radio waves from space using sensitive antennas and receivers, which give them precise information about what an astronomical object is and where it is in our night sky. And just like the signals that we send and receive here on Earth, we can convert these transmissions into sound using simple analog techniques. And therefore, it ’ s through listening that we ’ ve come to uncover some of the universe ’ s most important secrets — its scale, what it ’ s made of and even how old it is. So today, I ’ m going to tell you a short story of the history of the universe through listening. It ’ s punctuated by three quick anecdotes, which show how accidental encounters with strange noises gave us some of the most important information we have about space. Now this story doesn ’ t start with vast telescopes or futuristic spacecraft, but a rather more humble technology — and in fact, the very medium which gave us the telecommunications revolution that we ’ re all part of today : the telephone. It ’ s 1876, it ’ s in Boston, and this is Alexander Graham Bell who was working with Thomas Watson on the invention of the telephone. A key part of their technical set up was a half-mile long length of wire, which was thrown across the rooftops of several houses in Boston. The line carried the telephone signals that would later make Bell a household name. But like any long length of charged wire, it also inadvertently became an antenna. Thomas Watson spent hours listening to the strange crackles and hisses and chirps and whistles that his accidental antenna detected. Now you have to remember, this is 10 years before Heinrich Hertz proved the existence of radio waves — 15 years before Nikola Tesla ’ s four-tuned circuit — nearly 20 years before Marconi ’ s first broadcast. So

Thomas Watson wasn ’ t listening to us. We didn ’ t have the technology to transmit. So what were these strange noises? Watson was in fact listening to very low- frequency radio emissions caused by nature. Some of the crackles and **pops** were lightning, but the eerie whistles and curiously melodious chirps had a rather more exotic origin. Using the very first telephone, Watson was in fact dialed into the heavens. As he correctly guessed, some of these sounds were caused by activity on the surface of the Sun. It was a solar wind interacting with our ionosphere that he was listening to — a phenomena which we can see at the extreme northern and southern latitudes of our planet as the aurora. So whilst inventing the technology that would usher in the telecommunications revolution, Watson had discovered that the star at the center of our solar system emitted powerful radio waves. He had accidentally been the first person to tune in to them. Fast-forward 50 years, and Bell and Watson ’ s technology has completely transformed global communications. But going from slinging some wire across rooftops in Boston to laying thousands and thousands of miles of cable on the Atlantic Ocean seabed is no easy matter. And so before long, Bell were looking to new technologies to optimize their revolution. Radio could carry sound without wires. But the medium is lossy — it ’ s subject to a lot of noise and interference. So Bell employed an engineer to study those noises, to try and find out where they came from, with a view towards building the perfect hardware codec, which would get rid of them so they could think about using radio for the purposes of telephony. Most of the noises that the engineer, Karl Jansky, investigated were fairly prosaic in origin. They turned out to be lightning or sources of electrical power. But there was one persistent noise that Jansky couldn ’ t identify, and it seemed to appear in his radio headset four minutes earlier each day. Now any astronomer will tell you, this is the telltale sign of something that doesn ’ t originate from Earth. Jansky had made a historic discovery, that celestial objects could emit radio waves as well as light waves. Fifty years on from Watson ’ s accidental encounter with the Sun, Jansky ’ s careful listening ushered in a new age of space exploration : the radio astronomy age. Over the next few years, astronomers connected up their antennas to loudspeakers and learned about our radio sky, about Jupiter and the Sun, by listening. Let ’ s jump ahead again. It ’ s 1964, and we ’ re back at Bell Labs. And once again, two scientists have got a problem with noise. Arno Penzias and Robert Wilson were using the horn antenna at Bell ’ s Holmdel laboratory to study the Milky Way with extraordinary precision. They were really listening to the galaxy in high fidelity. There was a glitch in their soundtrack. A mysterious persistent noise was disrupting their research. It was in the microwave range, and it appeared to be coming from all directions simultaneously. Now this didn ’ t make any sense, and like any reasonable engineer or scientist, they assumed that the problem must be the technology itself, it must be the dish. There were pigeons roosting in the dish. And so perhaps once they cleaned up the pigeon droppings, get the disk kind of operational again, normal operations would resume. But the noise didn ’ t disappear. The mysterious noise that Penzias and Wilson were listening to turned out to be the oldest and most significant sound that anyone had ever heard. It was cosmic radiation left over from the very birth of the universe. This was the first experimental evidence that the Big Bang existed and the universe was born at a precise moment some 14. 7 billion years ago. So our story ends at the beginning — the beginning of all things, the Big Bang. This is the noise that Penzias and Wilson heard — the oldest sound that you ’ re ever going to hear, the cosmic microwave background radiation left over from the Big Bang. Thanks.

Text Sample 593: NEG Dim 4 – 2012 – Score -1.00 – 2012/t002401.txt

I call myself a body architect. I trained in classical ballet and have a background in architecture and fashion. As a body architect, I fascinate with the human body and explore how I can transform it. I worked at Philips Electronics in the far-future design research lab, looking 20 years into the future. I explored the human skin, and how technology can transform the body. I worked on concepts like an electronic tattoo, which is augmented by touch, or dresses that blushed and shivered with light. I started my own experiments. These were the low-tech approaches to the high-tech conversations I was having. These are Q-tips stuck to my roommate with wig glue. I started a collaboration with a friend of mine, Bart Hess — he doesn ’ t normally look like this — and we used ourselves as models. We transformed our apartments into our laboratories, and worked in a very spontaneous and immediate way. We were creating visual imagery provoking human evolution. Whilst I was at Philips, we discussed this idea of a maybe technology, something that wasn ’ t either switched on or off, but in between. A maybe that could take the form of a gas or a liquid. And I became obsessed with this idea of blurring the perimeter of the body, so you couldn ’ t see where the skin ended and the near environment started. I set up my studio in the red-light district and obsessively wrapped myself in plumbing tubing, and found a way to redefine the skin and create this dynamic textile. I was introduced to Robyn, the Swedish **pop** star, and she was also exploring how technology coexists with raw human emotion. And she talked about how technology with these new feathers, this new face paint, this punk, the way that we identify with the world, and we made this music video. I ’ m fascinated with the idea of what happens when you merge biology with technology, and I remember reading about this idea of being able to reprogram biology, in the future, away from disease and aging. And I thought about this concept of, imagine if we could reprogram our own body odor, modify and biologically enhance it, and how would that change the way that we communicate with each other? Or the way that we attract sexual partners? And would we revert back to being more like animals, more primal modes of communication? I worked with a synthetic biologist, and I created a swallowable perfume, which is a cosmetic pill that you eat and the fragrance comes out through the skin ’ s surface when you perspire. It completely blows apart the way that perfume is, and provides a whole new format. It ’ s perfume coming from the inside out. It redefines the role of skin, and our bodies become an atomizer. I ’ ve learned that there ’ s no boundaries, and if I look at the evolution of my work i can see threads and connections that make sense. But when I look towards the future, the next project is completely unknown and wide open. I feel like I have all these ideas existing embedded inside of me, and it ’ s these conversations and these experiences that connect these ideas, and they kind of instinctively come out. As a body architect, I ’ ve created this limitless and boundless platform for me to discover whatever I want. And I feel like I ’ ve just got started. So here ’ s to another day at the office. Thank you! Thank you!

Text Sample 594: NEG Dim 4 – 2012 – Score -1.00 – 2012/t002163.txt

Organic chemists make molecules, very complicated molecules, by chopping up a big molecule into small molecules and reverse engineering. And as a chemist, one of the things I wanted to ask my research group a couple of years ago is, could we make a really cool universal chemistry set? In essence, could we”app”chemistry? Now what would this mean, and how would we do it? Well to start to do this, we took a 3D printer and we started to print our beakers and our test tubes on one side and then print the molecule at the same time on the other side and combine them together in what we call reactionware. And so by printing the vessel and doing the chemistry at the same time, we may start to access this universal toolkit of chemistry. Now what could this mean? Well if we can embed biological and chemical networks like a search engine, so if you have a cell that ’ s ill that you need to cure or bacteria that you want to kill, if you have this embedded in your device at the same time, and you do the chemistry, you may be able to make drugs in a new way. So how are we doing this in the lab? Well it requires software, it requires hardware and it requires chemical **inks**. And so the really cool bit is, the idea is that we want to have a universal set of **inks** that we put out with the printer, and you download the blueprint, the organic chemistry for that molecule and you make it in the device. And so you can make your molecule in the printer using this software. So what could this mean? Well, ultimately, it could mean that you could print your own medicine. And this is what we ’ re doing in the lab at the moment. But to take baby steps to get there, first of all we want to look at drug design and production, or drug discovery and manufacturing. Because if we can manufacture it after we ’ ve discovered it, we could deploy it anywhere. You don ’ t need to go to the chemist anymore. We can print drugs at point of need. We can download new diagnostics. Say a new super bug has emerged. You put it in your search engine, and you create the drug to treat the threat. So this allows you on-the-fly molecular assembly. But perhaps for me the core bit going into the future is this idea of taking your own stem cells, with your genes and your environment, and you print your own personal medicine. And if that doesn ’ t seem fanciful enough, where do you think we ’ re going to go? Well, you ’ re going to have your own personal matter fabricator. Beam me up, Scotty.

Text Sample 595: NEG Dim 4 – 2012 – Score -1.00 – 2012/t002225.txt
Five years ago, I experienced a bit of what it must have been like to be Alice in Wonderland. Penn State asked me, a communications teacher, to teach a communications class for engineering students. And I was scared. Really scared. Scared of these students with their big **brains** and their big books and their big, unfamiliar words. But as these conversations unfolded, I experienced what Alice must have when she went down that rabbit hole and saw that door to a whole new world. That ’ s just how I felt as I had those conversations with the students. I was amazed at the ideas that they had, and I wanted others to experience this wonderland as well. And I believe the key to opening that door is great communication. We desperately need great communication from our scientists and engineers in order to change the world. Our scientists and engineers are the ones that are tackling our grandest challenges, from energy to environment to health care, among others, and if we don ’ t know about it and understand it, then the work isn ’ t done, and I believe it ’ s our responsibility as non-scientists to have these interactions. But these great conversations can ’ t occur if our scientists and engineers don ’ t invite us in to see their wonderland. So scientists and engineers, please, talk nerdy to us. I want to share a few keys on how you can do that to make sure that we can see that your science is sexy and that your engineering is engaging. First question to answer for us : so what? Tell us why your science is relevant to us. Don ’ t just tell me that you study trabeculae, but tell me that you study trabeculae, which is the mesh-like structure of our bones because it ’ s important to understanding and treating osteoporosis. And when you ’ re describing your science, beware

of jargon. Jargon is a barrier to our understanding of your ideas. Sure, you can say”spatial and temporal,”but why not just say”space and time,”which is so much more accessible to us? And making your ideas accessible is not the same as dumbing it down. Instead, as Einstein said, make everything as simple as possible, but no simpler. You can clearly communicate your science without compromising the ideas. A few things to consider are having examples, stories and analogies. Those are ways to engage and excite us about your content. And when presenting your work, drop the bullet points. Have you ever wondered why they ’ re called bullet points? What do bullets do? Bullets kill, and they will kill your presentation. A slide like this is not only boring, but it relies too much on the language area of our **brain**, and causes us to become overwhelmed. Instead, this example slide by Genevieve Brown is much more effective. It ’ s showing that the special structure of trabeculae are so strong that they actually inspired the unique design of the Eiffel Tower. And the trick here is to use a single, readable sentence that the audience can key into if they get a bit lost, and then provide visuals which appeal to our other senses and create a deeper sense of understanding of what ’ s being described. So I think these are just a few keys that can help the rest of us to open that door and see the wonderland that is science and engineering. And because the engineers that I ’ ve worked with have taught me to become really in touch with my inner nerd, I want to summarize with an equation. Take your science, subtract your bullet points and your jargon, divide by relevance, meaning share what ’ s relevant to the audience, and multiply it by the passion that you have for this incredible work that you ’ re doing, and that is going to equal incredible interactions that are full of understanding. And so, scientists and engineers, when you ’ ve solved this equation, by all means, talk nerdy to me. Thank you.

Text Sample 596: NEG Dim 4 – 2014 – Score -2.00 – 2014/t001631.txt

What ’ s the fastest growing threat to Americans ’ health? Cancer? Heart attacks? Diabetes? The answer is actually none of these ; it ’ s Alzheimer ’ s disease. Every 67 seconds, someone in the United States is diagnosed with Alzheimer ’ s. As the number of Alzheimer ’ s patients triples by the year 2050, caring for them, as well as the rest of the aging population, will become an overwhelming societal challenge. My family has experienced firsthand the struggles of caring for an Alzheimer ’ s patient. Growing up in a family with three generations, I ’ ve always been very close to my grandfather. When I was four years old, my grandfather and I were walking in a park in Japan when he suddenly got lost. It was one of the scariest moments I ’ ve ever experienced in my life, and it was also the first instance that informed us that my grandfather had Alzheimer ’ s disease. Over the past 12 years, his condition got worse and worse, and his wandering in particular caused my family a lot of stress. My aunt, his primary caregiver, really struggled to stay awake at night to keep an eye on him, and even then often failed to catch him leaving the bed. I became really concerned about my aunt ’ s well-being as well as my grandfather ’ s safety. I searched extensively for a solution that could help my family ’ s problems, but couldn ’ t find one. Then, one night about two years ago, I was looking after my grandfather and I saw him stepping out of the bed. The moment his foot landed on the floor, I thought, why don ’ t I put a pressure sensor on the heel of his foot? Once he stepped onto the floor and out of the bed, the pressure sensor would detect an increase in pressure caused by body weight and then wirelessly send an audible alert to the caregiver ’ s smartphone. That way, my aunt could sleep much better at night without having to worry about my grandfather ’ s wandering. So now I ’ d like to perform a demonstration of this sock. Could I please have my sock model on the stage? Great. So once the patient steps onto the floor — — an alert is sent to the caregiver ’ s smartphone. Thank you. Thank you, sock model. So this is a drawing of my preliminary design. My desire to create a sensor-based technology perhaps stemmed from my lifelong love for sensors and technology. When I was six years old, an elderly family friend fell down in the bathroom and suffered severe injuries. I became concerned about my own grandparents and decided to invent a smart bathroom system. Motion sensors would be installed inside the **tiles** of bathroom floors to detect the falls of elderly patients whenever they fell down in the bathroom. Since I was only six years old at the time and I hadn ’ t graduated from kindergarten yet, I didn ’ t have the necessary resources and tools to translate my idea into reality, but nonetheless, my research experience really implanted in me a firm desire to use sensors to help the elderly people. I really believe that sensors can improve the quality of life of the elderly. When I laid out my plan, I realized that I faced three main challenges : first, creating a sensor ; second, designing a circuit ; and third, coding a smartphone app. This made me realize that my project was actually much harder to realize than I initially had thought it to be. First, I had to create a wearable sensor that was thin and flexible enough to be worn comfortably on the bottom of the patient ’ s foot. After extensive research and testing of different materials like rubber, which I realized was too thick to be worn snugly on the bottom of the foot, I decided to print a film sensor with electrically conductive pressure-sensitive **ink** particles. Once pressure is applied, the connectivity between the particles increases. Therefore, I could design a circuit that would measure pressure by measuring electrical resistance. Next, I had to design a wearable wireless circuit, but wireless signal transmission consumes lots of power and requires heavy, bulky batteries. Thankfully, I was able to find out about the Bluetooth low energy technology, which consumes very little power and can be driven by a coin-sized battery. This prevented the system from dying in the middle of the night. Lastly, I had to code a smartphone app that would essentially transform the care-giver ’ s smartphone into a remote monitor. For this, I had to expand upon my knowledge of coding with Java and XCode and I also had to learn about how to code for Bluetooth low energy devices by watching YouTube tutorials and reading various textbooks. Integrating these components, I was able to successfully create two prototypes, one in which the sensor is embedded inside a sock, and another that ’ s a re- attachable sensor assembly that can be adhered anywhere that makes contact with the bottom of the patient ’ s foot. I ’ ve tested the device on my grandfather for about a year now, and it ’ s had a 100 percent success rate in detecting the over 900 known cases of his wandering. Last summer, I was able to beta test my device at several residential care facilities in California, and I ’ m currently incorporating the feedback to further improve the device into a marketable product. Testing the device on a number of patients made me realize that I needed to invent solutions for people who didn ’ t want to wear socks to sleep at night. So sensor data, collected on a vast number of patients, can be useful for improving patient care and also leading to a cure for the disease, possibly. For example, I ’ m currently examining correlations between the frequency of a patient ’ s nightly wandering and his or her daily activities and diet. One thing I ’ ll never forget is when my device first caught my grandfather ’ s wandering out of bed at night. At that moment, I was really struck by the power of technology to change lives for the better. People living happily and healthfully — that ’ s the world that I imagine. Thank you very much.

Text Sample 597: NEG Dim 4 – 2014 – Score -1.00 – 2014/t001717.txt

I know a man who soars above the city every night. In his dreams, he twirls and swirls with his toes kissing the Earth. Everything has motion, he claims, even a body as paralyzed as his own. This man is my father. Three years ago, when I found out that my father had suffered a severe stroke in his **brain** stem, I walked into his room in the ICU at the Montreal Neurological Institute and found him lying deathly still, tethered to a breathing machine. Paralysis had closed over his body slowly, beginning in his toes, then legs, torso, fingers and arms. It made its way up his neck, cutting off his ability to breathe, and stopped just beneath the eyes. He never lost consciousness. Rather, he watched from within as his body shut down, limb by limb, muscle by muscle. In that ICU room, I walked up to my father ' s body, and with a quivering voice and through tears, I began reciting the alphabet. A, B, C, D, E, F, G, H, I, J, K. At K, he blinked his eyes. I began again. A, B, C, D, E, F, G, H, I. He blinked again at the letter I, then at T, then at R, and A : Kitra. He said"Kitra, my beauty, don ' t cry. This is a blessing."There was no audible voice, but my father called out my name powerfully. Just 72 hours after his stroke, he had already embraced the totality of his condition. Despite his extreme physical state, he was completely present with me, guiding, nurturing, and being my father as much if not more than ever before. Locked-in syndrome is many people ' s worst nightmare. In French, it ' s sometimes called"maladie de l ' emmur vivant."Literally,"walled-in-alive disease."For many people, perhaps most, paralysis is an unspeakable horror, but my father ' s experience losing every system of his body was not an experience of feeling trapped, but rather of turning the psyche inwards, **dimming** down the external chatter, facing the recesses of his own mind, and in that place, falling in love with life and body anew. As a rabbi and spiritual man dangling between mind and body, life and death, the paralysis opened up a new awareness for him. He realized he no longer needed to look beyond the corporeal world in order to find the divine."Paradise is in this body. It ' s in this world,"he said. I slept by my father ' s side for the first four months, tending as much as I could to his every discomfort, understanding the deep human psychological fear of not being able to call out for help. My mother, sisters, brother and I, we surrounded him in a cocoon of healing. We became his mouthpiece, spending hours each day reciting the alphabet as he whispered back sermons and poetry with blinks of his eye. His room, it became our temple of healing. His bedside became a site for those seeking advice and spiritual counsel, and through us, my father was able to speak and uplift, letter by letter, blink by blink. Everything in our world became slow and tender as the din, drama and death of the hospital ward faded into the background. I want to read to you one of the first things that we transcribed in the week following the stroke. He composed a letter, addressing his synagogue congregation, and ended it with the following lines :"When my nape exploded, I entered another dimension : inchoate, sub-planetary, protozoan. Universes are opened and closed continually. There are many when low, who stop growing. Last week, I was brought so low, but I felt the hand of my father around me, and my father brought me back."When we weren ' t his voice, we were his legs and arms. I moved them like I know I would have wanted my own arms and legs to be moved were they still for all the hours of the day. I remember I ' d hold his fingers near my face, bending each joint to keep it soft and limber. I ' d ask him again and again to visualize the motion, to watch from within as the finger curled and extended, and to move along with it in his mind. Then, one day, from the corner of my eye, I saw his body slither like a snake, an involuntary spasm passing through the course of his limbs. At first, I thought it was my own hallucination, having spent so much time tending to this one body, so desperate to see anything react on its own. But he told me he felt tingles, sparks of electricity flickering on and off just beneath the surface of the skin. The following week, he began ever so slightly to show muscle resistance. Connections were being made. Body was slowly and gently reawakening, limb by limb, muscle by muscle, twitch by twitch. As a documentary photographer, I felt the need to photograph each of his first movements like a mother with her newborn. I photographed him taking his first unaided breath, the celebratory moment after he showed muscle resistance for the very first time, the new adapted technologies that allowed him to gain more and more independence. I photographed the care and the love that surrounded him. But my photographs only told the outside story of a man lying in a hospital bed attached to a breathing machine. I wasn ’ t able to portray his story from within, and so I began to search for a new visual language, one which strived to express the ephemeral quality of his spiritual experience. Finally, I want to share with you a video from a series that I ’ ve been working on that tries to express the slow, in-between existence that my father has experienced. As he began to regain his ability to breathe, I started recording his thoughts, and so the voice that you hear in this video is his voice. Ronnie Cahana : You have to believe you ’ re paralyzed to play the part of a quadriplegic. I don ’ t. In my mind, and in my dreams every night I Chagall-man float over the city twirl and swirl with my toes kissing the floor. I know nothing about the statement of man without motion. Everything has motion. The heart pumps. The body heaves. The mouth moves. We never stagnate. Life triumphs up and down. Kitra Cahana : For most of us, our muscles begin to twitch and move long before we are conscious, but my father tells me his privilege is living on the far periphery of the human experience. Like an astronaut who sees a perspective that very few of us will ever get to share, he wonders and watches as he takes his first breaths and dreams about **crawling** back home. So begins life at 57, he says. A toddler has no attitude in its being, but a man insists on his world every day. Few of us will ever have to face physical limitations to the degree that my father has, but we will all have moments of paralysis in our lives. I know I frequently confront walls that feel completely unscalable, but my father insists that there are no dead ends. Instead, he invites me into his space of co-healing to give the very best of myself, and for him to give the very best of himself to me. Paralysis was an opening for him. It was an opportunity to emerge, to rekindle life force, to sit still long enough with himself so as to fall in love with the full continuum of creation. Today, my father is no longer locked in. He moves his neck with ease, has had his feeding peg removed, breathes with his own lungs, speaks slowly with his own quiet voice, and works every day to gain more movement in his paralyzed body. But the work will never be finished. As he says,”I ’ m living in a broken world, and there is holy work to do.”Thank you.

Text Sample 598: NEG Dim 4 – 2014 – Score -1.00 – 2014/t001856.txt

The universe is teeming with planets. I want us, in the next decade, to build a space telescope that ’ ll be able to image an Earth about another star and figure out whether it can harbor life. My colleagues at the NASA Jet Propulsion Laboratory at Princeton and I are working on technology that will be able to do just that in the coming years. Astronomers now believe that every star in the galaxy has a planet, and they speculate that up to one fifth of them have an Earth-like planet that might be able to harbor life, but we haven ’ t seen any of them. We ’ ve only detected them indirectly. This is NASA ’ s famous picture of the pale blue dot. It was taken by the Voyager spacecraft in 1990, when they turned it around as it was exiting the solar system to take a picture of the Earth from six billion kilometers away. I want to take that of an Earth-like planet about another star. Why haven ' t we done that? Why is that hard? Well to see, let ' s imagine we take the Hubble Space Telescope and we turn it around and we move it out to the orbit of Mars. We ' ll see something like that, a slightly blurry picture of the Earth, because we ' re a fairly small telescope out at the orbit of Mars. Now let ' s move ten times further away. Here we are at the orbit of Uranus. It ' s gotten smaller, it ' s got less detail, less resolve. We can still see the little moon, but let ' s go ten times further away again. Here we are at the edge of the solar system, out at the Kuiper Belt. Now it ' s not resolved at all. It ' s that pale blue dot of Carl Sagan ' s. But let ' s move yet again ten times further away. Here we are out at the Oort Cloud, outside the solar system, and we ' re starting to see the sun move into the field of view and get into where the planet is. One more time, ten times further away. Now we ' re at Alpha Centauri, our nearest neighbor star, and the planet is gone. All we ' re seeing is the big beaming image of the star that ' s ten billion times brighter than the planet, which should be in that little red circle. That ' s what we want to see. That ' s why it ' s hard. The light from the star is diffracting. It ' s scattering inside the telescope, creating that very bright image that washes out the planet. So to see the planet, we have to do something about all of that light. We have to get rid of it. I have a lot of colleagues working on really amazing technologies to do that, but I want to tell you about one today that I think is the coolest, and probably the most likely to get us an Earth in the next decade. It was first suggested by Lyman Spitzer, the father of the space telescope, in 1962, and he took his inspiration from an eclipse. You ' ve all seen that. That ' s a solar eclipse. The moon has moved in front of the sun. It blocks out most of the light so we can see that **dim** corona around it. It would be the same thing if I put my thumb up and blocked that spotlight that ' s getting right in my eye, I can see you in the back row. Well, what ' s going on? Well the moon is casting a shadow down on the Earth. We put a telescope or a camera in that shadow, we look back at the sun, and most of the light ' s been removed and we can see that **dim**, fine structure in the corona. Spitzer ' s suggestion was we do this in space. We build a big screen, we fly it in space, we put it up in front of the star, we block out most of the light, we fly a space telescope in that shadow that ' s created, and boom, we get to see planets. Well that would look something like this. So there ' s that big screen, and there ' s no planets, because unfortunately it doesn ' t actually work very well, because the light waves of the light and waves diffracts around that screen the same way it did in the telescope. It ' s like water bending around a rock in a stream, and all that light just destroys the shadow. It ' s a terrible shadow. And we can ' t see planets. But Spitzer actually knew the answer. If we can feather the edges, soften those edges so we can control diffraction, well then we can see a planet, and in the last 10 years or so we ' ve come up with optimal solutions for doing that. It looks something like that. We call that our flower petal starshade. If we make the edges of those petals exactly right, if we control their shape, we can control diffraction, and now we have a great shadow. It ' s about 10 billion times **dimmer** than it was before, and we can see the planets beam out just like that. That, of course, has to be bigger than my thumb. That starshade is about the size of half a football field and it has to fly 50, 000 kilometers away from the telescope that has to be held right in its shadow, and then we can see those planets. This sounds formidable, but brilliant engineers, colleagues of mine at JPL, came up with a fabulous design for how to do that and it looks like this. It starts wrapped around a hub. It separates from the telescope. The petals unfurl, they open up, the telescope turns around. Then you ’ ll see it flip and fly out that 50, 000 kilometers away from the telescope. It ’ s going to move in front of the star just like that, creates a wonderful shadow. Boom, we get planets orbiting about it. Thank you. That ’ s not science fiction. We ’ ve been working on this for the last five or six years. Last summer, we did a really cool test out in California at Northrop Grumman. So those are four petals. This is a sub-scale star shade. It ’ s about half the size of the one you just saw. You ’ ll see the petals unfurl. Those four petals were built by four undergraduates doing a summer internship at JPL. Now you ’ re seeing it deploy. Those petals have to rotate into place. The base of those petals has to go to the same place every time to within a tenth of a millimeter. We ran this test 16 times, and 16 times it went into the exact same place to a tenth of a millimeter. This has to be done very precisely, but if we can do this, if we can build this technology, if we can get it into space, you might see something like this. That ’ s a picture of one our nearest neighbor stars taken with the Hubble Space Telescope. If we can take a similar space telescope, slightly larger, put it out there, fly an occulter in front of it, what we might see is something like that — that ’ s a family portrait of our solar system — but not ours. We ’ re hoping it ’ ll be someone else ’ s solar system as seen through an occulter, through a starshade like that. You can see Jupiter, you can see Saturn, Uranus, Neptune, and right there in the center, next to the residual light is that pale blue dot. That ’ s Earth. We want to see that, see if there ’ s water, oxygen, ozone, the things that might tell us that it could harbor life. I think this is the coolest possible science. That ’ s why I got into doing this, because I think that will change the world. That will change everything when we see that. Thank you.

Text Sample 599: NEG Dim 4 – 1994 – Score 6.00 – 1994/t002458.txt

Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ’ s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC microprocessors versus the power of local area networks. And the interesting thing about it is that this slide, like so many technology slides that we ’ re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ’ s basically what I ’ m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ’ m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well the answer is, if I drew it on a normal curve where, let ’ s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ’ m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ’ t tell us much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log curve, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as microprocessor technology, then the day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ’ s not moving like that. And there ’ s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ’ d like to ask is, if you look at these exponential curves, they don ’ t go on forever. Things just can ’ t possibly keep changing as fast as they are. One of two things is going to happen. Either it ’ s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ’ s going to do this. That ’ s about all it can do. Now I ’ m an optimist, so I sort of think it ’ s probably going to do something like that. If so, that means that what we ’ re in the middle of right now is a transition. We ’ re sort of on this line in a transition from the way the world used to be to some new way that the world is. And so what I ’ m trying to ask, what I ’ ve been asking myself, is what ’ s this new way that the world is? What ’ s that new state that the world is heading toward? Because the transition seems very, very confusing when we ’ re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ’ s a conference in which people talk about the future, and you notice that the future is still at about the year 2000. It ’ s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ’ s happening there. That transition is happening. We can all sense it. And we know that it just doesn ’ t make too much sense to think out 30, 50 years because everything ’ s going to be so different that a simple extrapolation of what we ’ re doing just doesn ’ t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ’ re going through. Now in order to do that I ’ m going to have to talk about a bunch of stuff that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step back and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, the Earth was this big, sterile hunk of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ’ s theories that are beginning to understand about how it started with RNA, but I ’ m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ’ t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ’ s sort of just a very simple chemical form of life, but when things got interesting was when these drops learned a trick about abstraction. Somehow by ways that we don ’ t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for amusement purposes is we can now write things in this code. And I ' ve got here a little 100 micrograms of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ' s right here. If I had really been a egotist, I would have put it into a virus and released it in the room. So what was the next step? Writing down the DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ' t know if you know this, but bacteria can actually exchange DNA. Now that ' s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the transition point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ' s interesting in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ' re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a **brain** cell and so on. So these communities began to evolve so that the interesting level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very special structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in the community and special structures that were responsible for recording, understanding, learning information. And that was the **brains** and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ' d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these special information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ’ re aware of is human language. It ’ s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ’ m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ’ re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your **brain**. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ’ ve done is we, humanity, have started abstracting out. We ’ re going through the same levels that multi-cellular organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. Telephony, computers, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ’ ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of microseconds. And you saw Ty ’ s little evolutionary example where he sort of did a little bit of evolution on the Convolution program right before your eyes. So now we ’ ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and **brains**, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ’ s what we ’ re seeing here in this explosion of curve. We ’ re seeing this process feeding back on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ’ s really impossible for me to design them in the traditional sense. I don ’ t know what every transistor in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of abstraction and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ’ t quite even understand. One method that ’ s particularly interesting that I ’ ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the microsecond time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,”Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted.”So in other words, I define what I wanted. Let ’ s say I want to sort numbers, as a simple example I ’ ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,”Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ’ s reproduce the ones that sorted numbers the best. And let ’ s reproduce them by a process of recombination analogous to sex.”Take two programs and they produce children by exchanging their subroutines, and the children inherit the traits of the subroutines of the two programs. So I ’ ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,”Please repeat that process.”Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ’ t tell you how they work. I ’ ve tried looking at them and telling you how they work. They ’ re obscure, weird programs. But they do the job. And in fact, I know, I ’ m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,”Oh look. Look at this. It says, ’ This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ’ Doesn ’ t that make you feel confident?”In fact, we know that the engineering process doesn ’ t work very well when it gets complicated. So we ’ re beginning to depend on computers to do a process that ’ s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ’ t quite understand the options of it. So in a sense, it ’ s getting ahead of us. We ’ re now using those programs to make much faster computers so that we ’ ll be able to run this process much faster. So it ’ s feeding back on itself. The thing is becoming faster and that ’ s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ’ re taking off. And what we are is we ’ re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ’ re the amoebas and we can ’ t quite figure out what the hell this thing is we ’ re creating. We ’ re right at that point of transition. But I think that there really is something coming along after us. I think it ’ s very haughty of us to think that we ’ re the end product of evolution. And I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

Text Sample 600: NEG Dim 4 – 2018 – Score -1.00 – 2018/t000703.txt

Greg Gage : If I asked you to think of a ferocious killer animal, you ’ d probably think of a lion, and for all the wonderful predatory skills that a lion has, it still only has about a 20 percent success rate at catching a meal. Now, one of the most successful hunters in the entire animal kingdom is surprising : the dragonfly. Now, dragonflies are killer flies, and when they see a smaller fly, they have about a 97 percent chance of catching it for a meal. And this is in mid-flight. But how can such a small insect be so precise? In this episode, we ’ re going to see how the dragonfly ’ s **brain** is highly specialized to be a deadly killer. [DIY Neuroscience] So what makes the dragonfly one of the most successful predators in the animal kingdom? One, it ’ s the eyes. It has near 360-degree vision. Two, the wings. With individual control of its wings, the dragonfly can move precisely in any direction. But the real secret to the dragonfly ’ s success is how its **brain** coordinates this complex information between the eyes and the wings and turns hunting into a simple reflex. To study this, Jaimie ’ s been spending a lot of time socializing with dragonflies. What do you need to do your experiments? Jaimie Spahr : First of all, you need dragonflies. Oliver : I have a mesh cage to catch the dragonflies. JS : The more I worked with them, the more terrified I got of them. They ’ re actually very scary, especially under a microscope. They have really sharp mandibles, are generally pretty aggressive, which I guess also helps them to be really good predators. GG : In order to learn what ’ s going on inside the dragonfly ’ s **brain** when it sees a prey, we ’ re going to eavesdrop in on a conversation between the eyes and the wings, and to do that, we need to anesthetize the dragonfly on ice and make sure we protect its wings so that we can release it afterwards. Now, the dragonfly ’ s **brain** is made up of specialized cells called neurons and these neurons are what allow the dragonfly to see and move so quickly. The individual neurons form circuits by connecting to each other via long, tiny threads called axons and the neurons communicate over these axons using electricity. In the dragonfly, we ’ re going to place little metal wires, or electrodes, along the axon tracks, and this is what ’ s really cool. In the dragonfly, there ’ s only 16 neurons ; that ’ s eight per eye that tell the wings exactly where the target is. We ’ ve placed the electrodes so that we can record from these neurons that connect the eyes to the wings. Whenever a message is being passed from the eye to the wing, our electrode intercepts that conversation in the form of an electrical current, and it amplifies it. Now, we can both hear it and see it in the form of a spike, which we also call an action potential. Now let ’ s listen in. Right now, we have the dragonfly flipped upside down, so he ’ s looking down towards the ground. We ’ re going to take a prey, or what we sometimes call a target. In this case, the target ’ s going to be a fake fly. We ’ re going to move it into the dragonfly ’ s sights. Oh! Oh, look at that. Look at that, but it ’ s only in one direction. Oh, yes! You don ’ t see any spikes when I go forward, but they ’ re all when I come back. In our experiments, we were able to see that the neurons of the dragonfly fired when we moved the target in one direction but not the other. Now, why is that? Remember when I said that the dragonfly had near 360-degree vision. Well, there ’ s a section of the eye called the fovea and this is the part that has the sharpest visual acuity, and you can think of it as its crosshairs. Remember when I told you the dragonfly had individual precise control of its wings? When a dragonfly sees its prey, it trains its crosshairs on it and along its axons it sends messages only to the neurons that control the parts of the wings that are needed to keep that dragonfly on target. So if the prey is on the left of the dragonfly, only the neurons that are tugging the wings to the left are fired. And if the prey moves to the right of the dragonfly, those same neurons are not needed, so they ’ re going to remain quiet. And the dragonfly speeds toward the prey at a fixed angle that ’ s communicated by this crosshairs to the wings, and then boom, dinner. Now, all this happens in a split second, and it ’ s effortless for the dragonfly. It ’ s almost like a reflex. And this whole incredibly efficient process is called fixation. But there ’ s one more story to this process. We saw how the neurons respond to movements, but how does the dragonfly know that something really is prey? This is where size matters. Let ’ s show the dragonfly a series of dots. Oh, yeah! JS : Yeah, it prefers that one. GG : Out of all the sizes, we found that the dragonfly responded to smaller targets over larger ones. In other words, the dragonfly was programmed to go after smaller flies versus something much larger, like a bird. And as soon as it recognizes something as prey, that poor little fly only has seconds to live. Today we got to see how the dragonfly ’ s **brain** works to make it a very efficient killer. And let ’ s be thankful that we didn ’ t live 300 million years ago when dragonflies were the size of cats.

Text Sample 601: NEG Dim 4 – 2018 – Score 1.00 – 2018/t000577.txt

I ’ d like you all to close your eyes, please... and imagine yourself sitting in the middle of a large, open field with the sun setting on your right. And as the sun sets, imagine that tonight you don ’ t just see the stars appear, but you ’ re able to hear the stars appear with the brightest stars being the loudest notes and the hotter, bluer stars producing the higher-pitched notes. And since each constellation is made up of different types of stars, they ’ ll each produce their own unique melody, such as Aries, the ram. Or Orion, the hunter. Or even Taurus, the bull. We live in a musical universe, and we can use that to experience it from a new perspective, and to share that perspective with a wider range of people. Let me show you what I mean. Now, when I tell people I ’ m an astrophysicist, they ’ re usually pretty impressed. And then I say I ’ m also a musician — they ’ re like,”Yeah, we know.”So everyone seems to know that there ’ s this deep connection between music and astronomy. And it ’ s actually a very old idea ; it goes back over 2, 000 years to Pythagoras. You might remember Pythagoras from such theorems as the Pythagorean theorem — And he said :”There is geometry in the humming of the strings, there is music in the spacing of the spheres.”And so he literally thought that the motions of the planets along the celestial sphere created harmonious music. And if you asked him,”Why don ’ t we hear anything?”he ’ d say you can ’ t hear it because you don ’ t know what it ’ s like to not hear it ; you don ’ t know what true silence is. It ’ s like how you have to wait for your power to go out to hear how annoying your refrigerator was. Maybe you buy that, but not everybody else was buying it, including such names as Aristotle. Exact words. So I ’ ll paraphrase his exact words. He said it ’ s a nice idea, but if something as large and vast as the heavens themselves were moving and making sounds, it wouldn ’ t just be audible, it would be earth-shatteringly loud. We exist, therefore there is no music of the spheres. He also thought that the **brain** ’ s only purpose was to cool down the blood, so there ’ s that... But I ’ d like to show you that in some way they were actually both right. And we ’ re going to start by understanding what makes music musical. It may sound like a silly question, but have you ever wondered why it is that certain notes, when played together, sound relatively pleasing or consonant, such as these two — while others are a lot more tense or dissonant, such as these two. Right? Why is that? Why are there notes at all? Why can you be in or out of tune? Well, the answer to that question was actually solved by Pythagoras himself. Take a look at the string on the far left. If you bow that string, it will produce a note as it oscillates very fast back and forth. But now if you cut the string in half, you ’ ll get two strings, each oscillating twice as fast. And that will produce a related note. Or three times as fast, or four times — And so the secret to musical harmony really is simple ratios : the simpler the ratio, the more pleasing or consonant those two notes will sound together. And the more complex the ratio, the more dissonant they will sound. And it ’ s this interplay between tension and release, or consonance and dissonance, that makes what we call music. Thank you. But there ’ s more. So the two features of music we like to think of as pitch and rhythms, they ’ re actually two versions of the same thing, and I can show you. That ’ s a rhythm right? Watch what happens when we speed it up. So once a rhythm starts happening more than about 20 times per second, your **brain** flips. It stops hearing it as a rhythm and starts hearing it as a pitch. So what does this have to do with astronomy? Well, that ’ s when we get to the TRAPPIST-1 system. This is an exoplanetary system discovered last February of 2017, and it got everyone excited because it is seven Earth-sized planets all orbiting a very near red dwarf star. And we think that three of the planets have the right temperature for liquid water. It ’ s also so close that in the next few years, we should be able to detect elements in their atmospheres such as oxygen and methane — potential signs of life. But one thing about the TRAPPIST system is that it is tiny. So here we have the orbits of the inner rocky planets in our solar system : Mercury, Venus, Earth and Mars, and all seven Earth-sized planets of TRAPPIST-1 are tucked well inside the orbit of Mercury. I have to expand this by 25 times for you to see the orbits of the TRAPPIST-1 planets. It ’ s actually much more similar in size to our planet Jupiter and its moons, even though it ’ s seven Earth-size planets orbiting a star. Another reason this got everyone excited was artist renderings like this. You got some liquid water, some ice, maybe some land, maybe you can go for a dive in this amazing orange sunset. It got everyone excited, and then a few months later, some other papers came out that said, actually, it probably looks more like this. So there were signs that some of the surfaces might actually be molten lava and that there were very damaging X-rays coming from the central star — X-rays that will sterilize the surface of life and even strip off atmospheres. Luckily, just a few months ago in 2018, some new papers came out with more refined measurements, and they found actually it does look something like that. So we now know that several of them have huge supplies of water — global oceans — and several of them have thick atmospheres, so it ’ s the right place to look for potential life. But there ’ s something even more exciting about this system, especially for me. And that ’ s that TRAPPIST-1 is a resonant chain. And so that means for every two orbits of the outer planet, the next one in orbits three times, and the next one in four, and then six, nine, 15 and 24. So you see a lot of very simple ratios among the orbits of these planets. Clearly, if you speed up their motion, you can get rhythms, right? One beat, say, for every time a planet goes around. But now we know if you speed that motion up even more, you ’ ll actually produce musical pitches, and in this case alone, those pitches will work together, making harmonious, even human-like harmony. So let ’ s hear TRAPPIST-1. The first thing you ’ ll hear will be a note for every orbit of each planet, and just keep in mind, this music is coming from the system itself. I ’ m not creating the pitches or rhythms, I ’ m just bringing them into the human hearing range. And after all seven planets have entered, you ’ re going to see — well, you ’ re going to hear a drum for every time two planets align. That ’ s when they kind of get close to each other and give each other a gravitational tug. And that ’ s the sound of the star itself — its light converted into sound. So you may wonder how this is even possible. And it ’ s good to think of the analogy of an orchestra. When everyone gets together to start playing in an orchestra, they can ’ t just dive into it, right? They have to all get in tune ; they have to make sure their instruments resonate with their neighbors ’ instruments, and something very similar happened to TRAPPIST-1 early in its existence. When the planets were first forming, they were orbiting within a disc of gas, and while inside that disc, they can actually slide around and adjust their orbits to their neighbors until they ’ re perfectly in tune. And it ’ s a good thing they did because this system is so compact — a lot of mass in a tight space — if every aspect of their orbits wasn ’ t very finely tuned, they would very quickly disrupt each other ’ s orbits, destroying the whole system. So it ’ s really music that is keeping this system alive — and any of its potential inhabitants. But what does our solar system sound like? I hate to be the one to show you this, but it ’ s not pretty. So for one thing, our solar system is on a much, much larger scale, and so to hear all eight planets, we have to start with Neptune near the bottom of our hearing range, and then Mercury ' s going to be all the way up near the very top of our hearing range. But also, since our planets are not very compact — they ' re very spread out — they didn ' t have to adjust their orbits to each other, so they ' re kind of just all playing their own random note at random times. So, I ' m sorry, but here it is. That ' s Neptune. Uranus. Saturn. Jupiter. And then tucked in, that ' s Mars. Earth. Venus. And that ' s Mercury — OK, OK, I ' ll stop. So this was actually Kepler ' s dream. Johannes Kepler is the person that figured out the laws of planetary motion. He was completely fascinated by this idea that there ' s a connection between music, astronomy and geometry. And so he actually spent an entire book just searching for any kind of musical harmony amongst the solar system ' s planets and it was really, really hard. It would have been much easier had he lived on TRAPPIST-1, or for that matter... K2- 138. This is a new system discovered in January of 2018 with five planets, and just like TRAPPIST, early on in their existence, they were all finely tuned. They were actually tuned into a tuning structure proposed by Pythagoras himself, over 2, 000 years before. But the system ' s actually named after Kepler, discovered by the Kepler space telescope. And so, in the last few billion years, they ' ve actually lost their tuning, quite a bit more than TRAPPIST has, and so what we ' re going to do is go back in time and imagine what they would ' ve sounded like just as they were forming. Thank you. Now, you may be wondering : How far does this go? How much music actually is out there? And that ' s what I was wondering last fall when I was working at U of T ' s planetarium, and I was contacted by an artist named Robyn Rennie and her daughter Erin. Robyn loves the night sky, but she hasn ' t been able to fully see it for 13 years because of vision loss. And so they wondered if there was anything I could do. So I collected all the sounds I could think of from the universe and packaged them into what became"Our Musical Universe."This is a sound-based planetarium show exploring the rhythm and harmony of the cosmos. And Robyn was so moved by this presentation that when she went home, she painted this gorgeous representation of her experience. And then I defaced it by putting Jupiter on it for the poster. So... in this show, I take people of all vision levels and bring them on an audio tour of the universe, from the night sky all the way out to the edge of the observable universe. But even this is just the start of a musical odyssey to experience the universe with new eyes and with new ears, and I hope you ' ll join me. Thank you.

Text Sample 602: NEG Dim 4 – 2018 – Score 1.00 – 2018/t000528.txt

Hi. My name is Aparna. I am a shopaholic â and I ' m addicted to online returns. Well, at least I was. At one time, I had two or three packages of clothing delivered to me every other day. I would intentionally buy the same item in a couple different sizes and many colors, because I did not know what I really wanted. So I overordered, I tried things on, and then I sent what didn ' t work back. Once my daughter was watching me return some of those packages back, and she said,"Mom, I think you have a problem."I didn ' t think so. I mean, it ' s free shipping and free returns, right? I didn ' t even think twice about it, until I heard a statistic at work that shocked me. You see, I ' m a global solutions director for top-tier retail, and we were in a meeting with one of my largest customers, discussing how to streamline costs. One of their biggest concerns was managing returns. Just this past holiday season alone, they had 7. 5 million pieces of clothing returned to them. I could not stop thinking about it. What happens to all these returned clothes? So I came home and researched. And I learned that every year, four billion pounds of returned clothing ends up in the landfill. That ’ s like every resident in the US did a load of laundry last night and decided to throw it in the trash today. I was horrified. I ’ m like,”Of all people, I should be able to help prevent this.”My job is to find solutions to logistical issues like these â not create them. So this issue became very personal to me. I said,”You know what? We have to solve this.”And we can, with some of the existing systems we already have in place. And then I started to wonder : How did we get here? I mean, it was only like six years ago when a study recommended that offering free online returns would drive customers to spend more. We started seeing companies offering free online returns to drive more sales and provide a better experience. What we didn ’ t realize is that this would lead to more items being returned as well. In the US, companies lost $351 billion in sales in 2017 alone. Retailers are scrambling to recover their losses. They try to place that returned item online to be sold again, or they ’ ll sell it to a discount partner or a liquidator. Basically, if companies cannot find a place for this item quickly and economically, its place becomes the trash. Suddenly, I felt very guilty for being that shopper, somebody who contributes to this. Who would have thought my innocent shopping behavior would be hurting not only me, but our planet as well? And as I thought about what to do, I kept thinking : Why does the item have to be returned to the retailer in the first place? What if there was another way, a win-win for everyone? What if when a person is trying to return something, it could go to the next shopper who wants it, and not the retailer? What if, instead of a return, they could do what I call a”green turn”? Consumers could use an app to take pictures of the item and verify the condition while returning it. Artificial intelligence systems could then sort these clothes by condition â mint condition or slightly used â and direct it to the next appropriate person. Mint-condition clothes could automatically go to the next buyer, while slightly used clothes could be marked down and offered online again. The retailer can decide the business rules on the number of times a particular item can be resold. All that the consumer would need to do is obtain a mobile code, take it to the nearest shipping place to be packed and shipped, and off it goes from one buyer to the next, not the landfill. Now you will ask,”Would people really go through all this trouble?”I think they would if they had incentives, like loyalty points or cash back. Let ’ s call it”green cash.”There would be a whole new opportunity to make money from this new customer base looking to buy these returns. This system would make a fun thing like shopping a spiritual experience that helps save our planet. This is doable and would probably take six months to weave some of our existing systems and run a pilot. Even before any of these logistical systems are in place, each of us shoppers can act now, if every single adult in the US made a few small changes to our shopping behavior. Take the extra time to research and think â Do I really need this item? No : Do I really want this item? â before making a purchase. And if every one of us adults in the US returned five less items this year, we would keep 240 million pounds of clothes out of the landfill. Six percent reduction, just like that. This environmental problem that we have created is not thousands of years away ; it ’ s happening today, and must stop now to prevent growing landfills across the globe. I want to leave my daughter and my daughter ’ s daughter a better and cleaner place than I found it, so I have not only stopped overordering, I recycle religiously as well. And you can, too. It ’ s not difficult. Before we fill our shopping carts and our landfills with extra items that we don ’ t want, let ’ s pause next time we are shopping online and think twice about what we all hopefully really do want : a beautiful Earth to call home. Thank you.

Text Sample 603: NEG Dim 4 – 2010 – Score -1.00 – 2010/t002876.txt

We are drowning in news. Reuters alone puts out three and a half million news stories a year. That ' s just one source. My question is : How many of those stories are actually going to matter in the long run? That ' s the idea behind The Long News. It ' s a project by The Long Now Foundation, which was founded by TEDsters including Kevin Kelly and Stewart Brand. And what we ' re looking for is news stories that might still matter 50 or 100 or 10, 000 years from now. And when you look at the news through that filter, a lot falls by the wayside. To take the top stories from the A. P. this last year, is this going to matter in a decade? Or this? Or this? Really? Is this going to matter in 50 or 100 years? Okay, that was kind of cool. But the top story of this past year was the economy, and I ' m just betting that, sooner or later, this particular recession is going to be old news. So, what kind of stories might make a difference for the future? Well, let ' s take science. Someday, little robots will go through our bloodstreams fixing things. That someday is already here if you ' re a mouse. Some recent stories : nanobees zap tumors with real bee venom ; they ' re sending genes into the **brain** ; a robot they built that can **crawl** through the human body. What about resources? How are we going to feed nine billion people? We ' re having trouble feeding six billion today. As we heard yesterday, there ' s over a billion people hungry. Britain will starve without genetically modified crops. Bill Gates, fortunately, has bet a billion on [agricultural] research. What about global politics? The world ' s going to be very different when and if China sets the agenda, and they may. They ' ve overtaken the U. S. as the world ' s biggest car market, they ' ve overtaken Germany as the largest exporter, and they ' ve started doing DNA tests on kids to choose their careers. We ' re finding all kinds of ways to push back the limits of what we know. Some recent discoveries : There ' s an ant colony from Argentina that has now spread to every continent but Antarctica ; there ' s a self-directed robot scientist that ' s made a discovery — soon, science may no longer need us, and life may no longer need us either ; a microbe wakes up after 120, 000 years. It seems that with or without us, life will go on. But my pick for the top Long News story of this past year was this one : water found on the moon. Makes it a lot easier to put a colony up there. And if NASA doesn ' t do it, China might, or somebody in this room might write a big check. My point is this : In the long run, some news stories are more important than others.

Text Sample 604: NEG Dim 4 – 2010 – Score 1.00 – 2010/t002824.txt

Why grow homes? Because we can. Right now, America is in an unremitting state of trauma. And there ' s a cause for that, all right. We ' ve got McPeople, McCars, McHouses. As an architect, I have to confront something like this. So what ' s a technology that will allow us to make ginormous houses? Well, it ' s been around for 2, 500 years. It ' s called pleaching, or grafting trees together, or grafting inosculate matter into one contiguous, vascular system. And we do something different than what we did in the past ; we add kind of a modicum of intelligence to that. We use CNC to make scaffolding to train semi-epithetic matter, plants, into a specific geometry that makes a home that we call a Fab Tree Hab. It fits into the environment. It is the environment. It is the landscape, right? And you can have a hundred million of these homes, and it ' s great because they suck carbon. They ' re perfect. You can have 100 million families, or take things out of the suburbs, because these are homes that are a part of the environment. Imagine pre-growing a village — it takes about seven to 10 years — and everything is green. So not only do we do the veggie house, we also do the in-vitro meat habitat, or homes that we ’ re doing research on now in Brooklyn, where, as an architecture office, we ’ re for the first of its kind to put in a molecular cell biology lab and start experimenting with regenerative medicine and tissue engineering and start thinking about what the future would be if architecture and biology became one. So we ’ ve been doing this for a couple of years, and that ’ s our lab. And what we do is we grow extracellular matrix from pigs. We use a modified inkjet printer, and we print geometry. We print geometry where we can make industrial design objects like, you know, shoes, leather belts, handbags, etc., where no sentient creature is harmed. It ’ s victimless. It ’ s meat from a test tube. So our theory is that eventually we should be doing this with homes. So here is a typical stud wall, an architectural construction, and this is a section of our proposal for a meat house, where you can see we use fatty cells as insulation, cilia for dealing with wind loads and sphincter muscles for the doors and windows. And we know it ’ s incredibly ugly. It could have been an English Tudor or Spanish Colonial, but we kind of chose this shape. And there it is kind of grown, at least one particular section of it. We had a big show in Prague, and we decided to put it in front of the cathedral so religion can confront the house of meat. That ’ s why we grow homes. Thanks very much.

Text Sample 605: NEG Dim 4 – 2010 – Score 1.00 – 2010/t002852.txt

Today, I want you to look at children who become suicide bombers through a completely different lens. In 2009, there were 500 bomb blasts across Pakistan. I spent the year working with children who were training to become suicide bombers and with Taliban recruiters, trying to understand how the Taliban were converting these children into live ammunition and why these children were actively signing up to their cause. I want you to watch a short video from my latest documentary film,”Children of the Taliban.”The Taliban now run their own schools. They target poor families and convince the parents to send their children. In return, they provide free food and shelter and sometimes pay the families a monthly stipend. We ’ ve obtained a propaganda video made by the Taliban. Young boys are taught justifications for suicide attacks and the execution of spies. I made contact with a child from Swat who studied in a madrassa like this. Hazrat Ali is from a poor farming family in Swat. He joined the Taliban a year ago when he was 13. How do the Taliban in your area get people to join them? Hazrat Ali : They first call us to the mosque and preach to us. Then they take us to a madrassa and teach us things from the Koran. Sharmeen Obaid Chinoy : He tells me that children are then given months of military training. HA : They teach us to use machine guns, Kalashnikov, rocket launchers, grenades, bombs. They ask us to use them only against the infidels. Then they teach us to do a suicide attack. SOC : Would you like to carry out a suicide attack? HA : If God gives me strength. SOC : I, in my research, have seen that the Taliban have perfected the way in which they recruit and train children, and I think it ’ s a five-step process. Step one is that the Taliban prey on families that are large, that are poor, that live in rural areas. They separate the parents from the children by promising to provide food, clothing, shelter to these children. Then they ship them off, hundreds of miles away to hard-line schools that run along the Taliban agenda. Step two : They teach the children the Koran, which is Islam ’ s holiest book, in Arabic, a language these children do not understand and cannot speak. They rely very heavily on teachers, who I have personally seen distort the message to these children as and when it suits their purpose to. These children are explicitly forbidden from reading newspapers, listening to radio, reading any books that the teachers do not prescribe them. If any child is found violating these rules, he is severely reprimanded. Effectively, the Taliban create a complete blackout of any other source of information for these children. Step three : The Taliban want these children to hate the world that they currently live in. So they beat these children — I have seen it ; they feed them twice a day dried bread and water ; they rarely allow them to play games ; they tell them that, for eight hours at a time, all they have to do is read the Koran. The children are virtual prisoners ; they cannot leave, they cannot go home. Their parents are so poor, they have no resources to get them back. Step four : The older members of the Taliban, the fighters, start talking to the younger boys about the glories of martyrdom. They talk to them about how when they die, they will be received up with lakes of honey and milk, how there will be 72 virgins waiting for them in paradise, how there will be unlimited food, and how this glory is going to propel them to become heroes in their neighborhoods. Effectively, this is the brainwashing process that has begun. Step five : I believe the Taliban have one of the most effective means of propaganda. Their videos that they use are intercut with photographs of men and women and children dying in Iraq and Afghanistan and in Pakistan. And the basic message is that the Western powers do not care about civilian deaths, so those people who live in areas and support governments that work with Western powers are fair game. That ’ s why Pakistani civilians, over 6, 000 of whom have been killed in the last two years alone, are fair game. Now these children are primed to become suicide bombers. They ’ re ready to go out and fight because they ’ ve been told that this is effectively their only way to glorify Islam. I want you to watch another excerpt from the film. This boy is called Zenola. He blew himself up, killing six. This boy is called Sadik. He killed 22. This boy is called Messoud. He killed 28. The Taliban are running suicide schools, preparing a generation of boys for atrocities against civilians. Do you want to carry out a suicide attack? Boy : I would love to. But only if I get permission from my dad. When I look at suicide bombers younger than me, or my age, I get so inspired by their terrific attacks. SOC : What blessing would you get from carrying out a suicide attack? Boy : On the day of judgment, God will ask me,”Why did you do that?”I will answer,”My Lord! Only to make you happy! I have laid down my life fighting the infidels.”Then God will look at my intention. If my intention was to eradicate evil for Islam, then I will be rewarded with paradise. Singer : On the day of judgment My God will call me My body will be put back together And God will ask me why I did this SOC : I leave you all with this thought : If you grew up in these circumstances, faced with these choices, would you choose to live in this world or in the glorious afterlife? As one Taliban recruiter told me,”There will always be sacrificial lambs in this war.”Thank you.

Text Sample 606: NEG Dim 4 – 2004 – Score 1.00 – 2004/t000431.txt

Imagine spending seven years at MIT and research laboratories, only to find out that you ’ re a performance artist. I ’ m also a software engineer, and I make lots of different kinds of art with the computer. And I think the main thing that I ’ m interested in is trying to find a way of making the computer into a personal mode of expression. And many of you out there are the heads of Macromedia and Microsoft, and in a way those are my bane : I think there ’ s a great homogenizing force that software imposes on people and limits the way they think about what ’ s possible on the computer. Of course, it ’ s also a great liberating force that makes possible, you know, publishing and so forth, and standards, and so on. But, in a way, the computer makes possible much more than what most people think, and my art has just been about trying to find a personal way of using the computer, and so I end up writing software to do that. Chris has asked me to do a short performance, and so I ’ m going to take just this time — maybe 10 minutes — to do that, and hopefully at the end have just a moment to show you a couple of my other projects in video form. Thank you. We ’ ve got about a minute left. I ’ d just like to show a clip from a most recent project. I did a performance with two singers who specialize in making strange noises with their mouths. And this just came off last September at ARS Electronica ; we repeated it in England. And the idea is to visualize their speech and song behind them with a large screen. We used a computer vision tracking system in order to know where they were. And since we know where their heads are, and we have a wireless mic on them that we ’ re processing the sound from, we ’ re able to create visualizations which are linked very tightly to what they ’ re doing with their speech. This will take about 30 seconds or so. He ’ s making a, kind of, cheek-flapping sound. Well, suffice it to say it ’ s not all like that, but that ’ s part of it. Thanks very much. There ’ s always lots more. I ’ m overtime, so I just wanted to say you can, if you ’ re in New York, you can check out my work at the Whitney Biennial next week, and also at Bitforms Gallery in Chelsea. And with that, I think I should give up the stage, so, thank you so much.

Text Sample 607: NEG Dim 4 – 2004 – Score 3.00 – 2004/t000248.txt

A few words about how I got started, and it has a lot to do with happiness, actually. When I was a very young child, I was extremely introverted and very much to myself. And, kind of as a way of surviving, I would go into my own very personal space, and I would make things. I would make things for people as a way of, you know, giving, showing them my love. I would go into these private places, and I would put my ideas and my passions into objects — and sort of learning how to speak with my hands. So, the whole activity of working with my hands and creating objects is very much connected with not only the idea realm, but also with very much the feeling realm. And the ideas are very disparate. I ’ m going to show you many different kinds of pieces, and there ’ s no real connection between one or the other, except that they sort of come out of my **brain**, and they ’ re all different sort of thoughts that are triggered by looking at life, and seeing nature and seeing objects, and just having kind of playful random thoughts about things. When I was a child, I started to explore motion. I fell in love with the way things moved, so I started to explore motion by making little flipbooks. And this is one that I did, probably like when I was around seventh grade, and I remember when I was doing this, I was thinking about that little rock there, and the pathway of the vehicles as they would fly through the air, and how the characters — — would come shooting out of the car, so, on my mind, I was thinking about the trajectory of the vehicles. And of course, when you ’ re a little kid, there ’ s always destruction. So, it has to end with this — — gratuitous violence. So that was how I first started to explore the way things moved, and expressed it. Now, when I went to college, I found myself making fairly complicated, fragile machines. And this really came about from having many different kinds of interests. When I was in high school, I loved to program computers, so I sort of liked the logical flow of events. I was also very interested in perhaps going into surgery and becoming a surgeon, because it meant working with my hands in a very focused, intense way. So, I started taking art courses, and I found a way to make sculpture that brought together my love for being very precise with my hands, with coming up with different kinds of logical flows of energy through a system. And also, working with wire — everything that I did was both a visual and a mechanical engineering decision at the same time. So, I was able to sort of exercise all of that. Now, this kind of machine is as close as I can get to painting. And it ' s full of many little trivial end points, like there ' s a little foot here that just drags around in circles and it doesn ' t really mean anything. It ' s really just for the sort of joy of its own triviality. The connection I have with engineering is the same as any other engineer, in that I love to solve problems. I love to figure things out, but the end result of what I ' m doing is really completely ambiguous. That ' s pretty ambiguous. The next piece that is going to come up is an example of a kind of machine that is fairly complex. I gave myself the problem. Since I ' m always liking to solve problems, I gave myself the problem of turning a crank in one direction, and solving all of the mechanical problems for getting this little man to walk back and forth. So, when I started this, I didn ' t have an overall plan for the machine, but I did have a sense of the gesture, and a sense of the shape and how it would occupy space. And then it was a matter of starting from one point and sort of building to that final point. That little gear there switches back and forth to change direction. And that ' s a little found object. So a lot of the pieces that I ' ve made, they involve found objects. And it really — it ' s almost like doing visual puns all the time. When I see objects, I imagine them in motion. I imagine what can be said with them. This next one here,"Machine with Wishbone,"it came about from playing with this wishbone after dinner. You know, they say, never play with your food — but I always play with things. So, I had this wishbone, and I thought, it ' s kind of like a cowboy who ' s been on his horse for too long. And I started to make him walk across the table, and I thought,"Oh, I can make a little machine that will do that."So, I made this device, linked it up, and the wishbone walks. And because the wishbone is bone — it ' s animal — it ' s sort of a point where I think we can enter into it. And that ' s the whole piece. That ' s about that big. This kind of work is also very much like puppetry, where the found object is, in a sense, the puppet, and I ' m the puppeteer at first, because I ' m playing with an object. But then I make the machine, which is sort of the stand-in for me, and it is able to achieve the action that I want. The next piece I ' ll show you is a much more conceptual thought, and it ' s a little piece called"Cory ' s Yellow Chair."I had this image in my mind, when I saw my son ' s little chair, and I saw it explode up and out. And — so the way I saw this in my mind at first, was that the pieces would explode up and out with infinite speed, and the pieces would move far out, and then they would begin to be pulled back with a kind of a gravitational feel, to the point where they would approach infinite speed back to the center. And they would coalesce for just a moment, so you could perceive that there was a chair there. For me, it ' s kind of a feeling about the fleetingness of the present moment, and I wanted to express that. Now, the machine is — in this case, it ' s a real approximation of that, because obviously you can ' t move physical matter infinitely with infinite speed and have it stop instantaneously. This whole thing is about four feet wide, and the chair itself is only about a few inches. Now, this is a funny sort of conceptual thing, and yesterday we were talking about Danny Hillis '"10, 000 Year Clock."So, we have a motor here on the left, and it goes through a gear train. There are 12 pairs of 50:1 reductions, so that means that the final speed of that gear on the end is so slow that it would take two trillion years to turn once. So I ' ve invented it in concrete, because it doesn ' t really matter. Because it could run all the time. Now, a completely different thought. I ' m always imagining myself in different situations. I ’ m imagining myself as a machine. What would I love? I would love to be bathed in oil. So, this machine does nothing but just bathe itself in oil. And it ’ s really, just sort of — for me, it was just really about the lusciousness of oil. And then, I got a call from a friend who wanted to have a show of erotic art, and I didn ’ t have any pieces. But when she suggested to be in the show, this piece came to mind. So, it ’ s sort of related, but you can see it ’ s much more overtly erotic. And this one I call”Machine with Grease.”It ’ s just continually ejaculating, and it ’ s — — this is a happy machine, I ’ ll tell you. It ’ s definitely happy. From an engineering point of view, this is just a little four-bar linkage. And then again, this is a found object, a little fan that I found. And I thought, what about the gesture of opening the fan, and how simply could I state something. And, in a case like this, I ’ m trying to make something which is clear but also not suggestive of any particular kind of animal or plant. For me, the process is very important, because I ’ m inventing machines, but I ’ m also inventing tools to make machines, and the whole thing is all sort of wrapped up from the beginning. So this is a little wire-bending tool. After many years of bending gears with a pair of pliers, I made that tool, and then I made this other tool for sort of centering gears very quickly — sort of developing my own little world of technology. My life completely changed when I found a spot welder. And that was that tool. It completely changed what I could do. Now here, I ’ m going to do a very poor job of silver soldering. This is not the way they teach you to silver solder when you ’ re in school. I just like, throw it in. I mean, real jewelers put little bits of solder in. So, that ’ s a finished gear. When I moved to Boston, I joined a group called the World Sculpture Racing Society. And the idea, their premise was that we wanted to show pieces of sculpture on the street, and there ’ d be no subjective decision about what was the best. It would be — whatever came across the finish line first would be the winner. So I made — this is my first racing sculpture, and I thought,”Oh, I ’ m going to make a cart, and I ’ m going to have it — I ’ m going to have my hand writing ’ faster, ’ so as I run down the street, the cart ’ s going to talk to me and it ’ s going to go, ’ Faster, faster! ’”So, that ’ s what it does. But then in the end, what I decided was every time you finish writing the word, I would stop and I would give the card to somebody on the side of the road. So I would never win the race because I ’ m always stopping. But I had a lot of fun. Now, I only have two and a half minutes — I ’ m going to play this. This is a piece that, for me, is in some ways the most complete kind of piece. Because when I was a kid, I also played a lot of guitar. And when I had this thought, I was imagining that I would make — I would have a whole machine theater evening, where I would — you would have an audience, the curtain would open, and you ’ d be entertained by machines on stage. So, I imagined a very simple gestural dance that would be between a machine and just a very simple chair, and... When I ’ m making these pieces, I ’ m always trying to find a point where I ’ m saying something very clearly and it ’ s very simple, but also at the same time it ’ s very ambiguous. And I think there ’ s a point between simplicity and ambiguity which can allow a viewer to perhaps take something from it. And that leads me to the thought that all of these pieces start off in my own mind, in my heart, and I do my best at finding ways to express them with materials, and it always feels really crude. It ’ s always a struggle, but somehow I manage to sort of get this thought out into an object, and then it ’ s there, OK. It means nothing at all. The object itself just means nothing. Once it ’ s perceived, and someone brings it into their own mind, then there ’ s a cycle that has been completed. And to me, that ’ s the most important thing because, ever since being a kid, I ’ ve wanted to communicate my passion and love. And that means the complete cycle of coming from inside, out to the physical, to someone perceiving it. So I

’ ll just let this chair come down. Thank you.

Text Sample 608: NEG Dim 4 – 2004 – Score 3.00 – 2004/t000064.txt

This is called Hooked on a Feeling : The Pursuit of Happiness and Human Design. I put up a somewhat dour Darwin, but a very happy chimp up there. My first point is that the pursuit of happiness is obligatory. Man wishes to be happy, only wishes to be happy, and cannot wish not to be so. We are wired to pursue happiness, not only to enjoy it, but to want more and more of it. So given that that ’ s true, how good are we at increasing our happiness? Well, we certainly try. If you look on the Amazon site, there are over 2, 000 titles with advice on the seven habits, the nine choices, the 10 secrets, the 14, 000 thoughts that are supposed to bring happiness. Now another way we try to increase our happiness is we medicate ourselves. And so there ’ s over 120 million prescriptions out there for antidepressants. Prozac was really the first absolute blockbuster drug. It was clean, efficient, there was no high, there was really no danger, it had no street value. In 1995, illegal drugs were a $400 billion business, representing eight percent of world trade, roughly the same as gas and oil. These routes to happiness haven ’ t really increased happiness very much. One problem that ’ s happening now is, although the rates of happiness are about as flat as the surface of the moon, depression and anxiety are rising. Some people say this is because we have better diagnosis, and more people are being found out. It isn ’ t just that. We ’ re seeing it all over the world. In the United States right now there are more suicides than homicides. There is a rash of suicide in China. And the World Health Organization predicts by the year 2020 that depression will be the second largest cause of disability. Now the good news here is that if you take surveys from around the world, we see that about three quarters of people will say they are at least pretty happy. But this does not follow any of the usual trends. For example, these two show great growth in income, absolutely flat happiness curves. My field, the field of psychology, hasn ’ t done a whole lot to help us move forward in understanding human happiness. In part, we have the legacy of Freud, who was a pessimist, who said that pursuit of happiness is a doomed quest, is propelled by infantile aspects of the individual that can never be met in reality. He said,”One feels inclined to say that the intention that man should be happy is not included in the plan of creation.”So the ultimate goal of psychoanalytic psychotherapy was really what Freud called ordinary misery. And Freud in part reflects the anatomy of the human emotion system — which is that we have both a positive and a negative system, and our negative system is extremely sensitive. So for example, we ’ re born loving the taste of something sweet and reacting aversively to the taste of something bitter. We also find that people are more averse to losing than they are happy to gain. The formula for a happy marriage is five positive remarks, or interactions, for every one negative. And that ’ s how powerful the one negative is. Especially expressions of contempt or disgust, well you really need a lot of positives to upset that. I also put in here the stress response. We ’ re wired for dangers that are immediate, that are physical, that are imminent, and so our body goes into an incredible reaction where endogenous opioids come in. We have a system that is really ancient, and really there for physical danger. And so over time, this becomes a stress response, which has enormous effects on the body. Cortisol floods the **brain** ; it destroys hippocampal cells and memory, and can lead to all kinds of health problems. But unfortunately, we need this system in part. If we were only governed by pleasure we would not survive. We really have two command posts. Emotions are short-lived intense responses to challenge and to opportunity. And each one of them allows us to click into alternate selves that tune in, turn on, drop out thoughts, perceptions, feelings and memories. We tend to think of emotions as just feelings. But in fact, emotions are an all-systems alert that change what we remember, what kind of decisions we make, and how we perceive things. So let me go forward to the new science of happiness. We ’ ve come away from the Freudian gloom, and people are now actively studying this. And one of the key points in the science of happiness is that happiness and unhappiness are not endpoints of a single continuum. The Freudian model is really one continuum that, as you get less miserable, you get happier. And that isn ’ t true — when you get less miserable, you get less miserable. And that happiness is a whole other end of the equation. And it ’ s been missing. It ’ s been missing from psychotherapy. So when people ’ s symptoms go away, they tend to recur, because there isn ’ t a sense of the other half — of what pleasure, happiness, compassion, gratitude, what are the positive emotions. And of course we know this intuitively, that happiness is not just the absence of misery. But somehow it was not put forward until very recently, seeing these as two parallel systems. So that the body can both look for opportunity and also protect itself from danger, at the same time. And they ’ re sort of two reciprocal and dynamically interacting systems. People have also wanted to deconstruct. We use this word”happy,”and it ’ s this very large umbrella of a term. And then three emotions for which there are no English words : fiero, which is the pride in accomplishment of a challenge ; schadenfreude, which is happiness in another ’ s misfortune, a malicious pleasure ; and naches, which is a pride and joy in one ’ s children. Absent from this list, and absent from any discussions of happiness, are happiness in another ’ s happiness. We don ’ t seem to have a word for that. We are very sensitive to the negative, but it is in part offset by the fact that we have a positivity. We ’ re also born pleasure-seekers. Babies love the taste of sweet and hate the taste of bitter. They love to touch smooth surfaces rather than rough ones. They like to look at beautiful faces rather than plain faces. They like to listen to consonant melodies instead of dissonant melodies. Babies really are born with a lot of innate pleasures. There was once a statement made by a psychologist that said that 80 percent of the pursuit of happiness is really just about the genes, and it ’ s as difficult to become happier as it is to become taller. That ’ s nonsense. There is a decent contribution to happiness from the genes — about 50 percent — but there is still that 50 percent that is unaccounted for. Let ’ s just go into the **brain** for a moment, and see where does happiness arise from in evolution. We have basically at least two systems here, and they both are very ancient. One is the reward system, and that ’ s fed by the chemical dopamine. And it starts in the ventral tegmental area. It goes to the nucleus accumbens, all the way up to the prefrontal cortex, orbital frontal cortex, where decisions are made, high level. This was originally seen as a system that was the pleasure system of the **brain**. In the 1950s, Olds and Milner put electrodes into the **brain** of a rat. And the rat would just keep pressing that bar thousands and thousands and thousands of times. It wouldn ’ t eat. It wouldn ’ t sleep. It wouldn ’ t have sex. It wouldn ’ t do anything but press this bar. So they assumed this must be, you know, the **brain** ’ s orgasmatron. It turned out that it wasn ’ t, that it really is a system of motivation, a system of wanting. It gives objects what ’ s called incentive salience. It makes something look so attractive that you just have to go after it. That ’ s something different from the system that is the pleasure system, which simply says,”I like this.”The pleasure system, as you see, which is the internal opiates, there is a hormone oxytocin, is widely spread throughout the **brain**. Dopamine system, the wanting system, is much more centralized. The other thing about positive emotions is that they have a universal signal. And we see here the smile. And the universal signal is not just raising the corner of the lips to the zygomatic major. It ' s also crinkling the outer corner of the eye, the orbicularis oculi. So you see, even 10-month-old babies, when they see their mother, will show this particular kind of smile. Extroverts use it more than introverts. People who are relieved of depression show it more after than before. So if you want to unmask a true look of happiness, you will look for this expression. Our pleasures are really ancient. And we learn, of course, many, many pleasures, but many of them are base. And one of them, of course, is biophilia — that we have a response to the natural world that ' s very profound. Very interesting studies done on people recovering from surgery, who found that people who faced a brick wall versus people who looked out on trees and nature, the people who looked out on the brick wall were in the hospital longer, needed more medication, and had more medical complications. There is something very restorative about nature, and it ' s part of how we are tuned. Humans, particularly so, we ' re very imitative creatures. And we imitate from almost the second we are born. Here is a three-week-old baby. And if you stick your tongue out at this baby, the baby will do the same. We are social beings from the beginning. And even studies of cooperation show that cooperation between individuals lights up reward centers of the **brain**. One problem that psychology has had is instead of looking at this intersubjectivity — or the importance of the social **brain** to humans who come into the world helpless and need each other tremendously — is that they focus instead on the self and self-esteem, and not self-other. It ' s sort of"me,"not"we."And I think this has been a really tremendous problem that goes against our biology and nature, and hasn ' t made us any happier at all. Because when you think about it, people are happiest when in flow, when they ' re absorbed in something out in the world, when they ' re with other people, when they ' re active, engaged in sports, focusing on a loved one, learning, having sex, whatever. They ' re not sitting in front of the mirror trying to figure themselves out, or thinking about themselves. These are not the periods when you feel happiest. The other thing is, that a piece of evidence is, is if you look at computerized text analysis of people who commit suicide, what you find there, and it ' s quite interesting, is use of the first person singular —"I,""me,""my,"not"we"and"us"— and the letters are less hopeless than they are really alone. And being alone is very unnatural to the human. There is a profound need to belong. But there are ways in which our evolutionary history can really trip us up. Because, for example, the genes don ' t care whether we ' re happy, they care that we replicate, that we pass our genes on. So for example we have three systems that underlie reproduction, because it ' s so important. There ' s lust, which is just wanting to have sex. And that ' s really mediated by the sex hormones. Romantic attraction, that gets into the desire system. And that ' s dopamine-fed. And that ' s,"I must have this one person."There ' s attachment, which is oxytocin, and the opiates, which says,"This is a long-term bond."See the problem is that, as humans, these three can separate. So a person can be in a long term attachment, become romantically infatuated with someone else, and want to have sex with a third person. The other way in which our genes can sometimes lead us astray is in social status. We are very acutely aware of our social status and always seek to further and increase it. Now in the animal world, there is only one way to increase status, and that ' s dominance. I seize command by physical prowess, and I keep it by beating my chest, and you make submissive gestures. Now, the human has a whole other way to rise to the top, and that is a prestige route, which is freely conferred. Someone has expertise and knowledge, and knows how to do things, and we give that person status. And that ' s clearly the way for us to create many more niches of status so that people don ' t have to be lower on the status hierarchy as they are in the animal world. The data isn ’ t terribly supportive of money buying happiness. But it ’ s not irrelevant. So if you look at questions like this, life satisfaction, you see life satisfaction going up with each rung of income. You see mental distress going up with lower income. So clearly there is some effect. But the effect is relatively small. And one of the problems with money is materialism. What happens when people pursue money too avidly, is they forget about the real basic pleasures of life. So we have here, this couple.”Do you think the less- fortunate are having better sex?”And then this kid over here is saying,”Leave me alone with my toys.”So one of the things is that it really takes over. That whole dopamine-wanting system takes over and derails from any of the pleasure system. Maslow had this idea back in the 1950s that as people rise above their biological needs, as the world becomes safer and we don ’ t have to worry about basic needs being met — our biological system, whatever motivates us, is being satisfied — we can rise above them, to think beyond ourselves toward self-actualization or transcendence, and rise above the materialist. So to just quickly conclude with some brief data that suggests this might be so. One is people who underwent what is called a quantum change : they felt their life and their whole values had changed. And sure enough, if you look at the kinds of values that come in, you see wealth, adventure, achievement, pleasure, fun, be respected, before the change, and much more post-materialist values after. Women had a whole different set of value shifts. But very similarly, the only one that survived there was happiness. They went from attractiveness and happiness and wealth and self-control to generosity and forgiveness. I end with a few quotes.”There is only one question : How to love this world?”And Rilke,”If your daily life seems poor, do not blame it ; blame yourself. Tell yourself that you are not poet enough to call forth its riches.””First, say to yourself what you would be. Then do what you have to do.”Thank you.

Text Sample 609: NEG Dim 4 – 2008 – Score -1.00 – 2008/t000158.txt

Unless we do something to prevent it, over the next 40 years weâ□□re facing an epidemic of neurologic diseases on a global scale. A cheery thought. On this map, every country thatâ□□s colored blue has more than 20 percent of its population over the age of 65. This is the world we live in. And this is the world your children will live in. For 12, 000 years, the distribution of ages in the human population has looked like a pyramid, with the oldest on top. Itâ□□s already flattening out. By 2050, itâ□□s going to be a column and will start to invert. This is why itâ□□s happening. The average lifespanâ□□s more than doubled since 1840, and itâ□□s increasing currently at the rate of about five hours every day. And this is why thatâ□□s not entirely a good thing : because over the age of 65, your risk of getting Alzheimerâ□□s or Parkinsonâ□□s disease will increase exponentially. By 2050, thereâ□□ll be about 32 million people in the United States over the age of 80, and unless we do something about it, half of them will have Alzheimerâ□□s disease and three million more will have Parkinsonâ□□s disease. Right now, those and other neurologic diseases â□□ for which we have no cure or prevention â□□ cost about a third of a trillion dollars a year. It will be well over a trillion dollars by 2050. Alzheimerâ□□s disease starts when a protein that should be folded up properly misfolds into a kind of demented origami. So one approach weâ□□re taking is to try to design drugs that function like molecular Scotch tape, to hold the protein into its proper shape. That would keep it from forming the tangles that seem to kill large sections of the **brain** when they do. Interestingly enough, other neurologic diseases which affect very different parts of the **brain** also show tangles of misfolded protein, which suggests that the approach might be a general one, and might be used to cure many neurologic diseases, not just Alzheimerâs disease. Thereâs also a fascinating connection to cancer here, because people with neurologic diseases have a very low incidence of most cancers. And this is a connection that most people arenât pursuing right now, but which weâre fascinated by. Most of the important and all of the creative work in this area is being funded by private philanthropies. And thereâs tremendous scope for additional private help here, because the government has dropped the ball on much of this, Iâm afraid. In the meantime, while weâre waiting for all these things to happen, hereâs what you can do for yourself. If you want to lower your risk of Parkinsonâs disease, caffeine is protective to some extent ; nobody knows why. Head injuries are bad for you. They lead to Parkinsonâs disease. And the Avian Flu is also not a good idea. As far as protecting yourself against Alzheimerâs disease, well, it turns out that fish oil has the effect of reducing your risk for Alzheimerâs disease. You should also keep your blood pressure down, because chronic high blood pressure is the biggest single risk factor for Alzheimerâs disease. Itâs also the biggest risk factor for glaucoma, which is just Alzheimerâs disease of the eye. And of course, when it comes to cognitive effects,"use it or lose it"applies, so you want to stay mentally stimulated. But hey, youâre listening to me. So youâve got that covered. And one final thing. Wish people like me luck, okay? Because the clock is ticking for all of us. Thank you.

Text Sample 610: NEG Dim 4 – 2008 – Score 1.00 – 2008/t000204.txt

I was here about four years ago, talking about the relationship of design and happiness. At the very end of it, I showed a list under that title. I learned very few things in addition since â but made a whole number of them into projects since. These are inflatable monkeys in every city in Scotland :"Everybody always thinks they are right."They were combined in the media."Drugs are fun in the beginning but become a drag later on."We ' re doing changing media. This is a projection that can see the viewer as the viewer walks by. You can ' t help but actually ripping that spider web apart. All of these things are pieces of graphic design. We do them for our clients. They are commissioned. I would never have the money to actually pay for the installment or pay for all the billboards or the production of these, so there ' s always a client attached to them. These are 65, 000 coat hangers in a street that ' s lined with fashion stores."Worrying solves nothing.""Money does not make me happy"appeared first as double-page spreads in a magazine. The printer lost the file, didn ' t tell us. When the magazine â actually, when I got the subscription â it was 12 following pages. It said,"Money does does make me happy."And a friend of mine in Austria felt so sorry for me that he talked the largest casino owner in Linz into letting us wrap his building. So this is the big pedestrian zone in Linz. It just says"Money,"and if you look down the side street, it says,"does not make me happy."We had a show that just came down last week in New York. We steamed up the windows permanently, and every hour we had a different designer come in and write these things that they ' ve learned into the steam in the window. Everybody participated â Milton Glaser, Massimo Vignelli. Singapore was quite in discussion. This is a little spot that we filmed there that ' s to be displayed on the large JumboTrons in Singapore. And, of course, it ' s one that ' s dear to my heart, because all of these sentiments â some banal, some a bit more profound â all originally had come out of my diary. And I do go often into the diary and check if I wanted to change something about the situation. If it ’ s â see it for a long enough time, I actually do something about it. And the very last one is a billboard. This is our roof in New York, the roof of the studio. This is newsprint plus stencils that lie on the newsprint. We let that lie around in the sun. As you all know, newsprint yellows significantly in the sun. After a week, we took the stencils and the leaves off, shipped the newsprints to Lisbon to a very sunny spot, so on day one the billboard said,”Complaining is silly. Either act or forget.”Three days later it faded, and a week later, no more complaining anywhere. Thank you so much.

Text Sample 611: NEG Dim 4 – 2008 – Score 2.00 – 2008/t000303.txt
Good morning. Let ’ s look for a minute at the greatest icon of all, Leonardo da Vinci. We ’ re all familiar with his fantastic work â his drawings, his paintings, his inventions, his writings. But we do not know his face. Thousands of books have been written about him, but there ’ s controversy, and it remains, about his looks. Even this well-known portrait is not accepted by many art historians. So, what do you think? Is this the face of Leonardo da Vinci or isn ’ t it? Let ’ s find out. Leonardo was a man that drew everything around him. He drew people, anatomy, plants, animals, landscapes, buildings, water, everything. But no faces? I find that hard to believe. His contemporaries made faces, like the ones you see here â en face or three-quarters. So, surely a passionate drawer like Leonardo must have made self-portraits from time to time. So let ’ s try to find them. I think that if we were to scan all of his work and look for self-portraits, we would find his face looking at us. So I looked at all of his drawings, more than 700, and looked for male portraits. There are about 120, you see them here. Which ones of these could be self-portraits? Well, for that they have to be done as we just saw, en face or three-quarters. So we can eliminate all the profiles. It also has to be sufficiently detailed. So we can also eliminate the ones that are very vague or very stylized. And we know from his contemporaries that Leonardo was a very handsome, even beautiful man. So we can also eliminate the ugly ones or the caricatures. And look what happens â only three candidates remain that fit the bill. And here they are. Yes, indeed, the old man is there, as is this famous pen drawing of the Homo Vitruvianus. And lastly, the only portrait of a male that Leonardo painted,”The Musician.”Before we go into these faces, I should explain why I have some right to talk about them. I ’ ve made more than 1, 100 portraits myself for newspapers, over the course of 300 â 30 years, sorry, 30 years only. But there are 1, 100, and very few artists have drawn so many faces. So I know a little about drawing and analyzing faces. OK, now let ’ s look at these three portraits. And hold onto your seats, because if we zoom in on those faces, remark how they have the same broad forehead, the horizontal eyebrows, the long nose, the curved lips and the small, well-developed chin. I couldn ’ t believe my eyes when I first saw that. There is no reason why these portraits should look alike. All we did was look for portraits that had the characteristics of a self-portrait, and look, they are very similar. Now, are they made in the right order? The young man should be made first. And as you see here from the years that they were created, it is indeed the case. They are made in the right order. What was the age of Leonardo at the time? Does that fit? Yes, it does. He was 33, 38 and 63 when these were made. So we have three pictures, potentially of the same person of the same age as Leonardo at the time. But how do we know it ’ s him, and not someone else? Well, we need a reference. And here ’ s the only picture of Leonardo that ’ s widely accepted. It ’ s a statue made by Verrocchio,

of David, for which Leonardo posed as a boy of 15. And if we now compare the face of the statue, with the face of the musician, you see the very same features again. The statue is the reference, and it connects the identity of Leonardo to those three faces. Ladies and gentlemen, this story has not yet been published. It ' s only proper that you here at TED hear and see it first. The icon of icons finally has a face. Here he is : Leonardo da Vinci.

Text Sample 612: NEG Dim 4 – 2015 – Score -3.00 – 2015/t001484.txt
As a singer-songwriter, people often ask me about my influences or, as I like to call them, my sonic lineages. And I could easily tell you that I was shaped by the jazz and hip hop that I grew up with, by the Ethiopian heritage of my ancestors, or by the 1980s **pop** on my childhood radio stations. But beyond genre, there is another question : how do the sounds we hear every day influence the music that we make? I believe that everyday soundscape can be the most unexpected inspiration for songwriting, and to look at this idea a little bit more closely, I ' m going to talk today about three things : nature, language and silence — or rather, the impossibility of true silence. And through this I hope to give you a sense of a world already alive with musical expression, with each of us serving as active participants, whether we know it or not. I ' m going to start today with nature, but before we do that, let ' s quickly listen to this snippet of an opera singer warming up. Here it is. It ' s beautiful, isn ' t it? Gotcha! That is actually not the sound of an opera singer warming up. That is the sound of a bird slowed down to a pace that the human ear mistakenly recognizes as its own. It was released as part of Peter Szke ' s 1987 Hungarian recording”The Unknown Music of Birds,”where he records many birds and slows down their pitches to reveal what ' s underneath. Let ' s listen to the full-speed recording. Now, let ' s hear the two of them together so your **brain** can juxtapose them. It ' s incredible. Perhaps the techniques of opera singing were inspired by birdsong. As humans, we intuitively understand birds to be our musical teachers. In Ethiopia, birds are considered an integral part of the origin of music itself. The story goes like this : 1, 500 years ago, a young man was born in the Empire of Aksum, a major trading center of the ancient world. His name was Yared. When Yared was seven years old his father died, and his mother sent him to go live with an uncle, who was a priest of the Ethiopian Orthodox tradition, one of the oldest churches in the world. Now, this tradition has an enormous amount of scholarship and learning, and Yared had to study and study and study and study, and one day he was studying under a tree, when three birds came to him. One by one, these birds became his teachers. They taught him music — scales, in fact. And Yared, eventually recognized as Saint Yared, used these scales to compose five volumes of chants and hymns for worship and celebration. And he used these scales to compose and to create an indigenous musical notation system. And these scales evolved into what is known as kiit, the unique, pentatonic, five-note, modal system that is very much alive and thriving and still evolving in Ethiopia today. Now, I love this story because it ' s true at multiple levels. Saint Yared was a real, historical figure, and the natural world can be our musical teacher. And we have so many examples of this : the Pygmies of the Congo tune their instruments to the pitches of the birds in the forest around them. Musician and natural soundscape expert Bernie Krause describes how a healthy environment has animals and insects taking up low, medium and high-frequency bands, in exactly the same way as a symphony does. And countless works of music were inspired by bird and forest song. Yes, the natural world can be our cultural teacher. So let ' s go now to the

uniquely human world of language. Every language communicates with pitch to varying degrees, whether it ’ s Mandarin Chinese, where a shift in melodic inflection gives the same phonetic syllable an entirely different meaning, to a language like English, where a raised pitch at the end of a sentence... implies a question? As an Ethiopian-American woman, I grew up around the language of Amharic, Amharia. It was my first language, the language of my parents, one of the main languages of Ethiopia. And there are a million reasons to fall in love with this language : its depth of poetics, its double entendres, its wax and gold, its humor, its proverbs that illuminate the wisdom and **follies** of life. But there ’ s also this melodicism, a musicality built right in. And I find this distilled most clearly in what I like to call emphatic language — language that ’ s meant to highlight or underline or that springs from surprise. Take, for example, the word :”indey.”Now, if there are Ethiopians in the audience, they ’ re probably chuckling to themselves, because the word means something like”No!”or”How could he?”or”No, he didn ’ t.”It kind of depends on the situation. But when I was a kid, this was my very favorite word, and I think it ’ s because it has a pitch. It has a melody. You can almost see the shape as it springs from someone ’ s mouth.”Indey”— it dips, and then raises again. And as a musician and composer, when I hear that word, something like this is floating through my mind. Or take, for example, the phrase for”It is right”or”It is correct”—”Lickih nehu... Lickih nehu.”It ’ s an affirmation, an agreement.”Lickih nehu.”When I hear that phrase, something like this starts rolling through my mind. And in both of those cases, what I did was I took the melody and the phrasing of those words and phrases and I turned them into musical parts to use in these short compositions. And I like to write bass lines, so they both ended up kind of as bass lines. Now, this is based on the work of Jason Moran and others who work intimately with music and language, but it ’ s also something I ’ ve had in my head since I was a kid, how musical my parents sounded when they were speaking to each other and to us. It was from them and from Amharia that I learned that we are awash in musical expression with every word, every sentence that we speak, every word, every sentence that we receive. Perhaps you can hear it in the words I ’ m speaking even now. Finally, we go to the 1950s United States and the most seminal work of 20th century avant-garde composition : John Cage ’ s”4:33,”written for any instrument or combination of instruments. The musician or musicians are invited to walk onto the stage with a stopwatch and open the score, which was actually purchased by the Museum of Modern Art — the score, that is. And this score has not a single note written and there is not a single note played for four minutes and 33 seconds. And, at once enraging and enrapturing, Cage shows us that even when there are no strings being plucked by fingers or hands hammering piano keys, still there is music, still there is music, still there is music. And what is this music? It was that sneeze in the back. It is the everyday soundscape that arises from the audience themselves : their coughs, their sighs, their rustles, their whispers, their sneezes, the room, the wood of the floors and the walls expanding and contracting, creaking and groaning with the heat and the cold, the pipes clanking and contributing. And controversial though it was, and even controversial though it remains, Cage ’ s point is that there is no such thing as true silence. Even in the most silent environments, we still hear and feel the sound of our own heartbeats. The world is alive with musical expression. We are already immersed. Now, I had my own moment of, let ’ s say, remixing John Cage a couple of months ago when I was standing in front of the stove cooking lentils. And it was late one night and it was time to stir, so I lifted the lid off the cooking pot, and I placed it onto the kitchen counter next to me, and it started to roll back and forth making this sound. And it stopped me cold. I thought,”What a weird, cool swing that cooking pan lid has.”So when the lentils were ready and eaten, I hightailed it to my backyard studio, and I made this. Now, John Cage wasn ’ t instructing musicians to mine the soundscape for sonic textures to turn into music. He was saying that on its own, the environment is musically generative, that it is generous, that it is fertile, that we are already immersed. Musician, music researcher, surgeon and human hearing expert Charles Limb is a professor at Johns Hopkins University and he studies music and the **brain**. And he has a theory that it is possible — it is possible — that the human auditory system actually evolved to hear music, because it is so much more complex than it needs to be for language alone. And if that ’ s true, it means that we ’ re hard-wired for music, that we can find it anywhere, that there is no such thing as a musical desert, that we are permanently hanging out at the oasis, and that is marvelous. We can add to the soundtrack, but it ’ s already playing. And it doesn ’ t mean don ’ t study music. Study music, trace your sonic lineages and enjoy that exploration. But there is a kind of sonic lineage to which we all belong. So the next time you are seeking percussion inspiration, look no further than your tires, as they roll over the unusual grooves of the freeway, or the top-right burner of your stove and that strange way that it clicks as it is preparing to light. When seeking melodic inspiration, look no further than dawn and dusk avian orchestras or to the natural lilt of emphatic language. We are the audience and we are the composers and we take from these pieces we are given. We make, we make, we make, we make, knowing that when it comes to nature or language or soundscape, there is no end to the inspiration — if we are listening. Thank you.

Text Sample 613: NEG Dim 4 – 2015 – Score 1.00 – 2015/t001604.txt

My first love was for the night sky. Love is complicated. You ’ re looking at a fly-through of the Hubble Space Telescope Ultra-Deep Field, one of the most distant images of our universe ever observed. Everything you see here is a galaxy, comprised of billions of stars each. And the farthest galaxy is a trillion, trillion kilometers away. As an astrophysicist, I have the awesome privilege of studying some of the most exotic objects in our universe. The objects that have captivated me from first crush throughout my career are supermassive, hyperactive black holes. Weighing one to 10 billion times the mass of our own sun, these galactic black holes are devouring material, at a rate of upwards of 1, 000 times more than your”average”supermassive black hole. These two characteristics, with a few others, make them quasars. At the same time, the objects I study are producing some of the most powerful particle streams ever observed. These narrow streams, called jets, are moving at 99. 99 percent of the speed of light, and are pointed directly at the Earth. These jetted, Earth-pointed, hyperactive and supermassive black holes are called blazars, or blazing quasars. What makes blazars so special is that they ’ re some of the universe ’ s most efficient particle accelerators, transporting incredible amounts of energy throughout a galaxy. Here, I ’ m showing an artist ’ s conception of a blazar. The dinner plate by which material falls onto the black hole is called the accretion disc, shown here in blue. Some of that material is slingshotted around the black hole and accelerated to insanely high speeds in the jet, shown here in white. Although the blazar system is rare, the process by which nature pulls in material via a disk, and then flings some of it out via a jet, is more common. We ’ ll eventually zoom out of the blazar system to show its approximate relationship to the larger galactic context. Beyond the cosmic accounting of what goes in to what goes out, one of the hot topics in blazar astrophysics right now is where the highest-energy jet emission comes from. In this image, I ’ m interested in where this white blob forms and if, as a result, there ’ s any relationship between the jet and the accretion disc material. Clear answers to this question were almost completely inaccessible until 2008, when NASA launched a new telescope that better detects gamma ray light — that is, light with energies a million times higher than your standard x-ray scan. I simultaneously compare variations between the gamma ray light data and the visible light data from day to day and year to year, to better localize these gamma ray blobs. My research shows that in some instances, these blobs form much closer to the black hole than we initially thought. As we more confidently localize where these gamma ray blobs are forming, we can better understand how jets are being accelerated, and ultimately reveal the dynamic processes by which some of the most fascinating objects in our universe are formed. This all started as a love story. And it still is. This love transformed me from a curious, stargazing young girl to a professional astrophysicist, hot on the heels of celestial discovery. Who knew that chasing after the universe would ground me so deeply to my mission here on Earth. Then again, when do we ever know where love ’ s first flutter will truly take us. Thank you.

Text Sample 614: NEG Dim 4 – 2015 – Score 1.00 – 2015/t001603.txt
The **brain** is an amazing and complex organ. And while many people are fascinated by the **brain**, they can ’ t really tell you that much about the properties about how the **brain** works because we don ’ t teach neuroscience in schools. And one of the reasons why is that the equipment is so complex and so expensive that it ’ s really only done at major universities and large institutions. And so in order to be able to access the **brain**, you really need to dedicate your life and spend six and a half years as a graduate student just to become a neuroscientist to get access to these tools. And that ’ s a shame because one out of five of us, that ’ s 20 percent of the entire world, will have a neurological disorder. And there are zero cures for these diseases. And so it seems that what we should be doing is reaching back earlier in the eduction process and teaching students about neuroscience so that in the future, they may be thinking about possibly becoming a **brain** scientist. When I was a graduate student, my lab mate Tim Marzullo and myself, decided that what if we took this complex equipment that we have for studying the **brain** and made it simple enough and affordable enough that anyone that you know, an amateur or a high school student, could learn and actually participate in the discovery of neuroscience. And so we did just that. A few years ago, we started a company called Backyard **Brains** and we make DIY neuroscience equipment and I brought some here tonight, and I want to do some demonstrations. You guys want to see some? So I need a volunteer. So right before — what is your name? Sam Kelly : Sam. Greg Gage : All right, Sam, I ’ m going to record from your **brain**. Have you had this before? SK : No. GG : I need you to stick out your arm for science, roll up your sleeve a bit, So what I ’ m going to do, I ’ m putting electrodes on your arm, and you ’ re probably wondering, I just said I ’ m going to record from your **brain**, what am I doing with your arm? Well, you have about 80 billion neurons inside your **brain** right now. They ’ re sending electrical messages back and forth, and chemical messages. But some of your neurons right here in your motor cortex are going to send messages down when you move your arm like this. They ’ re going to go down across your corpus callosum, down onto your spinal cord to your lower motor neuron out to your muscles here, and that electrical discharge is going to be picked up by these electrodes right here and we ’ re

going to be able to listen to exactly what your **brain** is going to be doing. So I ’ m going to turn this on for a second. Have you ever heard what your **brain** sounds like? SK : No. GG : Let ’ s try it out. So go ahead and squeeze your hand. So what you ’ re listening to, so this is your motor units happening right here. Let ’ s take a look at it as well. So I ’ m going to stand over here, and I ’ m going to open up our app here. So now I want you to squeeze. So right here, these are the motor units that are happening from her spinal cord out to her muscle right here, and as she ’ s doing it, you ’ re seeing the electrical activity that ’ s happening here. You can even click here and try to see one of them. So keep doing it really hard. So now we ’ ve paused on one motor action potential that ’ s happening right now inside of your **brain**. Do you guys want to see some more? That ’ s interesting, but let ’ s get it better. I need one more volunteer. What is your name, sir? Miguel Goncalves : Miguel. GG : Miguel, all right. You ’ re going to stand right here. So when you ’ re moving your arm like this, your **brain** is sending a signal down to your muscles right here. I want you to move your arm as well. So your **brain** is going to send a signal down to your muscles. And so it turns out that there is a nerve that ’ s right here that runs up here that innervates these three fingers, and it ’ s close enough to the skin that we might be able to stimulate that so that what we can do is copy your **brain** signals going out to your hand and inject it into your hand, so that your hand will move when your **brain** tells your hand to move. So in a sense, she will take away your free will and you will no longer have any control over this hand. You with me? So I just need to hook you up. So I ’ m going to find your ulnar nerve, which is probably right around here. You don ’ t know what you ’ re signing up for when you come up. So now I ’ m going to move away and we ’ re going to plug it in to our human-to-human interface over here. Okay, so Sam, I want you to squeeze your hand again. Do it again. Perfect. So now I ’ m going to hook you up over here so that you get the — It ’ s going to feel a little bit weird at first, this is going to feel like a — You know, when you lose your free will, and someone else becomes your agent, it does feel a bit strange. Now I want you to relax your hand. Sam, you ’ re with me? So you ’ re going to squeeze. I ’ m not going to turn it on yet, so go ahead and give it a squeeze. So now, are you ready, Miguel? MG : Ready as I ’ ll ever be. GG : I ’ ve turned it on, so go ahead and turn your hand. Do you feel that a little bit? MG : Nope. GG : Okay, do it again? MG : A little bit. GG : A little bit? So relax. So hit it again. Oh, perfect, perfect. So relax, do it again. All right, so right now, your **brain** is controlling your arm and it ’ s also controlling his arm, so go ahead and just do it one more time. All right, so it ’ s perfect. So now, what would happen if I took over my control of your hand? And so, just relax your hand. What happens? Ah, nothing. Why not? Because the **brain** has to do it. So you do it again. All right, that ’ s perfect. Thank you guys for being such a good sport. This is what ’ s happening all across the world — electrophysiology! We ’ re going to bring on the neuro-revolution. Thank you.

Text Sample 615: NEG Dim 4 – 2019 – Score 1.00 – 2019/t002074.txt

Chris Anderson : Shep, thank you so much for coming. I think your plane landed literally two hours ago in Vancouver. Such a treat to have you. So, talk us through how do you get from Einstein ’ s equation to a black hole? Sheperd Doeleman : Over 100 years ago, Einstein came up with this geometric theory of gravity which deforms space-time. So, matter deforms space-time, and then space-time tells matter in turn how to move around it. And you can get enough matter into a small enough region that it punctures space-time, and that even light can ’ t escape, the force of gravity keeps even light inside. CA : And so, before that, the reason the Earth moves around the Sun is not because the Sun is pulling the Earth as we think, but it ’ s literally changed the shape of space so that we just sort of fall around the Sun. SD : Exactly, the geometry of space-time tells the Earth how to move around the Sun. You ’ re almost seeing a black hole puncture through space-time, and when it goes so deeply in, then there ’ s a point at which light orbits the black hole. CA : And so that ’ s, I guess, is what ’ s happening here. This is not an image, this is a computer simulation of what we always thought, like, the event horizon around the black hole. SD : Until last week, we had no idea what a black hole really looked like. The best we could do were simulations like this in supercomputers, but even here you see this ring of light, which is the orbit of photons. That ’ s where photons literally move around the black hole, and around that is this hot gas that ’ s drawn to the black hole, and it ’ s hot because of friction. All this gas is trying to get into a very small volume, so it heats up. CA : A few years ago, you embarked on this mission to try and actually image one of these things. And I guess you took — you focused on this galaxy way out there. Tell us about this galaxy. SD : This is the galaxy — we ’ re going to zoom into the galaxy M87, it ’ s 55 million light-years away. CA : Fifty-five million. SD : Which is a long way. And at its heart, there ’ s a six-and-a-half-billion- solar-mass black hole. That ’ s hard for us to really fathom, right? Six and a half billion suns compressed into a single point. And it ’ s governing some of the energetics of the center of this galaxy. CA : But even though that thing is so huge, because it ’ s so far away, to actually dream of getting an image of it, that ’ s incredibly hard. The resolution would be incredible that you need. SD : Black holes are the smallest objects in the known universe. But they have these outsize effects on whole galaxies. But to see one, you would need to build a telescope as large as the Earth, because the black hole that we ’ re looking at gives off copious radio waves. It ’ s emitting all the time. CA : And that ’ s exactly what you did. SD : Exactly. What you ’ re seeing here is we used telescopes all around the world, we synchronized them perfectly with atomic clocks, so they received the light waves from this black hole, and then we stitched all of that data together to make an image. CA : To do that the weather had to be right in all of those locations at the same time, so you could actually get a clear view. SD : We had to get lucky in a lot of different ways. And sometimes, it ’ s better to be lucky than good. In this case, we were both, I like to think. But light had to come from the black hole. It had to come through intergalactic space, through the Earth ’ s atmosphere, where water vapor can absorb it, and everything worked out perfectly, the size of the Earth at that wavelength of light, one millimeter wavelength, was just right to resolve that black hole, 55 million light-years away. The universe was telling us what to do. CA : So you started capturing huge amounts of data. I think this is like half the data from just one telescope. SD : Yeah, this is one of the members of our team, Lindy Blackburn, and he ’ s sitting with half the data recorded at the Large Millimeter Telescope, which is atop a 15, 000-foot mountain in Mexico. And what he ’ s holding there is about half a petabyte. Which, to put it in terms that we might understand, it ’ s about 5, 000 people ’ s lifetime selfie budget. CA : It ’ s a lot of data. So this was all shipped, you couldn ’ t send this over the internet. All this data was shipped to one place and the massive computer effort began to try and analyze it. And you didn ’ t really know what you were going to see coming out of this. SD : The way this technique works that we used — imagine taking an optical mirror and smashing it and putting all the shards in different places. The way a normal mirror works is the light rays bounce off the surface, which is perfect, and they focus in a certain point at the same time. We take all these recordings, and with atomic clock precision we align them perfectly, later in a supercomputer. And we recreate kind of an Earth-sized lens. And the only way to do that is to bring the data back by plane. You can ’ t beat the bandwidth of a 747 filled with hard discs. CA : And so, I guess a few weeks or a few months ago, on a computer screen somewhere, this started to come into view. This moment. SD : Well, it took a long time. CA : I mean, look at this. That was it. That was the first image. So tell us what we ’ re really looking at there. SD : I still love it. So what you ’ re seeing is that last orbit of photons. You ’ re seeing Einstein ’ s geometry laid bare. The puncture in space-time is so deep that light moves around in orbit, so that light behind the black hole, as I think we ’ ll see soon, moves around and comes to us on these parallel lines at exactly that orbit. It turns out, that orbit is the square root of 27 times just a handful of fundamental constants. It ’ s extraordinary when you think about it. CA : When... In my head, initially, when I thought of black holes, I ’ m thinking that is the event horizon, there ’ s lots of matter and light whirling around in that shape. But it ’ s actually more complicated than that. Well, talk us through this animation, because it ’ s light being lensed around it. SD : You ’ ll see here that some light from behind it gets lensed, and some light does a loop-the-loop around the entire orbit of the black hole. But when you get enough light from all this hot gas swirling around the black hole, then you wind up seeing all of these light rays come together on this screen, which is a stand-in for where you and I are. And you see the definition of this ring begin to come into shape. And that ’ s what Einstein predicted over 100 years ago. CA : Yeah, that is amazing. So tell us more about what we ’ re actually looking at here. First of all, why is part of it brighter than the rest? SD : So what ’ s happening is that the black hole is spinning. And you wind up with some of the gas moving towards us below and receding from us on the top. And just as the train whistle has a higher pitch when it ’ s coming towards you, there ’ s more energy from the gas coming towards us than going away from us. You see the bottom part brighter because the light is actually being boosted in our direction. CA : And how physically big is that? SD : Our entire solar system would fit well within that dark region. And if I may, that dark region is the signature of the event horizon. The reason we don ’ t see light from there, is that the light that would come to us from that place was swallowed by the event horizon. So that — that ’ s it. CA : And so when we think of a black hole, you think of these huge rays jetting out of it, which are pointed directly in our direction. Why don ’ t we see them? SD : This is a very powerful black hole. Not by universal standards, it ’ s still powerful, and from the north and south poles of this black hole we think that jets are coming. Now, we ’ re too close to really see all the jet structure, but it ’ s the base of those jets that are illuminating the space-time. And that ’ s what ’ s being bent around the black hole. CA : And if you were in a spaceship whirling around that thing somehow, how long would it take to actually go around it? SD : First, I would give anything to be in that spaceship. Sign me up. There—s something called the — if I can get wonky for one moment — the innermost stable circular orbit, that ’ s the innermost orbit at which matter can move around a black hole before it spirals in. And for this black hole, it ’ s going to be between three days and about a month. CA : It ’ s so powerful, it ’ s weirdly slow at one level. I mean, you wouldn ’ t even notice falling into that event horizon if you were there. SD : So you may have heard of”spaghettification,”where you fall into a black hole and the gravitational field on your feet is much stronger than on your head, so you ’ re ripped apart. This black hole is so big that you ’ re not going to become a spaghetti noodle. You ’ re just going to drift right through that event horizon. CA : So, it ’ s like a giant tornado. When Dorothy was whipped by a tornado, she ended up in Oz. Where do you end up if you fall into a black hole? SD : Vancouver. CA : Oh, my God. It

’ s the red circle, that ’ s terrifying. No, really. SD : Black holes really are the central mystery of our age, because that ’ s where the quantum world and the gravitational world come together. What ’ s inside is a singularity. And that ’ s where all the forces become unified, because gravity finally is strong enough to compete with all the other forces. But it ’ s hidden from us, the universe has cloaked it in the ultimate invisibility cloak. So we don ’ t know what happens in there. CA : So there ’ s a smaller one of these in our own galaxy. Can we go back to our own beautiful galaxy? This is the Milky Way, this is home. And somewhere in the middle of that there ’ s another one, which you ’ re trying to find as well. SD : We already know it ’ s there, and we ’ ve already taken data on it. And we ’ re working on those data right now. So we hope to have something in the near future, I can ’ t say when. CA : It ’ s way closer but also a lot smaller, maybe the similar kind of size to what we saw? SD : Right. So it turns out that the black hole in M87, that we saw before, is six and a half billion solar masses. But it ’ s so far away that it appears a certain size. The black hole in the center of our galaxy is a thousand times less massive, but also a thousand times closer. So it looks the same angular size on the sky. CA : Finally, I guess, a nod to a remarkable group of people. Who are these guys? SD : So these are only some of the team. We marveled at the resonance that this image has had. If you told me that it would be above the fold in all of these newspapers, I ’ m not sure I would have believed you, but it was. Because this is a great mystery, and it ’ s inspiring for us, and I hope it ’ s inspiring to everyone. But the more important thing is that this is just a small number of the team. We ’ re 200 people strong with 60 institutes and 20 countries and regions. If you want to build a global telescope you need a global team. And this technique that we use of linking telescopes around the world kind of effortlessly sidesteps some of the issues that divide us. And as scientists, we naturally come together to do something like this. CA : Wow, boy, that ’ s inspiring for our whole team this week. Shep, thank you so much for what you did and for coming here. SD : Thank you.

Text Sample 616: NEG Dim 4 – 2019 – Score 1.00 – 2019/t001501.txt

I ’ m an astronomer who builds telescopes. I build telescopes because, number one, they are awesome. But number two, I believe if you want to discover a new thing about the universe, you have to look at the universe in a new way. New technologies in astronomy â things like lenses, photographic plates, all the way up to space telescopes â each gave us new ways to see the universe and directly led to a new understanding of our place in it. But those discoveries come with a cost. It took thousands of people and 44 years to get the Hubble Space Telescope from an idea into orbit. It takes time, it takes a tolerance for failure, it takes individual people choosing every day not to give up. I know how hard that choice is because I live it. The reality of my job is that I fail almost all the time and still keep going, because that ’ s how telescopes get built. The telescope I helped build is called the faint intergalactic-medium red- shifted emission balloon, which is a mouthful, so we call it”FIREBall.”And don ’ t worry, it is not going to explode at the end of this story. I ’ ve been working on FIREBall for more than 10 years and now lead the team of incredible people who built it. FIREBall is designed to observe some of the faintest structures known : huge clouds of hydrogen gas. These clouds are giant. They are even bigger than whatever you ’ re thinking of. They are huge, huge clouds of hydrogen that we think flow into and out of galaxies. I work on FIREBall because what I really want is to take our view of the universe from one with just light from stars to one where we can see and measure every atom that exists. That ’ s all that I want to do. But observing at least some of those atoms is crucial to our understanding of why galaxies look the way they do. I want to know how that hydrogen gas gets into a galaxy and creates a star. My work on FIREBall started in 2008, working not on the telescope but on the light sensor, which is the heart of any telescope. This new sensor was being developed by a team that I joined at NASA ’ s Jet Propulsion Laboratory. And our goal was to prove that this sensor would work really well to detect that hydrogen gas. In my work on this, I destroyed several very, very, very expensive sensors before realizing that the machine I was using created a plasma that shorted out anything electrical that we put in it. We used a different machine, there were other challenges, and it took years to get it right. But when that first sensor worked, it was glorious. And our sensors are now 10 times better than the previous state of the art and are getting put into all kinds of new telescopes. Our sensors will give us a new way to see the universe and our place in it. So, sensors done, time to build a telescope. And FIREBall is weird as far as telescopes go, because it ’ s not in space, and it ’ s not on the ground. Instead, it hangs on a cable from a giant balloon and observes for one night only from 130, 000 feet in the stratosphere, at the very edge of space. This is partly because the edge of space is much cheaper than actual space. So building it, of course, more failures : mirrors that failed, scratched mirrors that had to be remade ; cooling system failures, an entire system that had to be remade ; calibration failures, we ran tests again and again and again and again ; failures when you literally least expect them : we had an adorable but super angry baby falcon that landed on our spectrograph tank one day. Although to be fair, this was the greatest day in the history of this project. I really loved that falcon. But falcon damage fixed, we got it built for an August 2017 launch attempt â and then failed to launch, due to six weeks of continuous rain in the New Mexico desert. Our spirits dampened, we showed up again, August 2018, year 10. And on the morning of September 22nd, we finally got the telescope launched. I have put so much of myself â my whole life â into this project, and I, like, still can ’ t believe that that happened. And I have this picture that ’ s taken right around sunset on that day of our balloon, FIREBall hanging from it, and the nearly full moon. And I love this picture. God, I love it. But I look at it, and it makes me want to cry, because when fully inflated, these balloons are spherical, and this one isn ’ t. It ’ s shaped like a teardrop. And that ’ s because there is a hole in it. Sometimes balloons fail, too. FIREBall crash-landed in the New Mexico desert, and we didn ’ t get the data that we wanted. And at the end of that day, I thought to myself,”Why am I doing this?”And I ’ ve thought a lot about why since that day. And I ’ ve realized that all of my work has been full of things that break and fail, that we don ’ t understand and they fail, that we just get wrong the first time, and so they fail. I think about the thousands of people who built Hubble and how many failures they endured. There were countless failures, heartbreaking failures, even when it was in space. And none of those failures were a reason for them to give up. I think about why I love my job. I want to know what is happening in the universe. You all want to know what ’ s happening in the universe, too. I want to know what ’ s going on with that hydrogen. And so I ’ ve realized that discovery is mostly a process of finding things that don ’ t work, and failure is inevitable when you ’ re pushing the limits of knowledge. And that ’ s what I want to do. So I ’ m choosing to keep going. And our team is going to do what everyone who has ever built anything before us has done : we ’ re going to try again, in 2020. And it might feel like a failure today â and it really does â but it ’ s only going to stay a failure if I give up. Thank you very much.

Text Sample 617: NEG Dim 4 – 2019 – Score 1.00 – 2019/t000927.txt

Preparing for this talk has been scarier for me than preparing for LSD therapy.”Psychedelics are to the study of the mind what the microscope is to biology and the telescope is to astronomy.”Dr. Stanislav Grof spoke those words. He ’ s one of the leading psychedelic researchers in the world, and he ’ s also been my mentor. Today, I ’ d like to share with you how psychedelics, when used wisely, have the potential to help heal us, help inspire us, and perhaps even to help save us. In the 1950s and 60s, psychedelic research flourished all over the world and showed great promise for the fields of psychiatry, psychology and psychotherapy, neuroscience and the study of mystical experiences. But psychedelics leaked out of the research settings and began to be used by the counterculture, and by the anti-Vietnam War movement. And there was unwise use. And so there was a backlash. And in 1970, the US government criminalized all uses of psychedelics, and they began shutting down all psychedelic research. And this ban spread all over the world and lasted for decades. and it was tragic, since psychedelics are really just tools, and whether their outcomes are beneficial or harmful depends on how they ’ re used. Psychedelic means”mind-manifesting,”and it relates to drugs like LSD, psilocybin, mescaline, iboga and other drugs. When I was 18 years old, I was a college freshman, I was experimenting with LSD and mescaline, and these experiences brought me in touch with my emotions. And they helped me have a spiritual connection that unfortunately, my bar mitzvah did not produce. When I wanted to tease my parents, I would tell them that they drove me to psychedelics because my bar mitzvah had failed to turn me into a man. But most importantly, psychedelics gave me this feeling of our shared humanity, of our unity with all life. And other people reported that same thing as well. And I felt that these experiences had the potential to help be an antidote to tribalism, to fundamentalism, to genocide and environmental destruction. And so I decided to focus my life on changing the laws and becoming a legal psychedelic psychotherapist. Now, half a century after the ban, we ’ re in the midst of a global renaissance of psychedelic research. Psychedelic psychotherapy is showing great promise for the treatment of post-traumatic stress disorder, or PTSD, depression, social anxiety, substance abuse and alcoholism and suicide. Psychedelic psychotherapy is an attempt to go after the root causes of the problems, with just relatively few administrations, as contrasted to most of the psychiatric drugs used today that are mostly just reducing symptoms and are meant to be taken on a daily basis. Psychedelics are now also being used as tools for neuroscience to study **brain** function and to study the enduring mystery of human consciousness. And psychedelics and the mystical experiences they produce are being explored for their connections between meditation and mindfulness, including a paper just recently published about lifelong zen meditators taking psilocybin in the midst of a meditation retreat and showing long-term benefits and **brain** changes. Now, how do these drugs work? Modern neuroscience research has demonstrated that psychedelics reduce activity in what ’ s known as the **brain** ’ s default mode network. This is where we create our sense of self. It ’ s our equivalent to the ego, and it filters all incoming information according to our personal needs and priorities. When activity is reduced in the default mode network, our ego shifts from the foreground to the background, and we see that it ’ s just part of a larger field of awareness. It ’ s similar to the shift that Copernicus and Galileo were able to produce in humanity using the telescope to show that the earth was no longer the center of the universe, but was actually something that revolved around the sun, something bigger than itself. For some people, this shift in awareness is the most important and among the most important experiences of their lives. They feel more connected to the world bigger than themselves. They feel more altruistic, and they lose some of their fear of death. Not all drugs work this way. MDMA, also known as Ecstasy, or Molly, works fundamentally different. And I ’ ll be able to share with you the story of Marcela, who suffered from post-traumatic stress disorder from a violent sexual assault. Marcela and I were introduced in 1984, when MDMA was still legal, but it was beginning also to leak out of therapeutic circles. Marcela had tried MDMA in a recreational setting, and during that, her past trauma flooded her awareness and it intensified her suicidal feelings. During our first conversation, I shared that when MDMA is taken therapeutically, it can reduce the fear of difficult emotions, and she could help move forward past her trauma. I asked her to promise not to commit suicide if we were to work together. She agreed and made that promise. During her therapeutic sessions, Marcela was able to process her trauma more fluidly, more easily. And yet, she was able to tell that the rapist had told her that if she ever shared her story, he would kill her. And she realized that that was keeping her a prisoner in her own mind. So being able to share the story and experience the feelings and the thoughts in her mind freed her, and she was able to decide that she wanted to move forward with her life. And in that moment, I realized that MDMA could be very effective for treating PTSD. Now, 35 years later, after Marcela ’ s treatment, she ’ s actually a therapist, training other therapists to help people overcome PTSD with MDMA. Now, how does MDMA work? How did MDMA help Marcela? People who have PTSD have **brains** that are different from those of us who don ’ t have PTSD. They have a hyperactive amygdala, where we process fear. They have reduced activity in the prefrontal cortex, where we think logically. And they have reduced activity in the hippocampus, where we store memories into long-term storage. MDMA changes the **brain** in the opposite way. MDMA reduces activity in the amygdala, increases activity in the prefrontal cortex and increases connectivity between the amygdala and the hippocampus to remit traumatic memories to move into long-term storage. Recently, researchers at Johns Hopkins published a paper in”Nature,”in which they demonstrated that MDMA releases oxytocin, the hormone of love and nurturing. The same researchers also did studies in octopuses, who are normally asocial, unless it ’ s mating season. But lo and **behold**, you give them MDMA, and they become prosocial. Several months after Marcela and I worked together, the Drug Enforcement Administration moved to criminalize Ecstasy, having no knowledge of its therapeutic use. So I went to Washington, and I went into the headquarters of the Drug Enforcement Administration, and I filed a lawsuit demanding a hearing, at which psychiatrists and psychotherapists would be able to present information about therapeutic use of MDMA to try to keep it legal. And in the middle of the hearing, the DEA freaked out, declared an emergency and criminalized all uses of MDMA. And so the only way that I could see to bring it back was through science, through medicine and through the FDA drug development process. So in 1986, I started MAPS as a nonprofit psychedelic pharmaceutical company. It took us 30 years, till 2016, to develop the data that we needed to present to FDA to request permission to move into the large-scale Phase 3 studies that are required to prove safety and efficacy before you get approval for prescription use. Tony was a veteran in one of our pilot studies. According to the Veterans Administration, there ’ s over a million veterans now disabled with PTSD. And at least 20 veterans a day are committing suicide, many of them from PTSD. The treatment that Tony was to receive was three and a half months long. But during that period of time, he would only get MDMA on three occasions, separated by 12, 90-minute non-drug psychotherapy sessions, three before the first MDMA session for preparation and three after each MDMA session for integration. We call our treatment approach"inner-directed therapy,"in that we support the patient to experience whatever ' s emerging within their minds or their bodies. Even with MDMA, this is hard work. And a lot of our subjects have said,"I don ' t know why they call this Ecstasy."During Tony ' s first MDMA session, he lay on the couch, he had eyeshades on, he listened to music, and he would speak to the therapists, who were a male-female co-therapy team, whenever he felt that he needed to. After several hours, in a moment of calmness and clarity, Tony shared that he had realized his PTSD was a way of connecting him to his friends. It was a way of honoring the memory of his friends who had died. But he was able to shift and see himself through the eyes of his dead friends. And he realized that they would not want him to suffer, to squander his life. They would want him to live more fully, which they were unable to do. And so he realized that there was a new way to honor their memory, which was to live as fully as possible. He also realized that he was telling himself a story that he was taking opiates for pain. But actually, he realized, he was taking them for escape. So he decided he didn ' t need the opiates anymore, he didn ' t need the MDMA anymore, and he was dropping out of the study. That was seven years ago. Tony is still free of PTSD, has never returned to opiates and is helping others less fortunate than himself in Cambodia. The data that we presented to FDA from 107 people in our pilot studies, including Tony, showed that 23 percent of the people that received therapy without active MDMA no longer had PTSD at the end of treatment. This is really pretty good for this patient population. However, when you add MDMA, the results more than double, to 56 percent no longer having PTSD. But most importantly, once people learn that if they don ' t need to suppress their trauma, but they can process it, they keep getting better on their own. So at the 12-month follow-up one year after the last treatment session, two- thirds no longer have PTSD. And of the one-third that do, many have clinically significant reductions in symptoms. On the basis of this data, the FDA has declared MDMA-assisted psychotherapy for PTSD a breakthrough therapy. FDA has also declared psilocybin a breakthrough therapy for treatment-resistant depression and just recently approved esketamine for depression. I ' m proud to say that we have now initiated our Phase 3 studies. And if the results are as we hope, and if they ' re similar to the Phase 2 studies, by the end of 2021, FDA will approve MDMA-assisted psychotherapy for PTSD. If approved, the only therapists who will be able to directly administer it to patients are going to be therapists that have been through our training program, and they will only be able to administer MDMA under direct supervision in clinic settings. We anticipate that over the next several decades, there will be thousands of psychedelic clinics established, at which, therapists will be able to administer MDMA, psilocybin, ketamine and other psychedelics to potentially millions of patients. These clinics can also evolve into centers where people can come for psychedelic psychotherapy for personal growth, for couples therapy or for spiritual, mystical experiences. Humanity now is in a race between catastrophe and consciousness. The psychedelic renaissance is here to help consciousness triumph. And now, if you all just look under your seats... Just joking! Thank you. Thank you. Corey Hajim : You ' ve got to stay up here for a minute. Thank you so much, Rick. I guess it ' s a supportive audience. Rick Doblin : Yes, very. Many of them have also been to Burning Man. CH : There ' s some synergy. RD : CH : So, in your talk, you talked about using these drugs to address some pretty serious traumas. So what about some more common mental illnesses like anxiety and depression, and is that where microdosing comes in? RD : Well, microdosing can be helpful for depression, I do know someone that has been using it. But in general, for therapeutic purposes, we prefer macro-dosing rather than microdosing, in order to really help people deal with the root causes. Microdosing is more for creativity, for artistic inspiration, for focus... And it also does have a mood-elevation lift. But I think for serious illnesses, we ' d rather not get people thinking that they need a daily drug, but do more deeper, intense work. CH : And what about outside the United States and North America, is this research being done there? RD : Oh yeah, we ' re globalizing. Our Phase 3 studies are actually being done in Israel, Canada and the United States. So once we get approval in FDA, it will also become approved in Israel and in Canada. We ' re just starting research in Europe. And we ' re actually going to be training some therapists from China. CH : That ' s great. We were going to do an audience vote to see if people felt like this was a good idea to move forward with this research or not, but I have a feeling I know the answer to that, so... Thank you so much, Rick. RD : Thank you. Thank you all.

Text Sample 618: NEG Dim 4 – 2009 – Score -1.00 – 2009/t002974.txt

So I am a pediatric cancer doctor and stem-cell researcher at Stanford University where my clinical focus has been bone marrow transplantation. Now, inspired by Jill Bolte Taylor last year, I didn ' t bring a human **brain**, but I did bring a liter of bone marrow. And bone marrow is actually what we use to save the lives of tens of thousands of patients, most of whom have advanced malignancies like leukemia and lymphoma and some other diseases. So, a few years ago, I ' m doing my transplant fellowship at Stanford. I ' m in the operating room. We have Bob here, who is a volunteer donor. We ' re sending his marrow across the country to save the life of a child with leukemia. So actually how do we harvest this bone marrow? Well we have a whole O. R. team, general anesthesia, nurses, and another doctor across from me. Bob ' s on the table, and we take this sort of small needle, you know, not too big. And the way we do this is we basically place this through the soft tissue, and kind of punch it into the hard bone, into the tuchus — that ' s a technical term — and aspirate about 10 mls of bone marrow out, each time, with a syringe. And hand it off to the nurse. She squirts it into a tin. Hands it back to me. And we do that again and again. About 200 times usually. And by the end of this my arm is sore, I ' ve got a callus on my hand, let alone Bob, whose rear end looks something more like this, like Swiss cheese. So I ' m thinking, you know, this procedure hasn ' t changed in about 40 years. And there is probably a better way to do this. So I thought of a minimally invasive approach, and a new device that we call the Marrow Miner. This is it. And the Marrow Miner, the way it works is shown here. Our standard see-through patient. Instead of entering the bone dozens of times, we enter just once, into the front of the hip or the back of the hip. And we have a flexible, powered catheter with a special wire loop tip that stays inside the crunchy part of the marrow and follows the contours of the hip, as it moves around. So it enables you to very rapidly aspirate, or suck out, rich bone marrow very quickly through one hole. We can do multiple passes through that same entry. No robots required. And, so, very quickly, Bob can just get one puncture, local anesthesia, and do this harvest as an outpatient. So I did a few prototypes. I got a small little grant at Stanford. And played around with this a little bit. And our team members developed this technology. And eventually we got two large animals, and pig studies. And we found, to our surprise, that we not only got bone marrow out, but we got 10 times the stem cell activity in the marrow from the Marrow Miner, compared to the normal device. This device was just FDA approved in the last year. Here is a live patient. You can see it following the flexible curves around. There will be two passes here, in the same patient, from the same hole. This was done under local anesthesia, as an outpatient. And we got, again, about three to six times more stem cells than the standard approach done on the same patient. So why should you care? Bone marrow is a very rich source of adult stem cells. You all know about embryonic stem cells. They ' ve got great potential but haven ' t yet entered clinical trials. Adult stem cells are throughout our body, including the blood-forming stem cells in our bone marrow, which we ' ve been using as a form of stem-cell therapy for over 40 years. In the last decade there ' s been an explosion of use of bone marrow stem cells to treat the patient ' s other diseases such as heart disease, vascular disease, orthopedics, tissue engineering, even in neurology to treat Parkinson ' s and diabetes. We ' ve just come out, we ' re commercializing, this year, generation 2. 0 of the Marrow Miner. The hope is that this gets more stem cells out, which translates to better outcomes. It may encourage more people to sign up to be potential live-saving bone marrow donors. It may even enable you to bank your own marrow stem cells, when you ' re younger and healthier, to use in the future should you need it. And ultimately — and here ' s a picture of our bone marrow transplant survivors, who come together for a reunion each year at Stanford. Hopefully this technology will let us have more of these survivors in the future. Thanks.

Text Sample 619: NEG Dim 4 – 2009 – Score 1.00 – 2009/t000053.txt

My journey to coming here today started in 1974. That ' s me with the funny gloves. I was 17 and going on a peace walk. What I didn ' t know though, was most of those people, standing there with me, were Moonies. And within a week I had come to believe that the second coming of Christ had occurred, that it was Sun Myung Moon, and that I had been specially chosen and prepared by God to be his disciple. Now as cool as that sounds, my family was not that thrilled with this. And they tried everything they could to get me out of there. There was an underground railroad of sorts that was going on during those years. Maybe some of you remember it. They were called deprogrammers. And after about five long years my family had me deprogrammed. And I then became a deprogrammer. I started going out on cases. And after about five years of doing this, I was arrested for kidnapping. Most of the cases I went out on were called involuntary. What happened was that the family had to get their loved ones some safe place somehow. And so they took them to some safe place. And we would come in and talk to them, usually for about a week. And so after this happened, I decided it was a good time to turn my back on this work. And about 20 years went by. There was a burning question though that would not leave me. And that was,”How did this happen to me?”And in fact, what did happen to my **brain**? Because something did. And so I decided to write a book, a memoir, about this decade of my life. And toward the end of writing that book there was a documentary that came out. It was on Jonestown. And it had a chilling effect on me. These are the dead in Jonestown. About 900 people died that day, most of them taking their own lives. Women gave poison to their babies, and watched foam come from their mouths as they died. The top picture is a group of Moonies that have been blessed by their messiah. Their mates were chosen for them. The bottom picture is Hitler youth. This is the leg of a suicide bomber. The thing I had to admit to myself, with great repulsion, was that I get it. I understand how this could happen. I understand how someone ' s **brain**, how someone ' s mind can come to the place where it makes sense — in fact it would be wrong, when your **brain** is working like that — not to try to save the world through genocide. And so what is this? How does this work? And how I ’ ve come to view what happened to me is a viral, memetic infection. For those of you who aren ’ t familiar with memetics, a meme has been defined as an idea that replicates in the human **brain** and moves from **brain** to **brain** like a virus, much like a virus. The way a virus works is — it can infect and do the most damage to someone who has a compromised immune system. In 1974, I was young, I was naive, and I was pretty lost in my world. I was really idealistic. These easy ideas to complex questions are very appealing when you are emotionally vulnerable. What happens is that circular logic takes over.”Moon is one with God. God is going to fix all the problems in the world. All I have to do is humbly follow. Because God is going to stop war and hunger — all these things I wanted to do — all I have to do is humbly follow. Because after all, God is [working through] the messiah. He ’ s going to fix all this.”It becomes impenetrable. And the most dangerous part of this is that is creates”us”and”them,””right”and”wrong,””good”and”evil.”And it makes anything possible, makes anything rationalizable. And the thing is, though, if you looked at my **brain** during those years in the Moonies — neuroscience is expanding exponentially, as Ray Kurzweil said yesterday. Science is expanding. We ’ re beginning to look inside the **brain**. And so if you looked at my **brain**, or any **brain** that ’ s infected with a viral memetic infection like this, and compared it to anyone in this room, or anyone who uses critical thinking on a regular basis, I am convinced it would look very, very different. And that, strange as it may sound, gives me hope. And the reason that gives me hope is that the first thing is to admit that we have a problem. But it ’ s a human problem. It ’ s a scientific problem, if you will. It happens in the human **brain**. There is no evil force out there to get us. And so this is something that, through research and education, I believe that we can solve. And so the first step is to realize that we can do this together, and that there is no”us”and”them.”Thank you very much.

Text Sample 620: NEG Dim 4 – 2009 – Score 1.00 – 2009/t002964.txt

So, I am indeed going to talk about the spaces men create for themselves, but first I want to tell you why I ’ m here. I ’ m here for two reasons. These two guys are my two sons Ford and Wren. When Ford was about three years old, we shared a very small room together, in a very small space. My office was on one half of the bedroom, and his bedroom was on the other half. And you can imagine, if you ’ re a writer, that things would get really crowded around deadlines. So when Wren was on the way, I realized I needed to find a space of my own. There was no more space in the house. So I went out to the backyard, and without any previous building experience, and about 3, 000 dollars and some recycled materials, I built this space. It had everything I needed. It was quiet. There was enough space. And I had control, which was very important. As I was building this space, I thought to myself,”Surely I ’ m not the only guy to have to have carved out a space for his own.”So I did some research. And I found that there was an historic precedence. Hemingway had his writing space. Elvis had two or three manspaces, which is pretty unique because he lived with both his wife and his mother in Graceland. In the popular culture, Superman had the Fortress of Solitude, and there was, of course, the Batcave. So I realized then that I wanted to go out on a journey and see what guys were creating for themselves now. Here is one of the first spaces I found. It is in Austin, Texas, which is where I ’ m from. On the outside it looks like a very typical garage, a nice garage. But on the inside, it ’ s anything but. And this, to me, is a pretty classic manspace. It has neon concert posters, a bar and, of course, the leg lamp, which is very important. I soon realized that manspaces didn ’ t have to be only inside. This guy built a bowling alley in his backyard, out of landscaping timbers, astroturf. And he found the scoreboard in the trash. Here ’ s another outdoor space, a little bit more sophisticated. This a 1923 wooden tugboat, made completely out of Douglas fir. The guy did it all himself. And there is about 1, 000 square feet of hanging-out space inside. So, pretty early on in my investigations I realized what I was finding was not what I expected to find, which was, quite frankly, a lot of beer can pyramids and overstuffed couches and flat-screen TVs. There were definitely hang-out spots. But some were for working, some were for playing, some were for guys to collect their things. Most of all, I was just surprised with what I was finding. Take this place for example. On the outside it looks like a typical northeastern garage. This is in Long Island, New York. The only thing that might tip you off is the round window. On the inside it ’ s a recreation of a 16th century Japanese tea house. The man imported all the materials from Japan, and he hired a Japanese carpenter to build it in the traditional style. It has no nails or screws. All the joints are hand-carved and hand-scribed. Here is another pretty typical scene. This is a suburban Las Vegas neighborhood. But you open one of the garage doors and there is a professional- size boxing ring inside. And so there is a good reason for this. It was built by this man who is Wayne McCullough. He won the silver medal for Ireland in the 1992 Olympics, and he trains in this space. He trains other people. And right off the garage he has his own trophy room where he can sort of bask in his accomplishments, which is another sort of important part about a manspace. So, while this space represents someone ’ s profession, this one certainly represents a passion. It ’ s made to look like the inside of an English sailing ship. It ’ s a collection of nautical antiques from the 1700s and 1800s. Museum quality. So, as I came to the end of my journey, I found over 50 spaces. And they were unexpected and they were surprising. But they were also — I was really impressed by how personalized they were, and how much work went into them. And I realized that ’ s because the guys that I met were all very passionate about what they did. And they really loved their professions. And they were very passionate about their collections and their hobbies. And so they created these spaces to reflect what they love to do, and who they were. So if you don ’ t have a space of your own, I highly recommend finding one, and getting into it. Thank you very much.

Text Sample 621: NEG Dim 4 – 2017 – Score -1.00 – 2017/t001060.txt

Cancer. Many of us have lost family, friends or loved ones to this horrible disease. I know there are some of you in the audience who are cancer survivors, or who are fighting cancer at this moment. My heart goes out to you. While this word often conjures up emotions of sadness and anger and fear, I bring you good news from the front lines of cancer research. The fact is, we are starting to win the war on cancer. In fact, we lie at the intersection of the three of the most exciting developments within cancer research. The first is cancer genomics. The genome is a composition of all the genetic information encoded by DNA in an organism. In cancers, changes in the DNA called mutations are what drive these cancers to go out of control. Around 10 years ago, I was part of the team at Johns Hopkins that first mapped the mutations of cancers. We did this first for colorectal, breast, pancreatic and **brain** cancers. And since then, there have been over 90 projects in 70 countries all over the world, working to understand the genetic basis of these diseases. Today, tens of thousands of cancers are understood down to exquisite molecular detail. The second revolution is precision medicine, also known as”personalized medicine.”Instead of one-size-fits-all methods to be able to treat cancers, there is a whole new class of drugs that are able to target cancers based on their unique genetic profile. Today, there are a host of these tailor-made drugs, called targeted therapies, available to physicians even today to be able to personalize their therapy for their patients, and many others are in development. The third exciting revolution is immunotherapy, and this is really exciting. Scientists have been able to leverage the immune system in the fight against cancer. For example, there have been ways where we find the off switches of cancer, and new drugs have been able to turn the immune system back on, to be able to fight cancer. In addition, there are ways where you can take away immune cells from the body, train them, engineer them and put them back into the body to fight cancer. Almost sounds like science fiction, doesn ’ t it? While I was a researcher at the National Cancer Institute, I had the privilege of working with some of the pioneers of this field and watched the development firsthand. It ’ s been pretty amazing. Today, over 600 clinical trials are open, actively recruiting patients to explore all aspects in immunotherapy. While these three exciting revolutions are ongoing, unfortunately, this is only the beginning, and there are still many, many challenges. Let me illustrate with a patient. Here is a patient with a skin cancer called melanoma. It ’ s horrible ; the cancer has gone everywhere. However, scientists were able to map the mutations of this cancer and give a specific treatment that targets one of the mutations. And the result is almost miraculous. Tumors almost seem to melt away. Unfortunately, this is not the end of the story. A few months later, this picture is taken. The tumor has come back. The question is : Why? The answer is tumor heterogeneity. Let me explain. Even a cancer as small as one centimeter in diameter harbors over a hundred million different cells. While genetically similar, there are small differences in these different cancers that make them differently prone to different drugs. So even if you have a drug that ’ s highly effective, that kills almost all the cells, there is a chance that there ’ s a small population that ’ s resistant to the drug. This ultimately is the population that comes back, and takes over the patient. So then the question is : What do we do with this information? Well, the key, then, is to apply all these exciting advancements in cancer therapy earlier, as soon as we can, before these resistance clones emerge. The key to cancer and curing cancer is early detection. And we intuitively know this. Finding cancer early results in better outcomes, and the numbers show this as well. For example, in ovarian cancer, if you detect cancer in stage four, only 17 percent of the women survive at five years. However, if you are able to detect this cancer as early as stage one, over 92 percent of women will survive. But the sad fact is, only 15 percent of women are detected at stage one, whereas the vast majority, 70 percent, are detected in stages three and four. We desperately need better detection mechanisms for cancers. The current best ways to screen cancer fall into one of three categories. First is medical procedures, which is like colonoscopy for colon cancer. Second is protein biomarkers, like PSA for prostate cancer. Or third, imaging techniques, such as mammography for breast cancer. Medical procedures are the gold standard ; however, they are highly invasive and require a large infrastructure to implement. Protein markers, while effective in some populations, are not very specific in some circumstances, resulting in high numbers of false positives, which then results in unnecessary work-ups and unnecessary procedures. Imaging methods, while useful in some populations, expose patients to harmful radiation. In addition, it is not applicable to all patients. For example, mammography has problems in women with dense breasts. So what we need is a method that is noninvasive, that is light in infrastructure, that is highly specific, that also does not have false positives, does not use any radiation and is applicable to large populations. Even more importantly, we need a method to be able to detect cancers before they ’ re 100 million cells in size. Does such a technology exist? Well, I wouldn ’ t be up here giving a talk if it didn ’ t. I ’ m excited to tell you about this latest technology we ’ ve developed. Central to our technology is a simple blood test. The blood circulatory system, while seemingly mundane, is essential for you to survive, providing oxygen and nutrients to your cells, and removing waste and carbon dioxide. Here ’ s a key biological insight : Cancer cells grow and die faster than normal cells, and when they die, DNA is shed into the blood system. Since we know the signatures of these cancer cells from all the different cancer genome sequencing projects, we can look for those signals in the blood to be able to detect these cancers early. So instead of waiting for cancers to be large enough to cause symptoms, or for them to be dense enough to show up on imaging, or for them to be prominent enough for you to be able to visualize on medical procedures, we can start looking for cancers while they are relatively pretty small, by looking for these small amounts of DNA in the blood. So let me tell you how we do this. First, like I said, we start off with a simple blood test — no radiation, no complicated equipment — a simple blood test. Then the blood is shipped to us, and what we do is extract the DNA out of it. While your body is mostly healthy cells, most of the DNA that ’ s detected will be from healthy cells. However, there will be a small amount, less than one percent, that comes from the cancer cells. Then we use molecular biology methods to be able to enrich this DNA for areas of the genome which are known to be associated with cancer, based on the information from the cancer genomics projects. We ’ re able to then put this DNA into DNA-sequencing machines and are able to digitize the DNA into A ’ s, C ’ s, T ’ s and G ’ s and have this final readout. Ultimately, we have information of billions of letters that output from this run. We then apply statistical and computational methods to be able to find the small signal that ’ s present, indicative of the small amount of cancer DNA in the blood. So does this actually work in patients? Well, because there ’ s no way of really predicting right now which patients will get cancer, we use the next best population : cancers in remission ; specifically, lung cancer. The sad fact is, even with the best drugs that we have today, most lung cancers come back. The key, then, is to see whether we ’ re able to detect these recurrences of cancers earlier than with standard methods. We just finished a major trial with Professor Charles Swanton at University College London, examining this. Let me walk you through an example of one patient. Here ’ s an example of one patient who undergoes surgery at time point zero, and then undergoes chemotherapy. Then the patient is under remission. He is monitored using clinical exams and imaging methods. Around day 450, unfortunately, the cancer comes back. The question is : Are we able to catch this earlier? During this whole time, we ’ ve been collecting blood serially to be able to measure the amount of ctDNA in the blood. So at the initial time point, as expected, there ’ s a high level of cancer DNA in the blood. However, this goes away to zero in subsequent time points and remains negligible after subsequent points. However, around day 340, we see the rise of cancer DNA in the blood, and eventually, it goes up higher for days 400 and 450. Here ’ s the key, if you ’ ve missed it : At day 340, we see the rise in the cancer DNA in the blood. That means we are catching this cancer over a hundred days earlier than traditional methods. This is a hundred days earlier where we can give therapies, a hundred days earlier where we can do surgical interventions, or even a hundred days less for the cancer to grow or a hundred days less for resistance to occur. For some patients, this hundred days means the matter of life and death.

We ’ re really excited about this information. Because of this assignment, we ’ ve done additional studies now in other cancers, including breast cancer, lung cancer and ovarian cancer, and I can ’ t wait to see how much earlier we can find these cancers. Ultimately, I have a dream, a dream of two vials of blood, and that, in the future, as part of all of our standard physical exams, we ’ ll have two vials of blood drawn. And from these two vials of blood we will be able to compare the DNA from all known signatures of cancer, and hopefully then detect cancers months to even years earlier. Even with the therapies we have currently, this could mean that millions of lives could be saved. And if you add on to that recent advancements in immunotherapy and targeted therapies, the end of cancer is in sight. The next time you hear the word”cancer,”I want you to add to the emotions : hope. Hold on. Cancer researchers all around the world are working feverishly to beat this disease, and tremendous progress is being made. This is the beginning of the end. We will win the war on cancer. And to me, this is amazing news. Thank you.

Text Sample 622: NEG Dim 4 – 2017 – Score 1.00 – 2017/t000748.txt
Words matter. They can heal and they can kill... yet, they have a limit. When I was in eighth grade, my teacher gave me a vocabulary sheet with the word”genocide.”I hated it. The word genocide is clinical... overgeneral... bloodless... dehumanizing. No word can describe what this does to a nation. You need to know, in this kind of war, husbands kills wives, wives kill husbands, neighbors and friends kill each other. Someone in power says,”Those over there... they don ’ t belong. They ’ re not human.”And people believe it. I don ’ t want words to describe this kind of behavior. I want words to stop it. But where are the words to stop this? And how do we find the words? But I believe, truly, we have to keep trying. I was born in Kigali, Rwanda. I felt loved by my entire family and my neighbors. I was constantly being teased by everybody, especially my two older siblings. When I lost my front tooth, my brother looked at me and said,”Oh, it has happened to you, too? It will never grow back.”I enjoyed playing everywhere, especially my mother ’ s garden and my neighbor ’ s. I loved my kindergarten. We sang songs, we played everywhere and ate lunch. I had a childhood that I would wish for anyone. But when I was six, the adults in my family began to speak in whispers and shushed me any time that I asked a question. One night, my mom and dad came. They had this strange look when they woke us. They sent my older sister Claire and I to our grandparent ’ s, hoping whatever was happening would blow away. Soon we had to escape from there, too. We hid, we **crawled**, we sometimes ran. Sometimes I heard laughter and then screaming and crying and then noise that I had never heard. You see, I did not know what those noises were. They were neither human — and also at the same time, they were human. I saw people who were not breathing. I thought they were asleep. I still didn ’ t understand what death was, or killing in itself. When we would stop to rest for a little bit or search for food, I would close my eyes, hoping when I opened them, I would be awake. I had no idea which direction was home. Days were for hiding and night for walking. You go from a person who ’ s away from home to a person with no home. The place that is supposed to want you has pushed you out, and no one takes you in. You are unwanted by anyone. You are a refugee. From age six to 12, I lived in seven different countries, moving from one refugee camp to another, hoping we would be wanted. My older sister Claire, she became a young mother... and a master at getting things done. When I was 12, I came to America with Claire and her family on refugee status. And that ’ s only the beginning, because even though

I was 12 years old, sometimes I felt like three years old and sometimes 50 years old. My past receded, grew jumbled, distorted. Everything was too much and nothing. Time seemed like pages torn out of a book and scattered everywhere. This still happens to me standing right here. After I got to America, Claire and I did not talk about our past. In 2006, after 12 years being separated away from my family, and then seven years knowing that they were dead and them thinking that we were dead, we reunited... in the most dramatic, American way possible. Live, on television — on"The Oprah Show."I told you, I told you. But after the show, as I spent time with my mom and dad and my little sister and my two new siblings that I never met, I felt anger. I felt every deep pain in me. And I know that there is absolutely nothing, nothing, that could restore the time we lost with each other and the relationship we could ’ ve had. Soon, my parents moved to the United States, but like Claire, they don ’ t talk about our past. They live in never-ending present. Not asking too many questions, not allowing themselves to feel — moving in small steps. None of us, of course, can make sense of what happened to us. Though my family is alive — yes, we were broken, and yes, we are numb and we were silenced by our own experience. It ’ s not just my family. Rwanda is not the only country where people have turned on each other and murdered each other. The entire human race, in many ways, is like my family. Not dead ; yes, broken, numb and silenced by the violence of the world that has taken over. You see, the chaos of the violence continues inside in the words we use and the stories we create every single day. But also on the labels that we impose on ourselves and each other. Once we call someone"other,""less than,""one of them"or"better than,"believe me... under the right condition, it ’ s a short path to more destruction. More chaos and more noise that we will not understand. Words will never be enough to quantify and qualify the many magnitudes of human-caused destruction. In order for us to stop the violence that goes on in the world, I hope — at least I beg you — to pause. Let ’ s ask ourselves : Who are we without words? Who are we without labels? Who are we in our breath? Who are we in our heartbeat?

Text Sample 623: NEG Dim 4 – 2017 – Score 1.00 – 2017/t000380.txt

Almost 50 years ago, psychiatrists Richard Rahe and Thomas Holmes developed an inventory of the most distressing human experiences that we could have. Number one on the list? Death of a spouse. Number two, divorce. Three, marital separation. Now, generally, but not always, for those three to occur, we need what comes in number seven on the list, which is marriage. Fourth on the list is imprisonment in an institution. Now, some say number seven has been counted twice. I don ’ t believe that. When the life stress inventory was built, back then, a long-term relationship pretty much equated to a marriage. Not so now. So for the purposes of this talk, I ’ m going to be including de facto relationships, common-law marriages and same-sex marriages, or same-sex relationships soon hopefully to become marriages. And I can say from my work with same-sex couples, the principles I ’ m about to talk about are no different. They ’ re the same across all relationships. So in a modern society, we know that prevention is better than cure. We vaccinate against polio, diphtheria, tetanus, whooping cough, measles. We have awareness campaigns for melanoma, stroke, diabetes — all important campaigns. But none of those conditions come close to affecting 45 percent of us. Forty- five percent : that ’ s our current divorce rate. Why no prevention campaign for divorce? Well, I think it ’ s because our policymakers don ’ t believe that things like attraction and the way relationships are built is changeable or educable. Why? Well, our policymakers currently are Generation X. They ' re in their 30s to 50s. And when I ' m talking to these guys about these issues, I see their eyes glaze over, and I can see them thinking,"Doesn ' t this crazy psychiatrist get it? You can ' t control the way in which people attract other people and build relationships."Not so, our dear millennials. This is the most information- connected, analytical and skeptical generation, making the most informed decisions of any generation before them. And when I talk to millennials, I get a very different reaction. They actually want to hear about this. They want to know about how do we have relationships that last? So for those of you who want to embrace the post-"romantic destiny"era with me, let me talk about my three life hacks for preventing divorce. Now, we can intervene to prevent divorce at two points : later, once the cracks begin to appear in an established relationship ; or earlier, before we commit, before we have children. And that ' s where I ' m going to take us now. So my first life hack : millennials spend seven-plus hours on their devices a day. That ' s American data. And some say, probably not unreasonably, this has probably affected their face-to-face relationships. Indeed, and add to that the hookup culture, ergo apps like Tinder, and it ' s no great surprise that the 20-somethings that I work with will often talk to me about how it is often easier for them to have sex with somebody that they ' ve met than have a meaningful conversation. Now, some say this is a bad thing. I say this is a really good thing. It ' s a particularly good thing to be having sex outside of the institution of marriage. Now, before you go out and get all moral on me, remember that Generation X, in the American Public Report, they found that 91 percent of women had had premarital sex by the age of 30. Ninety-one percent. It ' s a particularly good thing that these relationships are happening later. See, boomers in the ' 60s — they were getting married at an average age for women of 20 and 23 for men. 2015 in Australia? That is now 30 for women and 32 for men. That ' s a good thing, because the older you are when you get married, the lower your divorce rate. Why? Why is it helpful to get married later? Three reasons. Firstly, getting married later allows the other two preventers of divorce to come into play. They are tertiary education and a higher income, which tends to go with tertiary education. So these three factors all kind of get mixed up together. Number two, neuroplasticity research tell us that the human **brain** is still growing until at least the age of 25. So that means how you ' re thinking and what you ' re thinking is still changing up until 25. And thirdly, and most importantly to my mind, is personality. Your personality at the age of 20 does not correlate with your personality at the age of 50. But your personality at the age of 30 does correlate with your personality at the age of 50. So when I ask somebody who got married young why they broke up, and they say,"We grew apart,"they ' re being surprisingly accurate, because the 20s is a decade of rapid change and maturation. So the first thing you want to get before you get married is older. Number two, John Gottman, psychologist and relationship researcher, can tell us many factors that correlate with a happy, successful marriage. But the one that I want to talk about is a big one : 81 percent of marriages implode, self- destruct, if this problem is present. And the second reason why I want to talk about it here is because it ' s something you can evaluate while you ' re dating. Gottman found that the relationships that were the most stable and happy over the longer term were relationships in which the couple shared power. They were influenceable : big decisions, like buying a house, overseas trips, buying a car, having children. But when Gottman drilled down on this data, what he found was that women were generally pretty influenceable. Guess where the problem lay? Yeah, there ' s only two options here, isn ' t there? Yeah, we men were to blame. The other thing that Gottman found is that men who are influenceable also tended to be"outstanding fathers."So women : How influenceable is your man? Men : you ' re with her because you respect her. Make sure that respect plays out in the decision-making process. Number three. I ' m often intrigued by why couples come in to see me after they ' ve been married for 30 or 40 years. This is a time when they ' re approaching the infirmities and illness of old age. It ' s a time when they ' re particularly focused on caring for each other. They ' ll forgive things that have bugged them for years. They ' ll forgive all betrayals, even infidelities, because they ' re focused on caring for each other. So what pulls them apart? The best word I have for this is reliability, or the lack thereof. Does your partner have your back? It takes two forms. Firstly, can you rely on your partner to do what they say they ' re going to do? Do they follow through? Secondly, if, for example, you ' re out and you ' re being verbally attacked by somebody, or you ' re suffering from a really disabling illness, does your partner step up and do what needs to be done to leave you feeling cared for and protected? And here ' s the rub : if you ' re facing old age, and your partner isn ' t doing that for you — in fact, you ' re having to do that for them — then in an already-fragile relationship, it can look a bit like you might be better off out of it rather than in it. So is your partner there for you when it really matters? Not all the time, 80 percent of the time, but particularly if it ' s important to you. On your side, think carefully before you commit to do something for your partner. It is much better to commit to as much as you can follow through than to commit to more sound-good-in-the-moment and then let them down. And if it ' s really important to your partner, and you commit to it, make sure you move hell and high water to follow through. Now, these are things that I ' m saying you can look for. Don ' t worry, these are also things that can be built in existing relationships. I believe that the most important decision that you can make is who you choose as a life partner, who you choose as the other parent of your children. And of course, romance has to be there. Romance is a grand and beautiful and quirky thing. But we need to add to a romantic, loving heart an informed, thoughtful mind, as we make the most important decision of our life. Thank you.

Text Sample 624: NEG Dim 4 – 2016 – Score 1.00 – 2016/t001137.txt

I am holding something remarkably old. It is older than any human artifact, older than life on Earth, older than the continents and the oceans between them. This was formed over four billion years ago in the earliest days of the solar system while the planets were still forming. This rusty lump of nickel and iron may not appear special, but when it is cut open... you can see that it is different from earthly metals. This pattern reveals metallic crystals that can only form out in space where molten metal can cool extremely slowly, a few degrees every million years. This was once part of a much larger object, one of millions left over after the planets formed. We call these objects asteroids. Asteroids are our oldest and most numerous cosmic neighbors. This graphic shows near-Earth asteroids orbiting around the Sun, shown in yellow, and swinging close to the Earth ' s orbit, shown in blue. The sizes of the Earth, Sun and asteroids have been greatly exaggerated so you can see them clearly. Teams of scientists across the globe are searching for these objects, discovering new ones every day, steadily mapping near-Earth space. Much of this work is funded by NASA. I think of the search for these asteroids as a giant public works project, but instead of building a highway, we ' re charting outer space, building an archive that will last for generations. These are the 1, 556 near-Earth asteroids discovered just last year. And these are all of the known near-Earth asteroids, which at last count was 13, 733. Each one has been imaged, cataloged and had its path around the Sun determined. Although it varies from asteroid to asteroid, the paths of most asteroids can be predicted for dozens of years. And the paths of some asteroids can be predicted with incredible precision. For example, scientists at the Jet Propulsion Laboratory predicted where the asteroid Toutatis was going to be four years in advance to within 30 kilometers. In those four years, Toutatis traveled 8. 5 billion kilometers. That ’ s a fractional precision of 0. 000000004. Now, the reason I have this beautiful asteroid fragment is because, like all neighbors, asteroids sometimes drop by unexpectedly. Three years ago today, a small asteroid exploded over the city of Chelyabinsk, Russia. That object was about 19 meters across, or about as big as a convenience store. Objects of this size hit the Earth every 50 years or so. 66 million years ago, a much larger object hit the Earth, causing a massive extinction. 75 percent of plant and animal species were lost, including, sadly, the dinosaurs. That object was about 10 kilometers across, and 10 kilometers is roughly the cruising altitude of a 747 jet. So the next time you ’ re in an airplane, snag a window seat, look out and imagine a rock so enormous that resting on the ground, it just grazes your wingtip. It ’ s so wide that it takes your plane one full minute to fly past it. That ’ s the size of the asteroid that hit the Earth. It has only been within my lifetime that asteroids have been considered a credible threat to our planet. And since then, there ’ s been a focused effort underway to discover and catalog these objects. I am lucky enough to be part of this effort. I ’ m part of a team of scientists that use NASA ’ s NEOWISE telescope. Now, NEOWISE was not designed to find asteroids. It was designed to orbit the earth and look far beyond our solar system to seek out the coldest stars and the most luminous galaxies. And it did that very well for its designed lifetime of seven months. But today, six years later, it ’ s still going. We ’ ve repurposed it to discover and study asteroids. And although it ’ s a wonderful little space robot, these days it ’ s kind of like a used car. The cryogen that used to refrigerate its sensors is long gone, so we joke that its air-conditioning is broken. It ’ s got 920 million miles on the odometer, but it still runs great and reliably takes a photograph of the sky every 11 seconds. It ’ s taken 23 photos since I began speaking to you. One of the reasons NEOWISE is so valuable is that it sees the sky in the thermal infrared. That means that instead of seeing the sunlight that asteroids reflect, NEOWISE sees the heat that they emit. This is a vital capability since some asteroids are as dark as coal and can be difficult or impossible to spot with other telescopes. But all asteroids, light or dark, shine brightly for NEOWISE. Astronomers are using every technique at their disposal to discover and study asteroids. In 2010, a historic milestone was reached. The community, together, discovered over 90 percent of asteroids bigger than one kilometer across — objects capable of massive destruction to Earth. But the job ’ s not done yet. An object 140 meters or bigger could decimate a medium-sized country. So far, we ’ ve only found 25 percent of those. We must keep searching the sky for near-Earth asteroids. We are the only species able to understand calculus or build telescopes. We know how to find these objects. This is our responsibility. If we found a hazardous asteroid with significant early warning, we could nudge it out of the way. Unlike earthquakes, hurricanes or volcanic eruptions, an asteroid impact can be precisely predicted and prevented. What we need to do now is map near-Earth space. We must keep searching the sky. Thank you.

Text Sample 625: NEG Dim 4 – 2016 – Score 1.00 – 2016/t001143.txt

Today I want to talk about the meaning of words, how we define them and how they, almost as revenge, define us. The English language is a magnificent sponge. I love the English language. I ’ m glad that I speak it. But for all that, it has a lot of holes. In Greek, there ’ s a word,”lachesism”which is the hunger for disaster. You know, when you see a thunderstorm on the horizon and you just find yourself rooting for the storm. In Mandarin, they have a word”y y”— I ’ m not pronouncing that correctly — which means the longing to feel intensely again the way you did when you were a kid. In Polish, they have a word”jouska”which is the kind of hypothetical conversation that you compulsively play out in your head. And finally, in German, of course in German, they have a word called”zielschmerz”which is the dread of getting what you want. Finally fulfilling a lifelong dream. I ’ m German myself, so I know exactly what that feels like. Now, I ’ m not sure if I would use any of these words as I go about my day, but I ’ m really glad they exist. But the only reason they exist is because I made them up. I am the author of”The Dictionary of Obscure Sorrows,”which I ’ ve been writing for the last seven years. And the whole mission of the project is to find holes in the language of emotion and try to fill them so that we have a way of talking about all those human peccadilloes and quirks of the human condition that we all feel but may not think to talk about because we don ’ t have the words to do it. And about halfway through this project, I defined”sonder,”the idea that we all think of ourselves as the main character and everyone else is just extras. But in reality, we ’ re all the main character, and you yourself are an extra in someone else ’ s story. And so as soon as I published that, I got a lot of response from people saying,”Thank you for giving voice to something I had felt all my life but there was no word for that.”So it made them feel less alone. That ’ s the power of words, to make us feel less alone. And it was not long after that that I started to notice sonder being used earnestly in conversations online, and not long after I actually noticed it, I caught it next to me in an actual conversation in person. There is no stranger feeling than making up a word and then seeing it take on a mind of its own. I don ’ t have a word for that yet, but I will. I ’ m working on it. I started to think about what makes words real, because a lot of people ask me, the most common thing I got from people is,”Well, are these words made up? I don ’ t really understand.”And I didn ’ t really know what to tell them because once sonder started to take off, who am I to say what words are real and what aren ’ t. And so I sort of felt like Steve Jobs, who described his epiphany as when he realized that most of us, as we go through the day, we just try to avoid bouncing against the walls too much and just sort of get on with things. But once you realize that people — that this world was built by people no smarter than you, then you can reach out and touch those walls and even put your hand through them and realize that you have the power to change it. And when people ask me,”Are these words real?”I had a variety of answers that I tried out. Some of them made sense. Some of them didn ’ t. But one of them I tried out was,”Well, a word is real if you want it to be real.”The way that this path is real because people wanted it to be there. It happens on college campuses all the time. It ’ s called a”desire path.”But then I decided, what people are really asking when they ’ re asking if a word is real, they ’ re really asking,”Well, how many **brains** will this give me access to?”Because I think that ’ s a lot of how we look at language. A word is essentially a key that gets us into certain people ’ s heads. And if it gets us into one **brain**, it ’ s not really worth it, not really worth knowing. Two **brains**, eh, it depends on who it is. A million **brains**, OK, now we ’ re talking. And so a real word is one that gets you access to as many **brains** as you can. That ’ s what makes it worth knowing. Incidentally, the realest word of all by this measure is this. [O. K.] That ’ s it. The realest word we have. That is the closest thing we have to a master key. That ’ s the most commonly understood word in the world, no matter where you are. The problem with that is, no one seems to know what those two letters stand for. Which is kind of weird, right? I mean, it could be a misspelling of”all correct,”I guess, or”old kinderhook.”No one really seems to know, but the fact that it doesn ’ t matter says something about how we add meaning to words. The meaning is not in the words themselves. We ’ re the ones that pour ourselves into it. And I think, when we ’ re all searching for meaning in our lives, and searching for the meaning of life, I think words have something to do with that. And I think if you ’ re looking for the meaning of something, the dictionary is a decent place to start. It brings a sense of order to a very chaotic universe. Our view of things is so limited that we have to come up with patterns and shorthands and try to figure out a way to interpret it and be able to get on with our day. We need words to contain us, to define ourselves. I think a lot of us feel boxed in by how we use these words. We forget that words are made up. It ’ s not just my words. All words are made up, but not all of them mean something. We ’ re all just sort of trapped in our own lexicons that don ’ t necessarily correlate with people who aren ’ t already like us, and so I think I feel us drifting apart a little more every year, the more seriously we take words. Because remember, words are not real. They don ’ t have meaning. We do. And I ’ d like to leave you with a reading from one of my favorite philosophers, Bill Watterson, who created”Calvin and Hobbes.”He said,”Creating a life that reflects your values and satisfies your soul is a rare achievement. To invent your own life ’ s meaning is not easy, but it is still allowed, and I think you ’ ll be happier for the trouble.”Thank you.

Text Sample 626: NEG Dim 4 – 2016 – Score 1.00 – 2016/t001352.txt

By raising your hand, how many of you know at least one person on the screen? Wow, it ’ s almost a full house. It ’ s true, they are very famous in their fields. And do you know what all of them have in common? They all died of pancreatic cancer. However, although it ’ s very, very sad this news, it ’ s also thanks to their personal stories that we have raised awareness of how lethal this disease can be. It ’ s become the third cause of cancer deaths, and only eight percent of the patients will survive beyond five years. That ’ s a very tiny number, especially if you compare it with breast cancer, where the survival rate is almost 90 percent. So it doesn ’ t really come as a surprise that being diagnosed with pancreatic cancer means facing an almost certain death sentence. What ’ s shocking, though, is that in the last 40 years, this number hasn ’ t changed a bit, while much more progress has been made with other types of tumors. So how can we make pancreatic cancer treatment more effective? As a biomedical entrepreneur, I like to work on problems that seem impossible, understanding their limitations and trying to find new, innovative solutions that can change their outcome. The first piece of bad news with pancreatic cancer is that your pancreas is in the middle of your belly, literally. It ’ s depicted in orange on the screen. But you can barely see it until I remove all the other organs in front. It ’ s also surrounded by many other vital organs, like the liver, the stomach, the bile duct. And the ability of the tumor to grow into those organs is the reason why pancreatic cancer is one of the most painful tumor types. The hard-to-reach location also prevents the doctor from surgically removing it, as is routinely done for breast cancer, for example. So all of these reasons leave chemotherapy as the only option for the pancreatic cancer patient. This brings us to the second piece of bad news. Pancreatic cancer tumors have very few blood vessels. Why should we care about the blood vessel of a tumor? Let ’ s think for a second how chemotherapy works. The drug is injected in the vein and it navigates throughout the body until it reaches the tumor site. It ’ s like driving on a highway, trying to reach a destination. But what if your destination doesn ’ t have an exit on the highway? You will never get there. And that ’ s exactly the same problem for chemotherapy and pancreatic cancer. The drugs navigate throughout all of your body. They will reach healthy organs, resulting in high toxic effect for the patients overall, but very little will go to the tumor. Therefore, the efficacy is very limited. To me, it seems very counterintuitive to have a whole-body treatment to target a specific organ. However, in the last 40 years, a lot of money, research and effort have gone towards finding new, powerful drugs to treat pancreatic cancer, but nothing has been done in changing the way we deliver them to the patient. So after two pieces of bad news, I ’ m going to give you good news, hopefully. With a collaborator at MIT and the Massachusetts General Hospital in Boston, we have revolutionized the way we treat cancer by making localized drug delivery a reality. We are basically parachuting you on top of your destination, avoiding your having to drive all around the highway. We have embedded the drug into devices that look like this one. They are flexible enough that they can be folded to fit into the catheter, so the doctor can implant it directly on top of the tumor with minimally invasive surgery. But they are solid enough that once they are positioned on top of the tumor, they will act as a cage. They will actually physically prevent the tumor from entering other organs, controlling the metastasis. The devices are also biodegradable. That means that once in the body, they start dissolving, delivering the drug only locally, slowly and more effectively than what is done with the current whole-body treatment. In pre-clinical study, we have demonstrated that this localized approach is able to improve by 12 times the response to treatment. So we took a drug that is already known and by just delivering it locally where it ’ s needed the most, we allow a response that is 12 times more powerful, reducing the systemic toxic effect. We are working relentlessly to bring this technology to the next level. We are finalizing the pre-clinical testing and the animal model required prior to asking the FDA for approval for clinical trials. Currently, the majority of patients will die from pancreatic cancer. We are hoping that one day, we can reduce their pain, extend their life and potentially make pancreatic cancer a curable disease. By rethinking the way we deliver the drug, we don ’ t only make it more powerful and less toxic, we are also opening the door to finding new innovative solutions for almost all other impossible problems in pancreatic cancer patients and beyond. Thank you very much.

Text Sample 627: NEG Dim 4 – 1984 – Score 7.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ’ ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a computer and really address the qualities of the human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, the intelligence of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ’ ve been able to collect, which I think illustrate this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a slide taken from a

TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ' s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ' s the distance you should sit away from the TV set. Now we ' ve put people 18 inches in front of a TV, and all the artifacts that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow mask, the scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ' m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ' m very interested in touch-sensitive displays. High-tech, high-touch. Isn ' t that what some of you said? It ' s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ' re not : it ' s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on the market these systems that have just a few light emitting diodes on the side and are very low resolution, it ' s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ' t have to pick them up, and people don ' t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ' re typing — what you have — you want to now put something — first of all, you ' ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ' ve got to sort of find that mouse. Then you find the mouse, and you ' re going to have to wiggle it a little bit to see where the cursor is on the screen. And then when you finally see where it is, then you ' ve got to move it to get the cursor over there, and then —"Bang"— you ' ve got to hit a button or do whatever. That ' s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ' m trying to do is just illustrate the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of the computer field, is when you have a bug that you can ' t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the substrate, which it usually is. So we found that that actually was a feature in the sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ’ s pressure-sensitive. And it ’ s pressure-sensitive to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ’ t figure out how to come in the other direction. But let me get rid of the slide, and let ’ s see if this comes on. OK. So there is the pressure-sensitive display in operation. The person ’ s just, if you will, pushing on the screen to make a curve. But this is the interesting part. I want to stop it for a second because the movie is very badly made. And the particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an anvil on a fuzzy rug and the other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across the screen, and yet you touch the ping-pong ball very lightly and it just scoots across the screen. And what you can do — oops, I didn ’ t mean to do that — what you can do is actually feed back to the user the feeling of the physical properties. So again, they don ’ t have to be weight ; they could be a general trying to move troops, and he ’ s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the interface there are physical properties in that transducer — in this case it ’ s pressure and touches — that allow you to present things to the user that you could never present before. So it ’ s not simply looking at the quality or, if you will, the luxury of that interface, but it ’ s actually looking at the idea of presenting things that previously couldn ’ t be presented before. I want to move on to another example, which is one of a different sort, where we ’ re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ’ re going to take this book, if you will, and it ’ s going to come alive. You ’ re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are the experts in monologue making : you make a film and it has a well-formed beginning, middle and end, and in some sense the art of it is that. And you then say,”There ’ s an opportunity for making conversational movies.”Well, what does that mean? And it sort of nibbles at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ’ ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the movie has information about itself. So it knows, or at least there is computer-readable information in the medium itself. It ’ s just not a static movie frame. That ’ s one thing. The other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the cookbook, the”Larousse Gastronomique.”And I think I use the example all too often, but it ’ s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to the end of the recipe and it says,”Cook until done.”Now, that would be, if you will, the top green track, which doesn ’ t mean too much. But you might have to elaborate for me or for somebody who isn ’ t an expert, and say,”Cook at 380 degrees for 45 minutes.”And then for a real beginner, you would go down even further and elaborate more — say,”Open the oven, preheat, wait for the light to go out, open the door, don ’ t leave it open too long, put the penguin in and shut the door...”whatever. And that ’ s a much more elaborate one than you dribble back. That ’ s one kind of use of random access. And the other is where you want to explain the same thing in different ways. If you ’ re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ’ s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ’ s a rather boring book, but I ’ m afraid sometimes you have to do boring books because your sponsors aren ’ t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ’ t even know what vintage the transmission is, but let me just show you very quickly some of it, and we ’ ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of the transmission, and as you rub your finger across the transmission it highlights the various parts. Narrator : When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, superimposed over the illustration. NN : This is about the oil pan, or the oil filter and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get back to the illustration, but in this case it ’ s not a single frame, but it ’ s actually a movie of someone coming into the frame and doing the repair that ’ s described in the text. The two-headed slider is a speed control that allows me to watch the movie at various speeds, in forward or reverse. And the movie is displayed as a full frame movie. I can go back to the beginning... and play the movie at full speed. Here ’ s another step-by-step procedure, only in this case — NN : Okay, this movie is... Everybody ’ s heard of sound-sync movies — this is text-sync movies, so as the movie plays, the text gets highlighted. We highlight the text as we go through the movie. Repairman :... Not too far out. Front poles, preferably. Don ’ t loosen them too far. If you loosen them too far, you ’ ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ’ m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ’ ve been doing — in this case, in Senegal — where we have tried to use personal computers as a pedagogical medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ’ s a story I ’ d like to tell. And that was when, before we did this in any developing countries — we ’ re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ’ ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ’ t read, didn ’ t even make it in the lowest section of the school ’ s classes — and was pretty much not in school, though physically there. But did hang around the, quote,”computer room,”where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very interesting. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,"Let me show you how this works,"and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ' t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ' d actually come to see — not the computer room — they went upstairs and they said,"This is absolutely remarkable! That child was very articulate and showed us and even dealt with the things he couldn ' t do automatically with that manual. It was just absolutely fantastic."The principal said,"There ' s a dreadful mistake, because that child can ' t read. And you obviously have been hoodwinked or you ' ve talked about somebody else."And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,"Can you read?"And the child said,"No, I can ' t."And then they said,"But wait a minute. You just looked through that manual and you found..."and he said,"Oh, but that ' s not reading."And so they said,"Well, what ' s reading then?"He says,"Well, reading is this junk they give me in little books to read. It ' s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return."And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ' m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ' s not, but we think speech —"My God, little children pick it up somehow, and by the age of two they ' re doing a mediocre job, and by three and four they ' re speaking reasonably well. And yet you ' ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder."Well, it ' s not harder. What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ' s going to help you. So what happens is you go to school and people say,"Just believe me, you ' re going to like it. You ' re going to like reading,"and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on the screen has a huge payoff, and it ' s a lot of fun. And in fact, it ' s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ' s the girl on the left leaning with her chalkboard, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ' s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example

— as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And the reason it remains a favorite project is that we were asked to do a teleconferencing system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ’ s faces. So if I wanted to call my friend Peter Sprague on the phone, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ’ s uncanny : there ’ s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ’ s something that didn ’ t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,”He or she wouldn ’ t agree,”and the empty chair is that person and the spatiality is crucial. So we said,”Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ’ s site you ’ d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ’ t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ’ ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ’ s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ’ s site, he ’ ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.

Text Sample 628: NEG Dim 4 – 2003 – Score 1.00 – 2003/t000192.txt
When I first arrived in beautiful Zimbabwe, it was difficult to understand that 35 percent of the population is HIV positive. It really wasn ’ t until I was invited to the homes of people that I started to understand the human toll of the epidemic. For instance, this is Herbert with his grandmother. When I first met him, he was sitting on his grandmother ’ s lap. He has been orphaned, as both of his parents died of AIDS, and his grandmother took care of him until he too died of AIDS. He liked to sit on her lap because he said that it was painful for him to lie in his own bed. When she got up to make tea, she placed him in my own lap and I had never felt a child that was that emaciated. Before I left, I actually asked him if I could get him something. I thought he would ask for a toy, or candy, and he asked me for slippers, because he said that his feet were cold. This is Joyce who ’ s — in this picture — 21. Single mother, HIV positive. I photographed

her before and after the birth of her beautiful baby girl, Issa. And I was last week walking on Lafayette Street in Manhattan and got a call from a woman who I didn ’ t know, but she called to tell me that Joyce had passed away at the age of 23. Joyce ’ s mother is now taking care of her daughter, like so many other Zimbabwean children who ’ ve been orphaned by the epidemic. So a few of the stories. With every picture, there are individuals who have full lives and stories that deserve to be told. All these pictures are from Zimbabwe. Chris Anderson : Kirsten, will you just take one minute, just to tell your own story of how you got to Africa? Kirsten Ashburn : Mmm, gosh. CA : Just — KA : Actually, I was working at the time, doing production for a fashion photographer. And I was constantly reading the New York Times, and stunned by the statistics, the numbers. It was just frightening. So I quit my job and decided that that ’ s the subject that I wanted to tackle. And I first actually went to Botswana, where I spent a month — this is in December 2000 — then went to Zimbabwe for a month and a half, and then went back again this March 2002 for another month and a half in Zimbabwe. CA : That ’ s an amazing story, thank you. KB : Thanks for letting me show these.

Text Sample 629: NEG Dim 4 – 2003 – Score 3.00 – 2003/t000110.txt

This is a sculpture I made, which is a way of, kind of, freeing a form into an object that has different degrees of freedom. So, it can balance on a point. This is a bronze ball, an aluminum arm here, and then this wooden disk. And the wooden disk was really thought about as something that you ’ d want to hold on to, and that would glide easily through your hands. The aluminum is because it ’ s very light. The bronze is nice hard, durable material that could roll on the ground. Inside of the bronze ball there ’ s a lead weight that is free-swinging on an axle that ’ s on two bearings that pass in between, across it, like this, that counterbalance this weight. So it allows it to roll. And the sphere has that balance property that it always sort of stays still and looks the same from every direction. But if you put something on top of it, it disbalances it. And so it would tip over. But in this case because the interior is free-swinging in relation to the sphere, it can stand up on one point. And then there was a second level to this object, which is that it — I wanted it to convey some proportions that I was interested in, which is the diameter of the Moon and the diameter of the Earth in proportion to each other. I was exploring, really early on, wanting to make things float in the air. And I thought up a lot of ideas. This is sculpture that I made that — it ’ s magnetically levitated. The thing is, is that it ’ s slightly dangerous. Normally it ’ s sort of cordoned off when it ’ s in a museum. But it ’ s uh — let ’ s see if I can manipulate it a little bit without, um — oops. So this is just floating, floating on a permanent magnetic field, which stabilizes it in all directions. Except there is a slight tether here, which keeps it from going over the top of its field. It ’ s sort of surfing on a magnetic field at the crest of a wave. And that ’ s what supports the object and keeps it stable. I think we could roll the tape, admin. I have a sort of a collection of videos that I took of different installations, which I could narrate. This is a sculpture of the Sun and the Earth, in proportion. Representing that eight and a half minutes that it takes light and gravity to connect the two. So here is the Earth. It ’ s a little less than a millimeter that was turned of solid bronze. And here is a similar sculpture. That ’ s the Sun at that end. And then in a series of 55 balls, it reduces, proportionately — each ball and the spaces between them reduce proportionately, until they get down to this little Earth. This is in a sculpture park in Taejon. This one is about the Moon and then the distance to the Earth, in proportion also. This is a little stone ball, floating. As you can see the little tether, that it ’ s also magnetically levitated. And then this is the first part of — this is 109 spheres, since the Sun is 109 times the diameter of the Earth. And so this is the size of the Sun. And then each of these little spheres is the size of the Earth in proportion to the Sun. It ’ s made up of 16 concentric shells. Each one has 92 spheres. This is in the courtyard of a twelfth-century alchemist. I was thinking that the Sun is kind of the ultimate alchemist. So this, again, is on the subject — a slice from the equator of the Earth. And then the Moon in the center, and it ’ s floating. And this is in France. This is in Sapporo. It ’ s balancing on a shaft and a ball, right at the center of gravity, or just slightly above the center of gravity, which means that the lower half of the object is just a little bit more weighty. So you can see it rotating here. It weighs about a ton or over a ton. It ’ s made of stainless steel, quite thick. But it ’ s being balanced like that in equilibrium. It ’ s susceptible to motion by the air currents. This is another species of work that I do. These are these arrays. These spheres are all suspended, but they have magnets horizontally in them that make them all like compasses. So all the red sides, for example, face one direction : south. And the blue side, the compliment, faces the other way. So as you turn around you ’ re seeing different colors. This is based on the structure of a diamond. It was a diamond cell structure was the point of departure. And then there were kind of large spaces in the hollows between the atoms. And so I placed one more element of each one of them. These were white spheres. Then I had video projectors that were projecting intermittently onto the spheres. So they would catch parts of the images, and make sort of three-dimensional color volumes, as you walk through it, through the object. This is something I did of a tactile communication system. It was the idea of isolating the tactile component of sculpture, and then putting it into a communication system. This is an idea of moving a sculpture, a ball, that would be directed around the room by a computer. This is a clock I designed. It has Buckminster Fuller ’ s Dymaxion Map edited here. It turns once per day in synchrony with the Earth. And then, this is like projects that are harder to build. This has a diamond-bottomed lake. So it ’ s a floating island with water, fresh water, that can fly from place to place. This would be grown, I suppose, with nanotechnology in the future sometime. In the course of doing my work I sort of have a broad range of interests. And some of it is just the idea of creating media — media as a sculpture, something that would keep the sculpture fresh and ever-changing, by just creating the media that the sculpture is made of. And I had a lot of — always interested in the concept of a crystal ball. And the idea that you could see things inside of a crystal ball and predict the future — or a television set, where it ’ s sort of like a magic box where anything can appear. I had thought about, a long time ago, in the late ’ 60s — when I was just starting out, I was under the influence of thinking about Buckminster Fuller ’ s grand project for an electric globe across from the United Nations — and other things that were happening, the space program at that time, and Whole Earth Catalog, things like that. I was thinking about mass produced spherical television sets that could be linked to orbiting camera satellites. So if we could roll the next film here. This has evolved over the years in a lot of different iterations. But this the current version of it, is a flying airship that is about 35 meters in diameter, about 110 feet in diameter. The whole surface of it is covered with 60 million diodes, red, blue, and green, that allow you to have a high-resolution picture, visible in daylight. I came with a plan. I brought it to Paul MacCready ’ s company AeroVironment to do a feasibility study, and they analyzed it, and came up with a lot of innovative ideas about how to propel it. So we have a physical plan of how to make this actually happen. This is the control room inside of the ship. The idea of this air genie is, it ’ s something that can just transform and become anything. It ’ s like a traveling show. It has speakers on it. And it has cameras over the surface of it. So it can see its environment, and then it can mimic its environment and disappear. Here the legs are retracting. The cabin is open or closed, as you like. It weighs about 20 tons. It has on-board generators. It can generate about a million kilowatts, in order to be bright enough to be visible in daylight. The idea of it is to make a kind of a traveling show. It really would be dedicated to the arts and to interacting. There would be on board a crew of artists, musicians, that would allow the thing to become actually kind of a conscious object that would respond to the moment, and to interact as an entity that was aware, that could communicate. It ’ s completely silent and nonpolluting. It has electric motors with a novel propulsion system. It could be interacted with large crowds in different ways. Primarily I would be interested in how it would interact with, say, going to a college campus, and then being used as a way of talking about the earth sciences, the world, the situation of the globe. The default image on the object would probably be a high-resolution Earth image. But then one could interact with that and show plate tectonics or global warming issues, or migrations — all of the things that we ’ re concerned with today. And then at night the idea is that it would be used as kind of a rave situation, where the people could cut loose, and the music and the lights, and everything. So it could land in a park, for example. Or this could represent a college green. And then it would have a corresponding website that would show the itinerary of this. And so interacting with the same kind of imagery. It would also be able to be an open code, so people could interact with it. It would be forum for people ’ s ideas about what they would like to see on a giant screen of this type. So that ’ s pretty much it. Okay. Thank you.

Text Sample 630: NEG Dim 4 – 2003 – Score 4.00 – 2003/t000124.txt
The new me is beauty. Yeah, people used to say,”Norman ’ s OK, but if you followed what he said, everything would be usable but it would be ugly.”Well, I didn ’ t have that in mind, so... This is neat. Thank you for setting up my display. I mean, it ’ s just wonderful. And I haven ’ t the slightest idea of what it does or what it ’ s good for, but I want it. And that ’ s my new life. My new life is trying to understand what beauty is about, and”pretty,”and”emotions.”The new me is all about making things kind of neat and fun. And so this is a Philippe Starck juicer, produced by Alessi. It ’ s just neat ; it ’ s fun. It ’ s so much fun I have it in my house — but I have it in the entryway, I don ’ t use it to make juice. In fact, I bought the gold-plated special edition and it comes with a little slip of paper that says,”Don ’ t use this juicer to make juice.”The acid will ruin the gold plating. So actually, I took a carton of orange juice and I poured it in the glass to take this picture. Beneath it is a wonderful knife. It ’ s a Global cutting knife made in Japan. First of all, look at the shape — it ’ s just wonderful to look at. Second of all, it ’ s really beautifully balanced : it holds well, it feels well. And third of all, it ’ s so sharp, it just cuts. It ’ s a delight to use. And so it ’ s got everything, right? It ’ s beautiful and it ’ s functional. And I can tell you stories about it, which makes it reflective, and so you ’ ll see I have a theory of emotion. And those are the three components. Hiroshi Ishii and his group at the MIT Media Lab took a ping-pong table and placed a projector above it, and on the ping-pong table they projected an image of water with fish swimming in it. And as you play ping-pong, whenever the ball hits part of the table, the ripples spread out and the fish run away. But of course, then the ball hits the other side, the ripples hit the — poor fish, they can ’ t find any peace and quiet.

Is that a good way to play ping-pong? No. But is it fun? Yeah! Yeah. Or look at Google. If you type in, oh say,"emotion and design,"you get 10 pages of results. So Google just took their logo and they spread it out. Instead of saying,"You got 73, 000 results. This is one through 20. Next,"they just give you as many o ' s as there are pages. It ' s really simple and subtle. I bet a lot of you have seen it and never noticed it. That ' s the subconscious mind that sort of notices it — it probably is kind of pleasant and you didn ' t know why. And it ' s just clever. And of course, what ' s especially good is, if you type"design and emotion,"the first response out of those 10 pages is my website. Now, the weird thing is Google lies, because if I type"design and emotion,"it says,"You don ' t need the ' and. ' We do it anyway."So, OK. So I type"design emotion"and my website wasn ' t first again. It was third. Oh well, different story. There was this wonderful review in The New York Times about the MINI Cooper automobile. It said,"You know, this is a car that has lots of faults. Buy it anyway. It ' s so much fun to drive."And if you look at the inside of the car — I mean, I loved it, I wanted to see it, I rented it, this is me taking a picture while my son is driving — and the inside of the car, the whole design is fun. It ' s round, it ' s neat. The controls work wonderfully. So that ' s my new life ; it ' s all about fun. I really have the feeling that pleasant things work better, and that never made any sense to me until I finally figured out — look... I ' m going to put a plank on the ground. So, imagine I have a plank about two feet wide and 30 feet long and I ' m going to walk on it, and you see I can walk on it without looking, I can go back and forth and I can jump up and down. No problem. Now I ' m going to put the plank 300 feet in the air — and I ' m not going to go near it, thank you. Intense fear paralyzes you. It actually affects the way the **brain** works. So, Paul Saffo, before his talk said that he didn ' t really have it down until just a few days or hours before the talk, and that anxiety was really helpful in causing him to focus. That ' s what fear and anxiety does ; it causes you to be — what ' s called depth-first processing — to focus, not be distracted. And I couldn ' t force myself across that. Now some people can — circus workers, steel workers. But it really changes the way you think. And then, a psychologist, Alice Isen, did this wonderful experiment. She brought students in to solve problems. So, she ' d bring people into the room, and there ' d be a string hanging down here and a string hanging down here. It was an empty room, except for a table with a bunch of crap on it — some papers and scissors and stuff. And she ' d bring them in, and she ' d say,"This is an IQ test and it determines how well you do in life. Would you tie those two strings together?"So they ' d take one string and they ' d pull it over here and they couldn ' t reach the other string. Still can ' t reach it. And, basically, none of them could solve it. You bring in a second group of people, and you say,"Oh, before we start, I got this box of candy, and I don ' t eat candy. Would you like the box of candy?"And turns out they liked it, and it made them happy — not very happy, but a little bit of happy. And guess what — they solved the problem. And it turns out that when you ' re anxious you squirt neural transmitters in the **brain**, which focuses you makes you depth-first. And when you ' re happy — what we call positive valence — you squirt dopamine into the prefrontal lobes, which makes you a breadth-first problem solver : you ' re more susceptible to interruption ; you do out-of-the-box thinking. That ' s what brainstorming is about, right? With brainstorming we make you happy, we play games, and we say,"No criticism,"and you get all these weird, neat ideas. But in fact, if that ' s how you always were you ' d never get any work done because you ' d be working along and say,"Oh, I got a new way of doing it."So to get work done, you ' ve got to set a deadline, right? You ' ve got be anxious. The **brain** works differently if you ' re happy. Things work better because you ' re more creative. You get a little problem, you say,”Ah, I ’ ll figure it out.”No big deal. There ’ s something I call the visceral level of processing, and there will be visceral-level design. Biology — we have co-adapted through biology to like bright colors. That ’ s especially good that mammals and primates like fruits and bright plants, because you eat the fruit and you thereby spread the seed. There ’ s an amazing amount of stuff that ’ s built into the **brain**. We dislike bitter tastes, we dislike loud sounds, we dislike hot temperatures, cold temperatures. We dislike scolding voices. We dislike frowning faces ; we like symmetrical faces, etc., etc. So that ’ s the visceral level. In design, you can express visceral in lots of ways, like the choice of type fonts and the red for hot, exciting. Or the 1963 Jaguar : It ’ s actually a crummy car, falls apart all the time, but the owners love it. And it ’ s beautiful — it ’ s in the Museum of Modern Art. A water bottle : You buy it because of the bottle, not because of the water. And when people are finished, they don ’ t throw it away. They keep it for — you know, it ’ s like the old wine bottles, you keep it for decoration or maybe fill it with water again, which proves it ’ s not the water. It ’ s all about the visceral experience. The middle level of processing is the behavioral level and that ’ s actually where most of our stuff gets done. Visceral is subconscious, you ’ re unaware of it. Behavioral is subconscious, you ’ re unaware of it. Almost everything we do is subconscious. I ’ m walking around the stage — I ’ m not attending to the control of my legs. I ’ m doing a lot ; most of my talk is subconscious ; it has been rehearsed and thought about a lot. Most of what we do is subconscious. Automatic behavior — skilled behavior — is subconscious, controlled by the behavioral side. And behavioral design is all about feeling in control, which includes usability, understanding — but also the feel and heft. That ’ s why the Global knives are so neat. They ’ re so nicely balanced, so sharp, that you really feel you ’ re in control of the cutting. Or, just driving a high- performance sports car over a demanding curb — again, feeling that you are in complete control of the environment. Or the sensual feeling. This is a Kohler shower, a waterfall shower, and actually, all those knobs beneath are also showerheads. It will squirt you all around and you can stay in that shower for hours — and not waste water, by the way, because it recirculates the same dirty water. Or this — this is a really neat teapot I found at high tea at The Four Seasons Hotel in Chicago. It ’ s a Ronnefeldt tilting teapot. That ’ s kind of what the teapot looks like but the way you use it is you lay it on its back, and you put tea in, and then you fill it with water. The water then seeps over the tea. And the tea is sitting in this stuff to the right — the tea is to the right of this line. There ’ s a little ledge inside, so the tea is sitting there and the water is filling it up like that. And when the tea is ready, or almost ready, you tilt it. And that means the tea is partially covered while it completes the brewing. And when it ’ s finished, you put it vertically, and now the tea is — you remember — above this line and the water only comes to here — and so it keeps the tea out. On top of that, it communicates, which is what emotion does. Emotion is all about acting ; emotion is really about acting. It ’ s being safe in the world. Cognition is about understanding the world, emotion is about interpreting it — saying good, bad, safe, dangerous, and getting us ready to act, which is why the muscles tense or relax. And that ’ s why we can tell the emotion of somebody else — because their muscles are acting, subconsciously, except that we ’ ve evolved to make the facial muscles really rich with emotion. Well, this has emotions if you like, because it signals the waiter that,”Hey, I ’ m finished. See — upright.”And the waiter can come by and say,”Would you like more water?”It ’ s kind of neat. What a wonderful design. And the third level is reflective, which is, if you like the superego, it ’ s a little part of the **brain** that has no control over what you do, no control over the — doesn ’ t see the senses, doesn ’ t control the muscles. It looks over what ’ s going on.

It ’ s that little voice in your head that ’ s watching and saying,”That ’ s good. That ’ s bad.”Or,”Why are you doing that? I don ’ t understand.”It ’ s that little voice in your head that ’ s the seat of consciousness. Here ’ s a great reflective product. Owners of the Hummer have said,”You know I ’ ve owned many cars in my life — all sorts of exotic cars, but never have I had a car that attracted so much attention.”It ’ s about attention. It ’ s about their image, not about the car. If you want a more positive model — this is the GM car. And the reason you might buy it now is because you care about the environment. And you ’ ll buy it to protect the environment, even though the first few cars are going to be really expensive and not perfected. But that ’ s reflective design as well. Or an expensive watch, so you can impress people —”Oh gee, I didn ’ t know you had that watch.”As opposed to this one, which is a pure behavioral watch, which probably keeps better time than the $13, 000 watch I just showed you. But it ’ s ugly. This is a clear Don Norman watch. And what ’ s neat is sometimes you pit one emotion against the other, the visceral fear of falling against the reflective state saying,”It ’ s OK. It ’ s OK. It ’ s safe. It ’ s safe.”If that amusement park were rusty and falling apart, you ’ d never go on the ride. So, it ’ s pitting one against the other. The other neat thing... So Jake Cress is this furniture maker, and he makes this unbelievable set of furniture. And this is his chair with claw, and the poor little chair has lost its ball and it ’ s trying to get it back before anybody notices. And what ’ s so neat about it is how you accept that story. And that ’ s what ’ s nice about emotion. So that ’ s the new me. I ’ m only saying positive things from now on.

Text Sample 631: NEG Dim 4 – 2021 – Score -1.00 – 2021/t003768.txt
I had a friend, a songwriting mentor and a great songwriter, John Stewart, who told me just before he died,”We are all just radios, hoping to pick up each other ’ s signals.”I found that to be true and profound. And I ’ ve spent my life trying to clear the static. The strongest signals that come to me are musical in the form of songs or stories told in rhythm and rhyme. Sometimes I can tune in to a revolution in the heart. Sometimes I hear a keen from my Celtic ancestors. And some songs have been postcards from my future. In my family, there was a song for every loss, every celebration, every unspoken need, every longing. But there was also so much chaos that often I was the only one who recognized what was being communicated. My grandmother, Carrie Cash, moved with my grandfather and their five children into a New Deal era colony created by FDR for poor families in the sunken lands of the Arkansas Delta in 1935. So they moved into this freshly painted cottage, and she gave birth to two more babies there with the assistance only of a doctor who came by in a horse and buggy and pulled two loose aspirin from his pocket to give to her. The same pocket in which he kept his fishing worms. That ’ s true. I read once that every time an old woman dies, a library disappears. And before her library disappeared, I tuned into my grandmother ’ s signals and gleaned her tenacity which I borrowed, and her long suffering in her life of constant work with seven children, six of whom made it to adulthood. In a house without electricity. In the sweltering cotton fields, and I wrote these words about her. Five cans of paint in the empty fields And the dust reveals And the children cry the work never ends There ’ s not a single friend Who will hold her hand in the sunken lands? And the mud and tears melt the cotton balls It ’ s a heavy toll Oh, oh His words are cruel and they sting like fire Like the devil ’ s choir Oh, oh Who will hold her hand in the sunken lands? The river rises and she sails away She could never stay Oh, oh Now her work is done in the sunken lands There ’ s

five empty cans In 1992, I was writing a song about my divorce, which hadn ’ t happened yet. But which I saw coming like a freight train. So I was stuck in this song like I was stuck in my life. And the same time across the country, my mother was going through a box of my school assignments and childhood drawings, and she sent it to me. So I started going through this box, and I came across this yellowed paper, this school assignment I had done in the seventh grade in Catholic school on metaphors and similes. Now I vividly remember the pleasure I took in that assignment because it was literally the only thing the nuns had ever given me to do that I enjoyed. So this line I had written **popped** out at me.”A lonely road is a bodyguard.”What did it mean? I had even pasted a picture of this empty road next to the line. It was a metaphor. It wasn ’ t”a lonely road is like a bodyguard,”a simile. It was the much more poetic :”A lonely road is a bodyguard.”As painful as that was then, and as it still can be painful now, I knew what she was telling me. That... Solitude can protect the seeds of creativity. And that loneliness contains a priceless gift if we can tolerate the initial discomfort and avoid the seduction of despair. So my 12-year-old waved at me across the decades, saying that who I was was who I would become. And I waved back and I dropped her line right in the song I was writing. I ’ ll send the angels to watch over you tonight And you send them right back to me A lonely road is a bodyguard If we really want it to be We ’ re all just radios hoping to pick up each other ’ s signals. And some of those signals have a backbeat and a melody, and they ’ re universal. And music can unlock a frozen memory that melts into the seeds of our creativity. And the reverse is also true. A memory can unlock a song that ’ s waiting to be written. It can reveal us to ourselves across time and generations, in community, in solitude. Some frequencies are sound, and some frequencies are light. And some take a half a lifetime to reach their destination. Light is particle and wave Our histories written large upon the page The star in middle age The love that fades to black Once revealed, won ’ t be taken back Light nothing can escape The ignorance we once forgave In the future, if we don ’ t decide to change The things we cannot save But it slows to shine upon your face We owe everything Everything Everything To this rainbow of suffering Light is particle and wave Refractions of this place Reflections of our grace It reveals what we hold dear And it ’ s slow so I can hold you near There ’ s one thing I know for certain. That we clear the static by being vulnerable and by telling the truth. And here we are doing just that. Thank you.

Text Sample 632: NEG Dim 4 – 2021 – Score -1.00 – 2021/t003815.txt

Breast cancer is one of the leading causes of cancer deaths globally. About one in eight US women will develop invasive breast cancer over the course of their lifetime. And globally, millions of women suffer from breast cancer every year. But it is quite treatable if detected early. Right now, actually, mammography is the gold standard for breast cancer diagnosis. But mammography has a 10 percent chance of missed detection. Thousands of lives could be lost each year because of this 10 percent. Today, I ’ m going to introduce a new technology : photoacoustic imaging. As you can see, it provides a much clearer image, leading to a more accurate diagnosis. It will be affordable, just like an ultrasound scan. It ’ s painless and fast, taking only 15 seconds to scan the entire breast in 3D. And immediate results will be delivered to the patients. Beyond breast imaging, this technology will broadly transform how we see inside our bodies – and, maybe one day, even allow us to diagnose cancer via a wearable watch-like device that monitors circulating tumor cells. So what is photoacoustic imaging? Based on a photoacoustic effect, it is a conversion of light energy into sound energy. We shot a gentle laser pulse onto the tissue. The light is absorbed, raising its temperature a bit. The rise in temperature leads to a tiny fraction of volume expansion, which, in turn, generates acoustic waves. Sensors process those sound signals, resulting in a high-resolution image whose level of clarity and detail far surpasses what you ’ ve got with traditional CT scans or ultrasound. Now, about me. I started out in industrial optics, but changed direction after my grandparents died of cancer and stroke. I realized that we needed better imaging technology to aid early diagnosis and to provide a better understanding of the diseases. So I decided to devote myself to biomedical optical imaging. I now research and develop next generation medical imaging with applications ranging from diagnosing cancer to mapping **brain** functions and navigating medical micro robots for drug delivery. Here are some examples showing what we can do. Take this mouse. The mouse has been virtually sliced into 600 pieces from head to toe. It took only 12 seconds to complete the whole body scan. It looks a little like a mouse carpaccio. But don ’ t worry, no mice were hurt during the imaging. In this next video, we hold the animal, another mouse, in position to image its liver. The liver has a lot of blood vessels inside. You can see them as a tree-like network. Because our imaging exposure time is too short, only 15 microseconds, there is no blur at all, despite the movement of the animal during the imaging. The mouse is breathing normally, and every frame in our video is clear. With each slice we can clearly see the internal structure and the blood vessel network. This enables us to differentiate a tumor from normal tissue. The light dose we use is well below the safety limit, and we don ’ t need to inject any contrast agents. It is totally non-invasive. Now, for an example, that is a little closer to home. This is a side by side comparison of human **brain** images. On the left, you see an image from an MRI. On the right, from photoacoustic imaging. Photoacoustic imaging can reveal detailed vasculature, but with even faster detection of the **brain** functions and without using the costly high-magnetic field. What you are seeing here is the **brain** ’ s activity, where a patient, now a human this time, taps his finger, puckers his lips, taps his tongue and is listening and thinking of words. Although I don ’ t have a visual for it, I ’ d like to share one more example. In science fiction, micro robots enter our bodies to cure diseases in hard-to-reach areas. However, in reality, locating, guiding and controlling them inside of the body is a big challenge. Just like the satellites in space guiding cars to their destinations, a photoacoustic imaging system outside the body can serve similarly as a GPS for the micro robots. Biomedical optics has come a long way. The microscope used every day in modern medical diagnosis was invented in the 17th century, which revolutionized 19th century medicine by letting us see into a cell. Then, optical coherence tomography developed in 1990s increased optical penetration to 1 millimeter, bringing huge benefits to clinical care for skin and eyes. Now, photoacoustic imaging, first adopted for medical use in the 2000s, allows us to see even more, allowing penetration by another order of magnitude, to several centimeters, allowing organ-level in vivo human imaging. Photoacoustic imaging is a highly-active and a fast-growing research field. Using microwaves instead of light, this imaging method holds promises for whole body penetration in humans in the future. We are hoping that the further advancement of this technology will aid early diagnosis of cancer and **brain** diseases, ultimately benefiting global health. I hope you all can share my excitement over this fast-growing field and hope you ’ ll join us in advancing the technology. Thank you.

Text Sample 633: NEG Dim 4 – 2021 – Score 1.00 – 2021/t003792.txt

Pasture is the single largest type of land on the surface of the Earth today, thanks to our taste for meat and dairy. Just over a quarter of all land is used for livestock. That ' s more than forests or farm fields or anything else. Most of that land is best for ruminants, such as cows, that can digest high-fiber feed such as grass and straw. However, the process by which grass and fiber is broken up in the stomachs of cows and other grazing ruminants has a byproduct, methane, a potent greenhouse gas. Despite what you might have heard about methane and cows, most of the methane is actually burped out, not through the back end. And that represents about two billion tons of carbon dioxide equivalent per year, or more than four percent of our annual global greenhouse gas emissions. We have a methane problem from cows. So how can you reduce these methane burps? My colleagues and I may have found a solution. Seaweed! Let me explain. A couple of years ago, an article was published by Rob Kinley and colleagues that showed almost complete elimination of methane when seaweed was added to chopped grass in the lab. Great. But as an agricultural researcher, I know lots of additives work well in the lab, but not in real animals. But there was something different about seaweed and the way in which it reduced methane. So we thought we should test this in live animals. In collaboration with Joan Salwen, an entrepreneur, and colleagues from James Cook University and CSIRO, we decided to conduct a small experiment to determine the amount of seaweed we might need to use. This was the first ever experiment in dairy cattle, and we had no idea how much to give them. So we started with about 60 grams per day, going up to 250 grams. Mind you, this was mixed in with 25 kilograms of their feed. One of the graduate students I work with, Breanna Roque, trapped their methane burps. In that first experiment, the emissions were reduced by up to 67 percent. And I thought at first, the equipment must have malfunctioned, but it was real. But we are left with more questions than answers. Would the microbes in the gut get used to it and start producing methane over time? Would the seaweed be stable over a long period of time in storage? Would the taste be affected and the cows turn up their noses? Or would the seaweed affect the cows ' s health or milk production? So we teamed up again to conduct another trial. Over a five-month period, we saw the seaweed that was harvested three years prior reduce emissions by over 80 percent. Our colleagues in Australia, they saw up to 98 percent reduction in a similar trial. That kind of reduction is simply staggering. And in the graph, you see methane emissions in three levels of intake. So the first line is for cattle with no seaweed, The second line is for cows that were supplemented with about 30 to 40 grams of seaweed. And the last one is for cows supplemented with about 60 to 80 grams of seaweed. As you can see, as you increase the amount of seaweed, you see a reduction in methane emissions. We have also seen an improvement in bulking up of the beef cattle with no adverse health effects. So it ' s a win for the environment ; it ' s a win for the farmers and consumers. A panel of 112 people got to taste steak made from steers offered seaweed and control. And they did not detect any difference. We also did a nutritional quality of the meat, and we found no difference between animals that were offered seaweed and the control. So how does it work? Some seaweeds contain ingredients that directly inhibit microbes in the cow ' s gut from forming methane without interfering with food digestion. The amount of methane produced is dependent on how much the animal is eating and what ' s in the diet. And as such, previous efforts to reduce methane emissions focused on changing their diets or improving forage quality. And we do have potential solutions other than seaweed to reduce methane emissions. We looked at additives to feed, such as 3-NOPs, reduce emissions by about 30 percent. Even garlic and citrus extract can reduce methane burps by over 20 percent without affecting animal health and productivity. Now you may ask : Why not stop eating beef and drinking milk? Audience member : Yeah! EK : That would be a good question. So adopting a plant-based diet with supplements may help shrink a person ' s carbon footprint in high-income countries like the US. But, you know, a lot of people are not going to do that. And for the rest of the world, these foods are needed to provide key nutrients, such as vitamin B12 and vitamin A, which are critical for **brain** function, for vision and immunity. These are found almost exclusively in beef and milk. In this graph, you see different countries with their annual meat consumption per capita on the bar graph, and then you see the dots representing the stunting rate in children below the age of five years old. And countries that have lower consumption of methane are associated with higher incidence of stunting. I know this firsthand. Growing up in Eritrea, I loved milk and meat when you could get it, which wasn ' t often. Even as a kid, I always wondered how a cow, eating just grass, produces nutritious milk I love to drink. And that wonder pushed me into a career to understand and improve livestock production so that people in low-income countries do not suffer with stunting and other nutrient-deficiency diseases. So what now? We know seaweed can work. But the cultivation of the specific seaweed has been a barrier. It ' s not so easy to farm asparagopsis taxiformis. But there are a number of efforts going on at the moment to scale up production. Blue Ocean Barns is growing seaweed in Hawaii already, and they estimate that there will be enough production to feed all cattle in the US by 2030. All we need now is for governments to step up and facilitate the use of these methane-busting feed additives. Cattle industries in some countries have already committed to climate neutrality by 2050. But if we can get these feed-additive innovations into cow stomachs earlier, we can cut methane burps significantly. Given that methane only lasts in the atmosphere for a decade or so, we could even slow global warming in the short term. Yes, we have a methane problem from cows, but we may have seaweed and other solutions for these methane burps, helping us provide meat and dairy while maintaining a safe climate. Thank you.

Text Sample 634: NEG Dim 4 – 2023 – Score 1.00 – 2023/t003290.txt

I was born and I grew up in Ivory Coast, and like so many other African kids, I used to resent my Blackness, including my hair. I longed for straight hair that could be easily managed and brushed. After years of using hair relaxer, I went back to natural hair at 16 after an incident with my hair that forced me to shave and start over, and it was a very hard new beginning. But at 18, while I was still in the process of learning how to love my natural hair, I came across a photo album showing the hair of African women in pre-colonial society, and I was instantly inspired. Seeing that photo album was so powerful to me. It made me see hair completely differently. Not just as a beauty tool, but as a way to communicate and to tell stories about who we are. And immediately, I started to experiment with my own hair. I wanted to see what I was able to create. And today I ' m an artist and hair sculpting is one of my media. I use my hair to advocate for what I believe in, particularly for issues that affect women and girls. But it didn ' t start that way. At the beginning, I was just doing beautiful shapes for the aesthetic aspects. I started with doing simple geometrical shapes just for the beauty of it, and when I was posting those shapes online, I was receiving so much love that it encouraged me to push my limit to create more complex sculpture. From 2017 when I first started until now, I ' ve made hundreds of sculptures. Some can be playful, others can be very serious. So here are just a few examples of what I can do with my hair. Animals. Random things, like an umbrella or car. Portraits. Audience : Wow! Text, like here where I write"perfect"with my hair because I do believe my hair is perfect. Cartoon. Sunflowers. Fashion accessories. Wings. Even zodiac-inspired hairstyle because my rising sign is Scorpio. I ' m another astrology freak. Medusa and even body parts. Thanks. The process to make the sculpture is actually pretty simple. The idea comes to my head, I do a little sketch and then I just sculpt, using my natural hair, some hair extension and wire. I just wrap the braid around the wire, and then it ' s very easy for me to shape as I want. And in general, you know, to make the sculpture being able to stand on top of my head, the base needs to be extremely tight. So if you ' re wondering, is it hurtful? Yes. Yes it is. But this is why, in general, I quickly undo it after the photo shoot. Most of the time I will take the picture by myself, but when I need help, my little sister, Florencia, who is the most supportive person on the planet, will help me. And depending on what I want to create, a sculpture can take me from five minutes to more than three, four or five hours. When I started to create, I was just doing, as I said before, for the beauty of it, no meanings. But it changed in 2018 when one of my photo series went completely viral. It was this one. I shaped my hair as a second pair of hands, very playful, in various configurations, and I started to receive a lot of messages from Black women around the world telling me that seeing my hair helped them to feel better about their hair, about their Blackness. And I realized, OK, actually, what I ' m doing is pretty powerful. So maybe it can just serve a greater purpose and I can use it to advocate for change. So I decided to use my hair as a platform to advocate for the equity of sexes. Why this specific subject, you may ask? Because I was born and I grew up in an environment that normalized the bad treatment of women. I have my experience, many women have their experience, so it touched me a lot. And when it comes to equality of sexes, I touch a lot of different subjects. Unfortunately, I can ' t share all of them with you. I can just give you today a glimpse of it. Bodily autonomy and self-love is very important for me. I went myself into a long process towards self love. Coming from a place where I used to dislike everything about myself to a place where I just love everything. So I love to inspire myself – So I love to inspire myself from our female body to create sculpture and shapes that are associated with uplifting words to empower those who need it. And with this sculpture, what I want to do is to encourage women to have agency over their body, but also to destigmatize all the classic taboos around our bodies, like, for example, periods. Aging, disabilities and much more. This sculpture, for example, was to – Was to address the big taboo of body hair on women. When I was 15, I posted a photograph of me on social media without realizing that my armpit hair was visible and immediately, the comments were wild."This is disgusting.""Go shave.""Ew, I can buy you a razor if you want."I was traumatized, so I deleted the picture and after that I started to shave way more regularly. Even if, for me, it ' s a very uncomfortable routine. It took me years to be able to love my body the way it is and to embrace everything, including my body hair. Besides all the topics related to bodies, I love to also address topics related to our intellect as women, because this world doesn ' t only encourage women to dislike the body, but also encourages women to dislike what they have inside. And I think just as we ' re supposed to love our bodies, we ' re supposed to also own our hopes, dreams, our ideas or opinion without shame. And for that, I think – And for that I think education is extremely important. This is why I did this sculpture. Unfortunately, in Ivory Coast, many women still can ' t read. When I was a child, many women were working at our home as nannies. They weren ' t educated. They would ask me to read the text messages because they couldn ' t. In rural Ivory Coast, many parents think that it ' s useless to put a girl in school because they are better at home learning how to cook or to take care of of the home for their future husbands. But when it comes to their son, they are allowed to have an education. So most of the time those women will end up marrying older men, and when those men abandon them, they are left with nothing, with very few career options, because they are not educated. And to me, this is revolting. Growing up in an environment where I wasn ’ t always encouraged to love who I was, my journey has been about becoming my own friend and advocate. And when I embraced the capacity I have to share my stories and the ones of others, I found the strength to advocate for what I believe in through my hair. Thank you.

Text Sample 635: NEG Dim 4 – 2023 – Score 1.00 – 2023/t003334.txt
When I was a medical student at UCLA, one of my patients was a young boy suffering from dozens of seizures daily. His family was going bankrupt, paying out of pocket for a cocktail of drugs with little benefit. Between the side effects and his constant seizures, this boy was wasting away. On a return visit, I was in disbelief to find that he was now seizure-free on most days. His mom beamed, telling me he was going to school for the first time, at the age of 10, and attributed this life-changing improvement to a regimen of natural supplements. It sounded incredible, but this was an anecdote. But it gave me a radical idea. What if it was easy to prove the effects of natural products? Maybe anyone could treat their own health affordably, more effectively, and with less side effects. No pharma or insurance needed. You see, for millennia, humans relied on plants as medicine. It was only a century ago that pharma created synthetic drugs, and for the first time ever, medicine could be patented and highly profitable. But to prove the safety and effectiveness of these never-before-seen compounds, slow, expensive clinical trials were invented. The natural medicines can ’ t be patented and can ’ t afford these trials. Ironically, nearly half of all FDA-approved drugs were first discovered in plants, but pharma created a synthetic version and only funded trials into what they could own and monopolize. And that ’ s why you see these natural products, a vast majority of them haven ’ t been able to prove they ’ re better than placebo. But AI, virtualization and the ubiquity of smartphones is changing all of that. What if it was easy to prove the right natural product, the right dose, for the right person and the right condition? You could bypass a broken healthcare system, go to the grocery store and buy the product proven for someone like you. It ’ s personalized and accessible and affordable for all. So to make this proof-generation easy, Pelin Thorogood and I created a B Corp called Radicle Science. We ’ re using AI-driven, crowdsourced trials so these natural products can generate proof at a fraction of the cost and time. Volunteers automatically get mailed product or placebo, and we collect data via smartphones and wearables. And since these are direct-to-consumer, anyone can join, including those historically excluded from trials – women, minorities, rural populations. You see, trials have studied mostly white, urban men. Last time I checked, I don ’ t think that ’ s me, and for my cofounder, that wasn ’ t her either. But direct-to-consumer crowdsourced trials can finally generate diverse data that ’ s relevant to all of us. So what happens when proof is easy? You get medicines that are natural, personalized, proven and democratized. Medicines that are directly available for all future generations. Because it turns out no one can patent nature. Thank you.

Text Sample 636: NEG Dim 4 – 2023 – Score 1.00 – 2023/t003338.txt
The fairytale industrial complex has been lying to you. Through a pervasive web of marketing, advertising, music, movies and more, it ’ s said that while you can ’ t buy love, you can buy your worthiness of being loved. Its comodofication of romantic love as a storybook fantasy can be summed up in three simple words :”Happily ever after.”Happiness, forever. If only you ’ re lucky enough and good enough and good-looking enough to be chosen by a high-status partner, ideally in a romantic comedy-worthy moment. Over the last 10 years, I have spoken to thousands of people about their romantic hopes and dreams, and I can report that for many, their vision of an ideal relationship is straight out of the fairytale fantasy playbook. Which is no wonder, given that”happily ever after”is used to sell us everything, from makeup to cars to chewing gum. We ’ ve incorporated this propaganda into our real-life approach to relationships, which disconnects us from love, which disconnects us from self-worth and causes genuine confusion about compatibility. Instead of making us feel that love is an abundant, infinitely renewable resource inside of us, which it is, it tries to convince us that love is external and scarce. It ’ s time to consider a new possibility for our collective romantic future. One that centers self-love, self-worth and prioritizes making romantic choices in alignment with our authentic values. No purchase required. Despite what you ’ ve been told, each of us holds the keys to true love. A true-love relationship, as I define it, has the foundational values of love, respect, intimacy, safety, commitment, adoration and joy. When you approach romantic love from this foundation, you empower yourself to transcend the shallow fantasy of love you ’ ve been sold for so long. With sincere intention, the courage to open your heart to be vulnerable and the cultivation of dating and love skills, true love can be a reality for all. How could it be any other way? You are a born love genius. You were born knowing how to love. All of the love that you ’ ve ever experienced and ever will experience is inside of you right now. The more that we cultivate the love within, the better it is for everyone. That ’ s because how we each individually approach love contributes to how we all collectively experience it. We have the opportunity to transcend the manipulations of the fairytale industrial complex and embrace a brighter, more intentional, more inclusive romantic future for all. One that sees us all as worthy without exception. Thank you.

Text Sample 637: NEG Dim 4 – 2022 – Score -1.00 – 2022/t003570.txt
About a decade ago, I met someone who had experienced a few episodes of schizophrenia. They had felt that their sense of self, of what it feels like to be them, changing somewhat. The boundaries of their body began to feel a bit nebulous. Even their psychological self felt a bit porous at times. They were experiencing what could be called an altered sense of self. Over the years, I met many such brave and insightful people who shared what it ’ s like to live with their altered selves. And by”altered,”I mean”different,”not”deficient,”while acknowledging that coping with altered selves can be a struggle at times. So speaking with them, and with theologians, philosophers, neuroscientists, I came to understand that this self that each one of us takes oneself to be is not as real as it seems. The self is a slippery subject. We all intuitively know what it means. It ’ s there when we wake up. It disappears when we fall asleep. It reappears in our dreams. It ’ s what makes us who we are. It seems solid, unchanging, permanent. And yet, we can examine aspects of the self that seem real to us, and ask,”Just how real are they?”Take, for instance, the question”Who am I?”The most likely answer you will get or give to such a

question will be in the form of a story. We tell others – and indeed, ourselves – stories about who we are. We take our stories to be sacrosanct. We are our stories. But a condition that most of us, sadly, will be familiar with – Alzheimer ' s disease – tells us something quite different. Alzheimer ' s begins by affecting short-term memory. Think about what that does to someone ' s story. In order for our stories to form, to grow, something that just happens to us has to first enter short-term memory, and then, get incorporated into what ' s called long-term episodic memory. It has to become an episode in our narrative. But what if the experience doesn ' t even enter short-term memory? That ' s exactly what Alzheimer ' s does. In the beginning, Alzheimer ' s impairs the formation of short-term memory. It impairs the growth of the narrative. It ' s as if our stories begin stalling upon the onset of the disease. Eventually, Alzheimer ' s eats away at all the long-term memories. So if you were to meet someone with mid-stage Alzheimer ' s, they will likely be able to tell you stories about who they are. But if you know their real stories, you ' ll be able to tell that they sometimes scramble up their narrative, that they sometimes mix up the sequence of episodes from their lives. It ' s as if they are recalling their own stories in ways that are not quite accurate. It ' s important, at this stage, to realize that there is still a person experiencing that scrambled narrative. Sadly, Alzheimer ' s goes on to destroy one ' s narrative, and so much more. And towards the end, it ' s unclear whether there is still someone experiencing something, because the person cannot communicate verbally anymore. And yet, Alzheimer ' s tells us that these stories that we take ourselves to be, what philosophers call the"narrative self,"these are spun by the **brain** and body. They are constructions. Sometimes, the constructions are disrupted, even destroyed. And while that is horrific for the person experiencing it, and for their caregivers, it is nonetheless a window onto the constructed nature of our narrative self. And when the construction goes wrong, we perceive our own stories in ways that are not quite real. From the narrative self, let ' s talk about our body. Let ' s take a very basic aspect of our bodily self. This feeling we all have, that we are owners of our body and body parts, that our bodies and body parts belong to us. It seems such a strange thing to think that it could even be otherwise. If I were to ask you,"Does your hand belong to you?"you ' re going to say,"Of course it does. What a foolish question."But not everyone would agree. Early on in my research, a neuropsychologist alerted me to a condition called xenomelia, or foreign limb syndrome. You may have heard of something called phantom limb syndrome, in which people who have had an amputation feel the presence of that limb, sometimes. Xenomelia is somewhat of an opposite condition, where people feel like some part of their body – usually the extremities, their hands or legs – don ' t belong to them. So this neuropsychologist talked of phantom limb syndrome as animation without incarnation. So the limb is gone, it ' s not incarnate anymore, but it ' s animated in your mind. And he talked of xenomelia as incarnation without animation. So the limb is present, healthy even, incarnate, and yet, in your own mind, it feels like it doesn ' t belong to you. So in xenomelia, the **brain** and bodily processes that give rise to our sense of ownership of our body parts, they ' re misfiring, so to speak, and the consequences can be serious. People with xenomelia will sometimes take extreme measures to get rid of, to amputate their foreign-seeming body parts. From the perspective of the self, though, xenomelia is telling us something very profound. It ' s telling us that something as basic as the sense of ownership of our own body parts is a construction. And sometimes, the construction goes wrong, and we perceive our own bodies in ways that are not quite real. Let ' s take another aspect of our bodily self. It ' s called the sense of agency. So when I do something like pick up a cup, I have this implicit feeling that I am the agent of that action, that I have willed that action into existence. That feeling is the sense of agency. But someone with schizophrenia may not have that feeling, always. Someone with schizophrenia might do something and not feel like they are the agent of that action. So schizophrenia tells us that it is possible to be someone who does something but doesn ’ t have an accompanying sense of agency. So just like the narrative self and the sense of ownership of body parts, the sense of agency is also a construction, and it, too, can fail. So you can see where this is going. Let me take one more example to drive home this point. Let ’ s talk of what it feels to be a body here and now. Not the feeling of being a story, but the feeling of being a body in the present moment. Psychologists estimate that about five percent of the general population will, at some point in their lives, have an out-of-body experience. Let ’ s assume that all of us right now are having an in-body experience. But what that means is having this feeling of being in a body, being anchored to a body, occupying a certain volume of space and looking at the world from behind our eyes. But if you are having an out-of-body experience, you could possibly be feeling that you ’ re up near the ceiling, looking down at your own body sitting in the chair below. People do report such experiences, and mild versions of this have been replicated in labs. But if you think, like I do, that out-of-body experiences are the outcome of **brain** processes that are misfiring, then it stands to reason that the experience of being in-body, of being embodied, is itself a construction, and that, too, can come apart. So what are these experiences of altered selves telling us? They ’ re telling us that just about everything we take to be real about ourselves –”real”in the sense that we think we are always experiencing undeniable truths about our bodies, our stories – well, that ’ s just not the case. So when theologians and philosophers tell us that the self is an illusion, this is partly what they mean. You may have realized by now that there still remains the question of who or what is doing the experiencing, even in the case of altered selves. This experiencing”I”in the question”Who am I?”is at the heart of the debate about the self. This experiencing”I”doesn ’ t go away if one or a few aspects of the self are disrupted. But what if all of the aspects of the self that comprise us were to be disrupted? Would the experiencing”I”disappear? We don ’ t have a satisfactory answer to that question, yet. It ’ s possible that the experiencing”I”is also an illusion, in the sense of being a construction, a construction without a constructor. That debate, however, is somewhat unresolved. Despite such doubts, I, personally – whatever I am – think that the self has no reality outside of the **brain** and body. I think that the experiencing”I”will not persist after the body is gone. So what does one make of such knowledge? Well, firstly, these ideas will feel liberating to some and might sit heavily upon others. Regardless, I think we can all attend to the stories that we think we are. Our feelings and emotions are modulated by our stories, and in turn, our feelings and emotions become part of our stories. And our stories, our narratives, are not just cognitive – they live in our bodies, and our bodies structure and shape our stories. So knowing all this, recognizing the constructive nature of it all, maybe we can hold on less tightly to our stories. Maybe we can learn to let go. But that ’ s easier said than done, because the thing that is doing the letting go is also the thing that has to be let go of. Maybe we can just marvel at the efforts of people over millennia, from the Buddha sitting under the Bodhi tree to the modern philosopher and neuroscientist who has asked themselves the question”Who am I?”But most of all, I think we owe a debt to those amongst us who bravely bear witness to our altered selves – whether we do so voluntarily, like monks and nuns do when they meditate, or whether it ’ s brought upon us by biology and circumstance. There is something remarkably robust about the processes that give rise to the totality of our sense of self. But there ’ s something frighteningly fragile about them too. They can crack. And any one of us, at any time in our lives, may have to confront such cracks. And that knowledge, I believe, should make us empathetic towards those of us dealing with altered selves. But I also believe that altered selves should not be seen as the outcome of deficits, or as the outcome of a lack of attributes considered normal. They are different ways of being, and it ’ s the willingness of some of us to confront the self ’ s constructed nature that is helping make sense of the self for all of us. Thank you. Thank you.

Text Sample 638: NEG Dim 4 – 2022 – Score -1.00 – 2022/t003583.txt
On a usual Saturday, you wouldn ’ t find me here. And I mean, of course, I don ’ t give TEDx Talks every Saturday. But what I mean is you probably wouldn ’ t find me in a museum. Further, if you were looking for me, you ’ d probably have to go to the mountains. And that is because I absolutely love hiking. And in addition to just being outside and being in the nature, what I really like is this feeling of physical exhaustion, sort of this satisfaction you can feel in every single muscle. And in the evening when it ’ s time to go to bed, I absolutely cherish this amazing sleepiness that fills you from head to toe. Now, you might say,”Well, of course she ’ s tired. She ’ s been hiking all day.”Plus, not to forget about the effects of supposedly fresh air. But there is one factor that most likely outweighs the effects of fresh air and perhaps even physical activity, and that is exposure to natural daylight. And today I hope to convince you that we should all appreciate daylight as sort of of a natural soporific. And hopefully by the end of the talk, you will also recognize that we should all pay more attention to our daily light diet. But let ’ s start with a question, why daylight is so important for our organism, for health and eventually for sleep. Now, throughout your body, in each and every single cell, there is a tiny molecular clock ticking inside it. And to keep these millions of clocks in sync, we have one central body clock that is located in the **brain** in an area which we call the hypothalamus. And like the conductor of an orchestra, it communicates the time of day to each of these molecular clocks. And this way it is able to regulate bodily processes in sync with the time of day. For example, it causes your body temperature to increase in the morning and to decrease in the evening. It choreographs the release of certain hormones at a proper time, and in the evening when it ’ s time to go to bed, it will also make you tired and sleepy. But given this precise synchronization between internal or biological time and external time or environmental time, it seems clear that the body clock cannot be blind or isolated or shut off from the environment, but rather it needs to receive information about the time of day from the environment for it to synchronize with sun time. And this is achieved by close connections between the internal biological clock in the **brain** and our eyes. And now you may know that in the human retina, there are different types of receptors, photoreceptors, so receptors that sense light. And classically, we distinguish two types : the rods and the cones. Now, the rods, they only contribute to a visual impression under very **dim** lighting conditions. And here in the background, you now see a picture that might more or less be brought about by only the involvement of the rods. What you can see is that it ’ s only shades of gray, it ’ s slightly blurred, and around the point of fixation, which is indicated by the gray dot here, you have a little scotoma, so an area where you can ’ t really see anything. Now, who recognizes what that is? Yeah? Yeah, excellent. But it ’ s going to be way easier, and the majority of you will recognize what it is when I now switch to the next slide, which is brought about by the involvement of the cones. Of course, it ’ s the town hall of Basel, but now you can appreciate the colors, fine details,

and if this was animated, you ’ d even see fine details in motion. But this is not the whole story. Because only fairly recently, only in the early 2000s, another type of cell has been discovered, and we call them retinal ganglion cells. Now, you might ask yourselves, well, What picture ’ s she going to show next? But I have to disappoint you. I actually can ’ t show another picture because from all we know, these cells do not contribute to a visual impression. But they ’ re exclusively designed to sense short wavelength proportions in daylight. Sometimes we also call this blue light. So they ’ re designed to extract important information about the time of day from the environment and pass this on to the internal biological clock in the **brain**. And I guess you ’ ve all experienced how well this biological timing system, this connection between our biological clock and the external world or our eyes works, when we, for example, travel across time zones, because with a speed of approximately one hour per day, you can fairly easily adapt to the new time zone. Now, how much light do we actually need? How much light is enough for the positive effects on, for instance, sleep to occur? And I have to admit, this is not so easy to answer. But I think what we have to keep in mind is that the biological timing system has evolved under the open sky and not in offices or museums. So it is also optimally tuned to the conditions we find outside. Office light or light in rooms is in no way comparable to what we find outside. And therefore, from a scientific perspective, I can only recommend you to spend as much time as you can under the open sky, but try to make it at least 30 minutes per day. Now, let ’ s finally talk about sleep. And I ’ ve repeatedly alluded to the fact that daylight is beneficial and important for sleep. And in this context, I ’ d like to share a little story : So last year I had to go to hospital for surgery. And generally the hospital environment is a very challenging one for sleep because you spend way too much time in bed not moving much, you might be in pain, now and then someone comes in to check on you even during the night, and if you ’ re as lucky as I was, you have a snoring roommate, and because you, of course, don ’ t get a lot of daylight. And that ’ s why many people, while they ’ re in hospital, ask for sleep medication. But as I consider myself to be a generally good sleeper, and also because I know how these drugs work, I didn ’ t want this. But I knew there was something I could do to help my body sleep as well as possible, even in this situation, and that was to maximize exposure to daylight. Because as a sleep scientist, of course I know about the research findings. I know that the more daylight that you get, the more tired you will be in the evening, the easier it will be for you to fall asleep and better the experienced sleep quality. And in addition, daylight exposure has also been shown to increase proportions of deep sleep, which again has been linked to processes of tissue repair, something not to be underestimated after surgery. And beyond the effects on sleep, we know that patients in brighter rooms experience less pain and less stress. And also the effects of daylight on mood are well-established. So I thought it might be time for me to put into practice what I often preach. And I have to confess that usually I ’ m much better at giving advice than following it myself. But here ’ s what I did. So first, when it became free, I asked the nurses to move my bed to the window place because that does not only give you power of a fresh air supply, which is highly recommended in the hospital, but it also maximizes daylight exposure even though you ’ re inside. Second, as soon as I could leave my bed again, I went to the park for short walks, and last, every meal I had, I tried to take on the little balcony. Did it work? Well, it ’ s a bit difficult to say because it, of course, was the single case study. But from patients suffering from insomnia, so one of the most common sleep disorders, we know that light therapy is beneficial. And therefore, I would like to encourage all of you, and perhaps especially those who might belong to the 25 percent suffering from sleep problems to start your own single case study. Thank you.

Text Sample 639: NEG Dim 4 – 2022 – Score -1.00 – 2022/t003020.txt
On a usual Saturday, you wouldn ’ t find me here. And I mean, of course, I don ’ t give TEDx Talks every Saturday. But what I mean is, you probably wouldn ’ t find me in a museum. Rather, if you were looking for me, you ’ d probably have to go to the mountains, and that is because I absolutely love hiking. And in addition to just being outside and being in nature, what I really like is this feeling of physical exhaustion, sort of this satisfaction you can feel in every single muscle. And in the evening, when it ’ s time to go to bed, I absolutely cherish this amazing sleepiness that fills you from head to toe. Now you might say,”Well, of course she ’ s tired. She ’ s been hiking all day. Plus, not to forget about the effects of supposedly fresh air.”But there is one factor that most likely outweighs the effects of fresh air and perhaps even physical activity, and that is exposure to natural daylight. And today, I hope to convince you that we should all appreciate daylight as sort of a natural soporific. And hopefully, by the end of the talk, you will also recognize that we should all pay more attention to our daily light diet. But let ’ s start with the question why daylight is so important for our organism, for health and, eventually, for sleep. Now throughout your body, in each and every single cell, there is a tiny molecular clock ticking inside it. And to keep these millions of clocks in sync, we have one central body clock that is located in the **brain**, in an area which we call the hypothalamus. And like the conductor of an orchestra, it communicates the time of day to each of these molecular clocks. And this way, it is able to regulate bodily processes in sync with the time of day. For example, it causes your body temperature to increase in the morning and to decrease in the evening. It choreographs the release of certain hormones at appropriate times. And in the evening, when it ’ s time to go to bed, it will also make you tired and sleepy. But given this precise synchronization between internal or biological time and external time or environmental time, it seems clear that the body clock cannot be blind or isolated or shut off from the environment, but rather, it needs to receive information about the time of day from the environment for it to synchronize with sun time. And this is achieved by close connections between the internal biological clock in the **brain** and our eyes. And now you may know that in the human retina, there are different types of receptors, so photoreceptors, so receptors that sense light. And classically, we distinguish two types, the rods and the cones. Now the rods only contribute to a visual impression under very **dim** lighting conditions. And here, in the background, you now see a picture that might, more or less, be brought about by only the involvement of the rods. And what you can see is that it ’ s only shades of gray, it ’ s slightly blurred, and around the point of fixation, which is indicated by the gray dot here, you have a little scotoma, so an area where you can ’ t really see anything. Now who recognizes what that is? Yeah, excellent. But it ’ s going to be way easier, and the majority of you will recognize what it is, when I now switch to the next slide, which is brought about by the involvement of the cones. Of course, it ’ s the town hall of Basel, but now you can appreciate the colors, fine details. And if this was animated, you ’ d even see fine details in motion. But this is not the whole story, because only fairly recently, only in the early 2000s, another type of cell has been discovered, and we call them retinal ganglion cells. Now, you might ask yourselves,”What picture is she going to show next?”But I ’ ll have to disappoint you... I ’ m not, and I actually can ’ t, show another picture. Because from all we know, these cells do not contribute to a visual impression.

But they are exclusively designed to sense short wavelength proportions in daylight. Sometimes, we also call this blue light. So they ’ re designed to extract important information about the time of day from the environment and pass this on to the internal biological clock in the **brain**. And I guess you ’ ve all experienced how well this biological timing system, this connection between our biological clock and the external world, or our eyes, works when we, for example, travel across time zones. Now how much light do we actually need? How much light is enough for the positive effects on, for instance, sleep to occur? And I have to admit, this is not so easy to answer. But I think what we have to keep in keep in mind is that the biological timing system has evolved under the open sky and not in offices or museums. So it is also optimally tuned to the conditions we find outside. Office light or light in rooms is in no way comparable to what we find outside. And therefore, from a scientific perspective, I can only recommend you to spend as much time as you can under the open sky, but try to make it at least 30 minutes per day. Now let ’ s finally talk about sleep. And I ’ ve repeatedly alluded to the fact that daylight is beneficial and important for sleep. And in this context, I ’ d like to share a little story. So last year, I had to go to hospital for surgery. And generally, the hospital environment is a very challenging one for sleep. Because you spend way too much time in bed, not moving much. You might be in pain. Now and then, someone comes in to check on you, even during the night. And if you ’ re as lucky as I was, you have a snoring roommate. And because you, of course, don ’ t get a lot of daylight. And that ’ s why many people, while they ’ re in hospital, ask for sleep medication. But as I consider myself to be a generally good sleeper, and also because I know how these drugs work, I didn ’ t want this. But I knew there was something I could do to help my body sleep as well as possible, even in this situation. And that was to maximize exposure to daylight. Because as a sleep scientist, of course, I know about the research findings. I know that the more daylight that you get, the more tired you will be in the evening, the easier it will be for you to fall asleep and the better the experienced sleep quality. And in addition, daylight exposure has also been shown to increase proportions of deep sleep, which again has been linked to processes of tissue repair, something not to be underestimated after surgery. And beyond the effects on sleep, we know that patients in brighter rooms experience less pain and less stress. And also the effects of daylight on mood are well established. So I thought it might be time for me to put into practice what I often preach. And I have to confess that usually I ’ m much better at giving advice than following it myself. But here ’ s what I did. So first, when it became free, I asked the nurses to move my bed to the window place, because that does not only give you power of the fresh air supply, which is highly recommended in the hospital, but it also maximizes daylight exposure even though you ’ re inside. Second, as soon as I could leave my bed again, I went to the park for short walks. And last, every meal I had, I tried to take on the little balcony. Did it work? Well, it ’ s a bit difficult to say, because it of course was a single-case study. But from patients suffering from insomnia, so one of the most common sleep disorders, we know that light therapy is beneficial. And therefore, I would like to encourage all of you, and perhaps especially those who might belong to the 25 percent suffering from sleep problems, to start your own single-case study. Thank you.

Text Sample 640: NEG Dim 4 – 2002 – Score 1.00 – 2002/t000333.txt

Thank you. Ooh, I ’ m like,”Phew, phew, calm down. Get back into my body now.”Usually when I play out, the first thing that happens is people scream out,”What ’ s she doing?!”I ’ ll play at these rock shows, be on stage standing completely still, and they ’ re like,”What ’ s she doing?! What ’ s she doing?!”And then I ’ ll kind of be like — — and then they ’ re like,”Whoa!”I ’ m sure you ’ re trying to figure out,”Well, how does this thing work?”Well, what I ’ m doing is controlling the pitch with my left hand. See, the closer I get to this antenna, the higher the note gets — — and you can get it really low. And with this hand I ’ m controlling the volume, so the further away my right hand gets, the louder it gets. So basically, with both of your hands you ’ re controlling pitch and volume and kind of trying to create the illusion that you ’ re doing separate notes, when really it ’ s continuously going... Sometimes I startle myself : I ’ ll forget that I have it on, and I ’ ll lean over to pick up something, and then it goes like — —”Oh!”And it ’ s like a funny sound effect that follows you around if you don ’ t turn the thing off. Maybe we ’ ll go into the next tune, because I totally lost where this is going. We ’ re going to do a song by David Mash called”Listen : the Words Are Gone,”and maybe I ’ ll have words come back into me afterwards if I can relax. So, I ’ m trying to think of some of the questions that are commonly asked ; there are so many. And... Well, I guess I could tell you a little of the history of the theremin. It was invented around the 1920s, and the inventor, Lon Theremin — he also was a musician besides an inventor — he came up with the idea for making the theremin, I think, when he was working on some shortwave radios. And there ’ d be that sound in the signal — it ’ s like — and he thought,”Oh, what if I could control that sound and turn it into an instrument, because there are pitches in it.”And so somehow through developing that, he eventually came to make the theremin the way it is now. And a lot of times, even kids nowadays, they ’ ll make reference to a theremin by going,”Whoo-hoo-hoo-hoo,”because in the ’ 50s it was used in the sci-fi horror movies, that sound that ’ s like... It ’ s kind of a funny, goofy sound to do. And sometimes if I have too much coffee, then my vibrato gets out of hand. You ’ re really sensitive to your body and its functions when you ’ re behind this thing. You have to stay so still if you want to have the most control. It reminds me of the balancing act earlier on — what Michael was doing — because you ’ re fighting so hard to keep the balance with what you ’ re playing with and stay in tune, and at the same time you don ’ t want to focus so much on being in tune all the time ; you want to be feeling the music. And then also, you ’ re trying to stay very, very, very still because little movements with other parts of your body will affect the pitch, or sometimes if you ’ re holding a low note — — and breathing will make it... If I pass out on the next song... I think of it almost like like a yoga instrument because it makes you so aware of every little crazy thing your body is doing, or just aware of what you don ’ t want it to be doing while you ’ re playing ; you don ’ t want to have any sudden movements. And if I go to a club and play a gig, people are like,”Here, have some drinks on us!”And it ’ s like,”Well, I ’ m about to go on soon ; I don ’ t want to be like — — you know?”It really does reflect the mood that you ’ re in also, if you ’ re... it ’ s similar to being a vocalist, except instead of it coming out of your throat, you ’ re controlling it just in the air and you don ’ t really have a point of reference ; you ’ re always relying on your ears and adjusting constantly. You just have to always adjust to what ’ s happening and realize you ’ ll have bummer notes come here and there and listen to it, adjust it, and just move on, or else you ’ ll get too tied up and go crazy. Like me. I think we will play another tune now. I ’ m going to do”Lush Life.”It ’ s one of my favorite tunes to play.

Text Sample 641: NEG Dim 4 – 2002 – Score 1.00 – 2002/t000079.txt

I ’ m going to read a few strips. These are, most of these are from a monthly page I do in and architecture and design magazine called Metropolis. And the first story is called”The Faulty Switch.”Another beautifully designed new building ruined by the sound of a common wall light switch. It ’ s fine during the day when the main rooms are flooded with sunlight. But at dusk everything changes. The architect spent hundreds of hours designing the burnished brass switchplates for his new office tower. And then left it to a contractor to install these 79-cent switches behind them. We know instinctively where to reach when we enter a dark room. We automatically throw the little nub of plastic upward. But the sound we are greeted with, as the room is bathed in the simulated glow of late-afternoon light, recalls to mind a dirty men ’ s room in the rear of a Greek coffee shop. This sound colors our first impression of any room ; it can ’ t be helped. But where does this sound, commonly described as a click, come from? Is it simply the byproduct of a crude mechanical action? Or is it an imitation of one half the set of sounds we make to express disappointment? The often dental consonant of no Indo-European language. Or is it the amplified sound of a synapse firing in the **brain** of a cockroach? In the 1950s they tried their best to muffle this sound with mercury switches and silent knob controls. But today these improvements seem somehow inauthentic. The click is the modern triumphal clarion proceeding us through life, announcing our entry into every lightless room. The sound made flicking a wall switch off is of a completely different nature. It has a deep melancholy ring. Children don ’ t like it. It ’ s why they leave lights on around the house. Adults find it comforting. But wouldn ’ t it be an easy matter to wire a wall switch so that it triggers the muted horn of a steam ship? Or the recorded crowing of a rooster? Or the distant peel of thunder? Thomas Edison went through thousands of unlikely substances before he came upon the right one for the filament of his electric light bulb. Why have we settled so quickly for the sound of its switch? That ’ s the end of that. The next story is called”In Praise of the Taxpayer.”That so many of the city ’ s most venerable taxpayers have survived yet another commercial building boom, is cause for celebration. These one or two story structures, designed to yield only enough income to cover the taxes on the land on which they stand, were not meant to be permanent buildings. Yet for one reason or another they have confounded the efforts of developers to be combined into lots suitable for high-rise construction. Although they make no claim to architectural beauty, they are, in their perfect temporariness, a delightful alternative to the large-scale structures that might someday take their place. The most perfect examples occupy corner lots. They offer a pleasant respite from the high-density development around them. A break of light and air, an architectural biding of time. So buried in signage are these structures, that it often takes a moment to distinguish the modern specially constructed taxpayer from its neighbor : the small commercial building from an earlier century, whose upper floors have been sealed, and whose groundfloor space now functions as a taxpayer. The few surfaces not covered by signs are often clad in a distinctive, dark green-gray, striated aluminum siding. Take-out sandwich shops, film processing drop-offs, peep-shows and necktie stores. Now these provisional structures have, in some cases, remained standing for the better part of a human lifetime. The temporary building is a triumph of modern industrial organization, a healthy sublimation of the urge to build, and proof that not every architectural idea need be set in stone. That ’ s the end. And the next story is called,”On the Human Lap.”For the ancient Egyptians the lap was a platform upon which to place the earthly possessions of the dead — 30 cubits from foot to knee. It was not until the 14th century that an Italian painter recognized the lap as a Grecian temple, upholstered in flesh and cloth. Over the next 200 years we see the infant Christ go from a sitting to a standing position on the Virgin ’ s lap, and then back again. Every child recapitulates this ascension, straddling one or both legs, sitting sideways, or leaning against the body. From there, to the modern ventriloquist ’ s dummy, is but a brief moment in history. You were late for school again this morning. The ventriloquist must first make us believe that a small boy is sitting on his lap. The illusion of speech follows incidentally. What have you got to say for yourself, Jimmy? As adults we admire the lap from a nostalgic distance. We have fading memories of that provisional temple, erected each time an adult sat down. On a crowded bus there was always a lap to sit on. It is children and teenage girls who are most keenly aware of its architectural beauty. They understand the structural integrity of a deep avuncular lap, as compared to the shaky arrangement of a neurotic niece in high heels. The relationship between the lap and its owner is direct and intimate. I envision a 36-story, 450-unit residential high-rise — a reason to consider the mental health of any architect before granting an important commission. The bathrooms and kitchens will, of course, have no windows. The lap of luxury is an architectural construct of childhood, which we seek, in vain, as adults, to employ. That ’ s the end. The next story is called”The Haverpiece Collection”A nondescript warehouse, visible for a moment from the northbound lanes of the Prykushko Expressway, serves as the temporary resting place for the Haverpiece collection of European dried fruit. The profound convolutions on the surface of a dried cherry. The foreboding sheen of an extra-large date. Do you remember wandering as a child through those dark wooden storefront galleries? Where everything was displayed in poorly labeled roach-proof bins. Pears dried in the form of genital organs. Apricot halves like the ears of cherubim. In 1962 the unsold stock was purchased by Maurice Haverpiece, a wealthy prune juice bottler, and consolidated to form the core collection. As an art form it lies somewhere between still-life painting and plumbing. Upon his death in 1967, a quarter of the items were sold off for compote to a high-class hotel restaurant. Unsuspecting guests were served stewed turn-of-the-century Turkish figs for breakfast. The rest of the collection remains here, stored in plain brown paper bags until funds can be raised to build a permanent museum and study center. A shoe made of apricot leather for the daughter of a czar. That ’ s the end. Thank you.

Text Sample 642: NEG Dim 4 – 2002 – Score 2.00 – 2002/t000397.txt

How many Creationists do we have in the room? Probably none. I think we ’ re all Darwinians. And yet many Darwinians are anxious, a little uneasy â would like to see some limits on just how far the Darwinism goes. It ’ s all right. You know spiderwebs? Sure, they are products of evolution. The World Wide Web? Not so sure. Beaver dams, yes. Hoover Dam, no. What do they think it is that prevents the products of human ingenuity from being themselves, fruits of the tree of life, and hence, in some sense, obeying evolutionary rules? And yet people are interestingly resistant to the idea of applying evolutionary thinking to thinking â to our thinking. And so I ’ m going to talk a little bit about that, keeping in mind that we have a lot on the program here. So you ’ re out in the woods, or you ’ re out in the pasture, and you see this ant **crawling** up this blade of grass. It climbs up to the top, and it falls, and it climbs, and it falls, and it climbs â trying to stay at the very top of the blade of grass. What is this ant doing? What is this in aid of? What goals is this ant trying to achieve by climbing this blade of grass? What ’ s in it for the ant? And the answer is : nothing. There ’ s nothing in it for the ant. Well then, why is it doing this? Is it just a fluke? Yeah, it ’ s just a fluke. It ’ s a lancet fluke. It ’ s a little **brain** worm. It ’ s a parasitic **brain** worm that has to get into the stomach of a sheep or a cow in order to continue its life cycle. Salmon swim upstream to get to their spawning grounds, and lancet flukes commandeer a passing ant, **crawl** into its **brain**, and drive it up a blade of grass like an all-terrain vehicle. So there ’ s nothing in it for the ant. The ant ’ s **brain** has been hijacked by a parasite that infects the **brain**, inducing suicidal behavior. Pretty scary. Well, does anything like that happen with human beings? This is all on behalf of a cause other than one ’ s own genetic fitness, of course. Well, it may already have occurred to you that Islam means”surrender,”or”submission of self-interest to the will of Allah.”Well, it ’ s ideas â not worms â that hijack our **brains**. Now, am I saying that a sizable minority of the world ’ s population has had their **brain** hijacked by parasitic ideas? No, it ’ s worse than that. Most people have. There are a lot of ideas to die for. Freedom, if you ’ re from New Hampshire. Justice. Truth. Communism. Many people have laid down their lives for communism, and many have laid down their lives for capitalism. And many for Catholicism. And many for Islam. These are just a few of the ideas that are to die for. They ’ re infectious. Yesterday, Amory Lovins spoke about”infectious repititis.”It was a term of abuse, in effect. This is unthinking engineering. Well, most of the cultural spread that goes on is not brilliant, new, out-of-the-box thinking. It ’ s”infectious repetitis,”and we might as well try to have a theory of what ’ s going on when that happens so that we can understand the conditions of infection. Hosts work hard to spread these ideas to others. I myself am a philosopher, and one of our occupational hazards is that people ask us what the meaning of life is. And you have to have a bumper sticker, you know. You have to have a statement. So, this is mine. The secret of happiness is : Find something more important than you are and dedicate your life to it. Most of us â now that the”Me Decade”is well in the past â now we actually do this. One set of ideas or another have simply replaced our biological imperatives in our own lives. This is what our summum bonum is. It ’ s not maximizing the number of grandchildren we have. Now, this is a profound biological effect. It ’ s the subordination of genetic interest to other interests. And no other species does anything at all like it. Well, how are we going to think about this? It is, on the one hand, a biological effect, and a very large one. Unmistakable. Now, what theories do we want to use to look at this? Well, many theories. But how could something tie them together? The idea of replicating ideas ; ideas that replicate by passing from **brain** to **brain**. Richard Dawkins, whom you ’ ll be hearing later in the day, invented the term”memes,”and put forward the first really clear and vivid version of this idea in his book”The Selfish Gene.”Now here am I talking about his idea. Well, you see, it ’ s not his. Yes â he started it. But it ’ s everybody ’ s idea now. And he ’ s not responsible for what I say about memes. I ’ m responsible for what I say about memes. Actually, I think we ’ re all responsible for not just the intended effects of our ideas, but for their likely misuses. So it is important, I think, to Richard, and to me, that these ideas not be abused and misused. They ’ re very easy to misuse. That ’ s why they ’ re dangerous. And it ’ s just about a full-time job trying to prevent people who are scared of these ideas from caricaturing them and then running off to one dire purpose or another. So we have to keep plugging away, trying to correct the misapprehensions so that only the benign and useful variants of our ideas continue to spread. But it is a problem. We don ’ t have much time, and I ’ m going to go over just a little bit of this and cut out, because there ’ s a lot of other things that are going to be said. So let me just point out : memes are like viruses. That ’ s what Richard said, back in ’ 93. And you might think,”Well, how can that be? I mean, a virus is â you know, it ' s stuff! What ' s a meme made of?"Yesterday, Negroponte was talking about viral telecommunications but â what ' s a virus? A virus is a string of nucleic acid with attitude. That is, there is something about it that tends to make it replicate better than the competition does. And that ' s what a meme is. It ' s an information packet with attitude. What ' s a meme made of? What are bits made of, Mom? Not silicon. They ' re made of information, and can be carried in any physical medium. What ' s a word made of? Sometimes when people say,"Do memes exist?"I say,"Well, do words exist? Are they in your ontology?"If they are, words are memes that can be pronounced. Then there ' s all the other memes that can ' t be pronounced. There are different species of memes. Remember the Shakers? Gift to be simple? Simple, beautiful furniture? And, of course, they ' re basically extinct now. And one of the reasons is that among the creed of Shaker-dom is that one should be celibate. Not just the priests. Everybody. Well, it ' s not so surprising that they ' ve gone extinct. But in fact that ' s not why they went extinct. They survived as long as they did at a time when the social safety nets weren ' t there. And there were lots of widows and orphans, people like that, who needed a foster home. And so they had a ready supply of converts. And they could keep it going. And, in principle, it could ' ve gone on forever, with perfect celibacy on the part of the hosts. The idea being passed on through proselytizing, instead of through the gene line. So the ideas can live on in spite of the fact that they ' re not being passed on genetically. A meme can flourish in spite of having a negative impact on genetic fitness. After all, the meme for Shaker-dom was essentially a sterilizing parasite. There are other parasites that do this â which render the host sterile. It ' s part of their plan. They don ' t have to have minds to have a plan. I ' m just going to draw your attention to just one of the many implications of the memetic perspective, which I recommend. I ' ve not time to go into more of it. In Jared Diamond ' s wonderful book,"Guns, Germs and Steel,"he talks about how it was germs, more than guns and steel, that conquered the new hemisphere â the Western hemisphere â that conquered the rest of the world. When European explorers and travelers spread out, they brought with them the germs that they had become essentially immune to, that they had learned how to tolerate over hundreds and hundreds of years, thousands of years, of living with domesticated animals who were the sources of those pathogens. And they just wiped out â these pathogens just wiped out the native people, who had no immunity to them at all. And we ' re doing it again. We ' re doing it this time with toxic ideas. Yesterday, a number of people â Nicholas Negroponte and others â spoke about all the wonderful things that are happening when our ideas get spread out, thanks to all the new technology all over the world. And I agree. It is largely wonderful. Largely wonderful. But among all those ideas that inevitably flow out into the whole world thanks to our technology, are a lot of toxic ideas. Now, this has been realized for some time. Sayyid Qutb is one of the founding fathers of fanatical Islam, one of the ideologues that inspired Osama bin Laden."One has only to glance at its press films, fashion shows, beauty contests, ballrooms, wine bars and broadcasting stations."Memes. These memes are spreading around the world and they are wiping out whole cultures. They are wiping out languages. They are wiping out traditions and practices. And it ' s not our fault, anymore than it ' s our fault when our germs lay waste to people that haven ' t developed the immunity. We have an immunity to all of the junk that lies around the edges of our culture. We ' re a free society, so we let pornography and all these things â we shrug them off. They ' re like a mild cold. They ' re not a big deal for us. But we should recognize that for many people in the world, they are a big deal. And we should be very alert to this. As we spread our education and our technology, one of the things that we are doing is we ’ re the vectors of memes that are correctly viewed by the hosts of many other memes as a dire threat to their favorite memes â the memes that they are prepared to die for. Well now, how are we going to tell the good memes from the bad memes? That is not the job of the science of memetics. Memetics is morally neutral. And so it should be. This is not the place for hate and anger. If you ’ ve had a friend who ’ s died of AIDS, then you hate HIV. But the way to deal with that is to do science, and understand how it spreads and why in a morally neutral perspective. Get the facts. Work out the implications. There ’ s plenty of room for moral passion once we ’ ve got the facts and can figure out the best thing to do. And, as with germs, the trick is not to try to annihilate them. You will never annihilate the germs. What you can do, however, is foster public health measures and the like that will encourage the evolution of avirulence. That will encourage the spread of relatively benign mutations of the most toxic varieties. That ’ s all the time I have, so thank you very much for your attention.

Text Sample 643: NEG Dim 4 – 2020 – Score -1.00 – 2020/t004230.txt

Sleep is perhaps the single most effective thing that we can do each and every day to reset the health of our **brain** and our body. And by understanding a little bit more about what sleep is, perhaps we can get the chance to improve both the quantity and the quality of our sleep. [Sleeping with Science] So, exactly what is sleep? Well, sleep, at least in human beings, is subdivided into two main types. On the one hand, we have non-rapid eye movement sleep, or non-REM sleep for short. But on the other hand, we have rapid eye movement sleep, or REM sleep. And non-REM sleep has been further subdivided into four separate stages, unimaginatively called stages one through four, increasing in their depth of sleep. And as we go into those light stages of non-REM sleep, your heart rate starts to decrease, your body temperature starts to drop and your electrical **brain** wave activity starts to slow down. But as we move into deeper non-rapid eye movement sleep, stages three and four, now all of a sudden the **brain** erupts with these huge, big, powerful **brain** waves. The body is actually recharged in terms of its immune system. We also get this beautiful overhaul of our cardiovascular system. And, in fact, upstairs in the **brain**, deep non-REM sleep will help consolidate memories and fixate them into the neural architecture of the **brain**. So that ’ s non-REM sleep. But let ’ s come on to REM sleep, which is the other main type of sleep. And it ’ s during REM sleep when we principally have the most vivid, the most hallucinogenic types of dreams. The **brain** wave activity actually starts to speed up again. It ’ s during REM sleep that we receive almost a form of emotional first aid. And it ’ s also during REM sleep where we get a boost for creativity, that it stitches information together so that we wake up with solutions to previously difficult problems that we were facing. Coming back to these two types of sleep, it turns out that non-REM and REM will play out in a battle for **brain** domination throughout the night, and that cerebral war is going to be won and lost every 90 minutes, and then it ’ s going to be replayed every 90 minutes. And what this produces is a standard cycling architecture of human sleep, a standard 90-minute cycle. But what ’ s different, however, is that the ratio of non-REM to REM within those 90-minute cycles changes as we move across the night, such that in the first half the night, the majority of those 90-minute cycles are comprised of lots of deep non-REM sleep, particularly stages three and four of non-REM sleep. But as we push through to the second half of the night, now that seesaw balance actually shifts over, and instead, most of those 90-minute cycles are comprised of a lot more rapid eye movement sleep, or dream sleep, as well as stage-two non-REM sleep, that lighter form of non-REM sleep. And it turns out that there are implications for understanding how sleep is structured in this way. Let ’ s take someone who typically goes to bed at 10pm, and they wake up at 6am, so they have an eight-hour sleep window. But this morning, they have to wake up early for an early morning meeting, or they want to get a jump start on the day to get to the gym. And as a consequence, they have to wake up at 4am in the morning, rather than 6am in the morning. How much sleep have they actually lost? Two hours out of an eight-hour night of sleep means that they ’ ve lost 25 percent of their sleep. Well, yes and no. They have lost 25 percent of all of their sleep, but because REM sleep comes mostly in the second half of the night and particularly in those last few hours, they may have lost perhaps 50, 60, maybe even 70 percent of all of their REM sleep. So there are real consequences to understanding what sleep is and how sleep is structured. And we ’ ll learn all about the benefits of these different stages of sleep and the detriments that happen when we don ’ t get enough of them in subsequent episodes.

Text Sample 644: NEG Dim 4 – 2020 – Score -1.00 – 2020/t004224.txt

Whether you ’ re cramming for an exam or trying to learn a new musical instrument or even trying to perfect a new sport, sleep may actually be your secret memory weapon. [Sleeping with Science] Studies have actually told us that sleep is critical for memory in at least three different ways. First, we know that you need sleep before learning to actually get your **brain** ready, almost like a dry sponge, ready to initially soak up new information. And without sleep, the memory circuits within the **brain** effectively become waterlogged, as it were, and we can ’ t absorb new information. We can ’ t effectively lay down those new memory traces. But it ’ s not only important that you sleep before learning, because we also know that you need sleep after learning to essentially hit the save button on those new memories so that we don ’ t forget. In fact, sleep will actually future-proof that information within the **brain**, cementing those memories into the architecture of those neural networks. And we ’ ve begun to discover exactly how sleep achieves this memory-consolidation benefit. The first mechanism is a file-transfer process. And here, we can speak about two different structures within the **brain**. The first is called the hippocampus and the hippocampus sits on the left and the right side of your **brain**. And you can think of the hippocampus almost like the informational inbox of your **brain**. It ’ s very good at receiving new memory files and holding onto them. The second structure that we can speak about is called the cortex. This wrinkled massive tissue that sits on top of your **brain**. And during deep sleep, there is this file-transfer mechanism. Think of the hippocampus like a USB stick and your cortex like the hard drive. And during the day, we ’ re going around and we ’ re gathering lots of files, but then during deep sleep at night, because of that limited storage capacity, we have to transfer those files from the hippocampus over to the hard drive of the **brain**, the cortex. And that ’ s exactly one of the mechanisms that deep sleep seems to provide. But there ’ s another mechanism that we ’ ve become aware of that helps cement those memories into the **brain**. And it ’ s called replay. Several years ago, scientists were looking at how rats learned as they would run around a maze. And they were recording the activity in the memory centers of these rats. And as the rat was running around the maze, different **brain** cells would code different parts of the maze. And so if you added a tone to each one of the **brain** cells what you would hear as the rat was starting to learn the maze was the signature of that memory. So it would sound a little bit like... It was this signature of learning that we could hear. But then they did something clever. They kept listening to the **brain** as these rats fell asleep, and what they heard was remarkable. The rat, as it was sleeping, started to replay that same memory signature. But now it started to replay it almost 10 times faster than it was doing when it was awake. So now instead you would start to hear... That seems to be the second way in which sleep can actually strengthen these memories. Sleep is actually replaying and scoring those memories into a new circuit within the **brain**, strengthening that memory representation. The final way in which sleep is beneficial for memory is integration and association. In fact, we ' re now learning that sleep is much more intelligent than we ever imagined. Sleep doesn ' t just simply strengthen individual memories, sleep will actually cleverly interconnect new memories together. And as a consequence, you can wake up the next day with a revised mind-wide web of associations, we can come up with solutions to previously impenetrable problems. And this is probably the reason that you ' ve never been told to stay awake on a problem. Instead, you ' re told to sleep on a problem, and that ' s exactly what the science teaching us.

Text Sample 645: NEG Dim 4 – 2020 – Score -1.00 – 2020/t004112.txt
It is the third of March, 2016, and I ' m anxiously waiting for my wife to deliver our firstborn son. Seconds turn into minutes, then hours, without a sign of a child coming through. Then a midwife emerges with a silent baby in her hands, and she runs past me as though I ' m not even there. Why is he not crying? I ' m gripped with chills in my spine as I run after her in terror. She puts the baby on a bench and begins a resuscitation procedure. Thirty minutes later, she tells me,"Don ' t worry, he will be fine, and thank you for staying calm."He was placed in the ICU, and though I cannot touch him, I repeatedly say,"Shine on, my son. Don ' t give up. I am here with you, and you don ' t have to be scared. Please pull through and let us go home. You do not belong here."Seven months later, he would be diagnosed with cerebral palsy, a nonprogressive **brain** injury which primarily affects body movement and muscle coordination. About two to three children out of 1, 000 in the United States have cerebral palsy. I do not know the statistics for my country and continent, because there ' s not much documentation. Maybe this could be the journey that changes everything. We named him Lubuto, a beautiful Zambian name from my Lunda tribe of the Bemba-speaking people, meaning"light."By the time he was seven months, Lubuto ' s physical impairment was predominant in the left part of his body. Both his left leg and arm were less responsive. He couldn ' t grasp items ; worse off, couldn ' t babble his first words because the cerebral palsy shackle affects the muscle in his lips. Rolling over and other milestones that come naturally in typical babies, couldn ' t be seen in our son. Lubuto was visibly unaware of his own body. And some specialists started preparing us for the worst by telling us that we were going to be very lucky if he ever sat upright and unsupported. Before us was a gigantic, seemingly immovable mountain. What do we do? For the past 15 years, I have worked as a computer programmer, and now I ' m a certified project management professional. After the denial, crying and partial depression was over, I began to wonder if we could put my programming and project management skills together to try and help the situation. Acceptance kicked in, and I searched from deep within for strength and any available knowledge to help with the challenge before us. I ordered two books online and spent countless sleepless nights researching neuroplasticity in a child ' s **brain**. My extensive research indicated that people who have

strokes are able to recover through assiduous rehabilitation programs that activates new parts in the better part of their **brains**. This left me with one big question : If this works for grown people, why should it not work for a baby? I also learned that human beings pick up fundamental patterns mainly between ages zero to five, and after that, consolidation of habits happens. It was scary to realize that we may just have five years to figure out the immobility of Lubuto. On such a tight timeline, we needed to build a support system around him, leveraging the limited resources available to us. This was a clear project before us which needed to be carefully executed, and we needed a capable, self-driven team, an agile team.”Agile”is a methodology that we use to execute projects with changing requirements to achieve progressive results in increments. We needed to deliver quick results, and in pieces, considering our work was largely dependent on Lubuto ’ s responsiveness and capability. The first team member I acquired was my beautiful wife Abigail, who is luckily a project manager, too. You know how rough that can be, right? Two project managers under one roof. We searched around Zambia for a neonatal physiotherapist, an occupational therapist and a speech therapist. It felt like mission impossible. We set road maps of one to three months, just enough planning and just in time. We then identified features like,”We want him to stand and walk independently,”under different themes like gross motor, fine motor, adaptive skills, communication, asymmetric movement and balance. Next, we created sprints to work on the stimulation of different parts of Lubuto ’ s body. When you ’ re working on an agile project, you do a series of lead-to tasks, collectively called”sprints,”which the team reviews after execution. We, for example, set a goal to stimulate his left arm. Say occupational therapists use different textures to rub on his arm. Physiotherapists make deliberate movements in his arm to build the muscles. And self-proclaimed general therapists, who was usually myself, engage in logical stimulations like slowly moving his favorite toy from his right hand across and by in front of him to his left side to prompt movement in his left arm. And at the end of each week, we would review our results as a team : How did OT go? How did physio go? How did stimulation go? Did we meet our goal? Because frequent communication is very important on an agile project, we created a WhatsApp group for quicker updates. Failing early and picking up is a special characteristic of agility, and we leveraged that because our work is largely dependent on his response. Luckily, Lubuto is a fighter, and his determination is out of this world. After we achieved the goal of activating his arm, we then moved to his leg. The activities were totally different, but followed the similar iterative process. I would come to learn in **brain** plasticity that Lubuto was better off learning certain skills when he was ready, even if it meant delaying him, because he had to learn it right. While working and managing Lubuto as an agile project, a new team member **popped** up. Oh! It ’ s Lubuto ’ s sister, Yawila. We had no idea how we were going to manage the process without disturbing him while not making the sister feel neglected because we were giving the brother a lot of attention. Our daily iterations continued, and now Lubuto was able to walk on his stiff legs. With me cheerleading from the front as I walked backwards, because I needed to keep eye contact with him, I sang his favorite songs, as we oscillated between our bedroom and the kitchen. We then traveled to South Africa and introduced a neuromovement therapist to the team, coupled with hyperbaric oxygen therapy. These sprints were much shorter and focused on his **brain**, teaching him about his own body through small body movements. Terry did a miraculous job. Lubuto started opening his knees in unison with his hips. And in our second week, he was able to run with better balance. He started making intentional sounds to communicate with us as a result of the new neuropaths firing. We returned to Zambia with amazing results. And guess who effectively picked up the therapy? The new team member. Lubuto started mimicking the sister, and soon he was learning more things good and bad from the sister than he was learning from his team of therapists. To make sure that he stays on track, we built a unique curriculum that incorporates all the therapies with teacher Goodson. We ' ve been blessed to have the knowledge before us and be able to practically apply it. Not all families with special needs children are as fortunate as we are. We still have backlog stories, which is a fancy agile term for pushing failure to a later date, in Lubuto ' s case, drooling and potty training. But in iterative, little daily activities, we managed to improve the entire left part of Lubuto ' s body – from the arm, to one finger to the other, from the leg to the toes. Lubuto began to roll over. He began to independently sit. He was able to **crawl**, stand, walk, run, and now he plays soccer with me in a more coordinated manner. This has left my wife ' s heart and mine melting, and we ' ve been blown away by the unbelievable results we ' ve witnessed as a result of this experimental methodology. And now, we proudly call ourselves"agile parents."You may be a parent with a special needs child like me, or you could be facing different types of limitation in your life : professionally, financially, academically or even physically. I want to remind you that, in striving for bigger goals, dare to take small sprints. These sprints are usually far from excellent themselves, but they add up to magnificent results. Thank you.

Text Sample 646: NEG Dim 4 – 2007 – Score 1.00 – 2007/t000328.txt
So I thought,"I will talk about death."Seemed to be the passion today. Actually, it ' s not about death. It ' s inevitable, terrible, but really what I want to talk about is, I ' m just fascinated by the legacy people leave when they die. That ' s what I want to talk about. So Art Buchwald left his legacy of humor with a video that appeared soon after he died, saying,"Hi! I ' m Art Buchwald, and I just died."And Mike, who I met at Galapagos, a trip which I won at TED, is leaving notes on cyberspace where he is chronicling his journey through cancer. And my father left me a legacy of his handwriting through letters and a notebook. In the last two years of his life, when he was sick, he filled a notebook with his thoughts about me. He wrote about my strengths, weaknesses, and gentle suggestions for improvement, quoting specific incidents, and held a mirror to my life. After he died, I realized that no one writes to me anymore. Handwriting is a disappearing art. I ' m all for email and thinking while typing, but why give up old habits for new? Why can ' t we have letter writing and email exchange in our lives? There are times when I want to trade all those years that I was too busy to sit with my dad and chat with him, and trade all those years for one hug. But too late. But that ' s when I take out his letters and I read them, and the paper that touched his hand is in mine, and I feel connected to him. So maybe we all need to leave our children with a value legacy, and not a financial one. A value for things with a personal touch — an autographed book, a soul-searching letter. If a fraction of this powerful TED audience could be inspired to buy a beautiful paper — John, it ' ll be a recycled one — and write a beautiful letter to someone they love, we actually may start a revolution where our children may go to penmanship classes. So what do I plan to leave for my son? I collect autographed books, and those of you authors in the audience know I hound you for them — and CDs too, Tracy. I plan to publish my own notebook. As I witnessed my father ' s body being swallowed by fire, I sat by his funeral pyre and wrote. I have no idea how I ' m going to do it, but I am committed to compiling his thoughts and mine into a book, and leave that published book

for my son. I ’ d like to end with a few verses of what I wrote at my father ’ s cremation. And those linguists, please pardon the grammar, because I ’ ve not looked at it in the last 10 years. I took it out for the first time to come here.”Picture in a frame, ashes in a bottle, boundless energy confined in the bottle, forcing me to deal with reality, forcing me to deal with being grown up. I hear you and I know that you would want me to be strong, but right now, I am being sucked down, surrounded and suffocated by these raging emotional waters, craving to cleanse my soul, trying to emerge on a firm footing one more time, to keep on fighting and flourishing just as you taught me. Your encouraging whispers in my whirlpool of despair, holding me and heaving me to shores of sanity, to live again and to love again.”Thank you.

Text Sample 647: NEG Dim 4 – 2007 – Score 2.00 – 2007/t000413.txt

What I want to talk to you about today is virtual worlds, digital globes, the 3-D Web, the Metaverse. What does this all mean for us? What it means is the Web is going to become an exciting place again. It ’ s going to become super exciting as we transform to this highly immersive and interactive world. With graphics, computing power, low latencies, these types of applications and possibilities are going to stream rich data into your lives. So the Virtual Earth initiative, and other types of these initiatives, are all about extending our current search metaphor. When you think about it, we ’ re so constrained by browsing the Web, remembering URLs, saving favorites. As we move to search, we rely on the relevance rankings, the Web matching, the index **crawling**. But we want to use our **brain**! We want to navigate, explore, discover information. In order to do that, we have to put you as a user back in the driver ’ s seat. We need cooperation between you and the computing network and the computer. So what better way to put you back in the driver ’ s seat than to put you in the real world that you interact in every day? Why not leverage the learnings that you ’ ve been learning your entire life? So Virtual Earth is about starting off creating the first digital representation, comprehensive, of the entire world. What we want to do is mix in all types of data. Tag it. Attribute it. Metadata. Get the community to add local depth, global perspective, local knowledge. So when you think about this problem, what an enormous undertaking. Where do you begin? Well, we collect data from satellites, from airplanes, from ground vehicles, from people. This process is an engineering problem, a mechanical problem, a logistical problem, an operational problem. Here is an example of our aerial camera. This is panchromatic. It ’ s actually four color cones. In addition, it ’ s multi-spectral. We collect four gigabits per second of data, if you can imagine that kind of data stream coming down. That ’ s equivalent to a constellation of 12 satellites at highest res capacity. We fly these airplanes at 5, 000 feet in the air. You can see the camera on the front. We collect multiple viewpoints, vantage points, angles, textures. We bring all that data back in. We sit here — you know, think about the ground vehicles, the human scale — what do you see in person? We need to capture that up close to establish that what it ’ s like-type experience. I bet many of you have seen the Apple commercials, kind of poking at the PC for their brilliance and simplicity. So a little unknown secret is — did you see the one with the guy, he ’ s got the Web cam? The poor PC guy. They ’ re duct taping his head. They ’ re just wrapping it on him. Well, a little unknown secret is his brother actually works on the Virtual Earth team.. So they ’ ve got a little bit of a sibling rivalry thing going on here. But let me tell you — it doesn ’ t affect his day job. We think a lot of good can come from this technology. This was after Katrina. We were the first commercial fleet of airplanes to be cleared into the disaster impact zone. We flew the area. We imaged it. We sent in people. We took pictures of interiors, disaster areas. We helped with the first responders, the search and rescue. Often the first time anyone saw what happened to their house was on Virtual Earth. We made it all freely available on the Web, just to — it was obviously our chance of helping out with the cause. When we think about how all this comes together, it ’ s all about software, algorithms and math. You know, we capture this imagery but to build the 3- D models we need to do geo-positioning. We need to do geo-registering of the images. We have to bundle adjust them. Find tie points. Extract geometry from the images. This process is a very calculated process. In fact, it was always done manual. Hollywood would spend millions of dollars to do a small urban corridor for a movie because they ’ d have to do it manually. They ’ d drive the streets with lasers called LIDAR. They ’ d collected information with photos. They ’ d manually build each building. We do this all through software, algorithms and math — a highly automated pipeline creating these cities. We took a decimal point off what it cost to build these cities, and that ’ s how we ’ re going to be able to scale this out and make this reality a dream. We think about the user interface. What does it mean to look at it from multiple perspectives? An ortho-view, a nadir-view. How do you keep the precision of the fidelity of the imagery while maintaining the fluidity of the model? I ’ ll wrap up by showing you the — this is a brand-new peek I haven ’ t really shown into the lab area of Virtual Earth. What we ’ re doing is — people like this a lot, this bird ’ s eye imagery we work with. It ’ s this high resolution data. But what we ’ ve found is they like the fluidity of the 3- D model. A child can navigate with an Xbox controller or a game controller. So here what we ’ re trying to do is we bring the picture and project it into the 3-D model space. You can see all types of resolution. From here, I can slowly pan the image over. I can get the next image. I can blend and transition. By doing this I don ’ t lose the original detail. In fact, I might be recording history. The freshness, the capacity. I can turn this image. I can look at it from multiple viewpoints and angles. What we ’ re trying to do is build a virtual world. We hope that we can make computing a user model you ’ re familiar with, and really derive insights from you, from all different directions. I thank you very much for your time.

Text Sample 648: NEG Dim 4 – 2007 – Score 2.00 – 2007/t002580.txt

You know, what I do is write for children, and I ’ m probably America ’ s most widely read children ’ s author, in fact. And I always tell people that I don ’ t want to show up looking like a scientist. You can have me as a farmer, or in leathers, and no one has ever chose farmer. I ’ m here today to talk to you about circles and epiphanies. And you know, an epiphany is usually something you find that you dropped someplace. You ’ ve just got to go around the block to see it as an epiphany. That ’ s a painting of a circle. A friend of mine did that — Richard Bollingbroke. It ’ s the kind of complicated circle that I ’ m going to tell you about. My circle began back in the ’ 60s in high school in Stow, Ohio where I was the class queer. I was the guy beaten up bloody every week in the boys ’ room, until one teacher saved my life. She saved my life by letting me go to the bathroom in the teachers ’ lounge. She did it in secret. She did it for three years. And I had to get out of town. I had a thumb, I had 85 dollars, and I ended up in San Francisco, California — met a lover — and back in the ’ 80s, found it necessary to begin work on AIDS organizations. About three or four years ago, I got a phone call in the middle of the night from that teacher, Mrs. Posten, who said,”I need to see you. I ’ m disappointed that we never got to know each other as adults. Could you please come to Ohio, and please bring that man that I know you have found by now. And I should mention that I have pancreatic cancer, and I ’ d like you to please be quick about this.”Well, the next day we were in Cleveland. We took a look at her, we laughed, we cried, and we knew that she needed to be in a hospice. We found her one, we got her there, and we took care of her and watched over her family, because it was necessary. It ’ s something we knew how to do. And just as the woman who wanted to know me as an adult got to know me, she turned into a box of ashes and was placed in my hands. And what had happened was the circle had closed, it had become a circle — and that epiphany I talked about presented itself. The epiphany is that death is a part of life. She saved my life ; I and my partner saved hers. And you know, that part of life needs everything that the rest of life does. It needs truth and beauty, and I ’ m so happy it ’ s been mentioned so much here today. It also needs — it needs dignity, love and pleasure, and it ’ s our job to hand those things out. Thank you.

Text Sample 649: NEG Dim 4 – 2005 – Score 1.00 – 2005/t002624.txt

I ’ m used to thinking of the TED audience as a wonderful collection of some of the most effective, intelligent, intellectual, savvy, worldly and innovative people in the world. And I think that ’ s true. However, I also have reason to believe that many, if not most, of you are actually tying your shoes incorrectly. Now I know that seems ludicrous. I know that seems ludicrous. And believe me, I lived the same sad life until about three years ago. And what happened to me was I bought, what was for me, a very expensive pair of shoes. But those shoes came with round nylon laces, and I couldn ’ t keep them tied. So I went back to the store and said to the owner,”I love the shoes, but I hate the laces.”He took a look and said,”Oh, you ’ re tying them wrong.”Now up until that moment, I would have thought that, by age 50, one of the life skills that I had really nailed was tying my shoes. But not so — let me demonstrate. This is the way that most of us were taught to tie our shoes. Now as it turns out — thank you. Wait, there ’ s more. As it turns out — there ’ s a strong form and a weak form of this knot, and we were taught the weak form. And here ’ s how to tell. If you pull the strands at the base of the knot, you will see that the bow will orient itself down the long axis of the shoe. That ’ s the weak form of the knot. But not to worry. If we start over and simply go the other direction around the bow, we get this, the strong form of the knot. And if you pull the cords under the knot, you will see that the bow orients itself along the transverse axis of the shoe. This is a stronger knot. It will come untied less often. It will let you down less, and not only that, it looks better. We ’ re going to do this one more time. Start as usual — go the other way around the loop. This is a little hard for children, but I think you can handle it. Pull the knot. There it is : the strong form of the shoe knot. Now, in keeping with today ’ s theme, I ’ d like to point out — something you already know — that sometimes a small advantage someplace in life can yield tremendous results someplace else. Live long and prosper.

Text Sample 650: NEG Dim 4 – 2005 – Score 4.00 – 2005/t000467.txt

I want to start with a story, a la Seth Godin, from when I was 12 years old. My uncle Ed gave me a beautiful blue sweater â at least I thought it was beautiful. And it had fuzzy zebras walking across the stomach, and Mount Kilimanjaro and Mount Meru were kind of right across the chest, that were also fuzzy. And I wore it whenever I could, thinking it was the most fabulous thing I owned.

Until one day in ninth grade, when I was standing with a number of the football players. And my body had clearly changed, and Matt, who was undeniably my nemesis in high school, said in a booming voice that we no longer had to go far away to go on ski trips, but we could all ski on Mount Novogratz. And I was so humiliated and mortified that I immediately ran home to my mother and chastised her for ever letting me wear the hideous sweater. We drove to the Goodwill and we threw the sweater away somewhat ceremoniously, my idea being that I would never have to think about the sweater nor see it ever again. Fast forward â 11 years later, I ’ m a 25-year-old kid. I ’ m working in Kigali, Rwanda, jogging through the steep slopes, when I see, 10 feet in front of me, a little boy â 11 years old â running toward me, wearing my sweater. And I ’ m thinking, no, this is not possible. But so, curious, I run up to the child â of course scaring the living bejesus out of him â grab him by the collar, turn it over, and there is my name written on the collar of this sweater. I tell that story, because it has served and continues to serve as a metaphor to me about the level of connectedness that we all have on this Earth. We so often don ’ t realize what our action and our inaction does to people we think we will never see and never know. I also tell it because it tells a larger contextual story of what aid is and can be. That this traveled into the Goodwill in Virginia, and moved its way into the larger industry, which at that point was giving millions of tons of secondhand clothing to Africa and Asia. Which was a very good thing, providing low cost clothing. And at the same time, certainly in Rwanda, it destroyed the local retailing industry. Not to say that it shouldn ’ t have, but that we have to get better at answering the questions that need to be considered when we think about consequences and responses. So, I ’ m going to stick in Rwanda, circa 1985, 1986, where I was doing two things. I had started a bakery with 20 unwed mothers. We were called the”Bad News Bears,”and our notion was we were going to corner the snack food business in Kigali, which was not hard because there were no snacks before us. And because we had a good business model, we actually did it, and I watched these women transform on a micro-level. But at the same time, I started a micro- finance bank, and tomorrow Iqbal Quadir is going to talk about Grameen, which is the grandfather of all micro-finance banks, which now is a worldwide movement â you talk about a meme â but then it was quite new, especially in an economy that was moving from barter into trade. We got a lot of things right. We focused on a business model ; we insisted on skin in the game. The women made their own decisions at the end of the day as to how they would use this access to credit to build their little businesses, earn more income so they could take care of their families better. What we didn ’ t understand, what was happening all around us, with the confluence of fear, ethnic strife and certainly an aid game, if you will, that was playing into this invisible but certainly palpable movement inside Rwanda, that at that time, 30 percent of the budget was all foreign aid. The genocide happened in 1994, seven years after these women all worked together to build this dream. And the good news was that the institution, the banking institution, lasted. In fact, it became the largest rehabilitation lender in the country. The bakery was completely wiped out, but the lessons for me were that accountability counts â got to build things with people on the ground, using business models where, as Steven Levitt would say, the incentives matter. Understand, however complex we may be, incentives matter. So when Chris raised to me how wonderful everything that was happening in the world, that we were seeing a shift in zeitgeist, on the one hand I absolutely agree with him, and I was so thrilled to see what happened with the G8 â that the world, because of people like Tony Blair and Bono and Bob Geldof â the world is talking about global poverty ; the world is talking about Africa in ways I have never seen in my life. It ' s thrilling. And at the same time, what keeps me up at night is a fear that we ' ll look at the victories of the G8 â 50 billion dollars in increased aid to Africa, 40 billion in reduced debt â as the victory, as more than chapter one, as our moral absolution. And in fact, what we need to do is see that as chapter one, celebrate it, close it, and recognize that we need a chapter two that is all about execution, all about the how-to. And if you remember one thing from what I want to talk about today, it ' s that the only way to end poverty, to make it history, is to build viable systems on the ground that deliver critical and affordable goods and services to the poor, in ways that are financially sustainable and scaleable. If we do that, we really can make poverty history. And it was that â that whole philosophy â that encouraged me to start my current endeavor called"Acumen Fund,"which is trying to build some mini- blueprints for how we might do that in water, health and housing in Pakistan, India, Kenya, Tanzania and Egypt. And I want to talk a little bit about that, and some of the examples, so you can see what it is that we ' re doing. But before I do this â and this is another one of my pet peeves â I want to talk a little bit about who the poor are. Because we too often talk about them as these strong, huge masses of people yearning to be free, when in fact, it ' s quite an amazing story. On a macro level, four billion people on Earth make less than four dollars a day. That ' s who we talk about when we think about"the poor."If you aggregate it, it ' s the third largest economy on Earth, and yet most of these people go invisible. Where we typically work, there ' s people making between one and three dollars a day. Who are these people? They are farmers and factory workers. They work in government offices. They ' re drivers. They are domestics. They typically pay for critical goods and services like water, like healthcare, like housing, and they pay 30 to 40 times what their middleclass counterparts pay â certainly where we work in Karachi and Nairobi. The poor also are willing to make, and do make, smart decisions, if you give them that opportunity. So, two examples. One is in India, where there are 240 million farmers, most of whom make less than two dollars a day. Where we work in Aurangabad, the land is extraordinarily parched. You see people on average making 60 cents to a dollar. This guy in pink is a social entrepreneur named Ami Tabar. What he did was see what was happening in Israel, larger approaches, and figure out how to do a drip irrigation, which is a way of bringing water directly to the plant stock. But previously it ' s only been created for large-scale farms, so Ami Tabar took this and modularized it down to an eighth of an acre. A couple of principles : build small. Make it infinitely expandable and affordable to the poor. This family, Sarita and her husband, bought a 15-dollar unit when they were living in a â literally a three-walled lean-to with a corrugated iron roof. After one harvest, they had increased their income enough to buy a second system to do their full quarter-acre. A couple of years later, I meet them. They now make four dollars a day, which is pretty much middle class for India, and they showed me the concrete foundation they had just laid to build their house. And I swear, you could see the future in that woman ' s eyes. Something I truly believe. You can ' t talk about poverty today without talking about malaria bed nets, and I again give Jeffrey Sachs of Harvard huge kudos for bringing to the world this notion of his rage â for five dollars you can save a life. Malaria is a disease that kills one to three million people a year. 300 to 500 million cases are reported. It ' s estimated that Africa loses about 13 billion dollars a year to the disease. Five dollars can save a life. We can send people to the moon ; we can see if there ' s life on Mars â why can ' t we get five-dollar nets to 500 million people? The question, though, is not"Why can ' t we?"The question is how can we help Africans do this for themselves? A lot of hurdles. One : production is too low. Two : price is too high. Three : this is a good road in â right near where our factory is located. Distribution is a nightmare, but not impossible. We started by making a 350, 000-dollar loan to the largest traditional bed net manufacturer in Africa so that they could transfer technology from Japan and build these long-lasting, five-year nets. Here are just some pictures of the factory. Today, three years later, the company has employed another thousand women. It contributes about 600, 000 dollars in wages to the economy of Tanzania. It ’ s the largest company in Tanzania. The throughput rate right now is 1. 5 million nets, three million by the end of the year. We hope to have seven million at the end of next year. So the production side is working. On the distribution side, though, as a world, we have a lot of work to do. Right now, 95 percent of these nets are being bought by the U. N., and then given primarily to people around Africa. We ’ re looking at building on some of the most precious resources of Africa : people. Their women. And so I want you to meet Jacqueline, my namesake, 21 years old. If she were born anywhere else but Tanzania, I ’ m telling you, she could run Wall Street. She runs two of the lines, and has already saved enough money to put a down payment on her house. She makes about two dollars a day, is creating an education fund, and told me she is not marrying nor having children until these things are completed. And so, when I told her about our idea â that maybe we could take a Tupperware model from the United States, and find a way for the women themselves to go out and sell these nets to others â she quickly started calculating what she herself could make and signed up. We took a lesson from IDEO, one of our favorite companies, and quickly did a prototyping on this, and took Jacqueline into the area where she lives. She brought 10 of the women with whom she interacts together to see if she could sell these nets, five dollars apiece, despite the fact that people say nobody will buy one, and we learned a lot about how you sell things. Not coming in with our own notions, because she didn ’ t even talk about malaria until the very end. First, she talked about comfort, status, beauty. These nets, she said, you put them on the floor, bugs leave your house. Children can sleep through the night ; the house looks beautiful ; you hang them in the window. And we ’ ve started making curtains, and not only is it beautiful, but people can see status â that you care about your children. Only then did she talk about saving your children ’ s lives. A lot of lessons to be learned in terms of how we sell goods and services to the poor. I want to end just by saying that there ’ s enormous opportunity to make poverty history. To do it right, we have to build business models that matter, that are scaleable and that work with Africans, Indians, people all over the developing world who fit in this category, to do it themselves. Because at the end of the day, it ’ s about engagement. It ’ s about understanding that people really don ’ t want handouts, that they want to make their own decisions ; they want to solve their own problems ; and that by engaging with them, not only do we create much more dignity for them, but for us as well. And so I urge all of you to think next time as to how to engage with this notion and this opportunity that we all have â to make poverty history â by really becoming part of the process and moving away from an us-and-them world, and realizing that it ’ s about all of us, and the kind of world that we, together, want to live in and share. Thank you.

Text Sample 651: NEG Dim 4 – 2005 – Score 4.00 – 2005/t000479.txt

This is really a two-hour presentation I give to high school students, cut down to three minutes. And it all started one day on a plane, on my way to TED, seven years ago. And in the seat next to me was a high school student, a teenager, and she came from a really poor family. And she wanted to make something of her life, and she asked me a simple little question. She said,”What leads to success?”And I felt really badly, because I couldn ’ t give her a good answer. So I get off the plane, and I come to TED. And I think, jeez, I ’ m in the middle of a room of successful people! So why don ’ t I ask them what helped them succeed, and pass it on to kids? So here we are, seven years, 500 interviews later, and I ’ m going to tell you what really leads to success and makes TEDsters tick. And the first thing is passion. Freeman Thomas says,”I ’ m driven by my passion.”TEDsters do it for love ; they don ’ t do it for money. Carol Coletta says,”I would pay someone to do what I do.”And the interesting thing is : if you do it for love, the money comes anyway. Work! Rupert Murdoch said to me,”It ’ s all hard work. Nothing comes easily. But I have a lot of fun.”Did he say fun? Rupert? Yes! TEDsters do have fun working. And they work hard. I figured, they ’ re not workaholics. They ’ re workafrolics. Good! Alex Garden says,”To be successful, put your nose down in something and get damn good at it.”There ’ s no magic ; it ’ s practice, practice, practice. And it ’ s focus. Norman Jewison said to me,”I think it all has to do with focusing yourself on one thing.”And push! David Gallo says,”Push yourself. Physically, mentally, you ’ ve got to push, push, push.”You ’ ve got to push through shyness and self-doubt. Goldie Hawn says,”I always had self-doubts. I wasn ’ t good enough ; I wasn ’ t smart enough. I didn ’ t think I ’ d make it.”Now it ’ s not always easy to push yourself, and that ’ s why they invented mothers. Frank Gehry said to me,”My mother pushed me.”Serve! Sherwin Nuland says,”It was a privilege to serve as a doctor.”A lot of kids want to be millionaires. The first thing I say is :”OK, well you can ’ t serve yourself ; you ’ ve got to serve others something of value. Because that ’ s the way people really get rich.”Ideas! TEDster Bill Gates says,”I had an idea : founding the first micro- computer software company.”I ’ d say it was a pretty good idea. And there ’ s no magic to creativity in coming up with ideas — it ’ s just doing some very simple things. And I give lots of evidence. Persist! Joe Kraus says,”Persistence is the number one reason for our success.”You ’ ve got to persist through failure. You ’ ve got to persist through crap! Which of course means”Criticism, Rejection, Assholes and Pressure.”So, the answer to this question is simple : Pay 4, 000 bucks and come to TED. Or failing that, do the eight things — and trust me, these are the big eight things that lead to success. Thank you TEDsters for all your interviews!

Text Sample 652: NEG Dim 4 – 1998 – Score 4.00 – 1998/t000134.txt

’ Theme and variations ’ is one of those forms that require a certain kind of intellectual activity, because you are always comparing the variation with the theme that you hold in your mind. You might say that the theme is nature and everything that follows is a variation on the subject. I was asked, I guess about six years ago, to do a series of paintings that in some way would celebrate the birth of Piero della Francesca. And it was very difficult for me to imagine how to paint pictures that were based on Piero until I realized that I could look at Piero as nature — that I would have the same attitude towards looking at Piero della Francesca as I would if I were looking out a window at a tree. And that was enormously liberating to me. Perhaps it ’ s not a very insightful observation, but that really started me on a path to be able to do a kind of theme and variations based on a work by Piero, in this case that remarkable painting that ’ s in the Uffizi,”The Duke of Montefeltro,”who faces his consort, Battista. Once I realized that I could take some liberties with the subject, I did the following series of drawings. That ’ s the real Piero della Francesca —

one of the greatest portraits in human history. And these, I ’ ll just show these without comment. It ’ s just a series of variations on the head of the Duke of Montefeltro, who ’ s a great, great figure in the Renaissance, and probably the basis for Machiavelli ’ s”The Prince.”He apparently lost an eye in battle, which is why he is always shown in profile. And this is Battista. And then I decided I could move them around a little bit — so that for the first time in history, they ’ re facing the same direction. Whoops! Passed each other. And then a visitor from another painting by Piero, this is from”The Resurrection of Christ”— as though the cast had just gotten of the set to have a chat. And now, four large panels : this is upper left ; upper right ; lower left ; lower right. Incidentally, I ’ ve never understood the conflict between abstraction and naturalism. Since all paintings are inherently abstract to begin with there doesn ’ t seem to be an argument there. On another subject — — I was driving in the country one day with my wife, and I saw this sign, and I said,”That is a fabulous piece of design.”And she said,”What are you talking about?”I said,”Well, it ’ s so persuasive, because the purpose of that sign is to get you into the garage, and since most people are so suspicious of garages, and know that they ’ re going to be ripped off, they use the word ’ reliable. ’ But everybody says they ’ re reliable. But, reliable Dutchman”— —”Fantastic!”Because as soon as you hear the word Dutchman — which is an archaic word, nobody calls Dutch people”Dutchmen”anymore — but as soon as you hear Dutchman, you get this picture of the kid with his finger in the dike, preventing the thing from falling and flooding Holland, and so on. And so the entire issue is detoxified by the use of”Dutchman.”Now, if you think I ’ m exaggerating at all in this, all you have to do is substitute something else, like”Indonesian.”Or even”French.”Now,”Swiss”works, but you know it ’ s going to cost a lot of money. I ’ m going to take you quickly through the actual process of doing a poster. I do a lot of work for the School of Visual Arts, where I teach, and the director of this school — a remarkable man named Silas Rhodes — often gives you a piece of text and he says,”Do something with this.”And so he did. And this was the text —”In words as fashion the same rule will hold / Alike fantastic if too new or old / Be not the first by whom the new are tried / Nor yet the last to lay the old aside.”I could make nothing of that. And I really struggled with this one. And the first thing I did, which was sort of in the absence of another idea, was say I ’ ll sort of write it out and make some words big, and I ’ ll have some kind of design on the back somehow, and I was hoping — as one often does — to stumble into something. So I took another crack at it — you ’ ve got to keep it moving — and I Xeroxed some words on pieces of colored paper and I pasted them down on an ugly board. I thought that something would come out of it, like”Words rule fantastic new old first last Pope”because it ’ s by Alexander Pope, but I sort of made a mess out of it, and then I thought I ’ d repeat it in some way so it was legible. So, it was going nowhere. Sometimes, in the middle of a resistant problem, I write down things that I know about it. But you can see the beginning of an idea there, because you can see the word”new”emerging from the”old.”That ’ s what happens. There ’ s a relationship between the old and the new ; the new emerges from the context of the old. And then I did some variations of it, but it still wasn ’ t coalescing graphically at all. I had this other version which had something interesting about it in terms of being able to put it together in your mind from clues. The W was clearly a W, the N was clearly an N, even though they were very fragmentary and there wasn ’ t a lot of information in it. Then I got the words”new”and”old”and now I had regressed back to a point where there seemed to be no return. I was really desperate at this point. And so, I do something I ’ m truthfully ashamed of, which is that I took two drawings I had made for another purpose and I put them together. It says”dreams”at the top. And I was going to do a thing, I say,"Well, change the copy. Let it say something about dreams, and come to SVA and you ' ll sort of fulfill your dreams."But, to my credit, I was so embarrassed about doing that that I never submitted this sketch. And, finally, I arrived at the following solution. Now, it doesn ' t look terribly interesting, but it does have something that distinguishes it from a lot of other posters. For one thing, it transgresses the idea of what a poster ' s supposed to be, which is to be understood and seen immediately, and not explained. I remember hearing all of you in the graphic arts —"If you have to explain it, it ain ' t working."And one day I woke up and I said,"Well, suppose that ' s not true?"So here ' s what it says in my explanation at the bottom left. It says,"Thoughts : This poem is impossible. Silas usually has a better touch with his choice of quotations. This one generates no imagery at all."I am now exposing myself to my audience, right? Which is something you never want to do professionally."Maybe the words can make the image without anything else happening. What ' s the heart of this poem? Don ' t be trendy if you want to be serious. Is doing the poster this way trendy in itself? I guess one could reduce the idea further by suggesting that the new emerges behind and through the old, like this."And then I show you a little drawing — you see, you remember that old thing I discarded? Well, I found a way to use it. So, there ' s that little alternative over there, and I say,"Not bad,"— criticizing myself —"but more didactic than visual. Maybe what wants to be said is that old and the new are locked in a dialectical embrace, a kind of dance where each defines the other."And then more self-questioning —"Am I being simple-minded? Is this the kind of simple that looks obvious, or the kind that looks profound? There ' s a significant difference. This could be embarrassing. Actually, I realize fear of embarrassment drives me as much as any ambition. Do you think this sort of thing could really attract a student to the school?"Well, I think there are two fresh things here — two fresh things. One is the sort of willingness to expose myself to a critical audience, and not to suggest that I am confident about what I ' m doing. And as you know, you have to have a front. I mean — you ' ve got to be confident ; if you don ' t believe in your work, who else is going to believe in it? So that ' s one thing, to introduce the idea of doubt into graphics. That can be a big contribution. The other thing is to actually give you two solutions for the price of one ; you get the big one and if you don ' t like that, how about the little one? And that too is a relatively new idea. And here ' s just a series of experiments where I ask the question of — does a poster have to be square? Now, this is a little illusion. That poster is not folded. It ' s not folded, that ' s a photograph and it ' s cut on the diagonal. Same cheap trick in the upper left- hand corner. And here, a very peculiar poster because, simply because of using the isometric perspective in the computer, it won ' t sit still in the space. At times, it seems to be wider at the back than the front, and then it shifts. And if you sit here long enough, it ' ll float off the page into the audience. But, we don ' t have time. And then an experiment — a little bit about the nature of perspective, where the outside shape is determined by the peculiarity of perspective, but the shape of the bottle — which is identical to the outside shape — is seen frontally. And another piece for the art directors ' club is"Anna Rees"casting long shadows. This is another poster from the School of Visual Arts. There were 10 artists invited to participate in it, and it was one of those things where it was extremely competitive and I didn ' t want to be embarrassed, so I worked very hard on this. The idea was — and it was a brilliant idea — was to have 10 posters distributed throughout the city ' s subway system so every time you got on the subway you ' d be passing a different poster, all of which had a different idea of what art is. But I was absolutely stuck on the idea of"art is"and trying to determine what art was.

But then I gave up and I said,”Well, art is whatever.”And as soon as I said that, I discovered that the word”hat”was hidden in the word”whatever,”and that led me to the inevitable conclusion. But then again, it ’ s on my list of didactic posters. My intent is to have a literary accompaniment that explains the poster, in case you don ’ t get it. Now this says,”Note to the viewer : I thought I might use a visual cliche of our time — Magritte ’ s everyman — to express the idea that art is mystery, continuity and history. I ’ m also convinced that, in an age of computer manipulation, surrealism has become banal, a shadow of its former self. The phrase ’ Art is whatever ’ expresses the current inclusiveness that surrounds art-making — a sort of ’ it ain ’ t what you do, it ’ s the way that you do it ’ notion. The shadow of Magritte falls across the central part of the poster a poetic event that occurs as the shadow man isolates the word ’ hat, ’ hidden in the word ’ whatever. ’ The four hats shown in the poster suggests how art might be defined : as a thing itself, the worth of the thing, the shadow of the thing, and the shape of the thing. Whatever.”OK. And the one that I did not submit, which I still like, I wanted to use the same phrase. There were wonderful experiments by Bruno Munari on letterforms some years ago — sort of, see how far you could go and still be able to read them. And that idea stuck in my head. But then I took the pieces that I had taken off and put them at the bottom. And, of course those are the remains, and they ’ re so labeled. But what really happens is that you read it as :”Art is whatever remains.”Thank you.

Text Sample 653: NEG Dim 4 – 1998 – Score 5.00 – 1998/t000216.txt
David Gallo : This is Bill Lange. I ’ m Dave Gallo. And we ’ re going to tell you some stories from the sea here in video. We ’ ve got some of the most incredible video of Titanic that ’ s ever been seen, and we ’ re not going to show you any of it. The truth of the matter is that the Titanic — even though it ’ s breaking all sorts of box office records — it ’ s not the most exciting story from the sea. And the problem, I think, is that we take the ocean for granted. When you think about it, the oceans are 75 percent of the planet. Most of the planet is ocean water. The average depth is about two miles. Part of the problem, I think, is we stand at the beach, or we see images like this of the ocean, and you look out at this great big blue expanse, and it ’ s shimmering and it ’ s moving and there ’ s waves and there ’ s surf and there ’ s tides, but you have no idea for what lies in there. And in the oceans, there are the longest mountain ranges on the planet. Most of the animals are in the oceans. Most of the earthquakes and volcanoes are in the sea, at the bottom of the sea. The biodiversity and the biodensity in the ocean is higher, in places, than it is in the rainforests. It ’ s mostly unexplored, and yet there are beautiful sights like this that captivate us and make us become familiar with it. But when you ’ re standing at the beach, I want you to think that you ’ re standing at the edge of a very unfamiliar world. We have to have a very special technology to get into that unfamiliar world. We use the submarine Alvin and we use cameras, and the cameras are something that Bill Lange has developed with the help of Sony. Marcel Proust said,”The true voyage of discovery is not so much in seeking new landscapes as in having new eyes.”People that have partnered with us have given us new eyes, not only on what exists — the new landscapes at the bottom of the sea — but also how we think about life on the planet itself. Here ’ s a jelly. It ’ s one of my favorites, because it ’ s got all sorts of working parts. This turns out to be the longest creature in the oceans. It gets up to about 150 feet long. But see all those different working things? I love that kind of stuff. It ’ s got

these fishing lures on the bottom. They ’ re going up and down. It ’ s got tentacles dangling, swirling around like that. It ’ s a colonial animal. These are all individual animals banding together to make this one creature. And it ’ s got these jet thrusters up in front that it ’ ll use in a moment, and a little light. If you take all the big fish and schooling fish and all that, put them on one side of the scale, put all the jelly-type of animals on the other side, those guys win hands down. Most of the biomass in the ocean is made out of creatures like this. Here ’ s the X-wing death jelly. The bioluminescence — they use the lights for attracting mates and attracting prey and communicating. We couldn ’ t begin to show you our archival stuff from the jellies. They come in all different sizes and shapes. Bill Lange : We tend to forget about the fact that the ocean is miles deep on average, and that we ’ re real familiar with the animals that are in the first 200 or 300 feet, but we ’ re not familiar with what exists from there all the way down to the bottom. And these are the types of animals that live in that three- dimensional space, that micro-gravity environment that we really haven ’ t explored. You hear about giant squid and things like that, but some of these animals get up to be approximately 140, 160 feet long. They ’ re very little understood. DG : This is one of them, another one of our favorites, because it ’ s a little octopod. You can actually see through his head. And here he is, flapping with his ears and very gracefully going up. We see those at all depths and even at the greatest depths. They go from a couple of inches to a couple of feet. They come right up to the submarine — they ’ ll put their eyes right up to the window and peek inside the sub. This is really a world within a world, and we ’ re going to show you two. In this case, we ’ re passing down through the mid-ocean and we see creatures like this. This is kind of like an undersea rooster. This guy, that looks incredibly formal, in a way. And then one of my favorites. What a face! This is basically scientific data that you ’ re looking at. It ’ s footage that we ’ ve collected for scientific purposes. And that ’ s one of the things that Bill ’ s been doing, is providing scientists with this first view of animals like this, in the world where they belong. They don ’ t catch them in a net. They ’ re actually looking at them down in that world. We ’ re going to take a joystick, sit in front of our computer, on the Earth, and press the joystick forward, and fly around the planet. We ’ re going to look at the mid-ocean ridge, a 40, 000-mile long mountain range. The average depth at the top of it is about a mile and a half. And we ’ re over the Atlantic — that ’ s the ridge right there — but we ’ re going to go across the Caribbean, Central America, and end up against the Pacific, nine degrees north. We make maps of these mountain ranges with sound, with sonar, and this is one of those mountain ranges. We ’ re coming around a cliff here on the right. The height of these mountains on either side of this valley is greater than the Alps in most cases. And there ’ s tens of thousands of those mountains out there that haven ’ t been mapped yet. This is a volcanic ridge. We ’ re getting down further and further in scale. And eventually, we can come up with something like this. This is an icon of our robot, Jason, it ’ s called. And you can sit in a room like this, with a joystick and a headset, and drive a robot like that around the bottom of the ocean in real time. One of the things we ’ re trying to do at Woods Hole with our partners is to bring this virtual world — this world, this unexplored region — back to the laboratory. Because we see it in bits and pieces right now. We see it either as sound, or we see it as video, or we see it as photographs, or we see it as chemical sensors, but we never have yet put it all together into one interesting picture. Here ’ s where Bill ’ s cameras really do shine. This is what ’ s called a hydrothermal vent. And what you ’ re seeing here is a cloud of densely packed, hydrogen-sulfide-rich water coming out of a volcanic axis on the sea floor. Gets up to 600, 700 degrees F, somewhere in that range. So that ’ s all water under the sea — a mile and a half, two miles, three miles down. And we knew it was volcanic back in the ’ 60s, ’ 70s. And then we had some hint that these things existed all along the axis of it, because if you ’ ve got volcanism, water ’ s going to get down from the sea into cracks in the sea floor, come in contact with magma, and come shooting out hot. We weren ’ t really aware that it would be so rich with sulfides, hydrogen sulfides. We didn ’ t have any idea about these things, which we call chimneys. This is one of these hydrothermal vents. Six hundred degree F water coming out of the Earth. On either side of us are mountain ranges that are higher than the Alps, so the setting here is very dramatic. BL : The white material is a type of bacteria that thrives at 180 degrees C. DG : I think that ’ s one of the greatest stories right now that we ’ re seeing from the bottom of the sea, is that the first thing we see coming out of the sea floor after a volcanic eruption is bacteria. And we started to wonder for a long time, how did it all get down there? What we find out now is that it ’ s probably coming from inside the Earth. Not only is it coming out of the Earth — so, biogenesis made from volcanic activity — but that bacteria supports these colonies of life. The pressure here is 4, 000 pounds per square inch. A mile and a half from the surface to two miles to three miles — no sun has ever gotten down here. All the energy to support these life forms is coming from inside the Earth — so, chemosynthesis. And you can see how dense the population is. These are called tube worms. BL : These worms have no digestive system. They have no mouth. But they have two types of gill structures. One for extracting oxygen out of the deep-sea water, another one which houses this chemosynthetic bacteria, which takes the hydrothermal fluid — that hot water that you saw coming out of the bottom — and converts that into simple sugars that the tube worm can digest. DG : You can see, here ’ s a crab that lives down there. He ’ s managed to grab a tip of these worms. Now, they normally retract as soon as a crab touches them. Oh! Good going. So, as soon as a crab touches them, they retract down into their shells, just like your fingernails. There ’ s a whole story being played out here that we ’ re just now beginning to have some idea of because of this new camera technology. BL : These worms live in a real temperature extreme. Their foot is at about 200 degrees C and their head is out at three degrees C, so it ’ s like having your hand in boiling water and your foot in freezing water. That ’ s how they like to live. DG : This is a female of this kind of worm. And here ’ s a male. You watch. It doesn ’ t take long before two guys here — this one and one that will show up over here — start to fight. Everything you see is played out in the pitch black of the deep sea. There are never any lights there, except the lights that we bring. Here they go. On one of the last dive series, we counted 200 species in these areas — 198 were new, new species. BL : One of the big problems is that for the biologists working at these sites, it ’ s rather difficult to collect these animals. And they disintegrate on the way up, so the imagery is critical for the science. DG : Two octopods at about two miles depth. This pressure thing really amazes me — that these animals can exist there at a depth with pressure enough to crush the Titanic like an empty Pepsi can. What we saw up till now was from the Pacific. This is from the Atlantic. Even greater depth. You can see this shrimp is harassing this poor little guy here, and he ’ ll bat it away with his claw. Whack! And the same thing ’ s going on over here. What they ’ re getting at is that — on the back of this crab — the foodstuff here is this very strange bacteria that lives on the backs of all these animals. And what these shrimp are trying to do is actually harvest the bacteria from the backs of these animals. And the crabs don ’ t like it at all. These long filaments that you see on the back of the crab are actually created by the product of that bacteria. So, the bacteria grows hair on the crab. On the back, you see this again. The red dot is the laser light of the submarine Alvin to give us an idea about how far away we are from the vents. Those are all shrimp. You see the hot water over here, here and here, coming out. They ’ re clinging to a rock face and actually scraping bacteria off that rock face. Here ’ s a tiny, little vent that ’ s come out of the side of that pillar. Those pillars get up to several stories. So here, you ’ ve got this valley with this incredible alien landscape of pillars and hot springs and volcanic eruptions and earthquakes, inhabited by these very strange animals that live only on chemical energy coming out of the ground. They don ’ t need the sun at all. BL : You see this white V-shaped mark on the back of the shrimp? It ’ s actually a light-sensing organ. It ’ s how they find the hydrothermal vents. The vents are emitting a black body radiation — an IR signature — and so they ’ re able to find these vents at considerable distances. DG : All this stuff is happening along that 40, 000-mile long mountain range that we ’ re calling the ribbon of life, because just even today, as we speak, there ’ s life being generated there from volcanic activity. This is the first time we ’ ve ever tried this any place. We ’ re going to try to show you high definition from the Pacific. We ’ re moving up one of these pillars. This one ’ s several stories tall. In it, you ’ ll see that it ’ s a habitat for a lot of different animals. There ’ s a funny kind of hot plate here, with vent water coming out of it. So all of these are individual homes for worms. Now here ’ s a closer view of that community. Here ’ s crabs here, worms here. There are smaller animals **crawling** around. Here ’ s pagoda structures. I think this is the neatest-looking thing. I just can ’ t get over this — that you ’ ve got these little chimneys sitting here smoking away. This stuff is toxic as hell, by the way. You could never get a permit to dump this in the ocean, and it ’ s coming out all from it. It ’ s unbelievable. It ’ s basically sulfuric acid, and it ’ s being just dumped out, at incredible rates. And animals are thriving — and we probably came from here. That ’ s probably where we evolved from. BL : This bacteria that we ’ ve been talking about turns out to be the most simplest form of life found. There are a number of groups that are proposing that life evolved at these vent sites. Although the vent sites are short-lived — an individual site may last only 10 years or so — as an ecosystem they ’ ve been stable for millions — well, billions — of years. DG : It works too well. You see there ’ re some fish inside here as well. There ’ s a fish sitting here. Here ’ s a crab with his claw right at the end of that tube worm, waiting for that worm to stick his head out. BL : The biologists right now cannot explain why these animals are so active. The worms are growing inches per week! DG : I already said that this site, from a human perspective, is toxic as hell. Not only that, but on top — the lifeblood — that plumbing system turns off every year or so. Their plumbing system turns off, so the sites have to move. And then there ’ s earthquakes, and then volcanic eruptions, on the order of one every five years, that completely wipes the area out. Despite that, these animals grow back in about a year ’ s time. You ’ re talking about biodensities and biodiversity, again, higher than the rainforest that just springs back to life. Is it sensitive? Yes. Is it fragile? No, it ’ s not really very fragile. I ’ ll end up with saying one thing. There ’ s a story in the sea, in the waters of the sea, in the sediments and the rocks of the sea floor. It ’ s an incredible story. What we see when we look back in time, in those sediments and rocks, is a record of Earth history. Everything on this planet — everything — works by cycles and rhythms. The continents move apart. They come back together. Oceans come and go. Mountains come and go. Glaciers come and go. El Nino comes and goes. It ’ s not a disaster, it ’ s rhythmic. What we ’ re learning now, it ’ s almost like a symphony. It ’ s just like music — it really is just like music. And what we ’ re learning now is that you can ’ t listen to a five-billion-year long symphony, get to today and say,”Stop! We want tomorrow ’ s note to be the same as it was today.”It ’ s absurd. It ’ s just absurd. So, what we ’ ve got to learn now is to find out where this planet ’ s going at all these different scales and work with it. Learn to manage it. The concept of preservation is futile. Conservation ’ s tougher, but we can probably get there. Thank you very much. Thank you.

Text Sample 654: NEG Dim 4 – 1998 – Score 8.00 – 1998/t000113.txt
Sheryl Shade : Hi, Aimee. Aimee Mullins : Hi. SS : Aimee and I thought we ’ d just talk a little bit, and I wanted her to tell all of you what makes her a distinctive athlete. AM : Well, for those of you who have seen the picture in the little bio â it might have given it away â I ’ m a double amputee, and I was born without fibulas in both legs. I was amputated at age one, and I ’ ve been running like hell ever since, all over the place. SS : Well, why don ’ t you tell them how you got to Georgetown â why don ’ t we start there? Why don ’ t we start there? AM : I ’ m a senior in Georgetown in the Foreign Service program. I won a full academic scholarship out of high school. They pick three students out of the nation every year to get involved in international affairs, and so I won a full ride to Georgetown and I ’ ve been there for four years. Love it. SS : When Aimee got there, she decided that she ’ s, kind of, curious about track and field, so she decided to call someone and start asking about it. So, why don ’ t you tell that story? AM : Yeah. Well, I guess I ’ ve always been involved in sports. I played softball for five years growing up. I skied competitively throughout high school, and I got a little restless in college because I wasn ’ t doing anything for about a year or two sports-wise. And I ’ d never competed on a disabled level, you know â I ’ d always competed against other able-bodied athletes. That ’ s all I ’ d ever known. In fact, I ’ d never even met another amputee until I was 17. And I heard that they do these track meets with all disabled runners, and I figured,”Oh, I don ’ t know about this, but before I judge it, let me go see what it ’ s all about.”So, I booked myself a flight to Boston in ’ 95, 19 years old and definitely the dark horse candidate at this race. I ’ d never done it before. I went out on a gravel track a couple of weeks before this meet to see how far I could run, and about 50 meters was enough for me, panting and heaving. And I had these legs that were made of a wood and plastic compound, attached with Velcro straps â big, thick, five-ply wool socks on â you know, not the most comfortable things, but all I ’ d ever known. And I ’ m up there in Boston against people wearing legs made of all things â carbon graphite and, you know, shock absorbers in them and all sorts of things â and they ’ re all looking at me like, OK, we know who ’ s not going to win this race. And, I mean, I went up there expecting â I don ’ t know what I was expecting â but, you know, when I saw a man who was missing an entire leg go up to the high jump, hop on one leg to the high jump and clear it at six feet, two inches... Dan O ’ Brien jumped 5 ’ 11”in ’ 96 in Atlanta, I mean, if it just gives you a comparison of â these are truly accomplished athletes, without qualifying that word”athlete.”And so I decided to give this a shot : heart pounding, I ran my first race and I beat the national record-holder by three hundredths of a second, and became the new national record-holder on my first try out. And, you know, people said,”Aimee, you know, you ’ ve got speed â you ’ ve got natural speed â but you don ’ t have any skill or finesse going down that track. You were all over the place. We all saw how hard you were working.”And so I decided to call the track coach at Georgetown. And I thank god I didn ’ t know just how huge this man is in the track and field world. He ’ s coached five Olympians, and the man ’ s office is lined from floor to ceiling with All America certificates of all these athletes he ’ s coached. He ’ s just a rather intimidating figure. And I called him up and said,”Listen, I ran

one race and I won...""I want to see if I can, you know â I need to just see if I can sit in on some of your practices, see what drills you do and whatever."That ' s all I wanted â just two practices."Can I just sit in and see what you do?"And he said,"Well, we should meet first, before we decide anything."You know, he ' s thinking,"What am I getting myself into?"So, I met the man, walked in his office, and saw these posters and magazine covers of people he has coached. And we got to talking, and it turned out to be a great partnership because he ' d never coached a disabled athlete, so therefore he had no preconceived notions of what I was or wasn ' t capable of, and I ' d never been coached before. So this was like,"Here we go â let ' s start on this trip."So he started giving me four days a week of his lunch break, his free time, and I would come up to the track and train with him. So that ' s how I met Frank. That was fall of ' 95. But then, by the time that winter was rolling around, he said,"You know, you ' re good enough. You can run on our women ' s track team here."And I said,"No, come on."And he said,"No, no, really. You can. You can run with our women ' s track team."In the spring of 1996, with my goal of making the U. S. Paralympic team that May coming up full speed, I joined the women ' s track team. And no disabled person had ever done that â run at a collegiate level. So I don ' t know, it started to become an interesting mix. SS : Well, on your way to the Olympics, a couple of memorable events happened at Georgetown. Why don ' t you just tell them? AM : Yes, well, you know, I ' d won everything as far as the disabled meets â everything I competed in â and, you know, training in Georgetown and knowing that I was going to have to get used to seeing the backs of all these women ' s shirts â you know, I ' m running against the next Flo-Jo â and they ' re all looking at me like,"Hmm, what ' s, you know, what ' s going on here?"And putting on my Georgetown uniform and going out there and knowing that, you know, in order to become better â and I ' m already the best in the country â you know, you have to train with people who are inherently better than you. And I went out there and made it to the Big East, which was sort of the championship race at the end of the season. It was really, really hot. And it ' s the first â I had just gotten these new sprinting legs that you see in that bio, and I didn ' t realize at that time that the amount of sweating I would be doing in the sock â it actually acted like a lubricant and I ' d be, kind of, pistoning in the socket. And at about 85 meters of my 100 meters sprint, in all my glory, I came out of my leg. Like, I almost came out of it, in front of, like, 5, 000 people. And I, I mean, was just mortified â because I was signed up for the 200, you know, which went off in a half hour. I went to my coach :"Please, don ' t make me do this."I can ' t do this in front of all those people. My legs will come off. And if it came off at 85 there ' s no way I ' m going 200 meters. And he just sat there like this. My pleas fell on deaf ears, thank god. Because you know, the man is from Brooklyn ; he ' s a big man. He says,"Aimee, so what if your leg falls off? You pick it up, you put the damn thing back on, and finish the goddamn race!"And I did. So, he kept me in line. He kept me on the right track. SS : So, then Aimee makes it to the 1996 Paralympics, and she ' s all excited. Her family ' s coming down â it ' s a big deal. It ' s now two years that you ' ve been running? AM : No, a year. SS : A year. And why don ' t you tell them what happened right before you go run your race? AM : Okay, well, Atlanta. The Paralympics, just for a little bit of clarification, are the Olympics for people with physical disabilities â amputees, persons with cerebral palsy, and wheelchair athletes â as opposed to the Special Olympics, which deals with people with mental disabilities. So, here we are, a week after the Olympics and down at Atlanta, and I ' m just blown away by the fact that just a year ago, I got out on a gravel track and couldn ' t run 50 meters. And so, here I am â never lost. I set new records at the U. S. Nationals â the

Olympic trials â that May, and was sure that I was coming home with the gold. I was also the only, what they call”bilateral BK”â below the knee. I was the only woman who would be doing the long jump. I had just done the long jump, and a guy who was missing two legs came up to me and says,”How do you do that? You know, we ’ re supposed to have a planar foot, so we can ’ t get off on the springboard.”I said,”Well, I just did it. No one told me that.”So, it ’ s funny â I ’ m three inches within the world record â and kept on from that point, you know, so I ’ m signed up in the long jump â signed up? No, I made it for the long jump and the 100-meter. And I ’ m sure of it, you know? I made the front page of my hometown paper that I delivered for six years, you know? It was, like, this is my time for shine. And we ’ re at the trainee warm-up track, which is a few blocks away from the Olympic stadium. These legs that I was on, which I ’ ll take out right now â I was the first person in the world on these legs. I was the guinea pig., I ’ m telling you, this was, like â talk about a tourist attraction. Everyone was taking pictures â”What is this girl running on?”And I ’ m always looking around, like, where is my competition? It ’ s my first international meet. I tried to get it out of anybody I could, you know,”Who am I running against here?””Oh, Aimee, we ’ ll have to get back to you on that one.”I wanted to find out times.”Don ’ t worry, you ’ re doing great.”This is 20 minutes before my race in the Olympic stadium, and they post the heat sheets. And I go over and look. And my fastest time, which was the world record, was 15. 77. Then I ’ m looking : the next lane, lane two, is 12. 8. Lane three is 12. 5. Lane four is 12. 2. I said,”What ’ s going on?”And they shove us all into the shuttle bus, and all the women there are missing a hand. So, I ’ m just, like â they ’ re all looking at me like ’ which one of these is not like the other, ’ you know? I ’ m sitting there, like,”Oh, my god. Oh, my god.”You know, I ’ d never lost anything, like, whether it would be the scholarship or, you know, I ’ d won five golds when I skied. In everything, I came in first. And Georgetown â that was great. I was losing, but it was the best training because this was Atlanta. Here we are, like, crĂ¨me de la crĂ¨me, and there is no doubt about it, that I ’ m going to lose big. And, you know, I ’ m just thinking,”Oh, my god, my whole family got in a van and drove down here from Pennsylvania.”And, you know, I was the only female U. S. sprinter. So they call us out and, you know â”Ladies, you have one minute.”And I remember putting my blocks in and just feeling horrified because there was just this murmur coming over the crowd, like, the ones who are close enough to the starting line to see. And I ’ m like,”I know! Look! This isn ’ t right.”And I ’ m thinking that ’ s my last card to play here ; if I ’ m not going to beat these girls, I ’ m going to mess their heads a little, you know? I mean, it was definitely the”Rocky IV”sensation of me versus Germany, and everyone else â Estonia and Poland â was in this heat. And the gun went off, and all I remember was finishing last and fighting back tears of frustration and incredible â incredible â this feeling of just being overwhelmed. And I had to think,”Why did I do this?”If I had won everything â but it was like, what was the point? All this training â I had transformed my life. I became a collegiate athlete, you know. I became an Olympic athlete. And it made me really think about how the achievement was getting there. I mean, the fact that I set my sights, just a year and three months before, on becoming an Olympic athlete and saying,”Here ’ s my life going in this direction â and I want to take it here for a while, and just seeing how far I could push it.”And the fact that I asked for help â how many people jumped on board? How many people gave of their time and their expertise, and their patience, to deal with me? And that was this collective glory â that there was, you know, 50 people behind me that had joined in this incredible experience of going to Atlanta. So, I apply this sort of philosophy now to everything I do : sitting back and realizing the progression, how far you ’ ve come at this day to this goal, you know. It ’ s important to focus on a goal, I think, but also recognize the progression on the way there and how you ’ ve grown as a person. That ’ s the achievement, I think. That ’ s the real achievement. SS : Why don ’ t you show them your legs? AM : Oh, sure. SS : You know, show us more than one set of legs. AM : Well, these are my pretty legs. No, these are my cosmetic legs, actually, and they ’ re absolutely beautiful. You ’ ve got to come up and see them. There are hair follicles on them, and I can paint my toenails. And, seriously, like, I can wear heels. Like, you guys don ’ t understand what that ’ s like to be able to just go into a shoe store and buy whatever you want. SS : You got to pick your height? AM : I got to pick my height, exactly. Patrick Ewing, who played for Georgetown in the ’ 80s, comes back every summer. And I had incessant fun making fun of him in the training room because he ’ d come in with foot injuries. I ’ m like,”Get it off! Don ’ t worry about it, you know. You can be eight feet tall. Just take them off.”He didn ’ t find it as humorous as I did, anyway. OK, now, these are my sprinting legs, made of carbon graphite, like I said, and I ’ ve got to make sure I ’ ve got the right socket. No, I ’ ve got so many legs in here. These are â do you want to hold that actually? That ’ s another leg I have for, like, tennis and softball. It has a shock absorber in it so it, like,”Shhhh,”makes this neat sound when you jump around on it. All right. And then this is the silicon sheath I roll over, to keep it on. Which, when I sweat, you know, I ’ m pistoning out of it. SS : Are you a different height? AM : In these? SS : In these. AM : I don ’ t know. I don ’ t think so. I may be a little taller. I actually can put both of them on. SS : She can ’ t really stand on these legs. She has to be moving, so... AM : Yeah, I definitely have to be moving, and balance is a little bit of an art in them. But without having the silicon sock, I ’ m just going to try slip in it. And so, I run on these, and have shocked half the world on these. These are supposed to simulate the actual form of a sprinter when they run. If you ever watch a sprinter, the ball of their foot is the only thing that ever hits the track. So when I stand in these legs, my hamstring and my glutes are contracted, as they would be had I had feet and were standing on the ball of my feet. AM : It ’ s a company in San Diego called Flex-Foot. And I was a guinea pig, as I hope to continue to be in every new form of prosthetic limbs that come out. But actually these, like I said, are still the actual prototype. I need to get some new ones because the last meet I was at, they were everywhere. You know, it ’ s like a big â it ’ s come full circle. Moderator : Aimee and the designer of them will be at TEDMED 2, and we ’ ll talk about the design of them. AM : Yes, we ’ ll do that. SS : Yes, there you go. AM : So, these are the sprint legs, and I can put my other... SS : Can you tell about who designed your other legs? AM : Yes. These I got in a place called Bournemouth, England, about two hours south of London, and I ’ m the only person in the United States with these, which is a crime because they are so beautiful. And I don ’ t even mean, like, because of the toes and everything. For me, while I ’ m such a serious athlete on the track, I want to be feminine off the track, and I think it ’ s so important not to be limited in any capacity, whether it ’ s, you know, your mobility or even fashion. I mean, I love the fact that I can go in anywhere and pick out what I want â the shoes I want, the skirts I want â and I ’ m hoping to try to bring these over here and make them accessible to a lot of people. They ’ re also silicon. This is a really basic, basic prosthetic limb under here. It ’ s like a Barbie foot under this. It is. It ’ s just stuck in this position, so I have to wear a two-inch heel. And, I mean, it ’ s really â let me take this off so you can see it. I don ’ t know how good you can see it, but, like, it really is. There ’ re veins on the feet, and then my heel is pink, and my Achilles ’ tendon â that moves a little bit. And it ’ s really an amazing store. I got them a year and two weeks ago. And this is just a silicon piece of skin. I mean, what happened was, two years ago this man in Belgium was saying,”God, if I can go to Madame Tussauds ’ wax museum and see Jerry Hall replicated down to the color of her eyes, looking so real as if she breathed, why can ’ t they build a limb for someone that looks like a leg, or an arm, or a hand?”I mean, they make ears for burn victims. They do amazing stuff with silicon. SS : Two weeks ago, Aimee was up for the Arthur Ashe award at the ESPYs. And she came into town and she rushed around and she said,”I have to buy some new shoes!”We ’ re an hour before the ESPYs, and she thought she ’ d gotten a two-inch heel but she ’ d actually bought a three-inch heel. AM : And this poses a problem for me, because it means I ’ m walking like that all night long. SS : For 45 minutes. Luckily, the hotel was terrific. They got someone to come in and saw off the shoes. AM : I said to the receptionist â I mean, I am just harried, and Sheryl ’ s at my side â I said,”Look, do you have anybody here who could help me? Because I have this problem...”You know, at first they were just going to write me off, like,”If you don ’ t like your shoes, sorry. It ’ s too late.””No, no, no, no. I ’ ve got these special feet that need a two-inch heel. I have a three-inch heel. I need a little bit off.”They didn ’ t even want to go there. They didn ’ t even want to touch that one. They just did it. No, these legs are great. I ’ m actually going back in a couple of weeks to get some improvements. I want to get legs like these made for flat feet so I can wear sneakers, because I can ’ t with these ones. So... Moderator : That ’ s it. SS : That ’ s Aimee Mullins.

Text Sample 655: NEG Dim 4 – 2024 – Score -1.00 – 2024/t003036.txt

I am a cognitive neuroscientist, and I ’ m trying to understand how we perceive time and how our perception of time arises from the workings of our **brain**. Why? Why do I want to understand that? Well, time is the ultimate master of our lives. We are all constantly faced with its fleeting nature. Yet how we feel the passing of time can be highly malleable. When we are bored, in pain, but also when we encounter something novel or extraordinary, time feels to be passing much slower than when we are busy or simply having fun. But what does it mean to feel time? And why does the feeling of time distort depending on the situation, our level of focus, our emotional state? Do these distortions serve some function, and can we gain some level of control over how we feel time? These are very big questions, and we do not have the answer to any of them. Not just yet. But I will tell you about a surprising discovery I believe will take us a step closer to unlocking the neural basis of time. And I will show you that time is not something that is created solely by the **brain**, but it is also intimately shaped by what is happening inside the rest of the body. I work in the Lab of Action and Body at Royal Holloway, University of London, and in our lab we look at the **brain** from an embodied point of view. What this means is that we believe we cannot fully understand the workings of the **brain** if we take it out of the body, because after all, the main reason for us to have a **brain** is to keep the body alive, so that we can act in the world. And for that, it is not enough for the **brain** to perceive what is happening in the world around us, but it also needs to perceive what is happening inside of our own body. It needs to understand what our body needs at any moment in time. That additional internal sense, the perception of the body from within is called interoception. And one example of interoception is the perception of our own heart. Yes, the heart. I think all of us know that the main function of the heart is to transport oxygen-rich blood all through the body, and like other bodily functions, it is controlled by the **brain**. So when I ’ m on the move, the heart should start beating faster to provide more oxygen. When I need to slow down and focus, it will also slow down to preserve the oxygen. But what many of you may not know is that the activity of the heart itself shapes the activity of the **brain**. But before I get into that, let me give you some very basic neuroscience. So the **brain** receives information from the outside world through our eyes, through our ears. But to make sense of that information, to perceive, it has specialized sensory areas, like the visual sensory area that allows us to see the redness of the TEDx sign, or the auditory-sensory areas that allow us to hear the sound of my voice. The information can be the same, but our perception can differ very much. But in order to act in response to what we perceive, the **brain** has separate movement control centers that are responsible for initiating and controlling movement. That ’ s what makes me move my hands constantly as I ’ m speaking. So how does the heart shape the **brain** then? Well, it turns out that there are sensory neurons in our heart that fire signals to our **brain** to inform it about the state of the body. But they fire only at a time when the heart contracts, and they remain silent when the heart is relaxed. So the **brain** and the heart are in a constant rhythmic dance. Every beat makes the **brain** step into an active mode, priming us for action. And between the beats, the **brain** steps back into perceptual mode, priming us to take in information. Now imagine what happens when the heart starts beating faster. There are more beats, so the **brain** is more likely to be in the active mode, whereas when the heart slows down, there are more periods of time between the beats, so the **brain** is more likely to be in the perceptual mode. This got me thinking. If the heart shapes perception in such a way, will it also shape the perception of time? Is there a causal relationship between our heart and how we experience time? Of course, we tested that in our lab. We invited 67 volunteers to participate in a study where we asked them to judge very brief durations of different stimuli. These could be sounds, images of simple shapes, images of people showing different emotions. We wanted to systematically determine whether what participants felt and sensed influenced how they perceived durations. But the key part of that study was that we hooked the volunteers to an ECG machine so we could see their heart beating in real time while they were doing this study. And this allowed us to flash the stimuli precisely at the moment when the heart squeezed tight or when it relaxed. And what we found was that stimuli that occurred during heart ’ s contraction were perceived to last shorter than stimuli that occurred between the beats, when the heart was relaxed. What this means is that the momentary state of the heart caused time to contract and expand within each heartbeat. Why am I so excited about this finding? Well because it shows that perception of time is an embodied experience. It might be constructed in the **brain**, but it is molded by the body. And I ’ m sure it seems intuitive, you know, that something rhythmic, like the heartbeat, would influence something rhythmic like time perception. But now we have a scientific evidence for that intuition. What ’ s more, I can use these heart-driven time distortions to look at what is happening in the **brain** just before duration is distorted by each beat. Could it be that distortions in time are connected to how each heartbeat affects the sensory and the movement centers of the **brain**? Remember how I told you that each beat momentarily suppressed perception, but between the beats, suppression was momentarily increased? So could it be... And we showed that time is contracted during the beat, but expanded in between the beats. So could it be that our experience of time arises from the way in which our **brain** takes in sensory information that is shaped by the heart? And if that is true, could the function of our highly distorted sense of time be to shift us between an active and perceptual mode? Time could contract when we need to move, but it may expand when we want to perceive. So maybe that slowing of time you feel when you ’ re bored is there to actually expand your perception and allow your mind to wander to something new. I hope I ’ ve shown you that one way to shape how we experience time is to work on the internal state of our body. So the next time time becomes a little bit too fleeting, like right now for me – Let ’ s maybe try to take a deep breath. Yeah? Feel the heart slow down, and let the **brain** expand the moment. And maybe that momentary expansion of time will also broaden our perception. Thank you very much for listening. Thank you.

Text Sample 656: NEG Dim 4 – 2024 – Score 1.00 – 2024/t003098.txt

When I was younger, I just did not heal the same way. I would see these scabs, and, like, I ’ d pick at them, they ’ d take forever to close, and my brother ’ s skin would just close right up. And then, you know, my sisters could have cramps, and they would just take ibuprofen, and it would be like nothing. And I would take ibuprofen, I would take a lot of ibuprofen, and I would get no response for my whole body. And so I just was like,”Why? Please help me.”At first, I felt like I got the short end of the biological stick, right? Why does my body not work the same way? There was a lot of frustration, because no one knows how they respond to something until they try it. But it was just I was different from them, that ’ s all. I ’ m Dr. Erika Moore and I ’ m an equity bioengineer. So every single time you get a paper cut or even a bruise, your immune system is there. Macrophage immune cells are present. They ’ re like the watchdog of the body. I think of them as, like, squishy balls, and they just kind of are always surveilling around the body. Sometimes, they ’ re eating, but other times, they ’ re just saying,”Hey, man, don ’ t be mad. This is cool. We can heal now, right?”Just imagine, like, a squishy ball that ’ s sometimes angry, sometimes happy. They ’ re super cute, I love them. I really study what function or state they adopt based on what environment they ’ re in and based on whose body they come from. There are a lot of other material scientists, bioengineers who ’ ve asked and developed really cool applications, but oftentimes, we don ’ t really consider who is studied. My work really focuses on the who of the disease, so I really want to study the diseases that particularly are health disparities, that only certain swaths of the population are affected by, because I think that by shining a light on those diseases, we can ensure better health equity for everyone in the world. I think I ’ m the first person or lab to really spearhead or cheerlead some of these endeavors, because there are lived experiences that a lot of other people in academia and in traditional PhD training don ’ t really have to consider, right? So there aren ’ t that many women of color who are assistant professors, who run research programs and do these other things. I knew so many women, growing up, who died from lupus. It was very common. When I went to try to study lupus, there were a lot of people who were like,”We ’ re agnostic. Like, we don ’ t care about the background of the patients.”But I was like,”We know that, like, 90 percent of lupus patients are women, and of that, about 70 percent are women of color.”So we really have to consider that, right? It should be a variable. Some women develop lupus, they don ’ t ever have flares. They ’ re fine. Other women develop lupus and have flares back to back, and pretty much develop major cardiovascular disease very early on. And so we want to take their patient cells and look at their interactions with blood vessels and see how they confer or propagate different inflammation in the system. And we do that all in this little jello construct. So it ’ s almost like a mini-tissue outside of the body. And we put the cells in it, and then we can study how they would respond in a tissue-like environment. What we found was really cool. The patient ' s background really directly affected the blood vessels in the system. We saw an increased number of blood vessel interactions between macrophages that were isolated from African-American women compared to those macrophages that were isolated from European women. These tissue models help us answer some really important questions. What if we could pinpoint why inflammation happens in certain bodies more violently than others? What if we could cure autoimmune illnesses by accounting for disease differences based on your background? What if we could prevent excessive fibrosis by tailoring your macrophages to respond differently to injury? I think my lived experiences made me more likely to ask these questions. And then I was just willing to try, you know, and I ' m still willing to try. I think by continuing to try, we innovate on what ' s the norm and set new standards for the future. If we pay a little bit more attention to the details, I think we can easily build a more equitable health care system for everyone. That ' s my goal.

Text Sample 657: NEG Dim 4 – 2024 – Score 1.00 – 2024/t003170.txt
Welcome, ladies and gentlemen, to the Alchemy of Pop. What else in the world can change your mind, can change your mood, can change your entire energetic frequency in 3. 5 minutes? For me, that ' s **pop** music. And for me, that ' s alchemy. I ' m Kesha, and I am a **pop** star. I ' m a singer, I ' m a songwriter, I ' m an activist, and I ' m a surprisingly good scuba diver, amongst many other things. But the thing that drives me the most in this life is to write **pop** songs. And let me tell you why. For me, songwriting and scuba diving are actually very similar. When I go diving, I like to go deep. I just jump right in that bitch. And I am a cave diver, so I go in and I see this looming darkness, and I swim right into it, and I look behind every corner, I leave no stone unturned. That ' s where the exciting shit happens. When I ' m writing a song, I also like to go deep. I like to explore my deepest, darkest thoughts. This is also where the exciting shit happens. This is where I find my truth. Songwriting is a direct line of communication with the truth. In song you can say things you can ' t say out loud to anybody else. Not your friends, not your therapist, not your mom. When I ' m feeling anything intensely, I know it ' s time for me to write a song. I don ' t hide from my emotions. I dive into them. You can write a song and you can not tell the truth, but your song will suck. So if you would like to write a good song, you have to be honest. You have to sit with yourself and let your truth come, whatever it is. This sounds simple, but where it can lead is profound. Songwriting teaches you that all emotions are valid, deserving of curiosity and deserving of a voice. The song called"Tik Tok,"I wrote that when I was 21. Yeah. It was way before the app. Don ' t blame me. I was feeling very playful. I was a young nihilist. I didn ' t have too many problems at the time. That was so fucking nice. And the dumber the lyrics got, the better the song got. And I thought these feelings, they were kind of frivolous. They were just a guilty pleasure. But those exact emotions, that youthful, playful, pure stupidity, made"Tik Tok"connect globally in ways I could have never imagined. Now, let me take you to a very different point in my life. It was 2017, and I had to make a boundary that affected every corner of my life. I found myself in a very ugly litigation with the same person who I had signed the rights to my recorded voice to. I had lost the rights to my voice, and in that I felt as if I had lost the rights to myself. It was very public. It was very painful. I was very angry. So I did what I do with all my emotions, and I wrote my way through it. And that led me to a song called"Praying."Someday, maybe you ' ll see the light

Oh, some say, in life, you ’ re gonna get what you give But some things only God can forgive I hope you ’ re somewhere prayin ’, prayin ’ I hope your soul is changin ’, changin ’ I hope you find your peace Falling on your knees prayin ’ Thank you, thank you. So after”Tik Tok,”I wondered if anybody would connect to these uglier emotions coming from me, like, I ’ m the girl who”brushes her teeth with a bottle of Jack,”you know? And also, who wants to listen to an angry woman? Turns out a lot of fucking people do. Hehehe That was the first time I was nominated for a Grammy ; it ’ s the first time I performed at the Grammys. And I ’ ve heard from many people going through their own struggles that this song helped them in some way. I dove deep into the pain. I didn ’ t hide from it. I alchemized it into something healing not only for me but, inadvertently, for other people. I ’ m now – probably going to cry – I ’ m now in a drastically different point in my life. On March 6, 2024, I ’ ve gained back legal rights over my own voice. And with it, a newly found freedom for the first time in nearly 20 years. It ’ s funny how you can lose the rights to your own voice, but you can never lose the rights to your truth. Throughout this journey, songwriting has been my therapist, my best friend, my lover, my drinking buddy, and my higher power. It ’ s where I can be truly honest. And I don ’ t claim to be an expert on much, but I ’ ve experienced over and over and over how you can create art out of emotion and through that process, not only heal yourself but help to heal others. And I believe there ’ s an artist in every single one of us. Creation is just exploring. It ’ s becoming childlike. It ’ s not judging yourself. So I encourage all of you to not be afraid of the intensity inside of you. The stupidity, the rage, the anger, the darkness. All of it. Dive into it. Explore it. And I want to remind you that whether or not you are free, you are entitled to your truth. Find your medium, whatever speaks to you, and let your truth come. You never know who else will need to hear it. You never know who else can heal from it. Please don ’ t take this for granted. I ’ d like to share a song I wrote recently. No one has heard this. This is the first time I have shared a new song since I have owned my voice as an adult woman. So this song is called”Cathedral.”Baby, I ’ ve been baptized Been bathing in the moonlight I surrender, so take your time I ’ m summoning my divine Tried it all to numb the pain But nobody ever stayed It could be too much to even take Look what you find when there ’ s nothing left to lose I kept on running, just gasping for breath Nothing to trust in ’ cause nothing was left I pray when I ’ m desperate I ’ m down on my knees Oh, my God, I ’ m the cathedral Finally coming home Life was so lethal I ’ m the savior, I ’ m the altar, I ’ m the Holy Ghost Pain was my ritual, fear my religion I was the one who needed forgiving, oh I ’ m the cathedral Let it in, sunlight It feels good, that ’ s alright Touch my skin, love my pain away And, God, forgive all my mistakes Maybe redemption lies in the dark My happy ending ’ s through my broken heart Hope is a madman that hides in my mind Oh, my God I ’ m the cathedral Finally coming home Life was so lethal I ’ m the savior, I ’ m the altar, I ’ m the Holy Ghost Pain was my ritual, fear my religion I was the one who needed forgiving, oh I ’ m the cathedral Who I once was, she seems so far I ’ ve had to sacrifice her from the start I ’ m born again with every scar I can ’ t control what ’ s written in the stars Been on my knees, begging for God She ’ s there inside of me, she was just lost Oh, my mind, look what it caused Darkness leads into light all along I ’ m the cathedral Finally coming home Oh, God, it feels good I ’ m the savior, I ’ m the altar, I ’ m the Holy Ghost Every minute is a new beginning I died in hell, and I ’ m finally living again In the cathedral I ’ m the cathedral Thank you so much.

Text Sample 658: NEG Dim 4 – 2006 – Score 2.00 – 2006/t000485.txt

Hi, everyone. I ’ m Sirena. I ’ m 11 years old and from Connecticut. Well, I ’ m not really sure why I ’ m here. I mean, what does this have to do with technology, entertainment and design? Well, I count my iPod, cellphone and computer as technology, but this has nothing to do with that. So I did a little research on it. Well, this is what I found. Of course, I hope I can memorize it. The violin is made of a wood box and four metal strings. By pulling a string, it vibrates and produces a sound wave, which passes through a piece of wood called a bridge, and goes down to the wood box and gets amplified, but... let me think. Placing your finger at different places on the fingerboard changes the string length, and that changes the frequency of the sound wave. Oh, my gosh! OK, this is sort of technology, but I can call it a 16th-century technology. But actually, the most fascinating thing that I found was that even the audio system or wave transmission nowadays are still based on the same principle of producing and projecting sound. Isn ’ t that cool? Design — I love its design. I remember when I was little, my mom asked me,”Would you like to play the violin or the piano?”I looked at that giant monster and said to myself —”I am not going to lock myself on that bench the whole day!”This is small and lightweight. I can play from standing, sitting or walking. And, you know what? The best of all is that if I don ’ t want to practice, I can hide it. The violin is very beautiful. Some people relate it as the shape of a lady. But whether you like it or not, it ’ s been so for more than 400 years, unlike modern stuff [that] easily looks dated. But I think it ’ s very personal and unique that, although each violin looks pretty similar, no two violins sound the same — even from the same maker or based on the same model. Entertainment — I love the entertainment. But actually, the instrument itself isn ’ t very entertaining. I mean, when I first got my violin and tried to play around on it, it was actually really bad, because it didn ’ t sound the way I ’ d heard from other kids — it was so horrible and so scratchy. So, it wasn ’ t entertaining at all. But besides, my brother found this very funny : Yuk! Yuk! Yuk! A few years later, I heard a joke about the greatest violinist, Jascha Heifetz. After Mr. Heifetz ’ s concert, a lady came over and complimented him :”Oh, Mr. Heifetz, your violin sounded so great tonight!”And Mr. Heifetz was a very cool person, so he picked up his violin and said,”Funny — I don ’ t hear anything.”Now I realize that as the musician, we human beings, with our great mind, artistic heart and skill, can change this 16th-century technology and a legendary design to a wonderful entertainment. Now I know why I ’ m here. At first, I thought I was just going to be here to perform, but unexpectedly, I learned and enjoyed much more. But... although some of the talks were quite up there for me. Like the multi-dimension stuff. I mean, honestly, I ’ d be happy enough if I could actually get my two dimensions correct in school. But actually, the most impressive thing to me is that — well, actually, I would also like to say this for all children is to say thank you to all adults, for actually caring for us a lot, and to make our future world much better. Thank you.

Text Sample 659: NEG Dim 4 – 2006 – Score 3.00 – 2006/t000461.txt

With all the legitimate concerns about AIDS and avian flu — and we ’ ll hear about that from the brilliant Dr. Brilliant later today — I want to talk about the other pandemic, which is cardiovascular disease, diabetes, hypertension — all of which are completely preventable for at least 95 percent of people just by changing diet and lifestyle. And what ’ s happening is that there ’ s a globalization of illness occurring, that people are starting to eat like us, and live like us, and die like us. And in one generation, for example, Asia ’ s gone from having one of the lowest rates of heart disease and obesity and diabetes to one of the highest. And in Africa, cardiovascular disease equals the HIV and AIDS deaths in most countries. So there ' s a critical window of opportunity we have to make an important difference that can affect the lives of literally millions of people, and practice preventive medicine on a global scale. Heart and blood vessel diseases still kill more people — not only in this country, but also worldwide — than everything else combined, and yet it ' s completely preventable for almost everybody. It ' s not only preventable ; it ' s actually reversible. And for the last almost 29 years, we ' ve been able to show that by simply changing diet and lifestyle, using these very high-tech, expensive, state-of-the-art measures to prove how powerful these very simple and low-tech and low-cost interventions can be like — quantitative arteriography, before and after a year, and cardiac PET scans. We showed a few months ago — we published the first study showing you can actually stop or reverse the progression of prostate cancer by making changes in diet and lifestyle, and 70 percent regression in the tumor growth, or inhibition of the tumor growth, compared to only nine percent in the control group. And in the MRI and MR spectroscopy here, the prostate tumor activity is shown in red — you can see it diminishing after a year. Now there is an epidemic of obesity : two-thirds of adults and 15 percent of kids. What ' s really concerning to me is that diabetes has increased 70 percent in the past 10 years, and this may be the first generation in which our kids live a shorter life span than we do. That ' s pitiful, and it ' s preventable. Now these are not election returns, these are the people — the number of the people who are obese by state, beginning in ' 85, ' 86, ' 87 — these are from the CDC website — ' 88, ' 89, ' 90, ' 91 — you get a new category — ' 92, ' 93, ' 94, ' 95, ' 96, ' 97, ' 98, ' 99, 2000, 2001 — it gets worse. We ' re kind of devolving. Now what can we do about this? Well, you know, the diet that we ' ve found that can reverse heart disease and cancer is an Asian diet. But the people in Asia are starting to eat like we are, which is why they ' re starting to get sick like we are. So I ' ve been working with a lot of the big food companies. They can make it fun and sexy and hip and crunchy and convenient to eat healthier foods, like — I chair the advisory boards to McDonald ' s, and PepsiCo, and ConAgra, and Safeway, and soon Del Monte, and they ' re finding that it ' s good business. The salads that you see at McDonald ' s came from the work — they ' re going to have an Asian salad. At Pepsi, two-thirds of their revenue growth came from their better foods. And so if we can do that, then we can free up resources for buying drugs that you really do need for treating AIDS and HIV and malaria and for preventing avian flu. Thank you.

Text Sample 660: NEG Dim 4 – 2006 – Score 3.00 – 2006/t000480.txt

I wrote this poem after hearing a pretty well known actress tell a very well known interviewer on television,"I ' m really getting into the Internet lately. I just wish it were more organized."So... If I controlled the Internet, you could auction your broken heart on eBay. Take the money ; go to Amazon ; buy a phonebook for a country you ' ve never been to — call folks at random until you find someone who flirts really well in a foreign language. If I were in charge of the Internet, you could Mapquest your lover ' s mood swings. Hang left at cranky, right at preoccupied, U-turn on silent treatment, all the way back to tongue kissing and good lovin '. You could navigate and understand every emotional intersection. Some days, I ' m as shallow as a baking pan, but I still stretch miles in all directions. If I owned the Internet, Napster, Monster and Friendster. com would be one big website. That way you could listen to cool music while you pretend to look for a job and you ’ re really just chattin ’ with your pals. Heck, if I ran the Web, you could email dead people. They would not email you back — but you ’ d get an automated reply. Their name in your inbox — it ’ s all you wanted anyway. And a message saying,”Hey, it ’ s me. I miss you. Listen, you ’ ll see being dead is dandy. Now you go back to raising kids and waging peace and craving candy.”If I designed the Internet, childhood. com would be a loop of a boy in an orchard, with a ski pole for a sword, trashcan lid for a shield, shouting,”I am the emperor of oranges. I am the emperor of oranges. I am the emperor of oranges.”Now follow me, OK? Grandma. com would be a recipe for biscuits and spit-bath instructions. One, two, three. That links with hotdiggitydog. com. That is my grandfather. They take you to gruff-ex-cop-on-his-fourth-marriage. dad. He forms an attachment to kind-of-ditzy-but-still-sends-ginger-snaps-for-Christmas. mom, who downloads the boy in the orchard, the emperor of oranges, who grows up to be me — the guy who usually goes too far. So if I were emperor of the Internet, I guess I ’ d still be mortal, huh? But at that point, I would probably already have the lowest possible mortgage and the most enlarged possible penis — so I would outlaw spam on my first day in office. I wouldn ’ t need it. I ’ d be like some kind of Internet genius, and me, I ’ d like to upgrade to deity and maybe just like that — **pop**! — I ’ d go wireless. Huh? Maybe Google would hire this. I could zip through your servers and firewalls like a virus until the World Wide Web is as wise, as wild and as organized as I think a modern-day miracle / oracle can get, but, ooh-eee, you want to bet just how whack and un-PC your Mac or PC is going to be when I ’ m rocking hot-shit-hot-shot-god. net? I guess it ’ s just like life. It is not a question of if you can — it ’ s : do ya? We can interfere with the interface. We can make”You ’ ve got Hallelujah”the national anthem of cyberspace every lucky time we log on. You don ’ t say a prayer. You don ’ t write a psalm. You don ’ t chant an”om.”You send one blessed email to whomever you ’ re thinking of at dah-da-la-dat-da-dah-da-la-dat. com. Thank you, TED.

Text Sample 661: NEG Dim 4 – 1990 – Score 10.00 – 1990/t000287.txt

I ’ m going to go right into the slides. And all I ’ m going to try and prove to you with these slides is that I do just very straight stuff. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the left-hand side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage the neighborhood â one made accommodations. I tried to be inclusive, to include the buildings in the neighborhood, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of stuff and folded it up to put it in the wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at postmodernism â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker. And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And the client said,”Why no fish?”And so I made a drawing with a fish, and I left Japan. Three weeks later, I received a complete set of drawings saying we ’ d won the competition. Now, it ’ s hard to do. It ’ s hard to translate a fish form, because they ’ re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ’ t do it, and so that made it even more exciting. But he was right â I couldn ’ t do the tail. I started to get the head OK, but the tail I couldn ’ t do. It was pretty hard. The thing on the right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with the context again. Now, if you saw a picture of this as it was published in Architectural Record â they didn ’ t show the context, so you would think,”God, what a pushy guy this is.”But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ’ t find it. So... As for craft and technology and all those things that you ’ ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent drawings to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made the drawings of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ’ s one of the nicest pieces I ’ ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ’ t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ' m getting good at temporary â and we put a conference room in that ' s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ' s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ' t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ' s offices and I made a pedestal for a sculpture. And he didn ' t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ' s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ' ve been friends for a long time. And we ' ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called"Il Corso del Coltello"â the Swiss Army knife. And most of the imagery is â Claes ', but those two little boys are my sons, and they were Claes ' assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ' s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ' ve been known to mess with things like chain link fencing. I do it because it ' s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ' s so much denial about them. People hate it. And I ' m fascinated with that, which, like the paper furniture â it ' s one of those materials. And I ' m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ' s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to the building on the left. And I still think the Time magazine picture will be of the binoculars, you know, leaving out the â what the hell. I use a lot of metal in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,"If I can ' t get the craft that I want, I ' ll use the craft I can get."There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the metal because it was a way of building a building that was a sculpture. And it was all of one material, and the metal could go on the roof as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and stuff. I was given an opportunity to design an exhibit for the metalworkers ' unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future metal buildings, etc. etc. And it ’ s working very well to have these people, these craftsmen, interested in it. I just tell the stories. It ’ s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The metal continued into a building â Herman Miller, in Sacramento. And it ’ s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ’ s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ’ s out in the middle of nowhere, and you approach it. It ’ s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ’ s aesthetic. Everybody ’ s trying to get the panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ’ s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ’ s all about [a] metaphor for a city, maybe. And so Stanley did the little dome thing. And we did it over the phone and by fax. He would send me a fax and show me something. He ’ d made a building with a dome and he had a little tower. I told him,”No, no, that ’ s too ongepotchket. I don ’ t want the tower.”So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going back and forth over a couple of weeks ’ period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,”David and Goliath.”And that ’ s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and cleaned it up and fixed it up and used the kind of â I thought â the language of the neighborhood, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ’ t normally think I would fit them. The detailing was very careful with the lead copper and making panels and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some film festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â a little shopping center. And this is a laser laboratory at the University of Iowa, in which the fish comes back as an abstraction in the back. It ’ s the support labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the curved part there ’ s all the mechanical equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ’ t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ’ ve been building it so long I don ’ t remember where it is. It ’ s in the West Valley. And we started with the stream and built the house along the stream â dammed it up to make a lake.

These are the models. The reality, with the lake â the workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ’ s hard to get perfect corners and stuff. That big metal thing is a passage, and in it is â you go downstairs into the living room and then down into the bedroom, which is on the right. It ’ s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â they asked Philip to do it. He was too busy. He didn ’ t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ’ t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ’ ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ’ s made of metal and the brown stuff is Fin-Ply â it ’ s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ’ t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ’ s Burnham Mall, on the left. It ’ s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ’ s a railroad track. This is the city hall over here somewhere, and the courthouse. And the centerline of the mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ’ re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this stuff at the bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the centerline of this thing â we ’ d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ’ s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ’ 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the back end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake. And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees.

That ’ s the living room. Servants ’ quarters. A guest bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is back in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a fun thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it ain ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the conference facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais des Sports, the Gare de Lyon over here. The Minister of Finance â the guy that moved from the Louvre â goes in here. There ’ s a new library across the river. And back in here, in this already treed park, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan coupe. And I struggled with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in French limestone, except for this metal piece. And it faces into a park. And the idea was to make this express the energy of this. On the side facing the street it ’ s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in the Hamptons. And it ’ s got a fish. And I keep thinking,”This is going to be the last fish.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s the living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in the budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of the guys that presented to you earlier. We ’ ve had a lot of fun working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ’ s perpendicular to the train station, to the route of the train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good slides to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â you see it as all of these pieces accrue and become part of an overall neighborhood. It ’ s plaster and just zinc. And you wonder, if this is a museum, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and stuff. And from the building in the back, you see it as an iceberg floating by in the hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is the Disney Hall â the concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a compositional relationship between us that would strengthen both of us. And the plan of this â it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â to MOCA, etc. The acoustician in the competition gave us criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But everybody liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were the three buildings that were the ideal â the Concertgebouw, Boston and Berlin. Everybody liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new acoustician, he told us this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an acoustical, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the sloping outside walls, which the acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on the way to being â and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with laser stuff, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, struggling with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the foot of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the foot of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th- century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ’ t as good as the stuff, and so it failed. It should have been the other way around â the food should have been good first. It didn ’ t work. Thank you very much.

Text Sample 662: NEG Dim 4 – 2025 – Score 5.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in the world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the air. In all my travels, I ’ ve been struck by so many things. I ’ ve observed how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ’ ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced us to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ’ s hard to see the big picture right now. It ’ s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think back to my travels. I ’ ve seen how different we are, how diverse and distinct. But I ’ ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ’ m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman”A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story is the dominant sound of our lives.”- Joseph Campbell I ’ ve loved telling stories all of my life. I ’ ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 663: NEG Dim 4 – 2025 – Score 7.00 – 2025/t003001.txt

- I don ’ t remember exactly, but... I use the most expensive hair color in the world. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color. I expect great color. What ’ s worth more to me is the way my hair feels. Feels good around my neck. I can ’ t remember it. I can ’ t finish it all. If only we had more time. - Why don ’ t we have more time? - I ’ m dying. I ’ m croaking! - It ’ s tough, coming home. But this is what you do for a parent that you love. But, you know? She ’ s not my mother. Technically. In my own family, there ’ s a lot of dysfunction, a lot of mental illness, a lot of substance abuse. Every generation. My parents divorced when I was about five. My mother moved us out of state, so we had to fly to New York to see my father ’ cause he was already working in advertising, and New York was the only place to be. - Many of the commercials you see on television begin life here. - Advertising is one of the main sources of messaging about who you should be. - Nowadays, most of you women lead two lives, combining homemaking with the role of glamorous hostess. - And in the ’ 60s, the ads were often made from a male point of view. - In the pre-production meeting, everybody who has anything to do with the commercial is there. - It was mostly men. These superstar art directors and writers. Like my father, Gene Case.”Mad Men”was like home movies, and he was Don Draper. He was very witty, but I don ’ t think my father realized how powerful his words were. He knew it professionally. But with his kids, he could be incredibly cutting. Flay you just with a phrase. But it ’ s the time with my mother that was the nightmare. She had a horrible childhood, but she sure as hell didn ’ t make any effort to overcome that in her relations with us. You know, it was really horrifying to my mother that I was not popular. But I was also really awkward because she had been undermining my confidence for all my life. She would pick apart your personality in detail. I was lazy, I looked bad, I acted bad, I was awkward. Emotionally, just out for blood. At a certain point in the night, I would know,”OK, the situation is dangerous. I need to get outta here.”I had this fantasy when I was in fourth grade that I vividly remember that some kind of outside agency was gonna come in and determine from all the salt on my pillow that I was crying myself to sleep every night and do something about it. No, it was awful. It was awful. And that was the point where Ilon came into my life. - Ilon, do you need anything to eat? - I don ’ t think so, but look up this wallpaper, Waverly Fabrics, and see if they have a rose pattern. - OK. Oh, what about this one here? - That ’ s not it. - No? - No, first of all, I was an art dir... I was a creative director at an advertising agency. The first thing I know is what I ’ m looking at. Ilon the girl. I don ’ t remember particularly being into advertising, but somehow I found myself in it. And I seemed to be very good at it. First of all, I ’ d always listened to everybody. And I noticed that I saw things differently than other people. - Women could get somewhere in advertising, but they were marginalized. - Women were given a lesser role. All the men were always arguing with you and always taking credit. - And one of the things that was coming through loud and clear is that women are for men. - Glory, glory, hallelujah Glory, glory, hallelujah Glory, glory, hallelujah His truth is marching on - Men were always the ones who everyone was pleasing. - What will your eyes say to him tonight? Wait till the fellas take a look at this plunging backline. - You look at ads from that period, and it ’ s jaw-dropping. - This is one of our nation ’ s most beautiful sights, the kind of girl who keeps her figure. - But that ’ s what feminism was pushing back against. - I was a feminist. I knew women were equal if not superior to men from the time I was a little girl. - And in advertising, she wanted to be a trailblazer that way. At one point, she worked in the same agency with my father, and that ’ s how they met. - He was a good writer, and I was a good writer. I liked the fact that he was good at what he did and cared about it. We got engaged, and then we got married. - I ’ m sitting there in the background. I ’ m grinning. I ’ m so thrilled because I adored her instantly. I thought she was super cool. And she treated me like I mattered. - She was a sweet little girl, and she deserved some caretaking. My mother was kinda terrible, so I knew that she needed support, and I gave her support. And that was what we did. - She was interested in me. She listened to me. Treated me like I was lovable. - She didn ’ t have anyone to recognize her being, to support her, to see who she could be, who she was. - She was very interested in what I would do with my life, and the assumption was that it would be something worthwhile. She just went way, way beyond the call of duty to make me feel like I was worth it. And she did that for a lot of other women, too. - I was working at McCann Erickson. We were given an assignment. L ’ Oréal, they wanted a campaign for Preference, it ’ s a hair color. - She ’ s in this room full of men brainstorming for this ad. - The men were busy flirting with the women, telling us we all looked like models and we didn ’ t look like advertising people, you know? - They ’ re like,”Oh, this woman is gonna be standing by an open window, and the wind will be blowing the curtains.”And she ’ s thinking, you know, this is just making the woman totally an object. - I was feeling angry. I ’ m not interested in writing anything about looking good for men. Fuck ’ em. Fuck you too. - Why fuck me? - Well, you ’ re a man. - Yeah, but I ’ m here trying to help you tell your story. - Yeah, that ’ s good. I like that part. I was pissed. We weren ’ t there to just dance for the men. - It is what she said. She just thought,”Fuck you.”And wrote this. - I just opened my eyes and knew. Because I ’ m worth it. The campaign got approved, but all the men were always telling you that you were doing something wrong or they knew how to do it better. - The agency insisted on a male voiceover. - She uses the most expensive hair color in the world, Preference by L ’ Oréal. - You know, it was just totally different. - Smooth and silky, but with body. It feels good against her neck. Actually, she doesn ’ t mind spending more for L ’ Oréal. Because she ’ s worth it. - It was wrong. This was not for men but for women and for other human beings. - I use the most expensive hair color in the world, Preference by L ’ Oréal. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color, I expect great color. What ’ s worth more to me is the way my hair feels. Smooth and silky, but with body. It feels good against my neck. Actually, I don ’ t mind spending more for L ’ Oréal. Because I ’ m worth it. - I wrote that. - It was everywhere. It was a national campaign. - Yes, it ’ s expensive, but I ’ m worth it. - What they realized after they started running it was that that was only the tip of the iceberg. - It was supporting women, finally. - And I ’ m worth it. And I ’ m worth it. I ’ m worth it. - It ’ s a four-word feminist manifesto, defying this social pressure to think about yourself in relation to men. - You ’ re worth it. I ’ m worth it. I ’ m worth it. - It was a very big success. - At a certain point, L ’ Oréal adopted it as the tagline for their entire business. - Parce que je le vaux bien. - Even today, that ’ s their motto. - Because I ’ m worth it. - ’ Cause you ’ re worth it. - You ’ re worth it. - We ’ re worth it. - And I ’ m worth it. - She wrote the most influential tagline in women ’ s beauty advertising. - I ’ m worth it. - It ’ s magic, that phrase. - Those words just get better with age, don ’ t they? - I ’ m worth it. Because you are and always will be. - She came up with that line because she believed it. She believed it about herself, and she believed that women should believe it. She did that for me. You know, even after they split up, she was still a big part of my life. I ’ ve always kind of felt like I dodged a bullet that I have watched take down other members of my family. She saved me. And today, I am a professor of English at Williams College. I teach Victorian literature, and I also teach women ’ s, gender and sexuality studies. She was my mother. She also taught me how to be a mother. I had been very uncertain about motherhood for myself because my role model for that was so nightmarish. But Ilon really taught me, I can do this. I can be a good mother. She made me believe that I was worth it. And I was not either the first or the last. - I have to go away now. Go to my room. - It ’ s been lovely. - It ’ s been lovely. Oh... I ’ m not interested in advertising. I don ’ t give a shit. It wasn ’ t like it was my whole life. It was just a portion. It ’ s about humans ; it ’ s not about advertising. It ’ s about caring for people because... we ’ re all worth it, or no one is worth it.

Text Sample 664: NEG Dim 4 – 2025 – Score 19.00 – 2025/t003466.txt

Can you paint with all the colors of the wind? I love this song from the first time I heard it. I was 14, growing up in public housing in western Sydney. So maybe a little too old for this Disney classic from”Pocahontas.”But it spoke to me. I saw something of myself in it. At this stage, I think I knew I was queer. I mean, come on, I was rocking out to Disney. And look at those glasses. Seriously, what a queer kid. But I was also from a Pacific Indigenous heritage with conservative views and values. I didn ’ t feel like I could fit into any of the boxes that were being presented to me. Straight, gay, Fijian, Australian. I couldn ’ t pick just one color to paint with. Prior to colonization, sexuality amongst many of our Pacific islands across the region was fluid. Men would have sex with men without the stigma or shame afforded it today. Sexual intimacy was a way of being able to connect socially and relationally. Sex wasn ’ t just for procreation. Or in marriage. It wasn ’ t exclusively monogamous. Just because you had sex with a man didn ’ t define your sexuality. It didn ’ t make you an outcast. But as Europeans started to colonize the Pacific region, they brought Christian missionaries with them. They told us to cover up our beautiful bodies. We were taught shame. We were told to stop practicing Pacifica cultures and values that were seen as provocative. To a Western and white perspective, our non-binary sexuality was judged as tribal, outdated, sinful and impossible to practice alongside a Christian faith. Still today, those who are Indigenous to the Pacific are negatively impacted by whiteness. That”white is right, West is best.”I was brought up to espouse strong Christian beliefs, views and values which would shape my burgeoning identity as an adult. From these staunch conservative views, I was taught to fear the queer. Those people that were seen as unusual and fruity. And in essence, I grew to despise a part of myself. I sought to separate my queerness from my identity. I could never see myself as part of that world. Growing up, biracial and bisexual constantly challenged the way I saw the world around me. I was unsure about my thinking, feelings and behaviors around liking males and females. I was unsure about my thinking, feelings and behaviors when it came to whether I needed to be a white Australian or a Fijian person. And whether this needed to be framed within a binary to choose one over the other. In this way, I don ’ t think that my story is unique. There ’ s plenty of people in the queer world that have escaped conservative upbringings, religions and cultures. But this is only part of my story. Another common story in the queer community is the coming-out story. When did you tell your parents? How did they react? How it liberated you or ultimately separated you from your family. But this didn ’ t feel like an option for me either. I believe that embedded in these particular stories, these tropes, is a part of dominant queer culture that made it very hard for me to access the queer world. For example, when I started to go to gay clubs, I would be lining up at the front of the club and often I would get door staff come up to me and go,”Sorry, you do know this is a gay club, right?”From these lived experiences, I feel that there is a whiteness that pervades queer culture. And I didn ’ t want to give up my identity and my appearance as a Pacific indigenous person to be queer. Pacifica cultures really, really value and esteem the collective above the individual. For Pacifica people, our families are inextricably connected to our own sense of self-agency, self-determination, self-esteem, self-love and self-worth. Our own individual identities are meshed and intertwined with our family ’ s reputation and standing in the community. Our mind, body and souls are inextricably connected to the broader community, to our families, to ourselves and to others. I didn ’ t want to come out to potentially ostracize myself, to make a point to separate myself from my Pacific and faith communities. I couldn ’ t then see my authentic self in any of these spaces. And a quick shout out to my son on the left-hand side. Round of applause for my son. He said to me this morning,”Daddy, make sure you mention me.”So I was like,”OK, son. Alright.”So there you go for the recording, son. And the rest are my other family, that ’ s up there as well. Across queer cultures and the broader community, we have learned to embrace our gender and sexual differences. We encourage individuals to be allies for each other and to support safe spaces for those that are seen to not fit in. We empower and really encourage individuals to really develop and express their self-agency, their self-determination, their self love, their self-worth. We boldly go into spaces striving to bring others with us to ensure that we can create inclusive societies for everyone. However, the dominant Western view in queer cultures is still based on the individual. Me, myself and I. And my individual pursuits towards happiness. As a queer Indigenous Pacific person, I feel that there are parts of queer culture that is white, feels very white. We may fail to see the importance of bringing our families and our cultural communities along our journey as queer people. It ’ s either the queer way or the highway. It ’ s one or the other. Young or old. Rich or poor. Smart or stupid. Masculine or feminine. Fit or fat. These binaries create a sense of us and them, those people. A binary approach and position that we generally in Western white societies use to determine the norm. We create, conflate, perpetuate binaries that create divisions in our society. Saints or sinner, straight or gay. I see it like this. Binaries are a very white and Western way of thinking, feeling and behaving. As individuals today, we live within these binaries and create identities that then create these labels. We use these labels to then define and confine people and places. We are one or the other. But what we fail to see is that our identities are more than a set of binaries. Our own identities are comprised of intersecting identifiers that contributes to who you are, including our age, class, color, education, gender, size, spirituality, sexuality, and so, so much more. We need to decolonize queerness, to live beyond”white is right, West is best.”Because if we don ’ t, we ’ re going to continue to create silos and divisions in our societies that don ’ t create inclusive and safe spaces for everyone. For Indigenous queer people, they shape our connection to community and culture, our families. If we don ’ t include that within our understanding of who we are, even as queer people, then we ’ re not able to be, again, our true and authentic selves. We ’ re not able to then be our fabulous selves as part of the queer community. And it ’ s really, really interesting because I believe that we can learn so much from pre-colonial cultures like those that were practiced in the Pacific, that provide a different way of looking at a society. A queer way that pre-dates all the privileges that go into binary ways of thinking, feeling and behaving. Across many of our Pacifica languages we have a lot of key concepts that help solidify and support a shared understanding of the collective. For example, in Fijian,”solesolevaki”is all about reciprocal living, where my wellbeing is your wellbeing. I will actively seek to ensure that your well-being and every part of your life is catered for, knowing that you will do the same for me. We strive to enact”veiqaravi,”as tatted here on my hand, to serve others. We strive to also embed and embold and empower people to practice”veirokovi,”respect for self and others where we are relationally driven, where that is so important to our spaces and places. My sense of ability to actually put this into practice, my ability to actually understand these Indigenous perspectives has greatly helped shape and reconcile my queer identity today. But what ’ s really interesting is that it ’ s not just queer communities that need to decolonize itself. It ’ s all of us as a broader society that needs to be part of a shared conversation to decolonize, deconstruct and disrupt binary ways of thinking, feeling and behaving. So how can we practically do that? Practice cultural humility, where you are constantly learning and embracing differences, where you get over yourselves and strive to learn about other people ’ s differences and how that can be meaningfully included in the way in which you create safe spaces for everyone. I also believe it ’ s possible to bring Indigenous perspectives into Western and white workplaces. For example, I ’ ve introduced the concept of”talanoa”in all of my work meetings. Talanoa is all about being able to create a safe space where everyone is able to contribute to a collective, collegial and collaborative conversation. Where people are able to bring their authentic selves into these spaces. Literally, I will chair meetings where I have no firm set agenda items and I literally would just hold space for participants just to bring themselves. So you can tell I ’ m a social worker, right? Safe space, safe space, safe space. It ’ s OK, it ’ s OK. But it ’ s through these safe spaces that people can bring their multiple, intersecting identifiers, including their age, their class, color, education, gender, size, spirituality, sexuality and so, so much more. I invite queer communities to come talanoa with Pacific Indigenous Communities to create robust and resilient queer cultures. I have been able to greatly reconcile my queer and Pacifica identities, which has now really been embraced to help shape how I see the world around me today. And it ’ s through these shared perspectives and approaches that we can truly embrace cultural diversity and its differences. And to know that people can live beyond those binaries and those labels we continue to perpetuate and perpetrate against others. It is a commitment from queer communities to decolonize queerness. It ’ s a commitment from me, from you and all of us to decolonize Western and white societies and to live beyond the binary way of thinking, feeling and behaving. I genuinely believe that it ’ s through this common goal that we can truly learn how to paint with all the colors of the wind.”Vinaka vakalevu”and thank you.

9 POS Dim 5

Text Sample 665: POS Dim 5 – 1990 – Score 77.00 – 1990/t000287.txt

I ’ m going to go right into the slides. And all I ’ m going to try and prove to you with these slides is that I do just very straight stuff. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a **porno** comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the **left-hand** side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take **pictures** of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this **avenue** that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t **upstage** the neighborhood â one made accommodations. I tried to be inclusive, to include the buildings in the neighborhood, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made **flooring**, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an **architect** â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of stuff and folded it up to put it in the **wastebasket**. And I put a piece of **tape** around it, as you see there, and realized you could sit on it, and it had a lot of **resilience** and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at **postmodernism** â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker. And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on **napkins**. And I made some sketches on **napkins** â little boxes and Morandi-like things that I used to do. And the client said,”Why no fish?”And so I made a **drawing** with a fish, and I left Japan. Three weeks later, I received a complete set of **drawings** saying we ’ d won the competition. Now, it ’ s hard to do. It ’ s hard to translate a fish form, because they ’ re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ’ t do it, and so that made it even more exciting. But he was right â I couldn ’ t do the tail. I started to get the head OK, but the tail I couldn ’ t do. It was pretty hard. The thing on the right is a snake form, a **ziggurat**. And I put them together, and you walk between them. It was a dialog with the context again. Now, if you saw a **picture** of this as it was published in Architectural Record â they didn ’ t show the context, so you would think,”God, what a pushy guy this is.”But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ’ t find it. So... As for craft and technology and all those things that you ’ ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent **drawings** to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing **scanned**. We made the **drawings** of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ’ s one of the nicest pieces I ’ ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ’ t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ’ m getting good at temporary â and we put a conference room in that ’ s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ’ s a real story â about my grandmother buying a **carp** on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ’ t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ’ s offices and I made a pedestal for a sculpture. And he didn ’ t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ’ s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ’ ve been friends for a long time. And we ’ ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called”Il Corso del Coltello”â the Swiss Army knife. And most of the imagery is â Claes ’, but those two little boys are my sons, and they were Claes ’ assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ’ s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ’ ve been known to mess with things like chain link **fencing**. I do it because it ’ s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ’ s so much denial about them. People hate it. And I ’ m **fascinated** with that, which, like the paper furniture â it ’ s one of those materials. And I ’ m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ’ s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this **three-piece** thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to the building on the left. And I still think the Time magazine **picture** will be of the binoculars, you know, leaving out the â what the **hell**. I use a lot of **metal** in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,”If I can ’ t get the craft that I want, I ’ ll use the craft I can get.”There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the **metal** because it was a way of building a building that was a sculpture. And it was all of one material, and the **metal** could go on the **roof** as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and stuff. I was given an opportunity to design an exhibit for the metalworkers ’ unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future **metal** buildings, etc. etc. And it ’ s working very well to have these people, these craftsmen, interested in it. I just tell the stories. It ’ s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The **metal** continued into a building â Herman Miller, in Sacramento. And it ’ s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ’ s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ’ s out in the middle of nowhere, and you approach it. It ’ s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ’ s aesthetic. Everybody ’ s trying to get the **panels** perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ’ s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ’ s all about [a] **metaphor** for a city, maybe. And so Stanley did the little dome thing. And we did it over the phone and by fax. He would send me a fax and show me something. He ’ d made a building with a dome and he had a little tower. I told him,”No, no, that ’ s too ongepotchket. I don ’ t want the tower.”So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going back and forth over a couple of weeks ’ period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,”David and Goliath.”And that ’ s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and **cleaned** it up and fixed it up and used the kind of â I thought â the language of the neighborhood, which had these **cornices**, projecting **cornices**. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ’ t normally think I would fit them. The detailing was very careful with the lead copper and making **panels** and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some **film** festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this **picture**. But they made it a third smaller than my model without telling me.

And then more **metal** and some chain link in Santa Monica â a little shopping center. And this is a **laser** laboratory at the University of Iowa, in which the fish comes back as an **abstraction** in the back. It ' s the support labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the **curved** part there ' s all the **mechanical** equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ' t have to have any **protruding** ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ' ve been building it so long I don ' t remember where it is. It ' s in the West Valley. And we started with the stream and built the house along the stream â dammed it up to make a lake. These are the models. The reality, with the lake â the **workmanship** is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ' s hard to get perfect corners and stuff. That big **metal** thing is a passage, and in it is â you go downstairs into the living room and then down into the bedroom, which is on the right. It ' s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â they asked Philip to do it. He was too busy. He didn ' t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ' t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ' ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ' s made of **metal** and the brown stuff is Fin-Ply â it ' s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ' t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ' s Burnham Mall, on the left. It ' s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ' s a **railroad** track. This is the city hall over here somewhere, and the courthouse. And the **centerline** of the mall goes out. Burnham had designed a **railroad** station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ' re two kind of **goalposts**. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ' s about a 10-story C-clamp. And all this stuff at the **bottom** is a **museum**, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the **centerline** of this thing â we ' d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ' s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ' 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the back end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake. And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ’ s the living room. Servants ’ quarters. A **guest** bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is back in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a **fun** thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it **ain** ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the conference facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais **des** Sports, the Gare de Lyon over here. The Minister of Finance â the guy that moved from the Louvre â goes in here. There ’ s a new library across the river. And back in here, in this already treed **park**, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art **museum**, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan **coupe**. And I struggled with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan **coupe** â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in French limestone, except for this **metal** piece. And it faces into a **park**. And the idea was to make this express the energy of this. On the side facing the street it ’ s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a **glutton** for punishment, and he hired me to do a house for him in the Hamptons. And it ’ s got a fish. And I keep thinking,”This is going to be the last fish.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s the living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in the budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of the guys that presented to you earlier. We ’ ve had a lot of **fun** working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ’ s perpendicular to the train station, to the route of the train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture **museum**. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good slides to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â you see it as all of these pieces accrue and become part of an overall neighborhood. It ’ s plaster and just zinc. And you wonder, if this is a **museum**, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and stuff. And from the building in the back, you see it as an iceberg floating by in the hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is the Disney Hall â the concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a **compositional** relationship between us that would strengthen both of us. And the plan of this â it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â to MOCA, etc. The **acoustician** in the competition gave us criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But everybody liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were the three buildings that were the **ideal** â the Concertgebouw, Boston and Berlin. Everybody liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new **acoustician**, he told us this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an **acoustical**, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the **sloping** outside walls, which the **acoustician** said were **crucial** to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on the way to being â and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with **laser** stuff, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, struggling with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the **foot** of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the **foot** of a Ritz-Carlton Tower being done by Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old **gas** tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th-century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ’ t as good as the stuff, and so it failed. It should have been the other way around â the food should have been good first. It didn ’ t work. Thank you very much.

Text Sample 666: POS Dim 5 – 2002 – Score 14.00 – 2002/t000320.txt

Frank Gehry : I listened to this scientist this morning. Dr. Mullis was talking about his experiments, and I realized that I almost became a scientist. When I was 14 my parents bought me a chemistry set and I decided to make water. So, I made a hydrogen generator and I made an oxygen generator, and I had the two pipes leading into a beaker and I threw a match in. And the glass — luckily I turned around — I had it all in my back and I was about 15 **feet** away. The wall was covered with... I had an explosion. Richard Saul Wurman : Really? FG : People on the street came and knocked on the door to see if I was okay. RSW :... huh. I ’ d like to start this session again. The gentleman to my left is the very famous, perhaps overly famous, Frank Gehry. And Frank, you ’ ve come to a place in your life, which is astonishing. I mean it is astonishing for an artist, for an **architect**, to become actually an icon and a legend in their own time. I mean you have become, whether you can giggle at it because it ’ s a funny... you know, it ’ s a strange thought, but your building is an icon — you can draw a little **picture** of that building, it can be used in ads — and you ’ ve had not rock star status, but celebrity status in doing what you wanted to do for most of your life. And I know the road was extremely difficult. And it didn ’ t seem, at least, that your sell outs, whatever they were, were very big. You kept moving ahead in a life where you ’ re dependent on working for somebody. But that ’ s an interesting thing for a creative person. A lot of us work for people ; we ’ re in the hands of other people. And that ’ s one of the great dilemmas — we ’ re in a creativity session — it ’ s one of the great dilemmas in creativity : how to do work that ’ s big enough and not sell out. And you ’ ve achieved that and that makes your win doubly big, triply big. It ’ s not quite a question but you can comment on it. It ’ s a big issue. FG : Well, I ’ ve always just... I ’ ve never really gone out looking for work. I always waited for it to sort of hit me on the head. And when I started out, I thought that architecture was a service business and that you had to please the clients and stuff. And I realized when I ’ d come into the meetings with these corrugated **metal** and chain link stuff, and people would just look at me like I ’ d just landed from Mars. But I couldn ’ t do anything else. That was my response to the people in the time. And actually, it was responding to clients that I had who didn ’ t have very much money, so they couldn ’ t afford very much. I think it was circumstantial. Until I got to my house, where the client was my wife. We bought this tiny little bungalow in Santa Monica and for like 50 grand I built a house around it. And a few people got excited about it. I was visiting with an artist, Michael Heizer, out in the desert near Las Vegas somewhere. He ’ s building this huge concrete place. And it was late in the evening. We ’ d had a lot to drink. We were standing out in the desert all alone and, thinking about my house, he said,”Did it ever occur to you if you built stuff more permanent, somewhere in 2000 years somebody ’ s going to like it?”So, I thought,”Yeah, that ’ s probably a good idea.”Luckily I started to get some clients that had a little more money, so the stuff was a little more permanent. But I just found out the world **ain** ’ t going to last that long, this guy was telling us the other day. So where do we go now? Back to — everything ’ s so temporary. I don ’ t see it the way you characterized it. For me, every day is a new thing. I approach each project with a new insecurity, almost like the first project I ever did, and I get the sweats, I go in and start working, I ’ m not sure where I ’ m going — if I knew where I was going, I wouldn ’ t do it. When I can predict or plan it, I don ’ t do it. I discard it. So I approach it with the same trepidation. Obviously, over time I have a lot more confidence that it ’ s going to be OK. I do run a kind of a business — I ’ ve got 120 people and you ’ ve got to pay them, so there ’ s a lot of responsibility involved — but the actual work on the project is with, I think, a healthy insecurity. And like the playwright said the other day — I could relate to him : you ’ re not sure. When Bilbao was finished and I looked at it, I saw all the mistakes, I saw... They weren ’ t mistakes ; I saw everything that I would have changed and I was embarrassed by it. I felt an embarrassment —”How could I have done that? How could I have made shapes like that or done stuff like that?”It ’ s taken several years to now look at it detached and say — as you walk around the corner and a piece of it works with the road and the street, and it appears to have a relationship — that I started to like it. RSW : What ’ s the status of the New York project? FG : I don ’ t really know. Tom Krens came to me with Bilbao and explained it all to me, and I thought he was nuts. I didn ’ t think he knew what he was doing, and he pulled it off. So, I think he ’ s Icarus and Phoenix all in one guy. He gets up there and then he... comes back up. They ’ re still talking about it. September 11 generated some interest in moving it over to Ground Zero, and I ’ m totally against that. I just feel uncomfortable talking about or building anything on Ground Zero I think for a long time. RSW : The **picture** on the screen, is that Disney? FG : Yeah. RSW : How much further along is it than that, and when will that be finished? FG : That will be finished in 2003 — September, October — and I ’ m hoping Kyu, and Herbie, and Yo-Yo and all those guys come play with us at that place. Luckily, today most of the people I ’ m working with are people I really like. Richard Koshalek is probably one of the main reasons that Disney Hall came to me. He ’ s been a cheerleader for quite a long time. There aren ’ t many people around that are really involved with architecture as clients. If you think about the world, and even just in this audience, most of us are involved with buildings. Nothing that you would call architecture, right? And so to find one, a guy like that, you hang on to him. He ’ s become the head of Art Center, and there ’ s a building by Craig Ellwood there. I knew Craig and respected him. They want to add to it and it ’ s hard to add to a building like that — it ’ s a beautiful, minimalist, black steel building — and Richard wants to add a library and more student stuff and it ’ s a lot of acreage. I convinced him to let me bring in another **architect** from Portugal : Alvaro Siza. RSW : Why did you want that? FG : I knew you ’ d ask that question. It was intuitive. Alvaro Siza grew up and lived in Portugal and is probably considered the Portuguese main guy in architecture. I visited with him a few years ago and he showed me his early work, and his early work had a resemblance to my early work. When I came out of college, I started to try to do things contextually in Southern California, and you got into the logic of Spanish colonial tile **roofs** and things like that. I tried to understand that language as a beginning, as a place to jump off, and there was so much of it being done by spec builders and it was trivialized so much that it wasn ’ t... I just stopped. I mean, Charlie Moore did a bunch of it, but it didn ’ t feel good to me. Siza, on the other hand, continued in Portugal where the real stuff was and evolved a modern language that relates to that historic language. And I always felt that he should come to Southern California and do a building. I tried to get him a couple of jobs and they didn ’ t pan out. I like the idea of collaboration with people like that because it pushes you. I ’ ve done it with Claes Oldenburg and with Richard Serra, who doesn ’ t think architecture is art. Did you see that thing? RSW : No. What did he say? FG : He calls architecture”plumbing.”FG : Anyway, the Siza thing. It ’ s a richer experience. It must be like that for Kyu doing things with musicians — it ’ s similar to that I would imagine — where you... huh? Audience : Liquid architecture. FG : Liquid architecture. Where you... It ’ s like jazz : you improvise, you work together, you play off each other, you make something, they make something. And I think for me, it ’ s a way of trying to understand the city and what might happen in the city. RSW : Is it going to be near the current campus? Or is it going to be down near... FG :

No, it ’ s near the current campus. Anyway, he ’ s that kind of patron. It ’ s not his money, of course. RSW : What ’ s his schedule on that? FG : I don ’ t know. What ’ s the schedule, Richard? Richard Koshalek : [Unclear] starts from 2004. FG : 2004. You can come to the opening. I ’ ll invite you. No, but the issue of city building in democracy is interesting because it creates chaos, right? Everybody doing their thing makes a very chaotic environment, and if you can figure out how to work off each other — if you can get a bunch of people who respect each other ’ s work and play off each other, you might be able to create models for how to build sections of the city without resorting to the one **architect**. Like the Rockefeller Center model, which is kind of from another era. RSW : I found the most remarkable thing. My preconception of Bilbao was this wonderful building, you go inside and there ’ d be extraordinary spaces. I ’ d seen **drawings** you had presented here at TED. The surprise of Bilbao was in its context to the city. That was the surprise of going across the river, of going on the highway around it, of walking down the street and finding it. That was the real surprise of Bilbao. FG : But you know, Richard, most **architects** when they present their work — most of the people we know, you get up and you talk about your work, and it ’ s almost like you tell everybody you ’ re a good guy by saying,”Look, I ’ m worried about the context, I ’ m worried about the city, I ’ m worried about my client, I worry about budget, that I ’ m on time.”Blah, blah, blah and all that stuff. And it ’ s like cleansing yourself so that you can... by saying all that, it means your work is good somehow. And I think everybody — I mean that should be a matter of fact, like gravity. You ’ re not going to defy gravity. You ’ ve got to work with the building department. If you don ’ t meet the budgets, you ’ re not going to get much work. If it leaks — Bilbao did not leak. I was so proud. The MIT project — they were interviewing me for MIT and they sent their facilities people to Bilbao. I met them in Bilbao. They came for three days. RSW : This is the computer building? FG : Yeah, the computer building. They were there three days and it rained every day and they kept walking around — I noticed they were looking under things and looking for things, and they wanted to know where the buckets were hidden, you know? People put buckets out... I was **clean**. There wasn ’ t a bloody leak in the place, it was just fantastic. But you ’ ve got to — yeah, well up until then every building leaked, so this... RSW : Frank had a sort of... FG : Ask Miriam! RW :... sort of had a fame. His fame was built on that in L. A. for a while. FG : You ’ ve all heard the Frank Lloyd Wright story, when the woman called and said,”Mr. Wright, I ’ m sitting on the couch and the water ’ s pouring in on my head.”And he said,”Madam, move your chair.”So, some years later I was doing a building, a little house on the beach for Norton Simon, and his secretary, who was kind of a **hell** on wheels type lady, called me and said,”Mr. Simon ’ s sitting at his desk and the water ’ s coming in on his head.”And I told her the Frank Lloyd Wright story. RSW : Didn ’ t get a laugh. FG : No. Not now either. But my point is that... and I call it the”then what?”OK, you solved all the problems, you did all the stuff, you made nice, you loved your clients, you loved the city, you ’ re a good guy, you ’ re a good person... and then what? What do you bring to it? And I think that ’ s what I ’ ve always been interested in, is that — which is a personal kind of expression. Bilbao, I think, shows that you can have that kind of personal expression and still touch all the bases that are necessary of fitting into the city. That ’ s what reminded me of it. And I think that ’ s the issue, you know ; it ’ s the”then what”that most clients who hire **architects** — most clients aren ’ t hiring **architects** for that. They ’ re hiring them to get it done, get it on budget, be polite, and they ’ re missing out on the real value of an **architect**. RSW : At a certain point a number of years ago, people — when Michael Graves was a fashion, before teapots... FG : I did a teapot and nobody bought it. RSW : Did it leak? FG : No. RSW :... people wanted a Michael Graves building. Is that a curse, that people want a Bilbao building? FG : Yeah. Since Bilbao opened, which is now four, five years, both Krens and I have been called with at least 100 opportunities — China, Brazil, other parts of Spain — to come in and do the Bilbao effect. And I ' ve met with some of these people. Usually I say no right away, but some of them come with pedigree and they sound well-intentioned and they get you for at least one or two meetings. In one case, I flew all the way to Malaga with a team because the thing was signed with seals and various very official seals from the city, and that they wanted me to come and do a building in their port. I asked them what kind of building it was."When you get here we ' ll explain it."Blah, blah, blah. So four of us went. And they took us — they put us up in a great hotel and we were looking over the bay, and then they took us in a boat out in the water and showed us all these sights in the harbor. Each one was more beautiful than the other. And then we were going to have lunch with the mayor and we were going to have dinner with the most important people in Malaga. Just before going to lunch with the mayor, we went to the harbor commissioner. It was a table as long as this carpet and the harbor commissioner was here, and I was here, and my guys. We sat down, and we had a drink of water and everybody was quiet. And the guy looked at me and said,"Now what can I do for you, Mr. Gehry?"RSW : Oh, my God. FG : So, I got up. I said to my team,"Let ' s get out of here."We stood up, we walked out. They followed — the guy that dragged us there followed us and he said,"You mean you ' re not going to have lunch with the mayor?"I said,"Nope.""You ' re not going to have dinner at all?"They just brought us there to hustle this group, you know, to create a project. And we get a lot of that. Luckily, I ' m old enough that I can complain I can ' t travel. I don ' t have my own plane yet. RSW : Well, I ' m going to wind this up and wind up the meeting because it ' s been very long. But let me just say a couple words. FG : Can I say something? Are you going to talk about me or you? RSW : Once a shit, always a shit! FG : Because I want to get a standing ovation like everybody, so... RSW : You ' re going to get one! You ' re going to get one! I ' m going to make it for you! FG : No, no. Wait a minute!

Text Sample 667: POS Dim 5 – 2002 – Score 11.00 – 2002/t000317.txt

I draw to better understand things. Sometimes I make a lot of **drawings** and I still don ' t understand what it is I ' m **drawing**. Those of you who are comfortable with digital stuff and even smug about that relationship might be amused to know that the guy who is best known for"The Way Things Work,"while preparing for part of a **panel** for called Understanding, spent two days trying to get his laptop to communicate with his new CD burner. Who knew about extension managers? I ' ve always managed my own extensions so it never even occurred to me to read the instructions, but I did finally figure it out. I had to figure it out, because along with the invitation came the frightening reminder that there would be no projector, so bringing those carousels would no longer be necessary but some alternate form of communication would. Now, I could talk about something that I ' m known for, something that would be particularly appropriate for many of the more technically minded people here, or I could talk about something I really care about. I decided to go with the latter. I ' m going to talk about Rome. Now, why would I care about Rome, particularly? Well, I went to Rhode Island School of Design in the second half of the ' 60s to study architecture. I was lucky enough to spend my last year, my fifth year, in Rome as a student. It changed my life. Not the least reason was the fact that I had spent those first four years living at home, driving into RISD everyday, driving back. I missed the ' 60s. I read about them ; I understand they were pretty interesting. I missed them, but I did spend that extraordinary year in Rome, and it ' s a place that is never far from my mind. So, whenever given an opportunity, I try to do something in it or with it or for it. I also make **drawings** to help people understand things. Things that I want them to believe I understand. And that ' s what I do as an illustrator, that ' s my job. So, I ' m going to show you some **pictures** of Rome. I ' ve made a lot of **drawings** of Rome over the years. These are just **drawings** of Rome. I get back as often as possible — I need to. All different materials, all different styles, all different times, **drawings** from sketchbooks looking at the details of Rome. Part of the reason I ' m showing you these is that it sort of helps illustrate this process I go through of trying to figure out what it is I feel about Rome and why I feel it. These are sketches of some of the little details. Rome is a city full of surprises. I mean, we ' re talking about unusual perspectives, we ' re talking about narrow little winding streets that suddenly open into vast, sun-drenched piazzas — never, though, piazzas that are not humanly scaled. Part of the reason for that is the fact that they grew up organically. That amazing juxtaposition of old and new, the bits of light that come down between the buildings that sort of create a map that ' s traveling above your head of usually blue — especially in the summer — compared to the map that you would normally expect to see of conventional streets. And I began to think about how I could communicate this in book form. How could I share my sense of Rome, my understanding of Rome? And I ' m going to show you a bunch of dead ends, basically. The primary reason for all these dead ends is when if you ' re not quite sure where you ' re going, you ' re certainly not going to get there with any kind of efficiency. Here ' s a little map. And I thought of maps at the beginning ; maybe I should just try and do a little atlas of my favorite streets and connections in Rome. And here ' s a line of text that actually evolves from the exhaust of a scooter zipping across the page. Here that same line of text wraps around a fountain in an illustration that can be turned upside down and read both ways. Maybe that line of text could be a story to help give some human aspect to this. Maybe I should get away from this map completely, and really be honest about wanting to show you my favorite bits and pieces of Rome and simply kick a soccer ball in the air — which happens in so many of the squares in the city — and let it bounce off of things. And I ' ll simply explain what each of those things is that the soccer ball hits. That seemed like a sort of a cheap shot. But even though I just started this presentation, this is not the first thing that I tried to do and I was getting sort of desperate. Eventually, I realized that I had really no content that I could count on, so I decided to move towards packaging. I mean, it seems to work for a lot of things. So I thought a little box set of four small books might do the trick. But one of the ideas that emerged from some of those sketches was the notion of traveling through Rome in different vehicles at different speeds in order to show the different aspects of Rome. Sort of an overview of Rome and the plan that you might see from a dirigible. Quick snapshots of things you might see from a speeding motor scooter, and very slow walking through Rome, you might be able to study in more detail some of the wonderful surfaces and whatnot that you come across. Anyways, I went back to the dirigible notion. Went to Alberto Santos-Dumont. Found one of his dirigibles that had enough dimensions so I could actually use it as a scale that I would then juxtapose with some of the things in Rome. This thing would be flying over or past or be **parked** in front of, but it would be like having a ruler — sort of travel through the pages without being a ruler. Not that you know how long number 11 actually is, but you would be able to compare number 11 against the Pantheon with number 11 against the Baths of Caracalla, and so on and so forth. If you were interested. This is Beatrix. She has a dog named Ajax, she has purchased a dirigible — a small dirigible — she ’ s assembling the structure, Ajax is sniffing for holes in the balloon before they set off. She launches this thing above the Spanish Steps and sets off for an aerial tour of the city. Over the Spanish Steps we go. A nice way to show that river, that stream sort of pouring down the hill. Unfortunately, just across the road from it or quite close by is the Column of Marcus Aurelius, and the diameter of the dirigible makes an impression, as you can see, as she starts trying to read the story that spirals around the Column of Marcus Aurelius — gets a little too close, nudges it. This gives me a chance to suggest to you the structure of the Column of Marcus Aurelius, which is really no more than a pile of quarters high — thick quarters. Over the Piazza of Saint Ignacio, completely ruining the symmetry, but that aside a spectacular place to visit. A spectacular framework, inside of which you see, usually, extraordinary blue sky. Over the Pantheon and the 26-foot diameter Oculus. She **parks** her dirigible, lowers the anchor rope and climbs down for a closer look inside. The text here is right side and upside down so that you are forced to turn the book around, and you can see it from ground point of view and from her point of view — looking in the hole, getting a different kind of perspective, moving you around the space. Particularly appropriate in a building that can contain perfectly a sphere dimensions of the diameter being the same as the distance from the center of the floor to the center of the Oculus. Unfortunately for her, the anchor line gets tangled around the **feet** of some Boy Scouts who are visiting the Pantheon, and they are immediately yanked out and given an extraordinary but terrifying tour of some of the domes of Rome, which would, from their point of view, naturally be hanging upside down. They bail out as soon as they get to the top of Saint Ivo, that little spiral structure you see there. She continues on her way over Piazza Navona. Notices a lot of activity at the Tre Scalini restaurant, is reminded that it is lunchtime and she ’ s hungry. They keep on motoring towards the Campo de ’ Fiori, which they soon reach. Ajax the dog is put in a basket and lowered with a list of food into the marketplace, which flourishes there until about one in the afternoon, and then is completely removed and doesn ’ t appear again until six or seven the following morning. Anyway, the pooch gets back to the dirigible with the stuff. Unfortunately, when she goes to unwrap the prosciutto, Ajax makes a lunge for it. She ’ s managed to save the prosciutto, but in the process she loses the tablecloth, which you can see flying away in the upper **left-hand** corner. They continue without their tablecloth, looking for a place to land this thing so that they can actually have lunch. They eventually discover a huge wall that ’ s filled with small holes, **ideal** for docking a dirigible because you ’ ve got a place to tie up. Turns out to be the exterior wall — that part of it that remains — of the Coliseum, so they **park** themselves there and have a terrific lunch and have a spectacular view. At the end of lunch, they untie the anchor, they set off through the Baths of Caracalla and over the walls of the city and then an abandoned gatehouse and decide to take one more look at the Pyramid of Cestius, which has this lightning rod on top. Unfortunately, that ’ s a problem : they get a little too close, and when you ’ re in a dirigible you have to be very careful about spikes. So that sort of brings her little story to a conclusion. Marcello, on the other hand, is sort of a lazy guy, but he ’ s not due at work until about noon. So, the alarm goes off and it ’ s five to 12 or so. He gets up, leaps onto his scooter, races through the city past the church of Santa Maria della Pace, down the alleys, through the streets that tourists may be wandering through, disturbing the quiet backstreet life of Rome at every turn. That speed with which he is moving, I hope I have suggested in this little image, which, again, can be turned around and read from both sides because there ’ s text on the **bottom** and text on the top, one of which is upside down in this image. So, he keeps on moving, approaching an unsuspecting waiter who is trying to deliver two plates of linguine in a delicate white wine clam sauce to diners who are sitting at a table just outside of a restaurant in the street. Waiter catches on, but it ’ s too late. And Marcello keeps moving in his scooter. Everything he sees from this point on is slightly affected by the linguine, but keeps on moving because this guy ’ s got a job to do. Removes some scaffolding. One of the reasons Rome remains the extraordinary place it is that because of scaffolding and the determination to maintain the fabric, it is a city that continues to grow and adapt to the needs of the particular time in which it finds itself, or we find it. Right through the Piazza della Rotonda, in front of the Pantheon — again wreaking havoc — and finally getting to work. Marcello, as it turns out, is the driver of the number 64 bus, and if you ’ ve been on the number 64 bus, you know that it ’ s driven with the same kind of exuberance as Marcello demonstrated on his scooter. And finally Carletto. You see his apartment in the upper **left-hand** corner. He ’ s looking at his table ; he ’ s planning to propose this evening to his girlfriend of 40 years, and he wants it to be perfect. He ’ s got candles out, he ’ s got flowers in the middle and he ’ s trying to figure out where to put the plates and the glasses. But he ’ s not happy ; something ’ s wrong. The phone rings anyway, he ’ s called to the palazzo. He saunters — he saunters at a good clip, but as compared to all the traveling we ’ ve just seen, he ’ s sauntering. Everybody knows Carletto, because he ’ s in entertainment, actually ; he ’ s in television. He ’ s actually in television repair, which is why people know him. So they all have his number. He arrives at the palazzo, arrives at the big front door. Enters the courtyard and talks to the custodian, who tells him that there ’ s been a disaster in the palazzo ; nobody ’ s TVs are working and there ’ s a big soccer game coming up, and the crowd is getting a little restless and a little nervous. He goes down to the basement and starts to check the wiring, and then gradually works his way up to the top of the building, apartment by apartment, checking every television, checking every connection, hoping to find out what this problem is. He works his way up, finally, the grand staircase and then a smaller staircase until he reaches the attic. He opens the window of the attic, of course, and there ’ s a tablecloth wrapped around the building ’ s television antenna. He removes it, the problems are solved, everybody in the palazzo is happy. And of course, he also solves his own problem. All he has to do now, with a perfect table, is wait for her to arrive. That was the first attempt, but it didn ’ t seem substantial enough to convey whatever it was I wanted to convey about Rome. So I thought, well, I ’ ll just do piazzas, and I ’ ll get inside and underneath and I ’ ll show these things growing and show why they ’ re shaped the way they are and so on. And then I thought, that ’ s too complicated. No, I ’ ll just take my favorite bits and pieces and I ’ ll put them inside the Pantheon but keep the scale, so you can see the top of Sant ’ Ivo and the Pyramid of Cestius and the Tempietto of Bramante all side by side in this amazing space. Now that ’ s one **drawing**, so I thought maybe it ’ s time for Piranesi to meet Escher. You see that I ’ m beginning to really lose control here and also hope. There ’ s a very thin blue line of exhaust that sort of runs through this thing that would be kind of the trail that holds it all together. Then I thought,”Wait a minute, what am I doing?”A book is not only a neat way of collecting and storing information, it ’ s a series of layers. I mean, you always peel one layer off another ; we think of them as pages, doing it a certain way. But think of them as layers. I mean, Rome is a place of layers — horizontal layers, vertical layers — and I thought, well just peeling off a page would allow me to — if I got you thinking about it the right way — would allow me to sort of show you the depth of layers. The stucco on the walls of most of the buildings in Rome covers the scars ; the scars of centuries of change as these structures have been adapted rather than being torn down. If I do a foldout page on the **left-hand** side and let you just unfold it, you see behind it what I mean by scar tissue. You can see that in 1635, it became essential to make smaller windows because the bad guys were coming or whatever. Adaptations all get buried under the stucco. I could peel out a page of this palazzo to show you what ' s going on inside of it. But more importantly, I could also show you what it looks like at the corner of one of those magnificent buildings with all the massive stone blocks, or the fake stone blocks done with brick and stucco, which is more often the case. So it becomes slightly three-dimensional. I could take you down one of those narrow little streets into one of those surprising piazzas by using a double gate fold — double foldout page — which, if you were like me reading a pop- up book as a child, you hopefully stick your head into. You wrap the pages around your head and are in that piazza for that brief period of time. And I ' ve really not done anything much more complicated than make foldout pages. But then I thought, maybe I could be simpler here. Let ' s look at the Pantheon and the Piazza della Rotonda in front of it. Here ' s a book completely wide open. OK, if I don ' t open the book the whole way, if I just open it 90 degrees, we ' re looking down the front of the Pantheon, and we ' re looking sort of at the top, more or less down on the square. And if I turn the book the other way, we ' re looking across the square at the front of the Pantheon. No foldouts, no tricks — just a book that isn ' t open the whole way. That seemed promising. I thought, maybe I ' ll do it inside and I can even combine the foldouts with the only partially opened book. So we get inside the Pantheon and it grows and so on and so forth. And I thought, maybe I ' m on the right track, but it sort of lost its human quality. So I went back to the notion of story, which is always a good thing to have if you ' re trying to get people to pay attention to a book and pick up information along the way."Pigeon ' s Progress"struck me as a catchy title. If it was a homing pigeon, it would be called"Homer ' s Odyssey."But it was the journey of the... I mean, if a title works, use it. But it would be a journey that went through Rome and showed all the things that I like about Rome. It ' s a pigeon sitting on top of a church. Goes off during the day and does normal pigeon stuff. Comes back, the whole place is covered with scaffolding and green netting and there ' s no way this pigeon can get home. So it ' s a homeless pigeon now and it ' s going to have to find another place to live, and that allows me to go through my catalog of favorite things, and we start with the tall ones and so on. Maybe it has to go back and live with family members ; that ' s not always a good thing, but it does sort of bring pigeons together again. And I thought, that ' s sort of interesting, but maybe there ' s a person who should be involved in this in some way. So I kind of came up with this old guy who spends his life looking after sick pigeons. He ' ll go anywhere to get them — dangerous places and whatnot — and they become really friends with this guy, and learn to do tricks for him and entertain him at lunchtime and stuff like that. There ' s a real bond that develops between this old man and these pigeons. But unfortunately he gets sick. He gets really sick at the end of the story. He ' s taught them to spell his name, which is Aldo. They show up one day after three or four days of not seeing him — he lives in this little garret — and they spell his name and fly around. And he finally gets enough strength together to climb up the ladder onto the **roof**, and all the pigeons, a la Red Balloon, are there waiting for him and they carry him off over the walls of the city. And I forgot to mention this : whenever he lost a pigeon, he would take that pigeon out beyond the walls of the city. In the old Roman custom, the dead were never buried within the walls. And I thought that ' s a really cheery story.. That ' s really going to go a long way. So anyway, I went through... And again, if packaging doesn ’ t work and if the stories aren ’ t going anywhere, I just come up with titles and hope that a title will sort of kick me off in the right direction. And sometimes it does focus me enough and I ’ ll even do a title page. So, these are all title pages that eventually led me to the solution I settled on, which is the story of a young woman who sends a message on a homing pigeon — she lives outside the walls of the city of Rome — to someone in the city. And the pigeon is flying down above the Appian Way here. You can see the tombs and pines and so on and so forth along the way. If you see the red line, you are seeing the trail of the pigeon ; if you don ’ t see the red line, you are the pigeon. And it becomes necessary and possible, at this point, to try to convey what that sense would be like of flying over the city without actually moving. Past the Pyramid of Cestius — these will seem very familiar to you, even if you haven ’ t been to Rome recently — past the gatehouse. This is something that ’ s a little bit unusual. This pigeon does something that most homing pigeons do not do : it takes the scenic route, which was a device that I felt was necessary to actually extend this book beyond about four pages. So, we circle around the Coliseum, past the Church of Santa Maria in Cosmedin and the Temple of Hercules towards the river. We almost collide with the **cornice** of the Palazzo Farnese — designed by Michelangelo, built of stone taken from the Coliseum — narrow escape. We swoop down over the Campo de ’ Fiori. This is one of those things I show to my students because it ’ s a complete bastardization — a denial of any rules of perspective. The only rule of perspective that I think matters is if it looks believable, you ’ ve succeeded. But you try and figure out where the vanishing points meet here ; a couple are on Mars and a couple of others in Cremona. But into the piazza in front of Santa Maria della Pace, where invariably a soccer game is going on, and we ’ re hit by a soccer ball. Now this is a terrible illustration of being hit by a soccer ball. I have all the pieces : there ’ s Santa Maria della Pace, there ’ s a soccer ball, there ’ s a little bit of a bird ’ s wing — nothing ’ s happening, so I had to rethink it. And if you do want to see Santa Maria della Pace, these books are really flexible, incredibly interactive — just turn it around and look at it the other way. Through the alley, we can see the impact is captured in the red line. And then bird manages to pull itself together past this medieval tower — one of the few remaining medieval towers — towards the church of Sant ’ Agnese and around the dome looking down into Piazza Navona — which we ’ ve already mentioned and seen and flown over a couple of times ; there ’ s the Bernini statue of the Four Rivers — and then past the wonderful Borromini Sant ’ Ivo, stopping just long enough on the 26-foot diameter Oculus of the Pantheon to catch our breath. And then we can swoop inside and around ; and because we ’ re flying, we don ’ t really have to worry about gravity at this particular moment in time, so this **drawing** can be oriented in any way on the page. We get a little exuberant as we pass Gesu ; it ’ s not surprising to sort of mimic the architecture in this way. Past the wonderful wall filled with the juxtaposition that I was talking about ; beautiful carvings set into the walls above the neon”Ristorante”sign, and so on. And eventually, we arrive at the courtyard of the palazzo, which is our destination. Straight up through the courtyard into a little window into the attic, where somebody is working at the **drawing** board. He removes the message from the leg of the bird ; this is what it says. As we look at the **drawing** board, we see what he ’ s working on is, in fact, a map of the journey that the pigeon has just taken, and the red line extends through all the sights. And if you want the information, so that we complete this cycle of understanding, all you have to do is read these paragraphs. Thank you very much.

Text Sample 668: POS Dim 5 – 2002 – Score 9.00 – 2002/t000312.txt

It ’ s a great honor to be here with you. The good news is I ’ m very aware of my responsibilities to get you out of here because I ’ m the only thing standing between you and the bar. And the good news is I don ’ t have a prepared speech, but I have a box of slides. I have some **pictures** that represent my life and what I do for a living. I ’ ve learned through experience that people remember **pictures** long after they ’ ve forgotten words, and so I hope you ’ ll remember some of the **pictures** I ’ m going to share with you for just a few minutes. The whole story really starts with me as a high school kid in Pittsburgh, Pennsylvania, in a tough neighborhood that everybody gave up on for dead. And on a Wednesday afternoon, I was walking down the corridor of my high school kind of minding my own business. And there was this artist teaching, who made a great big old ceramic vessel, and I happened to be looking in the door of the art room — and if you ’ ve ever seen clay done, it ’ s magic — and I ’ d never seen anything like that before in my life. So, I walked in the art room and I said,”What is that?”And he said,”Ceramics. And who are you?”And I said,”I ’ m Bill Strickland. I want you to teach me that.”And he said,”Well, get your homeroom teacher to sign a piece of paper that says you can come here, and I ’ ll teach it to you.”And so for the remaining two years of my high school, I cut all my classes. But I had the presence of mind to give the teachers ’ classes that I cut the pottery that I made, and they gave me passing grades. And that ’ s how I got out of high school. And Mr. Ross said,”You ’ re too smart to die and I don ’ t want it on my conscience, so I ’ m leaving this school and I ’ m taking you with me.”And he drove me out to the University of Pittsburgh where I filled out a college application and got in on probation. Well, I ’ m now a trustee of the university, and at my installation ceremony I said,”I ’ m the guy who came from the neighborhood who got into the place on probation. Don ’ t give up on the poor kids, because you never know what ’ s going to happen to those children in life.”What I ’ m going to show you for a couple of minutes is a facility that I built in the toughest neighborhood in Pittsburgh with the highest crime rate. One is called Bidwell Training Center ; it is a vocational school for ex-steel workers and single parents and welfare mothers. You remember we used to make steel in Pittsburgh? Well, we don ’ t make any steel anymore, and the people who used to make the steel are having a very tough time of it. And I rebuild them and give them new life. Manchester Craftsmen ’ s Guild is named after my neighborhood. I was adopted by the Bishop of the Episcopal Diocese during the riots, and he donated a row house. And in that row house I started Manchester Craftsmen ’ s Guild, and I learned very quickly that wherever there are Episcopalians, there ’ s money in very close proximity. And the Bishop adopted me as his kid. And last year I spoke at his memorial service and wished him well in this life. I went out and hired a student of Frank Lloyd Wright, the **architect**, and I asked him to build me a world class center in the worst neighborhood in Pittsburgh. And my building was a scale model for the Pittsburgh airport. And when you come to Pittsburgh — and you ’ re all invited — you ’ ll be flying into the blown-up version of my building. That ’ s the building. Built in a tough neighborhood where people have been given up for dead. My view is that if you want to involve yourself in the life of people who have been given up on, you have to look like the solution and not the problem. As you can see, it has a fountain in the courtyard. And the reason it has a fountain in the courtyard is I wanted one and I had the checkbook, so I bought one and put it there. And now that I ’ m giving speeches at conferences like TED, I got put on the board of the Carnegie Museum. At a reception in their courtyard, I noticed that they had a fountain because they think that the people who go to the **museum** deserve a fountain. Well, I think that welfare mothers and at-risk kids and ex-steel workers deserve a fountain in their life. And so the first thing that you see in my center in the springtime is water that greets you — water is life and water of human possibility — and it sets an attitude and expectation about how you feel about people before you ever give them a speech. So, from that fountain I built this building. As you can see, it has world class art, and it ' s all my taste because I raised all the money. I said to my boy,"When you raise the money, we ' ll put your taste on the wall."That we have quilts and clay and calligraphy and everywhere your eye turns, there ' s something beautiful looking back at you, that ' s deliberate. That ' s intentional. In my view, it is this kind of world that can redeem the soul of poor people. We also created a boardroom, and I hired a Japanese cabinetmaker from Kyoto, Japan, and commissioned him to do 60 pieces of furniture for our building. We have since spun him off into his own business. He ' s making a ton of money doing custom furniture for rich people. And I got 60 pieces out of it for my school because I felt that welfare moms and ex-steel workers and single parents deserved to come to a school where there was handcrafted furniture that greeted them every day. Because it sets a tone and an attitude about how you feel about people long before you give them the speech. We even have flowers in the hallway, and they ' re not plastic. Those are real and they ' re in my building every day. And now that I ' ve given lots of speeches, we had a bunch of high school principals come and see me, and they said,"Mr. Strickland, what an extraordinary story and what a great school. And we were particularly touched by the flowers and we were curious as to how the flowers got there."I said,"Well, I got in my car and I went out to the greenhouse and I bought them and I brought them back and I put them there."You don ' t need a task force or a study group to buy flowers for your kids. What you need to know is that the children and the adults deserve flowers in their life. The cost is incidental but the gesture is huge. And so in my building, which is full of sunlight and full of flowers, we believe in hope and human possibilities. That happens to be at Christmas time. And so the next thing you ' ll see is a million dollar kitchen that was built by the Heinz company — you ' ve heard of them? They did all right in the ketchup business. And I happen to know that company pretty well because John Heinz, who was our U. S. senator — who was tragically killed in a plane accident — he had heard about my desire to build a new building, because I had a cardboard box and I put it in a garbage bag and I walking all over Pittsburgh trying to raise money for this site. And he called me into his office — which is the equivalent of going to see the Wizard of Oz — and John Heinz had 600 million dollars, and at the time I had about 60 cents. And he said,"But we ' ve heard about you. We ' ve heard about your work with the kids and the ex- steel workers, and we ' re inclined to want to support your desire to build a new building. And you could do us a great service if you would add a culinary program to your program."Because back then, we were building a trades program. He said,"That way we could fulfill our affirmative action goals for the Heinz company."I said,"Senator, I ' m reluctant to go into a field that I don ' t know much about, but I promise you that if you ' ll support my school, I ' ll get it built and in a couple of years, I ' ll come back and weigh out that program that you desire."And Senator Heinz sat very quietly and he said,"Well, what would your reaction be if I said I ' d give you a million dollars?"I said,"Senator, it appears that we ' re going into the food training business."And John Heinz did give me a million bucks. And most importantly, he loaned me the head of research for the Heinz company. And we kind of borrowed the curriculum from the Culinary Institute of America, which in their mind is kind of the Harvard of cooking schools, and we created a gourmet cooks program for welfare mothers in this million dollar kitchen in the middle of the inner city. And we ' ve never looked back. I would like to show you now some of the food that these welfare mothers do in this million dollar kitchen. That happens to be our cafeteria line. That ' s puff pastry day. Why? Because the students made puff pastry and that ' s what the school ate every day. But the concept was that I wanted to take the stigma out of food. That good food ' s not for rich people — good food ' s for everybody on the planet, and there ' s no excuse why we all can ' t be eating it. So at my school, we subsidize a gourmet lunch program for welfare mothers in the middle of the inner city because we ' ve discovered that it ' s good for their stomachs, but it ' s better for their heads. Because I wanted to let them know every day of their life that they have value at this place I call my center. We have students who sit together, black kids and white kids, and what we ' ve discovered is you can solve the race problem by creating a world class environment, because people will have a tendency to show you world class behavior if you treat them in that way. These are examples of the food that welfare mothers are doing after six months in the training program. No sophistication, no class, no dignity, no history. What we ' ve discovered is the only thing wrong with poor people is they don ' t have any money, which happens to be a curable condition. It ' s all in the way that you think about people that often determines their behavior. That was done by a student after seven months in the program, done by a very brilliant young woman who was taught by our pastry chef. I ' ve actually eaten seven of those baskets and they ' re very good. They have no calories. That ' s our dining room. It looks like your average high school cafeteria in your average town in America. But this is my view of how students ought to be treated, particularly once they have been pushed aside. We train pharmaceutical technicians for the pharmacy industry, we train medical technicians for the medical industry, and we train chemical technicians for companies like Bayer and Calgon Carbon and Fisher Scientific and Exxon. And I will guarantee you that if you come to my center in Pittsburgh — and you ' re all invited — you ' ll see welfare mothers doing analytical chemistry with logarithmic calculators 10 months from enrolling in the program. There is absolutely no reason why poor people can ' t learn world class technology. What we ' ve discovered is you have to give them flowers and sunlight and food and expectations and Herbie ' s music, and you can cure a spiritual cancer every time. We train corporate travel agents for the travel industry. We even teach people how to read. The kid with the red stripe was in the program two years ago — he ' s now an instructor. And I have children with high school diplomas that they can ' t read. And so you must ask yourself the question : how is it possible in the 21st century that we graduate children from schools who can ' t read the diplomas that they have in their hands? The reason is that the system gets reimbursed for the kids they spit out the other end, not the children who read. I can take these children and in 20 weeks, demonstrated aptitude ; I can get them high school equivalent. No big deal. That ' s our library with more handcrafted furniture. And this is the arts program I started in 1968. Remember I ' m the black kid from the ' 60s who got his life saved with ceramics. Well, I went out and decided to reproduce my experience with other kids in the neighborhood, the theory being if you get kids flowers and you give them food and you give them sunshine and enthusiasm, you can bring them right back to life. I have 400 kids from the Pittsburgh public school system that come to me every day of the week for arts education. And these are children who are flunking out of public school. And last year I put 88 percent of those kids in college and I ' ve averaged over 80 percent for 15 years. We ' ve made a fascinating discovery : there ' s nothing wrong with the kids that affection and sunshine and food and enthusiasm and Herbie ' s music can ' t cure. For that I won a big old plaque — Man of the Year in Education. I beat out all the Ph. D. ' s because I figured that if you treat children like human beings, it increases the likelihood they ’ re going to behave that way. And why we can ’ t institute that policy in every school and in every city and every town remains a mystery to me. Let me show you what these people do. We have ceramics and photography and computer imaging. And these are all kids with no artistic ability, no talent, no imagination. And we bring in the world ’ s greatest artists — Gordon Parks has been there, Chester Higgins has been there — and what we ’ ve learned is that the children will become like the people who teach them. In fact, I brought in a mosaic artist from the Vatican, an African-American woman who had studied the old Vatican mosaic techniques, and let me show you what they did with the work. These were children who the whole world had given up on, who were flunking out of public school, and that ’ s what they ’ re capable of doing with affection and sunlight and food and good music and confidence. We teach photography. And these are examples of some of the kids ’ work. That boy won a four-year scholarship on the strength of that photograph. This is our gallery. We have a world class gallery because we believe that poor kids need a world class gallery, so I designed this thing. We have smoked salmon at the art openings, we have a formal printed invitation, and I even have figured out a way to get their parents to come. I couldn ’ t buy a parent 15 years ago so I hired a guy who got off on the Jesus big time. He was dragging guys out of bars and saving those lives for the Lord. And I said,”Bill, I want to hire you, man. You have to tone down the Jesus stuff a little bit, but keep the enthusiasm. I can ’ t get these parents to come to the school.”He said,”I ’ ll get them to come to the school.”So, he jumped in the van, he went to Miss Jones ’ house and said,”Miss Jones, I knew you wanted to come to your kid ’ s art opening but you probably didn ’ t have a ride. So, I came to give you a ride.”And he got 10 parents and then 20 parents. At the last show that we did, 200 parents showed up and we didn ’ t pick up one parent. Because now it ’ s become socially not acceptable not to show up to support your children at the Manchester Craftsmen ’ s Guild because people think you ’ re bad parents. And there is no statistical difference between the white parents and the black parents. Mothers will go where their children are being celebrated, every time, every town, every city. I wanted you to see this gallery because it ’ s as good as it gets. And by the time I cut these kids loose from high school, they ’ ve got four shows on their resume before they apply to college because it ’ s all up here. You have to change the way that people see themselves before you can change their behavior. And it ’ s worked out pretty good up to this day. I even stuck another room on the building, which I ’ d like to show you. This is brand new. We just got this slide done in time for the TED Conference. I gave this little slide show at a place called the Silicon Valley and I did all right. And the woman came out of the audience, she said,”That was a great story and I was very impressed with your presentation. My only criticism is your computers are getting a little bit old.”And I said,”Well, what do you do for a living?”She said,”Well, I work for a company called Hewlett-Packard.”And I said,”You ’ re in the computer business, is that right?”She said,”Yes, sir.”And I said,”Well, there ’ s an easy solution to that problem.”Well, I ’ m very pleased to announce to you that HP and a furniture company called Steelcase have adopted us as a demonstration model for all of their technology and all their furniture for the United States of America. And that ’ s the room that ’ s initiating the relationship. We got it just done in time to show you, so it ’ s kind of the world debut of our digital imaging center. I only have a couple more slides, and this is where the story gets kind of interesting. So, I just want you to listen up for a couple more minutes and you ’ ll understand why he ’ s there and I ’ m here. In 1986, I had the presence of mind to stick a music hall on the north end of the building while I was building it. And a guy named Dizzy Gillespie showed up to play there because he knew this man over here, Marty Ashby. And I stood on that stage with Dizzy Gillespie on sound check on a Wednesday afternoon, and I said,”Dizzy, why would you come to a black-run center in the middle of an industrial **park** with a high crime rate that doesn ’ t even have a reputation in music?”He said,”Because I heard you built the center and I didn ’ t believe that you did it, and I wanted to see for myself. And now that I have, I want to give you a gift.”I said,”You ’ re the gift.”He said,”No, sir. You ’ re the gift. And I ’ m going to allow you to record the concert and I ’ m going to give you the music, and if you ever choose to sell it, you must sign an agreement that says the money will come back and support the school.”And I recorded Dizzy. And he died a year later, but not before telling a fellow named McCoy Tyner what we were doing. And he showed up and said,”Dizzy talking about you all over the country, man, and I want to help you.”And then a guy named Wynton Marsalis showed up. Then a bass player named Ray Brown, and a fellow named Stanley Turrentine, and a piano player named Herbie Hancock, and a band called the Count Basie Orchestra, and a fellow named Tito Puente, and a guy named Gary Burton, and Shirley Horn, and Betty Carter, and Dakota Staton and Nancy Wilson all have come to this center in the middle of an industrial **park** to sold out audiences in the middle of the inner city. And I ’ m very pleased to tell you that, with their permission, I have now accumulated 600 recordings of the greatest artists in the world, including Joe Williams, who died, but not before his last recording was done at my school. And Joe Williams came up to me and he put his hand on my shoulder and he said,”God ’ s picked you, man, to do this work. And I want my music to be with you.”And that worked out all right. When the Basie band came, the band got so excited about the school they voted to give me the rights to the music. And I recorded it and we won something called a Grammy. And like a fool, I didn ’ t go to the ceremony because I didn ’ t think we were going to win. Well, we did win, and our name was literally in lights over Madison Square Garden. Then the U. N. Jazz Orchestra dropped by and we recorded them and got nominated for a second Grammy back to back. So, we ’ ve become one of the hot, young jazz recording studios in the United States of America in the middle of the inner city with a high crime rate. That ’ s the place all filled up with Republicans. If you ’ d have dropped a bomb on that room, you ’ d have wiped out all the money in Pennsylvania because it was all sitting there. Including my mother and father, who lived long enough to see their kid build that building. And there ’ s Dizzy, just like I told you. He was there. And he was there, Tito Puente. And Pat Metheny and Jim Hall were there and they recorded with us. And that was our first recording studio, which was the broom closet. We put the mops in the hallway and re-engineered the thing and that ’ s where we recorded the first Grammy. And this is our new facility, which is all video technology. And that is a room that was built for a woman named Nancy Wilson, who recorded that album at our school last Christmas. And any of you who happened to have been watching Oprah Winfrey on Christmas Day, he was there and Nancy was there singing excerpts from this album, the rights to which she donated to our school. And I can now tell you with absolute certainty that an appearance on Oprah Winfrey will sell 10, 000 CDs. We are currently number four on the Billboard Charts, right behind Tony Bennett. And I think we ’ re going to be fine. This was burned out during the riots — this is next to my building — and so I had another cardboard box built and I walked back out in the streets again. And that ’ s the building, and that ’ s the model, and on the right ’ s a high-tech greenhouse and in the middle ’ s the medical technology building. And I ’ m very pleased to tell you that the building ’ s done. It ’ s also full of anchor tenants at 20 dollars a **foot** — triple that in the middle of the inner city. And there ’ s the fountain. Every building has a fountain. And the University of Pittsburgh Medical Center are anchor tenants and they took half the building, and we now train medical technicians through all their system. And Mellon Bank ’ s a tenant. And I love them because they pay the rent on time. And as a result of the association, I ’ m now a director of the Mellon Financial Corporation that bought Dreyfus. And this is in the process of being built as we speak. Multiply that **picture** times four and you will see the greenhouse that ’ s going to open in October this year because we ’ re going to grow those flowers in the middle of the inner city. And we ’ re going to have high school kids growing Phalaenopsis orchids in the middle of the inner city. And we have a handshake with one of the large retail grocers to sell our orchids in all 240 stores in six states. And our partners are Zuma Canyon Orchids of Malibu, California, who are Hispanic. So, the Hispanics and the black folks have formed a partnership to grow high technology orchids in the middle of the inner city. And I told my United States senator that there was a very high probability that if he could find some funding for this, we would become a **left-hand** column in the Wall Street Journal, to which he readily agreed. And we got the funding and we open in the fall. And you ought to come and see it — it ’ s going to be a **hell** of a story. And this is what I want to do when I grow up. The brown building is the one you guys have been looking at and I ’ ll tell you where I made my big mistake. I had a chance to buy this whole industrial **park** — which is less than 1, 000 **feet** from the riverfront — for four million dollars and I didn ’ t do it. And I built the first building, and guess what happened? I appreciated the real estate values beyond everybody ’ s expectations and the owners of the **park** turned me down for eight million dollars last year, and said,”Mr. Strickland, you ought to get the Civic Leader of the Year Award because you ’ ve appreciated our property values beyond our wildest expectations. Thank you very much for that.”The moral of the story is you must be prepared to act on your dreams, just in case they do come true. And finally, there ’ s this **picture**. This is in a place called San Francisco. And the reason this **picture** ’ s in here is I did this slide show a couple years ago at a big economics summit, and there was a fellow in the audience who came up to me. He said,”Man, that ’ s a great story. I want one of those.”I said,”Well, I ’ m very flattered. What do you do for a living?”He says,”I run the city of San Francisco. My name ’ s Willie Brown.”And so I kind of accepted the flattery and the praise and put it out of my mind. And that weekend, I was going back home and Herbie Hancock was playing our center that night — first time I ’ d met him. And he walked in and he says,”What is this?”And I said,”Herbie, this is my concept of a training center for poor people.”And he said,”As God as my witness, I ’ ve had a center like this in my mind for 25 years and you ’ ve built it. And now I really want to build one.”I said,”Well, where would you build this thing?”He said,”San Francisco.”I said,”Any chance you know Willie Brown?”As a matter of fact he did know Willie Brown, and Willie Brown and Herbie and I had dinner four years ago, and we started drawing out that center on the tablecloth. And Willie Brown said,”As sure as I ’ m the mayor of San Francisco, I ’ m going to build this thing as a legacy to the poor people of this city.”And he got me five acres of land on San Francisco Bay and we got an **architect** and we got a general contractor and we got Herbie on the board, and our friends from HP, and our friends from Steelcase, and our friends from Cisco, and our friends from Wells Fargo and Genentech. And along the way, I met this real short guy at my slide show in the Silicon Valley. He came up to me afterwards, he said,”Man, that ’ s a fabulous story. I want to help you.”And I said,”Well, thank you very much for that. What do you do for a living?”He said,”Well, I built a company called eBay.”I said,”Well, that ’ s very nice. Thanks very much, and give me your card and sometime we ’ ll talk.”I didn ’ t know eBay from that jar of water sitting on that piano, but I had the presence of mind to go back and talk to one of the techie kids at my center. I said,"Hey man, what is eBay?"He said,"Well, that ' s the electronic commerce network."I said,"Well, I met the guy who built the thing and he left me his card."So, I called him up on the phone and I said,"Mr. Skoll, I ' ve come to have a much deeper appreciation of who you are and I ' d like to become your friend."And Jeff and I did become friends, and he ' s organized a team of people and we ' re going to build this center. And I went down into the neighborhood called Bayview-Hunters Point, and I said,"The mayor sent me down here to work with you and I want to build a center with you, but I ' m not going to build you anything if you don ' t want it. And all I ' ve got is a box of slides."And so I stood up in front of 200 very angry, very disappointed people on a summer night, and the air conditioner had broken and it was 100 degrees outside, and I started showing these **pictures**. And after about 10 **pictures** they all settled down. And I ran the story and I said,"What do you think?"And in the back of the room, a woman stood up and she said,"In 35 years of living in this God forsaken place, you ' re the only person that ' s come down here and treated us with dignity. I ' m going with you, man."And she turned that audience around on a pin. And I promised these people that I was going to build this thing, and we ' re going to build it all right. And I think we can get in the ground this year as the first replication of the center in Pittsburgh. But I met a guy named Quincy Jones along the way and I showed him the box of slides. And Quincy said,"I want to help you, man. Let ' s do one in L. A."And so he ' s assembled a group of people. And I ' ve fallen in love with him, as I have with Herbie and with his music. And Quincy said,"Where did the idea for centers like this come from?"And I said,"It came from your music, man. Because Mr. Ross used to bring in your albums when I was 16 years old in the pottery class, when the world was all dark, and your music got me to the sunlight."And I said,"If I can follow that music, I ' ll get out into the sunlight and I ' ll be OK. And if that ' s not true, how did I get here?"I want you all to know that I think the world is a place that ' s worth living. I believe in you. I believe in your hopes and your dreams, I believe in your intelligence and I believe in your enthusiasm. And I ' m tired of living like this, going into town after town with people standing around on corners with holes where eyes used to be, their spirits damaged. We won ' t make it as a country unless we can turn this thing around. In Pennsylvania it costs 60, 000 dollars to keep people in jail, most of whom look like me. It ' s 40, 000 dollars to build the University of Pittsburgh Medical School. It ' s 20, 000 dollars cheaper to build a medical school than to keep people in jail. Do the math — it will never work. I am banking on you and I ' m banking on guys like Herbie and Quincy and Hackett and Richard and very decent people who still believe in something. And I want to do this in my lifetime, in every city and in every town. And I don ' t think I ' m crazy. I think we can get home on this thing and I think we can build these all over the country for less money than we ' re spending on prisons. And I believe we can turn this whole story around to one of celebration and one of hope. In my business it ' s very difficult work. You ' re always fighting upstream like a salmon — never enough money, too much need — and so there is a tendency to have an occupational depression that accompanies my work. And so I ' ve figured out, over time, the solution to the depression : you make a friend in every town and you ' ll never be lonely. And my hope is that I ' ve made a few here tonight. And thanks for listening to what I had to say.

Text Sample 669: POS Dim 5 – 1984 – Score 5.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ' ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a computer and really address the qualities of the human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, the intelligence of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ' ve been able to collect, which I think illustrate this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,"My God! What a terrible image you get when you look at still **pictures** on TV."Well, it doesn ' t have to be terrible. And that is a slide taken from a TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ' s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ' s the distance you should sit away from the TV set. Now we ' ve put people 18 inches in front of a TV, and all the artifacts that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow mask, the **scan** lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful **pictures**. I ' m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ' m very interested in touch-sensitive displays. High-tech, high-touch. Isn ' t that what some of you said? It ' s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ' re not : it ' s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on the market these systems that have just a few light emitting diodes on the side and are very low resolution, it ' s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ' t have to pick them up, and people don ' t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ' re typing — what you have — you want to now put something — first of all, you ' ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ' ve got to sort of find that mouse. Then you find the mouse, and you ' re going to have to wiggle it a little bit to see where the cursor is on the screen. And then when you finally see where it is, then you ' ve got to move it to get the cursor over there, and then —"Bang"— you ' ve got to hit a button or do whatever. That ' s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ' m trying to do is just illustrate the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of the computer field, is when you have a bug that you can ' t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the substrate, which it usually is. So we found that that actually was a feature in the sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ' s pressure-sensitive. And it ' s pressure-sensitive to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ' t figure out how to come in the other direction. But let me get rid of the slide, and let ' s see if this comes on. OK. So there is the pressure-sensitive display in operation. The person ' s just, if you will, pushing on the screen to make a **curve**. But this is the interesting part. I want to stop it for a second because the movie is very badly made. And the particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an anvil on a fuzzy rug and the other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across the screen, and yet you touch the ping-pong ball very lightly and it just scoots across the screen. And what you can do — oops, I didn ' t mean to do that — what you can do is actually feed back to the user the feeling of the physical properties. So again, they don ' t have to be weight ; they could be a general trying to move troops, and he ' s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the interface there are physical properties in that transducer — in this case it ' s pressure and touches — that allow you to present things to the user that you could never present before. So it ' s not simply looking at the quality or, if you will, the luxury of that interface, but it ' s actually looking at the idea of presenting things that previously couldn ' t be presented before. I want to move on to another example, which is one of a different sort, where we ' re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ' re going to take this book, if you will, and it ' s going to come alive. You ' re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are the experts in monologue making : you make a **film** and it has a well-formed beginning, middle and end, and in some sense the art of it is that. And you then say,”There ' s an opportunity for making conversational movies.”Well, what does that mean? And it sort of nibbles at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ' ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the movie has information about itself. So it knows, or at least there is computer-readable information in the medium itself.

It ’ s just not a static movie frame. That ’ s one thing. The other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the cookbook, the”Larousse Gastronomique.”And I think I use the example all too often, but it ’ s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to the end of the recipe and it says,”Cook until done.”Now, that would be, if you will, the top green track, which doesn ’ t mean too much. But you might have to elaborate for me or for somebody who isn ’ t an expert, and say,”Cook at 380 degrees for 45 minutes.”And then for a real beginner, you would go down even further and elaborate more — say,”Open the oven, preheat, wait for the light to go out, open the door, don ’ t leave it open too long, put the penguin in and shut the door...”whatever. And that ’ s a much more elaborate one than you dribble back. That ’ s one kind of use of random access. And the other is where you want to explain the same thing in different ways. If you ’ re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ’ s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ’ s a rather boring book, but I ’ m afraid sometimes you have to do boring books because your sponsors aren ’ t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ’ t even know what vintage the transmission is, but let me just show you very quickly some of it, and we ’ ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a **picture** of the transmission, and as you rub your finger across the transmission it highlights the various parts. Narrator : When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, superimposed over the illustration. NN : This is about the oil pan, or the oil filter and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get back to the illustration, but in this case it ’ s not a single frame, but it ’ s actually a movie of someone coming into the frame and doing the repair that ’ s described in the text. The two-headed slider is a speed control that allows me to watch the movie at various speeds, in forward or reverse. And the movie is displayed as a full frame movie. I can go back to the beginning... and play the movie at full speed. Here ’ s another step-by-step procedure, only in this case — NN : Okay, this movie is... Everybody ’ s heard of sound-sync movies — this is text-sync movies, so as the movie plays, the text gets highlighted. We highlight the text as we go through the movie. Repairman :... Not too far out. Front poles, preferably. Don ’ t loosen them too far. If you loosen them too far, you ’ ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ’ m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ’ ve been doing — in this case, in Senegal — where we have tried to use personal computers as a pedagogical medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ' s a story I ' d like to tell. And that was when, before we did this in any developing countries — we ' re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ' ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ' t read, didn ' t even make it in the lowest section of the school ' s classes — and was pretty much not in school, though physically there. But did hang around the, quote,"computer room,"where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of **fun**, it was very interesting. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,"Let me show you how this works,"and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ' t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ' d actually come to see — not the computer room — they went upstairs and they said,"This is absolutely remarkable! That child was very articulate and showed us and even dealt with the things he couldn ' t do automatically with that manual. It was just absolutely fantastic."The principal said,"There ' s a dreadful mistake, because that child can ' t read. And you obviously have been hoodwinked or you ' ve talked about somebody else."And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,"Can you read?"And the child said,"No, I can ' t."And then they said,"But wait a minute. You just looked through that manual and you found..."and he said,"Oh, but that ' s not reading."And so they said,"Well, what ' s reading then?"He says,"Well, reading is this junk they give me in little books to read. It ' s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return."And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ' m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ' s not, but we think speech —"My God, little children pick it up somehow, and by the age of two they ' re doing a mediocre job, and by three and four they ' re speaking reasonably well. And yet you ' ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder."Well, it ' s not harder. What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ' s going to help you. So what happens is you go to school and people say,"Just believe me, you ' re going to like it. You ' re going to like reading,"and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on the screen has a huge payoff, and it ' s a lot of **fun**. And in fact, it ' s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ' s the girl on the left leaning with her chalkboard, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ' s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example — as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ' s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And the reason it remains a favorite project is that we were asked to do a teleconferencing system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ' s faces. So if I wanted to call my friend Peter Sprague on the phone, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ' s uncanny : there ' s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ' s something that didn ' t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,"He or she wouldn ' t agree,"and the empty chair is that person and the spatiality is **crucial**. So we said,"Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ' s site you ' d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ' t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ' ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ' s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ' s site, he ' ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.

Text Sample 670: POS Dim 5 – 1994 – Score 4.00 – 1994/t002458.txt

Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ’ s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC microprocessors versus the power of local area networks. And the interesting thing about it is that this slide, like so many technology slides that we ’ re used to, is a sort of a straight line on a semi-log **curve**. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log **curves**. Something really weird is going on here. And that ’ s basically what I ’ m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ’ m just going to use a piece of paper here. Now why do we draw technology **curves** in semi-log **curves**? Well the answer is, if I drew it on a normal **curve** where, let ’ s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ’ m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ’ t tell us much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log **curve**, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as microprocessor technology, then the day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ’ s not moving like that. And there ’ s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ’ d like to ask is, if you look at these exponential **curves**, they don ’ t go on forever. Things just can ’ t possibly keep changing as fast as they are. One of two things is going to happen. Either it ’ s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ’ s going to do this. That ’ s about all it can do. Now I ’ m an optimist, so I sort of think it ’ s probably going to do something like that. If so, that means that what we ’ re in the middle of right now is a transition. We ’ re sort of on this line in a transition from the way the world used to be to some new way that the world is. And so what I ’ m trying to ask, what I ’ ve been asking myself, is what ’ s this new way that the world is? What ’ s that new state that the world is heading toward? Because the transition seems very, very confusing when we ’ re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ’ s a conference in which people talk about the future, and you notice that the future is still at about the year 2000. It ’ s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ’ s happening there. That transition is happening. We can all sense it. And we know that it just doesn ’ t make too much sense to think out 30, 50 years because everything ’ s going to be so different that a simple extrapolation of what we ’ re doing just doesn ’ t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ’ re going through. Now in order to do that I ’ m going to have to talk about a bunch of stuff that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step back and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this **picture** makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, the Earth was this big, sterile hunk of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ' s theories that are beginning to understand about how it started with RNA, but I ' m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ' t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ' s sort of just a very simple chemical form of life, but when things got interesting was when these drops learned a trick about **abstraction**. Somehow by ways that we don ' t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for amusement purposes is we can now write things in this code. And I ' ve got here a little 100 micrograms of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ' s right here. If I had really been a egotist, I would have put it into a virus and released it in the room. So what was the next step? Writing down the DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ' t know if you know this, but bacteria can actually exchange DNA. Now that ' s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the transition point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ’ s interesting in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ’ re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to evolve so that the interesting level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very special structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in the community and special structures that were responsible for recording, understanding, learning information. And that was the brains and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ’ d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these special information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ’ re aware of is human language. It ’ s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ’ m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ’ re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ’ ve done is we, humanity, have started abstracting out. We ’ re going through the same levels that multi-cellular organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. Telephony, computers, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ’ ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of microseconds. And you saw Ty ’ s little evolutionary example where he sort of did a little bit of evolution on the Convolution program right before your eyes. So now we ’ ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and brains, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ' s what we ' re seeing here in this explosion of **curve**. We ' re seeing this process feeding back on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ' s really impossible for me to design them in the traditional sense. I don ' t know what every transistor in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of **abstraction** and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ' t quite even understand. One method that ' s particularly interesting that I ' ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the microsecond time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,"Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted."So in other words, I define what I wanted. Let ' s say I want to sort numbers, as a simple example I ' ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,"Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ' s reproduce the ones that sorted numbers the best. And let ' s reproduce them by a process of recombination analogous to sex."Take two programs and they produce children by exchanging their subroutines, and the children inherit the traits of the subroutines of the two programs. So I ' ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,"Please repeat that process."Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ' t tell you how they work. I ' ve tried looking at them and telling you how they work. They ' re obscure, weird programs. But they do the job. And in fact, I know, I ' m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,"Oh look. Look at this. It says, ' This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ' Doesn ' t that make you feel confident?"In fact, we know that the engineering process doesn ' t work very well when it gets complicated. So we ' re beginning to depend on computers to do a process that ' s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ' t quite understand the options of it. So in a sense, it ' s getting ahead of us. We ' re now using those programs to make much faster computers so that we ' ll be able to run this process much faster.

So it ’ s feeding back on itself. The thing is becoming faster and that ’ s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ’ re taking off. And what we are is we ’ re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ’ re the amoebas and we can ’ t quite figure out what the **hell** this thing is we ’ re creating. We ’ re right at that point of transition. But I think that there really is something coming along after us. I think it ’ s very haughty of us to think that we ’ re the end product of evolution. And I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

Text Sample 671: POS Dim 5 – 2005 – Score 10.00 – 2005/t000435.txt

It is a thrill to be here at a conference that ’ s devoted to”Inspired by Nature”â you can imagine. And I ’ m also thrilled to be in the foreplay section. Did you notice this section is foreplay? Because I get to talk about one of my favorite critters, which is the Western Grebe. You haven ’ t lived until you ’ ve seen these guys do their courtship dance. I was on Bowman Lake in Glacier National Park, which is a long, skinny lake with sort of mountains upside down in it, and my partner and I have a rowing shell. And so we were rowing, and one of these Western Grebes came along. And what they do for their courtship dance is, they go together, the two of them, the two mates, and they begin to run underwater. They paddle faster, and faster, and faster, until they ’ re going so fast that they literally lift up out of the water, and they ’ re standing upright, sort of paddling the top of the water. And one of these Grebes came along while we were rowing. And so we ’ re in a skull, and we ’ re moving really, really quickly. And this Grebe, I think, sort of, mistaked us for a prospect, and started to run along the water next to us, in a courtship dance â for miles. It would stop, and then start, and then stop, and then start. Now that is foreplay. I came this close to changing species at that moment. Obviously, life can teach us something in the entertainment section. Life has a lot to teach us. But what I ’ d like to talk about today is what life might teach us in technology and in design. What ’ s happened since the book came out â the book was mainly about research in biomimicry â and what ’ s happened since then is **architects**, designers, engineers â people who make our world â have started to call and say, we want a biologist to sit at the design table to help us, in real time, become inspired. Or â and this is the **fun** part for me â we want you to take us out into the natural world. We ’ ll come with a design challenge and we find the champion adapters in the natural world, who might inspire us. So this is a **picture** from a Galapagos trip that we took with some wastewater treatment engineers ; they purify wastewater. And some of them were very resistant, actually, to being there. What they said to us at first was, you know, we already do biomimicry. We use bacteria to **clean** our water. And we said, well, that ’ s not exactly being inspired by nature. That ’ s bioprocessing, you know ; that ’ s bio-assisted technology : using an organism to do your wastewater treatment is an old, old technology called”domestication.”This is learning something, learning an idea, from an organism and then applying it. And so they still weren ’ t getting it. So we went for a walk on the beach and I said, well, give me one of your big problems. Give me a design challenge, sustainability speed bump, that ’ s keeping you from being sustainable. And they said scaling, which is the build-up of minerals inside of pipes. And they said, you know what happens is, mineral â just like at your house â mineral builds up. And then the aperture closes, and we have to flush the pipes with toxins, or we have to dig them up. So if we had some way to stop this scaling â and so I picked up some shells on the beach. And I asked them, what is scaling? What ’ s inside your pipes? And they said, calcium carbonate. And I said, that ’ s what this is ; this is calcium carbonate. And they didn ’ t know that. They didn ’ t know that what a seashell is, it ’ s templated by proteins, and then ions from the seawater crystallize in place to create a shell. So the same sort of a process, without the proteins, is happening on the inside of their pipes. They didn ’ t know. This is not for lack of information ; it ’ s a lack of integration. You know, it ’ s a silo, people in silos. They didn ’ t know that the same thing was happening. So one of them thought about it and said, OK, well, if this is just crystallization that happens automatically out of seawater â self-assembly â then why aren ’ t shells infinite in size? What stops the scaling? Why don ’ t they just keep on going? And I said, well, in the same way that they exude a protein and it starts the crystallization â and then they all sort of leaned in â they let go of a protein that stops the crystallization. It literally adheres to the growing face of the crystal. And, in fact, there is a product called TPA that ’ s mimicked that protein â that stop-protein â and it ’ s an environmentally friendly way to stop scaling in pipes. That changed everything. From then on, you could not get these engineers back in the boat. The first day they would take a hike, and it was, click, click, click, click. Five minutes later they were back in the boat. We ’ re done. You know, I ’ ve seen that island. After this, they were crawling all over. They would snorkel for as long as we would let them snorkel. What had happened was that they realized that there were organisms out there that had already solved the problems that they had spent their careers trying to solve. Learning about the natural world is one thing ; learning from the natural world â that ’ s the switch. That ’ s the profound switch. What they realized was that the answers to their questions are everywhere ; they just needed to change the lenses with which they saw the world. 3. 8 billion years of field-testing. 10 to 30 â Craig Venter will probably tell you ; I think there ’ s a lot more than 30 million â well-adapted solutions. The important thing for me is that these are solutions solved in context. And the context is the Earth â the same context that we ’ re trying to solve our problems in. So it ’ s the conscious emulation of life ’ s genius. It ’ s not slavishly mimicking â although Al is trying to get the hairdo going â it ’ s not a slavish mimicry ; it ’ s taking the design principles, the genius of the natural world, and learning something from it. Now, in a group with so many IT people, I do have to mention what I ’ m not going to talk about, and that is that your field is one that has learned an enormous amount from living things, on the software side. So there ’ s computers that protect themselves, like an immune system, and we ’ re learning from gene regulation and biological development. And we ’ re learning from neural nets, genetic algorithms, evolutionary computing. That ’ s on the software side. But what ’ s interesting to me is that we haven ’ t looked at this, as much. I mean, these machines are really not very high tech in my estimation in the sense that there ’ s dozens and dozens of carcinogens in the water in Silicon Valley. So the hardware is not at all up to snuff in terms of what life would call a success. So what can we learn about making â not just computers, but everything? The plane you came in, cars, the seats that you ’ re sitting on. How do we redesign the world that we make, the human-made world? More importantly, what should we ask in the next 10 years? And there ’ s a lot of cool technologies out there that life has. What ’ s the syllabus? Three questions, for me, are key. How does life make things? This is the opposite ; this is how we make things. It ’ s called heat, beat and treat â that ’ s what material scientists call it. And it ’ s carving things down from the top, with 96 percent waste left over and only 4 percent product. You heat it up ; you beat it with high pressures ; you use chemicals. OK. Heat, beat and treat. Life can ’ t afford to do that. How does life make things? How does life make the most of things? That ’ s a geranium pollen. And its shape is what gives it the function of being able to tumble through air so easily. Look at that shape. Life adds information to matter. In other words : structure. It gives it information. By adding information to matter, it gives it a function that ’ s different than without that structure. And thirdly, how does life make things disappear into systems? Because life doesn ’ t really deal in things ; there are no things in the natural world divorced from their systems. Really quick syllabus. As I ’ m reading more and more now, and following the story, there are some amazing things coming up in the biological sciences. And at the same time, I ’ m listening to a lot of businesses and finding what their sort of grand challenges are. The two groups are not talking to each other. At all. What in the world of biology might be helpful at this juncture, to get us through this sort of evolutionary knothole that we ’ re in? I ’ m going to try to go through 12, really quickly. One that ’ s exciting to me is self-assembly. Now, you ’ ve heard about this in terms of nanotechnology. Back to that shell : the shell is a self-assembling material. On the lower left there is a **picture** of mother of pearl forming out of seawater. It ’ s a layered structure that ’ s mineral and then polymer, and it makes it very, very tough. It ’ s twice as tough as our high-tech ceramics. But what ’ s really interesting : unlike our ceramics that are in kilns, it happens in seawater. It happens near, in and near, the organism ’ s body. This is Sandia National Labs. A guy named Jeff Brinker has found a way to have a self- assembling coding process. Imagine being able to make ceramics at room temperature by simply dipping something into a liquid, lifting it out of the liquid, and having evaporation force the molecules in the liquid together, so that they jigsaw together in the same way as this crystallization works. Imagine making all of our hard materials that way. Imagine spraying the precursors to a PV cell, to a solar cell, onto a **roof**, and having it self- assemble into a layered structure that harvests light. Here ’ s an interesting one for the IT world : bio-silicon. This is a diatom, which is made of silicates. And so silicon, which we make right now â it ’ s part of our carcinogenic problem in the manufacture of our chips â this is a bio-mineralization process that ’ s now being mimicked. This is at UC Santa Barbara. Look at these diatoms. This is from Ernst Haeckel ’ s work. Imagine being able to â and, again, it ’ s a templated process, and it solidifies out of a liquid process â imagine being able to have that sort of structure coming out at room temperature. Imagine being able to make perfect lenses. On the left, this is a brittle star ; it ’ s covered with lenses that the people at Lucent Technologies have found have no distortion whatsoever. It ’ s one of the most distortion-free lenses we know of. And there ’ s many of them, all over its entire body. What ’ s interesting, again, is that it self-assembles. A woman named Joanna Aizenberg, at Lucent, is now learning to do this in a low- temperature process to create these sort of lenses. She ’ s also looking at fiber optics. That ’ s a sea sponge that has a fiber optic. Down at the very base of it, there ’ s fiber optics that work better than ours, actually, to move light, but you can tie them in a knot ; they ’ re incredibly flexible. Here ’ s another big idea : CO2 as a feedstock. A guy named Geoff Coates, at Cornell, said to himself, you know, plants do not see CO2 as the biggest poison of our time. We see it that way. Plants are busy making long chains of starches and glucose, right, out of CO2. He ’ s found a way â he ’ s found a catalyst â and he ’ s found a way to take CO2 and make it into polycarbonates. Biodegradable plastics out of CO2 â how plant-like. Solar transformations : the most exciting one. There are people who are mimicking the energy-harvesting device inside of purple bacterium, the people at ASU. Even more interesting, lately, in the last couple of weeks, people have seen that there ' s an enzyme called hydrogenase that ' s able to evolve hydrogen from proton and electrons, and is able to take hydrogen up â basically what ' s happening in a fuel cell, in the anode of a fuel cell and in a reversible fuel cell. In our fuel cells, we do it with platinum ; life does it with a very, very common iron. And a team has now just been able to mimic that hydrogen-juggling hydrogenase. That ' s very exciting for fuel cells â to be able to do that without platinum. Power of shape : here ' s a whale. We ' ve seen that the fins of this whale have tubercles on them. And those little bumps actually increase efficiency in, for instance, the edge of an airplane â increase efficiency by about 32 percent. Which is an amazing fossil fuel savings, if we were to just put that on the edge of a wing. Color without pigments : this peacock is creating color with shape. Light comes through, it bounces back off the layers ; it ' s called thin- **film** interference. Imagine being able to self-assemble products with the last few layers playing with light to create color. Imagine being able to create a shape on the outside of a surface, so that it ' s self-cleaning with just water. That ' s what a leaf does. See that up-close **picture**? That ' s a ball of water, and those are dirt particles. And that ' s an up-close **picture** of a lotus leaf. There ' s a company making a product called Lotusan, which mimics â when the building facade paint dries, it mimics the bumps in a self-cleaning leaf, and rainwater **cleans** the building. Water is going to be our big, grand challenge : quenching thirst. Here are two organisms that pull water. The one on the left is the Namibian beetle pulling water out of fog. The one on the right is a pill bug â pulls water out of air, does not drink fresh water. Pulling water out of Monterey fog and out of the sweaty air in Atlanta, before it gets into a building, are key technologies. Separation technologies are going to be extremely important. What if we were to say, no more hard rock mining? What if we were to separate out **metals** from waste streams, small amounts of **metals** in water? That ' s what microbes do ; they chelate **metals** out of water. There ' s a company here in San Francisco called MR3 that is embedding mimics of the microbes ' molecules on filters to mine waste streams. Green chemistry is chemistry in water. We do chemistry in organic solvents. This is a **picture** of the spinnerets coming out of a spider and the silk being formed from a spider. Isn ' t that beautiful? Green chemistry is replacing our industrial chemistry with nature ' s recipe book. It ' s not easy, because life uses only a subset of the elements in the periodic table. And we use all of them, even the toxic ones. To figure out the elegant recipes that would take the small subset of the periodic table, and create miracle materials like that cell, is the task of green chemistry. Timed degradation : packaging that is good until you don ' t want it to be good anymore, and dissolves on cue. That ' s a mussel you can find in the waters out here, and the threads holding it to a rock are timed ; at exactly two years, they begin to dissolve. Healing : this is a good one. That little guy over there is a tardigrade. There is a problem with vaccines around the world not getting to patients. And the reason is that the refrigeration somehow gets broken ; what ' s called the"cold chain"gets broken. A guy named Bruce Rosner looked at the tardigrade â which dries out completely, and yet stays alive for months and months and months, and is able to regenerate itself. And he found a way to dry out vaccines â encase them in the same sort of sugar capsules as the tardigrade has within its cells â meaning that vaccines no longer need to be refrigerated. They can be put in a glove compartment, OK. Learning from organisms. This is a session about water â learning about organisms that can do without water, in order to create a vaccine that lasts and lasts and lasts without refrigeration. I ' m not going to get to 12. But what I am going to do is tell you that the most important thing, besides all of these adaptations, is the fact that these organisms have figured out a way to do the amazing things they do while taking care of the place that ' s going to take care of their offspring. When they ' re involved in foreplay, they ' re thinking about something very, very important â and that ' s having their genetic material remain, 10, 000 generations from now. And that means finding a way to do what they do without destroying the place that ' ll take care of their offspring. That ' s the biggest design challenge. Luckily, there are millions and millions of geniuses willing to gift us with their best ideas. Good luck having a conversation with them. Thank you. Chris Anderson : Talk about foreplay, I â we need to get to 12, but really quickly. Janine Benyus : Oh really? CA : Yeah. Just like, you know, like the 10-second version of 10, 11 and 12. Because we just â your slides are so gorgeous, and the ideas are so big, I can ' t stand to let you go down without seeing 10, 11 and 12. JB : OK, put this â OK, I ' ll just hold this thing. OK, great. OK, so that ' s the healing one. Sensing and responding : feedback is a huge thing. This is a locust. There can be 80 million of them in a square kilometer, and yet they don ' t collide with one another. And yet we have 3. 6 million car collisions a year. Right. There ' s a person at Newcastle who has figured out that it ' s a very large neuron. And she ' s actually figuring out how to make a collision-avoidance circuitry based on this very large neuron in the locust. This is a huge and important one, number 11. And that ' s the growing fertility. That means, you know, net fertility farming. We should be growing fertility. And, oh yes â we get food, too. Because we have to grow the capacity of this planet to create more and more opportunities for life. And really, that ' s what other organisms do as well. In ensemble, that ' s what whole ecosystems do : they create more and more opportunities for life. Our farming has done the opposite. So, farming based on how a prairie builds soil, ranching based on how a native ungulate herd actually increases the health of the range, even wastewater treatment based on how a marsh not only **cleans** the water, but creates incredibly sparkling productivity. This is the simple design brief. I mean, it looks simple because the system, over 3. 8 billion years, has worked this out. That is, those organisms that have not been able to figure out how to enhance or sweeten their places, are not around to tell us about it. That ' s the twelfth one. Life â and this is the secret trick ; this is the magic trick â life creates conditions conducive to life. It builds soil ; it **cleans** air ; it **cleans** water ; it mixes the cocktail of **gases** that you and I need to live. And it does that in the middle of having great foreplay and meeting their needs. So it ' s not mutually exclusive. We have to find a way to meet our needs, while making of this place an Eden. CA : Janine, thank you so much.

Text Sample 672: POS Dim 5 – 2005 – Score 10.00 – 2005/t000418.txt

In 1962, with Rachel Carson ' s"Silent Spring,"I think for people like me in the world of the making of things, the canary in the mine wasn ' t singing. And so the question that we might not have birds became kind of fundamental to those of us wandering around looking for the meadowlarks that seemed to have all disappeared. And the question was, were the birds singing? Now, I ' m not a scientist, that ' ll be really clear. But, you know, we ' ve just come from this discussion of what a bird might be. What is a bird? Well, in my world, this is a rubber duck. It comes in California with a warning —"This product contains chemicals known by the State of California to cause cancer and birth defects or other reproductive harm."This is a bird. What kind of culture would produce a product of this kind and then label it and sell it to children? I think we have a design problem. Someone heard the six hours of talk that I gave called"The Monticello Dialogues"on NPR, and sent me this as a thank you note —"We realize that design is a signal of intention, but it also has to occur within a world, and we have to understand that world in order to imbue our designs with inherent intelligence, and so as we look back at the basic state of affairs in which we design, we, in a way, need to go to the primordial condition to understand the operating system and the frame conditions of a planet, and I think the exciting part of that is the good news that ' s there, because the news is the news of abundance, and not the news of limits, and I think as our culture tortures itself now with tyrannies and concerns over limits and fear, we can add this other dimension of abundance that is coherent, driven by the sun, and start to imagine what that would be like to share."That was a nice thing to get. That was one sentence. Henry James would be proud. This is — I put it down at the **bottom**, but that was extemporaneous, obviously. The fundamental issue is that, for me, design is the first signal of human intentions. So what are our intentions, and what would our intentions be — if we wake up in the morning, we have designs on the world — well, what would our intention be as a species now that we ' re the dominant species? And it ' s not just stewardship and dominion debate, because really, dominion is implicit in stewardship — because how could you dominate something you had killed? And stewardship ' s implicit in dominion, because you can ' t be steward of something if you can ' t dominate it. So the question is, what is the first question for designers? Now, as guardians — let ' s say the state, for example, which reserves the right to kill, the right to be duplicitous and so on — the question we ' re asking the guardian at this point is are we meant, how are we meant, to secure local societies, create world peace and save the environment? But I don ' t know that that ' s the common debate. Commerce, on the other hand, is relatively quick, essentially creative, highly effective and efficient, and fundamentally honest, because we can ' t exchange value for very long if we don ' t trust each other. So we use the tools of commerce primarily for our work, but the question we bring to it is, how do we love all the children of all species for all time? And so we start our designs with that question. Because what we realize today is that modern culture appears to have adopted a strategy of tragedy. If we come here and say,"Well, I didn ' t intend to cause global warming on the way here,"and we say,"That ' s not part of my plan,"then we realize it ' s part of our de facto plan. Because it ' s the thing that ' s happening because we have no other plan. And I was at the White House for President Bush, meeting with every federal department and agency, and I pointed out that they appear to have no plan. If the end game is global warming, they ' re doing great. If the end game is mercury toxification of our children downwind of coal fire plants as they scuttled the Clean Air Act, then I see that our education programs should be explicitly defined as,"Brain death for all children. No child left behind."So, the question is, how many federal officials are ready to move to Ohio and Pennsylvania with their families? So if you don ' t have an endgame of something delightful, then you ' re just moving chess pieces around, if you don ' t know you ' re taking the king. So perhaps we could develop a strategy of change, which requires humility. And in my business as an **architect**, it ' s unfortunate the word"humility"and the word"**architect**"have not appeared in the same paragraph since"The Fountainhead."So if anybody here has trouble with the concept of design humility, reflect on this — it took us 5, 000 years to put wheels on our luggage. So, as Kevin Kelly pointed out, there is no endgame. There is an infinite game, and we ' re playing in that infinite game. And so we call it"cradle to cradle,"and our goal is very simple. This is what I presented to the White House. Our goal is a delightfully diverse, safe, healthy and just world, with **clean** air, **clean** water, soil and power — economically, equitably, ecologically and elegantly enjoyed, period. What don ’ t you like about this? Which part of this don ’ t you like? So we realized we want full diversity, even though it can be difficult to remember what De Gaulle said when asked what it was like to be President of France. He said,”What do you think it ’ s like trying to run a country with 400 kinds of cheese?”But at the same time, we realize that our products are not safe and healthy. So we ’ ve designed products and we analyzed chemicals down to the parts per million. This is a baby blanket by Pendleton that will give your child nutrition instead of Alzheimer ’ s later in life. We can ask ourselves, what is justice, and is justice blind, or is justice blindness? And at what point did that uniform turn from white to black? Water has been declared a human right by the United Nations. Air quality is an obvious thing to anyone who breathes. Is there anybody here who doesn ’ t breathe? **Clean** soil is a critical problem — the nitrification, the dead zones in the Gulf of Mexico. A fundamental issue that ’ s not being addressed. We ’ ve seen the first form of solar energy that ’ s beat the hegemony of fossil fuels in the form of wind here in the Great Plains, and so that hegemony is leaving. And if we remember Sheikh Yamani when he formed OPEC, they asked him,”When will we see the end of the age of oil?”I don ’ t know if you remember his answer, but it was,”The Stone Age didn ’ t end because we ran out of stones.”We see that companies acting ethically in this world are outperforming those that don ’ t. We see the flows of materials in a rather terrifying prospect. This is a hospital monitor from Los Angeles, sent to China. This woman will expose herself to toxic phosphorous, release four pounds of toxic lead into her childrens ’ environment, which is from copper. On the other hand, we see great signs of hope. Here ’ s Dr. Venkataswamy in India, who ’ s figured out how to do mass-produced health. He has given eyesight to two million people for free. We see in our material flows that car steels don ’ t become car steel again because of the contaminants of the coatings — bismuth, antimony, copper and so on. They become building steel. On the other hand, we ’ re working with Berkshire Hathaway, Warren Buffett and Shaw Carpet, the largest carpet company in the world. We ’ ve developed a carpet that is continuously recyclable, down to the parts per million. The upper is Nylon 6 that can go back to caprolactam, the **bottom**, a polyolephine — infinitely recyclable thermoplastic. Now if I was a bird, the building on my left is a liability. The building on my right, which is our corporate campus for The Gap with an ancient meadow, is an asset — its nesting grounds. Here ’ s where I come from. I grew up in Hong Kong, with six million people in 40 square miles. During the dry season, we had four hours of water every fourth day. And the relationship to landscape was that of farmers who have been farming the same piece of ground for 40 centuries. You can ’ t farm the same piece of ground for 40 centuries without understanding nutrient flow. My childhood summers were in the Puget Sound of Washington, among the first growth and big growth. My grandfather had been a lumberjack in the Olympics, so I have a lot of tree karma I am working off. I went to Yale for graduate school, studied in a building of this style by Le Corbusier, affectionately known in our business as Brutalism. If we look at the world of architecture, we see with Mies ’ 1928 tower for Berlin, the question might be,”Well, where ’ s the sun?”And this might have worked in Berlin, but we built it in Houston, and the windows are all closed. And with most products appearing not to have been designed for indoor use, this is actually a vertical **gas** chamber. When I went to Yale, we had the first energy crisis, and I was designing the first solar-heated house in Ireland as a student, which I then built — which would give you a sense of my ambition. And Richard Meier, who was one of my teachers, kept coming over to my desk to give me criticism, and he would say,”Bill, you ’ ve got to understand- — solar energy has nothing to do with architecture.”I guess he didn ’ t read Vitruvius. In 1984, we did the first so- called”green office”in America for Environmental Defense. We started asking manufacturers what were in their materials. They said,”They ’ re proprietary, they ’ re legal, go away.”The only indoor quality work done in this country at that time was sponsored by R. J. Reynolds Tobacco Company, and it was to prove there was no danger from secondhand smoke in the workplace. So, all of a sudden, here I am, graduating from high school in 1969, and this happens, and we realize that”away”went away. Remember we used to throw things away, and we ’ d point to away? And yet, NOAA has now shown us, for example — you see that little blue thing above Hawaii? That ’ s the Pacific Gyre. It was recently dragged for plankton by scientists, and they found six times as much plastic as plankton. When asked, they said,”It ’ s kind of like a giant toilet that doesn ’ t flush.”Perhaps that ’ s away. So we ’ re looking for the design rules of this — this is the highest biodiversity of trees in the world, Irian Jaya, 259 species of tree, and we described this in the book,”Cradle to Cradle.”The book itself is a polymer. It is not a tree. That ’ s the name of the first chapter —”This Book is Not a Tree.”Because in poetics, as Margaret Atwood pointed out,”we write our history on the skin of fish with the blood of bears.”And with so much polymer, what we really need is technical nutrition, and to use something as elegant as a tree — imagine this design assignment : Design something that makes oxygen, sequesters carbon, fixes nitrogen, distills water, accrues solar energy as fuel, makes complex sugars and food, creates microclimates, changes colors with the seasons and self-replicates. Well, why don ’ t we knock that down and write on it? So, we ’ re looking at the same criteria as most people — you know, can I afford it? Does it work? Do I like it? We ’ re adding the Jeffersonian agenda, and I come from Charlottesville, where I ’ ve had the privilege of living in a house designed by Thomas Jefferson. We ’ re adding life, liberty and the pursuit of happiness. Now if we look at the word”competition,”I ’ m sure most of you ’ ve used it. You know, most people don ’ t realize it comes from the Latin competere, which means strive together. It means the way Olympic athletes train with each other. They get fit together, and then they compete. The Williams sisters compete — one wins Wimbledon. So we ’ ve been looking at the idea of competition as a way of cooperating in order to get fit together. And the Chinese government has now — I work with the Chinese government now — has taken this up. We ’ re also looking at survival of the fittest, not in just competition terms in our modern context of destroy the other or beat them to the ground, but really to fit together and build niches and have growth that is good. Now most environmentalists don ’ t say growth is good, because, in our lexicon, asphalt is two words : assigning blame. But if we look at asphalt as our growth, then we realize that all we ’ re doing is destroying the planetary ’ s fundamental underlying operating system. So when we see E equals mc squared come along, from a poet ’ s perspective, we see energy as physics, chemistry as mass, and all of a sudden, you get this biology. And we have plenty of energy, so we ’ ll solve that problem, but the biology problem ’ s tricky, because as we put through all these toxic materials that we disgorge, we will never be able to recover that. And as Francis Crick pointed out, nine years after discovering DNA with Mr. Watson, that life itself has to have growth as a precondition — it has to have free energy, sunlight and it needs to be an open system of chemicals. So we ’ re asking for human artifice to become a living thing, and we want growth, we want free energy from sunlight and we want an open metabolism for chemicals. Then, the question becomes not growth or no growth, but what do you want to grow? So instead of just growing destruction, we want to grow the things that we might enjoy, and someday the FDA will allow us to make French cheese. So therefore, we have these two metabolisms, and I worked with a German chemist, Michael Braungart, and we ' ve identified the two fundamental metabolisms. The biological one I ' m sure you understand, but also the technical one, where we take materials and put them into closed cycles. We call them biological nutrition and technical nutrition. Technical nutrition will be in an order of magnitude of biological nutrition. Biological nutrition can supply about 500 million humans, which means that if we all wore Birkenstocks and cotton, the world would run out of cork and dry up. So we need materials in closed cycles, but we need to analyze them down to the parts per million for cancer, birth defects, mutagenic effects, disruption of our immune systems, biodegradation, persistence, heavy **metal** content, knowledge of how we ' re making them and their production and so on. Our first product was a textile where we analyzed 8, 000 chemicals in the textile industry. Using those intellectual filters, we eliminated [7, 962.] We were left with 38 chemicals. We have since databased the 4000 most commonly used chemicals in human manufacturing, and we ' re releasing this database into the public in six weeks. So designers all over the world can analyze their products down to the parts per million for human and ecological health. We ' ve developed a protocol so that companies can send these same messages all the way through their supply chains, because when we asked most companies we work with — about a trillion dollars — and say,"Where does your stuff come from?"They say,"Suppliers.""And where does it go?""Customers."So we need some help there. So the biological nutrients, the first fabrics — the water coming out was **clean** enough to drink. Technical nutrients — this is for Shaw Carpet, infinitely reusable carpet. Here ' s nylon going back to caprolactam back to carpet. Biotechnical nutrients — the Model U for Ford Motor, a cradle to cradle car — concept car. Shoes for Nike, where the uppers are polyesters, infinitely recyclable, the **bottoms** are biodegradable soles. Wear your old shoes in, your new shoes out. There is no finish line. The idea here of the car is that some of the materials go back to the industry forever, some of the materials go back to soil — it ' s all solar-powered. Here ' s a building at Oberlin College we designed that makes more energy than it needs to operate and purifies its own water. Here ' s a building for The Gap, where the ancient grasses of San Bruno, California, are on the **roof**. And this is our project for Ford Motor Company. It ' s the revitalization of the River Rouge in Dearborn. This is obviously a color photograph. These are our tools. These are how we sold it to Ford. We saved Ford 35 million dollars doing it this way, day one, which is the equivalent of the Ford Taurus at a four percent margin of an order for 900 million dollars worth of cars. Here it is. It ' s the world ' s largest green **roof**, 10 and a half acres. This is the **roof**, saving money, and this is the first species to arrive here. These are killdeer. They showed up in five days. And we now have 350-pound auto workers learning bird songs on the Internet. We ' re developing now protocols for cities — that ' s the home of technical nutrients. The country — the home of biological. And putting them together. And so I will finish by showing you a new city we ' re designing for the Chinese government. We ' re doing 12 cities for China right now, based on cradle to cradle as templates. Our assignment is to develop protocols for the housing for 400 million people in 12 years. We did a mass energy balance — if they use brick, they will lose all their soil and burn all their coal. They ' ll have cities with no energy and no food. We signed a Memorandum of Understanding — here ' s Madam Deng Nan, Deng Xiaoping ' s daughter — for China to adopt cradle to cradle. Because if they toxify themselves, being the lowest-cost producer, send it to the lowest-cost distribution — Wal-Mart — and then we send them all our money, what we ' ll discover is that we have what, effectively, when I was a student, was called mutually assured destruction. Now we do it by molecule.

These are our cities. We ’ re building a new city next to this city ; look at that landscape. This is the site. We don ’ t normally do green fields, but this one is about to be built, so they brought us in to intercede. This is their plan. It ’ s a rubber stamp grid that they laid right on that landscape. And they brought us in and said,”What would you do?”This is what they would end up with, which is another color photograph. So this is the existing site, so this is what it looks like now, and here ’ s our proposal. So the way we approached this is we studied the hydrology very carefully. We studied the biota, the ancient biota, the current farming and the protocols. We studied the winds and the sun to make sure everybody in the city will have fresh air, fresh water and direct sunlight in every single apartment at some point during the day. We then take the **parks** and lay them out as ecological infrastructure. We lay out the building areas. We start to integrate commercial and mixed use so the people all have centers and places to be. The transportation is all very simple, everybody ’ s within a five-minute walk of mobility. We have a 24-hour street, so that there ’ s always a place that ’ s alive. The waste systems all connect. If you flush a toilet, your feces will go to the sewage treatment plants, which are sold as assets, not liabilities. Because who wants the fertilizer factory that makes natural **gas**? The waters are all taken in to construct the wetlands for habitat restorations. And then it makes natural **gas**, which then goes back into the city to power the fuel for the cooking for the city. So this is — these are fertilizer **gas** plants. And then the compost is all taken back to the **roofs** of the city, where we ’ ve got farming, because what we ’ ve done is lifted up the city, the landscape, into the air to — to restore the native landscape on the **roofs** of the buildings. The solar power of all the factory centers and all the industrial zones with their light **roofs** powers the city. And this is the concept for the top of the city. We ’ ve lifted the earth up onto the **roofs**. The farmers have little bridges to get from one **roof** to the next. We inhabit the city with work / live space on all the ground floors. And so this is the existing city, and this is the new city.

Text Sample 673: POS Dim 5 – 2005 – Score 10.00 – 2005/t000237.txt

My mission in life since I was a kid was, and is, to take the rest of you into space. It ’ s during our lifetime that we ’ re going to take the Earth, take the people of Earth and transition off, permanently. And that ’ s exciting. In fact, I think it is a moral imperative that we open the space frontier. You know, it ’ s the first time that we ’ re going to have a chance to have planetary redundancy, a chance to, if you would, back up the biosphere. And if you think about space, everything we hold of value on this planet — **metals** and minerals and real estate and energy — is in infinite quantities in space. In fact, the Earth is a crumb in a supermarket filled with resources. The analogy for me is Alaska. You know, we bought Alaska. We Americans bought Alaska in the 1850s. It ’ s called Seward ’ s folly. We valued it as the number of seal pelts we could kill. And then we discovered these things — gold and oil and fishing and timber — and it became, you know, a trillion-dollar economy, and now we take our honeymoons there. The same thing will happen in space. We are on the verge of the greatest exploration that the human race has ever known. We explore for three reasons, the weakest of which is curiosity. You know, it ’ s funded NASA ’ s budget up until now. Some images from Mars, 1997. In fact, I think in the next decade, without any question, we will discover life on Mars and find that it is literally ubiquitous under the soils and different parts of that planet. The stronger motivator, the much stronger motivator, is fear. It drove us to the moon. We — literally in fear — with the Soviet Union raced to the moon. And we have these huge rocks, you know, killer-sized rocks in the hundreds of thousands or millions out there, and while the probability is very small, the impact, figured in literally, of one of these hitting the Earth is so huge that to spend a small fraction looking, searching, preparing to defend, is not unreasonable. And of course, the third motivator, one near and dear to my heart as an entrepreneur, is wealth. In fact, the greatest wealth. If you think about these other asteroids, there ' s a class of the nickel iron, which in platinum-group **metal** markets alone are worth something like 20 trillion dollars, if you can go out and grab one of these rocks. My plan is to actually buy puts on the precious **metal** market, and then actually claim that I ' m going to go out and get one. And that will fund the actual mission to go and get one. But fear, curiosity and greed have driven us. And for me, this is — I ' m the short kid on the right. This was — my motivation was actually during Apollo. And Apollo was one of the greatest motivators ever. If you think about what happened at the turn of — early 1960s, on May 25, JFK said,"We ' re going to go to the moon."And people left their jobs and they went to obscure locations to go and be part of this amazing mission. And we knew nothing about going to space. We went from having literally put Alan Shepard in suborbital flight to going to the moon in eight years, and the average age of the people that got us there was 26 years old. They didn ' t know what couldn ' t be done. They had to make up everything. And that, my friend, is amazing motivation. This is Gene Cernan, a good friend of mine, saying,"If I can go to the moon"— this is the last human on the moon so far —"nothing, nothing is impossible."But of course, we ' ve thought about the government always as the person taking us there. But I put forward here, the government is not going to get us there. The government is unable to take the risks required to open up this precious frontier. The shuttle is costing a billion dollars a launch. That ' s a pathetic number. It ' s unreasonable. We shouldn ' t be happy in standing for that. One of the things that we did with the Ansari X PRIZE was take the challenge on that risk is OK, you know. As we are going out there and taking on a new frontier, we should be allowed to risk. In fact, anyone who says we shouldn ' t, you know, just needs to be put aside, because, as we go forward, in fact, the greatest discoveries we will ever know is ahead of us. The entrepreneurs in the space business are the furry mammals, and clearly the industrial-military complex — with Boeing and Lockheed and NASA — are the dinosaurs. The ability for us to access these resources to gain planetary redundancy — we can now gather all the information, the genetic codes, you know, everything stored on our databases, and back them up off the planet, in case there would be one of those disastrous situations. The difficulty is getting there, and clearly, the cost to orbit is key. Once you ' re in orbit, you are two thirds of the way, energetically, to anywhere — the moon, to Mars. And today, there ' s only three vehicles — the U. S. shuttle, the Russian Soyuz and the Chinese vehicle — that gets you there. Arguably, it ' s about 100 million dollars a person on the space shuttle. One of the companies I started, Space Adventures, will sell you a ticket. We ' ve done two so far. We ' ll be announcing two more on the Soyuz to go up to the space station for 20 million dollars. But that ' s expensive and to understand what the potential is — — it is expensive. But people are willing to pay that! You know, one — we have a very unique period in time today. For the first time ever, we have enough wealth concentrated in the hands of few individuals and the technology accessible that will allow us to really drive space exploration. But how cheap could it get? I want to give you the end point. We know — 20 million dollars today, you can go and buy a ticket, but how cheap could it get? Let ' s go back to high school physics here. If you calculate the amount of potential energy, mgh, to take you and your spacesuit up to a couple hundred miles, and then you accelerate yourself to 17, 500 miles per hour — remember, that one half MV squared — and you figure it out. It ’ s about 5. 7 gigajoules of energy. If you expended that over an hour, it ’ s about 1. 6 megawatts. If you go to one of Vijay ’ s micro-power sources, and they sell it to you for seven cents a kilowatt hour — anybody here fast in math? How much will it cost you and your spacesuit to go to orbit? 100 bucks. That ’ s the price-improvement **curve** that — we need some breakthroughs in physics along the way, I ’ ll grant you that. But guys, if history has taught us anything, it ’ s that if you can imagine it, you will get there eventually. I have no question that the physics, the engineering to get us down to the point where all of us can afford orbital space flight is around the corner. The difficulty is that there needs to be a real marketplace to drive the investment. Today, the Boeings and the Lockheeds don ’ t spend a dollar of their own money in R&D. It ’ s all government research dollars, and very few of those. And in fact, the large corporations, the governments, can ’ t take the risk. So we need what I call an exothermic economic reaction in space. Today ’ s commercial markets worldwide, global commercial launch market? 12 to 15 launches per year. Number of commercial companies out there? 12 to 15 companies. One per company. That ’ s not it. There ’ s only one marketplace, and I call them self-loading carbon payloads. They come with their own money. They ’ re easy to make. It ’ s people. The Ansari X PRIZE was my solution, reading about Lindbergh for creating the vehicles to get us there. We offered 10 million dollars in cash for the first reusable ship, carry three people up to 100 kilometers, come back down, and within two weeks, make the trip again. Twenty-six teams from seven countries entered the competition, spending between one to 25 million dollars each. And of course, we had beautiful SpaceShipOne, which made those two flights and won the competition. And I ’ d like to take you there, to that morning, for just a quick video. Pilot : Release our fire. Richard Searfoss : Good luck. RS : We ’ ve got an altitude call of 368, 000 **feet**. RS : So in my official capacity as the chief judge of the Ansari X PRIZE competition, I declare that Mojave Aerospace Ventures has indeed earned the Ansari X PRIZE. Peter Diamandis : Probably the most difficult thing that I had to do was raise the capital for this. It was literally impossible. We went — I went to 100, 200 CEOs, CMOs. No one believed it was done. Everyone said,”Oh, what does NASA think? Well, people are going to die, how can you possibly going to put this forward?”I found a visionary family, the Ansari family, and Champ Car, and raised part of the money, but not the full 10 million. And what I ended up doing was going out to the insurance industry and buying a hole-in-one insurance policy. See, the insurance companies went to Boeing and Lockheed, and said,”Are you going to compete?”No.”Are you going to compete?”No.”No one ’ s going to win this thing.”So, they took a bet that no one would win by January of ’ 05, and I took a bet that someone would win. So — and the best thing is they paid off and the check didn ’ t bounce. We ’ ve had a lot of accomplishments and it ’ s been a tremendous success. One of the things I ’ m most happy about is that the SpaceShipOne is going to hang in Air and Space Museum, next to the Spirit of St. Louis and the Wright Flyer. Isn ’ t that great? So a little bit about the future, steps to space, what ’ s available for you. Today, you can go and experience weightless flights. By ’ 08, suborbital flights, the price tag for that, you know, on Virgin, is going to be about 200, 000. There are three or four other serious efforts that will bring the price down very rapidly, I think, to about 25, 000 dollars for a suborbital flight. Orbital flights — we can take you to the space station. And then I truly believe, once a group is in orbit around the Earth — I know if they don ’ t do it, I am — we ’ re going to stockpile some fuel, make a beeline for the moon and grab some real estate. Quick moment for the designers in the audience. We spent 11 years getting FAA approval to do zero gravity flights. Here are some **fun** images. Here ’ s Burt Rutan and my good friend Greg Meronek inside a zero gravity — people think a zero gravity room, there ' s a switch on there that turns it off — but it ' s actually parabolic flight of an airplane. And turns out 7-Up has just done a little commercial that ' s airing this month. If we can get the audio up? Narrator : For a chance to win the first free ticket to space, look for specially marked packages of Diet 7-Up. When you want the taste that won ' t weigh you down, the only way to go is up. PD : That was **filmed** inside our airplane, and so, you can now do this. We ' re based down in Florida. Let me talk about the other thing I ' m excited about. The future of prizes. You know, prizes are a very old idea. I had the pleasure of borrowing from the Longitude Prize and the Orteig Prize that put Lindbergh forward. And we have made a decision in the X PRIZE Foundation to actually carry that concept forward into other technology areas, and we just took on a new mission statement :"to bring about radical breakthroughs in space and other technologies for the benefit of humanity."And this is something that we ' re very excited about. I showed this slide to Larry Page, who just joined our board. And you know, when you give to a nonprofit, you might have 50 cents on the dollar. If you have a matching grant, it ' s typically two or three to one. If you put up a prize, you can get literally a 50 to one leverage on your dollars. And that ' s huge. And then he turned around and said,"Well, if you back a prize institute that runs a 10 prize, you get 500 to one."I said,"Well, that ' s great."So, we have actually — are looking to turn the X PRIZE into a world- class prize institute. This is what happens when you put up a prize, when you announce it and teams start to begin doing trials. You get publicity increase, and when it ' s won, publicity shoots through the **roof** — if it ' s properly managed — and that ' s part of the benefits to a sponsor. Then, when the prize is actually won, after it ' s moving, you get societal benefits, you know, new technology, new capability. And the benefit to the sponsors is the sum of the publicity and societal benefits over the long term. That ' s our value proposition in a prize. If you were going to go and try to create SpaceShipOne, or any kind of a new technology, you have to fund that from the beginning and maintain that funding with an uncertain outcome. It may or may not happen. But if you put up a prize, the beautiful thing is, you know, it ' s a very small maintenance fee, and you pay on success. Orteig didn ' t pay a dime out to the nine teams that went across — tried to go across the Atlantic, and we didn ' t pay a dime until someone won the Ansari X PRIZE. So, prizes work great. You know, innovators, the entrepreneurs out there, you know that when you ' re going for a goal, the first thing you have to do is believe that you can do it yourself. Then, you ' ve got to, you know, face potential public ridicule of — that ' s a crazy idea, it ' ll never work. And then you have to convince others, so that they can, in fact, help you raise the funds, and then you ' ve got to deal with the fact that you ' ve got government bureaucracies and institutions that don ' t want you to move those things forward, and you have to deal with failures. What a prize does, what we ' ve experienced a prize doing, is literally help to short-circuit or support all of these things, because a prize credentials the idea that this is a good idea. Well, it must be a good idea. Someone ' s offering 10 million dollars to go and do this thing. And each of these areas was something that we found the Ansari X PRIZE helped short-circuit these for innovation. So, as an organization, we put together a prize discovery process of how to come up with prizes and write the rules, and we ' re actually looking at creating prizes in a number of different categories. We ' re looking at attacking energy, environment, nanotechnology — and I ' ll talk about those more in a moment. And the way we ' re doing that is we ' re creating prize teams within the X PRIZE. We have a space prize team. We ' re going after an orbital prize. We are looking at a number of energy prizes. Craig Venter has just joined our board and we ' re doing a rapid genome sequencing prize with him, we ’ ll be announcing later this fall, about — imagine being able to sequence anybody ’ s DNA for under 1, 000 dollars, revolutionize medicine. And **clean** water, education, medicine and even looking at social entrepreneurship. So my final slide here is, the most critical tool for solving humanity ’ s grand challenges — it isn ’ t technology, it isn ’ t money, it ’ s only one thing — it ’ s the committed, passionate human mind.

Text Sample 674: POS Dim 5 – 1998 – Score 6.00 – 1998/t000216.txt

David Gallo : This is Bill Lange. I ’ m Dave Gallo. And we ’ re going to tell you some stories from the sea here in video. We ’ ve got some of the most incredible video of Titanic that ’ s ever been seen, and we ’ re not going to show you any of it. The truth of the matter is that the Titanic — even though it ’ s breaking all sorts of box office records — it ’ s not the most exciting story from the sea. And the problem, I think, is that we take the ocean for granted. When you think about it, the oceans are 75 percent of the planet. Most of the planet is ocean water. The average depth is about two miles. Part of the problem, I think, is we stand at the beach, or we see images like this of the ocean, and you look out at this great big blue expanse, and it ’ s shimmering and it ’ s moving and there ’ s waves and there ’ s surf and there ’ s tides, but you have no idea for what lies in there. And in the oceans, there are the longest mountain ranges on the planet. Most of the animals are in the oceans. Most of the earthquakes and volcanoes are in the sea, at the **bottom** of the sea. The biodiversity and the biodensity in the ocean is higher, in places, than it is in the rainforests. It ’ s mostly unexplored, and yet there are beautiful sights like this that captivate us and make us become familiar with it. But when you ’ re standing at the beach, I want you to think that you ’ re standing at the edge of a very unfamiliar world. We have to have a very special technology to get into that unfamiliar world. We use the submarine Alvin and we use cameras, and the cameras are something that Bill Lange has developed with the help of Sony. Marcel Proust said,”The true voyage of discovery is not so much in seeking new landscapes as in having new eyes.”People that have partnered with us have given us new eyes, not only on what exists — the new landscapes at the **bottom** of the sea — but also how we think about life on the planet itself. Here ’ s a jelly. It ’ s one of my favorites, because it ’ s got all sorts of working parts. This turns out to be the longest creature in the oceans. It gets up to about 150 **feet** long. But see all those different working things? I love that kind of stuff. It ’ s got these fishing lures on the **bottom**. They ’ re going up and down. It ’ s got tentacles dangling, swirling around like that. It ’ s a colonial animal. These are all individual animals banding together to make this one creature. And it ’ s got these jet thrusters up in front that it ’ ll use in a moment, and a little light. If you take all the big fish and schooling fish and all that, put them on one side of the scale, put all the jelly-type of animals on the other side, those guys win hands down. Most of the biomass in the ocean is made out of creatures like this. Here ’ s the X-wing death jelly. The bioluminescence — they use the lights for attracting mates and attracting prey and communicating. We couldn ’ t begin to show you our archival stuff from the jellies. They come in all different sizes and shapes. Bill Lange : We tend to forget about the fact that the ocean is miles deep on average, and that we ’ re real familiar with the animals that are in the first 200 or 300 **feet**, but we ’ re not familiar with what exists from there all the way down to the **bottom**. And these are the types of animals that live in that three- dimensional space, that micro-gravity environment that we really haven ’ t explored. You hear about giant squid and things like that, but some of these animals get up to be approximately 140, 160 **feet** long. They ’ re very little understood. DG : This is one of them, another one of our favorites, because it ’ s a little octopod. You can actually see through his head. And here he is, flapping with his ears and very gracefully going up. We see those at all depths and even at the greatest depths. They go from a couple of inches to a couple of **feet**. They come right up to the submarine — they ’ ll put their eyes right up to the window and peek inside the sub. This is really a world within a world, and we ’ re going to show you two. In this case, we ’ re passing down through the mid-ocean and we see creatures like this. This is kind of like an undersea rooster. This guy, that looks incredibly formal, in a way. And then one of my favorites. What a face! This is basically scientific data that you ’ re looking at. It ’ s footage that we ’ ve collected for scientific purposes. And that ’ s one of the things that Bill ’ s been doing, is providing scientists with this first view of animals like this, in the world where they belong. They don ’ t catch them in a net. They ’ re actually looking at them down in that world. We ’ re going to take a joystick, sit in front of our computer, on the Earth, and press the joystick forward, and fly around the planet. We ’ re going to look at the mid-ocean ridge, a 40, 000-mile long mountain range. The average depth at the top of it is about a mile and a half. And we ’ re over the Atlantic — that ’ s the ridge right there — but we ’ re going to go across the Caribbean, Central America, and end up against the Pacific, nine degrees north. We make maps of these mountain ranges with sound, with sonar, and this is one of those mountain ranges. We ’ re coming around a cliff here on the right. The height of these mountains on either side of this **valley** is greater than the Alps in most cases. And there ’ s tens of thousands of those mountains out there that haven ’ t been mapped yet. This is a volcanic ridge. We ’ re getting down further and further in scale. And eventually, we can come up with something like this. This is an icon of our robot, Jason, it ’ s called. And you can sit in a room like this, with a joystick and a headset, and drive a robot like that around the **bottom** of the ocean in real time. One of the things we ’ re trying to do at Woods Hole with our partners is to bring this virtual world — this world, this unexplored region — back to the laboratory. Because we see it in bits and pieces right now. We see it either as sound, or we see it as video, or we see it as photographs, or we see it as chemical sensors, but we never have yet put it all together into one interesting **picture**. Here ’ s where Bill ’ s cameras really do shine. This is what ’ s called a hydrothermal vent. And what you ’ re seeing here is a cloud of densely packed, hydrogen-sulfide-rich water coming out of a volcanic axis on the sea floor. Gets up to 600, 700 degrees F, somewhere in that range. So that ’ s all water under the sea — a mile and a half, two miles, three miles down. And we knew it was volcanic back in the ’ 60s, ’ 70s. And then we had some hint that these things existed all along the axis of it, because if you ’ ve got volcanism, water ’ s going to get down from the sea into cracks in the sea floor, come in contact with magma, and come shooting out hot. We weren ’ t really aware that it would be so rich with sulfides, hydrogen sulfides. We didn ’ t have any idea about these things, which we call chimneys. This is one of these hydrothermal vents. Six hundred degree F water coming out of the Earth. On either side of us are mountain ranges that are higher than the Alps, so the setting here is very dramatic. BL : The white material is a type of bacteria that thrives at 180 degrees C. DG : I think that ’ s one of the greatest stories right now that we ’ re seeing from the **bottom** of the sea, is that the first thing we see coming out of the sea floor after a volcanic eruption is bacteria. And we started to wonder for a long time, how did it all get down there? What we find out now is that it ’ s probably coming from inside the Earth. Not only is it coming out of the Earth — so, biogenesis made from volcanic activity — but that bacteria supports these colonies of life. The pressure here is 4, 000 pounds per square inch. A mile and a half from the surface to two miles to three miles — no sun has ever gotten down here. All the energy to support these life forms is coming from inside the Earth — so, chemosynthesis. And you can see how dense the population is. These are called tube worms. BL : These worms have no digestive system. They have no mouth. But they have two types of gill structures. One for extracting oxygen out of the deep-sea water, another one which houses this chemosynthetic bacteria, which takes the hydrothermal fluid — that hot water that you saw coming out of the **bottom** — and converts that into simple sugars that the tube worm can digest. DG : You can see, here ' s a crab that lives down there. He ' s managed to grab a tip of these worms. Now, they normally retract as soon as a crab touches them. Oh! Good going. So, as soon as a crab touches them, they retract down into their shells, just like your fingernails. There ' s a whole story being played out here that we ' re just now beginning to have some idea of because of this new camera technology. BL : These worms live in a real temperature extreme. Their **foot** is at about 200 degrees C and their head is out at three degrees C, so it ' s like having your hand in boiling water and your **foot** in freezing water. That ' s how they like to live. DG : This is a female of this kind of worm. And here ' s a male. You watch. It doesn ' t take long before two guys here — this one and one that will show up over here — start to fight. Everything you see is played out in the pitch black of the deep sea. There are never any lights there, except the lights that we bring. Here they go. On one of the last dive series, we counted 200 species in these areas — 198 were new, new species. BL : One of the big problems is that for the biologists working at these sites, it ' s rather difficult to collect these animals. And they disintegrate on the way up, so the imagery is critical for the science. DG : Two octopods at about two miles depth. This pressure thing really amazes me — that these animals can exist there at a depth with pressure enough to crush the Titanic like an empty Pepsi can. What we saw up till now was from the Pacific. This is from the Atlantic. Even greater depth. You can see this shrimp is harassing this poor little guy here, and he ' ll bat it away with his claw. Whack! And the same thing ' s going on over here. What they ' re getting at is that — on the back of this crab — the foodstuff here is this very strange bacteria that lives on the backs of all these animals. And what these shrimp are trying to do is actually harvest the bacteria from the backs of these animals. And the crabs don ' t like it at all. These long filaments that you see on the back of the crab are actually created by the product of that bacteria. So, the bacteria grows hair on the crab. On the back, you see this again. The red dot is the **laser** light of the submarine Alvin to give us an idea about how far away we are from the vents. Those are all shrimp. You see the hot water over here, here and here, coming out. They ' re clinging to a rock face and actually scraping bacteria off that rock face. Here ' s a tiny, little vent that ' s come out of the side of that pillar. Those pillars get up to several stories. So here, you ' ve got this **valley** with this incredible alien landscape of pillars and hot springs and volcanic eruptions and earthquakes, inhabited by these very strange animals that live only on chemical energy coming out of the ground. They don ' t need the sun at all. BL : You see this white V-shaped mark on the back of the shrimp? It ' s actually a light-sensing organ. It ' s how they find the hydrothermal vents. The vents are emitting a black body radiation — an IR signature — and so they ' re able to find these vents at considerable distances. DG : All this stuff is happening along that 40, 000-mile long mountain range that we ' re calling the ribbon of life, because just even today, as we speak, there ' s life being generated there from volcanic activity. This is the first time we ' ve ever tried this any place. We ’ re going to try to show you high definition from the Pacific. We ’ re moving up one of these pillars. This one ’ s several stories tall. In it, you ’ ll see that it ’ s a habitat for a lot of different animals. There ’ s a funny kind of hot plate here, with vent water coming out of it. So all of these are individual homes for worms. Now here ’ s a closer view of that community. Here ’ s crabs here, worms here. There are smaller animals crawling around. Here ’ s pagoda structures. I think this is the neatest-looking thing. I just can ’ t get over this — that you ’ ve got these little chimneys sitting here smoking away. This stuff is toxic as **hell**, by the way. You could never get a permit to dump this in the ocean, and it ’ s coming out all from it. It ’ s unbelievable. It ’ s basically sulfuric acid, and it ’ s being just dumped out, at incredible rates. And animals are thriving — and we probably came from here. That ’ s probably where we evolved from. BL : This bacteria that we ’ ve been talking about turns out to be the most simplest form of life found. There are a number of groups that are proposing that life evolved at these vent sites. Although the vent sites are short-lived — an individual site may last only 10 years or so — as an ecosystem they ’ ve been stable for millions — well, billions — of years. DG : It works too well. You see there ’ re some fish inside here as well. There ’ s a fish sitting here. Here ’ s a crab with his claw right at the end of that tube worm, waiting for that worm to stick his head out. BL : The biologists right now cannot explain why these animals are so active. The worms are growing inches per week! DG : I already said that this site, from a human perspective, is toxic as **hell**. Not only that, but on top — the lifeblood — that plumbing system turns off every year or so. Their plumbing system turns off, so the sites have to move. And then there ’ s earthquakes, and then volcanic eruptions, on the order of one every five years, that completely wipes the area out. Despite that, these animals grow back in about a year ’ s time. You ’ re talking about biodensities and biodiversity, again, higher than the rainforest that just springs back to life. Is it sensitive? Yes. Is it fragile? No, it ’ s not really very fragile. I ’ ll end up with saying one thing. There ’ s a story in the sea, in the waters of the sea, in the sediments and the rocks of the sea floor. It ’ s an incredible story. What we see when we look back in time, in those sediments and rocks, is a record of Earth history. Everything on this planet — everything — works by cycles and rhythms. The continents move apart. They come back together. Oceans come and go. Mountains come and go. Glaciers come and go. El Nino comes and goes. It ’ s not a disaster, it ’ s rhythmic. What we ’ re learning now, it ’ s almost like a symphony. It ’ s just like music — it really is just like music. And what we ’ re learning now is that you can ’ t listen to a five-billion-year long symphony, get to today and say,”Stop! We want tomorrow ’ s note to be the same as it was today.”It ’ s absurd. It ’ s just absurd. So, what we ’ ve got to learn now is to find out where this planet ’ s going at all these different scales and work with it. Learn to manage it. The concept of preservation is futile. Conservation ’ s tougher, but we can probably get there. Thank you very much. Thank you.

Text Sample 675: POS Dim 5 – 1998 – Score 5.00 – 1998/t000134.txt

’ Theme and variations ’ is one of those forms that require a certain kind of intellectual activity, because you are always comparing the variation with the theme that you hold in your mind. You might say that the theme is nature and everything that follows is a variation on the subject. I was asked, I guess about six years ago, to do a series of paintings that in some way would celebrate the birth of Piero della Francesca. And it was very difficult for me to imagine how to paint **pictures** that were based on Piero until I realized that I could look at

Piero as nature — that I would have the same attitude towards looking at Piero della Francesca as I would if I were looking out a window at a tree. And that was enormously liberating to me. Perhaps it ' s not a very insightful observation, but that really started me on a path to be able to do a kind of theme and variations based on a work by Piero, in this case that remarkable painting that ' s in the Uffizi,"The Duke of Montefeltro,"who faces his consort, Battista. Once I realized that I could take some liberties with the subject, I did the following series of **drawings**. That ' s the real Piero della Francesca — one of the greatest portraits in human history. And these, I ' ll just show these without comment. It ' s just a series of variations on the head of the Duke of Montefeltro, who ' s a great, great figure in the Renaissance, and probably the basis for Machiavelli ' s"The Prince."He apparently lost an eye in battle, which is why he is always shown in profile. And this is Battista. And then I decided I could move them around a little bit — so that for the first time in history, they ' re facing the same direction. Whoops! Passed each other. And then a visitor from another painting by Piero, this is from"The Resurrection of Christ"— as though the cast had just gotten of the set to have a chat. And now, four large **panels** : this is upper left ; upper right ; lower left ; lower right. Incidentally, I ' ve never understood the conflict between **abstraction** and naturalism. Since all paintings are inherently abstract to begin with there doesn ' t seem to be an argument there. On another subject — — I was driving in the country one day with my wife, and I saw this sign, and I said,"That is a fabulous piece of design."And she said,"What are you talking about?"I said,"Well, it ' s so persuasive, because the purpose of that sign is to get you into the garage, and since most people are so suspicious of garages, and know that they ' re going to be ripped off, they use the word ' reliable. ' But everybody says they ' re reliable. But, reliable Dutchman"— —"Fantastic!"Because as soon as you hear the word Dutchman — which is an archaic word, nobody calls Dutch people"Dutchmen"anymore — but as soon as you hear Dutchman, you get this **picture** of the kid with his finger in the dike, preventing the thing from falling and flooding Holland, and so on. And so the entire issue is detoxified by the use of"Dutchman."Now, if you think I ' m exaggerating at all in this, all you have to do is substitute something else, like"Indonesian."Or even"French."Now,"Swiss"works, but you know it ' s going to cost a lot of money. I ' m going to take you quickly through the actual process of doing a poster. I do a lot of work for the School of Visual Arts, where I teach, and the director of this school — a remarkable man named Silas Rhodes — often gives you a piece of text and he says,"Do something with this."And so he did. And this was the text —"In words as fashion the same rule will hold / Alike fantastic if too new or old / Be not the first by whom the new are tried / Nor yet the last to lay the old aside."I could make nothing of that. And I really struggled with this one. And the first thing I did, which was sort of in the absence of another idea, was say I ' ll sort of write it out and make some words big, and I ' ll have some kind of design on the back somehow, and I was hoping — as one often does — to stumble into something. So I took another crack at it — you ' ve got to keep it moving — and I Xeroxed some words on pieces of colored paper and I pasted them down on an ugly board. I thought that something would come out of it, like"Words rule fantastic new old first last Pope"because it ' s by Alexander Pope, but I sort of made a mess out of it, and then I thought I ' d repeat it in some way so it was legible. So, it was going nowhere. Sometimes, in the middle of a resistant problem, I write down things that I know about it. But you can see the beginning of an idea there, because you can see the word"new"emerging from the"old."That ' s what happens. There ' s a relationship between the old and the new ; the new emerges from the context of the old. And then I did some variations of it, but it still wasn ’ t coalescing graphically at all. I had this other version which had something interesting about it in terms of being able to put it together in your mind from clues. The W was clearly a W, the N was clearly an N, even though they were very fragmentary and there wasn ’ t a lot of information in it. Then I got the words”new”and”old”and now I had regressed back to a point where there seemed to be no return. I was really desperate at this point. And so, I do something I ’ m truthfully ashamed of, which is that I took two **drawings** I had made for another purpose and I put them together. It says”dreams”at the top. And I was going to do a thing, I say,”Well, change the copy. Let it say something about dreams, and come to SVA and you ’ ll sort of fulfill your dreams.”But, to my credit, I was so embarrassed about doing that that I never submitted this sketch. And, finally, I arrived at the following solution. Now, it doesn ’ t look terribly interesting, but it does have something that distinguishes it from a lot of other posters. For one thing, it transgresses the idea of what a poster ’ s supposed to be, which is to be understood and seen immediately, and not explained. I remember hearing all of you in the graphic arts —”If you have to explain it, it **ain** ’ t working.”And one day I woke up and I said,”Well, suppose that ’ s not true?”So here ’ s what it says in my explanation at the **bottom** left. It says,”Thoughts : This poem is impossible. Silas usually has a better touch with his choice of quotations. This one generates no imagery at all.”I am now exposing myself to my audience, right? Which is something you never want to do professionally.”Maybe the words can make the image without anything else happening. What ’ s the heart of this poem? Don ’ t be trendy if you want to be serious. Is doing the poster this way trendy in itself? I guess one could reduce the idea further by suggesting that the new emerges behind and through the old, like this.”And then I show you a little **drawing** — you see, you remember that old thing I discarded? Well, I found a way to use it. So, there ’ s that little alternative over there, and I say,”Not bad,”— criticizing myself —”but more didactic than visual. Maybe what wants to be said is that old and the new are locked in a dialectical embrace, a kind of dance where each defines the other.”And then more self-questioning —”Am I being simple-minded? Is this the kind of simple that looks obvious, or the kind that looks profound? There ’ s a significant difference. This could be embarrassing. Actually, I realize fear of embarrassment drives me as much as any ambition. Do you think this sort of thing could really attract a student to the school?”Well, I think there are two fresh things here — two fresh things. One is the sort of willingness to expose myself to a critical audience, and not to suggest that I am confident about what I ’ m doing. And as you know, you have to have a front. I mean — you ’ ve got to be confident ; if you don ’ t believe in your work, who else is going to believe in it? So that ’ s one thing, to introduce the idea of doubt into graphics. That can be a big contribution. The other thing is to actually give you two solutions for the price of one ; you get the big one and if you don ’ t like that, how about the little one? And that too is a relatively new idea. And here ’ s just a series of experiments where I ask the question of — does a poster have to be square? Now, this is a little illusion. That poster is not folded. It ’ s not folded, that ’ s a photograph and it ’ s cut on the diagonal. Same cheap trick in the upper left-hand corner. And here, a very peculiar poster because, simply because of using the isometric perspective in the computer, it won ’ t sit still in the space. At times, it seems to be wider at the back than the front, and then it shifts. And if you sit here long enough, it ’ ll float off the page into the audience. But, we don ’ t have time. And then an experiment — a little bit about the nature of perspective, where the outside shape is determined by the peculiarity of perspective, but the shape of the bottle — which is identical to the outside shape — is seen frontally. And another piece for the art directors ’ club is”Anna Rees”casting long shadows. This is another poster from the School of Visual Arts. There were 10 artists invited to participate in it, and it was one of those things where it was extremely competitive and I didn ' t want to be embarrassed, so I worked very hard on this. The idea was — and it was a brilliant idea — was to have 10 posters distributed throughout the city ' s subway system so every time you got on the subway you ' d be passing a different poster, all of which had a different idea of what art is. But I was absolutely stuck on the idea of"art is"and trying to determine what art was. But then I gave up and I said,"Well, art is whatever."And as soon as I said that, I discovered that the word"hat"was hidden in the word"whatever,"and that led me to the inevitable conclusion. But then again, it ' s on my list of didactic posters. My intent is to have a literary accompaniment that explains the poster, in case you don ' t get it. Now this says,"Note to the viewer : I thought I might use a visual cliche of our time — Magritte ' s everyman — to express the idea that art is mystery, continuity and history. I ' m also convinced that, in an age of computer manipulation, surrealism has become banal, a shadow of its former self. The phrase ' Art is whatever ' expresses the current inclusiveness that surrounds art-making — a sort of ' it **ain** ' t what you do, it ' s the way that you do it ' notion. The shadow of Magritte falls across the central part of the poster a poetic event that occurs as the shadow man isolates the word ' hat, ' hidden in the word ' whatever. ' The four hats shown in the poster suggests how art might be defined : as a thing itself, the worth of the thing, the shadow of the thing, and the shape of the thing. Whatever."OK. And the one that I did not submit, which I still like, I wanted to use the same phrase. There were wonderful experiments by Bruno Munari on letterforms some years ago — sort of, see how far you could go and still be able to read them. And that idea stuck in my head. But then I took the pieces that I had taken off and put them at the **bottom**. And, of course those are the remains, and they ' re so labeled. But what really happens is that you read it as :"Art is whatever remains."Thank you.

Text Sample 676: POS Dim 5 – 1998 – Score 4.00 – 1998/t000200.txt

You hear that this is the era of environment â or biology, or information technology... Well, it ' s the era of a lot of different things that we ' re in right now. But one thing for sure : it ' s the era of change. There ' s more change going on than ever has occurred in the history of human life on earth. And you all sort of know it, but it ' s hard to get it so that you really understand it. And I ' ve tried to put together something that ' s a good start for this. I ' ve tried to show in this â though the color doesn ' t come out â that what I ' m concerned with is the little 50-year time bubble that you are in. You tend to be interested in a generation past, a generation future â your parents, your kids, things you can change over the next few decades â and this 50-year time bubble you kind of move along in. And in that 50 years, if you look at the population **curve**, you find the population of humans on the earth more than doubles and we ' re up three-and-a-half times since I was born. When you have a new baby, by the time that kid gets out of high school more people will be added than existed on earth when I was born. This is unprecedented, and it ' s big. Where it goes in the future is questioned. So that ' s the human part. Now, the human part related to animals : look at the left side of that. What I call the human portion â humans and their livestock and pets â versus the natural portion â all the other wild animals and just â these are vertebrates and all the birds, etc., in the land and air, not in the water. How does it balance? Certainly, 10, 000 years ago, the civilization ' s beginning, the human portion was less than one tenth of one percent. Let ’ s look at it now. You follow this **curve** and you see the whiter spot in the middle â that ’ s your 50- year time bubble. Humans, livestock and pets are now 97 percent of that integrated total mass on earth and all wild nature is three percent. We have won. The next generation doesn ’ t even have to worry about this game â it is over. And the biggest problem came in the last 25 years : it went from 25 percent up to that 97 percent. And this really is a sobering **picture** upon realizing that we, humans, are in charge of life on earth ; we ’ re like the capricious Gods of old Greek myths, kind of playing with life â and not a great deal of wisdom injected into it. Now, the third **curve** is information technology. This is Moore ’ s Law plotted here, which relates to density of information, but it has been pretty good for showing a lot of other things about information technology â computers, their use, Internet, etc. And what ’ s important is it just goes straight up through the top of the **curve**, and has no real limits to it. Now try and contrast these. This is the size of the earth going through that same â â frame. And to make it really clear, I ’ ve put all four on one graph. There ’ s no need to see the little detailed words on it. That first one is humans-versus-nature ; we ’ ve won, there ’ s no more gain. Human population. And so if you ’ re looking for growth industries to get into, that ’ s not a good one â protecting natural creatures. Human population is going up ; it ’ s going to continue for quite a while. Good business in obstetricians, morticians, and farming, housing, etc. â they all deal with human bodies, which require being fed, transported, housed and so on. And the information technology, which connects to our brains, has no limit â now, that is a wonderful field to be in. You ’ re looking for growth opportunity? It ’ s just going up through the **roof**. And then, the size of the Earth. Somehow making these all compatible with the Earth looks like a pretty bad industry to be involved with. So, that ’ s the stage out of all this. I find, for reasons I don ’ t understand, I really do have a goal. And the goal is that the world be desirable and sustainable when my kids reach my age â and I think that ’ s â in other words, the next generation. I think that ’ s a goal that we probably all share. I think it ’ s a hopeless goal. Technologically, it ’ s achievable ; economically, it ’ s achievable ; politically, it means sort of the habits, institutions of people â it ’ s impossible. The institutions of the past with all their inertia are just irrelevant for the future, except they ’ re there and we have to deal with them. I spend about 15 percent of my time trying to save the world, the other 85 percent, the usual â and whatever else we devote ourselves to. And in that 15 percent, the main focus is on human mind, thinking skills, somehow trying to unleash kids from the straightjacket of school, which is putting information and dogma into them, get them so they really think, ask tough questions, argue about serious subjects, don ’ t believe everything that ’ s in the book, think broadly or creative. They can be. Our school systems are very flawed and do not reward you for the things that are important in life or for the survival of civilization ; they reward you for a lot of learning and sopping up stuff. We can ’ t go into that today because there isn ’ t time â it ’ s a broad subject. One thing for sure, in the future there is an essential feature â necessary, but not sufficient â which is doing more with less. We ’ ve got to be doing things with more efficiency using less energy, less material. Your great-great grandparents got by on muscle power, and yet we all think there ’ s this huge power that ’ s essential for our lifestyle. And with all the wonderful technology we have we can do things that are much more efficient : conserve, recycle, etc. Let me just rush very quickly through things that we ’ ve done. Human-powered airplane â Gossamer Condor sort of started me in this direction in 1976 and 77, winning the Kremer prize in aviation history, followed by the Albatross. And we began making various odd planes and creatures. Here ’ s a giant flying replica of a pterosaur that has no tail. Trying to have it fly straight is like trying to shoot an arrow with the feathered end forward. It was a tough job, and boy it made me have a lot of respect for nature. This was the full size of the original creature. We did things on land, in the air, on water â vehicles of all different kinds, usually with some electronics or electric power systems in them. I find they ' re all the same, whether its land, air or water. I ' ll be focusing on the air here. This is a solar-powered airplane â 165 miles carrying a person from France to England as a symbol that solar power is going to be an important part of our future. Then we did the solar car for General Motors â the Sunracer â that won the race in Australia. We got a lot of people thinking about electric cars, what you could do with them. A few years later, when we suggested to GM that now is the time and we could do a thing called the Impact, they sponsored it, and here ' s the Impact that we developed with them on their programs. This is the demonstrator. And they put huge effort into turning it into a commercial product. With that preamble, let ' s show the first two-minute videotape, which shows a little airplane for surveillance and moving to a giant airplane. Narrator : A tiny airplane, the AV Pointer serves for surveillance â in effect, a pair of roving eyeglasses. A cutting-edge example of where miniaturization can lead if the operator is remote from the vehicle. It is convenient to carry, assemble and launch by hand. Battery-powered, it is silent and rarely noticed. It sends high-resolution video **pictures** back to the operator. With onboard GPS, it can navigate autonomously, and it is rugged enough to self-land without damage. The modern sailplane is superbly efficient. Some can glide as flat as 60 **feet** forward for every **foot** of descent. They are powered only by the energy they can extract from the atmosphere â an atmosphere nature stirs up by solar energy. Humans and soaring birds have found nature to be generous in providing replenishable energy. Sailplanes have flown over 1, 000 miles, and the altitude record is over 50, 000 **feet**. The Solar Challenger was made to serve as a symbol that photovoltaic cells can produce real power and will be part of the world ' s energy future. In 1981, it flew 163 miles from Paris to England, solely on the power of sunbeams, and established a basis for the Pathfinder. The message from all these vehicles is that ideas and technology can be harnessed to produce remarkable gains in doing more with less â gains that can help us attain a desirable balance between technology and nature. The stakes are high as we speed toward a challenging future. Buckminster Fuller said it clearly :"there are no passengers on spaceship Earth, only crew. We, the crew, can and must do more with less â much less."Paul MacCready : If we could have the second video, the one-minute, put in as quickly as you can, which â this will show the Pathfinder airplane in some flights this past year in Hawaii, and will show a sequence of some of the beauty behind it after it had just flown to 71, 530 **feet** â higher than any propeller airplane has ever flown. It ' s amazing : just on the puny power of the sun â by having a super lightweight plane, you ' re able to get it up there. It ' s part of a long-term program NASA sponsored. And we worked very closely with the whole thing being a team effort, and with wonderful results like that flight. And we ' re working on a bigger plane â 220-foot span â and an intermediate-size, one with a regenerative fuel cell that can store excess energy during the day, feed it back at night, and stay up 65, 000 **feet** for months at a time. Ray Morgan ' s voice will come in here. There he ' s the project manager. Anything they do is certainly a team effort. He ran this program. Here ' s... some things he showed as a celebration at the very end. Ray Morgan : We ' d just ended a seven-month deployment of Hawaii. For those who live on the mainland, it was tough being away from home. The friendly support, the quiet confidence, congenial hospitality shown by our Hawaiian and military hosts â this is starting â made the experience enjoyable and unforgettable. PM : We have real-time IR **scans** going out through the Internet while the plane is flying. And it ’ s exploring without polluting the stratosphere. That ’ s its goal : the stratosphere, the blanket that really controls the radiation of the earth and permits life on earth to be the success that it is â probing that is very important. And also we consider it as a sort of poor man ’ s stationary satellite, because it can stay right overhead for months at a time, 2, 000 times closer than the real GFC synchronous satellite. We couldn ’ t bring one here to fly it and show you. But now let ’ s look at the other end. In the video you saw that nine-pound or eight-pound Pointer airplane surveillance drone that Keenan has developed and just done a remarkable job. Where some have servos that have gotten down to, oh, 18 or 25 grams, his weigh one-third of a gram. And what he ’ s going to bring out here is a surveillance drone that weighs about 2 ounces â that includes the video camera, the batteries that run it, the telemetry, the receiver and so on. And we ’ ll fly it, we hope, with the same success that we had last night when we did the practice. So Matt Keenan, just any time you ’ re â all right â ready to let her go. But first, we ’ re going to make sure that it ’ s appearing on the screen, so you see what it sees. You can imagine yourself being a mouse or fly inside of it, looking out of its camera. Matt Keenan : It ’ s switched on. PM : But now we ’ re trying to get the video. There we go. MK : Can you bring up the house lights? PM : Yeah, the house lights and we ’ ll see you all better and be able to fly the plane better. MK : All right, we ’ ll try to do a few laps around and bring it back in. Here we go. PM : The video worked right for the first few and I don ’ t know why it â there it goes. Oh, that was only a minute, but I think you ’ d be safe to have that near the end of the flight, perhaps. We get to do the classic. All right. If this hits you, it will not hurt you. OK. Thank you very much. Thank you. But now, as they say in infomercials, we have something much better for you, which we ’ re working on : planes that are only six inches â 15 centimeters â in size. And Matt ’ s plane was on the cover of Popular Science last month, showing what this can lead to. And in a while, something this size will have GPS and a video camera in it. We ’ ve had one of these fly nine miles through the air at 35 miles an hour with just a little battery in it. But there ’ s a lot of technology going. There are just milestones along the way of some remarkable things. This one doesn ’ t have the video in it, but you get a little feel from what it can do. OK, here we go. MK : Sorry. OK. PM : If you can pass it down when you ’ re done. Yeah, I think â I lost a little orientation ; I looked up into this light. It hit the building. And the building was poorly placed, actually. But you ’ re beginning to see what can be done. We ’ re working on projects now â even wing-flapping things the size of hawk moths â DARPA contracts, working with Caltech, UCLA. Where all this leads, I don ’ t know. Is it practical? I don ’ t know. But like any basic research, when you ’ re really forced to do things that are way beyond existing technology, you can get there with micro-technology, nanotechnology. You can do amazing things when you realize what nature has been doing all along. As you get to these small scales, you realize we have a lot to learn from nature â not with 747s â but when you get down to the nature ’ s realm, nature has 200 million years of experience. It never makes a mistake. Because if you make a mistake, you don ’ t leave any progeny. We should have nothing but success stories from nature, for you or for birds, and we ’ re learning a lot from its fascinating subjects. In concluding, I want to get back to the big **picture** and I have just two final slides to try and put it in perspective. The first I ’ ll just read. At last, I put in three sentences and had it say what I wanted. Over billions of years on a unique sphere, chance has painted a thin covering of life â complex, improbable, wonderful and fragile. Suddenly, we humans â a recently arrived species, no longer subject to the checks and balances inherent in nature â have grown in population, technology and intelligence to a position of terrible power. We now wield the paintbrush. And that ’ s serious : we ’ re not very bright. We ’ re short on wisdom ; we ’ re high on technology. Where ’ s it going to lead? Well, inspired by the sentences, I decided to wield the paintbrush. Every 25 years I do a **picture**. Here ’ s the one â tries to show that the world isn ’ t getting any bigger. Sort of a timeline, very non-linear scale, nature rates and trilobites and dinosaurs, and eventually we saw some humans with caves... Birds were flying overhead, after pterosaurs. And then we get to the civilization above the little TV set with a gun on it. Then traffic jams, and power systems, and some dots for digital. Where it ’ s going to lead â I have no idea. And so I just put robotic and natural cockroaches out there, but you can fill in whatever you want. This is not a forecast. This is a warning, and we have to think seriously about it. And that time when this is happening is not 100 years or 500 years. Things are going on this decade, next decade ; it ’ s a very short time that we have to decide what we are going to do. And if we can get some agreement on where we want the world to be â desirable, sustainable when your kids reach your age â I think we actually can reach it. Now, I said this was a warning, not a forecast. That was before â I painted this before we started in on making robotic versions of hawk moths and cockroaches, and now I ’ m beginning to wonder seriously â was this more of a forecast than I wanted? I personally think the surviving intelligent life form on earth is not going to be carbon-based ; it ’ s going to be silicon-based. And so where it all goes, I don ’ t know. The one final bit of sparkle we ’ ll put in at the very end here is an utterly impractical flight vehicle, which is a little ornithopter wing-flapping device that â rubber-band powered â that we ’ ll show you. MK : 32 gram. Sorry, one gram. PM : Last night we gave it a few too many turns and it tried to bash the **roof** out also. It ’ s about a gram. The tube there ’ s hollow, about paper-thin. And if this lands on you, I assure you it will not hurt you. But if you reach out to grab it or hold it, you will destroy it. So, be gentle, just act like a wooden Indian or something. And when it comes down â and we ’ ll see how it goes. We consider this to be sort of the spirit of TED. And you wonder, is it practical? And it turns out if I had not been â Unfortunately, we have some light bulb changes. We can probably get it down, but it ’ s possible it ’ s gone up to a greater destiny up there â â than it ever had. And I wanted to make â just â But I want to make just two points. One is, you think it ’ s frivolous ; there ’ s nothing to it. And yet if I had not been making ornithopters like that, a little bit cruder, in 1939 â a long, long time ago â there wouldn ’ t have been a Gossamer Condor, there wouldn ’ t have been an Albatross, a Solar Challenger, there wouldn ’ t be an Impact car, there wouldn ’ t be a mandate on zero-emission vehicles in California. A lot of these things â or similar â would have happened some time, probably a decade later. I didn ’ t realize at the time I was doing inquiry-based, hands-on things with teams, like they ’ re trying to get in education systems. So I think that, as a symbol, it ’ s important. And I believe that also is important. You can think of it as a sort of a symbol for learning and TED that somehow gets you thinking of technology and nature, and puts it all together in things that are â that make this conference, I think, more important than any that ’ s taken place in this country in this decade. Thank you.

Text Sample 677: POS Dim 5 – 2008 – Score 10.00 – 2008/t000112.txt
About four years ago, the New Yorker published an article about a cache of

dodo bones that was found in a pit on the island of Mauritius. Now, the island of Mauritius is a small island off the east coast of Madagascar in the Indian Ocean, and it is the place where the dodo bird was discovered and extinguished, all within about 150 years. Everyone was very excited about this archaeological find, because it meant that they might finally be able to assemble a single dodo skeleton. See, while **museums** all over the world have dodo skeletons in their collection, nobody — not even the actual Natural History Museum on the island of Mauritius — has a skeleton that ' s made from the bones of a single dodo. Well, this isn ' t exactly true. The fact is, is that the British Museum had a complete specimen of a dodo in their collection up until the 18th century — it was actually mummified, skin and all — but in a fit of space-saving zeal, they actually cut off the head and they cut off the **feet** and they burned the rest in a bonfire. If you go look at their website today, they ' ll actually list these specimens, saying, the rest was lost in a fire. Not quite the whole truth. Anyway. The frontispiece of this article was this photo, and I ' m one of the people that thinks that Tina Brown was great for bringing photos to the New Yorker, because this photo completely rocked my world. I became obsessed with the object — not just the beautiful photograph itself, and the color, the shallow depth of field, the detail that ' s visible, the wire you can see on the beak there that the conservator used to put this skeleton together — there ' s an entire story here. And I thought to myself, wouldn ' t it be great if I had my own dodo skeleton? I want to point out here at this point that I ' ve spent my life obsessed by objects and the stories that they tell, and this was the very latest one. So I began looking around for — to see if anyone sold a kit, some kind of model that I could get, and I found lots of reference material, lots of lovely **pictures**. No dice : no dodo skeleton for me. But the damage had been done. I had saved a few hundred photos of dodo skeletons into my"Creative Projects"folder — it ' s a repository for my brain, everything that I could possibly be interested in. Any time I have an internet connection, there ' s a sluice of stuff moving into there, everything from beautiful rings to cockpit photos. The key that the Marquis du Lafayette sent to George Washington to celebrate the storming of the Bastille. Russian nuclear launch key : The one on the top is the **picture** of the one I found on eBay ; the one on the **bottom** is the one I made for myself, because I couldn ' t afford the one on eBay. Storm trooper costumes. Maps of Middle Earth — that ' s one I hand-drew myself. There ' s the dodo skeleton folder. This folder has 17, 000 photos — over 20 gigabytes of information — and it ' s growing constantly. And one day, a couple of weeks later, it might have been maybe a year later, I was in the art store with my kids, and I was buying some clay tools — we were going to have a craft day. I bought some Super Sculpeys, some armature wire, some various materials. And I looked down at this Sculpey, and I thought, maybe, yeah, maybe I could make my own dodo skull. I should point out at this time — I ' m not a sculptor ; I ' m a hard-edged model maker. You give me a **drawing**, you give me a prop to replicate, you give me a crane, scaffolding, parts from"Star Wars"— especially parts from"Star Wars"— I can do this stuff all day long. It ' s exactly how I made my living for 15 years. But you give me something like this — my friend Mike Murnane sculpted this ; it ' s a maquette for"Star Wars, Episode Two"— this is not my thing — this is something other people do — dragons, soft things. However, I felt like I had looked at enough photos of dodo skulls to actually be able to understand the topology and perhaps replicate it — I mean, it couldn ' t be that difficult. So, I started looking at the best photos I could find. I grabbed all the reference, and I found this lovely piece of reference. This is someone selling this on eBay ; it was clearly a woman—s hand, hopefully a woman ' s hand. Assuming it was roughly the size of my wife ' s hand, I made some measurements of her thumb, and I scaled them out to the size of the skull. I blew it up to the actual size, and I began using that, along with all the other reference that I had, comparing it to it as size reference for figuring out exactly how big the beak should be, exactly how long, etc. And over a few hours, I eventually achieved what was actually a pretty reasonable dodo skull. And I didn ' t mean to continue, I — it ' s kind of like, you know, you can only **clean** a super messy room by picking up one thing at a time ; you can ' t think about the totality. I wasn ' t thinking about a dodo skeleton ; I just noticed that as I finished this skull, the armature wire that I had been used to holding it up was sticking out of the back just where a spine would be. And one of the other things I ' d been interested in and obsessed with over the years is spines and skeletons, having collected a couple of hundred. I actually understood the mechanics of vertebrae enough to kind of start to imitate them. And so button by button, vertebrae by vertebrae, I built my way down. And actually, by the end of the day, I had a reasonable skull, a moderately good vertebrae and half of a pelvis. And again, I kept on going, looking for more reference, every bit of reference I could find — **drawings**, beautiful photos. This guy — I love this guy! He put a dodo leg bones on a scanner with a ruler. This is the kind of accuracy that I wanted, and I replicated every last bone and put it in. And after about six weeks, I finished, painted, mounted my own dodo skeleton. You can see that I even made a **museum** label for it that includes a brief history of the dodo. And TAP Plastics made me — although I didn ' t photograph it — a **museum** vitrine. I don ' t have the room for this in my house, but I had to finish what I had started. And this actually represented kind of a sea change to me. Again, like I said, my life has been about being **fascinated** by objects and the stories that they tell, and also making them for myself, obtaining them, appreciating them and diving into them. And in this folder,"Creative Projects,"there are tons of projects that I ' m currently working on, projects that I ' ve already worked on, things that I might want to work on some day, and things that I may just want to find and buy and have and look at and touch. But now there was potentially this new category of things that I could sculpt that was different, that I — you know, I have my own R2D2, but that ' s — honestly, relative to sculpting, to me, that ' s easy. And so I went back and looked through my"Creative Projects"folder, and I happened across the Maltese Falcon. Now, this is funny for me : to fall in love with an object from a Hammett novel, because if it ' s true that the world is divided into two types of people, Chandler people and Hammett people, I am absolutely a Chandler person. But in this case, it ' s not about the author, it ' s not about the book or the movie or the story, it ' s about the object in and of itself. And in this case, this object is — plays on a host of levels. First of all, there ' s the object in the world. This is the"Kniphausen Hawk."It is a ceremonial pouring vessel made around 1700 for a Swedish Count, and it is very likely the object from which Hammett drew his inspiration for the Maltese Falcon. Then there is the fictional bird, the one that Hammett created for the book. Built out of words, it is the engine that drives the plot of his book and also the movie, in which another object is created : a prop that has to represent the thing that Hammett created out of words, inspired by the Kniphausen Hawk, and this represents the falcon in the movie. And then there is this fourth level, which is a whole new object in the world : the prop made for the movie, the representative of the thing, becomes, in its own right, a whole other thing, a whole new object of desire. And so now it was time to do some research. I actually had done some research a few years before — it ' s why the folder was there. I ' d bought a replica, a really crappy replica, of the Maltese Falcon on eBay, and had downloaded enough **pictures** to actually have some reasonable reference. But I discovered, in researching further, really wanting precise reference, that one of the original lead birds had been sold at Christie ' s in 1994, and so I contacted an antiquarian bookseller who had the original Christie ' s catalogue, and in it I found this magnificent **picture**, which included a size reference. I was able to **scan** the **picture**, blow it up to exactly full size. I found other reference. Avi [Ara] Chekmayan, a New Jersey editor, actually found this resin Maltese Falcon at a flea market in 1991, although it took him five years to authenticate this bird to the auctioneers ' specifications, because there was a lot of controversy about it. It was made out of resin, which wasn ' t a common material for movie props about the time the movie was made. It ' s funny to me that it took a while to authenticate it, because I can see it compared to this thing, and I can tell you — it ' s real, it ' s the real thing, it ' s made from the exact same mold that this one is. In this one, because the auction was actually so controversial, Profiles in History, the auction house that sold this — I think in 1995 for about 100, 000 dollars — they actually included — you can see here on the **bottom** — not just a front elevation, but also a side, rear and other side elevation. So now, I had all the topology I needed to replicate the Maltese Falcon. What do they do, how do you start something like that? I really don ' t know. So what I did was, again, like I did with the dodo skull, I blew all my reference up to full size, and then I began cutting out the negatives and using those templates as shape references. So I took Sculpey, and I built a big block of it, and I passed it through until, you know, I got the right profiles. And then slowly, feather by feather, detail by detail, I worked out and achieved — working in front of the television and Super Sculpey — here ' s me sitting next to my wife — it ' s the only **picture** I took of the entire process. As I moved through, I achieved a very reasonable facsimile of the Maltese Falcon. But again, I am not a sculptor, and so I don ' t know a lot of the tricks, like, I don ' t know how my friend Mike gets beautiful, shiny surfaces with his Sculpey ; I certainly wasn ' t able to get it. So, I went down to my shop, and I molded it and I cast it in resin, because in the resin, then, I could absolutely get the glass smooth finished. Now there ' s a lot of ways to fill and get yourself a nice smooth finish. My preference is about 70 coats of this — matte black auto primer. I spray it on for about three or four days, it drips to **hell**, but it allows me a really, really nice gentle sanding surface and I can get it glass-smooth. Oh, finishing up with triple-zero steel wool. Now, the great thing about getting it to this point was that because in the movie, when they finally bring out the bird at the end, and they place it on the table, they actually spin it. So I was able to actually screen-shot and freeze-frame to make sure. And I ' m following all the light kicks on this thing and making sure that as I ' m holding the light in the same position, I ' m getting the same type of reflection on it — that ' s the level of detail I ' m going into this thing. I ended up with this : my Maltese Falcon. And it ' s beautiful. And I can state with authority at this point in time, when I ' d finished it, of all of the replicas out there — and there is a few — this is by far the most accurate representation of the original Maltese Falcon than anyone has sculpted. Now the original one, I should tell you, is sculpted by a guy named Fred Sexton. This is where it gets weird. Fred Sexton was a friend of this guy, George Hodel. Terrifying guy — agreed by many to be the killer of the Black Dahlia. Now, James Ellroy believes that Fred Sexton, the sculptor of the Maltese Falcon, killed James Elroy ' s mother. I ' ll go you one stranger than that : In 1974, during the production of a weird comedy sequel to"The Maltese Falcon,"called"The Black Bird,"starring George Segal, the Los Angeles County Museum of Art had a plaster original of the Maltese Falcon — one of the original six plasters, I think, made for the movie — stolen out of the **museum**. A lot of people thought it was a publicity stunt for the movie. John ' s Grill, which actually is seen briefly in"The Maltese Falcon,"is still a viable

San Francisco eatery, counted amongst its regular customers Elisha Cook, who played Wilmer Cook in the movie, and he gave them one of his original plasters of the Maltese Falcon. And they had it in their cabinet for about 15 years, until it got stolen in January of 2007. It would seem that the object of desire only comes into its own by disappearing repeatedly. So here I had this Falcon, and it was lovely. It looked really great, the light worked on it really well, it was better than anything that I could achieve or obtain out in the world. But there was a problem. And the problem was that : I wanted the entirety of the object, I wanted the weight behind the object. This thing was made of resin and it was too light. There ' s this group online that I frequent. It ' s a group of prop crazies just like me called the Replica Props Forum, and it ' s people who trade, make and travel in information about movie props. And it turned out that one of the guys there, a friend of mine that I never actually met, but befriended through some prop deals, was the manager of a local foundry. He took my master Falcon pattern, he actually did lost wax casting in bronze for me, and this is the bronze I got back. And this is, after some acid etching, the one that I ended up with. And this thing, it ' s deeply, deeply satisfying to me. Here, I ' m going to put it out there, later on tonight, and I want you to pick it up and handle it. You want to know how obsessed I am. This project ' s only for me, and yet I went so far as to buy on eBay a 1941 Chinese San Francisco-based newspaper, in order so that the bird could properly be wrapped... like it is in the movie. Yeah, I know! There you can see, it ' s weighing in at 27 and a half pounds. That ' s half the weight of my dog, Huxley. But there ' s a problem. Now, here ' s the most recent progression of Falcons. On the far left is a piece of crap — a replica I bought on eBay. There ' s my somewhat ruined Sculpey Falcon, because I had to get it back out of the mold. There ' s my first casting, there ' s my master and there ' s my bronze. There ' s a thing that happens when you mold and cast things, which is that every time you throw it into silicone and cast it in resin, you lose a little bit of volume, you lose a little bit of size. And when I held my bronze one up against my Sculpey one, it was shorter by three-quarters of an inch. Yeah, no, really, this was like aah — why didn ' t I remember this? Why didn ' t I start and make it bigger? So what do I do? I figure I have two options. One, I can fire a freaking **laser** at it, which I have already done, to do a 3D **scan** — there ' s a 3D **scan** of this Falcon. I had figured out the exact amount of shrinkage I achieved going from a wax master to a bronze master and blown this up big enough to make a 3D lithography master of this, which I will polish, then I will send to the mold maker and then I will have it done in bronze. Or : There are several people who own originals, and I have been attempting to contact them and reach them, hoping that they will let me spend a few minutes in the presence of one of the real birds, maybe to take a **picture**, or even to pull out the hand-held **laser** scanner that I happen to own that fits inside a cereal box, and could maybe, without even touching their bird, I swear, get a perfect 3D **scan**. And I ' m even willing to sign pages saying that I ' ll never let anyone else have it, except for me in my office, I promise. I ' ll give them one if they want it. And then, maybe, then I ' ll achieve the end of this exercise. But really, if we ' re all going to be honest with ourselves, I have to admit that achieving the end of the exercise was never the point of the exercise to begin with, was it. Thank you.

Text Sample 678: POS Dim 5 – 2008 – Score 9.00 – 2008/t000135.txt

I was raised in Seoul, Korea, and moved to New York City in 1999 to attend college. I was pre-med at the time, and I thought I would become a surgeon because I was interested in anatomy and dissecting animals really piqued my curiosity. At the same time, I fell in love with New York City. I started to realize that I could look at the whole city as a living organism. I wanted to dissect it and look into its unseen layers. And the way to it, for me, was through artistic means. So, eventually I decided to pursue an MFA instead of an M. D. and in grad school I became interested in creatures that dwell in the hidden corners of the city. In New York City, rats are part of commuters ’ daily lives. Most people ignore them or are frightened of them. But I took a liking to them because they dwell on the fringes of society. And even though they ’ re used in labs to promote human lives, they ’ re also considered pests. I also started looking around in the city and trying to photograph them. One day, in the subway, I was snapping **pictures** of the tracks hoping to catch a rat or two, and a man came up to me and said,”You can ’ t take photographs here. The MTA will confiscate your camera.”I was quite shocked by that, and thought to myself,”Well, OK then. I ’ ll follow the rats.”Then I started going into the tunnels, which made me realize that there ’ s a whole new dimension to the city that I never saw before and most people don ’ t get to see. Around the same time, I met like-minded individuals who call themselves urban explorers, adventurers, spelunkers, guerrilla historians, etc. I was welcomed into this loose, Internet-based network of people who regularly explore urban ruins such as abandoned subway stations, tunnels, sewers, aqueducts, factories, hospitals, shipyards and so on. When I took photographs in these locations, I felt there was something missing in the **pictures**. Simply documenting these soon-to-be-demolished structures wasn ’ t enough for me. So I wanted to create a fictional character or an animal that dwells in these underground spaces, and the simplest way to do it, at the time, was to model myself. I decided against clothing because I wanted the figure to be without any cultural implications or time-specific elements. I wanted a simple way to represent a living body inhabiting these decaying, derelict spaces. This was taken in the Riviera Sugar Factory in Red Hook, Brooklyn. It ’ s now an empty, six-acre lot waiting for a shopping mall right across from the new Ikea. I was very fond of this space because it ’ s the first massive industrial complex I found on my own that is abandoned. When I first went in, I was scared, because I heard dogs barking and I thought they were guard dogs. But they happened to be wild dogs living there, and it was right by the water, so there were swans and ducks swimming around and trees growing everywhere and bees nesting in the sugar barrels. The nature had really reclaimed the whole complex. And, in a way, I wanted the human figure in the **picture** to become a part of that nature. When I got comfortable in the space, it also felt like a big playground. I would climb up the tanks and hop across exposed beams as if I went back in time and became a child again. This was taken in the old Croton Aqueduct, which supplied fresh water to New York City for the first time. The construction began in 1837. It lasted about five years. It got abandoned when the new Croton Aqueducts opened in 1890. When you go into spaces like this, you ’ re directly accessing the past, because they sit untouched for decades. I love feeling the aura of a space that has so much history. Instead of looking at reproductions of it at home, you ’ re actually feeling the hand-laid bricks and shimmying up and down narrow cracks and getting wet and muddy and walking in a dark tunnel with a flashlight. This is a tunnel underneath Riverside Park. It was built in the 1930s by Robert Moses. The murals were done by a graffiti artist to commemorate the hundreds of homeless people that got relocated from the tunnel in 1991 when the tunnel reopened for trains. Walking in this tunnel is very peaceful. There ’ s nobody around you, and you hear the kids playing in the **park** above you, completely unaware of what ’ s underneath. When I was going out a lot to these places, I was feeling a lot of anxiety and isolation because I was in a solitary phase in my life, and I decided to title my series"Naked City Spleen,"which references Charles Baudelaire."Naked City"is a nickname for New York, and"Spleen"embodies the melancholia and inertia that come from feeling alienated in an urban environment. This is the same tunnel. You see the sunbeams coming from the ventilation ducts and the train approaching. This is a tunnel that ' s abandoned in **Hell** ' s Kitchen. I was there alone, setting up, and a homeless man approached. I was basically intruding in his living space. I was really frightened at first, but I calmly explained to him that I was working on an art project and he didn ' t seem to mind and so I went ahead and put my camera on self-timer and ran back and forth. And when I was done, he actually offered me his shirt to wipe off my **feet** and kindly walked me out. It must have been a very unusual day for him. One thing that struck me, after this incident, was that a space like that holds so many deleted memories of the city. That homeless man, to me, really represented an element of the unconscious of the city. He told me that he was abused above ground and was once in Riker ' s Island, and at last he found peace and quiet in that space. The tunnel was once built for the prosperity of the city, but is now a sanctuary for outcasts, who are completely forgotten in the average urban dweller ' s everyday life. This is underneath my alma mater, Columbia University. The tunnels are famous for having been used during the development of the Manhattan Project. This particular tunnel is interesting because it shows the original foundations of Bloomingdale Insane Asylum, which was demolished in 1890 when Columbia moved in. This is the New York City Farm Colony, which was a poorhouse in Staten Island from the 1890s to the 1930s. Most of my photos are set in places that have been abandoned for decades, but this is an exception. This children ' s hospital was closed in 1997 ; it ' s located in Newark. When I was there three years ago, the windows were broken and the walls were peeling, but everything was left there as it was. You see the autopsy table, morgue trays, x-ray machines and even used utensils, which you see on the autopsy table. After exploring recently-abandoned buildings, I felt that everything could fall into ruins very fast : your home, your office, a shopping mall, a church — any man-made structures around you. I was reminded of how fragile our sense of security is and how vulnerable people truly are. I love to travel, and Berlin has become one of my favorite cities. It ' s full of history, and also full of underground bunkers and ruins from the war. This was taken under a homeless asylum built in 1885 to house 1, 100 people. I saw the structure while I was on the train, and I got off at the next station and met people there that gave me access to their catacomb-like basement, which was used for ammunition storage during the war and also, at some point, to hide groups of Jewish refugees. This is the actual catacombs in Paris. I explored there extensively in the off-limits areas and fell in love right away. There are more than 185 miles of tunnels, and only about a mile is open to the public as a **museum**. The first tunnels date back to 60 B. C. They were consistently dug as limestone quarries and by the 18th century, the caving-in of some of these quarries posed safety threats, so the government ordered reinforcing of the existing quarries and dug new observation tunnels in order to monitor and map the whole place. As you can see, the system is very complex and vast. It ' s very dangerous to get lost in there. And at the same time, there was a problem in the city with overflowing cemeteries. So the bones were moved from the cemeteries into the quarries, making them into the catacombs. The remains of over six million people are housed in there, some over 1, 300 years old. This was taken under the Montparnasse Cemetery where most of the ossuaries are located. There are also phone cables that were used in the ' 50s and many bunkers from the World War II era. This is a German bunker. Nearby there ' s a French bunker, and the whole tunnel system is so complex that the two parties never met. The tunnels are famous for having been used by the Resistance, which Victor Hugo wrote about in”Les Miserables.”And I saw a lot of graffiti from the 1800s, like this one. After exploring the underground of Paris, I decided to climb up, and I climbed a Gothic monument that ’ s right in the middle of Paris. This is the Tower of Saint Jacques. It was built in the early 1500s. I don ’ t recommend sitting on a gargoyle in the middle of January, naked. It was not very comfortable. And all this time, I never saw a single rat in any of these places, until recently, when I was in the London sewers. This was probably the toughest place to explore. I had to wear a **gas** mask because of the toxic fumes — I guess, except for in this **picture**. And when the tides of waste matter come in it sounds as if a whole storm is approaching you. This is a still from a **film** I worked on recently, called”Blind Door.”I ’ ve become more interested in capturing movement and texture. And the 16mm black- and-white **film** gave a different feel to it. And this is the first theater project I worked on. I adapted and produced”A Dream Play”by August Strindberg. It was performed last September one time only in the Atlantic Avenue tunnel in Brooklyn, which is considered to be the oldest underground train tunnel in the world, built in 1844. I ’ ve been leaning towards more collaborative projects like these, lately. But whenever I get a chance I still work on my series. The last place I visited was the Mayan ruins of Copan, Honduras. This was taken inside an archaeological tunnel in the main temple. I like doing more than just exploring these spaces. I feel an obligation to animate and humanize these spaces continually in order to preserve their memories in a creative way — before they ’ re lost forever. Thank you.

Text Sample 679: POS Dim 5 – 2008 – Score 9.00 – 2008/t000180.txt

I thought I ’ d start with telling you or showing you the people who started [Jet Propulsion Lab]. When they were a bunch of kids, they were kind of very imaginative, very adventurous, as they were trying at Caltech to mix chemicals and see which one blows up more. Well, I don ’ t recommend that you try to do that now. Naturally, they blew up a shack, and Caltech, well, then, hey, you go to the Arroyo and really do all your tests in there. So, that ’ s what we call our first five employees during the tea break, you know, in here. As I said, they were adventurous people. As a matter of fact, one of them, who was, kind of, part of a cult which was not too far from here on Orange Grove, and unfortunately he blew up himself because he kept mixing chemicals and trying to figure out which ones were the best chemicals. So, that gives you a kind of flavor of the kind of people we have there. We try to avoid blowing ourselves up. This one I thought I ’ d show you. Guess which one is a JPL employee in the heart of this crowd. I tried to come like him this morning, but as I walked out, then it was too cold, and I said, I ’ d better put my shirt back on. But more importantly, the reason I wanted to show this **picture** : look where the other people are looking, and look where he is looking. Wherever anybody else looks, look somewhere else, and go do something different, you know, and doing that. And that ’ s kind of what has been the spirit of what we are doing. And I want to tell you a quote from Ralph Emerson that one of my colleagues, you know, put on my wall in my office, and it says,”Do not go where the path may lead. Go instead where there is no path, and leave a trail.”And that ’ s my recommendation to all of you : look what everybody is doing, what they are doing ; go do something completely different. Don ’ t try to improve a little bit on what somebody else is doing, because that doesn ’ t get you very far. In our early days we used to work a lot on rockets, but we also used to have a lot of parties, you know. As you can see, one of our parties, you know, a few years ago. But then a big difference happened about 50 years ago, after Sputnik was launched. We launched the first American satellite, and that ' s the one you see on the left in there. And here we made 180 degrees change : we changed from a rocket house to be an exploration house. And that was done over a period of a couple of years, and now we are the leading organization, you know, exploring space on all of your behalf. But even when we did that, we had to remind ourselves, sometimes there are setbacks. So you see, on the **bottom**, that rocket was supposed to go upward ; somehow it ended going sideways. So that ' s what we call the misguided missile. But then also, just to celebrate that, we started an event at JPL for"Miss Guided Missile."So, we used to have a celebration every year and select — there used to be competition and parades and so on. It ' s not very appropriate to do it now. Some people tell me to do it ; I think, well, that ' s not really proper, you know, these days. So, we do something a little bit more serious. And that ' s what you see in the last Rose Bowl, you know, when we entered one of the floats. That ' s more on the play side. And on the right side, that ' s the Rover just before we finished its testing to take it to the Cape to launch it. These are the Rovers up here that you have on Mars now. So that kind of tells you about, kind of, the **fun** things, you know, and the serious things that we try to do. But I said I ' m going to show you a short clip of one of our employees to kind of give you an idea about some of the talent that we have. Video : Morgan Hendry : Beware of Safety is an instrumental rock band. It branches on more the experimental side. There ' s the improvisational side of jazz. There ' s the heavy-hitting sound of rock. Being able to treat sound as an instrument, and be able to dig for more abstract sounds and things to play live, mixing electronics and acoustics. The music ' s half of me, but the other half — I landed probably the best gig of all. I work for the Jet Propulsion Lab. I ' m building the next Mars Rover. Some of the most brilliant engineers I know are the ones who have that sort of artistic quality about them. You ' ve got to do what you want to do. And anyone who tells you you can ' t, you don ' t listen to them. Maybe they ' re right - I doubt it. Tell them where to put it, and then just do what you want to do. I ' m Morgan Hendry. I am NASA. Charles Elachi : Now, moving from the play stuff to the serious stuff, always people ask, why do we explore? Why are we doing all of these missions and why are we exploring them? Well, the way I think about it is fairly simple. Somehow, 13 billion years ago there was a Big Bang, and you ' ve heard a little bit about, you know, the origin of the universe. But somehow what strikes everybody ' s imagination — or lots of people ' s imagination — somehow from that original Big Bang we have this beautiful world that we live in today. You look outside : you have all that beauty that you see, all that life that you see around you, and here we have intelligent people like you and I who are having a conversation here. All that started from that Big Bang. So, the question is : How did that happen? How did that evolve? How did the universe form? How did the galaxies form? How did the planets form? Why is there a planet on which there is life which have evolved? Is that very common? Is there life on every planet that you can see around the stars? So we literally are all made out of stardust. We started from those stars ; we are made of stardust. So, next time you are really depressed, look in the mirror and you can look and say, hi, I ' m looking at a star here. You can skip the dust part. But literally, we are all made of stardust. So, what we are trying to do in our exploration is effectively write the book of how things have came about as they are today. And one of the first, or the easiest, places we can go and explore that is to go towards Mars. And the reason Mars takes particular attention : it ' s not very far from us. You know, it ' ll take us only six months to get there. Six to nine months at the right time of the year. It

’ s a planet somewhat similar to Earth. It ’ s a little bit smaller, but the land mass on Mars is about the same as the land mass on Earth, you know, if you don ’ t take the oceans into account. It has polar caps. It has an atmosphere somewhat thinner than ours, so it has weather. So, it ’ s very similar to some extent, and you can see some of the features on it, like the Grand Canyon on Mars, or what we call the Grand Canyon on Mars. It is like the Grand Canyon on Earth, except a **hell** of a lot larger. So it ’ s about the size, you know, of the United States. It has volcanoes on it. And that ’ s Mount Olympus on Mars, which is a kind of huge volcanic shield on that planet. And if you look at the height of it and you compare it to Mount Everest, you see, it ’ ll give you an idea of how large that Mount Olympus, you know, is, relative to Mount Everest. So, it basically dwarfs, you know, Mount Everest here on Earth. So, that gives you an idea of the tectonic events or volcanic events which have happened on that planet. Recently from one of our satellites, this shows that it ’ s Earth-like — we caught a landslide occurring as it was happening. So it is a dynamic planet, and activity is going on as we speak today. And these Rovers, people wonder now, what are they doing today, so I thought I would show you a little bit what they are doing. This is one very large crater. Geologists love craters, because craters are like digging a big hole in the ground without really working at it, and you can see what ’ s below the surface. So, this is called Victoria Crater, which is about a few football fields in size. And if you look at the top left, you see a little teeny dark dot. This **picture** was taken from an orbiting satellite. If I zoom on it, you can see : that ’ s the Rover on the surface. So, that was taken from orbit ; we had the camera zoom on the surface, and we actually saw the Rover on the surface. And we actually used the combination of the satellite images and the Rover to actually conduct science, because we can observe large areas and then you can get those Rovers to move around and basically go to a certain location. So, specifically what we are doing now is that Rover is going down in that crater. As I told you, geologists love craters. And the reason is, many of you went to the Grand Canyon, and you see in the wall of the Grand Canyon, you see these layers. And what these layers — that ’ s what the surface used to be a million years ago, 10 million years ago, 100 million years ago, and you get deposits on top of them. So if you can read the layers it ’ s like reading your book, and you can learn the history of what happened in the past in that location. So what you are seeing here are the layers on the wall of that crater, and the Rover is going down now, measuring, you know, the properties and analyzing the rocks as it ’ s going down, you know, that canyon. Now, it ’ s kind of a little bit of a challenge driving down a slope like this. If you were there you wouldn ’ t do it yourself. But we really made sure we tested those Rovers before we got them down — or that Rover — and made sure that it ’ s all working well. Now, when I came last time, shortly after the landing — I think it was, like, a hundred days after the landing — I told you I was surprised that those Rovers are lasting even a hundred days. Well, here we are four years later, and they ’ re still working. Now you say, Charles, you are really lying to us, and so on, but that ’ s not true. We really believed they were going to last 90 days or 100 days, because they are solar powered, and Mars is a dusty planet, so we expected the dust would start accumulating on the surface, and after a while we wouldn ’ t have enough power, you know, to keep them warm. Well, I always say it ’ s important that you are smart, but every once in a while it ’ s good to be lucky. And that ’ s what we found out. It turned out that every once in a while there are dust devils which come by on Mars, as you are seeing here, and when the dust devil comes over the Rover, it just **cleans** it up. It is like a brand new car that you have, and that ’ s literally why they have lasted so long. And now we designed them reasonably well, but that ’ s exactly why they are lasting that long and still providing all the science data. Now, the two Rovers, each one of them is, kind of, getting old. You know, one of them, one of the wheels is stuck, is not working, one of the front wheels, so what we are doing, we are driving it backwards. And the other one has arthritis of the shoulder joint, you know, it ’ s not working very well, so it ’ s walking like this, and we can move the arm, you know, that way. But still they are producing a lot of scientific data. Now, during that whole period, a number of people got excited, you know, outside the science community about these Rovers, so I thought I ’ d show you a video just to give you a reflection about how these Rovers are being viewed by people other than the science community. So let me go on the next short video. By the way, this video is pretty accurate of how the landing took place, you know, about four years ago. Video : Okay, we have parachute aligned. Okay, deploy the airbags. Open. Camera. We have a **picture** right now. Yeah! CE : That ’ s about what happened in the Houston operation room. It ’ s exactly like this. Video : Now, if there is life, the Dutch will find it. What is he doing? What is that? CE : Not too bad. So anyway, let me continue on showing you a little bit about the beauty of that planet. As I said earlier, it looked very much like Earth, so you see sand dunes. It looks like I could have told you these are **pictures** taken from the Sahara Desert or somewhere, and you ’ d have believed me, but these are **pictures** taken from Mars. But one area which is particularly intriguing for us is the northern region, you know, of Mars, close to the North Pole, because we see ice caps, and we see the ice caps shrinking and expanding, so it ’ s very much like you have in northern Canada. And we wanted to find out — and we see all kinds of glacial features on it. So, we wanted to find out, actually, what is that ice made of, and could that have embedded in it some organic, you know, material. So we have a spacecraft which is heading towards Mars, called Phoenix, and that spacecraft will land 17 days, seven hours and 20 seconds from now, so you can adjust your watch. So it ’ s on May 25 around just before five o ’ clock our time here on the West Coast, actually we will be landing on another planet. And as you can see, this is a **picture** of the spacecraft put on Mars, but I thought that just in case you ’ re going to miss that show, you know, in 17 days, I ’ ll show you, kind of, a little bit of what ’ s going to happen. Video : That ’ s what we call the seven minutes of terror. So the plan is to dig in the soil and take samples that we put them in an oven and actually heat them and look what **gases** will come from it. So this was launched about nine months ago. We ’ ll be coming in at 12, 000 miles per hour, and in seven minutes we have to stop and touch the surface very softly so we don ’ t break that lander. Ben Cichy : Phoenix is the first Mars Scout mission. It ’ s the first mission that ’ s going to try to land near the North Pole of Mars, and it ’ s the first mission that ’ s actually going to try and reach out and touch water on the surface of another planet. Lynn Craig : Where there tends to be water, at least on Earth, there tends to be life, and so it ’ s potentially a place where life could have existed on the planet in the past. Erik Bailey : The main purpose of EDL is to take a spacecraft that is traveling at 12, 500 miles an hour and bring it to a screeching halt in a soft way in a very short amount of time. BC : We enter the Martian atmosphere. We ’ re 70 miles above the surface of Mars. And our lander is safely tucked inside what we call an aeroshell. EB : Looks kind of like an ice cream cone, more or less. BC : And on the front of it is this heat shield, this saucer-looking thing that has about a half-inch of essentially what ’ s cork on the front of it, which is our heat shield. Now, this is really special cork, and this cork is what ’ s going to protect us from the violent atmospheric entry that we ’ re about to experience. Rob Grover : Friction really starts to build up on the spacecraft, and we use the friction when it ’ s flying through the atmosphere to our advantage to slow us down. BC : From this point, we ’ re going to decelerate from 12, 500 miles an hour down to 900 miles an hour. EB : The outside can get almost as hot as the surface of the Sun. RG : The temperature of the heat shield can reach 2, 600 degrees Fahrenheit. EB : The inside doesn ’ t get very hot. It probably gets about room temperature. Richard Kornfeld : There is this window of opportunity within which we can deploy the parachute. EB : If you fire the ’ chute too early, the parachute itself could fail. The fabric and the stitching could just pull apart. And that would be bad. BC : In the first 15 seconds after we deploy the parachute, we ’ ll decelerate from 900 miles an hour to a relatively slow 250 miles an hour. We no longer need the heat shield to protect us from the force of atmospheric entry, so we jettison the heat shield, exposing for the first time our lander to the atmosphere of Mars. LC : After the heat shield has been jettisoned and the legs are deployed, the next step is to have the radar system begin to detect how far Phoenix really is from the ground. BC : We ’ ve lost 99 percent of our entry velocity. So, we ’ re 99 percent of the way to where we want to be. But that last one percent, as it always seems to be, is the tricky part. EB : Now the spacecraft actually has to decide when it ’ s going to get rid of its parachute. BC : We separate from the lander going 125 miles an hour at roughly a kilometer above the surface of Mars : 3, 200 **feet**. That ’ s like taking two Empire State Buildings and stacking them on top of one another. EB : That ’ s when we separate from the back shell, and we ’ re now in free-fall. It ’ s a very scary moment ; a lot has to happen in a very short amount of time. LC : So it ’ s in a free-fall, but it ’ s also trying to use all of its actuators to make sure that it ’ s in the right position to land. EB : And then it has to light up its engines, right itself, and then slowly slow itself down and touch down on the ground safely. BC : Earth and Mars are so far apart that it takes over ten minutes for a signal from Mars to get to Earth. And EDL itself is all over in a matter of seven minutes. So by the time you even hear from the lander that EDL has started it ’ ll already be over. EB : We have to build large amounts of autonomy into the spacecraft so that it can land itself safely. BC : EDL is this immense, technically challenging problem. It ’ s about getting a spacecraft that ’ s hurtling through deep space and using all this bag of tricks to somehow figure out how to get it down to the surface of Mars at zero miles an hour. It ’ s this immensely exciting and challenging problem. CE : Hopefully it all will happen the way you saw it in here. So it will be a very tense moment, you know, as we are watching that spacecraft landing on another planet. So now let me talk about the next things that we are doing. So we are in the process, as we speak, of actually designing the next Rover that we are going to be sending to Mars. So I thought I would go a little bit and tell you, kind of, the steps we go through. It ’ s very similar to what you do when you design your product. As you saw a little bit earlier, when we were doing the Phoenix one, we have to take into account the heat that we are going to be facing. So we have to study all kinds of different materials, the shape that we want to do. In general we don ’ t try to please the customer here. What we want to do is to make sure we have an effective, you know, an efficient kind of machine. First we start by we want to have our employees to be as imaginative as they can. And we really love being close to the art center, because we have, as a matter of fact, one of the alumni from the art center, Eric Nyquist, had put a series of displays, far-out displays, you know, in our what we call mission design or spacecraft design room, just to get people to think wildly about things. We have a bunch of Legos. So, as I said, this is a playground for adults, where they sit down and try to play with different shapes and different designs. Then we get a little bit more serious, so we have what we call our CAD / CAMs and all the engineers who are involved, or scientists who are involved, who know about thermal properties, know about design, know about atmospheric interaction, parachutes, all of these things, which they work in a team effort and actually design a spacecraft in a computer to some extent, so to see, does that meet the requirement that we need. On the right, also, we have to take into account the environment of the planet where we are going. If you are going to Jupiter, you have a very high-radiation, you know, environment. It ’ s about the same radiation environment close by Jupiter as inside a nuclear reactor. So just imagine : you take your P. C. and throw it into a nuclear reactor and it still has to work. So these are kind of some of the little challenges, you know, that we have to face. If we are doing entry, we have to do tests of parachutes. You saw in the video a parachute breaking. That would be a bad day, you know, if that happened, so we have to test, because we are deploying this parachute at supersonic speeds. We are coming at extremely high speeds, and we are deploying them to slow us down. So we have to do all kinds of tests. To give you an idea of the size, you know, of that parachute relative to the people standing there. Next step, we go and actually build some kind of test models and actually test them, you know, in the lab at JPL, in what we call our Mars Yard. We kick them, we hit them, we drop them, just to make sure we understand how, where would they break. And then we back off, you know, from that point. And then we actually do the actual building and the flight. And this next Rover that we ’ re flying is about the size of a car. That big shield that you see outside, that ’ s a heat shield which is going to protect it. And that will be basically built over the next year, and it will be launched June a year from now. Now, in that case, because it was a very big Rover, we couldn ’ t use airbags. And I know many of you, kind of, last time afterwards said well, that was a cool thing to have — those airbags. Unfortunately this Rover is, like, ten times the size of the, you know, mass-wise, of the other Rover, or three times the mass. So we can ’ t use airbags. So we have to come up with another ingenious idea of how do we land it. And we didn ’ t want to take it propulsively all the way to the surface because we didn ’ t want to contaminate the surface ; we wanted the Rover to immediately land on its legs. So we came up with this ingenious idea, which is used here on Earth for helicopters. Actually, the lander will come down to about 100 **feet** and hover above that surface for 100 **feet**, and then we have a sky crane which will take that Rover and land it down on the surface. Hopefully it all will work, you know, it will work that way. And that Rover will be more kind of like a chemist. What we are going to be doing with that Rover as it drives around, it ’ s going to go and analyze the chemical composition of rocks. So it will have an arm which will take samples, put them in an oven, crush and analyze them. But also, if there is something that we cannot reach because it is too high on a cliff, we have a little **laser** system which will actually zap the rock, evaporate some of it, and actually analyze what ’ s coming from that rock. So it ’ s a little bit like”Star Wars,”you know, but it ’ s real. It ’ s real stuff. And also to help you, to help the community so you can do ads on that Rover, we are going to train that Rover to actually in addition to do this, to actually serve cocktails, you know, also on Mars. So that ’ s kind of giving you an idea of the kind of, you know, **fun** things we are doing on Mars. I thought I ’ d go to”The Lord of the Rings”now and show you some of the things we have there. Now,”The Lord of the Rings”has two things played through it. One, it ’ s a very attractive planet — it just has the beauty of the rings and so on. But for scientists, also the rings have a special meaning, because we believe they represent, on a small scale, how the Solar System actually formed. Some of the scientists believe that the way the Solar System formed, that the Sun when it collapsed and actually created the Sun, a lot of the dust around it created rings and then the particles in those rings accumulated together, and they formed bigger rocks, and then that ’ s how the planets, you know, were formed. So, the idea is, by watching Saturn we ’ re actually watching our solar system in real time being formed on a smaller scale, so it ' s like a test bed for it. So, let me show you a little bit on what that Saturnian system looks like. First, I ' m going to fly you over the rings. By the way, all of this is real stuff. This is not animation or anything like this. This is actually taken from the satellite that we have in orbit around Saturn, the Cassini. And you see the amount of detail that is in those rings, which are the particles. Some of them are agglomerating together to form larger particles. So that ' s why you have these gaps, is because a small satellite, you know, is being formed in that location. Now, you think that those rings are very large objects. Yes, they are very large in one dimension ; in the other dimension they are paper thin. Very, very thin. What you are seeing here is the shadow of the ring on Saturn itself. And that ' s one of the satellites which was actually formed on that one. So, think about it as a paper-thin, huge area of many hundreds of thousands of miles, which is rotating. And we have a wide variety of kind of satellites which will form, each one looking very different and very odd, and that keeps scientists busy for tens of years trying to explain this, and telling NASA we need more money so we can explain what these things look like, or why they formed that way. Well, there were two satellites which were particularly interesting. One of them is called Enceladus. It ' s a satellite which was all made of ice, and we measured it from orbit. Made of ice. But there was something bizarre about it. If you look at these stripes in here, what we call tiger stripes, when we flew over them, all of a sudden we saw an increase in the temperature, which said that those stripes are warmer than the rest of the planet. So as we flew by away from it, we looked back. And guess what? We saw geysers coming out. So this is a Yellowstone, you know, of Saturn. We are seeing geysers of ice which are coming out of that planet, which indicate that most likely there is an ocean, you know, below the surface. And somehow, through some dynamic effect, we ' re having these geysers which are being, you know, emitted from it. And the reason I showed the little arrow there, I think that should say 30 miles, we decided a few months ago to actually fly the spacecraft through the plume of that geyser so we can actually measure the material that it is made of. That was [unclear] also — you know, because we were worried about the risk of it, but it worked pretty well. We flew at the top of it, and we found that there is a fair amount of organic material which is being emitted in combination with the ice. And over the next few years, as we keep orbiting, you know, Saturn, we are planning to get closer and closer down to the surface and make more accurate measurements. Now, another satellite also attracted a lot of attention, and that ' s Titan. And the reason Titan is particularly interesting, it ' s a satellite bigger than our moon, and it has an atmosphere. And that atmosphere is very — as dense as our own atmosphere. So if you were on Titan, you would feel the same pressure that you feel in here. Except it ' s a lot colder, and that atmosphere is heavily made of methane. Now, methane gets people all excited, because it ' s organic material, so immediately people start thinking, could life have evolved in that location, when you have a lot of organic material. So people believe now that Titan is most likely what we call a pre-biotic planet, because it ' s so cold organic material did not get to the stage of becoming biological material, and therefore life could have evolved on it. So it could be Earth, frozen three billion years ago before life actually started on it. So that ' s getting a lot of interest, and to show you some example of what we did in there, we actually dropped a probe, which was developed by our colleagues in Europe, we dropped a probe as we were orbiting Saturn. We dropped a probe in the atmosphere of Titan. And this is a **picture** of an area as we were coming down. Just looked like the coast of California for me. You see the rivers which are coming along the coast, and you see that white area which looks like Catalina Island, and that looks like an ocean. And then with an instrument we have on board, a radar instrument, we found there are lakes like the Great Lakes in here, so it looks very much like Earth. It looks like there are rivers on it, there are oceans or lakes, we know there are clouds. We think it ’ s raining also on it. So it ’ s very much like the cycle on Earth except because it ’ s so cold, it could not be water, you know, because water would have frozen. What it turned out, that all that we are seeing, all this liquid, [is made of] hydrocarbon and ethane and methane, similar to what you put in your car. So here we have a cycle of a planet which is like our Earth, but is all made of ethane and methane and organic material. So if you were on Mars — sorry, on Titan, you don ’ t have to worry about four-dollar gasoline. You just drive to the nearest lake, stick your hose in it, and you ’ ve got your car filled up. On the other hand, if you light a match the whole planet will blow up. So in closing, I said I want to close by a couple of **pictures**. And just to kind of put us in perspective, this is a **picture** of Saturn taken with a spacecraft from behind Saturn, looking towards the Sun. The Sun is behind Saturn, so we see what we call”forward scattering,”so it highlights all the rings. And I ’ m going to zoom. There is a — I ’ m not sure you can see it very well, but on the top left, around 10 o ’ clock, there is a little teeny dot, and that ’ s Earth. You barely can see ourselves. So what I did, I thought I ’ d zoom on it. So as you zoom in, you know, you can see Earth, you know, just in the middle here. So we zoomed all the way on the art center. So thank you very much.

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Welcome to 10, 000 **feet**. Let me explain why we are here and why some of you have a pine cone close to you. Once upon a time, I did a book called”How Buildings Learn.”Today ’ s event you might call”How Mountains Teach.”A little background : For 10 years I ’ ve been trying to figure out how to hack civilization so that we can get long-term thinking to be automatic and common instead of difficult and rare — or in some cases, non-existent. It would be helpful if humanity got into the habit of thinking of the now not just as next week or next quarter, but you know, next 10, 000 years and the last 10, 000 years — basically civilization ’ s story so far. So we have the Long Now Foundation in San Francisco. It ’ s an incubator for about a dozen projects, all having to do with continuity over the long term. Our core project is a rather ambitious folly — I suppose, a mythic undertaking : to build a 10, 000-year clock that can really keep good time for that long a period. And the design problems of a project like that are just absolutely delicious. Go to the clock. And what we have here is something many of you saw here three years ago. It ’ s the first working prototype of the clock. It ’ s about nine **feet** high. Designed by Danny Hillis and Alexander Rose. It ’ s presently in London, and is ticking away very deliberately at the science **museum** there. So the design problem for today is going to be, how do you house an eventual monumental clock like this so it can really tick, save time beautifully for 100 centuries? Well, this was the first solution. Alexander Rose came up with this idea of a spiraloid tower with continuous **sloping** ramps. And it looked like a way to go, until you start thinking about, what does deep time do to a building? Well, this is what deep time does to a building. This is the Parthenon. It ’ s only 2, 450 years old, and look what happened to it. Here ’ s a beautiful project. They really knew it ’ d last forever, because they ’ d build it out of absolutely huge stones. And now it ’ s a pathetic ruin and no one even knows what it was used for. That ’ s what happens to buildings. They ’ re vulnerable. Even the most durable and intactable buildings, like the pyramids of Giza, are in bad shape when you look up close. They ’ ve been looted inside and out. And they ’ re built to protect things but they don ’ t protect things. So we got to thinking, if you can ’ t put things safely in a building, where can you safely put them? We thought, OK, underground. How about underground with a view? Underground in a place that ’ s really solid. So the obvious answer was, we need a mountain. You don ’ t want just any mountain. You need absolutely the right mountain if you ’ re going to have a clock for 10, 000 years. So here ’ s an image of the long view of the search problem. And we got to thinking for various reasons it ought to be a desert mountain, so we got looking in the dry areas of the Southwest. We looked at mesas in New Mexico. We were looking at dead volcanoes in Arizona. Then Roger Kennedy, who was the director of the National Parks Service, led us to Eastern Nevada, to America ’ s newest and oldest national **park**, which is called Great Basin National Park. It ’ s right on the eastern border of Nevada. It ’ s the highest range in the state — over 13, 000 **feet**. And you ’ ll notice that on the left, on the left, on the west, it ’ s very steep, and on the right it ’ s gentle. This place is remote. It ’ s over 200 miles from any major city. It ’ s nowhere near any Interstate or **railroad**. And it ’ s — the only thing that goes by is what ’ s called America ’ s loneliest highway, U. S. 50. Now, inside the yellow line here, on the right is — that ’ s all national **park**. Inside the green line is national forest. And then over to the left is Bureau of Land Management land and some private land. Now, as it happened, that two- mile-long strip right in the middle, this vertical, was available because it was private land. And thanks to Jay Walker who was here and Mitch Kapor who was here, who started the process, Long Now was able to get that two-mile-long strip of land. And now let ’ s look at the grand truth of what ’ s there. We ’ re in Pole Canyon, looking west up the western escarpment of Mount Washington, which is 11, 600 **feet** on top. Those white cliffs are a dense Cambrian limestone. That ’ s a 2, 000-foot thick formation, and it might be a beautiful place to hide a clock. It would be a pilgrimage to get to it ; it would be a serious hike to get up to where the clock is. So last June, the Long Now board, some staff and some donors and advisors, made a two-week expedition to the mountain to explore it and investigate, one, if it ’ s the right mountain, and two, if it ’ s the right mountain, how it might actually work for us. Now Danny Hillis sort of framed the problem. He has a theory of how the overall clock experience should work. It ’ s what he calls the seven stages of a mythic adventure. It starts with the image. The image is a **picture** you have in your mind of the goal at the end of the journey. In this case it might well be an image of the clock. Then there ’ s the point of embarkation, that is, the point of transition from ordinary life to being a pilgrim on a quest. Then — this is a nice image of it, there ’ s the labyrinth. The labyrinth is a concept, it ’ s like a twilight zone, it ’ s a place where it ’ s difficult, where you get disoriented, maybe you get scared — but you have to go through it if you ’ re going to get to some kind of deep reintegration. Then there should always be in sight the draw — a kind of a beacon that draws you on through the labyrinth to finish the process of getting there. Now Brian Eno, who ’ s been in the thick of the Long Now process, spent two years making a C. D. called”January 7003,”and it ’ s”Bell Studies for the Clock of the Long Now.”Based on — parts of it are based on an algorithm that Danny Hillis developed, so that a peal of 10 bells makes a different peal every day for 10, 000 years. The Hillis algorithm. 10 factorial gives you that number. And in fact, pretty soon we ’ ll hear the sound. January 7003. There it is. OK, back to Danny ’ s list. Number five of the seven is the payoff. This is it. The climax. The goal. The main thing that you ’ re trying to get to. And then Danny says a really great journey will have a secret payoff. Something you didn ’ t expect that caps what you did expect. Then there ’ s the return. You ’ ve got to have a gradual return to the ordinary world, so you have time to assimilate what you ’ ve learned. And then, how about a memento? Number seven. At the end of it there ’ s something physical, a kind of reward that you take away. It might be a piece of a core drill of the mountain. Something that ’ s just yours. How do you study a mountain for the kinds of things we ’ re talking about? This is not a normal building project. What do you look for? What are the elements that will most affect your ideas and decisions? Start with borders. If you look on the left side of the cliffs here, that ’ s national **park**. That ’ s sacrosanct — you can ’ t do anything with that. To the right of it is national forest. There ’ s possibilities. The borders are important. Other elements were mines, weather, approaches and elevation. And especially trees. Look at those things up on top there. It turns out that Mount Washington is covered with bristlecone pines. They ’ re the world ’ s oldest living thing. People think they ’ re just the size of shrubs, but that ’ s not actually true. There are trees on that mountain that are 5, 000 years old and still living. The wood is so solid it ’ s like stone, and it lasts for a long time. So when you do tree ring studies of trunks that are on the mountain, some of them go back 10, 000 years. The stone itself is absolutely beautiful, sculpted by millennia of very tough winters up there. We had tree ring analysts from the University of Arizona join us on the expedition. Now, if you guys have a pine cone handy, now ’ s a good time to put it in your hand and feel it, especially on the end. That ’ s interesting. You ’ ll find out why it ’ s called a bristlecone pine. A little sensory experience. Here ’ s Danny Hillis in the midst of a bristlecone pine forest on Long Now land. I should say that the age of bristlecones was discovered, led by a theory. Edmund Schulman in the 1950s had been studying trees under great stress at Timberline, and came to the realization that he put in an article in Science magazine called,”Longevity under Adversity in Conifers.”And then, based on that principle, he started looking around at the various trees at Timberline, and realized that the bristlecone pines — he found some in the White Mountains that were over 4, 000 years old. Longevity under adversity is a pretty interesting design principle in its own right. OK, onto the mines. The first asking price for the property when we looked at it in 1998 was one billion dollars for 180 acres and a couple of mines. Because the owner said,”There ’ s one billion dollars of beryllium in that mountain.”And we said,”Wow, that ’ s great. Listen, we ’ ll counter. How about zero? And we ’ re a non-profit foundation, you can give us the property and take a **hell** of a tax deduction. All you have to do is prove to the government it ’ s worth a billion dollars.”Well, a few years went by and there was some kind of back and forth, and by and by, thanks to Mitch and Jay, we were able to buy the whole property for 140, 000 dollars. This is one of the mines. It doesn ’ t have any beryllium in it. It ’ s called the Pole Adit. And it does have tungsten, a little bit of tungsten, left over, that ’ s the kind of mine it was. But it goes a mile-and-a-half in a straight line, due east into the range, into very interesting territory — except that, as you ’ ll see when we go inside in a minute, we were hoping for limestone but in there is just shale. And shale is not quite completely competent rock. Competent rock is rock that will hold itself up without any shoring. The shale would like some shoring, and so parts of it are caved in in there. That ’ s Ben Roberts from — he ’ s the bat specialist from the National Park. But there are many wonders back in there, like this weird fungus on some of the collapsed timbers. OK, here ’ s another mine that ’ s up on top of the property, and it dates back to 1870. That ’ s what the property was originally built around — it was a set of mining claims. It was a very productive silver mine. In fact, it was the highest-operating mine in Nevada, and it ran year round. You can imagine what it was like in the winter at 10, 000 **feet**.

You may recognize a couple of the miners there. There ’ s Jeff Bezos on the right and Paul Saville on the left looking for galena, which is the lead-silver thing. They didn ’ t find any. They both kept their day jobs. Here ’ s the last mine. It ’ s called the Bonanza Adit. It ’ s down in a canyon. And Alexander Rose on the left there worked with a bunch of people from the National Park to survey the whole mine. It ’ s a mile deep. And they also found four species of bats in there. Now, almost all those mines, by the way, meet underneath the mountain. They don ’ t quite, but it ’ s something to think about. They don ’ t quite meet. Let ’ s go to weather. Mountains specialize in interesting weather. Way more interesting than Monterey even today. And so one Tuesday morning last June, there we were. Woke up in the morning — the mountain was covered with snow. That was a great time to go up and visit our weather station which again, thanks to Mitch Kapor, we ’ re building up there. And it ’ s a pretty interesting scene. This is, on the left there, the joyful lady is Pat Irwin, who ’ s the regional head of the National Forest Service, and they gave us the temporary use permit to be there. We want a temporary use permit for the clock, eventually — 10, 000-year temporary use permit. The weather station ’ s pretty interesting. Kurt Bollacker and Alexander Rose designed a radically wireless station. It runs on solar, and it sends a signal with that antenna and bounces it off of micrometeorite trails in the atmosphere to a place in Bozeman, Montana, where the data is taken down and then sent through landlines to San Francisco, where we put the data in real time up on our website. And there you see a week of weather at 9, 400 **feet** on Mount Washington. Let ’ s go to approaches. As it happens, there are no trails anywhere on Mount Washington, just a few old mining roads like this, so you have to bushwhack everywhere. But there ’ s no bears, and there ’ s no poison oak, and there ’ s basically no people because this place has been empty for a long time. You can hike for days and not encounter anybody. Well, here ’ s a potential approach. You need to come up the Lincoln Canyon. It ’ s this beautiful world all of its own, surrounded by cliffs, and it ’ s an easy hike to stroll up the canyon **bottom**, until you get to this barrier, and it actually presents a problem. So you can scratch Lincoln Canyon as an approach. Another possible approach is right up the western front of the mountain. You can see why we sometimes call it Long Mountain. And from where you ’ re standing at 6, 000 **feet** in the **valley**, it ’ s an easy hike up to the mature pinyon and juniper forest through that knoll at the front at 7, 600 **feet**. And you can carry right on up through meadows and steepening forest to the high base of the cliffs at 10, 500 **feet**, where there ’ s a bit of a problem. Now, Jeff Bezos advised us when he left at the end of the expedition,”Make the clock inaccessible. The harder it is to get to, the more people will value it.”And check — those are 600-foot vertical walls there. So Alexander Rose wanted to explore this route, and he started over here on the left from his pickup truck at 8, 900 **feet** and headed up the mountain. Now, as you gain elevation your IQ goes down — but your emotional affect goes up, which is great for having a mythic experience, whether you want to or not. In fact, Danny Hillis can estimate altitude by how much math he can ’ t do in his head. Now, I happened to be on the radio with Alexander when he got to this point at the base of the cliffs, and he said, quote,”There ’ s a hidden notch. I think I can get up a ways.”Now, he ’ s a rock climber, but you know, he ’ s our executive director. I don ’ t want him killed. I know he ’ s going to love cliffs. I ’ m saying,”Be careful, be careful, be careful.”Then he starts going up, and the next thing I hear is,”I ’ m half-way up. It ’ s like climbing stairs. I ’ m going up 60 degrees. It ’ s a secret passage. It ’ s like something from Tolkien.”And I ’ m going,”Careful, careful. Please be careful.”And then, of course, the next thing I hear is,”I ’ ve made it to the top. You can see all of creation from up here.”And he dashed across the top of the mountains. In fact, there he is. That ' s Alexander Rose. First ascent of the western face to Mount Washington, and a solo ascent at that. This discovery changed everything about our sense of these cliffs and what to do with them. We realized that we had to name this thing that Alexander discovered. How about Zander ' s Crevice? No. So we finally decided on Alexander ' s Siq. Zander ' s Siq is named after — some of you have been to Petra, there ' s this wonderful slot canyon that leads into Petra called the Siq, and so this is the Siq. And it really is hidden. I can ' t find it in this image, and I ' m not sure you can. Only when you get fresh snow can you see just along the rim there, and that brings it out. Now, Danny and I were up at this same area one day, and Danny looked over to the right and noticed something halfway up the cliffs, which is a kind of a porch or a cliff shelf with bristlecones on it, and supposed that people going up to the clock inside the mountain could come out onto that shelf and look down at the view. And the people toiling up the mountain could see them, these tiny little people up there, incredibly halfway up the cliff. How did they get there? Do I have to do that? And so that maybe becomes part of the draw and part of the labyrinth. You can get another angle on Danny ' s porch by going around to the south and looking north at the whole formation there. And you need to know that Danny ' s clock is to be kept accurate by a ray of sunshine, that perfect noon hitting it every sunny day, and the pulse of heat from that sets off a solar trigger which resets the clock to make it perfectly accurate. So even with the slowing of the rotation of the earth and so on, the clock will keep perfectly good time. So here we ' re looking from the south, look north. This is all Forest Service land. If you go up on top of those cliffs, that ' s some of the Long Now land in those trees. And if you go up there and look back, then you ' ll get a sense of what the view starts to be like from the top of the mountain. That ' s the long view. That ' s 80 miles to the horizon. And that ' s also timberline and those bristlecones really are shrubs. That ' s a different place to be. It ' s 11, 400 **feet** and it ' s exquisite. Now, if you go over to the right from this image to looking at the edge of the cliffs, it ' s 600 **foot**, just about a yard to the left of Kurt Bollacker ' s **foot**, there is a 600-foot drop. He ' s ambling on over to Zander ' s Siq. That ' s what it looks like looking down it. We should probably put in a rail or something. Over on the eastern side it ' s gentle, as you can see. And that ' s not snow — that ' s what the white limestone looks like. You also see there a bighorn sheep. Their herd was reintroduced from Wyoming. And they ' re doing pretty well, but they ' ve got a bit of trouble. This is Danny Hillis, and he ' s figuring out a design problem. he ' s trying to determine if where he is on a bit of Long Now land would appear from down in the **valley** to be the actual peak of the mountain. because the real peak is hidden around the corner. This is what in the infantry we used to call the military crest. And as it turned out the answer is, yes, that is from down below in the **valley** it does look like the peak, and that might be conjured with. We gradually realized we have three serious design domains to work on with this. One is the experience of the mountain. Another is the experience in the mountain. And the third is the experience from the mountain, which is really dominated by the view shed of the spring **valley** there behind Danny, and if you look off to the right, out there, 15 miles across to the Schell Creek range. In the front, there are 10 ranches strung right along the base of the mountains using the water from the mountains. In fact, there are artesian wells where water springs right into the air. One of the ranches is called the Kirkeby Ranch, and I ' ll take you there for a minute. It ' s a very nice ranch. Alfalfa and cattle, run by Paul and Ronnie Brenham, and it ' s pretty idyllic. It ' s also hard work. And most of these ranches are having trouble keeping going. This is their view to the west of the Schell Creek range. And if you go out to that line of trees at the far end, you ' ll see what the **valley** used to look like. This is Rocky Mountain junipers that have been there for thousands of years. And a scheme emerged that Long Now is looking to see if it might be possible to buy up the whole **valley**, because those 10 ranches with their 17, 000 acres dominate a 500 square mile **valley** with their grazing allotments and so on, and there ' s a possibility that you could get the whole thing for five million dollars and gradually restore it to its wild condition, and somewhere in the process turn it back over to the National Park, and it would double the size of Great Basin National Park. That would be swell. OK, let ' s take one more look at the mountain itself. The clock experience should be profound, but from the outside it should be invisible. Now, at the base of the high cliffs there ' s this natural cave. It ' s only about 12 **feet** deep, but what if it were deepened from inside? You excavated from somewhere, came up from inside and deepened it. And then you could have an entrance which was very rough and narrow as you first went in, that gradually becomes more refined and then actually quite exquisite. And this stone takes a perfect polish. You ' d have a polished set of passages and chambers in there eventually leading to the 10, 000 year clock. And it ' s not a mine. This would be a nuanced evocation of the basic structure of the mountain, and you would be appreciating it as much from inside as you do from outside. This is architecture not made by building, but by what you very carefully take away. So that ' s what the mountain taught us. Most of the amazingness of the clock we can borrow from the amazingness of the mountain. All we have to do is highlight its spectacular features and blend in with them. It ' s not a clock in a mountain — it ' s a mountain clock. Now, the Tewa Indians in the Southwest have a saying for what you need to do when you want to think long term about anything. They say,"pin peya obe"— welcome to the mountain. Thank you.

Text Sample 681: POS Dim 5 – 2004 – Score 8.00 – 2004/t000445.txt

If you ' d like to learn how to play the lobster, we have some here. And that ' s not a joke, we really do. So come up afterwards and I ' ll show you how to play a lobster. So, actually, I started working on what ' s called the mantis shrimp a few years ago because they make sound. This is a recording I made of a mantis shrimp that ' s found off the coast of California. And while that ' s an absolutely fascinating sound, it actually turns out to be a very difficult project. And while I was struggling to figure out how and why mantis shrimp, or stomatopods, make sound, I started to think about their appendages. And mantis shrimp are called"mantis shrimp"after the praying mantises, which also have a fast feeding appendage. And I started to think, well, maybe it will be interesting, while listening to their sounds, to figure out how these animals generate very fast feeding strikes. And so today I ' ll talk about the extreme stomatopod strike, work that I ' ve done in collaboration with Wyatt Korff and Roy Caldwell. So, mantis shrimp come in two varieties : there are spearers and smashers. And this is a spearing mantis shrimp, or stomatopod. And he lives in the sand, and he catches things that go by overhead. So, a quick strike like that. And if we slow it down a bit, this is the mantis shrimp — the same species — recorded at 1, 000 frames a second, played back at 15 frames per second. And you can see it ' s just a really spectacular extension of the limbs, exploding upward to actually just catch a dead piece of shrimp that I had offered it. Now, the other type of mantis shrimp is the smasher stomatopod, and these guys open up snails for a living. And so this guy gets the snail all set up and gives it a good whack. So, I ' ll play it one more time. He wiggles it in place, tugs it with his nose, and smash. And a few smashes later, the snail is broken open, and he ’ s got a good dinner. So, the smasher raptorial appendage can stab with a point at the end, or it can smash with the heel. And today I ’ ll talk about the smashing type of strike. And so the first question that came to mind was, well, how fast does this limb move? Because it ’ s moving pretty darn fast on that video. And I immediately came upon a problem. Every single high-speed video system in the biology department at Berkeley wasn ’ t fast enough to catch this movement. We simply couldn ’ t capture it on video. And so this had me stymied for quite a long period of time. And then a BBC crew came cruising through the biology department, looking for a story to do about new technologies in biology. And so we struck up a deal. I said,”Well, if you guys rent the high-speed video system that could capture these movements, you guys can **film** us collecting the data.”And believe it or not, they went for it. So we got this incredible video system. It ’ s very new technology — it just came out about a year ago — that allows you to **film** at extremely high speeds in low light. And low light is a critical issue with **filming** animals, because if it ’ s too high, you fry them. So this is a mantis shrimp. There are the eyes up here, and there ’ s that raptorial appendage, and there ’ s the heel. And that thing ’ s going to swing around and smash the snail. And the snail ’ s wired to a stick, so he ’ s a little bit easier to set up the shot. And — yeah. I hope there aren ’ t any snail rights activists around here. So this was **filmed** at 5, 000 frames per second, and I ’ m playing it back at 15. And so this is slowed down 333 times. And as you ’ ll notice, it ’ s still pretty gosh darn fast slowed down 333 times. It ’ s an incredibly powerful movement. The whole limb extends out. The body flexes backwards — just a spectacular movement. And so what we did is, we took a look at these videos, and we measured how fast the limb was moving to get back to that original question. And we were in for our first surprise. So what we calculated was that the limbs were moving at the peak speed ranging from 10 meters per second all the way up to 23 meters per second. And for those of you who prefer miles per hour, that ’ s over 45 miles per hour in water. And this is really darn fast. In fact, it ’ s so fast we were able to add a new point on the extreme animal movement spectrum. And mantis shrimp are officially the fastest measured feeding strike of any animal system. So our first surprise. So that was really cool and very unexpected. So, you might be wondering, well, how do they do it? And actually, this work was done in the 1960s by a famous biologist named Malcolm Burrows. And what he showed in mantis shrimp is that they use what ’ s called a”catch mechanism,”or”click mechanism.”And what this basically consists of is a large muscle that takes a good long time to contract, and a latch that prevents anything from moving. So the muscle contracts, and nothing happens. And once the muscle ’ s contracted completely, everything ’ s stored up — the latch flies upward, and you ’ ve got the movement. And that ’ s basically what ’ s called a”power amplification system.”It takes a long time for the muscle to contract, and a very short time for the limb to fly out. And so I thought that this was sort of the end of the story. This was how mantis shrimps make these very fast strikes. But then I took a trip to the National Museum of Natural History. And if any of you ever have a chance, backstage of the National Museum of Natural History is one of the world ’ s best collections of preserved mantis shrimp. And what — this is serious business for me. So, this — what I saw, on every single mantis shrimp limb, whether it ’ s a spearer or a smasher, is a beautiful saddle-shaped structure right on the top surface of the limb. And you can see it right here. It just looks like a saddle you ’ d put on a horse. It ’ s a very beautiful structure. And it ’ s surrounded by membranous areas. And those membranous areas suggested to me that maybe this is some kind of dynamically flexible structure. And this really sort of had me scratching my head for a while. And then we did a series of calculations, and what we were able to show is that these mantis shrimp have to have a spring. There needs to be some kind of spring-loaded mechanism in order to generate the amount of force that we observe, and the speed that we observe, and the output of the system. So we thought, OK, this must be a spring — the saddle could very well be a spring. And we went back to those high-speed videos again, and we could actually visualize the saddle compressing and extending. And I ' ll just do that one more time. And then if you take a look at the video — it ' s a little bit hard to see — it ' s outlined in yellow. The saddle is outlined in yellow. You can actually see it extending over the course of the strike, and actually hyperextending. So, we ' ve had very solid evidence showing that that saddle-shaped structure actually compresses and extends, and does, in fact, function as a spring. The saddle-shaped structure is also known as a"hyperbolic paraboloid surface,"or an"anticlastic surface."And this is very well known to engineers and **architects**, because it ' s a very strong surface in compression. It has **curves** in two directions, one **curve** upward and opposite transverse **curve** down the other, so any kind of perturbation spreads the forces over the surface of this type of shape. So it ' s very well known to engineers, not as well known to biologists. It ' s also known to quite a few people who make jewelry, because it requires very little material to build this type of surface, and it ' s very strong. So if you ' re going to build a thin gold structure, it ' s very nice to have it in a shape that ' s strong. Now, it ' s also known to **architects**. One of the most famous **architects** is Eduardo Catalano, who popularized this structure. And what ' s shown here is a saddle-shaped **roof** that he built that ' s 87 and a half **feet** spanwise. It ' s two and a half inches thick, and supported at two points. And one of the reasons why he designed **roofs** this way is because it ' s — he found it fascinating that you could build such a strong structure that ' s made of so few materials and can be supported by so few points. And all of these are the same principles that apply to the saddle-shaped spring in stomatopods. In biological systems it ' s important not to have a whole lot of extra material requirements for building it. So, very interesting parallels between the biological and the engineering worlds. And interestingly, this turns out — the stomatopod saddle turns out to be the first described biological hyperbolic paraboloid spring. That ' s a bit long, but it is sort of interesting. So the next and final question was, well, how much force does a mantis shrimp produce if they ' re able to break open snails? And so I wired up what ' s called a load cell. A load cell measures forces, and this is actually a piezoelectronic load cell that has a little crystal in it. And when this crystal is squeezed, the electrical properties change and it — which — in proportion to the forces that go in. So these animals are wonderfully aggressive, and are really hungry all the time. And so all I had to do was actually put a little shrimp paste on the front of the load cell, and they ' d smash away at it. And so this is just a regular video of the animal just smashing the heck out of this load cell. And we were able to get some force measurements out. And again, we were in for a surprise. I purchased a 100-pound load cell, thinking, no animal could produce more than 100 pounds at this size of an animal. And what do you know? They immediately overloaded the load cell. So these are actually some old data where I had to find the smallest animals in the lab, and we were able to measure forces of well over 100 pounds generated by an animal about this big. And actually, just last week I got a 300-pound load cell up and running, and I ' ve clocked these animals generating well over 200 pounds of force. And again, I think this will be a world record. I have to do a little bit more background reading, but I think this will be the largest amount of force produced by an animal of a given — per body mass. So, really incredible forces. And again, that brings us back to the importance of that spring in storing up and releasing so much energy in this system. But that was not the end of the story. Now, things — I ’ m making this sound very easy, this is actually a lot of work. And I got all these force measurements, and then I went and looked at the force output of the system. And this is just very simple — time is on the X- axis and the force is on the Y-axis. And you can see two peaks. And that was what really got me puzzled. The first peak, obviously, is the limb hitting the load cell. But there ’ s a really large second peak half a millisecond later, and I didn ’ t know what that was. So now, you ’ d expect a second peak for other reasons, but not half a millisecond later. Again, going back to those high-speed videos, there ’ s a pretty good hint of what might be going on. Here ’ s that same orientation that we saw earlier. There ’ s that raptorial appendage — there ’ s the heel, and it ’ s going to swing around and hit the load cell. And what I ’ d like you to do in this shot is keep your eye on this, on the surface of the load cell, as the limb comes flying through. And I hope what you are able to see is actually a flash of light. Audience : Wow. Sheila Patek : And so if we just take that one frame, what you can actually see there at the end of that yellow arrow is a vapor bubble. And what that is, is cavitation. And cavitation is an extremely potent fluid dynamic phenomenon which occurs when you have areas of water moving at extremely different speeds. And when this happens, it can cause areas of very low pressure, which results in the water literally vaporizing. And when that vapor bubble collapses, it emits sound, light and heat, and it ’ s a very destructive process. And so here it is in the stomatopod. And again, this is a situation where engineers are very familiar with this phenomenon, because it destroys boat propellers. People have been struggling for years to try and design a very fast rotating boat propeller that doesn ’ t cavitate and literally wear away the **metal** and put holes in it, just like these **pictures** show. So this is a potent force in fluid systems, and just to sort of take it one step further, I ’ m going to show you the mantis shrimp approaching the snail. This is taken at 20, 000 frames per second, and I have to give full credit to the BBC cameraman, Tim Green, for setting this shot up, because I could never have done this in a million years — one of the benefits of working with professional cameramen. You can see it coming in, and an incredible flash of light, and all this cavitation spreading over the surface of the snail. So really, just an amazing image, slowed down extremely, to extremely slow speeds. And again, we can see it in slightly different form there, with the bubble forming and collapsing between those two surfaces. In fact, you might have even seen some cavitation going up the edge of the limb. So to solve this quandary of the two force peaks : what I think was going on is : that first impact is actually the limb hitting the load cell, and the second impact is actually the collapse of the cavitation bubble. And these animals may very well be making use of not only the force and the energy stored with that specialized spring, but the extremes of the fluid dynamics. And they might actually be making use of fluid dynamics as a second force for breaking the snail. So, really fascinating double whammy, so to speak, from these animals. So, one question I often get after this talk — so I figured I ’ d answer it now — is, well, what happens to the animal? Because obviously, if it ’ s breaking snails, the poor limb must be disintegrating. And indeed it does. That ’ s the smashing part of the heel on both these images, and it gets worn away. In fact, I ’ ve seen them wear away their heel all the way to the flesh. But one of the convenient things about being an arthropod is that you have to molt. And every three months or so these animals molt, and they build a new limb and it ’ s no problem. Very, very convenient solution to that particular problem. So, I ’ d like to end on sort of a wacky note. Maybe this is all wacky to folks like you, I don ’ t know. So, the saddles — that saddle-shaped spring — has actually been well known to biologists for a long time, not as a spring but as a visual signal. And there ' s actually a spectacular colored dot in the center of the saddles of many species of stomatopods. And this is quite interesting, to find evolutionary origins of visual signals on what ' s really, in all species, their spring. And I think one explanation for this could be going back to the molting phenomenon. So these animals go into a molting period where they ' re unable to strike — their bodies become very soft. And they ' re literally unable to strike or they will self-destruct. This is for real. And what they do is, up until that time period when they can ' t strike, they become really obnoxious and awful, and they strike everything in sight ; it doesn ' t matter who or what. And the second they get into that time point when they can ' t strike any more, they just signal. They wave their legs around. And it ' s one of the classic examples in animal behavior of bluffing. It ' s a well-established fact of these animals that they actually bluff. They can ' t actually strike, but they pretend to. And so I ' m very curious about whether those colored dots in the center of the saddles are conveying some kind of information about their ability to strike, or their strike force, and something about the time period in the molting cycle. So sort of an interesting strange fact to find a visual structure right in the middle of their spring. So to conclude, I mostly want to acknowledge my two collaborators, Wyatt Korff and Roy Caldwell, who worked closely with me on this. And also the Miller Institute for Basic Research in Science, which gave me three years of funding to just do science all the time, and for that I ' m very grateful. Thank you very much.

Text Sample 682: POS Dim 5 – 2004 – Score 6.00 – 2004/t000486.txt

You ' ll be happy to know that I ' ll be talking not about my own tragedy, but other people ' s tragedy. It ' s a lot easier to be lighthearted about other people ' s tragedy than your own, and I want to keep it in the spirit of the conference. So, if you believe the media accounts, being a drug dealer in the height of the crack cocaine epidemic was a very glamorous life, in the words of Virginia Postrel. There was money, there was drugs, guns, women, you know, you name it — jewelry, bling-bling — it had it all. What I ' m going to tell you today is that, in fact, based on 10 years of research, a unique opportunity to go inside a gang — to see the actual books, the financial records of the gang — that the answer turns out not to be that being in the gang was a glamorous life. But I think, more realistically, that being in a gang — selling drugs for a gang — is perhaps the worst job in all of America. And that ' s what I ' d like to convince you of today. So there are three things I want to do. First, I want to explain how and why crack cocaine had such a profound influence on inner-city gangs. Secondly, I want to tell you how somebody like me came to be able to see the inner workings of a gang — an interesting story, I think. And then third, I want to tell you, in a very superficial way, about some of the things we found when we actually got to look at the financial records, the books, of the gang. So before I do that, just one warning, which is that this presentation has been rated ' R ' by the Motion Picture Association of America. It contains adult themes, adult language. Given who is up on the stage, you ' ll be delighted to know that, in fact, there ' ll be no nudity — Unexpected wardrobe malfunctions aside. So let me start by talking about crack cocaine, and how it transformed the gang. To do that, you have to actually go back to a time before crack cocaine, in the early ' 80s, and look at it from the perspective of a gang leader. Being a gang leader in the inner city wasn ' t such a bad deal in the mid- ' 80s — the early ' 80s, let me say. Now, you had a lot of power, and you got to beat people up — you got a lot of prestige, a lot of respect. But the thing is, there was no money in it. The gang had no way to make money. You couldn ' t charge dues to the people in the gang, because the people in the gang didn ' t have any money. You couldn ' t really make any money selling marijuana — marijuana ' s too cheap, it turns out. You can ' t get rich selling marijuana. You couldn ' t sell cocaine ; cocaine ' s a great product — powdered cocaine — but you ' ve got to know rich white people. And most of the inner-city gang members didn ' t know any rich white people, so couldn ' t sell to that market. You couldn ' t really do petty crime, either. Turns out, petty crime ' s a terrible way to make a living. As a result, as a gang leader, you had, you know, power — it ' s a pretty good life — but the thing was, in the end, you were living at home with your mother. And so it wasn ' t really a career. There were limits to how powerful and important you could be if you had to live at home with your mother. Then along comes crack cocaine. And in the words of Malcolm Gladwell, crack cocaine was the extra-chunky version of tomato sauce for the inner city. Because crack cocaine was an unbelievable innovation. I don ' t have time to talk about it today, but if you think about it, I would say that in the last 25 years, of every invention or innovation that ' s occurred in this country, the biggest one in terms of impact on the well-being of people who live in the inner city, was crack cocaine. And for the worse — not for the better, but for the worse. It had a huge impact on life. So what was it about crack cocaine? It was a brilliant way of getting the brain high. Because you could smoke crack cocaine — you can ' t smoke powdered cocaine — and smoking is a much more efficient mechanism of delivering a high than snorting it. And it turned out there was this audience that didn ' t know it wanted crack cocaine, but when it came, it really did. And it was a perfect drug ; you could buy the cocaine that went into it for a dollar, sell it for five dollars. Highly addictive — the high was very short. So for fifteen minutes, you get this great high, and then when you come down, all you want to do is get high again. It created a wonderful market. And for the people who were there running the gang, it was a great way, seemingly, to make a lot of money. At least for the people on the top. So this is where we enter the **picture**. Not really me — I ' m really a bit player in all this. My co-author, Sudhir Venkatesh, is the main character. He was a math major in college who had a good heart, and decided he wanted to get a sociology PhD, came to the University of Chicago. Now, the three months before he came to Chicago, he had spent following the Grateful Dead. And in his own words, he"looked like a freak."He ' s a South Asian — very dark-skinned South Asian. Big man, and he had hair, in his words,"down to his ass."Defied all kinds of boundaries : Was he black or white? Was he man or woman? He was really a curious sight to be seen. So he showed up at the University of Chicago, and the famous sociologist William Julius Wilson was doing a book that involved surveying people all across Chicago. He took one look at Sudhir, who was going to go do some surveys for him, and decided he knew exactly the place to send him, which was to one of the toughest, most notorious housing projects not just in Chicago, but in the entire United States. So Sudhir, the suburban boy who had never really been in the inner city, dutifully took his clipboard and walked down to this housing project, gets to the first building. The first building? Well, there ' s nobody there. But he hears some voices up in the stairwell, so he climbs up the stairwell, comes around the corner, and finds a group of young African-American men playing dice. This is about 1990, peak of the crack epidemic. This is a very dangerous job, being in a gang. You don ' t like to be surprised. You don ' t like to be surprised by people who come around the corner. And the mantra was : shoot first ; ask questions later. Now, Sudhir was lucky — he was such a freak, and that clipboard probably saved his life, because they figured no other rival gang member would be coming up to shoot at them with a clipboard. So his greeting was not particularly warm, but they did say, well, OK — let ' s hear your questions on your survey. So — I kid you not — the first question on the survey that he was sent to ask was :"How do you feel about being poor and Black in America?"Makes you wonder about academics. So the choice of answers were : [A) Very Good B) Good C) Bad D) Very Bad] What Sudhir found out is, in fact, that the real answer was the following : [A) Very Good B) Good C) Bad D) Very Bad E) Fuck you] The survey was not, in the end, going to be what got Sudhir off the hook. He was held hostage overnight in the stairwell. There was a lot of gunfire, there were a lot of philosophical discussions he had with the gang members. By morning, the gang leader arrived, checked out Sudhir, decided he was no threat, and they let him go home. So Sudhir went home, took a shower, took a nap. And you and I, probably, faced with the situation, would think,"I guess I ' m going to write my dissertation on The Grateful Dead, I ' ve been following them for the last three months."Sudhir, on the other hand, got right back, walked down to the housing project, went up to the second floor, and said :"Hey, guys, I had so much **fun** hanging out with you last night, I wonder if I could do it again tonight."And that was the beginning of what turned out to be a beautiful relationship that involved Sudhir living in the housing project on and off for 10 years, hanging out in crack houses, going to jail with the gang members, having the windows shot out of his car, having the police break into his apartment and steal his computer disks — you name it. But ultimately, the story has a happy ending for Sudhir, who became one of the most respected sociologists in the country. And especially for me, as I sat in my office with my Excel spreadsheet open, waiting for Sudhir to come and deliver to me the latest load of data that he would get from the gang. It was one of the most unequal co-authoring relationships ever — But I was glad to be the beneficiary of it. So what did we find? What did we find in the gang? Well, let me say one thing : We really got access to everybody in the gang. We got an inside look at the gang, from the very **bottom** up to the very top. They trusted Sudhir, in ways that really no academic has ever — or really anybody, any outsider — has ever earned the trust of these gangs, to the point where they actually opened up what was most interesting for me — their books, the financial records they kept. They made them available to us, and we not only could study them, but we could ask them questions about what was in them. So if I have to kind of summarize very quickly in the short time I have what the **bottom** line of what I take away from the gang is, it ' s that, if I had to draw a parallel between the gang and any other organization, it would be that the gang is just like McDonald ' s, in a lot of different respects — the restaurant McDonald ' s. So first, in one way, which isn ' t maybe the most interesting way, but it ' s a good way to start — is in the way it ' s organized, the hierarchy of the gang, the way it looks. So here ' s what the org chart of the gang looks like. I don ' t know if you know much about org charts, but if you were to assign a stripped- down and simplified McDonald ' s org chart, this is exactly what it would look like. It ' s amazing, but the top level of the gang, they actually call themselves the"Board of Directors."And Sudhir says it ' s not like these guys had a very sophisticated view of what happened in American corporate life, but they had seen movies like"Wall Street,"and they had learned a little bit about what it was like to be in the real world. Now, below that board of directors, you ' ve got essentially what are regional VPs — people who control, say, the South Side of Chicago, or the West Side of Chicago. Sudhir got to know very well the guy who had the unfortunate assignment of trying to take the Iowa franchise, which, it turned out, for this black gang, was not one of the more brilliant financial endeavors they undertook. But the thing that really makes the gang seem like McDonald ' s is its franchisees. The guys who are running the local gangs — the four-square-block by four-square-block areas — they ' re just like the guys, in some sense, who are running the McDonald ’ s. They are the entrepreneurs. They get the exclusive property rights to control the drug-selling. They get the name of the gang behind them, for merchandising and marketing. And they ’ re the ones who basically make the profit or lose a profit, depending on how good they are at running the business. Now, the group I really want you to think about, though, are the ones at the **bottom** — the **foot** soldiers. These are the teenagers, typically, who ’ d be standing out on the street corner, selling the drugs. Extremely dangerous work. And important to note is that almost all of the weight, all of the people in this organization are at the **bottom** — just like McDonald ’ s. So in some sense, the **foot** soldiers are a lot like the people who are taking your order at McDonald ’ s, and it ’ s not just by chance that they ’ re like them. In fact, in these neighborhoods, they ’ d be the same people. So the same kids who are working in the gang were actually, at the very same time, typically working part-time at a place like McDonald ’ s. Which already foreshadows the main result that I ’ ve talked about, about what a crappy job it was, being in the gang. Because obviously, if being in the gang were such a wonderful, lucrative job, why in the world would these guys moonlight at McDonald ’ s? So what do the wages look like? You might be surprised. But based on being able to talk to them and to see their records, this is what it looks like in terms of the wages. The hourly wage the **foot** soldiers were earning was $3. 50 an hour. It was below the minimum wage. And this is well-documented. It ’ s easy to see by the patterns of consumption they have. It really is not fiction — it ’ s fact. There was very little money in the gang, especially at the **bottom**. Now if you managed to rise up, say, and be that local leader, the guy who ’ s the equivalent of the McDonald ’ s franchisee, you ’ d be making 100, 000 dollars a year. And that, in some ways, was the best job you could hope to get if you were growing up in one of these neighborhoods as a young black male. If you managed to rise to the very top, 200, 000 or 400, 000 dollars a year is what you ’ d hope to make. Truly, you would be a great success story. And one of the sad parts of this is that, indeed, among the many other ramifications of crack cocaine is that the most talented individuals in these communities — this is what they were striving for. They weren ’ t trying to make it in legitimate ways, because there were no legitimate channels out. This was the best way out. And it actually was the right choice, probably, to try to make it out this way. You look at this, the relationship to McDonald ’ s breaks down here. The money looks about the same. Why is it such a bad job? Well, the reason it ’ s such a bad job is that there ’ s somebody shooting at you a lot of the time. So, with shooting at you, what are the death rates? We found, in our gang — and admittedly, this was not really a standard situation ; this was a time of intense violence, of a lot of gang wars, as this gang actually became quite successful. But there were costs. And so the death rate — not to mention the rate of being arrested, sent to prison, being wounded — the death rate in our sample was seven percent per person per year. You ’ re in the gang for four years, you expect to die with about a 25 percent likelihood. That is about as high as you can get. So for comparison ’ s purposes, let ’ s think about some other walk of life you may expect might be extremely risky. Let ’ s say that you were a murderer and you were convicted of murder, and you ’ re sent to death row. It turns out, the death rates on death row from all causes, including execution : two percent a year. So it ’ s a lot safer being on death row than it is selling drugs out on the street. That gives you some pause, for those of you who believe that a death penalty ’ s going to have an enormous deterrent effect on crime. To give you a sense of just how bad the inner city was during crack — and I ’ m not really focusing on the negatives, but really, there ’ s another story to tell you there — if you look at the death rates just of random, young black males growing up in the inner city in the United States, the death rates during crack were about one percent. That ’ s extremely high. And this is violent death — it ’ s unbelievable, in some sense. To put it into perspective : if you compare this to the soldiers in Iraq, for instance, right now fighting the war : 0. 5 percent. So in some very literal way, the young black men who were growing up in this country were living in a war zone, very much in the sense that the soldiers over in Iraq are fighting in a war. So why in the world, you might ask, would anybody be willing to stand out on a street corner selling drugs for $3. 50 an hour, with a 25 percent chance of dying over the next four years? Why would they do that? And I think there are a couple answers. I think the first one is that they got fooled by history. It used to be the gang was a rite of passage ; that the young people controlled the gang ; that as you got older, you dropped out of the gang. So what happened was, the people who happened to be in the right place at the right time — the people who happened to be leading the gang in the mid-to-late- ’ 80s — became very, very wealthy. And so the logical thing to think was that they are going to age out of the gang like everybody else has, and the next generation is going to take over and get the wealth. There are striking similarities, I think, to the Internet boom. The first set of people in Silicon Valley got very, very rich. And then all of my friends said,”Maybe I should go do that, too.”And they were willing to work very cheap for stock options that never came. In some sense, that ’ s what happened, exactly, to the set of people we were looking at. They were willing to start at the **bottom**, just like, say, a first-year lawyer at a law firm is willing to start at the **bottom**, work 80-hour weeks for not that much money, because they think they ’ re going to make partner. But the rules changed, and they never got to make partner. Indeed, the same people who were running all of the major gangs in the late 1980s are still running the major gangs in Chicago today. They never passed on any of the wealth, So everybody got stuck at that $3. 50-an-hour job, and it turned out to be a disaster. The other thing the gang was very good at was marketing and trickery. And so for instance, one thing the gang would do is — the gang leaders would have big entourages, and they ’ d drive fancy cars and have fancy jewelry. So what Sudhir eventually realized as he hung out with them more, is that, really, they didn ’ t own those cars — they just leased them, because they couldn ’ t afford to own the fancy cars. And they didn ’ t really have gold jewelry, they had gold- plated jewelry. It goes back to, you know, the real-real versus the fake-real. And really, they did all sorts of things to trick the young people into thinking what a great deal the gang was going to be. So for instance, they would give a 14-year-old kid a whole roll of bills to hold. That 14-year-old kid would say to his friends,”Hey, look at all the money I got in the gang.”It wasn ’ t his money — until he spent it, and then he was in debt to the gang, and was sort of an indentured servant for a while. So I have a couple minutes. Let me do one last thing I hadn ’ t thought I ’ d have time to do, which is to talk about what we learned more generally about economics, from the study of the gang. So, economists tend to talk in technical words. Often, our theories fail quite miserably when we over the data, but what ’ s kind of interesting is that in this setting, it turned out that some of the economic theories that worked not so well in the real economy worked very well in the drug economy, in some sense, because it ’ s unfettered capitalism. Here ’ s an economic principle. This is one of the basic ideas in labor economics, called a”compensating differential.”It ’ s the idea that the increment to wages that a worker requires to leave him indifferent between performing two tasks, one which is more unpleasant than the other. Compensating differential — it ’ s why we think garbagemen might be paid more than people who work in **parks**. The words of one of the members of the gang, I think, make this clear. So it turns out — I ’ m sort of getting ahead of myself — it turns out, in the gang, when there ’ s a war going on, they actually pay the **foot** soldiers twice as much money. It ’ s exactly this concept. Because they ’ re not willing to be at risk. And the words of a gang member capture it quite nicely, he says :”Would you stand around here when all this shit...”— the shooting —”... if all this shit ’ s going on? No, right? So if I gonna be asked to put my life on the line, then front me the cash, man.”I think the gang member says it much more articulately than the economist, about what ’ s going on. Here ’ s another one. Economists talk about game theory, that every two-person game has a Nash equilibrium. Here ’ s the translation you get from the gang member. They ’ re talking about the decision of why they don ’ t go shoot — One thing that turns out to be a great business tactic in the gang : if you go and just shoot guns in the air in the other gang ’ s territory — people are afraid to go buy drugs there, they ’ re going to come into your neighborhood. Here ’ s what he says about why they don ’ t do that :”If we start shooting around there, the other gang ’ s territory, nobody, I mean, you dig it, nobody gonna step on their turf. But we gotta be careful, ’ cause they can shoot around here too and then we all fucked.”So that ’ s the same concept. Then again, sometimes economists get it wrong. One thing we observed in the data is that it looked like — the gang leader always got paid. No matter how bad it was economically, he always got himself paid. We had some theories related to cash flow, and lack of access to capital markets, and things like that. Then we asked the gang member,”Why is it you always get paid and your workers don ’ t always get paid?”His response is,”You got all these niggers below you who want your job, you dig? If you start taking losses, they see you as weak and shit.”And I thought about it and said,”CEOs often pay themselves million-dollar bonuses, even when companies are losing a lot of money. And it never would really occur to an economist that this idea of ’ weak and shit ’ could really be important.”Maybe”weak and shit”is an important hypothesis that needs more analysis. Thank you very much.

Text Sample 683: POS Dim 5 – 2006 – Score 7.00 – 2006/t000417.txt

I just want to say, over the last few years I ’ ve been â had the opportunity to do this closing conference. And I ’ ve had some incredible warm-up acts. About eight years ago, Billy Graham opened for me. And I thought that there was â I thought that there was absolutely no way in **hell** to top that. But I just wanted to say â and I mean this without irony â I think I can speak for everybody in the audience when I say that I wish to God that you were the President of the United States. OK, this is the title of my talk today. I just want to give you a quick overview. First of all, please remember I ’ m completely politically correct, and I mean everything with great affection. If any of you have sensitive stomachs or are feeling queasy, now is the time to check your Blackberry. Just to review, this is my TEDTalk. We ’ re going to do some jokes, some gags, some little skits â and then we ’ re going to talk about the L1 point. So, one of the questions I ask myself is, was this the most distressing TED ever? Let ’ s try and sum things up, shall we? Images of limb regeneration and faces filled with smallpox : 21 percent of the conference. Mentions of polar bears drowning : four percent. Images of the earth being wiped out by flood or bird flu : 64 percent. And David Pogue singing show tunes. Because this is the most distressing TED ever, I ’ ve been working with Neil Gershenfeld on next year ’ s TED Bag. And if the â if the conference is anywhere near this distressing, then we ’ re going to have a scream bag next year. It ’ s going to be a cradle-to-cradle scream bag, of course. So you ’ re going to be able to go like this. Bring it over here and open it up. Aaaah! Meanwhile, back at TED University, this wonderful woman is teaching you how to chop Sun Chips. So Robert Wright â I don ’ t know, I felt like if there was anyone that Helen needed to give antidepressants to, it might have been him. I want to deliberately interfere with his dopamine levels. He was talking about morality. Economy class morality is, we want to bomb you back to the Stone Age. Business class morality is, don ’ t bomb Japan â they built my car. And first-class morality is, don ’ t bomb Mexico â they **clean** my house. Yes, it is politically incorrect. All right, now I want to do a little bit of a thing for you... All right, now these are the wha â I ’ d say, [mumble, mumble â mumble, mumble, mumble, mumble, mumble] â ahh! So I wanted to show you guys â I wanted to talk about a revolutionary new computer interface that lets you work with images just as easily as you â as a completely natural user interface. And you can â you can use really natural hand gestures to like, go like this. Now we had a Harvard professor here â she was from Harvard, I just wanted to mention and â and she was actually a professor from Harvard. And she was talking about seven-dimensional, inverted universes. With, you know, of course, there ’ s the gravity brain. There ’ s the weak brain. And then there ’ s my weak brain, which is too â too â absolutely too weak to understand what the fuck she was talking about. Now â one of the things that is very important to me is to try and figure out what on Earth am I here for. And that ’ s why I went out and I picked up a best-selling business book. You know, it basically uses as its central premise Greek mythology. And it ’ s by a guy named Pastor Rick Warren, and it ’ s called”The Porpoise Driven Life.”And Rick is as a pagan god, which I thought was kind of appropriate, in a certain way. And now we ’ re going to have kind of a little more visualization about Rick Warren. OK. All right. Now, red is Rick Warren, and green is Daniel Dennett, OK? The scales here are religiosity from zero percent, or atheist, to 100 percent, Bible literally true. And then this is books sold â the logarithmic scale. 30, 000, 300, 000, three million, 30 million, 300 million. OK, now they ’ re duking it out. Now they ’ re duking it out. And Rick Warren ’ s kind of pulling ahead, kind of pulling ahead. Yup, and his installed base is getting a little bigger. But Darwin ’ s dangerous idea is coming back. It ’ s coming back. Let me turn the trails on, so you can see that a little bit better. Now, one of the things that ’ s very important is, Nicholas Negroponte talked to us about one lap dance per â I ’ m sorry â about One Laptop Per Child. Now let ’ s talk about some of the characteristics that are important for this revolutionary device. I ’ ll tell you a little bit about the design parameters, and then I ’ ll show it to you in person. First of all, it needs to be small. It needs to be flat, so it ’ s transportable. Lightweight. Portable. Uses very, very little power. Very, very high resolution. Has to be visible in bright daylight. Will work anywhere. And broadly applicable across many platforms. Now, we ’ ve actually done some research â Neil Gershenfeld and the Fab Labs went out into the market. They did some research ; we came back ; and we think we have the perfect prototype of what the students in the field are actually asking for. And here it is, the $100 computer. OK, OK, OK, OK â excellent, excellent. Now, I bought this device from Clifford Stoll for about 900 bucks. And he and his team of junior-high school students were doing real science. So we ’ re trying to check and trying to douse here, and see who uses marijuana. See who uses marijuana. Are we going to be able to find any marijuana, Jim Young? Only if we open enough locker doors. OK, now smallpox is an extremely distressing illness. We had Dr. Larry Brilliant talking about how we eradicated smallpox. I wanted to show you the stages of smallpox. We start. This is day one. Day two. Day three, she gets a massively big pox on her shoulder. Day three. Day four. Day five and day six. Now the good news is, because I ’ m a trained medical professional, I know that even though she ’ ll be scarred for life, she ’ s going to make a full recovery. Now the good news about **Architects** for Humanity is they ’ re really kind of the most amazing group. They ’ ve been sponsoring a design competition to come up with innovative medical housing solutions, clinic solutions, in Africa, and they ’ ve had a design competition. Now the wonderful thing is, Larry Brilliant was just appointed the head of the Google Foundation, and so he decided that he would support â he would support Cameron ’ s work. And the way he decided to support that work was by shipping over 50, 000 shipping containers of Google snacks. So I want to show you some prototypes. The U. N. â you know, they took 20 years just to add a flap to a tent, but I think we have some more exciting things. This is a home made entirely out of Fruit Roll-Ups. And those roll-up cookies coated with white chocolate. And the really wonderful thing about this is, when you ’ re done, well â you can eat it. But the thing that I ’ m really, really excited about is this incredible granola house. And the granola house has a special Sun Chip **roof** to collect water and recycle it. And it ’ s â well, on this side it has regular Sour Patch Kids and Gummy Bears to let in the light. But on this side, it has sugary Gummy Bears, to diffuse the light more slightly. And we â we wanted just to show you what this might look like in situ. So, Einstein â Einstein, tell me â what ’ s your favorite song? No, I said what ’ s your favorite song? No, I said what ’ s your favorite song?”Free Bird.”OK, so, Einstein, what ’ s your favorite singing group? Could you say that again? What ’ s your favorite singing group? OK, one more time â I ’ m just going to give you a little help. Your favorite singing group â it ’ s Diana Ross and the â Audience : Supremes! Tom Reilly : Exactly. Could we have the sound up on the laptop, please?”Free Bird”kind of reminds me that if you â if you listen to”Free Bird”backwards, this is what you might hear. Computer : Satan. Satan. Satan. Satan. Satan. Satan. Satan. TR : Now it ’ s a little hard to hear the whole message, so I wanted to â so I wanted to help you a little bit. Computer : My sweet Satan. Dan Dennett worships Satan. Buy”The Purpose-Driven Life,”or Satan will take your soul. TR : So, we ’ ve talked a lot about global warming, but, you know, as Jill said, it sounds kind of nice â good weather in the wintertime, and New York City. And as Jay Walker pointed out, that is just not scary enough. So Al, I actually think I ’ m rather good at branding. So I ’ ve tried to figure out a good design process to come up with a new term to replace”global warming.”So we started with Babel Fish. We put in global warming. And then we decided that we ’ d change it from English to Dutch â into”Het globale Verwarmen.”From Dutch to [Korean], into”Hordahordaneecheewa.”[Korean] to Portuguese : Aquecer-se Global. Then Portuguese to Pig Latin. Aquecer-se ucked-fay. And then finally back into the English, which is, we ’ re totally fucked. Now I don ’ t know about you, but Michael Shermer talked about the willingness for human beings â evolutionarily, they ’ re designed to see patterns in things. For example, in cheese sandwiches. Now can you look at that carefully and see if you see the Virgin Mary? I tried to make it a little bit clearer. Is it the Virgin Mary? Or is it Mena Trott? So, I talked to Josh Prince-Ramus about the convention center and the conferences. It ’ s getting awfully big. It ’ s getting just a little bit too big. It ’ s bursting at the seams here a little bit. So we tried to come up with a program â how we could remake this structure to better accommodate TED. So first of all we decided â that we needed about one-third bookstore, one-third Google cafe, about 20 percent registration, 80 percent luxury hotel, about five percent for restrooms. And then of course, we wanted to have the simulcast lounge, the lobby and the Steinbeck forum. Now let me show you how that literally translated into the design program. So first, one of the problems with Monterey is that if there is global warming and

Greenland melts as you say, the ocean level is going to rise 20 **feet** and flood the **hell** out of the convention center. So we ’ re going to build this new building on stilts. So we build this building on stilts, then up here â is where we ’ re going to put the new Steinbeck auditorium. And the wonderful thing about the new bookstore is, it ’ s going to be shaped in a spiral that ’ s organized by the Dewey Decimal System. Then we ’ re going to make an escalator that helps you get up there. And finally, we ’ re going to put the Marriott Hotel and the Portola Plaza on the top. Now I don ’ t know about you, but sometimes I have these images in my head of separated at birth. I don ’ t know about you, but when I see Aubrey de Grey, I immediately go to Gandalf the Grey. OK. Now, we ’ ve heard, of course, that we ’ re all soldiers here. So what I ’ d really, really like you to do now is, pick up your white piece of paper. Does everybody have their white piece of paper? And I want you to get out a pen, and I want you to write a terrorist note. If we put up the ELMO for a moment â if we put up the ELMO, then we ’ ll get, you know, I ’ ll give you a model that you can work from, OK? And then I want you to fold that note into a paper airplane. And once you ’ ve folded it into a paper airplane, I want you to take some anthrax â and I want you to put that in the paper airplane. And then I want you to throw it on Jim Young. Luckily, I was the recipient of the TED Prize this year. And I wanted to see â I want to dedicate this **film** to my father, Homer. OK. Now this **film** isn ’ t really hard enough, so I wanted to make it a little bit harder. So I ’ m going to try and do this while reciting pi. 3. 1415, 2657, 753, 8567, 24972 â â 85871, 25871, 3928, 5657, 2592, 5624. Can we cue the music please? Now I wanted to use this talk to talk about global warming a little bit. Back in 1968, you can see that the mountain range of Brokeback Mountain was covered in 151 inches of snow pack. Parenthetically, over there on the slopes, I did want to show you that black men ski. But over the years, 10 years later, the snow packs eroded, and, if you notice, the trees have started turning yellow. The water level of the lake has started drying up. A few years later, there ’ s no snow left at all. And all the trees have turned brown. This year, unfortunately the lakebed ’ s turned into an absolute cracked dry bed. And I fear, if we do nothing for our planet, in 20 years, it ’ s going to look like this. Mr. Vice President, I wish I knew how to quit you. Thank you very much.

Text Sample 684: POS Dim 5 – 2006 – Score 6.00 – 2006/t000210.txt

My name is Ursus Wehrli, and I would like to talk to you this morning about my project, Tidying Up Art. First of all â any questions so far? First of all, I have to say I ’ m not from around here. I ’ m from a completely different cultural area, maybe you noticed? I mean, I ’ m wearing a tie, first. And then secondly, I ’ m a little bit nervous because I ’ m speaking in a foreign language, and I want to apologize in advance, for any mistakes I might make. Because I ’ m from Switzerland, and I just don ’ t hope you think this is Swiss German I ’ m speaking now here. This is just what it sounds like if we Swiss try to speak American. But don ’ t worry â I don ’ t have trouble with English, as such. I mean, it ’ s not my problem, it ’ s your language after all. I am fine. After this presentation here at TED, I can simply go back to Switzerland, and you have to go on talking like this all the time. So I ’ ve been asked by the organizers to read from my book. It ’ s called”Tidying Up Art”and it ’ s, as you can see, it ’ s more or less a **picture** book. So the reading would be over very quickly. But since I ’ m here at TED, I decided to hold my talk here in a more modern way, in the spirit of TED here, and I managed to do some slides here for you. I ’ d like to show them around so we can just, you know â Actually, I managed to prepare for you some enlarged **pictures** â even better. So Tidying Up Art, I mean, I have to say, that ’ s a relatively new term. You won ’ t be familiar with it. I mean, it ’ s a hobby of mine that I ’ ve been indulging in for the last few years, and it all started out with this **picture** of the American artist, Donald Baechler I had hanging at home. I had to look at it every day and after a while I just couldn ’ t stand the mess anymore this guy was looking at all day long. Yeah, I kind of felt sorry for him. And it seemed to me even he felt really bad facing these unorganized red squares day after day. So I decided to give him a little support, and brought some order into neatly stacking the blocks on top of each other. Yeah. And I think he looks now less miserable. And it was great. With this experience, I started to look more closely at modern art. Then I realized how, you know, the world of modern art is particularly topsy-turvy. And I can show here a very good example. It ’ s actually a simple one, but it ’ s a good one to start with. It ’ s a **picture** by Paul Klee. And we can see here very clearly, it ’ s a confusion of color. Yeah. The artist doesn ’ t really seem to know where to put the different colors. The various **pictures** here of the various elements of the **picture** â the whole thing is unstructured. We don ’ t know, maybe Mr. Klee was probably in a hurry, I mean â â maybe he had to catch a plane, or something. We can see here he started out with orange, and then he already ran out of orange, and here we can see he decided to take a break for a square. And I would like to show you here my tidied up version of this **picture**. We can see now what was barely recognizable in the original : 17 red and orange squares are juxtaposed with just two green squares. Yeah, that ’ s great. So I mean, that ’ s just tidying up for beginners. I would like to show you here a **picture** which is a bit more advanced. What can you say? What a mess. I mean, you see, everything seems to have been scattered aimlessly around the space. If my room back home had looked like this, my mother would have grounded me for three days. So I ’ d like to â I wanted to reintroduce some structure into that **picture**. And that ’ s really advanced tidying up. Yeah, you ’ re right. Sometimes people clap at this point, but that ’ s actually more in Switzerland. We Swiss are famous for chocolate and cheese. Our trains run on time. We are only happy when things are in order. But to go on, here is a very good example to see. This is a **picture** by Joan Miro. And yeah, we can see the artist has drawn a few lines and shapes and dropped them any old way onto a yellow background. And yeah, it ’ s the sort of thing you produce when you ’ re doodling on the phone. And this is my â â you can see now the whole thing takes up far less space. It ’ s more economical and also more efficient. With this method Mr. Miro could have saved canvas for another **picture**. But I can see in your faces that you ’ re still a little bit skeptical. So that you can just appreciate how serious I am about all this, I brought along the patents, the specifications for some of these works, because I ’ ve had my working methods patented at the EidgenĂśssische Amt fĂźr Geistiges Eigentum in Bern, Switzerland. I ’ ll just quote from the specification.”Laut den KunstprĂźfer Dr. Albrecht â”It ’ s not finished yet.”Laut den KunstprĂźfer Dr. Albrecht GĂśtz von Ohlenhusen wird die Verfahrensweise rechtlich geschĂźtzt welche die Kunst durch spezifisch aufgerĂ¤umte RegelmĂ¤ssigkeiten **des** allgemeinen Formenschatzes neue Wirkungen zu erzielen mĂśglich wird.”Ja, well I could have translated that, but you would have been none the wiser. I ’ m not sure myself what it means but it sounds good anyway. I just realized it ’ s important how one introduces new ideas to people, that ’ s why these patents are sometimes necessary. I would like to do a short test with you. Everyone is sitting in quite an orderly fashion here this morning. So I would like to ask you all to raise your right hand. Yeah. The right hand is the one we write with, apart from the left-handers. And now, I ’ ll count to three. I mean, it still looks very orderly to me. Now, I ’ ll count to three, and on the count of three I ’ d like you all to shake hands with the person behind you. OK? One, two, three. You can see now, that ’ s a good example : even behaving in an orderly, systematic way can sometimes lead to complete chaos. So we can also see that very clearly in this next painting. This is a painting by the artist, Niki de Saint Phalle. And I mean, in the original it ’ s completely unclear to see what this tangle of colors and shapes is supposed to depict. But in the tidied up version, it ’ s plain to see that it ’ s a sunburnt woman playing volleyball. Yeah, it ’ s a â this one here, that ’ s much better. That ’ s a **picture** by Keith Haring. I think it doesn ’ t matter. So, I mean, this **picture** has not even got a proper title. It ’ s called”Untitled”and I think that ’ s appropriate. So, in the tidied-up version we have a sort of Keith Haring spare parts shop. This is Keith Haring looked at statistically. One can see here quite clearly, you can see we have 25 pale green elements, of which one is in the form of a circle. Or here, for example, we have 27 pink squares with only one pink **curve**. I mean, that ’ s interesting. One could extend this sort of statistical analysis to cover all Mr. Haring ’ s various works, in order to establish in which period the artist favored pale green circles or pink squares. And the artist himself could also benefit from this sort of listing procedure by using it to estimate how many pots of paint he ’ s likely to need in the future. One can obviously also make combinations. For example, with the Keith Haring circles and Kandinsky ’ s dots. You can add them to all the squares of Paul Klee. In the end, one has a list with which one then can arrange. Then you categorize it, then you file it, put that file in a filing cabinet, put it in your office and you can make a living doing it. Yeah, from my own experience. So I ’ m â Actually, I mean, here we have some artists that are a bit more structured. It ’ s not too bad. This is Jasper Johns. We can see here he was practicing with his ruler. But I think it could still benefit from more discipline. And I think the whole thing adds up much better if you do it like this. And here, that ’ s one of my favorites. Tidying up Rene Magritte â this is really **fun**. You know, there is a â I ’ m always being asked what inspired me to embark on all this. It goes back to a time when I was very often staying in hotels. So once I had the opportunity to stay in a ritzy, five-star hotel. And you know, there you had this little sign â I put this little sign outside the door every morning that read,”Please tidy room.”I don ’ t know if you have them over here. So actually, my room there hasn ’ t been tidied once daily, but three times a day. So after a while I decided to have a little **fun**, and before leaving the room each day I ’ d scatter a few things around the space. Like books, clothes, toothbrush, etc. And it was great. By the time I returned everything had always been neatly returned to its place. But then one morning, I hang the same little sign onto that **picture** by Vincent van Gogh. And you have to say this room hadn ’ t been tidied up since 1888. And when I returned it looked like this. Yeah, at least it is now possible to do some vacuuming. OK, I mean, I can see there are always people that like reacting that one or another **picture** hasn ’ t been properly tidied up. So we can make a short test with you. This is a **picture** by Rene Magritte, and I ’ d like you all to inwardly â like in your head, that is â to tidy that up. So it ’ s possible that some of you would make it like this. Yeah? I would actually prefer to do it more this way. Some people would make apple pie out of it. But it ’ s a very good example to see that the whole work was more of a handicraft endeavor that involved the very time-consuming job of cutting out the various elements and sticking them back in new arrangements. And it ’ s not done, as many people imagine, with the computer, otherwise it would look like this. So now I ’ ve been able to tidy up **pictures** that I ’ ve wanted to tidy up for a long time. Here is a very good example. Take Jackson Pollock, for example. It ’ s â oh, no, it ’ s â that ’ s a really hard one. But after a while, I just decided here to go all the way and put the paint back into the cans. Or you could go into three-dimensional art. Here we have the fur cup by Meret Oppenheim. Here I just brought it back to its original state. But yeah, and it ’ s great, you can even go, you know â Or we have this pointillist movement for those of you who are into art. The pointillist movement is that kind of paintings where everything is broken down into dots and pixels. And then I â this sort of thing is **ideal** for tidying up. So I once applied myself to the work of the inventor of that method, Georges Seurat, and I collected together all his dots. And now they ’ re all in here. You can count them afterwards, if you like. You see, that ’ s the wonderful thing about the tidy up art idea : it ’ s new. So there is no existing tradition in it. There is no textbooks, I mean, not yet, anyway. I mean, it ’ s”the future we will create.”But to round things up I would like to show you just one more. This is the village square by Pieter Bruegel. That ’ s how it looks like when you send everyone home. Yeah, maybe you ’ re asking yourselves where old Bruegel ’ s people went? Of course, they ’ re not gone. They ’ re all here. I just piled them up. So I ’ m â yeah, actually I ’ m kind of finished at that moment. And for those who want to see more, I ’ ve got my book downstairs in the bookshop. And I ’ m happy to sign it for you with any name of any artist. But before leaving I would like to show you, I ’ m working right now on another â in a related field with my tidying up art method. I ’ m working in a related field. And I started to bring some order into some flags. Here â that ’ s just my new proposal here for the Union Jack. And then maybe before I leave you... yeah, I think, after you have seen that I have to leave anyway. Yeah, that was a hard one. I couldn ’ t find a way to tidy that up properly, so I just decided to make it a little bit more simpler. Thank you very much.

Text Sample 685: POS Dim 5 – 2006 – Score 6.00 – 2006/t000181.txt

I grew up in Northern Ireland, right up in the very, very north end of it there, where it ’ s absolutely freezing cold. This was me running around in the back garden mid-summer. I couldn ’ t pick a career. In Ireland the obvious choice is the military, but to be honest it actually kind of sucks. My mother wanted me to be a dentist. But the problem was that people kept blowing everything up. So I actually went to school in Belfast, which was where all the action happened. And this was a pretty common sight. The school I went to was pretty boring. They forced us to learn things like Latin. The school teachers weren ’ t having much **fun**, the sports were very dirty or very painful. So I cleverly chose rowing, which I got very good at. And I was actually rowing for my school here until this fateful day, and I flipped over right in front of the entire school. And that was the finishing post right there. So this was extremely embarrassing. But our school at that time got a grant from the government, and they got an incredible computer — the research machine 3DZ — and they left the programming manuals lying around. And so students like myself with nothing to do, we would learn how to program it. Also around this time, at home, this was the computer that people were buying. It was called the Sinclair ZX80. This was a 1K computer, and you ’ d buy your programs on cassette **tape**. Actually I ’ m just going to pause for one second, because I heard that there ’ s a prerequisite to speak here at TED — you had to have a **picture** of yourself from the old days with big hair. So I brought a **picture** with big hair.. I just want to get that out of the way. So after the Sinclair ZX80 came along the very cleverly named Sinclair ZX81. And — you see the **picture** at the **bottom**? There ’ s a **picture** of a guy doing homework with his son. That ’ s what they thought they had built it for. The reality is we got the programming manual and we started making games for it. We were programming in BASIC, which is a pretty awful language for games, so we ended up learning Assembly language so we could really take control of the hardware. This is the guy that invented it, Sir Clive Sinclair, and he ’ s showing his machine. You had this same thing in America, it was called the Timex Sinclair1000. To play a game in those days you had to have an imagination to believe that you were really playing”Battlestar Galactica.”The graphics were just horrible. You had to have an even better imagination to play this game,”Death Rider.”But of course the scientists couldn ’ t help themselves. They started making their own video games. This is one of my favorite ones here, where they have rabbit breeding, so males choose the lucky rabbit. It was around this time we went from 1K to 16K, which was quite the leap. And if you ’ re wondering how much 16K is, this eBay logo here is 16K. And in that amount of memory someone programmed a full flight simulation program. And that ’ s what it looked like. I spent ages flying this flight simulator, and I honestly believed I could fly airplanes by the end of it. Here ’ s Clive Sinclair now launching his color computer. He ’ s recognized as being the father of video games in Europe. He ’ s a multi-millionaire, and I think that ’ s why he ’ s smiling in this photograph. So I went on for the next 20 years or so making a lot of different games. Some of the highlights were things like”The Terminator,””Aladdin,”the”Teenage Mutant Hero Turtles.”Because I was from the United Kingdom, they thought the word ninja was a little too mean for children, so they decided to call it hero instead. I personally preferred the Spanish version, which was”Tortugas Ninja.”That was much better. Then the last game I did was based on trying to get the video game industry and Hollywood to actually work together on something — instead of licensing from each other, to actually work. Now, Chris did ask me to bring some statistics with me, so I ’ ve done that. The video game industry in 2005 became a 29 billion dollar business. It grows every year. Last year was the biggest year. By 2008, we ’ re going to kick the butt of the music industry. By 2010, we ’ re going to hit 42 billion. 43 percent of gamers are female. So there ’ s a lot more female gamers than people are really aware. The average age of gamers? Well, obviously it ’ s for children, right? Well, no, actually it ’ s 30 years old. And interestingly, the people who buy the most games are 37. So 37 is our target audience. All video games are violent. Of course the newspapers love to beat on this. But 83 percent of games don ’ t have any mature content whatsoever, so it ’ s just not true. Online gaming statistics. I brought some stuff on”World of Warcraft.”It ’ s 5. 5 million players. It makes about 80 million bucks a month in subscriptions. It costs 50 bucks just to install it on your computer, making the publisher about another 275 million. The game costs about 80 million dollars to make, so basically it pays for itself in about a month. A player in a game called”Project Entropia”actually bought his own island for 26, 500 dollars. You have to remember that this is not a real island. He didn ’ t actually buy anything, just some data. But he got great terms on it. This purchase included mining and hunting rights, ownership of all land on the island, and a castle with no furniture included. This market is now estimated at over 800 million dollars annually. And what ’ s interesting about it is the market was actually created by the gamers themselves. They found clever ways to trade items and to sell their accounts to each other so that they could make money while they were playing their games. I dove onto eBay a couple of days ago just to see what was gong on, typed in World of Warcraft, got 6, 000 items. I liked this one the best : a level 60 Warlock with lots of epics for 174, 000 dollars. It ’ s like that guy obviously had some pain while making it. So as far as popularity of games, what do you think these people are doing here? It turns out they ’ re actually in Hollywood Bowl in Los Angeles listening to the L. A. Philharmonic playing video game music. That

’ s what the show looks like. You would expect it to be cheesy, but it ’ s not. It ’ s very, very epic and a very beautiful concert. And the people that went there absolutely loved it. What do you think these people are doing? They ’ re actually bringing their computers so they can play games against each other. And this is happening in every city around the world. This is happening in your local cities too, you ’ re probably just not aware of it. Now, Chris told me that you had a timeline video a few years ago here just to show how video game graphics have been improving. I wanted to update that video and give you a new look at it. But what I want you to do is to try to understand it. We ’ re on this **curve**, and the graphics are getting so ridiculously better. And I ’ m going to show you up to maybe 2007. But I want you to try and think about what games could look like 10 years from now. So we ’ re going to start that video. Video : Throughout human history people have played games. As man ’ s intellect and technology have evolved so too have the games he plays. David Perry : The thing again I want you to think about is, don ’ t look at these graphics and think of that ’ s the way it is. Think about that ’ s where we are right now, and the **curve** that we ’ re on means that this is going to continue to get better. This is an example of the kind of graphics you need to be able to draw if you wanted to get a job in the video game industry today. You need to be really an incredible artist. And once we get enough of those guys, we ’ re going to want more fantasy artists that can create places we ’ ve never been to before, or characters that we ’ ve just never seen before. So the obvious thing for me to talk about today is graphics and audio. But if you were to go to a game developers conference, what they ’ re all talking about is emotion, purpose, meaning, understanding and feeling. You ’ ll hear about talks like, can a video game make you cry? And these are the kind of topics we really actually care about. I came across a student who ’ s absolutely excellent at expressing himself, and this student agreed that he would not show his video to anybody until you here at TED had seen it. So I ’ d like to play this video. So this is a student ’ s opinion on what his experience of games are. Video : I, like many of you, live somewhere between reality and video games. Some part of me — a true living, breathing person — has become programmed, electronic and virtual. The boundary of my brain that divides real from fantasy has finally begun to crumble. I ’ m a video game addict and this is my story. In the year of my birth the Nintendo Entertainment System also went into development. I played in the backyard, learned to read, and even ate some of my vegetables. Most of my childhood was spent playing with Legos. But as was the case for most of my generation, I spent a lot of time in front of the TV. Mr. Rogers, Walt Disney, Nick Junior, and roughly half a million commercials have undoubtedly left their mark on me. When my parents bought my sister and I our first Nintendo, whatever inherent addictive quality this early interactive electronic entertainment possessed quickly took hold of me. At some point something clicked. With the combination of simple, interactive stories and the warmth of the TV set, my simple 16-bit Nintendo became more than just an escape. It became an alternate existence, my virtual reality. I ’ m a video game addict, and it ’ s not because of a certain number of hours I have spent playing, or nights I have gone without sleep to finish the next level. It is because I have had life-altering experiences in virtual space, and video games had begun to erode my own understanding of what is real and what is not. I ’ m addicted, because even though I know I ’ m losing my grip on reality, I still crave more. From an early age I learned to invest myself emotionally in what unfolded before me on screen. Today, after 20 years of watching TV geared to make me emotional, even a decent insurance commercial can bring tears to my eyes. I am just one of a new generation that is growing up. A generation who may experience much more meaning through video games than they will through the real world. Video games are nearing an evolutionary leap, a point where game worlds will look and feel just as real as the **films** we see in theatres, or the news we watch on TV. And while my sense of free will in these virtual worlds may still be limited, what I do learn applies to my real life. Play enough video games and eventually you will really believe you can snowboard, fly a plane, drive a nine-second quarter mile, or kill a man. I know I can. Unlike any pop culture phenomenon before it, video games actually allow us to become part of the machine. They allow us to sublimate into the culture of interactive, downloaded, streaming, HD reality. We are interacting with our entertainment. I have come to expect this level of interaction. Without it, the problems faced in the real world — poverty, war, disease and genocide — lack the levity they should. Their importance blends into the sensationalized drama of prime time TV. But the beauty of video games today lies not in the lifelike graphics, the vibrating joysticks or virtual surround sound. It lies in that these games are beginning to make me emotional. I have fought in wars, feared for my own survival, watched my cohorts die on beaches and woods that look and feel more real than any textbook or any news story. The people who create these games are smart. They know what makes me scared, excited, panicked, proud or sad. Then they use these emotions to dimensionalize the worlds they create. A well-designed video game will seamlessly weave the user into the fabric of the virtual experience. As one becomes more experienced the awareness of physical control melts away. I know what I want and I do it. No buttons to push, no triggers to pull, just me and the game. My fate and the fate of the world around me lie inside my hands. I know violent video games make my mother worry. What troubles me is not that video game violence is becoming more and more like real life violence, but that real life violence is starting to look more and more like a video game. These are all troubles outside of myself. I, however, have a problem very close to home. Something has happened to my brain. Perhaps there is a single part of our brain that holds all of our gut instincts, the things we know to do before we even think. While some of these instincts may be innate, most are learned, and all of them are hardwired into our brains. These instincts are essential for survival in both real and virtual worlds. Only in recent years has the technology behind video games allowed for a true overlap in stimuli. As gamers we are now living by the same laws of physics in the same cities and doing many of the same things we once did in real life, only virtually. Consider this — my real life car has about 25, 000 miles on it. In all my driving games, I ’ ve driven a total of 31, 459 miles. To some degree I ’ ve learned how to drive from the game. The sensory cues are very similar. It ’ s a funny feeling when you have spent more time doing something on the TV than you have in real life. When I am driving down a road at sunset all I can think is, this is almost as beautiful as my games are. For my virtual worlds are perfect. More beautiful and rich than the real world around us. I ’ m not sure what the implications of my experience are, but the potential for using realistic video game stimuli in repetition on a vast number of loyal participants is frightening to me. Today I believe Big Brother would find much more success brainwashing the masses with video games rather than just simply TVs. Video games are **fun**, engaging, and leave your brain completely vulnerable to re-programming. But maybe brainwashing isn ’ t always bad. Imagine a game that teaches us to respect each other, or helps us to understand the problems we ’ re all facing in the real world. There is a potential to do good as well. It is critical, as these virtual worlds continue to mirror the real world we live in, that game developers realize that they have tremendous responsibilities before them. I ’ m not sure what the future of video games holds for our civilization. But as virtual and real world experiences increasingly overlap there is a greater and greater potential for other people to feel the same way I do. What I have only recently come to realize is that beyond the graphics, sound, game play and emotion it is the power to break down reality that is so fascinating and addictive to me. I know that I am losing my grip. Part of me is just waiting to let go. I know though, that no matter how amazing video games may become, or how flat the real world may seem to us, that we must stay aware of what our games are teaching us and how they leave us feeling when we finally do unplug. DP : Wow. I found that video very, very thought provoking, and that ' s why I wanted to bring it here for you guys to see. And what was interesting about it is the obvious choice for me to talk about was graphics and audio. But as you heard, Michael talked about all these other elements as well. Video games give an awful lot of other things too, and that ' s why people get so addicted. The most important one being **fun**. The name of this track is"The Magic To Come."Who is that going to come from? Is it going to come from the best directors in the world as we thought it probably would? I don ' t think so. I think it ' s going to come from the children who are growing up now that aren ' t stuck with all of the stuff that we remember from the past. They ' re going to do it their way, using the tools that we ' ve created. The same with students or highly creative people, writers and people like that. As far as colleges go, there ' s about 350 colleges around the world teaching video game courses. That means there ' s literally thousands of new ideas. Some of the ideas are really dreadful and some of them are great. There ' s nothing worse than having to listen to someone try and pitch you a really bad video game idea. Chris Anderson : You ' re off, you ' re off. That ' s it. He ' s out of time. DP : I ' ve just got a little tiny bit more if you ' ll indulge me. CA : Go ahead. I ' m going to stay right here though. DP : This is just a cool shot, because this is students coming to school after class. The school is closed ; they ' re coming back at midnight because they want to pitch their video game ideas. I ' m sitting at the front of the class, and they ' re actually pitching their ideas. So it ' s hard to get students to come back to class, but it is possible. This is my daughter, her name ' s Emma, she ' s 17 months old. And I ' ve been asking myself, what is Emma going to experience in the video game world? And as I ' ve shown here, we have the audience. She ' s never going to know a world where you can ' t press a button and have millions of people ready to play. You know, we have the technology. She ' s never going to know a world where the graphics just aren ' t stunning and really immersive. And as the student video showed, we can impact and move. She ' s never going to know a world where video games aren ' t incredibly emotional and will probably make her cry. I just hope she likes video games. So, my closing thought. Games on the surface seem simple entertainment, but for those that like to look a little deeper, the new paradigm of video games could open entirely new frontiers to creative minds that like to think big. Where better to challenge those minds than here at TED? Thank you. Chris Anderson : David Perry. That was awesome.

Text Sample 686: POS Dim 5 – 2003 – Score 8.00 – 2003/t000434.txt

Good morning everyone. First of all, it ' s been fantastic being here over these past few days. And secondly, I feel it ' s a great honor to kind of wind up this extraordinary gathering of people, these amazing talks that we ' ve had. I feel that I ' ve fitted in, in many ways, to some of the things that I ' ve heard. I came directly here from the deep, deep tropical rainforest in Ecuador, where I was out — you could only get there by a plane — with indigenous people with paint on their faces and parrot feathers on their headdresses, where these people are fighting to try and keep the oil companies, and keep the roads, out of their forests. They ’ re fighting to develop their own way of living within the forest in a world that ’ s **clean**, a world that isn ’ t contaminated, a world that isn ’ t polluted. And what was so amazing to me, and what fits right in with what we ’ re all talking about here at TED, is that there, right in the middle of this rainforest, was some solar **panels** — the first in that part of Ecuador — and that was mainly to bring water up by pump so that the women wouldn ’ t have to go down. The water was **cleaned**, but because they got a lot of batteries, they were able to store a lot of electricity. So every house — and there were, I think, eight houses in this little community — could have light for, I think it was about half an hour each evening. And there is the Chief, in all his regal finery, with a laptop computer. And this man, he has been outside, but he ’ s gone back, and he was saying,”You know, we have suddenly jumped into a whole new era, and we didn ’ t even know about the white man 50 years ago, and now here we are with laptop computers, and there are some things we want to learn from the modern world. We want to know about health care. We want to know about what other people do — we ’ re interested in it. And we want to learn other languages. We want to know English and French and perhaps Chinese, and we ’ re good at languages.”So there he is with his little laptop computer, but fighting against the might of the pressures — because of the debt, the foreign debt of Ecuador — fighting the pressure of World Bank, IMF, and of course the people who want to exploit the forests and take out the oil. And so, coming directly from there to here. But, of course, my real field of expertise lies in an even different kind of civilization — I can ’ t really call it a civilization. A different way of life, a different being. We ’ ve talked earlier — this wonderful talk by Wade Davis about the different cultures of the humans around the world — but the world is not composed only of human beings ; there are also other animal beings. And I propose to bring into this TED conference, as I always do around the world, the voice of the animal kingdom. Too often we just see a few slides, or a bit of **film**, but these beings have voices that mean something. And so, I want to give you a greeting, as from a chimpanzee in the forests of Tanzania — Ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh, ooh! I ’ ve been studying chimpanzees in Tanzania since 1960. During that time, there have been modern technologies that have really transformed the way that field biologists do their work. For example, for the first time, a few years ago, by simply collecting little fecal samples we were able to have them analyzed — to have DNA profiling done — so for the first time, we actually know which male chimps are the fathers of each individual infant. Because the chimps have a very promiscuous mating society. So this opens up a whole new **avenue** of research. And we use GSI — geographic whatever it is, GSI — to determine the range of the chimps. And we ’ re using — you can see that I ’ m not really into this kind of stuff — but we ’ re using satellite imagery to look at the deforestation in the area. And of course, there ’ s developments in infrared, so you can watch animals at night, and equipment for recording by video, and **tape** recording is getting lighter and better. So in many, many ways, we can do things today that we couldn ’ t do when I began in 1960. Especially when chimpanzees, and other animals with large brains, are studied in captivity, modern technology is helping us to search for the upper levels of cognition in some of these non-human animals. So that we know today, they ’ re capable of performances that would have been thought absolutely impossible by science when I began. I think the chimpanzee in captivity who is the most skilled in intellectual performance is one called Ai in Japan — her name means love — and she has a wonderfully sensitive partner working with her. She loves her computer — she ’ ll leave her big group, and her running water, and her trees and everything. And she ’ ll come in to sit at this computer — it ’ s like a video game for a kid ; she ’ s hooked. She ’ s 28, by the way, and she does things with her computer screen and a touch pad that she can do faster than most humans. She does very complex tasks, and I haven ’ t got time to go into them, but the amazing thing about this female is she doesn ’ t like making mistakes. If she has a bad run, and her score isn ’ t good, she ’ ll come and reach up and tap on the glass — because she can ’ t see the experimenter — which is asking to have another go. And her concentration — she ’ s already concentrated hard for 20 minutes or so, and now she wants to do it all over again, just for the satisfaction of having done it better. And the food is not important — she does get a tiny reward, like one raisin for a correct response — but she will do it for nothing, if you tell her beforehand. So here we are, a chimpanzee using a computer. Chimpanzees, gorillas, orangutans also learn human sign language. But the point is that when I was first in Gombe in 1960 — I remember so well, so vividly, as though it was yesterday — the first time, when I was going through the vegetation, the chimpanzees were still running away from me, for the most part, although some were a little bit acclimatized — and I saw this dark shape, hunched over a termite mound, and I peered with my binoculars. It was, fortunately, one adult male whom I ’ d named David Greybeard — and by the way, science at that time was telling me that I shouldn ’ t name the chimps ; they should all have numbers ; that was more scientific. Anyway, David Greybeard — and I saw that he was picking little pieces of grass and using them to fish termites from their underground nest. And not only that — he would sometimes pick a leafy twig and strip the leaves — modifying an object to make it suitable for a specific purpose — the beginning of tool-making. The reason this was so exciting and such a breakthrough is at that time, it was thought that humans, and only humans, used and made tools. When I was at school, we were defined as man, the toolmaker. So that when Louis Leakey, my mentor, heard this news, he said,”Ah, we must now redefine ’ man, ’ redefine ’ tool, ’ or accept chimpanzees as humans.”We now know that at Gombe alone, there are nine different ways in which chimpanzees use different objects for different purposes. Moreover, we know that in different parts of Africa, wherever chimps have been studied, there are completely different tool-using behaviors. And because it seems that these patterns are passed from one generation to the next, through observation, imitation and practice — that is a definition of human culture. What we find is that over these 40-odd years that I and others have been studying chimpanzees and the other great apes, and, as I say, other mammals with complex brains and social systems, we have found that after all, there isn ’ t a sharp line dividing humans from the rest of the animal kingdom. It ’ s a very wuzzy line. It ’ s getting wuzzier all the time as we find animals doing things that we, in our arrogance, used to think was just human. The chimps — there ’ s no time to discuss their fascinating lives — but they have this long childhood, five years of suckling and sleeping with the mother, and then another three, four or five years of emotional dependence on her, even when the next child is born. The importance of learning in that time, when behavior is flexible — and there ’ s an awful lot to learn in chimpanzee society. The long-term affectionate supportive bonds that develop throughout this long childhood with the mother, with the brothers and sisters, and which can last through a lifetime, which may be up to 60 years. They can actually live longer than 60 in captivity, so we ’ ve only done 40 years in the wild so far. And we find chimps are capable of true compassion and altruism. We find in their non-verbal communication — this is very rich — they have a lot of sounds, which they use in different circumstances, but they also use touch, posture, gesture, and what do they do? They kiss ; they embrace ; they hold hands.

They pat one another on the back ; they swagger ; they shake their fist — the kind of things that we do, and they do them in the same kind of context. They have very sophisticated cooperation. Sometimes they hunt — not that often, but when they hunt, they show sophisticated cooperation, and they share the prey. We find that they show emotions, similar to — maybe sometimes the same — as those that we describe in ourselves as happiness, sadness, fear, despair. They know mental as well as physical suffering. And I don ' t have time to go into the information that will prove some of these things to you, save to say that there are very bright students, in the best universities, studying emotions in animals, studying personalities in animals. We know that chimpanzees and some other creatures can recognize themselves in mirrors —"self"as opposed to"other."They have a sense of humor, and these are the kind of things which traditionally have been thought of as human prerogatives. But this teaches us a new respect — and it ' s a new respect not only for the chimpanzees, I suggest, but some of the other amazing animals with whom we share this planet. Once we ' re prepared to admit that after all, we ' re not the only beings with personalities, minds and above all feelings, and then we start to think about ways we use and abuse so many other sentient, sapient creatures on this planet, it really gives cause for deep shame, at least for me. So, the sad thing is that these chimpanzees — who ' ve perhaps taught us, more than any other creature, a little humility — are in the wild, disappearing very fast. They ' re disappearing for the reasons that all of you in this room know only too well. The deforestation, the growth of human populations, needing more land. They ' re disappearing because some timber companies go in with clear-cutting. They ' re disappearing in the heart of their range in Africa because the big multinational logging companies have come in and made roads — as they want to do in Ecuador and other parts where the forests remain untouched — to take out oil or timber. And this has led in Congo basin, and other parts of the world, to what is known as the bush- meat trade. This means that although for hundreds, perhaps thousands of years, people have lived in those forests, or whatever habitat it is, in harmony with their world, just killing the animals they need for themselves and their families — now, suddenly, because of the roads, the hunters can go in from the towns. They shoot everything, every single thing that moves that ' s bigger than a small rat ; they sun-dry it or smoke it. And now they ' ve got transport ; they take it on the logging trucks or the mining trucks into the towns where they sell it. And people will pay more for bush-meat, as it ' s called, than for domestic meat — it ' s culturally preferred. And it ' s not sustainable, and the huge logging camps in the forest are now demanding meat, so the Pygmy hunters in the Congo basin who ' ve lived there with their wonderful way of living for so many hundreds of years are now corrupted. They ' re given weapons ; they shoot for the logging camps ; they get money. Their culture is being destroyed, along with the animals upon whom they depend. So, when the logging camp moves, there ' s nothing left. We talked already about the loss of human cultural diversity, and I ' ve seen it happening with my own eyes. And the grim **picture** in Africa — I love Africa, and what do we see in Africa? We see deforestation ; we see the desert spreading ; we see massive hunger ; we see disease and we see population growth in areas where there are more people living on a certain piece of land than the land can possibly support, and they ' re too poor to buy food from elsewhere. Were the people that we heard about yesterday, on the Easter Island, who cut down their last tree — were they stupid? Didn ' t they know what was happening? Of course, but if you ' ve seen the crippling poverty in some of these parts of the world it isn ' t a question of"Let ' s leave the tree for tomorrow.""How am I going to feed my family today? Maybe I can get just a few dollars from this last tree which will keep us going a little bit longer, and then we ’ ll pray that something will happen to save us from the inevitable end.”So, this is a pretty grim **picture**. The one thing we have, which makes us so different from chimpanzees or other living creatures, is this sophisticated spoken language — a language with which we can tell children about things that aren ’ t here. We can talk about the distant past, plan for the distant future, discuss ideas with each other, so that the ideas can grow from the accumulated wisdom of a group. We can do it by talking to each other ; we can do it through video ; we can do it through the written word. And we are abusing this great power we have to be wise stewards, and we ’ re destroying the world. In the developed world, in a way, it ’ s worse, because we have so much access to knowledge of the stupidity of what we ’ re doing. Do you know, we ’ re bringing little babies into a world where, in many places, the water is poisoning them? And the air is harming them, and the food that ’ s grown from the contaminated land is poisoning them. And that ’ s not just in the far-away developing world ; that ’ s everywhere. Do you know we all have about 50 chemicals in our bodies we didn ’ t have about 50 years ago? And so many of these diseases, like asthma and certain kinds of cancers, are on the increase around places where our filthy toxic waste is dumped. We ’ re harming ourselves around the world, as well as harming the animals, as well as harming nature herself — Mother Nature, that brought us into being ; Mother Nature, where I believe we need to spend time, where there ’ s trees and flowers and birds for our good psychological development. And yet, there are hundreds and hundreds of children in the developed world who never see nature, because they ’ re growing up in concrete and all they know is virtual reality, with no opportunity to go and lie in the sun, or in the forest, with the dappled sun-specks coming down from the canopy above. As I was traveling around the world, you know, I had to leave the forest — that ’ s where I love to be. I had to leave these fascinating chimpanzees for my students and field staff to continue studying because, finding they dwindled from about two million 100 years ago to about 150, 000 now, I knew I had to leave the forest to do what I could to raise awareness around the world. And the more I talked about the chimpanzees ’ plight, the more I realized the fact that everything ’ s interconnected, and the problems of the developing world so often stem from the greed of the developed world, and everything was joining together, and making — not sense, hope lies in sense, you said — it ’ s making a nonsense. How can we do it? Somebody said that yesterday. And as I was traveling around, I kept meeting young people who ’ d lost hope. They were feeling despair, they were feeling,”Well, it doesn ’ t matter what we do ; eat, drink and be merry, for tomorrow we die. Everything is hopeless — we ’ re always being told so by the media.”And then I met some who were angry, and anger that can turn to violence, and we ’ re all familiar with that. And I have three little grandchildren, and when some of these students would say to me at high school or university, they ’ d say,”We ’ re angry,”or”We ’ re filled with despair, because we feel you ’ ve compromised our future, and there ’ s nothing we can do about it.”And I looked in the eyes of my little grandchildren, and think how much we ’ ve harmed this planet since I was their age. I feel this deep shame, and that ’ s why in 1991 in Tanzania, I started a program that ’ s called Roots and Shoots. There ’ s little brochures all around outside, and if any of you have anything to do with children and care about their future, I beg that you pick up that brochure. And Roots and Shoots is a program for hope. Roots make a firm foundation. Shoots seem tiny, but to reach the sun they can break through brick walls. See the brick walls as all the problems that we ’ ve inflicted on this planet. Then, you see, it is a message of hope. Hundreds and thousands of young people around the world can break through, and can make this a better world. And the most important message of Roots and Shoots is that every single individual makes a difference. Every individual has a role to play. Every one of us impacts the world around us everyday, and you scientists know that you can ' t actually — even if you stay in bed all day, you ' re breathing oxygen and giving out CO_2, and probably going to the loo, and things like that — you ' re making a difference in the world. So, the Roots and Shoots program involves youth in three kinds of projects. And these are projects to make the world around them a better place. One project to show care and concern for your own human community. One for animals, including domestic animals — and I have to say, I learned everything I know about animal behavior even before I got to Gombe and the chimps from my dog, Rusty, who was my childhood companion. And the third kind of project : something for the local environment. So what the kids do depends first of all, how old are they — and we go now from pre-school right through university. It ' s going to depend whether they ' re inner-city or rural. It ' s going to depend if they ' re wealthy or impoverished. It ' s going to depend which part, say, of America they ' re in. We ' re in every state now, and the problems in Florida are different from the problems in New York. It ' s going to depend on which country they ' re in — and we ' re already in 60-plus countries, with about 5, 000 active groups — and there are groups all over the place that I keep hearing about that I ' ve never even heard of, because the kids are taking the program and spreading it themselves. Why? Because they ' re buying into it, and they ' re the ones who get to decide what they ' re going to do. It isn ' t something that their parents tell them, or their teachers tell them. That ' s effective, but if they decide themselves,"We want to **clean** this river and put the fish back that used to be there. We want to clear away the toxic soil from this area and have an organic garden. We want to go and spend time with the old people and hear their stories and record their oral histories. We want to go and work in a dog shelter. We want to learn about animals. We want..."You know, it goes on and on, and this is very hopeful for me. As I travel around the world 300 days a year, everywhere there ' s a group of Roots and Shoots of different ages. Everywhere there are children with shining eyes saying,"Look at the difference we ' ve made."And now comes the technology into it, because with this new way of communicating electronically these kids can communicate with each other around the world. And if anyone is interested to help us, we ' ve got so many ideas but we need help — we need help to create the right kind of system that will help these young people to communicate their excitement. But also — and this is so important — to communicate their despair, to say,"We ' ve tried this and it doesn ' t work, and what shall we do?"And then, lo and behold, there ' s another group answering these kids who may be in America, or maybe this is a group in Israel, saying,"Yeah, you did it a little bit wrong. This is how you should do it."The philosophy is very simple. We do not believe in violence. No violence, no bombs, no guns. That ' s not the way to solve problems. Violence leads to violence, at least in my view. So how do we solve? The tools for solving the problems are knowledge and understanding. Know the facts, but see how they fit in the big **picture**. Hard work and persistence — don ' t give up — and love and compassion leading to respect for all life. How many more minutes? Two, one? Chris Anderson : One — one to two. Jane Goodall : Two, two, I ' m going to take two. Are you going to come and drag me off? Anyway — so basically, Roots and Shoots is beginning to change young people ' s lives. It ' s what I ' m devoting most of my energy to. And I believe that a group like this can have a very major impact, not just because you can share technology with us, but because so many of you have children. And if you take this program out, and give it to your children, they have such a good opportunity to go out and do good, because they ' ve got parents like you. And it ’ s been so clear how much you all care about trying to make this world a better place. It ’ s very encouraging. But the kids do ask me — and this won ’ t take more than two minutes, I promise — the kids say,”Dr. Jane, do you really have hope for the future? You travel, you see all these horrible things happening.”Firstly, the human brain — I don ’ t need to say anything about that. Now that we know what the problems are around the world, human brains like yours are rising to solve those problems. And we ’ ve talked a lot about that. Secondly, the **resilience** of nature. We can destroy a river, and we can bring it back to life. We can see a whole area desolated, and it can be brought back to bloom again, with time or a little help. And thirdly, the last speaker talked about — or the speaker before last, talked about the indomitable human spirit. We are surrounded by the most amazing people who do things that seem to be absolutely impossible. Nelson Mandela — I take a little piece of limestone from Robben Island Prison, where he labored for 27 years, and came out with so little bitterness, he could lead his people from the horror of apartheid without a bloodbath. Even after the 11th of September — and I was in New York and I felt the fear — nevertheless, there was so much human courage, so much love and so much compassion. And then as I went around the country after that and felt the fear — the fear that was leading to people feeling they couldn ’ t worry about the environment any more, in case they seemed not to be patriotic — and I was trying to encourage them, somebody came up with a little quotation from Mahatma Gandhi,”If you look back through human history, you see that every evil regime has been overcome by good.”And just after that a woman brought me this little bell, and I want to end on this note. She said,”If you ’ re talking about hope and peace, ring this. This bell is made from **metal** from a defused landmine, from the killing fields of Pol Pot — one of the most evil regimes in human history — where people are now beginning to put their lives back together after the regime has crumbled. So, yes, there is hope, and where is the hope? Is it out there with the politicians? It ’ s in our hands. It ’ s in your hands and my hands and those of our children. It ’ s really up to us. We ’ re the ones who can make a difference. If we lead lives where we consciously leave the lightest possible ecological footprints, if we buy the things that are ethical for us to buy and don ’ t buy the things that are not, we can change the world overnight. Thank you.

Text Sample 687: POS Dim 5 – 2003 – Score 7.00 – 2003/t000405.txt

When you think about **resilience** and technology it ’ s actually much easier. You ’ re going to see some other speakers today, I already know, who are going to talk about breaking-bones stuff, and, of course, with technology it never is. So it ’ s very easy, comparatively speaking, to be resilient. I think that, if we look at what happened on the Internet, with such an incredible last half a dozen years, that it ’ s hard to even get the right analogy for it. A lot of how we decide, how we ’ re supposed to react to things and what we ’ re supposed to expect about the future depends on how we bucket things and how we categorize them. And so I think the tempting analogy for the boom-bust that we just went through with the Internet is a gold rush. It ’ s easy to think of this analogy as very different from some of the other things you might pick. For one thing, both were very real. In 1849, in that Gold Rush, they took over $700 million worth of gold out of California. It was very real. The Internet was also very real. This is a real way for humans to communicate with each other. It ’ s a big deal. Huge boom. Huge boom. Huge bust. Huge bust. You keep going, and both things are lots of hype. I don ’ t have to remind you of all the hype that was involved with the Internet — like GetRich. com. But you had the same thing with the Gold Rush."Gold. Gold. Gold."Sixty-eight rich men on the Steamer Portland. Stacks of yellow **metal**. Some have 5, 000. Many have more. A few bring out 100, 000 dollars each. People would get very excited about this when they read these articles."The Eldorado of the United States of America : the discovery of inexhaustible gold mines in California."And the parallels between the Gold Rush and the Internet Rush continue very strongly. So many people left what they were doing. And what would happen is — and the Gold Rush went on for years. People on the East Coast in 1849, when they first started to get the news, they thought,"Ah, this isn ' t real."But they keep hearing about people getting rich, and then in 1850 they still hear that. And they think it ' s not real. By about 1852, they ' re thinking,"Am I the stupidest person on Earth by not rushing to California?"And they start to decide they are. These are community affairs, by the way. Local communities on the East Coast would get together and whole teams of 10, 20 people would caravan across the United States, and they would form companies. These were typically not solitary efforts. But no matter what, if you were a lawyer or a banker, people dropped what they were doing, no matter what skill set they had, to go pan for gold. This guy on the left, Dr. Richard Beverley Cole, he lived in Philadelphia and he took the Panama route. They would take a ship down to Panama, across the isthmus, and then take another ship north. This guy, Dr. Toland, went by covered wagon to California. This has its parallels, too. Doctors leaving their practices. These are both very successful — a physician in one case, a surgeon in the other. Same thing happened on the Internet. You get DrKoop. com. In the Gold Rush, people literally jumped ship. The San Francisco harbor was clogged with 600 ships at the peak because the ships would get there and the crews would abandon to go search for gold. So there were literally 600 captains and 600 ships. They turned the ships into hotels, because they couldn ' t sail them anywhere. You had dotcom fever. And you had gold fever. And you saw some of the excesses that the dotcom fever created and the same thing happened. The fort in San Francisco at the time had about 1, 300 soldiers. Half of them deserted to go look for gold. And they wouldn ' t let the other half out to go look for the first half because they were afraid they wouldn ' t come back. And one of the soldiers wrote home, and this is the sentence that he put :"The struggle between right and six dollars a month and wrong and 75 dollars a day is a rather severe one."They had bad burn rate in the Gold Rush. A very bad burn rate. This is actually from the Klondike Gold Rush. This is the White Pass Trail. They loaded up their mules and their horses. And they didn ' t plan right. And they didn ' t know how far they would really have to go, and they overloaded the horses with hundreds and hundreds of pounds of stuff. In fact it was so bad that most of the horses died before they could get where they were going. It got renamed the"Dead Horse Trail."And the Canadian Minister of the Interior wrote this at the time :"Thousands of pack horses lie dead along the way, sometimes in bunches under the cliffs, with pack saddles and packs where they ' ve fallen from the rock above, sometimes in tangled masses, filling the mud holes and furnishing the only footing for our poor pack animals on the march, often, I regret to say, exhausted, but still alive, a fact we were unaware of, until after the miserable wretches turned beneath the hooves of our cavalcade. The eyeless sockets of the pack animals everywhere account for the myriads of ravens along the road. The inhumanity which this trail has been witness to, the heartbreak and suffering which so many have undergone, cannot be imagined. They certainly cannot be described."And you know, without the smell that would have accompanied that, we had the same thing on the Internet : very bad burn rate calculations. I ' ll just play one of these and you

’ ll remember it. This is a commercial that was played on the Super Bowl in the year 2000. : Bride #1 : You said you had a large selection of invitations. Clerk : But we do. Bride #2 : Then why does she have my invitation? Announcer : What may be a little thing to some... Bride #3 : You are mine, little man. Announcer : Could be a really big deal to you. Husband #1 : Is that your wife? Husband #2 : Not for another 15 minutes. Announcer : After all, it ’ s your special day. OurBeginning. com. Life ’ s an event. Announce it to the world. Jeff Bezos : It ’ s very difficult to figure out what that ad is for. But they spent three and a half million dollars in the 2000 Super Bowl to air that ad, even though, at the time, they only had a million dollars in annual revenue. Now, here ’ s where our analogy with the Gold Rush starts to diverge, and I think rather severely. And that is, in a gold rush, when it ’ s over, it ’ s over. Here ’ s this guy :”There are many men in Dawson at the present time who feel keenly disappointed. They ’ ve come thousands of miles on a perilous trip, risked life, health and property, spent months of the most arduous labor a man can perform and at length with expectations raised to the highest pitch have reached the coveted goal only to discover the fact that there is nothing here for them.”And that was, of course, the very common story. Because when you take out that last piece of gold — and they did incredibly quickly. I mean, if you look at the 1849 Gold Rush — the entire American river region, within two years — every stone had been turned. And after that, only big companies who used more sophisticated mining technologies started to take gold out of there. So there ’ s a much better analogy that allows you to be incredibly optimistic and that analogy is the electric industry. And there are a lot of similarities between the Internet and the electric industry. With the electric industry you actually have to — one of them is that they ’ re both sort of thin, horizontal, enabling layers that go across lots of different industries. It ’ s not a specific thing. But electricity is also very, very broad, so you have to sort of narrow it down. You know, it can be used as an incredible means of transmitting power. It ’ s an incredible means of coordinating, in a very fine-grained way, information flows. There ’ s a bunch of things that are interesting about electricity. And the part of the electric revolution that I want to focus on is sort of the golden age of appliances. The killer app that got the world ready for appliances was the light bulb. So the light bulb is what wired the world. And they weren ’ t thinking about appliances when they wired the world. They were really thinking about — they weren ’ t putting electricity into the home ; they were putting lighting into the home. And, but it really — it got the electricity. It took a long time. This was a huge — as you would expect — a huge capital build out. All the streets had to be torn up. This is work going on down in lower Manhattan where they built some of the first electric power generating stations. And they ’ re tearing up all the streets. The Edison Electric Company, which became Edison General Electric, which became General Electric, paid for all of this digging up of the streets. It was incredibly expensive. But that is not the — and that ’ s not the part that ’ s really most similar to the Web. Because, remember, the Web got to stand on top of all this heavy infrastructure that had been put in place because of the long-distance phone network. So all of the cabling and all of the heavy infrastructure — I ’ m going back now to, sort of, the explosive part of the Web in 1994, when it was growing 2, 300 percent a year. How could it grow at 2, 300 percent a year in 1994 when people weren ’ t really investing in the Web? Well, it was because that heavy infrastructure had already been laid down. So the light bulb laid down the heavy infrastructure, and then home appliances started coming into being. And this was huge. The first one was the electric fan — this was the 1890 electric fan. And the appliances, the golden age of appliances really lasted — it depends how you want to measure it — but it ’ s anywhere from 40 to 60 years. It goes on a long time. It starts about 1890. And the electric fan was a big success. The electric iron, also very big. By the way, this is the beginning of the asbestos lawsuit. There ' s asbestos under that handle there. This is the first vacuum cleaner, the 1905 Skinner Vacuum, from the Hoover Company. And this one weighed 92 pounds and took two people to operate and cost a quarter of a car. So it wasn ' t a big seller. This was truly, truly an early-adopter product — the 1905 Skinner Vacuum. But three years later, by 1908, it weighed 40 pounds. Now, not all these things were highly successful. This is the electric tie press, which never really did catch on. People, I guess, decided that they would not wrinkle their ties. These never really caught on either : the electric shoe warmer and drier. Never a big seller. This came in, like, six different colors. I don ' t know why. But I thought, you know, sometimes it ' s just not the right time for an invention ; maybe it ' s time to give this one another shot. So I thought we could build a Super Bowl ad for this. We ' d need the right partner. And I thought that really — I thought that would really work, to give that another shot. Now, the toaster was huge because they used to make toast on open fires, and it took a lot of time and attention. I want to point out one thing. This is — you guys know what this is. They hadn ' t invented the electric socket yet. So this was — remember, they didn ' t wire the houses for electricity. They wired them for lighting. So your — your appliances would plug in. They would — each room typically had a light bulb socket at the top. And you ' d plug it in there. In fact, if you ' ve seen the Carousel of Progress at Disney World, you ' ve seen this. Here are the cables coming up into this light fixture. All the appliances plug in there. And you would just unscrew your light bulb if you wanted to plug in an appliance. The next thing that really was a big, big deal was the washing machine. Now, this was an object of much envy and lust. Everybody wanted one of these electric washing machines. On the **left-hand** side, this was the soapy water. And there ' s a rotor there — that this motor is spinning. And it would **clean** your clothes. This is the **clean** rinse-water. So you ' d take the clothes out of here, put them in here, and then you ' d run the clothes through this electric wringer. And this was a big deal. You ' d keep this on your porch. It was a little bit messy and kind of a pain. And you ' d run a long cord into the house where you could screw it into your light socket. And that ' s actually kind of an important point in my presentation, because they hadn ' t invented the off switch. That was to come much later — the off switch on appliances — because it didn ' t make any sense. I mean, you didn ' t want this thing clogging up a light socket. So you know, when you were done with it, you unscrewed it. That ' s what you did. You didn ' t turn it off. And as I said before, they hadn ' t invented the electric outlet either, so the washing machine was a particularly dangerous device. And there are — when you research this, there are gruesome descriptions of people getting their hair and clothes caught in these devices. And they couldn ' t yank the cord out because it was screwed into a light socket inside the house. And there was no off switch, so it wasn ' t very good. And you might think that that was incredibly stupid of our ancestors to be plugging things into a light socket like this. But, you know, before I get too far into condemning our ancestors, I thought I ' d show you : this is my conference room. This is a total kludge, if you ask me. First of all, this got installed upside down. This light socket — and so the cord keeps falling out, so I **taped** it in. This is supposed — don ' t even get me started. But that ' s not the worst one. This is what it looks like under my desk. I took this **picture** just two days ago. So we really haven ' t progressed that much since 1908. It ' s a total, total mess. And, you know, we think it ' s getting better, but have you tried to install 802. 11 yourself? I challenge you to try. It ' s very hard. I know Ph. D. s in Computer Science — this process has brought them to tears, absolute tears. And that ’ s assuming you already have DSL in your house. Try to get DSL installed in your house. The engineers who do it everyday can ’ t do it. They have to — typically, they come three times. And one friend of mine was telling me a story : not only did they get there and have to wait, but then the engineers, when they finally did get there, for the third time, they had to call somebody. And they were really happy that the guy had a speakerphone because then they had to wait on hold for an hour to talk to somebody to give them an access code after they got there. So we ’ re not — we ’ re pretty kludge-y ourselves. By the way, DSL is a kludge. I mean, this is a twisted pair of copper that was never designed for the purpose it ’ s being put to — you know it ’ s the whole thing — we ’ re very, very primitive. And that ’ s kind of the point. Because, you know, **resilience** — if you think of it in terms of the Gold Rush, then you ’ d be pretty depressed right now because the last nugget of gold would be gone. But the good thing is, with innovation, there isn ’ t a last nugget. Every new thing creates two new questions and two new opportunities. And if you believe that, then you believe that where we are — this is what I think — I believe that where we are with the incredible kludge — and I haven ’ t even talked about user interfaces on the Web — but there ’ s so much kludge, so much terrible stuff — we are at the 1908 Hurley washing machine stage with the Internet. That ’ s where we are. We don ’ t get our hair caught in it, but that ’ s the level of primitiveness of where we are. We ’ re in 1908. And if you believe that, then stuff like this doesn ’ t bother you. This is 1996 :”All the negatives add up to making the online experience not worth the trouble.”1998 :”Amazon. toast.”In 1999 :”Amazon. bomb.”My mom hates this **picture**. She — but you know, if you really do believe that it ’ s the very, very beginning, if you believe it ’ s the 1908 Hurley washing machine, then you ’ re incredibly optimistic. And I do think that that ’ s where we are. And I do think there ’ s more innovation ahead of us than there is behind us. And in 1917, Sears — I want to get this exactly right. This was the advertisement that they ran in 1917. It says :”Use your electricity for more than light.”And I think that ’ s where we are. We ’ re very, very early. Thank you very much.

Text Sample 688: POS Dim 5 – 2003 – Score 7.00 – 2003/t000364.txt

I am known best for human-powered flight, but that was just one thing that got me going in the sort of things that I ’ m working in now. As a youngster, I was very interested in model airplanes, ornithopters, autogyros, helicopters, gliders, power planes, indoor models, outdoor models, everything, which I just thought was a lot of **fun**, and wondered why most other people didn ’ t share my same enthusiasm with them. And then, navy pilot training, and, after college, I got into sailplane flying, power plane flying, and considered the sailplanes as a sort of hobby and **fun**, but got tangled up with some great professor types, who convinced me and everybody else in the field that this was a good way to get into really deep science. While this was all going on, I was in the field of weather modification, although getting a Ph. D. in aeronautics. The weather modification subject was getting started, and as a graduate student, I could go around to the various talks that were being given, on a hitchhiker ride to the East Coast, and so on. And everybody would talk to me, but all the professionals in the field hated each other, and they wouldn ’ t communicate. And as a result, I got the absolutely unique background in that field, and started a company, which did more research in weather modification than anybody, and there are a lot of things that I just can ’ t go into. But then, 1971 started AeroVironment, with no employees — then one or two, three, and sort of fumbled along on trying to get interesting projects. We had AirDynamisis, who, like I, did not want to work for aerospace companies on some big, many year project, and so we did our small projects, and the company slowly grew. The thing that is exciting was, in 1976, I suddenly got interested in the human-powered airplane because I ’ d made a made a loan to a friend of 100, 000 dollars, or I guaranteed the money at the bank. He needed them — he needed the money for starting a company. The company did not succeed, and he couldn ’ t pay the money back, and I was the guarantor of the note. So, I had a $100, 000 debt, and I noticed that the Kramer prize for human-powered flight, which had then been around for — — 17 years at the time, was 50, 000 pounds, which, at the exchange rate, was just about 100, 000 dollars. So suddenly, I was interested in human-powered flight — — and did not — the way I approached it, first, thinking about ways to make the planes, was just like they ’ d been doing in England, and not succeeding, and I gave it up. I figured, nah, there isn ’ t any simple, easy way. But then, got off on a vacation trip, and was studying bird flight, just for the **fun** of it, and you can watch a bird soaring around in circles, and measure the time, and estimate the bank angle, and immediately, figure out its speed, and the turning radius, and so on, which I could do in the car, as we ’ re driving along on a vacation trip — — with my three sons, young sons, helping me, but ridiculing the whole thing very much. But that began thinking about how birds went around, and then how airplanes would, how hang gliders would fly, and then other planes, and the idea of the Gossamer-Condor-type airplane quickly emerged, was so logical, one should have thought of it in the first place, but one didn ’ t. And it was just, keep the weight down — 70 pounds was all it weighed — but let the size swell up, like a hang glider, but three times the span, three times the cord. You ’ re down to a third of the speed, a third of the power, and a good bicyclist can put out that power, and that worked, and we won the prize a year later. We didn ’ t — a lot of flying, a lot of experiments, a lot of things that didn ’ t work, and ones that did work, and the plane kept getting a little better, a little better. Got a good pilot, Brian Allen, to operate it, and finally, succeeded. But unfortunately, about 65, 000 dollars was spent on the project. And there was only about 30 to help retire the debt. But fortunately, Henry Kramer, who put up the prize for — that was a one-mile flight — put up a new prize for flying the English Channel, 21 miles. And he thought it would take another 18 years for somebody to win that. We realized that if you just **cleaned** up our Gossamer Condor a little bit, the power to fly would be decreased a little bit, and if you decrease the power required a little, the pilot can fly a much longer period of time. And Brian Allen was able, in a miraculous flight, to get the Gossamer Albatross across the English Channel, and we won the 100, 000-pound, 200, 000-dollar prize for that. And when all expenses were paid, the debt was handled, and everything was fine. It turned out that giving the planes to the **museum** was worth much more than the debt, so for five years, six years, I only had to pay one third income tax. So, there were good economic reasons for the project, but — — that ’ s not, well, the project was done entirely for economic reasons, and we have not been involved in any human-powered flight since then — — because the prizes are all over. But that sure started me thinking about various things, and immediately, we began making a solar-powered plane because we felt solar power was going to be so important for the country and the world, we didn ’ t want the small funding in the government to be decreased, which is what the government was trying to do with it. And we thought a solar-powered plane wouldn ’ t really make sense, but you could do it and it would get a lot of publicity for solar power and maybe help that field. And that project continued, did succeed, and we then got into other projects in aviation and **mechanical** things and ground devices. But while this was going on, in 1982, I got a prize from the Lindbergh Foundation — their annual prize — and I had to prepare a paper on it, which collected all my varied thoughts and varied interests over the years. This was the one chance that I had to focus on what I, really, was after, and what was important. And to my surprise, I realized the importance of environmental issues, which Charles Lindbergh devoted the last third of his life to, and preparing that paper did me a lot of good. I thought back about if I was a space traveler, and came and visited Earth every 5, 000 years. And for a few thousand visits, I would see the same thing every time, the little differences in the Earth. But this last time, just coming round, right now, suddenly, there ’ d be huge changes in the environment, in the concentration of people, and it was just unbelievable, the amount of — all the change in it. I wanted to — well, one of the biggest changes is, 200 years ago, we began using coal from underground, which has a lot of pollution, and 100 years ago, began getting gasoline from underground, with a lot of pollution. And gasoline consumption, or production, will reach its limit in about ten years, and then go down, and we wonder what ’ s going to happen with transportation. I wanted to show the slide — this slide, I think, is the most important one any of you will see, ever, because — — it shows nature versus humans, and goes from 1850 to 2050. And so, the year 2000, you see there. And this is the weight of all air and land vertebrates. Humans and muskrats and giraffes and birds and so on, are — the red line goes up. That ’ s the humans and livestock and pets portion. The green line goes down. That ’ s the wild nature portion. Humans, livestock and pets are, now, 98 percent of the total world ’ s mass of vertebrates on land and air. And you don ’ t know what the future will hold, but it ’ s not going to get a lower percentage. Ten thousand years ago, the humans and livestock and pets were not even one tenth of one percent and wouldn ’ t even have been visible on such a **curve**. Now they are 98 percent, and it, I think, shows human domination of the Earth. I give a talk to some remarkable high school students each summer, and ask them, after they ’ ve asked me questions, and I give them a talk and so on. Then I ask them questions. What ’ s the population of the Earth? What ’ s the population of the Earth going to be when you ’ re the age of your parents? Which I ’ d never, really — they had never, really, thought about but, now, they think about it. And then, what population of the Earth would be an equilibrium that could continue on, and be for 2050, 2100, 2150? And they form little groups, all fighting with each other, and when I leave, two hours later, most of them are saying about 2 billion people, and they don ’ t have any clue about how to get down to 2 billion, nor do I, but I think they ’ re right and this is a serious problem. Rachel Carson was thinking of these, and came out with”Silent Spring,”way back.”Solar Manifesto”by Hermann Scheer, in Germany, claims all energy on Earth can be derived, for every country, from solar energy and water, and so on. You don ’ t need to dig down for these chemicals, and we can do things much more efficiently. Let ’ s have the next slide. So this just summarizes it.”Over billions of years, on a unique sphere, chance has painted a thin covering of life — complex and probable, wonderful and fragile. Suddenly, we humans, a recently arrived species, no longer subject to the checks and balances inherent in nature, have grown in population, technology and intelligence to a position of terrible power. We, now, wield the paintbrush.”We ’ re in charge. It ’ s frightening. And I do a painting every 20 or 25 years. This is the last one. And [it] shows the Earth in a time flag : on the right, in trilobites and dinosaurs and so on ; and over the triangle, we now get to civilization and TV and traffic jams and so on. I have no idea of what comes next, so I just used robotic and natural cockroaches as the future, as a little warning. And two weeks after this **drawing** was done, we actually had our first project contract, at AeroVironment, on robotic cockroaches, which was very frightening to me. Well, that ’ ll be all the slides. As time went on, we stopped our environmental programs. We focused more on the really serious energy problems of the future, and we produced products for the company. And we developed the impact car that General Motors made, the EV1, out of — and got the Air Resources Board to have the regulations that stimulated the electric cars, but they ’ ve since come apart. And we ’ ve done a lot of things, small drone airplanes and so on. I have a Helios. We have the first video. Narrator : With a wingspan of 247 **feet**, this makes her larger than a Boeing 747. Her designers ’ attention to detail and her construction gives Helios ’ structure the flexibility and strength to deal with the turbulence encountered in the atmosphere. This enables her to easily ride through the air currents as if she ’ s sliding along on the ocean waves. Paul MacCready : The wings could touch together on top and not break. We think. Narrator : And Helios now begins the process of turning her back to the sun, to maximize the power from her solar array. As the sky gets darker, and the outside air temperatures drop below minus 100 degrees Fahrenheit, the most environmentally hostile segment of Helios ’ s journey has gone by without notice, except for being recorded by specially designed data acquisition systems and their associated sensors. Approaching a peak radar altitude of 96, 863 **feet**, at 4:12 p. m., Helios is standing on top of 98 percent of the Earth ’ s atmosphere. This is more than 10, 000 **feet** higher than the previous world ’ s altitude record held by the SR-71 Blackbird. PM : That plane has many purposes, but it ’ s aimed for communications, and it can fly so slowly that it ’ ll just stay up at 65, 000 **feet**. Eventually, it will be able to have to stay up day, night, day, night, for six months at a time, acting like the synchronous satellite, but only ten miles above the Earth. Let ’ s have the next video. This shows the other end of the spectrum. Narrator : A tiny airplane, the AV Pointer serves for surveillance. In effect a pair of roving eyeglasses, a cutting-edge example of where miniaturization can lead if the operator is remote from the vehicle. It is convenient to carry, assemble, and launch by hand. Battery-powered, it is silent and rarely noticed. It sends high-resolution video **pictures** back to the operator. With onboard GPS, it can navigate autonomously, and it is rugged enough to self-land without damage. PM : Okay, and let ’ s have the next. That plane is widely used by the military, now, in all their operations. Let ’ s have the next video. Alan Alda : He ’ s got it, he ’ s got it, he ’ s got it on his head. We ’ re going to end our visit with Paul MacCready ’ s flying circus by meeting his son, Tyler, who, with his two brothers, helped build the Gossamer Condor, 25 years ago. Tyler MacCready : You can chase it, like this, for hours. AA : When they got bored with their father ’ s project, they invented an extraordinary little plane of their own. TM : And I can control it by putting the lift on one side of the wing, or on the other. AA : They called it their Walkalong Glider. I ’ ve never seen anything like that. How old were you when you invented that? TM : Oh, 10, 11. TM : 12, something like that. PM : And Tyler ’ s here to show you the Walkalong. TM : All right. You all got a couple of these in your gift bags, and one of the first things, the production version seemed to dive a little bit, and so I would just suggest you bend the wing tips up a little bit before you try flying it. I ’ ll give you a demonstration of how it works. The idea is that it soars on the lift over your body, like a seagull soaring on a cliff. As the wind comes up, it has to go over the cliff, so as you walk through the air, it goes around your body, some has to go over you. And so you just keep the glider positioned in that up current. The launch is the difficult part : you ’ ve got to hold it high up, over your head, and you start walking forward, and just let go of it, and you can control it like that. And then also, like it said in the video, you can turn it left or right just by putting the lift under one wing or another. So I can do it —

oops, that was going to be a right turn. Okay, this one will be a left turn. Here, but — — anyway. And that ' s it, so you can just control it, wherever you want, and it ' s just hours of **fun**. And these are no longer in production, so you have real collector ' s items. And this, we just wanted to show you — if we can get the video running on this, yeah — just an example of a little video surveillance. This was flying around in the party last night, and — — you can see how it just can fly around, and you can spy on anybody you want. And that ' s it. I was going to bring an airplane, but I was worried about hitting people in here, so I thought this would be a little bit more gentle. And that ' s it, yeah, just a few inventions. All right.

Text Sample 689: POS Dim 5 – 2007 – Score 9.00 – 2007/t000408.txt

So I really consider myself a storyteller. But I don ' t really tell stories in the usual way, in the sense that I don ' t usually tell my own stories. Instead, I ' m really interested in building tools that allow large numbers of other people to tell their stories, people all around the world. I do this because I think that people actually have a lot in common. I think people are very similar, but I also think that we have trouble seeing that. You know, as I look around the world I see a lot of gaps, and I think we all see a lot of gaps. And we define ourselves by our gaps. There ' s language gaps, there ' s ethnicity and racial gaps, there ' s age gaps, there ' s gender gaps, there ' s sexuality gaps, there ' s wealth and money gaps, there ' s education gaps, there ' s also religious gaps. You know, we have all these gaps and I think we like our gaps because they make us feel like we identify with something, some smaller community. But I think that actually, despite our gaps, we really have a lot in common. And I think one thing we have in common is a very deep need to express ourselves. I think this is a very old human desire. It ' s nothing new. But the thing about self-expression is that there ' s traditionally been this imbalance between the desire that we have to express ourselves and the number of sympathetic friends who are willing to stand around and listen. This, also, is nothing new. Since the dawn of human history, we ' ve tried to rectify this imbalance by making art, writing poems, singing songs, scripting editorials and sending them in to a newspaper, gossiping with friends. This is nothing new. What ' s new is that in the last several years a lot of these very traditional physical human activities, these acts of self-expression, have been moving onto the Internet. And as that ' s happened, people have been leaving behind footprints, footprints that tell stories of their moments of self-expression. And so what I do is, I write computer programs that study very large sets of these footprints, and then try to draw conclusions about the people who left them — what they feel, what they think, what ' s different in the world today than usual, these sorts of questions. One project that explores these ideas, which was made about a year ago, is a piece called We Feel Fine. This is a piece that every two or three minutes **scans** the world ' s newly-posted blog entries for occurrences of the phrases"I feel"or"I am feeling."And when it finds one of those phrases, it grabs the sentence up to the period, and then automatically tries to deduce the age, gender and geographical location of the person that wrote that sentence. Then, knowing the geographical location and the time, we can also then figure out the weather when that person wrote the sentence. All of this information is saved in a database that collects about 20, 000 feelings a day. It ' s been running for about a year and a half. It ' s reached about seven-and-a-half million human feelings now. And I ' ll show you a glimpse of how this information is then visualized. So this is We Feel Fine. What you see here is a madly swarming mass of particles, each of which represents a single human feeling that was stated in the last few hours. The color of each particle corresponds to the type of feeling inside — so that happy, positive feelings are brightly colored. And sad, negative feelings are darkly colored. The diameter of each dot represents the length of the sentence inside, so that the large dots contain large sentences, and the small dots contain small sentences. Any dot can be clicked and expanded. And we see here,"I would just feel so much better if I could curl up in his arms right now and feel his affection for me in the embrace of his body and the tenderness of his lips."So it gets pretty hot and steamy sometimes in the world of human emotions. And all of these are stated by people :"I know that objectively it really doesn ' t mean much, but after spending so many years as a small fish in a big pond, it ' s nice to feel bigger again."The dots exhibit human qualities. They kind of have their own physics, and they swarm wildly around, kind of exploring the world of life. And then they also exhibit curiosity. You can see a few of them are swarming around the cursor right now. You can see some other ones are swarming around the **bottom** left corner of the screen around six words. Those six words represent the six movements of We Feel Fine. We ' re currently seeing Madness. There ' s also Murmurs, Montage, Mobs, Metrics and Mounds. And I ' ll walk you through a few of those now. Murmurs causes all of the feelings to fly to the ceiling. And then, one by one, in reverse chronological order, they excuse themselves, entering the scrolling list of feelings."I feel a bit better now.""I feel confused and unsure of what the **hell** I want to do.""I feel gypped out of something awesome here.""I feel so free ; I feel so good.""I feel like I ' m in this fog of depression that I can ' t get out of."And you can click any of these to go out and visit the blog from which it was collected. And in that way, you can connect with the authors of these statements if you feel some degree of empathy. The next movement is called Montage. Montage causes all of the feelings that contain photographs to become extracted and display themselves in a grid. This grid is then said to represent the **picture** of the world ' s feelings in the last few hours, if you will. Each of these can be clicked and we can blow it up. We see,"I just feel like I ' m not going to have **fun** if it ' s not the both of us."That was from someone in Michigan. We see,"I feel like I have been at a computer all day."These are automatically constructed using the found objects :"I think I feel a little full."The next movement is called mobs. Mobs provides different statistical breakdowns of the population of the world ' s feelings in the last few hours. We see that"better"is the most frequent feeling right now, followed by"good,""bad,""guilty,""right,""down,""sick"and so on. We can also get a gender breakdown. And we see that women are slightly more prolific talking about their emotions in the last few hours than men. We can do an age breakdown, which gives us a histogram of the world ' s emotional distribution by age. We see people in their twenties are the most prolific, followed by teenagers, and then people in their thirties, and it dies out very quickly from there. In weather, the feelings assume the physical characteristics of the weather that they represent, so that the ones collected on a sunny day swirl around as if they ' re part of the sun. The cloudy ones float along as if they ' re on a breeze. The rainy ones fall down as if they ' re in a rainstorm, and the snowy ones kind of flutter to the ground. Finally, location causes the feelings to move to their positions on a world map showing the geographical distribution of feelings. Metrics provides more numerical views on the data. We see that the world is feeling"used"at 3. 3 times the normal level right now. They ' re feeling"warm"at 2. 9 times the normal level, and so on. Other views are also available. Here are gender, age, weather, location. The final movement is called Mounds. It ' s a bit different from the others. Mounds visualizes the entire dataset as large, gelatinous blobs which kind of jiggle. And if I hold down my cursor, they do a little dance. We see”better”is the most frequent feeling, followed by”bad.”And then if I go over here, the list begins to scroll, and there are actually thousands of feelings that have been collected. You can see the little pink cursor moving along, representing our position. Here we see people that feel”slipping,””nauseous,””responsible.”There ’ s also a search capability, if you ’ re interested in finding out about a certain population. For instance, you could find women who feel”addicted”in their 20s when it was cloudy in Bangladesh. But I ’ ll spare you that. So here are some of my favorite montages that have been collected :”I feel so much of my dad alive in me that there isn ’ t even room for me.””I feel very lonely.””I need to be in some backwoods redneck town so that I can feel beautiful.””I feel invisible to you.””I wouldn ’ t hide it if society didn ’ t make me feel like I needed to.””I feel in love with Carolyn.””I feel so naughty.””I feel these weirdoes are actually an asset to college life.””I love how I feel today.”So as you can see, We Feel Fine uses a technique that I call”passive observation.”What I mean by that is that it passively observes people as they live their lives. It **scans** the world ’ s blogs and looks at what people are writing, and these people don ’ t know they ’ re being watched or interviewed. And because of that, you end up getting very honest, candid, sincere responses that are often very moving. And this is a technique that I usually prefer in my work because people don ’ t know they ’ re being interviewed. They ’ re just living life, and they end up just acting like that. Another technique is directly questioning people. And this is a technique that I explored in a different project, the Yahoo! Time Capsule, which was designed to take a fingerprint of the world in 2006. It was divided into ten very simple themes — love, anger, sadness and so on — each of which contained a single, very open-ended question put to the world : What do you love? What makes you angry? What makes you sad? What do you believe in? And so on. The time capsule was available for one month online, translated into 10 languages, and this is what it looked like. It ’ s a spinning globe, the surface of which is entirely composed of the **pictures** and words and **drawings** of people that submitted to the time capsule. The ten themes radiate out and orbit the time capsule. You can sift through this data and see what people have submitted. This is in response to, What ’ s beautiful?”Miss World.”There are two modes to the time capsule. There ’ s One World, which presents the spinning globe, and Many Voices, which splits the data out into **film** strips and lets you sift through them one by one. So this project was punctuated by a really amazing event, which was held in the desert outside Albuquerque in New Mexico at the Jemez Pueblo, where for three consecutive nights, the contents of the capsule were projected onto the sides of the ancient Red Rock Canyon walls, which stand about 200 **feet** tall. It was really incredible. And we also projected the contents of the time capsule as binary code using a 35-watt **laser** into outer space. You can see the orange line leaving the desert floor at about a 45 degree angle there. This was amazing because the first night I looked at all this information and really started seeing the gaps that I talked about earlier — the differences in age, gender and wealth and so on. But, you know, as I looked at this more and more and more, and saw these images go across the rocks, I realized I was seeing the same archetypal events depicted again and again and again. You know : weddings, births, funerals, the first car, the first kiss, the first camel or horse — depending on the culture. And it was really moving. And this **picture** here was taken the final night from a distant cliff about two miles away, where the contents of the capsule were being beamed into space. And there was something very moving about all of this human expression being shot off into the night sky. And it started to make me think a lot about the night sky, and how humans have always used the night sky to project their great stories. You know, as a child in Vermont, on a farm where I grew up, I would often look up into the dark sky and see the three star belt of Orion, the Hunter. And as an adult, I ’ ve been more aware of the great Greek myths playing out in the sky overhead every night. You know, Orion facing the roaring bull. Perseus flying to the rescue of Andromeda. Zeus battling Chronos for control of Mount Olympus. I mean, these are the great tales of the Greeks. And it caused me to wonder about our world today. And it caused me to wonder specifically, if we could make new constellations today, what would those look like? What would those be? If we could make new **pictures** in the sky, what would we draw? What are the great stories of today? And those are the questions that inspired my new project, which is debuting here today at TED. Nobody ’ s seen this yet, publicly. It ’ s called Universe : Revealing Our Modern Mythology. And it uses this **metaphor** of an interactive night sky. So, it ’ s my great pleasure now to show this to you. So, Universe will open here. And you ’ ll see that it leads with a shifting star field, and there ’ s an Aurora Borealis in the background, kind of morphing with color. The color of the Aurora Borealis can be controlled using this single bar of color at the **bottom**, and we ’ ll put it down here to red. So you see this kind of — these stars moving along. Now, these aren ’ t just little points of light, little pixels. Each of those stars actually represents a specific event in the real world — a quote that was stated by somebody, an image, a news story, a person, a company. You know, some kind of heroic personality. And you might notice that as the cursor begins to touch some of these stars, that shapes begin to emerge. We see here there ’ s a little man walking along, or maybe a woman. And we see here a photograph with a head. You can start to see words emerging here. And those are the constellations of today. And I can turn them all on, and you can see them moving across the sky now. This is the universe of 2007, the last two months. The data from this is global news coverage from thousands of news sources around the world. It ’ s using the API of a really great company that I work with in New York, actually, called Daylife. And it ’ s kind of the zeitgeist view at this level of the world ’ s current mythology over the last couple of months. So we can see where it ’ s emerging here, like President Ford, Iraq, Bush. And we can actually isolate just the words — I call them secrets — and we can cause them to form an alphabetical list. And we see Anna Nicole Smith playing a big role recently. President Ford — this is Gerald Ford ’ s funeral. We can actually click anything in Universe and have it become the center of the universe, and everything else will enter its orbit. So, we ’ ll click Ford, and now that becomes the center. And the things that relate to Ford enter its orbit and swirl around it. We can isolate just the photographs, and we now see those. We can click on one of those and have the photograph be the center of the universe. Now the things that relate to it are swirling around. We can click on this and we see this iconic image of Betty Ford kissing her husband ’ s coffin. In Universe, there ’ s kind of no end. It just goes infinitely, and you can just kind of click on stuff. This is a photographic representation, called Snapshots. But we can actually be more specific in defining our universe. So, if we want to, let ’ s check out what Bill Clinton ’ s universe looks like. And let ’ s see, in the past week, what he ’ s been up to. So now, we have a new universe, which is just constrained to all things Bill Clinton. We can have his constellations emerge here. We can pull out his secrets, and we see that it has a lot to do with candidates, Hillary, presidential, Barack Obama. We can see the stories that Bill Clinton is taking part in right now. Any of those can be opened up. So we see Obama and the Clintons meet in Alabama. You can see that this is an important story ; there are a lot of things in its orbit. If we open this up, we get different perspectives on this story. You can click any of those to go out and read the article at the source. This one ’ s from Al Jazeera. We can also see the superstars. These would be the people that are kind of the looming heroes and heroines in the universe of Bill Clinton. So there ’ s Bill Clinton, Hillary, Iraq, George Bush, Barack Obama, Scooter Libby — these are kind of the people of Bill Clinton. We can also see a world map, so this shows us the geographic reach of Bill Clinton in the last week or so. We can see he ’ s been focused in America because he ’ s been campaigning, probably, but a little bit of action over here in the Middle East. And then we can also see a timeline. So we see that he was a bit quiet on Saturday, but he was back to work on Sunday morning, and actually been tapering off since then this week. And it ’ s not limited to just people or dates, but we can actually put in concepts also. So if I put in climate change for all of 2006, we ’ ll see what that universe looks like. Here we have our star field. Here we have our shapes. Here we have our secrets. So we see again, climate change is large : Nairobi, global conference, environmental. And there are also quotes that you can see, if you ’ re interested in reading about quotes on climate change. You know, this is really an infinite thing. The superstars of climate change in 2006 : United States, Britain, China. You know, these are the towering countries that kind of define this concept. So this is a piece that demands exploration. This will be online in several days, probably next Tuesday. And you ’ ll all be able to use it and kind of explore what your own personal mythology might be. You ’ ll notice that in Daylife — rather, in Universe — it supports both the notion of a global mythology, which is represented by something as broad as, say, 2007, and also a personal mythology. As you search for the things that are important to you in your world, and then see what the constellations of those might look like. So it ’ s been a pleasure. Thank you very much.

Text Sample 690: POS Dim 5 – 2007 – Score 8.00 – 2007/t000351.txt

Now, have any of y ’ all ever looked up this word? You know, in a dictionary? Yeah, that ’ s what I thought. How about this word? Here, I ’ ll show it to you. Lexicography : the practice of compiling dictionaries. Notice â we ’ re very specific â that word”compile.”The dictionary is not carved out of a piece of granite, out of a lump of rock. It ’ s made up of lots of little bits. It ’ s little discrete â that ’ s spelled D-I-S-C-R-E-T-E â bits. And those bits are words. Now one of the perks of being a lexicographer â besides getting to come to TED â is that you get to say really **fun** words, like lexicographical. Lexicographical has this great pattern : it ’ s called a double dactyl. And just by saying double dactyl, I ’ ve sent the geek needle all the way into the red. But”lexicographical”is the same pattern as”higgledy- piggledy.”Right? It ’ s a **fun** word to say, and I get to say it a lot. Now, one of the non-perks of being a lexicographer is that people don ’ t usually have a kind of warm, fuzzy, snuggly image of the dictionary. Right? Nobody hugs their dictionaries. But what people really often think about the dictionary is, they think more like this. Just to let you know, I do not have a lexicographical whistle. But people think that my job is to let the good words make that difficult **left-hand** turn into the dictionary, and keep the bad words out. But the thing is, I don ’ t want to be a traffic cop. For one thing, I just do not do uniforms. And for another, deciding what words are good and what words are bad is actually not very easy. And it ’ s not very **fun**. And when parts of your job are not easy or **fun**, you kind of look for an excuse not to do them. So if I had to think of some kind of occupation as a **metaphor** for my work, I would much rather be a fisherman. I want to throw my big net into the deep, blue ocean of English and see what marvelous creatures I can drag up from the **bottom**. But why do people want me to direct traffic, when I would much rather go fishing? Well, I blame the Queen. Why do

I blame the Queen? Well, first of all, I blame the Queen because it ’ s funny. But secondly, I blame the Queen because dictionaries have really not changed. Our idea of what a dictionary is has not changed since her reign. The only thing that Queen Victoria would not be amused by in modern dictionaries is our inclusion of the F-word, which has happened in American dictionaries since 1965. So, there ’ s this guy, right? Victorian era. James Murray, first editor of the Oxford English Dictionary. I do not have that hat. I wish I had that hat. So he ’ s really responsible for a lot of what we consider modern in dictionaries today. When a guy who looks like that, in that hat, is the face of modernity, you have a problem. And so, James Murray could get a job on any dictionary today. There ’ d be virtually no learning **curve**. And of course, a few of us are saying : okay, computers! Computers! What about computers? The thing about computers is, I love computers. I mean, I ’ m a huge geek, I love computers. I would go on a hunger strike before I let them take away Google Book Search from me. But computers don ’ t do much else other than speed up the process of compiling dictionaries. They don ’ t change the end result. Because what a dictionary is, is it ’ s Victorian design merged with a little bit of modern propulsion. It ’ s steampunk. What we have is an electric velocipede. You know, we have Victorian design with an engine on it. That ’ s all! The design has not changed. And OK, what about online dictionaries, right? Online dictionaries must be different. This is the Oxford English Dictionary Online, one of the best online dictionaries. This is my favorite word, by the way. Erinaceous : pertaining to the hedgehog family ; of the nature of a hedgehog. Very useful word. So, look at that. Online dictionaries right now are paper thrown up on a screen. This is flat. Look how many links there are in the actual entry : two! Right? Those little buttons, I had them all expanded except for the date chart. So there ’ s not very much going on here. There ’ s not a lot of clickiness. And in fact, online dictionaries replicate almost all the problems of print, except for searchability. And when you improve searchability, you actually take away the one advantage of print, which is serendipity. Serendipity is when you find things you weren ’ t looking for, because finding what you are looking for is so damned difficult. So â□□ â□□ now, when you think about this, what we have here is a ham butt problem. Does everyone know the ham butt problem? Woman ’ s making a ham for a big, family dinner. She goes to cut the butt off the ham and throw it away, and she looks at this piece of ham and she ’ s like,”This is a perfectly good piece of ham. Why am I throwing this away?”She thought,”Well, my mom always did this.”So she calls up mom, and she says,”Mom, why ’ d you cut the butt off the ham, when you ’ re making a ham?”She says,”I don ’ t know, my mom always did it!”So they call grandma, and grandma says,”My pan was too small!”So, it ’ s not that we have good words and bad words. We have a pan that ’ s too small! You know, that ham butt is delicious! There ’ s no reason to throw it away. The bad words â□□ see, when people think about a place and they don ’ t find a place on the map, they think,”This map sucks!”When they find a nightspot or a bar, and it ’ s not in the guidebook, they ’ re like,”Ooh, this place must be cool! It ’ s not in the guidebook.”When they find a word that ’ s not in the dictionary, they think,”This must be a bad word.”Why? It ’ s more likely to be a bad dictionary. Why are you blaming the ham for being too big for the pan? So, you can ’ t get a smaller ham. The English language is as big as it is. So, if you have a ham butt problem, and you ’ re thinking about the ham butt problem, the conclusion that it leads you to is inexorable and counterintuitive : paper is the enemy of words. How can this be? I mean, I love books. I really love books. Some of my best friends are books. But the book is not the best shape for the dictionary. Now they ’ re going to think”Oh, boy. People are going to take away my beautiful, paper dictionaries?”No. There will still be paper dictionaries. When we had cars â when cars became the dominant mode of transportation, we didn ’ t round up all the horses and shoot them. You know, there ’ re still going to be paper dictionaries, but it ’ s not going to be the dominant dictionary. The book-shaped dictionary is not going to be the only shape dictionaries come in. And it ’ s not going to be the prototype for the shapes dictionaries come in. So, think about it this way : if you ’ ve got an artificial constraint, artificial constraints lead to arbitrary distinctions and a skewed worldview. What if biologists could only study animals that made people go,”Aww.”Right? What if we made aesthetic judgments about animals, and only the ones we thought were cute were the ones that we could study? We ’ d know a whole lot about charismatic megafauna, and not very much about much else. And I think this is a problem. I think we should study all the words, because when you think about words, you can make beautiful expressions from very humble parts. Lexicography is really more about material science. We are studying the tolerances of the materials that you use to build the structure of your expression : your speeches and your writing. And then, often people say to me,”Well, OK, how do I know that this word is real?”They think,”OK, if we think words are the tools that we use to build the expressions of our thoughts, how can you say that screwdrivers are better than hammers? How can you say that a sledgehammer is better than a ball- peen hammer?”They ’ re just the right tools for the job. And so people say to me,”How do I know if a word is real?”You know, anybody who ’ s read a children ’ s book knows that love makes things real. If you love a word, use it. That makes it real. Being in the dictionary is an artificial distinction. It doesn ’ t make a word any more real than any other way. If you love a word, it becomes real. So if we ’ re not worrying about directing traffic, if we ’ ve transcended paper, if we are worrying less about control and more about description, then we can think of the English language as being this beautiful mobile. And any time one of those little parts of the mobile changes, is touched, any time you touch a word, you use it in a new context, you give it a new connotation, you verb it, you make the mobile move. You didn ’ t break it. It ’ s just in a new position, and that new position can be just as beautiful. Now, if you ’ re no longer a traffic cop â the problem with being a traffic cop is there can only be so many traffic cops in any one intersection, or the cars get confused. Right? But if your goal is no longer to direct the traffic, but maybe to count the cars that go by, then more eyeballs are better. You can ask for help! If you ask for help, you get more done. And we really need help. Library of Congress : 17 million books, of which half are in English. If only one out of every 10 of those books had a word that ’ s not in the dictionary in it, that would be equivalent to more than two unabridged dictionaries. And I find an un-dictionaried word â a word like”un-dictionaried,”for example â in almost every book I read. What about newspapers? Newspaper archive goes back to 1759, 58. 1 million newspaper pages. If only one in 100 of those pages had an un-dictionaried word on it, it would be an entire other OED. That ’ s 500, 000 more words. So that ’ s a lot. And I ’ m not even talking about magazines. I ’ m not talking about blogs â and I find more new words on BoingBoing in a given week than I do Newsweek or Time. There ’ s a lot going on there. And I ’ m not even talking about polysemy, which is the greedy habit some words have of taking more than one meaning for themselves. So if you think of the word”set,”a set can be a badger ’ s burrow, a set can be one of the pleats in an Elizabethan ruff, and there ’ s one numbered definition in the OED. The OED has 33 different numbered definitions for set. Tiny, little word, 33 numbered definitions. One of them is just labeled”miscellaneous technical senses.”Do you know what that says to me? That says to me, it was Friday afternoon and somebody wanted to go down the pub. That ’ s a lexicographical cop out, to say,”miscellaneous technical senses.”So, we have all these words, and we really need help! And the thing is, we could ask for help â asking for help ’ s not that hard. I mean, lexicography is not rocket science. See, I just gave you a lot of words and a lot of numbers, and this is more of a visual explanation. If we think of the dictionary as being the map of the English language, these bright spots are what we know about, and the dark spots are where we are in the dark. If that was the map of all the words in American English, we don ’ t know very much. And we don ’ t even know the shape of the language. If this was the dictionary â if this was the map of American English â look, we have a kind of lumpy idea of Florida, but there ’ s no California! We ’ re missing California from American English. We just don ’ t know enough, and we don ’ t even know that we ’ re missing California. We don ’ t even see that there ’ s a gap on the map. So again, lexicography is not rocket science. But even if it were, rocket science is being done by dedicated amateurs these days. You know? It can ’ t be that hard to find some words! So, enough scientists in other disciplines are really asking people to help, and they ’ re doing a good job of it. For instance, there ’ s eBird, where amateur birdwatchers can upload information about their bird sightings. And then, ornithologists can go and help track populations, migrations, etc. And there ’ s this guy, Mike Oates. Mike Oates lives in the U. K. He ’ s a director of an electroplating company. He ’ s found more than 140 comets. He ’ s found so many comets, they named a comet after him. It ’ s kind of out past Mars. It ’ s a hike. I don ’ t think he ’ s getting his **picture** taken there anytime soon. But he found 140 comets without a telescope. He downloaded data from the NASA SOHO satellite, and that ’ s how he found them. If we can find comets without a telescope, shouldn ’ t we be able to find words? Now, y ’ all know where I ’ m going with this. Because I ’ m going to the Internet, which is where everybody goes. And the Internet is great for collecting words, because the Internet ’ s full of collectors. And this is a little-known technological fact about the Internet, but the Internet is actually made up of words and enthusiasm. And words and enthusiasm actually happen to be the recipe for lexicography. Isn ’ t that great? So there are a lot of really good word- collecting sites out there right now, but the problem with some of them is that they ’ re not scientific enough. They show the word, but they don ’ t show any context. Where did it come from? Who said it? What newspaper was it in? What book? Because a word is like an archaeological artifact. If you don ’ t know the provenance or the source of the artifact, it ’ s not science, it ’ s a pretty thing to look at. So a word without its source is like a cut flower. You know, it ’ s pretty to look at for a while, but then it dies. It dies too fast. So, this whole time I ’ ve been saying,”The dictionary, the dictionary, the dictionary, the dictionary.”Not”a dictionary,”or”dictionaries.”And that ’ s because, well, people use the dictionary to stand for the whole language. They use it synecdochically. And one of the problems of knowing a word like”synecdochically”is that you really want an excuse to say”synecdochically.”This whole talk has just been an excuse to get me to the point where I could say”synecdochically”to all of you. So I ’ m really sorry. But when you use a part of something â like the dictionary is a part of the language, or a flag stands for the United States, it ’ s a symbol of the country â then you ’ re using it synecdochically. But the thing is, we could make the dictionary the whole language. If we get a bigger pan, then we can put all the words in. We can put in all the meanings. Doesn ’ t everyone want more meaning in their lives? And we can make the dictionary not just be a symbol of the language â we can make it be the whole language. You see, what I ’ m really hoping for is that my son, who turns seven this month â I want him to barely remember that this is the form factor that dictionaries used to come in. This is what dictionaries used to look like. I want him to think of this kind of dictionary as an eight-track **tape**. It ’ s a format that died because it wasn ’ t useful enough. It wasn ’ t really what people needed. And the thing is, if we can put in all the words, no longer have that artificial distinction between good and bad, we can really describe the language like scientists. We can leave the aesthetic judgments to the writers and the speakers. If we can do that, then I can spend all my time fishing, and I don ’ t have to be a traffic cop anymore. Thank you very much for your kind attention.

Text Sample 691: POS Dim 5 – 2007 – Score 8.00 – 2007/t000213.txt
We really need to put the best we have to offer within reach of our children. If we don ’ t do that, we ’ re going to get the generation we deserve. They ’ re going to learn from whatever it is they have around them. And we, as now the elite, parents, librarians, professionals, whatever it is, a bunch of our activities are, in fact, in trying to get the best we have to offer within reach of those around us, or as broadly as we can. I ’ m going to start and end this talk with a couple things that are carved in stone. One is what ’ s on the Boston Public Library. Carved above their door is,”Free to All.”It ’ s kind of an inspiring statement, and I ’ ll go back at the end of this. I ’ m a librarian, and what I ’ m trying to do is bring all of the works of knowledge to as many people as want to read it. And the idea of using technology is perfect for us. I think we have the opportunity to one-up the Greeks. It ’ s not easy to one-up the Greeks. But with the industriousness of the Egyptians, they were able to build the Library of Alexandria — the idea of a copy of every book of all the peoples of the world. The problem was you actually had to go to Alexandria to go to it. On the other hand, if you did, then great things happened. I think we can one-up the Greeks and achieve something. And I ’ m going to try to argue only one point today : that universal access to all knowledge is within our grasp. So if I ’ m successful, then you ’ ll actually come away thinking, yeah, we could actually achieve the great vision of everything ever published, everything that was ever meant for distribution, available to anybody in the world that ’ s ever wanted to have access to it. Yes, there ’ s issues about how money should be distributed, and that ’ s still being refigured out. But I ’ d say there ’ s plenty of money, and there ’ s plenty of demand, so we can actually achieve that. But I ’ m going to go over the technological, social and sort of where are we as a whole, trying to get to that particular vision. And the way I ’ m going to try to do this is do it like the Amazon. com website, the books, music, video and just go step — media type by media type, just go and say, all right, how ’ re we doing on this? So if we start with books, you know, sort of where are we? Well, first you have to, as an engineer, scope the problem. How big is it? If you wanted to put all of the published works online so that anybody could have it available, well, how big a problem is it? Well, we don ’ t really know, but the largest print library in the world is the Library of Congress. It ’ s 26 million volumes, 26 million volumes. It is, by far and away, the largest print library in the world. And a book, if you had a book, is about a megabyte, so — you know, if you had it in Microsoft Word. So a megabyte, 26 million megabytes is 26 terabytes — it goes mega-, giga-, tera-. 26 terabytes. 26 terabytes fits in a computer system that ’ s about this big, on spinning Linux drives, and it costs about 60, 000 dollars. So for the cost of a house — or around here, a garage — you can put, you can have spinning all of the words in the Library of Congress. That ’ s pretty neat. Then the question is, what do you get? You know, is it worth trying to get there? Do you actually want it online? Some of the first things that people do is they make book readers that allow you to search inside the books, and that ’ s kind of **fun**.

And you can download these things, and look around them in new and different ways. And you can get at them remotely, if you happen to have a laptop. There ’ s starting to be some of these sort of page turn-y interfaces that look a whole lot like books in certain ways, and you can search them, make little tabs, and it ’ s kind of cute — still very book-like — on your laptop. But I don ’ t know, reading things on a laptop — whenever I pull up my laptop, it always feels like work. I think that ’ s one of the reasons why the Kindle is so great. I don ’ t have to feel like I ’ m at work to read a Kindle. It ’ s starting to be a little bit more specified. But I have to say that there ’ s older technologies that I tend to like. I like the physical book. And I think we can go and use our technology to go and digitize things, put them on the Net, and then download, print them and bind them, and end up with books again. And we sort of said, well, how hard is this? And it turns out to not be very hard. We actually went off to make a bookmobile. And a bookmobile — the size of a van with a satellite dish, a printer, binder and cutter, and kids make their own books. It costs about three dollars to download, print and bind a normal, old book. And they actually come out kind of nice looking. You can actually get really good-looking books for on the order of one penny per page, sort of the parts cost for doing this. So the idea of — this technology actually may end up putting books back in people ’ s hands again. There are some other bookmobiles running around. This is Eric Eldred making books at Walden Pond — Thoreau ’ s works. This is just before he got kicked out by the Parks Services, for competing with the bookstore there. In India, they ’ ve got another couple bookmobiles running around. And this is the opening day at the Library of Alexandria, the new Library of Alexandria, in Egypt. It was quite popularly attended. And kids starting to make their own books, and a happy kid with the first book that he ’ s ever owned. So the idea of being able to use this technology to end up with paper where I can handle sort of sounds a little retro, but I think it still has its place. And being from the Silicon Valley, sort of utopian sort of world, we thought, if we can make this technology work in rural Uganda, we might have something. So we actually got some funding from the World Bank to try it out. And we found in about 30 days we could go and take a couple folks from Silicon Valley, fly them to Uganda, buy a car, set up the first Internet connection at the National Library of Uganda, figure out what they wanted, and get a program going making books in rural Uganda. And it actually — so technologically, it works. What we found out of this is we didn ’ t have the right books. So the books were in the library. We could get it to people, if they ’ re digitized, but we didn ’ t know how to quite get them digitized. Everybody thought the answer is, send things to India and China. And so we ’ ve tried that, and I ’ ll go over that in a moment. There are some newer technologies for delivering that have happened that are actually quite exciting as well. One is a print-on-demand machine that looks like a Rube Goldberg machine. We have one of these things now. It ’ s completely cool. It ’ s all conveyor belt, and it makes a book. And it ’ s called the”Espresso Book Machine,”and in about 10 minutes, you can press a button and make a book. Something else I ’ m quite excited about in this particular domain, beyond these sort of kiosk-y things where you can get books on demand, is some of these new little screens that are coming out. And one of my favorites in this is the $100 laptop. And I don ’ t mean to steal any thunder here, but we ’ ve gone and used one of these things to be an e-book reader. So here ’ s one of the beta units and you can — it actually turns out to be a really good-looking e-book reader. And we have a quick hack that we did to try to put one of our books on it, and it turns out that 200 dots per inch means that you can put **scanned** books on them that look really good. At 200 dots per inch, it ’ s kind of the equivalent of a 300 dot print **laser** printer. We ’ re in good enough shape. You actually can go and read **scanned** books quite easily. So the idea of electronic books is starting to come about. But how do you go about doing all this **scanning**? So we thought, okay, well, let ' s try out this send books to India thing. And there was a project with, funded by the National Science Foundation — sent a bunch of scanners, and the American libraries were supposed to send books. Well, they didn ' t. They didn ' t want to send their books. So we bought 100, 000 books and sent them to India. And then we learned why you don ' t want to send books to India. The lesson we learned out of this is, **scan** your own books. If you really care about books, you ' re going to **scan** them better, especially if they ' re valuable books. If they ' re new books and you can just, you know, butcher them, because you could just buy another one, that ' s not such a big deal in terms of doing high-quality **scanning**. But do things that you love. But the Indians have been **scanning** a lot of their own books — about 300, 000 now — doing very well. The Chinese did over a million, and the Egyptians are about 30, 000. But we sent — thought, OK, if we ' re going to need to do this, let ' s do it in- library. How do we go and do this, and how do we get it down so that it ' s a cost point that we could afford? And we sort of picked the price point of 10 cents a page. If it ' s basically the cost of xeroxing to basically digitize, OCR, package it up, make it so that you could download, print and bind it — the whole shebang — we would have achieved something. So we started out trying to figure out. How do we get to 10 cents? And we tried these robot things, and they worked pretty well — sort of these auto-page-turning things. If we can have Mars Rovers, you ' d think you could turn pages. But it actually turns out to be pretty hard to turn pages, and the volume isn ' t there. So anyway — so we ended up making our own book scanner, and with two digital, high-grade, professional digital cameras, controlled **museum** lighting, so even if it ' s a black and white book, you can go and get the proper intonation. So you basically do a beautiful, respectful job. This is not a fax, this is — the idea is to do a beautiful job as you ' re going through these libraries. And we ' ve been able to achieve 10 cents a page if we run things in volume. This is what it looks like at the University of Toronto. And actually, it turns out to, you know, pay a living wage. People seem to love it. Yes, it ' s a little boring, but some people kind of get into the Zen of it. And especially if it ' s kind of interesting books that you care about, in languages that you can read. We actually have been able to do a pretty good job of this, at getting 10 cents a page. So 10 cents a page, 300 pages in your average book, 30 dollars a book. The Library of Congress, if you did the whole darn thing — 26 million books — is about 750 million dollars, right? But a million books, I think, actually would be a pretty good start, and that would cost 30 million dollars. That ' s not that big a bill. And what we ' ve been able to do is get into libraries. We ' ve now got eight of these **scanning** centers in three countries, and libraries are up for having their books **scanned**. The Getty here is moving their books to the UCLA, which is where we have one these **scanning** centers, and **scanning** their out-of-copyright books, which is fabulous. So we ' re starting to get the institutional responsibility. The thing we ' re missing is the 10 cents. If we can get the 10 cents, all the rest of it flows. We ' ve **scanned** about 200, 000 books. Now we ' re **scanning** about 15, 000 books a month, and it ' s starting to gear up another factor of two from there. So all in all, that ' s going very well. And we ' re starting to move out of the just out-of-copyright into the out-of-print world. So I think of — we ' re kind of going from the out-of-copyright, library stuff, and Amazon. com is coming from the in-print world. And I think we ' ll meet in the middle some place, and have the classic thing that you have, which is a publishing system and a library system working in parallel. And so we ' re starting up a program to do out-of-print works, but loaning them. Exactly what loaning means, I ' m not quite sure. But anyway, loaning out-of-print works from the Boston Public Library, the Woods Hole Oceanographic Institute and a few other libraries that are starting to participate in this program, to try out this model of where does a library stop and where does the bookstore take over. So all in all, it ' s possible to do this in large scale. We ' re also going back over microfilm and getting that online. So, we can do 10 cents a page, we ' re going 15, 000 books a month and we ' ve got about 250, 000 books online, counting all the other projects that are starting to add in. So what I wanted to argue is, books are within our grasp. The idea of taking on the whole ball of wax is not that big a deal. Yes, it costs tens of millions, low hundreds of millions, but one time shot and we ' ve got basically the history of printed literature online. And then, there ' s business model issues about how to try to effectively market it and get it to people. But it is within our grasp, technologically and law-wise, at least for the out of print and out of copyright, we suggest, to be able to get the whole darn thing online. Now let ' s go for audio, and I ' m going to go through these. So how much is there? Well, as best we can tell, there are about two to three million disks having been published — so 78s, long-playing records and CDs — or at least that ' s the largest archives of published materials we ' ve been able to sort of point at. It costs about 10 dollars a piece to go and take a disk and put it online, if you ' re doing things in volume. But we ' ve found that the rights issues are really quite thorny. This is a fairly heavily litigated area, so we ' ve found that there are niches in the music world that aren ' t served terribly well by the classic commercial publishing system. And we ' ve been starting to make these available by going and offering shelf space on the Net. In the United States, it doesn ' t cost you to give something away. Right? If you give something to a charity or to the public, you get a pat on the back and a tax donation — except on the Net, where you can go broke. If you put up a video of your garage band, and it starts getting heavily accessed, you can lose your guitars or your house. This doesn ' t make any sense. So we ' ve offered unlimited storage, unlimited bandwidth, forever, for free, to anybody that has something to share that belongs in a library. And we ' ve been getting a lot of takers. One is the rock ' n ' rollers. The rock ' n ' rollers had a tradition of sharing, as long as nobody made any money. You could — concert recordings, it ' s not the commercial recordings, but concert recordings, started by the Grateful Dead. And we get about two or three bands a day signing up. They give permission, and we get about 40 or 50 concerts a day. We have about 40, 000 concerts, everything the Grateful Dead ever did, up on the Net, so that people can see it and listen to this material. So audio is possible to put up, but the rights issues are really pretty thorny. We ' ve got a lot of collections now — a couple hundred thousand items — and it ' s growing over time. Moving images : if you think of theatrical releases, there are not that many of them. As best we can tell, there are about 150, 000 to 200, 000 movies ever that are really meant for a large-scale theatrical distribution. It ' s just not that many. But half of those were Indian. But anyway, it ' s doable, but we ' ve only found about a thousand of these things that — to be out of copyright. So we ' ve digitized those and made those available. But we ' ve found that there ' s lots of other types of movies that haven ' t really seen the light of day — archival **films**. We ' ve found, also, a lot of political **films**, a lot of amateur **films**, all sorts of things that are basically needing a home, a permanent home. So we ' ve been starting to make these available and it ' s grown to be very popular. We ' re not quite a YouTube. We tended towards longer-term things and also things that people can reuse and make into new movies, which has just been great **fun**. Television comes quite a bit larger. We started recording 20 channels of television 24 hours a day. It ' s sort of the biggest TiVo box you ' ve ever seen. It ’ s about a petabyte, so far, of worldwide television — Russian, Chinese, Japanese, Iraqi, Al Jazeera, BBC, CNN, ABC, CBS, NBC — 24 hours a day. We only put one week up, which is mostly for cost reasons, which is the 9 / 11, sort of from 9 / 11 / 2001. For one week, what did the world see? CNN was saying that Palestinians were dancing in the streets. Were they? Let ’ s look at the Palestinian television and find out. How can we have critical thinking without being able to quote and being able to compare what happened in the past? And television is dreadfully unrecorded and unquotable, except by Jon Stewart, who does a fabulous job. So anyway, television is, I would suggest, within our grasp. So 15 dollars per video hour, and also about 100 dollars to 150 dollars per celluloid hour, we ’ re able to go and get materials online very inexpensively and have them up on the Net. And we ’ ve got, now, a lot of these materials. So we ’ ve got about 100, 000 pieces up there. So books, music, video, software. There ’ s only 50, 000 titles of it. Mostly the issues there are legal issues and breaking copy protections. But we ’ ve worked through some of those, but we ’ ve still got real problems in Washington. Well, we ’ re best known as the World Wide Web. We ’ ve been archiving the World Wide Web since 1996. We take a snapshot of every website and all of the pages on it, every two months. And actually, it ’ s really been pioneered by Alexa Internet, which donates this collection to the Internet Archive. And it ’ s been growing along for the last 11 years, and it ’ s a fantastic resource. And we ’ ve made a Wayback Machine that you can then go and see old websites kind of the way they were. If you go and search on something — this is Google. com, the different versions of it that we have, this is what it looks like when it was an alpha release, and this is what it looked like at Stanford. So anyway, you ’ ve got basically an idea of where things came from. Mostly, people want to see their old stuff out of this. If there ’ s one thing that we want to learn from the Library of Alexandria version one, which is probably best known for burning, is, don ’ t just have one copy. So we ’ ve started to — we ’ ve made another copy of all of this and we actually put it back in the Library of Alexandria. So this is a **picture** of the Internet Archive at the Library of Alexandria. And we now have also another copy building up in Amsterdam. So, we should put it in the San Andreas Fault Line in San Francisco, flood zone in Amsterdam and in the Middle East. Right, so anyway... so we ’ re hedging our bets here. If we go and put it in a couple more places, I think we ’ ll be in good shape. There ’ s a political and social question out of this. Is all of this, as we go digital, is it going to be public or private? There ’ s some large companies that have seen this vision, that are doing large-scale digitization, but they ’ re locking up the public domain. The question is, is that the world that we really want to live in? What ’ s the role of the public versus the private as things go forward? How do we go and have a world where we both have libraries and publishing in the future, just as we basically benefited as we were growing up? So universal access to all knowledge — I think it can be one of the greatest achievements of humankind, like the man on the moon, or the Gutenberg Bible, or the Library of Alexandria. It could be something that we ’ re remembered for, for millennia, for having achieved. And as I said before, I ’ ll end with something that ’ s carved above the door of the Carnegie Library. Carnegie — one of the great capitalists of this country — carved above his legacy,”Free to the People.”Thank you very much.

Text Sample 692: POS Dim 5 – 2010 – Score 9.00 – 2010/t002718.txt
Thank you very much. I have a few **pictures**, and I ’ ll talk a little bit about how

I ’ m able to do what I do. All these houses are built from between 70 and 80 percent recycled material, stuff that was headed to the mulcher, the landfill, the burn pile. It was all just gone. This is the first house I built. This double front door here with the three-light transom, that was headed to the landfill. Have a little turret there. And then these buttons on the corbels here — right there — those are hickory nuts. And these buttons there — those are chicken eggs. Of course, first you have breakfast, and then you fill the shell full of Bondo and paint it and nail it up, and you have an architectural button in just a fraction of the time. This is a look at the inside. You can see the three-light transom there with the eyebrow windows. Certainly an architectural antique headed to the landfill — even the lockset is probably worth 200 dollars. Everything in the kitchen was salvaged. There ’ s a 1952 O ’ Keefe & Merritt stove, if you like to cook — cool stove. This is going up into the turret. I got that staircase for 20 dollars, including delivery to my lot. Then, looking up in the turret, you see there are bulges and pokes and sags and so forth. Well, if that ruins your life, well, then, you shouldn ’ t live there. This is a laundry chute. And this right here is a shoe last — those are those cast-iron things you see at antique shops. So I had one of those, so I made some low-tech gadgetry, where you just stomp on the shoe last, and then the door flies open and you throw your laundry down. And then if you ’ re smart enough, it goes on a basket on top of the washer. If not, it goes into the toilet. This is a bathtub I made, made out of scrap two-by-four. Started with the rim, and then glued and nailed it up into a flat, corbeled it up and flipped it over, then did the two profiles on this side. It ’ s a two-person tub. After all, it ’ s not just a question of hygiene, but there ’ s a possibility of recreation as well. Then, this faucet here is just a piece of Osage orange. It looks a little phallic, but after all, it ’ s a bathroom. This is a house based on a Budweiser can. It doesn ’ t look like a can of beer, but the design take-offs are absolutely unmistakable : the barley hops design worked up into the eaves, then the dentil work comes directly off the can ’ s red, white, blue and silver. Then, these corbels going down underneath the eaves are that little design that comes off the can. I just put a can on a copier and kept enlarging it until I got the size I want. Then, on the can it says,”This is the famous Budweiser beer, we know of no other beer, blah, blah, blah.”So we changed that and put,”This is the famous Budweiser house. We don ’ t know of any other house...”and so forth and so on. This is a deadbolt. It ’ s a fence from a 1930s shaper, which is a very angry woodworking machine. And they gave me the fence, but they didn ’ t give me the shaper, so we made a deadbolt out of it. That ’ ll keep bull elephants out, I promise. And sure enough, we ’ ve had no problems with bull elephants. The shower is intended to simulate a glass of beer. We ’ ve got bubbles going up there, then suds at the top with lumpy tiles. Where do you get lumpy tiles? Well, of course, you don ’ t. But I get a lot of toilets, and so you just dispatch a toilet with a hammer, and then you have lumpy tiles. And then the faucet is a beer tap. Then, this **panel** of glass is the same **panel** of glass that occurs in every middle-class front door in America. We ’ re getting tired of it. It ’ s kind of clichd now. If you put it in the front door, your design fails. So don ’ t put it in the front door ; put it somewhere else. It ’ s a pretty **panel** of glass. But if you put it in the front door, people say,”Oh, you ’ re trying to be like those guys, and you didn ’ t make it.”So don ’ t put it there. Then, another bathroom upstairs. This light up here is the same light that occurs in every middle-class foyer in America. Don ’ t put it in the foyer. Put it in the shower, or in the closet, but not in the foyer. Then, somebody gave me a bidet, so it got a bidet. This little house here, those branches there are made out of Bois d ’ arc or Osage orange. These **pictures** will keep scrolling as I talk a little bit. In order to do what I do, you have to understand what causes waste in the building industry.

Our housing has become a commodity, and I ’ ll talk a little bit about that. But the first cause of waste is probably even buried in our DNA. Human beings have a need for maintaining consistency of the apperceptive mass. What does that mean? What it means is, for every perception we have, it needs to tally with the one like it before, or we don ’ t have continuity, and we become a little bit disoriented. So I can show you an object you ’ ve never seen before. Oh, that ’ s a cell phone. But you ’ ve never seen this one before. What you ’ re doing is sizing up the pattern of structural features, and then you go through your databanks : Cell phone. Oh! That ’ s a cell phone. If I took a bite out of it, you ’ d go,”Wait a second.”That ’ s not a cell phone. That ’ s one of those new chocolate cell phones.”You ’ d have to start a new category, right between cell phones and chocolate. That ’ s how we process information. You translate that to the building industry. If we have a wall of windowpanes and one pane is cracked, we go,”Oh, dear. That ’ s cracked. Let ’ s repair it. Let ’ s take it out and throw it away so nobody can use it and put a new one in.”Because that ’ s what you do with a cracked pane. Never mind that it doesn ’ t affect our lives at all. It only rattles that expected pattern and unity of structural features. However, if we took a small hammer, and we added cracks to all the other windows — then we have a pattern. Because Gestalt psychology emphasizes recognition of pattern over parts that comprise a pattern. We ’ ll go,”Ooh, that ’ s nice.”So, that serves me every day. Repetition creates pattern. If I have 100 of these, 100 of those, it makes no difference what these and those are. If I can repeat anything, I have the possibility of a pattern, from hickory nuts and chicken eggs, shards of glass, branches. It doesn ’ t make any difference. That causes a lot of waste in the building industry. The second cause is, Friedrich Nietzsche, along about 1885, wrote a book titled”The Birth of Tragedy.”And in there, he said cultures tend to swing between one of two perspectives : on the one hand, we have an Apollonian perspective, which is very crisp and premeditated and intellectualized and perfect. On the other end of the spectrum, we have a Dionysian perspective, which is more given to the passions and intuition, tolerant of organic texture and human gesture. So the way the Apollonian personality takes a **picture** or hangs a **picture** is, they ’ ll get out a transit and a **laser** level and a micrometer.”OK, honey. A thousandth of an inch to the left. That ’ s where we want the **picture**. Right. Perfect!”Predicated on plumb level, square and centered. The Dionysian personality takes the **picture** and goes : That ’ s the difference. I feature blemish. I feature organic process. Dead center John Dewey. Apollonian mindset creates mountains of waste. If something isn ’ t perfect, if it doesn ’ t line up with that premeditated model? Dumpster.”Oops. Scratch. Dumpster.””Oops”this,”oops”that. Landfill, landfill, landfill. The third thing is arguably — The Industrial Revolution started in the Renaissance with the rise of humanism, then got a little jump start along about the French Revolution. By the middle of the 19th century, it ’ s in full flower. And we have dumaflaches and gizmos and contraptions that will do anything that we, up to that point, had to do by hand. So now we have standardized materials. Well, trees don ’ t grow two inches by four inches, eight, ten and twelve **feet** tall. We create mountains of waste. And they ’ re doing a pretty good job there in the forest, working all the byproduct of their industry — with OSB and particle board and so forth and so on — but it does no good to be responsible at the point of harvest in the forest if consumers are wasting the harvest at the point of consumption. And that ’ s what ’ s happening. And so if something isn ’ t standard,”Oops, dumpster.””Oops”this.”Oops, warped.”If you buy a two-by- four and it ’ s not straight, you can take it back.”Oh, I ’ m so sorry, sir. We ’ ll get you a straight one.”Well, I feature all those warped things because repetition creates pattern, and it ’ s from a Dionysian perspective. The fourth thing is labor is disproportionately more expensive than materials. Well, that ' s just a myth. And there ' s a story : Jim Tulles, one of the guys I trained — I said,"Jim, it ' s time now. I got a job for you as a foreman on a framing crew. Time for you to go.""Dan, I just don ' t think I ' m ready.""Jim, now it ' s time. You ' re the down — oh!"So we hired on. And he was out there with a **tape** measure, going through the trash heap, looking for header material, or the board that goes over a door, thinking he ' d impress his boss — that ' s how we taught him to do it. The superintendent walked up and said,"What are you doing?""Oh, just looking for header material,"waiting for that kudos. He said,"I ' m not paying you to go through the trash. Get back to work."And Jim had the wherewithal to say,"You know, if you were paying me 300 dollars an hour, I can see how you might say that. But right now, I ' m saving you five dollars a minute. Do the math.""Good call, Tulles. From now on, you guys hit this pile first."And the irony is that he wasn ' t very good at math. But once in a while, you get access to the control room, and then you can kind of mess with the dials. And that ' s what happened there. The fifth thing is that maybe, after 2, 500 years, Plato is still having his way with us in his notion of perfect forms. He said that we have in our noggin the perfect idea of what we want, and we force environmental resources to accommodate that. So we all have in our head the perfect house, the American dream, which is a house, the dream house. The problem is we can ' t afford it. So we have the American dream look-alike, which is a mobile home. Now there ' s a blight on the planet. It ' s a chattel mortgage, just like furniture, just like a car. You write the check, and instantly, it depreciates 30 percent. After a year, you can ' t get insurance on everything you have in it, only on 70 percent. Wired with 14-Gauge wire, typically. Nothing wrong with that, unless you ask it to do what 12-Gauge wire ' s supposed to do, and that ' s what happens. It out-gasses formaldehyde — so much so that there is a federal law in place to warn new mobile home buyers of the formaldehyde atmosphere danger. Are we just being numbingly stupid? The walls are this thick. The whole thing has the structural value of corn."So... I thought Palm Harbor Village was over there.""No, no. We had a wind last night. It ' s gone now."Then when they degrade, what do you do with them? Now, all that — that Apollonian, Platonic model — is what the building industry is predicated on, and there are a number of things that exacerbate that. One is that all the professionals, all the tradesmen, vendors, inspectors, engineers, **architects** all think like this. And then it works its way back to the consumer, who demands the same model. It ' s a self-fulfilling prophecy. We can ' t get out of it. Then here come the marketeers and the advertisers."Woo. Woo-hoo."We buy stuff we didn ' t know we needed. All we have to do is look at what one company did with carbonated prune juice. How disgusting. But you know what they did? They hooked a **metaphor** into it and said,"I drink Dr. Pepper..."And pretty soon, we ' re swilling that stuff by the lake-ful, by the billions of gallons. It doesn ' t even have real prunes! Doesn ' t even keep you regular. My oh my, that makes it worse. And we get sucked into that faster than anything. Then, a man named Jean-Paul Sartre wrote a book titled"Being and Nothingness."It ' s a pretty quick read. You can snap through it in maybe — maybe two years, if you read eight hours a day. In there, he talked about the divided self. He said human beings act differently when they know they ' re alone than when they know somebody else is around. So if I ' m eating spaghetti, and I know I ' m alone, I can eat like a backhoe. I can wipe my mouth on my sleeve, **napkin** on the table, chew with my mouth open, make little noises, scratch wherever I want. But as soon as you walk in, I go,"Oops! Lil ' spaghetti sauce there."**Napkin** in my lap, half-bites, chew with my mouth closed, no scratching. Now, what I ' m doing is fulfilling your expectations of how I should live my life. I feel that expectation, and so I accommodate it, and I ’ m living my life according to what you expect me to do. That happens in the building industry as well. That ’ s why all subdivisions look the same. Sometimes, we even have these formalized cultural expectations. I ’ ll bet all your shoes match. Sure enough, we all buy into that... And with gated communities, we have a formalized expectation, with a homeowners ’ association. Sometimes those guys are Nazis, my oh my. That exacerbates and continues this model. The last thing is gregariousness. Human beings are a social species. We like to hang together in groups, just like wildebeests, just like lions. Wildebeests don ’ t hang with lions, because lions eat wildebeests. Human beings are like that. We do what that group does that we ’ re trying to identify with. You see this in junior high a lot. Those kids, they ’ ll work all summer long — kill themselves — so that they can afford one pair of designer jeans. So along about September, they can stride in and go,”I ’ m important today. See? Don ’ t touch my designer jeans! I see you don ’ t have designer jeans. You ’ re not one of the beautiful — See, I ’ m one of the beautiful people. See my jeans?”Right there is reason enough to have uniforms. And so that happens in the building industry as well. We have confused Maslow ’ s hierarchy of needs, just a little bit. On the **bottom** tier, we have basic needs : shelter, clothing, food, water, mating and so forth. Second : security. Third : relationships. Fourth : status, self-esteem — that is, vanity — and we ’ re taking vanity and shoving it down here. And so we end up with vain decisions, and we can ’ t even afford our mortgage. We can ’ t afford to eat anything except beans ; that is, our housing has become a commodity. And it takes a little bit of nerve to dive into those primal, terrifying parts of ourselves and make our own decisions and not make our housing a commodity, but make it something that bubbles up from seminal sources. That takes a little bit of nerve, and, darn it, once in a while, you fail. But that ’ s okay. If failure destroys you, then you can ’ t do this. I fail all the time, every day, and I ’ ve had some whopping failures, I promise — big, public, humiliating, embarrassing failures. Everybody points and laughs, and they say,”He tried it a fifth time, and it still didn ’ t work! What a moron!”Early on, contractors come by and say,”Dan, you ’ re a cute little bunny, but you know, this just isn ’ t going to work. What don ’ t you do this? Why don ’ t you do that?”And your instinct is to say,”Well, why don ’ t you suck an egg?”But you don ’ t say that, because they ’ re the guys you ’ re targeting. And so what we ’ ve done — and this isn ’ t just in housing ; it ’ s in clothing and food and our transportation needs, our energy — we sprawl just a little bit. And when I get a little bit of press, I hear from people all over the world. And we may have invented excess, but the problem of waste is worldwide. We ’ re in trouble. And I don ’ t wear ammo belts crisscrossing my chest and a red bandana. But we ’ re clearly in trouble. And what we need to do is reconnect with those really primal parts of ourselves and make some decisions and say,”You know, I think I would like to put CDs across the wall there. What do you think, honey?”If it doesn ’ t work, take it down. What we need to do is reconnect with who we really are, and that ’ s thrilling indeed. Thank you very much.

Text Sample 693: POS Dim 5 – 2010 – Score 7.00 – 2010/t002603.txt

My name is Arvind Gupta, and I ’ m a toymaker. I ’ ve been making toys for the last 30 years. The early ’ 70s, I was in college. It was a very revolutionary time. It was a political ferment, so to say â students out in the streets of Paris, revolting against authority. America was jolted by the anti-Vietnam movement, the Civil Rights movement. In India, we had the Naxalite movement, the [unclear] movement. But you know, when there is a political churning of society, it unleashes a lot of energy. The National Movement of India was testimony to that. Lots of people resigned from well-paid jobs and jumped into the National Movement. Now in the early ’ 70s, one of the great programs in India was to revitalize primary science in village schools. There was a person, Anil Sadgopal, did a Ph. D. from Caltech and returned back as a molecular biologist in India ’ s cutting-edge research institute, the TIFR. At 31, he was not able to relate the kind of [unclear] research, which he was doing with the lives of the ordinary people. So he designed and went and started a village science program. Many people were inspired by this. The slogan of the early ’ 70s was”Go to the people. Live with them ; love them. Start from what they know. Build on what they have.”This was kind of the defining slogan. Well I took one year. I joined Telco, made TATA trucks, pretty close to Pune. I worked there for two years, and I realized that I was not born to make trucks. Often one doesn ’ t know what one wants to do, but it ’ s good enough to know what you don ’ t want to do. So I took one year off, and I went to this village science program. And it was a turning point. It was a very small village â a weekly bazaar where people, just once in a week, they put in all the vats. So I said,”I ’ m going to spend a year over here.”So I just bought one specimen of everything which was sold on the roadside. And one thing which I found was this black rubber. This is called a cycle valve tube. When you pump in air in a bicycle, you use a bit of this. And some of these models â so you take a bit of this cycle valve tube, you can put two matchsticks inside this, and you make a flexible joint. It ’ s a joint of tubes. You start by teaching angles â an acute angle, a right angle, an obtuse angle, a straight angle. It ’ s like its own little coupling. If you have three of them, and you loop them together, well you make a triangle. With four, you make a square, you make a pentagon, you make a hexagon, you make all these kind of polygons. And they have some wonderful properties. If you look at the hexagon, for instance, it ’ s like an amoeba, which is constantly changing its own profile. You can just pull this out, this becomes a rectangle. You give it a push, this becomes a parallelogram. But this is very shaky. Look at the pentagon, for instance, pull this out â it becomes a boat shape trapezium. Push it and it becomes house shaped. This becomes an isosceles triangle â again, very shaky. This square might look very square and prim. Give it a little push â this becomes a rhombus. It becomes kite-shaped. But give a child a triangle, he can ’ t do a thing to it. Why use triangles? Because triangles are the only rigid structures. We can ’ t make a bridge with squares because the train would come, it would start doing a jig. Ordinary people know about this because if you go to a village in India, they might not have gone to engineering college, but no one makes a **roof** placed like this. Because if they put tiles on top, it ’ s just going to crash. They always make a triangular **roof**. Now this is people science. And if you were to just poke a hole over here and put a third matchstick, you ’ ll get a T joint. And if I were to poke all the three legs of this in the three vertices of this triangle, I would make a tetrahedron. So you make all these 3D shapes. You make a tetrahedron like this. And once you make these, you make a little house. Put this on top. You can make a joint of four. You can make a joint of six. You just need a ton. Now this was â you make a joint of six, you make an icosahedron. You can play around with it. This makes an igloo. Now this is in 1978. I was a 24-year-old young engineer. And I thought this was so much better than making trucks. If you, as a matter of fact, put four marbles inside, you simulate the molecular structure of methane, CH4. Four atoms of hydrogen, the four points of the tetrahedron, which means the little carbon atom. Well since then, I just thought that I ’ ve been really privileged to go to over 2, 000 schools in my country â village schools, government schools, municipal schools, Ivy League schools â I ' ve been invited by most of them. And every time I go to a school, I see a gleam in the eyes of the children. I see hope. I see happiness in their faces. Children want to make things. Children want to do things. Now this, we make lots and lots of pumps. Now this is a little pump with which you could inflate a balloon. It ' s a real pump. You could actually pop the balloon. And we have a slogan that the best thing a child can do with a toy is to break it. So all you do is â it ' s a very kind of provocative statement â this old bicycle tube and this old plastic [unclear] This filling cap will go very snugly into an old bicycle tube. And this is how you make a valve. You put a little sticky **tape**. This is one-way traffic. Well we make lots and lots of pumps. And this is the other one â that you just take a straw, and you just put a stick inside and you make two half-cuts. Now this is what you do, is you bend both these legs into a triangle, and you just wrap some **tape** around. And this is the pump. And now, if you have this pump, it ' s like a great, great sprinkler. It ' s like a centrifuge. If you spin something, it tends to fly out. Well in terms of â if you were in Andhra Pradesh, you would make this with the palmyra leaf. Many of our folk toys have great science principles. If you spin-top something, it tends to fly out. If I do it with both hands, you can see this **fun** Mr. Flying Man. Right. This is a toy which is made from paper. It ' s amazing. There are four **pictures**. You see insects, you see frogs, snakes, eagles, butterflies, frogs, snakes, eagles. Here ' s a paper which you could [unclear] â designed by a mathematician at Harvard in 1928, Arthur Stone, documented by Martin Gardner in many of his many books. But this is great **fun** for children. They all study about the food chain. The insects are eaten by the frogs ; the frogs are eaten by the snakes ; the snakes are eaten by the eagles. And this can be, if you had a whole photocopy paper â A4 size paper â you could be in a municipal school, you could be in a government school â a paper, a scale and a pencil â no glue, no scissors. In three minutes, you just fold this up. And what you could use it for is just limited by your imagination. If you take a smaller paper, you make a smaller flexagon. With a bigger one, you make a bigger one. Now this is a pencil with a few slots over here. And you put a little fan here. And this is a hundred-year-old toy. There have been six major research papers on this. There ' s some grooves over here, you can see. And if I take a reed â if I rub this, something very amazing happens. Six major research papers on this. As a matter of fact, Feynman, as a child, was very **fascinated** by this. He wrote a paper on this. And you don ' t need the three billion-dollar Hadron Collider for doing this. This is there for every child, and every child can enjoy this. If you want to put a colored disk, well all these seven colors coalesce. And this is what Newton talked about 400 years back, that white light ' s made of seven colors, just by spinning this around. This is a straw. What we ' ve done, we ' ve just sealed both the ends with **tape**, nipped the right corner and the **bottom** left corner, so there ' s holes in the opposite corners, there ' s a little hole over here. This is a kind of a blowing straw. I just put this inside this. There ' s a hole here, and I shut this. And this costs very little money to make â great **fun** for children to do. What we do is make a very simple electric motor. Now this is the simplest motor on Earth. The most expensive thing is the battery inside this. If you have a battery, it costs five cents to make it. This is an old bicycle tube, which gives you a broad rubber band, two safety pins. This is a permanent magnet. Whenever current flows through the coil, this becomes an electromagnet. It ' s the interaction of both these magnets which makes this motor spin. We made 30, 000. Teachers who have been teaching science for donkey years, they just muck up the definition and they spit it out. When teachers make it, children make it. You can see a gleam in their eye. They get a thrill of what science is all about. And this science is not a rich man ’ s game. In a democratic country, science must reach to our most oppressed, to the most marginalized children. This program started with 16 schools and spread to 1, 500 government schools. Over 100, 000 children learn science this way. And we ’ re just trying to see possibilities. Look, this is the tetrapak â awful materials from the point of view of the environment. There are six layers â three layers of plastic, aluminum â which are are sealed together. They are fused together, so you can ’ t separate them. Now you can just make a little network like this and fold them and stick them together and make an icosahedron. So something which is trash, which is choking all the seabirds, you could just recycle this into a very, very joyous â all the platonic solids can be made with things like this. This is a little straw, and what you do is you just nip two corners here, and this becomes like a baby crocodile ’ s mouth. You put this in your mouth, and you blow. It ’ s children ’ s delight, a teacher ’ s envy, as they say. You ’ re not able to see how the sound is produced, because the thing which is vibrating goes inside my mouth. I ’ m going to keep this outside, to blow out. I ’ m going to suck in air. So no one actually needs to muck up the production of sound with wire vibrations. The other is that you keep blowing at it, keep making the sound, and you keep cutting it. And something very, very nice happens. And when you get a very small one â This is what the kids teach you. You can also do this. Well before I go any further, this is something worth sharing. This is a touching slate meant for blind children. This is strips of Velcro, this is my **drawing** slate, and this is my **drawing** pen, which is basically a **film** box. It ’ s basically like a fisherman ’ s line, a fishing line. And this is wool over here. If I crank the handle, all the wool goes inside. And what a blind child can do is to just draw this. Wool sticks on Velcro. There are 12 million blind children in our country â who live in a world of darkness. And this has come as a great boon to them. There ’ s a factory out there making our children blind, not able to provide them with food, not able to provide them with vitamin A. But this has come as a great boon for them. There are no patents. Anyone can make it. This is very, very simple. You can see, this is the generator. It ’ s a crank generator. These are two magnets. This is a large pulley made by sandwiching rubber between two old CDs. Small pulley and two strong magnets. And this fiber turns a wire attached to an LED. If I spin this pulley, the small one ’ s going to spin much faster. There will be a spinning magnetic field. Lines, of course, would be cut, the force will be generated. And you can see, this LED is going to glow. So this is a small crank generator. Well, this is, again, it ’ s just a ring, a steel ring with steel nuts. And what you can do is just, if you give it a twirl, well they just keep going on. And imagine a bunch of kids standing in a circle and just waiting for the steel ring to be passed on. And they ’ d be absolutely joyous playing with this. Well in the end, what we can also do : we use a lot of old newspapers to make caps. This is worthy of Sachin Tendulkar. It ’ s a great cricket cap. When first you see Nehru and Gandhi, this is the Nehru cap â just half a newspaper. We make lots of toys with newspapers, and this is one of them. And this is â you can see â this is a flapping bird. All of our old newspapers, we cut them into little squares. And if you have one of these birds â children in Japan have been making this bird for many, many years. And you can see, this is a little fantail bird. Well in the end, I ’ ll just end with a story. This is called”The Captain ’ s Hat Story.”The captain was a captain of a sea-going ship. It goes very slowly. And there were lots of passengers on the ship, and they were getting bored, so the captain invited them on the deck.”Wear all your colorful clothes and sing and dance, and I ’ ll provide you with good food and drinks.”And the captain would wear a cap everyday and join in the regalia. The first day, it was a huge umbrella cap, like a captain ’ s cap. That night, when the passengers would be sleeping, he would give it one more fold, and the second day, he would be wearing a fireman ’ s cap â with a little shoot just like a designer cap, because it protects the spinal cord. And the second night, he would take the same cap and give it another fold. And the third day, it would be a Shikari cap â just like an adventurer ’ s cap. And the third night, he would give it two more folds â and this is a very, very famous cap. If you ’ ve seen any of our Bollywood **films**, this is what the policeman wears, it ’ s called a zapalu cap. It ’ s been catapulted to international glory. And we must not forget that he was the captain of the ship. So that ’ s a ship. And now the end : everyone was enjoying the journey very much. They were singing and dancing. Suddenly there was a storm and huge waves. And all the ship can do is to dance and pitch along with the waves. A huge wave comes and slaps the front and knocks it down. And another one comes and slaps the aft and knocks it down. And there ’ s a third one over here. This swallows the bridge and knocks it down. And the ship sinks, and the captain has lost everything, but for a life jacket. Thank you so much.

Text Sample 694: POS Dim 5 – 2010 – Score 7.00 – 2010/t002785.txt

So I ’ m going to talk to you about you about the political chemistry of oil spills and why this is an incredibly important, long, oily, hot summer, and why we need to keep ourselves from getting distracted. But before I talk about the political chemistry, I actually need to talk about the chemistry of oil. This is a photograph from when I visited Prudhoe Bay in Alaska in 2002 to watch the Minerals Management Service testing their ability to burn oil spills in ice. And what you see here is, you see a little bit of crude oil, you see some ice cubes, and you see two sandwich baggies of napalm. The napalm is burning there quite nicely. And the thing is, is that oil is really an **abstraction** for us as the American consumer. We ’ re four percent of the world ’ s population ; we use 25 percent of the world ’ s oil production. And we don ’ t really understand what oil is, until you check out its molecules, And you don ’ t really understand that until you see this stuff burn. So this is what happens as that burn gets going. It takes off. It ’ s a big woosh. I highly recommend that you get a chance to see crude oil burn someday, because you will never need to hear another poli sci lecture on the geopolitics of oil again. It ’ ll just bake your retinas. So there it is ; the retinas are baking. Let me tell you a little bit about this chemistry of oil. Oil is a stew of hydrocarbon molecules. It starts of with the very small ones, which are one carbon, four hydrogen — that ’ s methane — it just floats off. Then there ’ s all sorts of intermediate ones with middle amounts of carbon. You ’ ve probably heard of benzene rings ; they ’ re very carcinogenic. And it goes all the way over to these big, thick, galumphy ones that have hundreds of carbons, and they have thousands of hydrogens, and they have vanadium and heavy **metals** and sulfur and all kinds of craziness hanging off the sides of them. Those are called the asphaltenes ; they ’ re an ingredient in asphalt. They ’ re very important in oil spills. Let me tell you a little bit about the chemistry of oil in water. It is this chemistry that makes oil so disastrous. Oil doesn ’ t sink, it floats. If it sank, it would be a whole different story as far as an oil spill. And the other thing it does is it spreads out the moment it hits the water. It spreads out to be really thin, so you have a hard time corralling it. The next thing that happens is the light ends evaporate, and some of the toxic things float into the water column and kill fish eggs and smaller fish and things like that, and shrimp. And then the asphaltenes — and this is the **crucial** thing — the asphaltenes get whipped by the waves into a frothy emulsion, something like mayonnaise. It triples the amount of oily, messy goo that you have in the water, and it makes it very hard to handle. It also makes it very viscous. When the Prestige sank off the coast of Spain, there were big, floating cushions the size of sofa cushions of emulsified oil, with the consistency, or the viscosity, of chewing gum. It ' s incredibly hard to **clean** up. And every single oil is different when it hits water. When the chemistry of the oil and water also hits our politics, it ' s absolutely explosive. For the first time, American consumers will kind of see the oil supply chain in front of themselves. They have a"eureka!"moment, when we suddenly understand oil in a different context. So I ' m going to talk just a little bit about the origin of these politics, because it ' s really **crucial** to understanding why this summer is so important, why we need to stay focused. Nobody gets up in the morning and thinks,"Wow! I ' m going to go buy some three- carbon-to-12-carbon molecules to put in my tank and drive happily to work."No, they think,"Ugh. I have to go buy **gas**. I ' m so angry about it. The oil companies are ripping me off. They set the prices, and I don ' t even know. I am helpless over this."And this is what happens to us at the **gas** pump — and actually, **gas** pumps are specifically designed to diffuse that anger. You might notice that many **gas** pumps, including this one, are designed to look like ATMs. I ' ve talked to engineers. That ' s specifically to diffuse our anger, because supposedly we feel good about ATMs. That shows you how bad it is. But actually, I mean, this feeling of helplessness comes in because most Americans actually feel that oil prices are the result of a conspiracy, not of the vicissitudes of the world oil market. And the thing is, too, is that we also feel very helpless about the amount that we consume, which is somewhat reasonable, because in fact, we have designed this system where, if you want to get a job, it ' s much more important to have a car that runs, to have a job and keep a job, than to have a GED. And that ' s actually very perverse. Now there ' s another perverse thing about the way we buy **gas**, which is that we ' d rather be doing anything else. This is BP ' s **gas** station in downtown Los Angeles. It is green. It is a shrine to greenishness."Now,"you think,"why would something so lame work on people so smart?"Well, the reason is, is because, when we ' re buying **gas**, we ' re very invested in this sort of cognitive dissonance. I mean, we ' re angry at the one hand and we want to be somewhere else. We don ' t want to be buying oil ; we want to be doing something green. And we get kind of in on our own con. I mean — and this is funny, it looks funny here. But in fact, that ' s why the slogan"beyond petroleum"worked. But it ' s an inherent part of our energy policy, which is we don ' t talk about reducing the amount of oil that we use. We talk about energy independence. We talk about hydrogen cars. We talk about biofuels that haven ' t been invented yet. And so, cognitive dissonance is part and parcel of the way that we deal with oil, and it ' s really important to dealing with this oil spill. Okay, so the politics of oil are very moral in the United States. The oil industry is like a huge, gigantic octopus of engineering and finance and everything else, but we actually see it in very moral terms. This is an early- on photograph — you can see, we had these gushers. Early journalists looked at these spills, and they said,"This is a filthy industry."But they also saw in it that people were getting rich for doing nothing. They weren ' t farmers, they were just getting rich for stuff coming out of the ground. It ' s the"Beverly Hillbillies,"basically. But in the beginning, this was seen as a very morally problematic thing, long before it became funny. And then, of course, there was John D. Rockefeller. And the thing about John D. is that he went into this chaotic wild-east of oil industry, and he rationalized it into a vertically integrated company, a multinational. It was terrifying ; you think Walmart is a terrifying business model now, imagine what this looked like in the 1860s or 1870s. And it also the kind of root of how we see oil as a conspiracy. But what ’ s really amazing is that Ida Tarbell, the journalist, went in and did a big expos of Rockefeller and actually got the whole antitrust laws put in place. But in many ways, that image of the conspiracy still sticks with us. And here ’ s one of the things that Ida Tarbell said — she said,”He has a thin nose like a thorn. There were no lips. There were puffs under the little colorless eyes with creases running from them.”Okay, so that guy is actually still with us. I mean, this is a very pervasive — this is part of our DNA. And then there ’ s this guy, okay. So, you might be wondering why it is that, every time we have high oil prices or an oil spill, we call these CEOs down to Washington, and we sort of pepper them with questions in public and we try to shame them. And this is something that we ’ ve been doing since 1974, when we first asked them,”Why are there these obscene profits?”And we ’ ve sort of personalized the whole oil industry into these CEOs. And we take it as, you know — we look at it on a moral level, rather than looking at it on a legal and financial level. And so I ’ m not saying these guys aren ’ t liable to answer questions — I ’ m just saying that, when you focus on whether they are or are not a bunch of greedy bastards, you don ’ t actually get around to the point of making laws that are either going to either change the way they operate, or you ’ re going to get around to really reducing the amount of oil and reducing our dependence on oil. So I ’ m saying this is kind of a distraction. But it makes for good theater, and it ’ s powerfully cathartic as you probably saw last week. So the thing about water oil spills is that they are very politically galvanizing. I mean, these **pictures** — this is from the Santa Barbara spill. You have these **pictures** of birds. They really influence people. When the Santa Barbara spill happened in 1969, it formed the environmental movement in its modern form. It started Earth Day. It also put in place the National Environmental Policy Act, the Clean Air Act, the Clean Water Act. Everything that we are really stemmed from this period. I think it ’ s important to kind of look at these **pictures** of the birds and understand what happens to us. Here we are normally ; we ’ re standing at the **gas** pump, and we ’ re feeling kind of helpless. We look at these **pictures** and we understand, for the first time, our role in this supply chain. We connect the dots in the supply chain. And we have this kind of — as voters, we have kind of a”eureka!”moment. This is why these moments of these oil spills are so important. But it ’ s also really important that we don ’ t get distracted by the theater or the morals of it. We actually need to go in and work on the roots of the problem. One of the things that happened with the two previous oil spills was that we really worked on some of the symptoms. We were very reactive, as opposed to being proactive about what happened. And so what we did was, actually, we made moratoriums on the east and west coasts on drilling. We stopped drilling in ANWR, but we didn ’ t actually reduce the amount of oil that we consumed. In fact, it ’ s continued to increase. The only thing that really reduces the amount of oil that we consume is much higher prices. As you can see, our own production has fallen off as our reservoirs have gotten old and expensive to drill out. We only have two percent of the world ’ s oil reserves ; 65 percent of them are in the Persian Gulf. One of the things that ’ s happened because of this is that, since 1969, the country of Nigeria, or the part of Nigeria that pumps oil, which is the delta — which is two times the size of Maryland — has had thousands of oil spills a year. I mean, we ’ ve essentially been exporting oil spills when we import oil from places without tight environmental regulations. That has been the equivalent of an Exxon Valdez spill every year since 1969. And we can wrap our heads around the spills, because that ’ s what we see here, but in fact, these guys actually live in a war zone. There ’ s a thousand battle-related deaths a year in this area twice the size of Maryland, and it ’ s all related to the oil. And these guys, I mean, if they were in the U. S., they might be actually here in this room. They have degrees in political science, degrees in business — they ’ re entrepreneurs. They don ’ t actually want to be doing what they ’ re doing. And it ’ s sort of one of the other groups of people who pay a price for us. The other thing that we ’ ve done, as we ’ ve continued to increase demand, is that we kind of play a shell game with the costs. One of the places we put in a big oil project in Chad, with Exxon. So the U. S. taxpayer paid for it ; the World Bank, Exxon paid for it. We put it in. There was a tremendous banditry problem. I was there in 2003. We were driving along this dark, dark road, and the guy in the green stepped out, and I was just like,”Ahhh! This is it.”And then the guy in the Exxon uniform stepped out, and we realized it was okay. They have their own private sort of army around them at the oil fields. But at the same time, Chad has become much more unstable, and we are not paying for that price at the pump. We pay for it in our taxes on April 15th. We do the same thing with the price of policing the Persian Gulf and keeping the shipping lanes open. This is 1988 — we actually bombed two Iranian oil platforms that year. That was the beginning of an escalating U. S. involvement there that we do not pay for at the pump. We pay for it on April 15th, and we can ’ t even calculate the cost of this involvement. The other place that is sort of supporting our dependence on oil and our increased consumption is the Gulf of Mexico, which was not part of the moratoriums. Now what ’ s happened in the Gulf of Mexico — as you can see, this is the Minerals Management diagram of wells for **gas** and oil. It ’ s become this intense industrialized zone. It doesn ’ t have the same resonance for us that the Arctic National Wildlife Refuge has, but it should, I mean, it ’ s a bird sanctuary. Also, every time you buy gasoline in the United States, half of it is actually being refined along the coast, because the Gulf actually has about 50 percent of our refining capacity and a lot of our marine terminals as well. So the people of the Gulf have essentially been subsidizing the rest of us through a less-clean environment. And finally, American families also pay a price for oil. Now on the one hand, the price at the pump is not really very high when you consider the actual cost of the oil, but on the other hand, the fact that people have no other transit options means that they pay a large amount of their income into just getting back and forth to work, generally in a fairly crummy car. If you look at people who make $50, 000 a year, they have two kids, they might have three jobs or more, and then they have to really commute. They ’ re actually spending more on their car and fuel than they are on taxes or on health care. And the same thing happens at the 50th percentile, around 80, 000. Gasoline costs are a tremendous drain on the American economy, but they ’ re also a drain on individual families and it ’ s kind of terrifying to think about what happens when prices get higher. So, what I ’ m going to talk to you about now is : what do we have to do this time? What are the laws? What do we have to do to keep ourselves focused? One thing is — we need to stay away from the theater. We need to stay away from the moratoriums. We need to focus really back again on the molecules. The moratoriums are fine, but we do need to focus on the molecules on the oil. One of the things that we also need to do, is we need to try to not kind of fool ourselves into thinking that you can have a green world, before you reduce the amount of oil that we use. We need to focus on reducing the oil. What you see in this top **drawing** is a schematic of how petroleum gets used in the U. S. economy. It comes in on the side — the useful stuff is the dark gray, and the un-useful stuff, which is called the rejected energy — the waste, goes up to the top. Now you can see that the waste far outweighs the actually useful amount. And one of the things that we need to do is, not only fix the fuel efficiency of our vehicles and make them much more efficient, but we also need to fix the economy in general. We need to remove the perverse incentives to use more fuel. For example, we have an insurance system where the person who drives 20, 000 miles a year pays the same insurance as somebody who drives 3, 000. We actually encourage people to drive more. We have policies that reward sprawl — we have all kinds of policies. We need to have more mobility choices. We need to make the **gas** price better reflect the real cost of oil. And we need to shift subsidies from the oil industry, which is at least 10 billion dollars a year, into something that allows middle-class people to find better ways to commute. Whether that ' s getting a much more efficient car and also kind of building markets for new cars and new fuels down the road, this is where we need to be. We need to kind of rationalize this whole thing, and you can find more about this policy. It ' s called STRONG, which is"Secure Transportation Reducing Oil Needs Gradually,"and the idea is instead of being helpless, we need to be more strong. They ' re up at NewAmerica. net. What ' s important about these is that we try to move from feeling helpless at the pump, to actually being active and to really sort of thinking about who we are, having kind of that special moment, where we connect the dots actually at the pump. Now supposedly, oil taxes are the third rail of American politics — the no- fly zone. I actually — I agree that a dollar a gallon on oil is probably too much, but I think that if we started this year with three cents a gallon on gasoline, and upped it to six cents next year, nine cents the following year, all the way up to 30 cents by 2020, that we could actually significantly reduce our gasoline consumption, and at the same time we would give people time to prepare, time to respond, and we would be raising money and raising consciousness at the same time. Let me give you a little sense of how this would work. This is a **gas** receipt, hypothetically, for a year from now. The first thing that you have on the tax is — you have a tax for a stronger America — 33 cents. So you ' re not helpless at the pump. And the second thing that you have is a kind of warning sign, very similar to what you would find on a cigarette pack. And what it says is,"The National Academy of Sciences estimates that every gallon of **gas** you burn in your car creates 29 cents in health care costs."That ' s a lot. And so this — you can see that you ' re paying considerably less than the health care costs on the tax. And also, the hope is that you start to be connected to the whole greater system. And at the same time, you have a number that you can call to get more information on commuting, or a low-interest loan on a different kind of car, or whatever it is you ' re going to need to actually reduce your gasoline dependence. With this whole sort of suite of policies, we could actually reduce our gasoline consumption — or our oil consumption — by 20 percent by 2020. So, three million barrels a day. But in order to do this, one of the things we really need to do, is we need to remember we are people of the hydrocarbon. We need to keep or minds on the molecules and not get distracted by the theater, not get distracted by the cognitive dissonance of the green possibilities that are out there. We need to kind of get down and do the gritty work of reducing our dependence upon this fuel and these molecules. Thank you.

Text Sample 695: POS Dim 5 – 2011 – Score 8.00 – 2011/t002523.txt

I am a papercutter. I cut stories. So my process is very straightforward. I take a piece of paper, I visualize my story, sometimes I sketch, sometimes I don ' t. And as my image is already inside the paper, I just have to remove what ' s not from that story. So I didn ' t come to papercutting in a straight line. In fact, I see it more as a spiral. I was not born with a blade in my hand. And I don ' t remember papercutting as a child. As a teenager, I was sketching, drawing, and I wanted to be an artist. But I was also a rebel. And I left everything and went for a long series of odd jobs. So among them, I have been a shepherdess, a truck driver, a factory worker, a cleaning lady. I worked in tourism for one year in Mexico, one year in Egypt. I moved for two years in Taiwan. And then I settled in New York where I became a tour guide. And I still worked as a tour leader, traveled back and forth in China, Tibet and Central Asia. So of course, it took time, and I was nearly 40, and I decided it ' s time to start as an artist. I chose papercutting because paper is cheap, it ' s light, and you can use it in a lot of different ways. And I chose the language of silhouette because graphically it ' s very efficient. And it ' s also just getting to the essential of things. So the word"silhouette"comes from a minister of finance, Etienne de Silhouette. And he slashed so many budgets that people said they couldn ' t afford paintings anymore, and they needed to have their portrait"a la silhouette."So I made series of images, cuttings, and I assembled them in portfolios. And people told me — like these 36 views of the Empire State building — they told me,"You ' re making artist books."So artist books have a lot of definitions. They come in a lot of different shapes. But to me, they are **fascinating** objects to visually narrate a story. They can be with words or without words. And I have a passion for images and for words. I love pun and the relation to the unconscious. I love oddities of languages. And everywhere I lived, I learned the languages, but never mastered them. So I ' m always looking for the false cognates or identical words in different languages. So as you can guess, my mother tongue is French. And my daily language is English. So I did a series of work where it was identical words in French and in English. So one of these works is the"Spelling Spider."So the Spelling Spider is a cousin of the spelling bee. But it ' s much more connected to the Web. And this spider spins a bilingual alphabet. So you can read"architecture active"or"active architecture."So this spider goes through the whole alphabet with identical adjectives and substantives. So if you don ' t know one of these languages, it ' s instant learning. And one ancient form of the book is scrolls. So scrolls are very convenient, because you can create a large image on a very small table. So the unexpected consequences of that is that you only see one part of your image, so it makes a very freestyle architecture. And I ' m making all those kinds of windows. So it ' s to look beyond the surface. It ' s to have a look at different worlds. And very often I ' ve been an outsider. So I want to see how things work and what ' s happening. So each window is an image and is a world that I often revisit. And I revisit this world thinking about the image or clich about what we want to do, and what are the words, colloquialisms, that we have with the expressions. It ' s all if. So what if we were living in balloon houses? It would make a very uplifting world. And we would leave a very low footprint on the planet. It would be so light. So sometimes I view from the inside, like EgoCentriCity and the inner circles. Sometimes it ' s a global view, to see our common roots and how we can use them to catch dreams. And we can use them also as a safety net. And my inspirations are very eclectic. I ' m influenced by everything I read, everything I see. I have some stories that are humorous, like"Dead Beats."Other ones are historical. Here it ' s"CandyCity."It ' s a non-sugar- coated history of sugar. It goes from slave trade to over-consumption of sugar with some sweet moments in between. And sometimes I have an emotional response to news, such as the 2010 Haitian earthquake. Other times, it ' s not even my stories. People tell me their lives, their memories, their aspirations, and I create a mindscape. I channel their history [so that] they have a place to go back to look at their life and its possibilities. I call them Freudian cities. I cannot speak for all my images, so I ' ll just go through a few of my worlds just with the title."ModiCity.""ElectriCity.""MAD Growth on Columbus Circle.""ReefCity.""A Web of Time.""Chaos City.""Daily Battles.""FeliCity.""Floating Islands."And at one point, I had to do"The Whole Nine Yards."So it ' s actually a papercut that ' s nine yards long. So in life and in papercutting, everything is connected. One story leads to another. I was also interested in the physicality of this format, because you have to walk to see it. And parallel to my cutting is my running. I started with small images, I started with a few miles. Larger images, I started to run marathons. Then I went to run 50K, then 60K. Then I ran 50 miles — ultramarathons. And I still feel I ' m running, it ' s just the training to become a long-distance papercutter. And running gives me a lot of energy. Here is a three-week papercutting marathon at the Museum of Arts and Design in New York City. The result is"Hells and Heavens."It ' s two **panels** 13 ft. high. They were installed in the **museum** on two floors, but in fact, it ' s a continuous image. And I call it"Hells and Heavens"because it ' s daily **hells** and daily heavens. There is no border in between. Some people are born in **hells**, and against all odds, they make it to heavens. Other people make the opposite trip. That ' s the border. You have sweatshops in **hells**. You have people renting their wings in the heavens. And then you have all those individual stories where sometimes we even have the same action, and the result puts you in **hells** or in heavens. So the whole"Hells and Heavens"is about free will and determinism. And in papercutting, you have the **drawing** as the structure itself. So you can take it off the wall. Here it ' s an artist book installation called"Identity Project."It ' s not autobiographical identities. They are more our social identities. And then you can just walk behind them and try them on. So it ' s like the different layers of what we are made of and what we present to the world as an identity. That ' s another artist book project. In fact, in the **picture**, you have two of them. It ' s one I ' m wearing and one that ' s on exhibition at the Center for Books Arts in New York City. Why do I call it a book? It ' s called"Fashion Statement,"and there are quotes about fashion, so you can read it, and also, because the definition of artist book is very generous. So artist books, you take them off the wall. You take them for a walk. You can also install them as public art. Here it ' s in Scottsdale, Arizona, and it ' s called"Floating Memories."So it ' s regional memories, and they are just randomly moved by the wind. I love public art. And I entered competitions for a long time. After eight years of rejection, I was thrilled to get my first commission with the Percent for Art in New York City. It was for a merger station for emergency workers and firemen. I made an artist book that ' s in stainless steel instead of paper. I called it"Working in the Same Direction."But I added weathervanes on both sides to show that they cover all directions. With public art, I could also make cut glass. Here it ' s faceted glass in the Bronx. And each time I make public art, I want something that ' s really relevant to the place it ' s installed. So for the subway in New York, I saw a correspondence between riding the subway and reading. It is travel in time, travel on time. And Bronx literature, it ' s all about Bronx writers and their stories. Another glass project is in a public library in San Jose, California. So I made a vegetable point of view of the growth of San Jose. So I started in the center with the acorn for the Ohlone Indian civilization. Then I have the fruit from Europe for the ranchers. And then the fruit of the world for Silicon Valley today. And it ' s still growing. So the technique, it ' s cut, sandblasted, etched and printed glass into architectural glass. And outside the library, I wanted to make a place to cultivate your mind. I took library material that had fruit in their title and I used them to make an orchard walk with these fruits of knowledge. I also planted the bibliotree. So it ' s a tree, and in its trunk you have the roots of languages. And it ' s all about international writing systems. And on the branches you have library material growing. You can also have function and form with public art. So in Aurora, Colorado it ' s a bench. But you have a bonus with this bench. Because if you sit a long time in summer in shorts, you will walk away with temporary branding of the story element on your thighs.

Another functional work, it ’ s in the south side of Chicago for a subway station. And it ’ s called”Seeds of the Future are Planted Today.”It ’ s a story about transformation and connections. So it acts as a screen to protect the rail and the commuter, and not to have objects falling on the rails. To be able to change fences and window guards into flowers, it ’ s fantastic. And here I ’ ve been working for the last three years with a South Bronx developer to bring art to life to low-income buildings and affordable housing. So each building has its own personality. And sometimes it ’ s about a legacy of the neighborhood, like in Morrisania, about the jazz history. And for other projects, like in Paris, it ’ s about the name of the street. It ’ s called Rue **des** Prairies — Prairie Street. So I brought back the rabbit, the dragonfly, to stay in that street. And in 2009, I was asked to make a poster to be placed in the subway cars in New York City for a year. So that was a very captive audience. And I wanted to give them an escape. I created”All Around Town.”It is a papercutting, and then after, I added color on the computer. So I can call it techno-crafted. And along the way, I ’ m kind of making papercuttings and adding other techniques. But the result is always to have stories. So the stories, they have a lot of possibilities. They have a lot of scenarios. I don ’ t know the stories. I take images from our global imagination, from clich, from things we are thinking about, from history. And everybody ’ s a narrator, because everybody has a story to tell. But more important is everybody has to make a story to make sense of the world. And in all these universes, it ’ s like imagination is the vehicle to be transported with, but the destination is our minds and how we can reconnect with the essential and with the magic. And it ’ s what story cutting is all about.

Text Sample 696: POS Dim 5 – 2011 – Score 7.00 – 2011/t002595.txt
I ’ m a contemporary artist with a bit of an unexpected background. I was in my 20s before I ever went to an art **museum**. I grew up in the middle of nowhere on a dirt road in rural Arkansas, an hour from the nearest movie theater. And I think it was a great place to grow up as an artist because I grew up around quirky, colorful characters who were great at making with their hands. And my childhood is more hick than I could ever possibly relate to you, and also more intellectual than you would ever expect. For instance, me and my sister, when we were little, we would compete to see who could eat the most squirrel brains. But on the other side of that, though, we were big readers in our house. And if the TV was on, we were watching a documentary. And my dad is the most voracious reader I know. He can read a novel or two a day. But when I was little, I remember, he would kill flies in our house with my BB gun. And what was so amazing to me about that — well he would be in his recliner, would holler for me to fetch the BB gun, and I ’ d go get it. And what was amazing to me — well it was pretty kickass ; he was killing a fly in the house with a gun — but what was so amazing to me was that he knew just enough how to pump it. And he could shoot it from two rooms away and not damage what it was on because he knew how to pump it just enough to kill the fly and not damage what it landed on. So I should talk about art. Or we ’ ll be here all day with my childhood stories. I love contemporary art, but I ’ m often really frustrated with the contemporary art world and the contemporary art scene. A few years ago, I spent months in Europe to see the major international art exhibitions that have the pulse of what is supposed to be going on in the art world. And I was struck by going to so many, one after the other, with some clarity of what it was that I was longing for. And I was longing for several things that I wasn ’ t getting, or not getting enough of. But two of the main things :

one of it, I was longing for more work that was appealing to a broad public, that was accessible. And the second thing that I was longing for was some more exquisite craftsmanship and technique. So I started thinking and listing what all it was that I thought would make a perfect biennial. So I decided, I ' m going to start my own biennial. I ' m going to organize it and direct it and get it going in the world. So I thought, okay, I have to have some criteria of how to choose work. So amongst all the criteria I have, there ' s two main things. One of them, I call my Mimaw ' s Test. And what that is is I imagine explaining a work of art to my grandmother in five minutes, and if I can ' t explain it in five minutes, then it ' s too obtuse or esoteric and it hasn ' t been refined enough yet. It needs to worked on until it can speak fluently. And then my other second set of rules — I hate to say"rules"because it ' s art — my criteria would be the three H ' s, which is head, heart and hands. And great art would have"head": it would have interesting intellectual ideas and concepts. It would have"heart"in that it would have passion and heart and soul. And it would have"hand"in that it would be greatly crafted. So I started thinking about how am I going to do this biennial, how am I going to travel the world and find these artists? And then I realized one day, there ' s an easier solution to this. I ' m just going to make the whole thing myself. And so this is what I did. So I thought, a biennial needs artists. I ' m going to do an international biennial ; I need artists from all around the world. So what I did was I invented a hundred artists from around the world. I figured out their bios, their passions in life and their art styles, and I started making their work. I felt, oh this is the kind of project that I could spend my whole life doing. So I decided, I ' m going to make this a real biennial. It ' s going to be two years of studio work. And I ' m going to create this in two years, and I have. So I should start to talk about these guys. Well the range is quite a bit. And I ' m such a technician, so I loved this project, getting to play with all the techniques. So for example, in realist paintings, it ranges from this, which is kind of old masters style, to really realistic still-life, to this type of painting where I ' m painting with a single hair. And then at the other end, there ' s performance and short **films** and indoor installations like this indoor installation and this one, and outdoor installations like this one and this one. I know I should mention : I ' m making all these things. This isn ' t Photoshopped. I ' m under the river with those fish. So now let me introduce some of my fictional artists to you. This is Nell Remmel. Nell is interested in agricultural processes, and her work is based in these practices. This piece, which is called"Flipped Earth"— she was interested in taking the sky and using it to cleanse barren ground. And by taking giant mirrors — and here she ' s taking giant mirrors and pulling them into the dirt. And this is 22 **feet** long. And what I loved about her work is, when I would walk around it and look down into the sky, looking down to watch the sky, and it unfolded in a new way. And probably the best part of this piece is at dusk and dawn when the twilight wedge has fallen and the ground ' s dark, but there ' s still the light above, bright above. And so you ' re standing there and everything else is dark, but there ' s this portal that you want to jump in. This piece was great. This is in my parents ' backyard in Arkansas. And I love to dig a hole. So this piece was great **fun** because it was two days of digging in soft dirt. The next artist is Kay Overstry, and she ' s interested in ephemerality and transience. And in her most recent project, it ' s called"Weather I Made."And she ' s making weather on her body ' s scale. And this piece is"Frost."And what she did was she went out on a cold, dry night and breathed back and forth on the lawn to leave — to leave her life ' s mark, the mark of her life. And so this is five-foot, five-inches of frost that she left behind. The sun rises, and it melts away. And that was played by my mom. So the next artist, this is a group of Japanese artists, a collective of Japanese artists — in Tokyo. And they were interested in developing a new, alternative art space, and they needed funding for it, so they decided to come up with some interesting fundraising projects. One of these is scratch-off masterpieces. And so what they ' re doing — each of these artists on a nine-by-seven-inch card, which they sell for 10 bucks, they drew original works of art. And you buy one, and maybe you get a real piece, and maybe not. Well this has sparked a craze in Japan, because everyone ' s wanting a masterpiece. And the ones that are the most sought after are the ones that are only barely scratched off. And all these works, in some way, talk about luck or fate or chance. Those first two are portraits of mega- jackpot winners years before and after their win. And in this one it ' s called"Drawing the Short Stick."I love this piece because I have a little cousin at home who introduced me — which I think is such a great introduction — to a friend one day as,"This is my cousin Shea. He draws sticks real good."Which is one of the best compliments ever. This artist is Gus Weinmueller, and he ' s doing a project, a large project, called"Art for the Peoples."And within this project, he ' s doing a smaller project called"Artists in Residence."And what he does is — he spends a week at a time with a family. And he shows up on their porch, their doorstep, with a toothbrush and pajamas, and he ' s ready to spend the week with them. And using only what ' s present, he goes in and makes a little abode studio to work out of. And he spends that week talking to the family about what do they think great art is. He has all these discussions with their family, and he digs through everything they have, and he finds materials to make work. And he makes a work that answers what they think great art is. For this family, he made this still-life painting. And whatever he makes somehow references nesting and space and personal property. This next project, this is by Jaochim Parisvega, and he ' s interested in — he believes art is everywhere waiting — that it just needs a little bit of a push to happen. And he provides this push by harnessing natural forces, like in his series where he used rain to make paintings. This project is called"Love Nests."What he did was to get wild birds to make his art for him. So he put the material in places where the birds were going to collect them, and they crafted his nests for him. And this one ' s called"Lovelock ' s Nest."This one ' s called"Mixtape Love Song ' s Nest."And this one ' s called"Lovemaking Nest."Next is Sylvia Slater. Sylvia ' s interested in art training. She ' s a very serious Swiss artist. And she was thinking about her friends and family who work in chaos-ridden places and developing countries, and she was thinking, what can I make that would be of value to them, in case something bad happens and they have to buy their way across the border or pay off a gunman? And so she came up with creating these pocket-sized artworks that are portraits of the person that would carry them. And you would carry this around with you, and if everything went to **hell**, you could make payments and buy your life. So this life price is for an irrigation non-profit director. So hopefully what happens is you never use it, and it ' s an heirloom that you pass down. And she makes them so they could either be broken up into payments, or they could be like these, which are leaves that can be payments. And so they ' re valuable. This is precious **metals** and gemstones. And this one had to get broken up. He had to break off a piece to get out of Egypt recently. This is by a duo, Michael Abernathy and Bud Holland. And they ' re interested in creating culture, just tradition. So what they do is they move into an area and try to establish a new tradition in a small geographic area. So this is in Eastern Tennessee, and what they decided was that we need a positive tradition that goes with death. So they came up with"dig jigs."And a dig jig — a dig jig is where, for a milestone anniversary or a birthday, you gather all your friends and family together and you dance on where you ' re going to be buried. And we got a lot of attention when we did it. I talked my family into doing this, and they didn ’ t know what I was doing. And I was like,”Get dressed for a funeral. We ’ re going to go do some work.”And so we got to the grave and made this, which was hilarious — the attention that we got. So what happens is you dance on the grave, and after you ’ ve done your dance, everyone toasts you and tells you how great you are. And you in essence have a funeral that you get to be present for. That ’ s my mom and dad. This is by Jason Birdsong. He is interested in how we see as an animal, how we are interested in mimicry and camouflage. You know, we look down a dark alley or a jungle path, trying to make out a face or a creature. We just have that natural way of seeing. And he plays with this idea. And this piece : those aren ’ t actually leaves. They ’ re butterfly specimens who have a natural camouflage. So he pairs these up. There ’ s another pile of leaves. Those are actually all real butterfly specimens. And he pairs these up with paintings. Like this is a painting of a snake in a box. So you open the box and you think,”Whoa, there ’ s a snake in there.”But it ’ s actually a painting. So he makes these interesting conversations about realism and mimicry and our drive to be fooled by great camouflage. The next artist is Hazel Clausen. Hazel Clausen is an anthropologist who took a sabbatical and decided,”You know, I would learn a lot about culture if I created a culture that doesn ’ t exist from scratch.”So that ’ s what she did. She created the Swiss people named the Uvulites, and they have this distinctive yodeling song that they use the uvula for. And also they reference how the uvula — everything they say is fallen because of the forbidden fruit. And that ’ s the symbol of their culture. And this is from a documentary called”Sexual Practices and Populations Control Among the Uvulites.”This is a typical angora embroidery for them. This is one of their founders, Gert Schaeffer. And actually this is my Aunt Irene. It was so funny having a fake person who was making fake things. And I crack up at this piece, because when I see it I know that ’ s French angora and all antique German ribbons and wool that I got in a Nebraska mill and carried around for 10 years and then antique Chinese skirts. The next is a collective of artists called the Silver Dobermans, and their motto is to spread pragmatism one person at a time. And they ’ re really interested in how over-coddled we ’ ve become. So this is one of their comments on how over-coddled we ’ ve become. And what they ’ ve done is they put a warning sign on every single barb on this fence. And this is called”Horse Sense Fence.”The next artist is K. M. Yoon, a really interesting South Korean artist. And he ’ s reworking a Confucian art tradition of scholar stones. Next is Maynard Sipes. And I love Maynard Sipes, but he ’ s off in his own world, and, bless his heart, he ’ s so paranoid. Next is Roy Penig, a really interesting Kentucky artist, and he ’ s the nicest guy. He even once traded a work of art for a block of government cheese because the person wanted it so badly. Next is an Australian artist, Janeen Jackson, and this is from a project of hers called”What an Artwork Does When We ’ re Not Watching.”Next is by a Lithuanian fortune teller, Jurgi Petrauskas. Next is Ginger Cheshire. This is from a short **film** of hers called”The Last Person.”And that ’ s my cousin and my sister ’ s dog, Gabby. The next, this is by Sam Sandy. He ’ s an Australian Aboriginal elder, and he ’ s also an artist. And this is from a large traveling sculpture project that he ’ s doing. This is from Estelle Willoughsby. She heals with color. And she ’ s one of the most prolific of all these hundred artists, even though she ’ s going to be 90 next year. This is by Z. Zhou, and he ’ s interested in stasis. Next is by Hilda Singh, and she ’ s doing a whole project called”Social Outfits.”Next is by Vera Sokolova. And I have to say, Vera kind of scares me. You can ’ t look her directly in the eyes because she ’ s kind of scary. And it ’ s good that she ’ s not real ; she ’ d be mad that I said that. And she ’ s an optometrist in St. Petersburg, and she plays with optics. Next, this is by Thomas Swifton. This is from a short **film**,"Adventures with Skinny."And this is by Cicily Bennett, and it ' s from a series of short **films**. And after this one, there ' s 77 other artists. And all together with those other 77 you ' re not seeing, that ' s my biennial. Thank you. Thank you. Thanks. Thank you. Thanks.

Text Sample 697: POS Dim 5 – 2011 – Score 7.00 – 2011/t002591.txt

I ' d like to begin with a thought experiment. Imagine that it ' s 4, 000 years into the future. Civilization as we know it has ceased to exist â no books, no electronic devices, no Facebook or Twitter. All knowledge of the English language and the English alphabet has been lost. Now imagine archeologists digging through the rubble of one of our cities. What might they find? Well perhaps some rectangular pieces of plastic with strange symbols on them. Perhaps some circular pieces of **metal**. Maybe some cylindrical containers with some symbols on them. And perhaps one archeologist becomes an instant celebrity when she discovers â buried in the hills somewhere in North America â massive versions of these same symbols. Now let ' s ask ourselves, what could such artifacts say about us to people 4, 000 years into the future? This is no hypothetical question. In fact, this is exactly the kind of question we ' re faced with when we try to understand the Indus Valley civilization, which existed 4, 000 years ago. The Indus civilization was roughly contemporaneous with the much better known Egyptian and the Mesopotamian civilizations, but it was actually much larger than either of these two civilizations. It occupied the area of approximately one million square kilometers, covering what is now Pakistan, Northwestern India and parts of Afghanistan and Iran. Given that it was such a vast civilization, you might expect to find really powerful rulers, kings, and huge monuments glorifying these powerful kings. In fact, what archeologists have found is none of that. They ' ve found small objects such as these. Here ' s an example of one of these objects. Well obviously this is a replica. But who is this person? A king? A god? A priest? Or perhaps an ordinary person like you or me? We don ' t know. But the Indus people also left behind artifacts with writing on them. Well no, not pieces of plastic, but stone seals, copper tablets, pottery and, surprisingly, one large sign board, which was found buried near the gate of a city. Now we don ' t know if it says Hollywood, or even Bollywood for that matter. In fact, we don ' t even know what any of these objects say, and that ' s because the Indus script is undeciphered. We don ' t know what any of these symbols mean. The symbols are most commonly found on seals. So you see up there one such object. It ' s the square object with the unicorn-like animal on it. Now that ' s a magnificent piece of art. So how big do you think that is? Perhaps that big? Or maybe that big? Well let me show you. Here ' s a replica of one such seal. It ' s only about one inch by one inch in size â pretty tiny. So what were these used for? We know that these were used for stamping clay tags that were attached to bundles of goods that were sent from one place to the other. So you know those packing slips you get on your FedEx boxes? These were used to make those kinds of packing slips. You might wonder what these objects contain in terms of their text. Perhaps they ' re the name of the sender or some information about the goods that are being sent from one place to the other â we don ' t know. We need to decipher the script to answer that question. Deciphering the script is not just an intellectual puzzle ; it ' s actually become a question that ' s become deeply intertwined with the politics and the cultural history of South Asia. In fact, the script has become a battleground of sorts between three different groups of people. First, there ' s a group of people who are very passionate in their belief that the Indus script does not represent a language at all. These people believe that the symbols are very similar to the kind of symbols you find on traffic signs or the emblems you find on shields. There ’ s a second group of people who believe that the Indus script represents an Indo-European language. If you look at a map of India today, you ’ ll see that most of the languages spoken in North India belong to the Indo-European language family. So some people believe that the Indus script represents an ancient Indo-European language such as Sanskrit. There ’ s a last group of people who believe that the Indus people were the ancestors of people living in South India today. These people believe that the Indus script represents an ancient form of the Dravidian language family, which is the language family spoken in much of South India today. And the proponents of this theory point to that small pocket of Dravidian-speaking people in the North, actually near Afghanistan, and they say that perhaps, sometime in the past, Dravidian languages were spoken all over India and that this suggests that the Indus civilization is perhaps also Dravidian. Which of these hypotheses can be true? We don ’ t know, but perhaps if you deciphered the script, you would be able to answer this question. But deciphering the script is a very challenging task. First, there ’ s no Rosetta Stone. I don ’ t mean the software ; I mean an ancient artifact that contains in the same text both a known text and an unknown text. We don ’ t have such an artifact for the Indus script. And furthermore, we don ’ t even know what language they spoke. And to make matters even worse, most of the text that we have are extremely short. So as I showed you, they ’ re usually found on these seals that are very, very tiny. And so given these formidable obstacles, one might wonder and worry whether one will ever be able to decipher the Indus script. In the rest of my talk, I ’ d like to tell you about how I learned to stop worrying and love the challenge posed by the Indus script. I ’ ve always been **fascinated** by the Indus script ever since I read about it in a middle school textbook. And why was I **fascinated**? Well it ’ s the last major undeciphered script in the ancient world. My career path led me to become a computational neuroscientist, so in my day job, I create computer models of the brain to try to understand how the brain makes predictions, how the brain makes decisions, how the brain learns and so on. But in 2007, my path crossed again with the Indus script. That ’ s when I was in India, and I had the wonderful opportunity to meet with some Indian scientists who were using computer models to try to analyze the script. And so it was then that I realized there was an opportunity for me to collaborate with these scientists, and so I jumped at that opportunity. And I ’ d like to describe some of the results that we have found. Or better yet, let ’ s all collectively decipher. Are you ready? The first thing that you need to do when you have an undeciphered script is try to figure out the direction of writing. Here are two texts that contain some symbols on them. Can you tell me if the direction of writing is right to left or left to right? I ’ ll give you a couple of seconds. Okay. Right to left, how many? Okay. Okay. Left to right? Oh, it ’ s almost 50 / 50. Okay. The answer is : if you look at the **left-hand** side of the two texts, you ’ ll notice that there ’ s a cramping of signs, and it seems like 4, 000 years ago, when the scribe was writing from right to left, they ran out of space. And so they had to cram the sign. One of the signs is also below the text on the top. This suggests the direction of writing was probably from right to left, and so that ’ s one of the first things we know, that directionality is a very key aspect of linguistic scripts. And the Indus script now has this particular property. What other properties of language does the script show? Languages contain patterns. If I give you the letter Q and ask you to predict the next letter, what do you think that would be? Most of you said U, which is right. Now if I asked you to predict one more letter, what do you think that would be? Now there ’ s several thoughts. There ’ s E. It could be I. It could be A, but certainly not B, C or D, right? The Indus script also exhibits similar kinds of patterns. There ’ s a lot of text that start with this diamond-shaped symbol. And this in turn tends to be followed by this quotation marks-like symbol. And this is very similar to a Q and U example. This symbol can in turn be followed by these fish-like symbols and some other signs, but never by these other signs at the **bottom**. And furthermore, there ’ s some signs that really prefer the end of texts, such as this jar-shaped sign, and this sign, in fact, happens to be the most frequently occurring sign in the script. Given such patterns, here was our idea. The idea was to use a computer to learn these patterns, and so we gave the computer the existing texts. And the computer learned a statistical model of which symbols tend to occur together and which symbols tend to follow each other. Given the computer model, we can test the model by essentially quizzing it. So we could deliberately erase some symbols, and we can ask it to predict the missing symbols. Here are some examples. You may regard this as perhaps the most ancient game of Wheel of Fortune. What we found was that the computer was successful in 75 percent of the cases in predicting the correct symbol. In the rest of the cases, typically the second best guess or third best guess was the right answer. There ’ s also practical use for this particular procedure. There ’ s a lot of these texts that are damaged. Here ’ s an example of one such text. And we can use the computer model now to try to complete this text and make a best guess prediction. Here ’ s an example of a symbol that was predicted. And this could be really useful as we try to decipher the script by generating more data that we can analyze. Now here ’ s one other thing you can do with the computer model. So imagine a monkey sitting at a keyboard. I think you might get a random jumble of letters that looks like this. Such a random jumble of letters is said to have a very high entropy. This is a physics and information theory term. But just imagine it ’ s a really random jumble of letters. How many of you have ever spilled coffee on a keyboard? You might have encountered the stuck-key problem â so basically the same symbol being repeated over and over again. This kind of a sequence is said to have a very low entropy because there ’ s no variation at all. Language, on the other hand, has an intermediate level of entropy ; it ’ s neither too rigid, nor is it too random. What about the Indus script? Here ’ s a graph that plots the entropies of a whole bunch of sequences. At the very top you find the uniformly random sequence, which is a random jumble of letters â and interestingly, we also find the DNA sequence from the human genome and instrumental music. And both of these are very, very flexible, which is why you find them in the very high range. At the lower end of the scale, you find a rigid sequence, a sequence of all A ’ s, and you also find a computer program, in this case in the language Fortran, which obeys really strict rules. Linguistic scripts occupy the middle range. Now what about the Indus script? We found that the Indus script actually falls within the range of the linguistic scripts. When this result was first published, it was highly controversial. There were people who raised a hue and cry, and these people were the ones who believed that the Indus script does not represent language. I even started to get some hate mail. My students said that I should really seriously consider getting some protection. Who ’ d have thought that deciphering could be a dangerous profession? What does this result really show? It shows that the Indus script shares an important property of language. So, as the old saying goes, if it looks like a linguistic script and it acts like a linguistic script, then perhaps we may have a linguistic script on our hands. What other evidence is there that the script could actually encode language? Well linguistic scripts can actually encode multiple languages. So for example, here ’ s the same sentence written in English and the same sentence written in Dutch using the same letters of the alphabet. If you don ’ t know Dutch and you only know English and I give you some words in Dutch, you ’ ll tell me that these words contain some very unusual patterns. Some things are not right, and you ’ ll say these words are probably not English words. The same thing happens in the case of the Indus script. The computer found several texts â two of them are shown here â that have very unusual patterns. So for example the first text : there ’ s a doubling of this jar-shaped sign. This sign is the most frequently-occurring sign in the Indus script, and it ’ s only in this text that it occurs as a doubling pair. Why is that the case? We went back and looked at where these particular texts were found, and it turns out that they were found very, very far away from the Indus Valley. They were found in present day Iraq and Iran. And why were they found there? What I haven ’ t told you is that the Indus people were very, very enterprising. They used to trade with people pretty far away from where they lived, and so in this case, they were traveling by sea all the way to Mesopotamia, present-day Iraq. And what seems to have happened here is that the Indus traders, the merchants, were using this script to write a foreign language. It ’ s just like our English and Dutch example. And that would explain why we have these strange patterns that are very different from the kinds of patterns you see in the text that are found within the Indus Valley. This suggests that the same script, the Indus script, could be used to write different languages. The results we have so far seem to point to the conclusion that the Indus script probably does represent language. If it does represent language, then how do we read the symbols? That ’ s our next big challenge. So you ’ ll notice that many of the symbols look like **pictures** of humans, of insects, of fishes, of birds. Most ancient scripts use the rebus principle, which is, using **pictures** to represent words. So as an example, here ’ s a word. Can you write it using **pictures**? I ’ ll give you a couple seconds. Got it? Okay. Great. Here ’ s my solution. You could use the **picture** of a bee followed by a **picture** of a leaf â and that ’ s”belief,”right. There could be other solutions. In the case of the Indus script, the problem is the reverse. You have to figure out the sounds of each of these **pictures** such that the entire sequence makes sense. So this is just like a crossword puzzle, except that this is the mother of all crossword puzzles because the stakes are so high if you solve it. My colleagues, Iravatham Mahadevan and Asko Parpola, have been making some headway on this particular problem. And I ’ d like to give you a quick example of Parpola ’ s work. Here ’ s a really short text. It contains seven vertical strokes followed by this fish-like sign. And I want to mention that these seals were used for stamping clay tags that were attached to bundles of goods, so it ’ s quite likely that these tags, at least some of them, contain names of merchants. And it turns out that in India there ’ s a long tradition of names being based on horoscopes and star constellations present at the time of birth. In Dravidian languages, the word for fish is”meen”which happens to sound just like the word for star. And so seven stars would stand for”elu meen,”which is the Dravidian word for the Big Dipper star constellation. Similarly, there ’ s another sequence of six stars, and that translates to”aru meen,”which is the old Dravidian name for the star constellation Pleiades. And finally, there ’ s other combinations, such as this fish sign with something that looks like a **roof** on top of it. And that could be translated into”mey meen,”which is the old Dravidian name for the planet Saturn. So that was pretty exciting. It looks like we ’ re getting somewhere. But does this prove that these seals contain Dravidian names based on planets and star constellations? Well not yet. So we have no way of validating these particular readings, but if more and more of these readings start making sense, and if longer and longer sequences appear to be correct, then we know that we are on the right track. Today, we can write a word such as TED in Egyptian hieroglyphics and in cuneiform script, because both of these were deciphered in the 19th century. The decipherment of these two scripts enabled these civilizations to speak to us again directly. The Mayans started speaking to us in the 20th century, but the Indus civilization remains silent. Why should we care? The Indus civilization does not belong to just the South Indians or the North Indians or the Pakistanis ; it belongs to all of us. These are our ancestors â yours and mine. They were silenced by an unfortunate accident of history. If we decipher the script, we would enable them to speak to us again. What would they tell us? What would we find out about them? About us? I can ’ t wait to find out. Thank you.

Text Sample 698: POS Dim 5 – 2009 – Score 9.00 – 2009/t002971.txt

The public debate about architecture quite often just stays on contemplating the final result, the architectural object. Is the latest tower in London a gherkin or a sausage or a sex tool? So recently, we asked ourselves if we could invent a format that could actually tell the stories behind the projects, maybe combining images and **drawings** and words to actually sort of tell stories about architecture. And we discovered that we didn ’ t have to invent it, it already existed in the form of a comic book. So we basically copied the format of the comic book to actually tell the stories of behind the scenes, how our projects actually evolve through adaptation and improvisation. Sort of through the turmoil and the opportunities and the incidents of the real world. We call this comic book”Yes is More,”which is obviously a sort of evolution of the ideas of some of our heroes. In this case it ’ s Mies van der Rohe ’ s Less is More. He triggered the modernist revolution. After him followed the post-modern counter-revolution, Robert Venturi saying,”Less is a bore.”After him, Philip Johnson sort of introduced you could say promiscuity, or at least openness to new ideas with,”I am a whore.”Recently, Obama has introduced optimism at a sort of time of global financial crisis. And what we ’ d like to say with”Yes is More”is basically trying to question this idea that the architectural avant-garde is almost always negatively defined, as who or what we are against. The cliche of the radical **architect** is the sort of angry young man rebelling against the establishment. Or this idea of the misunderstood genius, frustrated that the world doesn ’ t fit in with his or her ideas. Rather than revolution, we ’ re much more interested in evolution, this idea that things gradually evolve by adapting and improvising to the changes of the world. In fact, I actually think that Darwin is one of the people who best explains our design process. His famous evolutionary tree could almost be a diagram of the way we work. As you can see, a project evolves through a series of generations of design meetings. At each meeting, there ’ s way too many ideas. Only the best ones can survive. And through a process of architectural selection, we might choose a really beautiful model or we might have a very functional model. We mate them. They have sort of mutant offspring. And through these sort of generations of design meetings we arrive at a design. A very literal way of showing it is a project we did for a library and a hotel in Copenhagen. The design process was really tough, almost like a struggle for survival, but gradually an idea evolved : this sort of idea of a rational tower that melts together with the surrounding city, sort of expanding the public space onto what we refer to as a Scandinavian version of the Spanish Steps in Rome, but sort of public on the outside, as well as on the inside, with the library. But Darwin doesn ’ t only explain the evolution of a single idea. As you can see, sometimes a subspecies branches off. And quite often we sit in a design meeting and we discover that there is this great idea. It doesn ’ t really work in this context. But for another client in another culture, it could really be the right answer to a different question. So as a result, we never throw anything out. We keep our office almost like an archive of architectural biodiversity. You never know when you might need it. And what I ’ d like to do now, in an act of warp-speed storytelling, is tell the story of how two projects evolved by adapting and improvising to the happenstance of the world. The first story starts last year when we went to Shanghai to do the competition for the Danish National Pavilion for the World Expo in 2010. And we saw this guy, Haibao. He ’ s the mascot of the expo, and he looks strangely familiar. In fact he looked like a building we had designed for a hotel in the north of Sweden. When we submitted it for the Swedish competition we thought it was a really cool scheme, but it didn ’ t exactly look like something from the north of Sweden. The Swedish jury didn ’ t think so either. So we lost. But then we had a meeting with a Chinese businessman who saw our design and said,”Wow, that ’ s the Chinese character for the word ’ people. ’”So, apparently this is how you write”people,”as in the People ’ s Republic of China. We even double checked. And at the same time, we got invited to exhibit at the Shanghai Creative Industry Week. So we thought like, this is too much of an opportunity, so we hired a feng shui master. We scaled the building up three times to Chinese proportions, and went to China. So the People ’ s Building, as we called it. This is our two interpreters, sort of reading the architecture. It went on the cover of the Wen Wei Po newspaper, which got Mr. Liangyu Chen, the mayor of Shanghai, to visit the exhibition. And we had the chance to explain the project. And he said,”Shanghai is the city in the world with most skyscrapers,”but to him it was as if the connection to the roots had been cut over. And with the People ’ s Building, he saw an architecture that could bridge the gap between the ancient wisdom of China and the progressive future of China. So we obviously profoundly agreed with him. Unfortunately, Mr. Chen is now in prison for corruption. But like I said, Haibao looked very familiar, because he is actually the Chinese character for”people.”And they chose this mascot because the theme of the expo is”Better City, Better Life.”Sustainability. And we thought, sustainability has grown into being this sort of neo-Protestant idea that it has to hurt in order to do good. You know, you ’ re not supposed to take long, warm showers. You ’ re not supposed to fly on holidays because it ’ s bad for the environment. Gradually, you get this idea that sustainable life is less **fun** than normal life. So we thought that maybe it could be interesting to focus on examples where a sustainable city actually increases the quality of life. We also asked ourselves, what could Denmark possibly show China that would be relevant? You know, it ’ s one of the biggest countries in the world, one of the smallest. China symbolized by the dragon. Denmark, we have a national bird, the swan. China has many great poets, but we discovered that in the People ’ s Republic public school curriculum, they have three fairy tales by An Tu Sheng, or Hans Christian Anderson, as we call him. So that means that all 1. 3 billion Chinese have grown up with”The Emperor ’ s New Clothes,””The Matchstick Girl”and”The Little Mermaid.”It ’ s almost like a fragment of Danish culture integrated into Chinese culture. The biggest tourist attraction in China is the Great Wall. The Great Wall is the only thing that can be seen from the moon. The big tourist attraction in Denmark is The Little Mermaid. That can actually hardly be seen from the canal tours. And it sort of shows the difference between these two cities. Copenhagen, Shanghai, modern, European. But then we looked at recent urban development, and we noticed that this is like a Shanghai street, 30 years ago. All bikes, no cars. This is how it looks today ; all traffic jam. Bicycles have become forbidden many places. Meanwhile, in Copenhagen we ’ re actually expanding the bicycle lanes. A third of all the people commute by bike. We have a free system of bicycles called the City Bike that you can borrow if you visit the city. So we thought, why don ’ t we reintroduce the bicycle in China? We donate 1, 000 bikes to Shanghai. So if you come to the expo, go straight to the Danish pavilion, get a Danish bike, and then continue on that to visit the other pavilions. Like I said, Shanghai and Copenhagen are both port cities, but in Copenhagen the water has gotten so **clean** that you can actually swim in it. One of the first projects we ever did was the harbor bath in Copenhagen, sort of continuing the public realm into the water. So we thought that these expos quite often have a lot of state financed propaganda, images, statements, but no real experience. So just like with a bike, we don ’ t talk about it. You can try it. Like with the water, instead of talking about it, we ’ re going to sail a million liters of harbor water from Copenhagen to Shanghai, so the Chinese who have the courage can actually dive in and feel how **clean** it is. This is where people normally object that it doesn ’ t sound very sustainable to sail water from Copenhagen to China. But in fact, the container ships go full of goods from China to Denmark, and then they sail empty back. So quite often you load water for ballast. So we can actually hitch a ride for free. And in the middle of this sort of harbor bath, we ’ re actually going to put the actual Little Mermaid. So the real Mermaid, the real water, and the real bikes. And when she ’ s gone, we ’ re going to invite a Chinese artist to reinterpret her. The architecture of the pavilion is this sort of loop of exhibition and bikes. When you go to the exhibition, you ’ ll see the Mermaid and the pool. You ’ ll walk around, start looking for a bicycle on the **roof**, jump on your ride and then continue out into the rest of the expo. So when we actually won the competition we had to do an exhibition in China explaining the project. And to our surprise we got one of our boards back with corrections from the Chinese state censorship. The first thing, the China map missed Taiwan. It ’ s a very serious political issue in China. We will add on. The second thing, we had compared the swan to the dragon, and then the Chinese state said,”Suggest change to panda.”So, when it came out in Denmark that we were actually going to move our national monument, the National People ’ s Party sort of rebelled against it. They tried to pass a law against moving the Mermaid. So for the first time, I got invited to speak at the National Parliament. It was kind of interesting because in the morning, from 9 to 11, they were discussing the bailout package — how many billions to invest in saving the Danish economy. And then at 11 o ’ clock they stopped talking about these little issues. And then from 11 to 1, they were debating whether or not to send the Mermaid to China. But to conclude, if you want to see the Mermaid from May to December next year, don ’ t come to Copenhagen, because she ’ s going to be in Shanghai. If you do come to Copenhagen, you will probably see an installation by Ai Weiwei, the Chinese artist. But if the Chinese government intervenes, it might even be a panda. So the second story that I ’ d like to tell is, actually starts in my own house. This is my apartment. This is the view from my apartment, over the sort of landscape of triangular balconies that our client called the Leonardo DiCaprio balcony. And they form this sort of vertical backyard where, on a nice summer day, you ’ ll actually get introduced to all your neighbors in a vertical radius of 10 meters. The house is sort of a distortion of a square block. Trying to zigzag it to make sure that all of the apartments look at the straight views, instead of into each other. Until recently, this was the view from my apartment, onto this place where our client actually bought the neighbor site. And he said that he was going to do an apartment block next to a parking structure. And we thought, rather than doing a traditional stack of apartments looking straight into a big boring block of cars, why don ’ t we turn all the apartments into penthouses, put them on a podium of cars. And because Copenhagen is completely flat, if you want to have a nice south- facing slope with a view, you basically have to do it yourself. Then we sort of cut up the volume, so we wouldn ’ t block the view from my apartment. And essentially the parking is sort of occupying the deep space underneath the apartments. And up in the sun, you have a single layer of apartments that combine all the splendors of a suburban lifestyle, like a house with a garden with a sort of metropolitan view, and a sort of dense urban location. This is our first architectural model. This is an aerial photo taken last summer. And essentially, the apartments cover the parking. They are accessed through this diagonal elevator. It ’ s actually a stand-up product from Switzerland, because in Switzerland they have a natural need for diagonal elevators. And the facade of the parking, we wanted to make the parking naturally ventilated, so we needed to perforate it. And we discovered that by controlling the size of the holes, we could actually turn the entire facade into a gigantic, naturally ventilated, rasterized image. And since we always refer to the project as The Mountain, we commissioned this Japanese Himalaya photographer to give us this beautiful photo of Mount Everest, making the entire building a 3, 000 square meter artwork. So if you go back into the parking, into the corridors, it ’ s almost like traveling into a parallel universe from cars and colors, into this sort of south-facing urban oasis. The wood of your apartment continues outside becoming the facades. If you go even further, it turns into this green garden. And all the rainwater that drops on the Mountain is actually accumulated. And there is an automatic irrigation system that makes sure that this sort of landscape of gardens, in one or two years it will sort of transform into a Cambodian temple ruin, completely covered in green. So, the Mountain is like our first built example of what we like to refer to as architectural alchemy. This idea that you can actually create, if not gold, then at least added value by mixing traditional ingredients, like normal apartments and normal parking, and in this case actually offer people the chance that they don ’ t have to choose between a life with a garden or a life in the city. They can actually have both. As an **architect**, it ’ s really hard to set the agenda. You can ’ t just say that now I ’ d like to do a sustainable city in central Asia, because that ’ s not really how you get commissions. You always have to sort of adapt and improvise to the opportunities and accidents that happen, and the sort of turmoil of the world. One last example is that recently we, like last summer, we won the competition to design a Nordic national bank. This was the director of the bank when he was still smiling. It was in the middle of the capital so we were really excited by this opportunity. Unfortunately, it was the national bank of Iceland. At the same time, we actually had a visitor — a minister from Azerbaijan came to our office. We took him to see the Mountain. And he got very excited by this idea that you could actually make mountains out of architecture, because Azerbaijan is known as the Alps of Central Asia. So he asked us if we could actually imagine an urban master plan on an island outside the capital that would recreate the silhouette of the seven most significant mountains of Azerbaijan. So we took the commission. And we made this small movie that I ’ d like to show. We quite often make little movies. We always argue a lot about the soundtrack, but in this case it was really easy to choose the song. So basically, Baku is this sort of crescent bay overlooking the island of Zira, the island that we are planning — almost like the diagram of their flag. And our main idea was to sort of sample the seven most significant mountains of the topography of Azerbaijan and reinterpret them into urban and architectural structures, inhabitable of human life. Then we place these mountains on the island, surrounding this sort of central green **valley**, almost like a central **park**. And what makes it interesting is that the island right now is just a piece of desert. It has no vegetation.

It has no water. It has no energy, no resources. So we actually sort of designed the entire island as a single ecosystem, exploiting wind energy to drive the desalination plants, and to use the thermal properties of water to heat and cool the buildings. And all the sort of excess freshwater wastewater is filtered organically into the landscape, gradually transforming the desert island into sort of a green, lush landscape. So, you can say where an urban development normally happens at the expense of nature, in this case it ’ s actually creating nature. And the buildings, they don ’ t only sort of invoke the imagery of the mountains, they also operate like mountains. They create shelter from the wind. They accumulate the solar energy. They accumulate the water. So they actually transform the entire island into a single ecosystem. So we recently presented the master plan, and it has gotten approved. And this summer we are starting the construction documents of the two first mountains, in what ’ s going to be the first carbon-neutral island in Central Asia. Yes, maybe just to round off. So in a way you can see how the Mountain in Copenhagen sort of evolved into the Seven Peaks of Azerbaijan. With a little luck and some more evolution, maybe in 10 years it could be the Five Mountains on Mars. Thank you.

Text Sample 699: POS Dim 5 – 2009 – Score 7.00 – 2009/t000114.txt

I was speaking to a group of about 300 kids, ages six to eight, at a children ’ s **museum**, and I brought with me a bag full of legs, similar to the kinds of things you see up here, and had them laid out on a table for the kids. And, from my experience, you know, kids are naturally curious about what they don ’ t know, or don ’ t understand, or is foreign to them. They only learn to be frightened of those differences when an adult influences them to behave that way, and maybe censors that natural curiosity, or you know, reins in the question-asking in the hopes of them being polite little kids. So I just **pictured** a first grade teacher out in the lobby with these unruly kids, saying,”Now, whatever you do, don ’ t stare at her legs.”But, of course, that ’ s the point. That ’ s why I was there, I wanted to invite them to look and explore. So I made a deal with the adults that the kids could come in without any adults for two minutes on their own. The doors open, the kids descend on this table of legs, and they are poking and prodding, and they ’ re wiggling toes, and they ’ re trying to put their full weight on the sprinting leg to see what happens with that. And I said,”Kids, really quickly — I woke up this morning, I decided I wanted to be able to jump over a house — nothing too big, two or three stories — but, if you could think of any animal, any superhero, any cartoon character, anything you can dream up right now, what kind of legs would you build me?”And immediately a voice shouted,”Kangaroo!””No, no, no! Should be a frog!””No. It should be Go Go Gadget!””No, no, no! It should be the Incredibles.”And other things that I don ’ t — aren ’ t familiar with. And then, one eight- year-old said,”Hey, why wouldn ’ t you want to fly too?”And the whole room, including me, was like,”Yeah.”And just like that, I went from being a woman that these kids would have been trained to see as”disabled”to somebody that had potential that their bodies didn ’ t have yet. Somebody that might even be super-abled. Interesting. So some of you actually saw me at TED, 11 years ago. And there ’ s been a lot of talk about how life-changing this conference is for both speakers and attendees, and I am no exception. TED literally was the launch pad to the next decade of my life ’ s exploration. At the time, the legs I presented were groundbreaking in prosthetics. I had woven carbon fiber sprinting legs modeled after the hind leg of a cheetah, which you may have seen on stage yesterday. And also these very life-like, intrinsically painted silicone legs. So at the time, it was my opportunity to put a call out to innovators outside the traditional medical prosthetic community to come bring their talent to the science and to the art of building legs. So that we can stop compartmentalizing form, function and aesthetic, and assigning them different values. Well, lucky for me, a lot of people answered that call. And the journey started, funny enough, with a TED conference attendee — Chee Pearlman, who hopefully is in the audience somewhere today. She was the editor then of a magazine called ID, and she gave me a cover story. This started an incredible journey. Curious encounters were happening to me at the time ; I ’ d been accepting numerous invitations to speak on the design of the cheetah legs around the world. And people would come up to me after the conference, after my talk, men and women. And the conversation would go something like this,”You know Aimee, you ’ re very attractive. You don ’ t look disabled.”I thought,”Well, that ’ s amazing, because I don ’ t feel disabled.”And it really opened my eyes to this conversation that could be explored, about beauty. What does a beautiful woman have to look like? What is a sexy body? And interestingly, from an identity standpoint, what does it mean to have a disability? I mean, people — Pamela Anderson has more prosthetic in her body than I do. Nobody calls her disabled. So this magazine, through the hands of graphic designer Peter Saville, went to fashion designer Alexander McQueen, and photographer Nick Knight, who were also interested in exploring that conversation. So, three months after TED I found myself on a plane to London, doing my first fashion shoot, which resulted in this cover —”Fashionable”? Three months after that, I did my first runway show for Alexander McQueen on a pair of hand-carved wooden legs made from solid ash. Nobody knew — everyone thought they were wooden boots. Actually, I have them on stage with me : grapevines, magnolias — truly stunning. Poetry matters. Poetry is what elevates the banal and neglected object to a realm of art. It can transform the thing that might have made people fearful into something that invites them to look, and look a little longer, and maybe even understand. I learned this firsthand with my next adventure. The artist Matthew Barney, in his **film** opus called the”The Cremaster Cycle.”This is where it really hit home for me — that my legs could be wearable sculpture. And even at this point, I started to move away from the need to replicate human-ness as the only aesthetic **ideal**. So we made what people lovingly referred to as glass legs even though they ’ re actually optically clear polyurethane, a. k. a. bowling ball material. Heavy! Then we made these legs that are cast in soil with a potato root system growing in them, and beetroots out the top, and a very lovely brass toe. That ’ s a good close-up of that one. Then another character was a half-woman, half-cheetah — a little homage to my life as an athlete. 14 hours of prosthetic make-up to get into a creature that had articulated paws, claws and a tail that whipped around, like a gecko. And then another pair of legs we collaborated on were these — look like jellyfish legs, also polyurethane. And the only purpose that these legs can serve, outside the context of the **film**, is to provoke the senses and ignite the imagination. So whimsy matters. Today, I have over a dozen pair of prosthetic legs that various people have made for me, and with them I have different negotiations of the terrain under my **feet**, and I can change my height — I have a variable of five different heights. Today, I ’ m 6 ’ 1”. And I had these legs made a little over a year ago at Dorset Orthopedic in England and when I brought them home to Manhattan, my first night out on the town, I went to a very fancy party. And a girl was there who has known me for years at my normal 5 ’ 8”. Her mouth dropped open when she saw me, and she went,”But you ’ re so tall!”And I said,”I know. Isn ’ t it **fun**?”I mean, it ’ s a little bit like wearing stilts on stilts, but I have an entirely new relationship to door jams that I never expected I would ever have. And I was having **fun** with it. And she looked at me, and she said,"But, Aimee, that ' s not fair."And the incredible thing was she really meant it. It ' s not fair that you can change your height, as you want it. And that ' s when I knew — that ' s when I knew that the conversation with society has changed profoundly in this last decade. It is no longer a conversation about overcoming deficiency. It ' s a conversation about augmentation. It ' s a conversation about potential. A prosthetic limb doesn ' t represent the need to replace loss anymore. It can stand as a symbol that the wearer has the power to create whatever it is that they want to create in that space. So people that society once considered to be disabled can now become the **architects** of their own identities and indeed continue to change those identities by designing their bodies from a place of empowerment. And what is exciting to me so much right now is that by combining cutting-edge technology — robotics, bionics — with the age-old poetry, we are moving closer to understanding our collective humanity. I think that if we want to discover the full potential in our humanity, we need to celebrate those heartbreaking strengths and those glorious disabilities that we all have. I think of Shakespeare ' s Shylock :"If you prick us, do we not bleed, and if you tickle us, do we not laugh?"It is our humanity, and all the potential within it, that makes us beautiful. Thank you.

Text Sample 700: POS Dim 5 – 2009 – Score 7.00 – 2009/t002945.txt

The substance of things unseen. Cities, past and future. In Oxford, perhaps we can use Lewis Carroll and look in the looking glass that is New York City to try and see our true selves, or perhaps pass through to another world. Or, in the words of F. Scott Fitzgerald,"As the moon rose higher, the inessential houses began to melt away until gradually I became aware of the old island here that once flowered for Dutch sailors ' eyes, a fresh green breast of the new world."My colleagues and I have been working for 10 years to rediscover this lost world in a project we call The Mannahatta Project. We ' re trying to discover what Henry Hudson would have seen on the afternoon of September 12th, 1609, when he sailed into New York harbor. And I ' d like to tell you the story in three acts, and if I have time still, an epilogue. So, Act I : A Map Found. So, I didn ' t grow up in New York. I grew up out west in the Sierra Nevada Mountains, like you see here, in the Red Rock Canyon. And from these early experiences as a child I learned to love landscapes. And so when it became time for me to do my graduate studies, I studied this emerging field of landscape ecology. Landscape ecology concerns itself with how the stream and the meadow and the forest and the cliffs make habitats for plants and animals. This experience and this training lead me to get a wonderful job with the Wildlife Conservation Society, which works to save wildlife and wild places all over the world. And over the last decade, I traveled to over 40 countries to see jaguars and bears and elephants and tigers and rhinos. But every time I would return from my trips I ' d return back to New York City. And on my weekends I would go up, just like all the other tourists, to the top of the Empire State Building, and I ' d look down on this landscape, on these ecosystems, and I ' d wonder,"How does this landscape work to make habitat for plants and animals? How does it work to make habitat for animals like me?"I ' d go to Times Square and I ' d look at the amazing ladies on the wall, and wonder why nobody is looking at the historical figures just behind them. I ' d go to Central Park and see the rolling topography of Central Park come up against the abrupt and sheer topography of midtown Manhattan. I started reading about the history and the geography in New York City. I read that New York City was the first mega-city, a city of 10 million people or more, in 1950. I started seeing paintings like this. For those of you who are from New York, this is 125th street under the West Side Highway. It was once a beach. And this painting has John James Audubon, the painter, sitting on the rock. And it ' s looking up on the wooded heights of Washington Heights to Jeffrey ' s Hook, where the George Washington Bridge goes across today. Or this painting, from the 1740s, from Greenwich Village. Those are two students at King ' s College — later Columbia University — sitting on a hill, overlooking a **valley**. And so I ' d go down to Greenwich Village and I ' d look for this hill, and I couldn ' t find it. And I couldn ' t find that palm tree. What ' s that palm tree doing there? So, it was in the course of these investigations that I ran into a map. And it ' s this map you see here. It ' s held in a geographic information system which allows me to zoom in. This map isn ' t from Hudson ' s time, but from the American Revolution, 170 years later, made by British military cartographers during the occupation of New York City. And it ' s a remarkable map. It ' s in the National Archives here in Kew. And it ' s 10 **feet** long and three and a half **feet** wide. And if I zoom in to lower Manhattan you can see the extent of New York City as it was, right at the end of the American Revolution. Here ' s Bowling Green. And here ' s Broadway. And this is City Hall Park. So the city basically extended to City Hall Park. And just beyond it you can see features that have vanished, things that have disappeared. This is the Collect Pond, which was the fresh water source for New York City for its first 200 years, and for the Native Americans for thousands of years before that. You can see the Lispenard Meadows draining down through here, through what is TriBeCa now, and the beaches that come up from the Battery, all the way to 42nd St. This map was made for military reasons. They ' re mapping the roads, the buildings, these fortifications that they built. But they ' re also mapping things of ecological interest, also military interest : the hills, the marshes, the streams. This is Richmond Hill, and Minetta Water, which used to run its way through Greenwich Village. Or the swamp at Gramercy Park, right here. Or Murray Hill. And this is the Murrays ' house on Murray Hill, 200 years ago. Here is Times Square, the two streams that came together to make a wetland in Times Square, as it was at the end of the American Revolution. So I saw this remarkable map in a book. And I thought to myself,"You know, if I could georeference this map, if I could place this map in the grid of the city today, I could find these lost features of the city, in the block-by-block geography that people know, the geography of where people go to work, and where they go to live, and where they like to eat."So, after some work we were able to georeference it, which allows us to put the modern streets on the city, and the buildings, and the open spaces, so that we can zoom in to where the Collect Pond is. We can digitize the Collect Pond and the streams, and see where they actually are in the geography of the city today. So this is **fun** for finding where things are relative to the old topography. But I had another idea about this map. If we take away the streets, and if we take away the buildings, and if we take away the open spaces, then we could take this map. If we pull off the 18th century features we could drive it back in time. We could drive it back to its ecological fundamentals : to the hills, to the streams, to the basic hydrology and shoreline, to the beaches, the basic aspects that make the ecological landscape. Then, if we added maps like the geology, the bedrock geology, and the surface geology, what the glaciers leave, if we make the soil map, with the 17 soil classes, that are defined by the National Conservation Service, if we make a digital elevation model of the topography that tells us how high the hills were, then we can calculate the slopes. We can calculate the aspect. We can calculate the winter wind exposure — so, which way the winter winds blow across the landscape.

The white areas on this map are the places protected from the winter winds. We compiled all the information about where the Native Americans were, the Lenape. And we built a probability map of where they might have been. So, the red areas on this map indicate the places that are best for human sustainability on Manhattan, places that are close to water, places that are near the harbor to fish, places protected from the winter winds. We know that there was a Lenape settlement down here by the Collect Pond. And we knew that they planted a kind of horticulture, that they grew these beautiful gardens of corn, beans, and squash, the"Three Sisters"garden. So, we built a model that explains where those fields might have been. And the old fields, the successional fields that go. And we might think of these as abandoned. But, in fact, they ' re grassland habitats for grassland birds and plants. And they have become successional shrub lands, and these then mix in to a map of all the ecological communities. And it turns out that Manhattan had 55 different ecosystem types. You can think of these as neighborhoods, as distinctive as TriBeCa and the Upper East Side and Inwood — that these are the forest and the wetlands and the marine communities, the beaches. And 55 is a lot. On a per-area basis, Manhattan had more ecological communities per acre than Yosemite does, than Yellowstone, than Amboseli. It was really an extraordinary landscape that was capable of supporting an extraordinary biodiversity. So, Act II : A Home Reconstructed. So, we studied the fish and the frogs and the birds and the bees, the 85 different kinds of fish that were on Manhattan, the Heath hens, the species that aren ' t there anymore, the beavers on all the streams, the black bears, and the Native Americans, to study how they used and thought about their landscape. We wanted to try and map these. And to do that what we did was we mapped their habitat needs. Where do they get their food? Where do they get their water? Where do they get their shelter? Where do they get their reproductive resources? To an ecologist, the intersection of these is habitat, but to most people, the intersection of these is their home. So, we would read in field guides, the standard field guides that maybe you have on your shelves, you know, what beavers need is,"A slowly meandering stream with aspen trees and alders and willows, near the water."That ' s the best thing for a beaver. So we just started making a list. Here is the beaver. And here is the stream, and the aspen and the alder and the willow. As if these were the maps that we would need to predict where you would find the beaver. Or the bog turtle, needing wet meadows and insects and sunny places. Or the bobcat, needing rabbits and beavers and den sites. And rapidly we started to realize that beavers can be something that a bobcat needs. But a beaver also needs things. And that having it on either side means that we can link it together, that we can create the network of the habitat relationships for these species. Moreover, we realized that you can start out as being a beaver specialist, but you can look up what an aspen needs. An aspen needs fire and dry soils. And you can look at what a wet meadow needs. And it need beavers to create the wetlands, and maybe some other things. But you can also talk about sunny places. So, what does a sunny place need? Not habitat per se. But what are the conditions that make it possible? Or fire. Or dry soils. And that you can put these on a grid that ' s 1, 000 columns long across the top and 1, 000 rows down the other way. And then we can visualize this data like a network, like a social network. And this is the network of all the habitat relationships of all the plants and animals on Manhattan, and everything they needed, going back to the geology, going back to time and space at the very core of the web. We call this the Muir Web. And if you zoom in on it it looks like this. Each point is a different species or a different stream or a different soil type. And those little gray lines are the connections that connect them together. They are the connections that actually make nature resilient.

And the structure of this is what makes nature work, seen with all its parts. We call these Muir Webs after the Scottish-American naturalist John Muir, who said,"When we try to pick out anything by itself, we find that it ' s bound fast by a thousand invisible cords that cannot be broken, to everything in the universe."So then we took the Muir webs and we took them back to the maps. So if we wanted to go between 85th and 86th, and Lex and Third, maybe there was a stream in that block. And these would be the kind of trees that might have been there, and the flowers and the lichens and the mosses, the butterflies, the fish in the stream, the birds in the trees. Maybe a timber rattlesnake lived there. And perhaps a black bear walked by. And maybe Native Americans were there. And then we took this data. You can see this for yourself on our website. You can zoom into any block on Manhattan, and see what might have been there 400 years ago. And we used it to try and reveal a landscape here in Act III. We used the tools they use in Hollywood to make these fantastic landscapes that we all see in the movies. And we tried to use it to visualize Third Avenue. So we would take the landscape and we would build up the topography. We ' d lay on top of that the soils and the waters, and illuminate the landscape. We would lay on top of that the map of the ecological communities. And feed into that the map of the species. So that we would actually take a photograph, flying above Times Square, looking toward the Hudson River, waiting for Hudson to come. Using this technology, we can make these fantastic georeferenced views. We can basically take a **picture** out of any window on Manhattan and see what that landscape looked like 400 years ago. This is the view from the East River, looking up Murray Hill at where the United Nations is today. This is the view looking down the Hudson River, with Manhattan on the left, and New Jersey out on the right, looking out toward the Atlantic Ocean. This is the view over Times Square, with the beaver pond there, looking out toward the east. So we can see the Collect Pond, and Lispenard Marshes back behind. We can see the fields that the Native Americans made. And we can see this in the geography of the city today. So when you ' re watching"Law and Order,"and the lawyers walk up the steps they could have walked back down those steps of the New York Court House, right into the Collect Pond, 400 years ago. So these images are the work of my friend and colleague, Mark Boyer, who is here in the audience today. And I ' d just like, if you would give him a hand, to call out for his fine work. There is such power in bringing science and visualization together, that we can create images like this, perhaps looking on either side of a looking glass. And even though I ' ve only had a brief time to speak, I hope you appreciate that Mannahatta was a very special place. The place that you see here on the left side was interconnected. It was based on this diversity. It had this **resilience** that is what we need in our modern world. But I wouldn ' t have you think that I don ' t like the place on the right, which I quite do. I ' ve come to love the city and its kind of diversity, and its **resilience**, and its dependence on density and how we ' re connected together. In fact, that I see them as reflections of each other, much as Lewis Carroll did in"Through the Looking Glass."We can compare these two and hold them in our minds at the same time, that they really are the same place, that there is no way that cities can escape from nature. And I think this is what we ' re learning about building cities in the future. So if you ' ll allow me a brief epilogue, not about the past, but about 400 years from now, what we ' re realizing is that cities are habitats for people, and need to supply what people need : a sense of home, food, water, shelter, reproductive resources, and a sense of meaning. This is the particular additional habitat requirement of humanity. And so many of the talks here at TED are about meaning, about bringing meaning to our lives in all kinds of different ways, through technology, through art, through science, so much so that I think we focus so much on that side of our lives, that we haven ’ t given enough attention to the food and the water and the shelter, and what we need to raise the kids. So, how can we envision the city of the future? Well, what if we go to Madison Square Park, and we imagine it without all the cars, and bicycles instead and large forests, and streams instead of sewers and storm drains? What if we imagined the Upper East Side with green **roofs**, and streams winding through the city, and windmills supplying the power we need? Or if we imagine the New York City metropolitan area, currently home to 12 million people, but 12 million people in the future, perhaps living at the density of Manhattan, in only 36 percent of the area, with the areas in between covered by farmland, covered by wetlands, covered by the marshes we need. This is the kind of future I think we need, is a future that has the same diversity and abundance and dynamism of Manhattan, but that learns from the sustainability of the past, of the ecology, the original ecology, of nature with all its parts. Thank you very much.

Text Sample 701: POS Dim 5 – 2012 – Score 8.00 – 2012/t002309.txt

When the Industrial Revolution started, the amount of carbon sitting underneath Britain in the form of coal was as big as the amount of carbon sitting under Saudi Arabia in the form of oil. This carbon powered the Industrial Revolution, it put the”Great”in Great Britain, and led to Britain ’ s temporary world domination. And then, in 1918, coal production in Britain peaked, and has declined ever since. In due course, Britain started using oil and **gas** from the North Sea, and in the year 2000, oil and **gas** production from the North Sea also peaked, and they ’ re now on the decline. These observations about the finiteness of easily accessible, local, secure fossil fuels, is a motivation for saying,”Well, what ’ s next? What is life after fossil fuels going to be like? Shouldn ’ t we be thinking hard about how to get off fossil fuels?”Another motivation, of course, is climate change. And when people talk about life after fossil fuels and climate change action, I think there ’ s a lot of fluff, a lot of greenwash, a lot of misleading advertising, and I feel a duty as a physicist to try to guide people around the claptrap and help people understand the actions that really make a difference, and to focus on ideas that do add up. Let me illustrate this with what physicists call a back-of-envelope calculation. We love back-of-envelope calculations. You ask a question, write down some numbers, and get an answer. It may not be very accurate, but it may make you say,”Hmm.”So here ’ s a question : Imagine if we said,”Oh yes, we can get off fossil fuels. We ’ ll use biofuels. Problem solved. Transport... We don ’ t need oil anymore.”Well, what if we grew the biofuels for a road on the grass verge at the edge of the road? How wide would the verge have to be for that to work out? OK, so let ’ s put in some numbers. Let ’ s have our cars go at 60 miles per hour. Let ’ s say they do 30 miles per gallon. That ’ s the European average for new cars. Let ’ s say the productivity of biofuel plantations is 1, 200 liters of biofuel per hectare per year. That ’ s true of European biofuels. And let ’ s imagine the cars are spaced 80 meters apart from each other, and they ’ re perpetually going along this road. The length of the road doesn ’ t matter, because the longer the road, the more biofuel plantation. What do we do with these numbers? Take the first number, divide by the other three, and get eight kilometers. And that ’ s the answer. That ’ s how wide the plantation would have to be, given these assumptions. And maybe that makes you say,”Hmm. Maybe this isn ’ t going to be quite so easy.”And it might make you think, perhaps there ’ s an issue to do with areas. And in this talk, I ’ d like to talk about land areas, and ask : Is there an issue about areas? The answer is going to be yes, but it depends which country you are in. So let ' s start in the United Kingdom, since that ' s where we are today. The energy consumption of the United Kingdom, the total energy consumption — not just transport, but everything — I like to quantify it in lightbulbs. It ' s as if we ' ve all got 125 lightbulbs on all the time, 125 kilowatt-hours per day per person is the energy consumption of the UK. So there ' s 40 lightbulbs ' worth for transport, 40 lightbulbs ' worth for heating, and 40 lightbulbs ' worth for making electricity, and other things are relatively small, compared to those three big fish. It ' s actually a bigger footprint if we take into account the embodied energy in the stuff we import into our country as well. And 90 percent of this energy, today, still comes from fossil fuels, and 10 percent, only, from other, greener — possibly greener — sources, like nuclear power and renewables. So. That ' s the UK. The population density of the UK is 250 people per square kilometer. I ' m now going to show you other countries by these same two measures. On the vertical axis, I ' m going to show you how many lightbulbs — what our energy consumption per person is. We ' re at 125 lightbulbs per person, and that little blue dot there is showing you the land area of the United Kingdom. The population density is on the horizontal axis, and we ' re 250 people per square kilometer. Let ' s add European countries in blue, and you can see there ' s quite a variety. I should emphasize, both of these axes are logarithmic ; as you go from one gray bar to the next gray bar, you ' re going up a factor of 10. Next, let ' s add Asia in red, the Middle East and North Africa in green, sub-Saharan Africa in blue, black is South America, purple is Central America, and then in pukey-yellow, we have North America, Australia and New Zealand. You can see the great diversity of population densities and of per capita consumptions. Countries are different from each other. Top left, we have Canada and Australia, with enormous land areas, very high per capita consumption — 200 or 300 lightbulbs per person — and very low population densities. Top right : Bahrain has the same energy consumption per person, roughly, as Canada — over 300 lightbulbs per person, but their population density is a factor of 300 times greater, 1, 000 people per square kilometer. **Bottom** right : Bangladesh has the same population density as Bahrain, but consumes 100 times less per person. **Bottom** left : well, there ' s no one. But there used to be a whole load of people. Here ' s another message from this diagram. I ' ve added on little blue tails behind Sudan, Libya, China, India, Bangladesh. That ' s 15 years of progress. Where were they 15 years ago, and where are they now? And the message is, most countries are going to the right, and they ' re going up. Up and to the right : bigger population density and higher per capita consumption. So, we may be off in the top right-hand corner, slightly unusual, the United Kingdom accompanied by Germany, Japan, South Korea, the Netherlands, and a bunch of other slightly odd countries, but many other countries are coming up and to the right to join us. So we ' re a **picture**, if you like, of what the future energy consumption might be looking like in other countries, too. I ' ve also added in this diagram now some pink lines that go down and to the right. Those are lines of equal power consumption per unit area, which I measure in watts per square meter. So, for example, the middle line there, 0. 1 watts per square meter, is the energy consumption per unit area of Saudi Arabia, Norway, Mexico in purple, and Bangladesh 15 years ago. Half of the world ' s population lives in countries that are already above that line. The United Kingdom is consuming 1. 25 watts per square meter. So is Germany, and Japan is consuming a bit more. So, let ' s now say why this is relevant. Why is it relevant? Well, we can measure renewables in the same units and other forms of power production in the same units. Renewables is one of the leading ideas for how we could get off our 90 percent fossil-fuel habit.

So here come some renewables. Energy crops deliver half a watt per square meter in European climates. What does that mean? You might have anticipated that result, given what I told you about the biofuel plantation a moment ago. Well, we consume 1. 25 watts per square meter. What this means is, even if you covered the whole of the United Kingdom with energy crops, you couldn ’ t match today ’ s energy consumption. Wind power produces a bit more — 2. 5 watts per square meter. But that ’ s only twice as big as 1. 25 watts per square meter. So that means if you wanted, literally, to produce total energy consumption in all forms, on average, from wind farms, you need wind farms half the area of the UK. I ’ ve got data to back up all these assertions, by the way. Next, let ’ s look at solar power. Solar **panels**, when you put them on a **roof**, deliver about 20 watts per square meter in England. If you really want to get a lot from solar **panels**, you need to adopt the traditional Bavarian farming method, where you leap off the **roof**, and coat the countryside with solar **panels**, too. Solar **parks**, because of the gaps between the **panels**, deliver less. They deliver about 5 watts per square meter of land area. And here ’ s a solar **park** in Vermont, with real data, delivering 4. 2 watts per square meter. Remember where we are, 1. 25 watts per square meter, wind farms 2. 5, solar **parks** about five. So whichever of those renewables you pick, the message is, whatever mix of those renewables you ’ re using, if you want to power the UK on them, you ’ re going to need to cover something like 20 percent or 25 percent of the country with those renewables. I ’ m not saying that ’ s a bad idea ; we just need to understand the numbers. I ’ m absolutely not anti-renewables. I love renewables. But I ’ m also pro-arithmetic. Concentrating solar power in deserts delivers larger powers per unit area, because you don ’ t have the problem of clouds. So, this facility delivers 14 watts per square meter ; this one 10 watts per square meter ; and this one in Spain, 5 watts per square meter. Being generous to concentrating solar power, I think it ’ s perfectly credible it could deliver 20 watts per square meter. So that ’ s nice. Of course, Britain doesn ’ t have any deserts. Yet. So here ’ s a summary so far : All renewables, much as I love them, are diffuse. They all have a small power per unit area, and we have to live with that fact. And that means, if you do want renewables to make a substantial difference for a country like the United Kingdom on the scale of today ’ s consumption, you need to be imagining renewable facilities that are country-sized. Not the entire country, but a fraction of the country, a substantial fraction. There are other options for generating power as well, which don ’ t involve fossil fuels. So there ’ s nuclear power, and on this ordinance survey map, you can see there ’ s a Sizewell B inside a blue square kilometer. That ’ s one gigawatt in a square kilometer, which works out to 1, 000 watts per square meter. So by this particular metric, nuclear power isn ’ t as intrusive as renewables. Of course, other metrics matter, too, and nuclear power has all sorts of popularity problems. But the same goes for renewables as well. Here ’ s a photograph of a consultation exercise in full swing in the little town of Penicuik just outside Edinburgh, and you can see the children of Penicuik celebrating the burning of the effigy of the windmill. So — People are anti-everything, and we ’ ve got to keep all the options on the table. What can a country like the UK do on the supply side? Well, the options are, I ’ d say, these three : power renewables, and recognizing that they need to be close to country-sized ; other people ’ s renewables, so we could go back and talk very politely to the people in the top **left-hand** side of the diagram and say,”Uh, we don ’ t want renewables in our backyard, but, um, please could we put them in yours instead?”And that ’ s a serious option. It ’ s a way for the world to handle this issue. So countries like Australia, Russia, Libya, Kazakhstan, could be our best friends for renewable production. And a third option is nuclear power. So that ’ s some supply-side options. In addition to the supply levers that we can push — and remember, we need large amounts, because at the moment, we get 90 percent of our energy from fossil fuels — in addition to those levers, we could talk about other ways of solving this issue. Namely, we could reduce demand, and that means reducing population — I ' m not sure how to do that — or reducing per capita consumption. So let ' s talk about three more big levers that could really help on the consumption side. First, transport. Here are the physics principles that tell you how to reduce the energy consumption of transport. People often say,"Technology can answer everything. We can make vehicles that are 100 times more efficient."And that ' s almost true. Let me show you. The energy consumption of this typical tank here is 80 kilowatt hours per hundred person kilometers. That ' s the average European car. Eighty kilowatt hours. Can we make something 100 times better by applying the physics principles I just listed? Yes. Here it is. It ' s the bicycle. It ' s 80 times better in energy consumption, and it ' s powered by biofuel, by Weetabix. And there are other options in between, because maybe the lady in the tank would say,"No, that ' s a lifestyle change. Don ' t change my lifestyle, please."We could persuade her to take a train, still a lot more efficient than a car, but that might be a lifestyle change. Or there ' s the EcoCAR, top-left. It comfortably accommodates one teenager and it ' s shorter than a traffic cone, and it ' s almost as efficient as a bicycle, as long as you drive it at 15 miles per hour. In between, perhaps some more realistic options on the transport lever are electric vehicles, so electric bikes and electric cars in the middle, perhaps four times as energy efficient as the standard petrol-powered tank. Next, there ' s the heating lever. Heating is a third of our energy consumption in Britain, and quite a lot of that is going into homes and other buildings, doing space heating and water heating. So here ' s a typical crappy British house. It ' s my house, with a Ferrari out front. What can we do to it? Well, the laws of physics are written up there, which describe how the power consumption for heating is driven by the things you can control. The things you can control are the temperature difference between the inside and the outside. There ' s this remarkable technology called a thermostat : you grasp it, rotate it to the left, and your energy consumption in the home will decrease. I ' ve tried it. It works. Some people call it a lifestyle change. You can also get the fluff men in to reduce the leakiness of your building — put fluff in the walls, fluff in the **roof**, a new front door, and so forth. The sad truth is, this will save you money. That ' s not sad, that ' s good. But the sad truth is, it ' ll only get about 25 percent of the leakiness of your building if you do these things, which are good ideas. If you really want to get a bit closer to Swedish building standards with a crappy house like this, you need to be putting external insulation on the building, as shown by this block of flats in London. You can also deliver heat more efficiently using heat pumps, which use a smaller bit of high-grade energy like electricity to move heat from your garden into your house. The third demand-side option I want to talk about, the third way to reduce energy consumption is : read your meters. People talk a lot about smart meters, but you can do it yourself. Use your own eyes and be smart. Read your meter, and if you ' re anything like me, it ' ll change your life. Here ' s a graph I made. I was writing a book about sustainable energy, and a friend asked me,"How much energy do you use at home?"I was embarrassed ; I didn ' t actually know. And so I started reading the meter every week. The old meter readings are shown in the top half of the graph, and then 2007 is shown in green at the **bottom**. That was when I was reading the meter every week. And my life changed, because I started doing experiments and seeing what made a difference. My **gas** consumption plummeted, because I started tinkering with the thermostat and the timing on the heating system, and I knocked more than half off my **gas** bills. There ’ s a similar story for my electricity consumption, where switching off the DVD players, the stereos, the computer peripherals that were on all the time, and just switching them on when I needed them, knocked another third off my electricity bills, too. So we need a plan that adds up. I ’ ve described for you six big levers. We need big action, because we get 90 percent of our energy from fossil fuels, and so you need to push hard on most, if not all, of these levers. Most of these levers have popularity problems, and if there is a lever you don ’ t like the use of, well, please do bear in mind that means you need even stronger effort on the other levers. So I ’ m a strong advocate of having grown-up conversations that are based on numbers and facts. And I want to close with this map that just visualizes for you the requirement of land and so forth in order to get just 16 lightbulbs per person from four of the big possible sources. So, if you wanted to get 16 lightbulbs — remember, today our total energy consumption is 125 lightbulbs ’ worth — if you wanted 16 from wind, this map visualizes a solution for the UK. It ’ s got 160 wind farms, each 100 square kilometers in size, and that would be a twentyfold increase over today ’ s amount of wind. Nuclear power : to get 16 lightbulbs per person, you ’ d need two gigawatts at each of the purple dots on the map. That ’ s a fourfold increase over today ’ s levels of nuclear power. Biomass : to get 16 lightbulbs per person, you ’ d need a land area something like three and a half Wales ’ worth, either in our country, or in someone else ’ s country, possibly Ireland, possibly somewhere else. And a fourth supply-side option : concentrating solar power in other people ’ s deserts. If you wanted to get 16 lightbulbs ’ worth, then we ’ re talking about these eight hexagons down at the **bottom** right. The total area of those hexagons is two Greater London ’ s worth of someone else ’ s Sahara, and you ’ ll need power lines all the way across Spain and France to bring the power from the Sahara to Surrey. We need a plan that adds up. We need to stop shouting and start talking. And if we can have a grown-up conversation, make a plan that adds up and get building, maybe this low-carbon revolution will actually be **fun**. Thank you very much for listening.

Text Sample 702: POS Dim 5 – 2012 – Score 7.00 – 2012/t002286.txt

My passions are music, technology and making things. And it ’ s the combination of these things that has led me to the hobby of sound visualization, and, on occasion, has led me to play with fire. This is a Rubens ’ tube. It ’ s one of many I ’ ve made over the years, and I have one here tonight. It ’ s about an 8-foot-long tube of **metal**, it ’ s got a hundred or so holes on top, on that side is the speaker, and here is some lab tubing, and it ’ s connected to this tank of propane. So, let ’ s fire it up and see what it does. So let ’ s play a 550-herz frequency and watch what happens. Thank you. It ’ s okay to applaud the laws of physics, but essentially what ’ s happening here — — is the energy from the sound via the air and **gas** molecules is influencing the combustion properties of propane, creating a visible waveform, and we can see the alternating regions of compression and rarefaction that we call frequency, and the height is showing us amplitude. So let ’ s change the frequency of the sound, and watch what happens to the fire. So every time we hit a resonant frequency we get a standing wave and that emergent sine **curve** of fire. So let ’ s turn that off. We ’ re indoors. Thank you. I also have with me a flame table. It ’ s very similar to a Rubens ’ tube, and it ’ s also used for visualizing the physical properties of sound, such as eigenmodes, so let ’ s fire it up and see what it does. Ooh. Okay. Now, while the table comes up to pressure, let me note here that the sound is not traveling in perfect lines. It ’ s actually traveling in all directions, and the Rubens ’ tube ’ s a little like bisecting those waves with a line, and the flame table ’ s a little like bisecting those waves with a plane, and it can show a little more subtle complexity, which is why I like to use it to watch Geoff Farina play guitar. All right, so it ’ s a delicate dance. If you watch closely — If you watch closely, you may have seen some of the eigenmodes, but also you may have seen that jazz music is better with fire. Actually, a lot of things are better with fire in my world, but the fire ’ s just a foundation. It shows very well that eyes can hear, and this is interesting to me because technology allows us to present sound to the eyes in ways that accentuate the strength of the eyes for seeing sound, such as the removal of time. So here, I ’ m using a rendering algorithm to paint the frequencies of the song”Smells Like Teen Spirit”in a way that the eyes can take them in as a single visual impression, and the technique will also show the strengths of the visual cortex for pattern recognition. So if I show you another song off this album, and another, your eyes will easily pick out the use of repetition by the band Nirvana, and in the frequency distribution, the colors, you can see the clean- dirty-clean sound that they are famous for, and here is the entire album as a single visual impression, and I think this impression is pretty powerful. At least, it ’ s powerful enough that if I show you these four songs, and I remind you that this is”Smells Like Teen Spirit,”you can probably correctly guess, without listening to any music at all, that the song a die hard Nirvana fan would enjoy is this song,”I ’ ll Stick Around”by the Foo Fighters, whose lead singer is Dave Grohl, who was the drummer in Nirvana. The songs are a little similar, but mostly I ’ m just interested in the idea that someday maybe we ’ ll buy a song because we like the way it looks. All right, now for some more sound data. This is data from a skate **park**, and this is Mabel Davis skate **park** in Austin, Texas. And the sounds you ’ re hearing came from eight microphones attached to obstacles around the **park**, and it sounds like chaos, but actually all the tricks start with a very distinct slap, but successful tricks end with a pop, whereas unsuccessful tricks more of a scratch and a tumble, and tricks on the rail will ring out like a gong, and voices occupy very unique frequencies in the skate **park**. So if we were to render these sounds visually, we might end up with something like this. This is all 40 minutes of the recording, and right away the algorithm tells us a lot more tricks are missed than are made, and also a trick on the rails is a lot more likely to produce a cheer, and if you look really closely, we can tease out traffic patterns. You see the skaters often trick in this direction. The obstacles are easier. And in the middle of the recording, the mics pick this up, but later in the recording, this kid shows up, and he starts using a line at the top of the **park** to do some very advanced tricks on something called the tall rail. And it ’ s fascinating. At this moment in time, all the rest of the skaters turn their lines 90 degrees to stay out of his way. You see, there ’ s a subtle etiquette in the skate **park**, and it ’ s led by key influencers, and they tend to be the kids who can do the best tricks, or wear red pants, and on this day the mics picked that up. All right, from skate physics to theoretical physics. I ’ m a big fan of Stephen Hawking, and I wanted to use all eight hours of his Cambridge lecture series to create an homage. Now, in this series he ’ s speaking with the aid of a computer, which actually makes identifying the ends of sentences fairly easy. So I wrote a steering algorithm. It listens to the lecture, and then it uses the amplitude of each word to move a point on the x-axis, and it uses the inflection of sentences to move a same point up and down on the y-axis. And these trend lines, you can see, there ’ s more questions than answers in the laws of physics, and when we reach the end of a sentence, we place a star at that location. So there ’ s a lot of sentences, so a lot of stars, and after rendering all of the audio, this is what we get. This is

Stephen Hawking ’ s universe. It ’ s all eight hours of the Cambridge lecture series taken in as a single visual impression, and I really like this image, but a lot of people think it ’ s fake. So I made a more interactive version, and the way I did that is I used their position in time in the lecture to place these stars into 3D space, and with some custom software and a Kinect, I can walk right into the lecture. I ’ m going to wave through the Kinect here and take control, and now I ’ m going to reach out and I ’ m going to touch a star, and when I do, it will play the sentence that generated that star. Stephen Hawking : There is one, and only one, arrangement in which the pieces make a complete **picture**. Jared Ficklin : Thank you. There are 1, 400 stars. It ’ s a really **fun** way to explore the lecture, and, I hope, a fitting homage. All right. Let me close with a work in progress. I think, after 30 years, the opportunity exists to create an enhanced version of closed captioning. Now, we ’ ve all seen a lot of TEDTalks online, so let ’ s watch one now with the sound turned off and the closed captioning turned on. There ’ s no closed captioning for the TED theme song, and we ’ re missing it, but if you ’ ve watched enough of these, you hear it in your mind ’ s ear, and then applause starts. It usually begins here, and it grows and then it falls. Sometimes you get a little star applause, and then I think even Bill Gates takes a nervous breath, and the talk begins. All right, so let ’ s watch this clip again. This time, I ’ m not going to talk at all. There ’ s still going to be no audio, but what I am going to do is I ’ m going to render the sound visually in real time at the **bottom** of the screen. So watch closely and see what your eyes can hear. This is fairly amazing to me. Even on the first view, your eyes will successfully pick out patterns, but on repeated views, your brain actually gets better at turning these patterns into information. You can get the tone and the timbre and the pace of the speech, things that you can ’ t get out of closed captioning. That famous scene in horror movies where someone is walking up from behind is something you can see, and I believe this information would be something that is useful at times when the audio is turned off or not heard at all, and I speculate that deaf audiences might actually even be better at seeing sound than hearing audiences. I don ’ t know. It ’ s a theory right now. Actually, it ’ s all just an idea. And let me end by saying that sound moves in all directions, and so do ideas. Thank you.

Text Sample 703: POS Dim 5 – 2012 – Score 7.00 – 2012/t002287.txt

So the machine I ’ m going to talk you about is what I call the greatest machine that never was. It was a machine that was never built, and yet, it will be built. It was a machine that was designed long before anyone thought about computers. If you know anything about the history of computers, you will know that in the ’ 30s and the ’ 40s, simple computers were created that started the computer revolution we have today, and you would be correct, except for you ’ d have the wrong century. The first computer was really designed in the 1830s and 1840s, not the 1930s and 1940s. It was designed, and parts of it were prototyped, and the bits of it that were built are here in South Kensington. That machine was built by this guy, Charles Babbage. Now, I have a great affinity for Charles Babbage because his hair is always completely unkempt like this in every single **picture**. He was a very wealthy man, and a sort of, part of the aristocracy of Britain, and on a Saturday night in Marylebone, were you part of the intelligentsia of that period, you would have been invited round to his house for a soiree — and he invited everybody : kings, the Duke of Wellington, many, many famous people — and he would have shown you one of his **mechanical** machines. I really miss that era, you know, where you could go around for a soiree and see a **mechanical** computer get demonstrated to you. But Babbage, Babbage himself was born at the end of the 18th century, and was a fairly famous mathematician. He held the post that Newton held at Cambridge, and that was recently held by Stephen Hawking. He ' s less well known than either of them because he got this idea to make **mechanical** computing devices and never made any of them. The reason he never made any of them, he ' s a classic nerd. Every time he had a good idea, he ' d think,"That ' s brilliant, I ' m going to start building that one. I ' ll spend a fortune on it. I ' ve got a better idea. I ' m going to work on this one. And I ' m going to do this one."He did this until Sir Robert Peel, then Prime Minister, basically kicked him out of Number 10 Downing Street, and kicking him out, in those days, that meant saying,"I bid you good day, sir."The thing he designed was this monstrosity here, the analytical engine. Now, just to give you an idea of this, this is a view from above. Every one of these circles is a cog, a stack of cogs, and this thing is as big as a steam locomotive. So as I go through this talk, I want you to imagine this gigantic machine. We heard those wonderful sounds of what this thing would have sounded like. And I ' m going to take you through the architecture of the machine — that ' s why it ' s computer architecture — and tell you about this machine, which is a computer. So let ' s talk about the memory. The memory is very like the memory of a computer today, except it was all made out of **metal**, stacks and stacks of cogs, 30 cogs high. Imagine a thing this high of cogs, hundreds and hundreds of them, and they ' ve got numbers on them. It ' s a decimal machine. Everything ' s done in decimal. And he thought about using binary. The problem with using binary is that the machine would have been so tall, it would have been ridiculous. As it is, it ' s enormous. So he ' s got memory. The memory is this bit over here. You see it all like this. This monstrosity over here is the CPU, the chip, if you like. Of course, it ' s this big. Completely **mechanical**. This whole machine is **mechanical**. This is a **picture** of a prototype for part of the CPU which is in the Science Museum. The CPU could do the four fundamental functions of arithmetic — so addition, multiplication, subtraction, division — which already is a bit of a feat in **metal**, but it could also do something that a computer does and a calculator doesn ' t : this machine could look at its own internal memory and make a decision. It could do the"if then"for basic programmers, and that fundamentally made it into a computer. It could compute. It couldn ' t just calculate. It could do more. Now, if we look at this, and we stop for a minute, and we think about chips today, we can ' t look inside a silicon chip. It ' s just so tiny. Yet if you did, you would see something very, very similar to this. There ' s this incredible complexity in the CPU, and this incredible regularity in the memory. If you ' ve ever seen an electron microscope **picture**, you ' ll see this. This all looks the same, then there ' s this bit over here which is incredibly complicated. All this cog wheel mechanism here is doing is what a computer does, but of course you need to program this thing, and of course, Babbage used the technology of the day and the technology that would reappear in the ' 50s, ' 60s and ' 70s, which is punch cards. This thing over here is one of three punch card readers in here, and this is a program in the Science Museum, just not far from here, created by Charles Babbage, that is sitting there — you can go see it — waiting for the machine to be built. And there ' s not just one of these, there ' s many of them. He prepared programs anticipating this would happen. Now, the reason they used punch cards was that Jacquard, in France, had created the Jacquard loom, which was weaving these incredible patterns controlled by punch cards, so he was just repurposing the technology of the day, and like everything else he did, he ' s using the technology of his era, so 1830s, 1840s, 1850s, cogs, steam, **mechanical** devices. Ironically, born the same year as Charles Babbage was Michael Faraday, who would completely revolutionize everything with the dynamo, transformers, all these sorts of things. Babbage, of course, wanted to use proven technology, so steam and things. Now, he needed accessories. Obviously, you ' ve got a computer now. You ' ve got punch cards, a CPU and memory. You need accessories you ' re going to come with. You ' re not just going to have that, So, first of all, you had sound. You had a bell, so if anything went wrong — — or the machine needed the attendant to come to it, there was a bell it could ring. And there ' s actually an instruction on the punch card which says"Ring the bell."So you can imagine this"Ting!"You know, just stop for a moment, imagine all those noises, this thing,"Click, clack click click click,"steam engine,"Ding,"right? You also need a printer, obviously, and everyone needs a printer. This is actually a **picture** of the printing mechanism for another machine of his, called the Difference Engine No. 2, which he never built, but which the Science Museum did build in the ' 80s and ' 90s. It ' s completely **mechanical**, again, a printer. It prints just numbers, because he was obsessed with numbers, but it does print onto paper, and it even does word wrapping, so if you get to the end of the line, it goes around like that. You also need graphics, right? I mean, if you ' re going to do anything with graphics, so he said,"Well, I need a plotter. I ' ve got a big piece of paper and an ink pen and I ' ll make it plot."So he designed a plotter as well, and, you know, at that point, I think he got pretty much a pretty good machine. Along comes this woman, Ada Lovelace. Now, imagine these soirees, all these great and good comes along. This lady is the daughter of the mad, bad and dangerous-to-know Lord Byron, and her mother, being a bit worried that she might have inherited some of Lord Byron ' s madness and badness, thought,"I know the solution : Mathematics is the solution. We ' ll teach her mathematics. That ' ll calm her down."Because of course, there ' s never been a mathematician that ' s gone crazy, so, you know, that ' ll be fine. Everything ' ll be fine. So she ' s got this mathematical training, and she goes to one of these soirees with her mother, and Charles Babbage, you know, gets out his machine. The Duke of Wellington is there, you know, get out the machine, obviously demonstrates it, and she gets it. She ' s the only person in his lifetime, really, who said,"I understand what this does, and I understand the future of this machine."And we owe to her an enormous amount because we know a lot about the machine that Babbage was intending to build because of her. Now, some people call her the first programmer. This is actually from one of — the paper that she translated. This is a program written in a particular style. It ' s not, historically, totally accurate that she ' s the first programmer, and actually, she did something more amazing. Rather than just being a programmer, she saw something that Babbage didn ' t. Babbage was totally obsessed with mathematics. He was building a machine to do mathematics, and Lovelace said,"You could do more than mathematics on this machine."And just as you do, everyone in this room already ' s got a computer on them right now, because they ' ve got a phone. If you go into that phone, every single thing in that phone or computer or any other computing device is mathematics. It ' s all numbers at the **bottom**. Whether it ' s video or text or music or voice, it ' s all numbers, it ' s all, underlying it, mathematical functions happening, and Lovelace said,"Just because you ' re doing mathematical functions and symbols doesn ' t mean these things can ' t represent other things in the real world, such as music."This was a huge leap, because Babbage is there saying,"We could compute these amazing functions and print out tables of numbers and draw graphs,"— — and Lovelace is there and she says,"Look, this thing could even compose music if you told it a representation of music numerically."So this is what I call Lovelace ' s Leap. When you say she ' s a programmer, she did do some, but the real thing is to have said the future is going to be much, much more than this. Now, a hundred years later, this guy comes along, Alan Turing, and in 1936, and invents the computer all over again. Now, of course, Babbage ’ s machine was entirely **mechanical**. Turing ’ s machine was entirely theoretical. Both of these guys were coming from a mathematical perspective, but Turing told us something very important. He laid down the mathematical foundations for computer science, and said,”It doesn ’ t matter how you make a computer.”It doesn ’ t matter if your computer ’ s **mechanical**, like Babbage ’ s was, or electronic, like computers are today, or perhaps in the future, cells, or, again, **mechanical** again, once we get into nanotechnology. We could go back to Babbage ’ s machine and just make it tiny. All those things are computers. There is in a sense a computing essence. This is called the Church—Turing thesis. And so suddenly, you get this link where you say this thing Babbage had built really was a computer. In fact, it was capable of doing everything we do today with computers, only really slowly. To give you an idea of how slowly, it had about 1k of memory. It used punch cards, which were being fed in, and it ran about 10, 000 times slower the first ZX81. It did have a RAM pack. You could add on a lot of extra memory if you wanted to. So, where does that bring us today? So there are plans. Over in Swindon, the Science Museum archives, there are hundreds of plans and thousands of pages of notes written by Charles Babbage about this analytical engine. One of those is a set of plans that we call Plan 28, and that is also the name of a charity that I started with Doron Swade, who was the curator of computing at the Science Museum, and also the person who drove the project to build a difference engine, and our plan is to build it. Here in South Kensington, we will build the analytical engine. The project has a number of parts to it. One was the **scanning** of Babbage ’ s archive. That ’ s been done. The second is now the study of all of those plans to determine what to build. The third part is a computer simulation of that machine, and the last part is to physically build it at the Science Museum. When it ’ s built, you ’ ll finally be able to understand how a computer works, because rather than having a tiny chip in front of you, you ’ ve got to look at this humongous thing and say,”Ah, I see the memory operating, I see the CPU operating, I hear it operating. I probably smell it operating.”But in between that we ’ re going to do a simulation. Babbage himself wrote, he said, as soon as the analytical engine exists, it will surely guide the future course of science. Of course, he never built it, because he was always fiddling with new plans, but when it did get built, of course, in the 1940s, everything changed. Now, I ’ ll just give you a little taste of what it looks like in motion with a video which shows just one part of the CPU mechanism working. So this is just three sets of cogs, and it ’ s going to add. This is the adding mechanism in action, so you imagine this gigantic machine. So, give me five years. Before the 2030s happen, we ’ ll have it. Thank you very much.

Text Sample 704: POS Dim 5 – 2013 – Score 7.00 – 2013/t002089.txt

I ’ m almost like a crazy evangelical. I ’ ve always known that the age of design is upon us, almost like a rapture. If the day is sunny, I think,”Oh, the gods have had a good design day.”Or, I go to a show and I see a beautiful piece by an artist, particularly beautiful, I say he ’ s so good because he clearly looked to design to understand what he needed to do. So I really do believe that design is the highest form of creative expression. That ’ s why I ’ m talking to you today about the age of design, and the age of design is the age in which design is still cute furniture, is still posters, is still fast cars, what you see at MoMA today. But in truth, what I really would like to explain to the public and to the audiences of MoMA is that the most interesting chairs are the ones that are actually made by a robot, like this beautiful chair by Dirk Vander Kooij, where a robot deposits a toothpaste-like slur of recycled refrigerator parts, as if he were a big candy, and makes a chair out of it. Or good design is digital fonts that we use all the time and that become part of our identity. I want people to understand that design is so much more than cute chairs, that it is first and foremost everything that is around us in our life. And it ' s interesting how so much of what we ' re talking about tonight is not simply design but interaction design. And in fact, interaction design is what I ' ve been trying to insert in the collection of the Museum of Modern Art for a few years, starting not very timidly but just pointedly with works, for instance, by Martin Wattenberg — the way a machine plays chess with itself, that you see here, or Lisa Strausfeld and her partners, the Sugar interface for One Laptop Per Child, Toshio Iwai ' s Tenori-On musical instruments, and Philip Worthington ' s Shadow Monsters, and John Maeda ' s Reactive Books, and also Jonathan Harris and Sep Kamvar ' s I Want You To Want Me. These were some of the first acquisitions that really introduced the idea of interaction design to the public. But more recently, I ' ve been trying really to go even deeper into interaction design with examples that are emotionally really suggestive and that really explain interaction design at a level that is almost undeniable. The Wind Map, by Wattenberg and Fernanda Vigas, I don ' t know if you ' ve ever seen it — it ' s really fantastic. It looks at the territory of the United States as if it were a wheat field that is procured by the winds and that is really giving you a pictorial image of what ' s going on with the winds in the United States. But also, more recently, we started acquiring video games, and that ' s where all **hell** broke loose in a really interesting way. There are still people that believe that there ' s a high and there ' s a low. And that ' s really what I find so intriguing about the reactions that we ' ve had to the anointment of video games in the MoMA collection. We ' ve — No, first of all, New York Magazine always gets it. I love them. So we are in the right quadrant. We are in the Highbrow — that ' s daring, that ' s courageous — and Brilliant, which is great. Timidly, we ' ve been higher on the diagonal in other situations, but it ' s okay. It ' s good. It ' s good. It ' s good. But here comes the art critic. Oh, that was fantastic. So the first was Jonathan Jones from The Guardian."Sorry, MoMA, video games are not art."Did I ever say they were art? I was talking about interaction design. Excuse me."Exhibiting Pac-Man and Tetris alongside Picasso and Van Gogh"— They ' re two floors away. —"will mean game over for any real understanding of art."I ' m bringing in the end of the world. You know? We were talking about the rapture? It ' s coming. And Jonathan Jones is making it happen. So the same Guardian rebuts,"Are video games art : the debate that shouldn ' t be. Last week, Guardian art critic blah blah suggested that games cannot qualify as art. But is he right? And does it matter?"Thank you. Does it matter? You know, it ' s like once again there ' s this whole problem of design being often misunderstood for art, or the idea that is so diffuse that designers want to aspire to, would like to be called, artists. No. Designers aspire to be really great designers. Thank you very much. And that ' s more than enough. So my knight in shining armor, John Maeda, without any prompt, came out with this big declaration on why video games belong in the MoMA. And that was fantastic. And I thought that was it. But then there was another wonderfully pretentious article that came out in The New Republic, so pretentious, by Liel Leibovitz, and it said,"MoMA has mistaken video games for art."Again."The **museum** is putting Pac-Man alongside Picasso."Again."That misses the point."Excuse me. You ' re missing the point. And here, look, the above question is put bluntly :"Are video games art? No. Video games aren ' t art because they are quite thoroughly something else

: code.”Oh, so Picasso is not art because it ’ s oil paint. Right? So it ’ s so fantastic to see how these feathers that were ruffled, and these reactions, were so vehement. And you know what? The International Cat Video Film Festival didn ’ t have that much of a reaction. I think this was truly fantastic. We were talking about dancing ponies, but I was really jealous of the Walker Arts Center for putting up this festival, because it ’ s very, very wonderful. And there ’ s this Flaubert quote that I love :”I have always tried to live in an ivory tower, but a tide of shit is beating at its walls, threatening to undermine it.”I consider myself the tide of shit. You know, we have to go through that. Even in the 1930s, my colleagues that were trying to put together an abstract art show had all of these works stopped by the customs officers that decided they were not art. So it ’ s happened before, and it will happen in the future, but right now I can tell you that I am so, so proud to be able to call Pac-Man part of the MoMA collection. And the same with, for instance, Tetris, original version, the Soviet one. And you know, the amount of work — yeah, Alexey Pajitnov was working for the Soviet government and that ’ s how he developed Tetris, and Alexey himself reconstructed the whole game and even gave us a simulation of the cathode ray tube that makes it look slightly bombed. And it ’ s fantastic. So behind these acquisitions is an enormous amount of work, because we ’ re still the Museum of Modern Art, so even when we tackle popular culture, we tackle it as a form of interaction design and as something that has to go into the collection at MoMA, therefore, has to be researched. So to get to choosing Eric Chahi ’ s wonderful Another World, amongst others, we put together a **panel** of experts, and we worked on this acquisition, and it ’ s mostly myself and Kate Carmody and Paul Galloway. We worked on it for a year and a half. So many people helped us — designers of games, you might know Jamin Warren and his collaborators at Kill Screen magazine, and you know, Kevin Slavin. You name it. We bugged everybody, because we knew that we were ignorant. We were not real gamers enough, so we had to really talk to them. And so we decided, of course, to have Sim City 2000, not the other Sim City, that one in particular, so the criteria that we developed along the way were really strong, and were not only criteria of selection. They were also criteria of exhibition and of preservation. That ’ s what makes this acquisition more than a little game or a little joke. It ’ s truly a way to think of how to preserve and show artifacts that will more and more become part of our lives in the future. We live today, as you know very well, not in the digital, not in the physical, but in the kind of minestrone that our mind makes of the two. And that ’ s really where interaction lies, and that ’ s the importance of interaction. And in order to explain interaction, we need to really bring people in and make them realize how interaction is part of their lives. So when I talk about it, I don ’ t talk only about video games, which are in a way the purest form of interaction, unadulterated by any kind of function or finality. I also talk about the MetroCard vending machine, which I consider a masterpiece of interaction. I mean, that interface is beautiful. It looks like a burly MTA guy coming out of the tunnel. You know, with your mitt you can actually paw the MetroCard, and I talk about how bad ATM machines usually are. So I let people understand that it ’ s up to them to know how to judge interaction so as to know when it ’ s good or when it ’ s bad. So when I show The Sims, I try to make people really feel what it meant to have an interaction with The Sims, not only the **fun** but also the responsibility that came with the Tamagotchi. You know, video games can be truly deep even when they ’ re completely mindless. I ’ m sure that all of you know Katamari Damacy. It ’ s about rolling a ball and picking up as many objects as you can in a finite amount of time and hopefully you ’ ll be able to make it into a planet. I ’ ve never made it into a planet, but that ’ s it. Or, you know, Vib-Ribbon was not distributed here in the United States. It was a PlayStation game, but mostly for Japan. And it was one of the first video games in which you could choose your own music. So you would put into the PlayStation, you would put your own CD, and then the game would change alongside your music. So really fantastic. Not to mention Eve Online. Eve Online is an artificial universe, if you wish, but one of the diplomats that was killed in Benghazi, not Ambassador Stevens, but one of his collaborators, was a really big shot in Eve Online, so here you have a diplomat in the real world that spends his time in Eve Online to kind of test, maybe, all of his ideas about diplomacy and about universe-building, and to the point that the first announcement of the bombing was actually given on Eve Online, and after his death, several parts of the universe were named after him. And I was just recently at the Eve Online fan festival in Reykjavk that was quite amazing. I mean, we ' re talking about an experience that of course can seem weird to many, but that is very educational. Of course, there are games that are even more educational. Dwarf Fortress is like the holy grail of this kind of massive multiplayer online game, and in fact the two Adams brothers were in Reykjavk, and they were greeted by a standing ovation by all the Eve Online fans. It was amazing to see. And it ' s a beautiful game. So you start seeing here that the aesthetics that are so important to a **museum** collection like MoMA ' s are kept alive also by the selection of these games. And you know, Valve — you know, Portal — is an example of a video game in which you have a certain type of violence which also leads me to talk about one of the biggest issues that we had to discuss when we acquired the video games, what to do with violence. Right? We had to make decisions. At MoMA, interestingly, there ' s a lot of violence depicted in the art part of the collection, but when I came to MoMA 19 years ago, and as an Italian, I said,"You know what, we need a Beretta."And I was told,"No. No guns in the design collection."And I was like,"Why?"Interestingly, I learned that it ' s considered that in design and in the design collection, what you see is what you get. So when you see a gun, it ' s an instrument for killing in the design collection. If it ' s in the art collection, it might be a critique of the killing instrument. So it ' s very interesting. But we are acquiring our critical dimension also in design, so maybe one day we ' ll be able to acquire also the guns. But here, in this particular case, we decided, you know, with Kate and Paul, that we would have no gratuitous violence. So we have Portal because you shoot walls in order to create new spaces. We have Street Fighter II, because martial arts are good. But we don ' t have GTA because, maybe it ' s my own reflection, I ' ve never been able to do anything but crashing cars and shooting prostitutes and pimps. So it was not very constructive. So, I ' m making **fun** of it, but we discussed this for so many days. You have no idea. And to this day, I am ambivalent, but when you have instead games like Flow, there ' s no doubt. It ' s like, it ' s about serenity and it ' s about sublime. It ' s about experiencing what it means to be a sea creature. Then we have a few also side-scrollers — classical ones. So it ' s quite a hefty collection. And right now, we started with the first 14, but we have several that are coming up, and the reason why we haven ' t acquired them yet is because you don ' t acquire just the game. You acquire the relationship with the company. What we want, what we aspire to, is the code. It ' s very hard to get, of course. But that ' s what would enable us to preserve the video games for a really long time, and that ' s what **museums** do. They also preserve artifacts for posterity. In absence of the code, because, you know, video game companies are not very forthcoming in some cases, in absence of that, we acquire the relationship with the company. We ' re going to stay with them forever. They ' re not going to get rid of us. And one day, we ' ll get that code. But I want to explain to you the criteria that we chose for interaction design. Aesthetics are really important. And I ' m showing you Core War here, which is an early game that takes advantage aesthetically of the limitations of the processor. So the kind of interferences that you see here that look like beautiful barriers in the game are actually a consequence of the processor ' s limitedness, which is fantastic. So aesthetics is always important. And so is space, the spatial aspect of games. You know, I feel that the best video games are the ones that have really savvy **architects** that are behind them, and if they ' re not **architects**, bona fide trained in architecture, they have that feeling. But the spatial evolution in video games is extremely important. Time. The way we experience time in video games, as in other forms of interaction design, is really quite amazing. It can be real time or it can be the time within the game, as is in Animal Crossing, where seasons follow each other at their own pace. So time, space, aesthetics, and then, most important, behavior. The real core issue of interaction design is behavior. Designers that deal with interaction design behaviors that go to influence the rest of our lives. They ' re not just limited to our interaction with the screen. In this case, I ' m showing you Marble Madness, which is a beautiful game in which the controller is a big sphere that vibrates with you, so you have a sphere that ' s moving in this landscape, and the sphere, the controller itself, gives you a sense of the movement. In a way, you can see how video games are the purest aspect of interaction design and are very useful to explain what interaction is. We don ' t want to show the video games with the paraphernalia. No arcade nostalgia. If anything, we want to show the code, and here you see Ben Fry ' s distellamap of Pac-Man, of the Pac-Man code. So the way we acquired the games is very interesting and very unorthodox. You see them here displayed alongside other examples of design, furniture and other parts, but there ' s no paraphernalia, no nostalagia, only the screen and a little shelf with the controllers. The controllers are, of course, part of the experience, so you cannot do away with it. But interestingly, this choice was not condemned too vehemently by gamers. I was afraid that they would kill us, and instead they understood, especially when I told them that I was trying to apply the same stratagem that Philip Johnson applied in 1934 when he wanted to make people understand the importance of design, and he took propeller blades and pieces of machinery and in the MoMA galleries he put them on white pedestals against white walls, as if they were Brancusi sculptures. He created this strange distance, this shock, that made people realize how gorgeous formally, and also important functionally, design pieces were. I would like to do the same with video games. By getting rid of the sticky carpets and the cigarette butts and everything else that we might remember from our childhood, I want people to understand that those are important forms of design. And in a way, the video games, the fonts and everything else lead us to make people understand a wider meaning for design. One of my dream acquisitions, which has been on hold for a few years but now will come back on the front burner, is a 747. I would like to acquire it, but without owning it. I don ' t want it to be at MoMA and possessed by MoMA. I want it to keep flying. So it ' s an acquisition where MoMA makes an arrangement with an airline and keeps the Boeing 747 flying. And the same with the"@"sign that we acquired a few years ago. It was the first example of an acquisition of something that is in the public domain. And what I say to people, it ' s almost as if a butterfly were flying by and we captured the shadow on the wall, and just we ' re showing the shadow. So in a way, we ' re showing a manifestation of something that is truly important and that is part of our identity but that nobody can have. And it ' s too long to explain the acquisition, but if you want to go on the MoMA blog, there ' s a long post where I explain why it ' s such a great example of design. Along the way, I ' ve had to burn a few chairs. You know? I ' ve had to do away with a few concepts of design past. But I see that people are coming along, that the audiences, paradoxically, are much more responsive and much more understanding of this expansion of design than some of my colleagues are. Design is truly everywhere, and design is as important as anything, and I ’ m so glad that, because of its diversity and because of its centrality to our lives, many more people are coming to it as a profession, as a passion, and as, very simply, part of their own culture. Thank you very much.

Text Sample 705: POS Dim 5 – 2013 – Score 5.00 – 2013/t001434.txt
Announcer : 10 seconds. Five, four, three, two, one. Official top. Plus one, two, three, four, five six, seven, eight, nine, ten. Guillaume NĂŠry, France. Constant weight, 123 meters, three minutes and 25 seconds. National record attempt. 70 meters. [123 meters] Judge : White card. Guillaume NĂŠry! National record! Guillaume NĂŠry : Thank you. Thank you very much, thanks for the warm welcome. That dive you just watched is a journey â a journey between two breaths. A journey that takes place between two breaths â the last one before diving into the water, and the first one, coming back to the surface. That dive is a journey to the very limits of human possibility, a journey into the unknown. But it ’ s also, and above all, an inner journey, where a number of things happen, physiologically as well as mentally. And that ’ s why I ’ m here today, to share my journey with you and to take you along with me. So, we start with the last breath. As you noticed, that last breath in is slow, deep and intense. It ends with a special technique called the **carp**, which allows me to store one to two extra liters of air in my lungs by compressing it. When I leave the surface, I have about 10 liters of air in my lungs. As soon as I leave the surface the first mechanism kicks in : the diving reflex. The first thing the diving reflex does is make your heart rate drop. My heart beat will drop from about 60-70 per minute to about 30-40 beats per minute in a matter of seconds ; almost immediately. Next, the diving reflex causes peripheral vasoconstriction, which means that the blood flow will leave the body ’ s extremities to feed the most important organs : the lungs, the heart and the brain. This mechanism is innate. I cannot control it. If you go underwater, even if you ’ ve never done it before, you ’ ll experience the exact same effects. All human beings share this characteristic. And what ’ s extraordinary is that we share this instinct with marine mammals â all marine mammals : dolphins, whales, sea lions, etc. When they dive deep into the ocean, these mechanisms become activated, but to a greater extent. And, of course, it works much better for them. It ’ s absolutely fascinating. Right as I leave the surface, nature gives me a push in the right direction, allowing me to descend with confidence. So as I dive deeper into the blue, the pressure slowly starts to squeeze my lungs. And since it ’ s the amount of air in my lungs that makes me float, the farther down I go, the more pressure there is on my lungs, the less air they contain and the easier it is for my body to fall. And at one point, around 35 or 40 meters down, I don ’ t even need to swim. My body is dense and heavy enough to fall into the depths by itself, and I enter what ’ s called the free fall phase. The free fall phase is the best part of the dive. It ’ s the reason I still dive. Because it feels like you ’ re being pulled down and you don ’ t need to do anything. I can go from 35 meters to 123 meters without making a single movement. I let myself be pulled by the depths, and it feels like I ’ m flying underwater. It ’ s truly an amazing feeling â an extraordinary feeling of freedom. And so I slowly continue sliding to the **bottom**. 40 meters down, 50 meters down, and between 50 and 60 meters, a second physiological response kicks in. My lungs reach residual volume, below which they ’ re not supposed to be compressed, in theory. And this second

response is called blood shift, or"pulmonary erection"in French. I prefer"blood shift."So blood shift â how does it work? The capillaries in the lungs become engorged with blood â which is caused by the suction â so the lungs can harden and protect the whole chest cavity from being crushed. It prevents the two walls of the lungs from collapsing, from sticking together and caving in. Thanks to this phenomenon, which we also share with marine mammals, I ' m able to continue with my dive. 60, 70 meters down, I keep falling, faster and faster, because the pressure is crushing my body more and more. Below 80 meters, the pressure becomes a lot stronger, and I start to feel it physically. I really start to feel the suffocation. You can see what it looks like â not pretty at all. The diaphragm is completely collapsed, the rib cage is squeezed in, and mentally, there is something going on as well. You may be thinking,"This doesn ' t look enjoyable. How do you do it?"If I relied on my earthly reflexes â what do we do above water when there ' s a problem? We resist, we go against it. We fight. Underwater, that doesn ' t work. If you try that underwater, you might tear your lungs, spit up blood, develop an edema and you ' ll have to stop diving for a good amount of time. So what you need to do, mentally, is to tell yourself that nature and the elements are stronger than you. And so I let the water crush me. I accept the pressure and go with it. At this point, my body receives this information, and my lungs start relaxing. I relinquish all control, and relax completely. The pressure starts crushing me, and it doesn ' t feel bad at all. I even feel like I ' m in a cocoon, protected. And the dive continues. 80, 85 meters down, 90, 100. 100 meters â the magic number. In every sport, it ' s a magic number. For swimmers and athletes and also for us, free divers, it ' s a number everyone dreams of. Everyone wishes one day to be able to get to 100 meters. And it ' s a symbolic number for us, because in the 1970s, doctors and physiologists did their math, and predicted that the human body would not be able to go below 100 meters. Below that, they said, the human body would implode. And then the Frenchman, Jacques Mayol â you all know him as the hero in"The Big Blue"â came along and dived down to 100 meters. He even reached 105 meters. At that time, he was doing"no limits."He ' d use weights to descend faster and come back up with a balloon, just like in the movie. Today, we go down 200 meters in no limit free diving. I can do 123 meters by simply using muscle strength. And in a way, it ' s all thanks to him, because he challenged known facts, and with a sweep of his hand, got rid of the theoretical beliefs and all the mental limits that we like to impose on ourselves. He showed us that the human body has an infinite ability to adapt. So I carry on with my dive. 105, 110, 115. The **bottom** is getting closer. 120, 123 meters. I ' m at the **bottom**. And now, I ' d like to ask you to join me and put yourself in my place. Close your eyes. Imagine you get to 123 meters. The surface is far, far away. You ' re alone. There ' s hardly any light. It ' s cold â freezing cold. The pressure is crushing you completely â 13 times stronger than on the surface. And I know what you ' re thinking :"This is horrible. What the **hell** am I doing here? He ' s insane."But no. That ' s not what I think when I ' m down there. When I ' m at the **bottom**, I feel good. I get this extraordinary feeling of well-being. Maybe it ' s because I ' ve completely released all tensions and let myself go. I feel great, without the need to breathe. Although, you ' d agree, I should be worried. I feel like a tiny dot, a little drop of water, floating in the middle of the ocean. And each time, I **picture** the same image. [The Pale Blue Dot] It ' s that small dot the arrow is pointing to. Do you know what it is? It ' s planet Earth. Planet Earth, photographed by the Voyager probe, from 4 billion kilometers away. And it shows that our home is that small dot over there, floating in the middle of nothing. That ' s how I feel when I ' m at the **bottom**, at 123 meters. I feel like a small dot, a speck of dust, stardust, floating in the middle of the cosmos, in the middle of nothing, in the immensity of space. It ’ s a fascinating sensation, because when I look up, down, left, right, in front, behind, I see the same thing : the infinite deep blue. Nowhere else on Earth you can experience this â looking all around you, and seeing the same thing. It ’ s extraordinary. And at that moment, I still get that feeling each time, building up inside of me â the feeling of humility. Looking at this **picture**, I feel very humble â just like when I ’ m all the way down at the **bottom** â because I ’ m nothing, I ’ m a little speck of nothingness lost in all of time and space. And it still is absolutely fascinating. I decide to go back to the surface, because this is not where I belong. I belong up there, on the surface. So I start heading back up. I get something of a shock at the very moment when I decide to go up. First, because it takes a huge effort to tear yourself away from the **bottom**. It pulled you down on the way in, and will do the same on the way up. You have to swim twice as hard. Then, I ’ m hit with another phenomenon known as narcosis. I don ’ t know if you ’ ve heard of that. It ’ s called nitrogen narcosis. It ’ s something that happens to scuba divers, but it can happen to free divers. It ’ s caused by nitrogen dissolving in the blood, which causes confusion between the conscious and unconscious mind. A flurry of thoughts goes spinning through your head. You can ’ t control them, and you shouldn ’ t try to â you have to let it happen. The more you try to control it, the harder it is to manage. Then, a third thing happens : the desire to breathe. I ’ m not a fish, I ’ m a human being, and the desire to breathe reminds me of that fact. Around 60, 70 meters, you start to feel the need to breathe. And with everything else that ’ s going on, you can very easily lose your ground and start to panic. When that happens, you think,”Where ’ s the surface? I want to go up. I want to breathe now.”You should not do that. Never look up to the surface â not with your eyes, or your mind. You should never **picture** yourself up there. You have to stay in the present. I look at the rope right in front of me, leading me back to the surface. And I focus on that, on the present moment. Because if I think about the surface, I panic. And if I panic, it ’ s over. Time goes faster this way. And at 30 meters : deliverance. I ’ m not alone any more. The safety divers, my guardian angels, join me. They leave the surface, we meet at 30 meters, and they escort me for the final few meters, where potential problems could arise. Every time I see them, I think to myself,”It ’ s thanks to you.”It ’ s thanks to them, my team, that I ’ m here. It brings back the sense of humility. Without my team, without all the people around me, the adventure into the deep would be impossible. A journey into the deep is above all a group effort. So I ’ m happy to finish my journey with them, because I wouldn ’ t be here if it weren ’ t for them. 20 meters, 10 meters, my lungs slowly return to their normal volume. Buoyancy pushes me up to the surface. Five meters below the surface, I start to breathe out, so that as soon as I get to the surface all I do is breathe in. And so I arrive at the surface. Air floods into my lungs. It ’ s like being born again, a relief. It feels good. Though the journey was extraordinary, I do need to feel those small oxygen molecules fueling my body. It ’ s an extraordinary sensation, but at the same time it ’ s traumatizing. It ’ s a shock to the system, as you can you imagine. I go from complete darkness to the light of day, from the near-silence of the depths to the commotion up top. In terms of touch, I go from the soft, velvety feeling of the water, to air rubbing across my face. In terms of smell, there is air rushing into my lungs. And in return, my lungs open up. They were completely squashed just 90 seconds ago, and now, they ’ ve opened up again. So all of this affects quite a lot of things. I need a few seconds to come back, and to feel”all there”again. But that needs to happen quickly, because the judges are there to verify my performance ; I need to show them I ’ m in perfect physical condition.

You saw in the video, I was doing a so-called exit protocol. Once at the surface, I have 15 seconds to take off my nose clip, give this signal and say"I am OK."Plus, you need to be bilingual. On top of everything â that ' s not very nice. Once the protocol is completed, the judges show me a white card, and that ' s when the joy starts. I can finally celebrate what has just happened. So, the journey I ' ve just described to you is a more extreme version of free diving. Luckily, it ' s far from just that. For the past few years, I ' ve been trying to show another side of free diving, because the media mainly talks about competitions and records. But free diving is more than just that. It ' s about being at ease in the water. It ' s extremely beautiful, very poetic and artistic. So my wife and I decided to **film** it and try to show another side of it, mostly to make people want to go into the water. Let me show you some images to finish my story. It ' s a mix of beautiful underwater photos. I ' d like you to know that if one day you try to stop breathing, you ' ll realize that when you stop breathing, you stop thinking, too. It calms your mind. Today, in the 21st century, we ' re under so much pressure. Our minds are overworked, we think at a million miles an hour, we ' re always stressed. Being able to free dive lets you, just for a moment, relax your mind. Holding your breath underwater means giving yourself the chance to experience weightlessness. It means being underwater, floating, with your body completely relaxed, letting go of all your tensions. This is our plight in the 21st century : our backs hurt, our necks hurt, everything hurts, because we ' re stressed and tense all the time. But when you ' re in the water, you let yourself float, as if you were in space. You let yourself go completely. It ' s an extraordinary feeling. You can finally get in touch with your body, mind and spirit. Everything feels better, all at once. Learning how to free dive is also about learning to breathe correctly. We breathe with our first breath at birth, up until our last one. Breathing gives rhythm to our lives. Learning how to breathe better is learning how to live better. Holding your breath in the sea, not necessarily at 100 meters, but maybe at two or three, putting on your goggles, a pair of flippers, means you can go see another world, another universe, completely magical. You can see little fish, seaweed, the flora and fauna, you can watch it all discreetly, sliding underwater, looking around, and coming back to the surface, leaving no trace. It ' s an amazing feeling to become one with nature like that. And if I may say one more thing, holding your breath, being in the water, finding this underwater world â it ' s all about connecting with yourself. You heard me talk a lot about the body ' s memory that dates back millions of years, to our marine origins. The day you get back into the water, when you hold your breath for a few seconds, you will reconnect with those origins. And I guarantee it ' s absolute magic. I encourage you to try it out. Thank you.

Text Sample 706: POS Dim 5 – 2013 – Score 5.00 – 2013/t001925.txt

I would like to show you how architecture has helped to change the life of my community and has opened opportunities to hope. I am a native of Burkina Faso. According to the World Bank, Burkina Faso is one of the poorest countries in the world, but what does it look like to grow up in a place like that? I am an example of that. I was born in a little village called Gando. In Gando, there was no electricity, no access to **clean** drinking water, and no school. But my father wanted me to learn how to read and write. For this reason, I had to leave my family when I was seven and to stay in a city far away from my village with no contact with my family. In this place I sat in a class like that with more than 150 other kids, and for six years. In this time, it just happened to me to come to school to realize that my classmate died. Today, not so much has changed. There is still no electricity in my village. People still are dying in Burkina Faso, and access to **clean** drinking water is still a big problem. I had luck. I was lucky, because this is a fact of life when you grow up in a place like that. But I was lucky. I had a scholarship. I could go to Germany to study. So now, I suppose, I don ’ t need to explain to you how great a privilege it is for me to be standing before you today. From Gando, my home village in Burkina Faso, to Berlin in Germany to become an **architect** is a big, big step. But what to do with this privilege? Since I was a student, I wanted to open up better opportunities to other kids in Gando. I just wanted to use my skills and build a school. But how do you do it when you ’ re still a student and you don ’ t have money? Oh yes, I started to make **drawings** and asked for money. Fundraising was not an easy task. I even asked my classmates to spend less money on coffee and cigarettes, but to sponsor my school project. In real wonder, two years later, I was able to collect 50, 000 U. S. dollars. When I came home to Gando to bring the good news, my people were over the moon, but when they realized that I was planning to use clay, they were shocked.”A clay building is not able to stand a rainy season, and Francis wants us to use it and build a school. Is this the reason why he spent so much time in Europe studying instead of working in the field with us?”My people build all the time with clay, but they don ’ t see any innovation with mud. So I had to convince everybody. I started to speak with the community, and I could convince everybody, and we could start to work. And the women, the men, everybody from the village, was part of this building process. I was allowed to use even traditional techniques. So clay floor for example, the young men come and stand like that, beating, hours for hours, and then their mothers came, and they are beating in this position, for hours, giving water and beating. And then the polishers come. They start polishing it with a stone for hours. And then you have this result, very fine, like a baby **bottom**. It ’ s not photoshopped. This is the school, built with the community. The walls are totally made out of compressed clay blocks from Gando. The **roof** structure is made with cheap steel bars normally hiding inside concrete. And the classroom, the ceiling is made out of both of them used together. In this school, there was a simple idea : to create comfort in a classroom. Don ’ t forget, it can be 45 degrees in Burkina Faso, so with simple ventilation, I wanted to make the classroom good for teaching and learning. And this is the project today, 12 years old, still in best condition. And the kids, they love it. And for me and my community, this project was a huge success. It has opened up opportunities to do more projects in Gando. So I could do a lot of projects, and here I am going to share with you only three of them. The first one is the school extension, of course. How do you explain **drawings** and engineering to people who are neither able to read nor write? I started to build a prototype like that. The innovation was to build a clay vault. So then, I jumped on the top like that, with my team, and it works. The community is looking. It still works. So we can build. And we kept building, and that is the result. The kids are happy, and they love it. The community is very proud. We made it. And even animals, like these donkeys, love our buildings. The next project is the library in Gando. And see now, we tried to introduce different ideas in our buildings, but we often don ’ t have so much material. Something we have in Gando are clay pots. We wanted to use them to create openings. So we just bring them like you can see to the building site. we start cutting them, and then we place them on top of the **roof** before we pour the concrete, and you have this result. The openings are letting the hot air out and light in. Very simple. My most recent project in Gando is a high school project. I would like to share with you this. The innovation in this project is to cast mud like you cast concrete. How do you cast mud? We start making a lot of mortars, like you can see, and when everything is ready, when you know what is the best recipe and the best form, you start working with the community. And sometimes I can leave. They will do it themselves. I came to speak to you like that. Another factor in Gando is rain. When the rains come, we hurry up to protect our fragile walls against the rain. Don ’ t confound with Christo and Jeanne- Claude. It is simply how we protect our walls. The rain in Burkina comes very fast, and after that, you have floods everywhere in the country. But for us, the rain is good. It brings sand and gravel to the river we need to use to build. We just wait for the rain to go. We take the sand, we mix it with clay, and we keep building. That is it. The Gando project was always connected to training the people, because I just wanted, one day when I fall down and die, that at least one person from Gando keeps doing this work. But you will be surprised. I ’ m still alive. And my people now can use their skills to earn money themselves. Usually, for a young man from Gando to earn money, you have to leave the country to the city, sometimes leave the country and some never come back, making the community weaker. But now they can stay in the country and work on different building sites and earn money to feed their family. There ’ s a new quality in this work. Yes, you know it. I have won a lot of awards through this work. For sure, it has opened opportunities. I have become myself known. But the reason why I do what I do is my community. When I was a kid, I was going to school, I was coming back every holiday to Gando. By the end of every holidays, I had to say goodbye to the community, going from one compound to another one. All women in Gando will open their clothes like that and give me the last penny. In my culture, this is a symbol of deep affection. As a seven-year-old guy, I was impressed. I just asked my mother one day,”Why do all these women love me so much?”She just answered,”They are contributing to pay for your education hoping that you will be successful and one day come back and help improve the quality of life of the community.”I hope now that I was able to make my community proud through this work, and I hope I was able to prove you the power of community, and to show you that architecture can be inspiring for communities to shape their own future. Merci beaucoup. Thank you. Thank you. Thank you. Thank you. Thank you. Thank you.

Text Sample 707: POS Dim 5 – 2014 – Score 7.00 – 2014/t001677.txt
Now, I ’ m an ethnobotanist. That ’ s a scientist who works in the rainforest to document how people use local plants. I ’ ve been doing this for a long time, and I want to tell you, these people know these forests and these medicinal treasures better than we do and better than we ever will. But also, these cultures, these indigenous cultures, are disappearing much faster than the forests themselves. And the greatest and most endangered species in the Amazon Rainforest is not the jaguar, it ’ s not the harpy eagle, it ’ s the isolated and uncontacted tribes. Now four years ago, I injured my **foot** in a climbing accident and I went to the doctor. She gave me heat, she gave me cold, aspirin, narcotic painkillers, anti-inflammatories, cortisone shots. It didn ’ t work. Several months later, I was in the northeast Amazon, walked into a village, and the shaman said,”You ’ re limping.”And I ’ ll never forget this as long as I live. He looked me in the face and he said,”Take off your shoe and give me your machete.”He walked over to a palm tree and carved off a fern, threw it in the fire, applied it to my **foot**, threw it in a pot of water, and had me drink the tea. The pain disappeared for seven months. When it came back, I went to see the shaman again. He gave me the same treatment, and I ’ ve been cured for three years now. Who would

you rather be treated by? Now, make no mistake — Western medicine is the most successful system of healing ever devised, but there ' s plenty of holes in it. Where ' s the cure for breast cancer? Where ' s the cure for schizophrenia? Where ' s the cure for acid reflux? Where ' s the cure for insomnia? The fact is that these people can sometimes, sometimes, sometimes cure things we cannot. Here you see a medicine man in the northeast Amazon treating leishmaniasis, a really nasty protozoal disease that afflicts 12 million people around the world. Western treatment are injections of antimony. They ' re painful, they ' re expensive, and they ' re probably not good for your heart ; it ' s a heavy **metal**. This man cures it with three plants from the Amazon Rainforest. This is the magic frog. My colleague, the late great Loren McIntyre, discoverer of the source lake of the Amazon, Laguna McIntyre in the Peruvian Andes, was lost on the Peru-Brazil border about 30 years ago. He was rescued by a group of isolated Indians called the Matss. They beckoned for him to follow them into the forest, which he did. There, they took out palm leaf baskets. There, they took out these green monkey frogs — these are big suckers, they ' re like this — and they began licking them. It turns out, they ' re highly hallucinogenic. McIntyre wrote about this and it was read by the editor of High Times magazine. You see that ethnobotanists have friends in all sorts of strange cultures. This guy decided he would go down to the Amazon and give it a whirl, or give it a lick, and he did, and he wrote,"My blood pressure went through the **roof**, I lost full control of my bodily functions, I passed out in a heap, I woke up in a hammock six hours later, felt like God for two days."An Italian chemist read this and said,"I ' m not really interested in the theological aspects of the green monkey frog. What ' s this about the change in blood pressure?"Now, this is an Italian chemist who ' s working on a new treatment for high blood pressure based on peptides in the skin of the green monkey frog, and other scientists are looking at a cure for drug-resistant Staph aureus. How ironic if these isolated Indians and their magic frog prove to be one of the cures. Here ' s an ayahuasca shaman in the northwest Amazon, in the middle of a yage ceremony. I took him to Los Angeles to meet a foundation officer looking for support for monies to protect their culture. This fellow looked at the medicine man, and he said,"You didn ' t go to medical school, did you?"The shaman said,"No, I did not."He said,"Well, then what can you know about healing?"The shaman looked at him and he said,"You know what? If you have an infection, go to a doctor. But many human afflictions are diseases of the heart, the mind and the spirit. Western medicine can ' t touch those. I cure them."But all is not rosy in learning from nature about new medicines. This is a viper from Brazil, the venom of which was studied at the Universidade de So Paulo here. It was later developed into ACE inhibitors. This is a frontline treatment for hypertension. Hypertension causes over 10 percent of all deaths on the planet every day. This is a $4 billion industry based on venom from a Brazilian snake, and the Brazilians did not get a nickel. This is not an acceptable way of doing business. The rainforest has been called the greatest expression of life on Earth. There ' s a saying in Suriname that I dearly love :"The rainforests hold answers to questions we have yet to ask."But as you all know, it ' s rapidly disappearing. Here in Brazil, in the Amazon, around the world. I took this **picture** from a small plane flying over the eastern border of the Xingu indigenous reserve in the state of Mato Grosso to the northwest of here. The top half of the **picture**, you see where the Indians live. The line through the middle is the eastern border of the reserve. Top half Indians, **bottom** half white guys. Top half wonder drugs, **bottom** half just a bunch of skinny-ass cows. Top half carbon sequestered in the forest where it belongs, **bottom** half carbon in the atmosphere where it ' s driving climate change. In fact, the number two cause of carbon being released into the atmosphere is forest destruction. But in talking about destruction, it ' s important to keep in mind that the Amazon is the mightiest landscape of all. It ' s a place of beauty and wonder. The biggest anteater in the world lives in the rain forest, tips the scale at 90 pounds. The goliath bird-eating spider is the world ' s largest spider. It ' s found in the Amazon as well. The harpy eagle wingspan is over seven **feet**. And the black cayman — these monsters can tip the scale at over half a ton. They ' re known to be man-eaters. The anaconda, the largest snake, the capybara, the largest rodent. A specimen from here in Brazil tipped the scale at 201 pounds. Let ' s visit where these creatures live, the northeast Amazon, home to the Akuriyo tribe. Uncontacted peoples hold a mystical and iconic role in our imagination. These are the people who know nature best. These are the people who truly live in total harmony with nature. By our standards, some would dismiss these people as primitive."They don ' t know how to make fire, or they didn ' t when they were first contacted."But they know the forest far better than we do. The Akuriyos have 35 words for honey, and other Indians look up to them as being the true masters of the emerald realm. Here you see the face of my friend Pohnay. When I was a teenager rocking out to the Rolling Stones in my hometown of New Orleans, Pohnay was a forest nomad roaming the jungles of the northeast Amazon in a small band, looking for game, looking for medicinal plants, looking for a wife, in other small nomadic bands. But it ' s people like these that know things that we don ' t, and they have lots of lessons to teach us. However, if you go into most of the forests of the Amazon, there are no indigenous peoples. This is what you find : rock carvings which indigenous peoples, uncontacted peoples, used to sharpen the edge of the stone axe. These cultures that once danced, made love, sang to the gods, worshipped the forest, all that ' s left is an imprint in stone, as you see here. Let ' s move to the western Amazon, which is really the epicenter of isolated peoples. Each of these dots represents a small, uncontacted tribe, and the big reveal today is we believe there are 14 or 15 isolated groups in the Colombian Amazon alone. Why are these people isolated? They know we exist, they know there ' s an outside world. This is a form of resistance. They have chosen to remain isolated, and I think it is their human right to remain so. Why are these the tribes that hide from man? Here ' s why. Obviously, some of this was set off in 1492. But at the turn of the last century was the rubber trade. The demand for natural rubber, which came from the Amazon, set off the botanical equivalent of a gold rush. Rubber for bicycle tires, rubber for automobile tires, rubber for zeppelins. It was a mad race to get that rubber, and the man on the left, Julio Arana, is one of the true thugs of the story. His people, his company, and other companies like them killed, massacred, tortured, butchered Indians like the Witotos you see on the right hand side of the slide. Even today, when people come out of the forest, the story seldom has a happy ending. These are Nukaks. They were contacted in the ' 80s. Within a year, everybody over 40 was dead. And remember, these are preliterate societies. The elders are the libraries. Every time a shaman dies, it ' s as if a library has burned down. They have been forced off their lands. The drug traffickers have taken over the Nukak lands, and the Nukaks live as beggars in public **parks** in eastern Colombia. From the Nukak lands, I want to take you to the southwest, to the most spectacular landscape in the world : Chiribiquete National Park. It was surrounded by three isolated tribes and thanks to the Colombian government and Colombian colleagues, it has now expanded. It ' s bigger than the state of Maryland. It is a treasure trove of botanical diversity. It was first explored botanically in 1943 by my mentor, Richard Schultes, seen here atop the Bell Mountain, the sacred mountains of the Karijonas. And let me show you what it looks like today. Flying over Chiribiquete, realize that these lost world mountains are still lost. No scientist has been atop them. In fact, nobody has been atop the Bell Mountain since Schultes in ' 43. And we ' ll end up here with the Bell Mountain just to the east of the **picture**. Let me show you what it looks like today. Not only is this a treasure trove of botanical diversity, not only is it home to three isolated tribes, but it ' s the greatest treasure trove of pre-Colombian art in the world : over 200, 000 paintings. The Dutch scientist Thomas van der Hammen described this as the Sistine Chapel of the Amazon Rainforest. But move from Chiribiquete down to the southeast, again in the Colombian Amazon. Remember, the Colombian Amazon is bigger than New England. The Amazon ' s a big forest, and Brazil ' s got a big part of it, but not all of it. Moving down to these two national **parks**, Cahuinari and Pur in the Colombian Amazon — that ' s the Brazilian border to the right — it ' s home to several groups of isolated and uncontacted peoples. To the trained eye, you can look at the **roofs** of these malocas, these longhouses, and see that there ' s cultural diversity. These are, in fact, different tribes. As isolated as these areas are, let me show you how the outside world is crowding in. Here we see trade and transport increased in Putumayo. With the diminishment of the Civil War in Colombia, the outside world is showing up. To the north, we have illegal gold mining, also from the east, from Brazil. There ' s increased hunting and fishing for commercial purposes. We see illegal logging coming from the south, and drug runners are trying to move through the **park** and get into Brazil. This, in the past, is why you didn ' t mess with isolated Indians. And if it looks like this **picture** is out of focus because it was taken in a hurry, here ' s why. This looks like — This looks like a hangar from the Brazilian Amazon. This is an art exhibit in Havana, Cuba. A group called Los Carpinteros. This is their perception of why you shouldn ' t mess with uncontacted Indians. But the world is changing. These are Mashco-Piros on the Brazil-Peru border who stumbled out of the jungle because they were essentially chased out by drug runners and timber people. And in Peru, there ' s a very nasty business. It ' s called human safaris. They will take you in to isolated groups to take their **picture**. Of course, when you give them clothes, when you give them tools, you also give them diseases. We call these"inhuman safaris."These are Indians again on the Peru border, who were overflown by flights sponsored by missionaries. They want to get in there and turn them into Christians. We know how that turns out. What ' s to be done? Introduce technology to the contacted tribes, not the uncontacted tribes, in a culturally sensitive way. This is the perfect marriage of ancient shamanic wisdom and 21st century technology. We ' ve done this now with over 30 tribes, mapped, managed and increased protection of over 70 million acres of ancestral rainforest. So this allows the Indians to take control of their environmental and cultural destiny. They also then set up guard houses to keep outsiders out. These are Indians, trained as indigenous **park** rangers, patrolling the borders and keeping the outside world at bay. This is a **picture** of actual contact. These are Chitonahua Indians on the Brazil-Peru border. They ' ve come out of the jungle asking for help. They were shot at, their malocas, their longhouses, were burned. Some of them were massacred. Using automatic weapons to slaughter uncontacted peoples is the single most despicable and disgusting human rights abuse on our planet today, and it has to stop. But let me conclude by saying, this work can be spiritually rewarding, but it ' s difficult and it can be dangerous. Two colleagues of mine passed away recently in the crash of a small plane. They were serving the forest to protect those uncontacted tribes. So the question is, in conclusion, is what the future holds. These are the Uray people in Brazil. What does the future hold for them, and what does the future hold for us? Let ' s think differently. Let ' s make a better world. If the climate ' s going to change, let ' s have a climate that changes for the better rather than the worse. Let ' s live on a planet full of luxuriant vegetation, in which isolated peoples can remain in isolation, can maintain that mystery and that knowledge if they so choose. Let ' s live in a world where the shamans live in these forests and heal themselves and us with their mystical plants and their sacred frogs. Thanks again.

Text Sample 708: POS Dim 5 – 2014 – Score 7.00 – 2014/t000045.txt

Hello, Manchester. I ' m so happy to be here. My name is Sophia Wallace. I ' m an artist. And I ' m here today to share with you a project that I hope will empower you personally, and by extension those that you love, especially if they have a clitoris. To make this work, I have to talk about independent female desire. I have to speak about a universal taboo - female genitals. This is not easy. It taints the speaker. I want to thank TEDxSalford for having me here and hosting this conversation. All bodies are entitled to experience the pleasure that they are capable of. This is a core pillar of cliteracy. In making this work, I had to say that the clitoris, first, as an organ, has a right to being, and that this right is not just about not being cut off. Sadly, to this day, over a 140 million women have had their external clitorises cut off. This doesn ' t make it into the news very often, and this doesn ' t come up in foreign policy discussion. So number one, the clitoris has a right to exist, free of harm, like any other organ. But secondly, I argue with cliteracy that the clitoris has a right to pleasure, and this is part of its primary right of being. How is it possible that we landed on the Moon and walked around 29 years before we discovered the anatomy of the clitoris? We actually cloned sheep, identified the Higgs boson particle, and only discovered the clitoris 29 years ago. Unfortunately, this discovery has not been adopted, so most people don ' t know the actual anatomy of the clitoris. The clitoris is not a button. It is an iceberg. Like many in the room who are hearing this for the first time, I was shocked to find out that I didn ' t know the actual anatomy of half of the population, that I didn ' t know my own anatomy. In fact, the clitoris is not a button, it is like an iceberg. Most of the organ is internal. This slide is an anatomical example of the penis and the clitoris side by side. Now, we ' ve all been taught that male bodies and female bodies are opposites : the male body sort of sticks out, the female body is solely internal. Well, in fact, there ' re so many similarities between the penis and the clitoris. So, if you ' ll see, both the glans and - The glans of the penis and the glans of the clitoris - both organs have a glans. There are 3000 nerves in the glans of the penis. There are 8000 nerves in the glans of the clitoris. Both organs have a corpus cavernosum. Both organs have crura, like two little legs or wings. Both organs have bulbs of erectile tissue. Both organs get erect. The penis is outside of the body mostly, and the clitoris is inside the body mostly. That ' s the biggest difference. In fact, they ' re very similar. Actually, fetuses have the same tissue, and in boys it develops as a penis, in girls it develops as a clitoris. Some people have small penises, some people have very large clitorises. If one has a clitoris and takes testosterone, their clitoris can expand. What I ' m saying is that these organs are actually quite similar. And, while we are different, while we are unique as men and women, our differences are not a sign of opposition. In fact, we ' re related to each other, we ' re connected. And that ' s an exciting fact. With Cliteracy, I started with language. Language has been a place that so much of sort of the division of men and women and the subjugation of women has been entrenched with the language itself. ' Vagina ' - the single most misused word in the English language. This is one of the laws of Cliteracy. It ' s intentionally hyperbolic. But unfortunately, it ' s more true than I wish it was. ’ Vagina ’ is a Latin word. It means ’ sword holder ’. Vagina, medically, technically, only includes the opening. This term is used almost universally in doctor ’ s offices. It ’ s also used in feminism to sort of advocate. But it ’ s a term that ignores the clitoris, which is the female sexual organ. And secondly, it reduces the female body to being a receptacle, a sword holder. If you want to use a term that addresses all the female genitals, both the reproductive and the sexual parts of it, the word is ’ vulva ’. This is a word that almost no one uses, but this is the word if you want to talk about female genitals - vulva. If you want to talk about pleasure, ’ clitoris ’ is the term. The clitoris is both internal and external. So, when the clitoris is engaged, internal stimulation feels great. If the clitoris is not engaged, it can feel not great, or it can feel painful. It ’ s all about what ’ s happening with the clitoris. If it was about the vagina, there would be so many nerves that childbirth would be impossible. There ’ re very few nerves in the vagina. All of the nerves inside are from the internal clitoris, which gets stimulated both from external and internal portions. With Cliteracy I felt that - Yes, language has been this way of restricting and confining the female body, but if language can do these thing, it can also liberate, it can also be expansive. It ’ s also an opportunity to come together. And so I sought to use language as a way to shift the discourse and create new space and more alignment. With Cliteracy, I first began with the term, and then a definition of the term, and then an eye chart. The aesthetics of this project were also extremely important and intentional. I avoided any kind of pink and purple ; I didn ’ t use flowers ; I didn ’ t use any fabric or yarn, anything soft and fluffy. I also didn ’ t make any small works that you could hold in your hand or eat off of like plates. I also intentionally avoided any kind of sexual imagery, any kind of graphic, close-up, literal depictions. I think that everywhere we see the exposed female body, and yet we don ’ t know the actual female sexual organ, the clitoris. So showing it is not the point, right? Understanding it is the point. Literacy of it, knowledge of it is the point. [The hole is not the whole] The hole is not the whole. With Cliteracy I had a lot of **fun** with word-play. There was just so much material to work with and so much to talk about. I also am making the radical claim with Cliteracy that we can ’ t truly be free if our bodies are assailed. We can ’ t truly be fully enjoying our democracy when half of the population can ’ t speak about their own body, is censored when they say the words because these words are taboo, or are regularly having sex without orgasms. And we don ’ t talk about this. I ’ m making the radical claim that freedom in society can also be measured by the distribution of orgasms. This could be one indicator that we use when we look at education, access to health care, economics. We could also say : How are orgasms being distributed? That tells us something about a society. There is no lack. Truly. Freud invented the paradigm of the phallus versus the lack. He said that men have the phallus, they have the penis, they have agency ; women have a lack, they have a vagina, they have a void. In fact, Freud was wrong, the vagina is not the female sexual organ, it is the clitoris. There is no lack, none of us have lack, none of us are lacking, we ’ re all whole. And none of us need to be depicted in terms solely of a void. [Girls are taught it ’ s normal for sex to hurt. Phallusy.] In many ways, all of us have had a psychological clitoridectomy because the clitoris is never taught. In sex education it is taught that boys are both sexual and reproductive, boys have erections, boys have wet dreams, boys ejaculate, and then the semen fertilizes the egg. Girls, we ’ re taught, have reproductive organs, they menstruate, menstruation is painful. Girls should not get pregnant if they don ’ t mean to. Girls should not get sexually transmitted diseases. We never learn about the clitoris. We never learn that girls have desire, that this is natural, that girls have sexual dreams, that girls have fantasies. So already, as a culture, I would say we all have clitoridectomies. ’ Clitoris, say my name, say my name. ’ I really enjoyed using word-play and putting the clitoris into popular culture and song lyrics. So much of popular culture, music - the female genitals are kind of riffed on but almost always in a negative way. If you want to humiliate a man, call him a word for female genitals, right? But this is also an opportunity, you can just pop in the clit, and suddenly the whole song changes ; it ’ s very powerful. So, this is a Destiny ’ s Child song, there ’ re many more in the Laws of Cliteracy like ’ **Ain** ’ t no half-stepping to the clit, ’ ’ Sleeping on the clit? That shit cray ’, which it is. Here ’ s an example from looking below at the 100 Laws of Cliteracy. So this work spans 13 **feet** long by 10 **feet** tall. It dwarfs anyone ’ s body. When I first showed this work, I had no idea what the response would be. Of course, I was hoping it would be positive, but I didn ’ t know. And I have to say, I was overwhelmed with the way audiences responded. They really wanted to have this conversation. They would stay with the work for 15 minutes. I would come back, they would still be there, they would have friends with them. And I felt like this work was needed. I was invited to speak about it a lot, and I was also contacted privately. People shared secrets with me, people told me for the first time they didn ’ t feel ashamed about their body, or they went home with this knowledge and are having a great time with their girlfriend, and wanted to thank me for that. So this was extremely gratifying. I think for me, what made me feel like I was onto something was that such a diverse group of people supported this project : men and women, young and old, religious and secular, queer and straight. So many people came together to support this project. And I was contacted by people from as far as New Zealand, Egypt, Brazil, saying : ’ How can we help you with this project? Can you translate this project into Arabic or into Portuguese, like, we need this here in our country. ’ I wanted Cliteracy to go everywhere, but I didn ’ t know how to do that yet, and I ’ m still figuring that out. But I know that it ’ s needed. And I know that it can ’ t just stay within the walls of the art world. I don ’ t consider myself a street artist, but I began making work on the street for the same reason that I think a lot of artist do - I wanted to communicate more broadly. This is an example from a documentary of just me putting up some pieces. These are prints that I made on a news print, then I ’ m putting them up with wheat paste just like old posters were put up. I ’ m just doing this in Brooklyn. Some people like to do street art at the night time, I do it in the day because I feel like it ’ s a little safer, the anonymity of the crowd in New York. So far so good. And one of the cool things about doing street art is that people will comment on your work. Sometimes they cross it out or destroy it, but other times they put their art on top of it. And in this case that happened. Again, like when I showed the work in the gallery, on the street there was a big response, people would photograph it, they would post it on Twitter and Instagram. And it felt like I was talking about something that needed to be talked about, that the project was needed and people were grateful to find it and to keep pushing it out there more. Doing the street work emboldened me to take on even crazier ideas that I never thought I would find myself doing. And I actually, together with an artist named Clit Eastwood, or Ken Thomas, held the world ’ s first-ever clit rodeo last summer. We created a rideable golden clit. And we held the first clit rodeo. And, you know, there were two rules to the rodeo. Basically, one : respect the clit. Respect it. Two : have **fun**. Those were the rules. We had so many riders who wanted to ride, more than we could host for our event. But the riders were judged on three categories : dexterity, style and generosity. And they were really good riders, I have to say. I was worried that it might get a little boring because as you saw on the previous slide, it was just a spring. But people were reading erotica to the clit, someone did a striptease for the clit, someone surfed the clit, someone offered a cigarette to the clit, and then, like - There was a couple where the woman was nine-months-pregnant ; she was riding, and her husband was in the background as her backup dancer, dancing around... So, it was way better than I ever expected. I just was thrilled because the clit was the star of the show. Finally! I ’ ve always wanted Cliteracy to be in the public space, to be at large scale, to be seen over time, not to have to be hidden away or be a secret. And I had the opportunity last fall. Together with Center and Santa Fe we put up a 35 **feet** billboard of Cliteracy, or 11 metre billboard. The text says : ’ Democracy without cliteracy - phallusy ’. And I was thrilled to do this, especially because where it was on this highway is travelled by such a broad range of people, from long-haul truck drivers to art collectors and everyone in between. The billboard company was a little bit less psyched about how much feedback they got about the billboard, but I thought this was great. A lot of people were like : ’ What are you selling? I don ’ t understand. ’ I actually got a call that I ’ ll never forget from a mom, saying : ’ I have to drive this route with my son every day, and I don ’ t know what to tell him. ’ But I was thrilled because she ’ s going to talk to him about cliteracy, and this is something that he needs to know about. Cliteracy needs more than text though, and I always knew that I wanted to explore the form as well. None of us know this form, right? I didn ’ t know this form. So I set about making the world ’ s first anatomically correct sculpture of the clitoris. And this was something that was actually quite hard to do because there are so few accurate representations of the anatomy. And when you find these very few **drawings** or **scans**, they contradict each other, they don ’ t make sense. So, it was actually not that easy to do, but I set about making this form. With the form I wanted to not only explore the anatomy and get it accurate, but I also wanted to show the gesture of beauty of this organ and the gracefulness of it. Here is the first sculpture that I know of of the anatomically correct clitoris. It ’ s six **feet** tall and five **feet** wide. And I wanted to create an iconic form of this unseen organ half of us have, all of us were born through the body of someone that has a clitoris. Everyone in this room was born through the body of someone that has a clitoris. So all of us have been touched by the clitoris. This is universal and yet we don ’ t know about it. So I wanted to create an iconic form that ’ s memorable, that puts this into our consciousness. And I hope that finally this form would be treated with honour and respect and not be treated as obscene. I think it ’ s a beautiful form, and I didn ’ t know it, but once I saw it, it started to feel familiar in this way. And I started to see it around in the natural world, in plants. I also saw it in engravings, on architectural sites. I saw it in weavings of oriental tapestries. I started to see it around. And that was very exciting. And the form is interesting not strictly as a sculpture, but in patterns. There ’ s something very exciting about looking into the power of the small, the power of multiplicity. Instead of creating just this singular superior object, what about putting all these tiny beautiful forms that together form a baby-clit army. The one on the right, I call it ’ Fleur-de-clits ’, and the one on the left was later used in the intervention at the Whitney Museum. So here ’ s a sort of subversive clit army coming together to make the ’ Fleur-de-clit ’, which is this beautiful pattern. But unfortunately, if some people knew what it was, it wouldn ’ t be allowed to be. And that ’ s sort of the rub of it. Here is an example of more clit forms and patterns that I created. This is a clit damask pattern with clit forms burnished onto wood. On the left is a new sculpture. So, this is the first sketch of an invisible clit sculpture. It ’ s the same digital form that I used to make the gold sculpture I just showed to you. But this exists in the negative space. So I used the **laser** to actually cut out the form from clear plexiglass. And so this invisible sculpture addresses the fact that this is omnipresent, and yet it ’ s negated, it ’ s invisible, it ’ s not allowed to be spoken of. I also continued with this idea of negation and using the **laser** to burn away with laser-cut works on paper. And I developed a brand new technology. You might not have heard about it, but it ’ s very cutting edge. I ’ ll try to explain. It ’ s called ’ clitglass ’, and the way that it works - anyone can wear it ; anyone who wants neutral vision can wear a clitglass. So you put on the clitglass, and you look through the perspective of the clit. And the clit refracts any kind of phallocentricacy that ’ s coming back at you. And so you obtain neutral vision, or what I like to call ’ normal ’ vision. Now, you can use clitglass at the Whitney Museum, or you can use it at work, in front of the TV, even at a family reunion. This is an example of looking through this cutting edge technology. So those forms that I showed earlier, I also played a game at my intervention at the Whitney Museum, called ’ put a clit on it ’, or - ’ clit-dazzle the Whitney ’. So basically I handed out these unknown clit forms, and I said : ’ Put the clit wherever you think it needs to go - put it as a subject in art history, put into the designs, put it on the American flag. Whole country has a problem with illcliteracy. Help America out. Just take the clit where it needs to go. ’ So you can see on the lower left, that ’ s a Clitchtenstein. On the lower right, Clitsper Johns. This is the family at the Whitney Museum during the intervention, and the boy on the left who looks to be about 11 years old, at one point asks his mom - they ’ d been wearing the clitglasses for about 15 minutes, having a great time. He was like, ’ Mom, what ’ s a clit? ’ And she said : ’ Oh, it ’ s a really sensitive part of the woman ’ s body. ’ And he was like, ’ Okay, cool. ’ And I was thrilled because one : he felt comfortable asking the question ; two : his mom was supportive and answered the question. And it was totally normal - nothing obscene, nothing secret, no one had to be dragged out of the room, no one had to be ashamed. And that ’ s what I ’ m hoping that Cliteracy can continue to do. Overwhelmingly, the response to Cliteracy has been positive. So many people have supported the project and wanted to help with it. And there ’ ve been a few institutions who have courageously started showing it. But there ’ s so much more that needs to be done. My dream is to radically change the way that we think about bodies so that everyone ’ s body is respected. I want to do this by creating large scale permanent public sculptures that exist for thousands of years. I want to work with **metals** and stone so that these forms don ’ t disappear in future generations, and we don ’ t have to have this conversation again and again. Democracy without cliteracy is a phallusy. I want cliteracy to be taught in schools so that no child has an unnameable part of their body. The clit should be a starring role in any bedroom that it ’ s in. And it shouldn ’ t be censored in the Parliament. So, in closing, I want to ask you to see the clit. See it everywhere. Don ’ t stop seeing it. And if you need help, you can borrow this pair of glasses from me. And don ’ t just stop with seeing it. Say it. Say its name. Thank you.

Text Sample 709: POS Dim 5 – 2014 – Score 6.00 – 2014/t001635.txt

Today I ’ m going to speak to you about the last 30 years of architectural history. That ’ s a lot to pack into 18 minutes. It ’ s a complex topic, so we ’ re just going to dive right in at a complex place : New Jersey. Because 30 years ago, I ’ m from Jersey, and I was six, and I lived there in my parents ’ house in a town called Livingston, and this was my childhood bedroom. Around the corner from my bedroom was the bathroom that I used to share with my sister. And in between my bedroom and the bathroom was a balcony that overlooked the family room. And that ’ s where everyone would hang out and watch TV, so that every time that I walked from my bedroom to the bathroom, everyone would see me, and every time I took a shower and would come back in a towel, everyone would see me. And I looked like this. I was awkward, insecure, and I hated it. I hated that walk, I hated that balcony, I hated that room, and I hated that house. And that ' s architecture. Done. That feeling, those emotions that I felt, that ' s the power of architecture, because architecture is not about math and it ' s not about zoning, it ' s about those visceral, emotional connections that we feel to the places that we occupy. And it ' s no surprise that we feel that way, because according to the EPA, Americans spend 90 percent of their time indoors. That ' s 90 percent of our time surrounded by architecture. That ' s huge. That means that architecture is shaping us in ways that we didn ' t even realize. That makes us a little bit gullible and very, very predictable. It means that when I show you a building like this, I know what you think : You think"power"and"stability"and"democracy."And I know you think that because it ' s based on a building that was build 2, 500 years ago by the Greeks. This is a trick. This is a trigger that **architects** use to get you to create an emotional connection to the forms that we build our buildings out of. It ' s a predictable emotional connection, and we ' ve been using this trick for a long, long time. We used it [200] years ago to build banks. We used it in the 19th century to build art **museums**. And in the 20th century in America, we used it to build houses. And look at these solid, stable little soldiers facing the ocean and keeping away the elements. This is really, really useful, because building things is terrifying. It ' s expensive, it takes a long time, and it ' s very complicated. And the people that build things — developers and governments — they ' re naturally afraid of innovation, and they ' d rather just use those forms that they know you ' ll respond to. That ' s how we end up with buildings like this. This is a nice building. This is the Livingston Public Library that was completed in 2004 in my hometown, and, you know, it ' s got a dome and it ' s got this round thing and columns, red brick, and you can kind of guess what Livingston is trying to say with this building : children, property values and history. But it doesn ' t have much to do with what a library actually does today. That same year, in 2004, on the other side of the country, another library was completed, and it looks like this. It ' s in Seattle. This library is about how we consume media in a digital age. It ' s about a new kind of public amenity for the city, a place to gather and read and share. So how is it possible that in the same year, in the same country, two buildings, both called libraries, look so completely different? And the answer is that architecture works on the principle of a pendulum. On the one side is innovation, and **architects** are constantly pushing, pushing for new technologies, new typologies, new solutions for the way that we live today. And we push and we push and we push until we completely alienate all of you. We wear all black, we get very depressed, you think we ' re adorable, we ' re dead inside because we ' ve got no choice. We have to go to the other side and reengage those symbols that we know you love. So we do that, and you ' re happy, we feel like sellouts, so we start experimenting again and we push the pendulum back and back and forth and back and forth we ' ve gone for the last 300 years, and certainly for the last 30 years. Okay, 30 years ago we were coming out of the ' 70s. **Architects** had been busy experimenting with something called brutalism. It ' s about concrete. You can guess this. Small windows, dehumanizing scale. This is really tough stuff. So as we get closer to the ' 80s, we start to reengage those symbols. We push the pendulum back into the other direction. We take these forms that we know you love and we update them. We add neon and we add pastels and we use new materials. And you love it. And we can ' t give you enough of it. We take Chippendale armoires and we turned those into skyscrapers, and skyscrapers can be medieval castles made out of glass. Forms got big, forms got bold and colorful. Dwarves became columns. Swans grew to the size of buildings. It was crazy. But it ’ s the ’ 80s, it ’ s cool. We ’ re all hanging out in malls and we ’ re all moving to the suburbs, and out there, out in the suburbs, we can create our own architectural fantasies. And those fantasies, they can be Mediterranean or French or Italian. Possibly with endless breadsticks. This is the thing about **postmodernism**. This is the thing about symbols. They ’ re easy, they ’ re cheap, because instead of making places, we ’ re making memories of places. Because I know, and I know all of you know, this isn ’ t Tuscany. This is Ohio. So **architects** get frustrated, and we start pushing the pendulum back into the other direction. In the late ’ 80s and early ’ 90s, we start experimenting with something called deconstructivism. We throw out historical symbols, we rely on new, computer-aided design techniques, and we come up with new compositions, forms crashing into forms. This is academic and heady stuff, it ’ s super unpopular, we totally alienate you. Ordinarily, the pendulum would just swing back into the other direction. And then, something amazing happened. In 1997, this building opened. This is the Guggenheim Bilbao, by Frank Gehry. And this building fundamentally changes the world ’ s relationship to architecture. Paul Goldberger said that Bilbao was one of those rare moments when critics, academics, and the general public were completely united around a building. The New York Times called this building a miracle. Tourism in Bilbao increased 2, 500 percent after this building was completed. So all of a sudden, everybody wants one of these buildings : L. A., Seattle, Chicago, New York, Cleveland, Springfield. Everybody wants one, and Gehry is everywhere. He is our very first starchitect. Now, how is it possible that these forms — they ’ re wild and radical — how is it possible that they become so ubiquitous throughout the world? And it happened because media so successfully galvanized around them that they quickly taught us that these forms mean culture and tourism. We created an emotional reaction to these forms. So did every mayor in the world. So every mayor knew that if they had these forms, they had culture and tourism. This phenomenon at the turn of the new millennium happened to a few other starchitects. It happened to Zaha and it happened to Libeskind, and what happened to these elite few **architects** at the turn of the new millennium could actually start to happen to the entire field of architecture, as digital media starts to increase the speed with which we consume information. Because think about how you consume architecture. A thousand years ago, you would have had to have walked to the village next door to see a building. Transportation speeds up : You can take a boat, you can take a plane, you can be a tourist. Technology speeds up : You can see it in a newspaper, on TV, until finally, we are all architectural photographers, and the building has become disembodied from the site. Architecture is everywhere now, and that means that the speed of communication has finally caught up to the speed of architecture. Because architecture actually moves quite quickly. It doesn ’ t take long to think about a building. It takes a long time to build a building, three or four years, and in the interim, an **architect** will design two or eight or a hundred other buildings before they know if that building that they designed four years ago was a success or not. That ’ s because there ’ s never been a good feedback loop in architecture. That ’ s how we end up with buildings like this. Brutalism wasn ’ t a two-year movement, it was a 20-year movement. For 20 years, we were producing buildings like this because we had no idea how much you hated it. It ’ s never going to happen again, I think, because we are living on the verge of the greatest revolution in architecture since the invention of concrete, of steel, or of the elevator, and it ’ s a media revolution. So my theory is that when you apply media to this pendulum, it starts swinging faster and faster, until it ’ s at both extremes nearly simultaneously, and that effectively blurs the difference between innovation and symbol, between us, the **architects**, and you, the public. Now we can make nearly instantaneous, emotionally charged symbols out of something that ' s brand new. Let me show you how this plays out in a project that my firm recently completed. We were hired to replace this building, which burned down. This is the center of a town called the Pines in Fire Island in New York State. It ' s a vacation community. We proposed a building that was audacious, that was different than any of the forms that the community was used to, and we were scared and our client was scared and the community was scared, so we created a series of photorealistic renderings that we put onto Facebook and we put onto Instagram, and we let people start to do what they do : share it, comment, like it, hate it. But that meant that two years before the building was complete, it was already a part of the community, so that when the renderings looked exactly like the finished product, there were no surprises. This building was already a part of this community, and then that first summer, when people started arriving and sharing the building on social media, the building ceased to be just an edifice and it became media, because these, these are not just **pictures** of a building, they ' re your **pictures** of a building. And as you use them to tell your story, they become part of your personal narrative, and what you ' re doing is you ' re short-circuiting all of our collective memory, and you ' re making these charged symbols for us to understand. That means we don ' t need the Greeks anymore to tell us what to think about architecture. We can tell each other what we think about architecture, because digital media hasn ' t just changed the relationship between all of us, it ' s changed the relationship between us and buildings. Think for a second about those librarians back in Livingston. If that building was going to be built today, the first thing they would do is go online and search"new libraries."They would be bombarded by examples of experimentation, of innovation, of pushing at the envelope of what a library can be. That ' s ammunition. That ' s ammunition that they can take with them to the mayor of Livingston, to the people of Livingston, and say, there ' s no one answer to what a library is today. Let ' s be a part of this. This abundance of experimentation gives them the freedom to run their own experiment. Everything is different now. **Architects** are no longer these mysterious creatures that use big words and complicated **drawings**, and you aren ' t the hapless public, the consumer that won ' t accept anything that they haven ' t seen anymore. **Architects** can hear you, and you ' re not intimidated by architecture. That means that that pendulum swinging back and forth from style to style, from movement to movement, is irrelevant. We can actually move forward and find relevant solutions to the problems that our society faces. This is the end of architectural history, and it means that the buildings of tomorrow are going to look a lot different than the buildings of today. It means that a public space in the ancient city of Seville can be unique and tailored to the way that a modern city works. It means that a stadium in Brooklyn can be a stadium in Brooklyn, not some red-brick historical pastiche of what we think a stadium ought to be. It means that robots are going to build our buildings, because we ' re finally ready for the forms that they ' re going to produce. And it means that buildings will twist to the whims of nature instead of the other way around. It means that a parking garage in Miami Beach, Florida, can also be a place for sports and for yoga and you can even get married there late at night. It means that three **architects** can dream about swimming in the East River of New York, and then raise nearly half a million dollars from a community that gathered around their cause, no one client anymore. It means that no building is too small for innovation, like this little reindeer pavilion that ' s as muscly and sinewy as the animals it ' s designed to observe. And it means that a building doesn ' t have to be beautiful to be lovable, like this ugly little building in Spain, where the **architects** dug a hole, packed it with hay, and then poured concrete around it, and when the concrete dried, they invited someone to come and **clean** that hay out so that all that ’ s left when it ’ s done is this hideous little room that ’ s filled with the imprints and scratches of how that place was made, and that becomes the most sublime place to watch a Spanish sunset. Because it doesn ’ t matter if a cow builds our buildings or a robot builds our buildings. It doesn ’ t matter how we build, it matters what we build. **Architects** already know how to make buildings that are greener and smarter and friendlier. We ’ ve just been waiting for all of you to want them. And finally, we ’ re not on opposite sides anymore. Find an **architect**, hire an **architect**, work with us to design better buildings, better cities, and a better world, because the stakes are high. Buildings don ’ t just reflect our society, they shape our society down to the smallest spaces : the local libraries, the homes where we raise our children, and the walk that they take from the bedroom to the bathroom. Thank you.

Text Sample 710: POS Dim 5 – 2022 – Score 8.00 – 2022/t003710.txt

Welcome to the gates of **hell**. Now depending on your frame of mind, that is either a bizarrely morbid or entirely appropriate way to start a talk about climate action in the year 2022. Behind me is a **picture** from the **Hell** ’ s Gate National Park in the town of Naivasha, in the Great Rift Valley in my home country, Kenya. Now its name may not scream”tourist trap,”but believe me, it is a beautiful part of the world and you should all try and visit sometime. But more importantly, it could play – It has the potential to play a **crucial** role in the fight against global climate catastrophe. The most recent IPCC reports are clear. Humanity has left cutting emissions too late. Any realistic path to avoiding unacceptable levels of warming now requires us to not only drastically cut emissions, at least halving them by 2030, but also undertake an equally massive effort to remove greenhouse **gases** from the atmosphere at an accelerating rate. Now, let ’ s be clear. Greenhouse **gas** removal is not and cannot be an excuse for continuing to emit. Just as installing seat belts and airbags is not an excuse for deliberately ramming your car into a wall. Indeed, current estimates suggest that even with drastic emissions reductions, the world will need to be removing between five and 16 billion tons of carbon dioxide from the atmosphere every single year by 2050. Now to give you a sense of the scale of that, the low end of that range, five billion tons, that ’ s bigger than the size of the global petroleum industry in 2020. So let ’ s not kid ourselves that carbon removal, at anywhere close to the scale that we will need in order to survive, is some sort of easy way out. It is going to be damn difficult to do. So how do we do it? Well, the first and most familiar measures would be interventions such as reforestation and landscape restoration. Essentially giving Mother Nature the time and space to heal herself. In addition, we can increase the amount of carbon held in our soils through the widespread application of biochar and enhanced weathering of chemically suitable rocks. We estimate that in Africa alone, something like 100 million to 680 million additional tons of carbon dioxide could be drawn from the atmosphere using these types of methods. However, they do require a lot of land, a lot of water and a lot of other natural resources that may limit the extent to which we can scale them. Moreover, they are subject to some of the feedback loops from the climate change that we are already experiencing, such as more frequent and intense wildfires. And all of that means we are going to need to supplement them with technologies that accelerate and amplify natural processes to remove carbon dioxide from the atmosphere. Enter the members of my new favorite boy band. DAC, BECCS and BiCRS. These are a set of engineered approaches that use physical, chemical and biological processes to gather and concentrate carbon dioxide from the atmosphere before safely sequestering it, usually underground. As more people run the climate math, you ' re seeing growing levels of interest and investment in these technologies, with billions of dollars already being committed to early pilots and installations in various parts of the world, particularly in Europe and North America. But the reality is they have a very long way to go. To date, engineered removals around the world have accounted for something like 100, 000 tons of carbon dioxide removed total. To get to the multi-billion-ton scale we ' re going to need by 2050 is going to take a truly epic process of exponential scaling. Probably means we need to get to something – If we want to have a realistic shot at it, we need to get to something like 100 million tons per year by 2030. For those of you running the calculators, that ' s a thousand-fold increase in less than a decade. And guess what? We will have to continue that insane rate of growth for another two decades after that. And here ' s the really bad news. Anything close to that level of scaling of this industry in the places where it ' s currently being piloted presents some really difficult climate action trade offs. For that, let me take the example of DAC or direct air capture. The best known DAC facility in the world is in Iceland. It ' s the Orca plant in Iceland, it was inaugurated last year, 2021. It uses plentiful green geothermal energy to capture carbon dioxide, dissolve it in water and inject it into porous basalt deep underground, where it chemically reacts to create a stable solid that can stay there for centuries. It takes the equivalent of between two and three megawatt hours of energy to take a single ton of carbon dioxide today and render it in that way. To get to the hundred million number in 2030, on that track, would entail something like 200 to 300 terawatt hours of electricity. Again, that ' s about half the electricity usage of Germany. And all of that power would need to be renewable, otherwise, we would be taking two steps forward and one and a half steps back. Now it ' s reasonable to expect and assume that we are going to see substantial improvements in energy efficiency of these technologies as we deploy them and learn to use them better. However, keep in mind that probably the most urgent thing we can do to slow climate change right now is stop current emissions. And so scaling these technologies in places where we do have fossil fuel energy emissions that we could be curtailing does not make sense. Essentially, every unit of renewable energy that we are bringing on stream in places like North America and Europe should be going towards displacing and retiring existing fossil fuel capacity. And so the world is kind of stuck. Right? We need to scale this technology. We need to get DAC down the cost **curve** and up the efficiency **curve** urgently. Our lives literally depend on it. But at the same time, we cannot do it except at the expense of other equally urgent climate imperatives. So we need places in the world that somehow have three characteristics. A, they need to have the right geophysical conditions. You know, plenty of porous basalt rock in a geothermally active zone is one such example. Two, they need to have plenty of renewable energy potential. And three, they need to have no current proximate emissions that that renewable energy could be used to displace. And that brings us back to **Hell** ' s Gate National Park. Here ' s another view of the **park** from an angle that may explain its potential. That is one of the power plants that together constitute the Olkaria Geothermal Energy Plant, which provides about a third of Kenya ' s electricity. That ' s right. My home country not only has 92 percent renewable electricity being dispatched on its grids today, but its largest single-energy installation is seamlessly integrated into an honest-to-goodness national **park**. Literally, between the different plants you can see herds of zebra peacefully grazing all times of the day. It ' s amazing. Now at just under 1, 000 megawatts, Olkaria is nothing to sneeze at. It ' s one of the largest geothermal electricity installations in the world. But it ' s barely scratching the surface of the potential in Kenya. There ' s 10 gigawatts of proven, high-quality geothermal resource in the country, widely recognized, ready to be tapped. And in addition, Kenya is endowed with excellent wind and solar resources that have also barely been exploited. We are on the equator, after all. We estimate conservatively that there ' s about 50 gigawatts of potential deployable renewable energy in Kenya that can be readily accessed with the right level of investment. And yet, Kenya remains an energy-poor country where, despite a lot of progress in recent years, more than a quarter of the population still does not have access to basic electricity. And those that do often pay prices that are almost three times as much as much as their counterparts in countries like India and China. Now you might be sitting there wondering,"Well, all right, James, if this is true, if Kenya has all of this renewable energy potential and all of these people in need of energy, well, before we have this whole conversation about fancy climate tech, shouldn ' t we first have a TED Talk about affordable energy access?"And you would be right. Were it not for a particularly cruel paradox of energy economics in countries like Kenya. You see, part of the reason why energy is so expensive in the country is those consumers who are on the grid have to pay for capacity that is not currently being used. There ' s something like 1, 000 megawatt hours every day that goes begging because there isn ' t sufficient industrial demand. At the same time, those very same high energy prices make the country unattractive and uncompetitive for manufacturers and other users of energy looking for places to site their industries. So to get this straight, the reason why the average Kenyan cannot get affordable access to **clean**, renewable energy despite all of this natural bounty, is this tremendously frustrating feedback loop where firstly, we would have all of that energy if someone invested in renewable power plants. People would invest in those power plants if there was a lot of available industry to use the energy. Available industry would come if energy costs weren ' t so high. And energy costs wouldn ' t be so high if there was enough demand. It ' s enough to drive you crazy. But it also points the way to a potential huge triple opportunity. Firstly, introducing DAC and other energy-hungry climate technology into places like the Rift Valley would give them the space and capacity they need to really scale to planetary levels. With no competition, with none of the trade-offs they would face in other parts of the world. At the same time, having that energy-hungry anchor industry available suddenly creates the basis on which people are willing to invest in expanding the country ' s renewable energy potential. Actually creating the business case for providing tens of millions of people with the productive energy they need to improve the quality of their lives. And thirdly, introducing these new and exciting technologies on the continent with the world ' s youngest and fastest-growing workforce could potentially activate their imaginations and their energies towards becoming climate innovators and solution builders themselves, basically building an army from the world ' s largest workforce to solve the world ' s biggest problem. I call it the"Great Carbon Valley."And it ' s just one of the ways in which Africa, as the continent, which, per capita, is the closest to net-zero and has contributed the least to climate change, can play a role in helping the planet avert climate disaster. But in addition, it can do more and be the first continent to go substantially net-negative. We ' re used to thinking about the continent in terms of its forests, its peatlands, its grasslands, its wetlands that need to be preserved. And we should definitely continue to invest in the Indigenous communities, the smallholder farmers and the local innovators who are protecting and expanding natural carbon sinks. But that should not blind us to the fact that Africa also provides an **ideal** potential home for scaling the latest and most ambitious of climate technologies. Whichever of these narratives most speaks to you, one thing should be clear. We need to shake the old, tired idea that Africa is a poor, hapless, helpless climate change victim. Instead, Africa and its people have the potential. They can, and they should, be the world ’ s climate vanguard. Thank you.

Text Sample 711: POS Dim 5 – 2022 – Score 7.00 – 2022/t003702.txt
Recently, I spent several days exploring Kashgar, a city in Xinjiang, northwest China. I got to wander the streets of the old town and visit the bazaar and several mosques and take in the sights. I ’ ve never been to Kashgar personally, but through the YouTube videos and Instagram posts of tourists, I was able to experience the city at a key moment in time : October 2017, just as the mass detention campaign in the region was gathering pace. These videos could help us investigate the visual signs of the crackdown. The checkpoints at each intersection with their **metal** detectors, ID checks and iris **scans**, the CCTV cameras which pervade the city and the riot police on every corner. Over the past decade, online and open-source investigations have taken off in the fields of journalism and human rights monitoring, using photographs, videos and the digital traces we leave behind as we use the internet to conduct investigations. Social media data is combined with tools like satellite imagery and 3D modelling, as well as more traditional journalistic techniques like interviews and searches of government documents. It ’ s also brought new kinds of people to journalism. Software developers, animators, archaeologists, or, like me, an **architect**. I got involved in investigating Xinjiang in the summer of 2018 when I met Megha Rajagopalan, an American journalist who had been working in China for several years. Over the past few years, China has been carrying out a campaign of oppression in Xinjiang against Turkic Muslims, including the largest group, the Uyghurs. It ’ s part of a campaign of forcible assimilation, and several nations have described it as a genocide. It ’ s estimated that over a million people have been disappeared into detention camps. And while the Chinese government claims that these are part of a benign program of reeducation, dozens of former detainees describe being tortured and abused and women being forcibly sterilized. And yet, for a long time, we lacked information about what was happening in Xinjiang, because the Chinese government controls the internet tightly and restricts journalists ’ work in the region. Journalists would be followed or detained, and the authorities occasionally even went so far as to set up fake roadworks or stage car crashes to prevent access to certain roads. Local people who did speak to journalists face the risk of being sent to a detention camp for doing so. Megha had been the first journalist to visit one of the camps. But shortly after publishing her article, the Chinese authorities declined to renew her visa, and she had to leave. Other journalists had managed to visit a handful of the camps, but this still represented a fraction of what we believed was out there, and no one knew where the others were. But Megha was keen to find the rest. She just needed to find a way to work effectively from outside China. Another challenge was that Xinjiang is huge. It ’ s four times the size of California, and that made it difficult to look for a network of camps that was spread across the region. Satellite imagery could help to solve both of those problems. But more importantly, satellite imagery was a source of information that the Chinese government couldn ’ t control because the satellites and the imagery they produce was owned by US and European organizations. But that still left us with the question of where in that

huge amount of satellite imagery to look. And then I heard about something strange that was happening in Baidu Total View, which is the Chinese equivalent of Google Street View. Photographer Jonathan Browning had discovered that buildings and facilities like industrial estates were being photoshopped out of ground level imagery, often very clumsily. Yeah, it ' s bizarre, right? At the time, it wasn ' t clear why this was happening, but I realized that if industrial estates in eastern China were being obscured, then probably the same thing was happening with detention camps in Xinjiang. And I went to look at the imagery there to see what I could find. There were a handful of camps which had been visited by journalists. And so I went to those locations in Baidu to see what the platform showed. There was no street level imagery. But as I zoomed in on the satellite images, this weird thing happened. A light gray square suddenly appeared above the location of the camp and then disappeared just as quickly as I zoomed in further. It was a bit like the map wasn ' t loading properly, but then I zoomed out and in again only for the same thing to happen. I realized it couldn ' t be a problem with the map loading because the tiles would have been in the browser ' s cache. And when I found the same thing happening at the other locations we knew to be camps, I realized that we had a technique we could use to find the rest of the network. It ' s quite rare for maps and satellite images to have these blank spots because blank areas tend to draw attention to themselves. But here we got lucky. Obscuring the camps had inadvertently revealed all of their locations. We worked with developer Christo Buschek, who specializes in documenting human rights issues and building tools for open-source researchers to map the masked-tile locations. We had to work quickly and secretively to map the masked tiles before anyone found out what we were doing and removed them because our investigation relied on access to that information. The idea was that we could go and look at the masked-tile locations and then look at that same location in other unaltered satellite imagery and see what was there. And this is what we saw. This is a former high school that became Kashgar Vocational Skills Education and Training Center. Zooming in on the satellite imagery, we can see the barbed wire in the courtyards that creates exercise pens for the detainees adjacent to the buildings. In other images, we can even see people, all wearing red uniforms, lined up in the courtyard. These features could help us decide whether a location was a camp or not. As we investigated further, we realized that the camp ' s program had evolved away from the early days of makeshift camps in former schools and hospitals, and had become more permanent, that the camps were now larger, higher-security and purpose-built. This is the largest camp that we know of. It ' s in Dabancheng. The complex is two miles long, and it would cover a quarter of New York ' s Central Park. In the satellite images, we can see the thick perimeter walls, the guard towers and these blueish buildings, which we believe to be factories. We estimate that this complex can hold over 40, 000 people without overcrowding. We corroborated these locations using government documents, many of which mention the camps address, the few media reports which did exist on the camps and our own interviews with former detainees who had managed to leave Xinjiang and are now living in Kazakhstan, Turkey or Europe. In total, we found 348 locations bearing the hallmarks of camps and prisons. And we believe that this is close to being the full network. We estimate that these facilities have been built to hold more than a million people. That ' s enough space to detain one in every 25 of Xinjiang ' s residents. And that doesn ' t take into account the overcrowding that so many former detainees have described. So that number could be even higher. And then one morning, a few months after we had published our map, I woke up to a series of messages about a YouTube video that was doing the rounds on Chinese social media. A Chinese vlogger, who goes by the name Guanguan, had taken our map and traveled to Xinjiang. In his video, we see him driving down a main road past a compound with barbed wire on top of the perimeter wall and bars on the windows. Next, he pretends to take a wrong turn down a side street so that he can **film** the facility at the end. The sign on the gate says"13th Division Detention Center."And then he hurriedly turns his car and drives away. Later, he hangs his camera from his backpack as he walks past this huge prison complex in Ürümqi. From Ürümqi he drove to the Dabancheng, that small town with the enormous detention facility that I showed earlier. He turned off the main road and drove up a gravel track, then got out of his car and climbed up on an earth berm overlooking the new compound. This was a recklessly brave thing to do because, as he notes in the video, tourists don ' t go to that place. He had no plausible deniability for being there. But this is the view from the top, and it ' s the first image that I ' m aware of of the new camp at Dabancheng. This video showed us places from ground level that previously we had only seen from above, indicating that our interpretations were correct. Seeing the signs at the gates of the facility, which told us the name and the type of facility, added further evidence that these places were camps. This video helped us to corroborate a series of locations where previously all we had had was satellite imagery. In Xinjiang, open sources have allowed us to examine and counter the Chinese government ' s claims about what ' s happening in the region. But this isn ' t the only time that open-source data has led to a government losing control of their narrative. At the time, the civil war in Syria was probably the most documented conflict ever, as people **filmed** bombings and their aftermath and uploaded the videos to social media. Researchers like Bellingcat then used that material to investigate allegations of war crimes, such as the use of chlorine **gas** against civilians. Open-source data has allowed journalistic work that previously would have been really difficult, either because it happened in a place that you can ' t safely go to or because often there previously wouldn ' t have been adequate evidence to examine. Now researchers are using these same tools and techniques to monitor the most recent Russian invasion of Ukraine. One of the first signs of the invasion came in Google Maps with a traffic jam created by Russian artillery moving across the border that blocked the roads for civilian traffic. TikTok videos have given away Russian troop movements. Researchers are investigating potential war crimes and aiming to fact-check claims about the war in close to real time. To do this work, satellite imagery is essential. In Xinjiang, we were lucky enough to have satellite imagery, high-resolution, up-to-date, often taken every month or so and available to us for free. This allowed us to verify potential camp locations and to follow the progress of the camp ' s construction closely. But this isn ' t true of everywhere that journalists would want to investigate, and we need affordable access to imagery of those places as well. We also rely on access to other forms of data. We not only need people to take photos and videos, we need them to upload them to a platform where researchers can access them. And then we need that material to be preserved. Often social media platforms have removed material showing violence, even when it ' s providing key evidence of human rights violations. Civil society actors such as the Syrian Archive have stepped in to download and preserve that material. With social media data and satellite imagery, we can provide evidence of human rights abuses in a way that wasn ' t possible before. We can move beyond looking at individual instances of human rights violations to show the scale of what ' s happened. We can corroborate the testimony of eyewitnesses and provide further proof of their stories. We can build a more detailed **picture** of what ' s happening to inform policymakers or to provide evidence that can be presented in court. With open-source data, we can provide the evidence needed for accountability and then, hopefully, action.

Thank you.

Text Sample 712: POS Dim 5 – 2022 – Score 6.00 – 2022/t003703.txt
Hi. It ’ s about as intimidating as I thought it would be. And yet you ’ d think or I ’ d think that I ’ d be accustomed to more stressful situations. You see, I work in international security and in conflict resolution. And today, I ’ m here to talk about some of our blind spots related to decarbonization. Now, what does one have to do with the other, you may ask. Good question. We often hear that a climate-safe future is a necessary condition for peace. That ’ s true. We also often hear that renewables could be the energy of peace. Less true. To understand, I need to tell you about the materials that we need in order to decarbonize. They ’ re pretty. And they can be deadly. Looking into their story tells us that confronting conflict and building new forms of international peace are going to be critical foundations to build a climate-safe future. So let me tell you about them, starting with where we stand now. When we talk about a decarbonized future, we generally have in mind the possibility of decoupling economic growth from greenhouse **gas** emissions. That ’ s what we call”green growth.”What we tend to think about less often is that to get there we need to recouple economic growth with intensive mineral extraction. To harness renewables or renewable energy like the sun and the wind, we obviously need to build technologies such as solar **panels**, windmills, batteries, right? And to build those, we need to mine huge quantities of non-renewable materials such as these. Knowing that it takes mines as big as these to produce that much amount of usable material. Our ticket to green growth, in other words, is digging deep in the environment. Now we know that mining can have grave impacts for local ecosystems and populations. I ’ ve seen it myself, and it really isn ’ t pretty. But what I want to talk about today is about how much and where we ’ re going to have to dig, and what that means for planetary security and for geopolitics. I ’ ll start from there. History tells us that when the dominant source of energy changes, power relations change as well. Countries that can transform energy to their own advantage, can gain the upper hand economically and politically, and then can put themselves at the center of the global order. Think of the United Kingdom and coal, for instance, or how oil determined the ascendance of the US to a global superpower. What that tells us is that the access to and processing of energy literally materializes into the ability to shape geopolitical power dynamics. And today, we ’ re facing the challenge of implementing the biggest energy transition in the history of humankind under a ticking climate clock. The race is on for a new generation of power. At the heart of which you have all of the critical materials that we need to decarbonize on the one hand and digitalize on the other. So what ’ s happening with them? On the demand side we ’ re at the beginning of an exponential demand **curve**. If you take lithium as a proxy, a key component for [batteries], global production already increased by just short of 300 percent between 2010 and 2020. I ’ m going to pause here for a sec. This is really good news. It means that decarbonization is in motion. The not so good news is that our”**clean**”future is going to be more materially intensive than before. If you take a simple measure for it, the International Energy Agency tells us that with the current level of innovation, an electric car requires six times more mineral inputs than a conventional car. And this is only the start. The World Bank tells us that with the current projections, global production for minerals such as graphite and cobalt will increase by 500 percent by 2050, only to meet the demand for **clean** energy technologies. Now let ’ s look on the supply side. That ’

s where a lot of really interesting things are happening. Who currently exploits and processes minerals and where deposits to meet future demand are located tell us exactly how the transition is going to change geopolitics. So if you look at a material such as lithium, countries like Chile and Australia tend to dominate extraction while China dominates processing. For cobalt, the Democratic Republic of Congo dominates extraction while China dominates processing. For nickel, countries like Indonesia and the Philippines tend to dominate extraction, while China, you guessed it, thank you, dominates processing. And for rare earths, China dominates extraction while China dominates processing. I ' ve just said China a lot, didn ' t I? Well, that ' s because China skillfully leveraged its geo-economic rise to power over the last two decades on the back of integrating supply chains for rare earths from extraction to processing to export. We tend to point fingers at China today for not going fast enough on its own domestic energy transition, but the truth is that China understood already long ago that it would play a central role in other countries ' transitions. And it is. The European Union, for instance, is 98 percent dependent on China for rare earths. Needless to say, this puts China in a prime position to redesign the global balance of power. Now, you may argue that this is a good thing because the global balance of power needs a rehaul anyway. And you know what? I can totally roll with that. But – and this applies to China, the United States, and any other big player – we need to make sure that the redesigning process doesn ' t compromise on human rights or open societies. And that it doesn ' t lead to the weaponization of supply chains at a time of international instability, and more importantly, at a time of complete climate breakdown. Unfortunately, we ' re already seeing signs of this happening. China is currently trying to gain access to more mineral resources through its Belt and Road Initiative. The United States and Europe are both thinking of reshoring critical mining and processing and orienting some of their international partnerships to facilitate access to more mineral resources. Japan is exploring some of its oceanic marine reserves to build strategic reserves. I ' m also speaking in the shadow of a war on the European continent. Now at first sight, Russia ' s invasion of Ukraine has nothing to do with what I ' ve been talking about. But Ukraine happens to be mineral rich. It also happens to be one of only two countries that had struck a partnership with the European Union to diversify and develop supply chains for critical raw materials. That partnership was specifically designed to help the EU decarbonize and in the process to better integrate with Ukraine from a political and economic perspective. Eight months after the partnership was struck, the invasion took place. Now, mineral resources may not explain everything about the war. But they certainly can ' t be ignored in analyzing the events. Because when it comes to the race for critical raw materials, what ' s actually happening is that we ' re headed right back into a new scramble for resources, at the heart of which you find all of the big players eyeing countries with vast mineral deposits. And yet it ' s so obvious many of these countries that are located, for the most part, in Africa, in Latin America, in Central Asia and in the Indo-Pacific. Economists will tell you that this is a great thing, because these countries, or at least a lot of them, need economic resources and many, to accelerate their development pathway and climate adaptation. But. Many of these countries also have very real overlapping risk profiles. The International Institute for Sustainable Development first produced this map back in 2018. Can you see the green dots on the map? They represent all of the different materials that we need in order to decarbonize, their geographic location and their deposit size. As it so happens, a lot of the deposits are located in countries that rank fairly high on corruption indices. They are represented essentially by the shades of brown and red on the map. And as it so happens, a lot of the materials are also located in countries that are fragile, such as Sri Lanka, or downright conflict affected, like Myanmar and the Central African Republic. That ’ s not all. The Notre Dame Institute tells us, with this map, in which you see, again, some red and orange, that countries that are climate vulnerable are also the ones that are resource endowed. And one final thing. You know those big ecosystems that we need to protect and regenerate in order to stabilize the global climate regime? To reboot the hydrological cycle and to protect biodiversity? They ’ re also represented in orange and red on this map. Many of these big ecosystems are located in the same fragile countries that I was mentioning before. They also happen to sit on vast mineral deposits. Changing or eliminating these ecosystems through mining, through deforestation or anything else would undermine planetary security. Not just international security. Planetary security. It ’ s essentially like a perfect storm in the making. Corruption, institutional and socioeconomic fragility, climate disruptions and environmental plundering, all acting as a backdrop to a competition to gain access to the minerals that we need in order to decarbonize. All of these factors will be magnified if we don ’ t rein in the scramble for resources. All of them will reinforce one another. And I want to make something very clear here. The countries at the heart of the resource scrambling may suffer the most direct consequences in terms of their ability to develop, to adapt to climate change and to avoid violence. But their fate is not isolated. Their problems are not geographically distant. Our big blind spot here is that we ’ re headed towards a decarbonization trajectory that may end up undermining ecological integrity and heighten the risks of conflict and insecurity whose consequences would reverberate worldwide. I know that this is not a particularly encouraging **picture**. And that it comes on top of layers of **pictures** that are not particularly encouraging. Our modern economies have advanced and grown for two centuries through the gigantic blind spot of fossil fuel exploitation and its unintended consequences. The big lesson here is that we can ’ t afford to just shift to a different set of energies, technologies and materials without paying attention to the unintended consequences. The stakes are too high. They involve our future. That we know. But they also involve our humanity. And they involve our nature, by which I mean the nature that we choose for ourselves. Decarbonization is the way forward. There ’ s not one single doubt allowedd about this. But the way forward also demands of us that we start imagining our future beyond decarbonization already. Remember what I said at the beginning? A climate-safe future is a necessary condition for peace. But we won ’ t achieve a climate-safe future without peace. And to build peace, we need to shake things up in international politics and in the way that we do business and economics. So where do we start? I ’ d like to offer the scaffolding of a plan in four different baskets. First, science. Science can tell us exactly where it is safe to mine and where it isn ’ t, from an ecological perspective. Where it is not safe to mine, we need to act as though these minerals did not exist and establish protected areas under which no mining licensing can take place. Where mining does take place, we can integrate socioeconomic and ecological regeneration within business models. Second, a global public good regime. If decarbonization is a matter of human survival, then the materials that we need in order to decarbonize should be managed collectively under a global public good regime. The alternative is conflict and planetary breakdown. So while we figure out exactly how to design this regime, the countries at the heart of the scramble for resources should receive adequate support, competent and coherent support to face off the joint challenges of geopolitical competition and climate disruptions on the other hand. In other words, investing into conflict resolution, into the fight against corruption and into context-specific **resilience**, should be top priorities of our global energy transition. Third, changing the way that we do business and economics. We can ’ t just switch from one energy system to another. I ’ ve made that pretty clear, right? What we need instead is to reduce our need for energy and for materials. And that starts with massive public and private investments into circular economic models that favor recyclability and material substitution. Now, here ’ s the thing. We know that this is a necessary step, but not a sufficient one. So what we also need to do is to develop ecological assessments for supply chains that account for greenhouse **gas** emissions, but also for water, soil, biodiversity, material and energy footprint all at once. Only on this all-encompassing basis will we understand how supply and distribution chains need to change and therefore how globalization needs to transform. Fourth, innovation. All of this can only happen if we start shifting our thinking about innovation. Innovation in our times is about bringing back economic footprint within planetary boundaries. Anything else, even the coolest of new products, if it isn ’ t aligned with that goal, it ’ s not innovation, it ’ s business as usual. In our little corner of the world, my team and I at Carnegie Europe have been working really hard to identify what regenerative foreign policy looks like and what it aims for. There are two things that we know by now. One is obvious, we need to tackle fundamental issues around economic redistribution on a global scale. The other thing is that we need a geopolitical de-escalation around decarbonisation and regeneration. We ’ ve translated that into a concept we ’ ve called ecological diplomacy. And we ’ re pushing really hard for the European Union to adopt this framework within their foreign policy. Because if there is one thing that we ’ ve understood, it ’ s that ecological integrity is the foundation for all types of security. Which makes it the one common denominator that we can work on rebuilding collectively. And we can manage. Truly, I believe that we can. As long as we shed light on our transition blind spots and take them as our guiding companions to identify what truly systemic, truly peaceful and truly safe solution pathways look like for the age of climate-disrupted futures. Thank you so much.

Text Sample 713: POS Dim 5 – 2001 – Score 2.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,”Are you still doing this or that?”I don ’ t answer because I ’ m not doing things still, I ’ m doing it like I always did. I still have — or did I use the word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years. But, of course, I ’ m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ’ ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called”Innovation.”Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,"The playful search for beauty was Man ' s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty."These are tiles. The word,"playful"is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ' t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our **feet** when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ' t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 714: POS Dim 5 – 2001 – Score 2.00 – 2001/t000367.txt

I ’ d like to do pretty much what I did the first time, which is to choose a lighthearted theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, exorcise those evil spirits, and this is still going on, as you know. But it wasn ’ t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ’ re feeling depressed? It ’ s better than Prozac, but I wouldn ’ t recommend it. So what we see in the seventeenth, eighteenth century is the continued search for medications other than camphor that ’ ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we fast-forward to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a row — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very interested in producing convulsions, measured types of convulsions. And they thought,"Well, we ’ ve got electricity, we ’ ll plug somebody into the wall. That always makes hair stand up and people shake a lot."So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,"We know that at the Rome **railroad** station, there are all these lost souls wandering around, muttering gibberish. Can you bring one of them to us?"Someone who is, as the Italians say,"cagoots."So they found this"cagoots"guy, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,"Well, we ’ ll try 55 volts, two-tenths of a second. That ’ s not going to do anything terrible to him."So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,"This fellow"— remember, he wasn ’ t even put to sleep —"after this major grand mal convulsion, sat right up, looked at these three fellas and said, ’ What the fuck are you assholes trying to do? ’"If I could only say that in Italian. Well, they were happy as could be, because he hadn ’ t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and everybody in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ’ t work very well on the schizophrenics, but it was pretty clear in the ’ 30s and by the middle of the ’ 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ’ t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the brain waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ’ 60s, the first antidepressants came out. Tofranil was the first. In the late ’ 70s, early ’ 80s, there were others, and they were very effective. And patients ’ rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ’ m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ’ ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ’ m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ’ clock in the afternoon, because I couldn ’ t get out of bed before about 11 o ’ clock. And anybody who ’ s been depressed here knows what that ’ s like. I couldn ’ t even pull the covers off myself. Well, you ’ re in a university medical center, where everybody knows everybody, and it ’ s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ’ t work anymore. And, in fact, it didn ’ t make any difference because I didn ’ t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,”Don ’ t worry, chap. Six weeks, you ’ re back in the operating room. Everything ’ s going to be great.”Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ’ t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five **feet** in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ’ t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had.

They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the senior staff. All the senior staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful stuff. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it got so there was a throbbing, there was a ferocious fear in my head. You ' ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ' t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case. He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the senior staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,"No. I know this man better than any of you. I have met with him over and over again. You ' ve just seen him from time to time. You ' ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ' ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go back to surgery, he will never be the loving father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ' One Flew Over the Cuckoo ' s Nest. ' And you know about that, just essentially in a stupor the rest of his life."Well, he said,"Can ' t we try a course of electroshock therapy?"And you know why they agreed? They agreed to humor him. They just thought,"Well, we ' ll give a course of 10. And so we ' ll lose a little time. Big deal. It doesn ' t make any difference."So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ' t work. Seven didn ' t work. Eight didn ' t work. At nine, I noticed — and it ' s wonderful that I could notice anything — I noticed a change. And at 10, I noticed a real change. And he went back to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, lo and behold, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could blow the obsessional thinking away. I could blow the depression away. And I ' ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,"I ' ve got the strength now to do this."It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a formula. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember"Ruddigore,"and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,"We must have a word to bring you back to reality, and the word, my dear, will be ' Basingstoke. '"So every time she got a little nuts, he would say,"Basingstoke!"And she would say,"Basingstoke, it is."And she would be fine for a little while. Well, you know, I ' m from the Bronx. I can ' t say"Basingstoke."But I had something better. And it was very simple. It was,"Ah, fuck it!"Much better than"Basingstoke,"at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,"Ah, fuck it."And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went back to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came back to live with us. We had two more children after that. Resuscitated the career, even better than it had been before. Went right back into the university and began to write books. Well, you know, it ' s been a wonderful life. It ' s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ' s been very exciting. It ' s been very happy. Every once in a while, I have to say,"Ah, fuck it."Every once in a while, I get somewhat depressed and a little obsessional. So, I ' m not free of all of this. But it ' s worked. It ' s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ' ve gotten thousands of letters about them by people who do think this — that based on my life ' s history as I ' ve portrayed in the books, my early life ' s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the bitter dregs of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ' ve always felt guilty about that. I ' ve always felt that somehow I was an impostor because my readers don ' t know what I have just told you. It ' s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an untroubled mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes back to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ’ t happen, there will be adversities. If I, with the bleakness of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way back from this, believe me, anybody can find their way back from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

Text Sample 715: POS Dim 5 – 2001 – Score 2.00 – 2001/t000092.txt

I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ’ children, because they were all average. But they weren ’ t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ’ s not right. The good Lord in his infinite wisdom didn ’ t create us all equal as far as intelligence is concerned, any more than we ’ re equal for size, appearance. Not everybody could earn an A or a B, and I didn ’ t like that way of judging, and I did know how the alumni of various schools back in the ’ 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ’ t lose a game. But it seemed that we didn ’ t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ’ t like it, I didn ’ t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ’ m sure at the time he did that, I didn ’ t — it didn ’ t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ’ s under your control. If you get too engrossed and involved and concerned in regard to the things over which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,”At God ’ s footstool to confess, a poor soul knelt, and bowed his head. ’ I failed! ’ he cried. The Master said, ’ Thou didst thy best, that is success. ’”From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ’ re capable. I believe that ’ s true. If you make the effort to do the best of which you ’ re capable, trying to improve the situation that exists for you, I think that ’ s success, and I don ’ t think others can judge that ; it ’ s like character and reputation — your reputation is what you ’ re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ’ s what the teachers are themselves.”That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by coal oil lamp — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I reply, ’ Where could I find such splendid company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ’ s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ’ Where could I find such splendid company? ’”And I believe the teaching profession — it ’ s true, you have so many youngsters, and I ’ ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ’ re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ’ re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was”Never be late.”Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and **clean**. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ’ s not right for me to keep this other [rule] so I let them just — they had to be neat and **clean**. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ’ t **clean** and neat, so I wouldn ’ t let him go. He couldn ’ t get on the bus, he had to go home and get **cleaned** up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ’ t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ’ ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,”Don ’ t run practices late, because you ’ ll go home in a bad mood, and that ’ s not good, for a young married man to go home in a bad mood. When you get older, it doesn ’ t make any difference, but —”So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ’ re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ’ t want that. I used to tell them I was paid to do that. That ’ s my job. I ’ m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ’ s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ’ s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ’ t have the time to go on that. But that helped me, I think, become a better teacher. It ’ s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ’ re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ’ re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ’ t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ’ i ’. I ’ d never seen that before, but he did. Big league baseball players — they ’ re very perceptive about those things, and they noticed he had only one ’ i ’ in his name. You ’ d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it”The Road Ahead, or the Road Behind.”He said,”Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ’ t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ’ s game. You and I know deeper down, there ’ s always a chance to win the crown. But when we fail to give our best, we simply haven ’ t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ' s bearing down that wins the cup. Of dreaming there ' s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ' s you and I who make our fates — we open up or close the gates on the road ahead or the road behind."Reminds me of another set of threes that my dad tried to get across to us : Don ' t whine. Don ' t complain. Don ' t make excuses. Just get out there, and whatever you ' re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ' re outscored. I ' ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ' t know the outcome, I hope they couldn ' t tell by your actions whether you outscored an opponent or the opponent outscored you. That ' s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ' d want them to be but they ' ll be about what they should ; only you will know whether you can do that. And that ' s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,"The journey is better than the end."And I like that. I think that it is — it ' s getting there. Sometimes when you get there, there ' s almost a let down. But it ' s the getting there that ' s the **fun**. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ' d done a decent job during the week. There again, it ' s getting the players to get that self-satisfaction, in knowing that they ' d made the effort to do the best of which they are capable. Sometimes I ' m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,"Suppose that you, in some way, could make the perfect player. What would you want?"And I said,"Well, I ' d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ' d want one that could play, too. I ' d want one to realize that defense usually wins championships, and who would work hard on defense. But I ' d want one who would play offense, too. I ' d want him to be unselfish, and look for the pass first and not shoot all the time. And I ' d want one that could pass and would pass. I ' ve had some that could and wouldn ' t, and I ' ve had some that would and could. So, yeah, I ' d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ' d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,"They Call Me Coach,"two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ' t play varsity when I taught. I thought,"Oh gracious, if these two players, either one of them"—

they were different years, but I thought about each one at the time he was there —"Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ' s good enough to make it."And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ' d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ' t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ' s taken, they assumed would be missed. I ' ve had too many stand around and wait to see if it ' s missed, then they go and it ' s too late, somebody else is in there ahead of them. They weren ' t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 716: POS Dim 5 – 2015 – Score 7.00 – 2015/t001601.txt

I ' d like to take you on the epic quest of the Rosetta spacecraft. To escort and land the probe on a comet, this has been my passion for the past two years. In order to do that, I need to explain to you something about the origin of the solar system. When we go back four and a half billion years, there was a cloud of **gas** and dust. In the center of this cloud, our sun formed and ignited. Along with that, what we now know as planets, comets and asteroids formed. What then happened, according to theory, is that when the Earth had cooled down a bit after its formation, comets massively impacted the Earth and delivered water to Earth. They probably also delivered complex organic material to Earth, and that may have bootstrapped the emergence of life. You can compare this to having to solve a 250-piece puzzle and not a 2, 000-piece puzzle. Afterwards, the big planets like Jupiter and Saturn, they were not in their place where they are now, and they interacted gravitationally, and they swept the whole interior of the solar system **clean**, and what we now know as comets ended up in something called the Kuiper Belt, which is a belt of objects beyond the orbit of Neptune. And sometimes these objects run into each other, and they gravitationally deflect, and then the gravity of Jupiter pulls them back into the solar system. And they then become the comets as we see them in the sky. The important thing here to note is that in the meantime, the four and a half billion years, these comets have been sitting on the outside of the solar system, and haven ' t changed — deep, frozen versions of our solar system. In the sky, they look like this. We know them for their tails. There are actually two tails. One is a dust tail, which is blown away by the solar wind. The other one is an ion tail, which is charged particles, and they follow the magnetic field in the solar system. There ' s the coma, and then there is the nucleus, which here is too small to see, and you have to remember that in the case of Rosetta, the spacecraft is in that center pixel. We are only 20, 30, 40 kilometers away from the comet. So what ' s important to remember? Comets contain the original material from which our solar system was formed, so they ' re **ideal** to study the components that were present at the time when Earth, and life, started. Comets are also suspected of having brought the elements which may have bootstrapped life. In 1983, ESA set up its long-term Horizon 2000 program, which contained one cornerstone, which would be a mission to a comet. In parallel, a small mission to a comet, what you see here, Giotto, was launched, and in 1986, flew by the comet of Halley with an armada of other spacecraft. From the results of that mission, it became immediately clear that comets were **ideal** bodies to study to understand our solar system. And thus, the Rosetta mission was approved in 1993, and originally it was supposed to be launched in 2003, but a problem arose with an Ariane rocket. However, our P. R. department, in its enthusiasm, had already made 1, 000 Delft Blue plates with the name of the wrong comets. So I ’ ve never had to buy any china since. That ’ s the positive part. Once the whole problem was solved, we left Earth in 2004 to the newly selected comet, Churyumov-Gerasimenko. This comet had to be specially selected because A, you have to be able to get to it, and B, it shouldn ’ t have been in the solar system too long. This particular comet has been in the solar system since 1959. That ’ s the first time when it was deflected by Jupiter, and it got close enough to the sun to start changing. So it ’ s a very fresh comet. Rosetta made a few historic firsts. It ’ s the first satellite to orbit a comet, and to escort it throughout its whole tour through the solar system — closest approach to the sun, as we will see in August, and then away again to the exterior. It ’ s the first ever landing on a comet. We actually orbit the comet using something which is not normally done with spacecraft. Normally, you look at the sky and you know where you point and where you are. In this case, that ’ s not enough. We navigated by looking at landmarks on the comet. We recognized features — boulders, craters — and that ’ s how we know where we are respective to the comet. And, of course, it ’ s the first satellite to go beyond the orbit of Jupiter on solar cells. Now, this sounds more heroic than it actually is, because the technology to use radio isotope thermal generators wasn ’ t available in Europe at that time, so there was no choice. But these solar arrays are big. This is one wing, and these are not specially selected small people. They ’ re just like you and me. We have two of these wings, 65 square meters. Now later on, of course, when we got to the comet, you find out that 65 square meters of sail close to a body which is outgassing is not always a very handy choice. Now, how did we get to the comet? Because we had to go there for the Rosetta scientific objectives very far away — four times the distance of the Earth to the sun — and also at a much higher velocity than we could achieve with fuel, because we ’ d have to take six times as much fuel as the whole spacecraft weighed. So what do you do? You use gravitational flybys, slingshots, where you pass by a planet at very low altitude, a few thousand kilometers, and then you get the velocity of that planet around the sun for free. We did that a few times. We did Earth, we did Mars, we did twice Earth again, and we also flew by two asteroids, Lutetia and Steins. Then in 2011, we got so far from the sun that if the spacecraft got into trouble, we couldn ’ t actually save the spacecraft anymore, so we went into hibernation. Everything was switched off except for one clock. Here you see in white the trajectory, and the way this works. You see that from the circle where we started, the white line, actually you get more and more and more elliptical, and then finally we approached the comet in May 2014, and we had to start doing the rendezvous maneuvers. On the way there, we flew by Earth and we took a few **pictures** to test our cameras. This is the moon rising over Earth, and this is what we now call a selfie, which at that time, by the way, that word didn ’ t exist. It ’ s at Mars. It was taken by the CIVA camera. That ’ s one of the cameras on the lander, and it just looks under the solar arrays, and you see the planet Mars and the solar array in the distance. Now, when we got out of hibernation in January 2014, we started arriving at a distance of two million kilometers from the comet in May. However, the velocity the spacecraft had was much too fast. We were going 2, 800 kilometers an hour faster than the comet, so we had to brake. We had to do eight maneuvers, and you see here, some of them were really big. We had to brake the first one by a few hundred kilometers per hour, and actually, the duration of that was seven hours, and it used 218 kilos of fuel, and those were seven nerve-wracking hours, because in 2007, there was a leak in the system of the propulsion of Rosetta, and we had to close off a branch, so the system was actually operating at a pressure which it was never designed or qualified for. Then we got in the vicinity of the comet, and these were the first **pictures** we saw. The true comet rotation period is 12 and a half hours, so this is accelerated, but you will understand that our flight dynamics engineers thought, this is not going to be an easy thing to land on. We had hoped for some kind of spud-like thing where you could easily land. But we had one hope : maybe it was smooth. No. That didn ’ t work either. So at that point in time, it was clearly unavoidable : we had to map this body in all the detail you could get, because we had to find an area which is 500 meters in diameter and flat. Why 500 meters? That ’ s the error we have on landing the probe. So we went through this process, and we mapped the comet. We used a technique called photoclinometry. You use shadows thrown by the sun. What you see here is a rock sitting on the surface of the comet, and the sun shines from above. From the shadow, we, with our brain, can immediately determine roughly what the shape of that rock is. You can program that in a computer, you then cover the whole comet, and you can map the comet. For that, we flew special trajectories starting in August. First, a triangle of 100 kilometers on a side at 100 kilometers ’ distance, and we repeated the whole thing at 50 kilometers. At that time, we had seen the comet at all kinds of angles, and we could use this technique to map the whole thing. Now, this led to a selection of landing sites. This whole process we had to do, to go from the mapping of the comet to actually finding the final landing site, was 60 days. We didn ’ t have more. To give you an idea, the average Mars mission takes hundreds of scientists for years to meet about where shall we go? We had 60 days, and that was it. We finally selected the final landing site and the commands were prepared for Rosetta to launch Philae. The way this works is that Rosetta has to be at the right point in space, and aiming towards the comet, because the lander is passive. The lander is then pushed out and moves towards the comet. Rosetta had to turn around to get its cameras to actually look at Philae while it was departing and to be able to communicate with it. Now, the landing duration of the whole trajectory was seven hours. Now do a simple calculation : if the velocity of Rosetta is off by one centimeter per second, seven hours is 25, 000 seconds. That means 252 meters wrong on the comet. So we had to know the velocity of Rosetta much better than one centimeter per second, and its location in space better than 100 meters at 500 million kilometers from Earth. That ’ s no mean feat. Let me quickly take you through some of the science and the instruments. I won ’ t bore you with all the details of all the instruments, but it ’ s got everything. We can sniff **gas**, we can measure dust particles, the shape of them, the composition, there are magnetometers, everything. This is one of the results from an instrument which measures **gas** density at the position of Rosetta, so it ’ s **gas** which has left the comet. The **bottom** graph is September of last year. There is a long-term variation, which in itself is not surprising, but you see the sharp peaks. This is a comet day. You can see the effect of the sun on the evaporation of **gas** and the fact that the comet is rotating. So there is one spot, apparently, where there is a lot of stuff coming from, it gets heated in the Sun, and then cools down on the back side. And we can see the density variations of this. These are the **gases** and the organic compounds that we already have measured. You will see it ’ s an impressive list, and there is much, much, much more to come, because there are more measurements. Actually, there is a conference going on in Houston at the moment where many of these results are presented. Also, we measured dust particles. Now, for you, this will not look very impressive, but the scientists were thrilled when they saw this. Two dust particles : the right one they call Boris, and they shot it with tantalum in order to be able to analyze it. Now, we found sodium and magnesium. What this tells you is this is the concentration of these two materials at the time the solar system was formed, so we learned things about which materials were there when the planet was made. Of course, one of the important elements is the imaging. This is one of the cameras of Rosetta, the OSIRIS camera, and this actually was the cover of Science magazine on January 23 of this year. Nobody had expected this body to look like this. Boulders, rocks — if anything, it looks more like the Half Dome in Yosemite than anything else. We also saw things like this : dunes, and what look to be, on the righthand side, wind-blown shadows. Now we know these from Mars, but this comet doesn ’ t have an atmosphere, so it ’ s a bit difficult to create a wind-blown shadow. It may be local outgassing, stuff which goes up and comes back. We don ’ t know, so there is a lot to investigate. Here, you see the same image twice. On the **left-hand** side, you see in the middle a pit. On the right-hand side, if you carefully look, there are three jets coming out of the **bottom** of that pit. So this is the activity of the comet. Apparently, at the **bottom** of these pits is where the active regions are, and where the material evaporates into space. There is a very intriguing crack in the neck of the comet. You see it on the right-hand side. It ’ s a kilometer long, and it ’ s two and a half meters wide. Some people suggest that actually, when we get close to the sun, the comet may split in two, and then we ’ ll have to choose, which comet do we go for? The lander — again, lots of instruments, mostly comparable except for the things which hammer in the ground and drill, etc. But much the same as Rosetta, and that is because you want to compare what you find in space with what you find on the comet. These are called ground truth measurements. These are the landing descent images that were taken by the OSIRIS camera. You see the lander getting further and further away from Rosetta. On the top right, you see an image taken at 60 meters by the lander, 60 meters above the surface of the comet. The boulder there is some 10 meters. So this is one of the last images we took before we landed on the comet. Here, you see the whole sequence again, but from a different perspective, and you see three blown-ups from the bottom-left to the middle of the lander traveling over the surface of the comet. Then, at the top, there is a before and an after image of the landing. The only problem with the after image is, there is no lander. But if you carefully look at the right-hand side of this image, we saw the lander still there, but it had bounced. It had departed again. Now, on a bit of a comical note here is that originally Rosetta was designed to have a lander which would bounce. That was discarded because it was way too expensive. Now, we forgot, but the lander knew. During the first bounce, in the magnetometers, you see here the data from them, from the three axes, x, y and z. Halfway through, you see a red line. At that red line, there is a change. What happened, apparently, is during the first bounce, somewhere, we hit the edge of a crater with one of the legs of the lander, and the rotation velocity of the lander changed. So we ’ ve been rather lucky that we are where we are. This is one of the iconic images of Rosetta. It ’ s a man-made object, a leg of the lander, standing on a comet. This, for me, is one of the very best images of space science I have ever seen. One of the things we still have to do is to actually find the lander. The blue area here is where we know it must be. We haven ’ t been able to find it yet, but the search is continuing, as are our efforts to start getting the lander to work again. We listen every day, and we hope that between now and somewhere in April, the lander will wake up again. The findings of what we found on the comet : This thing would float in water. It ’ s half the density of water. So it looks like a very big rock, but it ’ s not. The activity increase we saw in June, July, August last year was a four-fold activity increase. By the time we will be at the sun, there will be 100 kilos a second leaving this comet : **gas**, dust, whatever. That ’ s 100 million kilos a day. Then, finally, the landing day. I will never forget — absolute madness, 250 TV crews in Germany. The BBC was interviewing me, and another TV crew who was following me all day were **filming** me being interviewed, and it went on like that for the whole day. The Discovery Channel crew actually caught me when leaving the control room, and they asked the right question, and man, I got into tears, and I still feel this. For a month and a half, I couldn ’ t think about landing day without crying, and I still have the emotion in me. With this image of the comet, I would like to leave you. Thank you.

Text Sample 717: POS Dim 5 – 2015 – Score 7.00 – 2015/t001481.txt

So I ’ ve had the great privilege of traveling to some incredible places, photographing these distant landscapes and remote cultures all over the world. I love my job. But people think it ’ s this string of epiphanies and sunrises and rainbows, when in reality, it looks more something like this. This is my office. We can ’ t afford the fanciest places to stay at night, so we tend to sleep a lot outdoors. As long as we can stay dry, that ’ s a bonus. We also can ’ t afford the fanciest restaurants. So we tend to eat whatever ’ s on the local menu. And if you ’ re in the Ecuadorian Pramo, you ’ re going to eat a large rodent called a cuy. But what makes our experiences perhaps a little bit different and a little more unique than that of the average person is that we have this gnawing thing in the back of our mind that even in our darkest moments, and those times of despair, we think,”Hey, there might be an image to be made here, there might be a story to be told.”And why is storytelling important? Well, it helps us to connect with our cultural and our natural heritage. And in the Southeast, there ’ s an alarming disconnect between the public and the natural areas that allow us to be here in the first place. We ’ re visual creatures, so we use what we see to teach us what we know. Now the majority of us aren ’ t going to willingly go way down to a swamp. So how can we still expect those same people to then advocate on behalf of their protection? We can ’ t. So my job, then, is to use photography as a communication tool, to help bridge the gap between the science and the aesthetics, to get people talking, to get them thinking, and to hopefully, ultimately, get them caring. I started doing this 15 years ago right here in Gainesville, right here in my backyard. And I fell in love with adventure and discovery, going to explore all these different places that were just minutes from my front doorstep. There are a lot of beautiful places to find. Despite all these years that have passed, I still see the world through the eyes of a child and I try to incorporate that sense of wonderment and that sense of curiosity into my photography as often as I can. And we ’ re pretty lucky because here in the South, we ’ re still blessed with a relatively blank canvas that we can fill with the most fanciful adventures and incredible experiences. It ’ s just a matter of how far our imagination will take us. See, a lot of people look at this and they say,”Oh yeah, wow, that ’ s a pretty tree.”But I don ’ t just see a tree — I look at this and I see opportunity. I see an entire weekend. Because when I was a kid, these were the types of images that got me off the sofa and dared me to explore, dared me to go find the woods and put my head underwater and see what we have. And folks, I ’ ve been photographing all over the world and I promise you, what we have here in the South, what we have in the Sunshine State, rivals anything else that I ’ ve seen. But yet our tourism industry is busy promoting all the wrong things. Before most kids are 12, they ’ ll have been to Disney World more times than they ’ ve been in a canoe or camping under a starry sky. And I have nothing against Disney or Mickey ; I used to go there, too. But they ’ re missing out on those fundamental connections that create a real sense of pride and ownership for the place that they call home. And this is compounded by the issue that the landscapes that define our natural heritage and fuel our aquifer for our drinking water have been deemed as scary and dangerous and spooky. When our ancestors first came here, they warned,”Stay out of these areas, they ’ re haunted. They ’ re full of evil spirits and ghosts.”I don ’ t know where they came up with that idea. But it ’ s actually led to a very real disconnect, a very real negative mentality that has kept the public disinterested, silent, and ultimately, our environment at risk. We ’ re a state that ’ s surrounded and defined by water, and yet for centuries, swamps and wetlands have been regarded as these obstacles to overcome. And so we ’ ve treated them as these second-class ecosystems, because they have very little monetary value and of course, they ’ re known to harbor alligators and snakes — which, I ’ ll admit, these aren ’ t the most cuddly of ambassadors. So it became assumed, then, that the only good swamp was a drained swamp. And in fact, draining a swamp to make way for agriculture and development was considered the very essence of conservation not too long ago. But now we ’ re backpedaling, because the more we come to learn about these sodden landscapes, the more secrets we ’ re starting to unlock about interspecies relationships and the connectivity of habitats, watersheds and flyways. Take this bird, for example : this is the prothonotary warbler. I love this bird because it ’ s a swamp bird, through and through, a swamp bird. They nest and they mate and they breed in these old-growth swamps in these flooded forests. And so after the spring, after they raise their young, they then fly thousand of miles over the Gulf of Mexico into Central and South America. And then after the winter, the spring rolls around and they come back. They fly thousands of miles over the Gulf of Mexico. And where do they go? Where do they land? Right back in the same tree. That ’ s nuts. This is a bird the size of a tennis ball — I mean, that ’ s crazy! I used a GPS to get here today, and this is my hometown. It ’ s crazy. So what happens, then, when this bird flies over the Gulf of Mexico into Central America for the winter and then the spring rolls around and it flies back, and it comes back to this : a freshly sodded golf course? This is a narrative that ’ s all too commonly unraveling here in this state. And this is a natural process that ’ s occurred for thousands of years and we ’ re just now learning about it. So you can imagine all else we have to learn about these landscapes if we just preserve them first. Now despite all this rich life that abounds in these swamps, they still have a bad name. Many people feel uncomfortable with the idea of wading into Florida ’ s blackwater. I can understand that. But what I loved about growing up in the Sunshine State is that for so many of us, we live with this latent but very palpable fear that when we put our toes into the water, there might be something much more ancient and much more adapted than we are. Knowing that you ’ re not top dog is a welcomed discomfort, I think. How often in this modern and urban and digital age do you actually get the chance to feel vulnerable, or consider that the world may not have been made for just us? So for the last decade, I began seeking out these areas where the concrete yields to forest and the pines turn to cypress, and I viewed all these mosquitoes and reptiles, all these discomforts, as affirmations that I ’ d found true wilderness, and I embrace them wholly. Now as a conservation photographer obsessed with blackwater, it ’ s only fitting that I ’ d eventually end up in the most famous swamp of all : the Everglades. Growing up here in North Central Florida, it always had these enchanted names, places like Loxahatchee and Fakahatchee, Corkscrew, Big Cypress. I started what turned into a five-year project to hopefully reintroduce the Everglades in a new light, in a more inspired light. But I knew this would be a tall order, because here you have an area that ’ s roughly a third the size the state of Florida, it ’ s huge. And when I say Everglades, most people are like,”Oh, yeah, the national **park**.”But the Everglades is not just a **park** ; it ’ s an entire watershed, starting with the Kissimmee chain of lakes in the north, and then as the rains would fall in the summer, these downpours would flow into Lake Okeechobee, and Lake Okeechobee would fill up and it would overflow its banks and spill southward, ever slowly, with the topography, and get into the river of grass, the Sawgrass Prairies, before meting into the cypress slews, until going further south into the mangrove swamps, and then finally — finally — reaching Florida Bay, the emerald gem of the Everglades, the great estuary, the 850 square-mile estuary. So sure, the national **park** is the southern end of this system, but all the things that make it unique are these inputs that come in, the fresh water that starts 100 miles north. So no manner of these political or invisible boundaries protect the **park** from polluted water or insufficient water. And unfortunately, that ’ s precisely what we ’ ve done. Over the last 60 years, we have drained, we have dammed, we have dredged the Everglades to where now only one third of the water that used to reach the bay now reaches the bay today. So this story is not all sunshine and rainbows, unfortunately. For better or for worse, the story of the Everglades is intrinsically tied to the peaks and the **valleys** of mankind ’ s relationship with the natural world. But I ’ ll show you these beautiful **pictures**, because it gets you on board. And while I have your attention, I can tell you the real story. It ’ s that we ’ re taking this, and we ’ re trading it for this, at an alarming rate. And what ’ s lost on so many people is the sheer scale of which we ’ re discussing. Because the Everglades is not just responsible for the drinking water for 7 million Floridians ; today it also provides the agricultural fields for the year-round tomatoes and oranges for over 300 million Americans. And it ’ s that same seasonal pulse of water in the summer that built the river of grass 6, 000 years ago. Ironically, today, it ’ s also responsible for the over half a million acres of the endless river of sugarcane. These are the same fields that are responsible for dumping exceedingly high levels of fertilizers into the watershed, forever changing the system. But in order for you to not just understand how this system works, but to also get personally connected to it, I decided to break the story down into several different narratives. And I wanted that story to start in Lake Okeechobee, the beating heart of the Everglade system. And to do that, I picked an ambassador, an iconic species. This is the Everglade snail kite. It ’ s a great bird, and they used to nest in the thousands, thousands in the northern Everglades. And then they ’ ve gone down to about 400 nesting pairs today. And why is that? Well, it ’ s because they eat one source of food, an apple snail, about the size of a ping-pong ball, an aquatic gastropod. So as we started damming up the Everglades, as we started diking Lake Okeechobee and draining the wetlands, we lost the habitat for the snail. And thus, the population of the kites declined. And so, I wanted a photo that would not only communicate this relationship between wetland, snail and bird, but I also wanted a photo that would communicate how incredible this relationship was, and how very important it is that they ’ ve come to depend on each other, this healthy wetland and this bird. And to do that, I brainstormed this idea. I started sketching out these plans to make a photo, and I sent it to the wildlife biologist down in Okeechobee — this is an endangered bird, so it takes special permission to do. So I built this submerged platform that would hold snails just right under the water. And I spent months planning this crazy idea. And I took this platform down to Lake Okeechobee and I spent over a week in the water, wading waist-deep, 9-hour shifts from dawn until dusk, to get one image that I thought might communicate this. And here ' s the day that it finally worked : [Video : After setting up the platform, I look off and I see a kite coming over the cattails. And I see him **scanning** and searching. And he gets right over the trap, and I see that he ' s seen it. And he beelines, he goes straight for the trap. And in that moment, all those months of planning, waiting, all the sunburn, mosquito bites — suddenly, they ' re all worth it. Oh my gosh, I can ' t believe it!] You can believe how excited I was when that happened. But what the idea was, is that for someone who ' s never seen this bird and has no reason to care about it, these photos, these new perspectives, will help shed a little new light on just one species that makes this watershed so incredible, so valuable, so important. Now, I know I can ' t come here to Gainesville and talk to you about animals in the Everglades without talking about gators. I love gators, I grew up loving gators. My parents always said I had an unhealthy relationship with gators. But what I like about them is, they ' re like the freshwater equivalent of sharks. They ' re feared, they ' re hated, and they are tragically misunderstood. Because these are a unique species, they ' re not just apex predators. In the Everglades, they are the very **architects** of the Everglades, because as the water drops down in the winter during the dry season, they start excavating these holes called gator holes. And they do this because as the water drops down, they ' ll be able to stay wet and they ' ll be able to forage. And now this isn ' t just affecting them, other animals also depend on this relationship, so they become a keystone species as well. So how do you make an apex predator, an ancient reptile, at once look like it dominates the system, but at the same time, look vulnerable? Well, you wade into a pit of about 120 of them, then you hope that you ' ve made the right decision. I still have all my fingers, it ' s cool. But I understand, I know I ' m not going to rally you guys, I ' m not going to rally the troops to"Save the Everglades for the gators!"It won ' t happen because they ' re so ubiquitous, we see them now, they ' re one of the great conservation success stories of the US. But there is one species in the Everglades that no matter who you are, you can ' t help but love, too, and that ' s the roseate spoonbill. These birds are great, but they ' ve had a really tough time in the Everglades, because they started out with thousands of nesting pairs in Florida Bay, and at the turn of the 20th century, they got down to two — two nesting pairs. And why? That ' s because women thought they looked better on their hats then they did flying in the sky. Then we banned the plume trade, and their numbers started rebounding. And as their numbers started rebounding, scientists began to pay attention, they started studying these birds. And what they found out is that these birds ' behavior is intrinsically tied to the annual draw-down cycle of water in the Everglades, the thing that defines the Everglades watershed. What they found out is that these birds started nesting in the winter as the water drew down, because they ' re tactile feeders, so they have to touch whatever they eat. And so they wait for these concentrated pools of fish to be able to feed enough to feed their young. So these birds became the very icon of the Everglades — an indicator species of the overall health of the system. And just as their numbers were rebounding in the mid-20th century — shooting up to 900, 1, 000, 1, 100, 1, 200 — just as that started happening, we started draining the southern Everglades. And we stopped two-thirds of that water from moving south. And it had drastic consequences. And just as those numbers started reaching their peak, unfortunately, today, the real spoonbill story, the real photo of what it looks like is more something like this. And we ' re down to less than 70 nesting pairs in Florida Bay today, because we ' ve disrupted the system so much. So all these different organizations are shouting, they ' re screaming,"The Everglades is fragile! It ' s fragile!"It is not. It is resilient. Because despite all we ' ve taken, despite all we ' ve done and we ' ve drained and we ' ve dammed and we ' ve dredged it, pieces of it are still here, waiting to be put back together. And this is what I ' ve loved about South Florida, that in one place, you have this unstoppable force of mankind meeting the immovable object of tropical nature. And it ' s at this new frontier that we are forced with a new appraisal. What is wilderness worth? What is the value of biodiversity, or our drinking water? And fortunately, after decades of debate, we ' re finally starting to act on those questions. We ' re slowly undertaking these projects to bring more freshwater back to the bay. But it ' s up to us as citizens, as residents, as stewards to hold our elected officials to their promises. What can you do to help? It ' s so easy. Just get outside, get out there. Take your friends out, take your kids out, take your family out. Hire a fishing guide. Show the state that protecting wilderness not only makes ecological sense, but economic sense as well. It ' s a lot of **fun**, just do it — put your **feet** in the water. The swamp will change you, I promise. Over the years, we ' ve been so generous with these other landscapes around the country, cloaking them with this American pride, places that we now consider to define us : Grand Canyon, Yosemite, Yellowstone. And we use these **parks** and these natural areas as beacons and as cultural compasses. And sadly, the Everglades is very commonly left out of that conversation. But I believe it ' s every bit as iconic and emblematic of who we are as a country as any of these other wildernesses. It ' s just a different kind of wild. But I ' m encouraged, because maybe we ' re finally starting to come around, because what was once deemed this swampy wasteland, today is a World Heritage site. It ' s a wetland of international importance. And we ' ve come a long way in the last 60 years. And as the world ' s largest and most ambitious wetland restoration project, the international spotlight is on us in the Sunshine State. Because if we can heal this system, it ' s going to become an icon for wetland restoration all over the world. But it ' s up to us to decide which legacy we want to attach our flag to. They say that the Everglades is our greatest test. If we pass it, we get to keep the planet. I love that quote, because it ' s a challenge, it ' s a prod. Can we do it? Will we do it? We have to, we must. But the Everglades is not just a test. It ' s also a gift, and ultimately, our responsibility. Thank you.

Text Sample 718: POS Dim 5 – 2015 – Score 6.00 – 2015/t001534.txt

Blah blah blah blah blah. Blah blah blah blah, blah blah, blah blah blah blah blah blah. Blah blah blah, blah. So what the **hell** was that? Well, you don ' t know because you couldn ' t understand it. It wasn ' t clear. But hopefully, it was said with enough conviction that it was at least alluringly mysterious. Clarity or mystery? I ' m balancing these two things in my daily work as a graphic designer, as well as my daily life as a New Yorker every day, and there are two elements that absolutely **fascinate** me. Here ' s an example. Now, how many people know what this is? Okay. Now how many people know what this is? Okay. Thanks to two more deft strokes by the genius Charles M. Schulz, we now have seven deft strokes that in and of themselves create an entire emotional life, one that has enthralled hundreds of millions of fans for over 50 years. This is actually a cover of a book that I designed about the work of Schulz and his art, which will be coming out this fall, and that is the entire cover. There is no other typographic information or visual information on the front, and the name of the book is”Only What ’ s Necessary.”So this is sort of symbolic about the decisions I have to make every day about the design that I ’ m perceiving, and the design I ’ m creating. So clarity. Clarity gets to the point. It ’ s blunt. It ’ s honest. It ’ s sincere. We ask ourselves this. [”When should you be clear?”] Now, something like this, whether we can read it or not, needs to be really, really clear. Is it? This is a rather recent example of urban clarity that I just love, mainly because I ’ m always late and I am always in a hurry. So when these meters started showing up a couple of years ago on street corners, I was thrilled, because now I finally knew how many seconds I had to get across the street before I got run over by a car. Six? I can do that. So let ’ s look at the yin to the clarity yang, and that is mystery. Mystery is a lot more complicated by its very definition. Mystery demands to be decoded, and when it ’ s done right, we really, really want to. [”When should you be mysterious?”] In World War II, the Germans really, really wanted to decode this, and they couldn ’ t. Here ’ s an example of a design that I ’ ve done recently for a novel by Haruki Murakami, who I ’ ve done design work for for over 20 years now, and this is a novel about a young man who has four dear friends who all of a sudden, after their freshman year of college, completely cut him off with no explanation, and he is devastated. And the friends ’ names each have a connotation in Japanese to a color. So there ’ s Mr. Red, there ’ s Mr. Blue, there ’ s Ms. White, and Ms. Black. Tsukuru Tazaki, his name does not correspond to a color, so his nickname is Colorless, and as he ’ s looking back on their friendship, he recalls that they were like five fingers on a hand. So I created this sort of abstract representation of this, but there ’ s a lot more going on underneath the surface of the story, and there ’ s more going on underneath the surface of the jacket. The four fingers are now four train lines in the Tokyo subway system, which has significance within the story. And then you have the colorless subway line intersecting with each of the other colors, which basically he does later on in the story. He catches up with each of these people to find out why they treated him the way they did. And so this is the three-dimensional finished product sitting on my desk in my office, and what I was hoping for here is that you ’ ll simply be allured by the mystery of what this looks like, and will want to read it to decode and find out and make more clear why it looks the way it does. [”The Visual Vernacular.”] This is a way to use a more familiar kind of mystery. What does this mean? This is what it means. [”Make it look like something else.”] The visual vernacular is the way we are used to seeing a certain thing applied to something else so that we see it in a different way. This is an approach I wanted to take to a book of essays by David Sedaris that had this title at the time. [”All the Beauty You Will Ever Need”] Now, the challenge here was that this title actually means nothing. It ’ s not connected to any of the essays in the book. It came to the author ’ s boyfriend in a dream. Thank you very much, so — — so usually, I am creating a design that is in some way based on the text, but this is all the text there is. So you ’ ve got this mysterious title that really doesn ’ t mean anything, so I was trying to think : Where might I see a bit of mysterious text that seems to mean something but doesn ’ t? And sure enough, not long after, one evening after a Chinese meal, this arrived, and I thought,”Ah, bing, ideagasm!”I ’ ve always loved the hilariously mysterious tropes of fortune cookies that seem to mean something extremely deep but when you think about them — if you think about them — they really don ’ t. This says,”Hardly anyone knows how much is gained by ignoring the future.”Thank you. But we can take this visual vernacular and apply it to Mr. Sedaris, and we are so familiar with how fortune cookie fortunes look that we don ’ t even need the bits of the cookie anymore. We ’ re just seeing this strange thing and we know we love David Sedaris, and so we ’ re hoping that we ’ re in for a good time. [”’ Fraud ’ Essays by David Rakoff”] David

Rakoff was a wonderful writer and he called his first book”Fraud”because he was getting sent on assignments by magazines to do things that he was not equipped to do. So he was this skinny little urban guy and GQ magazine would send him down the Colorado River whitewater rafting to see if he would survive. And then he would write about it, and he felt that he was a fraud and that he was misrepresenting himself. And so I wanted the cover of this book to also misrepresent itself and then somehow show a reader reacting to it. This led me to graffiti. I ’ m **fascinated** by graffiti. I think anybody who lives in an urban environment encounters graffiti all the time, and there ’ s all different sorts of it. This is a **picture** I took on the Lower East Side of just a transformer box on the sidewalk and it ’ s been tagged like crazy. Now whether you look at this and think,”Oh, that ’ s a charming urban affectation,”or you look at it and say,”That ’ s illegal abuse of property,”the one thing I think we can all agree on is that you cannot read it. Right? There is no clear message here. There is another kind of graffiti that I find far more interesting, which I call editorial graffiti. This is a **picture** I took recently in the subway, and sometimes you see lots of prurient, stupid stuff, but I thought this was interesting, and this is a poster that is saying rah-rah Airbnb, and someone has taken a Magic Marker and has editorialized about what they think about it. And it got my attention. So I was thinking, how do we apply this to this book? So I get the book by this person, and I start reading it, and I ’ m thinking, this guy is not who he says he is ; he ’ s a fraud. And I get out a red Magic Marker, and out of frustration just scribble this across the front. Design done. And they went for it! Author liked it, publisher liked it, and that is how the book went out into the world, and it was really **fun** to see people reading this on the subway and walking around with it and what have you, and they all sort of looked like they were crazy. [”’ Perfidia ’ a novel by James Ellroy”] Okay, James Ellroy, amazing crime writer, a good friend, I ’ ve worked with him for many years. He is probably best known as the author of”The Black Dahlia”and”L. A. Confidential.”His most recent novel was called this, which is a very mysterious name that I ’ m sure a lot of people know what it means, but a lot of people don ’ t. And it ’ s a story about a Japanese-American detective in Los Angeles in 1941 investigating a murder. And then Pearl Harbor happens, and as if his life wasn ’ t difficult enough, now the race relations have really ratcheted up, and then the Japanese- American internment camps are quickly created, and there ’ s lots of tension and horrible stuff as he ’ s still trying to solve this murder. And so I did at first think very literally about this in terms of all right, we ’ ll take Pearl Harbor and we ’ ll add it to Los Angeles and we ’ ll make this apocalyptic dawn on the horizon of the city. And so that ’ s a **picture** from Pearl Harbor just grafted onto Los Angeles. My editor in chief said,”You know, it ’ s interesting but I think you can do better and I think you can make it simpler.”And so I went back to the **drawing** board, as I often do. But also, being alive to my surroundings, I work in a high-rise in Midtown, and every night, before I leave the office, I have to push this button to get out, and the big heavy glass doors open and I can get onto the elevator. And one night, all of a sudden, I looked at this and I saw it in a way that I hadn ’ t really noticed it before. Big red circle, danger. And I thought this was so obvious that it had to have been done a zillion times, and so I did a Google image search, and I couldn ’ t find another book cover that looked quite like this, and so this is really what solved the problem, and graphically it ’ s more interesting and creates a bigger tension between the idea of a certain kind of sunrise coming up over L. A. and America. [”’ Gulp ’ A tour of the human digestive system by Mary Roach.”] Mary Roach is an amazing writer who takes potentially mundane scientific subjects and makes them not mundane at all ; she makes them really **fun**. So in this particular case, it ’ s about the human digestive system. So I

’ m trying to figure out what is the cover of this book going to be. This is a self-portrait. Every morning I look at myself in the medicine cabinet mirror to see if my tongue is black. And if it ’ s not, I ’ m good to go. I recommend you all do this. But I also started thinking, here ’ s our introduction. Right? Into the human digestive system. But I think what we can all agree on is that actual photographs of human mouths, at least based on this, are off-putting. So for the cover, then, I had this illustration done which is literally more palatable and reminds us that it ’ s best to approach the digestive system from this end. I don ’ t even have to complete the sentence. All right. [”Unuseful mystery”] What happens when clarity and mystery get mixed up? And we see this all the time. This is what I call unuseful mystery. I go down into the subway — I take the subway a lot — and this piece of paper is **taped** to a girder. Right? And now I ’ m thinking, uh-oh, and the train ’ s about to come and I ’ m trying to figure out what this means, and thanks a lot. Part of the problem here is that they ’ ve compartmentalized the information in a way they think is helpful, and frankly, I don ’ t think it is at all. So this is mystery we do not need. What we need is useful clarity, so just for **fun**, I redesigned this. This is using all the same elements. Thank you. I am still waiting for a call from the MTA. You know, I ’ m actually not even using more colors than they use. They just didn ’ t even bother to make the 4 and the 5 green, those idiots. So the first thing we see is that there is a service change, and then, in two complete sentences with a beginning, a middle and an end, it tells us what the change is and what ’ s going to be happening. Call me crazy! [”Useful mystery”] All right. Now, here is a piece of mystery that I love : packaging. This redesign of the Diet Coke can by Turner Duckworth is to me truly a piece of art. It ’ s a work of art. It ’ s beautiful. But part of what makes it so heartening to me as a designer is that he ’ s taken the visual vernacular of Diet Coke — the typefaces, the colors, the silver background — and he ’ s reduced them to their most essential parts, so it ’ s like going back to the Charlie Brown face. It ’ s like, how can you give them just enough information so they know what it is but giving them the credit for the knowledge that they already have about this thing? It looks great, and you would go into a delicatessen and all of a sudden see that on the shelf, and it ’ s wonderful. Which makes the next thing — [”Unuseful clarity”] — all the more disheartening, at least to me. So okay, again, going back down into the subway, after this came out, these are **pictures** that I took. Times Square subway station : Coca-Cola has bought out the entire thing for advertising. Okay? And maybe some of you know where this is going. Ahem.”You moved to New York with the clothes on your back, the cash in your pocket, and your eyes on the prize. You ’ re on Coke.””You moved to New York with an MBA, one **clean** suit, and an extremely firm handshake. You ’ re on Coke.”These are real! Not even the support beams were spared, except they switched into Yoda mode.”Coke you ’ re on.”[”Excuse me, I ’ m on WHAT??”] This campaign was a huge misstep. It was pulled almost instantly due to consumer backlash and all sorts of unflattering parodies on the web — — and also that dot next to”You ’ re on,”that ’ s not a period, that ’ s a trademark. So thanks a lot. So to me, this was just so bizarre about how they could get the packaging so mysteriously beautiful and perfect and the message so unbearably, clearly wrong. It was just incredible to me. So I just hope that I ’ ve been able to share with you some of my insights on the uses of clarity and mystery in my work, and maybe how you might decide to be more clear in your life, or maybe to be a bit more mysterious and not so over- sharing. And if there ’ s just one thing that I leave you with from this talk, I hope it ’ s this : Blih blih blih blah. Blah blah blih blih. [”’ Judge This, ’ Chip Kidd”] Blih blih blah blah blah. Blah blah blah. Blah blah.

Text Sample 719: POS Dim 5 – 2016 – Score 8.00 – 2016/t001286.txt

There ’ s this fact that I love that I read somewhere once, that one of the things that ’ s contributed to homo sapiens ’ success as a species is our lack of body hair â that our hairlessness, our nakedness combined with our invention of clothing, gives us the ability to modulate our body temperature and thus be able to survive in any climate we choose. And now we ’ ve evolved to the point where we can ’ t survive without clothing. And it ’ s more than just utility, now it ’ s a communication. Everything that we choose to put on is a narrative, a story about where we ’ ve been, what we ’ re doing, who we want to be. I was a lonely kid. I didn ’ t have an easy time finding friends to play with, and I ended up making a lot of my own play. I made a lot of my own toys. It began with ice cream. There was a Baskin-Robbins in my hometown, and they served ice cream from behind the counter in these giant, five-gallon, cardboard tubs. And someone told me â I was eight years old â someone told me that when they were done with those tubs, they washed them out and kept them in the back, and if you asked they would give you one. It took me a couple of weeks to work up the courage, but I did, and they did. They gave me one â I went home with this beautiful cardboard tub. I was trying to figure out what I could do with this exotic material â **metal** ring, top and **bottom**. I started turning it over in my head, and I realized,”Wait a minute â my head actually fits inside this thing.”Yeah, I cut a hole out, I put some acetate in there and I made myself a space helmet. I needed a place to wear the space helmet, so I found a refrigerator box a couple blocks from home. I pushed it home, and in my parents ’ **guest** room closet, I turned it into a spaceship. I started with a control **panel** out of cardboard. I cut a hole for a radar screen and put a flashlight underneath it to light it. I put a view screen up, which I offset off the back wall â and this is where I thought I was being really clever â without permission, I painted the back wall of the closet black and put a star field, which I lit up with some Christmas lights I found in the attic, and I went on some space missions. A couple years later, the movie”Jaws”came out. I was way too young to see it, but I was caught up in”Jaws”fever, like everyone else in America at the time. There was a store in my town that had a”Jaws”costume in their window, and my mom must have overheard me talking to someone about how awesome I thought this costume was, because a couple days before Halloween, she blew my freaking mind by giving me this”Jaws”costume. Now, I recognize it ’ s a bit of a trope for people of a certain age to complain that kids these days have no idea how good they have it, but let me just show you a random sampling of entry-level kids ’ costumes you can buy online right now...... and this is the”Jaws”costume my mom bought for me. This is a paper-thin shark face and a vinyl bib with the poster of”Jaws”on it. And I loved it. A couple years later, my dad took me to a **film** called”Excalibur.”I actually got him to take me to it twice, which is no small thing, because it is a hard, R-rated **film**. But it wasn ’ t the blood and guts or the boobs that made me want to go see it again. They helped â It was the armor. The armor in”Excalibur”was intoxicatingly beautiful to me. These were literally knights in shining, mirror-polished armor. And moreover, the knights in”Excalibur”wear their armor everywhere. All the time â they wear it at dinner, they wear it to bed. I was like,”Are they reading my mind? I want to wear armor all the time!”So I went back to my favorite material, the gateway drug for making, corrugated cardboard, and I made myself a suit of armor, replete with the neck shields and a white horse. Now that I ’ ve oversold it, here ’ s a **picture** of the armor that I made. Now, this is only the first suit of armor I made inspired by”Excalibur.”A couple of years later, I convinced my dad to embark on making me a proper suit of armor. Over about a month, he graduated me from cardboard to roofing aluminum called flashing and still, one of my all-time favorite attachment materials, POP rivets. We carefully, over that month, constructed an articulated suit of aluminum armor with compound **curves**. We drilled holes in the helmet so that I could breathe, and I finished just in time for Halloween and wore it to school. Now, this is the one thing in this talk that I don ’ t have a slide to show you, because no photo exists of this armor. I did wear it to school, there was a yearbook photographer patrolling the halls, but he never found me, for reasons that are about to become clear. There were things I didn ’ t anticipate about wearing a complete suit of aluminum armor to school. In third period math, I was standing in the back of class, and I ’ m standing in the back of class because the armor did not allow me to sit down. This is the first thing I didn ’ t anticipate. And then my teacher looks at me sort of concerned about halfway through the class and says,”Are you feeling OK?”I ’ m thinking,”Are you kidding? Am I feeling OK? I ’ m wearing a suit of armor! I am having the time of my â”And I ’ m just about to tell her how great I feel, when the classroom starts to list to the left and disappear down this long tunnel, and then I woke up in the nurse ’ s office. I had passed out from heat exhaustion, wearing the armor. And when I woke up, I wasn ’ t embarrassed about having passed out in front of my class, I was wondering,”Who took my armor? Where ’ s my armor?”OK, fast-forward a whole bunch of years, some colleagues and I get hired to make a show for Discovery Channel, called”MythBusters.”And over 14 years, I learn on the job how to build experimental methodologies and how to tell stories about them for television. I also learn early on that costuming can play a key role in this storytelling. I use costumes to add humor, comedy, color and narrative clarity to the stories we ’ re telling. And then we do an episode called”Dumpster Diving,”and I learn a little bit more about the deeper implications of what costuming means to me. In the episode”Dumpster Diving,”the question we were trying to answer is : Is jumping into a dumpster as safe as the movies would lead you to believe? The episode was going to have two distinct parts to it. One was where we get trained to jump off buildings by a stuntman into an air bag. And the second was the graduation to the experiment : we ’ d fill a dumpster full of material and we ’ d jump into it. I wanted to visually separate these two elements, and I thought,”Well, for the first part we ’ re training, so we should wear sweatsuits â Oh! Let ’ s put ’ Stunt Trainee ’ on the back of the sweatsuits. That ’ s for the training.”But for the second part, I wanted something really visually striking â”I know! I ’ ll dress as Neo from ’ The Matrix. ’”So I went to Haight Street. I bought some beautiful knee-high, buckle boots. I found a long, flowing coat on eBay. I got sunglasses, which I had to wear contact lenses in order to wear. The day of the experiment shoot comes up, and I step out of my car in this costume, and my crew takes a look at me... and start suppressing their church giggles. They ’ re like,”.”And I feel two distinct things at this moment. I feel total embarrassment over the fact that it ’ s so nakedly clear to my crew that I ’ m completely into wearing this costume. But the producer in my mind reminds myself that in the high-speed shot in slow- mo, that flowing coat is going to look beautiful behind me. Five years into the”MythBusters”run, we got invited to appear at San Diego Comic-Con. I ’ d known about Comic-Con for years and never had time to go. This was the big leagues â this was costuming mecca. People fly in from all over the world to show their amazing creations on the floor in San Diego. And I wanted to participate. I decided that I would put together an elaborate costume that covered me completely, and I would walk the floor of San Diego Comic-Con anonymously. The costume I chose? Hellboy. That ’ s not my costume, that ’ s actually Hellboy. But I spent months assembling the most screen-accurate Hellboy costume I could, from the boots to the belt to the pants to the right hand of doom. I found a guy who made a prosthetic Hellboy head and chest and I put them on. I even had contact lenses made in my prescription. I wore it onto the floor at Comic-Con and I can ' t even tell you how balls hot it was in that costume. Sweating! I should ' ve remembered this. I ' m sweating buckets and the contact lenses hurt my eyes, and none of it matters because I ' m totally in love. Not just with the process of putting on this costume and walking the floor, but also with the community of other costumers. It ' s not called costuming at Cons, it ' s called"cosplay."Now ostensibly, cosplay means people who dress up as their favorite characters from **film** and television and especially anime, but it is so much more than that. These aren ' t just people who find a costume and put it on â they mash them up. They bend them to their will. They change them to be the characters they want to be in those productions. They ' re super clever and genius. They let their freak flag fly and it ' s beautiful. But more than that, they rehearse their costumes. At Comic-Con or any other Con, you don ' t just take **pictures** of people walking around. You go up and say,"Hey, I like your costume, can I take your **picture**?"And then you give them time to get into their pose. They ' ve worked hard on their pose to make their costume look great for your camera. And it ' s so beautiful to watch. And I take this to heart. At subsequent Cons, I learn Heath Ledger ' s shambling walk as the Joker from"The Dark Knight."I learn how to be a scary Ringwraith from"Lord of the Rings,"and I actually frighten some children. I learned that"hrr hrr hrr"â that head laugh that Chewbacca does. And then I dressed up as No-Face from"Spirited Away."If you don ' t know about"Spirited Away"and its director, Hayao Miyazaki, first of all, you ' re welcome. This is a masterpiece, and one of my all-time favorite **films**. It ' s about a young girl named Chihiro who gets lost in the spirit world in an abandoned Japanese theme **park**. And she finds her way back out again with the help of a couple of friends she makes â a captured dragon named Haku and a lonely demon named No-Face. No-Face is lonely and he wants to make friends, and he thinks the way to do it is by luring them to him and producing gold in his hand. But this doesn ' t go very well, and so he ends up going on kind of a rampage until Chihiro saves him, rescues him. So I put together a No-Face costume, and I wore it on the floor at Comic-Con. And I very carefully practiced No-Face ' s gestures. I resolved I would not speak in this costume at all. When people asked to take my **picture**, I would nod and I would shyly stand next to them. They would take the **picture** and then I would secret out from behind my robe a chocolate gold coin. And at the end of the photo process, I ' d make it appear for them. Ah, ah ah! â like that. And people were freaking out."Holy crap! Gold from No-Face! Oh my god, this is so cool!"And I ' m feeling and I ' m walking the floor and it ' s fantastic. And about 15 minutes in something happens. Somebody grabs my hand, and they put a coin back into it. And I think maybe they ' re giving me a coin as a return gift, but no, this is one of the coins that I ' d given away. I don ' t know why. And I keep on going, I take some more **pictures**. And then it happens again. Understand, I can ' t see anything inside this costume. I can see through the mouth â I can see people ' s shoes. I can hear what they ' re saying and I can see their **feet**. But the third time someone gives me back a coin, I want to know what ' s going on. So I sort of tilt my head back to get a better view, and what I see is someone walking away from me going like this. And then it hits me : it ' s bad luck to take gold from No-Face. In the **film**"Spirited Away,"bad luck befalls those who take gold from No-Face. This isn ' t a performer-audience relationship ; this is cosplay. We are, all of us on that floor, injecting ourselves into a narrative that meant something to us. And we ' re making it our own. We ' re connecting with something important inside of us. And the costumes are how we reveal ourselves to each other. Thank you.

Text Sample 720: POS Dim 5 – 2016 – Score 8.00 – 2016/t001333.txt

I ’ m driven by pure passion to create photographs that tell stories. Photography can be described as the recording of a single moment frozen within a fraction of time. Each moment or photograph represents a tangible piece of our memories as time passes. But what if you could capture more than one moment in a photograph? What if a photograph could actually collapse time, compressing the best moments of the day and the night seamlessly into one single image? I ’ ve created a concept called”Day to Night”and I believe it ’ s going to change the way you look at the world. I know it has for me. My process begins by photographing iconic locations, places that are part of what I call our collective memory. I photograph from a fixed vantage point, and I never move. I capture the fleeting moments of humanity and light as time passes. Photographing for anywhere from 15 to 30 hours and shooting over 1, 500 images, I then choose the best moments of the day and night. Using time as a guide, I seamlessly blend those best moments into one single photograph, visualizing our conscious journey with time. I can take you to Paris for a view from the Tournelle Bridge. And I can show you the early morning rowers along the River Seine. And simultaneously, I can show you Notre Dame aglow at night. And in between, I can show you the romance of the City of Light. I am essentially a street photographer from 50 **feet** in the air, and every single thing you see in this photograph actually happened on this day. Day to Night is a global project, and my work has always been about history. I ’ m **fascinated** by the concept of going to a place like Venice and actually seeing it during a specific event. And I decided I wanted to see the historical Regata, an event that ’ s actually been taking place since 1498. The boats and the costumes look exactly as they did then. And an important element that I really want you guys to understand is : this is not a timelapse, this is me photographing throughout the day and the night. I am a relentless collector of magical moments. And the thing that drives me is the fear of just missing one of them. The entire concept came about in 1996. LIFE Magazine commissioned me to create a panoramic photograph of the cast and crew of Baz Luhrmann ’ s **film** Romeo + Juliet. I got to the set and realized : it ’ s a square. So the only way I could actually create a panoramic was to shoot a collage of 250 single images. So I had DiCaprio and Claire Danes embracing. And as I pan my camera to the right, I noticed there was a mirror on the wall and I saw they were actually reflecting in it. And for that one moment, that one image I asked them,”Would you guys just kiss for this one **picture**?”And then I came back to my studio in New York, and I hand-glued these 250 images together and stood back and went,”Wow, this is so cool! I ’ m changing time in a photograph.”And that concept actually stayed with me for 13 years until technology finally has caught up to my dreams. This is an image I created of the Santa Monica Pier, Day to Night. And I ’ m going to show you a little video that gives you an idea of what it ’ s like being with me when I do these **pictures**. To start with, you have to understand that to get views like this, most of my time is spent up high, and I ’ m usually in a cherry picker or a crane. So this is a typical day, 12-18 hours, non-stop capturing the entire day unfold. One of the things that ’ s great is I love to people-watch. And trust me when I tell you, this is the greatest seat in the house to have. But this is really how I go about creating these photographs. So once I decide on my view and the location, I have to decide where day begins and night ends. And that ’ s what I call the time vector. Einstein described time as a fabric. Think of the surface of a trampoline : it warps and stretches with gravity. I see time as a fabric as well, except I take that fabric and flatten it, compress it into single plane. One of the unique aspects of this work is also, if you look at all my **pictures**, the time vector changes : sometimes I ' ll go left to right, sometimes front to back, up or down, even diagonally. I am exploring the space-time continuum within a two-dimensional still photograph. Now when I do these **pictures**, it ' s literally like a real-time puzzle going on in my mind. I build a photograph based on time, and this is what I call the master plate. This can take us several months to complete. The **fun** thing about this work is I have absolutely zero control when I get up there on any given day and capture photographs. So I never know who ' s going to be in the **picture**, if it ' s going to be a great sunrise or sunset — no control. It ' s at the end of the process, if I ' ve had a really great day and everything remained the same, that I then decide who ' s in and who ' s out, and it ' s all based on time. I ' ll take those best moments that I pick over a month of editing and they get seamlessly blended into the master plate. I ' m compressing the day and night as I saw it, creating a unique harmony between these two very discordant worlds. Painting has always been a really important influence in all my work and I ' ve always been a huge fan of Albert Bierstadt, the great Hudson River School painter. He inspired a recent series that I did on the National Parks. This is Bierstadt ' s Yosemite Valley. So this is the photograph I created of Yosemite. This is actually the cover story of the 2016 January issue of National Geographic. I photographed for over 30 hours in this **picture**. I was literally on the side of a cliff, capturing the stars and the moonlight as it transitions, the moonlight lighting El Capitan. And I also captured this transition of time throughout the landscape. The best part is obviously seeing the magical moments of humanity as time changed — from day into night. And on a personal note, I actually had a photocopy of Bierstadt ' s painting in my pocket. And when that sun started to rise in the **valley**, I started to literally shake with excitement because I looked at the painting and I go,"Oh my god, I ' m getting Bierstadt ' s exact same lighting 100 years earlier."Day to Night is about all the things, it ' s like a compilation of all the things I love about the medium of photography. It ' s about landscape, it ' s about street photography, it ' s about color, it ' s about architecture, perspective, scale — and, especially, history. This is one of the most historical moments I ' ve been able to photograph, the 2013 Presidential Inauguration of Barack Obama. And if you look closely in this **picture**, you can actually see time changing in those large television sets. You can see Michelle waiting with the children, the president now greets the crowd, he takes his oath, and now he ' s speaking to the people. There ' s so many challenging aspects when I create photographs like this. For this particular photograph, I was in a 50-foot scissor lift up in the air and it was not very stable. So every time my assistant and I shifted our weight, our horizon line shifted. So for every **picture** you see, and there were about 1, 800 in this **picture**, we both had to **tape** our **feet** into position every time I clicked the shutter. I ' ve learned so many extraordinary things doing this work. I think the two most important are patience and the power of observation. When you photograph a city like New York from above, I discovered that those people in cars that I sort of live with everyday, they don ' t look like people in cars anymore. They feel like a giant school of fish, it was a form of emergent behavior. And when people describe the energy of New York, I think this photograph begins to really capture that. When you look closer in my work, you can see there ' s stories going on. You realize that Times Square is a canyon, it ' s shadow and it ' s sunlight. So I decided, in this photograph, I would checkerboard time. So wherever the shadows are, it ' s night and wherever the sun is, it ' s actually day. Time is this extraordinary thing that we never can really wrap our heads around. But in a very unique and special way, I believe these photographs begin to put a face on time. They embody a new metaphysical visual reality. When you spend 15 hours looking at a place, you ’ re going to see things a little differently than if you or I walked up with our camera, took a **picture**, and then walked away. This was a perfect example. I call it”Sacr-Coeur Selfie.”I watched over 15 hours all these people not even look at Sacr-Coeur. They were more interested in using it as a backdrop. They would walk up, take a **picture**, and then walk away. And I found this to be an absolutely extraordinary example, a powerful disconnect between what we think the human experience is versus what the human experience is evolving into. The act of sharing has suddenly become more important than the experience itself. And finally, my most recent image, which has such a special meaning for me personally : this is the Serengeti National Park in Tanzania. And this is photographed in the middle of the Seronera, this is not a reserve. I went specifically during the peak migration to hopefully capture the most diverse range of animals. Unfortunately, when we got there, there was a drought going on during the peak migration, a five-week drought. So all the animals were drawn to the water. I found this one watering hole, and felt if everything remained the same way it was behaving, I had a real opportunity to capture something unique. We spent three days studying it, and nothing could have prepared me for what I witnessed during our shoot day. I photographed for 26 hours in a sealed crocodile blind, 18 **feet** in the air. What I witnessed was unimaginable. Frankly, it was Biblical. We saw, for 26 hours, all these competitive species share a single resource called water. The same resource that humanity is supposed to have wars over during the next 50 years. The animals never even grunted at each other. They seem to understand something that we humans don ’ t. That this precious resource called water is something we all have to share. When I created this **picture**, I realized that Day to Night is really a new way of seeing, compressing time, exploring the space-time continuum within a photograph. As technology evolves along with photography, photographs will not only communicate a deeper meaning of time and memory, but they will compose a new narrative of untold stories, creating a timeless window into our world. Thank you.

Text Sample 721: POS Dim 5 – 2016 – Score 7.00 – 2016/t001399.txt

I was excited to be a part of the”Dream”theme, and then I found out I ’ m leading off the”Nightmare?”section of it. And certainly there are things about the climate crisis that qualify. And I have some bad news, but I have a lot more good news. I ’ m going to propose three questions and the answer to the first one necessarily involves a little bad news. But — hang on, because the answers to the second and third questions really are very positive. So the first question is,”Do we really have to change?”And of course, the Apollo Mission, among other things changed the environmental movement, really launched the modern environmental movement. 18 months after this Earthrise **picture** was first seen on earth, the first Earth Day was organized. And we learned a lot about ourselves looking back at our planet from space. And one of the things that we learned confirmed what the scientists have long told us. One of the most essential facts about the climate crisis has to do with the sky. As this **picture** illustrates, the sky is not the vast and limitless expanse that appears when we look up from the ground. It is a very thin shell of atmosphere surrounding the planet. That right now is the open sewer for our industrial civilization as it ’ s currently organized. We are spewing 110 million tons of heat-trapping global warming pollution into it every 24 hours, free of charge, go ahead. And there are many sources of the greenhouse **gases**, I ’ m certainly not going to go through them all. I ’ m going to focus on the main one, but agriculture is involved, diet is involved, population is involved. Management of forests, transportation, the oceans, the melting of the permafrost. But I ’ m going to focus on the heart of the problem, which is the fact that we still rely on dirty, carbon-based fuels for 85 percent of all the energy that our world burns every year. And you can see from this image that after World War II, the emission rates started really accelerating. And the accumulated amount of man- made, global warming pollution that is up in the atmosphere now traps as much extra heat energy as would be released by 400, 000 Hiroshima-class atomic bombs exploding every 24 hours, 365 days a year. Fact-checked over and over again, conservative, it ’ s the truth. Now it ’ s a big planet, but — that is a lot of energy, particularly when you multiply it 400, 000 times per day. And all that extra heat energy is heating up the atmosphere, the whole earth system. Let ’ s look at the atmosphere. This is a depiction of what we used to think of as the normal distribution of temperatures. The white represents normal temperature days ; 1951-1980 are arbitrarily chosen. The blue are cooler than average days, the red are warmer than average days. But the entire **curve** has moved to the right in the 1980s. And you ’ ll see in the lower right-hand corner the appearance of statistically significant numbers of extremely hot days. In the 90s, the **curve** shifted further. And in the last 10 years, you see the extremely hot days are now more numerous than the cooler than average days. In fact, they are 150 times more common on the surface of the earth than they were just 30 years ago. So we ’ re having record-breaking temperatures. Fourteen of the 15 of the hottest years ever measured with instruments have been in this young century. The hottest of all was last year. Last month was the 371st month in a row warmer than the 20th-century average. And for the first time, not only the warmest January, but for the first time, it was more than two degrees Fahrenheit warmer than the average. These higher temperatures are having an effect on animals, plants, people, ecosystems. But on a global basis, 93 percent of all the extra heat energy is trapped in the oceans. And the scientists can measure the heat buildup much more precisely now at all depths : deep, mid-ocean, the first few hundred meters. And this, too, is accelerating. It goes back more than a century. And more than half of the increase has been in the last 19 years. This has consequences. The first order of consequence : the ocean-based storms get stronger. Super Typhoon Haiyan went over areas of the Pacific five and a half degrees Fahrenheit warmer than normal before it slammed into Tacloban, as the most destructive storm ever to make landfall. Pope Francis, who has made such a difference to this whole issue, visited Tacloban right after that. Superstorm Sandy went over areas of the Atlantic nine degrees warmer than normal before slamming into New York and New Jersey. The second order of consequences are affecting all of us right now. The warmer oceans are evaporating much more water vapor into the skies. Average humidity worldwide has gone up four percent. And it creates these atmospheric rivers. The Brazilian scientists call them”flying rivers.”And they funnel all of that extra water vapor over the land where storm conditions trigger these massive record-breaking downpours. This is from Montana. Take a look at this storm last August. As it moves over Tucson, Arizona. It literally splashes off the city. These downpours are really unusual. Last July in Houston, Texas, it rained for two days, 162 billion gallons. That represents more than two days of the full flow of Niagara Falls in the middle of the city, which was, of course, paralyzed. These record downpours are creating historic floods and mudslides. This one is from Chile last year. And you ’ ll see that warehouse going by. There are oil tankers cars going by. This is from Spain last September, you could call this the running of the cars and trucks, I guess. Every night on the TV news now is like a nature hike through the Book of Revelation. I mean, really. The insurance industry has certainly noticed, the losses have been mounting up. They ' re not under any illusions about what ' s happening. And the causality requires a moment of discussion. We ' re used to thinking of linear cause and linear effect — one cause, one effect. This is systemic causation. As the great Kevin Trenberth says,"All storms are different now. There ' s so much extra energy in the atmosphere, there ' s so much extra water vapor. Every storm is different now."So, the same extra heat pulls the soil moisture out of the ground and causes these deeper, longer, more pervasive droughts and many of them are underway right now. It dries out the vegetation and causes more fires in the western part of North America. There ' s certainly been evidence of that, a lot of them. More lightning, as the heat energy builds up, there ' s a considerable amount of additional lightning also. These climate-related disasters also have geopolitical consequences and create instability. The climate-related historic drought that started in Syria in 2006 destroyed 60 percent of the farms in Syria, killed 80 percent of the livestock, and drove 1. 5 million climate refugees into the cities of Syria, where they collided with another 1. 5 million refugees from the Iraq War. And along with other factors, that opened the gates of **Hell** that people are trying to close now. The US Defense Department has long warned of consequences from the climate crisis, including refugees, food and water shortages and pandemic disease. Right now we ' re seeing microbial diseases from the tropics spread to the higher latitudes ; the transportation revolution has had a lot to do with this. But the changing conditions change the latitudes and the areas where these microbial diseases can become endemic and change the range of the vectors, like mosquitoes and ticks that carry them. The Zika epidemic now — we ' re better positioned in North America because it ' s still a little too cool and we have a better public health system. But when women in some regions of South and Central America are advised not to get pregnant for two years — that ' s something new, that ought to get our attention. The Lancet, one of the two greatest medical journals in the world, last summer labeled this a medical emergency now. And there are many factors because of it. This is also connected to the extinction crisis. We ' re in danger of losing 50 percent of all the living species on earth by the end of this century. And already, land-based plants and animals are now moving towards the poles at an average rate of 15 **feet** per day. Speaking of the North Pole, last December 29, the same storm that caused historic flooding in the American Midwest, raised temperatures at the North Pole 50 degrees Fahrenheit warmer than normal, causing the thawing of the North Pole in the middle of the long, dark, winter, polar night. And when the land- based ice of the Arctic melts, it raises sea level. Paul Nicklen ' s beautiful photograph from Svalbard illustrates this. It ' s more dangerous coming off Greenland and particularly, Antarctica. The 10 largest risk cities for sea-level rise by population are mostly in South and Southeast Asia. When you measure it by assets at risk, number one is Miami : three and a half trillion dollars at risk. Number three : New York and Newark. I was in Miami last fall during the supermoon, one of the highest high-tide days. And there were fish from the ocean swimming in some of the streets of Miami Beach and Fort Lauderdale and Del Rey. And this happens regularly during the highest-tide tides now. Not with rain — they call it"sunny-day flooding."It comes up through the storm sewers. And the Mayor of Miami speaks for many when he says it is long past time this can be viewed through a partisan lens. This is a crisis that ' s getting worse day by day. We have to move beyond partisanship. And I want to take a moment to honor these House Republicans — who had the courage last fall to step out and take a political risk, by telling the truth about the climate crisis. So the cost of the climate crisis is mounting up, there are many of these aspects I haven ’ t even mentioned. It ’ s an enormous burden. I ’ ll mention just one more, because the World Economic Forum last month in Davos, after their annual survey of 750 economists, said the climate crisis is now the number one risk to the global economy. So you get central bankers like Mark Carney, the head of the UK Central Bank, saying the vast majority of the carbon reserves are unburnable. Subprime carbon. I ’ m not going to remind you what happened with subprime mortgages, but it ’ s the same thing. If you look at all of the carbon fuels that were burned since the beginning of the industrial revolution, this is the quantity burned in the last 16 years. Here are all the ones that are proven and left on the books, 28 trillion dollars. The International Energy Agency says only this amount can be burned. So the rest, 22 trillion dollars — unburnable. Risk to the global economy. That ’ s why divestment movement makes practical sense and is not just a moral imperative. So the answer to the first question,”Must we change?”is yes, we have to change. Second question,”Can we change?”This is the exciting news! The best projections in the world 16 years ago were that by 2010, the world would be able to install 30 gigawatts of wind capacity. We beat that mark by 14 and a half times over. We see an exponential **curve** for wind installations now. We see the cost coming down dramatically. Some countries — take Germany, an industrial powerhouse with a climate not that different from Vancouver ’ s, by the way — one day last December, got 81 percent of all its energy from renewable resources, mainly solar and wind. A lot of countries are getting more than half on an average basis. More good news : energy storage, from batteries particularly, is now beginning to take off because the cost has been coming down very dramatically to solve the intermittency problem. With solar, the news is even more exciting! The best projections 14 years ago were that we would install one gigawatt per year by 2010. When 2010 came around, we beat that mark by 17 times over. Last year, we beat it by 58 times over. This year, we ’ re on track to beat it 68 times over. We ’ re going to win this. We are going to prevail. The exponential **curve** on solar is even steeper and more dramatic. When I came to this stage 10 years ago, this is where it was. We have seen a revolutionary breakthrough in the emergence of these exponential **curves**. And the cost has come down 10 percent per year for 30 years. And it ’ s continuing to come down. Now, the business community has certainly noticed this, because it ’ s crossing the grid parity point. Cheaper solar penetration rates are beginning to rise. Grid parity is understood as that line, that threshold, below which renewable electricity is cheaper than electricity from burning fossil fuels. That threshold is a little bit like the difference between 32 degrees Fahrenheit and 33 degrees Fahrenheit, or zero and one Celsius. It ’ s a difference of more than one degree, it ’ s the difference between ice and water. And it ’ s the difference between markets that are frozen up, and liquid flows of capital into new opportunities for investment. This is the biggest new business opportunity in the history of the world, and two-thirds of it is in the private sector. We are seeing an explosion of new investment. Starting in 2010, investments globally in renewable electricity generation surpassed fossils. The gap has been growing ever since. The projections for the future are even more dramatic, even though fossil energy is now still subsidized at a rate 40 times larger than renewables. And by the way, if you add the projections for nuclear on here, particularly if you assume that the work many are doing to try to break through to safer and more acceptable, more affordable forms of nuclear, this could change even more dramatically. So is there any precedent for such a rapid adoption of a new technology? Well, there are many, but let ’ s look at cell phones. In 1980, AT&T, then Ma Bell, commissioned McKinsey to do a global market survey of those clunky new mobile phones that appeared then.”How many can we sell by the year 2000?”they asked. McKinsey came back and said,”900, 000.”And sure enough, when the year 2000 arrived, they did sell 900, 000 — in the first three days. And for the balance of the year, they sold 120 times more. And now there are more cell connections than there are people in the world. So, why were they not only wrong, but way wrong? I ’ ve asked that question myself,”Why?”And I think the answer is in three parts. First, the cost came down much faster than anybody expected, even as the quality went up. And low-income countries, places that did not have a landline grid — they leap-frogged to the new technology. The big expansion has been in the developing counties. So what about the electricity grids in the developing world? Well, not so hot. And in many areas, they don ’ t exist. There are more people without any electricity at all in India than the entire population of the United States of America. So now we ’ re getting this : solar **panels** on grass huts and new business models that make it affordable. Muhammad Yunus financed this one in Bangladesh with micro- credit. This is a village market. Bangladesh is now the fastest-deploying country in the world : two systems per minute on average, night and day. And we have all we need : enough energy from the Sun comes to the earth every hour to supply the full world ’ s energy needs for an entire year. It ’ s actually a little bit less than an hour. So the answer to the second question,”Can we change?”is clearly”Yes.”And it ’ s an ever-firmer”yes.”Last question,”Will we change?”Paris really was a breakthrough, some of the provisions are binding and the regular reviews will matter a lot. But nations aren ’ t waiting, they ’ re going ahead. China has already announced that starting next year, they ’ re adopting a nationwide cap and trade system. They will likely link up with the European Union. The United States has already been changing. All of these coal plants were proposed in the next 10 years and canceled. All of these existing coal plants were retired. All of these coal plants have had their retirement announced. All of them — canceled. We are moving forward. Last year — if you look at all of the investment in new electricity generation in the United States, almost three-quarters was from renewable energy, mostly wind and solar. We are solving this crisis. The only question is : how long will it take to get there? So, it matters that a lot of people are organizing to insist on this change. Almost 400, 000 people marched in New York City before the UN special session on this. Many thousands, tens of thousands, marched in cities around the world. And so, I am extremely optimistic. As I said before, we are going to win this. I ’ ll finish with this story. When I was 13 years old, I heard that proposal by President Kennedy to land a person on the Moon and bring him back safely in 10 years. And I heard adults of that day and time say,”That ’ s reckless, expensive, may well fail.”But eight years and two months later, in the moment that Neil Armstrong set **foot** on the Moon, there was great cheer that went up in NASA ’ s mission control in Houston. Here ’ s a little-known fact about that : the average age of the systems engineers, the controllers in the room that day, was 26, which means, among other things, their age, when they heard that challenge, was 18. We now have a moral challenge that is in the tradition of others that we have faced. One of the greatest poets of the last century in the US, Wallace Stevens, wrote a line that has stayed with me :”After the final ’ no, ’ there comes a ’ yes, ’ and on that ’ yes ’, the future world depends.”When the abolitionists started their movement, they met with no after no after no. And then came a yes. The Women ’ s Suffrage and Women ’ s Rights Movement met endless no ’ s, until finally, there was a yes. The Civil Rights Movement, the movement against apartheid, and more recently, the movement for gay and lesbian rights here in the United States and elsewhere. After the final”no”comes a”yes.”When any great moral challenge is ultimately resolved into a binary choice between what is right and what is wrong, the outcome is fore-ordained because of who we are as human beings. Ninety-nine percent of us, that is where we are now and it is why we ’ re going to win this. We have everything we need. Some still doubt that we have the will to act, but I say the will to act is itself a renewable resource. Thank you very much. Chris Anderson : You ’ ve got this incredible combination of skills. You ’ ve got this scientist mind that can understand the full range of issues, and the ability to turn it into the most vivid language. No one else can do that, that ’ s why you led this thing. It was amazing to see it 10 years ago, it was amazing to see it now. Al Gore : Well, you ’ re nice to say that, Chris. But honestly, I have a lot of really good friends in the scientific community who are incredibly patient and who will sit there and explain this stuff to me over and over and over again until I can get it into simple enough language that I can understand it. And that ’ s the key to trying to communicate. CA : So, your talk. First part : terrifying, second part : incredibly hopeful. How do we know that all those graphs, all that progress, is enough to solve what you showed in the first part? AG : I think that the crossing — you know, I ’ ve only been in the business world for 15 years. But one of the things I ’ ve learned is that apparently it matters if a new product or service is more expensive than the incumbent, or cheaper than. Turns out, it makes a difference if it ’ s cheaper than. And when it crosses that line, then a lot of things really change. We are regularly surprised by these developments. The late Rudi Dornbusch, the great economist said,”Things take longer to happen then you think they will, and then they happen much faster than you thought they could.”I really think that ’ s where we are. Some people are using the phrase”The Solar Singularity”now, meaning when it gets below the grid parity, unsubsidized in most places, then it ’ s the default choice. Now, in one of the presentations yesterday, the jitney thing, there is an effort to use regulations to slow this down. And I just don ’ t think it ’ s going to work. There ’ s a woman in Atlanta, Debbie Dooley, who ’ s the Chairman of the Atlanta Tea Party. They enlisted her in this effort to put a tax on solar **panels** and regulations. And she had just put solar **panels** on her **roof** and she didn ’ t understand the request. And so she went and formed an alliance with the Sierra Club and they formed a new organization called the Green Tea Party. And they defeated the proposal. So, finally, the answer to your question is, this sounds a little corny and maybe it ’ s a clich, but 10 years ago — and Christiana referred to this — there are people in this audience who played an incredibly significant role in generating those exponential **curves**. And it didn ’ t work out economically for some of them, but it kick-started this global revolution. And what people in this audience do now with the knowledge that we are going to win this. But it matters a lot how fast we win it. CA : Al Gore, that was incredibly powerful. If this turns out to be the year, that the partisan thing changes, as you said, it ’ s no longer a partisan issue, but you bring along people from the other side together, backed by science, backed by these kinds of investment opportunities, backed by reason that you win the day — boy, that ’ s really exciting. Thank you so much. AG : Thank you so much for bringing me back to TED. Thank you!

Text Sample 722: POS Dim 5 – 2017 – Score 9.00 – 2017/t001061.txt

Imagine that when you walked in here this evening, you discovered that everybody in the room looked almost exactly the same : ageless, raceless, generically good-looking. That person sitting right next to you might have the most idiosyncratic inner life, but you don ’ t have a clue because we ’ re all wearing the same blank expression all the time. That is the kind of creepy transformation that is taking over cities, only it applies to buildings, not people. Cities are full of roughness and shadow, texture and color. You can still find architectural surfaces of great individuality and character in apartment buildings in Riga and Yemen, social housing in Vienna, Hopi villages in Arizona, brownstones in New York, wooden houses in San Francisco. These aren ' t palaces or cathedrals. These are just ordinary residences expressing the ordinary splendor of cities. And the reason they ' re like that is that the need for shelter is so bound up with the human desire for beauty. Their rough surfaces give us a touchable city. Right? Streets that you can read by running your fingers over brick and stone. But that ' s getting harder to do, because cities are becoming smooth. New downtowns sprout towers that are almost always made of concrete and steel and covered in glass. You can look at skylines all over the world — Houston, Guangzhou, Frankfurt — and you see the same army of high-gloss robots marching over the horizon. Now, just think of everything we lose when **architects** stop using the full range of available materials. When we reject granite and limestone and sandstone and wood and copper and terra-cotta and brick and wattle and plaster, we simplify architecture and we impoverish cities. It ' s as if you reduced all of the world ' s cuisines down to airline food. Chicken or pasta? But worse still, assemblies of glass towers like this one in Moscow suggest a disdain for the civic and communal aspects of urban living. Right? Buildings like these are intended to enrich their owners and tenants, but not necessarily the lives of the rest of us, those of us who navigate the spaces between the buildings. And we expect to do so for free. Shiny towers are an invasive species and they are choking our cities and killing off public space. We tend to think of a facade as being like makeup, a decorative layer applied at the end to a building that ' s effectively complete. But just because a facade is superficial doesn ' t mean it ' s not also deep. Let me give you an example of how a city ' s surfaces affect the way we live in it. When I visited Salamanca in Spain, I gravitated to the Plaza Mayor at all hours of the day. Early in the morning, sunlight rakes the facades, sharpening shadows, and at night, lamplight segments the buildings into hundreds of distinct areas, balconies and windows and arcades, each one a separate pocket of visual activity. That detail and depth, that glamour gives the plaza a theatrical quality. It becomes a stage where the generations can meet. You have teenagers sprawling on the pavers, seniors monopolizing the benches, and real life starts to look like an opera set. The curtain goes up on Salamanca. So just because I ' m talking about the exteriors of buildings, not form, not function, not structure, even so those surfaces give texture to our lives, because buildings create the spaces around them, and those spaces can draw people in or push them away. And the difference often has to do with the quality of those exteriors. So one contemporary equivalent of the Plaza Mayor in Salamanca is the Place de la Dfense in Paris, a windswept, glass-walled open space that office workers hurry through on the way from the metro to their cubicles but otherwise spend as little time in as possible. In the early 1980s, the **architect** Philip Johnson tried to recreate a gracious European plaza in Pittsburgh. This is PPG Place, a half acre of open space encircled by commercial buildings made of mirrored glass. And he ornamented those buildings with **metal** trim and bays and Gothic turrets which really pop on the skyline. But at ground level, the plaza feels like a black glass cage. I mean, sure, in summertime kids are running back and forth through the fountain and there ' s ice-skating in the winter, but it lacks the informality of a leisurely hangout. It ' s just not the sort of place you really want to just hang out and chat. Public spaces thrive or fail for many different reasons. Architecture is only one, but it ' s an important one. Some recent plazas like Federation Square in Melbourne or Superkilen in Copenhagen succeed because they combine old and new, rough and smooth, neutral and bright colors, and because they don ’ t rely excessively on glass. Now, I ’ m not against glass. It ’ s an ancient and versatile material. It ’ s easy to manufacture and transport and install and replace and **clean**. It comes in everything from enormous, ultraclear sheets to translucent bricks. New coatings make it change mood in the shifting light. In expensive cities like New York, it has the magical power of being able to multiply real estate values by allowing views, which is really the only commodity that developers have to offer to justify those surreal prices. In the middle of the 19th century, with the construction of the Crystal Palace in London, glass leapt to the top of the list of quintessentially modern substances. By the mid-20th century, it had come to dominate the downtowns of some American cities, largely through some really spectacular office buildings like Lever House in midtown Manhattan, designed by Skidmore, Owings and Merrill. Eventually, the technology advanced to the point where **architects** could design structures so transparent they practically disappear. And along the way, glass became the default material of the high-rise city, and there ’ s a very powerful reason for that. Because as the world ’ s populations converge on cities, the least fortunate pack into jerry-built shantytowns. But hundreds of millions of people need apartments and places to work in ever-larger buildings, so it makes economic sense to put up towers and wrap them in cheap and practical curtain walls. But glass has a limited ability to be expressive. This is a section of wall framing a plaza in the pre-Hispanic city of Mitla, in southern Mexico. Those 2, 000-year-old carvings make it clear that this was a place of high ritual significance. Today we look at those and we can see a historical and textural continuity between those carvings, the mountains all around and that church which is built on top of the ruins using stone plundered from the site. In nearby Oaxaca, even ordinary plaster buildings become canvasses for bright colors, political murals and sophisticated graphic arts. It ’ s an intricate, communicative language that an epidemic of glass would simply wipe out. The good news is that **architects** and developers have begun to rediscover the joys of texture without backing away from modernity. Some find innovative uses for old materials like brick and terra-cotta. Others invent new products like the molded **panels** that Snhetta used to give the San Francisco Museum of Modern Art that crinkly, sculptural quality. The **architect** Stefano Boeri even created living facades. This is his Vertical Forest, a pair of apartment towers in Milan, whose most visible feature is greenery. And Boeri is designing a version of this for Nanjing in China. And imagine if green facades were as ubiquitous as glass ones how much **cleaner** the air in Chinese cities would become. But the truth is that these are mostly one-offs, boutique projects, not easily reproduced at a global scale. And that is the point. When you use materials that have a local significance, you prevent cities from all looking the same. Copper has a long history in New York — the Statue of Liberty, the crown of the Woolworth Building — but it fell out of fashion for a long time until SHoP **Architects** used it to cover the American Copper Building, a pair of twisting towers on the East River. It ’ s not even finished and you can see the way sunset lights up that metallic facade, which will weather to green as it ages. Buildings can be like people. Their faces broadcast their experience. And that ’ s an important point, because when glass ages, you just replace it, and the building looks pretty much the same way it did before until eventually it ’ s demolished. Almost all other materials have the ability to absorb infusions of history and memory, and project it into the present. The firm Ennead clad the Utah Natural History Museum in Salt Lake City in copper and zinc, ores that have been mined in the area for 150 years and that also camouflage the building against the ochre hills so that you have a natural history

museum that reflects the region ’ s natural history. And when the Chinese Pritzker Prize winner Wang Shu was building a history **museum** in Ningbo, he didn ’ t just create a wrapper for the past, he built memory right into the walls by using brick and stones and shingles salvaged from villages that had been demolished. Now, **architects** can use glass in equally lyrical and inventive ways. Here in New York, two buildings, one by Jean Nouvel and this one by Frank Gehry face off across West 19th Street, and the play of reflections that they toss back and forth is like a symphony in light. But when a city defaults to glass as it grows, it becomes a hall of mirrors, disquieting and cold. After all, cities are places of concentrated variety where the world ’ s cultures and languages and lifestyles come together and mingle. So rather than encase all that variety and diversity in buildings of crushing sameness, we should have an architecture that honors the full range of the urban experience. Thank you.

Text Sample 723: POS Dim 5 – 2017 – Score 7.00 – 2017/t001083.txt

Chris Anderson : Elon, hey, welcome back to TED. It ’ s great to have you here. Elon Musk : Thanks for having me. CA : So, in the next half hour or so, we ’ re going to spend some time exploring your vision for what an exciting future might look like, which I guess makes the first question a little ironic : Why are you boring? EM : Yeah. I ask myself that frequently. We ’ re trying to dig a hole under LA, and this is to create the beginning of what will hopefully be a 3D network of tunnels to alleviate congestion. So right now, one of the most soul-destroying things is traffic. It affects people in every part of the world. It takes away so much of your life. It ’ s horrible. It ’ s particularly horrible in LA. CA : I think you ’ ve brought with you the first visualization that ’ s been shown of this. Can I show this? EM : Yeah, absolutely. So this is the first time — Just to show what we ’ re talking about. So a couple of key things that are important in having a 3D tunnel network. First of all, you have to be able to integrate the entrance and exit of the tunnel seamlessly into the fabric of the city. So by having an elevator, sort of a car skate, that ’ s on an elevator, you can integrate the entrance and exits to the tunnel network just by using two parking spaces. And then the car gets on a skate. There ’ s no speed limit here, so we ’ re designing this to be able to operate at 200 kilometers an hour. CA : How much? EM : 200 kilometers an hour, or about 130 miles per hour. So you should be able to get from, say, Westwood to LAX in six minutes — five, six minutes. CA : So possibly, initially done, it ’ s like on a sort of toll road-type basis. EM : Yeah. CA : Which, I guess, alleviates some traffic from the surface streets as well. EM : So, I don ’ t know if people noticed it in the video, but there ’ s no real limit to how many levels of tunnel you can have. You can go much further deep than you can go up. The deepest mines are much deeper than the tallest buildings are tall, so you can alleviate any arbitrary level of urban congestion with a 3D tunnel network. This is a very important point. So a key rebuttal to the tunnels is that if you add one layer of tunnels, that will simply alleviate congestion, it will get used up, and then you ’ ll be back where you started, back with congestion. But you can go to any arbitrary number of tunnels, any number of levels. CA : But people — seen traditionally, it ’ s incredibly expensive to dig, and that would block this idea. EM : Yeah. Well, they ’ re right. To give you an example, the LA subway extension, which is — I think it ’ s a two-and-a-half mile extension that was just completed for two billion dollars. So it ’ s roughly a billion dollars a mile to do the subway extension in LA. And this is not the highest utility subway in the world. So yeah, it ’ s quite difficult to dig tunnels normally. I think we need to have at least a tenfold improvement in the cost per mile of tunneling. CA :

And how could you achieve that? EM : Actually, if you just do two things, you can get to approximately an order of magnitude improvement, and I think you can go beyond that. So the first thing to do is to cut the tunnel diameter by a factor of two or more. So a single road lane tunnel according to regulations has to be 26 **feet**, maybe 28 **feet** in diameter to allow for crashes and emergency vehicles and sufficient ventilation for combustion engine cars. But if you shrink that diameter to what we ' re attempting, which is 12 **feet**, which is plenty to get an electric skate through, you drop the diameter by a factor of two and the cross-sectional area by a factor of four, and the tunneling cost scales with the cross-sectional area. So that ' s roughly a half-order of magnitude improvement right there. Then tunneling machines currently tunnel for half the time, then they stop, and then the rest of the time is putting in reinforcements for the tunnel wall. So if you design the machine instead to do continuous tunneling and reinforcing, that will give you a factor of two improvement. Combine that and that ' s a factor of eight. Also these machines are far from being at their power or thermal limits, so you can jack up the power to the machine substantially. I think you can get at least a factor of two, maybe a factor of four or five improvement on top of that. So I think there ' s a fairly straightforward series of steps to get somewhere in excess of an order of magnitude improvement in the cost per mile, and our target actually is — we ' ve got a pet snail called Gary, this is from Gary the snail from"South Park,"I mean, sorry,"SpongeBob SquarePants."So Gary is capable of — currently he ' s capable of going 14 times faster than a tunnel-boring machine. CA : You want to beat Gary. EM : We want to beat Gary. He ' s not a patient little fellow, and that will be victory. Victory is beating the snail. CA : But a lot of people imagining, dreaming about future cities, they imagine that actually the solution is flying cars, drones, etc. You go aboveground. Why isn ' t that a better solution? You save all that tunneling cost. EM : Right. I ' m in favor of flying things. Obviously, I do rockets, so I like things that fly. This is not some inherent bias against flying things, but there is a challenge with flying cars in that they ' ll be quite noisy, the wind force generated will be very high. Let ' s just say that if something ' s flying over your head, a whole bunch of flying cars going all over the place, that is not an anxiety-reducing situation. You don ' t think to yourself,"Well, I feel better about today."You ' re thinking,"Did they service their hubcap, or is it going to come off and guillotine me?"Things like that. CA : So you ' ve got this vision of future cities with these rich, 3D networks of tunnels underneath. Is there a tie-in here with Hyperloop? Could you apply these tunnels to use for this Hyperloop idea you released a few years ago. EM : Yeah, so we ' ve been sort of puttering around with the Hyperloop stuff for a while. We built a Hyperloop test track adjacent to SpaceX, just for a student competition, to encourage innovative ideas in transport. And it actually ends up being the biggest vacuum chamber in the world after the Large Hadron Collider, by volume. So it was quite **fun** to do that, but it was kind of a hobby thing, and then we think we might — so we ' ve built a little pusher car to push the student pods, but we ' re going to try seeing how fast we can make the pusher go if it ' s not pushing something. So we ' re cautiously optimistic we ' ll be able to be faster than the world ' s fastest bullet train even in a. 8-mile stretch. CA : Whoa. Good brakes. EM : Yeah, I mean, it ' s — yeah. It ' s either going to smash into tiny pieces or go quite fast. CA : But you can **picture**, then, a Hyperloop in a tunnel running quite long distances. EM : Exactly. And looking at tunneling technology, it turns out that in order to make a tunnel, you have to — In order to seal against the water table, you ' ve got to typically design a tunnel wall to be good to about five or six atmospheres. So to go to vacuum is only one atmosphere, or near-vacuum. So actually, it sort of turns out that automatically, if you build a tunnel that is good enough to resist the water table, it is automatically capable of holding vacuum. CA : Huh. EM : So, yeah. CA : And so you could actually **picture**, what kind of length tunnel is in Elon ' s future to running Hyperloop? EM : I think there ' s no real length limit. You could dig as much as you want. I think if you were to do something like a DC-to-New York Hyperloop, I think you ' d probably want to go underground the entire way because it ' s a high- density area. You ' re going under a lot of buildings and houses, and if you go deep enough, you cannot detect the tunnel. Sometimes people think, well, it ' s going to be pretty annoying to have a tunnel dug under my house. Like, if that tunnel is dug more than about three or four tunnel diameters beneath your house, you will not be able to detect it being dug at all. In fact, if you ' re able to detect the tunnel being dug, whatever device you are using, you can get a lot of money for that device from the Israeli military, who is trying to detect tunnels from Hamas, and from the US Customs and Border patrol that try and detect drug tunnels. So the reality is that earth is incredibly good at absorbing vibrations, and once the tunnel depth is below a certain level, it is undetectable. Maybe if you have a very sensitive seismic instrument, you might be able to detect it. CA : So you ' ve started a new company to do this called The Boring Company. Very nice. Very funny. EM : What ' s funny about that? CA : How much of your time is this? EM : It ' s maybe... two or three percent. CA : You ' ve called it a hobby. This is what an Elon Musk hobby looks like. EM : I mean, it really is, like — This is basically interns and people doing it part time. We bought some second-hand machinery. It ' s kind of puttering along, but it ' s making good progress, so — CA : So an even bigger part of your time is being spent on electrifying cars and transport through Tesla. Is one of the motivations for the tunneling project the realization that actually, in a world where cars are electric and where they ' re self-driving, there may end up being more cars on the roads on any given hour than there are now? EM : Yeah, exactly. A lot of people think that when you make cars autonomous, they ' ll be able to go faster and that will alleviate congestion. And to some degree that will be true, but once you have shared autonomy where it ' s much cheaper to go by car and you can go point to point, the affordability of going in a car will be better than that of a bus. Like, it will cost less than a bus ticket. So the amount of driving that will occur will be much greater with shared autonomy, and actually traffic will get far worse. CA : You started Tesla with the goal of persuading the world that electrification was the future of cars, and a few years ago, people were laughing at you. Now, not so much. EM : OK. I don ' t know. I don ' t know. CA : But isn ' t it true that pretty much every auto manufacturer has announced serious electrification plans for the short- to medium-term future? EM : Yeah. Yeah. I think almost every automaker has some electric vehicle program. They vary in seriousness. Some are very serious about transitioning entirely to electric, and some are just dabbling in it. And some, amazingly, are still pursuing fuel cells, but I think that won ' t last much longer. CA : But isn ' t there a sense, though, Elon, where you can now just declare victory and say, you know,"We did it."Let the world electrify, and you go on and focus on other stuff? EM : Yeah. I intend to stay with Tesla as far into the future as I can imagine, and there are a lot of exciting things that we have coming. Obviously the Model 3 is coming soon. We ' ll be unveiling the Tesla Semi truck. CA : OK, we ' re going to come to this. So Model 3, it ' s supposed to be coming in July-ish. EM : Yeah, it ' s looking quite good for starting production in July. CA : Wow. One of the things that people are so excited about is the fact that it ' s got autopilot. And you put out this video a while back showing what that technology would look like. EM : Yeah. CA : There ' s obviously autopilot in Model S right now. What are we seeing here? EM : Yeah, so this is using only cameras and GPS. So there ' s no

LIDAR or radar being used here. This is just using passive optical, which is essentially what a person uses. The whole road system is meant to be navigated with passive optical, or cameras, and so once you solve cameras or vision, then autonomy is solved. If you don ’ t solve vision, it ’ s not solved. So that ’ s why our focus is so heavily on having a vision neural net that ’ s very effective for road conditions. CA : Right. Many other people are going the LIDAR route. You want cameras plus radar is most of it. EM : You can absolutely be superhuman with just cameras. Like, you can probably do it ten times better than humans would, just cameras. CA : So the new cars being sold right now have eight cameras in them. They can ’ t yet do what that showed. When will they be able to? EM : I think we ’ re still on track for being able to go cross-country from LA to New York by the end of the year, fully autonomous. CA : OK, so by the end of the year, you ’ re saying, someone ’ s going to sit in a Tesla without touching the steering wheel, tap in”New York,”off it goes. EM : Yeah. CA : Won ’ t ever have to touch the wheel — by the end of 2017. EM : Yeah. Essentially, November or December of this year, we should be able to go all the way from a parking lot in California to a parking lot in New York, no controls touched at any point during the entire journey. CA : Amazing. But part of that is possible because you ’ ve already got a fleet of Teslas driving all these roads. You ’ re accumulating a huge amount of data of that national road system. EM : Yes, but the thing that will be interesting is that I ’ m actually fairly confident it will be able to do that route even if you change the route dynamically. So, it ’ s fairly easy — If you say I ’ m going to be really good at one specific route, that ’ s one thing, but it should be able to go, really be very good, certainly once you enter a highway, to go anywhere on the highway system in a given country. So it ’ s not sort of limited to LA to New York. We could change it and make it Seattle-Florida, that day, in real time. So you were going from LA to New York. Now go from LA to Toronto. CA : So leaving aside regulation for a second, in terms of the technology alone, the time when someone will be able to buy one of your cars and literally just take the hands off the wheel and go to sleep and wake up and find that they ’ ve arrived, how far away is that, to do that safely? EM : I think that ’ s about two years. So the real trick of it is not how do you make it work say 99. 9 percent of the time, because, like, if a car crashes one in a thousand times, then you ’ re probably still not going to be comfortable falling asleep. You shouldn ’ t be, certainly. It ’ s never going to be perfect. No system is going to be perfect, but if you say it ’ s perhaps — the car is unlikely to crash in a hundred lifetimes, or a thousand lifetimes, then people are like, OK, wow, if I were to live a thousand lives, I would still most likely never experience a crash, then that ’ s probably OK. CA : To sleep. I guess the big concern of yours is that people may actually get seduced too early to think that this is safe, and that you ’ ll have some horrible incident happen that puts things back. EM : Well, I think that the autonomy system is likely to at least mitigate the crash, except in rare circumstances. The thing to appreciate about vehicle safety is this is probabilistic. I mean, there ’ s some chance that any time a human driver gets in a car, that they will have an accident that is their fault. It ’ s never zero. So really the key threshold for autonomy is how much better does autonomy need to be than a person before you can rely on it? CA : But once you get literally safe hands-off driving, the power to disrupt the whole industry seems massive, because at that point you ’ ve spoken of people being able to buy a car, drops you off at work, and then you let it go and provide a sort of Uber-like service to other people, earn you money, maybe even cover the cost of your lease of that car, so you can kind of get a car for free. Is that really likely? EM : Yeah. Absolutely this is what will happen. So there will be a shared autonomy fleet where you buy your car and you can choose to use that car exclusively, you could choose to have it be used only by friends and family, only by other drivers who are rated five star, you can choose to share it sometimes but not other times. That ’ s 100 percent what will occur. It ’ s just a question of when. CA : Wow. So you mentioned the Semi and I think you ’ re planning to announce this in September, but I ’ m curious whether there ’ s anything you could show us today? EM : I will show you a teaser shot of the truck. It ’ s alive. CA : OK. EM : That ’ s definitely a case where we want to be cautious about the autonomy features. Yeah. CA : We can ’ t see that much of it, but it doesn ’ t look like just a little friendly neighborhood truck. It looks kind of badass. What sort of semi is this? EM : So this is a heavy duty, long-range semitruck. So it ’ s the highest weight capability and with long range. So essentially it ’ s meant to alleviate the heavy-duty trucking loads. And this is something which people do not today think is possible. They think the truck doesn ’ t have enough power or it doesn ’ t have enough range, and then with the Tesla Semi we want to show that no, an electric truck actually can out-torque any diesel semi. And if you had a tug-of-war competition, the Tesla Semi will tug the diesel semi uphill. CA : That ’ s pretty cool. And short term, these aren ’ t driverless. These are going to be trucks that truck drivers want to drive. EM : Yes. So what will be really **fun** about this is you have a flat torque RPM **curve** with an electric motor, whereas with a diesel motor or any kind of internal combustion engine car, you ’ ve got a torque RPM **curve** that looks like a hill. So this will be a very spry truck. You can drive this around like a sports car. There ’ s no gears. It ’ s, like, single speed. CA : There ’ s a great movie to be made here somewhere. I don ’ t know what it is and I don ’ t know that it ends well, but it ’ s a great movie. EM : It ’ s quite bizarre test-driving. When I was driving the test prototype for the first truck. It ’ s really weird, because you ’ re driving around and you ’ re just so nimble, and you ’ re in this giant truck. CA : Wait, you ’ ve already driven a prototype? EM : Yeah, I drove it around the parking lot, and I was like, this is crazy. CA : Wow. This is no vaporware. EM : It ’ s just like, driving this giant truck and making these mad maneuvers. CA : This is cool. OK, from a really badass **picture** to a kind of less badass **picture**. This is just a cute house from”Desperate Housewives”or something. What on earth is going on here? EM : Well, this illustrates the **picture** of the future that I think is how things will evolve. You ’ ve got an electric car in the driveway. If you look in between the electric car and the house, there are actually three Powerwalls stacked up against the side of the house, and then that house **roof** is a solar **roof**. So that ’ s an actual solar glass **roof**. CA : OK. EM : That ’ s a **picture** of a real — well, admittedly, it ’ s a real fake house. That ’ s a real fake house. CA : So these **roof** tiles, some of them have in them basically solar power, the ability to — EM : Yeah. Solar glass tiles where you can adjust the texture and the color to a very fine-grained level, and then there ’ s sort of microlouvers in the glass, such that when you ’ re looking at the **roof** from street level or close to street level, all the tiles look the same whether there is a solar cell behind it or not. So you have an even color from the ground level. If you were to look at it from a helicopter, you would be actually able to look through and see that some of the glass tiles have a solar cell behind them and some do not. You can ’ t tell from street level. CA : You put them in the ones that are likely to see a lot of sun, and that makes these **roofs** super affordable, right? They ’ re not that much more expensive than just tiling the **roof**. EM : Yeah. We ’ re very confident that the cost of the **roof** plus the cost of electricity — A solar glass **roof** will be less than the cost of a normal **roof** plus the cost of electricity. So in other words, this will be economically a no-brainer, we think it will look great, and it will last — We thought about having the warranty be infinity, but then people thought, well, that might sound like were just talking rubbish, but actually this is toughened glass. Well after the house has collapsed and there ' s nothing there, the glass tiles will still be there. CA : I mean, this is cool. So you ' re rolling this out in a couple week ' s time, I think, with four different roofing types. EM : Yeah, we ' re starting off with two, two initially, and the second two will be introduced early next year. CA : And what ' s the scale of ambition here? How many houses do you believe could end up having this type of roofing? EM : I think eventually almost all houses will have a solar **roof**. The thing is to consider the time scale here to be probably on the order of 40 or 50 years. So on average, a **roof** is replaced every 20 to 25 years. But you don ' t start replacing all **roofs** immediately. But eventually, if you say were to fast- forward to say 15 years from now, it will be unusual to have a **roof** that does not have solar. CA : Is there a mental model thing that people don ' t get here that because of the shift in the cost, the economics of solar power, most houses actually have enough sunlight on their **roof** pretty much to power all of their needs. If you could capture the power, it could pretty much power all their needs. You could go off-grid, kind of. EM : It depends on where you are and what the house size is relative to the **roof** area, but it ' s a fair statement to say that most houses in the US have enough **roof** area to power all the needs of the house. CA : So the key to the economics of the cars, the Semi, of these houses is the falling price of lithium-ion batteries, which you ' ve made a huge bet on as Tesla. In many ways, that ' s almost the core competency. And you ' ve decided that to really, like, own that competency, you just have to build the world ' s largest manufacturing plant to double the world ' s supply of lithium-ion batteries, with this guy. What is this? EM : Yeah, so that ' s the Gigafactory, progress so far on the Gigafactory. Eventually, you can sort of roughly see that there ' s sort of a diamond shape overall, and when it ' s fully done, it ' ll look like a giant diamond, or that ' s the idea behind it, and it ' s aligned on true north. It ' s a small detail. CA : And capable of producing, eventually, like a hundred gigawatt hours of batteries a year. EM : A hundred gigawatt hours. We think probably more, but yeah. CA : And they ' re actually being produced right now. EM : They ' re in production already. CA : You guys put out this video. I mean, is that speeded up? EM : That ' s the slowed down version. CA : How fast does it actually go? EM : Well, when it ' s running at full speed, you can ' t actually see the cells without a strobe light. It ' s just blur. CA : One of your core ideas, Elon, about what makes an exciting future is a future where we no longer feel guilty about energy. Help us **picture** this. How many Gigafactories, if you like, does it take to get us there? EM : It ' s about a hundred, roughly. It ' s not 10, it ' s not a thousand. Most likely a hundred. CA : See, I find this amazing. You can **picture** what it would take to move the world off this vast fossil fuel thing. It ' s like you ' re building one, it costs five billion dollars, or whatever, five to 10 billion dollars. Like, it ' s kind of cool that you can **picture** that project. And you ' re planning to do, at Tesla — announce another two this year. EM : I think we ' ll announce locations for somewhere between two and four Gigafactories later this year. Yeah, probably four. CA : Whoa. No more teasing from you for here? Like — where, continent? You can say no. EM : We need to address a global market. CA : OK. This is cool. I think we should talk for — Actually, global market. I ' m going to ask you one question about politics, only one. I ' m kind of sick of politics, but I do want to ask you this. You ' re on a body now giving advice to a guy — EM : Who? CA : Who has said he doesn ' t really believe in climate change, and there ' s a lot of people out there who think you shouldn ' t be doing that. They ' d like you to walk away from that. What would you say to them? EM : Well, I think that first of all, I ' m just on two advisory councils where the format consists of going around the room and asking people ' s opinion on things, and so there ' s like a meeting every month or two. That ' s the sum total of my contribution.

But I think to the degree that there are people in the room who are arguing in favor of doing something about climate change, or social issues, I ’ ve used the meetings I ’ ve had thus far to argue in favor of immigration and in favor of climate change. And if I hadn ’ t done that, that wasn ’ t on the agenda before. So maybe nothing will happen, but at least the words were said. CA : OK. So let ’ s talk SpaceX and Mars. Last time you were here, you spoke about what seemed like a kind of incredibly ambitious dream to develop rockets that were actually reusable. And you ’ ve only gone and done it. EM : Finally. It took a long time. CA : Talk us through this. What are we looking at here? EM : So this is one of our rocket boosters coming back from very high and fast in space. So just delivered the upper stage at high velocity. I think this might have been at sort of Mach 7 or so, delivery of the upper stage. CA : So that was a sped-up — EM : That was the slowed down version. CA : I thought that was the sped-up version. But I mean, that ’ s amazing, and several of these failed before you finally figured out how to do it, but now you ’ ve done this, what, five or six times? EM : We ’ re at eight or nine. CA : And for the first time, you ’ ve actually reflown one of the rockets that landed. EM : Yeah, so we landed the rocket booster and then prepped it for flight again and flew it again, so it ’ s the first reflight of an orbital booster where that reflight is relevant. So it ’ s important to appreciate that reusability is only relevant if it is rapid and complete. So like an aircraft or a car, the reusability is rapid and complete. You do not send your aircraft to Boeing in- between flights. CA : Right. So this is allowing you to dream of this really ambitious idea of sending many, many, many people to Mars in, what, 10 or 20 years time, I guess. EM : Yeah. CA : And you ’ ve designed this outrageous rocket to do it. Help us understand the scale of this thing. EM : Well, visually you can see that ’ s a person. Yeah, and that ’ s the vehicle. CA : So if that was a skyscraper, that ’ s like, did I read that, a 40-story skyscraper? EM : Probably a little more, yeah. The thrust level of this is really — This configuration is about four times the thrust of the Saturn V moon rocket. CA : Four times the thrust of the biggest rocket humanity ever created before. EM : Yeah. Yeah. CA : As one does. EM : Yeah. In units of 747, a 747 is only about a quarter of a million pounds of thrust, so for every 10 million pounds of thrust, there ’ s 40 747s. So this would be the thrust equivalent of 120 747s, with all engines blazing. CA : And so even with a machine designed to escape Earth ’ s gravity, I think you told me last time this thing could actually take a fully loaded 747, people, cargo, everything, into orbit. EM : Exactly. This can take a fully loaded 747 with maximum fuel, maximum passengers, maximum cargo on the 747 — this can take it as cargo. CA : So based on this, you presented recently this Interplanetary Transport System which is visualized this way. This is a scene you **picture** in, what, 30 years time? 20 years time? People walking into this rocket. EM : I ’ m hopeful it ’ s sort of an eight- to 10-year time frame. Aspirationally, that ’ s our target. Our internal targets are more aggressive, but I think — CA : OK. EM : While vehicle seems quite large and is large by comparison with other rockets, I think the future spacecraft will make this look like a rowboat. The future spaceships will be truly enormous. CA : Why, Elon? Why do we need to build a city on Mars with a million people on it in your lifetime, which I think is kind of what you ’ ve said you ’ d love to do? EM : I think it ’ s important to have a future that is inspiring and appealing. I just think there have to be reasons that you get up in the morning and you want to live. Like, why do you want to live? What ’ s the point? What inspires you? What do you love about the future? And if we ’ re not out there, if the future does not include being out there among the stars and being a multiplanet species, I find that it ’ s incredibly depressing if that ’ s not the future that we ’ re going to have. CA : People want to position this as an either or, that there are so many desperate things happening on the planet now from climate to poverty to, you know, you pick your issue. And this feels like a distraction. You shouldn ’ t be thinking about this. You should be solving what ’ s here and now. And to be fair, you ’ ve done a fair old bit to actually do that with your work on sustainable energy. But why not just do that? EM : I think there ’ s — I look at the future from the standpoint of probabilities. It ’ s like a branching stream of probabilities, and there are actions that we can take that affect those probabilities or that accelerate one thing or slow down another thing. I may introduce something new to the probability stream. Sustainable energy will happen no matter what. If there was no Tesla, if Tesla never existed, it would have to happen out of necessity. It ’ s tautological. If you don ’ t have sustainable energy, it means you have unsustainable energy. Eventually you will run out, and the laws of economics will drive civilization towards sustainable energy, inevitably. The fundamental value of a company like Tesla is the degree to which it accelerates the advent of sustainable energy, faster than it would otherwise occur. So when I think, like, what is the fundamental good of a company like Tesla, I would say, hopefully, if it accelerated that by a decade, potentially more than a decade, that would be quite a good thing to occur. That ’ s what I consider to be the fundamental aspirational good of Tesla. Then there ’ s becoming a multiplanet species and space-faring civilization. This is not inevitable. It ’ s very important to appreciate this is not inevitable. The sustainable energy future I think is largely inevitable, but being a space- faring civilization is definitely not inevitable. If you look at the progress in space, in 1969 you were able to send somebody to the moon. 1969. Then we had the Space Shuttle. The Space Shuttle could only take people to low Earth orbit. Then the Space Shuttle retired, and the United States could take no one to orbit. So that ’ s the trend. The trend is like down to nothing. People are mistaken when they think that technology just automatically improves. It does not automatically improve. It only improves if a lot of people work very hard to make it better, and actually it will, I think, by itself degrade, actually. You look at great civilizations like Ancient Egypt, and they were able to make the pyramids, and they forgot how to do that. And then the Romans, they built these incredible aqueducts. They forgot how to do it. CA : Elon, it almost seems, listening to you and looking at the different things you ’ ve done, that you ’ ve got this unique double motivation on everything that I find so interesting. One is this desire to work for humanity ’ s long-term good. The other is the desire to do something exciting. And often it feels like you feel like you need the one to drive the other. With Tesla, you want to have sustainable energy, so you made these super sexy, exciting cars to do it. Solar energy, we need to get there, so we need to make these beautiful **roofs**. We haven ’ t even spoken about your newest thing, which we don ’ t have time to do, but you want to save humanity from bad AI, and so you ’ re going to create this really cool brain-machine interface to give us all infinite memory and telepathy and so forth. And on Mars, it feels like what you ’ re saying is, yeah, we need to save humanity and have a backup plan, but also we need to inspire humanity, and this is a way to inspire. EM : I think the value of beauty and inspiration is very much underrated, no question. But I want to be clear. I ’ m not trying to be anyone ’ s savior. That is not the — I ’ m just trying to think about the future and not be sad. CA : Beautiful statement. I think everyone here would agree that it is not — None of this is going to happen inevitably. The fact that in your mind, you dream this stuff, you dream stuff that no one else would dare dream, or no one else would be capable of dreaming at the level of complexity that you do. The fact that you do that, Elon Musk, is a really remarkable thing. Thank you for helping us all to dream a bit bigger. EM : But you ’ ll tell me if it ever starts getting genuinely insane, right? CA : Thank you, Elon Musk. That was really, really fantastic. That was really fantastic.

Text Sample 724: POS Dim 5 – 2017 – Score 7.00 – 2017/t000756.txt

You may have noticed that I ’ m wearing two different shoes. It probably looks funny â it definitely feels funny â but I wanted to make a point. Let ’ s say my left shoe corresponds to a sustainable footprint, meaning we humans consume less natural resources than our planet can regenerate, and emit less carbon dioxide than our forests and oceans can reabsorb. That ’ s a stable and healthy condition. Today ’ s situation is more like my other shoe. It ’ s way oversized. At the second of August in 2017, we had already consumed all resources our planet can regenerate this year. This is like spending all your money until the 18th of a month and then needing a credit from the bank for the rest of the time. For sure, you can do this for some months in a row, but if you don ’ t change your behavior, sooner or later, you will run into big problems. We all know the devastating effects of this excessive exploitation : global warming, rising of the sea levels, melting of the glaciers and polar ice, increasingly extreme climate patterns and more. The enormity of this problem really frustrates me. What frustrates me even more is that there are solutions to this, but we keep doing things like we always did. Today I want to share with you how a new solar technology can contribute to a sustainable future of buildings. Buildings consume about 40 percent of our total energy demand, so tackling this consumption would significantly reduce our climate emissions. A building designed along sustainable principles can produce all the power it needs by itself. To achieve this, you first have to reduce the consumption as much as possible, by using well-insulated walls or windows, for instance. These technologies are commercially available. Then you need energy for warm water and heating. You can get this in a renewable way from the sun through solar-thermal installations or from the ground and air, with heat pumps. All of these technologies are available. Then you are left with the need for electricity. In principle, there are several ways to get renewable electricity, but how many buildings do you know which have a windmill on the **roof** or a water power plant in the garden? Probably not so many, because usually, it doesn ’ t make sense. But the sun provides abundant energy to our **roofs** and facades. The potential to harvest this energy at our buildings ’ surfaces is enormous. Let ’ s take Europe as an example. If you would utilize all areas which have a nice orientation to the sun and they ’ re not overly shaded, the power generated by photovoltaics would correspond to about 30 percent of our total energy demand. But today ’ s photovoltaics have some issues. They do offer a good cost-performance ratio, but they aren ’ t really flexible in terms of their design, and this makes aesthetics a challenge. People often imagine **pictures** like this when thinking about solar cells on buildings. This may work for solar farms, but when you think of buildings, of streets, of architecture, aesthetics does matter. This is the reason why we don ’ t see many solar cells on buildings today. They just don ’ t match. Our team is working on a totally different solar-cell technology, which is called organic photovoltaics or OPV. The term organic describes that the material used for light absorption and charge transport are mainly based on the element carbon, and not on **metals**. We utilize the mixture of a polymer which is set up by different repeating units, like the pearls in a pearl chain, and a small molecule which has the shape of a football and is called fullerene. These two compounds are mixed and dissolved to become an ink. And like ink, they can be printed with simple printing techniques like slot-die coating in a continuous roll-to-roll process on flexible substrates. The resulting thin layer is the active layer, absorbing the energy of the sun. This active layer is extremely effective. You only need a layer thickness of 0. 2 micrometers to absorb the energy of the sun. This is 100 times thinner than a human hair. To give you another example, take one kilogram of the basic polymer and use it to formulate the active ink. With this amount of ink, you can print a solar cell the size of a complete football field. So OPV is extremely material efficient, which I think is a **crucial** thing when talking about sustainability. After the printing process, you can have a solar module which could look like this... It looks a bit like a plastic foil and actually has many of its features. It ’ s lightweight... it ’ s bendable... and it ’ s semi-transparent. But it can harvest the energy of the sun outdoors and also of this indoor light, as you can see with this small, illuminated LED. You can use it in its plastic form and take advantage of its low weight and its bendability. The first is important when thinking about buildings in warmer regions. Here, the **roofs** are not designed to bear additionally heavy loads. They aren ’ t designed for snow in winter, for instance, so heavy silicon solar cells cannot be used for light harvesting, but these lightweight solar foils are very well suited. The bendability is important if you want to combine the solar cell with membrane architecture. Imagine the sails of the Sydney Opera as power plants. Alternatively, you can combine the solar foils with conventional construction materials like glass. Many glass facade elements contain a foil anyway, to create laminated safety glass. It ’ s not a big deal to add a second foil in the production process, but then the facade element contains the solar cell and can produce electricity. Besides looking nice, these integrated solar cells come along with two more important benefits. Do you remember the solar cell attached to a **roof** I showed before? In this case, we install the **roof** first, and as a second layer, the solar cell. This is adding on the installation costs. In the case of integrated solar cells, at the site of construction, only one element is installed, being at the same time the envelope of the building and the solar cell. Besides saving on the installation costs, this also saves resources, because the two functions are combined into one element. Earlier, I ’ ve talked about optics. I really like this solar **panel** â maybe you have different taste or different design needs... No problem. With the printing process, the solar cell can change its shape and design very easily. This will give the flexibility to **architects**, to planners and building owners, to integrate this electricity-producing technology as they wish. I want to stress that this is not just happening in the labs. It will take several more years to get to mass adoption, but we are at the edge of commercialization, meaning there are several companies out there with production lines. They are scaling up their capacities, and so are we, with the inks. This smaller footprint is much more comfortable. It is the right size, the right scale. We have to come back to the right scale when it comes to energy consumption. And making buildings carbon-neutral is an important part here. In Europe, we have the goal to decarbonize our building stock [by] 2050. I hope organic photovoltaics will be a big part of this. Here are a couple of examples. This is the first commercial installation of fully printed organic solar cells.”Commercial”means that the solar cells were printed on industrial equipment. The so-called”solar trees”were part of the German pavilion at the World Expo in Milan in 2015. They provided shading during the day and electricity for the lighting in the evening. You may wonder why this hexagonal shape was chosen for the solar cells. Easy answer : the **architects** wanted to have a specific shading pattern on the floor and asked for it, and then it was printed as requested. Being far from a real product, this free-form installation hooked the imagination of the visiting **architects** much more than we expected. This other application is closer to the projects and applications we are targeting. In an office building in SĂŁo Paulo, Brazil, semitransparent OPV **panels** are integrated into the glass facade, serving different needs. First, they provided shading for the meeting rooms behind. Second, the logo of the company is displayed in an innovative way. And of course, electricity is produced, reducing the energy footprint of the building. This is pointing towards a future where buildings are no longer energy consumers, but energy providers. I want to see solar cells seamlessly integrated into our building shells to be both resource-efficient and a pleasure to look at. For **roofs**, silicon solar cells will often continue to be a good solution. But to exploit the potential of all facades and other areas, such as semitransparent areas, **curved** surfaces and shadings, I believe organic photovoltaics can offer a significant contribution, and they can be made in any form **architects** and planners will want them to. Thank you.

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Arthur Brooks : Starting when I was eight or nine years old, I literally wanted to be the greatest French horn player in the world. That was what I always wanted to do. That was my whole life dream. Adam Grant : What was behind that? AB : I was ambitious. I wanted to play for a lot of people. And I guess that was just simply expressed in the only thing that I really knew how to do, the only thing I really cared about, which was music. AG : Arthur Brooks was a natural. He could read music before he could actually read. He played in youth orchestras as a kid, prestigious ones. Eventually he dropped out of college to go on tour. AB : I went on the road playing classical music. I did that for six years. I saw all 50 states from the back of a van. AG : Arthur recorded an album that played on classical radio stations across the country. But then something started to change. AB : I realized, about age 21, that things were starting to go south, and I couldn ’ t explain it. Ordinarily, classical musicians, they will see their skills increase through their 20s, and sometimes even their 30s. So whereas if I were a baseball pitcher, it would be totally normal that I would be losing heat off my fastball by my early 20s. As a French horn player, I shouldn ’ t have been seeing that, but I was. AG : To turn things around he practiced his heart out. Finally, he got his big break, his debut at Carnegie Hall. AB : So I got ready for this concert, and the first half of the concert, I have to say, was phenomenal. It was really one of the best concerts I ’ d ever given. And in the second half, I had this experience where I had to actually talk to the audience and I walked toward the front of the stage to address the crowd, and I wasn ’ t watching my **feet** and I slipped, or something collapsed, I still don ’ t know, and I fell five **feet** into the audience. AG : You literally fell off the stage? AB : I literally fell off the stage. It was just ridiculous. It was... I wrecked my instrument, I fell on my instrument, I hurt my arm. And, of course, I did what you would do when you ’ re 22 years old. I jumped up and said,”I ’ m okay, folks.”And I obviously was not okay. AG : Oh, no. AB : It was just so terrible, Adam. It was so terrible. AG : Arthur tried to shake off the feeling that falling off the stage was a bad omen. He spent the next decade practicing. He studied with the best tutors in Europe. He got a job at a symphony in Barcelona, but – AB : It was a wreck. It was a slow-moving wreck in my career itself. I was in decline. Finally had to pack it in. I had to admit it to myself that I wasn ’ t the horn player I had always dreamed of being. And right now, I ’ m talking about it with you and, you know, we ’ re laughing and all that, but I ’ ve got to tell you, it kills me, I still hate it. AG : Arthur didn ’ t plan to quit music so early. He simply stopped being great at his job. And in the years since, it ’ s made him wonder, does career decline happened to all of us eventually? And is there anything we can do about it? I ’ m Adam Grant, and this is WorkLife, my podcast with TED. I ’ m an

organizational psychologist. I study how to make work not suck. In this show, I ’ m inviting myself inside the minds of some truly unusual people, because they mastered something I wish everyone knew about work. Today, career staying power. It ’ s more about the choices you make than the age you reach. Thanks to Accenture for sponsoring this episode. Let ’ s be honest, we all have peaks and **valleys** in our careers, times when we hit our stride and do our best work, and times when we ’ re in a slump. Most of us are worried that as we age, our physical and mental skills will decline and we might enter into a permanent slump. And that fear is compounded by the fact that age is seldom seen as an asset in the modern workplace. Far too often, our older colleagues don ’ t get the respect they deserve. Arthur ’ s slump happened at a particularly early age, so he changed careers and became a public policy expert. AB : Basically I ’ m just an old fashioned, number-crunching, quantitative social scientist. AG : Given what he ’ s gone through, it ’ s not entirely surprising that his big area of focus now is career decline. AB : Career decline is a very personal and personalized phenomenon in which your skills are not what they once were. So you might be making as much, or more, money than you ever have before, but you know that you are not at the same level of technical adeptness that you had once attained, and / or you ’ re not on the trajectory to attain what you had hoped. AG : Maybe you felt your stomach lurch when someone younger than you has a smashing success in your field or your workplace. Or maybe you ’ ve been humming along feeling good about your work, until one day you noticed your skills don ’ t seem quite as sharp as they used to. In 2019, Arthur wrote a major story about this phenomenon in”The Atlantic.”The headline was”Your Professional Decline Is Coming Sooner Than You Think.”As in, long before you even start to see retirement on the horizon. AB : I mean the number one myth is that, you know, particularly in certain professions, like, let ’ s just say yours and mine, which is coming up with big ideas and sharing them, that that ’ ll never go into decline. Why? Because it doesn ’ t require, you know, strong biceps, and yet there ’ s overwhelming evidence that in idea professions people experience decline as well. They just don ’ t expect it. AG : What happens to our cognitive abilities as we age is not straightforward. It actually depends on what kind of mental skills we ’ re talking about. Psychologists have long distinguished between two kinds of intelligence : fluid and crystallized. Fluid intelligence is your raw processing power. It ’ s basically your IQ, your innate capacity for learning and problem solving. AB : That ’ s a big part of the modern idea economy. I ’ m going to come in here, I ’ m going to be a whizzbang person who comes up with brand new ideas. That ’ s actually what ’ s rewarded for young people because young people have high levels of fluid intelligence. It increases through your teens and 20s, and maybe through your 30s, although it tends to start declining for most people in their 30s and 40s, and it makes this original idea generation harder. Now that doesn ’ t mean that you ’ re going to be unsuccessful, but it means if you ’ re relying entirely on this fluid intelligence, you ’ re going to struggle. AG : A common refrain in Silicon Valley is that young people are just smarter. When it comes to fluid intelligence, that ’ s generally true. But the story changes with crystallized intelligence, your acquired ability to solve problems by applying your knowledge and experience. AB : Your ability to synthesize things that you didn ’ t necessarily invent, that ’ s crystallized intelligence and that tends to happen later. AG : Think about how crystallized intelligence comes in handy in your job. If you ’ re in sales, you ’ ve used it to tailor your pitches to different clients based on what you ’ ve learned about them in the past. If you ’ re in government, you ’ ve relied on it to get around red **tape**. If you ’ re in finance, maybe you ’ ve applied it to predict how markets are affected by crises. Crystallized intelligence peaks much later than fluid intelligence, in large part because our body of knowledge keeps growing. AB : Think of it like a library. Think of the brain as basically filling up with a lot of volumes. And that means you have a lot of crystallized intelligence. AG : Depth and breadth of knowledge aren ’ t the only benefits of aging. Evidence suggests that self-control and socio-emotional skills often keep climbing, too. Many of us continue gaining willpower and emotional intelligence well into our 40s and 50s. We ’ re still getting better at delaying gratification, reading other people ’ s emotions and regulating our own moods. Even into our 70s and 80s, we ’ re still improving at ignoring sunk costs, all of which can enhance our job performance. But Arthur doesn ’ t think that ’ s enough to keep us from declining. AB : Everybody falls in their game at some particular point. AG : At what age do you think the average person is going to start to face significant career decline? AB : I ’ ve got to give you the social scientist answer – it depends but if I were to guess, I ’ ve got to say that in your mid-40s is when you ’ re likely to see the peak of your originality and creativity, and you should be making plans in your 40s to how you can be the best creator with your crystallized intelligence, how you can be the best instructor, the best synthesizer, you can possibly be. AG : OK, so this is where I want to start disagreeing with you. So my read of the evidence is that people don ’ t inherently become less creative. One of the things that happens as people ’ s careers advance is they tend to produce less, just, you know, in terms of overall volume. I think it ’ s not ability. I think it ’ s motivation that ’ s dropping. I think, you know, a lot of people become complacent and they feel like they ’ ve achieved enough success, and so they may scale back a little bit. It may be that priorities outside of work become more important to them. And if we can maintain that motivation, there ’ s no inherent reason why the quality of creative output would decline. What do you think? AB : Hmm, I ’ m willing to entertain your hypothesis. AG : OK, cool. So there ’ s a study I really like of scientists, filmmakers and artists, showing that as they advance from their 20s and 30s toward their 50s and 60s, they do less work. The scientists run fewer experiments, the filmmakers don ’ t make as many movies and the artists produce fewer paintings. But those who stay productive, their hit rate was every bit as high in their 50s and 60s as it was in their 20s and 30s. So, you know, I think about Frank Lloyd Wright designing Fallingwater at 68 years old, his crowning architectural achievement. Or I ’ m sure you ’ ve come across Raymond Davis Jr, the scientists who measured neutrinos from the sun. He didn ’ t build the key detector until he was 51 and kept running his experiments until he was 88, the year he won the Nobel Prize. I ’ d be willing to bet that we would either uncover, or even create more Frank Lloyd Wrights and more Raymond Davises if we had ways of keeping more people motivated to produce into and past mid career. Agree or disagree? AB : I think that if there is a natural cadence to the average person ’ s life, and you ’ re saying it doesn ’ t have to be this way, then the societal goal will be one in which everybody is above average. And you know, that becomes an exercise in frustration. AG : In other words, that ’ s a nice theory about encouraging everyone to keep producing at maximum capacity for decades, but it might not be realistic if you ’ re not a superstar in your field. The rest of us are going to do what people naturally do. And Arthur wants there to be space for that path, too. He wants people to be excellent in their own way, which made me wonder if he thinks career decline is more inevitable for people who achieve excellence. AB : What goes up must come down ultimately, and the higher up you go, the more obvious it is that you ’ re not there anymore. That ’ s what I call the Law of Psychoprofessional Gravitation. You notice a lot when you were great and you ’ re not as great as you once were. And if you can do something to get on your crystallized intelligence **curve**, you get a different kind of productivity, and you can be successful in a different way. That ' s the knowledge I want to get across. AG : He ' s arguing that regardless of what field you ' re in, as your fluid intelligence declines, you should make a change. Stop trying to innovate. Take what you ' ve learned and accomplished and shift into a mode where you can transmit your knowledge and skills to other people. Become a teacher or a mentor. AB : And frankly, the world needs those people. And we don ' t have enough of them. When I ' m at a certain point in my career, and I see that I ' m not on the top of my game, I need to start thinking, is what got me here, whether it ' s the insight or the innovation or whatever it happens to be that got me here, is it something that I can sustain? If it ' s not, start looking for something else. Specifically, start looking for a way that you can move into instruction mode. AG : In Arthur ' s case, after giving up on music, he spent years running a conservative think tank. And then having learned the lessons of his fall at Carnegie Hall, he took his own advice. AB : And I had somebody I really respect, and he said,"You know, when you are a chief executive and you have to be on the razor ' s edge every single day, you have two choices on how to step out of your chief executive job."And I said,"Well, good, two choices. I like two choices. You know, it ' s like the Monty Hall game. You know, door number one and door number two. So what ' s behind door number one?"And he said,"Leave before you ' re ready."And I said,"Oh, I don ' t like that. What ' s behind door number two?"And he says,"Leave on someone else ' s terms."I said,"Ooh, I really hate that one."AG : What did Arthur decide to do then? Exactly what he ' s suggesting you do. He went into teaching. AB : The **bottom** line is, when the stakes are high, it ' s worth leaving things before you ' re ready. AG : Arthur is content with how his career choices worked out. He wants you to walk off the stage before you fall off of it. I understand where he ' s coming from, but I ' d rather see you find your balance, or get on a different stage altogether. You do have choices beyond aging out and opting out. More on those after the break. OK, this is going to be a different kind of ad. I played a personal role in selecting the sponsors for this podcast, because they all have interesting cultures of their own. Today, we ' re going inside the workplace at Accenture. Gonzalo Adriazola : I ' m from Arequipa, which is the second largest city in Peru. We have the coast, we have the Andes and we have a bit of jungle. AG : This is Gonzalo Adriazola. When he was 19, he and his mother left Peru and moved to rural Ohio. GA : I don ' t think I fully processed that I was leaving a whole life behind. I think I was just excited for the future. AG : As a young immigrant to a new country, Gonzalo didn ' t know exactly what the future held, but his mother became a tremendous source of inspiration. From early on, she taught him about the importance of education. GA : She would always tell me,"I might not leave you anything, but I ' m going to leave you a good education."AG : Gonzalo got a job at a local restaurant while he applied to colleges. GA : When I got the acceptance letter, my mom and I were just, like, jumping and dancing and just so happy. AG : He headed to the Ohio State University, and it didn ' t take him long to fit in. He was crowned homecoming king, started working full time at a bank and found a community within the Hispanic Business Student Association. They sponsored a trip to a conference in Washington. GA : It ' s the largest gathering of professional Latinos in the US. I saw this one speaker and he was so good. He was so passionate, energetic. Besides being phenomenal, he looked like me, and this was the first time I had seen someone in person that looked like me that was a total boss. AG : At the end of the talk, Gonzalo went up to the speaker and asked for advice. GA : I asked him like,"I want to be a CEO one day. How do I get to be like you?"And he laughed and looked at me and told me that professional services firms, all they have is people. So they ' re going to do their best to make the best of you. And that ' s how I found out about consulting. AG : Research shows that role models can elevate our aspirations. When we see someone similar to us do something inspiring, it builds our confidence that we can follow in their footsteps. When Gonzalo graduated from college, he followed his role model ’ s advice and went to work as a consultant at Accenture. GA : I was really happy to join the firm. And I mean, it has only been great experiences. AG : Gonzalo had accomplished a big goal, and for four years, he helped companies grow and perform better. But he had a nagging feeling that something was missing from his life. GA : As soon as I got to Accenture, I wanted to give back to Latin America. That was kind of, like, my life objective. I saw that we had worked with this microlending institution called Caja Arequipa, and I ’ m like,”What are the odds?”AG : He reached out to the leadership team running the project. GA : They told me about, like, everything we were doing in regards to financial inclusion in Latin America. I was really excited to know that I was from that same city. Very welcoming to my help. AG : Alongside his consulting work, Gonzalo has been a leader in the Hispanic Employee Resource Group and volunteered his extra time to help the team in Latin America. His dedication paid off. Earlier this year, Accenture development partnerships moved him to Colombia where he joined the project team full time. GA : It was a dream come true. I called my mom. We laughed, we smiled, we danced. AG : This isn ’ t the only way Gonzalo gives back. Every year he ’ s gone back to the conference where he was first inspired by someone he felt he could relate to. GA : It feels like I ’ m doing my mission. I was in their shoes one day and now I ’ m the person lifting up, just as others lift me up. And there ’ s people lifting me up right now. There ’ s going to be a brighter future for my community. AG : Accenture is working to become one of the most truly human companies in the digital age. Learn more at Accenture. com / careers. Lyn Slater : I ’ m not a Pollyanna about aging. There ’ s certainly changes in my body. You know, I have my wrinkles, I have my white hair, but I think that I could generate content as much as someone who ’ s younger than me. In fact, most of the peers who are at my level are all younger than me. AG : Lyn Slater is 65. She recently quit her career as a professor to become a full-time social media influencer. Lots of people call themselves influencers, but Lyn ’ s version is a real job. LS : I ’ m also known as The Accidental Icon. AG : The Accidental Icon has more than 700, 000 followers on Instagram, and she gets paid to run ads on her account for brands like Kate Spade, Visa and Teva. Her feed is filled with stylized photos of herself posing around New York City in designer clothes and large sunglasses. She ’ s been featured in magazines like”Cosmo”and”In Style.”And she ’ s walked the runway at New York Fashion Week. Actually, she ’ s walked more than one runway. LS : I have no background in the fashion industry, but I was in the runway show last spring with Kate Spade. I ’ ve worked with them quite a bit. AG : Some people will tell you that most of us do our best work when we ’ re younger, that when we approach retirement age, we should slow down, take up a hobby, play with the grandkids. But excellence later in your career is possible, and there are a few ways you can get there. Lyn is living proof that you don ’ t have to avoid career decline by shifting from doing into teaching. You can actually reinvigorate your career by experimenting, by doing something new. In Lyn ’ s case, she did something completely new. We saw her in action one Saturday near her home in Manhattan. LS : We ’ re on Fifth Avenue in New York City. It ’ s called Museum Mile because everywhere is a **museum**. I ’ m going to be shooting my outfit outside. AG : She ’ s with her partner who doubles as her photographer. Calvin : Your bangs are good. LS : Thank you. AG : Lyn is petite with a short white bob and bangs that stop just above her huge sunglasses. She takes off a long velvet coat and **scans** the brick facades of the Museum of the City of New York for her next spot of inspiration.

LS : And right now, that little nook over there is really calling to me because it does look like a place... AG : Before becoming a social media influencer, she had an entirely different decades-long career in social work. She was an expert for New York City family courts on child abuse issues. She wrote a seminal book on social work and law, and she spent the last 20 years as a professor of social work at Fordham University. She was quite an authority in her field, but she told me she ’ d become frustrated with the glacial pace of academia. Like, how long it takes to write and publish journal articles. LS : The whole process could take almost two years and maybe a handful of people would read it. And so, as I began to get more followers on Instagram and I began to do more things, you know, for example, I did a short **film** and within a couple of weeks, that was in front of, you know, millions of people. AG : Lyn went about as far from her original field of work as she could go. So what leads to a spectacular pivot like that? A dose of novelty. Research shows that curiosity is a key factor that can help you find creative peaks as you age, rather than decline. This is one way to stay productive and maintain motivation later in your career. Find something new to explore. For you, novelty might mean switching to a new team or a new market, getting certified in a fresh skill set, or mastering a hobby completely outside your experience. For Lyn, that hobby was fashion. LS : I ’ ve always had, sort of, what I call a very performative relationship with my clothes. I resist like crazy social categories or constructions of who I should be. AG : In this case, she was resisting the idea that a long time professor of social work wouldn ’ t also be interested in high fashion. She ’ d been toying with the idea of starting a fashion blog. And one day, unexpectedly, the name came to her. LS : It was the first week of school, and I always used to dress up for that. And I was meeting a friend for lunch and I had on a Yohji Yamamoto suit, and a couple of photographers started taking my **picture** and then a bunch of other photographers saw them do it so they came over, and my friend was making a joke like,”You ’ re an accidental icon.”AG : If you ’ re going to experiment with something new, it helps to fully immerse yourself in that world. Lyn was creative in where she looked for her mentors. LS : I found out that I could get a press pass to what is called the market shows, and I would spend hours talking with these young designers. They taught me a great deal about how clothing is made, what the processes are. That ’ s how I really began to understand fashion. AG : Instead of being intimidated by new technology and seeing it as something only young people can master, she went straight to the experts she knew. LS : My students helped me to figure out Instagram and Snapchat, and I started to take classes in social media and fashion and just random things, like, how to start your own vintage store. AG : In a study in the fashion industry, the more time designers and directors spend abroad, the more creative they became over their careers, but it wasn ’ t traveling or even living abroad that mattered. It was the amount of time they spent working abroad. They couldn ’ t just observe it from the sidelines. They had to be in the arena. That ’ s how they really internalize the novel elements of a foreign culture. For Lyn, seeking out new perspectives included small steps, too. LS : I do things that I don ’ t normally do. And so that could be very simple things, like taking a different route to work. It ’ s really, put yourself into different experiences that are creative and see where that takes you. AG : You may not aspire to quit your job and become an Instagram star, or to switch fields at all. And you don ’ t have to. There are ways to reach new heights while staying within your career path. Think about whether you ’ re a sprinter or a marathoner, not in terms of running, but how you go about generating creative ideas. Economists call the sprinters conceptual innovators. They typically start with a big idea and they tend to peak early, because they have eureka moments when they ’ re relatively new to a domain. Like Einstein at 16 imagining himself chasing a beam of light, or Mary Shelley at 18 starting to write"Frankenstein."But as they accumulate knowledge, they ' re more likely to get stuck in their ways and stop gaining new perspectives. There ' s a term for that : cognitive entrenchment. The longer they stay in a field, the more likely they are to take for granted assumptions that need to be questioned. Marathoners, on the other hand, are experimental innovators. They usually discover their ideas through trial and error. They often peak later because it takes time to test out different approaches, like Darwin, spending decades tracing the origins of species, or da Vinci experimenting for years with different ways of modeling light in his paintings. But those habits allow them to sustain their peaks longer and peak again later as they explore new areas. That kind of experimentation is familiar to Lynda Weinman. Lynda Weinman : I ' m an autodidactic person. I always love learning new things. AG : She got her start in the 1980s as an animator at a **film** studio. She often compared herself to her higher achieving peers who were sprinting ahead. LW : I remember when I was in my 20s watching some of my friends who were uber successful and I, you know, at my 30th birthday was in tears that I hadn ' t accomplished what I wanted to accomplish yet with my life. AG : But she didn ' t rush it. Instead, for the next 10 years, she kept chugging along, testing out different ways to build and use her skills. Finally, the year of her next big birthday, she created a website. Lynda called her site Lynda. com. LW : Well, when we started Lynda. com I was 40 and when we sold it I was 60. AG : She sold it in 2015 to LinkedIn, for, oh, 1. 5 billion dollars. Lynda. com was one of the first companies to offer professional learning online. Maybe you ' ve heard of their courses in animation, design and software. Maybe you ' ve even taken classes there that advanced your career. Many founders who are trying to grow a unicorn company, think they need to generate one brilliant idea and scale it rapidly. They think they need to be sprinters, but Lynda approached her career more like a marathoner. Her online learning company didn ' t look anything like a success early on. Instead, she was constantly trying out new ideas and innovating over time. In some ways, Lynda was primed for this kind of scrappy test-and-learn approach. LW : I actually had a pretty troubled childhood. I moved in with my dad and my stepmom who weren ' t expecting to raise me and my siblings. And, you know, it was just this feeling of having to find my way because all the adults in my life were, you know, having their own problems. And I think what I probably am is a very resourceful person who realized at an early age that it was on me to figure out my life. AG : During her early days as an animator, Lynda encountered her first computer. She was unimpressed, at first. LW : It was a Apple II Plus clone that my boyfriend brought home, and so to please him, I turned it on and I started pecking at the keys and I caught the bug that so many people since have caught. And he eventually called himself a computer widow, because I liked the computer more than him. AG : LW : I mean, I was just amazed at what it could do. AG : What it could do was draw. So Lynda got her own Macintosh and brought it into work. People started asking her to do projects for them. LW : Somehow they thought I knew more than they did, which was, you know, basically I ' d had five minutes longer with something than they had. AG : Lynda was onto something. By experimenting with new tools and following her curiosities to uncharted places, she could carve out a niche and continue iterating instead of stagnating. LW : Oh, do I want to do what everybody else is doing, or I ' m being asked to do something that only I know how to do, and maybe that ' s more **fun**. AG : Lynda ran two more experiments. First, she started teaching in-person college classes in web design. Then when she saw a lack of good manuals for students, she wrote one herself. LW : Then it became a huge success, way bigger than any of us could have imagined. AG : Well, you ’ re being very modest, but can I get you to describe a little bit just how successful the book was? LW : I ’ m not sure how many millions it sold, but it was translated into many, many languages and it was the de facto web-design book for all colleges. It was the”it”book. AG : From there, Lynda continued to test out new methods of teaching. She started selling instructional videos on VHS **tapes**. Then, as technology changed, she switched to CDs, then DVDs. Then she launched the website Lynda. com. In the late ’ 90s, she and her business partner and husband tried launching memberships on Lynda. com. They weren ’ t exactly an instant hit. LW : In the beginning it was just an abject failure. It was cannibalizing some of our other work. AG : But they didn ’ t give up. Not every experiment succeeds, and they don ’ t all succeed right away. Memberships weren ’ t growing like crazy, but there was steady growth. Lynda paid attention to that. Eventually they hit an inflection point. While Lynda. com was quietly adding hours of material to their catalog, more people were using the internet for online learning. Their online audience doubled each year. And in time that grew into a big number. By 2012, they had over a million paying users and over a hundred million dollars in revenue. LW : It was literally a very organic growth process. AG : Lynda could ’ ve simply continued to develop her skills as an expert animator and would have had a nice career. She could ’ ve kept teaching, too, but her greatest success came when her experimentation led her into entrepreneurship, turning her various projects into a business. She didn ’ t listen to the narrative that as you get older, you should pull over into the slow lane. She saw some benefits of building something later in her career. LW : By the time you ’ re 40, you believe in yourself more. You ’ re a more seasoned human being that has better judgment and better experience and better confidence. AG : This might be one of the reasons why older founders are actually more successful than younger ones. In a study of several million entrepreneurs, 40-year-olds were more than twice as likely as 25-year-olds to launch a successful start-up. And they were also nearly a third more likely to be in the top 0. 1 percent of fastest-growing tech start-ups. Yes, sprinters sometimes get lucky right out of the gate, but marathoners know how to persevere. LW : I think everybody would probably like to be instantly successful and be revered and acknowledged and acclaimed and all those things, but when it happens to you too young, it actually can be very damaging. At least the people that I ’ ve seen, who, you know, they burned bright and they burned all the way out. They didn ’ t understand that it wasn ’ t going to last forever and that they had to keep reinventing and they had to earn it again. AG : If you ’ re worried that your career will decline, don ’ t let your output decline, and don ’ t let your curiosity stagnate either. It ’ s never too late to invent or to reinvent yourself. LW : I think the older you are, the more valuable you can be. AG : I love that because I don ’ t ever want to retire. Next time on”WorkLife.”Working remotely is full of challenges that have made our work lives harder. Lizet Ocampo : So we turn our cameras on and there I am as something else. At first, we didn ’ t realize it was a potato. We ’ re like,”What is going on?”I just essentially was a potato for the meeting. AG : But some of the solutions might be easier than you think.”WorkLife”is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell, Constanza Gallardo, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelin. This episode was produced by Jessica Glazer. Our show is mixed by Rick Kwan. Original music by Hahnsdale Hsu and Allison Layton-Brown. Ad stories produced by Pineapple Street Studios. Special thanks to our sponsors, Accenture, BetterUp, Hilton and SAP. For their research thanks to Dean Simonton on creative productivity, Lu Liu and colleagues on career hot streaks, Ben Jones and colleagues on the success of older entrepreneurs, David Galenson on conceptual and experimental innovators, Erik Dane on cognitive entrenchment and Will Maddux, Adam Galinsky and colleagues on working abroad. In some ways, Lynda was primed for this kind of crap... Excuse me. In some ways, Lynda was primed for this kind of crap... It ’ s not crappy, it ’ s scrappy. In some ways, Lynda was primed for this kind of crappy... Sorry.

Text Sample 726: POS Dim 5 – 2020 – Score 8.00 – 2020/t004270.txt

Lizet Ocampo : I have a team meeting with my team. It ’ s Monday morning, so we kind of miss each other.”Let ’ s turn our cameras on.”Adam Grant : For Lizet Ocampo, it seemed like a regular day in quarantine. But when they turned on their cameras, something was off. Her face was a potato, a flat image of a potato, with Lizet ’ s eyes and lips moving on it. LO : And I just remember thinking,”Why am I a potato?”and of all things, why a potato? Like, I couldn ’ t go back to a regular camera, so I just essentially was a potato for the meeting. AG : The previous week, Lizet had been in a virtual happy hour for Latino leaders. LO : And I wanted to make them laugh, and so I thought,”You know what? Next week, I ’ m going to try to do some funny filters,”and so I download these Snap filters, but it didn ’ t really download all the way. There was an error message that kept popping up, so I kind of gave up on it and forgot about it. AG : Turns out the filters had worked a little too well and a little too late. Now Lizet was stuck as a potato. Her colleagues laughed and went on with the meeting. Lizet didn ’ t think much of it until she got a text. LO : Someone that ’ s on my staff took a **picture** of it and posted it on her personal Twitter account, so I had no idea. AG : She wrote,”My boss turned herself into a potato and can ’ t figure out how to turn the setting off.”LO : And then my colleagues ’ faces are hilarious because it looks like they ’ re trying to be serious, but they ’ re trying not to laugh at the same time. AG : It is amazing because you see her, you see the guy, and then there ’ s just a potato. LO : Yes. AG : Yeah, this poor potato is like,”How do I stop being a potato?”LO : Forty million people have seen it, and it ’ s become sort of a cultural moment in this quarantine world that we ’ re in right now. AG : You have a new identity now, don ’ t you, though? LO : That ’ s true. I ’ m known to the world as”potato boss.”I have potato appearances requests – AG : LO : and so I ’ ve showed up as a potato to some meetings for friends at other organizations and things like that. AG : The coronavirus pandemic has thrown many of us into the deep end of working remotely. While some people have found creative ways to swim, many others are struggling just to stay afloat. The good news is that remote work is not uncharted waters. We ’ ve actually been doing it and studying it for a long time, and with those lessons in hand, you might actually be more prepared than you think to make remote work work. AG : I ’ m Adam Grant, and this is”WorkLife,”my podcast with TED. I ’ m an organizational psychologist. I study how to make work not suck. In this show, I ’ m inviting myself inside the minds of some truly unusual people, because they ’ ve mastered something I wish everyone knew about work. Today : remote work, how we can communicate better with and without a camera, no potatoes required, and how, even during this unprecedented pandemic, we can stay resilient. This episode is sponsored by Hilton. As the world has gone into lockdown, many of us have experienced the challenges of working from home. But some jobs are hard to do remotely. If you work in health care or public safety, you need to be on the front lines. If you ’ re in manufacturing, production or distribution, you have to be on-site at a farm, factory or warehouse. We ’ re all depending on the people who are risking their health and safety to go to work. Yet, in the knowledge and service industries, many jobs can be done with online tools. And that ' s not as new as it seems. Half a century ago, futurists started predicting the rise of remote work. In 1993, management guru Peter Drucker wrote that commuting to an office is obsolete. But a quarter-century later, nearly half of global companies still weren ' t allowing remote work at all. Not anymore. Around the world, many companies are now requiring remote work. This forced remote work has amplified many of the challenges we had when we worked face-to-face. And even in that hard-to-imagine future beyond the pandemic, at least some of this shift to remote work is probably here to stay. So, how do we do it well? As teams, we need to figure out how to stay connected and collaborative. As individuals, we ' re all seeking guidance on how to stay sane when we ' re working alone. Recorded caller 1 : So, I ' m incredibly vulnerable and scared of losing some of the relationships that have taken a long time to build and not having the edge of my personality and my ability to communicate clearly face-to-face. Recorded caller 2 : Working from home has been a nightmare. AG : Interruptions have become a fact of life. We ' re all"BBC Dad"now, with kids or pets dancing right into our video conferences. Recorded caller 3 : We get on a Zoom call with our region, and in runs my four-year-old, completely naked, and I was speaking, so I wasn ' t on mute, and all you hear is me screaming,"No!"Recorded caller 4 : I finally made a rule that said,"If you interrupt a Zoom meeting, you ' ll have to stay for the whole thing."The first offender was my 10-year-old daughter. She ended up sitting there for 45 minutes with nothing to play with, and I can remember her telling all the other kids,"Wow, don ' t go down there, because you ' re going to have to stay for one of those boring meetings."Recorded caller 5 : On one Zoom call where I was pleading with government officials to please listen to the plight of small business owners, my five-year-old daughter walked behind me and said,"Mom, if you die of the virus, I ' ll run your company,"and everyone went silent. Try to come back from that when you ' re campaigning. Martine Haas : I think in some ways, we are adapting very, very fast right now. We are the experiment that will create the data. AG : Martine Haas is my colleague at Wharton and a renowned expert on how teams collaborate at a distance. Now she isn ' t just studying it. She ' s living it. MH : I ' ve been working to kind of get my setup really functional, and I invested in a screen to separate my workspace from the rest of the room that I ' m in. I open the screen ceremoniously in the morning, and then I close it ceremoniously. I literally move it so that it kind of covers the desk just as a way to kind of switch off. AG : One of my favorite insights about teams is that they ' re kind of like amplifiers : whatever you put in comes out louder. When we go remote, that ' s even more true. All of our microphones are blasting at once, and that can cause problems. MH : I sometimes think about it as kind of like an iceberg, right? At the top of the iceberg, the stuff that ' s more obvious – you know, it ' s just difficult to communicate, partly because of the technology, right? AG : Think about how much time you ' ve wasted on tech problems. First man : Hey, Paul. Can you hear me? Paul : Hey, guys. Hey, Tyler. Tyler : Sorry I ' m late. Having a hard time connecting. FM : One second. Paul ' s having a sound issue. P : I can ' t hear you. T : Try adjusting your output settings. P : Can you hear me? T : It ' s the gear icon. P : Never mind. I got it. I just had to change a few settings. FM : Great. Great. I think we can get started then started then then. Oh, great. Oh, great. AG : Another obvious issue is power. MH : And it ' s very easy to fall into a very hierarchical form of communication. AG : You know, where the person with the most authority or status hijacks the conversation. In a classic study of airline crews, it was possible to predict a flight ' s timeliness and fuel efficiency just by watching the first 15 minutes of a captain

’ s meeting with the crew. The ineffective captains tended to monopolize the conversation. The effective ones did more two-way communication. They spent more time engaging their crews and listening to them. They also encouraged their crews to talk to one another about the task. Creating open channels of communication is even more important in virtual teams. You ’ ve probably had video calls that were dominated by one talking head, but that ’ s only the tip of the iceberg. MH : So the less obvious things, there are two big buckets of things called”shared identity”problems and”shared understanding”problems. AG : Shared identity and shared understanding. Beneath the everyday communication challenges you ’ ve seen, these are probably the bigger issues at play. In fact, these are the key ingredients that Martine has found for making remote teams effective. But how do you build them among teammates who can only see the dimly lit, poorly framed video conference versions of each other? Let ’ s start with shared identity. It ’ s the sense that we ’ re all in this together. When teams lose that sense of unity... MH :... it ’ s really easy to get an us-versus-them kind of dynamic going.”Here in Philadelphia, we work really hard, but that part of the team that ’ s over there in Belgium, they really don ’ t work so hard, they don ’ t come in on time. We don ’ t know what they ’ re up to.”So it ’ s very easy to get those kind of psychological distancing dynamics that come into play, and it ’ s a basic psychological process. AG : In the study of airline crews, the effective captains created a shared identity. They talked about everyone as”we,”not just the flight attendants, but also the maintenance crew, the gate agents and even the air traffic controllers communicating remotely with them. They saw themselves as leading one team that was responsible for the safety of the flight. The ineffective captains reinforced fault lines between subgroups. They called the pilots”we”but everyone else”you.”This lack of shared identity hurts the work of face-to-face colleagues but creates an even bigger problem for remote teams. MH : It reduces trust. It reduces psychological safety, that sense that it ’ s safe to kind of speak up in the group. It can lead to people withholding information rather than sharing what they know, it can increase conflict. And so it ’ s really important of the team leader to articulate for an overarching goal. AG : Goal-setting isn ’ t usually the first thing that comes to mind when thinking about team identity. What most people try instead is team-building activities. I ’ m guessing you ’ ve suffered through Zoom calls where you ’ re asked to name your favorite pizza topping or to share highlights from your totally boring weekend in quarantine. Those superficial strategies overlook that we bond best when our individual actions contribute to a common purpose. Research reveals that it ’ s critical to establish team goals and clarify roles, especially in big teams. Having team goals creates a sense of shared identity. Having clear roles prevents social loafing and free-riding, helping people feel that their individual responsibilities matter. The other key challenge of virtual teams is shared understanding, being able to get on the same page, not just about what you ’ re doing, but also about who you are and what you value. MH : It ’ s basically the problem that you kind of have incomplete information. You don ’ t have a good understanding of each other ’ s contexts, right? So this is actually why it ’ s really valuable to ask members of your remote team,”What does your workplace look like? Who else is there? Do you have a dog? Are you watching your kid? What else is going on?”AG : You know you ’ ve reached shared understanding when a colleague does something that would be puzzling to most people but you get it right away. The project manager who cc ’ s the boss isn ’ t trying to make everyone look bad. She ’ s anxious about your team ’ s work being overlooked. The guy in accounts payable isn ’ t signing off at 3pm because he ’ s a slacker. He ’ s taking care of his mom, who ’ s ill. MH : Because if you don ’ t understand people ’ s contexts, you can make attributions

– and they ’ re often negative attributions – about what other people are doing or whether they ’ re working hard enough or whether they ’ re distracted that are not really fair, right? Because you don ’ t really get what their situation is in the same way as you would if you were working face-to-face. AG : Think about your virtual team. Shared identity and shared understanding are pivotal for trust. How do you make that happen when you ’ re not in the same physical space? You might strengthen shared identity by circulating feedback from customers on why your team ’ s work is more important than ever. And you could get to some shared understanding by talking about your favorite video conference bloopers. Some of you have been experimenting with other strategies. Woman to baby : Hey, bud, can you say”welcome”? Baby : Welcome. Woman : To. Baby : To. Woman : Principles. Baby : Woman : Of. Baby : Of. Woman : Management! Baby : Man : My little sister, she had found a sign that said”Wall Street”on it. She put the Wall Street sign behind me so that when I get on my Zoom calls, my colleagues can see it, and she said,”Look, now you ’ re on Wall Street. Stop complaining. Act like it.”Woman : Can you say”my”? Baby : My Woman : Mama. Baby : Mama. Woman : Misses. Baby : Misses. Woman : All. Baby : All. Woman : Of. Baby : Of. Woman : You. Baby : You. Man : I love asking folks to give me a quick tour of their makeshift home office if they ’ re comfortable with it. I ’ ve found it often leads to a conversation about hobbies or life outside of work, which has been really, really cool for building closer relationships and kind of humanizing the other folks on the team. Woman : OK, one more thing. If you ’ re happy and you know it shout”hooray”! Baby : Hooray! Woman : All right! AG : I ’ ve been doing the majority of my work remotely for the past 15 years. Some of the key members of our”WorkLife”team have never even met face-to-face, so reaching shared understanding has been key in our collaborations. If you work with me, we have a shared understanding that my biggest vice is being chronically late. I have a hard time disengaging from a task before it ’ s done, and I try to squeeze too much into each day, but I often have calls with people who don ’ t have that context. Eventually, I had to start managing expectations. Now my calendar invite automatically notifies people that I ’ m usually running five to 10 minutes late. It helps them understand my schedule. But there ’ s something that helps even more. Alex Cornell : Well, I ’ ve been sittin ’ here all day / I ’ ve been sittin ’ in this waiting room. AG : Hold music. But not just any hold music. AC : Waitin ’ on my friends / Yes, I ’ m waitin ’ on this conference call / All alone. AG : Now when I jump on the line late, people aren ’ t disappointed or frustrated. They ’ re often amused. AC : Guess I ’ m on hold / I hope it ’ s not all day. AG : So when I scheduled a call with the guy who wrote this song, for once, I was deliberately late. I let him call in first, so he could hear his own hold music. AC : Here ’ s my voice. Y ’ all sing along. AG : Alex Cornell wrote this song for UberConference, because he hates hold music. AC : Whenever I call into an airline or anything like that, and you hear the same song over and over and over again, you ’ re now sort of already in this like, low-level tortured state of just been listening to this terrible song over and over again. AG : So he decided to make something better. AC : Tell me where can they be... AC : If you could imagine being on hold, and while you ’ re on hold, there ’ s just actually a person there that ’ s sort of like a doorman, so to speak. I ’ m just smiling as I ’ m saying it because it ’ s such a weird thing to **picture**, this, like, virtual doorman saying like,”Oh, hey, like, here you are, too, calling into this conference call,”but then also, it acts as a great icebreaker. AG : Alex, didn ’ t just write the hold music for UberConference. He cofounded the company, and then went on to be the lead product designer of Facebook Live. So he ’ s spent a lot of time working on virtual communication and has a lot of experience building shared identity and shared understanding. Alex ’ s new start-up, Cocoon, has developed an app for communicating privately with our closest friends and families. Before the pandemic, the team was six people in the same physical space while one person was remote. Now they ' re all remote. AC : We have brought on at Cocoon two new team members since this began, and they need to be onboarded to the company, to the product and to what we do and what we believe in. AG : He ' s instinctively put into place some of the core strategies that Martine Haas recommends. To strengthen shared identity, Alex has created written memos to share the company ' s history and values. Spelling it out forced him to get clear on the team ' s goals and culture and helped newcomers get up to speed from a distance. It ' s something they can all revisit occasionally to remember their common purpose and norms. And to foster shared understanding, Alex has found that it helps to open up a little more about his home life. AC : I ' ve found that it can be pretty helpful for this situation too, even just for our coworkers, to have a little bit more awareness of what we ' re all doing day to day. I wouldn ' t normally have shared with them that I am building a gigantic LEGO with my stepson, and that ' s something that they know, and I think we ' re actually, in some ways, more in touch with one another than we ever have been. AG : There ' s evidence that when we gain this kind of personal knowledge about our colleagues ' lives outside work, it humanizes them. We trust them more and feel more similar to them. If we only talk about work, we miss out on that information. So Alex has been experimenting with unstructured interaction to create shared understanding. AC : We ' ve definitely done virtual lunches. I think one of the things that is missing most instantly when you go full remote is the sort of just camaraderie associated with being able to just kind of have a conversation about nothing around the lunch table, and it ' s weird to set a meeting and say,"The agenda is we ' re not going to talk about work, and everybody please have **fun**."It ' s like that is, that ' s a weird agenda. It ' s hard to do. I think there ' s all kinds of these new norms that are being established just as we go. AG : When you communicate virtually, what norms have you been setting to facilitate shared understanding? Maybe you ' ve been encouraging people to use the chat function so the quieter voices get heard. Or you might be turning off your self-view so you ' re not distracted by all your weird facial expressions. But if you ' re like most people, you want to keep seeing your team members ' videos. You prefer that to audio-only because you value having the rich visual cues of facial expressions and body language. That ' s how you read their emotions, right? Wrong. It turns out that if you want to know what someone is feeling, you might be better off just hearing their voice. In one experiment, people read another person ' s emotions more accurately when the lights were turned off. In another experiment, they were more accurate when the video feed was turned off. Follow-up studies showed that when we don ' t have visual cues, we pay more attention to the content and tone of voice. AC : I think that there is definitely something that is pure about when all you can hear somebody ' s voice. Then when you switch to audio only, it ' s like that actually is something that we ' re all familiar with and are somewhat used to, and there ' s no sort of missing piece. AG : Plus, if you ' re on video all day, you could be losing contributions from one group of colleagues in particular : introverts. Why? Because of screen-out. AC : Even though I really want to see my friends, if I ' ve been on Zoom all day, I don ' t really want to spend another couple hours on Zoom just because, socially, I ' m tired of interacting, but I ' m a bit of an introvert, I suppose, so my energy is just depleted, usually, by the end of the day so. AG : Introverts are more sensitive to stimulation than extroverts, not just bright lights and loud noises, but also eye contact, and video calls seem to be magnifying that overload. AC : And you ' re staring at somebody ' s eyes for, in some cases, like an hour, and I don ' t know about you, but when I talk to somebody else, I ’ m very rarely staring at their face for all 60 minutes. AG : There ’ s a **fun** study that tracked people ’ s reactions to seeing a large virtual face on their screen. The most common response was to flinch. AG : We know face-to-face interaction is critical for establishing trust, for building that shared understanding that Martine Haas has found is so important. But once trust exists, I think we should all have the license to turn off our video feeds from time to time. That ’ s what I ’ ve been doing. When people ask what video conference platform I prefer, I often tell them”audio.”After all, if you know what I look like, and you trust me, you don ’ t need to see my face every minute. But for times when you don ’ t have that option, Alex recently created a video conference version of his hold song. AC : My version of this on video conferencing is to have a looping version of myself who ’ s sitting there – AG : AC :... seemingly paying attention. Technically – I ’ m not saying I would do this – but you could have the looping version of yourself just on the call the whole time. This video one, it actually is me, so it ’ s sort of like a”Westworld”-type version of it. AG : If we were robots, it would be easy to be resilient. Just turn your emotions off until this crisis is over. Since we can ’ t do that, the next best thing is to learn from someone who ’ s done more remote work than almost anyone in human history. After the break. OK, this is going to be a different kind of ad. I ’ ve played a personal role in selecting the sponsors for this podcast because they all have interesting cultures of their own. Today, we ’ re going inside the workplace at Hilton. We had recorded a story back in February. Then the world changed, so we decided to do something different, to speak to the moment we ’ re living in. Charles Hill : I was in denial. We weren ’ t going to be impacted. We ’ re making it out of this. And it was that moment, it was 4:30 on a Friday, March 6. I knew that we were in for something else. AG : Meet Charles Hill. He ’ s the General Manager of DoubleTree by Hilton in Crystal City, Virginia, just outside Washington, DC. That Friday, his hotel was hit with a wave of cancellations. As the reality of how severely the coronavirus would affect them set in, he was called into nonstop meetings to answer an impossible question : How do you manage a hotel in the middle of a pandemic? CH : What is it that ’ s happening around us? How big is this? And what do we need to do? There was nothing coming in. It was all going out. But we still had these hotel rooms. AG : That ’ s when Hilton came up with a plan. They would do what they do best : offer hospitality. CH : Hospitality is caring for others and spreading light and warmth. AG : In partnership with American Express and its hotel owners, Hilton donated a million rooms across the US to frontline healthcare professionals. CH : For us to be able to provide accommodations for people who are literally working on the front lines to take care of those in need, it ’ s such a great way to impact our community. AG : They took in nurses, doctors, EMTs – the people working tireless hours who needed a safe and **clean** place to stay. When they arrived, Charles was there to welcome them at the front desk. CH : I think there ’ s a sense of surprise that they don ’ t have to pay. There isn ’ t a fee. We ’ re not going to charge you. It ’ s OK. We want you here. We ’ re thankful for what you ’ re doing. AG : For Charles and his staff, it felt good to be able to do something, and they were inspired to bring more light and warmth – literally. CH : The lights in our north tower have this warm glow, and we came up with the idea of putting messages in that north tower so that anybody who passes would be able to see something from us. AG : With no one staying in the north tower of the hotel, the dark windows were like a blank canvas. They could turn on the lights inside certain rooms to beam shapes to the outside. CH : The first message that we put up was this big heart. Think about these warm bulbs in 52 rooms that just emanates from the building to create this kind of glow outside. AG : A few days later, they changed the lights to form another message. This time, they beamed out one word. CH : We did”hope,”we put H-O-P-E, so you can see very clearly”hope”glowing in the night. AG : Because the hotel is so close to DC, anyone going in or out of the city could see the lights. That night, Charles got a message from a police officer who works at the Pentagon. CH : He had just finished his shift. He was walking back. He saw it, and he just wanted to say,”Thanks.”He said,”This was the first positive thing I ’ ve seen in a while.”AG : Their community started to take notice. Other hotels followed suit. The local news picked up the story, and people started calling the hotel, hoping for a sneak preview of what shape the lights would take next. CH : It turned into this moment where we all had something to be happy about again and something to celebrate, and it was small, but it was a win, and it brought joy. AG : Psychologists have discovered an alternative to post-traumatic stress. It ’ s called”post-traumatic growth.”It ’ s the experience of not just bouncing back, but bouncing forward from adversity. Often, it takes the form of a greater sense of strength. CH : This may be the toughest crisis any of us have seen in our lifetimes, but if we can make it through, anything else is a piece of cake. We ’ re going to be stronger for making our way through this. AG : COVID-19 has created unprecedented challenges for the travel industry. In times like these, hospitality is needed more than ever. Hilton hotels all around the world are coming together to be the bright light for those who need it most. Read more inspiring stories of Hilton hospitality at newsroom. hilton. com / actsofhospitality. JaBarie Anderson : We ’ re going to make magic today. OK, so – JA : This is”WorkLife ’ s”first live virtual haircut, and we ’ re here with”You Probably Need a Haircut”on webcam, because this is the new normal. You ’ re going to need a mirror, a comb and scissors. Coco : I have scissors, they ’ re the dog scissors. AG : Since I don ’ t have hair, our producer Coco volunteered to be a guinea pig. JA : So cut straight across, a little bit higher. Coco : A little bit higher? OK. AG : Before the pandemic, JaBarie Anderson was a freelance stylist. So when he tried doing virtual haircuts from home about a month ago, he wasn ’ t sure what to expect. JA : I ’ ve literally been in so many bathrooms and showers with people now, and people turn on their camera, and it ’ s just like people in their boxes getting their hair cut. I ’ m like,”Oh, OK. This is happening.”AG : Wait. That happens? JA : Yes, people are shirtless, or the whole family is in there watching, talking like,”This is crazy. I want to see how this is going to turn out.”I ’ m like,”Turn your head this way. Turn your head that way,”and you ’ re trying to get the hand positions right. Coco : OK, cutting. JA : OK. Now I ’ ll ask you to go in and point-cut it. Coco : Got it. OK, so now I ’ m point-cutting this. So basically... Ohhhh. JA : People being so willing and trusting – that was one thing that shocked me the most. Coco : A little bit higher? OK. JA : Right there. Coco : So, like, here? JA : Yep. Coco : OK. Cutting. JA : OK, now I want you to... AG : So far, there haven ’ t been any major hair disasters, and by moving his job online, JaBarie ’ s been able to get clients in San Francisco, Texas and Germany. JA : This is absolutely amazing now. It gives me a platform to really showcase my talent. AG : Wait. Did you just say this is amazing? JA : Yes, it allows you to interact with your clients in a different way, and it builds trust, and it builds a bond amongst people in your household. So, say your wife is cutting your hair. That ’ s another level of trust because it ’ s your hair, you know? AG : Wow, I would not have expected that. So it ’ s gotta be harder, though, to do your job in some ways, because you can ’ t touch the person or even control the scissors or the clippers. JA : Well, that ’ s where the education comes in. It ’ s truly amazing to see what people can do from a video chat call and their home tools. That line is kind of shattered. Coco : OK, I think I am shattering that line. It ’ s still, like, a little – JA : Shattering like broken glass! AG : JaBarie ’ s enthusiasm for this new platform has helped him maintain his livelihood during the pandemic. This kind of ingenuity is visible in many service jobs. Some physical therapists are doing virtual appointments. So are plumbers. Plumber : You know, as far as your toilet, what have you been flushing down there? Did you use a plunger for this, because you really don ' t want to. Oh, I told him what tools he needed, and I stayed on the video with him while he fixed it himself. It ended up being a very simple fix. AG : And people doing more independent work have found creative ways to stay engaged while working from home. Woman : I fake my commute, which means I get on my stationary bike and listen to a podcast in the same way I do when I drive to work for roughly 30 minutes before going to actually sit at my desk. Man : To stay productive, I could only use my phone in the living room, so pretty much my cell phone never leaves that area, and if I have to pick up a call, I ' ll have to go to the living room. If I want to check Facebook, I have to go to the living room in order to do so. Scott Kelly : I have a little bit of a mission control setup, so I got an iPad, a second iPad. I have a little TV next, over my right shoulder, so I can monitor the situation. AG : You might have your own personal mission control center as you navigate working remotely. But"mission control"has an entirely different meaning to Scott Kelly, because he ' s an astronaut. SK : So, five years ago today, I launched into space for my mission that wound up being 340 days on the ISS. AG : Three-hundred and forty consecutive days in space. It ' s an American record for a single mission. The rest of us weren ' t schooled in how to stay at home that long in sustained isolation, so I turned to Scott for ideas on building **resilience**. Other than the few fellow space travelers in isolation with him, communication with his family, friends and coworkers was entirely through video, e-mail and radio calls. He couldn ' t go outside without the right equipment, and he didn ' t even know when his mission was going to end. Sound familiar? SK : I knew I was going to come home about a year later. I didn ' t know exactly when. I had to make this mental kind of change to my whole reference system with regards to my life. AG : He ' d been to space three times before, but he ' d never been out there for so long away from everyone he loved. SK : And I felt like I needed to pace myself, as you can wind up working too much when you go to sleep in your office, and you wake up, and you ' re still in your office. The walls could be closing in a little bit, and I definitely felt that on my six-month flight. AG : Under normal circumstances, conventional wisdom is to be in the present, live in the moment, seize the day. But in isolation, sometimes the day sucks, especially in outer space. SK : I also missed nature, going outside, sun, rain, wind, fresh air. People become depressed without that sun. AG : I ' m imagining it ' s something like wake up in the morning and then wait for the Earth to rise? SK : Yeah, so – Yeah,"earthrise."Never heard it that way, put it that way, but yeah, that ' s a good way to put it. AG : For Scott, every day started feeling like"Groundhog Day."SK : I would generally wake up at around 6:15, have breakfast, work for a few hours, take time for lunch, schedule time for exercise, schedule time to connect with people. AG : It sounds a lot like what many of us are experiencing now. Time becomes meaningless. You ' ve probably already heard the most popular tips for coping with that :"Make a daily schedule.""Wear different clothes for different parts of the day.""Have Zoom calls with family and friends."But one of the lesser-known ways to give this time meaning is to master the art of time travel. More specifically, mental time travel. In psychology, it ' s a skill. Rewinding to think about the past and fast-forwarding to imagine the future. It ' s a distinctively human skill, and if we use it thoughtfully, it gives us the capacity to find meaning in the mundane and happiness in the midst of sadness, and make time pass faster or slower at will. The first destination for your mental time machine is the future. That ' s where Scott started. Months before he set **foot** in the rocket to leave Earth, Scott started imagining how he wanted his trip to end. He didn ’ t just focus on what he wanted to accomplish. He imagined how he wanted to feel when his mission was wrapping up. SK : My goal was to get to the end of this with the same enthusiasm and ability and energy as I had in the beginning, and I always wanted to convince myself that this is just my life. I ’ m here for a purpose, and my purpose was to be an astronaut and complete our mission objectives. AG : It helped Scott stay focused, and made the days pass faster. Psychologists find that a mental trip to the future can help us think less about the monotonous how of our days and more on the meaningful why. We ’ re able to get out of the dull weeds of the process and shift our attention to an exciting purpose. The farther ahead we look, the easier it becomes to tell a coherent story about our experience. One of the challenges of the current pandemic is that we don ’ t know when our mission will end, but we do know it isn ’ t endless. It will end. So think about how you want to feel on the day this crisis is over. How will you want to have gone through it? You might still find yourself worrying about what will happen in between. It ’ s inevitable. But here ’ s the thing : worrying isn ’ t necessarily a bad thing. SK : One of the things that NASA does very well is, and how we operate in space is, we ’ re always having to think about,”OK, what is the next worst thing that can happen to us right now? And how do we react and respond to that?”You know, in some ways that can cause stress and anxiety. AG : There ’ s a distinction that we often make in psychology between worrying and rumination, where worrying is basically an attempt at problem-solving and trying to figure out,”OK, what could go wrong? What ’ s that next worst thing? How do I avoid it?”And rumination is when you get stuck in a loop, just worrying about the situation, but not trying to act in the face of it. Did you have strategies for making sure that the stuff you were worried about didn ’ t cause you to ruminate? SK : Oh yeah, I think so, and I think that ’ s part of our nature as astronauts is to understand that bad things can happen. We don ’ t have control over them, but we still have to be prepared for them. AG : Some psychologists recommend a simple strategy for preventing rumination : adding worry time to your daily schedule. SK : Schedule time for your rumination, 15 minutes, maybe. Doing that will make the days go by so much quicker rather than just sitting there thinking about what you ’ re going to do next and bingeing on Netflix all day. AG : When I went through this exercise, I started thinking about silver linings, aspects of being fully remote that I ’ ll miss, like having more family time and flex time, not having to travel or commute, not having to change out of pajamas or at least pajama **bottoms**. Psychologists find that imagining these silver linings fading in the future makes them feel scarce. That helps us appreciate them more in the present. Next, you can direct your mental DeLorean to the past. This is something Scott has been doing during the current crisis : thinking about memories from his days in space. SK : When people ask me,”Well, what do you miss about space?”It ’ s not launching in the rocket. It ’ s not doing a space walk. It ’ s not floating around. It ’ s not the view. It is the people I was there with and also the sense of purpose. AG : What I heard there was nostalgia. The Greek roots of nostalgia translate to”return”and”pain.”The literal definition is : the pain of being unable to return to the past. But here ’ s the strange thing : that ’ s not how it feels. After people are randomly assigned to reflect on a nostalgic event in their lives, they end up feeling happier. They have a stronger sense of connection to others and become more willing to give and seek help. They gain a deeper sense of meaning and purpose in life. Think about a nostalgic event in your life, something you had before the pandemic but no longer do, like celebrating a holiday with your extended family, going on vacation, even going to a restaurant. Reminiscing about those moments might seem bittersweet, but the aftertaste is mostly sweet. It motivates you to make the most of the present. But it doesn ’ t just help to remember pleasant memories from the past. There are also benefits to reflecting on painful ones, too. SK : It was the worst day of my 520 days in space by far and probably one of the worst days of my life was January 8, 2011. I was on the space station about halfway through my six-month mission, and I get a call from the control center in Houston, and then they say,”Scott, I don ’ t know how to tell you this, so I ’ m just going to tell you. Your sister-in-law Gabby was shot in Tucson, Arizona.”And I quickly got on the phone with my brother. He was packing his bags to head to Tucson. You know, I ’ m on the space station now, and I can ’ t be there physically to be supportive to my family. Couple hours later, the CAPCOM called up on the privatized channel, and they told me Gabby had passed away because that ’ s what the news was reporting. Later, I found out she was still alive, thankfully. AG : That experience taught Scott a powerful lesson. SK : Oh absolutely. I mean, I always realize things could be worse. AG : This brings us to a third destination to transport yourself to alternative time lines, things that could have happened but didn ’ t. It ’ s called”counterfactual thinking.”Psychologists have shown that imagining how things could be worse helps us find gratitude. There ’ s evidence, for example, that people who graduate from college during a recession end up being happier with their jobs a decade later. They can easily imagine a world in which they ’ re unemployed, so they don ’ t take it for granted that they have jobs. Counterfactual thinking has come in handy for Scott over the past month. As bad as the coronavirus situation is, he ’ s been reflecting on how things could be worse. SK : You know, if given the choice of spending a year in my apartment or a year in space, the apartment wins hands-down every time. AG : Even if you ’ ve never been to space, you know the current circumstances could be a lot worse. My wife and I have been talking with our kids about how lucky we are that grocery stores are still open and utilities are still on. We feel deep appreciation for the people who are hard at work keeping those services going. I ’ ve been thinking about how much harder it would be to do my job without the internet or a phone. Recording this podcast would be pretty much impossible. As it does with many astronauts, being in space broadened Scott ’ s idea of who his neighbors were. SK : When we ’ re in space, and you look down at planet Earth, the planet is incredibly beautiful. During the daytime, you don ’ t see political borders, and it makes it look like humanity is all part of one big team, and when you ’ re part of a team, you need to work together to solve our problems. And I think this virus has shown us that we ’ re really, for better or worse, more interconnected than I really realized. If you consider the last pandemic, which was like 100 years ago, and if you consider what we have been able to accomplish as a species in a hundred years, it ’ s mind-blowing. I am absolutely confident that we will get through this, but it ’ s going to take all of us working together, making the right decisions, doing the right things. Teamwork is absolutely critical when you ’ re trying to do something like this. We ’ re really fighting, right now, the greatest battle for our lives that we ’ ve ever had to fight.”WorkLife”is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ’ Donnell, Jessica Glazer, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelan. This episode was produced by Constanza Gallardo. Our show is mixed by Rick Kwan. Original music by Hahnsdale Hsu and Allison Leyton-Brown. Ad stories produced by Pineapple Street Studios. Special thanks to our sponsors, Accenture, BetterUp, Hilton and SAP. For the conference call in real life, thanks to Tripp Crosby. For the snippets of what it ’ s like to work from home, thanks to Jason Gates, Ryan Smith, Amanda Monday, Danielle Tussing, Erin Kahana, Tiffany Chang, John Mi, Aaron Guo, Next Plumbing and everyone else who called in to share their stories with us. For their research, thanks to Bob Ginnett on airline crews, Cameron Klein and colleagues on team building, Ashley Hardin on personal knowledge, Jeremy Bailenson on giant virtual heads, Yaacov Trope and Nira Liberman on mental time travel, Kate Sweeny and Michael Dooley on worrying versus rumination, Constantine Sedikides and colleagues on nostalgia, Laura Kray and colleagues on counterfactual thinking, Emily Bianchi on starting your career in a recession, and my late advisor, Richard Hackman, on teams as amplifiers. For all their help this season, we thank Nicole Bode, Valentina Bojanini, Sammy Case, Micah Eames, Will Hennessy, Dian Lofton, Jen Michalski, Sarah Jane Souther and Emma Taubner. That ' s a wrap for season three of"WorkLife."Thank you so much for joining us. If you like what you heard, please rate and review the show. It helps people find us. And are you still a potato? LO : I ' m sorry – am I still a potato? AG : Yeah, I mean, you ' re potato boss now. You have to be a potato, right? LO : The funny thing is, the first time I got the request to be a potato, I couldn ' t figure out how to be a potato. It took me – it was like the opposite problem. It took me hours to get the potato back up and running again.

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Mm hmm. This is Design Matters with Debbie Millman, you could say that Cheryl Strayed is very adaptable. Her memoir, Wild, was adapted into a movie starring Reese Witherspoon. Another book, Tiny, Beautiful Things, was adapted for the stage by Nia Vardalos and Thomas Kael. Tiny, Beautiful Things itself was adapted from an advice column she once wrote called Dear Sugar. And that advice column has ultimately been adapted into a New York Times podcast called Sugar Calling. I ' m joined by Cheryl Strayed, who is recording herself in her home in Portland, Oregon. Cheryl Strayed, welcome to Design Matters. Hi, Debbie. I am so thrilled to be here. I ' m a fan of the podcast and I ' ve been dying to talk to you for ages. Oh, I ' m so glad. I ' m so excited. I listen to a recent podcast where you said that this pandemic has made it clear to you that the first thing you are is a writer. Was that ever really in doubt? No, no. It was never in doubt inside myself. But what I meant by that is one of the things that happened after a while became a bestseller, is suddenly I had so many opportunities that were not writing. They were writing adjacent. For example, I now have a really active career as a as a paid public speaker. I never I mean, I do unpaid public talks, too. But what I mean is I never in my wildest dreams imagined that I would be, you know, traveling the world, giving talks. And I and I am like I actually have a whole career of that in addition to my writing. And that was born out of a combination of, you know, wild success. And then also my you know, much to my surprise, I ' m good at it and I enjoy it. I understand that you ' re a huge planner, as am I. And it ' s been hard for you not to know what you ' re doing next month or July or August. How are you managing your schedule? And I ' m mostly asking this for my own sake to really get a sense of how you ' re managing. It ' s really interesting, isn ' t I mean, for me, I realized that planning has always made me feel safe and it ' s always been for me the vehicle of my ambitions. What I mean by that is that setting intentions has always been really important for me in terms of like if I ' m in a place, whether it be emotionally or professionally or financially, that that that it ' s not a good place, I think, OK, my intention is to go there and I make a plan and I see it on the horizon and I think about the steps I need to take to get there. And so on the deepest level, planning has been actually an incredibly healing act for me. It ' s also been just I get pleasure from knowing the logistics of everything. I '

m a detail person. I love maybe that sense of control that I have when I look at my calendar, I go, OK, we ’ re going to do this in June and that in July and that and August. And this time next year will be here, you know, and I love that. It gives me a kind of. Pleasure. I mean, I even joke with my husband, our long running argument, so let ’ s see, we ’ ve met we met in 1995, so I guess it ’ ll be 25 years since we met this fall, this September. My running joke with him, you know, we ’ ve been fighting for for decades about him not putting things on the calendar and not being a planner. And then every once in a while, he ’ ll do it like he ’ ll put something on the calendar. And I ’ m like, that is the sexiest thing you ever did for me. Yeah. Yeah. I totally lost my love language. Yeah. What about you? Well, I. I know that you ’ re a list maker and you ’ re not only a list speaker, but you make sublist hear lists. And I do the same thing and I am so I don ’ t know what the word would be attached to my calendar. I have a paper calendar, I ’ ve had a paper calendar for decades, and my dad used to send me the American Express book calendar, which I used to use, and then he passed away. So I needed another vehicle. And so I just have this little paper calendar. It ’ s actually little, but it ’ s two year calendar and I am attached to this thing. It goes with me everywhere. So I didn ’ t even answer your question. So how am I dealing with it? The first thing I was kind of in denial. Like a lot of people, what I decided is the pandemic would last about eight weeks at least. Its impact on my life would be, you know, that was the first thing like, OK, everything is canceled in March and April and maybe May, but June and July and August are totally on. Right. And then as the weeks passed and I realized, oh, my gosh, you know, and I finally, you know, I had to do. The thing I have many times advise others to do when they feel powerless is to surrender and to accept that ’ s about accepting what ’ s true, accepting what ’ s true. Really one of the most radical acts of my life. And I think any life, even if what ’ s true isn ’ t what you want to be true. Right. Because then you can work from a place of reality rather than delusion. And so I ’ m trying to accept it and let go of. The future, or at least my sense of knowing what ’ s going to happen in the future, let ’ s go into the past a little bit. I ’ d love to talk to you a little bit about how you have navigated the arc of your life. You were born Cheryl Niland in Spangler, Pennsylvania. Yeah. And moved to Chaska, Minnesota, when you were six years old. Shortly thereafter, your parents got divorced. And in addition to the time after your mother died, it seems as if those years were some of the darkest years of your life. Mm hmm. Did you realize that at the time? Gosh, that ’ s such a great question. I did I did realize it to the extent that a child can, which is somewhat limited. I was born into a house of really extremes. On one hand, I had this mother who was very loving and very kind hearted and warm and optimistic and in so many ways communicated to me and my my older. I have an older sister who ’ s three years older and a younger brother who ’ s about three years younger. And she always communicated to us that sense of of wonder and love and light and the beautiful things. But we were living in a house that was, you know, frankly, terrifying. My father was violent and abusive. He was emotionally abusive to all of us. He was physically violent to my mother to an extreme degree. And we were terrified of him. And also, you know, we witnessed I witnessed my brother and sister and I all witnessed horrifying things, things that I that I never witnessed beyond that, you know, as an adult. I mean, I saw as a little child my first some of the first things I saw were really, you know, my mother being beaten by my father, my mother almost being killed before my eyes and my father my mother being raped by my father. And so my memory, my perception of what I understood in those years. It ’ s definitely one of fear and and sorrow and terror and darkness. But because that was my life, it wasn ’ t really until my mother finally escaped my dad that I realized, oh, this is what happiness is, this is how it should be. So, yeah, I mean, I really think I had these kind of two childhoods, really three childhoods. But the first one was terror and darkness and violence and abuse. And the second one was my mom is a single mom and we were very poor. We were poor with my dad too, but really living in poverty with a single mom and three kids and a lot of chaos and disarray, but but also a lot of light and joy and **fun** and no longer being under the the sort of weight of that fear that you have when you live with somebody who ' s abusive. I completely understand in many ways. I had a lot of similar experiences so that a lot of the questions I ' m going to ask you are really not only for my listeners benefit, but also for my own in terms of really being so curious about how somebody can emerge from that kind of darkness to be able to. Say, this is happiness, you know, this is happiness. Mm hmm. You ' ve written this about what a father ' s role is in his children ' s life. The father ' s job is to teach his children how to be warriors, to give them the confidence to get on the horse, to ride into battle when it ' s necessary to do so. If you don ' t get that from your father, you have to teach yourself. Well, that ' s so resonated with me. What do you think you had to teach yourself? Like, what is the biggest thing you think you had to teach yourself? Oh, you don ' t ask little questions, do you? You asked me questions. Big questions, sorry. Big questions. You know, I think that the biggest thing is that I ' m OK in this world. I have the strength and the courage and the **resilience** and the heart to be OK, to be safe within myself. And I think that that ' s what I mean. When I said to be a warrior, I mean, I think we are very often think of this in terms of battle. And years ago I wrote I wrote about this in a while, too. But so right after my mom died, I was living in Minneapolis. I was twenty two and a friend of mine gave me a gift certificate to see an astrologer. And I was like, OK, well, I don ' t know, like what says astrology stuff, you know? And but I thought, OK, I ' ll go. And I went and I talked to this woman, Pat Kaluza, and she had this like hippie sort of place in Minneapolis, and she read my birth chart. And it was astounding and amazing. And one of the things she said to me, she kept going to the father. She kept saying, your father, he ' s a Vietnam vet or he ' s troubled or he ' s you know. And I kept saying, like, oh, yeah, my father is not in my life. He ' s nothing. He ' s nothing. He ' s not anything. And she said to me, well, you were wounded. Your father was wounded. And when you have a parent who ' s wounded and who hasn ' t healed his or her wounds, you you as the child, you ' re wounded in the same place. And so you ' re going to have to heal that wounds. And the way she talked to me about it is that there will be times in your life that you need to ride into battle for yourself and you need to teach yourself how to do that. And, you know, I would say that that extends beyond necessarily the father. I think that, you know, if we didn ' t get that essential sense of self-worth from both parents, we need to reckon with that in our adult lives. And so with my father, I had to heal many things. But but the most the biggest one you ask what the most important one I think it was that sense is that. That I ' m secure and safe in the world and that I ' m strong enough to face anything really and to really step into that knowledge, not that you ' ll be like always brave or always do the right thing or always accept what ' s happening in in a sort of graceful way or courageous way. But that at my deepest, deepest, deepest place within me, I believe in the power of my own **resilience** and ability to survive and persist. And I think that ' s what the parents give us if they love us well and they love us. Right. And if we don ' t get that, we have to find it ourselves and the world. I think that as I was rereading Wilde and as I watch the movie again, too, which was really wonderful. I also got the sense that that your journey was one of finding out if you could rely on yourself, if you could take care of yourself, pretty extreme way. Yeah. Testing yourself. But but I got the sense that that ability to do that was was also underneath everything else that you were doing. I think so, too. And, you know, I think I want to say to like I think though we all need to do that. You know, obviously somebody like me who had a father who was abusive and not, you know, not the father that any one wants. And then and then a mother who died, I was really an orphan. And, you know, I had to go and find those things, as you say. And yet I think part of maybe the human journey is that like that I even think of my own kids, teenagers right now who are loved and secure and living in a very happy home and have wonderful lives. And yet what I know about them is that that part of their journey is going to be finding their way and finding their strength and finding their courage and and also finding their path, you know, and all of those things are made more difficult when we have difficult parents or dead parents or abusive parents. But they ’ re all it ’ s part of what we have to do as humans. And and that ’ s why, you know, I think so often it wasn ’ t until after I actually wrote Wild that I understood what I had done on that hike is that I had given myself my own rite of passage. And, you know, I said, like, you have to go test yourself to see who you are. And that ’ s those rituals of rites of passage are what what we ’ ve done as humans throughout all time, across every culture and, you know, continents and so on and so forth. We don ’ t do that so much anymore. And I think it ’ s a loss. I think most of us would benefit from being asked to find out who we really are by being put in uncomfortable circumstances or challenging circumstances. When I was doing my research on your childhood and adolescence, they came across a couple of little facts that really were wonderfully surprising. I know that when you were 13, you moved to Aitkin County. It was very narrowly lived in a house with your mom and your stepfather. They built the house and for many years the house didn ’ t have electricity or running water, didn ’ t have indoor plumbing until after you went to college. But despite all of this, Cheryl Strayed, you were a high school cheerleader and the homecoming queen. And so you you were an overachiever from day one. Aha. I was. I was. So let me explain. My stepfather, who was a carpenter, he was seven years younger than my mom. They they married when I was like 11 and he was working under the table for this roofing contractor. And it was the middle of the winter in Minnesota. There was ice on the **roof**. He slid off the **roof** and broke his back. And as I said, we were always flat broke and he was injured and out of work for more than a year. And my mom at the time was working as an administrative assistant for the like the small town attorney in Chaska, Minnesota. And he said, you know, I ’ ll represent you pro bono. It ’ s not fair because my stepfather was working under the table. His boss said, oh, no, I don ’ t need to pay you anything. So by the end of the year, they got a twelve thousand dollar check. That was the payment for a broken back back in 1980 or so. And my mom said, you know, this is our only chance we ’ ll ever have our own home that we own and let ’ s not buy a home. Let ’ s buy land. So they went to northern Minnesota. Yeah. And we moved to 40 acres of land. We lived really in a tarpaper shack, one room tarpaper shack for the first six months. And we built the house ourselves. And it was a lot of work and it was incredibly difficult. And I was a teenager and I wanted to be pretty and popular and not associated with going to the bathroom in an outhouse or taking a bath in a pond, which is what I did. Or taking a bath in a bucket, which is what I did. So, yeah, my my rebellion in my teen years was to seem to be a version of myself that thought that I wanted to project a sense of success and grace and togetherness. And I you know, I wanted to be popular because to be popular is to be loved. I wanted people to love me. Now, let ’ s talk a little bit about books, because you ’ ve written about how, as you were growing up, books were your religion and you cited the experience of reading Dalton Trumbo novel Johnny Got His Gun is a book that first exposed you to the power of inhabiting the life of another human. Yeah. What was that like for you? How did how did that infuse who you were? Johnny got his gun. Dalton Trumbo, really, really powerfully important book. I think I was about 14 when I read it. And it ’ s just, you know, you ’ re inside the mind of a man who ’ s had, you know, been deeply injured in the war and lost his limbs. And he ’ s, you know, you ’ re just living in his head and and having his memories and his delusions and his his sorrows and his rages. And you ’ re right there inside of him. And I think it was this maybe the first book that the material was so utterly dark and painful and true that it was the it was the first time that I understood what war was, what grief was. You know, I ’ d learned about things from a distance. And what that novel taught me is how you can inhabit an experience that is so not your own. And and, you know, I loved books long before that, but that was the first time I stepped into one and thought, this is a kind of magic, if you will. This is a kind of portal that that I guess I ’ ve been longing to enter for a long time. Another piece of this that goes way back is always as a young child, I always wanted to know what was happening inside of other people ’ s minds. Like really like what did they really feel? What did they really think? What was their actual experience of being human? And so in this book, I was like, wow, I ’ ve finally been let in to that to that secret. You also started working at 13. You had a variety of jobs. You were a janitor ’ s assistant at your high school waxing floors. You were a waitress at the Dairy Queen. And I understand you can put a on on the top of a soft serve ice cream cone like a pro. Of course, I worked there. So, yeah. I mean, that ’ s the thing growing up poor. What you quickly realize is if you ever want anything, you have to earn the money yourself. Because even though my mom provided for us to the best of her abilities, you know, I wanted things like brand name shoes or Levi jeans, like we would go to Kmart and at the beginning the school year, and we ’ d each get like a certain amount of money we could spend. And then that was it. And I was like, I don ’ t want to, but I want to wear the brands, you know? And my mom would say, I can ’ t afford it. So as soon as I could, you know, I babysat before that. But but honestly, as soon as I could, I got myself a job and I was like thirteen and a half. I sort of fudged my age. I think you had to be 14 to actually work. But by 13, I worked as a full time job as a janitor ’ s assistant and in my school, **cleaning** the books that the shelves and the drawers and the desk getting gum off of things and painting. It was through this program for low income kids. You know, I worked and I earned my money and I bought my stuff. That ’ s that was part of the whole plan, you know, that I would get it myself if it couldn ’ t be provided for me. Sometimes I. I talk to my peers who were like going to camp or going to Paris or whatever, and I envy them. And yet I also think, wow, the best education I ever had was being a janitor in my high school and then going from that job straight to my job at the Dairy Queen. And I did that all summer and they were minimum wage jobs. But but they were the first lesson I had and really how to be self-sufficient and making it happen, like not expecting others to make it happen for you. And I treasure that. Like, I think I learned more doing that than I would have going to a lovely summer camp. But, you know, we all learned we all find our way. I read that it never occurred to you to attend college outside of Minnesota and you only applied to one school, the University of St. Thomas in St. Paul. How come how come it didn ’ t occur to you that you could go out outside of Minnesota? It never occurred to me. It absolutely never occurred to me that it was possible to leave my state to go to college. Like to me, the furthest even going to college seemed like going a very far. Our way and the reason I didn ’ t know to apply to more than one is because nobody told me. I wasn ’ t folded into anyone ’ s arms when it came to like, well, let ’ s talk about college, let ’ s talk about your options, let ’ s talk about the process. I figured out by reading something that I had to take the ACT test. Nobody talked to me about it. I paid for it myself. I drove myself there. I took the test. I didn ’ t study for it because I was told, you can ’ t study for that test. You just go. And it ’ s like an aptitude test. I don ’ t know what score I got because it didn ’ t even matter to me. I just took the test, did my best, and I went to I put in my application to one school and and I applied to this one school because it was a small school. I was overwhelmed of thinking, like going to like the University of Minnesota. It just seemed too big. So, yeah, you know, I look back at it and think, what was I thinking? And all I can say is I just didn ’ t have that information available. And I think we think this is so unique. But it ’ s not I mean, it ’ s the reality for a lot of kids living in poverty that they don ’ t know the way to college. For your sophomore year, you transferred to the University of Minnesota in Minneapolis and you studied English and women ’ s studies. What did you think you wanted to do professionally at that point as a poor kid getting an education? The first goal and really at first the only goal was like, you get a job so you can make money. And I first thought that the only path to that would be journalism. So when I was a freshman, I majored in journalism. And when I transferred to the University of Minnesota, I was like journalism, but I quit. I took a class really in that first year or so with Michael Dennis Browne, the poet Michael Dennis Browne and my eyes, I just everything, oh, you know, absolutely exploded. And I thought. I have to trust this, I have to you know, I thought that I could sort of funnel my my desire to write into the channel of journalism because it ’ s the job, you know, that you can actually get paid to write. But I don ’ t want to do that. Like, I want to really put all of my heart and my faith in and creative writing. So I switched majors and became an English major. And what I thought I ’ d do with that is become a great American writer. That ’ s what I thought I ’ d do with that I was absolutely relentlessly, ruthlessly committed to following through and being, you know, just really, really, you know, holding hard onto that thread. That was my writing and doing it and doing it and doing it until I succeeded. And you know, it ’ s funny, as I say, these words, you know, the female in me, the woman who was raised as a girl, is like, Oh, don ’ t say that. Don ’ t say you want to be a great American writer because that seems cocky or that you ’ re bragging or but I ’ ll tell you, like, that ’ s the thing that got me through. Is that that like, again, the intention, the plan, the ambition, if I sort of dithered around and said, well, you know, I hope that this turns out and I hope I can, you know, publish a story like I would never, ever, ever be talking to you right now. Yeah. And it reminds me of that little mantra you had while you were on your hike. I ’ m not afraid. I ’ m not afraid. I will not be afraid. Yeah, well, and of course. And of course, when did I say I ’ m not afraid? And I was. Yeah. And, you know, even to this day, like, writing is so hard for me. It ’ s so hard for me. I still have to say, Cheryl, you can do this. You ’ re going to do this and you are not going to give up. You ’ re not going to be second. You ’ re going to go you ’ re going to, you know, go all the way to the finish line. In March of 1991, when your mom was 45, you were 22, your mom died of cancer. And you ’ ve said that your mother ’ s death was in many ways your Genesis story and the start of what you called your wild years. And you said that for you, using drugs or having a lot of sex or any sort of reckless behavior was about love, was about trying to find love in this weird way, trying to show the world this woman ’ s life meant so much that I ’ m going to ruin mine to honor her. Mm hmm. And I wanted to ask you about that destructive thing that we do. Why do you think we hurt ourselves when we ’ re hurt? Mm. Well, again, you you with the big questions, Debbie, I kind of want to say I ’ m sorry, but I ’ m sorry. It ’ s so it ’ s so it ’ s so deep and so big that it layered the answers. Why do we hurt ourselves when we ’ re suffering? Why do we self-destruct when. Yeah, when we feel like we ’ ve been ruined? I think it ’ s a couple of things. I think one of the things is it ’ s a signal. It ’ s a signal to the people around us that we ’ re saying help me. Even if we with our words are saying like, oh, no, I ’ m fine, just leave me alone. And in so many ways, that ’ s, you know, when you turn to drugs, for example, you know, that ’ s a way of of pushing others away from you. Right. And yet what I was so clearly trying to do, I can see, is to be like, help me, help me, help me. And it ’ s also a kind of test. So it ’ s a signal. I need help. It ’ s a test. Is there anyone out there who loves me enough to help me? I also think in my case, there was this sort of division within me or this polarity that was the that ’ s almost like it ’ s almost like mythic in it ’ s you know, when I think about it and I interpret it this way is like the mother, the good mother who ’ s been taken from me and the and the bad father. The dark father who abandoned me. You know, if I can ’ t be the woman my mother raised me to be that ambitious, generous life, you know, light filled person, maybe I can be the junk, the pile of shit, the darkness that my father nurtured in me. There was something that I had to figure out about those primal relationships. That I had to rage against and he ’ ll. And understand and revise, and I think that a lot of us have to do that, you know, I think that a lot of people who are suffering and certainly people who have written to me as sugar, you know, they have a problem. They write with the problem right there. Like, this is my question. This is my thing. But really, the problem is, is that deep, deep river that ’ s flowing beneath all the troubles that that subterranean channel, that that is your parents, that is those early stories you received, your losses and your gains and your wounds and your sorrows that you have to you have to heal them. And sometimes, you know, healing is an ugly thing. Sometimes healing is destruction. Sometimes healing is turning away. Sometimes healing is a kind of rage and anger, you know? And I think that for me, it was just like I had to pass through. Everything so the image that always comes to mind to me is one of total destruction, when I saw that I was going to lose everything after my mom died and I did, my family also really fell apart and was lost when I understood that that was what was going to happen and that I couldn ’ t make it not happen. That ’ s when I really turned to heroin. That ’ s when I was like, OK, if if the house is going to burn down, I ’ m going to go like the the piece of this that I in some ways I can have control over is I ’ m going to actually burn the whole the whole, you know, the whole land, like the whole homestead. The hard thing about that is, of course, some people stay there, they get lost there. They ’ re walking through the ashes forever and luck. And I ’ m so grateful that that wasn ’ t my fate, you know, that that I had to do that stuff in order to realize that. That I wasn ’ t the person my father raised me to be, my father didn ’ t raise me, I was the person my mother raised me to be. And the best thing I could do, and this is why I said that so much of that stuff was about love, as I realized, like I was trying to show the world, listen, this amazing woman is gone and I am suffering. I wanted to, you know, with my own life, demonstrate how how how gigantic that loss was. And what I realized is the only way I could do that, the absolute only way I could do that was to make good on my intentions, to make good on my ambitions, to be the woman my mother raised me to be, as I said, and wild to become, to become. It ’ s interesting you brought her to life through your words and, you know, she brought you to life through her life. It ’ s a really nice symmetry there, it ’ s crazy, you know, I lost her, I was the same age when she died as she was when she was pregnant with me. So I lost her at the same age that I that I came into her life. Right. You know, I wish that I wish that I didn ’ t have to go into the darkness. But, you know, I was always trying to move in the direction of love.

And I felt so alone in my grief. And then when I wrote about it and told the truth about it, how savage it was, I felt like, OK, everyone ' s going to think I ' m crazy. But instead, what everyone thought was me too to this day, you know, so this day, really now, you know, hundreds of people, hundreds of thousands of people around the world, maybe millions of people around the world are saying, I know how you feel because I felt that way, too. And and I ' m suddenly not alone in my grief. Yeah. And I ' m always shocked by that. I am very grateful looking back on my life. And maybe this is synthesized happiness, I ' m not entirely sure in in terms of what it has given me in terms of my ambition or my creativity or even just my sense of the world. But I also deeply, deeply regret the pain that I caused others with my own self destructiveness. And one of the biggest regrets that I have. You know what what I put other people through in that journey to be who I am and where I am now. But I also know that I couldn ' t have survived in many ways without that destructiveness and that testing of of who I was as I revised myself, so to speak. Yeah. Do you think that part of that revision for you was to change your name? Oh, absolutely. Mm hmm. Yeah. So I was born Cheryl Milind, as you said, and then I got married young. I was Cheryl Milind Littig. We we both hyphenated our names. And because we were trying to be radical feminist and cool, which it was a weird thing to do back then. And so, yeah, when I got divorced right before I went to hike on the Pacific Crest Trail in 1995, I was getting divorced like 94, 95. I realized that. Yeah. So. I had to set my life on a new course, which was really the old course, which was the course, you know, way back, you know, when I was like hiding Jane Eyre in 10th grade or whatever. I think now those plans for myself, I took a little detour, did some other things, and then I was like, OK, this is, you know, life. My mother ' s dead. My first marriage is over. This man I loved but got married too young and I ' m an orphan and I need to make my life I ' m going to make my life. And part of that, obviously, for a writer is about language. And Cheryl Niland just didn ' t feel like me. And so I came up with my own last name, Cheryl Strayed. What ' s so funny to me now? So I ' ve been Cheryl strayed longer than I was anything else. Now I ' m 51. People still say like, oh, but Cheryl Strayed is not your real name. And I ' m like, Cheryl Strayed is the realest name I ever had. So you talked about your hike along the Pacific Crest Trail. You were 26 when you embarked on this solo three month, eleven hundred mile hike. If you knew that there were people listening who are considering taking the same hike, what would you share with them? What would you tell them? What advice might you give them? Well, absolutely go if you have any, whether it ' s the same hike as mine or any long hike. If you have any desire to do this, do it because it is walking, especially walking along way for many days on end day after day. It ' s it ' s it ' s a deeply, deeply challenging thing. So you get you you gain your sense of your own strength and your own ability to endure difficulty, monotony, pain. And of course, what happens on the outside, one **foot** in front of the other in front of the other, also happens on the inside that, you know, turns out for me, you know, the way to heal. Anything is to keep going. And to keep going with humility and faith and a sense of optimism, even when it looks and feels really hard and, you know, I just I love this idea of the body teaching us what what the soul and the spirit and the heart needs to know. And that ' s what happens on a long walk. That ' s exactly what happens on a long walk. I mean, I can ' t say enough what a powerfully humbling and healing act that was. And it seems unimaginable to me to consider hiking 11 hundred miles in near solitude without a phone on the Internet. Well, the phone off the entire time, I mean, it ' s just not even comprehensible. What would people say? Well, that ' s really this was 1995. And I didn ' t realize until, you know, I was sort of midway through wild that I realized, oh, I ' m I ' m actually writing a kind of historical. Memoir about a world that is no longer that, a world that is now past, that our experience of the wilderness is now one where, first of all, we can just research everything online, you know, where does that trail begin and end? Where is the water? Where ’ s the you know, I had this right, you know, and that was one of one of the things and of course, and wild, I did make comic of like unpreparedness or whatever. But the other piece of this I want to say is that I prepared to the extent that I could that there wasn ’ t the Internet. You know, I went to the Minneapolis Public Library and said, you know, what books do you have on the Pacific Crest Trail? And they had one. And it was the book I already purchased at. It wasn ’ t it wasn ’ t like things were available. You know, you just had to go and see how it was. Yeah, I was absolutely alone. I was the first eight days of my hike. I didn ’ t even see another person. What I learned is that that would be, you know, a regular thing like that. I would many, many times go three, four or five days without another seeing another person. There was no way to contact anyone except if I came upon a payphone or if I sent them a letter. Right. I ’ m so grateful for that world. I mean, I ’ m so grateful that I took my hike during that time because I think had I not done that, I would have spent a lot of time. Tweeting at people in town, right, connecting great **pictures**, the fox, I mean, yeah, and getting feedback from people. Right. And not just sitting in my solitude. I mean, that ’ s the thing about that. That kind of deep solitude is it ’ s just like it ’ s just you and there ’ s nothing to do but reckon with yourself. There ’ s nothing to do but have that conversation with yourself in your head. There ’ s nothing to do but allow those memories to emerge. By the time I was finished with my hike, I honestly felt like I had thought about everything I remembered in my whole life, every relationship, every person. What a therapeutic experience. Monster was the name of your backpack, which at its heaviest weighed nearly 70 pounds, even if even at its lightest it was 50 pounds. And making yourself suffer in a physical way kind of feels like the opposite of **fun**. And I read that you said that the act of remembering your suffering can become pleasure afterwards. And I wanted to know, like, how so? How does that become pleasure? I ’ m such a believer. I call it retrospective **fun**. OK, and this is the advice too. I ’ d give someone wanted to take a long hike is you just have to or really any kind of journey. You just have to acknowledge that, that very often the best things we do are painful and complicated and difficult and exhausting and require us to be out of our comfort zone and to accept difficult things. Right. If the journeys we take are just like exactly how we imagined they ’ d be and plan they ’ d be and and everything was idyllic and blissful. And there was no there was no sort of difficulty. We would be like, yeah, that that was that was **fun**. But, you know, there ’ s there ’ s nothing about it. Right. There ’ s no texture to it. Yeah. No, no. Great. And I think that the the grittier an experience is, the more it teaches us, we never, ever, ever forget the lessons we learned the hard way. I began on a backpacking novice. I became a backpacking expert. I thought that I couldn ’ t do that. And I did. I over and over and over again said to myself, I can ’ t go on. I can ’ t. And I always did. And then that becomes part of who you are. It becomes part of the story you tell yourself. So then, you know, 10 years later, you ’ re in labor. As I was trying to give birth to my eleven pound baby boy and I was thinking, I can ’ t do this. And I want what the deepest voice in me said, you know, you can you know, you can. Nine days after your hike and a few days after your 27th birthday, you met your husband through someone who came to the yard sale you had selling your hiking gear to raise a little bit of money to live on. Several years later, you married him, you had two children. Do you think that this sort of new life, big quotes was the gift you got at the end of a long struggle? Do you think it was just luck? Tell me about how you view that sort of moment in time where you come back and everything changes? Well, you know, I think it ’ s it ’ s a combination of things, right? Like luck is always luck is always a factor in everything. You know, how how do we ever know that we ’ re going to be standing there when that person walks up and you say hello and then something that ’ s born of that? Right. How I think of Brian ’ s life in my life and thinking about how how did our paths ever get to cross? And and of course, a lot of it is that by the time I met Brian, I was OK. You know, I was in a place that I was able to see more clearly again and be more kind to myself and to be truer to my nature and truer to my ambition and my vision. I was doing, again, what I ’ m here to do. And when you ’ re doing that and then you meet somebody else who ’ s doing that, it ’ s like really good timing, you know. So in so many ways, it is the gift. You know, getting, Brian, is the gift of. Doing all of that laundry and searching and finding and it ’ s not the end of the story, of course, it ’ s like it wasn ’ t like, well, now I ’ m just like all great and everything. And and then, you know, 25 years later, we ’ re just here we are in the same place. I mean, we both struggled and grown and gone through all kinds of things. Right. But, you know, to make yourself ready for that kind of relationship, I wasn ’ t walking the trail to do that. But that was one of the outcomes of it. Yeah. You moved to Portland, you got a job waiting tables. You started working full time, writing your first novel, Torch, and you then went to graduate school to Syracuse University and got a master of fine arts in fiction writing so you could actually finish the book. Why? What made you decide to do that? What made you feel like that would help you get to that moment that were where the book would be finished? Yeah, it was it was mostly financial. And some also just like the logistics of of giving myself real time to write. So all through my 20s, I was a waitress, I was a youth advocate. I was a vegetable picker in an organic farm. I like I was an EMT. I did all kinds of things. But what I was really doing was writing. That was my real work. And it was I had this big student loan to pay off. It was exhausting always to be just living hand-to-mouth and struggling financially while trying to write. And and I sort of always thought like. OK, my my first novel will, of course, be published and done and everything by the time I ’ m 30, but I, I just said, OK, as I was approaching 30, I was like, OK, this hasn ’ t been done yet. And here are the reasons why. One is I ’ m always having to work full time to pay the bills and to you know, it ’ s hard to have that kind of time and focus when you ’ re working full time. So graduate school kind of solve that problem for me. And I only looked into I only applied to graduate schools that, you know, with the idea that I would only go if I were offered a full a full fellowship or scholarship and tuition remission and that it wouldn ’ t put me further in student loan debt Torch was published in 2006. At that time, you had two children under the age of two, 18 months apart. You started writing Wild in 2008, 13 years after you completed the hike. Initially, you thought it would be a collection of essays. Your second book, What? What Changed? Yeah, well, you know, the reason I thought it would be a collection of essays is I thought there is no way in **hell** that I can write a book while having two little babies and being in financial stress. And, you know, all of this. I just thought, I can ’ t write another book. And what happened is I started writing and then it went on and on and on and on. And I was like, oh, I have a bigger story to tell. And it really I didn ’ t know that until I was doing the writing because I found the bigger story when I was writing and buy the bigger story. I mean, the story that exceeds the kind of like, oh, here ’ s you know, here ’ s my interesting journey or here ’ s the big loss I suffered, you know, that that I knew that I needed to find a universal thread that the my journey and my loss and my experience would be had the potential to be expressed in a way that other people would see themselves in it. And it took some writing for me to to figure out how to do that or to figure out that that was there. You said that being a memoirist is about learning how to re-enter previous versions of yourself. Yeah. How do you go back into that previous version while still maintaining who you are? Well, you know, you enter the magic of writing, that ' s that ' s what ' s so cool about it is so what I mean by that is this is the only way for me to write. About my real heartbreak over the decision to end my first marriage is to abandon the woman I am now who says, oh my gosh, I was too young to get married and he was great and everything, but it ' s a good thing we broke up because now we ' re both married to other people and we ' re happy, you know, so leave that person at the door and and start writing your way into the person who was in love with this man. Truly. And who also felt like she could not stay with him, who had to break his heart and her own in order to live the life that she, for whatever inexplicable reason that was still kind of, you know, beyond her explanation, had to trust herself to do that. And so I went in and reinhabited that and and as I was writing, for example, that scene in Wild, where my husband and I are decided to get divorced and then we get divorced and we ' re saying goodbye to each other. You know, I was just sobbing as I was writing it. And even though it ' s actually not sad, like it ' s actually not sad to me now. It ' s a memory of sadness that I that I reinhabited, this was really made a life to me. I was on the set a lot when we were making the movie. So I was there like every day and, you know, really involved in everything. And there ' s this scene in the movie and in the book where me and my ex-husband, we ' ve just mailed off, you know, our divorce papers and we ' re standing on the street and we ' re talking to each other and crying and embracing for the last time. And it ' s very emotional in the book. I ' m crying, you know, I ' m like, it ' s sad. And then, you know, I ' m standing there with Reese Witherspoon right before she ' s going to walk into the street with Thomas Sadowski, who plays my ex-husband. And they ' re going to begin shooting the scene. And she suddenly looks at me and she just just starts sobbing. She was getting ready for that scene and they go into the street and they shoot this, and I ' m standing next to the director, I ' m watching this on the camera. And I noticed that some people on the crew were kind of gathered around me and a couple of them sort of put their hand on my back and put their hand on my shoulder to comfort me. Wow. And I thought, oh, OK. I, I it was so clear, crystal clear to me because I was watching that scene and I wasn ' t crying because it ' s not sad anymore. But when I watch the scene of my mom dying. I cried, Oh, that ' s so sad, and it doesn ' t mean that one thing isn ' t sad and one thing is it means that there are different kinds of sorrow and some of them are sorrow. Our sad in the moment and others are sad forever. And, you know, so I think that ' s a really important thing that I try to remember still in my life, that it ' s like, is this a sorrow that I ' m going to carry with me forever, or is it a sorrow that is like a crucible and I have to endure it and then I will be better for it that I you know, that ' s what I when I wrote as sugar, you ' ve got to be brave enough to break your own heart. I was talking about exactly that thing where, you know, I had to make a decision that caused me and another person pain, but I ' m better for it. And it ' s not sad anymore. It was it was actually the the golden key that opened the door to to my liberation. And there was loss in that. But but there was more gain. You know, it became a gift in the end. Like, very often I think how memoir writing is almost like the process of therapy, right. Where you go back there and you say, well, who was that person and why did she do this and think this and love these people and leave these people. And there ' s always an answer if you ' re willing to dig for it. While you were working on Wilde, you also started writing your column, Dear Sugar for the Rumpus. You were mentoring students at the Attic Institute in Portland. You were teaching workshops at universities, writing for magazines. But you and Brian, your husband, who ' s a documentary filmmaker, were, as you put it, epically broke. You were, as they say, the classic starving artists. Yes. And I also thought it was interesting that, well, you were writing Dear Sugar, you were giving people advice and in wild you weren ' t. But people have read it that way. And I ' m wondering how you feel about that, that whole sort of notion of advice giving. I know you ' ve referred to self-help work as intellectually mushy and and I think I think that, too. But I do think that people look for that in work even when it ' s not there for their own needs. Yeah. When I was writing, while it never occurred to me that anyone would be experienced as inspiring, you know, I was really just trying to write the truest Rozz realist. Story about that experience, about my grief, about my finding my way on this long walk, about the experience I had in the wilderness, and so, yeah, people do experience that in a in a like, well, what ' s the message of wild? I ' m like, oh, you know, the message is whatever you think, whatever truth you find in yourself when you read it. That ' s what I meant when books were my religion as I felt saved by them, like I felt seen by them. So, you know, to me, I ' ve always said, like in airports, there are many of these kind of little convenience stores and airports will have these little turnstiles that are there, like inspirational or self-help like that. And there are never books. There ' s never like novels there. There ' s never, you know, tiny, beautiful things there. It ' s all these very specific things that are very instructive, like, here ' s your problem. Here ' s what you need to do. And I find that the most helpful literature when it comes to like what real self-help is, is things like, I don ' t know, Jane Eyre, you know, Alice Munro. Short stories. There I am. And Alice Munro, short stories. There I am. And Toni Morrison ' s beloved. There I am. And you know, Mary Oliver ' s poem. So, yeah, you know, I didn ' t intend for those things to be self-help and frankly, even sugar when when terrible, beautiful things came out, I was like it was in the self-help section of many books or so I was like, what? What? So I think of myself as an accidental self-help writer because of course, of course, dear sugar columns are self-help. And yet what they also are is literature. Yes, I understand you ' ve been at a sort of crossroads now is different opportunities have come your way weighing the reasons to do it, weighing the reasons not to do it. And you ' ve stated it ' s not about the number of things on the list. It ' s about the weight of those things. And almost always you think that the things that mean and matter the most really come down to one question. And that question is, what do you really want to do? And I wanted to see if you can help me understand how to know when something is the thing you really want to do. Archives again, with the heart of the system. So for me, that Deep is wanting, it ' s not that it ' s the easiest thing to do. It ' s the thing about which you feel like I can make I can create something that feels larger than me if I decide to do this. So if I if I write a book and not only is it a deep and true expression of some deep and true things, I want to put into the world, if it ' s not only that and that, it becomes also something that is meaningful to others, that ' s a big thing to contribute to the world. And I think that for me. That sense of like rightness or a sense of like if this mission is fulfilled, will it extend beyond my small little life? Like I feel that I feel that as a sort of powerful call. Cheryl, I have to ask questions for you. OK. The first is something that I was really heartened to read about. I understand that sandwiches are problematic. Of course they are. Sandwiches are just like chaos. You know, machines, right? They ' re just like willy nilly. See, I know you ' re my sister. I know you that I know you also feel this way like I to the orderly. Absolutely. Everybody has to be perfect. Not only do they have to be orderly. Yes. That ' s the whole goal of a sandwich for me is to make every bit as much like the previous bite as possible. OK, consistency. So you put the you know, whatever whatever you ’ re putting on it, it has to be uniformly applied everywhere. Burritos also sometimes have this problem. If people don ’ t do the burrito correctly, it ’ s just tacos. Yeah, absolutely. Yeah. So this is my last question, Cheryl. What is the thing you want to do most next? Finish my next book, I really am ready to do that. So wild and tiny, beautiful things were published within four months of each other in 2012, and basically my life was just an absolute you don ’ t like it was just like a volcano. And I was, you know, I movies. I was a little girly and. Yeah, and the play and I was involved in the podcast and the public speaking career and also the kids. You know, during all this time I got your mom these, you know, little records, you know, and who now are 14 and 16. And I just feel like, wow, OK, I really now I ’ m ready to go back. Go back to that basic go back to that sit there and write your next book. And so that ’ s what I ’ m doing. And I really want to do it and I ’ m really excited about it. I ’ m also afraid and doubtful as scared at all the things I am when I ’ m writing, which means I ’ m writing my next book. I can ’ t wait to read it, cannot wait to read it. Thank you, Cheryl Strayed. Thank you so much for joining me today. And design matters. Such an open for so many wonderful things to say. Thank you, Debbie. You are. You are a woman after my own heart, I swear. We have to meet in real life someday. Have sandwiches. Absolutely. I would love that. I would love that. Thank you, my dear. This is the 16th year broadcasting design matters. And I ’ d like to thank you for listening. And remember, we can talk about making a difference. We could make a difference. Well, we can do both. I ’ m Debbie Millman and I look forward to talking with you again soon. Special thanks to the sponsor of this episode. Hoffler and Co Design Matters is produced by Curtis Fox Productions and on Pandemic Times. The show was recorded at the School of Visual Arts Masters in Branding Program in New York City, the first and longest running branding program in the world. The editor in chief of Design Matters is Zachary Petitt, and the art director is Emily Weiland.

Text Sample 728: POS Dim 5 – 2018 – Score 7.00 – 2018/t000514.txt

It was my first year as an atmospheric science professor at Texas Tech University. We had just moved to Lubbock, Texas, which had recently been named the second most conservative city in the entire United States. A colleague asked me to **guest** teach his undergraduate geology class. I said,”Sure.”But when I showed up, the lecture hall was cavernous and dark. As I tracked the history of the carbon cycle through geologic time to present day, most of the students were slumped over, dozing or looking at their phones. I ended my talk with a hopeful request for any questions. And one hand shot up right away. I looked encouraging, he stood up, and in a loud voice, he said,”You ’ re a democrat, aren ’ t you?””No,”I said,”I ’ m Canadian.”That was my baptism by fire into what has now become a sad fact of life here in the United States and increasingly across Canada as well. The fact that the number one predictor of whether we agree that climate is changing, humans are responsible and the impacts are increasingly serious and even dangerous, has nothing to do with how much we know about science or even how smart we are but simply where we fall on the political spectrum. Does the thermometer give us a different answer depending on if we ’ re liberal or conservative? Of course not. But if that thermometer tells us that the planet is warming, that humans are responsible and that to fix this thing, we have to wean ourselves off fossil fuels as soon as possible — well, some people would rather cut off their arm than give the government any further excuse to disrupt their comfortable lives and tell them what to do. But saying,”Yes, it ’ s a real problem, but I don ’ t want to fix it,”that makes us the bad guy, and nobody wants to be the bad guy. So instead, we use arguments like,"It ' s just a natural cycle.""It ' s the sun."Or my favorite,"Those climate scientists are just in it for the money."I get that at least once a week. But these are just sciencey-sounding smoke screens, that are designed to hide the real reason for our objections, which have nothing to do with the science and everything to do with our ideology and our identity. So when we turn on the TV these days, it seems like pundit X is saying,"It ' s cold outside. Where is global warming now?"And politician Y is saying,"For every scientist who says this thing is real, I can find one who says it isn ' t."So it ' s no surprise that sometimes we feel like everybody is saying these myths. But when we look at the data — and the Yale Program on Climate [Change] Communication has done public opinion polling across the country now for a number of years — the data shows that actually 70 percent of people in the United States agree that the climate is changing. And 70 percent also agree that it will harm plants and animals, and it will harm future generations. But then when we dig down a bit deeper, the rubber starts to hit the road. Only about 60 percent of people think it will affect people in the United States. Only 40 percent of people think it will affect us personally. And then when you ask people,"Do you ever talk about this?"two-thirds of people in the entire United States say,"Never."And even worse, when you say,"Do you hear the media talk about this?"Over three-quarters of people say no. So it ' s a vicious cycle. The planet warms. Heat waves get stronger. Heavy precipitation gets more frequent. Hurricanes get more intense. Scientists release yet another doom-filled report. Politicians push back even more strongly, repeating the same sciencey-sounding myths. What can we do to break this vicious cycle? The number one thing we can do is the exact thing that we ' re not doing : talk about it. But you might say,"I ' m not a scientist. How am I supposed to talk about radiative forcing or cloud parametrization in climate models?"We don ' t need to be talking about more science ; we ' ve been talking about the science for over 150 years. Did you know that it ' s been 150 years or more since the 1850s, when climate scientists first discovered that digging up and burning coal and **gas** and oil is producing heat- trapping **gases** that is wrapping an extra blanket around the planet? That ' s how long we ' ve known. It ' s been 50 years since scientists first formally warned a US president of the dangers of a changing climate, and that president was Lyndon B. Johnson. And what ' s more, the social science has taught us that if people have built their identity on rejecting a certain set of facts, then arguing over those facts is a personal attack. It causes them to dig in deeper, and it digs a trench, rather than building a bridge. So if we aren ' t supposed to talk about more science, or if we don ' t need to talk about more science, then what should we be talking about? The most important thing to do is, instead of starting up with your head, with all the data and facts in our head, to start from the heart, to start by talking about why it matters to us, to begin with genuinely shared values. Are we both parents? Do we live in the same community? Do we enjoy the same outdoor activities : hiking, biking, fishing, even hunting? Do we care about the economy or national security? For me, one of the most foundational ways I found to connect with people is through my faith. As a Christian, I believe that God created this incredible planet that we live on and gave us responsibility over every living thing on it. And I furthermore believe that we are to care for and love the least fortunate among us, those who are already suffering the impacts of poverty, hunger, disease and more. If you don ' t know what the values are that someone has, have a conversation, get to know them, figure out what makes them tick. And then once we have, all we have to do is connect the dots between the values they already have and why they would care about a changing climate. I truly believe, after thousands of conversations that I ' ve had over the past decade and more, that just about every single person in the world already has the values they need to care about a changing climate. They just haven ' t connected the dots. And that ' s what we can do through our conversation with them. The only reason why I care about a changing climate is because of who I already am. I ' m a mother, so I care about the future of my child. I live in West Texas, where water is already scarce, and climate change is impacting the availability of that water. I ' m a Christian, I care about a changing climate because it is, as the military calls it, a"threat multiplier."It takes those issues, like poverty and hunger and disease and lack of access to **clean** water and even political crises that lead to refugee crises — it takes all of these issues and it exacerbates them, it makes them worse. I ' m not a Rotarian. But when I gave my first talk at a Rotary Club, I walked in and they had this giant banner that had the Four-Way Test on it. Is it the truth? Absolutely. Is it fair? Heck, no, that ' s why I care most about climate change, because it is absolutely unfair. Those who have contributed the least to the problem are bearing the brunt of the impacts. It went on to ask : Would it be beneficial to all, would it build goodwill? Well, to fix it certainly would. So I took my talk, and I reorganized it into the Four-Way Test, and then I gave it to this group of conservative businesspeople in West Texas. And I will never forget at the end, a local bank owner came up to me with the most bemused look on his face. And he said,"You know, I wasn ' t sure about this whole global warming thing, but it passed the Four-Way Test."These values, though — they have to be genuine. I was giving a talk at a Christian college a number of years ago, and after my talk, a fellow scientist came up and he said,"I need some help. I ' ve been really trying hard to get my **foot** in the door with our local churches, but I can ' t seem to get any traction. I want to talk to them about why climate change matters."So I said,"Well, the best thing to do is to start with the denomination that you ' re part of, because you share the most values with those people. What type of church do you attend?""Oh, I don ' t attend any church, I ' m an atheist,"he said. I said,"Well, in that case, starting with a faith community is probably not the best idea. Let ' s talk about what you do enjoy doing, what you are involved in."And we were able to identify a community group that he was part of, that he could start with. The **bottom** line is, we don ' t have to be a liberal tree hugger to care about a changing climate. All we have to be is a human living on this planet. Because no matter where we live, climate change is already affecting us today. If we live along the coasts, in many places, we ' re already seeing"sunny-day flooding."If we live in western North America, we ' re seeing much greater area being burned by wildfires. If we live in many coastal locations, from the Gulf of Mexico to the South Pacific, we are seeing stronger hurricanes, typhoons and cyclones, powered by a warming ocean. If we live in Texas or if we live in Syria, we ' re seeing climate change supersize our droughts, making them more frequent and more severe. Wherever we live, we ' re already being affected by a changing climate. So you might say,"OK, that ' s good. We can talk impacts. We can scare the pants off people, because this thing is serious."And it is, believe me. I ' m a scientist, I know. But fear is not what is going to motivate us for the long-term, sustained change that we need to fix this thing. Fear is designed to help us run away from the bear. Or just run faster than the person beside us. What we need to fix this thing is rational hope. Yes, we absolutely do need to recognize what ' s at stake. Of course we do. But we need a vision of a better future — a future with abundant energy, with a stable economy, with resources available to all, where our lives are not worse but better than they are today. There are solutions. And that ' s why the second important thing that we have to talk about is solutions — practical, viable, accessible, attractive solutions. Like what? Well, there ' s no silver bullet, as they say, but there ' s plenty of silver buckshot. There ' s simple solutions that save us money and reduce our carbon footprint at the same time. Yes, light bulbs. I love my plug-in car. I ' d like some solar shingles. But imagine if every home came with a switch beside the front door, that when you left the house, you could turn off everything except your fridge. And maybe the DVR. Lifestyle choices : eating local, eating lower down the food chain and reducing food waste, which at the global scale, is one of the most important things that we can do to fix this problem. I ' m a climate scientist, so the irony of traveling around to talk to people about a changing climate is not lost on me. The biggest part of my personal carbon footprint is my travel. And that ' s why I carefully collect my invitations. I usually don ' t go anywhere unless I have a critical mass of invitations in one place — anywhere from three to four to sometimes even as many as 10 or 15 talks in a given place — so I can minimize the impact of my carbon footprint as much as possible. And I ' ve transitioned nearly three-quarters of the talks I give to video. Often, people will say,"Well, we ' ve never done that before."But I say,"Well, let ' s give it a try, I think it could work."Most of all, though, we need to talk about what ' s already happening today around the world and what could happen in the future. Now, I live in Texas, and Texas has the highest carbon emissions of any state in the United States. You might say,"Well, what can you talk about in Texas?"The answer is : a lot. Did you know that in Texas there ' s over 25, 000 jobs in the wind energy industry? We are almost up to 20 percent of our electricity from **clean**, renewable sources, most of that wind, though solar is growing quickly. The largest army base in the United States, Fort Hood, is, of course, in Texas. And they ' ve been powered by wind and solar energy now, because it ' s saving taxpayers over 150 million dollars. Yes. What about those who don ' t have the resources that we have? In sub-Saharan Africa, there are hundreds of millions of people who don ' t have access to any type of energy except kerosine, and it ' s very expensive. Around the entire world, the fastest-growing type of new energy today is solar. And they have plenty of solar. So social impact investors, nonprofits, even corporations are going in and using innovative new microfinancing schemes, like, pay-as-you-go solar, so that people can buy the power they need in increments, sometimes even on their cell phone. One company, Azuri, has distributed tens of thousands of units across 11 countries, from Rwanda to Uganda. They estimate that they ' ve powered over 30 million hours of electricity and over 10 million hours of cell phone charging. What about the giant growing economies of China and India? Well, climate impacts might seem a little further down the road, but air quality impacts are right here today. And they know that **clean** energy is essential to powering their future. So China is investing hundreds of billions of dollars in **clean** energy. They ' re flooding coal mines, and they ' re putting floating solar **panels** on the surface. They also have a panda-shaped solar farm. Yes, they ' re still burning coal. But they ' ve shut down all the coal plants around Beijing. And in India, they ' re looking to replace a quarter of a billion incandescent light bulbs with LEDs, which will save them seven billion dollars in energy costs. They ' re investing in green jobs, and they ' re looking to decarbonize their entire vehicle fleet. India may be the first country to industrialize without relying primarily on fossil fuels. The world is changing. But it just isn ' t changing fast enough. Too often, we **picture** this problem as a giant boulder sitting at the **bottom** of a hill, with only a few hands on it, trying to roll it up the hill. But in reality, that boulder is already at the top of the hill. And it ' s got hundreds of millions of hands, maybe even billions on it, pushing it down. It just isn ' t going fast enough. So how do we speed up that giant boulder so we can fix climate change in time? You guessed it. The number one way is by talking about it. The **bottom** line is this : climate change is affecting you and me right here, right now, in the places where we live. But by working together, we can fix it. Sure, it ’ s a daunting problem. Nobody knows that more than us climate scientists. But we can ’ t give in to despair. We have to go out and actively look for the hope that we need, that will inspire us to act. And that hope begins with a conversation today. Thank you.

Text Sample 729: POS Dim 5 – 2018 – Score 6.00 – 2018/t000383.txt

At this very moment, with every breath we take, major delta cities across the globe are sinking, including New York, London, Tokyo, Shanghai, New Orleans, and as well as my city, Bangkok. Here is the usual version of climate change. This is mine. Nothing much, just a crocodile on the street. This is an urgent impact of climate change : over sinking cities. Here, you can see the urbanization of Bangkok, growing in every direction, shifting from porous, agricultural land — the land that can breathe and absorb water — to a concrete jungle. This is what parts of it look like after 30 minutes of rainfall. And every time it rains, I wish my car could turn into a boat. This land has no room for water. It has lost its absorbent capacity. The reality of Bangkok ’ s metropolitan region is a city of 15 million people living, working and commuting on top of a shifting, muddy river delta. Bangkok is sinking more than one centimeter per year, which is four times faster than the rate of predicted sea level rise. And we could be below sea level by 2030, which will be here too soon. There is no coincidence that I am here as a landscape **architect**. As a child, I grew up in a row house next to the busy road always filled with traffic. In front of my house, there was a concrete parking lot, and that was my playground. The only living creature I would find, and had **fun** with, were these sneaky little plants trying to grow through the crack of the concrete pavement. My favorite game with friends was to dig a bigger and bigger hole through this crack to let this little plant creep out — sneak out more and more. And yes, landscape architecture gives me the opportunity to continue my cracking ambition — to connect this concrete land back to nature. Before, Thais — my people — we were adapted to the cycle of the wet and dry season, and you could call us amphibious. We lived both on land and on water. We were adapted to both. And flooding was a happy event, when the water fertilized our land. But now, flooding means... disaster. In 2011, Thailand was hit by the most damaging and the most expensive flood disaster in our history. Flooding has turned central Thailand into an enormous lake. Here, you can see the scale of the flood in the center of the image, to the scale of Bangkok, outlined in yellow. The water was overflowing from the north, making its way across several provinces. Millions of my people, including me and my family, were displaced and homeless. Some had to escape the city. Many were terrified of losing their home and their belongings, so they stayed back in the flood with no electricity and **clean** water. For me, this flood reflects clearly that our modern infrastructure, and especially our notion of fighting flood with concrete, had made us so extremely vulnerable to the climate uncertainty. But in the heart of this disaster, I found my calling. I cannot just sit and wait as my city continues to sink. The city needed me, and I had the ability to fix this problem. Six years ago, I started my project. My teams and I won the design competition for Chulalongkorn Centenary Park. This was the big, bold mission of the first university in Thailand for celebrating its hundredth anniversary by giving this piece of land as a public **park** to our city. Having a **park** sounds very normal to many other cities, but not in Bangkok, which has one of the lowest public green space per capita among megacities in Asia. Our project ’ s become the first new public **park** in almost 30 years. The

11-acre **park** — a big green crack at the heart of Bangkok — opened just last year. Thank you. For four years, we have pushed through countless meetings to convince and never give up to convincing that this **park** isn ’ t just for beautification or recreation : it must help the city deal with water, it must help the city confront climate change. And here is how it works. Bangkok is a flat city, so we harnessed the power of gravity by inclining the whole **park** to collect every drop of rain. The gravity force pulls down the runoff from the highest point to the lowest point. This **park** has three main elements that work as one system. The first — the green **roof**. This is the biggest green **roof** in Thailand, with the rainwater tanks and **museum** underneath. In the dry season, the collected rain can be used to water the **park** for up to a month. The runoff on the green **roof** then falls through wetlands with the native water plants that can help filter and help **clean** water. And at the lower end, the retention pond collects all of the water. At this pond, there are water bikes. People can pedal and help **clean** water. Their exercise becomes an active part of the **park** water system. When life gives you a flood, you have **fun** with the water. Centenary Park gives room for people and room for water, which is exactly what we and our cities need. This is an amphibious design. This **park** is not about getting rid of flood. It ’ s about creating a way to live with it. And not a single drop of rain is wasted in this **park**. This **park** can hold and collect a million gallons of water. Thank you. Every given project, for me, is an opportunity to create more green cracks through this concrete jungle by using landscape architecture as a solution, like turning this concrete **roof** into an urban farm, which can help absorb rain ; reduce urban heat island and grow food in the middle of the city ; reuse the abandoned concrete structure to become a green pedestrian bridge ; and another flood-proof **park** at Thammasat University, which nearly completes the biggest green **roof** on an academic campus yet in Southeast Asia. Severe flooding is our new normal, putting the southeast Asian region — the region with the most coastline — at extreme risk. Creating a **park** is just one solution. The awareness of climate change means we, in every profession we are involved, are increasingly obligated to understand the climate risk and put whatever we are working on as part of the solution. Because if our cities continue the way they are now, a similar catastrophe will happen again... and again. Creating a solution in these sinking cities is like making the impossible possible. And for that, I would like to share one word that I always keep in mind, that is,”tangjai.”The literal translation for”tang”is”to firmly stand,”and”jai”means”heart.”Firmly stand your heart at your goal. In Thai language, when you commit to do something, you put tangjai in front of your word, so your heart will be in your action. No matter how rough the path, how big the crack, you push through to your goal, because that ’ s where your heart is. And yes, Thailand is home. This land is my only home, and that ’ s where I firmly stand my heart. Where do you stand yours? Thank you. Thank you. Kp kun ka.

Text Sample 730: POS Dim 5 – 2018 – Score 6.00 – 2018/t000502.txt

It ’ s 4am, you ’ ve been awake for forty hours, when you unlock a puzzle containing this video of some kind of dance-off between a chicken and a roller- skating beaver. The confusion and delight you ’ re experiencing is a typical moment at the MIT Mystery Hunt, which is basically the Olympics meets Burning Man for a specific type of nerd. Today, I ’ m going to take you inside this strange, intellectually masochistic and incredibly joyful world. But first, I have to explain what I mean when I say”puzzle.”A puzzle-hunt- style puzzle is a data set.

It can be a grid of letters, a sudoku, a video, an audio — it can be anything that contains hidden information that can eventually resolve into an answer that is a word or a phrase. So, to give you an example, this is a puzzle called”Master Pieces.”It consists of 10 images of LEGO people looking at piles of LEGOs. And to save us some time, I ’ m going to explain what ’ s going on here. Each of the piles of LEGOs is a deconstructed work of art in the style of a famous artist. So, does anybody recognize the artist on the left? They used a lot of red. I heard”Rothko,”yeah. The second one? Mondrian. Alex Rosenthal : Yeah, well done. And the third one? This is the hardest one — Yeah, Klimt, I heard it. Well done, the color is the biggest clue there. So the puzzle has various clues that tell you what matters here are the artists, not the specific works of art. And what you need to do is then look at what you haven ’ t used yet, which is the number of LEGO people in each painting. And you can count them and then count into the artists ’ last names by the same number of letters. So there ’ s three people in front of the Rothko on the left, so you take the third letter, which is a T. There ’ s only one in front of the Mondrian, so you take the first letter, M. And there ’ s three again in front of Klimt, so you take the third letter, I. You do that for all 10 of the original artists and put them in the order, and you get the answer, which is”illuminate.”Puzzles like this are about communicating an idea. But where I ’ m trying to be as clear as possible for you now, puzzles have to navigate the line between **abstraction** and clarity. They have to be obtuse enough to make you work for it, but elegant enough so you can get to the aha moment, where everything clicks into place. Puzzle solvers are junkies for this aha moment — it feels like a brief high and an instant of pristine clarity. And there ’ s also a deeper fulfillment at play here, which is that humans are innate problem-solvers. That ’ s why we love crosswords and escape rooms and figuring out how to explore the **bottom** of the ocean. Solving deviously difficult puzzles expands our minds in new directions, and it also helps us come at problems from diverse perspectives. These puzzles come in various puzzle hunts, which come in various shapes and sizes. There ’ s one-hour ones designed for novices, 24-hour road rallies, and the puzzle hunt of puzzle hunts, the MIT Mystery Hunt. This is an event that takes place once a year and has around 2, 000 people descending on MIT ’ s campus and solving puzzles in teams that range from a single person to over 100. My team has 60 people on it — that includes a national crossword puzzle tournament champion, a particle physicist, a composer, an actual deep-sea explorer, and me, feeling like”Mr. Bean goes to Bletchley Park.”That ’ s actually an apt comparison, because one year involved a puzzle where you had to construct a working Enigma machine out of pieces of cardboard. Each Mystery Hunt has a theme. Past ones have included”The Matrix”and”Alice in Wonderland.”It ’ s often pop culture- and literary-based themes. And the goal is to find the coin that ’ s been hidden somewhere on MIT ’ s campus. And in order to get there, you have to solve around 150 puzzles and do various events and challenges. I had done this for about 10 years without ever dreaming of winning, until January of 2016, where 53 hours into a hunt whose theme is the movie”Inception,”we haven ’ t slept in days, so everything is hilarious... The tables are covered in piles of papers, of our notes and completed puzzles. The whiteboards are an unintelligible mess of three days ’ worth of insights. And we ’ re stuck on two puzzles. If we could crack them, we would get into the endgame, and after hours of work, in a magical moment, they both fall within 10 seconds of each other, and soon, we ’ re on the final runaround, a series of clues that will lead us to the coin, and we ’ re racing through the halls of MIT, trying not to knock over or terrify tour groups, when we realize we ’ re not alone, there ’ s another team on the runaround as well, and we don ’ t know who ’ s ahead. So, we ’ re a mess of anxiety, anticipation, exhilaration and sleep deprivation, when we arrive at the Alchemist, a sculpture in which we find... this coin. Yeah. And in claiming it, we win the MIT Mystery Hunt by a tiny margin of five minutes. What I didn ’ t mention before is that the prize for winning is that you get to construct the whole hunt for the following year. The punishment for winning is that you have to construct the whole hunt for the following year. At the beginning of 2016, I had never constructed a puzzle before — I had solved plenty of puzzles, but constructing and solving are entirely different beasts. But once again, I was lucky to be on a team full of brilliant mentors and collaborators. So, from a constructor ’ s point of view, a puzzle is where I have an idea, and instead of telling you what it is, I ’ m going to leave a trail of breadcrumbs so you can figure it out for yourself, and have the joy and experience of the aha moment. This is another way of looking at the aha moment. And what ’ s incredible to me is that this experience, which is very emotional and kind of almost physical, is something that can be carefully designed. So, to show you what I mean, this is a puzzle I co-constructed with my friend Matt Gruskin. It ’ s a text adventure, which is the old-school adventure game format, where you ’ re exploring, going north, east, south and west, picking up items and using them. And you could get to the end of the game part, but you won ’ t have solved the puzzle. In order to do so, you have to recognize a hidden layer of information, and the easiest way of seeing it is by mapping the game out. That looks something like this. Does anybody recognize what this is? Yeah, exactly. This text adventure takes place within”Settlers of Catan.”Who here knows what”Settlers”is? Nerds. If you don ’ t know,”Settlers”is a board game where you ’ re competing against other people to collect resources and use them to build structures. And within the text adventure, we hid information in various ways, with which you could reconstruct an entire game. You could figure out the roads, the cities, the towns, the resources, the numbers on the tiles, even the dice rolls. You put all that information together and you could extract an answer in a way that ’ s too complicated to explain right now. But find me afterwards if you really want to know. But what this puzzle emphasized for me is the value of perspective shifts in inspiring an aha. So, in this puzzle, you go from experiencing the world on the ground, as a character, to looking down on it from above as if you ’ re playing a board game, and in that shift, you completely reframe all the information you ’ ve been given. The hardest part of construction for me is coming up with a great idea for an aha. Fortunately, the world is a torrent of ideas and information. I ’ ve seen fantastic puzzles constructed out of the waggle dances of bees, and the remarkable coincidence that the 88 keys of a piano can be perfectly mapped to the 88 constellations in the sky. Once you find that out, you can ’ t not construct the puzzle, and it ’ s going to be about having the solvers make that connection in their own minds. Whether you give them stars on a keyboard or play the celestial music of the cosmos, you ’ re getting them there, one way or another. Before long, you find yourself staring at a turtle, and asking yourself,”Is this a puzzle?”And also, staring at a turtle and saying,”I never appreciated what multitudes this contains in its shell alone.”This might be a familiar experience to you, if you ’ ve ever been watching a TED Talk and asked yourself,”Is this a puzzle?”I ’ m not telling. But what I will say is that puzzles can be found in the most unexpected of places. That brings us back to one of my favorite puzzles of all time, which was constructed by Trip Payne. And this time, I ’ m going to play it for you with the sound on, so get ready to name that tune. Who knows what that is? Yeah,”You Make Me Feel Like a Natural Woman.”So you can identify that and seven other songs and clips, and then look at the videos themselves for clues, where the way that they are **filmed** and edited together plus things like the cutaways to the **panel** of five people sitting at a table, which is reminiscent of a **panel** of judges, all of this can suggest”reality competition show.”And either through internet research, or from just recognizing this, you can get to the aha, which is that these clips are shot- for-shot recreations of lip-synch battles from”RuPaul ’ s Drag Race.”So, why do we do this? You tell me, I don ’ t know. So, first of all, it ’ s really **fun**. But I think it also improves our lives in various ways. Being able to solve puzzles, when I ’ m confronted with a challenge, has allowed me to explore it from multiple perspectives before I lock in an approach. Also, the process of solving is great training for working with a team, knowing when to listen, when to share, and how to recognize and celebrate insight and being able to construct ahas is a very powerful tool. Think of how powerful and exciting and convincing an idea is that comes from your own mind, where you make all the connections yourself. So in January of 2017, after tens of thousands of hours of work, we finally run our Mystery Hunt. And it ’ s a different sort of satisfaction than the quick high of an aha moment. Instead, it ’ s the slow burn of saying something through perplexing **abstraction**, yet being understood. And when it was all over, in our exhaustion, we turned to each other and the world, and we said,”We ’ re never doing this again. It ’ s too much work. It ’ s really **fun**, but no more winning.”One year later, in January of 2018, we won the MIT Mystery Hunt again. So, we ’ re currently I don ’ t know how many tens of thousands of hours of work in, and we ’ re two months out from the 2019 Hunt. So, thank you for listening, I have to go write a puzzle.

Text Sample 731: POS Dim 5 – 2024 – Score 7.00 – 2024/t003231.txt

Auntie 1 : Why are you still single? Auntie 2 : How much do you earn? Auntie 3 : You look fat. Niceaunties : I just love aunties. Some of my aunties, I only see during Chinese New Year. I used to brace myself for inquisition prior to family gatherings. They call you fat but present you with platters of delicious food, made with love and great skill. This is exasperating and endearing. I ’ m not alone in this. This behavior and line of questioning is so common we even have memes for it. I grew up in a big family, where the women, my late grandmother, my mother and some of my 11 aunties, were expected to take care of children. They were big influences in my life, and I witnessed great personalities, strength, eccentricity and humor. They held back from fully expressing themselves because of societal pressures, family expectations and lack of support. My grandma spent most of her life at home. She did not travel and had little contact with the outside world. In the last 20 years of her life, she had dementia and was bedridden. I often imagined her in an alternate reality, where she ’ s out and about, having **fun**, laughing, chatting, enjoying life. But aunties are not just about family. Anyone of any age, of any culture, can be labelled an auntie. It is not something you really want to be associated with. As it could mean you ’ re old-fashioned, rigid in your thinking or naggy. So it is rather negative to be called one. Women of my generation live in fear of being labeled an auntie. But what if you could change this perception, turning it from dread to something **fun** and positive? Thank you. I ’ m a designer from Singapore, and in the last year, also an artist. I came across some intriguing and beautiful images on social media and then found out they were made with AI. Curious about what AI could do, I found a program, I signed up, I keyed in some words, and within seconds, images appeared. I was amazed and mesmerized. Never before have I found a medium that felt like the shortest path between my ideas and the visual. It became addictive. Rather than just making images, I wanted to do more with this tool, to tell stories about auntie culture. My mother engages in many social activities. She is very active, a result of her mother ’ s immobility and limited life. She dances and performs just like the aunties do in Singapore ’ s **parks** every day. This was a huge inspiration for me. So”Niceaunties”was born, a project about freedom, exuberance, self-expression and **fun**. It is a lighthearted perspective on the joyful side of auntie life. Also a surreal narrative, loosely based on the women in my family, their social lives, their relationships, their quirks, humor, and, of course, food. In the last 20 years, I worked in architecture, a profession of discipline, rules and clients. I always worked in a team. It was about fulfilling the client ’ s brief and budgets. With AI, it ’ s just me. It ’ s a breath of fresh air. This lack of restraint has transported me to an imaginary world, where my aunties can live freely as themselves, where anything **fun** can and does happen. For instance, aunties in public places, taking photos with sculptures in absurd ways. This is a common sight. A typical holiday photo becomes aunties trying to kiss the moon, but failing badly. Over the course of a year – instead of designing buildings, I developed”Niceaunties”into a world-building project : the”Auntieverse.”Just like in any world, there are cities, citizens, their lives and their culture. Auntieverse is partly inspired by Singapore, where acronyms are used for everything. A sentence would typically sound like :”I ’ m going to take the MRT to the NTUC near my mom ’ s HDB to avoid the ERP.”In the Auntieverse, we have Tofu-Engineered Sushi Luxury Autos, aka TESLA. It ’ s a transportation factory – where aunties assemble food and legs to create vehicles running on leg power. It ’ s very sustainable. In Chinese dialects,”legs”have the same sound as”cars,”so a common joke would be to say,”Let ’ s get around on our own cars.”Meaning we can move around on our own legs, implying self-reliance. So I imagine at TESLA, we have the usual car assembly, a car wash and, of course, a car junkyard. But legs don ’ t just move cars around in the Auntieverse. Buildings use them too, as do animals and vegetables. Another destination in the Auntieverse is MoMA, the Museum of Modern Aunties... Where modern auntie culture is celebrated. This is a social space where aunties gather, hang out, eat, chitchat, have **fun**, chillax. So one day, I said to my mother that she should show others how to make, and then sell, her delicious food. She said”No lah, too tired.”So I imagined NASA, Nice Auntie Sushi Academy – A culinary school on the moon, where aunties share their cooking skills. Here, amazing food is made available for all, 24-7. Aunties in training learn how to forage fresh ingredients on the moon, how to slice fish, and the cat will ensure all lessons go according to plan. But food isn ’ t just used for eating – In the Auntieverse. It ’ s used for all kinds of relaxing activities. I would love to soak in this ramen hot tub with my aunties. Well, sushi is used extensively in Auntie Spa for all kinds of beauty treatments and relaxing body wraps, a reference to many aunties ’ beauty tips, like putting cucumbers and other foods on their faces and bodies. Personally, I have been rubbed down in coffee grounds, and then wrapped in cling **film**. It ’ s very unsettling. Along with many beauty treatments, like **lasers** and acupuncture, I felt like I had to try, under the promise of youth and beauty. So many women undergo the pain of these beauty treatments to maintain body **ideals** and notions of beauty standards. Sometimes, it feels like having construction sites on our faces and bodies. Does it actually work? I don ’ t know. On the lighter end of the spectrum, Aunties ’ Nail Spa provides many beautiful options for our fingers, including very useful tools for everyday life. Cute and practical. Meanwhile, on Auntlantis, a parallel world inspired by Plato and the Great Pacific Garbage Patch, my aunties take a break on the beach, after a day ’ s work **cleaning** out the ocean floor, in a world filled with plastic waste as we know it. My Auntieverse is an ever-expanding narrative of food, culture, family, relationships and the world we live in. I could not have imagined having created this without the help of AI. Before AI, I have never edited a video. This technology lowered the barrier to learning and allowed me to bring my imagination to life in only 12 months. But this isn ’ t just about AI. The Auntieverse is about honoring the memory of extraordinary women who nurtured one another and their families. It is about celebrating personalities and bringing into the spotlight the mundane and the bizarre, like some aunties ’ big permed hair.”The taller the hair, the closer to God,”some said. [Auntie ’ s Hair Spray] [smells nice] [holds shape] [animal testing] [long lasting] [while stocks last!] Thank you.

Text Sample 732: POS Dim 5 – 2024 – Score 6.00 – 2024/t003159.txt

So have you ever walked in front of a glass facade and see yourself in the reflection and try to check yourself out? Oh wait, wait, I forgot – the glass facade is there not for you, but it ’ s for the people on the inside to get a great view out. And they ’ re probably looking at you picking the spinach from your teeth. Have you ever walked in a city, come upon a very large commercial building, and in the very hot heat of the summer, the doors fly open, you feel that air-conditioned air come out and blast you, and you just sit there and wallow in that cool air? Oh wait, I forgot, that cool air is for the people on the inside. It ’ s a private building. They won ’ t even let you in to use a bathroom. Have you ever wondered why cities get hotter and hotter every year? Oh, wait, I forgot. In order to cool these buildings, the air conditioning systems have to give off air and emit it to the atmosphere, making it hotter and hotter every year, contributing to the heat island effect. So I ’ m an **architect**, a building tech innovator, and a 10, 000-step-a-day pedestrian. And I want to talk about how buildings can not only benefit the people on the inside, but they can actually improve life for those on the outside. So if you go back into history, things were different, right? In the cave, humans had to only worry about the interior, the outside facade was designed by nature. Once leaving the cave, humans then started to stack rocks to make shelter and in doing so inadvertently made some the earliest facades. From there, culture refined, building construction techniques refined, they discovered that the outside facade that ’ s very public-facing can actually tell stories and narratives. It could be used for public service. So they can tell stories about the family, the histories, the conquests, the wealth, the status, all these types of things appeared on facades, and facades became a thing. So fast forward, we have new technologies, new materials. We ’ re able to thin out the building so it ’ s transparent, the facade itself. But the problem is, glass is a very poor insulator. And that glass then can let the heat in, the cool in, and we had to develop big air-conditioning systems, heating systems to make the interiors even tolerable to stay in. And more recently, we want to keep that transparency. We want the glass, the view. But in order to do that and make the systems efficient, we had to expand that facade system again. So it ’ s highly engineered now to keep out the heat, the cold, the moisture and glare. So all these systems, in order to make it happen, even have four panes of glass to do it. It ’ s comfortable, it works. But what do we do next? So I would like to propose that that thickness of the facade still be used to make people on the inside comfortable. But can we think about the outside portion of the facade benefiting people on the outside? Can we actually think about those thousands of square **feet** of surface area? No, millions of square **feet** of surface area inside the city and use it for the public? Can we use it for infrastructure and for public good? So I ’ m going to show you a few technologies of other people that do that. Here, we can make habitats for animals, ecosystems like these for bees, birds, bats, all kinds of animals are even important to cities. This is a habitat for microorganisms. They ’ re invisible, but they ’ re essential to our livelihood, just like the microbiomes inside your gut. Think about the probiotic qualities of it. The same thing has to happen to cities. Water is an important resource. And so in this system it not only collects water, but it also can purify it as well as store it in a hyperlocalized way. So that places where water security is an issue starts to eliminate and reduce. Think about areas that have drought. And we can also modulate the surfaces, perforate the different types of materials in order to reduce the amount of noise pollution in streets. So think about garbage trucks, sirens, honking horns. That type of stuff can actually diminish with new technologies. In my own work, I look at architecture as something that can contribute positively to sustainability, to equity and to wellness. I ' m going to show you two technologies at different levels of development that I use and apply to public health issues. So the first one is a smog-eating **panel** for architecture. And in this case, it actually starts to filter particulate matter and noxious waste in cities in urban-canyon areas. So an urban canyon is when a street is lined by two very tall buildings on both sides, right, forming a canyon-like scenario. And what happens is in these areas, the air movement is very predictable. It comes from prevailing winds, from thermal convection, from the movement of vehicles in there. And it moves in a circular pattern where it makes it very difficult for the heavy particulate matter to escape. So therefore it just keeps going through and accumulates in the **bottom** over time, more and more. So what we ' re trying to do is modulate the surface so the surface can allow those prevailing winds to flow into these tubes, to filter that smog and move out as fresh air. So you can see here we use models and wind tunnels that we build as well as simulation modeling to test it. We pack the tubes together so that we can maximize the efficacy of it, as well as minimize the material it takes to make it. And we put them in such a position of where that wind flow is actually going, and so that it would have an easy time going into the tubes, **cleaning** the air and coming out as fresh air. We also use the thermal convection model, where in the case the pavement is heating the air, that hot air wants to rise up the face of the building. They can go into these tubes also to filter and then send out fresh air. So it ' s done in a very passive way. And not only do we try to apply this to building facades so that the facades are doing the cleaning, we want to actually design these bus shelters that can use this kind of technology, bringing the freshest pockets of air to the people who spend the most amount of time on the streets breathing the worst air. Another technology that I ' m working on is a self-shading window system. Here you can see it. Basically what it does is it shades the interior automatically so that you can use less energy on air conditioning. The idea came from using these thermo bimetal material, which is a material that curls when heated. And we make these very low-tech robots that we do for **fun** in our office. I know, everybody chuckles when they see it, and it ' s important that you know that because what happens is people get connected to them as if they ' re little pets, as if they are alive. And even we name them. This one ' s actually called Roly Poly. So from there, the idea was that we use this thermo bimetal on the facade of a building so that it actually can react to the outdoor environment. And through the course of a day, through the course of a year, as the temperature changes all the time, the building would also change to follow that. And if we can animate a building, then possibly we can get people to also feel more connected to the building, like their pets. And if we do that, we can make people smile more, they can cherish their buildings and they can even feel protective of their cities. So therefore the happiness level of the city can go up as well as mental health. So we embrace this idea of putting bimetal in the facade. In this case, it ' s actually in the cavity of a double-glazed standard window system. And we allow these little critters, again like butterflies, to flutter around and flip. So when the heat comes in... You can see simulated like the sun, these pieces will flip over and respond to any changes outside. What happens, too, is when they flip, you can still see pretty well through the system, it doesn ’ t block your view so that it turns like into a blackout situation. And then when the sun goes away, they go back to their original position. So you know, what happens, too, is because it ’ s a pixilated system, we can put graphics on it. It opens and closes whenever it wants to so that you can start to see the graphics. The result is awe-inspiring. It ’ s biophilic, and it ’ s magical. And I ’ m happy to say that it ’ s also safe for birds, right? Birds on the outside. So it ’ s actually a great thing. We ’ re hoping to put this technology onto buildings as soon as the end of this year, so stay tuned for that. So how can we get a lot of these cool technologies on buildings? This is the big question. I have three suggestions. The first one is change policy. Can we require owners and and contractors and real estate developers to spend a percentage of their construction cost on public-centric technologies for facades? Can we do that? Two, think entrepreneurially. Can owners actually rent, lease, sell this facade to others for infrastructure? Can it be to agriculture? Can it be to even not-for-profit situations? Can it be for utility companies? There ’ s a lot of possibilities that can happen. And three, advocate for change. Renters inside the buildings must start to demand that their building facades are much more magnanimous. Because think about it : everybody in these buildings must come out to the streets, too. And together we can actually enjoy the improvements to our society and public health. So next time you go out into a city and you ’ re walking around, look carefully at the buildings. They don ’ t have to be just a pretty face. They can serve a bigger purpose. Thank you.

Text Sample 733: POS Dim 5 – 2024 – Score 6.00 – 2024/t003035.txt

You ’ ve probably all heard that solar modules are cheap, but I ’ m not sure you appreciate just how cheap. You can buy a solar module for the price of about eight cups of the posh coffee you get here in Brussels. You can make solar **panels** into a fence, and it ’ s not significantly more expensive than ordinary **fencing** material. But this isn ’ t just great news for rich Europeans who drink posh coffee. Solar power is starting to displace fossil fuels in the countries that most need cheap and **clean** electricity. Now, I have been an analyst of this sector for nearly 20 years and the growth is really impressive. This chart shows new solar built every year, and last year, 444 gigawatts of solar modules got installed worldwide. More than half of that was in China, actually. Four hundred and forty-four gigawatts is quite a lot. It ’ s more than the power capacity of Japan. And even more is being installed this year. It ’ s probably just under 600 gigawatts. But I have to take you off on a diversion now to complain about how hard my job is. Back in 2007, there were about 12 countries installing solar, and the official data coming out of them was good. Now my team is trying to cover 146 solar markets, and some of the data is really bad. Even Germany and Spain are installing more solar than is in their official data. So what hope is there for good data about countries in Africa or Southeast Asia? Take, for example, Pakistan. Now this chart shows the exports of solar modules from China to Pakistan according to China Customs. It reports in dollars, but we convert to gigawatts using average prices. So the official data on how much solar is installed in Pakistan is that there ’ s less than three gigawatts in the country. But more solar modules left China for Pakistan in 2022 alone. Now, we did think that substantial amounts of those were being re-exported to Afghanistan to run irrigation pumps, to grow poppies, to make heroin. And there ’ s also a money laundering scheme which distorts the data. But these don ’ t seem to be that big. And in 2024, things got ridiculous. Sixteen gigawatts of solar modules have left China for Pakistan in the first eight months of this year. What is going on here? I worked with a machine-learning and satellite data firm called Atlas Maps to detect the solar modules in Pakistan. Now, the great thing about detecting solar modules using satellite data is you can ’ t really hide solar modules. The bad thing is that they are distributed, they ’ re quite small installations, and sometimes they look a lot like greenhouses or skylights. But Atlas Maps found 443 installations in Pakistan, many of which we had no idea existed, and looking a lot like the ones on the right of this map, that ’ s some rooftop solar. It also missed some, like, you can see some on the left of this map that the machine-learning algorithm just did not detect. But you can see them. You would see them better if the **picture** was better, but they ’ re there. What the satellite data did confirm is that solar modules in Pakistan are installed near the industrial clusters of Lahore, Karachi and Islamabad. And actually, if you go on Google Earth and just look at a random spot somewhere in Pakistan, in these cities, you can see that loads of the **roofs** have solar **panels** on them. It ’ s really incredible. So the solar boom in Pakistan is really happening. It ’ s not just an artifact of China customs data. So why? So unfortunately, Pakistan is vulnerable to extreme temperatures. In 2024, 500 people have died in Pakistan in heat waves and temperatures reached 52 degrees Celsius So this is a country where people really need fans and air conditioning to survive. But power is expensive in Pakistan, increasingly expensive, and the total power capacity per person in Pakistan is 1 / 18 as much as it is in the United States. This is a country that really needs power. Now, the good thing about solar **panels** is that they generate in the daytime. So if you want to run air conditioning and fans, they ’ re generating at the right time to do that. And we ’ re pretty sure that this is what businesses and homes in Pakistan are doing. They ’ re installing solar **panels** to reduce how much they have to pay the grid and to run their cooling systems, as well as the other stuff that they do. But this is a problem for the largely coal-fired grid in Pakistan because it ’ s losing its best customers, which means it has to put up the prices for its remaining customers to fund the continuing operation of its coal plants. Which means that those remaining customers have higher prices and even more reason to go solar. And the power demand in Pakistan fell nine percent in 2023 as tracked by the grid. And this is probably partly because power is expensive, but also because of all these solar **panels** that are not being seen by the grid. So... As analysts – obviously, this is complicated, but we feel generally glad that Pakistanis are getting their power from cheap and **clean** sources. It seems in general a good thing. But you can ’ t always expect solar build to just go steadily up. Solar markets are more like a roller coaster. Actually, we call it the solar coaster. Take, for example, South Africa. This is data on rooftop solar new build by month from state utility Eskom, which has a really interesting way of tracking this data. It estimates rooftop solar build by the power demand it doesn ’ t see on the grid at particular times. There was a boom in solar installations in South Africa because it was a really bad blackout season, and people literally built solar, so they had power. And as analysts, we got really excited about this. A gigawatt of rooftop solar was built in two months. A gigawatt is the size of a medium-sized coal-fired power plant. OK, it only runs in the daytime, but it works. So we got really excited. We thought that South Africans would copy their neighbors, and the South African solar market would keep going up and up and up. We were wrong about that. The solar market has fallen back actually in South Africa in 2024, and this is because there were fewer blackouts. And this is partly because Eskom fixed its coal-fired power plants. Yay! But also because the solar is helping to support the grid. So it ’ s not always simple, but cheaper solar power is helping poorer countries to meet their power demand without increasing fossil fuel use. And this is what makes my job interesting. But what we need is to stop burning fossil fuels. And it ’ s most obvious that solar is helping there in markets with less unmet need for power. Take California, for example. This is an average day of power supply on each year from 2012 to 2023. In 2012, California got 43 percent of its power from **gas** burned in state, and it imported 29 percent. By 2015, you can see the little yellow bit in this chart, that ’ s large-scale solar eating into power imports in the middle of the day. And by 2023, on an average day, there is so much solar that the state is exporting power around noon. So this is having an impact. What ’ s next? Well, do you see the little pink bit in 2022 and 2023. That ’ s batteries. Batteries are doing exactly what they ’ re meant to do. They ’ re charging on solar in the daytime, and they ’ re discharging in the evening so that the gas-fired power plants don ’ t have to be ramped up. And thanks to all these trends, California carbon emissions per unit of electricity generation have fallen over 30 percent since 2012. So solar power is pushing out fossil fuels in both richer and poorer countries. And the next step is probably batteries working with solar and working with wind and other renewables to continue to push out fossil fuels. This is actually working. And that is why I have hope that we can beat climate change. Thank you.

Text Sample 734: POS Dim 5 – 2023 – Score 9.00 – 2023/t003333.txt

Christiana Figueres : There ’ s no doubt that we are facing exponential impacts of climate change. There is also, however, no doubt that that is being met with exponential progress of the technologies that can help to address climate change. So what we have here is, frankly, a race between two exponential **curves**. By 2030, we collectively will have chosen between door number one. And door number two. Julio Friedmann : The question I am most commonly asked about climate change is :”Should I be optimistic or pessimistic?”Cynthia Williams : Here ’ s what we need to do to make sure that we ’ re ready for an all-electric future. Anika Goss : Being financially secure and climate resilient should be the most important priority. Faustine Delasalle : It ’ s really easy when you look at where we are today and where we ought to be by 2030, to feel like there ’ s a huge gap that will never be filled. But we have good reasons to say that that gap is actually fixable. Nigel Topping : If you ’ re not worried or anxious, you ’ re probably not paying attention. But we should also not be defeatist. We shouldn ’ t say,”We ’ ve done nothing, we ’ re failing.”We ’ ve really started 20 years too late, but it takes a long time to get the flywheel of momentum going. Kingsmill Bond : The tide of change is coming. The costs keep falling. The political pressure keeps increasing. JF : We ’ re going to build a thriving, exciting world. AG : We can do this. JF : Full of potential. AG : We have to do this. [TED Explores A New Climate Vision] Manoush Zamorodi : When it comes to climate change, are you an optimist or a pessimist? News about the climate can be overwhelming and emotional, to the point that many of us just feel numb. I ’ m Manoush Zomorodi, a longtime public media journalist and host of the TED Radio Hour. And today, we want to bring you something special : an hour of nuance, of real people and clarity about what is happening to our planet and what we ’ re doing about it. Because we all know there ’ s plenty of bad news when it comes to the climate crisis. But guess what? There ’ s also some really good news. That ’ s why we ’ re here in Detroit at the TED Countdown Summit, where a cross-section of people who know the most and who are doing the most have gathered to share their ideas and plans with each other and with you. You ’ re going to hear from some incredible speakers about their personal stories and tested solutions. But first, let ’ s get a quick reminder on the basics : climate change in a minute or so. [Prologue : What is Climate Change?] David Biello : The air we breathe has changed. The mix of **gases** is shifting, with more and more greenhouse **gases** like carbon dioxide. And the shift is happening faster each year. In just a few hundred years, fossil fuels formed over eons have been burned as coal, oil and **gas**. The exhaust has transformed the entire atmosphere and ocean. It ’ s like a pollution blanket. And the result we know as climate change. The planet has already warmed more than one degree Celsius and is on a path to heat up even more. That may not seem like a lot, but it ’ s already caused major destruction across the globe. To give us some room to breathe, the world must reduce greenhouse **gas** pollution by more than seven percent each year, every year of this decade. There are a number of ways to do that. Let ’ s start by looking at the greenhouse **gases** themselves. CO2 makes up nearly 75 percent of the pollution emitted each year. And then there ’ s methane or natural **gas**, which makes up 17 percent. Finally, there ’ s nitrous oxide, making up six percent of the problem. There are other greenhouse **gases**, but these three make up the bulk of the climate challenge. Where do they come from? There are several ways to break down the sources of greenhouse **gas** emissions. Here we ’ re using the public Climate Watch data. Start with energy. The majority of modern energy comes from burning fossil fuels, which makes the energy sector 76 percent of the climate challenge, including the fuels used for transportation, industrial processes and agricultural production. Farming and animal raising also have to change, as agriculture accounts for 12 percent of greenhouse **gas** pollution. The remaining 12 percent comes from a grab bag of human activities like industrial heating, clearing forests and more. This is the climate challenge we face : going from adding billions of metric tons of greenhouse **gases** to the air each year to adding zero. Will it be easy? No. Can we do it? Yes, if we choose to. MZ : OK, so that ’ s the situation. Where do we go from here? Well, we ’ ve all heard of a tipping point, that moment when an idea or concept finally takes hold and then changes the world. Well, the UN ’ s climate chief, Simon Stiell, believes we are almost at a tipping point for massive action on climate change. And to get you ready, he wants to take you back to the early ’ 90s, when he was an executive at Nokia, and the world wasn ’ t sending text messages yet. [Chapter 1 : Exponential Change] Simon Stiell : I want you to imagine it ’ s 1992, and you ’ re talking to a telecoms expert. She tells you that mobile phone text messaging is going to fundamentally change the way we communicate. You think, why on Earth am I going to go from this to tapping on a screen with my fingers and my thumbs? But then, after a slow start, text-based services exploded. Today, over 23 billion messages are sent every single day. This paved the way for the smartphone revolution. I know how much we struggle to comprehend and predict what exponential change means, because I was at Nokia in 1993 when that technological revolution started. And now, as head of UN climate change, I ’ m here to tell you that what was true for mobile phones is also true for climate action today. We ’ re here in Detroit, the heart of car manufacturing. Take a look at this graph. Electric vehicle sales are expected to increase to become 70 percent by 2030. Another example, solar energy. Between 2010 and 2020, we moved from 20 gigawatts of solar energy installed to 150 gigawatts. And by 2030, that number is expected to increase again, to 1, 000 gigawatts per year. Change can come fast. FD : So exponential change means that change happens very slowly at first and then comes a tipping point. And suddenly things accelerate and accelerate and accelerate. NT : It ’ s a mathematical term, but it means basically stuff happens very slowly and then it goes really fast. Tessa Kahn : Governments and experts have consistently underestimated the pace and scale at which we can deploy renewable energy and how quickly renewable energy and associated technologies become affordable. Ramez Naam : I predicted in 2011 that we ' d have solar as cheap as coal in some parts of the world by 2015 and half the cost of coal by 2020, and everyone thought I was crazy, I ' ve got to say, almost. And in fact, I was wrong. The actual decline in the cost of solar happened twice as fast as I thought it would. Solar prices today are a century ahead of where the world ' s leading energy experts thought they ' d be in 2010. KB : It ' s now spreading from solar, so it ' s happening also in the wind sector, it ' s happening in the battery sector, it ' s happening in the electric vehicle sector. CF : I don ' t think that we do a good job at communicating the fact that we are progressing more than is commonly known. Nili Gilbert : The human brain is trained best at thinking about linear change. But when you think about an exponential **curve**, next to a line, at first the exponential **curve** is below the line, and so it ' s moving more slowly than you would think. And then it crosses the line and it ' s moving faster than you would think. Habiba Daggash : We call it cautious optimism. We ' re very hopeful about the trajectories that we ' ve seen. But we shouldn ' t let up yet. KB : So we can make this change happen slow or we can make it happen fast. The key point to say is we have agency. MZ : This idea of exponential growth sounds awesome. It sounds amazing, but is it actually happening? Well, I ' m headed into a workshop with industry and climate leaders to find out. The moderators of the workshop challenged us to hold two competing ideas in our minds at the same time. NT : What we ' re really going to explore today is two apparently opposing realities, and they ' re often two opposing narratives. The one which is true, right, is that in terms of the Paris goals, we ' re way off track. We ' re deep into the emergency. But what we ' re going to explore today is the other reality that we know what to do. Exponential change is happening, but it doesn ' t happen by magic. MZ : So what are we seeing in different industries? Well, let ' s start with one that ' s known to be really difficult to decarbonize : cement. Vicente Saiso Alva : I think everything converges into making climate a huge urgent matter for society. We started to feel the pressure from all different types of stakeholders, and in one year of running this program, we realized how fast we could move and how fast we were going in reducing CO2 emissions, which means now to go from 40 to close to 50 percent reduction by 2030. MZ : Encouraging, but to tackle that remaining 50 percent may require technologies that don ' t exist at scale yet. OK, what about shipping? Narin Phol : In Maersk, a few years back, we came out with a net-zero ambition by 2050, and in a few years later, based on the data that we have got, we ' ve really seen a path that we can really accelerate that one. That ' s why the revised target of 2040 came in. MZ : The news in offshore wind was quite dramatic. Jennie Dodson : So my moment of mindset shift was in 2019. The government put out a request to bid for offshore wind permits. And that year the price that came in was two thirds less than just four years before. And really critically, it was the same price as the mainstream market. And that was transformative. Nobody expected that when they started. MZ : The conversation got pretty technical and a bit wonky. NP : So to drive exponential growth, you focus on a reinforcing loop to make sure it works and you slow down on the balancing loop. MZ : But the takeaway is that certain sectors, like EVs and renewables, are being upended, while others, like cement and heavy industry are further behind. But generally across every industry – JD : It ' s amazing stuff. MZ : There ' s an understanding that they all need to work together. And the message? Big change can happen. It ' s going to be hard, but some of us are already doing it, so learn from us. [Chapter 2 : The Electric Vehicle Revolution] OK, we are in an exponential mindset. And as you heard, one example of exponential growth might be sitting in your driveway right now. Electric vehicles are finally really happening, which means that people here in Detroit are rethinking the auto industry. They ’ re rethinking their jobs, their identity. Take Cynthia Williams, head of sustainability at Ford. I sat down with her at Lucky Detroit, a local coffee and barber shop, where she told me that there are three things that need to happen for electric vehicles to go to the next level, from this tipping point to exponential growth. CW : The three things for me are change, collaboration and capacity. So the change piece of it is changing the way folks think about the vehicle, making sure that they understand we ’ re bringing a better vehicle to the customers. MZ : So you don ’ t have to make a sacrifice. CW : I think one of the things we need to do is actually get people in the cars and show them how **fun** it is. And it ’ s no different, it ’ s just powered by a battery. We are at a tipping point. We ’ re at a point of acceleration of electric vehicles. Many are calling it the electric vehicle revolution. Now with that comes unprecedented change that will require us to build new capacity and collaborate in ways that we ’ ve never seen before. To make things happen faster, you have to think differently. You can ’ t just collaborate with the same people because you get the same answer, right? You need to broaden your collaboration space. Go outside of your industry, understand what worked in Norway, for example, to get them to where they are. MZ : Where ’ s the weak link, Cynthia? What ’ s the thing that you ’ re like, that if we don ’ t push through that, we may not get to this full adoption? CW : The critical things that we hear from the consumers is they need infrastructure. That ’ s what it ’ s taking for us to work with governments to make sure that we bring more charging to everyone, whether it ’ s rural, urban, wealthy, low-income, we all need chargers. And I ’ m not saying we need one on every corner, but we do need many, many more. When we start building out the charging infrastructure, we need lighting. We need, you know, security, shelter, we need restrooms. We need other amenities so that people know where to go, where to charge. Next comes capacity. We have to invest in new facilities and talent. Now, that ’ s true for any industry transitioning to greener goals and strategies. And we ’ re investing in talent, we ’ re investing in education and training, and we ’ re also listening to the communities where we live and operate. MZ : So a young Cynthia Williams, now, your counterpart who ’ s studying engineering, will she be learning something very different if she wants to go and work in the automotive industry in Detroit? CW : There will be different options, right. And so, I ’ m a **mechanical** engineer by trade, but I think there will be opportunities for electrical engineering, software engineering. And there ’ s even certificates that students can get instead of going through a four-year college. It ’ s the proper training to get them right, to be dedicated, to build the vehicles that we need for the future. At the end of this year, globally, we ’ ll have the ability to produce 600, 000 electric vehicles. And in 2026, we ’ ll have a global goal to produce two million electric vehicles. And it just goes up from there. MZ : What is this going to look like, this growth for me in the next couple of years, in five years, in 20 years? CW : **Clean** air. That ’ s what I look at. If you think about when we were, during the pandemic, when everybody stopped driving and they were at home, clear skies and moving to zero tailpipe emission vehicles. And if you chose an electric vehicle, that ’ s you participating and you contributing to a better world, to a better future. We have come a long way. It ’ s so gratifying to see electric vehicles on the road, to move from aspiration to reality. To see plants dedicated to building electric vehicles. We have the technology, we have the leadership, we have the ingenuity to respond to the environmental crisis with courage. And I ’ ll tell you, as a person who ’ s dedicated my entire career to sustainability, the road ahead is paved with promise. TED-Ed : If you were buying a car in 1899, you would have had three major options to choose from. You could buy a steam-powered car, typically relying on gas-powered boilers. These could drive as far as you wanted, provided you also wanted to lug around extra water to refuel and didn ’ t mind waiting 30 minutes for your engine to heat up. Alternatively, you could buy a car powered by gasoline. However, the internal combustion engines in these models required dangerous hand cranking to start and emitted loud noises and foul smelling exhaust while driving. So your best bet was probably option number three : a battery-powered electric vehicle. These cars were quick to start, **clean** and quiet to run, and if you lived somewhere with access to electricity, easy to refuel overnight. If this seems like an easy choice, you ’ re not alone. By the end of the 19th century, nearly 40 percent of American cars were electric. In cities with early electric systems, battery-powered cars were a popular and reliable alternative to their occasionally explosive competitors. But electric vehicles had one major problem. Batteries. MZ : Flash forward to the present and batteries are still on everyone ’ s mind. So I visited Mujeeb Ijaz, the founder of the energy storage company One, who also happens to be an electric vehicle history buff. Mujeeb Ijaz : So this is a 1922 Detroit Electric. MZ : OK, so 100-year-old vehicle. MI : That ’ s right, electric vehicle. And then this was Walter Baker. He developed the Baker Electric. And he made the range run 201 miles on a single charge. He also felt strongly that you should be able to wear your top hat, you know, so he has a taller carriage. These had lead acid batteries to begin with. And then Thomas Edison introduced the nickel-iron battery. So this is the battery compartment, one of the battery compartments. And this is where you would charge. MZ : Right there? MI : And we ’ re building on what these original pioneers did. And I think it ’ s important to not think about us starting from scratch now. But go backwards, think about how they solved the problem, what disrupted their ability to advance this to the market, and then pick those problems up and make sure we don ’ t get stuck again. MZ : So we ’ ve been here before, a moment when electric vehicles could have taken off but didn ’ t. So what do we need to do differently this time? MI : Electric vehicles were able to deal with mobility in urban areas, but there were charging deserts where rural driving resulted in no way to get back because you couldn ’ t find electric charging everywhere. And that was a problem. MZ : And that ’ s what people are worried about right now. We ’ re at the same thing. We ’ re worried about range, we ’ re worried about accessibility. All of the things are the same. But the key difference being that we now know that our planet can ’ t keep going like this. MI : That ’ s right. Had the vision of the Model T assembly line and mass manufacturing been applied to electric car, the electric car would have also dropped in price, but range and charging deserts would have still been the obstruction. MZ : OK. We didn ’ t have the technology, but we do now. MI : We do now. MZ : In the US, there hasn ’ t been a mass move to electric vehicles... yet. Mujeeb ’ s company is one of many who want to change that by building batteries that give drivers longer range. And these batteries aren ’ t just for cars. MI : So we produce battery packs for transportation, but we ’ ve also started developing the same battery products into grid batteries. So micro-grids are those technologies that combine solar and storage and replace a coal plant or a natural **gas** plant for generating electricity. So buildings and industry and transportation all working together. And that ’ s an exciting future, is all these industries working together, that makes me more confident that we ’ re at the time in history where this transition will take place. MZ : How many years till we get to that? MI : I think it ’ s within 10 years. MZ : No way, 10 years? MI : We ’ re in a place where that transition is going to be much more visible, and then we just need technologies to drive it up to the mass markets. MZ : I am so excited. Like, genuinely so excited. Everyone everywhere driving electric sounds great, but that also means that the power to charge all those vehicles needs to be green. And that brings us to the energy sector, which is also changing at an incredible pace, even though renewable energy has been years in the making. [Chapter 3 : Renewables] TED-Ed : In the spring of 1954, the press excitedly gathered around Bell Laboratories ' latest invention : a silicon-based solar cell that could efficiently convert the sun ' s energy into electrical current. The creation was celebrated as the dawn of a new era, as reporters touted that civilization would soon run on the sun ' s limitless energy. But the dream had a catch. As this first commercially sold solar cell cost around 300 dollars per watt, meaning at its current rate it would cost well over a million dollars to buy a unit large enough to power a single home. But today, in many countries, solar is the cheapest form of energy to produce, surpassing fossil fuel alternatives like coal and natural **gas**. Millions of homes are equipped with rooftop solar, with most units paying for themselves in their first seven to 12 years and then generating further savings. So how did solar become so affordable? A turning point in solar ' s price history occurred on the floor of Germany ' s parliament, where in 2000, Hermann Scheer introduced the Renewable Energy Sources Act. This legislation laid out a vision for the country ' s energy future in solar and wind. It incentivized citizens to personally invest in rooftop solar **panels** by guaranteeing payment to homeowners for the renewable energy they generated and sold to the grid. The pay rate for this electricity was highly subsidized, at times reaching four times the market price. Several other countries soon followed Germany ' s example, implementing similar policies and incentives to drive their country ' s solar use. This created unprecedented demand for solar **panels** worldwide. Manufacturers were able to scale up production and innovate in ways that cut costs. As a result, solar **panel** prices dropped while efficiency grew. HD : We see the cost reduction in solar and other renewable energy technologies has been so steep, but a lot of it is because of the enabling regulatory and policy environments created in places like the European Union and the US. Akil Callender : We cannot divert catastrophic climate change without addressing emissions coming from our energy systems. Currently, emissions from the energy system account for around two thirds of global greenhouse **gas** emissions. So the good news is, yes, we are seeing tremendous progress across the globe in renewable energy. It ' s rolling out faster than ever before. Luisa Neubauer : The brilliant thing about energy supply in the world is that we have alternatives. So we don ' t need fossil fuels to provide safe and equitable access to energy systems. Ramón Méndez Galain : Almost 90 percent of the total new installed capacity all over the world is just renewable. This is a tremendous change. AC : There ' s more finance than ever before going into **clean** energy. In many countries, it is the largest contributor to their electricity system for the first time ever. KB : And also because China is the leading manufacturing nation in the world, change is happening there much faster. And those innovations are then spreading around the world. Hongquiao Liu : China ' s coal demand might not have peaked, and China will continue to consume quite a lot of coal in the next few years. But the decarbonization is happening. It ' s happening faster than anyone has expected, and China is adding an astonishing amount of wind and solar to its grid every year. Eduarda Zoghbi : I think one of the biggest obstacles today are natural resources, in the sense that we ' re talking more and more about critical minerals and the role that they have. And batteries are demanding a lot of critical minerals that are present in most of the developing countries. So I think that even though costs are decreasing, we have to be more and more creative when it comes to recycling the batteries. We also have to think of how they ’ re impacting local communities. KB : So to give you the numbers, we extract, at the moment, 15, 000 million tons of fossil fuels every year. The International Energy Agency says that we will need about 43 million tons of critical materials in order to build out the **clean** energy economy. So that ’ s 300 times less stuff that is required. It ’ s worth saying that the impact of extracting 300 times less stuff upon nature is dramatically lower. And I know that it ’ s often argued that the renewable economy may be damaging for nature. Again, this is a completely false statement. It ’ s in fact, our only chance of reducing the pressure upon our natural resources, because there ’ s far less of it. MZ : Of course, transitioning to using **clean** energy, upending industries, it has to happen across the world, not just here in the US. And it can be hard to even imagine what life will be like. But carbon scientist and futurist Julio Friedmann can help us. Julio Friedmann : I think we need a compelling vision of the future that gets people excited about the energy transition and solving climate change. Right now, it ’ s like we ’ re at a restaurant and there ’ s two options : burnt toast, unflavored oatmeal. And people are not excited about either, right? The burnt toast option for me is what I call the big switch. Everything ’ s the same, it ’ s just **clean**, but it costs a lot of money and it ’ s super hard to do. The other vision is kind of like,”Thou shalt not.”That vision is a narrower sort of,”eat your peas,”hair shirt kind of vision of the future. And a lot of people don ’ t want that either. And there ’ s nothing in the physics, chemistry or biology that says we can ’ t have a super interesting, exciting, vibrant world. And that is compelling in a way that we have not heard yet. MZ : Help me out here, what does that even look like? JF : So, for lack of a better word, like, I like the Marvel cinematic universe. MZ : OK, I ’ ll go with you. JF : So a place like Wakanda, right? It ’ s got flying cars, it ’ s got maglev trains, and it ’ s a totally fabulous place to live. Like, they ’ ve got lots of food. That is built on abundant, sustainable, cheap energy. MZ : OK, so like, we ’ re not Wakanda, we ’ re not in the Marvel universe. This is real life, sort of. How do we make this possible? JF : So what can you actually do here on Earth, constrained by economics and engineering and all the rest of it, right? And the answer is we can do an incredible amount. Abundant, available where you want it, when you want it. Everybody should have energy, including the three billion people who use less electricity than my refrigerator uses. Well, the good news is, every day, the Earth receives 163, 000 terawatts of energy from the sun. About half of that bounces back to space, but about 80, 000 terawatts arrive at the Earth in a form we can use. For reference, today, the world uses about 26 terawatts of energy, and we got more than solar and wind. We have geothermal, we have hydro, we have nuclear. There ’ s other kinds of **clean** energies. And some of the best resources are, in fact, in the Global South. These places are not simply future climate victims. These places are latent energy superpowers. My favorite example of this is Chile. Almost eight years ago now their government said, wait a second, we can make the cheapest green electricity on Earth. We have hydropower, we have solar power, we have wind power and the best resources anywhere in the world. So let ’ s start by making a lot of green electricity. And then they said, because of that, we can also make green hydrogen cheaper than anywhere else. And then Japan was like, hello, we want your green hydrogen. Can you ship it to us? MZ : Is that possible? JF : Totally. And Japan is like, we will also give you money to pay for these projects. We ’ ll loan you money to pay for these projects. Could you please use Japanese technology? So they ’ re using Japanese turbines, electrolyzers, they ’ re using Japanese ships to haul the ammonia around. So Japan gets rich on this, Chile gets rich on this. MZ : Win-win. JF : Totally win-win. MZ : Win-win-win, planet, too. JF : And that model, other countries are looking at that, going, I want some of that. And that ’ s why I like this vision of abundance. 15 years ago, solar was the most expensive power. Now it is the cheapest form of electricity, right? Which is astonishing, it ’ s totally amazing. And a whole bunch of people are like, now we ’ re done. And I ’ m like, no, we can do that with everything. We can do that with wind. We can do that with nuclear, we can do that with geothermal. But these projects take time, money and people. We need to develop the human capital as part of these investment projects for decades. And innovation, we need much more energy in many places, much cheaper. That ’ s an innovation agenda. For solar, that was the United States, Germany, China and others acting together. Well, if that ’ s the recipe, we can do that again. We ’ re already doing it with electric vehicles and **clean** hydrogen production. We can certainly do it by turning electricity into fuels. MZ : It sounds like the climate discussion needs to be kind of like improv, that it ’ s a”yes and”sort of thing. JF : I ’ m so glad you said that. So many people don ’ t understand my love of improv comedy. Yes, it is a”yes and”circumstance. Because mostly it is about room for agreement. Where do you find the point where everybody is like, I can agree on that? One of the things I love about the infrastructure bill, one of the things I love about the Inflation Reduction Act, neither of them had the word climate in it. They were both massive climate bills, right? That was a way to get everybody on the same page. Nationwide, they basically tripled the **clean** energy research budget. That ’ s a good start. But that made it all of one percent of GDP. And for comparison, we spend twice that on pharmaceuticals. So I think it ’ s fair to say we should get going. We should spend a lot more money. We ’ ve done stuff like this before. We ’ ve done World War II, we ’ ve done the Marshall Plan, like, we ’ ve done the space shot, like we actually know how to move really quickly as a nation. We did this for COVID. Collective action, building together is what makes the difficult possible and nourishes the soul through mission and purpose. We know what to do and we can act, not out of anger or fear, but out of generosity and common purpose, bringing aspiration and humility together. We ’ re going to build a thriving, vibrant, exciting world full of potential that ’ s going to be built on the back of infrastructure, innovation and investment that will harness the abundant, sustainable, cheap energy that is our planet ’ s endowment. Thank you. [Chapter 4 : A Just Transition] FD : There is a high risk that when those exponential kick in, we see businesses running after those opportunities. But then the distribution of the benefits being very unequal. EZ : Many people like discussing the importance of the just energy transition. And a lot of people don ’ t understand what justice really means. If we are to reach a revolution, a renewable energy future has to be inclusive. AC : We have this push for what we ’ re calling a just and equitable energy transition. We need to make sure that we are not exacerbating inequality, but rather we are ensuring that that gap is closed and that the groups that have been historically marginalized are not once again being marginalized in a new way. NT : Most of emissions come from wealthy, energy-guzzling citizens, but a lot of the economic benefit and human development benefit will come not from decarbonizing high-carbon lives and lifestyles, but from providing **clean**, distributed energy and tools and solutions to those who don ’ t have access to any of them. LN : When we speak of countries in need of economic development, we mostly speak of countries that are hit by the climate crisis the hardest right now. It should be the biggest wish and job and task for all those countries who have burned so much fossil fuel already to support any other country in the so-called Global South, to get an economy running without fossil fuels, to not repeat all the mistakes that we have made. HD : To ensure that the solar revolution that has been seen around the world comes to places like Africa, we need to be really conscious that the history and the lived experience when it comes to energy is very different for poor countries. Rebekah Shirley : We love to remind Global South communities, especially Africa, about the vast renewable energy potential that they have, which is true. But the finance flows to deliver on that potential remain very scarce. Africa, despite being 90 percent of those across the entire globe that still do not have access to electricity and being 17 percent of the global population, we ' re actually two percent of international finance. Money doesn ' t actually flow to these spaces naturally, and we need to do a lot of work to get global finance to these spaces. MZ : A world of abundant **clean** energy for everyone. That is the goal. But what about in the short term? Take Detroit. This once-wealthy city has been through some seriously hard times. How is it rebuilding itself so that everyone here can thrive while dealing with extreme heat, flooding, and other climate problems? Anika Goss is looking to Detroit ' s past to plan its future. AG : I am a third-generation Detroiter. My grandmother moved to Detroit in 1936 during the Great Migration and brought all of her southern ways with her. She owned a home and knew that home ownership would create wealth and opportunity for her growing family. Up until the late 1950s, Detroit was a haven for middle-class families living in neighborhoods where there was green space and community connectivity and opportunity. My grandmother, she had this amazing garden, and it was just abundant. There were all kinds of flowers and food. You know, she grew vegetables back there, and it was just a magical place for a little girl. MZ : I mean, what you ' re describing is a Detroit that, yes, has huge problems, segregation. But at the same time, a woman, she can have a job, she can own a house, she can raise her family, she can have fresh fruits and vegetables. AG : It was certainly a place where you could gain economic opportunity faster than you can now. My grandmother ' s Detroit is not the Detroit that I live in today. All of those sites where there was manufacturing and industrial sites that led to our economic boom, many stand vacant and abandoned. These industrial sites have led to dangerous contamination to our land and our water and our air. MZ : So, just lay it out for us. For people who maybe don ' t see, like, how is there a link between economic inequality and climate change? AG : You know, I ' m so surprised that it feels like separate problems. But I ' m also really glad that people are asking. Because if you ' re living around and working in a community that is low-income, where there are fewer jobs, the housing is deteriorated, this is a neighborhood that ' s also more likely to be closer to a factory that ' s emitting pollutants, likely to have water that ' s contaminated, probably has fewer **parks** and fewer intentional green spaces. The people who are living in those communities are at risk for climate impacts first. So we have to do something in those neighborhoods right away. MZ : So what can this healthier future look like? Anika showed me some of the work she ' s done in partnership with different neighborhoods in Detroit, to make them both more enjoyable and more resilient. First, she took me to the Circle Forest. It ' s an oasis right in the middle of the city. AG : This is six lots that were vacant, and we all worked together with this community to build a forest. MZ : Beautiful and a great example of how nonprofits can work with communities to transform them into sustainable, welcoming places. AG : We might have a good idea about gardens and tree planting, but this is where people live. And so you ask first. This is a new effort, of this idea of a resilient neighborhood or calling it that. So these are large tracts of vacancy that can be used to provide environmental sustainability, community **resilience** and beauty. Watch for the poison ivy. It ' s not pretend. MZ : Next we met up with Katrina Watkins, the founder and CEO of the Bailey Park Neighborhood Development Corporation, a group revitalizing this historic area by turning vacant lots into **parks** that can stand up to extreme weather. Katrina Watkins : My family has been in this neighborhood since 1946. I grew up hearing the stories from my dad, him telling me about how people thought it was a ghetto, but it wasn ' t. Those were families, and people worked and had businesses, and people looked out for each other. And it was those type of stories that made me say, wow, we just really need to bring that back. MZ : So then briefly fill us in about how things changed. KW : Things deteriorated in this neighborhood. The homes got torn down, left just a lot of vacant lots, a lot of overgrowth. It wasn ' t safe. I would see my little nieces and nephews. They would just be on their little skateboards, you know, not paying attention, the stop sign is covered. And me and my dad was like, we just have to do something. So we started out as Bailey Park Project, focusing on the **park** and green infrastructure and trees and flowers. And then over the course of the years, we started doing more community development. MZ : I wonder, like, how do you describe it to people when you explain what ' s happening, what you ' re trying to do for your neighborhood, the changes that you ' re making and how it relates to climate change? KW : One really great example, when people come to the **park**, like right now, this is like, literally a heat island because our trees haven ' t grown. So it is very hot. So sometimes it ' s so hot that the kids don ' t come out and play until the evening. That education of knowing the benefit of trees and cooling down an environment, how it makes the air cleaner. There ' s a lot of people that suffer from asthma. So trees can make a difference. And if you go to places like Grosse Pointe, you will see this beautiful tree canopy down. Like their homes are probably 20 degrees cooler you know, than our homes, AG : Having a lot of green space actually contributes to the climate resiliency for this neighborhood. So all of this grass actually can manage stormwater better. The idea of leading new development in this neighborhood with green, what it says to me is that the residents are really envisioning the neighborhood that they want to live in. And that I just – I ' m really excited about that. MZ : Turning great loss and upheaval into even greater opportunities. These Detroiters are creating a blueprint for an urban future that every city can learn from. [Chapter 5 : Agriculture] TED-Ed : About 10, 000 years ago, humans began to farm. This agricultural revolution was a turning point in our history that enabled people to settle, build and create. In short, agriculture enabled the existence of civilization. Today, approximately 40 percent of our planet is farmland. Spread all over the world, these agricultural lands are the pieces to a global puzzle we are all facing. In the future, how can we feed every member of a growing population a healthy diet? Meeting this goal will require nothing short of a second agricultural revolution. The first agricultural revolution was characterized by expansion and exploitation, feeding people at the expense of forests, wildlife and water, and destabilizing the climate in the process. That ' s not an option the next time around. Agriculture depends on a stable climate with predictable seasons and weather patterns. This means we can ' t keep expanding our agricultural lands, because doing so will undermine the environmental conditions that make agriculture possible in the first place. Instead, the next agricultural revolution will have to increase the output of our existing farmland for the long term while protecting biodiversity, conserving water, and reducing pollution and greenhouse **gas** emissions. So what will the future farms look like? In Bangladesh, Cambodia and Nepal, new approaches to rice production may dramatically decrease greenhouse **gas** emissions in the future. Rice is a staple food for three billion people and the main source of livelihood for millions of households. More than 90 percent of rice is grown in flooded paddies, which use a lot of water and release 11 percent of annual methane emissions, which accounts for one to two percent of total annual greenhouse **gas** emissions globally. By experimenting with new strains of rice, irrigating less and adopting less labor-intensive ways of planting seeds, farmers in these countries have already increased their incomes and crop yields while cutting down on greenhouse **gas** emissions. MZ : Related technologies are also taking root in the US. Father and daughter farmers, Jim and Jessica Whitaker, have come up with their own strategies to reduce methane that are working on their farm in Arkansas. But convincing everyone else to change is their next big challenge. So let ' s go back to how you got to be such a big supplier of rice in this country. You tell a story about how you were 22 and you were like, I ' m going to start my own farm, but – not that easy. Jim Whitaker : No, it ' s not. Let me tell you something about renting a farm when you ' re 22 years old. No one rents you a farm unless no one else wants to rent it. It was a big piece of land, it was cash rent, I mean, the stakes were set. We were doomed to fail. And we weren ' t focused on environmental sustainability back then. We were focused on economic sustainability. How do we make higher yield? How do we use less fertilizer? How do we get to the next year and feed our family? It happens that economic and environmental sustainability go hand in hand along with social sustainability. As we used less fertilizer, used less water, our yield started going up. MZ : Can you spell out for me what the problem is with rice when it comes to the environment? JW : Soil is full of microbes. And what they ' re doing in the soil on a basic level is they ' re down there chewing away, eating up the biomass that ' s left over from the crop before. And out of that comes methane, the same way it happens in cattle operations or whatever. MZ : But you decided to do it a different way. JW : We do it a little bit differently. Arkansas is a very water-rich state, but we have a dry season, right in the middle of summer is a dry season. So when we dry the soil, that anaerobic microbial that ' s living in that soup, that mushy soil, either dies or goes dormant. So we break his life cycle. MZ : Oh! JW : And then it stops emitting methane. But it doesn ' t hurt the rice. That actually helps rice, when you use less water. We have less runoff, less erosion, less nutrients leaving our field. Nothing leaves our field unless we want it to. Jessica Whitaker Allen : We started tracking this. We saw that our water use was down and we thought, OK, well, we need to add to that, let ' s add more data points to that. Let ' s add our fertilizer, let ' s add our fuel usage. And so we really started recording all this stuff by hand in just an Excel spreadsheet. We didn ' t know what we were doing. MZ : But you knew you were on to something. JWA : Yes, so we started recording everything. MZ : Do you remember if there was a moment when the two of you thought, like, what we ' re doing, it shouldn ' t be special to us. Why aren ' t more people – JW : To collect the data, have the equipment and the cloud-based technology and the stuff in the field to record it, it ' s just massively time-consuming, and consumers don ' t pay for it. JWA : Well and people like us, why would you want to do it? I mean, take a walk through your supermarket. Everything ' s sustainable, everything ' s greenwashed. You can put whatever you want to on a package, there ' s no rules, there ' s no standards. So we ' re trying to make that standard. We live and work in southeast Arkansas. Our town has 4, 000 residents and one stoplight. The nearest airport, Starbucks shopping mall, Whole Foods, is two hours away in any direction. Without places like this and farmers like us, you ' d be hungry, naked and sober. I ' m a farmer, but not in the way you might think. I don ' t drive the tractors, and I don ' t plant the rice. But I know how to take what my dad has learned and what he is doing and share that with others. We are going to work with farmers in southeast Arkansas to educate them about the benefits of growing sustainable rice. JW : Let me tell you why that ' s important. There are 400 million acres of rice grown globally. And our methods, if used, can reduce greenhouse **gas** by 50 percent, reduce water use by 50 percent, increase yields to feed a hungry world. MZ : We hear a lot about how weather has gotten more extreme, flooding, heat. What ' s it been like in Arkansas? JW : So in 2016, we had a 500-year flood. We were... We get a **picture** of our place flooded. 14 inches of rain. That was a 500-year flood. Hurt a lot of people. And then we had a 1, 000-year flood. And we lost so many acres. And you know, when houses get flooded in rural communities, they don ' t come back. I mean, when you have whole neighborhoods get flooded, they don ' t rebuild them, they don ' t come back. JWA : I feel like almost every day I ' m getting a notification from the weather Channel of an excessive heat warning or thunderstorm warning or hail threat. You know, you don ' t really know what you ' re going to expect. MZ : When you think about what it will be like, I don ' t know, five years, 10 years, for someone to go into the supermarket and think, oh, right, I need to get rice off my grocery list. What ideally would you like to see on the shelf there? JW : So I think it ' s going to change, because I like to study what companies want. I mean, if you figure out what they want, then you know what to give them. And when you read their sustainability goals, they ' re all saying they ' re going to be net zero. The only way they can be net zero is they bring the farmer into the loop. So if we ' re able to do this, in five years, everybody ' s doing these practices, the US rice industry will be the most sustainable industry in the world. MZ : Can you do it? JW : Oh, yeah. It ' s about to happen. MZ : You ' ve just heard a lot of ideas, a lot of stories and a lot of big plans. So we want to wrap things up by asking our experts : What if we actually achieved our climate goals? What would that future even look like? They, of course, had a lot of thoughts. We ' ll let you pick your favorites. Thanks for watching. NT : If we are successful in addressing the climate crisis, we massively reduce the vulnerability and improve the economic and the creative prospects of billions of people around the world who currently are basically excluded from all of the opportunities which we have, which come from having abundant material and energy all around us. That ' s the real prize. FD : A world where we have had that exponential would be a world in which we have **clean**, cheap energy for all. KB : At the end of this decade, the International Energy Agency believes that we will have enough capacity to be producing 10 billion, 10 billion batteries a year. It ' s enough for every person on the planet to have their own battery. AC : I ' m constantly surrounded by young people just like me who are really, really, really passionate about the world that we live in. For obvious reasons. It ' s the world that we will inherit. EZ : Women are very important for the energy transition. So if we ' re talking about a world where we have, you know, where we tackle climate change, all I want to see is, you know, women in power. I want to see more solutions being driven by young people. And I want our politicians to be truly and meaningfully engaged in making change. NG : We always tend to underestimate the pace of ultimate adoption and change. Today, there are more than 150 pathways, scientific pathways that we could take to limit global warming. Al Gore : If we get to true net zero, astonishingly, global temperatures will stop going up with a lag time of as little as three to five years. Colin Averill : We ' ve been able to accelerate tree growth and carbon capture and tree stems by 30 to 70 percent. Dan Jørgensen : Offshore wind has the potential to cover the current electricity demand of the entire world not once, not twice, 18 times. NG : We need to decarbonize the old economy that we have and invest to create the new economy that we need Paul Hawken : Regenerative farming is fairly simple in concept. It means creating the conditions for more life. Josephine Phillips : We need to buy less stuff, and we need to look after what we buy. Valuing the things that we own is a climate solution. Vishaan Chakrabarti : We can all live in this transit-rich, carbon-negative, affordable way and leave the vast majority of the planet for nature, for agriculture, for **clean** oceans. We can do this. Susan Ruffo : The ocean is a powerful source of solutions that we ’ ve overlooked for far too long. Al Roker : You have to be engaged. Let your elected officials know that this is important to you. You have to vote, you have to go out there and support politicians who are going to support our planet. Peggy Shepard : We can create a legacy of environmental quality and climate **resilience** for all. Fehinti Balogun : You are not powerless that the”oh, there ’ s nothing we can do, just keep going”is symptom, not cure. But you are needed. Avinash Persaud : The power of what we can do when it matters to us is unlimited. CF : It ’ s not that we ’ re simply going to avoid the worst impacts of climate change. It ’ s that we ’ re going to improve the quality of life for humans, both in the urban and in the rural sector, and for non-humans.

Text Sample 735: POS Dim 5 – 2023 – Score 8.00 – 2023/t003445.txt

The construction industry accounts for 30 to 40 percent of the world ’ s total energy and resources. It also accounts for 40 percent of the world ’ s total carbon emissions. A man who lived long before us gave us a very simple solution. And he is Mahatma Gandhi, the father of our nation, India. He said that the **ideal** house should be built with materials found in a five-mile radius around the house. So you think, OK, what is that material? And it is right below our **feet**. It ’ s mud. Now imagine, if we can build a house with mud, there would be nothing more sustainable or eco-friendly than that. But then, you might ask, how strong is mud? This compressed five percent cement-added mud block, made by my alma mater, Auroville Earth Institute, is twice as strong as a country fired brick. In fact, it has a dry compressive strength of 6. 5 megapascal, as compared to a normal fired-brick, which just has 3 to 3. 5 megapascal. But then you may again ask, what about water? Wouldn ’ t water destroy mud brick? This is a brick from the same institute kept in water since 1995, a little less than 30 years, and you can see that it can still hold a significant amount of load. So what does this tell us? Can we change our perception of construction? Can we imagine a new way of building? Or can we bring beauty to dirt? I demoted myself from being an architecture graduate to being that of a mason. Or maybe I promoted myself, I don ’ t know. I joined hands with local workers and started teaching them mud techniques. Very soon we started to change every aspect of our construction. Our foundations became poured earth or rammed earth, our walls became mud blocks or compressed earth bricks. Why, even our **roofs** became domes and vaults made of mud bricks. This is a brick wall residence called Pirouette House, which we made in a very crowded community. So it goes in a zigzag pattern in need of natural ventilation. Now, please remember that this wonderful parametric form, or **curves**, were made not with any computational technology, AI or robotic arms. These were actually made with coconut twigs, thread and the masons ’ bear arms. At a time when the building industry is fast moving towards a future that exceeds mankind, we brought the value of human resources back into it. But way back in 2012, I was posed with a very life-altering question. I was asked to build a residence, a low-budget, eco-friendly residence for a primary school teacher called Biju Mathew. And when I reached the site, this was what I saw. A mountain of waste and debris dumped by the neighbors after they finished their construction. And because there was no other empty plot. So what do I do? Do I stand back and allow this waste to be dumped elsewhere? Or do I reimagine the philosophy of building with materials found in my vicinity? Neither me nor the future generations can unsee the waste that is getting piled in our vicinity. Therefore, we decided to use this waste as a part of our construction technique. We hand-pulverized this material, added a bit of soil and again, five-percent cement. And using the techniques we learned so far, we kind of poured it in between two meshes and poured the entire mixture into it. The resultant product was not just a formidable partition. It was load-bearing, means these walls took the load of the entire **roof** as well. But this came at a price. I was nicknamed"scrap engineer"by locals who found it amusing to see an engineer search through scrap in the junkyard. But what I found in the junkyard was pure gold : this is discarded electric meter boxes, washing machine wheels, decaying wood, all became part of a new philosophy that included more and more waste. Thank you. Now, emboldened by this, our new aim was to build multi-storied mud and waste construction. Is that possible? Yes. This is Shikhara, a three-storeyed multi-story, mud and waste construction. We decided to further modify what we did in the Biju Mathew residence. So instead of using two meshes, we put two wooden shutters and poured the same mixture which we did in the earlier project. And the resultant product was as strong as before and economical too. This wall had a dry compressive strength of 4. 5 megapascal. And mind you, this building, this residence, we completed in a meager budget of 90, 000 dollars as compared to a general budget of 200, 000 dollars or more. Very soon – Very soon, many practices in the country started adopting this technique, not just because of its eco-friendly part, but also because it was cost-effective. For us Indians, born to traditional societies, very often, our dreams culminate in lavish weddings and festivals. This, by the way, is a wedding stage. But the very next day, this too, is a common scene. India discards 3. 6 million tons of plastic every year. Therefore, we decided to utilize plastic bottles as a part of our construction for a new project called Chuzhi. It ' s a housing project. And in this project, we started to build circular beams using plastic bottles and reinforced concrete and of course, mud. We started to build it in circles. To be honest, we didn ' t know how this would eventually look, but we still kept on building the circles. We made it stable and put a **roof** on top of it. You can see how weird this looks. Chuzhi is a project that resides in a rocky terrain. And it ' s populated by a lot of trees and shrubs. Now, generally, these two factors are attributed as very bad things for construction. People generally cut trees or clear them to make spaces for their rooms or buildings or whatever. But in here, the trees not only form shelter but also cool the building in a tropical climate. And as far as the terrain was concerned, generally, people would fill it up, level it, clear it and build on top of it. But here, the building is perfectly camouflaged in the rocky terrain, that 10 metres away from this project, you will not see it. That is how good it is camouflaged. In this project – We managed to use 4, 300 plastic bottles which would have otherwise ended up as a landfill or even worse, in the ocean. One day, we all got sick working at the site. We had breathing difficulty and we kind of went to the doctor. And the doctor very calmly said,"Ah, it must be the burning tires used in road construction. Stay away."And we were shocked. There are hundreds of kids who travel these roads every day to their schools. There are thousands of families who traverse these roads every day to their jobs, why some of us even stay on these roads. And the answer is stay away. Upon further investigation, adding to our worst nightmare, we found out that our country discards half a million tires in a day. No, no, I ' m not done yet. And the rest of the world discards another half a million tires to India on the same day. Yeah. This is an aerial pic of a tire dump, seen from space. This is in a GCC country, that collects the tires from all of you guys, and eventually, dumps it in countries like India. Now you should remember, all of these tires, most of them at least, eventually end up either in the road tarring illegal process or we end up burning it to make again fuel oil, because this is a petroleum product. Now, one way to neutralize this huge toxic threat is to use it as part of construction. Now inspired from **architects** like Michael Reynolds, we introduced mud and tire to form a new building construction technique. And now we are building residences, we are building **museums**, schools. And this is the best way to neutralize this threat and create beautiful construction. We believe that tomorrow this may also be the alternative for the housing crisis that is a big reality for many of our underdeveloped countries. The heart of every project of ours lies with the problems that we created amongst us. Discarded tires, plastic bottles, construction debris. None of these things should be dumped thoughtlessly or carelessly. The world can not just find a solution, but we can even find beauty in this disaster that we created for ourselves. I remember a single line by the poet of my native language, Ulloor, and it goes like this. And it translates to this : It is our choice whether to build heaven or **hell** on Earth. Thank you, thank you.

Text Sample 736: POS Dim 5 – 2023 – Score 6.00 – 2023/t003361.txt

So my wife and I have two young children. And like many of you, we are terrified about what the worst impacts of the climate crisis are going to be for them in their lives. Do we honestly have a plan that ' s specific that ' s achievable, that ' s actionable, that ' s going to allow us to reduce emissions at scale with the speed that ' s necessary to respond to the catastrophe in time to improve the quality of life for our kids? We ' ve decided to focus and devote our time and attention to the challenge of greening buildings in America. And I ' ll tell you a little bit about why. There are 125 million buildings in America, coast to coast. These buildings are where we cook and eat and work and pray and raise our kids. A lot of us spend almost 90 percent of our time indoors in buildings, and these buildings are powered by fossil fuels. We burn coal and **gas** at power plants, we burn oil and **gas** in buildings. This is a furnace that produces hot water and heating. This is in an apartment complex in Harlem that we visited back in 2014. And I remember visiting it, I was a little worried because we could smell **gas** kind of, leaking. And a few blocks away, a few weeks before this visit, an apartment building had a **gas** furnace that exploded. It collapsed the entire apartment building, it collapsed the apartment building next door, and eight people died. So I remember looking at this boiler and taking a **picture** and thinking,"Get me the heck out of here."So because fossil fuels power our 125 million buildings, it ' s 30 percent of emissions, 30 percent of American emissions are tied to our buildings. And there is no actionable path to responding to the climate crisis without going building by building and greening all of the existing buildings in America. So I ' m Donnel Baird, I ' m the CEO of BlocPower. We are a technology company that focuses on moving buildings off of fossil fuels. We want to make buildings smarter, healthier, greener, and we want to do that by making buildings 100-percent electric. We ' re here in Detroit. Detroit is learning how to rip fossil fuel engines out of vehicles and replacing them with smarter, all-electric engines. We want to do the same thing for buildings. We want to rip out all the fossil fuel equipment and replace it with smart, green, all-electric equipment. And we want that equipment to be powered by electricity that ' s **clean**. Stanford professor Mark Jacobson has written a plan for all 50 states to be powered by wind, by hydroelectric power, by solar, state by state, so that our entire grid can be **clean**. I got focused on the dangers, the perils of fossil fuel systems. Growing up as a kid with my family in a one-bedroom apartment in Brooklyn, we didn ' t have a heating system that worked in our apartment building in Brooklyn. So when it got really cold in the winters, my parents, sometimes they worked nights, they would tell me,”Look, if it gets too cold, light a match, turn on the **gas** oven, open up the oven door, and don ’ t forget to open up a window because heat ’ s going to come into the apartment, but so is carbon monoxide, and we don ’ t want you to suffocate.”Then they ’ d go to work. And millions of Americans across the country heat their buildings this way in the winter and worse. But it ’ s not just the low-income buildings that are super dangerous with fossil fuel infrastructure. Recent studies indicate that **gas** ovens, **gas** heating systems in our home leak benzene, they leak methane, they leak nitrogen dioxide. There ’ s so much nitrogen dioxide emitted into American homes that it causes about 13 percent of chronic childhood asthma. There ’ s so much methane leaked into our homes that it ’ s equal to the emissions of 500, 000 gas-powered vehicles. And benzene, benzene is a carcinogen. And the latest studies from Stanford indicate that benzene is emitted into our homes, it spreads throughout the home, it lingers, we breathe it in. And it actually can be worse than breathing in secondhand cigarette smoke. And so for any parent, the thought of our children sleeping and breathing all this stuff in, it ’ s horrific. But good news, it ’ s 2023. We don ’ t live in ancient Mesopotamia. We no longer have to dig up dead dinosaurs from the ground and burn them in our buildings for energy. We ’ ve got options. This is an electric induction oven that uses electricity and induction power of magnets. It ’ s hyper-efficient, it ’ s safe. The finest chefs in the world are starting to use these ovens and systems. They can boil hot water in less than 60 seconds. Pretty awesome. And one of the new technologies that I ’ m really excited about is the cold climate heat pump. Cold climate heat pumps are like a silver bullet for reducing emissions in buildings. We can now do things that we could not do five years ago because of this technology. They use electricity and refrigerant and compressors to pump hot air into your home in the winter, to pump hot air out of your home during the summer, to heat up hot water. But most importantly, it allows us to fundamentally unhook buildings from fossil fuels. We can unplug all of our buildings and homes from fossil fuels entirely due to these systems. So let ’ s go back to my building in Brooklyn. We now have a mix of technologies that we can install in our building : solar **panels** on the **roof**, high-efficiency lighting, electric hot water heating systems, electric heat pumps for heating and cooling, the electric oven. We can make these buildings all electric. If we ’ re in Detroit or LA or in Houston, one of the cities where people love their cars, we ’ ll of course, install some electric-vehicle charging stations. We ’ ll have some backup battery systems because all three electricity and energy grids are collapsing under the weight of heat waves and the climate crisis. But we can electrify buildings. Now, is it simple? Is it easy to electrify buildings? No, there ’ s challenges and opportunities. We have to solve the finance problem, we have to solve a workforce problem, we have to solve a data problem. Electrification can cost 10, 20, 30, 000 dollars per building. And that ’ s not affordable for the average American. But what we can do is most Americans can afford to buy a home on their own. And so we have a mortgage. And so we studied the mortgage market and found ways to amortize and spread out the upfront cost of building electrification over 15 years so that the monthly payment is affordable and lower in some instances than what families pay for fossil fuels. We have a workforce problem in that, like Europe, there ’ s a shortage of skilled electricians and technicians across America to deliver on these projects. Mayor Eric Adams in New York City has asked us to train and hire 3, 000 vulnerable and low-income individuals, ex-offenders, gang members, military veterans with PTSD, formerly homeless folks, to train them in building electrification. We had a gang, a street gang kind of come to us and say,”Look, if you guys hire us and give us jobs, we will shut down our street gang.”We had a crew of workers who had been locked up in Rikers Island, our

New York City jail, and we trained them, we hired them, we took them back to Rikers Island, They installed solar **panels** in the parking lot. We have a data problem. Every building is unique, like a snowflake. So in order to make sure we have the right size of the electrification equipment for that building, to unhook it from fossil fuels, we have to collect data and make sure we specify and recommend the right equipment. We ’ ve been working with the Bezos Earth Fund to aggregate all of the data on all 125 million buildings across America so that families can search for their address, get an electrification work plan, hand it to a local, well-trained electrician who we ’ ve trained and hired. These are good jobs that can ’ t be outsourced, so that every family has access to the data that they need. We ’ ve recently started working with Ithaca, New York, where Cornell is located, and San Jose and Menlo Park. These are cities that have committed to decarbonize 100 percent by the year 2030, in six and a half years. So we ’ re going to go building to building and move each building off of fossil fuels and make them all-electric. So we ’ ve got six and a half years, but there ’ s dozens of cities across the country where we ’ re working with partners : Portland, Oregon, Denver, Atlanta, Chicago to electrify buildings at scale. Electrification is possible. It ’ s not simple, it ’ s not easy, but it is possible. And what that means is if we can electrify one building, it means that we can electrify a whole block of buildings. If we can electrify a block of buildings, it means that we can electrify a neighborhood. If we can electrify a neighborhood, it means that we can electrify a city. And if we can electrify a city, that means we can electrify a country. There ’ s a saying : you have to be the change that you wish to see in the world. By doing this, we ’ re going to create great jobs. There ’ s going to be massive economic and public health outcomes. But we also are going to inspire cities and countries around the world to electrify. And we ’ ll be able to look our children in the eyes when they ask us and say that we ’ re doing everything we can to reduce emissions at speed and scale, to fight the most damaging impacts of the climate catastrophe. Thank you.

Text Sample 737: POS Dim 5 – 2021 – Score 8.00 – 2021/t004073.txt
Hey, work lifers, this is J. J. Abrams, we ’ re trying something new on the podcast today, I am going to be conducting an interview on the show and my **guest** is Adam Grant. We ’ re going be talking about his new book, Think Again Automatical. Oh, JJ, it ’ s great to hear your voice, but this is my show. I don ’ t know if it is. I can ’ t just let you take my job and ask all the questions. I ’ m insisting on asking you some questions. Can you promise me that I can ask one Star Wars question? Yes, you can ask anything you want. You can ask. Fantastic. Jumping in here as host of my show in the off chance. You don ’ t know who J. J. Abrams is. You absolutely know his work. He ’ s directed, produced and written some of the most beloved **films** and shows of the last 30 years. If you love Lost alias Felicity, the more recent incarnations of Star Wars or Star Trek, you have to think as a bonus episode. While we ’ re hard at work on Season four, J. J. kindly agreed to turn the microphone around and host me for a chat about my new book, Think Again The Power of Knowing What You Don ’ t Know. It ’ s a book about the science of rethinking our opinions, opening other people ’ s minds and creating cultures of learning. It officially launches February 2nd, but I wanted to give you a sneak preview. So I sent JJ an early copy and told him he was in charge of our interview. Thanks to Viking Penguin, my publisher, for Think Again for supporting this episode. I have read this book and I just truly think again, your new book available now is so compelling and I just I ’ m thrilled to get to talk to you about it. I feel like you ’ re my book fairy godfather.

That ’ s such a thing. That ’ s my dream. Thank you. I ’ ve got to say that I had so many moments of kind of gasping when I read Think again, because you provided the kind of optimism that there is a way to find common ground, to find, to embrace nuance, to reexamine how we, you know, we go through the world. What motivated you to write the book? Well, there ’ s a part of me that that thinks I was probably motivated to write this book by one of my best friends from growing up, Khan, who I remember this was in seventh grade here. We ’ re in the middle of an argument, I think, during a commercial on Seinfeld. And he he hung up on me and I called him back and he said, Shut up, Adam, I won ’ t talk to you until you admit you ’ re wrong. And that was sort of a it really it was a moment that stuck with me. I didn ’ t have the language for it at the time, but I spent a lot of time afterward thinking, why is it so important to me to be right? And, you know, why do I have such a hard time admitting when I ’ m not? And that, you know, I think I was I was sort of a defining moment. And then, you know, much later I feel like my job as an organizational psychologist is, you know, is to really think again about how we work, how we lead, how we live, and to motivate other people to do that, too. And my most frustrating experiences as an author, as a speaker, as a researcher, as a teacher, as a consultant are always when people are not willing to rethink assumptions that I think they ought to be questioning. And so, you know, just over the past few years, the number of conversations I ’ ve had with with founders and CEOs where I you know, I give them the best evidence I have available. And they say, well, that ’ s that ’ s not how we do things around here or that ’ s not the way I ’ ve always done things. All right. And I just I just want to come back to them and say, you know, Blockbuster, BlackBerry, Kodak, Sears, you know what? You should definitely not rethink anything in your vision, strategy or business model. And so I guess, J. J., when you know, when I have both personal and professional experiences that drive me insane on the same topic, I feel like there might be a problem with tackling or a question worth writing about. Hmm. That ’ s great. Answered in reading the book. It made me wonder if people ’ s not bandwidth, but their comfort level is so low that there are people finding comfort in belief systems. And that ’ s the thing that they can rely on because there ’ s so little to rely on. And and so rethinking is a threat to one of the few comforts left. That ’ s such an interesting question, I don ’ t know is my first thought, my second thought is it reminds me of some research that I loved by sociologists that started in the 70s. They did. This study is where they were interested in the effects of being micromanaged at work and not just how we thought about our jobs, but how we behaved in the rest of our lives. And they found that if you had a really controlling boss, that you actually became more authoritarian as a parent with your children, which was just kind of a stunning spillover for me to, you know, to think that the way you get treated at work might affect the way that you raise your children who just never would have occurred to me. And I think that body of research and a whole bunch of other studies that followed kind of that they made me start to think about this basic truth, that when we lack control in one domain of our lives, we become motivated to seek it and try to regain it in other parts of our lives. And I think, you know, especially in the wake of this pandemic and a global recession where we ’ re facing a lot of threats to control right now. And I wonder if if part of what people are doing is they ’ re grasping at the things they do have control over. And one of the things we all have control over is what we believe. And so, you know, holding fast to the things that I think are true, they make that sort of makes the world feel more predictable, less uncertain. It also gives me a sense of belonging with whoever my tribe is. Right. It validates my worldview. And I ’ ve even wondered if, you know, if that ’ s part of why the the George Floyd protests, you know, really rose up when they did. I feel like before the pandemic, people felt like, you know, there ' s nothing that I personally can do to fight systemic racism, whereas, you know, juxtaposed against a global disease like covid, which I really can ' t fight. Well, you know what? I could I could show up in protest. And so I guess there ' s there ' s something about both holding fast to your beliefs that helps render an uncontrollable world more controllable. But also, there ' s something about being in an extremely uncontrollable world that can lead you to think again about where you do have some control areas where you can make a difference. And so I wonder if that cuts both ways. Hmmm, that ' s interesting, I wanted to ask you about the value of humility, because it seems like. Humility is also at the center of what allows for someone to. Pursue a truth or information or a point of view that might not be theirs, but I think so many of us take a more defensive posture and our are more closed off and declarative about things that we think we know. And I just wanted to ask you kind of what role you feel humility plays and why it might not be such a given that the actual value of it is the truth and is enlightenment and is wisdom and why that doesn ' t seem so appealing in the moment. Yeah, interesting. So if pride is what fuels overconfidence, I think humility is what gives us the freedom and the flexibility to rethink the opinions and assumptions we should be questioning. And I think that part of the problem is that we live in a culture where people are rewarded for being the smartest person in the room and they learn to wield their knowledge and expertise like a weapon, as opposed to treating it as a resource to share. And so, you know, I think in most workplaces, in many schools and frankly, in too many families and communities, too, you know, the currency of status is knowing more than everybody else. And so knowing that, you know, admitting what you don ' t know, questioning yourself, showing, you know, any hint of uncertainty is like admitting weakness and setting yourself up for defeat in battle. And obviously, I think that ' s a problem. And I think that maybe the first way to get people past that fear is we need people in positions of power to model it. Right. We know that if you ' ve been validated by your power or your status or your authority, that people are much more comfortable letting you say you don ' t know. There ' s evidence, for example, that when experts say, I ' m not sure or I don ' t know, they actually become more persuasive, in part because we respect their humility and in part because we ' re surprised that we actually pay more attention to the substance of their argument, which, you know, is likely to be persuasive if they are knowledgeable. So I guess what I want to do is I want to let people know that humility is a potential source of strength, not just a sign of weakness. And I don ' t I don ' t entirely know how to get people there. And I want to turn this around on you because I feel like this is one of the the themes that that really cuts through a lot of your work is every time I watch a show or a movie that was directed or produced by you, I know it ' s going to lead me to question an assumption. And it might be an assumption I ' ve held for a long time. And I ' m curious about is this something you do intentionally and how do you get your audiences to confront that kind of humility? Well, thanks for saying so. I feel like the the way into a story for me is usually finding a character who is vulnerable, who is in a place where they ' re out of their depth and they ' re desperate for something. And I think that desperation is like the greatest motivator. And it also it forces you as a storyteller to know what a character ' s desperate for and therefore what you are either afraid of or hopeful for and ideally both the idea of desperation as a motivator. I think that ' s incredibly powerful. And it ' s something that that not only strikes me as relevant to, you know, to thinking about characters in a story, but also to understanding the people that I work with and the people that I love. And I ' m wondering is, is that a lens that you apply? And I guess if you turn it inward a little bit, when you ’ ve achieved the level of success that you have, how do you stay desperate to rethink some of the stories that you tell? Well, I have to do is go on Twitter. I feel like, you know, I ’ m always desperate to do something, to tell a story that that moves people. And sometimes I ’ m more successful than others. And, you know, I never go into anything with a kind of template for what it should or shouldn ’ t be. I feel like I ’ ve learned a ton of lessons along the way. And by the way, I always feel like we ’ re carrying, you know, whatever a thousand lessons with us that we all need to, you know, remember at all times. But we can only really carry 900 of them. And we ’ re constantly dropping lessons and picking up the ones that we ’ ve dropped in by picking those up. We ’ ve dropped some other ones and we keep relearning the same thing. So I think, you know, rethinking is really important and relearning is a weird Sisyphean inevitability that I don ’ t think we can avoid. But I feel like, you know, the drive to do. What I do like I ’ m guessing the drive for you to do what you do, you do it because you feel compelled to share a point of view, an idea, a feeling, an insight, a question, something you might not even understand the answer to. And you just feel compelled to do it. And I think I start everything desperately, and I think they probably all end that way, too. I love that. I want to ask you, because your book is all about rethinking. I was just curious, do you feel like there is a fundamental barrier to rethinking and if so, what it is and I do have a follow up. Yeah, I think so. So, yeah, I feel like one of the things that I ’ ve rethought twice now as I ’ ve been writing books is, you know, I wrote my first book, Give and Take with with a whole framework to organize the world in terms of givers, takers and matters. And then I didn ’ t want to be constrained by a framework like that when I wrote originals and I felt like I overcorrected and there wasn ’ t enough connective tissue between the different chapters. I felt like I had a lot of interesting trees, but the forest was not clear enough. And so I decided when I was writing Think again that I was going to be open to finding an overarching framework, but I was not going to be attached to it. And that that felt like the sweet spot. And about halfway through the writing, it hit me and I had to rewrite the book from scratch. The framework was my my colleague, Phil Tetlock, where he observed that whatever your job or career is, that you spend a disproportionate amount of your time thinking like certain professions. He said, look, you know, there are moments when we think like preachers, prosecutors and politicians. And I think that that all three of these mindsets can stand in the way of rethinking, because when you ’ re preaching, you ’ re already convinced that you found the truth. And so you don ’ t need to question any of your assumptions when you ’ re prosecuting. You are trying to win your case. That means the other side is wrong. And so you ’ ve got to get them to do all the rethinking, but you get to stand still. And when you ’ re thinking more like a politician, you ’ re basically trying to cater to your audience and campaign for their approval or get their likes or their votes. And so you may be doing a little bit of flexing in the way that you present, but you ’ re not rethinking your underlying beliefs. Right. You ’ re just you ’ re just trying to appease your constituents. And I think we spend way too much time thinking in these modes. I know, you know, as somebody who ’ s been called a logic bully once or twice, maybe seven times at this point, by my count, I spend too much time in prosecutor mode. And when I think that somebody is selling snake oil or trying to, you know, indoctrinate their followers into some kind of idea cult, I feel like it ’ s my job to debunk those myths and bring the better evidence to the table. And the problem is then that I get too close minded. And so I ’ m trying to get out of prosecutor mode and spend more time thinking like a scientist, which is something I think we could all learn to do better. It ’ s funny, in the book, you say how I think the first time you were called The Logic Bully, it actually was something that you liked, that at first it appealed to you. I was proud. I thought, yeah, that ’ s my job as a social scientist. I want to decimate your bad arguments with rigorous evidence and airtight logic. And I ’ m good. And then, yeah, I didn ’ t like the bully part so much as I thought about it more. Given that, what are some of your favorite practices for questioning your own assumptions and for rethinking yourself? I think the first one for me is just the basic idea of thinking like a scientist to say that, you know, what scientists do is instead of forming beliefs, they actually form hypotheses. And so instead of having an opinion that set in stone, that means, OK, this is a hunch, how would I go and tested? And, you know, I don ’ t think that we have to always operate like we ’ re in a lab. Right, with a bunch of test tubes. But I do think that we should all be running experiments in our lives. So during the pandemic, for example, I have instead of assuming there ’ s a routine that ’ s going to work for me, I ’ ve run experiments to say, OK, what if I shift my creative brainstorming from the morning to the night? What does that do, you know, to to the number of original ideas that I generate? What if I turn my camera off during a zoo meeting? Do I actually have, you know, a more a more reflective, more thoughtful discussion? And all of those you know, there ’ s those little changes and adjustments that we make. We could think about them as experiments and then we could say, OK, well, you know what what information what data do I need in order to find out if that was a successful or failed experiment? And I I think that ’ s a huge step. And I guess the other thing I would I would just put on the table is when I was writing the book again, I made a list of all the things that I know I ’ m completely ignorant about. And my goal was was actually not to reduce the list, it was to expand the list. Mm hmm. Because I feel like the more I know about what I don ’ t know, the more curious I am and the more I ’ m going to learn from people who are actually are knowledgeable in those areas. So, you know, my list, I think it started out I immediately said I know nothing about music. I don ’ t understand financial markets. I ’ m clueless about chemistry. And, you know, the list just kept growing from there. And I ’ ve actually been just keeping that list. I have a file on my desktop and my goal is to add something new to it every week. And if I stay clear about what I don ’ t know, then I hope I ’ m going to stay more humble, more curious mindset. That ’ s great. When I was reading, I think, again, it was occurring to me, you know, that you tell such great stories, including like the story of the BlackBerry company and, you know, Mike ’ s inability to rethink the way perhaps he could have and what might have been. And I, too, was a huge fan of that keyboard. The question I had is, how do you find partnerships where there is a thinker and then a thinker, or is it always or typically whether it ’ s one or two or more people that are able to rethink and think, oh, that ’ s such a cool question. I think my **ideal** partnership is one where. There are clear norms that make sure rethinking happens. The question of where it comes from is something that I don ’ t have good data on, which really mildly annoys me right now and I need to go gather some. So stay tuned. But I mean, I can think of examples of both. One of the examples that that I ’ ve spent probably too much time thinking about now. Is the, you know, the Jerry Seinfeld Larry David example from Seinfeld. And one of the most interesting things that I learned from studying that collaboration was that they were both ruthless free thinkers, but they applied it to their own work, probably even more so than they did to each other ’ s work. And so one would not even bring a script or a joke to the other until he had thought it was, you know, really good. Mm hmm. And I can see how those kinds of norms could be really effective in partnerships. I ’ ve also seen examples, though, where, you know, one person is basically the kind of a creative dreamer and the other person is the really critical evaluator. And so, you know, the first person does the thinking, the second does the rethinking, and then it ends up being kind of this beautiful symphony. Mm hmm. I imagine, J. J., you ’ ve seen a lot of this in in the world of Hollywood. Is there is there one or the other that you think is more common or more effective? And are there are there are people who do more of the thinking versus rethinking in your collaborations? Well, obviously, I think, you know, everyone is different. And I know I have worked with people who who go to a place and not that they ’ re stubborn about where they go, but they arrive at an answer very, very quickly. And that ’ s sort of their thing and that they ’ re perhaps a bit reluctant to be as flexible or rethink things. Then, you know, I ’ ve worked people like, for example, with Matt Reeves, with whom I did Felicity, and we did the Cloverfield movie together. And he ’ s someone who you know, and I ’ ve known him since we were kids. And he ’ s always rethinking. And I find that I ’ m maybe more, you know, someone who would want to try something and put it out there and sort of commit to it for now and sort of see where it goes. And Matt. To his great credit, and it ’ s one of the reasons I love working with him, is he ’ s someone who really wants to stop and examine and consider and reconsider it and look at it from all sides and see if there ’ s a better way. And can it go deeper and can it be, you know, can it be more emotional? And I think that that, you know, part of that is probably, you know, less about thinking and rethinking and more about my being more impatient. And when we work together, we had our offices right next to each other. And in fact, our offices were in the location where they actually **filmed** the office, the show, the pilot for the American Office Show. Oh, and weirdly, his office in my office were Matt ’ s office with Michael Scott ’ s office. And my office was the the conference room, which years later, when I went and directed an episode of The Office, I was like, why do I know this place? And then I realized, oh, my God, it ’ s literally based on our old offices. But Matt, we had a little little sign in between is a **picture** of the two of us in between both of our offices. And someone had put up sticky notes over each of us and overmatch it, said, how do we make it more emotional and undermines lives, how to make it funnier? And there is a sort of like, you know, clearly everyone brings their whatever their interest is, whatever their their point of view is to the table. And even though sometimes it can be frustrating when someone says, well, wait a minute, let ’ s reconsider this, you know, it ’ s almost always a better thing because there ’ s nothing wrong with kicking the tires and interrogating something. But the best thing and this is what I found, you know, recently I co-wrote a pilot with a writer named Latoya Morgan, and she ’ s a wonderful writer. She ’ s a black woman who brings a point of view that is clearly going to be a life experience, very different from mine. And when I bring up an idea and she says, you know, hold on, let ’ s talk about this or that or vice versa, I feel like we almost invariably end up in a place where there ’ s something that we are both, like, beaming about that we wouldn ’ t have arrived at had it just been one of us in that room. And in all likelihood, you know, someone who looked like us, you know, either just me or just her. So I feel like there ’ s a lot of talk of how diversity is good and how you say in your book, diversity is good but hard. But, you know, I have personally felt like, you know, working with people who don ’ t look like you, especially when they are, you know, smart and freethinkers. It ’ s incredible how much better the conversation is, how much richer it is, and I think how much better the work ends up being. And I ’ m grateful for any time I work with someone, whether it ’ s Matt, Latoya, I can name a huge list of people who are really willing to rethink things that I might just assume need to stay. And the only caveat I would say here is that I can also think of times when there were paths we were on and the ability to switch it up and rethink something might not have been for the best. Yeah, this is one of the hardest questions that

I really wrestled with while writing the book and I ' m still grappling with now, which is how much rethinking is too much and how do I know whether when I ' ve started rethinking something, I should actually change my mind or go back to my original first answer and the only coherent place that I ' ve landed on this so far as to say, well, if this is a **curve** and there ' s an optimal amount of rethinking, I think most of us don ' t do enough of it. Agree or disagree? You ' re asking me, yeah, yeah, it ' s a question, do you agree or disagree? You know, my guess is I would agree. Got it. I have to take a sidebar here, JJ, because you mentioned the office. This is work life. I read once that you did not like the office when it premiered in the U. S. and that you weren ' t even a fan of the entire first season. Is it true? And what led you to rethink that view? I think that what I felt was when I first watched it, because I became a huge fan, which is why I reached out to them, essentially begging them to hire me as a director. I was such a fan of the original and in fact, to a degree that I reached out to Ricky Gervais and wrote a part for him in Alias. And I was such a fan of that show and the specificity of it that it felt. For me, I remember when I first saw it, I thought, oh, no, like it didn ' t matter how good everyone was because they were great. It felt like it was somehow sacrilege that it was redoing exactly the same thing in essentially the same time period, you know, putting it through the American TV mill. And then you realize, oh, they ' re doing it perfectly. And it ' s it ' s exactly what it should be. And it ' s its own thing. It was the quality of the show that that won me over. OK, next question. I ' m sure you hate getting asked about Star Wars in particular, because it might what might be the most beloved universe ever created a minimum. It ' s got a lot of passionate fans. And yeah, I loved The Force Awakens. I love the rise of Skywalker. I especially love the points where you rethought some moments and characters. And I wanted to ask you about one specific one, which is I think my biggest moment of shock in the new trilogy was when Emperor Palpatine came back to life because I thought he ' d been dead since 1983. And so I was wondering how you went about rethinking his demise. Well, that was something that came in and out and then back in again. But that idea was something that that felt like it was sitting there, set up in the prequels. And when we were looking at the story, the idea of a character who is desperate to find immortality and it felt like it was sitting there and that it was a set up potential set up. And I kept imagining it as something that could work. You know, if I ' m a kid watching all nine movies years from now, it felt like something that was a possible setup to be examined later. I will say that it ' s something that I loved the possibility of it and the promise of it. And it was one of, you know, so many things that we were rethinking and trying to find ways to sort of fit the pieces of the puzzle together. And I know that for some people, it they clicked in and it worked and was **fun** and surprising. And for other people it was, you know, understandably infuriating and sacrilege and stupid and wrong. The reactions were were vast, but it was also something that was, you know, in various moments felt like it was the right move. I imagine that happens a lot to that. You you rethink a classic world or story, whether it ' s Star Wars or Star Trek or even characters that we ' ve fallen in love with. Right. I still feel like, you know, with Lost in particular, I still feel like, you know, Sawyer and Sayid and Kate and some of the others are long lost friends. How often do you rethink the stories you ' ve told versus once you put it out into the world, you ' re you ' re happy with the creative choices you ' ve made, regardless of the feedback? Well, all I see when I work on something is the thing that we could have and should have done to make this better or that different. The rethinking doesn ' t stop because it ' s it ' s out there. And I do think that one of the enemies of certainly movies and in some cases TV is the sort of, you know, the announced release date where the time that one normally should have to rethink doesn ' t exist. And it ' s very difficult to be. In a healthy way, rethinking when you are under the gun, impractical material and often very expensive ways. For example, when Damon and I started working on Lost, it was a very particular, you know, crazy ticking clock where the head of ABC at the time, Lloyd Braun, wanted a show about people who survived a plane crash. And so in Damini, who had never met, we wrote that outline. It was in a week based on just pure instinct. And we went off and we shot this to our pilot that we were, you know, writing, casting, shooting, doing post and everything in nine weeks. And it was a crazy crunched period. Wow. And it ended up working as a pilot. And then Damon and Carlton Cuse wrote the series. But I, I feel like it ' s one of those lessons that not having an ending, it just makes it harder. I know it sounds so obvious and so easy, but when you take that leap of faith, when Stephen King starts writing a novel, he gets to finish that novel and decide, does anyone need to see this? And I ' m sure he ' s got drawers full of things that people have not read because he feels like it ' s not the thing. And one of the crazy things that this, you know, pandemic quarantine time has done for TV and for movies is it ' s given people time to write, not just a pilot or a rough outline, but write entire seasons of a series before it even goes, which, you know, you want to be flexible and you want to be able to cast a role and realize, oh, my God, we ' ve got to do this with this character because we didn ' t realize it or this thing isn ' t working the way I thought it would. But this time has allowed writers to, you know, write without the pressure of we have to be casting this thing right now. We have to be location scouting. We have to be, you know, getting the all these episodes edited and posted and ready for broadcast. And I think that it in no way do I feel like, oh, great, we had this time to do this. You know, I ' d much rather life be normal and covid not exist. But I will say that if there is any kind of creative benefit to this time, it ' s that suddenly release dates and broadcast dates are not being, you know, as aggressively announced. And time is available for some writers to be able to actually work on the stories in a more, you know, I think human pace. And that is no small thing. Is there creative practice that you ' ve rethought during the past year? If you had said to me that we could be breaking stories without ever being in the same room together, I would have thought it was probably impossible, if not wildly unlikely. And I think that online, not just the Zoom ' s, but also the sort of online whiteboard apps that exist that allow people to collaborate remotely has, I think, probably profoundly changed the way stories, at least on television and writers rooms, will be broken. Wow. Does that does that mean in writers rooms won ' t always be physical even after the pandemic? They definitely will. But I think the that someone can now be part of it who might not be in town. I think that that the norm will always seem to be better to be together. But there ' s something great about the focus and the, you know, the access that you have from anywhere. So I just think that that that ' s something that, you know, clearly there will be infinitely fewer business trips to have meetings that could now clearly be done on Zoome. Will people go back to it? You know, the way we were before? More than not, I believe they will, but. This is now accelerated the evidence that this is a valid and really productive way to work. I ' ve never said those words. I ' ve never said this before. We ' re not going to hold you accountable. Thank you. Next question from one magician to another. How does all the time you ' ve spent learning magic tricks and performing them affect your job? Has it spilled over into your work life in any way? I think the thing about magic, as much as some might see it as the geekiest sort of, you know, silliest thing to me, it ' s incredibly powerful. We had a magician in our office and this was like two years ago, and he was in my office, just the two of us. And he did a few tricks and he did one that literally made me, as a 50 year old man, think to myself in my head, he might be magical, like, you know, because I know how magic tricks are done. I know how a lot of I know bunch of gimmicks. I know a bunch of tricks. He did something where I was like, I wonder if he ' s maybe magical. Does he have? And my whole point is the fact that someone who ' s a half a century old can still have a moment of thinking. There might be powers beyond that, which I know is so profound. And the the wow and the gasping and the amazement that any audience has, big or small, it ' s there ' s something about the the sense of possibility that the world is more than it seems and that we want to see things that we can ' t imagine and that we don ' t expect. And so I feel like that is a natural. Aspect to telling a story, and I just think that whether it ' s writing a a book or making a show or a movie, it ' s all a bit of a magic trick. You are so much more comfortable not knowing than I am. I immediately want to answer. I have to figure out the secret, but then you can never see the trick again. You can admire the the skills and the genius of the performer who executed it, right? Yes. OK, next favorite character from the office. It has to be Michael Scott, not because just because I think he is the hub of the wheel. And so everyone is kind of it feels like they ' re all kind of in his orbit, to mix **metaphors**. So interesting. He wasn ' t even in my top five. I love that. And then can I ask you the same question about Star Wars? I think my favorite my favorite character in in Star Wars was Han. Just because I mean, it ' s very hard to think of a better scene in a movie ever than when you meet Han Solo and then in Empire Strikes Back. I mean, like there ' s just there are some of the greatest, truest larger than life, but completely relatable and grounded moments. And, you know, when we got to shoot in Force Awakens, you know, Harrison and Chewie coming into the, you know, the Falcon and saying, you know, we ' re home because it just there was something so crazy about it. And I feel super lucky to have, you know, been part of that. I had tears in my eyes when when Nonintuitive had their reunion. It just felt like we just couldn ' t wait to show it to people because it just felt like, oh, my God, you should see this. It ' s just so cool having them back on the ship. It was a beautiful moment. I ' d love to talk to you about as a Red Sox fan, the Red Sox versus Yankees. And you have this great chapter about just how far fans of these teams are willing to go to torture the opposing team. And what you tried, what didn ' t work and what finally seemed to break the ice of having them sort of find the humanity in each other. So my question to you is, what was what surprised you in your attempts to build those bridges between those two fan bases? And what were your findings? In summary, I ' m so glad that resonated. You know, I was I was when thinking about the prejudice that so many people hold toward other groups, I was looking for a context where we could study it, where the actual stakes are minimal, but the emotional stakes feel very high. And sports fan allegiances, it just it felt like the perfect place to go. I mean, a few years ago, I got to know an amazing woman in her 70s who helps Holocaust survivors. And one day she mentioned that she went to Ohio State. And my first reaction was, yuck. Yeah, I ' m a Michigan Wolverine. We ' re rivals, but who cares where she went to school half a century ago? Yeah, it just bothered me at a fundamental level. And so I decided, you know, like any self respecting social scientist, I should study that. And so Tim and I and all these experiments with the Yankees and Red Sox fans, which was such an easy rivalry to choose. And I think my my biggest surprise was that two of the steps that were supposed to work based on all the existing science, didn ' t you know one of those was was just creating a common identity as baseball fans? And, you know, people would say, yeah, you know, I would help a Yankees fan if they were in an emergency. But otherwise, you know, they ' re rooting for an evil team. So no, thank you. And then the other one was to to try to humanize the individual fan, where basically what happened was people said, all right, you know what? There ’ s this one Yankees fan who ’ s OK, but he ’ s not like the other ones. And I still hate all the rest of them. So what ultimately ended up working and in a way that I was I was surprised by the power of it actually was saying, look, you know, maybe we just need to get people to rethink how they became fans in the first place. So imagine if you had grown up in New York. Would you possibly root for the now you ’ re probably root for the Mets. But the point is, you might not be a Red Sox fan. I ’ m not crazy. Exactly. I ’ m joking. I ’ m joking. All my Mets fans, friends, I ’ m joking. I don ’ t know if you are, but go on. I ’ m joking anyway. I know I keep going, please. Well, no. So I thought this idea of just, you know, considering the possibility that we could hold different views all of a sudden, it it just destabilizes a lot of the stereotypes and prejudices that people hold. And so, as you know, we ran these really entertaining experiments where we gave people a chance to to punish fans of the opposing team by giving them extra spicy hot sauce or by giving them really difficult math problems, which were actually going to hurt their payment for participating in the study. And we found that, lo and behold, after we just got fans to think about how they might root for a different team if they had been born in a different city, that was enough to get them to show less animosity toward fans of the opposing team, which I thought was amazing. And this did not make it into the book. But we just finished a couple of experiments with with gun control and gun rights advocates where we found that the same process worked for them. That if we got somebody who is extremely concerned about about gun safety, to imagine having been born in a hunting family, they actually were less nasty than to somebody on the other side of the aisle. So I think there might be something to this. Fantastic. So I can ’ t thank you enough for talking to me on your show about you, you should do the the goodbye because it ’ s your show. But I just I wanted to thank you for the the honor and opportunity. Well, I can ’ t thank you enough for coming. I ’ m so grateful that you read the book. You liked it that you ’ re willing to talk about it. It ’ s a huge gift to me and our listeners. And I loved getting a little bit inside of your mind and your work life. And most importantly, the work you do has has always been just a huge source of inspiration to me. But over the past year, you have been a source of light and hope and entertainment and creative inspiration. And I just want to thank you for for everything that you do to entertain us and to get us to rethink so many of the things that we think we know. Holy crap. That was unbelievable. Way too generous, but your your words are incredibly appreciated. And and again, thank you for this amazing book and your kind words. WorkLife is hosted by me, Adam Grant. Our team includes Colin Helmes, Greg Achen, Dan O ’ Donnell, Constanza Guardo, Joanne DeLuna, Grace Rubenstein, Michelle Quint, Angela Chang and Anna Felin. This episode was produced by Jessica Glaser and supported by Viking Penguin. Think again. The power of knowing what you don ’ t know is available in whatever format you like print, electronic stone, tablet or audio narrated by me.

Text Sample 738: POS Dim 5 – 2021 – Score 7.00 – 2021/t004052.txt

Have you ever in your life met someone who felt like everything at their job and in their industry just worked perfectly, no room for improvement? I definitely have not. In fact, if I was talking to someone and they started to express anything even remotely similar to that view, I would be like, OK, take up the disguise. You ’ re my boss. Undercover. I caught you ripping. My point is, whatever you do for work, there is clearly room for improvement, whether it ’ s making hiring practices more inclusive or limiting the plastic waste and packaging materials work, stopping the spread of misinformation. We all have a role to play in today ’ s episode is all about how to catalyze change. How do you get people to try something new when they ’ re already very familiar and very comfortable with these well-worn paths? Well, Franklin Leonard managed to do just that in Hollywood. He created The Blacklist. It ’ s a list of the unproduced screenplays that Hollywood insiders love the most. And in doing so, he changed the way that Hollywood worked. Once a script made that list, it made the blacklist. And then powerful people started to see that there was consensus that the script was actually amazing. Well, then, these previously unsellable projects, they started getting sold and getting made and winning awards. And here ’ s how Franklin describes the importance of that in his talk at Ted Venice Beach. Simply put, the conventional wisdom about screenwriting merit where it was and where it could be found was wrong. And this is notable because, as I mentioned before, in the triage of finding movies to make and making them, there ’ s a lot of relying on conventional wisdom and that conventional wisdom maybe, just maybe might be wrong to even greater consequence. **Films** about black people don ’ t sell overseas. Female driven action movies don ’ t work because women will see themselves in men, but men won ’ t see themselves and women that no one wants to see. Movies about women over 40 that are on screen heroes have to conform to a very narrow idea about beauty that we consider conventional. What does that mean when those images are projected thirty **feet** high and the lights go down for a kid that looks like me in Columbus, Georgia, or Muslim girl in Cardiff, Wales, or a gay kid in Chennai. What does it mean for how we see ourselves and how we see the world and for how the world sees us? We live in very strange times, but I think for the most part, we all live in a state of constant rage. There ’ s just too much information, too much stuff to contend with. And so as a rule, we tend to default to conventional wisdom. I think it ’ s important that we ask ourselves constantly how much of that conventional wisdom is all convention and no wisdom and at what cost. As a writer myself, I think that there is something really amazing here. Normally what makes a script hot is if there ’ s a huge celebrity attached or if it ’ s a remake of something beloved, or if your last movie won six Academy Awards and grossed a billion dollars, you know, and not that those will stop getting scripts attention, I ’ m sure those guys will keep getting sold. But what ’ s really amazing about what Franklin did is he managed to find another way to get scripts attention. If enough of the people who read scripts all day say that this one, this one deserves attention. Well, now all of a sudden, people actually read it and people would take it seriously. And whatever industry you work in, whatever you do. The question that Franklin ’ s experience with the blacklist raises is, I think central to all progress. How can you challenge conventional wisdom today on how to be a better human? We ’ ve got Franklin here to answer that question and so many more. This is Franklin Leonard, founder of The Blacklist. The blacklist has gone from being just a list of the most beloved scripts to so much more than that. So I ’ m wondering, just in your own words, how do you now think of it and describe what the blacklist is? Yeah, I mean, I think of our North Star as being in identifying and celebrating great screenwriting and the people who do it. And that can take many forms. It ’ s everything from giving folks who are trying to become better screenwriters reasonably priced feedback that from reputable sources. It is when that feedback returns good telling people in the industry that can help their careers and help their movies get made. Hey, this is a really good script. It ’ s providing workshops for the best among those writers, oftentimes in collaboration with other organizations. It ’ s the annual survey of the industry ’ s most liked screenplays. It ’ s the partnered list that we do with GLADD, IMPAC and other sort of affinity groups, you know, for the Muslim community, the Asian Pacific Islander community, etc., all the way up to and including making some of those scripts and movies. So we ’ re producing a lot of these things now. It ’ s more about how can we be supportive of the Hollywood community at large and especially screenwriters within it. And I think that that as a general guide would sort of be the guiding principle for for everything we do. What ’ s so cool about the blacklist is you basically found a way to give people an excuse to trust their actual taste and to say like this thing that we really love, we actually can make. And I think that ’ s a really powerful thing across industries, not even just in Hollywood. I think that ’ s right. Look, and I don ’ t think that it ’ s that Hollywood lacks imagination. I mean, I can say concretely, having worked in the business for now, for coming up on 18 years, the people that work in Hollywood are wildly imaginative and wildly talented. And it is a joy to be able to work with them. I think that the the difficulty and the frustration is that the industry, you know, people are running scared at all times. And the decisions that are made about the economics of the business are made based on a set of conventional wisdom. That is all conventional wisdom that has been passed down through generations. And implicit in that sort of passed down, conventional wisdom is a ton of bias, some of which is, you know, sort of innocuous. And a lot of it is is terribly dangerous. Right. So it can be something as simple as, you know, certain kinds of action sequences don ’ t work right now. Does it really matter about like, you know, a certain kind of car chases work or don ’ t work in movies? Probably not really. Doesn ’ t matter when we decide, as the industry had for years, that female driven action movies don ’ t work commercially. And the consequences of that we see in our gender relationships, in our daily lives when people assume, oh, well, you can ’ t sell black actors abroad outside of the US, the consequences of that are apocalyptic in terms of like the actual valuing of black lives in America and around the world. Because we make fewer black movies, we don ’ t market those movies abroad, you know, and it ’ s just fundamentally not true. Stacy Smith, a professor at USC, ran the numbers and found that basically when you support movies with diversity in at the same level, that you support movies that don ’ t have that diversity, guess what? They make the same amount of money. People don ’ t have a problem seeing diverse actors on screen or seeing diverse stories. What they want more than anything is for those movies to be good. And what ’ s the blacklist, I hope has done is created more of a true meritocracy where the focus is not who ’ s in the movie, what ’ s the movie about it simply is this a good script? And probably one of the most gratifying things about the sort of 15 year history of the blacklist and. Up on 16 years is that last year, the Harvard Business School did a study on the economics of the Black List and found that movies on the black list, when controlling for every other factor, movies made from scripts on the Black List made 90 percent more in revenue than movies made from scripts, not on the black list. And I want to say it again, because I think that it can ’ t be emphasized enough that movies on the planet that were made from scripts on the Black List made 90 percent more than movies that were made from scripts not on the Blacklist. And there ’ s one reason why, which is if you start with a great screenplay, you have a better chance of making a great movie. And if you make a great movie, you have a better chance of making a profitable one. And so, you know, I think that that ’ s a lesson that everybody instinctively knows. But it ’ s not one that has been the guiding principle of the **film** industry for a very long time, if ever. So they ’ ve kind of worked both artistically and profitably. What lessons do you think you ’ ve learned that apply to people who don ’ t work in entertainment or maybe even in a creative field at all? Because it seems like so much of what you ’ ve learned here is that challenging the conventional wisdom is not just good for diversity and equity and inclusion. It ’ s also good for the **bottom** line. That ’ s exactly right. And I think that ’ s probably a number one. Increasing diversity is good for the **bottom** line, like it ’ s good morally and ethically, but it ’ s also good capitalistically. If we can use that probably neologism. No, look, I think the other the other thing that I ’ ve learned. Is that conventional wisdom is more often than not convention and not wisdom? You know, I think that in a world, especially over the last, let ’ s say, 20, 25 years or the amount of information that we ’ re expected to sort of keep in our brain and the analytics that we have to do on a daily basis to do our job and to process the world and to interact with other people, we are inclined to create these rustics that we just take for granted. And a lot of those critics are deeply, deeply, deeply flawed. And we as individuals and as organizations have to do a better job of aggressively interrogating them both for the good of the world, but also for our own individual self-interest. That means that I have to do that as well. Right. Like, this is not just me giving advice to other people and saying, why aren ’ t you doing better? It ’ s me looking in the mirror every day and saying, are you doing better? When you look at your business, are you just saying, well, I ’ m a black guy from the south, so I ’ m sure I ’ m doing fine? Or am I saying, you know, are we good on gender or are we making sure that everybody has a seat at the table? Are we making sure that we ’ re deconstructing the table and deconstructing the house and allowing everybody to rebuild it? And if we ’ re not, then I have to make changes. And I think that ’ s probably the biggest thing is trying to build a mirror for myself that actually presents an image of me as I am and not as I want to imagine myself. If that makes sense, that totally makes sense. So for everyone listening who may not know, last year the Academy issued some new rules for **films** to be considered for an Oscar. The rules had minimum requirements for diversity and inclusion, and there ’ s been a mixed response as to what the effects of those rules might be. Some people think it ’ s going to make a huge difference. Some people think it doesn ’ t go far enough and some people are angry about it, frankly. And you have really publicly said that you think that the new rules are a good start and you ’ re optimistic. I ’ m curious, though, if you think they ’ re going to make a real tangible difference in the kinds of movies that are getting produced. But again, because of the way in which the sort of thresholds are structured, if you just hire one like a woman of color in a senior role at your distribution company and like have an internship program with two interns, you ’ re fine. And so the way I read the academy ’ s sort of announcement is a public statement that in order to be a responsible corporate citizen of the **film** industry, you have to be trying to expand the pipeline ever so slightly. And if you ’ re not doing that, then we ’ re not going to give you the chance of winning an Oscar. But they did not prevent anyone who has made a movie from getting, you know, the sort of laurels that their artistic accomplishment may have earned them. And that ’ s the thing like, look, for me personally, I don ’ t need for any individual movie to include black people. Right. Or any other group. If you want to make a movie with all like made by and about all straight white men over the age of 50 who grew up upper middle class, like more power to you. I just want to make sure that if somebody wants to make a movie about trans women who are black and poor, that they have just as much likelihood of getting that movie made as the white dudes did. And then, you know, best movie wins. The problem is not that we need all of these movies to be super diverse and for all of these groups to be diverse when they make them, though, that would be nice. The problem is, is that for the entire history of Hollywood, we ’ ve had massive amounts of affirmative action for one group, white, upper middle class, straight says men and everybody else has to not only make something good, but also do it and overcome all of these obstacles to just getting their movie made or even being in a position where they can make a movie. So I would like to focus on the the access to resources and the access to distribution problem far more than I would. Hey, who ’ s eligible for an Oscar? But I do think that because the Academy Awards are, you know, the time every year when most people are thinking about the the ecosystem of the **film** industry, it ’ s critical that we have that conversation about the Oscars as part of a broader conversation that should be tackling year round. I also have to say, shout out to Epuron, who came up with the hashtag Oscars so white, there ’ s very little chance that we ’ d be having this conversation right now if it wasn ’ t for her. And I think it ’ s really important that we remember that Oscars so white is not just about black actors, it is about all non-white men and making sure that everyone is represented in the culture because we have a better culture when that ’ s true and we all make more money when that ’ s true. And I think that, you know, I ’ m really just in awe of what she built with that with something very, very simple that had the power to change the world. Yeah. And the fact that she did make such a huge impact with that. And she ’ s not at the very top of the power structure and the money. She ’ s not the person. Greenlining. The **films**, I think, does speak to the fact that. Anyone can actually have a real impact on the **films** that are getting made in the culture that is being spread around the world. That is the power that all of us have in a world where social media exists. Again, that is a sword that cuts both ways as well. But it is something that that power exists for all of us if we want to become advocates on behalf of any ideas, you know, diversifying Hollywood or diversifying Congress or making sure that people have enough food to eat and a **roof** over their heads. We ’ ll be right back with more from Franklin Leonard after this break. Here we are, we ’ re back. How do you think people who maybe don ’ t see themselves as having that kind of power, how can they think about the the creative force that they can they can create change? And I think it ’ s really about just modeling your values in your day to day actions. You don ’ t have to be an advocate to to change the way a person sees the world or somebody else. But I think that if you are in a position where you see somebody mistreating somebody else or you see somebody being disrespectful to somebody else or you you hear somebody say something that ’ s maybe not even disrespectful to anybody who ’ s in the room, but maybe tell them, hey. Not cool. Have you considered this? Do you realize that when you say this, you also mean this? That ’ s one way, but also than just modeling kindness? Like, again, it ’ s super simple. It ’ s a super it ’ s a very cliched idea, but on a fundamental basis, you don ’ t know the effect that your actions will have on someone else who may be watching you and you never know who may be watching you. We all fail to live up to our highest **ideals**. We all do. I know I do. But aspiring towards them has effects that we can never anticipate. And so. You may never even know what the consequences, but you can ’ t really go wrong by trying. Hmm. That ’ s literally saccharin, but true, you know, it ’ s weird. So what can audiences both in the U. S. and abroad, what can audiences do to kind of help support systemic change or broader representation? Ironically, because I think a lot of people in the **film** and television industry are very uncomfortable with these sort of review aggregators. But Rotten Tomatoes and Metacritic are a great place to start. You know, look, we are all in a time of sort of super abundance of content, right? There ’ s more TV shows to watch than any human being could ever watch. There ’ s more movies to watch than any human being could ever watch. And we all want to watch the good stuff. Right. And by good, I want to be clear. I don ’ t mean pretentious. I don ’ t mean Oscar winning. I just mean best in class. Right. Like, if you want to watch a weird comedy, you want to watch the best week comedy. You want to watch the bad one. Right. **Film** critics, television critics. There are deep problems with those communities. They tend to be, you know, overrepresented by it, by white older men. But seek out critics who who consistently have opinions that mirror your own right. If you love a movie, go find a critic who felt similarly to you that wrote about the movie in a way that you found compelling and go see what else they liked. Right. Because odds are you will find other movies that you will be intrigued by. And then you can be the critic that shares information about those movies with other people in your community. And I know that sounds like a very elaborate thing to do in order to find a good movie or television show. But I promise you two things. One, you will enjoy the process of looking because you will learn about things that you would not otherwise learn about. And if people are reviewing things in a way that is compelling, that process alone will be entertaining. And too, you will find better things to watch. You will have fewer nights where you made the decision to watch something for two hours and at the end of the two hours you ' re like, that ' s two hours of my life. I ' m never getting back. So there ' s there ' s obviously a huge portion of the movie going audience that mainly watches things like superhero movies or big franchise **films**. Do you not believe that that ' s a problem? I think people should watch what they like. And if that ' s superhero movies, it ' s all good. Right? There are a lot of really good superhero movies out there. Black Panther, excellent **film**. Thor Ragnarok, excellent **film**. Right. Thor Ragnarok is a meditation on refugees and the displacement of peoples Black Panther. There ' s a reason why Immigrant Song is the song they play over the climactic battle scene. Black Panther is about many things, but it is fundamentally about this tension between, you know, the black community wanting to sort of shutter itself off and sort of integrate into the world despite the tortures that the rest of the world has put us through. Right. As Martin versus Malcolm and literally the climactic fight scene happens on a literal underground **railroad**. There ' s a tendency for a lot of people to sort of tut tut about, you know, these big studio action movies and act like they ' re somehow like a diminution of the art form. And I just have never believe that that ' s true. Now, some of them are not good, but there are many indie pretentious movies that are not good either. So what I would say is, is look for things that you love. And if you loved that thing right, if you love Black Panther, maybe check out Creed by the same director, Ryan Coogler. And if you love Creed, maybe check out Fruitvale Station also by that director. You know, if you loved Thor Ragnarok, there ' s a reason Tycho Waititi, right. An indigenous New Zealander, got the job for Thor. Why don ' t you go watch the stuff that he made that got him that job? There ' s a good chance you ' re going to like that, too. And the thing about it is, is you ' re the only person losing by not checking those things out. Right. Like they got your money for Thor, they got your money for Black Panther. The industry is going to be fine. You have an opportunity and the world is going to open up to you and you ' re going to have these moments of joy in these moments of sadness and these moments of exhilaration that you haven ' t gotten to have yet. And that is fundamentally, for me at least, the beauty of **film** when the beauty of art and the beauty of a cultural world in which we live. You know, we ' ve been talking about movies and cinema, but obviously the experience of watching a **film** has changed dramatically with theaters being closed. I guess even if they ' re open, people being scared to go. I even think about that a little bit personally, because there ' s a movie theater right down the block from where I live here in Los Angeles. And on their big marquee rather than new movie title, it says to be continued. But it ' s said that for months now and their doors still haven ' t reopened. So what at first was kind of just like charming and even funny sign is now a real open question, right? Like, will that theater ever actually reopen? And I hope they do. I hope they do, because I think that there ’ s something really powerful about seeing movies in person. That classic experience, which you describe so beautifully in your talk from a few years ago. Here ’ s a clip of that this weekend. Tens of millions of people in the United States and tens of millions of. All around the world, in Columbus, Georgia, in Cardiff, Wales, in Chongqing, China, in Chennai, India, will leave their homes, they ’ ll get in their cars or they ’ ll take public transportation, or they will carry themselves by **foot** and they ’ ll step into a room and sit down next to someone they don ’ t know or maybe someone they do. And the lights will go down and they ’ ll watch a movie. They watch movies about aliens or robots or robot aliens or regular people, but they will all be movies about what it means to be human. Millions will feel all or fear, millions will laugh and millions will cry, and then the lights will come back on and they ’ ll reemerge into the world they knew several hours prior. And millions of people will look at the world a little bit differently than they did when they went in by going to temple or a mosque or a church or any other religious institution. Movie going is in many ways a sacred ritual, repeated week after week after week. I ’ ll be there this weekend, just like I was on most weekends between the years of 1996 and 1990 at the multiplex near the shopping mall, about five miles from my childhood home in Columbus, Georgia. The funny thing is that somewhere between then and now, I accidentally changed part of the conversation about which of those movies get made. You obviously gave that talk well before the pandemic or any of the current concerns about movie theaters and public health existed. But I imagine you must be thinking about that a lot during this time right now. So do you have any new perspective on why movies matter and why this experience matters? Well, you know, I think the absence of these communal environments wherein we learn about what it means to be human and right. And that was sort of a link that I was making between religion and movies, is that, you know, um, but I think what ’ s interesting to me about movies and I would include television and really any storytelling in this regard or art more generally, the movies as a popular medium is that, you know, fortunately we have these virtual spaces where we can sort of commune around them and it ’ s not quite the same, but it still ends up being a common language and a common touch point for humankind. Right. You know, I think Netflix just put out that they had seventy eight million people watched Gina Prince Bythewood movie The Old Guard. And when I meet somebody and they ’ ve watched it also, we will have a really positive conversation about Gina Prince bywords, brilliant work, and we will feel closer as a consequence. And that has nothing to do with us being both black or both men or whatever it is. It ’ s just that like we saw this thing about these people and we bonded over it. I don ’ t know. I ’ m really appreciative that that exists. Now, that ’ s the positive side. There is also a negative side, which is and I think that the sort of moment of racial reckoning that we ’ re seeing around the globe is in large part connected to the movie industry, because when we go into a room and we sit with a lot of people we don ’ t know and we learn about the world and what we learn about the world is a lie in terms of race, in terms of gender, in terms of sexuality, in terms of religion. Those lies being projected 40 **feet** high in front of tens of millions, if not hundreds of millions of people have real human substantive consequences, particularly for black lives. You know, I ’ ve increasingly, over the last few months been struck by the notion that the first ever Hollywood blockbuster was birth of a nation. And, you know, we ’ re seeing the consequences of it now. So I think it cuts both ways. I ’ m and part of the reason why I ’ m so attracted to **film** as an art form is because it does cut both ways incredibly sharply and with an incredibly large sword. Hmm. What is one movie or book or cultural artifact or idea that ’ s made you a better human? I mean. Look, I ’ m very lucky in that I have two parents who. Very clearly communicated to me and my two younger siblings that we could do anything and as black kids in the Deep South in the 80s, that probably wasn ’ t true. But they convinced us of that anyway, and I think between that and their very clear expectation that the obligation that we had was not just to do whatever we wanted to do and aspire to whatever we wanted to aspire towards, it was to make sure that we made it more likely that anybody had more of a chance of doing it. Somehow they managed to convince us that, like we could do anything and also explain to us that the world was organized so that not everybody could and that we it was our responsibility to make sure that everybody could. And that ’ s not a cultural artifact, but it ’ s the thing that for me, I ’ m most thankful for and it ’ s the thing that I hope I ’ m able to incorporate from a values perspective and all of my work and the arts that I contribute to. I don ’ t know if that answers your question, but it ’ s something that I been greatly on my mind of late. And a related question right now, in this point in your life, what is something that you ’ re trying to be a better human at? I ’ m trying to have more patience with people. I ’ m trying to be better at recognizing that the world is on fire, figuratively and literally, and that everybody is going through a lot. And that moment when I feel the need to judge or they feel the need to cast disapproval on, I need to take a moment and realize that there may be other explanations than that, which I would assume. Well, Franklin, Leonard, thank you so much for talking with us. It ’ s been an absolute honor and a pleasure being a pleasure. Thank you for having me. Thanks so much for listening to this episode of How to be a Better Human. That ’ s our show for today. Thank you to our **guest**, Franklin Leonard. You can find the black list at B. L. Seek Elstein Dotcom. I am your host, Chris Duffie. This show is produced by Abby MONITUS, Danielle Arezzo, Frederica Elizabeth Yosfiah and Karen Newman at Ted and Jocelyn Gonzalez, Pedro Rafael Rosado and Sandra Lopez among from Peru Productions. For more on how to be a better human visit, ideas dot. com. We ’ ll see you next week.

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So the climate crisis, in my way of thinking about it, is the most serious manifestation of an underlying collision between human civilization as we ’ ve presently organized it and the Earth ’ s ecological systems. And the system most in jeopardy is the very thin shell of atmosphere surrounding our planet, because we ’ re spewing 162 million tons of human-caused global warming pollution into it every single day, as if this is an open sewer. It ’ s not an open sewer. It ’ s so thin that if you could drive an automobile at autobahn speeds to the top of that blue shell, you ’ d reach it in about five minutes. And that ’ s where all the greenhouse **gases** are. The accumulated amount, coming from the burning of fossil fuels primarily, CO2 – fastest-growing source of methane is from fossil fuels as well, also, agriculture as you have heard – the accumulated amount traps as much heat now as would be released by 600, 000 Hiroshima-class atomic bombs exploding every day. So the temperatures are going up at almost record levels every year. Last year was the hottest year in recorded history, according to NASA, and the scientists say it ’ s absolutely unequivocal that we are the cause of that. And we ’ re hearing Mother Nature and seeing the extreme events. The most anomalous extreme event since records began 200 years ago was in the Pacific Northwest of North America. Forty nine and a half degrees, 121 degrees Fahrenheit in British Columbia. In the tropics and subtropics, the combination of temperature and humidity is now making larger areas literally uninhabitable, the area so described is relatively small now, but if we continue over the next few decades, these areas are predicted to expand, and billions of people will be in areas where it is not safe to stay outside for more than a couple of hours. Already, the climate refugees and migrants are four times more from the climate crisis than from all the wars and conflicts going on right now. And the respected”Lancet”commission has predicted as many as a billion climate refugees in this century. That ’ s one of the reasons why we are seeing this wave of populist authoritarianism. Sea level rise is also contributing, and of course, 93 percent of the extra heat is absorbed in the oceans, and the ocean temperatures are reaching record levels as well. And that means more water vapor comes off the oceans, and the warmer ocean temperature makes the cyclonic storms, like hurricanes and typhoons much stronger. Hurricane Ida struck the Gulf Coast as a category four, and as is so often the case, communities of color and poor people were disproportionately victimized. This storm continued on north and northeast across North America to New York, New Jersey and Pennsylvania, killed a lot of people and dropped rain bombs. And in New York City some basement apartments were flooded. And I appreciate this family giving me the right to show this video. He ’ s safe, the guy is screaming for his mother. He and his mother were both rescued by his brother, who punched a hole through the **roof** from the floor above. These rain bombs are becoming much more frequent and much more extreme. And actually the water coming off the oceans is held in much greater volumes by the warmer air. Atmospheric rivers are becoming larger. Here ’ s one, from Hawaii in the lower left to Silicon Valley in the upper right, 2, 300 kilometers. What was happening in Silicon Valley when this satellite **picture** was made? Well, that ’ s what happens when these rain bombs drop, and the average atmospheric river now contains as much moisture as 25 Mississippi Rivers. They ’ re creating what you could call”atmospheric tsunamis”in some areas. The downpours get much bigger and much more frequent, four times more frequent now than in 1980. One of them dropped on Germany and neighboring countries in July, killed more than 200 people. Look at the before and after. This is an example of what we ’ re doing to our planet. Ten days ago, in part of Italy, there were 74 centimeters that fell in 12 hours, 29 inches of rain in 12 hours. One year ago, October 3rd, as much rain fell here in the UK as there is water in Loch Ness. And as a result, the insurance industry is now seeing record recoveries, and those recoveries I showed you from last year may get a tiny bit larger, depending on the outcome of this lawsuit. The owners and operators of this giant replica of Noah ’ s Ark have sued their insurance company for a million dollars in flood damages. Hard to make some of this stuff up. The same extra heat is pulling the moisture out of the first meter of the soil, creating the worst droughts in memory. 93 percent of the American West is in drought now, 100 percent of California, half of California is in the most extreme form of drought, and the same heat is also draining the reservoirs, evaporating the reservoirs. Lake Mead is down to levels not seen since they started filling it in the 1930s. In Brazil, the same thing is happening. In Eastern Europe, the Czech Republic has been going through the worst drought in at least 500 years. And in the southern cone of Africa, 100 million people are experiencing food insecurity primarily because of the extended droughts. And when the temperatures go up, the fires get a lot worse and the worst fires in California and western US history have been in the last two years. Also in Siberia, in Australia, in southern Europe. It doesn ’ t have to ruin your golf game. And this is a serious point, because we can ’ t let these conditions become the new normal. It ’ s not fine. And this is an example of why it ’ s not fine. A lightning strike hitting a natural **gas** leak in the middle of the Gulf of Mexico. Lightning gets more common with the climate crisis. We ’ re relying on dead plants and animals, leaving their residue in the atmosphere to threaten the species that are still alive, and 50 percent of them are in danger of extinction in this century. And as we push into previously wild areas of the world, we encounter millions of new viruses that we have not dealt with in the past. Five new infectious diseases every year, emerging, three quarters of them from animals, like the COVID-19 pandemic. Which raises some of the same questions that are raised by the climate crisis. When the world ’ s leading scientists are setting their hair on fire to get our attention, should we listen to them? Check. Can our interconnected global civilization suddenly be turned upside down? Check. Are the poor and marginalized populations of the world the most affected? Check. Can science and technology give us nearly miraculous solutions in record time? Check. Will we deploy those solutions in time? That is the question. We have inequitable vaccine distribution in the world, and it threatens everyone in the world. We also have solutions to the climate crisis, but they ’ re not equitably distributed. Worldwide, 90 percent of all of the new electricity generation installed last year was renewables. Almost all of it from solar and wind. Just one year before the Paris Agreement, solar and wind electricity was cheaper than fossil electricity in only one percent of the world. Five years later, two thirds of the world. Three years from now, it will be the cheapest source in 100 percent of the world. Coal is not getting cheaper. **Gas** is not getting cheaper. Nuclear has been getting more expensive. Wind is cheaper than all three, and solar is continuing to plummet in cost and is the cheapest of all now because the cost of making the solar **panels** and the windmills continues to come down. By the way, the famous coal **museum** in Kentucky just installed solar **panels** on its **roof** in order to save in its operating budget. Now we ’ re getting battery storage. I ’ ll rescale this graph to show you the predicted emergence of a new one-trillion-dollar industry in the next few decades and green hydrogen on top of that. The car batteries are getting cheaper as demand continues to increase. And within less than two years in some model categories, the EVs will be cheaper than their internal combustion engine counterparts and within four years in all model categories. So, the oil and **gas** industry has been the worst investment in global markets for more than a decade, while the **clean** energy companies are really becoming more profitable. Now the oil and **gas** companies say that they ’ re going to invest a lot more in renewables and CCS. Yes, it ’ s the fastest-growing – they ’ ve tripled their investments all the way up to four percent. Ninety-six percent of the money they ’ re spending is still going into oil and **gas** and coal. The impression they ’ ve given us is not an honest impression, not for the first time. And they ’ re telling a different story to Wall Street. They still have planes, trains, a few automobiles and ships, but they ’ re telling Wall Street and financial markets they ’ re going to make it up in plastics, which is 75 percent of their third-largest market, petrochemicals. How ’ s that working out for us? Not so well. Banning single-use plastics is one of the many things that we have to do. We have to shift to sustainability. And here ’ s the good news, the sustainability revolution is the single biggest business opportunity in the history of the world. It has the scale of the industrial revolution coupled with the speed of the digital revolution. And what we ’ re seeing is that we have to seize this opportunity, but we need reforms in the current version of capitalism. Capitalism does a lot of things great. It balances supply and demand and allocates resources efficiently and often unlocks a higher fraction of the human potential. And by the way, the alternatives on the left and right explored in the 20th century didn ’ t work out very well. But in order to see how we need to change capitalism, we have to pull out our focus. And let me present an analogy. The electromagnetic spectrum from the long radio waves to the short gamma rays and all the things in between, the portion of visible light that we can see with our eyes makes up only 0. 1 percent of the total. For the eight years I worked in the White House, I got a daily report from the intelligence community that collected from all parts of the spectrum, and it was a much more accurate **picture**. Here ' s the analogy. The value spectrum is something we too frequently look at through the very narrow aperture of short-term profits for one stakeholder, the shareholders. But this leaves out negative externalities, as the economists call them, like the pollution. That ' s why we need a price on carbon and a price on plastic pollution. It also leaves out positive externalities. So we ' re chronically underinvesting in education and health care and environmental protection, pandemic preparedness. We ' re ignoring the depletion of resources like topsoil and underground water aquifers. And we ' re ignoring the distribution of incomes and net worths to the point where one percent of the people have almost half of the world ' s wealth. That ' s one of the other factors that ' s driving populist authoritarianism. We have to take account of the environmental effects and the effects on people and their families and the communities where they live and the communities in their supply chain and the ethics in the C-suites. And we have to realize that hyper inequality is a threat to both capitalism and to democracy. There is an emergent form called multistakeholder capitalism, and it is driving a lot of new decisions. For example, in asset management, almost half of all the assets in the world under management are now in portfolios committed to net-zero. One reason is that the Paris Agreement set the direction of travel, where every country in the world committed to net-zero. So just last May, the G7 nations banned financing of coal plants overseas. And just three weeks ago, China did, which is good, because they were financing most of them. I hope that they will cut down on the coal chain in China domestically as well. And now more than half of all the greenhouse **gas** emissions and two thirds of global GDP is coming from countries that have set net-zero targets. So here ' s the hope. Once we reach net-zero, then with a lag time of as little as three to five years, the temperatures in the world will stop going up. And once we reach net-zero within as little as 25 to 30 years, half of the human-caused CO2 will fall out of the atmosphere. It is as if we have a switch that we can flip in order to stop the climate crisis. Regrettably, some damage has been done, but we can stop the temperatures from going up and start the healing process. But we all have to flip this switch known as reaching net-zero. The young people are telling us that we have to, and they ' re marching in every country in the world. And what ' s the response? As recently as this morning we heard,"Well, it may be impossible."Well, it ' s not impossible. Nelson Mandela said,"It always seems impossible until it ' s done."That ' s what they told the abolitionists."It ' s impossible to eliminate slavery."That ' s what they told the women ' s suffragettes."It ' s impossible to have equal rights for women."In the civil rights movement in my country and the anti-apartheid movement in South Africa and more recently, the lesbian and gay liberation and equal rights movement. We can do this. This is the biggest emergent social movement in all of history. We can do this. And if anybody thinks that we don ' t have the political will, remember, political will is itself a renewable resource. Thank you very much. Thank you.

Text Sample 740: POS Dim 5 – 2019 – Score 8.00 – 2019/t001498.txt

My mom has always reminded me that I have the same proportions as a LEGO man. And she does actually have a point. LEGO is a company that has succeeded in making everybody believe that LEGO is from their home country. But it ’ s not, it ’ s from my home country. So you can imagine my excitement when the LEGO family called me and asked us to work with them to design the Home of the Brick. This is the architectural model — we built it out of LEGO, obviously. This is the final result. And what we tried to do was to design a building that would be as interactive and as engaging and as playful as LEGO is itself, with these kind of interconnected playgrounds on the roofscape. You can enter a square on the ground where the citizens of Billund can roam around freely without a ticket. And it ’ s probably one of the only **museums** in the world where you ’ re allowed to touch all the artifacts. But the Danish word for design is”formgivning,”which literally means to give form to that which has not yet been given form. In other words, to give form to the future. And what I love about LEGO is that LEGO is not a toy. It ’ s a tool that empowers the child to build his or her own world, and then to inhabit that world through play and to invite her friends to join her in cohabiting and cocreating that world. And that is exactly what formgivning is. As human beings, we have the power to give form to our future. Inspired by LEGO, we ’ ve built a social housing project in Copenhagen, where we stacked blocks of wood next to each other. Between them, they leave spaces with extra ceiling heights and balconies. And by gently wiggling the blocks, we can actually create **curves** or any organic form, adapting to any urban context. Because adaptability is probably one of the strongest drivers of architecture. Another example is here in Vancouver. We were asked to look at the site where Granville bridge triforks as it touches downtown. And we started, like, mapping the different constraints. There ’ s like a 100-foot setback from the bridge because the city want to make sure that no one looks into the traffic on the bridge. There ’ s a **park** where we can ’ t cast any shadows. So finally, we ’ re left with a tiny triangular footprint, almost too small to build. But then we thought, like, what if the 100-foot minimum distance is really about minimum distance — once we get 100 **feet** up in the air, we can grow the building back out. And so we did. When you drive over the bridge, it ’ s as if someone is pulling a curtain aback, welcoming you to Vancouver. Or a like a weed growing through the cracks in the pavement and blossoming as it gets light and air. Underneath the bridge, we ’ ve worked with Rodney Graham and a handful of Vancouver artists, to create what we called the Sistine Chapel of street art, an art gallery turned upside down, that tries to turn the negative impact of the bridge into a positive. So even if it looks like this kind of surreal architecture, it ’ s highly adapted to its surroundings. So if a bridge can become a **museum**, a **museum** can also serve as a bridge. In Norway, we are building a **museum** that spans across a river and allows people to sort of journey through the exhibitions as they cross from one side of a sculpture **park** to the other. An architecture sort of adapted to its landscape. In China, we built a headquarters for an energy company and we designed the facade like an Issey Miyake fabric. It ’ s rippled, so that facing the predominant direction of the sun, it ’ s all opaque ; facing away from the sun, it ’ s all glass. On average, it sort of transitions from solid to clear. And this very simple idea without any moving parts or any sort of technology, purely because of the geometry of the facade, reduces the energy consumption on cooling by 30 percent. So you can say what makes the building look elegant is also what makes it perform elegantly. It ’ s an architecture that is adapted to its climate. You can also adapt one culture to another, like in Manhattan, we took the Copenhagen courtyard building with a social space where people can hang out in this kind of oasis in the middle of a city, and we combined it with the density and the verticality of an American skyscraper, creating what we ’ ve called a”courtscraper.”From New York to Copenhagen. On the waterfront of Copenhagen, we are right now finishing this waste-to-energy power plant. It ’ s going to be the **cleanest** waste-to-energy power plant in the world, there are no toxins coming out of the chimney. An amazing marvel of engineering that is completely invisible. So we thought, how can we express this? And in Copenhagen we have snow, as you can see, but we have absolutely no mountains. We have to go six hours by bus to get to Sweden, to get alpine skiing. So we thought, let ’ s put an alpine ski slope on the **roof** of the power plant. So this is the first test run we did a few months ago. And what I like about this is that it also show you the sort of world-changing power of formgivning. I have a five-month-old son, and he ’ s going to grow up in a world not knowing that there was ever a time when you couldn ’ t ski on the **roof** of the power plant. So imagine for him and his generation, that ’ s their baseline. Imagine how far they can leap, what kind of wild ideas they can put forward for their future. So right in front of it, we ’ re building our smallest project. It ’ s basically nine containers that we have stacked in a shipyard in Poland, then we ’ ve schlepped it across the Baltic sea and docked it in the port of Copenhagen, where it is now the home of 12 students. Each student has a view to the water, they can jump out the window into the **clean** port of Copenhagen, and they can get back in. All of the heat comes from the thermal mass of the sea, all the power comes from the sun. This is the first 12 units in Copenhagen, another 60 on their way, another 200 are going to Gothenburg, and we ’ re speaking with the Paris Olympics to put a small floating village on the Seine. So very much this kind of, almost like nomadic, impermanent architecture. And the waterfronts of our cities are experiencing a lot of change. Economic change, industrial change and climate change. This is Manhattan before Hurricane Sandy, and this is Manhattan after Sandy. We got invited by the city of New York to look if we could make the necessary flood protection for Manhattan without building a seawall that would segregate the life of the city from the water around it. And we got inspired by the High Line. You probably know the High Line — it ’ s this amazing new **park** in New York. It ’ s basically decommissioned train tracks that now have become one of the most popular promenades in the city. So we thought, could we design the necessary flood protection for Manhattan so we don ’ t have to wait until we shut it down before it gets nice? So we sat down with the citizens living along the waterfront of New York, and we worked with them to try to design the necessary flood protection in such a way that it only makes their waterfront more accessible and more enjoyable. Underneath the FDR, we are putting, like, pavilions with pocket walls that can slide out and protect from the water. We are creating little stepped terraces that are going to make the underside more enjoyable, but also protect from flooding. Further north in the East River Park, we are creating rolling hills that protect the **park** from the noise of the highway, but in turn also become the necessary flood protection that can stop the waves during an incoming storm surge. So in a way, this project that we have called the Dryline, it ’ s essentially the High Line — The High Line that ’ s going to keep Manhattan dry. It ’ s scheduled to break ground on the first East River portion at the end of this year. But it has essentially been codesigned with the citizens of Lower Manhattan to take all of the necessary infrastructure for **resilience** and give it positive social and environmental side effects. So, New York is not alone in facing this situation. In fact, by 2050, 90 percent of the major cities in the world are going to be dealing with rising seas. In Hamburg, they ’ ve created a whole neighborhood where the **bottom** floors are designed to withstand the inevitable flood. In Sweden, they ’ ve designed a city where all of the **parks** are wet gardens, designed to deal with storm water and waste water. So we thought, could we perhaps — Actually, today, three million people are already permanently living on the sea. So we thought, could we actually imagine a floating city designed to incorporate all of the Sustainable Development Goals of the United Nations into a whole new human-made ecosystem. And of course, we have to design it so it can produce its own power, harvesting the thermal mass of the oceans, the force of the tides, of the currents, of the waves, the power of the wind, the heat and the energy of the sun. Also, we are going to collect all of the rain water that drops on this man-made archipelago and deal with it organically and mechanically and store it and **clean** it. We have to grow all of our food locally, it has to be fish- and plant-based, because you won ’ t have the space or the resources for a dairy diet. And finally, we are going to deal with all the waste locally, with compost, recycling, and turning the waste into energy. So imagine where a traditional urban master plan, you typically draw the street grid where the cars can drive and the building plots where you can put some buildings. This master plan, we sat down with a handful of scientists and basically started with all of the renewable, available natural resources, and then we started channeling the flow of resources through this kind of human- made ecosystem or this kind of urban metabolism. So it ’ s going to be modular, it ’ s going to be buoyant, it ’ s going to be designed to resist a tropical storm. You can prefabricate it at scale, and tow it to dock with others, to form a small community. We ’ re designing these kind of coastal additions, so that even if it ’ s modular and rational, each island can be unique with its own coastal landscape. The architecture has to remain relatively low to keep the center of gravity buoyant. We ’ re going to take all of the agriculture and use it to also create social space so you can actually enjoy the permaculture gardens. We ’ re designing it for the tropics, so all of the **roofs** are maximized to harvest solar power and to shade from the sun. All the materials are going to be light and renewable, like bamboo and wood, which is also going to create this charming, warm environment. And any architecture is supposed to be able to fit on this platform. Underneath we have all the storage inside the pontoon, almost like a mega version of the student housings that we ’ ve already worked with. We have all the storage for the energy that ’ s produced, all of the water storage and remediation. We are sort of dealing with all of the waste and the composting. And we also have some backup farming with aeroponics and hydroponics. So imagine almost like a vertical section through this landscape that goes from the air above, where we have vertical farms ; below, we have the aeroponics and the aquaponics. Even further below, we have the ocean farms and where we tie the island to the ground, we ’ re using biorock to create new reefs to regenerate habitat. So think of this small island for 300 people. It can then group together to form a cluster or a neighborhood that then can sort of group together to form an entire city for 10, 000 people. And you can imagine if this floating city flourishes, it can sort of grow like a culture in a petri dish. So one of the first places we are looking at placing this, or anchoring this floating city, is in the Pearl River delta. So imagine this kind of canopy of photovoltaics on this archipelago floating in the sea. As you sail towards the island, you will see the maritime residents moving around on alternative forms of aquatic transportation. You come into this kind of community port. You can roam around in the permaculture gardens that are productive landscapes, but also social landscapes. The greenhouses also become orangeries for the cultural life of the city, and below, under the sea, it ’ s teeming with life of farming and science and social spaces. So in a way, you can imagine this community port is where people gather, both by day and by night. And even if the first one is designed for the tropics, we also imagine that the architecture can adapt to any culture, so imagine, like, a Middle Eastern floating city or Southeast Asian floating city or maybe a Scandinavian floating city one day. So maybe just to conclude. The human body is 70 percent water. And the surface of our planet is 70 percent water. And it ’ s rising. And even if the whole world woke up tomorrow and became carbon-neutral over night, there are still island nations that are destined to sink in the seas, unless we also develop alternate forms of floating human habitats. And the only constant in the universe is change. Our world is always changing, and right now, our climate is changing. No matter how critical the crisis is, and it is, this is also our collective human superpower. That we have the power to adapt to change and we have the power to give form to our future.

Text Sample 741: POS Dim 5 – 2019 – Score 5.00 – 2019/t001219.txt

My day starts just like yours. When I wake up in the morning, I check my phone, and then I have a cup of coffee. But then my day truly starts. It may not be like yours, because I live my life as an artwork. **Picture** yourself in a giant jewelry box with all the beautiful things that you have ever seen in your life. Then imagine that your body is a canvas. And on that canvas, you have a mission to create a masterpiece using the contents of your giant jewelry box. Once you ’ ve created your masterpiece, you might think,”Wow, I created that. This is who I am today.”Then you would pick up your house keys, walk out the door into the real world, maybe take public transport to the center of the town... Possibly walk along the streets or even go shopping. That ’ s my life, every day. When I walk out the door, these artworks are me. I am art. I have lived as art my entire adult life. Living as art is how I became myself. I was brought up in a small village called Fillongley, in England, and it was last mentioned in the”Domesday Book,”so that ’ s the mentality. I was raised by my grandparents, and they were antiques dealers, so I grew up surrounded by history and beautiful things. I had the most amazing dress-up box. So as you can imagine, it started then. I moved to London when I was 17 to become a model. And then I went to study photography. I wasn ’ t really happy with myself at the time, so I was always looking for escapism. I studied the works of David LaChapelle and Steven Arnold, photographers who both curated and created worlds that were mind-blowing to me. So I decided one day to cross over from the superficial fashion world to the superficial art world. I decided to live my life as a work of art. I spend hours, sometimes months, making things. My go-to tool is a safety pin, like this — They ’ re never big enough. And I use my fabrics time and time again, so I recycle everything that I use. When I get dressed I ’ m guided by color, texture and shape. I rarely have a theme. I find beautiful objects from all over the world, and I curate them into 3-D tapestries over a base layer that covers my whole body shape... because I ’ m not very happy with my body. I ask myself,”Should I take something off or should I put something on? 100 pieces, maybe?”And sometimes, I do that. I promise you it ’ s not too uncomfortable — well, just a little — I might have a safety pin poking at me sometimes when I ’ m having a conversation with you, so I ’ ll kind of go off — It usually takes me about 20 minutes to get ready, which nobody ever believes. It ’ s true — sometimes. So, it ’ s my version of a t-shirt and jeans. When I get dressed, I build like an **architect**. I carefully place things till I feel they belong. Then, I get a lot of my ideas from lucid dreaming. I actually go to sleep to come up with my ideas, and I ’ ve taught myself to wake up to write them down. I wear things till they fall apart, and then, I give them a new life. The gold outfit, for example — it was the outfit that I wore to the Houses of Parliament in London. It ’ s made of armor, sequins and broken jewelry, and I was the first person to wear armor to Parliament since Oliver Cromwell banned it in the 17th century. Things don ’ t need to be expensive to be beautiful. Try making outfits out of bin liners or trash you found out on the streets. You never know, they might end up on the pages of”Vogue.”There ’ s over 6, 000 pieces in my collection, ranging from 2, 000-year-old Roman rings to ancient Buddhist artifacts. I believe in sharing what I do and what I have with others, so I decided to create an art exhibition, which is currently traveling to **museums** around the world. It contains an army of me — life-size sculptures as you can see behind me, they ’ re here — they are my life, really. They ’ re kind of like 3-D tapestries of my existence as living as art. They contain plastic crystals mixed with diamonds, beer cans and royal silks all in one look. I like the fact that the viewer can never make the assumption about what ’ s real and what ’ s fake. I find it important to explore and share cultures through my works. I use clothing as a means to investigate and appreciate people from all over the world. Sometimes, people think I ’ m a performer or a drag queen. I ’ m not. Although my life appears to be a performance, it ’ s not. It ’ s very real. People respond to me as they would any other type of artwork. Many people are **fascinated** and engaged. Some people walk around me, staring, shy at first. Then they come up to me and they say they love or absolutely hate what I do. I sometimes respond, and other times I let the art talk for itself. The most annoying thing in the world is when people want to touch the artwork. But I understand. But like a lot of contemporary art, many people are dismissive. Some people are critical, others are abusive. I think it comes from the fear of the different — the unknown. There are so many reactions to what I do, and I ’ ve just learned not to take them personally. I ’ ve never lived as Daniel Lismore, the person. I ’ ve lived as Daniel Lismore, the artwork. And I ’ ve faced every obstacle as an artwork. It can be hard... especially if your wardrobe takes up a 40-foot container, three storage units and 30 boxes from IKEA — and sometimes, it can be very difficult, getting into cars, and sometimes — well, this morning I didn ’ t fit through my bathroom door, so that was a problem. What does it mean to be yourself? People say it all the time, but what does it truly mean, and why does it matter? How does life change when you choose to be unapologetically yourself? I ’ ve had to face struggles and triumphs whilst living my life as art. I ’ ve been put on private jets and flown around the world. My work ’ s been displayed in prestigious **museums**, and I ’ ve had the opportunity — that is my grandparents, by the way, they ’ re the people that raised me, and there I am — So I ’ ve been put on private jets, flown around the world, and yet, it ’ s not been that easy because at times, I ’ ve been homeless, I ’ ve been spat at, I ’ ve been abused, sometimes daily, bullied my entire life, rejected by countless individuals, and I ’ ve been stabbed. But what hurt the most was being put on the”Worst Dressed”list. It can be hard, being yourself, but I ’ ve found it ’ s the best way. There ’ s the”Worst Dressed.”As the quote goes,”Everyone else is already taken.”I ’ ve come to realize that confidence is a concept you can choose. I ’ ve come to realize that authenticity is necessary, and it ’ s powerful. I ’ ve tried to spend time being like other people. It didn ’ t work. It ’ s a lot of hard work, not being yourself. I have a few questions for you all. Who are you? How many versions of you are there? And I have one final question : Are you using them all to your advantage? In reality, everyone is capable of creating their own masterpiece. You should try it sometime. It ’ s quite **fun**. Thank you.

Text Sample 742: POS Dim 5 – 2019 – Score 5.00 – 2019/t002371.txt

Hi. I ’ m Maisie Williams. And I ’ m kind of just waiting for someone to come on stage and tell me that there ’ s been some sort of miscommunication, and that I should probably leave. No? Damn it. So, some of you may know me as an actress. Some of you may know me for my really average tweets. Oh, yeah. And some of you may be finding out who I am for the first time right now. Hello. Whether you knew me before or not, you ’ re probably wondering what I ’ m going to talk to you about today. And I would be lying if I said it didn ’ t take me one or two sleepless nights, trying to figure that out, too. At last, here I am. Upon finding out the news that I would be giving a TEDx Talk, I did what I think most people do and watched about 50 TED talks back-to-back, and read”Talk like TED”by Carmine Gallo for some inspiration. Was I inspired? Yes and no. Did it make me want to go out and change the world? **Hell** yeah. Did it make me feel like a totally inadequate public speaker with absolutely no point to make, who was definitely in need of a big thesaurus if she wants to keep up? Indeed. What could I possibly say that would have any impact? What point am I trying to make? And who the **hell** thought it was a good idea to give me a TEDx talk? So here ’ s the part where I tell you what I know : I ’ m the youngest of four siblings. My parents divorced when I was four months old. I really was the icing on the cake of a terrible marriage. I have two step siblings who are younger than me and a half brother who ’ s older than all of us. I grew up in a three-bedroom council house with four of my six siblings just outside of Bristol. I went to a very ordinary school. I got very ordinary grades. I wasn ’ t quite good enough to get a gold star, and I also wasn ’ t quite bad enough to be kept after school. I walked that nice center line where if I kept my mouth shut in class, then I could probably get away with not being spoken to you by teachers for weeks on end. Everything about me was pretty damn ordinary, except for how I felt on the inside. I had big dreams. Shock. From as young as I can remember, I have dreamed of becoming a professional dancer. There are certain memories from my childhood that I would really rather forget. But during those times of immense pain, I found myself instinctively walking over to my mother ’ s CD player, cranking up the volume to drown out the noise and letting my body move to the beat. It ’ s hard to describe how it felt. I was harnessing emotions that I didn ’ t even really know the names of yet. I was summoning all of this energy and feeling it flow through my body and out of my fingertips. I was alone in my own head, and I felt the most alive. I didn ’ t really know much about the big wide world then, but I knew that this feeling was addictive ; and I was going to stop at nothing until I made it my profession. At eight years old, I was enrolled in dance class. And by ten, I informed my mother that I didn ’ t want to go to school anymore. I wanted to be like Billy Elliot and go to stage school. This was the first opportunity or challenge I was presented with. Even as young as ten, I was willing to give up all of my friends and go away to board at a private school, away from my siblings, away from my mom. She would repeatedly ask me,”Are you sure this is what you want?”And to me, it was a no-brainer. I didn ’ t just want this ; I needed it. My grubby knees and crooked teeth were not on the list of requirements for becoming a professional dancer. And when I look back now, both myself and my mother looked severely out of place. But at the time, I was just too young and naive to feel inadequate. I didn ’ t care. If Billy Elliot could do it, so could I. Once my audition was done, I returned home for two weeks of staring out the window, waiting for the postman, waiting for my ticket out of my sleepy village and into a world of jazz hands and dorm rooms. It was good news followed by bad news : I had got in, but the fees to attend a school like this were not cheap, and despite my best efforts, I had not received any government funding. I auditioned again the following year. And this time, I received 40 % funding, but this was still just money that we didn ’ t have, and it broke my heart. I was good enough. I made the cut. But I wasn ’ t going anywhere. It was a blessing in disguise, although if anyone had said that to me back then, I probably would ' ve given them the finger and told them to jog on. I wasn ' t willing to give up that easily. So at age 11, I was bursting with excitement when my dance teacher informed me of a talent show which boasted opportunities of making you a star. This was the second opportunity I was faced with. I entered into singing, acting, dancing and modeling. The talent show consisted of workshops and seminars with specialists who would help train you up for your performance at the end of the week. After meeting a woman called Louise Johnston in an improvisation acting workshop, she gave me the words"bowling ball,"and asked me to create a short scene inspired by these words. After making her laugh with a fictional story, of how I threw a bowling ball at my brother and it bounced, she asked me to join her acting agency. I didn ' t really know what this meant. I knew that I would do auditions for **films** and maybe become an actor, but I still had big dreams of becoming a professional dancer, so this woman was going to have to work a lot harder than that if she was going to convince eleven-year-old me that I was going to become an actress. Was this going to take time away from the 30 hours of dancing I was doing a week? And what if I didn ' t get the part? Was this going to be too upsetting? And do actresses have teeth like mine? Because if they do, I ' m yet to watch any of their movies. After meeting Louise in the February of 2009 and trying but failing to land the part in the hit sequel"Nanny McPhee"to"The Big Bang,"my second audition was for a show called"Game of Thrones."This was the third opportunity or challenge I was presented with. I climbed the steps to the Methodist Church with my mother ' s hand in mine. I perched my tiny **bottom** in one of the seats outside the audition room and listened to an annoying girl with her even more annoying mother tell me all about the number of auditions she had done prior to this one. And also about her pet fish. My name was called, then I stepped inside. I had a hard Bristolian accent and dark rings around my eyes that were so big they took up half my face and a hole in the knee of my trousers which I tried to cover with my left hand as I was talking to the kind lady who **taped** my audition. But as soon as she pressed record, it all drifted away. Much like when I was dancing in my mother ' s living room, I harnessed all of my insecurities and self-doubt and let it flow through the words that came out of my mouth. I was cheeky. I was loud. I was angry. And for this, I was perfect. After getting the part and shooting the pilot episode, the show slowly grew to become one of the biggest shows in television history. To this day, we ' ve smashed previous HBO viewing records. We ' ve been nominated for over 130 Emmys, making us the most Emmy-nominated show to ever exist. We ' ve recently finished shooting our eighth and final season, which is predicted to smash records that we ' ve already broken. And nearly a decade to the day since my first audition, I ' m still wondering, when am I going to get to be Billy Elliot? I joke, but in all seriousness, I have absolutely no plans of slowing down. Throughout my time in this industry, it has been a minefield. I have grown from a child into an adult, and from four **feet** tall into a whopping five **feet** tall. I have constantly been trying to say the right thing, accidentally saying the wrong thing, trying not to swear too much and trying to stop saying"like, like"all of the time. In February of 2017, a friend of mine, Dom, and I were swigging beers in my kitchen, and he confessed to me that there is a huge problem with the creative industries. I agreed. The series of events that had got me to that point were based mainly on luck and timing and were unable to be recreated. He suggested to me that we create a social media, but just for artists to be able to collaborate with one another and create a career. This was the fourth opportunity or challenge I was presented with."Great,"I thought."How the **hell** do we do that?"And daisie was born. Of course, everyone who I spoke to about my latest endeavor thought that

I was mad ; however, I know that this is something that I can help change. This last year in the industry, we ’ ve seen a huge shift with the Me Too movement. The industry is built with gatekeepers holding all of the power and selecting who they deem talented enough to advance to the next level. More often than not, it ’ s easier to catch the attention of those people if you have graduated from an expensive school. But even then, I have so many friends who are fresh out of art school, having trained for years and are still no closer to creating a career. Now, I ’ m not claiming that with daisie I can make everybody a star, but I do believe that the key to success within creative industries is collaborating. Actors are only as good as their writers. Musicians are only as strong as their producers. And designers need their teams. To start the company, we self-funded. I had a pot of cash from”Game of Thrones”that I was free to invest wherever I liked. Dom had a series of businesses from the age of 16, which meant he was also left with a pot of cash. We threw our money together 50-50, and we built a team. Now, Lady Gaga has repeatedly said that there could be a room of 100 people, and 99 don ’ t believe in you, but it just takes that one person to believe in you, and they can change your life. Well, now we have a team of six. Over the next 16 months, we built our MVP. Now, if you ’ re wondering what an MVP is, I only found out what it is about six months ago. And from what I can gather, it ’ s a product which proves as a problem worth solving with the minimum team effort. So basically from my point of view, you ’ re marketing something which you know is going to be good one day, but is a little bit bad right now. And for us, that was an iOS app. The six of us made an office in Dom ’ s garden, and on August 1, 2018, we released our version one. We had over 30, 000 downloads in the first 24 hours and over 30, 000 comments asking when the Android version was going to be coming. Despite our app being imperfect, buggy and literally built by one man alone, this was exactly what we needed for people to invest. We learned a lot from our angry users and our scary investors. And over the last six months, we have grown our team to 16 people. From then till now, we ’ ve been building version two, which we will be launching in April. Within the industry, there is a common phrase which I think we ’ re all pretty familiar with. And that is,”It ’ s not what you know, it ’ s who you know.”And with daisie, I hope to give that power back to the creator. I want to encourage people to create a list of contacts that they will work with and support as they take their first steps into the fickle and often challenging creative world. I am of the generation who grew up with the Internet. I ’ ve never known anything else. We are connected, we are aware, and we are the future. I hope daisie can breathe new life into the slightly dystopian, ad-riddled hellscapes that social media platforms have become. I hope to create a space where people can boast their art and creativity rather than what car they are driving and whether or not they bought it in cash or on finance. In a world where literally anyone can be famous, I hope to inspire people to be talented instead. Talent will carry you so much further than your 15 minutes of fame. So why am I telling you all this? The very fact that I ’ m here giving a TEDx talk right now is so far from anything I thought that I was capable of. Even writing the bio for my speech made me realize that in a decade, everything in my life has changed. I am an Emmy-nominated actress, an entrepreneur and an activist ; yet I have no formal qualifications to my name. When I left school about seven years ago, I made it my mission to continue learning even though I never wanted to set **foot** in a classroom again. Who knows what ’ s going to happen to my life in the next 10 years? I surely have no idea. I ’ ve never had an end goal. It ’ s working out okay so far. So trust that you ’ re good enough. If there ’ s one thing that I ’ ve learned is that there truly is a place for everyone. Ask questions, and laugh in the face of people who say that they ’ re stupid questions. Be open to learning and admitting when you don ’ t know what the **hell** is going on. Refuse to hold yourself back, and dare to dream big. Thank you for listening.

Text Sample 743: POS Dim 5 – 2025 – Score 1.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in the world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the air. In all my travels, I ’ ve been struck by so many things. I ’ ve observed how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ’ ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced us to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ’ s hard to see the big **picture** right now. It ’ s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think back to my travels. I ’ ve seen how different we are, how diverse and distinct. But I ’ ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ’ m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman”A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story is the dominant sound of our lives.”- Joseph Campbell I ’ ve loved telling stories all of my life. I ’ ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 744: POS Dim 5 – 2025 – Score 1.00 – 2025/t003466.txt

Can you paint with all the colors of the wind? I love this song from the first time I heard it. I was 14, growing up in public housing in western Sydney. So maybe a little too old for this Disney classic from”Pocahontas.”But it spoke to me. I saw something of myself in it. At this stage, I think I knew I was queer. I mean, come on, I was rocking out to Disney. And look at those glasses. Seriously, what a queer kid. But I was also from a Pacific Indigenous heritage with conservative views and values. I didn ’ t feel like I could fit into any of the boxes that were being presented to me. Straight, gay, Fijian, Australian. I couldn ’ t pick just one color to paint with. Prior to colonization, sexuality amongst many of our Pacific islands across the region was fluid. Men would have sex with men without the stigma or shame afforded it today. Sexual intimacy was a way of being able to connect socially and relationally. Sex wasn ’ t just for procreation. Or in marriage. It wasn ’ t exclusively monogamous. Just because you had sex with a man didn ’ t define your sexuality. It didn ’ t make you an outcast. But as Europeans started to colonize the Pacific region, they brought Christian missionaries with them. They told us to cover up our beautiful bodies. We were taught shame. We were told to stop practicing Pacifica cultures and values that were seen as provocative. To a Western and white perspective, our non-binary sexuality was judged as tribal, outdated, sinful and impossible to practice alongside a Christian faith. Still today, those who are Indigenous to the Pacific are negatively impacted by whiteness. That”white is right, West is best.”I was brought up to espouse strong Christian beliefs, views and values which would shape my burgeoning identity as an adult. From these staunch conservative views, I was taught to fear the queer. Those people that were seen as unusual and fruity. And in essence, I grew to despise a part of myself. I sought to separate my queerness from my identity. I could never see myself as part of that world. Growing up, biracial and bisexual constantly challenged the way I saw the world around me. I was unsure about my thinking, feelings and behaviors around liking males and females. I was unsure about my thinking, feelings and behaviors when it came to whether I needed to be a white Australian or a Fijian person. And whether this needed to be framed within a binary to choose one over the other. In this way, I don ’ t think that my story is unique. There ’ s plenty of people in the queer world that have escaped conservative upbringings, religions and cultures. But this is only part of my story. Another common story in the queer community is the coming-out story. When did you tell your parents? How did they react? How it liberated you or ultimately separated you from your family. But this didn ’ t feel like an option for me either. I believe that embedded in these particular stories, these tropes, is a part of dominant queer culture that made it very hard for me to access the queer world. For example, when I started to go to gay clubs, I would be lining up at the front of the club and often I would get door staff come up to me and go,"Sorry, you do know this is a gay club, right?"From these lived experiences, I feel that there is a whiteness that pervades queer culture. And I didn ' t want to give up my identity and my appearance as a Pacific indigenous person to be queer. Pacifica cultures really, really value and esteem the collective above the individual. For Pacifica people, our families are inextricably connected to our own sense of self-agency, self-determination, self-esteem, self-love and self-worth. Our own individual identities are meshed and intertwined with our family ' s reputation and standing in the community. Our mind, body and souls are inextricably connected to the broader community, to our families, to ourselves and to others. I didn ' t want to come out to potentially ostracize myself, to make a point to separate myself from my Pacific and faith communities. I couldn ' t then see my authentic self in any of these spaces. And a quick shout out to my son on the **left-hand** side. Round of applause for my son. He said to me this morning,"Daddy, make sure you mention me."So I was like,"OK, son. Alright."So there you go for the recording, son. And the rest are my other family, that ' s up there as well. Across queer cultures and the broader community, we have learned to embrace our gender and sexual differences. We encourage individuals to be allies for each other and to support safe spaces for those that are seen to not fit in. We empower and really encourage individuals to really develop and express their self-agency, their self-determination, their self love, their self-worth. We boldly go into spaces striving to bring others with us to ensure that we can create inclusive societies for everyone. However, the dominant Western view in queer cultures is still based on the individual. Me, myself and I. And my individual pursuits towards happiness. As a queer Indigenous Pacific person, I feel that there are parts of queer culture that is white, feels very white. We may fail to see the importance of bringing our families and our cultural communities along our journey as queer people. It ' s either the queer way or the highway. It ' s one or the other. Young or old. Rich or poor. Smart or stupid. Masculine or feminine. Fit or fat. These binaries create a sense of us and them, those people. A binary approach and position that we generally in Western white societies use to determine the norm. We create, conflate, perpetuate binaries that create divisions in our society. Saints or sinner, straight or gay. I see it like this. Binaries are a very white and Western way of thinking, feeling and behaving. As individuals today, we live within these binaries and create identities that then create these labels. We use these labels to then define and confine people and places. We are one or the other. But what we fail to see is that our identities are more than a set of binaries. Our own identities are comprised of intersecting identifiers that contributes to who you are, including our age, class, color, education, gender, size, spirituality, sexuality, and so, so much more. We need to decolonize queerness, to live beyond"white is right, West is best."Because if we don ' t, we ' re going to continue to create silos and divisions in our societies that don ' t create inclusive and safe spaces for everyone. For Indigenous queer people, they shape our connection to community and culture, our families. If we don ' t include that within our understanding of who we are, even as queer people, then we ' re not able to be, again, our true and authentic selves. We ' re not able to then be our fabulous selves as part of the queer community. And it ' s really, really interesting because I believe that we can learn so much from pre-colonial cultures like those that were practiced in the Pacific, that provide a different way of looking at a society. A queer way that pre-dates all the privileges that go into binary ways of thinking, feeling and behaving. Across many of our Pacifica languages we have a lot of key concepts that help solidify and support a shared understanding of the collective. For example, in Fijian,”solesolevaki”is all about reciprocal living, where my wellbeing is your wellbeing. I will actively seek to ensure that your well-being and every part of your life is catered for, knowing that you will do the same for me. We strive to enact”veiqaravi,”as tatted here on my hand, to serve others. We strive to also embed and embold and empower people to practice”veirokovi,”respect for self and others where we are relationally driven, where that is so important to our spaces and places. My sense of ability to actually put this into practice, my ability to actually understand these Indigenous perspectives has greatly helped shape and reconcile my queer identity today. But what ’ s really interesting is that it ’ s not just queer communities that need to decolonize itself. It ’ s all of us as a broader society that needs to be part of a shared conversation to decolonize, deconstruct and disrupt binary ways of thinking, feeling and behaving. So how can we practically do that? Practice cultural humility, where you are constantly learning and embracing differences, where you get over yourselves and strive to learn about other people ’ s differences and how that can be meaningfully included in the way in which you create safe spaces for everyone. I also believe it ’ s possible to bring Indigenous perspectives into Western and white workplaces. For example, I ’ ve introduced the concept of”talanoa”in all of my work meetings. Talanoa is all about being able to create a safe space where everyone is able to contribute to a collective, collegial and collaborative conversation. Where people are able to bring their authentic selves into these spaces. Literally, I will chair meetings where I have no firm set agenda items and I literally would just hold space for participants just to bring themselves. So you can tell I ’ m a social worker, right? Safe space, safe space, safe space. It ’ s OK, it ’ s OK. But it ’ s through these safe spaces that people can bring their multiple, intersecting identifiers, including their age, their class, color, education, gender, size, spirituality, sexuality and so, so much more. I invite queer communities to come talanoa with Pacific Indigenous Communities to create robust and resilient queer cultures. I have been able to greatly reconcile my queer and Pacifica identities, which has now really been embraced to help shape how I see the world around me today. And it ’ s through these shared perspectives and approaches that we can truly embrace cultural diversity and its differences. And to know that people can live beyond those binaries and those labels we continue to perpetuate and perpetrate against others. It is a commitment from queer communities to decolonize queerness. It ’ s a commitment from me, from you and all of us to decolonize Western and white societies and to live beyond the binary way of thinking, feeling and behaving. I genuinely believe that it ’ s through this common goal that we can truly learn how to paint with all the colors of the wind.”Vinaka vakalevu”and thank you.

Text Sample 745: POS Dim 5 – 2025 – Score 1.00 – 2025/t003002.txt

HW : Hi everyone, I am Helen Walters. I am head of Media and Curation here at TED. Welcome to another episode of TED Explains the World with the one and only, Ian Bremmer. It is Monday, February 24, a little more than a month since President Trump was inaugurated once more, and safe to say, a lot has been going on. So we figured we ’ d check in with Ian to determine what we should really be paying attention to amid the extreme noise. Ian, hi. Ian Bremmer : Helen, great to be back with you. HW : So we asked the TED community to share their questions for you, and we wanted them to share what they ’ re most curious to know. And we really got some amazing questions that I plan to pepper this conversation with. So let ’ s start with one right now. Two months into 2025, where does the US stand? IB : Very powerfully, in the sense that the US economy, of course, is performing considerably better than any of the other G7 economies coming still out of the pandemic. Technologically, only close competitor is China, ahead of the US in some areas, behind the US and other critical areas. But compared to every other country in the world, the US is head and shoulders and neck or waist above them. Militarily, of course, the US is the only country with global ability to project power. No one else is close. And the US dollar is still the global reserve currency, no one is approximate challenger. I say all of those things because those are the things that haven ' t really changed over the course of the last few months, but I suspect that isn ' t what the questioner was asking. What they were really asking is what ' s happening politically. And politically, the United States is unwinding its own global order. It is no longer particularly interested in promoting collective security or NATO. It ' s no longer really interested in promoting involvement and leadership of multilateral institutions, of consistent rule of law and free trade that ' s well regulated, certainly not very interested in promoting democracy around the world. I mean, this is an environment where other countries around the world have to figure out how to adapt to the United States, and that the US has become the principal driver of geopolitical risk and uncertainty in the world today, unlike any other time in my lifetime and yours, that ' s perhaps the biggest change. HW : That is quite a statement. Alright, you are a geopolitical expert. Let ' s talk about Europe. So Defense Secretary Pete Hegseth stated that America ' s foreign policy focus no longer lies in Europe. Vice President JD Vance caused some raised eyebrows, dropped jaws and, I suspect, some choice expletives after his recent speech at the Munich Security Conference, in which he accused European leaders of suppressing free speech and warned of the threat from within. So you were in Munich. Was it as dramatic as the stories we read had it? And what happens next? IB : It was. I was in the room. I was maybe 30 **feet** away from him when he gave that speech. Standing right in the front, right next to me was the president of Czechia, the president of Finland, the prime minister of Sweden, watching all of them and their reactions. What he said, let ' s keep in mind, this is the Munich Security Conference. It ' s been going on now for something like 61 years. And he was the head of the US delegation. And every year the head of the US delegation gives a big speech on the state of the transatlantic alliance and on global security. And he didn ' t do that. I mean, the people in the audience were expecting a challenging speech. They were expecting that the Americans would be less committed to the alignment with the Europeans on Ukraine. The US had just had that direct Trump phone call with Vladimir Putin, 90 minutes long. Hadn ' t coordinated that with the Europeans, never mind the Ukrainians, in advance. So, I mean, they were definitely prepared for a very challenging plenary. But Vance did none of that. Vance said,"I ' m not going to talk about Ukraine or Russia or China, because the biggest problem is actually what ' s happening inside Europe. The biggest problem, essentially, are you guys sitting in front of me, that I ' m speaking to, because you ' re suppressing free speech. You ' re suppressing the far right. You ' ve been infected by the woke mind virus, and you aren ' t real democracies as a consequence."And specifically he said, and this is something that I think is underappreciated in the United States, he attacked the German firewall. And said that – And let me explain what the German firewall is. That is a principle by the leaders of all of the mainstream German parties and their supporters that they will not, under any circumstances, work with, enter into coalition with the Alternative for Deutschland party, the AfD. Because the AfD, under surveillance from German intelligence agencies, is considered to be a neo-Nazi party. And the United States, of course, after World War II, led de-nazification of Germany.

The day before, Vance had actually gone to Dachau and visited a concentration camp, and then came to speak in Germany about the firewall needing to be ended. And at that point someone yelled out from the crowd,"This is unacceptable!"And it was right in the front. And people in the room, about 1, 500 people in the room, standing room only, couldn ' t see who it was. They ' d just see the back of his head. I saw who it was. It was the German Defense Minister, Pistorius. And in 15 years of me going to the Munich Security Conference, I have never seen anything remotely like that. And the reason I mention all of this, and of course, on the back of that, let me be clear, Vance refused to meet with the German chancellor because he ' s basically, well, he ' s not relevant. He ' s only going to be there for a couple more months until a new government, so why should I bother? But does meet with the leader of this AfD, goes and meets with her directly, does a bilateral that evening. So that is a very important backdrop for this last weekend ' s German elections, where the victor, Friedrich Mertz, who is going to be the next chancellor, actually referred specifically to America ' s intervention in German democracy and support of the AfD being every bit as bad as what the Kremlin is doing inside Germany, and also said that the Europeans need to develop a defense policy which is independent of the United States, in other words, essentially saying that it will be the end of NATO. Again, staggering comments, unheard of, impossible to imagine even two weeks ago. And that ' s why it ' s very critical that you have the context of what happened between Vance, Elon Musk over the past couple of months, Trump, with the Germans, with the Europeans. So you can understand why the incoming German chancellor would make a statement like that. HW : What should we make of all of the conversation about the rise of Nazism, the return of Nazism? Obviously, the AfD just won 20 percent in the German election. We ' ve also seen people allegedly giving Nazi salutes at various US conventions and speeches. Do you think this is overblown? Is this a fear that people have rationally? And what should we make of that? IB : In Germany, the Alternative for Deutschland, are winning all of former East Germany. This is a group that increasingly understands that they ' re never going to really catch up to the rest of Germany. Remember, Germany hasn ' t really grown in five years now. They ' ve been in recession for the last two. And the average citizen, the average voter, living in former East Germany no longer are just angry about not catching up. They ' re basically saying,"I ' m done with this system."So that ' s why you ' re getting this very strident, revanchist nationalism that is, you know, it ' s not just taking a hard line on illegal immigrants. The AfD even supports removing citizens of Germany that they claim have not adequately integrated into German culture and nationality. And so, I mean, this is some very, very strong stuff, stuff that really does feel like what they were doing back in Nazi Germany. But in former East Germany, they win. They get the largest number of votes, where, in much of the wealthier part of West Germany, including, you know, where I was, in Munich, the AfD can ' t break out of single digits. So it ' s a very, very divided country. And also the AfD is doing very, very well among young men. There ' s a clear gender divide here as well, just as there is in the United States. In the US, I mean, I, of course, saw Elon Musk and that, you know, sort of grabbing his heart, saying,"My heart goes out to you"and then doing what appeared to be mimicking a Nazi salute. And then Steve Bannon doing the same without the initial heartfelt comment. It clearly was trolling. Clearly was trying to get a reaction from folks. It ' s very performative. And it ' s meant to also, you know, attack and attach, make the left lose their minds on something that isn ' t very critical to what the Trump administration is trying to do compared to the revolutionary changes inside the US government that they are very much prioritizing right now. So I ' m very concerned about the rise of neo-Nazism in Germany. I am very much not concerned about that in the US. I ’ m concerned about other things I ’ m thinking is being used as gaslighting for those that can use distractions or can be distracted by them. And I would also say that this election in Germany wasn ’ t surprising. It was exactly as the polls have expected for several months now. But the key elections in Germany and the key elections in Europe are the next cycle. I ’ ve spoken with a lot of people around the Trump administration that believe that the Europeans are one electoral cycle behind the United States. So in other words, Trumpism is coming, just not quite yet. And so in the UK, you ’ ve got Labour for a full term. But the Reform Party is increasingly the most popular. And just with a little nudge, maybe a little external money, and Elon said he ’ s already thinking about giving them 100 million dollars, which is a massive amount in UK politics, not so much in the US, that the Conservatives would fragment, a rump group would join with Reform, and they could win the next elections. In France, you ’ ve had all of these governments continue to collapse. Macron, very unpopular. Marine Le Pen and the National Rally party could easily win upcoming elections in 2027. In 2029, in Germany, if the Germans are incapable of turning their economy around, incapable of unwinding their debt break and spending some of the capital that ’ s just sitting there and as their debt to GDP goes down, but their economy contracts at the same time, then AfD could easily win in 2029, and then the firewall is no more. And then, of course, in the European Union elections overall, in five years ’ time, now four and a half, you could easily see the Patriots front, which is the equivalent to the far right grouping that these parties are a part of, they could all come together and be the dominant party. And then the EU as we know it is really a thing of the past. So these are existential changes that are happening not just for NATO right now, but also for the EU and quite plausibly, for the future of democracy as we know it. HW : Alright, let ’ s turn to Ukraine. So President Trump accused President Zelenskyy of starting the war with Russia and called him”a dictator without elections.”Last week, senior American and Russian officials met in Riyadh to discuss a potential cease-fire with no Ukrainians present. Now Zelenskyy has offered to step down in return for Ukrainian admission to NATO, although apparently he doesn ’ t really mean this. So what happens to Ukraine now? What should we be paying attention to? IB : Well, I mean, I think he would mean it if NATO membership happened. I ’ d love to see Trump call his bluff on that. But of course it ’ s not going to. And he wasn ’ t saying that six months or a year ago. He ’ s saying it now that NATO is truly being pulled off the table. I think it ’ s the right thing for him to do. It is, again, performative and helps to remind those around the world that Ukraine is fighting for self-determination. They ’ re fighting for the ability to join their own alliance, as any country in the world should be able to do. But that is not Trump ’ s belief. Trump believes that what you get to do is determined by how powerful you are, not by the fact that you happen to run a country. And indeed, the territorial integrity is something that is afforded to powerful countries, but not so much for weak countries. And that ’ s the way he feels about Panama, and that ’ s the way he feels about Denmark and Greenland. And that ’ s certainly the way he feels about Ukraine. So right now, as you and I are speaking, the United States is putting forward a new UN resolution at the General Assembly that says that the war has to end as fast as possible but does not recognize Ukraine ’ s territorial integrity, which has been a core precept of the international order that the United States has supported since creating the United Nations and since the end of World War II in 1945. The Americans are now throwing that out. And the Russians, of course, are fully supportive of that. America ’ s allies in Europe are not. But the US has gotten the Saudis to agree, and they ’ re getting a unified Arab block. The Americans have privately worked with a lot of poorer countries in Africa, in the Asia Pacific and others, smaller countries, and saying,"You ' re not going to get any aid from the US unless you vote in favor of this new resolution."I fully expect the resolution is going to pass. But of course, what that means is that the Americans are preparing to cut a deal on Ukraine over the heads of the Ukrainians. In other words, the US and Russia together will figure out what a ceasefire should and should not entail as part of a broader US-Russia rapprochement. This is obviously a disaster for the Ukrainians and the Ukrainians who initially refuse to accept this so-called deal, on Ukrainian natural resources that would grant the Americans $500 billion in access to such resources in return for"paying off"aid that the Americans had granted to the Ukrainians without conditions or strings, but apparently no longer. And so, you know, what does this all mean? Well it means it ' s yet another case of why people shouldn ' t trust the United States from one administration to the next, because foreign policy will change completely. And what you thought was an American commitment won ' t stand. It shows that there is a fundamental rift between the Americans and the Europeans and indeed, Emmanuel Macron in the United States right now, as you and I speak, hoping that he can do something that will keep Trump from, as Macron believes, essentially surrendering to, conceding to, the Russian position and maintaining some level of coordination with the Europeans. But he doesn ' t have a lot to offer. And the Ukrainians, you know, now in a position of desperation where maybe they will end up signing over a lot of their natural resources to the Americans. But what, if anything, will they get in return for that? And how empowered will the Russians be in any deal that is cut? And will an independent Ukraine even be able to survive, and what will the terms that will be required of them? Of course, Putin is saying things like, no European troops of any sort, whether in a NATO formulation or not, would be allowed as peacekeepers in Ukraine. Well, then, how would you guarantee, what kind of security commitments would you give to the Ukrainians? These are all open questions, but apparently questions that President Trump is prepared to largely resolve on Russia ' s terms. And again, this is such a dramatic change from where the United States and where the Western order was just a month or two ago. HW : So let ' s take some quick-fire questions from our community related to this topic. And I think the first one is particularly pertinent to what you just said, so it ' s simple. What does Putin have over Trump? IB : I don ' t think that Putin"has"anything over Trump. If he did, then why wouldn ' t Trump have granted Russia more in his first term? In his first term he gave the Ukrainians those javelin anti-tank missiles that Obama refused to give. And, you know, he thought they were too risky. He increased sanctions against Russia. Now that was his administration with a lot of hawkish in orientation advisors around him that he doesn ' t have this time around. They ' re all directly loyal to him. But, I mean, if it was a priority for him, he could have stopped it. And if the Russians had something over him, then why wouldn ' t they have used it in that case? It feels like a conspiracy theory. I don ' t buy it. I think very differently. We can explain what ' s happening because number one, Trump wants to end wars. It ' s very popular with his base. He ' s said this on Gaza. He ' s said this on Iran, where Israel wants him to support attacks on Iran ' s nuclear capabilities, and Trump has said no. And he ' s showing this on Ukraine. He doesn ' t want to spend money on the Ukrainians going forward after spending over 100 billion dollars under the last three years from the Biden administration. He doesn ' t want to spend that money going forward. And he also doesn ' t value a strong Europe. He actually thinks that a weak Europe is in America ' s interests, where you have more Brexits. He used to always talk to the French about that during his first term,"Why won ' t you do Brexit?"He wants all of these

MAGA-type populist parties, including the AfD, to win in Europe because that legitimizes his own popularity. Just like he loves Milei in Argentina or Bukele in El Salvador. And so I think if you put those things together, you actually get to why Trump is doing what he ' s doing with the Russians without needing to create a story about Trump somehow being threatened by some secret, shadowy information that the Russians have on Trump that they ' ve let him know about but nobody else knows. I just don ' t buy that. HW : Great. OK, another community question. If the US hands Ukraine to Russia, will Russia keep going? Will they attempt to take over Europe? IB : It ' s interesting that Trump just met with the Polish president, President Duda. It was supposed to be an hour meeting, it was only 10 minutes. So he kind of embarrassed him at home in Poland. But he did say that the US is still committed to maintaining a troop presence on the ground in Poland. And so it ' s very hard to imagine the Russians rolling through Ukraine and then into Poland, even though there ' s been lots of asymmetric warfare, for example, and even there have been, you know, Russian missiles launched into Lviv, Ukrainian air defense, missile defense, and with Polish villagers getting killed because of the fighting. So, I mean, there ' s been spillover. But America is still going to be in Poland. Why is that? They get along pretty well, and the Poles are heading towards five percent of GDP defense spend this year, which has been the high water mark of American demands of defense spend. So he ' s, you know, been consistent on that over the past months. Baltic states. I mean, they ' re all spending a lot more money on their defense. And so, I mean, in principle, you could imagine the United States maintaining these rotational deployments in the Baltics, which would prevent the Russians from invading. But might Trump say, as part of a deal,"Why do we have all those troops there?"You mentioned Pete Hegseth saying the Americans need to get troops out of Europe and get them to Asia, which is where their principal competitive, you know, sort of, strategy is going to be oriented. And the US hasn ' t been pressing the Japanese or the Indians, the principal partners, allies in Asia vis-à-vis China, the way they ' ve been pressing the French, the Brits, the Germans in Europe. So I think it is certainly plausible that the Russians will do more, especially with those countries that refuse to align with a Trump rapprochement towards Russia. The countries I would be most worried about are Moldova, are Georgia. I ' d be most worried about the countries that are in the so-called"near abroad"that aren ' t a part of NATO, and aren ' t going to become a part of NATO, and therefore are much more vulnerable. But certainly in terms of Russia ' s willingness to engage in espionage, to fund actors on the ground for arson attacks, assassinations, critical infrastructure attacks on fiber networks, on pipelines, I think all of those things are likely to increase against European states across the board, on the back of this US-Russia rapprochement. HW : OK, final community question for this segment. What happens if the US no longer supports NATO? IB : It ' s possible that the United States is moving towards a similar posture in Europe that they presently have in Asia, which is strong individual bilateral deals with a number of countries : South Korea, Japan, Australia, but not a collective security agreement, where those countries together have more of a call on the United States and there ' s a lot more free riding. So I could imagine that NATO falls apart and becomes that. The danger is that even though the Europeans now understand the nature of the threat, that it is very plausible and maybe even baseline, that the Americans are going to leave Europe collectively on its own. That that seriousness does not equate to the ability to increase spending adequately in the near term. The Germans did manage to pull together an electoral outcome that will allow for a so-called grand coalition, just barely. So you ' ll have a two-party government in all likelihood instead of a three. But you also have a blocking capacity on the part of the far right and the far left that will prevent the Germans from really blowing out their spending, including their defense spending, unless they can pull something off before that new government is formed. Which is possible, but it ' s unlikely to be very big. The Brits are now talking about maybe moving the 2. 5 percent of GDP defense spend over years with a very challenging fiscal environment. The Spaniards, not even close, the Italians, not even close. So what does a European military capability really look like? And without the Americans, the answer is it ' s not there. It ' s not there. So I think that if NATO falls apart, the reality is that most of Europe will be incredibly vulnerable to external attack, especially Russian attack, And there won ' t be a way for them to defend themselves, and some of them will break and therefore want to work more directly and individually with the United States. And that could be the end, not only of NATO, but it could be the end of the European Union. I think this is, the stakes are incredibly high, this is an existential challenge for European governments that needed to take this seriously 30 years ago, 20 years ago, certainly in 2014, when the Russians invaded Ukraine, certainly, certainly in 2016 when Trump was elected, and certainly, certainly, certainly in 2022 when the Russians then invaded all of Ukraine. And at every point the Europeans were basically saying,"We ' re still fine with the Americans basically doing this."And now they are in very serious trouble. HW : No time like the present, I suppose. Alright, let ' s move on. So I mentioned the discussions in Saudi and obviously also on the agenda there with the discussions about Gaza and Trump ' s proposal to build a Gaza Riviera. What should we make of that? Or as one of our community members asked, what is the actual solution for Gaza? And surely it isn ' t what the US is proposing. IB : Well, look, here ' s what ' s interesting. You know, Trump says a lot of things, and when they don ' t work [out], he backs out fairly quickly. And you know, I thought – just stick with Ukraine for one second on this. It was interesting to me that Trump has said that he doesn ' t want to talk to Zelenskyy because he has no cards, he has no leverage. And so why should he let him at the table? And so Trump ' s going to cut the deal. Very interesting that if that is true, then why is it that the Secretary of Treasury brings this deal and says,"You ' ve got to sign this right now, Zelenskyy"And Zelenskyy refuses. And then a few days later, the Americans come back with a new deal with altered terms that are not quite as predatory vis-à-vis Ukraine. Why would the Americans change their tune for someone that has no leverage? Because, I don ' t think it ' s because Trump is a nice guy. He ' s not going soft at 78. I think it ' s because he overstated how little leverage Ukraine has. Ukraine is very popular among the GOP in the House and Senate, far more than Putin. Zelenskyy is far more popular in the United States among the voting population than Putin. And of course, he ' s also much more popular among the Europeans and even globally than Putin. And so it turns out that maybe there is a little bit of leverage that Zelenskyy has, and maybe Trump does have to give a little bit more. So this is relevant in the context of Gaza, because Trump was very annoyed that, you know, despite his ability to get a cease fire between Hamas and Israel over the goal line, that his friends in the Middle East were not coming up with a plan for Gaza. Where they were supposed to provide security and provide investment and lead reconstruction and deal with the Palestinian populations in the interim. And that wasn ' t happening. And so Trump then meeting with Prime Minister Netanyahu from Israel in Washington, then after that, summoning the Jordanian king, who was the ally that is most reliant on the United States and so, therefore, most needing to say yes to whatever Trump orders of him. Basically says, as you mentioned, Helen, that Gaza is going to be a Riviera, that the US is going to turn this into an incredible place. They ' re going to build wonderful homes for the Palestinians someplace else. And the Palestinians will all volunteer to leave Gaza and go to those homes, and Gaza will be depopulated and new people will come in and live there. That was the plan. And when he was asked, standing there with King Abdullah from Jordan, under whose authority? Trump ' s response was :"Under my authority."And that went over like a lead balloon among America ' s allies in the Middle East. They all said,"We ' re not up for this. We ' re not resettling Palestinian populations. They won ' t actually leave voluntarily. This will would actually be a huge problem for Israel ' s own national security long-term. And we ' re going to come up with another plan. And our plan will keep the Palestinians on the ground in Gaza."At which point Trump pivots to work with the new plan. And his advisers say that Trump was only trying to start the conversation. And what he said was his plan is not actually what he ' s going to do. So, what ' s sustainable? What would be sustainable would be a lot of Gulf money going in and maybe some European money too, though they ' re going to be really pressed, given everything you and I just talked about, to try to reconstruct a completely devastated Gaza that eventually some two million-plus Palestinians will be able to live normal lives in, and the governance will be provided by some technocratic group of a Palestinian Authority that will be working with, selected by, those that are involved in the reconstruction and the security. Namely the Egyptians, the Jordanians, the Saudis and the Emiratis. That is the best possible deal that they ' re going to get. But that ' s happening as Israel has just evacuated forcibly another 40, 000 Palestinians from refugee camps in the West Bank, sending tanks in for the first time in decades and probably precipitating more Israeli settlers taking more land on the ground in the West Bank. And so, as we are possibly taking a small step towards a sustainable solution in Gaza, we are taking a slightly larger step away from a sustainable solution for the Palestinians in the West Bank, twas ever thus over the past decades in this incredibly horrible problem and conflict. HW : And why is Israel doing that? I mean, are they feeling emboldened by the US, or why the incursions in the West Bank now? IB : Well, I think they ' re feeling emboldened by their own successes. They have shown that they are the dominant military power in the Middle East, and they have the ability to determine the level of escalation unilaterally, and their adversaries and enemies can ' t really do any damage to them. They proved that after October 7, in their response to Hamas. They proved that in decapitating and taking out the military capabilities of Hezbollah. And they ' ve proved that also with overturning – they didn ' t do it, but the fact that the Assad regime in Syria is gone, which means that the Iranians no longer have a land bridge to get additional weapons to what had been the strongest adversary of Israel militarily, Hezbollah. So all of that has shown that Israel can decide outcomes. They ' re the ones that have all the power here. And also that depopulation of Gaza is very popular among Israelis. There was a poll recently in the Jerusalem Post, which is pretty mainstream in Israel. 80 percent of Israeli respondents said that they wanted to depopulate Gaza of all Palestinians. Taking more land is very popular, especially in the West Bank, especially among Netanyahu ' s present right-wing coalition, that he would like to keep together, to continue to govern. And this is a sop to them as he is looking to potentially extend the ceasefire in Gaza and move from phase one to phase two, which is facing some delays and open questions right now. So, yes, certainly, the United States has been underpinning that military capability of Israel. And I think it is instructive to remember that not only does the US provide more military aid in peacetime to any country in the world, to Israel, but that unlike in Ukraine, where the Trump administration is saying,"No, I ' m not happy with the grants, you ' ve got to pay that back,"The billions and billions that the United States sends to

Israel, nobody is suggesting that that should be alone or that the Americans should get technology rights or mineral rights or anything else from Israel, despite the fact that Trump ' s policy is America First. Israel is a very clear exception to America First in Trump ' s visioning of it. HW : Alright, so we talked about the fact that the foreign policy focus of America is shifting to Asia. So what are you hearing about how China is feeling in all of this? And what should we be paying attention to on that front? IB : China has had a couple of good weeks economically and specifically on the back of that huge announcement of DeepSeek, which has, you know, far less money behind it than the comparable American chatbots but is performing at an extremely high level. And, you know, given the fact that there are all of these export controls on semiconductors and other parts of advanced technology, that this is giving people more of a belief that the Chinese really are capable of driving innovation even in that space. And they ' re, of course, dominating the post-carbon energy space. We see that with electric vehicles and batteries. But we ' re also seeing that with massive investment now in green hydrogen, for example, we ' re seeing it in advanced nuclear capabilities, all of these things. And so there is more investment in that space that the Chinese are not only driving as a state, but that they ' re also increasingly able to raise and move in their own private sector. So the story that you and I have been discussing for the last couple of years is the Chinese economy is radically underperforming. That is still true, but there is an emerging counter-narrative here that I would be remiss not to mention. Now more broadly, what the Chinese here are seeing is that Trump ' s direct policies are going to hurt them economically. His tariffs that he ' s already announced on China, the 10 percent, on a whole bunch of Chinese goods for export, the reciprocal tariffs that are global, that will clearly hurt China and that they ' ll have a hard time negotiating out of, and also the willingness of the United States to expand their export controls, their sanctions on China, especially in areas that are considered relevant, even loosely relevant to American national security. All of that is going to constrain Chinese growth. All of that is going to negatively impact the Chinese economy. That ' s the downside. And the Americans are doing that not only in bilateral relations with China but also pressuring other countries to get Chinese tech out of their supply chain. And you see that with Americans pushing other countries that produce semiconductors to take the Chinese out. They ' re pushing American allies and trade partners to get China out of the export chain. So Mexico being pushed very hard by Trump to stop allowing China to pass goods through Mexico into the United States. India, same conversation. Vietnam, same conversation. A lot of that going on. So baseline, the US-China relationship is getting worse. But the Chinese see huge advantages in America ' s reversion to unilateralism that we haven ' t really seen since the original America First movement in the run up to World War II. Charles Lindbergh saying,"Don ' t get into the war, don ' t fight the Nazis. The US is far away. This isn ' t our our problem."And now you have the Americans not only doing that with Russia and Ukraine with the new America First, but you also see that happening with the Americans pulling out of the Paris Climate Accord, again, pulling out of the World Health Organization. You see it with the Americans shutting down USAID. And America has historically been responsible for about 40 percent of global humanitarian support and aid. And absolutely, some of that is corrupt. Absolutely, some of that is on woke programs that the vast majority of American taxpayers would never support. But the majority of that money is actually spent on programs that are really important in advancing American soft power and influence over countries in the Global South, all over the world. The Chinese know that. That ' s why they ' ve started humanitarian programs. It ' s not out of some great love for, you know, providing foreign aid for these countries. It ’ s more because they see it creates influence. Now the Americans are leaving a vacuum that the Chinese can absolutely take advantage of. And we ’ ll see this across, you know, sort of, microstates in the Pacific. We ’ ll see this across sub-Saharan Africa. We ’ ll see this across South America. The Chinese see huge opportunities from the United States creating a leadership vacuum. And the place they ’ re most capable of doing that is if the US stops paying dues in the United Nations, where China is number two, and they ’ ll suddenly have the most influence over all the key positions, which will give them a role in global governance that they ’ ve really been constrained from having over the past decades. HW : What do you think are the chances that the US stops paying its dues to the UN? IB : You know, if it were up to Congress and only Congress and Trump doesn ’ t lean on them, I ’ d say fairly low. I ’ d say that they ’ d probably, you know, cut back on some programs. But in terms of baseline dues, they ’ d keep paying it. But Trump personally, I think increasingly sees the UN as hostile to the United States. That ’ s particularly true in terms of Israel policy, which Trump is leaning in on. And, you know – let me put it this way. Tulsi Gabbard did not have the votes to get confirmed. There were Republican senators that were going to vote against her. And Elon called those individual senators and threatened them. And this was, you know, fully aligned with what Trump wanted Elon to do. And those votes went away. And the only person that ended up voting against Tulsi was McConnell. And as a consequence, it went through. So look, I think that we should not underestimate the level of power and control that Trump and Elon have over the Republican Party this time around, compared to 2017, where it was really the Republican Party that was much stronger than Trump, both in his administration creating a lot of guardrails, but also in Congress. This time around, Trump has all the chips and he ’ s using them. So if he decides he wants to stop paying dues, I think that ’ s what ’ s going to happen. This is a much more revolutionary presidency, domestically, where last time around it was much more transactional. HW : Alright. Let ’ s talk a little bit more about what ’ s actually happening inside the US. You bring up Elon, who obviously with DOGE has been sending out emails and asking for information from people and kind of trying to do a lot very quickly. Some of these high-profile moves are – I think what ’ s interesting is that no one would really argue that there isn ’ t inefficiency in government, that that ’ s an understood reality. But the way that some of these moves are being made are maybe going to cause irreparable damage. And that ’ s to people from red states, that ’ s to scientists, that ’ s to veterans, that ’ s to people who voted for Trump and that ’ s to people around the world. You talk about soft power and the United States, but what about actually what ’ s happening inside the United States? And why do you think that the administration is willing to go so quickly, to go so fast and to cause such damage? IB : Well, and you saw Trump this weekend saying that he wants Elon to move faster, to be more aggressive. And I don ’ t think that was just a troll. I think that ’ s true. Trump is 78 years old, and with his friends he talks about the fact that he doesn ’ t have much time. That in two years ’ time, the Democrats could take the House in four years ’ time, you know, that ’ s it. He ’ s not talking privately of”I ’ m going to be president for life.”He also knows he was almost killed. He was shot in the head. And do I think that changes somebody? Absolutely. I think that he believes that he was saved by God, and that his level of confidence that what he is doing is fundamentally right and necessary now is so much higher than the insecurity that I frequently saw in him in his first presidency. So I think his willingness to push, push, push, and his view that Elon is a great activator and executor on that intention, and someone that he had nowhere close to someone with that kind of capability and energy and execution ability in his first term. And now he does. So first of all, I think that relationship between Trump and Elon isn ' t going anywhere. I think it ' s actually very stable. People that say it is are people that want it to go away. But hope is not a strategy. And I also think that the intention of revolutionary politics domestically is to shoot first and ask questions later. It ' s to break things. It ' s a view that the bureaucracy has been weaponized against Trump first and foremost, and also against the will of the American people. And therefore it must be broken. And you ' re going to break eggs when you want to make that omelet. Now your point is that there are a lot of Republicans that work in government. There are a lot of Republicans and Trump voters that are losing their jobs and that are not being treated with a level of care and compassion as a result of what Trump and Elon are doing. And I think there will be a backlash. But I ' m not so sure that a one-term, 78-year-old-Trump presidency cares all that much about that. And, you know, if that means that his popularity slips to like, the high 30s, frankly, you know, in a very, very polarized country, high 30s isn ' t that bad. It still means 80, 90 percent of the people that voted for him still support him, and I think he ' s probably more comfortable with that this time around. So I think a lot more has to happen before we can start talking about whether there ' s really going to be significant pushback against what Trump is trying to accomplish at home. HW : Do you expect to see any of Congress standing up to Trump? We saw the governor of Maine recently kind of take him on. Do you expect to see more of that, or do you feel like for the next two years at least, people will just go where Trump takes them? IB : Well, sure, but, you know, in the case of Maine, that ' s a Democrat. So I don ' t know how much that means. I don ' t expect to see a lot of Republicans standing up against him. And of course, since the Republicans control the House and the Senate, that ' s the only thing that really matters. I mean, you know, you saw Kash Patel got through, Pete Hegseth got through. Now look, I actually think that a lot of the members of cabinet are very capable. I mean, they ' ve been picked for their loyalty and they will be loyal to Trump, but they ' re very capable. I think JD Vance is very capable. Mike Waltz, I know quite well. Marco Rubio, over the years. These are people that are very well respected, have been by their colleagues and could have served under any Republican administration. But there are some people that are completely incapable for the roles that they have been selected, completely incapable. They would not be selected in a role in any cabinet, in any democracy. And yet here they are, serving in critical roles in the most powerful country in the world. And yes, I ' m talking about the Secretary of Defense. I ' m talking about the Secretary of Health and Human Services. I ' m talking about the Director of National Intelligence. I ' m talking about the Director of FBI. These are important roles. And they were all confirmed. And if you ' re a Republican senator – and senator – I ' m not talking about the House here where, you know, it ' s a lot more diffuse and there are a lot more wackos and it ' s easier to get in. But Senate has always been like, responsible and more oriented towards bipartisanship and the rest, and they are utterly petrified of what it means if they publicly get crosswise with President Trump and number two, Elon. And they ' re not willing to do it. So I think that there ' s very little belief that you are going to see Congress step up. I think that you will see justices step up because those justices are independent. The problem is that many of those justices are progressive and have a known progressive lean. And so when cases come up and justices that have that political lean oppose something Trump wants, his willingness, his administration ' s willingness to attack them and refuse to execute on that judicial decision, unconstitutionally, is real. Now the case will of course go then up to the Supreme Court eventually. And I don ' t feel the same way about Trump refusing to listen to adhere to the rulings of the Supreme Court. But if there ’ s 100 rulings like that and you overwhelm the judicial process, you quickly see how you might erode a core check on executive power in the United States. And Trump and Elon clearly intend to every day test those checks and balances and break them if they can. I mean, that is the intent, that is the intent. HW : So you mentioned democracy. When we last spoke, you said very clearly that the US is not Hungary, meaning that democracy is not threatened and that the US is not in danger of turning into an autocracy. Do you stand by that? IB : Yes, I do. I mean, in the sense that I think in two years we ’ re going to have midterm elections, and they will be largely free and fair. And in four years we ’ ll have presidential elections. And US elections are held at the federal level, right? They ’ re held state by state. They make the rules. It ’ s very easy to make people believe that the election has been stolen. It ’ s extremely hard to actually rig an American election. And I don ’ t believe that that is very likely at all. It ’ s very unlikely. The American military, despite the firings that we ’ ve seen just last Friday, is still an independent and professional military that I believe will carry out its orders and its duties professionally. I see the judiciary in the United States as independent and a clear check on the executive power in the United States. Where I see a problem, is that the US is becoming more kleptocratic every day. And the US was already much more of a kleptocracy over the past 10 years under Democrats and Republicans than any other advanced industrial democracy in the world. Clearly, money buys power in America in a way that it doesn ’ t in any other major democracy, wealthy democracy. That is getting dramatically worse. I saw specifically that Elon Musk, who is a walking conflict of interest at the highest level, met with Narendra Modi, the Prime Minister of India, in Blair House, so part of the official White House complex, right after Trump met with him. And Trump was asked later in the day, was Modi meeting with Elon in Elon ’ s private-sector capacity to advance his business interests or his official capacity to advance the policies of the United States? And Trump answered honestly. He said,”I don ’ t know.”And how could he know, right? Because, I mean, those two things are completely intertwined. But that is not remotely acceptable for rule of law in a functional democracy, and yet nothing is being done about it. In fact, it ’ s getting more and more entrenched every day. So the single place where the United States was least functional as an advanced democracy is now getting far, far worse. And I don ’ t see any checks and balances on that. I think that ’ s a very, very serious problem. HW : Alright, Ian, thank you, as always for all of your insights. Let ’ s close out with a couple more questions from our community. And this one indeed relates to the kleptocratic conversation that you were just raising. So why are the billionaires enabling fascism? What do they get out of it? IB : Fascism is a very politically loaded argument. That ’ s a political system where the cruelty against a subgroup or a perceived subgroup in the society is the intention, the expressed intention of the policy. And I certainly wouldn ’ t define the United States as a fascist system today. But I do believe that billionaires in the United States, at least a significant subset of them, are really under-interested in the well-being of their fellow countrymen and women. And this is a country that really makes heroic the individual. And it undermines the community. We ’ ve seen a significant erosion of civic engagement in the United States and civic associations. The church is not what it was in the US. Fewer people go, fewer people trust the church. Public education is not what it was. And wealthy people increasingly opt out. The family is not what it was. It ’ s increasingly atomized and people spend much more, much more of their time being, you know, isolated and atomized and intermediated by algorithms. And you and I have spoken about that problem in the past. We are increasingly a society of individuals as opposed to a nation together. I will tell you that that is a real problem when you have large numbers of billionaires that believe that they built it themselves, that they ’ re not responsible to a collective whole for any of their success and therefore they ’ re not accountable for their fellow citizens. And that ’ s what I think we have. I think we have a lot of winners in the United States and not a lot of leaders. And when you have winners, you have losers. When you have winners, you don ’ t have a collectivity. I will say that in this anti-DEI thing, I have a lot of sympathy for the people that don ’ t like DEI. I thought it was rammed down the throats of a lot of people And I ’ ve said this before, it was well beyond what the average American was willing to tolerate, and also it came with a level of high-handedness that that basically said, if you ’ re not with us in supporting these programs, you ’ re evil, you ’ re stupid, you ’ re uneducated. That ’ s unacceptable, too. It was a completely, you know, sort of oppositional way to try to have what ’ s a very challenging conversation. But I ’ ve never been a champion of diversity for diversity ’ s sake. What ’ s amazing about the history of the United States is despite all of our diversity, we find commonalities with each other. We find connectedness. That ’ s what ’ s amazing, is the connectedness, is that, you know, on this world, we ’ ve got eight billion people. And despite how different we all are, we all actually connect as human beings. That is what we need to focus on, not the diversity, the connectedness. And I think that ’ s what the billionaires today in the United States are losing. They ’ re doing incredibly well themselves. They ’ ve got theirs, they ’ ve got access to power, they ’ re paying good money for it, they ’ re going to get even more. But they ’ re forgetting that for us to succeed as a society and as a planet, we have to be connected to each other. You can ’ t throw out diversity and forget connectedness, and that ’ s what we ’ re forgetting right now. That ’ s why the country feels so much trauma and pain right now. And I think why the United States has given up on a lot of its really aspirational values that made such a difference to the world, that almost lost World War II for many generations. And now we ’ re giving it all away. HW : I mean, I think that ’ s why you find that diversity and inclusion go hand in hand, that that ’ s actually speaking to that. But in some of the moves that are being made to kind of throw out any type of diversity, to **scan** documents, to find words in order to highlight programs that are wasteful, that we ’ re actually going to lose what many, many studies actually highlight as being the benefits of diversity and the fact that you need diversity in order to be actually successful. IB : Shooting first and asking questions later, you know, moving fast and breaking things, those things make sense in a turbocharged venture capital technology environment. They do not work in government. They don ’ t work in government at all. In government, we have something called checks and balances, and we need them. And moving fast and breaking things undermines those checks and balances. So we ’ re going to do a lot of damage unintentionally. A lot of people are going to get hurt. A lot of folks don ’ t care about that in power right now, but they should because it would be much, much better. I know people want to move fast, but actually to think a little bit, do a little bit of your own research before you decide to kill that program, before you decide to fire that person. We ’ d be in a lot better shape as a government right now. HW : OK, the final question that I have for you actually comes from my 10-year-old son, and it ’ s a bit of a miserable question. We have a very lovely house, I promise. But his question nonetheless is are we heading into World War III? So, Ian, what should I tell him? IB : No, no, I don ’ t think we ’ re heading into World War III but we are absolutely heading into a world of much greater conflict. I think it ’ s much more likely, for example, in the next five, 10 years, that we ’ ll have a nuclear weapon go off. And I thought that was pretty unlikely since the Cuban Missile

Crisis. In other words, over the course of my life. I think we ’ re going to see a lot more terrorism at scale, because a lot more people are going to feel angry and disaffected, I think. You know, you saw that assassination of the CEO of UnitedHealthcare just a few dozen blocks from my house. I think you ’ re going to see more things like that. And so, no, not World War III because the fact is that outside of the US and China, all the other countries know that they need to work with everybody. And there are even forces inside US-China that are trying to make sure that you don ’ t throw the baby out with the bathwater. But underneath that World War III risk, there ’ s an awful lot of damage that ’ s going to get done, an awful lot of crisis that I think we ’ re going to have to live through before we start to see what we create on the other side. A lot of defense as well, that we ’ re going to be playing in this environment. HW : Well, Ian, it is always a pleasure to talk to you, and we ’ re so glad that you are **scanning** the world to understand what is happening and then you come here and explain it to us all. Couldn ’ t be more grateful and thank you for everything. IB : Thank you.

10 NEG Dim 5

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HW : Hi everyone, I am Helen Walters. I am head of Media and Curation here at TED. Welcome to another episode of TED Explains the World with the one and only, Ian Bremmer. It is Monday, February 24, a little more than a month since President Trump was inaugurated once more, and safe to say, a lot has been going on. So we figured we ’ d check in with Ian to determine what we should really be paying attention to amid the extreme noise. Ian, hi. Ian Bremmer : Helen, great to be back with you. HW : So we asked the TED community to share their questions for you, and we wanted them to share what they ’ re most curious to know. And we really got some amazing questions that I plan to pepper this conversation with. So let ’ s start with one right now. Two months into 2025, where does the US stand? IB : Very powerfully, in the sense that the US economy, of course, is performing considerably better than any of the other G7 economies coming still out of the pandemic. Technologically, only close competitor is China, ahead of the US in some areas, behind the US and other critical areas. But compared to every other country in the world, the US is head and shoulders and neck or waist above them. Militarily, of course, the US is the only country with global ability to project power. No one else is close. And the US dollar is still the global reserve currency, no one is approximate challenger. I say all of those things because those are the things that haven ’ t really changed over the course of the last few months, but I suspect that isn ’ t what the questioner was asking. What they were really asking is what ’ s happening politically. And politically, the United States is unwinding its own global order. It is no longer particularly interested in promoting collective security or NATO. It ’ s no longer really interested in promoting involvement and leadership of multilateral institutions, of consistent rule of law and free trade that ’ s well regulated, certainly not very interested in promoting democracy around the world. I mean, this is an environment where other countries around the world have to figure out how to adapt to the United States, and that the US has become the principal driver of geopolitical risk and uncertainty in the world today, unlike any other time in my lifetime and yours, that ’ s perhaps the biggest

change. HW : That is quite a statement. Alright, you are a geopolitical expert. Let ’ s talk about Europe. So Defense Secretary Pete Hegseth stated that America ’ s foreign policy focus no longer lies in Europe. Vice President JD Vance caused some raised eyebrows, dropped jaws and, I suspect, some choice expletives after his recent speech at the Munich Security Conference, in which he accused European leaders of suppressing free speech and warned of the threat from within. So you were in Munich. Was it as dramatic as the stories we read had it? And what happens next? IB : It was. I was in the room. I was maybe 30 feet away from him when he gave that speech. Standing right in the front, right next to me was the president of Czechia, the president of Finland, the prime minister of Sweden, watching all of them and their reactions. What he said, let ’ s keep in mind, this is the Munich Security Conference. It ’ s been going on now for something like 61 years. And he was the head of the US delegation. And every year the head of the US delegation gives a big speech on the state of the transatlantic alliance and on global security. And he didn ’ t do that. I mean, the people in the audience were expecting a challenging speech. They were expecting that the Americans would be less committed to the alignment with the Europeans on Ukraine. The US had just had that direct Trump phone call with Vladimir Putin, 90 minutes long. Hadn ’ t coordinated that with the Europeans, never mind the Ukrainians, in advance. So, I mean, they were definitely prepared for a very challenging plenary. But Vance did none of that. Vance said,”I ’ m not going to talk about Ukraine or Russia or China, because the biggest problem is actually what ’ s happening inside Europe. The biggest problem, essentially, are you guys sitting in front of me, that I ’ m speaking to, because you ’ re suppressing free speech. You ’ re suppressing the far right. You ’ ve been infected by the woke mind virus, and you aren ’ t real democracies as a consequence.”And specifically he said, and this is something that I think is underappreciated in the United States, he attacked the German firewall. And said that – And let me explain what the German firewall is. That is a principle by the leaders of all of the mainstream German parties and their supporters that they will not, under any circumstances, work with, enter into coalition with the Alternative for Deutschland party, the AfD. Because the AfD, under surveillance from German intelligence agencies, is considered to be a neo-Nazi party. And the United States, of course, after World War II, led de-nazification of Germany. The day before, Vance had actually gone to Dachau and visited a concentration camp, and then came to speak in Germany about the firewall needing to be ended. And at that point someone yelled out from the crowd,”This is unacceptable!”And it was right in the front. And people in the room, about 1, 500 people in the room, standing room only, couldn ’ t see who it was. They ’ d just see the back of his head. I saw who it was. It was the German Defense Minister, Pistorius. And in 15 years of me going to the Munich Security Conference, I have never seen anything remotely like that. And the reason I mention all of this, and of course, on the back of that, let me be clear, Vance refused to meet with the German chancellor because he ’ s basically, well, he ’ s not relevant. He ’ s only going to be there for a couple more months until a new government, so why should I bother? But does meet with the leader of this AfD, goes and meets with her directly, does a bilateral that evening. So that is a very important backdrop for this last weekend ’ s German elections, where the victor, Friedrich Mertz, who is going to be the next chancellor, actually referred specifically to America ’ s intervention in German democracy and support of the AfD being every bit as bad as what the Kremlin is doing inside Germany, and also said that the Europeans need to develop a defense policy which is independent of the United States, in other words, essentially saying that it will be the end of NATO. Again, staggering comments, unheard of, impossible to imagine even two weeks ago. And that ’ s why it ’ s very critical that you have the context of what happened between Vance, Elon Musk over the past couple of months, Trump, with the Germans, with the Europeans. So you can understand why the incoming German chancellor would make a statement like that. HW : What should we make of all of the conversation about the rise of Nazism, the return of Nazism? Obviously, the AfD just won 20 percent in the German election. We ’ ve also seen people allegedly giving Nazi salutes at various US conventions and speeches. Do you think this is overblown? Is this a fear that people have rationally? And what should we make of that? IB : In Germany, the Alternative for Deutschland, are winning all of former East Germany. This is a group that increasingly understands that they ’ re never going to really catch up to the rest of Germany. Remember, Germany hasn ’ t really grown in five years now. They ’ ve been in recession for the last two. And the average citizen, the average voter, living in former East Germany no longer are just angry about not catching up. They ’ re basically saying,”I ’ m done with this system.”So that ’ s why you ’ re getting this very strident, revanchist nationalism that is, you know, it ’ s not just taking a hard line on illegal immigrants. The AfD even supports removing citizens of Germany that they claim have not adequately integrated into German culture and nationality. And so, I mean, this is some very, very strong stuff, stuff that really does feel like what they were doing back in Nazi Germany. But in former East Germany, they win. They get the largest number of votes, where, in much of the wealthier part of West Germany, including, you know, where I was, in Munich, the AfD can ’ t break out of single digits. So it ’ s a very, very divided country. And also the AfD is doing very, very well among young men. There ’ s a clear gender divide here as well, just as there is in the United States. In the US, I mean, I, of course, saw Elon Musk and that, you know, sort of grabbing his heart, saying,”My heart goes out to you”and then doing what appeared to be mimicking a Nazi salute. And then Steve Bannon doing the same without the initial heartfelt comment. It clearly was trolling. Clearly was trying to get a reaction from folks. It ’ s very performative. And it ’ s meant to also, you know, attack and attach, make the left lose their minds on something that isn ’ t very critical to what the Trump administration is trying to do compared to the revolutionary changes inside the US government that they are very much prioritizing right now. So I ’ m very concerned about the rise of neo-Nazism in Germany. I am very much not concerned about that in the US. I ’ m concerned about other things I ’ m thinking is being used as gaslighting for those that can use distractions or can be distracted by them. And I would also say that this election in Germany wasn ’ t surprising. It was exactly as the polls have expected for several months now. But the key elections in Germany and the key elections in Europe are the next cycle. I ’ ve spoken with a lot of people around the Trump administration that believe that the Europeans are one electoral cycle behind the United States. So in other words, Trumpism is coming, just not quite yet. And so in the UK, you ’ ve got Labour for a full term. But the Reform Party is increasingly the most popular. And just with a little nudge, maybe a little external money, and Elon said he ’ s already thinking about giving them 100 million dollars, which is a massive amount in UK politics, not so much in the US, that the Conservatives would fragment, a rump group would join with Reform, and they could win the next elections. In France, you ’ ve had all of these governments continue to collapse. Macron, very unpopular. Marine Le Pen and the National Rally party could easily win upcoming elections in 2027. In 2029, in Germany, if the Germans are incapable of turning their economy around, incapable of unwinding their debt break and spending some of the capital that ’ s just sitting there and as their debt to GDP goes down, but their economy contracts at the same time, then AfD could easily win in 2029, and then the firewall is no more. And then, of course, in the European Union elections overall, in five years ’ time, now four and a half, you could easily see the Patriots front, which is the equivalent to the far right grouping that these parties are a part of, they could all come together and be the dominant party. And then the EU as we know it is really a thing of the past. So these are existential changes that are happening not just for NATO right now, but also for the EU and quite plausibly, for the future of democracy as we know it. HW : Alright, let ’ s turn to Ukraine. So President Trump accused President Zelenskyy of starting the war with Russia and called him”a dictator without elections.”Last week, senior American and Russian officials met in Riyadh to discuss a potential cease-fire with no Ukrainians present. Now Zelenskyy has offered to step down in return for Ukrainian admission to NATO, although apparently he doesn ’ t really mean this. So what happens to Ukraine now? What should we be paying attention to? IB : Well, I mean, I think he would mean it if NATO membership happened. I ’ d love to see Trump call his bluff on that. But of course it ’ s not going to. And he wasn ’ t saying that six months or a year ago. He ’ s saying it now that NATO is truly being pulled off the table. I think it ’ s the right thing for him to do. It is, again, performative and helps to remind those around the world that Ukraine is fighting for self-determination. They ’ re fighting for the ability to join their own alliance, as any country in the world should be able to do. But that is not Trump ’ s belief. Trump believes that what you get to do is determined by how powerful you are, not by the fact that you happen to run a country. And indeed, the territorial integrity is something that is afforded to powerful countries, but not so much for weak countries. And that ’ s the way he feels about Panama, and that ’ s the way he feels about Denmark and Greenland. And that ’ s certainly the way he feels about Ukraine. So right now, as you and I are speaking, the United States is putting forward a new UN resolution at the General Assembly that says that the war has to end as fast as possible but does not recognize Ukraine ’ s territorial integrity, which has been a core precept of the international order that the United States has supported since creating the United Nations and since the end of World War II in 1945. The Americans are now throwing that out. And the Russians, of course, are fully supportive of that. America ’ s allies in Europe are not. But the US has gotten the Saudis to agree, and they ’ re getting a unified Arab block. The Americans have privately worked with a lot of poorer countries in Africa, in the Asia Pacific and others, smaller countries, and saying,”You ’ re not going to get any aid from the US unless you vote in favor of this new resolution.”I fully expect the resolution is going to pass. But of course, what that means is that the Americans are preparing to cut a deal on Ukraine over the heads of the Ukrainians. In other words, the US and Russia together will figure out what a ceasefire should and should not entail as part of a broader US-Russia rapprochement. This is obviously a disaster for the Ukrainians and the Ukrainians who initially refuse to accept this so-called deal, on Ukrainian natural resources that would grant the Americans $500 billion in access to such resources in return for”paying off”aid that the Americans had granted to the Ukrainians without conditions or strings, but apparently no longer. And so, you know, what does this all mean? Well it means it ’ s yet another case of why people shouldn ’ t trust the United States from one administration to the next, because foreign policy will change completely. And what you thought was an American commitment won ’ t stand. It shows that there is a fundamental rift between the Americans and the Europeans and indeed, Emmanuel Macron in the United States right now, as you and I speak, hoping that he can do something that will keep Trump from, as Macron believes, essentially surrendering to, conceding to, the Russian position and maintaining some level of coordination with the Europeans. But he doesn ’ t have a lot to offer. And the Ukrainians, you know, now in a position of desperation where maybe they will end up signing over a lot of their natural resources to the Americans. But what, if anything, will they get in return for that? And how empowered will the Russians be in any deal that is cut? And will an independent Ukraine even be able to survive, and what will the terms that will be required of them? Of course, Putin is saying things like, no European troops of any sort, whether in a NATO formulation or not, would be allowed as peacekeepers in Ukraine. Well, then, how would you guarantee, what kind of security commitments would you give to the Ukrainians? These are all open questions, but apparently questions that President Trump is prepared to largely resolve on Russia ’ s terms. And again, this is such a dramatic change from where the United States and where the Western order was just a month or two ago. HW : So let ’ s take some quick-fire questions from our community related to this topic. And I think the first one is particularly pertinent to what you just said, so it ’ s simple. What does Putin have over Trump? IB : I don ’ t think that Putin”has”anything over Trump. If he did, then why wouldn ’ t Trump have granted Russia more in his first term? In his first term he gave the Ukrainians those javelin anti-tank missiles that Obama refused to give. And, you know, he thought they were too risky. He increased sanctions against Russia. Now that was his administration with a lot of hawkish in orientation advisors around him that he doesn ’ t have this time around. They ’ re all directly loyal to him. But, I mean, if it was a priority for him, he could have stopped it. And if the Russians had something over him, then why wouldn ’ t they have used it in that case? It feels like a conspiracy theory. I don ’ t buy it. I think very differently. We can explain what ’ s happening because number one, Trump wants to end wars. It ’ s very popular with his base. He ’ s said this on Gaza. He ’ s said this on Iran, where Israel wants him to support attacks on Iran ’ s nuclear capabilities, and Trump has said no. And he ’ s showing this on Ukraine. He doesn ’ t want to spend money on the Ukrainians going forward after spending over 100 billion dollars under the last three years from the Biden administration. He doesn ’ t want to spend that money going forward. And he also doesn ’ t value a strong Europe. He actually thinks that a weak Europe is in America ’ s interests, where you have more Brexits. He used to always talk to the French about that during his first term,”Why won ’ t you do Brexit?”He wants all of these MAGA-type populist parties, including the AfD, to win in Europe because that legitimizes his own popularity. Just like he loves Milei in Argentina or Bukele in El Salvador. And so I think if you put those things together, you actually get to why Trump is doing what he ’ s doing with the Russians without needing to create a story about Trump somehow being threatened by some secret, shadowy information that the Russians have on Trump that they ’ ve let him know about but nobody else knows. I just don ’ t buy that. HW : Great. OK, another community question. If the US hands Ukraine to Russia, will Russia keep going? Will they attempt to take over Europe? IB : It ’ s interesting that Trump just met with the Polish president, President Duda. It was supposed to be an hour meeting, it was only 10 minutes. So he kind of embarrassed him at home in Poland. But he did say that the US is still committed to maintaining a troop presence on the ground in Poland. And so it ’ s very hard to imagine the Russians rolling through Ukraine and then into Poland, even though there ’ s been lots of asymmetric warfare, for example, and even there have been, you know, Russian missiles launched into Lviv, Ukrainian air defense, missile defense, and with Polish villagers getting killed because of the fighting. So, I mean, there ’ s been spillover. But America is still going to be in Poland. Why is that? They get along pretty well, and the Poles are heading towards five percent of GDP defense spend this year, which has been the high water mark of American demands of defense spend. So he ’ s, you know, been consistent on that over the past months. Baltic states. I mean, they ’ re all spending a lot more money on their defense. And so, I mean, in principle, you could imagine the United States maintaining these rotational deployments in the Baltics, which would prevent the Russians from invading. But might Trump say, as part of a deal,”Why do we have all those troops there?”You mentioned Pete Hegseth saying the Americans need to get troops out of Europe and get them to Asia, which is where their principal competitive, you know, sort of, strategy is going to be oriented. And the US hasn ’ t been pressing the Japanese or the Indians, the principal partners, allies in Asia vis-à-vis China, the way they ’ ve been pressing the French, the Brits, the Germans in Europe. So I think it is certainly plausible that the Russians will do more, especially with those countries that refuse to align with a Trump rapprochement towards Russia. The countries I would be most worried about are Moldova, are Georgia. I ’ d be most worried about the countries that are in the so-called”near abroad”that aren ’ t a part of NATO, and aren ’ t going to become a part of NATO, and therefore are much more vulnerable. But certainly in terms of Russia ’ s willingness to engage in espionage, to fund actors on the ground for arson attacks, assassinations, critical infrastructure attacks on fiber networks, on pipelines, I think all of those things are likely to increase against European states across the board, on the back of this US-Russia rapprochement. HW : OK, final community question for this segment. What happens if the US no longer supports NATO? IB : It ’ s possible that the United States is moving towards a similar posture in Europe that they presently have in Asia, which is strong individual bilateral deals with a number of countries : South Korea, Japan, Australia, but not a collective security agreement, where those countries together have more of a call on the United States and there ’ s a lot more free riding. So I could imagine that NATO falls apart and becomes that. The danger is that even though the Europeans now understand the nature of the threat, that it is very plausible and maybe even baseline, that the Americans are going to leave Europe collectively on its own. That that seriousness does not equate to the ability to increase spending adequately in the near term. The Germans did manage to pull together an electoral outcome that will allow for a so-called grand coalition, just barely. So you ’ ll have a two-party government in all likelihood instead of a three. But you also have a blocking capacity on the part of the far right and the far left that will prevent the Germans from really **blowing** out their spending, including their defense spending, unless they can pull something off before that new government is formed. Which is possible, but it ’ s unlikely to be very big. The Brits are now talking about maybe moving the 2. 5 percent of GDP defense spend over years with a very challenging fiscal environment. The Spaniards, not even close, the Italians, not even close. So what does a European military capability really look like? And without the Americans, the answer is it ’ s not there. It ’ s not there. So I think that if NATO falls apart, the reality is that most of Europe will be incredibly vulnerable to external attack, especially Russian attack, And there won ’ t be a way for them to defend themselves, and some of them will break and therefore want to work more directly and individually with the United States. And that could be the end, not only of NATO, but it could be the end of the European Union. I think this is, the stakes are incredibly high, this is an existential challenge for European governments that needed to take this seriously 30 years ago, 20 years ago, certainly in 2014, when the Russians invaded Ukraine, certainly, certainly in 2016 when Trump was elected, and certainly, certainly, certainly in 2022 when the Russians then invaded all of Ukraine. And at every point the Europeans were basically saying,”We ’ re still fine with the Americans basically doing this.”And now they are in very serious trouble. HW : No time like the present, I suppose. Alright, let ’ s move on. So I mentioned the discussions in Saudi and obviously also on the agenda there with the discussions about Gaza and Trump ’ s proposal to build a Gaza Riviera. What should we make of that? Or as one of our community members asked, what is the actual solution for Gaza? And surely it isn ’ t what the US is proposing. IB : Well, look, here ’ s what ’ s interesting. You know, Trump says a lot of things, and when they don ’ t work [out], he backs out fairly quickly. And you know, I thought – just stick with Ukraine for one second on this. It was interesting to me that Trump has said that he doesn ’ t want to talk to Zelenskyy because he has no cards, he has no leverage. And so why should he let him at the table? And so Trump ’ s going to cut the deal. Very interesting that if that is true, then why is it that the Secretary of Treasury brings this deal and says,”You ’ ve got to sign this right now, Zelenskyy”And Zelenskyy refuses. And then a few days later, the Americans come back with a new deal with altered terms that are not quite as predatory vis-à-vis Ukraine. Why would the Americans change their tune for someone that has no leverage? Because, I don ’ t think it ’ s because Trump is a nice guy. He ’ s not going soft at 78. I think it ’ s because he overstated how little leverage Ukraine has. Ukraine is very popular among the GOP in the House and Senate, far more than Putin. Zelenskyy is far more popular in the United States among the voting population than Putin. And of course, he ’ s also much more popular among the Europeans and even globally than Putin. And so it turns out that maybe there is a little bit of leverage that Zelenskyy has, and maybe Trump does have to give a little bit more. So this is relevant in the context of Gaza, because Trump was very annoyed that, you know, despite his ability to get a cease fire between Hamas and Israel over the goal line, that his friends in the Middle East were not coming up with a plan for Gaza. Where they were supposed to provide security and provide investment and lead reconstruction and deal with the Palestinian populations in the interim. And that wasn ’ t happening. And so Trump then meeting with Prime Minister Netanyahu from Israel in Washington, then after that, summoning the Jordanian king, who was the ally that is most reliant on the United States and so, therefore, most needing to say yes to whatever Trump orders of him. Basically says, as you mentioned, Helen, that Gaza is going to be a Riviera, that the US is going to turn this into an incredible place. They ’ re going to build wonderful homes for the Palestinians someplace else. And the Palestinians will all volunteer to leave Gaza and go to those homes, and Gaza will be depopulated and new people will come in and live there. That was the plan. And when he was asked, standing there with King Abdullah from Jordan, under whose authority? Trump ’ s response was :”Under my authority.”And that went over like a lead balloon among America ’ s allies in the Middle East. They all said,”We ’ re not up for this. We ’ re not resettling Palestinian populations. They won ’ t actually leave voluntarily. This will would actually be a huge problem for Israel ’ s own national security long-term. And we ’ re going to come up with another plan. And our plan will keep the Palestinians on the ground in Gaza.”At which point Trump pivots to work with the new plan. And his advisers say that Trump was only trying to start the conversation. And what he said was his plan is not actually what he ’ s going to do. So, what ’ s sustainable? What would be sustainable would be a lot of Gulf money going in and maybe some European money too, though they ’ re going to be really pressed, given everything you and I just talked about, to try to reconstruct a completely devastated Gaza that eventually some two million-plus Palestinians will be able to live normal lives in, and the governance will be provided by some technocratic group of a Palestinian Authority that will be working with, selected by, those that are involved in the reconstruction and the security. Namely the Egyptians, the Jordanians, the Saudis and the Emiratis. That is the best possible deal that they ’ re going to get. But that ’ s happening as Israel has just evacuated forcibly another 40, 000 Palestinians from refugee camps in the West Bank, sending tanks in for the first time in decades and probably precipitating more Israeli settlers taking more land on the ground in the West Bank. And so, as we are possibly taking a small step towards a sustainable solution in Gaza, we are taking a slightly larger step away from a sustainable solution for the Palestinians in the West Bank, twas ever thus over the past decades in this incredibly horrible problem and conflict. HW : And why is Israel doing that? I mean, are they feeling emboldened by the US, or why the incursions in the West Bank now? IB : Well, I think they ’ re feeling emboldened by their own successes. They have shown that they are the dominant military power in the Middle East, and they have the ability to determine the level of escalation unilaterally, and their adversaries and enemies can ’ t really do any damage to them. They proved that after October 7, in their response to Hamas. They proved that in decapitating and taking out the military capabilities of Hezbollah. And they ’ ve proved that also with overturning – they didn ’ t do it, but the fact that the Assad regime in Syria is gone, which means that the Iranians no longer have a land bridge to get additional weapons to what had been the strongest adversary of Israel militarily, Hezbollah. So all of that has shown that Israel can decide outcomes. They ’ re the ones that have all the power here. And also that depopulation of Gaza is very popular among Israelis. There was a poll recently in the Jerusalem Post, which is pretty mainstream in Israel. 80 percent of Israeli respondents said that they wanted to depopulate Gaza of all Palestinians. Taking more land is very popular, especially in the West Bank, especially among Netanyahu ’ s present right-wing coalition, that he would like to keep together, to continue to govern. And this is a sop to them as he is looking to potentially extend the ceasefire in Gaza and move from phase one to phase two, which is facing some delays and open questions right now. So, yes, certainly, the United States has been underpinning that military capability of Israel. And I think it is instructive to remember that not only does the US provide more military aid in peacetime to any country in the world, to Israel, but that unlike in Ukraine, where the Trump administration is saying,”No, I ’ m not happy with the grants, you ’ ve got to pay that back,”The billions and billions that the United States sends to Israel, nobody is suggesting that that should be alone or that the Americans should get technology rights or mineral rights or anything else from Israel, despite the fact that Trump ’ s policy is America First. Israel is a very clear exception to America First in Trump ’ s visioning of it. HW : Alright, so we talked about the fact that the foreign policy focus of America is shifting to Asia. So what are you hearing about how China is feeling in all of this? And what should we be paying attention to on that front? IB : China has had a couple of good weeks economically and specifically on the back of that huge announcement of DeepSeek, which has, you know, far less money behind it than the comparable American chatbots but is performing at an extremely high level. And, you know, given the fact that there are all of these export controls on semiconductors and other parts of advanced technology, that this is giving people more of a belief that the Chinese really are capable of driving innovation even in that space. And they ’ re, of course, dominating the post-carbon energy space. We see that with electric vehicles and batteries. But we ’ re also seeing that with massive investment now in green hydrogen, for example, we ’ re seeing it in advanced nuclear capabilities, all of these things. And so there is more investment in that space that the Chinese are not only driving as a state, but that they ’ re also increasingly able to raise and move in their own private sector. So the story that you and I have been discussing for the last couple of years is the Chinese economy is radically underperforming. That is still true, but there is an emerging counter-narrative here that I would be remiss not to mention. Now more broadly, what the Chinese here are seeing is that Trump ' s direct policies are going to hurt them economically. His tariffs that he ' s already announced on China, the 10 percent, on a whole bunch of Chinese goods for export, the reciprocal tariffs that are global, that will clearly hurt China and that they ' ll have a hard time negotiating out of, and also the willingness of the United States to expand their export controls, their sanctions on China, especially in areas that are considered relevant, even loosely relevant to American national security. All of that is going to constrain Chinese growth. All of that is going to negatively impact the Chinese economy. That ' s the downside. And the Americans are doing that not only in bilateral relations with China but also pressuring other countries to get Chinese tech out of their supply chain. And you see that with Americans pushing other countries that produce semiconductors to take the Chinese out. They ' re pushing American allies and trade partners to get China out of the export chain. So Mexico being pushed very hard by Trump to stop allowing China to pass goods through Mexico into the United States. India, same conversation. Vietnam, same conversation. A lot of that going on. So baseline, the US-China relationship is getting worse. But the Chinese see huge advantages in America ' s reversion to unilateralism that we haven ' t really seen since the original America First movement in the run up to World War II. Charles Lindbergh saying,"Don ' t get into the war, don ' t fight the Nazis. The US is far away. This isn ' t our our problem."And now you have the Americans not only doing that with Russia and Ukraine with the new America First, but you also see that happening with the Americans pulling out of the Paris Climate Accord, again, pulling out of the World Health Organization. You see it with the Americans shutting down USAID. And America has historically been responsible for about 40 percent of global humanitarian support and aid. And absolutely, some of that is corrupt. Absolutely, some of that is on woke programs that the vast majority of American taxpayers would never support. But the majority of that money is actually spent on programs that are really important in advancing American soft power and influence over countries in the Global South, all over the world. The Chinese know that. That ' s why they ' ve started humanitarian programs. It ' s not out of some great love for, you know, providing foreign aid for these countries. It ' s more because they see it creates influence. Now the Americans are leaving a vacuum that the Chinese can absolutely take advantage of. And we ' ll see this across, you know, sort of, microstates in the Pacific. We ' ll see this across sub-Saharan Africa. We ' ll see this across South America. The Chinese see huge opportunities from the United States creating a leadership vacuum. And the place they ' re most capable of doing that is if the US stops paying dues in the United Nations, where China is number two, and they ' ll suddenly have the most influence over all the key positions, which will give them a role in global governance that they ' ve really been constrained from having over the past decades. HW : What do you think are the chances that the US stops paying its dues to the UN? IB : You know, if it were up to Congress and only Congress and Trump doesn ' t lean on them, I ' d say fairly low. I ' d say that they ' d probably, you know, cut back on some programs. But in terms of baseline dues, they ' d keep paying it. But Trump personally, I think increasingly sees the UN as hostile to the United States. That ' s particularly true in terms of Israel policy, which Trump is leaning in on. And, you know – let me put it this way. Tulsi Gabbard did not have the votes to get confirmed. There were Republican senators that were going to vote against her. And Elon called those individual senators and threatened them. And this was, you know, fully aligned with what Trump wanted Elon to do. And those votes went away. And the only person that ended up voting against Tulsi was McConnell. And as a consequence, it went through. So look, I think that we should not underestimate the level of power and control that Trump and Elon have over the Republican Party this time around, compared to 2017, where it was really the Republican Party that was much stronger than Trump, both in his administration creating a lot of guardrails, but also in Congress. This time around, Trump has all the chips and he ’ s using them. So if he decides he wants to stop paying dues, I think that ’ s what ’ s going to happen. This is a much more revolutionary presidency, domestically, where last time around it was much more transactional. HW : Alright. Let ’ s talk a little bit more about what ’ s actually happening inside the US. You bring up Elon, who obviously with DOGE has been sending out emails and asking for information from people and kind of trying to do a lot very quickly. Some of these high-profile moves are – I think what ’ s interesting is that no one would really argue that there isn ’ t inefficiency in government, that that ’ s an understood reality. But the way that some of these moves are being made are maybe going to cause irreparable damage. And that ’ s to people from red states, that ’ s to scientists, that ’ s to veterans, that ’ s to people who voted for Trump and that ’ s to people around the world. You talk about soft power and the United States, but what about actually what ’ s happening inside the United States? And why do you think that the administration is willing to go so quickly, to go so fast and to cause such damage? IB : Well, and you saw Trump this weekend saying that he wants Elon to move faster, to be more aggressive. And I don ’ t think that was just a troll. I think that ’ s true. Trump is 78 years old, and with his friends he talks about the fact that he doesn ’ t have much time. That in two years ’ time, the Democrats could take the House in four years ’ time, you know, that ’ s it. He ’ s not talking privately of”I ’ m going to be president for life.”He also knows he was almost killed. He was shot in the head. And do I think that changes somebody? Absolutely. I think that he believes that he was saved by God, and that his level of confidence that what he is doing is fundamentally right and necessary now is so much higher than the insecurity that I frequently saw in him in his first presidency. So I think his willingness to push, push, push, and his view that Elon is a great activator and executor on that intention, and someone that he had nowhere close to someone with that kind of capability and energy and execution ability in his first term. And now he does. So first of all, I think that relationship between Trump and Elon isn ’ t going anywhere. I think it ’ s actually very stable. People that say it is are people that want it to go away. But hope is not a strategy. And I also think that the intention of revolutionary politics domestically is to shoot first and ask questions later. It ’ s to break things. It ’ s a view that the bureaucracy has been weaponized against Trump first and foremost, and also against the will of the American people. And therefore it must be broken. And you ’ re going to break eggs when you want to make that omelet. Now your point is that there are a lot of Republicans that work in government. There are a lot of Republicans and Trump voters that are losing their jobs and that are not being treated with a level of care and compassion as a result of what Trump and Elon are doing. And I think there will be a backlash. But I ’ m not so sure that a one-term, 78-year-old-Trump presidency cares all that much about that. And, you know, if that means that his popularity slips to like, the high 30s, frankly, you know, in a very, very polarized country, high 30s isn ’ t that bad. It still means 80, 90 percent of the people that voted for him still support him, and I think he ’ s probably more comfortable with that this time around. So I think a lot more has to happen before we can start talking about whether there ’ s really going to be significant pushback against what Trump is trying to accomplish at home. HW : Do you expect to see any of Congress standing up to Trump? We saw the governor of Maine recently kind of take him on. Do you expect to see more of that, or do you feel like for the next two years at least, people will just go where Trump takes them? IB : Well, sure, but, you know, in the case of Maine, that ’ s a Democrat. So I don ’ t know how much that means. I don ’ t expect to see a lot of Republicans standing up against him. And of course, since the Republicans control the House and the Senate, that ’ s the only thing that really matters. I mean, you know, you saw Kash Patel got through, Pete Hegseth got through. Now look, I actually think that a lot of the members of cabinet are very capable. I mean, they ’ ve been picked for their loyalty and they will be loyal to Trump, but they ’ re very capable. I think JD Vance is very capable. Mike Waltz, I know quite well. Marco Rubio, over the years. These are people that are very well respected, have been by their colleagues and could have served under any Republican administration. But there are some people that are completely incapable for the roles that they have been selected, completely incapable. They would not be selected in a role in any cabinet, in any democracy. And yet here they are, serving in critical roles in the most powerful country in the world. And yes, I ’ m talking about the Secretary of Defense. I ’ m talking about the Secretary of Health and Human Services. I ’ m talking about the Director of National Intelligence. I ’ m talking about the Director of FBI. These are important roles. And they were all confirmed. And if you ’ re a Republican senator – and senator – I ’ m not talking about the House here where, you know, it ’ s a lot more diffuse and there are a lot more wackos and it ’ s easier to get in. But Senate has always been like, responsible and more oriented towards bipartisanship and the rest, and they are utterly petrified of what it means if they publicly get crosswise with President Trump and number two, Elon. And they ’ re not willing to do it. So I think that there ’ s very little belief that you are going to see Congress step up. I think that you will see justices step up because those justices are independent. The problem is that many of those justices are progressive and have a known progressive lean. And so when cases come up and justices that have that political lean oppose something Trump wants, his willingness, his administration ’ s willingness to attack them and refuse to execute on that judicial decision, unconstitutionally, is real. Now the case will of course go then up to the Supreme Court eventually. And I don ’ t feel the same way about Trump refusing to listen to adhere to the rulings of the Supreme Court. But if there ’ s 100 rulings like that and you overwhelm the judicial process, you quickly see how you might erode a core check on executive power in the United States. And Trump and Elon clearly intend to every day test those checks and balances and break them if they can. I mean, that is the intent, that is the intent. HW : So you mentioned democracy. When we last spoke, you said very clearly that the US is not Hungary, meaning that democracy is not threatened and that the US is not in danger of turning into an autocracy. Do you stand by that? IB : Yes, I do. I mean, in the sense that I think in two years we ’ re going to have midterm elections, and they will be largely free and fair. And in four years we ’ ll have presidential elections. And US elections are held at the federal level, right? They ’ re held state by state. They make the rules. It ’ s very easy to make people believe that the election has been stolen. It ’ s extremely hard to actually rig an American election. And I don ’ t believe that that is very likely at all. It ’ s very unlikely. The American military, despite the firings that we ’ ve seen just last Friday, is still an independent and professional military that I believe will carry out its orders and its duties professionally. I see the judiciary in the United States as independent and a clear check on the executive power in the United States. Where I see a problem, is that the US is becoming more kleptocratic every day. And the US was already much more of a kleptocracy over the past 10 years under

Democrats and Republicans than any other advanced industrial democracy in the world. Clearly, money buys power in America in a way that it doesn ' t in any other major democracy, wealthy democracy. That is getting dramatically worse. I saw specifically that Elon Musk, who is a walking conflict of interest at the highest level, met with Narendra Modi, the Prime Minister of India, in Blair House, so part of the official White House complex, right after Trump met with him. And Trump was asked later in the day, was Modi meeting with Elon in Elon ' s private-sector capacity to advance his business interests or his official capacity to advance the policies of the United States? And Trump answered honestly. He said,"I don ' t know."And how could he know, right? Because, I mean, those two things are completely intertwined. But that is not remotely acceptable for rule of law in a functional democracy, and yet nothing is being done about it. In fact, it ' s getting more and more entrenched every day. So the single place where the United States was least functional as an advanced democracy is now getting far, far worse. And I don ' t see any checks and balances on that. I think that ' s a very, very serious problem. HW : Alright, Ian, thank you, as always for all of your insights. Let ' s close out with a couple more questions from our community. And this one indeed relates to the kleptocratic conversation that you were just raising. So why are the billionaires enabling fascism? What do they get out of it? IB : Fascism is a very politically loaded argument. That ' s a political system where the cruelty against a subgroup or a perceived subgroup in the society is the intention, the expressed intention of the policy. And I certainly wouldn ' t define the United States as a fascist system today. But I do believe that billionaires in the United States, at least a significant subset of them, are really under-interested in the well-being of their fellow countrymen and women. And this is a country that really makes heroic the individual. And it undermines the community. We ' ve seen a significant erosion of civic engagement in the United States and civic associations. The church is not what it was in the US. Fewer people go, fewer people trust the church. Public education is not what it was. And wealthy people increasingly opt out. The family is not what it was. It ' s increasingly atomized and people spend much more, much more of their time being, you know, isolated and atomized and intermediated by algorithms. And you and I have spoken about that problem in the past. We are increasingly a society of individuals as opposed to a nation together. I will tell you that that is a real problem when you have large numbers of billionaires that believe that they built it themselves, that they ' re not responsible to a collective whole for any of their success and therefore they ' re not accountable for their fellow citizens. And that ' s what I think we have. I think we have a lot of winners in the United States and not a lot of leaders. And when you have winners, you have losers. When you have winners, you don ' t have a collectivity. I will say that in this anti-DEI thing, I have a lot of sympathy for the people that don ' t like DEI. I thought it was rammed down the throats of a lot of people And I ' ve said this before, it was well beyond what the average American was willing to tolerate, and also it came with a level of high-handedness that that basically said, if you ' re not with us in supporting these programs, you ' re evil, you ' re stupid, you ' re uneducated. That ' s unacceptable, too. It was a completely, you know, sort of oppositional way to try to have what ' s a very challenging conversation. But I ' ve never been a champion of diversity for diversity ' s sake. What ' s amazing about the history of the United States is despite all of our diversity, we find commonalities with each other. We find connectedness. That ' s what ' s amazing, is the connectedness, is that, you know, on this world, we ' ve got eight billion people. And despite how different we all are, we all actually connect as human beings. That is what we need to focus on, not the diversity, the connectedness. And I think that ’ s what the billionaires today in the United States are losing. They ’ re doing incredibly well themselves. They ’ ve got theirs, they ’ ve got access to power, they ’ re paying good money for it, they ’ re going to get even more. But they ’ re forgetting that for us to succeed as a society and as a planet, we have to be connected to each other. You can ’ t throw out diversity and forget connectedness, and that ’ s what we ’ re forgetting right now. That ’ s why the country feels so much trauma and pain right now. And I think why the United States has given up on a lot of its really aspirational values that made such a difference to the world, that almost lost World War II for many generations. And now we ’ re giving it all away. HW : I mean, I think that ’ s why you find that diversity and inclusion go hand in hand, that that ’ s actually speaking to that. But in some of the moves that are being made to kind of throw out any type of diversity, to scan documents, to find words in order to highlight programs that are wasteful, that we ’ re actually going to lose what many, many studies actually highlight as being the benefits of diversity and the fact that you need diversity in order to be actually successful. IB : Shooting first and asking questions later, you know, moving fast and breaking things, those things make sense in a turbocharged venture capital technology environment. They do not work in government. They don ’ t work in government at all. In government, we have something called checks and balances, and we need them. And moving fast and breaking things undermines those checks and balances. So we ’ re going to do a lot of damage unintentionally. A lot of people are going to get hurt. A lot of folks don ’ t care about that in power right now, but they should because it would be much, much better. I know people want to move fast, but actually to think a little bit, do a little bit of your own research before you decide to kill that program, before you decide to fire that person. We ’ d be in a lot better shape as a government right now. HW : OK, the final question that I have for you actually comes from my 10-year-old son, and it ’ s a bit of a miserable question. We have a very lovely house, I promise. But his question nonetheless is are we heading into World War III? So, Ian, what should I tell him? IB : No, no, I don ’ t think we ’ re heading into World War III but we are absolutely heading into a world of much greater conflict. I think it ’ s much more likely, for example, in the next five, 10 years, that we ’ ll have a nuclear weapon go off. And I thought that was pretty unlikely since the Cuban Missile Crisis. In other words, over the course of my life. I think we ’ re going to see a lot more terrorism at scale, because a lot more people are going to feel angry and disaffected, I think. You know, you saw that assassination of the CEO of UnitedHealthcare just a few dozen blocks from my house. I think you ’ re going to see more things like that. And so, no, not World War III because the fact is that outside of the US and China, all the other countries know that they need to work with everybody. And there are even forces inside US-China that are trying to make sure that you don ’ t throw the baby out with the bathwater. But underneath that World War III risk, there ’ s an awful lot of damage that ’ s going to get done, an awful lot of crisis that I think we ’ re going to have to live through before we start to see what we create on the other side. A lot of defense as well, that we ’ re going to be playing in this environment. HW : Well, Ian, it is always a pleasure to talk to you, and we ’ re so glad that you are scanning the world to understand what is happening and then you come here and explain it to us all. Couldn ’ t be more grateful and thank you for everything. IB : Thank you.

Text Sample 747: NEG Dim 5 – 2025 – Score 1.00 – 2025/t003466.txt

Can you paint with all the colors of the wind? I love this song from the first time I heard it. I was 14, growing up in public housing in western Sydney. So maybe a little too old for this Disney classic from"Pocahontas."But it spoke to me. I saw something of myself in it. At this stage, I think I knew I was queer. I mean, come on, I was rocking out to Disney. And look at those glasses. Seriously, what a queer kid. But I was also from a Pacific Indigenous heritage with conservative views and values. I didn ' t feel like I could fit into any of the boxes that were being presented to me. Straight, gay, Fijian, Australian. I couldn ' t pick just one color to paint with. Prior to colonization, sexuality amongst many of our Pacific islands across the region was fluid. Men would have sex with men without the stigma or shame afforded it today. Sexual intimacy was a way of being able to connect socially and relationally. Sex wasn ' t just for procreation. Or in marriage. It wasn ' t exclusively monogamous. Just because you had sex with a man didn ' t define your sexuality. It didn ' t make you an outcast. But as Europeans started to colonize the Pacific region, they brought Christian missionaries with them. They told us to cover up our beautiful bodies. We were taught shame. We were told to stop practicing Pacifica cultures and values that were seen as provocative. To a Western and white perspective, our non-binary sexuality was judged as tribal, outdated, sinful and impossible to practice alongside a Christian faith. Still today, those who are Indigenous to the Pacific are negatively impacted by whiteness. That"white is right, West is best."I was brought up to espouse strong Christian beliefs, views and values which would shape my burgeoning identity as an adult. From these staunch conservative views, I was taught to fear the queer. Those people that were seen as unusual and fruity. And in essence, I grew to despise a part of myself. I sought to separate my queerness from my identity. I could never see myself as part of that world. Growing up, biracial and bisexual constantly challenged the way I saw the world around me. I was unsure about my thinking, feelings and behaviors around liking males and females. I was unsure about my thinking, feelings and behaviors when it came to whether I needed to be a white Australian or a Fijian person. And whether this needed to be framed within a binary to choose one over the other. In this way, I don ' t think that my story is unique. There ' s plenty of people in the queer world that have escaped conservative upbringings, religions and cultures. But this is only part of my story. Another common story in the queer community is the coming-out story. When did you tell your parents? How did they react? How it liberated you or ultimately separated you from your family. But this didn ' t feel like an option for me either. I believe that embedded in these particular stories, these tropes, is a part of dominant queer culture that made it very hard for me to access the queer world. For example, when I started to go to gay clubs, I would be lining up at the front of the club and often I would get door staff come up to me and go,"Sorry, you do know this is a gay club, right?"From these lived experiences, I feel that there is a whiteness that pervades queer culture. And I didn ' t want to give up my identity and my appearance as a Pacific indigenous person to be queer. Pacifica cultures really, really value and esteem the collective above the individual. For Pacifica people, our families are inextricably connected to our own sense of self-agency, self-determination, self-esteem, self-love and self-worth. Our own individual identities are meshed and intertwined with our family ' s reputation and standing in the community. Our mind, body and souls are inextricably connected to the broader community, to our families, to ourselves and to others. I didn ' t want to come out to potentially ostracize myself, to make a point to separate myself from my Pacific and faith communities. I couldn ' t then see my authentic self in any of these spaces. And a quick shout out to my son on the left-hand side. Round of applause for my son. He said to me this morning,”Daddy, make sure you mention me.”So I was like,”OK, son. Alright.”So there you go for the recording, son. And the rest are my other family, that ’ s up there as well. Across queer cultures and the broader community, we have learned to embrace our gender and sexual differences. We encourage individuals to be allies for each other and to support safe spaces for those that are seen to not fit in. We empower and really encourage individuals to really develop and express their self-agency, their self-determination, their self love, their self-worth. We boldly go into spaces striving to bring others with us to ensure that we can create inclusive societies for everyone. However, the dominant Western view in queer cultures is still based on the individual. Me, myself and I. And my individual pursuits towards happiness. As a queer Indigenous Pacific person, I feel that there are parts of queer culture that is white, feels very white. We may fail to see the importance of bringing our families and our cultural communities along our journey as queer people. It ’ s either the queer way or the highway. It ’ s one or the other. Young or old. Rich or poor. Smart or stupid. Masculine or feminine. Fit or fat. These binaries create a sense of us and them, those people. A binary approach and position that we generally in Western white societies use to determine the norm. We create, conflate, perpetuate binaries that create divisions in our society. Saints or sinner, straight or gay. I see it like this. Binaries are a very white and Western way of thinking, feeling and behaving. As individuals today, we live within these binaries and create identities that then create these labels. We use these labels to then define and confine people and places. We are one or the other. But what we fail to see is that our identities are more than a set of binaries. Our own identities are comprised of intersecting identifiers that contributes to who you are, including our age, class, color, education, gender, size, spirituality, sexuality, and so, so much more. We need to decolonize queerness, to live beyond”white is right, West is best.”Because if we don ’ t, we ’ re going to continue to create silos and divisions in our societies that don ’ t create inclusive and safe spaces for everyone. For Indigenous queer people, they shape our connection to community and culture, our families. If we don ’ t include that within our understanding of who we are, even as queer people, then we ’ re not able to be, again, our true and authentic selves. We ’ re not able to then be our fabulous selves as part of the queer community. And it ’ s really, really interesting because I believe that we can learn so much from pre-colonial cultures like those that were practiced in the Pacific, that provide a different way of looking at a society. A queer way that pre-dates all the privileges that go into binary ways of thinking, feeling and behaving. Across many of our Pacifica languages we have a lot of key concepts that help solidify and support a shared understanding of the collective. For example, in Fijian,”solesolevaki”is all about reciprocal living, where my wellbeing is your wellbeing. I will actively seek to ensure that your well-being and every part of your life is catered for, knowing that you will do the same for me. We strive to enact”veiqaravi,”as tatted here on my hand, to serve others. We strive to also embed and embold and empower people to practice”veirokovi,”respect for self and others where we are relationally driven, where that is so important to our spaces and places. My sense of ability to actually put this into practice, my ability to actually understand these Indigenous perspectives has greatly helped shape and reconcile my queer identity today. But what ’ s really interesting is that it ’ s not just queer communities that need to decolonize itself. It ’ s all of us as a broader society that needs to be part of a shared conversation to decolonize, deconstruct and disrupt binary ways of thinking, feeling and behaving. So how can we practically do that? Practice cultural humility, where you are constantly learning and embracing differences, where you get over yourselves and strive to learn about other people ’ s differences and how that can be meaningfully included in the way in which you create safe spaces for everyone. I also believe it ’ s possible to bring Indigenous perspectives into Western and white workplaces. For example, I ’ ve introduced the concept of”talanoa”in all of my work meetings. Talanoa is all about being able to create a safe space where everyone is able to contribute to a collective, collegial and collaborative conversation. Where people are able to bring their authentic selves into these spaces. Literally, I will chair meetings where I have no firm set agenda items and I literally would just hold space for participants just to bring themselves. So you can tell I ’ m a social worker, right? Safe space, safe space, safe space. It ’ s OK, it ’ s OK. But it ’ s through these safe spaces that people can bring their multiple, intersecting identifiers, including their age, their class, color, education, gender, size, spirituality, sexuality and so, so much more. I invite queer communities to come talanoa with Pacific Indigenous Communities to create robust and resilient queer cultures. I have been able to greatly reconcile my queer and Pacifica identities, which has now really been embraced to help shape how I see the world around me today. And it ’ s through these shared perspectives and approaches that we can truly embrace cultural diversity and its differences. And to know that people can live beyond those binaries and those labels we continue to perpetuate and perpetrate against others. It is a commitment from queer communities to decolonize queerness. It ’ s a commitment from me, from you and all of us to decolonize Western and white societies and to live beyond the binary way of thinking, feeling and behaving. I genuinely believe that it ’ s through this common goal that we can truly learn how to paint with all the colors of the wind.”Vinaka vakalevu”and thank you.

Text Sample 748: NEG Dim 5 – 2025 – Score 1.00 – 2025/t004225.txt

A Love Letter to my Garden by Debbie Millman In the beginning of time, there was no time. The universe began with every bit of energy contained in a very tiny, incredibly dense point. It was even smaller than this point. The point exploded with an unimaginable force and created everything we know and are made of. Seeds are a bit like that too. They ’ re tiny and densely packed with their entire existence. Time feels different now, while we shelter at home. Time is scary and uncertain. So we decided to plant some mint. We actually started a little garden a while ago, but now we have more time. We thought that these flowers had died, but they miraculously came back to life. A hopeful sign about the future, perhaps? We planted foxglove and it thrived. And then we went a little crazy. We planted tomatoes, carrots, beans, corn, lettuce and cucumbers, strawberries, blueberries and limes. It ’ s such a strange time in the world. There is so much beauty amidst so much fear. It ’ s so hard to know what to do. In the meantime, we watch and wait and remain grateful we ’ re able to make things with our hands. A love letter to New York City by Debbie Millman I ’ m a native New Yorker. I ’ ve lived here my entire life, yet the city never ceases to surprise me. Everywhere you look, there ’ s something to see. There is Tom Otterness underground and Lawrence Weiner underfoot. There are so many of us. Everyone wants to be someone. Everyone seeks to be heard. Everyone has an opinion, A point of view. And seemingly something to sell. But life is different now as we shelter at home. Under the same sky, at the same moment, at the same time. I ’ m thinking about how tough New York can be, how rough and unrelenting. But maybe that intensity is what fuels our tenacity. As we navigate through this crisis, I am hopeful we will find some solace together. A Love Letter to Travel by Debbie Millman My work has taken me all over the world. There have been many early departures, and many evenings in the air.

In all my travels, I ’ ve been struck by so many things. I ’ ve observed how people lived in some of the most ancient places on the planet. This is Machu Picchu. This is Easter Island. This is Samoa. This is Angkor Wat. This is Petra in Jordan. The ancient city contained a church, a theater and tombs for the dead. Families lived in two and three-bedroom apartments. We ’ ve come a long way. Some of the structures that remain are rather mysterious, some are otherworldly, some are glorious odes to love, some house the souls of the dead. Some have forced us to choose sides. Some have succumbed to the power of nature. And some have not. The view from the sky is so abstract. It reveals our connections, our continuity and our scale. Nevertheless, it ’ s hard to see the big picture right now. It ’ s such a difficult time in the world as we pause and dismantle and rebuild our culture. I think back to my travels. I ’ ve seen how different we are, how diverse and distinct. But I ’ ve also seen our commonalities. How much worship means in Tibet, in Pakistan, in Italy, in China, in Cambodia and in Peru. How much peace means to our future. How much we regret the mistakes of our past. And now, insisting on what matters no matter what. There is still so much beauty in the world. So much love. I ’ m hopeful for the next generation. For every creature, large and small. And for the planet we call home. A Love Letter to Storytelling by Debbie Millman”A need to tell and hear stories is essential to the species Homo Sapiens. The sound of story is the dominant sound of our lives.”- Joseph Campbell I ’ ve loved telling stories all of my life. I ’ ve loved writing them and drawing them, and I love teaching my students how to make them. Now I conduct visual storytelling workshops all over the world. Over the years, the students have invented many new ways of expressing their stories. They tell their truths. They reveal secrets. They bare their souls. They share stories of pain, of feeling seen, of not being smart enough. Stories of being scared, feeling pretty or not. No matter where I teach, I ’ ve come to realize how many of their stories are universal. They convey so many shared experiences : rejection, judgment, insecurity, prejudice, forgetting and remembering. But there are also stories of hope, strength, awareness, courage and realizations. Their stories reflect who they really are, how deeply they want to feel connected. And now, as they share their visions for a more optimistic future, their stories are becoming the stories of our times.

Text Sample 749: NEG Dim 5 – 2019 – Score -1.00 – 2019/t001497.txt

My parents gave me an extraordinary name : Baratunde Rafiq Thurston. Now, Baratunde is based on a Yoruba name from Nigeria, but we ’ re not Nigerian. That ’ s just how black my mama was.”Get this boy the blackest name possible. What does the book say?”Rafiq is an Arabic name, but we are not Arabs. My mom just wanted me to have difficulty boarding planes in the 21st century. She foresaw America ’ s turn toward nativism. She was a black futurist. Thurston is a British name, but we are not British. Shoutout to the multigenerational, dehumanizing economic institution of American chattel slavery, though. Also, Thurston makes for a great Starbucks name. Really expedites the process. My mother was a renaissance woman. Arnita Lorraine Thurston was a computer programmer, former domestic worker, survivor of sexual assault, an artist and an activist. She prepared me for this world with lessons in black history, in martial arts, in urban farming, and then she sent me in the seventh grade to the private Sidwell Friends School, where US presidents send their daughters, and where she sent me looking like this. I had two key tasks going to that school : don ’ t lose your blackness and don ’ t lose your glasses. This accomplished both. Sidwell was a great place to learn the arts and the sciences, but also the art of living amongst whiteness. That would prepare me for life later at Harvard, or doing corporate consulting, or for my jobs at"The Daily Show"and"The Onion."I would write down many of these lessons in my memoir,"How to Be Black,"which if you haven ' t read yet, makes you a racist, because — you ' ve had plenty of time to read the book. But America insists on reminding me and teaching me what it means to be black in America. It ' s December 2018, I ' m with my fianc in the suburbs of Wisconsin. We are visiting her parents, both of whom are white, which makes her white. That ' s how it works. I don ' t make the rules. She ' s had some drinks, so I drive us in her parents ' car, and we get pulled over by the police. I ' m scared. I turn on the flashing lights to indicate compliance. I pull over slowly under the brightest streetlight I can find in case I need witnesses or dashcam footage. We get out my identification, the car registration, lay it out in the open, roll down the windows, my hands are placed on the steering wheel, all before the officer exits the vehicle. This is how to stay alive. As we wait, I think about these headlines —"Police shoot another unarmed black person"— and I don ' t want to join them. The good news is, our officer was friendly. She told us our tags were expired. So to all the white parents out there, if your child is involved with a person whose skin tone is rated Dwayne"The Rock"Johnson or darker — you need to get that car inspected, update the paperwork every time we visit. That ' s just common courtesy. I got lucky. I got a law enforcement professional. I survived something that should not require survival. And I think about this series of stories —"Police shoot another unarmed black person"— and that season when those stories popped up everywhere. I would scroll through my feed and I would see a baby announcement photo. I ' d see an ad for a product I had just whispered to a friend about yesterday. I would see a video of a police officer gunning down someone who looked just like me. And I ' d see a think piece about how millennials have replaced sex with avocado toast. It was a confusing time. Those stories kept popping up, but in 2018, those stories got changed out for a different type of story, stories like,"White Woman Calls Cops On Black Woman Waiting For An Uber."That was Brooklyn Becky. Then there was,"White Woman Calls Police On Eight-Year-Old Black Girl Selling Water."That was Permit Patty. Then there was,"Woman Calls Police On Black Family BBQing At Lake In Oakland."That was now infamous BBQ Becky. And I contend that these stories of living while black are actually progress. We used to find out after the extrajudicial police killings. Now, we ' re getting video of people calling 911. We ' re moving upstream, closer to the problem and closer to the solution. So I started a collection of as many of these stories as I could find. I built an evolving, still-growing database at baratunde. com / livingwhileblack. Seeking understanding, I realized the process was really diagramming sentences to understand these headlines. And I want to thank my Sidwell English teacher Erica Berry and all English teachers. You have given us tools to fight for our own freedom. What I found was a process to break down the headline and understand the consistent layers in each one : a **subject** takes an action against a target engaged in some activity, so that"White Woman Calls Police On Eight-Year-Old Black Girl"is the same as"White Man Calls Police On Black Woman Using Neighborhood Pool"is the same as"Woman Calls Cops On Black Oregon Lawmaker Campaigning In Her District."They ' re the same. Diagramming the sentences allowed me to diagram the white supremacy which allowed such sentences to be true, and I will pause to define my terms. When I say"white supremacy,"I ' m not just talking about Nazis or white power activists, and I ' m definitely not saying that all white people are racist. What I ' m referring to is a system of structural advantage that favors white people over others in social, economic and political arenas. It ' s what Bryan Stevenson at the Equal Justice

Initiative calls the narrative of racial difference, the story we told ourselves to justify slavery and Jim Crow and mass incarceration and beyond. So when I saw this pattern repeating, I got angry, but I also got inspired to create a game, a game of words that would allow me to transform this traumatic exposure into more of a healing experience. I ’ m going to talk you through the game. The first level is a training level, and I need your participation. Our objective : to determine if this is real or fake. Did this happen or not? Here is the example :”Catholic University Law Librarian Calls Police On Student For ’ Being Argumentative. ’”Clap your hands if you think this is real. Clap your hands if you think this is fake. The reals have it, unfortunately, and a point of information, being argumentative in a law library is the exact right place to do that. This student should be promoted to professor. Training level complete, so we move on to the real levels. Level one, our objective is simple : reverse the roles. That means”Woman Calls Cops On Black Oregon Lawmaker”becomes”Black Oregon Lawmaker Calls Cops On Woman.”That means”White Man Calls Police On Black Woman Using Neighborhood Pool”becomes”Black Woman Calls Police On White Man Using Neighborhood Pool.”How do you like them reverse racist apples? That ’ s it, level one complete, and so we level up to level two, where our objective is to increase the believability of the reversal. Let ’ s face it, a black woman calling police on a white man using a pool isn ’ t absurd enough, but what if that white man was trying to touch her hair without asking, or maybe he was making oat milk while riding a unicycle, or maybe he ’ s just talking over everyone in a meeting. We ’ ve all been there, right? Seriously, we ’ ve all been there. So that ’ s it, level two complete. But it comes with a warning : simply reversing the flow of injustice is not justice. That is vengeance, that is not our mission, that ’ s a different game so we level up to level three, where the objective is to change the action, also known as”calling the police is not your only option OMG, what is wrong with you people!”And I need to pause the game to remind us of the structure. A **subject** takes an action against a target engaged in some activity.”White Woman Calls Police On Black Real Estate Investor Inspecting His Own Property.””California Safeway Calls Cops On Black Woman Donating Food To The Homeless.””Gold Club Twice Calls Cops On Black Women For Playing Too Slow.”In all these cases, the **subject** is usually white, the target is usually black, and the activities are anything, from sitting in a Starbucks to using the wrong type of barbecue to napping to walking”agitated”on the way to work, which I just call”walking to work.”And, my personal favorite, not stopping his dog from humping her dog, which is clearly a case for dog police, not people police. All of these activities add up to living. Our existence is being interpreted as crime. Now, this is the obligatory moment in the presentation where I have to say, not everything is about race. Crime is a thing, should be reported, but ask yourself, do we need armed men to show up and resolve this situation, because when they show up for me, it ’ s different. We know that police officers use force more with black people than with white people, and we are learning the role of 911 calls in this. Thanks to preliminary research from the Center for Policing Equity, we ’ re learning that in some cities, most of the interactions between cops and citizens is due to 911 calls, not officer-initiated stops, and most of the violence, the use of force by police on citizens, is in response to those calls. Further, when those officers responding to calls use force, that increases in areas where the percentage of the white population has also increased, aka gentrification, aka unicycles and oat milk, aka when BBQ Becky feels threatened, she becomes a threat to me in my own neighborhood, which forces me and people like me to police ourselves. We quiet ourselves, we walk on eggshells, we maybe pull over to the side of the road under the brightest light we can find so that our murder might be caught cleanly on camera, and we do this because we live in a system in which white people can too easily call on deadly force to ensure their comfort. The California Safeway didn ’ t just call cops on black woman donating food to homeless. They ordered armed, unaccountable men upon her. They essentially called in a drone strike. This is weaponized discomfort, and it is not new. From 1877 to 1950, there were at least 4, 400 documented racial terror lynchings of black people in the United States. They had headlines as well.”Rev. T. A. Allen was lynched in Hernando, Mississippi for organizing local sharecroppers.””Oliver Moore was lynched in Edgecomb County, North Carolina, for frightening a white girl.””Nathan Bird was lynched near Luling, Texas, for refusing to turn his son over to a mob.”We need to change the action, whether that action is”lynches”or”calls police.”And now that I have shortened the distance between those two, let ’ s get back to our game, to our mission. Our objective in level three is to change the action. So what if, instead of”Calls Cops On Black Woman Donating Food To Homeless,”that California Safeway simply thanks her. Thanking is far cheaper than bringing law enforcement to the scene. Or, instead, they could give the food they would have wasted to her, upped their civic cred. Or, the white woman who called the police on the eight-year- old black girl, she could have bought all the inventory from that little black girl, support a small business. And the white woman who called the police on the black real estate investor, we would all be better off, the cops agree, if she had simply ignored him and minded her own damn business. Minding one ’ s own damn business is an excellent choice, excellent choice. Choose it more often. Level three is complete, but there is a final bonus level, where the objective is inclusion. We have also seen headlines like this :”Powerful Man Masturbates In Front Of Young Women Visiting His Office.”What an odd choice for powerful man to make. So many other actions available to him. Like, such as,”listens to,””mentors,””inspired by, starts joint venture, everybody rich now.”I want to live in that world of everybody rich now, but because of his poor choice, we are all in a poorer world. Doesn ’ t have to be this way. This word game reminded me that there is a structure to white supremacy, as there is to misogyny, as there is to all systemic abuses of power. Structure is what makes them systemic. I ’ m asking people here to see the structure, where the power is in it, and even more importantly to see the humanity of those of us made targets by this structure. I am here because I was loved and invested in and protected and lucky, because I went to the right schools, I ’ m semifamous, mostly happy, meditate twice a day, and yet, I walk around in fear, because I know that someone seeing me as a threat can become a threat to my life, and I am tired. I am tired of carrying this invisible burden of other people ’ s fears, and many of us are, and we shouldn ’ t have to, because we can change this, because we can change the action, which changes the story, which changes the system that allows those stories to happen. Systems are just collective stories we all buy into. When we change them, we write a better reality for us all to be a part of. I am asking us to use our power to choose. I am asking us to level up. Thank you. I am Baratunde Rafiq Thurston.

Text Sample 750: NEG Dim 5 – 2019 – Score -1.00 – 2019/t001503.txt

When I look in the mirror today, I see a justice and education scholar at Columbia University, a youth mentor, an activist and a future New York state senator. I see all of that and a man who spent a quarter of his life in state prison â six years, to be exact, starting as a teenager on Rikers Island for an act that nearly cost a man his life. But what got me from there to here wasn ’ t the punishment I faced as a teenager in adult prison or the harshness of our legal system. Instead, it was a learning environment of a classroom that introduced me to something I didn ’ t think was possible for me or our justice system as a whole. A few weeks before my release on parole, a counselor encouraged me to enroll in a new college course being offered in the prison. It was called Inside Criminal Justice. That seems pretty straightforward, though, right? Well, it turns out, the class would be made up of eight incarcerated men and eight assistant district attorneys. Columbia University psychology professor Geraldine Downey and Manhattan Assistant DA Lucy Lang co-taught the course, and it was the first of its kind. I can honestly say this wasn ’ t how I imagined starting college. My mind was **blown** from day one. I assumed all the prosecutors in the room would be white. But I remember walking into the room on the first day of class and seeing three black prosecutors and thinking to myself,”Wow, being a black prosecutor â that ’ s a thing!”By the end of the first session, I was all in. In fact, a few weeks after my release, I found myself doing something I prayed I wouldn ’ t. I walked right back into prison. But thankfully, this time it was just as a student, to join my fellow classmates. And this time, I got to go home when class was over. In the next session, we talked about what had brought each of us to this point of our lives and into the classroom together. I eventually got comfortable enough to reveal my truth to everyone in the room about where I came from. I talked about how my sisters and I watched our mother suffer years of abuse at the hands of our stepfather, escaping, only to find ourselves living in a shelter. I talked about how I swore an oath to my family to keep them safe. I even explained how I didn ’ t feel like a teenager at 13, but more like a soldier on a mission. And like any soldier, this meant carrying an emotional burden on my shoulders, and I hate to say it, but a gun on my waist. And just a few days after my 17th birthday, that mission completely failed. As my sister and I were walking to the laundromat, a crowd stopped in front of us. Two girls out of nowhere attacked my sister. Still confused about what was happening, I tried to pull one girl away, and just as I did, I felt something brush across my face. With my adrenaline rushing, I didn ’ t realize a man had leaped out of the crowd and cut me. As I felt warm blood ooze down my face, and watching him raise his knife toward me again, I turned to defend myself and pulled that gun from my waistband and squeezed the trigger. Thankfully, he didn ’ t lose his life that day. My hands shaking and heart racing, I was paralyzed in fear. From that moment, I felt regret that would never leave me. I learned later on they attacked my sister in a case of mistaken identity, thinking she was someone else. It was terrifying, but clear that I wasn ’ t trained, nor was I qualified, to be the soldier that I thought I needed to be. But in my neighborhood, I only felt safe carrying a weapon. Now, back in the classroom, after hearing my story, the prosecutors could tell I never wanted to hurt anyone. I just wanted us to make it home. I could literally see the gradual change in each of their faces as they heard story after story from the other incarcerated men in the room. Stories that have trapped many of us within the vicious cycle of incarceration, that most haven ’ t been able to break free of. And sure â there are people who commit terrible crimes. But the stories of these individuals ’ lives before they commit those acts were the kinds of stories these prosecutors had never heard. And when it was their turn to speak â the prosecutors â I was surprised, too. They weren ’ t emotionless drones or robocops, preprogrammed to send people to prison. They were sons and daughters, brothers and sisters. But most of all, they were good students. They were ambitious and motivated. And they believed that they could use the power of law to protect people. They were on a mission that I could definitely understand. Midway through the course, Nick, a fellow incarcerated student, poured out his concern that the prosecutors were tiptoeing around the racial bias and discrimination within our criminal justice system. Now, if you ’ ve ever been to prison, you would know it ’ s impossible to talk about justice reform without talking about race. So we silently cheered for Nick and were eager to hear the prosecutors ’ response. And no, I don ’ t remember who spoke first, but when Chauncey Parker, a senior prosecutor, agreed with Nick and said he was committed to ending the mass incarceration of people of color, I believed him. And I knew we were headed in the right direction. We now started to move as a team. We started exploring new possibilities and uncovering truths about our justice system and how real change happens for us. For me, it wasn ’ t the mandatory programs inside of the prison. Instead, it was listening to the advice of elders â men who have been sentenced to spend the rest of their lives in prison. These men helped me reframe my mindset around manhood. And they instilled in me all of their aspirations and goals, in the hopes that I would never return to prison, and that I would serve as their ambassador to the free world. As I talked, I could see the lights turning on for one prosecutor, who said something I thought was obvious : that I had transformed despite my incarceration and not because of it. It was clear these prosecutors hadn ’ t thought much about what happens to us after they win a conviction. But through the simple process of sitting in a classroom, these lawyers started to see that keeping us locked up didn ’ t benefit our community or us. Toward the end of the course, the prosecutors were excited, as we talked about our plans for life after being released. But they hadn ’ t realized how rough it was actually going to be. I can literally still see the shock on one of the junior ADA ’ s face when it hit her : the temporary ID given to us with our freedom displayed that we were just released from prison. She hadn ’ t imagined how many barriers this would create for us as we reenter society. But I could also see her genuine empathy for the choice we had to make between coming home to a bed in a shelter or a couch in a relative ’ s overcrowded apartment. What we learned in the class worked its way into concrete policy recommendations. We presented our proposals to the state Department of Corrections commissioner and to the Manhattan DA, at our graduation in a packed Columbia auditorium. As a team, I couldn ’ t have imagined a more memorable way to conclude our eight weeks together. And just 10 months after coming home from prison, I again found myself in a strange room, invited by the commissioner of NYPD to share my perspective at a policing summit. And while speaking, I recognized a familiar face in the audience. It was the attorney who prosecuted my case. Seeing him, I thought about our days in the courtroom seven years earlier, as I listened to him recommend a long prison sentence, as if my young life was meaningless and had no potential. But this time, the circumstances were different. I shook off my thoughts and walked over to shake his hand. He looked happy to see me. Surprised, but happy. He acknowledged how proud he was about being in that room with me, and we began a conversation about working together to improve the conditions of our community. And so today, I carry all of these experiences with me, as I develop the Justice Ambassadors Youth Council at Columbia University, bringing young New Yorkers â some who have already spent time locked up and others who are still enrolled in high school â together with city officials. And in this classroom, everyone will brainstorm ideas about improving the lives of our city ’ s most vulnerable youth before they get tried within the criminal justice system. This is possible if we do the work. Our society and justice system has convinced us that we can lock up our problems and punish our way out of social challenges. But that ’ s not real. Imagine with me for a second a future where no one can become a prosecutor, a judge, a cop or even a parole officer without first sitting in a classroom to learn from and connect with the very people whose lives will be in their hands. I ’ m doing my part to promote the power of conversations and the need for collaborations. It is through education that we will arrive at a truth that is inclusive and unites us all in the pursuit of justice. For me, it was a brand- new conversation and a new kind of classroom that showed me how both my mindset and our criminal justice system could be transformed. They say the truth shall set you free. But I believe it ’ s education and communication. Thank you.

Text Sample 751: NEG Dim 5 – 2019 – Score -1.00 – 2019/t000927.txt

Preparing for this talk has been scarier for me than preparing for LSD therapy.”Psychedelics are to the study of the mind what the microscope is to biology and the telescope is to astronomy.”Dr. Stanislav Grof spoke those words. He ’ s one of the leading psychedelic researchers in the world, and he ’ s also been my mentor. Today, I ’ d like to share with you how psychedelics, when used wisely, have the potential to help heal us, help inspire us, and perhaps even to help save us. In the 1950s and 60s, psychedelic research flourished all over the world and showed great promise for the fields of psychiatry, psychology and psychotherapy, neuroscience and the study of mystical experiences. But psychedelics leaked out of the research settings and began to be used by the counterculture, and by the anti-Vietnam War movement. And there was unwise use. And so there was a backlash. And in 1970, the US government criminalized all uses of psychedelics, and they began shutting down all psychedelic research. And this ban spread all over the world and lasted for decades. and it was tragic, since psychedelics are really just tools, and whether their outcomes are beneficial or harmful depends on how they ’ re used. Psychedelic means”mind-manifesting,”and it relates to drugs like LSD, psilocybin, mescaline, iboga and other drugs. When I was 18 years old, I was a college freshman, I was experimenting with LSD and mescaline, and these experiences brought me in touch with my emotions. And they helped me have a spiritual connection that unfortunately, my bar mitzvah did not produce. When I wanted to tease my parents, I would tell them that they drove me to psychedelics because my bar mitzvah had failed to turn me into a man. But most importantly, psychedelics gave me this feeling of our shared humanity, of our unity with all life. And other people reported that same thing as well. And I felt that these experiences had the potential to help be an antidote to tribalism, to fundamentalism, to genocide and environmental destruction. And so I decided to focus my life on changing the laws and becoming a legal psychedelic psychotherapist. Now, half a century after the ban, we ’ re in the midst of a global renaissance of psychedelic research. Psychedelic psychotherapy is showing great promise for the treatment of post-traumatic stress disorder, or PTSD, depression, social anxiety, substance abuse and alcoholism and suicide. Psychedelic psychotherapy is an attempt to go after the root causes of the problems, with just relatively few administrations, as contrasted to most of the psychiatric drugs used today that are mostly just reducing symptoms and are meant to be taken on a daily basis. Psychedelics are now also being used as tools for neuroscience to study brain function and to study the enduring mystery of human consciousness. And psychedelics and the mystical experiences they produce are being explored for their connections between meditation and mindfulness, including a paper just recently published about lifelong zen meditators taking psilocybin in the midst of a meditation retreat and showing long-term benefits and brain changes. Now, how do these drugs work? Modern neuroscience research has demonstrated that psychedelics reduce activity in what ’ s known as the brain ’ s default mode network. This is where we create our sense of self. It ’ s our equivalent to the ego, and it filters all incoming information according to our personal needs and priorities. When activity is reduced in the default mode network, our ego shifts from the foreground to the background, and we see that it ' s just part of a larger field of awareness. It ' s similar to the shift that Copernicus and Galileo were able to produce in humanity using the telescope to show that the earth was no longer the center of the universe, but was actually something that revolved around the sun, something bigger than itself. For some people, this shift in awareness is the most important and among the most important experiences of their lives. They feel more connected to the world bigger than themselves. They feel more altruistic, and they lose some of their fear of death. Not all drugs work this way. MDMA, also known as Ecstasy, or Molly, works fundamentally different. And I ' ll be able to share with you the story of Marcela, who suffered from post-traumatic stress disorder from a violent sexual assault. Marcela and I were introduced in 1984, when MDMA was still legal, but it was beginning also to leak out of therapeutic circles. Marcela had tried MDMA in a recreational setting, and during that, her past trauma flooded her awareness and it intensified her suicidal feelings. During our first conversation, I shared that when MDMA is taken therapeutically, it can reduce the fear of difficult emotions, and she could help move forward past her trauma. I asked her to promise not to commit suicide if we were to work together. She agreed and made that promise. During her therapeutic sessions, Marcela was able to process her trauma more fluidly, more easily. And yet, she was able to tell that the rapist had told her that if she ever shared her story, he would kill her. And she realized that that was keeping her a prisoner in her own mind. So being able to share the story and experience the feelings and the thoughts in her mind freed her, and she was able to decide that she wanted to move forward with her life. And in that moment, I realized that MDMA could be very effective for treating PTSD. Now, 35 years later, after Marcela ' s treatment, she ' s actually a therapist, training other therapists to help people overcome PTSD with MDMA. Now, how does MDMA work? How did MDMA help Marcela? People who have PTSD have brains that are different from those of us who don ' t have PTSD. They have a hyperactive amygdala, where we process fear. They have reduced activity in the prefrontal cortex, where we think logically. And they have reduced activity in the hippocampus, where we store memories into long-term storage. MDMA changes the brain in the opposite way. MDMA reduces activity in the amygdala, increases activity in the prefrontal cortex and increases connectivity between the amygdala and the hippocampus to remit traumatic memories to move into long-term storage. Recently, researchers at Johns Hopkins published a paper in"Nature,"in which they demonstrated that MDMA releases oxytocin, the hormone of love and nurturing. The same researchers also did studies in octopuses, who are normally asocial, unless it ' s mating season. But lo and behold, you give them MDMA, and they become prosocial. Several months after Marcela and I worked together, the Drug Enforcement Administration moved to criminalize Ecstasy, having no knowledge of its therapeutic use. So I went to Washington, and I went into the headquarters of the Drug Enforcement Administration, and I filed a lawsuit demanding a hearing, at which psychiatrists and psychotherapists would be able to present information about therapeutic use of MDMA to try to keep it legal. And in the middle of the hearing, the DEA freaked out, declared an emergency and criminalized all uses of MDMA. And so the only way that I could see to bring it back was through science, through medicine and through the FDA drug development process. So in 1986, I started MAPS as a nonprofit psychedelic pharmaceutical company. It took us 30 years, till 2016, to develop the data that we needed to present to FDA to request permission to move into the large-scale Phase 3 studies that are required to prove safety and efficacy before you get approval for prescription use. Tony was a veteran in one of our pilot studies. According to the Veterans Administration, there ’ s over a million veterans now disabled with PTSD. And at least 20 veterans a day are committing suicide, many of them from PTSD. The treatment that Tony was to receive was three and a half months long. But during that period of time, he would only get MDMA on three occasions, separated by 12, 90-minute non-drug psychotherapy sessions, three before the first MDMA session for preparation and three after each MDMA session for integration. We call our treatment approach”inner-directed therapy,”in that we support the patient to experience whatever ’ s emerging within their minds or their bodies. Even with MDMA, this is hard work. And a lot of our **subjects** have said,”I don ’ t know why they call this Ecstasy.”During Tony ’ s first MDMA session, he lay on the couch, he had eyeshades on, he listened to music, and he would speak to the therapists, who were a male-female co-therapy team, whenever he felt that he needed to. After several hours, in a moment of calmness and clarity, Tony shared that he had realized his PTSD was a way of connecting him to his friends. It was a way of honoring the memory of his friends who had died. But he was able to shift and see himself through the eyes of his dead friends. And he realized that they would not want him to suffer, to squander his life. They would want him to live more fully, which they were unable to do. And so he realized that there was a new way to honor their memory, which was to live as fully as possible. He also realized that he was telling himself a story that he was taking opiates for pain. But actually, he realized, he was taking them for escape. So he decided he didn ’ t need the opiates anymore, he didn ’ t need the MDMA anymore, and he was dropping out of the study. That was seven years ago. Tony is still free of PTSD, has never returned to opiates and is helping others less fortunate than himself in Cambodia. The data that we presented to FDA from 107 people in our pilot studies, including Tony, showed that 23 percent of the people that received therapy without active MDMA no longer had PTSD at the end of treatment. This is really pretty good for this patient population. However, when you add MDMA, the results more than double, to 56 percent no longer having PTSD. But most importantly, once people learn that if they don ’ t need to suppress their trauma, but they can process it, they keep getting better on their own. So at the 12-month follow-up one year after the last treatment session, two- thirds no longer have PTSD. And of the one-third that do, many have clinically significant reductions in symptoms. On the basis of this data, the FDA has declared MDMA-assisted psychotherapy for PTSD a breakthrough therapy. FDA has also declared psilocybin a breakthrough therapy for treatment-resistant depression and just recently approved esketamine for depression. I ’ m proud to say that we have now initiated our Phase 3 studies. And if the results are as we hope, and if they ’ re similar to the Phase 2 studies, by the end of 2021, FDA will approve MDMA-assisted psychotherapy for PTSD. If approved, the only therapists who will be able to directly administer it to patients are going to be therapists that have been through our training program, and they will only be able to administer MDMA under direct supervision in clinic settings. We anticipate that over the next several decades, there will be thousands of psychedelic clinics established, at which, therapists will be able to administer MDMA, psilocybin, ketamine and other psychedelics to potentially millions of patients. These clinics can also evolve into centers where people can come for psychedelic psychotherapy for personal growth, for couples therapy or for spiritual, mystical experiences. Humanity now is in a race between catastrophe and consciousness. The psychedelic renaissance is here to help consciousness triumph. And now, if you all just look under your seats... Just joking! Thank you. Thank you. Corey Hajim : You ’ ve got to stay up here for a minute. Thank you so much, Rick. I guess it ' s a supportive audience. Rick Doblin : Yes, very. Many of them have also been to Burning Man. CH : There ' s some synergy. RD : CH : So, in your talk, you talked about using these drugs to address some pretty serious traumas. So what about some more common mental illnesses like anxiety and depression, and is that where microdosing comes in? RD : Well, microdosing can be helpful for depression, I do know someone that has been using it. But in general, for therapeutic purposes, we prefer macro-dosing rather than microdosing, in order to really help people deal with the root causes. Microdosing is more for creativity, for artistic inspiration, for focus... And it also does have a mood-elevation lift. But I think for serious illnesses, we ' d rather not get people thinking that they need a daily drug, but do more deeper, intense work. CH : And what about outside the United States and North America, is this research being done there? RD : Oh yeah, we ' re globalizing. Our Phase 3 studies are actually being done in Israel, Canada and the United States. So once we get approval in FDA, it will also become approved in Israel and in Canada. We ' re just starting research in Europe. And we ' re actually going to be training some therapists from China. CH : That ' s great. We were going to do an audience vote to see if people felt like this was a good idea to move forward with this research or not, but I have a feeling I know the answer to that, so... Thank you so much, Rick. RD : Thank you. Thank you all.

Text Sample 752: NEG Dim 5 – 2021 – Score -1.00 – 2021/t003544.txt

I love bugs. I think of them as nature ' s tiny engineers because they come up with the most extraordinary and incredible solutions to life ' s problems. And I just love observing them because they ' re so full of surprises and curiosities. During my career, I studied spiders that use their webs as a slingshot to capture prey deep in the Amazon rainforest, worms that tangle up with each other and form knots to form these shape-shifting blobs to survive in harsh environments such as sewers and caves, and tiny aquatic beetles that bring their own scuba gear when they dive, on their butts. But today, I ' m going to tell you a story about one of nature ' s most extraordinary engineers that pushes the limits of fluid mechanics and bioengineering and, arguably, solves their number one problem – how insects pee. A few years ago, my student, Elio Challita, and I observed this tiny insect having a private moment in our own backyards in Atlanta, and we couldn ' t believe our eyes. This insect was peeing for hours, and it was so quick we could barely see it. And we were **blown** away – we had never seen anything like this. We had never seen an insect pee. So today, I ' m going to tell you how insects pee, why they pee so much and in this way, and finally, why you should care about insect pee. So sit back and relax – you ' re in for a treat. So one of the first things we had to discuss is"Wait a second – insects pee?"And it turns out that almost all insects pee, in one form or another. It ' s a closed system – what goes in must come out. But the protagonist of today ' s story, Homalodisca vitripennis, or a glassy-winged sharpshooter – isn ' t it gorgeous? You can see where it gets its name from, it ' s got these transparent wings. This insect specializes in feeding on xylem fluid from plants. It ' s a sapsucker. But I think of it as a plant ' s mosquito. And just after a mosquito sucks your blood, it leaves behind a parting gift – so do these sharpshooters that spread bacteria into these plants, and it causes devastating and deadly diseases in plants, killing them. This is a huge problem in agriculture, including here in California, where it caused millions of dollars in damage to vineyards. And you can appreciate what they lack in size, they make up in numbers. This is what millions of insects feeding and peeing looks like. If only this woman knew where that water was coming from. Now, don ’ t worry, this insect ’ s pee is just water. And so for the first time, I ’ m going to show you this behavior, slowed down with our high-speed cameras, and this is what we discovered. We realized that this insect forms a droplet of pee, and then it flings them at extreme accelerations, of 40 g-forces. That ’ s 40 times faster than the sprint of a cheetah. These insects are really packing a punch from their butts. And we wanted to under – We wanted to take a closer look at this flicker, so we put this insect and took a look under a microscope at its business end, and this beautiful structure has a scientific name : it ’ s called a butt flicker. And this is what we discovered. We realized that this insect had evolved springs and latches, just like a catapult, so that it could efficiently hurl its droplets of pee, repeatedly, at these high accelerations. Now, we wanted to measure the speed at which this flicker was moving, and the droplets, so we measured the speed of both the droplets and the flicker. And this is where we made a puzzling observation. The speed of the droplets in the air was faster than the flicker. So if you take a ratio of that, we were expecting it to be 100 percent, but turns out that the speed of the droplets are about 150 to 200 [percent] faster than the flicker itself. This is why this is counterintuitive : imagine a Yankees baseball player throwing a ball at 100 miles an hour. At some point during that throw, their hands and fingers are moving at 100 miles an hour. Say an amateur ballplayer throws a ball at 50 miles an hour, and if you measure the speed of the ball and it ’ s 100 miles an hour in midair, we ’ d be surprised – where would that extra speed come from, right? This is exactly what these insects are doing. So to solve this puzzle, we went back and looked at our videos. And we ’ ll play this a couple of times – see if you can figure this out. Did you catch it? Let me grab a frame. We realized that unlike a baseball that ’ s rigid... due to surface tension, these tiny droplets are squishy, and we had an”aha”moment. We were wondering if this insect is storing energy due to the surface tension just before launch. And to test this, we did, naturally, what any of us would do – we converted our kitchen tables into a lab to test this. So now, we ’ re going to place droplets on a speaker, to squish them at high speeds, and this is what we discovered. We realized that water that flows in our faucets like a liquid, at these tiny scales, due to surface tension, with the right timing, can get a kick, store energy, and if you time it just right, you can launch these off at extremely high speeds, just like a child on a trampoline. And to test this idea of timing, we even built a catapult. Our should I say a”cata-pee”? And it reinforced this idea : too slow, the droplets don ’ t go off, and then if you move at the right speed, you ’ ve flung this. So I ’ ve told you today how these insects use this catapulting structure to store energy and surface tension and throw these droplets at record-breaking speeds to be, really, number one... at number one. But why have these insects evolved this remarkable ability? And there are two reasons for this. Number one is they are on a zero-calorie diet. These tiny bugs are feeding from the xylem fluid that comes up through the roots, through the soil, and goes to the rest of the plant. It ’ s very, very poor in nutrients – it ’ s just water and some minerals. This is unlike the phloem fluid, which is rich in sugar, coming through photosynthesis at the leaves and going through the rest of the plant. This is akin to a human purely sustaining themselves on a diet of diet lemonade. Very low energy source – you ’ d have to constantly drink, but you ’ d have to constantly pee as well. This still doesn ’ t explain why these insects pee in droplets, and not in jets, like you and I would, if you were to take a bio break. Let ’ s take a look at cicadas. These are the cousins of these sharpshooters. They ’ re also xylem fluid feeders... and they pee in jets. I love this video. This shows cicadas doing two of their favorite things when they come out of hibernation – singing at the top of their lungs and peeing in the wind. And the second reason why these insects pee in droplets is size – these things are tiny, they ’ re smaller than my pinky. In fact, the surface tension that enables them to store energy in these droplets to launch is actually an impediment, because gravity doesn ’ t matter, and surface tension sticks these droplets to their bodies. So they actually have to flick these droplets away – it ’ s actually very difficult for these tiny bugs to pee. And that ’ s why I just love studying bugs. This tiny engineer has figured out how to survive on barely just water through the xylem fluid. And it ’ s figured out to do so, it has to drink a lot, and pee a lot. In that sense, it ’ s not so different from other engineers I know on a Friday night at a bar. But to do so, it ’ s figured out it had to evolve this catapulting structure and fling these droplets at high speeds. And that ’ s why I always tell my students when we explore new bugs,”It ’ s not a bug, it ’ s a feature.”OK, so why should you care about insect pee, and what ’ s the practical application of this work? Maybe, maybe one day, these ideas, these principles could help us design more efficient water-ejector systems for our smartphones, our watches and hearing aids. But I have to be honest, that ’ s not really why I ’ ve obsessed over insect pee for the last four years. I do this because when I share my work with kids, it ignites their curiosity. When I talk about insect pee, it makes their eyes light up. They laugh, they run around in their backyards, looking at bugs and asking questions. It reminds them and us that we can discover marvelous in the mundane, even in our own backyards. We just have to look closely. Thank you so much. I ’ ve got to go pee.

Text Sample 753: NEG Dim 5 – 2021 – Score -1.00 – 2021/t003790.txt

So when I was born on September 26, 1988, my grandparents and great grandparents back in India didn ’ t find out for two weeks, which is a shame because I mean, look how perfect I was. And it ’ s not because the phone lines were down or because they weren ’ t available. It ’ s because there was a complication with my birth. And that complication was being assigned female at birth. You see, because my mom had been told that if she gave birth to a daughter, it wasn ’ t worth phoning home about. After all, she ’ d already given birth to my older sister, and this time everyone had high hopes that she would do right and have a son. But she didn ’ t. She had me. And so there were no congratulations or Indian sweets sent our way. Just the reality that from the moment I came into this world, I was already a disappointment to so many people. It ’ s as if they had a time machine and already knew the trajectory of my entire career and life and decided that I had less to offer. And it sucked. So why am I telling you this heavy story? I ’ m supposed to be a funny person. I have the nerve to come out here and hit you right in the feels. How dare I? I ’ m telling you this because although this is my lived experience, it ’ s also the reality that millions of girls face every day across every culture and in every country. And I ’ m telling you this because being born into this reality set me on a lifetime mission of trying to prove myself and just feel like I was enough. What did I want to be when I grow up? I wanted to be treated equally. And I ’ m not alone in this mission. In fact, us girls, what we desperately want is a seat at the table. It ’ s what every motivational poster, Tumblr post, Instagram account you follow, business card tells us : Success is a seat at the table. And if they want to be extra spicy, they say,”If there is no seat, drag your own seat.”I ’ m sure you ’ ve heard this, right? And so my marching orders were clear. Get a seat at this coveted table by any means necessary. And that ’ s been the driving force behind my entire career. Now, in 2010, I noticed that no one on YouTube looked like me. There was no South Asian woman who ' s very loud and uses her hands a lot, giving her take on the world. There was no me in front of a camera. I saw a seat up for grabs. So I got to work, and I started a channel under the name"Superwoman."Yeah, because although I ' m smart enough to do a TED Talk, I ' m not smart enough to understand copyright. I taught myself how to write, shoot and edit my own content. And I worked really hard. When I finally got the hang of it, I committed to posting two comedy videos a week. And I found success. With a backwards snapback on my head, I gave my take on relationships, pop culture, taboo **subjects** and, most popularly, dressed up like my parents. I can ' t tell you how many times I forgot to wipe that chest hair off. A lot of times. Now fast forward to 2015, and I ' m on stage in India announcing my first world tour. As fate would have it, the day after this monumental milestone, I was set to fly to Punjab, India to visit my grandfather for the first time in my adult life. And whoo, nothing could have prepared me for what was about to happen. I vividly remember it. I was in the car driving to his house. He was standing outside. I nervously got out of the car, walked up to him. He walked up to me, looked me right in the eyes, and he raised his hand and decorated me with a flower garland : a gesture fit for people of importance. He then proceeded to welcome me into his home, my mom by my side, and proceed to show me all the newspaper clippings he had saved with my name and face on them. He said the words he was wrong. Words I had never heard a man say before to me. He said that I had done what no one else could have done, and I had made the family name proud. Me, Lilly, the baby born a girl. That ' s right. Now in that moment I truly felt like Superwoman. I did. You know, through my YouTube videos, I ' ve amassed almost 15 million subscribers and three billion views. But more important than all of that, I managed to change one view. I challenged my grandfather ' s entrenched gender beliefs. And for the first time in my life, I remember thinking in that moment,"Oh, I finally got a seat at the table..."Hello, props."... alongside the men in the industry."I felt like that. And encouraged by my grandfather ' s approval, I became more confident in my influence. I remember thinking,"Oh, I ' m going to talk at this table, I ' m going to join the dinner conversation."You know, a lot of my male mentors make comments and posts about box office numbers and salaries and titles and those dollar dollar bills. So I thought, I ' m going to chime in here. I learned very quickly that whenever I spoke of money, people got a little uncomfortable. Like the time I pointed out the gender gap in the Forbes list for online creators, a list I ' d previously been on. I remember wanting to start a critical conversation because I saw this article and I was heartbroken. You know, the digital space had always been a place that I thought was without gatekeepers. And here it was looking just like old Hollywood. Well, let me tell you, the internet was not interested. I don ' t know how it ' s possible, but it literally felt as if Twitter leaped through my screen and body slammed me onto my desk. The message was clear : you can be on this list, but don ' t try to start any conversations about the inequality on this list. I have thousands and thousands of videos. One of my most disliked videos is why I ' m not in a relationship. Yeah. A lot of the men at the table did not like me telling them why I didn ' t need a boyfriend. I quickly learned that there ' s an invisible gatekeeper called culture, and the table is smack dab in the middle of it. Now, in 2019, I made history with my late-night show,"A Little Late with Lilly Singh."Thank you, thank you. There I was, Lilly, the baby born a brown girl, rubbing elbows, or at least time slots, with comedy royalty. And I got to give a huge shout out to NBC for boldly trying to break late-night tradition. I remember when the show came out, I remember all the articles because they looked practically identical."Bisexual Woman of Color Gets Late-Night Show."I almost legally changed my name to"bisexual woman of color”because that ’ s what people called me so often. And you know, as strange as that sentiment was, I thought, OK, the silver lining is that we ’ ll finally get a different perspective in late-night. A little bit of melanin, a dash of queer, a different take on things. Let ’ s do this! And I remember thinking,”Now, oh, now I ’ ve been invited to the big table. And now things will be different.”So I took my seat. Now, unfortunately, the budget wasn ’ t based on the importance or significance or historic nature of the show. It was based on the 1:30am time-slot that we had. So to say the budget was small, the writing staff even smaller, and to do the first season, I had to shoot 96 episodes of late-night television in three months. Audience : Whoo. LS : Whoo is right. To put that into perspective, that is shooting two to three episodes a day versus the network standard of one a day, maybe two on Thursday. We did it all with a writing staff of about half a dozen writers versus the network standard that ’ s about double that. Words cannot explain to you how exhausting, emotionally and spiritually challenging that was. And I started to feel like,”Hm, I think this chair ’ s a little wobbly.”Now, I think we can all agree that the beauty and magic of late-night is its timeliness. You know that no matter what ’ s happening in the world, you can turn on late-night television and hear all about it. But when you shoot 96 episodes in three months, you kind of lose that magic. I was the only show talking about hooking up, partying, cuddling, traveling, in front of a live audience during a literal global pandemic. Now, still, I thought, if the budget doesn ’ t celebrate the historicness of the show, then the creative can. I can bring some much needed spice to late-night. And sometimes I was successful. But other times, I would receive notes like,”Don ’ t be so loud.””Don ’ t be so big.””Don ’ t be so angry, smile more.”And my all time favorite,”Don ’ t overindex on the South Asian stuff.”After all, everyone else at the table who ’ s been sitting there for years, people are used to them. I might be a little jarring to audiences. Now, during season two of my show, I remember I went into overdrive. I found all the loopholes, I did all the necessary jobs to try to make the show more timely. And I was excited to. And I felt compelled to because for the first time in history, we had a woman, not to mention half South Asian woman, become vice president of the United States. Now we witnessed one of the greatest protests in human history with the farmer protests in India. And I was excited to finally give my take on these things. But my take was almost never included in topical media news coverage round-ups. You know, we still got the same voices, the same perspectives, even though someone and something different was literally in the next time slot. I kept trying to pull up my seat. I kept trying to join the dinner conversation. I kept trying to ask for a more supportive seat. But every time I would be told that I should be grateful to have a seat in the first place. After all, everyone else that looks like me is still waiting outside the restaurant in the cold. You know, the strange thing about having a wobbly seat is that you spend so much time trying to keep it upright that you can never bring your full self to the table. So now, why am I telling you all this? Well, because my therapist costs 200 dollars an hour and this is way cheaper. But also because I just experienced one of the most notorious boys ’ clubs ever in late-night television. And I ’ m here to offer solutions. I don ’ t always follow up a venting session with solutions, but when I do, it ’ s a TED Talk. That ’ s right. You see, my goal was always a seat at the table. It ’ s what women are conditioned to believe success is. And when the chair doesn ’ t fit, when it doesn ’ t reach the table, when it ’ s wobbly, when it ’ s full of splinters, we don ’ t have the luxury of fixing it or finding another one. But we try anyways. We take on that responsibility, and we shoulder that burden. Now, I ’ ve been fortunate enough to sit at a few seats, at a few different tables. And what I ’ ve learned is, when you get the seat, trying to fix the seat won ’ t fix the problem. Why? Because the table was never built for us in the first place. The solution? Build better tables. So, allow me to be your very own IKEA manual. I would like to present to you a set of guidelines I very eloquently call :"How to Build a Table that Doesn ' t Suck". I ' ve been told I ' m very literal. Now, right off the bat, let me tell you, this assembly is going to take more than one person or group of people. It ' s going to take everyone. Are you ready? Should we dive in? Let ' s do it. Up first, don ' t weaponize gratitude. Now, don ' t get me wrong, gratitude is a great word. It ' s nice, it ' s fluffy, a solid 11 points in Scrabble. However, let ' s be clear. Although gratitude feels warm and fuzzy, it ' s not a form of currency. Women are assigned 10 percent more work and spend more time on unrewarded, unrecognized and non-promotable tasks. Basically, what this means is all the things men don ' t want to do are being handed to women, and a lot of those things largely include things that advance inclusivity, equity and diversity in the workplace. So hear me when I say, a woman shouldn ' t be grateful to sit at a table. She should be paid to sit at a table. Especially ones she largely helped build. And a woman ' s seat shouldn ' t be threatened if she doesn ' t seem"grateful"enough. In other words, corporations, this step involves a woman doing a job and being paid in money, opportunity and promotion, not just gratitude. And women – yeah, go ahead, live it up, live your life. And women, a moment of real talk, trust me, I ' ve been there and I know it ' s so tough, but we have to understand and remember that being grateful and being treated fairly are not mutually exclusive. I can be grateful but still know exactly what I deserve. And that ' s the way to do it. Up next, invest in potential. When investing in women, don ' t invest in the 1:30am time slot. Invest in empowering something different. Invest in a new voice. Give them the support they actually need. Cultural change takes time and money. Heck, it took my grandfather 25 years to see that I was worthy of more. So a true investment is one that values potential over proof. Because so often that proof doesn ' t exist for women. Not because we aren ' t qualified, but because we haven ' t been given the opportunity. In other words, if you ' re trying to be inclusive, don ' t give someone new a seat made of straw until they prove they deserve a better one. Don ' t hold something called a"prove it again"bias, which requires less privileged people to constantly keep proving themselves, even though white men tend to get by on just their potential. Give them a seat that they can thrive in, that they can do the job you hired them to do in. Allow them to contribute to the table, and they will make it better. Up next, this is my favorite one. My favorite one, it ' s quite common sense, actually. Make space for us. You know, for every three men at a table, there ' s only one place setting for a woman. People are so used to more men showing up that they plan for it. There ' s an extra seat in the corner, there ' s a steak under the heat lamp. When more men show up, the table gets longer. But when that extra RSVP is a woman, more often than not she ' s encouraged to compete against the only other woman that was invited to the table. Instead, we need to build multiple seats for multiple women, not just one or two, so that women are not sitting on top of each other ' s laps, fighting for one meal. We already know that more diverse teams perform better. A recent study shows that corporations that have more gender diversity on their executive teams, were 25 percent more likely to experience above average profitability. And more racially diverse companies had 36 percent more profit. So really, no matter how you look at it, it ' s time to build longer tables and more seats. And I want to say something, and I want to admit something, I want to be vulnerable for a second. Because I ' ve fallen victim to this so many times, and women, let me know if you ' ve experienced this. We have to get rid of the scarcity mindset and champion each other, you know, because I ' ve learned what ' s the better win? Me sitting at a table or us sitting at a table?

Don ’ t be convinced to fight for one spot. Instead fight for multiple spots. And let me lead by example right now and say, I know there ’ s many other women that are going to come on this stage, and I hope they all nail it, and I will be cheering you all on because we can all win. And I ’ m going to be your biggest cheerleader when you ’ re up here. Last, it ’ s time to upgrade the table talk. Now, I believe stories make the world go around. You thought it had something to do the solar system? Joke ’ s on you, it ’ s stories. Stories are how we understand ourselves, how we understand others and how we understand the world. And arguably the most important stories are those we see in the media. Because we ’ ve seen time and time that they control the narrative and impact culture. Now, when it comes to genre, you can argue that certain genres have certain target demographics. When it comes to the world news, the target demographic is the world, and we know half of the world is female. Yet women and girls make up only a quarter of the people interviewed or that the news is even about in the first place. Instead, when it comes to issues that impact women, we not only need to be included in the coverage, we need to be driving those stories and dimensionalizing our own experience. Inviting everyone in on the table talk isn ’ t just a nice gesture. It makes for better, more productive, smarter conversation with more than one point of view. And that ’ s how you get better. So this all sounds like a lot of work. And it is. But I ’ m going to tell you why it ’ s necessary and worth it. To be honest, this is about so much more than just women in the workplace. In fact, I could probably come up with many more guidelines across many other industries. This is about creating a world where half of the population can thrive. You see, because the work we do today can create a world where future generations of girls can have equitable access and opportunity. And here ’ s the best part. Are you ready for it? Everyone listening today, all the men, the women, everyone in between, the big companies, the small ones, the media outlets, the people that snuck into the back, all of you, you can help create this future. A future where we have longer tables and more seats that actually work instead of fighting for a seat at the old ones that don ’ t. A future where everyone is seated at the table equally. And a future where being assigned female at birth is not a disappointment or a disadvantage, because girls are encouraged, empowered and expected to do great things. And I can ’ t wait to make that a reality. Thank you so much.

Text Sample 754: NEG Dim 5 – 2021 – Score -1.00 – 2021/t003745.txt

My name, Ndidi, means patience in Igbo. But I ’ m probably one of the most impatient people you ’ ve ever met. I was in such a rush to enter the world that I was born in the parking lot of the University of Nigeria Teaching Hospital. True story. Agriculture was my favorite **subject** in school. My mom remembers me squealing for joy when I ran home from school and saw green beans ready for harvesting. Or when I sold a bag of avocados in our local market. I considered studying agriculture in school, but instead opted to study business and started in corporate America. But many twists and turns led me back to my first love : consulting, working with entrepreneurs and then starting businesses. A defining moment for me was in 2007, during the first world food crisis in my adult life. I recognized how connected we were. Oil price shocks led to cereal price hikes, which affected bread prices all over the world. And the most vulnerable were hurt. Fourteen years later, my impatience has grown. We ’ re even more connected, and our food ecosystem is even more broken. Thankfully, the Rockefeller Foundation has some pioneering research on the true cost of food in the United States, which indicates that even though we spend 1. 1 trillion on our annual food expenditure, we actually spend 2. 1 trillion on costs linked to health and climate, all because of our broken food ecosystem. These same statistics have been replicated all over the world. In fact, according to the UN Food Systems Scientific Group, our food system is value-destroying. There ’ s a need for urgent action. From at least three lenses. A health lens, an equity lens and a climate lens. Starting with the climate lens, we must modify how we grow food and reduce food waste. Our food ecosystem is one of the largest contributors to climate change. We keep cutting down trees to grow more food, and we keep wasting food which ends up in landfills and rots, generates methane. The good news is that we have the technology and the science today to grow enough food to feed the world and to address our food waste problems. We have the knowledge, but we ’ re not using it. There are exciting examples from my ecosystem where we ’ re seeing dramatic impact. The Songhai Center in Benin Republic educates young Africans on regenerative agriculture and zero-waste total production. And one of these young people is doing just that. Through his company, Bioloop, he ’ s feeding waste, cassava peels, yam peels, to black soldier fly larva. They ’ re growing really quickly and becoming wonderful fish feed. The byproducts and the residue from this process is wonderful soil supplements. And the entire production process is run on renewable energy. Through my work with Sahel Consulting, we ’ re demonstrating that farmers can double and triple yields without hurting the environment. We ’ re using technology and science from our local research institutions, and we ’ re proving that in Africa, there are great examples for the rest of the world to emulate. Now we need to scale these interventions. And we need to ensure that our governments, our private sector, our farmers, are incentivized to change behavior and to improve lives. If you ’ re as impatient as I am, you also have a role to play, by reducing the food waste in your home. Every single person here can take up a policy to ensure that their schools, their companies, their civil society groups have a sustainability policy and a food waste policy. And please don ’ t tell your children to finish their dinner because there are hungry children anywhere. Tell them to finish their dinner because it ’ s good for the environment, and healthy food is good for them. Second, we must ensure that healthy food is affordable and accessible for the most vulnerable. This is a huge challenge. Unhealthy food kills. And we know this. One out of every five deaths is linked to unhealthy food. And yet, one third of our world ’ s population cannot afford a healthy diet. This is a big challenge. Now food is medicine. Healthy food gives us long and productive lives. And during COVID, we ’ ve seen the impact of closures, shipping challenges that have affected food prices, and the most vulnerable have had to shift from healthy diets to unhealthy diets because they are cheaper. This has caused more damage to lives all over the world. We must take a stand on this, and we can learn a lot from Africa. There Hadzabe people in Tanzania live in harmony with the land, and through their lives, we ’ ve seen the ability to have a healthy diet. They eat ten times more fiber than the average American. Oftentimes we don ’ t realize that even in our own traditional communities, we have so much to learn. In urban communities, we also have exciting social enterprises like mDoc that ’ s using digital technology, cell phones and training to get urban populations that are struggling with diabetes, high blood pressure and so many other health challenges linked to eating ultra-processed food to shift to more traditional diets, and they ’ re seeing measurable outcomes. We must scale these type of interventions, but we must also hold our private sector companies responsible for the amount of sugar and salt contained in food. We must set standards for what healthy food is and define healthy food according to plant-based diet, low salt, low sugar, and keep all of us accountable. We must also encourage our governments and ensure that at the local, state and federal level, our school feeding programs prioritize healthy food, our public procurement programs prioritize healthy food, and collectively, we ensure that we keep the standards high for everybody, every child. If you are as impatient as me, you must also set standards and hold those in your spheres of influence accountable for delivering unhealthy food for the most vulnerable. Third, we must support small- and medium-sized enterprises. In the food ecosystem, small- and medium-sized enterprises are the bedrock of our economy. They create jobs, they are innovative, and they can pivot very quickly. But during the pandemic, we ' ve seen something. They are most affected by shocks. The mortality rates of small- and medium-sized enterprises in the food ecosystem has been quite high. Now through my work, I ' ve seen the value and the power of small- and medium-sized enterprises. I ' m the co-founder of a food company called AACE Foods. Over 50 percent of our staff are women. Fifty percent of our board are women. And we produce healthy food, sourced from over 10, 000 farmers. Through this company, we ' re demonstrating that when you empower women, you empower communities. One product that we sell has a ripple effect through the entire ecosystem. Another company that ' s worthy of emulation is Twiga, using mobile money and cell phones to connect farmers to urban retailers. Now their efficiency and removing the middlemen, creating shorter value chains, ensures that not only the farmer benefits, the retailer benefits, but the consumer has access to healthier food. And finally, I ' m most inspired by these dynamic women entrepreneurs. Through Nourishing Africa, I work with enterprises in 37 African countries who are scaling sustainable businesses, building healthy food companies and demonstrating that strong, small- and medium-sized enterprises, committed to sustainable agriculture and healthy food, can become change agents. We need to support our small- and medium-sized enterprises, creating an enabling environment for them, providing catalytic financing, enabling them to scale, and supporting women-owned businesses, especially businesses run by young women. In the food ecosystem, we ' re often told to be patient when we plant the seed to let it grow. We ' re often told to let the vegetables simmer so that the juices will flow. We ' re told that”ndidiamaka,”patience is a virtue. But a wise woman once said,”For the dreamer, impatience is a virtue.”I am impatient about the current pace of change in the food ecosystem. And I think we all must be courageous and bold to transform this landscape. The next time you eat a meal, ask yourself a few questions. Who grew this food? Where was it grown? When was it grown? How many steps and stops did it make before it got to your table? How much food waste was generated because of this meal? Let your answers influence your next meal. The fact that you have a choice gives you privilege and also gives you a voice to demand that the solutions to the food ecosystem align with what works for people and planet. We must collectively create a food ecosystem that works not just for us but for future generations. Our children and grandchildren will hold us accountable for what we chose to do today to transform the food ecosystem. Thank you so much.

Text Sample 755: NEG Dim 5 – 2023 – Score -1.00 – 2023/t003532.txt
Whenever I tell people I work at a middle school, they often lean back and suck their teeth. It ' s like they ' re having a visceral reaction to the mere mention of those years, and it makes sense. Middle school is a time like no other. It ' s when significant biological, neurological and emotional changes are happening all at the same time. So how do middle schoolers respond to these changes? Well, some might ignore deodorant but overuse Axe body spray. You can find

them holding up the walls during the school dance. And there ’ s usually a desire to be treated like an adult, but they can ’ t quite let go of their action figures. You might imagine that it looks a little like this. This was me in sixth grade, and like many middle school students, I was earnest, I was goofy, and I was just discovering who I wanted to be. Now I had no idea that I ’ d go on to teach middle school. They say the grade levels you teach are most reflective of your personality. So I ’ m not quite sure what it says about me that I later went on to found a middle school for boys. But in all seriousness, it didn ’ t take long as a teacher to realize that my male students were acting kind of strange. I remember there was this one year, we were doing a get-to-know-you activity where students would use old magazines to create collages representing who they were. And many of the collages had all of the things that typical middle schoolers like : the outdoors, sports, the latest fashion, the hottest shoes, you know, all the important stuff. However, there were some that were not exactly what I had in mind. A group of middle school boys created these collages that were comical, if not concerning. It was almost as if they had made templates of who they thought that they should be. Girls in bikinis, fast cars, professional wrestlers, first-person shooter video games. You get the idea. One collage actually had to have had at least 25 different images of Kim Kardashian. And this wasn ’ t an isolated incident. Whether it was going down a somewhat sketchy YouTube rabbit hole or mindlessly indulging in meme culture, which we know can get really hairy really fast, I was noticing a pattern with my boys. Instead of chalking it up as mindless activities or typical middle school behavior, I decided to investigate. I became a mentor for an afterschool program called My Brother ’ s Keeper. And in this space, we could have more in-depth conversations. Inspired by the 2016 presidential debates, I asked this group of boys an age-old question :”Does absolute power corrupt absolutely?”Students began to discuss amongst themselves. And then I asked,”Now what would you do if you had this kind of power? And what if it was unchecked?”Students continued talking and then they shared out. Many of them said that they would use their power for good or even share it. And as I listened, I felt hopeful. Realizing that young men could take a different path. And then I brought the group together and I just said,”Does it have to be this way?”Their collective light bulbs lit up, realizing that they could reject this version of masculinity. And at the same time, I too had an”aha!”moment. It became clear to me that middle school boys are so impressionable and so full of potential. But what if I told you those same middle school boys could lead us to a more just and equitable society by redefining masculinity? Now in the days and weeks that followed, I continued to reflect on this idea. What actually is masculinity? If we reject the gender binary and affirm people of all genders, how does masculinity fit into that? What are the expectations of masculinity when it comes to race, class and other social factors? I knew that middle school is fertile ground for this work. And my reflection led me to identify three critical skills that middle schoolers can practice to redefine masculinity. I call them the three C ’ s. The first one is confidence, the second is communication, and the third is community. Now these three C ’ s stand as the pillars of my school to show people that middle schoolers can redefine masculinity. Now number one, confidence is essential to teach in middle school. Students are exploring their identities, and they ’ re more open to abstract thinking. I believe that having a healthy and balanced confidence allows boys to feel good about who they are rather than feeling uncomfortable for trying to be someone they ’ re not. It ’ s different than simply being praised or rewarded for achievements but rather rooted in a deep sense of self. And so what my school does is move away from either-or thinking. Instead of boys believing that they have to choose between being smart or athletic, poetic or pragmatic, we guide our boys to a more holistic version of masculinity that includes both-and. And as bell hooks and Olga Silverstein said, we need men who are empathetic and strong ; autonomous and connected ; responsible to self, friends, family, to community ; and capable of understanding how those responsibilities are ultimately inseparable. And from a purely academic standpoint, we provide opportunities to teach confidence through cross-curricular work and projects that include math, science, the humanities, art, home ec, sports to show them that deep learning and critical thinking often require an integration of all of these **subjects**. Teaching confidence allows boys to understand that there ' s an entire spectrum of how they can express themselves, and they can feel good and value the complexity of their identity and stand firm in it with confidence. The second C represents communication. Now communication is key. To counter the messages that society tells boys to constrict their emotions, my school practices a variety of communication methods, both intrapersonally and interpersonally. Now intrapersonally is how you communicate with yourself, and interpersonally is how you communicate with others. One example of our intrapersonal communication methods is we ' ll have students arrive at their desks at the beginning of the day. They ' ll close their eyes, breathe evenly for about a minute so they can just check in with themselves, see how they ' re doing, what they ' re thinking, how they ' re feeling. It allows them to put a frame around their thoughts and emotions so they can focus on it a little more deeply throughout the day. Students also keep gratitude journals. Research shows that when students express gratitude on a regular basis, it increases positive emotions toward themselves and toward others. An interpersonal practice that we like to do is at the end of the day, we ' ll have students gather in a circle to offer an appreciation for someone or something. Open it up to an apology or talk about a social issue that might be on their hearts. And we normalize these forms of communication to show boys that it is perfectly human to open your minds and your hearts to your community. Now the third C represents community, to counter this false sense of individualism and having to pull yourself up by your bootstraps. While we also know that there is great value in teaching our boys the importance of independence, it can be stymied when taken to an extreme. And so what we do is we engage in an inquiry process where we observe what ' s going on in our communities, either locally, nationally or globally, and then pose an essential question. One year we posed an essential question to our students that asked,"How can we create a community where everyone feels a sense of belonging?"Students took this question. They generated initial thoughts. They sought multiple perspectives from their peers, their teachers, community leaders. And then they came back to us and said,"We ' d like to address homelessness in Seattle."We loved this idea. And so we partnered with a local construction company to design, build and donate a tiny house. Now what we realized is through this community learning process, students felt a greater sense of satisfaction with studying, taking academic risks and just valuing the overall learning process. In addition, students felt more comfortable taking on leadership roles inside and outside of the classroom. One thing to note that in each of these three C ' s the adults involved modeled this new version of masculinity to prove to students that they don ' t have to fit into a stereotype. And while I ' m often impressed by the vulnerability and kindness of each of my students, it ' s still a middle school. It ' s this liminal space between childhood and adulthood. And amongst our sophisticated conversations, there ' s a lot of nonsensical humor. And very few of them have taken up regular usage of deodorant. And I know that this is lifelong work and there ' s no quick fix, but they show me that a better future is possible. What if masculinity meant having a healthy and balanced confidence, communicating clearly, being connected to your community? Just imagine how different our world would be. My students don ’ t even show me that this future is possible, but this future is here, with middle school boys leading the way. Thank you.

Text Sample 756: NEG Dim 5 – 2023 – Score -1.00 – 2023/t003545.txt
My name is Sharif El-Mekki. I ’ ve been an educator for almost 30 years. 10 as a classroom teacher, 16 as a principal. In the last few years, I ’ ve been a proud CEO for the Center for Black Educator Development. Between my time as a student and my career as an educator, I noticed a deep disconnect between the education of Black students — or what we should continue to call their miseducation — and their daily lived experiences. I grew up in the great city of Philadelphia. But I was fortunate because I did not experience the traditional American schooling system. My earliest memories of school was in a pre-K my mother started or helped to start, in Masjid Mujahideen. She insisted that faith-based institutions should not just be for the nourishment of adults, but they should also support the intellectual development of children. From this pre-K, I transitioned to Nidhamu Sasa, an African free school in the historic Germantown section of Philadelphia. From there we transitioned to Qom, Iran, where I attended Tabatabai Middle School, before returning for high school at Overbrook, an iconic and legendary school in West Philadelphia. The partnership between my home and my elementary school, Nidhamu Sasa, was grounded in the histories that stretched from the freedom schools of the north to the deep south, to enslavement, to reconstruction, to Jim Crow and redlining, to the Civil Rights, Black Power, and independent Black school movements. But it went deeper than that. My educational experience was rooted in how people on the continent of Africa viewed their relationship between teaching and learning, scholars and scholarship, and self-discipline, self-determination, and education. After Nidhamu Sasa, we went to Iran. Qom, Iran. And from there I gained an international perspective of how people viewed education and their children. From there, we returned to the States. And I enrolled at Overbrook High School. I loved my high school. And it was the first time I experienced low expectations as a student. It started with my placement test. Based on my score, my mother was encouraged to allow me to skip two grades. She declined — but she did let me skip one. I remember my sophomore year and cutting English class for almost a month. And sometimes I would run into my teacher in the hallway. And I was so nervous. We would lock eyes and she would look away and go about her way. But I also had Mr. Charles Mosley. One of the few Black teachers that I had in high school. He would connect history to his literature class. He had the audacity to believe that an English class should include the works of Black authors. One of the few times in high school that I experienced seeing myself in the curriculum. It was my high school experience that convinced me that there was something deeply troubling with the American education system and how I and other Black students were being educated. But it also helped propel me into a career and mission. My teaching career was grounded in the intellectual genealogy of Black educators who were immersed in Black history and Black culture, and Black social and political perspectives. I knew from intimate experiences the connection between activism and teaching Black students superbly ; the inextricable linkages between education and racial justice. You see today, in schools and classrooms around the country, educators use the foundations that were provided to them by educator prep programs. But most of these educator prep programs derive their understanding of the art and science of teaching and learning from white,

educational and behavioral theorists — Horace Mann, John Dewey, B. F. Skinner, Piaget, and Freud. Theorists who couldn ’ t even fathom that the majority of public school students today would be made up of students of color. So it goes without saying that they did not create these practices and educational frameworks with Black students ’ well-being and learning needs in mind. Today, almost 80 % of public school teachers are white. And often they enter America ’ s schools trained and armed with theories and practices that do not address Black students ’ needs or their lived experiences. But I believe there ’ s a better way. And it resembles how I was taught : with methods and mindsets that stretch from pre-colonial Africa to today. Dr. Greg Carr has taught about this extensively, and my friend Dr. Akosua Lesesne wrote her dissertation on it. She coined it the”Black Teaching Tradition.”The Black Teaching Tradition is a lineage of practices, values, and belief systems that stretch across time and centers how Black people have always taught and learned, regardless of the time and space. The Black Teaching Tradition is grounded in the humanity of students and is meant to support them specifically, but not exclusively. Because these practices and mindsets can be applied to the humanity of all students. In our work at the Center for Black Educator Development, we have 20 competencies that we believe fuel the healthy mindsets that educators must have in order to be effective educators of children and be able to serve and partner with communities. Behind these competencies are three core tenets : communities centered on care, content that is rich, rigorous, and accurate, and the courage to implement both, in often hostile environments. So, let ’ s start with community. This educator is not well known. But she should be considered one of the matriarchs of Black education and Black school founders. Her intellectual genealogy can be traced through the work of Black Educator Hall of Famers, who she mentored and inspired. Educators like Nannie Helen Burroughs, Charlotte Hawkins, and Mary McLeod Bethune. Lucy Craft Laney would convene with other educators to evolve and advance the education of Black children in America. And during that time, this was no easy feat. They weren ’ t hopping in Zoom rooms and DM-ing each other on social media, or hopping on jets. This was a slow and deliberate process. Lucy and her colleagues would come together to plot and plan for their people. For the education of the children and their students ’ grandchildren. Lucy and other educators were relying on word of mouth and handwritten letters to come together to do this very important work. And it ’ s because of her understanding of and proximity to the Black community, that Lucy and her colleagues understood the racial stress that Black children were often **subjected** to in school. And from this understanding they tailor their teaching, and design schools to accelerate student achievement. From Lucy Craft Laney and her protégés, hundreds of schools for Black children were opened and thousands of Black students were taught. From Lucy Craft Laney we can learn the importance and impact of communities centered on care, but also the ripple effect that it can have on generations to come. This next educator may be a little bit more well-known. His dedication to Black history led to Black History Month, formally known as Negro History Week. But what many people don ’ t realize is that this educator was fired from Howard University, an HBCU. At that time, talking about Black history and centering it in education was looked at as too radical. He should be considered as the father of Black history : none other than Dr. Carter G. Woodson. Dr. Carter G. Woodson believed that in order to be an effective educator of Black youth, you had to have both content mastery and understand the sociopolitical conditions that undermine their education. He believed that the content mastery and understanding of the historical importance of their contributions is what will allow Black students to thrive. The first part, content mastery, is pretty straightforward. Because how can you teach effectively if you ’ ve not mastered the content that you ’ re trying to teach students? But the second part, understanding the sociopolitical conditions and accurate Black history, is equally as important. Because how can you teach with an affirming lens if your understanding of their historical identity and contributions is minimized to enslavement? And how can you teach them and lead them effectively if you ’ re ignorant of the barriers that fence in their success? Dr. Carter G. Woodson teaches us that you can only be an effective educator if you have **subject** mastery and deep understanding of the context in which students are situated. Last, but definitely not least, Septima Clark was born to a laundry woman and a formerly enslaved father. Her parents were not educated, but they insisted on education, and Septima eventually became a well-respected teacher-activist in South Carolina. Her activism included work with the NAACP. And she participated in a class-action lawsuit against disparate pay between Black and white teachers. This eventually raised the ire of her employers, and they passed a policy where teachers could not belong to a civil rights organization. Septima and many Black educators had to decide between their activism and their employment. Septima chose courage. And she was eventually fired. But that did not stop her teaching. She continued teaching at the Highlander School in Tennessee. Rosa Parks participated in Septima ’ s workshops mere months before heading back to Montgomery to help lead the Montgomery Bus Boycott. In 1976, 20 years after being unjustly fired, Septima ’ s pension was reinstated by the governor of South Carolina. Septima teaches us today that courage may be educating despite anti-CRT legislation. And it may be at the risk of losing our teacher certification or our livelihood. And it may come at a great cost. But we also cannot continue to abide by the ramifications of staying silent. I invite everyone to join us in the Black Teaching Tradition. We know that teaching and centering communities of care, mastering content in the sociopolitical context that our students grow up in, can lead to better educational outcomes. And not just for Black students, but for all. Thank you.

Text Sample 757: NEG Dim 5 – 2023 – Score -1.00 – 2023/t003523.txt

Many years ago, I was finishing college and I was in that,”so what ’ s next?”phase of life, and I decided I wanted to see the world. And that was all well and good, but to do that I needed money and a job. So I headed off to my college ’ s advising center, thumbed through a Transitions Abroad magazine, and there it was. It was this ad that said something like,”Do you want to teach English abroad?”And I thought,”well, if that ’ ll get me abroad, then yeah, I guess perhaps I do.”So fast forward to several months later, I had a fellowship and I was off to Finland for a year to teach English to elementary school students. Now, before I got to Finland, I already knew that it was a really cool place, but what I didn ’ t realize was how linguistically diverse a place it was. Because I was on my way to teach elementary school students English, but those same students, unbeknownst to me, were already studying Finnish and Swedish and sometimes other languages in school. At the time I thought Finland must be some sort of multilingual anomaly. It couldn ’ t be the norm for elementary school students to be learning multiple languages, could it? Well, it turns out, globally, it often is. America really is more of a global standout, and not for good reason in this case. The US stands out due to its monolingualism in schools, which is in sharp contrast to the fact that this country is an incredibly linguistically diverse place. So, how do I know? Well, I ’ ve been working with and studying multilingual learners for over 20 years. I ’ ve been a language learner, I ’ ve taught English as a second language and English as a foreign language. And I ’ ve carried out language policy research in several different countries. And this is what I ’ ve realized : here in the US, we ’ re doing it wrong. America is not tapping into what I see as our greatest asset, which is our linguistic diversity. So you might be thinking, okay, but does this even matter? And if it matters, what do we do about it? So, spoiler alert : it matters. And I ’ m going to talk about what we can do about it. But first, I ’ d like you to come with me on a journey to an American public elementary school. Feel free to imagine your own from growing up, or maybe one near where you currently live, or maybe even one from TV. Now, let ’ s go inside. Once we step in, it ’ s immediately evident that this is a multilingual place. So, how do you know? Maybe there ’ s posters and decor in multiple languages. Maybe you hear conversations happening with adults and students in different languages, and maybe there ’ s even signs that explicitly say that multiple languages are welcomed and can be supported. Now let ’ s walk down the hall into a classroom. Inside, students are doing a math lesson in English. They ’ re working in small groups on word problems as the teacher walks around the class supporting the learning. And then, in Spanish, the teacher says it ’ s time to transition to a science lesson. She talks about the science content, hands out some papers for the day ’ s assignment. The students attentively listen, and then they begin to interact with each other and the new science content — all in Spanish. So one **subject** in English and the next **subject** in Spanish. Whoa. How does this feel? Full disclosure, to me, this is like a school-based multilingual dream come true. But I bet that this also might feel kind of unnerving or surprising or a little bit chaotic to some. And I think that those feelings may stem from the fact that this true embracing of multilingualism is very, very unusual in the US and it ’ s very uncommon to see in American schools. But here ’ s the thing, here in the US, we have tons of raw potential when it comes to language diversity. In fact, in 2022, Ethnologue ranked the US 5th of all countries in the world in terms of language diversity with 337 spoken languages. Similarly, nearly one quarter of students in American schools speak a home language other than English. This means that millions of students possess rich language skills, but we ’ re not encouraging the development of those skills alongside the development of English. And therefore we are not tapping into the linguistic diversity that exists among us and is already present. When students enter schools, their language use and English proficiency is screened, and if they ’ re in need of support, they ’ re provided it. But their home and heritage languages are often set aside due to long standing policies and practices that focus heavily on English. Now, don ’ t get me wrong — I am not here to tell you that students educated in America shouldn ’ t learn English. Well, there ’ s no official language in this country. That ’ s right. There is no official language in the US. English absolutely is a lingua franca and it ’ s an important sociocultural and economic tool for communicating in various settings. However, students ’ home and heritage languages are incredibly valuable too. And that is what we ’ re missing. Students are entering schools with rich language backgrounds and cultural experience and knowledge. And schools are often ill-equipped to support that, which means that these languages often go uncelebrated, or even worse, over time, they ’ re lost and forgotten. So it ’ s time that we stop leveraging dated policies and operating schools in a monolingual way. But there ’ s good news here. Because research shows that there ’ s a language development program model called dual language that is highly effective for all students. And by all, I mean all. There are different kinds of dual language programs, but the general gist is that participating students receive instruction in two languages, meaning that over time they become bilingual. Historically in the US, language policies and programs have often been subtractive, but dual language programs are additive, which means that they ’ re focused on developing, cultivating, and sustaining two languages. And research shows that these programs are really good for all kids. Not only are they the most effective type of instruction for students who speak a language other than English, but they also benefit native English speakers, and students from diverse racial and ethnic backgrounds, and students with special education needs and exceptionalities. Students enter these programs speaking one language and they come out of these programs speaking, listening, reading, and writing in two languages proficiently. Dual language programs work and we could implement them around the country. Because the fact of the matter is that being multilingual matters, both for the individual and for our society. Speaking multiple languages expands our worldview. It connects us to our own identity and links us to our communities, our cultures, and our families. And research shows that there ’ s these incredible cognitive benefits too. Brains function better and can do more when they ’ re multilingual. For instance, research indicates that bilingual brains are better than their monolingual counterparts at conflict management, and at executive functioning activities, like multitasking. I mean, the effects of this are incredibly far-reaching and impactful. It really is amazing, isn ’ t it? And it doesn ’ t end there, because there ’ s these economic and geopolitical benefits too. More and more jobs seek bilingual candidates, and being multilingual often leads to more lucrative job opportunities with global potential. Around the world, schools are producing bi- and trilingual individuals. But that ’ s notably uncommon here, which means that students educated in America are at a disadvantage globally in this way. As technology shrinks our world and leads to faster communication, and commerce, and political action, being multilingual can lead to increased opportunities for travel, study, work, and personal growth. So let ’ s go back to that journey we took. Because the fact of the matter is that doesn ’ t have to be a figment of our imagination. That journey took us to a dual language program, and what we need to do is build and grow these programs in order to build and grow multilingualism. Through dual language, we have the opportunity to cultivate languages, honor cultural and linguistic capital, and foster connections within schools and communities. So instead of thinking of it as a journey, let ’ s think of it as a foundation upon which to build better school programs that honor the language diversity that exists among us. It ’ s the first step towards celebrating and elevating multilingualism. Together, we can help students develop skills in more than one language. Students who will then go on to be multilingual adults, workers, travelers, community members, and global citizens. Thank you.

Text Sample 758: NEG Dim 5 – 2024 – Score -2.00 – 2024/t003155.txt

Chris Anderson : Cai is incredible, Cai ’ s language is Mandarin. Many of us do not speak Mandarin. But Cai agreed to do a wild experiment. For this talk, we collaborated with another TED speaker, Tom Graham, and his team at Metaphysic, who built an AI model to make Cai ’ s translator ’ s voice sound just like Cai speaking – just like he is speaking in English. So Sang is in a sound booth now. Sang, Hello. Sang : Hi, everyone. CA : OK, so it ’ s great to see you. Now let ’ s hear you sounding as if you were Cai. Sang : My name is Cai Guo-Qiang. CA : OK, nothing ’ s going to go wrong here. Nothing ’ s going to go wrong. The stage is yours. Cai Guo-Qiang : Thank you. More than 1, 000 years ago, when Chinese alchemists were developing elixir of immortality, one recipe caused an explosion. Boom. Boom. They named their discovery”fire medicine,”the Chinese word for gunpowder. From the very beginning, gunpowder has been about accidents, loss of control and destruction. But it ’ s also been about the healing power and unseen energies. I dreamed of becoming an artist when I was little, but like my father, who was an avid painter himself, I was cautious and timid. Caution is a fine quality in life, but not so in art. And Chinese society was also very controlling when I was young, so I longed for an artistic medium that could help me free myself and lose control. I came from an ancient city in southeast China called Quanzhou. The city had many firecracker factories when I was young, so it was easy to get gunpowder. When I first began using gunpowder to create art, I would lay out a canvas in the living room and set up small explosions on it. Seeing the canvas on fire one day, my grandmother threw a linen rag over the flame and put it out with a small puff. It was my grandma who taught me that while it ’ s important to light fires, it ’ s more important to know how to put them out. Over the decades, I ’ ve grown closer to gunpowder and mastered more techniques. My creations forever oscillate between destruction and construction, control and freedom, dictatorship and democracy. For example, here I first painted my imagination of Paradise, a mirage of temptations. I exploded colored gunpowder to create a sensual and dazzling garden, so beautiful that I didn ’ t want to lay a finger on it. However, I picked up my courage and scattered black powder all over this beauty. Covered it with a blank canvas and ignited again. Bang! When I removed the canvas on top, the once enchanting garden was now forever sealed beneath the black. However, what shocked me the most was the canvas on top, which now looked like an apparition of that heavenly garden. At the end of 1986, I moved to Japan. My cosmology, which till then was a simple one, developed by stargazing and studying feng shui in Quanzhou, suddenly expanded to include the latest developments in modern astrophysics. As a young artist from China, my grueling experiences with getting visas around the world inspired me to explode a chain of big footprints that traversed the Earth. The footprints would evoke extraterrestrials racing across several kilometers, bam bam bam, in only a few strides, ignoring artificial borders and disappearing into the distance. After decades of attempts around the world, this concept was finally realized as 29 footprint fireworks at the opening ceremony of the 2008 Beijing Olympics. The big footprints walked across the 15-kilometer central axis of Beijing, like an invisible giant in the sky. Witnessed by 1. 5... Witnessed by 1. 5 billion people in person and through live broadcasts, this work symbolized the era ’ s idealistic delusion of globalization. In the early ’ 90s, I conceived a work titled Sky ladder, a ladder made of fireworks that would connect heaven and Earth. I made many failed attempts to realize the idea over 21 years. The difficulty of the project lay in its technical requirements. We needed a helium balloon of over 6, 000 cubic meters in order to raise a ladder as high as the World Trade Center. Once up, the balloon could easily be **blown** away. And because it would carry lots of dangerous materials, we had to acquire numerous permits for land, sea and air. In my hometown, it ’ s said that 500 meters is the height of the clouds, so the ladder symbolized a useful dream of reaching for the stars and touching the clouds. I ’ ve created so much art around the world, but my grandma has never seen any of them in person, so I was determined to do something awesome for her to see. One! Fire! Cai Guo-Qiang : One morning in 2015, at the crack of dawn, a golden ladder rose into the sky. It was a birthday present for my grandmother, who turned 100 that year. She passed away one month later. In 1995, I moved from Japan to New York with my wife and daughter. After I came to New York, my work became more site-specific, addressing more socio-political themes and reflecting the changes I ’ ve developed from living in the West. This transition also allowed me to better thrive in different cultures around the world. Some have asked why I never deal with the **subject** of sex. I would often say,"Isn ' t explosion sexy in itself?"Ten years ago, I was invited to create an artwork in Paris. I decided to invite 50 couples from around the world to a sightseeing boat on the Seine, where they would first enjoy a 12-minute firework display that simulates the process of lovemaking. Why 12 minutes? Because that seems to be the average duration of French lovemaking. According to the internet. Excited by the passionate climax of the fireworks, the couples then entered individual tents to do whatever they wanted. When satisfied, they could press a button and trigger fireworks from a small boat nearby. I had prepared 300 shots of fireworks. However, the couples didn ' t use them all. In the end, the fireworks spelled out the words :"Sorry, gotta go."Nighttime fireworks are visible because of light and are more focused on the explosions themselves. Daytime fireworks rely on smoke. They are like a painter ' s brush moving across the sunlit sky in real time. In Shanghai, I realized the daytime fireworks"Elegy,"lamenting the severe environmental problems China faces. On the day, clouds loomed low over the Huangpu River, and the firework smoke lingered in the air long afterwards, like an ink painting with its gentle sorrow. For my solo exhibition at Uffizi Galleries, I created fireworks in the shapes of flowers and plants from Renaissance paintings. People from across the city could see the fireworks as they launched from the Michelangelo Square, reigniting the spirit of the Renaissance. Last year I realized the project"When the Sky Blooms with Sakura"in Fukushima, which suffered the earthquake and tsunami 12 years ago. On the June day, we had a rare collaboration of the wind and the waves to realize these daytime fireworks like a symphony of reverence for nature. I ' ve realized over 600 solo exhibitions and projects worldwide, often facing numerous challenges such as weather conditions, legal regulations and social-political hurdles. But such is the nature of my art. Behind the momentary magic lies countless unknown factors. Gunpowder and I have been travel companions on a 40-year-long, fantastical journey. Yet, I have never grown tired of it, thanks to its uncertainty and uncontrollability. And it ' s the same fascination with the unforeseeable that led to my research in artificial intelligence that began in 2017. This led to the launch of my AI Cai, my custom AI model. AI Cai deep learns from my artworks, archives and areas of interest. It also mimics contemporary and historical figures I admire developing distinct personas that can debate with each other, forming an independent and free community. AI Cai is my artwork, but it ' s also a partner for dialogue and collaboration. In the future, it may even create art by itself. Recently, we also enabled AI Cai to sprinkle and ignite gunpowder on canvases. Nowadays, if I burn through a canvas, it ' s usually an intentional loss of control for effect. But when AI Cai burns something, it ' s a genuine accident. Perhaps AI Cai is that rash, clumsy boy Cai. People often assume that I like fireworks, but what I really like are explosions. I like their energy and magic. Over the years, the goal of my gunpowder creations has never been political, but their results do carry political significance. At a recent Nobel Prize event, I said,"Using explosions to create beauty rather than warfare and violence provides a sliver of hope for our shared human future."Gunpowder helped me set my timid personality and liberate myself in a repressive society. Its uncertainty makes me both uneasy and exhilarated. That ' s similar to my interactions with AI. The unknown and uncontrollable aspects of AI are indeed unsettling. But today, as contemporary art seems weak and conservative, I hope that AI can help me unleash creativity, transcending the current dimensions of human cognition. Can AI reveal heavenly secrets and open a door to interspecies civilizations for us? If the disruptive nature of gunpowder can bring hope to people through the beauty of explosions, then can the unsettling power of AI do the same, bringing hope to mankind ' s future by expanding the unknown world? I ’ m always at the beginning of the next great journey. Thank you everyone, and thank you AI.

Text Sample 759: NEG Dim 5 – 2024 – Score -1.00 – 2024/t003118.txt

I love being a cartoonist because I can travel anywhere. I can visit historical artifacts and make improvements. I can voyage to mythical lands and solve problems. I can bring objects to life, and I can make those objects think and talk. And I can send those objects wherever I want them to go. I became a cartoonist to travel through space and time, and I became a graphic memoirist because the place I wanted to go was the past. I come from a legacy of dramatic stories and lost characters. My grandmother, Lily, on my mother ’ s side, was born in Warsaw, Poland, the oldest of four sisters. She was 13 in 1939, when Nazi bombs razed her home and her family was sealed to starve inside the Warsaw Ghetto. Eventually, her father encouraged her to slip through a hole in the wall, and she survived the Holocaust on her own, hiding her Jewish identity. This is the **subject** of my first book. I wondered : What did my grandmother ’ s lost home and lost family look like? Her parents, her grandmother and her sisters, they are all gone without a trace. My father ’ s parents were luckier. They were also Jewish, and they both fled Austria at the start of the war. My father ’ s father, Fred, was a pianist and conductor. In 1937, the year before the Nazis marched into Austria, he was 26, and he conducted a magnificent choral concert at a music hall in Vienna. A wealthy American woman in the audience was so impressed with his performance that she later agreed to sponsor his visa to the US. So music saved his life. But three decades later, Fred died of heart disease. I never met him. While alive, Fred meticulously preserved the documents of his life, a response to the threat of erasure he fled in Europe. And for decades after his father ’ s death, my father continued this preservation project. This is the **subject** of my second book. You might know my father, Ray Kurzweil, as an inventor and futurist. You should also know that he ’ s a person with an extraordinary sense of humor. [”Can I call you an Uber?””Sure.”] [”You ’ re an Uber.”] Good one dad. And although he ’ s dedicated his mind to the future, his life is full of the past. My father has worked for decades on natural language processing. And several years ago, he realized that if we married AI with my grandfather ’ s writing, we could build a chatbot that writes in my grandfather ’ s voice. Back in 2018, this seemed very sci-fi. But rather than ushering in our demise, this project helped me realize that AI could actually help us ward off annihilation by animating the legacies of our families and our cultures. I wanted to talk to my grandfather because he, like me, was an artist. I wondered : Could I get to know him? Could I even come to love him, even though our lifespans didn ’ t overlap? So I got involved. This chatbot needed language from my grandfather, as much as could be found. So I, with some assistance, set about finding his words and transcribing them. This was a selective chatbot, meaning it responded to questions with answers from the pool of sentences that Fred actually wrote at some point in his life. The more examples of Fred ’ s writing we could find, the more dynamic the experience of chatting with the bot would feel. Sometimes this transcription task proved challenging. But the more time I spent with the symbols of my grandfather ’ s life, the more easily I could decode them. Finally, after much anticipation, I sat down to chat with this new intelligence : an algorithm commanding over 600 typed pages of letters, lectures, notes, essays and other written documents from the grandfather I never met. When I asked about Fred ’ s dreams, he told me about the challenge of keeping his new orchestra afloat. When I asked about Fred ’ s anxieties, I learned about the stress of being a new father while working so hard. When I asked about the meaning of life, Fred wrote about the joy of working with other musicians in pursuit of beauty, and he wrote about the highest aims of art. I asked again about the meaning of life because isn ’ t that really the best question for a robot? And Fred ’ s second answer was much simpler, but even better. [”Love.”] Some of these answers felt familiar to me. I remembered seeing them in the archive, but the words gained impact through surprise and the role-play of conversation. I could identify patterns in my grandfather ’ s life and patterns across generations, because I was also an artist trying to make it in New York City. And I also believe the meaning of life is art and connection and love. I had wondered if this project would feel like a resurrection. But rather than bringing my grandfather from the past into the present, it felt like I was the one time traveling, visiting him for a moment at different points in his life. And this kind of time travel didn ’ t feel like sci-fi. It felt like the kind of imaginative travel I do when I ’ m cartooning. When I ’ m cartooning, I ’ m always thinking about how I could possibly represent a person fully. And the answer is : I can ’ t. Similarly, I know how many aspects of my grandfather can ’ t be captured by digital text alone. There ’ s all those quivers in his handwriting and what they denote about the sensations in his body. There ’ s his body, how it moved and how it felt. There ’ s his music and all the ineffable aspects of his performance. And of course, there ’ s everything he thought but didn ’ t write down. What would we have to do to be able to capture all of this? I may fail as an artist to fully represent a person ’ s constantly evolving complexity, but I can ask what features of a person are essential to who they are across a lifetime. The puzzle of personal identity is one of our oldest philosophical questions, so I ’ m not here to solve that one for you, I ’ m just a cartoonist after all. [Robot cat passes Turing test] I do believe that we, maybe not cats, but we are more than our bodies. That the projects and impressions we leave behind are a part of our essential selves. And I think AI has a special role to play in the mission of memory. I did not come to see the chatbot of my grandfather as replacing my grandfather. I came to see it as one way to interact with his legacy. As somebody who has spent their whole life trying to document people, I can assure you that people are much bigger and weirder than any one depiction or any one moment in time can possibly evoke. And I can also assure you that people don ’ t just disappear when they die. AI swirls our conception of time and space. It can remix and extend our identities. Our own digital archives are growing beyond belief, and we need a framework for understanding technologies of representation. So I offer you mine. Just like the comics I ’ ve drawn, about the characters in my life, these technologies are animated portraits. They are one part of our true immortal selves. Seen this way, AI, like cartooning and all good artistic endeavors, could help us appreciate the vastness of humanity – if we let it. Thank you.

Text Sample 760: NEG Dim 5 – 2024 – Score -1.00 – 2024/t003070.txt

I ’ m a couples therapist and an absolute romance fiend. I ’ m talking about everything from”The Notebook”to”Twilight”to a show some of you may remember called”The Flavor of Love.”It ’ s a reality competition show where the prize was the love of Flavor Flav. I think about relationships a lot, and something that comes up a lot in my work is this belief that relationships are hard. And we believe that due to one primary reason : our metric of success is based on what we ’ ve seen everyone else do. Imagine how you would honestly feel if you heard the following about another couple. OK, I said honestly. OK?”I heard they don ’ t even sleep in the same bed anymore.””They claim they never want to get married.””I don ’ t think they ever plan to live together.”Would you think to yourself”it sounds like they have some serious issues?”If we ’ re honest, a lot of us would. And it ’ s not because we ’ re not open-minded, but we ’ ve been taught that these are warning signs for a relationship in trouble. And while they can be for a lot of people, that is not always the case. Relationship experts have found that one of the primary obstacles that couples face are their own expectations. When we compare ourselves to societal norms, we can develop a sense of resentment toward our partner, as well as a sense of shame for how we ourselves are coming up short. Now, before we really get into this, I have to say that some of us have to reckon with the fact that we may be with the wrong person. And that will be clear if your deepest desire is that your partner change fundamental aspects of who they are. You really want them to be a different person. But if you ’ re confident that you ’ re with the right person and you just still feel frustrated and dissatisfied, we may find that rejecting everything we ’ ve known about good relationships is the key to actually having one. I work with couples every day, and I help them through relational crises. I remember I was working with an engaged couple for about a year, and when they first came to me, they said,”We ’ re 95 percent good. We just want to address the five percent.”And I hear something to this effect often when I first meet a couple. It turns out that five percent was more like 75 percent and increasing. They were struggling to make a blended family work. One partner had kids, the other one had never lived with kids before, and they moved in together after only knowing each other for three months. One time I went on vacation, and by the time I got back, they ’ d called off their wedding. But why? Their love was, honestly, it was evident, and they were not cruel to each other. Their issue was figuring out how to continue building their romantic relationship while also figuring out how to raise teenagers, who, to be honest, already had two very involved parents, they weren ’ t really in need of a third. After a particularly big **blow** up over chores and responsibilities, I finally asked a dangerous question. I said,”Do you think that living together has hurt or helped your relationship more?”We took a few weeks to explore that question, and they decided to test it out. They got a short-term lease on an apartment nearby for the partner who didn ’ t have kids. And we were really strategic. We made a contract. Let ’ s talk about dates. Let ’ s talk about expectations while you guys are living separately. And by the time they came back to me, I ’ d never seen them communicate so well. They said that they were looking forward to every weekend that they got to spend together. It felt like a vacation because they would spend the entire week planning their time together and savoring every moment they had together. They also found that their individual relationships with the kids drastically improved, without the pressure of trying to transition them into an entirely new household dynamic, especially when they only had a couple years left in the house. So at this point, some of you may be asking yourselves,”What kind of couples therapist recommends that couples live apart?”That ’ s a fair question. And to be honest, for a majority of my clients, this solution would not work. And that is the point. When we ’ re thinking about our relationships, we have to avoid focusing on what is normal. There ’ s no such thing as normal when we ’ re talking about two unique individuals with their own backgrounds and their own values. For this particular couple, they had to figure out a way to separate their romantic relationship from what really boiled down to roommate issues. And they had a circumstance that supported the option to live apart. One conflict that comes up a lot in my work is the difference in values between arriving on time and arriving looking and feeling your best. Neither one is wrong. But I had a great model for this with my parents. When I was growing up, we drove absolutely everywhere separately. Everywhere. You know, if you ’ re going to be a little bit late, you would ride with my mom and if you ’ re arriving on time, you ’ d go with my dad. They had two minivans for only two kids. OK. We didn ’ t go anywhere together. And one time when I was about 12 years old, one of my closest friends finally worked up the courage to ask me about it. And I could tell she was so nervous. Like I was about to reveal to her that my parents were secretly separated, and she just figured it out. You know, now that I think about it, I bet her mom put her up to this. What ’ s interesting is that her parents did go on to get divorced, and my parents stayed together for 23 years before my mom passed away. Now do I think that ’ s due to them commuting separately? Of course not. But I think it shows us two things. First, it shows us that any deviation from the norm can be met with curiosity and even judgment. It also shows us that sometimes, when we decide to do things a little differently, we can avoid the difference between having a really challenging day as a couple or a smooth day, by simply accepting our differences not as a couple, but as individuals. Instead of trying to change our partners, what if we instead embraced their differences, our difference in values, and release the pressure of doing what everyone else is doing? It ’ s OK to be a stay-at-home dad. It ’ s OK if you prefer to travel without each other. It ’ s OK if you need to have your own bedrooms so you can maintain personal space and be sane for each other. It ’ s OK if you want to break tradition and create a new last name. It ’ s OK if you want to share your love on social media, but it ’ s also OK if you want to protect it from public opinion. It ’ s OK if you ’ re in a season of life where you both just cannot prioritize sex. It ’ s OK if people are confused about your relationship. It was never theirs to understand in the first place. If we continue to accept the narrative that relationships are hard, then we ’ ll continue to do nothing about it. If our relationships feel hard, I encourage us to reflect on what is hard about it. Is it really the relationship or is it external factors like our own personal trauma histories or work stress? If it really is your relationship, let ’ s really think about what you and your partner are willing to do differently to enjoy it again. I want us to reject everything we ’ ve ever known about relationships, and challenge ourselves to create a relationship that not only defies expectations but honors the peculiarities that make us, us. Thank you.

Text Sample 761: NEG Dim 5 – 2018 – Score -1.00 – 2018/t000033.txt

I have a tendency to assume the worst, and once in a while, this habit plays tricks on me. For example, if I feel unexpected pain in my body that I have not experienced before and that I cannot attribute, then all of a sudden, my mind might turn a tense back into heart disease or calf muscle pain into deep vein thrombosis. But so far, I haven ’ t been diagnosed with any deadly or incurable disease. Sometimes things just hurt for no clear reason. But not everyone is as lucky as me. Every year, more than 50 million people die worldwide. Especially in high-income economies like ours, a large fraction of deaths is caused by slowly progressing diseases : heart disease, chronic lung disease, cancer, Alzheimer ’ s, diabetes, just to name a few. Now, humanity has made tremendous progress in diagnosing and treating many of these. But we are at a stage where further advancement in health cannot be achieved only by developing new treatments. And this becomes evident when we look at one aspect that many of these diseases have in common : the probability for successful treatment strongly depends on when treatment is started. But a disease is typically only detected once symptoms occur. The problem here is that, in fact, many diseases can remain asymptomatic, hence undetected, for a long period of time. Because of this, there is a persisting need for new ways of detecting disease at early stage, way before any symptoms occur. In health care, this is called screening. And as defined by the World Health Organization, screening is"the presumptive identification of unrecognized disease in an apparently healthy person, by means of tests... that can be applied rapidly and easily..."That ' s a long definition, so let me repeat it : identification of unrecognized disease in an apparently healthy person by means of tests that can be applied both rapidly and easily. And I want to put special emphasis on the words"rapidly"and"easily"because many of the existing screening methods are exactly the opposite. And those of you who have undergone colonoscopy as part of a screening program for colorectal cancer will know what I mean. Obviously, there ' s a variety of medical tools available to perform screening tests. This ranges from imaging techniques such as radiography or magnetic resonance imaging to the analysis of blood or tissue. We have all had such tests. But there ' s one medium that for long has been overlooked : a medium that is easily accessible, basically nondepletable, and it holds tremendous promise for medical analysis. And that is our breath. Human breath is essentially composed of five components : nitrogen, oxygen, carbon dioxide, water and argon. But besides these five, there are hundreds of other components that are present in very low quantity. These are called volatile organic compounds, and we release hundreds, even thousands of them every time we exhale. The analysis of these volatile organic compounds in our breath is called breath analysis. In fact, I believe that many of you have already experienced breath analysis. Imagine : you ' re driving home late at night, when suddenly, there ' s a friendly police officer who asks you kindly but firmly to pull over and **blow** into a device like this one. This is an alcohol breath tester that is used to measure the ethanol concentration in your breath and determine whether driving in your condition is a clever idea. Now, I ' d say my driving was pretty good, but let me check. 0. 0, so nothing to worry about, all fine. Now imagine a device like this one, that does not only measure alcohol levels in your breath, but that detects diseases like the ones I ' ve shown you and potentially many more. The concept of correlating the smell of a person ' s breath with certain medical conditions, in fact, dates back to Ancient Greece. But only recently, research efforts on breath analysis have skyrocketed, and what once was a dream is now becoming reality. And let me pull up this list again that I showed you earlier. For the majority of diseases listed here, there ' s substantial scientific evidence suggesting that the disease could be detected by breath analysis. But how does it work, exactly? The essential part is a sensor device that detects the volatile organic compounds in our breath. Simply put : when exposed to a breath sample, the sensor outputs a complex signature that results from the mixture of volatile organic compounds that we exhale. Now, this signature represents a fingerprint of your metabolism, your microbiome and the biochemical processes that occur in your body. If you have a disease, your organism will change, and so will the composition of your exhaled breath. And then the only thing that is left to do is to correlate a certain signature with the presence or absence of certain medical conditions. The technology promises several undeniable benefits. Firstly, the sensor can be miniaturized and integrated into small, handheld devices like this alcohol breath tester. This would allow the test to be used in many different settings and even at home, so that a visit at the doctor ' s office is not needed each time a test shall be performed. Secondly, breath analysis is noninvasive and can be as simple as **blowing** into an alcohol breath tester. Such simplicity and ease of use would reduce patient burden and provide an incentive for broad adoption of the technology. And thirdly, the technology is so flexible that the same device could be used to detect a broad range of medical conditions. Breath analysis could be used to screen for multiple diseases at the same time. Nowadays, each disease typically requires a different medical tool to perform a screening test. But this means you can only find what you ' re looking for. With all of these features, breath analysis is predestined to deliver what many traditional screening tests are lacking. And most importantly, all of these features should eventually provide us with a platform for medical analysis that can operate at attractively low cost per test. On the contrary, existing medical tools often lead to rather high cost per test. Then, in order to keep costs down, the number of tests needs to be restricted, and this means that the tests can only be performed on a narrow part of the population, for example, the high-risk population ; and that the number of tests per person needs to be kept at a minimum. But wouldn ' t it actually be beneficial if the test was performed on a larger group of people, and more often and over a longer period of time for each individual? Especially the latter would give access to something very valuable that is called longitudinal data. Longitudinal data is a data set that tracks the same patient over the course of many months or years. Nowadays, medical decisions are often based on a limited data set, where only a glimpse of a patient ' s medical history is available for decision-making. In such a case, abnormalities are typically detected by comparing a patient ' s health profile to the average health profile of a reference population. Longitudinal data would open up a new dimension and allow abnormalities to be detected based on a patient ' s own medical history. This will pave the way for personalized treatment. Sounds pretty great, right? Now you will certainly have a question that is something like,"If the technology is as great as he says, then why aren ' t we using it today?"And the only answer I can give you is : not everything is as easy as it sounds. There are technical challenges, for example. There ' s the need for extremely reliable sensors that can detect mixtures of volatile organic compounds with sufficient reproducibility. And another technical challenge is this : How do you sample a person ' s breath in a very defined manner so that the sampling process itself does not alter the result of the analysis? And there ' s the need for data. Breath analysis needs to be validated in clinical trials, and enough data needs to be collected so that individual conditions can be measured against baselines. Breath analysis can only succeed if a large enough data set can be generated and made available for broad use. If breath analysis holds up to its promises, this is a technology that could truly aid us to transform our health care system — transform it from a reactive system where treatment is triggered by symptoms of disease to a proactive system, where disease detection, diagnosis and treatment can happen at early stage, way before any symptoms occur. Now this brings me to my last point, and it ' s a fundamental one. What exactly is a disease? Imagine that breath analysis can be commercialized as I describe it, and early detection becomes routine. A problem that remains is, in fact, a problem that any screening activity has to face because, for many diseases, it is often impossible to predict with sufficient certainty whether the disease would ever cause any symptoms or put a person ' s life at risk. This is called overdiagnosis, and it leads to a dilemma. If a disease is identified, you could decide not to treat it because there ' s a certain probability that you would never suffer from it. But how much would you suffer just from knowing that you have a potentially deadly disease? And wouldn ' t you actually regret that the disease was detected in the first place? Your second option is to undergo early treatment with the hope for curing it. But often, this would not come without side effects. To be precise : the bigger problem is not overdiagnosis, it ' s overtreatment, because not every disease has to be treated immediately just because a treatment is available. The increasing adoption of routine screening will raise the question

: What do we call a disease that can rationalize treatment, and what is just an abnormality that should not be a source of concern? My hopes are that routine screening using breath analysis can provide enough data and insight so that at some point, we ’ ll be able to break this dilemma and predict with sufficient certainty whether and when to treat at early stage. Our breath and the mixture of volatile organic compounds that we exhale hold tremendous amounts of information on our physiological condition. With what we know today, we have only scratched the surface. As we collect more and more data and breath profiles across the population, including all varieties of gender, age, origin and lifestyle, the power of breath analysis should increase. And eventually, breath analysis should provide us with a powerful tool not only to proactively detect specific diseases but to predict and ultimately prevent them. And this should be enough motivation to embrace the opportunities and challenges that breath analysis can provide, even for people that are not part-time hypochondriacs like me. Thank you.

Text Sample 762: NEG Dim 5 – 2018 – Score -1.00 – 2018/t000037.txt

Do you hear that? Do you know what that is? Silence. The sound of silence. Simon and Garfunkel wrote a song about it. But silence is a pretty rare commodity these days, and we ’ re all paying a price for it in terms of our health â a surprisingly big price, as it turns out. Luckily, there are things we can do right now, both individually and as a society, to better protect our health and give us more of the benefits of the sounds of silence. I assume that most of you know that too much noise is bad for your hearing. Whenever you leave a concert or a bar and you have that ringing in your ears, you can be certain that you have done some damage to your hearing, likely permanent. And that ’ s very important. However, noise affects our health in many different ways beyond hearing. They ’ re less well-known, but they ’ re just as dangerous as the auditory effects. So what do we mean when we talk about noise? Well, noise is defined as unwanted sound, and as such, both has a physical component, the sound, and a psychological component, the circumstances that make the sound unwanted. A very good example is a rock concert. A person attending the rock concert, being exposed to 100 decibels, does not think of the music as noise. This person likes the band, and even paid a hundred dollars for the ticket, so no matter how loud the music, this person doesn ’ t think of it as noise. In contrast, think of a person living three blocks away from the concert hall. That person is trying to read a book, but cannot concentrate because of the music. And although the sound pressure levels are much lower in this situation, this person still thinks of the music as noise, and it may trigger reactions that can, in the long run, have health consequences. So why are quiet spaces so important? Because noise affects our health in so many ways beyond hearing. However, it ’ s becoming increasingly difficult to find quiet spaces in times of constantly increasing traffic, growing urbanization, construction sites, air-conditioning units, leaf blowers, lawnmowers, outdoor concerts and bars, personal music players, and your neighbors partying until 3am. Whew! In 2011, the World Health Organization estimated that 1. 6 million healthy life years are lost every year due to exposure to environmental noise in the Western European member states alone. One important effect of noise is that it disturbs communication. You may have to raise your voice to be understood. In extreme cases, you may even have to pause the conversation. It ’ s also more likely to be misunderstood in a noisy environment. These are all likely reasons why studies have found that children who attend schools in noisy areas are more likely to lag behind their peers in academic performance. Another very important health effect of noise is the increased risk for cardiovascular disease in those who are exposed to relevant noise levels for prolonged periods of time. Noise is stress, especially if we have little or no control over it. Our body excretes stress hormones like adrenaline and cortisol that lead to changes in the composition of our blood and in the structure of our blood vessels, which have been shown to be stiffer after a single night of noise exposure. Epidemiological studies show associations between the noise exposure and an increased risk for high blood pressure, heart attacks and stroke, and although the overall risk increases are relatively small, this still constitutes a major public health problem because noise is so ubiquitous, and so many people are exposed to relevant noise levels. A recent study found that US society could save 3. 9 billion dollars each year by lowering environmental noise exposure by five decibels, just by saving costs for treating cardiovascular disease. There are other diseases like cancer, diabetes and obesity that have been linked to noise exposure, but we do not have enough evidence yet to, in fact, conclude that these diseases are caused by the noise. Yet another important effect of noise is sleep disturbance. Sleep is a very active mechanism that recuperates us and prepares us for the next wake period. A quiet bedroom is a cornerstone of what sleep researchers call"a good sleep hygiene."And our auditory system has a watchman function. It ' s constantly monitoring our environment for threats, even while we ' re sleeping. So noise in the bedroom can cause a delay in the time it takes us to fall asleep, it can wake us up during the night, and it can prevent our blood pressure from going down during the night. We have the hypothesis that if these noise-induced sleep disturbances continue for months and years, then an increased risk for cardiovascular disease is likely the consequence. However, we are often not aware of these noise-induced sleep disturbances, because we are unconscious while we ' re sleeping. In the past, we ' ve done studies on the effects of traffic noise on sleep, and research **subjects** would often wake up in the morning and say,"Ah, I had a wonderful night, I fell asleep right away, never really woke up."When we would go back to the physiological signals we had recorded during the night, we would often see numerous awakenings and a severely fragmented sleep structure. These awakenings were too brief for the **subjects** to regain consciousness and to remember them during the next morning, but they may nevertheless have a profound impact on how restful our sleep is. So when is loud too loud? A good sign of too loud is once you start changing your behavior. You may have to raise your voice to be understood, or you increase the volume of your TV. You ' re avoiding outside areas, or you ' re closing your window. You ' re moving your bedroom to the basement of the house, or you even have sound insulation installed. Many people will move away to less noisy areas, but obviously not everybody can afford that. So what can we do right now to improve our sound environment and to better protect our health? Well, first of all, if something ' s too loud, speak up. For example, many owners of movie theaters seem to think that only people hard of hearing are still going to the movies. If you complain about the noise and nothing happens, demand a refund and leave. That ' s the language that managers typically do understand. Also, talk to your children about the health effects of noise and that listening to loud music today will have consequences when they ' re older. You can also move your bedroom to the quiet side of the house, where your own building shields you from road traffic noise. If you ' re looking to rent or buy a new place, make low noise a priority. Visit the property during different times of the day and talk to the neighbors about noise. You can wear noise-canceling headphones when you ' re traveling or if your office has high background noise levels. In general, seek out quiet spaces, especially on the weekend or when you ’ re on vacation. Allow your system to wind down. I, very appropriately for this talk, attended a noise conference in Japan four years ago. When I returned to the United States and entered the airport, a wall of sound hit me. This tells you that we don ’ t realize anymore the constant degree of noise pollution we ’ re exposed to and how much we could profit from more quiet spaces. What else can we do about noise? Well, very much like a carbon footprint, we all have a noise footprint, and there are things we can do to make that noise footprint smaller. For example, don ’ t start mowing your lawn at 7am on a Saturday morning. Your neighbors will thank you. Or use a rake instead of a leaf blower. In general, noise reduction at the source makes the most sense, so whenever you ’ re looking to buy a new car, air-conditioning unit, blender, you name it, make low noise a priority. Many manufacturers will list the noise levels their devices generate, and some even advertise with them. Use that information. Many people think that stronger noise regulation and enforcement are good ideas, even obvious solutions, perhaps, but it ’ s not as easy as you may think, because many of the activities that generate noise also generate revenue. Think about an airport and all the business that is associated with it. Our research tells politicians at what noise level they can expect a certain health effect, and that helps inform better noise policy. Robert Koch supposedly once said,”One day, mankind will fight noise as relentlessly as cholera and the pest.”I think we ’ re there, and I hope that we will win this fight, and when we do, we can all have a nice, quiet celebration. Thank you.

Text Sample 763: NEG Dim 5 – 2018 – Score -1.00 – 2018/t000041.txt

I want to share with you a moment in my life when the hurt and wounds of racism were both deadly and paralyzing for me. And I think what I ’ ve learned can be a source of healing for all of us. When I was 17 years old, I was a college student at Tuskegee University, and I was a worker in the Southern freedom movement, which we call the Civil Rights Movement. During this time, I met another young 26-year-old, white seminary and college student named Jonathan Daniels, from Cambridge, Massachusetts. He and I were both part of a generation of idealistic young people, whose life has been ignited by the freedom fire that ordinary black people were spreading around the nation and throughout the South. We had come to Lowndes County to work in the movement. And it was a nonviolent movement to redeem the souls of America. We believe that everyone, both black and white, people in the South, could find a redemptive pathway out of the stranglehold of racism that had gripped them for more than 400 years. And on a hot, summer day in August, Jonathan and I joined a demonstration of local young black people, who were protesting the exploitation [of] black sharecroppers by rich land holders who cheated them out of their money. We decided to demonstrate alongside them. And on the morning that we showed up for the demonstration, we were met with a mob of howling white men with baseball bats, shotguns and any weapon that you could imagine. And they were threatening to kill us. And the sheriff, seeing the danger that we faced, arrested us and put us on a garbage truck and took us to the local jail, where we were put in cells with the most inhumane conditions you can imagine. And we were threatened by the jailers with drinking water that came from toilets. We were finally released on the sixth day, without any knowledge, without any forewarning. Just out of the clear blue sky, we were made to leave. And we knew that this was a dangerous sign, because Goodman, Schwerner and Chaney had also been forced to leave jail and were murdered because no one knew what had happened to them. And so, despite our fervent resistance, the sheriff made us leave the jail, and of course, nobody was waiting for us. It was hot, one of those Southern days where you could literally feel the pavement — the vapor seeping out of the pavement. And the group of about 14 of us selected Jonathan Daniels, Father Morrisroe, who had recently come to the county, Joyce Bailey, a local 17-year-old girl and I to go and get the drinks. When we got to the door, a white man was standing in the doorway with a shotgun, and he said,"Bitch, I ' ll **blow** your brains out!"And before I could even react, before I could even process what was going on, Jonathan intentionally pulled my blouse, and I fell back, thinking that I was dead. And in that instant, when I looked up, Jonathan Daniels was standing in the line of fire, and he took the blast, and he saved my life. I was so traumatized and paralyzed by that event, where Tom Coleman deliberately, with malicious intent, killed my beloved friend and colleague, Jonathan Myrick Daniels. On that day, which was one of the most important days in my life, I saw both love and hate coming from two very different white men that represented the best and the worst of white America. So deep was my hurt at seeing Tom Coleman murder Jonathan before my eyes, that I became a silent person, and I did not speak for six months. I finally learned to touch that hurt in me as I became older and began to talk about the Southern freedom movement, and began to connect my stories with the stories of my other colleagues and freedom fighters, who, like me, had faced deadly trauma of racism, and who had lost friends along the way, and who themselves have been beaten and thrown in jail. It is 50 years later. Many people were beaten and thrown in jail. Others were murdered like Jonathan Daniels. And yet, we are still, as a nation, mired down in the quicksand of racism. And everywhere I go around the nation, I see and hear the hurt. And I ask people everywhere,"Tell me, where does it hurt?"Do you see and feel the hurt that I see and feel? I feel and see the hurt in black and brown people who every day feel the vicious volley of racism and every day have their civil and human rights stripped away. And the people who do this use stereotypes and myths to justify doing it. Everywhere I go, I see and hear women who speak out against — who speak out against men who invade our bodies. These same men who then turn around — the same men who promote racism and then turn around and steal our labor and pay us unequal wages. I hear and feel the hurt of white men at the betrayal by the same powerful white men who tell them that their skin color is their ticket to a good life and power, only to discover, as the circle of whiteness narrows, that their tickets have expired and no longer carry first- class status. Now that we ' ve touched the hurt, we must ask ourselves,"Where does it hurt and what is the source of the hurt?"I propose that we must look deeply into the culture of whiteness. That is a river that drowns out all of our identities and drowns us in false uniformity to protect the status quo. Notice, everybody, I said culture of whiteness, and not white people. Because in my estimation, the problem is not white people. Instead, it is the culture of whiteness. And by culture of whiteness, I mean a systemic and organized set of beliefs, values, canonized knowledge and even religion, to maintain a hierarchical, over-and-against power structure based on skin color, against people of color. It is a culture where white people are seen as necessary and friendly insiders, while people of color, especially black people, are seen as dangerous and threatening outsiders, who pose a clear and present danger to the safety and the efficacy of the culture of whiteness. Listen to me and see if you can imagine the culture of whiteness as a dehumanizing process that melts away all of our multiple and interlocking identities, such as race, class, gender and sexualities, so that... so that unity is maintained for power. I believe, because I know and believe that the culture of whiteness is a social construct. Each of us, from birth to death, are socialized in this culture. And it marks people of color also. And it makes people of color, like white people, vote against our interests. Some of you might ask — and my students always tell me I give hard assignments — some of you might ask, and rightfully so,"How do we fix this? It seems so all-powerful and overwhelming."I believe that we must fix it, because we cannot humanize our future if we continue to be complicit with the culture of whiteness. Each of us must connect with our authentic selves, with our authentic ethnic selves. And we must connect with the other aspects of our identities. And we must move out of the constructs of whiteness, brownness and blackness to become who we are at our fullest. How do we do this? I believe that we do this through our collective narratives. And our collective narratives must contain our individual stories, the arts, spiritual reflections, literature, and yes, even drumming. It must be a collective telling, because individual stories just create a paradigm where we are pitting one story against another story. These different models that I have talked about tonight I think are essential to providing us a pathway out of the quagmire of racism. And I want to talk about another very important model. And that is redemption. I believe that movements for racial justice must be redemptive rather than punitive. And yes, I believe that we must provide the possibility of redemption for everyone. And we must be willing, despite some of the vitriolic language that might come from those very people who oppress us, I think that we must listen to them and try to figure out where do they hurt. We must do this, I believe, because our redemption is tied into their redemption, And we will not be free until we ’ ve all been redeemed from unredemptive anger. The challenge is not easy. And in a technological society, it grows even more complicated, because often we use technologies to perpetuate the very values of racism that we indulge in every day. We use technology to bully, to perpetuate hate speech and to degrade each other ’ s humanities. And so I believe that if we ’ re going to humanize the future, we must design ways to use technology not to degrade us, but to elevate us so that we can live into the fullest of our capacities. And I believe that technology must provide us larger vistas so that we might engage with each other and move beyond the segregated spaces that we live in, every day of our lives. I believe that we can achieve this if we set our minds and hopes on the prize. The question before us tonight is very serious. It is :"Do you want to be healed? Do you want to be healed?"Do you want to become whole and live into all of your identities? Or do you want to continue to cannibalize your multiple identities and privilege one identity over the other? Do you want to join a long line of generations of people who believed in the promise of America and had the faith to upbuild democracy? Do you want to live into the fullest of your potential? I certainly do. And I believe you do, too. Let me just say, quite seriously, I believe in you. And despite everything, I still believe in America. I hope that this offering that I ’ ve given to you tonight, that I ’ ve shared with you tonight, will provide redemptive pathways so that you might claim the fullest of your identity and become a major participant in humanizing not only the future for yourselves, but also for our democracy. Thank you.

Text Sample 764: NEG Dim 5 – 2020 – Score -1.00 – 2020/t004137.txt

María Teresa Kumar : Much has been made of the 2020 US election. Right now, just over a week later, pollsters are issuing mea culpas, Democrats are tentatively celebrating, Republicans are **blowing** their collective tops, lawyers are busier than ever, ballot-counters are still hard at it, and demographers are desperately trying to understand who voted, for whom, where and why. Much has been said of the Latino vote in this election, which is something I know a little bit about, having been working obsessively over it for the last 16 years. Latinos are the fastest-growing demographic, with the largest voter registration cap in America. A Latino youth turns 18 every 30 seconds. While the mode for whites in America is 58, the mode for Latinx is 11 years old. You heard that right. And it ’ s these new voters and the youth who are translating America for their immigrant families who are leading the charge for audacious change. An estimated 73 percent of Latinx youth voted for Biden. As members of the largest generation globally, these Latino youth mirror their peers, seeking intervention for climate equity, racial justice and gender parity. What we ’ re hearing right now in America and around the globe is a demand for a massive reset on how we will govern in the 21st century for a world that is livable, equitable and just. Too many young people are drowning in student debt here in America, their families have been ravaged by the pandemic, who have lost jobs, lives and housing, and still, in 2020, they showed up for an America to believe in. Many say that 1914, the eve of World War I, defined the 20th century in America. That meant FDR ’ s New Deal that doubled down on its citizens by nation-building, offering pathways to the middle class through public works, education and sponsoring artists and musicians, building roads to provide jobs and sponsoring science-driven blueprints that allowed a man almost 40 years later to look up at the Moon and say that he wanted to go there. And we did that with less technology than the smartphone feeding this talk. So my hope is that the 21st century will be remembered as starting February 2020, not because that was when COVID ravaged us and in doing so, exposed the real, deep socioeconomic and racial disparities that ail us, but because that was when Americans cast a ballot for the future that believes in addressing the climate crisis, that health care is a right, that racial inequities hinder us all. We have a window to meet the precedent set by the Greatest Generation and define our century as one that is equitable and sustained. I, for one, am excited to get to work. I hope you ’ ll join me to usher in this audacious change together. Bianca DeJesus : María Teresa, thank you so much for that. MTK : Thank you, Bianca. Thank you for this conversation. BD : It is an honor. So, some commentators seem to be confounded that in certain places, Republicans received meaningful numbers of Latinx votes. Of course, it ’ s kind of silly to imagine that any demographic is a monolith, and within our community, there are so many differences. So what is the most productive way to think about heterogeneity within the Latinx, and really, within any community? MTK : If we don ’ t have public elected officials talking to our community, especially a new community, that is coming of age, that is relatively new to the democratic process, someone else will fill that vacuum. But I can share with you one of the things that we knew at Voto Latino was that young Latinos are navigating America for their families. Those youth turned up to protect their families, and it was not just in Arizona, but we also saw it Nevada, we also saw it in Pennsylvania, we saw it in Georgia and in North Carolina. And if you want to have an inclusive America, you have to fight for the vote, and that is basically what we need to see right now. But when we talk to young people, they voted disproportionately because they wanted climate change, they wanted access to health care, and they wanted to talk about the real racial inequities. When George Floyd sadly was murdered tragically, Latinos were side by side with the African American community because we recognize that that is something that truly plagues our American existence and that we have to address it if we want to move forward. BD : Absolutely. So do you see evidence that patterns change regarding first- and second- generation Latinx voters, and how does assimilation play out in terms of long-term voting trends? MTK : That ’ s interesting. So at Voto Latino, we don ’ t believe that there ’ s an assimilation. Right? What we want is an enhancement of American culture. Just like we celebrate St. Patrick ’ s Day, we want to be able to celebrate our roots and recognize the importance of that richness. We are in a very unique moment in America, where we have the most diverse population in the world, and one can argue that that is why some people don ’ t want us to succeed, because it ’ s our human capital, our vision, our ability to move forward and our diversity that prepares us for this century. And so when we talk about the differences in the Latino community, it ’ s also the differences in America that makes us so much richer with our imagination, with our ability to have entrepreneurship, and we have to use that and harness it for good. Some people will say race is what is our Achilles ’ heel. I actually believe that it ’ s the diversity of our races and our cultures that actually prepares us to battle the 21st century that it ’ s already interglobal. And the more that we harness that beauty of that diversity, that is what prepares us to compete and define the 21st century. BD : Wow. Yeah. I think that ’ s beautiful and totally agree. So how can we make first-time voters repeat voters who are engaged in future elections and not just for presidential elections but for local government as well? MTK : One of the things that we are seeing is that we ’ re seeing more young people run for office, and the more people start running for office, they realize that local government is what makes the most impact, at least here in America. So if you want, for example, some racial reform in your judicial system, vote for your district attorney, vote for your city councilman. If you think that there ’ s disparities in our education system, run for your school board. So that ’ s one. But the other thing to send very clearly to politicians is that when young – Americans voted their heart out. Young Latinos, youth in general, outvoted the people before them, but they ’ re voting on making a bet that their life will change, because the last four years could not have been rockier. And if the folks that are elected don ’ t meet the challenges of addressing climate change, addressing racial equity and gender parity and health care for all, they run the risk of not having those people vote again in 2024, and we need everybody on deck. And so our job as citizens is to ensure that we give the people that we just voted into office the courage to do the right thing, and that means to continue the rallies, continue calling our members of Congress, writing those letters and running for office ourselves. BD : So one question that speaks to the theme of this year ’ s TED Women,”Fearless,”I think it ’ s accurate to say that there ’ s been a lot of fear within the Latinx community over the last few years. How does that begin to change now? MTK : I will share with you, the day after Donald Trump was elected, all of our worst nightmares came to fruition. We saw family separation, one of the cruelest forms of our nation ’ s history came back to haunt us, because we ’ ve done it before, and everyone lived in fear. And the day after Joe Biden ’ s declaration on Saturday, I can tell that there was a collective – we ’ ve been holding our breath for so long, there was a collective release of not only that are we going to be OK but that fellow Americans stood up as allies and said,”Not one more.”And so that is what gives me hope, is that this was a collective America who outvoted their hearts out, because we see that in our celebration of our country ’ s future is believing in democracy, believing in a transition of power, believing that the most votes won and the electoral college was on our side, and more importantly, that these issues that Trump tried to ascend his presidency for the second time that were based on racism, that were based on the callousness of treating people and women differently, that they were not going to withstand. And so we do have to rebuild, but we have to rebuild not because of the four years of Donald Trump. If anything, I think he just exposed a lot of our fractures. We have to rebuild based on the last, I would say, 20 years. But the great thing is that the voters are here for it, and young people are here for it. I don ’ t have to change a young person ’ s mind that we are in a climate crisis. They get it. Cultural change is the hardest to do, but we have generations there with us, because they ’ re there and they get it. BD : That ’ s a relief. So, you yourself have been fearlessly outspoken. What drives you forward personally? MTK : I deeply – I don ’ t know if I ’ ve been fearless – I deeply believe in our country, and I deeply believe in us, and I deeply believe that when we are present, there ’ s nothing we can ’ t do. And when I say that, we... As a generation, we will not have an opportunity to reimagine what our country looks like, our systems of governments look like, and there will be people – you know, my children are six and eight, who will ask me 15 years from now,”What did you do?”And I want to say that I was alongside allies and the American people to rebuild better and to reimagine better. And we have always been a country of entrepreneurship, design and imagination, and what a perfect place to start when the majority of Americans are with us. BD : Absolutely. Well, thank you so, so much, María Teresa. MTK : Thank you.

Text Sample 765: NEG Dim 5 – 2020 – Score -1.00 – 2020/t004186.txt

Right now, there ’ s a lot happening with the Moon. China has announced plans for an inhabited South Pole station by the 2030s, and the United States has an official road map seeking an increasing number of people living and working in space. This will start with NASA ’ s Artemis program, an international program to send the first woman and the next man to the Moon this decade. Billionaires and the private sector are getting involved in unprecedented ways. There are over a hundred launch companies around the world and roughly a dozen private lunar transportation companies readying robotic missions to the lunar surface. We have reusable rockets for the first time in human history. This will enable the development of infrastructure and utilization of resources. While estimates vary, scientists think there could be up to a billion metric tons of water ice on the Moon. That ’ s greater than the size of Lake Erie, and enough water to support perhaps hundreds of thousands of people living and working on the Moon. So although official plans are always evolving, there ’ s real reason to think that we could see people starting to live and work on the Moon in the next decade. However, the Moon is roughly the size of the continent of Africa, and we ’ re starting to see that the key resources may be concentrated in small areas near the poles. This raises important questions about coordinating access to scarce resources. And there are also legitimate questions about going to the Moon : colonialism, cultural heritage and reproducing the systemic inequalities of today ’ s capitalism. And more to the point : Don ’ t we have enough big challenges here on Earth? Internet governance, pandemics, terrorism and, perhaps most importantly, climate crisis and biodiversity loss. In some senses, the idea of the Moon as just a destination embodies these problematic qualities. It conjures a frontier attitude of conquest, big rockets and expensive projects, competition and winning. But what ’ s most interesting about the Moon isn ’ t the billionaires with their rockets or the same old power struggle between states. In fact, it ’ s not the hardware at all. It ’ s the software. It ’ s the norms, customs and laws. It ’ s our social technologies. And it ’ s the opportunity to update our democratic institutions and the rule of law to respond to a new era of planetary-scale challenges. I ’ m going to tell you about how the Moon can be a canvas for solving some of our biggest challenges here on Earth. I ’ ve been kind of obsessed with this topic since I was a teenager. I ’ ve spent the last two decades working on international space policy, but also on small community projects with bottom-up governance design. When I was 17, I went to a UN conference on the peaceful uses of outer space in Vienna. Over two weeks, 160 young people from over 60 countries were crammed into a big hotel next to the UN building. We were invited to make recommendations to Member States about the role of space in humanity ' s future. After the conference, some of us were so inspired that we actually decided to keep living together. Now, living with 20 people might sound kind of crazy, but over the years, it enabled us to create a high-trust group that allowed us to experiment with these social technologies. We designed governance systems ranging from assigning a CEO to using a jury process. And as we grew into our careers, and we moved from DC think tanks to working for NASA to starting our own companies, these experiments enabled us to see how even small groups could be a petri dish for important societal questions such as representation, sustainability or opportunity. People often talk about the Moon as a petri dish or even a blank slate. But because of the legal agreements that govern the Moon, it actually has something very important in common with our global challenges here on Earth. They both involve issues that require us to think beyond territory and borders, meaning the Moon is actually more of a template than a blank slate. Signed in 1967, the Outer Space Treaty is the defining treaty governing activities in outer space, including the Moon. And it has two key ingredients that radically alter the basis on which laws can be constructed. The first is a requirement for free access to all areas of a celestial body. And the second is that the Moon and other celestial bodies are not **subject** to national appropriation. Now, this is crazy, because the entire earthly international system – the United Nations, the system of treaties and international agreements – is built on the idea of state sovereignty, on the appropriation of land and resources within borders and the autonomy to control free access within those borders. By doing away with both of these, we create the conditions for what are called the"commons."Based on the work of Nobel Prize-winning economist Elinor Ostrom, global commons are those resources that we all share that require us to work together to manage and protect important aspects of our survival and well-being, like climate or the oceans. Commons-based approaches offer a greenfield for institution design that ' s only beginning to be explored at the global and interplanetary level. What do property rights look like? And how do we manage resources when the traditional tools of external authority and private property don ' t apply? Though we don ' t have all the answers, climate, internet governance, authoritarianism – these are all deeply existential threats that we have failed to address with our current ways of thinking. Successful paths forward will require us to develop new tools. So how do we incorporate commons-based logic into our global and space institutions? Well, here ' s one attempt that came from an unlikely source. As a young activist in World War II, Arvid Pardo was arrested for anti-fascist organizing and held under death sentence by the Gestapo. After the war, he worked his way into the diplomatic corps, eventually becoming the first permanent representative of Malta to the United Nations. Pardo saw that international law did not have the tools to address management of shared global resources, such as the high seas. He also saw an opportunity to advocate for equitable sharing between nations. In 1967, Pardo gave a famous speech to the United Nations, introducing the idea that the oceans and their resources were the"common heritage of mankind."The phrase was eventually adopted as part of the Law of the Sea Treaty, probably the most sophisticated commons-management regime on the planet today. It was seen as a watershed moment, a constitution for the seas. But the language proved so controversial that it took over 12 years to gain enough signatures for the treaty to enter into force, and some states still refuse to sign it. The objection was not so much about sharing per se, but the obligation to share. States felt that the principle of equality undermined their autonomy and state sovereignty, the same autonomy and state sovereignty that underpins international law. So in many ways, the story of the common heritage principle is a tragedy. But it ’ s powerful because it makes plain the ways in which the current world order will put up antibodies and defenses and resist attempts at structural reform. But here ’ s the thing : the Outer Space Treaty has already made these structural reforms. At the height of the Cold War, terrified that each would get to the Moon first, the United States and the USSR made the Westphalian equivalent of a deal with the devil. By requiring free access and preventing territorial appropriation, we are required to redesign our most basic institutions, and perhaps in doing so, learn something new we can apply here on Earth. So although the Moon might seem a little far away sometimes, how we answer basic questions now will set precedent for who has a seat at the table and what consent looks like. And these are questions of social technology, not rockets and hardware. In fact, these conversations are starting to happen right now. The space community is discussing basic shared agreements, such as how do we designate lunar areas as heritage sites, and how do we get permission for where to land when traditional external authority doesn ’ t apply? How do we enforce requirements for coordination when it ’ s against the rules to tell people where to go? And how do we manage access to scarce resources such as water, minerals or even the peaks of eternal light – craters that sit at just the right latitude to receive near-constant exposure to sunlight – and therefore, power? Now, some people think that the lack of rules on the Moon is terrifying. And there are legitimately some terrifying elements of it. If there are no rules on the Moon, then won ’ t we end up in a first-come, first-served situation? And we might, if we dismiss this moment. But not if we ’ re willing to be bold and to engage the challenge. As we learned in our communities of self-governance, it ’ s easier to create something new than trying to dismantle the old. And where else but the Moon can we prototype new institutions at global scale in a self-contained environment with the exact design constraints needed for our biggest challenges here on Earth? Back in 1999, the United Nations taught a group of young space geeks that we could think bigger, that we could impact nations if we chose to. Today, the stage is set for the next step : to envision what comes after territory and borders. Thank you.

Text Sample 766: NEG Dim 5 – 2020 – Score 1.00 – 2020/t003961.txt

When I ’ m out at the grocery store or maybe a restaurant or the park with my son – he ’ s six and a half – people will stop us and mention that they think that he ’ s handsome. I agree. They ’ ll use that opportunity to chop it up with him, and often when they ’ re done talking with him, they ’ ll mention that they think he ’ s a smart and engaging little guy. When those people walk away, the thought that comes to my mind is that I hope they remember meeting him as a child when they see him again as a grown man. This thought comes to my mind because I ’ ve written two books about race and racism in the United States, and this kind of work can produce feelings of pessimism. One of the things that I ’ ve learned is that Americans have an orientation toward progress. In this context, what that means is that we often celebrate the distance between where we were and where we are now. But that same orientation can blind us from the gap between where we are and where we could or should be. The other thing I ’ ve learned about Americans is that we have a very, very narrow understanding of racism, mostly in the minds and hearts of people, usually old people – old people from the South. And this really narrow definition can constrain our opportunities to produce a more racially egalitarian society. We like to hunt for races and distance ourselves from people who say mean things about whole groups of people or who idealize the 1950s. But the fact of the matter is that we might just need to look in the mirror. Now, I ’ m not saying that everyone here is a racist, but what I am saying is that everyone here has the capacity and perhaps even the propensity to live their life in a way, to make decisions, to rely on biases that reproduce racial inequality. Some people say,”Well, you do all this work about racism. What ’ s the answer?”And I say that the first thing we might need to do is to come to a shared understanding about what racism is in the first place. History shows that racists have had the upper hand in deciding who the racists are and what racism is, and it ’ s never them or the things that they do. But maybe if we come together and come to a shared and perhaps a precise definition of what racism is, we can work toward creating a society where mothers like me aren ’ t in constant fear of their children ’ s lives. I ’ d like to dispel three myths about racism on our trek toward mutual understanding. First : it ’ s true that the South has done its work to earn its reputation as the most racist region. But there are other states and regions that are competing for the title. For example, if we look at the most segregated states in terms of where Black kids go to school, we ’ ll see, sure, some are in the South. There are some out west, in the Midwest and in the Northeast. They ’ re where we live. Or if we look at states with the biggest racial disparities in terms of prison populations, we see that none of them are in the South. They ’ re where we live. My colleague Rebecca Kreitzer and I looked at a standard battery of racial attitudes of prejudice, and we found that in the 1990s, states in the South dominated the most racially negative attitudes. But this geography has evolved, and things have changed. By 2016, we found that the Dakotas, Nebraska, states in the Midwest, in the Northeast, were competing for the”most prejudiced population”titles. Now, I ’ m not saying that one state is more racist than another, but what I am saying is that every state might have its own special brand of racism. And it doesn ’ t have to be like this. Most of the inequalities that we see in our day-to-day lives happen at the state and local level. What that means is that we don ’ t have to go all the way to Congress to make change in our communities. We can simply hold our city, our county, our state legislators to task to produce more equitable outcomes. Myth two : we ’ re not that good at hunting for racists. Remember that time when the governor of Virginia did blackface, and people were like,”Oh, that ’ s bad. I need to get that racist out of here”? I was giving y ’ all the side-eye, and here ’ s why. While people were going back to yearbooks to look for things that were obviously racist, fewer people were looking into the current-day policy stances of legislators who probably did blackface but didn ’ t get caught. So, how many of us might have supported a candidate who is willing to let neighborhoods secede from their district so that kids could go to all-white schools – in the 21st century? Or how many of us might have supported a ballot measure that systematically reduced some groups ’ chances of voting? Or how many of us might have focused on the behavior of Black mothers rather than doctors or health care systems and policies when we learn about the huge racial disparities in maternal and infant mortality? It doesn ’ t have to be like this. We could do something different. We could scrutinize the behaviors of the rule makers. We could orient ourselves toward a more just society, and on our way there, we can ’ t mystify practical policy solutions. Myth three : If you believe that when all the grandmas in Mississippi die that racism is going to go with them, you are in for a big disappointment. We like to think that young people are going to do the hard work of eradicating racism, but there are some things that we should note. We know that young folks, young white folks especially, like diversity, they appreciate it, they ’ re looking for it. But we also know that they don ’ t live diverse lives. Research shows that the average white American literally has one black friend. And what that means is that most don ’ t have any at all. Sociologists like Sarah Mayorga show that even when well-meaning white folks move to diverse neighborhoods, they don ’ t necessarily have positive interactions, no less any with their neighbors who aren ’ t white. My research with Professor Christopher DeSante shows that when we ask white millennials their racial attitudes and policy preferences, that they ’ re sometimes, just as in other times, even more racially conservative than boomers. When we ask them about the things that are important to them, they don ’ t have any particular sense of urgency around questions of racial inequality. How did we get like this? Well, one of the things we might think about is how we raise our kids and equip them to solve the problems that we want them to solve. Research shows that white parents in particular will either choose to not talk about issues of racism to their kids in order to protect them from a harsh racial reality or they instill colorblind lessons, and that can actually reinforce negative racial attitudes. So it ’ s kind of like how some of your parents might have given you books about puberty so they didn ’ t have to talk about the birds and the bees, and then you tried to connect all the dots and then you did it all wrong. It ’ s like that. It doesn ’ t have to be like this. We can do better. We can have hard conversations with our kids so that they don ’ t grow up like many of us did, thinking that talking about racism makes you a racist – it doesn ’ t – and so that we can prevent them from making the same mistakes that we ’ ve seen in the past. Remember a long, long time ago in 2008, when we were all pining to live in a post-racial world? Well, I say that it ’ s time for us to think bigger and dream bigger and think about what it would be like to live in a post-racist world. But in order to do that, we ’ d have to come together to have a shared definition of racism – not just in the matter of hearts and minds, but in systems, policies, rules, decisions made over and over again to marginalize some people – and agree to become anti-racists – people who learn more and do better. So we could ask harder questions of candidates about their stances on racial inequality before we throw our full weight behind them. We could buycott or boycott businesses whose practices don ’ t align with our values. We could talk to our kids about racism. We could figure out our state ’ s special brand of racism and work to eradicate it. People made racial disparities, and people can unmake them. And sure, it ’ ll be hard, but the fact of the matter is, someone is depending on us to do nothing at all. Thank you.

Text Sample 767: NEG Dim 5 – 2017 – Score -1.00 – 2017/t000753.txt

You ’ ve all been in a bar, right? But have you ever gone to a bar and come out with a $200 million business? That ’ s what happened to us about 10 years ago. We ’ d had a terrible day. We had this huge client that was killing us. We ’ re a software consulting firm, and we couldn ’ t find a very specific programming skill to help this client deploy a cutting-edge cloud system. We have a bunch of engineers, but none of them could please this client. And we were about to be fired. So we go out to the bar, and we ’ re hanging out with our bartender friend Jeff, and he ’ s doing what all good bartenders do : he ’ s commiserating with us, making us feel better, relating to our pain, saying,”Hey, these guys are overblowing it. Don ’ t worry about it.”And finally, he deadpans us and says,”Why don ’ t you send me in there? I can figure it out.”So the next morning, we ’ re hanging out in our team meeting, and we ’ re all a little hazy... and I half-jokingly throw it out there. I say,”Hey, I mean, we ’ re about to be fired.”So I say,”Why don ’ t we send in Jeff, the bartender?”And there ’ s some silence, some quizzical looks. Finally, my chief of staff says,”That is a great idea.””Jeff is wicked smart. He ’ s brilliant. He ’ ll figure it out. Let ’ s send him in there.”Now, Jeff was not a programmer. In fact, he had dropped out of Penn as a philosophy major. But he was brilliant, and he could go deep on topics, and we were about to be fired. So we sent him in. After a couple days of suspense, Jeff was still there. They hadn ’ t sent him home. I couldn ’ t believe it. What was he doing? Here ’ s what I learned. He had completely disarmed their fixation on the programming skill. And he had changed the conversation, even changing what we were building. The conversation was now about what we were going to build and why. And yes, Jeff figured out how to program the solution, and the client became one of our best references. Back then, we were 200 people, and half of our company was made up of computer science majors or engineers, but our experience with Jeff left us wondering : Could we repeat this through our business? So we changed the way we recruited and trained. And while we still sought after computer engineers and computer science majors, we sprinkled in artists, musicians, writers... and Jeff ’ s story started to multiply itself throughout our company. Our chief technology officer is an English major, and he was a bike messenger in Manhattan. And today, we ’ re a thousand people, yet still less than a hundred have degrees in computer science or engineering. And yes, we ’ re still a computer consulting firm. We ’ re the number one player in our market. We work with the fastest- growing software package to ever reach 10 billion dollars in annual sales. So it ’ s working. Meanwhile, the push for STEM-based education in this country — science, technology, engineering, mathematics — is fierce. It ’ s in all of our faces. And this is a colossal mistake. Since 2009, STEM majors in the United States have increased by 43 percent, while the humanities have stayed flat. Our past president dedicated over a billion dollars towards STEM education at the expense of other **subjects**, and our current president recently redirected 200 million dollars of Department of Education funding into computer science. And CEOs are continually complaining about an engineering-starved workforce. These campaigns, coupled with the undeniable success of the tech economy — I mean, let ’ s face it, seven out of the 10 most valuable companies in the world by market cap are technology firms — these things create an assumption that the path of our future workforce will be dominated by STEM. I get it. On paper, it makes sense. It ’ s tempting. But it ’ s totally overblown. It ’ s like, the entire soccer team chases the ball into the corner, because that ’ s where the ball is. We shouldn ’ t overvalue STEM. We shouldn ’ t value the sciences any more than we value the humanities. And there are a couple of reasons. Number one, today ’ s technologies are incredibly intuitive. The reason we ’ ve been able to recruit from all disciplines and swivel into specialized skills is because modern systems can be manipulated without writing code. They ’ re like LEGO : easy to put together, easy to learn, even easy to program, given the vast amounts of information that are available for learning. Yes, our workforce needs specialized skill, but that skill requires a far less rigorous and formalized education than it did in the past. Number two, the skills that are imperative and differentiated in a world with intuitive technology are the skills that help us to work together as humans, where the hard work is envisioning the end product and its usefulness, which requires real-world experience and judgment and historical context. What Jeff ’ s story taught us is that the customer was focused on the wrong thing. It ’ s the classic case : the technologist struggling to communicate with the business and the end user, and the business failing to articulate their needs. I see it every day. We are scratching the surface in our ability as humans to communicate and invent together, and while the sciences teach us how to build things, it ’ s the humanities that teach us what to build and why to build them. And they ’ re equally as important, and they ’ re just as hard. It irks me... when I hear people treat the humanities as a lesser path, as the easier path. Come on! The humanities give us the context of our world. They teach us how to think critically. They are purposely unstructured, while the sciences are purposely structured. They teach us to persuade, they give us our language, which we use to convert our emotions to thought and action. And they need to be on equal footing with the sciences. And yes, you can hire a bunch of artists and build a tech company and have an incredible outcome. Now, I ’ m not here today to tell you that STEM ’ s bad. I ’ m not here today to tell you that girls shouldn ’ t code. Please. And that next bridge I drive over or that next elevator we all jump into — let ’ s make sure there ’ s an engineer behind it. But to fall into this paranoia that our future jobs will be dominated by STEM, that ’ s just folly. If you have friends or kids or relatives or grandchildren or nieces or nephews... encourage them to be whatever they want to be. The jobs will be there. Those tech CEOs that are clamoring for STEM grads, you know what they ’ re hiring for? Google, Apple, Facebook. Sixty-five percent of their open job opportunities are non-technical : marketers, designers, project managers, program managers, product managers, lawyers, HR specialists, trainers, coaches, sellers, buyers, on and on. These are the jobs they ’ re hiring for. And if there ’ s one thing that our future workforce needs — and I think we can all agree on this — it ’ s diversity. But that diversity shouldn ’ t end with gender or race. We need a diversity of backgrounds and skills, with introverts and extroverts and leaders and followers. That is our future workforce. And the fact that the technology is getting easier and more accessible frees that workforce up to study whatever they damn well please. Thank you.

Text Sample 768: NEG Dim 5 – 2017 – Score -1.00 – 2017/t000748.txt
Words matter. They can heal and they can kill... yet, they have a limit. When I was in eighth grade, my teacher gave me a vocabulary sheet with the word”genocide.”I hated it. The word genocide is clinical... overgeneral... bloodless... dehumanizing. No word can describe what this does to a nation. You need to know, in this kind of war, husbands kills wives, wives kill husbands, neighbors and friends kill each other. Someone in power says,”Those over there... they don ’ t belong. They ’ re not human.”And people believe it. I don ’ t want words to describe this kind of behavior. I want words to stop it. But where are the words to stop this? And how do we find the words? But I believe, truly, we have to keep trying. I was born in Kigali, Rwanda. I felt loved by my entire family and my neighbors. I was constantly being teased by everybody, especially my two older siblings. When I lost my front tooth, my brother looked at me and said,”Oh, it has happened to you, too? It will never grow back.”I enjoyed playing everywhere, especially my mother ’ s garden and my neighbor ’ s. I loved my kindergarten. We sang songs, we played everywhere and ate lunch. I had a childhood that I would wish for anyone. But when I was six, the adults in my family began to speak in whispers and shushed me any time that I asked a question. One night, my mom and dad came. They had this strange look when they woke us. They sent my older sister Claire and I to our grandparent ’ s, hoping whatever was happening would **blow** away. Soon we had to escape from there, too. We hid, we crawled, we sometimes ran. Sometimes I heard laughter and then screaming and crying and then noise that I had never heard. You see, I did not know what those noises were. They were neither human — and also at the same time, they were human. I saw people who were not breathing. I

thought they were asleep. I still didn ’ t understand what death was, or killing in itself. When we would stop to rest for a little bit or search for food, I would close my eyes, hoping when I opened them, I would be awake. I had no idea which direction was home. Days were for hiding and night for walking. You go from a person who ’ s away from home to a person with no home. The place that is supposed to want you has pushed you out, and no one takes you in. You are unwanted by anyone. You are a refugee. From age six to 12, I lived in seven different countries, moving from one refugee camp to another, hoping we would be wanted. My older sister Claire, she became a young mother... and a master at getting things done. When I was 12, I came to America with Claire and her family on refugee status. And that ’ s only the beginning, because even though I was 12 years old, sometimes I felt like three years old and sometimes 50 years old. My past receded, grew jumbled, distorted. Everything was too much and nothing. Time seemed like pages torn out of a book and scattered everywhere. This still happens to me standing right here. After I got to America, Claire and I did not talk about our past. In 2006, after 12 years being separated away from my family, and then seven years knowing that they were dead and them thinking that we were dead, we reunited... in the most dramatic, American way possible. Live, on television — on”The Oprah Show.”I told you, I told you. But after the show, as I spent time with my mom and dad and my little sister and my two new siblings that I never met, I felt anger. I felt every deep pain in me. And I know that there is absolutely nothing, nothing, that could restore the time we lost with each other and the relationship we could ’ ve had. Soon, my parents moved to the United States, but like Claire, they don ’ t talk about our past. They live in never-ending present. Not asking too many questions, not allowing themselves to feel — moving in small steps. None of us, of course, can make sense of what happened to us. Though my family is alive — yes, we were broken, and yes, we are numb and we were silenced by our own experience. It ’ s not just my family. Rwanda is not the only country where people have turned on each other and murdered each other. The entire human race, in many ways, is like my family. Not dead ; yes, broken, numb and silenced by the violence of the world that has taken over. You see, the chaos of the violence continues inside in the words we use and the stories we create every single day. But also on the labels that we impose on ourselves and each other. Once we call someone”other,””less than,””one of them”or”better than,”believe me... under the right condition, it ’ s a short path to more destruction. More chaos and more noise that we will not understand. Words will never be enough to quantify and qualify the many magnitudes of human-caused destruction. In order for us to stop the violence that goes on in the world, I hope — at least I beg you — to pause. Let ’ s ask ourselves : Who are we without words? Who are we without labels? Who are we in our breath? Who are we in our heartbeat?

Text Sample 769: NEG Dim 5 – 2017 – Score -1.00 – 2017/t000845.txt

Nina DÃ¸lvik Brochmann : We grew up believing that the hymen is a proof of virginity. But it turns out, we were wrong. What we discovered is that the popular story we ’ re told about female virginity is based on two anatomical myths. The truth has been known in medical communities for over 100 years, yet somehow these two myths continue to make life difficult for women around the world. Ellen StÃ¸kken Dahl : The first myth is about blood. It tells us that the hymen breaks and bleeds the first time a woman has vaginal sex. In other words, if there is no blood on the sheets afterwards, then the woman was simply not a virgin. The second myth is a logical consequence of the first. Since the hymen is thought to break and bleed, people also believe that it actually disappears or is in some way radically altered during a woman ' s first intercourse. If that were true, one would easily be able to determine if a woman is a virgin or not by examining her genitals, by doing a virginity check. NDB : So that ' s our two myths : virgins bleed, and hymens are lost forever. Now, this may sound like a minor issue to you. Why should you care about an obscure little skin fold on the female body? But the truth is, this is about so much more than an anatomical misunderstanding. The myths about the hymen have lived on for centuries because they have cultural significance. They have been used as a powerful tool in the effort to control women ' s sexuality in about every culture, religion and historical decade. Women are still mistrusted, shamed, harmed and, in the worst cases, **subjected** to honor killings if they don ' t bleed on their wedding night. Other women are forced through degrading virginity checks, simply to obtain a job, to save their reputation or to get married. ESD : Like in Indonesia, where women are systematically examined to enter military service. After the Egyptian uprisings in 2011, a group of female protesters were forced to undergo virginity checks by their military. In Oslo, doctors are examining the hymens of young girls to reassure parents that their children are not ruined. And sadly, the list goes on. Women are so afraid not to live up to the myths about the hymen that they choose to use different virginity quick fixes to assure a bleeding. That could be plastic surgery, known as"revirgination,"it could be vials of blood poured on the sheets after sex or fake hymens bought online, complete with theater blood and a promise to"kiss your deep, dark secret goodbye."NDB : By telling girls that no deed can be kept secret, that their bodies will reveal them no matter what, we have endowed them with fear. Girls are afraid of ruining themselves, either through sport, play, tampon use or a sexual activity. We have curtailed their opportunities and their freedoms. It ' s time we put an end to the virginity fraud. It ' s time we break the myths about the hymen once and for all. ESD : We are medical students, sexual health workers and the authors of"The Wonder Down Under."That ' s a popular science book about the female genitals. And in our experience, people seem to believe that the hymen is some kind of a seal covering the vaginal opening. In Norwegian, it is even called"the virgin membrane."And with this, we picture something fragile, something easily destructible, something you can rip through, perhaps like a sheet of plastic wrapping. You may have wondered why we brought a hula hoop onstage today. We ' ll show you. Now, it is very hard to hide that something has happened to this hoop, right? It is different before and after I punched it. The seal is broken, and unless we change the plastic, it won ' t get back to its intact state. So if we wanted to do a virginity check on this hoop right here, right now, that would be very easy. It ' s easy to say that this hoop is not a virgin anymore. NDB : But the hymen is nothing like a piece of plastic you can wrap around your food, or a seal. In fact... it ' s more like this â a scrunchie or a rubber band. The hymen is a rim of tissue at the outer opening of the vagina. And usually, it has a doughnut or a half-moon shape with a large, central hole. But this varies a lot, and sometimes hymens can have fringes, it can have several holes, or it can consist of lobes. In other words, hymens naturally vary a lot in looks, and that is what makes it so hard to do a virginity check. ESD : Now that we know a bit more about the hymen ' s anatomy, it ' s time to get back to our two myths : virgins bleed, hymens are lost forever. But the hymen doesn ' t have to break at all. The hymen is like a scrunchie in function as well as in looks. And you can stretch a scrunchie, right? You can stretch a hymen, too. In fact, it ' s very elastic. And for a lot of women, the hymen will be elastic enough to handle a vaginal intercourse without sustaining any damage. For other women, the hymen may tear a bit to make room for the penis, but that won ’ t make it disappear. But it may look a bit different from before. It naturally follows that you can ’ t examine the hymen to check for virginity status. This was noted over 100 years ago in 1906 by the Norwegian doctor Marie Jeancet. She examined a middle-aged sex worker and concluded that her genitalia were reminiscent of a teenage virgin. But that makes sense, right? Because if her hymen was never damaged during sex, then what were we expecting to see? ESD : Since hymens come in every shape and form, it is difficult to know if a dent or a fold in it is there because of previous damage or if it ’ s just a normal anatomical variant. The absurdity of virgin testing is illustrated in a study done on 36 pregnant teenagers. When doctors examined their hymens, they could only find clear signs of penetration in two out of the 36 girls. So unless you believe in 34 cases of virgin births â we must all agree that also our second myth has taken a vital **blow**. You simply cannot look a woman between her legs and read her sexual story. NDB : Like most myths, the myths about the hymen are untrue. There is no virgin seal that magically disappears after sex, and half of virgins can easily have sex without bleeding. We wish we could say that by removing these myths, everything would be OK, that shame, harm and honor killings would all just disappear. But of course, it ’ s not that simple. Sexual oppression of women comes from something much deeper than a simple anatomical misunderstanding about the properties of the hymen. It ’ s a question of cultural and religious control of women ’ s sexuality. And that is much harder to change. But we must try. ESD : As medical professionals, this is our contribution. We want every girl, parent and [future] husband to know what the hymen is and how it works. We want them to know that the hymen can ’ t be used as a proof of virginity. And that way, we can remove one of the most powerful tools used to control young women today. After telling you this, you may wonder what the alternative is, for if we cannot use the hymen as a proof of virginity for women, then what should we use? We opt for using nothing. If you â If you really want to know if a woman is a virgin or not, ask her. But how she answers that question is her choice. Thank you.

Text Sample 770: NEG Dim 5 – 2016 – Score -1.00 – 2016/t000952.txt

A few years ago, about seven years ago, I found myself hiding in a festival toilet, a music festival toilet, and if anyone ’ s been to a music festival, yeah, you ’ ll know that by the third day, it ’ s pretty nasty. I was standing in the toilet because I couldn ’ t even sit down, because the toilet roll had run out, there was mud everywhere, and it smelled pretty bad. And I stood there thinking,”What am I doing? I don ’ t even need the toilet.”But the reason I went was because I was volunteering for a large charity on climate justice, and it was seven years ago, when lots of people didn ’ t believe in climate change, people were very cynical about activism, and my role, with all of my teammates, was to get people to sign petitions on climate justice and educate them a bit more about the issue. And I cared deeply about climate change and lots of inequality, so I ’ d go and I ’ d talk to lots of people, which made me nervous and drained me of energy, but I did it because I cared, but I would hide in the toilets, because I ’ d be exhausted, and I didn ’ t want my teammates doubting my commitment to the cause, thinking that I was slacking. And we ’ d go and meet at the end of our shift, and we ’ d count how many petitions had been signed, and often I ’ d win the amount of petitions signed even though I had my little breaks in the toilet. But I was always very jealous of the other activists, because either they had the same amount of energy as they had when they began the shift of getting people to sign petitions, or often they had more energy, and they ’ d be really excited about going to watch the bands in the evening and having a dance. And even if I loved the bands, all I wanted to do was to go back to my tent and have a sleep, because I ' d just feel completely wiped out, and I was really jealous of people that had the energy to go and party hard at the festivals. But it also made me really angry, as well, inside. I thought,"This isn ' t fair, I ' m an introvert, and all of the offline campaigning seems to be favoring extroverts."I would go on marches which drained me. That was the other option. Or I ' d go and join campaigns outside embassies or shops. The only thing that was on offer was around lots of people, it was very loud activism, it always involved lots of people, it was performing. None of it was for introverts, and I not only thought that that wasn ' t fair, because a third to a half of the world ' s population are introverts, which isn ' t fair on them, because we burn out, or we ' d be put off by activism and not do it, and everyone needs to be an activist in this world. And also, I didn ' t think it was particularly clever, but I could see that a lot of the activism that worked wasn ' t only extrovert activism. It wasn ' t only the loud stuff. It wasn ' t about people performing all the time. A lot of the work that was needed was in the background, was hidden, wasn ' t seen. And when I ended up just being a campaigner, because it ' s the only job I can do, really — I was campaigning at university, and for the last 10 years, I ' ve been a professional campaigner for large charities, and now I ' m a creative campaigner consultant for different charities as well as other work I do — but I knew that there were other forms of activism that were needed. I started tinkering about seven years ago to see what quieter forms of activism I could engage with so I didn ' t burn out as an activist, but also to look at some of the issues I was concerned about in campaigning. I was very lucky that, when I worked for Oxfam and other big charities, I could read lots of big reports on what influenced politicians and businesses and the general public, what campaigns worked really well, which ones didn ' t. And I ' m a bit of a geek, so I look at all of that stuff, and I wanted to tinker around to see how I could engage people in social change in a different way, because I think if we want the world to be more beautiful, kind and just, then our activism should be beautiful, kind and just, and often it ' s not. And today, I just want to talk about three ways that I think activism needs introverts. I think there ' s lot of other ways, but I ' m just going to talk about three. And the first one is : activism is often very quick, and it ' s about doing, so extroverts, often their immediate response to injustice is, we ' ve got to do stuff now, we ' ve got to react really quickly — and yes, we do need to react, but we need to be strategic in our campaigning, and if we just act on anger, often we do the wrong things. I use craft, like needlework — like this guy behind me is doing — as a way to not only slow down those extrovert doers, but also to bring in nervous, quiet introverts into activism. By doing repetitive actions, like handicraft, you can ' t do it fast, you have to do it slowly. And those repetitive stitches help you meditate on the big, complex, messy social change issues and figure out what we can do as a citizen, as a consumer, as a constituent, and all of those different things. It helps you think critically while you ' re stitching away, and it helps you be more mindful of what are your motives. Are you that Barbie aid worker that was mentioned before? Are you about joining people in solidarity, or do you want to be the savior, which often isn ' t very ethical? But doing needle work together, as well, extroverts and introverts and ambivert — everyone ' s on the scale in different places — because it ' s a quiet, slow form of activism, it really helps introverts be heard in other areas, where they are often not heard. It sounds odd, but while you ' re stitching, you don ' t need eye contact with people. So, for nervous introverts, it means that you can stitch away next to someone or a group of people and ask questions that you ' re thinking that often you don ' t get time to ask people, or you ' re too nervous to ask if you give them eye contact. So you can get introverts, who are those big, deep thinkers, saying,"That ' s really interesting that you want to do that extrovert form of activism that ' s about shaming people or quickly going out somewhere, but who are you trying to target and how, and is that the best way to do it?"So it means you could have these discussions in a very slow way, which is great for the extrovert to slow down and think deeply, but it ' s really good for the introvert as well, to be heard and to feel part of that movement for change, in a good way. Some ways we do it is stitch cards about what values we thread through our activism, and making sure that we don ' t just react in unethical ways. One, sometimes we work with art institutions where we will get over 150 people at the V&A who can come for hours, sit and stitch together on a particular issue, and then tweet what they ' re thinking or how it went, like this one. Also, I always think that activism needs introverts because we ' re really good at intimate activism. So we ' re good at slow activism, and we ' re really good at intimate activism, and if this year has told us anything, it ' s told us that we need to, when we ' re engaging power holders, we need to engage them by listening to people we disagree with, by building bridges not walls — walls or wars — and by being critical friends, not aggressive enemies. And one example that I do a lot with introverts, but with lots of people, is make gifts for people in power, so not be outside screaming at them, but to give them something like a bespoke handkerchief saying,"Don ' t **blow** it. Use your power for good. We know you ' ve got a difficult job in your position of power. How can we help you?"And what ' s great is, for the introverts, we can write letters while we ' re making these gifts, so for us, Marks and Spencer, we tried to campaign to get them to implement the living wage. So we made all the 14 board members bespoke handkerchiefs. We wrote them letters, we boxed them up, and we went to the AGM to hand-deliver our gifts and to have that form of intimate activism where we had discussions with them. And what was brilliant was that the chair of the board told us how amazing our campaign was, how heartfelt it was. The board members, like Martha Lane Fox, who has hundreds of thousands of followers on Twitter, and highly influential in business, tweeted how impressed she was, and within 10 months, we ' d had meetings with Marks and Spencer to say,"We know this is difficult to be a living wage employer, but if you can be one, the rest of the sector will look at it, and it ' s not right that some of your amazing workers are working full time and still can ' t pay their bills. And we love Marks and Spencer. How can you be the role model that we want you to be?"So that was that intimate form of activism. We had lots of meetings with them. We then gave them Christmas cards and Valentine ' s cards to say,"We really want to encourage you to implement the living wage, and within 10 months, they ' d announced to the media that they were going to pay the independent living wage, and now — Thank you. And now we ' re trying to work with them to be accredited, which is really important, and we went back to the last AGM this June and we had these amazing one-to-one discussions with the board members, who told us how much they loved their hankies and how it really moved them, what we were doing, and they all told us that if we were standing outside screaming at them and not being gentle in our protest, they wouldn ' t have even listened to us, never mind had those discussions with us. And I think introverts are really good at intimate activism because we like to listen, we like one-to-ones, we don ' t like small talks, we like those big, juicy issues to discuss with people, we don ' t like conflict, so we avoid it at all costs, which is really important when we ' re trying to engage power holders, not to be conflicting with them all the time. The third way I think activists are really missing out if they don ' t engage introverts is that introverts, like I said, can be half of the world ' s population, and most of us won ' t say that we ' re introvert, or we get embarrassed by saying what overwhelms us. So for me, a few years ago, my mom used to send me texts in capital letters — and she can now do emojis and everything, she ’ s fine — but as soon as I ’ d see this text, I ’ d wince and think,”Ooh, it ’ s capital letters, it ’ s too much.”And I ’ d have to ignore it to read the lovely text she sent me. And that ’ s a bit embarrassing, to tell people that capital letters overwhelm you, but we really need introverts to help us do intriguing activism that attracts them rather than puts them off. We ’ re put off by big and brash giant posters and capital letters and explanation marks telling us what to do and vying for our attention. So some of the things I do with people around the world who take part is make small bits of provocative street art which are hung off eye level, very small, and they ’ re provocative messages. They ’ re not preaching at people or telling them what to do. They ’ re just getting people to engage in different ways, and think for themselves, because we don ’ t like to be told what to do. It might be wearing a green heart on your sleeve saying what you love and how climate change will affect it, and we ’ ll wear it, and if people say,”Why are you wearing a green heart with the word ’ chocolate ’ on?”and we can have those one-to-one intimate conversations and say,”I love chocolate. Climate change is going to affect it, and I think there ’ s lot of other things that climate change will affect, and I really want to make sure I ’ m part of the solution, not the problem.”And then we deflect, because we don ’ t like to be the center of attention, and say,”What do you love and how will climate change affect it?”Or it might be shop-dropping instead of shop-lifting, where we ’ ll make little mini-scrolls with lovely stories on about what ’ s the story behind your clothes. Is it a joyful story of how it ’ s made, or is it a torturous one? And we ’ ll just drop them in little pockets in shops, all lowercase, all handwritten, with kisses and smiley faces in ribbon, and then people are excited that they found it. And we often drop them in unethical shops or in front pockets, and it ’ s a way that we can do offline campaigning that engages us and doesn ’ t burn us out, but also engages other people in an intriguing way online and offline. So I ’ ve got two calls to action, for the introverts and for the extroverts. For the ambivert, you ’ re involved in all of it. For the extroverts, I want to say that when you ’ re planning a campaign, think about introverts. Think about how valuable our skills are, just as much as extroverts ’. We ’ re good at slowing down and thinking deeply, and the detail of issues, we ’ re really good at bringing them out. We ’ re good at intimate activism, so use us in that way. And we ’ re good at intriguing people by doing strange little things that help create conversations and thought. Introverts, my call to action for you is, I know you like being on your own, I know you like being in your head, but activism needs you, so sometimes you ’ ve got to get out there. It doesn ’ t mean that you ’ ve got to turn into an extrovert and burn out, because that ’ s no use for anyone, but what it does mean is that you should value the skills and the traits that you have that activism needs. So for everyone in this room, whether you ’ re an extrovert or an introvert or an ambivert, the world needs you now more than ever, and you ’ ve got no excuse not to get involved. Thanks.

Text Sample 771: NEG Dim 5 – 2016 – Score -1.00 – 2016/t000580.txt

I love football. I started playing when I was 12 years old. It has been a dominant presence in my life ever since. From youth football, to high-school, and college, and now as a coach. Football has played a huge role in shaping me as a person. Recently, I achieved my dream job of becoming varsity line coach at my high-school alma mater. Last March I was at a coaching clinic. It was late in the afternoon and I was tired. These coaching clinics can tend to be a bit of a drag at times. Hours and hours, and days and days of the presentations on the x ’ s and o ’ s of football. There was one presentation, though, that looked interesting. It was titled”Disneyland.”This presentation changed my life. This talk was not about the x ’ s and o ’ s of football. But rather about the emotional truth of why we coaches coach. We are coaches so that we can have a meaningful impact on young people ’ s lives, and help them become better people. As coaches, we want to have the kind of impact that assure that one day our players can hold hands with their children and walk into Disneyland. The moral of this incredible message was this : as coaches, we can preach to our players that it is more important for them be better people than it is great football players. But if we are not honest with them and do not practice what we preach, it will never work. In other words, teenagers are adept at sensing bullshit. As coaches, we must be willing to share our truth. And sharing our truth is the only way we can have the kind of impact we truly want to have. The coach who gave this presentation, his truth was that his son became a drug addict due to the bullying and pressure he was **subjected** to from his dad being the head football coach at his school in Eagle, Idaho, a program that at the time was not very succesful. I went from a presentation on how to run the power against a 3-4 defense to the most honest, emotional and inspirational talks on coaching philosophy I ’ d ever heard. So it got me thinking. What ’ s my bullshit? What is unique about me that I bring to the table that will have a meaningful impact? What is my truth? My truth is this. I ’ m a former college football player. I ’ m a current high-school football coach. And I am gay. I battled with this truth for a long time. Personally, professionally and certainly in my coaching and athletic career. It was an identity crisis that had a tremendous impact on my life. Long before I heard this presentation on Disneyland I thought about coming out in football. In the beginning, I told myself”You ’ re gay and you will take it to the grave.”Now, I ’ ve come a long way since that day. Thank you. Thank you. I ’ ve come a long way since that day, but the process almost cost me my life. I chose to come out to family and friends starting in August of 2014. I told my immediate family first, and then I carefully picked my way through family and friends, trying to choose the right time and place. Only being part way out of the closet meant that I had to be two different people and always be very aware of where I was and whom I was with. It caused a constant state of panic and anxiety. So the question became,”Is it important for me to come out publicly?”Yes it is, and here is why. My time in-and-out of the closet helped my realize that there are other people in my situation. And it is not a pretty one. It also helped me realize that coming out is not about sharing my bedroom habits, but about giving myself permission to live my life in its entirety. Let me explain. After talking to one of my fellow coaches, he told me :”I don ’ t talk about my sex life with players, why would you?”And it ’ s a legitimate question, but a common misconception about coming out. Again, I ’ m not talking about my sex life. I ’ m talking about my life. It ’ s often said that your private life is your private life. But imagine you are put in a situation where going virtually anywhere and doing very normal, healthy, human activities with your significant other could have substantial consequences on your life, reputation and career. You gain a very new perspective on what your private life actually is. That ’ s how being in the closet affected my life. It was quite literally a closet. No space, no freedom and no comfort. It ’ s a suffocating lifestyle with measurable effects. It wore me down until eventually I was abusing alcohol and prescription medications to keep my anxiety in check. It was a horrible time in my life. When you are put in a situation of having to live a double life, it strips you off dignity and normal coping mechanisms. At that time the only place I felt somewhat safe was at home. I was living with my parents, who had been nothing but supportive, but I did not feel comfortable bringing guys around them yet. So there I was. No safe place, no place to be myself, facing a new lifestyle I did not know how to navigate, the anxiety and depression were inescapable. And I responded the only way I felt I could at the time. Drugs and alcohol. Football teaches so many great life lessons to those who play or coach. Perseverance, toughness, respect, self-esteem. But the one negative thing that it does teach is that being gay is not OK. To be frank, the word"faggot"is used almost as much as the word"football."There is a misconception about the prominence of gay men in football, and it has serious consequences. I ' m the perfect example of this issue. An all-state player in high-school. A two-year varsity captain among a select few in school history to go play for a top tier division 1 football program. The youngest line coach in the history of the school. And I was ready to kill myself. Because I thought that even though this program was like a second family to me, I feared they would shun me. It was a crushing weight that I was carrying with me all the time. Months and months of sleep deprivation, severe anxiety and depression. And honestly, a lack of the will to live began to catch up with me. It was too much. I was desperate for a way out. Any way out. One night I reached for a bottle of vodka and a couple of pills. I didn ' t see a way out. I just wanted it to be over. If I couldn ' t be me and still live my life, what was the point? I passed out on the bathroom floor, and my mom found me. I was fortunate to wake up the next day. I escaped an overdose. It was the scariest moment of my life. When I woke up the next day, I knew I really wasn ' t ready to give up. And when I heard that talk on Disneyland, I knew how, when and why it was important for me to share my story. I worried for so long about how the football community would react. And while at this point only time will tell, my experience has given me a theory. It ' s simple. One day, being gay in football will be normal. But in order for that to happen, those of us who are gay need to stand up and own it. The coaches I know are perfect examples of this. I have been met with nothing but love and support from my fellow coaching friends. But now the challenge is to change this. And not just on the private level. The odds of a gay teen or young adult abusing drugs, alcohol, or experiencing anxiety, depression, or even attempting suicide are drastically high. When it comes to football, the social norm that we ' ve created leaves many without hope. We ' ve made incredible progress in the equal rights movement under the law. But now we must tackle a different problem. Social equality and the message we send to young people who play football. Continue to use sports to inspire, connect and to share a positive message. A healthy person cannot live life in the dark. And if you are out there, stand up and own it. Thank you.

Text Sample 772: NEG Dim 5 – 2016 – Score -1.00 – 2016/t001161.txt

For the past few years, I ' ve been spending my summers in the marine biological laboratory in Woods Hole, Massachusetts. And there, what I ' ve been doing is essentially renting a boat. What I would like to do is ask you to come on a boat ride with me tonight. So, we ride off from Eel Pond into Vineyard Sound, right off the coast of Martha ' s Vineyard, equipped with a drone to identify potential spots from which to peer into the Atlantic. Earlier, I was going to say into the depths of the Atlantic, but we don ' t have to go too deep to reach the unknown. Here, barely two miles away from what is arguably the greatest marine biology lab in the world, we lower a simple plankton net into the water and bring up to the surface things that humanity rarely pays any attention to, and oftentimes has never seen before. Here ' s one of the organisms that we caught in our net. This is a jellyfish. But look closely, and living inside of this animal is another organism that is very likely entirely new to science. A complete new species. Or how about this other transparent beauty with a beating heart, asexually growing on top of its head, progeny that will move on to reproduce sexually. Let me say that again : this animal is growing asexually on top of its head, progeny that is going to reproduce sexually in the next generation. A weird jellyfish? Not quite. This is an ascidian. This is a group of animals that now we know we share extensive genomic ancestry with, and it is perhaps the closest invertebrate species to our own. Meet your cousin, Thalia democratica. I ' m pretty sure you didn ' t save a spot at your last family reunion for Thalia, but let me tell you, these animals are profoundly related to us in ways that we ' re just beginning to understand. So, next time you hear anybody derisively telling you that this type of research is a simple fishing expedition, I hope that you ' ll remember the trip that we just took. Today, many of the biological sciences only see value in studying deeper what we already know — in mapping already-discovered continents. But some of us are much more interested in the unknown. We want to discover completely new continents, and gaze at magnificent vistas of ignorance. We crave the experience of being completely baffled by something we ' ve never seen before. And yes, I agree there ' s a lot of little ego satisfaction in being able to say,"Hey, I was the first one to discover that."But this is not a self- aggrandizing enterprise, because in this type of discovery research, if you don ' t feel like a complete idiot most of the time, you ' re just not sciencing hard enough. So every summer I bring onto the deck of this little boat of ours more and more things that we know very little about. I would like tonight to tell you a story about life that rarely gets told in an environment like this. From the vantage point of our 21st-century biological laboratories, we have begun to illuminate many mysteries of life with knowledge. We sense that after centuries of scientific research, we ' re beginning to make significant inroads into understanding some of the most fundamental principles of life. Our collective optimism is reflected by the growth of biotechnology across the globe, striving to utilize scientific knowledge to cure human diseases. Things like cancer, aging, degenerative diseases ; these are but some of the undesirables we wish to tame. I often wonder : Why is it that we are having so much trouble trying to solve the problem of cancer? Is it that we ' re trying to solve the problem of cancer, and not trying to understand life? Life on this planet shares a common origin, and I can summarize 3. 5 billion years of the history of life on this planet in a single slide. What you see here are representatives of all known species in our planet. In this immensity of life and biodiversity, we occupy a rather unremarkable position. Homo sapiens. The last of our kind. And though I don ' t really want to disparage at all the accomplishments of our species, as much as we wish it to be so and often pretend that it is, we are not the measure of all things. We are, however, the measurers of many things. We relentlessly quantify, analyze and compare, and some of this is absolutely invaluable and indeed necessary. But this emphasis today on forcing biological research to specialize and to produce practical outcomes is actually restricting our ability to interrogate life to unacceptably narrow confines and unsatisfying depths. We are measuring an astonishingly narrow sliver of life, and hoping that those numbers will save all of our lives. How narrow do you ask? Well, let me give you a number. The National Oceanic and Atmospheric Administration recently estimated that about 95 percent of our oceans remain unexplored. Now let that sink in for a second. 95 percent of our oceans remain unexplored. I think it ' s very safe to say that we don ' t even know how much about life we do not know. So, it ' s not surprising that every week in my field we begin to see the addition of more and more new species to this amazing tree of life. This one for example

— discovered earlier this summer, new to science, and now occupying its lonely branch in our family tree. What is even more tragic is that we know about a bunch of other species of animals out there, but their biology remains sorely under-studied. I ’ m sure some of you have heard about the fact that a starfish can actually regenerate its arm after it ’ s lost. But some of you might not know that the arm itself can actually regenerate a complete starfish. And there are animals out there that do truly astounding things. I ’ m almost willing to bet that many of you have never heard of the flatworm, Schmidtea mediterranea. This little guy right here does things that essentially just **blow** my mind. You can grab one of these animals and cut it into 18 different fragments, and each and every one of those fragments will go on to regenerate a complete animal in under two weeks. 18 heads, 18 bodies, 18 mysteries. For the past decade and a half or so, I ’ ve been trying to figure out how these little dudes do what they do, and how they pull this magic trick off. But like all good magicians, they ’ re not really releasing their secrets readily to me. So here we are, after 20 years of essentially studying these animals, genome mapping, chin scratching, and thousands of amputations and thousands of regenerations, we still don ’ t fully understand how these animals do what they do. Each planarian an ocean unto itself, full of unknowns. One of the common characteristics of all of these animals I ’ ve been talking to you about is that they did not appear to have received the memo that they need to behave according to the rules that we have derived from a handful of randomly selected animals that currently populate the vast majority of biomedical laboratories across the world. Meet our Nobel Prize winners. Seven species, essentially, that have produced for us the brunt of our understanding of biological behavior today. This little guy right here — three Nobel Prizes in 12 years. And yet, after all the attention they have garnered, and all the knowledge they have generated, as well as the lion ’ s share of the funding, here we are standing [before] the same litany of intractable problems and many new challenges. And that ’ s because, unfortunately, these seven animals essentially correspond to 0. 0009 percent of all of the species that inhabit the planet. So I ’ m beginning to suspect that our specialization is beginning to impede our progress at best, and at worst, is leading us astray. That ’ s because life on this planet and its history is the history of rule breakers. Life started on the face of this planet as single-cell organisms, swimming for millions of years in the ocean, until one of those creatures decided,”I ’ m going to do things differently today ; today I would like to invent something called multicellularity, and I ’ m going to do this.”And I ’ m sure it wasn ’ t a popular decision at the time — but somehow, it managed to do it. And then, multicellular organisms began to populate all these ancestral oceans, and they thrived. And we have them here today. Land masses began to emerge from the surface of the oceans, and another creature thought,”Hey, that looks like a really nice piece of real estate. I ’ d like to move there.””Are you crazy? You ’ re going to desiccate out there. Nothing can live out of water.”But life found a way, and there are organisms now that live on land. Once on land, they may have looked up into the sky and said,”It would be nice to go to the clouds, I ’ m going to fly.””You can ’ t break the law of gravity, there ’ s no way you can fly.”And yet, nature has invented — multiple and independent times — ways to fly. I love to study these animals that break the rules, because every time they break a rule, they invent something new that made it possible for us to be able to be here today. These animals did not get the memo. They break the rules. So if we ’ re going to study animals that break the rules, shouldn ’ t how we study them also break the rules? I think we need to renew our spirit of exploration. Rather than bring nature into our laboratories and interrogate it there, we need to bring our science into the majestic laboratory that is nature, and there, with our modern technological armamentarium, interrogate every new form of life we find, and any new biological attribute that we may find. We actually need to bring all of our intelligence to becoming stupid again — clueless [before] the immensity of the unknown. Because after all, science is not really about knowledge. Science is about ignorance. That ’ s what we do. Once, Antoine de Saint-Exupry wrote,”If you want to build a ship, don ’ t drum up people to collect wood and don ’ t assign them tasks and work, but rather teach them to long for the endless immensity of the sea...”As a scientist and a teacher, I like to paraphrase this to read that we scientists need to teach our students to long for the endless immensity of the sea that is our ignorance. We Homo sapiens are the only species we know of that is driven to scientific inquiry. We, like all other species on this planet, are inextricably woven into the history of life on this planet. And I think I ’ m a little wrong when I say that life is a mystery, because I think that life is actually an open secret that has been beckoning our species for millennia to understand it. So I ask you : Aren ’ t we the best chance that life has to know itself? And if so, what the heck are we waiting for? Thank you.

Text Sample 773: NEG Dim 5 – 2015 – Score -1.00 – 2015/t001347.txt

I ’ m turning 44 next month, and I have the sense that 44 is going to be a very good year, a year of fulfillment, realization. I have that sense, not because of anything particular in store for me, but because I read it would be a good year in a 1968 book by Norman Mailer.”He felt his own age, forty-four...”wrote Mailer in”The Armies of the Night,””... felt as if he were a solid embodiment of bone, muscle, heart, mind, and sentiment to be a man, as if he had arrived.”Yes, I know Mailer wasn ’ t writing about me. But I also know that he was ; for all of us — you, me, the **subject** of his book, age more or less in step, proceed from birth along the same great sequence : through the wonders and confinements of childhood ; the emancipations and frustrations of adolescence ; the empowerments and millstones of adulthood ; the recognitions and resignations of old age. There are patterns to life, and they are shared. As Thomas Mann wrote :”It will happen to me as to them.”We don ’ t simply live these patterns. We record them, too. We write them down in books, where they become narratives that we can then read and recognize. Books tell us who we ’ ve been, who we are, who we will be, too. So they have for millennia. As James Salter wrote,”Life passes into pages if it passes into anything.”And so six years ago, a thought leapt to mind : if life passed into pages, there were, somewhere, passages written about every age. If I could find them, I could assemble them into a narrative. I could assemble them into a life, a long life, a hundred-year life, the entirety of that same great sequence through which the luckiest among us pass. I was then 37 years old,”an age of discretion,”wrote William Trevor. I was prone to meditating on time and age. An illness in the family and later an injury to me had long made clear that growing old could not be assumed. And besides, growing old only postponed the inevitable, time seeing through what circumstance did not. It was all a bit disheartening. A list, though, would last. To chronicle a life year by vulnerable year would be to clasp and to ground what was fleeting, would be to provide myself and others a glimpse into the future, whether we made it there or not. And when I then began to compile my list, I was quickly obsessed, searching pages and pages for ages and ages. Here we were at every annual step through our first hundred years.”Twenty-seven... a time of sudden revelations,””sixty-two,... of subtle diminishments.”I was mindful, of course, that such insights were relative. For starters, we now live longer, and so age more slowly. Christopher Isherwood used the phrase"the yellow leaf"to describe a man at 53, only one century after Lord Byron used it to describe himself at 36. I was mindful, too, that life can swing wildly and unpredictably from one year to the next, and that people may experience the same age differently. But even so, as the list coalesced, so, too, on the page, clear as the reflection in the mirror, did the life that I had been living : finding at 20 that"... one is less and less sure of who one is ;"emerging at 30 from the"... wasteland of preparation into active life ;"learning at 40"... to close softly the doors to rooms [I would] not be coming back to."There I was. Of course, there we all are. Milton Glaser, the great graphic designer whose beautiful visualizations you see here, and who today is 85 — all those years"... a ripening and an apotheosis,"wrote Nabokov — noted to me that, like art and like color, literature helps us to remember what we ' ve experienced. And indeed, when I shared the list with my grandfather, he nodded in recognition. He was then 95 and soon to die, which, wrote Roberto Bolao,"... is the same as never dying."And looking back, he said to me that, yes, Proust was right that at 22, we are sure we will not die, just as a thanatologist named Edwin Shneidman was right that at 90, we are sure we will. It had happened to him, as to them. Now the list is done : a hundred years. And looking back over it, I know that I am not done. I still have my life to live, still have many more pages to pass into. And mindful of Mailer, I await 44. Thank you.

Text Sample 774: NEG Dim 5 – 2015 – Score -1.00 – 2015/t001375.txt

People returning to work after a career break : I call them relaunchers. These are people who have taken career breaks for elder care, for childcare reasons, pursuing a personal interest or a personal health issue. Closely related are career transitioners of all kinds : veterans, military spouses, retirees coming out of retirement or repatriating expats. Returning to work after a career break is hard because of a disconnect between the employers and the relaunchers. Employers can view hiring people with a gap on their resume as a high-risk proposition, and individuals on career break can have doubts about their abilities to relaunch their careers, especially if they ' ve been out for a long time. This disconnect is a problem that I ' m trying to help solve. Now, successful relaunchers are everywhere and in every field. This is Sami Kafala. He ' s a nuclear physicist in the UK who took a five-year career break to be home with his five children. The Singapore press recently wrote about nurses returning to work after long career breaks. And speaking of long career breaks, this is Mimi Kahn. She ' s a social worker in Orange County, California, who returned to work in a social services organization after a 25-year career break. That ' s the longest career break that I ' m aware of. Supreme Court Justice Sandra Day O ' Connor took a five-year career break early in her career. And this is Tracy Shapiro, who took a 13-year career break. Tracy answered a call for essays by the Today Show from people who were trying to return to work but having a difficult time of it. Tracy wrote in that she was a mom of five who loved her time at home, but she had gone through a divorce and needed to return to work, plus she really wanted to bring work back into her life because she loved working. Tracy was doing what so many of us do when we feel like we ' ve put in a good day in the job search. She was looking for a finance or accounting role, and she had just spent the last nine months very diligently researching companies online and applying for jobs with no results. I met Tracy in June of 2011, when the Today Show asked me if I could work with her to see if I could help her turn things around. The first thing I told Tracy was she had to get out of the house. I told her she had to go public with her job search and tell everyone she knew about her interest in returning to work. I also told her,"You are going to have a lot of conversations that don ' t go anywhere. Expect that, and don ' t be discouraged by it. There will be a handful that ultimately lead to a job opportunity."I ' ll tell you what happened with Tracy in a little bit, but I want to share with you a discovery that I made when I was returning to work after my own career break of 11 years out of the full-time workforce. And that is, that people ' s view of you is frozen in time. What I mean by this is, when you start to get in touch with people and you get back in touch with those people from the past, the people with whom you worked or went to school, they are going to remember you as you were before your career break. And that ' s even if your sense of self has diminished over time, as happens with so many of us the farther removed we are from our professional identities. So for example, you might think of yourself as someone who looks like this. This is me, crazy after a day of driving around in my minivan. Or here I am in the kitchen. But those people from the past, they don ' t know about any of this. They only remember you as you were, and it ' s a great confidence boost to be back in touch with these people and hear their enthusiasm about your interest in returning to work. There ' s one more thing I remember vividly from my own career break. And that was that I hardly kept up with the business news. My background is in finance, and I hardly kept up with any news when I was home caring for my four young children. So I was afraid I ' d go into an interview and start talking about a company that didn ' t exist anymore. So I had to resubscribe to the Wall Street Journal and read it for a good six months cover to cover before I felt like I had a handle on what was going on in the business world again. I believe relaunchers are a gem of the workforce, and here ' s why. Think about our life stage : for those of us who took career breaks for childcare reasons, we have fewer or no maternity leaves. We did that already. We have fewer spousal or partner job relocations. We ' re in a more settled time of life. We have great work experience. We have a more mature perspective. We ' re not trying to find ourselves at an employer ' s expense. Plus we have an energy, an enthusiasm about returning to work precisely because we ' ve been away from it for a while. On the flip side, I speak with employers, and here are two concerns that employers have about hiring relaunchers. The first one is, employers are worried that relaunchers are technologically obsolete. Now, I can tell you, having been technologically obsolete myself at one point, that it ' s a temporary condition. I had done my financial analysis so long ago that I used Lotus 1-2-3. I don ' t know if anyone can even remember back that far, but I had to relearn it on Excel. It actually wasn ' t that hard. A lot of the commands are the same. I found PowerPoint much more challenging, but now I use PowerPoint all the time. I tell relaunchers that employers expect them to come to the table with a working knowledge of basic office management software. And if they ' re not up to speed, then it ' s their responsibility to get there. And they do. The second area of concern that employers have about relaunchers is they ' re worried that relaunchers don ' t know what they want to do. I tell relaunchers that they need to do the hard work to figure out whether their interests and skills have changed or have not changed while they have been on career break. That ' s not the employer ' s job. It ' s the relauncher ' s responsibility to demonstrate to the employer where they can add the most value. Back in 2010 I started noticing something. I had been tracking return to work programs since 2008, and in 2010, I started noticing the use of a short-term paid work opportunity, whether it was called an internship or not, but an internship-like experience, as a way for professionals to return to work. I saw Goldman Sachs and Sara Lee start corporate reentry internship programs. I saw a returning engineer, a nontraditional reentry candidate, apply for an entry- level internship program in the military, and then get a permanent job afterward. I saw two universities integrate internships into mid-career executive education programs. So I wrote a report about what I was seeing, and it became this article for Harvard Business Review called"The 40-Year-Old Intern."I have to thank the editors there for that title, and also for this artwork where you can see the 40-year-old intern in the midst of all the college interns. And then, courtesy of Fox Business News, they called the concept"The 50-Year-Old Intern."So five of the biggest financial services companies have reentry internship programs for returning finance professionals. And at this point, hundreds of people have participated. These internships are paid, and the people who move on to permanent roles are commanding competitive salaries. And now, seven of the biggest engineering companies are piloting reentry internship programs for returning engineers as part of an initiative with the Society of Women Engineers. Now, why are companies embracing the reentry internship? Because the internship allows the employer to base their hiring decision on an actual work sample instead of a series of interviews, and the employer does not have to make that permanent hiring decision until the internship period is over. This testing out period removes the perceived risk that some managers attach to hiring relaunchers, and they are attracting excellent candidates who are turning into great hires. Think about how far we have come. Before this, most employers were not interested in engaging with relaunchers at all. But now, not only are programs being developed specifically with relaunchers in mind, but you can ' t even apply for these programs unless you have a gap on your rsum. This is the mark of real change, of true institutional shift, because if we can solve this problem for relaunchers, we can solve it for other career transitioners too. In fact, an employer just told me that their veterans return to work program is based on their reentry internship program. And there ' s no reason why there can ' t be a retiree internship program. Different pool, same concept. So let me tell you what happened with Tracy Shapiro. Remember that she had to tell everyone she knew about her interest in returning to work. Well, one critical conversation with another parent in her community led to a job offer for Tracy, and it was an accounting job in a finance department. But it was a temp job. The company told her there was a possibility it could turn into something more, but no guarantees. This was in the fall of 2011. Tracy loved this company, and she loved the people and the office was less than 10 minutes from her house. So even though she had a second job offer at another company for a permanent full-time role, she decided to take her chances with this internship and hope for the best. Well, she ended up **blowing** away all of their expectations, and the company not only made her a permanent offer at the beginning of 2012, but they made it even more interesting and challenging, because they knew what Tracy could handle. Fast forward to 2015, Tracy ' s been promoted. They ' ve paid for her to get her MBA at night. She ' s even hired another relauncher to work for her. Tracy ' s temp job was a tryout, just like an internship, and it ended up being a win for both Tracy and her employer. Now, my goal is to bring the reentry internship concept to more and more employers. But in the meantime, if you are returning to work after a career break, don ' t hesitate to suggest an internship or an internship-like arrangement to an employer that does not have a formal reentry internship program. Be their first success story, and you can be the example for more relaunchers to come. Thank you.

Text Sample 775: NEG Dim 5 – 2015 – Score -1.00 – 2015/t001422.txt

So you go to the doctor and get some tests. The doctor determines that you have high cholesterol and you would benefit from medication to treat it. So you get a pillbox. You have some confidence, your physician has some confidence that this is going to work. The company that invented it did a lot of studies, submitted it to the FDA. They studied it very carefully, skeptically, they approved it. They have a rough idea of how it works, they have a rough idea of what the side effects are. It should be OK. You have a little more of a conversation with your physician and the physician is a little worried because you ’ ve been blue, haven ’ t felt like yourself, you haven ’ t been able to enjoy things in life quite as much as you usually do. Your physician says,”You know, I think you have some depression. I ’ m going to have to give you another pill.”So now we ’ re talking about two medications. This pill also — millions of people have taken it, the company did studies, the FDA looked at it — all good. Think things should go OK. Think things should go OK. Well, wait a minute. How much have we studied these two together? Well, it ’ s very hard to do that. In fact, it ’ s not traditionally done. We totally depend on what we call”post-marketing surveillance,”after the drugs hit the market. How can we figure out if bad things are happening between two medications? Three? Five? Seven? Ask your favorite person who has several diagnoses how many medications they ’ re on. Why do I care about this problem? I care about it deeply. I ’ m an informatics and data science guy and really, in my opinion, the only hope — only hope — to understand these interactions is to leverage lots of different sources of data in order to figure out when drugs can be used together safely and when it ’ s not so safe. So let me tell you a data science story. And it begins with my student Nick. Let ’ s call him”Nick,”because that ’ s his name. Nick was a young student. I said,”You know, Nick, we have to understand how drugs work and how they work together and how they work separately, and we don ’ t have a great understanding. But the FDA has made available an amazing database. It ’ s a database of adverse events. They literally put on the web — publicly available, you could all download it right now — hundreds of thousands of adverse event reports from patients, doctors, companies, pharmacists. And these reports are pretty simple : it has all the diseases that the patient has, all the drugs that they ’ re on, and all the adverse events, or side effects, that they experience. It is not all of the adverse events that are occurring in America today, but it ’ s hundreds and hundreds of thousands of drugs. So I said to Nick,”Let ’ s think about glucose. Glucose is very important, and we know it ’ s involved with diabetes. Let ’ s see if we can understand glucose response. I sent Nick off. Nick came back.”Russ,”he said,”I ’ ve created a classifier that can look at the side effects of a drug based on looking at this database, and can tell you whether that drug is likely to change glucose or not.”He did it. It was very simple, in a way. He took all the drugs that were known to change glucose and a bunch of drugs that don ’ t change glucose, and said,”What ’ s the difference in their side effects? Differences in fatigue? In appetite? In urination habits?”All those things conspired to give him a really good predictor. He said,”Russ, I can predict with 93 percent accuracy when a drug will change glucose.”I said,”Nick, that ’ s great.”He ’ s a young student, you have to build his confidence.”But Nick, there ’ s a problem. It ’ s that every physician in the world knows all the drugs that change glucose, because it ’ s core to our practice. So it ’ s great, good job, but not really that interesting, definitely not publishable.”He said,”I know, Russ. I thought you might say that.”Nick is smart.”I thought you might say that, so I did one other experiment. I looked at people in this database who were on two drugs, and I looked for signals similar, glucose-changing signals, for people taking two drugs, where each drug alone did not change glucose, but together I saw a strong signal.”And I said,”Oh! You ’ re clever. Good idea.

Show me the list."And there ' s a bunch of drugs, not very exciting. But what caught my eye was, on the list there were two drugs : paroxetine, or Paxil, an antidepressant ; and pravastatin, or Pravachol, a cholesterol medication. And I said,"Huh. There are millions of Americans on those two drugs."In fact, we learned later, 15 million Americans on paroxetine at the time, 15 million on pravastatin, and a million, we estimated, on both. So that ' s a million people who might be having some problems with their glucose if this machine-learning mumbo jumbo that he did in the FDA database actually holds up. But I said,"It ' s still not publishable, because I love what you did with the mumbo jumbo, with the machine learning, but it ' s not really standard-of-proof evidence that we have."So we have to do something else. Let ' s go into the Stanford electronic medical record. We have a copy of it that ' s OK for research, we removed identifying information. And I said,"Let ' s see if people on these two drugs have problems with their glucose."Now there are thousands and thousands of people in the Stanford medical records that take paroxetine and pravastatin. But we needed special patients. We needed patients who were on one of them and had a glucose measurement, then got the second one and had another glucose measurement, all within a reasonable period of time — something like two months. And when we did that, we found 10 patients. However, eight out of the 10 had a bump in their glucose when they got the second P — we call this P and P — when they got the second P. Either one could be first, the second one comes up, glucose went up 20 milligrams per deciliter. Just as a reminder, you walk around normally, if you ' re not diabetic, with a glucose of around 90. And if it gets up to 120, 125, your doctor begins to think about a potential diagnosis of diabetes. So a 20 bump — pretty significant. I said,"Nick, this is very cool. But, I ' m sorry, we still don ' t have a paper, because this is 10 patients and — give me a break — it ' s not enough patients."So we said, what can we do? And we said, let ' s call our friends at Harvard and Vanderbilt, who also — Harvard in Boston, Vanderbilt in Nashville, who also have electronic medical records similar to ours. Let ' s see if they can find similar patients with the one P, the other P, the glucose measurements in that range that we need. God bless them, Vanderbilt in one week found 40 such patients, same trend. Harvard found 100 patients, same trend. So at the end, we had 150 patients from three diverse medical centers that were telling us that patients getting these two drugs were having their glucose bump somewhat significantly. More interestingly, we had left out diabetics, because diabetics already have messed up glucose. When we looked at the glucose of diabetics, it was going up 60 milligrams per deciliter, not just 20. This was a big deal, and we said,"We ' ve got to publish this."We submitted the paper. It was all data evidence, data from the FDA, data from Stanford, data from Vanderbilt, data from Harvard. We had not done a single real experiment. But we were nervous. So Nick, while the paper was in review, went to the lab. We found somebody who knew about lab stuff. I don ' t do that. I take care of patients, but I don ' t do pipettes. They taught us how to feed mice drugs. We took mice and we gave them one P, paroxetine. We gave some other mice pravastatin. And we gave a third group of mice both of them. And lo and behold, glucose went up 20 to 60 milligrams per deciliter in the mice. So the paper was accepted based on the informatics evidence alone, but we added a little note at the end, saying, oh by the way, if you give these to mice, it goes up. That was great, and the story could have ended there. But I still have six and a half minutes. So we were sitting around thinking about all of this, and I don ' t remember who thought of it, but somebody said,"I wonder if patients who are taking these two drugs are noticing side effects of hyperglycemia. They could and they should. How would we ever determine that?"We said, well, what do you do? You ' re taking a medication, one new medication or two, and you get a funny feeling. What do you do? You go to Google and type in the two drugs you ' re taking or the one drug you ' re taking, and you type in"side effects."What are you experiencing? So we said OK, let ' s ask Google if they will share their search logs with us, so that we can look at the search logs and see if patients are doing these kinds of searches. Google, I am sorry to say, denied our request. So I was bummed. I was at a dinner with a colleague who works at Microsoft Research and I said,"We wanted to do this study, Google said no, it ' s kind of a bummer."He said,"Well, we have the Bing searches."Yeah. That ' s great. Now I felt like I was — I felt like I was talking to Nick again. He works for one of the largest companies in the world, and I ' m already trying to make him feel better. But he said,"No, Russ — you might not understand. We not only have Bing searches, but if you use Internet Explorer to do searches at Google, Yahoo, Bing, any... Then, for 18 months, we keep that data for research purposes only."I said,"Now you ' re talking!"This was Eric Horvitz, my friend at Microsoft. So we did a study where we defined 50 words that a regular person might type in if they ' re having hyperglycemia, like"fatigue,""loss of appetite,""urinating a lot,""peeing a lot"— forgive me, but that ' s one of the things you might type in. So we had 50 phrases that we called the"diabetes words."And we did first a baseline. And it turns out that about. 5 to one percent of all searches on the Internet involve one of those words. So that ' s our baseline rate. If people type in"paroxetine"or"Paxil"— those are synonyms — and one of those words, the rate goes up to about two percent of diabetes-type words, if you already know that there ' s that"paroxetine"word. If it ' s"pravastatin,"the rate goes up to about three percent from the baseline. If both"paroxetine"and"pravastatin"are present in the query, it goes up to 10 percent, a huge three- to four-fold increase in those searches with the two drugs that we were interested in, and diabetes-type words or hyperglycemia-type words. We published this, and it got some attention. The reason it deserves attention is that patients are telling us their side effects indirectly through their searches. We brought this to the attention of the FDA. They were interested. They have set up social media surveillance programs to collaborate with Microsoft, which had a nice infrastructure for doing this, and others, to look at Twitter feeds, to look at Facebook feeds, to look at search logs, to try to see early signs that drugs, either individually or together, are causing problems. What do I take from this? Why tell this story? Well, first of all, we have now the promise of big data and medium-sized data to help us understand drug interactions and really, fundamentally, drug actions. How do drugs work? This will create and has created a new ecosystem for understanding how drugs work and to optimize their use. Nick went on ; he ' s a professor at Columbia now. He did this in his PhD for hundreds of pairs of drugs. He found several very important interactions, and so we replicated this and we showed that this is a way that really works for finding drug-drug interactions. However, there ' s a couple of things. We don ' t just use pairs of drugs at a time. As I said before, there are patients on three, five, seven, nine drugs. Have they been studied with respect to their nine-way interaction? Yes, we can do pair-wise, A and B, A and C, A and D, but what about A, B, C, D, E, F, G all together, being taken by the same patient, perhaps interacting with each other in ways that either makes them more effective or less effective or causes side effects that are unexpected? We really have no idea. It ' s a blue sky, open field for us to use data to try to understand the interaction of drugs. Two more lessons : I want you to think about the power that we were able to generate with the data from people who had volunteered their adverse reactions through their pharmacists, through themselves, through their doctors, the people who allowed the databases at Stanford, Harvard, Vanderbilt, to be used for research. People are worried about data. They ' re worried about their privacy and security — they should be. We need secure systems. But we can ’ t have a system that closes that data off, because it is too rich of a source of inspiration, innovation and discovery for new things in medicine. And the final thing I want to say is, in this case we found two drugs and it was a little bit of a sad story. The two drugs actually caused problems. They increased glucose. They could throw somebody into diabetes who would otherwise not be in diabetes, and so you would want to use the two drugs very carefully together, perhaps not together, make different choices when you ’ re prescribing. But there was another possibility. We could have found two drugs or three drugs that were interacting in a beneficial way. We could have found new effects of drugs that neither of them has alone, but together, instead of causing a side effect, they could be a new and novel treatment for diseases that don ’ t have treatments or where the treatments are not effective. If we think about drug treatment today, all the major breakthroughs — for HIV, for tuberculosis, for depression, for diabetes — it ’ s always a cocktail of drugs. And so the upside here, and the **subject** for a different TED Talk on a different day, is how can we use the same data sources to find good effects of drugs in combination that will provide us new treatments, new insights into how drugs work and enable us to take care of our patients even better? Thank you very much.

Text Sample 776: NEG Dim 5 – 2001 – Score 1.00 – 2001/t000083.txt

I thought I would read poems I have that relate to the **subject** of youth and age. I was sort of astonished to find out how many I have actually. The first one is dedicated to Spencer, and his grandmother, who was shocked by his work. My poem is called”Dirt.”My grandmother is washing my mouth out with soap ; half a long century gone and still she comes at me with that thick cruel yellow bar. All because of a word I said, not even said really, only repeated. But”Open,”she says,”open up!”her hand clawing at my head. I know now her life was hard ; she lost three children as babies, then her husband died too, leaving young sons, and no money. She ’ d stand me in the sink to pee because there was never room in the toilet. But oh, her soap! Might its bitter burning have been what made me a poet? The street she lived on was unpaved, her flat, two cramped rooms and a fetid kitchen where she stalked and caught me. Dare I admit that after she did it I never really loved her again? She lived to a hundred, even then. All along it was the sadness, the squalor, but I never, until now loved her again. When that was published in a magazine I got an irate letter from my uncle.”You have maligned a great woman.”It took some diplomacy. This is called”The Dress.”It ’ s a longer poem. In those days, those days which exist for me only as the most elusive memory now, when often the first sound you ’ d hear in the morning would be a storm of birdsong, then the soft clop of the hooves of the horse hauling a milk wagon down your block, and the last sound at night as likely as not would be your father pulling up in his car, having worked late again, always late, and going heavily down to the cellar, to the furnace, to shake out the ashes and damp the draft before he came upstairs to fall into bed — in those long-ago days, women, my mother, my friends ’ mothers, our neighbors, all the women I knew — wore, often much of the day, what were called housedresses, cheap, printed, pulpy, seemingly purposefully shapeless light cotton shifts that you wore over your nightgown and, when you had to go look for a child, hang wash on the line, or run down to the grocery store on the corner, under a coat, the twisted hem of the nightgown always lank and yellowed, dangling beneath. More than the curlers some of the women seemed constantly to have in their hair in preparation for some great event — a ball, one would think — that never came to pass ; more than the way most women ’ s faces not only were never made up during the day, but seemed scraped, bleached, and, with their plucked eyebrows, scarily masklike ; more than all that it was those dresses that made women so unknowable and forbidding, adepts of enigmas to which men could have no access, and boys no conception. Only later would I see the dresses also as a proclamation : that in your dim kitchen, your laundry, your bleak concrete yard, what you revealed of yourself was a fabulation ; your real sensual nature, veiled in those sexless vestments, was utterly your dominion. In those days, one hid much else as well : grown men didn ’ t embrace one another, unless someone had died, and not always then ; you shook hands or, at a ball game, thumped your friend ’ s back and exchanged **blows** meant to be codes for affection ; once out of childhood you ’ d never again know the shock of your father ’ s whiskers on your cheek, not until mores at last had evolved, and you could hug another man, then hold on for a moment, then even kiss. What release finally, the embrace : though we were wary — it seemed so audacious — how much unspoken joy there was in that affirmation of equality and communion, no matter how much misunderstanding and pain had passed between you by then. We knew so little in those days, as little as now, I suppose about healing those hurts : even the women, in their best dresses, with beads and sequins sewn on the bodices, even in lipstick and mascara, their hair aflow, could only stand wringing their hands, begging for peace, while father and son, like thugs, like thieves, like Romans, simmered and hissed and hated, inflicting sorrows that endured, the worst anyway, through the kiss and embrace, bleeding from brother to brother, into the generations. In those days there was still countryside close to the city, farms, cornfields, cows ; even not far from our building with its blurred brick and long shadowy hallway you could find tracts with hills and trees you could pretend were mountains and forests. Or you could go out by yourself even to a half-block-long empty lot, into the bushes : like a creature of leaves you ’ d lurk, crouched, crawling, simplified, savage, alone ; already there was wanting to be simpler, wanting, when they called you, never to go back. This is another longish one, about the old and the young. It actually happened right at the time we met. Part of the poem takes place in space we shared and time we shared. It ’ s called”The Neighbor.”Her five horrid, deformed little dogs who incessantly yap on the roof under my window. Her cats, God knows how many, who must piss on her rugs — her landing ’ s a sickening reek. Her shadow once, fumbling the chain on her door, then the door slamming fearfully shut, only the barking and the music — jazz — filtering as it does, day and night into the hall. The time it was Chris Connor singing”Lush Life”— how it brought back my college sweetheart, my first real love, who — till I left her — played the same record. And head on my shoulder, hand on my thigh, sang sweetly along, of regrets and depletions she was too young for, as I was too young, later, to believe in her pain. It startled, then bored, then repelled me. My starting to fancy she ’ d ended up in this fire-trap in the Village, that my neighbor was her. My thinking we ’ d meet, recognize one another, become friends, that I ’ d accomplish a penance. My seeing her, it wasn ’ t her, at the mailbox. Gray-yellow hair, army pants under a nightgown, her turning away, hiding her ravaged face in her hands, muttering an inappropriate”Hi.”Sometimes there are frightening goings-on in the stairwell. A man shouting,”Shut up!”The dogs frantically snarling, claws scrabbling, then her — her voice hoarse, harsh, hollow, almost only a tone, incoherent, a note, a squawk, bone on metal, metal gone molten, calling them back,”Come back darlings, come back dear ones. My sweet angels, come back.”Medea she was, next time I saw her. Sorceress, tranced, ecstatic, stock-still on the sidewalk ragged coat hanging agape, passersby flowing around her, her mouth torn suddenly open as though in a scream, silently though, as though only in her brain or breast had it erupted. A cry so pure, practiced, detached, it had no need of a voice, or could no longer bear one. These invisible links that allure, these transfigurations, even of anguish, that hold us. The girl, my old love, the last lost time I saw her when she came to find me at a party, her drunkenly stumbling, falling, sprawling, skirt hiked, eyes veined red, swollen with tears, her shame, her dishonor. My ignorant, arrogant coarseness, my secret pride, my turning away. Still life on a rooftop, dead trees in barrels, a bench broken, dogs, excrement, sky. What pathways through pain, what junctures of vulnerability, what crossings and counterings? Too many lives in our lives already, too many chances for sorrow, too many unaccounted-for pasts.”Behold me,”the god of frenzied, inexhaustible love says, rising in bloody splendor,”Behold me.”Her making her way down the littered vestibule stairs, one agonized step at a time. My holding the door. Her crossing the fragmented tiles, faltering at the step to the street, droning, not looking at me,”Can you help me?”Taking my arm, leaning lightly against me. Her wavering step into the world. Her whispering,”Thanks love.”Lightly, lightly against me. I think I ’ ll lighten up a little. Another, different kind of poem of youth and age. It ’ s called”Gas.”Wouldn ’ t it be nice, I think, when the blue-haired lady in the doctor ’ s waiting room bends over the magazine table and farts, just a little, and violently blushes. Wouldn ’ t it be nice if intestinal gas came embodied in visible clouds, so she could see that her really quite inoffensive pop had only barely grazed my face before it drifted away. Besides, for this to have happened now is a nice coincidence. Because not an hour ago, while we were on our walk, my dog was startled by a backfire and jumped straight up like a horse bucking. And that brought back to me the stable I worked on weekends when I was 12, and a splendid piebald stallion, who whenever he was mounted would buck just like that, though more hugely of course, enormous, gleaming, resplendent. And the woman, her face abashedly buried in her”Elle”now, reminded me — I ’ d forgotten that not the least part of my awe consisted of the fact that with every jump he took the horse would powerfully fart. Phwap! Phwap! Phwap! Something never mentioned in the dozens of books about horses and their riders I devoured in those days. All that savage grandeur, the steely glinting hooves, the eruptions driven from the creature ’ s mighty innards, breath stopped, heart stopped, nostrils madly flared, I didn ’ t know if I wanted to break him, or be him. This is called”Thirst.”Many — most of my poems actually are urban poems. I happen to be reading a bunch that aren ’ t.”Thirst.”Here was my relation with the woman who lived all last autumn and winter, day and night, on a bench in the 103rd Street subway station, until finally one day she vanished. We regarded each other, scrutinized one another. Me shyly, obliquely, trying not to be furtive. She boldly, unblinkingly, even pugnaciously, wrathfully even, when her bottle was empty. I was frightened of her. I felt like a child. I was afraid some repressed part of myself would go out of control, and I ’ d be forever entrapped in the shocking seethe of her stench. Not excrement merely, not merely surface and orifice going unwashed, rediffusion of rum, there was will in it, and intention, power and purpose — a social, ethical rage and rebellion — despair too, though, grief, loss. Sometimes I ’ d think I should take her home with me, bathe her, comfort her, dress her. She wouldn ’ t have wanted me to, I would think. Instead, I ’ d step into my train. How rich I would think, is the lexicon of our self-absolving. How enduring, our bland fatal assurance that reflection is righteousness being accomplished. The dance of our glances, the clash, pulling each other through our perceptual punctures, then holocaust, holocaust, host on host of ill, injured presences, squandered, consumed. Her vigil somewhere I know continues. Her occupancy, her absolute, faithful attendance. The dance of our glances, challenge, abdication, effacement, the perfume of our consternation. This is a newer poem, a brand new poem. The title is”This Happened.”A student, a young woman in a fourth-floor hallway of her lycee, perched on the ledge of an open window chatting with friends between classes ; a teacher passes and chides her,”Be careful, you might fall,”almost banteringly chides her,”You might fall,”and the young woman, 18, a girl really, though she wouldn ’ t think that, as brilliant as she is, first in her class, and”Beautiful, too,”she ’ s often told, smiles back, and leans into the open window, which wouldn ’ t even be open if it were winter — if it were winter someone would have closed it — leans into the window, farther, still smiling, farther and farther, though it takes less time than this, really an instant, and lets herself fall. Herself fall. A casual impulse, a fancy, never thought of until now, hardly thought of even now... No, more than impulse or fancy, the girl knows what she ’ s doing, the girl means something, the girl means to mean, because it occurs to her in that instant, that beautiful or not, bright yes or no, she ’ s not who she is, she ’ s not the person she is, and the reason, she suddenly knows, is that there ’ s been so much premeditation where she is, so much plotting and planning, there ’ s hardly a person where she is, or if there is, it ’ s not her, or not wholly her, it ’ s a self inhabited, lived in by her, and seemingly even as she thinks it she knows what ’ s been missing : grace, not premeditation but grace, a kind of being in the world spontaneously, with grace. Weightfully upon me was the world. Weightfully this self which graced the world yet never wholly itself. Weightfully this self which weighed upon me, the release from which is what I desire and what I achieve. And the girl remembers, in this infinite instant already now so many times divided, the sadness she felt once, hardly knowing she felt it, to merely inhabit herself. Yes, the girl falls, absurd to fall, even the earth with its compulsion to take unto itself all that falls must know that falling is absurd, yet the girl falling isn ’ t myself, or she is myself, but a self I took of my own volition unto myself. Forever. With grace. This happened. I ’ ll read just one more. I don ’ t usually say that. I like to just end. But I ’ m afraid that Ricky will come out here and shake his fist at me. This is called”Old Man,”appropriately enough.”Special : big tits,”Says the advertisement for a soft-core magazine on our neighborhood newsstand. But forget her breasts. A lush, fresh-lipped blond, skin glowing gold, sprawls there, resplendent. 60 nearly, yet these hardly tangible, hardly better than harlots, can still stir me. Maybe a coming of age in the American sensual darkness, never seeing an unsmudged nipple, an uncensored vagina, has left me forever infected with an unquenchable lust of the eye. Always that erotic murmur, I ’ m hardly myself if I ’ m not in a state of incipient desire. God knows though, there are worse twists your obsessions can take. Last year in Israel, a young ultra-orthodox Rabbi guiding some teenage girls through the Shrine of the Shoah forbade them to look in one room. Because there were images in it he said were licentious. The display was a photo. Men and women stripped naked, some trying to cover their genitals, others too frightened to bother, lined up in snow waiting to be shot and thrown into a ditch. The girls, to my horror, averted their gaze. What carnal mistrust had their teacher taught them. Even that though. Another confession : Once in a book on pre-war Poland, a studio portrait, an absolute angel, an absolute angel with tormented, tormenting eyes. I kept finding myself at her page. That she died in the camps made her — I didn ’ t dare wonder why — more present, more precious. Died in the camps, that too people — or Jews anyway — kept from their children back then. But it was like sex, you didn ’ t have to be told. Sex and death, how close they can seem. So constantly conscious now of death moving towards me, sometimes I think I confound them. My wife ’ s loveliness almost consumes me. My passion for her goes beyond reasonable bounds. When we make love, her holding me everywhere all around me, I ’ m there and not there. My mind teems, jumbles of faces, voices, impressions, I live my life over, as though I were drowning. Then I am drowning, in despair at having to leave her, this, everything, all, unbearable, awful. Still, to be able to die with no special contrition, not having been slaughtered, or enslaved. And not having to know history ’ s next mad rage or regression, it might be a relief. No. Again, no. I don ’ t mean that for a moment. What I mean is the world holds me so tightly — the good and the bad — my own follies and weakness that even this counterfeit Venus with her sham heat, and her bosom probably plumped with gel, so moves me my breath catches. Vamp. Siren. Seductress. How much more she reveals in her glare of ink than she knows. How she incarnates our desperate human need for regard, our passion to live in beauty, to be beauty, to be cherished by glances, if by no more, of something like love, or love. Thank you.

Text Sample 777: NEG Dim 5 – 2001 – Score 2.00 – 2001/t000367.txt

I ’ d like to do pretty much what I did the first time, which is to choose a lighthearted theme. Last time, I talked about death and dying. This time, I ’ m going to talk about mental illness. But it has to be technological, so I ’ ll talk about electroshock therapy. You know, ever since man had any notion that some of his other people, his colleagues, could be different, could be strange, could be severely depressed or what we now recognize as schizophrenia, he was certain that this kind of illness had to come from evil spirits getting into the body. So, the way of treating these diseases in early times was to, in some way or other, exorcise those evil spirits, and this is still going on, as you know. But it wasn ’ t enough to use the priests. When medicine became somewhat scientific, in about 450 BC, with Hippocrates and those boys, they tried to look for herbs, plants that would literally shake the bad spirits out. So, they found certain plants that could cause convulsions. And the herbals, the botanical books of up to the late Middle Ages, the Renaissance are filled with prescriptions for causing convulsions to shake the evil spirits out. Finally, in about the sixteenth century, a physician whose name was Theophrastus Bombastus Aureolus von Hohenheim, called Paracelsus, a name probably familiar to some people here — — good, old Paracelsus found that he could predict the degree of convulsion by using a measured amount of camphor to produce the convulsion. Can you imagine going to your closet, pulling out a mothball, and chewing on it if you ’ re feeling depressed? It ’ s better than Prozac, but I wouldn ’ t recommend it. So what we see in the seventeenth, eighteenth century is the continued search for medications other than camphor that ’ ll do the trick. Well, along comes Benjamin Franklin, and he comes close to convulsing himself with a bolt of electricity off the end of his kite. And so people begin thinking in terms of electricity to produce convulsions. And then, we fast-forward to about 1932, when three Italian psychiatrists, who were largely treating depression, began to notice among their patients, who were also epileptics, that if they had an epileptic — a series of epileptic fits, a lot of them in a row — the depression would very frequently lift. Not only would it lift, but it might never return. So they got very interested in producing convulsions, measured types of convulsions. And they thought,”Well, we ’ ve got electricity, we ’ ll plug somebody into the wall. That always makes hair stand up and people shake a lot.”So, they tried it on a few pigs, and none of the pigs were killed. So, they went to the police and they said,”We know that at the Rome railroad station, there are all these lost souls wandering around, muttering gibberish. Can you bring one of them to us?”Someone who is, as the Italians say,”cagoots.”So they found this”cagoots”guy, a 39-year-old man who was really hopelessly schizophrenic, who was known, had been known for months, to be literally defecating on himself, talking nothing that made any sense, and they brought him into the hospital. So these three psychiatrists, after about two or three weeks of observation, laid him down on a table, connected his temples to a very small source of current. They thought,”Well, we ’ ll try 55 volts, two-tenths of a second. That ’ s not going to do anything terrible to him.”So they did that. Well, I have the following from a firsthand observer, who told me this about 35 years ago, when I was thinking about these things for some research project of mine. He said,”This fellow”— remember, he wasn ’ t even put to sleep —”after this major grand mal convulsion, sat right up, looked at these three fellas and said, ’ What the fuck are you assholes trying to do? ’”If I could only say that in Italian. Well, they were happy as could be, because he hadn ’ t said a rational word in the weeks of observation. So they plugged him in again, and this time they used 110 volts for half a second. And to their amazement, after it was over, he began speaking like he was perfectly well. He relapsed a little bit, they gave him a series of treatments, and he was essentially cured. But of course, having schizophrenia, within a few months, it returned. But they wrote a paper about this, and everybody in the Western world began using electricity to convulse people who were either schizophrenic or severely depressed. It didn ’ t work very well on the schizophrenics, but it was pretty clear in the ’ 30s and by the middle of the ’ 40s that electroconvulsive therapy was very, very effective in the treatment of depression. And of course, in those days, there were no antidepressant drugs, and it became very, very popular. They would anesthetize people, convulse them, but the real difficulty was that there was no way to paralyze muscles. So people would have a real grand mal seizure. Bones were broken. Especially in old, fragile people, you couldn ’ t use it. And then in the 1950s, late 1950s, the so-called muscle relaxants were developed by pharmacologists, and it got so that you could induce a complete convulsion, an electroencephalographic convulsion — you could see it on the brain waves — without causing any convulsion in the body except a little bit of twitching of the toes. So again, it was very, very popular and very, very useful. Well, you know, in the middle ’ 60s, the first antidepressants came out. Tofranil was the first. In the late ’ 70s, early ’ 80s, there were others, and they were very effective. And patients ’ rights groups seemed to get very upset about the kinds of things that they would witness. And so the whole idea of electroconvulsive, electroshock therapy disappeared, but has had a renaissance in the last 10 years. And the reason that it has had a renaissance is that probably about 10 percent of the people, severe depressives, do not respond, regardless of what is done for them. Now, why am I telling you this story at this meeting? I ’ m telling you this story, because actually ever since Richard called me and asked me to talk about — as he asked all of his speakers — to talk about something that would be new to this audience, that we had never talked about, never written about, I ’ ve been planning this moment. This reason really is that I am a man who, almost 30 years ago, had his life saved by two long courses of electroshock therapy. And let me tell you this story. I was, in the 1960s, in a marriage. To use the word bad would be perhaps the understatement of the year. It was dreadful. There are, I ’ m sure, enough divorced people in this room to know about the hostility, the anger, who knows what. Being someone who had had a very difficult childhood, a very difficult adolescence — it had to do with not quite poverty but close. It had to do with being brought up in a family where no one spoke English, no one could read or write English. It had to do with death and disease and lots of other things. I was a little prone to depression. So, as things got worse, as we really began to hate each other, I became progressively depressed over a period of a couple of years, trying to save this marriage, which was inevitably not to be saved. Finally, I would schedule — all my major surgical cases, I was scheduling them for 12, one o ’ clock in the afternoon, because I couldn ’ t get out of bed before about 11 o ’ clock. And anybody who ’ s been depressed here knows what that ’ s like. I couldn ’ t even pull the covers off myself. Well, you ’ re in a university medical center, where everybody knows everybody, and it ’ s perfectly clear to my colleagues, so my referrals began to decrease. As my referrals began to decrease, I clearly became increasingly depressed until I thought, my God, I can ’ t work anymore. And, in fact, it didn ’ t make any difference because I didn ’ t have any patients anymore. So, with the advice of my physician, I had myself admitted to the acute care psychiatric unit of our university hospital. And my colleagues, who had known me since medical school in that place, said,”Don ’ t worry, chap. Six weeks, you ’ re back in the operating room. Everything ’ s going to be great.”Well, you know what bovine stercus is? That proved to be a lot of bovine stercus. I know some people who got tenure in that place with lies like that. So I was one of their failures. But it wasn ’ t that simple. Because by the time I got out of that unit, I was not functional at all. I could hardly see five feet in front of myself. I shuffled when I walked. I was bowed over. I rarely bathed. I sometimes didn ’ t shave. It was dreadful. And it was clear — not to me, because nothing was clear to me at that time anymore — that I would need long-term hospitalization in that awful place called a mental hospital. So I was admitted, in 1973, in the spring of 1973, to the Institute of Living, which used to be called the Hartford Retreat. It was founded in the eighteenth century, the largest psychiatric hospital in the state of Connecticut, other than the huge public hospitals that existed at that time. And they tried everything they had. They tried the usual psychotherapy. They tried every medication available in those days. And they did have Tofranil and other things — Mellaril, who knows what. Nothing happened except that I got jaundiced from one of these things. And finally, because I was well known in Connecticut, they decided they better have a meeting of the senior staff. All the senior staff got together, and I later found out what happened. They put all their heads together and they decided that there was nothing that could be done for this surgeon who had essentially separated himself from the world, who by that time had become so overwhelmed, not just with depression and feelings of worthlessness and inadequacy, but with obsessional thinking, obsessional thinking about coincidences. And there were particular numbers that every time I saw them, just got me dreadfully upset — all kinds of ritualistic observances, just awful, awful stuff. Remember when you were a kid, and you had to step on every line? Well, I was a grown man who had all of these rituals, and it got so there was a throbbing, there was a ferocious fear in my head. You ’ ve seen this painting by Edvard Munch, The Scream. Every moment was a scream. It was impossible. So they decided there was no therapy, there was no treatment. But there was one treatment, which actually had been pioneered at the Hartford hospital in the early 1940s, and you can imagine what it was. It was pre-frontal lobotomy. So they decided — I didn ’ t know this, again, I found this out later — that the only thing that could be done was for this 43-year-old man to have a pre-frontal lobotomy. Well, as in all hospitals, there was a resident assigned to my case. He was 27 years old, and he would meet with me two or three times a week. And of course, I had been there, what, three or four months at the time. And he asked to meet with the senior staff, and they agreed to meet with him because he was very well thought of in that place. They thought he had a really extraordinary future. And he dug in his heels and said,”No. I know this man better than any of you. I have met with him over and over again. You ’ ve just seen him from time to time. You ’ ve read reports and so forth. I really honestly believe that the basic problem here is pure depression, and all of the obsessional thinking comes out of it. And you know, of course, what ’ ll happen if you do a pre- frontal lobotomy. Any of the results along the spectrum, from pretty bad to terrible, terrible, terrible is going to happen. If he does the best he can, he will have no further obsessions, probably no depression, but his affect will be dulled, he will never go back to surgery, he will never be the loving father that he was to his two children, his life will be changed. If he has the usual result, he will end up like ’ One Flew Over the Cuckoo ’ s Nest. ’ And you know about that, just essentially in a stupor the rest of his life.”Well, he said,”Can ’ t we try a course of electroshock therapy?”And you know why they agreed? They agreed to humor him. They just thought,”Well, we ’ ll give a course of 10. And so we ’ ll lose a little time. Big deal. It doesn ’ t make any difference.”So they gave the course of 10, and the first — the usual course, incidentally, was six to eight and still is six to eight. Plugged me into the wires, put me to sleep, gave me the muscle relaxant. Six didn ’ t work. Seven didn ’ t work. Eight didn ’ t work. At nine, I noticed — and it ’ s wonderful that I could notice anything — I noticed a change. And at 10, I noticed a real change. And he went back to them, and they agreed to do another 10. Again, not a single one of them — I think there are about seven or eight of them — thought this would do any good. They thought this was a temporary change. But, lo and behold, by 16, by 17, there were demonstrable differences in the way I felt. By 18 and 19, I was sleeping through the night. And by 20, I had the sense, I really had the sense that I could overcome this, that I was now strong enough that by an act of will, I could **blow** the obsessional thinking away. I could **blow** the depression away. And I ’ ve never forgotten — I never will forget — standing in the kitchen of the unit, it was a Sunday morning in January of 1974, standing in the kitchen by myself and thinking,”I ’ ve got the strength now to do this.”It was as though those tightly coiled wires in my head had been disconnected and I could think clearly. But I need a formula. I need some thing to say to myself when I begin thinking obsessionally, obsessively. Well, the Gilbert and Sullivan fans in this room will remember”Ruddigore,”and they will remember Mad Margaret, and they will remember that she was married to a fellow named Sir Despard Murgatroyd. And she used to go nuts, every five minutes or so in the play, and he said to her,”We must have a word to bring you back to reality, and the word, my dear, will be ’ Basingstoke. ’”So every time she got a little nuts, he would say,”Basingstoke!”And she would say,”Basingstoke, it is.”And she would be fine for a little while. Well, you know, I ’ m from the Bronx. I can ’ t say”Basingstoke.”But I had something better. And it was very simple. It was,”Ah, fuck it!”Much better than”Basingstoke,”at least for me. And it worked — my God, it worked. Every time I would begin thinking obsessionally — again, once more, after 20 shock treatments — I would say,”Ah, fuck it.”And things got better and better, and within three or four months, I was discharged from that hospital, and I joined a group of surgeons where I could work with other people in the community, not in New Haven, but fairly close by. I stayed there for three years. At the end of three years, I went back to New Haven, had remarried by that time. I brought my wife with me, actually, to make sure I could get through this. My children came back to live with us. We had two more children after that. Resuscitated the career, even better than it had been before. Went right back into the university and began to write books. Well, you know, it ’ s been a wonderful life. It ’ s been, as I said, close to 30 years. I stopped doing surgery about six years ago and became a full-time writer, as many people know. But it ’ s been very exciting. It ’ s been very happy. Every once in a while, I have to say,”Ah, fuck it.”Every once in a while, I get somewhat depressed and a little obsessional. So, I ’ m not free of all of this. But it ’ s worked. It ’ s always worked. Why have I chosen, after never, ever talking about this, to talk about it now? Well, those of you who know some of these books know that one is about death and dying, one is about the human body and the human spirit, one is about the way mystical thoughts are constantly in our minds, and they have always to do with my own personal experiences. One might think reading these books — and I ’ ve gotten thousands of letters about them by people who do think this — that based on my life ’ s history as I ’ ve portrayed in the books, my early life ’ s history, I am someone who has overcome adversity. That I am someone who has drunk, drank, drunk of the bitter dregs of near-disaster in childhood and emerged not just unscathed but strengthened. I really have it figured out, so that I can advise people about death and dying, so that I can talk about mysticism and the human spirit. And I ’ ve always felt guilty about that. I ’ ve always felt that somehow I was an impostor because my readers don ’ t know what I have just told you. It ’ s known by some people in New Haven, obviously, but it is not generally known. So one of the reasons that I have come here to talk about this today is to — frankly, selfishly — unburden myself and let it be known that this is not an untroubled mind that has written all of these books. But more importantly, I think, is the fact that a very significant proportion of people in this audience are under 30, and there are many, of course, who are well over 30. For people under 30, and it looks to me like almost all of you — I would say all of you — are either on the cusp of a magnificent and exciting career or right into a magnificent and exciting career : anything can happen to you. Things change. Accidents happen. Something from childhood comes back to haunt you. You can be thrown off the track. I hope it happens to none of you, but it will probably happen to a small percentage of you. To those to whom it doesn ’ t happen, there will be adversities. If I, with the bleakness of spirit, with no spirit, that I had in the 1970s and no possibility of recovery, as far as that group of very experienced psychiatrists thought, if I can find my way back from this, believe me, anybody can find their way back from any adversity that exists in their lives. And for those who are older, who have lived through perhaps not something as bad as this, but who have lived through difficult times, perhaps where they lost everything, as I did, and started out all over again, some of these things will seem very familiar. There is recovery. There is redemption. And there is resurrection. There are resurrection themes in every society that has ever been studied, and it is because not just only do we fantasize about the possibility of resurrection and recovery, but it actually happens. And it happens a lot. Perhaps the most popular resurrection theme, outside of specifically religious ones, is the one about the phoenix, the ancient story of the phoenix, who, every 500 years, resurrects itself from its own ashes to go on to live a life that is even more beautiful than it was before. Richard, thanks very much.

Text Sample 778: NEG Dim 5 – 2001 – Score 2.00 – 2001/t000092.txt

I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ’ children, because they were all average. But they weren ’ t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ’ s not right. The good Lord in his infinite wisdom didn ’ t create us all equal as far as intelligence is concerned, any more than we

’ re equal for size, appearance. Not everybody could earn an A or a B, and I didn ’ t like that way of judging, and I did know how the alumni of various schools back in the ’ 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ’ t lose a game. But it seemed that we didn ’ t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ’ t like it, I didn ’ t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ’ m sure at the time he did that, I didn ’ t — it didn ’ t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ’ s under your control. If you get too engrossed and involved and concerned in regard to the things over which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,”At God ’ s footstool to confess, a poor soul knelt, and bowed his head. ’ I failed! ’ he cried. The Master said, ’ Thou didst thy best, that is success. ’”From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ’ re capable. I believe that ’ s true. If you make the effort to do the best of which you ’ re capable, trying to improve the situation that exists for you, I think that ’ s success, and I don ’ t think others can judge that ; it ’ s like character and reputation — your reputation is what you ’ re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ’ s what the teachers are themselves.”That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by coal oil lamp — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I reply,

’ Where could I find such splendid company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ’ s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ’ Where could I find such splendid company? ’”And I believe the teaching profession — it ’ s true, you have so many youngsters, and I ’ ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ’ re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ’ re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was”Never be late.”Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ’ s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ’ t clean and neat, so I wouldn ’ t let him go. He couldn ’ t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ’ t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ’ ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,”Don ’ t run practices late, because you ’ ll go home in a bad mood, and that ’ s not good, for a young married man to go home in a bad mood. When you get older, it doesn ’ t make any difference, but —”So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ’ re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ’ t want that. I used to tell them I was paid to do that. That ’ s my job. I ’ m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ’ s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ’ s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ’ t have the time to go on that. But that helped me, I think, become a better teacher. It ’ s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ’ re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ' re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ' t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ' i '. I ' d never seen that before, but he did. Big league baseball players — they ' re very perceptive about those things, and they noticed he had only one ' i ' in his name. You ' d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it"The Road Ahead, or the Road Behind."He said,"Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ' t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ' s game. You and I know deeper down, there ' s always a chance to win the crown. But when we fail to give our best, we simply haven ' t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ' s bearing down that wins the cup. Of dreaming there ' s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ' s you and I who make our fates — we open up or close the gates on the road ahead or the road behind."Reminds me of another set of threes that my dad tried to get across to us : Don ' t whine. Don ' t complain. Don ' t make excuses. Just get out there, and whatever you ' re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ' re outscored. I ' ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ' t know the outcome, I hope they couldn ' t tell by your actions whether you outscored an opponent or the opponent outscored you. That ' s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ' d want them to be but they ' ll be about what they should ; only you will know whether you can do that. And that ' s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,"The journey is better than the end."And I like that. I think that it is — it ' s getting there. Sometimes when you get there, there ' s almost a let down. But it ' s the getting there that ' s the fun. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ’ d done a decent job during the week. There again, it ’ s getting the players to get that self-satisfaction, in knowing that they ’ d made the effort to do the best of which they are capable. Sometimes I ’ m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,”Suppose that you, in some way, could make the perfect player. What would you want?”And I said,”Well, I ’ d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ’ d want one that could play, too. I ’ d want one to realize that defense usually wins championships, and who would work hard on defense. But I ’ d want one who would play offense, too. I ’ d want him to be unselfish, and look for the pass first and not shoot all the time. And I ’ d want one that could pass and would pass. I ’ ve had some that could and wouldn ’ t, and I ’ ve had some that would and could. So, yeah, I ’ d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ’ d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,”They Call Me Coach,”two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ’ t play varsity when I taught. I thought,”Oh gracious, if these two players, either one of them”— they were different years, but I thought about each one at the time he was there —”Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ’ s good enough to make it.”And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ’ d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 779: NEG Dim 5 – 2022 – Score -1.00 – 2022/t003570.txt

About a decade ago, I met someone who had experienced a few episodes of schizophrenia. They had felt that their sense of self, of what it feels like to be them, changing somewhat. The boundaries of their body began to feel a bit nebulous. Even their psychological self felt a bit porous at times. They were experiencing what could be called an altered sense of self. Over the years, I met many such brave and insightful people who shared what it ’ s like to live with their altered selves. And by”altered,”I mean”different,”not”deficient,”while acknowledging that coping with altered selves can be a struggle at times. So speaking with them, and with theologians, philosophers, neuroscientists, I came to understand that this self that each one of us takes oneself to be is not as real as it seems. The self is a slippery **subject**. We all intuitively know what it means. It ’ s there when we wake up. It disappears when we fall asleep. It reappears in our dreams. It ’ s what makes us who we are. It seems solid, unchanging, permanent. And yet, we can examine aspects of the self that seem real to us, and ask,”Just how real are they?”Take, for instance, the question”Who am I?”The most likely answer you will get or give to such a question will be in the form of a story. We tell others – and indeed, ourselves – stories about who we are. We take our stories to be sacrosanct. We are our stories. But a condition that most of us, sadly, will be familiar with – Alzheimer ’ s disease – tells us something quite different. Alzheimer ’ s begins by affecting short-term memory. Think about what that does to someone ’ s story. In order for our stories to form, to grow, something that just happens to us has to first enter short-term memory, and then, get incorporated into what ’ s called long-term episodic memory. It has to become an episode in our narrative. But what if the experience doesn ’ t even enter short-term memory? That ’ s exactly what Alzheimer ’ s does. In the beginning, Alzheimer ’ s impairs the formation of short-term memory. It impairs the growth of the narrative. It ’ s as if our stories begin stalling upon the onset of the disease. Eventually, Alzheimer ’ s eats away at all the long-term memories. So if you were to meet someone with mid-stage Alzheimer ’ s, they will likely be able to tell you stories about who they are. But if you know their real stories, you ’ ll be able to tell that they sometimes scramble up their narrative, that they sometimes mix up the sequence of episodes from their lives. It ’ s as if they are recalling their own stories in ways that are not quite accurate. It ’ s important, at this stage, to realize that there is still a person experiencing that scrambled narrative. Sadly, Alzheimer ’ s goes on to destroy one ’ s narrative, and so much more. And towards the end, it ’ s unclear whether there is still someone experiencing something, because the person cannot communicate verbally anymore. And yet, Alzheimer ’ s tells us that these stories that we take ourselves to be, what philosophers call the”narrative self,”these are spun by the brain and body. They are constructions. Sometimes, the constructions are disrupted, even destroyed. And while that is horrific for the person experiencing it, and for their caregivers, it is nonetheless a window onto the constructed nature of our narrative self. And when the construction goes wrong, we perceive our own stories in ways that are not quite real. From the narrative self, let ’ s talk about our body. Let ’ s take a very basic aspect of our bodily self. This feeling we all have, that we are owners of our body and body parts, that our bodies and body parts belong to us. It seems such a strange thing to think that it could even be otherwise. If I were to ask you,”Does your hand belong to you?”you ’ re going to say,”Of course it does. What a foolish question.”But not everyone would agree. Early on in my research, a neuropsychologist alerted me to a condition called xenomelia, or foreign limb syndrome. You may have heard of something called phantom limb syndrome, in which people who have had an amputation feel the presence of that limb, sometimes. Xenomelia is somewhat of an opposite condition, where people feel like some part of their body – usually the extremities, their hands or legs – don ’ t belong to them. So this neuropsychologist talked of phantom limb syndrome as animation without incarnation. So the limb is gone, it ’ s not incarnate anymore, but it ’ s animated in your mind. And he talked of xenomelia as incarnation without animation. So the limb is present, healthy even, incarnate, and yet, in your own mind, it feels like it doesn ’ t belong to you. So in xenomelia, the brain and bodily processes that give rise to our sense of ownership of our body parts, they ’ re misfiring, so to speak, and the consequences can be serious. People with xenomelia will sometimes take extreme measures to get rid of, to amputate their foreign-seeming body parts. From the perspective of the self, though, xenomelia is telling us something very profound. It ’ s telling us that something as basic as the sense of ownership of our own body parts is a construction. And sometimes, the construction goes wrong, and we perceive our own bodies in ways that are not quite real. Let ’ s take another aspect of our bodily self. It ’ s called the sense of agency. So when I do something like pick up a cup, I have this implicit feeling that I am the agent of that action, that I have willed that action into existence. That feeling is the sense of agency. But someone with schizophrenia may not have that feeling, always. Someone with schizophrenia might do something and not feel like they are the agent of that action. So schizophrenia tells us that it is possible to be someone who does something but doesn ’ t have an accompanying sense of agency. So just like the narrative self and the sense of ownership of body parts, the sense of agency is also a construction, and it, too, can fail. So you can see where this is going. Let me take one more example to drive home this point. Let ’ s talk of what it feels to be a body here and now. Not the feeling of being a story, but the feeling of being a body in the present moment. Psychologists estimate that about five percent of the general population will, at some point in their lives, have an out-of-body experience. Let ’ s assume that all of us right now are having an in-body experience. But what that means is having this feeling of being in a body, being anchored to a body, occupying a certain volume of space and looking at the world from behind our eyes. But if you are having an out-of-body experience, you could possibly be feeling that you ’ re up near the ceiling, looking down at your own body sitting in the chair below. People do report such experiences, and mild versions of this have been replicated in labs. But if you think, like I do, that out-of-body experiences are the outcome of brain processes that are misfiring, then it stands to reason that the experience of being in-body, of being embodied, is itself a construction, and that, too, can come apart. So what are these experiences of altered selves telling us? They ’ re telling us that just about everything we take to be real about ourselves –”real”in the sense that we think we are always experiencing undeniable truths about our bodies, our stories – well, that ’ s just not the case. So when theologians and philosophers tell us that the self is an illusion, this is partly what they mean. You may have realized by now that there still remains the question of who or what is doing the experiencing, even in the case of altered selves. This experiencing”I”in the question”Who am I?”is at the heart of the debate about the self. This experiencing”I”doesn ’ t go away if one or a few aspects of the self are disrupted. But what if all of the aspects of the self that comprise us were to be disrupted? Would the experiencing”I”disappear? We don ’ t have a satisfactory answer to that question, yet. It ’ s possible that the experiencing”I”is also an illusion, in the sense of being a construction, a construction without a constructor. That debate, however, is somewhat unresolved. Despite such doubts, I, personally – whatever I am – think that the self has no reality outside of the brain and body. I think that the experiencing”I”will not persist after the body is gone. So what does one make of such knowledge? Well, firstly, these ideas will feel liberating to some and might sit heavily upon others. Regardless, I think we can all attend to the stories that we think we are. Our feelings and emotions are modulated by our stories, and in turn, our feelings and emotions become part of our stories. And our stories, our narratives, are not just cognitive – they live in our bodies, and our bodies structure and shape our stories. So knowing all this, recognizing the constructive nature of it all, maybe we can hold on less tightly to our stories. Maybe we can learn to let go. But that ’ s easier said than done, because the thing that is doing the letting go is also the thing that has to be let go of. Maybe we can just marvel at the efforts of people over millennia, from the Buddha sitting under the Bodhi tree to the modern philosopher and neuroscientist who has asked themselves the question”Who am I?”But most of all, I think we owe a debt to those amongst us who bravely bear witness to our altered selves – whether we do so voluntarily, like monks and nuns do when they meditate, or whether it ’ s brought upon us by biology and circumstance. There is something remarkably robust about the processes that give rise to the totality of our sense of self. But there ’ s something frighteningly fragile about them too. They can crack. And any one of us, at any time in our lives, may have to confront such cracks. And that knowledge, I believe, should make us empathetic towards those of us dealing with altered selves. But I also believe that altered selves should not be seen as the outcome of deficits, or as the outcome of a lack of attributes considered normal. They are different ways of being, and it ’ s the willingness of some of us to confront the self ’ s constructed nature that is helping make sense of the self for all of us. Thank you. Thank you.

Text Sample 780: NEG Dim 5 – 2022 – Score -1.00 – 2022/t003596.txt

So I would like to discuss about meditation. But first of all, I ’ d like to ask you a very simple question. Can you see my hand? Yes, raise your hand. Audience : Yes. Yongey Mingyur Rinpoche : OK, can you hear me? Audience : Yes. YMR : Yes? Great. That is the meditation. So, finished. My TED Talk is finished. Of course, I ’ m just kidding. But in a way that is true. Why? What we call the essence of meditation is awareness. And what is awareness? Knows what you are thinking, feeling, doing, seeing, hearing. That ’ s all. So actually meditation is very easy, but many people find it difficult. Why? There are two misunderstandings about meditation. So the first is many people think meditation means, think of nothing, stop thinking, concentration. Ah. Shh! I ’ m meditating, keep quiet. So the more, when you try to stop thinking what happens, you will think more. So we will do a small experiment, OK? Now, please don ’ t think about pizza. No pizza. No pizza. What happened? Did you think about pizza or not? Yes, raise your hand. I know. Actually, we don ’ t need to stop thinking. We just need to connect with awareness. And another misunderstanding about meditation is what we call”blissing out.”Looking for peace, calm, joy, relax. The more you look for relax, then peace, calm, joy, relax run away. So... Let me share [with] you my own experience. When I was young, I had panic attacks. Although I was born right in the middle of the Himalayan mountains, the area, the village, wonderful, but panic followed me like a shadow. And I have so much fear [of] strangers, I cannot go out and meet people. And lot of storms in the Himalayan mountains, thunderstorm, snowstorm. These storms drive me crazy. And when I was nine years old, I asked my father to teach me meditation. Luckily, he was a great meditation teacher. And the first thing what he said is,”Don ’ t try to fight with the panic. Don ’ t try to get rid of panic.””And actually,”he said,”you don ’ t have to.”Why? Awareness is like sky in the mountain, and the panic is like storm in the mountain, like cloud. And no matter how strong a storm is, it doesn ’ t change the nature of sky. So sky is always present, pure, calm. Similarly, our fundamental quality of mind, awareness is always present, pure, calm. But the problem is we don ’ t know how to connect with awareness, what we see, only a thought, emotion, that ’ s all. So he said, there are three steps of practice to connect with awareness. The first one, we have to use an object. Support, to connect with awareness. So this is one of my first meditation techniques that I learned from my father. So you can join and you can relax your muscles in your body. If you cannot relax, also OK, allowed. And close your eyes, and please listen to sound. And when you hear the sound, through ear and mind together... That is the meditation. And let panic come and go. Let pizza come and go. And maybe two pizzas, three pizzas, ten pizzas. As long as you remember the sound, you can have pizza. OK, how was it? Did you hear the sound? Yes, raise your hand. Great. That is the meditation. Very easy. Just hear, that ’ s all. You don ’ t have to do anything. So if panic comes, let panic come and go, don ’ t care. Just listen to sound. Monkey mind comes blah, blah, blah. Let it come and go, just listen to sound. So I did that. But I had a big problem. The problem is lazy. I ’ m lazy boy, I love the idea of meditation, but I don ’ t like the practice of meditation. So on and off like that for five years. When I was 13 years old in India, there ’ s traditional three-year retreat going to start. I thought I should join this because it ’ s good for my laziness. And I join. The first month, wonderful, no lazy. Second month, lazy come back. Now what happened? My lazy and my panic, they two become good friends. The life in the retreat becomes disaster. And I thought I should leave. But I feel embarrassed to leave because I told all my childhood friends that I can do the retreat, you know. I don ’ t want to lose my face. But if I stay, almost three years to go. Then I thought, what should I do? In the end, I decided to learn how to live with the panic. So now we have the second step what we call, actually, we can meditate everywhere, anytime with anything. So you can meditate with panic. How do you do that? Just like listen to sound, when you listen to sound, sound becomes support for your meditation. Now you ’ re going to watch panic. If you see the panic, great. What we call, when you see the river, you ’ re out of the river. When you see the mountain, you ’ re out of the montain. So now awareness becomes more than panic. More than depression, stress. Monkey mind, whatever. Let them come, let them go. So that is the first benefit and second benefit of what we call, there ’ s a wisdom comes. So when you look at the panic, panic is not solid stone anymore. Panic becomes pieces. Sensation here, frightening image, voice, background belief. And if you take one of these away, you cannot find panic. So what I call panic becomes like shaving foam, looks like a piece of rock but inside full of bubbles. Then three, what I call this is acceptance, like self-kind, self-love, self-compassion. You let panic come and go, that ’ s the real acceptance, isn ’ t it? So three in one : awareness, love and compassion, wisdom. Sometimes what I call buy one, get two free. Big deal, right? And all this because of panic. So now panic becomes your teacher, your best friend. So I did this practice and in the end, me and my panic become very good friends. And few weeks later, panic is gone. I miss my friend. And I finished my retreat, my retreat went very well. And after that I have been eager to share this wonderful technique with the world. So I taught meditation many places, wrote three books, become bestseller, and then students and become abbot of a few monasteries. And what happened? Kind of new ego emerged within me. I thought, oh, I have to watch out, this so. Then I decided to do something very special, what we call wandering retreat. Meaning you leave everything behind, go on the street with nothing. So I decided to do that. In 2011, I left my monastery, my students, my wonderful cozy bed. Everything, and got on street with only a few thousand Indian rupees, and that finished within a few weeks. Now I have to beg for food. And I got food poisoning. Vomiting, diarrhea. I ’ m alone on the street, now I ’ m going to die. Then I thought, what should I do? So now we have the third step, what we call open awareness meditation. Awareness, be with itself. Sky, be with itself. Now, no need to have support. Just be awareness itself. I did that practice. Then what happened? My body become decayed, I cannot see, I cannot hear. But my mind becomes so present, beyond free. And I was in that state for a few hours. Luckily, I didn ’ t die, I come back. So when I come back, the street becomes like my home. When I look at a tree, tree becomes like tree of love. And the wind **blows** to my face become a joyful experience. And the rest of my retreat went very well. I learned a lot from my retreat. So I like to share this open awareness meditation, but it is very difficult to explain. But I want to do something drama. And this is what I learned from my father. So what we call, this mala is the crazy monkey mind, blah, blah, blah, yada yada. And the open awareness meditation means you don ’ t have to do anything. Just be. That ’ s all, and you don ’ t need to meditate. The sense of presence, being, but not loss. Be free. Be present. Thank you very much.

Text Sample 781: NEG Dim 5 – 2022 – Score 1.00 – 2022/t003808.txt

Let ’ s talk about regret. It is, to my mind, our most misunderstood emotion, and so I decided to spend a couple of years studying it. And one of the things that I did is I went back and I looked at about 50 years of social science on regret. And here ’ s what it tells you. I ’ ll save you the trouble of reading a half century of social science. The research tells us that everybody has regrets, regrets make us human. Truly, the only people without regrets are five-year-olds, people with brain damage and sociopaths. The rest of us, we have regrets, and if we treat our regrets right, and that ’ s a big if, but there are ways to do it, regrets can actually make us better. They can improve our decision-making skills, improve our negotiation skills, make us better strategists, make us better problem solvers, enhance our sense of meaning if we treat them right. And the good news is that there ’ s a systematic way to do that. But I want to take just a few minutes to tell you about another aspect of regret that I think is really, really just super interesting. As part of the research here, I decided to ask people for their regrets, and to my surprise, I ended up collecting about 16, 000 regrets from people in 105 countries. It ’ s an extraordinary trove. And what I realized when I went through this incredible database of human longing and aspiration is that around the world, and there ’ s very little national difference here, people kept expressing the same four regrets. Around the world, there are the same four regrets that keep coming up over and over and over again. So what I want to do is just quickly tell you about these four core regrets, because I think they reveal something incredibly important and interesting. So the four core regrets that I ’ m going to cover. Number one, what I call foundation regrets. Foundation regrets. These are people who regret things like this : not saving enough money, which would be like, you know, financial regret, not taking care of their health and not eating right, health regret. But they ’ re the same. Those kinds of regrets are about making choices that didn ’ t allow you to have some stability, a stable platform for their life. I have a lot of people who regret not working hard enough in school. A lot of people who regret – I got a lot of regrets about not saving money. And it reminds me a little bit of Aesop ’ s fable of the ant and the grasshopper, where earlier in their life they acted like a grasshopper instead of the ant, and now it ’ s catching up with them. So foundation regrets sound like this :”If only”– And that ’ s the catch phrase of regret,”if only”–”If only I ’ d done the work.”Second category. I love this category, it ’ s fascinating. Boldness regrets. I have hundreds of regrets around the world that go like this :”X years ago, there was a man / woman whom I really liked. I wanted to ask him / her out on a date, but I was too scared to do it and I ’ ve regretted it ever since.”I also have hundreds of regrets by people who said :”Oh, I always wanted to start a business, but I never had the guts to do that.”People who said :”Oh, I wish I ’ d spoken up more. I wish I ’ d said something and asserted myself.”These are, as I said before, what I call boldness regrets. And we get to a juncture in our life and we have a choice. We can play it safe or we can take the chance. And what I found is overwhelmingly people regret not taking the chance. Even people who took the chance and it didn ’ t work out don ’ t really have many regrets about that. It ’ s the people who didn ’ t take the chance. So this is boldness regrets. Boldness regrets sound like this :”If only I ’ d taken the chance.”Third category. Moral regrets. Very interesting, very interesting category. These are people who, again, a lot of these regrets begin at a juncture. You ’ re at a juncture, you can do the right thing or you can do the wrong thing. People do the wrong thing, and they regret it. I mean, one of the ones that really stuck with me, I ’ m going to try to pull it up here, is this one here, this woman. She ’ s a 71-year-old woman in New Jersey.”When I was a kid, my mother would send me to a small local store for a few items. I frequently would steal a candy bar when the grocer wasn ’ t looking. That ’ s bothered me for about 60 years.”So 71-year-old woman in New Jersey, for 60 years, she ’ s been bugged by this moral breach. So moral regrets. We have people regretting bullying, we have people regretting marital infidelity. All kinds of things. Moral regrets sound like this,”If only I ’ d done the right thing.”And finally, our fourth category, or what I call connection regrets. Connection regrets are like this : You have a relationship or ought to have a relationship. And it doesn ’ t matter what the relationship is. Kids, parents, siblings, cousins, friends, colleagues, but you have a relationship or ought to have had a relationship, and the relationship comes apart. And what ’ s interesting is that what these 16, 000 people were telling me is that the way these relationships come apart is often not very dramatic, not very dramatic at all. They often come apart by drifting apart rather than through some kind of explosive rift. And what happens is that people don ’ t want to reach out because they say it ’ s going to be awkward to reach out, and the other side is not going to care. One of the lessons that I learned from this book for myself is always reach out. So that ’ s what connection regrets are.”If only I ’ d reached out.”And so over and over and over again, we see these same regrets : Foundation regrets :”If only I ’ d done the work.”Boldness regrets :”If only I ’ d taken the chance.”Moral regrets :”If only I ’ d done the right thing.”And connection regrets :”If only I ’ d reached out.”And when we look at these regrets, so that ’ s interesting in itself, but what I realized is that these four core regrets operate as a kind of photographic negative of the good life. Because if we understand what people regret the most, we actually can understand what they value the most. And each of these regrets, to my mind, reveals something fundamental about humanity and about what we need. We need stability. Nobody wants to have an unstable life. We want a chance to learn and grow and do something. We recognize that we are not here forever, and we want to do something and try something. And at least feel the exhilaration of being bold. Moral regrets, I think most of us, almost all of us want to do the right thing. At some level, these moral regrets are very heartening. The idea that people are bugged for years, decades, by these moral breaches earlier in their life. I think most of us want to do the right thing. And then connection regrets. We want love, not love only in the romantic sense, but love in the broader sense of connection and relationship and affinity with other people. And so in a weird way, this negative emotion of regret points the way to a good life. By studying regret, we know what constitutes a good life, a life of stability, a life where you have a chance to take a few risks, a life where you ’ re doing the right thing and a life where you have people who love you and whom you love. And so to me, I started out saying,”Oh, boy, is this book going to be a downer, studying regret?”And it ended up being very uplifting. And so, those are the four core regrets. Regret points us to the good life. And so I hope that you ’ ll begin to reckon with your own regrets because I think they ’ re going to give you direction to a life well lived. Whitney Pennigton Rogers : Well, thank you for that, Dan I was clapping behind the scenes when you couldn ’ t actually see me, for everyone who I know also really appreciated what you shared there. First, Dan, you mentioned, you know, this big takeaway about how thinking about regret can help us figure out what is the recipe for the good life. I guess what has been your biggest takeaway from doing this work beyond that? DHP : I found it really interesting how much people want to talk about this, and that ’ s what got me on it in the first place. That is, I had an experience in my life where one of my kids graduated from college, and that sort of marker in my life made me start thinking about what regrets that I had. And I just mentioned it to a few people, and I found them, like, leaning in to the conversation. So I was amazed at how much people want to talk about this and how much this taboo of like,”Oh, I don ’ t have any regrets,”is so ridiculous. I mean, it ’ s absurd. And that if we actually start talking about it, we ’ re going to be better off. For me personally, I think that the biggest takeaway was the... Was the connection regrets. Because I had so many people who had the same story where they had a friendship, some kind of relationship, and it comes apart, and they want to reach out and they say,”Oh no, it ’ s going to be really awkward. And the other side ’ s not going to care.”And we ’ re so wrong about that. It ’ s not awkward, and the other side almost always appreciates it. And so for me, I guess the takeaway is if I ’ m at a juncture in my life where I ’ m thinking,”Should I reach out or should I not reach out?”I ’ ve answered the question. That the answer to that question at that juncture, if you reach that juncture, the answer is, always reach out. You know, especially coming out a time like this, Whitney, we need that sense of of connection. And so the ethic of always reaching out, to me, is one of the best life lessons that I ’ ve learned. WPR : Well, we ’ re going to do something a little interesting next, Dan, which is have some Members share their own regrets. And so I want to, I guess, hand things over to you right now so that you can bring in our first Member and we can explore more what this process of thinking about making our regrets help us live the good life actually looks like. DHP : Sure, sure. And so let ’ s bring on Lily. I don ’ t want it to sound like a magic act, but Lily and I don ’ t know each other. We haven ’ t gone through this before, but what I want to try to do is actually, the hearing of the stories of people ’ s regrets I think is super interesting and revealing. We ’ re going to hear Lily ’ s regret, and we ’ re going to talk through what science says might be some appropriate responses to that. So, Lily, welcome. Lily : Thank you, hi everyone. DHP : And tell us where you are. L : I ’ m currently in Brooklyn, New York. DHP : Brooklyn is in the house here at TED Membership. So Lily, tell us you regret. L : Yeah. So my regret that I want to share is that for most of my young adult life from kindergarten, really, straight through high school is that I was painfully, painfully shy with really low self-confidence. As I was thinking about this, I was remembering, and there were times where I just wanted to close my eyes and be invisible. And I think that, you know, my... Like, I didn ’ t really come into my own until I got to college, where I found a really great group of friends, really, like, I was confident in expressing myself and, you know, just being myself. And I think that, you know, my regret is that I just really wish I had taken a little bit more effort to build my confidence to fight this a little bit more, because I worry about what opportunities I might have missed. So ever since then, I feel like I try to counteract it now. And if ever I meet someone who might be going through, especially if they ’ re younger, like going through the same thing I did, I try to make them feel seen and try to empathize with how they ’ re feeling. So that ’ s kind of a takeaway, I guess, from that regret. DHP : But I mean, it sounds like... So is this a regret that ’ s still with you? L : I think... DHP : It sounds to me like you might have sort of begun the process of resolving it a little bit. L : Absolutely. But I think that, you know, even just, you know, prepping for this, I start to think about like, you know, there could have been more things that I could have done if I had just put myself a little bit out there, if I didn ’ t, just try to hide so much. DHP : OK, alright. This is fascinating, Lily, and I have to say, I have this database of regrets. And you can search the database. And if I were to search the database for the phrase”speak up,””spoke up,””spoken up,”I would get huge, huge numbers of of people. It is one of the most common regrets, is that people regret not speaking up. The important thing about our regrets that comes from the science is this : it ’ s how we deal with them. So we can take that regret and say,”You know what? It doesn ’ t matter that I feel terrible and I have this regret, because I ’ m just going to ignore it,”right? That ’ s like the blithe”no regrets”philosophy. That ’ s a bad idea, alright? The other way at it is to say,”Oh my God, I have all these regrets, it ’ s so terrible, I ’ m going to wallow in them.”That ’ s a bad idea too. What we want to do, and I think that you ’ ve already done a really brilliant job of it is use these regrets as signals. Signals for our thinking ; what is it teaching me? And so there are a few things in the research that give us some clues about what to do. So one of them is this. So we start with sort of, reframing the regret and how we think about it in ourselves. So do you think that you are the only person with this kind of regret? L : I don ’ t know. DHP : Not at all. I ’ m watching the chat, man. Somebody said,”Lily, you ’ re telling my story.”So one of the things that we can do with our regret is treat ourselves with self-compassion, alright? Not boost our self-esteem, that ’ s sometimes dangerous. Not rip ourselves down with self-criticism, but actually say, treat ourselves with kindness rather than contempt and recognize that what we ’ re going through is part of the shared human experience. That ’ s one thing. The second thing that we can do is we can disclose our regret. And there are few things that are interesting about disclosure. There ’ s something amazing why 16, 000 people were willing to share their regrets with me. I mean, like, what ’ s going on there, right? And the reason is that when we disclose our regrets, we relieve some of the burden. That ’ s one thing. The second thing that we do is that when we actually talk about our regrets, converting these kind of blobby mental abstractions into concrete words, whether it ’ s spoken or writing, defangs them. It begins the sense-making process. And the other thing about disclosure, which is a dirty little secret that I ’ ll reveal to all of you that comes out in the research very clearly, is that when we disclose our vulnerabilities and our weaknesses, people don ’ t like us less, they actually like us more. Because they empathize with us. They respect our courage. And the final thing is to actually try to extract a lesson from it, to use this regret. So what would you say, Lily, is the lesson that you ’ ve learned from this regret? L : I think that... What would have gone wrong if I... DHP : That ’ s interesting. L : If I were more open about expressing myself, like, people might discover I ’ m a little weird, or they might think that maybe I ’ m nice and hopefully maybe a little funny. So I think, like, maybe that ’ s one thing that jumps to mind. Like, what could have gone wrong? You know. DHP : So what ’ s the lesson that you have applied going forward? Taking this regret, OK, so you ’ ve sort of treated yourself with kindness rather than contempt. You ’ ve disclosed it to all these people here. You ’ ve begun the sense-making process by talking about it and writing about it. What ’ s a lesson that you can extract from this?

L : I think that... I ’ m not sure. DHP : Well, then let me tell you. I think that the lesson is to... Next time, speak up. Next time, speak up. Next time when you are at a juncture,”Should I speak up or not,”think about this. Think about this and speak up. Do you have any kind of work meetings or anything coming on where you ’ re going to be confronted with this? L : Yeah, and I think that happens all the time. Like, you know, I have an idea. Oh, but someone starts talking, and then like, you just sort of fade back into the background, and that ’ s something I want to counteract more often. Because, more often than not, you know that idea is a contribution, and why am I hesitating so much? DHP : Yeah. So you have a lesson. The lesson is, speak up. So why don ’ t you – So how about the next meeting you ’ re in, when you have something to say, don ’ t hesitate and speak up. L : Done, I ’ ll do it. DHP : OK, but here ’ s the thing, what I like about this is you ’ ve just made a promise to 300 people. So you ’ re on the hook. L : I ’ m on the hook. DHP : So this is it. So, Lily has this regret. She ’ s looking backwards, saying,”Oh, if only I ’ d spoken up,”and instead of beating herself up, she is divulging it, she ’ s extracting a lesson from it, and she ’ s taking that and applying it to some next interaction. So this is what we do. This is how, again, looking backward can move you forward. Lily, that ’ s such a fantastic – People in the chat are saying,”We will hold Lily accountable,”which I love. So Lily, thanks for that. We ’ re going to bring – I really, really appreciate your sharing that with us, and I want you to report back that you did speak up. L : I will, thank you. DHP : Thanks, Lily. WPR : Well, we have a question here from Claudia, who asks,”Can you speak to the issue of painful life regrets? Major opportunities lost? Do you have some advice on how to avoid being paralyzed by fear or further regret?”DHP : Yeah. It ’ s interesting that Claudia said”opportunities lost,”and let me pick up on that phrase right here. Because one of the things I saw in my own research, because I also did a huge survey of the American population where we surveyed a representative sample of 4, 489 Americans about regret and how it worked. But one of the things you see widespread is that there are, in the architecture of regret, there are often two kinds of regrets. One are regrets of action and one are regrets of inaction. Regrets about what we did. Regrets about what we didn ’ t do. And overwhelmingly, inaction regrets predominate. And that ’ s what an opportunity lost is. With action regrets, we can try to undo them. We can make amends. We can look for the silver lining, and we can reduce the sting. For inaction regrets, it is harder. And so the key here on the opportunities lost is to think about, you know, really like, what are you going to do? You sort of reduce the level of abstraction and say,”What are you going to do next time?”Not an abstraction of like,”Oh, I ’ m going to be more bold.”It ’ s like, what are you going to do next time? This is what we were talking about with Lily. What are you going to do next time? All regrets begin at the juncture. You can go this way or you can go that way. And so for Claudia, I would say, the next time you ’ re at this juncture, take the opportunity, play it safe. Stop, think about your regret, and make the decision there. Or another thing that you could do. I ’ ll give you another, sort of, decision-making heuristic. Two of them, in fact. When you ’ re at that juncture, Claudia, next time, go forward five years. This is called self-distancing. Be Claudia five years from now, look back on Claudia today. What decision do you want? What decision does Claudia of 2027 want Claudia of 2022 to make? It ’ s very clear. Or the best decision-making heuristic there is : You ’ re at a juncture, what would you tell your best friend to do? When you ask people that when they ’ re trying to make a decision, say,”What would you tell your best friend to do?”Everybody always knows. So, I think that ’ s it. Remember, the main thing, though, is don ’ t let it bog you down. Use it as a tool for thinking – not as a tool for wallowing, not as a tool for ignoring

– but as a tool for thinking. WPR : A question from Kim, she ’ s asking – She says you ’ re talking as if any bad decision or mistake is also a regret, and I ’ m not sure that that ’ s always the case. Can you share your definition of regret? Especially after doing this project. What is your definition? DHP : There ’ s a difference between a regret and a mistake, alright? So you can make a mistake and not regret it because you say, you immediately learn something from it or it was a worthy mistake. A regret is something where you look backward at something that where you had control, where you had some agency... Where you had some agency, you did something that, well, you did something wrong, and it sticks with you. It doesn ’ t go away. And it sticks with you for a very long time. So there ’ s a big difference, for instance, between... I can make a mistake and actually not regret it because it ’ s not significant enough to me to linger, right? So that ’ s the difference between a regret and a mistake. It ’ s basically the duration, essentially, the half-life of the negative emotion. There ’ s a huge difference between regret and disappointment. Huge difference between regret and disappointment. Because with disappointment, you don ’ t have any kind of control. The great example of that is from Janet Landman, a former professor at the University of Michigan who, to me, is like, she tells the story of like, OK, so a kid loses her third tooth. A seven-year-old loses her third tooth. She loses her tooth, she goes to sleep, before she goes to sleep, she puts the tooth under the pillow. When she wakes up, the tooth is still there. The kid is disappointed, but the parents regret not leaving that – So you have to have some agency and it has to have a... It has to have enough significance that it stays with you. And once again, going back to these four core regrets, it ends up being the same kinds of... it ends up being the same kinds of things. If you said,”Oh, I shouldn ’ t have bought that kind of car,”it might sting for a little bit, but the half-life is very, very short. But other kinds of things stick with us and stick with us, and those are the things often of significance. WPR : Thank you so much, Dan, for chatting with us, and I love ending there,”If not now when?”And we ’ ll see you soon. Thank you, Dan. DHP : Thanks a lot, what a pleasure. [Get access to thought-provoking events you won ’ t want to miss.] [Become a TED Member at ted. com / membership]

Text Sample 782: NEG Dim 5 – 2014 – Score -1.00 – 2014/t001690.txt

I just met you on a bus, and we would really like to get to know each other, but I ’ ve got to get off at the next stop, so you ’ re going to tell me three things about yourself that just define you as a person, three things about yourself that will help me understand who you are, three things that just get to your very essence. And what I ’ m wondering is, of those three things, is any one of them surviving some kind of trauma? Cancer survivor, rape survivor, Holocaust survivor, incest survivor. Ever notice how we tend to identify ourselves by our wounds? And where I have seen this survivor identity have the most consequences is in the cancer community. And I ’ ve been around this community for a long time, because I ’ ve been a hospice and a hospital chaplain for almost 30 years. And in 2005, I was working at a big cancer center when I received the news that my mother had breast cancer. And then five days later, I received the news that I had breast cancer. My mother and I can be competitive — — but I was really not trying to compete with her on this one. And in fact, I thought, well, if you have to have cancer, it ’ s pretty convenient to be working at a place that treats it. But this is what I heard from a lot of outraged people. What? You ’ re the chaplain. You should be immune. Like, maybe I should have just gotten off with a warning instead of an actual ticket, because I ’ m on the force. So I did get my treatment at the cancer center where I worked, which was amazingly convenient, and I had chemotherapy and a mastectomy, and a saline implant put in, and so before I say another word, let me just say right now, this is the fake one. I have found that I need to get that out of the way, because I ’ ll see somebody go”Oh, I know it ’ s this one.”And then I ’ ll move or I ’ ll gesture and they ’ ll go,”No, it ’ s that one.”So now you know. I learned a lot being a patient, and one of the surprising things was that only a small part of the cancer experience is about medicine. Most of it is about feelings and faith and losing and finding your identity and discovering strength and flexibility you never even knew you had. It ’ s about realizing that the most important things in life are not things at all, but relationships, and it ’ s about laughing in the face of uncertainty and learning that the way to get out of almost anything is to say,”I have cancer.”So the other thing I learned was that I don ’ t have to take on”cancer survivor”as my identity, but, boy, are there powerful forces pushing me to do just that. Now, don ’ t, please, misunderstand me. Cancer organizations and the drive for early screening and cancer awareness and cancer research have normalized cancer, and this is a wonderful thing. We can now talk about cancer without whispering. We can talk about cancer and we can support one another. But sometimes, it feels like people go a little overboard and they start telling us how we ’ re going to feel. So about a week after my surgery, we had a houseguest. That was probably our first mistake. And keep in mind that at this point in my life I had been a chaplain for over 20 years, and issues like dying and death and the meaning of life, these are all things I ’ d been yakking about forever. So at dinner that night, our houseguest proceeds to stretch his arms up over his head, and say,”You know, Deb, now you ’ re really going to learn what ’ s important. Yes, you are going to make some big changes in your life, and now you ’ re going to start thinking about your death. Yep, this cancer is your wakeup call.”Now, these are golden words coming from someone who is speaking about their own experience, but when someone is telling you how you are going to feel, it ’ s instant crap. The only reason I did not kill him with my bare hands was because I could not lift my right arm. But I did say a really bad word to him, followed by a regular word, that — — made my husband say,”She ’ s on narcotics.”And then after my treatment, it just felt like everyone was telling me what my experience was going to mean.”Oh, this means you ’ re going to be doing the walk.””Oh, this means you ’ re coming to the luncheon.””This means you ’ re going to be wearing the pink ribbon and the pink t-shirt and the headband and the earrings and the bracelet and the panties.”Panties. No, seriously, google it. How is that raising awareness? Only my husband should be seeing my panties. He ’ s pretty aware of cancer already. It was at that point where I felt like, oh my God, this is just taking over my life. And that ’ s when I told myself, claim your experience. Don ’ t let it claim you. We all know that the way to cope with trauma, with loss, with any life- changing experience, is to find meaning. But here ’ s the thing : No one can tell us what our experience means. We have to decide what it means. And it doesn ’ t have to be some gigantic, extroverted meaning. We don ’ t all have to start a foundation or an organization or write a book or make a documentary. Meaning can be quiet and introverted. Maybe we make one small decision about our lives that can bring about big change. Many years ago, I had a patient, just a wonderful young man who was loved by the staff, and so it was something of a shock to us to realize that he had no friends. He lived by himself, he would come in for chemotherapy by himself, he would receive his treatment, and then he ’ d walk home alone. And I even asked him. I said,”Hey, how come you never bring a friend with you?”And he said,”I don ’ t really have any friends.”But he had tons of friends on the infusion floor. We all loved him, and people were going in and out of his room all the time. So at his last chemo, we sang him the song and we put the crown on his head and we **blew** the bubbles, and then I asked him, I said,”So what are you going to do now?”And he answered,”Make friends.”And he did. He started volunteering and he made friends there, and he began going to a church and he made friends there, and at Christmas he invited my husband and me to a party in his apartment, and the place was filled with his friends. Claim your experience. Don ’ t let it claim you. He decided that the meaning of his experience was to know the joy of friendship, and then learn to make friends. So what about you? How are you going to find meaning in your crappy experience? It could be a recent one, or it could be one that you ’ ve been carrying around for a really long time. It ’ s never too late to change what it means, because meaning is dynamic. What it means today may not be what it means a year from now, or 10 years from now. It ’ s never too late to become someone other than simply a survivor. Hear how static that word sounds? Survivor. No movement, no growth. Claim your experience. Don ’ t let it claim you, because if you do, I believe you will become trapped, you will not grow, you will not evolve. But of course, sometimes it ’ s not outside pressures that cause us to take on that identity of survivor. Sometimes we just like the perks. Sometimes there ’ s a payoff. But then we get stuck. Now, one of the first things I learned as a chaplain intern was the three C ’ s of the chaplain ’ s job : Comfort, clarify and, when necessary, confront or challenge. Now, we all pretty much love the comforting and the clarifying. The confronting, not so much. One of the other things that I loved about being a chaplain was seeing patients a year, or even several years after their treatment, because it was just really cool to see how they had changed and how their lives had evolved and what had happened to them. So I was thrilled one day to get a page down into the lobby of the clinic from a patient who I had seen the year before, and she was there with her two adult daughters, who I also knew, for her one year follow-up exam. So I got down to the lobby, and they were ecstatic because she had just gotten all of her test results back and she was NED : No Evidence of Disease. Which I used to think meant Not Entirely Dead. So they were ecstatic, we sat down to visit, and it was so weird, because within two minutes, she started retelling me the story of her diagnosis and her surgery and her chemo, even though, as her chaplain, I saw her every week, and so I knew this story. And she was using words like suffering, agony, struggle. And she ended her story with,”I felt crucified.”And at that point, her two daughters got up and said,”We ’ re going to go get coffee.”And they left. Tell me three things about yourself before the next stop. People were leaving the bus before she even got to number two or number three. So I handed her a tissue, and I gave her a hug, and then, because I really cared for this woman, I said,”Get down off your cross.”And she said,”What?”And I repeated,”Get down off your cross.”And to her credit, she could talk about her reasons for embracing and then clinging to this identity. It got her a lot of attention. People were taking care of her for a change. But now, it was having the opposite effect. It was pushing people away. People kept leaving to get coffee. She felt crucified by her experience, but she didn ’ t want to let that crucified self die. Now, perhaps you are thinking I was a little harsh with her, so I must tell you that I was speaking out of my own experience. Many, many years before, I had been fired from a job that I loved, and I would not stop talking about my innocence and the injustice and the betrayal and the deceipt, until finally, just like this woman, people were walking away from me, until I finally realized I wasn ’ t just processing my feelings, I was feeding them. I didn ’ t want to let that crucified self die. But we all know that with any resurrection story, you have to die first. The Christian story, Jesus was dead a whole day in the tomb before he was resurrected. And I believe that for us, being in the tomb means doing our own deep inner work around our wounds and allowing ourselves to be healed. We have to let that crucified self die so that a new self, a truer self, is born. We have to let that old story go so that a new story, a truer story, can be told. Claim your experience. Don ' t let it claim you. What if there were no survivors, meaning, what if people decided to just claim their trauma as an experience instead of taking it on as an identity? Maybe it would be the end of being trapped in our wounds and the beginning of amazing self-exploration and discovery and growth. Maybe it would be the start of defining ourselves by who we have become and who we are becoming. So perhaps survivor was not one of the three things that you would tell me. No matter. I just want you all to know that I am really glad that we are on this bus together, and this is my stop.

Text Sample 783: NEG Dim 5 – 2014 – Score -1.00 – 2014/t001633.txt
As an Arab female photographer, I have always found ample inspiration for my projects in personal experiences. The passion I developed for knowledge, which allowed me to break barriers towards a better life was the motivation for my project I Read I Write. Pushed by my own experience, as I was not allowed initially to pursue my higher education, I decided to explore and document stories of other women who changed their lives through education, while exposing and questioning the barriers they face. I covered a range of topics that concern women ' s education, keeping in mind the differences among Arab countries due to economic and social factors. These issues include female illiteracy, which is quite high in the region ; educational reforms ; programs for dropout students ; and political activism among university students. As I started this work, it was not always easy to convince the women to participate. Only after explaining to them how their stories might influence other women ' s lives, how they would become role models for their own community, did some agree. Seeking a collaborative and reflexive approach, I asked them to write their own words and ideas on prints of their own images. Those images were then shared in some of the classrooms, and worked to inspire and motivate other women going through similar educations and situations. Aisha, a teacher from Yemen, wrote,"I sought education in order to be independent and to not count on men with everything."One of my first **subjects** was Umm El-Saad from Egypt. When we first met, she was barely able to write her name. She was attending a nine-month literacy program run by a local NGO in the Cairo suburbs. Months later, she was joking that her husband had threatened to pull her out of the classes, as he found out that his now literate wife was going through his phone text messages. Naughty Umm El-Saad. Of course, that ' s not why Umm El-Saad joined the program. I saw how she was longing to gain control over her simple daily routines, small details that we take for granted, from counting money at the market to helping her kids in homework. Despite her poverty and her community ' s mindset, which belittles women ' s education, Umm El-Saad, along with her Egyptian classmates, was eager to learn how to read and write. In Tunisia, I met Asma, one of the four activist women I interviewed. The secular bioengineering student is quite active on social media. Regarding her country, which treasured what has been called the Arab Spring, she said,"I ' ve always dreamt of discovering a new bacteria. Now, after the revolution, we have a new one every single day."Asma was referring to the rise of religious fundamentalism in the region, which is another obstacle to women in particular. Out of all the women I met, Fayza from Yemen affected me the most. Fayza was forced to drop out of school at the age of eight when she was married. That marriage lasted for a year. At 14, she became the third wife of a 60-year-old man, and by

the time she was 18, she was a divorced mother of three. Despite her poverty, despite her social status as a divorce in an ultra-conservative society, and despite the opposition of her parents to her going back to school, Fayza knew that her only way to control her life was through education. She is now 26. She received a grant from a local NGO to fund her business studies at the university. Her goal is to find a job, rent a place to live in, and bring her kids back with her. The Arab states are going through tremendous change, and the struggles women face are overwhelming. Just like the women I photographed, I had to overcome many barriers to becoming the photographer I am today, many people along the way telling me what I can and cannot do. Umm El-Saad, Asma and Fayza, and many women across the Arab world, show that it is possible to overcome barriers to education, which they know is the best means to a better future. And here I would like to end with a quote by Yasmine, one of the four activist women I interviewed in Tunisia. Yasmine wrote,"Question your convictions. Be who you to want to be, not who they want you to be. Don ' t accept their enslavement, for your mother birthed you free."Thank you.

Text Sample 784: NEG Dim 5 – 2014 – Score -1.00 – 2014/t001738.txt
When my first children ' s book was published in 2001, I returned to my old elementary school to talk to the students about being an author and an illustrator, and when I was setting up my slide projector in the cafetorium, I looked across the room, and there she was : my old lunch lady. She was still there at the school and she was busily preparing lunches for the day. So I approached her to say hello, and I said,"Hi, Jeannie! How are you?"And she looked at me, and I could tell that she recognized me, but she couldn ' t quite place me, and she looked at me and she said,"Stephen Krosoczka?"And I was amazed that she knew I was a Krosoczka, but Stephen is my uncle who is 20 years older than I am, and she had been his lunch lady when he was a kid. And she started telling me about her grandkids, and that **blew** my mind. My lunch lady had grandkids, and therefore kids, and therefore left school at the end of the day? I thought she lived in the cafeteria with the serving spoons. I had never thought about any of that before. Well, that chance encounter inspired my imagination, and I created the Lunch Lady graphic novel series, a series of comics about a lunch lady who uses her fish stick nunchucks to fight off evil cyborg substitutes, a school bus monster, and mutant mathletes, and the end of every book, they get the bad guy with their hairnet, and they proclaim,"Justice is served!"And it ' s been amazing, because the series was so welcomed into the reading lives of children, and they sent me the most amazing letters and cards and artwork. And I would notice as I would visit schools, the lunch staff would be involved in the programming in a very meaningful way. And coast to coast, all of the lunch ladies told me the same thing :"Thank you for making a superhero in our likeness."Because the lunch lady has not been treated very kindly in popular culture over time. But it meant the most to Jeannie. When the books were first published, I invited her to the book launch party, and in front of everyone there, everyone she had fed over the years, I gave her a piece of artwork and some books. And two years after this photo was taken, she passed away, and I attended her wake, and nothing could have prepared me for what I saw there, because next to her casket was this painting, and her husband told me it meant so much to her that I had acknowledged her hard work, I had validated what she did. And that inspired me to create a day where we could recreate that feeling in cafeterias across the country : School Lunch Hero Day, a day where kids can make creative projects for their lunch staff. And I partnered with the

School Nutrition Association, and did you know that a little over 30 million kids participate in school lunch programs every day. That equals up to a little over five billion lunches made every school year. And the stories of heroism go well beyond just a kid getting a few extra chicken nuggets on their lunch tray. There is Ms. Brenda in California, who keeps a close eye on every student that comes through her line and then reports back to the guidance counselor if anything is amiss. There are the lunch ladies in Kentucky who realized that 67 percent of their students relied on those meals every day, and they were going without food over the summer, so they retrofitted a school bus to create a mobile feeding unit, and they traveled around the neighborhoods feedings 500 kids a day during the summer. And kids made the most amazing projects. I knew they would. Kids made hamburger cards that were made out of construction paper. They took photos of their lunch lady ' s head and plastered it onto my cartoon lunch lady and fixed that to a milk carton and presented them with flowers. And they made their own comics, starring the cartoon lunch lady alongside their actual lunch ladies. And they made thank you pizzas, where every kid signed a different topping of a construction paper pizza. For me, I was so moved by the response that came from the lunch ladies, because one woman said to me, she said,"Before this day, I felt like I was at the end of the planet at this school. I didn ' t think that anyone noticed us down here."Another woman said to me,"You know, what I got out of this is that what I do is important."And of course what she does is important. What they all do is important. They ' re feeding our children every single day, and before a child can learn, their belly needs to be full, and these women and men are working on the front lines to create an educated society. So I hope that you don ' t wait for School Lunch Hero Day to say thank you to your lunch staff, and I hope that you remember how powerful a thank you can be. A thank you can change a life. It changes the life of the person who receives it, and it changes the life of the person who expresses it. Thank you.

Text Sample 785: NEG Dim 5 – 2013 – Score -1.00 – 2013/t001950.txt

I would like to tell you a story connecting the notorious privacy incident involving Adam and Eve, and the remarkable shift in the boundaries between public and private which has occurred in the past 10 years. You know the incident. Adam and Eve one day in the Garden of Eden realize they are naked. They freak out. And the rest is history. Nowadays, Adam and Eve would probably act differently. [@Adam Last nite was a blast! loved dat apple LOL] [@Eve yep.. babe, know what happened to my pants tho?] We do reveal so much more information about ourselves online than ever before, and so much information about us is being collected by organizations. Now there is much to gain and benefit from this massive analysis of personal information, or big data, but there are also complex tradeoffs that come from giving away our privacy. And my story is about these tradeoffs. We start with an observation which, in my mind, has become clearer and clearer in the past few years, that any personal information can become sensitive information. Back in the year 2000, about 100 billion photos were shot worldwide, but only a minuscule proportion of them were actually uploaded online. In 2010, only on Facebook, in a single month, 2. 5 billion photos were uploaded, most of them identified. In the same span of time, computers ' ability to recognize people in photos improved by three orders of magnitude. What happens when you combine these technologies together : increasing availability of facial data ; improving facial recognizing ability by computers ; but also cloud computing, which gives anyone in this theater the kind of computational power which a few years ago was only the domain of three-letter agencies ; and ubiquitous computing, which allows my phone, which is not a supercomputer, to connect to the Internet and do there hundreds of thousands of face metrics in a few seconds? Well, we conjecture that the result of this combination of technologies will be a radical change in our very notions of privacy and anonymity. To test that, we did an experiment on Carnegie Mellon University campus. We asked students who were walking by to participate in a study, and we took a shot with a webcam, and we asked them to fill out a survey on a laptop. While they were filling out the survey, we uploaded their shot to a cloud-computing cluster, and we started using a facial recognizer to match that shot to a database of some hundreds of thousands of images which we had downloaded from Facebook profiles. By the time the **subject** reached the last page on the survey, the page had been dynamically updated with the 10 best matching photos which the recognizer had found, and we asked the **subjects** to indicate whether he or she found themselves in the photo. Do you see the **subject**? Well, the computer did, and in fact did so for one out of three **subjects**. So essentially, we can start from an anonymous face, offline or online, and we can use facial recognition to give a name to that anonymous face thanks to social media data. But a few years back, we did something else. We started from social media data, we combined it statistically with data from U. S. government social security, and we ended up predicting social security numbers, which in the United States are extremely sensitive information. Do you see where I ' m going with this? So if you combine the two studies together, then the question becomes, can you start from a face and, using facial recognition, find a name and publicly available information about that name and that person, and from that publicly available information infer non- publicly available information, much more sensitive ones which you link back to the face? And the answer is, yes, we can, and we did. Of course, the accuracy keeps getting worse. [27 % of **subjects** ' first 5 SSN digits identified] But in fact, we even decided to develop an iPhone app which uses the phone ' s internal camera to take a shot of a **subject** and then upload it to a cloud and then do what I just described to you in real time : looking for a match, finding public information, trying to infer sensitive information, and then sending back to the phone so that it is overlaid on the face of the **subject**, an example of augmented reality, probably a creepy example of augmented reality. In fact, we didn ' t develop the app to make it available, just as a proof of concept. In fact, take these technologies and push them to their logical extreme. Imagine a future in which strangers around you will look at you through their Google Glasses or, one day, their contact lenses, and use seven or eight data points about you to infer anything else which may be known about you. What will this future without secrets look like? And should we care? We may like to believe that the future with so much wealth of data would be a future with no more biases, but in fact, having so much information doesn ' t mean that we will make decisions which are more objective. In another experiment, we presented to our **subjects** information about a potential job candidate. We included in this information some references to some funny, absolutely legal, but perhaps slightly embarrassing information that the **subject** had posted online. Now interestingly, among our **subjects**, some had posted comparable information, and some had not. Which group do you think was more likely to judge harshly our **subject**? Paradoxically, it was the group who had posted similar information, an example of moral dissonance. Now you may be thinking, this does not apply to me, because I have nothing to hide. But in fact, privacy is not about having something negative to hide. Imagine that you are the H. R. director of a certain organization, and you receive rsums, and you decide to find more information about the candidates. Therefore, you Google their names and in a certain universe, you find this information. Or in a parallel universe, you find this information. Do you think that you would be equally likely to call either candidate for an interview? If you think so, then you are not like the U. S. employers who are, in fact, part of our experiment, meaning we did exactly that. We created Facebook profiles, manipulating traits, then we started sending out rsums to companies in the U. S., and we detected, we monitored, whether they were searching for our candidates, and whether they were acting on the information they found on social media. And they were. Discrimination was happening through social media for equally skilled candidates. Now marketers like us to believe that all information about us will always be used in a manner which is in our favor. But think again. Why should that be always the case? In a movie which came out a few years ago,"Minority Report,"a famous scene had Tom Cruise walk in a mall and holographic personalized advertising would appear around him. Now, that movie is set in 2054, about 40 years from now, and as exciting as that technology looks, it already vastly underestimates the amount of information that organizations can gather about you, and how they can use it to influence you in a way that you will not even detect. So as an example, this is another experiment actually we are running, not yet completed. Imagine that an organization has access to your list of Facebook friends, and through some kind of algorithm they can detect the two friends that you like the most. And then they create, in real time, a facial composite of these two friends. Now studies prior to ours have shown that people don ' t recognize any longer even themselves in facial composites, but they react to those composites in a positive manner. So next time you are looking for a certain product, and there is an ad suggesting you to buy it, it will not be just a standard spokesperson. It will be one of your friends, and you will not even know that this is happening. Now the problem is that the current policy mechanisms we have to protect ourselves from the abuses of personal information are like bringing a knife to a gunfight. One of these mechanisms is transparency, telling people what you are going to do with their data. And in principle, that ' s a very good thing. It ' s necessary, but it is not sufficient. Transparency can be misdirected. You can tell people what you are going to do, and then you still nudge them to disclose arbitrary amounts of personal information. So in yet another experiment, this one with students, we asked them to provide information about their campus behavior, including pretty sensitive questions, such as this one. [Have you ever cheated in an exam?] Now to one group of **subjects**, we told them,"Only other students will see your answers."To another group of **subjects**, we told them,"Students and faculty will see your answers."Transparency. Notification. And sure enough, this worked, in the sense that the first group of **subjects** were much more likely to disclose than the second. It makes sense, right? But then we added the misdirection. We repeated the experiment with the same two groups, this time adding a delay between the time we told **subjects** how we would use their data and the time we actually started answering the questions. How long a delay do you think we had to add in order to nullify the inhibitory effect of knowing that faculty would see your answers? Ten minutes? Five minutes? One minute? How about 15 seconds? Fifteen seconds were sufficient to have the two groups disclose the same amount of information, as if the second group now no longer cares for faculty reading their answers. Now I have to admit that this talk so far may sound exceedingly gloomy, but that is not my point. In fact, I want to share with you the fact that there are alternatives. The way we are doing things now is not the only way they can done, and certainly not the best way they can be done. When someone tells you,"People don ' t care about privacy,"consider whether the game has been designed and rigged so that they cannot care about privacy, and coming to the realization that these manipulations occur is already halfway through the process of being able to protect yourself. When someone tells you that privacy is incompatible with the benefits of big data, consider that in the last 20 years, researchers have created technologies to allow virtually any electronic transactions to take place in a more privacy-preserving manner. We can browse the Internet anonymously. We can send emails that can only be read by the intended recipient, not even the NSA. We can have even privacy-preserving data mining. In other words, we can have the benefits of big data while protecting privacy. Of course, these technologies imply a shifting of cost and revenues between data holders and data **subjects**, which is why, perhaps, you don ’ t hear more about them. Which brings me back to the Garden of Eden. There is a second privacy interpretation of the story of the Garden of Eden which doesn ’ t have to do with the issue of Adam and Eve feeling naked and feeling ashamed. You can find echoes of this interpretation in John Milton ’ s”Paradise Lost.”In the garden, Adam and Eve are materially content. They ’ re happy. They are satisfied. However, they also lack knowledge and self-awareness. The moment they eat the aptly named fruit of knowledge, that ’ s when they discover themselves. They become aware. They achieve autonomy. The price to pay, however, is leaving the garden. So privacy, in a way, is both the means and the price to pay for freedom. Again, marketers tell us that big data and social media are not just a paradise of profit for them, but a Garden of Eden for the rest of us. We get free content. We get to play Angry Birds. We get targeted apps. But in fact, in a few years, organizations will know so much about us, they will be able to infer our desires before we even form them, and perhaps buy products on our behalf before we even know we need them. Now there was one English author who anticipated this kind of future where we would trade away our autonomy and freedom for comfort. Even more so than George Orwell, the author is, of course, Aldous Huxley. In”Brave New World,”he imagines a society where technologies that we created originally for freedom end up coercing us. However, in the book, he also offers us a way out of that society, similar to the path that Adam and Eve had to follow to leave the garden. In the words of the Savage, regaining autonomy and freedom is possible, although the price to pay is steep. So I do believe that one of the defining fights of our times will be the fight for the control over personal information, the fight over whether big data will become a force for freedom, rather than a force which will hiddenly manipulate us. Right now, many of us do not even know that the fight is going on, but it is, whether you like it or not. And at the risk of playing the serpent, I will tell you that the tools for the fight are here, the awareness of what is going on, and in your hands, just a few clicks away. Thank you.

Text Sample 786: NEG Dim 5 – 2013 – Score -1.00 – 2013/t001914.txt

So when I was in Morocco, in Casablanca, not so long ago, I met a young unmarried mother called Faiza. Faiza showed me photos of her infant son and she told me the story of his conception, pregnancy, and delivery. It was a remarkable tale, but Faiza saved the best for last.”You know, I am a virgin,”she told me.”I have two medical certificates to prove it.”This is the modern Middle East, where two millennia after the coming of Christ, virgin births are still a fact of life. Faiza ’ s story is just one of hundreds I ’ ve heard over the years, traveling across the Arab region talking to people about sex. Now, I know this might sound like a dream job, or possibly a highly dubious occupation, but for me, it ’ s something else altogether. I ’ m half Egyptian, and I ’ m Muslim. But I grew up in Canada, far from my Arab roots. Like so many who straddle East and West, I ’ ve been drawn, over the years, to try to better understand my origins. That I chose to look at sex comes from my background in HIV / AIDS, as a writer and a researcher and an activist. Sex lies at the heart of an emerging epidemic in the Middle East and North Africa, which is one of only two regions in the world where HIV / AIDS is still on the rise. Now sexuality is an incredibly powerful lens with which to study any society, because what happens in our intimate lives is reflected by forces on a bigger stage : in politics and economics, in religion and tradition, in gender and generations. As I found, if you really want to know a people, you start by looking inside their bedrooms. Now to be sure, the Arab world is vast and varied. But running across it are three red lines — these are topics you are not supposed to challenge in word or deed. The first of these is politics. But the Arab Spring has changed all that, in uprisings which have blossomed across the region since 2011. Now while those in power, old and new, continue to cling to business as usual, millions are still pushing back, and pushing forward to what they hope will be a better life. That second red line is religion. But now religion and politics are connected, with the rise of such groups as the Muslim Brotherhood. And some people, at least, are starting to ask questions about the role of Islam in public and private life. You know, as for that third red line, that off-limits **subject**, what do you think it might be? Audience : Sex. Shereen El Feki : Louder, I can ’ t hear you. Audience : Sex. SEF : Again, please don ’ t be shy. Audience : Sex. SEF : Absolutely, that ’ s right, it ’ s sex. Across the Arab region, the only accepted context for sex is marriage — approved by your parents, sanctioned by religion and registered by the state. Marriage is your ticket to adulthood. If you don ’ t tie the knot, you can ’ t move out of your parents ’ place, and you ’ re not supposed to be having sex, and you ’ re definitely not supposed to be having children. It ’ s a social citadel ; it ’ s an impregnable fortress which resists any assault, any alternative. And around the fortress is this vast field of taboo against premarital sex, against condoms, against abortion, against homosexuality, you name it. Faiza was living proof of this. Her virginity statement was not a piece of wishful thinking. Although the major religions of the region extoll premarital chastity, in a patriarchy, boys will be boys. Men have sex before marriage, and people more or less turn a blind eye. Not so for women, who are expected to be virgins on their wedding night — that is, to turn up with your hymen intact. This is not a question of individual concern, this is a matter of family honor, and in particular, men ’ s honor. And so women and their relatives will go to great lengths to preserve this tiny piece of anatomy — from female genital mutilation, to virginity testing, to hymen repair surgery. Faiza chose a different route : non-vaginal sex. Only she became pregnant all the same. But Faiza didn ’ t actually realize this, because there ’ s so little sexuality education in schools, and so little communication in the family. When her condition became hard to hide, Faiza ’ s mother helped her flee her father and brothers. This is because honor killings are a real threat for untold numbers of women in the Arab region. And so when Faiza eventually fetched up at a hospital in Casablanca, the man who offered to help her, instead tried to rape her. Sadly, Faiza is not alone. In Egypt, where my research is focused, I have seen plenty of trouble in and out of the citadel. There are legions of young men who can ’ t afford to get married, because marriage has become a very expensive proposition. They are expected to bear the burden of costs in married life, but they can ’ t find jobs. This is one of the major drivers of the recent uprisings, and it is one of the reasons for the rising age of marriage in much of the Arab region. There are career women who want to get married, but can ’ t find a husband, because they defy gender expectations, or as one young female doctor in Tunisia put it to me,”The women, they are becoming more and more open. But the man, he is still at the prehistoric stage.”And then there are men and women who cross the heterosexual line, who have sex with their own sex, or who have a different gender identity. They are on the receiving end of laws which punish their activities, even their appearance. And they face a daily struggle with social stigma, with family despair, and with religious fire and brimstone. Now, it ’ s not as if it ’ s all rosy in the marital bed either. Couples who are looking for greater happiness, greater sexual happiness in their married lives, but are at a loss of how to achieve it, especially wives, who are afraid of being seen as bad women if they show some spark in the bedroom. And then there are those whose marriages are actually a veil for prostitution. They have been sold by their families, often to wealthy Arab tourists. This is just one face of a booming sex trade across the Arab region. Now raise your hand if any of this is sounding familiar to you, from your part of the world. Yeah. It ’ s not as if the Arab world has a monopoly on sexual hangups. And although we don ’ t yet have an Arab Kinsey Report to tell us exactly what ’ s happening inside bedrooms across the Arab region, It ’ s pretty clear that something is not right. Double standards for men and women, sex as a source of shame, family control limiting individual choices, and a vast gulf between appearance and reality : what people are doing and what they ’ re willing to admit to, and a general reluctance to move beyond private whispers to a serious and sustained public discussion. As one doctor in Cairo summed it up for me,”Here, sex is the opposite of sport. Football, everybody talks about it, but hardly anyone plays. But sex, everybody is doing it, but nobody wants to talk about it.”SEF : I want to give you a piece of advice, which if you follow it, will make you happy in life. When your husband reaches out to you, when he seizes a part of your body, sigh deeply and look at him lustily. When he penetrates you with his penis, try to talk flirtatiously and move yourself in harmony with him. Hot stuff! And it might sound that these handy hints come from”The Joy of Sex”or YouPorn. But in fact, they come from a 10th-century Arabic book called”The Encyclopedia of Pleasure,”which covers sex from aphrodisiacs to zoophilia, and everything in between. The Encyclopedia is just one in a long line of Arabic erotica, much of it written by religious scholars. Going right back to the Prophet Muhammad, there is a rich tradition in Islam of talking frankly about sex : not just its problems, but also its pleasures, and not just for men, but also for women. A thousand years ago, we used to have whole dictionaries of sex in Arabic. Words to cover every conceivable sexual feature, position and preference, a body of language that was rich enough to make up the body of the woman you see on this page. Today, this history is largely unknown in the Arab region. Even by educated people, who often feel more comfortable talking about sex in a foreign language than they do in their own tongue. Today ’ s sexual landscape looks a lot like Europe and America on the brink of the sexual revolution. But while the West has opened on sex, what we found is that Arab societies appear to have been moving in the opposite direction. In Egypt and many of its neighbors, this closing down is part of a wider closing in political, social and cultural thought. And it is the product of a complex historical process, one which has gained ground with the rise of Islamic conservatism since the late 1970s.”Just say no”is what conservatives around the world say to any challenge to the sexual status quo. In the Arab region, they brand these attempts as a Western conspiracy to undermine traditional Arab and Islamic values. But what ’ s really at stake here is one of their most powerful tools of control : sex wrapped up in religion. But history shows us that even as recently as our fathers ’ and grandfathers ’ day, there have been times of greater pragmatism, and tolerance, and a willingness to consider other interpretations : be it abortion, or masturbation, or even the incendiary topic of homosexuality. It is not black and white, as conservatives would have us believe. In these, as in so many other matters, Islam offers us at least 50 shades of gray. Over my travels, I ’ ve met men and women across the Arab region who ’ ve been exploring that spectrum — sexologists who are trying to help couples find greater happiness in their marriages, innovators who are managing to get sexuality education into schools, small groups of men and women, lesbian, gay, transgendered, transsexual, who are reaching out to their peers with online initiatives and real-world support. Women, and increasingly men, who are starting to speak out and push back against sexual violence on the streets and in the home. Groups that are trying to help sex workers protect themselves against HIV and other occupational hazards, and NGOs that are helping unwed mothers like Faiza find a place in society, and critically, stay with their kids. Now these efforts are small, they ’ re often underfunded, and they face formidable opposition. But I am optimistic that, in the long run, times are changing, and they and their ideas will gain ground. Social change doesn ’ t happen in the Arab region through dramatic confrontation, beating or indeed baring of breasts, but rather through negotiation. What we ’ re talking here is not about a sexual revolution, but a sexual evolution, learning from other parts of the world, adapting to local conditions, forging our own path, not following one blazed by another. That path, I hope, will one day lead us to the right to control our own bodies, and to access the information and services we need to lead satisfying and safe sexual lives. The right to express our ideas freely, to marry whom we choose, to choose our own partners, to be sexually active or not, to decide whether to have children and when, all this without violence or force or discrimination. Now we are very far from this across the Arab region, and so much needs to change : law, education, media, the economy, the list goes on and on, and it is the work of a generation, at least. But it begins with a journey that I myself have made, asking hard questions of received wisdoms in sexual life. And it is a journey which has only served to strengthen my faith, and my appreciation of local histories and cultures by showing me possibilities where I once only saw absolutes. Now given the turmoil in many countries in the Arab region, talking about sex, challenging the taboos, seeking alternatives might sound like something of a luxury. But at this critical moment in history, if we do not anchor freedom and justice, dignity and equality, privacy and autonomy in our personal lives, in our sexual lives, we will find it very hard to achieve in public life. The political and the sexual are intimate bedfellows, and that is true for us all. no matter where we live and love. Thank you.

Text Sample 787: NEG Dim 5 – 2013 – Score -1.00 – 2013/t001915.txt

Pat Mitchell : Your first time back on the TEDWomen stage. Sheryl Sandberg : First time back. Nice to see everyone. It ’ s always so nice to look out and see so many women. It ’ s so not my regular experience, as I know anyone else ’ s. PM : So when we first started talking about, maybe the **subject** wouldn ’ t be social media, which we assumed it would be, but that you had very much on your mind the missing leadership positions, particularly in the sector of technology and social media. But how did that evolve for you as a thought, and end up being the TED Talk that you gave? SS : So I was really scared to get on this stage and talk about women, because I grew up in the business world, as I think so many of us did. You never talk about being a woman, because someone might notice that you ’ re a woman, right? They might notice. Or worse, if you say”woman,”people on the other end of the table think you ’ re asking for special treatment, or complaining. Or worse, about to sue them.

And so I went through — Right? I went through my entire business career, and never spoke about being a woman, never spoke about it publicly. But I also had noticed that it wasn ’ t working. I came out of college over 20 years ago, and I thought that all of my peers were men and women, all the people above me were all men, but that would change, because your generation had done such an amazing job fighting for equality, equality was now ours for the taking. And it wasn ’ t. Because year after year, I was one of fewer and fewer, and now, often the only woman in a room. And I talked to a bunch of people about, should I give a speech at TEDWomen about women, and they said, oh no, no. It will end your business career. You cannot be a serious business executive and speak about being a woman. You ’ ll never be taken seriously again. But fortunately, there were the few, the proud — like you — who told me I should give the speech, and I asked myself the question Mark Zuckerberg might — the founder of Facebook and my boss — asks all of us, which is, what would I do if I wasn ’ t afraid? And the answer to what would I do if I wasn ’ t afraid is I would get on the TED stage, and talk about women, and leadership. And I did, and survived. PM : I would say, not only survived. I ’ m thinking of that moment, Sheryl, when you and I were standing backstage together, and you turned to me, and you told me a story. And I said — very last minute — you know, you really should share that story. SS : Oh, yeah. PM : What was that story? SS : Well, it ’ s an important part of the journey. So I had — TEDWomen — the original one was in D. C. — so I live here, so I had gotten on a plane the day before, and my daughter was three, she was clinging to my leg :”Mommy, don ’ t go.”And Pat ’ s a friend, and so, not related to the speech I was planning on giving, which was chock full of facts and figures, and nothing personal, I told Pat the story. I said, well, I ’ m having a hard day. Yesterday my daughter was clinging to my leg, and”Don ’ t go.”And you looked at me and said, you have to tell that story. I said, on the TED stage? Are you kidding? I ’ m going to get on a stage and admit my daughter was clinging to my leg? And you said yes, because if you want to talk about getting more women into leadership roles, you have to be honest about how hard it is. And I did. And I think that ’ s a really important part of the journey. The same thing happened when I wrote my book. I started writing the book. I wrote a first chapter, I thought it was fabulous. It was chock-full of data and figures, I had three pages on matrilineal Maasai tribes, and their sociological patterns. My husband read it and he was like, this is like eating your Wheaties. No one — and I apologize to Wheaties if there ’ s someone — no one, no one will read this book. And I realized through the process that I had to be more honest and more open, and I had to tell my stories. My stories of still not feeling as self-confident as I should, in many situations. My first and failed marriage. Crying at work. Felling like I didn ’ t belong there, feeling guilty to this day. And part of my journey, starting on this stage, going to”Lean In,”going to the foundation, is all about being more open and honest about those challenges, so that other women can be more open and honest, and all of us can work together towards real equality. PM : I think that one of the most striking parts about the book, and in my opinion, one of the reasons it ’ s hit such a nerve and is resonating around the world, is that you are personal in the book, and that you do make it clear that, while you ’ ve observed some things that are very important for other women to know, that you ’ ve had the same challenges that many others of us have, as you faced the hurdles and the barriers and possibly the people who don ’ t believe the same. So talk about that process : deciding you ’ d go public with the private part, and then you would also put yourself in the position of something of an expert on how to resolve those challenges. SS : After I did the TED Talk, what happened was — you know, I never really expected to write a book, I ’ m not an author, I

’ m not a writer, and it was viewed a lot, and it really started impacting people ’ s lives. I got this great — - one of the first letters I got was from a woman who said that she was offered a really big promotion at work, and she turned it down, and she told her best friend she turned it down, and her best friend said, you really need to watch this TED Talk. And so she watched this TED Talk, and she went back the next day, she took the job, she went home, and she handed her husband the grocery list. And she said, I can do this. And what really mattered to me — it wasn ’ t only women in the corporate world, even though I did hear from a lot of them, and it did impact a lot of them, it was also people of all different circumstances. There was a doctor I met who was an attending physician at Johns Hopkins, and he said that until he saw my TED Talk, it never really occurred to him that even though half the students in his med school classes were women, they weren ’ t speaking as much as the men as he did his rounds. So he started paying attention, and as he waited for raised hands, he realized the men ’ s hands were up. So he started encouraging the women to raise their hands more, and it still didn ’ t work. So he told everyone, no more hand raising, I ’ m cold-calling. So he could call evenly on men and women. And what he proved to himself was that the women knew the answers just as well or better, and he was able to go back to them and tell them that. And then there was the woman, stay-at-home mom, lives in a really difficult neighborhood, with not a great school, she said that TED Talk — she ’ s never had a corporate job, but that TED Talk inspired her to go to her school and fight for a better teacher for her child. And I guess it was part of was finding my own voice. And I realized that other women and men could find their voice through it, which is why I went from the talk to the book. PM : And in the book, you not only found your voice, which is clear and strong in the book, but you also share what you ’ ve learned — the experiences of other people in the lessons. And that ’ s what I ’ m thinking about in terms of putting yourself in a — you became a sort of expert in how you lean in. So what did that feel like, and become like in your life? To launch not just a book, not just a best-selling, best-viewed talk, but a movement, where people began to literally describe their actions at work as, I ’ m leaning in. SS : I mean, I ’ m grateful, I ’ m honored, I ’ m happy, and it ’ s the very beginning. So I don ’ t know if I ’ m an expert, or if anyone is an expert. I certainly have done a lot of research. I have read every study, I have pored over the materials, and the lessons are very clear. Because here ’ s what we know : What we know is that stereotypes are holding women back from leadership roles all over the world. It ’ s so striking.”Lean In”is very global, I ’ ve been all over the world, talking about it, and — cultures are so different. Even within our own country, to Japan, to Korea, to China, to Asia, Europe, they ’ re so different. Except for one thing : gender. All over the world, no matter what our cultures are, we think men should be strong, assertive, aggressive, have voice ; we think women should speak when spoken to, help others. Now we have, all over the world, women are called”bossy.”There is a word for”bossy,”for little girls, in every language there ’ s one. It ’ s a word that ’ s pretty much not used for little boys, because if a little boy leads, there ’ s no negative word for it, it ’ s expected. But if a little girl leads, she ’ s bossy. Now I know there aren ’ t a lot of men here, but bear with me. If you ’ re a man, you ’ ll have to represent your gender. Please raise your hand if you ’ ve been told you ’ re too aggressive at work. There ’ s always a few, it runs about five percent. Okay, get ready, gentlemen. If you ’ re a woman, please raise your hand if you ’ ve ever been told you ’ re too aggressive at work. That is what audiences have said in every country in the world, and it ’ s deeply supported by the data. Now, do we think women are more aggressive than men? Of course not. It ’ s just that we judge them through a different lens, and a lot of the character traits that you must exhibit to perform at work, to get results, to lead, are ones that we think, in a man, he ’ s a boss, and in a woman, she ’ s bossy. And the good news about this is that we can change this by acknowledging it. One of the happiest moments I had in this whole journey is, after the book came out, I stood on a stage with John Chambers, the CEO of Cisco. He read the book. He stood on a stage with me, he invited me in front of his whole management team, men and women, and he said, I thought we were good at this. I thought I was good at this. And then I read this book, and I realized that we — my company — we have called all of our senior women too aggressive, and I ’ m standing on this stage, and I ’ m sorry. And I want you to know we ’ re never going to do it again. PM : Can we send that to a lot of other people that we know? SS : And so John is doing that because he believes it ’ s good for his company, and so this kind of acknowledgement of these biases can change it. And so next time you all see someone call a little girl”bossy,”you walk right up to that person, big smile, and you say,”That little girl ’ s not bossy. That little girl has executive leadership skills.”PM : I know that ’ s what you ’ re telling your daughter. SS : Absolutely. PM : And you did focus in the book — and the reason, as you said, in writing it, was to create a dialogue about this. I mean, let ’ s just put it out there, face the fact that women are — in a time when we have more open doors, and more opportunities — are still not getting to the leadership positions. So in the months that have come since the book, in which”Lean In”focused on that and said, here are some of the challenges that remain, and many of them we have to own within ourselves and look at ourselves. What has changed? Have you seen changes? SS : Well, there ’ s certainly more dialogue, which is great. But what really matters to me, and I think all of us, is action. So everywhere I go, CEOs, they ’ re mostly men, say to me, you ’ re costing me so much money because all the women want to be paid as much as the men. And to them I say, I ’ m not sorry at all. At all. I mean, the women should be paid as much as the men. Everywhere I go, women tell me they ask for raises. Everywhere I go, women say they ’ re getting better relationships with their spouses, asking for more help at home, asking for the promotions they should be getting at work, and importantly, believing it themselves. Even little things. One of the governors of one of the states told me that he didn ’ t realize that more women were, in fact, literally sitting on the side of the room, which they are, and now he made a rule that all the women on his staff need to sit at the table. The foundation I started along with the book”Lean In”helps women, or men, start circles — small groups, it can be 10, it can be however many you want, which meet once a month. I would have hoped that by now, we ’ d have about 500 circles. That would ’ ve been great. You know, 500 times roughly 10. There are over 12, 000 circles in 50 countries in the world. PM : Wow, that ’ s amazing. SS : And these are people who are meeting every single month. I met one of them, I was in Beijing. A group of women, they ’ re all about 29 or 30, they started the first Lean In circle in Beijing, several of them grew up in very poor, rural China. These women are 29, they are told by their society that they are”left over,”because they are not yet married, and the process of coming together once a month at a meeting is helping them define who they are for themselves. What they want in their careers. The kind of partners they want, if at all. I looked at them, we went around and introduced ourselves, and they all said their names and where they ’ re from, and I said, I ’ m Sheryl Sandberg, and this was my dream. And I kind of just started crying. Right, which, I admit, I do. Right? I ’ ve talked about it before. But the fact that a woman so far away out in the world, who grew up in a rural village, who ’ s being told to marry someone she doesn ’ t want to marry, can now go meet once a month with a group of people and refuse that, and find life on her own terms. That

’ s the kind of change we have to hope for. PM : Have you been surprised by the global nature of the message? Because I think when the book first came out, many people thought, well, this is a really important handbook for young women on their way up. They need to look at this, anticipate the barriers, and recognize them, put them out in the open, have the dialogue about it, but that it ’ s really for women who are that. Doing that. Pursuing the corporate world. And yet the book is being read, as you say, in rural and developing countries. What part of that has surprised you, and perhaps led to a new perspective on your part? SS : The book is about self-confidence, and about equality. And it turns out, everywhere in the world, women need more self-confidence, because the world tells us we ’ re not equal to men. Everywhere in the world, we live in a world where the men get”and,”and women get”or.”I ’ ve never met a man who ’ s been asked how he does it all. Again, I ’ m going to turn to the men in the audience : Please raise your hand if you ’ ve been asked, how do you do it all? Men only. Women, women. Please raise your hand if you ’ ve been asked how you do it all? We assume men can do it all, slash — have jobs and children. We assume women can ’ t, and that ’ s ridiculous, because the great majority of women everywhere in the world, including the United States, work full time and have children. And I think people don ’ t fully understand how broad the message is. There is a circle that ’ s been started for rescued sex workers in Miami. They ’ re using”Lean In”to help people make the transition back to what would be a fair life, really rescuing them from their pimps, and using it. There are dress-for- success groups in Texas which are using the book, for women who have never been to college. And we know there are groups all the way to Ethiopia. And so these messages of equality — of how women are told they can ’ t have what men can have — how we assume that leadership is for men, how we assume that voice is for men, these affect all of us, and I think they are very universal. And it ’ s part of what TEDWomen does. It unites all of us in a cause we have to believe in, which is more women, more voice, more equality. PM : If you were invited now to make another TEDWomen talk, what would you say that is a result of this experience, for you personally, and what you ’ ve learned about women, and men, as you ’ ve made this journey? SS : I think I would say — I tried to say this strongly, but I think I can say it more strongly — I want to say that the status quo is not enough. That it ’ s not good enough, that it ’ s not changing quickly enough. Since I gave my TED Talk and published my book, another year of data came out from the U. S. Census. And you know what we found? No movement in the wage gap for women in the United States. Seventy-seven cents to the dollar. If you are a black woman, 64 cents. If you are a Latina, we ’ re at 54 cents. Do you know when the last time those numbers went up? 2002. We are stagnating, we are stagnating in so many ways. And I think we are not really being honest about that, for so many reasons. It ’ s so hard to talk about gender. We shy away from the word”feminist,”a word I really think we need to embrace. We have to get rid of the word bossy and bring back — I think I would say in a louder voice, we need to get rid of the word”bossy”and bring back the word”feminist,”because we need it. PM : And we all need to do a lot more leaning in. SS : A lot more leaning in. PM : Thank you, Sheryl. Thanks for leaning in and saying yes. SS : Thank you.

Text Sample 788: NEG Dim 5 – 2012 – Score -1.00 – 2012/t002169.txt

So I want to talk to you about two things tonight. Number one : Teaching surgery and doing surgery is really hard. And second, that language is one of the most profound things that separate us all over the world. And in my little corner of the world, these two things are actually related, and I want to tell you how tonight. Now, nobody wants an operation. Who here has had surgery? Did you want it? Keep your hands up if you wanted an operation. Nobody wants an operation. In particular, nobody wants an operation with tools like these through large incisions that cause a lot of pain, that cause a lot of time out of work or out of school, that leave a big scar. But if you have to have an operation, what you really want is a minimally invasive operation. That ' s what I want to talk to you about tonight — how doing and teaching this type of surgery led us on a search for a better universal translator. Now, this type of surgery is hard, and it starts by putting people to sleep, putting carbon dioxide in their abdomen, **blowing** them up like a balloon, sticking one of these sharp pointy things into their abdomen — it ' s dangerous stuff — and taking instruments and watching it on a TV screen. So let ' s see what it looks like. So this is gallbladder surgery. We perform a million of these a year in the United States alone. This is the real thing. There ' s no blood. And you can see how focused the surgeons are, how much concentration it takes. You can see it in their faces. It ' s hard to teach, and it ' s not all that easy to learn. We do about five million of these in the United States and maybe 20 million of these worldwide. All right, you ' ve all heard the term :”He ' s a born surgeon.”Let me tell you, surgeons are not born. Surgeons are not made either. There are no little tanks where we ' re making surgeons. Surgeons are trained one step at a time. It starts with a foundation, basic skills. We build on that and we take people, hopefully, to the operating room where they learn to be an assistant. Then we teach them to be a surgeon in training. And when they do all of that for about five years, they get the coveted board certification. If you need surgery, you want to be operated on by a board-certified surgeon. You get your board certificate, and you can go out into practice. And eventually, if you ' re lucky, you achieve mastery. Now that foundation is so important that a number of us from the largest general surgery society in the United States, SAGES, started in the late 1990s a training program that would assure that every surgeon who practices minimally invasive surgery would have a strong foundation of knowledge and skills necessary to go on and do procedures. Now the science behind this is so potent that it became required by the American Board of Surgery in order for a young surgeon to become board certified. It ' s not a lecture, it ' s not a course, it ' s all of that plus a high-stakes assessment. It ' s hard. Now just this past year, one of our partners, the American College of Surgeons, teamed up with us to make an announcement that all surgeons should be FLS -certified before they do minimally invasive surgery. And are we talking about just people here in the U. S. and Canada? No, we just said all surgeons. So to lift this education and training worldwide is a very large task, something I ' m very personally excited about as we travel around the world. SAGES does surgery all over the world, teaching and educating surgeons. So we have a problem, and one of the problems is distance. We can ' t travel everywhere. We need to make the world a smaller place. And I think that we can develop some tools to do so. And one of the tools I like personally is using video. So I was inspired by a friend. This is Allan Okrainec from Toronto. And he proved that you could actually teach people to do surgery using video conferencing. So here ' s Allan teaching an English-speaking surgeon in Africa these basic fundamental skills necessary to do minimally invasive surgery. Very inspiring. But for this examination, which is really hard, we have a problem. Even people who say they speak English, only 14 percent pass. Because for them it ' s not a surgery test, it ' s an English test. Let me bring it to you locally. I work at the Cambridge Hospital. It ' s the primary Harvard Medical School teaching facility. We have more than 100 translators covering 63 languages, and we spend millions of dollars just in our little hospital. It ’ s a big labor-intensive effort. If you think about the worldwide burden of trying to talk to your patients — not just teaching surgeons, just trying to talk to your patients — there aren ’ t enough translators in the world. We need to employ technology to assist us in this quest. At our hospital we see everybody from Harvard professors to people who just got here last week. And you have no idea how hard it is to talk to somebody or take care of somebody you can ’ t talk to. And there isn ’ t always a translator available. So we need tools. We need a universal translator. One of the things that I want to leave you with as you think about this talk is that this talk is not just about us preaching to the world. It ’ s really about setting up a dialogue. We have a lot to learn. Here in the United States we spend more money per person for outcomes that are not better than many countries in the world. Maybe we have something to learn as well. So I ’ m passionate about teaching these FLS skills all over the world. This past year I ’ ve been in Latin America, I ’ ve been in China, talking about the fundamentals of laparoscopic surgery. And everywhere I go the barrier is :”We want this, but we need it in our language.”So here ’ s what we think we want to do : Imagine giving a lecture and being able to talk to people in their own native language simultaneously. I want to talk to the people in Asia, Latin America, Africa, Europe seamlessly, accurately and in a cost-effective fashion using technology. And it has to be bi-directional. They have to be able to teach us something as well. It ’ s a big task. So we looked for a universal translator ; I thought there would be one out there. Your webpage has translation, your cellphone has translation, but nothing that ’ s good enough to teach surgery. Because we need a lexicon. What is a lexicon? A lexicon is a body of words that describes a domain. I need to have a health care lexicon. And in that I need a surgery lexicon. That ’ s a tall order. We have to work at it. So let me show you what we ’ re doing. This is research — can ’ t buy it. We ’ re working with the folks at IBM Research from the Accessibility Center to string together technologies to work towards the universal translator. It starts with a framework system where when the surgeon delivers the lecture using a framework of captioning technology, we then add another technology to do video conferencing. But we don ’ t have the words yet, so we add a third technology. And now we ’ ve got the words, and we can apply the special sauce : the translation. We get the words up in a window and then apply the magic. We work with a fourth technology. And we currently have access to eleven language pairs. More to come as we think about trying to make the world a smaller place. And I ’ d like to show you our prototype of stringing all of these technologies that don ’ t necessarily always talk to each other to become something useful. Narrator : Fundamentals of Laparoscopic Surgery. Module five : manual skills practice. Students may display captions in their native language. Steven Schwaitzberg : If you ’ re in Latin America, you click the”I want it in Spanish”button and out it comes in real time in Spanish. But if you happen to be sitting in Beijing at the same time, by using technology in a constructive fashion, you could get it in Mandarin or you could get it in Russian — on and on and on, simultaneously without the use of human translators. But that ’ s the lectures. If you remember what I told you about FLS at the beginning, it ’ s knowledge and skills. The difference in an operation between doing something successfully and not may be moving your hand this much. So we ’ re going to take it one step further ; we ’ ve brought my friend Allan back. Allan Okrainec : Today we ’ re going to practice suturing. This is how you hold the needle. Grab the needle at the tip. It ’ s important to be accurate. Aim for the black dots. Orient your loop this way. Now go ahead and cut. Very good Oscar. I ’ ll see you next week. SS : So that ’ s what we ’ re working on in our quest for the universal translator. We want it to be bi-directional. We have a need to learn as well as to teach. I can think of a million uses for a tool like this. As we think about intersecting technologies — everybody has a cell phone with a camera — we could use this everywhere, whether it be health care, patient care, engineering, law, conferencing, translating videos. This is a ubiquitous tool. In order to break down our barriers, we have to learn to talk to people, to demand that people work on translation. We need it for our everyday life, in order to make the world a smaller place. Thank you very much.

Text Sample 789: NEG Dim 5 – 2012 – Score -1.00 – 2012/t002146.txt
I ’ d like to share with you the story of one of my patients called Celine. Celine is a housewife and lives in a rural district of Cameroon in west Central Africa. Six years ago, at the time of her HIV diagnosis, she was recruited to participate in the clinical trial which was running in her health district at the time. When I first met Celine, a little over a year ago, she had gone for 18 months without any antiretroviral therapy, and she was very ill. She told me that she stopped coming to the clinic when the trial ended because she had no money for the bus fare and was too ill to walk the 35-kilometer distance. Now during the clinical trial, she ’ d been given all her antiretroviral drugs free of charge, and her transportation costs had been covered by the research funds. All of these ended once the trial was completed, leaving Celine with no alternatives. She was unable to tell me the names of the drugs she ’ d received during the trial, or even what the trial had been about. I didn ’ t bother to ask her what the results of the trial were because it seemed obvious to me that she would have no clue. Yet what puzzled me most was Celine had given her informed consent to be a part of this trial, yet she clearly did not understand the implications of being a participant or what would happen to her once the trial had been completed. Now, I have shared this story with you as an example of what can happen to participants in the clinical trial when it is poorly conducted. Maybe this particular trial yielded exciting results. Maybe it even got published in a high-profile scientific journal. Maybe it would inform clinicians around the world on how to improve on the clinical management of HIV patients. But it would have done so at a price to hundreds of patients who, like Celine, were left to their own devices once the research had been completed. I do not stand here today to suggest in any way that conducting HIV clinical trials in developing countries is bad. On the contrary, clinical trials are extremely useful tools, and are much needed to address the burden of disease in developing countries. However, the inequalities that exist between richer countries and developing countries in terms of funding pose a real risk for exploitation, especially in the context of externally-funded research. Sadly enough, the fact remains that a lot of the studies that are conducted in developing countries could never be authorized in the richer countries which fund the research. I ’ m sure you must be asking yourselves what makes developing countries, especially those in sub-Saharan Africa, so attractive for these HIV clinical trials? Well, in order for a clinical trial to generate valid and widely applicable results, they need to be conducted with large numbers of study participants and preferably on a population with a high incidence of new HIV infections. Sub-Saharan Africa largely fits this description, with 22 million people living with HIV, an estimated 70 percent of the 30 million people who are infected worldwide. Also, research within the continent is a lot easier to conduct due to widespread poverty, endemic diseases and inadequate health care systems. A clinical trial that is considered to be potentially beneficial to the population is more likely to be authorized, and in the

absence of good health care systems, almost any offer of medical assistance is accepted as better than nothing. Even more problematic reasons include lower risk of litigation, less rigorous ethical reviews, and populations that are willing to participate in almost any study that hints at a cure. As funding for HIV research increases in developing countries and ethical review in richer countries become more strict, you can see why this context becomes very, very attractive. The high prevalence of HIV drives researchers to conduct research that is sometimes scientifically acceptable but on many levels ethically questionable. How then can we ensure that, in our search for the cure, we do not take an unfair advantage of those who are already most affected by the pandemic? I invite you to consider four areas I think we can focus on in order to improve the way in which things are done. The first of these is informed consent. Now, in order for a clinical trial to be considered ethically acceptable, participants must be given the relevant information in a way in which they can understand, and must freely consent to participate in the trial. This is especially important in developing countries, where a lot of participants consent to research because they believe it is the only way in which they can receive medical care or other benefits. Consent procedures that are used in richer countries are often inappropriate or ineffective in a lot of developing countries. For example, it is counterintuitive to have an illiterate study participant, like Celine, sign a lengthy consent form that they are unable to read, let alone understand. Local communities need to be more involved in establishing the criteria for recruiting participants in clinical trials, as well as the incentives for participation. The information in these trials needs to be given to the potential participants in linguistically and culturally acceptable formats. The second point I would like for you to consider is the standard of care that is provided to participants within any clinical trial. Now, this is **subject** to a lot of debate and controversy. Should the control group in the clinical trial be given the best current treatment which is available anywhere in the world? Or should they be given an alternative standard of care, such as the best current treatment available in the country in which the research is being conducted? Is it fair to evaluate a treatment regimen which may not be affordable or accessible to the study participants once the research has been completed? Now, in a situation where the best current treatment is inexpensive and simple to deliver, the answer is straightforward. However, the best current treatment available anywhere in the world is often very difficult to provide in developing countries. It is important to assess the potential risks and benefits of the standard of care which is to be provided to participants in any clinical trial, and establish one which is relevant for the context of the study and most beneficial for the participants within the study. That brings us to the third point I want you think about : the ethical review of research. An effective system for reviewing the ethical suitability of clinical trials is primordial to safeguard participants within any clinical trial. Unfortunately, this is often lacking or inefficient in a lot of developing countries. Local governments need to set up effective systems for reviewing the ethical issues around the clinical trials which are authorized in different developing countries, and they need to do this by setting up ethical review committees that are independent of the government and research sponsors. Public accountability needs to be promoted through transparency and independent review by non-governmental and international organizations as appropriate. The final point I would like for you to consider tonight is what happens to participants in the clinical trial once the research has been completed. I think it is absolutely wrong for research to begin in the first place without a clear plan for what would happen to the participants once the trial has ended. Now, researchers need to make every effort to ensure that an intervention that has been shown to be beneficial during a clinical trial is accessible to the participants of the trial once the trial has been completed. In addition, they should be able to consider the possibility of introducing and maintaining effective treatments in the wider community once the trial ends. If, for any reason, they feel that this might not be possible, then I think they should have to ethically justify why the clinical trial should be conducted in the first place. Now, fortunately for Celine, our meeting did not end in my office. I was able to get her enrolled into a free HIV treatment program closer to her home, and with a support group to help her cope. Her story has a positive ending, but there are thousands of others in similar situations who are much less fortunate. Although she may not know this, my encounter with Celine has completely changed the way in which I view HIV clinical trials in developing countries, and made me even more determined to be part of the movement to change the way in which things are done. I believe that every single person listening to me tonight can be part of that change. If you are a researcher, I hold you to a higher standard of moral conscience, to remain ethical in your research, and not compromise human welfare in your search for answers. If you work for a funding agency or pharmaceutical company, I challenge you to hold your employers to fund research that is ethically sound. If you come from a developing country like myself, I urge you to hold your government to a more thorough review of the clinical trials which are authorized in your country. Yes, there is a need for us to find a cure for HIV, to find an effective vaccine for malaria, to find a diagnostic tool that works for T. B., but I believe that we owe it to those who willingly and selflessly consent to participate in these clinical trials to do this in a humane way. Thank you.

Text Sample 790: NEG Dim 5 – 2012 – Score 1.00 – 2012/t001893.txt

["Rebecca Newberger Goldstein"] ["Steven Pinker"] ["The Long Reach of Reason"] Cabbie : Twenty-two dollars. Steven Pinker : Okay. Rebecca Newberger Goldstein : Reason appears to have fallen on hard times : Popular culture plumbs new depths of dumbth and political discourse has become a race to the bottom. We ' re living in an era of scientific creationism, 9 / 11 conspiracy theories, psychic hotlines, and a resurgence of religious fundamentalism. People who think too well are often accused of elitism, and even in the academy, there are attacks on logocentrism, the crime of letting logic dominate our thinking. SP : But is this necessarily a bad thing? Perhaps reason is overrated. Many pundits have argued that a good heart and steadfast moral clarity are superior to triangulations of overeducated policy wonks, like the best and brightest and that dragged us into the quagmire of Vietnam. And wasn ' t it reason that gave us the means to despoil the planet and threaten our species with weapons of mass destruction? In this way of thinking, it ' s character and conscience, not cold- hearted calculation, that will save us. Besides, a human being is not a brain on a stick. My fellow psychologists have shown that we ' re led by our bodies and our emotions and use our puny powers of reason merely to rationalize our gut feelings after the fact. RNG : How could a reasoned argument logically entail the ineffectiveness of reasoned arguments? Look, you ' re trying to persuade us of reason ' s impotence. You ' re not threatening us or bribing us, suggesting that we resolve the issue with a show of hands or a beauty contest. By the very act of trying to reason us into your position, you ' re conceding reason ' s potency. Reason isn ' t up for grabs here. It can ' t be. You show up for that debate and you ' ve already lost it. SP : But can reason lead us in directions that are good or decent or moral? After all, you pointed out that reason is just a means to an end, and the end depends on the reasoner ' s passions. Reason can lay out a road map to peace and harmony if the reasoner wants peace and harmony, but it can also lay out a road map to conflict and strife if the reasoner delights in conflict and strife. Can reason force the reasoner to want less cruelty and waste? RNG : All on its own, the answer is no, but it doesn ’ t take much to switch it to yes. You need two conditions : The first is that reasoners all care about their own well-being. That ’ s one of the passions that has to be present in order for reason to go to work, and it ’ s obviously present in all of us. We all care passionately about our own well-being. The second condition is that reasoners are members of a community of reasoners who can affect one another ’ s well- being, can exchange messages, and comprehend each other ’ s reasoning. And that ’ s certainly true of our gregarious and loquatious species, well endowed with the instinct for language. SP : Well, that sounds good in theory, but has it worked that way in practice? In particular, can it explain a momentous historical development that I spoke about five years ago here at TED? Namely, we seem to be getting more humane. Centuries ago, our ancestors would burn cats alive as a form of popular entertainment. Knights waged constant war on each other by trying to kill as many of each other ’ s peasants as possible. Governments executed people for frivolous reasons, like stealing a cabbage or criticizing the royal garden. The executions were designed to be as prolonged and as painful as possible, like crucifixion, disembowelment, breaking on the wheel. Respectable people kept slaves. For all our flaws, we have abandoned these barbaric practices. RNG : So, do you think it ’ s human nature that ’ s changed? SP : Not exactly. I think we still harbor instincts that can erupt in violence, like greed, tribalism, revenge, dominance, sadism. But we also have instincts that can steer us away, like self-control, empathy, a sense of fairness, what Abraham Lincoln called the better angels of our nature. RNG : So if human nature didn ’ t change, what invigorated those better angels? SP : Well, among other things, our circle of empathy expanded. Years ago, our ancestors would feel the pain only of their family and people in their village. But with the expansion of literacy and travel, people started to sympathize with wider and wider circles, the clan, the tribe, the nation, the race, and perhaps eventually, all of humanity. RNG : Can hard-headed scientists really give so much credit to soft-hearted empathy? SP : They can and do. Neurophysiologists have found neurons in the brain that respond to other people ’ s actions the same way they respond to our own. Empathy emerges early in life, perhaps before the age of one. Books on empathy have become bestsellers, like”The Empathic Civilization”and”The Age of Empathy.”RNG : I ’ m all for empathy. I mean, who isn ’ t? But all on its own, it ’ s a feeble instrument for making moral progress. For one thing, it ’ s innately biased toward blood relations, babies and warm, fuzzy animals. As far as empathy is concerned, ugly outsiders can go to hell. And even our best attempts to work up sympathy for those who are unconnected with us fall miserably short, a sad truth about human nature that was pointed out by Adam Smith. Adam Smith : Let us suppose that the great empire of China was suddenly swallowed up by an earthquake, and let us consider how a man of humanity in Europe would react on receiving intelligence of this dreadful calamity. He would, I imagine, first of all express very strongly his sorrow for the misfortune of that unhappy people. He would make many melancholy reflections upon the precariousness of human life, and when all these humane sentiments had been once fairly expressed, he would pursue his business or his pleasure with the same ease and tranquility as if no such accident had happened. If he was to lose his little finger tomorrow, he would not sleep tonight, but provided he never saw them, he would snore with the most profound security over the ruin of a hundred million of his brethren. SP : But if empathy wasn ’ t enough to make us more humane, what else was there? RNG

: Well, you didn ' t mention what might be one of our most effective better angels : reason. Reason has muscle. It ' s reason that provides the push to widen that circle of empathy. Every one of the humanitarian developments that you mentioned originated with thinkers who gave reasons for why some practice was indefensible. They demonstrated that the way people treated some particular group of others was logically inconsistent with the way they insisted on being treated themselves. SP : Are you saying that reason can actually change people ' s minds? Don ' t people just stick with whatever conviction serves their interests or conforms to the culture that they grew up in? RNG : Here ' s a fascinating fact about us : Contradictions bother us, at least when we ' re forced to confront them, which is just another way of saying that we are susceptible to reason. And if you look at the history of moral progress, you can trace a direct pathway from reasoned arguments to changes in the way that we actually feel. Time and again, a thinker would lay out an argument as to why some practice was indefensible, irrational, inconsistent with values already held. Their essay would go viral, get translated into many languages, get debated at pubs and coffee houses and salons, and at dinner parties, and influence leaders, legislators, popular opinion. Eventually their conclusions get absorbed into the common sense of decency, erasing the tracks of the original argument that had gotten us there. Few of us today feel any need to put forth a rigorous philosophical argument as to why slavery is wrong or public hangings or beating children. By now, these things just feel wrong. But just those arguments had to be made, and they were, in centuries past. SP : Are you saying that people needed a step-by-step argument to grasp why something might be a wee bit wrong with burning heretics at the stake? RNG : Oh, they did. Here ' s the French theologian Sebastian Castellio making the case. Sebastian Castellio : Calvin says that he ' s certain, and other sects say that they are. Who shall be judge? If the matter is certain, to whom is it so? To Calvin? But then, why does he write so many books about manifest truth? In view of the uncertainty, we must define heretics simply as one with whom we disagree. And if then we are going to kill heretics, the logical outcome will be a war of extermination, since each is sure of himself. SP : Or with hideous punishments like breaking on the wheel? RNG : The prohibition in our constitution of cruel and unusual punishments was a response to a pamphlet circulated in 1764 by the Italian jurist Cesare Beccaria. Cesare Beccaria : As punishments become more cruel, the minds of men, which like fluids always adjust to the level of the objects that surround them, become hardened, and after a hundred years of cruel punishments, breaking on the wheel causes no more fear than imprisonment previously did. For a punishment to achieve its objective, it is only necessary that the harm that it inflicts outweighs the benefit that derives from the crime, and into this calculation ought to be factored the certainty of punishment and the loss of the good that the commission of the crime will produce. Everything beyond this is superfluous, and therefore tyrannical. SP : But surely antiwar movements depended on mass demonstrations and catchy tunes by folk singers and wrenching photographs of the human costs of war. RNG : No doubt, but modern anti-war movements reach back to a long chain of thinkers who had argued as to why we ought to mobilize our emotions against war, such as the father of modernity, Erasmus. Erasmus : The advantages derived from peace diffuse themselves far and wide, and reach great numbers, while in war, if anything turns out happily, the advantage redounds only to a few, and those unworthy of reaping it. One man ' s safety is owing to the destruction of another. One man ' s prize is derived from the plunder of another. The cause of rejoicings made by one side is to the other a cause of mourning. Whatever is unfortunate in war, is severely so indeed, and whatever, on the contrary, is called good fortune, is a savage and a cruel good fortune, an ungenerous happiness deriving its existence from another ' s woe. SP : But everyone knows that the movement to abolish slavery depended on faith and emotion. It was a movement spearheaded by the Quakers, and it only became popular when Harriet Beecher Stowe ' s novel"Uncle Tom ' s Cabin"became a bestseller. RNG : But the ball got rolling a century before. John Locke bucked the tide of millennia that had regarded the practice as perfectly natural. He argued that it was inconsistent with the principles of rational government. John Locke : Freedom of men under government is to have a standing rule to live by common to everyone of that society and made by the legislative power erected in it, a liberty to follow my own will in all things where that rule prescribes not, not to be **subject** to the inconstant, uncertain, unknown, arbitrary will of another man, as freedom of nature is to be under no other restraint but the law of nature. SP : Those words sound familiar. Where have I read them before? Ah, yes. Mary Astell : If absolute sovereignty be not necessary in a state, how comes it to be so in a family? Or if in a family, why not in a state? Since no reason can be alleged for the one that will not hold more strongly for the other, if all men are born free, how is it that all women are born slaves, as they must be if being **subjected** to the inconstant, uncertain, unknown, arbitrary will of men be the perfect condition of slavery? RNG : That sort of co-option is all in the job description of reason. One movement for the expansion of rights inspires another because the logic is the same, and once that ' s hammered home, it becomes increasingly uncomfortable to ignore the inconsistency. In the 1960s, the Civil Rights Movement inspired the movements for women ' s rights, children ' s rights, gay rights and even animal rights. But fully two centuries before, the Enlightenment thinker Jeremy Bentham had exposed the indefensibility of customary practices such as the cruelty to animals. Jeremy Bentham : The question is not, can they reason, nor can they talk, but can they suffer? RNG : And the persecution of homosexuals. JB : As to any primary mischief, it ' s evident that it produces no pain in anyone. On the contrary, it produces pleasure. The partners are both willing. If either of them be unwilling, the act is an offense, totally different in its nature of effects. It ' s a personal injury. It ' s a kind of rape. As to the any danger exclusive of pain, the danger, if any, much consist in the tendency of the example. But what is the tendency of this example? To dispose others to engage in the same practices. But this practice produces not pain of any kind to anyone. SP : Still, in every case, it took at least a century for the arguments of these great thinkers to trickle down and infiltrate the population as a whole. It kind of makes you wonder about our own time. Are there practices that we engage in where the arguments against them are there for all to see but nonetheless we persist in them? RNG : When our great grandchildren look back at us, will they be as appalled by some of our practices as we are by our slave-owning, heretic-burning, wife- beating, gay-bashing ancestors? SP : I ' m sure everyone here could think of an example. RNG : I opt for the mistreatment of animals in factory farms. SP : The imprisonment of nonviolent drug offenders and the toleration of rape in our nation ' s prisons. RNG : Scrimping on donations to life-saving charities in the developing world. SP : The possession of nuclear weapons. RNG : The appeal to religion to justify the otherwise unjustifiable, such as the ban on contraception. SP : What about religious faith in general? RNG : Eh, I ' m not holding my breath. SP : Still, I have become convinced that reason is a better angel that deserves the greatest credit for the moral progress our species has enjoyed and that holds out the greatest hope for continuing moral progress in the future. RNG : And if, our friends, you detect a flaw in this argument, just remember you ' ll be depending on reason to point it out. Thank you. SP : Thank you.

Text Sample 791: NEG Dim 5 – 2009 – Score -1.00 – 2009/t000207.txt

A human child is born, and for quite a long time is a consumer. It cannot be consciously a contributor. It is helpless. It doesn ’ t know how to survive, even though it is endowed with an instinct to survive. It needs the help of mother, or a foster mother, to survive. It can ’ t afford to doubt the person who tends the child. It has to totally surrender, as one surrenders to an anesthesiologist. It has to totally surrender. That implies a lot of trust. That implies the trusted person won ’ t violate the trust. As the child grows, it begins to discover that the person trusted is violating the trust. It doesn ’ t know even the word”violation.”Therefore, it has to blame itself, a wordless blame, which is more difficult to really resolve — the wordless self-blame. As the child grows to become an adult : so far, it has been a consumer, but the growth of a human being lies in his or her capacity to contribute, to be a contributor. One cannot contribute unless one feels secure, one feels big, one feels : I have enough. To be compassionate is not a joke. It ’ s not that simple. One has to discover a certain bigness in oneself. That bigness should be centered on oneself, not in terms of money, not in terms of power you wield, not in terms of any status that you can command in the society, but it should be centered on oneself. The self : you are self-aware. On that self, it should be centered — a bigness, a wholeness. Otherwise, compassion is just a word and a dream. You can be compassionate occasionally, more moved by empathy than by compassion. Thank God we are empathetic. When somebody ’ s in pain, we pick up the pain. In a Wimbledon final match, these two guys fight it out. Each one has got two games. It can be anybody ’ s game. What they have sweated so far has no meaning. One person wins. The tennis etiquette is, both the players have to come to the net and shake hands. The winner boxes the air and kisses the ground, throws his shirt as though somebody is waiting for it. And this guy has to come to the net. When he comes to the net, you see, his whole face changes. It looks as though he ’ s wishing that he didn ’ t win. Why? Empathy. That ’ s human heart. No human heart is denied of that empathy. No religion can demolish that by indoctrination. No culture, no nation and nationalism — nothing can touch it because it is empathy. And that capacity to empathize is the window through which you reach out to people, you do something that makes a difference in somebody ’ s life — even words, even time. Compassion is not defined in one form. There ’ s no Indian compassion. There ’ s no American compassion. It transcends nation, the gender, the age. Why? Because it is there in everybody. It ’ s experienced by people occasionally. Then this occasional compassion, we are not talking about — it will never remain occasional. By mandate, you cannot make a person compassionate. You can ’ t say,”Please love me.”Love is something you discover. It ’ s not an action, but in the English language, it is also an action. I will come to it later. So one has got to discover a certain wholeness. I am going to cite the possibility of being whole, which is within our experience, everybody ’ s experience. In spite of a very tragic life, one is happy in moments which are very few and far between. And the one who is happy, even for a slapstick joke, accepts himself and also the scheme of things in which one finds oneself. That means the whole universe, known things and unknown things. All of them are totally accepted because you discover your wholeness in yourself. The **subject** —”me”— and the object — the scheme of things — fuse into oneness, an experience nobody can say,”I am denied of,”an experience common to all and sundry. That experience confirms that, in spite of all your limitations — all your wants, desires, unfulfilled, and the credit cards and layoffs and, finally, baldness — you can be happy. But the extension of the logic is that you don ’ t need to fulfill your desire to be happy. You are the very happiness, the wholeness that you want to be. There ’ s no choice in this : that only confirms the reality that the wholeness cannot be different from you, cannot be minus you. It has got to be you. You cannot be a part of wholeness and still be whole. Your moment of happiness reveals that reality, that realization, that recognition :”Maybe I am the whole. Maybe the swami is right. Maybe the swami is right.”You start your new life. Then everything becomes meaningful. I have no more reason to blame myself. If one has to blame oneself, one has a million reasons plus many. But if I say, in spite of my body being limited — if it is black it is not white, if it is white it is not black : body is limited any which way you look at it. Limited. Your knowledge is limited, health is limited, and power is therefore limited, and the cheerfulness is going to be limited. Compassion is going to be limited. Everything is going to be limitless. You cannot command compassion unless you become limitless, and nobody can become limitless, either you are or you are not. Period. And there is no way of your being not limitless too. Your own experience reveals, in spite of all limitations, you are the whole. And the wholeness is the reality of you when you relate to the world. It is love first. When you relate to the world, the dynamic manifestation of the wholeness is, what we say, love. And itself becomes compassion if the object that you relate to evokes that emotion. Then that again transforms into giving, into sharing. You express yourself because you have compassion. To discover compassion, you need to be compassionate. To discover the capacity to give and share, you need to be giving and sharing. There is no shortcut : it is like swimming by swimming. You learn swimming by swimming. You cannot learn swimming on a foam mattress and enter into water. You learn swimming by swimming. You learn cycling by cycling. You learn cooking by cooking, having some sympathetic people around you to eat what you cook. And, therefore, what I say, you have to fake it and make it. You need to. My predecessor meant that. You have to act it out. You have to act compassionately. There is no verb for compassion, but you have an adverb for compassion. That ’ s interesting to me. You act compassionately. But then, how to act compassionately if you don ’ t have compassion? That is where you fake. You fake it and make it. This is the mantra of the United States of America. You fake it and make it. You act compassionately as though you have compassion : grind your teeth, take all the support system. If you know how to pray, pray. Ask for compassion. Let me act compassionately. Do it. You ’ ll discover compassion and also slowly a relative compassion, and slowly, perhaps if you get the right teaching, you ’ ll discover compassion is a dynamic manifestation of the reality of yourself, which is oneness, wholeness, and that ’ s what you are. With these words, thank you very much.

Text Sample 792: NEG Dim 5 – 2009 – Score -1.00 – 2009/t000106.txt

We look around the media, as we see on the news from Iraq, Afghanistan, Sierra Leone, and the conflict seems incomprehensible to us. And that ’ s certainly how it seemed to me when I started this project. But as a physicist, I thought, well if you give me some data, I could maybe understand this. You know, give us a go. So as a naive New Zealander I thought, well I ’ ll go to the Pentagon. Can you get me some information? No. So I had to think a little harder. And I was watching the news one night in Oxford. And I looked down at the chattering heads on my channel of choice. And I saw that there was information there. There was data within the streams of news that we consume. All this noise around us actually has information. So what I started thinking was, perhaps there is something like open source intelligence here. If we can get enough of these streams of information together, we can perhaps start to understand the war. So this is exactly what I did. We started bringing a team together, an interdisciplinary team of scientists, of economists, mathematicians. We brought these guys together and we started to try and solve this. We did it in three steps. The first step we did was to collect. We did 130 different sources of information — from NGO reports to newspapers and cable news. We brought this raw data in and we filtered it. We extracted the key bits on information to build the database. That database contained the timing of attacks, the location, the size and the weapons used. It ' s all in the streams of information we consume daily, we just have to know how to pull it out. And once we had this we could start doing some cool stuff. What if we were to look at the distribution of the sizes of attacks? What would that tell us? So we started doing this. And you can see here on the horizontal axis you ' ve got the number of people killed in an attack or the size of the attack. And on the vertical axis you ' ve got the number of attacks. So we plot data for sample on this. You see some sort of random distribution — perhaps 67 attacks, one person was killed, or 47 attacks where seven people were killed. We did this exact same thing for Iraq. And we didn ' t know, for Iraq what we were going to find. It turns out what we found was pretty surprising. You take all of the conflict, all of the chaos, all of the noise, and out of that comes this precise mathematical distribution of the way attacks are ordered in this conflict. This **blew** our mind. Why should a conflict like Iraq have this as its fundamental signature? Why should there be order in war? We didn ' t really understand that. We thought maybe there is something special about Iraq. So we looked at a few more conflicts. We looked at Colombia, we looked at Afghanistan, and we looked at Senegal. And the same pattern emerged in each conflict. This wasn ' t supposed to happen. These are different wars, with different religious factions, different political factions, and different socioeconomic problems. And yet the fundamental patterns underlying them are the same. So we went a little wider. We looked around the world at all the data we could get our hands on. From Peru to Indonesia, we studied this same pattern again. And we found that not only were the distributions these straight lines, but the slope of these lines, they clustered around this value of alpha equals 2. 5. And we could generate an equation that could predict the likelihood of an attack. What we ' re saying here is the probability of an attack killing X number of people in a country like Iraq is equal to a constant, times the size of that attack, raised to the power of negative alpha. And negative alpha is the slope of that line I showed you before. So what? This is data, statistics. What does it tell us about these conflicts? That was a challenge we had to face as physicists. How do we explain this? And what we really found was that alpha, if we think about it, is the organizational structure of the insurgency. Alpha is the distribution of the sizes of attacks, which is really the distribution of the group strength carrying out the attacks. So we look at a process of group dynamics : coalescence and fragmentation, groups coming together, groups breaking apart. And we start running the numbers on this. Can we simulate it? Can we create the kind of patterns that we ' re seeing in places like Iraq? Turns out we kind of do a reasonable job. We can run these simulations. We can recreate this using a process of group dynamics to explain the patterns that we see all around the conflicts around the world. So what ' s going on? Why should these different — seemingly different conflicts have the same patterns? Now what I believe is going on is that the insurgent forces, they evolve over time. They adapt. And it turns out there is only one solution to fight a much stronger enemy. And if you don ' t find that solution as an insurgent force, you don ' t exist. So every insurgent force that is ongoing, every conflict that is ongoing, it ' s going to look something like this. And that is what we think is happening. Taking it forward, how do we change it? How do we end a war like Iraq? What does it look like? Alpha is the structure. It ’ s got a stable state at 2. 5. This is what wars look like when they continue. We ’ ve got to change that. We can push it up : the forces become more fragmented ; there is more of them, but they are weaker. Or we push it down : they ’ re more robust ; there is less groups ; but perhaps you can sit and talk to them. So this graph here, I ’ m going to show you now. No one has seen this before. This is literally stuff that we ’ ve come through last week. And we see the evolution of Alpha through time. We see it start. And we see it grow up to the stable state the wars around the world look like. And it stays there through the invasion of Fallujah until the Samarra bombings in the Iraqi elections of ’ 06. And the system gets perturbed. It moves upwards to a fragmented state. This is when the surge happens. And depending on who you ask, the surge was supposed to push it up even further. The opposite happened. The groups became stronger. They became more robust. And so I ’ m thinking, right, great, it ’ s going to keep going down. We can talk to them. We can get a solution. The opposite happened. It ’ s moved up again. The groups are more fragmented. And this tells me one of two things. Either we ’ re back where we started and the surge has had no effect ; or finally the groups have been fragmented to the extent that we can start to think about maybe moving out. I don ’ t know what the answer is to that. But I know that we should be looking at the structure of the insurgency to answer that question. Thank you.

Text Sample 793: NEG Dim 5 – 2009 – Score 1.00 – 2009/t000107.txt

So sometimes I get invited to give weird talks. I got invited to speak to the people who dress up in big stuffed animal costumes to perform at sporting events. Unfortunately I couldn ’ t go. But it got me thinking about the fact that these guys, at least most of them, know what it is that they do for a living. What they do is they dress up as stuffed animals and entertain people at sporting events. Shortly after that I got invited to speak at the convention of the people who make balloon animals. And again, I couldn ’ t go. But it ’ s a fascinating group. They make balloon animals. There is a big schism between the ones who make gospel animals and porn animals, but — they do a lot of really cool stuff with balloons. Sometimes they get in trouble, but not often. And the other thing about these guys is, they also know what they do for a living. They make balloon animals. But what do we do for a living? What exactly to the people watching this do every day? And I want to argue that what we do is we try to change everything. That we try to find a piece of the status quo — something that bothers us, something that needs to be improved, something that is itching to be changed — and we change it. We try to make big, permanent, important change. But we don ’ t think about it that way. And we haven ’ t spent a lot of time talking about what that process is like. And I ’ ve been studying it for a couple years. And I want to share a couple stories with you today. First, about a guy named Nathan Winograd. Nathan was the number two person at the San Francisco SPCA. And what you may not know about the history of the SPCA is, it was founded to kill dogs and cats. Cities gave them a charter to get rid of the stray animals on the street and destroy them. In a typical year four million dogs and cats were killed, most of them within 24 hours of being scooped off of the street. Nathan and his boss saw this, and they could not tolerate it. So they set out to make San Francisco a no-kill city : create an entire city where every dog and cat, unless it was ill or dangerous, would be adopted, not killed. And everyone said it was impossible. Nathan and his boss went to the city council to get a change in the ordinance. And people from SPCAs and humane shelters around the country flew to San Francisco to testify against them — to say it would hurt the movement and it was inhumane. They persisted. And Nathan went directly to the community. He connected with people who cared about this : nonprofessionals, people with passion. And within just a couple years, San Francisco became the first no-kill city, running no deficit, completely supported by the community. Nathan left and went to Tompkins County, New York — a place as different from San Francisco as you can be and still be in the United States. And he did it again. He went from being a glorified dogcatcher to completely transforming the community. And then he went to North Carolina and did it again. And he went to Reno and he did it again. And when I think about what Nathan did, and when I think about what people here do, I think about ideas. And I think about the idea that creating an idea, spreading an idea has a lot behind it. I don ' t know if you ' ve ever been to a Jewish wedding, but what they do is, they take a light bulb and they smash it. Now there is a bunch of reasons for that, and stories about it. But one reason is because it indicates a change, from before to after. It is a moment in time. And I want to argue that we are living through and are right at the key moment of a change in the way ideas are created and spread and implemented. We started with the factory idea : that you could change the whole world if you had an efficient factory that could churn out change. We then went to the TV idea, that said if you had a big enough mouthpiece, if you could get on TV enough times, if you could buy enough ads, you could win. And now we ' re in this new model of leadership, where the way we make change is not by using money or power to lever a system, but by leading. So let me tell you about the three cycles. The first one is the factory cycle. Henry Ford comes up with a really cool idea. It enables him to hire men who used to get paid 50 cents a day and pay them five dollars a day. Because he ' s got an efficient enough factory. Well with that sort of advantage you can churn out a lot of cars. You can make a lot of change. You can get roads built. You can change the fabric of an entire country. That the essence of what you ' re doing is you need ever-cheaper labor, and ever-faster machines. And the problem we ' ve run into is, we ' re running out of both. Ever-cheaper labor and ever- faster machines. So we shift gears for a minute, and say,"I know : television ; advertising. Push push. Take a good idea and push it on the world. I have a better mousetrap. And if I can just get enough money to tell enough people, I ' ll sell enough."And you can build an entire industry on that. If necessary you can put babies in your ads. If necessary you can use babies to sell other stuff. And if babies don ' t work, you can use doctors. But be careful. Because you don ' t want to get an unfortunate juxtaposition, where you ' re talking about one thing instead of the other. This model requires you to act like the king, like the person in the front of the room throwing things to the peons in the back. That you are in charge, and you ' re going to tell people what to do next. The quick little diagram of it is, you ' re up here, and you are pushing it out to the world. This method — mass marketing — requires average ideas, because you ' re going to the masses, and plenty of ads. What we ' ve done as spammers is tried to hypnotize everyone into buying our idea, hypnotize everyone into donating to our cause, hypnotize everyone into voting for our candidate. And, unfortunately, it doesn ' t work so well anymore either. But there is good news around the corner — really good news. I call it the idea of tribes. What tribes are, is a very simple concept that goes back 50, 000 years. It ' s about leading and connecting people and ideas. And it ' s something that people have wanted forever. Lots of people are used to having a spiritual tribe, or a church tribe, having a work tribe, having a community tribe. But now, thanks to the internet, thanks to the explosion of mass media, thanks to a lot of other things that are bubbling through our society around the world, tribes are everywhere. The Internet was supposed to homogenize everyone by connecting us all. Instead what it ' s allowed is silos of interest. So you ' ve got the red-hat ladies over here. You ' ve got the red-hat triathletes over there. You ' ve got the organized armies over here. You ' ve got the disorganized rebels over here. You ' ve got people in white hats making food. And people in white hats sailing boats. The point is that you can find Ukrainian folk dancers and connect with them, because you want to be connected. That people on the fringes can find each other, connect and go somewhere. Every town that has a volunteer fire department understands this way of thinking. Now it turns out this is a legitimate non-photoshopped photo. People I know who are firemen told me that this is not uncommon. And that what firemen do to train sometimes is they take a house that is going to be torn down, and they burn it down instead, and practice putting it out. But they always stop and take a picture. You know the pirate tribe is a fascinating one. They ' ve got their own flag. They ' ve got the eye patches. You can tell when you ' re running into someone in a tribe. And it turns out that it ' s tribes — not money, not factories — that can change our world, that can change politics, that can align large numbers of people. Not because you force them to do something against their will, but because they wanted to connect. That what we do for a living now, all of us, I think, is find something worth changing, and then assemble tribes that assemble tribes that spread the idea and spread the idea. And it becomes something far bigger than ourselves, it becomes a movement. So when Al Gore set out to change the world again, he didn ' t do it by himself. And he didn ' t do it by buying a lot of ads. He did it by creating a movement. Thousands of people around the country who could give his presentation for him, because he can ' t be in 100 or 200 or 500 cities in each night. You don ' t need everyone. What Kevin Kelley has taught us is you just need, I don ' t know, a thousand true fans — a thousand people who care enough that they will get you the next round and the next round and the next round. And that means that the idea you create, the product you create, the movement you create isn ' t for everyone, it ' s not a mass thing. That ' s not what this is about. What it ' s about instead is finding the true believers. It ' s easy to look at what I ' ve said so far, and say,"Wait a minute, I don ' t have what it takes to be that kind of leader."So here are two leaders. They don ' t have a lot in common. They ' re about the same age. But that ' s about it. What they did, though, is each in their own way, created a different way of navigating your way through technology. So some people will go out and get people to be on one team. And some people will get people to be on the other team. It also informs the decisions you make when you make products or services. You know, this is one of my favorite devices. But what a shame that it ' s not organized to help authors create movements. What would happen if, when you ' re using your Kindle, you could see the comments and quotes and notes from all the other people reading the same book as you in that moment. Or from your book group. Or from your friends, or from the circle you want. What would happen if authors, or people with ideas could use version two, which comes out on Monday, and use it to organize people who want to talk about something. Now there is a million things I could share with you about the mechanics here. But let me just try a couple. The Beatles did not invent teenagers. They merely decided to lead them. That most movements, most leadership that we ' re doing is about finding a group that ' s disconnected but already has a yearning — not persuading people to want something they don ' t have yet. When Diane Hatz worked on"The Meatrix,"her video that spread all across the internet about the way farm animals are treated, she didn ' t invent the idea of being a vegan. She didn

’ t invent the idea of caring about this issue. But she helped organize people, and helped turn it into a movement. Hugo Chavez did not invent the disaffected middle and lower class of Venezuela. He merely led them. Bob Marley did not invent Rastafarians. He just stepped up and said,”Follow me.”Derek Sivers invented CD Baby, which allowed independent musicians to have a place to sell their music without selling out to the man — to have place to take the mission they already wanted to go to, and connect with each other. What all these people have in common is that they are heretics. That heretics look at the status quo and say,”This will not stand. I can ’ t abide this status quo. I am willing to stand up and be counted and move things forward. I see what the status quo is ; I don ’ t like it.”That instead of looking at all the little rules and following each one of them, that instead of being what I call a sheepwalker — somebody who ’ s half asleep, following instructions, keeping their head down, fitting in — every once in a while someone stands up and says,”Not me.”Someone stands up and says,”This one is important. We need to organize around it.”And not everyone will. But you don ’ t need everyone. You just need a few people — — who will look at the rules, realize they make no sense, and realize how much they want to be connected. So Tony Hsieh does not run a shoe store. Zappos isn ’ t a shoe store. Zappos is the one, the only, the best-there-ever-was place for people who are into shoes to find each other, to talk about their passion, to connect with people who care more about customer service than making a nickel tomorrow. It can be something as prosaic as shoes, and something as complicated as overthrowing a government. It ’ s exactly the same behavior though. What it requires, as Geraldine Carter has discovered, is to be able to say,”I can ’ t do this by myself. But if I can get other people to join my Climb and Ride, then together we can get something that we all want. We ’ re just waiting for someone to lead us.”Michelle Kaufman has pioneered new ways of thinking about environmental architecture. She doesn ’ t do it by quietly building one house at a time. She does it by telling a story to people who want to hear it. By connecting a tribe of people who are desperate to be connected to each other. By leading a movement and making change. And around and around and around it goes. So three questions I ’ d offer you. The first one is, who exactly are you upsetting? Because if you ’ re not upsetting anyone, you ’ re not changing the status quo. The second question is, who are you connecting? Because for a lot of people, that ’ s what they ’ re in it for : the connections that are being made, one to the other. And the third one is, who are you leading? Because focusing on that part of it — not the mechanics of what you ’ re building, but the who, and the leading part — is where change comes. So Blake, at Tom ’ s Shoes, had a very simple idea.”What would happen if every time someone bought a pair of these shoes I gave exactly the same pair to someone who doesn ’ t even own a pair of shoes?”This is not the story of how you get shelf space at Neiman Marcus. It ’ s a story of a product that tells a story. And as you walk around with this remarkable pair of shoes and someone says,”What are those?”You get to tell the story on Blake ’ s behalf, on behalf of the people who got the shoes. And suddenly it ’ s not one pair of shoes or 100 pairs of shoes. It ’ s tens of thousands of pairs of shoes. My friend Red Maxwell has spent the last 10 years fighting against juvenile diabetes. Not fighting the organization that ’ s fighting it — fighting with them, leading them, connecting them, challenging the status quo because it ’ s important to him. And the people he surrounds himself with need the connection. They need the leadership. It makes a difference. You don ’ t need permission from people to lead them. But in case you do, here it is : they ’ re waiting, we ’ re waiting for you to show us where to go next. So here is what leaders have in common. The first thing is, they challenge the status quo. They challenge what ’ s currently there. The second thing is, they build a culture. A secret language, a seven-second handshake, a way of knowing that you ’ re in or out. They have curiosity. Curiosity about people in the tribe, curiosity about outsiders. They ’ re asking questions. They connect people to one another. Do you know what people want more than anything? They want to be missed. They want to be missed the day they don ’ t show up. They want to be missed when they ’ re gone. And tribe leaders can do that. It ’ s fascinating, because all tribe leaders have charisma, but you don ’ t need charisma to become a leader. Being a leader gives you charisma. If you look and study the leaders who have succeeded, that ’ s where charisma comes from — from the leading. Finally, they commit. They commit to the cause. They commit to the tribe. They commit to the people who are there. So I ’ d like you to do something for me. And I hope you ’ ll think about it before you reject it out-of-hand. What I want you to do, it only takes 24 hours, is : create a movement. Something that matters. Start. Do it. We need it. Thank you very much. I appreciate it.

Text Sample 794: NEG Dim 5 – 2011 – Score -1.00 – 2011/t002541.txt

The story I want to tell you about, started to me, in 1996 when I was studying in England. One day, I was glancing through my bank statement and I ’ ve noticed that in the upper corner there was a symbol, that I ’ m going to show you, saying that document had been made in plain language, for me to understand it. The idea interested me ; I tried to find out what that was and I found out that there was a campaign for the simplification of language, which was the”Plain English Campaign”. I thought it was a fabulous idea, for about one day or two, and I ’ ve never thought about it since. When I came back to Portugal, for good, I came across with several documents - for example, my work contract, the papers I had to sign to buy a house, the electricity bill — a series of documents that reminded me of that bank statement. Not because they were equally clear and simple, but because they were the exact opposite. Because I had to read everything twice or thrice, to begin to understand what was written there. And then, that little seed that had been sown in 1996, started germinating. And one day, I found myself entering through my boss ’ office and quitting my job to dedicate myself to this. And so, what did I find out, right off the start? I found out that it was a much more severe problem than I thought. It wasn ’ t only about these documents being complex and annoying, it was the fact that Portuguese people ’ s literacy — literacy is the hability to understand written documents — is extremely low. I ’ m going to show you a chart about Portuguese people ’ s literacy [rate]. You know that about 10, 11 % of people, in Portugal, don ’ t know how to read nor write at all, yet. Over there are those who know, or say they know, how to read and write. So, what do we have? We have that group of red people in level 1, which is the lowest in literacy [rate]. They are persons who are able to join letters, but they can ’ t, actually, understand. For example, if a person, from that red group, has to pick up the package leaflet of a medicinal product, to give a dose of medicine to his / her child, he / she can ’ t, can ’ t understand the information. 50 % of the Portuguese. Then we have 30 % more, those yellow ones, over there, people who get by on the daily basis. That is, if they don ’ t have to read anything too new or too different, they will manage. But, for example, if they work in a factory and a new machine arrives and they have to read the machine ’ s manual to be able to work with it, they can ’ t do it anymore. And there they go, 80 % of the Portuguese people. Then we have a few more that can handle documents, as long as they are not too complex, and we have 5 % of the population that can handle really complex documents.

Now, just so you don ’ t think this is normal, that over there is Sweden. While we have 20 % of people with the literacy considered essential for a daily basis, Sweden has 75 %. And looking at that, what did I realize? I realized we live in an apartheid of information. I realized that there ’ s a small minority of people who has indeed access to information and can use it to their advantage and a huge majority that can ’ t. And because they can ’ t, they are excluded and they are impaired. Let me give you an example. That one is Mr. Domingos, my building ’ s doorkeeper. Mr. Domingos started to read at the age of 27, so, he falls in that yellow group that we have seen a little while ago. From time to time, I ’ m arriving home and he says :”Miss Sandra!”;”Yes, Mr. Domingos.””Here is a little letter.”So, the deal is : when Mr. Domingos or someone in the family or in the neighbourhood receives something that they don ’ t understand, they come to me and I help them to translate it. And so, that time he said to me like this :”Oh, Miss Sandra, I ’ m about to throw this away, but check it if this is important.”It was very important. He had been waiting, for quite a while, to have a knee surgery and that was the letter of the famous surgery-bank checks. When a person is waiting for a long time, they get a bank check and with that bank check they can have that surgery in the private sector. It was almost thrown away into the garbage, Mr. Domingos ’ letter. In that same year, I ’ ve found out, only 20 % of people had used these surgery-bank checks. The other 80 %, I don ’ t believe they had been cured while they were waiting. They most likely did the same as Mr. Domingos :”Ah! What ’ s this? I ’ m not understanding, it ’ s going to the garbage”. They lost the opportunity to have the surgery they needed. What happens here? When people don ’ t understand this has severe consequences — to the individual, but also for the country. When I don ’ t understand which are my rights, the benefits I can have access to, I can ’ t understand my duties and I ’ m not an active and participative citizen either. Now, maybe, you are sitting there and thinking :”Yeah, poor Mr. Domingos... Bad luck, isn ’ t it? Me?! I ’ m in the green ones.””I ’ m absolutely sure that I ’ m one of the green ones. I was selected to come to TED! Hum?”So the deal is : I have here some texts to read to you and I ’ m going to see what color are you when I finish reading. So, come on, this is an automobile insurance contract. It says this :”Unless contrary stipulation, the decease of the ensured person, the ensured capital is provided, in case of predeceasing of the beneficiary relatively to the ensured person, to the heirs of the last, in case of the simultaneous decease of the ensured person and of the beneficiary, to the heirs of the last”. Hum? Next, there ’ s another one, from the medicines ’ package leaflet, that says the following :”Warning! It may also occur erythema, edema, vesiculation, keratolysis and urticaria.”Hum? Are you clear? And this one is really good. This one, I signed a renting contract to the new office on Friday and I laughed my head off. So, it said :”I, as a cosigner, assume the opportune payment of the rents, waiving the benefits of the division and prior foreclosure.”When I heard”opportune payment”[”storm payment”in Portuguese], I imagined myself bursting in there, slamming the door and saying :”Here ’ s the money!!!”But it ’ s not! But it ’ s not! So, it ’ s this : when we leave our area of expertise — and it only takes a little ; there is no need to go to the string theory or something like that, it only takes a little... What happens? We get as lost as Mr. Domingos. And these documents are not written by experts to experts, as the ones from the string theory. No! These are documents written for me, these are the public documents, the public documents I need to understand in my daily life, to govern myself, to live my life. These are the renting contracts, the package leaflet of the medicines. It ’ s all this. The electricity bills. This has to be clear, so that I can understand. Because, if I can ’ t understand, what happens? I make mistakes, I get wrong. I ’ m going to give you an example of mistakes that were committed, bad decisions made for not being able to understand documents. Do you remember the subprime crisis that took place in the United States? What happened? People signed those loans to buy houses, without truly understanding what they were signing. Because if they knew what they were signing, they ’ d know that as soon as the interest rate started to rise, the monthly payment would also increase, they wouldn ’ t be able to pay anymore and they would end up without a house. So, the rest, after that, next, everything crumbled down and we know the rest. Do you think that if there was a culture of clarity in the financial sector things would have got up to where they ’ ve got? I don ’ t think so. So, how do you solve this problem, this such big difference between the literacy [rate] of the Portuguese, that is down here, and the complexity of the public documents, those ones we need to understand in our daily lives, sorry, that are up here? So, the first think that comes to mind is,”If literacy is down here, let ’ s make it rise”— isn ’ t it? Let ’ s educate people. Let ’ s, let ’ s... of course we will, of course we will educate people. The thing is, it ’ s hard and it ’ s slow. And I don ’ t even want to imagine how many generations it will take until we are at Sweden ’ s level. But, besides that, it ’ s not only because it is slow. There is another problem. If the language of the documents isn ’ t simpler, we ’ ve already seen that people, even with a high literacy level, like you, if the language is hard, they don ’ t get cleared, they continue to be unable to understand these documents. So, besides increasing literacy [rate], and for now, it ’ s much more important to reduce the complexity of documents and simplify the language. I ’ m going to show you an example. When I talk about simplifying the language. Notice! On the left side of a contract :”It is agreed that the insurance company, bla bla bla, bla bla bla, bla bla...”This is a before and after example. What do we mean with simplifying language? It ’ s communicating in a simple and clear way, enabling our reader to understand it at first glance. What do you prefer? Before or after? There isn ’ t much doubt, is there? So, how is this achieved? How can you make the State and companies communicate with citizens in a language they can understand at first glance? There are several ways. There are countries that are going by the path of legislation. For example, Sweden and the United States introduced, last year, a legislation that forces the State to communicate with people in a language they can understand. And you think,”Well, that ’ s normal. They are countries that are a bit ahead.”Sweden, maybe, much more than the United States.”But they are countries that are a bit ahead.”It would be nice if in our country there was also legislation towards that, wouldn ’ t it be?”So it would. And there is! Since 1999. The Law of the Administrative Modernization says that the communication between the State and the people should be simple, clear, concise, meaningful, without acronyms, bla bla bla... But the thing is : it is not applied. So, the question, the path of going through legislation, enforcing through legislation, works in those countries where laws are made to be applied. Now, there is another way, the way of marketing. How does that work? It ’ s like this : private companies change their language ; communicate in a clearer and simpler way, they make a big fuss about it, consumers love it, the sales rise, it works beautifully. But it works for the private sector. Now, what is the third path, and to me, the most important one? It ’ s the path of civil movements, that are based in a mentality shift. In countries where this really went forward — do you remember the English label, of the”Plain English Campaign”? — this is all based on a consumers ’ movement. So, what does it take for that civic movement to happen? We need to understand two very important things : First, wanting to understand these public documents is not a whim, is not an intellectual curiosity. It is a necessity that I have in my daily life. And above all, it is a right, it is everyone ’ s right. On this side we have : understanding is a right. And we have to understand another thing too, which is : he who writes, has to write in order to be understood. How do we get there? First of all : we have to become demanding consumers and citizens. Think it this way : next time someone gives you a document that you just simply don ’ t understand, don ’ t be shy, don ’ t keep quiet, pretending that you are understanding everything. No! Demand to understand. Ask. I know that this is not easy. Turn to the gentleman in the little suite and say :”Look here, this contract, this part, what does this mean?”It ’ s not easy, and perhaps it ’ s even a bit embarrassing, isn ’ t it? But it is not. It ’ s a sign of intelligence. What do you teach your children when they have doubts in school?”Don ’ t say nothing to anyone, shut up really good and act smart.”No, you don ’ t, right? You say :”Look, when you don ’ t understand, you put your hand up in the air and ask the teacher until you are cleared.”And that is exactly what we have to do, as consumers and citizens. So, next time you come across with a document you don ’ t understand, demand to understand, put aside your pride and ask until you get completely cleared. Then, there ’ s the other side, which is :”Okay, these awful documents”which are in circulation, they don ’ t grow on trees.”Someone wrote them.”Hum? Maybe, among this big group, some persons... I don ’ t know, some lawyers, some public employees — I ’ m not asking you to identify yourselves, but I ask you to search your own conscience. How many times do you write documents for the general public, for people that don ’ t share your language, and you write them in a language that only you will understand? And you ’ ll say :”Ah, that ’ s because there is a reason, here!”Dear friends, I already heard all the reasons. There are thousands of reasons. They go like these :”Ah, it ’ s the house ’ s culture”;”Oh, my boss!””Oh, the judge! And what if this goes to court?””What you want is to destroy language. We have to educate people.””We can ’ t lower the level”; bla, bla, bla... I ’ ve seen all of this. These are excuses. A while ago, they told us about Einstein. Einstein said this :”If you can ’ t write about a **subject**, in a simple way, it ’ s because, in fact, you don ’ t understand it.”So, if you... To Einstein... To Einstein... So — ah, ah, I don ’ t have time! — so, if you do know what you want to say — don ’ t you? — you are aware of what you want to say, you just have to do one thing : believe that it is possible to write in a simple way. And how do you do that? It ’ s very easy. You write to your grandmother. Hum? You write to your grandmother. I ’ m going to show grandma to you. You write with respect and without paternalism. And you use three techniques that I ’ m going to teach you. First of all : Start by the most important. Grandma has a lot to do. She is not going to read three pages to get to the main idea. Start by the most important. Next, use short sentences, because grandma, like anyone of us, if you make very long sentences, she gets to the end and she can ’ t remember what you said in the beginning, anymore. And lastly, the third, use plain words, those ones that grandma knows. Alright? It ’ s easy! Before I leave, I would like to talk to you about”Clara”.”Clara”is a project of social responsibility. It has this mission : to change the way how public communication is made. What do we do? We are going to launch this year a collection of”Claro”guides, that is, we are going to pick very, very, very complex **subjects** and put them simple. We are going to start with the”Claro”Guide of Justice. I think it is an area... that comes in handy. And now I ’ m going to follow Manuel [Forjaz] ’ s advise. If it ’ s time to ask, whoever wants to sponsor the”Claro”Guide of Justice — I don ’ t know... for example, the EDP Foundation, or... ha? — come talk to me later. Another thing we will do is to grant prizes to the worse and best documents, because, indeed, there are people who are working to communicate in a clearer way, and that have to be rewarded, and there are lazy people that do nothing about it and need to be humiliated. So, we ’ ll give prizes to the worst and to the best. I ’ m counting on your help. We are going to launch... we are going to launch the campaign via Clara ’ s Facebook, so, send your friendship request and you ’ ll be up to date. But, lastly, above all, what does”Clara”want? It wants to put two little things in your heads. The first one : demand to understand. Don ’ t be shy. And write in order to be understood. Write to your grandma. And if you don ’ t have a grandmother, write to Mr. Domingos, he will like it. Thank you.

Text Sample 795: NEG Dim 5 – 2011 – Score -1.00 – 2011/t002473.txt
There ’ s a poem written by a very famous English poet at the end of the 19th century. It was said to echo in Churchill ’ s brain in the 1930s. And the poem goes :”On the idle hill of summer, lazy with the flow of streams, hark I hear a distant drummer, drumming like a sound in dreams, far and near and low and louder on the roads of earth go by, dear to friend and food to powder, soldiers marching, soon to die.”Those who are interested in poetry, the poem is”A Shropshire Lad”written by A. E. Housman. But what Housman understood, and you hear it in the symphonies of Nielsen too, was that the long, hot, silvan summers of stability of the 19th century were coming to a close, and that we were about to move into one of those terrifying periods of history when power changes. And these are always periods, ladies and gentlemen, accompanied by turbulence, and all too often by blood. And my message for you is that I believe we are condemned, if you like, to live at just one of those moments in history when the gimbals upon which the established order of power is beginning to change and the new look of the world, the new powers that exist in the world, are beginning to take form. And these are — and we see it very clearly today — nearly always highly turbulent times, highly difficult times, and all too often very bloody times. By the way, it happens about once every century. You might argue that the last time it happened — and that ’ s what Housman felt coming and what Churchill felt too — was that when power passed from the old nations, the old powers of Europe, across the Atlantic to the new emerging power of the United States of America — the beginning of the American century. And of course, into the vacuum where the too-old European powers used to be were played the two bloody catastrophes of the last century — the one in the first part and the one in the second part : the two great World Wars. Mao Zedong used to refer to them as the European civil wars, and it ’ s probably a more accurate way of describing them. Well, ladies and gentlemen, we live at one of those times. But for us, I want to talk about three factors today. And the first of these, the first two of these, is about a shift in power. And the second is about some new dimension which I want to refer to, which has never quite happened in the way it ’ s happening now. But let ’ s talk about the shifts of power that are occurring to the world. And what is happening today is, in one sense, frightening because it ’ s never happened before. We have seen lateral shifts of power — the power of Greece passed to Rome and the power shifts that occurred during the European civilizations — but we are seeing something slightly different. For power is not just moving laterally from nation to nation. It ’ s also moving vertically. What ’ s happening today is that the power that was encased, held to accountability, held to the rule of law, within the institution of the nation state has now migrated in very large measure onto the global stage. The globalization of power — we talk about the globalization of markets, but actually it ’ s the globalization of real power. And where, at the nation state level that power is held to accountability **subject** to the rule of law, on the international stage it is not. The international stage and the global stage where power now resides :

the power of the Internet, the power of the satellite broadcasters, the power of the money changers — this vast money-go-round that circulates now 32 times the amount of money necessary for the trade it ' s supposed to be there to finance — the money changers, if you like, the financial speculators that have brought us all to our knees quite recently, the power of the multinational corporations now developing budgets often bigger than medium-sized countries. These live in a global space which is largely unregulated, not **subject** to the rule of law, and in which people may act free of constraint. Now that suits the powerful up to a moment. It ' s always suitable for those who have the most power to operate in spaces without constraint, but the lesson of history is that, sooner or later, unregulated space — space not **subject** to the rule of law — becomes populated, not just by the things you wanted — international trade, the Internet, etc. — but also by the things you don ' t want — international criminality, international terrorism. The revelation of 9 / 11 is that even if you are the most powerful nation on earth, nevertheless, those who inhabit that space can attack you even in your most iconic of cities one bright September morning. It ' s said that something like 60 percent of the four million dollars that was taken to fund 9 / 11 actually passed through the institutions of the Twin Towers which 9 / 11 destroyed. You see, our enemies also use this space — the space of mass travel, the Internet, satellite broadcasters — to be able to get around their poison, which is about destroying our systems and our ways. Sooner or later, sooner or later, the rule of history is that where power goes governance must follow. And if it is therefore the case, as I believe it is, that one of the phenomenon of our time is the globalization of power, then it follows that one of the challenges of our time is to bring governance to the global space. And I believe that the decades ahead of us now will be to a greater or lesser extent turbulent the more or less we are able to achieve that aim : to bring governance to the global space. Now notice, I ' m not talking about government. I ' m not talking about setting up some global democratic institution. My own view, by the way, ladies and gentlemen, is that this is unlikely to be done by spawning more U. N. institutions. If we didn ' t have the U. N., we ' d have to invent it. The world needs an international forum. It needs a means by which you can legitimize international action. But when it comes to governance of the global space, my guess is this won ' t happen through the creation of more U. N. institutions. It will actually happen by the powerful coming together and making treaty-based systems, treaty-based agreements, to govern that global space. And if you look, you can see them happening, already beginning to emerge. The World Trade Organization : treaty-based organization, entirely treaty-based, and yet, powerful enough to hold even the most powerful, the United States, to account if necessary. Kyoto : the beginnings of struggling to create a treaty- based organization. The G20 : we know now that we have to put together an institution which is capable of bringing governance to that financial space for financial speculation. And that ' s what the G20 is, a treaty-based institution. Now there ' s a problem there, and we ' ll come back to it in a minute, which is that if you bring the most powerful together to make the rules in treaty-based institutions, to fill that governance space, then what happens to the weak who are left out? And that ' s a big problem, and we ' ll return to it in just a second. So there ' s my first message, that if you are to pass through these turbulent times more or less turbulently, then our success in doing that will in large measure depend on our capacity to bring sensible governance to the global space. And watch that beginning to happen. My second point is, and I know I don ' t have to talk to an audience like this about such a thing, but power is not just shifting vertically, it ' s also shifting horizontally. You might argue that the story, the history of civilizations, has been civilizations gathered around seas — with the first ones around the Mediterranean, the more recent ones in the ascendents of Western power around the Atlantic. Well it seems to me that we ’ re now seeing a fundamental shift of power, broadly speaking, away from nations gathered around the Atlantic [seaboard] to the nations gathered around the Pacific rim. Now that begins with economic power, but that ’ s the way it always begins. You already begin to see the development of foreign policies, the augmentation of military budgets occurring in the other growing powers in the world. I think actually this is not so much a shift from the West to the East ; something different is happening. My guess is, for what it ’ s worth, is that the United States will remain the most powerful nation on earth for the next 10 years, 15, but the context in which she holds her power has now radically altered ; it has radically changed. We are coming out of 50 years, most unusual years, of history in which we have had a totally mono-polar world, in which every compass needle for or against has to be referenced by its position to Washington — a world bestrode by a single colossus. But that ’ s not a usual case in history. In fact, what ’ s now emerging is the much more normal case of history. You ’ re beginning to see the emergence of a multi-polar world. Up until now, the United States has been the dominant feature of our world. They will remain the most powerful nation, but they will be the most powerful nation in an increasingly multi-polar world. And you begin to see the alternative centers of power building up — in China, of course, though my own guess is that China ’ s ascent to greatness is not smooth. It ’ s going to be quite grumpy as China begins to democratize her society after liberalizing her economy. But that ’ s a **subject** of a different discussion. You see India, you see Brazil. You see increasingly that the world now looks actually, for us Europeans, much more like Europe in the 19th century. Europe in the 19th century : a great British foreign secretary, Lord Canning, used to describe it as the”European concert of powers.”There was a balance, a five-sided balance. Britain always played to the balance. If Paris got together with Berlin, Britain got together with Vienna and Rome to provide a counterbalance. Now notice, in a period which is dominated by a mono-polar world, you have fixed alliances — NATO, the Warsaw Pact. A fixed polarity of power means fixed alliances. But a multiple polarity of power means shifting and changing alliances. And that ’ s the world we ’ re coming into, in which we will increasingly see that our alliances are not fixed. Canning, the great British foreign secretary once said,”Britain has a common interest, but no common allies.”And we will see increasingly that even we in the West will reach out, have to reach out, beyond the cozy circle of the Atlantic powers to make alliances with others if we want to get things done in the world. Note, that when we went into Libya, it was not good enough for the West to do it alone ; we had to bring others in. We had to bring, in this case, the Arab League in. My guess is Iraq and Afghanistan are the last times when the West has tried to do it themselves, and we haven ’ t succeeded. My guess is that we ’ re reaching the beginning of the end of 400 years — I say 400 years because it ’ s the end of the Ottoman Empire — of the hegemony of Western power, Western institutions and Western values. You know, up until now, if the West got its act together, it could propose and dispose in every corner of the world. But that ’ s no longer true. Take the last financial crisis after the Second World War. The West got together — the Bretton Woods Institution, World Bank, International Monetary Fund — the problem solved. Now we have to call in others. Now we have to create the G20. Now we have to reach beyond the cozy circle of our Western friends. Let me make a prediction for you, which is probably even more startling. I suspect we are now reaching the end of 400 years when Western power was enough. People say to me,”The Chinese, of course, they ’ ll never get themselves involved in peace-making, multilateral peace-making around the world.”Oh yes? Why not? How many Chinese troops are serving under the blue beret, serving under the blue flag, serving under the U. N. command in the world today? 3, 700. How many Americans? 11. What is the largest naval contingent tackling the issue of Somali pirates? The Chinese naval contingent. Of course they are, they are a mercantilist nation. They want to keep the sea lanes open. Increasingly, we are going to have to do business with people with whom we do not share values, but with whom, for the moment, we share common interests. It ’ s a whole new different way of looking at the world that is now emerging. And here ’ s the third factor, which is totally different. Today in our modern world, because of the Internet, because of the kinds of things people have been talking about here, everything is connected to everything. We are now interdependent. We are now interlocked, as nations, as individuals, in a way which has never been the case before, never been the case before. The interrelationship of nations, well it ’ s always existed. Diplomacy is about managing the interrelationship of nations. But now we are intimately locked together. You get swine flu in Mexico, it ’ s a problem for Charles de Gaulle Airport 24 hours later. Lehman Brothers goes down, the whole lot collapses. There are fires in the steppes of Russia, food riots in Africa. We are all now deeply, deeply, deeply interconnected. And what that means is the idea of a nation state acting alone, not connected with others, not working with others, is no longer a viable proposition. Because the actions of a nation state are neither confined to itself, nor is it sufficient for the nation state itself to control its own territory, because the effects outside the nation state are now beginning to affect what happens inside them. I was a young soldier in the last of the small empire wars of Britain. At that time, the defense of my country was about one thing and one thing only : how strong was our army, how strong was our air force, how strong was our navy and how strong were our allies. That was when the enemy was outside the walls. Now the enemy is inside the walls. Now if I want to talk about the defense of my country, I have to speak to the Minister of Health because pandemic disease is a threat to my security, I have to speak to the Minister of Agriculture because food security is a threat to my security, I have to speak to the Minister of Industry because the fragility of our hi-tech infrastructure is now a point of attack for our enemies — as we see from cyber warfare — I have to speak to the Minister of Home Affairs because who has entered my country, who lives in that terraced house in that inner city has a direct effect on what happens in my country — as we in London saw in the 7 / 7 bombings. It ’ s no longer the case that the security of a country is simply a matter for its soldiers and its ministry of defense. It ’ s its capacity to lock together its institutions. And this tells you something very important. It tells you that, in fact, our governments, vertically constructed, constructed on the economic model of the Industrial Revolution — vertical hierarchy, specialization of tasks, command structures — have got the wrong structures completely. You in business know that the paradigm structure of our time, ladies and gentlemen, is the network. It ’ s your capacity to network that matters, both within your governments and externally. So here is Ashdown ’ s third law. By the way, don ’ t ask me about Ashdown ’ s first law and second law because I haven ’ t invented those yet ; it always sounds better if there ’ s a third law, doesn ’ t it? Ashdown ’ s third law is that in the modern age, where everything is connected to everything, the most important thing about what you can do is what you can do with others. The most important bit about your structure — whether you ’ re a government, whether you ’ re an army regiment, whether you ’ re a business — is your docking points, your interconnectors, your capacity to network with others. You understand that in industry ; governments don ’ t. But now one final thing. If it is the case, ladies and gentlemen — and it is — that we are now locked together in a way that has never been quite the same before, then it ’ s also the case that we share a destiny with each other. Suddenly and for the very first time, collective defense, the thing that has dominated us as the concept of securing our nations, is no longer enough. It used to be the case that if my tribe was more powerful than their tribe, I was safe ; if my country was more powerful than their country, I was safe ; my alliance, like NATO, was more powerful than their alliance, I was safe. It is no longer the case. The advent of the interconnectedness and of the weapons of mass destruction means that, increasingly, I share a destiny with my enemy. When I was a diplomat negotiating the disarmament treaties with the Soviet Union in Geneva in the 1970s, we succeeded because we understood we shared a destiny with them. Collective security is not enough. Peace has come to Northern Ireland because both sides realized that the zero-sum game couldn ’ t work. They shared a destiny with their enemies. One of the great barriers to peace in the Middle East is that both sides, both Israel and, I think, the Palestinians, do not understand that they share a collective destiny. And so suddenly, ladies and gentlemen, what has been the proposition of visionaries and poets down the ages becomes something we have to take seriously as a matter of public policy. I started with a poem, I ’ ll end with one. The great poem of John Donne ’ s.”Send not for whom the bell tolls.”The poem is called”No Man is an Island.”And it goes :”Every man ’ s death affected me, for I am involved in mankind, send not to ask for whom the bell tolls, it tolls for thee.”For John Donne, a recommendation of morality. For us, I think, part of the equation for our survival. Thank you very much.

Text Sample 796: NEG Dim 5 – 2011 – Score -1.00 – 2011/t002558.txt

So the type of magic I like, and I ’ m a magician, is magic that uses technology to create illusions. So I would like to show you something I ’ ve been working on. It ’ s an application that I think will be useful for artists â multimedia artists in particular. It synchronizes videos across multiple screens of mobile devices. I borrowed these three iPods from people here in the audience to show you what I mean. And I ’ m going to use them to tell you a little bit about my favorite **subject** : deception. One of my favorite magicians is Karl Germain. He had this wonderful trick where a rosebush would bloom right in front of your eyes. But it was his production of a butterfly that was the most beautiful. Announcer : Ladies and gentlemen, the creation of life. Marco Tempest : When asked about deception, he said this : Announcer : Magic is the only honest profession. A magician promises to deceive you â and he does. MT : I like to think of myself as an honest magician. I use a lot of tricks, which means that sometimes I have to lie to you. Now I feel bad about that. But people lie every day. Hold on. Phone : Hey, where are you? MT : Stuck in traffic. I ’ ll be there soon. You ’ ve all done it. Right : I ’ ll be ready in just a minute, darling. Center : It ’ s just what I ’ ve always wanted. Left : You were great. MT : Deception, it ’ s a fundamental part of life. Now polls show that men tell twice as many lies as women â assuming the women they asked told the truth. We deceive to gain advantage and to hide our weaknesses. The Chinese general Sun Tzu said that all war was based on deception. Oscar Wilde said the same thing of romance. Some people deceive for money. Let ’ s play a game. Three cards, three chances. Announcer : One five will get you 10, 10 will get you 20. Now, where ’ s the lady? Where is the queen? MT : This one? Sorry. You lose. Well, I didn ’ t deceive you. You deceived yourself. Self-deception. That ’ s when we convince ourselves that a lie is the truth. Sometimes it ’ s hard to tell the two apart. Compulsive gamblers are experts at self-deception. They believe they can win. They forget the times they lose. The brain is very good at forgetting. Bad experiences are quickly forgotten. Bad experiences quickly disappear. Which is why in this vast and lonely cosmos, we are so wonderfully optimistic. Our self-deception becomes a positive illusion â why movies are able to take us onto extraordinary adventures ; why we believe Romeo when he says he loves Juliet ; and why single notes of music, when played together, become a sonata and conjure up meaning. That ’ s”Clair De lune.”Its composer, called Debussy, said that art was the greatest deception of all. Art is a deception that creates real emotions â a lie that creates a truth. And when you give yourself over to that deception, it becomes magic. [MAGIC] Thank you. Thank you very much.

Text Sample 797: NEG Dim 5 – 2010 – Score -1.00 – 2010/t002838.txt

The greatest irony in global health is that the poorest countries carry the largest disease burden. If we resize the countries of the globe in proportion to the **subject** of interest, we see that Sub-Saharan Africa is the worst hit region by HIV / AIDS. This is the most devastating epidemic of our time. We also see that this region has the least capability in terms of dealing with the disease. There are very few doctors and, quite frankly, these countries do not have the resources that are needed to cope with such epidemics. So what the Western countries, developed countries, have generously done is they have proposed to provide free drugs to all people in Third World countries who actually can ’ t afford these medications. And this has already saved millions of lives, and it has prevented entire economies from capsizing in Sub- Saharan Africa. But there is a fundamental problem that is killing the efforts in fighting this disease, because if you keep throwing drugs out at people who don ’ t have diagnostic services, you end up creating a problem of drug resistance. This is already beginning to happen in Sub-Saharan Africa. The problem is that, what begins as a tragedy in the Third World could easily become a global problem. And the last thing we want to see is drug-resistant strains of HIV popping up all over the world, because it will make treatment more expensive and it could also restore the pre-ARV carnage of HIV / AIDS. I experienced this firsthand as a high school student in Uganda. This was in the 90s during the peak of the HIV epidemic, before there were any ARVs in Sub- Saharan Africa. And during that time, I actually lost more relatives, as well as the teachers who taught me, to HIV / AIDS. So this became one of the driving passions of my life, to help find real solutions that could address these kinds of problems. We all know about the miracle of miniaturization. Back in the day, computers used to fill this entire room, and people actually used to work inside the computers. But what electronic miniaturization has done is that it has allowed people to shrink technology into a cell phone. And I ’ m sure everyone here enjoys cell phones that can actually be used in the remote areas of the world, in the Third World countries. The good news is that the same technology that allowed miniaturization of electronics is now allowing us to miniaturize biological laboratories. So, right now, we can actually miniaturize biological and chemistry laboratories onto microfluidic chips. I was very lucky to come to the US right after high school, and was able to work on this technology and develop some devices. This is a microfluidic chip that I developed. A close look at how the technology works : These are channels that are about the size of a human hair — so you have integrated valves, pumps, mixers and injectors — so you can fit entire diagnostic experiments onto a microfluidic system. So what I plan to do with this technology is to actually take the current state of the technology and build an HIV kit in a microfluidic system. So, with one microfluidic chip, which is the size of an iPhone, you can actually diagnose 100 patients at the same time. For each patient, we will be able to do up to 100 different viral loads per patient. And this is only done in four hours, 50 times faster than the current state of the art, at a cost that will be five to 500 times cheaper than the current options. So this will allow us to create personalized medicines in the Third World at a cost that is actually achievable and make the world a safer place. I invite your interest as well as your involvement in driving this vision to a point of practical reality. Thank you very much.

Text Sample 798: NEG Dim 5 – 2010 – Score -1.00 – 2010/t002772.txt
So I work in marketing, which I love, but my first passion was physics, a passion brought to me by a wonderful school teacher, when I had a little less gray hair. So he taught me that physics is cool because it teaches us so much about the world around us. And I ’ m going to spend the next few minutes trying to convince you that physics can teach us something about marketing. So quick show of hands — who studied some marketing in university? Who studied some physics in university? Pretty good. And at school? Okay, lots of you. So, hopefully this will bring back some happy, or possibly some slightly disturbing memories. So, physics and marketing. We ’ ll start with something very simple — Newton ’ s Law :”The force equals mass times acceleration.”This is something that perhaps Turkish Airlines should have studied a bit more carefully before they ran this campaign. But if we rearrange this formula quickly, we can get to acceleration equals force over mass, which means that for a larger particle — a larger mass — it requires more force to change its direction. It ’ s the same with brands : the more massive a brand, the more baggage it has, the more force is needed to change its positioning. And that ’ s one of the reasons why Arthur Andersen chose to launch Accenture rather than try to persuade the world that Andersen ’ s could stand for something other than accountancy. It explains why Hoover found it very difficult to persuade the world that it was more than vacuum cleaners, and why companies like Unilever and P&G keep brands separate, like Ariel and Pringles and Dove rather than having one giant parent brand. So the physics is that the bigger the mass of an object the more force is needed to change its direction. The marketing is, the bigger a brand, the more difficult it is to reposition it. So think about a portfolio of brands or maybe new brands for new ventures. Now, who remembers Heisenberg ’ s uncertainty principle? Getting a little more technical now. So this says that it ’ s impossible, by definition, to measure exactly the state — i. e., the position — and the momentum of a particle, because the act of measuring it, by definition, changes it. So to explain that — if you ’ ve got an elementary particle and you shine a light on it, then the photon of light has momentum, which knocks the particle, so you don ’ t know where it was before you looked at it. By measuring it, the act of measurement changes it. The act of observation changes it. It ’ s the same in marketing. So with the act of observing consumers, changes their behavior. Think about the group of moms who are talking about their wonderful children in a focus group, and almost none of them buy lots of junk food. And yet, McDonald ’ s sells hundreds of millions of burgers every year. Think about the people who are on accompanied shops in supermarkets, who stuff their trolleys full of fresh green vegetables and fruit, but don ’ t shop like that any other day. And if you think about the number of people who claim in surveys to regularly look for porn on the Web, it ’ s very few. Yet, at Google, we know

it ’ s the number-one searched for category. So luckily, the science — no, sorry — the marketing is getting easier. Luckily, with now better point-of-sale tracking, more digital media consumption, you can measure more what consumers actually do, rather than what they say they do. So the physics is you can never accurately and exactly measure a particle, because the observation changes it. The marketing is — the message for marketing is — that try to measure what consumers actually do, rather than what they say they ’ ll do or anticipate they ’ ll do. So next, the scientific method — an axiom of physics, of all science — says you cannot prove a hypothesis through observation, you can only disprove it. What this means is you can gather more and more data around a hypothesis or a positioning, and it will strengthen it, but it will not conclusively prove it. And only one contrary data point can **blow** your theory out of the water. So if we take an example — Ptolemy had dozens of data points to support his theory that the planets would rotate around the Earth. It only took one robust observation from Copernicus to **blow** that idea out of the water. And there are parallels for marketing — you can invest for a long time in a brand, but a single contrary observation of that positioning will destroy consumers ’ belief. Take BP — they spent millions of pounds over many years building up its credentials as an environmentally friendly brand, but then one little accident. Think about Toyota. It was, for a long time, revered as the most reliable of cars, and then they had the big recall incident. And Tiger Woods, for a long time, the perfect brand ambassador. Well, you know the story. So the physics is that you cannot prove a hypothesis, but it ’ s easy to disprove it — any hypothesis is shaky. And the marketing is that not matter how much you ’ ve invested in your brand, one bad week can undermine decades of good work. So be really careful to try and avoid the screw-ups that can undermine your brand. And lastly, to the slightly obscure world of entropy — the second law of thermodynamics. This says that entropy, which is a measure of the disorder of a system, will always increase. The same is true of marketing. If we go back 20 years, the one message pretty much controlled by one marketing manager could pretty much define a brand. But where we are today, things have changed. You can get a strong brand image or a message and put it out there like the Conservative Party did earlier this year with their election poster. But then you lose control of it. With the kind of digital comment creation and distribution tools that are available now to every consumer, it ’ s impossible to control where it goes. Your brand starts being dispersed, it gets more chaotic. It ’ s out of your control. I actually saw him speak — he did a good job. But while this may be unsettling for marketers, it ’ s actually a good thing. This distribution of brand energy gets your brand closer to the people, more in with the people. It makes this distribution of energy a democratizing force, which is ultimately good for your brand. So, the lesson from physics is that entropy will always increase ; it ’ s a fundamental law. The message for marketing is that your brand is more dispersed. You can ’ t fight it, so embrace it and find a way to work with it. So to close, my teacher, Mr. Vutter, told me that physics is cool, and hopefully, I ’ ve convinced you that physics can teach all of us, even in the world of marketing, something special. Thank you.

Text Sample 799: NEG Dim 5 – 2010 – Score -1.00 – 2010/t002669.txt

Some of the greatest innovations and developments in the world often happen at the intersection of two fields. So tonight I ’ d like to tell you about the intersection that I ’ m most excited about at this very moment, which is entertainment and robotics. So if we ’ re trying to make robots that can be more expressive and that can connect better with us in society, maybe we should look to some of the human professionals of artificial emotion and personality that occur in the dramatic arts. I ’ m also interested in creating new technologies for the arts and to attract people to science and technology. Some people in the last decade or two have started creating artwork with technology. With my new venture, Marilyn Monrobot, I would like to use art to create tech. So we ’ re based in New York City. And if you ’ re a performer that wants to collaborate with an adorable robot, or if you have a robot that needs entertainment representation, please contact me, the Bot-Agent. The bot, our rising celebrity, also has his own Twitter account : @robotinthewild. I ’ d like to introduce you to one of our first robots, Data. He ’ s named after the Star Trek character. I think he ’ s going to be super popular. We ’ ve got the robot — in his head is a database of a lot of jokes. Now each of these jokes is labeled with certain attributes. So it knows something about the **subject** ; it knows about the length. It knows how much it ’ s moving. And so it ’ s going to try to watch your response. I actually have no idea what my robot is going to do today. It can also learn from you about the quality of its jokes and cater things, sort of like Netflix-style, over longer-term to different communities or audiences, children versus adults, different cultures. You can learn something from the robot about the community that you ’ re in. And also I can use each one of you as the acting coach to our future robot companions. Some of you in this middle section — you have red / green paddles. If you like what ’ s going on, show the green. If you don ’ t like the **subject** or the performance, you can hold the red. Now don ’ t be shy. It ’ s just a robot. It doesn ’ t have feelings... yet. And the rest of you, you still count, you still matter. There ’ s also a microphone that ’ s listening to the aggregate laughter and applause and booing — I hope not — to help make some of its next decisions. Right, so, let the robot stand-up comedy begin. Data : Hello TEDWomen. It ’ s an honor to be here. You guys are looking good out there. Ready for some jokes? Audience : Yeah. Data : Here ’ s the first one. Right, so, a doctor says to his patient,”I have bad news and worse news. The bad news is that you only have 24 hours to live.””That ’ s terrible,”said the patient.”How can the news possibly be worse?””I ’ ve been trying to contact you since yesterday.”The Swiss have an interesting army. Five hundred years without a war. Heather Knight : He ’ s talking about the Swiss. Data : Pretty lucky for them. Ever see that little Swiss Army knife they have to fight with?”Come on buddy, I have the toe clippers right here. You get past me, the guy behind me has a spoon.”HK : He ’ s a French robot. Data : A couple of New Jersey hunters are out in the woods. One of them falls to the ground. He does not seem to be breathing. The other guy whips out his cell phone and calls 911. He gasps to the operator,”My friend is dead. What can I do?”The operator says,”Just take it easy. I can help. First, let ’ s make sure he ’ s dead.”There is a silence, and the operator hears a shot. The guy ’ s voice comes back on the line,”Okay, now what?”Question : Why is television called a medium? Anyone? Because it ’ s neither rare nor well done. But to be completely honest with you, I kind of love television. Any of you like television? Audience : Yes. Data : I find it incredibly educational. Actually, as soon as someone turns it on, I go into the other room and read. That ’ s all for now. Was that okay for my first time? You ’ ve been a great audience. Thank you. HK : Yay. So this is actually the first time we ’ ve ever done live audience feedback to a performance. So thank you all for being a part of it. There ’ s a lot more to come. And we hope to learn a lot about robot expression. Thank you very much.

Text Sample 800: NEG Dim 5 – 2007 – Score -1.00 – 2007/t000344.txt

I am very, very happy to be amidst some of the most â the lights are really disturbing my eyes and they ’ re reflecting on my glasses. I am very happy and honored to be amidst very, very innovative and intelligent people. I have listened to the three previous speakers, and guess what happened? Every single thing I planned to say, they have said it here, and it looks and sounds like I have nothing else to say. But there is a saying in my culture that if a bud leaves a tree without saying something, that bud is a young one. So, I will â since I am not young and am very old, I still will say something. We are hosting this conference at a very opportune moment, because another conference is taking place in Berlin. It is the G8 Summit. The G8 Summit proposes that the solution to Africa ’ s problems should be a massive increase in aid, something akin to the Marshall Plan. Unfortunately, I personally do not believe in the Marshall Plan. One, because the benefits of the Marshall Plan have been overstated. Its largest recipients were Germany and France, and it was only 2. 5 percent of their GDP. An average African country receives foreign aid to the tune of 13, 15 percent of its GDP, and that is an unprecedented transfer of financial resources from rich countries to poor countries. But I want to say that there are two things we need to connect. How the media covers Africa in the West, and the consequences of that. By displaying despair, helplessness and hopelessness, the media is telling the truth about Africa, and nothing but the truth. However, the media is not telling us the whole truth. Because despair, civil war, hunger and famine, although they ’ re part and parcel of our African reality, they are not the only reality. And secondly, they are the smallest reality. Africa has 53 nations. We have civil wars only in six countries, which means that the media are covering only six countries. Africa has immense opportunities that never navigate through the web of despair and helplessness that the Western media largely presents to its audience. But the effect of that presentation is, it appeals to sympathy. It appeals to pity. It appeals to something called charity. And, as a consequence, the Western view of Africa ’ s economic dilemma is framed wrongly. The wrong framing is a product of thinking that Africa is a place of despair. What should we do with it? We should give food to the hungry. We should deliver medicines to those who are ill. We should send peacekeeping troops to serve those who are facing a civil war. And in the process, Africa has been stripped of self-initiative. I want to say that it is important to recognize that Africa has fundamental weaknesses. But equally, it has opportunities and a lot of potential. We need to reframe the challenge that is facing Africa, from a challenge of despair, which is called poverty reduction, to a challenge of hope. We frame it as a challenge of hope, and that is worth creation. The challenge facing all those who are interested in Africa is not the challenge of reducing poverty. It should be a challenge of creating wealth. Once we change those two things â if you say the Africans are poor and they need poverty reduction, you have the international cartel of good intentions moving onto the continent, with what? Medicines for the poor, food relief for those who are hungry, and peacekeepers for those who are facing civil war. And in the process, none of these things really are productive because you are treating the symptoms, not the causes of Africa ’ s fundamental problems. Sending somebody to school and giving them medicines, ladies and gentlemen, does not create wealth for them. Wealth is a function of income, and income comes from you finding a profitable trading opportunity or a well-paying job. Now, once we begin to talk about wealth creation in Africa, our second challenge will be, who are the wealth-creating agents in any society? They are entrepreneurs. [Unclear] told us they are always about four percent of the population, but 16 percent are imitators. But they also succeed at the job of entrepreneurship. So, where should we be putting the money? We need to put money where it can productively grow. Support private investment in Africa, both domestic and foreign. Support research institutions, because knowledge is an important part of wealth creation. But what is the international aid community doing with Africa today? They are throwing large sums of money for primary health, for primary education, for food relief. The entire continent has been turned into a place of despair, in need of charity. Ladies and gentlemen, can any one of you tell me a neighbor, a friend, a relative that you know, who became rich by receiving charity? By holding the begging bowl and receiving alms? Does any one of you in the audience have that person? Does any one of you know a country that developed because of the generosity and kindness of another? Well, since I ’ m not seeing the hand, it appears that what I ’ m stating is true. Andrew Mwenda : I can see Bono says he knows the country. Which country is that? AM : Thank you very much. But let me tell you this. External actors can only present to you an opportunity. The ability to utilize that opportunity and turn it into an advantage depends on your internal capacity. Africa has received many opportunities. Many of them we haven ’ t benefited much. Why? Because we lack the internal, institutional framework and policy framework that can make it possible for us to benefit from our external relations. I ’ ll give you an example. Under the Cotonou Agreement, formerly known as the Lome Convention, African countries have been given an opportunity by Europe to export goods, duty-free, to the European Union market. My own country, Uganda, has a quota to export 50, 000 metric tons of sugar to the European Union market. We haven ’ t exported one kilogram yet. We import 50, 000 metric tons of sugar from Brazil and Cuba. Secondly, under the beef protocol of that agreement, African countries that produce beef have quotas to export beef duty-free to the European Union market. None of those countries, including Africa ’ s most successful nation, Botswana, has ever met its quota. So, I want to argue today that the fundamental source of Africa ’ s inability to engage the rest of the world in a more productive relationship is because it has a poor institutional and policy framework. And all forms of intervention need support, the evolution of the kinds of institutions that create wealth, the kinds of institutions that increase productivity. How do we begin to do that, and why is aid the bad instrument? Aid is the bad instrument, and do you know why? Because all governments across the world need money to survive. Money is needed for a simple thing like keeping law and order. You have to pay the army and the police to show law and order. And because many of our governments are quite dictatorial, they need really to have the army clobber the opposition. The second thing you need to do is pay your political hangers-on. Why should people support their government? Well, because it gives them good, paying jobs, or, in many African countries, unofficial opportunities to profit from corruption. The fact is no government in the world, with the exception of a few, like that of Idi Amin, can seek to depend entirely on force as an instrument of rule. Many countries in the [unclear], they need legitimacy. To get legitimacy, governments often need to deliver things like primary education, primary health, roads, build hospitals and clinics. If the government ’ s fiscal survival depends on it having to raise money from its own people, such a government is driven by self-interest to govern in a more enlightened fashion. It will sit with those who create wealth. Talk to them about the kind of policies and institutions that are necessary for them to expand a scale and scope of business so that it can collect more tax revenues from them. The problem with the African continent and the problem with the aid industry is that it has distorted the structure of incentives facing the governments in Africa. The productive margin in our governments ’ search for revenue does not lie in the domestic economy, it lies with international donors. Rather than sit with Ugandan â â rather than sit with Ugandan entrepreneurs, Ghanaian businessmen, South African enterprising leaders, our governments find it more productive to talk to the IMF and the World Bank. I can tell you, even if you have ten Ph. Ds., you can never beat Bill Gates in understanding the computer industry. Why? Because the knowledge that is required for you to understand the incentives necessary to expand a business â it requires that you listen to the people, the private sector actors in that industry. Governments in Africa have therefore been given an opportunity, by the international community, to avoid building productive arrangements with your own citizens, and therefore allowed to begin endless negotiations with the IMF and the World Bank, and then it is the IMF and the World Bank that tell them what its citizens need. In the process, we, the African people, have been sidelined from the policy-making, policy-orientation, and policy- implementation process in our countries. We have limited input, because he who pays the piper calls the tune. The IMF, the World Bank, and the cartel of good intentions in the world has taken over our rights as citizens, and therefore what our governments are doing, because they depend on aid, is to listen to international creditors rather than their own citizens. But I want to put a caveat on my argument, and that caveat is that it is not true that aid is always destructive. Some aid may have built a hospital, fed a hungry village. It may have built a road, and that road may have served a very good role. The mistake of the international aid industry is to pick these isolated incidents of success, generalize them, pour billions and trillions of dollars into them, and then spread them across the whole world, ignoring the specific and unique circumstances in a given village, the skills, the practices, the norms and habits that allowed that small aid project to succeed â like in Sauri village, in Kenya, where Jeffrey Sachs is working â and therefore generalize this experience as the experience of everybody. Aid increases the resources available to governments, and that makes working in a government the most profitable thing you can have, as a person in Africa seeking a career. By increasing the political attractiveness of the state, especially in our ethnically fragmented societies in Africa, aid tends to accentuate ethnic tensions as every single ethnic group now begins struggling to enter the state in order to get access to the foreign aid pie. Ladies and gentlemen, the most enterprising people in Africa cannot find opportunities to trade and to work in the private sector because the institutional and policy environment is hostile to business. Governments are not changing it. Why? Because they don ’ t need to talk to their own citizens. They talk to international donors. So, the most enterprising Africans end up going to work for government, and that has increased the political tensions in our countries precisely because we depend on aid. I also want to say that it is important for us to note that, over the last 50 years, Africa has been receiving increasing aid from the international community, in the form of technical assistance, and financial aid, and all other forms of aid. Between 1960 and 2003, our continent received 600 billion dollars of aid, and we are still told that there is a lot of poverty in Africa. Where has all the aid gone? I want to use the example of my own country, called Uganda, and the kind of structure of incentives that aid has brought there. In the 2006-2007 budget, expected revenue : 2. 5 trillion shillings. The expected foreign aid : 1. 9 trillion. Uganda ’ s recurrent expenditure â by recurrent what do I mean? Hand- to-mouth is 2. 6 trillion. Why does the government of Uganda budget spend 110 percent of its own revenue? It ’ s because there ’ s somebody there called foreign aid, who contributes for it. But this shows you that the government of Uganda is not committed to spending its own revenue to invest in productive investments, but rather it devotes this revenue to paying structure of public expenditure. Public administration, which is largely patronage, takes 690 billion. The military, 380 billion. Agriculture, which employs 18 percent of our poverty-stricken citizens, takes only 18 billion. Trade and industry takes 43 billion. And let me show you, what does public expenditure â rather, public administration expenditure â in Uganda constitute? There you go. 70 cabinet ministers, 114 presidential advisers, by the way, who never see the president, except on television. And when they see him physically, it is at public functions like this, and even there, it is him who advises them. We have 81 units of local government. Each local government is organized like the central government â a bureaucracy, a cabinet, a parliament, and so many jobs for the political hangers-on. There were 56, and when our president wanted to amend the constitution and remove term limits, he had to create 25 new districts, and now there are 81. Three hundred thirty-three members of parliament. You need Wembley Stadium to host our parliament. One hundred thirty-four commissions and semi-autonomous government bodies, all of which have directors and the cars. And the final thing, this is addressed to Mr. Bono. In his work, he may help us on this. A recent government of Uganda study found that there are 3, 000 four-wheel drive motor vehicles at the Minister of Health headquarters. Uganda has 961 sub- counties, each of them with a dispensary, none of which has an ambulance. So, the four-wheel drive vehicles at the headquarters drive the ministers, the permanent secretaries, the bureaucrats and the international aid bureaucrats who work in aid projects, while the poor die without ambulances and medicine. Finally, I want to say that before I came to speak here, I was told that the principle of TEDGlobal is that the good speech should be like a miniskirt. It should be short enough to arouse interest, but long enough to cover the **subject**. I hope I have achieved that. Thank you very much.

Text Sample 801: NEG Dim 5 – 2007 – Score 1.00 – 2007/t002580.txt

You know, what I do is write for children, and I ’ m probably America ’ s most widely read children ’ s author, in fact. And I always tell people that I don ’ t want to show up looking like a scientist. You can have me as a farmer, or in leathers, and no one has ever chose farmer. I ’ m here today to talk to you about circles and epiphanies. And you know, an epiphany is usually something you find that you dropped someplace. You ’ ve just got to go around the block to see it as an epiphany. That ’ s a painting of a circle. A friend of mine did that — Richard Bollingbroke. It ’ s the kind of complicated circle that I ’ m going to tell you about. My circle began back in the ’ 60s in high school in Stow, Ohio where I was the class queer. I was the guy beaten up bloody every week in the boys ’ room, until one teacher saved my life. She saved my life by letting me go to the bathroom in the teachers ’ lounge. She did it in secret. She did it for three years. And I had to get out of town. I had a thumb, I had 85 dollars, and I ended up in San Francisco, California — met a lover — and back in the ’ 80s, found it necessary to begin work on AIDS organizations. About three or four years ago, I got a phone call in the middle of the night from that teacher, Mrs. Posten, who said,”I need to see you. I ’ m disappointed that we never got to know each other as adults. Could you please come to Ohio, and please bring that man that I know you have found by now. And I should mention that I have pancreatic cancer, and I ’ d like you to please be quick about this.”Well, the next day we were in Cleveland. We took a look at her, we laughed, we cried, and we knew that she needed to be in a hospice. We found her one, we got her there, and we took care of her and watched over her family, because it was necessary. It

’ s something we knew how to do. And just as the woman who wanted to know me as an adult got to know me, she turned into a box of ashes and was placed in my hands. And what had happened was the circle had closed, it had become a circle — and that epiphany I talked about presented itself. The epiphany is that death is a part of life. She saved my life ; I and my partner saved hers. And you know, that part of life needs everything that the rest of life does. It needs truth and beauty, and I ’ m so happy it ’ s been mentioned so much here today. It also needs — it needs dignity, love and pleasure, and it ’ s our job to hand those things out. Thank you.

Text Sample 802: NEG Dim 5 – 2007 – Score 1.00 – 2007/t000392.txt

So, I kind of believe that we ’ re in like the”cave-painting”era of computer interfaces. Like, they ’ re very kind of â they don ’ t go as deep or as emotionally engaging as they possibly could be and I ’ d like to change all that. Hit me. OK. So I mean, this is the kind of status quo interface, right? It ’ s very flat, kind of rigid. And OK, so you could sex it up and like go to a much more lickable Mac, you know, but really it ’ s the kind of same old crap we ’ ve had for the last, you know, 30 years. Like I think we really put up with a lot of crap with our computers. I mean it ’ s point and click, it ’ s like the menus, icons, it ’ s all the kind of same thing. And so one kind of information space that I take inspiration from is my real desk. It ’ s so much more subtle, so much more visceral â you know, what ’ s visible, what ’ s not. And I ’ d like to bring that experience to the desktop. So I kind of have a â this is BumpTop. It ’ s kind of like a new approach to desktop computing. So you can bump things â they ’ re all physically, you know, manipulable and stuff. And instead of that point and click, it ’ s like a push and pull, things collide as you ’ d expect them. Just like on my real desk, I can â let me just grab these guys â I can turn things into piles instead of just the folders that we have. And once things are in a pile I can browse them by throwing them into a grid, or you know, flip through them like a book or I can lay them out like a deck of cards. When they ’ re laid out, I can pull things to new locations or delete things or just quickly sort a whole pile, you know, just immediately, right? And then, it ’ s all smoothly animated, instead of these jarring changes you see in today ’ s interfaces. Also, if I want to add something to a pile, well, how do I do that? I just toss it to the pile, and it ’ s added right to the top. It ’ s a kind of nice way. Also some of the stuff we can do is, for these individual icons we thought â I mean, how can we play with the idea of an icon, and push that further? And one of the things I can do is make it bigger if I want to emphasize it and make it more important. But what ’ s really cool is that since there ’ s a physics simulation running under this, it ’ s actually heavier. So the lighter stuff doesn ’ t really move but if I throw it at the lighter guys, right? So it ’ s cute, but it ’ s also like a subtle channel of conveying information, right? This is heavy so it feels more important. So it ’ s kind of cool. Despite computers everywhere paper really hasn ’ t disappeared, because it has a lot of, I think, valuable properties. And some of those we wanted to transfer to the icons in our system. So one of the things you can do to our icons, just like paper, is crease them and fold them, just like paper. Remember, you know, something for later. Or if you want to be destructive, you can just crumple it up and, you know, toss it to the corner. Also just like paper, around our workspace we ’ ll pin things up to the wall to remember them later, and I can do the same thing here, and you know, you ’ ll see post-it notes and things like that around people ’ s offices. And I can pull them off when I want to work with them. So, one of the criticisms of this kind of approach to organization is that, you know,”Okay, well my real desk is really messy. I don ’ t want that mess on my computer.”So one thing we have for that is like a grid align, kind of â so you get that more traditional desktop. Things are kind of grid aligned. More boring, but you still have that kind of colliding and bumping. And you can still do fun things like make shelves on your desktop. Let ’ s just break this shelf. Okay, that shelf broke. I think beyond the icons, I think another really cool domain for this software â I think it applies to more than just icons and your desktop â but browsing photographs. I think you can really enrich the way we browse our photographs and bring it to that kind of shoebox of, you know, photos with your family on the kitchen table kind of thing. I can toss these things around. They ’ re so much more tangible and touchable â and you know I can double-click on something to take a look at it. And I can do all that kind of same stuff I showed you before. So I can pile things up, I can flip through it, I can, you know â okay, let ’ s move this photo to the back, let ’ s delete this guy here, and I think it ’ s just a much more rich kind of way of interacting with your information. And that ’ s BumpTop. Thanks!

Text Sample 803: NEG Dim 5 – 2003 – Score 1.00 – 2003/t000176.txt

You know, I am so bad at tech that my daughter â who is now 41 â when she was five, was overheard by me to say to a friend of hers, If it doesn ’ t bleed when you cut it, my daddy doesn ’ t understand it. So, the assignment I ’ ve been given may be an insuperable obstacle for me, but I ’ m certainly going to try. What have I heard during these last four days? This is my third visit to TED. One was to TEDMED, and one, as you ’ ve heard, was a regular TED two years ago. I ’ ve heard what I consider an extraordinary thing that I ’ ve only heard a little bit in the two previous TEDs, and what that is is an interweaving and an interlarding, an intermixing, of a sense of social responsibility in so many of the talks â global responsibility, in fact, appealing to enlightened self- interest, but it goes far beyond enlightened self-interest. One of the most impressive things about what some, perhaps 10, of the speakers have been talking about is the realization, as you listen to them carefully, that they ’ re not saying : Well, this is what we should do ; this is what I would like you to do. It ’ s : This is what I have done because I ’ m excited by it, because it ’ s a wonderful thing, and it ’ s done something for me and, of course, it ’ s accomplished a great deal. It ’ s the old concept, the real Greek concept, of philanthropy in its original sense : philanthropy, the love of humankind. And the only explanation I can have for some of what you ’ ve been hearing in the last four days is that it arises, in fact, out of a form of love. And this gives me enormous hope. And hope, of course, is the topic that I ’ m supposed to be speaking about, which I ’ d completely forgotten about until I arrived. And when I did, I thought, well, I ’ d better look this word up in the dictionary. So, Sarah and I â my wife â walked over to the public library, which is four blocks away, on Pacific Street, and we got the OED, and we looked in there, and there are 14 definitions of hope, none of which really hits you between the eyes as being the appropriate one. And, of course, that makes sense, because hope is an abstract phenomenon ; it ’ s an abstract idea, it ’ s not a concrete word. Well, it reminds me a little bit of surgery. If there ’ s one operation for a disease, you know it works. If there are 15 operations, you know that none of them work. And that ’ s the way it is with definitions of words. If you have appendicitis, they take your appendix out, and you ’ re cured. If you ’ ve got reflux oesophagitis, there are 15 procedures, and Joe Schmo does it one way and Will Blow does it another way, and none of them work, and that ’ s the way it is with this word, hope. They all come down to the idea of an expectation of something good that is due to happen. And you know what I found out? The Indo- European root of the word hope is a stem, K-E-U â we would spell it K-E-U ; it ’ s pronounced koy â and it is the same root from which the word curve comes from. But what it means in the original Indo-European is a change in direction, going in a different way. And I find that very interesting and very provocative, because what you ’ ve been hearing in the last couple of days is the sense of going in different directions : directions that are specific and unique to problems. There are different paradigms. You ’ ve heard that word several times in the last four days, and everyone ’ s familiar with Kuhnian paradigms. So, when we think of hope now, we have to think of looking in other directions than we have been looking. There ’ s another â not definition, but description, of hope that has always appealed to me, and it was one by VĂĄclav Havel in his perfectly spectacular book”Breaking the Peace,”in which he says that hope does not consist of the expectation that things will come out exactly right, but the expectation that they will make sense regardless of how they come out. I can ’ t tell you how reassured I was by the very last sentence in that glorious presentation by Dean Kamen a few days ago. I wasn ’ t sure I heard it right, so I found him in one of the inter-sessions. He was talking to a very large man, but I didn ’ t care. I interrupted, and I said,”Did you say this?”He said,”I think so.”So, here ’ s what it is : I ’ ll repeat it.”The world will not be saved by the Internet.”It ’ s wonderful. Do you know what the world will be saved by? I ’ ll tell you. It ’ ll be saved by the human spirit. And by the human spirit, I don ’ t mean anything divine, I don ’ t mean anything supernatural â certainly not coming from this skeptic. What I mean is this ability that each of us has to be something greater than herself or himself ; to arise out of our ordinary selves and achieve something that at the beginning we thought perhaps we were not capable of. On an elemental level, we have all felt that spirituality at the time of childbirth. Some of you have felt it in laboratories ; some of you have felt it at the workbench. We feel it at concerts. I ’ ve felt it in the operating room, at the bedside. It is an elevation of us beyond ourselves. And I think that it ’ s going to be, in time, the elements of the human spirit that we ’ ve been hearing about bit by bit by bit from so many of the speakers in the last few days. And if there ’ s anything that has permeated this room, it is precisely that. I ’ m intrigued by a concept that was brought to life in the early part of the 19th century â actually, in the second decade of the 19th century â by a 27-year-old poet whose name was Percy Shelley. Now, we all think that Shelley obviously is the great romantic poet that he was ; many of us tend to forget that he wrote some perfectly wonderful essays, too, and the most well- remembered essay is one called”A Defence of Poetry.”Now, it ’ s about five, six, seven, eight pages long, and it gets kind of deep and difficult after about the third page, but somewhere on the second page he begins talking about the notion that he calls”moral imagination.”And here ’ s what he says, roughly translated : A man â generic man â a man, to be greatly good, must imagine clearly. He must see himself and the world through the eyes of another, and of many others. See himself and the world â not just the world, but see himself. What is it that is expected of us by the billions of people who live in what Laurie Garrett the other day so appropriately called despair and disparity? What is it that they have every right to ask of us? What is it that we have every right to ask of ourselves, out of our shared humanity and out of the human spirit? Well, you know precisely what it is. There ’ s a great deal of argument about whether we, as the great nation that we are, should be the policeman of the world, the world ’ s constabulary, but there should be virtually no argument about whether we should be the world ’ s healer. There has certainly been no argument about that in this room in the past four days. So, if we are to be the world ’ s healer, every disadvantaged person in this world â including in the United States â becomes our patient. Every disadvantaged nation, and perhaps our own nation, becomes our patient. So, it ’ s fun to think about the etymology of the word”patient.”It comes initially from the Latin patior, to endure, or to suffer. So, you go back to the old Indo- European root again, and what do you find? The Indo-European stem is pronounced payen â we would spell it P-A-E-N â and, lo and behold, mirabile dictu, it is the same root as the word compassion comes from, P-A-E-N. So, the lesson is very clear. The lesson is that our patient â the world, and the disadvantaged of the world â that patient deserves our compassion. But beyond our compassion, and far greater than compassion, is our moral imagination and our identification with each individual who lives in that world, not to think of them as a huge forest, but as individual trees. Of course, in this day and age, the trick is not to let each tree be obscured by that Bush in Washington that can get â can get in the way. So, here we are. We are, should be, morally committed to being the healer of the world. And we have had examples over and over and over again â you ’ ve just heard one in the last 15 minutes â of people who have not only had that commitment, but had the charisma, the brilliance â and I think in this room it ’ s easy to use the word brilliant, my God â the brilliance to succeed at least at the beginning of their quest, and who no doubt will continue to succeed, as long as more and more of us enlist ourselves in their cause. Now, if we ’ re talking about medicine, and we ’ re talking about healing, I ’ d like to quote someone who hasn ’ t been quoted. It seems to me everybody in the world ’ s been quoted here : Pogo ’ s been quoted ; Shakespeare ’ s been quoted backwards, forwards, inside out. I would like to quote one of my own household gods. I suspect he never really said this, because we don ’ t know what Hippocrates really said, but we do know for sure that one of the great Greek physicians said the following, and it has been recorded in one of the books attributed to Hippocrates, and the book is called”Precepts.”And I ’ ll read you what it is. Remember, I have been talking about, essentially philanthropy : the love of humankind, the individual humankind and the individual humankind that can bring that kind of love translated into action, translated, in some cases, into enlightened self-interest. And here he is, 2, 400 years ago :”Where there is love of humankind, there is love of healing.”We have seen that here today with the sense, with the sensitivity â and in the last three days, and with the power of the indomitable human spirit. Thank you very much.

Text Sample 804: NEG Dim 5 – 2003 – Score 1.00 – 2003/t000192.txt

When I first arrived in beautiful Zimbabwe, it was difficult to understand that 35 percent of the population is HIV positive. It really wasn ’ t until I was invited to the homes of people that I started to understand the human toll of the epidemic. For instance, this is Herbert with his grandmother. When I first met him, he was sitting on his grandmother ’ s lap. He has been orphaned, as both of his parents died of AIDS, and his grandmother took care of him until he too died of AIDS. He liked to sit on her lap because he said that it was painful for him to lie in his own bed. When she got up to make tea, she placed him in my own lap and I had never felt a child that was that emaciated. Before I left, I actually asked him if I could get him something. I thought he would ask for a toy, or candy, and he asked me for slippers, because he said that his feet were cold. This is Joyce who ’ s — in this picture — 21. Single mother, HIV positive. I photographed her before and after the birth of her beautiful baby girl, Issa. And I was last week walking on Lafayette Street in Manhattan and got a call from a woman who I didn ’ t know, but she called to tell me that Joyce had passed away at the age of 23. Joyce ’ s mother is now taking care of her daughter, like so many other Zimbabwean children who ’ ve been orphaned by the epidemic. So a few of the stories. With every picture, there are individuals who have full lives and stories that deserve to be told. All these pictures are from Zimbabwe. Chris Anderson : Kirsten, will you just take one minute, just to tell your own story of how you got to Africa? Kirsten Ashburn : Mmm, gosh. CA : Just — KA : Actually, I was working at the time, doing production for a fashion photographer. And I was constantly reading the New York Times, and stunned by the statistics, the numbers. It was just frightening. So I quit my job and decided that that ’ s the **subject** that I wanted to tackle. And I first actually went to Botswana, where I spent a month — this is in December 2000 — then went to Zimbabwe for a month and a half, and then went back again this March 2002 for another month and a half in Zimbabwe. CA : That ’ s an amazing story, thank you. KB : Thanks for letting me show these.

Text Sample 805: NEG Dim 5 – 2003 – Score 1.00 – 2003/t000193.txt

So, about three years ago I was in London, and somebody called Howard Burton came to me and said, I represent a group of people, and we want to start an institute in theoretical physics. We have about 120 million dollars, and we want to do it well. We want to be in the forefront fields, and we want to do it differently. We want to get out of this thing where the young people have all the ideas, and the old people have all the power and decide what science gets done. It took me about 25 seconds to decide that that was a good idea. Three years later, we have the Perimeter Institute for Theoretical Physics in Waterloo, Ontario. It—s the most exciting job I—ve ever had. And it—s the first time I—ve had a job where I—m afraid to go away because of everything that—s going to happen in this week when I—m here. But in any case, what I—m going to do in my little bit of time is take you on a quick tour of some of the things that we talk about and we think about. So, we think a lot about what really makes science work? The first thing that anybody who knows science, and has been around science, is that the stuff you learn in school as a scientific method is wrong. There is no method. On the other hand, somehow we manage to reason together as a community, from incomplete evidence to conclusions that we all agree about. And this is, by the way, something that a democratic society also has to do. So how does it work? Well, my belief is that it works because scientists are a community bound together by an ethics. And here are some of the ethical principles. I—m not going to read them all to you because I—m not in teacher mode. I—m in entertain, amaze mode. But one of the principles is that everybody who is part of the community gets to fight and argue as hard as they can for what they believe. But we—re all disciplined by the understanding that the only people who are going to decide, you know, whether I—m right or somebody else is right, are the people in our community in the next generation, in 30 and 50 years. So it—s this combination of respect for the tradition and community we—re in, and rebellion that the community requires to get anywhere, that makes science work. And being in this process of being in a community that reasons from shared evidence to conclusions, I believe, teaches us about democracy. Not only is there a relationship between the ethics of science and the ethics of being a citizen in democracy, but there has been, historically, a relationship between how people think about space and time, and what the cosmos is, and how people think about the society that they live in. And I want to talk about three stages in that evolution. The first science of cosmology that was anything like science was Aristotelian science, and that was hierarchical. The earth is in the center, then there are these crystal spheres, the sun, the moon, the planets and finally the celestial sphere, where the stars are. And everything in this universe has a place. And Aristotle—s law of motion was that everything goes to its natural place, which was of course, the rule of the society that Aristotle lived in, and more importantly, the medieval society that, through Christianity, embraced Aristotle and blessed it. And the idea is that everything is defined. Where something is, is defined with respect to this last sphere, the celestial sphere, outside of which is this eternal, perfect realm, where lives God, who is the ultimate judge of everything. So that is both Aristotelian cosmology, and in a certain sense, medieval society. Now, in the 17th century there was a revolution in thinking about space and time and motion and so forth of Newton. And at the same time there was a revolution in social thought of John Locke and his collaborators. And they were very closely associated. In fact, Newton and Locke were friends. Their way of thinking about space and time and motion on the one hand, and a society on the other hand, were closely related. And let me show you. In a Newtonian universe, there—s no center — thank you. There are particles and they move around with respect to a fixed, absolute framework of space and time. It—s meaningful to say absolutely where something is in space, because that—s defined, not with respect to say, where other things are, but with respect to this absolute notion of space, which for Newton was God. Now, similarly, in Locke—s society there are individuals who have certain rights, properties in a formal sense, and those are defined with respect to some absolute, abstract notions of rights and justice, and so forth, which are independent of what else has happened in the society. Of who else there is, of the history and so forth. There is also an omniscient observer who knows everything, who is God, who is in a certain sense outside the universe, because he has no role in anything that happens, but is in a certain sense everywhere, because space is just the way that God knows where everything is, according to Newton, OK? So this is the foundations of what—s called, traditionally, liberal political theory and Newtonian physics. Now, in the 20th century we had a revolution that was initiated at the beginning of the 20th century, and which is still going on. It was begun with the invention of relativity theory and quantum theory. And merging them together to make the final quantum theory of space and time and gravity, is the culmination of that, something that—s going on right now. And in this universe there—s nothing fixed and absolute. Zilch, OK. This universe is described by being a network of relationships. Space is just one aspect, so there—s no meaning to say absolutely where something is. There—s only where it is relative to everything else that is. And this network of relations is ever-evolving. So we call it a relational universe. All properties of things are about these kinds of relationships. And also, if you—re embedded in such a network of relationships, your view of the world has to do with what information comes to you through the network of relations. And there—s no place for an omniscient observer or an outside intelligence knowing everything and making everything. So this is general relativity, this is quantum theory. This is also, if you talk to legal scholars, the foundations of new ideas in legal thought. They—re thinking about the same things. And not only that, they make the analogy to relativity theory and cosmology often. So there—s an interesting discussion going on there. This last view of cosmology is called the relational view. So the main slogan here is that there—s nothing outside the universe, which means that there—s no place to put an explanation for something outside. So in such a relational universe, if you come upon something that—s ordered and structured, like this device here, or that device there, or something beautiful, like all the living things, all of you guys in the room —”guys”in physics, by the way, is a generic term : men and women. Then you want to know, you—re a person, you want to know how is it made. And in a relational universe the only possible explanation was, somehow it made itself. There must be mechanisms of self-organization inside the universe that make things. Because there—s no place to put a maker outside, as there was in the Aristotelian and the Newtonian universe. So in a relational universe we must have processes of self-organization. Now, Darwin taught us that there are processes of self-organization that suffice to explain all of us and everything we see. So it works. But not only that, if you think about how natural selection works, then it turns out that natural selection would only make sense in such a relational universe. That is, natural selection works on properties, like fitness, which are about relationships of some species to some other species. Darwin wouldn—t make sense in an Aristotelian universe, and wouldn—t really make sense in a Newtonian universe. So a theory of biology based on natural selection requires a relational notion of what are the properties of biological systems. And if you push that all the way down, really, it makes the best sense in a relational universe where all properties are relational. Now, not only that, but Einstein taught us that gravity is the result of the world being relational. If it wasn—t for gravity, there wouldn—t be life, because gravity causes stars to form and live for a very long time, keeping pieces of the world, like the surface of the Earth, out of thermal equilibrium for billions of years so life can evolve. In the 20th century, we saw the independent development of two big themes in science. In the biological sciences, they explored the implications of the notion that order and complexity and structure arise in a self-organized way. That was the triumph of Neo-Darwinism and so forth. And slowly, that idea is leaking out to the cognitive sciences, the human sciences, economics, et cetera. At the same time, we physicists have been busy trying to make sense of and build on and integrate the discoveries of quantum theory and relativity. And what we—ve been working out is the implications, really, of the idea that the universe is made up of relations. 21st-century science is going to be driven by the integration of these two ideas : the triumph of relational ways of thinking about the world, on the one hand, and self-organization or Darwinian ways of thinking about the world, on the other hand. And also, is that in the 21st century our thinking about space and time and cosmology, and our thinking about society are both going to continue to evolve. And what they—re evolving towards is the union of these two big ideas, Darwinism and relationalism. Now, if you think about democracy from this perspective, a new pluralistic notion of democracy would be one that recognizes that there are many different interests, many different agendas, many different individuals, many different points of view. Each one is incomplete, because you—re embedded in a network of relationships. Any actor in a democracy is embedded in a network of relationships. And you understand some things better than other things, and because of that there—s a continual jostling and give and take, which is politics. And politics is, in the ideal sense, the way in which we continually address our network of relations in order to achieve a better life and a better society. And I also think that science will never go away and — I—m finishing on this line. In fact, I—m finished. Science will never go away.

Text Sample 806: NEG Dim 5 – 2006 – Score -1.00 – 2006/t000488.txt

Thank you so much, Chris. And it ’ s truly a great honor to have the opportunity to come to this stage twice ; I ’ m extremely grateful. I have been **blown** away by this conference, and I want to thank all of you for the many nice comments about what I had to say the other night. And I say that sincerely, partly because I need that. Put yourselves in my position. I flew on Air Force Two for eight years. Now I have to take off my shoes or boots to get on an airplane! I ' ll tell you one quick story to illustrate what that ' s been like for me. It ' s a true story — every bit of this is true. Soon after Tipper and I left the — White House — we were driving from our home in Nashville to a little farm we have 50 miles east of Nashville. Driving ourselves. I know it sounds like a little thing to you, but — I looked in the rear-view mirror and all of a sudden it just hit me. There was no motorcade back there. You ' ve heard of phantom limb pain? This was a rented Ford Taurus. It was dinnertime, and we started looking for a place to eat. We were on I-40. We got to Exit 238, Lebanon, Tennessee. We got off the exit, we found a Shoney ' s restaurant. Low-cost family restaurant chain, for those of you who don ' t know it. We went in and sat down at the booth, and the waitress came over, made a big commotion over Tipper. She took our order, and then went to the couple in the booth next to us, and she lowered her voice so much, I had to really strain to hear what she was saying. And she said"Yes, that ' s former Vice President Al Gore and his wife, Tipper."And the man said,"He ' s come down a long way, hasn ' t he?"There ' s been kind of a series of epiphanies. The very next day, continuing the totally true story, I got on a G-V to fly to Africa to make a speech in Nigeria, in the city of Lagos, on the topic of energy. And I began the speech by telling them the story of what had just happened the day before in Nashville. And I told it pretty much the same way I ' ve just shared it with you : Tipper and I were driving ourselves, Shoney ' s, low-cost family restaurant chain, what the man said — they laughed. I gave my speech, then went back out to the airport to fly back home. I fell asleep on the plane until, during the middle of the night, we landed on the Azores Islands for refueling. I woke up, they opened the door, I went out to get some fresh air, and I looked, and there was a man running across the runway. And he was waving a piece of paper, and he was yelling,"Call Washington! Call Washington!"And I thought to myself, in the middle of the night, in the middle of the Atlantic, what in the world could be wrong in Washington? Then I remembered it could be a bunch of things. But what it turned out to be, was that my staff was extremely upset because one of the wire services in Nigeria had already written a story about my speech, and it had already been printed in cities all across the United States of America. It was printed in Monterey, I checked. And the story began,"Former Vice President Al Gore announced in Nigeria yesterday,"quote : ' My wife Tipper and I have opened a low-cost family restaurant '"—"' named Shoney ' s, and we are running it ourselves. '"Before I could get back to U. S. soil, David Letterman and Jay Leno had already started in on — one of them had me in a big white chef ' s hat, Tipper was saying,"One more burger with fries!"Three days later, I got a nice, long, handwritten letter from my friend and partner and colleague Bill Clinton, saying,"Congratulations on the new restaurant, Al!"We like to celebrate each other ' s successes in life. I was going to talk about information ecology. But I was thinking that, since I plan to make a lifelong habit of coming back to TED, that maybe I could talk about that another time. Chris Anderson : It ' s a deal! Al Gore : I want to focus on what many of you have said you would like me to elaborate on : What can you do about the climate crisis? I want to start with a couple of — I ' m going to show some new images, and I ' m going to recapitulate just four or five. Now, the slide show. I update the slide show every time I give it. I add new images, because I learn more about it every time I give it. It ' s like beach-combing, you know? Every time the tide comes in and out, you find some more shells. Just in the last two days, we got the new temperature records in January. This is just for the United States of America. Historical average for

Januarys is 31 degrees ; last month was 39. 5 degrees. Now, I know that you wanted some more bad news about the environment — I ’ m kidding. But these are the recapitulation slides, and then I ’ m going to go into new material about what you can do. But I wanted to elaborate on a couple of these. First of all, this is where we ’ re projected to go with the U. S. contribution to global warming, under business as usual. Efficiency in end-use electricity and end-use of all energy is the low-hanging fruit. Efficiency and conservation — it ’ s not a cost ; it ’ s a profit. The sign is wrong. It ’ s not negative ; it ’ s positive. These are investments that pay for themselves. But they are also very effective in deflecting our path. Cars and trucks — I talked about that in the slideshow, but I want you to put it in perspective. It ’ s an easy, visible target of concern — and it should be — but there is more global warming pollution that comes from buildings than from cars and trucks. Cars and trucks are very significant, and we have the lowest standards in the world. And so we should address that. But it ’ s part of the puzzle. Other transportation efficiency is as important as cars and trucks. Renewables at the current levels of technological efficiency can make this much difference. And with what Vinod, and John Doerr and others, many of you here — there are a lot of people directly involved in this — this wedge is going to grow much more rapidly than the current projection shows it. Carbon Capture and Sequestration — that ’ s what CCS stands for — is likely to become the killer app that will enable us to continue to use fossil fuels in a way that is safe. Not quite there yet. OK. Now, what can you do? Reduce emissions in your home. Most of these expenditures are also profitable. Insulation, better design. Buy green electricity where you can. I mentioned automobiles — buy a hybrid. Use light rail. Figure out some of the other options that are much better. It ’ s important. Be a green consumer. You have choices with everything you buy, between things that have a harsh effect, or a much less harsh effect on the global climate crisis. Consider this : Make a decision to live a carbon-neutral life. Those of you who are good at branding, I ’ d love to get your advice and help on how to say this in a way that connects with the most people. It is easier than you think. It really is. A lot of us in here have made that decision, and it is really pretty easy. It means reduce your carbon dioxide emissions with the full range of choices that you make, and then purchase or acquire offsets for the remainder that you have not completely reduced. And what it means is elaborated at climatecrisis. net. There is a carbon calculator. Participant Productions convened — with my active involvement — the leading software writers in the world, on this arcane science of carbon calculation, to construct a consumer-friendly carbon calculator. You can very precisely calculate what your CO2 emissions are, and then you will be given options to reduce. And by the time the movie comes out in May, this will be updated to 2. 0, and we will have click-through purchases of offsets. Next, consider making your business carbon-neutral. Again, some of us have done that, and it ’ s not as hard as you think. Integrate climate solutions into all of your innovations, whether you are from the technology, or entertainment, or design and architecture community. Invest sustainably. Majora mentioned this. Listen, if you have invested money with managers who you compensate on the basis of their annual performance, don ’ t ever again complain about quarterly report CEO management. Over time, people do what you pay them to do. And if they judge how much they ’ re going to get paid on your capital that they ’ ve invested, based on the short-term returns, you ’ re going to get short-term decisions. A lot more to be said about that. Become a catalyst of change. Teach others, learn about it, talk about it. The movie is a movie version of the slideshow I gave two nights ago, except it ’ s a lot more entertaining. And it comes out in May. Many of you here have the opportunity to ensure that a lot of people see it. Consider sending somebody to Nashville. Pick well. And I am personally going to train people to give this slideshow — re-purposed, with some of the personal stories obviously replaced with a generic approach, and it ’ s not just the slides, it ’ s what they mean. And it ’ s how they link together. And so I ’ m going to be conducting a course this summer for a group of people that are nominated by different folks to come and then give it en masse, in communities all across the country, and we ’ re going to update the slideshow for all of them every single week, to keep it right on the cutting edge. Working with Larry Lessig, it will be, somewhere in that process, posted with tools and limited-use copyrights, so that young people can remix it and do it in their own way. Where did anybody get the idea that you ought to stay arm ’ s length from politics? It doesn ’ t mean that if you ’ re a Republican, that I ’ m trying to convince you to be a Democrat. We need Republicans as well. This used to be a bipartisan issue, and I know that in this group it really is. Become politically active. Make our democracy work the way it ’ s supposed to work. Support the idea of capping carbon dioxide emissions — global warming pollution — and trading it. Here ’ s why : as long as the United States is out of the world system, it ’ s not a closed system. Once it becomes a closed system, with U. S. participation, then everybody who ’ s on a board of directors — how many people here serve on the board of directors of a corporation? Once it ’ s a closed system, you will have legal liability if you do not urge your CEO to get the maximum income from reducing and trading the carbon emissions that can be avoided. The market will work to solve this problem — if we can accomplish this. Help with the mass persuasion campaign that will start this spring. We have to change the minds of the American people. Because presently, the politicians do not have permission to do what needs to be done. And in our modern country, the role of logic and reason no longer includes mediating between wealth and power the way it once did. It ’ s now repetition of short, hot-button, 30-second, 28-second television ads. We have to buy a lot of those ads. Let ’ s re-brand global warming, as many of you have suggested. I like”climate crisis”instead of”climate collapse,”but again, those of you who are good at branding, I need your help on this. Somebody said the test we ’ re facing now, a scientist told me, is whether the combination of an opposable thumb and a neocortex is a viable combination. That ’ s really true. I said the other night, and I ’ ll repeat now : this is not a political issue. Again, the Republicans here — this shouldn ’ t be partisan. You have more influence than some of us who are Democrats do. This is an opportunity. Not just this, but connected to the ideas that are here, to bring more coherence to them. We are one. Thank you very much, I appreciate it.

Text Sample 807: NEG Dim 5 – 2006 – Score 1.00 – 2006/t000442.txt

I ’ ve got apparently 18 minutes to convince you that history has a direction, an arrow ; that in some fundamental sense, it ’ s good ; that the arrow points to something positive. Now, when the TED people first approached me about giving this upbeat talk â□□ â□□ that was before the cartoon of Muhammad had triggered global rioting. It was before the avian flu had reached Europe. It was before Hamas had won the Palestinian election, eliciting various counter- measures by Israel. And to be honest, if I had known when I was asked to give this upbeat talk that even as I was giving the upbeat talk, the apocalypse would be unfolding â□□ â□□ I might have said,”Is it okay if I talk about something else?”But I didn ’ t, OK. So we ’ re here. I ’ ll do what I can. I ’ ll do what I can. I ’ ve got to warn you : the sense in which my worldview is upbeat has always been kind of subtle, sometimes even elusive. The sense in which I can be uplifting and inspiring â I mean, there ' s always been a kind of a certain grim dimension to the way I try to uplift, so if grim inspiration â â if grim inspiration is not a contradiction in terms, that is, I ' m afraid, the most you can hope for. OK, today â that ' s if I succeed. I ' ll see what I can do. OK? Now, in one sense, the claim that history has a direction is not that controversial. If you ' re just talking about social structure, OK, clearly that ' s gotten more complex a little over the last 10, 000 years â has reached higher and higher levels. And in fact, that ' s actually sustaining a long- standing trend that predates human beings, OK, that biological evolution was doing for us. Because what happened in the beginning, this stuff encases itself in a cell, then cells start hanging out together in societies. Eventually they get so close, they form multicellular organisms, then you get complex multicellular organisms ; they form societies. But then at some point, one of these multicellular organisms does something completely amazing with this stuff, which is it launches a whole second kind of evolution : cultural evolution. And amazingly, that evolution sustains the trajectory that biological evolution had established toward greater complexity. By cultural evolution we mean the evolution of ideas. A lot of you have heard the term"memes."The evolution of technology, I pay a lot of attention to, so, you know, one of the first things you got was a little hand axe. Generations go by, somebody says, hey, why don ' t we put it on a stick? Just absolutely delights the little ones. Next best thing to a video game. This may not seem to impress, but technological evolution is progressive, so another 10, 20, 000 years, and armaments technology takes you here. Impressive. And the rate of technological evolution speeds up, so a mere quarter of a century after this, you get this, OK. And this. I ' m sorry â it was a cheap laugh, but I wanted to find a way to transition back to this idea of the unfolding apocalypse, and I thought that might do it. So, what threatens to happen with this unfolding apocalypse is the collapse of global social organization. Now, first let me remind you how much work it took to get us where we are, to be on the brink of true global social organization. Originally, you had the most complex societies, the hunter-gatherer village. Stonehenge is the remnant of a chiefdom, which is what you get with the invention of agriculture : multi-village polity with centralized rule. With the invention of writing, you start getting cities. This is blurry. I kind of like that because it makes it look like a one-celled organism and reminds you how many levels organic organization has already moved through to get to this point. And then you get to, you know, you get empires. I want to stress, you know, social organization can transcend political bounds. This is the Silk Road connecting the Chinese Empire and the Roman Empire. So you had social complexity spanning the whole continent, even if no polity did similarly. Today, you ' ve got nation states. Point is : there ' s obviously collaboration and organization going on beyond national bounds. This is actually just a picture of the earth at night, and I ' m just putting it up because I think it ' s pretty. Does kind of convey the sense that this is an integrated system. Now, I explained this growth of complexity by reference to something called"non-zero sumness."Assuming that a few of you did not do the assigned reading, very quickly, the key idea is the distinction between zero-sum games, in which correlations are inverse : always a winner and a loser. Non-zero-sum games in which correlations can be positive, OK. So like in tennis, usually it ' s win- lose ; it always adds up to zero-zero-sum. But if you ' re playing doubles, the person on your side of the net, they ' re in the same boat as you, so you ' re playing a non-zero-sum game with them. It ' s either for the better or for the worse, OK. A lot of forms of non-zero-sum behavior in the realm of economics and so on in everyday life often leads to cooperation. The argument I make is basically that, well, non-zero-sum games have always been part of life. You have them in hunter-gatherer societies, but then through technological evolution, new forms of technology arise that facilitate or encourage the playing of non-zero-sum games, involving more people over larger territory. Social structure adapts to accommodate this possibility and to harness this productive potential, so you get cities, you know, and you get all the non-zero-sum games you don ' t think about that are being played across the world. Like, have you ever thought when you buy a car, how many people on how many different continents contributed to the manufacture of that car? Those are people in effect you ' re playing a non-zero-sum game with. I mean, there are certainly plenty of them around. Now, this sounds like an intrinsically upbeat worldview in a way, because when you think of non-zero, you think win-win, you know, that ' s good. Well, there are a few reasons that actually it ' s not intrinsically upbeat. First of all, it can accommodate ; it doesn ' t deny the existence of inequality exploitation war. But there ' s a more fundamental reason that it ' s not intrinsically upbeat, because a non-zero-sum game, all it tells you for sure is that the fortunes will be correlated for better or worse. It doesn ' t necessarily predict a win- win outcome. So, in a way, the question is : on what grounds am I upbeat at all about history? And the answer is, first of all, on balance I would say people have played their games to more win-win outcomes than lose-lose outcomes. On balance, I think history is a net positive in the non-zero-sum game department. And a testament to this is the thing that most amazes me, most impresses me, and most uplifts me, which is that there is a moral dimension to history ; there is a moral arrow. We have seen moral progress over time. 2, 500 years ago, members of one Greek city-state considered members of another Greek city-state subhuman and treated them that way. And then this moral revolution arrived, and they decided that actually, no, Greeks are human beings. It ' s just the Persians who aren ' t fully human and don ' t deserve to be treated very nicely. But this was progress â you know, give them credit. And now today, we ' ve seen more progress. I think â I hope â most people here would say that all people everywhere are human beings, deserve to be treated decently, unless they do something horrendous, regardless of race or religion. And you have to read your ancient history to realize what a revolution that has been, OK. This was not a prevalent view, few thousand years ago, and I attribute it to this non- zero-sum dynamic. I think that ' s the reason there is as much tolerance toward nationalities, ethnicities, religions as there is today. If you asked me, you know, why am I not in favor of bombing Japan, well, I ' m only half-joking when I say they built my car. We have this non-zero-sum relationship, and I think that does lead to a kind of a tolerance to the extent that you realize that someone else ' s welfare is positively correlated with yours â you ' re more likely to cut them a break. I kind of think this is a kind of a business-class morality. Unfortunately, I don ' t fly trans-Atlantic business class often enough to know, or any other kind of business class really, but I assume that in business class, you don ' t hear many expressions of, you know, bigotry about racial groups or ethnic groups, because the people who are flying trans-Atlantic business class are doing business with all these people ; they ' re making money off all these people. And I really do think that, in that sense at least, capitalism has been a constructive force, and more fundamentally, it ' s a non-zero-sumness that has been a constructive force in expanding people ' s realm of moral awareness. I think the non-zero-sum dynamic, which is not only economic by any means â it ' s not always commerce â but it has driven us to the verge of a moral truth, which is the fundamental equality of everyone. It has done that. As it has moved global, moved us toward a global level of social organization, it has driven us toward moral truth. I think that ' s wonderful. Now, back to the unfolding apocalypse. And you may wonder, OK, that ' s all fine, sounds great â moral direction in history â but what about this so-called clash of civilizations? Well, first of all, I would emphasize that it fits into the non-zero-sum framework, OK. If you look at the relationship between the so- called Muslim world and Western world â two terms I don ' t like, but can ' t really avoid ; in such a short span of time, they ' re efficient if nothing else â it is non-zero-sum. And by that I mean, if people in the Muslim world get more hateful, more resentful, less happy with their place in the world, it ' ll be bad for the West. If they get more happy, it ' ll be good for the West. So that is a non-zero-sum dynamic. And I would say the non-zero-sum dynamic is only going to grow more intense over time because of technological trends, but more intense in a kind of negative way. It ' s the downside correlation of their fortunes that will become more and more possible. And one reason is because of something I call the"growing lethality of hatred."More and more, it ' s possible for grassroots hatred abroad to manifest itself in the form of organized violence on American soil. And that ' s pretty new, and I think it ' s probably going to get a lot worse â this capacity â because of trends in information technology, in technologies that can be used for purposes of munitions like biotechnology and nanotechnology. We may be hearing more about that today. And there ' s something I worry about especially, which is that this dynamic will lead to a kind of a feedback cycle that puts us on a slippery slope. What I have in mind is : terrorism happens here ; we overreact to it. That, you know, we ' re not sufficiently surgical in our retaliation leads to more hatred abroad, more terrorism. We overreact because being human, we feel like retaliating, and it gets worse and worse and worse. You could call this the positive feedback of negative vibes, but I think in something so spooky, we really shouldn ' t have the word positive there at all, even in a technical sense. So let ' s call it the death spiral of negativity. I assure you if it happens, at the end, both the West and the Muslim world will have suffered. So, what do we do? Well, first of all, we can do a lot more with arms control, the international regulation of dangerous technologies. I have a whole global governance sermon that I will spare you right now, because I don ' t think that ' s going to be enough anyway, although it ' s essential. I think we ' re going to have to have a major round of moral progress in the world. I think you ' re just going to have to see less hatred among groups, less bigotry, and, you know, racial groups, religious groups, whatever. I ' ve got to admit I feel silly saying that. It sounds so kind of Pollyannaish. I feel like Rodney King, you know, saying, why can ' t we all just get along? But hey, I don ' t really see any alternative, given the way I read the situation. There ' s going to have to be moral progress. There ' s going to have to be a lessening of the amount of hatred in the world, given how dangerous it ' s becoming. In my defense, I ' d say, as naive as this may sound, it ' s ultimately grounded in cynicism. That is to say â â thank you, thank you. That is to say, remember : my whole view of morality is that it boils down to self-interest. It ' s when people ' s fortunes are correlated. It ' s when your welfare conduces to mine, that I decide, oh yeah, I ' m all in favor of your welfare. That ' s what ' s responsible for this growth of this moral progress so far, and I ' m saying we once again have a correlation of fortunes, and if people respond to it intelligently, we will see the development of tolerance and so on â the norms that we need, you know. We will see the further evolution of this kind of business-class morality. So, these two things, you know, if they get people ' s attention and drive home the positive correlation and people do what ' s in their self-interests, which is further the moral evolution, then they could actually have a constructive effect. And that ' s why I lump growing lethality of hatred and death spiral of negativity under the general rubric, reasons to be cheerful. Doing the best I can, OK. I never called myself Mr. Uplift. I ' m just doing what I can here. Now, launching a moral revolution has got to be hard, right? I mean, what do you do? And I think the answer is a lot of different people are going to have to do a lot of different things. We all start where we are. Speaking as an American who has children whose security 10, 20, 30 years down the road I worry about â what I personally want to start out doing is figuring out why so many people around the world hate us, OK. I think that ’ s a worthy research project myself. I also like it because it ’ s an intrinsically kind of morally redeeming exercise. Because to understand why somebody in a very different culture does something â somebody you ’ re kind of viewing as alien, who ’ s doing things you consider strange in a culture you consider strange â to really understand why they do the things they do is a morally redeeming accomplishment, because you ’ ve got to relate their experience to yours. To really understand it, you ’ ve got to say,”Oh, I get it. So when they feel resentful, it ’ s kind of like the way I feel resentful when this happens, and for somewhat the same reasons.”That ’ s true understanding. And I think that is an expansion of your moral compass when you manage to do that. It ’ s especially hard to do when people hate you, OK, because you don ’ t really, in a sense, want to completely understand why people hate you. I mean, you want to hear the reason, but you don ’ t want to be able to relate to it. You don ’ t want it to make sense, right? You don ’ t want to say,”Well, yeah, I can kind of understand how a human being in those circumstances would hate the country I live in.”That ’ s not a pleasant thing, but I think it ’ s something that we ’ re going to have to get used to and work on. Now, I want to stress that to understand, you know â there are people who don ’ t like this whole business of understanding the grassroots, the root causes of things ; they don ’ t want to know why people hate us. I want to understand it. The reason you ’ re trying to understand why they hate us, is to get them to quit hating us. The idea when you go through this moral exercise of really coming to appreciate their humanity and better understand them, is part of an effort to get them to appreciate your humanity in the long run. I think it ’ s the first step toward that. That ’ s the long-term goal. There are people who worry about this, and in fact, I, myself, apparently, was denounced on national TV a couple of nights ago because of an op-ed I ’ d written. It was kind of along these lines, and the allegation was that I have, quote,”affection for terrorists.”Now, the good news is that the person who said it was Ann Coulter. I mean, if you ’ ve got to have an enemy, do make it Ann Coulter. But it ’ s not a crazy concern, OK, because understanding behavior can lead to a kind of empathy, and it can make it a little harder to deliver tough love, and so on. But I think we ’ re a lot closer to erring on the side of not comprehending the situation clearly enough, than in comprehending it so clearly that we just can ’ t, you know, get the army out to kill terrorists. So I ’ m not really worried about it. So â â I mean, we ’ re going to have to work on a lot of fronts, but if we succeed â if we succeed â then once again, non-zero-sumness and the recognition of non-zero-sum dynamics will have forced us to a higher moral level. And a kind of saving higher moral level, something that kind of literally saves the world. If you look at the word”salvation”in the Bible â the Christian usage that we ’ re familiar with â saving souls, that people go to heaven â that ’ s actually a latecomer. The original meaning of the word”salvation”in the Bible is about saving the social system.”Yahweh is our Savior”means”He has saved the nation of Israel,”which at the time, was a pretty high-level social organization. Now, social organization has reached the global level, and I guess, if there ’ s good news I can say I ’ m bringing you, it ’ s just that all the salvation of the world requires is the intelligent pursuit of self-interests in a disciplined and careful way. It ’ s going to be hard. I say we give it a shot anyway because we ’ ve just come too far to screw it up now. Thanks.

Text Sample 808: NEG Dim 5 – 2006 – Score 1.00 – 2006/t000447.txt

If you haven ’ t ordered yet, I generally find the rigatoni with the spicy tomato sauce goes best with diseases of the small intestine. So, sorry â it just feels like I should be doing stand-up up here because of the setting. No, what I want to do is take you back to 1854 in London for the next few minutes, and tell the story â in brief â of this outbreak, which in many ways, I think, helped create the world that we live in today, and particularly the kind of city that we live in today. This period in 1854, in the middle part of the 19th century, in London ’ s history, is incredibly interesting for a number of reasons. But I think the most important one is that London was this city of 2. 5 million people, and it was the largest city on the face of the planet at that point. But it was also the largest city that had ever been built. And so the Victorians were trying to live through and simultaneously invent a whole new scale of living : this scale of living that we, you know, now call”metropolitan living.”And it was in many ways, at this point in the mid-1850s, a complete disaster. They were basically a city living with a modern kind of industrial metropolis with an Elizabethan public infrastructure. So people, for instance, just to gross you out for a second, had cesspools of human waste in their basement. Like, a foot to two feet deep. And they would just kind of throw the buckets down there and hope that it would somehow go away, and of course it never really would go away. And all of this stuff, basically, had accumulated to the point where the city was incredibly offensive to just walk around in. It was an amazingly smelly city. Not just because of the cesspools, but also the sheer number of livestock in the city would shock people. Not just the horses, but people had cows in their attics that they would use for milk, that they would hoist up there and keep them in the attic until literally their milk ran out and they died, and then they would drag them off to the bone boilers down the street. So, you would just walk around London at this point and just be overwhelmed with this stench. And what ended up happening is that an entire emerging public health system became convinced that it was the smell that was killing everybody, that was creating these diseases that would wipe through the city every three or four years. And cholera was really the great killer of this period. It arrived in London in 1832, and every four or five years another epidemic would take 10, 000, 20, 000 people in London and throughout the U. K. And so the authorities became convinced that this smell was this problem. We had to get rid of the smell. And so, in fact, they concocted a couple of early, you know, founding public-health interventions in the system of the city, one of which was called the”Nuisances Act,”which they got everybody as far as they could to empty out their cesspools and just pour all that waste into the river. Because if we get it out of the streets, it ’ ll smell much better, and â oh right, we drink from the river. So what ended up happening, actually, is they ended up increasing the outbreaks of cholera because, as we now know, cholera is actually in the water. It ’ s a waterborne disease, not something that ’ s in the air. It ’ s not something you smell or inhale ; it ’ s something you ingest. And so one of the founding moments of public health in the 19th century effectively poisoned the water supply of London much more effectively than any modern day bioterrorist could have ever dreamed of doing. So this was the state of London in 1854, and in the middle of all this carnage and offensive conditions, and in the midst of all this scientific confusion about what was actually killing people, it was a very talented classic 19th century multi- disciplinarian named John Snow, who was a local doctor in Soho in London, who had been arguing for about four or five years that cholera was, in fact, a waterborne disease, and had basically convinced nobody of this. The public health authorities had largely ignored what he had to say. And he ' d made the case in a number of papers and done a number of studies, but nothing had really stuck. And part of â what ' s so interesting about this story to me is that in some ways, it ' s a great case study in how cultural change happens, how a good idea eventually comes to win out over much worse ideas. And Snow labored for a long time with this great insight that everybody ignored. And then on one day, August 28th of 1854, a young child, a five-month-old girl whose first name we don ' t know, we know her only as Baby Lewis, somehow contracted cholera, came down with cholera at 40 Broad Street. You can ' t really see it in this map, but this is the map that becomes the central focus in the second half of my book. It ' s in the middle of Soho, in this working class neighborhood, this little girl becomes sick and it turns out that the cesspool, that they still continue to have, despite the Nuisances Act, bordered on an extremely popular water pump, local watering hole that was well known for the best water in all of Soho, that all the residents from Soho and the surrounding neighborhoods would go to. And so this little girl inadvertently ended up contaminating the water in this popular pump, and one of the most terrifying outbreaks in the history of England erupted about two or three days later. Literally, 10 percent of the neighborhood died in seven days, and much more would have died if people hadn ' t fled after the initial outbreak kicked in. So it was this incredibly terrifying event. You had these scenes of entire families dying over the course of 48 hours of cholera, alone in their one-room apartments, in their little flats. Just an extraordinary, terrifying scene. Snow lived near there, heard about the outbreak, and in this amazing act of courage went directly into the belly of the beast because he thought an outbreak that concentrated could actually potentially end up convincing people that, in fact, the real menace of cholera was in the water supply and not in the air. He suspected an outbreak that concentrated would probably involve a single point source. One single thing that everybody was going to because it didn ' t have the traditional slower path of infections that you might expect. And so he went right in there and started interviewing people. He eventually enlisted the help of this amazing other figure, who ' s kind of the other protagonist of the book â this guy, Henry Whitehead, who was a local minister, who was not at all a man of science, but was incredibly socially connected ; he knew everybody in the neighborhood. And he managed to track down, Whitehead did, many of the cases of people who had drunk water from the pump, or who hadn ' t drunk water from the pump. And eventually Snow made a map of the outbreak. He found increasingly that people who drank from the pump were getting sick. People who hadn ' t drunk from the pump were not getting sick. And he thought about representing that as a kind of a table of statistics of people living in different neighborhoods, people who hadn ' t, you know, percentages of people who hadn ' t, but eventually he hit upon the idea that what he needed was something that you could see. Something that would take in a sense a higher-level view of all this activity that had been happening in the neighborhood. And so he created this map, which basically ended up representing all the deaths in the neighborhoods as black bars at each address. And you can see in this map, the pump right at the center of it and you can see that one of the residences down the way had about 15 people dead. And the map is actually a little bit bigger. As you get further and further away from the pump, the deaths begin to grow less and less frequent. And so you can see this something poisonous emanating out of this pump that you could see in a glance. And so, with the help of this map, and with the help of more evangelizing that he did over the next few years and that Whitehead did, eventually, actually, the authorities slowly started to come around. It took much longer than sometimes we like to think in this story, but by 1866, when the next big cholera outbreak came to London, the authorities had been convinced â in part because of this story, in part because of this map â that in fact the water was the problem. And they had already started building the sewers in London, and they immediately went to this outbreak and they told everybody to start boiling their water. And that was the last time that London has seen a cholera outbreak since. So, part of this story, I think â well, it ' s a terrifying story, it ' s a very dark story and it ' s a story that continues on in many of the developing cities of the world. It ' s also a story really that is fundamentally optimistic, which is to say that it ' s possible to solve these problems if we listen to reason, if we listen to the kind of wisdom of these kinds of maps, if we listen to people like Snow and Whitehead, if we listen to the locals who understand what ' s going on in these kinds of situations. And what it ended up doing is making the idea of large-scale metropolitan living a sustainable one. When people were looking at 10 percent of their neighborhoods dying in the space of seven days, there was a widespread consensus that this couldn ' t go on, that people weren ' t meant to live in cities of 2. 5 million people. But because of what Snow did, because of this map, because of the whole series of reforms that happened in the wake of this map, we now take for granted that cities have 10 million people, cities like this one are in fact sustainable things. We don ' t worry that New York City is going to collapse in on itself quite the way that, you know, Rome did, and be 10 percent of its size in 100 years or 200 years. And so that in a way is the ultimate legacy of this map. It ' s a map of deaths that ended up creating a whole new way of life, the life that we ' re enjoying here today. Thank you very much.

Text Sample 809: NEG Dim 5 – 2004 – Score -1.00 – 2004/t000459.txt

When you have 21 minutes to speak, two million years seems like a really long time. But evolutionarily, two million years is nothing. And yet, in two million years, the human brain has nearly tripled in mass, going from the one-and-a-quarter-pound brain of our ancestor here, Habilis, to the almost three-pound meatloaf everybody here has between their ears. What is it about a big brain that nature was so eager for every one of us to have one? Well, it turns out when brains triple in size, they don ' t just get three times bigger ; they gain new structures. And one of the main reasons our brain got so big is because it got a new part, called the"frontal lobe,"particularly, a part called the"pre-frontal cortex."What does a prefrontal cortex do for you that should justify the entire architectural overhaul of the human skull in the blink of evolutionary time? Well, it turns out the prefrontal cortex does lots of things, but one of the most important things it does is it ' s an experience simulator. Pilots practice in flight simulators so that they don ' t make real mistakes in planes. Human beings have this marvelous adaptation that they can actually have experiences in their heads before they try them out in real life. This is a trick that none of our ancestors could do, and that no other animal can do quite like we can. It ' s a marvelous adaptation. It ' s up there with opposable thumbs and standing upright and language as one of the things that got our species out of the trees and into the shopping mall. All of you have done this. Ben and Jerry ' s doesn ' t have"liver and onion"ice cream, and it ' s not because they whipped some up, tried it and went,"Yuck!"It ' s because, without leaving your armchair, you can simulate that flavor and say"yuck"before you make it. Let ' s see how your experience simulators are working. Let ' s just run a quick diagnostic before I proceed with the rest of the talk. Here ' s two different futures that I invite you to contemplate. You can try to simulate them and tell me which one you think you might prefer. One of them is winning the lottery. This is about 314 million dollars. And the other is becoming paraplegic. Just give it a moment of thought. You probably don ’ t feel like you need a moment of thought. Interestingly, there are data on these two groups of people, data on how happy they are. And this is exactly what you expected, isn ’ t it? But these aren ’ t the data. I made these up! These are the data. You failed the pop quiz, and you ’ re hardly five minutes into the lecture. Because the fact is that a year after losing the use of their legs and a year after winning the lotto, lottery winners and paraplegics are equally happy with their lives. Don ’ t feel too bad about failing the first pop quiz, because everybody fails all of the pop quizzes all of the time. The research that my laboratory has been doing, that economists and psychologists around the country have been doing, has revealed something really quite startling to us, something we call the”impact bias,”which is the tendency for the simulator to work badly, for the simulator to make you believe that different outcomes are more different than, in fact, they really are. From field studies to laboratory studies, we see that winning or losing an election, gaining or losing a romantic partner, getting or not getting a promotion, passing or not passing a college test, on and on, have far less impact, less intensity and much less duration than people expect them to have. A recent study — this almost floors me — a recent study showing how major life traumas affect people suggests that if it happened over three months ago, with only a few exceptions, it has no impact whatsoever on your happiness. Why? Because happiness can be synthesized. Sir Thomas Brown wrote in 1642,”I am the happiest man alive. I have that in me that can convert poverty to riches, adversity to prosperity. I am more invulnerable than Achilles ; fortune hath not one place to hit me.”What kind of remarkable machinery does this guy have in his head? Well, it turns out it ’ s precisely the same remarkable machinery that all of us have. Human beings have something that we might think of as a”psychological immune system,”a system of cognitive processes, largely nonconscious cognitive processes, that help them change their views of the world, so that they can feel better about the worlds in which they find themselves. Like Sir Thomas, you have this machine. Unlike Sir Thomas, you seem not to know it. We synthesize happiness, but we think happiness is a thing to be found. Now, you don ’ t need me to give you too many examples of people synthesizing happiness, I suspect, though I ’ m going to show you some experimental evidence. You don ’ t have to look very far for evidence. I took a copy of the”New York Times”and tried to find some instances of people synthesizing happiness. Here are three guys synthesizing happiness.”I ’ m better off physically, financially, mentally...””I don ’ t have one minute ’ s regret. It was a glorious experience.””I believe it turned out for the best.”Who are these characters who are so damn happy? The first one is Jim Wright. Some of you are old enough to remember : he was the chairman of the House of Representatives, and he resigned in disgrace when this young Republican named Newt Gingrich found out about a shady book deal that he had done. He lost everything. The most powerful Democrat in the country lost everything : he lost his money, he lost his power. What does he have to say all these years later about it?”I am so much better off physically, financially, mentally and in almost every other way.”What other way would there be to be better off? Vegetably? Minerally? Animally? He ’ s pretty much covered them there. Moreese Bickham is somebody you ’ ve never heard of. Moreese Bickham uttered these words upon being released. He was 78 years old. He ’ d spent 37 years in Louisiana State Penitentiary for a crime he didn ’ t commit. He was ultimately [released for good behavior halfway through his sentence.] What did he have to say about his experience?”I don ’ t have one minute ’ s regret. It was a glorious experience.”Glorious! This guy ’ s not saying,”There were some nice guys. They had a gym.””Glorious”— a word we usually reserve for something like a religious experience. Harry S. Langerman uttered these words. He ’ s somebody you might have known but didn ’ t, because in 1949, he read a little article in the paper about a hamburger stand owned by these two brothers named McDonald. And he thought,”That ’ s a really neat idea!”So he went to find them. They said,”We can give you a franchise on this for 3, 000 bucks.”Harry went back to New York, asked his brother, an investment banker, to loan him 3, 000 dollars, and his brother ’ s immortal words were,”You idiot, nobody eats hamburgers.”He wouldn ’ t lend him the money. Of course, six months later, Ray Kroc had exactly the same idea. It turns out, people do eat hamburgers, and Ray Kroc, for a while, became the richest man in America. And then, finally, some of you recognize this young photo of Pete Best, who was the original drummer for the Beatles, until they, you know, sent him out on an errand and snuck away and picked up Ringo on a tour. Well, in 1994, when Pete Best was interviewed — yes, he ’ s still a drummer ; yes, he ’ s a studio musician — he had this to say :”I ’ m happier than I would have been with the Beatles.”OK, there ’ s something important to be learned from these people, and it is the secret of happiness. Here it is, finally to be revealed. First : accrue wealth, power and prestige, then lose it. Second : spend as much of your life in prison as you possibly can. Third : make somebody else really, really rich. And finally : never, ever join the Beatles. Yeah, right. Because when people synthesize happiness, as these gentlemen seem to have done, we all smile at them, but we kind of roll our eyes and say,”Yeah, right, you never really wanted the job.””Oh yeah, right — you really didn ’ t have that much in common with her, and you figured that out just about the time she threw the engagement ring in your face.”We smirk, because we believe that synthetic happiness is not of the same quality as what we might call”natural happiness.”What are these terms? Natural happiness is what we get when we get what we wanted, and synthetic happiness is what we make when we don ’ t get what we wanted. And in our society, we have a strong belief that synthetic happiness is of an inferior kind. Why do we have that belief? Well, it ’ s very simple. What kind of economic engine would keep churning if we believed that not getting what we want could make us just as happy as getting it? With all apologies to my friend Matthieu Ricard, a shopping mall full of Zen monks is not going to be particularly profitable, because they don ’ t want stuff enough. I want to suggest to you that synthetic happiness is every bit as real and enduring as the kind of happiness you stumble upon when you get exactly what you were aiming for. Now, I ’ m a scientist, so I ’ m going to do this not with rhetoric, but by marinating you in a little bit of data. Let me first show you an experimental paradigm that ’ s used to demonstrate the synthesis of happiness among regular old folks. This isn ’ t mine, it ’ s a 50- year-old paradigm called the”free choice paradigm.”It ’ s very simple. You bring in, say, six objects, and you ask a **subject** to rank them from the most to the least liked. In this case, because this experiment uses them, these are Monet prints. Everybody ranks these Monet prints from the one they like the most to the one they like the least. Now we give you a choice :”We happen to have some extra prints in the closet. We ’ re going to give you one as your prize to take home. We happen to have number three and number four,”we tell the **subject**. This is a bit of a difficult choice, because neither one is preferred strongly to the other, but naturally, people tend to pick number three, because they liked it a little better than number four. Sometime later — it could be 15 minutes, it could be 15 days — the same stimuli are put before the **subject**, and the **subject** is asked to re-rank the stimuli.”Tell us how much you like them now.”What happens? Watch as happiness is synthesized.

This is the result that ' s been replicated over and over again. You ' re watching happiness be synthesized. Would you like to see it again? Happiness!"The one I got is really better than I thought! That other one I didn ' t get sucks!"That ' s the synthesis of happiness. Now, what ' s the right response to that?"Yeah, right!"Now, here ' s the experiment we did, and I hope this is going to convince you that"Yeah, right!"was not the right response. We did this experiment with a group of patients who had anterograde amnesia. These are hospitalized patients. Most of them have Korsakoff syndrome, a polyneuritic psychosis. They drank way too much, and they can ' t make new memories. They remember their childhood, but if you walk in and introduce yourself and then leave the room, when you come back, they don ' t know who you are. We took our Monet prints to the hospital. And we asked these patients to rank them from the one they liked the most to the one they liked the least. We then gave them the choice between number three and number four. Like everybody else, they said,"Gee, thanks Doc! That ' s great! I could use a new print. I ' ll take number three."We explained we would have number three mailed to them. We gathered up our materials, and we went out of the room and counted to a half hour. Back into the room, we say,"Hi, we ' re back."The patients, bless them, say,"Ah, Doc, I ' m sorry, I ' ve got a memory problem ; that ' s why I ' m here. If I ' ve met you before, I don ' t remember.""Really, Jim, you don ' t remember? I was just here with the Monet prints?""Sorry, Doc, I just don ' t have a clue.""No problem, Jim. All I want you to do is rank these for me, from the one you like the most to the one you like the least."What do they do? Well, let ' s first check and make sure they ' re really amnesiac. We ask these amnesiac patients to tell us which one they own, which one they chose last time, which one is theirs. And what we find is, amnesiac patients just guess. These are normal controls, where if I did this with you, all of you would know which print you chose. But if I do this with amnesiac patients, they don ' t have a clue. They can ' t pick their print out of a lineup. Here ' s what normal controls do : they synthesize happiness. Right? This is the change in liking score, the change from the first time they ranked to the second time they ranked. Normal controls show — that was the magic I showed you ; now I ' m showing it to you in graphical form —"The one I own is better than I thought. The one I didn ' t own, the one I left behind, is not as good as I thought."Amnesiacs do exactly the same thing. Think about this result. These people like better the one they own, but they don ' t know they own it."Yeah, right"is not the right response! What these people did when they synthesized happiness is they really, truly changed their affective, hedonic, aesthetic reactions to that poster. They ' re not just saying it because they own it, because they don ' t know they own it. When psychologists show you bars, you know that they are showing you averages of lots of people. And yet, all of us have this psychological immune system, this capacity to synthesize happiness, but some of us do this trick better than others. And some situations allow anybody to do it more effectively than other situations do. It turns out that freedom, the ability to make up your mind and change your mind, is the friend of natural happiness, because it allows you to choose among all those delicious futures and find the one that you would most enjoy. But freedom to choose, to change and make up your mind, is the enemy of synthetic happiness, and I ' m going to show you why. Dilbert already knows, of course."Dogbert ' s tech support. How may I abuse you?""My printer prints a blank page after every document.""Why complain about getting free paper?""Free? Aren ' t you just giving me my own paper?""Look at the quality of the free paper compared to your lousy regular paper! Only a fool or a liar would say that they look the same!""Now that you mention it, it does seem a little silkier!""What are you doing?""I ' m helping people accept the things they cannot change."Indeed. The psychological immune system works best when we are totally stuck, when we are trapped. This is the difference between dating and marriage. You go out on a date with a guy, and he picks his nose ; you don ’ t go out on another date. You ’ re married to a guy and he picks his nose? He has a heart of gold. Don ’ t touch the fruitcake! You find a way to be happy with what ’ s happened. Now, what I want to show you is that people don ’ t know this about themselves, and not knowing this can work to our supreme disadvantage. Here ’ s an experiment we did at Harvard. We created a black-and-white photography course, and we allowed students to come in and learn how to use a darkroom. So we gave them cameras, they went around campus, they took 12 pictures of their favorite professors and their dorm room and their dog, and all the other things they wanted to have Harvard memories of. They bring us the camera, we make up a contact sheet, they figure out which are the two best pictures. We now spend six hours teaching them about darkrooms, and they **blow** two of them up. They have two gorgeous 8 x 10 glossies of meaningful things to them, and we say,”Which one would you like to give up?””I have to give one up?””Yes, we need one as evidence of the class project. So you have to give me one. You have to make a choice. You get to keep one, and I get to keep one.”Now, there are two conditions in this experiment. In one case, the students are told,”But you know, if you want to change your mind, I ’ ll always have the other one here, and in the next four days, before I actually mail it to headquarters”— yeah,”headquarters”—”I ’ ll be glad to swap it out with you. In fact, I ’ ll come to your dorm room, just give me an email. Better yet, I ’ ll check with you. You ever want to change your mind, it ’ s totally returnable.”The other half of the students are told exactly the opposite :”Make your choice, and by the way, the mail is going out, gosh, in two minutes, to England. Your picture will be winging its way over the Atlantic. You will never see it again.”Half of the students in each of these conditions are asked to make predictions about how much they ’ re going to come to like the picture that they keep and the picture they leave behind. Other students are just sent back to their little dorm rooms and they are measured over the next three to six days on their satisfaction with the pictures. Look at what we find. First of all, here ’ s what students think is going to happen. They think they ’ re going to maybe come to like the picture they chose a little more than the one they left behind. But these are not statistically significant differences. It ’ s a very small increase, and it doesn ’ t much matter whether they were in the reversible or irreversible condition. Wrong-o. Bad simulators. Because here ’ s what ’ s really happening. Both right before the swap and five days later, people who are stuck with that picture, who have no choice, who can never change their mind, like it a lot. And people who are deliberating —”Should I return it? Have I gotten the right one? Maybe this isn ’ t the good one. Maybe I left the good one?”— have killed themselves. They don ’ t like their picture. In fact, even after the opportunity to swap has expired, they still don ’ t like their picture. Why? Because the [reversible] condition is not conducive to the synthesis of happiness. So here ’ s the final piece of this experiment. We bring in a whole new group of naive Harvard students and we say,”You know, we ’ re doing a photography course, and we can do it one of two ways. We could do it so that when you take the two pictures, you ’ d have four days to change your mind, or we ’ re doing another course where you take the two pictures and you make up your mind right away and you can never change it. Which course would you like to be in?”Duh! Sixty-six percent of the students, two-thirds, prefer to be in the course where they have the opportunity to change their mind. Hello? Sixty-six percent of the students choose to be in the course in which they will ultimately be deeply dissatisfied with the picture — because they do not know the conditions under which synthetic happiness grows. The Bard said everything best, of course, and he ’ s making my point here but he ’ s making it hyperbolically :”’ Tis nothing good or bad But thinking makes it so.”It ’ s nice poetry, but that can ’ t exactly be right. Is there really nothing good or bad? Is it really the case that gall bladder surgery and a trip to Paris are just the same thing? That seems like a one-question IQ test. They can ’ t be exactly the same. In more turgid prose, but closer to the truth, was the father of modern capitalism, Adam Smith, and he said this. This is worth contemplating :”The great source of both the misery and disorders of human life seems to arise from overrating the difference between one permanent situation and another. Some of these situations may, no doubt, deserve to be preferred to others, but none of them can deserve to be pursued with that passionate ardor which drives us to violate the rules either of prudence or of justice, or to corrupt the future tranquility of our minds, either by shame from the remembrance of our own folly, or by remorse for the horror of our own injustice.”In other words : yes, some things are better than others. We should have preferences that lead us into one future over another. But when those preferences drive us too hard and too fast because we have overrated the difference between these futures, we are at risk. When our ambition is bounded, it leads us to work joyfully. When our ambition is unbounded, it leads us to lie, to cheat, to steal, to hurt others, to sacrifice things of real value. When our fears are bounded, we ’ re prudent, we ’ re cautious, we ’ re thoughtful. When our fears are unbounded and overblown, we ’ re reckless, and we ’ re cowardly. The lesson I want to leave you with, from these data, is that our longings and our worries are both to some degree overblown, because we have within us the capacity to manufacture the very commodity we are constantly chasing when we choose experience. Thank you.

Text Sample 810: NEG Dim 5 – 2004 – Score 1.00 – 2004/t000098.txt

This machine, which we all have residing in our skulls, reminds me of an aphorism, of a comment of Woody Allen to ask about what is the very best thing to have within your skull. And it ’ s this machine. And it ’ s constructed for change. It ’ s all about change. It confers on us the ability to do things tomorrow that we can ’ t do today, things today that we couldn ’ t do yesterday. And of course it ’ s born stupid. The last time you were in the presence of a baby — this happens to be my granddaughter, Mitra. Isn ’ t she fabulous? But nonetheless when she popped out despite the fact that her brain had actually been progressing in its development for several months before on the basis of her experiences in the womb — nonetheless she had very limited abilities, as does every infant at the time of normal, natural full-term birth. If we were to assay her perceptual abilities, they would be crude. There is no real indication that there is any real thinking going on. In fact there is little evidence that there is any cognitive ability in a very young infant. Infants don ’ t respond to much. There is not really much of an indication in fact that there is a person on board. And they can only in a very primitive way, and in a very limited way control their movements. It would be several months before this infant could do something as simple as reach out and grasp under voluntary control an object and retrieve it, usually to the mouth. And it will be some months beforeward, and we see a long steady progression of the evolution from the first wiggles, to rolling over, and sitting up, and crawling, standing, walking, before we get to that magical point in which we can motate in the world. And yet, when we look forward in the brain we see really remarkable advance. By this age the brain can actually store. It has stored, recorded, can fastly retrieve the meanings of thousands, tens of thousands of objects, actions, and their relationships in the world. And those relationships can in fact be constructed in hundreds of thousands, potentially millions of ways. By this age the brain controls very refined perceptual abilities. And it actually has a growing repertoire of cognitive skills. This brain is very much a thinking machine. And by this age there is absolutely no question that this brain, it has a person on board. And in fact at this age it is substantially controlling its own self-development. And by this age we see a remarkable evolution in its capacity to control movement. Now movement has advanced to the point where it can actually control movement simultaneously, in a complex sequence, in complex ways as would be required for example for playing a complicated game, like soccer. Now this boy can bounce a soccer ball on his head. And where this boy comes from, Sao Paulo, Brazil, about 40 percent of boys of his age have this ability. You could go out into the community in Monterey, and you ' d have difficulty finding a boy that has this ability. And if you did he ' d probably be from Sao Paulo. That ' s all another way of saying that our individual skills and abilities are very much shaped by our environments. That environment extends into our contemporary culture, the thing our brain is challenged with. Because what we ' ve done in our personal evolutions is build up a large repertoire of specific skills and abilities that are specific to our own individual histories. And in fact they result in a wonderful differentiation in humankind, in the way that, in fact, no two of us are quite alike. Every one of us has a different set of acquired skills and abilities that all derive out of the plasticity, the adaptability of this really remarkable adaptive machine. In an adult brain of course we ' ve built up a large repertoire of mastered skills and abilities that we can perform more or less automatically from memory, and that define us as acting, moving, thinking creatures. Now we study this, as the nerdy, laboratory, university-based scientists that we are, by engaging the brains of animals like rats, or monkeys, or of this particularly curious creature — one of the more bizarre forms of life on earth — to engage them in learning new skills and abilities. And we try to track the changes that occur as the new skill or ability is acquired. In fact we do this in individuals of any age, in these different species — that is to say from infancies, infancy up to adulthood and old age. So we might engage a rat, for example, to acquire a new skill or ability that might involve the rat using its paw to master particular manual grasp behaviors just like we might examine a child and their ability to acquire the sub-skills, or the general overall skill of accomplishing something like mastering the ability to read. Or you might look in an older individual who has mastered a complex set of abilities that might relate to reading musical notation or performing the mechanical acts of performance that apply to musical performance. From these studies we defined two great epochs of the plastic history of the brain. The first great epoch is commonly called the"Critical Period."And that is the period in which the brain is setting up in its initial form its basic processing machinery. This is actually a period of dramatic change in which it doesn ' t take learning, per se, to drive the initial differentiation of the machinery of the brain. All it takes for example in the sound domain, is exposure to sound. And the brain actually is at the mercy of the sound environment in which it is reared. So for example I can rear an animal in an environment in which there is meaningless dumb sound, a repertoire of sound that I make up, that I make, just by exposure, artificially important to the animal and its young brain. And what I see is that the animal ' s brain sets up its initial processing of that sound in a form that ' s idealized, within the limits of its processing achievements to represent it in an organized and orderly way. The sound doesn ' t have to be valuable to the animal : I could raise the animal in something that could be hypothetically valuable, like the sounds that simulate the sounds of a native language of a child. And I see the brain actually develop a processor that is specialized — specialized for that complex array, a repertoire of sounds. It actually exaggerates their separateness of representation, in multi-dimensional neuronal representational terms. Or I can expose the animal to a completely meaningless and destructive sound. I can raise an animal under conditions that would be equivalent to raising a baby under a moderately loud ceiling fan, in the presence of continuous noise. And when I do that I actually specialize the brain to be a master processor for that meaningless sound. And I frustrate its ability to represent any meaningful sound as a consequence. Such things in the early history of babies occur in real babies. And they account for, for example the beautiful evolution of a language-specific processor in every normally developing baby. And so they also account for development of defective processing in a substantial population of children who are more limited, as a consequence, in their language abilities at an older age. Now in this early period of plasticity the brain actually changes outside of a learning context. I don ’ t have to be paying attention to what I hear. The input doesn ’ t really have to be meaningful. I don ’ t have to be in a behavioral context. This is required so the brain sets up it ’ s processing so that it can act differentially, so that it can act selectively, so that the creature that wears it, that carries it, can begin to operate on it in a selective way. In the next great epoch of life, which applies for most of life, the brain is actually refining its machinery as it masters a wide repertoire of skills and abilities. And in this epoch, which extends from late in the first year of life to death ; it ’ s actually doing this under behavioral control. And that ’ s another way of saying the brain has strategies that define the significance of the input to the brain. And it ’ s focusing on skill after skill, or ability after ability, under specific attentional control. It ’ s a function of whether a goal in a behavior is achieved or whether the individual is rewarded in the behavior. This is actually very powerful. This lifelong capacity for plasticity, for brain change, is powerfully expressed. It is the basis of our real differentiation, one individual from another. You can look down in the brain of an animal that ’ s engaged in a specific skill, and you can witness or document this change on a variety of levels. So here is a very simple experiment. It was actually conducted about five years ago in collaboration with scientists from the University of Provence in Marseilles. It ’ s a very simple experiment where a monkey has been trained in a task that involves it manipulating a tool that ’ s equivalent in its difficulty to a child learning to manipulate or handle a spoon. The monkey actually mastered the task in about 700 practice tries. So in the beginning the monkey could not perform this task at all. It had a success rate of about one in eight tries. Those tries were elaborate. Each attempt was substantially different from the other. But the monkey gradually developed a strategy. And 700 or so tries later the monkey is performing it flawlessly — never fails. He ’ s successful in his retrieval of food with this tool every time. At this point the task is being performed in a beautifully stereotyped way : very beautifully regulated and highly repeated, trial to trial. We can look down in the brain of the monkey. And we see that it ’ s distorted. We can track these changes, and have tracked these changes in many such behaviors across time. And here we see the distortion reflected in the map of the skin surfaces of the hand of the monkey. Now this is a map, down in the surface of the brain, in which, in a very elaborate experiment we ’ ve reconstructed the responses, location by location, in a highly detailed response mapping of the responses of its neurons. We see here a reconstruction of how the hand is represented in the brain. We ’ ve actually distorted the map by the exercise. And that is indicated in the pink. We have a couple fingertip surfaces that are larger. These are the surfaces the monkey is using to manipulate the tool. If we look at the selectivity of responses in the cortex of the monkey, we see that the monkey has actually changed the filter characteristics which represents input from the skin of the fingertips that are engaged. In other words there is still a single, simple representation of the fingertips in this most organized of cortical areas of the surface of the skin of the body. Monkey has like you have. And yet now it ' s represented in substantially finer grain. The monkey is getting more detailed information from these surfaces. And that is an unknown — unsuspected, maybe, by you — part of acquiring the skill or ability. Now actually we ' ve looked in several different cortical areas in the monkey learning this task. And each one of them changes in ways that are specific to the skill or ability. So for example we can look to the cortical area that represents input that ' s controlling the posture of the monkey. We look in cortical areas that control specific movements, and the sequences of movements that are required in the behavior, and so forth. They are all remodeled. They all become specialized for the task at hand. There are 15 or 20 cortical areas that are changed specifically when you learn a simple skill like this. And that represents in your brain, really massive change. It represents the change in a reliable way of the responses of tens of millions, possibly hundreds of millions of neurons in your brain. It represents changes of hundreds of millions, possibly billions of synaptic connections in your brain. This is constructed by physical change. And the level of construction that occurs is massive. Think about the changes that occur in the brain of a child through the course of acquiring their movement behavior abilities in general. Or acquiring their native language abilities. The changes are massive. What it ' s all about is the selective representations of things that are important to the brain. Because in most of the life of the brain this is under control of behavioral context. It ' s what you pay attention to. It ' s what ' s rewarding to you. It ' s what the brain regards, itself, as positive and important to you. It ' s all about cortical processing and forebrain specialization. And that underlies your specialization. That is why you, in your many skills and abilities, are a unique specialist : a specialist that ' s vastly different in your physical brain in detail than the brain of an individual 100 years ago ; enormously different in the details from the brain of the average individual 1, 000 years ago. Now, one of the characteristics of this change process is that information is always related to other inputs or information that is occurring in immediate time, in context. And that ' s because the brain is constructing representations of things that are correlated in little moments of time and that relate to one another in little moments of successive time. The brain is recording all information and driving all change in temporal context. Now overwhelmingly the most powerful context that ' s occurred in your brain is you. Billions of events have occurred in your history that are related in time to yourself as the receiver, or yourself as the actor, yourself as the thinker, yourself as the mover. Billions of times little pieces of sensation have come in from the surface of your body that are always associated with you as the receiver, and that result in the embodiment of you. You are constructed, your self is constructed from these billions of events. It ' s constructed. It ' s created in your brain. And it ' s created in the brain via physical change. This is a marvelously constructed thing that results in individual form because each one of us has vastly different histories, and vastly different experiences, that drive in to us this marvelous differentiation of self, of personhood. Now we ' ve used this research to try to understand not just how a normal person develops, and elaborates their skills and abilities, but also try to understand the origins of impairment, and the origins of differences or variations that might limit the capacities of a child, or an adult. I ' m going to talk about using these strategies to actually design brain plasticity-based approach to drive corrections in the machinery of a child that increases the competence of the child as a language receiver and user and, thereafter, as a reader. And I ’ m going to talk about experiments that involve actually using this brain science, first of all to understand how it contributes to the loss of function as we age. And then, by using it in a targeted approach we ’ re going to try to differentiate the machinery to recover function in old age. So the first example I ’ m going to talk about relates to children with learning impairments. We now have a large body of literature that demonstrates that the fundamental problem that occurs in the majority of children that have early language impairments, and that are going to struggle to learn to read, is that their language processor is created in a defective form. And the reason that it rises in a defective form is because early in the baby ’ s brain ’ s life the machine process is noisy. It ’ s that simple. It ’ s a signal-to-noise problem. Okay? And there are a lot of things that contribute to that. There are numerous inherited faults that could make the machine process noisier. Now I might say the noise problem could also occur on the basis of information provided in the world from the ears. If any — those of you who are older in the audience know that when I was a child we understood that a child born with a cleft palate was born with what we called mental retardation. We knew that they were going to be slow cognitively ; we knew they were going to struggle to learn to develop normal language abilities ; and we knew that they were going to struggle to learn to read. Most of them would be intellectual and academic failures. That ’ s disappeared. That no longer applies. That inherited weakness, that inherited condition has evaporated. We don ’ t hear about that anymore. Where did it go? Well, it was understood by a Dutch surgeon, about 35 years ago, that if you simply fix the problem early enough, when the brain is still in this initial plastic period so it can set up this machinery adequately, in this initial set up time in the critical period, none of that happens. What are you doing by operating on the cleft palate to correct it? You ’ re basically opening up the tubes that drain fluid from the middle ears, which have had them reliably full. Every sound the child hears uncorrected is muffled. It ’ s degraded. The child ’ s native language is such a case is not English. It ’ s not Japanese. It ’ s muffled English. It ’ s degraded Japanese. It ’ s crap. And the brain specializes for it. It creates a representation of language crap. And then the child is stuck with it. Now the crap doesn ’ t just happen in the ear. It can also happen in the brain. The brain itself can be noisy. It ’ s commonly noisy. There are many inherited faults that can make it noisier. And the native language for a child with such a brain is degraded. It ’ s not English. It ’ s noisy English. And that results in defective representations of sounds of words — not normal — a different strategy, by a machine that has different time constants and different space constants. And you can look in the brain of such a child and record those time constants. They are about an order of magnitude longer, about 11 times longer in duration on average, than in a normal child. Space constants are about three times greater. Such a child will have memory and cognitive deficits in this domain. Of course they will. Because as a receiver of language, they are receiving it and representing it, and in information it ’ s representing crap. And they are going to have poor reading skills. Because reading is dependent upon the translation of word sounds into this orthographic or visual representational form. If you don ’ t have a brain representation of word sounds that translation makes no sense. And you are going to have corresponding abnormal neurology. Then these children increasingly in evaluation after evaluation, in their operations in language, and their operations in reading — we document that abnormal neurology. The point is is that you can train the brain out of this. A way to think about this is you can actually re-refine the processing capacity of the machinery by changing it. Changing it in detail. It takes about 30 hours on the average. And we ’ ve accomplished that in about 430, 000 kids today. Actually, probably about 15, 000 children are being trained as we speak. And actually when you look at the impacts, the impacts are substantial. So here we ’ re looking at the normal distribution. What we ’ re most interested in is these kids on the left side of the distribution. This is from about 3, 000 children. You can see that most of the children on the left side of the distribution are moving into the middle or the right. This is in a broad assessment of their language abilities. This is like an IQ test for language. The impact in the distribution, if you trained every child in the United States, would be to shift the whole distribution to the right and narrow the distribution. This is a substantially large impact. Think of a classroom of children in the language arts. Think of the children on the slow side of the class. We have the potential to move most of those children to the middle or to the right side. In addition to accurate language training it also fixes memory and cognition speech fluency and speech production. And an important language dependent skill is enabled by this training — that is to say reading. And to a large extent it fixes the brain. You can look down in the brain of a child in a variety of tasks that scientists have at Stanford, and MIT, and UCSF, and UCLA, and a number of other institutions. And children operating in various language behaviors, or in various reading behaviors, you see for the most extent, for most children, their neuronal responses, complexly abnormal before you start, are normalized by the training. Now you can also take the same approach to address problems in aging. Where again the machinery is deteriorating now from competent machinery, it ’ s going south. Noise is increasing in the brain. And learning modulation and control is deteriorating. And you can actually look down on the brain of such an individual and witness a change in the time constants and space constants with which, for example, the brain is representing language again. Just as the brain came out of chaos at the beginning, it ’ s going back into chaos in the end. This results in declines in memory in cognition, and in postural ability and agility. It turns out you can train the brain of such an individual — this is a small population of such individuals — train equally intensively for about 30 hours. These are 80- to 90-year-olds. And what you see are substantial improvements of their immediate memory, of their ability to remember things after a delay, of their ability to control their attention, their language abilities and visual-spatial abilities. The overall neuropsychological index of these trained individuals in this population is about two standard deviations. That means that if you sit at the left side of the distribution, and I ’ m looking at your neuropyschological abilities, the average person has moved to the middle or the right side of the distribution. It means that most people who are at risk for senility, more or less immediately, are now in a protected position. My issues are to try to get to rescuing older citizens more completely and in larger numbers, because I think this can be done in this arena on a vast scale — and the same for kids. My main interest is how to elaborate this science to address other maladies. I ’ m specifically interested in things like autism, and cerebral palsy, these great childhood catastrophes. And in older age conditions like Parkinsonism, and in other acquired impairments like schizophrenia. Your issues as it relates to this science, is how to maintain your own high- functioning learning machine. And of course, a well-ordered life in which learning is a continuous part of it, is key. But also in your future is brain aerobics. Get ready for it. It ’ s going to be a part of every life not too far in the future, just like physical exercise is a part of every well organized life in the contemporary period. The other way that we will ultimately come to consider this literature and the science that is important to you is in a consideration of how to nurture yourself. Now that you know, now that science is telling us that you are in charge, that it ’ s under your control, that your happiness, your well-being, your abilities, your capacities, are capable of continuous modification, continuous improvement, and you ’ re the responsible agent and party. Of course a lot of people will ignore this advice. It will be a long time before they really understand it. Now that ’ s another issue and not my fault. Okay. Thank you.

Text Sample 811: NEG Dim 5 – 2004 – Score 1.00 – 2004/t000196.txt

I grew up in Europe, and World War II caught me when I was between seven and 10 years old. And I realized how few of the grown-ups that I knew were able to withstand the tragedies that the war visited on them â how few of them could even resemble a normal, contented, satisfied, happy life once their job, their home, their security was destroyed by the war. So I became interested in understanding what contributed to a life that was worth living. And I tried, as a child, as a teenager, to read philosophy and to get involved in art and religion and many other ways that I could see as a possible answer to that question. And finally I ended up encountering psychology by chance. I was at a ski resort in Switzerland without any money to actually enjoy myself, because the snow had melted and I didn ’ t have money to go to a movie. But I found that on the â I read in the newspapers that there was to be a presentation by someone in a place that I ’ d seen in the center of Zurich, and it was about flying saucers [that] he was going to talk. And I thought, well, since I can ’ t go to the movies, at least I will go for free to listen to flying saucers. And the man who talked at that evening lecture was very interesting. Instead of talking about little green men, he talked about how the psyche of the Europeans had been traumatized by the war, and now they ’ re projecting flying saucers into the sky. He talked about how the mandalas of ancient Hindu religion were kind of projected into the sky as an attempt to regain some sense of order after the chaos of war. And this seemed very interesting to me. And I started reading his books after that lecture. And that was Carl Jung, whose name or work I had no idea about. Then I came to this country to study psychology and I started trying to understand the roots of happiness. This is a typical result that many people have presented, and there are many variations on it. But this, for instance, shows that about 30 percent of the people surveyed in the United States since 1956 say that their life is very happy. And that hasn ’ t changed at all. Whereas the personal income, on a scale that has been held constant to accommodate for inflation, has more than doubled, almost tripled, in that period. But you find essentially the same results, namely, that after a certain basic point â which corresponds more or less to just a few 1, 000 dollars above the minimum poverty level â increases in material well-being don ’ t seem to affect how happy people are. In fact, you can find that the lack of basic resources, material resources, contributes to unhappiness, but the increase in material resources does not increase happiness. So my research has been focused more on â after finding out these things that actually corresponded to my own experience, I tried to understand : where â in everyday life, in our normal experience â do we feel really happy? And to start those studies about 40 years ago, I began to look at creative people â first artists and scientists, and so forth â trying to understand what made them feel that it was worth essentially spending their life doing things for which many of them didn ’ t expect either fame or fortune, but which made their life meaningful and worth doing. This was one of the leading composers of American music back in the ’ 70s. And the interview was 40 pages long. But this little excerpt is a very good summary of what he was saying during the interview. And it describes how he feels when composing is going well. And he says by describing it as an ecstatic state. Now,"ecstasy"in Greek meant simply to stand to the side of something. And then it became essentially an analogy for a mental state where you feel that you are not doing your ordinary everyday routines. So ecstasy is essentially a stepping into an alternative reality. And it ' s interesting, if you think about it, how, when we think about the civilizations that we look up to as having been pinnacles of human achievement â whether it ' s China, Greece, the Hindu civilization, or the Mayas, or Egyptians â what we know about them is really about their ecstasies, not about their everyday life. We know the temples they built, where people could come to experience a different reality. We know about the circuses, the arenas, the theaters. These are the remains of civilizations and they are the places that people went to experience life in a more concentrated, more ordered form. Now, this man doesn ' t need to go to a place like this, which is also â this place, this arena, which is built like a Greek amphitheatre, is a place for ecstasy also. We are participating in a reality that is different from that of the everyday life that we ' re used to. But this man doesn ' t need to go there. He needs just a piece of paper where he can put down little marks, and as he does that, he can imagine sounds that had not existed before in that particular combination. So once he gets to that point of beginning to create, like Jennifer did in her improvisation, a new reality â that is, a moment of ecstasy â he enters that different reality. Now he says also that this is so intense an experience that it feels almost as if he didn ' t exist. And that sounds like a kind of a romantic exaggeration. But actually, our nervous system is incapable of processing more than about 110 bits of information per second. And in order to hear me and understand what I ' m saying, you need to process about 60 bits per second. That ' s why you can ' t hear more than two people. You can ' t understand more than two people talking to you. Well, when you are really involved in this completely engaging process of creating something new, as this man is, he doesn ' t have enough attention left over to monitor how his body feels, or his problems at home. He can ' t feel even that he ' s hungry or tired. His body disappears, his identity disappears from his consciousness, because he doesn ' t have enough attention, like none of us do, to really do well something that requires a lot of concentration, and at the same time to feel that he exists. So existence is temporarily suspended. And he says that his hand seems to be moving by itself. Now, I could look at my hand for two weeks, and I wouldn ' t feel any awe or wonder, because I can ' t compose. So what it ' s telling you here is that obviously this automatic, spontaneous process that he ' s describing can only happen to someone who is very well trained and who has developed technique. And it has become a kind of a truism in the study of creativity that you can ' t be creating anything with less than 10 years of technical-knowledge immersion in a particular field. Whether it ' s mathematics or music, it takes that long to be able to begin to change something in a way that it ' s better than what was there before. Now, when that happens, he says the music just flows out. And because all of these people I started interviewing â this was an interview which is over 30 years old â so many of the people described this as a spontaneous flow that I called this type of experience the"flow experience."And it happens in different realms. For instance, a poet describes it in this form. This is by a student of mine who interviewed some of the leading writers and poets in the United States. And it describes the same effortless, spontaneous feeling that you get when you enter into this ecstatic state. This poet describes it as opening a door that floats in the sky â a very similar description to what Albert Einstein gave as to how he imagined the forces of relativity, when he was struggling with trying to understand how it worked. But it happens in other activities. For instance, this is another student of mine, Susan Jackson from Australia, who did work with some of the leading athletes in the world. And you see here in this description of an Olympic skater, the same essential description of the phenomenology of the inner state of the person. You don ’ t think ; it goes automatically, if you merge yourself with the music, and so forth. It happens also, actually, in the most recent book I wrote, called”Good Business,”where I interviewed some of the CEOs who had been nominated by their peers as being both very successful and very ethical, very socially responsible. You see that these people define success as something that helps others and at the same time makes you feel happy as you are working at it. And like all of these successful and responsible CEOs say, you can ’ t have just one of these things be successful if you want a meaningful and successful job. Anita Roddick is another one of these CEOs we interviewed. She is the founder of Body Shop, the natural cosmetics king. It ’ s kind of a passion that comes from doing the best and having flow while you ’ re working. This is an interesting little quote from Masaru Ibuka, who was at that time starting out Sony without any money, without a product â they didn ’ t have a product, they didn ’ t have anything, but they had an idea. And the idea he had was to establish a place of work where engineers can feel the joy of technological innovation, be aware of their mission to society and work to their heart ’ s content. I couldn ’ t improve on this as a good example of how flow enters the workplace. Now, when we do studies â we have, with other colleagues around the world, done over 8, 000 interviews of people â from Dominican monks, to blind nuns, to Himalayan climbers, to Navajo shepherds â who enjoy their work. And regardless of the culture, regardless of education or whatever, there are these seven conditions that seem to be there when a person is in flow. There ’ s this focus that, once it becomes intense, leads to a sense of ecstasy, a sense of clarity : you know exactly what you want to do from one moment to the other ; you get immediate feedback. You know that what you need to do is possible to do, even though difficult, and sense of time disappears, you forget yourself, you feel part of something larger. And once the conditions are present, what you are doing becomes worth doing for its own sake. In our studies, we represent the everyday life of people in this simple scheme. And we can measure this very precisely, actually, because we give people electronic pagers that go off 10 times a day, and whenever they go off you say what you ’ re doing, how you feel, where you are, what you ’ re thinking about. And two things that we measure is the amount of challenge people experience at that moment and the amount of skill that they feel they have at that moment. So for each person we can establish an average, which is the center of the diagram. That would be your mean level of challenge and skill, which will be different from that of anybody else. But you have a kind of a set point there, which would be in the middle. If we know what that set point is, we can predict fairly accurately when you will be in flow, and it will be when your challenges are higher than average and skills are higher than average. And you may be doing things very differently from other people, but for everyone that flow channel, that area there, will be when you are doing what you really like to do â play the piano, be with your best friend, perhaps work, if work is what provides flow for you. And then the other areas become less and less positive. Arousal is still good because you are over-challenged there. Your skills are not quite as high as they should be, but you can move into flow fairly easily by just developing a little more skill. So, arousal is the area where most people learn from, because that ’ s where they ’ re pushed beyond their comfort zone and to enter that â going back to flow â then they develop higher skills. Control is also a good place to be, because there you feel comfortable, but not very excited. It ’ s not very challenging any more. And if you want to enter flow from control, you have to increase the challenges. So those two are ideal and complementary areas from which flow is easy to go into. The other combinations of challenge and skill become progressively less optimal. Relaxation is fine â you still feel OK. Boredom begins to be very aversive and apathy becomes very negative : you don ’ t feel that you ’ re doing anything, you don ’ t use your skills, there ’ s no challenge. Unfortunately, a lot of people ’ s experience is in apathy. The largest single contributor to that experience is watching television ; the next one is being in the bathroom, sitting. Even though sometimes watching television about seven to eight percent of the time is in flow, but that ’ s when you choose a program you really want to watch and you get feedback from it. So the question we are trying to address â and I ’ m way over time â is how to put more and more of everyday life in that flow channel. And that is the kind of challenge that we ’ re trying to understand. And some of you obviously know how to do that spontaneously without any advice, but unfortunately a lot of people don ’ t. And that ’ s what our mandate is, in a way, to do. Thank you.

Text Sample 812: NEG Dim 5 – 2008 – Score -1.00 – 2008/t000203.txt

I spent the better part of a decade looking at American responses to mass atrocity and genocide. And I ’ d like to start by sharing with you one moment that to me sums up what there is to know about American and democratic responses to mass atrocity. And that moment came on April 21, 1994. So 14 years ago, almost, in the middle of the Rwandan genocide, in which 800, 000 people would be systematically exterminated by the Rwandan government and some extremist militia. On April 21, in the New York Times, the paper reported that somewhere between 200, 000 and 300, 000 people had already been killed in the genocide. It was in the paper — not on the front page. It was a lot like the Holocaust coverage, it was buried in the paper. Rwanda itself was not seen as newsworthy, and amazingly, genocide itself was not seen as newsworthy. But on April 21, a wonderfully honest moment occurred. And that was that an American congresswoman named Patricia Schroeder from Colorado met with a group of journalists. And one of the journalists said to her, what ’ s up? What ’ s going on in the U. S. government? Two to 300, 000 people have just been exterminated in the last couple of weeks in Rwanda. It ’ s two weeks into the genocide at that time, but of course, at that time you don ’ t know how long it ’ s going to last. And the journalist said, why is there so little response out of Washington? Why no hearings, no denunciations, no people getting arrested in front of the Rwandan embassy or in front of the White House? What ’ s the deal? And she said — she was so honest — she said,”It ’ s a great question. All I can tell you is that in my congressional office in Colorado and my office in Washington, we ’ re getting hundreds and hundreds of calls about the endangered ape and gorilla population in Rwanda, but nobody is calling about the people. The phones just aren ’ t ringing about the people.”And the reason I give you this moment is there ’ s a deep truth in it. And that truth is, or was, in the 20th century, that while we were beginning to develop endangered species movements, we didn ’ t have an endangered people ’ s movement. We had Holocaust education in the schools. Most of us were groomed not only on images of nuclear catastrophe, but also on images and knowledge of the Holocaust. There ’ s a museum, of course, on the Mall in Washington, right next to Lincoln and Jefferson. I mean, we have owned Never Again culturally, appropriately, interestingly. And yet the politicization of Never Again, the operationalization of Never Again, had never occurred in the 20th century. And that ’ s what that moment with Patricia Schroeder I think shows : that if we are to bring about an end to the world ’ s worst atrocities, we have to make it such. There has to be a role — there has to be the creation of political noise and political costs in response to massive crimes against humanity, and so forth. So that was the 20th century. Now here — and this will be a relief to you at this point in the afternoon — there is good news, amazing news, in the 21st century, and that is that, almost out of nowhere, there has come into being an anti-genocide movement, an anti-genocide constituency, and one that looks destined, in fact, to be permanent. It grew up in response to the atrocities in Darfur. It is comprised of students. There are something like 300 anti-genocide chapters on college campuses around the country. It ’ s bigger than the anti-apartheid movement. There are something like 500 high school chapters devoted to stopping the genocide in Darfur. Evangelicals have joined it. Jewish groups have joined it.”Hotel Rwanda”watchers have joined it. It is a cacophonous movement. To call it a movement, as with all movements, perhaps, is a little misleading. It ’ s diverse. It ’ s got a lot of different approaches. It ’ s got all the ups and the downs of movements. But it has been amazingly successful in one regard, in that it has become, it has congealed into this endangered people ’ s movement that was missing in the 20th century. It sees itself, such as it is, the it, as something that will create the impression that there will be political cost, there will be a political price to be paid, for allowing genocide, for not having an heroic imagination, for not being an upstander but for being, in fact, a bystander. Now because it ’ s student-driven, there ’ s some amazing things that the movement has done. They have launched a divestment campaign that has now convinced, I think, 55 universities in 22 states to divest their holdings of stocks with regard to companies doing business in Sudan. They have a 1-800-GENOCIDE number — this is going to sound very kitsch, but for those of you who may not be, I mean, may be apolitical, but interested in doing something about genocide, you dial 1-800-GENOCIDE and you type in your zip code, and you don ’ t even have to know who your congressperson is. It will refer you directly to your congressperson, to your U. S. senator, to your governor where divestment legislation is pending. They ’ ve lowered the transaction costs of stopping genocide. I think the most innovative thing they ’ ve introduced recently are genocide grades. And it takes students to introduce genocide grades. So what you now have when a Congress is in session is members of Congress calling up these 19-year-olds or 24-year-olds and saying, I ’ m just told I have a D minus on genocide ; what do I do to get a C? I just want to get a C. Help me. And the students and the others who are part of this incredibly energized base are there to answer that, and there ’ s always something to do. Now, what this movement has done is it has extracted from the Bush administration from the United States, at a time of massive over-stretch — military, financial, diplomatic — a whole series of commitments to Darfur that no other country in the world is making. For instance, the referral of the crimes in Darfur to the International Criminal Court, which the Bush administration doesn ’ t like. The expenditure of 3 billion dollars in refugee camps to try to keep, basically, the people who ’ ve been displaced from their homes by the Sudanese government, by the so-called Janjaweed, the militia, to keep those people alive until something more durable can be achieved. And recently, or now not that recently, about six months ago, the authorization of a peacekeeping force of 26, 000 that will go. And that ’ s all the Bush administration ’ s leadership, and it ’ s all because of this bottom-up pressure and the fact that the phones haven ’ t stopped ringing from the beginning of this crisis. The bad news, however, to this question of will evil prevail, is that evil lives on. The people in those camps are surrounded on all sides by so-called Janjaweed, these men on horseback with spears and

Kalashnikovs. Women who go to get firewood in order to heat the humanitarian aid in order to feed their families — humanitarian aid, the dirty secret of it is it has to be heated, really, to be edible — are themselves **subjected** to rape, which is a tool of the genocide that is being used. And the peacekeepers I ’ ve mentioned, the force has been authorized, but almost no country on Earth has stepped forward since the authorization to actually put its troops or its police in harm ’ s way. So we have achieved an awful lot relative to the 20th century, and yet far too little relative to the gravity of the crime that is unfolding as we sit here, as we speak. Why the limits to the movement? Why is what has been achieved, or what the movement has done, been necessary but not sufficient to the crime? I think there are a couple — there are many reasons — but a couple just to focus on briefly. The first is that the movement, such as it is, stops at America ’ s borders. It is not a global movement. It does not have too many compatriots abroad who themselves are asking their governments to do more to stop genocide. And the Holocaust culture that we have in this country makes Americans, sort of, more prone to, I think, want to bring Never Again to life. The guilt that the Clinton administration expressed, that Bill Clinton expressed over Rwanda, created a space in our society for a consensus that Rwanda was bad and wrong and we wish we had done more, and that is something that the movement has taken advantage of. European governments, for the most part, haven ’ t acknowledged responsibility, and there ’ s nothing to kind of to push back and up against. So this movement, if it ’ s to be durable and global, will have to cross borders, and you will have to see other citizens in democracies, not simply resting on the assumption that their government would do something in the face of genocide, but actually making it such. Governments will never gravitate towards crimes of this magnitude naturally or eagerly. As we saw, they haven ’ t even gravitated towards protecting our ports or reigning in loose nukes. Why would we expect in a bureaucracy that it would orient itself towards distant suffering? So one reason is it hasn ’ t gone global. The second is, of course, that at this time in particular in America ’ s history, we have a credibility problem, a legitimacy problem in international institutions. It is structurally really, really hard to do, as the Bush administration rightly does, which is to denounce genocide on a Monday and then describe water boarding on a Tuesday as a no-brainer and then turn up on Wednesday and look for troop commitments. Now, other countries have their own reasons for not wanting to get involved. Let me be clear. They ’ re in some ways using the Bush administration as an alibi. But it is essential for us to be a leader in this sphere, of course to restore our standing and our leadership in the world. The recovery ’ s going to take some time. We have to ask ourselves, what now? What do we do going forward as a country and as citizens in relationship to the world ’ s worst places, the world ’ s worst suffering, killers, and the kinds of killers that could come home to roost sometime in the future? The place that I turned to answer that question was to a man that many of you may not have ever heard of, and that is a Brazilian named Sergio Vieira de Mello who, as Chris said, was **blown** up in Iraq in 2003. He was the victim of the first-ever suicide bomb in Iraq. It ’ s hard to remember, but there was actually a time in the summer of 2003, even after the U. S. invasion, where, apart from looting, civilians were relatively safe in Iraq. Now, who was Sergio? Sergio Vieira de Mello was his name. In addition to being Brazilian, he was described to me before I met him in 1994 as someone who was a cross between James Bond on the one hand and Bobby Kennedy on the other. And in the U. N., you don ’ t get that many people who actually manage to merge those qualities. He was James Bond-like in that he was ingenious. He was drawn to the flames, he chased the flames, he was like a moth to the flames. Something of an adrenalin junkie. He was successful with women. He was Bobby Kennedy-like because in some ways one could never tell if he was a realist masquerading as an idealist or an idealist masquerading as a realist, as people always wondered about Bobby Kennedy and John Kennedy in that way. What he was was a decathlete of nation-building, of problem-solving, of troubleshooting in the world ’ s worst places and in the world ’ s most broken places. In failing states, genocidal states, under-governed states, precisely the kinds of places that threats to this country exist on the horizon, and precisely the kinds of places where most of the world ’ s suffering tends to get concentrated. These are the places he was drawn to. He moved with the headlines. He was in the U. N. for 34 years. He joined at the age of 21. Started off when the causes in the wars du jour in the ’ 70s were wars of independence and decolonization. He was there in Bangladesh dealing with the outflow of millions of refugees — the largest refugee flow in history up to that point. He was in Sudan when the civil war broke out there. He was in Cyprus right after the Turkish invasion. He was in Mozambique for the War of Independence. He was in Lebanon. Amazingly, he was in Lebanon — the U. N. base was used — Palestinians staged attacks out from behind the U. N. base. Israel then invaded and overran the U. N. base. Sergio was in Beirut when the U. S. Embassy was hit by the first-ever suicide attack against the United States. People date the beginning of this new era to 9 / 11, but surely 1983, with the attack on the US Embassy and the Marine barracks — which Sergio witnessed — those are, in fact, in some ways, the dawning of the era that we find ourselves in today. From Lebanon he went to Bosnia in the ’ 90s. The issues were, of course, ethnic sectarian violence. He was the first person to negotiate with the Khmer Rouge. Talk about evil prevailing. I mean, here he was in the room with the embodiment of evil in Cambodia. He negotiates with the Serbs. He actually crosses so far into this realm of talking to evil and trying to convince evil that it doesn ’ t need to prevail that he earns the nickname — not Sergio but Serbio while he ’ s living in the Balkans and conducting these kinds of negotiations. He then goes to Rwanda and to Congo in the aftermath of the genocide, and he ’ s the guy who has to decide — huh, OK, the genocide is over ; 800, 000 people have been killed ; the people responsible are fleeing into neighboring countries — into Congo, into Tanzania. I ’ m Sergio, I ’ m a humanitarian, and I want to feed those — well, I don ’ t want to feed the killers but I want to feed the two million people who are with them, so we ’ re going to go, we ’ re going to set up camps, and we ’ re going to supply humanitarian aid. But, uh-oh, the killers are within the camps. Well, I ’ d like to separate the sheep from the wolves. Let me go door-to-door to the international community and see if anybody will give me police or troops to do the separation. And their response, of course, was no more than we wanted to stop the genocide and put our troops in harm ’ s way to do that, nor do we now want to get in the way and pluck genocidaires from camps. So then you have to make the decision. Do you turn off the international spigot of life support and risk two million civilian lives? Or do you continue feeding the civilians, knowing that the genocidaires are in the camps, literally sharpening their knives for future battle? What do you do? It ’ s all lesser-evil terrain in these broken places. Late ’ 90s : nation-building is the cause du jour. He ’ s the guy put in charge. He ’ s the Paul Bremer or the Jerry Bremer of first Kosovo and then East Timor. He governs the places. He ’ s the viceroy. He has to decide on tax policy, on currency, on border patrol, on policing. He has to make all these judgments. He ’ s a Brazilian in these places. He speaks seven languages. He ’ s been up to that point in 14 war zones so he ’ s positioned to make better judgments, perhaps, than people who have never done that kind of work. But nonetheless, he is the cutting edge of our experimentation with doing good with very few resources being brought to bear in, again, the world ’ s worst places. And then after Timor, 9 / 11 has happened, he ’ s named U. N. Human Rights Commissioner, and he has to balance liberty and security and figure out, what do you do when the most powerful country in the United Nations is bowing out of the Geneva Conventions, bowing out of international law? Do you denounce? Well, if you denounce, you ’ re probably never going to get back in the room. Maybe you stay reticent. Maybe you try to charm President Bush — and that ’ s what he did. And in so doing he earned himself, unfortunately, his final and tragic appointment to Iraq — the one that resulted in his death. One note on his death, which is so devastating, is that despite predicating the war on Iraq on a link between Saddam Hussein and terrorism in 9 / 11, believe it or not, the Bush administration or the invaders did no planning, no pre-war planning, to respond to terrorism. So Sergio — this receptacle of all of this learning on how to deal with evil and how to deal with brokenness, lay under the rubble for three and a half hours without rescue. Stateless. The guy who tried to help the stateless people his whole career. Like a refugee. Because he represents the U. N. If you represent everyone, in some ways you represent no one. You ’ re un-owned. And what the American — the most powerful military in the history of mankind was able to muster for his rescue, believe it or not, was literally these heroic two American soldiers went into the shaft. Building was shaking. One of them had been at 9 / 11 and lost his buddies on September 11th, and yet went in and risked his life in order to save Sergio. But all they had was a woman ’ s handbag — literally one of those basket handbags — and they tied it to a curtain rope from one of the offices at U. N. headquarters, and created a pulley system into this shaft in this quivering building in the interests of rescuing this person, the person we most need to turn to now, this shepherd, at a time when so many of us feel like we ’ re lacking guidance. And this was the pulley system. This was what we were able to muster for Sergio. The good news, for what it ’ s worth, is after Sergio and 21 others were killed that day in the attack on the U. N., the military created a search and rescue unit that had the cutting equipment, the shoring wood, the cranes, the things that you would have needed to do the rescue. But it was too late for Sergio. I want to wrap up, but I want to close with what I take to be the four lessons from Sergio ’ s life on this question of how do we prevent evil from prevailing, which is how I would have framed the question. Here ’ s this guy who got a 34- year head start thinking about the kinds of questions we as a country are grappling with, we as citizens are grappling with now. What do we take away? First, I think, is his relationship to, in fact, evil is something to learn from. He, over the course of his career, changed a great deal. He had a lot of flaws, but he was very adaptive. I think that was his greatest quality. He started as somebody who would denounce harmdoers, he would charge up to people who were violating international law, and he would say, you ’ re violating, this is the U. N. Charter. Don ’ t you see it ’ s unacceptable what you ’ re doing? And they would laugh at him because he didn ’ t have the power of states, the power of any military or police. He just had the rules, he had the norms, and he tried to use them. And in Lebanon, Southern Lebanon in ’ 82, he said to himself and to everybody else, I will never use the word”unacceptable”again. I will never use it. I will try to make it such, but I will never use that word again. But he lunged in the opposite direction. He started, as I mentioned, to get in the room with evil, to not denounce, and became almost obsequious when he won the nickname Serbio, for instance, and even when he negotiated with the Khmer Rouge would black-box what had occurred prior to entering the room. But by the end of his life, I think he had struck a balance that we as a country can learn from. Be in the room, don ’ t be afraid of talking to your adversaries, but don ’ t bracket what happened before you entered the room.

Don ’ t black-box history. Don ’ t check your principles at the door. And I think that ’ s something that we have to be in the room, whether it ’ s Nixon going to China or Khrushchev and Kennedy or Reagan and Gorbachev. All the great progress in this country with relation to our adversaries has come by going into the room. And it doesn ’ t have to be an act of weakness. You can actually do far more to build an international coalition against a harmdoer or a wrongdoer by being in the room and showing to the rest of the world that that person, that regime, is the problem and that you, the United States, are not the problem. Second take-away from Sergio ’ s life, briefly. What I take away, and this in some ways is the most important, he espoused and exhibited a reverence for dignity that was really, really unusual. At a micro level, the individuals around him were visible. He saw them. At a macro level, he thought, you know, we talk about democracy promotion, but we do it in a way sometimes that ’ s an affront to people ’ s dignity. We put people on humanitarian aid and we boast about it because we ’ ve spent three billion. It ’ s incredibly important, those people would no longer be alive if the United States, for instance, hadn ’ t spent that money in Darfur, but it ’ s not a way to live. If we think about dignity in our conduct as citizens and as individuals with relation to the people around us, and as a country, if we could inject a regard for dignity into our dealings with other countries, it would be something of a revolution. Third point, very briefly. He talked a lot about freedom from fear. And I recognize there is so much to be afraid of. There are so many genuine threats in the world. But what Sergio was talking about is, let ’ s calibrate our relationship to the threat. Let ’ s not hype the threat ; let ’ s actually see it clearly. We have reason to be afraid of melting ice caps. We have reason to be afraid that we haven ’ t secured loose nuclear material in the former Soviet Union. Let ’ s focus on what are the legitimate challenges and threats, but not lunge into bad decisions because of a panic, of a fear. In times of fear, for instance, one of the things Sergio used to say is, fear is a bad advisor. We lunge towards the extremes when we aren ’ t operating and trying to, again, calibrate our relationship to the world around us. Fourth and final point : he somehow, because he was working in all the world ’ s worst places and all lesser evils, had a humility, of course, and an awareness of the complexity of the world around him. I mean, such an acute awareness of how hard it was. How Sisyphean this task was of mending, and yet aware of that complexity, humbled by it, he wasn ’ t paralyzed by it. And we as citizens, as we go through this experience of the kind of, the crisis of confidence, crisis of competence, crisis of legitimacy, I think there ’ s a temptation to pull back from the world and say, ah, Katrina, Iraq — we don ’ t know what we ’ re doing. We can ’ t afford to pull back from the world. It ’ s a question of how to be in the world. And the lesson, I think, of the anti-genocide movement that I mentioned, that is a partial success but by no means has it achieved what it has set out to do — it ’ ll be many decades, probably, before that happens — but is that if we want to see change, we have to become the change. We can ’ t rely upon our institutions to do the work of necessarily talking to adversaries on their own without us creating a space for that to happen, for having respect for dignity, and for bringing that combination of humility and a sort of emboldened sense of responsibility to our dealings with the rest of the world. So will evil prevail? Is that the question? I think the short answer is : no, not unless we let it. Thank you.

Text Sample 813: NEG Dim 5 – 2008 – Score 1.00 – 2008/t000289.txt
I grew up to study the brain because I have a brother who has been diagnosed

with a brain disorder, schizophrenia. And as a sister and later, as a scientist, I wanted to understand, why is it that I can take my dreams, I can connect them to my reality, and I can make my dreams come true? What is it about my brother ' s brain and his schizophrenia that he cannot connect his dreams to a common and shared reality, so they instead become delusion? So I dedicated my career to research into the severe mental illnesses. And I moved from my home state of Indiana to Boston, where I was working in the lab of Dr. Francine Benes, in the Harvard Department of Psychiatry. And in the lab, we were asking the question,"What are the biological differences between the brains of individuals who would be diagnosed as normal control, as compared with the brains of individuals diagnosed with schizophrenia, schizoaffective or bipolar disorder?"So we were essentially mapping the microcircuitry of the brain : which cells are communicating with which cells, with which chemicals, and then in what quantities of those chemicals? So there was a lot of meaning in my life because I was performing this type of research during the day, but then in the evenings and on the weekends, I traveled as an advocate for NAMI, the National Alliance on Mental Illness. But on the morning of December 10, 1996, I woke up to discover that I had a brain disorder of my own. A blood vessel exploded in the left half of my brain. And in the course of four hours, I watched my brain completely deteriorate in its ability to process all information. On the morning of the hemorrhage, I could not walk, talk, read, write or recall any of my life. I essentially became an infant in a woman ' s body. If you ' ve ever seen a human brain, it ' s obvious that the two hemispheres are completely separate from one another. And I have brought for you a real human brain. So this is a real human brain. This is the front of the brain, the back of brain with the spinal cord hanging down, and this is how it would be positioned inside of my head. And when you look at the brain, it ' s obvious that the two cerebral cortices are completely separate from one another. For those of you who understand computers, our right hemisphere functions like a parallel processor, while our left hemisphere functions like a serial processor. The two hemispheres do communicate with one another through the corpus callosum, which is made up of some 300 million axonal fibers. But other than that, the two hemispheres are completely separate. Because they process information differently, each of our hemispheres think about different things, they care about different things, and, dare I say, they have very different personalities. Excuse me. Thank you. It ' s been a joy. Assistant : It has been. Our right human hemisphere is all about this present moment. It ' s all about"right here, right now."Our right hemisphere, it thinks in pictures and it learns kinesthetically through the movement of our bodies. Information, in the form of energy, streams in simultaneously through all of our sensory systems and then it explodes into this enormous collage of what this present moment looks like, what this present moment smells like and tastes like, what it feels like and what it sounds like. I am an energy-being connected to the energy all around me through the consciousness of my right hemisphere. We are energy- beings connected to one another through the consciousness of our right hemispheres as one human family. And right here, right now, we are brothers and sisters on this planet, here to make the world a better place. And in this moment we are perfect, we are whole and we are beautiful. My left hemisphere, our left hemisphere, is a very different place. Our left hemisphere thinks linearly and methodically. Our left hemisphere is all about the past and it ' s all about the future. Our left hemisphere is designed to take that enormous collage of the present moment and start picking out details, and more details about those details. It then categorizes and organizes all that information, associates it with everything in the past we ' ve ever learned, and projects into the future all of our possibilities. And our left hemisphere thinks in language. It ' s that ongoing brain chatter that connects me and my internal world to my external world. It ’ s that little voice that says to me,”Hey, you ’ ve got to remember to pick up bananas on your way home. I need them in the morning.”It ’ s that calculating intelligence that reminds me when I have to do my laundry. But perhaps most important, it ’ s that little voice that says to me,”I am. I am.”And as soon as my left hemisphere says to me”I am,”I become separate. I become a single solid individual, separate from the energy flow around me and separate from you. And this was the portion of my brain that I lost on the morning of my stroke. On the morning of the stroke, I woke up to a pounding pain behind my left eye. And it was the kind of caustic pain that you get when you bite into ice cream. And it just gripped me — and then it released me. And then it just gripped me — and then it released me. And it was very unusual for me to ever experience any kind of pain, so I thought,”OK, I ’ ll just start my normal routine.”So I got up and I jumped onto my cardio glider, which is a full-body, full-exercise machine. And I ’ m jamming away on this thing, and I ’ m realizing that my hands look like primitive claws grasping onto the bar. And I thought,”That ’ s very peculiar.”And I looked down at my body and I thought,”Whoa, I ’ m a weird- looking thing.”And it was as though my consciousness had shifted away from my normal perception of reality, where I ’ m the person on the machine having the experience, to some esoteric space where I ’ m witnessing myself having this experience. And it was all very peculiar, and my headache was just getting worse. So I get off the machine, and I ’ m walking across my living room floor, and I realize that everything inside of my body has slowed way down. And every step is very rigid and very deliberate. There ’ s no fluidity to my pace, and there ’ s this constriction in my area of perception, so I ’ m just focused on internal systems. And I ’ m standing in my bathroom getting ready to step into the shower, and I could actually hear the dialogue inside of my body. I heard a little voice saying,”OK. You muscles, you ’ ve got to contract. You muscles, you relax.”And then I lost my balance, and I ’ m propped up against the wall. And I look down at my arm and I realize that I can no longer define the boundaries of my body. I can ’ t define where I begin and where I end, because the atoms and the molecules of my arm blended with the atoms and molecules of the wall. And all I could detect was this energy — energy. And I ’ m asking myself,”What is wrong with me? What is going on?”And in that moment, my left hemisphere brain chatter went totally silent. Just like someone took a remote control and pushed the mute button. Total silence. And at first I was shocked to find myself inside of a silent mind. But then I was immediately captivated by the magnificence of the energy around me. And because I could no longer identify the boundaries of my body, I felt enormous and expansive. I felt at one with all the energy that was, and it was beautiful there. Then all of a sudden my left hemisphere comes back online and it says to me,”Hey! We ’ ve got a problem! We ’ ve got to get some help.”And I ’ m going,”Ahh! I ’ ve got a problem!”So it ’ s like,”OK, I ’ ve got a problem.”But then I immediately drifted right back out into the consciousness — and I affectionately refer to this space as La La Land. But it was beautiful there. Imagine what it would be like to be totally disconnected from your brain chatter that connects you to the external world. So here I am in this space, and my job, and any stress related to my job — it was gone. And I felt lighter in my body. And imagine all of the relationships in the external world and any stressors related to any of those — they were gone. And I felt this sense of peacefulness. And imagine what it would feel like to lose 37 years of emotional baggage! Oh! I felt euphoria — euphoria. It was beautiful. And again, my left hemisphere comes online and it says,”Hey! You ’ ve got to pay attention. We ’ ve got to get help.”And I ’ m thinking,”I ’ ve got to get help. I ’ ve got to focus.”So I get out of the shower and I mechanically dress and I ’ m walking around my apartment, and I ’ m thinking,”I ’ ve got to get to work. Can I drive?”And in that moment, my right arm went totally paralyzed by my side. Then I realized,”Oh my gosh! I ’ m having a stroke!”And the next thing my brain says to me is, Wow! This is so cool! This is so cool! How many brain scientists have the opportunity to study their own brain from the inside out?”And then it crosses my mind,”But I ’ m a very busy woman!””I don ’ t have time for a stroke!”So I ’ m like,”OK, I can ’ t stop the stroke from happening, so I ’ ll do this for a week or two, and then I ’ ll get back to my routine. OK. So I ’ ve got to call help. I ’ ve got to call work.”I couldn ’ t remember the number at work, so I remembered, in my office I had a business card with my number. So I go into my business room, I pull out a three-inch stack of business cards. And I ’ m looking at the card on top and even though I could see clearly in my mind ’ s eye what my business card looked like, I couldn ’ t tell if this was my card or not, because all I could see were pixels. And the pixels of the words blended with the pixels of the background and the pixels of the symbols, and I just couldn ’ t tell. And then I would wait for what I call a wave of clarity. And in that moment, I would be able to reattach to normal reality and I could tell that ’ s not the card... that ’ s not the card. It took me 45 minutes to get one inch down inside of that stack of cards. In the meantime, for 45 minutes, the hemorrhage is getting bigger in my left hemisphere. I do not understand numbers, I do not understand the telephone, but it ’ s the only plan I have. So I take the phone pad and I put it right here. I take the business card, I put it right here, and I ’ m matching the shape of the squiggles on the card to the shape of the squiggles on the phone pad. But then I would drift back out into La La Land, and not remember when I came back if I ’ d already dialed those numbers. So I had to wield my paralyzed arm like a stump and cover the numbers as I went along and pushed them, so that as I would come back to normal reality, I ’ d be able to tell,”Yes, I ’ ve already dialed that number.”Eventually, the whole number gets dialed and I ’ m listening to the phone, and my colleague picks up the phone and he says to me,”Woo woo woo woo.”And I think to myself,”Oh my gosh, he sounds like a Golden Retriever!”And so I say to him — clear in my mind, I say to him :”This is Jill! I need help!”And what comes out of my voice is,”Woo woo woo woo woo.”I ’ m thinking,”Oh my gosh, I sound like a Golden Retriever.”So I couldn ’ t know — I didn ’ t know that I couldn ’ t speak or understand language until I tried. So he recognizes that I need help and he gets me help. And a little while later, I am riding in an ambulance from one hospital across Boston to [Massachusetts] General Hospital. And I curl up into a little fetal ball. And just like a balloon with the last bit of air, just right out of the balloon, I just felt my energy lift and just I felt my spirit surrender. And in that moment, I knew that I was no longer the choreographer of my life. And either the doctors rescue my body and give me a second chance at life, or this was perhaps my moment of transition. When I woke later that afternoon, I was shocked to discover that I was still alive. When I felt my spirit surrender, I said goodbye to my life. And my mind was now suspended between two very opposite planes of reality. Stimulation coming in through my sensory systems felt like pure pain. Light burned my brain like wildfire, and sounds were so loud and chaotic that I could not pick a voice out from the background noise, and I just wanted to escape. Because I could not identify the position of my body in space, I felt enormous and expansive, like a genie just liberated from her bottle. And my spirit soared free, like a great whale gliding through the sea of silent euphoria. Nirvana. I found Nirvana. And I remember thinking, there ’ s no way I would ever be able to squeeze the enormousness of myself back inside this tiny little body. But then I realized,”But I ’ m still alive! I ’ m still alive, and I have found Nirvana. And if I have found Nirvana and I ’ m still alive, then everyone who is alive can find Nirvana."And I pictured a world filled with beautiful, peaceful, compassionate, loving people who knew that they could come to this space at any time. And that they could purposely choose to step to the right of their left hemispheres — and find this peace. And then I realized what a tremendous gift this experience could be, what a stroke of insight this could be to how we live our lives. And it motivated me to recover. Two and a half weeks after the hemorrhage, the surgeons went in, and they removed a blood clot the size of a golf ball that was pushing on my language centers. Here I am with my mama, who is a true angel in my life. It took me eight years to completely recover. So who are we? We are the life-force power of the universe, with manual dexterity and two cognitive minds. And we have the power to choose, moment by moment, who and how we want to be in the world. Right here, right now, I can step into the consciousness of my right hemisphere, where we are. I am the life-force power of the universe. I am the life-force power of the 50 trillion beautiful molecular geniuses that make up my form, at one with all that is. Or, I can choose to step into the consciousness of my left hemisphere, where I become a single individual, a solid. Separate from the flow, separate from you. I am Dr. Jill Bolte Taylor : intellectual, neuroanatomist. These are the"we"inside of me. Which would you choose? Which do you choose? And when? I believe that the more time we spend choosing to run the deep inner-peace circuitry of our right hemispheres, the more peace we will project into the world, and the more peaceful our planet will be. And I thought that was an idea worth spreading. Thank you.

Text Sample 814: NEG Dim 5 – 2008 – Score 1.00 – 2008/t000292.txt
How many of you have seen the Alfred Hitchcock film"The Birds"? Any of you get really freaked out by that? You might want to leave now. So this is a vending machine for crows. Over the past few days, many of you have been asking,"How did you come to this? How did you get started doing this?"It started, as with many great ideas, or many ideas you can ’ t get rid of, anyway, at a cocktail party. About 10 years ago, I was at a cocktail party with a friend of mine. We were sitting there, and he was complaining about the crows that were all over his yard and making a big mess. And he was telling me we ought to eradicate these things, kill them, because they ’ re making a mess. I said that was stupid, maybe we should just train them to do something useful. And he said that was impossible. And I ’ m sure I ’ m in good company in finding that tremendously annoying, when someone tells you it ’ s impossible. So I spent the next 10 years reading about crows in my spare time. And after 10 years of this, my wife said,"You ’ ve got to do this thing you ’ ve been talking about, and build the vending machine."So I did. But part of the reason I found this interesting is, I started noticing that we ’ re very aware of all the species that are going extinct on the planet as a result of human habitation expansion, and no one seems to be paying attention to all the species that are actually living ; they ’ re surviving. And I ’ m talking specifically about synanthropic species, which have adapted specifically for human ecologies, species like rats and cockroaches and crows. And as I started looking at them, I was finding that they had hyper-adapted. They ’ d become extremely adept at living with us. And in return, we just tried to kill them all the time. And in doing so, we were breeding them for parasitism. We were giving them all sorts of reasons to adapt new ways. So, for example, rats are incredibly responsive breeders. And cockroaches, as anyone who ’ s tried to get rid of them knows, have become really immune to the poisons that we ’ re using. So I thought, let ’ s build something that

’ s mutually beneficial ; something that we can both benefit from, and find some way to make a new relationship with these species. So I built the vending machine. But the story of the vending machine is a little more interesting if you know more about crows. It turns out, crows aren ’ t just surviving with human beings ; they ’ re actually thriving. They ’ re found everywhere on the planet except for the Arctic and the southern tip of South America. And in all that area, they ’ re only rarely found breeding more than five kilometers away from human beings. So we may not think about them, but they ’ re always around. And not surprisingly, given the human population growth, more than half of the human population is living in cities now. And out of those, nine-tenths of the human growth population is occurring in cities. We ’ re seeing a population boom with crows. So bird counts are indicating that we might be seeing up to exponential growth in their numbers. So that ’ s no great surprise. But what was really interesting to me was to find out that the birds were adapting in a pretty unusual way. And I ’ ll give you an example of that. This is Betty. She ’ s a New Caledonian crow. And these crows use sticks in the wild to get insects and whatnot out of pieces of wood. Here, she ’ s trying to get a piece of meat out of a tube. But the researchers had a problem. They messed up and left just a stick of wire in there. And she hadn ’ t had the opportunity to do this before. You see, it wasn ’ t working very well. So she adapted. Now, this is completely unprompted ; she had never seen this done before. No one taught her to bend this into a hook or had shown her how it could happen. But she did it all on her own. So keep in mind — she ’ s never seen this done. Right. Yeah. All right. So that ’ s the part where the researchers freak out. It turns out, we ’ ve been finding more and more that crows are really intelligent. Their brains are in the same proportion as chimpanzee brains are. There ’ s all kinds of anecdotes for the different kinds of intelligence they have. For example, in Sweden, crows will wait for fishermen to drop lines through holes in the ice. And when the fishermen move off, the crows fly down, reel up the lines, and eat the fish or the bait. It ’ s pretty annoying for the fishermen. On an entirely different tack, at University of Washington a few years ago, they were doing an experiment where they captured some crows on campus. Some students went out, netted some crows, brought them in, weighed and measured them, and let them back out again. And they were entertained to discover that for the rest of the week, whenever these particular students walked around campus, these crows would caw at them and run around, and make their life kind of miserable. They were significantly less entertained when this went on for the next week. And the next month. And after summer break. Until they finally graduated and left campus, and — glad to get away, I ’ m sure — came back sometime later, and found the crows still remembered them. So, the moral being : don ’ t piss off crows. So now, students at the University of Washington that are studying these crows, do so with a giant wig and a big mask. It ’ s fairly interesting. So we know these crows are really smart, but the more I dug into this, the more I found that they actually have an even more significant adaptation. Video : Crows have become highly skilled at making a living in these new urban environments. In this Japanese city, they have devised a way of eating a food that normally they can ’ t manage : drop it among the traffic. The problem now is collecting the bits, without getting run over. Wait for the light to stop the traffic. Then, collect your cracked nut in safety. Joshua Klein : Yeah, pretty interesting. What ’ s significant about this isn ’ t that crows are using cars to crack nuts. In fact, that ’ s old hat for crows. This happened about 10 years ago in a place called Sendai City, at a driving school in the suburbs of Tokyo. And since that time, all the crows in the neighborhood are picking up this behavior. Now every crow within five kilometers is standing by a sidewalk, waiting to collect its lunch. So they ’ re learning from each other. And research bears this out. Parents seem to be teaching their young. They learn from their peers, they learn from their enemies. If I have a little extra time, I ’ ll tell you about a case of crow infidelity that illustrates that nicely. The point being, they ’ ve developed cultural adaptation. And as we heard yesterday, that ’ s the Pandora ’ s box that ’ s getting human beings in trouble, and we ’ re starting to see it with them. They ’ re able to very quickly and very flexibly adapt to new challenges and new resources in their environment, which is really useful if you live in a city. So we know that there ’ s lots of crows. We found out they ’ re really smart and they can teach each other. When all this became clear, I realized the only obvious thing to do is build a vending machine. So that ’ s what we did. This is a vending machine for crows. And it uses Skinnerian training to shape their behavior over four stages. It ’ s pretty simple. Basically, what happens is that we put this out in a field or someplace where there ’ s lots of crows. We put coins and peanuts all around the base of the machine. Crows eventually come by, eat the peanuts, and get used to the machine being there. Eventually, they eat all the peanuts. Then they see peanuts here on the feeder tray, and hop up and help themselves. Then they leave, the machine spits up more coins and peanuts, and life is dandy if you ’ re a crow — you can come back anytime and get yourself a peanut. So when they get really used to that, we move on to the crows coming back. Now they ’ re used to the sound of the machine ; they keep coming back and digging out peanuts from the pile of coins that ’ s there. When they get really happy about this, we stymie them. We move to the third stage, where we only give them a coin. Now, like most of us who have gotten used to a good thing, this really pisses them off. So they do what they do in nature when they ’ re looking for something : sweep things out of the way with their beak. They do that here, and that knocks the coins down the slot. When that happens, they get a peanut. This goes on for some time. The crows learn that all they have to do is show up, wait for the coin to come out, put it in the slot, then get their peanut. When they ’ re good and comfortable with that, we move to the final stage, where they show up and nothing happens. This is where we see the difference between crows and other animals. Squirrels, for example, would show up, look for the peanut, go away. Come back, look for the peanut, go away. They do this maybe half a dozen times before they get bored, and then they go off and play in traffic. Crows, on the other hand, show up and they try and figure it out. They know this machine has been messing with them through three different stages of behavior. They figure there must be more to it. So they poke at it and peck at it. And eventually some crow gets a bright idea :”Hey, there ’ s lots of coins lying around from the first stage, hops down, picks it up, drops it in the slot, and we ’ re off to the races. That crow enjoys a temporary monopoly on peanuts, until his friends figure out how to do it, and then there we go. So, what ’ s significant about this to me isn ’ t that we can train crows to pick up peanuts. Mind you, there ’ s 216 million dollars ’ worth of change lost every year, but I ’ m not sure I can depend on that ROI from crows. Instead, I think we should look a little bit larger. I think crows can be trained to do other things. For example, why not train them to pick up garbage after stadium events? Or find expensive components from discarded electronics? Or maybe do search and rescue? The main point of all this for me is, we can find mutually beneficial systems for these species. We can find ways to interact with these other species that doesn ’ t involve exterminating them, but involves finding an equilibrium with them that ’ s a useful balance. Thanks very much.

Text Sample 815: NEG Dim 5 – 1998 – Score 2.00 – 1998/t000118.txt

Back in 1992, I started working for a company called Interval Research, which was just then being founded by David Lidell and Paul Allen as a for-profit research enterprise in Silicon Valley. I met with David to talk about what I might do in his company. I was just coming out of a failed virtual reality business and supporting myself by being on the speaking circuit and writing books — after twenty years or so in the computer game industry having ideas that people didn ' t think they could sell. And David and I discovered that we had a question in common, that we really wanted the answer to, and that was,"Why hasn ' t anybody built any computer games for little girls?"Why is that? It can ' t just be a giant sexist conspiracy. These people aren ' t that smart. There ' s six billion dollars on the table. They would go for it if they could figure out how. So, what is the deal here? And as we thought about our goals — I should say that Interval is really a humanistic institution, in the classical sense that humanism, at its best, finds a way to combine clear-eyed empirical research with a set of core values that fundamentally love and respect people. The basic idea of humanism is the improvable quality of life ; that we can do good things, that there are things worth doing because they ' re good things to do, and that clear-eyed empiricism can help us figure out how to do them. So, contrary to popular belief, there is not a conflict of interest between empiricism and values. And Interval Research is kind of the living example of how that can be true. So David and I decided to go find out, through the best research we could muster, what it would take to get a little girl to put her hands on a computer, to achieve the level of comfort and ease with the technology that little boys have because they play video games. We spent two and a half years conducting research ; we spent another year and a half in advance development. Then we formed a spin-off company. In the research phase of the project at Interval, we partnered with a company called Cheskin Research, and these people — Davis Masten and Christopher Ireland — changed my mind entirely about what market research was and what it could be. They taught me how to look and see, and they did not do the incredibly stupid thing of saying to a child,"Of all these things we already make you, which do you like best?"— which gives you zero answers that are usable. So, what we did for the first two and a half years was four things : We did an extensive review of the literature in related fields, like cognitive psychology, spatial cognition, gender studies, play theory, sociology, primatology. Thank you Frans de Waal, wherever you are, I love you and I ' d give anything to meet you. After we had done that with a pretty large team of people and discovered what we thought the salient issues were with girls and boys and playing — because, after all, that ' s really what this is about — we moved to the second phase of our work, where we interviewed adult experts in academia, some of the people who ' d produced the literature that we found relevant. Also, we did focus groups with people who were on the ground with kids every day, like playground supervisors. We talked to them, confirmed some hypotheses and identified some serious questions about gender difference and play. Then we did what I consider to be the heart of the work : interviewed 1, 100 children, boys and girls, ages seven to 12, all over the United States — except for Silicon Valley, Boston and Austin because we knew that their little families would have millions of computers in them and they wouldn ' t be a representative sample. And at the end of those remarkable conversations with kids and their best friends across the United States, after two years, we pulled together some survey data from another 10, 000 children, drew up a set up of what we thought were the key findings of our research, and spent another year transforming them into design heuristics, for designing computer-based products — and, in fact, any kind of products — for little girls, ages eight to 12. And we spent that time designing interactive prototypes for computer software and testing them with little girls. In 1996, in November, we formed the company Purple Moon which was a spinoff of Interval Research, and our chief investors were Interval Research, Vulcan Northwest, Institutional Venture Partners and Allen and Company. We launched a website on September 2nd that has now served 25 million pages, and has 42, 000 registered young girl users. They visit an average of one and a half times a day, spend an average of 35 minutes a visit, and look at 50 pages a visit. So we feel that we ’ ve formed a successful online community with girls. We launched two titles in October —”Rockett ’ s New School”— the first of a series of products — is about a character called Rockett beginning her first day of school in eighth grade at a brand new place, with a blank slate, which allows girls to play with the question of,”What will I be like when I ’ m older?””What ’ s it going to be like to be in high school or junior high school? Who are my friends?”; to exercise the love of social complexity and the narrative intelligence that drives most of their play behavior ; and which embeds in it values about noticing that we have lots of choices in our lives and the ways that we conduct ourselves. The other title that we launched is called”Secret Paths in the Forest,”which addresses the more fantasy-oriented, inner lives of girls. These two titles both showed up in the top 50 entertainment titles in PC Data — entertainment titles in PC Data in December, right up there with”John Madden Football,”which thrills me to death. So, we ’ re real, and we ’ ve touched several hundreds of thousands of little girls. We ’ ve made half-a-billion impressions with marketing and PR for this brand, Purple Moon. Ninety-six percent of them, roughly, have been positive ; four percent of them have been”other.”I want to talk about the other, because the politics of this enterprise, in a way, have been the most fascinating part of it, for me. There are really two kinds of negative reviews that we ’ ve received. One kind of reviewer is a male gamer who thinks he knows what games ought to be, and won ’ t show the product to little girls. The other kind of reviewer is a certain flavor of feminist who thinks they know what little girls ought to be. And so it ’ s funny to me that these interesting, odd bedfellows have one thing in common : they don ’ t listen to little girls. They haven ’ t looked at children and they ’ re certainly not demonstrating any love for them. I ’ d like to play you some voices of little girls from the two-and-a-half years of research that we did — actually, some of the voices are more recent. And these voices will be accompanied by photographs that they took for us of their lives, of the things that they value and care about. These are pictures the girls themselves never saw, but they gave to us This is the stuff those reviewers don ’ t know about and aren ’ t listening to and this is the kind of research I recommend to those who want to do humanistic work. Girl 1 : Yeah, my character is usually a tomboy. Hers is more into boys. Girl 2 : Uh, yeah. Girl 1 : We have — in the very beginning of the whole game, always we do this : we each have a piece of paper ; we write down our name, our age — are we rich, very rich, not rich, poor, medium, wealthy, boyfriends, dogs, pets — what else — sisters, brothers, and all those. Girl 2 : Divorced — parents divorced, maybe. Girl 3 : This is my pretend [unclear] one. Girl 4 : We make a school newspaper on the computer. Girl 5 : For a girl ’ s game also usually they ’ ll have really pretty scenery with clouds and flowers. Girl 6 : Like, if you were a girl and you were really adventurous and a real big tomboy, you would think that girls ’ games were kinda sissy. Girl 7 : I run track, I played soccer, I play basketball, and I love a lot of things to do. And sometimes I feel like I can ’ t really enjoy myself unless it ’ s like a vacation, like when I get Mondays and all those days off. Girl 8 : Well, sometimes there is a lot of stuff going on because I have music lessons and I ’ m on swim team — all this different stuff that I have to do, and sometimes it gets overwhelming. Girl 9 :

My friend Justine kinda took my friend Kelly, and now they ’ re being mean to me. Girl 10 : Well, sometimes it gets annoying when your brothers and sisters, or brother or sister, when they copy you and you get your idea first and they take your idea and they do it themselves. Girl 11 : Because my older sister, she gets everything and, like, when I ask my mom for something, she ’ ll say,”No”—all the time. But she gives my sister everything. Brenda Laurel : I want to show you, real quickly, just a minute of”Rockett ’ s Tricky Decision,”which went gold two days ago. Let ’ s hope it ’ s really stable. This is the second day in Rockett ’ s life. The reason I ’ m showing you this is I ’ m hoping that the scene that I ’ m going to show you will look familiar and sound familiar, now that you ’ ve listened to some girls ’ voices. And you can see how we ’ ve tried to incorporate the issues that matter to them in the game that we ’ ve created. Miko : Hey Rockett! C ’ mere! Rockett : Hi Miko! What ’ s going on? Miko : Did you hear about Nakilia ’ s big Halloween party this weekend? She asked me to make sure you knew about it. Nakilia invited Reuben too, but — Rockett : But what? Isn ’ t he coming? Miko : I don ’ t think so. I mean, I heard his band is playing at another party the same night. Rockett : Really? What other party? Girl : Max ’ s party is going to be so cool, Whitney. He ’ s invited all the best people. BL : I ’ m going to fast-forward to the decision point because I know I don ’ t have a lot of time. After this awful event occurs, Rocket gets to decide how she feels about it. Rockett : Who ’ d want to show up at that party anyway? I could get invited to that party any day if I wanted to. Gee, I doubt I ’ ll make Max ’ s best people list. BL : OK, so we ’ re going to emotionally navigate. If we were playing the game, that ’ s what we ’ d do. If at any time during the game we want to learn more about the characters, we can go into this hidden hallway, and I ’ ll quickly just show you the interface. We can, for example, go find Miko ’ s locker and get some more information about her. Oops, I turned the wrong way. But you get the general idea of the product. I wanted to show you the ways, innocuous as they seem, in which we ’ re incorporating what we ’ ve learned about girls — their desires to experience greater emotional flexibility, and to play around with the social complexity of their lives. I want to make the point that what we ’ re giving girls, I think, through this effort, is a kind of validation, a sense of being seen. And a sense of the choices that are available in their lives. We love them. We see them. We ’ re not trying to tell them who they ought to be. But we ’ re really, really happy about who they are. It turns out they ’ re really great. I want to close by showing you a videotape that ’ s a version of a future game in the Rockett series that our graphic artists and design people put together, that we feel would please that four percent of reviewers.”Rockett 28!”Rockett : It ’ s like I ’ m just waking up, you know? BL : Thanks.

Text Sample 816: NEG Dim 5 – 1998 – Score 2.00 – 1998/t000250.txt

As a clergyman, you can imagine how out of place I feel. I feel like a fish out of water, or maybe an owl out of the air. I was preaching in San Jose some time ago, and my friend Mark Kvamme, who helped introduce me to this conference, brought several CEOs and leaders of some of the companies here in the Silicon Valley to have breakfast with me, or I with them. And I was so stimulated. And had such — it was an eye-opening experience to hear them talk about the world that is yet to come through technology and science. I know that we ’ re near the end of this conference, and some of you may be wondering why they have a speaker from the field of religion. Richard can answer that, because he made that decision. But some years ago I was on an elevator in Philadelphia, coming down. I was to address a conference at a hotel. And on that elevator a man said,"I hear Billy Graham is staying in this hotel."And another man looked in my direction and said,"Yes, there he is. He ' s on this elevator with us."And this man looked me up and down for about 10 seconds, and he said,"My, what an anticlimax!"I hope that you won ' t feel that these few moments with me is not a — is an anticlimax, after all these tremendous talks that you ' ve heard, and addresses, which I intend to listen to every one of them. But I was on an airplane in the east some years ago, and the man sitting across the aisle from me was the mayor of Charlotte, North Carolina. His name was John Belk. Some of you will probably know him. And there was a drunk man on there, and he got up out of his seat two or three times, and he was making everybody upset by what he was trying to do. And he was slapping the stewardess and pinching her as she went by, and everybody was upset with him. And finally, John Belk said,"Do you know who ' s sitting here?"And the man said,"No, who?"He said,"It ' s Billy Graham, the preacher."He said,"You don ' t say!"And he turned to me, and he said,"Put her there!"He said,"Your sermons have certainly helped me."And I suppose that that ' s true with thousands of people. I know that as you have been peering into the future, and as we ' ve heard some of it here tonight, I would like to live in that age and see what is going to be. But I won ' t, because I ' m 80 years old. This is my eightieth year, and I know that my time is brief. I have phlebitis at the moment, in both legs, and that ' s the reason that I had to have a little help in getting up here, because I have Parkinson ' s disease in addition to that, and some other problems that I won ' t talk about. But this is not the first time that we ' ve had a technological revolution. We ' ve had others. And there ' s one that I want to talk about. In one generation, the nation of the people of Israel had a tremendous and dramatic change that made them a great power in the Near East. A man by the name of David came to the throne, and King David became one of the great leaders of his generation. He was a man of tremendous leadership. He had the favor of God with him. He was a brilliant poet, philosopher, writer, soldier — with strategies in battle and conflict that people study even today. But about two centuries before David, the Hittites had discovered the secret of smelting and processing of iron, and, slowly, that skill spread. But they wouldn ' t allow the Israelis to look into it, or to have any. But David changed all of that, and he introduced the Iron Age to Israel. And the Bible says that David laid up great stores of iron, and which archaeologists have found, that in present-day Palestine, there are evidences of that generation. Now, instead of crude tools made of sticks and stones, Israel now had iron plows, and sickles, and hoes and military weapons. And in the course of one generation, Israel was completely changed. The introduction of iron, in some ways, had an impact a little bit like the microchip has had on our generation. And David found that there were many problems that technology could not solve. There were many problems still left. And they ' re still with us, and you haven ' t solved them, and I haven ' t heard anybody here speak to that. How do we solve these three problems that I ' d like to mention? The first one that David saw was human evil. Where does it come from? How do we solve it? Over again and again in the Psalms, which Gladstone said was the greatest book in the world, David describes the evils of the human race. And yet he says,"He restores my soul."Have you ever thought about what a contradiction we are? On one hand, we can probe the deepest secrets of the universe and dramatically push back the frontiers of technology, as this conference vividly demonstrates. We ' ve seen under the sea, three miles down, or galaxies hundreds of billions of years out in the future. But on the other hand, something is wrong. Our battleships, our soldiers, are on a frontier now, almost ready to go to war with Iraq. Now, what causes this? Why do we have these wars in every generation, and in every part of the world? And revolutions? We can ’ t get along with other people, even in our own families. We find ourselves in the paralyzing grip of self-destructive habits we can ’ t break. Racism and injustice and violence sweep our world, bringing a tragic harvest of heartache and death. Even the most sophisticated among us seem powerless to break this cycle. I would like to see Oracle take up that, or some other technological geniuses work on this. How do we change man, so that he doesn ’ t lie and cheat, and our newspapers are not filled with stories of fraud in business or labor or athletics or wherever? The Bible says the problem is within us, within our hearts and our souls. Our problem is that we are separated from our Creator, which we call God, and we need to have our souls restored, something only God can do. Jesus said,”For out of the heart come evil thoughts : murders, sexual immorality, theft, false testimonies, slander.”The British philosopher Bertrand Russell was not a religious man, but he said,”It ’ s in our hearts that the evil lies, and it ’ s from our hearts that it must be plucked out.”Albert Einstein — I was just talking to someone, when I was speaking at Princeton, and I met Mr. Einstein. He didn ’ t have a doctor ’ s degree, because he said nobody was qualified to give him one. But he made this statement. He said,”It ’ s easier to denature plutonium than to denature the evil spirit of man.”And many of you, I ’ m sure, have thought about that and puzzled over it. You ’ ve seen people take beneficial technological advances, such as the Internet we ’ ve heard about tonight, and twist them into something corrupting. You ’ ve seen brilliant people devise computer viruses that bring down whole systems. The Oklahoma City bombing was simple technology, horribly used. The problem is not technology. The problem is the person or persons using it. King David said that he knew the depths of his own soul. He couldn ’ t free himself from personal problems and personal evils that included murder and adultery. Yet King David sought God ’ s forgiveness, and said,”You can restore my soul.”You see, the Bible teaches that we ’ re more than a body and a mind. We are a soul. And there ’ s something inside of us that is beyond our understanding. That ’ s the part of us that yearns for God, or something more than we find in technology. Your soul is that part of you that yearns for meaning in life, and which seeks for something beyond this life. It ’ s the part of you that yearns, really, for God. I find [that] young people all over the world are searching for something. They don ’ t know what it is. I speak at many universities, and I have many questions and answer periods, and whether it ’ s Cambridge, or Harvard, or Oxford — I ’ ve spoken at all of those universities. I ’ m going to Harvard in about three or four — no, it ’ s about two months from now — to give a lecture. And I ’ ll be asked the same questions that I was asked the last few times I ’ ve been there. And it ’ ll be on these questions : where did I come from? Why am I here? Where am I going? What ’ s life all about? Why am I here? Even if you have no religious belief, there are times when you wonder that there ’ s something else. Thomas Edison also said,”When you see everything that happens in the world of science, and in the working of the universe, you cannot deny that there ’ s a captain on the bridge.”I remember once, I sat beside Mrs. Gorbachev at a White House dinner. I went to Ambassador Dobrynin, whom I knew very well. And I ’ d been to Russia several times under the Communists, and they ’ d given me marvelous freedom that I didn ’ t expect. And I knew Mr. Dobrynin very well, and I said,”I ’ m going to sit beside Mrs. Gorbachev tonight. What shall I talk to her about?”And he surprised me with the answer. He said,”Talk to her about religion and philosophy. That ’ s what she ’ s really interested in.”I was a little bit surprised, but that evening that ’ s what we talked about, and it was a stimulating conversation. And afterward, she said,”You know, I ’ m an atheist, but I know that there ’ s something up there higher than we are.”The second problem that King David realized he could not solve was the problem of human suffering. Writing the oldest book in the world was Job, and he said,"Man is born unto trouble as the sparks fly upward."Yes, to be sure, science has done much to push back certain types of human suffering. But I ' m — in a few months, I ' ll be 80 years of age. I admit that I ' m very grateful for all the medical advances that have kept me in relatively good health all these years. My doctors at the Mayo Clinic urged me not to take this trip out here to this — to be here. I haven ' t given a talk in nearly four months. And when you speak as much as I do, three or four times a day, you get rusty. That ' s the reason I ' m using this podium and using these notes. Every time you ever hear me on the television or somewhere, I ' m ad-libbing. I ' m not reading. I never read an address. I never read a speech or a talk or a lecture. I talk ad lib. But tonight, I ' ve got some notes here so that if I begin to forget, which I do sometimes, I ' ve got something I can turn to. But even here among us, most — in the most advanced society in the world, we have poverty. We have families that self-destruct, friends that betray us. Unbearable psychological pressures bear down on us. I ' ve never met a person in the world that didn ' t have a problem or a worry. Why do we suffer? It ' s an age- old question that we haven ' t answered. Yet David again and again said that he would turn to God. He said,"The Lord is my shepherd."The final problem that David knew he could not solve was death. Many commentators have said that death is the forbidden **subject** of our generation. Most people live as if they ' re never going to die. Technology projects the myth of control over our mortality. We see people on our screens. Marilyn Monroe is just as beautiful on the screen as she was in person, and our — many young people think she ' s still alive. They don ' t know that she ' s dead. Or Clark Gable, or whoever it is. The old stars, they come to life. And they ' re — they ' re just as great on that screen as they were in person. But death is inevitable. I spoke some time ago to a joint session of Congress, last year. And we were meeting in that room, the statue room. About 300 of them were there. And I said,"There ' s one thing that we have in common in this room, all of us together, whether Republican or Democrat, or whoever."I said,"We ' re all going to die. And we have that in common with all these great men of the past that are staring down at us."And it ' s often difficult for young people to understand that. It ' s difficult for them to understand that they ' re going to die. As the ancient writer of Ecclesiastes wrote, he said, there ' s every activity under heaven. There ' s a time to be born, and there ' s a time to die. I ' ve stood at the deathbed of several famous people, whom you would know. I ' ve talked to them. I ' ve seen them in those agonizing moments when they were scared to death. And yet, a few years earlier, death never crossed their mind. I talked to a woman this past week whose father was a famous doctor. She said he never thought of God, never talked about God, didn ' t believe in God. He was an atheist. But she said, as he came to die, he sat up on the side of the bed one day, and he asked the nurse if he could see the chaplain. And he said, for the first time in his life he ' d thought about the inevitable, and about God. Was there a God? A few years ago, a university student asked me,"What is the greatest surprise in your life?"And I said,"The greatest surprise in my life is the brevity of life. It passes so fast."But it does not need to have to be that way. Wernher von Braun, in the aftermath of World War II concluded, quote :"science and religion are not antagonists. On the contrary, they ' re sisters."He put it on a personal basis. I knew Dr. von Braun very well. And he said,"Speaking for myself, I can only say that the grandeur of the cosmos serves only to confirm a belief in the certainty of a creator."He also said,"In our search to know God, I ' ve come to believe that the life of Jesus Christ should be the focus of our efforts and inspiration. The reality of this life and His resurrection is the hope of mankind."I ' ve done a lot of speaking in Germany and in France, and in different parts of the world — 105 countries it ' s been my privilege to speak in. And I was invited one day to visit Chancellor Adenauer, who was looked upon as sort of the founder of modern Germany, since the war. And he once — and he said to me, he said,"Young man."He said,"Do you believe in the resurrection of Jesus Christ?"And I said,"Sir, I do."He said,"So do I."He said,"When I leave office, I ' m going to spend my time writing a book on why Jesus Christ rose again, and why it ' s so important to believe that."In one of his plays, Alexander Solzhenitsyn depicts a man dying, who says to those gathered around his bed,"The moment when it ' s terrible to feel regret is when one is dying."How should one live in order not to feel regret when one is dying? Blaise Pascal asked exactly that question in seventeenth-century France. Pascal has been called the architect of modern civilization. He was a brilliant scientist at the frontiers of mathematics, even as a teenager. He is viewed by many as the founder of the probability theory, and a creator of the first model of a computer. And of course, you are all familiar with the computer language named for him. Pascal explored in depth our human dilemmas of evil, suffering and death. He was astounded at the phenomenon we ' ve been considering : that people can achieve extraordinary heights in science, the arts and human enterprise, yet they also are full of anger, hypocrisy and have — and self- hatreds. Pascal saw us as a remarkable mixture of genius and self-delusion. On November 23, 1654, Pascal had a profound religious experience. He wrote in his journal these words :"I submit myself, absolutely, to Jesus Christ, my redeemer."A French historian said, two centuries later,"Seldom has so mighty an intellect submitted with such humility to the authority of Jesus Christ."Pascal came to believe not only the love and the grace of God could bring us back into harmony, but he believed that his own sins and failures could be forgiven, and that when he died he would go to a place called heaven. He experienced it in a way that went beyond scientific observation and reason. It was he who penned the well-known words,"The heart has its reasons, which reason knows not of."Equally well known is Pascal ' s Wager. Essentially, he said this :"if you bet on God, and open yourself to his love, you lose nothing, even if you ' re wrong. But if instead you bet that there is no God, then you can lose it all, in this life and the life to come."For Pascal, scientific knowledge paled beside the knowledge of God. The knowledge of God was far beyond anything that ever crossed his mind. He was ready to face him when he died at the age of 39. King David lived to be 70, a long time in his era. Yet he too had to face death, and he wrote these words :"even though I walk through the valley of the shadow of death, I will fear no evil, for you are with me."This was David ' s answer to three dilemmas of evil, suffering and death. It can be yours, as well, as you seek the living God and allow him to fill your life and give you hope for the future. When I was 17 years of age, I was born and reared on a farm in North Carolina. I milked cows every morning, and I had to milk the same cows every evening when I came home from school. And there were 20 of them that I had — that I was responsible for, and I worked on the farm and tried to keep up with my studies. I didn ' t make good grades in high school. I didn ' t make them in college, until something happened in my heart. One day, I was faced face-to-face with Christ. He said,"I am the way, the truth and the life."Can you imagine that?"I am the truth. I ' m the embodiment of all truth."He was a liar. Or he was insane. Or he was what he claimed to be. Which was he? I had to make that decision. I couldn ' t prove it. I couldn ' t take it to a laboratory and experiment with it. But by faith I said, I believe him, and he came into my heart and changed my life. And now I ' m ready, when I hear that call, to go into the presence of God. Thank you, and God bless all of you. Thank you for the privilege. It was great. Richard Wurman : You did it. Thanks.

Text Sample 817: NEG Dim 5 – 1998 – Score 4.00 – 1998/t000113.txt

Sheryl Shade : Hi, Aimee. Aimee Mullins : Hi. SS : Aimee and I thought we ’ d just talk a little bit, and I wanted her to tell all of you what makes her a distinctive athlete. AM : Well, for those of you who have seen the picture in the little bio â it might have given it away â I ’ m a double amputee, and I was born without fibulas in both legs. I was amputated at age one, and I ’ ve been running like hell ever since, all over the place. SS : Well, why don ’ t you tell them how you got to Georgetown â why don ’ t we start there? Why don ’ t we start there? AM : I ’ m a senior in Georgetown in the Foreign Service program. I won a full academic scholarship out of high school. They pick three students out of the nation every year to get involved in international affairs, and so I won a full ride to Georgetown and I ’ ve been there for four years. Love it. SS : When Aimee got there, she decided that she ’ s, kind of, curious about track and field, so she decided to call someone and start asking about it. So, why don ’ t you tell that story? AM : Yeah. Well, I guess I ’ ve always been involved in sports. I played softball for five years growing up. I skied competitively throughout high school, and I got a little restless in college because I wasn ’ t doing anything for about a year or two sports-wise. And I ’ d never competed on a disabled level, you know â I ’ d always competed against other able-bodied athletes. That ’ s all I ’ d ever known. In fact, I ’ d never even met another amputee until I was 17. And I heard that they do these track meets with all disabled runners, and I figured,”Oh, I don ’ t know about this, but before I judge it, let me go see what it ’ s all about.”So, I booked myself a flight to Boston in ’ 95, 19 years old and definitely the dark horse candidate at this race. I ’ d never done it before. I went out on a gravel track a couple of weeks before this meet to see how far I could run, and about 50 meters was enough for me, panting and heaving. And I had these legs that were made of a wood and plastic compound, attached with Velcro straps â big, thick, five-ply wool socks on â you know, not the most comfortable things, but all I ’ d ever known. And I ’ m up there in Boston against people wearing legs made of all things â carbon graphite and, you know, shock absorbers in them and all sorts of things â and they ’ re all looking at me like, OK, we know who ’ s not going to win this race. And, I mean, I went up there expecting â I don ’ t know what I was expecting â but, you know, when I saw a man who was missing an entire leg go up to the high jump, hop on one leg to the high jump and clear it at six feet, two inches... Dan O ’ Brien jumped 5 ’ 11”in ’ 96 in Atlanta, I mean, if it just gives you a comparison of â these are truly accomplished athletes, without qualifying that word”athlete.”And so I decided to give this a shot : heart pounding, I ran my first race and I beat the national record-holder by three hundredths of a second, and became the new national record-holder on my first try out. And, you know, people said,”Aimee, you know, you ’ ve got speed â you ’ ve got natural speed â but you don ’ t have any skill or finesse going down that track. You were all over the place. We all saw how hard you were working.”And so I decided to call the track coach at Georgetown. And I thank god I didn ’ t know just how huge this man is in the track and field world. He ’ s coached five Olympians, and the man ’ s office is lined from floor to ceiling with All America certificates of all these athletes he ’ s coached. He ’ s just a rather intimidating figure. And I called him up and said,”Listen, I ran one race and I won...””I want to see if I can, you know â I need to just see if I can sit in on some of your practices, see what drills you do and whatever.”That ’ s all I wanted â just two practices.”Can I just sit in and see what you do?”And he said,”Well, we should meet first, before we decide anything.”You know, he ’ s thinking,”What am I getting myself into?”So, I met the man, walked in his office, and saw these posters and magazine covers of people he has coached. And we got to talking, and it turned out to be a great partnership because he ’ d never coached a disabled athlete, so therefore he had no preconceived notions of what I was or wasn ’ t capable of, and I ’ d never been coached before. So this was like,”Here we go â let ’ s start on this trip.”So he started giving me four days a week of his lunch break, his free time, and I would come up to the track and train with him. So that ’ s how I met Frank. That was fall of ’ 95. But then, by the time that winter was rolling around, he said,”You know, you ’ re good enough. You can run on our women ’ s track team here.”And I said,”No, come on.”And he said,”No, no, really. You can. You can run with our women ’ s track team.”In the spring of 1996, with my goal of making the U. S. Paralympic team that May coming up full speed, I joined the women ’ s track team. And no disabled person had ever done that â run at a collegiate level. So I don ’ t know, it started to become an interesting mix. SS : Well, on your way to the Olympics, a couple of memorable events happened at Georgetown. Why don ’ t you just tell them? AM : Yes, well, you know, I ’ d won everything as far as the disabled meets â everything I competed in â and, you know, training in Georgetown and knowing that I was going to have to get used to seeing the backs of all these women ’ s shirts â you know, I ’ m running against the next Flo-Jo â and they ’ re all looking at me like,”Hmm, what ’ s, you know, what ’ s going on here?”And putting on my Georgetown uniform and going out there and knowing that, you know, in order to become better â and I ’ m already the best in the country â you know, you have to train with people who are inherently better than you. And I went out there and made it to the Big East, which was sort of the championship race at the end of the season. It was really, really hot. And it ’ s the first â I had just gotten these new sprinting legs that you see in that bio, and I didn ’ t realize at that time that the amount of sweating I would be doing in the sock â it actually acted like a lubricant and I ’ d be, kind of, pistoning in the socket. And at about 85 meters of my 100 meters sprint, in all my glory, I came out of my leg. Like, I almost came out of it, in front of, like, 5, 000 people. And I, I mean, was just mortified â because I was signed up for the 200, you know, which went off in a half hour. I went to my coach :”Please, don ’ t make me do this.”I can ’ t do this in front of all those people. My legs will come off. And if it came off at 85 there ’ s no way I ’ m going 200 meters. And he just sat there like this. My pleas fell on deaf ears, thank god. Because you know, the man is from Brooklyn ; he ’ s a big man. He says,”Aimee, so what if your leg falls off? You pick it up, you put the damn thing back on, and finish the goddamn race!”And I did. So, he kept me in line. He kept me on the right track. SS : So, then Aimee makes it to the 1996 Paralympics, and she ’ s all excited. Her family ’ s coming down â it ’ s a big deal. It ’ s now two years that you ’ ve been running? AM : No, a year. SS : A year. And why don ’ t you tell them what happened right before you go run your race? AM : Okay, well, Atlanta. The Paralympics, just for a little bit of clarification, are the Olympics for people with physical disabilities â amputees, persons with cerebral palsy, and wheelchair athletes â as opposed to the Special Olympics, which deals with people with mental disabilities. So, here we are, a week after the Olympics and down at Atlanta, and I ’ m just **blown** away by the fact that just a year ago, I got out on a gravel track and couldn ’ t run 50 meters. And so, here I am â never lost. I set new records at the U. S. Nationals â the Olympic trials â that May, and was sure that I was coming home with the gold. I was also the only, what they call”bilateral BK”â below the knee. I was the only woman who would be doing the long jump. I had just done the long jump, and a guy who was missing two legs came up to me and says,"How do you do that? You know, we ' re supposed to have a planar foot, so we can ' t get off on the springboard."I said,"Well, I just did it. No one told me that."So, it ' s funny â I ' m three inches within the world record â and kept on from that point, you know, so I ' m signed up in the long jump â signed up? No, I made it for the long jump and the 100-meter. And I ' m sure of it, you know? I made the front page of my hometown paper that I delivered for six years, you know? It was, like, this is my time for shine. And we ' re at the trainee warm-up track, which is a few blocks away from the Olympic stadium. These legs that I was on, which I ' ll take out right now â I was the first person in the world on these legs. I was the guinea pig., I ' m telling you, this was, like â talk about a tourist attraction. Everyone was taking pictures â"What is this girl running on?"And I ' m always looking around, like, where is my competition? It ' s my first international meet. I tried to get it out of anybody I could, you know,"Who am I running against here?""Oh, Aimee, we ' ll have to get back to you on that one."I wanted to find out times."Don ' t worry, you ' re doing great."This is 20 minutes before my race in the Olympic stadium, and they post the heat sheets. And I go over and look. And my fastest time, which was the world record, was 15. 77. Then I ' m looking : the next lane, lane two, is 12. 8. Lane three is 12. 5. Lane four is 12. 2. I said,"What ' s going on?"And they shove us all into the shuttle bus, and all the women there are missing a hand. So, I ' m just, like â they ' re all looking at me like ' which one of these is not like the other, ' you know? I ' m sitting there, like,"Oh, my god. Oh, my god."You know, I ' d never lost anything, like, whether it would be the scholarship or, you know, I ' d won five golds when I skied. In everything, I came in first. And Georgetown â that was great. I was losing, but it was the best training because this was Atlanta. Here we are, like, crĂ¨me de la crĂ¨me, and there is no doubt about it, that I ' m going to lose big. And, you know, I ' m just thinking,"Oh, my god, my whole family got in a van and drove down here from Pennsylvania."And, you know, I was the only female U. S. sprinter. So they call us out and, you know â"Ladies, you have one minute."And I remember putting my blocks in and just feeling horrified because there was just this murmur coming over the crowd, like, the ones who are close enough to the starting line to see. And I ' m like,"I know! Look! This isn ' t right."And I ' m thinking that ' s my last card to play here ; if I ' m not going to beat these girls, I ' m going to mess their heads a little, you know? I mean, it was definitely the"Rocky IV"sensation of me versus Germany, and everyone else â Estonia and Poland â was in this heat. And the gun went off, and all I remember was finishing last and fighting back tears of frustration and incredible â incredible â this feeling of just being overwhelmed. And I had to think,"Why did I do this?"If I had won everything â but it was like, what was the point? All this training â I had transformed my life. I became a collegiate athlete, you know. I became an Olympic athlete. And it made me really think about how the achievement was getting there. I mean, the fact that I set my sights, just a year and three months before, on becoming an Olympic athlete and saying,"Here ' s my life going in this direction â and I want to take it here for a while, and just seeing how far I could push it."And the fact that I asked for help â how many people jumped on board? How many people gave of their time and their expertise, and their patience, to deal with me? And that was this collective glory â that there was, you know, 50 people behind me that had joined in this incredible experience of going to Atlanta. So, I apply this sort of philosophy now to everything I do : sitting back and realizing the progression, how far you ' ve come at this day to this goal, you know. It ' s important to focus on a goal, I think, but also recognize the progression on the way there and how you ’ ve grown as a person. That ’ s the achievement, I think. That ’ s the real achievement. SS : Why don ’ t you show them your legs? AM : Oh, sure. SS : You know, show us more than one set of legs. AM : Well, these are my pretty legs. No, these are my cosmetic legs, actually, and they ’ re absolutely beautiful. You ’ ve got to come up and see them. There are hair follicles on them, and I can paint my toenails. And, seriously, like, I can wear heels. Like, you guys don ’ t understand what that ’ s like to be able to just go into a shoe store and buy whatever you want. SS : You got to pick your height? AM : I got to pick my height, exactly. Patrick Ewing, who played for Georgetown in the ’ 80s, comes back every summer. And I had incessant fun making fun of him in the training room because he ’ d come in with foot injuries. I ’ m like,”Get it off! Don ’ t worry about it, you know. You can be eight feet tall. Just take them off.”He didn ’ t find it as humorous as I did, anyway. OK, now, these are my sprinting legs, made of carbon graphite, like I said, and I ’ ve got to make sure I ’ ve got the right socket. No, I ’ ve got so many legs in here. These are â do you want to hold that actually? That ’ s another leg I have for, like, tennis and softball. It has a shock absorber in it so it, like,”Shhhh,”makes this neat sound when you jump around on it. All right. And then this is the silicon sheath I roll over, to keep it on. Which, when I sweat, you know, I ’ m pistoning out of it. SS : Are you a different height? AM : In these? SS : In these. AM : I don ’ t know. I don ’ t think so. I may be a little taller. I actually can put both of them on. SS : She can ’ t really stand on these legs. She has to be moving, so... AM : Yeah, I definitely have to be moving, and balance is a little bit of an art in them. But without having the silicon sock, I ’ m just going to try slip in it. And so, I run on these, and have shocked half the world on these. These are supposed to simulate the actual form of a sprinter when they run. If you ever watch a sprinter, the ball of their foot is the only thing that ever hits the track. So when I stand in these legs, my hamstring and my glutes are contracted, as they would be had I had feet and were standing on the ball of my feet. AM : It ’ s a company in San Diego called Flex-Foot. And I was a guinea pig, as I hope to continue to be in every new form of prosthetic limbs that come out. But actually these, like I said, are still the actual prototype. I need to get some new ones because the last meet I was at, they were everywhere. You know, it ’ s like a big â it ’ s come full circle. Moderator : Aimee and the designer of them will be at TEDMED 2, and we ’ ll talk about the design of them. AM : Yes, we ’ ll do that. SS : Yes, there you go. AM : So, these are the sprint legs, and I can put my other... SS : Can you tell about who designed your other legs? AM : Yes. These I got in a place called Bournemouth, England, about two hours south of London, and I ’ m the only person in the United States with these, which is a crime because they are so beautiful. And I don ’ t even mean, like, because of the toes and everything. For me, while I ’ m such a serious athlete on the track, I want to be feminine off the track, and I think it ’ s so important not to be limited in any capacity, whether it ’ s, you know, your mobility or even fashion. I mean, I love the fact that I can go in anywhere and pick out what I want â the shoes I want, the skirts I want â and I ’ m hoping to try to bring these over here and make them accessible to a lot of people. They ’ re also silicon. This is a really basic, basic prosthetic limb under here. It ’ s like a Barbie foot under this. It is. It ’ s just stuck in this position, so I have to wear a two-inch heel. And, I mean, it ’ s really â let me take this off so you can see it. I don ’ t know how good you can see it, but, like, it really is. There ’ re veins on the feet, and then my heel is pink, and my Achilles ’ tendon â that moves a little bit. And it ’ s really an amazing store. I got them a year and two weeks ago. And this is just a silicon piece of skin. I mean, what happened was, two years ago this man in Belgium was saying,”God, if I can go to Madame Tussauds ’ wax museum and see Jerry Hall replicated down to the color of her eyes, looking so real as if she breathed, why can ’ t they build a limb for someone that looks like a leg, or an arm, or a hand?”I mean, they make ears for burn victims. They do amazing stuff with silicon. SS : Two weeks ago, Aimee was up for the Arthur Ashe award at the ESPYs. And she came into town and she rushed around and she said,”I have to buy some new shoes!”We ’ re an hour before the ESPYs, and she thought she ’ d gotten a two-inch heel but she ’ d actually bought a three-inch heel. AM : And this poses a problem for me, because it means I ’ m walking like that all night long. SS : For 45 minutes. Luckily, the hotel was terrific. They got someone to come in and saw off the shoes. AM : I said to the receptionist â I mean, I am just harried, and Sheryl ’ s at my side â I said,”Look, do you have anybody here who could help me? Because I have this problem...”You know, at first they were just going to write me off, like,”If you don ’ t like your shoes, sorry. It ’ s too late.””No, no, no, no. I ’ ve got these special feet that need a two-inch heel. I have a three-inch heel. I need a little bit off.”They didn ’ t even want to go there. They didn ’ t even want to touch that one. They just did it. No, these legs are great. I ’ m actually going back in a couple of weeks to get some improvements. I want to get legs like these made for flat feet so I can wear sneakers, because I can ’ t with these ones. So... Moderator : That ’ s it. SS : That ’ s Aimee Mullins.

Text Sample 818: NEG Dim 5 – 2005 – Score 1.00 – 2005/t000220.txt

I think it ’ ll be a relief to some people and a disappointment to others that I ’ m not going to talk about vaginas today. I began”The Vagina Monologues”because I was worried about vaginas. I ’ m very worried today about this notion, this world, this prevailing kind of force of security. I see this word, hear this word, feel this word everywhere. Real security, security checks, security watch, security clearance. Why has all this focus on security made me feel so much more insecure? What does anyone mean when they talk about real security? And why have we, as Americans particularly, become a nation that strives for security above all else? In fact, I think that security is elusive. It ’ s impossible. We all die. We all get old. We all get sick. People leave us. People change us. Nothing is secure. And that ’ s actually the good news. This is, of course, unless your whole life is about being secure. I think that when that is the focus of your life, these are the things that happen. You can ’ t travel very far or venture too far outside a certain circle. You can ’ t allow too many conflicting ideas into your mind at one time, as they might confuse you or challenge you. You can ’ t open yourself to new experiences, new people, new ways of doing things — they might take you off course. You can ’ t not know who you are, so you cling to hard-matter identity. You become a Christian, Muslim, Jew. You ’ re an Indian, Egyptian, Italian, American. You ’ re a heterosexual or a homosexual, or you never have sex. Or at least, that ’ s what you say when you identify yourself. You become part of an”us.”In order to be secure, you defend against”them.”You cling to your land because it is your secure place. You must fight anyone who encroaches upon it. You become your nation. You become your religion. You become whatever it is that will freeze you, numb you and protect you from doubt or change. But all this does, actually, is shut down your mind. In reality, it does not really make you safer. I was in Sri Lanka, for example, three days after the tsunami, and I was standing on the beaches and it was absolutely clear that, in a matter of five minutes, a 30-foot wave could rise up and desecrate a people, a population and lives. All this striving for security, in fact, has made you much more insecure because now you have to watch out all the time. There are people not like you — people who you now call enemies. You have places you cannot go, thoughts you cannot think, worlds that you can no longer inhabit. And so you spend your days fighting things off, defending your territory and becoming more entrenched in your fundamental thinking. Your days become devoted to protecting yourself. This becomes your mission. That is all you do. Ideas get shorter. They become sound bytes. There are evildoers and saints, criminals and victims. There are those who, if they ' re not with us, are against us. It gets easier to hurt people because you do not feel what ' s inside them. It gets easier to lock them up, force them to be naked, humiliate them, occupy them, invade them and kill them, because they are only obstacles now to your security. In six years, I ' ve had the extraordinary privilege through V-Day, a global movement against [violence against] women, to travel probably to 60 countries, and spend a great deal of time in different portions. I ' ve met women and men all over this planet, who through various circumstances — war, poverty, racism, multiple forms of violence — have never known security, or have had their illusion of security forever devastated. I ' ve spent time with women in Afghanistan under the Taliban, who were essentially brutalized and censored. I ' ve been in Bosnian refugee camps. I was with women in Pakistan who have had their faces melted off with acid. I ' ve been with girls all across America who were date-raped, or raped by their best friends when they were drugged one night. One of the amazing things that I ' ve discovered in my travels is that there is this emerging species. I loved when he was talking about this other world that ' s right next to this world. I ' ve discovered these people, who, in V-Day world, we call Vagina Warriors. These particular people, rather than getting AK-47s, or weapons of mass destruction, or machetes, in the spirit of the warrior, have gone into the center, the heart of pain, of loss. They have grieved it, they have died into it, and allowed and encouraged poison to turn into medicine. They have used the fuel of their pain to begin to redirect that energy towards another mission and another trajectory. These warriors now devote themselves and their lives to making sure what happened to them doesn ' t happen to anyone else. There are thousands if not millions of them on the planet. I venture there are many in this room. They have a fierceness and a freedom that I believe is the bedrock of a new paradigm. They have broken out of the existing frame of victim and perpetrator. Their own personal security is not their end goal, and because of that, because, rather than worrying about security, because the transformation of suffering is their end goal, I actually believe they are creating real safety and a whole new idea of security. I want to talk about a few of these people that I ' ve met. Tomorrow, I am going to Cairo, and I ' m so moved that I will be with women in Cairo who are V-Day women, who are opening the first safe house for battered women in the Middle East. That will happen because women in Cairo made a decision to stand up and put themselves on the line, and talk about the degree of violence that is happening in Egypt, and were willing to be attacked and criticized. And through their work over the last years, this is not only happening that this house is opening, but it ' s being supported by many factions of the society who never would have supported it. Women in Uganda this year, who put on"The Vagina Monologues"during V-Day, actually evoked the wrath of the government. And, I love this story so much. There was a cabinet meeting and a meeting of the presidents to talk about whether"Vaginas"could come to Uganda. And in this meeting — it went on for weeks in the press, two weeks where there was huge discussion. The government finally made a decision that"The Vagina Monologues"could not be performed in Uganda. But the amazing news was that because they had stood up, these women, and because they had been willing to risk their security, it began a discussion that not only happened in Uganda, but all of Africa. As a result, this production, which had already sold out, every single person in that 800-seat audience, except for 10 people, made a decision to keep the money. They raised 10, 000 dollars on a production that never occurred. There ’ s a young woman named Carrie Rethlefsen in Minnesota. She ’ s a high school student. She had seen”The Vagina Monologues”and she was really moved. And as a result, she wore an”I heart my vagina”button to her high school in Minnesota. She was basically threatened to be expelled from school. They told her she couldn ’ t love her vagina in high school, that it was not a legal thing, that it was not a moral thing, that it was not a good thing. So she really struggled with this, what to do, because she was a senior and she was doing well in her school and she was threatened expulsion. So what she did is she got all her friends together — I believe it was 100, 150 students all wore”I love my vagina”T-shirts, and the boys wore”I love her vagina”T-shirts to school. Now this seems like a fairly, you know, frivolous, but what happened as a result of that, is that that school now is forming a sex education class. It ’ s beginning to talk about sex, it ’ s beginning to look at why it would be wrong for a young high school girl to talk about her vagina publicly or to say that she loved her vagina publicly. I know I ’ ve talked about Agnes here before, but I want to give you an update on Agnes. I met Agnes three years ago in the Rift Valley. When she was a young girl, she had been mutilated against her will. That mutilation of her clitoris had actually obviously impacted her life and changed it in a way that was devastating. She made a decision not to go and get a razor or a glass shard, but to devote her life to stopping that happening to other girls. For eight years, she walked through the Rift Valley. She had this amazing box that she carried and it had a torso of a woman ’ s body in it, a half a torso, and she would teach people, everywhere she went, what a healthy vagina looked like and what a mutilated vagina looked like. In the years that she walked, she educated parents, mothers, fathers. She saved 1, 500 girls from being cut. When V-Day met her, we asked her how we could support her and she said,”Well, if you got me a Jeep, I could get around a lot faster.”So, we bought her a Jeep. In the year she had the Jeep, she saved 4, 500 girls from being cut. So, we said, what else could we do? She said,”If you help me get money, I could open a house.”Three years ago, Agnes opened a safe house in Africa to stop mutilation. When she began her mission eight years ago, she was reviled, she was detested, she was completely slandered in her community. I am proud to tell you that six months ago, she was elected the deputy mayor of Narok. I think what I ’ m trying to say here is that if your end goal is security, and if that ’ s all you ’ re focusing on, what ends up happening is that you create not only more insecurity in other people, but you make yourself far more insecure. Real security is contemplating death, not pretending it doesn ’ t exist. Not running from loss, but entering grief, surrendering to sorrow. Real security is not knowing something, when you don ’ t know it. Real security is hungering for connection rather than power. It cannot be bought or arranged or made with bombs. It is deeper, it is a process, it is acute awareness that we are all utterly inter-bended, and one action by one being in one tiny town has consequences everywhere. Real security is not only being able to tolerate mystery, complexity, ambiguity, but hungering for them and only trusting a situation when they are present. Something happened when I began traveling in V-Day, eight years ago. I got lost. I remember being on a plane going from Kenya to South Africa, and I had no idea where I was. I didn ’ t know where I was going, where I ’ d come from, and I panicked. I had a total anxiety attack. And then I suddenly realized that it absolutely didn ’ t matter where I was going, or where I had come from because we are all essentially permanently displaced people. All of us are refugees. We come from somewhere and we are hopefully traveling all the time, moving towards a new place. Freedom means I may not be identified as any one group, but that I can visit and find myself in every group. It does not mean that I don ’ t have values or beliefs, but it does mean I am not hardened around them. I do not use them as weapons. In the shared future, it will be just that, shared. The end goal will [be] becoming vulnerable, realizing the place of our connection to one another, rather than becoming secure, in control and alone. Thank you very much. Chris Anderson : And how are you doing? Are you exhausted? On a typical day, do you wake up with hope or gloom? Eve Ensler : You know, I think Carl Jung once said that in order to survive the twentieth century, we have to live with two existing thoughts, opposite thoughts, at the same time. And I think part of what I ’ m learning in this process is that one must allow oneself to feel grief. And I think as long as I keep grieving, and weeping, and then moving on, I ’ m fine. When I start to pretend that what I ’ m seeing isn ’ t impacting me, and isn ’ t changing my heart, then I get in trouble. Because when you spend a lot of time going from place to place, country to country, and city to city, the degree to which women, for example, are violated, and the epidemic of it, and the kind of ordinariness of it, is so devastating to one ’ s soul that you have to take the time, or I have to take the time now, to process that. CA : There are a lot of causes out there in the world that have been talked about, you know, poverty, sickness and so on. You spent eight years on this one. Why this one? EE : I think that if you think about women, women are the primary resource of the planet. They give birth, we come from them, they are mothers, they are visionaries, they are the future. If you think that the U. N. now says that one out of three women on the planet will be raped or beaten in their lifetime, we ’ re talking about the desecration of the primary resource of the planet, we ’ re talking about the place where we come from, we ’ re talking about parenting. Imagine that you ’ ve been raped and you ’ re bringing up a boy child. How does it impact your ability to work, or envision a future, or thrive, as opposed to just survive? What I believe is if we could figure out how to make women safe and honor women, it would be parallel or equal to honoring life itself.

Text Sample 819: NEG Dim 5 – 2005 – Score 1.00 – 2005/t000310.txt

Good morning, ladies and gentlemen. My name is Art Benjamin, and I am a”mathemagician.”What that means is, I combine my loves of math and magic to do something I call”mathemagics.”But before I get started, I have a quick question for the audience. By any chance, did anyone happen to bring with them this morning a calculator? Seriously, if you have a calculator with you, raise your hand. Raise your hand. Did your hand go up? Now bring it out, bring it out. Anybody else? I see, I see one way in the back. You sir, that ’ s three. And anybody on this side here? OK, over there on the aisle. Would the four of you please bring out your calculators, then join me up on stage. Let ’ s give them a nice round of applause. That ’ s right. Now, since I haven ’ t had the chance to work with these calculators, I need to make sure that they are all working properly. Would somebody get us started by giving us a two-digit number, please? How about a two-digit number? Audience : 22. AB : 22. And another two-digit number, sir? Audience : 47. AB : Multiply 22 times 47, make sure you get 1, 034, or the calculators are not working. Do all of you get 1, 034? 1, 034? Volunteer : No. AB : 594. Let ’ s give three of them a nice round of applause there. Would you like to try a more standard calculator, just in case? OK, great. What I ’ m going to try and do then — I notice it took some of you a little bit of time to get your answer. That ’ s OK. I ’ ll give you a shortcut for multiplying even faster on the calculator. There is something called the square of a number, which most of you know is taking a number and multiplying it by itself. For instance, five squared would be? Audience : 25. AB : 25. The way we can square on most calculators — let me demonstrate with this one — is by taking the number, such as five, hitting”times”and then”equals,”and on most calculators that will give you the square. On some of these ancient RPN calculators, you ’ ve got an”x squared”button on it, will allow you to do the calculation even faster. What I ’ m going to try and do now is to square, in my head, four two-digit numbers faster than they can do on their calculators, even using the shortcut method. What I ’ ll use is the second row this time, and I ’ ll get four of you to each yell out a two-digit number, and if you would square the first number, and if you would square the second, the third and the fourth, I will try and race you to the answer. OK? So quickly, a two-digit number please. Audience : 37. Arthur Benjamin : 37 squared, OK. Audience : 23. AB : 23 squared, OK. Audience : 59. AB : 59 squared, OK, and finally? Audience : 93. AB : 93 squared. Would you call out your answers, please? Volunteer : 1369. AB : 1369. Volunteer : 529. AB : 529. Volunteer : 3481. AB : 3481. Volunteer : 8649. AB : Thank you very much. Let me try to take this one step further. I ’ m going to try to square some three-digit numbers this time. I won ’ t even write these down — I ’ ll just call them out as they ’ re called out to me. Anyone I point to, call out a three-digit number. Anyone on our panel, verify the answer. Just give some indication if it ’ s right. A three-digit number, sir, yes? Audience : 987. AB : 987 squared is 974, 169. AB : Yes? Good. Another three-digit — — another three-digit number, sir? Audience : 457. AB : 457 squared is 205, 849. 205, 849? AB : Yes? OK, another, another three-digit number, sir? Audience : 321. AB : 321 is 103, 041. 103, 041. Yes? One more three-digit number please. Audience : Oh, 722. AB : 722 is 500, that ’ s a harder one. Is that 513, 284? Volunteer : Yes. AB : Yes? Oh, one more, one more three-digit number please. Audience : 162. 162 squared is 26, 244. Volunteer : Yes. Thank you very much. Let me try to take this one step further. I ’ m going to try to square a four-digit number this time. You can all take your time on this ; I will not beat you to the answer on this one, but I will try to get the answer right. To make this a little bit more random, let ’ s take the fourth row this time, let ’ s say, one, two, three, four. If each of you would call out a single digit between zero and nine, that will be the four-digit number that I ’ ll square. Nine. Seven. Five. Eight. 9, 758, this will take me a little bit of time, so bear with me. 95 million — 218, 564? Volunteer : Yes! AB : Thank you very much. Now, I would attempt to square a five-digit number — and I can — but unfortunately, most calculators cannot. Eight-digit capacity — don ’ t you hate that? So, since we ’ ve reached the limits of our calculators — what ’ s that? Does yours go higher? Volunteer : I don ’ t know. AB : Oh, yours does? Volunteer : I can probably do it. AB : I ’ ll talk to you later. In the meanwhile, let me conclude the first part of my show by doing something a little trickier. Let ’ s take the largest number on the board here, 8649. Would you each enter that on your calculator? And instead of squaring it this time, I want you to take that number and multiply it by any three-digit number that you want, but don ’ t tell me what you ’ re multiplying by — just multiply it by any random three-digit number. So you should have as an answer either a six-digit or probably a seven-digit number. How many digits do you have, six or seven? Seven, and yours? Seven? Seven? And, uncertain. Seven. Is there any possible way that I could know what seven-digit numbers you have? Say”No.”Good, then I shall attempt the impossible — or at least the improbable. What I ’ d like each of you to do is to call out for me any six of your seven digits, any six of them, in any order you ’ d like. One digit at a time, I shall try and determine the digit you ' ve left out. Starting with your seven-digit number, call out any six of them please. Volunteer : 1, 9, 7, 0, 4, 2. AB : Did you leave out the number 6? Good, OK, that ' s one. You have a seven-digit number, call out any six of them please. Volunteer : 4, 4, 8, 7, 5. I think I only heard five numbers. I — wait — 44875 — did you leave out the number 6? Same as she did, OK. You ' ve got a seven-digit number — call out any six of them loud and clear. Volunteer : 0, 7, 9, 0, 4, 4. I think you left out the number 3? AB : That ' s three. The odds of me getting all four of these right by random guessing would be one in 10, 000 : 10 to the fourth power. OK, any six of them. Really scramble them up this time, please. Volunteer : 2, 6, 3, 9, 7, 2. Did you leave out the number 7? And let ' s give all four of these people a nice round of applause. Thank you very much. For my next number — while I mentally recharge my batteries, I have one more question for the audience. By any chance, does anybody here happen to know the day of the week that they were born on? If you think you know your birth day, raise your hand. Let ' s see, starting with — let ' s start with a gentleman first. What year was it, first of all? That ' s why I pick a gentleman first. Audience : 1953. 1953, and the month? November what? 23rd — was that a Monday? Audience : Yes. Good. Somebody else? I haven ' t seen any women ' s hands up. OK, how about you, what year? 1949, and the month? October what? Fifth — was that a Wednesday? Yes! I ' ll go way to the back right now, how about you? Yell it out, what year? Audience : 1959. 1959, OK — and the month? Audience : February. February what? Sixth — was that a Friday? Audience : Yes. Good, how about the person behind her? Call out, what year was it? Audience : 1947. AB : 1947, and the month? Audience : May. AB : May what? Seventh — would that be a Wednesday? Audience : Yes. AB : Thank you very much. Anybody here who ' d like to know the day of the week they were born? We can do it that way. Of course, I could just make up an answer and you wouldn ' t know, so I come prepared for that. I brought with me a book of calendars. It goes as far back into the past as 1800, because you never know. I didn ' t mean to look at you, sir — you were just sitting there. Anyway, Chris, you can help me out here, if you wouldn ' t mind. This is a book of calendars. Who wanted to know their birth day? What year was it, first of all? Audience : 1966. 66 — turn to the calendar with 1966. And what month? Audience : April. AB : April what? Audience : 17th. I believe that was a Sunday. Can you confirm, Chris? Chris Anderson : Yes. AB : I ' ll tell you what, Chris : as long as you have that book in front of you, do me a favor, turn to a year outside of the 1900s, either into the 1800s or way into the 2000s — that ' ll be a much greater challenge for me. AB : What year would you like? CA : 1824. AB : 1824, OK. AB : And what month? CA : June. AB : June what? CA : Sixth. AB : Was that a Sunday? CA : It was. AB : And it was cloudy. Good, thank you very much. But I ' d like to wrap things up now by alluding to something from earlier in the presentation. There was a gentleman up here who had a 10-digit calculator. Where is he, would you stand up, 10-digit guy? OK, stand up for me just for a second, so I can see where you are. You have a 10-digit calculator, sir, as well? OK, what I ' m going to try and do, is to square in my head a five-digit number requiring a 10-digit calculator. But to make my job more interesting for you, as well as for me, I ' m going to do this problem thinking out loud. So you can actually, honestly hear what ' s going on in my mind while I do a calculation of this size. Now, I have to apologize to our magician friend Lennart Green. I know as a magician we ' re not supposed to reveal our secrets, but I ' m not too afraid that people are going to start doing my show next week, so — I think we ' re OK. So, let ' s see, let ' s take a different row of people, starting with you. I ' ll get five digits : one, two, three, four. Oh, I did this row already. Let ' s do the row before you, starting with you : one, two, three, four, five. Call out a single digit — that will be the five-digit number that I will try to square, go ahead. Five. Seven. Six. Eight. Three. 57, 683 — squared. Yuck. Let me explain to you how I ’ m going to attempt this problem. I ’ m going to break the problem down into three parts. I ’ ll do 57, 000 squared, plus 683 squared, plus 57, 000 times 683 times two. Add all those numbers together, and with any luck, arrive at the answer. Now, let me recap. Thank you. While I explain something else — — I know, that you can use, right? While I do these calculations, you might hear certain words, as opposed to numbers, creep into the calculation. Let me explain what that is. This is a phonetic code, a mnemonic device that I use, that allows me to convert numbers into words. I store them as words, and later on retrieve them as numbers. I know it sounds complicated ; it ’ s not. I don ’ t want you to think you ’ re seeing something out of”Rain Man.”There ’ s definitely a method to my madness — definitely, definitely. Sorry. If you want to talk to me about ADHD afterwards, you can talk to me then. By the way, one last instruction, for my judges with the calculators — you know who you are — there is at least a 50 percent chance that I will make a mistake here. If I do, don ’ t tell me what the mistake is ; just say,”you ’ re close,”or something like that, and I ’ ll try and figure out the answer — which could be pretty entertaining in itself. If, however, I am right, whatever you do, don ’ t keep it to yourself, OK? Make sure everybody knows that I got the answer right, because this is my big finish, OK. So, without any more stalling, here we go. I ’ ll start the problem in the middle, with 57 times 683. 57 times 68 is 3, 400, plus 476 is 3876, that ’ s 38, 760 plus 171, 38, 760 plus 171 is 38, 931. 38, 931 ; double that to get 77, 862. 77, 862 becomes cookie fission, cookie fission is 77, 822. That seems right, I ’ ll go on. Cookie fission, OK. Next, I do 57 squared, which is 3, 249, so I can say, three billion. Take the 249, add that to cookie, 249, oops, but I see a carry coming — 249 — add that to cookie, 250 plus 77, is 327 million — fission, fission, OK, finally, we do 683 squared, that ’ s 700 times 666, plus 17 squared is 466, 489, rev up if I need it, rev up, take the 466, add that to fission, to get, oh gee — 328, 489. Audience : Yeah! AB : Good. Thank you very much. I hope you enjoyed mathemagics. Thank you.

Text Sample 820: NEG Dim 5 – 2005 – Score 1.00 – 2005/t000454.txt

18 minutes is an absolutely brutal time limit, so I ’ m going to dive straight in, right at the point where I get this thing to work. Here we go. I ’ m going to talk about five different things. I ’ m going to talk about why defeating aging is desirable. I ’ m going to talk about why we have to get our shit together, and actually talk about this a bit more than we do. I ’ m going to talk about feasibility as well, of course. I ’ m going to talk about why we are so fatalistic about doing anything about aging. And then I ’ m going spend perhaps the second half of the talk talking about, you know, how we might actually be able to prove that fatalism is wrong, namely, by actually doing something about it. I ’ m going to do that in two steps. The first one I ’ m going to talk about is how to get from a relatively modest amount of life extension — which I ’ m going to define as 30 years, applied to people who are already in middle-age when you start — to a point which can genuinely be called defeating aging. Namely, essentially an elimination of the relationship between how old you are and how likely you are to die in the next year — or indeed, to get sick in the first place. And of course, the last thing I ’ m going to talk about is how to reach that intermediate step, that point of maybe 30 years life extension. So I ’ m going to start with why we should. Now, I want to ask a question. Hands

up : anyone in the audience who is in favor of malaria? That was easy. OK. OK. Hands up : anyone in the audience who ' s not sure whether malaria is a good thing or a bad thing? OK. So we all think malaria is a bad thing. That ' s very good news, because I thought that was what the answer would be. Now the thing is, I would like to put it to you that the main reason why we think that malaria is a bad thing is because of a characteristic of malaria that it shares with aging. And here is that characteristic. The only real difference is that aging kills considerably more people than malaria does. Now, I like in an audience, in Britain especially, to talk about the comparison with foxhunting, which is something that was banned after a long struggle, by the government not very many months ago. I mean, I know I ' m with a sympathetic audience here, but, as we know, a lot of people are not entirely persuaded by this logic. And this is actually a rather good comparison, it seems to me. You know, a lot of people said,"Well, you know, city boys have no business telling us rural types what to do with our time. It ' s a traditional part of the way of life, and we should be allowed to carry on doing it. It ' s ecologically sound ; it stops the population explosion of foxes."But ultimately, the government prevailed in the end, because the majority of the British public, and certainly the majority of members of Parliament, came to the conclusion that it was really something that should not be tolerated in a civilized society. And I think that human aging shares all of these characteristics in spades. What part of this do people not understand? It ' s not just about life, of course — — it ' s about healthy life, you know — getting frail and miserable and dependent is no fun, whether or not dying may be fun. So really, this is how I would like to describe it. It ' s a global trance. These are the sorts of unbelievable excuses that people give for aging. And, I mean, OK, I ' m not actually saying that these excuses are completely valueless. There are some good points to be made here, things that we ought to be thinking about, forward planning so that nothing goes too — well, so that we minimize the turbulence when we actually figure out how to fix aging. But these are completely crazy, when you actually remember your sense of proportion. You know, these are arguments ; these are things that would be legitimate to be concerned about. But the question is, are they so dangerous — these risks of doing something about aging — that they outweigh the downside of doing the opposite, namely, leaving aging as it is? Are these so bad that they outweigh condemning 100, 000 people a day to an unnecessarily early death? You know, if you haven ' t got an argument that ' s that strong, then just don ' t waste my time, is what I say. Now, there is one argument that some people do think really is that strong, and here it is. People worry about overpopulation ; they say,"Well, if we fix aging, no one ' s going to die to speak of, or at least the death toll is going to be much lower, only from crossing St. Giles carelessly. And therefore, we ' re not going to be able to have many kids, and kids are really important to most people."And that ' s true. And you know, a lot of people try to fudge this question, and give answers like this. I don ' t agree with those answers. I think they basically don ' t work. I think it ' s true, that we will face a dilemma in this respect. We will have to decide whether to have a low birth rate, or a high death rate. A high death rate will, of course, arise from simply rejecting these therapies, in favor of carrying on having a lot of kids. And, I say that that ' s fine — the future of humanity is entitled to make that choice. What ' s not fine is for us to make that choice on behalf of the future. If we vacillate, hesitate, and do not actually develop these therapies, then we are condemning a whole cohort of people — who would have been young enough and healthy enough to benefit from those therapies, but will not be, because we haven ' t developed them as quickly as we could — we ' ll be denying those people an indefinite life span, and I consider that that is immoral. That ' s my answer to the overpopulation question. Right. So the next thing is, now why should we get a little bit more active on this? And the fundamental answer is that the pro-aging trance is not as dumb as it looks. It ' s actually a sensible way of coping with the inevitability of aging. Aging is ghastly, but it ' s inevitable, so, you know, we ' ve got to find some way to put it out of our minds, and it ' s rational to do anything that we might want to do, to do that. Like, for example, making up these ridiculous reasons why aging is actually a good thing after all. But of course, that only works when we have both of these components. And as soon as the inevitability bit becomes a little bit unclear — and we might be in range of doing something about aging — this becomes part of the problem. This pro-aging trance is what stops us from agitating about these things. And that ' s why we have to really talk about this a lot — evangelize, I will go so far as to say, quite a lot — in order to get people ' s attention, and make people realize that they are in a trance in this regard. So that ' s all I ' m going to say about that. I ' m now going to talk about feasibility. And the fundamental reason, I think, why we feel that aging is inevitable is summed up in a definition of aging that I ' m giving here. A very simple definition. Aging is a side effect of being alive in the first place, which is to say, metabolism. This is not a completely tautological statement ; it ' s a reasonable statement. Aging is basically a process that happens to inanimate objects like cars, and it also happens to us, despite the fact that we have a lot of clever self-repair mechanisms, because those self-repair mechanisms are not perfect. So basically, metabolism, which is defined as basically everything that keeps us alive from one day to the next, has side effects. Those side effects accumulate and eventually cause pathology. That ' s a fine definition. So we can put it this way : we can say that, you know, we have this chain of events. And there are really two games in town, according to most people, with regard to postponing aging. They ' re what I ' m calling here the"gerontology approach"and the"geriatrics approach."The geriatrician will intervene late in the day, when pathology is becoming evident, and the geriatrician will try and hold back the sands of time, and stop the accumulation of side effects from causing the pathology quite so soon. Of course, it ' s a very short-term-ist strategy ; it ' s a losing battle, because the things that are causing the pathology are becoming more abundant as time goes on. The gerontology approach looks much more promising on the surface, because, you know, prevention is better than cure. But unfortunately the thing is that we don ' t understand metabolism very well. In fact, we have a pitifully poor understanding of how organisms work — even cells we ' re not really too good on yet. We ' ve discovered things like, for example, RNA interference only a few years ago, and this is a really fundamental component of how cells work. Basically, gerontology is a fine approach in the end, but it is not an approach whose time has come when we ' re talking about intervention. So then, what do we do about that? I mean, that ' s a fine logic, that sounds pretty convincing, pretty ironclad, doesn ' t it? But it isn ' t. Before I tell you why it isn ' t, I ' m going to go a little bit into what I ' m calling step two. Just suppose, as I said, that we do acquire — let ' s say we do it today for the sake of argument — the ability to confer 30 extra years of healthy life on people who are already in middle age, let ' s say 55. I ' m going to call that"robust human rejuvenation."OK. What would that actually mean for how long people of various ages today — or equivalently, of various ages at the time that these therapies arrive — would actually live? In order to answer that question — you might think it ' s simple, but it ' s not simple. We can ' t just say,"Well, if they ' re young enough to benefit from these therapies, then they ' ll live 30 years longer."That ' s the wrong answer. And the reason it ' s the wrong answer is because of progress. There are two sorts of technological progress really, for this purpose. There are fundamental, major breakthroughs, and there are incremental refinements of those breakthroughs. Now, they differ a great deal in terms of the predictability of time frames. Fundamental breakthroughs : very hard to predict how long it ’ s going to take to make a fundamental breakthrough. It was a very long time ago that we decided that flying would be fun, and it took us until 1903 to actually work out how to do it. But after that, things were pretty steady and pretty uniform. I think this is a reasonable sequence of events that happened in the progression of the technology of powered flight. We can think, really, that each one is sort of beyond the imagination of the inventor of the previous one, if you like. The incremental advances have added up to something which is not incremental anymore. This is the sort of thing you see after a fundamental breakthrough. And you see it in all sorts of technologies. Computers : you can look at a more or less parallel time line, happening of course a bit later. You can look at medical care. I mean, hygiene, vaccines, antibiotics — you know, the same sort of time frame. So I think that actually step two, that I called a step a moment ago, isn ’ t a step at all. That in fact, the people who are young enough to benefit from these first therapies that give this moderate amount of life extension, even though those people are already middle-aged when the therapies arrive, will be at some sort of cusp. They will mostly survive long enough to receive improved treatments that will give them a further 30 or maybe 50 years. In other words, they will be staying ahead of the game. The therapies will be improving faster than the remaining imperfections in the therapies are catching up with us. This is a very important point for me to get across. Because, you know, most people, when they hear that I predict that a lot of people alive today are going to live to 1, 000 or more, they think that I ’ m saying that we ’ re going to invent therapies in the next few decades that are so thoroughly eliminating aging that those therapies will let us live to 1, 000 or more. I ’ m not saying that at all. I ’ m saying that the rate of improvement of those therapies will be enough. They ’ ll never be perfect, but we ’ ll be able to fix the things that 200- year-olds die of, before we have any 200-year-olds. And the same for 300 and 400 and so on. I decided to give this a little name, which is”longevity escape velocity.”Well, it seems to get the point across. So, these trajectories here are basically how we would expect people to live, in terms of remaining life expectancy, as measured by their health, for given ages that they were at the time that these therapies arrive. If you ’ re already 100, or even if you ’ re 80 — and an average 80-year-old, we probably can ’ t do a lot for you with these therapies, because you ’ re too close to death ’ s door for the really initial, experimental therapies to be good enough for you. You won ’ t be able to withstand them. But if you ’ re only 50, then there ’ s a chance that you might be able to pull out of the dive and, you know — — eventually get through this and start becoming biologically younger in a meaningful sense, in terms of your youthfulness, both physical and mental, and in terms of your risk of death from age-related causes. And of course, if you ’ re a bit younger than that, then you ’ re never really even going to get near to being fragile enough to die of age-related causes. So this is a genuine conclusion that I come to, that the first 150-year-old — we don ’ t know how old that person is today, because we don ’ t know how long it ’ s going to take to get these first-generation therapies. But irrespective of that age, I ’ m claiming that the first person to live to 1, 000 — **subject** of course, to, you know, global catastrophes — is actually, probably, only about 10 years younger than the first 150-year-old. And that ’ s quite a thought. Alright, so finally I ’ m going to spend the rest of the talk, my last seven-and- a-half minutes, on step one ; namely, how do we actually get to this moderate amount of life extension that will allow us to get to escape velocity? And in order to do that, I need to talk about mice a little bit. I have a corresponding milestone to robust human rejuvenation. I ' m calling it"robust mouse rejuvenation,"not very imaginatively. And this is what it is. I say we ' re going to take a long-lived strain of mouse, which basically means mice that live about three years on average. We do exactly nothing to them until they ' re already two years old. And then we do a whole bunch of stuff to them, and with those therapies, we get them to live, on average, to their fifth birthday. So, in other words, we add two years — we treble their remaining lifespan, starting from the point that we started the therapies. The question then is, what would that actually mean for the time frame until we get to the milestone I talked about earlier for humans? Which we can now, as I ' ve explained, equivalently call either robust human rejuvenation or longevity escape velocity. Secondly, what does it mean for the public ' s perception of how long it ' s going to take for us to get to those things, starting from the time we get the mice? And thirdly, the question is, what will it do to actually how much people want it? And it seems to me that the first question is entirely a biology question, and it ' s extremely hard to answer. One has to be very speculative, and many of my colleagues would say that we should not do this speculation, that we should simply keep our counsel until we know more. I say that ' s nonsense. I say we absolutely are irresponsible if we stay silent on this. We need to give our best guess as to the time frame, in order to give people a sense of proportion so that they can assess their priorities. So, I say that we have a 50 / 50 chance of reaching this RHR milestone, robust human rejuvenation, within 15 years from the point that we get to robust mouse rejuvenation. 15 years from the robust mouse. The public ' s perception will probably be somewhat better than that. The public tends to underestimate how difficult scientific things are. So they ' ll probably think it ' s five years away. They ' ll be wrong, but that actually won ' t matter too much. And finally, of course, I think it ' s fair to say that a large part of the reason why the public is so ambivalent about aging now is the global trance I spoke about earlier, the coping strategy. That will be history at this point, because it will no longer be possible to believe that aging is inevitable in humans, since it ' s been postponed so very effectively in mice. So we ' re likely to end up with a very strong change in people ' s attitudes, and of course that has enormous implications. So in order to tell you now how we ' re going to get these mice, I ' m going to add a little bit to my description of aging. I ' m going to use this word"damage"to denote these intermediate things that are caused by metabolism and that eventually cause pathology. Because the critical thing about this is that even though the damage only eventually causes pathology, the damage itself is caused ongoing-ly throughout life, starting before we ' re born. But it is not part of metabolism itself. And this turns out to be useful. Because we can re-draw our original diagram this way. We can say that, fundamentally, the difference between gerontology and geriatrics is that gerontology tries to inhibit the rate at which metabolism lays down this damage. And I ' m going to explain exactly what damage is in concrete biological terms in a moment. And geriatricians try to hold back the sands of time by stopping the damage converting into pathology. And the reason it ' s a losing battle is because the damage is continuing to accumulate. So there ' s a third approach, if we look at it this way. We can call it the"engineering approach,"and I claim that the engineering approach is within range. The engineering approach does not intervene in any processes. It does not intervene in this process or this one. And that ' s good because it means that it ' s not a losing battle, and it ' s something that we are within range of being able to do, because it doesn ' t involve improving on evolution. The engineering approach simply says,"Let ' s go and periodically repair all of these various types of damage — not necessarily repair them completely, but repair them quite a lot, so that we keep the level of damage down below the threshold that must exist, that causes it to be pathogenic."We know that this threshold exists, because we don ' t get age-related diseases until we ' re in middle age, even though the damage has been accumulating since before we were born. Why do I say that we ' re in range? Well, this is basically it. The point about this slide is actually the bottom. If we try to say which bits of metabolism are important for aging, we will be here all night, because basically all of metabolism is important for aging in one way or another. This list is just for illustration ; it is incomplete. The list on the right is also incomplete. It ' s a list of types of pathology that are age-related, and it ' s just an incomplete list. But I would like to claim to you that this list in the middle is actually complete — this is the list of types of thing that qualify as damage, side effects of metabolism that cause pathology in the end, or that might cause pathology. And there are only seven of them. They ' re categories of things, of course, but there ' s only seven of them. Cell loss, mutations in chromosomes, mutations in the mitochondria and so on. First of all, I ' d like to give you an argument for why that list is complete. Of course one can make a biological argument. One can say,"OK, what are we made of?"We ' re made of cells and stuff between cells. What can damage accumulate in? The answer is : long-lived molecules, because if a short-lived molecule undergoes damage, but then the molecule is destroyed — like by a protein being destroyed by proteolysis — then the damage is gone, too. It ' s got to be long-lived molecules. So, these seven things were all under discussion in gerontology a long time ago and that is pretty good news, because it means that, you know, we ' ve come a long way in biology in these 20 years, so the fact that we haven ' t extended this list is a pretty good indication that there ' s no extension to be done. However, it ' s better than that ; we actually know how to fix them all, in mice, in principle — and what I mean by in principle is, we probably can actually implement these fixes within a decade. Some of them are partially implemented already, the ones at the top. I haven ' t got time to go through them at all, but my conclusion is that, if we can actually get suitable funding for this, then we can probably develop robust mouse rejuvenation in only 10 years, but we do need to get serious about it. We do need to really start trying. So of course, there are some biologists in the audience, and I want to give some answers to some of the questions that you may have. You may have been dissatisfied with this talk, but fundamentally you have to go and read this stuff. I ' ve published a great deal on this ; I cite the experimental work on which my optimism is based, and there ' s quite a lot of detail there. The detail is what makes me confident of my rather aggressive time frames that I ' m predicting here. So if you think that I ' m wrong, you ' d better damn well go and find out why you think I ' m wrong. And of course the main thing is that you shouldn ' t trust people who call themselves gerontologists because, as with any radical departure from previous thinking within a particular field, you know, you expect people in the mainstream to be a bit resistant and not really to take it seriously. So, you know, you ' ve got to actually do your homework, in order to understand whether this is true. And we ' ll just end with a few things. One thing is, you know, you ' ll be hearing from a guy in the next session who said some time ago that he could sequence the human genome in half no time, and everyone said,"Well, it ' s obviously impossible."And you know what happened. So, you know, this does happen. We have various strategies — there ' s the Methuselah Mouse Prize, which is basically an incentive to innovate, and to do what you think is going to work, and you get money for it if you win. There ' s a proposal to actually put together an institute. This is what ' s going to take a bit of money. But, I mean, look — how long does it take to spend that on the war in Iraq? Not very long. OK. It ' s got to be philanthropic, because profits distract biotech, but it ' s basically got a 90 percent chance, I think, of succeeding in this. And I think we know how to do it. And I ’ ll stop there. Thank you. Chris Anderson : OK. I don ’ t know if there ’ s going to be any questions but I thought I would give people the chance. Audience : Since you ’ ve been talking about aging and trying to defeat it, why is it that you make yourself appear like an old man? AG : Because I am an old man. I am actually 158. Audience : Species on this planet have evolved with immune systems to fight off all the diseases so that individuals live long enough to procreate. However, as far as I know, all the species have evolved to actually die, so when cells divide, the telomerase get shorter, and eventually species die. So, why does — evolution has — seems to have selected against immortality, when it is so advantageous, or is evolution just incomplete? AG : Brilliant. Thank you for asking a question that I can answer with an uncontroversial answer. I ’ m going to tell you the genuine mainstream answer to your question, which I happen to agree with, which is that, no, aging is not a product of selection, evolution ; [aging] is simply a product of evolutionary neglect. In other words, we have aging because it ’ s hard work not to have aging ; you need more genetic pathways, more sophistication in your genes in order to age more slowly, and that carries on being true the longer you push it out. So, to the extent that evolution doesn ’ t matter, doesn ’ t care whether genes are passed on by individuals, living a long time or by procreation, there ’ s a certain amount of modulation of that, which is why different species have different lifespans, but that ’ s why there are no immortal species. CA : The genes don ’ t care but we do? AG : That ’ s right. Audience : Hello. I read somewhere that in the last 20 years, the average lifespan of basically anyone on the planet has grown by 10 years. If I project that, that would make me think that I would live until 120 if I don ’ t crash on my motorbike. That means that I ’ m one of your **subjects** to become a 1, 000-year- old? AG : If you lose a bit of weight. Your numbers are a bit out. The standard numbers are that lifespans have been growing at between one and two years per decade. So, it ’ s not quite as good as you might think, you might hope. But I intend to move it up to one year per year as soon as possible. Audience : I was told that many of the brain cells we have as adults are actually in the human embryo, and that the brain cells last 80 years or so. If that is indeed true, biologically are there implications in the world of rejuvenation? If there are cells in my body that live all 80 years, as opposed to a typical, you know, couple of months? AG : There are technical implications certainly. Basically what we need to do is replace cells in those few areas of the brain that lose cells at a respectable rate, especially neurons, but we don ’ t want to replace them any faster than that — or not much faster anyway, because replacing them too fast would degrade cognitive function. What I said about there being no non-aging species earlier on was a little bit of an oversimplification. There are species that have no aging — Hydra for example — but they do it by not having a nervous system — and not having any tissues in fact that rely for their function on very long-lived cells.

Text Sample 821: NEG Dim 5 – 1994 – Score 4.00 – 1994/t002458.txt

Because I usually take the role of trying to explain to people how wonderful the new technologies that are coming along are going to be, and I thought that, since I was among friends here, I would tell you what I really think and try to look back and try to understand what is really going on here with these amazing jumps in technology that seem so fast that we can barely keep on top of it. So I ’ m going to start out by showing just one very boring technology slide. And then, so if you can just turn on the slide that ’ s on. This is just a random slide that I picked out of my file. What I want to show you is not so much the details of the slide, but the general form of it. This happens to be a slide of some analysis that we were doing about the power of RISC microprocessors versus the power of local area networks. And the interesting thing about it is that this slide, like so many technology slides that we ’ re used to, is a sort of a straight line on a semi-log curve. In other words, every step here represents an order of magnitude in performance scale. And this is a new thing that we talk about technology on semi-log curves. Something really weird is going on here. And that ’ s basically what I ’ m going to be talking about. So, if you could bring up the lights. If you could bring up the lights higher, because I ’ m just going to use a piece of paper here. Now why do we draw technology curves in semi-log curves? Well the answer is, if I drew it on a normal curve where, let ’ s say, this is years, this is time of some sort, and this is whatever measure of the technology that I ’ m trying to graph, the graphs look sort of silly. They sort of go like this. And they don ’ t tell us much. Now if I graph, for instance, some other technology, say transportation technology, on a semi-log curve, it would look very stupid, it would look like a flat line. But when something like this happens, things are qualitatively changing. So if transportation technology was moving along as fast as microprocessor technology, then the day after tomorrow, I would be able to get in a taxi cab and be in Tokyo in 30 seconds. It ’ s not moving like that. And there ’ s nothing precedented in the history of technology development of this kind of self- feeding growth where you go by orders of magnitude every few years. Now the question that I ’ d like to ask is, if you look at these exponential curves, they don ’ t go on forever. Things just can ’ t possibly keep changing as fast as they are. One of two things is going to happen. Either it ’ s going to turn into a sort of classical S-curve like this, until something totally different comes along, or maybe it ’ s going to do this. That ’ s about all it can do. Now I ’ m an optimist, so I sort of think it ’ s probably going to do something like that. If so, that means that what we ’ re in the middle of right now is a transition. We ’ re sort of on this line in a transition from the way the world used to be to some new way that the world is. And so what I ’ m trying to ask, what I ’ ve been asking myself, is what ’ s this new way that the world is? What ’ s that new state that the world is heading toward? Because the transition seems very, very confusing when we ’ re right in the middle of it. Now when I was a kid growing up, the future was kind of the year 2000, and people used to talk about what would happen in the year 2000. Now here ’ s a conference in which people talk about the future, and you notice that the future is still at about the year 2000. It ’ s about as far as we go out. So in other words, the future has kind of been shrinking one year per year for my whole lifetime. Now I think that the reason is because we all feel that something ’ s happening there. That transition is happening. We can all sense it. And we know that it just doesn ’ t make too much sense to think out 30, 50 years because everything ’ s going to be so different that a simple extrapolation of what we ’ re doing just doesn ’ t make any sense at all. So what I would like to talk about is what that could be, what that transition could be that we ’ re going through. Now in order to do that I ’ m going to have to talk about a bunch of stuff that really has nothing to do with technology and computers. Because I think the only way to understand this is to really step back and take a long time scale look at things. So the time scale that I would like to look at this on is the time scale of life on Earth. So I think this picture makes sense if you look at it a few billion years at a time. So if you go back about two and a half billion years, the Earth was this big, sterile hunk of rock with a lot of chemicals floating around on it. And if you look at the way that the chemicals got organized, we begin to get a pretty good idea of how they do it. And I think that there ’ s theories that are beginning to understand about how it started with RNA, but I ’ m going to tell a sort of simple story of it, which is that, at that time, there were little drops of oil floating around with all kinds of different recipes of chemicals in them. And some of those drops of oil had a particular combination of chemicals in them which caused them to incorporate chemicals from the outside and grow the drops of oil. And those that were like that started to split and divide. And those were the most primitive forms of cells in a sense, those little drops of oil. But now those drops of oil weren ’ t really alive, as we say it now, because every one of them was a little random recipe of chemicals. And every time it divided, they got sort of unequal division of the chemicals within them. And so every drop was a little bit different. In fact, the drops that were different in a way that caused them to be better at incorporating chemicals around them, grew more and incorporated more chemicals and divided more. So those tended to live longer, get expressed more. Now that ’ s sort of just a very simple chemical form of life, but when things got interesting was when these drops learned a trick about abstraction. Somehow by ways that we don ’ t quite understand, these little drops learned to write down information. They learned to record the information that was the recipe of the cell onto a particular kind of chemical called DNA. So in other words, they worked out, in this mindless sort of evolutionary way, a form of writing that let them write down what they were, so that that way of writing it down could get copied. The amazing thing is that that way of writing seems to have stayed steady since it evolved two and a half billion years ago. In fact the recipe for us, our genes, is exactly that same code and that same way of writing. In fact, every living creature is written in exactly the same set of letters and the same code. In fact, one of the things that I did just for amusement purposes is we can now write things in this code. And I ’ ve got here a little 100 micrograms of white powder, which I try not to let the security people see at airports. But this has in it — what I did is I took this code — the code has standard letters that we use for symbolizing it — and I wrote my business card onto a piece of DNA and amplified it 10 to the 22 times. So if anyone would like a hundred million copies of my business card, I have plenty for everyone in the room, and, in fact, everyone in the world, and it ’ s right here. If I had really been a egotist, I would have put it into a virus and released it in the room. So what was the next step? Writing down the DNA was an interesting step. And that caused these cells — that kept them happy for another billion years. But then there was another really interesting step where things became completely different, which is these cells started exchanging and communicating information, so that they began to get communities of cells. I don ’ t know if you know this, but bacteria can actually exchange DNA. Now that ’ s why, for instance, antibiotic resistance has evolved. Some bacteria figured out how to stay away from penicillin, and it went around sort of creating its little DNA information with other bacteria, and now we have a lot of bacteria that are resistant to penicillin, because bacteria communicate. Now what this communication allowed was communities to form that, in some sense, were in the same boat together ; they were synergistic. So they survived or they failed together, which means that if a community was very successful, all the individuals in that community were repeated more and they were favored by evolution. Now the transition point happened when these communities got so close that, in fact, they got together and decided to write down the whole recipe for the community together on one string of DNA. And so the next stage that ’ s interesting in life took about another billion years. And at that stage, we have multi-cellular communities, communities of lots of different types of cells, working together as a single organism. And in fact, we ’ re such a multi- cellular community. We have lots of cells that are not out for themselves anymore. Your skin cell is really useless without a heart cell, muscle cell, a brain cell and so on. So these communities began to evolve so that the interesting level on which evolution was taking place was no longer a cell, but a community which we call an organism. Now the next step that happened is within these communities. These communities of cells, again, began to abstract information. And they began building very special structures that did nothing but process information within the community. And those are the neural structures. So neurons are the information processing apparatus that those communities of cells built up. And in fact, they began to get specialists in the community and special structures that were responsible for recording, understanding, learning information. And that was the brains and the nervous system of those communities. And that gave them an evolutionary advantage. Because at that point, an individual — learning could happen within the time span of a single organism, instead of over this evolutionary time span. So an organism could, for instance, learn not to eat a certain kind of fruit because it tasted bad and it got sick last time it ate it. That could happen within the lifetime of a single organism, whereas before they ' d built these special information processing structures, that would have had to be learned evolutionarily over hundreds of thousands of years by the individuals dying off that ate that kind of fruit. So that nervous system, the fact that they built these special information structures, tremendously sped up the whole process of evolution. Because evolution could now happen within an individual. It could happen in learning time scales. But then what happened was the individuals worked out, of course, tricks of communicating. And for example, the most sophisticated version that we ' re aware of is human language. It ' s really a pretty amazing invention if you think about it. Here I have a very complicated, messy, confused idea in my head. I ' m sitting here making grunting sounds basically, and hopefully constructing a similar messy, confused idea in your head that bears some analogy to it. But we ' re taking something very complicated, turning it into sound, sequences of sounds, and producing something very complicated in your brain. So this allows us now to begin to start functioning as a single organism. And so, in fact, what we ' ve done is we, humanity, have started abstracting out. We ' re going through the same levels that multi-cellular organisms have gone through — abstracting out our methods of recording, presenting, processing information. So for example, the invention of language was a tiny step in that direction. Telephony, computers, videotapes, CD-ROMs and so on are all our specialized mechanisms that we ' ve now built within our society for handling that information. And it all connects us together into something that is much bigger and much faster and able to evolve than what we were before. So now, evolution can take place on a scale of microseconds. And you saw Ty ' s little evolutionary example where he sort of did a little bit of evolution on the Convolution program right before your eyes. So now we ' ve speeded up the time scales once again. So the first steps of the story that I told you about took a billion years a piece. And the next steps, like nervous systems and brains, took a few hundred million years. Then the next steps, like language and so on, took less than a million years. And these next steps, like electronics, seem to be taking only a few decades. The process is feeding on itself and becoming, I guess, autocatalytic is the word for it — when something reinforces its rate of change. The more it changes, the faster it changes. And I think that that ' s what we ' re seeing here in this explosion of curve. We ' re seeing this process feeding back on itself. Now I design computers for a living, and I know that the mechanisms that I use to design computers would be impossible without recent advances in computers. So right now, what I do is I design objects at such complexity that it ' s really impossible for me to design them in the traditional sense. I don ' t know what every transistor in the connection machine does. There are billions of them. Instead, what I do and what the designers at Thinking Machines do is we think at some level of abstraction and then we hand it to the machine and the machine takes it beyond what we could ever do, much farther and faster than we could ever do. And in fact, sometimes it takes it by methods that we don ’ t quite even understand. One method that ’ s particularly interesting that I ’ ve been using a lot lately is evolution itself. So what we do is we put inside the machine a process of evolution that takes place on the microsecond time scale. So for example, in the most extreme cases, we can actually evolve a program by starting out with random sequences of instructions. Say,”Computer, would you please make a hundred million random sequences of instructions. Now would you please run all of those random sequences of instructions, run all of those programs, and pick out the ones that came closest to doing what I wanted.”So in other words, I define what I wanted. Let ’ s say I want to sort numbers, as a simple example I ’ ve done it with. So find the programs that come closest to sorting numbers. So of course, random sequences of instructions are very unlikely to sort numbers, so none of them will really do it. But one of them, by luck, may put two numbers in the right order. And I say,”Computer, would you please now take the 10 percent of those random sequences that did the best job. Save those. Kill off the rest. And now let ’ s reproduce the ones that sorted numbers the best. And let ’ s reproduce them by a process of recombination analogous to sex.”Take two programs and they produce children by exchanging their subroutines, and the children inherit the traits of the subroutines of the two programs. So I ’ ve got now a new generation of programs that are produced by combinations of the programs that did a little bit better job. Say,”Please repeat that process.”Score them again. Introduce some mutations perhaps. And try that again and do that for another generation. Well every one of those generations just takes a few milliseconds. So I can do the equivalent of millions of years of evolution on that within the computer in a few minutes, or in the complicated cases, in a few hours. At the end of that, I end up with programs that are absolutely perfect at sorting numbers. In fact, they are programs that are much more efficient than programs I could have ever written by hand. Now if I look at those programs, I can ’ t tell you how they work. I ’ ve tried looking at them and telling you how they work. They ’ re obscure, weird programs. But they do the job. And in fact, I know, I ’ m very confident that they do the job because they come from a line of hundreds of thousands of programs that did the job. In fact, their life depended on doing the job. I was riding in a 747 with Marvin Minsky once, and he pulls out this card and says,”Oh look. Look at this. It says, ’ This plane has hundreds of thousands of tiny parts working together to make you a safe flight. ’ Doesn ’ t that make you feel confident?”In fact, we know that the engineering process doesn ’ t work very well when it gets complicated. So we ’ re beginning to depend on computers to do a process that ’ s very different than engineering. And it lets us produce things of much more complexity than normal engineering lets us produce. And yet, we don ’ t quite understand the options of it. So in a sense, it ’ s getting ahead of us. We ’ re now using those programs to make much faster computers so that we ’ ll be able to run this process much faster. So it ’ s feeding back on itself. The thing is becoming faster and that ’ s why I think it seems so confusing. Because all of these technologies are feeding back on themselves. We ’ re taking off. And what we are is we ’ re at a point in time which is analogous to when single- celled organisms were turning into multi-celled organisms. So we ’ re the amoebas and we can ’ t quite figure out what the hell this thing is we ’ re creating. We ’ re right at that point of transition. But I think that there really is something coming along after us. I think it ’ s very haughty of us to think that we ’ re the end product of evolution. And

I think all of us here are a part of producing whatever that next thing is. So lunch is coming along, and I think I will stop at that point, before I get selected out.

Text Sample 822: NEG Dim 5 – 1984 – Score 5.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ’ ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a computer and really address the qualities of the human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, the intelligence of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ’ ve been able to collect, which I think illustrate this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a slide taken from a TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ’ s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ’ s the distance you should sit away from the TV set. Now we ’ ve put people 18 inches in front of a TV, and all the artifacts that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow mask, the scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ’ m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ’ m very interested in touch-sensitive displays. High-tech, high-touch. Isn ’ t that what some of you said? It ’ s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ’ re not : it ’ s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on the market these systems that have just a few light emitting diodes on the side and are very low resolution, it ’ s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ’ t have to pick them up, and people don ’ t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ’ re typing — what you have — you want to now put something — first of all, you ’ ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ’ ve got to sort of find that mouse. Then you find the mouse, and you ’ re going to have to wiggle it a little bit to see where the cursor is on the screen. And then when you finally see where it is, then you ’ ve got to move it to get the cursor over there, and then —"Bang"— you ' ve got to hit a button or do whatever. That ' s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ' m trying to do is just illustrate the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of the computer field, is when you have a bug that you can ' t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the substrate, which it usually is. So we found that that actually was a feature in the sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ' s pressure-sensitive. And it ' s pressure-sensitive to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ' t figure out how to come in the other direction. But let me get rid of the slide, and let ' s see if this comes on. OK. So there is the pressure-sensitive display in operation. The person ' s just, if you will, pushing on the screen to make a curve. But this is the interesting part. I want to stop it for a second because the movie is very badly made. And the particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an anvil on a fuzzy rug and the other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across the screen, and yet you touch the ping-pong ball very lightly and it just scoots across the screen. And what you can do — oops, I didn ' t mean to do that — what you can do is actually feed back to the user the feeling of the physical properties. So again, they don ' t have to be weight ; they could be a general trying to move troops, and he ' s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the interface there are physical properties in that transducer — in this case it ' s pressure and touches — that allow you to present things to the user that you could never present before. So it ' s not simply looking at the quality or, if you will, the luxury of that interface, but it ' s actually looking at the idea of presenting things that previously couldn ' t be presented before. I want to move on to another example, which is one of a different sort, where we ' re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ' re going to take this book, if you will, and it ' s going to come alive. You ' re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are the experts in monologue making : you make a film and it has a well-formed beginning, middle and end, and in some sense the art of it is that. And you then say,"There ' s an opportunity for making conversational movies.”Well, what does that mean? And it sort of nibbles at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ’ ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the movie has information about itself. So it knows, or at least there is computer-readable information in the medium itself. It ’ s just not a static movie frame. That ’ s one thing. The other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the cookbook, the”Larousse Gastronomique.”And I think I use the example all too often, but it ’ s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to the end of the recipe and it says,”Cook until done.”Now, that would be, if you will, the top green track, which doesn ’ t mean too much. But you might have to elaborate for me or for somebody who isn ’ t an expert, and say,”Cook at 380 degrees for 45 minutes.”And then for a real beginner, you would go down even further and elaborate more — say,”Open the oven, preheat, wait for the light to go out, open the door, don ’ t leave it open too long, put the penguin in and shut the door...”whatever. And that ’ s a much more elaborate one than you dribble back. That ’ s one kind of use of random access. And the other is where you want to explain the same thing in different ways. If you ’ re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ’ s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ’ s a rather boring book, but I ’ m afraid sometimes you have to do boring books because your sponsors aren ’ t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ’ t even know what vintage the transmission is, but let me just show you very quickly some of it, and we ’ ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of the transmission, and as you rub your finger across the transmission it highlights the various parts. Narrator : When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, superimposed over the illustration. NN : This is about the oil pan, or the oil filter and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get back to the illustration, but in this case it ’ s not a single frame, but it ’ s actually a movie of someone coming into the frame and doing the repair that ’ s described in the text. The two-headed slider is a speed control that allows me to watch the movie at various speeds, in forward or reverse. And the movie is displayed as a full frame movie. I can go back to the beginning... and play the movie at full speed. Here ’ s another step-by-step procedure, only in this case — NN : Okay, this movie is... Everybody ’ s heard of sound-sync movies — this is text-sync movies, so as the movie plays, the text gets highlighted. We highlight the text as we go through the movie. Repairman :... Not too far out. Front poles, preferably. Don ’ t loosen them too far. If you loosen them too far, you ’ ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ’ m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ’ ve been doing — in this case, in Senegal — where we have tried to use personal computers as a pedagogical medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ’ s a story I ’ d like to tell. And that was when, before we did this in any developing countries — we ’ re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ’ ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ’ t read, didn ’ t even make it in the lowest section of the school ’ s classes — and was pretty much not in school, though physically there. But did hang around the, quote,”computer room,”where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very interesting. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,”Let me show you how this works,”and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ’ t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ’ d actually come to see — not the computer room — they went upstairs and they said,”This is absolutely remarkable! That child was very articulate and showed us and even dealt with the things he couldn ’ t do automatically with that manual. It was just absolutely fantastic.”The principal said,”There ’ s a dreadful mistake, because that child can ’ t read. And you obviously have been hoodwinked or you ’ ve talked about somebody else.”And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,”Can you read?”And the child said,”No, I can ’ t.”And then they said,”But wait a minute. You just looked through that manual and you found...”and he said,”Oh, but that ’ s not reading.”And so they said,”Well, what ’ s reading then?”He says,”Well, reading is this junk they give me in little books to read. It ’ s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return.”And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ’ m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ’ s not, but we think speech —”My God, little children pick it up somehow, and by the age of two they ’ re doing a mediocre job, and by three and four they ’ re speaking reasonably well. And yet you ’ ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder.”Well, it ’ s not harder.

What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ’ s going to help you. So what happens is you go to school and people say,”Just believe me, you ’ re going to like it. You ’ re going to like reading,”and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on the screen has a huge payoff, and it ’ s a lot of fun. And in fact, it ’ s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ’ s the girl on the left leaning with her chalkboard, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ’ s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this **subject**. I will come to my last example and then quit. And my last example — as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And the reason it remains a favorite project is that we were asked to do a teleconferencing system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ’ s faces. So if I wanted to call my friend Peter Sprague on the phone, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ’ s uncanny : there ’ s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ’ s something that didn ’ t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,”He or she wouldn ’ t agree,”and the empty chair is that person and the spatiality is crucial. So we said,”Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ’ s site you ’ d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ’ t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ’ ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ’ s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ' s site, he ' ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.

Text Sample 823: NEG Dim 5 – 2002 – Score 1.00 – 2002/t000333.txt

Thank you. Ooh, I ' m like,"Phew, phew, calm down. Get back into my body now."Usually when I play out, the first thing that happens is people scream out,"What ' s she doing?!"I ' ll play at these rock shows, be on stage standing completely still, and they ' re like,"What ' s she doing?! What ' s she doing?!"And then I ' ll kind of be like — — and then they ' re like,"Whoa!"I ' m sure you ' re trying to figure out,"Well, how does this thing work?"Well, what I ' m doing is controlling the pitch with my left hand. See, the closer I get to this antenna, the higher the note gets — — and you can get it really low. And with this hand I ' m controlling the volume, so the further away my right hand gets, the louder it gets. So basically, with both of your hands you ' re controlling pitch and volume and kind of trying to create the illusion that you ' re doing separate notes, when really it ' s continuously going... Sometimes I startle myself : I ' ll forget that I have it on, and I ' ll lean over to pick up something, and then it goes like — —"Oh!"And it ' s like a funny sound effect that follows you around if you don ' t turn the thing off. Maybe we ' ll go into the next tune, because I totally lost where this is going. We ' re going to do a song by David Mash called"Listen : the Words Are Gone,"and maybe I ' ll have words come back into me afterwards if I can relax. So, I ' m trying to think of some of the questions that are commonly asked ; there are so many. And... Well, I guess I could tell you a little of the history of the theremin. It was invented around the 1920s, and the inventor, Lon Theremin — he also was a musician besides an inventor — he came up with the idea for making the theremin, I think, when he was working on some shortwave radios. And there ' d be that sound in the signal — it ' s like — and he thought,"Oh, what if I could control that sound and turn it into an instrument, because there are pitches in it."And so somehow through developing that, he eventually came to make the theremin the way it is now. And a lot of times, even kids nowadays, they ' ll make reference to a theremin by going,"Whoo-hoo-hoo-hoo,"because in the ' 50s it was used in the sci-fi horror movies, that sound that ' s like... It ' s kind of a funny, goofy sound to do. And sometimes if I have too much coffee, then my vibrato gets out of hand. You ' re really sensitive to your body and its functions when you ' re behind this thing. You have to stay so still if you want to have the most control. It reminds me of the balancing act earlier on — what Michael was doing — because you ' re fighting so hard to keep the balance with what you ' re playing with and stay in tune, and at the same time you don ' t want to focus so much on being in tune all the time ; you want to be feeling the music. And then also, you ' re trying to stay very, very, very still because little movements with other parts of your body will affect the pitch, or sometimes if you ' re holding a low note — — and breathing will make it... If I pass out on the next song... I think of it almost like like a yoga instrument because it makes you so aware of every little crazy thing your body is doing, or just aware of what you don ' t want it to be doing while you ' re playing ; you don ' t want to have any sudden movements. And if I go to a club and play a gig, people are like,"Here, have some drinks on us!"And it ' s like,"Well, I ' m about to go on soon ; I don ' t want to be like — — you know?"It really does reflect the mood that you ' re in also, if you ' re... it ' s similar to being a vocalist, except instead of it coming out of your throat, you

’ re controlling it just in the air and you don ’ t really have a point of reference ; you ’ re always relying on your ears and adjusting constantly. You just have to always adjust to what ’ s happening and realize you ’ ll have bummer notes come here and there and listen to it, adjust it, and just move on, or else you ’ ll get too tied up and go crazy. Like me. I think we will play another tune now. I ’ m going to do”Lush Life.”It ’ s one of my favorite tunes to play.

Text Sample 824: NEG Dim 5 – 2002 – Score 2.00 – 2002/t000397.txt
How many Creationists do we have in the room? Probably none. I think we ’ re all Darwinians. And yet many Darwinians are anxious, a little uneasy â would like to see some limits on just how far the Darwinism goes. It ’ s all right. You know spiderwebs? Sure, they are products of evolution. The World Wide Web? Not so sure. Beaver dams, yes. Hoover Dam, no. What do they think it is that prevents the products of human ingenuity from being themselves, fruits of the tree of life, and hence, in some sense, obeying evolutionary rules? And yet people are interestingly resistant to the idea of applying evolutionary thinking to thinking â to our thinking. And so I ’ m going to talk a little bit about that, keeping in mind that we have a lot on the program here. So you ’ re out in the woods, or you ’ re out in the pasture, and you see this ant crawling up this blade of grass. It climbs up to the top, and it falls, and it climbs, and it falls, and it climbs â trying to stay at the very top of the blade of grass. What is this ant doing? What is this in aid of? What goals is this ant trying to achieve by climbing this blade of grass? What ’ s in it for the ant? And the answer is : nothing. There ’ s nothing in it for the ant. Well then, why is it doing this? Is it just a fluke? Yeah, it ’ s just a fluke. It ’ s a lancet fluke. It ’ s a little brain worm. It ’ s a parasitic brain worm that has to get into the stomach of a sheep or a cow in order to continue its life cycle. Salmon swim upstream to get to their spawning grounds, and lancet flukes commandeer a passing ant, crawl into its brain, and drive it up a blade of grass like an all-terrain vehicle. So there ’ s nothing in it for the ant. The ant ’ s brain has been hijacked by a parasite that infects the brain, inducing suicidal behavior. Pretty scary. Well, does anything like that happen with human beings? This is all on behalf of a cause other than one ’ s own genetic fitness, of course. Well, it may already have occurred to you that Islam means”surrender,”or”submission of self-interest to the will of Allah.”Well, it ’ s ideas â not worms â that hijack our brains. Now, am I saying that a sizable minority of the world ’ s population has had their brain hijacked by parasitic ideas? No, it ’ s worse than that. Most people have. There are a lot of ideas to die for. Freedom, if you ’ re from New Hampshire. Justice. Truth. Communism. Many people have laid down their lives for communism, and many have laid down their lives for capitalism. And many for Catholicism. And many for Islam. These are just a few of the ideas that are to die for. They ’ re infectious. Yesterday, Amory Lovins spoke about”infectious repititis.”It was a term of abuse, in effect. This is unthinking engineering. Well, most of the cultural spread that goes on is not brilliant, new, out-of-the-box thinking. It ’ s”infectious repetitis,”and we might as well try to have a theory of what ’ s going on when that happens so that we can understand the conditions of infection. Hosts work hard to spread these ideas to others. I myself am a philosopher, and one of our occupational hazards is that people ask us what the meaning of life is. And you have to have a bumper sticker, you know. You have to have a statement. So, this is mine. The secret of happiness is : Find something more important than you are and dedicate your life to it. Most of us â now that the”Me Decade”is well

in the past â now we actually do this. One set of ideas or another have simply replaced our biological imperatives in our own lives. This is what our summum bonum is. It ’ s not maximizing the number of grandchildren we have. Now, this is a profound biological effect. It ’ s the subordination of genetic interest to other interests. And no other species does anything at all like it. Well, how are we going to think about this? It is, on the one hand, a biological effect, and a very large one. Unmistakable. Now, what theories do we want to use to look at this? Well, many theories. But how could something tie them together? The idea of replicating ideas ; ideas that replicate by passing from brain to brain. Richard Dawkins, whom you ’ ll be hearing later in the day, invented the term”memes,”and put forward the first really clear and vivid version of this idea in his book”The Selfish Gene.”Now here am I talking about his idea. Well, you see, it ’ s not his. Yes â he started it. But it ’ s everybody ’ s idea now. And he ’ s not responsible for what I say about memes. I ’ m responsible for what I say about memes. Actually, I think we ’ re all responsible for not just the intended effects of our ideas, but for their likely misuses. So it is important, I think, to Richard, and to me, that these ideas not be abused and misused. They ’ re very easy to misuse. That ’ s why they ’ re dangerous. And it ’ s just about a full-time job trying to prevent people who are scared of these ideas from caricaturing them and then running off to one dire purpose or another. So we have to keep plugging away, trying to correct the misapprehensions so that only the benign and useful variants of our ideas continue to spread. But it is a problem. We don ’ t have much time, and I ’ m going to go over just a little bit of this and cut out, because there ’ s a lot of other things that are going to be said. So let me just point out : memes are like viruses. That ’ s what Richard said, back in ’ 93. And you might think,”Well, how can that be? I mean, a virus is â you know, it ’ s stuff! What ’ s a meme made of?”Yesterday, Negroponte was talking about viral telecommunications but â what ’ s a virus? A virus is a string of nucleic acid with attitude. That is, there is something about it that tends to make it replicate better than the competition does. And that ’ s what a meme is. It ’ s an information packet with attitude. What ’ s a meme made of? What are bits made of, Mom? Not silicon. They ’ re made of information, and can be carried in any physical medium. What ’ s a word made of? Sometimes when people say,”Do memes exist?”I say,”Well, do words exist? Are they in your ontology?”If they are, words are memes that can be pronounced. Then there ’ s all the other memes that can ’ t be pronounced. There are different species of memes. Remember the Shakers? Gift to be simple? Simple, beautiful furniture? And, of course, they ’ re basically extinct now. And one of the reasons is that among the creed of Shaker-dom is that one should be celibate. Not just the priests. Everybody. Well, it ’ s not so surprising that they ’ ve gone extinct. But in fact that ’ s not why they went extinct. They survived as long as they did at a time when the social safety nets weren ’ t there. And there were lots of widows and orphans, people like that, who needed a foster home. And so they had a ready supply of converts. And they could keep it going. And, in principle, it could ’ ve gone on forever, with perfect celibacy on the part of the hosts. The idea being passed on through proselytizing, instead of through the gene line. So the ideas can live on in spite of the fact that they ’ re not being passed on genetically. A meme can flourish in spite of having a negative impact on genetic fitness. After all, the meme for Shaker-dom was essentially a sterilizing parasite. There are other parasites that do this â which render the host sterile. It ’ s part of their plan. They don ’ t have to have minds to have a plan. I ’ m just going to draw your attention to just one of the many implications of the memetic perspective, which I recommend. I ’ ve not time to go into more of it. In Jared Diamond ’ s wonderful book,”Guns, Germs and Steel,”he talks about how it was germs, more than guns and steel, that conquered the new hemisphere â the Western hemisphere â that conquered the rest of the world. When European explorers and travelers spread out, they brought with them the germs that they had become essentially immune to, that they had learned how to tolerate over hundreds and hundreds of years, thousands of years, of living with domesticated animals who were the sources of those pathogens. And they just wiped out â these pathogens just wiped out the native people, who had no immunity to them at all. And we ’ re doing it again. We ’ re doing it this time with toxic ideas. Yesterday, a number of people â Nicholas Negroponte and others â spoke about all the wonderful things that are happening when our ideas get spread out, thanks to all the new technology all over the world. And I agree. It is largely wonderful. Largely wonderful. But among all those ideas that inevitably flow out into the whole world thanks to our technology, are a lot of toxic ideas. Now, this has been realized for some time. Sayyid Qutb is one of the founding fathers of fanatical Islam, one of the ideologues that inspired Osama bin Laden.”One has only to glance at its press films, fashion shows, beauty contests, ballrooms, wine bars and broadcasting stations.”Memes. These memes are spreading around the world and they are wiping out whole cultures. They are wiping out languages. They are wiping out traditions and practices. And it ’ s not our fault, anymore than it ’ s our fault when our germs lay waste to people that haven ’ t developed the immunity. We have an immunity to all of the junk that lies around the edges of our culture. We ’ re a free society, so we let pornography and all these things â we shrug them off. They ’ re like a mild cold. They ’ re not a big deal for us. But we should recognize that for many people in the world, they are a big deal. And we should be very alert to this. As we spread our education and our technology, one of the things that we are doing is we ’ re the vectors of memes that are correctly viewed by the hosts of many other memes as a dire threat to their favorite memes â the memes that they are prepared to die for. Well now, how are we going to tell the good memes from the bad memes? That is not the job of the science of memetics. Memetics is morally neutral. And so it should be. This is not the place for hate and anger. If you ’ ve had a friend who ’ s died of AIDS, then you hate HIV. But the way to deal with that is to do science, and understand how it spreads and why in a morally neutral perspective. Get the facts. Work out the implications. There ’ s plenty of room for moral passion once we ’ ve got the facts and can figure out the best thing to do. And, as with germs, the trick is not to try to annihilate them. You will never annihilate the germs. What you can do, however, is foster public health measures and the like that will encourage the evolution of avirulence. That will encourage the spread of relatively benign mutations of the most toxic varieties. That ’ s all the time I have, so thank you very much for your attention.

Text Sample 825: NEG Dim 5 – 2002 – Score 3.00 – 2002/t000313.txt

This is your conference, and I think you have a right to know a little bit right now, in this transition period, about this guy who ’ s going to be looking after it for you for a bit. So, I ’ m just going to grab a chair here. Two years ago at TED, I think â I ’ ve come to this conclusion â I think I may have been suffering from a strange delusion. I think that I may have believed unconsciously, then, that I was kind of a business hero. I had this company that I ’ d spent 15 years building. It ’ s called Future ; it was a magazine publishing company. It had recently gone public and the market said that it was apparently worth two billion dollars, a number I didn ’ t really understand. A magazine I ’ d recently launched called Business 2. 0 was fatter than a telephone directory, busy pumping hot air into the bubble. And I was the 40 percent owner of a dotcom that was about to go public and no doubt be worth billions more. And all this had come from nothing. Fifteen years earlier, I was a science journalist who people just laughed at when I said,"I really would like to start my own computer magazine."And 15 years later, there are 100 of them and 2, 000 people on staff and it was just such heady times. The date was February 2000. I thought the little graph of my business life that kind of looked a bit like Moore ' s Law â ever upward and to the right â it was going to go on forever. I mean, it had to. Right? I was in for quite a surprise. The dotcom, ironically called Snowball, was the very last consumer web company to go public the next month before NASDAQ exploded, and I entered 18 months of business hell. I watched everything that I ' d built crumbling, and it looked like all this stuff was going to die and 15 years work would have come for nothing. And it was gut wrenching. It took eight years of blood, sweat and tears to reach 350 employees, something which I was very proud of in the business. February 2001 â in one day we laid off 350 people, and before the bloodshed was finished, 1, 000 people had lost their jobs from my companies. I felt sick. I watched my own net worth falling by about a million dollars a day, every day, for 18 months. And worse than that, far worse than that, my sense of self-worth was kind of evaporating. I was going around with this big sign on my forehead :"LOSER."And I think what disgusts me more than anything, looking back, is how the hell did I let my personal happiness get so tied up with this business thing? Well, in the end, we were able to save Future and Snowball, but I was, at that point, ready to move on. And to cut a long story short, here ' s where I came to. And the reason I ' m telling this story is that I believe, from many conversations, that a lot of people in this room have been through a similar kind of rollercoaster â emotional rollercoaster â in the last couple years. This has been a big, big transition time, and I believe that this conference can play a big part for all of us in taking us forward to the next stage to whatever ' s next. The theme next year is re-birth. It was at the same TED two years ago when Richard and I reached an agreement on the future of TED. And at about the same time, and I think partly because of that, I started doing something that I ' d forgotten about in my business focus : I started to read again. And I discovered that while I ' d been busy playing business games, there ' d been this incredible revolution in so many areas of interest : cosmology to psychology to evolutionary psychology to anthropology to... all this stuff had changed. And the way in which you could think about us as a species and us as a planet had just changed so much, and it was incredibly exciting. And what was really most exciting â and I think Richard Wurman discovered this at least 20 years before I did â was that all this stuff is connected. It ' s connected ; it all hooks into each other. We talk about this a lot, and I thought about trying to give an example of this. So, just one example : Madame de Gaulle, the wife of the French president, was famously asked once,"What do you most desire?"And she answered,"A penis."And when you think about it, it ' s very true : what we all most desire is a penis â or"happiness"as we say in English. And something... good luck with that one in the Japanese translation room. But something as basic as happiness, which 20 years ago would have been just something for discussion in the church or mosque or synagogue, today it turns out that there ' s dozens of TED-like questions that you can ask about it, which are really interesting. You can ask about what causes it biochemically : neuroscience, serotonin, all that stuff. You can ask what are the psychological causes of it : nature? Nurture? Current circumstance? Turns out that the research done on that is absolutely mind-blowing. You can view it as a computing problem, an artificial intelligence problem : do you need to incorporate some sort of analog of happiness into a computer brain to make it work properly? You can view it in sort of geopolitical terms and say, why is it that a billion people on this planet are so desperately needy that they have no possibility of happiness, and whereas almost all the rest of them, regardless of how much money they have â whether it ’ s two dollars a day or whatever â are almost equally happy on average? Or you can view it as an evolutionary psychology kind of thing : did our genes invent this as a kind of trick to get us to behave in certain ways? The ant ’ s brain, parasitized, to make us behave in certain ways so that our genes would propagate? Are we the victims of a mass delusion? And so on, and so on. To understand even something as important to us as happiness, you kind of have to branch off in all these different directions, and there ’ s nowhere that I ’ ve discovered â other than TED â where you can ask that many questions in that many different directions. And so, it ’ s the profound thing that Richard talks about : to understand anything, you just need to understand the little bits ; a little bit about everything that surrounds it. And so, gradually over these three days, you start off kind of trying to figure out,”Why am I listening to all this irrelevant stuff?”And at the end of the four days, your brain is humming and you feel energized, alive and excited, and it ’ s because all these different bits have been put together. It ’ s the total brain experience, we ’ re going to... it ’ s the mental equivalent of the full body massage. Every mental organ addressed. It really is. Enough of the theory, Chris. Tell us what you ’ re actually going to do, all right? So, I will. Here ’ s the vision for TED. Number one : do nothing. This thing ain ’ t broke, so I ain ’ t gonna fix it. Jeff Bezos kindly remarked to me,”Chris, TED is a really great conference. You ’ re going to have to fuck up really badly to make it bad.”So, I gave myself the job title of TED Custodian for a reason, and I will promise you right here and now that the core values that make TED special are not going to be interfered with. Truth, curiosity, diversity, no selling, no corporate bullshit, no bandwagoning, no platforms. Just the pursuit of interest, wherever it lies, across all the disciplines that are represented here. That ’ s not going to be changed at all. Number two : I am going to put together an incredible line up of speakers for next year. The time scale on which TED operates is just fantastic after coming out of a magazine business with monthly deadlines. There ’ s a year to do this, and already â I hope to show you a bit later â there ’ s 25 or so terrific speakers signed up for next year. And I ’ m getting fantastic help from the community ; this is just such a great community. And combined, our contacts reach pretty much everyone who ’ s interesting in the country, if not the planet. It ’ s true. Number three : I do want to, if I can, find a way of extending the TED experience throughout the year a little bit. And one key way that we ’ re going to do this is to introduce this book club. Books kind of saved me in the last couple years, and that ’ s a gift that I would like to pass on. So, when you sign up for TED2003, every six weeks you ’ ll get a care package with a book or two and a reason why they ’ re linked to TED. They may well be by a TED speaker, and so we can get the conversation going during the year and come back next year having had the same intellectual, emotional journey. I think it will be great. And then, fourthly : I want to mention the Sapling Foundation, which is the new owner of TED. What Sapling ’ s ownership means is that all of the proceeds of TED will go towards the causes that Sapling stands for. And more important, I think, the ideas that are exhibited and realized here are ideas that the foundation can use, because there ’ s fantastic synergy. Already, just in the last few days, we ’ ve had so many people talking about stuff that they care about, that they ’ re passionate about, that can make a difference in the world, and the idea of getting this group of people together â some of the causes that we believe in, the money that this conference can raise and the ideas â I really believe that that combination will, over time, make a difference. I ’ m incredibly excited about that. In fact, I don ’ t think, overall, that I ’ ve been as excited by anything ever in my life. I ’ m in this for the long run, and I would be greatly honored and excited if you ’ ll come on this journey with me.

Text Sample 826: NEG Dim 5 – 1990 – Score 77.00 – 1990/t000287.txt

I ’ m going to go right into the slides. And all I ’ m going to try and prove to you with these slides is that I do just very straight stuff. And my ideas are â in my head, anyway â they ’ re very logical and relate to what ’ s going on and problem solving for clients. I either convince clients at the end that I solve their problems, or I really do solve their problems, because usually they seem to like it. Let me go right into the slides. Can you turn off the light? Down. I like to be in the dark. I don ’ t want you to see what I ’ m doing up here. Anyway, I did this house in Santa Monica, and it got a lot of notoriety. In fact, it appeared in a porno comic book, which is the slide on the right. This is in Venice. I just show it because I want you to know I ’ m concerned about context. On the left-hand side, I had the context of those little houses, and I tried to build a building that fit into that context. When people take pictures of these buildings out of that context they look really weird, and my premise is that they make a lot more sense when they ’ re photographed or seen in that space. And then, once I deal with the context, I then try to make a place that ’ s comfortable and private and fairly serene, as I hope you ’ ll find that slide on the right. And then I did a law school for Loyola in downtown L. A. I was concerned about making a place for the study of law. And we continue to work with this client. The building on the right at the top is now under construction. The garage on the right â the gray structure â will be torn down, finally, and several small classrooms will be placed along this avenue that we ’ ve created, this campus. And it all related to the clients and the students from the very first meeting saying they felt denied a place. They wanted a sense of place. And so the whole idea here was to create that kind of space in downtown, in a neighborhood that was difficult to fit into. And it was my theory, or my point of view, that one didn ’ t upstage the neighborhood â one made accommodations. I tried to be inclusive, to include the buildings in the neighborhood, whether they were buildings I liked or not. In the ’ 60s I started working with paper furniture and made a bunch of stuff that was very successful in Bloomingdale ’ s. We even made flooring, walls and everything, out of cardboard. And the success of it threw me for a loop. I couldn ’ t deal with the success of furniture â I wasn ’ t secure enough as an architect â and so I closed it all up and made furniture that nobody would like. So, nobody would like this. And it was in this, preliminary to these pieces of furniture, that Ricky and I worked on furniture by the slice. And after we failed, I just kept failing. The piece on the left â and that ultimately led to the piece on the right â happened when the kid that was working on this took one of those long strings of stuff and folded it up to put it in the wastebasket. And I put a piece of tape around it, as you see there, and realized you could sit on it, and it had a lot of resilience and strength and so on. So, it was an accidental discovery. I got into fish. I mean, the story I tell is that I got mad at postmodernism â at po-mo â and said that fish were 500 million years earlier than man, and if you ’ re going to go back, we might as well go back to the beginning. And so I started making these funny things. And they started to have a life of their own and got bigger â as the one glass at the Walker. And then, I sliced off the head and the tail and everything and tried to translate what I was learning about the form of the fish and the movement. And a lot of my architectural ideas that came from it â accidental, again â it was an intuitive kind of thing, and I just kept going with it, and made this proposal for a building, which was only a proposal. I did this building in Japan. I was taken out to dinner after the contract for this little restaurant was signed. And I love sake and Kobe and all that stuff. And after I got â I was really drunk â I was asked to do some sketches on napkins. And I made some sketches on napkins â little boxes and Morandi-like things that I used to do. And the client said,"Why no fish?"And so I made a drawing with a fish, and I left Japan. Three weeks later, I received a complete set of drawings saying we ' d won the competition. Now, it ' s hard to do. It ' s hard to translate a fish form, because they ' re so beautiful â perfect â into a building or object like this. And Oldenburg, who I work with a little once in a while, told me I couldn ' t do it, and so that made it even more exciting. But he was right â I couldn ' t do the tail. I started to get the head OK, but the tail I couldn ' t do. It was pretty hard. The thing on the right is a snake form, a ziggurat. And I put them together, and you walk between them. It was a dialog with the context again. Now, if you saw a picture of this as it was published in Architectural Record â they didn ' t show the context, so you would think,"God, what a pushy guy this is."But a friend of mine spent four hours wandering around here looking for this restaurant. Couldn ' t find it. So... As for craft and technology and all those things that you ' ve all been talking about, I was thrown for a complete loop. This was built in six months. The way we sent drawings to Japan : we used the magic computer in Michigan that does carved models, and we used to make foam models, which that thing scanned. We made the drawings of the fish and the scales. And when I got there, everything was perfect â except the tail. So, I decided to cut off the head and the tail. And I made the object on the left for my show at the Walker. And it ' s one of the nicest pieces I ' ve ever made, I think. And then Jay Chiat, a friend and client, asked me to do his headquarters building in L. A. For reasons we don ' t want to talk about, it got delayed. Toxic waste, I guess, is the key clue to that one. And so we built a temporary building â I ' m getting good at temporary â and we put a conference room in that ' s a fish. And, finally, Jay dragged me to my hometown, Toronto, Canada. And there is a story â it ' s a real story â about my grandmother buying a carp on Thursday, bringing it home, putting it in the bathtub when I was a kid. I played with it in the evening. When I went to sleep, the next day it wasn ' t there. And the next night, we had gefilte fish. And so I set up this interior for Jay ' s offices and I made a pedestal for a sculpture. And he didn ' t buy a sculpture, so I made one. I went around Toronto and found a bathtub like my grandmother ' s, and I put the fish in. It was a joke. I play with funny people like [Claes] Oldenburg. We ' ve been friends for a long time. And we ' ve started to work on things. A few years ago, we did a performance piece in Venice, Italy, called"Il Corso del Coltello"â the Swiss Army knife. And most of the imagery is â Claes ', but those two little boys are my sons, and they were Claes ' assistants in the play. He was the Swiss Army knife. He was a souvenir salesman who always wanted to be a painter, and I was Frankie P. Toronto. P for Palladio. Dressed up like the AT&T building by Claes â with a fish hat. The highlight of the performance was at the end. This beautiful object, the Swiss Army knife, which I get credit for participating in. And I can tell you â it ' s totally an Oldenburg. I had nothing to do with it. The only thing I did was, I made it possible for them to turn those blades so you could sail this thing in the canal, because I love sailing. We made it into a sailing craft. I ' ve been known to mess with things like chain link fencing. I do it because it ' s a curious thing in the culture, when things are made in such great quantities, absorbed in such great quantities, and there ’ s so much denial about them. People hate it. And I ’ m fascinated with that, which, like the paper furniture â it ’ s one of those materials. And I ’ m always drawn to that. And so I did a lot of dirty things with chain link, which nobody will forgive me for. But Claes made homage to it in the Loyola Law School. And that chain link is really expensive. It ’ s in perspective and everything. And then we did a camp together for children with cancer. And you can see, we started making a building together. Of course, the milk can is his. But we were trying to collide our ideas, to put objects next to each other. Like a Morandi â like the little bottles â composing them like a still life. And it seemed to work as a way to put he and I together. Then Jay Chiat asked me to do this building on this funny lot in Venice, and I started with this three-piece thing, and you entered in the middle. And Jay asked me what I was going to do with the piece in the middle. And he pushed that. And one day I had a â oh, well, the other way. I had the binoculars from Claes, and I put them there, and I could never get rid of them after that. Oldenburg made the binoculars incredible when he sent me the first model of the real proposal. It made my building look sick. And it was this interaction between that kind of, up-the-ante stuff that became pretty interesting. It led to the building on the left. And I still think the Time magazine picture will be of the binoculars, you know, leaving out the â what the hell. I use a lot of metal in my work, and I have a hard time connecting with the craft. The whole thing about my house, the whole use of rough carpentry and everything, was the frustration with the crafts available. I said,”If I can ’ t get the craft that I want, I ’ ll use the craft I can get.”There were plenty of models for that, in Rauschenberg and Jasper Johns, and many artists who were making beautiful art and sculpture with junk materials. I went into the metal because it was a way of building a building that was a sculpture. And it was all of one material, and the metal could go on the roof as well as the walls. The metalworkers, for the most part, do ducts behind the ceilings and stuff. I was given an opportunity to design an exhibit for the metalworkers ’ unions of America and Canada in Washington, and I did it on the condition that they become my partners in the future and help me with all future metal buildings, etc. etc. And it ’ s working very well to have these people, these craftsmen, interested in it. I just tell the stories. It ’ s a way of connecting, at least, with some of those people that are so important to the realization of architecture. The metal continued into a building â Herman Miller, in Sacramento. And it ’ s just a complex of factory buildings. And Herman Miller has this philosophy of having a place â a people place. I mean, it ’ s kind of a trite thing to say, but it is real that they wanted to have a central place where the cafeteria would be, where the people would come and where the people working would interact. So it ’ s out in the middle of nowhere, and you approach it. It ’ s copper and galvanize. I used the galvanize and copper in a very light gauge, so it would buckle. I spent a lot of time undoing Richard Meier ’ s aesthetic. Everybody ’ s trying to get the panels perfect, and I always try to get them sloppy and fuzzy. And they end up looking like stone. This is the central area. There ’ s a ramp. And that little dome in there is a building by Stanley Tigerman. Stanley was instrumental in my getting this job. And when I was awarded the contract I, at the very beginning, asked the client if they would let Stanley do a cameo piece with me. Because these were ideas that we were talking about, building things next to each other, making â it ’ s all about [a] metaphor for a city, maybe. And so Stanley did the little dome thing. And we did it over the phone and by fax. He would send me a fax and show me something. He ’ d made a building with a dome and he had a little tower. I told him,”No, no, that ’ s too ongepotchket. I don ’ t want the tower.”So he came back with a simpler building, but he put some funny details on it, and he moved it closer to my building. And so I decided to put him in a depression. I put him in a hole and made a kind of a hole that he sits in. And so then he put two bridges â this all happened on the fax, going back and forth over a couple of weeks ’ period. And he put these two bridges with pink guardrails on it. And so then I put this big billboard behind it. And I call it,”David and Goliath.”And that ’ s my cafeteria. In Boston, we had that old building on the left. It was a very prominent building off the freeway, and we added a floor and cleaned it up and fixed it up and used the kind of â I thought â the language of the neighborhood, which had these cornices, projecting cornices. Mine got a little exuberant, but I used lead copper, which is a beautiful material, and it turns green in 100 years. Instead of, like, copper in 10 or 15. We redid the side of the building and re-proportioned the windows so it sort of fit into the space. And it surprised both Boston and myself that we got it approved, because they have very strict kind of design guideline, and they wouldn ’ t normally think I would fit them. The detailing was very careful with the lead copper and making panels and fitting it tightly into the fabric of the existing building. In Barcelona, on Las Ramblas for some film festival, I did the Hollywood sign going and coming, made a building out of it, and they built it. I flew in one night and took this picture. But they made it a third smaller than my model without telling me. And then more metal and some chain link in Santa Monica â a little shopping center. And this is a laser laboratory at the University of Iowa, in which the fish comes back as an abstraction in the back. It ’ s the support labs, which, by some coincidence, required no windows. And the shape fit perfectly. I just joined the points. In the curved part there ’ s all the mechanical equipment. That solid wall behind it is a pipe chase â a pipe canyon â and so it was an opportunity that I seized, because I didn ’ t have to have any protruding ducts or vents or things in this form. It gave me an opportunity to make a sculpture out of it. This is a small house somewhere. They ’ ve been building it so long I don ’ t remember where it is. It ’ s in the West Valley. And we started with the stream and built the house along the stream â dammed it up to make a lake. These are the models. The reality, with the lake â the workmanship is pretty bad. And it reminded me why I play defensively in things like my house. When you have to do something really cheaply, it ’ s hard to get perfect corners and stuff. That big metal thing is a passage, and in it is â you go downstairs into the living room and then down into the bedroom, which is on the right. It ’ s kind of like a whole built town. I was asked to do a hospital for schizophrenic adolescents at Yale. I thought it was fitting for me to be doing that. This is a house next to a Philip Johnson house in Minnesota. The owners had a dilemma â they asked Philip to do it. He was too busy. He didn ’ t recommend me, by the way. We ended up having to make it a sculpture, because the dilemma was, how do you build a building that doesn ’ t look like the language? Is it going to look like this beautiful estate is sub-divided? Etc. etc. You ’ ve got the idea. And so we finally ended up making it. These people are art collectors. And we finally made it so it appears very sculptural from the main house and all the windows are on the other side. And the building is very sculptural as you walk around it. It ’ s made of metal and the brown stuff is Fin-Ply â it ’ s that formed lumber from Finland. We used it at Loyola on the chapel, and it didn ’ t work. I keep trying to make it work. In this case we learned how to detail it. In Cleveland, there ’ s Burnham Mall, on the left. It ’ s never been finished. Going out to the lake, you can see all those new buildings we built. And we had the opportunity to build a building on this site. There ’ s a railroad track. This is the city hall over here somewhere, and the courthouse. And the centerline of the mall goes out. Burnham had designed a railroad station that was never built, and so we followed. Sohio is on the axis here, and we followed the axis, and they ’ re two kind of goalposts. And this is our building, which is a corporate headquarters for an insurance company. We collaborated with Oldenburg and put the newspaper on top, folded. The health club is fastened to the garage with a C-clamp, for Cleveland. You drive down. So it ’ s about a 10-story C-clamp. And all this stuff at the bottom is a museum, and an idea for a very fancy automobile entry. This owner has a pet peeve about bad automobile entries. And this would be a hotel. So, the centerline of this thing â we ’ d preserve it, and it would start to work with the scale of the new buildings by Pelli and Kohn Pederson Fox, etc., that are underway. It ’ s hard to do high-rise. I feel much more comfortable down here. This is a piece of property in Brentwood. And a long time ago, about ’ 82 or something, after my house â I designed a house for myself that would be a village of several pavilions around a courtyard â and the owner of this lot worked for me and built that actual model on the left. And she came back, I guess wealthier or something â something happened â and asked me to design a house for her on this site. And following that basic idea of the village, we changed it as we got into it. I locked the house into the site by cutting the back end â here you see on the photographs of the site â slicing into it and putting all the bathrooms and dressing rooms like a retaining wall, creating a lower level zone for the master bedroom, which I designed like a kind of a barge, looking like a boat. And that ’ s it, built. The dome was a request from the client. She wanted a dome somewhere in the house. She didn ’ t care where. When you sleep in this bedroom, I hope â I mean, I haven ’ t slept in it yet. I ’ ve offered to marry her so I could sleep there, but she said I didn ’ t have to do that. But when you ’ re in that room, you feel like you ’ re on a kind of barge on some kind of lake. And it ’ s very private. The landscape is being built around to create a private garden. And then up above there ’ s a garden on this side of the living room, and one on the other side. These aren ’ t focused very well. I don ’ t know how to do it from here. Focus the one on the right. It ’ s up there. Left â it ’ s my right. Anyway, you enter into a garden with a beautiful grove of trees. That ’ s the living room. Servants ’ quarters. A guest bedroom, which has this dome with marble on it. And then you enter into the living room and then so on. This is the bedroom. You come down from this level along the stairway, and you enter the bedroom here, going into the lake. And the bed is back in this space, with windows looking out onto the lake. These Stonehenge things were designed to give foreground and to create a greater depth in this shallow lot. The material is lead copper, like in the building in Boston. And so it was an intent to make this small piece of land â it ’ s 100 by 250 â into a kind of an estate by separating these areas and making the living room and dining room into this pavilion with a high space in it. And this happened by accident that I got this right on axis with the dining room table. It looks like I got a Baldessari painting for free. But the idea is, the windows are all placed so you see pieces of the house outside. Eventually this will be screened â these trees will come up â and it will be very private. And you feel like you ’ re in your own kind of village. This is for Michael Eisner â Disney. We ’ re doing some work for him. And this is in Anaheim, California, and it ’ s a freeway building. You go under this bridge at about 65 miles an hour, and there ’ s another bridge here. And you ’ re through this room in a split second, and the building will sort of reflect that. On the backside, it ’ s much more humane â entrance, dining hall, etc. And then this thing here â I ’ m hoping as you drive by you ’ ll hear the picket fence effect of the sound hitting it. Kind of a fun thing to do. I ’ m doing a building in Switzerland, Basel, which is an office building for a furniture company. And we struggled with the image. These are the early studies, but they have to sell furniture to normal people, so if I did the building and it was too fancy, then people might say,”Well, the furniture looks OK in his thing, but no, it ain ’ t going to look good in my normal building.”So we ’ ve made a kind of pragmatic slab in the second phase here, and we ’ ve taken the conference facilities and made a villa out of them so that the communal space is very sculptural and separate. And you ’ re looking at it from the offices and you create a kind of interaction between these pieces. This is in Paris, along the Seine. Palais des Sports, the Gare de Lyon over here. The Minister of Finance â the guy that moved from the Louvre â goes in here. There ’ s a new library across the river. And back in here, in this already treed park, we ’ re doing a very dense building called the American Center, which has a theater, apartments, dance school, an art museum, restaurants and all kinds of â it ’ s a very dense program â bookstores, etc. In a very tight, small â this is the ground level. And the French have this extraordinary way of screwing things up by taking a beautiful site and cutting the corner off. They call it the plan coupe. And I struggled with that thing â how to get around the corner. These are the models for it. I showed you the other model, the one â this is the way I organized myself so I could make the drawing â so I understood the problem. I was trying to get around this plan coupe â how do you do it? Apartments, etc. And these are the kind of study models we did. And the one on the left is pretty awful. You can see why I was ready to commit suicide when this one was built. But out of it came finally this resolution, where the elevator piece worked frontally to this, parallel to this street, and also parallel to here. And then this kind of twist, with this balcony and the skirt, kind of like a ballerina lifting her skirt to let you into the foyer. The restaurants here â the apartments and the theater, etc. So it would all be built in stone, in French limestone, except for this metal piece. And it faces into a park. And the idea was to make this express the energy of this. On the side facing the street it ’ s much more normal, except I slipped a few mansards down, so that coming on the point, these housing units made a gesture to the corner. And this will be some kind of high-tech billboard. If any of you guys have any ideas for it, please contact me. I don ’ t know what to do. Jay Chiat is a glutton for punishment, and he hired me to do a house for him in the Hamptons. And it ’ s got a fish. And I keep thinking,”This is going to be the last fish.”It ’ s like a drug addict. I say,”I ’ m not going to do it anymore â I don ’ t want to do it anymore â I ’ m not going to do it.”And then I do it. There it is. But it ’ s the living room. And this piece here is â I don ’ t know what it is. I just added it so that we ’ d have enough money in the budget so we could take something out. This is Euro Disney, and I ’ ve worked with all of the guys that presented to you earlier. We ’ ve had a lot of fun working together. I think I ’ m from Mars for them, and they are for me, but somehow we all manage to work together, and I think, productively. So far. This is a shopping thing. You come into the Magic Kingdom and the hotel that Tony Baxter ’ s group is doing out here. And then this is a kind of a shopping mall, with a rodeo and restaurants. And another restaurant. What I did â because of the Paris skies being quite dull, I made a light grid that ’ s perpendicular to the train station, to the route of the train. It looks like it ’ s kind of been there, and then crashed all these simpler forms into it. The light grid will have a light, be lit up at night and give a kind of light ceiling. In Switzerland â Germany, actually â on the Rhine across from Basel, we did a furniture factory and a furniture museum. And I tried to â there ’ s a Nick Grimshaw building over here, there ’ s an Oldenburg sculpture over here â I tried to make a relationship urbanistically. And I don ’ t gave good slides to show â it ’ s just been completed â but this piece here is this building, and these pieces here and here. And as you pass by it ’ s always part â you see it as all of these pieces accrue and become part of an overall neighborhood. It ’ s plaster and just zinc. And you wonder, if this is a museum, what it ’ s going to be like inside? If it ’ s going to be so busy and crazy that you wouldn ’ t show anything, and just wait. I ’ m so cunning and clever â I made it quiet and wonderful. But on the outside it does scream out at you a bit. It ’ s actually basically three square rooms with a couple of skylights and stuff. And from the building in the back, you see it as an iceberg floating by in the hills. I know I ’ m over time. See, that skylight goes down and becomes that one. So it ’ s pretty quiet inside. This is the Disney Hall â the concert hall. It ’ s a complicated project. It has a chamber hall. It ’ s related to an existing Chandler Pavilion that was built with a lot of love and tears and caring. And it ’ s not a great building, but I approached it optimistically, that we would make a compositional relationship between us that would strengthen both of us. And the plan of this â it ’ s a concert hall. This is the foyer, which is kind of a garden structure. There ’ s commercial at the ground floor. These are offices, which, really, in the competition, we didn ’ t have to design. But finally, there ’ s a hotel there. These were the kind of relationships made to the Chandler, composing these elevations together and relating them to the buildings that existed â to MOCA, etc. The acoustician in the competition gave us criteria, which led to this compartmentalized scheme, which we found out after the competition would not work at all. But everybody liked these forms and liked the space, and so that ’ s one of the problems of a competition. You have to then try and get that back in some way. And we studied many models. This was our original model. These were the three buildings that were the ideal â the Concertgebouw, Boston and Berlin. Everybody liked the surround. Actually, this is the smallest hall in size, and it has more seats than any of these because it has double balconies. Our client doesn ’ t want balconies, so â and when we met our new acoustician, he told us this was the right shape or this was the right shape. And we tried many shapes, trying to get the energy of the original design within an acoustical, acceptable format. We finally settled on a shape that was the proportion of the Concertgebouw with the sloping outside walls, which the acoustician said were crucial to this and later decided they weren ’ t, but now we have them. And our idea is to make the seating carriage very sculptural and out of wood and like a big boat sitting in this plaster room. That ’ s the idea. And the corners would have skylights and these columns would be structural. And the nice thing about introducing columns is they give you a kind of sense of proscenium from wherever you sit, and create intimacy. Now, this is not a final design â these are just on the way to being â and so I wouldn ’ t take it literally, except the feeling of the space. We studied the acoustics with laser stuff, and they bounce them off this and see where it all works. But you get the sense of the hall in section. Most halls come straight down into a proscenium. In this case we ’ re opening it back up and getting skylights in the four corners. And so it will be quite a different shape. The original building, because it was frog-like, fit nicely on the site and cranked itself well. When you get into a box, it ’ s harder to do it â and here we are, struggling with how to put the hotel in. And this is a teapot I designed for Alessi. I just stuck it on there. But this is how I do work. I do take pieces and bits and look at it and struggle with it and cut it away. And of course it ’ s not going to look like that, but it is the crazy way I tend to work. And then finally, in L. A. I was asked to do a sculpture at the foot of Interstate Bank Tower, the highest building in L. A. Larry Halprin is doing the stairs. And I was asked to do a fish, and so I did a snake. It ’ s a public space, and I made it kind of a garden structure, and you can go in it. It ’ s a kiva, and Larry ’ s putting some water in there, and it works much better than a fish. In Barcelona I was asked to do a fish, and we ’ re working on that, at the foot of a Ritz-Carlton Tower being done by

Skidmore, Owings and Merrill. And the Ritz-Carlton Tower is being designed with exposed steel, non-fire proof, much like those old gas tanks. And so we took the language of this exposed steel and used it, perverted it, into the form of the fish, and created a kind of a 19th- century contraption that looks like, that will sit â this is the beach and the harbor out in front, and this is really a shopping center with department stores. And we split these bridges. Originally, this was all solid with a hole in it. We cut them loose and made several bridges and created a kind of a foreground for this hotel. We showed this to the hotel people the other day, and they were terrified and said that nobody would come to the Ritz-Carlton anymore, because of this fish. And finally, I just threw these in â Lou Danziger. I didn ’ t expect Lou Danziger to be here, but this is a building I did for him in 1964, I think. A little studio â and it ’ s sadly for sale. Time goes on. And this is my son working with me on a small fast-food thing. He designed the robot as the cashier, and the head moves, and I did the rest of it. And the food wasn ’ t as good as the stuff, and so it failed. It should have been the other way around â the food should have been good first. It didn ’ t work. Thank you very much.

11 POS Dim 6

Text Sample 827: POS Dim 6 – 1984 – Score 58.00 – 1984/t000297.txt

In this rather long sort of **marathon presentation**, I ’ ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more **pleasurable** to deal with a computer and really address the qualities of the human **interface**. And these will be some simple design qualities and they will also be some qualities of, if you will, the intelligence of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ’ ve been able to collect, which I think **illustrate** this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a slide taken from a TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ’ s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the **diagonal**. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ’ s the distance you should sit away from the TV set. Now we ’ ve put people 18 inches in front of a TV, and all the **artifacts** that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow **mask**, the scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ’ m talking here a little bit about **display** technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ’ m very interested in **touch-sensitive displays**. High-tech, **high-touch**. Isn ’ t that what some of you said? It ’ s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of **stylus** for inputting to a **display**. In fact, they ' re not : it ' s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a **cursor** with great accuracy. And so when you see on the market these systems that have just a few light emitting **diodes** on the side and are very low resolution, it ' s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ' t have to pick them up, and people don ' t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ' re typing — what you have — you want to now put something — first of all, you ' ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ' ve got to sort of find that mouse. Then you find the mouse, and you ' re going to have to wiggle it a little bit to see where the **cursor** is on the screen. And then when you finally see where it is, then you ' ve got to move it to get the **cursor** over there, and then —"Bang"— you ' ve got to hit a button or do whatever. That ' s four separate steps versus typing and then touching and typing and just doing it all in one motion — or **one-and-a-half**, depending on how you want to count. Again, what I ' m trying to do is just **illustrate** the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that **interface**. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is **typical** of the computer field, is when you have a bug that you can ' t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in **touch-sensitive displays** : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the **substrate**, which it usually is. So we found that that actually was a feature in the sense you could build a **pressure-sensitive display**. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only **touch-sensitive** now, it ' s **pressure-sensitive**. And it ' s **pressure-sensitive** to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ' t figure out how to come in the other direction. But let me get rid of the slide, and let ' s see if this comes on. OK. So there is the **pressure-sensitive display** in operation. The person ' s just, if you will, pushing on the screen to make a curve. But this is the interesting part. I want to stop it for a second because the **movie** is very badly made. And the particular **display** was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an **anvil** on a fuzzy rug and the other one is a **ping-pong ball** on a sheet of glass. And when you touch it, you have to really push very hard to move that **anvil** across the screen, and yet you touch the **ping-pong ball** very lightly and it just **scoots** across the screen. And what you can do — oops, I didn ’ t mean to do that — what you can do is actually feed back to the user the feeling of the physical properties. So again, they don ’ t have to be weight ; they could be a general trying to move **troops**, and he ’ s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the **interface** there are physical properties in that **transducer** — in this case it ’ s pressure and touches — that allow you to present things to the user that you could never present before. So it ’ s not simply looking at the quality or, if you will, the luxury of that **interface**, but it ’ s actually looking at the idea of presenting things that previously couldn ’ t be presented before. I want to move on to another example, which is one of a different sort, where we ’ re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ’ re going to take this book, if you will, and it ’ s going to come alive. You ’ re going to sort of breathe life into it. We are so used to doing monologues. **Filmmakers**, for example, are the experts in monologue making : you make a film and it has a **well-formed** beginning, middle and end, and in some sense the art of it is that. And you then say,”There ’ s an opportunity for making conversational **movies**.”Well, what does that mean? And it sort of **nibbles** at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ’ ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the **movie** has information about itself. So it knows, or at least there is computer-readable information in the medium itself. It ’ s just not a static **movie** frame. That ’ s one thing. The other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the **cookbook**, the”Larousse Gastronomique.”And I think I use the example all too often, but it ’ s a great one because there is a classic ending in that little encyclopedia-style **cookbook** that tells you how to do something like penguin, and you get to the end of the recipe and it says,”Cook until done.”Now, that would be, if you will, the top green track, which doesn ’ t mean too much. But you might have to elaborate for me or for somebody who isn ’ t an expert, and say,”Cook at 380 degrees for 45 minutes.”And then for a real **beginner**, you would go down even further and elaborate more — say,”Open the oven, preheat, wait for the light to go out, open the door, don ’ t leave it open too long, put the penguin in and shut the door...”whatever. And that ’ s a much more elaborate one than you **dribble** back. That ’ s one kind of use of random access. And the other is where you want to explain the same thing in different ways. If you ’ re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ’ s cognitive style, then you might say it in a way that you think would have a good **impedance** match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ’ s a rather boring book, but I ’ m afraid sometimes you have to do boring books because your sponsors aren ’ t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ’ t even know what **vintage** the transmission is, but let me just show you very quickly some of it, and we ’ ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of the transmission, and as you rub your finger across the transmission it highlights the **various** parts. Narrator :

When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, **superimposed** over the illustration. NN : This is about the oil pan, or the oil **filter** and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get back to the illustration, but in this case it ' s not a single frame, but it ' s actually a **movie** of someone coming into the frame and doing the repair that ' s described in the text. The **two-headed** slider is a speed control that allows me to watch the **movie** at **various** speeds, in forward or reverse. And the **movie** is **displayed** as a full frame **movie**. I can go back to the beginning... and play the **movie** at full speed. Here ' s another step-by-step procedure, only in this case — NN : Okay, this **movie** is... Everybody ' s heard of sound-sync **movies** — this is text-sync **movies**, so as the **movie** plays, the text gets highlighted. We highlight the text as we go through the **movie**. Repairman :... Not too far out. Front **poles**, preferably. Don ' t **loosen** them too far. If you **loosen** them too far, you ' ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ' m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ' ve been doing — in this case, in Senegal — where we have tried to use personal computers as a **pedagogical** medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ' s a story I ' d like to tell. And that was when, before we did this in any developing countries — we ' re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ' ve forgotten, was about seven or eight years old, absolutely considered mentally **handicapped** — couldn ' t read, didn ' t even make it in the lowest section of the school ' s classes — and was pretty much not in school, though physically there. But did hang around the, quote,"computer room,"where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very interesting. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,"Let me show you how this works,"and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ' t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ' d actually come to see — not the computer room — they went upstairs and they said,"This is absolutely **remarkable**! That child was very **articulate** and showed us and even dealt with the things he couldn ' t do automatically with that manual. It was just absolutely fantastic."The principal said,"There ' s a dreadful mistake, because that child can ' t read. And you obviously have been

hoodwinked or you ’ ve talked about somebody else.”And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,”Can you read?”And the child said,”No, I can ’ t.”And then they said,”But wait a minute. You just looked through that manual and you found...”and he said,”Oh, but that ’ s not reading.”And so they said,”Well, what ’ s reading then?”He says,”Well, reading is this junk they give me in little books to read. It ’ s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return.”And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ’ m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ’ s not, but we think speech —”My God, little children pick it up somehow, and by the age of two they ’ re doing a mediocre job, and by three and four they ’ re speaking reasonably well. And yet you ’ ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder.”Well, it ’ s not harder. What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ’ s going to help you. So what happens is you go to school and people say,”Just believe me, you ’ re going to like it. You ’ re going to like reading,”and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — **Poof**! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on the screen has a huge payoff, and it ’ s a lot of fun. And in fact, it ’ s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ’ s the girl on the left **leaning** with her **chalkboard**, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ’ s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example — as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of **teleconferencing**. And the reason it remains a favorite project is that we were asked to do a **teleconferencing** system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in **teleconference**, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently **zany** that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ’ s faces. So if I wanted to call my friend Peter Sprague on the phone, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ’ s uncanny : there ’ s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be **spatial** correspondence, which is straightforward, but again, it ’ s something that didn ’ t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or **spatial** concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,”He or she wouldn ’ t agree,”and the empty chair is that person and the spatiality is crucial. So we said,”Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ’ s site you ’ d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ’ t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ’ ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ’ s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ’ s site, he ’ ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, **teleconferencing**.

Text Sample 828: POS Dim 6 – 2005 – Score 6.00 – 2005/t000175.txt

Of the five senses, vision is the one that I appreciate the most, and it ’ s the one that I can least take for granted. I think this is partially due to my father, who was blind. It was a fact that he didn ’ t make much of a fuss about, usually. One time in Nova Scotia, when we went to see a total eclipse of the sun — Yeah, same one as in the Carly Simon song, which may or may not refer to James Taylor, Warren Beatty or Mick Jagger ; we ’ re not really sure. They handed out these dark plastic viewers that allowed us to look directly at the sun without damaging our eyes. But Dad got really scared ; he didn ’ t want us doing that. He wanted us instead to use these cheap cardboard viewers, so that there was no chance at all that our eyes would be damaged. I thought this was a little strange at the time. What I didn ’ t know at the time was that my father had actually been born with perfect eyesight. When he and his sister Martha were just very little, their mom took them out to see a total eclipse — or actually, a solar eclipse — and not long after that, both of them started losing their eyesight. Decades later, it turned out that the source of their blindness was most likely some sort of bacterial infection. As near as we can tell, it had nothing whatsoever to do with that solar eclipse, but by then my grandmother had already gone to her grave thinking it was her fault. So, Dad graduated Harvard in 1946, married my mom, and bought a house in Lexington, Massachusetts, where the first shots were fired against the British in 1775, although we didn ’ t actually hit any of them until Concord. He got a job working for Raytheon designing guidance systems, which was part of the Route 128 high-tech axis in those days — so, the equivalent of Silicon Valley in the ’ 70s. Dad wasn ’ t a real militaristic kind of guy ; he just felt bad that he wasn ’ t able to fight in World War II on account of his handicap, although they did let him get through the several-hour-long army physical exam before they got to the very last test, which was for vision. So Dad started racking up all of these patents and gaining a reputation as a blind genius, rocket scientist, inventor. But to us he was just Dad, and our home life was pretty normal. As a kid, I watched a lot of television and had lots of nerdy hobbies like mineralogy and microbiology and the space program and a little bit of politics. I played a lot of chess. But at the age of 14, a friend got me interested in comic books, and I decided that was what I wanted to do for a living. So, here ' s my dad : he ' s a scientist, he ' s an engineer and he ' s a military contractor. So, he has four kids, right? One grows up to become a computer scientist, one grows up to join the Navy, one grows up to become an engineer... And then there ' s me : the comic book artist. Which, incidentally, makes me the opposite of Dean Kamen, because I ' m a comic book artist, son of an inventor, and he ' s an inventor, son of a comic book artist. Right? It ' s true. The funny thing is, Dad had a lot of faith in me. He had faith in my abilities as a cartoonist, even though he had no direct evidence that I was any good whatsoever ; everything he saw was just a blur. Now, this gives a real meaning to the term"blind faith,"which doesn ' t have the same negative connotation for me that it does for other people. Now, faith in things which cannot be seen, which cannot be proved, is not the sort of faith that I ' ve ever really related to all that much. I tend to like science, where what we see and can ascertain are the foundation of what we know. But there ' s a middle ground, too — a middle ground tread by people like poor old Charles Babbage and his steam-driven computers that were never built. Nobody really understood what it was that he had in mind except for Ada Lovelace, and he went to his grave trying to pursue that dream. Vannevar Bush with his memex — this idea of all of human knowledge at your fingertips — he had this vision. And I think a lot of people in his day probably thought he was a bit of a kook. And, yeah, we can look back in retrospect and say,"Yeah, ha-ha, it ' s all microfilm — But that ' s not the point ; he understood the shape of the future. So did J. C. R. Licklider and his notions for computer-human interaction. Same thing : he understood the shape of the future, even though it was something that would only be implemented by people much later. Or Paul Baran, and his vision for packet switching. Hardly anybody listened to him in his day. Or even the people who actually pulled it off, the people at Bolt, Beranek and Newman in Boston, who just would sketch out these structures of what would eventually become a worldwide network, and sketching things on the back of napkins and on note papers and arguing over dinner at Howard Johnson ' s — on Route 128 in Lexington, Massachusetts, just two miles from where I was studying the Queen ' s Gambit Deferred and listening to Gladys Knight & The Pips singing"Midnight Train to Georgia"— in my dad ' s big easy chair, you know? So, three types of vision, right? Vision based on what one cannot see, the vision of that unseen and unknowable. The vision of that which has already been proven or can be ascertained. And this third kind, a vision of something which can be, which may be, based on knowledge but is, as yet, unproven. Now, we ' ve seen a lot of examples of people who are pursuing that sort of vision in science, but I think it ' s also true in the arts, it ' s true in politics, it ' s even true in personal endeavors. What it comes down to, really, is four basic principles : learn from everyone ; follow no one ; watch for patterns ; and work like hell. I think these are the four principles that go into this. And it ' s that third one, especially, where visions of the future begin to manifest themselves. What ' s interesting is that this particular way of looking at the world, is, I think, only one of four different ways that manifest themselves in different fields of endeavor. In comics, I know that it results in sort of a formalist attitude towards trying to understand how it works. Then there ' s another, more classical attitude which embraces beauty and craft ; another one which believes in the pure transparency of content ; and then another, which emphasizes the authenticity of human experience and honesty and rawness. These are four very different ways of looking at the world. I even gave them names : the classicist, the animist, the formalist and iconoclast. Interestingly, they seem to correspond more or less to Jung ' s four subdivisions of human thought. And they reflect a dichotomy of art and delight on left and the right ; tradition and revolution on the top and the bottom. And if you go on the **diagonal**, you get content and form, and then beauty and truth. And it probably applies just as much to music and **movies** and fine art, which has nothing whatsoever to do with vision at all, or, for that matter, nothing to do with our conference theme of"Inspired by Nature,"except to the extent of the fable of the frog who gives a ride to the scorpion on his back to get across the river because the scorpion promises not to sting him, but the scorpion stings him anyway and they both die, but not before the frog asks him why, and the scorpion says,"Because it ' s my nature."In that sense, yes. So this was my nature. The thing was, I saw that the route I took to discovering this focus in my work and who I was — I saw it as just this road to discovery. Actually, it was just me embracing my nature, which means that I didn ' t actually fall that far from the tree, after all. So what does a"scientific mind"do in the arts? I started making comics, but I also started trying to understand them, almost immediately. One of the most important things about comics that I discovered was that comics are a visual medium, but they try to embrace all of the senses within it. So, the different elements of comics, like pictures and words, and the different symbols and everything in between that comics presents, are all funneled through the single conduit, a vision. So we have things like resemblance, where something which resembles the physical world can be abstracted in a couple of different directions : abstracted from resemblance, but still retaining the complete meaning, or abstracted away from both resemblance and meaning towards the picture plane. Put all these three together, and you have a nice little map of the entire boundary of visual iconography, which comics can embrace. And if you move to the right you also get language, because that ' s abstracting even further from resemblance, but still maintaining meaning. Vision is called upon to represent sound and to understand the common properties of those two and their common heritage as well ; also, to try to represent the texture of sound to capture its essential character through visuals. There ' s also a balance between the visible and the invisible in comics. Comics is a kind of call and response, in which the artist gives you something to see within the panels, and then gives you something to imagine between the panels. Also, another sense which comics ' vision represents, and that ' s time. Sequence is a very important aspect of comics. Comics presents a kind of temporal map. And this temporal map was something that energizes modern comics, but I was wondering if perhaps it also energizes other sorts of forms, and I found some in history. You can see this same principle operating in these ancient versions of the same idea. What ' s happening is, an art form is colliding with a given technology, whether it ' s paint on stone, like the Tomb of Menna the Scribe in ancient Egypt, or a bas-relief sculpture rising up a stone column, or a 200- foot-long embroidery, or painted deerskin and tree bark running across 88 accordion-folded pages. What ' s interesting is, once you hit"print"— and this is from 1450, by the way — all of the **artifacts** of modern comics start to present themselves : rectilinear panel arrangements, simple line drawings without tone, and a left- to-right reading sequence. And within 100 years, you already start to see word balloons and captions, and it ' s really just a hop, skip and a jump from here to here. So I wrote a book about this in ' 93, but as I was finishing the book, I had to do a little bit of typesetting, and I was tired of going to my local copy shop to do it, so I bought a computer. And it was just a little thing — it wasn ' t good for much except text entry — but my father had told me about Moore ' s law back in the ’ 70s, and I knew what was coming. And so, I kept my eyes peeled to see if the sort of changes that happened when we went from pre-print comics to print comics would happen when we went beyond, to post-print comics. So, one of the first things proposed was that we could mix the visuals of comics with the sound, motion and interactivity of the CD-ROMs being made in those days. This was even before the Web. And one of the first things they did was, they tried to take the comics page as is and transplant it to monitors, which was a classic McLuhanesque mistake of appropriating the shape of the previous technology as the content of the new technology. And so, what they would do is have these comic pages that resemble print comics pages, and they would introduce all this sound and motion. The problem was that if you go with this basic idea that space equals time in comics, what happens is that when you introduce sound and motion, which are temporal phenomena that can only be represented through time, they break with that continuity of **presentation**. Interactivity was another thing. There were hypertext comics, but the thing about hypertext is that everything in hypertext is either here, not here, or connected to here ; it ’ s profoundly nonspatial. The distance from Abraham Lincoln to a Lincoln penny to Penny Marshall to the Marshall Plan to”Plan 9”to nine lives : it ’ s all the same. But in comics, every aspect of the work, every element of the work, has a **spatial** relationship to every other element at all times. So the question was : Was there any way to preserve that **spatial** relationship while still taking advantage of all of the things that digital had to offer us? And I found my personal answer for this in those ancient comics that I was showing you. Each of them has a single unbroken reading line, whether it ’ s going zigzag across the walls or spiraling up a column or just straight left to right, or even going in a backwards zigzag across those 88 accordion-folded pages, the same thing is happening ; that is, that the basic idea that as you move through space you move through time, is being carried out without any compromise, but there were compromises when print hit. Adjacent spaces were no longer adjacent moments, so the basic idea of comics was being broken again and again and again and again. And I thought, OK, well, if that ’ s true, is there any way, when we go beyond today ’ s print, to somehow bring that back? Now, the monitor is just as limited as the page, technically, right? It ’ s a different shape, but other than that, it ’ s the same basic limitation. But that ’ s only if you look at the monitor as a page, but not if you look at the monitor as a window. And that ’ s what I propose, that perhaps we could create these comics on an infinite canvas, along the X axis and the Y axis and staircases. We could do circular narratives that were literally circular. We could do a turn in a story that was literally a turn. Parallel narratives could be literally parallel. X, Y and also Z. So I had all these notions. This was back in the late ’ 90s, and other people in my business thought I was pretty crazy, but a lot of people then went on and actually did it. I ’ m going to show you a couple now. This was an early collage comic by a fellow named Jasen Lex. And notice what ’ s going on here. What I ’ m searching for is a durable mutation — that ’ s what all of us are searching for. As media head into this new era, we are looking for mutations that are durable, that have some sort of staying power. Now, we ’ re taking this basic idea of presenting comics in a visual medium, and we ’ re carrying it through all the way from beginning to end. That ’ s that entire comic you just saw, up on the screen right now. But even though we ’ re only experiencing it one piece at a time, that ’ s just where the technology is right now. As the technology evolves, as you get full immersive **displays** and whatnot, this sort of thing will only grow ; it will adapt. It will adapt to its environment ; it ’ s a durable mutation. Here ’ s another one. This is by Drew Weing ; this is called”’ Pup ’ Ponders the Heat Death of the Universe.”See what ’ s going on here as we draw these stories on an infinite canvas is you ' re creating a more pure expression of what this medium is all about. We ' ll go by this a little quickly. You get the idea. I just want to get to the last panel. [Cat 1 : Pup! Earth to Pup! Cat 2 : Come play baseball with us!] [Pup : Did either of you realize that eventually the universe will be nothing but a thin, cold gas spread across infinite, lonely space?] [Cat 1 : Oh... Cat 2 : We ' d better hurry, then!] Just one more. Talk about your infinite canvas. It ' s by a guy named Daniel Merlin Goodbrey, in Britain. Why is this important? I think this is important because media — all media — provide us a window back into our world. Now, it could be that motion pictures and eventually, virtual reality, or something equivalent to it, some sort of immersive **display**, is going to provide us with our most efficient escape from the world that we ' re in. That ' s why most people turn to storytelling, to escape. But media provides us with a window back into the world we live in. And when media evolve so that the identity of the media becomes increasingly unique — because what you ' re looking at is comics cubed, you ' re looking at comics that are more comics-like than they ' ve ever been before — when that happens, you provide people with multiple ways of reentering the world through different windows. And when you do that, it allows them to triangulate the world they live in and see its shape. That ' s why I think this is important. One of many reasons, but I ' ve got to go now. Thank you for having me.

Text Sample 829: POS Dim 6 – 2005 – Score 4.00 – 2005/t000369.txt

This is a picture of Maurice Druon, the Honorary Perpetual Secretary of L ' Academie francaise, the French Academy. He is splendidly attired in his 68, 000-dollar uniform, befitting the role of the French Academy as legislating the correct usage in French and perpetuating the language. The French Academy has two main tasks : it compiles a dictionary of official French. They ' re now working on their ninth edition, which they began in 1930, and they ' ve reached the letter P. They also legislate on correct usage, such as the proper term for what the French call"email,"which ought to be"courriel."The World Wide Web, the French are told, ought to be referred to as"la toile d ' araignee mondiale"— the Global Spider Web — recommendations that the French gaily ignore. Now, this is one model of how language comes to be : namely, it ' s legislated by an academy. But anyone who looks at language realizes that this is a rather silly conceit, that language, rather, emerges from human minds interacting from one another. And this is visible in the unstoppable change in language — the fact that by the time the Academy finishes their dictionary, it will already be well out of date. We see it in the constant appearance of slang and jargon, of the historical change in languages, in divergence of dialects and the formation of new languages. So language is not so much a creator or shaper of human nature, so much as a window onto human nature. In a book that I ' m currently working on, I hope to use language to shed light on a number of aspects of human nature, including the cognitive machinery with which humans conceptualize the world and the relationship types that govern human interaction. And I ' m going to say a few words about each one this morning. Let me start off with a technical problem in language that I ' ve worried about for quite some time — and indulge me in my passion for verbs and how they ' re used. The problem is, which verbs go in which constructions? The verb is the chassis of the sentence. It ' s the framework onto which the other parts are bolted. Let me give you a quick reminder of something that you ' ve long forgotten. An intransitive verb, such as"dine,"for example, can ' t take a direct object. You have to say,"Sam dined,"not,"Sam dined the pizza."A transitive verb mandates that there has to be an object there :”Sam devoured the pizza.”You can ’ t just say,”Sam devoured.”There are dozens or scores of verbs of this type, each of which shapes its sentence. So, a problem in explaining how children learn language, a problem in teaching language to adults so that they don ’ t make grammatical errors, and a problem in programming computers to use language is which verbs go in which constructions. For example, the dative construction in English. You can say,”Give a muffin to a mouse,”the prepositional dative. Or,”Give a mouse a muffin,”the double- object dative.”Promise anything to her,””Promise her anything,”and so on. Hundreds of verbs can go both ways. So a tempting generalization for a child, for an adult, for a computer is that any verb that can appear in the construction,”subject-verb-thing-to-a-recipient”can also be expressed as”subject-verb-recipient-thing.”A handy thing to have, because language is infinite, and you can ’ t just parrot back the sentences that you ’ ve heard. You ’ ve got to extract generalizations so you can produce and understand new sentences. This would be an example of how to do that. Unfortunately, there appear to be idiosyncratic exceptions. You can say,”Biff drove the car to Chicago,”but not,”Biff drove Chicago the car.”You can say,”Sal gave Jason a headache,”but it ’ s a bit odd to say,”Sal gave a headache to Jason.”The solution is that these constructions, despite initial appearance, are not synonymous, that when you crank up the microscope on human cognition, you see that there ’ s a subtle difference in meaning between them. So,”give the X to the Y,”that construction corresponds to the thought”cause X to go to Y.”Whereas”give the Y the X”corresponds to the thought”cause Y to have X.”Now, many events can be subject to either construal, kind of like the classic figure-ground reversal illusions, in which you can either pay attention to the particular object, in which case the space around it recedes from attention, or you can see the faces in the empty space, in which case the object recedes out of consciousness. How are these construals reflected in language? Well, in both cases, the thing that is construed as being affected is expressed as the direct object, the noun after the verb. So, when you think of the event as causing the muffin to go somewhere — where you ’ re doing something to the muffin — you say,”Give the muffin to the mouse.”When you construe it as”cause the mouse to have something,”you ’ re doing something to the mouse, and therefore you express it as,”Give the mouse the muffin.”So which verbs go in which construction — the problem with which I began — depends on whether the verb specifies a kind of motion or a kind of possession change. To give something involves both causing something to go and causing someone to have. To drive the car only causes something to go, because Chicago ’ s not the kind of thing that can possess something. Only humans can possess things. And to give someone a headache causes them to have the headache, but it ’ s not as if you ’ re taking the headache out of your head and causing it to go to the other person, and implanting it in them. You may just be loud or obnoxious, or some other way causing them to have the headache. So, that ’ s an example of the kind of thing that I do in my day job. So why should anyone care? Well, there are a number of interesting conclusions, I think, from this and many similar kinds of analyses of hundreds of English verbs. First, there ’ s a level of fine-grained conceptual structure, which we automatically and unconsciously compute every time we produce or utter a sentence, that governs our use of language. You can think of this as the language of thought, or”mentalese.”It seems to be based on a fixed set of concepts, which govern dozens of constructions and thousands of verbs — not only in English, but in all other languages — fundamental concepts such as space, time, causation and human intention, such as, what is the means and what is the ends? These are reminiscent of the kinds of categories that Immanuel Kant argued are the basic framework for human thought, and it ' s interesting that our unconscious use of language seems to reflect these Kantian categories. Doesn ' t care about perceptual qualities, such as color, texture, weight and speed, which virtually never differentiate the use of verbs in different constructions. An additional twist is that all of the constructions in English are used not only literally, but in a quasi-metaphorical way. For example, this construction, the dative, is used not only to transfer things, but also for the metaphorical transfer of ideas, as when we say,"She told a story to me"or"told me a story,""Max taught Spanish to the students"or"taught the students Spanish."It ' s exactly the same construction, but no muffins, no mice, nothing moving at all. It evokes the container metaphor of communication, in which we conceive of ideas as objects, sentences as containers, and communication as a kind of sending. As when we say we"gather"our ideas, to"put"them"into"words, and if our words aren ' t"empty"or"hollow,"we might get these ideas"across"to a listener, who can"unpack"our words to"extract"their"content."And indeed, this kind of verbiage is not the exception, but the rule. It ' s very hard to find any example of abstract language that is not based on some concrete metaphor. For example, you can use the verb"go"and the prepositions"to"and"from"in a literal, **spatial** sense."The messenger went from Paris to Istanbul."You can also say,"Biff went from sick to well."He needn ' t go anywhere. He could have been in bed the whole time, but it ' s as if his health is a point in state space that you conceptualize as moving. Or,"The meeting went from three to four,"in which we conceive of time as stretched along a line. Likewise, we use"force"to indicate not only physical force, as in,"Rose forced the door to open,"but also interpersonal force, as in,"Rose forced Sadie to go,"not necessarily by manhandling her, but by issuing a threat. Or,"Rose forced herself to go,"as if there were two entities inside Rose ' s head, engaged in a tug of a war. Second conclusion is that the ability to conceive of a given event in two different ways, such as"cause something to go to someone"and"causing someone to have something,"I think is a fundamental feature of human thought, and it ' s the basis for much human argumentation, in which people don ' t differ so much on the facts as on how they ought to be construed. Just to give you a few examples :"ending a pregnancy"versus"killing a fetus ;""a **ball** of cells"versus"an unborn child ;""invading Iraq"versus"liberating Iraq ;""redistributing wealth"versus"confiscating earnings."And I think the biggest picture of all would take seriously the fact that so much of our verbiage about abstract events is based on a concrete metaphor and see human intelligence itself as consisting of a repertoire of concepts — such as objects, space, time, causation and intention — which are useful in a social, knowledge- intensive species, whose evolution you can well imagine, and a process of metaphorical abstraction that allows us to bleach these concepts of their original conceptual content — space, time and force — and apply them to new abstract domains, therefore allowing a species that evolved to deal with rocks and tools and animals, to conceptualize mathematics, physics, law and other abstract domains. Well, I said I ' d talk about two windows on human nature — the cognitive machinery with which we conceptualize the world, and now I ' m going to say a few words about the relationship types that govern human social interaction, again, as reflected in language. And I ' ll start out with a puzzle, the puzzle of indirect speech acts. Now, I ' m sure most of you have seen the **movie**"Fargo."And you might remember the scene in which the kidnapper is pulled over by a police officer, is asked to show his driver ' s license and holds his wallet out with a 50-dollar bill extending at a slight angle out of the wallet. And he says,"I was just thinking that maybe we could take care of it here in Fargo,"which everyone, including the audience, interprets as a veiled bribe. This kind of indirect speech is rampant in language. For example, in polite requests, if someone says,”If you could pass the guacamole, that would be awesome,”we know exactly what he means, even though that ’ s a rather bizarre concept being expressed.”Would you like to come up and see my etchings?”I think most people understand the intent behind that. And likewise, if someone says,”Nice store you ’ ve got there. It would be a real shame if something happened to it”— — we understand that as a veiled threat, rather than a musing of hypothetical possibilities. So the puzzle is, why are bribes, polite requests, solicitations and threats so often veiled? No one ’ s fooled. Both parties know exactly what the speaker means, and the speaker knows the listener knows that the speaker knows that the listener knows, etc., etc. So what ’ s going on? I think the key idea is that language is a way of negotiating relationships, and human relationships fall into a number of types. There ’ s an influential taxonomy by the anthropologist Alan Fiske, in which relationships can be categorized, more or less, into communality, which works on the principle”what ’ s mine is thine, what ’ s thine is mine,”the kind of mindset that operates within a family, for example ; dominance, whose principle is”don ’ t mess with me ;”reciprocity,”you scratch my back, I ’ ll scratch yours ;”and sexuality, in the immortal words of Cole Porter,”Let ’ s do it.”Now, relationship types can be negotiated. Even though there are default situations in which one of these mindsets can be applied, they can be stretched and extended. For example, communality applies most naturally within family or friends, but it can be used to try to transfer the mentality of sharing to groups that ordinarily would not be disposed to exercise it. For example, in brotherhoods, fraternal organizations, sororities, locutions like”the family of man,”you try to get people who are not related to use the relationship type that would ordinarily be appropriate to close kin. Now, mismatches — when one person assumes one relationship type, and another assumes a different one — can be awkward. If you went over and you helped yourself to a shrimp off your boss ’ plate, for example, that would be an awkward situation. Or if a dinner guest after the meal pulled out his wallet and offered to pay you for the meal, that would be rather awkward as well. In less blatant cases, there ’ s still a kind of negotiation that often goes on. In the workplace, for example, there ’ s often a tension over whether an employee can socialize with the boss, or refer to him or her on a first-name basis. If two friends have a reciprocal transaction, like selling a car, it ’ s well known that this can be a source of tension or awkwardness. In dating, the transition from friendship to sex can lead to, notoriously, **various** forms of awkwardness, and as can sex in the workplace, in which we call the conflict between a dominant and a sexual relationship”sexual harassment.”Well, what does this have to do with language? Well, language, as a social interaction, has to satisfy two conditions. You have to convey the actual content — here we get back to the container metaphor. You want to express the bribe, the command, the promise, the solicitation and so on, but you also have to negotiate and maintain the kind of relationship you have with the other person. The solution, I think, is that we use language at two levels : the literal form signals the safest relationship with the listener, whereas the implicated content — the reading between the lines that we count on the listener to perform — allows the listener to derive the interpretation which is most relevant in context, which possibly initiates a changed relationship. The simplest example of this is in the polite request. If you express your request as a conditional —”if you could open the window, that would be great”— even though the content is an imperative, the fact that you ’ re not using the imperative voice means that you ’ re not acting as if you ’ re in a relationship of dominance, where you could presuppose the compliance of the other person. On the other hand, you want the damn guacamole. By expressing it as an if-then statement, you can get the message across without appearing to boss another person around. And in a more subtle way, I think, this works for all of the veiled speech acts involving plausible deniability : the bribes, threats, propositions, solicitations and so on. One way of thinking about it is to imagine what it would be like if language — where it could only be used literally. And you can think of it in terms of a game-theoretic payoff matrix. Put yourself in the position of the kidnapper wanting to bribe the officer. There ' s a high stakes in the two possibilities of having a dishonest officer or an honest officer. If you don ' t bribe the officer, then you will get a traffic ticket — or, as is the case of"Fargo,"worse — whether the honest officer is honest or dishonest. Nothing ventured, nothing gained. In that case, the consequences are rather severe. On the other hand, if you extend the bribe, if the officer is dishonest, you get a huge payoff of going free. If the officer is honest, you get a huge penalty of being arrested for bribery. So this is a rather fraught situation. On the other hand, with indirect language, if you issue a veiled bribe, then the dishonest officer could interpret it as a bribe, in which case you get the payoff of going free. The honest officer can ' t hold you to it as being a bribe, and therefore, you get the nuisance of the traffic ticket. So you get the best of both worlds. And a similar analysis, I think, can apply to the potential awkwardness of a sexual solicitation, and other cases where plausible deniability is an asset. I think this affirms something that ' s long been known by diplomats — namely, that the vagueness of language, far from being a bug or an imperfection, actually might be a feature of language, one that we use to our advantage in social interactions. So to sum up : language is a collective human creation, reflecting human nature, how we conceptualize reality, how we relate to one another. And then by analyzing the **various** quirks and complexities of language, I think we can get a window onto what makes us tick. Thank you very much.

Text Sample 830: POS Dim 6 – 2005 – Score 4.00 – 2005/t000437.txt

If you take 10, 000 people at random, 9, 999 have something in common : their interests in business lie on or near the Earth ' s surface. The odd one out is an astronomer, and I am one of that strange breed. My talk will be in two parts. I ' ll talk first as an astronomer, and then as a worried member of the human race. But let ' s start off by remembering that Darwin showed how we ' re the outcome of four billion years of evolution. And what we try to do in astronomy and cosmology is to go back before Darwin ' s simple beginning, to set our Earth in a cosmic context. And let me just run through a few slides. This was the impact that happened last week on a comet. If they ' d sent a nuke, it would have been rather more spectacular than what actually happened last Monday. So that ' s another project for NASA. That ' s Mars from the European Mars Express, and at New Year. This artist ' s impression turned into reality when a parachute landed on Titan, Saturn ' s giant moon. It landed on the surface. This is pictures taken on the way down. That looks like a coastline. It is indeed, but the ocean is liquid methane — the temperature minus 170 degrees centigrade. If we go beyond our solar system, we ' ve learned that the stars aren ' t twinkly points of light. Each one is like a sun with a retinue of planets orbiting around it. And we can see places where stars are forming, like the Eagle Nebula. We see stars dying. In six billion years, the sun will look like that. And some stars die spectacularly in a supernova explosion, leaving remnants like that. On a still bigger scale, we see entire galaxies of stars. We see entire ecosystems where gas is being recycled. And to the cosmologist, these galaxies are just the atoms, as it were, of the large-scale universe. This picture shows a patch of sky so small that it would take about 100 patches like it to cover the full moon in the sky. Through a small telescope, this would look quite blank, but you see here hundreds of little, faint smudges. Each is a galaxy, fully like ours or Andromeda, which looks so small and faint because its light has taken 10 billion light-years to get to us. The stars in those galaxies probably don ’ t have planets around them. There ’ s scant chance of life there — that ’ s because there ’ s been no time for the nuclear fusion in stars to make silicon and carbon and iron, the building blocks of planets and of life. We believe that all of this emerged from a Big Bang — a hot, dense state. So how did that amorphous Big Bang turn into our complex cosmos? I ’ m going to show you a **movie** simulation 16 powers of 10 faster than real time, which shows a patch of the universe where the expansions have subtracted out. But you see, as time goes on in gigayears at the bottom, you will see structures evolve as gravity feeds on small, dense irregularities, and structures develop. And we ’ ll end up after 13 billion years with something looking rather like our own universe. And we compare simulated universes like that — I ’ ll show you a better simulation at the end of my talk — with what we actually see in the sky. Well, we can trace things back to the earlier stages of the Big Bang, but we still don ’ t know what **banged** and why it **banged**. That ’ s a challenge for 21st-century science. If my research group had a logo, it would be this picture here : an ouroboros, where you see the micro-world on the left — the world of the quantum — and on the right the large-scale universe of planets, stars and galaxies. We know our universes are united though — links between left and right. The everyday world is determined by atoms, how they stick together to make molecules. Stars are fueled by how the nuclei in those atoms react together. And, as we ’ ve learned in the last few years, galaxies are held together by the gravitational pull of so-called dark matter : particles in huge swarms, far smaller even than atomic nuclei. But we ’ d like to know the synthesis symbolized at the very top. The micro-world of the quantum is understood. On the right hand side, gravity holds sway. Einstein explained that. But the unfinished business for 21st-century science is to link together cosmos and micro-world with a unified theory — symbolized, as it were, gastronomically at the top of that picture. And until we have that synthesis, we won ’ t be able to understand the very beginning of our universe because when our universe was itself the size of an atom, quantum effects could shake everything. And so we need a theory that unifies the very large and the very small, which we don ’ t yet have. One idea, incidentally — and I had this hazard sign to say I ’ m going to speculate from now on — is that our Big Bang was not the only one. One idea is that our three-dimensional universe may be embedded in a high- dimensional space, just as you can imagine on these sheets of paper. You can imagine ants on one of them thinking it ’ s a two-dimensional universe, not being aware of another population of ants on the other. So there could be another universe just a millimeter away from ours, but we ’ re not aware of it because that millimeter is measured in some fourth **spatial** dimension, and we ’ re imprisoned in our three. And so we believe that there may be a lot more to physical reality than what we ’ ve normally called our universe — the aftermath of our Big Bang. And here ’ s another picture. Bottom right depicts our universe, which on the horizon is not beyond that, but even that is just one bubble, as it were, in some vaster reality. Many people suspect that just as we ’ ve gone from believing in one solar system to zillions of solar systems, one galaxy to many galaxies, we have to go to many Big Bangs from one Big Bang, perhaps these many Big Bangs **displaying** an immense variety of properties. Well, let ’ s go back to this picture. There ’ s one challenge symbolized at the top, but there ’ s another challenge to science symbolized at the bottom. You want to not only synthesize the very large and the very small, but we want to understand the very complex. And the most complex things are ourselves, midway between atoms and stars. We depend on stars to make the atoms we ’ re made of. We depend on chemistry to determine our complex structure. We clearly have to be large, compared to atoms, to have layer upon layer of complex structure. We clearly have to be small, compared to stars and planets — otherwise we ’ d be crushed by gravity. And in fact, we are midway. It would take as many human bodies to make up the sun as there are atoms in each of us. The geometric mean of the mass of a proton and the mass of the sun is 50 kilograms, within a factor of two of the mass of each person here. Well, most of you anyway. The science of complexity is probably the greatest challenge of all, greater than that of the very small on the left and the very large on the right. And it ’ s this science, which is not only enlightening our understanding of the biological world, but also transforming our world faster than ever. And more than that, it ’ s engendering new kinds of change. And I now move on to the second part of my talk, and the book”Our Final Century”was mentioned. If I was not a self-effacing Brit, I would mention the book myself, and I would add that it ’ s available in paperback. And in America it was called”Our Final Hour”because Americans like instant gratification. But my theme is that in this century, not only has science changed the world faster than ever, but in new and different ways. Targeted drugs, genetic modification, artificial intelligence, perhaps even implants into our brains, may change human beings themselves. And human beings, their physique and character, has not changed for thousands of years. It may change this century. It ’ s new in our history. And the human impact on the global environment — greenhouse warming, mass extinctions and so forth — is unprecedented, too. And so, this makes this coming century a challenge. Bio- and cybertechnologies are environmentally benign in that they offer marvelous prospects, while, nonetheless, reducing pressure on energy and resources. But they will have a dark side. In our interconnected world, novel technology could empower just one fanatic, or some weirdo with a mindset of those who now design computer viruses, to trigger some kind on disaster. Indeed, catastrophe could arise simply from technical misadventure — error rather than terror. And even a tiny probability of catastrophe is unacceptable when the downside could be of global consequence. In fact, some years ago, Bill Joy wrote an article expressing tremendous concern about robots taking us over, etc. I don ’ t go along with all that, but it ’ s interesting that he had a simple solution. It was what he called”fine-grained relinquishment.”He wanted to give up the dangerous kind of science and keep the good bits. Now, that ’ s absurdly naive for two reasons. First, any scientific discovery has benign consequences as well as dangerous ones. And also, when a scientist makes a discovery, he or she normally has no clue what the applications are going to be. And so what this means is that we have to accept the risks if we are going to enjoy the benefits of science. We have to accept that there will be hazards. And I think we have to go back to what happened in the post-War era, post-World War II, when the nuclear scientists who ’ d been involved in making the atomic bomb, in many cases were concerned that they should do all they could to alert the world to the dangers. And they were inspired not by the young Einstein, who did the great work in relativity, but by the old Einstein, the icon of poster and t-shirt, who failed in his scientific efforts to unify the physical laws. He was premature. But he was a moral compass — an inspiration to scientists who were concerned with arms control. And perhaps the greatest living person is someone I ’ m privileged to know, Joe Rothblatt. Equally untidy office there, as you can see. He ’ s 96 years old, and he founded the Pugwash movement. He persuaded Einstein, as his last act, to sign the famous memorandum of Bertrand Russell. And he sets an example of the concerned scientist. And I think to harness science optimally, to choose which doors to open and which to leave closed, we need latter-day counterparts of people like Joseph Rothblatt. We need not just campaigning physicists, but we need biologists, computer experts and environmentalists as well. And I think academics and independent entrepreneurs have a special obligation because they have more freedom than those in government service, or company employees subject to commercial pressure. I wrote my book,"Our Final Century,"as a scientist, just a general scientist. But there ' s one respect, I think, in which being a cosmologist offered a special perspective, and that ' s that it offers an awareness of the immense future. The stupendous time spans of the evolutionary past are now part of common culture — outside the American Bible Belt, anyway — but most people, even those who are familiar with evolution, aren ' t mindful that even more time lies ahead. The sun has been shining for four and a half billion years, but it ' ll be another six billion years before its fuel runs out. On that schematic picture, a sort of time-lapse picture, we ' re halfway. And it ' ll be another six billion before that happens, and any remaining life on Earth is vaporized. There ' s an unthinking tendency to imagine that humans will be there, experiencing the sun ' s demise, but any life and intelligence that exists then will be as different from us as we are from bacteria. The unfolding of intelligence and complexity still has immensely far to go, here on Earth and probably far beyond. So we are still at the beginning of the emergence of complexity in our Earth and beyond. If you represent the Earth ' s lifetime by a single year, say from January when it was made to December, the 21st-century would be a quarter of a second in June — a tiny fraction of the year. But even in this concertinaed cosmic perspective, our century is very, very special, the first when humans can change themselves and their home planet. As I should have shown this earlier, it will not be humans who witness the end point of the sun ; it will be creatures as different from us as we are from bacteria. When Einstein died in 1955, one striking tribute to his global status was this cartoon by Herblock in the Washington Post. The plaque reads,"Albert Einstein lived here."And I ' d like to end with a vignette, as it were, inspired by this image. We ' ve been familiar for 40 years with this image : the fragile beauty of land, ocean and clouds, contrasted with the sterile moonscape on which the astronauts left their footprints. But let ' s suppose some aliens had been watching our pale blue dot in the cosmos from afar, not just for 40 years, but for the entire 4. 5 billion-year history of our Earth. What would they have seen? Over nearly all that immense time, Earth ' s appearance would have changed very gradually. The only abrupt worldwide change would have been major asteroid impacts or volcanic super-eruptions. Apart from those brief traumas, nothing happens suddenly. The continental landmasses drifted around. Ice cover waxed and waned. Successions of new species emerged, evolved and became extinct. But in just a tiny sliver of the Earth ' s history, the last one-millionth part, a few thousand years, the patterns of vegetation altered much faster than before. This signaled the start of agriculture. Change has accelerated as human populations rose. Then other things happened even more abruptly. Within just 50 years — that ' s one hundredth of one millionth of the Earth ' s age — the amount of carbon dioxide in the atmosphere started to rise, and ominously fast. The planet became an intense emitter of radio waves — the total output from all TV and cell phones and radar transmissions. And something else happened. Metallic objects — albeit very small ones, a few tons at most — escaped into orbit around the Earth. Some journeyed to the moons and planets. A race of advanced extraterrestrials watching our solar system from afar could confidently predict Earth ' s final doom in another six billion years. But could they have predicted this unprecedented spike less than halfway through the Earth ' s life? These human-induced alterations occupying overall less than a millionth of the elapsed lifetime and seemingly occurring with runaway speed? If they continued their vigil, what might these hypothetical aliens witness in the next hundred years? Will some spasm foreclose Earth ' s future? Or will the biosphere stabilize? Or will some of the metallic objects launched from the Earth spawn new oases, a post-human life elsewhere? The science done by the young Einstein will continue as long as our civilization, but for civilization to survive, we ' ll need the wisdom of the old Einstein — humane, global and farseeing. And whatever happens in this uniquely crucial century will resonate into the remote future and perhaps far beyond the Earth, far beyond the Earth as depicted here. Thank you very much.

Text Sample 831: POS Dim 6 – 2002 – Score 5.00 – 2002/t000095.txt

What I want to do today is spend some time talking about some stuff that ' s giving me a little bit of existential angst, for lack of a better word, over the past couple of years. And basically, these three quotes tell what ' s going on."When God made the color purple, God was just showing off,"Alice Walker wrote in"The Color Purple."And Zora Neale Hurston wrote in"Dust Tracks On A Road,""Research is a formalized curiosity. It ' s poking and prying with a purpose."And then finally, when I think about the near future, we have this attitude,"Well, whatever happens, happens."Right? So that goes along with the Cheshire Cat saying,"If you don ' t care much where you want to get to, it doesn ' t much matter which way you go."But I think it does matter which way we go and what road we take, because when I think about design in the near future, what I think are the most important issues, what ' s really crucial and vital, is that we need to revitalize the arts and sciences right now, in 2002. If we describe the near future as 10, 20, 15 years from now, that means that what we do today is going to be critically important, because in the year 2015, in the year 2020, 2025, the world our society is going to be building on, the basic knowledge and abstract ideas, the discoveries that we came up with today, just as all these wonderful things we ' re hearing about here at the TED conference that we take for granted in the world right now, were really knowledge and ideas that came up in the 50s, the 60s and the 70s. That ' s the **substrate** that we ' re exploiting today. Whether it ' s the internet, genetic engineering, laser scanners, guided missiles, fiber optics, high-definition television, remote sensing from space and the wonderful remote-sensing photos that we see in 3D weaving, TV programs like Tracker and Enterprise, CD-rewrite drives, flat-screen, Alvin Ailey ' s"Suite Otis,"or Sarah Jones ' s"Your Revolution Will Not [Happen] Between These Thighs,"which, by the way, was banned by the FCC, or ska — all of these things, without question, almost without exception, are really based on ideas and abstract and creativity from years before. So we have to ask ourselves : What are we contributing to that legacy right now? And when I think about it, I ' m really worried. To be quite frank, I ' m concerned. I ' m skeptical that we ' re doing very much of anything. We ' re, in a sense, failing to act in the future. We ' re purposefully, consciously being laggards. We ' re lagging behind. Frantz Fanon, who was a psychiatrist from Martinique, said,"Each generation must, out of relative obscurity, discover its mission and fulfill or betray it."What is our mission? What do we have to do? I think our mission is to reconcile, to reintegrate science and the arts, because right now, there ' s a schism that exists in popular culture. People have this idea that science and the arts are really separate ; we think of them as separate and different things. And this idea was probably introduced centuries ago, but it ' s really becoming critical now, because we ' re making decisions about our society every day that, if we keep thinking that the arts are separate from the sciences, and we keep thinking it ’ s cute to say,”I don ’ t understand anything about this one, I don ’ t understand anything about the other one,”then we ’ re going to have problems. Now, I know no one here at TED thinks this. All of us, we already know that they ’ re very connected. But I ’ m going to let you know that some folks in the outside world, believe it or not, think it ’ s neat when they say,”Scientists and science is not creative. Maybe scientists are ingenious, but they ’ re not creative.”And then we have this tendency, the career counselors and **various** people say things like,”Artists are not analytical. They ’ re ingenious, perhaps, but not analytical.”And when these concepts underlie our teaching and what we think about the world, then we have a problem, because we stymie support for everything. By accepting this dichotomy, whether it ’ s tongue-in- cheek, when we attempt to accommodate it in our world, and we try to build our foundation for the world, we ’ re messing up the future, because : Who wants to be uncreative? Who wants to be illogical? Talent would run from either of these fields if you said you had to choose either. Then they ’ ll go to something where they think,”Well, I can be creative and logical at the same time.”Now, I grew up in the ’ 60s and I ’ ll admit it — actually, my childhood spanned the ’ 60s, and I was a wannabe hippie, and I always resented the fact that I wasn ’ t old enough to be a hippie. And I know there are people here, the younger generation, who want to be hippies. People talk about the ’ 60s all the time. And they talk about the anarchy that was there. But when I think about the ’ 60s, what I took away from it was that there was hope for the future. We thought everyone could participate. There were wonderful, incredible ideas that were always percolating, and so much of what ’ s cool or hot today is really based on some of those concepts, whether it ’ s people trying to use the Prime Directive from Star Trek, being involved in things, or, again, that three- dimensional weaving and fax machines that I read about in my weekly readers that the technology and engineering was just getting started. But the ’ 60s left me with a problem. You see, I always assumed I would go into space, because I followed all of this. But I also loved the arts and sciences. You see, when I was growing up as a little girl and as a teenager, I loved designing and making doll clothes and wanting to be a fashion designer. I took art and ceramics. I loved dance : Lola Falana, Alvin Ailey, Jerome Robbins. And I also avidly followed the Gemini and the Apollo programs. I had science projects and tons of astronomy books. I took calculus and philosophy. I wondered about infinity and the Big Bang theory. And when I was at Stanford, I found myself, my senior year, chemical engineering major, half the folks thought I was a political science and performing arts major, which was sort of true, because I was Black Student Union President, and I did major in some other things. And I found myself the last quarter juggling chemical engineering separation processes, logic classes, nuclear magnetic resonance spectroscopy, and also producing and choreographing a dance production. And I had to do the lighting and the design work, and I was trying to figure out : Do I go to New York City to try to become a professional dancer, or to go to medical school? Now, my mother helped me figure that one out. But when I went into space, I carried a number of things up with me. I carried a poster by Alvin Ailey — you can figure out now, I love the dance company — an Alvin Ailey poster of Judith Jamison performing the dance”Cry,”dedicated to all black women everywhere ; a Bundu statue, which was from the women ’ s society in Sierra Leone ; and a certificate for the Chicago Public School students to work to improve their science and math. And folks asked me,”Why did you take up what you took up?”And I had to say,”Because it represents human creativity ; the creativity that allowed us, that we were required to have to conceive and build and launch the space shuttle, which springs from the same source as the imagination and analysis that it took to carve a Bundu statue, or the ingenuity it took to design, choreograph and stage"Cry."Each one of them are different manifestations, incarnations, of creativity — avatars of human creativity. And that ' s what we have to reconcile in our minds, how these things fit together. The difference between arts and sciences is not analytical versus intuitive. Right? E = mc2 required an intuitive leap, and then you had to do the analysis afterwards. Einstein said, in fact,"The most beautiful thing we can experience is the mysterious. It is the source of all true art and science."Dance requires us to express and want to express the jubilation in life, but then you have to figure out : Exactly what movement do I do to make sure it comes across correctly? The difference between arts and sciences is also not constructive versus deconstructive. A lot of people think of the sciences as deconstructive, you have to pull things apart. And yeah, subatomic physics is deconstructive — you literally try to tear atoms apart to understand what ' s inside of them. But sculpture, from what I understand from great sculptors, is deconstructive, because you see a piece and you remove what doesn ' t need to be there. Biotechnology is constructive. Orchestral arranging is constructive. So, in fact, we use constructive and deconstructive techniques in everything. The difference between science and the arts is not that they are different sides of the same coin, even, or even different parts of the same continuum, but rather, they ' re manifestations of the same thing. Different quantum states of an atom? Or maybe if I want to be more 21st century, I could say that they ' re different harmonic resonances of a superstring. But we ' ll leave that alone. They spring from the same source. The arts and sciences are avatars of human creativity. It ' s our attempt as humans to build an understanding of the universe, the world around us. It ' s our attempt to influence things, the universe internal to ourselves and external to us. The sciences, to me, are manifestations of our attempt to express or share our understanding, our experience, to influence the universe external to ourselves. It doesn ' t rely on us as individuals. It ' s the universe, as experienced by everyone. The arts manifest our desire, our attempt to share or influence others through experiences that are peculiar to us as individuals. Let me say it again another way : science provides an understanding of a universal experience, and arts provide a universal understanding of a personal experience. That ' s what we have to think about, that they ' re all part of us, they ' re all part of a continuum. It ' s not just the tools, it ' s not just the sciences, the mathematics and the numerical stuff and the statistics, because we heard, very much on this stage, people talked about music being mathematical. Arts don ' t just use clay, aren ' t the only ones that use clay, light and sound and movement. They use analysis as well. So people might say,"Well, I still like that intuitive versus analytical thing,"because everybody wants to do the right brain, left brain thing. We ' ve all been accused of being right-brained or left-brained at some point in time, depending on who we disagreed with. You know, people say"intuitive"— that ' s like you ' re in touch with nature, in touch with yourself and relationships ; analytical, you put your mind to work. I ' m going to tell you a little secret. You all know this, though. But sometimes people use this analysis idea, that things are outside of ourselves, to say, this is what we ' re going to elevate as the true, most important sciences, right? Then you have artists — and you all know this is true as well — artists will say things about scientists because they say they ' re too concrete, they ' re disconnected from the world. But, we ' ve even had that here on stage, so don ' t act like you don ' t know what I ' m talking about. We had folks talking about the Flat Earth Society and flower arrangers, so there ' s this whole dichotomy that we continue to carry along, even when we know better. And folks say we need to choose either-or.

But it would really be foolish to choose either one, intuitive versus analytical. That ’ s a foolish choice. It ’ s foolish just like trying to choose between being realistic or idealistic. You need both in life. Why do people do this? I ’ m going to quote a molecular biologist, Sydney Brenner, who ’ s 70 years old, so he can say this. He said,”It ’ s always important to distinguish between chastity and impotence.”Now — I want to share with you a little equation, OK? How does understanding science and the arts fit into our lives and what ’ s going on and the things we ’ re talking about here at the design conference? And this is a little thing I came up with : understanding and our resources and our will cause us to have outcomes. Our understanding is our science, our arts, our religion ; how we see the universe around us ; our resources, our money, our labor, our minerals — those things that are out there in the world we have to work with. But more importantly, there ’ s our will. This is our vision, our aspirations of the future, our hopes, our dreams, our struggles and our fears. Our successes and our failures influence what we do with all of those. And to me, design and engineering, craftsmanship and skilled labor, are all the things that work on this to have our outcome, which is our human quality of life. Where do we want the world to be? And guess what? Regardless of how we look at this, whether we look at arts and sciences as separate or different, they ’ re both being influenced now and they ’ re both having problems. I did a project called S. E. E. ing the Future : Science, Engineering and Education. It was looking at how to shed light on the most effective use of government funding. We got a bunch of scientists in all stages of their careers. They came to Dartmouth College, where I was teaching. And they talked about, with theologians and financiers : What are some of the issues of public funding for science and engineering research? What ’ s most important about it? There are some ideas that emerged that I think have really powerful parallels to the arts. The first thing they said was that the circumstances that we find ourselves in today in the sciences and engineering that made us world leaders are very different than the ’ 40s, the ’ 50s, and the ’ 60s and the ’ 70s, when we emerged as world leaders, because we ’ re no longer in competition with fascism, with Soviet-style communism. And by the way, that competition wasn ’ t just military ; it included social competition and political competition as well, that allowed us to look at space as one of those platforms to prove that our social system was better. Another thing they talked about was that the infrastructure that supports the sciences is becoming obsolete. We look at universities and colleges — small, mid-sized community colleges across the country — their laboratories are becoming obsolete. And this is where we train most of our science workers and our researchers — and our teachers, by the way. And there ’ s a media that doesn ’ t support the dissemination of any more than the most mundane and inane of information. There ’ s pseudoscience, crop circles, alien autopsy, haunted houses, or disasters. And that ’ s what we see. This isn ’ t really the information you need to operate in everyday life and figure out how to participate in this democracy and determine what ’ s going on. They also said there ’ s a change in the corporate mentality. Whereas government money had always been there for basic science and engineering research, we also counted on some companies to do some basic research. But what ’ s happened now is companies put more energy into short-term product development than they do in basic engineering and science research. And education is not keeping up. In K through 12, people are taking out wet labs. They think if we put a computer in the room, it ’ s going to take the place of actually mixing the acids or growing the potatoes. And government funding is decreasing in spending, and then they ’ re saying, let ’ s have corporations take over, and that ’ s not true. Government funding should at least do things like recognize cost benefits of basic science and engineering research. We have to know that we have a responsibility as global citizens in this world. We have to look at the education of humans. We need to build our resources today to make sure that they ’ re trained so they understand the importance of these things. And we have to support the vitality of science. That doesn ’ t mean that everything has to have one thing that ’ s going to go on, or that we know exactly what ’ s going to be the outcome of it, but that we support the vitality and the intellectual curiosity that goes along [with it]. And if you think about those parallels to the arts — the competition with the Bolshoi Ballet spurred the Joffrey and the New York City Ballet to become better. Infrastructure, museums, theaters, **movie** houses across the country are disappearing. We have more television stations with less to watch, we have more money spent on rewrites to get old television programs in the **movies**. We have corporate funding now that, when it goes to support the arts, it almost requires that the product be part of the picture that the artist draws. We have stadiums that are named over and over again by corporations. In Houston, we ’ re trying to figure out what to do with that Enron Stadium thing. Fine arts and education in the schools is disappearing, And we have a government that seems like it ’ s gutting the NEA and other programs. So we have to really stop and think : What are we trying to do with the sciences and the arts? There ’ s a need to revitalize them. We have to pay attention to it. I just want to tell you quickly what I ’ m doing — I want to tell you what I ’ ve been doing a little bit since... I feel this need to sort of integrate some of the ideas that I ’ ve had and run across over time. One of the things that I found out is that there ’ s a need to repair the dichotomy between the mind and body as well. My mother always told me, you have to be observant, know what ’ s going on in your mind and your body. And as a dancer, I had this tremendous faith in my ability to know my body, just as I knew how to sense colors. Then I went to medical school, and I was supposed to just go on what the machine said about bodies. You know, you would ask patients questions and some people would tell you,”Don ’ t listen to what the patient said.”We know that patients know and understand their bodies better, but these days we ’ re trying to divorce them from that idea. We have to reconcile the patient ’ s knowledge of their body with physicians ’ measurements. We had someone talk about measuring emotions and getting machines to figure out what to keep us from acting crazy. No, we shouldn ’ t measure. We shouldn ’ t use machines to measure road rage and then do something to keep us from engaging in it. Maybe we can have machines help us to recognize that we have road rage, and then we need to know how to control that without the machines. We even need to be able to recognize that without the machines. What I ’ m very concerned about is : How do we bolster our self-awareness as humans, as biological organisms? Michael Moschen spoke of having to teach and learn how to feel with my eyes, to see with my hands. We have all kinds of possibilities to use our senses by, and that ’ s what we have to do. That ’ s what I want to do — to try to use bioinstrumentation, those kind of things, to help our senses in what we do. That ’ s the work I ’ ve been doing now, as a company called BioSentient Corporation. I figured I ’ d have to do that ad, because I ’ m an entrepreneur, and”entrepreneur”says”somebody who does what they want to do, because they ’ re not broke enough that they have to get a real job.”But that ’ s the work I ’ m doing, BioSentient Corporation, trying to figure out : How do we integrate these things? Let me finish by saying that my personal design issue for the future is really about integrating ; to think about that intuitive and that analytical. The arts and sciences are not separate. High school physics lesson before you leave : high school physics teacher used to hold up a **ball**. She would say,”This **ball** has potential energy. But nothing will happen to it, it can ’ t do any work, until I drop it and it changes states.”I like to think of ideas as potential energy. They ’ re really wonderful, but nothing will happen until we risk putting them into action. This conference is filled with wonderful ideas. We ’ re going to share lots of things with people. But nothing ’ s going to happen until we risk putting those ideas into action. We need to revitalize the arts and sciences today. We need to take responsibility for the future. We can ’ t hide behind saying it ’ s just for company profits, or it ’ s just a business, or I ’ m an artist or an academician. Here ’ s how you judge what you ’ re doing : I talked about that balance between intuitive, analytical. Fran Lebowitz, my favorite cynic, said,”The three questions of greatest concern...”— now I ’ m going to add on to design —”... are : Is it attractive?”That ’ s the intuitive.”Is it amusing?”— the analytical, and,”Does it know its place?”— the balance. Thank you very much.

Text Sample 832: POS Dim 6 – 2002 – Score 4.00 – 2002/t000269.txt

Welcome. If I could have the first slide, please? Contrary to calculations made by some engineers, bees can fly, dolphins can swim, and geckos can even climb up the smoothest surfaces. Now, what I want to do, in the short time I have, is to try to allow each of you to experience the thrill of revealing nature ’ s design. I get to do this all the time, and it ’ s just incredible. I want to try to share just a little bit of that with you in this **presentation**. The challenge of looking at nature ’ s designs — and I ’ ll tell you the way that we perceive it, and the way we ’ ve used it. The challenge, of course, is to answer this question : what permits this extraordinary performance of animals that allows them basically to go anywhere? And if we could figure that out, how can we implement those designs? Well, many biologists will tell engineers, and others, organisms have millions of years to get it right ; they ’ re spectacular ; they can do everything wonderfully well. So, the answer is bio-mimicry : just copy nature directly. We know from working on animals that the truth is that ’ s exactly what you don ’ t want to do — because evolution works on the just-good-enough principle, not on a perfecting principle. And the constraints in building any organism, when you look at it, are really severe. Natural technologies have incredible constraints. Think about it. If you were an engineer and I told you that you had to build an automobile, but it had to start off to be this big, then it had to grow to be full size and had to work every step along the way. Or think about the fact that if you build an automobile, I ’ ll tell you that you also — inside it — have to put a factory that allows you to make another automobile. And you can absolutely never, absolutely never, because of history and the inherited plan, start with a clean slate. So, organisms have this important history. Really evolution works more like a tinkerer than an engineer. And this is really important when you begin to look at animals. Instead, we believe you need to be inspired by biology. You need to discover the general principles of nature, and then use these analogies when they ’ re advantageous. This is a real challenge to do this, because animals, when you start to really look inside them — how they work — appear hopelessly complex. There ’ s no detailed history of the design plans, you can ’ t go look it up anywhere. They have way too many motions for their joints, too many muscles. Even the simplest animal we think of, something like an insect, and they have more neurons and connections than you can imagine. How can you make sense of this? Well, we believed — and we hypothesized — that one way animals could work simply, is if the control of their movements tended to be built into their bodies themselves. What we discovered was that two-, four-, six- and eight-legged animals all produce the same forces on the ground when they move. They all work like this kangaroo, they bounce. And they can be modeled by a spring-mass system that we call the spring mass system because we ’ re biomechanists. It ’ s actually a pogo stick. They all produce the pattern of a pogo stick. How is that true? Well, a human, one of your legs works like two legs of a trotting dog, or works like three legs, together as one, of a trotting insect, or four legs as one of a trotting crab. And then they alternate in their propulsion, but the patterns are all the same. Almost every organism we ’ ve looked at this way — you ’ ll see next week, I ’ ll give you a hint, there ’ ll be an article coming out that says that really big things like T. rex probably couldn ’ t do this, but you ’ ll see that next week. Now, what ’ s interesting is the animals, then — we said — bounce along the vertical plane this way, and in our collaborations with Pixar, in”A Bug ’ s Life,”we discussed the bipedal nature of the characters of the ants. And we told them, of course, they move in another plane as well. And they asked us this question. They say,”Why model just in the sagittal plane or the vertical plane, when you ’ re telling us these animals are moving in the horizontal plane?”This is a good question. Nobody in biology ever modeled it this way. We took their advice and we modeled the animals moving in the horizontal plane as well. We took their three legs, we collapsed them down as one. We got some of the best mathematicians in the world from Princeton to work on this problem. And we were able to create a model where animals are not only bouncing up and down, but they ’ re also bouncing side to side at the same time. And many organisms fit this kind of pattern. Now, why is this important to have this model? Because it ’ s very interesting. When you take this model and you perturb it, you give it a push, as it bumps into something, it self-stabilizes, with no brain or no reflexes, just by the structure alone. It ’ s a beautiful model. Let ’ s look at the mathematics. That ’ s enough! The animals, when you look at them running, appear to be self-stabilizing like this, using basically springy legs. That is, the legs can do computations on their own ; the control algorithms, in a sense, are embedded in the form of the animal itself. Why haven ’ t we been more inspired by nature and these kinds of discoveries? Well, I would argue that human technologies are really different from natural technologies, at least they have been so far. Think about the **typical** kind of robot that you see. Human technologies have tended to be large, flat, with right angles, stiff, made of metal. They have rolling devices and axles. There are very few motors, very few sensors. Whereas nature tends to be small, and curved, and it bends and twists, and has legs instead, and appendages, and has many muscles and many, many sensors. So it ’ s a very different design. However, what ’ s changing, what ’ s really exciting — and I ’ ll show you some of that next — is that as human technology takes on more of the characteristics of nature, then nature really can become a much more useful teacher. And here ’ s one example that ’ s really exciting. This is a collaboration we have with Stanford. And they developed this new technique, called Shape Deposition Manufacturing. It ’ s a technique where they can mix materials together and mold any shape that they like, and put in the material properties. They can embed sensors and actuators right in the form itself. For example, here ’ s a leg : the clear part is stiff, the white part is compliant, and you don ’ t need any axles there or anything. It just bends by itself beautifully. So, you can put those properties in. It inspired them to show off this design by producing a little robot they named Sprawl. Our work has also inspired another robot, a biologically inspired bouncing robot, from the University of Michigan and McGill named RHex, for robot hexapod, and this one ’ s autonomous. Let ’ s go to the video, and let me show you some of these animals moving and then some of the simple robots that have been inspired by our discoveries. Here ’ s what some of you did this morning, although you did it outside, not on a treadmill. Here ’ s what we do. This is a death ’ s head cockroach. This is an American cockroach you think you don ’ t have in your kitchen. This is an eight-legged scorpion, six-legged ant, forty-four-legged centipede. Now, I said all these animals are sort of working like pogo sticks — they ’ re bouncing along as they move. And you can see that in this ghost crab, from the beaches of Panama and North Carolina. It goes up to four meters per second when it runs. It actually leaps into the air, and has aerial phases when it does it, like a horse, and you ’ ll see it ’ s bouncing here. What we discovered is whether you look at the leg of a human like Richard, or a cockroach, or a crab, or a kangaroo, the relative leg stiffness of that spring is the same for everything we ’ ve seen so far. Now, what good are springy legs then? What can they do? Well, we wanted to see if they allowed the animals to have greater stability and maneuverability. So, we built a terrain that had obstacles three times the hip height of the animals that we ’ re looking at. And we were certain they couldn ’ t do this. And here ’ s what they did. The animal ran over it and it didn ’ t even slow down! It didn ’ t decrease its preferred speed at all. We couldn ’ t believe that it could do this. It said to us that if you could build a robot with very simple, springy legs, you could make it as maneuverable as any that ’ s ever been built. Here ’ s the first example of that. This is the Stanford Shape Deposition Manufactured robot, named Sprawl. It has six legs — there are the tuned, springy legs. It moves in a gait that an insect uses, and here it is going on the treadmill. Now, what ’ s important about this robot, compared to other robots, is that it can ’ t see anything, it can ’ t feel anything, it doesn ’ t have a brain, yet it can maneuver over these obstacles without any difficulty whatsoever. It ’ s this technique of building the properties into the form. This is a graduate student. This is what he ’ s doing to his thesis project — very robust, if a graduate student does that to his thesis project. This is from McGill and University of Michigan. This is the RHex, making its first outing in a demo. Same principle : it only has six moving parts, six motors, but it has springy, tuned legs. It moves in the gait of the insect. It has the middle leg moving in synchrony with the front, and the hind leg on the other side. Sort of an alternating tripod, and they can negotiate obstacles just like the animal. Robert Full : It ’ ll go on different surfaces — here ’ s sand — although we haven ’ t perfected the feet yet, but I ’ ll talk about that later. Here ’ s RHex entering the woods. Again, this robot can ’ t see anything, it can ’ t feel anything, it has no brain. It ’ s just working with a tuned mechanical system, with very simple parts, but inspired from the fundamental dynamics of the animal. RF : Here ’ s it going down a pathway. I presented this to the jet propulsion lab at NASA, and they said that they had no ability to go down craters to look for ice, and life, ultimately, on Mars. And he said — especially with legged-robots, because they ’ re way too complicated. Nothing can do that. And I talk next. I showed them this video with the simple design of RHex here. And just to convince them we should go to Mars in 2011, I tinted the video orange just to give them the sense of being on Mars. Another reason why animals have extraordinary performance, and can go anywhere, is because they have an effective interaction with the environment. The animal I ’ m going to show you, that we studied to look at this, is the gecko. We have one here and notice its position. It ’ s holding on. Now I ’ m going to challenge you. I ’ m going show you a video. One of the animals is going to be running on the level, and the other one ’ s going to be running up a wall. Which one ’ s which? They ’ re going at a meter a second. How many think the one on the left is running up the wall? Okay. The point is it ’ s really hard to tell, isn ’ t it? It ’ s incredible, we looked at students do this and they couldn ’ t tell. They can run up a wall at a meter a second, 15 steps per second, and they look like they ’ re running on the level. How do they do this? It ’ s just phenomenal. The one on the right was going up the hill. How do they do this? They have bizarre toes. They have toes that uncurl like party favors when you blow them out, and then peel off the surface, like tape. Like if we had a piece of tape now, we ’ d peel it this way. They do this with their toes. It ’ s bizarre! This peeling inspired iRobot — that we work with — to build Mecho-Geckos. Here ’ s a legged version and a tractor version, or a bulldozer version. Let ’ s see some of the geckos move with some video, and then I ’ ll show you a little bit of a clip of the robots. Here ’ s the gecko running up a vertical surface. There it goes, in real time. There it goes again. Obviously, we have to slow this down a little bit. You can ’ t use regular cameras. You have to take 1, 000 pictures per second to see this. And here ’ s some video at 1, 000 frames per second. Now, I want you to look at the animal ’ s back. Do you see how much it ’ s bending like that? We can ’ t figure that out — that ’ s an unsolved mystery. We don ’ t know how it works. If you have a son or a daughter that wants to come to Berkeley, come to my lab and we ’ ll figure this out. Okay, send them to Berkeley because that ’ s the next thing I want to do. Here ’ s the gecko mill. It ’ s a see-through treadmill with a see-through treadmill belt, so we can watch the animal ’ s feet, and videotape them through the treadmill belt, to see how they move. Here ’ s the animal that we have here, running on a vertical surface. Pick a foot and try to watch a toe, and see if you can see what the animal ’ s doing. See it uncurl and then peel these toes. It can do this in 14 milliseconds. It ’ s unbelievable. Here are the robots that they inspire, the Mecho-Geckos from iRobot. First we ’ ll see the animals toes peeling — look at that. And here ’ s the peeling action of the Mecho-Gecko. It uses a pressure- sensitive adhesive to do it. Peeling in the animal. Peeling in the Mecho-Gecko — that allows them climb autonomously. Can go on the flat surface, transition to a wall, and then go onto a ceiling. There ’ s the bulldozer version. Now, it doesn ’ t use **pressure-sensitive** glue. The animal does not use that. But that ’ s what we ’ re limited to, at the moment. What does the animal do? The animal has weird toes. And if you look at the toes, they have these little leaves there, and if you blow them up and zoom in, you ’ ll see that ’ s there ’ s little striations in these leaves. And if you zoom in 270 times, you ’ ll see it looks like a rug. And if you blow that up, and zoom in 900 times, you see there are hairs there, tiny hairs. And if you look carefully, those tiny hairs have striations. And if you zoom in on those 30, 000 times, you ’ ll see each hair has split ends. And if you blow those up, they have these little structures on the end. The smallest branch of the hairs looks like spatulae, and an animal like that has one billion of these nano-size split ends, to get very close to the surface. In fact, there ’ s the diameter of your hair — a gecko has two million of these, and each hair has 100 to 1, 000 split ends. Think of the contact of that that ’ s possible. We were fortunate to work with another group at Stanford that built us a special manned sensor, that we were able to measure the force of an individual hair. Here ’ s an individual hair with a little split end there. When we measured the forces, they were enormous. They were so large that a patch of hairs about this size — the gecko ’ s foot could support the weight of a small child, about 40 pounds, easily. Now, how do they do it? We ’ ve recently discovered this. Do they do it by friction? No, force is too low. Do they do it by electrostatics? No, you can change the charge — they still hold on. Do they do it by interlocking? That ’ s kind of a like a Velcro-like thing. No, you can put them on molecular smooth surfaces — they don ’ t do it. How about suction? They stick on in a vacuum. How about wet adhesion? Or capillary adhesion? They don ’ t have any glue, and they even stick under water just fine. If you put their foot under water, they grab on. How do they do it then? Believe it or not, they grab on by intermolecular forces, by Van der Waals forces. You know, you probably had this a long time ago in chemistry, where you had these two atoms, they ’ re close together, and the electrons are moving around. That tiny force is sufficient to allow them to do that because it ' s added up so many times with these small structures. What we ' re doing is, we ' re taking that inspiration of the hairs, and with another colleague at Berkeley, we ' re manufacturing them. And just recently we ' ve made a breakthrough, where we now believe we ' re going to be able to create the first synthetic, self-cleaning, dry adhesive. Many companies are interested in this. We also presented to Nike even. We ' ll see where this goes. We were so excited about this that we realized that that small-size scale — and where everything gets sticky, and gravity doesn ' t matter anymore — we needed to look at ants and their feet, because one of my other colleagues at Berkeley has built a six-millimeter silicone robot with legs. But it gets stuck. It doesn ' t move very well. But the ants do, and we ' ll figure out why, so that ultimately we ' ll make this move. And imagine : you ' re going to be able to have swarms of these six-millimeter robots available to run around. Where ' s this going? I think you can see it already. Clearly, the Internet is already having eyes and ears, you have web cams and so forth. But it ' s going to also have legs and hands. You ' re going to be able to do programmable work through these kinds of robots, so that you can run, fly and swim anywhere. We saw David Kelly is at the beginning of that with his fish. So, in conclusion, I think the message is clear. If you need a message, if nature ' s not enough, if you care about search and rescue, or mine clearance, or medicine, or the **various** things we ' re working on, we must preserve nature ' s designs, otherwise these secrets will be lost forever. Thank you.

Text Sample 833: POS Dim 6 – 2002 – Score 3.00 – 2002/t000410.txt

That splendid music, the coming-in music,"The Elephant March"from"Aida,"is the music I ' ve chosen for my funeral. And you can see why. It ' s triumphal. I won ' t feel anything, but if I could, I would feel triumphal at having lived at all, and at having lived on this splendid planet, and having been given the opportunity to understand something about why I was here in the first place, before not being here. Can you understand my quaint English accent? Like everybody else, I was entranced yesterday by the animal session. Robert Full and Frans Lanting and others ; the beauty of the things that they showed. The only slight jarring note was when Jeffrey Katzenberg said of the mustang,"the most splendid creatures that God put on this earth."Now of course, we know that he didn ' t really mean that, but in this country at the moment, you can ' t be too careful. I ' m a biologist, and the central theorem of our subject : the theory of design, Darwin ' s theory of evolution by natural selection. In professional circles everywhere, it ' s of course universally accepted. In non-professional circles outside America, it ' s largely ignored. But in non-professional circles within America, it arouses so much hostility — it ' s fair to say that American biologists are in a state of war. The war is so worrying at present, with court cases coming up in one state after another, that I felt I had to say something about it. If you want to know what I have to say about Darwinism itself, I ' m afraid you ' re going to have to look at my books, which you won ' t find in the bookstore outside. Contemporary court cases often concern an allegedly new version of creationism, called"Intelligent Design,"or ID. Don ' t be fooled. There ' s nothing new about ID. It ' s just creationism under another name, rechristened — I choose the word advisedly — for tactical, political reasons. The arguments of so-called ID theorists are the same old arguments that had been refuted again and again, since Darwin down to the present day. There is an effective evolution lobby coordinating the fight on behalf of science, and I try to do all I can to help them, but they get quite upset when people like me dare to mention that we happen to be atheists as well as evolutionists. They see us as rocking the boat, and you can understand why. Creationists, lacking any coherent scientific argument for their case, fall back on the popular phobia against atheism : Teach your children evolution in biology class, and they ' ll soon move on to drugs, grand larceny and sexual"pre-version."In fact, of course, educated theologians from the Pope down are firm in their support of evolution. This book,"Finding Darwin ' s God,"by Kenneth Miller, is one of the most effective attacks on Intelligent Design that I know and it ' s all the more effective because it ' s written by a devout Christian. People like Kenneth Miller could be called a"godsend"to the evolution lobby, because they expose the lie that evolutionism is, as a matter of fact, tantamount to atheism. People like me, on the other hand, rock the boat. But here, I want to say something nice about creationists. It ' s not a thing I often do, so listen carefully. I think they ' re right about one thing. I think they ' re right that evolution is fundamentally hostile to religion. I ' ve already said that many individual evolutionists, like the Pope, are also religious, but I think they ' re deluding themselves. I believe a true understanding of Darwinism is deeply corrosive to religious faith. Now, it may sound as though I ' m about to preach atheism, and I want to reassure you that that ' s not what I ' m going to do. In an audience as sophisticated as this one, that would be preaching to the choir. No, what I want to urge upon you — Instead, what I want to urge upon you is militant atheism. But that ' s putting it too negatively. If I was a person who were interested in preserving religious faith, I would be very afraid of the positive power of evolutionary science, and indeed science generally, but evolution in particular, to inspire and enthrall, precisely because it is atheistic. Now, the difficult problem for any theory of biological design is to explain the massive statistical improbability of living things. Statistical improbability in the direction of good design —"complexity"is another word for this. The standard creationist argument — there is only one ; they ' re all reduced to this one — takes off from a statistical improbability. Living creatures are too complex to have come about by chance ; therefore, they must have had a designer. This argument of course, shoots itself in the foot. Any designer capable of designing something really complex has to be even more complex himself, and that ' s before we even start on the other things he ' s expected to do, like forgive sins, bless marriages, listen to prayers — favor our side in a war — disapprove of our sex lives, and so on. Complexity is the problem that any theory of biology has to solve, and you can ' t solve it by postulating an agent that is even more complex, thereby simply compounding the problem. Darwinian natural selection is so stunningly elegant because it solves the problem of explaining complexity in terms of nothing but simplicity. Essentially, it does it by providing a smooth ramp of gradual, step-by-step increment. But here, I only want to make the point that the elegance of Darwinism is corrosive to religion, precisely because it is so elegant, so parsimonious, so powerful, so economically powerful. It has the sinewy economy of a beautiful suspension bridge. The God theory is not just a bad theory. It turns out to be — in principle — incapable of doing the job required of it. So, returning to tactics and the evolution lobby, I want to argue that rocking the boat may be just the right thing to do. My approach to attacking creationism is — unlike the evolution lobby — my approach to attacking creationism is to attack religion as a whole. And at this point I need to acknowledge the **remarkable** taboo against speaking ill of religion, and I ' m going to do so in the words of the late Douglas Adams, a dear friend who, if he never came to TED, certainly should have been invited. Richard Dawkins : He was. Good. I thought he must have been. He begins this speech, which was tape recorded in Cambridge shortly before he died — he begins by explaining how science works through the testing of hypotheses that are framed to be vulnerable to disproof, and then he goes on. I quote,"Religion doesn ' t seem to work like that. It has certain ideas at the heart of it, which we call ' sacred ' or ' holy. ' What it means is : here is an idea or a notion that you ' re not allowed to say anything bad about. You ' re just not. Why not? Because you ' re not.""Why should it be that it ' s perfectly legitimate to support the Republicans or Democrats, this model of economics versus that, Macintosh instead of Windows, but to have an opinion about how the universe began, about who created the universe — no, that ' s holy. So, we ' re used to not challenging religious ideas, and it ' s very interesting how much of a furor Richard creates when he does it."— He meant me, not that one."Everybody gets absolutely frantic about it, because you ' re not allowed to say these things. Yet when you look at it rationally, there ' s no reason why those ideas shouldn ' t be as open to debate as any other, except that we ' ve agreed somehow between us that they shouldn ' t be."And that ' s the end of the quote from Douglas. In my view, not only is science corrosive to religion ; religion is corrosive to science. It teaches people to be satisfied with trivial, supernatural non- explanations, and blinds them to the wonderful, real explanations that we have within our grasp. It teaches them to accept authority, revelation and faith, instead of always insisting on evidence. There ' s Douglas Adams, magnificent picture from his book,"Last Chance to See."Now, there ' s a **typical** scientific journal, The Quarterly Review of Biology. And I ' m going to put together, as guest editor, a special issue on the question,"Did an asteroid kill the dinosaurs?"And the first paper is a standard scientific paper, presenting evidence,"Iridium layer at the K-T boundary, and potassium argon dated crater in Yucatan, indicate that an asteroid killed the dinosaurs."Perfectly ordinary scientific paper. Now, the next one."The President of the Royal Society has been vouchsafed a strong inner conviction that an asteroid killed the dinosaurs.""It has been privately revealed to Professor Huxtane that an asteroid killed the dinosaurs.""Professor Hordley was brought up to have total and unquestioning faith"— —"that an asteroid killed the dinosaurs.""Professor Hawkins has promulgated an official dogma binding on all loyal Hawkinsians that an asteroid killed the dinosaurs."That ' s inconceivable, of course. But suppose — [Supporters of the Asteroid Theory cannot be patriotic citizens] In 1987, a reporter asked George Bush, Sr. whether he recognized the equal citizenship and patriotism of Americans who are atheists. Mr. Bush ' s reply has become infamous."No, I don ' t know that atheists should be considered citizens, nor should they be considered patriots. This is one nation under God."Bush ' s bigotry was not an isolated mistake, blurted out in the heat of the moment and later retracted. He stood by it in the face of repeated calls for clarification or withdrawal. He really meant it. More to the point, he knew it posed no threat to his election — quite the contrary. Democrats as well as Republicans parade their religiousness if they want to get elected. Both parties invoke"one nation under God."What would Thomas Jefferson have said? [In every country and in every age, the priest has been hostile to liberty] Incidentally, I ' m not usually very proud of being British, but you can ' t help making the comparison. In practice, what is an atheist? An atheist is just somebody who feels about Yahweh the way any decent Christian feels about Thor or Baal or the golden calf. As has been said before, we are all atheists about most of the gods that humanity has ever believed in. Some of us just go one god further. And however we define atheism, it ' s surely the kind of academic belief that a person is entitled to hold without being vilified as an unpatriotic, unelectable non-citizen. Nevertheless, it ' s an undeniable fact that to own up to being an atheist is tantamount to introducing yourself as Mr. Hitler or Miss Beelzebub. And that all stems from the perception of atheists as some kind of weird, way-out minority. Natalie Angier wrote a rather sad piece in the New Yorker, saying how lonely she felt as an atheist. She clearly feels in a beleaguered minority. But actually, how do American atheists stack up numerically? The latest survey makes surprisingly encouraging reading. Christianity, of course, takes a massive lion ’ s share of the population, with nearly 160 million. But what would you think was the second largest group, convincingly outnumbering Jews with 2. 8 million, Muslims at 1. 1 million, Hindus, Buddhists and all other religions put together? The second largest group, with nearly 30 million, is the one described as non-religious or secular. You can ’ t help wondering why vote-seeking politicians are so proverbially overawed by the power of, for example, the Jewish lobby — the state of Israel seems to owe its very existence to the American Jewish vote — while at the same time, consigning the non-religious to political oblivion. This secular non-religious vote, if properly mobilized, is nine times as numerous as the Jewish vote. Why does this far more substantial minority not make a move to exercise its political muscle? Well, so much for quantity. How about quality? Is there any correlation, positive or negative, between intelligence and tendency to be religious? [Them folks misunderestimated me] The survey that I quoted, which is the ARIS survey, didn ’ t break down its data by socio-economic class or education, IQ or anything else. But a recent article by Paul G. Bell in the Mensa magazine provides some straws in the wind. Mensa, as you know, is an international organization for people with very high IQ. And from a meta-analysis of the literature, Bell concludes that, I quote —”Of 43 studies carried out since 1927 on the relationship between religious belief, and one ’ s intelligence or educational level, all but four found an inverse connection. That is, the higher one ’ s intelligence or educational level, the less one is likely to be religious.”Well, I haven ’ t seen the original 42 studies, and I can ’ t comment on that meta-analysis, but I would like to see more studies done along those lines. And I know that there are — if I could put a little plug here — there are people in this audience easily capable of financing a massive research survey to settle the question, and I put the suggestion up, for what it ’ s worth. But let me know show you some data that have been properly published and analyzed, on one special group — namely, top scientists. In 1998, Larson and Witham polled the cream of American scientists, those who ’ d been honored by election to the National Academy of Sciences, and among this select group, belief in a personal God dropped to a shattering seven percent. About 20 percent are agnostic ; the rest could fairly be called atheists. Similar figures obtained for belief in personal immortality. Among biological scientists, the figure is even lower : 5. 5 percent, only, believe in God. Physical scientists, it ’ s 7. 5 percent. I ’ ve not seen corresponding figures for elite scholars in other fields, such as history or philosophy, but I ’ d be surprised if they were different. So, we ’ ve reached a truly **remarkable** situation, a grotesque mismatch between the American intelligentsia and the American electorate. A philosophical opinion about the nature of the universe, which is held by the vast majority of top American scientists and probably the majority of the intelligentsia generally, is so abhorrent to the American electorate that no candidate for popular election dare affirm it in public. If I ’ m right, this means that high office in the greatest country in the world is barred to the very people best qualified to hold it — the intelligentsia — unless they are prepared to lie about their beliefs. To put it bluntly : American political opportunities are heavily loaded against those who are simultaneously intelligent and honest. I ’ m not a citizen of this country, so I hope it won ’ t be thought unbecoming if I suggest that something needs to be done. And I ’ ve already hinted what that something is. From what I ’ ve seen of TED, I think this may be the ideal place to launch it. Again, I fear it will cost money. We need a consciousness-raising, coming-out campaign for American atheists. This could be similar to the campaign organized by homosexuals a few years ago, although heaven forbid that we should stoop to public outing of people against their will. In most cases, people who out themselves will help to destroy the myth that there is something wrong with atheists. On the contrary, they ' ll demonstrate that atheists are often the kinds of people who could serve as decent role models for your children, the kinds of people an advertising agent could use to recommend a product, the kinds of people who are sitting in this room. There should be a snowball effect, a positive feedback, such that the more names we have, the more we get. There could be non-linearities, threshold effects. When a critical mass has been obtained, there ' s an abrupt acceleration in recruitment. And again, it will need money. I suspect that the word"atheist"itself contains or remains a stumbling block far out of proportion to what it actually means, and a stumbling block to people who otherwise might be happy to out themselves. So, what other words might be used to smooth the path, oil the wheels, sugar the pill? Darwin himself preferred"agnostic"— and not only out of loyalty to his friend Huxley, who coined the term. Darwin said,"I have never been an atheist in the same sense of denying the existence of a God. I think that generally an ' agnostic ' would be the most correct description of my state of mind."He even became uncharacteristically tetchy with Edward Aveling. Aveling was a militant atheist who failed to persuade Darwin to accept the dedication of his book on atheism — incidentally, giving rise to a fascinating myth that Karl Marx tried to dedicate"Das Kapital"to Darwin, which he didn ' t, it was actually Edward Aveling. What happened was that Aveling ' s mistress was Marx ' s daughter, and when both Darwin and Marx were dead, Marx ' s papers became muddled up with Aveling ' s papers, and a letter from Darwin saying,"My dear sir, thank you very much but I don ' t want you to dedicate your book to me,"was mistakenly supposed to be addressed to Marx, and that gave rise to this whole myth, which you ' ve probably heard. It ' s a sort of urban myth, that Marx tried to dedicate"Kapital"to Darwin. Anyway, it was Aveling, and when they met, Darwin challenged Aveling."Why do you call yourselves atheists?""' Agnostic, '"retorted Aveling,"was simply ' atheist ' writ respectable, and ' atheist ' was simply ' agnostic ' writ aggressive."Darwin complained,"But why should you be so aggressive?"Darwin thought that atheism might be well and good for the intelligentsia, but that ordinary people were not, quote,"ripe for it."Which is, of course, our old friend, the"don ' t rock the boat"argument. It ' s not recorded whether Aveling told Darwin to come down off his high horse. But in any case, that was more than 100 years ago. You ' d think we might have grown up since then. Now, a friend, an intelligent lapsed Jew, who, incidentally, observes the Sabbath for reasons of cultural solidarity, describes himself as a"tooth-fairy agnostic."He won ' t call himself an atheist because it ' s, in principle, impossible to prove a negative, but"agnostic"on its own might suggest that God ' s existence was therefore on equal terms of likelihood as his non-existence. So, my friend is strictly agnostic about the tooth fairy, but it isn ' t very likely, is it? Like God. Hence the phrase,"tooth-fairy agnostic."Bertrand Russell made the same point using a hypothetical teapot in orbit about Mars. You would strictly have to be agnostic about whether there is a teapot in orbit about Mars, but that doesn ' t mean you treat the likelihood of its existence as on all fours with its non-existence. The list of things which we strictly have to be agnostic about doesn ' t stop at tooth fairies and teapots ; it ' s infinite. If you want to believe one particular one of them — unicorns or tooth fairies or teapots or Yahweh — the onus is on you to say why. The onus is not on the rest of us to say why not. We, who are atheists, are also a-fairyists and a-teapotists. But we don ' t bother to say so. And this is why my friend uses"tooth-fairy agnostic”as a label for what most people would call atheist. Nonetheless, if we want to attract deep-down atheists to come out publicly, we ’ re going to have find something better to stick on our banner than”tooth-fairy”or”teapot agnostic.”So, how about”humanist”? This has the advantage of a worldwide network of well-organized associations and journals and things already in place. My problem with it is only its apparent anthropocentrism. One of the things we ’ ve learned from Darwin is that the human species is only one among millions of cousins, some close, some distant. And there are other possibilities, like”naturalist,”but that also has problems of confusion, because Darwin would have thought naturalist —”Naturalist”means, of course, as opposed to”supernaturalist”— and it is used sometimes — Darwin would have been confused by the other sense of”naturalist,”which he was, of course, and I suppose there might be others who would confuse it with”nudism”. Such people might be those belonging to the British lynch mob, which last year attacked a pediatrician in mistake for a pedophile. I think the best of the available alternatives for”atheist”is simply”non- theist.”It lacks the strong connotation that there ’ s definitely no God, and it could therefore easily be embraced by teapot or tooth-fairy agnostics. It ’ s completely compatible with the God of the physicists. When atheists like Stephen Hawking and Albert Einstein use the word”God,”they use it of course as a metaphorical shorthand for that deep, mysterious part of physics which we don ’ t yet understand.”Non-theist”will do for all that, yet unlike”atheist,”it doesn ’ t have the same phobic, hysterical responses. But I think, actually, the alternative is to grasp the nettle of the word”atheism”itself, precisely because it is a taboo word, carrying frissons of hysterical phobia. Critical mass may be harder to achieve with the word”atheist”than with the word”non- theist,”or some other non-confrontational word. But if we did achieve it with that dread word”atheist”itself, the political impact would be even greater. Now, I said that if I were religious, I ’ d be very afraid of evolution — I ’ d go further : I would fear science in general, if properly understood. And this is because the scientific worldview is so much more exciting, more poetic, more filled with sheer wonder than anything in the poverty-stricken arsenals of the religious imagination. As Carl Sagan, another recently dead hero, put it,”How is it that hardly any major religion has looked at science and concluded, ’ This is better than we thought! The universe is much bigger than our prophet said, grander, more subtle, more elegant ’? Instead they say, ’ No, no, no! My god is a little god, and I want him to stay that way. ’ A religion, old or new, that stressed the magnificence of the universe as revealed by modern science, might be able to draw forth reserves of reverence and awe hardly tapped by the conventional faiths.”Now, this is an elite audience, and I would therefore expect about 10 percent of you to be religious. Many of you probably subscribe to our polite cultural belief that we should respect religion. But I also suspect that a fair number of those secretly despise religion as much as I do. If you ’ re one of them, and of course many of you may not be, but if you are one of them, I ’ m asking you to stop being polite, come out, and say so. And if you happen to be rich, give some thought to ways in which you might make a difference. The religious lobby in this country is massively financed by foundations — to say nothing of all the tax benefits — by foundations, such as the Templeton Foundation and the Discovery Institute. We need an anti- Templeton to step forward. If my books sold as well as Stephen Hawking ’ s books, instead of only as well as Richard Dawkins ’ books, I ’ d do it myself. People are always going on about,”How did September the 11th change you?”Well, here ’ s how it changed me. Let ’ s all stop being so damned respectful. Thank you very much.

Text Sample 834: POS Dim 6 – 2006 – Score 5.00 – 2006/t000500.txt

I ’ ve been at MIT for 44 years. I went to TED I. There ’ s only one other person here, I think, who did that. All the other TEDs â and I went to them all, under Ricky ’ s regime â I talked about what the Media Lab was doing, which today has almost 500 people in it. And if you read the press, last week it actually said I quit the Media Lab. I didn ’ t quit the Media Lab, I stepped down as chairman â which was a kind of ridiculous title, but someone else has taken it on â and one of the things you can do as a professor is you stay on as a professor. And I will now do for the rest of my life the One Laptop Per Child, which I ’ ve sort of been doing for a year and a half, anyway. So I ’ m going to tell you about this, use my 18 minutes to tell you why we ’ re doing it, how we ’ re doing it and then what we ’ re doing. And at some point I ’ ll even pass around what the $100 laptop might be like. I was asked by Chris to talk about some of the big issues, and so I figured I ’ d start with the three that at least drove me to do this. And the first is pretty obvious. It ’ s amazing when you meet a head of state, and you say,”What is your most precious natural resource?”They will not say”children”at first, and then when you say,”children,”they will pretty quickly agree with you. And so that isn ’ t very hard. Everybody agrees that whatever the solutions are to the big problems, they include education, sometimes can be just education and can never be without some element of education. So that ’ s certainly part of it. And the third is a little bit less obvious. And that is that we all in this room learned how to walk, how to talk, not by being taught how to talk, or taught how to walk, but by interacting with the world, by having certain results as a consequence of being able to ask for something, or being able to stand up and reach it. Whereas at about the age six, we were told to stop learning that way, and that all learning from then on would happen through teaching, whether it ’ s people standing up, like I ’ m doing now, or a book, or something. But it was really through teaching. And one of the things in general that computers have provided to learning is that it now includes a kind of learning which is a little bit more like walking and talking, in the sense that a lot of it is driven by the learner himself or herself. So with those as the principles â some of you may know Seymour Papert. This is back in 1982, when we were working in Senegal. Because some people think that the $100 laptop just happened a year ago, or two years ago, or we were struck by lightning â this actually has gone back a long time, and in fact, back to the ’ 60s. Here we ’ re in the ’ 80s. Steve Jobs had given us some laptops. We were in Senegal. It didn ’ t scale but it at least was bringing computers to developing countries and learning pretty quickly that these kids, even though English wasn ’ t their language, the Latin alphabet barely was their language, but they could just swim like fish. They could play these like pianos. A little bit more recently, I got involved personally. And these are two anecdotes â one was in Cambodia, in a village that has no electricity, no water, no television, no telephone, but has broadband Internet now. And these kids, their first English word is”Google”and they only know Skype. They ’ ve never heard of telephony. They just use Skype. And they go home at night â they ’ ve got a broadband connection in a hut that doesn ’ t have electricity. The parents love it, because when they open up the laptops, it ’ s the brightest light source in the house. And talk about where metaphors and reality mix â this is the actual school. In parallel with this, Seymour Papert got the governor of Maine to legislate one laptop per child in the year 2002. Now at the time, I think it ’ s fair to say that 80 percent of the teachers were â let me say, apprehensive. Really, they were actually against it. And they really preferred that the money would be used for higher salaries, more schools, whatever. And now, three and a half years later, guess what? They ’ re reporting five things : drop of truancy to almost zero, attending parent-teacher meetings â which nobody did and now almost everybody does â drop in discipline problems, increase in student participation. Teachers are now saying it ' s kind of fun to teach. Kids are engaged â they have laptops! â and then the fifth, which interests me the most, is that the servers have to be turned off at certain times at night because the teachers are getting too much email from the kids asking them for help. So when you see that kind of thing â this is not something that you have to test. The days of pilot projects are over, when people say,"We ' d like to do three or four thousand in our country to see how it works."Screw you. Go to the back of the line and someone else will do it, and then when you figure out that this works, you can join as well. And this is what we ' re doing. So, One Laptop Per Child was formed about a year and a half ago. It ' s a nonprofit association. It raised about 20 million dollars to do the engineering to just get this built, and then have it produced afterwards. Scale is truly important. And it ' s not important because you can buy components at a lower price, OK? It ' s because you can go to a manufacturer â and I will leave the name out â but we wanted a small **display**, doesn ' t have to have perfect color uniformity. It can even have a pixel or two missing. It doesn ' t have to be that bright. And this particular manufacturer said,"We ' re not interested in that. We ' re interested in the living room. We ' re interested in perfect color uniformity. We ' re interested in big **displays**, bright **displays**. You ' re not part of our strategic plan."And I said,"That ' s kind of too bad, because we need 100 million units a year."And they said,"Oh, well, maybe we could become part of your strategic plan."And that ' s why scale counts. And that ' s why we will not launch this without five to 10 million units in the first run. And the idea is to launch with enough scale that the scale itself helps bring the price down, and that ' s why I said seven to 10 million there. And we ' re doing it without a sales-and- marketing team. I mean, you ' re looking at the sales-and-marketing team. We will do it by going to seven large countries and getting them to agree and launch it, and then the others can follow. We have partners. It ' s not hard to guess Google would be one. The others are all playing to pending. And this has been in the press a great deal. It ' s the so-called Green Machine that we introduced with Kofi Annan in November at the World Summit that was held in Tunisia. Now once people start looking at this, they say,"Ah, this is a laptop project."Well, no, it ' s not a laptop project. It ' s an education project. And the fun part â and I ' m quite focused on it â I tell people I used to be a light bulb, but now I ' m a laser â I ' m just going to get that thing built, and it turns out it ' s not so hard. Because laptop economics are the following : I say 50 percent here â it ' s more like 60, 60 percent of the cost of your laptop is sales, marketing, distribution and profit. Now we have none of those, OK? None of those figure into our cost, because first of all, we sell it at cost, and the governments distribute it. It gets distributed to the school system like a textbook. So that piece disappears. Then you have **display** and everything else. Now the **display** on your laptop costs, in rough numbers, 10 dollars a **diagonal** inch. That can drop to eight ; it can drop to seven but it ' s not going to drop to two, or to one and a half, unless we do some pretty clever things. It ' s the rest â that little brown box â that is pretty fascinating, because the rest of your laptop is devoted to itself. It ' s a little bit like an obese person having to use most of their energy to move their obesity. And we have a situation today which is incredible. I ' ve been using laptops since their inception. And my laptop runs slower, less reliably and less pleasantly than it ever has before. And this year is worse. People clap, sometimes you even get standing ovations, and I say,"What the hell ' s wrong with you? Why are we all sitting there?"And somebody â to remain nameless â called our laptop a"gadget"recently. And I said,"God, our laptop ' s going to go like a bat out of hell. When you open it up, it ’ s going to go ’ bing. ’”It ’ ll be on. It ’ ll be just like it was in 1985, when you bought an Apple Macintosh 512. It worked really well. And we ’ ve been going steadily downhill. Now, people ask all the time what it is. That ’ s what it is. The two pieces that are probably notable : it ’ ll be a mesh network, so when the kids open up their laptops, they all become a network, and then just need one or two points of backhaul. You can serve a couple of thousand kids with two megabits. So you really can bring into a village, and then the villages can connect themselves, and you really can do it quite well. The dual mode **display** â the idea is to have a **display** that both works outdoors â isn ’ t it fun using your cell phone outdoors in the sunlight? Well, you can ’ t see it. And one of the reasons you can ’ t see it is because it ’ s backlighting most of the time, most cell phones. Now, what we ’ re doing is, we ’ re doing one that will be both frontlit and backlit. And whether you manually switch it or you do it in the software is to be seen. But when it ’ s backlit, it ’ s color. And when it ’ s frontlit, it ’ s black and white at three times the resolution. Is it all worked out? No. That ’ s why a lot of our people are more or less living in Taiwan right now. And in about 30 days, we ’ ll know for sure whether this works. Probably the most important piece there is that the kids really can do the maintenance. And this is again something that people don ’ t believe, but I really think it ’ s quite true. That ’ s the machine we showed in Tunis. This is more the direction that we ’ re going to go. And it ’ s something that we didn ’ t think was possible. Now, I ’ m going to pass this around. This isn ’ t a design, OK? So this is just a mechanical engineering sort of embodiment of it for you to play with. And it ’ s clearly just a model. The working one is at MIT. I ’ m going to pass it to this handsome gentleman. At least you can decide whether it goes left or â Chris Anderson : Before you do it, for the people down in simulcast â Nicholas Negroponte : Sorry! I forgot. CA : Just show it off a bit. So wherever the camera is â OK, good point. Thank you, Chris. The idea was that it would be not only a laptop, but that it could transform into an electronic book. So it ’ s sort of an electronic book. This is where when you go outside, it ’ s in black and white. The games buttons are missing, but it ’ ll also be a games machine, book machine. Set it up this way, and it ’ s a television set. Etc., etc. â is that enough for simulcast? OK, sorry. I ’ ll let Jim decide which way to send it afterwards. OK. Seven countries. I say”maybe”for Massachusetts, because they actually have to do a bid. By law you ’ ve got to bid, and so on and so forth. So I can ’ t quite name them. In the other cases, they don ’ t have to do bids. They can decide â it ’ s the federal government in each case. It ’ s kind of agonizing, because a lot of people say,”Let ’ s do it at the state level,”because states are more nimble than the feds, just because of size. And yet we count. We ’ re really dealing with the federal government. We ’ re really dealing with ministries of education. And if you look at governments around the world, ministries of education tend to be the most conservative, and also the ones that have huge payrolls. Everybody thinks they know about education, a lot of culture is built into it as well. It ’ s really hard. And so it ’ s certainly the hard road. If you look at the countries, they ’ re pretty geoculturally distributed. Have they all agreed? No, not completely. Probably Thailand, Brazil and Nigeria are the three that are the most active and most agreed. We ’ re purposely not signing anything with anybody until we actually have the working ones. And since I visit each one of those countries within at least every three months, I ’ m just going around the world every three weeks. Here ’ s sort of the schedule and I put at the bottom we might give some away free in two years at this meeting. Everybody says it ’ s a $100 laptop â you can ’ t do it. Well, guess what, we ’ re not. We ’ re coming in probably at 135, to start, then drift down. And that ’ s very important, because so many things hit the market at a price and then drift up. It ’ s kind of the loss leader, and then as soon as it looks interesting, it can ’ t be afforded, or it can ’ t be scaled out. So we ’ re targeting 50 dollars in 2010. The gray market ’ s a big issue. And one of the ways â just one â but one of the ways to help in the case of the gray market is to make something that is so utterly unique â It ’ s a little bit like the fact that automobiles â thousands of automobiles are stolen every day in the United States. Not one single post-office truck is stolen. And why? Because there ’ s no market for post-office trucks. It looks like a post-office truck. You can spray paint it. You can do anything you want. I just learned recently : in South Africa, no white Volvos are stolen. Period. None. Zero. So we want to make it very much like a white Volvo. Each government has a task force. This perhaps is less interesting, but we ’ re trying to get the governments to all work together and it ’ s not easy. The economics of this is to start with the federal governments and then later, to subsequently go to other â whether it ’ s child-to-child funding, so a child in this country buys one for a child in the developing world, maybe of the same gender, maybe of the same age. An uncle gives a niece or a nephew that as a birthday present. I mean, there are all sorts of things that will happen, and they ’ ll be very, very exciting. And everybody says â I say â it ’ s an education project. Are we providing the software? The answer is : The system certainly has software, but no, we ’ re not providing the education content. That is really done in the countries. But we are certainly constructionists. And we certainly believe in learning by doing and everything from Logo, which was started in 1968, to more modern things, like Scratch, if you ’ ve ever even heard of it, are very, very much part of it. And that ’ s the rollout. Are we dreaming? Is this real? It actually is real. The only criticism, and people really don ’ t want to criticize this, because it is a humanitarian effort, a nonprofit effort and to criticize it is a little bit stupid, actually. But the one thing that people could criticize was,”Great idea, but these guys can ’ t do it.”And that could either mean these guys, professors and so on couldn ’ t do it, or that it ’ s not possible. Well, on December 12, a company called Quanta agreed to build it, and since they make about one-third of all the laptops on the planet today, that question disappeared. So it ’ s not a matter of whether it ’ s going to happen. It is going to happen. And if it comes out at 138 dollars, so what? If it comes out six months late, so what? That ’ s a pretty soft landing. Thank you.

Text Sample 835: POS Dim 6 – 2006 – Score 4.00 – 2006/t000498.txt

I ’ m the luckiest guy in the world. I got to see the last case of killer smallpox in the world. I was in India this past year, and I may have seen the last cases of polio in the world. There ’ s nothing that makes you feel more â the blessing and the honor of working in a program like that â than to know that something that horrible no longer exists. So I ’ m going to tell you â so I ’ m going to show you some dirty pictures. They are difficult to watch, but you should look at them with optimism, because the horror of these pictures will be matched by the uplifting quality of knowing that they no longer exist. But first, I ’ m going to tell you a little bit about my own journey. My background is not exactly the conventional medical education that you might expect. When I was an intern in San Francisco, I heard about a group of Native Americans who had taken over Alcatraz Island, and a Native American who wanted to give birth on that island, and no other doctor wanted to go and help her give birth. I went out to Alcatraz, and I lived on the island for several weeks. She gave birth ; I caught the baby ; I got off the island ; I landed in San Francisco ; and all the press wanted to talk to me, because my three weeks on the island made me an expert in Indian affairs. I wound up on every television show. Someone saw me on television ; they called me up ; and they asked me if I ’ d like to be in a **movie** and to play a young doctor for a bunch of rock and roll stars who were traveling in a bus ride from San Francisco to England. And I said, yes, I would do that, so I became the doctor in an absolutely awful **movie** called”Medicine Ball Caravan.”Now, you know from the ’ 60s, you ’ re either on the bus or you ’ re off the bus ; I was on the bus. My wife of 37 years and I joined the bus. Our bus ride took us from San Francisco to London, then we switched buses at the big pond. We then got on two more buses and we drove through Turkey and Iran, Afghanistan, over the Khyber Pass into Pakistan, like every other young doctor. This is us at the Khyber Pass, and that ’ s our bus. We had some difficulty getting over the Khyber Pass. But we wound up in India. And then, like everyone else in our generation, we went to live in a Himalayan monastery. This is just like a residency program, for those of you that are in medical school. And we studied with a wise man, a guru named Karoli Baba, who then told me to get rid of the dress, put on a three-piece suit, go join the United Nations as a diplomat and work for the World Health Organization. And he made an outrageous prediction that smallpox would be eradicated, and that this was God ’ s gift to humanity, because of the hard work of dedicated scientists. And that prediction came true. This little girl is Rahima Banu, and she was the last case of killer smallpox in the world. And this document is the certificate that the global commission signed, certifying the world to have eradicated the first disease in history. The key to eradicating smallpox was early detection, early response. I ’ m going to ask you to repeat that : early detection, early response. Can you say that? Audience : Early detection, early response. Larry Brilliant : Smallpox was the worst disease in history. It killed more people than all the wars in history. In the last century, it killed 500 million people. You ’ re reading about Larry Page already. Somebody reads very fast. In the year that Larry Page and Sergey Brin â with whom I have a certain affection and a new affiliation â in the year in which they were born, two million people died of smallpox. We declared smallpox eradicated in 1980. This is the most important slide that I ’ ve ever seen in public health, [Sovereigns killed by smallpox] because it shows you to be the richest and the strongest, and to be kings and queens of the world, did not protect you from dying of smallpox. Never can you doubt that we are all in this together. But to see smallpox from the perspective of a sovereign is the wrong perspective. You should see it from the perspective of a mother, watching her child develop this disease and standing by helplessly. Day one, day two, day three, day four, day five, day six. You ’ re a mother and you ’ re watching your child, and on day six, you see pustules that become hard. Day seven, they show the classic scars of smallpox umbilication. Day eight. And Al Gore said earlier that the most photographed image in the world, the most printed image in the world, was that of the Earth. But this was in 1974, and as of that moment, this photograph was the photograph that was the most widely printed, because we printed two billion copies of this photograph, and we took them hand to hand, door to door, to show people and ask them if there was smallpox in their house, because that was our surveillance system. We didn ’ t have Google, we didn ’ t have web crawlers, we didn ’ t have computers. By day nine â you look at this picture and you ’ re horrified ; I look at this picture and I say,”Thank God,”because it ’ s clear that this is only an ordinary case of smallpox, and I know this child will live. And by day 13, the lesions are scabbing, his eyelids are swollen, but you know this child has no other secondary infection. And by day 20, while he will be scarred for life, he will live. There are other kinds of smallpox that are not like that. This is confluent smallpox, in which there isn ’ t a single place on the body where you could put a finger and not be covered by lesions. Flat smallpox, which killed 100 percent of people who got it. And hemorrhagic smallpox, the most cruel of all, which had a predilection for pregnant women. I ’ ve probably had 50 women die. They all had hemorrhagic smallpox. I ’ ve never seen anybody die from it who wasn ’ t a pregnant woman. In 1967, the WHO embarked on what was an outrageous program to eradicate a disease. In that year, there were 34 countries affected with smallpox. By 1970, we were down to 18 countries. 1974, we were down to five countries. But in that year, smallpox exploded throughout India. And India was the place where smallpox made its last stand. In 1974, India had a population of 600 million. There are 21 linguistic states in India, which is like saying 21 different countries. There are 20 million people on the road at any time, in buses and trains, walking ; 500, 000 villages, 120 million households, and none of them wanted to report if they had a case of smallpox in their house, because they thought that smallpox was the visitation of a deity, Shitala Mata, the cooling mother, and it was wrong to bring strangers into your house when the deity was in the house. No incentive to report smallpox. It wasn ’ t just India that had smallpox deities ; smallpox deities were prevalent all over the world. So, how we eradicated smallpox was â mass vaccination wouldn ’ t work. You could vaccinate everybody in India, but one year later there ’ d be 21 million new babies, which was then the population of Canada. It wouldn ’ t do just to vaccinate everyone. You had to find every single case of smallpox in the world at the same time, and draw a circle of immunity around it. And that ’ s what we did. In India alone, my 150, 000 best friends and I went door to door, with that same picture, to every single house in India. We made over one billion house calls. And in the process, I learned something very important. Every time we did a house-to-house search, we had a spike in the number of reports of smallpox. When we didn ’ t search, we had the illusion that there was no disease. When we did search, we had the illusion that there was more disease. A surveillance system was necessary, because what we needed was early detection, early response. So we searched and we searched, and we found every case of smallpox in India. We had a reward. We raised the reward. We continued to increase the reward. We had a scorecard that we wrote on every house. And as we did that, the number of reported cases in the world dropped to zero. And in 1980, we declared the globe free of smallpox. It was the largest campaign in United Nations history, until the Iraq war. 150, 000 people from all over the world â doctors of every race, religion, culture and nation, who fought side by side, brothers and sisters, with each other, not against each other, in a common cause to make the world better. But smallpox was the fourth disease that was intended for eradication. We failed three other times. We failed against malaria, yellow fever and yaws. But soon we may see polio eradicated. But the key to eradicating polio is early detection, early response. This may be the year we eradicate polio. That will make it the second disease in history. And David Heymann, who ’ s watching this on the webcast â David, keep on going. We ’ re close! We ’ re down to four countries. I feel like Hank Aaron. Barry Bonds can replace me any time. Let ’ s get another disease off the list of terrible things to worry about. I was just in India working on the polio program. The polio surveillance program is four million people going door to door. That is the surveillance system. But we need to have early detection, early response. Blindness, the same thing. The key to discovering blindness is doing epidemiological surveys and finding out the causes of blindness, so you can mount the correct response. The Seva Foundation was started by a group of alumni of the Smallpox Eradication Programme, who, having climbed the highest mountain, tasted the elixir of the success of eradicating a disease, wanted to do it again. And over the last 27 years, Seva ’ s programs in 15 countries have given back sight to more than two million blind people. Seva got started because we wanted to apply these lessons of surveillance and epidemiology to something which nobody else was looking at as a public health issue : blindness, which heretofore had been thought of only as a clinical disease. In 1980, Steve Jobs gave me that computer, which is Apple number 12, and it ’ s still in Kathmandu, and it ’ s still working, and we ought to go get it and auction it off and make more money for Seva. And we conducted the first Nepal survey ever done for health, and the first nationwide blindness survey ever done, and we had astonishing results. Instead of finding out what we thought was the case â that blindness was caused mostly by glaucoma and trachoma â we were astounded to find out that blindness was caused instead by cataract. You can ’ t cure or prevent what you don ’ t know is there. In your TED packages there ’ s a DVD,”Infinite Vision,”about Dr. V and the Aravind Eye Hospital. I hope that you will take a look at it. Aravind, which started as a Seva project, is now the world ’ s largest and best eye hospital. This year, that one hospital will give back sight to more than 300, 000 people in Tamil Nadu, India. Bird flu. I stand here as a representative of all terrible things â this might be the worst. The key to preventing or mitigating pandemic bird flu is early detection and rapid response. We will not have a vaccine or adequate supplies of an antiviral to combat bird flu if it occurs in the next three years. WHO stages the progress of a pandemic. We are now at stage three on the pandemic alert stage, with just a little bit of human-to-human transmission, but no human-to-human-to-human sustained transmission. The moment WHO says we ’ ve moved to category four â this will not be like Katrina. The world as we know it will stop. There ’ ll be no airplanes flying. Would you get in an airplane with 250 people you didn ’ t know, coughing and sneezing, when you knew that some of them might carry a disease that could kill you, for which you had no antivirals or vaccine? I did a study of the top epidemiologists in the world in October. I asked them â these are all fluologists and specialists in influenza â and I asked them the questions you ’ d like to ask them : What do you think the likelihood is that there ’ ll be a pandemic? If it happens, how bad do you think it will be? Fifteen percent said they thought there ’ d be a pandemic within three years. But much worse than that, 90 percent said they thought there ’ d be a pandemic within your children or your grandchildren ’ s lifetime. And they thought that if there was a pandemic, a billion people would get sick. As many as 165 million people would die. There would be a global recession and depression as our just-in-time inventory system and the tight rubber band of globalization broke, and the cost to our economy of one to three trillion dollars would be far worse for everyone than merely 100 million people dying, because so many more people would lose their job and their healthcare benefits, that the consequences are almost unthinkable. And it ’ s getting worse, because travel is getting so much better. Let me show you a simulation of what a pandemic looks like. So we know what we ’ re talking about. Let ’ s assume, for example, that the first case occurs in South Asia. It initially goes quite slowly. You get two or three discrete locations. Then there ’ ll be secondary outbreaks, and the disease will spread from country to country so fast that you won ’ t know what hit you. Within three weeks it will be everywhere in the world. Now, if we had an”undo”button, and we could go back and isolate it and grab it when it first started â if we could find it early, and we had early detection and early response, and we could put each one of those viruses in jail â that ’ s the only way to deal with something like a pandemic. And let me show you why that is. We have a joke. This is an epidemic curve, and everyone in medicine, I think, ultimately gets to know what it is. But the joke is, an epidemiologist likes to arrive at an epidemic right here and ride to glory on the downhill curve. But you don ’ t get to do that usually. You usually arrive right about here. What we really want is to arrive right here, so we can stop the epidemic. But you can ’ t always do that. But there ’ s an organization that has been able to find a way to learn when the first cases occur, and that is called GPHIN ; it ’ s the Global Public Health Information Network. And that simulation that I showed you that you thought was bird flu â that was SARS. And SARS is the pandemic that did not occur. And it didn ’ t occur because GPHIN found the pandemic-to-be of SARS three months before WHO actually announced it, and because of that, we were able to stop the SARS pandemic. And I think we owe a great debt of gratitude to GPHIN and to Ron St. John, who I hope is in the audience some place â over there â who ’ s the founder of GPHIN. Hello, Ron! And TED has flown Ron here from Ottawa, where GPHIN is located, because not only did GPHIN find SARS early, but you may have seen last week that Iran announced that they had bird flu in Iran, but GPHIN found the bird flu in Iran not February 14 â but last September. We need an early-warning system to protect us against the things that are humanity ’ s worst nightmare. And so my TED wish is based on the common denominator of these experiences. Smallpox â early detection, early response. Blindness, polio â early detection, early response. Pandemic bird flu â early detection, early response. It is a litany. It is so obvious that our only way of dealing with these new diseases is to find them early and to kill them before they spread. So, my TED wish is for you to help build a global system â an early-warning system â to protect us against humanity ’ s worst nightmares. And what I thought I would call it is”Early Detection,”But it should really be called...”Total Early Detection.”[TED] What? What? But in all seriousness, because this idea is birthed in TED, I would like it to be a legacy of TED, and I ’ d like to call it the”International System for Total Early Disease Detection.”[INSTEDD] And INSTEDD then becomes our mantra. So instead of a hidden pandemic of bird flu, we find it and immediately contain it. Instead of a novel virus caused by bio-terror or bio-error, or shift or drift, we find it and we contain it. Instead of industrial accidents like oil spills or the catastrophe in Bhopal, we find them, and we respond to them. Instead of famine, hidden until it is too late, we detect it, and we respond. And instead of a system which is owned by a government, and hidden in the bowels of government, let ’ s build an early detection system that ’ s freely available to anyone in the world in their own language. Let ’ s make it transparent, non-governmental, not owned by any single country or company, housed in a neutral country, with redundant backup in a different time zone and a different continent. And let ’ s build it on GPHIN. Let ’ s start with GPHIN. Let ’ s increase the websites that they crawl from 20, 000 to 20 million. Let ’ s increase the languages they crawl from seven to 70, or more. Let ’ s build in outbound confirmation messages, using text messages or SMS or instant messaging to find out from people who are within 100 meters of the rumor that you hear, if it is, in fact, valid. And let ’ s add satellite confirmation. And we ’ ll add Gapminder ’ s amazing graphics to the front end. And we ’ ll grow it as a moral force in the world, finding out those terrible things before anybody else knows about them, and sending our response to them, so that next year, instead of us meeting here, lamenting how many terrible things there are in the world, we will have pulled together, used the unique skills and the magic of this community, and be proud that we have done everything we can to stop pandemics, other catastrophes, and change the world, beginning right now. Chris Anderson : An amazing **presentation**. First of all, just so everyone understands : you ’ re saying that by creating web crawlers, looking on the Internet for patterns, they can detect something suspicious before WHO, before anyone else can see it? Give an example of how that could possibly be true. Larry Brilliant : You ’ re not mad about the copyright violation? CA : No. I love it. LB : Well, as Ron St. John â I hope you ’ ll go and meet him in the dinner afterwards and talk to him. When he started GPHIN â In 1997, there was an outbreak of bird flu â H5N1. It was in Hong Kong. And a **remarkable** doctor in Hong Kong responded immediately, by slaughtering 1. 5 million chickens and birds, and they stopped that outbreak in its tracks. Immediate detection, immediate response. Then a number of years went by, and there were a lot of rumors about bird flu. Ron and his team in Ottawa began to crawl the web â only crawling 20, 000 different websites, mostly periodicals â and they read about and heard about a concern, of a lot of children who had high fever and symptoms of bird flu. They reported this to WHO. WHO took a little while taking action, because WHO will only receive a report from a government, because it ’ s the United Nations. But they were able to point to WHO and let them know that there was this surprising and unexplained cluster of illnesses that looked like bird flu. That turned out to be SARS. That ’ s how the world found out about SARS. And because of that, we were able to stop SARS. Now, what ’ s really important is that, before there was GPHIN, 100 percent of all the world ’ s reports of bad things â whether you ’ re talking about famine or you ’ re talking about bird flu or you ’ re talking about Ebola â 100 percent of all those reports came from nations. The moment these guys in Ottawa â on a budget of 800, 000 dollars a year â got cracking, 75 percent of all the reports in the world came from GPHIN, 25 percent of all the reports in the world came from all the other 180 nations. Now, here ’ s what ’ s really interesting : after they ’ d been working for a couple years, what do you think happened to those nations? They felt pretty stupid. So they started sending in their reports early. And now, their reporting percentage is down to 50 percent, because other nations have started to report. So, can you find diseases early by crawling the web? Of course you can. Can you find it even earlier than GPHIN does now? Of course you can. You saw that they found SARS using their Chinese web crawler a full six weeks before they found it using their English web crawler. Well, they ’ re only crawling in seven languages. These bad viruses really don ’ t have any intention of showing up first in English or Spanish or French. So yes, I want to take GPHIN, I want to build on it. I want to add all the languages of the world that we possibly can. I want to make this open to everybody, so that the health officer in Nairobi or in Patna, Bihar will have as much access to it as the folks in Ottawa or in CDC. And I want to make it part of our culture that there is a community of people who are watching out for the worst nightmares of humanity, and that it ’ s accessible to everyone.

Text Sample 836: POS Dim 6 – 2006 – Score 4.00 – 2006/t000160.txt

The future that we will create can be a future that we ’ ll be proud of. I think about this every day ; it ’ s quite literally my job. I ’ m co-founder and senior columnist at Worldchanging. com. Alex Steffen and I founded Worldchanging in late 2003, and since then we and our growing global team of contributors have documented the ever-expanding variety of solutions that are out there, right now and on the near horizon. In a little over two years, we ’ ve written up about 4, 000 items — replicable models, technological tools, emerging ideas — all providing a path to a future that ’ s more sustainable, more equitable and more desirable. Our emphasis on solutions is quite intentional. There are tons of places to go, online and off, if what you want to find is the latest bit of news about just how quickly our hell-bound handbasket is moving. We want to offer people an idea of what they can do about it. We focus primarily on the planet ’ s environment, but we also address issues of global development, international conflict, responsible use of emerging technologies, even the rise of the so-called Second Superpower and much, much more. The scope of solutions that we discuss is actually pretty broad, but that reflects both the range of challenges that need to be met and the kinds of innovations that will allow us to do so. A quick sampling really can barely scratch the surface, but to give you a sense of what we cover : tools for rapid disaster relief, such as this inflatable concrete shelter ; innovative uses of bioscience, such as a flower that changes color in the presence of landmines ; ultra high-efficiency designs for homes and offices ; distributed power generation using solar power, wind power, ocean power, other clean energy sources ; ultra, ultra high-efficiency vehicles of the future ; ultra high- efficiency vehicles you can get right now ; and better urban design, so you don ' t need to drive as much in the first place ; bio-mimetic approaches to design that take advantage of the efficiencies of natural models in both vehicles and buildings ; distributed computing projects that will help us model the future of the climate. Also, a number of the topics that we ' ve been talking about this week at TED are things that we ' ve addressed in the past on Worldchanging : cradle-to-cradle design, MIT ' s Fab Labs, the consequences of extreme longevity, the One Laptop per Child project, even Gapminder. As a born-in-the-mid-1960s Gen X-er, hurtling all too quickly to my fortieth birthday, I ' m naturally inclined to pessimism. But working at Worldchanging has convinced me, much to my own surprise, that successful responses to the world ' s problems are nonetheless possible. Moreover, I ' ve come to realize that focusing only on negative outcomes can really blind you to the very possibility of success. As Norwegian social scientist Evelin Lindner has observed,"Pessimism is a luxury of good times... In difficult times, pessimism is a self- fulfilling, self-inflicted death sentence."The truth is, we can build a better world, and we can do so right now. We have the tools : we saw a hint of that a moment ago, and we ' re coming up with new ones all the time. We have the knowledge, and our understanding of the planet improves every day. Most importantly, we have the motive : we have a world that needs fixing, and nobody ' s going to do it for us. Many of the solutions that I and my colleagues seek out and write up every day have some important aspects in common : transparency, collaboration, a willingness to experiment, and an appreciation of science — or, more appropriately, science! The majority of models, tools and ideas on Worldchanging encompass combinations of these characteristics, so I want to give you a few concrete examples of how these principles combine in world- changing ways. We can see world-changing values in the emergence of tools to make the invisible visible — that is, to make apparent the conditions of the world around us that would otherwise be largely imperceptible. We know that people often change their behavior when they can see and understand the impact of their actions. As a small example, many of us have experienced the change in driving behavior that comes from having a real time **display** of mileage showing precisely how one ' s driving habits affect the vehicle ' s efficiency. The last few years have all seen the rise of innovations in how we measure and **display** aspects of the world that can be too big, or too intangible, or too slippery to grasp easily. Simple technologies, like wall-mounted devices that **display** how much power your household is using, and what kind of results you ' ll get if you turn off a few lights — these can actually have a direct positive impact on your energy footprint. Community tools, like text messaging, that can tell you when pollen counts are up or smog levels are rising or a natural disaster is unfolding, can give you the information you need to act in a timely fashion. Data-rich **displays** like maps of campaign contributions, or maps of the disappearing polar ice caps, allow us to better understand the context and the flow of processes that affect us all.

We can see world-changing values in research projects that seek to meet the world ’ s medical needs through open access to data and collaborative action. Now, some people emphasize the risks of knowledge-enabled dangers, but I ’ m convinced that the benefits of knowledge-enabled solutions are far more important. For example, open-access journals, like the Public Library of Science, make cutting-edge scientific research free to all — everyone in the world. And actually, a growing number of science publishers are adopting this model. Last year, hundreds of volunteer biology and chemistry researchers around the world worked together to sequence the genome of the parasite responsible for some of the developing world ’ s worst diseases : African sleeping sickness, leishmaniasis and Chagas disease. That genome data can now be found on open-access genetic data banks around the world, and it ’ s an enormous boon to researchers trying to come up with treatments. But my favorite example has to be the global response to the SARS epidemic in 2003, 2004, which relied on worldwide access to the full gene sequence of the SARS virus. The U. S. National Research Council in its follow-up report on the outbreak specifically cited this open availability of the sequence as a key reason why the treatment for SARS could be developed so quickly. And we can see world-changing values in something as humble as a cell phone. I can probably count on my fingers the number of people in this room who do not use a mobile phone — and where is Aubrey, because I know he doesn ’ t? For many of us, cell phones have really become almost an extension of ourselves, and we ’ re really now beginning to see the social changes that mobile phones can bring about. You may already know some of the big-picture aspects : globally, more camera phones were sold last year than any other kind of camera, and a growing number of people live lives mediated through the lens, and over the network — and sometimes enter history books. In the developing world, mobile phones have become economic drivers. A study last year showed a direct correlation between the growth of mobile phone use and subsequent GDP increases across Africa. In Kenya, mobile phone minutes have actually become an alternative currency. The political aspects of mobile phones can ’ t be ignored either, from text message swarms in Korea helping to bring down a government, to the Blairwatch Project in the UK, keeping tabs on politicians who try to avoid the press. And it ’ s just going to get more wild. Pervasive, always-on networks, high quality sound and video, even devices made to be worn instead of carried in the pocket, will transform how we live on a scale that few really appreciate. It ’ s no exaggeration to say that the mobile phone may be among the world ’ s most important technologies. And in this rapidly evolving context, it ’ s possible to imagine a world in which the mobile phone becomes something far more than a medium for social interaction. I ’ ve long admired the Witness project, and Peter Gabriel told us more details about it on Wednesday, in his profoundly moving **presentation**. And I ’ m just incredibly happy to see the news that Witness is going to be opening up a Web portal to enable users of digital cameras and camera phones to send in their recordings over the Internet, rather than just hand-carrying the videotape. Not only does this add a new and potentially safer avenue for documenting abuses, it opens up the program to the growing global digital generation. Now, imagine a similar model for networking environmentalists. Imagine a Web portal collecting recordings and evidence of what ’ s happening to the planet : putting news and data at the fingertips of people of all kinds, from activists and researchers to businesspeople and political figures. It would highlight the changes that are underway, but would more importantly give voice to the people who are willing to work to see a new world, a better world, come about. It would give everyday citizens a chance to play a role in the protection of the planet. It would be, in essence, an”Earth Witness”project. Now, just to be clear, in this talk I ’ m using the name”Earth Witness”as part of the scenario, simply as a shorthand, for what this imaginary project could aspire to, not to piggyback on the wonderful work of the Witness organization. It could just as easily be called,”Environmental Transparency Project,””Smart Mobs for Natural Security”— but Earth Witness is a lot easier to say. Now, many of the people who participate in Earth Witness would focus on ecological problems, human-caused or otherwise, especially environmental crimes and significant sources of greenhouse gases and emissions. That ’ s understandable and important. We need better documentation of what ’ s happening to the planet if we ’ re ever going to have a chance of repairing the damage. But the Earth Witness project wouldn ’ t need to be limited to problems. In the best Worldchanging tradition, it might also serve as a showcase for good ideas, successful projects and efforts to make a difference that deserve much more visibility. Earth Witness would show us two worlds : the world we ’ re leaving behind, and the world we ’ re building for generations to come. And what makes this scenario particularly appealing to me is we could do it today. The key components are already widely available. Camera phones, of course, would be fundamental to the project. And for a lot of us, they ’ re as close as we have yet to always-on, widely available information tools. We may not remember to bring our digital cameras with us wherever we go, but very few of us forget our phones. You could even imagine a version of this scenario in which people actually build their own phones. Over the course of last year, open-source hardware hackers have come up with multiple models for usable, Linux-based mobile phones, and the Earth Phone could spin off from this kind of project. At the other end of the network, there ’ d be a server for people to send photos and messages to, accessible over the Web, combining a photo-sharing service, social networking platforms and a collaborative **filtering** system. Now, you Web 2. 0 folks in the audience know what I ’ m talking about, but for those of you for whom that last sentence was in a crazy moon language, I mean simply this : the online part of the Earth Witness project would be created by the users, working together and working openly. That ’ s enough right there to start to build a compelling chronicle of what ’ s now happening to our planet, but we could do more. An Earth Witness site could also serve as a collection spot for all sorts of data about conditions around the planet picked up by environmental sensors that attach to your cell phone. Now, you don ’ t see these devices as add-ons for phones yet, but students and engineers around the world have attached atmospheric sensors to bicycles and handheld computers and cheap robots and the backs of pigeons — that being a project that ’ s actually underway right now at U. C. Irvine, using bird-mounted sensors as a way of measuring smog-forming pollution. It ’ s hardly a stretch to imagine putting the same thing on a phone carried by a person. Now, the idea of connecting a sensor to your phone is not new : phone-makers around the world offer phones that sniff for bad breath, or tell you to worry about too much sun exposure. Swedish firm Uppsala Biomedical, more seriously, makes a mobile phone add-on that can process blood tests in the field, uploading the data, **displaying** the results. Even the Lawrence Livermore National Labs have gotten into the act, designing a prototype phone that has radiation sensors to find dirty bombs. Now, there ’ s an enormous variety of tiny, inexpensive sensors on the market, and you can easily imagine someone putting together a phone that could measure temperature, CO2 or methane levels, the presence of some biotoxins — potentially, in a few years, maybe even H5N1 avian flu virus. You could see that some kind of system like this would actually be a really good fit with Larry Brilliant ’ s InSTEDD project. Now, all of this data could be tagged with geographic information and mashed up with online maps for easy viewing and analysis. And that ’ s worth noting in particular. The impact of open-access online maps over the last year or two has been simply phenomenal. Developers around the world have come up with an amazing variety of ways to layer useful data on top of the maps, from bus routes and crime statistics to the global progress of avian flu. Earth Witness would take this further, linking what you see with what thousands or millions of other people see around the world. It ’ s kind of exciting to think about what might be accomplished if something like this ever existed. We ’ d have a far better — far better knowledge of what ’ s happening on our planet environmentally than could be gathered with satellites and a handful of government sensor nets alone. It would be a collaborative, bottom-up approach to environmental awareness and protection, able to respond to emerging concerns in a smart mobs kind of way — and if you need greater sensor density, just have more people show up. And most important, you can ’ t ignore how important mobile phones are to global youth. This is a system that could put the next generation at the front lines of gathering environmental data. And as we work to figure out ways to mitigate the worst effects of climate disruption, every little bit of information matters. A system like Earth Witness would be a tool for all of us to participate in the improvement of our knowledge and, ultimately, the improvement of the planet itself. Now, as I suggested at the outset, there are thousands upon thousands of good ideas out there, so why have I spent the bulk of my time telling you about something that doesn ’ t exist? Because this is what tomorrow could look like : bottom-up, technology-enabled global collaboration to handle the biggest crisis our civilization has ever faced. We can save the planet, but we can ’ t do it alone — we need each other. Nobody ’ s going to fix the world for us, but working together, making use of technological innovations and human communities alike, we might just be able to fix it ourselves. We have at our fingertips a cornucopia of compelling models, powerful tools, and innovative ideas that can make a meaningful difference in our planet ’ s future. We don ’ t need to wait for a magic bullet to save us all ; we already have an arsenal of solutions just waiting to be used. There ’ s a staggering array of wonders out there, across diverse disciplines, all telling us the same thing : success can be ours if we ’ re willing to try. And as we say at Worldchanging, another world isn ’ t just possible ; another world is here. We just need to open our eyes. Thank you very much.

Text Sample 837: POS Dim 6 – 2008 – Score 4.00 – 2008/t000156.txt

I thought I would think about changing your perspective on the world a bit, and showing you some of the designs that we have in nature. And so, I have my first slide to talk about the dawning of the universe and what I call the cosmic scene investigation, that is, looking at the relics of creation and inferring what happened at the beginning, and then following it up and trying to understand it. And so one of the questions that I asked you is, when you look around, what do you see? Well, you see this space that ’ s created by designers and by the work of people, but what you actually see is a lot of material that was already here, being reshaped in a certain form. And so the question is : how did that material get here? How did it get into the form that it had before it got reshaped, and so forth? It ’ s a question of what ’ s the continuity? So one of the things I look at is, how did the universe begin and shape? What was the whole process in the creation and the evolution of the universe to getting to the point that we have these kinds of materials? So that ’ s sort of the part, and let me move on then and show you the Hubble Ultra Deep Field. If you look at this picture, what you will see is a lot of dark with some light objects in it. And everything but — four of these light objects are stars, and you can see them there — little pluses. This is a star, this is a star, everything else is a galaxy, OK? So there ' s a couple of thousand galaxies you can see easily with your eye in here. And when I look out at particularly this galaxy, which looks a lot like ours, I wonder if there ' s an art design college conference going on, and intelligent beings there are thinking about, you know, what designs they might do, and there might be a few cosmologists trying to understand where the universe itself came from, and there might even be some in that galaxy looking at ours trying to figure out what ' s going on over here. But there ' s a lot of other galaxies, and some are nearby, and they ' re kind of the color of the Sun, and some are further away and they ' re a little bluer, and so forth. But one of the questions is — this should be, to you — how come there are so many galaxies? Because this represents a very clean fraction of the sky. This is only 1, 000 galaxies. We think there ' s on the order — visible to the Hubble Space Telescope, if you had the time to scan it around — about 100 billion galaxies. Right? It ' s a very large number of galaxies. And that ' s roughly how many stars there are in our own galaxy. But when you look at some of these regions like this, you ' ll see more galaxies than stars, which is kind of a conundrum. So the question should come to your mind is, what kind of design, you know, what kind of creative process and what kind of design produced the world like that? And then I ' m going to show you it ' s actually a lot more complicated. We ' re going to try and follow it up. We have a tool that actually helps us out in this study, and that ' s the fact that the universe is so incredibly big that it ' s a time machine, in a certain sense. We draw this set of nested spheres cut away so you see it. Put the Earth at the center of the nested spheres, just because that ' s where we ' re making observations. And the moon is only two seconds away, so if you take a picture of the moon using ordinary light, it ' s the moon two seconds ago, and who cares. Two seconds is like the present. The Sun is eight minutes ago. That ' s not such a big deal, right, unless there ' s solar flares coming then you want to get out the way. You ' d like to have a little advance warning. But you get out to Jupiter and it ' s 40 minutes away. It ' s a problem. You hear about Mars, it ' s a problem communicating to Mars because it takes light long enough to go there. But if you look out to the nearest set of stars, to the nearest 40 or 50 stars, it ' s about 10 years. So if you take a picture of what ' s going on, it ' s 10 years ago. But you go and look to the center of the galaxy, it ' s thousands of years ago. If you look at Andromeda, which is the nearest big galaxy, and it ' s two million years ago. If you took a picture of the Earth two million years ago, there ' d be no evidence of humans at all, because we don ' t think there were humans yet. I mean, it just gives you the scale. With the Hubble Space Telescope, we ' re looking at hundreds of millions of years to a billion years. But if we were capable to come up with an idea of how to look even further — there ' s some things even further, and that was what I did in a lot of my work, was to develop the techniques — we could look out back to even earlier epochs before there were stars and before there were galaxies, back to when the universe was hot and dense and very different. And so that ' s the sort of sequence, and so I have a more artistic impression of this. There ' s the galaxy in the middle, which is the Milky Way, and around that are the Hubble — you know, nearby kind of galaxies, and there ' s a sphere that marks the different times. And behind that are some more modern galaxies. You see the whole big picture? The beginning of time is funny — it ' s on the outside, right? And then there ' s a part of the universe we can ' t see because it ' s so dense and so hot, light can ' t escape. It ' s like you can ' t see to the center of the Sun ; you have to use other techniques to know what ' s going on inside the Sun. But you can see the edge of the Sun, and the universe gets that way, and you can see that. And then you see this sort of model area around the outside, and that is the radiation coming from the Big Bang, which is actually incredibly uniform. The universe is almost a perfect sphere, but there are these very tiny variations which we show here in great exaggeration. And from them in the time sequence we ’ re going to have to go from these tiny variations to these irregular galaxies and first stars to these more advanced galaxies, and eventually the solar system, and so forth. So it ’ s a big design job, but we ’ ll see about how things are going on. So the way these measurements were done, there ’ s been a set of satellites, and this is where you get to see. So there was the COBE satellite, which was launched in 1989, and we discovered these variations. And then in 2000, the MAP satellite was launched — the WMAP — and it made somewhat better pictures. And later this year — this is the cool stealth version, the one that actually has some beautiful design features to it, and you should look — the Planck satellite will be launched, and it will make very high-resolution maps. And that will be the sequence of understanding the very beginning of the universe. And what we saw was, we saw these variations, and then they told us the secrets, both about the structure of space-time, and about the contents of the universe, and about how the universe started in its original motions. So we have this picture, which is quite a spectacular picture, and I ’ ll come back to the beginning, where we ’ re going to have some mysterious process that kicks the universe off at the beginning. And we go through a period of accelerating expansion, and the universe expands and cools until it gets to the point where it becomes transparent, then to the Dark Ages, and then the first stars turn on, and they evolve into galaxies, and then later they get to the more expansive galaxies. And somewhere around this period is when our solar system started forming. And it ’ s maturing up to the present time. And there ’ s some spectacular things. And this wastebasket part, that ’ s to represent what the structure of space-time itself is doing during this period. And so this is a pretty weird model, right? What kind of evidence do we have for that? So let me show you some of nature ’ s patterns that are the result of this. I always think of space-time as being the real substance of space, and the galaxies and the stars just like the foam on the ocean. It ’ s a marker of where the interesting waves are and whatever went on. So here is the Sloan Digital Sky Survey showing the location of a million galaxies. So there ’ s a dot on here for every galaxy. They go out and point a telescope at the sky, take a picture, identify what are stars and throw them away, look at the galaxies, estimate how far away they are, and plot them up. And just put radially they ’ re going out that way. And you see these structures, this thing we call the Great Wall, but there are voids and those kinds of stuff, and they kind of fade out because the telescope isn ’ t sensitive enough to do it. Now I ’ m going to show you this in 3D. What happens is, you take pictures as the Earth rotates, you get a fan across the sky. There are some places you can ’ t look because of our own galaxy, or because there are no telescopes available to do it. So the next picture shows you the three-dimensional version of this rotating around. Do you see the fan-like scans made across the sky? Remember, every spot on here is a galaxy, and you see the galaxies, you know, sort of in our neighborhood, and you sort of see the structure. And you see this thing we call the Great Wall, and you see the complicated structure, and you see these voids. There are places where there are no galaxies and there are places where there are thousands of galaxies clumped together, right. So there ’ s an interesting pattern, but we don ’ t have enough data here to actually see the pattern. We only have a million galaxies, right? So we ’ re keeping, like, a million **balls** in the air but, what ’ s going on? There ’ s another survey which is very similar to this, called the Two-degree Field of View Galaxy Redshift Survey. Now we ’ re going to fly through it at warp a million. And every time there ’ s a galaxy — at its location there ’ s a galaxy — and if we know anything about the galaxy, which we do, because there ’ s a redshift measurement and everything, you put in the type of galaxy and the color, so this is the real representation. And when you ’ re in the middle of the galaxies it ’ s hard to see the pattern ; it ’ s like being in the middle of life. It ’ s hard to see the pattern in the middle of the audience, it ’ s hard to see the pattern of this. So we ’ re going to go out and swing around and look back at this. And you ’ ll see, first, the structure of the survey, and then you ’ ll start seeing the structure of the galaxies that we see out there. So again, you can see the extension of this Great Wall of galaxies showing up here. But you can see the voids, you can see the complicated structure, and you say, well, how did this happen? Suppose you ’ re the cosmic designer. How are you going to put galaxies out there in a pattern like that? It ’ s not just throwing them out at random. There ’ s a more complicated process going on here. How are you going to end up doing that? And so now we ’ re in for some serious play. That is, we have to seriously play God, not just change people ’ s lives, but make the universe, right. So if that ’ s your responsibility, how are you going to do that? What ’ s the kind of technique? What ’ s the kind of thing you ’ re going to do? So I ’ m going to show you the results of a very large-scale simulation of what we think the universe might be like, using, essentially, some of the play principles and some of the design principles that, you know, humans have labored so hard to pick up, but apparently nature knew how to do at the beginning. And that is, you start out with very simple ingredients and some simple rules, but you have to have enough ingredients to make it complicated. And then you put in some randomness, some fluctuations and some randomness, and realize a whole bunch of different representations. So what I ’ m going to do is show you the distribution of matter as a function of scales. We ’ re going to zoom in, but this is a plot of what it is. And we had to add one more thing to make the universe come out right. It ’ s called dark matter. That is matter that doesn ’ t interact with light the **typical** way that ordinary matter does, the way the light ’ s shining on me or on the stage. It ’ s transparent to light, but in order for you to see it, we ’ re going to make it white. OK? So the stuff that ’ s in this picture that ’ s white, that is the dark matter. It should be called invisible matter, but the dark matter we ’ ve made visible. And the stuff that is in the yellow color, that is the ordinary kind of matter that ’ s turned into stars and galaxies. So I ’ ll show you the next **movie**. So this — we ’ re going to zoom in. Notice this pattern and pay attention to this pattern. We ’ re going to zoom in and zoom in. And you ’ ll see there are all these filaments and structures and voids. And when a number of filaments come together in a knot, that makes a supercluster of galaxies. This one we ’ re zooming in on is somewhere between 100, 000 and a million galaxies in that small region. So we live in the boonies. We don ’ t live in the center of the solar system, we don ’ t live in the center of the galaxy and our galaxy ’ s not in the center of the cluster. So we ’ re zooming in. This is a region which probably has more than 100, 000, on the order of a million galaxies in that region. We ’ re going to keep zooming in. OK. And so I forgot to tell you the scale. A parsec is 3. 26 light years. So a gigaparsec is three billion light years — that ’ s the scale. So it takes light three billion years to travel over that distance. Now we ’ re into a distance sort of between here and here. That ’ s the distance between us and Andromeda, right? These little specks that you ’ re seeing in here, they ’ re galaxies. Now we ’ re going to zoom back out, and you can see this structure that, when we get very far out, looks very regular, but it ’ s made up of a lot of irregular variations. So they ’ re simple building blocks. There ’ s a very simple fluid to begin with. It ’ s got dark matter, it ’ s got ordinary matter, it ’ s got photons and it ’ s got neutrinos, which don ’ t play much role in the later part of the universe. And it ’ s just a simple fluid and it, over time, develops into this complicated structure. And so you know when you first saw this picture, it didn ’ t mean quite so much to you. Here you ’ re looking across one percent of the volume of the visible universe and you ’ re seeing billions of galaxies, right, and nodes, but you realize they ’ re not even the main structure. There ’ s a framework, which is the dark matter, the invisible matter, that ’ s out there that ’ s actually holding it all together. So let ’ s fly through it, and you can see how much harder it is when you ’ re in the middle of something to figure this out. So here ’ s that same end result. You see a filament, you see the light is the invisible matter, and the yellow is the stars or the galaxies showing up. And we ’ re going to fly around, and we ’ ll fly around, and you ’ ll see occasionally a couple of filaments intersect, and you get a large cluster of galaxies. And then we ’ ll fly in to where the very large cluster is, and you can see what it looks like. And so from inside, it doesn ’ t look very complicated, right? It ’ s only when you look at it at a very large scale, and explore it and so forth, you realize it ’ s a very intricate, complicated kind of a design, right? And it ’ s grown up in some kind of way. So the question is, how hard would it be to assemble this, right? How big a contractor team would you need to put this universe together, right? That ’ s the issue, right? And so here we are. You see how the filament — you see how several filaments are coming together, therefore making this supercluster of galaxies. And you have to understand, this is not how it would actually look if you — first, you can ’ t travel this fast, everything would be distorted, but this is using simple rendering and graphic arts kind of stuff. This is how, if you took billions of years to go around, it might look to you, right? And if you could see invisible matter, too. And so the idea is, you know, how would you put together the universe in a very simple way? We ’ re going to start and realize that the entire visible universe, everything we can see in every direction with the Hubble Space Telescope plus our other instruments, was once in a region that was smaller than an atom. It started with tiny quantum mechanical fluctuations, but expanding at a tremendous rate. And those fluctuations were stretched to astronomical sizes, and those fluctuations eventually are the things we see in the cosmic microwave background. And then we needed some way to turn those fluctuations into galaxies and clusters of galaxies and make these kinds of structures go on. So I ’ m going to show you a smaller simulation. This simulation was run on 1, 000 processors for a month in order to make just this simple visible one. So I ’ m going to show you one that can be run on a desktop in two days in the next picture. So you start out with teeny fluctuations when the universe was at this point, now four times smaller, and so forth. And you start seeing these networks, this cosmic web of structure forming. And this is a simple one, because it doesn ’ t have the ordinary matter and it just has the dark matter in it. And you see how the dark matter lumps up, and the ordinary matter just trails along behind. So there it is. At the beginning it ’ s very uniform. The fluctuations are a part in 100, 000. There are a few peaks that are a part in 10, 000, and then over billions of years, gravity just pulls in. This is light over density, pulls the material around in. That pulls in more material and pulls in more material. But the distances on the universe are so large and the time scales are so large that it takes a long time for this to form. And it keeps forming until the universe is roughly about half the size it is now, in terms of its expansion. And at that point, the universe mysteriously starts accelerating its expansion and cuts off the formation of larger-scale structure. So we ’ re just seeing as large a scale structure as we can see, and then only things that have started forming already are going to form, and then from then on it ’ s going to go on. So we ’ re able to do the simulation, but this is two days on a desktop. We need, you know, 30 days on 1, 000 processors to do the kind of simulation that I showed you before. So we have an idea of how to play seriously, creating the universe by starting with essentially less than an eyedrop full of material, and we create everything we can see in any direction, right, from almost nothing — that is, something extremely tiny, extremely small — and it is almost perfect, except it has these tiny fluctuations at a part in 100, 000 level, which turn out to produce the interesting patterns and designs we see, that is, galaxies and stars and so forth. So we have a model, and we can calculate it, and we can use it to make designs of what we think the universe really looks like. And that design is sort of way beyond what our original imagination ever was. So this is what we started with 15 years ago, with the Cosmic Background Explorer — made the map on the upper right, which basically showed us that there were large-scale fluctuations, and actually fluctuations on several scales. You can kind of see that. Since then we ’ ve had WMAP, which just gives us higher angular resolution. We see the same large-scale structure, but we see additional small-scale structure. And on the bottom right is if the satellite had flipped upside down and mapped the Earth, what kind of a map we would have got of the Earth. You can see, well, you can, kind of pick out all the major continents, but that ’ s about it. But what we ’ re hoping when we get to Planck, we ’ ll have resolution about equivalent to the resolution you see of the Earth there, where you can really see the complicated pattern that exists on the Earth. And you can also tell, because of the sharp edges and the way things fit together, there are some non- linear processes. Geology has these effects, which is moving the plates around and so forth. You can see that just from the map alone. We want to get to the point in our maps of the early universe we can see whether there are any non- linear effects that are starting to move, to modify, and are giving us a hint about how space-time itself was actually created at the beginning moments. So that ’ s where we are today, and that ’ s what I wanted to give you a flavor of. Give you a different view about what the design and what everything else looks like. Thank you.

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You know, we ’ re going to do things a little differently. I ’ m not going to show you a **presentation**. I ’ m going to talk to you. And at the same time, we ’ re going to look at just images from a photo stream that is pretty close to live of things that â snapshots from Second Life. So hopefully this will be fascinating. You can â I can compete for your attention with the strange pictures that you see on screen that come from there. I thought I ’ d talk a little bit about some just big ideas about this, and then get John back out here so we can talk interactively a little bit more and think and ask questions. You know, I guess the first question is, why build a virtual world at all? And I think the answer to that is always going to be at least driven to a certain extent by the people initially crazy enough to start the project, you know. So I can give you a little bit of first background just on me and what moved me as a â really going back as far as a teenager and then an adult, to actually try and build this kind of thing. I was a very creative kid who read a lot, and got into electronics first, and then later, programming computers, when I was really young. I was just always trying to make things. I was just obsessed with taking things apart and building things, and just anything I could do with my hands or with wood or electronics or metal or anything else. And so, for example â and it ’ s a great Second Life thing â I had a bedroom. And every kid, you know, as a teenager, has got his bedroom he retreats to â but I wanted my door, I thought it would be cool if my door went up rather than opened, like on Star Trek. I thought it would be neat to do that. And so I got up in the ceiling and I cut through the ceiling joists, much to my parents ’ delight, and put the door, you know, being pulled up through the ceiling. I built â I put a garage-door opener up in the attic that would pull this door up. You can imagine the amount of time that it took me to do this to the house and the displeasure of my parents. The thing that was always striking to me was that we as people could have so many really amazing ideas about things we ’ d like to do, but are so often unable, in the real world, to actually do those things â to actually cobble together the materials and go through the actual execution phase of building something that you imagine from a design perspective. And so for me, I know that when the Internet came around and I was doing computer programming and just, you know, just generally trying to run my own little company and figure out what to do with the Internet and with computers, I was just immediately struck by how the ultimate thing that you would really want to do with the Internet and with computers would be to use the Internet and connected computers to simulate a world to sort of recreate the laws of physics and the rules of how things went together â the sort of â the idea of atoms and how to make things, and do that inside a computer so that we could all get in there and make stuff. And so for me that was the thing that was so enticing. I just wanted this place where you could build things. And so I think you see that in the genesis of what has happened with Second Life, and I think it ’ s important. I also think that more generally, the use of the Internet and technology as a kind of a space between us for creativity and design is a general trend. It is a â sort of a great human progress. Technology is just generally being used to allow us to create in as shared and social a way as possible. And I think that Second Life and virtual worlds more generally represent the best we can do to achieve that right now. You know, another way to look at that, and related to the content and, you know, thinking about space, is to connect sort of virtual worlds to space. I thought that might be a fun thing to talk about for a second. If you think about going into space, it ’ s a fascinating thing. So many **movies**, so many kids, we all sort of dream about exploring space. Now, why is that? Stop for a moment and ask, why that conceit? Why do we as people want to do that? I think there ’ s a couple of things. It ’ s what we see in the **movies** â you know, it ’ s this dream that we all share. One is that if you went into space you ’ d be able to begin again. In some sense, you would become someone else in that journey, because there wouldn ’ t be â you ’ d leave society and life as you know it, behind. And so inevitably, you would transform yourself â irreversibly, in all likelihood â as you began this exploration. And then the second thing is that there ’ s this tangible sense that if you travel far enough, you can find out there â oh, yeah â you have no idea what you ’ re going to find once you get there, into space. It ’ s going to be different than here. And in fact, it ’ s going to be so different than what we see here on earth that anything is going to be possible. So that ’ s kind of the idea â we as humans crave the idea of creating a new identity and going into a place where anything is possible. And I think that if you really sit and think about it, virtual worlds, and where we ’ re going with more and more computing technology, represent essentially the likely, really tactically possible version of space exploration. We are moved by the idea of virtual worlds because, like space, they allow us to reinvent ourselves and they contain anything and everything, and probably anything could happen there. You know, to give you a size idea about scale, you know, comparing space to Second Life, most people don ’ t realize, kind of â and then this is just like the Internet in the early ’ 90s. In fact, Second Life virtual worlds are a lot like the Internet in the early ’ 90s today : everybody ’ s very excited, there ’ s a lot of hype and excitement about one idea or the next from moment to moment, and then there ’ s despair and everybody thinks the whole thing ’ s not going to work. Everything that ’ s happening with Second Life and more broadly with virtual worlds, all happened in the early ’ 90s. We always play a game at the office where you can take any article and find the same article where you just replace the words”Second Life”with”Web,”and”virtual reality”with”Internet.”You can find exactly the same articles written about everything that people are observing. To give you an idea of scale, Second Life is about 20, 000 CPUs at this point. It ’ s about 20, 000 computers connected together in three facilities in the United States right now, that are simulating this virtual space. And the virtual space itself â there ’ s about 250, 000 people a day that are wandering around in there, so the kind of, active population is something like a smallish city. The space itself is about 10 times the size of San Francisco, and it ’ s about as densely built out. So it gives you an idea of scale. Now, it ’ s expanding very rapidly â about five percent a month or so right now, in terms of new servers being added. And so of course, radically unlike the real world, and like the Internet, the whole thing is expanding very, very quickly, and historically exponentially. So that sort of space exploration thing is matched up here by the amount of content that ’ s in there, and I think that amount is critical. It was critical with the virtual world that it be this space of truly infinite possibility. We ’ re very sensitive to that as humans. You know, you know when you see it. You know when you can do anything in a space and you know when you can ’ t. Second Life today is this 20, 000 machines, and it ’ s about 100 million or so user-created objects where, you know, an object would be something like this, possibly interactive. Tens of millions of them are thinking all the time ; they have code attached to them. So it ’ s a really large world already, in terms of the amount of stuff that ’ s there and that ’ s very important. If anybody plays, like, World of Warcraft, World of Warcraft comes on, like, four DVDs. Second Life, by comparison, has about 100 terabytes of user-created data, making it about 25, 000 times larger. So again, like the Internet compared to AOL, and the sort of chat rooms and content on AOL at the time, what ’ s happening here is something very different, because the sheer scale of what people can do when they ’ re enabled to do anything they want is pretty amazing. The last big thought is that it is almost certainly true that whatever this is going to evolve into is going to be bigger in total usage than the Web itself. And let me justify that with two statements. Generically, what we use the Web for is to organize, exchange, create and consume information. It ’ s kind of like Irene talking about Google being data-driven. I ’ d say I kind of think about the world as being information. Everything that we interact with, all the experiences that we have, is kind of us flowing through a sea of information and interacting with it in different ways. The Web puts information in the form of text and images. The topology, the geography of the Web is text-to-text links for the most part. That ’ s one way of organizing information, but there are two things about the way you access information in a virtual world that I think are the important ways that they ’ re very different and much better than what we ’ ve been able to do to date with the Web. The first is that, as I said, the â well, the first difference for virtual worlds is that information is presented to you in the virtual world using the most powerful iconic symbols that you can possibly use with human beings. So for example, C-H-A-I-R is the English word for that, but a picture of this is a universal symbol. Everybody knows what it means. There ’ s no need to translate it. It ’ s also more memorable if I show you that picture, and I show you C-H-A- I-R on a piece of paper. You can do tests that show that you ’ ll remember that I was talking about a chair a couple of days later a lot better. So when you organize information using the symbols of our memory, using the most common symbols that we ’ ve been immersed in all our lives, you maximally both excite, stimulate, are able to remember, transfer and manipulate data. And so virtual worlds are the best way for us to essentially organize and experience information. And I think that ’ s something that people have talked about for 20 years â you know, that 3D, that lifelike environments are really important in some magical way to us. But the second thing â and I think this one is less obvious â is that the experience of creating, consuming, exploring that information is in the virtual world implicitly and inherently social. You are always there with other people. And we as humans are social creatures and must, or are aided by, or enjoy more, the consumption of information in the presence of others. It ’ s essential to us. You can ’ t escape it. When you ’ re on Amazon. com and you ’ re looking for digital cameras or whatever, you ’ re on there right now, when you ’ re on the site, with like 5, 000 other people, but you can ’ t talk to them. You can ’ t just turn to the people that are browsing digital cameras on the same page as you, and ask them,”Hey, have you seen one of these before? Because I ’ m thinking about buying it.”That experience of like, shopping together, just as a simple example, is an example of how as social creatures we want to experience information in that way. So that second point, that we inherently experience information together or want to experience it together, is critical to essentially, kind of, this trend of where we ’ re going to use technology to connect us. And so I think, again, that it ’ s likely that in the next decade or so these virtual worlds are going to be the most common way as human beings that we kind of use the electronics of the Internet, if you will, to be together, to consume information. You know, mapping in India â that ’ s such a great example. Maybe the solution there involves talking to other people in real time. Asking for advice, rather than any possible way that you could just statically organize a map. So I think that ’ s another big point. I think that wherever this is all going, whether it ’ s Second Life or its descendants, or something broader that happens all around the world at a lot of different points â this is what we ’ re going to see the Internet used for, and total traffic and total unique users is going to invert, so that the Web and its bibliographic set of text and graphical information is going to become a tool or a part of that consumption pattern, but the pattern itself is going to happen mostly in this type of an environment. Big idea, but I think highly defensible. So let me stop there and bring John back, and maybe we can just have a longer conversation. Thank you. John. That ’ s great. John Hockenberry : Why is the creation, the impulse to create Second Life, not a utopian impulse? Like for example, in the 19th century, any number of works of literature that imagined alternative worlds were explicitly utopian. Philip Rosedale : I think that ’ s great. That ’ s such a deep question. Yeah. Is a virtual world likely to be a utopia, would be one way I ’ d say it. The answer is no, and I think the reason why is because the Web itself as a good example is profoundly bottoms-up. That idea of infinite possibility, that magic of anything can happen, only happens in an environment where you really know that there ’ s a fundamental freedom at the level of the individual actor, at the level of the Lego blocks, if you will, that make up the virtual world. You have to have that level of freedom, and so I ’ m often asked that, you know, is there a, kind of, utopian or, is there a utopian tendency to Second Life and things like it, that you would create a world that has a grand scheme to it? Those top-down schemes are alienating to just about everybody, even if you mean well when you build them. And what ’ s more, human society, when it ’ s controlled, when you set out a grand scheme of rules, a new way of people interacting, or a new way of laying out a city, or whatever, that stuff historically has never scaled much beyond, you know â I always laughingly say â the Mall of America, you know, which is like, the largest piece of centrally designed architecture that, you know, has been built.

JH : The Kremlin was pretty big. PR : The Kremlin, yeah. That ’ s true. The whole complex. JH : Give me a story of a tool you created at the beginning in Second Life that you were pretty sure people would want to use in the creation of their avatars or in communicating that people actually in practice said, no, I ’ m not interested in that at all, and name something that you didn ’ t come up with that almost immediately people began to demand. PR : I ’ m sure I can think of multiple examples of both of those. One of my favorites. I had this feature that I built into Second Life â I was really passionate about it. It was an ability to kind of walk up close to somebody and have a more private conversation, but it wasn ’ t instant messaging because you had to sort of befriend somebody. It was just this idea that you could kind of have a private chat. I just remember it was one of those examples of data- driven design. I thought it was such a good idea from my perspective, and it was just absolutely never used, and we ultimately â I think we ’ ve now turned it off, if I remember. We finally gave up, took it out of the code. But more generally, you know, one other example I think about this, which is great relative to the utopian idea. Second Life originally had 16 simulators. It now has 20, 000. So when it only had 16, it was only about as big as this college campus. And we had â we zoned it, you know : we put a nightclub, we put a disco where you could dance, and then we had a place where you could fight with guns if you wanted to, and we had another place that was like a boardwalk, kind of a Coney Island. And we laid out the zoning, but of course, people could build all around it however they wanted to. And what was so amazing right from the start was that the idea that we had put out in the zoning concept, basically, was instantly and thoroughly ignored, and like, two months into the whole thing, â which is really a small amount of time, even in Second Life time â I remember the users, the people who were then using Second Life, the residents came to me and said, we want to buy the disco â because I had built it â we want to buy that land and raze it and put houses on it. And I sold it to them â I mean, we transferred ownership and they had a big party and blew up the entire building. And I remember that that was just so telling, you know, that you didn ’ t know exactly what was going to happen. When you think about stuff that people have built that ’ s popular â JH : CBGB ’ s has to close eventually, you know. That ’ s the rule. PR : Exactly. And it â but it closed on day one, basically, in Internet time. You know, an example of something â pregnancy. You can have a baby in Second Life. This is done entirely using, kind of, the tools that are built into Second Life, so the innate concept of becoming pregnant and having a baby, of course â Second Life is, at the platform level, at the level of the company â at Linden Lab â Second Life has no game properties to it whatsoever. There is no attempt to structure the experience, to make it utopian in that sense that we put into it. So of course, we never would have put a mechanism for having babies or, you know, taking two avatars and merging them, or something. But people built the ability to have babies and care for babies as a purchasable experience that you can have in Second Life and so â I mean, that ’ s a pretty fascinating example of, you know, what goes on in the overall economy. And of course, the existence of an economy is another idea. I didn ’ t talk about it, but it ’ s a critical feature. When people are given the opportunity to create in the world, there ’ s really two things they want. One is fair ownership of the things they create. And then the second one is â if they feel like it, and they ’ re not going to do it in every case, but in many they are â they want to actually be able to sell that creation as a way of providing for their own livelihood. True on the Web â also true in Second Life. And so the existence of an economy is critical. JH : Questions for Philip Rosedale? Right here. JH : The observation is, Philip has been accused of looking like a character, an avatar, in

Second Life. Respond, and then we ’ ll get the rest of your question. PR : But I don ’ t look like my avatar. How many people here know what my avatar looks like? That ’ s probably not very many. JH : Are you ripping off somebody else ’ s avatar with that, sort of â PR : No, no. I didn ’ t. One of the other guys at work had a fantastic avatar â a female avatar â that I used to be once in a while. But my avatar is a guy wearing chaps. Spiky hair â spikier than this. Kind of orange hair. Handlebar mustache. Kind of a Village People sort of a character. So, very cool. JH : And your question? JH : The question is, there appears to be a lack of cultural fine-tuning in Second Life. It doesn ’ t seem to have its own culture, and the sort of differences that exist in the real world aren ’ t translated into the Second Life map. PR : Well, first of all, we ’ re very early, so this has only been going on for a few years. And so part of what we see is the same evolution of human behavior that you see in emerging societies. So a fair criticism â is what it is â of Second Life today is that it ’ s more like the Wild West than it is like Rome, from a cultural standpoint. That said, the evolution of, and the nuanced interaction that creates culture, is happening at 10 times the speed of the real world, and in an environment where, if you walk into a bar in Second Life, 65 percent of the people there are not in the United States, and in fact are speaking their, you know, **various** and different languages. In fact, one of the ways to make money in Second Life is to make really cool translators that you drag onto your body and they basically, kind of, pop up on your screen and allow you to use Google or Babel Fish or one of the other online text translators to on-the-fly translate spoken â I ’ m sorry â typed text between individuals. And so, the multicultural nature and the sort of cultural melting pot that ’ s happening inside Second Life is quite â I think, quite **remarkable** relative to what in real human terms in the real world we ’ ve ever been able to achieve. So, I think that culture will fine-tune, it will emerge, but we still have some years to wait while that happens, as you would naturally expect. JH : Other questions? Right here. JH : What ’ s your demographic? PR : So, the question is, what ’ s the demographic. So, the average age of a person in Second Life is 32, however, the use of Second Life increases dramatically as your physical age increases. So as you go from age 30 to age 60 â and there are many people in their sixties using Second Life â this is also not a sharp curve â it ’ s very, very distributed â usage goes up in terms of, like, hours per week by 40 percent as you go from age 30 to age 60 in real life, so there ’ s not â many people make the mistake of believing that Second Life is some kind of an online game. Actually it ’ s generally unappealing â I ’ m just speaking broadly and critically â it ’ s not very appealing to people that play online video games, because the graphics are not yet equivalent to â I mean, these are very nice pictures, but in general the graphics are not quite equivalent to the fine-tuned graphics that you see in a Grand Theft Auto 4. So average age : 32. I mentioned 65 percent of the users are not in the United States. The distribution amongst countries is extremely broad. There ’ s users from, you know, virtually every country in the world now in Second Life. The dominant ones are â if you take the UK and Europe, together they make up about 55 percent of the usage base in Second Life. In terms of psychographic â oh, men and women : men and women are almost equally matched in Second Life, so about 45 percent of the people online right now on Second Life are women. Women use Second Life, though, about 30 to 40 percent more, on an hours basis, than men do, meaning that more men sign up than women, and more women stay and use it than men. So that ’ s another demographic fact. In terms of psychographic, you know, the people in Second Life are remarkably dissimilar relative to what you might think, when you go in and talk to them and meet them, and I would, you know, challenge you to just do this and find out. But it ’ s not a bunch of programmers. It ’ s not easy to describe as a demographic. If I had to just sort of paint a broad picture, I ’ d say, remember the people who were really getting into eBay in the first few years of eBay? Maybe a little bit like that : in other words, people who are early adopters. They tend to be creative. They tend to be entrepreneurial. A lot of them â about 55, 000 people so far â are cash-flow positive : they ’ re making money from what â I mean, real-world money â from what they ’ re doing in Second Life, so it ’ s a very build â still a creative, building things, build-your-own-business type of an orientation. So, that ’ s it. JH : You describe yourself, Philip, as someone who was really creative when you were young and, you know, liked to make things. I mean, it ’ s not often that you hear somebody describe themselves as really creative. I suspect that ’ s possibly a euphemism for C student who spent a lot of time in his room? Is it possible? PR : I was a â there were times I was a C student. You know, it ’ s funny. When I got to college â I studied physics in college â and I got really â it was funny, because I was definitely a more antisocial kid. I read all the time. I was shy. I don ’ t seem like it now, but I was very shy. Moved around a bunch â had that experience too. So I did, kind of, I think, live in my own world, and obviously that helps, you know, engage your real interest in something. JH : So you ’ re on your fifth life at this point? PR : If you count, yeah, cities. So â but I did â and I didn ’ t do â I think I didn ’ t do as well in school as I could have. I think you ’ re right. I wasn ’ t, like, an obsessed â you know, get A ’ s kind of guy. I was going to say, I had a great social experience when I went to college that I hadn ’ t had before, a more fraternal experience, where I met six or seven other guys who I studied physics with, and I was very competitive with them, so then I started to get A ’ s. But you ’ re right : I wasn ’ t an A student. JH : Last question. Right here. JH : You want to paraphrase that? PR : Yeah, so let me restate that. So, you ’ re saying that in the pamphlet there ’ s a statement that we may come to prefer our digital selves to our real ones â our more malleable or manageable digital identities to our real identities â and that in fact, much of human life and human experience may move into the digital realm. And then that ’ s kind of a horrifying thought, of course. That ’ s a frightening change, frightening disruption. I guess, and you ’ re asking, what do I think about that? How do I â JH : What ’ s your response to the people who would say, that ’ s horrifying? PR : Well, I ’ d say a couple of things. One is, it ’ s disturbing like the Internet or electricity was. That is to say, it ’ s a big change, but it isn ’ t avoidable. So, no amount of backpedaling or intentional behavior or political behavior is going to keep these technology changes from connecting us together, because the basic motive that people have â to be creative and entrepreneurial â is going to drive energy into these virtual worlds in the same way that it has with the Web. So this change, I believe, is a huge disruptive change. Obviously, I ’ m the optimist and a big believer in what ’ s going on here, but I think that as â even a sober, you know, the most sober, disconnected thinker about this, looking at it from the side, has to conclude, based on the data, that with those kinds of economic forces at play, there is definitely going to be a sea change, and that change is going to be intensely disruptive relative to our concept of our very lives and being, and our identities, as well. I don ’ t think we can get away from those changes. I think generally, we were talking about this â I think that generally being present in a virtual world and being challenged by it, being â surviving there, having a good life there, so to speak, is a challenge because of the multiculturality of it, because of the languages, because of the entrepreneurial richness of it, the sort of flea market nature, if you will, of the virtual world today. It puts challenges on us to rise to. We must be better than ourselves, in many ways. We must learn things and, you know, be more tolerant, and be smarter and learn faster and be more creative, perhaps, than we are typically in our real lives. And I think that if that is true of virtual worlds, then these changes, though scary â and, I say, inevitable â are ultimately for the better, and therefore something that we should ride out. But I would say that â and many other authors and speakers about this, other than me, have said, you know, fasten your seat belts because the change is coming. There are going to be big changes. JH : Philip Rosedale, thank you very much.

Text Sample 839: POS Dim 6 – 2008 – Score 4.00 – 2008/t000133.txt

The"Dirty Jobs"crew and I were called to a little town in Colorado, called Craig. It ' s only a couple dozen square miles. It ' s in the Rockies. And the job in question was sheep rancher. My role on the show, for those of you who haven ' t seen it â it ' s pretty simple. I ' m an apprentice, and I work with the people who do the jobs in question. And my responsibilities are to simply try and keep up, and give an honest account of what it ' s like to be these people for one day in their life. The job in question : herding sheep. Great. We go to Craig and we check into a hotel, and I realize the next day that castration is going to be an absolute part of this work. Normally, I never do any research at all. But this is a touchy subject, and I work for the Discovery Channel, and we want to portray accurately whatever it is we do. And we certainly want to do it with a lot of respect for the animals. So I call the Humane Society and I say,"Look, I ' m going to be castrating some lambs. Can you tell me the deal?"And they ' re like,"Yeah, it ' s pretty straightforward."They use a band, basically, a rubber band, like this, only a little smaller. This one was actually around the playing cards I got yesterday â But it had a certain familiarity to it. And I said,"Well, what exactly is the process?"And they said,"The band is applied to the tail, tightly. And then another band is applied to the scrotum, tightly. Blood flow is slowly retarded ; a week later the parts in question fall off."Great â got it."OK, I call the SPCA to confirm this. They confirm it. I also call PETA just for fun, and they don ' t like it, but they confirm it. OK, that ' s basically how you do it. So the next day I go out. And I ' m given a horse and we go get the lambs and we take them to a pen that we built, and we go about the business of animal husbandry. Melanie is the wife of Albert. Albert is the shepherd in question. Melanie picks up the lamb, one hand on both legs on the right, likewise on the left. Lamb goes on the post, she opens it up. Alright. Great. Albert goes in, I follow Albert, the crew is around. I always watch the process done the first time before I try it. Being an apprentice, you know, you do that. Albert reaches in his pocket to pull out, you know, this black rubber band, but what comes out instead is a knife. And I ' m like,"Hmm, that ' s not rubber at all,"you know? And he kind of flicked it open in a way that caught the sun that was just coming over the Rockies, it was very â It was... it was impressive. In the space of about two seconds, Albert had the knife between the cartilage of the tail, right next to the butt of the lamb, and very quickly, the tail was gone and in the bucket that I was holding. A second later, with a big thumb and a well-calloused forefinger, he had the scrotum firmly in his grasp. And he pulled it toward him, like so, and he took the knife and he put it on the tip."Now, you think you know what ' s coming, Michael, You don ' t, OK?"He snips it, throws the tip over his shoulder, and then grabs the scrotum and pushes it upward, and then his head dips down, obscuring my view. But what I hear is a slurping sound, and a noise that sounds like Velcro being yanked off a sticky wall, and I am not even kidding. Can we roll the video? No, I ' m kidding, we don ' t â I thought it best to talk in pictures. I do something now I ' ve never, ever done on a”Dirty Jobs”shoot, ever. I say,”Time out. Stop.”You guys know the show, we use take one ; we don ’ t do take two. There ’ s no writing, there ’ s no scripting, there ’ s no nonsense. We don ’ t fool around, we don ’ t rehearse â we shoot what we get! I said,”Stop. This is nuts.”I mean â”This is crazy. We can ’ t do this.”And Albert ’ s like,”What?”And I ’ m like,”I don ’ t know what just happened, but there are testicles in this bucket, and that ’ s not how we do it.”He said”Well, that ’ s how we do it.”I said,”Why would you do it this way?”And before I even let him explain, I said,”I want to do it the right way, with the rubber bands.”And he says,”Like the Humane Society?”I said,”Yes, like the Humane Society. Let ’ s do something that doesn ’ t make the lamb squeal and bleed. We ’ re on in five continents, dude! We ’ re on twice a day on the Discovery â we can ’ t do this.”He says,”OK.”He goes to his box and pulls out a bag of these little rubber bands. Melanie picks up another lamb, puts it on the post, band goes on the tail, band goes on the scrotum. Lamb goes on the ground, lamb takes two steps, falls down, gets up, shakes a little, takes another couple steps, falls down. I ’ m like, this is not a good sign for this lamb, at all. Gets up, walks to the corner. It ’ s quivering, and it lies down and it ’ s in obvious distress. And I ’ m looking at the lamb and I say,”Albert, how long? When does he get up?”He ’ s like,”A day?”I said,”A day! How long does it take them to fall off?””A week.”Meanwhile, the lamb that he had just done his little procedure on is, you know, he ’ s just prancing around, bleeding stopped. He ’ s, you know, **nibbling** on some grass, frolicking. And I was just so blown away at how completely wrong I was, in that second. And I was reminded how utterly wrong I am, so much of the time. And I was especially reminded of what a ridiculously short straw I had that day, because now I had to do what Albert had just done, and there are like 100 of these lambs in the pen. And suddenly, this whole thing ’ s starting to feel like a German porno, and I ’ m like â Melanie picks up the lamb, puts it on the post, opens it up. Albert hands me the knife. I go in, tail comes off. I go in, I grab the scrotum, tip comes off. Albert instructs,”Push it way up there.”I do.”Push it further.”I do. The testicles emerge. They look like thumbs, coming right at you. And he says,”Bite ’ em. Just bite ’ em off.”And I heard him, I heard all the words â Like, how did I get here? How did â I mean â how did I get here? It ’ s just â it ’ s one of those moments where the brain goes off on its own, and suddenly, I ’ m standing there in the Rockies, and all I can think of is the Aristotelian definition of a tragedy. You know, Aristotle says a tragedy is that moment when the hero comes face to face with his true identity. And I ’ m like,”What is this jacked-up metaphor? I don ’ t like what I ’ m thinking right now.”And I can ’ t get this thought out of my head, and I can ’ t get that vision out of my sight, so I did what I had to do. I went in and I took them. I took them like this, and I yanked my head back. And I ’ m standing there with two testicles on my chin. And now I can ’ t get â I can ’ t shake the metaphor. I ’ m still in”Poetics,”in Aristotle, and I ’ m thinking â out of nowhere, two terms come crashing into my head, that I hadn ’ t heard since my classics professor in college drilled them there. And they are”anagnorisis”and”peripeteia.”Anagnorisis and peripeteia. Anagnorisis is the Greek word for discovery. Literally, the transition from ignorance to knowledge is anagnorisis. It ’ s what our network does ; it ’ s what”Dirty Jobs”is. And I ’ m up to my neck in anagnorises every single day. Great. The other word, peripeteia, that ’ s the moment in the great tragedies â Euripides and Sophocles. That ’ s the moment where Oedipus has his moment, where he suddenly realizes that hot chick he ’ s been sleeping with and having babies with is his mother. That ’ s peripety, or peripeteia. And this metaphor in my head â I ’ ve got anagnorisis and peripeteia on my chin â I ’ ve got to tell you, it ’ s such a great device, though. When you start to look for peripeteia, you find it everywhere. I mean, Bruce Willis in"The Sixth Sense,"right? Spends the whole **movie** trying to help the little kid who sees dead people, and then â boom! â"Oh, I ' m dead."Peripeteia. You know? It ' s crushing when the audience sees it the right way. Neo in"The Matrix,"you know?"Oh, I ' m living in a computer program. That ' s weird."These discoveries that lead to sudden realizations. And I ' ve been having them, over 200 dirty jobs, I have them all the time, but that one â that one drilled something home in a way that I just wasn ' t prepared for. And, as I stood there, looking at the happy lamb that I had just defiled â but it looked OK ; looking at that poor other little thing that I ' d done it the right way on, and I just was struck by â if I ' m wrong about that, and if I ' m wrong so often, in a literal way, what other peripatetic misconceptions might I be able to comment upon? Because, look â I ' m not a social anthropologist, but I have a friend who is. And I talk to him. And he says,"Hey Mike, look. I don ' t know if your brain is interested in this sort of thing or not, but do you realize you ' ve shot in every state? You ' ve worked in mining, you ' ve worked in fishing, you ' ve worked in steel, you ' ve worked in every major industry. You ' ve had your back shoulder to shoulder with these guys that our politicians are desperate to relate to every four years, right?"I can still see Hillary doing the shots of rye, **dribbling** down her chin, with the steel workers. I mean, these are the people that I work with every single day."And if you have something to say about their thoughts, collectively, it might be time to think about it. Because, dude, you know, four years."So, that ' s in my head, testicles are on my chin, thoughts are bouncing around. And, after that shoot,"Dirty Jobs"really didn ' t change, in terms of what the show is, but it changed for me, personally. And now, when I talk about the show, I no longer just tell the story you heard and 190 like it. I do, but I also start to talk about some of the other things I got wrong ; some of the other notions of work that I ' ve just been assuming are sacrosanct, and they ' re not. People with dirty jobs are happier than you think. As a group, they ' re the happiest people I know. And I don ' t want to start whistling"Look for the Union Label,"and all that happy-worker crap. I ' m just telling you that these are balanced people who do unthinkable work. Roadkill picker-uppers whistle while they work, I swear to God â I did it with them. They ' ve got this amazing sort of symmetry to their life. And I see it over and over and over again. So I started to wonder what would happen if we challenged some of these sacred cows? Follow your passion â we ' ve been talking about it here for the last 36 hours. Follow your passion â what could possibly be wrong with that? It ' s probably the worst advice I ever got. Follow your dreams and go broke, right? I mean, that ' s all I heard growing up. I didn ' t know what to do with my life, but I was told if you follow your passion, it ' s going to work out. I can give you 30 examples right now. Bob Combs, the pig farmer in Las Vegas who collects the uneaten scraps of food from the casinos and feeds them to his swine. Why? Because there ' s so much protein in the stuff we don ' t eat, his pigs grow at twice the normal speed, and he ' s one rich pig farmer. He ' s good for the environment, he spends his days doing this incredible service, and he smells like hell, but God bless him. He ' s making a great living. You ask him,"Did you follow your passion here?"and he ' d laugh at you. The guy ' s worth â he just got offered like 60 million dollars for his farm and turned it down, outside of Vegas. He didn ' t follow his passion. He stepped back and he watched where everybody was going, and he went the other way. And I hear that story over and over. Matt Freund, a dairy farmer in New Canaan, Connecticut, who woke up one day and realized the crap from his cows was worth more than their milk, if he could use it to make these biodegradable flowerpots. Now he ' s selling them to Walmart, right? Follow his passion? The guy ' s â come on. So I started to look at passion, I started to look at efficiency vs. effectiveness. As Tim talked about earlier, that ' s a huge distinction. I started to look at teamwork and determination. And basically, all those platitudes they call"successories"that hang with that schmaltzy art in boardrooms around the world right now, that stuff â it ' s suddenly all been turned on its head. Safety. Safety first is... Going back to OSHA and PETA and the Humane Society : What if OSHA got it wrong? I mean â this is heresy, what I ' m about to say â but what if it ' s really safety third? Right? No, I mean, really. What I mean to say is : I value my safety on these crazy jobs as much as the people that I ' m working with, but the ones who really get it done â they ' re not out there talking about safety first. They know that other things come first â the business of doing the work comes first, the business of getting it done. And I ' ll never forget, up in the Bering Sea, I was on a crab boat with the"Deadliest Catch"guys â which I also work on in the first season. We were about 100 miles off the coast of Russia : 50-foot seas, big waves, green water coming over the wheelhouse, right? Most hazardous environment I ' d ever seen, and I was back with a guy, lashing the pots down. So I ' m 40 feet off the deck, which is like looking down at the top of your shoe, you know, and it ' s doing this in the ocean. Unspeakably dangerous. I scamper down, I go into the wheelhouse and I say, with some level of incredulity,"Captain â OSHA?"And he says,"OSHA? Ocean."And he points out there. But in that moment, what he said next can ' t be repeated in the Lower 48. It can ' t be repeated on any factory floor or any construction site. But he looked at me and said,"Son,"â he ' s my age, by the way, he calls me"son,"I love that â he says,"Son, I ' m the captain of a crab boat. My responsibility is not to get you home alive. My responsibility is to get you home rich."You want to get home alive, that ' s on you."And for the rest of that day â safety first. I mean, I was like â So, the idea that we create this sense of complacency when all we do is talk about somebody else ' s responsibility as though it ' s our own, and vice versa. Anyhow, a whole lot of things. I could talk at length about the many little distinctions we made and the endless list of ways that I got it wrong. But what it all comes down to is this : I ' ve formed a theory, and I ' m going to share it now in my remaining 2 minutes and 30 seconds. It goes like this : we ' ve declared war on work, as a society â all of us. It ' s a civil war. It ' s a cold war, really. We didn ' t set out to do it and we didn ' t twist our mustache in some Machiavellian way, but we ' ve done it. And we ' ve waged this war on at least four fronts, certainly in Hollywood. The way we portray working people on TV â it ' s laughable. If there ' s a plumber, he ' s 300 pounds and he ' s got a giant butt crack, admit it. You see him all the time. That ' s what plumbers look like, right? We turn them into heroes, or we turn them into punch lines. That ' s what TV does. We try hard on"Dirty Jobs"not to do that, which is why I do the work and I don ' t cheat. But, we ' ve waged this war on Madison Avenue. So many of the commercials that come out there in the way of a message â what ' s really being said?"Your life would be better if you could work a little less, didn ' t have to work so hard, got home a little earlier, could retire a little faster, punch out a little sooner."It ' s all in there, over and over, again and again. Washington? I can ' t even begin to talk about the deals and policies in place that affect the bottom-line reality of the available jobs, because I don ' t really know ; I just know that that ' s a front in this war. And right here, guys â Silicon Valley. I mean â how many people have an iPhone on them right now? How many people have their BlackBerry? We ' re plugged in ; we ' re connected. I would never suggest for a second that something bad has come out of the tech revolution. Good grief, not to this crowd. But I would suggest that innovation without imitation is a complete waste of time. And nobody celebrates imitation the way"Dirty

Jobs”guys know it has to be done. Your iPhone without those people making the same **interface**, the same circuitry, the same board, over and over â all of that â that ’ s what makes it equally as possible as the genius that goes inside of it. So, we ’ ve got this new toolbox. You know? Our tools today don ’ t look like shovels and picks. They look like the stuff we walk around with. And so the collective effect of all of that has been this marginalization of lots and lots of jobs. And I realized, probably too late in this game â I hope not, because I don ’ t know if I can do 200 more of these things â but we ’ re going to do as many as we can. And to me, the most important thing to know and to really come face to face with, is that fact that I got it wrong about a lot of things, not just the testicles on my chin. I got a lot wrong. So, we ’ re thinking â by”we,”I mean me â that the thing to do is to talk about a PR campaign for work â manual labor, skilled labor. Somebody needs to be out there, talking about the forgotten benefits. I ’ m talking about grandfather stuff, the stuff a lot us probably grew up with but we ’ ve kind of â you know, kind of lost a little. Barack wants to create two and a half million jobs. The infrastructure is a huge deal. This war on work that I suppose exists, has casualties like any other war. The infrastructure is the first one, declining trade school enrollments are the second one. Every single year, fewer electricians, fewer carpenters, fewer plumbers, fewer welders, fewer pipe fitters, fewer steam fitters. The infrastructure jobs that everybody is talking about creating are those guys â the ones that have been in decline, over and over. Meanwhile, we ’ ve got two trillion dollars, at a minimum, according to the American Society of Civil Engineers, that we need to expend to even make a dent in the infrastructure, which is currently rated at a D minus. So, if I were running for anything â and I ’ m not â I would simply say that the jobs we hope to make and the jobs we hope to create aren ’ t going to stick unless they ’ re jobs that people want. And I know the point of this conference is to celebrate things that are near and dear to us, but I also know that clean and dirty aren ’ t opposites. They ’ re two sides of the same coin, just like innovation and imitation, like risk and responsibility, like peripeteia and anagnorisis, like that poor little lamb, who I hope isn ’ t quivering anymore, and like my time that ’ s gone. It ’ s been great talking to you. And get back to work, will you?

Text Sample 840: POS Dim 6 – 1998 – Score 3.00 – 1998/t000118.txt

Back in 1992, I started working for a company called Interval Research, which was just then being founded by David Lidell and Paul Allen as a for-profit research enterprise in Silicon Valley. I met with David to talk about what I might do in his company. I was just coming out of a failed virtual reality business and supporting myself by being on the speaking circuit and writing books — after twenty years or so in the computer game industry having ideas that people didn ’ t think they could sell. And David and I discovered that we had a question in common, that we really wanted the answer to, and that was,”Why hasn ’ t anybody built any computer games for little girls?”Why is that? It can ’ t just be a giant sexist conspiracy. These people aren ’ t that smart. There ’ s six billion dollars on the table. They would go for it if they could figure out how. So, what is the deal here? And as we thought about our goals — I should say that Interval is really a humanistic institution, in the classical sense that humanism, at its best, finds a way to combine clear-eyed empirical research with a set of core values that fundamentally love and respect people. The basic idea of humanism is the improvable quality of life ; that we can do good things, that there are things worth doing because they ’ re good things to do, and that clear-eyed empiricism can help us figure out how to do them. So, contrary to popular belief, there is not a conflict of interest between empiricism and values. And Interval Research is kind of the living example of how that can be true. So David and I decided to go find out, through the best research we could muster, what it would take to get a little girl to put her hands on a computer, to achieve the level of comfort and ease with the technology that little boys have because they play video games. We spent two and a half years conducting research ; we spent another year and a half in advance development. Then we formed a spin-off company. In the research phase of the project at Interval, we partnered with a company called Cheskin Research, and these people — Davis Masten and Christopher Ireland — changed my mind entirely about what market research was and what it could be. They taught me how to look and see, and they did not do the incredibly stupid thing of saying to a child,"Of all these things we already make you, which do you like best?"— which gives you zero answers that are usable. So, what we did for the first two and a half years was four things : We did an extensive review of the literature in related fields, like cognitive psychology, **spatial** cognition, gender studies, play theory, sociology, primatology. Thank you Frans de Waal, wherever you are, I love you and I ' d give anything to meet you. After we had done that with a pretty large team of people and discovered what we thought the salient issues were with girls and boys and playing — because, after all, that ' s really what this is about — we moved to the second phase of our work, where we interviewed adult experts in academia, some of the people who ' d produced the literature that we found relevant. Also, we did focus groups with people who were on the ground with kids every day, like playground supervisors. We talked to them, confirmed some hypotheses and identified some serious questions about gender difference and play. Then we did what I consider to be the heart of the work : interviewed 1, 100 children, boys and girls, ages seven to 12, all over the United States — except for Silicon Valley, Boston and Austin because we knew that their little families would have millions of computers in them and they wouldn ' t be a representative sample. And at the end of those **remarkable** conversations with kids and their best friends across the United States, after two years, we pulled together some survey data from another 10, 000 children, drew up a set up of what we thought were the key findings of our research, and spent another year transforming them into design heuristics, for designing computer-based products — and, in fact, any kind of products — for little girls, ages eight to 12. And we spent that time designing interactive prototypes for computer software and testing them with little girls. In 1996, in November, we formed the company Purple Moon which was a spinoff of Interval Research, and our chief investors were Interval Research, Vulcan Northwest, Institutional Venture Partners and Allen and Company. We launched a website on September 2nd that has now served 25 million pages, and has 42, 000 registered young girl users. They visit an average of one and a half times a day, spend an average of 35 minutes a visit, and look at 50 pages a visit. So we feel that we ' ve formed a successful online community with girls. We launched two titles in October —"Rockett ' s New School"— the first of a series of products — is about a character called Rockett beginning her first day of school in eighth grade at a brand new place, with a blank slate, which allows girls to play with the question of,"What will I be like when I ' m older?""What ' s it going to be like to be in high school or junior high school? Who are my friends?"; to exercise the love of social complexity and the narrative intelligence that drives most of their play behavior ; and which embeds in it values about noticing that we have lots of choices in our lives and the ways that we conduct ourselves. The other title that we launched is called"Secret Paths in the Forest,"which addresses the more fantasy-oriented, inner lives of girls. These two titles both showed up in the top 50 entertainment titles in PC Data — entertainment titles in PC Data in December, right up there with"John Madden Football,"which thrills me to death. So, we ' re real, and we ' ve touched several hundreds of thousands of little girls. We ' ve made half-a-billion impressions with marketing and PR for this brand, Purple Moon. Ninety-six percent of them, roughly, have been positive ; four percent of them have been"other."I want to talk about the other, because the politics of this enterprise, in a way, have been the most fascinating part of it, for me. There are really two kinds of negative reviews that we ' ve received. One kind of reviewer is a male gamer who thinks he knows what games ought to be, and won ' t show the product to little girls. The other kind of reviewer is a certain flavor of feminist who thinks they know what little girls ought to be. And so it ' s funny to me that these interesting, odd bedfellows have one thing in common : they don ' t listen to little girls. They haven ' t looked at children and they ' re certainly not demonstrating any love for them. I ' d like to play you some voices of little girls from the two-and-a-half years of research that we did — actually, some of the voices are more recent. And these voices will be accompanied by photographs that they took for us of their lives, of the things that they value and care about. These are pictures the girls themselves never saw, but they gave to us This is the stuff those reviewers don ' t know about and aren ' t listening to and this is the kind of research I recommend to those who want to do humanistic work. Girl 1 : Yeah, my character is usually a tomboy. Hers is more into boys. Girl 2 : Uh, yeah. Girl 1 : We have — in the very beginning of the whole game, always we do this : we each have a piece of paper ; we write down our name, our age — are we rich, very rich, not rich, poor, medium, wealthy, boyfriends, dogs, pets — what else — sisters, brothers, and all those. Girl 2 : Divorced — parents divorced, maybe. Girl 3 : This is my pretend [unclear] one. Girl 4 : We make a school newspaper on the computer. Girl 5 : For a girl ' s game also usually they ' ll have really pretty scenery with clouds and flowers. Girl 6 : Like, if you were a girl and you were really adventurous and a real big tomboy, you would think that girls ' games were kinda sissy. Girl 7 : I run track, I played soccer, I play basketball, and I love a lot of things to do. And sometimes I feel like I can ' t really enjoy myself unless it ' s like a vacation, like when I get Mondays and all those days off. Girl 8 : Well, sometimes there is a lot of stuff going on because I have music lessons and I ' m on swim team — all this different stuff that I have to do, and sometimes it gets overwhelming. Girl 9 : My friend Justine kinda took my friend Kelly, and now they ' re being mean to me. Girl 10 : Well, sometimes it gets annoying when your brothers and sisters, or brother or sister, when they copy you and you get your idea first and they take your idea and they do it themselves. Girl 11 : Because my older sister, she gets everything and, like, when I ask my mom for something, she ' ll say,"No"— all the time. But she gives my sister everything. Brenda Laurel : I want to show you, real quickly, just a minute of"Rockett ' s Tricky Decision,"which went gold two days ago. Let ' s hope it ' s really stable. This is the second day in Rockett ' s life. The reason I ' m showing you this is I ' m hoping that the scene that I ' m going to show you will look familiar and sound familiar, now that you ' ve listened to some girls ' voices. And you can see how we ' ve tried to incorporate the issues that matter to them in the game that we ' ve created. Miko : Hey Rockett! C ' mere! Rockett : Hi Miko! What ' s going on? Miko : Did you hear about Nakilia ' s big Halloween party this weekend? She asked me to make sure you knew about it. Nakilia invited Reuben too, but — Rockett : But what? Isn ' t he coming? Miko : I don ' t think so. I mean, I heard his band is playing at another party the same night. Rockett : Really? What other party? Girl : Max ' s party is going to be so cool, Whitney. He ' s invited all the best people.

BL : I ’ m going to fast-forward to the decision point because I know I don ’ t have a lot of time. After this awful event occurs, Rocket gets to decide how she feels about it. Rockett : Who ’ d want to show up at that party anyway? I could get invited to that party any day if I wanted to. Gee, I doubt I ’ ll make Max ’ s best people list. BL : OK, so we ’ re going to emotionally navigate. If we were playing the game, that ’ s what we ’ d do. If at any time during the game we want to learn more about the characters, we can go into this hidden hallway, and I ’ ll quickly just show you the **interface**. We can, for example, go find Miko ’ s locker and get some more information about her. Oops, I turned the wrong way. But you get the general idea of the product. I wanted to show you the ways, innocuous as they seem, in which we ’ re incorporating what we ’ ve learned about girls — their desires to experience greater emotional flexibility, and to play around with the social complexity of their lives. I want to make the point that what we ’ re giving girls, I think, through this effort, is a kind of validation, a sense of being seen. And a sense of the choices that are available in their lives. We love them. We see them. We ’ re not trying to tell them who they ought to be. But we ’ re really, really happy about who they are. It turns out they ’ re really great. I want to close by showing you a videotape that ’ s a version of a future game in the Rockett series that our graphic artists and design people put together, that we feel would please that four percent of reviewers.”Rockett 28!”Rockett : It ’ s like I ’ m just waking up, you know? BL : Thanks.

Text Sample 841: POS Dim 6 – 1998 – Score 2.00 – 1998/t000134.txt

’ Theme and variations ’ is one of those forms that require a certain kind of intellectual activity, because you are always comparing the variation with the theme that you hold in your mind. You might say that the theme is nature and everything that follows is a variation on the subject. I was asked, I guess about six years ago, to do a series of paintings that in some way would celebrate the birth of Piero della Francesca. And it was very difficult for me to imagine how to paint pictures that were based on Piero until I realized that I could look at Piero as nature — that I would have the same attitude towards looking at Piero della Francesca as I would if I were looking out a window at a tree. And that was enormously liberating to me. Perhaps it ’ s not a very insightful observation, but that really started me on a path to be able to do a kind of theme and variations based on a work by Piero, in this case that **remarkable** painting that ’ s in the Uffizi,”The Duke of Montefeltro,”who faces his consort, Battista. Once I realized that I could take some liberties with the subject, I did the following series of drawings. That ’ s the real Piero della Francesca — one of the greatest portraits in human history. And these, I ’ ll just show these without comment. It ’ s just a series of variations on the head of the Duke of Montefeltro, who ’ s a great, great figure in the Renaissance, and probably the basis for Machiavelli ’ s”The Prince.”He apparently lost an eye in battle, which is why he is always shown in profile. And this is Battista. And then I decided I could move them around a little bit — so that for the first time in history, they ’ re facing the same direction. Whoops! Passed each other. And then a visitor from another painting by Piero, this is from”The Resurrection of Christ”— as though the cast had just gotten of the set to have a chat. And now, four large panels : this is upper left ; upper right ; lower left ; lower right. Incidentally, I ’ ve never understood the conflict between abstraction and naturalism. Since all paintings are inherently abstract to begin with there doesn ’ t seem to be an argument there. On another subject — — I was driving in the country one day with my wife, and I saw this sign, and I said,”That is a fabulous piece of design.”And she said,”What are you talking about?”I said,”Well, it ’ s so persuasive, because the purpose of that sign is to get you into the garage, and since most people are so suspicious of garages, and know that they ’ re going to be ripped off, they use the word ’ reliable. ’ But everybody says they ’ re reliable. But, reliable Dutchman”— —”Fantastic!”Because as soon as you hear the word Dutchman — which is an archaic word, nobody calls Dutch people”Dutchmen”anymore — but as soon as you hear Dutchman, you get this picture of the kid with his finger in the dike, preventing the thing from falling and flooding Holland, and so on. And so the entire issue is detoxified by the use of”Dutchman.”Now, if you think I ’ m exaggerating at all in this, all you have to do is substitute something else, like”Indonesian.”Or even”French.”Now,”Swiss”works, but you know it ’ s going to cost a lot of money. I ’ m going to take you quickly through the actual process of doing a poster. I do a lot of work for the School of Visual Arts, where I teach, and the director of this school — a **remarkable** man named Silas Rhodes — often gives you a piece of text and he says,”Do something with this.”And so he did. And this was the text —”In words as fashion the same rule will hold / Alike fantastic if too new or old / Be not the first by whom the new are tried / Nor yet the last to lay the old aside.”I could make nothing of that. And I really struggled with this one. And the first thing I did, which was sort of in the absence of another idea, was say I ’ ll sort of write it out and make some words big, and I ’ ll have some kind of design on the back somehow, and I was hoping — as one often does — to stumble into something. So I took another crack at it — you ’ ve got to keep it moving — and I Xeroxed some words on pieces of colored paper and I pasted them down on an ugly board. I thought that something would come out of it, like”Words rule fantastic new old first last Pope”because it ’ s by Alexander Pope, but I sort of made a mess out of it, and then I thought I ’ d repeat it in some way so it was legible. So, it was going nowhere. Sometimes, in the middle of a resistant problem, I write down things that I know about it. But you can see the beginning of an idea there, because you can see the word”new”emerging from the”old.”That ’ s what happens. There ’ s a relationship between the old and the new ; the new emerges from the context of the old. And then I did some variations of it, but it still wasn ’ t coalescing graphically at all. I had this other version which had something interesting about it in terms of being able to put it together in your mind from clues. The W was clearly a W, the N was clearly an N, even though they were very fragmentary and there wasn ’ t a lot of information in it. Then I got the words”new”and”old”and now I had regressed back to a point where there seemed to be no return. I was really desperate at this point. And so, I do something I ’ m truthfully ashamed of, which is that I took two drawings I had made for another purpose and I put them together. It says”dreams”at the top. And I was going to do a thing, I say,”Well, change the copy. Let it say something about dreams, and come to SVA and you ’ ll sort of fulfill your dreams.”But, to my credit, I was so embarrassed about doing that that I never submitted this sketch. And, finally, I arrived at the following solution. Now, it doesn ’ t look terribly interesting, but it does have something that distinguishes it from a lot of other posters. For one thing, it transgresses the idea of what a poster ’ s supposed to be, which is to be understood and seen immediately, and not explained. I remember hearing all of you in the graphic arts —”If you have to explain it, it ain ’ t working.”And one day I woke up and I said,”Well, suppose that ’ s not true?”So here ’ s what it says in my explanation at the bottom left. It says,”Thoughts : This poem is impossible. Silas usually has a better touch with his choice of quotations. This one generates no imagery at all.”I am now exposing myself to my audience, right? Which is something you never want to do professionally.”Maybe the words can make the image without anything else happening. What ’ s the heart of this poem? Don ’ t be trendy if you want to be serious. Is doing the poster this way trendy in itself? I guess one could reduce the idea further by suggesting that the new emerges behind and through the old, like this.”And then I show you a little drawing — you see, you remember that old thing I discarded? Well, I found a way to use it. So, there ’ s that little alternative over there, and I say,”Not bad,”— criticizing myself —”but more didactic than visual. Maybe what wants to be said is that old and the new are locked in a dialectical embrace, a kind of dance where each defines the other.”And then more self-questioning —”Am I being simple-minded? Is this the kind of simple that looks obvious, or the kind that looks profound? There ’ s a significant difference. This could be embarrassing. Actually, I realize fear of embarrassment drives me as much as any ambition. Do you think this sort of thing could really attract a student to the school?”Well, I think there are two fresh things here — two fresh things. One is the sort of willingness to expose myself to a critical audience, and not to suggest that I am confident about what I ’ m doing. And as you know, you have to have a front. I mean — you ’ ve got to be confident ; if you don ’ t believe in your work, who else is going to believe in it? So that ’ s one thing, to introduce the idea of doubt into graphics. That can be a big contribution. The other thing is to actually give you two solutions for the price of one ; you get the big one and if you don ’ t like that, how about the little one? And that too is a relatively new idea. And here ’ s just a series of experiments where I ask the question of — does a poster have to be square? Now, this is a little illusion. That poster is not folded. It ’ s not folded, that ’ s a photograph and it ’ s cut on the **diagonal**. Same cheap trick in the upper left- hand corner. And here, a very peculiar poster because, simply because of using the isometric perspective in the computer, it won ’ t sit still in the space. At times, it seems to be wider at the back than the front, and then it shifts. And if you sit here long enough, it ’ ll float off the page into the audience. But, we don ’ t have time. And then an experiment — a little bit about the nature of perspective, where the outside shape is determined by the peculiarity of perspective, but the shape of the bottle — which is identical to the outside shape — is seen frontally. And another piece for the art directors ’ club is”Anna Rees”casting long shadows. This is another poster from the School of Visual Arts. There were 10 artists invited to participate in it, and it was one of those things where it was extremely competitive and I didn ’ t want to be embarrassed, so I worked very hard on this. The idea was — and it was a brilliant idea — was to have 10 posters distributed throughout the city ’ s subway system so every time you got on the subway you ’ d be passing a different poster, all of which had a different idea of what art is. But I was absolutely stuck on the idea of”art is”and trying to determine what art was. But then I gave up and I said,”Well, art is whatever.”And as soon as I said that, I discovered that the word”hat”was hidden in the word”whatever,”and that led me to the inevitable conclusion. But then again, it ’ s on my list of didactic posters. My intent is to have a literary accompaniment that explains the poster, in case you don ’ t get it. Now this says,”Note to the viewer : I thought I might use a visual cliche of our time — Magritte ’ s everyman — to express the idea that art is mystery, continuity and history. I ’ m also convinced that, in an age of computer manipulation, surrealism has become banal, a shadow of its former self. The phrase ’ Art is whatever ’ expresses the current inclusiveness that surrounds art-making — a sort of ’ it ain ’ t what you do, it ’ s the way that you do it ’ notion. The shadow of Magritte falls across the central part of the poster a poetic event that occurs as the shadow man isolates the word ’ hat, ’ hidden in the word ’ whatever. ’ The four hats shown in the poster suggests how art might be defined : as a thing itself, the worth of the thing, the shadow of the thing, and the shape of the thing. Whatever.”OK. And the one that I did not submit, which I still like, I wanted to use the same phrase. There were wonderful experiments by Bruno Munari on letterforms some years ago — sort of, see how far you could go and still be able to read them. And that idea stuck in my head. But then I took the pieces that I had taken off and put them at the bottom. And, of course those are the remains, and they ’ re so labeled. But what really happens is that you read it as :”Art is whatever remains.”Thank you.

Text Sample 842: POS Dim 6 – 1998 – Score 2.00 – 1998/t000200.txt

You hear that this is the era of environment â or biology, or information technology... Well, it ’ s the era of a lot of different things that we ’ re in right now. But one thing for sure : it ’ s the era of change. There ’ s more change going on than ever has occurred in the history of human life on earth. And you all sort of know it, but it ’ s hard to get it so that you really understand it. And I ’ ve tried to put together something that ’ s a good start for this. I ’ ve tried to show in this â though the color doesn ’ t come out â that what I ’ m concerned with is the little 50-year time bubble that you are in. You tend to be interested in a generation past, a generation future â your parents, your kids, things you can change over the next few decades â and this 50-year time bubble you kind of move along in. And in that 50 years, if you look at the population curve, you find the population of humans on the earth more than doubles and we ’ re up three-and-a-half times since I was born. When you have a new baby, by the time that kid gets out of high school more people will be added than existed on earth when I was born. This is unprecedented, and it ’ s big. Where it goes in the future is questioned. So that ’ s the human part. Now, the human part related to animals : look at the left side of that. What I call the human portion â humans and their livestock and pets â versus the natural portion â all the other wild animals and just â these are vertebrates and all the birds, etc., in the land and air, not in the water. How does it balance? Certainly, 10, 000 years ago, the civilization ’ s beginning, the human portion was less than one tenth of one percent. Let ’ s look at it now. You follow this curve and you see the whiter spot in the middle â that ’ s your 50- year time bubble. Humans, livestock and pets are now 97 percent of that integrated total mass on earth and all wild nature is three percent. We have won. The next generation doesn ’ t even have to worry about this game â it is over. And the biggest problem came in the last 25 years : it went from 25 percent up to that 97 percent. And this really is a sobering picture upon realizing that we, humans, are in charge of life on earth ; we ’ re like the capricious Gods of old Greek myths, kind of playing with life â and not a great deal of wisdom injected into it. Now, the third curve is information technology. This is Moore ’ s Law plotted here, which relates to density of information, but it has been pretty good for showing a lot of other things about information technology â computers, their use, Internet, etc. And what ’ s important is it just goes straight up through the top of the curve, and has no real limits to it. Now try and contrast these. This is the size of the earth going through that same â â frame. And to make it really clear, I ’ ve put all four on one graph. There ’ s no need to see the little detailed words on it. That first one is humans-versus-nature ; we ’ ve won, there ’ s no more gain. Human population. And so if you ’ re looking for growth industries to get into, that ’ s not a good one â protecting natural creatures. Human population is going up ; it ’ s going to continue for quite a while. Good business in obstetricians, morticians, and farming, housing, etc. â they all deal with human bodies, which require being fed, transported, housed and so on. And the information technology, which connects to our brains, has no limit â now, that is a wonderful field to be in. You ’ re looking for growth opportunity? It ’ s just going up through the roof. And then, the size of the Earth. Somehow making these all compatible with the Earth looks like a pretty bad industry to be involved with. So, that ’ s the stage out of all this. I find, for reasons I don ’ t understand, I really do have a goal. And the goal is that the world be desirable and sustainable when my kids reach my age â and I think that ’ s â in other words, the next generation. I think that ’ s a goal that we probably all share. I think it ’ s a hopeless goal. Technologically, it ’ s achievable ; economically, it ’ s achievable ; politically, it means sort of the habits, institutions of people â it ’ s impossible. The institutions of the past with all their inertia are just irrelevant for the future, except they ’ re there and we have to deal with them. I spend about 15 percent of my time trying to save the world, the other 85 percent, the usual â and whatever else we devote ourselves to. And in that 15 percent, the main focus is on human mind, thinking skills, somehow trying to unleash kids from the straightjacket of school, which is putting information and dogma into them, get them so they really think, ask tough questions, argue about serious subjects, don ’ t believe everything that ’ s in the book, think broadly or creative. They can be. Our school systems are very flawed and do not reward you for the things that are important in life or for the survival of civilization ; they reward you for a lot of learning and sopping up stuff. We can ’ t go into that today because there isn ’ t time â it ’ s a broad subject. One thing for sure, in the future there is an essential feature â necessary, but not sufficient â which is doing more with less. We ’ ve got to be doing things with more efficiency using less energy, less material. Your great-great grandparents got by on muscle power, and yet we all think there ’ s this huge power that ’ s essential for our lifestyle. And with all the wonderful technology we have we can do things that are much more efficient : conserve, recycle, etc. Let me just rush very quickly through things that we ’ ve done. Human-powered airplane â Gossamer Condor sort of started me in this direction in 1976 and 77, winning the Kremer prize in aviation history, followed by the Albatross. And we began making **various** odd planes and creatures. Here ’ s a giant flying replica of a pterosaur that has no tail. Trying to have it fly straight is like trying to shoot an arrow with the feathered end forward. It was a tough job, and boy it made me have a lot of respect for nature. This was the full size of the original creature. We did things on land, in the air, on water â vehicles of all different kinds, usually with some electronics or electric power systems in them. I find they ’ re all the same, whether its land, air or water. I ’ ll be focusing on the air here. This is a solar-powered airplane â 165 miles carrying a person from France to England as a symbol that solar power is going to be an important part of our future. Then we did the solar car for General Motors â the Sunracer â that won the race in Australia. We got a lot of people thinking about electric cars, what you could do with them. A few years later, when we suggested to GM that now is the time and we could do a thing called the Impact, they sponsored it, and here ’ s the Impact that we developed with them on their programs. This is the demonstrator. And they put huge effort into turning it into a commercial product. With that preamble, let ’ s show the first two-minute videotape, which shows a little airplane for surveillance and moving to a giant airplane. Narrator : A tiny airplane, the AV Pointer serves for surveillance â in effect, a pair of roving eyeglasses. A cutting-edge example of where miniaturization can lead if the operator is remote from the vehicle. It is convenient to carry, assemble and launch by hand. Battery-powered, it is silent and rarely noticed. It sends high-resolution video pictures back to the operator. With onboard GPS, it can navigate autonomously, and it is rugged enough to self-land without damage. The modern sailplane is superbly efficient. Some can glide as flat as 60 feet forward for every foot of descent. They are powered only by the energy they can extract from the atmosphere â an atmosphere nature stirs up by solar energy. Humans and soaring birds have found nature to be generous in providing replenishable energy. Sailplanes have flown over 1, 000 miles, and the altitude record is over 50, 000 feet. The Solar Challenger was made to serve as a symbol that photovoltaic cells can produce real power and will be part of the world ’ s energy future. In 1981, it flew 163 miles from Paris to England, solely on the power of sunbeams, and established a basis for the Pathfinder. The message from all these vehicles is that ideas and technology can be harnessed to produce **remarkable** gains in doing more with less â gains that can help us attain a desirable balance between technology and nature. The stakes are high as we speed toward a challenging future. Buckminster Fuller said it clearly :”there are no passengers on spaceship Earth, only crew. We, the crew, can and must do more with less â much less.”Paul MacCready : If we could have the second video, the one-minute, put in as quickly as you can, which â this will show the Pathfinder airplane in some flights this past year in Hawaii, and will show a sequence of some of the beauty behind it after it had just flown to 71, 530 feet â higher than any propeller airplane has ever flown. It ’ s amazing : just on the puny power of the sun â by having a super lightweight plane, you ’ re able to get it up there. It ’ s part of a long-term program NASA sponsored. And we worked very closely with the whole thing being a team effort, and with wonderful results like that flight. And we ’ re working on a bigger plane â 220-foot span â and an intermediate-size, one with a regenerative fuel cell that can store excess energy during the day, feed it back at night, and stay up 65, 000 feet for months at a time. Ray Morgan ’ s voice will come in here. There he ’ s the project manager. Anything they do is certainly a team effort. He ran this program. Here ’ s... some things he showed as a celebration at the very end. Ray Morgan : We ’ d just ended a seven-month deployment of Hawaii. For those who live on the mainland, it was tough being away from home. The friendly support, the quiet confidence, congenial hospitality shown by our Hawaiian and military hosts â this is starting â made the experience enjoyable and unforgettable. PM : We have real-time IR scans going out through the Internet while the plane is flying. And it ’ s exploring without polluting the stratosphere. That ’ s its goal : the stratosphere, the blanket that really controls the radiation of the earth and permits life on earth to be the success that it is â probing that is very important. And also we consider it as a sort of poor man ’ s stationary satellite, because it can stay right overhead for months at a time, 2, 000 times closer than the real GFC synchronous satellite. We couldn ’ t bring one here to fly it and show you. But now let ’ s look at the other end. In the video you saw that nine-pound or eight-pound Pointer airplane surveillance drone that Keenan has developed and just done a **remarkable** job. Where some have servos that have gotten down to, oh, 18 or 25 grams, his weigh one-third of a gram. And what he ’ s going to bring out here is a surveillance drone that weighs about 2 ounces â that includes the video camera, the batteries that run it, the telemetry, the receiver and so on. And we ’ ll fly it, we hope, with the same success that we had last night when we did the practice. So Matt Keenan, just any time you ’ re â all right â ready to let her go. But first, we ’ re going to make sure that it ’ s appearing on the screen, so you see what it sees. You can imagine yourself being a mouse or fly inside of it, looking out of its camera. Matt Keenan : It ’ s switched on. PM : But now we ’ re trying to get the video. There we go.

MK : Can you bring up the house lights? PM : Yeah, the house lights and we ’ ll see you all better and be able to fly the plane better. MK : All right, we ’ ll try to do a few laps around and bring it back in. Here we go. PM : The video worked right for the first few and I don ’ t know why it â there it goes. Oh, that was only a minute, but I think you ’ d be safe to have that near the end of the flight, perhaps. We get to do the classic. All right. If this hits you, it will not hurt you. OK. Thank you very much. Thank you. But now, as they say in infomercials, we have something much better for you, which we ’ re working on : planes that are only six inches â 15 centimeters â in size. And Matt ’ s plane was on the cover of Popular Science last month, showing what this can lead to. And in a while, something this size will have GPS and a video camera in it. We ’ ve had one of these fly nine miles through the air at 35 miles an hour with just a little battery in it. But there ’ s a lot of technology going. There are just milestones along the way of some **remarkable** things. This one doesn ’ t have the video in it, but you get a little feel from what it can do. OK, here we go. MK : Sorry. OK. PM : If you can pass it down when you ’ re done. Yeah, I think â I lost a little orientation ; I looked up into this light. It hit the building. And the building was poorly placed, actually. But you ’ re beginning to see what can be done. We ’ re working on projects now â even wing-flapping things the size of hawk moths â DARPA contracts, working with Caltech, UCLA. Where all this leads, I don ’ t know. Is it practical? I don ’ t know. But like any basic research, when you ’ re really forced to do things that are way beyond existing technology, you can get there with micro-technology, nanotechnology. You can do amazing things when you realize what nature has been doing all along. As you get to these small scales, you realize we have a lot to learn from nature â not with 747s â but when you get down to the nature ’ s realm, nature has 200 million years of experience. It never makes a mistake. Because if you make a mistake, you don ’ t leave any progeny. We should have nothing but success stories from nature, for you or for birds, and we ’ re learning a lot from its fascinating subjects. In concluding, I want to get back to the big picture and I have just two final slides to try and put it in perspective. The first I ’ ll just read. At last, I put in three sentences and had it say what I wanted. Over billions of years on a unique sphere, chance has painted a thin covering of life â complex, improbable, wonderful and fragile. Suddenly, we humans â a recently arrived species, no longer subject to the checks and balances inherent in nature â have grown in population, technology and intelligence to a position of terrible power. We now wield the paintbrush. And that ’ s serious : we ’ re not very bright. We ’ re short on wisdom ; we ’ re high on technology. Where ’ s it going to lead? Well, inspired by the sentences, I decided to wield the paintbrush. Every 25 years I do a picture. Here ’ s the one â tries to show that the world isn ’ t getting any bigger. Sort of a timeline, very non-linear scale, nature rates and trilobites and dinosaurs, and eventually we saw some humans with caves... Birds were flying overhead, after pterosaurs. And then we get to the civilization above the little TV set with a gun on it. Then traffic jams, and power systems, and some dots for digital. Where it ’ s going to lead â I have no idea. And so I just put robotic and natural cockroaches out there, but you can fill in whatever you want. This is not a forecast. This is a warning, and we have to think seriously about it. And that time when this is happening is not 100 years or 500 years. Things are going on this decade, next decade ; it ’ s a very short time that we have to decide what we are going to do. And if we can get some agreement on where we want the world to be â desirable, sustainable when your kids reach your age â I think we actually can reach it. Now, I said this was a warning, not a forecast. That was before â I painted this before we started in on making robotic versions of hawk moths and cockroaches, and now I ’ m beginning to wonder seriously â was this more of a forecast than I wanted? I personally think the surviving intelligent life form on earth is not going to be carbon-based ; it ’ s going to be silicon-based. And so where it all goes, I don ’ t know. The one final bit of sparkle we ’ ll put in at the very end here is an utterly impractical flight vehicle, which is a little ornithopter wing-flapping device that â rubber-band powered â that we ’ ll show you. MK : 32 gram. Sorry, one gram. PM : Last night we gave it a few too many turns and it tried to bash the roof out also. It ’ s about a gram. The tube there ’ s hollow, about paper-thin. And if this lands on you, I assure you it will not hurt you. But if you reach out to grab it or hold it, you will destroy it. So, be gentle, just act like a wooden Indian or something. And when it comes down â and we ’ ll see how it goes. We consider this to be sort of the spirit of TED. And you wonder, is it practical? And it turns out if I had not been â Unfortunately, we have some light bulb changes. We can probably get it down, but it ’ s possible it ’ s gone up to a greater destiny up there â â than it ever had. And I wanted to make â just â But I want to make just two points. One is, you think it ’ s frivolous ; there ’ s nothing to it. And yet if I had not been making ornithopters like that, a little bit cruder, in 1939 â a long, long time ago â there wouldn ’ t have been a Gossamer Condor, there wouldn ’ t have been an Albatross, a Solar Challenger, there wouldn ’ t be an Impact car, there wouldn ’ t be a mandate on zero-emission vehicles in California. A lot of these things â or similar â would have happened some time, probably a decade later. I didn ’ t realize at the time I was doing inquiry-based, hands-on things with teams, like they ’ re trying to get in education systems. So I think that, as a symbol, it ’ s important. And I believe that also is important. You can think of it as a sort of a symbol for learning and TED that somehow gets you thinking of technology and nature, and puts it all together in things that are â that make this conference, I think, more important than any that ’ s taken place in this country in this decade. Thank you.

Text Sample 843: POS Dim 6 – 2007 – Score 6.00 – 2007/t000199.txt

I dabble in design. I ’ m a curator of architecture and design ; I happen to be at the Museum of Modern Art. But what we ’ re going to talk about today is really design. Really good designers are like sponges : they really are curious and absorb every kind of information that comes their way, and transform it so that it can be used by people like us. And so that gives me an opportunity, because every design show that I curate kind of looks at a different world. And it ’ s great, because it seems like every time I change jobs. And what I ’ m going to do today is I ’ m going to give you a preview of the next exhibition that I ’ m working on, which is called”Design and the Elastic Mind.”The world that I decided to focus on this particular time is the world of science and the world of technology. Technology always comes into play when design is involved, but science does a little less. But designers are great at taking big revolutions that happen and transforming them so that we can use them. And this is what this exhibition looks at. If you think about your life today, you go every day through many different scales, many different changes of rhythm and pace. You work over different time zones, you talk to very different people, you multitask. We all know it, and we do it kind of automatically. Some of the minds in this audience are super elastic, others are a little slower, others have a few stretch marks, but nonetheless this is a quite exceptional audience from that viewpoint. Other people are not as elastic. I can ’ t get my father in Italy to work on the Internet. He doesn ’ t want to put high-speed Internet at home. And that ’ s because there ’ s some little bit of fear, little bit of resistance or just clogged mechanisms. So designers work on this particular malaise that we have, these kinds of discomforts that we have, and try to make life easier for us. Elasticity of mind is something that we really need, you know, we really need, we really cherish and we really work on. And this exhibition is about the work of designers that help us be more elastic, and also of designers that really work on this elasticity as an opportunity. And one last thing is that it ’ s not only designers, but it ’ s also scientists. And before I launch into the **display** of some of the slides and into the preview, I would like to point out this beautiful detail about scientists and design. You can say that the relationship between science and design goes back centuries. You can of course talk about Leonardo da Vinci and many other Renaissance men and women — and there ’ s a gigantic history behind it. But according to a really great science historian you might know, Peter Galison — he teaches at Harvard — what nanotechnology in particular and quantum physics have brought to designers is this renewed interest, this real passion for design. So basically, the idea of being able to build things bottom up, atom by atom, has made them all into tinkerers. And all of a sudden scientists are seeking designers, just like designers are seeking scientists. It ’ s a brand-new love affair that we ’ re trying to cultivate at MOMA. Together with Adam Bly, who is the founder of Seed magazine — that ’ s now a multimedia company, you might know it — we founded about a year ago a monthly salon for designers and scientists, and it ’ s quite beautiful. And Keith has come, and also Jonathan has come and many others. And it was great, because at the beginning was this apology fest — you know, scientists would tell designers, you know, I don ’ t know what style is, I ’ m not really elegant. And designers would like, oh, I don ’ t know how to do an equation, I don ’ t understand what you ’ re saying. And then all of a sudden they really started talking each others ’ language, and now we ’ re already at the point that they collaborate. Paul Steinhardt, a physicist from New York, and Aranda / Lasch, architects, collaborated in an installation in London at the Serpentine. And it ’ s really interesting to see how this happens. The exhibition will talk about the work of both designers and scientists, and show how they ’ re presenting the possibilities of the future to us. I ’ m showing to you different sections of the show right now, just to give you a taste of it. Nanophysics and nanotechnology, for instance, have really opened the designer ’ s mind. In this case I ’ m showing more the designers ’ work, because they ’ re the ones that have really been stimulated. A lot of the objects in the show are concepts, not objects that exist already. But what you ’ re looking at here is the work of some scientists from UCLA. This kind of alphabet soup is a new way to mark proteins — not only by color but literally by alphabet letters. So they construct it, and they can construct all kinds of forms at the nanoscale. This is the work of design students from the Royal College of Arts in London that have been working together with their tutor, Tony Dunne, and with a bunch of scientists around Great Britain on the possibilities of nanotechnology for design in the future. New sensing elements on the body — you can grow hairs on your nails, and therefore grab some of the particles from another person. They seem very, very obsessed with finding out more about the ideal mate. So they ’ re working on enhancing everything : touch, smell — everything they can, in order to find the perfect mate. Very interesting. This is a typeface designer from Israel who has designed — he calls them”typosperma.”He ’ s thinking — of course it ’ s all a concept — of injecting typefaces into spermatozoa, I don ’ t know how to say it in English — spermatazoi — in order to make them become — to almost have a song or a whole poem written with every ejaculation. I tell you, designers are quite fantastic, you know. So, tissue design. In this case too, you have a mixture of scientists and designers. This here is part of the same lab at the Royal College of Arts. The RCA is really quite an amazing school from that viewpoint. One of the assignments for a year was to work with in-vitro meat. You know that already you can grow meat in vitro. In Australia they did it — this research company, called SymbioticA. But the problem is that it ' s a really ugly patty. And so, the assignment to the students was, how should the steak of tomorrow be? When you don ' t have to kill cows and it can have any shape, what should it be like? So this particular student, James King, went around the beautiful English countryside, picked the best, best cow that he could see, and then put her in the MRI machine. Then, he took the scans of the best organs and made the meat — of course, this is done with a Japanese resins food maker, but you know, in the future it could be made better — which reproduces the best MRI scan of the best cow he could find. Instead, this element here is much more banal. Something that you know can be done already is to grow bone tissue, so that you can make a wedding ring out of the bone tissue of your loved one — literally. So, this is indeed made of human bone tissue. This is SymbioticA, and they ' ve been working — they were the first ones to do this in-vitro meat — and now they ' ve also done an in-vitro coat, a leather coat. It ' s miniscule, but it ' s a real coat. It ' s shaped like one. So, we ' ll be able to really not have any excuse to be wearing everything leather in the future. One of the most important topics of the show — you know, as anything in our life today, we can look at it from many, many different viewpoints, and at different levels. One of the most interesting and most important concepts is the idea of scale. We change scale very often : we change resolution of screens, and we ' re not really fazed by it, we do it very comfortably. So you go, even in the exhibition, from the idea of nanotechnology and the nanoscale to the manipulation of really great amounts of data — the mapping and tagging of the universe and of the world. In this particular case a section will be devoted to information design. You see here the work of Ben Fry. This is"Human vs. Chimps"— the few chromosomes that distinguish us from chimps. It was a beautiful visualization that he did for Seed magazine. And here ' s the whole code of Pac-Man, visualized with all the go-to, go-back-to, also made into a beautiful choreography. And then also graphs by scientists, this beautiful diagraph of protein homology. Scientists are starting to also consider aesthetics. We were discussing with Keith Shrubb* this morning the fact that many scientists tend not to use anything beautiful in their **presentations**, otherwise they ' re afraid of being considered dumb blondes. So they pick the worst background from any kind of PowerPoint **presentation**, the worst typeface. It ' s only recently that this kind of marriage between design and science is producing some of the first"pretty"— if we can say so — scientific **presentations**. Another aspect of contemporary design that I think is mind-opening, promising and will really be the future of design, is the idea of collective design. You know, the whole XO laptop, from One Laptop per Child, is based on the idea of collaboration and mash and networking. So, the more the merrier. The more computers, the stronger the signal, and children work on the **interface** so that it ' s all based on doing things together, tasks together. So the idea of collective design is something that will become even bigger in the future, and this is chosen as an example. Related to the idea of collective design and to the new balance between the individual and the collectiveness, collectivity is the idea of existence maximum. That ' s a term that I coined a few years ago while I was thinking of how pressed we are together, and at the same time how these small objects, like the Walkman first and then the iPod, create bubbles of space around us that enable us to have a metaphysical space that is much bigger than our physical space. You can be in the subway and you can be completely isolated and have your own room in your iPod. And this is the work of several designers that really enhance the idea of solitude and expansion by means of **various** techniques. This is a spa telephone. The idea is that it ’ s become so difficult to have a private conversation anywhere that you go to the spa, you have a massage, you have a facial, maybe a rub, and then you have this beautiful pool with this perfect temperature, and you can have this isolation tank phone conversation with whomever you ’ ve been wanting to talk with for a long time. And same thing here, Social Tele- presence. It ’ s actually already used by the military a little bit, but it ’ s the idea of being able to be somewhere else with your senses while you ’ re removed from it physically. And this is called Blind Date. It ’ s a [unclear], so if you ’ re too shy to be really at the date, you can stay at a distance with your flowers and somebody else reenacts the date for you. Rapid manufacturing is another big area in which technology and design are, I think, bound to change the world. You ’ ve heard about it before many times. Rapid manufacturing is a computer file sent directly from the computer to the manufacturing machine. It used to be called rapid prototyping, rapid modeling. It started out in the ’ 80s, but at the beginning it was machines carving out of a foam block a model that was very, very fragile, and could not have any real use. Slowly but surely, the materials became better — better resins. Techniques became better — not only carving but also stereolithography and laser — solidifying all kinds of resins, whether in powder or in liquid form. And the vats became bigger, to the point that now we can have actual chairs made by rapid manufacturing. It takes seven days today to manufacture a chair, but you know what? One day it will take seven hours. And then the dream is that you ’ ll be able to, from home, customize your chair. You know, companies and designers will be designing the matrix or the margins that respect both solidity and brand, and design identity. And then you can send it to the Kinko ’ s store at the corner and go get your chair. Now, the implications of this are enormous, not only regarding the participation of the final buyer in the design process, but also no tracking, no warehousing, no wasted materials. Also, I can imagine many design manufacturers will have to retool their own business plans and maybe invest in this Kinko ’ s store. But it really is a big change. And here I ’ m showing a picture that was in Wired Magazine — you know, the **Artifacts** of the Future section that I love so much — that shows you can have your desktop 3D printer and print your own basketball. But here instead are examples, you can already 3D-print textiles, which is very interesting. This is just a really nice touch — it ’ s called slow prototyping. It ’ s a designer that put 10, 000 bees at work and they built this vase. They had a particular shape that they had to stay in. Mapping and tagging. As the capacity of computers becomes really, really big, and the capacity of our mind not that much bigger, we find that we need to tag as much as we can what we do in order to then retrace our path. Also, we do it in order to share with other people. Again, this communal sense of experience that seems to be so important today. So, **various** ways to map and tag are also the work of many designers nowadays. Also, the senses — designers and scientists all work on trying to expand our senses capabilities so that we can achieve more. And also animal senses in a way. This particular object that many people love so much is actually based on kind of a scientific experiment — the fact that bees have a very strong olfactory sense, and so — much like dogs that can smell certain kinds of skin cancer — bees can be trained by Pavlovian reflex to detect one type of cancer, and also pregnancy. And so this student at the RCA designed this beautiful blown- glass object where the bees move from one chamber to the other if they detect that particular smell that signifies, in this case, pregnancy. Another shape is made for cancer. Design for Debate is a very interesting new endeavor that designers have really shaped for themselves. Some designers don ’ t design objects, products, things that we ’ re going to actually use, but rather, they design scenarios that are object-based. They ’ re still very useful. They help companies and other designers think better about the future. And usually they are accompanied by videos. This is quite beautiful. It ’ s Dunne and Raby,”All the Robots.”Those are a series of robots that are meant to be taken care of. We always think that robots will take care of us, and instead they designed these robots that are very, very needy. You need to take one in your arms and look at it in the eyes for about five minutes before it does something. Another one gets really, really nervous if you get in to the room, and starts shaking, so you have to calm it down. So it ’ s really a way to make us think more about what robots mean to us. Noam Toran and”Accessories for Lonely Men”: the idea is that when you lose your loved one or you go through a bad breakup, what you miss the most are those annoying things that you used to hate when you were with the other person. So he designed all these series of accessories. This one is something that takes away the sheets from you at night. Then there ’ s another one that breathes on your neck. There ’ s another one that throws plates and breaks them. So it ’ s just this idea of what we really miss in life. Elio Caccavale : he took the idea of those dolls that explain leukemia. He ’ s working on dolls that explain xenotransplantation, and also the spider gene into the goat, from a few years ago. He ’ s working for the exhibition on a whole series of dolls that explain to children where babies come from today. Because it ’ s not anymore Mom, Dad, the flowers and the bees, and then there ’ s the baby. No, it can be two moms, three dads, in-vitro — there ’ s the whole idea of how babies can be made today that has changed. So it ’ s a series of dolls that he ’ s working on right now. One of the most beautiful things is that designers really work on life, even though they take technology into account. And many designers have been working recently on the idea of death and mourning, and what we can do about it today with new technologies. Or how we should behave about it with new technologies. These three objects over there are hard drives with a Bluetooth connection. But they ’ re in reality very, very beautiful sculpted **artifacts** that contain the whole desktop and computer memory of somebody who passed away. So instead of having only the pictures, you will be able to put this object next to the computer and all of a sudden have, you know, Gertrude ’ s whole life and all of her files and her address book come alive. And this is even better. This is Auger-Loizeau,”AfterLife.”It ’ s the idea that some people don ’ t believe in an afterlife. So to give them something tangible that shows that there is something after death, they take the gastric juices of people who passed away and concentrate them, and put them into a battery that can actually be used to power flashlights. They also go — you know, sex toys, whatever. It ’ s quite amazing how these things can make you smile, can make you laugh, can make you cry sometimes. But I ’ m hoping that this particular exhibition will be able to trace a new portrait of where design is going — which is always, hopefully, a portrait a few years in advance of where the world is going. Thank you very much.

Text Sample 844: POS Dim 6 – 2007 – Score 5.00 – 2007/t000238.txt

Well, good morning. You know, the computer and television both recently turned 60, and today I ’ d like to talk about their relationship. Despite their middle age, if you ’ ve been following the themes of this conference or the entertainment industry, it ’ s pretty clear that one has been picking on the other. So it ’ s about time that we talked about how the computer ambushed television, or why the invention of the atomic bomb unleashed forces that lead to the writers ’ strike. And it ’ s not just what these are doing to each other, but it ’ s what the audience thinks that really frames this matter. To get a sense of this, and it ’ s been a theme we ’ ve talked about all week, I recently talked to a bunch of tweeners. I wrote on cards :”television,””radio,””MySpace,””Internet,””PC.”And I said, just arrange these, from what ’ s important to you and what ’ s not, and then tell me why. Let ’ s listen to what happens when they get to the portion of the discussion on television. Girl 1 : Well, I think it ’ s important but, like, not necessary because you can do a lot of other stuff with your free time than watch programs. Peter Hirshberg : Which is more fun, Internet or TV? Girls : Internet. Girl 2 : I think we — the reasons, one of the reasons we put computer before TV is because nowadays, like, we have TV shows on the computer. Girl 2 : And then you can download onto your iPod. PH : Would you like to be the president of a TV network? Girl 4 : I wouldn ’ t like it. Girl 2 : That would be so stressful. Girl 5 : No. PH : How come? Girl 5 : Because they ’ re going to lose all their money eventually. Girl 3 : Like the stock market, it goes up and down and stuff. I think right now the computers will be at the top and everything will be kind of going down and stuff. PH : There ’ s been an uneasy relationship between the TV business and the tech business, really ever since they both turned about 30. We go through periods of enthrallment, followed by reactions in boardrooms, in the finance community best characterized as, what ’ s the finance term? Ick pooey. Let me give you an example of this. The year is 1976, and Warner buys Atari because video games are on the rise. The next year they march forward and they introduce Qube, the first interactive cable TV system, and the New York Times heralds this as telecommunications moving to the home, convergence, great things are happening. Everybody in the East Coast gets in the pictures — Citicorp, Penney, RCA — all getting into this big vision. By the way, this is about when I enter the picture. I ’ m going to do a summer internship at Time Warner. That summer I ’ m all — I ’ m at Warner that summer — I ’ m all excited to work on convergence, and then the bottom falls out. Doesn ’ t work out too well for them, they lose money. And I had a happy brush with convergence until, kind of, Warner basically has to liquidate the whole thing. That ’ s when I leave graduate school, and I can ’ t work in New York on kind of entertainment and technology because I have to be exiled to California, where the remaining jobs are, almost to the sea, to go to work for Apple Computer. Warner, of course, writes off more than 400 million dollars. Four hundred million dollars, which was real money back in the ’ 70s. But they were onto something and they got better at it. By the year 2000, the process was perfected. They merged with AOL, and in just four years, managed to shed about 200 billion dollars of market capitalization, showing that they ’ d actually mastered the art of applying Moore ’ s law of successive miniaturization to their balance sheet. Now, I think that one reason that the media and the entertainment communities, or the media community, is driven so crazy by the tech community is that tech folks talk differently. You know, for 50 years, we ’ ve talked about changing the world, about total transformation. For 50 years, it ’ s been about hopes and fears and promises of a better world. And I got to thinking, you know, who else talks that way? And the answer is pretty clearly — it ’ s people in religion and in politics. And so I realized that actually the tech world is best understood, not as a business cycle, but as a messianic movement. We promise something great, we evangelize it, we ’ re going to change the world. It doesn ’ t work out too well, and so we actually go back to the well and start all over again, as the people in New York and L. A. look on in absolute, morbid astonishment. But it ’ s this irrational view of things that drives us on to the next thing. So, what I ’ d like to ask is, if the computer is becoming a principal tool of media and entertainment, how did we get here? I mean, how did a machine that was built for accounting and artillery morph into media? Of course, the first computer was built just after World War II to solve military problems, but things got really interesting just a couple of years later — 1949 with Whirlwind, built at MIT ’ s Lincoln Lab. Jay Forrester was building this for the Navy, but you can ’ t help but see that the creator of this machine had in mind a machine that might actually be a potential media star. So take a look at what happens when the foremost journalist of early television meets one of the foremost computer pioneers, and the computer begins to express itself. Journalist : It ’ s a Whirlwind electronic computer. With considerable trepidation, we undertake to interview this new machine. Jay Forrester : Hello New York, this is Cambridge. And this is the oscilloscope of the Whirlwind electronic computer. Would you like if I used the machine? Journalist : Yes, of course. But I have an idea, Mr. Forrester. Since this computer was made in conjunction with the Office of Naval Research, why don ’ t we switch down to the Pentagon in Washington and let the Navy ’ s research chief, Admiral Bolster, give Whirlwind the workout? Calvin Bolster : Well, Ed, this problem concerns the Navy ’ s Viking rocket. This rocket goes up 135 miles into the sky. Now, at the standard rate of fuel consumption, I would like to see the computer trace the flight path of this rocket and see how it can determine, at any instant, say at the end of 40 seconds, the amount of fuel remaining, and the velocity at that set instant. JF : Over on the left-hand side, you will notice fuel consumption decreasing as the rocket takes off. And on the right-hand side, there ’ s a scale that shows the rocket ’ s velocity. The rocket ’ s position is shown by the trajectory that we ’ re now looking at. And as it reaches the peak of its trajectory, the velocity, you will notice, has dropped off to a minimum. Then, as the rocket dives down, velocity picks up again toward a maximum velocity and the rocket hits the ground. How ’ s that? Journalist : What about that, Admiral? CB : Looks very good to me. JF : And before leaving, we would like to show you another kind of mathematical problem that some of the boys have worked out in their spare time, in a less serious vein, for a Sunday afternoon. Journalist : Thank you very much indeed, Mr. Forrester and the MIT lab. PH : You know, so much was worked out : the first real-time interaction, the video **display**, pointing a gun. It lead to the microcomputer, but unfortunately, it was too pricey for the Navy, and all of this would have been lost if it weren ’ t for a happy coincidence. Enter the atomic bomb. We ’ re threatened by the greatest weapon ever, and knowing a good thing when it sees it, the Air Force decides it needs the biggest computer ever to protect us. They adapt Whirlwind to a massive air defense system, deploy it all across the frozen north, and spend nearly three times as much on this computer as was spent on the Manhattan Project building the A-Bomb in the first place. Talk about a shot in the arm for the computer industry. And you can imagine that the Air Force became a pretty good salesman. Here ’ s their marketing video. Narrator : In a mass raid, high-speed bombers could be in on us before we could determine their tracks. And then it would be too late to act. We cannot afford to take that chance. It is to meet this threat that the Air Force has been developing SAGE, the Semi-Automatic Ground Environment system, to strengthen our air defenses. This new computer, built to become the nerve center of a defense network, is able to perform all the complex mathematical problems involved in countering a mass enemy raid. It is provided with its own powerhouse containing large diesel-driven generators, air-conditioning equipment, and cooling towers required to cool the thousands of vacuum tubes in the computer. PH : You know, that one computer was huge. There ’ s an interesting marketing lesson from it, which is basically, when you market a product, you can either say, this is going to be wonderful, it will make you feel better and enliven you. Or there ’ s one other marketing proposition

: if you don ’ t use our product, you ’ ll die. This is a really good example of that. This had the first pointing device. It was distributed, so it worked out — distributed computing and modems — so all these things could talk to each other. About 20 percent of all the nation ’ s programmers were wrapped up in this thing, and it led to an awful lot of what we have today. It also used vacuum tubes. You saw how huge it was, and to give you a sense for this — because we ’ ve talked a lot about Moore ’ s law and making things small at this conference, so let ’ s talk about making things large. If we took Whirlwind and put it in a place that you all know, say, Century City, it would fit beautifully. You ’ d kind of have to take Century City out, but it could fit in there. But like, let ’ s imagine we took the latest Pentium processor, the latest Core 2 Extreme, which is a four-core processor that Intel ’ s working on, it will be our laptop tomorrow. To build that, what we ’ d do with Whirlwind technology is we ’ d have to take up roughly from the 10 to Mulholland, and from the 405 to La Cienega just with those Whirlwinds. And then, the 92 nuclear power plants that it would take to provide the power would fill up the rest of Los Angeles. That ’ s roughly a third more nuclear power than all of France creates. So, the next time they tell you they ’ re on to something, clearly they ’ re not. So — and we haven ’ t even worked out the cooling needs. But it gives you the kind of power that people have, that the audience has, and the reasons these transformations are happening. All of this stuff starts moving into industry. DEC kind of reduces all this and makes the first mini-computer. It shows up at places like MIT, and then a mutation happens. Spacewar! is built, the first computer game, and all of a sudden, interactivity and involvement and passion is worked out. Actually, many MIT students stayed up all night long working on this thing, and many of the principles of gaming today were worked out. DEC knew a good thing about wasting time. It shipped every one of its computers with that game. Meanwhile, as all of this is happening, by the mid- ’ 50s, the business model of traditional broadcasting and cinema has been busted completely. A new technology has confounded radio men and **movie** moguls and they ’ re quite certain that television is about to do them in. In fact, despair is in the air. And a quote that sounds largely reminiscent from everything I ’ ve been reading all week. RCA had David Sarnoff, who basically commercialized radio, said this,”I don ’ t say that radio networks must die. Every effort has been made and will continue to be made to find a new pattern, new selling arrangements and new types of programs that may arrest the declining revenues. It may yet be possible to eke out a poor existence for radio, but I don ’ t know how.”And of course, as the computer industry develops interactively, producers in the emerging TV business actually hit on the same idea. And they fake it. Jack Berry : Boys and girls, I think you all know how to get your magic windows up on the set, you just get them out. First of all, get your Winky Dink kits out. Put out your Magic Window and your erasing glove, and rub it like this. That ’ s the way we get some of the magic into it, boys and girls. Then take it and put it right up against the screen of your own television set, and rub it out from the center to the corners, like this. Make sure you keep your magic crayons handy, your Winky Dink crayons and your erasing glove, because you ’ ll be using them during the show to draw like that. You all set? OK, let ’ s get right to the first story about Dusty Man. Come on into the secret lab. PH : It was the dawn of interactive TV, and you may have noticed they wanted to sell you the Winky Dink kits. Those are the Winky Dink crayons. I know what you ’ re saying.”Pete, I could use any ordinary open-source crayon, why do I have to buy theirs?”I assure you, that ’ s not the case. Turns out they told us directly that these are the only crayons you should ever use with your Winky Dink Magic Window, other crayons may discolor or hurt the window. This proprietary principle of vendor lock-in would go on to be perfected with great success as one of the enduring principles of windowing systems everywhere. It led to lawsuits — — federal investigations, and lots of repercussions, and that ’ s a scandal we won ’ t discuss today. But we will discuss this scandal, because this man, Jack Berry, the host of”Winky Dink,”went on to become the host of”Twenty One,”one of the most important quiz shows ever. And it was rigged, and it became unraveled when this man, Charles van Doren, was outed after an unnatural winning streak, ending Berry ’ s career. And actually, ending the career of a lot of people at CBS. It turns out there was a lot to learn about how this new medium worked. And 50 years ago, if you ’ d been at a meeting like this and were trying to understand the media, there was one prophet and only but one you wanted to hear from, Professor Marshall McLuhan. He actually understood something about a theme that we ’ ve been discussing all week. It ’ s the role of the audience in an era of pervasive electronic communications. Here he is talking from the 1960s. Marshall McLuhan : If the audience can become involved in the actual process of making the ad, then it ’ s happy. It ’ s like the old quiz shows. They were great TV because it gave the audience a role, something to do. They were horrified when they discovered they ’ d really been left out all the time because the shows were rigged. Now, then, this was a horrible misunderstanding of TV on the part of the programmers. PH : You know, McLuhan talked about the global village. If you substitute the word blogosphere, of the Internet today, it is very true that his understanding is probably very enlightening now. Let ’ s listen in to him. MM : The global village is a world in which you don ’ t necessarily have harmony. You have extreme concern with everybody else ’ s business and much involvement in everybody else ’ s life. It ’ s a sort of Ann Landers ’ column writ large. And it doesn ’ t necessarily mean harmony and peace and quiet, but it does mean huge involvement in everybody else ’ s affairs. And so the global village is as big as a planet, and as small as a village post office. PH : We ’ ll talk a little bit more about him later. We ’ re now right into the 1960s. It ’ s the era of big business and data centers for computing. But all that was about to change. You know, the expression of technology reflects the people and the time of the culture it was built in. And when I say that code expresses our hopes and aspirations, it ’ s not just a joke about messianism, it ’ s actually what we do. But for this part of the story, I ’ d actually like to throw it to America ’ s leading technology correspondent, John Markoff. John Markoff : Do you want to know what the counterculture in drugs, sex, rock ’ n ’ roll and the anti-war movement had to do with computing? Everything. It all happened within five miles of where I ’ m standing, at Stanford University, between 1960 and 1975. In the midst of revolution in the streets and rock and roll concerts in the parks, a group of researchers led by people like John McCarthy, a computer scientist at the Stanford Artificial Intelligence Lab, and Doug Engelbart, a computer scientist at SRI, changed the world. Engelbart came out of a pretty dry engineering culture, but while he was beginning to do his work, all of this stuff was bubbling on the mid-peninsula. There was LSD leaking out of Kesey ’ s Veterans ’ Hospital experiments and other areas around the campus, and there was music literally in the streets. The Grateful Dead was playing in the pizza parlors. People were leaving to go back to the land. There was the Vietnam War. There was black liberation. There was women ’ s liberation. This was a **remarkable** place, at a **remarkable** time. And into that ferment came the microprocessor. I think it was that interaction that led to personal computing. They saw these tools that were controlled by the establishment as ones that could actually be liberated and put to use by these communities that they were trying to build. And most importantly, they had this ethos of sharing information. I think these ideas are difficult to understand, because when you

’ re trapped in one paradigm, the next paradigm is always like a science fiction universe — it makes no sense. The stories were so compelling that I decided to write a book about them. The title of the book is,”What the Dormouse Said : How the ’ 60s Counterculture Shaped the Personal Computer Industry.”The title was taken from the lyrics to a Jefferson Airplane song. The lyrics go,”Remember what the dormouse said. Feed your head, feed your head, feed your head.”PH : By this time, computing had kind of leapt into media territory, and in short order much of what we ’ re doing today was imagined in Cambridge and Silicon Valley. Here ’ s the Architecture Machine Group, the predecessor of the Media Lab, in 1981. Meanwhile, in California, we were trying to commercialize a lot of this stuff. HyperCard was the first program to introduce the public to hyperlinks, where you could randomly hook to any kind of picture, or piece of text, or data across a file system, and we had no way of explaining it. There was no metaphor. Was it a database? A prototyping tool? A scripted language? Heck, it was everything. So we ended up writing a marketing brochure. We asked a question about how the mind works, and we let our customers play the role of so many blind men filling out the elephant. A few years later, we then hit on the idea of explaining to people the secret of, how do you get the content you want, the way you want it and the easy way? Here ’ s the Apple marketing video. James Burke : You ’ ll be pleased to know, I ’ m sure, that there are several ways to create a HyperCard interactive video. The most involved method is to go ahead and produce your own videodisc as well as build your own HyperCard stacks. By far the simplest method is to buy a pre-made videodisc and HyperCard stacks from a commercial supplier. The method we **illustrate** in this video uses a pre-made videodisc but creates custom HyperCard stacks. This method allows you to use existing videodisc materials in ways which suit your specific needs and interests. PH : I hope you realize how subversive that is. That ’ s like a Dick Cheney speech. You think he ’ s a nice balding guy, but he ’ s just declared war on the content business. Find the commercial stuff, mash it up, tell the story your way. Now, as long as we confine this to the education market, and a personal matter between the computer and the file system, that ’ s fine, but as you can see, it was about to leap out and upset Jack Valenti and a lot of other people. By the way, speaking of the filing system, it never occurred to us that these hyperlinks could go beyond the local area network. A few years later, Tim Berners-Lee worked that out. It became a killer app of links, and today, of course, we call that the World Wide Web. Now, not only was I instrumental in helping Apple miss the Internet, but a couple of years later, I helped Bill Gates do the same thing. The year is 1993 and he was working on a book and I was working on a video to help him kind of explain where we were all heading and how to popularize all this. We were plenty aware that we were messing with media, and on the surface, it looks like we predicted a lot of the right things, but we also missed an awful lot. Let ’ s take a look. Narrator : The pyramids, the Colosseum, the New York subway system and TV dinners, ancient and modern wonders of the man-made world all. Yet each pales to insignificance with the completion of that magnificent accomplishment of twenty-first-century technology, the Digital Superhighway. Once it was only a dream of technoids and a few long-forgotten politicians. The Digital Highway arrived in America ’ s living rooms late in the twentieth century. Let us recall the pioneers who made this technical marvel possible. The Digital Highway would follow the rutted trail first blazed by Alexander Graham Bell. Though some were incredulous... Man 1 : The phone company! Narrator : Stirred by the prospects of mass communication and making big bucks on advertising, David Sarnoff commercializes radio. Man 2 : Never had scientists been put under such pressure and demand. Narrator : The medium introduced America to new products. Voice 1 : Say, mom, Windows for Radio means more enjoyment and greater ease of use for the whole family. Be sure to enjoy Windows for Radio at home and at work. Narrator : In 1939, the Radio Corporation of America introduced television. Man 2 : Never had scientists been put under such pressure and demand. Narrator : Eventually, the race to the future took on added momentum with the breakup of the telephone company. And further stimulus came with the deregulation of the cable television industry, and the re-regulation of the cable television industry. Ted Turner : We did the work to build this, this cable industry, now the broadcasters want some of our money. I mean, it ' s ridiculous. Narrator : Computers, once the unwieldy tools of accountants and other geeks, escaped the backrooms to enter the media fracas. The world and all its culture reduced to bits, the lingua franca of all media. And the forces of convergence exploded. Finally, four great industrial sectors combined. Telecommunications, entertainment, computing and everything else. Man 3 : We ' ll see channels for the gourmet and we ' ll see channels for the pet lover. Voice 2 : Next on the gourmet pet channel, decorating birthday cakes for your schnauzer. Narrator : All of industry was in play, as investors flocked to place their bets. At stake : the battle for you, the consumer, and the right to spend billions to send a lot of information into the parlors of America. PH : We missed a lot. You know, you missed, we missed the Internet, the long tail, the role of the audience, open systems, social networks. It just goes to show how tough it is to come up with the right uses of media. Thomas Edison had the same problem. He wrote a list of what the phonograph might be good for when he invented it, and kind of only one of his ideas turned out to have been the right early idea. Well, you know where we ' re going on from here. We come into the era of the dotcom, the World Wide Web, and I don ' t need to tell you about that because we all went through that bubble together. But when we emerge from this and what we call Web 2. 0, things actually are quite different. And I think it ' s the reason that TV ' s so challenged. If Internet one was about pages, now it ' s about people. It ' s a customer, it ' s an audience, it ' s a person who ' s participating. It ' s the formidable thing that is changing entertainment now. MM : Because it gave the audience a role, something to do. PH : In my own company, Technorati, we see something like 67, 000 blog posts an hour come in. That ' s about 2, 700 fresh, connective links across about 112 million blogs that are out there. And it ' s no wonder that as we head into the writers ' strike, odd things happen. You know, it reminds me of that old saw in Hollywood, that a producer is anyone who knows a writer. I now think a network boss is anyone who has a cable modem. But it ' s not a joke. This is a real headline."Websites attract striking writers : operators of sites like MyDamnChannel. com could benefit from labor disputes."Meanwhile, you have the TV bloggers going out on strike, in sympathy with the television writers. And then you have TV Guide, a Fox property, which is about to sponsor the online video awards — but cancels it out of sympathy with traditional television, not appearing to gloat. To show you how schizophrenic this all is, here ' s the head of MySpace, or Fox Interactive, a News Corp company, being asked, well, with the writers ' strike, isn ' t this going to hurt News Corp and help you online? Man : But I, yeah, I think there ' s an opportunity. As the strike continues, there ' s an opportunity for more people to experience video on places like MySpace TV. PH : Oh, but then he remembers he works for Rupert Murdoch. Man : Yes, well, first, you know, I ' m part of News Corporation as part of Fox Entertainment Group. Obviously, we hope that the strike is — that the issues are resolved as quickly as possible. PH : One of the great things that ' s going on here is the globalization of content really is happening. Here is a clip from a video, from a piece of animation that was written by a writer in Hollywood, animation worked out in

Israel, farmed out to Croatia and India, and it ’ s now an international series. Narrator : The following takes place between the minutes of 2:15 p. m. and 2:18 p. m., in the months preceding the presidential primaries. Voice 1 : You ’ ll have to stay here in the safe house until we get word the terrorist threat is over. Voice 2 : You mean we ’ ll have to live here, together? Voices 2, 3 and 4 : With her? Voice 2 : Well, there goes the neighborhood. PH : The company that created this, Aniboom, is an interesting example of where this is headed. Traditional TV animation costs, say, between 80, 000 and 10, 000 dollars a minute. They ’ re producing things for between 1, 500 and 800 dollars a minute. And they ’ re offering their creators 30 percent of the back end, in a much more entrepreneurial manner. So, it ’ s a different model. What the entertainment business is struggling with, the world of brands is figuring out. For example, Nike now understands that Nike Plus is not just a device in its shoe, it ’ s a network to hook its customers together. And the head of marketing at Nike says,”People are coming to our site an average of three times a week. We don ’ t have to go to them.”Which means television advertising is down 57 percent for Nike. Or, as Nike ’ s head of marketing says,”We ’ re not in the business of keeping media companies alive. We ’ re in the business of connecting with consumers.”And media companies realize the audience is important also. Here ’ s a man announcing the new Market Watch from Dow Jones, powered 100 percent by the user experience on the home page — user-generated content married up with traditional content. It turns out you have a bigger audience and more interest if you hook up with them. Or, as Geoffrey Moore once told me, it ’ s intellectual curiosity that ’ s the trade that brands need in the age of the blogosphere. And I think this is beginning to happen in the entertainment business. One of my heroes is songwriter, Ally Willis, who just wrote”The Color Purple”and has been an R and — rhythm and blues writer, and this is what she said about where songwriting ’ s going. Ally Willis : Where millions of collaborators wanted the song, because to look at them strictly as spam is missing what this medium is about. PH : So, to wrap up, I ’ d love to throw it back to Marshall McLuhan, who, 40 years ago, was dealing with audiences that were going through just as much change, and I think that, today, traditional Hollywood and the writers are framing this perhaps in the way that it was being framed before. But I don ’ t need to tell you this, let ’ s throw it back to him. Narrator : We are in the middle of a tremendous clash between the old and the new. MM : The medium does things to people and they are always completely unaware of this. They don ’ t really notice the new medium that is wrapping them up. They think of the old medium, because the old medium is always the content of the new medium, as **movies** tend to be the content of TV, and as books used to be the content, novels used to be the content of **movies**. And so every time a new medium arrives, the old medium is the content, and it is highly observable, highly noticeable, but the real, real roughing up and massaging is done by the new medium, and it is ignored. PH : I think it ’ s a great time of enthrallment. There ’ s been more raw DNA of communications and media thrown out there. Content is moving from shows to particles that are batted back and forth, and part of social communications, and I think this is going to be a time of great renaissance and opportunity. And whereas television may have gotten beat up, what ’ s getting built is a really exciting new form of communication, and we kind of have the merger of the two industries and a new way of thinking to look at it. Thanks very much.

Text Sample 845: POS Dim 6 – 2007 – Score 5.00 – 2007/t000276.txt

A great way to start, I think, with my view of simplicity is to take a look at TED. Here you are, understanding why we ’ re here, what ’ s going on with no difficulty at all. The best A. I. in the planet would find it complex and confusing, and my little dog Watson would find it simple and understandable but would miss the point. He would have a great time. And of course, if you ’ re a speaker here, like Hans Rosling, a speaker finds this complex, tricky. But in Hans Rosling ’ s case, he had a secret weapon yesterday, literally, in his sword swallowing act. And I must say, I thought of quite a few objects that I might try to swallow today and finally gave up on, but he just did it and that was a wonderful thing. So Puck meant not only are we fools in the pejorative sense, but that we ’ re easily fooled. In fact, what Shakespeare was pointing out is we go to the theater in order to be fooled, so we ’ re actually looking forward to it. We go to magic shows in order to be fooled. And this makes many things fun, but it makes it difficult to actually get any kind of picture on the world we live in or on ourselves. And our friend, Betty Edwards, the”Drawing on the Right Side of the Brain”lady, shows these two tables to her drawing class and says,”The problem you have with learning to draw is not that you can ’ t move your hand, but that the way your brain perceives images is faulty. It ’ s trying to perceive images into objects rather than seeing what ’ s there.”And to prove it, she says,”The exact size and shape of these tabletops is the same, and I ’ m going to prove it to you.”She does this with cardboard, but since I have an expensive computer here I ’ ll just rotate this little guy around and... Now having seen that — and I ’ ve seen it hundreds of times, because I use this in every talk I give — I still can ’ t see that they ’ re the same size and shape, and I doubt that you can either. So what do artists do? Well, what artists do is to measure. They measure very, very carefully. And if you measure very, very carefully with a stiff arm and a straight edge, you ’ ll see that those two shapes are exactly the same size. And the Talmud saw this a long time ago, saying,”We see things not as they are, but as we are.”I certainly would like to know what happened to the person who had that insight back then, if they actually followed it to its ultimate conclusion. So if the world is not as it seems and we see things as we are, then what we call reality is a kind of hallucination happening inside here. It ’ s a waking dream, and understanding that that is what we actually exist in is one of the biggest epistemological barriers in human history. And what that means :”simple and understandable”might not be actually simple or understandable, and things we think are”complex”might be made simple and understandable. Somehow we have to understand ourselves to get around our flaws. We can think of ourselves as kind of a noisy channel. The way I think of it is, we can ’ t learn to see until we admit we ’ re blind. Once you start down at this very humble level, then you can start finding ways to see things. And what ’ s happened, over the last 400 years in particular, is that human beings have invented”brainlets”— little additional parts for our brain — made out of powerful ideas that help us see the world in different ways. And these are in the form of sensory apparatus — telescopes, microscopes — reasoning apparatus — **various** ways of thinking — and, most importantly, in the ability to change perspective on things. I ’ ll talk about that a little bit. It ’ s this change in perspective on what it is we think we ’ re perceiving that has helped us make more progress in the last 400 years than we have in the rest of human history. And yet, it is not taught in any K through 12 curriculum in America that I ’ m aware of. So one of the things that goes from simple to complex is when we do more. We like more. If we do more in a kind of a stupid way, the simplicity gets complex and, in fact, we can keep on doing it for a very long time. But Murray Gell- Mann yesterday talked about emergent properties ; another name for them could be”architecture”as a metaphor for taking the same old material and thinking about non-obvious, non-simple ways of combining it. And in fact, what Murray was talking about yesterday in the fractal beauty of nature — of having the descriptions at **various** levels be rather similar — all goes down to the idea that the elementary particles are both sticky and standoffish, and they ’ re in violent motion. Those three things give rise to all the different levels of what seem to be complexity in our world. But how simple? So, when I saw Roslings ’ Gapminder stuff a few years ago, I just thought it was the greatest thing I ’ d seen in conveying complex ideas simply. But then I had a thought of,”Boy, maybe it ’ s too simple.”And I put some effort in to try and check to see how well these simple portrayals of trends over time actually matched up with some ideas and investigations from the side, and I found that they matched up very well. So the Roslings have been able to do simplicity without removing what ’ s important about the data. Whereas the film yesterday that we saw of the simulation of the inside of a cell, as a former molecular biologist, I didn ’ t like that at all. Not because it wasn ’ t beautiful or anything, but because it misses the thing that most students fail to understand about molecular biology, and that is : why is there any probability at all of two complex shapes finding each other just the right way so they combine together and be catalyzed? And what we saw yesterday was every reaction was fortuitous ; they just swooped in the air and bound, and something happened. But in fact, those molecules are spinning at the rate of about a million revolutions per second ; they ’ re agitating back and forth their size every two nanoseconds ; they ’ re completely crowded together, they ’ re jammed, they ’ re bashing up against each other. And if you don ’ t understand that in your mental model of this stuff, what happens inside of a cell seems completely mysterious and fortuitous, and I think that ’ s exactly the wrong image for when you ’ re trying to teach science. So, another thing that we do is to confuse adult sophistication with the actual understanding of some principle. So a kid who ’ s 14 in high school gets this version of the Pythagorean theorem, which is a truly subtle and interesting proof, but in fact it ’ s not a good way to start learning about mathematics. So a more direct one, one that gives you more of the feeling of math, is something closer to Pythagoras ’ own proof, which goes like this : so here we have this triangle, and if we surround that C square with three more triangles and we copy that, notice that we can move those triangles down like this. And that leaves two open areas that are kind of suspicious... and bingo. That is all you have to do. And this kind of proof is the kind of proof that you need to learn when you ’ re learning mathematics in order to get an idea of what it means before you look into the, literally, 1, 200 or 1, 500 proofs of Pythagoras ’ theorem that have been discovered. Now let ’ s go to young children. This is a very unusual teacher who was a kindergarten and first-grade teacher, but was a natural mathematician. So she was like that jazz musician friend you have who never studied music but is a terrific musician ; she just had a feeling for math. And here are her six-year- olds, and she ’ s got them making shapes out of a shape. So they pick a shape they like — like a diamond, or a square, or a triangle, or a trapezoid — and then they try and make the next larger shape of that same shape, and the next larger shape. You can see the trapezoids are a little challenging there. And what this teacher did on every project was to have the children act like first it was a creative arts project, and then something like science. So they had created these **artifacts**. Now she had them look at them and do this... laborious, which I thought for a long time, until she explained to me was to slow them down so they ’ ll think. So they ’ re cutting out the little pieces of cardboard here and pasting them up. But the whole point of this thing is for them to look at this chart and fill it out.”What have you noticed about what you did?”And so six-year-old Lauren there noticed that the first one took one, and the second one took three more and the total was four on that one, the third one took five more and the total was nine on that one, and then the next one. She saw right away that the additional tiles that you had to add around the edges was always going to grow by two, so she was very confident about how she made those numbers there. And she could see that these were the square numbers up until about six, where she wasn ’ t sure what six times six was and what seven times seven was, but then she was confident again. So that ’ s what Lauren did. And then the teacher, Gillian Ishijima, had the kids bring all of their projects up to the front of the room and put them on the floor, and everybody went batshit :”Holy shit! They ’ re the same!”No matter what the shapes were, the growth law is the same. And the mathematicians and scientists in the crowd will recognize these two progressions as a first-order discrete differential equation and a second-order discrete differential equation, derived by six- year-olds. Well, that ’ s pretty amazing. That isn ’ t what we usually try to teach six-year-olds. So, let ’ s take a look now at how we might use the computer for some of this. And so the first idea here is just to show you the kind of things that children do. I ’ m using the software that we ’ re putting on the $100 laptop. So I ’ d like to draw a little car here — I ’ ll just do this very quickly — and put a big tire on him. And I get a little object here and I can look inside this object, I ’ ll call it a car. And here ’ s a little behavior : car forward. Each time I click it, car turn. If I want to make a little script to do this over and over again, I just drag these guys out and set them going. And I can try steering the car here by... See the car turn by five here? So what if I click this down to zero? It goes straight. That ’ s a big revelation for nine-year-olds. Make it go in the other direction. But of course, that ’ s a little bit like kissing your sister as far as driving a car, so the kids want to do a steering wheel ; so they draw a steering wheel. And we ’ ll call this a wheel. See this wheel ’ s heading here? If I turn this wheel, you can see that number over there going minus and positive. That ’ s kind of an invitation to pick up this name of those numbers coming out there and to just drop it into the script here, and now I can steer the car with the steering wheel. And it ’ s interesting. You know how much trouble the children have with variables, but by learning it this way, in a situated fashion, they never forget from this single trial what a variable is and how to use it. And we can reflect here the way Gillian Ishijima did. So if you look at the little script here, the speed is always going to be 30. We ’ re going to move the car according to that over and over again. And I ’ m dropping a little dot for each one of these things ; they ’ re evenly spaced because they ’ re 30 apart. And what if I do this progression that the six-year-olds did of saying,”OK, I ’ m going to increase the speed by two each time, and then I ’ m going to increase the distance by the speed each time? What do I get there?”We get a visual pattern of what these nine-year-olds called acceleration. So how do the children do science? Teacher : [Choose] objects that you think will fall to the Earth at the same time. Student 1 : Ooh, this is nice. Teacher : Do not pay any attention to what anybody else is doing. Who ’ s got the apple? Alan Kay : They ’ ve got little stopwatches. Student 2 : What did you get? What did you get? AK : Stopwatches aren ’ t accurate enough. Student 3 : 0. 99 seconds. Teacher : So put”sponge **ball**”... Student 4l : [I decided to] do the shot put and the sponge **ball** because they ’ re two totally different weights, and if you drop them at the same time, maybe they ’ ll drop at the same speed. Teacher : Drop. Class : Whoa! AK : So obviously, Aristotle never asked a child about this particular point because, of course, he didn ’ t bother doing the experiment, and neither did St. Thomas Aquinas. And it was not until Galileo actually did it that an adult thought like a child, only 400 years ago. We get one child like that about every classroom of 30 kids who will actually cut straight to the chase. Now, what if we want to look at this more closely? We can take a **movie** of what ’ s going on, but even if we single stepped this **movie**, it ’ s tricky to see what ’ s going on. And so what we can do is we can lay out the frames side by side or stack them up. So when the children see this, they say,”Ah! Acceleration,”remembering back four months when they did their cars sideways, and they start measuring to find out what kind of acceleration it is. So what I ’ m doing is measuring from the bottom of one image to the bottom of the next image, about a fifth of a second later, like that. And they ’ re getting faster and faster each time, and if I stack these guys up, then we see the differences ; the increase in the speed is constant. And they say,”Oh, yeah. Constant acceleration. We ’ ve done that already.”And how shall we look and verify that we actually have it? So you can ’ t tell much from just making the **ball** drop there, but if we drop the **ball** and run the **movie** at the same time, we can see that we have come up with an accurate physical model. Galileo, by the way, did this very cleverly by running a **ball** backwards down the strings of his lute. I pulled out those apples to remind myself to tell you that this is actually probably a Newton and the apple type story, but it ’ s a great story. And I thought I would do just one thing on the $100 laptop here just to prove that this stuff works here. So once you have gravity, here ’ s this — increase the speed by something, increase the ship ’ s speed. If I start the little game here that the kids have done, it ’ ll crash the space ship. But if I oppose gravity, here we go... Oops! One more. Yeah, there we go. Yeah, OK? I guess the best way to end this is with two quotes : Marshall McLuhan said,”Children are the messages that we send to the future,”but in fact, if you think of it, children are the future we send to the future. Forget about messages ; children are the future, and children in the first and second world and, most especially, in the third world need mentors. And this summer, we ’ re going to build five million of these $100 laptops, and maybe 50 million next year. But we couldn ’ t create 1, 000 new teachers this summer to save our life. That means that we, once again, have a thing where we can put technology out, but the mentoring that is required to go from a simple new iChat instant messaging system to something with depth is missing. I believe this has to be done with a new kind of user **interface**, and this new kind of user **interface** could be done with an expenditure of about 100 million dollars. It sounds like a lot, but it is literally 18 minutes of what we ’ re spending in Iraq — we ’ re spending 8 billion dollars a month ; 18 minutes is 100 million dollars — so this is actually cheap. And Einstein said,”Things should be as simple as possible, but not simpler.”Thank you.

Text Sample 846: POS Dim 6 – 2003 – Score 4.00 – 2003/t000205.txt

I want to take you back basically to my hometown, and to a picture of my hometown of the week that”Emergence”came out. And it ’ s a picture we ’ ve seen several times. Basically,”Emergence”was published on 9 / 11. I live right there in the West Village, so the plume was luckily blowing west, away from us. We had a two-and-a-half-day-old baby in the house that was ours — we hadn ’ t taken it from somebody else. And one of the thoughts that I had dealing with these two separate emergences of a book and a baby, and having this event happen so close — that my first thought, when I was still kind of in the apartment looking out at it all or walking out on the street and looking out on it just in front of our building, was that I ’ d made a terrible miscalculation in the book that I ’ d just written. Because so much of that book was a celebration of the power and creative potential of density, of largely urban density, of connecting people and putting them together in one place, and putting them on sidewalks together and having them share ideas and share physical space together. And it seemed to me looking at that — that tower burning and then falling, those towers burning and falling — that in fact, one of the lessons here was that density kills. And that of all the technologies that were exploited to make that carnage come into being, probably the single group of technologies that cost the most lives were those that enable 50, 000 people to live in two buildings 110 stories above the ground. If they hadn ’ t been crowded — you compare the loss of life at the Pentagon to the Twin Towers, and you can see that very powerfully. And so I started to think, well, you know, density, density — I ’ m not sure if, you know, this was the right call. And I kind of ruminated on that for a couple of days. And then about two days later, the wind started to change a little bit, and you could sense that the air was not healthy. And so even though there were no cars still in the West Village where we lived, my wife sent me out to buy a, you know, a large air **filter** at the Bed Bath and Beyond, which was located about 20 blocks away, north. And so I went out. And obviously I ’ m physically a very strong person, as you can tell — — so I wasn ’ t worried about carrying this thing 20 blocks. And I walked out, and this really miraculous thing happened to me as I was walking north to buy this air **filter**, which was that the streets were completely alive with people. There was an incredible — it was, you know, a beautiful day, as it was for about a week after, and the West Village had never seemed more lively. I walked up along Hudson Street — where Jane Jacobs had lived and written her great book that so influenced what I was writing in"Emergence"— past the White Horse Tavern, that great old bar where Dylan Thomas drank himself to death, and the Bleecker Street playground was filled with kids. And all the people who lived in the neighborhood, who owned restaurants and bars in the neighborhood, were all out there — had them all open. People were out. There were no cars, so it seemed even better, in some ways. And it was a beautiful urban day, and the incredible thing about it was that the city was working. The city was there. All the things that make a great city successful and all the things that make a great city stimulating — they were all on **display** there on those streets. And I thought, well, this is the power of a city. I mean, the power of the city — we talked about cities as being centralized in space, but what makes them so strong most of the time is they ’ re decentralized in function. They don ’ t have a center executive branch that you can take out and cause the whole thing to fail. If they did, it probably was right there at Ground Zero. I mean, you know, the emergency bunker was right there, was destroyed by the attacks, and obviously the damage done to the building and the lives. But nonetheless, just 20 blocks north, two days later, the city had never looked more alive. If you ’ d gone into the minds of the people, well, you would have seen a lot of trauma, and you would have seen a lot of heartache, and you would have seen a lot of things that would take a long time to recover. But the system itself of this city was thriving. So I took heart in seeing that. So I wanted to talk a little bit about the reasons why that works so well, and how some of those reasons kind of map on to where the Web is going right now. The question that I found myself asking to people when I was talking about the book afterwards is — when you ’ ve talked about emergent behavior, when you ’ ve talked about collective intelligence, the best way to get people to kind of wrap their heads around that is to ask, who builds a neighborhood? Who decides that Soho should have this personality and that the Latin Quarter should have this personality? Well, there are some kind of executive decisions, but mostly the answer is — everybody and nobody. Everybody contributes a little bit. No single person is really the ultimate actor behind the personality of a neighborhood. Same thing to the question of, who was keeping the streets alive post-9 / 11 in my neighborhood? Well, it was the whole city. The whole system kind of working on it, and everybody contributing a small little part. And this is increasingly what we ' re starting to see on the Web in a bunch of interesting ways — most of which weren ' t around, actually, except in very experimental things, when I was writing"Emergence"and when the book came out. So it ' s been a very optimistic time, I think, and I want to just talk about a few of those things. I think that there is effectively a new kind of model of interactivity that ' s starting to emerge online right now. And the old one looked like this. This is not the future King of England, although it looks like it. It ' s some guy, it ' s a GeoCities homepage of some guy that I found online who ' s interested, if you look at the bottom, in soccer and Jesus and Garth Brooks and Clint Beckham and"my hometown"— those are his links. But nothing really says this model of interactivity — which was so exciting and captures the real, the Web Zeitgeist of 1995 — than"Click here for a picture of my dog."That is — you know, there ' s no sentence that kind of conjures up that period better than that, I think, which is that you suddenly have the power to put up a picture of your dog and link to it, and somebody reading the page has the power to click on that link or not click on that link. And, you know, I don ' t want to belittle that. That, in a sense — to reference what Jeff was talking about yesterday — that was, in a sense, the kind of **interface** electricity that powered a lot of the explosion of interest in the Web : that you could put up a link, and somebody could click on it, and it could take you anywhere you wanted to go. But it ' s still a very one-to-one kind of relationship. There ' s one person putting up the link, and there ' s another person on the other end trying to decide whether to click on it or not. The new model is much more like this, and we ' ve already seen a couple of references to this. This is what happens when you search"Steven Johnson"on Google. About two months ago, I had the great breakthrough — one of my great, kind of shining achievements — which is that my website finally became a top result for"Steven Johnson."There ' s some theoretical physicist at MIT named Steven Johnson who has dropped two spots, I ' m happy to say. And, you know, I mean, I ' ll look at a couple of things like this, but Google is obviously the greatest technology ever invented for navel gazing. It ' s just that there are so many other people in your navel when you gaze. Because effectively, what ' s happening here, what ' s creating this page, obviously — and we all know this, but it ' s worth just thinking about it — is not some person deciding that I am the number one answer for Steven Johnson, but rather somehow the entire web of people putting up pages and deciding to link to my page or not link to it, and Google just sitting there and running the numbers. So there ' s this collective decision-making that ' s going on. This page is effectively, collectively authored by the Web, and Google is just helping us kind of to put the authorship in one kind of coherent place. Now, they ' re more innovative — well, Google ' s pretty innovative — but there are some new twists on this. There ' s this incredibly interesting new site — Technorati — that ' s filled with lots of little widgets that are expanding on these. And these are looking in the blog world and the world of weblogs. He ' s analyzed basically all the weblogs out there that he ' s tracking. And he ' s tracking how many other weblogs linked to those weblogs, and so you have kind of an authority — a weblog that has a lot of links to it is more authoritative than a weblog that has few links to it. And so at any given time, on any given page on the Web, actually, you can say, what does the weblog community think about this page? And you can get a list. This is what they think about my site ; it ' s ranked by blog authority. You can also rank it by the latest posts. So when I was talking in"Emergence,"I talked about the limitations of the one-way linking architecture that, basically, you could link to somebody else but they wouldn ' t necessarily know that you were pointing to them. And that was one of the reasons why the web wasn ' t quite as emergent as it could be because you needed two-way linking, you needed that kind of feedback mechanism to be able to really do interesting things. Well, something like Technorati is supplying that. Now what ' s interesting here is that this is a quote from Dave Weinberger, where he talks about everything being purposive in the Web — there ' s nothing artificial. He has this line where he says, you know, you ' re going to put up a link there, if you see a link, somebody decided to put it there. And he says, the link to one site didn ' t just grow on the other page"like a tree fungus."And in fact, I think that ' s not entirely true anymore. I could put up a feed of all those links generated by Technorati on the right-hand side of my page, and they would change as the overall ecology of the Web changes. That little list there would change. I wouldn ' t really be directly in control of it. So it ' s much closer, in a way, to a data fungus, in a sense, wrapped around that page, than it is to a deliberate link that I ' ve placed there. Now, what you ' re having here is basically a global brain that you ' re able to do lots of kind of experiments on to see what it ' s thinking. And there are all these interesting tools. Google does the Google Zeitgeist, which looks at search requests to test what ' s going on, what people are interested in, and they publish it with lots of fun graphs. And I ' m saying a lot of nice things about Google, so I ' ll be I ' ll be saying one little critical thing. There ' s a problem with the Google Zeitgeist, which is it often comes back with news that a lot of people are searching for Britney Spears pictures, which is not necessarily news. The Columbia blows up, suddenly there are a lot of searches on Columbia. Well, you know, we should expect to see that. That ' s not necessarily something we didn ' t know already. So the key thing in terms of these new tools that are kind of plumbing the depths of the global brain, that are sending kind of trace dyes through that whole bloodstream — the question is, are you finding out something new? And one of the things that I experimented with is this thing called Google Share which is basically, you take an abstract term, and you search Google for that term, and then you search the results that you get back for somebody ' s name. So basically, the number of pages that mention this term, that also mention this page, the percentage of those pages is that person ' s Google Share of that term. So you can do kind of interesting contests. Like for instance, this is a Google Share of the TED Conference. So Richard Saul Wurman has about a 15 percent Google Share of the TED conference. Our good friend Chris has about a six percent — but with a bullet, I might add. But the interesting thing is, you can broaden the search a little bit. And it turns out, actually, that 42 percent is the Mola mola fish. I had no idea. No, that ' s not true. I made that up because I just wanted to put up a slide of the Mola mola fish. I also did — and I don ' t want to start a little fight in the next panel — but I did a Google Share analysis of evolution and natural selection. So right here — now this is a big category, you have smaller percentages, so this is 0. 7 percent — Dan Dennett, who ' ll be speaking shortly. Right below him, 0. 5 percent, Steven Pinker. So Dennett ' s in the lead a little bit there. But what ' s interesting is you can then broaden the search and actually see interesting things and get a sense of what else is out there. So Gary Bauer is not too far behind — has slightly different theories about evolution and natural selection. And right behind him is L. Ron Hubbard. So — you can see we ' re in the ascot, which is always good. And by the way, Chris, that would ' ve been a really good panel, I think, right there. Hubbard apparently started to reach, but besides that, I think it would be good next year. Another quick thing — this is a slightly different thing, but this analysis some of you may have seen. It just came out. This is bursty words, looking at the historical record of State of the Union Addresses. So these are words that suddenly start to appear out of nowhere, so they ' re kind of, you know, memes that start taking off, that didn ’ t have a lot of historical precedent before. So the first one is — these are the bursty words around 1860s — slaves, emancipation, slavery, rebellion, Kansas. That ’ s Britney Spears. I mean, you know, OK, interesting. They ’ re talking about slavery in 1860. 1935 — relief, depression, recovery banks. And OK, I didn ’ t learn anything new there as well — that ’ s pretty obvious. 1985, right at the center of the Reagan years — that ’ s, we ’ re, there ’ s, we ’ ve, it ’ s. Now, there ’ s one way to interpret this, which is to say that”emancipation”and”depression”and”recovery”all have a lot of syllables. So you know, you can actually download — it ’ s hard to remember those. But seriously, actually, what you can see there, in a way that would be very hard to detect otherwise, is Reagan reinventing the political language of the country and shifting to a much more intimate, much more folksy, much more telegenic — contracting all those verbs. You know, 20 years before it was still,”Ask not what you can do,”but with Reagan, it ’ s,”that ’ s where, there ’ s Nancy and I,”that kind of language. And so something we kind of knew, but you didn ’ t actually notice syntactically what he was doing. I ’ ll go very quickly. The question now — and this is the really interesting question — is, what kind of higher-level shape is emerging right now in the overall Web ecosystem — and particularly in the ecosystem of the blogs because they are really kind of at the cutting edge. And I think what happens there will also happen in the wider system. Now there was a very interesting article by Clay Shirky that got a lot of attention about a month ago, and this is basically the distribution of links on the web to all these **various** different blogs. It follows a power law, so that there are a few extremely well-linked to, popular blogs, and a long tail of blogs with very few links. So 20 percent of the blogs get 80 percent of the links. Now this is a very interesting thing. It ’ s caused a lot of controversy because people thought that this was the ultimate kind of one man, one modem democracy, where anybody can get out there and get their voice heard. And so the question is,”Why is this happening?”It ’ s not being imposed by fiat from above. It ’ s an emergent property of the blogosphere right now. Now, what ’ s great about it is that people are working on — within seconds of Clay publishing this piece, people started working on changing the underlying rules of the system so that a different shape would start appearing. And basically, the shape appears largely because of a kind of a first-mover advantage. if you ’ re the first site there, everybody links to you. If you ’ re the second site there, most people link to you. And so very quickly you can accumulate a bunch of links, and it makes it more likely for newcomers to link to you in the future, and then you get this kind of shape. And so what Dave Sifry at Technorati started working on, literally as Shirky started — after he published his piece — was something that basically just gave a new kind of priority to newcomers. And he started looking at interesting newcomers that don ’ t have a lot of links, that suddenly get a bunch of links in the last 24 hours. So in a sense, bursty weblogs coming from new voices. So he ’ s working on a tool right there that can actually change the overall system. And it creates a kind of planned emergence. You ’ re not totally in control, but you ’ re changing the underlying rules in interesting ways because you have an end result which is maybe a more democratic spread of voices. So the most amazing thing about this — and I ’ ll end on this note — is, most emergent systems, most self- organizing systems are not made up of component parts that are capable of looking at the overall pattern and changing their behavior based on whether they like the pattern or not. So the most wonderful thing, I think, about this whole debate about power laws and software that could change it is the fact that we ’ re having the conversation. I hope it continues here. Thanks a lot.

Text Sample 847: POS Dim 6 – 2003 – Score 3.00 – 2003/t000244.txt

A year ago, I spoke to you about a book that I was just in the process of completing, that has come out in the interim, and I would like to talk to you today about some of the controversies that that book inspired. The book is called”The Blank Slate,”based on the popular idea that the human mind is a blank slate, and that all of its structure comes from socialization, culture, parenting, experience. The”blank slate”was an influential idea in the 20th century. Here are a few quotes indicating that :”Man has no nature,”from the historian Jose Ortega y Gasset ;”Man has no instincts,”from the anthropologist Ashley Montagu ;”The human brain is capable of a full range of behaviors and predisposed to none,”from the late scientist Stephen Jay Gould. There are a number of reasons to doubt that the human mind is a blank slate, and some of them just come from common sense. As many people have told me over the years, anyone who ’ s had more than one child knows that kids come into the world with certain temperaments and talents ; it doesn ’ t all come from the outside. Oh, and anyone who has both a child and a house pet has surely noticed that the child, exposed to speech, will acquire a human language, whereas the house pet won ’ t, presumably because of some innate different between them. And anyone who ’ s ever been in a heterosexual relationship knows that the minds of men and the minds of women are not indistinguishable. There are also, I think, increasing results from the scientific study of humans that, indeed, we ’ re not born blank slates. One of them, from anthropology, is the study of human universals. If you ’ ve ever taken anthropology, you know that it ’ s a — kind of an occupational pleasure of anthropologists to show how exotic other cultures can be, and that there are places out there where, supposedly, everything is the opposite to the way it is here. But if you instead look at what is common to the world ’ s cultures, you find that there is an enormously rich set of behaviors and emotions and ways of construing the world that can be found in all of the world ’ s 6, 000-odd cultures. The anthropologist Donald Brown has tried to list them all, and they range from aesthetics, affection and age statuses all the way down to weaning, weapons, weather, attempts to control, the color white and a worldview. Also, genetics and neuroscience are increasingly showing that the brain is intricately structured. This is a recent study by the neurobiologist Paul Thompson and his colleagues in which they — using MRI — measured the distribution of gray matter — that is, the outer layer of the cortex — in a large sample of pairs of people. They coded correlations in the thickness of gray matter in different parts of the brain using a false color scheme, in which no difference is coded as purple, and any color other than purple indicates a statistically significant correlation. Well, this is what happens when you pair people up at random. By definition, two people picked at random can ’ t have correlations in the distribution of gray matter in the cortex. This is what happens in people who share half of their DNA — fraternal twins. And as you can see, large amounts of the brain are not purple, showing that if one person has a thicker bit of cortex in that region, so does his fraternal twin. And here ’ s what happens if you get a pair of people who share all their DNA — namely, clones or identical twins. And you can see huge areas of cortex where there are massive correlations in the distribution of gray matter. Now, these aren ’ t just differences in anatomy, like the shape of your ear lobes, but they have consequences in thought and behavior that are well **illustrated** in this famous cartoon by Charles Addams :”Separated at birth, the Mallifert twins meet accidentally.”As you can see, there are two inventors with identical contraptions in their lap, meeting in the waiting room of a patent attorney. Now, the cartoon is not such an exaggeration, because studies of identical twins who were separated at birth and then tested in adulthood show that they have astonishing similarities. And this happens in every pair of identical twins separated at birth ever studied — but much less so with fraternal twins separated at birth. My favorite example is a pair of twins, one of whom was brought up as a Catholic in a Nazi family in Germany, the other brought up in a Jewish family in Trinidad. When they walked into the lab in Minnesota, they were wearing identical navy blue shirts with epaulettes ; both of them liked to dip buttered toast in coffee, both of them kept rubber bands around their wrists, both of them flushed the toilet before using it as well as after, and both of them liked to surprise people by sneezing in crowded elevators to watch them jump. Now — the story might seem to good to be true, but when you administer batteries of psychological tests, you get the same results — namely, identical twins separated at birth show quite astonishing similarities. Now, given both the common sense and scientific data calling the doctrine of the blank slate into question, why should it have been such an appealing notion? Well, there are a number of political reasons why people have found it congenial. The foremost is that if we ’ re blank slates, then, by definition, we are equal, because zero equals zero equals zero. But if something is written on the slate, then some people could have more of it than others, and according to this line of thinking, that would justify discrimination and inequality. Another political fear of human nature is that if we are blank slates, we can perfect mankind — the age-old dream of the perfectibility of our species through social engineering. Whereas, if we ’ re born with certain instincts, then perhaps some of them might condemn us to selfishness, prejudice and violence. Well, in the book, I argue that these are, in fact, non sequiturs. And just to make a long story short : first of all, the concept of fairness is not the same as the concept of sameness. And so when Thomas Jefferson wrote in the Declaration of Independence,”We hold these truths to be self-evident, that all men are created equal,”he did not mean”We hold these truths to be self- evident, that all men are clones.”Rather, that all men are equal in terms of their rights, and that every person ought to be treated as an individual, and not prejudged by the statistics of particular groups that they may belong to. Also, even if we were born with certain ignoble motives, they don ’ t automatically lead to ignoble behavior. That is because the human mind is a complex system with many parts, and some of them can inhibit others. For example, there ’ s excellent reason to believe that virtually all humans are born with a moral sense, and that we have cognitive abilities that allow us to profit from the lessons of history. So even if people did have impulses towards selfishness or greed, that ’ s not the only thing in the skull, and there are other parts of the mind that can counteract them. In the book, I go over controversies such as this one, and a number of other hot buttons, hot zones, Chernobyls, third rails, and so on — including the arts, cloning, crime, free will, education, evolution, gender differences, God, homosexuality, infanticide, inequality, Marxism, morality, Nazism, parenting, politics, race, rape, religion, resource depletion, social engineering, technological risk and war. And needless to say, there were certain risks in taking on these subjects. When I wrote a first draft of the book, I circulated it to a number of colleagues for comments, and here are some of the reactions that I got :”Better get a security camera for your house.””Don ’ t expect to get any more awards, job offers or positions in scholarly societies.””Tell your publisher not to list your hometown in your author bio.””Do you have tenure?”Well, the book came out in October, and nothing terrible has happened. I — I like — There was indeed reason to be nervous, and there were moments in which I did feel nervous, knowing the history of what has happened to people who ’ ve taken controversial stands or discovered disquieting findings in the behavioral sciences. There are many cases, some of which I talk about in the book, of people who have been slandered, called Nazis, physically assaulted, threatened with criminal prosecution for stumbling across or arguing about controversial findings. And you never know when you ' re going to come across one of these booby traps. My favorite example is a pair of psychologists who did research on left-handers, and published some data showing that left-handers are, on average, more susceptible to disease, more prone to accidents and have a shorter lifespan. It ' s not clear, by the way, since then, whether that is an accurate generalization, but the data at the time seemed to support that. Well, pretty soon they were barraged with enraged letters, death threats, ban on the topic in a number of scientific journals, coming from irate left-handers and their advocates, and they were literally afraid to open their mail because of the venom and vituperation that they had inadvertently inspired. Well, the night is young, but the book has been out for half a year, and nothing terrible has happened. None of the dire professional consequences has taken place — I haven ' t been exiled from the city of Cambridge. But what I wanted to talk about are two of these hot buttons that have aroused the strongest response in the 80-odd reviews that The Blank Slate has received. I ' ll just put that list up for a few seconds, and see if you can guess which two — I would estimate that probably two of these topics inspired probably 90 percent of the reaction in the **various** reviews and radio interviews. It ' s not violence and war, it ' s not race, it ' s not gender, it ' s not Marxism, it ' s not Nazism. They are : the arts and parenting. So let me tell you what aroused such irate responses, and I ' ll let you decide if whether they — the claims are really that outrageous. Let me start with the arts. I note that among the long list of human universals that I presented a few slides ago are art. There is no society ever discovered in the remotest corner of the world that has not had something that we would consider the arts. Visual arts — decoration of surfaces and bodies — appears to be a human universal. The telling of stories, music, dance, poetry — found in all cultures, and many of the motifs and themes that give us pleasure in the arts can be found in all human societies : a preference for symmetrical forms, the use of repetition and variation, even things as specific as the fact that in poetry all over the world, you have lines that are very close to three seconds long, separated by pauses. Now, on the other hand, in the second half of the 20th century, the arts are frequently said to be in decline. And I have a collection, probably 10 or 15 headlines, from highbrow magazines deploring the fact that the arts are in decline in our time. I ' ll give you a couple of representative quotes :"We can assert with some confidence that our own period is one of decline, that the standards of culture are lower than they were 50 years ago, and that the evidences of this decline are visible in every department of human activity."That ' s a quote from T. S. Eliot, a little more than 50 years ago. And a more recent one :"The possibility of sustaining high culture in our time is becoming increasing problematical. Serious book stores are losing their franchise, nonprofit theaters are surviving primarily by commercializing their repertory, symphony orchestras are diluting their programs, public television is increasing its dependence on reruns of British sitcoms, classical radio stations are dwindling, museums are resorting to blockbuster shows, dance is dying."That ' s from Robert Brustein, the famous drama critic and director, in The New Republic about five years ago. Well, in fact, the arts are not in decline. I don ' t think this will as a surprise to anyone in this room, but by any standard they have never been flourishing to a greater extent. There are, of course, entirely new art forms and new media, many of which you ' ve heard over these few days. By any economic standard, the demand for art of all forms is skyrocketing, as you can tell from the price of opera tickets, by the number of books sold, by the number of books published, the number of musical titles released, the number of new albums and so on. The only grain of truth to this complaint that the arts are in decline come from three spheres. One of them is in elite art since the 1930s — say, the kinds of works performed by major symphony orchestras, where most of the repertory is before 1930, or the works shown in major galleries and prestigious museums. In literary criticism and analysis, probably 40 or 50 years ago, literary critics were a kind of cultural hero ; now they ' re kind of a national joke. And the humanities and arts programs in the universities, which by many measures, indeed are in decline. Students are staying away in droves, universities are disinvesting in the arts and humanities. Well, here ' s a diagnosis. They didn ' t ask me, but by their own admission, they need all the help that they can get. And I would like to suggest that it ' s not a coincidence that this supposed decline in the elite arts and criticism occurred in the same point in history in which there was a widespread denial of human nature. A famous quotation can be found — if you look on the web, you can find it in literally scores of English core syllabuses —"In or about December 1910, human nature changed."A paraphrase of a quote by Virginia Woolf, and there ' s some debate as to what she actually meant by that. But it ' s very clear, looking at these syllabuses, that — it ' s used now as a way of saying that all forms of appreciation of art that were in place for centuries, or millennia, in the 20th century were discarded. The beauty and pleasure in art — probably a human universal — were — began to be considered saccharine, or kitsch, or commercial. Barnett Newman had a famous quote that"the impulse of modern art is the desire to destroy beauty"— which was considered bourgeois or tacky. And here ' s just one example. I mean, this is perhaps a representative example of the visual depiction of the female form in the 15th century ; here is a representative example of the depiction of the female form in the 20th century. And, as you can see, there — something has changed in the way the elite arts appeal to the senses. Indeed, in movements of modernism and post-modernism, there was visual art without beauty, literature without narrative and plot, poetry without meter and rhyme, architecture and planning without ornament, human scale, green space and natural light, music without melody and rhythm, and criticism without clarity, attention to aesthetics and insight into the human condition. Let me give just you an example to back up that last statement. But here, there — one of the most famous literary English scholars of our time is the Berkeley professor, Judith Butler. And here is an example of one of her analyses :"The move from a structuralist account in which capital is understood to structure social relations in relatively homologous ways to a view of hegemony in which power relations are subject to repetition, convergence and rearticulation brought the question of temporality into the thinking of structure, and marked a shift from the form of Althusserian theory that takes structural totalities as theoretical objects..."Well, you get the idea. By the way, this is one sentence — you can actually parse it. Well, the argument in"The Blank Slate"was that elite art and criticism in the 20th century, although not the arts in general, have disdained beauty, pleasure, clarity, insight and style. People are staying away from elite art and criticism. What a puzzle — I wonder why. Well, this turned out to be probably the most controversial claim in the book. Someone asked me whether I stuck it in in order to deflect ire from discussions of gender and Nazism and race and so on. I won ' t comment on that. But it certainly inspired an energetic reaction from many university professors. Well, the other hot button is parenting. And the starting point is the — for that discussion was the fact that we have all been subject to the advice of the parenting industrial complex. Now, here is — here is a representative quote from a besieged mother :"I ' m overwhelmed with parenting advice. I ' m supposed to do lots of physical activity with my kids so I can instill in them a physical fitness habit so they ' ll grow up to be healthy adults. And I ' m supposed to do all kinds of intellectual play so they

’ ll grow up smart. And there are all kinds of play — clay for finger dexterity, word games for reading success, large motor play, small motor play. I feel like I could devote my life to figuring out what to play with my kids.”I think anyone who ’ s recently been a parent can sympathize with this mother. Well, here ’ s some sobering facts about parenting. Most studies of parenting on which this advice is based are useless. They ’ re useless because they don ’ t control for heritability. They measure some correlation between what the parents do, how the children turn out and assume a causal relation : that the parenting shaped the child. Parents who talk a lot to their kids have kids who grow up to be **articulate**, parents who spank their kids have kids who grow up to be violent and so on. And very few of them control for the possibility that parents pass on genes for — that increase the chances a child will be **articulate** or violent and so on. Until the studies are redone with adoptive children, who provide an environment but not genes to their kids, we have no way of knowing whether these conclusions are valid. The genetically controlled studies have some sobering results. Remember the Mallifert twins : separated at birth, then they meet in the patent office — remarkably similar. Well, what would have happened if the Mallifert twins had grown up together? You might think, well, then they ’ d be even more similar, because not only would they share their genes, but they would also share their environment. That would make them super-similar, right? Wrong. Identical twins, or any siblings, who are separated at birth are no less similar than if they had grown up together. Everything that happens to you in a given home over all of those years appears to leave no permanent stamp on your personality or intellect. A complementary finding, from a completely different methodology, is that adopted siblings reared together — the mirror image of identical twins reared apart, they share their parents, their home, their neighborhood, don ’ t share their genes — end up not similar at all. OK — two different bodies of research with a similar finding. What it suggests is that children are shaped not by their parents over the long run, but in part — only in part — by their genes, in part by their culture — the culture of the country at large and the children ’ s own culture, namely their peer group — as we heard from Jill Sobule earlier today, that ’ s what kids care about — and, to a very large extent, larger than most people are prepared to acknowledge, by chance : chance events in the wiring of the brain in utero ; chance events as you live your life. So let me conclude with just a remark to bring it back to the theme of choices. I think that the sciences of human nature — behavioral genetics, evolutionary psychology, neuroscience, cognitive science — are going to, increasingly in the years to come, upset **various** dogmas, careers and deeply-held political belief systems. And that presents us with a choice. The choice is whether certain facts about humans, or topics, are to be considered taboos, forbidden knowledge, where we shouldn ’ t go there because no good can come from it, or whether we should explore them honestly. I have my own answer to that question, which comes from a great artist of the 19th century, Anton Chekhov, who said,”Man will become better when you show him what he is like.”And I think that the argument can ’ t be put any more eloquently than that. Thank you very much.

Text Sample 848: POS Dim 6 – 2003 – Score 3.00 – 2003/t000242.txt

What I want to tell you about today is how I see robots invading our lives at multiple levels, over multiple timescales. And when I look out in the future, I can ’ t imagine a world, 500 years from now, where we don ’ t have robots everywhere. Assuming â despite all the dire predictions from many people about our future â assuming we ’ re still around, I can ’ t imagine the world not being populated with robots. And then the question is, well, if they ’ re going to be here in 500 years, are they going to be everywhere sooner than that? Are they going to be around in 50 years? Yeah, I think that ’ s pretty likely â there ’ s going to be lots of robots everywhere. And in fact I think that ’ s going to be a lot sooner than that. I think we ’ re sort of on the cusp of robots becoming common, and I think we ’ re sort of around 1978 or 1980 in personal computer years, where the first few robots are starting to appear. Computers sort of came around through games and toys. And you know, the first computer most people had in the house may have been a computer to play Pong, a little microprocessor embedded, and then other games that came after that. And we ’ re starting to see that same sort of thing with robots : LEGO Mindstorms, Furbies â who here â did anyone here have a Furby? Yeah, there ’ s 38 million of them sold worldwide. They are pretty common. And they ’ re a little tiny robot, a simple robot with some sensors, a little bit of processing actuation. On the right there is another robot doll, who you could get a couple of years ago. And just as in the early days, when there was a lot of sort of amateur interaction over computers, you can now get **various** hacking kits, how-to-hack books. And on the left there is a platform from Evolution Robotics, where you put a PC on, and you program this thing with a GUI to wander around your house and do **various** stuff. And then there ’ s a higher price point sort of robot toys â the Sony Aibo. And on the right there, is one that the NEC developed, the PaPeRo, which I don ’ t think they ’ re going to release. But nevertheless, those sorts of things are out there. And we ’ ve seen, over the last two or three years, lawn-mowing robots, Husqvarna on the bottom, Friendly Robotics on top there, an Israeli company. And then in the last 12 months or so we ’ ve started to see a bunch of home-cleaning robots appear. The top left one is a very nice home-cleaning robot from a company called Dyson, in the U. K. Except it was so expensive â 3, 500 dollars â they didn ’ t release it. But at the bottom left, you see Electrolux, which is on sale. Another one from Karcher. At the bottom right is one that I built in my lab about 10 years ago, and we finally turned that into a product. And let me just show you that. We ’ re going to give this away I think, Chris said, after the talk. This is a robot that you can go out and buy, and that will clean up your floor. And it starts off sort of just going around in ever-increasing circles. If it hits something â you people see that? Now it ’ s doing wall-following, it ’ s following around my feet to clean up around me. Let ’ s see, let ’ s â oh, who stole my Rice Krispies? They stole my Rice Krispies! Don ’ t worry, relax, no, relax, it ’ s a robot, it ’ s smart! See, the three-year-old kids, they don ’ t worry about it. It ’ s grown-ups that get really upset. We ’ ll just put some crap here. Okay. I don ’ t know if you see â so, I put a bunch of Rice Krispies there, I put some pennies, let ’ s just shoot it at that, see if it cleans up. Yeah, OK. So â we ’ ll leave that for later. Part of the trick was building a better cleaning mechanism, actually ; the intelligence on board was fairly simple. And that ’ s true with a lot of robots. We ’ ve all, I think, become, sort of computational chauvinists, and think that computation is everything, but the mechanics still matter. Here ’ s another robot, the PackBot, that we ’ ve been building for a bunch of years. It ’ s a military surveillance robot, to go in ahead of **troops** â looking at caves, for instance. But we had to make it fairly robust, much more robust than the robots we build in our labs. On board that robot is a PC running Linux. It can withstand a 400G shock. The robot has local intelligence : it can flip itself over, can get itself into communication range, can go upstairs by itself, et cetera. Okay, so it ’ s doing local navigation there. A soldier gives it a command to go upstairs, and it does. That was not a controlled descent. Now it ’ s going to head off. And the big breakthrough for these robots, really, was September 11th. We had the robots down at the World Trade Center late that evening. Couldn ’ t do a lot in the main rubble pile, things were just too â there was nothing left to do. But we did go into all the surrounding buildings that had been evacuated, and searched for possible survivors in the buildings that were too dangerous to go into. Let ’ s run this video. Reporter :... battlefield companions are helping to reduce the combat risks. Nick Robertson has that story. Rodney Brooks : Can we have another one of these? Okay, good. So, this is a corporal who had seen a robot two weeks previously. He ’ s sending robots into caves, looking at what ’ s going on. The robot ’ s being totally autonomous. The worst thing that ’ s happened in the cave so far was one of the robots fell down ten meters. So one year ago, the US military didn ’ t have these robots. Now they ’ re on active duty in Afghanistan every day. And that ’ s one of the reasons they say a robot invasion is happening. There ’ s a sea change happening in how â where technology ’ s going. Thanks. And over the next couple of months, we ’ re going to be sending robots in production down producing oil wells to get that last few years of oil out of the ground. Very hostile environments, 150Ë C, 10, 000 PSI. Autonomous robots going down, doing this sort of work. But robots like this, they ’ re a little hard to program. How, in the future, are we going to program our robots and make them easier to use? And I want to actually use a robot here â a robot named Chris â stand up. Yeah. Okay. Come over here. Now notice, he thinks robots have to be a bit stiff. He sort of does that. But I ’ m going to â Chris Anderson : I ’ m just British. RB : Oh. I ’ m going to show this robot a task. It ’ s a very complex task. Now notice, he nodded there, he was giving me some indication he was understanding the flow of communication. And if I ’ d said something completely bizarre he would have looked askance at me, and regulated the conversation. So now I brought this up in front of him. I ’ d looked at his eyes, and I saw his eyes looked at this bottle top. And I ’ m doing this task here, and he ’ s checking up. His eyes are going back and forth up to me, to see what I ’ m looking at â so we ’ ve got shared attention. And so I do this task, and he looks, and he looks to me to see what ’ s happening next. And now I ’ ll give him the bottle, and we ’ ll see if he can do the task. Can you do that? Okay. He ’ s pretty good. Yeah. Good, good, good. I didn ’ t show you how to do that. Now see if you can put it back together. And he thinks a robot has to be really slow. Good robot, that ’ s good. So we saw a bunch of things there. We saw when we ’ re interacting, we ’ re trying to show someone how to do something, we direct their visual attention. The other thing communicates their internal state to us, whether he ’ s understanding or not, regulates a social interaction. There was shared attention looking at the same sort of thing, and recognizing socially communicated reinforcement at the end. And we ’ ve been trying to put that into our lab robots because we think this is how you ’ re going to want to interact with robots in the future. I just want to show you one technical diagram here. The most important thing for building a robot that you can interact with socially is its visual attention system. Because what it pays attention to is what it ’ s seeing and interacting with, and what you ’ re understanding what it ’ s doing. So in the videos I ’ m about to show you, you ’ re going to see a visual attention system on a robot which has â it looks for skin tone in HSV space, so it works across all human colorings. It looks for highly saturated colors, from toys. And it looks for things that move around. And it weights those together into an attention window, and it looks for the highest-scoring place â the stuff where the most interesting stuff is happening â and that is what its eyes then segue to. And it looks right at that. At the same time, some top-down sort of stuff : might decide that it ’ s lonely and look for skin tone, or might decide that it ’ s bored and look for a toy to play with. And so these weights change. And over here on the right, this is what we call the Steven Spielberg memorial module. Did people see the **movie**”AI”? RB : Yeah, it was really bad, but â remember, especially when Haley Joel Osment, the little robot, looked at the blue fairy for 2, 000 years without taking his eyes off it? Well, this gets rid of that, because this is a habituation Gaussian that gets negative, and more and more intense as it looks at one thing. And it gets bored, so it will then look away at something else. So, once you ’ ve got that â and here ’ s a robot, here ’ s Kismet, looking around for a toy. You can tell what it ’ s looking at. You can estimate its gaze direction from those eyeballs covering its camera, and you can tell when it ’ s actually seeing the toy. And it ’ s got a little bit of an emotional response here. But it ’ s still going to pay attention if something more significant comes into its field of view â such as Cynthia Breazeal, the builder of this robot, from the right. It sees her, pays attention to her. Kismet has an underlying, three-dimensional emotional space, a vector space, of where it is emotionally. And at different places in that space, it expresses â can we have the volume on here? Can you hear that now, out there? Kismet : Do you really think so? Do you really think so? Do you really think so? RB : So it ’ s expressing its emotion through its face and the prosody in its voice. And when I was dealing with my robot over here, Chris, the robot, was measuring the prosody in my voice, and so we have the robot measure prosody for four basic messages that mothers give their children pre-linguistically. Here we ’ ve got naive subjects praising the robot : Voice : Nice robot. You ’ re such a cute little robot. RB : And the robot ’ s reacting appropriately. Voice :... very good, Kismet. Voice : Look at my smile. RB : It smiles. She imitates the smile. This happens a lot. These are naive subjects. Here we asked them to get the robot ’ s attention and indicate when they have the robot ’ s attention. Voice : Hey, Kismet, ah, there it is. RB : So she realizes she has the robot ’ s attention. Voice : Kismet, do you like the toy? Oh. RB : Now, here they ’ re asked to prohibit the robot, and this first woman really pushes the robot into an emotional corner. Voice : No. No. You ’ re not to do that. No. Not appropriate. No. No. RB : I ’ m going to leave it at that. We put that together. Then we put in turn taking. When we talk to someone, we talk. Then we sort of raise our eyebrows, move our eyes, give the other person the idea it ’ s their turn to talk. And then they talk, and then we pass the baton back and forth between each other. So we put this in the robot. We got a bunch of naive subjects in, we didn ’ t tell them anything about the robot, sat them down in front of the robot and said, talk to the robot. Now what they didn ’ t know was, the robot wasn ’ t understanding a word they said, and that the robot wasn ’ t speaking English. It was just saying random English phonemes. And I want you to watch carefully, at the beginning of this, where this person, Ritchie, who happened to talk to the robot for 25 minutes â â says,”I want to show you something. I want to show you my watch.”And he brings the watch center, into the robot ’ s field of vision, points to it, gives it a motion cue, and the robot looks at the watch quite successfully. We don ’ t know whether he understood or not that the robot â Notice the turn-taking. Ritchie : OK, I want to show you something. OK, this is a watch that my girlfriend gave me. Robot : Oh, cool. Ritchie : Yeah, look, it ’ s got a little blue light in it too. I almost lost it this week. RB : So it ’ s making eye contact with him, following his eyes. Ritchie : Can you do the same thing? Robot : Yeah, sure. RB : And they successfully have that sort of communication. And here ’ s another aspect of the sorts of things that Chris and I were doing. This is another robot, Cog. They first make eye contact, and then, when Christie looks over at this toy, the robot estimates her gaze direction and looks at the same thing that she ’ s looking at. So we ’ re going to see more and more of this sort of robot over the next few years in labs. But then the big questions, two big questions that people ask me are : if we make these robots more and more human-like, will we accept them, will we â will they need rights eventually? And the other question people ask me is, will they want to take over? And on the first â you know, this has been a very Hollywood theme with lots of **movies**. You probably recognize these characters here â where in each of these cases, the robots want more respect. Well, do you ever need to give robots respect? They ’ re just machines, after all. But I think, you know, we have to accept that we are just machines. After all, that ’ s certainly what modern molecular biology says about us. You don ’ t see a description of how, you know, Molecule A, you know, comes up and docks with this other molecule. And it ’ s moving forward, you know, propelled by **various** charges, and then the soul steps in and tweaks those molecules so that they connect. It ’ s all mechanistic. We are mechanism. If we are machines, then in principle at least, we should be able to build machines out of other stuff, which are just as alive as we are. But I think for us to admit that, we have to give up on our special-ness, in a certain way. And we ’ ve had the retreat from special-ness under the barrage of science and technology many times over the last few hundred years, at least. 500 years ago we had to give up the idea that we are the center of the universe when the earth started to go around the sun ; 150 years ago, with Darwin, we had to give up the idea we were different from animals. And to imagine â you know, it ’ s always hard for us. Recently we ’ ve been battered with the idea that maybe we didn ’ t even have our own creation event, here on earth, which people didn ’ t like much. And then the human genome said, maybe we only have 35, 000 genes. And that was really â people didn ’ t like that, we ’ ve got more genes than that. We don ’ t like to give up our special-ness, so, you know, having the idea that robots could really have emotions, or that robots could be living creatures â I think is going to be hard for us to accept. But we ’ re going to come to accept it over the next 50 years or so. And the second question is, will the machines want to take over? And here the standard scenario is that we create these things, they grow, we nurture them, they learn a lot from us, and then they start to decide that we ’ re pretty boring, slow. They want to take over from us. And for those of you that have teenagers, you know what that ’ s like. But Hollywood extends it to the robots. And the question is, you know, will someone accidentally build a robot that takes over from us? And that ’ s sort of like this lone guy in the backyard, you know â”I accidentally built a 747.”I don ’ t think that ’ s going to happen. And I don ’ t think â â I don ’ t think we ’ re going to deliberately build robots that we ’ re uncomfortable with. We ’ ll â you know, they ’ re not going to have a super bad robot. Before that has to come to be a mildly bad robot, and before that a not so bad robot. And we ’ re just not going to let it go that way. So, I think I ’ m going to leave it at that : the robots are coming, we don ’ t have too much to worry about, it ’ s going to be a lot of fun, and I hope you all enjoy the journey over the next 50 years.

Text Sample 849: POS Dim 6 – 2001 – Score 3.00 – 2001/t000083.txt

I thought I would read poems I have that relate to the subject of youth and age. I was sort of astonished to find out how many I have actually. The first one is dedicated to Spencer, and his grandmother, who was shocked by his work. My poem is called”Dirt.”My grandmother is washing my mouth out with soap ; half a long century gone and still she comes at me with that thick cruel yellow bar. All because of a word I said, not even said really, only repeated. But”Open,”she says,”open up!”her hand clawing at my head. I know now her life was hard ; she lost three children as babies, then her husband died too, leaving young sons, and no money. She ' d stand me in the sink to pee because there was never room in the toilet. But oh, her soap! Might its bitter burning have been what made me a poet? The street she lived on was unpaved, her flat, two cramped rooms and a fetid kitchen where she stalked and caught me. Dare I admit that after she did it I never really loved her again? She lived to a hundred, even then. All along it was the sadness, the squalor, but I never, until now loved her again. When that was published in a magazine I got an irate letter from my uncle."You have maligned a great woman."It took some diplomacy. This is called"The Dress."It ' s a longer poem. In those days, those days which exist for me only as the most elusive memory now, when often the first sound you ' d hear in the morning would be a storm of birdsong, then the soft clop of the hooves of the horse hauling a milk wagon down your block, and the last sound at night as likely as not would be your father pulling up in his car, having worked late again, always late, and going heavily down to the cellar, to the furnace, to shake out the ashes and damp the draft before he came upstairs to fall into bed — in those long-ago days, women, my mother, my friends ' mothers, our neighbors, all the women I knew — wore, often much of the day, what were called house-dresses, cheap, printed, pulpy, seemingly purposefully shapeless light cotton shifts that you wore over your nightgown and, when you had to go look for a child, hang wash on the line, or run down to the grocery store on the corner, under a coat, the twisted hem of the nightgown always lank and yellowed, dangling beneath. More than the curlers some of the women seemed constantly to have in their hair in preparation for some great event — a **ball**, one would think — that never came to pass ; more than the way most women ' s faces not only were never made up during the day, but seemed scraped, bleached, and, with their plucked eyebrows, scarily masklike ; more than all that it was those dresses that made women so unknowable and forbidding, adepts of enigmas to which men could have no access, and boys no conception. Only later would I see the dresses also as a proclamation : that in your dim kitchen, your laundry, your bleak concrete yard, what you revealed of yourself was a fabulation ; your real sensual nature, veiled in those sexless vestments, was utterly your dominion. In those days, one hid much else as well : grown men didn ' t embrace one another, unless someone had died, and not always then ; you shook hands or, at a **ball** game, thumped your friend ' s back and exchanged blows meant to be codes for affection ; once out of childhood you ' d never again know the shock of your father ' s whiskers on your cheek, not until mores at last had evolved, and you could hug another man, then hold on for a moment, then even kiss. What release finally, the embrace : though we were wary — it seemed so audacious — how much unspoken joy there was in that affirmation of equality and communion, no matter how much misunderstanding and pain had passed between you by then. We knew so little in those days, as little as now, I suppose about healing those hurts : even the women, in their best dresses, with beads and sequins sewn on the bodices, even in lipstick and mascara, their hair aflow, could only stand wringing their hands, begging for peace, while father and son, like thugs, like thieves, like Romans, simmered and hissed and hated, inflicting sorrows that endured, the worst anyway, through the kiss and embrace, bleeding from brother to brother, into the generations. In those days there was still countryside close to the city, farms, cornfields, cows ; even not far from our building with its blurred brick and long shadowy hallway you could find tracts with hills and trees you could pretend were mountains and forests. Or you could go out by yourself even to a half-block-long empty lot, into the bushes : like a creature of leaves you ' d lurk, crouched, crawling, simplified, savage, alone ; already there was wanting to be simpler, wanting, when they called you, never to go back. This is another longish one, about the old and the young. It actually happened right at the time we met. Part of the poem takes place in space we shared and time we shared. It ’ s called”The Neighbor.”Her five horrid, deformed little dogs who incessantly yap on the roof under my window. Her cats, God knows how many, who must piss on her rugs — her landing ’ s a sickening reek. Her shadow once, fumbling the chain on her door, then the door slamming fearfully shut, only the barking and the music — jazz — **filtering** as it does, day and night into the hall. The time it was Chris Connor singing”Lush Life”— how it brought back my college sweetheart, my first real love, who — till I left her — played the same record. And head on my shoulder, hand on my thigh, sang sweetly along, of regrets and depletions she was too young for, as I was too young, later, to believe in her pain. It startled, then bored, then repelled me. My starting to fancy she ’ d ended up in this fire-trap in the Village, that my neighbor was her. My thinking we ’ d meet, recognize one another, become friends, that I ’ d accomplish a penance. My seeing her, it wasn ’ t her, at the mailbox. Gray-yellow hair, army pants under a nightgown, her turning away, hiding her ravaged face in her hands, muttering an inappropriate”Hi.”Sometimes there are frightening goings-on in the stairwell. A man shouting,”Shut up!”The dogs frantically snarling, claws scrabbling, then her — her voice hoarse, harsh, hollow, almost only a tone, incoherent, a note, a squawk, bone on metal, metal gone molten, calling them back,”Come back darlings, come back dear ones. My sweet angels, come back.”Medea she was, next time I saw her. Sorceress, tranced, ecstatic, stock-still on the sidewalk ragged coat hanging agape, passersby flowing around her, her mouth torn suddenly open as though in a scream, silently though, as though only in her brain or breast had it erupted. A cry so pure, practiced, detached, it had no need of a voice, or could no longer bear one. These invisible links that allure, these transfigurations, even of anguish, that hold us. The girl, my old love, the last lost time I saw her when she came to find me at a party, her drunkenly stumbling, falling, sprawling, skirt hiked, eyes veined red, swollen with tears, her shame, her dishonor. My ignorant, arrogant coarseness, my secret pride, my turning away. Still life on a rooftop, dead trees in barrels, a bench broken, dogs, excrement, sky. What pathways through pain, what junctures of vulnerability, what crossings and counterings? Too many lives in our lives already, too many chances for sorrow, too many unaccounted-for pasts.”Behold me,”the god of frenzied, inexhaustible love says, rising in bloody splendor,”Behold me.”Her making her way down the littered vestibule stairs, one agonized step at a time. My holding the door. Her crossing the fragmented tiles, faltering at the step to the street, droning, not looking at me,”Can you help me?”Taking my arm, leaning lightly against me. Her wavering step into the world. Her whispering,”Thanks love.”Lightly, lightly against me. I think I ’ ll lighten up a little. Another, different kind of poem of youth and age. It ’ s called”Gas.”Wouldn ’ t it be nice, I think, when the blue-haired lady in the doctor ’ s waiting room bends over the magazine table and farts, just a little, and violently blushes. Wouldn ’ t it be nice if intestinal gas came embodied in visible clouds, so she could see that her really quite inoffensive pop had only barely grazed my face before it drifted away. Besides, for this to have happened now is a nice coincidence. Because not an hour ago, while we were on our walk, my dog was startled by a backfire and jumped straight up like a horse bucking. And that brought back to me the stable I worked on weekends when I was 12, and a splendid piebald stallion, who whenever he was mounted would buck just like that, though more hugely of course, enormous, gleaming, resplendent. And the woman, her face abashedly buried in her”Elle”now, reminded me — I ’ d forgotten that not the least part of my awe consisted of the fact that with every jump he took the horse would powerfully fart. Phwap! Phwap! Phwap! Something never mentioned in the dozens of books about horses and their riders I devoured in those days. All that savage grandeur, the steely glinting hooves, the eruptions driven from the creature ’ s mighty innards, breath stopped, heart stopped, nostrils madly flared, I didn ’ t know if I wanted to break him, or be him. This is called”Thirst.”Many — most of my poems actually are urban poems. I happen to be reading a bunch that aren ’ t.”Thirst.”Here was my relation with the woman who lived all last autumn and winter, day and night, on a bench in the 103rd Street subway station, until finally one day she vanished. We regarded each other, scrutinized one another. Me shyly, obliquely, trying not to be furtive. She boldly, unblinkingly, even pugnaciously, wrathfully even, when her bottle was empty. I was frightened of her. I felt like a child. I was afraid some repressed part of myself would go out of control, and I ’ d be forever entrapped in the shocking seethe of her stench. Not excrement merely, not merely surface and orifice going unwashed, rediffusion of rum, there was will in it, and intention, power and purpose — a social, ethical rage and rebellion — despair too, though, grief, loss. Sometimes I ’ d think I should take her home with me, bathe her, comfort her, dress her. She wouldn ’ t have wanted me to, I would think. Instead, I ’ d step into my train. How rich I would think, is the lexicon of our self-absolving. How enduring, our bland fatal assurance that reflection is righteousness being accomplished. The dance of our glances, the clash, pulling each other through our perceptual punctures, then holocaust, holocaust, host on host of ill, injured presences, squandered, consumed. Her vigil somewhere I know continues. Her occupancy, her absolute, faithful attendance. The dance of our glances, challenge, abdication, effacement, the perfume of our consternation. This is a newer poem, a brand new poem. The title is”This Happened.”A student, a young woman in a fourth-floor hallway of her lycee, perched on the ledge of an open window chatting with friends between classes ; a teacher passes and chides her,”Be careful, you might fall,”almost banteringly chides her,”You might fall,”and the young woman, 18, a girl really, though she wouldn ’ t think that, as brilliant as she is, first in her class, and”Beautiful, too,”she ’ s often told, smiles back, and leans into the open window, which wouldn ’ t even be open if it were winter — if it were winter someone would have closed it — leans into the window, farther, still smiling, farther and farther, though it takes less time than this, really an instant, and lets herself fall. Herself fall. A casual impulse, a fancy, never thought of until now, hardly thought of even now... No, more than impulse or fancy, the girl knows what she ’ s doing, the girl means something, the girl means to mean, because it occurs to her in that instant, that beautiful or not, bright yes or no, she ’ s not who she is, she ’ s not the person she is, and the reason, she suddenly knows, is that there ’ s been so much premeditation where she is, so much plotting and planning, there ’ s hardly a person where she is, or if there is, it ’ s not her, or not wholly her, it ’ s a self inhabited, lived in by her, and seemingly even as she thinks it she knows what ’ s been missing : grace, not premeditation but grace, a kind of being in the world spontaneously, with grace. Weightfully upon me was the world. Weightfully this self which graced the world yet never wholly itself. Weightfully this self which weighed upon me, the release from which is what I desire and what I achieve. And the girl remembers, in this infinite instant already now so many times divided, the sadness she felt once, hardly knowing she felt it, to merely inhabit herself. Yes, the girl falls, absurd to fall, even the earth with its compulsion to take unto itself all that falls must know that falling is absurd, yet the girl falling isn ’ t myself, or she is myself, but a self I took of my own volition unto myself. Forever. With grace. This happened. I ’ ll read just one more. I don ’ t usually say that. I like to just end. But I ’ m afraid that

Ricky will come out here and shake his fist at me. This is called”Old Man,”appropriately enough.”Special : big tits,”Says the advertisement for a soft-core magazine on our neighborhood newsstand. But forget her breasts. A lush, fresh-lipped blond, skin glowing gold, sprawls there, resplendent. 60 nearly, yet these hardly tangible, hardly better than harlots, can still stir me. Maybe a coming of age in the American sensual darkness, never seeing an unsmudged nipple, an uncensored vagina, has left me forever infected with an unquenchable lust of the eye. Always that erotic murmur, I ’ m hardly myself if I ’ m not in a state of incipient desire. God knows though, there are worse twists your obsessions can take. Last year in Israel, a young ultra-orthodox Rabbi guiding some teenage girls through the Shrine of the Shoah forbade them to look in one room. Because there were images in it he said were licentious. The **display** was a photo. Men and women stripped naked, some trying to cover their genitals, others too frightened to bother, lined up in snow waiting to be shot and thrown into a ditch. The girls, to my horror, averted their gaze. What carnal mistrust had their teacher taught them. Even that though. Another confession : Once in a book on pre-war Poland, a studio portrait, an absolute angel, an absolute angel with tormented, tormenting eyes. I kept finding myself at her page. That she died in the camps made her — I didn ’ t dare wonder why — more present, more precious. Died in the camps, that too people — or Jews anyway — kept from their children back then. But it was like sex, you didn ’ t have to be told. Sex and death, how close they can seem. So constantly conscious now of death moving towards me, sometimes I think I confound them. My wife ’ s loveliness almost consumes me. My passion for her goes beyond reasonable bounds. When we make love, her holding me everywhere all around me, I ’ m there and not there. My mind teems, jumbles of faces, voices, impressions, I live my life over, as though I were drowning. Then I am drowning, in despair at having to leave her, this, everything, all, unbearable, awful. Still, to be able to die with no special contrition, not having been slaughtered, or enslaved. And not having to know history ’ s next mad rage or regression, it might be a relief. No. Again, no. I don ’ t mean that for a moment. What I mean is the world holds me so tightly — the good and the bad — my own follies and weakness that even this counterfeit Venus with her sham heat, and her bosom probably plumped with gel, so moves me my breath catches. Vamp. Siren. Seductress. How much more she reveals in her glare of ink than she knows. How she incarnates our desperate human need for regard, our passion to live in beauty, to be beauty, to be cherished by glances, if by no more, of something like love, or love. Thank you.

Text Sample 850: POS Dim 6 – 2001 – Score 1.00 – 2001/t000092.txt

I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ’ children, because they were all average. But they weren ’ t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ’ s not right. The good Lord in his infinite wisdom didn ’ t create us all equal as far as intelligence is concerned, any more than we ’ re equal for size, appearance. Not everybody could earn an A or a B, and I didn ’ t like that way of judging, and I did know how the alumni of **various** schools back in the ’ 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ’ t lose a game. But it seemed that we didn ’ t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ’ t like it, I didn ’ t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ’ m sure at the time he did that, I didn ’ t — it didn ’ t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ’ s under your control. If you get too engrossed and involved and concerned in regard to the things over which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,”At God ’ s footstool to confess, a poor soul knelt, and bowed his head. ’ I failed! ’ he cried. The Master said, ’ Thou didst thy best, that is success. ’”From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ’ re capable. I believe that ’ s true. If you make the effort to do the best of which you ’ re capable, trying to improve the situation that exists for you, I think that ’ s success, and I don ’ t think others can judge that ; it ’ s like character and reputation — your reputation is what you ’ re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ’ s what the teachers are themselves.”That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by coal oil lamp — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I reply, ’ Where could I find such splendid company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ’ s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ' Where could I find such splendid company? '"And I believe the teaching profession — it ' s true, you have so many youngsters, and I ' ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ' re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ' re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ' d learned these prior to coming to UCLA, and I decided they were very important. One was"Never be late."Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ' s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ' t clean and neat, so I wouldn ' t let him go. He couldn ' t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ' t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ' ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,"Don ' t run practices late, because you ' ll go home in a bad mood, and that ' s not good, for a young married man to go home in a bad mood. When you get older, it doesn ' t make any difference, but —"So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ' re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ' t want that. I used to tell them I was paid to do that. That ' s my job. I ' m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ' s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ' s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ' t have the time to go on that. But that helped me, I think, become a better teacher. It ' s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ' re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ' re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ’ t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ’ i ’. I ’ d never seen that before, but he did. Big league baseball players — they ’ re very perceptive about those things, and they noticed he had only one ’ i ’ in his name. You ’ d be surprised how many also told him that that was one more than he had in his head at **various** times. But he wrote something where I think he did what I tried to do in this pyramid. He called it”The Road Ahead, or the Road Behind.”He said,”Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ’ t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ’ s game. You and I know deeper down, there ’ s always a chance to win the crown. But when we fail to give our best, we simply haven ’ t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ’ s bearing down that wins the cup. Of dreaming there ’ s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ’ s you and I who make our fates — we open up or close the gates on the road ahead or the road behind.”Reminds me of another set of threes that my dad tried to get across to us : Don ’ t whine. Don ’ t complain. Don ’ t make excuses. Just get out there, and whatever you ’ re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ’ re outscored. I ’ ve felt that way on certain occasions, at **various** times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ’ t know the outcome, I hope they couldn ’ t tell by your actions whether you outscored an opponent or the opponent outscored you. That ’ s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ’ d want them to be but they ’ ll be about what they should ; only you will know whether you can do that. And that ’ s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,”The journey is better than the end.”And I like that. I think that it is — it ’ s getting there. Sometimes when you get there, there ’ s almost a let down. But it ’ s the getting there that ’ s the fun. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ’ d done a decent job during the week. There again, it ’ s getting the players to get that self-satisfaction, in knowing that they

’ d made the effort to do the best of which they are capable. Sometimes I ’ m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,”Suppose that you, in some way, could make the perfect player. What would you want?”And I said,”Well, I ’ d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ’ d want one that could play, too. I ’ d want one to realize that defense usually wins championships, and who would work hard on defense. But I ’ d want one who would play offense, too. I ’ d want him to be unselfish, and look for the pass first and not shoot all the time. And I ’ d want one that could pass and would pass. I ’ ve had some that could and wouldn ’ t, and I ’ ve had some that would and could. So, yeah, I ’ d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ’ d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,”They Call Me Coach,”two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ’ t play varsity when I taught. I thought,”Oh gracious, if these two players, either one of them”— they were different years, but I thought about each one at the time he was there —”Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ’ s good enough to make it.”And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ’ d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 851: POS Dim 6 – 2001 – Score 1.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am **illustrating** Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,”Are you still doing this or that?”I don ’ t answer because I ’ m not doing things still, I ’ m doing it like I always did. I still have — or did I use the word still? I didn ’ t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here.

This was about 1925. All of these things were made during the last 75 years. But, of course, I ’ m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ’ ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ’ t call myself an industrial designer because I ’ m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called”Innovation.”Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,”The playful search for beauty was Man ’ s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty.”These are tiles. The word,”playful”is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ’ t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master. In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ’ t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ’ s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ’ s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 852: POS Dim 6 – 2004 – Score 4.00 – 2004/t000098.txt

This machine, which we all have residing in our skulls, reminds me of an aphorism, of a comment of Woody Allen to ask about what is the very best thing to have within your skull. And it ’ s this machine. And it ’ s constructed for change. It ’ s all about change. It confers on us the ability to do things tomorrow that we can ’ t do today, things today that we couldn ’ t do yesterday. And of course it ’ s born stupid. The last time you were in the presence of a baby — this happens to be my granddaughter, Mitra. Isn ’ t she fabulous? But nonetheless when she popped out despite the fact that her brain had actually been progressing in its development for several months before on the basis of her experiences in the womb — nonetheless she had very limited abilities, as does every infant at the time of normal, natural full-term birth. If we were to assay her perceptual abilities, they would be crude. There is no real indication that there is any real thinking going on. In fact there is little evidence that there is any cognitive ability in a very young infant. Infants don ’ t respond to much. There is not really much of an indication in fact that there is a person on board. And they can only in a very primitive way, and in a very limited way control their movements. It would be several months before this infant could do something as simple as reach out and grasp under voluntary control an object and retrieve it, usually to the mouth. And it will be some months beforeward, and we see a long steady progression of the evolution from the first wiggles, to rolling over, and sitting up, and crawling, standing, walking, before we get to that magical point in which we can motate in the world. And yet, when we look forward in the brain we see really **remarkable** advance. By this age the brain can actually store. It has stored, recorded, can fastly retrieve the meanings of thousands, tens of thousands of objects, actions, and their relationships in the world. And those relationships can in fact be constructed in hundreds of thousands, potentially millions of ways. By this age the brain controls very refined perceptual abilities. And it actually has a growing repertoire of cognitive skills. This brain is very much a thinking machine. And by this age there is absolutely no question that this brain, it has a person on board. And in fact at this age it is substantially controlling its own self-development. And by this age we see a **remarkable** evolution in its capacity to control movement. Now movement has advanced to the point where it can actually control movement simultaneously, in a complex sequence, in complex ways as would be required for example for playing a complicated game, like soccer. Now this boy can bounce a soccer **ball** on his head. And where this boy comes from, Sao Paulo, Brazil, about 40 percent of boys of his age have this ability. You could go out into the community in Monterey, and you ’ d have difficulty finding a boy that has this ability. And if you did he ’ d probably be from Sao Paulo. That ’ s all another way of saying that our individual skills and abilities are very much shaped by our environments. That environment extends into our contemporary culture, the thing our brain is challenged with. Because what we ’ ve done in our personal evolutions is build up a large repertoire of specific skills and abilities that are specific to our own individual histories. And in fact they result in a wonderful differentiation in humankind, in the way that, in fact, no two of us are quite alike. Every one of us has a different set of acquired skills and abilities that all derive out of the plasticity, the adaptability of this really **remarkable** adaptive machine. In an adult brain of course we ’ ve built up a large repertoire of mastered skills and abilities that we can perform more or less automatically from memory, and that define us as acting, moving, thinking creatures. Now we study this, as the nerdy, laboratory, university-based scientists that we are, by engaging the brains of animals like rats, or monkeys, or of this particularly curious creature — one of the more bizarre forms of life on earth — to engage them in learning new skills and abilities. And we try to track the changes that occur as the new skill or ability is acquired. In fact we do this in individuals of any age, in these different species — that is to say from infancies, infancy up to adulthood and old age. So we might engage a rat, for example, to acquire a new skill or ability that might involve the rat using its paw to master particular manual grasp behaviors just like we might examine a child and their ability to acquire the sub-skills, or the general overall skill of accomplishing something like mastering the ability to read. Or you might look in an older individual who has mastered a complex set of abilities that might relate to reading musical notation or performing the mechanical acts of performance that apply to musical performance. From these studies we defined two great epochs of the plastic history of the brain. The first great epoch is commonly called the"Critical Period."And that is the period in which the brain is setting up in its initial form its basic processing machinery. This is actually a period of dramatic change in which it doesn ' t take learning, per se, to drive the initial differentiation of the machinery of the brain. All it takes for example in the sound domain, is exposure to sound. And the brain actually is at the mercy of the sound environment in which it is reared. So for example I can rear an animal in an environment in which there is meaningless dumb sound, a repertoire of sound that I make up, that I make, just by exposure, artificially important to the animal and its young brain. And what I see is that the animal ' s brain sets up its initial processing of that sound in a form that ' s idealized, within the limits of its processing achievements to represent it in an organized and orderly way. The sound doesn ' t have to be valuable to the animal : I could raise the animal in something that could be hypothetically valuable, like the sounds that simulate the sounds of a native language of a child. And I see the brain actually develop a processor that is specialized — specialized for that complex array, a repertoire of sounds. It actually exaggerates their separateness of representation, in multi-dimensional neuronal representational terms. Or I can expose the animal to a completely meaningless and destructive sound. I can raise an animal under conditions that would be equivalent to raising a baby under a moderately loud ceiling fan, in the presence of continuous noise. And when I do that I actually specialize the brain to be a master processor for that meaningless sound. And I frustrate its ability to represent any meaningful sound as a consequence. Such things in the early history of babies occur in real babies. And they account for, for example the beautiful evolution of a language-specific processor in every normally developing baby. And so they also account for development of defective processing in a substantial population of children who are more limited, as a consequence, in their language abilities at an older age. Now in this early period of plasticity the brain actually changes outside of a learning context. I don ' t have to be paying attention to what I hear. The input doesn ' t really have to be meaningful. I don ' t have to be in a behavioral context. This is required so the brain sets up it ' s processing so that it can act differentially, so that it can act selectively, so that the creature that wears it, that carries it, can begin to operate on it in a selective way. In the next great epoch of life, which applies for most of life, the brain is actually refining its machinery as it masters a wide repertoire of skills and abilities. And in this epoch, which extends from late in the first year of life to death ; it ' s actually doing this under behavioral control. And that ' s another way of saying the brain has strategies that define the significance of the input to the brain. And it ' s focusing on skill after skill, or ability after ability, under specific attentional control. It ' s a function of whether a goal in a behavior is achieved or whether the individual is rewarded in the behavior. This is actually very powerful. This lifelong capacity for plasticity, for brain change, is powerfully expressed. It is the basis of our real differentiation, one individual from another. You can look down in the brain of an animal that ’ s engaged in a specific skill, and you can witness or document this change on a variety of levels. So here is a very simple experiment. It was actually conducted about five years ago in collaboration with scientists from the University of Provence in Marseilles. It ’ s a very simple experiment where a monkey has been trained in a task that involves it manipulating a tool that ’ s equivalent in its difficulty to a child learning to manipulate or handle a spoon. The monkey actually mastered the task in about 700 practice tries. So in the beginning the monkey could not perform this task at all. It had a success rate of about one in eight tries. Those tries were elaborate. Each attempt was substantially different from the other. But the monkey gradually developed a strategy. And 700 or so tries later the monkey is performing it flawlessly — never fails. He ’ s successful in his retrieval of food with this tool every time. At this point the task is being performed in a beautifully stereotyped way : very beautifully regulated and highly repeated, trial to trial. We can look down in the brain of the monkey. And we see that it ’ s distorted. We can track these changes, and have tracked these changes in many such behaviors across time. And here we see the distortion reflected in the map of the skin surfaces of the hand of the monkey. Now this is a map, down in the surface of the brain, in which, in a very elaborate experiment we ’ ve reconstructed the responses, location by location, in a highly detailed response mapping of the responses of its neurons. We see here a reconstruction of how the hand is represented in the brain. We ’ ve actually distorted the map by the exercise. And that is indicated in the pink. We have a couple fingertip surfaces that are larger. These are the surfaces the monkey is using to manipulate the tool. If we look at the selectivity of responses in the cortex of the monkey, we see that the monkey has actually changed the **filter** characteristics which represents input from the skin of the fingertips that are engaged. In other words there is still a single, simple representation of the fingertips in this most organized of cortical areas of the surface of the skin of the body. Monkey has like you have. And yet now it ’ s represented in substantially finer grain. The monkey is getting more detailed information from these surfaces. And that is an unknown — unsuspected, maybe, by you — part of acquiring the skill or ability. Now actually we ’ ve looked in several different cortical areas in the monkey learning this task. And each one of them changes in ways that are specific to the skill or ability. So for example we can look to the cortical area that represents input that ’ s controlling the posture of the monkey. We look in cortical areas that control specific movements, and the sequences of movements that are required in the behavior, and so forth. They are all remodeled. They all become specialized for the task at hand. There are 15 or 20 cortical areas that are changed specifically when you learn a simple skill like this. And that represents in your brain, really massive change. It represents the change in a reliable way of the responses of tens of millions, possibly hundreds of millions of neurons in your brain. It represents changes of hundreds of millions, possibly billions of synaptic connections in your brain. This is constructed by physical change. And the level of construction that occurs is massive. Think about the changes that occur in the brain of a child through the course of acquiring their movement behavior abilities in general. Or acquiring their native language abilities. The changes are massive. What it ’ s all about is the selective representations of things that are important to the brain. Because in most of the life of the brain this is under control of behavioral context. It ’ s what you pay attention to. It ’ s what ’ s rewarding to you. It ’ s what the brain regards, itself, as positive and important to you. It ’ s all about cortical processing and forebrain specialization. And that underlies your specialization. That is why you, in your many skills and abilities, are a unique specialist : a specialist that ' s vastly different in your physical brain in detail than the brain of an individual 100 years ago ; enormously different in the details from the brain of the average individual 1, 000 years ago. Now, one of the characteristics of this change process is that information is always related to other inputs or information that is occurring in immediate time, in context. And that ' s because the brain is constructing representations of things that are correlated in little moments of time and that relate to one another in little moments of successive time. The brain is recording all information and driving all change in temporal context. Now overwhelmingly the most powerful context that ' s occurred in your brain is you. Billions of events have occurred in your history that are related in time to yourself as the receiver, or yourself as the actor, yourself as the thinker, yourself as the mover. Billions of times little pieces of sensation have come in from the surface of your body that are always associated with you as the receiver, and that result in the embodiment of you. You are constructed, your self is constructed from these billions of events. It ' s constructed. It ' s created in your brain. And it ' s created in the brain via physical change. This is a marvelously constructed thing that results in individual form because each one of us has vastly different histories, and vastly different experiences, that drive in to us this marvelous differentiation of self, of personhood. Now we ' ve used this research to try to understand not just how a normal person develops, and elaborates their skills and abilities, but also try to understand the origins of impairment, and the origins of differences or variations that might limit the capacities of a child, or an adult. I ' m going to talk about using these strategies to actually design brain plasticity-based approach to drive corrections in the machinery of a child that increases the competence of the child as a language receiver and user and, thereafter, as a reader. And I ' m going to talk about experiments that involve actually using this brain science, first of all to understand how it contributes to the loss of function as we age. And then, by using it in a targeted approach we ' re going to try to differentiate the machinery to recover function in old age. So the first example I ' m going to talk about relates to children with learning impairments. We now have a large body of literature that demonstrates that the fundamental problem that occurs in the majority of children that have early language impairments, and that are going to struggle to learn to read, is that their language processor is created in a defective form. And the reason that it rises in a defective form is because early in the baby ' s brain ' s life the machine process is noisy. It ' s that simple. It ' s a signal-to-noise problem. Okay? And there are a lot of things that contribute to that. There are numerous inherited faults that could make the machine process noisier. Now I might say the noise problem could also occur on the basis of information provided in the world from the ears. If any — those of you who are older in the audience know that when I was a child we understood that a child born with a cleft palate was born with what we called mental retardation. We knew that they were going to be slow cognitively ; we knew they were going to struggle to learn to develop normal language abilities ; and we knew that they were going to struggle to learn to read. Most of them would be intellectual and academic failures. That ' s disappeared. That no longer applies. That inherited weakness, that inherited condition has evaporated. We don ' t hear about that anymore. Where did it go? Well, it was understood by a Dutch surgeon, about 35 years ago, that if you simply fix the problem early enough, when the brain is still in this initial plastic period so it can set up this machinery adequately, in this initial set up time in the critical period, none of that happens. What are you doing by operating on the cleft palate to correct it? You ' re basically opening up the tubes that drain fluid from the middle ears, which have had them reliably full. Every sound the child hears uncorrected is muffled. It ' s degraded. The child ' s native language is such a case is not English. It ' s not Japanese. It ' s muffled English. It ' s degraded Japanese. It ' s crap. And the brain specializes for it. It creates a representation of language crap. And then the child is stuck with it. Now the crap doesn ' t just happen in the ear. It can also happen in the brain. The brain itself can be noisy. It ' s commonly noisy. There are many inherited faults that can make it noisier. And the native language for a child with such a brain is degraded. It ' s not English. It ' s noisy English. And that results in defective representations of sounds of words — not normal — a different strategy, by a machine that has different time constants and different space constants. And you can look in the brain of such a child and record those time constants. They are about an order of magnitude longer, about 11 times longer in duration on average, than in a normal child. Space constants are about three times greater. Such a child will have memory and cognitive deficits in this domain. Of course they will. Because as a receiver of language, they are receiving it and representing it, and in information it ' s representing crap. And they are going to have poor reading skills. Because reading is dependent upon the translation of word sounds into this orthographic or visual representational form. If you don ' t have a brain representation of word sounds that translation makes no sense. And you are going to have corresponding abnormal neurology. Then these children increasingly in evaluation after evaluation, in their operations in language, and their operations in reading — we document that abnormal neurology. The point is is that you can train the brain out of this. A way to think about this is you can actually re-refine the processing capacity of the machinery by changing it. Changing it in detail. It takes about 30 hours on the average. And we ' ve accomplished that in about 430, 000 kids today. Actually, probably about 15, 000 children are being trained as we speak. And actually when you look at the impacts, the impacts are substantial. So here we ' re looking at the normal distribution. What we ' re most interested in is these kids on the left side of the distribution. This is from about 3, 000 children. You can see that most of the children on the left side of the distribution are moving into the middle or the right. This is in a broad assessment of their language abilities. This is like an IQ test for language. The impact in the distribution, if you trained every child in the United States, would be to shift the whole distribution to the right and narrow the distribution. This is a substantially large impact. Think of a classroom of children in the language arts. Think of the children on the slow side of the class. We have the potential to move most of those children to the middle or to the right side. In addition to accurate language training it also fixes memory and cognition speech fluency and speech production. And an important language dependent skill is enabled by this training — that is to say reading. And to a large extent it fixes the brain. You can look down in the brain of a child in a variety of tasks that scientists have at Stanford, and MIT, and UCSF, and UCLA, and a number of other institutions. And children operating in **various** language behaviors, or in **various** reading behaviors, you see for the most extent, for most children, their neuronal responses, complexly abnormal before you start, are normalized by the training. Now you can also take the same approach to address problems in aging. Where again the machinery is deteriorating now from competent machinery, it ' s going south. Noise is increasing in the brain. And learning modulation and control is deteriorating. And you can actually look down on the brain of such an individual and witness a change in the time constants and space constants with which, for example, the brain is representing language again. Just as the brain came out of chaos at the beginning, it ' s going back into chaos in the end. This results in declines in memory in cognition, and in postural ability and agility. It turns out you can train the brain of such an individual — this is a small population of such individuals — train equally intensively for about 30 hours. These are 80- to 90-year-olds. And what you see are substantial improvements of their immediate memory, of their ability to remember things after a delay, of their ability to control their attention, their language abilities and visual-spatial abilities. The overall neuropsychological index of these trained individuals in this population is about two standard deviations. That means that if you sit at the left side of the distribution, and I ' m looking at your neuropyschological abilities, the average person has moved to the middle or the right side of the distribution. It means that most people who are at risk for senility, more or less immediately, are now in a protected position. My issues are to try to get to rescuing older citizens more completely and in larger numbers, because I think this can be done in this arena on a vast scale — and the same for kids. My main interest is how to elaborate this science to address other maladies. I ' m specifically interested in things like autism, and cerebral palsy, these great childhood catastrophes. And in older age conditions like Parkinsonism, and in other acquired impairments like schizophrenia. Your issues as it relates to this science, is how to maintain your own high- functioning learning machine. And of course, a well-ordered life in which learning is a continuous part of it, is key. But also in your future is brain aerobics. Get ready for it. It ' s going to be a part of every life not too far in the future, just like physical exercise is a part of every well organized life in the contemporary period. The other way that we will ultimately come to consider this literature and the science that is important to you is in a consideration of how to nurture yourself. Now that you know, now that science is telling us that you are in charge, that it ' s under your control, that your happiness, your well-being, your abilities, your capacities, are capable of continuous modification, continuous improvement, and you ' re the responsible agent and party. Of course a lot of people will ignore this advice. It will be a long time before they really understand it. Now that ' s another issue and not my fault. Okay. Thank you.

Text Sample 853: POS Dim 6 – 2004 – Score 4.00 – 2004/t000361.txt

So, I guess it is a result of globalization that you can find Coca-Cola tins on top of Everest and a Buddhist monk in Monterey. And so I just came, two days ago, from the Himalayas to your kind invitation. So I would like to invite you, also, for a while, to the Himalayas themselves. And to show the place where meditators, like me, who began with being a molecular biologist in Pasteur Institute, and found their way to the mountains. So these are a few images I was lucky to take and be there. There ' s Mount Kailash in Eastern Tibet — wonderful setting. This is from Marlboro country. This is a turquoise lake. A meditator. This is the hottest day of the year somewhere in Eastern Tibet, on August 1. And the night before, we camped, and my Tibetan friends said,"We are going to sleep outside."And I said,"Why? We have enough space in the tent."They said,"Yes, but it ' s summertime."So now, we are going to speak of happiness. As a Frenchman, I must say that there are a lot of French intellectuals that think happiness is not at all interesting. I just wrote an essay on happiness, and there was a controversy. And someone wrote an article saying,"Don ' t impose on us the dirty work of happiness.""We don ' t care about being happy. We need to live with passion. We like the ups and downs of life. We like our suffering because it ' s so good when it ceases for a while."This is what I see from the balcony of my hermitage in the Himalayas. It ’ s about two meters by three, and you are all welcome any time. Now, let ’ s come to happiness or well-being. And first of all, you know, despite what the French intellectuals say, it seems that no one wakes up in the morning thinking,”May I suffer the whole day?”Which means that somehow, consciously or not, directly or indirectly, in the short or the long term, whatever we do, whatever we hope, whatever we dream — somehow, is related to a deep, profound desire for well-being or happiness. As Pascal said, even the one who hangs himself, somehow, is looking for cessation of suffering. He finds no other way. But then, if you look in the literature, East and West, you can find incredible diversity of definition of happiness. Some people say, I only believed in remembering the past, imagining the future, never the present. Some people say happiness is right now ; it ’ s the quality of the freshness of the present moment. And that led Henri Bergson, the French philosopher, to say,”All the great thinkers of humanity have left happiness in the vague so that each of them could define their own terms.”Well, that would be fine if it was just a secondary preoccupation in life. But now, if it is something that is going to determine the quality of every instant of our life, then we better know what it is, have some clearer idea. And probably, the fact that we don ’ t know that is why, so often, although we seek happiness, it seems we turn our back to it. Although we want to avoid suffering, it seems we are running somewhat towards it. And that can also come from some kind of confusions. One of the most common ones is happiness and pleasure. But if you look at the characteristics of those two, pleasure is contingent upon time, upon its object, upon the place. It is something that — changes of nature. Beautiful chocolate cake : first serving is delicious, second one not so much, then we feel disgust. That ’ s the nature of things. We get tired. I used to be a fan of Bach. I used to play it on the guitar, you know. I can hear it two, three, five times. If I had to hear it 24 hours, non-stop, it might be very tiring. If you are feeling very cold, you come near a fire, it ’ s so wonderful. After some moments, you just go a little back, and then it starts burning. It sort of uses itself as you experience it. And also, again, it can — also, it ’ s something that you — it is not something that is radiating outside. Like, you can feel intense pleasure and some others around you can be suffering a lot. Now, what, then, will be happiness? And happiness, of course, is such a vague word, so let ’ s say well-being. And so, I think the best definition, according to the Buddhist view, is that well-being is not just a mere **pleasurable** sensation. It is a deep sense of serenity and fulfillment. A state that actually pervades and underlies all emotional states, and all the joys and sorrows that can come one ’ s way. For you, that might be surprising. Can we have this kind of well-being while being sad? In a way, why not? Because we are speaking of a different level. Look at the waves coming near the shore. When you are at the bottom of the wave, you hit the bottom. You hit the solid rock. When you are surfing on the top, you are all elated. So you go from elation to depression — there ’ s no depth. Now, if you look at the high sea, there might be beautiful, calm ocean, like a mirror. There might be storms, but the depth of the ocean is still there, unchanged. So now, how is that? It can only be a state of being, not just a fleeting emotion, sensation. Even joy — that can be the spring of happiness. But there ’ s also wicked joy, you can rejoice in someone ’ s suffering. So how do we proceed in our quest for happiness? Very often, we look outside. We think that if we could gather this and that, all the conditions, something that we say,”Everything to be happy — to have everything to be happy.”That very sentence already reveals the doom, destruction of happiness. To have everything. If we miss something, it collapses. And also, when things go wrong, we try to fix the outside so much, but our control of the outer world is limited, temporary, and often, illusory. So now, look at inner conditions. Aren ’ t they stronger? Isn ’ t it the mind that translates the outer condition into happiness and suffering? And isn ’ t that stronger? We know, by experience, that we can be what we call”a little paradise,”and yet, be completely unhappy within. The Dalai Lama was once in Portugal, and there was a lot of construction going on everywhere. So one evening, he said,”Look, you are doing all these things, but isn ’ t it nice, also, to build something within?”And he said,”[Without] that — even if you get a high-tech flat on the 100th floor of a super-modern and comfortable building, if you are deeply unhappy within, all you are going to look for is a window from which to jump.”So now, at the opposite, we know a lot of people who, in very difficult circumstances, manage to keep serenity, inner strength, inner freedom, confidence. So now, if the inner conditions are stronger — of course, the outer conditions do influence, and it ’ s wonderful to live longer, healthier, to have access to information, education, to be able to travel, to have freedom. It ’ s highly desirable. However, this is not enough. Those are just auxiliary, help conditions. The experience that translates everything is within the mind. So then, when we ask oneself how to nurture the condition for happiness, the inner conditions, and which are those which will undermine happiness. So then, this just needs to have some experience. We have to know from ourselves, there are certain states of mind that are conducive to this flourishing, to this well-being, what the Greeks called eudaimonia, flourishing. There are some which are adverse to this well-being. And so, if we look from our own experience, anger, hatred, jealousy, arrogance, obsessive desire, strong grasping, they don ’ t leave us in such a good state after we have experienced it. And also, they are detrimental to others ’ happiness. So we may consider that the more those are invading our mind, and, like a chain reaction, the more we feel miserable, we feel tormented. At the opposite, everyone knows deep within that an act of selfless generosity, if from the distance, without anyone knowing anything about it, we could save a child ’ s life, make someone happy. We don ’ t need the recognition. We don ’ t need any gratitude. Just the mere fact of doing that fills such a sense of adequation with our deep nature. And we would like to be like that all the time. So is that possible, to change our way of being, to transform one ’ s mind? Aren ’ t those negative emotions, or destructive emotions, inherent to the nature of mind? Is change possible in our emotions, in our traits, in our moods? For that we have to ask, what is the nature of mind? And if we look from the experiential point of view, there is a primary quality of consciousness that ’ s just the mere fact to be cognitive, to be aware. Consciousness is like a mirror that allows all images to rise on it. You can have ugly faces, beautiful faces in the mirror. The mirror allows that, but the mirror is not tainted, is not modified, is not altered by those images. Likewise, behind every single thought there is the bare consciousness, pure awareness. This is the nature. It cannot be tainted intrinsically with hatred or jealousy because then, if it was always there — like a dye that would permeate the whole cloth — then it would be found all the time, somewhere. We know we ’ re not always angry, always jealous, always generous. So, because the basic fabric of consciousness is this pure cognitive quality that differentiates it from a stone, there is a possibility for change because all emotions are fleeting. That is the ground for mind training. Mind training is based on the idea that two opposite mental factors cannot happen at the same time. You could go from love to hate. But you cannot, at the same time, toward the same object, the same person, want to harm and want to do good. You cannot, in the same gesture, shake hand and give a blow. So, there are natural antidotes to emotions that are destructive to our inner well-being. So that ’ s the way to proceed. Rejoicing compared to jealousy. A kind of sense of inner freedom as opposite to intense grasping and obsession. Benevolence, loving kindness against hatred. But, of course, each emotion then would need a particular antidote. Another way is to try to find a general antidote to all emotions, and that ' s by looking at the very nature. Usually, when we feel annoyed, hatred or upset with someone, or obsessed with something, the mind goes again and again to that object. Each time it goes to the object, it reinforces that obsession or that annoyance. So then, it ' s a self-perpetuating process. So what we need to look for now is, instead of looking outward, we look inward. Look at anger itself. It looks very menacing, like a billowing monsoon cloud or thunderstorm. We think we could sit on the cloud, but if you go there, it ' s just mist. Likewise, if you look at the thought of anger, it will vanish like frost under the morning sun. If you do this again and again, the propensity, the tendencies for anger to arise again will be less and less each time you dissolve it. And, at the end, although it may rise, it will just cross the mind, like a bird crossing the sky without leaving any track. So this is the principal of mind training. Now, it takes time, because it took time for all those faults in our mind, the tendencies, to build up, so it will take time to unfold them as well. But that ' s the only way to go. Mind transformation — that is the very meaning of meditation. It means familiarization with a new way of being, new way of perceiving things, which is more in adequation with reality, with interdependence, with the stream and continuous transformation, which our being and our consciousness is. So, the **interface** with cognitive science, since we need to come to that, it was, I suppose, the subject of — we have to deal in such a short time — with brain plasticity. The brain was thought to be more or less fixed. All the nominal connections, in numbers and quantities, were thought, until the last 20 years, to be more or less fixed when we reached adult age. Now, recently, it has been found that it can change a lot. A violinist, as we heard, who has done 10, 000 hours of violin practice, some area that controls the movements of fingers in the brain changes a lot, increasing reinforcement of the synaptic connections. So can we do that with human qualities? With loving kindness, with patience, with openness? So that ' s what those great meditators have been doing. Some of them who came to the labs, like in Madison, Wisconsin, or in Berkeley, did 20 to 40, 000 hours of meditation. They do, like, three years ' retreat, where they do meditate 12 hours a day. And then, the rest of their life, they will do three or four hours a day. They are real Olympic champions of mind training. This is the place where the meditators — you can see it ' s kind of inspiring. Now, here with 256 electrodes. So what did they find? Of course, same thing. The scientific embargo — a paper has been submitted to"Nature,"hopefully, it will be accepted. It deals with the state of compassion, unconditional compassion. We asked meditators, who have been doing that for years and years, to put their mind in a state where there ' s nothing but loving kindness, total availability to sentient being. Of course, during the training, we do that with objects. We think of people suffering, of people we love, but at some point, it can be a state which is all pervading. Here is the preliminary result, which I can show because it ' s already been shown. The bell curve shows 150 controls, and what is being looked at is the difference between the right and the left frontal lobe. In very short, people who have more activity in the right side of the prefrontal cortex are more depressed, withdrawn. They don ' t describe a lot of positive affect. It ' s the opposite on the left side : more tendency to altruism, to happiness, to express, and curiosity and so forth. So there ' s a basic line for people. And also, it can be changed. If you see a comic **movie**, you go off to the left side. If you are happy about something, you ' ll go more to the left side. If you have a bout of depression, you ' ll go to the right side. Here, the -0. 5 is the full standard deviation of a meditator who meditated on compassion. It ' s something that is totally out of the bell curve. So, I ' ve no time to go into all the different scientific results. Hopefully, they will come. But they found that — this is after three and a half hours in an fMRI, it ' s like coming out of a space ship. Also, it has been shown in other labs — for instance, Paul Ekman ' s labs in Berkeley — that some meditators are able, also, to control their emotional response more than it could be thought. Like the startle experiments, for example. If you sit a guy on a chair with all this apparatus measuring your physiology, and there ' s kind of a bomb that goes off, it ' s such an instinctive response that, in 20 years, they never saw anyone who would not jump. Some meditators, without trying to stop it, but simply by being completely open, thinking that that **bang** is just going to be a small event like a shooting star, they are able not to move at all. So the whole point of that is not, sort of, to make, like, a circus thing of showing exceptional beings who can jump, or whatever. It ' s more to say that mind training matters. That this is not just a luxury. This is not a supplementary vitamin for the soul. This is something that ' s going to determine the quality of every instant of our lives. We are ready to spend 15 years achieving education. We love to do jogging, fitness. We do all kinds of things to remain beautiful. Yet, we spend surprisingly little time taking care of what matters most — the way our mind functions — which, again, is the ultimate thing that determines the quality of our experience. Now, compassion is supposed to be put in action. That ' s what we try to do in different places. Just this one example is worth a lot of work. This lady with bone TB, left alone in a tent, was going to die with her only daughter. One year later, how she is. Different schools and clinics we ' ve been doing in Tibet. And just, I leave you with the beauty of those looks that tells more about happiness than I could ever say. And jumping monks of Tibet. Flying monks. Thank you very much.

Text Sample 854: POS Dim 6 – 2004 – Score 3.00 – 2004/t000196.txt

I grew up in Europe, and World War II caught me when I was between seven and 10 years old. And I realized how few of the grown-ups that I knew were able to withstand the tragedies that the war visited on them â how few of them could even resemble a normal, contented, satisfied, happy life once their job, their home, their security was destroyed by the war. So I became interested in understanding what contributed to a life that was worth living. And I tried, as a child, as a teenager, to read philosophy and to get involved in art and religion and many other ways that I could see as a possible answer to that question. And finally I ended up encountering psychology by chance. I was at a ski resort in Switzerland without any money to actually enjoy myself, because the snow had melted and I didn ' t have money to go to a **movie**. But I found that on the â I read in the newspapers that there was to be a **presentation** by someone in a place that I ' d seen in the center of Zurich, and it was about flying saucers [that] he was going to talk. And I thought, well, since I can ' t go to the **movies**, at least I will go for free to listen to flying saucers. And the man who talked at that evening lecture was very interesting. Instead of talking about little green men, he talked about how the psyche of the Europeans had been traumatized by the war, and now they ' re projecting flying saucers into the sky. He talked about how the mandalas of ancient Hindu religion were kind of projected into the sky as an attempt to regain some sense of order after the chaos of war. And this seemed very interesting to me. And I started reading his books after that lecture. And that was Carl Jung, whose name or work I had no idea about. Then I came to this country to study psychology and I started trying to understand the roots of happiness. This is a **typical** result that many people have presented, and there are many variations on it. But this, for instance, shows that about 30 percent of the people surveyed in the United States since 1956 say that their life is very happy. And that hasn ’ t changed at all. Whereas the personal income, on a scale that has been held constant to accommodate for inflation, has more than doubled, almost tripled, in that period. But you find essentially the same results, namely, that after a certain basic point â which corresponds more or less to just a few 1, 000 dollars above the minimum poverty level â increases in material well-being don ’ t seem to affect how happy people are. In fact, you can find that the lack of basic resources, material resources, contributes to unhappiness, but the increase in material resources does not increase happiness. So my research has been focused more on â after finding out these things that actually corresponded to my own experience, I tried to understand : where â in everyday life, in our normal experience â do we feel really happy? And to start those studies about 40 years ago, I began to look at creative people â first artists and scientists, and so forth â trying to understand what made them feel that it was worth essentially spending their life doing things for which many of them didn ’ t expect either fame or fortune, but which made their life meaningful and worth doing. This was one of the leading composers of American music back in the ’ 70s. And the interview was 40 pages long. But this little excerpt is a very good summary of what he was saying during the interview. And it describes how he feels when composing is going well. And he says by describing it as an ecstatic state. Now,”ecstasy”in Greek meant simply to stand to the side of something. And then it became essentially an analogy for a mental state where you feel that you are not doing your ordinary everyday routines. So ecstasy is essentially a stepping into an alternative reality. And it ’ s interesting, if you think about it, how, when we think about the civilizations that we look up to as having been pinnacles of human achievement â whether it ’ s China, Greece, the Hindu civilization, or the Mayas, or Egyptians â what we know about them is really about their ecstasies, not about their everyday life. We know the temples they built, where people could come to experience a different reality. We know about the circuses, the arenas, the theaters. These are the remains of civilizations and they are the places that people went to experience life in a more concentrated, more ordered form. Now, this man doesn ’ t need to go to a place like this, which is also â this place, this arena, which is built like a Greek amphitheatre, is a place for ecstasy also. We are participating in a reality that is different from that of the everyday life that we ’ re used to. But this man doesn ’ t need to go there. He needs just a piece of paper where he can put down little marks, and as he does that, he can imagine sounds that had not existed before in that particular combination. So once he gets to that point of beginning to create, like Jennifer did in her improvisation, a new reality â that is, a moment of ecstasy â he enters that different reality. Now he says also that this is so intense an experience that it feels almost as if he didn ’ t exist. And that sounds like a kind of a romantic exaggeration. But actually, our nervous system is incapable of processing more than about 110 bits of information per second. And in order to hear me and understand what I ’ m saying, you need to process about 60 bits per second. That ’ s why you can ’ t hear more than two people. You can ’ t understand more than two people talking to you. Well, when you are really involved in this completely engaging process of creating something new, as this man is, he doesn ’ t have enough attention left over to monitor how his body feels, or his problems at home. He can ’ t feel even that he ’ s hungry or tired. His body disappears, his identity disappears from his consciousness, because he doesn ’ t have enough attention, like none of us do, to really do well something that requires a lot of concentration, and at the same time to feel that he exists. So existence is temporarily suspended. And he says that his hand seems to be moving by itself. Now, I could look at my hand for two weeks, and I wouldn ’ t feel any awe or wonder, because I can ’ t compose. So what it ’ s telling you here is that obviously this automatic, spontaneous process that he ’ s describing can only happen to someone who is very well trained and who has developed technique. And it has become a kind of a truism in the study of creativity that you can ’ t be creating anything with less than 10 years of technical-knowledge immersion in a particular field. Whether it ’ s mathematics or music, it takes that long to be able to begin to change something in a way that it ’ s better than what was there before. Now, when that happens, he says the music just flows out. And because all of these people I started interviewing â this was an interview which is over 30 years old â so many of the people described this as a spontaneous flow that I called this type of experience the”flow experience.”And it happens in different realms. For instance, a poet describes it in this form. This is by a student of mine who interviewed some of the leading writers and poets in the United States. And it describes the same effortless, spontaneous feeling that you get when you enter into this ecstatic state. This poet describes it as opening a door that floats in the sky â a very similar description to what Albert Einstein gave as to how he imagined the forces of relativity, when he was struggling with trying to understand how it worked. But it happens in other activities. For instance, this is another student of mine, Susan Jackson from Australia, who did work with some of the leading athletes in the world. And you see here in this description of an Olympic skater, the same essential description of the phenomenology of the inner state of the person. You don ’ t think ; it goes automatically, if you merge yourself with the music, and so forth. It happens also, actually, in the most recent book I wrote, called”Good Business,”where I interviewed some of the CEOs who had been nominated by their peers as being both very successful and very ethical, very socially responsible. You see that these people define success as something that helps others and at the same time makes you feel happy as you are working at it. And like all of these successful and responsible CEOs say, you can ’ t have just one of these things be successful if you want a meaningful and successful job. Anita Roddick is another one of these CEOs we interviewed. She is the founder of Body Shop, the natural cosmetics king. It ’ s kind of a passion that comes from doing the best and having flow while you ’ re working. This is an interesting little quote from Masaru Ibuka, who was at that time starting out Sony without any money, without a product â they didn ’ t have a product, they didn ’ t have anything, but they had an idea. And the idea he had was to establish a place of work where engineers can feel the joy of technological innovation, be aware of their mission to society and work to their heart ’ s content. I couldn ’ t improve on this as a good example of how flow enters the workplace. Now, when we do studies â we have, with other colleagues around the world, done over 8, 000 interviews of people â from Dominican monks, to blind nuns, to Himalayan climbers, to Navajo shepherds â who enjoy their work. And regardless of the culture, regardless of education or whatever, there are these seven conditions that seem to be there when a person is in flow. There ’ s this focus that, once it becomes intense, leads to a sense of ecstasy, a sense of clarity : you know exactly what you want to do from one moment to the other ; you get immediate feedback. You know that what you need to do is possible to do, even though difficult, and sense of time disappears, you forget yourself, you feel part of something larger. And once the conditions are present, what you are doing becomes worth doing for its own sake. In our studies, we represent the everyday life of people in this simple scheme. And we can measure this very precisely, actually, because we give people electronic pagers that go off 10 times a day, and whenever they go off you say what you ’ re doing, how you feel, where you are, what you ’ re thinking about. And two things that we measure is the amount of challenge people experience at that moment and the amount of skill that they feel they have at that moment. So for each person we can establish an average, which is the center of the diagram. That would be your mean level of challenge and skill, which will be different from that of anybody else. But you have a kind of a set point there, which would be in the middle. If we know what that set point is, we can predict fairly accurately when you will be in flow, and it will be when your challenges are higher than average and skills are higher than average. And you may be doing things very differently from other people, but for everyone that flow channel, that area there, will be when you are doing what you really like to do â play the piano, be with your best friend, perhaps work, if work is what provides flow for you. And then the other areas become less and less positive. Arousal is still good because you are over-challenged there. Your skills are not quite as high as they should be, but you can move into flow fairly easily by just developing a little more skill. So, arousal is the area where most people learn from, because that ’ s where they ’ re pushed beyond their comfort zone and to enter that â going back to flow â then they develop higher skills. Control is also a good place to be, because there you feel comfortable, but not very excited. It ’ s not very challenging any more. And if you want to enter flow from control, you have to increase the challenges. So those two are ideal and complementary areas from which flow is easy to go into. The other combinations of challenge and skill become progressively less optimal. Relaxation is fine â you still feel OK. Boredom begins to be very aversive and apathy becomes very negative : you don ’ t feel that you ’ re doing anything, you don ’ t use your skills, there ’ s no challenge. Unfortunately, a lot of people ’ s experience is in apathy. The largest single contributor to that experience is watching television ; the next one is being in the bathroom, sitting. Even though sometimes watching television about seven to eight percent of the time is in flow, but that ’ s when you choose a program you really want to watch and you get feedback from it. So the question we are trying to address â and I ’ m way over time â is how to put more and more of everyday life in that flow channel. And that is the kind of challenge that we ’ re trying to understand. And some of you obviously know how to do that spontaneously without any advice, but unfortunately a lot of people don ’ t. And that ’ s what our mandate is, in a way, to do. Thank you.

Text Sample 855: POS Dim 6 – 2009 – Score 5.00 – 2009/t002924.txt

I want you to put off your preconceptions, your preconceived fears and thoughts about reptiles. Because that is the only way I ’ m going to get my story across to you. And by the way, if I come across as a sort of rabid, hippie conservationist, it ’ s purely a figment of your imagination. Okay. We are actually the first species on Earth to be so prolific to actually threaten our own survival. And I know we ’ ve all seen images enough to make us numb, of the tragedies that we ’ re perpetrating on the planet. We ’ re kind of like greedy kids, using it all up, aren ’ t we? And today is a time for me to talk to you about water. It ’ s not only because we like to drink lots of it, and its marvelous derivatives, beer, wine, etc. And, of course, watch it fall from the sky and flow in our wonderful rivers, but for several other reasons as well. When I was a kid, growing up in New York, I was smitten by snakes, the same way most kids are smitten by tops, marbles, cars, trains, cricket **balls**. And my mother, brave lady, was partly to blame, taking me to the New York Natural History Museum, buying me books on snakes, and then starting this infamous career of mine, which has culminated in of course, arriving in India 60 years ago, brought by my mother, Doris Norden, and my stepfather, Rama Chattopadhyaya. It ' s been a roller coaster ride. Two animals, two iconic reptiles really captivated me very early on. One of them was the **remarkable** gharial. This crocodile, which grows to almost 20 feet long in the northern rivers, and this charismatic snake, the king cobra. What my purpose of the talk today really is, is to sort of indelibly scar your minds with these charismatic and majestic creatures. Because this is what you will take away from here, a reconnection with nature, I hope. The king cobra is quite **remarkable** for several reasons. What you ' re seeing here is very recently shot images in a forest nearby here, of a female king cobra making her nest. Here is a limbless animal, capable of gathering a huge mound of leaves, and then laying her eggs inside, to withstand 5 to 10 [meters of rainfall], in order that the eggs can incubate over the next 90 days, and hatch into little baby king cobras. So, she protects her eggs, and after three months, the babies finally do hatch out. A majority of them will die, of course. There is very high mortality in little baby reptiles who are just 10 to 12 inches long. My first experience with king cobras was in ' 72 at a magical place called Agumbe, in Karnataka, this state. And it is a marvelous rain forest. This first encounter was kind of like the Maasai boy who kills the lion to become a warrior. It really changed my life totally. And it brought me straight into the conservation fray. I ended up starting this research and education station in Agumbe, which you are all of course invited to visit. This is basically a base wherein we are trying to gather and learn virtually everything about the biodiversity of this incredibly complex forest system, and try to hang on to what ' s there, make sure the water sources are protected and kept clean, and of course, having a good time too. You can almost hear the drums throbbing back in that little cottage where we stay when we ' re there. It was very important for us to get through to the people. And through the children is usually the way to go. They are fascinated with snakes. They haven ' t got that steely thing that you end up either fearing or hating or despising or loathing them in some way. They are interested. And it really works to start with them. This gives you an idea of the size of some of these snakes. This is an average size king cobra, about 12 feet long. And it actually crawled into somebody ' s bathroom, and was hanging around there for two or three days. The people of this part of India worship the king cobra. And they didn ' t kill it. They called us to catch it. Now we ' ve caught more than 100 king cobras over the last three years, and relocated them in nearby forests. But in order to find out the real secrets of these creatures [it was necessary] for us to actually insert a small radio transmitter inside [each] snake. Now we are able to follow them and find out their secrets, where the babies go after they hatch, and **remarkable** things like this you ' re about to see. This was just a few days ago in Agumbe. I had the pleasure of being close to this large king cobra who had caught a venomous pit viper. And it does it in such a way that it doesn ' t get bitten itself. And king cobras feed only on snakes. This [little snake] was kind of a tid-bit for it, what we ' d call a"vadai"or a donut or something like that. Usually they eat something a bit larger. In this case a rather strange and inexplicable activity happened over the last breeding season, wherein a large male king cobra actually grabbed a female king cobra, didn ' t mate with it, actually killed it and swallowed it. We ' re still trying to explain and come to terms with what is the evolutionary advantage of this. But they do also a lot of other **remarkable** things. This is again, something [we were able to see] by virtue of the fact that we had a radio transmitter in one of the snakes. This male snake, 12 feet long, met another male king cobra. And they did this incredible ritual combat dance. It ' s very much like the rutting of mammals, including humans, you know, sorting out our differences, but gentler, no biting allowed. It ' s just a wresting match, but a **remarkable** activity. Now, what are we doing with all this information? What ' s the point of all this? Well, the king cobra is literally a keystone species in these rainforests. And our job is to convince the authorities that these forests have to be protected. And this is one of the ways we do it, by learning as much as we can about something so **remarkable** and so iconic in the rainforests there, in order to help protect trees, animals and of course the water sources. You ' ve all heard, perhaps, of Project Tiger which started back in the early ' 70s, which was, in fact, a very dynamic time for conservation. We were piloted, I could say, by a highly autocratic stateswoman, but who also had an incredible passion for environment. And this is the time when Project Tiger emerged. And, just like Project Tiger, our activities with the king cobra is to look at a species of animal so that we protect its habitat and everything within it. So, the tiger is the icon. And now the king cobra is a new one. All the major rivers in south India are sourced in the Western Ghats, the chain of hills running along the west coast of India. It pours out millions of gallons every hour, and supplies drinking water to at least 300 million people, and washes many, many babies, and of course feeds many, many animals, both domestic and wild, produces thousands of tons of rice. And what do we do? How do we respond to this? Well, basically, we dam it, we pollute it, we pour in pesticides, weedicides, fungicides. You drink it in peril of your life. And the thing is, it ' s not just big industry. It ' s not misguided river engineers who are doing all this ; it ' s us. It seems that our citizens find the best way to dispose of garbage are in water sources. Okay. Now we ' re going north, very far north. North central India, the Chambal River is where we have our base. This is the home of the gharial, this incredible crocodile. It is an animal which has been on the Earth for just about 100 million years. It survived even during the time that the dinosaurs died off. It has **remarkable** features. Even though it grows to 20 feet long, since it eats only fish it ' s not dangerous to human beings. It does have big teeth, however, and it ' s kind of hard to convince people if an animal has big teeth, that it ' s a harmless creature. But we, actually, back in the early ' 70s, did surveys, and found that gharial were extremely rare. In fact, if you see the map, the range of their original habitat was all the way from the Indus in Pakistan to the Irrawaddy in Burma. And now it ' s just limited to a couple of spots in Nepal and India. So, in fact at this point there are only 200 breeding gharial left in the wild. So, starting in the mid- ' 70s when conservation was at the fore, we were actually able to start projects which were basically government supported to collect eggs from the wild from the few remaining nests and release 5, 000 baby gharial back to the wild. And pretty soon we were seeing sights like this. I mean, just incredible to see bunches of gharial basking on the river again. But complacency does have a tendency to breed contempt. And, sure enough, with all the other pressures on the river, like sand mining, for example, very, very heavy cultivation all the way down to the river ' s edge, not allowing the animals to breed anymore, we ' re looking at even more problems building up for the gharial, despite the early good intentions. Their nests hatching along the riverside producing hundreds of hatchlings. It ' s just an amazing sight. This was actually just taken last year. But then the monsoon arrives, and unfortunately downriver there is always a dam or there is always a barrage, and, shoop, they get washed down to their doom. Luckily there is still a lot of interest. My pals in the Crocodile Specialist Group of the IUCN, the [Madras Crocodile Bank], an NGO, the World Wildlife Fund, the Wildlife Institute of India, State Forest Departments, and the Ministry of Environment, we all work together on stuff. But it ' s possibly, and definitely not enough. For example, in the winter of 2007 and 2008, there was this incredible die-off of gharial, in the Chambal River. Suddenly dozens of gharial appearing on the river, dead. Why? How could it happen? This is a relatively clean river. The Chambal, if you look at it, has clear water. People scoop water out of the Chambal and drink it, something you wouldn ’ t do in most north Indian rivers. So, in order to try to find out the answer to this, we got veterinarians from all over the world working with Indian vets to try to figure out what was happening. I was there for a lot of the necropsies on the riverside. And we actually looked through all their organs and tried to figure out what was going on. And it came down to something called gout, which, as a result of kidney breakdown is actually uric acid crystals throughout the body, and worse in the joints, which made the gharial unable to swim. And it ’ s a horribly painful death. Just downriver from the Chambal is the filthy Yamuna river, the sacred Yamuna river. And I hate to be so ironic and sarcastic about it but it ’ s the truth. It ’ s just one of the filthiest cesspools you can imagine. It flows down through Delhi, Mathura, Agra, and gets just about every bit of effluent you can imagine. So, it seemed that the toxin that was killing the gharial was something in the food chain, something in the fish they were eating. And, you know, once a toxin is in the food chain everything is affected, including us. Because these rivers are the lifeblood of people all along their course. In order to try to answer some of these questions, we again turn to technology, to biological technology, in this case, again, telemetry, putting radios on 10 gharial, and actually following their movements. They ’ re being watched everyday as we speak, to try to find out what this mysterious toxin is. The Chambal river is an absolutely incredible place. It ’ s a place that ’ s famous to a lot of you who know about the bandits, the dacoits who used to work up there. And there still are quite a few around. But Poolan Devi was one [of them]. Which actually Shekhar Kapur made an incredible **movie**,”The Bandit Queen,”which I urge you to see. You ’ ll get to see the incredible [Chambal] landscape as well. But, again, heavy fishing pressures. This is one of the last repositories of the Ganges river dolphin, **various** species of turtles, thousands of migratory birds, and fishing is causing problems like this. And now [these] new elements of human intolerance for river creatures like the gharial means that if they don ’ t drown in the net, then they simply cut their beaks off. Animals like the Ganges river dolphin which is just down to a few left, and it is also critically endangered. So, who is next? Us? Because we are all dependent on these water sources. So, we all know about the Narmada river, the tragedies of dams, the tragedies of huge projects which displace people and wreck river systems without providing livelihoods. And development just basically going berserk, for a double figure growth index, basically. So, we ’ re not sure where this story is going to end, whether it ’ s got a happy or sad ending. And climate change is certainly going to turn all of our theories and predictions on their heads. We ’ re still working hard at it. We ’ ve got a lot of a good team of people working up there. And the thing is, you know, the decision makers, the folks in power, they ’ re up in their bungalows and so on in Delhi, in the city capitals. They are all supplied with plenty of water. It ’ s cool. But out on the rivers there are still millions of people who are in really bad shape. And it ’ s a bleak future for them. So, we have our Ganges and Yamuna cleanup project. We ’ ve spent hundreds of millions of dollars on it, and nothing to show for it. Incredible. So, people talk about political will. During the die-off of the gharial we did galvanize a lot of action. Government cut through all the red tape, we got foreign vets on it. It was great. So, we can do it. But if you stroll down to the Yamuna or to the Gomati in Lucknow, or to the Adyar river in Chennai, or the Mula-Mutha river in Pune, just see what we ’ re capable of doing to a river. It ’ s sad. But I think the final note really is that we can do it. The corporates, the artists, the wildlife nuts, the good old everyday folks can actually bring these rivers back. And the final word is that there is a king cobra looking over our shoulders. And there is a gharial looking at us from the river. And these are powerful water totems. And they are going to disturb our dreams until we do the right thing. Namaste. Chris Anderson : Thanks, Rom. Thanks a lot. You know, most people are terrified of snakes. And there might be quite a few people here who would be very glad to see the last king cobra bite the dust. Do you have those conversations with people? How do you really get them to care? Romulus Whitaker : I take the sort of humble approach, I guess you could say. I don ’ t say that snakes are huggable exactly. It ’ s not like the teddy bear. But I sort of — there is an innocence in these animals. And when the average person looks at a cobra going”Ssssss!”like that, they say,”My god, look at that angry, dangerous creature.”I look at it as a creature who is totally frightened of something so dangerous as a human being. And that is the truth. And that ’ s what I try to get out. CA : Now, incredible footage you showed of the viper being killed. You were saying that that hasn ’ t been filmed before. RW : Yes, this is actually the first time anyone of us knew about it, for one thing. As I said, it ’ s just like a little snack for him, you know? Usually they eat larger snakes like rat snakes, or even cobras. But this guy who we ’ re following right now is in the deep jungle. Whereas other king cobras very often come into the human **interface**, you know, the plantations, to find big rat snakes and stuff. This guy specializes in pit vipers. And the guy who is working there with them, he ’ s from Maharashtra, he said,”I think he ’ s after the nusha.”Now, the nusha means the high. Whenever he eats the pit viper he gets this little venom rush. CA : Thanks Rom. Thank you.

Text Sample 856: POS Dim 6 – 2009 – Score 5.00 – 2009/t002933.txt

As technology progresses, and as it advances, many of us assume that these advances make us more intelligent, make us smarter and more connected to the world. And what I ’ d like to argue is that that ’ s not necessarily the case, as progress is simply a word for change, and with change you gain something, but you also lose something. And to really **illustrate** this point, what I ’ d like to do is to show you how technology has dealt with a very simple, a very common, an everyday question. And that question is this. What time is it? What time is it? If you glance at your iPhone, it ’ s so simple to tell the time. But, I ’ d like to ask you, how would you tell the time if you didn ’ t have an iPhone? How would you tell the time, say, 600 years ago? How would you do it? Well, the way you would do it is by using a device that ’ s called an astrolabe. So, an astrolabe is relatively unknown in today ’ s world. But, at the time, in the 13th century, it was the gadget of the day. It was the world ’ s first popular computer. And it was a device that, in fact, is a model of the sky. So, the different parts of the astrolabe, in this particular type, the rete corresponds to the positions of the stars. The plate corresponds to a coordinate system. And the mater has some scales and puts it all together. If you were an educated child, you would know how to not only use the astrolabe, you would also know how to make an astrolabe. And we know this because the first treatise on the astrolabe, the first technical manual in the English language, was written by Geoffrey Chaucer. Yes, that Geoffrey Chaucer, in 1391, to his little Lewis, his 11-year-old son. And in this book, little Lewis would know the big idea. And the central idea that makes this computer work is this thing called stereographic projection. And basically, the concept is, how do you represent the three-dimensional image of the night sky that surrounds us onto a flat, portable, two-dimensional surface. The idea is actually relatively simple. Imagine that that Earth is at the center of the universe, and surrounding it is the sky projected onto a sphere. Each point on the surface of the sphere is mapped through the bottom **pole**, onto a flat surface, where it is then recorded. So the North Star corresponds to the center of the device. The ecliptic, which is the path of the sun, moon, and planets correspond to an offset circle. The bright stars correspond to little daggers on the rete. And the altitude corresponds to the plate system. Now, the real genius of the astrolabe is not just the projection. The real genius is that it brings together two coordinate systems so they fit perfectly. There is the position of the sun, moon and planets on the movable rete. And then there is their location on the sky as seen from a certain latitude on the back plate. Okay? So how would you use this device? Well, let me first back up for a moment. This is an astrolabe. Pretty impressive, isn ' t it? And so, this astrolabe is on loan from us from the Oxford School of — Museum of History. And you can see the different components. This is the mater, the scales on the back. This is the rete. Okay. Do you see that? That ' s the movable part of the sky. And in the back you can see a spider web pattern. And that spider web pattern corresponds to the local coordinates in the sky. This is a rule device. And on the back are some other devices, measuring tools and scales, to be able to make some calculations. Okay? You know, I ' ve always wanted one of these. For my thesis I actually built one of these out of paper. And this one, this is a replica from a 15th-century device. And it ' s worth probably about three MacBook Pros. But a real one would cost about as much as my house, and the house next to it, and actually every house on the block, on both sides of the street, maybe a school thrown in, and some — you know, a church. They are just incredibly expensive. But let me show you how to work this device. So let ' s go to step one. First thing that you do is you select a star in the night sky, if you ' re telling time at night. So, tonight, if it ' s clear you ' ll be able to see the summer triangle. And there is a bright star called Deneb. So let ' s select Deneb. Second, is you measure the altitude of Deneb. So, step two, I hold the device up, and then I sight its altitude there so I can see it clearly now. And then I measure its altitude. So, it ' s about 26 degrees. You can ' t see it from over there. Step three is identify the star on the front of the device. Deneb is there. I can tell. Step four is I then move the rete, move the sky, so the altitude of the star corresponds to the scale on the back. Okay, so when that happens everything lines up. I have here a model of the sky that corresponds to the real sky. Okay? So, it is, in a sense, holding a model of the universe in my hands. And then finally, I take a rule, and move the rule to a date line which then tells me the time here. Right. So, that ' s how the device is used. So, I know what you ' re thinking :"That ' s a lot of work, isn ' t it? Isn ' t it a ton of work to be able to tell the time?"as you glance at your iPod to just check out the time. But there is a difference between the two, because with your iPod you can tell — or your iPhone, you can tell exactly what the time is, with precision. The way little Lewis would tell the time is by a picture of the sky. He would know where things would fit in the sky. He would not only know what time it was, he would also know where the sun would rise, and how it would move across the sky. He would know what time the sun would rise, and what time it would set. And he would know that for essentially every celestial object in the heavens. So, in computer graphics and computer user **interface** design, there is a term called affordances. So, affordances are the qualities of an object that allow us to perform an action with it. And what the astrolabe does is it allows us, it affords us, to connect to the night sky, to look up into the night sky and be much more — to see the visible and the invisible together. So, that ' s just one use. Incredible, there is probably 350, 400 uses. In fact, there is a text, and that has over a thousand uses of this first computer. On the back there is scales and measurements for terrestrial navigation. You can survey with it. The city of Baghdad was surveyed with it. It can be used for calculating mathematical equations of all different types. And it would take a full university course to **illustrate** it. Astrolabes have an incredible history. They are over 2, 000 years old. The concept of stereographic projection originated in 330 B. C. And the astrolabes come in many different sizes and shapes and forms. There is portable ones. There is large **display** ones. And I think what is common to all astrolabes is that they are beautiful works of art. There is a quality of craftsmanship and precision that is just astonishing and **remarkable**. Astrolabes, like every technology, do evolve over time. So, the earliest retes, for example, were very simple and primitive. And advancing retes became cultural emblems. This is one from Oxford. And I find this one really extraordinary because the rete pattern is completely symmetrical, and it accurately maps a completely asymmetrical, or random sky. How cool is that? This is just amazing. So, would little Lewis have an astrolabe? Probably not one made of brass. He would have one made out of wood, or paper. And the vast majority of this first computer was a portable device that you could keep in the back of your pocket. So, what does the astrolabe inspire? Well, I think the first thing is that it reminds us just how resourceful people were, our forebears were, years and years ago. It ’ s just an incredible device. Every technology advances. Every technology is transformed and moved by others. And what we gain with a new technology, of course, is precision and accuracy. But what we lose, I think, is an accurate — a felt sense of the sky, a sense of context. Knowing the sky, knowing your relationship with the sky, is the center of the real answer to knowing what time it is. So, it ’ s — I think astrolabes are just **remarkable** devices. And so, what can you learn from these devices? Well, primarily that there is a subtle knowledge that we can connect with the world. And astrolabes return us to this subtle sense of how things all fit together, and also how we connect to the world. Thanks very much.

Text Sample 857: POS Dim 6 – 2009 – Score 4.00 – 2009/t000123.txt
I want to start out by asking you to think back to when you were a kid, playing with blocks. As you figured out how to reach out and grasp, pick them up and move them around, you were actually learning how to think and solve problems by understanding and manipulating **spatial** relationships. **Spatial** reasoning is deeply connected to how we understand a lot of the world around us. So, as a computer scientist inspired by this utility of our interactions with physical objects — along with my adviser Pattie, and my collaborator Jeevan Kalanithi — I started to wonder — what if when we used a computer, instead of having this one mouse **cursor** that was a like a digital fingertip moving around a flat desktop, what if we could reach in with both hands and grasp information physically, arranging it the way we wanted? This question was so compelling that we decided to explore the answer, by building Siftables. In a nutshell, a Siftable is an interactive computer the size of a cookie. They ’ re able to be moved around by hand, they can sense each other, they can sense their motion, and they have a screen and a wireless radio. Most importantly, they ’ re physical, so like the blocks, you can move them just by reaching out and grasping. And Siftables are an example of a new ecosystem of tools for manipulating digital information. And as these tools become more physical, more aware of their motion, aware of each other, and aware of the nuance of how we move them, we can start to explore some new and fun interaction styles. So, I ’ m going to start with some simple examples. This Siftable is configured to show video, and if I tilt it in one direction, it ’ ll roll the video this way ; if I tilt it the other way it

rolls it backwards. And these interactive portraits are aware of each other. So if I put them next to each other, they get interested. If they get surrounded, they notice that too, they might get a little flustered. And they can also sense their motion and tilt. One of the interesting implications on interaction, we started to realize, was that we could use everyday gestures on data, like pouring a color the way we might pour a liquid. So in this case, we ’ ve got three Siftables configured to be paint buckets and I can use them to pour color into that central one, where they get mixed. If we overshoot, we can pour a little bit back. There are also some neat possibilities for education, like language, math and logic games where we want to give people the ability to try things quickly, and view the results immediately. So here I ’ m — This is a Fibonacci sequence that I ’ m making with a simple equation program. Here we have a word game that ’ s kind of like a mash-up between Scrabble and Boggle. Basically, in every round you get a randomly assigned letter on each Siftable, and as you try to make words it checks against a dictionary. Then, after about 30 seconds, it reshuffles, and you have a new set of letters and new possibilities to try. Thank you. So these are some kids that came on a field trip to the Media Lab, and I managed to get them to try it out, and shoot a video. They really loved it. And, one of the interesting things about this kind of application is that you don ’ t have to give people many instructions. All you have to say is,”Make words,”and they know exactly what to do. So here ’ s another few people trying it out. That ’ s our youngest beta tester, down there on the right. Turns out, all he wanted to do was to stack the Siftables up. So to him, they were just blocks. Now, this is an interactive cartoon application. And we wanted to build a learning tool for language learners. And this is Felix, actually. And he can bring new characters into the scene, just by lifting the Siftables off the table that have that character shown on them. Here, he ’ s bringing the sun out. Video : The sun is rising. David Merrill : Now he ’ s brought a tractor into the scene. Video : The orange tractor. Good job! Yeah! DM : So by shaking the Siftables and putting them next to each other he can make the characters interact — Video : Woof! DM : inventing his own narrative. Video : Hello! DM : It ’ s an open-ended story, and he gets to decide how it unfolds. Video : Fly away, cat. DM : So, the last example I have time to show you today is a music sequencing and live performance tool that we ’ ve built recently, in which Siftables act as sounds like lead, bass and drums. Each of these has four different variations, you get to choose which one you want to use. And you can inject these sounds into a sequence that you can assemble into the pattern that you want. And you inject it by just bumping up the sound Siftable against a sequence Siftable. There are effects that you can control live, like reverb and **filter**. You attach it to a particular sound and then tilt to adjust it. And then, overall effects like tempo and volume that apply to the entire sequence. So let ’ s have a look. Video : DM : We ’ ll start by putting a lead into two sequence Siftables, arrange them into a series, extend it, add a little more lead. Now I put a bass line in. Video : DM : Now I ’ ll put some percussion in. Video : DM : And now I ’ ll attach the **filter** to the drums, so I can control the effect live. Video : DM : I can speed up the whole sequence by tilting the tempo one way or the other. Video : DM : And now I ’ ll attach the **filter** to the bass for some more expression. Video : DM : I can rearrange the sequence while it plays. So I don ’ t have to plan it out in advance, but I can improvise, making it longer or shorter as I go. And now, finally, I can fade the whole sequence out using the volume Siftable, tilted to the left. Thank you. So, as you can see, my passion is for making new human-computer **interfaces** that are a better match to the ways our brains and bodies work. And today, I had time to show you one point in this new design space, and a few of the possibilities that we ’ re working to bring out of the laboratory. So the thought

I want to leave you with is that we ’ re on the cusp of this new generation of tools for interacting with digital media that are going to bring information into our world on our terms. Thank you very much. I look forward to talking with all of you.

Text Sample 858: POS Dim 6 – 2011 – Score 6.00 – 2011/t002610.txt

Thank you. I ’ m thrilled to be here. I ’ m going to talk about a new, old material that still continues to amaze us, and that might impact the way we think about material science, high technology â and maybe, along the way, also do some stuff for medicine and for global health and help reforestation. So that ’ s kind of a bold statement. I ’ ll tell you a little bit more. This material actually has some traits that make it seem almost too good to be true. It ’ s sustainable ; it ’ s a sustainable material that is processed all in water and at room temperature â and is biodegradable with a clock, so you can watch it dissolve instantaneously in a glass of water or have it stable for years. It ’ s edible ; it ’ s implantable in the human body without causing any immune response. It actually gets reintegrated in the body. And it ’ s technological, so it can do things like microelectronics, and maybe photonics do. And the material looks something like this. In fact, this material you see is clear and transparent. The components of this material are just water and protein. So this material is silk. So it ’ s kind of different from what we ’ re used to thinking about silk. So the question is, how do you reinvent something that has been around for five millennia? The process of discovery, generally, is inspired by nature. And so we marvel at silk worms â the silk worm you see here spinning its fiber. The silk worm does a **remarkable** thing : it uses these two ingredients, protein and water, that are in its gland, to make a material that is exceptionally tough for protection â so comparable to technical fibers like Kevlar. And so in the reverse engineering process that we know about, and that we ’ re familiar with, for the textile industry, the textile industry goes and unwinds the cocoon and then weaves glamorous things. We want to know how you go from water and protein to this liquid Kevlar, to this natural Kevlar. So the insight is how do you actually reverse engineer this and go from cocoon to gland and get water and protein that is your starting material. And this is an insight that came, about two decades ago, from a person that I ’ m very fortunate to work with, David Kaplan. And so we get this starting material. And so this starting material is back to the basic building block. And then we use this to do a variety of things â like, for example, this film. And we take advantage of something that is very simple. The recipe to make those films is to take advantage of the fact that proteins are extremely smart at what they do. They find their way to self-assemble. So the recipe is simple : you take the silk solution, you pour it, and you wait for the protein to self-assemble. And then you detach the protein and you get this film, as the proteins find each other as the water evaporates. But I mentioned that the film is also technological. And so what does that mean? It means that you can **interface** it with some of the things that are **typical** of technology, like microelectronics and nanoscale technology. And the image of the DVD here is just to **illustrate** a point that silk follows very subtle topographies of the surface, which means that it can replicate features on the nanoscale. So it would be able to replicate the information that is on the DVD. And we can store information that ’ s film with water and protein. So we tried something out, and we wrote a message in a piece of silk, which is right here, and the message is over there. And much like in the DVD, you can read it out optically. And this requires a stable hand, so this is why I decided to do it onstage in front of a thousand people. So let me see. So as you see the film go in transparently through there, and then... And the most **remarkable** feat is that my hand actually stayed still long enough to do that. So once you have these attributes of this material, then you can do a lot of things. It ’ s actually not limited to films. And so the material can assume a lot of formats. And then you go a little crazy, and so you do **various** optical components or you do microprism arrays, like the reflective tape that you have on your running shoes. Or you can do beautiful things that, if the camera can capture, you can make. You can add a third dimensionality to the film. And if the angle is right, you can actually see a hologram appear in this film of silk. But you can do other things. You can imagine that then maybe you can use a pure protein to guide light, and so we ’ ve made optical fibers. But silk is versatile and it goes beyond optics. And you can think of different formats. So for instance, if you ’ re afraid of going to the doctor and getting stuck with a needle, we do microneedle arrays. What you see there on the screen is a human hair **superimposed** on the needle that ’ s made of silk â just to give you a sense of size. You can do bigger things. You can do gears and nuts and bolts â that you can buy at Whole Foods. And the gears work in water as well. So you think of alternative mechanical parts. And maybe you can use that liquid Kevlar if you need something strong to replace peripheral veins, for example, or maybe an entire bone. And so you have here a little example of a small skull â what we call mini Yorick. But you can do things like cups, for example, and so, if you add a little bit of gold, if you add a little bit of semiconductors you could do sensors that stick on the surfaces of foods. You can do electronic pieces that fold and wrap. Or if you ’ re fashion forward, some silk LED tattoos. So there ’ s versatility, as you see, in the material formats, that you can do with silk. But there are still some unique traits. I mean, why would you want to do all these things for real? I mentioned it briefly at the beginning ; the protein is biodegradable and biocompatible. And you see here a picture of a tissue section. And so what does that mean, that it ’ s biodegradable and biocompatible? You can implant it in the body without needing to retrieve what is implanted. Which means that all the devices that you ’ ve seen before and all the formats, in principle, can be implanted and disappear. And what you see there in that tissue section, in fact, is you see that reflector tape. So, much like you ’ re seen at night by a car, then the idea is that you can see, if you illuminate tissue, you can see deeper parts of tissue because there is that reflective tape there that is made out of silk. And you see there, it gets reintegrated in tissue. And reintegration in the human body is not the only thing, but reintegration in the environment is important. So you have a clock, you have protein, and now a silk cup like this can be thrown away without guilt â unlike the polystyrene cups that unfortunately fill our landfills everyday. It ’ s edible, so you can do smart packaging around food that you can cook with the food. It doesn ’ t taste good, so I ’ m going to need some help with that. But probably the most **remarkable** thing is that it comes full circle. Silk, during its self-assembly process, acts like a cocoon for biological matter. And so if you change the recipe, and you add things when you pour â so you add things to your liquid silk solution â where these things are enzymes or antibodies or vaccines, the self-assembly process preserves the biological function of these dopants. So it makes the materials environmentally active and interactive. So that screw that you thought about beforehand can actually be used to screw a bone together â a fractured bone together â and deliver drugs at the same, while your bone is healing, for example. Or you could put drugs in your wallet and not in your fridge. So we ’ ve made a silk card with penicillin in it. And we stored penicillin at 60 degrees C, so 140 degrees Fahrenheit, for two months without loss of efficacy of the penicillin. And so that could be â- that could be potentially a good alternative to solar powered refrigerated camels. And of course, there ’ s no use in storage if you can ’ t use [it]. And so there is this other unique material trait that these materials have, that they ’ re programmably degradable. And so what you see there is the difference. In the top, you have a film that has been programmed not to degrade, and in the bottom, a film that has been programmed to degrade in water. And what you see is that the film on the bottom releases what is inside it. So it allows for the recovery of what we ’ ve stored before. And so this allows for a controlled delivery of drugs and for reintegration in the environment in all of these formats that you ’ ve seen. So the thread of discovery that we have really is a thread. We ’ re impassioned with this idea that whatever you want to do, whether you want to replace a vein or a bone, or maybe be more sustainable in microelectronics, perhaps drink a coffee in a cup and throw it away without guilt, maybe carry your drugs in your pocket, deliver them inside your body or deliver them across the desert, the answer may be in a thread of silk. Thank you.

Text Sample 859: POS Dim 6 – 2011 – Score 5.00 – 2011/t002429.txt
Hello, everyone. My name is Keith Nolan. Iâm a cadet private. My talk today is on the topic of the military. How many of you out there thought youâd ever like to join the military? I see a number of you nodding. And I thought the same thing ever since I was young. Growing up, Iâd always wanted to join the military. I loved military history and Iâve read a great deal on the subject. Also, I have **various** family members, such as my grandfather and great uncle, who fought in World War II. And like them, I wanted the same thing : to serve my country. So the question is : Can I? No, I canât. Why? Simply because Iâm deaf. Regardless of that fact, I still had that longing to join the military. For example, after I graduated from high school, three months before 9 / 11 occurred, I went to a naval recruiting center with high hopes of joining the navy. I went in and a strapping naval man stood up and addressed me. As he was speaking to me, it was impossible for me to read his lips, so I said, âI ’ m sorry, Iâm deaf. â He tore off a little piece of paper and wrote down three words : âBad ear. Disqual. â He didn ’ t even fully spell out”Disqualified,”just : âBad ear. Disqual. â So I went on my way. I tried **various** locations a number of different times, trying to join, but over and over again, I got the same response : âSorry, youâre deaf. We canât accept you. â So I shifted gears and decided to become a teacher. I completed a master ’ s in deaf education and taught for almost two years, until this past spring, when three things occurred that changed that course, the first of which, while I was teaching a high school history class. Iâd lectured on the Mexican-American War. The bell had rung, and I was seated at my desk, when one of my students, who is deaf, approached me and said that heâd like to join the military. I said, âAh, sorry. You canât. You ’ re deaf. â Then I caught myself. It struck me that all along I had been told no, I canât, and now I was perpetuating that same message to the next generation, to my own student. That realization had a large impact that really resonated with me. Now, the second thing that happened, my friend had just moved to Israel. Did you know that in Israel they accept deaf people into the military? How can deaf people be in the military, right? Could this really be true? Come on! Well, I went to Israel last summer to see for myself. I interviewed 10 deaf Israeli soldiers, all of those video interviews and questions Iâve compiled, and the findings, Iâll share with you later. Thirdly, CSUN here, my alma mater, had recently started up an Army ROTC program. ROTC, which stands for Reserve Officer Training Corps,

allows students working on their college majors to concurrently participate in the ROTC program. Upon graduation, ROTC students have a military career ready and waiting for them. So if one joined the army, one could commission as a second lieutenant. That ' s generally the ROTC program here at CSUN. Having learned that, I was intrigued. I already had a profession as a teacher, but I went ahead anyway and sent an email off to the program, explaining that I was a teacher of the deaf, wondering if I could take a few classes with them and perhaps share their lessons with my students. I got an email response back, and surprisingly, it was the first time that I wasn ' t told,"Sorry, no, you canâ□□t. Youâ□□re deaf."It said,"Well, that ' s interesting. I think maybe we can work something out and you can take a few classes with us."This was unprecedented. So naturally, I was shocked. Although I was teaching, I decided I had to grab the opportunity right away and get my foot in the door. Altogether, that ' s how it transpired. Now, with all my life experiences, having talked with all the people I had, and given everything Iâ□□ve read, I decided to write a research paper called â□□Deaf in the Military. â□□ I ' ll share with you what those 98 pages entail. Here in America, weâ□□ve actually had deaf soldiers serving in the past. In fact, during the Texas War of Independence, there was a key character named Deaf Smith, who made a large contribution to that war effort. For the American Civil War, Gallaudet University actually has archived a list of deaf soldiers in that war from the North and the South, showing that deaf soldiers were even fighting against each other. During World War II, there are a few rare examples of deaf people who made it into the military at that time and were able to serve their country as well. History **illustrates** the fact that America has had deaf soldiers, in contrast to today. In my paper, I also discuss the deaf Israeli soldiers. I learned that they serve in non-combat roles. The deaf soldiers are not on the front lines engaged in fire, but rather, are behind the lines serving in supportive roles. There are a plethora of **various** non-combat jobs accessible to the deaf : intelligence, computer technology, map drawing, supply, military dog training â□□ the list goes on. The communication between deaf Israeli soldiers and other soldiers who are hearing is carried out with the same approaches deaf people in general use with the hearing public on a daily basis. You can use your voice, lip-read, gestures, sometimes another soldier knows sign language and that can be utilized, pen and paper, texts, computers, emails â□□ seriously, thereâ□□s no magic wand necessary. It ' s the same thing we do every day. Interpreters are used there primarily for boot camp training. For the average work, itâ□□s not necessary to have an interpreter by your side. The Israeli Army is comprised of small groups. Each of these units with deaf soldiers have developed their own way of communicating with each other, so thereâ□□s no need for interpreters. The top picture is of one soldier I met. The bottom photo is of Prime Minister Begin with a deaf soldier in Israel. Another part of my paper touches on disabled soldiers in the US Military. Obviously, military work can be dangerous and involve injury. One example here is Captain Luckett. Due to an explosion, he lost his leg. Heâ□□s recovered and currently has a prosthetic leg. Now that heâ□□s strong, he ' s back in combat, still fighting in Afghanistan. Itâ□□s **remarkable**. And guess what? Heâ□□s not the only one. There are 40 other soldiers like him, amputees who are serving in combat zones. Incredible. Also, we have a blind soldier here. While he was in Iraq, an explosion from a suicide car bomber destroyed his eyesight. Heâ□□s recovered and hasnâ□□t left the army. The army has retained him on active duty, and heâ□□s currently running a hospital for wounded soldiers. I also found out online about another soldier, who is deaf in one ear. Heâ□□s developed civil programs in Iraq, one of which actually started a school for the deaf in Iraq. All of this is incredible. But I am going to ask all of you : If the US Military can retain their disabled soldiers, why canâ□□t they accept disabled citizens as well? Moreover, out of all the US Military jobs, 80 % are non-combat positions. There are many jobs that we in the Deaf community can do. If I were to be in the military, Iâ□□d like to do intelligence work. But there is an array of other things we can do, such as mechanics, finance, medicine, etc. So to summarize, Iâ□□ve presented three premises to support my argument, the first being, Israeli defense openly accepts deaf soldiers. If you have the qualities and skills required, theyâ□□ll take you. Secondly, the US Military has accommodations for retaining their disabled soldiers. And lastly, 80 % of occupations in the military are non-combat. Now, can we Deaf Americans serve our country? Yes! Of course! Absolutely, without a doubt! Now Iâ□□ll explain a bit about my experience in the Army ROTC, which began last fall. I have been involved with that thus far and itâ□□s still going on now. Really, I need to preface this by saying that this is the first time my battalion had ever had a deaf cadet. They had never experienced that before. So of course, they were taken aback, wondering, initially, how I would do this or that, how would we communicate and such, which is a natural reaction, considering that many of them had never interacted with a deaf person prior to me. Plus, I was taken aback by this â□□ it was the real thing, the army. I had to learn a whole new world, full of military jargon, with its own culture and everything. So we started out slow, getting to know each other and learning how to work together, progressively. For example, on the first day of class, I had no uniform. So I showed up in regular clothes, while the other cadets were all in uniform. I found out that every morning at 5:30, there was physical training, PT. On Fridays, there would be field training â□□ labs â□□ off-campus, and occasionally, we would have weekend training at a military base. So I showed up, ready, each morning at 5 : 30, with all the cadets in uniform and me in civilian clothes. They told me, â□□Hey, you know, you don ’ t need to work out. You can just simply take classes. â□□ I told them I wanted to, anyway. They acknowledged that, and I continued to show up every morning to train. When Friday came, I asked if I could do the field training. I was told no, just stick with class. I insisted that I wanted to try. Somewhat reluctantly, they let me attend the lab, but only as an observer ; I would only be allowed to sit and watch, not participate. Alright, so, I showed up on Friday, and watched as the cadets learned marching drill commands, like how to stand at attention, how to properly salute, and all the basics. I had to ask again if I could join. Finally, I got the go-ahead. I went to get in formation. I figured I better stand in the second line, so I could watch what the cadets were doing in the row ahead of me. But the officer who opened the door for me to join the ROTC program spotted me in the back and said, â□□Hey! Uh-uh. I want you in the front. You want to be a soldier? You ’ ve got to learn the commands just like the rest of them. Youâ□□re not going to follow other people. Learn it yourself! â□□ I thought,”Wow. Heâ□□s viewing me like any person, giving dignity to who I am.”I was impressed by that. So as the weeks went by, I still didnâ□□t have a uniform. I asked if it would be possible to get one, but I was told it wasn ’ t. So I continued on that way, until one day, I was informed that Iâ□□d be getting a uniform.”Please!”I said, â□□Really? Why? What changed? â□□ I was told, â□□We see your motivation, you show up every morning, dedicated, and always gave a 110 % effort. â□□ They wanted to give me the uniform. It was **remarkable**. We went to the warehouse to get my uniform. I assumed Iâ□□d just get a uniform and a pair of boots, nothing more. But they filled two duffel bags chock-full of gear : helmet, ammo vest, shovel, sleeping bag â□□ the whole nine yards. I was astonished. And I have to tell you, each morning that I get up and put on my uniform, I feel privileged. Itâ□□s truly an honor to wear the uniform. So, moving along, when it came time to train at the garrison base, at first, I was told I couldnâ€™t go. There was concern on the ROTC ’ s part that if the interpreter were to get injured during the training, it would be a liability issue. So we had to figure out all those issues and confusion, but we worked it out, and in the end, they let me go. That ’ s how events were unfolding ; I was permitted to do more and more. Once, at the garrison base, during one of the training days, a huge Chinook helicopter with its tandem rotors landed right down to us, forcefully spinning exhilaration in the air. All of us cadets were supposed to be getting on board. Everyone was geared up and ready. However, the cadre had decided I wasnâ€™t going to be able to ride the Chinook. They were afraid if the pilot shouted out orders, how would I be able to follow the instructions? Iâ€™d potentially cause a disruption. So I had to stand aside, while the others were filing toward the helicopter. I could see the cadre huddled up, discussing, mulling it over. At the last minute, one of them said, â€œCome on! Get on the helicopter! â€ I rushed over and got in. It was such a thrill. And that was the spirit of learning about and supporting one another that carried over. And since then, Iâ€™ve been involved in everything they do, without any separation. This is where my passion lies. I love them. Iâ€™ll show you some pictures here. Bruin Battalion, Bravo Company â€" thatâ€™s the name of the group I belong to. The cadre are the officers and sergeants who oversee the ROTC program. In the beginning, you can see, it was a bit of an awkward phase. But once they learned more about me and what Iâ€™m capable of doing, there ’ s been tremendous support and unity. The cadets, my fellow peers â€" well, when you train and sweat together, you feel the bond of camaraderie right away. A brother- or sisterly cohesiveness makes them like family. In training and military science classes, we learn theories of warfare, how to lead soldiers, how to do reconnaissance, strategies, how to knock out a bunker and land navigation, where youâ€™re finding your way out in the mountains. As far as accommodations, Iâ€™ve been provided with interpreters through the National Center on Deafness, NCOD, here at CSUN. And I really have to thank them, because it ’ s hard to find interpreters who are willing to wake up at 4:30 in the morning, or sometimes even 3:30 in the morning. That ’ s the officer who emailed me back, saying I think you can have a few classes with us. That ’ s Lieutenant Mendoza. Thatâ€™s my interpreter there, before class starts. This is a picture from last fall, when we were new to training. This is Lieutenant Colonel Phelps, this being his name sign. He ’ s the commanding officer of the entire Bruin Battalion. Every time I see him walk by, itâ€™s rather inspiring. I mean, the way he presents himself, you can see heâ€™s the epitome of a soldier. Plus, he doesn ’ t view me as a deaf person. He looks at my skills and capabilities instead. Heâ€™s really pushed for me, and I respect him for all that. Thatâ€™s me during one of the exercises. This is that Chinook helicopter I almost didnâ€™t get on. Every cadet has a mentor. My mentor is Cinatl. Heâ€™s a really sharp soldier. He teaches me all the finer points and how to execute them ideally. This top picture is when a group of us went to Las Vegas to compete in a test, to see if we could match the German **troops** ’ physical training standards. It involved swimming, timed sprints, marksmanship and numerous fitness events. I passed them and satisfied the requirements to be awarded the gold German Armed Forces Proficiency Badge right here. This is one of the sergeants, Sergeant Richardson. I love this guy. He doesn ’ t take baloney from any of us cadets. Here I am one morning, when we trekked seven and a half miles with a 40-pound rucksack in less than two hours. Here are a few of my fellow cadets. I ’ ve been with them long enough that Iâ€™ve developed name signs for them. On the right, here, is Trinidad. I gave him this name sign because he ’ s always very sarcastic. He ’ s a veteran, having served in Iraq and Afghanistan. The female is Frigo, whose nickname is”Refrigerator,”hence her name sign. Weâre always competing intellectually in class. The cadet on the end is Jarvy. He ’ s a top athlete. I ’ ve given him this sign because of the scar he has here. Do you know who this is? This is the Chairman of the Joint Chiefs of Staff. He is the highest-ranking military officer and principal military advisor to President Obama and Secretary of Defense Gates. He gave a talk at UCLA to a full house. Afterwards, I lined up to shake his hand. Having done so, I greeted him, âItâs a great pleasure to meet you. â I signed and my interpreter voiced for me. Admiral Mullen turned to the interpreter and said, âItâs nice to meet you, â addressing the interpreter, who refrained to clarify. He seemed a little confused and just quickly moved on to shake hands with the rest of the soldiers. So Iâm not sure whether he really knew that I ’ m deaf or not. So everythingâs been moving along, gung-ho, full speed ahead, until two weeks ago, when something occurred. Well, the ROTC has four levels. Iâm currently doing the first two levels, which finishes up this May. The third level will begin in the fall. But in order to move up, you need to pass a medical exam. Obviously, Iâm deaf, so Iâd fail a hearing test. So we sat down, and I was told that if I wanted to continue to the third level, I couldnât do any of the PT workouts in the morning, nor the Friday lab field trainings, nor the army base trainings. My uniform, I would have to give back as well. I could take the classes, audit them, and thatâs all. It really hit me. It was a huge blow. Many of the officers and cadets have empathized with this sudden shock of disappointment, and are wondering why this has to be the case. Colonel Phelps has tried to speak with the higher-ups in the chain of command and explain to them that Iâm one of the top cadets, having passed all the events and receiving high marks on my exams. But their response is unwavering : policy is policy, and if youâre deaf, youâre disqualified. I know that the cadre has tried to find **various** ways. They found out that there ’ s a deaf cadet at The Citadel, a military college in South Carolina. That particular cadet will be completing his fourth year there and graduating this May. Yet, he ’ s in the same predicament that I ’ m in â unable to join the army because heâs deaf. Yet, all of my fellow cadets and the officers have told me not to give up ; the policy must change. I was advised to talk with my congressman. And I ’ ve brought this issue to Henry Waxman, the district congressman here in LA, to get the **ball** rolling with his advocacy for my cause. However, I need your help and support to lobby. All of us, you know? If you remember back in US history, African-Americans were told they couldnât join the military, and now they serve. Women as well were banned, but now they ’ ve been allowed. The military has and is changing. Today is our time. Now itâs our turn. Hooah!

Text Sample 860: POS Dim 6 – 2011 – Score 5.00 – 2011/t002436.txt

The things we make have one supreme quality â they live longer than us. We perish, they survive ; we have one life, they have many lives, and in each life they can mean different things. Which means that, while we all have one biography, they have many. I want this morning to talk about the story, the biography â or rather the biographies â of one particular object, one **remarkable** thing. It doesn ’ t, I agree, look very much. It ’ s about the size of a rugby **ball**. It ’ s made of clay, and it ’ s been fashioned into a cylinder shape, covered with close writing and then baked dry in the sun. And as you can see, it ’ s been knocked about a bit, which is not surprising because it was made two and a half thousand years ago and was dug up in 1879. But today, this thing is, I believe, a major player in the politics of the Middle East. And it ’ s an object with fascinating stories and stories that are by no means over yet. The story begins in the Iran-Iraq war and that series of events that culminated in the invasion of Iraq by foreign forces, the removal of a despotic ruler and instant regime change. And I want to begin with one episode from that sequence of events that most of you would be very familiar with, Belshazzar ' s feast â because we ' re talking about the Iran-Iraq war of 539 BC. And the parallels between the events of 539 BC and 2003 and in between are startling. What you ' re looking at is Rembrandt ' s painting, now in the National Gallery in London, **illustrating** the text from the prophet Daniel in the Hebrew scriptures. And you all know roughly the story. Belshazzar, the son of Nebuchadnezzar, Nebuchadnezzar who ' d conquered Israel, sacked Jerusalem and captured the people and taken the Jews back to Babylon. Not only the Jews, he ' d taken the temple vessels. He ' d ransacked, desecrated the temple. And the great gold vessels of the temple in Jerusalem had been taken to Babylon. Belshazzar, his son, decides to have a feast. And in order to make it even more exciting, he added a bit of sacrilege to the rest of the fun, and he brings out the temple vessels. He ' s already at war with the Iranians, with the king of Persia. And that night, Daniel tells us, at the height of the festivities a hand appeared and wrote on the wall,"You are weighed in the balance and found wanting, and your kingdom is handed over to the Medes and the Persians."And that very night Cyrus, king of the Persians, entered Babylon and the whole regime of Belshazzar fell. It is, of course, a great moment in the history of the Jewish people. It ' s a great story. It ' s story we all know."The writing on the wall"is part of our everyday language. What happened next was **remarkable**, and it ' s where our cylinder enters the story. Cyrus, king of the Persians, has entered Babylon without a fight â the great empire of Babylon, which ran from central southern Iraq to the Mediterranean, falls to Cyrus. And Cyrus makes a declaration. And that is what this cylinder is, the declaration made by the ruler guided by God who had toppled the Iraqi despot and was going to bring freedom to the people. In ringing Babylonian â it was written in Babylonian â he says,"I am Cyrus, king of all the universe, the great king, the powerful king, king of Babylon, king of the four quarters of the world."They ' re not shy of hyperbole as you can see. This is probably the first real press release by a victorious army that we ' ve got. And it ' s written, as we ' ll see in due course, by very skilled P. R. consultants. So the hyperbole is not actually surprising. And what is the great king, the powerful king, the king of the four quarters of the world going to do? He goes on to say that, having conquered Babylon, he will at once let all the peoples that the Babylonians â Nebuchadnezzar and Belshazzar â have captured and enslaved go free. He ' ll let them return to their countries. And more important, he will let them all recover the gods, the statues, the temple vessels that had been confiscated. All the peoples that the Babylonians had repressed and removed will go home, and they ' ll take with them their gods. And they ' ll be able to restore their altars and to worship their gods in their own way, in their own place. This is the decree, this object is the evidence for the fact that the Jews, after the exile in Babylon, the years they ' d spent sitting by the waters of Babylon, weeping when they remembered Jerusalem, those Jews were allowed to go home. They were allowed to return to Jerusalem and to rebuild the temple. It ' s a central document in Jewish history. And the Book of Chronicles, the Book of Ezra in the Hebrew scriptures reported in ringing terms. This is the Jewish version of the same story."Thus said Cyrus, king of Persia, ' All the kingdoms of the earth have the Lord God of heaven given thee, and he has charged me to build him a house in Jerusalem. Who is there among you of his people? The Lord God be with him, and let him go up. '""Go up"â aaleh. The central element, still, of the notion of return, a central part of the life of Judaism. As you all know, that return from exile, the second temple, reshaped Judaism. And that change, that great historic moment, was made possible by Cyrus, the king of Persia, reported for us in Hebrew in scripture and in Babylonian in clay. Two great texts, what about the politics? What was going on was the fundamental shift in Middle Eastern history. The empire of Iran, the Medes and the Persians, united under Cyrus, became the first great world empire. Cyrus begins in the 530s BC. And by the time of his son Darius, the whole of the eastern Mediterranean is under Persian control. This empire is, in fact, the Middle East as we now know it, and it ’ s what shapes the Middle East as we now know it. It was the largest empire the world had known until then. Much more important, it was the first multicultural, multifaith state on a huge scale. And it had to be run in a quite new way. It had to be run in different languages. The fact that this decree is in Babylonian says one thing. And it had to recognize their different habits, different peoples, different religions, different faiths. All of those are respected by Cyrus. Cyrus sets up a model of how you run a great multinational, multifaith, multicultural society. And the result of that was an empire that included the areas you see on the screen, and which survived for 200 years of stability until it was shattered by Alexander. It left a dream of the Middle East as a unit, and a unit where people of different faiths could live together. The Greek invasions ended that. And of course, Alexander couldn ’ t sustain a government and it fragmented. But what Cyrus represented remained absolutely central. The Greek historian Xenophon wrote his book”Cyropaedia”promoting Cyrus as the great ruler. And throughout European culture afterward, Cyrus remained the model. This is a 16th century image to show you how widespread his veneration actually was. And Xenophon ’ s book on Cyrus on how you ran a diverse society was one of the great textbooks that inspired the Founding Fathers of the American Revolution. Jefferson was a great admirer â the ideals of Cyrus obviously speaking to those 18th century ideals of how you create religious tolerance in a new state. Meanwhile, back in Babylon, things had not been going well. After Alexander, the other empires, Babylon declines, falls into ruins, and all the traces of the great Babylonian empire are lost â until 1879 when the cylinder is discovered by a British Museum exhibition digging in Babylon. And it enters now another story. It enters that great debate in the middle of the 19th century : Are the scriptures reliable? Can we trust them? We only knew about the return of the Jews and the decree of Cyrus from the Hebrew scriptures. No other evidence. Suddenly, this appeared. And great excitement to a world where those who believed in the scriptures had had their faith in creation shaken by evolution, by geology, here was evidence that the scriptures were historically true. It ’ s a great 19th century moment. But â and this, of course, is where it becomes complicated â the facts were true, hurrah for archeology, but the interpretation was rather more complicated. Because the cylinder account and the Hebrew Bible account differ in one key respect. The Babylonian cylinder is written by the priests of the great god of Bablyon, Marduk. And, not surprisingly, they tell you that all this was done by Marduk.”Marduk, we hold, called Cyrus by his name.”Marduk takes Cyrus by the hand, calls him to shepherd his people and gives him the rule of Babylon. Marduk tells Cyrus that he will do these great, generous things of setting the people free. And this is why we should all be grateful to and worship Marduk. The Hebrew writers in the Old Testament, you will not be surprised to learn, take a rather different view of this. For them, of course, it can ’ t possibly by Marduk that made all this happen. It can only be Jehovah. And so in Isaiah, we have the wonderful texts giving all the credit of this, not to Marduk but to the Lord God of Israel â the Lord God of Israel who also called Cyrus by name, also takes Cyrus by the hand and talks of him shepherding his people. It ’ s a **remarkable** example of two different priestly appropriations of the same event, two different religious takeovers of a political fact. God, we know, is usually on the side of the big battalions. The question is, which god was it? And the debate unsettles everybody in the 19th century to realize that the Hebrew scriptures are part of a much wider world of religion. And it ' s quite clear the cylinder is older than the text of Isaiah, and yet, Jehovah is speaking in words very similar to those used by Marduk. And there ' s a slight sense that Isaiah knows this, because he says, this is God speaking, of course,"I have called thee by thy name though thou hast not known me."I think it ' s recognized that Cyrus doesn ' t realize that he ' s acting under orders from Jehovah. And equally, he ' d have been surprised that he was acting under orders from Marduk. Because interestingly, of course, Cyrus is a good Iranian with a totally different set of gods who are not mentioned in any of these texts. That ' s 1879. 40 years on and we ' re in 1917, and the cylinder enters a different world. This time, the real politics of the contemporary world â the year of the Balfour Declaration, the year when the new imperial power in the Middle East, Britain, decides that it will declare a Jewish national home, it will allow the Jews to return. And the response to this by the Jewish population in Eastern Europe is rhapsodic. And across Eastern Europe, Jews **display** pictures of Cyrus and of George V side by side â the two great rulers who have allowed the return to Jerusalem. And the Cyrus cylinder comes back into public view and the text of this as a demonstration of why what is going to happen after the war is over in 1918 is part of a divine plan. You all know what happened. The state of Israel is setup, and 50 years later, in the late 60s, it ' s clear that Britain ' s role as the imperial power is over. And another story of the cylinder begins. The region, the U. K. and the U. S. decide, has to be kept safe from communism, and the superpower that will be created to do this would be Iran, the Shah. And so the Shah invents an Iranian history, or a return to Iranian history, that puts him in the center of a great tradition and produces coins showing himself with the Cyrus cylinder. When he has his great celebrations in Persepolis, he summons the cylinder and the cylinder is lent by the British Museum, goes to Tehran, and is part of those great celebrations of the Pahlavi dynasty. Cyrus cylinder : guarantor of the Shah. 10 years later, another story : Iranian Revolution, 1979. Islamic revolution, no more Cyrus ; we ' re not interested in that history, we ' re interested in Islamic Iran â until Iraq, the new superpower that we ' ve all decided should be in the region, attacks. Then another Iran-Iraq war. And it becomes critical for the Iranians to remember their great past, their great past when they fought Iraq and won. It becomes critical to find a symbol that will pull together all Iranians â Muslims and non-Muslims, Christians, Zoroastrians, Jews living in Iran, people who are devout, not devout. And the obvious emblem is Cyrus. So when the British Museum and Tehran National Musuem cooperate and work together, as we ' ve been doing, the Iranians ask for one thing only as a loan. It ' s the only object they want. They want to borrow the Cyrus cylinder. And last year, the Cyrus cylinder went to Tehran for the second time. It ' s shown being presented here, put into its case by the director of the National Museum of Tehran, one of the many women in Iran in very senior positions, Mrs. Ardakani. It was a huge event. This is the other side of that same picture. It ' s seen in Tehran by between one and two million people in the space of a few months. This is beyond any blockbuster exhibition in the West. And it ' s the subject of a huge debate about what this cylinder means, what Cyrus means, but above all, Cyrus as **articulated** through this cylinder â Cyrus as the defender of the homeland, the champion, of course, of Iranian identity and of the Iranian peoples, tolerant of all faiths. And in the current Iran, Zoroastrians and Christians have guaranteed places in the Iranian parliament, something to be very, very proud of. To see this object in Tehran, thousands of

Jews living in Iran came to Tehran to see it. It became a great emblem, a great subject of debate about what Iran is at home and abroad. Is Iran still to be the defender of the oppressed? Will Iran set free the people that the tyrants have enslaved and expropriated? This is heady national rhetoric, and it was all put together in a great pageant launching the return. Here you see this out-sized Cyrus cylinder on the stage with great figures from Iranian history gathering to take their place in the heritage of Iran. It was a narrative presented by the president himself. And for me, to take this object to Iran, to be allowed to take this object to Iran was to be allowed to be part of an extraordinary debate led at the highest levels about what Iran is, what different Irans there are and how the different histories of Iran might shape the world today. It ’ s a debate that ’ s still continuing, and it will continue to rumble, because this object is one of the great declarations of a human aspiration. It stands with the American constitution. It certainly says far more about real freedoms than Magna Carta. It is a document that can mean so many things, for Iran and for the region. A replica of this is at the United Nations. In New York this autumn, it will be present when the great debates about the future of the Middle East take place. And I want to finish by asking you what the next story will be in which this object figures. It will appear, certainly, in many more Middle Eastern stories. And what story of the Middle East, what story of the world, do you want to see reflecting what is said, what is expressed in this cylinder? The right of peoples to live together in the same state, worshiping differently, freely â a Middle East, a world, in which religion is not the subject of division or of debate. In the world of the Middle East at the moment, the debates are, as you know, shrill. But I think it ’ s possible that the most powerful and the wisest voice of all of them may well be the voice of this mute thing, the Cyrus cylinder. Thank you.

Text Sample 861: POS Dim 6 – 2013 – Score 6.00 – 2013/t002003.txt

When I first began recording wild soundscapes 45 years ago, I had no idea that ants, insect larvae, sea anemones and viruses created a sound signature. But they do. And so does every wild habitat on the planet, like the Amazon rainforest you ’ re hearing behind me. In fact, temperate and tropical rainforests each produce a vibrant animal orchestra, that instantaneous and organized expression of insects, reptiles, amphibians, birds and mammals. And every soundscape that springs from a wild habitat generates its own unique signature, one that contains incredible amounts of information, and it ’ s some of that information I want to share with you today. The soundscape is made up of three basic sources. The first is the geophony, or the nonbiological sounds that occur in any given habitat, like wind in the trees, water in a stream, waves at the ocean shore, movement of the Earth. The second of these is the biophony. The biophony is all of the sound that ’ s generated by organisms in a given habitat at one time and in one place. And the third is all of the sound that we humans generate that ’ s called anthrophony. Some of it is controlled, like music or theater, but most of it is chaotic and incoherent, which some of us refer to as noise. There was a time when I considered wild soundscapes to be a worthless **artifact**. They were just there, but they had no significance. Well, I was wrong. What I learned from these encounters was that careful listening gives us incredibly valuable tools by which to evaluate the health of a habitat across the entire spectrum of life. When I began recording in the late ’ 60s, the **typical** methods of recording were limited to the fragmented capture of individual species like birds mostly, in the beginning, but later animals like mammals and amphibians. To me, this was a little like trying to understand the magnificence of Beethoven ’ s Fifth Symphony by abstracting the sound of a single violin player out of the context of the orchestra and hearing just that one part. Fortunately, more and more institutions are implementing the more holistic models that I and a few of my colleagues have introduced to the field of soundscape ecology. When I began recording over four decades ago, I could record for 10 hours and capture one hour of usable material, good enough for an album or a film soundtrack or a museum installation. Now, because of global warming, resource extraction, and human noise, among many other factors, it can take up to 1, 000 hours or more to capture the same thing. Fully 50 percent of my archive comes from habitats so radically altered that they ’ re either altogether silent or can no longer be heard in any of their original form. The usual methods of evaluating a habitat have been done by visually counting the numbers of species and the numbers of individuals within each species in a given area. However, by comparing data that ties together both density and diversity from what we hear, I ’ m able to arrive at much more precise fitness outcomes. And I want to show you some examples that typify the possibilities unlocked by diving into this universe. This is Lincoln Meadow. Lincoln Meadow ’ s a three-and-a-half-hour drive east of San Francisco in the Sierra Nevada Mountains, at about 2, 000 meters altitude, and I ’ ve been recording there for many years. In 1988, a logging company convinced local residents that there would be absolutely no environmental impact from a new method they were trying called”selective logging,”taking out a tree here and there rather than clear-cutting a whole area. With permission granted to record both before and after the operation, I set up my gear and captured a large number of dawn choruses to very strict protocol and calibrated recordings, because I wanted a really good baseline. This is an example of a spectrogram. A spectrogram is a graphic illustration of sound with time from left to right across the page — 15 seconds in this case is represented — and frequency from the bottom of the page to the top, lowest to highest. And you can see that the signature of a stream is represented here in the bottom third or half of the page, while birds that were once in that meadow are represented in the signature across the top. There were a lot of them. And here ’ s Lincoln Meadow before selective logging. Well, a year later I returned, and using the same protocols and recording under the same conditions, I recorded a number of examples of the same dawn choruses, and now this is what we ’ ve got. This is after selective logging. You can see that the stream is still represented in the bottom third of the page, but notice what ’ s missing in the top two thirds. Coming up is the sound of a woodpecker. Well, I ’ ve returned to Lincoln Meadow 15 times in the last 25 years, and I can tell you that the biophony, the density and diversity of that biophony, has not yet returned to anything like it was before the operation. But here ’ s a picture of Lincoln Meadow taken after, and you can see that from the perspective of the camera or the human eye, hardly a stick or a tree appears to be out of place, which would confirm the logging company ’ s contention that there ’ s nothing of environmental impact. However, our ears tell us a very different story. Young students are always asking me what these animals are saying, and really I ’ ve got no idea. But I can tell you that they do express themselves. Whether or not we understand it is a different story. I was walking along the shore in Alaska, and I came across this tide pool filled with a colony of sea anemones, these wonderful eating machines, relatives of coral and jellyfish. And curious to see if any of them made any noise, I dropped a hydrophone, an underwater microphone covered in rubber, down the mouth part, and immediately the critter began to absorb the microphone into its belly, and the tentacles were searching out of the surface for something of nutritional value. The static-like sounds that are very low, that you ’ re going to hear right now. Yeah, but watch. When it didn ’ t find anything to eat — I think that ’ s an expression that can be understood in any language. At the end of its breeding cycle, the Great Basin Spadefoot toad digs itself down about a meter under the hard-panned desert soil of the American West, where it can stay for many seasons until conditions are just right for it to emerge again. And when there ’ s enough moisture in the soil in the spring, frogs will dig themselves to the surface and gather around these large, vernal pools in great numbers. And they vocalize in a chorus that ’ s absolutely in sync with one another. And they do that for two reasons. The first is competitive, because they ’ re looking for mates, and the second is cooperative, because if they ’ re all vocalizing in sync together, it makes it really difficult for predators like coyotes, foxes and owls to single out any individual for a meal. This is a spectrogram of what the frog chorusing looks like when it ’ s in a very healthy pattern. Mono Lake is just to the east of Yosemite National Park in California, and it ’ s a favorite habitat of these toads, and it ’ s also favored by U. S. Navy jet pilots, who train in their fighters flying them at speeds exceeding 1, 100 kilometers an hour and altitudes only a couple hundred meters above ground level of the Mono Basin, very fast, very low, and so loud that the anthrophony, the human noise, even though it ’ s six and a half kilometers from the frog pond you just heard a second ago, it **masked** the sound of the chorusing toads. You can see in this spectrogram that all of the energy that was once in the first spectrogram is gone from the top end of the spectrogram, and that there ’ s breaks in the chorusing at two and a half, four and a half, and six and a half seconds, and then the sound of the jet, the signature, is in yellow at the very bottom of the page. Now at the end of that flyby, it took the frogs fully 45 minutes to regain their chorusing synchronicity, during which time, and under a full moon, we watched as two coyotes and a great horned owl came in to pick off a few of their numbers. The good news is that, with a little bit of habitat restoration and fewer flights, the frog populations, once diminishing during the 1980s and early ’ 90s, have pretty much returned to normal. I want to end with a story told by a beaver. It ’ s a very sad story, but it really **illustrates** how animals can sometimes show emotion, a very controversial subject among some older biologists. A colleague of mine was recording in the American Midwest around this pond that had been formed maybe 16, 000 years ago at the end of the last ice age. It was also formed in part by a beaver dam at one end that held that whole ecosystem together in a very delicate balance. And one afternoon, while he was recording, there suddenly appeared from out of nowhere a couple of game wardens, who for no apparent reason, walked over to the beaver dam, dropped a stick of dynamite down it, blowing it up, killing the female and her young babies. Horrified, my colleagues remained behind to gather his thoughts and to record whatever he could the rest of the afternoon, and that evening, he captured a **remarkable** event : the lone surviving male beaver swimming in slow circles crying out inconsolably for its lost mate and offspring. This is probably the saddest sound I ’ ve ever heard coming from any organism, human or other. Yeah. Well. There are many facets to soundscapes, among them the ways in which animals taught us to dance and sing, which I ’ ll save for another time. But you have heard how biophonies help clarify our understanding of the natural world. You ’ ve heard the impact of resource extraction, human noise and habitat destruction. And where environmental sciences have typically tried to understand the world from what we see, a much fuller understanding can be got from what we hear. Biophonies and geophonies are the signature voices of the natural world, and as we hear them, we ’ re endowed with a sense of place, the true story of the world we live in. In a matter of seconds, a soundscape reveals much more information from many perspectives, from quantifiable data to cultural inspiration. Visual capture implicitly frames a limited frontal perspective of a given **spatial** context, while soundscapes widen that scope to a full 360 degrees, completely enveloping us. And while a picture may be worth 1, 000 words, a soundscape is worth 1, 000 pictures. And our ears tell us that the whisper of every leaf and creature speaks to the natural sources of our lives, which indeed may hold the secrets of love for all things, especially our own humanity, and the last word goes to a jaguar from the Amazon. Thank you for listening.

Text Sample 862: POS Dim 6 – 2013 – Score 6.00 – 2013/t002089.txt
I ' m almost like a crazy evangelical. I ' ve always known that the age of design is upon us, almost like a rapture. If the day is sunny, I think,"Oh, the gods have had a good design day."Or, I go to a show and I see a beautiful piece by an artist, particularly beautiful, I say he ' s so good because he clearly looked to design to understand what he needed to do. So I really do believe that design is the highest form of creative expression. That ' s why I ' m talking to you today about the age of design, and the age of design is the age in which design is still cute furniture, is still posters, is still fast cars, what you see at MoMA today. But in truth, what I really would like to explain to the public and to the audiences of MoMA is that the most interesting chairs are the ones that are actually made by a robot, like this beautiful chair by Dirk Vander Kooij, where a robot deposits a toothpaste-like slur of recycled refrigerator parts, as if he were a big candy, and makes a chair out of it. Or good design is digital fonts that we use all the time and that become part of our identity. I want people to understand that design is so much more than cute chairs, that it is first and foremost everything that is around us in our life. And it ' s interesting how so much of what we ' re talking about tonight is not simply design but interaction design. And in fact, interaction design is what I ' ve been trying to insert in the collection of the Museum of Modern Art for a few years, starting not very timidly but just pointedly with works, for instance, by Martin Wattenberg — the way a machine plays chess with itself, that you see here, or Lisa Strausfeld and her partners, the Sugar **interface** for One Laptop Per Child, Toshio Iwai ' s Tenori-On musical instruments, and Philip Worthington ' s Shadow Monsters, and John Maeda ' s Reactive Books, and also Jonathan Harris and Sep Kamvar ' s I Want You To Want Me. These were some of the first acquisitions that really introduced the idea of interaction design to the public. But more recently, I ' ve been trying really to go even deeper into interaction design with examples that are emotionally really suggestive and that really explain interaction design at a level that is almost undeniable. The Wind Map, by Wattenberg and Fernanda Vigas, I don ' t know if you ' ve ever seen it — it ' s really fantastic. It looks at the territory of the United States as if it were a wheat field that is procured by the winds and that is really giving you a pictorial image of what ' s going on with the winds in the United States. But also, more recently, we started acquiring video games, and that ' s where all hell broke loose in a really interesting way. There are still people that believe that there ' s a high and there ' s a low. And that ' s really what I find so intriguing about the reactions that we ' ve had to the anointment of video games in the MoMA collection. We ' ve — No, first of all, New York Magazine always gets it. I love them. So we are in the right quadrant. We are in the Highbrow — that ' s daring, that ' s courageous — and Brilliant, which is great. Timidly, we ' ve been higher on the **diagonal** in other situations, but it ' s okay. It ' s good. It ' s good. It ' s good. But here comes the art critic. Oh, that was fantastic. So the first was Jonathan Jones from The Guardian."Sorry, MoMA, video games are not art."Did I ever say they were

art? I was talking about interaction design. Excuse me.”Exhibiting Pac-Man and Tetris alongside Picasso and Van Gogh”— They ’ re two floors away. —”will mean game over for any real understanding of art.”I ’ m bringing in the end of the world. You know? We were talking about the rapture? It ’ s coming. And Jonathan Jones is making it happen. So the same Guardian rebuts,”Are video games art : the debate that shouldn ’ t be. Last week, Guardian art critic blah blah suggested that games cannot qualify as art. But is he right? And does it matter?”Thank you. Does it matter? You know, it ’ s like once again there ’ s this whole problem of design being often misunderstood for art, or the idea that is so diffuse that designers want to aspire to, would like to be called, artists. No. Designers aspire to be really great designers. Thank you very much. And that ’ s more than enough. So my knight in shining armor, John Maeda, without any prompt, came out with this big declaration on why video games belong in the MoMA. And that was fantastic. And I thought that was it. But then there was another wonderfully pretentious article that came out in The New Republic, so pretentious, by Liel Leibovitz, and it said,”MoMA has mistaken video games for art.”Again.”The museum is putting Pac-Man alongside Picasso.”Again.”That misses the point.”Excuse me. You ’ re missing the point. And here, look, the above question is put bluntly :”Are video games art? No. Video games aren ’ t art because they are quite thoroughly something else : code.”Oh, so Picasso is not art because it ’ s oil paint. Right? So it ’ s so fantastic to see how these feathers that were ruffled, and these reactions, were so vehement. And you know what? The International Cat Video Film Festival didn ’ t have that much of a reaction. I think this was truly fantastic. We were talking about dancing ponies, but I was really jealous of the Walker Arts Center for putting up this festival, because it ’ s very, very wonderful. And there ’ s this Flaubert quote that I love :”I have always tried to live in an ivory tower, but a tide of shit is beating at its walls, threatening to undermine it.”I consider myself the tide of shit. You know, we have to go through that. Even in the 1930s, my colleagues that were trying to put together an abstract art show had all of these works stopped by the customs officers that decided they were not art. So it ’ s happened before, and it will happen in the future, but right now I can tell you that I am so, so proud to be able to call Pac-Man part of the MoMA collection. And the same with, for instance, Tetris, original version, the Soviet one. And you know, the amount of work — yeah, Alexey Pajitnov was working for the Soviet government and that ’ s how he developed Tetris, and Alexey himself reconstructed the whole game and even gave us a simulation of the cathode ray tube that makes it look slightly bombed. And it ’ s fantastic. So behind these acquisitions is an enormous amount of work, because we ’ re still the Museum of Modern Art, so even when we tackle popular culture, we tackle it as a form of interaction design and as something that has to go into the collection at MoMA, therefore, has to be researched. So to get to choosing Eric Chahi ’ s wonderful Another World, amongst others, we put together a panel of experts, and we worked on this acquisition, and it ’ s mostly myself and Kate Carmody and Paul Galloway. We worked on it for a year and a half. So many people helped us — designers of games, you might know Jamin Warren and his collaborators at Kill Screen magazine, and you know, Kevin Slavin. You name it. We bugged everybody, because we knew that we were ignorant. We were not real gamers enough, so we had to really talk to them. And so we decided, of course, to have Sim City 2000, not the other Sim City, that one in particular, so the criteria that we developed along the way were really strong, and were not only criteria of selection. They were also criteria of exhibition and of preservation. That ’ s what makes this acquisition more than a little game or a little joke. It ’ s truly a way to think of how to preserve and show **artifacts** that will more and more become part of our lives in the future. We live today, as you know very well, not in the digital, not in the physical, but in the kind of minestrone that our mind makes of the two. And that ' s really where interaction lies, and that ' s the importance of interaction. And in order to explain interaction, we need to really bring people in and make them realize how interaction is part of their lives. So when I talk about it, I don ' t talk only about video games, which are in a way the purest form of interaction, unadulterated by any kind of function or finality. I also talk about the MetroCard vending machine, which I consider a masterpiece of interaction. I mean, that **interface** is beautiful. It looks like a burly MTA guy coming out of the tunnel. You know, with your mitt you can actually paw the MetroCard, and I talk about how bad ATM machines usually are. So I let people understand that it ' s up to them to know how to judge interaction so as to know when it ' s good or when it ' s bad. So when I show The Sims, I try to make people really feel what it meant to have an interaction with The Sims, not only the fun but also the responsibility that came with the Tamagotchi. You know, video games can be truly deep even when they ' re completely mindless. I ' m sure that all of you know Katamari Damacy. It ' s about rolling a **ball** and picking up as many objects as you can in a finite amount of time and hopefully you ' ll be able to make it into a planet. I ' ve never made it into a planet, but that ' s it. Or, you know, Vib-Ribbon was not distributed here in the United States. It was a PlayStation game, but mostly for Japan. And it was one of the first video games in which you could choose your own music. So you would put into the PlayStation, you would put your own CD, and then the game would change alongside your music. So really fantastic. Not to mention Eve Online. Eve Online is an artificial universe, if you wish, but one of the diplomats that was killed in Benghazi, not Ambassador Stevens, but one of his collaborators, was a really big shot in Eve Online, so here you have a diplomat in the real world that spends his time in Eve Online to kind of test, maybe, all of his ideas about diplomacy and about universe-building, and to the point that the first announcement of the bombing was actually given on Eve Online, and after his death, several parts of the universe were named after him. And I was just recently at the Eve Online fan festival in Reykjavk that was quite amazing. I mean, we ' re talking about an experience that of course can seem weird to many, but that is very educational. Of course, there are games that are even more educational. Dwarf Fortress is like the holy grail of this kind of massive multiplayer online game, and in fact the two Adams brothers were in Reykjavk, and they were greeted by a standing ovation by all the Eve Online fans. It was amazing to see. And it ' s a beautiful game. So you start seeing here that the aesthetics that are so important to a museum collection like MoMA ' s are kept alive also by the selection of these games. And you know, Valve — you know, Portal — is an example of a video game in which you have a certain type of violence which also leads me to talk about one of the biggest issues that we had to discuss when we acquired the video games, what to do with violence. Right? We had to make decisions. At MoMA, interestingly, there ' s a lot of violence depicted in the art part of the collection, but when I came to MoMA 19 years ago, and as an Italian, I said,"You know what, we need a Beretta."And I was told,"No. No guns in the design collection."And I was like,"Why?"Interestingly, I learned that it ' s considered that in design and in the design collection, what you see is what you get. So when you see a gun, it ' s an instrument for killing in the design collection. If it ' s in the art collection, it might be a critique of the killing instrument. So it ' s very interesting. But we are acquiring our critical dimension also in design, so maybe one day we ' ll be able to acquire also the guns. But here, in this particular case, we decided, you know, with Kate and Paul, that we would have no gratuitous violence. So we have Portal because you shoot walls in order to create new spaces. We have Street Fighter II, because martial arts are good. But we don ’ t have GTA because, maybe it ’ s my own reflection, I ’ ve never been able to do anything but crashing cars and shooting prostitutes and pimps. So it was not very constructive. So, I ’ m making fun of it, but we discussed this for so many days. You have no idea. And to this day, I am ambivalent, but when you have instead games like Flow, there ’ s no doubt. It ’ s like, it ’ s about serenity and it ’ s about sublime. It ’ s about experiencing what it means to be a sea creature. Then we have a few also side-scrollers — classical ones. So it ’ s quite a hefty collection. And right now, we started with the first 14, but we have several that are coming up, and the reason why we haven ’ t acquired them yet is because you don ’ t acquire just the game. You acquire the relationship with the company. What we want, what we aspire to, is the code. It ’ s very hard to get, of course. But that ’ s what would enable us to preserve the video games for a really long time, and that ’ s what museums do. They also preserve **artifacts** for posterity. In absence of the code, because, you know, video game companies are not very forthcoming in some cases, in absence of that, we acquire the relationship with the company. We ’ re going to stay with them forever. They ’ re not going to get rid of us. And one day, we ’ ll get that code. But I want to explain to you the criteria that we chose for interaction design. Aesthetics are really important. And I ’ m showing you Core War here, which is an early game that takes advantage aesthetically of the limitations of the processor. So the kind of interferences that you see here that look like beautiful barriers in the game are actually a consequence of the processor ’ s limitedness, which is fantastic. So aesthetics is always important. And so is space, the **spatial** aspect of games. You know, I feel that the best video games are the ones that have really savvy architects that are behind them, and if they ’ re not architects, bona fide trained in architecture, they have that feeling. But the **spatial** evolution in video games is extremely important. Time. The way we experience time in video games, as in other forms of interaction design, is really quite amazing. It can be real time or it can be the time within the game, as is in Animal Crossing, where seasons follow each other at their own pace. So time, space, aesthetics, and then, most important, behavior. The real core issue of interaction design is behavior. Designers that deal with interaction design behaviors that go to influence the rest of our lives. They ’ re not just limited to our interaction with the screen. In this case, I ’ m showing you Marble Madness, which is a beautiful game in which the controller is a big sphere that vibrates with you, so you have a sphere that ’ s moving in this landscape, and the sphere, the controller itself, gives you a sense of the movement. In a way, you can see how video games are the purest aspect of interaction design and are very useful to explain what interaction is. We don ’ t want to show the video games with the paraphernalia. No arcade nostalgia. If anything, we want to show the code, and here you see Ben Fry ’ s distellamap of Pac-Man, of the Pac-Man code. So the way we acquired the games is very interesting and very unorthodox. You see them here **displayed** alongside other examples of design, furniture and other parts, but there ’ s no paraphernalia, no nostalagia, only the screen and a little shelf with the controllers. The controllers are, of course, part of the experience, so you cannot do away with it. But interestingly, this choice was not condemned too vehemently by gamers. I was afraid that they would kill us, and instead they understood, especially when I told them that I was trying to apply the same stratagem that Philip Johnson applied in 1934 when he wanted to make people understand the importance of design, and he took propeller blades and pieces of machinery and in the MoMA galleries he put them on white pedestals against white walls, as if they were Brancusi sculptures. He created this strange distance, this shock, that made people realize how gorgeous formally, and also important functionally, design pieces were. I would like to do the same with video games. By getting rid of the sticky carpets and the cigarette butts and everything else that we might remember from our childhood, I want people to understand that those are important forms of design. And in a way, the video games, the fonts and everything else lead us to make people understand a wider meaning for design. One of my dream acquisitions, which has been on hold for a few years but now will come back on the front burner, is a 747. I would like to acquire it, but without owning it. I don ’ t want it to be at MoMA and possessed by MoMA. I want it to keep flying. So it ’ s an acquisition where MoMA makes an arrangement with an airline and keeps the Boeing 747 flying. And the same with the”@”sign that we acquired a few years ago. It was the first example of an acquisition of something that is in the public domain. And what I say to people, it ’ s almost as if a butterfly were flying by and we captured the shadow on the wall, and just we ’ re showing the shadow. So in a way, we ’ re showing a manifestation of something that is truly important and that is part of our identity but that nobody can have. And it ’ s too long to explain the acquisition, but if you want to go on the MoMA blog, there ’ s a long post where I explain why it ’ s such a great example of design. Along the way, I ’ ve had to burn a few chairs. You know? I ’ ve had to do away with a few concepts of design past. But I see that people are coming along, that the audiences, paradoxically, are much more responsive and much more understanding of this expansion of design than some of my colleagues are. Design is truly everywhere, and design is as important as anything, and I ’ m so glad that, because of its diversity and because of its centrality to our lives, many more people are coming to it as a profession, as a passion, and as, very simply, part of their own culture. Thank you very much.

Text Sample 863: POS Dim 6 – 2013 – Score 5.00 – 2013/t002027.txt

Diana Reiss : You may think you ’ re looking through a window at a dolphin spinning playfully, but what you ’ re actually looking through is a two-way mirror at a dolphin looking at itself spinning playfully. This is a dolphin that is self-aware. This dolphin has self-awareness. It ’ s a young dolphin named Bayley. I ’ ve been very interested in understanding the nature of the intelligence of dolphins for the past 30 years. How do we explore intelligence in this animal that ’ s so different from us? And what I ’ ve used is a very simple research tool, a mirror, and we ’ ve gained great information, reflections of these animal minds. Dolphins aren ’ t the only animals, the only non-human animals, to show mirror self-recognition. We used to think this was a uniquely human ability, but we learned that the great apes, our closest relatives, also show this ability. Then we showed it in dolphins, and then later in elephants. We did this work in my lab with the dolphins and elephants, and it ’ s been recently shown in the magpie. Now, it ’ s interesting, because we ’ ve embraced this Darwinian view of a continuity in physical evolution, this physical continuity. But we ’ ve been much more reticent, much slower at recognizing this continuity in cognition, in emotion, in consciousness in other animals. Other animals are conscious. They ’ re emotional. They ’ re aware. There have been multitudes of studies with many species over the years that have given us exquisite evidence for thinking and consciousness in other animals, other animals that are quite different than we are in form. We are not alone. We are not alone in these abilities. And I hope, and one of my biggest dreams, is that, with our growing awareness about the consciousness of others and our relationship with the rest of the animal world, that we ’ ll give them the respect and protection that they deserve. So that ’ s a wish I ’ m throwing out here for everybody, and I hope I can really engage you in this idea. Now, I want to return to dolphins, because these are the animals that I feel like I ’ ve been working up closely and personal with for over 30 years. And these are real personalities. They are not persons, but they ’ re personalities in every sense of the word. And you can ’ t get more alien than the dolphin. They are very different from us in body form. They ’ re radically different. They come from a radically different environment. In fact, we ’ re separated by 95 million years of divergent evolution. Look at this body. And in every sense of making a pun here, these are true non-terrestrials. I wondered how we might **interface** with these animals. In the 1980s, I developed an underwater keyboard. This was a custom-made touch-screen keyboard. What I wanted to do was give the dolphins choice and control. These are big brains, highly social animals, and I thought, well, if we give them choice and control, if they can hit a symbol on this keyboard — and by the way, it was **interfaced** by fiber optic cables from Hewlett-Packard with an Apple II computer. This seems prehistoric now, but this was where we were with technology. So the dolphins could hit a key, a symbol, they heard a computer-generated whistle, and they got an object or activity. Now here ’ s a little video. This is Delphi and Pan, and you ’ re going to see Delphi hitting a key, he hears a computer-generated whistle — — and gets a **ball**, so they can actually ask for things they want. What was **remarkable** is, they explored this keyboard on their own. There was no intervention on our part. They explored the keyboard. They played around with it. They figured out how it worked. And they started to quickly imitate the sounds they were hearing on the keyboard. They imitated on their own. Beyond that, though, they started learning associations between the symbols, the sounds and the objects. What we saw was self-organized learning, and now I ’ m imagining, what can we do with new technologies? How can we create **interfaces**, new windows into the minds of animals, with the technologies that exist today? So I was thinking about this, and then, one day, I got a call from Peter. Peter Gabriel : I make noises for a living. On a good day, it ’ s music, and I want to talk a little bit about the most amazing music-making experience I ever had. I ’ m a farm boy. I grew up surrounded by animals, and I would look in these eyes and wonder what was going on there? So as an adult, when I started to read about the amazing breakthroughs with Penny Patterson and Koko, with Sue Savage-Rumbaugh and Kanzi, Panbanisha, Irene Pepperberg, Alex the parrot, I got all excited. What was amazing to me also was they seemed a lot more adept at getting a handle on our language than we were on getting a handle on theirs. I work with a lot of musicians from around the world, and often we don ’ t have any common language at all, but we sit down behind our instruments, and suddenly there ’ s a way for us to connect and emote. So I started cold-calling, and eventually got through to Sue Savage-Rumbaugh, and she invited me down. I went down, and the bonobos had had access to percussion instruments, musical toys, but never before to a keyboard. At first they did what infants do, just bashed it with their fists, and then I asked, through Sue, if Panbanisha could try with one finger only. Sue Savage-Rumbaugh : Can you play a grooming song? I want to hear a grooming song. Play a real quiet grooming song. PG : So groom was the subject of the piece. So I ’ m just behind, jamming, yeah, this is what we started with. Sue ’ s encouraging her to continue a little more. She discovers a note she likes, finds the octave. She ’ d never sat at a keyboard before. Nice triplets. SSR : You did good. That was very good. PG : She hit good. So that night, we began to dream, and we thought, perhaps the most amazing tool that man ’ s created is the Internet, and what would happen if we could somehow find new **interfaces**, visual-audio **interfaces** that would allow these **remarkable** sentient beings that we share the planet with access? And Sue

Savage-Rumbaugh got excited about that, called her friend Steve Woodruff, and we began hustling all sorts of people whose work related or was inspiring, which led us to Diana, and led us to Neil. Neil Gershenfeld : Thanks, Peter. PG : Thank you. NG : So Peter approached me. I lost it when I saw that clip. He approached me with a vision of doing these things not for people, for animals. And then I was struck in the history of the Internet. This is what the Internet looked like when it was born and you can call that the Internet of middle-aged white men, mostly middle-aged white men. Vint Cerf : NG : Speaking as one. Then, when I first came to TED, which was where I met Peter, I showed this. This is a $1 web server, and at the time that was radical. And the possibility of making a web server for a dollar grew into what became known as the Internet of Things, which is literally an industry now with tremendous implications for health care, energy efficiency. And we were happy with ourselves. And then when Peter showed me that, I realized we had missed something, which is the rest of the planet. So we started up this interspecies Internet project. Now we started talking with TED about how you bring dolphins and great apes and elephants to TED, and we realized that wouldn ' t work. So we ' re going to bring you to them. So if we could switch to the audio from this computer, we ' ve been video conferencing with cognitive animals, and we ' re going to have each of them just briefly introduce them. And so if we could also have this up, great. So the first site we ' re going to meet is Cameron Park Zoo in Waco, with orangutans. In the daytime they live outside. It ' s nighttime there now. So can you please go ahead? Terri Cox : Hi, I ' m Terri Cox with the Cameron Park Zoo in Waco, Texas, and with me I have KeraJaan and Mei, two of our Bornean orangutans. During the day, they have a beautiful, large outdoor habitat, and at night, they come into this habitat, into their night quarters, where they can have a climate-controlled and secure environment to sleep in. We participate in the Apps for Apes program Orangutan Outreach, and we use iPads to help stimulate and enrich the animals, and also help raise awareness for these critically endangered animals. And they share 97 percent of our DNA and are incredibly intelligent, so it ' s so exciting to think of all the opportunities that we have via technology and the Internet to really enrich their lives and open up their world. We ' re really excited about the possibility of an interspecies Internet, and K. J. has been enjoying the conference very much. NG : That ' s great. When we were rehearsing last night, he had fun watching the elephants. Next user group are the dolphins at the National Aquarium. Please go ahead. Allison Ginsburg : Good evening. Well, my name is Allison Ginsburg, and we ' re live in Baltimore at the National Aquarium. Joining me are three of our eight Atlantic bottlenose dolphins : 20-year-old Chesapeake, who was our first dolphin born here, her four-year-old daughter Bayley, and her half sister, 11-year-old Maya. Now, here at the National Aquarium we are committed to excellence in animal care, to research, and to conservation. The dolphins are pretty intrigued as to what ' s going on here tonight. They ' re not really used to having cameras here at 8 o ' clock at night. In addition, we are very committed to doing different types of research. As Diana mentioned, our animals are involved in many different research studies. NG : Those are for you. Okay, that ' s great, thank you. And the third user group, in Thailand, is Think Elephants. Go ahead, Josh. Josh Plotnik : Hi, my name is Josh Plotnik, and I ' m with Think Elephants International, and we ' re here in the Golden Triangle of Thailand with the Golden Triangle Asian Elephant Foundation elephants. And we have 26 elephants here, and our research is focused on the evolution of intelligence with elephants, but our foundation Think Elephants is focused on bringing elephants into classrooms around the world virtually like this and showing people how incredible these animals are. So we ' re able to bring the camera right up to the elephant, put food into the elephant ’ s mouth, show people what ’ s going on inside their mouths, and show everyone around the world how incredible these animals really are. NG : Okay, that ’ s great. Thanks Josh. And once again, we ’ ve been building great relationships among them just since we ’ ve been rehearsing. So at that point, if we can go back to the other computer, we were starting to think about how you integrate the rest of the biomass of the planet into the Internet, and we went to the best possible person I can think of, which is Vint Cerf, who is one of the founders who gave us the Internet. Vint? VC : Thank you, Neil. A long time ago in a galaxy — oops, wrong script. Forty years ago, Bob Kahn and I did the design of the Internet. Thirty years ago, we turned it on. Just last year, we turned on the production Internet. You ’ ve been using the experimental version for the last 30 years. The production version, it uses IP version 6. It has 3. 4 times 10 to the 38th possible terminations. That ’ s a number only that Congress can appreciate. But it leads to what is coming next. When Bob and I did this design, we thought we were building a system to connect computers together. What we very quickly discovered is that this was a system for connecting people together. And what you ’ ve seen tonight tells you that we should not restrict this network to one species, that these other intelligent, sentient species should be part of the system too. This is the system as it looks today, by the way. This is what the Internet looks like to a computer that ’ s trying to figure out where the traffic is supposed to go. This is generated by a program that ’ s looking at the connectivity of the Internet, and how all the **various** networks are connected together. There are about 400, 000 networks, interconnected, run independently by 400, 000 different operating agencies, and the only reason this works is that they all use the same standard TCP / IP protocols. Well, you know where this is headed. The Internet of Things tell us that a lot of computer-enabled appliances and devices are going to become part of this system too : appliances that you use around the house, that you use in your office, that you carry around with yourself or in the car. That ’ s the Internet of Things that ’ s coming. Now, what ’ s important about what these people are doing is that they ’ re beginning to learn how to communicate with species that are not us but share a common sensory environment. We ’ re beginning to explore what it means to communicate with something that isn ’ t just another person. Well, you can see what ’ s coming next. All kinds of possible sentient beings may be interconnected through this system, and I can ’ t wait to see these experiments unfold. What happens after that? Well, let ’ s see. There are machines that need to talk to machines and that we need to talk to, and so as time goes on, we ’ re going to have to learn how to communicate with computers and how to get computers to communicate with us in the way that we ’ re accustomed to, not with keyboards, not with mice, but with speech and gestures and all the natural human language that we ’ re accustomed to. So we ’ ll need something like C3PO to become a translator between ourselves and some of the other machines we live with. Now, there is a project that ’ s underway called the interplanetary Internet. It ’ s in operation between Earth and Mars. It ’ s operating on the International Space Station. It ’ s part of the spacecraft that ’ s in orbit around the Sun that ’ s rendezvoused with two planets. So the interplanetary system is on its way, but there ’ s a last project, which the Defense Advanced Research Projects Agency, which funded the original ARPANET, funded the Internet, funded the interplanetary architecture, is now funding a project to design a spacecraft to get to the nearest star in 100 years ’ time. What that means is that what we ’ re learning with these interactions with other species will teach us, ultimately, how we might interact with an alien from another world. I can hardly wait. June Cohen : So first of all, thank you, and I would like to acknowledge that four people who could talk to us for full four days actually managed to stay to four minutes each, and we thank you for that. I have so many questions, but maybe a few practical things that the audience might want to know. You ’ re launching this idea here at TED — PG : Today. JC : Today. This is the first time you ’ re talking about it. Tell me a little bit about where you ’ re going to take the idea. What ’ s next? PG : I think we want to engage as many people here as possible in helping us think of smart **interfaces** that will make all this possible. NG : And just mechanically, there ’ s a 501 and web infrastructure and all of that, but it ’ s not quite ready to turn on, so we ’ ll roll that out, and contact us if you want the information on it. The idea is this will be — much like the Internet functions as a network of networks, which is Vint ’ s core contribution, this will be a wrapper around all of these initiatives, that are wonderful individually, to link them globally. JC : Right, and do you have a web address that we might look for yet? NG : Shortly. JC : Shortly. We will come back to you on that. And very quickly, just to clarify. Some people might have looked at the video that you showed and thought, well, that ’ s just a webcam. What ’ s special about it? If you could talk for just a moment about how you want to go past that? NG : So this is scalable video infrastructure, not for a few to a few but many to many, so that it scales to symmetrical video sharing and content sharing across these sites around the planet. So there ’ s a lot of back-end signal processing, not for one to many, but for many to many. JC : Right, and then on a practical level, which technologies are you looking at first? I know you mentioned that a keyboard is a really key part of this. DR : We ’ re trying to develop an interactive touch screen for dolphins. This is sort of a continuation of some of the earlier work, and we just got our first seed money today towards that, so it ’ s our first project. JC : Before the talk, even. DR : Yeah. JC : Wow. Well done. All right, well thank you all so much for joining us. It ’ s such a delight to have you on the stage. DR : Thank you. VC : Thank you.

Text Sample 864: POS Dim 6 – 2010 – Score 8.00 – 2010/t002842.txt

We ’ re 25, 26 years after the advent of the Macintosh, which was an astoundingly seminal event in the history of human-machine **interface** and in computation in general. It fundamentally changed the way that people thought about computation, thought about computers, how they used them and who and how many people were able to use them. It was such a radical change, in fact, that the early Macintosh development team in ’ 82, ’ 83, ’ 84 had to write an entirely new operating system from the ground up. Now, this is an interesting little message, and it ’ s a lesson that has since, I think, been forgotten or lost or something, and that is, namely, that the OS is the **interface**. The **interface** is the OS. It ’ s like the land and the king they ’ re inseparable, they are one. And to write a new operating system was not a capricious matter. It wasn ’ t just a matter of tuning up some graphics routines. There were no graphics routines. There were no mouse drivers. So it was a necessity. But in the quarter-century since then, we ’ ve seen all of the fundamental supporting technologies go berserk. So memory capacity and disk capacity have been multiplied by something between 10, 000 and a million. Same thing for processor speeds. Networks, we didn ’ t have networks at all at the time of the Macintosh ’ s introduction, and that has become the single most salient aspect of how we live with computers. And, of course, graphics : Today 84 dollars and 97 cents at Best Buy buys you more graphics power than you could have gotten for a million bucks from SGI only a decade ago. So we ’ ve got that incredible ramp-up. Then, on the side, we ’ ve got the Web and, increasingly, the cloud, which is fantastic, but also — in the regard in which an **interface** is fundamental — kind of a distraction. So we ’ ve forgotten to invent new **interfaces**. Certainly we ’ ve seen in recent years a lot of change in that regard, and people are starting to wake up about that. So what happens next? Where do we go from there? The problem, as we see it, has to do with a single, simple word :”space,”or a single, simple phrase :”real world geometry.”Computers and the programming languages that we talk to them in, that we teach them in, are hideously insensate when it comes to space. They don ’ t understand real world space. It ’ s a funny thing because the rest of us occupy it quite frequently and quite well. They also don ’ t understand time, but that ’ s a matter for a separate talk. So what happens if you start to explain space to them? One thing you might get is something like the Luminous Room. The Luminous Room is a system in which it ’ s considered that input and output spaces are co-located. That ’ s a strangely simple, and yet unexplored idea, right? When you use a mouse, your hand is down here on the mouse pad. It ’ s not even on the same plane as what you ’ re talking about : The pixels are up on the **display**. So here was a room in which all the walls, floors, ceilings, pets, potted plants, whatever was in there, were capable, not only of **display** but of sensing as well. And that means input and output are in the same space enabling stuff like this. That ’ s a digital storage in a physical container. The contract is the same as with real word objects in real world containers. Has to come back out, whatever you put in. This little design experiment that was a small office here knew a few other tricks as well. If you presented it with a chess board, it tried to figure out what you might mean by that. And if there was nothing for them to do, the chess pieces eventually got bored and hopped away. The academics who were overseeing this work thought that that was too frivolous, so we built deadly serious applications like this optics prototyping workbench in which a toothpaste cap on a cardboard box becomes a laser. The beam splitters and lenses are represented by physical objects, and the system projects down the laser beam path. So you ’ ve got an **interface** that has no **interface**. You operate the world as you operate the real world, which is to say, with your hands. Similarly, a digital wind tunnel with digital wind flowing from right to left — not that **remarkable** in a sense ; we didn ’ t invent the mathematics. But if you **displayed** that on a CRT or flat panel **display**, it would be meaningless to hold up an arbitrary object, a real world object in that. Here, the real world merges with the simulation. And finally, to pull out all the stops, this is a system called Urp, for urban planners, in which we give architects and urban planners back the models that we confiscated when we insisted that they use CAD systems. And we make the machine meet them half way. It projects down digital shadows, as you see here. And if you introduce tools like this inverse clock, then you can control the sun ’ s position in the sky. That ’ s 8 a. m. shadows. They get a little shorter at 9 a. m. There you are, swinging the sun around. Short shadows at noon and so forth. And we built up a series of tools like this. There are inter-shadowing studies that children can operate, even though they don ’ t know anything about urban planning : To move a building, you simply reach out your hand and you move the building. A material wand makes the building into a sort of Frank Gehry thing that reflects light in all directions. Are you blinding passers by and motorists on the freeways? A zoning tool connects distant structures, a building and a roadway. Are you going to get sued by the zoning commission? And so forth. Now, if these ideas seem familiar or perhaps even a little dated, that ’ s great ; they should seem familiar. This work is 15 years old. This stuff was undertaken at MIT and the Media Lab under the incredible direction of Professor Hiroshi Ishii, director of the Tangible Media Group. But it was that work that was seen by Alex McDowell, one of the world ’ s legendary production designers. But Alex was preparing a little, sort of obscure, indie, arthouse film called”Minority Report”for Steven Spielberg, and invited us to come out from MIT and design the **interfaces** that would appear in that film. And the great thing about it was that Alex was so dedicated to the idea of verisimilitude, the idea that the putative 2054 that we were painting in the film be believable, that he allowed us to take on that design work as if it were an R&D effort. And the result is sort of gratifyingly perpetual. People still reference those sequences in”Minority Report”when they talk about new UI design. So this led full circle, in a strange way, to build these ideas into what we believe is the necessary future of human machine **interface** : the **Spatial** Operating Environment, we call it. So here we have a bunch of stuff, some images. And, using a hand, we can actually exercise six degrees of freedom, six degrees of navigational control. And it ’ s fun to fly through Mr. Beckett ’ s eye. And you can come back out through the scary orangutan. And that ’ s all well and good. Let ’ s do something a little more difficult. Here, we have a whole bunch of disparate images. We can fly around them. So navigation is a fundamental issue. You have to be able to navigate in 3D. Much of what we want computers to help us with in the first place is inherently **spatial**. And the part that isn ’ t **spatial** can often be spatialized to allow our wetware to make greater sense of it. Now we can distribute this stuff in many different ways. So we can throw it out like that. Let ’ s reset it. We can organize it this way. And, of course, it ’ s not just about navigation, but about manipulation as well. So if we don ’ t like stuff, or we ’ re intensely curious about Ernst Haeckel ’ s scientific falsifications, we can pull them out like that. And then if it ’ s time for analysis, we can pull back a little bit and ask for a different distribution. Let ’ s just come down a bit and fly around. So that ’ s a different way to look at stuff. If you ’ re of a more analytical nature then you might want, actually, to look at this as a color histogram. So now we ’ ve got the stuff color-sorted, angle maps onto color. And now, if we want to select stuff, 3D, space, the idea that we ’ re tracking hands in real space becomes really important because we can reach in, not in 2D, not in fake 2D, but in actual 3D. Here are some selection planes. And we ’ ll perform this Boolean operation because we really love yellow and tapirs on green grass. So, from there to the world of real work. Here ’ s a logistics system, a small piece of one that we ’ re currently building. There ’ re a lot of elements. And one thing that ’ s very important is to combine traditional tabular data with three-dimensional and geospatial information. So here ’ s a familiar place. And we ’ ll bring this back here for a second. Maybe select a little bit of that. And bring out this graph. And we should, now, be able to fly in here and have a closer look. These are logistics elements that are scattered across the United States. One thing that three-dimensional interactions and the general idea of imbuing computation with space affords you is a final destruction of that unfortunate one-to-one pairing between human beings and computers. That ’ s the old way, that ’ s the old mantra : one machine, one human, one mouse, one screen. Well, that doesn ’ t really cut it anymore. In the real world, we have people who collaborate ; we have people who have to work together, and we have many different **displays**. And we might want to look at these **various** images. We might want to ask for some help. The author of this new pointing device is sitting over there, so I can pull this from there to there. These are unrelated machines, right? So the computation is space soluble and network soluble. So I ’ m going to leave that over there because I have a question for Paul. Paul is the designer of this wand, and maybe its easiest for him to come over here and tell me in person what ’ s going on. So let me get some of these out of the way. Let ’ s pull this apart : I ’ ll go ahead and explode it. Kevin, can you help? Let me see if I can help us find the circuit board. Mind you, it ’ s a sort of gratuitous field- stripping exercise, but we do it in the lab all the time.

All right. So collaborative work, whether it ' s immediately co-located or distant and distinct, is always important. And again, that stuff needs to be undertaken in the context of space. And finally, I ' d like to leave you with a glimpse that takes us back to the world of imagery. This is a system called TAMPER, which is a slightly whimsical look at what the future of editing and media manipulation systems might be. We at Oblong believe that media should be accessible in much more fine-grained form. So we have a large number of **movies** stuck inside here. And let ' s just pick out a few elements. We can zip through them as a possibility. We can grab elements off the front, where upon they reanimate, come to life, and drag them down onto the table here. We ' ll go over to Jacques Tati here and grab our blue friend and put him down on the table as well. We may need more than one. And we probably need, well, we probably need a cowboy to be quite honest. Yeah, let ' s take that one. You see, cowboys and French farce people don ' t go well together, and the system knows that. Let me leave with one final thought, and that is that one of the greatest English language writers of the last three decades suggested that great art is always a gift. And he wasn ' t talking about whether the novel costs 24. 95 [dollars], or whether you have to spring 70 million bucks to buy the stolen Vermeer ; he was talking about the circumstances of its creation and of its existence. And I think that it ' s time that we asked for the same from technology. Technology is capable of expressing and being imbued with a certain generosity, and we need to demand that, in fact. For some of this kind of technology, ground center is a combination of design, which is crucially important. We can ' t have advances in technology any longer unless design is integrated from the very start. And, as well, as of efficacy, agency. We ' re, as human beings, the creatures that create, and we should make sure that our machines aid us in that task and are built in that same image. So I will leave you with that. Thank you. Chris Anderson : So to ask the obvious question — actually this is from Bill Gates — when? CA : When real? When for us, not just in a lab and on a stage? Can it be for every man, or is this just for corporations and **movie** producers? JU : No, it has to be for every human being. That ' s our goal entirely. We won ' t have succeeded unless we take that next big step. I mean it ' s been 25 years. Can there really be only one **interface**? There can ' t. CA : But does that mean that, at your desk or in your home, you need projectors, cameras? You know, how can it work? JU : No, this stuff will be built into the bezel of every **display**. It ' ll be built into architecture. The gloves go away in a matter of months or years. So this is the inevitability about it. CA : So, in your mind, five years time, someone can buy this as part of a standard computer **interface**? JU : I think in five years time when you buy a computer, you ' ll get this. CA : Well that ' s cool. The world has a habit of surprising us as to how these things are actually used. What do you think, what in your mind is the first killer app for this? JU : That ' s a good question, and we ask ourselves that every day. At the moment, our early-adopter customers — and these systems are deployed out in the real world — do all the big data intensive, data heavy problems with it. So, whether it ' s logistics and supply chain management or natural gas and resource extraction, financial services, pharmaceuticals, bioinformatics, those are the topics right now, but that ' s not a killer app. And I understand what you ' re asking. CA : C ' mon, c ' mon. Martial arts, games. C ' mon. John, thank you for making science-fiction real. JU : It ' s been a great pleasure. Thank you to you all.

Text Sample 865: POS Dim 6 – 2010 – Score 4.00 – 2010/t002780.txt

We live on a human-dominated planet, putting unprecedented pressure on the systems on Earth. This is bad news, but perhaps surprising to you, it ’ s also part of the good news. We ’ re the first generation â thanks to science â to be informed that we may be undermining the stability and the ability of planet Earth to support human development as we know it. It ’ s also good news, because the planetary risks we ’ re facing are so large, that business as usual is not an option. In fact, we ’ re in a phase where transformative change is necessary, which opens the window for innovation, for new ideas and new paradigms. This is a scientific journey on the challenges facing humanity in the global phase of sustainability. On this journey, I ’ d like to bring, apart from yourselves, a good friend, a stakeholder, who ’ s always absent when we deal with the negotiations on environmental issues, a stakeholder who refuses to compromise â planet Earth. So I thought I ’ d bring her with me today on stage, to have her as a witness of a **remarkable** journey, which humbly reminds us of the period of grace we ’ ve had over the past 10, 000 years. This is the living conditions on the planet over the last 100, 000 years. It ’ s a very important period â it ’ s roughly half the period when we ’ ve been fully modern humans on the planet. We ’ ve had the same, roughly, abilities that developed civilizations as we know it. This is the environmental conditions on the planet. Here, used as a proxy, temperature variability. It was a jumpy ride. 80, 000 years back in a crisis, we leave Africa, we colonize Australia in another crisis, 60, 000 years back, we leave Asia for Europe in another crisis, 40, 000 years back, and then we enter the remarkably stable Holocene phase, the only period in the whole history of the planet, that we know of, that can support human development. A thousand years into this period, we abandon our hunting and gathering patterns. We go from a couple of million people to the seven billion people we are today. The Mesopotamian culture : we invent agriculture, we domesticate animals and plants. You have the Roman, the Greek and the story as you know it. The only phase, as we know it that can support humanity. The trouble is we ’ re putting a quadruple sqeeze on this poor planet, a quadruple sqeeze, which, as its first squeeze, has population growth of course. Now, this is not only about numbers ; this is not only about the fact that we ’ re seven billion people committed to nine billion people, it ’ s an equity issue as well. The majority of the environmental impacts on the planet have been caused by the rich minority, the 20 percent that jumped onto the industrial bandwagon in the mid-18th century. The majority of the planet, aspiring for development, having the right for development, are in large aspiring for an unsustainable lifestyle, a momentous pressure. The second pressure on the planet is, of course the climate agenda â the big issue â where the policy interpretation of science is that it would be enough to stabilize greenhouse gases at 450 ppm to avoid average temperatures exceeding two degrees, to avoid the risk that we may be destabilizing the West Antarctic Ice Sheet, holding six meters â level rising, the risk of destabilizing the Greenland Ice Sheet, holding another seven meters â sea level rising. Now, you would have wished the climate pressure to hit a strong planet, a resilient planet, but unfortunately, the third pressure is the ecosystem decline. Never have we seen, in the past 50 years, such a sharp decline of ecosystem functions and services on the planet, one of them being the ability to regulate climate on the long term, in our forests, land and biodiversity. The forth pressure is surprise, the notion and the evidence that we need to abandon our old paradigm, that ecosystems behave linearly, predictably, controllably in our â so to say â linear systems, and that in fact, surprise is universal, as systems tip over very rapidly, abruptly and often irreversibly. This, dear friends, poses a human pressure on the planet of momentous scale. We may, in fact, have entered a new geological era â the Anthropocene, where humans are the predominant driver of change at a planetary level. Now, as a scientist, what ' s the evidence for this? Well, the evidence is, unfortunately, ample. It ' s not only carbon dioxide that has this hockey stick pattern of accelerated change. You can take virtually any parameter that matters for human well-being â nitrous oxide, methane, deforestation, overfishing land degredation, loss of species â they all show the same pattern over the past 200 years. Simultaneously, they branch off in the mid- 50s, 10 years after the Second World War, showing very clearly that the great acceleration of the human enterprise starts in the mid-50s. You see, for the first time, an imprint on the global level. And I can tell you, you enter the disciplinary research in each of these, you find something remarkably important, the conclusion that we may have come to the point where we have to bend the curves, that we may have entered the most challenging and exciting decade in the history humanity on the planet, the decade when we have to bend the curves. Now, as if this was not enough â to just bend the curves and understanding the accelerated pressure on the planet â we also have to recognize the fact that systems do have multiple stable states, separated by thresholds â **illustrated** here by this **ball** and cup diagram, where the depth of the cup is the resilience of the system. Now, the system may gradually â under pressure of climate change, erosion, biodiversity loss â lose the depth of the cup, the resilience, but appear to be healthy and appear to suddenly, under a threshold, be tipping over. Upff. Sorry. Changing state and literally ending up in an undesired situation, where new biophysical logic takes over, new species take over, and the system gets locked. Do we have evidence of this? Yes, coral reef systems. Biodiverse, low-nutrient, hard coral systems under multiple pressures of overfishing, unsustainable tourism, climate change. A trigger and the system tips over, loses its resilience, soft corals take over, and we get undesired systems that cannot support economic and social development. The Arctic â a beautiful system â a regulating biome at the planetary level, taking the knock after knock on climate change, appearing to be in a good state. No scientist could predict that in 2007, suddenly, what could be crossing a threshold. The system suddenly, very surprisingly, loses 30 to 40 percent of its summer ice cover. And the drama is, of course, that when the system does this, the logic may change. It may get locked in an undesired state, because it changes color, absorbs more energy, and the system may get stuck. In my mind, the largest red flag warning for humanity that we are in a precarious situation. As a sideline, you know that the only red flag that popped up here was a submarine from an unnamed country that planted a red flag at the bottom of the Arctic to be able to control the oil resources. Now, if we have evidence, which we now have, that wetlands, forests, [unclear] monsoon system, the rainforests, behave in this nonlinear way. 30 or so scientists around the world gathered and asked a question for the first time,"Do we have to put the planet into the the pot?"So we have to ask ourselves : are we threatening this extraordinarily stable Holocene state? Are we in fact putting ourselves in a situation where we ' re coming too close to thresholds that could lead to deleterious and very undesired, if now catastrophic, change for human development? You know, you don ' t want to stand there. In fact, you ' re not even allowed to stand where this gentleman is standing, at the foaming, slippery waters at the threshold. In fact, there ' s a fence quite upstream of this threshold, beyond which you are in a danger zone. And this is the new paradigm, which we gathered two, three years back, recognizing that our old paradigm of just analyzing and pushing and predicting parameters into the future, aiming at minimalizing environmental impacts, is of the past. Now we to ask ourselves : which are the large environmental processes that we have to be stewards of to keep ourselves safe in the Holocene? And could we even, thanks to major advancements in Earth systems science, identify the thresholds, the points where we may expect non-linear change? And could we even define a planetary boundary, a fence, within which we then have a safe operating space for humanity? This work, which was published in”Nature,”late 2009, after a number of years of analysis, led to the final proposition that we can only find nine planetary boundaries with which, under active stewardship, would allow ourselves to have a safe operating space. These include, of course, climate. It may surprise you that it ’ s not only climate. But it shows that we are interconnected, among many systems on the planet, with the three big systems, climate change, stratospheric ozone depletion and ocean acidification being the three big systems, where the scientific evidence of large-scale thresholds in the paleo-record of the history of the planet. But we also include, what we call, the slow variables, the systems that, under the hood, regulate and buffer the capacity of the resilience of the planet â the interference of the big nitrogen and phosphorus cycles on the planet, land use change, rate of biodiversity loss, freshwater use, functions which regulate biomass on the planet, carbon sequestration, diversity. And then we have two parameters which we were not able to quantify â air pollution, including warming gases and air-polluting sulfates and nitrates, but also chemical pollution. Together, these form an integrated whole for guiding human development in the Anthropocene, understanding that the planet is a complex self-regulating system. In fact, most evidence indicates that these nine may behave as three Musketeers,”One for all. All for one.”You degrade forests, you go beyond the boundary on land, you undermine the ability of the climate system to stay stable. The drama here is, in fact, that it may show that the climate challenge is the easy one, if you consider the whole challenge of sustainable development. Now this is the Big Bang equivalent then of human development within the safe operating space of the planetary boundaries. What you see here in black line is the safe operating space, the quantified boundaries, as suggested by this analysis. The yellow dot in the middle here is our starting point, the pre-industrial point, where we ’ re very safely in the safe operating space. In the ’ 50s, we start branching out. In the ’ 60s already, through the green revolution and the Haber-Bosch process of fixing nitrogen from the atmosphere â you know, human ’ s today take out more nitrogen from the atmosphere than the whole biosphere does naturally as a whole. We don ’ t transgress the climate boundary until the early ’ 90s, actually, right after Rio. And today, we are in a situation where we estimate that we ’ ve transgressed three boundaries, the rate of biodiversity loss, which is the sixth extinction period in the history of humanity â one of them being the extinctions of the dinosaurs â nitrogen and climate change. But we still have some degrees of freedom on the others, but we are approaching fast on land, water, phosphorus and oceans. But this gives a new paradigm to guide humanity, to put the light on our, so far over-powered industrial vehicle, which operates as if we ’ re only on a dark, straight highway. Now the question then is : how gloomy is this? Is then sustainable development utopia? Well, there ’ s no science to suggest. In fact, there is ample science to indicate that we can do this transformative change, that we have the ability to now move into a new innovative, a transformative gear, across scales. The drama is, of course, is that 200 countries on this planet have to simultaneously start moving in the same direction. But it changes fundamentally our governance and management paradigm, from the current linear, command and control thinking, looking at efficiencies and optimization towards a much more flexible, a much more adaptive approach, where we recognize that redundancy, both in social and environmental systems, is key to be able to deal with a turbulent era of global change. We have to invest in persistence, in the ability of social systems and ecological systems to withstand shocks and still remain in that desired cup. We have to invest in transformations capability, moving from crisis into innovation and the ability to rise after a crisis, and of course to adapt to unavoidable change. This is a new paradigm. We ’ re not doing that at any scale on governance. But is it happening anywhere? Do we have any examples of success on this mind shift being applied at the local level? Well, yes, in fact we do and the list can start becoming longer and longer. There ’ s good news here, for example, from Latin America, where plow-based farming systems of the ’ 50s and ’ 60s led farming basically to a dead-end, with lower and lower yields, degrading the organic matter and fundamental problems at the livelihood levels in Paraguay, Uruguay and a number of countries, Brazil, leading to innovation and entrepreneurship among farmers in partnership with scientists into an agricultural revolution of zero tillage systems combined with mulch farming with locally adapted technologies, which today, for example, in some countries, have led to a tremendous increase in area under mulch, zero till farming which, not only produces more food, but also sequesters carbon. The Australian Great Barrier Reef is another success story. Under the realization from tourist operators, fishermen, the Australian Great Barrier Reef Authority and scientists that the Great Barrier Reef is doomed under the current governance regime. Global change, beautification rack culture, overfishing and unsustainable tourism, all together placing this system in the realization of crisis. But the window of opportunity was innovation and new mindset, which today has led to a completely new governance strategy to build resilience, acknowledge redundancy and invest in the whole system as an integrated whole, and then allow for much more redundancy in the system. Sweden, the country I come from, has other examples, where wetlands in southern Sweden were seen as â as in many countries â as flood-prone polluted nuisance in the peri-urban regions. But again, a crisis, new partnerships, actors locally, transforming these into a key component of sustainable urban planning. So crisis leading into opportunities. Now, what about the future? Well, the future, of course, has one massive challenge, which is feeding a world of nine billion people. We need nothing less than a new green revolution, and the planet boundaries shows that agriculture has to go from a source of greenhouse gases to a sink. It has to basically do this on current land. We cannot expand anymore, because it erodes the planetary boundaries. We cannot continue consuming water as we do today, with 25 percent of world rivers not even reaching the ocean. And we need a transformation. Well, interestingly, and based on my work and others in Africa, for example, we ’ ve shown that even the most vulnerable small-scale rainfall farming systems, with innovations and supplementary irrigation to bridge dry spells and droughts, sustainable sanitation systems to close the loop on nutrients from toilets back to farmers ’ fields, and innovations in tillage systems, we can triple, quadruple, yield levels on current land. Elinor Ostrom, the latest Nobel laureate of economics, clearly shows empirically across the world that we can govern the commons if we invest in trust, local, action-based partnerships and cross-scale institutional innovations, where local actors, together, can deal with the global commons at a large scale. But even on the hard policy area we have innovations. We know that we have to move from our fossil dependence very quickly into a low-carbon economy in record time. And what shall we do? Everybody talks about carbon taxes â it won ’ t work â emission schemes, but for example, one policy measure, feed-in tariffs on the energy system, which is already applied, from China doing it on offshore wind systems, all the way to the U. S. where you give the guaranteed price for investment in renewable energy, but you can subsidize electricity to poor people. You get people out of poverty. You solve the climate issue with regards to the energy sector, while at the same time, stimulating innovation â examples of things that can be out scaled quickly at the planetary level. So there is â no doubt â opportunity here, and we can list many, many examples of transformative opportunities around the planet. The key though in all of these, the red thread, is the shift in mindset, moving away from a situation where we simply are pushing ourselves into a dark future, where we instead backcast our future, and we say,"What is the playing field on the planet? What are the planetary boundaries within which we can safely operate?"and then backtrack innovations within that. But of course, the drama is, it clearly shows that incremental change is not an option. So, there is scientific evidence. They sort of say the harsh news, that we are facing the largest transformative development since the industrialization. In fact, what we have to do over the next 40 years is much more dramatic and more exciting than what we did when we moved into the situation we ' re in today. Now, science indicates that, yes, we can achieve a prosperous future within the safe operating space, if we move simultaneously, collaborating on a global level, from local to global scale, in transformative options, which build resilience on a finite planet. Thank you.

Text Sample 866: POS Dim 6 – 2010 – Score 4.00 – 2010/t002822.txt

One thing the world needs, one thing this country desperately needs is a better way of conducting our political debates. We need to rediscover the lost art of democratic argument. If you think about the arguments we have, most of the time it ' s shouting matches on cable television, ideological food fights on the floor of Congress. I have a suggestion. Look at all the arguments we have these days over health care, over bonuses and bailouts on Wall Street, over the gap between rich and poor, over affirmative action and same-sex marriage. Lying just beneath the surface of those arguments, with passions raging on all sides, are big questions of moral philosophy, big questions of justice. But we too rarely **articulate** and defend and argue about those big moral questions in our politics. So what I would like to do today is have something of a discussion. First, let me take a famous philosopher who wrote about those questions of justice and morality, give you a very short lecture on Aristotle of ancient Athens, Aristotle ' s theory of justice, and then have a discussion here to see whether Aristotle ' s ideas actually inform the way we think and argue about questions today. So, are you ready for the lecture? According to Aristotle, justice means giving people what they deserve. That ' s it ; that ' s the lecture. Now, you may say, well, that ' s obvious enough. The real questions begin when it comes to arguing about who deserves what and why. Take the example of flutes. Suppose we ' re distributing flutes. Who should get the best ones? Let ' s see what people — What would you say? Who should get the best flute? You can just call it out. Michael Sandel : At random. You would do it by lottery. Or by the first person to rush into the hall to get them. Who else? MS : The best flute players. MS : The worst flute players. How many say the best flute players? Why? Actually, that was Aristotle ' s answer too. But here ' s a harder question. Why do you think, those of you who voted this way, that the best flutes should go to the best flute players? Peter : The greatest benefit to all. MS : The greatest benefit to all. We ' ll hear better music if the best flutes should go to the best flute players. That ' s Peter? MS : All right. Well, it ' s a good reason. We ' ll all be better off if good music is played rather than terrible music. But Peter, Aristotle doesn ' t agree with you that that ' s the reason. That ' s all right. Aristotle had a different reason for saying the best flutes should go to the best flute players. He said, that ' s what flutes are for — to be played well. He says that to reason about just distribution of a thing, we have to reason about, and sometimes argue about, the purpose of the thing, or the social activity — in this case, musical performance. And the point, the essential nature, of musical performance is to produce excellent music. It ’ ll be a happy byproduct that we ’ ll all benefit. But when we think about justice, Aristotle says, what we really need to think about is the essential nature of the activity in question and the qualities that are worth honoring and admiring and recognizing. One of the reasons that the best flute players should get the best flutes is that musical performance is not only to make the rest of us happy, but to honor and recognize the excellence of the best musicians. Now, flutes may seem... the distribution of flutes may seem a trivial case. Let ’ s take a contemporary example of the dispute about justice. It had to do with golf. Casey Martin — a few years ago, Casey Martin — did any of you hear about him? He was a very good golfer, but he had a disability. He had a bad leg, a circulatory problem, that made it very painful for him to walk the course. In fact, it carried risk of injury. He asked the PGA, the Professional Golfers ’ Association, for permission to use a golf cart in the PGA tournaments. They said,”No. Now that would give you an unfair advantage.”He sued, and his case went all the way to the Supreme Court, believe it or not, the case over the golf cart, because the law says that the disabled must be accommodated, provided the accommodation does not change the essential nature of the activity. He says,”I ’ m a great golfer. I want to compete. But I need a golf cart to get from one hole to the next.”Suppose you were on the Supreme Court. Suppose you were deciding the justice of this case. How many here would say that Casey Martin does have a right to use a golf cart? And how many say, no, he doesn ’ t? All right, let ’ s take a poll, show of hands. How many would rule in favor of Casey Martin? And how many would not? How many would say he doesn ’ t? All right, we have a good division of opinion here. Someone who would not grant Casey Martin the right to a golf cart, what would be your reason? Raise your hand, and we ’ ll try to get you a microphone. What would be your reason? MS : It would be an unfair advantage if he gets to ride in a golf cart. All right, those of you, I imagine most of you who would not give him the golf cart worry about an unfair advantage. What about those of you who say he should be given a golf cart? How would you answer the objection? Yes, all right. Audience : The cart ’ s not part of the game. MS : What ’ s your name? MS : Charlie says — We ’ ll get Charlie a microphone in case someone wants to reply. Tell us, Charlie, why would you say he should be able to use a golf cart? Charlie : The cart ’ s not part of the game. MS : But what about walking from hole to hole? Charlie : It doesn ’ t matter ; it ’ s not part of the game. MS : Walking the course is not part of the game of golf? Charlie : Not in my book, it isn ’ t. MS : All right. Stay there, Charlie. Who has an answer for Charlie? All right, who has an answer for Charlie? What would you say? Audience : The endurance element is a very important part of the game, walking all those holes. MS : Walking all those holes? That ’ s part of the game of golf? MS : What ’ s your name? MS : Warren. Charlie, what do you say to Warren? Charley : I ’ ll stick to my original thesis. MS : Warren, are you a golfer? Warren : I am not a golfer. Charley : And I am. You know, it ’ s interesting. In the case, in the lower court, they brought in golfing greats to testify on this very issue. Is walking the course essential to the game? And they brought in Jack Nicklaus and Arnold Palmer. And what do you suppose they all said? Yes. They agreed with Warren. They said, yes, walking the course is strenuous physical exercise. The fatigue factor is an important part of golf. And so it would change the fundamental nature of the game to give him the golf cart. Now, notice, something interesting — Well, I should tell you about the Supreme Court first. The Supreme Court decided. What do you suppose they said? They said yes, that Casey Martin must be provided a golf cart. Seven to two, they ruled. What was interesting about their ruling and about the discussion we ' ve just had is that the discussion about the right, the justice, of the matter depended on figuring out what is the essential nature of golf. And the Supreme Court justices wrestled with that question. And Justice Stevens, writing for the majority, said he had read all about the history of golf, and the essential point of the game is to get very small **ball** from one place into a hole in as few strokes as possible, and that walking was not essential, but incidental. Now, there were two dissenters, one of whom was Justice Scalia. He wouldn ' t have granted the cart, and he had a very interesting dissent. It ' s interesting because he rejected the Aristotelian premise underlying the majority ' s opinion. He said it ' s not possible to determine the essential nature of a game like golf. Here ' s how he put it."To say that something is essential is ordinarily to say that it is necessary to the achievement of a certain object. But since it is the very nature of a game to have no object except amusement, that is, what distinguishes games from productive activity, it is quite impossible to say that any of a game ' s arbitrary rules is essential."So there you have Justice Scalia taking on the Aristotelian premise of the majority ' s opinion. Justice Scalia ' s opinion is questionable for two reasons. First, no real sports fan would talk that way. If we had thought that the rules of the sports we care about are merely arbitrary, rather than designed to call forth the virtues and the excellences that we think are worthy of admiring, we wouldn ' t care about the outcome of the game. It ' s also objectionable on a second ground. On the face of it, it seemed to be — this debate about the golf cart — an argument about fairness, what ' s an unfair advantage. But if fairness were the only thing at stake, there would have been an easy and obvious solution. What would it be? Let everyone ride in a golf cart if they want to. Then the fairness objection goes away. But letting everyone ride in a cart would have been, I suspect, more anathema to the golfing greats and to the PGA, even than making an exception for Casey Martin. Why? Because what was at stake in the dispute over the golf cart was not only the essential nature of golf, but, relatedly, the question : What abilities are worthy of honor and recognition as athletic talents? Let me put the point as delicately as possible : Golfers are a little sensitive about the athletic status of their game. After all, there ' s no running or jumping, and the **ball** stands still. So if golfing is the kind of game that can be played while riding around in a golf cart, it would be hard to confer on the golfing greats the status that we confer, the honor and recognition that goes to truly great athletes. That **illustrates** that with golf, as with flutes, it ' s hard to decide the question of what justice requires, without grappling with the question,"What is the essential nature of the activity in question, and what qualities, what excellences connected with that activity, are worthy of honor and recognition?"Let ' s take a final example that ' s prominent in contemporary political debate : same-sex marriage. There are those who favor state recognition only of traditional marriage between one man and one woman, and there are those who favor state recognition of same-sex marriage. How many here favor the first policy : the state should recognize traditional marriage only? And how many favor the second, same-sex marriage? Now, put it this way : What ways of thinking about justice and morality underlie the arguments we have over marriage? The opponents of same-sex marriage say that the purpose of marriage, fundamentally, is procreation, and that ' s what ' s worthy of honoring and recognizing and encouraging. And the defenders of same-sex marriage say no, procreation is not the only purpose of marriage ; what about a lifelong, mutual, loving commitment? That ' s really what marriage is about. So with flutes, with golf carts, and even with a fiercely contested question like same-sex marriage, Aristotle has a point. Very hard to argue about justice without first arguing about the purpose of social institutions and about what qualities are worthy of honor and recognition. So let ’ s step back from these cases and see how they shed light on the way we might improve, elevate, the terms of political discourse in the United States, and for that matter, around the world. There is a tendency to think that if we engage too directly with moral questions in politics, that ’ s a recipe for disagreement, and for that matter, a recipe for intolerance and coercion. So better to shy away from, to ignore, the moral and the religious convictions that people bring to civic life. It seems to me that our discussion reflects the opposite, that a better way to mutual respect is to engage directly with the moral convictions citizens bring to public life, rather than to require that people leave their deepest moral convictions outside politics before they enter. That, it seems to me, is a way to begin to restore the art of democratic argument. Thank you very much. Thank you. Thank you. Thank you very much. Thanks. Thank you. Chris. Thanks, Chris. Chris Anderson : From flutes to golf courses to same-sex marriage — that was a genius link. Now look, you ’ re a pioneer of open education. Your lecture series was one of the first to do it big. What ’ s your vision for the next phase of this? MS : Well, I think that it is possible. In the classroom, we have arguments on some of the most fiercely held moral convictions that students have about big public questions. And I think we can do that in public life more generally. And so my real dream would be to take the public television series that we ’ ve created of the course — it ’ s available now, online, free for everyone anywhere in the world — and to see whether we can partner with institutions, at universities in China, in India, in Africa, around the world, to try to promote civic education and also a richer kind of democratic debate. CA : So you picture, at some point, live, in real time, you could have this kind of conversation, inviting questions, but with people from China and India joining in? MS : Right. We did a little bit of it here with 1, 500 people in Long Beach, and we do it in a classroom at Harvard with about 1, 000 students. Wouldn ’ t it be interesting to take this way of thinking and arguing, engaging seriously with big moral questions, exploring cultural differences and connect through a live video hookup, students in Beijing and Mumbai and in Cambridge, Massachusetts and create a global classroom. That ’ s what I would love to do. CA : So, I would imagine that there are a lot of people who would love to join you in that endeavor. Michael Sandel. Thank you so much.

Text Sample 867: POS Dim 6 – 2012 – Score 5.00 – 2012/t002322.txt

So a few weeks ago, a friend of mine gave this toy car to his 8-year-old son. But instead of going into a store and buying one, like we do normally, he went to this website and he downloaded a file, and then he printed it on this printer. So this idea that you can manufacture objects digitally using these machines is something that The Economist magazine defined as the Third Industrial Revolution. Actually, I argue that there is another revolution going on, and it ’ s the one that has to do with open-source hardware and the maker ’ s movement, because the printer that my friend used to print the toy is actually open-source. So you go to the same website, you can download all the files that you need in order to make that printer : the construction files, the hardware, the software, all the instruction is there. And also this is part of a large community where there are thousands of people around the world that are actually making these kinds of printers, and there ’ s a lot of innovation happening because it ’ s all open-source. You don ’ t need anybody ’ s permission to create something great. And that space is like the personal computer in 1976, like the Apples with the other companies are fighting, and we will see in a few years, there will be the Apple of this kind of market come out. Well, there ' s also another interesting thing. I said the electronics are open- source, because at the heart of this printer there is something I ' m really attached to : these Arduino boards, the motherboard that sort of powers this printer, is a project I ' ve been working on for the past seven years. It ' s an open-source project. I worked with these friends of mine that I have here. So the five of us, two Americans, two Italians and a Spaniard, we — You know, it ' s a worldwide project. So we came together in this design institute called the Interaction Design Institute Ivrea, which was teaching interaction design, this idea that you can take design from the simple shape of an object and you can move it forward to design the way you interact with things. Well, when you design an object that ' s supposed to interact with a human being, if you make a foam model of a mobile phone, it doesn ' t make any sense. You have to have something that actually interacts with people. So, we worked on Arduino and a lot of other projects there to create platforms that would be simple for our students to use, so that our students could just build things that worked, but they don ' t have five years to become an electronics engineer. We have one month. So how do I make something that even a kid can use? And actually, with Arduino, we have kids like Sylvia that you see here, that actually make projects with Arduino. I have 11-year-old kids stop me and show me stuff they built for Arduino that ' s really scary to see the capabilities that kids have when you give them the tools. So let ' s look at what happens when you make a tool that anybody can just pick up and build something quickly, so one of the examples that I like to sort of kick off this discussion is this example of this cat feeder. The gentleman who made this project had two cats. One was sick and the other one was healthy, so he had to make sure they ate the proper food. So he made this thing that recognizes the cat from a chip mounted inside on the collar of the cat, and opens the door and the cat can eat the food. This is made by recycling an old CD player that you can get from an old computer, some cardboard, tape, couple of sensors, a few blinking LEDs, and then suddenly you have a tool. You build something that you cannot find on the market. And I like this phrase :"Scratch your own itch."If you have an idea, you just go and you make it. This is the equivalent of sketching on paper done with electronics. So one of the features that I think is important about our work is that our hardware, on top of being made with love in Italy — as you can see from the back of the circuit — is that it ' s open, so we publish all the design files for the circuit online, so you can download it and you can actually use it to make something, or to modify, to learn. You know, when I was learning about programming, I learned by looking at other people ' s code, or looking at other people ' s circuits in magazines. And this is a good way to learn, by looking at other people ' s work. So the different elements of the project are all open, so the hardware is released with a Creative Commons license. So, you know, I like this idea that hardware becomes like a piece of culture that you share and you build upon, like it was a song or a poem with Creative Commons. Or, the software is GPL, so it ' s open-source as well. The documentation and the hands-on teaching methodology is also open-source and released as the Creative Commons. Just the name is protected so that we can make sure that we can tell people what is Arduino and what isn ' t. Now, Arduino itself is made of a lot of different open-source components that maybe individually are hard to use for a 12-year-old kid, so Arduino wraps everything together into a mashup of open-source technologies where we try to give them the best user experience to get something done quickly. So you have situations like this, where some people in Chile decided to make their own boards instead of buying them, to organize a workshop and to save money. Or there are companies that make their own variations of Arduino that fit in a certain market, and there ’ s probably, maybe like a 150 of them or something at the moment. This one is made by a company called Adafruit, which is run by this woman called Limor Fried, also known as Ladyada, who is one of the heroes of the open-source hardware movement and the maker movement. So, this idea that you have a new, sort of turbo-charged DIY community that believes in open-source, in collaboration, collaborates online, collaborates in different spaces. There is this magazine called Make that sort of gathered all these people and sort of put them together as a community, and you see a very technical project explained in a very simple language, beautifully typeset. Or you have websites, like this one, like Instructables, where people actually teach each other about anything. So this one is about Arduino projects, the page you see on the screen, but effectively here you can learn how to make a cake and everything else. So let ’ s look at some projects. So this one is a quadcopter. It ’ s a small model helicopter. In a way, it ’ s a toy, no? And so this one was military technology a few years ago, and now it ’ s open-source, easy to use, you can buy it online. DIY Drones is the community ; they do this thing called ArduCopter. But then somebody actually launched this start-up called Matternet, where they figured out that you could use this to actually transport things from one village to another in Africa, and the fact that this was easy to find, open-source, easy to hack, enabled them to prototype their company really quickly. Or, other projects. Matt Richardson : I ’ m getting a little sick of hearing about the same people on TV over and over and over again, so I decided to do something about it. This Arduino project, which I call the Enough Already, will mute the TV anytime any of these overexposed personalities is mentioned. I ’ ll show you how I made it. MB : Check this out. MR : Our producers caught up with Kim Kardashian earlier today to find out what she was planning on wearing to her — MB : Eh? MR : It should do a pretty good job of protecting our ears from having to hear about the details of Kim Kardashian ’ s wedding. MB : Okay. So, you know, again, what is interesting here is that Matt found this module that lets Arduino process TV signals, he found some code written by somebody else that generates infrared signals for the TV, put it together and then created this great project. It ’ s also used, Arduino ’ s used, in serious places like, you know, the Large Hadron Collider. There ’ s some Arduino **balls** collecting data and sort of measuring some parameters. Or it ’ s used for — So this is a musical **interface** built by a student from Italy, and he ’ s now turning this into a product. Because it was a student project becoming a product. Or it can be used to make an assistive device. This is a glove that understands the sign language and transforms the gestures you make into sounds and writes the words that you ’ re signing on a **display** And again, this is made of all different parts you can find on all the websites that sell Arduino-compatible parts, and you assemble it into a project. Or this is a project from the ITP part of NYU, where they met with this boy who has a severe disability, cannot play with the PS3, so they built this device that allows the kid to play baseball although he has limited movement capability. Or you can find it in arts projects. So this is the txtBomber. So you put a message into this device and then you roll it on the wall, and it basically has all these solenoids pressing the buttons on spray cans, so you just pull it over a wall and it just writes on the wall all the political messages. So, yeah. Then we have this plant here. This is called Botanicalls, because there ’ s an Arduino **ball** with a Wi-Fi module in the plant, and it ’ s measuring the well-being of the plant, and it ’ s creating a Twitter account where you can actually interact with the plant. So, you know, this plant will start to say,”This is really hot,”or there ’ s a lot of, you know,”I need water right now.”So it just gives a personality to your plant. Or this is something that twitters when the baby inside the belly of a pregnant woman kicks. Or this is a 14-year-old kid in Chile who made a system that detects earthquakes and publishes on Twitter. He has 280, 000 followers. He ’ s 14 and he anticipated a governmental project by one year. Or again, another project where, by analyzing the Twitter feed of a family, you can basically point where they are, like in the”Harry Potter”**movie**. So you can find out everything about this project on the website. Or somebody made a chair that twitters when somebody farts. It ’ s interesting how, in 2009, Gizmodo basically defined, said that this project actually gives a meaning to Twitter, so it was — a lot changed in between. So very serious project. When the Fukushima disaster happened, a bunch of people in Japan, they realized that the information that the government was giving wasn ’ t really open and really reliable, so they built this Geiger counter, plus Arduino, plus network **interface**. They made 100 of them and gave them to people around Japan, and essentially the data that they gathered gets published on this website called Cosm, another website they built, so you can actually get reliable real-time information from the field, and you can get unbiased information. Or this machine here, it ’ s from the DIY bio movement, and it ’ s one of the steps that you need in order to process DNA, and again, it ’ s completely open-source from the ground up. Or you have students in developing countries making replicas of scientific instruments that cost a lot of money to make. Actually they just build them themselves for a lot less using Arduino and a few parts. This is a pH probe. Or you get kids, like these kids, they ’ re from Spain. They learned how to program and to make robots when they were probably, like, 11, and then they started to use Arduino to make these robots that play football. They became world champions by making an Arduino-based robot. And so when we had to make our own educational robot, we just went to them and said, you know,”You design it, because you know exactly what is needed to make a great robot that excites kids.”Not me. I ’ m an old guy. What am I supposed to excite, huh? But as I — in terms of educational assets. There ’ s also companies like Google that are using the technology to create **interfaces** between mobile phones, tablets and the real world. So the Accessory Development Kit from Google is open-source and based on Arduino, as opposed to the one from Apple which is closed-source, NDA, sign your life to Apple. Here you are. There ’ s a giant maze, and Joey ’ s sitting there, and the maze is moving when you tilt the tablet. Also, I come from Italy, and the design is important in Italy, and yet very conservative. So we worked with a design studio called Habits, in Milan, to make this mirror, which is completely open-source. This doubles also as an iPod speaker. So the idea is that the hardware, the software, the design of the object, the fabrication, everything about this project is open-source and you can make it yourself. So we want other designers to pick this up and learn how to make great devices, to learn how to make interactive products by starting from something real. But when you have this idea, you know, what happens to all these ideas? There ’ s, like, thousands of ideas that I — You know, it would take seven hours for me to do all the **presentations**. I will not take all the seven hours. Thank you. But let ’ s start from this example : So, the group of people that started this company called Pebble, they prototyped a watch that communicates via Bluetooth with your phone, and you can **display** information on it. And they prototyped with an old LCD screen from a Nokia mobile phone and an Arduino. And then, when they had a final project, they actually went to Kickstarter and they were asking for 100, 000 dollars to make a few of them to sell. They got 10 million dollars. They got a completely fully funded start-up, and they don ’ t have to, you know, get VCs involved or anything, just excite the people with their great project. The last project I want to show you is this : It ’ s called ArduSat. It ’ s currently on Kickstarter, so if you want to contribute, please do it. It ’ s a satellite that goes into space, which is probably the least open-source thing you can imagine, and it contains an Arduino connected to a bunch of sensors. So if you know how to use Arduino, you can actually upload your experiments into this satellite and run them. So imagine, if you as a high school can have the satellite for a week and do satellite space experiments like that. So, as I said, there ’ s lots of examples, and I ’ m going to stop here. And I just want to thank the Arduino community for being the best, and just every day making lots of projects. Thank you. And thanks to the community. Chris Anderson : Massimo, you told me earlier today that you had no idea, of course, that it would take off like this. MB : No. CA : I mean, how must you feel when you read this stuff and you see what you ’ ve unlocked? MB : Well, it ’ s the work of a lot of people, so we as a community are enabling people to make great stuff, and I just feel overwhelmed. It ’ s just, it ’ s difficult to describe this. Every morning, I wake up and I look at all the stuff that Google Alerts sends me, and it ’ s just amazing. It ’ s just going into every field that you can imagine. CA : Thank you so much.

Text Sample 868: POS Dim 6 – 2012 – Score 4.00 – 2012/t002265.txt

I ’ d like to tell you about two games of chess. The first happened in 1997, in which Garry Kasparov, a human, lost to Deep Blue, a machine. To many, this was the dawn of a new era, one where man would be dominated by machine. But here we are, 20 years on, and the greatest change in how we relate to computers is the iPad, not HAL. The second game was a freestyle chess tournament in 2005, in which man and machine could enter together as partners, rather than adversaries, if they so chose. At first, the results were predictable. Even a supercomputer was beaten by a grandmaster with a relatively weak laptop. The surprise came at the end. Who won? Not a grandmaster with a supercomputer, but actually two American amateurs using three relatively weak laptops. Their ability to coach and manipulate their computers to deeply explore specific positions effectively counteracted the superior chess knowledge of the grandmasters and the superior computational power of other adversaries. This is an astonishing result : average men, average machines beating the best man, the best machine. And anyways, isn ’ t it supposed to be man versus machine? Instead, it ’ s about cooperation, and the right type of cooperation. We ’ ve been paying a lot of attention to Marvin Minsky ’ s vision for artificial intelligence over the last 50 years. It ’ s a sexy vision, for sure. Many have embraced it. It ’ s become the dominant school of thought in computer science. But as we enter the era of big data, of network systems, of open platforms, and embedded technology, I ’ d like to suggest it ’ s time to reevaluate an alternative vision that was actually developed around the same time. I ’ m talking about J. C. R. Licklider ’ s human-computer symbiosis, perhaps better termed”intelligence augmentation,”I. A. Licklider was a computer science titan who had a profound effect on the development of technology and the Internet. His vision was to enable man and machine to cooperate in making decisions, controlling complex situations without the inflexible dependence on predetermined programs. Note that word”cooperate.”Licklider encourages us not to take a toaster and make it Data from”Star Trek,”but to take a human and make her more capable. Humans are so amazing — how we think, our non-linear approaches, our creativity, iterative hypotheses, all very difficult if possible at all for computers to do. Licklider intuitively realized this, contemplating humans setting the goals, formulating the hypotheses, determining the criteria, and performing the evaluation. Of course, in other ways, humans are so limited. We ’ re terrible at scale, computation and volume. We require high-end talent management to keep the rock band together and playing. Licklider foresaw computers doing all the routinizable work that was required to prepare the way for insights and decision making. Silently, without much fanfare, this approach has been compiling victories beyond chess. Protein folding, a topic that shares the incredible expansiveness of chess — there are more ways of folding a protein than there are atoms in the universe. This is a world-changing problem with huge implications for our ability to understand and treat disease. And for this task, supercomputer field brute force simply isn ’ t enough. Foldit, a game created by computer scientists, **illustrates** the value of the approach. Non-technical, non-biologist amateurs play a video game in which they visually rearrange the structure of the protein, allowing the computer to manage the atomic forces and interactions and identify structural issues. This approach beat supercomputers 50 percent of the time and tied 30 percent of the time. Foldit recently made a notable and major scientific discovery by deciphering the structure of the Mason-Pfizer monkey virus. A protease that had eluded determination for over 10 years was solved was by three players in a matter of days, perhaps the first major scientific advance to come from playing a video game. Last year, on the site of the Twin Towers, the 9 / 11 memorial opened. It **displays** the names of the thousands of victims using a beautiful concept called”meaningful adjacency.”It places the names next to each other based on their relationships to one another : friends, families, coworkers. When you put it all together, it ’ s quite a computational challenge : 3, 500 victims, 1, 800 adjacency requests, the importance of the overall physical specifications and the final aesthetics. When first reported by the media, full credit for such a feat was given to an algorithm from the New York City design firm Local Projects. The truth is a bit more nuanced. While an algorithm was used to develop the underlying framework, humans used that framework to design the final result. So in this case, a computer had evaluated millions of possible layouts, managed a complex relational system, and kept track of a very large set of measurements and variables, allowing the humans to focus on design and compositional choices. So the more you look around you, the more you see Licklider ’ s vision everywhere. Whether it ’ s augmented reality in your iPhone or GPS in your car, human-computer symbiosis is making us more capable. So if you want to improve human-computer symbiosis, what can you do? You can start by designing the human into the process. Instead of thinking about what a computer will do to solve the problem, design the solution around what the human will do as well. When you do this, you ’ ll quickly realize that you spent all of your time on the **interface** between man and machine, specifically on designing away the friction in the interaction. In fact, this friction is more important than the power of the man or the power of the machine in determining overall capability. That ’ s why two amateurs with a few laptops handily beat a supercomputer and a grandmaster. What Kasparov calls process is a byproduct of friction. The better the process, the less the friction. And minimizing friction turns out to be the decisive variable. Or take another example : big data. Every interaction we have in the world is recorded by an ever growing array of sensors : your phone, your credit card, your computer. The result is big data, and it actually presents us with an opportunity to more deeply understand the human condition. The major emphasis of most approaches to big data focus on,”How do I store this data? How do I search this data? How do I process this data?”These are necessary but insufficient questions. The imperative is not to figure out how to compute, but what to compute. How do you impose human intuition on data at this scale? Again, we start by designing the human into the process. When PayPal was first starting as a business, their biggest challenge was not,”How do I send money back and forth online?”It was,”How do I do that without being defrauded by organized crime?”Why so challenging? Because while computers can learn to detect and identify fraud based on patterns, they can ’ t learn to do that based on patterns they ’ ve never seen before, and organized crime has a lot in common with this audience : brilliant people, relentlessly resourceful, entrepreneurial spirit — — and one huge and important difference : purpose. And so while computers alone can catch all but the cleverest fraudsters, catching the cleverest is the difference between success and failure. There ’ s a whole class of problems like this, ones with adaptive adversaries. They rarely if ever present with a repeatable pattern that ’ s discernable to computers. Instead, there ’ s some inherent component of innovation or disruption, and increasingly these problems are buried in big data. For example, terrorism. Terrorists are always adapting in minor and major ways to new circumstances, and despite what you might see on TV, these adaptations, and the detection of them, are fundamentally human. Computers don ’ t detect novel patterns and new behaviors, but humans do. Humans, using technology, testing hypotheses, searching for insight by asking machines to do things for them. Osama bin Laden was not caught by artificial intelligence. He was caught by dedicated, resourceful, brilliant people in partnerships with **various** technologies. As appealing as it might sound, you cannot algorithmically data mine your way to the answer. There is no”Find Terrorist”button, and the more data we integrate from a vast variety of sources across a wide variety of data formats from very disparate systems, the less effective data mining can be. Instead, people will have to look at data and search for insight, and as Licklider foresaw long ago, the key to great results here is the right type of cooperation, and as Kasparov realized, that means minimizing friction at the **interface**. Now this approach makes possible things like combing through all available data from very different sources, identifying key relationships and putting them in one place, something that ’ s been nearly impossible to do before. To some, this has terrifying privacy and civil liberties implications. To others it foretells of an era of greater privacy and civil liberties protections, but privacy and civil liberties are of fundamental importance. That must be acknowledged, and they can ’ t be swept aside, even with the best of intents. So let ’ s explore, through a couple of examples, the impact that technologies built to drive human-computer symbiosis have had in recent time. In October, 2007, U. S. and coalition forces raided an al Qaeda safe house in the city of Sinjar on the Syrian border of Iraq. They found a treasure trove of documents : 700 biographical sketches of foreign fighters. These foreign fighters had left their families in the Gulf, the Levant and North Africa to join al Qaeda in Iraq. These records were human resource forms. The foreign fighters filled them out as they joined the organization. It turns out that al Qaeda, too, is not without its bureaucracy. They answered questions like,”Who recruited you?””What ’ s your hometown?””What occupation do you seek?”In that last question, a surprising insight was revealed. The vast majority of foreign fighters were seeking to become suicide bombers for martyrdom — hugely important, since between 2003 and 2007, Iraq had 1, 382 suicide bombings, a major source of instability. Analyzing this data was hard. The originals were sheets of paper in Arabic that had to be scanned and translated. The friction in the process did not allow for meaningful results in an operational time frame using humans, PDFs and tenacity alone. The researchers had to lever up their human minds with technology to dive deeper, to explore non-obvious hypotheses, and in fact, insights emerged. Twenty percent of the foreign fighters were from Libya, 50 percent of those from a single town in Libya, hugely important since prior statistics put that figure at three percent. It also helped to hone in on a figure of rising importance in al Qaeda, Abu Yahya al-Libi, a senior cleric in the Libyan Islamic fighting group. In March of 2007, he gave a speech, after which there was a surge in participation amongst Libyan foreign fighters. Perhaps most clever of all, though, and least obvious, by flipping the data on its head, the researchers were able to deeply explore the coordination networks in Syria that were ultimately responsible for receiving and transporting the foreign fighters to the border. These were networks of mercenaries, not ideologues, who were in the coordination business for profit. For example, they charged Saudi foreign fighters substantially more than Libyans, money that would have otherwise gone to al Qaeda. Perhaps the adversary would disrupt their own network if they knew they cheating would-be jihadists. In January, 2010, a devastating 7. 0 earthquake struck Haiti, third deadliest earthquake of all time, left one million people, 10 percent of the population, homeless. One seemingly small aspect of the overall relief effort became increasingly important as the delivery of food and water started rolling. January and February are the dry months in Haiti, yet many of the camps had developed standing water. The only institution with detailed knowledge of Haiti ’ s floodplains had been leveled in the earthquake, leadership inside. So the question is, which camps are at risk, how many people are in these camps, what ’ s the timeline for flooding, and given very limited resources and infrastructure, how do we prioritize the relocation? The data was incredibly disparate. The U. S. Army had detailed knowledge for only a small section of the country. There was data online from a 2006 environmental risk conference, other geospatial data, none of it integrated. The human goal here was to identify camps for relocation based on priority need. The computer had to integrate a vast amount of geospacial information, social media data and relief organization information to answer this question. By implementing a superior process, what was otherwise a task for 40 people over three months became a simple job for three people in 40 hours, all victories for human-computer symbiosis. We ’ re more than 50 years into Licklider ’ s vision for the future, and the data suggests that we should be quite excited about tackling this century ’ s hardest problems, man and machine in cooperation together. Thank you.

Text Sample 869: POS Dim 6 – 2012 – Score 4.00 – 2012/t002165.txt

The kind of neuroscience that I do and my colleagues do is almost like the weatherman. We are always chasing storms. We want to see and measure storms â brainstorms, that is. And we all talk about brainstorms in our daily lives, but we rarely see or listen to one. So I always like to start these talks by actually introducing you to one of them. Actually, the first time we recorded more than one neuron â a hundred brain cells simultaneously â we could measure the electrical sparks of a hundred cells in the same animal, this is the first image we got, the first 10 seconds of this recording. So we got a little snippet of a thought, and we could see it in front of us. I always tell the students that we could also call neuroscientists some sort of astronomer, because we are dealing with a system that is only comparable in terms of number of cells to the number of galaxies that we have in the universe. And here we are, out of billions of neurons, just recording, 10 years ago, a hundred. We are doing a thousand now. And we hope to understand something fundamental about our human nature. Because, if you don ’ t know yet, everything that we use to define what human nature is comes from these storms, comes from these storms that roll over the hills and valleys of our brains and define our memories, our beliefs, our feelings, our plans for the future. Everything that we ever do, everything that every human has ever done, do or will do, requires the toil of populations of neurons producing these kinds of storms. And the sound of a brainstorm, if you ’ ve never heard one, is somewhat like this. You can put it louder if you can. My son calls this”making popcorn while listening to a badly-tuned A. M. station.”This is a brain. This is what happens when you route these electrical storms to a loudspeaker and you listen to a hundred brain cells firing, your brain will sound like this â my brain, any brain. And what we want to do as neuroscientists in this time is to actually listen to these symphonies, these brain symphonies, and try to extract from them the messages they carry. In particular, about 12 years ago we created a preparation that we named brain- machine **interfaces**. And you have a scheme here that describes how it works. The idea is, let ’ s have some sensors that listen to these storms, this electrical firing, and see if you can, in the same time that it takes for this storm to leave the brain and reach the legs or the arms of an animal â about half a second â let ’ s see if we can read these signals, extract the motor messages that are embedded in it, translate it into digital commands and send it to an artificial device that will reproduce the voluntary motor wheel of that brain in real time. And see if we can measure how well we can translate that message when we compare to the way the body does that. And if we can actually provide feedback, sensory signals that go back from this robotic, mechanical, computational actuator that is now under the control of the brain, back to the brain, how the brain deals with that, of receiving messages from an artificial piece of machinery. And that ’ s exactly what we did 10 years ago. We started with a superstar monkey called Aurora that became one of the superstars of this field. And Aurora liked to play video games. As you can see here, she likes to use a joystick, like any one of us, any of our kids, to play this game. And as a good primate, she even tries to cheat before she gets the right answer. So even before a target appears that she ’ s supposed to cross with the **cursor** that she ’ s controlling with this joystick, Aurora is trying to find the target, no matter where it is. And if she ’ s doing that, because every time she crosses that target with the little **cursor**, she gets a drop of Brazilian orange juice. And I can tell you, any monkey will do anything for you if you get a little drop of Brazilian orange juice. Actually any primate will do that. Think about that. Well, while Aurora was playing this game, as you saw, and doing a thousand trials a day and getting 97 percent correct and 350 milliliters of orange juice, we are recording the brainstorms that are produced in her head and sending them to a robotic arm that was learning to reproduce the movements that Aurora was making. Because the idea was to actually turn on this brain-machine **interface** and have Aurora play the game just by thinking, without interference of her body. Her brainstorms would control an arm that would move the **cursor** and cross the target. And to our shock, that ’ s exactly what Aurora did. She played the game without moving her body. So every trajectory that you see of the **cursor** now, this is the exact first moment she got that. That ’ s the exact first moment a brain intention was liberated from the physical domains of a body of a primate and could act outside, in that outside world, just by controlling an artificial device. And Aurora kept playing the game, kept finding the little target and getting the orange juice that she wanted to get, that she craved for. Well, she did that because she, at that time, had acquired a new arm. The robotic arm that you see moving here 30 days later, after the first video that I showed to you, is under the control of Aurora ’ s brain and is moving the **cursor** to get to the target. And Aurora now knows that she can play the game with this robotic arm, but she has not lost the ability to use her biological arms to do what she pleases. She can scratch her back, she can scratch one of us, she can play another game. By all purposes and means, Aurora ’ s brain has incorporated that artificial device as an extension of her body. The model of the self that Aurora had in her mind has been expanded to get one more arm. Well, we did that 10 years ago. Just fast forward 10 years. Just last year we realized that you don ’ t even need to have a robotic device. You can just build a computational body, an avatar, a monkey avatar. And you can actually use it for our monkeys to either interact with them, or you can train them to assume in a virtual world the first-person perspective of that avatar and use her brain activity to control the movements of the avatar ’ s arms or legs. And what we did basically was to train the animals to learn how to control these avatars and explore objects that appear in the virtual world. And these objects are visually identical, but when the avatar crosses the surface of these objects, they send an electrical message that is proportional to the microtactile texture of the object that goes back directly to the monkey ’ s brain, informing the brain what it is the avatar is touching. And in just four weeks, the brain learns to process this new sensation and acquires a new sensory pathway â like a new sense. And you truly liberate the brain now because you are allowing the brain to send motor commands to move this avatar. And the feedback that comes from the avatar is being processed directly by the brain without the interference of the skin. So what you see here is this is the design of the task. You ’ re going to see an animal basically touching these three targets. And he has to select one because only one carries the reward, the orange juice that they want to get. And he has to select it by touch using a virtual arm, an arm that doesn ’ t exist. And that ’ s exactly what they do. This is a complete liberation of the brain from the physical constraints of the body and the motor in a perceptual task. The animal is controlling the avatar to touch the targets. And he ’ s sensing the texture by receiving an electrical message directly in the brain. And the brain is deciding what is the texture associated with the reward. The legends that you see in the **movie** don ’ t appear for the monkey. And by the way, they don ’ t read English anyway, so they are here just for you to know that the correct target is shifting position. And yet, they can find them by tactile discrimination, and they can press it and select it. So when we look at the brains of these animals, on the top panel you see the alignment of 125 cells showing what happens with the brain activity, the electrical storms, of this sample of neurons in the brain when the animal is using a joystick. And that ’ s a picture that every neurophysiologist knows. The basic alignment shows that these cells are coding for all possible directions. The bottom picture is what happens when the body stops moving and the animal starts controlling either a robotic device or a computational avatar. As fast as we can reset our computers, the brain activity shifts to start representing this new tool, as if this too was a part of that primate ’ s body. The brain is assimilating that too, as fast as we can measure. So that suggests to us that our sense of self does not end at the last layer of the epithelium of our bodies, but it ends at the last layer of electrons of the tools that we ’ re commanding with our brains. Our violins, our cars, our bicycles, our soccer **balls**, our clothing â they all become assimilated by this voracious, amazing, dynamic system called the brain. How far can we take it? Well, in an experiment that we ran a few years ago, we took this to the limit. We had an animal running on a treadmill at Duke University on the East Coast of the United States, producing the brainstorms necessary to move. And we had a robotic device, a humanoid robot, in Kyoto, Japan at ATR Laboratories that was dreaming its entire life to be controlled by a brain, a human brain, or a primate brain. What happens here is that the brain activity that generated the movements in the monkey was transmitted to Japan and made this robot walk while footage of this walking was sent back to Duke, so that the monkey could see the legs of this robot walking in front of her. So she could be rewarded, not by what her body was doing but for every correct step of the robot on the other side of the planet controlled by her brain activity.

Funny thing, that round trip around the globe took 20 milliseconds less than it takes for that brainstorm to leave its head, the head of the monkey, and reach its own muscle. The monkey was moving a robot that was six times bigger, across the planet. This is one of the experiments in which that robot was able to walk autonomously. This is CB1 fulfilling its dream in Japan under the control of the brain activity of a primate. So where are we taking all this? What are we going to do with all this research, besides studying the properties of this dynamic universe that we have between our ears? Well the idea is to take all this knowledge and technology and try to restore one of the most severe neurological problems that we have in the world. Millions of people have lost the ability to translate these brainstorms into action, into movement. Although their brains continue to produce those storms and code for movements, they cannot cross a barrier that was created by a lesion on the spinal cord. So our idea is to create a bypass, is to use these brain-machine **interfaces** to read these signals, larger-scale brainstorms that contain the desire to move again, bypass the lesion using computational microengineering and send it to a new body, a whole body called an exoskeleton, a whole robotic suit that will become the new body of these patients. And you can see an image produced by this consortium. This is a nonprofit consortium called the Walk Again Project that is putting together scientists from Europe, from here in the United States, and in Brazil together to work to actually get this new body built â a body that we believe, through the same plastic mechanisms that allow Aurora and other monkeys to use these tools through a brain-machine **interface** and that allows us to incorporate the tools that we produce and use in our daily life. This same mechanism, we hope, will allow these patients, not only to imagine again the movements that they want to make and translate them into movements of this new body, but for this body to be assimilated as the new body that the brain controls. So I was told about 10 years ago that this would never happen, that this was close to impossible. And I can only tell you that as a scientist, I grew up in southern Brazil in the mid- ’ 60s watching a few crazy guys telling [us] that they would go to the Moon. And I was five years old, and I never understood why NASA didn ’ t hire Captain Kirk and Spock to do the job ; after all, they were very proficient â but just seeing that as a kid made me believe, as my grandmother used to tell me, that”impossible is just the possible that someone has not put in enough effort to make it come true.”So they told me that it ’ s impossible to make someone walk. I think I ’ m going to follow my grandmother ’ s advice. Thank you.

Text Sample 870: POS Dim 6 – 2016 – Score 5.00 – 2016/t001311.txt

The world is filled with incredible objects and rich cultural heritage. And when we get access to them, we are blown away, we fall in love. But most of the time, the world ’ s population is living without real access to arts and culture. What might the connections be when we start exploring our heritage, the beautiful locations and the art in this world? Before we get started in this **presentation**, I just want to take care of a few housekeeping points. First, I am no expert in art or culture. I fell into this by mistake, but I ’ m loving it. Secondly, all of what I ’ m going to show you belongs to the amazing museums, archives and foundations that we partner with. None of this belongs to Google. And finally, what you see behind me is available right now on your mobile phones, on your laptops. This is our current platform, where you can explore thousands of museums and objects at your fingertips, in extremely high-definition detail. The diversity of the content is what ’ s amazing. If we just had European paintings, if we just had modern art, I think it gets a bit boring. For example, this month, we launched the"Black History"channel with 82 curated exhibitions, which talk about arts and culture in that community. We also have some amazing objects from Japan, centered around craftsmanship, called"Made in Japan."And one of my favorite exhibitions, which actually is the idea of my talk, is — I didn ' t expect to become a fan of Japanese dolls. But I am, thanks to this exhibition, that has really taught me about the craftsmanship behind the soul of a Japanese doll. Trust me, it ' s very exciting. Take my word for it. So, moving on swiftly. One quick thing I wanted to showcase in this platform, which you can share with your kids and your friends right now, is you can travel to all these amazing institutions virtually, as well. One of our recent ideas was with The Guggenheim Museum in New York, where you can get a taste of what it might feel like to actually be there. You can go to the ground floor and obviously, most of you, I assume, have been there. And you can see the architectural masterpiece that it is. But imagine this accessibility for a kid in Bombay who ' s studying architecture, who hasn ' t had a chance to go to The Guggenheim as yet. You can obviously look at objects in the Guggenheim Museum, you can obviously get into them and so on and so forth. There ' s a lot of information here. But this is not the purpose of my talk today. This exists right now. What we now have are the building blocks to a very exciting future, when it comes to arts and culture and accessibility to arts and culture. So I am joined today onstage by my good friend and artist in residence at our office in Paris, Cyril Diagne, who is the professor of interactive design at ECAL University in Lausanne, Switzerland. What Cyril and our team of engineers have been doing is trying to find these connections and visualize a few of these. So I ' m going to go quite quick now. This object you see behind me — oh, just clarification : Always, seeing the real thing is better. In case people think I ' m trying to replicate the real thing. So, moving on. This object you see behind me is the Venus of Berekhat Ram. It ' s one of the oldest objects in the world, found in the Golan Heights around 233, 000 years ago, and currently residing at the Israel Museum in Jerusalem. It is also one of the oldest objects on our platform. So let ' s zoom. We start from this one object. What if we zoomed out and actually tried to experience our own cultural big **bang**? What might that look like? This is what we deal with on a daily basis at the Cultural Institute — over six million cultural **artifacts** curated and given to us by institutions, to actually make these connections. You can travel through time, you can understand more about our society through these. You can look at it from the perspective of our planet, and try to see how it might look without borders, if we just organized art and culture. We can also then plot it by time, which obviously, for the data geek in me, is very fascinating. You can spend hours looking at every decade and the contributions in that decade and in those years for art, history and cultures. We would love to spend hours showing you each and every decade, but we don ' t have the time right now. So you can go on your phone and actually do it yourself. But if you don ' t mind and can hold your applause till later, I don ' t want to run out of time, because I want to show you a lot of cool stuff. So, just very quickly : you can move on from here to another very interesting idea. Beyond the pretty picture, beyond the nice visualization, what is the purpose, how is this useful? This next idea comes from discussions with curators that we ' ve been having at museums, who, by the way, I ' ve fallen in love with, because they dedicate their whole life to try to tell these stories. One of the curators told me,"Amit, what would it be like if you could create a virtual curator ' s table where all these six million objects are **displayed** in a way for us to look at the connections between them?"You can spend a lot of time, trust me, looking at different objects and understanding where they come from. It ' s a crazy Matrix experience. Just moving on, let ' s take the world-famous Vincent Van Gogh, who is very well- represented on this platform. Thanks to the diversity of the institutions we have, we have over 211 high-definition, amazing artworks by this artist, now organized in one beautiful view. And as it resolves, and as Cyril goes deeper, you can see all the self-portraits, you can see still life. But I just wanted to highlight one very quick example, which is very timely :"The Bedroom."This is an artwork where three copies exist — one at the Van Gogh Museum in Amsterdam, one at the Orsay in Paris and one at the Art Institute of Chicago, which, actually, currently is hosting a reunion of all three artworks physically, I think only for the second time ever. But, it is united digitally and virtually for anybody to look at in a very different way, and you won ' t get pushed in the line in the crowd. So let ' s take you and let ' s travel through"The Bedroom"very quickly, so you can experience what we are doing for every single object. We want the image to speak as much as it can on a digital platform. And all you need is an internet connection and a computer And, Cyril, if you can go deeper, quickly. I ' m sorry, this is all live, so you have to give Cyril a little bit of — and this is available for every object : modern art, contemporary art, Renaissance — you name it, even sculpture. Sometimes, you don ' t know what can attract you to an artwork or to a museum or to a cultural discovery. So for me, personally, it was quite a challenge because when I decided to make this my full-time job at Google, my mother was not very supportive. I love my mother, but she thought I was wasting my life with this museum stuff. And for her, a museum is what you do when you go on vacation and you tick-mark and it ' s over, right? And it took around four and a half years for me to convince my lovely Indian mother that actually, this is worthwhile. And the way I did it was, I realized one day that she loves gold. So I started showing her all objects that have the material gold in them. And the first thing my mom asks me is,"How can we buy these?"And obviously, my salary is not that high, so I was like,"We can ' t actually do that, mom. But you can explore them virtually."And so now my mom — every time I meet her, she asks me,"Any more gold, any more silver in your project? Can you show me?"And that ' s the idea I ' m trying to **illustrate**. It does not matter how you get in, as long as you get in. Once you get in, you ' re hooked. Moving on from here very quickly, there is kind of a playful idea, actually, to **illustrate** the point of access, and I ' m going to go quite quickly on this one. We all know that seeing the artwork in person is amazing. But we also know that most of us can ' t do it, and the ones that can afford to do it, it ' s complicated. So — Cyril, can we load up our art trip, what do we call it? We don ' t have a good name for this. But essentially, we have around 1, 000 amazing institutions, 68 countries. But let ' s start with Rembrandt. We might have time for only one example. But thanks to the diversity, we ' ve got around 500 amazing Rembrandt object artworks from 46 institutions and 17 countries. Let ' s say that on your next vacation, you want to go see every single one of them. That is your itinerary, you will probably travel 53, 000 kilometers, visit around, I think, 46 institutions, and just FYI, you might release 10 tons of CO2 emissions. But remember, it ' s art, so you can justify it, perhaps, in some way. Moving on swiftly from here, is something a little bit more technical and more interesting. All that we ' ve shown you so far uses metadata to make the connections. But obviously we have something cool nowadays that everyone likes to talk about, which is machine learning. So what we thought is, let ' s strip out all the metadata, let ' s look at what machine learning can do based purely on visual recognition of this entire collection. What we ended up with is this very interesting map, these clusters that have no reference point information, but has just used visuals to cluster things together. Each cluster is an art to us by itself of discovery. But one of the clusters we want to show you very quickly is this amazing cluster of portraits that we found from museums around the world. If you could zoom in a little bit more, Cyril. Just to show you, you can just travel through portraits. And essentially, you can do nature, you can do horses and clusters galore. When we saw all these portraits, we were like,”Hey, can we do something fun for kids, or can we do something playful to get people interested in portraits?”Because I haven ’ t really seen young kids really excited to go to a portrait gallery. I wanted to try to figure something out. So we created something called the portrait matcher. It ’ s quite self-explanatory, so I ’ m just going to let Cyril show his beautiful face. And essentially what ’ s happening is, with the movement of his head, we are matching different portraits around the world from museums. And I don ’ t know about you, but I ’ ve shown it to my nephew and sister, and the reaction is just phenomenal. All they ask me is,”When can we go see this?”And by the way, if we ’ re nice, maybe, Cyril, you can smile and find a happy one? Oh, perfect. By the way, this is not rehearsed. Congrats, Cyril. Great stuff. Oh wow. OK, let ’ s move on ; otherwise, this will just take the whole time. So, art and culture can be fun also, right? For our last quick experiment — we call all of these”experiments”— our last quick experiment comes back to machine learning. We show you clusters, visual clusters, but what if we could ask the machine to also name these clusters? What if it could automatically tag them, using no actual metadata? So what we have is this kind of explorer, where we have managed to match, I think, around 4, 000 labels. And we haven ’ t really done anything special here, just fed the collection. And we found interesting categories. We can start with horses, a very straightforward category. You would expect to see that the machine has put images of horses, right? And it has, but you also notice, right over there, that it has a very abstract image that it has still managed to recognize and cluster as horses. We also have an amazing head in terms of a horse. And each one has the tags as to why it got categorized in this. So let ’ s move to another one which I found very funny and interesting, because I don ’ t understand how this category came up. It ’ s called”Lady in Waiting.”If, Cyril, you do it very quickly, you will see that we have these amazing images of ladies, I guess, in waiting or posing. I don ’ t really understand it. But I ’ ve been trying to ask my museum contacts, you know,”What is this? What ’ s going on here?”And it ’ s fascinating. Coming back to gold very quickly, I wanted to search for gold and see how the machine tagged all the gold. But, actually, it doesn ’ t tag it as gold. We are living in popular times. It tags it as”bling-bling.”I ’ m being hard on Cyril, because I ’ m moving too fast. Essentially, here you have all the bling-bling of the world ’ s museums organized for you. And finally, to end this talk and these experiments, what I hope you feel after this talk is happiness and emotion. And what would we see when we see happiness? If we actually look at all the objects that have been tagged under”happiness,”you would expect happiness, I guess. But there was one that came up that was very fascinating and interesting, which was this artwork by Douglas Coupland, our friend and artist in residence as well, called,”I Miss My Pre- Internet Brain.”I don ’ t know why the machine feels like it misses its pre- Internet brain and it ’ s been tagged here, but it ’ s a very interesting thought. I sometimes do miss my pre-Internet brain, but not when it comes to exploring arts and culture online. So take out your phones, take out your computers, go visit museums. And just a quick call-out to all the amazing archivists, historians, curators, who are sitting in museums, preserving all this culture. And the least we can do is get our daily dose of art and culture for ourselves and our kids. Thank you.

Text Sample 871: POS Dim 6 – 2016 – Score 5.00 – 2016/t001424.txt

So a while ago, I tried an experiment. For one year, I would say yes to all the things that scared me. Anything that made me nervous, took me out of my comfort zone, I forced myself to say yes to. Did I want to speak in public? No, but yes. Did I want to be on live TV? No, but yes. Did I want to try acting? No, no, no, but yes, yes, yes. And a crazy thing happened : the very act of doing the thing that scared me undid the fear, made it not scary. My fear of public speaking, my social anxiety, **poof**, gone. It ’ s amazing, the power of one word.”Yes”changed my life.”Yes”changed me. But there was one particular yes that affected my life in the most profound way, in a way I never imagined, and it started with a question from my toddler. I have these three amazing daughters, Harper, Beckett and Emerson, and Emerson is a toddler who inexplicably refers to everyone as”honey.”as though she ’ s a Southern waitress.”Honey, I ’ m gonna need some milk for my sippy cup.”The Southern waitress asked me to play with her one evening when I was on my way somewhere, and I said,”Yes.”And that yes was the beginning of a new way of life for my family. I made a vow that from now on, every time one of my children asks me to play, no matter what I ’ m doing or where I ’ m going, I say yes, every single time. Almost. I ’ m not perfect at it, but I try hard to practice it. And it ’ s had a magical effect on me, on my children, on our family. But it ’ s also had a stunning side effect, and it wasn ’ t until recently that I fully understood it, that I understood that saying yes to playing with my children likely saved my career. See, I have what most people would call a dream job. I ’ m a writer. I imagine. I make stuff up for a living. Dream job. No. I ’ m a titan. Dream job. I create television. I executive produce television. I make television, a great deal of television. In one way or another, this TV season, I ’ m responsible for bringing about 70 hours of programming to the world. Four television programs, 70 hours of TV — Three shows in production at a time, sometimes four. Each show creates hundreds of jobs that didn ’ t exist before. The budget for one episode of network television can be anywhere from three to six million dollars. Let ’ s just say five. A new episode made every nine days times four shows, so every nine days that ’ s 20 million dollars worth of television, four television programs, 70 hours of TV, three shows in production at a time, sometimes four, 16 episodes going on at all times : 24 episodes of”Grey ’ s,”21 episodes of”Scandal,”15 episodes of”How To Get Away With Murder,”10 episodes of”The Catch,”that ’ s 70 hours of TV, that ’ s 350 million dollars for a season. In America, my television shows are back to back to back on Thursday night. Around the world, my shows air in 256 territories in 67 languages for an audience of 30 million people. My brain is global, and 45 hours of that 70 hours of TV are shows I personally created and not just produced, so on top of everything else, I need to find time, real quiet, creative time, to gather my fans around the campfire and tell my stories. Four television programs, 70 hours of TV, three shows in production at a time, sometimes four, 350 million dollars, campfires burning all over the world. You know who else is doing that? Nobody, so like I said, I ’ m a titan. Dream job. Now, I don ’ t tell you this to impress you. I tell you this because I know what you think of when you hear the word”writer.”I tell you this so that all of you out there who work so hard, whether you run a company or a country or a classroom or a store or a home, take me seriously when I talk about working, so you ’ ll get that I don ’ t peck at a computer and imagine all day, so you ’ ll hear me when I say that I understand that a dream job is not about dreaming. It ’ s all job, all work, all reality, all blood, all sweat, no tears. I work a lot, very hard, and I love it. When I ’ m hard at work, when I ’ m deep in it, there is no other feeling. For me, my work is at all times building a nation out of thin air. It is manning the **troops**. It is painting a canvas. It is hitting every high note. It is running a **marathon**. It is being Beyonc. And it is all of those things at the same time. I love working. It is creative and mechanical and exhausting and exhilarating and hilarious and disturbing and clinical and maternal and cruel and judicious, and what makes it all so good is the hum. There is some kind of shift inside me when the work gets good. A hum begins in my brain, and it grows and it grows and that hum sounds like the open road, and I could drive it forever. And a lot of people, when I try to explain the hum, they assume that I ' m talking about the writing, that my writing brings me joy. And don ' t get me wrong, it does. But the hum — it wasn ' t until I started making television that I started working, working and making and building and creating and collaborating, that I discovered this thing, this buzz, this rush, this hum. The hum is more than writing. The hum is action and activity. The hum is a drug. The hum is music. The hum is light and air. The hum is God ' s whisper right in my ear. And when you have a hum like that, you can ' t help but strive for greatness. That feeling, you can ' t help but strive for greatness at any cost. That ' s called the hum. Or, maybe it ' s called being a workaholic. Maybe it ' s called genius. Maybe it ' s called ego. Maybe it ' s just fear of failure. I don ' t know. I just know that I ' m not built for failure, and I just know that I love the hum. I just know that I want to tell you I ' m a titan, and I know that I don ' t want to question it. But here ' s the thing : the more successful I become, the more shows, the more episodes, the more barriers broken, the more work there is to do, the more **balls** in the air, the more eyes on me, the more history stares, the more expectations there are. The more I work to be successful, the more I need to work. And what did I say about work? I love working, right? The nation I ' m building, the **marathon** I ' m running, the **troops**, the canvas, the high note, the hum, the hum, the hum. I like that hum. I love that hum. I need that hum. I am that hum. Am I nothing but that hum? And then the hum stopped. Overworked, overused, overdone, burned out. The hum stopped. Now, my three daughters are used to the truth that their mother is a single working titan. Harper tells people,”My mom won ' t be there, but you can text my nanny.”And Emerson says,”Honey, I ' m wanting to go to ShondaLand.”They ' re children of a titan. They ' re baby titans. They were 12, 3, and 1 when the hum stopped. The hum of the engine died. I stopped loving work. I couldn ' t restart the engine. The hum would not come back. My hum was broken. I was doing the same things I always did, all the same titan work, 15-hour days, working straight through the weekends, no regrets, never surrender, a titan never sleeps, a titan never quits, full hearts, clear eyes, yada, whatever. But there was no hum. Inside me was silence. Four television programs, 70 hours of TV, three shows in production at a time, sometimes four. Four television programs, 70 hours of TV, three shows in production at a time... I was the perfect titan. I was a titan you could take home to your mother. All the colors were the same, and I was no longer having any fun. And it was my life. It was all I did. I was the hum, and the hum was me. So what do you do when the thing you do, the work you love, starts to taste like dust? Now, I know somebody ' s out there thinking,”Cry me a river, stupid writer titan lady.”But you know, you do, if you make, if you work, if you love what you do, being a teacher, being a banker, being a mother, being a painter, being Bill Gates, if you simply love another person and that gives you the hum, if you know the hum, if you know what the hum feels like, if you have been to the hum, when the hum stops, who are you? What are you? What am I? Am I still a titan? If the song of my heart ceases to play, can I survive in the silence? And then my Southern waitress toddler asks me a question. I ' m on my way out the door, I ' m late, and she says,”Momma, wanna play?”And I ' m just about to say no, when I realize two things. One, I ' m supposed to say yes to everything, and two, my Southern waitress didn ' t call me”honey.”She ' s not calling everyone”honey”anymore. When did that happen? I ’ m missing it, being a titan and mourning my hum, and here she is changing right before my eyes. And so she says,”Momma, wanna play?”And I say,”Yes.”There ’ s nothing special about it. We play, and we ’ re joined by her sisters, and there ’ s a lot of laughing, and I give a dramatic reading from the book Everybody Poops. Nothing out of the ordinary. And yet, it is extraordinary, because in my pain and my panic, in the homelessness of my humlessness, I have nothing to do but pay attention. I focus. I am still. The nation I ’ m building, the **marathon** I ’ m running, the **troops**, the canvas, the high note does not exist. All that exists are sticky fingers and gooey kisses and tiny voices and crayons and that song about letting go of whatever it is that Frozen girl needs to let go of. It ’ s all peace and simplicity. The air is so rare in this place for me that I can barely breathe. I can barely believe I ’ m breathing. Play is the opposite of work. And I am happy. Something in me **loosens**. A door in my brain swings open, and a rush of energy comes. And it ’ s not instantaneous, but it happens, it does happen. I feel it. A hum creeps back. Not at full volume, barely there, it ’ s quiet, and I have to stay very still to hear it, but it is there. Not the hum, but a hum. And now I feel like I know a very magical secret. Well, let ’ s not get carried away. It ’ s just love. That ’ s all it is. No magic. No secret. It ’ s just love. It ’ s just something we forgot. The hum, the work hum, the hum of the titan, that ’ s just a replacement. If I have to ask you who I am, if I have to tell you who I am, if I describe myself in terms of shows and hours of television and how globally badass my brain is, I have forgotten what the real hum is. The hum is not power and the hum is not work-specific. The hum is joy-specific. The real hum is love-specific. The hum is the electricity that comes from being excited by life. The real hum is confidence and peace. The real hum ignores the stare of history, and the **balls** in the air, and the expectation, and the pressure. The real hum is singular and original. The real hum is God ’ s whisper in my ear, but maybe God was whispering the wrong words, because which one of the gods was telling me I was the titan? It ’ s just love. We could all use a little more love, a lot more love. Any time my child asks me to play, I will say yes. I make it a firm rule for one reason, to give myself permission, to free me from all of my workaholic guilt. It ’ s a law, so I don ’ t have a choice, and I don ’ t have a choice, not if I want to feel the hum. I wish it were that easy, but I ’ m not good at playing. I don ’ t like it. I ’ m not interested in doing it the way I ’ m interested in doing work. The truth is incredibly humbling and humiliating to face. I don ’ t like playing. I work all the time because I like working. I like working more than I like being at home. Facing that fact is incredibly difficult to handle, because what kind of person likes working more than being at home? Well, me. I mean, let ’ s be honest, I call myself a titan. I ’ ve got issues. And one of those issues isn ’ t that I am too relaxed. We run around the yard, up and back and up and back. We have 30-second dance parties. We sing show tunes. We play with **balls**. I blow bubbles and they pop them. And I feel stiff and delirious and confused most of the time. I itch for my cell phone always. But it is OK. My tiny humans show me how to live and the hum of the universe fills me up. I play and I play until I begin to wonder why we ever stop playing in the first place. You can do it too, say yes every time your child asks you to play. Are you thinking that maybe I ’ m an idiot in diamond shoes? You ’ re right, but you can still do this. You have time. You know why? Because you ’ re not Rihanna and you ’ re not a Muppet. Your child does not think you ’ re that interesting. You only need 15 minutes. My two- and four-year-old only ever want to play with me for about 15 minutes or so before they think to themselves they want to do something else. It ’ s an amazing 15 minutes, but it ’ s 15 minutes. If I ’ m not a ladybug or a piece of candy, I ’ m invisible after 15 minutes. And my 13-year-old, if I can get a 13-year-old to talk to me for 15 minutes I ' m Parent of the Year. 15 minutes is all you need. I can totally pull off 15 minutes of uninterrupted time on my worst day. Uninterrupted is the key. No cell phone, no laundry, no anything. You have a busy life. You have to get dinner on the table. You have to force them to bathe. But you can do 15 minutes. My kids are my happy place, they ' re my world, but it doesn ' t have to be your kids, the fuel that feeds your hum, the place where life feels more good than not good. It ' s not about playing with your kids, it ' s about joy. It ' s about playing in general. Give yourself the 15 minutes. Find what makes you feel good. Just figure it out and play in that arena. I ' m not perfect at it. In fact, I fail as often as I succeed, seeing friends, reading books, staring into space."Wanna play?"starts to become shorthand for indulging myself in ways I ' d given up on right around the time I got my first TV show, right around the time I became a titan-in-training, right around the time I started competing with myself for ways unknown. 15 minutes? What could be wrong with giving myself my full attention for 15 minutes? Turns out, nothing. The very act of not working has made it possible for the hum to return, as if the hum ' s engine could only refuel while I was away. Work doesn ' t work without play. It takes a little time, but after a few months, one day the floodgates open and there ' s a rush, and I find myself standing in my office filled with an unfamiliar melody, full on groove inside me, and around me, and it sends me spinning with ideas, and the humming road is open, and I can drive it and drive it, and I love working again. But now, I like that hum, but I don ' t love that hum. I don ' t need that hum. I am not that hum. That hum is not me, not anymore. I am bubbles and sticky fingers and dinners with friends. I am that hum. Life ' s hum. Love ' s hum. Work ' s hum is still a piece of me, it is just no longer all of me, and I am so grateful. And I don ' t give a crap about being a titan, because I have never once seen a titan play Red Rover, Red Rover. I said yes to less work and more play, and somehow I still run my world. My brain is still global. My campfires still burn. The more I play, the happier I am, and the happier my kids are. The more I play, the more I feel like a good mother. The more I play, the freer my mind becomes. The more I play, the better I work. The more I play, the more I feel the hum, the nation I ' m building, the **marathon** I ' m running, the **troops**, the canvas, the high note, the hum, the hum, the other hum, the real hum, life ' s hum. The more I feel that hum, the more this strange, quivering, uncocooned, awkward, brand new, alive non-titan feels like me. The more I feel that hum, the more I know who I am. I ' m a writer, I make stuff up, I imagine. That part of the job, that ' s living the dream. That ' s the dream of the job. Because a dream job should be a little bit dreamy. I said yes to less work and more play. Titans need not apply. Wanna play? Thank you.

Text Sample 872: POS Dim 6 – 2016 – Score 4.00 – 2016/t001140.txt

In the last couple of years, I have produced what I call"The Dead Mall Series,"32 short films and counting about dead malls. Now, for those of you who are not familiar with what a dead mall is, it ' s basically a shopping mall that has fallen into hard times. So it either has few shops and fewer shoppers, or it ' s abandoned and crumbling into ruin. No sale at Penny ' s. I started producing this series in early 2015 after going through kind of a dark period in my life where I just didn ' t want to create films anymore. I put my camera away and I just stopped. So in 2015, I decided to make a short film about the Owings Mills Mall. Owings Mills Mall opened in 1986. I should know because I was there on opening day. I was there with my family, along with every other family in

Baltimore, and you had to drive around for 45 minutes just to find a parking spot. So if you can imagine, that ’ s not happening at the malls today. My first mall job that I had as a teenager was at a sporting goods store called Herman ’ s World of Sports. Maybe you remember. Herman ’ s World of Sports. You guys remember that? Yeah, so I worked in a lady ’ s shoe store. I worked in a leather goods store, and I also worked in a video store, and not being one who was very fond of the retail arts — I got fired from every single job. In between these low-paying retail jobs, I did what any normal teenager did in the 1990s. I shoplifted. I ’ m just kidding. I hung out with my friends at the mall. Everyone ’ s like,”Oh my God, what kind of talk is this?”Hanging out at the mall could be fun, but it could be really lame, too, like sharing a cigarette with a 40-year-old unemployed mall rat who has put on black lipstick for the night while you ’ re on your break from your crappy minimum wage job. As I stand here today, Owings Mills has been gutted and it ’ s ready for the wrecking **ball**. The last time I was there, it was in the evening, and it was about three days before they closed the mall for good. And you kind of felt — they never announced the mall was closing, but you had this sort of feeling, this ominous feeling, that something big was going to happen, like it was the end of the road. It was a very creepy walk through the mall. Let me show you. So when I started producing”The Dead Mall Series,”I put the videos up onto YouTube, and while I thought they were interesting, frankly I didn ’ t think others would share the enthusiasm for such a drab and depressing topic. But apparently I was wrong, because a lot of people started to comment. And at first the comments were like — basically like,”Oh my God, that ’ s the mall from my childhood. What happened?”And then I would get comments from people who were like,”There ’ s a dead mall in my town. You should come and film it.”So I started to travel around the mid-Atlantic region filming these dead malls. Some were open. Some were abandoned. It was kind of always hard to get into the ones that were abandoned, but I somehow always found a way in. The malls that are still open, they always do this weird thing — like the dead malls. They ’ ll have three stores left, but they try to spruce it up to make it appear like things are on the up-and-up. For example, you ’ ll have an empty store and they bring the gate down. So at Owings Mills, for example, they put this tarp over the gate. Right? And it ’ s got a stock photo of a woman who is so happy and she ’ s holding a blouse, and she ’ s like — And then there ’ s a guy standing next to her, with, like, an espresso cup, and he ’ s like — And it says,”What brings you today?”I wanted to be scared and depressed. Thank you. So the comments just kept pouring in on the videos, from all over the country, and then all over the world. And I started to think, this could really be something, but I had to get creative, because I ’ m like, how long are people going to sit and watch me waddling through an empty mall? So the original episodes I filmed with an iPhone. So I ’ d walk through the mall with an iPhone, and, you know. Like that. And security — because malls, they don ’ t like photography — so the security would come up and be like,”Put that away,”and I ’ m like,”OK.”So I had to get creative and sneaky, so I started using a hidden camera and different techniques to get the footage that I needed, and basically what I wanted to do was make the video like it was a first-person experience, like you are sitting — put your headphones on watching the screen — it ’ s like, you ’ re there in the video, like a video game, basically. I also started to use music, collaborating with artists who create music called vaporwave. And vaporwave is a music genre that emerged in the early 2010s among internet communities. Here ’ s an example. That ’ s by an artist named Disconscious from an album he did called”Hologram Plaza.”So if you look that up, you can hear more of those tunes. Vaporwave is more than an art form. It ’ s like a movement. It ’ s nihilistic, it ’ s angsty, but it ’ s somehow comforting. The whole aesthetic is a way of dealing with things you can ' t do anything about, like no jobs, or sitting in your parents ' basement eating ramen noodles. Vaporwave came out of this generation ' s desire to express their hopelessness, the same way that the pre-internet generation did sitting around in the food court. One of my favorite malls I ' ve been to is in Corpus Christi, and it ' s called the Sunrise Mall. When I was a kid, my favorite thing to do was watch **movies**, and I used to watch **movies** over and over and over again. And one of my favorite films was"The Legend of Billie Jean."Now, for those of you who have seen"The Legend of Billie Jean,"you ' ll know that it ' s a great film. I love it. And Helen Slater and Christian Slater — and if you didn ' t know, they are not related. Many people thought that they were brother and sister. They ' re not. But anyway, Sunrise Mall was used in the film as a filming location. The mall is exactly the same as it was in 1984. We ' re talking 32 years later. Let me show you. Dan Bell : And here ' s Billie Jean running across the fountain, being chased by Hubie Pyatt ' s friends. And she jumps over here. And you can see the shot right here is what it looks like today. It ' s pretty incredible. I mean, honestly, it ' s exactly the same. And there they are falling in the fountain, and she runs up the stairs. This is a nice shot of the whole thing here. Dan Bell : I love that so much. I always think in my head, if I owned a dead mall — why don ' t they embrace their **vintage** look? Put in a bar, like, put vegan food in the food court and invite millennials and hipsters to come and drink and eat, and I guarantee you within three weeks H&M and Levi ' s will be **banging** on the door trying to get space. I don ' t know why they don ' t do this, but apparently, it ' s only in my mind, it goes all day. Anyway, in closing — When they first asked me to do this talk, I said,"Do you have the right person?"These talks are supposed to be kind of inspiring and — I remembered something, though. I put my camera down three or four years ago, and it took going to these malls for me to be inspired again. And to see my audience and people from all over the world writing me and saying,"God, I love your videos,"is incredible. I don ' t know how to even explain it, as an artist, how fulfilling that is. If you would have told me a year ago that I would be standing on this stage talking to all of you wonderful people, I would have never believed it. I am humbled and so appreciative. Thank you very much.

Text Sample 873: POS Dim 6 – 2014 – Score 9.00 – 2014/t001795.txt

Nicholas Negroponte : Can we switch to the video disc, which is in play mode? I ' m really interested in how you put people and computers together. We will be using the TV screens or their equivalents for electronic books of the future. Very interested in **touch-sensitive displays**, high-tech, **high-touch**, not having to pick up your fingers to use them. There is another way where computers touch people : wearing, physically wearing. Suddenly on September 11th, the world got bigger. NN : Thank you. Thank you. When I was asked to do this, I was also asked to look at all 14 TED Talks that I had given, chronologically. The first one was actually two hours. The second one was an hour, and then they became half hours, and all I noticed was my bald spot getting bigger. Imagine seeing your life, 30 years of it, go by, and it was, to say the least, for me, quite a shocking experience. So what I ' m going to do in my time is try and share with you what happened during the 30 years, and then also make a prediction, and then tell you a little bit about what I ' m doing next. And I put on a slide where TED 1 happened in my life. And it ' s rather important because I had done 15 years of research before it, so I had a backlog, so it was easy. It ' s not that I was Fidel Castro and I could talk for two hours, or Bucky Fuller. I had 15 years of stuff, and the Media Lab was about to start. So that was easy. But there are a couple of things about that period and about what happened that are really quite important. One is that it was a period when computers weren ’ t yet for people. And the other thing that sort of happened during that time is that we were considered sissy computer scientists. We weren ’ t considered the real thing. So what I ’ m going to show you is, in retrospect, a lot more interesting and a lot more accepted than it was at the time. So I ’ m going to characterize the years and I ’ m even going to go back to some very early work of mine, and this was the kind of stuff I was doing in the ’ 60s : very direct manipulation, very influenced as I studied architecture by the architect Moshe Safdie, and you can see that we even built robotic things that could build habitat-like structures. And this for me was not yet the Media Lab, but was the beginning of what I ’ ll call sensory computing, and I pick fingers partly because everybody thought it was ridiculous. Papers were published about how stupid it was to use fingers. Three reasons : One was they were low-resolution. The other is your hand would occlude what you wanted to see, and the third, which was the winner, was that your fingers would get the screen dirty, and hence, fingers would never be a device that you ’ d use. And this was a device we built in the ’ 70s, which has never even been picked up. It ’ s not just touch sensitive, it ’ s pressure sensitive. Voice : Put a yellow circle there. NN : Later work, and again this was before TED 1 — Voice : Move that west of the diamond. Create a large green circle there. Man : Aw, shit. NN : — was to sort of do **interface** concurrently, so when you talked and you pointed and you had, if you will, multiple channels. Entebbe happened. 1976, Air France was hijacked, taken to Entebbe, and the Israelis not only did an extraordinary rescue, they did it partly because they had practiced on a physical model of the airport, because they had built the airport, so they built a model in the desert, and when they arrived at Entebbe, they knew where to go because they had actually been there. The U. S. government asked some of us, ’ 76, if we could replicate that computationally, and of course somebody like myself says yes. Immediately, you get a contract, Department of Defense, and we built this truck and this rig. We did sort of a simulation, because you had video discs, and again, this is ’ 76. And then many years later, you get this truck, and so you have Google Maps. Still people thought, no, that was not serious computer science, and it was a man named Jerry Wiesner, who happened to be the president of MIT, who did think it was computer science. And one of the keys for anybody who wants to start something in life : Make sure your president is part of it. So when I was doing the Media Lab, it was like having a gorilla in the front seat. If you were stopped for speeding and the officer looked in the window and saw who was in the passenger seat, then,”Oh, continue on, sir.”And so we were able, and this is a cute, actually, device, parenthetically. This was a lenticular photograph of Jerry Wiesner where the only thing that changed in the photograph were the lips. So when you oscillated that little piece of lenticular sheet with his photograph, it would be in lip sync with zero bandwidth. It was a zero- bandwidth **teleconferencing** system at the time. So this was the Media Lab ’ s — this is what we said we ’ d do, that the world of computers, publishing, and so on would come together. Again, not generally accepted, but very much part of TED in the early days. And this is really where we were headed. And that created the Media Lab. One of the things about age is that I can tell you with great confidence, I ’ ve been to the future. I ’ ve been there, actually, many times. And the reason I say that is, how many times in my life have I said,”Oh, in 10 years, this will happen,”and then 10 years comes. And then you say,”Oh, in five years, this will happen.”And then five years comes. So I say this a little bit with having felt that I ’ d been there a number of times, and one of the things that is most quoted that I ’ ve ever said is that computing is not about computers, and that didn ’ t quite get enough traction, and then it started to. It started to because people caught on that the medium wasn ’ t the message. And the reason I show this car in actually a rather ugly slide is just again to tell you the kind of story that characterized a little bit of my life. This is a student of mine who had done a Ph. D. called”Backseat Driver.”It was in the early days of GPS, the car knew where it was, and it would give audio instructions to the driver, when to turn right, when to turn left and so on. Turns out, there are a lot of things in those instructions that back in that period were pretty challenging, like what does it mean, take the next right? Well, if you ’ re coming up on a street, the next right ’ s probably the one after, and there are lots of issues, and the student did a wonderful thesis, and the MIT patent office said”Don ’ t patent it. It ’ ll never be accepted. The liabilities are too large. There will be insurance issues. Don ’ t patent it.”So we didn ’ t, but it shows you how people, again, at times, don ’ t really look at what ’ s happening. Some work, and I ’ ll just go through these very quickly, a lot of sensory stuff. You might recognize a young Yo-Yo Ma and tracking his body for playing the cello or the hypercello. These fellows literally walked around like that at the time. It ’ s now a little bit more discreet and more commonplace. And then there are at least three heroes I want to quickly mention. Marvin Minsky, who taught me a lot about common sense, and I will talk briefly about Muriel Cooper, who was very important to Ricky Wurman and to TED, and in fact, when she got onstage, she said, the first thing she said was,”I introduced Ricky to Nicky.”And nobody calls me Nicky and nobody calls Richard Ricky, so nobody knew who she was talking about. And then, of course, Seymour Papert, who is the person who said,”You can ’ t think about thinking unless you think about thinking about something.”And that ’ s actually — you can unpack that later. It ’ s a pretty profound statement. I ’ m showing some slides that were from TED 2, a little silly as slides, perhaps. Then I felt television really was about **displays**. Again, now we ’ re past TED 1, but just around the time of TED 2, and what I ’ d like to mention here is, even though you could imagine intelligence in the device, I look today at some of the work being done about the Internet of Things, and I think it ’ s kind of tragically pathetic, because what has happened is people take the oven panel and put it on your cell phone, or the door key onto your cell phone, just taking it and bringing it to you, and in fact that ’ s actually what you don ’ t want. You want to put a chicken in the oven, and the oven says,”Aha, it ’ s a chicken,”and it cooks the chicken.”Oh, it ’ s cooking the chicken for Nicholas, and he likes it this way and that way.”So the intelligence, instead of being in the device, we have started today to move it back onto the cell phone or closer to the user, not a particularly enlightened view of the Internet of Things. Television, again, television what I said today, that was back in 1990, and the television of tomorrow would look something like that. Again, people, but they laughed cynically, they didn ’ t laugh with much appreciation. Telecommunications in the 1990s, George Gilder decided that he would call this diagram the Negroponte switch. I ’ m probably much less famous than George, so when he called it the Negroponte switch, it stuck, but the idea of things that came in the ground would go in the air and stuff in the air would go into the ground has played itself out. That is the original slide from that year, and it has worked in lockstep obedience. We started Wired magazine. Some people, I remember we shared the reception desk periodically, and some parent called up irate that his son had given up Sports Illustrated to subscribe for Wired, and he said,”Are you some porno magazine or something?”and couldn ’ t understand why his son would be interested in Wired, at any rate. I will go through this a little quicker. This is my favorite, 1995, back page of Newsweek magazine. Okay.

Read it. [”Nicholas Negroponte, director of the MIT Media Lab, predicts that we ’ ll soon buy books and newspapers straight over the Internet. Uh, sure.”—Clifford Stoll, Newsweek, 1995] You must admit that gives you, at least it gives me pleasure when somebody says how dead wrong you are.”Being Digital”came out. For me, it gave me an opportunity to be more in the trade press and get this out to the public, and it also allowed us to build the new Media Lab, which if you haven ’ t been to, visit, because it ’ s a beautiful piece of architecture aside from being a wonderful place to work. So these are the things we were saying in those TEDs. [Today, multimedia is a desktop or living room experience, because the apparatus is so clunky. This will change dramatically with small, bright, thin, high-resolution **displays**. — 1995] We came to them. I looked forward to it every year. It was the party that Ricky Wurman never had in the sense that he invited many of his old friends, including myself. And then something for me changed pretty profoundly. I became more involved with computers and learning and influenced by Seymour, but particularly looking at learning as something that is best approximated by computer programming. When you write a computer program, you ’ ve got to not just list things out and sort of take an algorithm and translate it into a set of instructions, but when there ’ s a bug, and all programs have bugs, you ’ ve got to de-bug it. You ’ ve got to go in, change it, and then re-execute, and you iterate, and that iteration is really a very, very good approximation of learning. So that led to my own work with Seymour in places like Cambodia and the starting of One Laptop per Child. Enough TED Talks on One Laptop per Child, so I ’ ll go through it very fast, but it did give us the chance to do something at a relatively large scale in the area of learning, development and computing. Very few people know that One Laptop per Child was a $1 billion project, and it was, at least over the seven years I ran it, but even more important, the World Bank contributed zero, USAID zero. It was mostly the countries using their own treasuries, which is very interesting, at least to me it was very interesting in terms of what I plan to do next. So these are the **various** places it happened. I then tried an experiment, and the experiment happened in Ethiopia. And here ’ s the experiment. The experiment is, can learning happen where there are no schools. And we dropped off tablets with no instructions and let the children figure it out. And in a short period of time, they not only turned them on and were using 50 apps per child within five days, they were singing”ABC”songs within two weeks, but they hacked Android within six months. And so that seemed sufficiently interesting. This is perhaps the best picture I have. The kid on your right has sort of nominated himself as teacher. Look at the kid on the left, and so on. There are no adults involved in this at all. So I said, well can we do this at a larger scale? And what is it that ’ s missing? The kids are giving a press conference at this point, and sort of writing in the dirt. And the answer is, what is missing? And I ’ m going to skip over my prediction, actually, because I ’ m running out of time, and here ’ s the question, is what ’ s going to happen? I think the challenge is to connect the last billion people, and connecting the last billion is very different than connecting the next billion, and the reason it ’ s different is that the next billion are sort of low-hanging fruit, but the last billion are rural. Being rural and being poor are very different. Poverty tends to be created by our society, and the people in that community are not poor in the same way at all. They may be primitive, but the way to approach it and to connect them, the history of One Laptop per Child, and the experiment in Ethiopia, lead me to believe that we can in fact do this in a very short period of time. And so my plan, and unfortunately I haven ’ t been able to get my partners at this point to let me announce them, but is to do this with a stationary satellite. There are many reasons that stationary satellites aren ’ t the best things, but there are a lot of reasons why they are, and for two billion dollars, you can connect a lot more than 100 million people, but the reason I picked two, and I will leave this as my last slide, is two billion dollars is what we were spending in Afghanistan every week. So surely if we can connect Africa and the last billion people for numbers like that, we should be doing it. Thank you very much. Chris Anderson : Stay up there. Stay up there. NN : You ’ re going to give me extra time? CA : No. That was wickedly clever, wickedly clever. You gamed it beautifully. Nicholas, what is your prediction? NN : Thank you for asking. I ’ ll tell you what my prediction is, and my prediction, and this is a prediction, because it ’ ll be 30 years. I won ’ t be here. But one of the things about learning how to read, we have been doing a lot of consuming of information going through our eyes, and so that may be a very inefficient channel. So my prediction is that we are going to ingest information You ’ re going to swallow a pill and know English. You ’ re going to swallow a pill and know Shakespeare. And the way to do it is through the bloodstream. So once it ’ s in your bloodstream, it basically goes through it and gets into the brain, and when it knows that it ’ s in the brain in the different pieces, it deposits it in the right places. So it ’ s ingesting. CA : Have you been hanging out with Ray Kurzweil by any chance? NN : No, but I ’ ve been hanging around with Ed Boyden and hanging around with one of the speakers who is here, Hugh Herr, and there are a number of people. This isn ’ t quite as far-fetched, so 30 years from now. CA : We will check it out. We ’ re going to be back and we ’ re going to play this clip 30 years from now, and then all eat the red pill. Well thank you for that. Nicholas Negroponte. NN : Thank you.

Text Sample 874: POS Dim 6 – 2014 – Score 6.00 – 2014/t001860.txt

Type is something we consume in enormous quantities. In much of the world, it ’ s completely inescapable. But few consumers are concerned to know where a particular typeface came from or when or who designed it, if, indeed, there was any human agency involved in its creation, if it didn ’ t just sort of materialize out of the software ether. But I do have to be concerned with those things. It ’ s my job. I ’ m one of the tiny handful of people who gets badly bent out of shape by the bad spacing of the T and the E that you see there. I ’ ve got to take that slide off. I can ’ t stand it. Nor can Chris. There. Good. So my talk is about the connection between technology and design of type. The technology has changed a number of times since I started work : photo, digital, desktop, screen, web. I ’ ve had to survive those changes and try to understand their implications for what I do for design. This slide is about the effect of tools on form. The two letters, the two K ’ s, the one on your left, my right, is modern, made on a computer. All straight lines are dead straight. The curves have that kind of mathematical smoothness that the Bzier formula imposes. On the right, ancient Gothic, cut in the resistant material of steel by hand. None of the straight lines are actually straight. The curves are kind of subtle. It has that spark of life from the human hand that the machine or the program can never capture. What a contrast. Well, I tell a lie. A lie at TED. I ’ m really sorry. Both of these were made on a computer, same software, same Bzier curves, same font format. The one on your left was made by Zuzana Licko at Emigre, and I did the other one. The tool is the same, yet the letters are different. The letters are different because the designers are different. That ’ s all. Zuzana wanted hers to look like that. I wanted mine to look like that. End of story. Type is very adaptable. Unlike a fine art, such as sculpture or architecture, type hides its methods. I think of myself as an industrial designer. The thing I design is manufactured, and it has a function : to be read, to convey meaning. But there is a bit more to it than that. There ' s the sort of aesthetic element. What makes these two letters different from different interpretations by different designers? What gives the work of some designers sort of characteristic personal style, as you might find in the work of a fashion designer, an automobile designer, whatever? There have been some cases, I admit, where I as a designer did feel the influence of technology. This is from the mid- ' 60s, the change from metal type to photo, hot to cold. This brought some benefits but also one particular drawback : a spacing system that only provided 18 discrete units for letters to be accommodated on. I was asked at this time to design a series of condensed sans serif types with as many different variants as possible within this 18- unit box. Quickly looking at the arithmetic, I realized I could only actually make three of related design. Here you see them. In Helvetica Compressed, Extra Compressed, and Ultra Compressed, this rigid 18-unit system really boxed me in. It kind of determined the proportions of the design. Here are the typefaces, at least the lower cases. So do you look at these and say,"Poor Matthew, he had to submit to a problem, and by God it shows in the results."I hope not. If I were doing this same job today, instead of having 18 spacing units, I would have 1, 000. Clearly I could make more variants, but would these three members of the family be better? It ' s hard to say without actually doing it, but they would not be better in the proportion of 1, 000 to 18, I can tell you that. My instinct tells you that any improvement would be rather slight, because they were designed as functions of the system they were designed to fit, and as I said, type is very adaptable. It does hide its methods. All industrial designers work within constraints. This is not fine art. The question is, does a constraint force a compromise? By accepting a constraint, are you working to a lower standard? I don ' t believe so, and I ' ve always been encouraged by something that Charles Eames said. He said he was conscious of working within constraints, but not of making compromises. The distinction between a constraint and a compromise is obviously very subtle, but it ' s very central to my attitude to work. Remember this reading experience? The phone book. I ' ll hold the slide so you can enjoy the nostalgia. This is from the mid- ' 70s early trials of Bell Centennial typeface I designed for the U. S. phone books, and it was my first experience of digital type, and quite a baptism. Designed for the phone books, as I said, to be printed at tiny size on newsprint on very high-speed rotary presses with ink that was kerosene and lampblack. This is not a hospitable environment for a typographic designer. So the challenge for me was to design type that performed as well as possible in these very adverse production conditions. As I say, we were in the infancy of digital type. I had to draw every character by hand on quadrille graph paper — there were four weights of Bell Centennial — pixel by pixel, then encode them raster line by raster line for the keyboard. It took two years, but I learned a lot. These letters look as though they ' ve been chewed by the dog or something or other, but the missing pixels at the intersections of strokes or in the crotches are the result of my studying the effects of ink spread on cheap paper and reacting, revising the font accordingly. These strange **artifacts** are designed to compensate for the undesirable effects of scale and production process. At the outset, AT&T had wanted to set the phone books in Helvetica, but as my friend Erik Spiekermann said in the Helvetica **movie**, if you ' ve seen that, the letters in Helvetica were designed to be as similar to one another as possible. This is not the recipe for legibility at small size. It looks very elegant up on a slide. I had to disambiguate these forms of the figures as much as possible in Bell Centennial by sort of opening the shapes up, as you can see in the bottom part of that slide. So now we ' re on to the mid- ' 80s, the early days of digital outline fonts, vector technology.

There was an issue at that time with the size of the fonts, the amount of data that was required to find and store a font in computer memory. It limited the number of fonts you could get on your typesetting system at any one time. I did an analysis of the data, and found that a **typical** serif face you see on the left needed nearly twice as much data as a sans serif in the middle because of all the points required to define the elegantly curved serif brackets. The numbers at the bottom of the slide, by the way, they represent the amount of data needed to store each of the fonts. So the sans serif, in the middle, sans the serifs, was much more economical, 81 to 151.”Aha,”I thought.”The engineers have a problem. Designer to the rescue.”I made a serif type, you can see it on the right, without curved serifs. I made them polygonal, out of straight line segments, chamfered brackets. And look, as economical in data as a sans serif. We call it Charter, on the right. So I went to the head of engineering with my numbers, and I said proudly,”I have solved your problem.””Oh,”he said.”What problem?”And I said,”Well, you know, the problem of the huge data you require for serif fonts and so on.””Oh,”he said.”We solved that problem last week. We wrote a compaction routine that reduces the size of all fonts by an order of magnitude. You can have as many fonts on your system as you like.””Well, thank you for letting me know,”I said. Foiled again. I was left with a design solution for a nonexistent technical problem. But here is where the story sort of gets interesting for me. I didn ’ t just throw my design away in a fit of pique. I persevered. What had started as a technical exercise became an aesthetic exercise, really. In other words, I had come to like this typeface. Forget its origins. Screw that. I liked the design for its own sake. The simplified forms of Charter gave it a sort of plain- spoken quality and unfussy spareness that sort of pleased me. You know, at times of technical innovation, designers want to be influenced by what ’ s in the air. We want to respond. We want to be pushed into exploring something new. So Charter is a sort of parable for me, really. In the end, there was no hard and fast causal link between the technology and the design of Charter. I had really misunderstood the technology. The technology did suggest something to me, but it did not force my hand, and I think this happens very often. You know, engineers are very smart, and despite occasional frustrations because I ’ m less smart, I ’ ve always enjoyed working with them and learning from them. Apropos, in the mid- ’ 90s, I started talking to Microsoft about screen fonts. Up to that point, all the fonts on screen had been adapted from previously existing printing fonts, of course. But Microsoft foresaw correctly the movement, the stampede towards electronic communication, to reading and writing onscreen with the printed output as being sort of secondary in importance. So the priorities were just tipping at that point. They wanted a small core set of fonts that were not adapted but designed for the screen to face up to the problems of screen, which were their coarse resolution **displays**. I said to Microsoft, a typeface designed for a particular technology is a self-obsoleting typeface. I ’ ve designed too many faces in the past that were intended to mitigate technical problems. Thanks to the engineers, the technical problems went away. So did my typeface. It was only a stopgap. Microsoft came back to say that affordable computer monitors with better resolutions were at least a decade away. So I thought, well, a decade, that ’ s not bad, that ’ s more than a stopgap. So I was persuaded, I was convinced, and we went to work on what became Verdana and Georgia, for the first time working not on paper but directly onto the screen from the pixel up. At that time, screens were binary. The pixel was either on or it was off. Here you see the outline of a letter, the cap H, which is the thin black line, the contour, which is how it is stored in memory, **superimposed** on the bitmap, which is the grey area, which is how it ’ s **displayed** on the screen. The bitmap is rasterized from the outline.

Here in a cap H, which is all straight lines, the two are in almost perfect sync on the Cartesian grid. Not so with an O. This looks more like bricklaying than type design, but believe me, this is a good bitmap O, for the simple reason that it ’ s symmetrical in both x and y axes. In a binary bitmap, you actually can ’ t ask for more than that. I would sometimes make, I don ’ t know, three or four different versions of a difficult letter like a lowercase A, and then stand back to choose which was the best. Well, there was no best, so the designer ’ s judgment comes in in trying to decide which is the least bad. Is that a compromise? Not to me, if you are working at the highest standard the technology will allow, although that standard may be well short of the ideal. You may be able to see on this slide two different bitmap fonts there. The”a”in the upper one, I think, is better than the”a”in the lower one, but it still ain ’ t great. You can maybe see the effect better if it ’ s reduced. Well, maybe not. So I ’ m a pragmatist, not an idealist, out of necessity. For a certain kind of temperament, there is a certain kind of satisfaction in doing something that cannot be perfect but can still be done to the best of your ability. Here ’ s the lowercase H from Georgia Italic. The bitmap looks jagged and rough. It is jagged and rough. But I discovered, by experiment, that there is an optimum slant for an italic on a screen so the strokes break well at the pixel boundaries. Look in this example how, rough as it is, how the left and right legs actually break at the same level. That ’ s a victory. That ’ s good, right there. And of course, at the lower depths, you don ’ t get much choice. This is an S, in case you were wondering. Well, it ’ s been 18 years now since Verdana and Georgia were released. Microsoft were absolutely right, it took a good 10 years, but screen **displays** now do have improved **spatial** resolution, and very much improved photometric resolution thanks to anti-aliasing and so on. So now that their mission is accomplished, has that meant the demise of the screen fonts that I designed for coarser **displays** back then? Will they outlive the now-obsolete screens and the flood of new web fonts coming on to the market? Or have they established their own sort of evolutionary niche that is independent of technology? In other words, have they been absorbed into the typographic mainstream? I ’ m not sure, but they ’ ve had a good run so far. Hey, 18 is a good age for anything with present-day rates of attrition, so I ’ m not complaining. Thank you.

Text Sample 875: POS Dim 6 – 2014 – Score 4.00 – 2014/t001699.txt

George and Charlotte Blonsky, who were a married couple living in the Bronx in New York City, invented something. They got a patent in 1965 for what they call,”a device to assist women in giving birth.”This device consists of a large, round table and some machinery. When the woman is ready to deliver her child, she lies on her back, she is strapped down to the table, and the table is rotated at high speed. The child comes flying out through centrifugal force. If you look at their patent carefully, especially if you have any engineering background or talent, you may decide that you see one or two points where the design is not perfectly adequate. Doctor Ivan Schwab in California is one of the people, one of the main people, who helped answer the question,”Why don ’ t woodpeckers get headaches?”And it turns out the answer to that is because their brains are packaged inside their skulls in a way different from the way our brains, we being human beings, true, have our brains packaged. They, the woodpeckers, typically will peck, they will **bang** their head on a piece of wood thousands of times every day. Every day! And as far as anyone knows, that doesn ’ t bother them in the slightest. How does this happen? Their brain does not slosh around like ours does. Their brain is packed in very tightly, at least for blows coming right from the front. Not too many people paid attention to this research until the last few years when, in this country especially, people are becoming curious about what happens to the brains of football players who **bang** their heads repeatedly. And the woodpecker maybe relates to that. There was a paper published in the medical journal The Lancet in England a few years ago called"A man who pricked his finger and smelled putrid for 5 years."Dr. Caroline Mills and her team received this patient and didn ' t really know what to do about it. The man had cut his finger, he worked processing chickens, and then he started to smell really, really bad. So bad that when he got in a room with the doctors and the nurses, they couldn ' t stand being in the room with him. It was intolerable. They tried every drug, every other treatment they could think of. After a year, he still smelled putrid. After two years, still smelled putrid. Three years, four years, still smelled putrid. After five years, it went away on its own. It ' s a mystery. In New Zealand, Dr. Lianne Parkin and her team tested an old tradition in her city. They live in a city that has huge hills, San Francisco-grade hills. And in the winter there, it gets very cold and very icy. There are lots of injuries. The tradition that they tested, they tested by asking people who were on their way to work in the morning, to stop and try something out. Try one of two conditions. The tradition is that in the winter, in that city, you wear your socks on the outside of your boots. And what they discovered by experiment, and it was quite graphic when they saw it, was that it ' s true. That if you wear your socks on the outside rather than the inside, you ' re much more likely to survive and not slip and fall. Now, I hope you will agree with me that these things I ' ve just described to you, each of them, deserves some kind of prize. And that ' s what they got, each of them got an Ig Nobel prize. In 1991, I, together with bunch of other people, started the Ig Nobel prize ceremony. Every year we give out 10 prizes. The prizes are based on just one criteria. It ' s very simple. It ' s that you ' ve done something that makes people laugh and then think. What you ' ve done makes people laugh and then think. Whatever it is, there ' s something about it that when people encounter it at first, their only possible reaction is to laugh. And then a week later, it ' s still rattling around in their heads and all they want to do is tell their friends about it. That ' s the quality we look for. Every year, we get in the neighborhood of 9, 000 new nominations for the Ig Nobel prize. Of those, consistently between 10 percent and 20 percent of those nominations are people who nominate themselves. Those self-nominees almost never win. It ' s very difficult, numerically, to win a prize if you want to. Even if you don ' t want to, it ' s very difficult numerically. You should know that when we choose somebody to win an Ig Nobel prize, We get in touch with that person, very quietly. We offer them the chance to decline this great honor if they want to. Happily for us, almost everyone who ' s offered a prize decides to accept. What do you get if you win an Ig Nobel prize? Well, you get several things. You get an Ig Nobel prize. The design is different every year. These are always handmade from extremely cheap materials. You ' re looking at a picture of the prize we gave last year, 2013. Most prizes in the world also give their winners some cash, some money. We don ' t have any money, so we can ' t give them. In fact, the winners have to pay their own way to come to the Ig Nobel ceremony, which most of them do. Last year, though, we did manage to scrape up some money. Last year, each of the 10 Ig Nobel prize winners received from us 10 trillion dollars. A $10 trillion bill from Zimbabwe. You may remember that Zimbabwe had a little adventure for a few years there of inflation. They ended up printing bills that were in denominations as large as 100 trillion dollars. The man responsible, who runs the national bank there, by the way, won an Ig Nobel prize in mathematics. The other thing you win is an invitation to come to the ceremony, which happens at Harvard University. And when you get there, you come to Harvard ' s biggest meeting place and classroom. It fits 1, 100 people, it ' s jammed to the gills, and up on the stage, waiting to shake your hand, waiting to hand you your Ig Nobel prize, are a bunch of Nobel prize winners. That ' s the heart of the ceremony. The winners are kept secret until that moment, even the Nobel laureates who will shake their hand don ' t know who they are until they ' re announced. I am going to tell you about just a very few of the other medical-related prizes we ' ve given. Keep in mind, we ' ve given 230 prizes. There are lots of these people who walk among you. Maybe you have one. A paper was published about 30 years ago called"Injuries due to Falling Coconuts."It was written by Dr. Peter Barss, who is Canadian. Dr. Barss came to the ceremony and explained that as a young doctor, he wanted to see the world. So he went to Papua New Guinea. When he got there, he went to work in a hospital, and he was curious what kinds of things happen to people that bring them to the hospital. He looked through the records, and he discovered that a surprisingly large number of people in that hospital were there because of injuries due to falling coconuts. One **typical** thing that happens is people will come from the highlands, where there are not many coconut trees, down to visit their relatives on the coast, where there are lots. And they ' ll think that a coconut tree is a fine place to stand and maybe lie down. A coconut tree that is 90 feet tall, and has coconuts that weigh two pounds that can drop off at any time. A team of doctors in Europe published a series of papers about colonoscopies. You ' re all familiar with colonoscopies, one way or another. Or in some cases, one way and another. They, in these papers, explained to their fellow doctors who perform colonoscopies, how to minimize the chance that when you perform a colonoscopy, your patient will explode. Dr. Emmanuel Ben-Soussan one of the authors, flew in from Paris to the ceremony, where he explained the history of this, that in the 1950s, when colonoscopies were becoming a common technique for the first time, people were figuring out how to do it well. And there were some difficulties at first. The basic problem, I ' m sure you ' re familiar with, that you ' re looking inside a long, narrow, dark place. And so, you want to have a larger space. You add some gas to inflate it so you have room to look around. Now, that ' s added to the gas, the methane gas, that ' s already inside. The gas that they used at first, in many cases, was oxygen. So they added oxygen to methane gas. And then they wanted to be able to see, they needed light, so they ' d put in a light source, which in the 1950s was very hot. So you had methane gas, which is flammable, oxygen and heat. They stopped using oxygen pretty quickly. Now it ' s rare that patients will explode, but it does still happen. The final thing that I want to tell you about is a prize we gave to Dr. Elena Bodnar. Dr. Elena Bodnar invented a brassiere that in an emergency can be quickly separated into a pair of protective face **masks**. One to save your life, one to save the life of some lucky bystander. Why would someone do this, you might wonder. Dr. Bodnar came to the ceremony and she explained that she grew up in Ukraine. She was one of the doctors who treated victims of the Chernobyl power plant meltdown. And they later discovered that a lot of the worst medical problems came from the particles people breathed in. So she was always thinking after that about could there be some simple **mask** that was available everywhere when the unexpected happens. Years later, she moved to America. She had a baby, One day she looked, and on the floor, her infant son had picked up her bra, and had her bra on his face. And that ' s where the idea came from. She came to the Ig Nobel ceremony with the first prototype of the bra and she demonstrated : ["Paul Krugman, Nobel laureate in economics"] ["Wolfgang Ketterle, Nobel laureate in physics"] I myself own an emergency bra. It ' s my favorite bra, but

I would be happy to share it with any of you, should the need arise. Thank you.

Text Sample 876: POS Dim 6 – 2015 – Score 4.00 – 2015/t001578.txt

This is a play called”Sell / Buy / Date.”It ’ s my first since”Bridge and Tunnel,”which I did on Broadway, and this one, I â thank you â I ’ ve excerpted it just for you, so here we go. Right. Class, let ’ s be absolutely certain all electronic devices are switched off before we begin. So class, hopefully you ’ ll recognize what you just heard me say as the â? Very good, the cellular phone announcement. Right? This was also known as a mobile phone. So you ’ ll remember, people of that era would have had an external electronic device, right, something like this, and they all would have carried one of these around with them, and amongst their biggest fears was the sheer mortification that one of these might ring at some inopportune moment. Right? So a bit of trivia about that era for you. So the format of today ’ s class is I will be presenting multiple BERT modules today from that period in history, right, so starting circa 2016. And remember, this was the very first year of the BERT program. So we ’ ve got quite a few of these to get through. Bear in mind, I will be living into **various** different bodies, different ages, also what were then called races, or ethnic groups, as you ’ ll remember from Unit 1. And â â and along the gender continuum, I will be living into males as well. It was quite binary at that time. Also, don ’ t forget, we are reading the book module for next week ’ s focus on gender. Now, I know some of you have requested the book in pill form. I know people still believe ingesting it is better for retention, but since we are trying to experience what our forebears did, right, let ’ s please just consider doing the actual ocular reading, okay? And also, how many people have your emotional shunts engaged? Right. Please toggle them off. Okay? I know it ’ s challenging, but I want you to be able to feel the entire natural emo range, all right? It is essential to this part of the syllabus. Yes, Macy? All right. I understand. If you ’ re unwilling to â All right, well, we can discuss that after class. All right, we will discuss your concerns. Just relax. Nobody ’ s died and gone to composting. Okay. After class. Okay? After class. Let ’ s just get started, okay. This first subject identified as a middle-class homemaker. Remember, these early modules in these people ’ s full identities were protected, and this allowed them to speak more freely on our topic, which for many of them was taboo. Okay honey, now, I ’ m ready when you are. No, sweetheart, I said, I ’ m ready when you are. I ’ m freezing. It ’ s like a meat locker in here in this recording studio. I should have brought a shmata. All this fancy technology but they can ’ t afford heat. What is he saying? I can ’ t hear you! I can ’ t hear you through the glass, honey! There you are in my ear. Oh, you can hear me? The whole time. Oh, yes, I am a little chilly. Yes, oh the cold is for the machines, the new technology. Okay. Yes, now remind me again, you ’ re recording not only my voice but my feelings and my memories? Right. Yes, BERT, yes, I read about it. Bio-Empathetic Resonant Technology. Right, right, so people will be able to feel my experience and my memory? Okay. No, right, I ’ m ready. I just thought you were going to give me a test to see how my memory ’ s doing. I was going to tell you you ’ re too late, it ’ s already bad news. No, no, go ahead, honey. Oh, that ’ s the first question? What do I think of prostitution? Are you soliciting me, young man? I ’ ve heard of May-December romances, but what are you, about 20 years old? Eighteen? Eighteen years. I think I have candies in my purse older than 18 years old. I ’ m teasing you, sweetheart. No, I ’ m comfortable with any question. Sure. So about the prostitution â oh, sex worker, sex worker. No, just in my day, they called it prostitution, not sex work. Oh, because it includes pornography also?

Okay. No, well, I guess when I was a girl, we didn ’ t really have a name for that either. We would have said dirty magazines, I suppose, or dirty **movies**. Well, it ’ s not like what you have with the Internet. No, well, I don ’ t mind sharing. My late husband and I, we were a very romantic couple. Lots of tenderness, you understand. Well, as you get older, you know, at one point I thought my husband might be helped by using some of the pills men can take, but he wasn ’ t interested in those, so I thought, what about maybe watching an adult **movie** on the Internet? Just for inspiration, you understand. Well, at the time, neither of us were very good on the computer, so usually, if we needed help with the Internet, we would just call our children or our grandchildren, but obviously, in this case, that wasn ’ t an option, so I thought, I ’ ll have a look myself, just to see. How difficult could it be? You search for certain key words and you look â Oh wow is right, young man. You can ’ t imagine what I saw. Well, first of all, I was just trying to find, you know, couples, normal couples making love, but this, so many people together at one time. You couldn ’ t tell which part belonged to which body. How they even got the cameras to capture some of this, I couldn ’ t tell you. But the one thing they didn ’ t capture was making love. There was lots of making of something, but they took the love part right out of it, you know, the fun. It was all very extreme, you know? Like you would say, with the extreme sports. Lots of endurance, but never tenderness. So anyway, needless to say, that was $19. 95 I ’ ll never get back again, but it only showed up on the credit card as”entertainment services,”so my husband was never the wiser, and after all of that, well, you could say it turned out he didn ’ t need the extra inspiration after all. Right, so next subject is a young woman â â Next subject, class, is a young woman called Bella, a university student interviewed in 2016 during what was called an Intro to Feminist Porn class as part of her major in sex work at a college in the Bay Area. Yeah, I just want to, like, get a recording of, like, you guys recording me, like a meta recording, or whatever. It ’ s just like this whole experience is just, like, really amazing, and I ’ d like to capture that for, like, Instagram and my Tumblr. So, like, hi guys, it ’ s me, Bella, and I am, like, being interviewed right now for this, like, really amazing Bio-Empathetic Resonance Technology, which is, like, basically where they are, like, recording, as you can see from these, whatever, like, electrodes, the formation of, like, neuropeptides in my hippocampus, or whatever. They will later be able to reconstitute these as, like, my own actual memory, like actual experiences, so other people can, like, actually feel what I ’ m feeling right now. Okay. Okay. So, like, hello, BERT person of the future who is experiencing me. This is what it feels like to be, like, a college freshman, and also the, like, headache that you are experiencing through me is the, like, residual effect of the Jell- O shots which I had last night at the bi-weekly feminist **pole** dancing party which I cohost on Wednesdays. It ’ s called”Don ’ t Get All Polemical”â â and it ’ s in Beekman Hall, and, what else, like, non-Jell-O shots are also available for vegans, and, oh, okay, yeah, totally, yeah, we should also focus on your questions also. So for your record, I am, like, a sex work studies major but minoring in social media with a concentration on notable YouTube memes. Yes, well, of course, like, I consider myself to be, like, obviously, like, a feminist. I was named for Bella Abzug, who was, like, a famous, like, feminist from history, and, like, also I feel that it is, like, important to, like, represent women who are, like, sex-positive feminists. What is sex-negative? Well, like, I guess I would ask, like, what do you think sex-negative is? Yeah, because, like, the terms that we use are, like, so important, because, like, we call it sex work because it helps people understand that, like, it ’ s work, and, like, you know, just like there are, like, healthcare providers and, like, insurance providers, like, we think of these workers as, like, sex care providers. Yeah, but like, I don

’ t think of myself like, providing direct sex care services per se as, like, being a requirement for me to be, like, an advocate. Like, I support other women ’ s right to choose it voluntarily, like, if they enjoy it. Yeah, but, like, I see myself going forward as more likely, like, protecting sex workers ’, like, legal freedoms and rights. Yeah, so, like, basically, I ’ m planning on becoming a lawyer. Right, class. So these next two modules are also circa 2016. One subject is an Irishwoman with a particularly noteworthy relationship to this issue, but first will be a West Indian woman, a self-described escort who was recorded at a sex workers ’ rights rally and parade. She was interviewed whilst marching in full carnival headdress and very little else. All right, you want me to start talking now. Yeah, I told you, you can put those wires anywhere you want to as long as it don ’ t get in the way. Yeah, no, but, tell me again what the name of â BERT? BERT. Yeah, I was telling you, you know, I think I have in all my time I have had at least one client with that name, so this won ’ t be the first time I had BERT all over me. Oh, I ’ m sorry, but you got to get into the spirit of it if you ’ re going to interview me. All right? You can say it. No justice, no piece! No justice, no piece! But you see the sign? You get it? P-I-E-C-E. No justice, no piece of us. You understand? Right, so that ’ s the part where I was telling you is that when I first came to this country, I worked every job I could find. I was a nanny ; I was a home care attendant for all these different old people, and then I said, child, if I have to touch another white man ’ s backside, I might as well get paid a lot more money for it than this, you understand? Pshh, you know how hard it is being a domestic worker? Some of these men, they ’ re heavy. You have to pick them up and flip them over. Now, I let them pick me up and flip me over, you understand? Well, you have to have a sense of humor about it, that ’ s what I think. No, but see, listen, you find me somebody who don ’ t hate some part of their job. I mean, there ’ s a lot of things about this job that I hate, but the money is not one of them, and I will tell you, as long as this is the best possibility for me to make real money, I am going to be Jamaican-No-Fakin ’ if that ’ s what they want to call me. No, I ’ m not even from Jamaica. That ’ s how they market me. My family is from Trinidad and the Virgin Islands. They don ’ t know what I do, but you know what? My children, they know that their school fees are paid, they have their books and their computer, and this way, I know that they have a chance. So I ’ m not going to tell you that what I do, it ’ s easy, I ’ m not going to tell you that I feel â what ’ s that you said, liberated? But I ’ m going to tell you that I feel paid. Right. Thanks, that ’ s lovely, and just the cup of tea, love, and just a splash of the whiskey. It ’ s perfect, that ’ s grand. Just a drop more. A splash. Perfect. What was your name? Peter? Is that right, so, Peter? Right. So that, that is the unique part of it for me, right, is that I ended up in both, first in the convent, and then in the prostitution after. That ’ s right. So one woman at the university here in Dublin, she wrote about me. She said, Maureen Fitzroy is the living embodiment of the whore-virgin dichotomy. Right? Doesn ’ t it sound like something you need to go into hospital? Well, I ’ ve got this terrible dichotomy. Doesn ’ t it. Right. Well, for me though, it was, as a girl, it started with me dad. I mean, half the time, when he spoke to us, it was just a sort of tell us we were all useless rotten idiots and we had no morals, that type of thing. And I certainly didn ’ t do myself any favors. By the time I was 16, I had started messing about with this older fella, and he wanted it to be our little secret, and I did as I was told, didn ’ t I, and when that got back to me dad, he had me sent straightaway to the convent. Well no, that older fella, he would still come to find me in the convent. Yeah, he ’ d leave me notes tucked into the holes in the brick at the back of the charity shop so we could meet. And he ’ d tell me how he ’ s leaving his wife, and I believed him, until I got pregnant. I did, Peter, and I left him a note about it in our special place there, and I never did hear from him again. No, I gave it up for adoption so it could have a decent life, and then they wouldn ’ t let me back into the convent. No, my one sister Virginia gave me a fiver for the coach to Dublin, and that ’ s how I ended up here. Well, surprise, surprise, I fell in love with another fella much older than me, and I always say I was just so happy because he didn ’ t drink, I married the bastard. Well, he didn ’ t drink, but he did have just the wee heroin problem, didn ’ t he, and â That ’ s right, and before I knew it, he was the one who turned me on to the prostitution, my own husband. He had me supporting the both of us. I was 18. Well, it wasn ’ t Pretty Woman, I can tell you that. That Julia Roberts, if she ’ d ever had to sleep with a man to put a few pounds in her pocket, I don ’ t think she ’ d ever have made that film. Well, for your record, my opinion of the legalization, I ’ d say I ’ m against it. I just, I don ’ t care what these young girls say. You know, living like that, you ’ re just lost, and, you know, I ’ m 63 years old. I ’ m still trying to find who I am. You know, I never was a wife or a nun, or a prostitute even, really, not really. Nobody ever asked who I wanted to be. They just told me, and if you legalize it, then you ’ re really telling these girls,”Go on and get lost for a living,”and a lot of them, they ’ ll do as they ’ re told. All right, so four perspectives from four quite â â four quite different voices there, right? One woman saying sex itself is natural but the sex industry seems to mechanize or industrialize it. Then the second woman considered sex work to be empowering, liberating, and feminist, though she, herself, notably, did not seem keen to do it. The third woman, who actually was a so-called sex worker did not agree that it was liberating but she wanted the right to the economic empowerment, and then we hear the fourth woman saying not only prostitution itself but proscribed roles for women in general prevented her from ever finding who she was, right? So another fact most people did not know was the average age of an at-risk girl being introduced to the sex industry was 12 or 13. Also consider that the age when all girls in that society first became exposed to sexualized images of women was quite a bit earlier, right? This was a doll called Barbie, right? I initially thought she was an educational tool for anorexia prevention â â but actually she was considered by many to be a wholesome symbol of femininity, and often young girls began what was called dieting. Remember this? This was restricting food intake on purpose by the age of six, and defining themselves based on attractiveness by around that same time. Right? Yes? Right, Bradley, okay, excellent point. So there was a lucrative market in that society in convincing all people they had to look a certain way to even have a sex life, right? But girls, especially, were expected to be”sexy”while avoiding being perceived as”sluts”for being sexual. Right? So there ’ s that shame piece we ’ ve heard about. Yes. Valerie, right? Okay, very good. Of course, men were having sex as well, but you ’ ll remember from the reading, what were male sluts called? Very good, they were called men. So not easy living in a world like that, right? Though it was not all bad news either. Most women in the early 2000s considered themselves empowered, and men generally felt they were also evolved in this area, and, in fact, most people would have been aware of issues like human trafficking, for example, but they would have seen that as quite separate from more recreational adult entertainment. And so we ’ ll just very briefly, class â we don ’ t have a lot of time â we ’ ll just very briefly hear from a man on our topic at this stage. So this next subject was interviewed on the night of his bachelor party. Dude, can you, all right, can you just keep it down? I ’ m trying to talk to BERT right now. Oh, your name ’ s not BERT. BERT ’ s the name of the, oh, all right. No, no, no, totally, it ’ s totally fine. I ’ m mostly sober, so I just want to be helpful. Yeah, and I totally believe in causes, yeah, like, all that stuff. And actually, I ’ m wearing Toms right now. Yeah, Toms, like, the shoes, like, you buy a pair and then a kid in Africa gets clean water. Yeah. Totally. But what was the question again? Sorry. Of course I believe in women ’ s rights. I ’ m marrying a woman. No, but I mean, like, just because I ’ m in a strip club parking lot doesn ’ t mean that I ’ m, like, a sexist or whatever. My fiancee is totally amazing, she ’ s totally a strong girl, woman, smart woman, like, the whole thing. Yeah, she knows I ’ m here. She ’ s probably at a strip club herself right now, like, as a joke, same as me. My best man, I told him he could surprise me, and he thought this would be hilarious, but this is not something. Yeah, we all went to B school together. Wharton. Yeah, so, dude, can you guys â All right, but it ’ s my bachelor party, and I can spend it in the parking lot with Anderson Cooper if I want to. All right, I ’ ll see you in there. All right, okay, so Anderson, so, like, first of all, stripping, but then, like, all the other things you ’ re talking about, prostitution and all that stuff, that ’ s, like, not the same thing at all. You know? Like, you keep calling it the sex industry or whatever, but it ’ s like, if the girl wants to be an exotic dancer and she ’ s 18, like, that ’ s her right. Whoa, whoa, I hear what you ’ re saying, but I just feel like people, they just want to make it seem like all dudes are just, like, predators, that we would just automatically go to a prostitute, or whatever. Even, like, when I pledged, you know, like when I rushed my fraternity. My brothers who I ’ m close to, those guys, they ’ re all like me. We ’ re just normal people, but, like, there ’ s this myth that you must be that guy who is kind of an asshole, and like, all bros before hos or whatever. And actually, like, bros before hos, it doesn ’ t mean like what it sounds like. It ’ s actually just like a joking way of saying that you care about your brothers and you put them first. Yeah, but, you can ’ t blame the media, either. I mean, like, if you go watch”Hangover 2,”and you think that ’ s an instruction manual for your life, like, I don ’ t know what to tell you. You know? You don ’ t watch”Bourne Identity”and go drive your car over a gondola in Venice. Well, yeah, okay, like, if you ’ re a little kid or whatever, of course it ’ s different, but â Yeah, all right, I remember one thing like that. I was at this kid ’ s house one time playing GTA, uh, Grand Theft Auto? Dude, are you from Canada? So, like, whatever, with Grand Theft Auto, you ’ re this kid, like, you ’ re this guy walking around or whatever, and you can basically, like, the more cops you kill, the more points you get, and stuff like that. But also, you can find prostitutes and obviously you can do sexual stuff with them, but you can, like, kill them and take your money back. Yeah, this kid, I remember he ran over a couple of them a few times with his car and he got all these points. We were, like, 10, I think. It felt pretty terrible, actually. No, I don ’ t think I said anything, I just finished playing and went home. All right class, so then there were men who had more than just a passing relationship to this issue. The next subject described himself as a reformed and remorseful pimp turned motivational speaker, life coach and therapist, but if you want to know more about him, you ’ ll have to come to the entire play. Thank you so much, you beautiful TED audience. I will see you for”Sell / Buy / Date.”

Text Sample 877: POS Dim 6 – 2015 – Score 4.00 – 2015/t001563.txt

I love a great mystery, and I ’ m fascinated by the greatest unsolved mystery in science, perhaps because it ’ s personal. It ’ s about who we are, and I can ’ t help but be curious. The mystery is this : What is the relationship between your brain and your conscious experiences, such as your experience of the taste of chocolate or the feeling of velvet? Now, this mystery is not new. In 1868, Thomas Huxley wrote,”How it is that anything so **remarkable** as a state of consciousness comes about as the result of irritating nervous tissue is just as unaccountable as the appearance of the genie when Aladdin rubbed his lamp."Now, Huxley knew that brain activity and conscious experiences are correlated, but he didn ' t know why. To the science of his day, it was a mystery. In the years since Huxley, science has learned a lot about brain activity, but the relationship between brain activity and conscious experiences is still a mystery. Why? Why have we made so little progress? Well, some experts think that we can ' t solve this problem because we lack the necessary concepts and intelligence. We don ' t expect monkeys to solve problems in quantum mechanics, and as it happens, we can ' t expect our species to solve this problem either. Well, I disagree. I ' m more optimistic. I think we ' ve simply made a false assumption. Once we fix it, we just might solve this problem. Today, I ' d like tell you what that assumption is, why it ' s false, and how to fix it. Let ' s begin with a question : Do we see reality as it is? I open my eyes and I have an experience that I describe as a red tomato a meter away. As a result, I come to believe that in reality, there ' s a red tomato a meter away. I then close my eyes, and my experience changes to a gray field, but is it still the case that in reality, there ' s a red tomato a meter away? I think so, but could I be wrong? Could I be misinterpreting the nature of my perceptions? We have misinterpreted our perceptions before. We used to think the Earth is flat, because it looks that way. Pythagorus discovered that we were wrong. Then we thought that the Earth is the unmoving center of the Universe, again because it looks that way. Copernicus and Galileo discovered, again, that we were wrong. Galileo then wondered if we might be misinterpreting our experiences in other ways. He wrote :"I think that tastes, odors, colors, and so on reside in consciousness. Hence if the living creature were removed, all these qualities would be annihilated."Now, that ' s a stunning claim. Could Galileo be right? Could we really be misinterpreting our experiences that badly? What does modern science have to say about this? Well, neuroscientists tell us that about a third of the brain ' s cortex is engaged in vision. When you simply open your eyes and look about this room, billions of neurons and trillions of synapses are engaged. Now, this is a bit surprising, because to the extent that we think about vision at all, we think of it as like a camera. It just takes a picture of objective reality as it is. Now, there is a part of vision that ' s like a camera : the eye has a lens that focuses an image on the back of the eye where there are 130 million photoreceptors, so the eye is like a 130-megapixel camera. But that doesn ' t explain the billions of neurons and trillions of synapses that are engaged in vision. What are these neurons up to? Well, neuroscientists tell us that they are creating, in real time, all the shapes, objects, colors, and motions that we see. It feels like we ' re just taking a snapshot of this room the way it is, but in fact, we ' re constructing everything that we see. We don ' t construct the whole world at once. We construct what we need in the moment. Now, there are many demonstrations that are quite compelling that we construct what we see. I ' ll just show you two. In this example, you see some red discs with bits cut out of them, but if I just rotate the disks a little bit, suddenly, you see a 3D cube pop out of the screen. Now, the screen of course is flat, so the three-dimensional cube that you ' re experiencing must be your construction. In this next example, you see glowing blue bars with pretty sharp edges moving across a field of dots. In fact, no dots move. All I ' m doing from frame to frame is changing the colors of dots from blue to black or black to blue. But when I do this quickly, your visual system creates the glowing blue bars with the sharp edges and the motion. There are many more examples, but these are just two that you construct what you see. But neuroscientists go further. They say that we reconstruct reality. So, when I have an experience that I describe as a red tomato, that experience is actually an accurate reconstruction of the properties of a real red tomato that would exist even if I weren ' t looking. Now, why would neuroscientists say that we don ' t just construct, we reconstruct? Well, the standard argument given is usually an evolutionary one. Those of our ancestors who saw more accurately had a competitive advantage compared to those who saw less accurately, and therefore they were more likely to pass on their genes. We are the offspring of those who saw more accurately, and so we can be confident that, in the normal case, our perceptions are accurate. You see this in the standard textbooks. One textbook says, for example,"Evolutionarily speaking, vision is useful precisely because it is so accurate."So the idea is that accurate perceptions are fitter perceptions. They give you a survival advantage. Now, is this correct? Is this the right interpretation of evolutionary theory? Well, let ' s first look at a couple of examples in nature. The Australian jewel beetle is dimpled, glossy and brown. The female is flightless. The male flies, looking, of course, for a hot female. When he finds one, he alights and mates. There ' s another species in the outback, Homo sapiens. The male of this species has a massive brain that he uses to hunt for cold beer. And when he finds one, he drains it, and sometimes throws the bottle into the outback. Now, as it happens, these bottles are dimpled, glossy, and just the right shade of brown to tickle the fancy of these beetles. The males swarm all over the bottles trying to mate. They lose all interest in the real females. Classic case of the male leaving the female for the bottle. The species almost went extinct. Australia had to change its bottles to save its beetles. Now, the males had successfully found females for thousands, perhaps millions of years. It looked like they saw reality as it is, but apparently not. Evolution had given them a hack. A female is anything dimpled, glossy and brown, the bigger the better. Even when crawling all over the bottle, the male couldn ' t discover his mistake. Now, you might say, beetles, sure, they ' re very simple creatures, but surely not mammals. Mammals don ' t rely on tricks. Well, I won ' t dwell on this, but you get the idea. So this raises an important technical question : Does natural selection really favor seeing reality as it is? Fortunately, we don ' t have to wave our hands and guess ; evolution is a mathematically precise theory. We can use the equations of evolution to check this out. We can have **various** organisms in artificial worlds compete and see which survive and which thrive, which sensory systems are more fit. A key notion in those equations is fitness. Consider this steak : What does this steak do for the fitness of an animal? Well, for a hungry lion looking to eat, it enhances fitness. For a well-fed lion looking to mate, it doesn ' t enhance fitness. And for a rabbit in any state, it doesn ' t enhance fitness, so fitness does depend on reality as it is, yes, but also on the organism, its state and its action. Fitness is not the same thing as reality as it is, and it ' s fitness, and not reality as it is, that figures centrally in the equations of evolution. So, in my lab, we have run hundreds of thousands of evolutionary game simulations with lots of different randomly chosen worlds and organisms that compete for resources in those worlds. Some of the organisms see all of the reality, others see just part of the reality, and some see none of the reality, only fitness. Who wins? Well, I hate to break it to you, but perception of reality goes extinct. In almost every simulation, organisms that see none of reality but are just tuned to fitness drive to extinction all the organisms that perceive reality as it is. So the bottom line is, evolution does not favor veridical, or accurate perceptions. Those perceptions of reality go extinct. Now, this is a bit stunning. How can it be that not seeing the world accurately gives us a survival advantage? That is a bit counterintuitive. But remember the jewel beetle. The jewel beetle survived for thousands, perhaps millions of years, using simple tricks and hacks. What the equations of evolution are telling us is that all organisms, including us, are in the same boat as the jewel beetle. We do not see reality as it is. We

’ re shaped with tricks and hacks that keep us alive. Still, we need some help with our intuitions. How can not perceiving reality as it is be useful? Well, fortunately, we have a very helpful metaphor : the desktop **interface** on your computer. Consider that blue icon for a TED Talk that you ’ re writing. Now, the icon is blue and rectangular and in the lower right corner of the desktop. Does that mean that the text file itself in the computer is blue, rectangular, and in the lower right-hand corner of the computer? Of course not. Anyone who thought that misinterprets the purpose of the **interface**. It ’ s not there to show you the reality of the computer. In fact, it ’ s there to hide that reality. You don ’ t want to know about the **diodes** and resistors and all the megabytes of software. If you had to deal with that, you could never write your text file or edit your photo. So the idea is that evolution has given us an **interface** that hides reality and guides adaptive behavior. Space and time, as you perceive them right now, are your desktop. Physical objects are simply icons in that desktop. There ’ s an obvious objection. Hoffman, if you think that train coming down the track at 200 MPH is just an icon of your desktop, why don ’ t you step in front of it? And after you ’ re gone, and your theory with you, we ’ ll know that there ’ s more to that train than just an icon. Well, I wouldn ’ t step in front of that train for the same reason that I wouldn ’ t carelessly drag that icon to the trash can : not because I take the icon literally — the file is not literally blue or rectangular — but I do take it seriously. I could lose weeks of work. Similarly, evolution has shaped us with perceptual symbols that are designed to keep us alive. We ’ d better take them seriously. If you see a snake, don ’ t pick it up. If you see a cliff, don ’ t jump off. They ’ re designed to keep us safe, and we should take them seriously. That does not mean that we should take them literally. That ’ s a logical error. Another objection : There ’ s nothing really new here. Physicists have told us for a long time that the metal of that train looks solid but really it ’ s mostly empty space with microscopic particles zipping around. There ’ s nothing new here. Well, not exactly. It ’ s like saying, I know that that blue icon on the desktop is not the reality of the computer, but if I pull out my trusty magnifying glass and look really closely, I see little pixels, and that ’ s the reality of the computer. Well, not really — you ’ re still on the desktop, and that ’ s the point. Those microscopic particles are still in space and time : they ’ re still in the user **interface**. So I ’ m saying something far more radical than those physicists. Finally, you might object, look, we all see the train, therefore none of us constructs the train. But remember this example. In this example, we all see a cube, but the screen is flat, so the cube that you see is the cube that you construct. We all see a cube because we all, each one of us, constructs the cube that we see. The same is true of the train. We all see a train because we each see the train that we construct, and the same is true of all physical objects. We ’ re inclined to think that perception is like a window on reality as it is. The theory of evolution is telling us that this is an incorrect interpretation of our perceptions. Instead, reality is more like a 3D desktop that ’ s designed to hide the complexity of the real world and guide adaptive behavior. Space as you perceive it is your desktop. Physical objects are just the icons in that desktop. We used to think that the Earth is flat because it looks that way. Then we thought that the Earth is the unmoving center of reality because it looks that way. We were wrong. We had misinterpreted our perceptions. Now we believe that spacetime and objects are the nature of reality as it is. The theory of evolution is telling us that once again, we ’ re wrong. We ’ re misinterpreting the content of our perceptual experiences. There ’ s something that exists when you don ’ t look, but it ’ s not spacetime and physical objects. It ’ s as hard for us to let go of spacetime and objects as it is for the jewel beetle to let go of its bottle. Why? Because we ’ re blind to our own blindnesses. But we have an advantage over the jewel beetle : our science and technology. By peering through the lens of a telescope we discovered that the Earth is not the unmoving center of reality, and by peering through the lens of the theory of evolution we discovered that spacetime and objects are not the nature of reality. When I have a perceptual experience that I describe as a red tomato, I am interacting with reality, but that reality is not a red tomato and is nothing like a red tomato. Similarly, when I have an experience that I describe as a lion or a steak, I ' m interacting with reality, but that reality is not a lion or a steak. And here ' s the kicker : When I have a perceptual experience that I describe as a brain, or neurons, I am interacting with reality, but that reality is not a brain or neurons and is nothing like a brain or neurons. And that reality, whatever it is, is the real source of cause and effect in the world — not brains, not neurons. Brains and neurons have no causal powers. They cause none of our perceptual experiences, and none of our behavior. Brains and neurons are a species-specific set of symbols, a hack. What does this mean for the mystery of consciousness? Well, it opens up new possibilities. For instance, perhaps reality is some vast machine that causes our conscious experiences. I doubt this, but it ' s worth exploring. Perhaps reality is some vast, interacting network of conscious agents, simple and complex, that cause each other ' s conscious experiences. Actually, this isn ' t as crazy an idea as it seems, and I ' m currently exploring it. But here ' s the point : Once we let go of our massively intuitive but massively false assumption about the nature of reality, it opens up new ways to think about life ' s greatest mystery. I bet that reality will end up turning out to be more fascinating and unexpected than we ' ve ever imagined. The theory of evolution presents us with the ultimate dare : Dare to recognize that perception is not about seeing truth, it ' s about having kids. And by the way, even this TED is just in your head. Thank you very much. Chris Anderson : If that ' s really you there, thank you. So there ' s so much from this. I mean, first of all, some people may just be profoundly depressed at the thought that, if evolution does not favor reality, I mean, doesn ' t that to some extent undermine all our endeavors here, all our ability to think that we can think the truth, possibly even including your own theory, if you go there? Donald Hoffman : Well, this does not stop us from a successful science. What we have is one theory that turned out to be false, that perception is like reality and reality is like our perceptions. That theory turns out to be false. Okay, throw that theory away. That doesn ' t stop us from now postulating all sorts of other theories about the nature of reality, so it ' s actually progress to recognize that one of our theories was false. So science continues as normal. There ' s no problem here. CA : So you think it ' s possible — — This is cool, but what you ' re saying I think is it ' s possible that evolution can still get you to reason. DH : Yes. Now that ' s a very, very good point. The evolutionary game simulations that I showed were specifically about perception, and they do show that our perceptions have been shaped not to show us reality as it is, but that does not mean the same thing about our logic or mathematics. We haven ' t done these simulations, but my bet is that we ' ll find that there are some selection pressures for our logic and our mathematics to be at least in the direction of truth. I mean, if you ' re like me, math and logic is not easy. We don ' t get it all right, but at least the selection pressures are not uniformly away from true math and logic. So I think that we ' ll find that we have to look at each cognitive faculty one at a time and see what evolution does to it. What ' s true about perception may not be true about math and logic. CA : I mean, really what you ' re proposing is a kind of modern-day Bishop Berkeley interpretation of the world : consciousness causes matter, not the other way around. DH : Well, it ' s slightly different than Berkeley. Berkeley thought that, he was a deist, and he thought that the ultimate nature of reality is God and so forth, and I don ’ t need to go where Berkeley ’ s going, so it ’ s quite a bit different from Berkeley. I call this conscious realism. It ’ s actually a very different approach. CA : Don, I could literally talk with you for hours, and I hope to do that. Thanks so much for that. DH : Thank you.

Text Sample 878: POS Dim 6 – 2015 – Score 4.00 – 2015/t001423.txt

We ’ ve evolved with tools, and tools have evolved with us. Our ancestors created these hand axes 1. 5 million years ago, shaping them to not only fit the task at hand but also their hand. However, over the years, tools have become more and more specialized. These sculpting tools have evolved through their use, and each one has a different form which matches its function. And they leverage the dexterity of our hands in order to manipulate things with much more precision. But as tools have become more and more complex, we need more complex controls to control them. And so designers have become very adept at creating **interfaces** that allow you to manipulate parameters while you ’ re attending to other things, such as taking a photograph and changing the focus or the aperture. But the computer has fundamentally changed the way we think about tools because computation is dynamic. So it can do a million different things and run a million different applications. However, computers have the same static physical form for all of these different applications and the same static **interface** elements as well. And I believe that this is fundamentally a problem, because it doesn ’ t really allow us to interact with our hands and capture the rich dexterity that we have in our bodies. And my belief is that, then, we must need new types of **interfaces** that can capture these rich abilities that we have and that can physically adapt to us and allow us to interact in new ways. And so that ’ s what I ’ ve been doing at the MIT Media Lab and now at Stanford. So with my colleagues, Daniel Leithinger and Hiroshi Ishii, we created inFORM, where the **interface** can actually come off the screen and you can physically manipulate it. Or you can visualize 3D information physically and touch it and feel it to understand it in new ways. Or you can interact through gestures and direct deformations to sculpt digital clay. Or **interface** elements can arise out of the surface and change on demand. And the idea is that for each individual application, the physical form can be matched to the application. And I believe this represents a new way that we can interact with information, by making it physical. So the question is, how can we use this? Traditionally, urban planners and architects build physical models of cities and buildings to better understand them. So with Tony Tang at the Media Lab, we created an **interface** built on inFORM to allow urban planners to design and view entire cities. And now you can walk around it, but it ’ s dynamic, it ’ s physical, and you can also interact directly. Or you can look at different views, such as population or traffic information, but it ’ s made physical. We also believe that these dynamic shape **displays** can really change the ways that we remotely collaborate with people. So when we ’ re working together in person, I ’ m not only looking at your face but I ’ m also gesturing and manipulating objects, and that ’ s really hard to do when you ’ re using tools like Skype. And so using inFORM, you can reach out from the screen and manipulate things at a distance. So we used the pins of the **display** to represent people ’ s hands, allowing them to actually touch and manipulate objects at a distance. And you can also manipulate and collaborate on 3D data sets as well, so you can gesture around them as well as manipulate them. And that allows people to collaborate on these new types of 3D information in a richer way than might be possible with traditional tools. And so you can also bring in existing objects, and those will be captured on one side and transmitted to the other. Or you can have an object that ’ s linked between two places, so as I move a **ball** on one side, the **ball** moves on the other as well. And so we do this by capturing the remote user using a depth-sensing camera like a Microsoft Kinect. Now, you might be wondering how does this all work, and essentially, what it is, is 900 linear actuators that are connected to these mechanical linkages that allow motion down here to be propagated in these pins above. So it ’ s not that complex compared to what ’ s going on at CERN, but it did take a long time for us to build it. And so we started with a single motor, a single linear actuator, and then we had to design a custom circuit board to control them. And then we had to make a lot of them. And so the problem with having 900 of something is that you have to do every step 900 times. And so that meant that we had a lot of work to do. So we sort of set up a mini-sweatshop in the Media Lab and brought undergrads in and convinced them to do”research”— and had late nights watching **movies**, eating pizza and screwing in thousands of screws. You know — research. But anyway, I think that we were really excited by the things that inFORM allowed us to do. Increasingly, we ’ re using mobile devices, and we interact on the go. But mobile devices, just like computers, are used for so many different applications. So you use them to talk on the phone, to surf the web, to play games, to take pictures or even a million different things. But again, they have the same static physical form for each of these applications. And so we wanted to know how can we take some of the same interactions that we developed for inFORM and bring them to mobile devices. So at Stanford, we created this haptic edge **display**, which is a mobile device with an array of linear actuators that can change shape, so you can feel in your hand where you are as you ’ re reading a book. Or you can feel in your pocket new types of tactile sensations that are richer than the vibration. Or buttons can emerge from the side that allow you to interact where you want them to be. Or you can play games and have actual buttons. And so we were able to do this by embedding 40 small, tiny linear actuators inside the device, and that allow you not only to touch them but also back-drive them as well. But we ’ ve also looked at other ways to create more complex shape change. So we ’ ve used pneumatic actuation to create a morphing device where you can go from something that looks a lot like a phone... to a wristband on the go. And so together with Ken Nakagaki at the Media Lab, we created this new high- resolution version that uses an array of servomotors to change from interactive wristband to a touch-input device to a phone. And we ’ re also interested in looking at ways that users can actually deform the **interfaces** to shape them into the devices that they want to use. So you can make something like a game controller, and then the system will understand what shape it ’ s in and change to that mode. So, where does this point? How do we move forward from here? I think, really, where we are today is in this new age of the Internet of Things, where we have computers everywhere — they ’ re in our pockets, they ’ re in our walls, they ’ re in almost every device that you ’ ll buy in the next five years. But what if we stopped thinking about devices and think instead about environments? And so how can we have smart furniture or smart rooms or smart environments or cities that can adapt to us physically, and allow us to do new ways of collaborating with people and doing new types of tasks? So for the Milan Design Week, we created TRANSFORM, which is an interactive table-scale version of these shape **displays**, which can move physical objects on the surface ; for example, reminding you to take your keys. But it can also transform to fit different ways of interacting. So if you want to work, then it can change to sort of set up your work system. And so as you bring a device over, it creates all the affordances you need and brings other objects to help you accomplish those goals. So, in conclusion, I really think that we need to think about a new, fundamentally different way of interacting with computers. We need computers that can physically adapt to us and adapt to the ways that we want to use them and really harness the rich dexterity that we have of our hands, and our ability to think spatially about information by making it physical. But looking forward, I think we need to go beyond this, beyond devices, to really think about new ways that we can bring people together, and bring our information into the world, and think about smart environments that can adapt to us physically. So with that, I will leave you. Thank you very much.

Text Sample 879: POS Dim 6 – 2020 – Score 6.00 – 2020/t004276.txt

Arthur Brooks : Starting when I was eight or nine years old, I literally wanted to be the greatest French horn player in the world. That was what I always wanted to do. That was my whole life dream. Adam Grant : What was behind that? AB : I was ambitious. I wanted to play for a lot of people. And I guess that was just simply expressed in the only thing that I really knew how to do, the only thing I really cared about, which was music. AG : Arthur Brooks was a natural. He could read music before he could actually read. He played in youth orchestras as a kid, prestigious ones. Eventually he dropped out of college to go on tour. AB : I went on the road playing classical music. I did that for six years. I saw all 50 states from the back of a van. AG : Arthur recorded an album that played on classical radio stations across the country. But then something started to change. AB : I realized, about age 21, that things were starting to go south, and I couldn ’ t explain it. Ordinarily, classical musicians, they will see their skills increase through their 20s, and sometimes even their 30s. So whereas if I were a baseball pitcher, it would be totally normal that I would be losing heat off my fastball by my early 20s. As a French horn player, I shouldn ’ t have been seeing that, but I was. AG : To turn things around he practiced his heart out. Finally, he got his big break, his debut at Carnegie Hall. AB : So I got ready for this concert, and the first half of the concert, I have to say, was phenomenal. It was really one of the best concerts I ’ d ever given. And in the second half, I had this experience where I had to actually talk to the audience and I walked toward the front of the stage to address the crowd, and I wasn ’ t watching my feet and I slipped, or something collapsed, I still don ’ t know, and I fell five feet into the audience. AG : You literally fell off the stage? AB : I literally fell off the stage. It was just ridiculous. It was... I wrecked my instrument, I fell on my instrument, I hurt my arm. And, of course, I did what you would do when you ’ re 22 years old. I jumped up and said,”I ’ m okay, folks.”And I obviously was not okay. AG : Oh, no. AB : It was just so terrible, Adam. It was so terrible. AG : Arthur tried to shake off the feeling that falling off the stage was a bad omen. He spent the next decade practicing. He studied with the best tutors in Europe. He got a job at a symphony in Barcelona, but – AB : It was a wreck. It was a slow-moving wreck in my career itself. I was in decline. Finally had to pack it in. I had to admit it to myself that I wasn ’ t the horn player I had always dreamed of being. And right now, I ’ m talking about it with you and, you know, we ’ re laughing and all that, but I ’ ve got to tell you, it kills me, I still hate it. AG : Arthur didn ’ t plan to quit music so early. He simply stopped being great at his job. And in the years since, it ’ s made him wonder, does career decline happened to all of us eventually? And is there anything we can do about it? I ’ m Adam Grant, and this is WorkLife, my podcast with TED. I ’ m an organizational psychologist. I study how to make work not suck. In this show, I ’ m inviting myself inside the minds of some truly unusual people, because they mastered something I wish everyone knew about work. Today, career staying power. It ’ s more about the choices you make than the age you reach. Thanks to Accenture for sponsoring this episode. Let ’ s be honest, we all have peaks and valleys in our careers, times when we hit our stride and do our best work, and times when we ’ re in a slump. Most of us are worried that as we age, our physical and mental skills will decline and we might enter into a permanent slump. And that fear is compounded by the fact that age is seldom seen as an asset in the modern workplace. Far too often, our older colleagues don ’ t get the respect they deserve. Arthur ’ s slump happened at a particularly early age, so he changed careers and became a public policy expert. AB : Basically I ’ m just an old fashioned, number-crunching, quantitative social scientist. AG : Given what he ’ s gone through, it ’ s not entirely surprising that his big area of focus now is career decline. AB : Career decline is a very personal and personalized phenomenon in which your skills are not what they once were. So you might be making as much, or more, money than you ever have before, but you know that you are not at the same level of technical adeptness that you had once attained, and / or you ’ re not on the trajectory to attain what you had hoped. AG : Maybe you felt your stomach lurch when someone younger than you has a smashing success in your field or your workplace. Or maybe you ’ ve been humming along feeling good about your work, until one day you noticed your skills don ’ t seem quite as sharp as they used to. In 2019, Arthur wrote a major story about this phenomenon in”The Atlantic.”The headline was”Your Professional Decline Is Coming Sooner Than You Think.”As in, long before you even start to see retirement on the horizon. AB : I mean the number one myth is that, you know, particularly in certain professions, like, let ’ s just say yours and mine, which is coming up with big ideas and sharing them, that that ’ ll never go into decline. Why? Because it doesn ’ t require, you know, strong biceps, and yet there ’ s overwhelming evidence that in idea professions people experience decline as well. They just don ’ t expect it. AG : What happens to our cognitive abilities as we age is not straightforward. It actually depends on what kind of mental skills we ’ re talking about. Psychologists have long distinguished between two kinds of intelligence : fluid and crystallized. Fluid intelligence is your raw processing power. It ’ s basically your IQ, your innate capacity for learning and problem solving. AB : That ’ s a big part of the modern idea economy. I ’ m going to come in here, I ’ m going to be a whizzbang person who comes up with brand new ideas. That ’ s actually what ’ s rewarded for young people because young people have high levels of fluid intelligence. It increases through your teens and 20s, and maybe through your 30s, although it tends to start declining for most people in their 30s and 40s, and it makes this original idea generation harder. Now that doesn ’ t mean that you ’ re going to be unsuccessful, but it means if you ’ re relying entirely on this fluid intelligence, you ’ re going to struggle. AG : A common refrain in Silicon Valley is that young people are just smarter. When it comes to fluid intelligence, that ’ s generally true. But the story changes with crystallized intelligence, your acquired ability to solve problems by applying your knowledge and experience. AB : Your ability to synthesize things that you didn ’ t necessarily invent, that ’ s crystallized intelligence and that tends to happen later. AG : Think about how crystallized intelligence comes in handy in your job. If you ’ re in sales, you ’ ve used it to tailor your pitches to different clients based on what you ’ ve learned about them in the past. If you ’ re in government, you ’ ve relied on it to get around red tape. If you ’ re in finance, maybe you ’ ve applied it to predict how markets are affected by crises. Crystallized intelligence peaks much later than fluid intelligence, in large part because our body of knowledge keeps growing. AB : Think of it like a library. Think of the brain as basically filling up with a lot of volumes. And that means you have a lot of crystallized intelligence. AG :

Depth and breadth of knowledge aren ’ t the only benefits of aging. Evidence suggests that self-control and socio-emotional skills often keep climbing, too. Many of us continue gaining willpower and emotional intelligence well into our 40s and 50s. We ’ re still getting better at delaying gratification, reading other people ’ s emotions and regulating our own moods. Even into our 70s and 80s, we ’ re still improving at ignoring sunk costs, all of which can enhance our job performance. But Arthur doesn ’ t think that ’ s enough to keep us from declining. AB : Everybody falls in their game at some particular point. AG : At what age do you think the average person is going to start to face significant career decline? AB : I ’ ve got to give you the social scientist answer – it depends but if I were to guess, I ’ ve got to say that in your mid-40s is when you ’ re likely to see the peak of your originality and creativity, and you should be making plans in your 40s to how you can be the best creator with your crystallized intelligence, how you can be the best instructor, the best synthesizer, you can possibly be. AG : OK, so this is where I want to start disagreeing with you. So my read of the evidence is that people don ’ t inherently become less creative. One of the things that happens as people ’ s careers advance is they tend to produce less, just, you know, in terms of overall volume. I think it ’ s not ability. I think it ’ s motivation that ’ s dropping. I think, you know, a lot of people become complacent and they feel like they ’ ve achieved enough success, and so they may scale back a little bit. It may be that priorities outside of work become more important to them. And if we can maintain that motivation, there ’ s no inherent reason why the quality of creative output would decline. What do you think? AB : Hmm, I ’ m willing to entertain your hypothesis. AG : OK, cool. So there ’ s a study I really like of scientists, filmmakers and artists, showing that as they advance from their 20s and 30s toward their 50s and 60s, they do less work. The scientists run fewer experiments, the **filmmakers** don ’ t make as many **movies** and the artists produce fewer paintings. But those who stay productive, their hit rate was every bit as high in their 50s and 60s as it was in their 20s and 30s. So, you know, I think about Frank Lloyd Wright designing Fallingwater at 68 years old, his crowning architectural achievement. Or I ’ m sure you ’ ve come across Raymond Davis Jr, the scientists who measured neutrinos from the sun. He didn ’ t build the key detector until he was 51 and kept running his experiments until he was 88, the year he won the Nobel Prize. I ’ d be willing to bet that we would either uncover, or even create more Frank Lloyd Wrights and more Raymond Davises if we had ways of keeping more people motivated to produce into and past mid career. Agree or disagree? AB : I think that if there is a natural cadence to the average person ’ s life, and you ’ re saying it doesn ’ t have to be this way, then the societal goal will be one in which everybody is above average. And you know, that becomes an exercise in frustration. AG : In other words, that ’ s a nice theory about encouraging everyone to keep producing at maximum capacity for decades, but it might not be realistic if you ’ re not a superstar in your field. The rest of us are going to do what people naturally do. And Arthur wants there to be space for that path, too. He wants people to be excellent in their own way, which made me wonder if he thinks career decline is more inevitable for people who achieve excellence. AB : What goes up must come down ultimately, and the higher up you go, the more obvious it is that you ’ re not there anymore. That ’ s what I call the Law of Psychoprofessional Gravitation. You notice a lot when you were great and you ’ re not as great as you once were. And if you can do something to get on your crystallized intelligence curve, you get a different kind of productivity, and you can be successful in a different way. That ’ s the knowledge I want to get across. AG : He ’ s arguing that regardless of what field you ’ re in, as your fluid intelligence declines, you should make a change. Stop trying to innovate. Take what you ' ve learned and accomplished and shift into a mode where you can transmit your knowledge and skills to other people. Become a teacher or a mentor. AB : And frankly, the world needs those people. And we don ' t have enough of them. When I ' m at a certain point in my career, and I see that I ' m not on the top of my game, I need to start thinking, is what got me here, whether it ' s the insight or the innovation or whatever it happens to be that got me here, is it something that I can sustain? If it ' s not, start looking for something else. Specifically, start looking for a way that you can move into instruction mode. AG : In Arthur ' s case, after giving up on music, he spent years running a conservative think tank. And then having learned the lessons of his fall at Carnegie Hall, he took his own advice. AB : And I had somebody I really respect, and he said,"You know, when you are a chief executive and you have to be on the razor ' s edge every single day, you have two choices on how to step out of your chief executive job."And I said,"Well, good, two choices. I like two choices. You know, it ' s like the Monty Hall game. You know, door number one and door number two. So what ' s behind door number one?"And he said,"Leave before you ' re ready."And I said,"Oh, I don ' t like that. What ' s behind door number two?"And he says,"Leave on someone else ' s terms."I said,"Ooh, I really hate that one."AG : What did Arthur decide to do then? Exactly what he ' s suggesting you do. He went into teaching. AB : The bottom line is, when the stakes are high, it ' s worth leaving things before you ' re ready. AG : Arthur is content with how his career choices worked out. He wants you to walk off the stage before you fall off of it. I understand where he ' s coming from, but I ' d rather see you find your balance, or get on a different stage altogether. You do have choices beyond aging out and opting out. More on those after the break. OK, this is going to be a different kind of ad. I played a personal role in selecting the sponsors for this podcast, because they all have interesting cultures of their own. Today, we ' re going inside the workplace at Accenture. Gonzalo Adriazola : I ' m from Arequipa, which is the second largest city in Peru. We have the coast, we have the Andes and we have a bit of jungle. AG : This is Gonzalo Adriazola. When he was 19, he and his mother left Peru and moved to rural Ohio. GA : I don ' t think I fully processed that I was leaving a whole life behind. I think I was just excited for the future. AG : As a young immigrant to a new country, Gonzalo didn ' t know exactly what the future held, but his mother became a tremendous source of inspiration. From early on, she taught him about the importance of education. GA : She would always tell me,"I might not leave you anything, but I ' m going to leave you a good education."AG : Gonzalo got a job at a local restaurant while he applied to colleges. GA : When I got the acceptance letter, my mom and I were just, like, jumping and dancing and just so happy. AG : He headed to the Ohio State University, and it didn ' t take him long to fit in. He was crowned homecoming king, started working full time at a bank and found a community within the Hispanic Business Student Association. They sponsored a trip to a conference in Washington. GA : It ' s the largest gathering of professional Latinos in the US. I saw this one speaker and he was so good. He was so passionate, energetic. Besides being phenomenal, he looked like me, and this was the first time I had seen someone in person that looked like me that was a total boss. AG : At the end of the talk, Gonzalo went up to the speaker and asked for advice. GA : I asked him like,"I want to be a CEO one day. How do I get to be like you?"And he laughed and looked at me and told me that professional services firms, all they have is people. So they ' re going to do their best to make the best of you. And that ' s how I found out about consulting. AG : Research shows that role models can elevate our aspirations. When we see someone similar to us do something inspiring, it builds our confidence that we can follow in their footsteps. When Gonzalo graduated from college, he followed his role model ' s advice and went to work as a consultant at Accenture. GA : I was really happy to join the firm. And I mean, it has only been great experiences. AG : Gonzalo had accomplished a big goal, and for four years, he helped companies grow and perform better. But he had a nagging feeling that something was missing from his life. GA : As soon as I got to Accenture, I wanted to give back to Latin America. That was kind of, like, my life objective. I saw that we had worked with this microlending institution called Caja Arequipa, and I ' m like,"What are the odds?"AG : He reached out to the leadership team running the project. GA : They told me about, like, everything we were doing in regards to financial inclusion in Latin America. I was really excited to know that I was from that same city. Very welcoming to my help. AG : Alongside his consulting work, Gonzalo has been a leader in the Hispanic Employee Resource Group and volunteered his extra time to help the team in Latin America. His dedication paid off. Earlier this year, Accenture development partnerships moved him to Colombia where he joined the project team full time. GA : It was a dream come true. I called my mom. We laughed, we smiled, we danced. AG : This isn ' t the only way Gonzalo gives back. Every year he ' s gone back to the conference where he was first inspired by someone he felt he could relate to. GA : It feels like I ' m doing my mission. I was in their shoes one day and now I ' m the person lifting up, just as others lift me up. And there ' s people lifting me up right now. There ' s going to be a brighter future for my community. AG : Accenture is working to become one of the most truly human companies in the digital age. Learn more at Accenture. com / careers. Lyn Slater : I ' m not a Pollyanna about aging. There ' s certainly changes in my body. You know, I have my wrinkles, I have my white hair, but I think that I could generate content as much as someone who ' s younger than me. In fact, most of the peers who are at my level are all younger than me. AG : Lyn Slater is 65. She recently quit her career as a professor to become a full-time social media influencer. Lots of people call themselves influencers, but Lyn ' s version is a real job. LS : I ' m also known as The Accidental Icon. AG : The Accidental Icon has more than 700, 000 followers on Instagram, and she gets paid to run ads on her account for brands like Kate Spade, Visa and Teva. Her feed is filled with stylized photos of herself posing around New York City in designer clothes and large sunglasses. She ' s been featured in magazines like"Cosmo"and"In Style."And she ' s walked the runway at New York Fashion Week. Actually, she ' s walked more than one runway. LS : I have no background in the fashion industry, but I was in the runway show last spring with Kate Spade. I ' ve worked with them quite a bit. AG : Some people will tell you that most of us do our best work when we ' re younger, that when we approach retirement age, we should slow down, take up a hobby, play with the grandkids. But excellence later in your career is possible, and there are a few ways you can get there. Lyn is living proof that you don ' t have to avoid career decline by shifting from doing into teaching. You can actually reinvigorate your career by experimenting, by doing something new. In Lyn ' s case, she did something completely new. We saw her in action one Saturday near her home in Manhattan. LS : We ' re on Fifth Avenue in New York City. It ' s called Museum Mile because everywhere is a museum. I ' m going to be shooting my outfit outside. AG : She ' s with her partner who doubles as her photographer. Calvin : Your **bangs** are good. LS : Thank you. AG : Lyn is petite with a short white bob and **bangs** that stop just above her huge sunglasses. She takes off a long velvet coat and scans the brick facades of the Museum of the City of New York for her next spot of inspiration. LS : And right now, that little nook over there is really calling to me because it does look like a place... AG : Before becoming a social media influencer, she had an entirely different decades-long career in social work. She was an expert for

New York City family courts on child abuse issues. She wrote a seminal book on social work and law, and she spent the last 20 years as a professor of social work at Fordham University. She was quite an authority in her field, but she told me she ’ d become frustrated with the glacial pace of academia. Like, how long it takes to write and publish journal articles. LS : The whole process could take almost two years and maybe a handful of people would read it. And so, as I began to get more followers on Instagram and I began to do more things, you know, for example, I did a short film and within a couple of weeks, that was in front of, you know, millions of people. AG : Lyn went about as far from her original field of work as she could go. So what leads to a spectacular pivot like that? A dose of novelty. Research shows that curiosity is a key factor that can help you find creative peaks as you age, rather than decline. This is one way to stay productive and maintain motivation later in your career. Find something new to explore. For you, novelty might mean switching to a new team or a new market, getting certified in a fresh skill set, or mastering a hobby completely outside your experience. For Lyn, that hobby was fashion. LS : I ’ ve always had, sort of, what I call a very performative relationship with my clothes. I resist like crazy social categories or constructions of who I should be. AG : In this case, she was resisting the idea that a long time professor of social work wouldn ’ t also be interested in high fashion. She ’ d been toying with the idea of starting a fashion blog. And one day, unexpectedly, the name came to her. LS : It was the first week of school, and I always used to dress up for that. And I was meeting a friend for lunch and I had on a Yohji Yamamoto suit, and a couple of photographers started taking my picture and then a bunch of other photographers saw them do it so they came over, and my friend was making a joke like,”You ’ re an accidental icon.”AG : If you ’ re going to experiment with something new, it helps to fully immerse yourself in that world. Lyn was creative in where she looked for her mentors. LS : I found out that I could get a press pass to what is called the market shows, and I would spend hours talking with these young designers. They taught me a great deal about how clothing is made, what the processes are. That ’ s how I really began to understand fashion. AG : Instead of being intimidated by new technology and seeing it as something only young people can master, she went straight to the experts she knew. LS : My students helped me to figure out Instagram and Snapchat, and I started to take classes in social media and fashion and just random things, like, how to start your own **vintage** store. AG : In a study in the fashion industry, the more time designers and directors spend abroad, the more creative they became over their careers, but it wasn ’ t traveling or even living abroad that mattered. It was the amount of time they spent working abroad. They couldn ’ t just observe it from the sidelines. They had to be in the arena. That ’ s how they really internalize the novel elements of a foreign culture. For Lyn, seeking out new perspectives included small steps, too. LS : I do things that I don ’ t normally do. And so that could be very simple things, like taking a different route to work. It ’ s really, put yourself into different experiences that are creative and see where that takes you. AG : You may not aspire to quit your job and become an Instagram star, or to switch fields at all. And you don ’ t have to. There are ways to reach new heights while staying within your career path. Think about whether you ’ re a sprinter or a marathoner, not in terms of running, but how you go about generating creative ideas. Economists call the sprinters conceptual innovators. They typically start with a big idea and they tend to peak early, because they have eureka moments when they ’ re relatively new to a domain. Like Einstein at 16 imagining himself chasing a beam of light, or Mary Shelley at 18 starting to write”Frankenstein.”But as they accumulate knowledge, they ’ re more likely to get stuck in their ways and stop gaining new perspectives. There ’ s a term for that : cognitive entrenchment. The longer they stay in a field, the more likely they are to take for granted assumptions that need to be questioned. Marathoners, on the other hand, are experimental innovators. They usually discover their ideas through trial and error. They often peak later because it takes time to test out different approaches, like Darwin, spending decades tracing the origins of species, or da Vinci experimenting for years with different ways of modeling light in his paintings. But those habits allow them to sustain their peaks longer and peak again later as they explore new areas. That kind of experimentation is familiar to Lynda Weinman. Lynda Weinman : I ’ m an autodidactic person. I always love learning new things. AG : She got her start in the 1980s as an animator at a film studio. She often compared herself to her higher achieving peers who were sprinting ahead. LW : I remember when I was in my 20s watching some of my friends who were uber successful and I, you know, at my 30th birthday was in tears that I hadn ’ t accomplished what I wanted to accomplish yet with my life. AG : But she didn ’ t rush it. Instead, for the next 10 years, she kept chugging along, testing out different ways to build and use her skills. Finally, the year of her next big birthday, she created a website. Lynda called her site Lynda. com. LW : Well, when we started Lynda. com I was 40 and when we sold it I was 60. AG : She sold it in 2015 to LinkedIn, for, oh, 1. 5 billion dollars. Lynda. com was one of the first companies to offer professional learning online. Maybe you ’ ve heard of their courses in animation, design and software. Maybe you ’ ve even taken classes there that advanced your career. Many founders who are trying to grow a unicorn company, think they need to generate one brilliant idea and scale it rapidly. They think they need to be sprinters, but Lynda approached her career more like a marathoner. Her online learning company didn ’ t look anything like a success early on. Instead, she was constantly trying out new ideas and innovating over time. In some ways, Lynda was primed for this kind of scrappy test-and-learn approach. LW : I actually had a pretty troubled childhood. I moved in with my dad and my stepmom who weren ’ t expecting to raise me and my siblings. And, you know, it was just this feeling of having to find my way because all the adults in my life were, you know, having their own problems. And I think what I probably am is a very resourceful person who realized at an early age that it was on me to figure out my life. AG : During her early days as an animator, Lynda encountered her first computer. She was unimpressed, at first. LW : It was a Apple II Plus clone that my boyfriend brought home, and so to please him, I turned it on and I started pecking at the keys and I caught the bug that so many people since have caught. And he eventually called himself a computer widow, because I liked the computer more than him. AG : LW : I mean, I was just amazed at what it could do. AG : What it could do was draw. So Lynda got her own Macintosh and brought it into work. People started asking her to do projects for them. LW : Somehow they thought I knew more than they did, which was, you know, basically I ’ d had five minutes longer with something than they had. AG : Lynda was onto something. By experimenting with new tools and following her curiosities to uncharted places, she could carve out a niche and continue iterating instead of stagnating. LW : Oh, do I want to do what everybody else is doing, or I ’ m being asked to do something that only I know how to do, and maybe that ’ s more fun. AG : Lynda ran two more experiments. First, she started teaching in-person college classes in web design. Then when she saw a lack of good manuals for students, she wrote one herself. LW : Then it became a huge success, way bigger than any of us could have imagined. AG : Well, you ’ re being very modest, but can I get you to describe a little bit just how successful the book was? LW : I ’ m not sure how many millions it sold, but it was translated into many, many languages and it was the de facto web-design book for all colleges. It was the"it"book. AG : From there, Lynda continued to test out new methods of teaching. She started selling instructional videos on VHS tapes. Then, as technology changed, she switched to CDs, then DVDs. Then she launched the website Lynda. com. In the late ' 90s, she and her business partner and husband tried launching memberships on Lynda. com. They weren ' t exactly an instant hit. LW : In the beginning it was just an abject failure. It was cannibalizing some of our other work. AG : But they didn ' t give up. Not every experiment succeeds, and they don ' t all succeed right away. Memberships weren ' t growing like crazy, but there was steady growth. Lynda paid attention to that. Eventually they hit an inflection point. While Lynda. com was quietly adding hours of material to their catalog, more people were using the internet for online learning. Their online audience doubled each year. And in time that grew into a big number. By 2012, they had over a million paying users and over a hundred million dollars in revenue. LW : It was literally a very organic growth process. AG : Lynda could ' ve simply continued to develop her skills as an expert animator and would have had a nice career. She could ' ve kept teaching, too, but her greatest success came when her experimentation led her into entrepreneurship, turning her **various** projects into a business. She didn ' t listen to the narrative that as you get older, you should pull over into the slow lane. She saw some benefits of building something later in her career. LW : By the time you ' re 40, you believe in yourself more. You ' re a more seasoned human being that has better judgment and better experience and better confidence. AG : This might be one of the reasons why older founders are actually more successful than younger ones. In a study of several million entrepreneurs, 40-year-olds were more than twice as likely as 25-year-olds to launch a successful start-up. And they were also nearly a third more likely to be in the top 0. 1 percent of fastest-growing tech start-ups. Yes, sprinters sometimes get lucky right out of the gate, but marathoners know how to persevere. LW : I think everybody would probably like to be instantly successful and be revered and acknowledged and acclaimed and all those things, but when it happens to you too young, it actually can be very damaging. At least the people that I ' ve seen, who, you know, they burned bright and they burned all the way out. They didn ' t understand that it wasn ' t going to last forever and that they had to keep reinventing and they had to earn it again. AG : If you ' re worried that your career will decline, don ' t let your output decline, and don ' t let your curiosity stagnate either. It ' s never too late to invent or to reinvent yourself. LW : I think the older you are, the more valuable you can be. AG : I love that because I don ' t ever want to retire. Next time on"WorkLife."Working remotely is full of challenges that have made our work lives harder. Lizet Ocampo : So we turn our cameras on and there I am as something else. At first, we didn ' t realize it was a potato. We ' re like,"What is going on?"I just essentially was a potato for the meeting. AG : But some of the solutions might be easier than you think."WorkLife"is hosted by me, Adam Grant. The show is produced by TED with Transmitter Media. Our team includes Colin Helms, Gretta Cohn, Dan O ' Donnell, Constanza Gallardo, Grace Rubenstein, Michelle Quint, Angela Cheng and Anna Phelin. This episode was produced by Jessica Glazer. Our show is mixed by Rick Kwan. Original music by Hahnsdale Hsu and Allison Layton-Brown. Ad stories produced by Pineapple Street Studios. Special thanks to our sponsors, Accenture, BetterUp, Hilton and SAP. For their research thanks to Dean Simonton on creative productivity, Lu Liu and colleagues on career hot streaks, Ben Jones and colleagues on the success of older entrepreneurs, David Galenson on conceptual and experimental innovators, Erik Dane on cognitive entrenchment and Will Maddux, Adam Galinsky and colleagues on working abroad. In some ways, Lynda was primed for this kind of crap... Excuse me. In some ways, Lynda was primed for this kind of crap... It ’ s not crappy, it ’ s scrappy. In some ways, Lynda was primed for this kind of crappy... Sorry.

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Chris Anderson : Well, hello, Helen. Very nice to see you. CA : You staying well? Helen Walters : How ’ s it going? CA : These are mad, mad, mad, mad, mad days. So many emotions. Not all bad, happily, but I ’ m just so aware that, among the people listening to this, some are in really tough times right now. I hope this is going to be a beautiful hour of therapy and help in its own way, because we have with us just an extraordinary author, an extraordinary mind, Elizabeth Gilbert, obviously known for her astonishing best-selling success with”Eat, Pray, Love,”although her favorite book from my point of view is called”Big Magic,”where the subtitle is,”Creative Living Beyond Fear.””Creative Living Beyond Fear.”Now when you think about it, that is a pretty good agenda for today ’ s conversation, I think. Liz describes the emotional landscape of our lives, I think like no one else I ’ ve read, and I ’ m not even her target audience. She ’ s really extraordinary in doing that. She gave an amazing TED Talk 11 years ago now,”In pursuit of your creative genius.”It really reframed how to think of creativity. It ’ s been seen, like, 19 million times or something, and it ’ s really changed how a lot of people – they ’ re just open to the creative genius coming from the outside. So it ’ s a delight to welcome to the TED Connects stage Elizabeth Gilbert. Elizabeth Gilbert : Hey, Chris. CA : Great to see you. How are you? Where are you? Who are you living with or staying with? What ’ s up? EG : I ’ m fine. I don ’ t want to brag, but I ’ m in New Jersey, where anybody would want to be. I ’ m by myself. I ’ ve got a little house out in the country, and I think I ’ m on day 17 of no human contact other than virtually, and I ’ m well. I ’ m not anybody you need to be worrying about right now. So I ’ m good. CA : Wow. Well, so in a way, you ’ re having a related experience to what so many people are having. I mean, these are days of isolation for many people, and that brings with it lots of difficult emotions, in a way. And we ’ re going to go through many of them, I hope, in the next hour. So I ’ m hoping to talk with you about – I wrote down a list here : about anxiety, loneliness, curiosity, creativity, procrastination, grief, connection and hope. How about that? That ’ s our agenda. Are you up for that? EG : I think that ’ s the whole buffet. Just a little light tasting menu of all the mass of human emotions. Let ’ s do it. Absolutely. CA : I think it ’ s probably good to dive straight in with the anxiety that I know a lot of people are feeling right now. So many reasons to be anxious, both for yourself, your loved ones, and just for this time and for the world and how we all get through this. Have you been feeling anxiety, Liz? And how do you think of it? What can you say to us? EG : I have been, and I think you would have to be either a sociopath or totally enlightened not to be feeling anxiety at a moment like this. So I would say that the first thing that I would want to encourage everybody to do is to give themselves a measure of mercy and compassion for the difficult emotions that you ’ re feeling right now. They ’ re extremely understandable. I think sometimes our emotions about our emotions become a bigger problem, so if you ’ re feeling frightened and anxious, and then you ’ re layering shame on top of that because you feel like you should be handling it better, or you should be doing your isolation better, or you should be creating more while you ’ re alone, or you should be serving the world in some better way, now you ’ ve just multiplied the suffering, right? So I think that the antidote for that, first of all, is just a really warm, loving dose of compassion and mercy towards yourself, because if you ’ re in anxiety, you ’ re a person who is suffering right now, and that deserves a show of mercy. The second thing that I would say about anxiety is this, that here ’ s what I think is the central paradox of the human emotional landscape that I ’ m finding particularly fascinating right this moment, and it ’ s really come to light for me. So there are these two aspects of humanity that don ’ t match – hence the word paradox – but they really define us. And the first is that there is no species on earth more anxious than humans. It ’ s a hallmark of our species, because we have the ability slash curse to imagine a future. And also, once you ’ ve lived on earth for a little while, you have the experience to recognize this terrifying piece of information, which is that literally anything can happen at literally any moment to literally any person. And because we have these vast, rich, colorful imaginations, we can see all sorts of terrifying **movies** in our heads about all of the possibilities and all of the scariest things that could occur. And actually, one of the scariest things that could occur is occurring. It ’ s something that people have imagined in fiction and imagined in science, and it ’ s actually happening right now, so that ’ s quite terrifying. The paradox is that, in that level, we ’ re very bad, emotionally, at fear and anxiety, because we stir ourselves up to a very heated degree because of our imaginations about how horrible it can get, and it get can get very horrible, but we can imagine it even worse. The paradox is that we ’ re also the most capable, resourceful and resilient species that has ever lived on earth. So history has shown that when change comes to humanity – either on the global level, like it ’ s happening now, or on the personal level – we ’ re really good at it. We ’ re really good at adaptation. And I think that if we can remember that, it can help to actually mitigate the fear. And you can remember it in a historical perspective, by looking at what humanity has gone through, and what we have not only survived but figured out how to thrive through. And you can also look at it at a personal level, where you can make an inventory of what you yourself have survived, and notice, as I often notice, my panic and my anxiety about the imagined future is deadly on my nervous system, but I actually have discovered that when there ’ s an actual emergency in the moment, I tend to be pretty good at it. And I think most of us are like that. You ’ ll see that repeated in history in so many examples. I think about those heartbreaking and devastating phone messages that people were leaving for their loved ones from the towers on September 11th, and you can hear the calm, the calm in peoples ’ voices. The biggest emergency ever was happening, and in that moment, intuition told them what to do. The important thing to do now is to make this phone call. And I think if you can trust that when the point of emergency actually arrives, you ’ ll be able to meet it, and then when the world changes, you ’ ll be able to adapt to it – it certainly helps me calm down. CA : I mean, I guess there ’ s a reason why fear is there. It didn ’ t just evolve by accident. It ’ s supposed to direct our behavior and help us avoid danger, and it ’ s just that sometimes, it gets out of control and actually gets in our way and damages us. I mean, any specific advice on how someone could turn their fear into something useful, at this moment? EG : Can I tell you a story that I ’ m using as a touchstone for myself right now and drawing wonder and inspiration from? So, some of you may have heard of a young woman named Amanda Eller. She was in the news recently, because she got lost in Hawaii in the wilderness for 17 days, and there was a massive, massive hunt for her, because she had left her car, she ’ d gone for a simple hike, had left her phone in her car, went up into the woods, took a wrong turn, and then had this disastrous 17 days, fell off a cliff, broke her leg, walked for 40 miles on a broken knee, lost her shoes in a flash flood. She had to sleep packed in mud in order to protect herself from the cold and the mosquitoes. She was eating moths. I mean, just a harrowing story of survival. I met her recently, and she was so lit and radiant with this kind of serenity and this kind of wonder and joy, and I said,”How are you like this? You went through one of the most traumatizing things that a person could go through.”She said,”First of all, I discovered that I can survive anything,”going back to this idea of how resourceful and adaptive humans actually are. But the piece of her story that I am using like a life raft right now is that she said, on her second day in the jungle, when she realized that she was truly and very much in trouble – she ’ d already spent one night in the woods and she was completely lost and she was totally alone and no one knew where she was, and she was full of terror – she said she closed her eyes and she prayed or asked or requested, she made a wish to herself, to consciousness, to the universe, and she said,”Please take my fear away, and when I open my eyes, have it be gone, and have it be gone and have it not come back.”And she opened her eyes, and it was gone, and it was replaced by intuition. And I think intuition is a little bit the opposite of fear, because fear is the terror that you feel about a frightening imagined future. Intuition can only happen when you ’ re in the moment. And so, from that point forward, she did not experience fear for the rest of the time she was in the woods. She just was guided by some deep intuitive sense, located somewhere between her sternum and her navel, and in every moment, she would ask it,”Right or left?””Up or down?””Eat this? Don ’ t eat this?”And just trust it. Complete, absolute surrender to the intuition of the moment. And she said it hasn ’ t returned, the fear hasn ’ t returned, and she still guides her life that way. So it ’ s a return to some sense that there ’ s a navigational system within you that will, if you stay present in this actual moment, tell you what to do one moment to the next. Now, if you want to suffer, pop out of the moment and imagine a future, and then you can suffer indefinitely. So it is almost like a spiritual or meditation practice, and anybody out there who ’ s done any spiritual or meditation practices, this is what you were practicing for. You were practicing for this moment, and those of you who haven ’ t tried that, this might be a really [inaudible] to be centered in the instant. CA : Wow, that ’ s a **remarkable** story, and I guess what I ’ m hearing is two things. It ’ s one, just the reaching out to the universe there, but specifically, there was a decision to let go of the future and just to focus on the moment. EG : That ’ s it, yeah. Nothing will bring you more pain than the future, and what I ’ m seeing happening right now – I said this to you the other day, Chris – is there ’ s a relatively small percentage of the population who will suffer physically from this disease, and there ’ s a larger percentage who are going to suffer economically from it. But then there ’ s this massive, uncountable number of people who will and are suffering from it emotionally, and right now, those people are my concern, because they ’ re really in pain, and there ’ s millions and millions of them. CA : So you ’ re living there by yourself, Liz, and many others are in that same circumstance right now. I suspect some are feeling, like, crushing loneliness. Talk about that. How do you handle loneliness in a situation like this, when it ’ s so alien to everything that we as a social species are usually about? We crave other people. We crave touch. We crave hugs. We want to be there with people. How can we avoid this being a period of crushing loneliness? EG : I don ’ t think you can avoid it, but I think you can walk toward it. And I think that, for me, I ’ ve deliberately, many times in my life, gone off into isolation in order to face those things. I ’ ve gone on long meditation retreats. This year, I was in India, and I spent 17 days alone with no contact with anybody, which was a weird practice run for what ’ s happening right now. And as I see people really losing it and feeling like they ’ re crawling out of their skin, either from anxiety, fear, boredom, anger, blame, loneliness, depression, all of these things that come up when you are forced to just be in your own presence. I know all of those feelings because, as a meditator, I ’ ve experienced all of those, in stillness. The hardest person in the entire world to be with is yourself, and so the only way that I learned, as a meditator, to be able to survive and endure my own company was with universal human compassion toward me, and to recognize that this is a person who is suffering right now from loneliness, and this person needs kindness from self towards self. And it ’ s a very high teaching, but I think that it ’ s a very interesting moment to practice that. And so what I would suggest to people – and again, this takes a certain amount of resolve and it takes a certain amount of curiosity about learning more about the human experience – what I ’ m seeing people do is people are spinning away from that isolation because they ’ re so terrified of it. What happened with the world right now is that basically all of our pacifiers were yanked out of our mouths. Everything that we ever can do and reach for that can get us out of having to be in the existential crisis of being alone with ourselves was taken away. And I see people rushing to fill it, I mean, constant Zoom meetings and constant parties online and constant interaction, and all of that is lovely, but from a spiritual and psychological standpoint, from a creative standpoint, I would say if you have any curiosity about this, don ’ t be in such a hurry to rush away from an experience that could actually transform your life. I think sometimes the experiences that can transform us the most intimately are the ones that we want to run away from, and I think of a story that the Dalai Lama told about one of his teachers. When the Chinese invaded Tibet, and all the monks were running into India for safety, one of his teachers, who was one of the great masters, the last glimpse that the Dalai Lama had of him was that he was walking into China – very patiently and very slowly, toward it. Everybody else was running away from it – he was walking toward it, and I think there ’ s a level at which first responders do that, and in the real world, in an intimate way, they go into the emergency, they go toward the emergency. All the people who are trying to solve this now in worldly ways are walking toward the emergency. But there ’ s a way that you can do it emotionally as well and that is to walk with curiosity and with an open mind toward your most difficult and painful emotions without resistance, and say,”What is it like for a person to feel like they don ’ t have something to do for an hour?”And you can open up your compassion in that. There ’ s so many lessons in compassion that can be found here. How about a general universal mercy that we can all feel toward people who are in solitary confinement. Let ’ s have that be part of the conversation. Now you ’ ve experienced it for two days in your own house. Maybe it ’ s time to change the prison system? You see how hard this is. Or you can have compassion toward people who have lost a loved one and they ’ re alone. By feeling your own feelings, you can open up your feelings more universally toward the world. So I think there ’ s a great opportunity here for growth on the personal level, but you have to have almost a whimsical curiosity to be the one walking into China rather than the one away from it. And that ’ s how I ’ m doing it right now. CA : So let ’ s follow up on that word”curiosity”that you ’ ve used a few times there. I mean, a lot of wisdom that I ’ ve heard sort of thrown around online right now is,”This is a great time to follow your passion and dive deep into whatever it is you ’ ve most been wanting to do.”I mean, in”Big Magic,”you made an argument that following your passion isn ’ t necessarily the wisest strategy. You argued, no, don ’ t do that, follow curiosity. Does that apply now? Make that case. EG : Yeah, you know, I ’ ve been on a personal crusade to rid the world of the world”passion”as an instruction for people on how as they should be living, because I know that in my case, it brings me nothing but anxiety.”Purpose”is another one that has become a cudgel that we use to bludgeon ourselves into thinking that we ’ re not doing enough or that we ’ re doing life right or that you ’ re supposed to be more useful, more productive, you ’ re supposed to be changing the world, or uncovering some particular talent that only you have and with it, you ’ re supposed to transform everybody and monetize it, no pressure. I start to get hives even repeating that, but that ’ s what we ’ ve been taught, that purpose and passion are everything. I would like to replace it with a far gentler word, and I think”curiosity”is very gentle, because the stakes are so much lower. The stakes of passion say you have to shave your head and move to India and get rid of all your possessions and start up, like – It ’ s so intense. But curiosity is a very simple, universal experience that causes you to want to look at something just a tiny bit closer, and you don ’ t have to change your life around it. You just look, and it might be taking a weekend to try something new for a little while. It ’ s almost so easily missed, and I think so many times, we ’ re looking way up at the sky for the sign from God of what our passion and what our purpose is supposed to be, and meanwhile, there ’ s this lovely little trail of breadcrumbs of curiosity that if you can slow down – and again, this is about not rushing out of the experience of being silent, still and alone – if you can slow down, you might be able to see them. But if I could say one thing I ’ m noticing is an obstacle right now – because I think a lot of people thought,”Isolation, great, this is the perfect time for me to learn Italian and take that calligraphy class and start writing that novel,”and they find that they ’ re actually in a paralysis of anxiety and they ’ re not creating anything or doing anything. First of all, again, like, a blanket of mercy on you. These are hard times, and it might take you a minute in your nervous system and your mind to adjust to the new reality. But the second thing I would say is that when people are saying they ’ re having trouble with their creativity because they ’ re in isolation, I might daringly suggest that perhaps you ’ re not in enough isolation. And by that I mean, are you monitoring how much external stimulus you ’ re bringing of this disaster into your home? So if you ’ re sitting watching the news all day, what you ’ re doing is you ’ re bringing the disaster into your work space. You ’ re bringing it into your soul. You ’ re bringing it into your mind. And you ’ re going to create the opposite of a creative environment, an environment of fear, panic and urgency. So I think if you ’ re going to be a good steward of your creativity right now, you have to isolate a little bit from the news. And that doesn ’ t mean disconnecting, it means I get up every morning and after I ’ ve meditated, I read the New York Times and I give myself 40 minutes with it, and then that ’ s it for the day, because I know that if I bring in any more, I ’ m going to go into a traumatized state and then I won ’ t be able to follow my intuition, I won ’ t be able to help people, because I myself will be suffering, and I won ’ t be able to be present for this very interesting moment in my life and in history, and I want to remain present for it as much as I can. So there ’ s a discipline of being a good steward of your senses and deciding what you ’ re going to put your senses in front of. CA : Helen. HW : Liz, there ’ s an outpouring on Facebook of gratitude for you. People are so grateful, and grateful for the calm that you are instilling in us all, so thank you, from them and from me. We ’ ve also had a number of questions about grief. We ’ re kind of dealing with grief at a different scale at the moment. One person has already lost five people to coronavirus. And so any thoughts of how to manage grief at this scale or how to process this in a way that honors both them and yourself? EG : First of all, my condolences. And I think any words that I would say about somebody who just lost five family members could only be inadequate. Grief is bigger than us. It ’ s bigger than your efforts to manage it, and if you want to hold yourself and your family members compassionately through grief, you have to allow that it cannot be managed. And I think that grief management is something that we ’ ve kind of created in our very Western idea that if we can figure out something, we can avoid suffering from it, so if we can figure out how to translate grief and if we can figure out how to walk through grief, then we won ’ t have to experience the magnitude of it. Many of you know that I lost the love of my life two years ago from pancreatic and liver cancer, and I was with her when she died, and I ’ ve been walking through my own path of grief, so I know what it feels like to lose the person in the world who is the most important to you, which is of course the biggest fear that we all have. I know that you can survive it, but I know that you survive it by allowing yourself to feel it. And again, to go back to the metaphor of the monk walking directly into China, into conflict rather than away from it, do you have the courage to let it break over you like waves? I wish I could remember her name. There ’ s this extraordinary woman who wrote a book called”Here If I Need You,”and she ’ s a chaplain for the police department in Maine, and she ’ s in charge of knocking on people ’ s doors and giving them the worst news they ’ re ever going to hear in their life, when she goes with the police when something happens. And she told a story once that I found very moving and very helpful for me in my grief. She said what she ’ d witnessed through years and years of sitting with people through what is literally the worst moment of their life, the nightmare of that loss, is that when she knocks on that door and tells that person, your daughter, your family member, your husband, your mother has been killed, there ’ s this universal collapse where the person will just be – it is the tidal wave that comes and just takes you down and you lose all civilization, you lose all your attainments, all your wisdom. Nothing can stand up to that. You literally go to the floor. And you sob and you grieve, and she holds them through that. And then she said that what she ’ s learned is the most astonishing thing, that that never lasts more than a half an hour, that first wave. It can ’ t. You actually physiologically can ’ t sustain that, and if you let it break over you and you just allow it, then within a half an hour, usually sooner – and she said this has happened every single time she ’ s been with somebody with a loved one ’ s death – the very next thing that happens is that that person calms down, they catch a breath, and the next question they ask is a very reasonable question.”Where is the body? What do we do next? When can we have the funeral? Who else was in the car?”And with that question, she says, they start to rebuild their new life already with this new piece of information that even an hour ago would have seemed unsurvivable. And she uses that as an example of, once again, the tremendous psychological resilience of a human being. And it doesn ’ t mean that they will never grieve again. It doesn ’ t mean that their grieving journey is over. It just means that, somewhere in their mind, that it ’ s landed, and now, already, they ’ re making a plan about,”OK, who do we need to notify, what ’ s the next thing we need to do.”And again, if you can remember this as you go through your panic, if you can remember that in the moment of emergency, there will be an intuitive, deep sense that will tell you there ’ s going to be some next steps and it ’ s time for us to take those next steps, and if you can also remember that resilience is our shared genetic and psychological inheritance – we are, each and every one of us, no matter how anxious you feel you are, no matter how ridden by fear you feel you are, every single one of us is the genetic survivor of hundreds of thousands of years of survivors. Each one of us came from a line of people who made the next correct intuitive move, survived incredibly difficult things, and were able to pass their genes on. So almost to the biological level, you can relax into a trust that when the moment comes where you will be faced with the biggest challenge, you will be able to draw on a deep reservoir of shared human consciousness that will say,”Now it ’ s time to make the next move, and we can do this.”HW : So beautiful. So many more questions. I will be back. EG : Thanks, Helen. CA : Thank you, Liz. I think the author ’ s name was Kate

Braestrup. EG : That ’ s it. CA : I guess that ’ s a book, if you need a book right now,”Here If You Need Me”by Kate Braestrup. EG : Very good, yeah. CA : Liz, you and I got to have a conversation a few months after Rayya passed away. It was actually the first-ever episode of the TED Interview podcast we did. And I found that it was probably my favorite episode ever of the TED Interview. And it was so moving how you spoke about your grief then. And I feel like that ’ s a potential resource to people. I know we were both sitting there shedding tears, and I found that an extraordinary experience personally, to be sure. But somehow... in this moment, if you follow this journey of curiosity, if you walk towards some of the harder moments, do you think that this actually can be a creative time for people if they ’ re willing to do that? EG : Absolutely, and I don ’ t think creativity in this case has to necessarily mean that you write the Great American Novel or start that business you always intended to start. It doesn ’ t need to be so literal. We ’ re going to be creating new worlds and new lives on the other side of this, and we ’ re going to be doing that individually and we ’ re going to be doing that collectively. I think of the shoots of small trees that can only come up after massive forest fires, where seed pods have to explode under great heat. We ’ re in a kind of crucible moment right now, and I wouldn ’ t begin to have the hubris to predict what sort of creativity will come, but look, if history is any measure, what we ’ ll probably see is people at their best and people at their worst. But I think we ’ ll see more of people at their best, because that ’ s typically how it works. CA : I mean, your model of how creativity happens is that it doesn ’ t all come from within. It ’ s not like you have to sit there, saying,”OK, this is my moment to be creative. Come on. Be creative. Be creative.”It involves, fundamentally, an openness to something coming to you, to be open, to be curious, listening, but then just to be open to that moment. Perhaps that could apply even more now than ever, just because we have this huge distraction of the news, some other distractions are taken away. Is there a chance that if people listened, they actually can receive more at this moment? EG : I think so, and I think, again, if you stop thinking about your self-isolation and your social distancing as quarantine and you start thinking of it as a retreat, you ’ ll find that you can ’ t really tell the difference between quarantine and retreat. You know, a lot of you out there have dreamed, I ’ ve heard you, because I talked about going to India to an ashram for four months and God, I can ’ t tell you how many people I ’ ve heard say,”I wish I could do that.”I ’ m like,”Well, you got it.”And by the way, this is what it felt like. This is what it felt like to learn how to be present with yourself. I think my screen needs to move a bit. To learn how to be present with yourself means sitting in a lot of terror, sitting in a lot of anxiety, sitting in a lot of fear, sitting in a lot of shame, and being able to allow that without having to resist it, without having to reach outside yourself for something to numb yourself with. I also want to tell a story that a friend of mine, Martha Beck, told me about when she goes to South Africa and teaches animal-tracking courses. And she works with all these great African animal trackers, and these old men who have had these skills passed down for generations. And she was using it as an example of the difference between focus and openness. So I think sometimes the mistake people make when they want to be creative is they think they have to get really focused, and focus is an anxiety-producing energy as well. You ’ ve got to drill down and you feel your whole body tense. But what she described witnessing in these animal trackers is when they go out to hunt the lions, these old, old men, the very first thing they do is they sit down against a tree and they appear to go to sleep. They drop into a state that she calls and that the mystics call”wordless oneness.”And wordless oneness, you can also call meditation. You can also call it the zone. But it ’ s a stillness where you actually can drop your nervous system into such a quiet place that you have 360-degree awareness of your senses and of presence. And they ’ ll sit like that, apparently doing nothing, for an extremely long time, just looking through half-lidded eyes at the world. And then, maybe an hour, two hours in, all of the sudden, they ’ ll say,”The lion ’ s over there.”And so for me, I ’ ve learned to hold my creative wishes lightly in that same way. I ’ m between books right now and I don ’ t have an idea for a book and in the past, that would have made me really anxious, but now I know – take a lot of naps, go for a lot of walks, do a lot of drawings. I ’ m doing weird little art projects as I ’ m sitting here, to distract my mind. CA : Wait, wait. Bring that back. Hold that up. EG : Owls. CA : Aww. EG : Aren ’ t they dear? CA : They ’ re beautiful. Goodness me. EG : Well, I ’ m just playing with color and texture because it calms me, and I think if you can ’ t think of what to do right now, I would suggest doing what you used to do when you were 10 years old that made you feel happy and relaxed, and that ’ s often creativity and play. And for many of us who were anxious children – and I was an anxious child – we learned at an early age that we could sedate ourselves with our curiosity and with our play, and then, usually around adolescence, the world taught us that there were faster and more immediate ways to bump out of that anxiety through sex or substances or distraction or workaholism or whatever we did and not have to sit with ourselves. And I think right now is a really good opportunity – You actually were on the right track when you were 10, whatever it was. So, you know, get some LEGOs. Get some LEGOs, get some coloring books, just get your hands in the mud, do whatever it is that will actually ground you into this, again, to take you out of the futurizing and the future-tripping that ’ s going to cause you nothing but anxiety and not going to make you be of service. There ’ s such a thing, too, that I just want to touch on if I can, for a minute, about empathetic overload and empathetic meltdown. We ’ re taught that empathy is a good thing. I would suggest that in a case this traumatic, what you want to talk about replacing empathy with is compassion, and the difference is extremely important. So compassion means”I ’ m actually not suffering right now, you are, I see your suffering, and I want to help you.”That ’ s what compassion is. Empathy is”You ’ re suffering, and now I ’ m suffering because you ’ re suffering.”So now we have two people suffering and nobody who can serve, and nobody who can be of help, and if you knew how your empathetic suffering actually makes you into another patient who needs assistance, you would be more willing to dip into compassion. And what underlies compassion is the virtual courage, the courage to be able to sit with and witness somebody else ’ s pain without inhabiting it yourself so much that you become another person who is suffering and now, there are no helpers. And it takes an enormous amount of courage to be able to watch that without diving into it and joining it and becoming sick yourself. CA : I mean, if empathy is just a feeling, does compassion, your use of compassion imply that it ’ s turning that feeling into something potentially practical to actually do something, if you can, for that person? EG : It ’ s recognizing that if I feel your pain, I can ’ t help you in your pain, because now my pain has taken over me, and sometimes, I think all you need to do is know that and it makes you turn the ship. Right? One of my favorite teachers, Byron Katie, says,”My favorite thing about my suffering is that it isn ’ t yours.””My favorite thing about my suffering is that it isn ’ t yours. My favorite thing about your suffering is that it isn ’ t mine.”So it will be, eventually, we all take a turn suffering. You cannot move through this earth without it. When it ’ s your turn, you ’ ll know. When it ’ s not your turn, stay out of that field of somebody else ’ s pain, because you can ’ t help them when you ’ re in pain yourself. And then see if you can find the inner resolve and courage. And I think some of that is just based on accepting the Buddhist First Noble Truth, which is that suffering is an unavoidable aspect of life on earth. We ' re all going to be in it at some point. We ' ve all been in it at some point. And now, how can I help? I ' m not saying this is easy. I ' m just saying, also, if you ' re suffering from empathetic overload and empathetic meltdown, which means your adrenals are up, your stress is up, your endorphins are down, you ' re going into a parasympathetic collapse – this would be another time to discipline yourself to stay away from the news, because you actually will have a breakdown, and you won ' t be able to help the people around you who are the people who need help. CA : But have you seen any signs that if someone takes that empathy and compassion, let ' s say, and decides to act in some way, big or small, on behalf of someone, that actually shifts how they feel, that there ' s a healthiness to that? Or is that the language of just inducing more guilt in people? EG : No, I think there ' s a beautiful healthiness that can come from being of service, and that ' s also how I ' ve been medicating my anxiety through this, by showing up in ways that I can with whatever resources I ' ve got. Here ' s what you have to keep in mind, though, and this is what I keep reminding people. Right now, in my own personal sphere, there is more need than I have resources to fix. So I have to begin with that reality, and I have to have the courage to sit in that reality soberly and acknowledge that that ' s the case. The second thing I think emotional sobriety would require of me right now is to recognize that this is going to be a **marathon**, not a sprint. And so the first week of the crisis, I had this deluge of all my really energized, let ' s-save-the-world friends, all my creative friends, everybody was e-mailing, texting, Zooming, and they all had a response."Let ' s do this! Let ' s do this! Let ' s fix it this way!"And I found myself joining with some of them and not joining with others, just, again, based on my intuition, but I also found myself cautioning them,"Guys, this is a **marathon**."We ' re in mile one of what ' s going to be a very long **marathon**. So pace yourselves, and pace your resources. Don ' t overgive to the point where you collapse, because we ' re still going to need helpers two months from now, and we ' re still going to need helpers six months from now. And so, find a steady pace and be willing to be in it for the entire long haul. CA : Yeah. Helen. HW : Such great advice, Liz, and so many questions pouring in. One of them is from a therapist who confesses that she, and many of her clients, are having trouble with the letting go of control in this moment, and wonders if you have any advice on how to let go of control in order to be willing to feel everything that we ' re enduring. EG : Just this... This sense that you had that you had control was a myth to begin with. And that may not be comforting, except that I find it very comforting. You know, control is an illusion, and there are times where we ' re able to fool ourselves because we ' re so good at technology, we ' re so good at creating safe worlds where we ' re able to trick ourselves into believing that we ' re in control of any of this. But we ' re not, and the paradox, for me, of surrender is how relaxing it is. Nobody ever wants to surrender, because nobody wants to lose control, but if you recognize that you never had control, all you ever had was anxiety, and then you let go of the myth of control, you ' ll find that, I find that if I even say that sentence,"I ' m losing control,"and then I remind myself,"You never had control, all you had was anxiety, and that ' s what you ' re having right now."So you ' re not letting go of anything. Surrender means letting go of something you never even had. So there ' s an awakening that ' s happening right now, where what ' s happening is not that you ' re losing control. What ' s happening is that, for the first time, you ' re noticing that you never had it. And the world is doing its job. The job of the world is to change, constantly, and sometimes radically, and sometimes immediately, and it ' s doing its job, and that is also the norm of things. And again, we are adaptive and we ' re resilient and we can handle it. But I don ' t kid myself for a minute to think that I ' m in control of anything that ' s ever happening. My realm of control is extremely small. It ' s usually about, like, might be able to go get a glass of water right now. Like, there ' s not a lot that I ' m in control of. And I ' m actually I ' ve ever been. HW : One more question from online, if I may, and then I will jump off again. You know, Chris, you and I, we ' re all in a pretty privileged position. TED has been able to go remote. We ' re able to work remotely. But many, many, many millions of people in the US and beyond are not able to do that, and people are really suffering. How can we help? What are your thoughts about people who are not able to socially distance, who are losing their jobs, the global catastrophe that is unveiling? How can we think about that in a humane and compassionate way? EG : It ' s crushing, and again, as with the same case of the person who said they had lost five family members, I can ' t, sitting in this position of comfort and safety, say anything that I think is going to be accurate and appropriate to that, other than to say that I just think of this Indian proverb that I keep going back to, which is,"I store my grain in the belly of my neighbor."Western, capitalistic society has taught and trained us to hoard long before this, long before this happened and people were hoarding toilet paper and canned goods. Advertising and the whole capitalist model has taught us scarcity, it ' s taught us that you have to be surrounded by abundance in order to safe. The disconnect between those who have and those who have not has never been bigger, and never in my lifetime, and probably in any of our lifetimes, has there been an invitation, again, to release the stranglehold on your hoarding. This is not the time for hoarding. This is the time to store your grain in the belly of your neighbor, in a way that is emotionally sober and accurate to what you can give, and to look at that in a really honest way, to not put your own family in danger, to not put yourself in crisis, but to be able to say,"What can I offer in the immediacy?"And then, in the longer term, a conversation about redistribution of resources, and why do so few have so much and why do so many have so little? But that ' s not a conversation I can fix today. That ' s, again, outside of my realm of control. But what I can do is unleash the white-knuckled grip that I have on what ' s mine and make sure that I ' m going into the world with an open hand – again, not a panicked open hand, where I ' m going to destroy myself to save somebody else, because then there will be no helper left, but in a reasonable way. I cannot save everybody. I can save a few. And that ' s the tragic, but, I think, sobering reality that I can offer right now, and again, underlying all of that, undergirding all of that is a recognition that anything that I have to say about people who are in extraordinary suffering right now is not enough. HW : I ' ll be back. Thank you. EG : Thanks. CA : Liz, talk to me a minute about anger. Like, I think a lot of, just from the conversation we just had, or just listening to you there, there are so many reasons to feel angry right now about what ' s going on. And part of me feels we should be. That ' s what anger is for. It ' s to highlight things that are unjust and unfair and that we must pay attention to, and yet part of me is honestly scared of it. I think there could be an eruption of anger that ' s dangerous, both personally and for society. Have you felt anger? What are you doing with it? EG : I feel anger at every White House press conference, and I think all thinking people do. I feel angry that this wasn ' t taken more seriously early on. I feel angry at myself that I didn ' t take it more seriously early on. As much as I feel contempt and disgust for government officials who I feel were slow to recognize how serious this is, I also have to be really candid that three weeks ago, I was one of the people walking around saying,"Why is everybody overreacting to this so much?"So I think we also have to own our own piece of that, and I think there are rolling waves of awakening that are happening in people, and so a lot of the anger I feel right now is for people who aren ' t taking this seriously enough, who aren ' t quarantining themselves, who are putting other people in danger. But a month ago, that was me. I was in the Hong Kong airport, sallying through the Hong Kong airport while everybody was scurrying around in **masks** and gloves, and I was like,"What ' s the big deal?"It takes people as long as it takes them to come to awakening, and some people, we have to also acknowledge, never will. Anger has its place, and I think that righteous anger, which is the kind of anger that says a violation has occurred here, a humanitarian violation is occurring here, can be very stirring for transformation. Again, it ' s how comfortable can you be, sitting with these discomforting emotions, and what are you going to do with your anger? CA : Umh. EG : Are you going to lash out at the people you ' re quarantined with? Are you going to go on Twitter rants? Is that useful? Is that productive? And so I think – again, I keep using the words"emotional sobriety,"but the emotional sobriety that would be required is to feel that anger, acknowledge it, to show yourself mercy for how uncomfortable it is, and then to steadily, recognizing, again, that this is a **marathon** not a sprint, do what you reasonably can do to change the situation. CA : I mean, the part of me that ' s constantly looking for the better narrative hopes that the anger we feel now could almost displace some of – I mean, the world ' s been an angry place for the last couple years. There ' s been so much anger inflamed online. We ' ve made each other angry, often, probably, unnecessarily – outrage sparking outrage, disgust, etc. I mean, is there any hope that this is a massive societal shaking up? It ' s like, don ' t be so silly. Look at what actually matters here. And we can at least focus more attention onto the things that, yes, some things that we really should be angry about, but other things that maybe... you know, could lead people to say human connection really matters in this moment. People from all sides, we need each other. We just have to use this as a moment when we come together. How do you think about that? Like, how do we turn some of these negative emotions into a force for good that at least gives us some permission to hope that something special comes out of all this. EG : Well, I think you have to give yourself permission to hope, and I don ' t think it ' s unreasonable to give yourself permission to hope, because, again, our resilience, our resourcefulness, and the way that history has shown how catastrophe can lead to transformation, gives us, actually, I think, reasonable cause to hope. One thing that I ' m noticing that I ' m, like, a little bit amused by is that when people start predicting what the post-pandemic world is going to be, I notice that their predictions seem to be, suspiciously, in exact alignment with their personal worldview. So my friends who are utopians are already living in this utopian future where this is going to be the big change. My friends who are dystopians are already predicting that this is the official beginning of the police state and the disastrous new world order. I think there ' s a lot of hubris in trying to imagine what that new world could be. A quote that I love that a friend of mine always says is"When people aren ' t busy being the worst, they ' re the best."And I think that gives me hope. And it ' s true the other way, too. When people aren ' t busy being the best, they ' re the worst. I ' m terrible at social engineering, Chris, and you know this, and you have great, better minds than mine who can come on and talk about this on the global scale. The only world that I have a really intimate, familiar engagement with is this one, and on the individual level, what I understand is that the only world that any of us are ever going to live in is this one. And so minding this, and learning how to calm this, how to open this, how to get on the other side of the emotions that are causing harm to you and others, that ' s my work, you know? Personally, whatever role I have in the public sphere. CA : You ' re an extraordinary storyteller and you already told us one amazing story earlier on. Have you come across any other recent stories that have given you reason for hope, perhaps? EG : Well, I ' ll give you one, and this one, I delight in. Years ago, 20 years ago in New York City – 30 years ago, I was in my 20s – I was friends with a woman named Winifred, who was in her 90s. She was this really cool West Village bohemian artist who had lived in Greenwich Village for her entire life, had had a very storied and checkered and wild life, surrounding herself with intellectuals and poets and artists and adventure, and she ’ d had a lot of loss and a lot of gain, and she was this extremely passionate person who had friends of all ages, which was something I admired about her. I was friends with her. I was 25, she was 95. But I would call her my very good friend, and she had a lot of friends. She was so open to everything. And at her 95th birthday party, I asked her,”What have you learned, more than anything else?”Because she was such a creature of learning. I wrote about her in”Big Magic.”I said to her one time,”What ’ s your favorite book that you ’ ve ever read?”And she said,”I can ’ t say my favorite book because there ’ s been so many, but I can tell you my favorite subject, the history of ancient Mesopotamia, which I started learning when I was 80 and it changed my life.”And it did. She ’ d gone on these expeditions to Jordan and Iraq. She was just so full of living, you know? And I said to her,”What have you learned in all of your experiences? What is the most central thing that you ’ ve learned?”And she said,”Human beings can adapt to anything. Human beings can adapt to absolutely anything.”And then she said this great line :”If Martians landed on Earth tomorrow, it would be off the front pages of the newspaper by next Tuesday. We would already be used to it.”Right? And there ’ s a level at which I ’ m seeing this adaptation happening. And that is both a good and a bad thing. Right? We can get used to totalitarianism, but we can also get used to – I ’ ve gotten used to a world without the love of my life in it. We can adapt. And I keep using that line as a touchstone for myself, because I don ’ t know, nor do I presume to know, what the world is going to be after this. I know that it will be different from the one before. I also just have to point out that all y ’ all had a lot of complaints about the world we had before, and I do a lot of talking, I do a lot of going around the world, and [I don ’ t remember] any one of you raising your hand in any of the seminars I ’ ve taught over the last years, and saying,”We are living in a golden age and I ’ m so grateful and appreciative for all that I have,”now you want that world back, right? So let ’ s actually remember that as we go forward, that this moment, for some of us, that we ’ re in right now, might be one that we look back later and say,”Wow, actually, that was pretty good, and I didn ’ t have any gratitude for it.”So personally, I ’ m just hoping that at an intimate level – and again, this is not a socioeconomic, global political level, but it ’ s an invitation to actually be grateful for the safety that you have and the people that you have, and maybe carry that forward a little bit. Maybe. We ’ re really good at forgetting. Once a crisis is over, we ’ re really good at forgetting our gratitude. It ’ s one of our great gifts. But you might want to make a note to actually try to be grateful for what you have. CA : Thank you, Liz. I think we have a last question from our online friends. HW : Yeah, what crisis, right? So Liz, just a request for a concrete strategy to try and reduce the fear or the shame that is coming at this moment. EG : I ’ ll give you mine, and it may feel weird and out of reach and woo-woo, but I ’ m beyond that at this point, and it has been a game changer and a life changer for me. I have a 20-year-long practice of writing myself, every day, a letter from Love. Now this may not feel concrete. It may feel very airy. But what it does is that it helps me through my anxiety, and I need it every single day, because I ’ m anxious every single day. I wake up frightened every day. I wake up shamed every day. I wake up angry every day. All of the difficult emotions that run through the software of a human consciousness are running through my software all the time, and they cause me pain and they cause me fear, and they cause me distress, and they make me sick. So, 20 years ago, when I was going through a very bad divorce and a depression, I began this tactic, and the tactic is that I will sit down with a notebook and I will write to myself, from myself, a letter from Love. And what I mean by"Love"is not romantic love. It ' s the infinite, bottomlessly merciful source of all human compassion. And every single one of these letters begins the same way. It starts with me saying,"I need you."It ' s a dialogue. It starts with me saying,"I need you,"and Love saying,"I ' m right here."And then I say what I ' m going through."I ' m really angry right now. I ' m terrified. I ' m spinning. I can ' t sleep. I ' m anxious."And then I just allow to come through my hand whatever, if you could imagine the most loving, compassionate, merciful voice in the world, if they were in the room with you, what would you want them to say? And you say that to yourself. And so for me, that usually is a combination of these sorts of phrases :"I ' ve got you. I ' m right here. I see how distressed you are. It ' s all right. I don ' t need you to feel better."I think a lot of our anxiety is that we want to get out of that feeling as fast as we can, and what Love always says to me is,"It makes no difference to me whether you ' re anxious or afraid or angry or hurt. I ' m with you, and I ' ll be with you through this entire thing for however long it takes. I ' m not going anywhere. I ' ve got nowhere better to be right now than sitting with you, loving you. I ' ll be with you at the moment of your death. I was here with you at the moment of your birth. There ' s nothing you can do to lose me. You can ' t fail. You can ' t do this wrong. You are infinitely, bottomlessly loved."And it ' s so interesting to me that the opposite of fear in my life, in my emotional landscape, on the color palette, the opposite of fear isn ' t courage, the opposite of fear is love. And that presence, a sense of,"I ' ve got you,"right? Which is the thing that we ' d all want somebody to say."I ' ve got you, and it ' s going to be all right."I would love to know, neurologically, what actually happens in my mind when I do this, but what happens to me physiologically is that my mind, just hearing those words and seeing those words, settles, and then from there, I ' m able to take the next intuitive right action the best that I can. CA : Liz, you can say no to this, it may be a totally inappropriate thing to ask, but you don ' t happen to have a letter from the last day or two that you ' d consider reading, all or in part of? I don ' t know how long they are. EG : You ' re putting me on the spot. Let me see what we ' ve got. Let ' s see. OK, so here ' s one. So I was panicking because I want to offer my apartment in New York to a woman who is a COVID-19 nurse who ' s volunteered to come into New York City to help, and I ' m afraid that I ' ll infect my neighbors if I let her come and stay there. So I was up in the middle of the night, thinking, ethically, is it appropriate for me to do this? So I wrote,"I need you."And Love said,"I ' m right here."And then I said,"I want to offer that COVID-19 nurse my apartment, but I ' m afraid that my neighbors will get infected, and I ' m scared, and I don ' t know what the right move is. Help me."And Love said,"I don ' t actually know what the right answer to that is, but I ' m with you."And I said,"But what do you think I should do?"And Love said,"Why don ' t you just sit with me right here for a minute and be with me and know that you ' re held no matter what, that you cannot make the wrong choice, that it doesn ' t matter in the grand scheme of things. You ' re my beloved, I ' ve got you. I can see how much you ' re spinning, I can see how tired you are, and it doesn ' t matter to me whether you make this decision in the next minute, in the next day, or not at all. I ' m with you, and I ' ll sit with you through this entire thing, and I ' ll love you no matter what you decide to do at the end of this. I will be just as much with you at the end of this decision as am I with you now."And then, I said,"So what do you think I should do?"And Love says,"I think you should go get a glass of water, and I think you should lie down and get some rest, and we ' ll talk about it some more in the morning.”What I have found over the years of writing myself these letters from Love is that Love never gives advice. This is actually really good for all of you who love to give unsolicited advice to people. Love never gives advice beyond,”Why don ’ t you get a glass of water? Why don ’ t you rest? We ’ ll try this again tomorrow. You ’ re doing your best, this is a hard time, and I ’ ve got you.”So I ’ ve got 20 years of those journals, and I ’ m assuming that I ’ m going to need it for the rest of my life. CA : Wow. I don ’ t know, Helen, I think we might be done. I think I ’ m done. I can ’ t ask any more after that. HW : How beautiful. Good grief. CA : Liz, you ’ re really phenomenal. You ’ ve just got this unique way of **articulating** what others can ’ t **articulate**, and you ’ ve brought all of us to a very tender, intimate place, and thank you for that. EG : Thank you, Chris. HW : Thank you so much. EG : And thank you, Helen. Take care of yourselves, everybody. We ’ re right here with each other through this. We can do this. CA : Thank you, Liz. Goodbye. HW : Thank you. CA : Oof. HW : Oof. HW : Deep breaths. CA : Yeah. No. That was special. That was special to me. I know that you are all in different circumstances online, and that there are so many elements to this thing. There ’ s the problems that those of us who are isolated have, and in many ways, those are the luxurious problems, and we ’ re really aware of that. But they ’ re still problems, and we ’ re going to give space on these TED Connects to many other voices as well. I think we ’ re hoping to hear next week from a doctor at the front line, a voice from India, we hope, on some of the horrifying things that are happening there, and also some pretty amazing proposals for how the world could come out of this, like specific proposals on how we get past this period of lockdown to bring back the economy. All of this matters. So I guess we want Helen and everyone to come back, calendar this, share with friends, and help us figure out how to use this time best. HW : I also wanted to flag that I don ’ t know if you were able to tune in for Susan David ’ s conversation earlier in the week. We have launched a new podcast with Susan that launched on Monday. We ’ re calling it”Checking In with Susan David,”and she is going to be sharing daily tips on how to deal with this pandemic. And so you can find that wherever you find podcasts in this day and age. For this conversation, we will be archiving it. It will be on Facebook, and we ’ ll also put it onto TED. com. You can find the TED Interview podcast that Chris and Liz did last year, which I confess just made me weep for... too long. You can find that at go. ted. com / tedconnects. But that ’ s it from us. And tomorrow, I want to flag that we have a very special treat, which is less chat, more beauty. We will be joined by the unbelievably talented Butterscotch, who is a beatboxer and a singer and a musician and a sage, and an all-around delight, and she is going to be giving us a glimpse into her world and delighting us all with some sonic deliciousness. So do tune in tomorrow. CA : Thanks so much, everyone. We ’ re in this together. Stay safe. See you soon. HW : See you soon. Be well.

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Chris Anderson : Al, welcome. So look, just six months ago – it seems a lifetime ago, but it really was just six months ago – climate seemed to be on the lips of every thinking person on the planet. Recent events seem to have swept it all away from our attention. How worried are you about that? Al Gore : Well, first of all Chris, thank you so much for inviting me to have this conversation. People are reacting differently to the climate crisis in the midst of these other great challenges that have taken over our awareness, appropriately. One reason is something that you mentioned. People get the fact that when scientists are warning us in ever more dire terms and setting their hair on fire, so to speak, it ’ s best to listen to what they ’ re saying, and I think that lesson has begun to sink in in a new way. Another similarity, by the way, is that the climate crisis, like the COVID-19 pandemic, has revealed in a new way the shocking injustices and inequalities and disparities that affect communities of color and low-income communities. There are differences. The climate crisis has effects that are not measured in years, as the pandemic is, but consequences that are measured in centuries and even longer. And the other difference is that instead of depressing economic activity to deal with the climate crisis, as nations around the world have had to do with COVID-19, we have the opportunity to create tens of millions of new jobs. That sounds like a political phrasing, but it ’ s literally true. For the last five years, the fastest-growing job in the US has been solar installer. The second-fastest has been wind turbine technician. And the”Oxford Review of Economics,”just a few weeks ago, pointed the way to a very jobs-rich recovery if we emphasize renewable energy and sustainability technology. So I think we are crossing a tipping point, and you need only look at the recovery plans that are being presented in nations around the world to see that they ’ re very much focused on a green recovery. CA : I mean, one obvious impact of the pandemic is that it ’ s brought the world ’ s economy to a shuddering halt, thereby reducing greenhouse gas emissions. I mean, how big an effect has that been, and is it unambiguously good news? AG : Well, it ’ s a little bit of an illusion, Chris, and you need only look back to the Great Recession in 2008 and ’ 09, when there was a one percent decline in emissions, but then in 2010, they came roaring back during the recovery with a four percent increase. The latest estimates are that emissions will go down by at least five percent during this induced coma, as the economist Paul Krugman perceptively described it, but whether it goes back the way it did after the Great Recession is in part up to us, and if these green recovery plans are actually implemented, and I know many countries are determined to implement them, then we need not repeat that pattern. After all, this whole process is occurring during a period when the cost of renewable energy and electric vehicles, batteries and a range of other sustainability approaches are continuing to fall in price, and they ’ re becoming much more competitive. Just a quick reference to how fast this is : five years ago, electricity from solar and wind was cheaper than electricity from fossil fuels in only one percent of the world. This year, it ’ s cheaper in two-thirds of the world, and five years from now, it will be cheaper in virtually 100 percent of the world. EVs will be cost-competitive within two years, and then will continue falling in price. And so there are changes underway that could interrupt the pattern we saw after the Great Recession. CA : The reason those pricing differentials happen in different parts of the world is obviously because there ’ s different amounts of sunshine and wind there and different building costs and so forth. AG : Well, yes, and government policies also account for a lot. The world is continuing to subsidize fossil fuels at a ridiculous amount, more so in many developing countries than in the US and developed countries, but it ’ s subsidized here as well. But everywhere in the world, wind and solar will be cheaper as a source of electricity than fossil fuels, within a few years. CA : I think I ’ ve heard it said that the fall in emissions caused by the pandemic isn ’ t that much more than, actually, the fall that we will need every single year if we ’ re to meet emissions targets. Is that true, and, if so, doesn ’ t that seem impossibly daunting? AG : It does seem daunting, but first look at the number. That number came from a study a little over a year ago released by the IPCC as to what it would take to keep the Earth ’ s temperatures from increasing more than 1. 5 degrees Celsius. And yes, the annual reductions would be significant, on the order of what we ’ ve seen with the pandemic. And yes, that does seem daunting. However, we do have the opportunity to make some fairly dramatic changes, and the plan is not a mystery. You start with the two sectors that are closest to an effective transition – electricity generation, as I mentioned – and last year, 2019, if you look at all of the new electricity generation built all around the world, 72 percent of it was from solar and wind. And already, without the continuing subsidies for fossil fuels, we would see many more of these plants being shut down. There are some new fossil plants being built, but many more are being shut down. And where transportation is concerned, the second sector ready to go, in addition to the cheaper prices for EVs that I made reference to before, there are some 45 jurisdictions around the world – national, regional and municipal – where laws have been passed beginning a phaseout of internal combustion engines. Even India said that by 2030, less than 10 years from now, it will be illegal to sell any new internal combustion engines in India. There are many other examples. So the past small reductions may not be an accurate guide to the kind we can achieve with serious national plans and a focused global effort. CA : So help us understand just the big picture here, Al. I think before the pandemic, the world was emitting about 55 gigatons of what they call”CO2 equivalent,”so that includes other greenhouse gases like methane dialed up to be the equivalent of CO2. And am I right in saying that the IPCC, which is the global organization of scientists, is recommending that the only way to fix this crisis is to get that number from 55 to zero by 2050 at the very latest, and that even then, there ' s a chance that we will end up with temperature rises more like two degrees Celsius rather than 1. 5? I mean, is that approximately the big picture of what the IPCC is recommending? AG : That ' s correct. The global goal established in the Paris Conference is to get to net zero on a global basis by 2050, and many people quickly add that that really means a 45 to 50 percent reduction by 2030 to make that pathway to net zero feasible. CA : And that kind of timeline is the kind of timeline where people couldn ' t even imagine it. It ' s just hard to think of policy over 30 years. So that ' s actually a very good shorthand, that humanity ' s task is to cut emissions in half by 2030, approximately speaking, which I think boils down to about a seven or eight percent reduction a year, something like that, if I ' m not wrong. AG : Not quite. Not quite that large but close, yes. CA : So it is something like the effect that we ' ve experienced this year may be necessary. This year, we ' ve done it by basically shutting down the economy. You ' re talking about a way of doing it over the coming years that actually gives some economic growth and new jobs. So talk more about that. You ' ve referred to changing our energy sources, changing how we transport. If we did those things, how much of the problem does that solve? AG : Well, we can get to – well, in addition to doing the two sectors that I mentioned, we also have to deal with manufacturing and all the use cases that require temperatures of a thousand degrees Celsius, and there are solutions there as well. I ' ll come back and mention an exciting one that Germany has just embarked upon. We also have to tackle regenerative agriculture. There is the opportunity to sequester a great deal of carbon in topsoils around the world by changing the agricultural techniques. There is a farmer-led movement to do that. We need to also retrofit buildings. We need to change our management of forests and the ocean. But let me just mention two things briefly. First of all, the high temperature use cases. Angela Merkel, just 10 days ago, with the leadership of her minister Peter Altmaier, who is a good friend and a great public servant, have just embarked on a green hydrogen strategy to make hydrogen with zero marginal cost renewable energy. And just a word on that, Chris : you ' ve heard about the intermittency of wind and solar – solar doesn ' t produce electricity when the sun ' s not shining, and wind doesn ' t when the wind ' s not blowing – but batteries are getting better, and these technologies are becoming much more efficient and powerful, so that for an increasing number of hours of each day, they ' re producing often way more electricity than can be used. So what to do with it? The marginal cost for the next kilowatt-hour is zero. So all of a sudden, the very energy-intensive process of cracking hydrogen from water becomes economically feasible, and it can be substituted for coal and gas, and that ' s already being done. There ' s a Swedish company already making steel with green hydrogen, and, as I say, Germany has just embarked on a major new initiative to do that. I think they ' re pointing the way for the rest of the world. Now, where building retrofits are concerned, just a moment on this, because about 20 to 25 percent of the global warming pollution in the world and in the US comes from inefficient buildings that were constructed by companies and individuals who were trying to be competitive in the marketplace and keep their margins acceptably high and thereby skimping on insulation and the right windows and LEDs and the rest. And yet the person or company that buys that building or leases that building, they want their monthly utility bills much lower. So there are now ways to close that so-called agent-principal divide, the differing incentives for the builder and occupier, and we can retrofit buildings with a program that literally pays for itself over three to five years, and we could put tens of millions of people to work in jobs that by definition cannot be outsourced because they exist in every single community. And we really ought to get serious about doing this, because we ' re going to need all those jobs to get sustainable prosperity in the aftermath of this pandemic. CA : Just going back to the hydrogen economy that you referred to there, when some people hear that, they think,"Oh, are you talking about hydrogen-fueled cars?"And they ' ve heard that that probably won ' t be a winning strategy. But you ' re thinking much more broadly than that, I think, that it ' s not just hydrogen as a kind of storage mechanism to act as a buffer for renewable energy, but also hydrogen could be essential for some of the other processes in the economy like making steel, making cement, that are fundamentally carbon-intensive processes right now but could be transformed if we had much cheaper sources of hydrogen. Is that right? AG : Yes, I was always skeptical about hydrogen, Chris, principally because it ' s been so expensive to make it, to"crack it out of water,"as they say. But the game-changer has been the incredible abundance of solar and wind electricity in volumes and amounts that people didn ' t expect, and all of a sudden, it ' s cheap enough to use for these very energy-intensive processes like creating green hydrogen. I ' m still a bit skeptical about using it in vehicles. Toyota ' s been betting on that for 25 years and it hasn ' t really worked for them. Never say never, maybe it will, but I think it ' s most useful for these high-temperature industrial processes, and we already have a pathway for decarbonizing transportation with electricity that ' s working extremely well. Tesla ' s going to be soon the most valuable automobile company in the world, already in the US, and they ' re about to overtake Toyota. There is now a semitruck company that ' s been stood up by Tesla and another that is going to be a hybrid with electricity and green hydrogen, so we ' ll see whether or not they can make it work in that application. But I think electricity is preferable for cars and trucks. CA : We ' re coming to some community questions in a minute. Let me ask you, though, about nuclear. Some environmentalists believe that nuclear, or maybe new generation nuclear power is an essential part of the equation if we ' re to get to a truly clean future, a clean energy future. Are you still pretty skeptical on nuclear, Al? AG : Well, the market ' s skeptical about it, Chris. It ' s been a crushing disappointment for me and for so many. I used to represent Oak Ridge, where nuclear energy began, and when I was a young congressman, I was a booster. I was very enthusiastic about it. But the cost overruns and the problems in building these plants have become so severe that utilities just don ’ t have an appetite for them. It ’ s become the most expensive source of electricity. Now, let me hasten to add that there are some older nuclear reactors that have more useful time that could be added onto their lifetimes. And like a lot of environmentalists, I ’ ve come to the view that if they can be determined to be safe, they should be allowed to continue operating for a time. But where new nuclear power plants are concerned, here ’ s a way to look at it. If you are – you ’ ve been a CEO, Chris. If you were the CEO of – I guess you still are. If you were the CEO of an electric utility, and you told your executive team,”I want to build a nuclear power plant,”two of the first questions you would ask are, number one : How much will it cost? And there ’ s not a single engineering consulting firm that I ’ ve been able to find anywhere in the world that will put their name on an opinion giving you a cost estimate. They just don ’ t know. A second question you would ask is : How long will it take to build it, so we can start selling the electricity? And again, the answer you will get is,”We have no idea.”So if you don ’ t know how much it ’ s going to cost, and you don ’ t know when it ’ s going to be finished, and you already know that the electricity is more expensive than the alternate ways to produce it, that ’ s going to be a little discouraging, and, in fact, that ’ s been the case for utilities around the world. CA : OK. So there ’ s definitely an interesting debate there, but we ’ re going to come on to some community questions. Let ’ s have the first of those questions up, please. From Prosanta Chakrabarty :”People who are skeptical of COVID and of climate change seem to be skeptical of science in general. It may be that the singular message from scientists gets diluted and convoluted. How do we fix that?”AG : Yeah, that ’ s a great question, Prosanta. Boy, I ’ m trying to put this succinctly and shortly. I think that there has been a feeling that experts in general have kind of let the US down, and that feeling is much more pronounced in the US than in most other countries. And I think that the considered opinion of what we call experts has been diluted over the last few decades by the unhealthy dominance of big money in our political system, which has found ways to really twist economic policy to benefit elites. And this sounds a little radical, but it ’ s actually what has happened. And we have gone for more than 40 years without any meaningful increase in middle-income pay, and where the injustice experienced by African Americans and other communities of color are concerned, the differential in pay between African Americans and majority Americans is the same as it was in 1968, and the family wealth, the net worth – it takes 11 and a half so-called”**typical**”African American families to make up the net worth of one so-called”**typical**”White American family. And you look at the soaring incomes in the top one or the top one-tenth of one percent, and people say,”Wait a minute. Whoever the experts were that designed these policies, they haven ’ t been doing a good job for me.”A final point, Chris : there has been an assault on reason. There has been a war against truth. There has been a strategy, maybe it was best known as a strategy decades ago by the tobacco companies who hired actors and dressed them up as doctors to falsely reassure people that there were no health consequences from smoking cigarettes, and a hundred million people died as a result. That same strategy of diminishing the significance of truth, diminishing, as someone said, the authority of knowledge, I think that has made it kind of open season on any inconvenient truth – forgive another buzz phrase, but it is apt. We cannot abandon our devotion to the best available evidence tested in reasoned discourse and used as the basis for the best policies we can form. CA : Is it possible, Al, that one consequence of the pandemic is actually a growing number of people have revisited their opinions on scientists? I mean, you ’ ve had a chance in the last few months to say,”Do I trust my political leader or do I trust this scientist in terms of what they ’ re saying about this virus?”Maybe lessons from that could be carried forward? AG : Well, you know, I think if the polling is accurate, people do trust their doctors a lot more than some of the politicians who seem to have a vested interest in pretending the pandemic isn ’ t real. And if you look at the incredible bust at President Trump ’ s rally in Tulsa, a stadium of 19, 000 people with less than one-third filled, according to the fire marshal, you saw all the empty seats if you saw the news clips, so even the most loyal Trump supporters must have decided to trust their doctors and the medical advice rather than Dr. Donald Trump. CA : With a little help from the TikTok generation, perchance. AG : Well, but that didn ’ t affect the turnout. What they did, very cleverly, and I ’ m cheering them on, what they did was affect the Trump White House ’ s expectations. They ’ re the reason why he went out a couple days beforehand and said,”We ’ ve had a million people sign up.”But they didn ’ t prevent – they didn ’ t take seats that others could have otherwise taken. They didn ’ t affect the turnout, just the expectations. CA : OK, let ’ s have our next question here.”Are you concerned the world will rush back to the use of the private car out of fear of using shared public transportation?”AG : Well, that could actually be one of the consequences, absolutely. Now, the trends on mass transit were already inching in the wrong direction because of Uber and Lyft and the ridesharing services, and if autonomy ever reaches the goals that its advocates have hoped for then that may also have a similar effect. But there ’ s no doubt that some people are going to be probably a little more reluctant to take mass transportation until the fear of this pandemic is well and truly gone. CA : Yeah. Might need a vaccine on that one. AG : Yeah. CA : Next question. Sonaar Luthra, thank you for this question from LA.”Given the temperature rise in the Arctic this past week, seems like the rate we are losing our carbon sinks like permafrost or forests is accelerating faster than we predicted. Are our models too focused on human emissions?”Interesting question. AG : Well, the models are focused on the factors that have led to these incredible temperature spikes in the north of the Arctic Circle. They were predicted, they have been predicted, and one of the reasons for it is that as the snow and ice cover melts, the sun ’ s incoming rays are no longer reflected back into space at a 90 percent rate, and instead, when they fall on the dark tundra or the dark ocean, they ’ re absorbed at a 90 percent rate. So that ’ s a magnifier of the warming in the Arctic, and this has been predicted. There are a number of other consequences that are also in the models, but some of them may have to be recalibrated. The scientists are freshly concerned that the emissions of both CO2 and methane from the thawing tundra could be larger than they had hoped they would be. There ’ s also just been a brand-new study. I won ’ t spend time on this, because it deals with a kind of geeky term called”climate sensitivity,”which has been a factor in the models with large error bars because it ’ s so hard to pin down. But the latest evidence indicates, worryingly, that the sensitivity may be greater than they had thought, and we will have an even more daunting task. That shouldn ’ t discourage us. I truly believe that once we cross this tipping point, and I do believe we ’ re doing it now, as I ’ ve said, then I think we ’ re going to find a lot of ways to speed up the emissions reductions. CA : We ’ ll take one more question from the community. Haha.”Geoengineering is making extraordinary progress. Exxon is investing in technology from Global Thermostat that seems promising. What do you think of these air and water carbon capture technologies?”Stephen Petranek. AG : Yeah. Well, you and I have talked about this before, Chris. I ’ ve been strongly opposed to conducting an unplanned global experiment that could go wildly wrong, and most are really scared of that approach. However, the term”geoengineering”is a nuanced term that covers a lot. If you want to paint roofs white to reflect more energy from the cityscapes, that ’ s not going to bring a danger of a runaway effect, and there are some other things that are loosely called”geoengineering”like that, which are fine. But the idea of blocking out the sun ’ s rays – that ’ s insane in my opinion. Turns out plants need sunlight for photosynthesis and solar panels need sunlight for producing electricity from the sun ’ s rays. And the consequences of changing everything we know and pretending that the consequences are going to precisely cancel out the unplanned experiment of global warming that we already have underway, you know, there are glitches in our thinking. One of them is called the”single solution bias,”and there are people who just have a hunger to say,”Well, that one solution, we just need to latch on to that and do that, and damn the consequences.”Well, it ’ s nuts. CA : But let me push back on this just a little bit. So let ’ s say that we agree that a single solution, all-or-nothing attempt at geoengineering is crazy. But there are scenarios where the world looks at emissions and just sees, in 10 years ’ time, let ’ s say, that they are just not coming down fast enough and that we are at risk of several other liftoff events where this train will just get away from us, and we will see temperature rises of three, four, five, six, seven degrees, and all of civilization is at risk. Surely, there is an approach to geoengineering that could be modeled, in a way, on the way that we approach medicine. Like, for hundreds of years, we don ’ t really understand the human body, people would try interventions, and some of them would work, and some of them wouldn ’ t. No one says in medicine,”You know, go in and take an all-or-nothing decision on someone ’ s life,”but they do say,”Let ’ s try some stuff.”If an experiment can be reversible, if it ’ s plausible in the first place, if there ’ s reason to think that it might work, we actually owe it to the future health of humanity to try at least some types of tests to see what could work. So, small tests to see whether, for example, seeding of something in the ocean might create, in a nonthreatening way, carbon sinks. Or maybe, rather than filling the atmosphere with sulfur dioxide, a smaller experiment that was not that big a deal to see whether, cost-effectively, you could reduce the temperature a little bit. Surely, that isn ’ t completely crazy and is at least something we should be thinking about in case these other measures don ’ t work? AG : Well, there ’ ve already been such experiments to seed the ocean to see if that can increase the uptake of CO_2. And the experiments were an unmitigated failure, as many predicted they would be. But that, again, is the kind of approach that ’ s very different from putting tinfoil strips in the atmosphere orbiting the Earth. That was the way that solar geoengineering proposal started. Now they ’ re focusing on chalk, so we have chalk dust all over everything. But more serious than that is the fact that it might not be reversible. CA : But, Al, that ’ s the rhetoric response. The amount of dust that you need to drop by a degree or two wouldn ’ t result in chalk dust over everything. It would be unbelievably – like, it would be less than the dust that people experience every day, anyway. I mean, I just – AG : First of all, I don ’ t know how you do a small experiment in the atmosphere. And secondly, if we were to take that approach, we would have to steadily increase the amount of whatever substance they decided. We ’ d have to increase it every single year, and if we ever stopped, then there would be a sudden snapback, like”The Picture of Dorian Gray,”that old book and **movie**, where suddenly all of the things caught up with you at once. The fact that anyone is even considering these approaches, Chris, is a measure of a feeling of desperation that some have begun to feel, which I understand, but I don ’ t think it should drive us toward these reckless experiments. And by the way, using your analogy to experimental cancer treatments, for example, you usually get informed consent from the patient. Getting informed consent from 7. 8 billion people who have no voice and no say, who are subject to the potentially catastrophic consequences of this wackadoodle proposal that somebody comes up with to try to rearrange the entire Earth ' s atmosphere and hope and pretend that it ' s going to cancel out, the fact that we ' re putting 152 million tons of heat-trapping, manmade global warming pollution into the sky every day. That ' s what ' s really insane. A scientist decades ago compared it this way. He said, if you had two people on a sinking boat and one of them says,"You know, we could probably use some mirrors to signal to shore to get them to build a sophisticated wave-generating machine that will cancel out the rocking of the boat by these guys in the back of the boat."Or you could get them to stop rocking the boat. And that ' s what we need to do. We need to stop what ' s causing the crisis. CA : Yeah, that ' s a great story, but if the effort to stop the people rocking in the back of the boat is as complex as the scientific proposal you just outlined, whereas the experiment to stop the waves is actually as simple as telling the people to stop rocking the boat, that story changes. And I think you ' re right that the issue of informed consent is a really challenging one, but, I mean, no one gave informed consent to do all of the other things we ' re doing to the atmosphere. And I agree that the moral hazard issue is worrying, that if we became dependent on geoengineering and took away our efforts to do the rest, that would be tragic. It just seems like, I wish it was possible to have a nuanced debate of people saying, you know what, there ' s multiple dials to a very complex problem. We ' re going to have to adjust several of them very, very carefully and keep talking to each other. Wouldn ' t that be a goal to just try and have a more nuanced debate about this, rather than all of that geoengineering can ' t work? AG : Well, I ' ve said some of it, you know, the benign forms that I ' ve mentioned, I ' m not ruling those out. But blocking the Sun ' s rays from the Earth, not only do you affect 7. 8 billion people, you affect the plants and the animals and the ocean currents and the wind currents and natural processes that we ' re in danger of disrupting even more. Techno-optimism is something I ' ve engaged in in the past, but to latch on to some brand-new technological solution to rework the entire Earth ' s natural system because somebody thinks he ' s clever enough to do it in a way that precisely cancels out the consequences of using the atmosphere as an open sewer for heat-trapping manmade gases. It ' s much more important to stop using the atmosphere as an open sewer. That ' s what the problem is. CA : All right, well, we ' ll agree that that is the most important thing, for sure, and speaking of which, do you believe the world needs carbon pricing, and is there any prospect for getting there? AG : Yes. Yes to both questions. For decades, almost every economist who is asked about the climate crisis says,"Well, we just need to put a price on carbon."And I have certainly been in favor of that approach. But it is daunting. Nevertheless, there are 43 jurisdictions around the world that already have a price on carbon. We ' re seeing it in Europe. They finally straightened out their carbon pricing mechanism. It ' s an emissions trading version of it. We have places that have put a tax on carbon. That ' s the approach the economists prefer. China is beginning to implement its national emissions trading program. California and quite a few other states in the US are already doing it. It can be given back to people in a revenue-neutral way. But the opposition to it, Chris, which you ' ve noted, is impressive enough that we do have to take other approaches, and I would say most climate activists are now saying, look, let ' s don ' t make the best the enemy of the better. There are other ways to do this as well. We need every solution we can rationally employ, including by regulation. And often, when the political difficulty of a proposal becomes too difficult in a market-oriented approach, the fallback is with regulation, and it ' s been given a bad name, regulation, but many places are doing it. I mentioned phasing out internal combustion engines. That ' s an example. There are 160 cities in the US that have already by regulation ordered that within a date certain, 100 percent of all their electricity will have to come from renewable sources. And again, the market forces that are driving the cost of renewable energy and sustainability solutions ever downward, that gives us the wind at our back. This is working in our favor. CA : I mean, the pushback on carbon pricing often goes further from parts of the environmental movement, which is to a pushback on the role of business in general. Business is actually – well, capitalism – is blamed for the climate crisis because of unrelenting growth, to the point where many people don ' t trust business to be part of the solution. The only way to go forward is to regulate, to force businesses to do the right thing. Do you think that business has to be part of the solution? AG : Well, definitely, because the allocation of capital needed to solve this crisis is greater than what governments can handle. And businesses are beginning, many businesses are beginning to play a very constructive role. They ' re getting a demand that they do so from their customers, from their investors, from their boards, from their executive teams, from their families. And by the way, the rising generation is demanding a brighter future, and when CEOs interview potential new hires, they find that the new hires are interviewing them. They want to make a nice income, but they want to be able to tell their family and friends and peers that they ' re doing something more than just making money. One illustration of how this new generation is changing, Chris : there are 65 colleges in the US right now where the College Young Republican Clubs have joined together to jointly demand that the Republican National Committee change its policy on climate, lest they lose that entire generation. This is a global phenomenon. The Greta Generation is now leading this in so many ways, and if you look at the polling, again, the vast majority of young Republicans are demanding a change on climate policy. This is really a movement that is building still. CA : I was going to ask you about that, because one of the most painful things over the last 20 years has just been how climate has been politicized, certainly in the US. You ' ve probably felt yourself at the heart of that a lot of the time, with people attacking you personally in the most merciless, and unfair ways, often. Do you really see signs that that might be changing, led by the next generation? AG : Yeah, there ' s no question about it. I don ' t want to rely on polls too much. I ' ve mentioned them already. But there was a new one that came out that looked at the wavering Trump supporters, those who supported him strongly in the past and want to do so again. The number one issue, surprisingly to some, that is giving them pause, is the craziness of President Trump and his administration on climate. We ' re seeing big majorities of the Republican Party overall saying that they ' re ready to start exploring some real solutions to the climate crisis. I think that we ' re really getting there, no question about it. CA : I mean, you ' ve been the figurehead for raising this issue, and you happen to be a Democrat. Is there anything that you can personally do to – I don ' t know – to open the tent, to welcome people, to try and say,"This is beyond politics, dear friends"? AG : Yeah. Well, I ' ve tried all of those things, and maybe it ' s made a little positive difference. I ' ve worked with the Republicans extensively. And, you know, well after I left the White House, I had Newt Gingrich and Pat Robertson and other prominent Republicans appear on national TV ads with me saying we ' ve got to solve the climate crisis. But the petroleum industry has really doubled down enforcing discipline within the Republican Party. I mean, look at the attacks they ' ve launched against the Pope when he came out with his encyclical and was demonized, not by all for sure, but there were hawks in the anti-climate movement who immediately started training their guns on Pope Francis, and there are many other examples. They enforce discipline and try to make it a partisan issue, even as Democrats reach out to try to make it bipartisan. I totally agree with you that it should not be a partisan issue. It didn ’ t use to be, but it ’ s been artificially weaponized as an issue. CA : I mean, the CEOs of oil companies also have kids who are talking to them. It feels like some of them are moving and are trying to invest and trying to find ways of being part of the future. Do you see signs of that? AG : Yeah. I think that business leaders, including in the oil and gas companies, are hearing from their families. They ’ re hearing from their friends. They ’ re hearing from their employees. And, by the way, we ’ ve seen in the tech industry some mass walkouts by employees who are demanding that some of the tech companies do more and get serious. I ’ m so proud of Apple. Forgive me for parenthetically praising Apple. You know, I ’ m on the board, but I ’ m such a big fan of Tim Cook and my colleagues at Apple. It ’ s an example of a tech company that ’ s really doing fantastic things. And there ’ s some others as well. There are others in many industries. But the pressures on the oil and gas companies are quite extraordinary. You know, BP just wrote down 12 and a half billion dollars ’ worth of oil and gas assets and said that they ’ re never going to see the light of day. Two-thirds of the fossil fuels that have already been discovered cannot be burned and will not be burned. And so that ’ s a big economic risk to the global economy, like the subprime mortgage crisis. We ’ ve got 22 trillion dollars of subprime carbon assets, and just yesterday, there was a major report that the fracking industry in the US is seeing now a wave of bankruptcies because the price of the fracked gas and oil has fallen below levels that make them economic. CA : Is the shorthand of what ’ s happened there that electric cars and electric technologies and solar and so forth have helped drive down the price of oil to the point where huge amounts of the reserves just can ’ t be developed profitably? AG : Yes, that ’ s it. That ’ s mainly it. The projections for energy sources in the next several years uniformly predict that electricity from wind and solar is going to continue to plummet in price, and therefore using gas or coal to make steam to turn the turbines is just not going to be economical. Similarly, the electrification of the transportation sector is having the same effect. Some are also looking at the trend in national, regional and local governance. I mentioned this before, but they ’ re predicting a very different energy future. But let me come back, Chris, because we talked about business leaders. I think you were getting in a question a moment ago about capitalism itself, and I do want to say a word on that, because there are a lot of people who say maybe capitalism is the basic problem. I think the current form of capitalism we have is desperately in need of reform. The short-term outlook is often mentioned, but the way we measure what is of value to us is also at the heart of the crisis of modern capitalism. Now, capitalism is at the base of every successful economy, and it balances supply and demand, unlocks a higher fraction of the human potential, and it ’ s not going anywhere, but it needs to be reformed, because the way we measure what ’ s valuable now ignores so-called negative externalities like pollution. It also ignores positive externalities like investments in education and health care, mental health care, family services. It ignores the depletion of resources like groundwater and topsoil and the web of living species. And it ignores the distribution of incomes and net worths, so when GDP goes up, people cheer, two percent, three percent – wow! – four percent, and they think,”Great!”But it ’ s accompanied by vast increases in pollution, chronic underinvestment in public goods, the depletion of irreplaceable natural resources, and the worst inequality crisis we ’ ve seen in more than a hundred years that is threatening the future of both capitalism and democracy. So we have to change it. We have to reform it. CA : So reform capitalism, but don ’ t throw it out. We ’ re going to need it as a tool as we go forward if we ’ re to solve this. AG : Yeah, I think that ’ s right, and just one other point : the worst environmental abuses in the last hundred years have been in jurisdictions that experimented during the 20th century with the alternatives to capitalism on the left and right. CA : Interesting. All right. Two last community questions quickly. Chadburn Blomquist :”As you are reading the tea leaves of the impact of the current pandemic, what do you think in regard to our response to combatting climate change will be the most impactful lesson learned?”AG : Boy, that ’ s a very thoughtful question, and I wish my answer could rise to the same level on short notice. I would say first, don ’ t ignore the scientists. When there is virtual unanimity among the scientific and medical experts, pay attention. Don ’ t let some politician dissuade you. I think President Trump is slowly learning that ’ s it ’ s kind of difficult to gaslight a virus. He tried to gaslight the virus in Tulsa. It didn ’ t come off very well, and tragically, he decided to recklessly roll the dice a month ago and ignore the recommendations for people to wear **masks** and to socially distance and to do the other things, and I think that lesson is beginning to take hold in a much stronger way. But beyond that, Chris, I think that this period of time has been characterized by one of the most profound opportunities for people to rethink the patterns of their lives and to consider whether or not we can ’ t do a lot of things better and differently. And I think that this rising generation I mentioned before has been even more profoundly affected by this interlude, which I hope ends soon, but I hope the lessons endure. I expect they will. CA : Yeah, it ’ s amazing how many things you can do without emitting carbon, that we ’ ve been forced to do. Let ’ s have one more question here. Frank Hennessy :”Are you encouraged by the ability of people to quickly adapt to the new normal due to COVID-19 as evidence that people can and will change their habits to respond to climate change?”AG : Yes, but I think we have to keep in mind that there is a crisis within this crisis. The impact on the African American community, which I mentioned before, on the Latinx community, Indigenous peoples. The highest infection rate is in the Navajo Nation right now. So some of these questions appear differently to those who are really getting the brunt of this crisis, and it is unacceptable that we allow this to continue. It feels one way to you and me and perhaps to many in our audience today, but for low-income communities of color, it ’ s an entirely different crisis, and we owe it to them and to all of us to get busy and to start using the best science and solve this pandemic. You know the phrase”pandemic economics.”Somebody said, the first principle of pandemic economics is take care of the pandemic, and we ’ re not doing that yet. We ’ re seeing the president try to goose the economy for his reelection, never mind the prediction of tens of thousands of additional American deaths, and that is just unforgivable in my opinion. CA : Thank you, Frank. So Al, you, along with others in the community played a key role in encouraging TED to launch this initiative called”Countdown.”Thank you for that, and I guess this conversation is continuing among many of us. If you ’ re interested in climate, watching this, check out the Countdown website, countdown. ted. com, and be part of 10 / 10 / 2020, when we are trying to put out an alert to the world that climate can ’ t wait, that it really matters, and there ’ s going to be some amazing content free to the world on that day. Thank you, Al, for your inspiration and support in doing that. I wonder whether you could end today ’ s session just by painting us a picture, like how might things roll out over the next decade or so? Just tell us whether there is still a story of hope here. AG : I ’ d be glad to. I ’ ve got to get one plug in. I ’ ll make it brief. July 18 through July 26, The Climate Reality Project is having a global training. We ’ ve already had 8, 000 people register. You can go to climatereality. com. Now, a bright future. It begins with all of the kinds of efforts that you ’ ve thrown yourself into in organizing Countdown. Chris, you and your team have been amazing to work with, and I ’ m so excited about the Countdown project. TED has an unparalleled ability to spread ideas that are worth spreading, to raise consciousness, to enlighten people around the world, and it ’ s needed for climate and the solutions to the climate crisis like it ’ s never been needed before, and I just want to thank you for what you personally are doing to organize this fantastic Countdown program. CA : Thank you. And the world? Are we going to do this? Do you think that humanity is going to pull this off and that our grandchildren are going to have beautiful lives where they can celebrate nature and not spend every day in fear of the next tornado or tsunami? AG : I am optimistic that we will do it, but the answer is in our hands. We have seen dark times in periods of the past, and we have risen to meet the challenge. We have limitations of our long evolutionary heritage and elements of our culture, but we also have the ability to transcend our limitations, and when the chips are down, and when survival is at stake and when our children and future generations are at stake, we ’ re capable of more than we sometimes allow ourselves to think we can do. This is such a time. I believe we will rise to the occasion, and we will create a bright, clean, prosperous, just and fair future. I believe it with all my heart. CA : Al Gore, thank you for your life of work, for all you ’ ve done to elevate this issue and for spending this time with us now. Thank you. AG : Back at you. Thank you.

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HW : Hi everyone, I am Helen Walters. I am head of Media and Curation here at TED. Welcome to another episode of TED Explains the World with the one and only, Ian Bremmer. It is Monday, February 24, a little more than a month since President Trump was inaugurated once more, and safe to say, a lot has been going on. So we figured we ’ d check in with Ian to determine what we should really be paying attention to amid the extreme noise. Ian, hi. Ian Bremmer : Helen, great to be back with you. HW : So we asked the TED community to share their questions for you, and we wanted them to share what they ’ re most curious to know. And we really got some amazing questions that I plan to pepper this conversation with. So let ’ s start with one right now. Two months into 2025, where does the US stand? IB : Very powerfully, in the sense that the US economy, of course, is performing considerably better than any of the other G7 economies coming still out of the pandemic. Technologically, only close competitor is China, ahead of the US in some areas, behind the US and other critical areas. But compared to every other country in the world, the US is head and shoulders and neck or waist above them. Militarily, of course, the US is the only country with global ability to project power. No one else is close. And the US dollar is still the global reserve currency, no one is approximate challenger. I say all of those things because those are the things that haven ’ t really changed over the course of the last few months, but I suspect that isn ’ t what the questioner was asking. What they were really asking is what ’ s happening politically. And politically, the United States is unwinding its own global order. It is no longer particularly interested in promoting collective security or NATO. It ’ s no longer really interested in promoting involvement and leadership of multilateral institutions, of consistent rule of law and free trade that ’ s well regulated, certainly not very interested in promoting democracy around the world. I mean, this is an environment where other countries around the world have to figure out how to adapt to the United States, and that the US has become the principal driver of geopolitical risk and uncertainty in the world today, unlike any other time in my lifetime and yours, that ’ s perhaps the biggest change. HW : That is quite a statement. Alright, you are a geopolitical expert.

Let ’ s talk about Europe. So Defense Secretary Pete Hegseth stated that America ’ s foreign policy focus no longer lies in Europe. Vice President JD Vance caused some raised eyebrows, dropped jaws and, I suspect, some choice expletives after his recent speech at the Munich Security Conference, in which he accused European leaders of suppressing free speech and warned of the threat from within. So you were in Munich. Was it as dramatic as the stories we read had it? And what happens next? IB : It was. I was in the room. I was maybe 30 feet away from him when he gave that speech. Standing right in the front, right next to me was the president of Czechia, the president of Finland, the prime minister of Sweden, watching all of them and their reactions. What he said, let ’ s keep in mind, this is the Munich Security Conference. It ’ s been going on now for something like 61 years. And he was the head of the US delegation. And every year the head of the US delegation gives a big speech on the state of the transatlantic alliance and on global security. And he didn ’ t do that. I mean, the people in the audience were expecting a challenging speech. They were expecting that the Americans would be less committed to the alignment with the Europeans on Ukraine. The US had just had that direct Trump phone call with Vladimir Putin, 90 minutes long. Hadn ’ t coordinated that with the Europeans, never mind the Ukrainians, in advance. So, I mean, they were definitely prepared for a very challenging plenary. But Vance did none of that. Vance said,”I ’ m not going to talk about Ukraine or Russia or China, because the biggest problem is actually what ’ s happening inside Europe. The biggest problem, essentially, are you guys sitting in front of me, that I ’ m speaking to, because you ’ re suppressing free speech. You ’ re suppressing the far right. You ’ ve been infected by the woke mind virus, and you aren ’ t real democracies as a consequence.”And specifically he said, and this is something that I think is underappreciated in the United States, he attacked the German firewall. And said that – And let me explain what the German firewall is. That is a principle by the leaders of all of the mainstream German parties and their supporters that they will not, under any circumstances, work with, enter into coalition with the Alternative for Deutschland party, the AfD. Because the AfD, under surveillance from German intelligence agencies, is considered to be a neo-Nazi party. And the United States, of course, after World War II, led de-nazification of Germany. The day before, Vance had actually gone to Dachau and visited a concentration camp, and then came to speak in Germany about the firewall needing to be ended. And at that point someone yelled out from the crowd,”This is unacceptable!”And it was right in the front. And people in the room, about 1, 500 people in the room, standing room only, couldn ’ t see who it was. They ’ d just see the back of his head. I saw who it was. It was the German Defense Minister, Pistorius. And in 15 years of me going to the Munich Security Conference, I have never seen anything remotely like that. And the reason I mention all of this, and of course, on the back of that, let me be clear, Vance refused to meet with the German chancellor because he ’ s basically, well, he ’ s not relevant. He ’ s only going to be there for a couple more months until a new government, so why should I bother? But does meet with the leader of this AfD, goes and meets with her directly, does a bilateral that evening. So that is a very important backdrop for this last weekend ’ s German elections, where the victor, Friedrich Mertz, who is going to be the next chancellor, actually referred specifically to America ’ s intervention in German democracy and support of the AfD being every bit as bad as what the Kremlin is doing inside Germany, and also said that the Europeans need to develop a defense policy which is independent of the United States, in other words, essentially saying that it will be the end of NATO. Again, staggering comments, unheard of, impossible to imagine even two weeks ago. And that ’ s why it ’ s very critical that you have the context of what happened between Vance, Elon Musk over the past couple of months, Trump, with the Germans, with the Europeans. So you can understand why the incoming German chancellor would make a statement like that. HW : What should we make of all of the conversation about the rise of Nazism, the return of Nazism? Obviously, the AfD just won 20 percent in the German election. We ’ ve also seen people allegedly giving Nazi salutes at **various** US conventions and speeches. Do you think this is overblown? Is this a fear that people have rationally? And what should we make of that? IB : In Germany, the Alternative for Deutschland, are winning all of former East Germany. This is a group that increasingly understands that they ’ re never going to really catch up to the rest of Germany. Remember, Germany hasn ’ t really grown in five years now. They ’ ve been in recession for the last two. And the average citizen, the average voter, living in former East Germany no longer are just angry about not catching up. They ’ re basically saying,”I ’ m done with this system.”So that ’ s why you ’ re getting this very strident, revanchist nationalism that is, you know, it ’ s not just taking a hard line on illegal immigrants. The AfD even supports removing citizens of Germany that they claim have not adequately integrated into German culture and nationality. And so, I mean, this is some very, very strong stuff, stuff that really does feel like what they were doing back in Nazi Germany. But in former East Germany, they win. They get the largest number of votes, where, in much of the wealthier part of West Germany, including, you know, where I was, in Munich, the AfD can ’ t break out of single digits. So it ’ s a very, very divided country. And also the AfD is doing very, very well among young men. There ’ s a clear gender divide here as well, just as there is in the United States. In the US, I mean, I, of course, saw Elon Musk and that, you know, sort of grabbing his heart, saying,”My heart goes out to you”and then doing what appeared to be mimicking a Nazi salute. And then Steve Bannon doing the same without the initial heartfelt comment. It clearly was trolling. Clearly was trying to get a reaction from folks. It ’ s very performative. And it ’ s meant to also, you know, attack and attach, make the left lose their minds on something that isn ’ t very critical to what the Trump administration is trying to do compared to the revolutionary changes inside the US government that they are very much prioritizing right now. So I ’ m very concerned about the rise of neo-Nazism in Germany. I am very much not concerned about that in the US. I ’ m concerned about other things I ’ m thinking is being used as gaslighting for those that can use distractions or can be distracted by them. And I would also say that this election in Germany wasn ’ t surprising. It was exactly as the polls have expected for several months now. But the key elections in Germany and the key elections in Europe are the next cycle. I ’ ve spoken with a lot of people around the Trump administration that believe that the Europeans are one electoral cycle behind the United States. So in other words, Trumpism is coming, just not quite yet. And so in the UK, you ’ ve got Labour for a full term. But the Reform Party is increasingly the most popular. And just with a little nudge, maybe a little external money, and Elon said he ’ s already thinking about giving them 100 million dollars, which is a massive amount in UK politics, not so much in the US, that the Conservatives would fragment, a rump group would join with Reform, and they could win the next elections. In France, you ’ ve had all of these governments continue to collapse. Macron, very unpopular. Marine Le Pen and the National Rally party could easily win upcoming elections in 2027. In 2029, in Germany, if the Germans are incapable of turning their economy around, incapable of unwinding their debt break and spending some of the capital that ’ s just sitting there and as their debt to GDP goes down, but their economy contracts at the same time, then AfD could easily win in 2029, and then the firewall is no more. And then, of course, in the European Union elections overall, in five years ’ time, now four and a half, you could easily see the Patriots front, which is the equivalent to the far right grouping that these parties are a part of, they could all come together and be the dominant party. And then the EU as we know it is really a thing of the past. So these are existential changes that are happening not just for NATO right now, but also for the EU and quite plausibly, for the future of democracy as we know it. HW : Alright, let ’ s turn to Ukraine. So President Trump accused President Zelenskyy of starting the war with Russia and called him”a dictator without elections.”Last week, senior American and Russian officials met in Riyadh to discuss a potential cease-fire with no Ukrainians present. Now Zelenskyy has offered to step down in return for Ukrainian admission to NATO, although apparently he doesn ’ t really mean this. So what happens to Ukraine now? What should we be paying attention to? IB : Well, I mean, I think he would mean it if NATO membership happened. I ’ d love to see Trump call his bluff on that. But of course it ’ s not going to. And he wasn ’ t saying that six months or a year ago. He ’ s saying it now that NATO is truly being pulled off the table. I think it ’ s the right thing for him to do. It is, again, performative and helps to remind those around the world that Ukraine is fighting for self-determination. They ’ re fighting for the ability to join their own alliance, as any country in the world should be able to do. But that is not Trump ’ s belief. Trump believes that what you get to do is determined by how powerful you are, not by the fact that you happen to run a country. And indeed, the territorial integrity is something that is afforded to powerful countries, but not so much for weak countries. And that ’ s the way he feels about Panama, and that ’ s the way he feels about Denmark and Greenland. And that ’ s certainly the way he feels about Ukraine. So right now, as you and I are speaking, the United States is putting forward a new UN resolution at the General Assembly that says that the war has to end as fast as possible but does not recognize Ukraine ’ s territorial integrity, which has been a core precept of the international order that the United States has supported since creating the United Nations and since the end of World War II in 1945. The Americans are now throwing that out. And the Russians, of course, are fully supportive of that. America ’ s allies in Europe are not. But the US has gotten the Saudis to agree, and they ’ re getting a unified Arab block. The Americans have privately worked with a lot of poorer countries in Africa, in the Asia Pacific and others, smaller countries, and saying,”You ’ re not going to get any aid from the US unless you vote in favor of this new resolution.”I fully expect the resolution is going to pass. But of course, what that means is that the Americans are preparing to cut a deal on Ukraine over the heads of the Ukrainians. In other words, the US and Russia together will figure out what a ceasefire should and should not entail as part of a broader US-Russia rapprochement. This is obviously a disaster for the Ukrainians and the Ukrainians who initially refuse to accept this so-called deal, on Ukrainian natural resources that would grant the Americans $500 billion in access to such resources in return for”paying off”aid that the Americans had granted to the Ukrainians without conditions or strings, but apparently no longer. And so, you know, what does this all mean? Well it means it ’ s yet another case of why people shouldn ’ t trust the United States from one administration to the next, because foreign policy will change completely. And what you thought was an American commitment won ’ t stand. It shows that there is a fundamental rift between the Americans and the Europeans and indeed, Emmanuel Macron in the United States right now, as you and I speak, hoping that he can do something that will keep Trump from, as Macron believes, essentially surrendering to, conceding to, the Russian position and maintaining some level of coordination with the Europeans. But he doesn ’ t have a lot to offer. And the Ukrainians, you know, now in a position of desperation where maybe they will end up signing over a lot of their natural resources to the Americans. But what, if anything, will they get in return for that? And how empowered will the Russians be in any deal that is cut? And will an independent Ukraine even be able to survive, and what will the terms that will be required of them? Of course, Putin is saying things like, no European **troops** of any sort, whether in a NATO formulation or not, would be allowed as peacekeepers in Ukraine. Well, then, how would you guarantee, what kind of security commitments would you give to the Ukrainians? These are all open questions, but apparently questions that President Trump is prepared to largely resolve on Russia ' s terms. And again, this is such a dramatic change from where the United States and where the Western order was just a month or two ago. HW : So let ' s take some quick-fire questions from our community related to this topic. And I think the first one is particularly pertinent to what you just said, so it ' s simple. What does Putin have over Trump? IB : I don ' t think that Putin"has"anything over Trump. If he did, then why wouldn ' t Trump have granted Russia more in his first term? In his first term he gave the Ukrainians those javelin anti-tank missiles that Obama refused to give. And, you know, he thought they were too risky. He increased sanctions against Russia. Now that was his administration with a lot of hawkish in orientation advisors around him that he doesn ' t have this time around. They ' re all directly loyal to him. But, I mean, if it was a priority for him, he could have stopped it. And if the Russians had something over him, then why wouldn ' t they have used it in that case? It feels like a conspiracy theory. I don ' t buy it. I think very differently. We can explain what ' s happening because number one, Trump wants to end wars. It ' s very popular with his base. He ' s said this on Gaza. He ' s said this on Iran, where Israel wants him to support attacks on Iran ' s nuclear capabilities, and Trump has said no. And he ' s showing this on Ukraine. He doesn ' t want to spend money on the Ukrainians going forward after spending over 100 billion dollars under the last three years from the Biden administration. He doesn ' t want to spend that money going forward. And he also doesn ' t value a strong Europe. He actually thinks that a weak Europe is in America ' s interests, where you have more Brexits. He used to always talk to the French about that during his first term,"Why won ' t you do Brexit?"He wants all of these MAGA-type populist parties, including the AfD, to win in Europe because that legitimizes his own popularity. Just like he loves Milei in Argentina or Bukele in El Salvador. And so I think if you put those things together, you actually get to why Trump is doing what he ' s doing with the Russians without needing to create a story about Trump somehow being threatened by some secret, shadowy information that the Russians have on Trump that they ' ve let him know about but nobody else knows. I just don ' t buy that. HW : Great. OK, another community question. If the US hands Ukraine to Russia, will Russia keep going? Will they attempt to take over Europe? IB : It ' s interesting that Trump just met with the Polish president, President Duda. It was supposed to be an hour meeting, it was only 10 minutes. So he kind of embarrassed him at home in Poland. But he did say that the US is still committed to maintaining a **troop** presence on the ground in Poland. And so it ' s very hard to imagine the Russians rolling through Ukraine and then into Poland, even though there ' s been lots of asymmetric warfare, for example, and even there have been, you know, Russian missiles launched into Lviv, Ukrainian air defense, missile defense, and with Polish villagers getting killed because of the fighting. So, I mean, there ' s been spillover. But America is still going to be in Poland. Why is that? They get along pretty well, and the Poles are heading towards five percent of GDP defense spend this year, which has been the high water mark of American demands of defense spend. So he ' s, you know, been consistent on that over the past months. Baltic states. I mean, they ’ re all spending a lot more money on their defense. And so, I mean, in principle, you could imagine the United States maintaining these rotational deployments in the Baltics, which would prevent the Russians from invading. But might Trump say, as part of a deal,”Why do we have all those **troops** there?”You mentioned Pete Hegseth saying the Americans need to get **troops** out of Europe and get them to Asia, which is where their principal competitive, you know, sort of, strategy is going to be oriented. And the US hasn ’ t been pressing the Japanese or the Indians, the principal partners, allies in Asia vis-à-vis China, the way they ’ ve been pressing the French, the Brits, the Germans in Europe. So I think it is certainly plausible that the Russians will do more, especially with those countries that refuse to align with a Trump rapprochement towards Russia. The countries I would be most worried about are Moldova, are Georgia. I ’ d be most worried about the countries that are in the so-called”near abroad”that aren ’ t a part of NATO, and aren ’ t going to become a part of NATO, and therefore are much more vulnerable. But certainly in terms of Russia ’ s willingness to engage in espionage, to fund actors on the ground for arson attacks, assassinations, critical infrastructure attacks on fiber networks, on pipelines, I think all of those things are likely to increase against European states across the board, on the back of this US-Russia rapprochement. HW : OK, final community question for this segment. What happens if the US no longer supports NATO? IB : It ’ s possible that the United States is moving towards a similar posture in Europe that they presently have in Asia, which is strong individual bilateral deals with a number of countries : South Korea, Japan, Australia, but not a collective security agreement, where those countries together have more of a call on the United States and there ’ s a lot more free riding. So I could imagine that NATO falls apart and becomes that. The danger is that even though the Europeans now understand the nature of the threat, that it is very plausible and maybe even baseline, that the Americans are going to leave Europe collectively on its own. That that seriousness does not equate to the ability to increase spending adequately in the near term. The Germans did manage to pull together an electoral outcome that will allow for a so-called grand coalition, just barely. So you ’ ll have a two-party government in all likelihood instead of a three. But you also have a blocking capacity on the part of the far right and the far left that will prevent the Germans from really blowing out their spending, including their defense spending, unless they can pull something off before that new government is formed. Which is possible, but it ’ s unlikely to be very big. The Brits are now talking about maybe moving the 2. 5 percent of GDP defense spend over years with a very challenging fiscal environment. The Spaniards, not even close, the Italians, not even close. So what does a European military capability really look like? And without the Americans, the answer is it ’ s not there. It ’ s not there. So I think that if NATO falls apart, the reality is that most of Europe will be incredibly vulnerable to external attack, especially Russian attack, And there won ’ t be a way for them to defend themselves, and some of them will break and therefore want to work more directly and individually with the United States. And that could be the end, not only of NATO, but it could be the end of the European Union. I think this is, the stakes are incredibly high, this is an existential challenge for European governments that needed to take this seriously 30 years ago, 20 years ago, certainly in 2014, when the Russians invaded Ukraine, certainly, certainly in 2016 when Trump was elected, and certainly, certainly, certainly in 2022 when the Russians then invaded all of Ukraine. And at every point the Europeans were basically saying,”We ’ re still fine with the Americans basically doing this.”And now they are in very serious trouble. HW : No time like the present, I suppose. Alright, let ’ s move on. So I mentioned the discussions in Saudi and obviously also on the agenda there with the discussions about Gaza and Trump ' s proposal to build a Gaza Riviera. What should we make of that? Or as one of our community members asked, what is the actual solution for Gaza? And surely it isn ' t what the US is proposing. IB : Well, look, here ' s what ' s interesting. You know, Trump says a lot of things, and when they don ' t work [out], he backs out fairly quickly. And you know, I thought – just stick with Ukraine for one second on this. It was interesting to me that Trump has said that he doesn ' t want to talk to Zelenskyy because he has no cards, he has no leverage. And so why should he let him at the table? And so Trump ' s going to cut the deal. Very interesting that if that is true, then why is it that the Secretary of Treasury brings this deal and says,"You ' ve got to sign this right now, Zelenskyy"And Zelenskyy refuses. And then a few days later, the Americans come back with a new deal with altered terms that are not quite as predatory vis-à-vis Ukraine. Why would the Americans change their tune for someone that has no leverage? Because, I don ' t think it ' s because Trump is a nice guy. He ' s not going soft at 78. I think it ' s because he overstated how little leverage Ukraine has. Ukraine is very popular among the GOP in the House and Senate, far more than Putin. Zelenskyy is far more popular in the United States among the voting population than Putin. And of course, he ' s also much more popular among the Europeans and even globally than Putin. And so it turns out that maybe there is a little bit of leverage that Zelenskyy has, and maybe Trump does have to give a little bit more. So this is relevant in the context of Gaza, because Trump was very annoyed that, you know, despite his ability to get a cease fire between Hamas and Israel over the goal line, that his friends in the Middle East were not coming up with a plan for Gaza. Where they were supposed to provide security and provide investment and lead reconstruction and deal with the Palestinian populations in the interim. And that wasn ' t happening. And so Trump then meeting with Prime Minister Netanyahu from Israel in Washington, then after that, summoning the Jordanian king, who was the ally that is most reliant on the United States and so, therefore, most needing to say yes to whatever Trump orders of him. Basically says, as you mentioned, Helen, that Gaza is going to be a Riviera, that the US is going to turn this into an incredible place. They ' re going to build wonderful homes for the Palestinians someplace else. And the Palestinians will all volunteer to leave Gaza and go to those homes, and Gaza will be depopulated and new people will come in and live there. That was the plan. And when he was asked, standing there with King Abdullah from Jordan, under whose authority? Trump ' s response was :"Under my authority."And that went over like a lead balloon among America ' s allies in the Middle East. They all said,"We ' re not up for this. We ' re not resettling Palestinian populations. They won ' t actually leave voluntarily. This will would actually be a huge problem for Israel ' s own national security long-term. And we ' re going to come up with another plan. And our plan will keep the Palestinians on the ground in Gaza."At which point Trump pivots to work with the new plan. And his advisers say that Trump was only trying to start the conversation. And what he said was his plan is not actually what he ' s going to do. So, what ' s sustainable? What would be sustainable would be a lot of Gulf money going in and maybe some European money too, though they ' re going to be really pressed, given everything you and I just talked about, to try to reconstruct a completely devastated Gaza that eventually some two million-plus Palestinians will be able to live normal lives in, and the governance will be provided by some technocratic group of a Palestinian Authority that will be working with, selected by, those that are involved in the reconstruction and the security. Namely the Egyptians, the Jordanians, the Saudis and the Emiratis. That is the best possible deal that they ' re going to get. But that ’ s happening as Israel has just evacuated forcibly another 40, 000 Palestinians from refugee camps in the West Bank, sending tanks in for the first time in decades and probably precipitating more Israeli settlers taking more land on the ground in the West Bank. And so, as we are possibly taking a small step towards a sustainable solution in Gaza, we are taking a slightly larger step away from a sustainable solution for the Palestinians in the West Bank, twas ever thus over the past decades in this incredibly horrible problem and conflict. HW : And why is Israel doing that? I mean, are they feeling emboldened by the US, or why the incursions in the West Bank now? IB : Well, I think they ’ re feeling emboldened by their own successes. They have shown that they are the dominant military power in the Middle East, and they have the ability to determine the level of escalation unilaterally, and their adversaries and enemies can ’ t really do any damage to them. They proved that after October 7, in their response to Hamas. They proved that in decapitating and taking out the military capabilities of Hezbollah. And they ’ ve proved that also with overturning – they didn ’ t do it, but the fact that the Assad regime in Syria is gone, which means that the Iranians no longer have a land bridge to get additional weapons to what had been the strongest adversary of Israel militarily, Hezbollah. So all of that has shown that Israel can decide outcomes. They ’ re the ones that have all the power here. And also that depopulation of Gaza is very popular among Israelis. There was a poll recently in the Jerusalem Post, which is pretty mainstream in Israel. 80 percent of Israeli respondents said that they wanted to depopulate Gaza of all Palestinians. Taking more land is very popular, especially in the West Bank, especially among Netanyahu ’ s present right-wing coalition, that he would like to keep together, to continue to govern. And this is a sop to them as he is looking to potentially extend the ceasefire in Gaza and move from phase one to phase two, which is facing some delays and open questions right now. So, yes, certainly, the United States has been underpinning that military capability of Israel. And I think it is instructive to remember that not only does the US provide more military aid in peacetime to any country in the world, to Israel, but that unlike in Ukraine, where the Trump administration is saying,”No, I ’ m not happy with the grants, you ’ ve got to pay that back,”The billions and billions that the United States sends to Israel, nobody is suggesting that that should be alone or that the Americans should get technology rights or mineral rights or anything else from Israel, despite the fact that Trump ’ s policy is America First. Israel is a very clear exception to America First in Trump ’ s visioning of it. HW : Alright, so we talked about the fact that the foreign policy focus of America is shifting to Asia. So what are you hearing about how China is feeling in all of this? And what should we be paying attention to on that front? IB : China has had a couple of good weeks economically and specifically on the back of that huge announcement of DeepSeek, which has, you know, far less money behind it than the comparable American chatbots but is performing at an extremely high level. And, you know, given the fact that there are all of these export controls on semiconductors and other parts of advanced technology, that this is giving people more of a belief that the Chinese really are capable of driving innovation even in that space. And they ’ re, of course, dominating the post-carbon energy space. We see that with electric vehicles and batteries. But we ’ re also seeing that with massive investment now in green hydrogen, for example, we ’ re seeing it in advanced nuclear capabilities, all of these things. And so there is more investment in that space that the Chinese are not only driving as a state, but that they ’ re also increasingly able to raise and move in their own private sector. So the story that you and I have been discussing for the last couple of years is the Chinese economy is radically underperforming. That is still true, but there is an emerging counter-narrative here that I would be remiss not to mention. Now more broadly, what the Chinese here are seeing is that Trump ’ s direct policies are going to hurt them economically. His tariffs that he ’ s already announced on China, the 10 percent, on a whole bunch of Chinese goods for export, the reciprocal tariffs that are global, that will clearly hurt China and that they ’ ll have a hard time negotiating out of, and also the willingness of the United States to expand their export controls, their sanctions on China, especially in areas that are considered relevant, even loosely relevant to American national security. All of that is going to constrain Chinese growth. All of that is going to negatively impact the Chinese economy. That ’ s the downside. And the Americans are doing that not only in bilateral relations with China but also pressuring other countries to get Chinese tech out of their supply chain. And you see that with Americans pushing other countries that produce semiconductors to take the Chinese out. They ’ re pushing American allies and trade partners to get China out of the export chain. So Mexico being pushed very hard by Trump to stop allowing China to pass goods through Mexico into the United States. India, same conversation. Vietnam, same conversation. A lot of that going on. So baseline, the US-China relationship is getting worse. But the Chinese see huge advantages in America ’ s reversion to unilateralism that we haven ’ t really seen since the original America First movement in the run up to World War II. Charles Lindbergh saying,”Don ’ t get into the war, don ’ t fight the Nazis. The US is far away. This isn ’ t our our problem.”And now you have the Americans not only doing that with Russia and Ukraine with the new America First, but you also see that happening with the Americans pulling out of the Paris Climate Accord, again, pulling out of the World Health Organization. You see it with the Americans shutting down USAID. And America has historically been responsible for about 40 percent of global humanitarian support and aid. And absolutely, some of that is corrupt. Absolutely, some of that is on woke programs that the vast majority of American taxpayers would never support. But the majority of that money is actually spent on programs that are really important in advancing American soft power and influence over countries in the Global South, all over the world. The Chinese know that. That ’ s why they ’ ve started humanitarian programs. It ’ s not out of some great love for, you know, providing foreign aid for these countries. It ’ s more because they see it creates influence. Now the Americans are leaving a vacuum that the Chinese can absolutely take advantage of. And we ’ ll see this across, you know, sort of, microstates in the Pacific. We ’ ll see this across sub-Saharan Africa. We ’ ll see this across South America. The Chinese see huge opportunities from the United States creating a leadership vacuum. And the place they ’ re most capable of doing that is if the US stops paying dues in the United Nations, where China is number two, and they ’ ll suddenly have the most influence over all the key positions, which will give them a role in global governance that they ’ ve really been constrained from having over the past decades. HW : What do you think are the chances that the US stops paying its dues to the UN? IB : You know, if it were up to Congress and only Congress and Trump doesn ’ t lean on them, I ’ d say fairly low. I ’ d say that they ’ d probably, you know, cut back on some programs. But in terms of baseline dues, they ’ d keep paying it. But Trump personally, I think increasingly sees the UN as hostile to the United States. That ’ s particularly true in terms of Israel policy, which Trump is leaning in on. And, you know – let me put it this way. Tulsi Gabbard did not have the votes to get confirmed. There were Republican senators that were going to vote against her. And Elon called those individual senators and threatened them. And this was, you know, fully aligned with what Trump wanted Elon to do. And those votes went away. And the only person that ended up voting against Tulsi was

McConnell. And as a consequence, it went through. So look, I think that we should not underestimate the level of power and control that Trump and Elon have over the Republican Party this time around, compared to 2017, where it was really the Republican Party that was much stronger than Trump, both in his administration creating a lot of guardrails, but also in Congress. This time around, Trump has all the chips and he ’ s using them. So if he decides he wants to stop paying dues, I think that ’ s what ’ s going to happen. This is a much more revolutionary presidency, domestically, where last time around it was much more transactional. HW : Alright. Let ’ s talk a little bit more about what ’ s actually happening inside the US. You bring up Elon, who obviously with DOGE has been sending out emails and asking for information from people and kind of trying to do a lot very quickly. Some of these high-profile moves are – I think what ’ s interesting is that no one would really argue that there isn ’ t inefficiency in government, that that ’ s an understood reality. But the way that some of these moves are being made are maybe going to cause irreparable damage. And that ’ s to people from red states, that ’ s to scientists, that ’ s to veterans, that ’ s to people who voted for Trump and that ’ s to people around the world. You talk about soft power and the United States, but what about actually what ’ s happening inside the United States? And why do you think that the administration is willing to go so quickly, to go so fast and to cause such damage? IB : Well, and you saw Trump this weekend saying that he wants Elon to move faster, to be more aggressive. And I don ’ t think that was just a troll. I think that ’ s true. Trump is 78 years old, and with his friends he talks about the fact that he doesn ’ t have much time. That in two years ’ time, the Democrats could take the House in four years ’ time, you know, that ’ s it. He ’ s not talking privately of”I ’ m going to be president for life.”He also knows he was almost killed. He was shot in the head. And do I think that changes somebody? Absolutely. I think that he believes that he was saved by God, and that his level of confidence that what he is doing is fundamentally right and necessary now is so much higher than the insecurity that I frequently saw in him in his first presidency. So I think his willingness to push, push, push, and his view that Elon is a great activator and executor on that intention, and someone that he had nowhere close to someone with that kind of capability and energy and execution ability in his first term. And now he does. So first of all, I think that relationship between Trump and Elon isn ’ t going anywhere. I think it ’ s actually very stable. People that say it is are people that want it to go away. But hope is not a strategy. And I also think that the intention of revolutionary politics domestically is to shoot first and ask questions later. It ’ s to break things. It ’ s a view that the bureaucracy has been weaponized against Trump first and foremost, and also against the will of the American people. And therefore it must be broken. And you ’ re going to break eggs when you want to make that omelet. Now your point is that there are a lot of Republicans that work in government. There are a lot of Republicans and Trump voters that are losing their jobs and that are not being treated with a level of care and compassion as a result of what Trump and Elon are doing. And I think there will be a backlash. But I ’ m not so sure that a one-term, 78-year-old-Trump presidency cares all that much about that. And, you know, if that means that his popularity slips to like, the high 30s, frankly, you know, in a very, very polarized country, high 30s isn ’ t that bad. It still means 80, 90 percent of the people that voted for him still support him, and I think he ’ s probably more comfortable with that this time around. So I think a lot more has to happen before we can start talking about whether there ’ s really going to be significant pushback against what Trump is trying to accomplish at home. HW : Do you expect to see any of Congress standing up to Trump? We saw the governor of Maine recently kind of take him on. Do you expect to see more of that, or do you feel like for the next two years at least, people will just go where Trump takes them? IB : Well, sure, but, you know, in the case of Maine, that ' s a Democrat. So I don ' t know how much that means. I don ' t expect to see a lot of Republicans standing up against him. And of course, since the Republicans control the House and the Senate, that ' s the only thing that really matters. I mean, you know, you saw Kash Patel got through, Pete Hegseth got through. Now look, I actually think that a lot of the members of cabinet are very capable. I mean, they ' ve been picked for their loyalty and they will be loyal to Trump, but they ' re very capable. I think JD Vance is very capable. Mike Waltz, I know quite well. Marco Rubio, over the years. These are people that are very well respected, have been by their colleagues and could have served under any Republican administration. But there are some people that are completely incapable for the roles that they have been selected, completely incapable. They would not be selected in a role in any cabinet, in any democracy. And yet here they are, serving in critical roles in the most powerful country in the world. And yes, I ' m talking about the Secretary of Defense. I ' m talking about the Secretary of Health and Human Services. I ' m talking about the Director of National Intelligence. I ' m talking about the Director of FBI. These are important roles. And they were all confirmed. And if you ' re a Republican senator – and senator – I ' m not talking about the House here where, you know, it ' s a lot more diffuse and there are a lot more wackos and it ' s easier to get in. But Senate has always been like, responsible and more oriented towards bipartisanship and the rest, and they are utterly petrified of what it means if they publicly get crosswise with President Trump and number two, Elon. And they ' re not willing to do it. So I think that there ' s very little belief that you are going to see Congress step up. I think that you will see justices step up because those justices are independent. The problem is that many of those justices are progressive and have a known progressive lean. And so when cases come up and justices that have that political lean oppose something Trump wants, his willingness, his administration ' s willingness to attack them and refuse to execute on that judicial decision, unconstitutionally, is real. Now the case will of course go then up to the Supreme Court eventually. And I don ' t feel the same way about Trump refusing to listen to adhere to the rulings of the Supreme Court. But if there ' s 100 rulings like that and you overwhelm the judicial process, you quickly see how you might erode a core check on executive power in the United States. And Trump and Elon clearly intend to every day test those checks and balances and break them if they can. I mean, that is the intent, that is the intent. HW : So you mentioned democracy. When we last spoke, you said very clearly that the US is not Hungary, meaning that democracy is not threatened and that the US is not in danger of turning into an autocracy. Do you stand by that? IB : Yes, I do. I mean, in the sense that I think in two years we ' re going to have midterm elections, and they will be largely free and fair. And in four years we ' ll have presidential elections. And US elections are held at the federal level, right? They ' re held state by state. They make the rules. It ' s very easy to make people believe that the election has been stolen. It ' s extremely hard to actually rig an American election. And I don ' t believe that that is very likely at all. It ' s very unlikely. The American military, despite the firings that we ' ve seen just last Friday, is still an independent and professional military that I believe will carry out its orders and its duties professionally. I see the judiciary in the United States as independent and a clear check on the executive power in the United States. Where I see a problem, is that the US is becoming more kleptocratic every day. And the US was already much more of a kleptocracy over the past 10 years under Democrats and Republicans than any other advanced industrial democracy in the world. Clearly, money buys power in America in a way that it doesn ’ t in any other major democracy, wealthy democracy. That is getting dramatically worse. I saw specifically that Elon Musk, who is a walking conflict of interest at the highest level, met with Narendra Modi, the Prime Minister of India, in Blair House, so part of the official White House complex, right after Trump met with him. And Trump was asked later in the day, was Modi meeting with Elon in Elon ’ s private-sector capacity to advance his business interests or his official capacity to advance the policies of the United States? And Trump answered honestly. He said,”I don ’ t know.”And how could he know, right? Because, I mean, those two things are completely intertwined. But that is not remotely acceptable for rule of law in a functional democracy, and yet nothing is being done about it. In fact, it ’ s getting more and more entrenched every day. So the single place where the United States was least functional as an advanced democracy is now getting far, far worse. And I don ’ t see any checks and balances on that. I think that ’ s a very, very serious problem. HW : Alright, Ian, thank you, as always for all of your insights. Let ’ s close out with a couple more questions from our community. And this one indeed relates to the kleptocratic conversation that you were just raising. So why are the billionaires enabling fascism? What do they get out of it? IB : Fascism is a very politically loaded argument. That ’ s a political system where the cruelty against a subgroup or a perceived subgroup in the society is the intention, the expressed intention of the policy. And I certainly wouldn ’ t define the United States as a fascist system today. But I do believe that billionaires in the United States, at least a significant subset of them, are really under-interested in the well-being of their fellow countrymen and women. And this is a country that really makes heroic the individual. And it undermines the community. We ’ ve seen a significant erosion of civic engagement in the United States and civic associations. The church is not what it was in the US. Fewer people go, fewer people trust the church. Public education is not what it was. And wealthy people increasingly opt out. The family is not what it was. It ’ s increasingly atomized and people spend much more, much more of their time being, you know, isolated and atomized and intermediated by algorithms. And you and I have spoken about that problem in the past. We are increasingly a society of individuals as opposed to a nation together. I will tell you that that is a real problem when you have large numbers of billionaires that believe that they built it themselves, that they ’ re not responsible to a collective whole for any of their success and therefore they ’ re not accountable for their fellow citizens. And that ’ s what I think we have. I think we have a lot of winners in the United States and not a lot of leaders. And when you have winners, you have losers. When you have winners, you don ’ t have a collectivity. I will say that in this anti-DEI thing, I have a lot of sympathy for the people that don ’ t like DEI. I thought it was rammed down the throats of a lot of people And I ’ ve said this before, it was well beyond what the average American was willing to tolerate, and also it came with a level of high-handedness that that basically said, if you ’ re not with us in supporting these programs, you ’ re evil, you ’ re stupid, you ’ re uneducated. That ’ s unacceptable, too. It was a completely, you know, sort of oppositional way to try to have what ’ s a very challenging conversation. But I ’ ve never been a champion of diversity for diversity ’ s sake. What ’ s amazing about the history of the United States is despite all of our diversity, we find commonalities with each other. We find connectedness. That ’ s what ’ s amazing, is the connectedness, is that, you know, on this world, we ’ ve got eight billion people. And despite how different we all are, we all actually connect as human beings. That is what we need to focus on, not the diversity, the connectedness. And I think that ’ s what the billionaires today in the United States are losing. They ’ re doing incredibly well themselves. They ’ ve got theirs, they ’ ve got access to power, they ’ re paying good money for it, they ’ re going to get even more. But they ’ re forgetting that for us to succeed as a society and as a planet, we have to be connected to each other. You can ’ t throw out diversity and forget connectedness, and that ’ s what we ’ re forgetting right now. That ’ s why the country feels so much trauma and pain right now. And I think why the United States has given up on a lot of its really aspirational values that made such a difference to the world, that almost lost World War II for many generations. And now we ’ re giving it all away. HW : I mean, I think that ’ s why you find that diversity and inclusion go hand in hand, that that ’ s actually speaking to that. But in some of the moves that are being made to kind of throw out any type of diversity, to scan documents, to find words in order to highlight programs that are wasteful, that we ’ re actually going to lose what many, many studies actually highlight as being the benefits of diversity and the fact that you need diversity in order to be actually successful. IB : Shooting first and asking questions later, you know, moving fast and breaking things, those things make sense in a turbocharged venture capital technology environment. They do not work in government. They don ’ t work in government at all. In government, we have something called checks and balances, and we need them. And moving fast and breaking things undermines those checks and balances. So we ’ re going to do a lot of damage unintentionally. A lot of people are going to get hurt. A lot of folks don ’ t care about that in power right now, but they should because it would be much, much better. I know people want to move fast, but actually to think a little bit, do a little bit of your own research before you decide to kill that program, before you decide to fire that person. We ’ d be in a lot better shape as a government right now. HW : OK, the final question that I have for you actually comes from my 10-year-old son, and it ’ s a bit of a miserable question. We have a very lovely house, I promise. But his question nonetheless is are we heading into World War III? So, Ian, what should I tell him? IB : No, no, I don ’ t think we ’ re heading into World War III but we are absolutely heading into a world of much greater conflict. I think it ’ s much more likely, for example, in the next five, 10 years, that we ’ ll have a nuclear weapon go off. And I thought that was pretty unlikely since the Cuban Missile Crisis. In other words, over the course of my life. I think we ’ re going to see a lot more terrorism at scale, because a lot more people are going to feel angry and disaffected, I think. You know, you saw that assassination of the CEO of UnitedHealthcare just a few dozen blocks from my house. I think you ’ re going to see more things like that. And so, no, not World War III because the fact is that outside of the US and China, all the other countries know that they need to work with everybody. And there are even forces inside US-China that are trying to make sure that you don ’ t throw the baby out with the bathwater. But underneath that World War III risk, there ’ s an awful lot of damage that ’ s going to get done, an awful lot of crisis that I think we ’ re going to have to live through before we start to see what we create on the other side. A lot of defense as well, that we ’ re going to be playing in this environment. HW : Well, Ian, it is always a pleasure to talk to you, and we ’ re so glad that you are scanning the world to understand what is happening and then you come here and explain it to us all. Couldn ’ t be more grateful and thank you for everything. IB : Thank you.

Text Sample 883: POS Dim 6 – 2025 – Score 1.00 – 2025/t003001.txt

- I don ’ t remember exactly, but... I use the most expensive hair color in the world. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color. I expect great color. What ’ s worth more to me is the way my hair feels. Feels good around my neck. I can ’ t remember it. I can ’ t finish it all. If only we had more time. - Why don ’ t we have more time? - I ’ m dying. I ’ m croaking! - It ’ s tough, coming home. But this is what you do for a parent that you love. But, you know? She ’ s not my mother. Technically. In my own family, there ’ s a lot of dysfunction, a lot of mental illness, a lot of substance abuse. Every generation. My parents divorced when I was about five. My mother moved us out of state, so we had to fly to New York to see my father ’ cause he was already working in advertising, and New York was the only place to be. - Many of the commercials you see on television begin life here. - Advertising is one of the main sources of messaging about who you should be. - Nowadays, most of you women lead two lives, combining homemaking with the role of glamorous hostess. - And in the ’ 60s, the ads were often made from a male point of view. - In the pre-production meeting, everybody who has anything to do with the commercial is there. - It was mostly men. These superstar art directors and writers. Like my father, Gene Case.”Mad Men”was like home **movies**, and he was Don Draper. He was very witty, but I don ’ t think my father realized how powerful his words were. He knew it professionally. But with his kids, he could be incredibly cutting. Flay you just with a phrase. But it ’ s the time with my mother that was the nightmare. She had a horrible childhood, but she sure as hell didn ’ t make any effort to overcome that in her relations with us. You know, it was really horrifying to my mother that I was not popular. But I was also really awkward because she had been undermining my confidence for all my life. She would pick apart your personality in detail. I was lazy, I looked bad, I acted bad, I was awkward. Emotionally, just out for blood. At a certain point in the night, I would know,”OK, the situation is dangerous. I need to get outta here.”I had this fantasy when I was in fourth grade that I vividly remember that some kind of outside agency was gonna come in and determine from all the salt on my pillow that I was crying myself to sleep every night and do something about it. No, it was awful. It was awful. And that was the point where Ilon came into my life. - Ilon, do you need anything to eat? - I don ’ t think so, but look up this wallpaper, Waverly Fabrics, and see if they have a rose pattern. - OK. Oh, what about this one here? - That ’ s not it. - No? - No, first of all, I was an art dir... I was a creative director at an advertising agency. The first thing I know is what I ’ m looking at. Ilon the girl. I don ’ t remember particularly being into advertising, but somehow I found myself in it. And I seemed to be very good at it. First of all, I ’ d always listened to everybody. And I noticed that I saw things differently than other people. - Women could get somewhere in advertising, but they were marginalized. - Women were given a lesser role. All the men were always arguing with you and always taking credit. - And one of the things that was coming through loud and clear is that women are for men. - Glory, glory, hallelujah Glory, glory, hallelujah Glory, glory, hallelujah His truth is marching on - Men were always the ones who everyone was pleasing. - What will your eyes say to him tonight? Wait till the fellas take a look at this plunging backline. - You look at ads from that period, and it ’ s jaw-dropping. - This is one of our nation ’ s most beautiful sights, the kind of girl who keeps her figure. - But that ’ s what feminism was pushing back against. - I was a feminist. I knew women were equal if not superior to men from the time I was a little girl. - And in advertising, she wanted to be a trailblazer that way. At one point, she worked in the same agency with my father, and that ’ s how they met. - He was a good writer, and I was a good writer. I liked the fact that he was good at what he did and cared about it. We got engaged, and then we got married. - I ’ m sitting there in the background. I ’ m grinning. I ’ m so thrilled because I adored her instantly. I thought she was super cool. And she treated me like I mattered. - She was a sweet little girl, and she deserved some caretaking. My mother was kinda terrible, so I knew that she needed support, and I gave her support. And that was what we did. - She was interested in me. She listened to me. Treated me like I was lovable. - She didn ’ t have anyone to recognize her being, to support her, to see who she could be, who she was. - She was very interested in what I would do with my life, and the assumption was that it would be something worthwhile. She just went way, way beyond the call of duty to make me feel like I was worth it. And she did that for a lot of other women, too. - I was working at McCann Erickson. We were given an assignment. L ’ Oréal, they wanted a campaign for Preference, it ’ s a hair color. - She ’ s in this room full of men brainstorming for this ad. - The men were busy flirting with the women, telling us we all looked like models and we didn ’ t look like advertising people, you know? - They ’ re like,”Oh, this woman is gonna be standing by an open window, and the wind will be blowing the curtains.”And she ’ s thinking, you know, this is just making the woman totally an object. - I was feeling angry. I ’ m not interested in writing anything about looking good for men. Fuck ’ em. Fuck you too. - Why fuck me? - Well, you ’ re a man. - Yeah, but I ’ m here trying to help you tell your story. - Yeah, that ’ s good. I like that part. I was pissed. We weren ’ t there to just dance for the men. - It is what she said. She just thought,”Fuck you.”And wrote this. - I just opened my eyes and knew. Because I ’ m worth it. The campaign got approved, but all the men were always telling you that you were doing something wrong or they knew how to do it better. - The agency insisted on a male voiceover. - She uses the most expensive hair color in the world, Preference by L ’ Oréal. - You know, it was just totally different. - Smooth and silky, but with body. It feels good against her neck. Actually, she doesn ’ t mind spending more for L ’ Oréal. Because she ’ s worth it. - It was wrong. This was not for men but for women and for other human beings. - I use the most expensive hair color in the world, Preference by L ’ Oréal. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color, I expect great color. What ’ s worth more to me is the way my hair feels. Smooth and silky, but with body. It feels good against my neck. Actually, I don ’ t mind spending more for L ’ Oréal. Because I ’ m worth it. - I wrote that. - It was everywhere. It was a national campaign. - Yes, it ’ s expensive, but I ’ m worth it. - What they realized after they started running it was that that was only the tip of the iceberg. - It was supporting women, finally. - And I ’ m worth it. And I ’ m worth it. I ’ m worth it. - It ’ s a four-word feminist manifesto, defying this social pressure to think about yourself in relation to men. - You ’ re worth it. I ’ m worth it. I ’ m worth it. - It was a very big success. - At a certain point, L ’ Oréal adopted it as the tagline for their entire business. - Parce que je le vaux bien. - Even today, that ’ s their motto. - Because I ’ m worth it. - ’ Cause you ’ re worth it. - You ’ re worth it. - We ’ re worth it. - And I ’ m worth it. - She wrote the most influential tagline in women ’ s beauty advertising. - I ’ m worth it. - It ’ s magic, that phrase. - Those words just get better with age, don ’ t they? - I ’ m worth it. Because you are and always will be. - She came up with that line because she believed it. She believed it about herself, and she believed that women should believe it. She did that for me. You know, even after they split up, she was still a big part of my life. I ’ ve always kind of felt like I dodged a bullet that I have watched take down other members of my family. She saved me. And today, I am a professor of English at Williams College. I teach Victorian literature, and I also teach women ’ s, gender and sexuality studies. She was my mother. She also taught me how to be a mother. I had been very uncertain about motherhood for myself because my role model for that was so nightmarish. But Ilon really taught me, I can do this. I can be a good mother. She made me believe that I was worth it. And I was not either the first or the last. - I have to go away now. Go to my room. - It ’ s been lovely. - It ’ s been lovely. Oh... I ’ m not interested in advertising. I don ’ t give a shit. It wasn ’ t like it was my whole life. It was just a portion. It ’ s about humans ; it ’ s not about advertising. It ’ s about caring for people because... we ’ re all worth it, or no one is worth it.

Text Sample 884: POS Dim 6 – 2017 – Score 5.00 – 2017/t000541.txt
We know more about other planets than our own, and today, I want to show you a new type of robot designed to help us better understand our own planet. It belongs to a category known in the oceanographic community as an unmanned surface vehicle, or USV. And it uses no fuel. Instead, it relies on wind power for propulsion. And yet, it can sail around the globe for months at a time. So I want to share with you why we built it, and what it means for you. A few years ago, I was on a sailboat making its way across the Pacific, from San Francisco to Hawaii. I had just spent the past 10 years working nonstop, developing video games for hundreds of millions of users, and I wanted to take a step back and look at the big picture and get some much-needed thinking time. I was the navigator on board, and one evening, after a long session analyzing weather data and plotting our course, I came up on deck and saw this beautiful sunset. And a thought occurred to me : How much do we really know about our oceans? The Pacific was stretching all around me as far as the eye could see, and the waves were rocking our boat forcefully, a sort of constant reminder of its untold power. How much do we really know about our oceans? I decided to find out. What I quickly learned is that we don ’ t know very much. The first reason is just how vast oceans are, covering 70 percent of the planet, and yet we know they drive complex planetary systems like global weather, which affect all of us on a daily basis, sometimes dramatically. And yet, those activities are mostly invisible to us. Ocean data is scarce by any standard. Back on land, I had grown used to accessing lots of sensors — billions of them, actually. But at sea, in situ data is scarce and expensive. Why? Because it relies on a small number of ships and buoys. How small a number was actually a great surprise. Our National Oceanic and Atmospheric Administration, better known as NOAA, only has 16 ships, and there are less than 200 buoys offshore globally. It is easy to understand why : the oceans are an unforgiving place, and to collect in situ data, you need a big ship, capable of carrying a vast amount of fuel and large crews, costing hundreds of millions of dollars each, or, big buoys tethered to the ocean floor with a four-mile-long cable and weighted down by a set of train wheels, which is both dangerous to deploy and expensive to maintain. What about satellites, you might ask? Well, satellites are fantastic, and they have taught us so much about the big picture over the past few decades. However, the problem with satellites is they can only see through one micron of the surface of the ocean. They have relatively poor **spatial** and temporal resolution, and their signal needs to be corrected for cloud cover and land effects and other factors. So what is going on in the oceans? And what are we trying to measure? And how could a robot be of any use? Let ’ s zoom in on a small cube in the ocean. One of the key things we want to understand is the surface, because the surface, if you think about it, is the nexus of all air-sea interaction. It is the **interface** through which all energy and gases must flow. Our sun radiates energy, which is absorbed by oceans as heat and then partially released into the atmosphere. Gases in our atmosphere like CO2 get dissolved into our oceans. Actually, about 30 percent of all global CO2 gets absorbed. Plankton and microorganisms release oxygen

into the atmosphere, so much so that every other breath you take comes from the ocean. Some of that heat generates evaporation, which creates clouds and then eventually leads to precipitation. And pressure gradients create surface wind, which moves the moisture through the atmosphere. Some of the heat radiates down into the deep ocean and gets stored in different layers, the ocean acting as some kind of planetary-scale boiler to store all that energy, which later might be released in short-term events like hurricanes or long-term phenomena like El Nio. These layers can get mixed up by vertical upwelling currents or horizontal currents, which are key in transporting heat from the tropics to the **poles**. And of course, there is marine life, occupying the largest ecosystem in volume on the planet, from microorganisms to fish to marine mammals, like seals, dolphins and whales. But all of these are mostly invisible to us. The challenge in studying those ocean variables at scale is one of energy, the energy that it takes to deploy sensors into the deep ocean. And of course, many solutions have been tried — from wave-actuated devices to surface drifters to sun-powered electrical drives — each with their own compromises. Our team breakthrough came from an unlikely source — the pursuit of the world speed record in a wind-powered land yacht. It took 10 years of research and development to come up with a novel wing concept that only uses three watts of power to control and yet can propel a vehicle all around the globe with seemingly unlimited autonomy. By adapting this wing concept into a marine vehicle, we had the genesis of an ocean drone. Now, these are larger than they appear. They are about 15 feet high, 23 feet long, seven feet deep. Think of them as surface satellites. They ’ re laden with an array of science-grade sensors that measure all key variables, both oceanographic and atmospheric, and a live satellite link transmits this high- resolution data back to shore in real time. Our team has been hard at work over the past few years, conducting missions in some of the toughest ocean conditions on the planet, from the Arctic to the tropical Pacific. We have sailed all the way to the polar ice shelf. We have sailed into Atlantic hurricanes. We have rounded Cape Horn, and we have slalomed between the oil rigs of the Gulf of Mexico. This is one tough robot. Let me share with you recent work that we did around the Pribilof Islands. This is a small group of islands deep in the cold Bering Sea between the US and Russia. Now, the Bering Sea is the home of the walleye pollock, which is a whitefish you might not recognize, but you might likely have tasted if you enjoy fish sticks or surimi. Yes, surimi looks like crabmeat, but it ’ s actually pollock. And the pollock fishery is the largest fishery in the nation, both in terms of value and volume — about 3. 1 billion pounds of fish caught every year. So over the past few years, a fleet of ocean drones has been hard at work in the Bering Sea with the goal to help assess the size of the pollock fish stock. This helps improve the quota system that ’ s used to manage the fishery and help prevent a collapse of the fish stock and protects this fragile ecosystem. Now, the drones survey the fishing ground using acoustics, i. e., a sonar. This sends a sound wave downwards, and then the reflection, the echo from the sound wave from the seabed or schools of fish, gives us an idea of what ’ s happening below the surface. Our ocean drones are actually pretty good at this repetitive task, so they have been gridding the Bering Sea day in, day out. Now, the Pribilof Islands are also the home of a large colony of fur seals. In the 1950s, there were about two million individuals in that colony. Sadly, these days, the population has rapidly declined. There ’ s less than 50 percent of that number left, and the population continues to fall rapidly. So to understand why, our science partner at the National Marine Mammal Laboratory has fitted a GPS tag on some of the mother seals, glued to their furs. And this tag measures location and depth and also has a really cool little camera that ’ s triggered by sudden acceleration. Here is a **movie** taken by an artistically inclined seal, giving us unprecedented insight into an underwater hunt deep in the Arctic, and the shot of this pollock prey just seconds before it gets devoured. Now, doing work in the Arctic is very tough, even for a robot. They had to survive a snowstorm in August and interferences from bystanders — that little spotted seal enjoying a ride. Now, the seal tags have recorded over 200, 000 dives over the season, and upon a closer look, we get to see the individual seal tracks and the repetitive dives. We are on our way to decode what is really happening over that foraging ground, and it ' s quite beautiful. Once you **superimpose** the acoustic data collected by the drones, a picture starts to emerge. As the seals leave the islands and swim from left to right, they are observed to dive at a relatively shallow depth of about 20 meters, which the drone identifies is populated by small young pollock with low calorific content. The seals then swim much greater distance and start to dive deeper to a place where the drone identifies larger, more adult pollock, which are more nutritious as fish. Unfortunately, the calories expended by the mother seals to swim this extra distance don ' t leave them with enough energy to lactate their pups back on the island, leading to the population decline. Further, the drones identify that the water temperature around the island has significantly warmed. It might be one of the driving forces that ' s pushing the pollock north, and to spread in search of colder regions. So the data analysis is ongoing, but already we can see that some of the pieces of the puzzle from the fur seal mystery are coming into focus. But if you look back at the big picture, we are mammals, too. And actually, the oceans provide up to 20 kilos of fish per human per year. As we deplete our fish stocks, what can we humans learn from the fur seal story? And beyond fish, the oceans affect all of us daily as they drive global weather systems, which affect things like global agricultural output or can lead to devastating destruction of lives and property through hurricanes, extreme heat and floods. Our oceans are pretty much unexplored and undersampled, and today, we still know more about other planets than our own. But if you divide this vast ocean in six-by-six-degree squares, each about 400 miles long, you ' d get about 1, 000 such squares. So little by little, working with our partners, we are deploying one ocean drone in each of those boxes, the hope being that achieving planetary coverage will give us better insights into those planetary systems that affect humanity. We have been using robots to study distant worlds in our solar system for a while now. Now it is time to quantify our own planet, because we cannot fix what we cannot measure, and we cannot prepare for what we don ' t know. Thank you.

Text Sample 885: POS Dim 6 – 2017 – Score 4.00 – 2017/t001083.txt

Chris Anderson : Elon, hey, welcome back to TED. It ' s great to have you here. Elon Musk : Thanks for having me. CA : So, in the next half hour or so, we ' re going to spend some time exploring your vision for what an exciting future might look like, which I guess makes the first question a little ironic : Why are you boring? EM : Yeah. I ask myself that frequently. We ' re trying to dig a hole under LA, and this is to create the beginning of what will hopefully be a 3D network of tunnels to alleviate congestion. So right now, one of the most soul-destroying things is traffic. It affects people in every part of the world. It takes away so much of your life. It ' s horrible. It ' s particularly horrible in LA. CA : I think you ' ve brought with you the first visualization that ' s been shown of this. Can I show this? EM : Yeah, absolutely. So this is the first time — Just to show what we ' re talking about. So a couple of key things that are important in having a 3D tunnel network. First of all, you have to be able to integrate the entrance and exit of the tunnel seamlessly into the fabric of the city. So by having an elevator, sort of a car skate, that ' s on an elevator, you can integrate the entrance and exits to the tunnel network just by using two parking spaces. And then the car gets on a skate. There ' s no speed limit here, so we ' re designing this to be able to operate at 200 kilometers an hour. CA : How much? EM : 200 kilometers an hour, or about 130 miles per hour. So you should be able to get from, say, Westwood to LAX in six minutes — five, six minutes. CA : So possibly, initially done, it ' s like on a sort of toll road-type basis. EM : Yeah. CA : Which, I guess, alleviates some traffic from the surface streets as well. EM : So, I don ' t know if people noticed it in the video, but there ' s no real limit to how many levels of tunnel you can have. You can go much further deep than you can go up. The deepest mines are much deeper than the tallest buildings are tall, so you can alleviate any arbitrary level of urban congestion with a 3D tunnel network. This is a very important point. So a key rebuttal to the tunnels is that if you add one layer of tunnels, that will simply alleviate congestion, it will get used up, and then you ' ll be back where you started, back with congestion. But you can go to any arbitrary number of tunnels, any number of levels. CA : But people — seen traditionally, it ' s incredibly expensive to dig, and that would block this idea. EM : Yeah. Well, they ' re right. To give you an example, the LA subway extension, which is — I think it ' s a two-and-a-half mile extension that was just completed for two billion dollars. So it ' s roughly a billion dollars a mile to do the subway extension in LA. And this is not the highest utility subway in the world. So yeah, it ' s quite difficult to dig tunnels normally. I think we need to have at least a tenfold improvement in the cost per mile of tunneling. CA : And how could you achieve that? EM : Actually, if you just do two things, you can get to approximately an order of magnitude improvement, and I think you can go beyond that. So the first thing to do is to cut the tunnel diameter by a factor of two or more. So a single road lane tunnel according to regulations has to be 26 feet, maybe 28 feet in diameter to allow for crashes and emergency vehicles and sufficient ventilation for combustion engine cars. But if you shrink that diameter to what we ' re attempting, which is 12 feet, which is plenty to get an electric skate through, you drop the diameter by a factor of two and the cross-sectional area by a factor of four, and the tunneling cost scales with the cross-sectional area. So that ' s roughly a half-order of magnitude improvement right there. Then tunneling machines currently tunnel for half the time, then they stop, and then the rest of the time is putting in reinforcements for the tunnel wall. So if you design the machine instead to do continuous tunneling and reinforcing, that will give you a factor of two improvement. Combine that and that ' s a factor of eight. Also these machines are far from being at their power or thermal limits, so you can jack up the power to the machine substantially. I think you can get at least a factor of two, maybe a factor of four or five improvement on top of that. So I think there ' s a fairly straightforward series of steps to get somewhere in excess of an order of magnitude improvement in the cost per mile, and our target actually is — we ' ve got a pet snail called Gary, this is from Gary the snail from"South Park,"I mean, sorry,"SpongeBob SquarePants."So Gary is capable of — currently he ' s capable of going 14 times faster than a tunnel-boring machine. CA : You want to beat Gary. EM : We want to beat Gary. He ' s not a patient little fellow, and that will be victory. Victory is beating the snail. CA : But a lot of people imagining, dreaming about future cities, they imagine that actually the solution is flying cars, drones, etc. You go aboveground. Why isn ' t that a better solution? You save all that tunneling cost. EM : Right. I ' m in favor of flying things. Obviously, I do rockets, so I like things that fly. This is not some inherent bias against flying things, but there is a challenge with flying cars in that they ' ll be quite noisy, the wind force generated will be very high. Let ' s just say that if something ' s flying over your head, a whole bunch of flying cars going all over the place, that is not an anxiety-reducing situation. You don ' t think to yourself,"Well, I feel better about today."You ' re thinking,"Did they service their hubcap, or is it going to come off and guillotine me?"Things like that. CA : So you ' ve got this vision of future cities with these rich, 3D networks of tunnels underneath. Is there a tie-in here with Hyperloop? Could you apply these tunnels to use for this Hyperloop idea you released a few years ago. EM : Yeah, so we ' ve been sort of puttering around with the Hyperloop stuff for a while. We built a Hyperloop test track adjacent to SpaceX, just for a student competition, to encourage innovative ideas in transport. And it actually ends up being the biggest vacuum chamber in the world after the Large Hadron Collider, by volume. So it was quite fun to do that, but it was kind of a hobby thing, and then we think we might — so we ' ve built a little pusher car to push the student pods, but we ' re going to try seeing how fast we can make the pusher go if it ' s not pushing something. So we ' re cautiously optimistic we ' ll be able to be faster than the world ' s fastest bullet train even in a. 8-mile stretch. CA : Whoa. Good brakes. EM : Yeah, I mean, it ' s — yeah. It ' s either going to smash into tiny pieces or go quite fast. CA : But you can picture, then, a Hyperloop in a tunnel running quite long distances. EM : Exactly. And looking at tunneling technology, it turns out that in order to make a tunnel, you have to — In order to seal against the water table, you ' ve got to typically design a tunnel wall to be good to about five or six atmospheres. So to go to vacuum is only one atmosphere, or near-vacuum. So actually, it sort of turns out that automatically, if you build a tunnel that is good enough to resist the water table, it is automatically capable of holding vacuum. CA : Huh. EM : So, yeah. CA : And so you could actually picture, what kind of length tunnel is in Elon ' s future to running Hyperloop? EM : I think there ' s no real length limit. You could dig as much as you want. I think if you were to do something like a DC-to-New York Hyperloop, I think you ' d probably want to go underground the entire way because it ' s a high- density area. You ' re going under a lot of buildings and houses, and if you go deep enough, you cannot detect the tunnel. Sometimes people think, well, it ' s going to be pretty annoying to have a tunnel dug under my house. Like, if that tunnel is dug more than about three or four tunnel diameters beneath your house, you will not be able to detect it being dug at all. In fact, if you ' re able to detect the tunnel being dug, whatever device you are using, you can get a lot of money for that device from the Israeli military, who is trying to detect tunnels from Hamas, and from the US Customs and Border patrol that try and detect drug tunnels. So the reality is that earth is incredibly good at absorbing vibrations, and once the tunnel depth is below a certain level, it is undetectable. Maybe if you have a very sensitive seismic instrument, you might be able to detect it. CA : So you ' ve started a new company to do this called The Boring Company. Very nice. Very funny. EM : What ' s funny about that? CA : How much of your time is this? EM : It ' s maybe... two or three percent. CA : You ' ve called it a hobby. This is what an Elon Musk hobby looks like. EM : I mean, it really is, like — This is basically interns and people doing it part time. We bought some second-hand machinery. It ' s kind of puttering along, but it ' s making good progress, so — CA : So an even bigger part of your time is being spent on electrifying cars and transport through Tesla. Is one of the motivations for the tunneling project the realization that actually, in a world where cars are electric and where they ' re self-driving, there may end up being more cars on the roads on any given hour than there are now? EM : Yeah, exactly. A lot of people think that when you make cars autonomous, they ' ll be able to go faster and that will alleviate congestion. And to some degree that will be true, but once you have shared autonomy where it ’ s much cheaper to go by car and you can go point to point, the affordability of going in a car will be better than that of a bus. Like, it will cost less than a bus ticket. So the amount of driving that will occur will be much greater with shared autonomy, and actually traffic will get far worse. CA : You started Tesla with the goal of persuading the world that electrification was the future of cars, and a few years ago, people were laughing at you. Now, not so much. EM : OK. I don ’ t know. I don ’ t know. CA : But isn ’ t it true that pretty much every auto manufacturer has announced serious electrification plans for the short- to medium-term future? EM : Yeah. Yeah. I think almost every automaker has some electric vehicle program. They vary in seriousness. Some are very serious about transitioning entirely to electric, and some are just dabbling in it. And some, amazingly, are still pursuing fuel cells, but I think that won ’ t last much longer. CA : But isn ’ t there a sense, though, Elon, where you can now just declare victory and say, you know,”We did it.”Let the world electrify, and you go on and focus on other stuff? EM : Yeah. I intend to stay with Tesla as far into the future as I can imagine, and there are a lot of exciting things that we have coming. Obviously the Model 3 is coming soon. We ’ ll be unveiling the Tesla Semi truck. CA : OK, we ’ re going to come to this. So Model 3, it ’ s supposed to be coming in July-ish. EM : Yeah, it ’ s looking quite good for starting production in July. CA : Wow. One of the things that people are so excited about is the fact that it ’ s got autopilot. And you put out this video a while back showing what that technology would look like. EM : Yeah. CA : There ’ s obviously autopilot in Model S right now. What are we seeing here? EM : Yeah, so this is using only cameras and GPS. So there ’ s no LIDAR or radar being used here. This is just using passive optical, which is essentially what a person uses. The whole road system is meant to be navigated with passive optical, or cameras, and so once you solve cameras or vision, then autonomy is solved. If you don ’ t solve vision, it ’ s not solved. So that ’ s why our focus is so heavily on having a vision neural net that ’ s very effective for road conditions. CA : Right. Many other people are going the LIDAR route. You want cameras plus radar is most of it. EM : You can absolutely be superhuman with just cameras. Like, you can probably do it ten times better than humans would, just cameras. CA : So the new cars being sold right now have eight cameras in them. They can ’ t yet do what that showed. When will they be able to? EM : I think we ’ re still on track for being able to go cross-country from LA to New York by the end of the year, fully autonomous. CA : OK, so by the end of the year, you ’ re saying, someone ’ s going to sit in a Tesla without touching the steering wheel, tap in”New York,”off it goes. EM : Yeah. CA : Won ’ t ever have to touch the wheel — by the end of 2017. EM : Yeah. Essentially, November or December of this year, we should be able to go all the way from a parking lot in California to a parking lot in New York, no controls touched at any point during the entire journey. CA : Amazing. But part of that is possible because you ’ ve already got a fleet of Teslas driving all these roads. You ’ re accumulating a huge amount of data of that national road system. EM : Yes, but the thing that will be interesting is that I ’ m actually fairly confident it will be able to do that route even if you change the route dynamically. So, it ’ s fairly easy — If you say I ’ m going to be really good at one specific route, that ’ s one thing, but it should be able to go, really be very good, certainly once you enter a highway, to go anywhere on the highway system in a given country. So it ’ s not sort of limited to LA to New York. We could change it and make it Seattle-Florida, that day, in real time. So you were going from LA to New York. Now go from LA to Toronto. CA : So leaving aside regulation for a second, in terms of the technology alone, the time when someone will be able to buy one of your cars and literally just take the hands off the wheel and go to sleep and wake up and find that they ’ ve arrived, how far away is that, to do that safely? EM : I think that ’ s about two years. So the real trick of it is not how do you make it work say 99. 9 percent of the time, because, like, if a car crashes one in a thousand times, then you ’ re probably still not going to be comfortable falling asleep. You shouldn ’ t be, certainly. It ’ s never going to be perfect. No system is going to be perfect, but if you say it ’ s perhaps — the car is unlikely to crash in a hundred lifetimes, or a thousand lifetimes, then people are like, OK, wow, if I were to live a thousand lives, I would still most likely never experience a crash, then that ’ s probably OK. CA : To sleep. I guess the big concern of yours is that people may actually get seduced too early to think that this is safe, and that you ’ ll have some horrible incident happen that puts things back. EM : Well, I think that the autonomy system is likely to at least mitigate the crash, except in rare circumstances. The thing to appreciate about vehicle safety is this is probabilistic. I mean, there ’ s some chance that any time a human driver gets in a car, that they will have an accident that is their fault. It ’ s never zero. So really the key threshold for autonomy is how much better does autonomy need to be than a person before you can rely on it? CA : But once you get literally safe hands-off driving, the power to disrupt the whole industry seems massive, because at that point you ’ ve spoken of people being able to buy a car, drops you off at work, and then you let it go and provide a sort of Uber-like service to other people, earn you money, maybe even cover the cost of your lease of that car, so you can kind of get a car for free. Is that really likely? EM : Yeah. Absolutely this is what will happen. So there will be a shared autonomy fleet where you buy your car and you can choose to use that car exclusively, you could choose to have it be used only by friends and family, only by other drivers who are rated five star, you can choose to share it sometimes but not other times. That ’ s 100 percent what will occur. It ’ s just a question of when. CA : Wow. So you mentioned the Semi and I think you ’ re planning to announce this in September, but I ’ m curious whether there ’ s anything you could show us today? EM : I will show you a teaser shot of the truck. It ’ s alive. CA : OK. EM : That ’ s definitely a case where we want to be cautious about the autonomy features. Yeah. CA : We can ’ t see that much of it, but it doesn ’ t look like just a little friendly neighborhood truck. It looks kind of badass. What sort of semi is this? EM : So this is a heavy duty, long-range semitruck. So it ’ s the highest weight capability and with long range. So essentially it ’ s meant to alleviate the heavy-duty trucking loads. And this is something which people do not today think is possible. They think the truck doesn ’ t have enough power or it doesn ’ t have enough range, and then with the Tesla Semi we want to show that no, an electric truck actually can out-torque any diesel semi. And if you had a tug-of-war competition, the Tesla Semi will tug the diesel semi uphill. CA : That ’ s pretty cool. And short term, these aren ’ t driverless. These are going to be trucks that truck drivers want to drive. EM : Yes. So what will be really fun about this is you have a flat torque RPM curve with an electric motor, whereas with a diesel motor or any kind of internal combustion engine car, you ’ ve got a torque RPM curve that looks like a hill. So this will be a very spry truck. You can drive this around like a sports car. There ’ s no gears. It ’ s, like, single speed. CA : There ’ s a great **movie** to be made here somewhere. I don ’ t know what it is and I don ’ t know that it ends well, but it ’ s a great **movie**. EM : It ’ s quite bizarre test-driving. When I was driving the test prototype for the first truck. It ’ s really weird, because you ’ re driving around and you ’ re just so nimble, and you ’ re in this giant truck. CA : Wait, you ’ ve already driven a prototype? EM : Yeah, I drove it around the parking lot, and I was like, this is crazy. CA : Wow. This is no vaporware. EM : It ’ s just like, driving this giant truck and making these mad maneuvers. CA : This is cool. OK, from a really badass picture to a kind of less badass picture. This is just a cute house from”Desperate Housewives”or something. What on earth is going on here? EM : Well, this **illustrates** the picture of the future that I think is how things will evolve. You ’ ve got an electric car in the driveway. If you look in between the electric car and the house, there are actually three Powerwalls stacked up against the side of the house, and then that house roof is a solar roof. So that ’ s an actual solar glass roof. CA : OK. EM : That ’ s a picture of a real — well, admittedly, it ’ s a real fake house. That ’ s a real fake house. CA : So these roof tiles, some of them have in them basically solar power, the ability to — EM : Yeah. Solar glass tiles where you can adjust the texture and the color to a very fine-grained level, and then there ’ s sort of microlouvers in the glass, such that when you ’ re looking at the roof from street level or close to street level, all the tiles look the same whether there is a solar cell behind it or not. So you have an even color from the ground level. If you were to look at it from a helicopter, you would be actually able to look through and see that some of the glass tiles have a solar cell behind them and some do not. You can ’ t tell from street level. CA : You put them in the ones that are likely to see a lot of sun, and that makes these roofs super affordable, right? They ’ re not that much more expensive than just tiling the roof. EM : Yeah. We ’ re very confident that the cost of the roof plus the cost of electricity — A solar glass roof will be less than the cost of a normal roof plus the cost of electricity. So in other words, this will be economically a no-brainer, we think it will look great, and it will last — We thought about having the warranty be infinity, but then people thought, well, that might sound like were just talking rubbish, but actually this is toughened glass. Well after the house has collapsed and there ’ s nothing there, the glass tiles will still be there. CA : I mean, this is cool. So you ’ re rolling this out in a couple week ’ s time, I think, with four different roofing types. EM : Yeah, we ’ re starting off with two, two initially, and the second two will be introduced early next year. CA : And what ’ s the scale of ambition here? How many houses do you believe could end up having this type of roofing? EM : I think eventually almost all houses will have a solar roof. The thing is to consider the time scale here to be probably on the order of 40 or 50 years. So on average, a roof is replaced every 20 to 25 years. But you don ’ t start replacing all roofs immediately. But eventually, if you say were to fast- forward to say 15 years from now, it will be unusual to have a roof that does not have solar. CA : Is there a mental model thing that people don ’ t get here that because of the shift in the cost, the economics of solar power, most houses actually have enough sunlight on their roof pretty much to power all of their needs. If you could capture the power, it could pretty much power all their needs. You could go off-grid, kind of. EM : It depends on where you are and what the house size is relative to the roof area, but it ’ s a fair statement to say that most houses in the US have enough roof area to power all the needs of the house. CA : So the key to the economics of the cars, the Semi, of these houses is the falling price of lithium-ion batteries, which you ’ ve made a huge bet on as Tesla. In many ways, that ’ s almost the core competency. And you ’ ve decided that to really, like, own that competency, you just have to build the world ’ s largest manufacturing plant to double the world ’ s supply of lithium-ion batteries, with this guy. What is this? EM : Yeah, so that ’ s the Gigafactory, progress so far on the Gigafactory. Eventually, you can sort of roughly see that there ’ s sort of a diamond shape overall, and when it ’ s fully done, it ’ ll look like a giant diamond, or that ’ s the idea behind it, and it ’ s aligned on true north. It ’ s a small detail. CA : And capable of producing, eventually, like a hundred gigawatt hours of batteries a year. EM : A hundred gigawatt hours. We think probably more, but yeah. CA : And they ’ re actually being produced right now. EM : They ’ re in production already.

CA : You guys put out this video. I mean, is that speeded up? EM : That ’ s the slowed down version. CA : How fast does it actually go? EM : Well, when it ’ s running at full speed, you can ’ t actually see the cells without a strobe light. It ’ s just blur. CA : One of your core ideas, Elon, about what makes an exciting future is a future where we no longer feel guilty about energy. Help us picture this. How many Gigafactories, if you like, does it take to get us there? EM : It ’ s about a hundred, roughly. It ’ s not 10, it ’ s not a thousand. Most likely a hundred. CA : See, I find this amazing. You can picture what it would take to move the world off this vast fossil fuel thing. It ’ s like you ’ re building one, it costs five billion dollars, or whatever, five to 10 billion dollars. Like, it ’ s kind of cool that you can picture that project. And you ’ re planning to do, at Tesla — announce another two this year. EM : I think we ’ ll announce locations for somewhere between two and four Gigafactories later this year. Yeah, probably four. CA : Whoa. No more teasing from you for here? Like — where, continent? You can say no. EM : We need to address a global market. CA : OK. This is cool. I think we should talk for — Actually, global market. I ’ m going to ask you one question about politics, only one. I ’ m kind of sick of politics, but I do want to ask you this. You ’ re on a body now giving advice to a guy — EM : Who? CA : Who has said he doesn ’ t really believe in climate change, and there ’ s a lot of people out there who think you shouldn ’ t be doing that. They ’ d like you to walk away from that. What would you say to them? EM : Well, I think that first of all, I ’ m just on two advisory councils where the format consists of going around the room and asking people ’ s opinion on things, and so there ’ s like a meeting every month or two. That ’ s the sum total of my contribution. But I think to the degree that there are people in the room who are arguing in favor of doing something about climate change, or social issues, I ’ ve used the meetings I ’ ve had thus far to argue in favor of immigration and in favor of climate change. And if I hadn ’ t done that, that wasn ’ t on the agenda before. So maybe nothing will happen, but at least the words were said. CA : OK. So let ’ s talk SpaceX and Mars. Last time you were here, you spoke about what seemed like a kind of incredibly ambitious dream to develop rockets that were actually reusable. And you ’ ve only gone and done it. EM : Finally. It took a long time. CA : Talk us through this. What are we looking at here? EM : So this is one of our rocket boosters coming back from very high and fast in space. So just delivered the upper stage at high velocity. I think this might have been at sort of Mach 7 or so, delivery of the upper stage. CA : So that was a sped-up — EM : That was the slowed down version. CA : I thought that was the sped-up version. But I mean, that ’ s amazing, and several of these failed before you finally figured out how to do it, but now you ’ ve done this, what, five or six times? EM : We ’ re at eight or nine. CA : And for the first time, you ’ ve actually reflown one of the rockets that landed. EM : Yeah, so we landed the rocket booster and then prepped it for flight again and flew it again, so it ’ s the first reflight of an orbital booster where that reflight is relevant. So it ’ s important to appreciate that reusability is only relevant if it is rapid and complete. So like an aircraft or a car, the reusability is rapid and complete. You do not send your aircraft to Boeing in- between flights. CA : Right. So this is allowing you to dream of this really ambitious idea of sending many, many, many people to Mars in, what, 10 or 20 years time, I guess. EM : Yeah. CA : And you ’ ve designed this outrageous rocket to do it. Help us understand the scale of this thing. EM : Well, visually you can see that ’ s a person. Yeah, and that ’ s the vehicle. CA : So if that was a skyscraper, that ’ s like, did I read that, a 40-story skyscraper? EM : Probably a little more, yeah. The thrust level of this is really — This configuration is about four times the thrust of the Saturn V moon rocket. CA : Four times the thrust of the biggest rocket humanity ever created before. EM : Yeah. Yeah. CA : As one does. EM : Yeah. In units of 747, a 747 is only about a quarter of a million pounds of thrust, so for every 10 million pounds of thrust, there ' s 40 747s. So this would be the thrust equivalent of 120 747s, with all engines blazing. CA : And so even with a machine designed to escape Earth ' s gravity, I think you told me last time this thing could actually take a fully loaded 747, people, cargo, everything, into orbit. EM : Exactly. This can take a fully loaded 747 with maximum fuel, maximum passengers, maximum cargo on the 747 — this can take it as cargo. CA : So based on this, you presented recently this Interplanetary Transport System which is visualized this way. This is a scene you picture in, what, 30 years time? 20 years time? People walking into this rocket. EM : I ' m hopeful it ' s sort of an eight- to 10-year time frame. Aspirationally, that ' s our target. Our internal targets are more aggressive, but I think — CA : OK. EM : While vehicle seems quite large and is large by comparison with other rockets, I think the future spacecraft will make this look like a rowboat. The future spaceships will be truly enormous. CA : Why, Elon? Why do we need to build a city on Mars with a million people on it in your lifetime, which I think is kind of what you ' ve said you ' d love to do? EM : I think it ' s important to have a future that is inspiring and appealing. I just think there have to be reasons that you get up in the morning and you want to live. Like, why do you want to live? What ' s the point? What inspires you? What do you love about the future? And if we ' re not out there, if the future does not include being out there among the stars and being a multiplanet species, I find that it ' s incredibly depressing if that ' s not the future that we ' re going to have. CA : People want to position this as an either or, that there are so many desperate things happening on the planet now from climate to poverty to, you know, you pick your issue. And this feels like a distraction. You shouldn ' t be thinking about this. You should be solving what ' s here and now. And to be fair, you ' ve done a fair old bit to actually do that with your work on sustainable energy. But why not just do that? EM : I think there ' s — I look at the future from the standpoint of probabilities. It ' s like a branching stream of probabilities, and there are actions that we can take that affect those probabilities or that accelerate one thing or slow down another thing. I may introduce something new to the probability stream. Sustainable energy will happen no matter what. If there was no Tesla, if Tesla never existed, it would have to happen out of necessity. It ' s tautological. If you don ' t have sustainable energy, it means you have unsustainable energy. Eventually you will run out, and the laws of economics will drive civilization towards sustainable energy, inevitably. The fundamental value of a company like Tesla is the degree to which it accelerates the advent of sustainable energy, faster than it would otherwise occur. So when I think, like, what is the fundamental good of a company like Tesla, I would say, hopefully, if it accelerated that by a decade, potentially more than a decade, that would be quite a good thing to occur. That ' s what I consider to be the fundamental aspirational good of Tesla. Then there ' s becoming a multiplanet species and space-faring civilization. This is not inevitable. It ' s very important to appreciate this is not inevitable. The sustainable energy future I think is largely inevitable, but being a space- faring civilization is definitely not inevitable. If you look at the progress in space, in 1969 you were able to send somebody to the moon. 1969. Then we had the Space Shuttle. The Space Shuttle could only take people to low Earth orbit. Then the Space Shuttle retired, and the United States could take no one to orbit. So that ' s the trend. The trend is like down to nothing. People are mistaken when they think that technology just automatically improves. It does not automatically improve. It only improves if a lot of people work very hard to make it better, and actually it will, I think, by itself degrade, actually. You look at great civilizations like Ancient Egypt, and they were able to make the pyramids, and they forgot how to do that. And then the Romans, they built these incredible aqueducts. They forgot how to do it. CA : Elon, it almost seems, listening to you and looking at the different things you ’ ve done, that you ’ ve got this unique double motivation on everything that I find so interesting. One is this desire to work for humanity ’ s long-term good. The other is the desire to do something exciting. And often it feels like you feel like you need the one to drive the other. With Tesla, you want to have sustainable energy, so you made these super sexy, exciting cars to do it. Solar energy, we need to get there, so we need to make these beautiful roofs. We haven ’ t even spoken about your newest thing, which we don ’ t have time to do, but you want to save humanity from bad AI, and so you ’ re going to create this really cool brain-machine **interface** to give us all infinite memory and telepathy and so forth. And on Mars, it feels like what you ’ re saying is, yeah, we need to save humanity and have a backup plan, but also we need to inspire humanity, and this is a way to inspire. EM : I think the value of beauty and inspiration is very much underrated, no question. But I want to be clear. I ’ m not trying to be anyone ’ s savior. That is not the — I ’ m just trying to think about the future and not be sad. CA : Beautiful statement. I think everyone here would agree that it is not — None of this is going to happen inevitably. The fact that in your mind, you dream this stuff, you dream stuff that no one else would dare dream, or no one else would be capable of dreaming at the level of complexity that you do. The fact that you do that, Elon Musk, is a really **remarkable** thing. Thank you for helping us all to dream a bit bigger. EM : But you ’ ll tell me if it ever starts getting genuinely insane, right? CA : Thank you, Elon Musk. That was really, really fantastic. That was really fantastic.

Text Sample 886: POS Dim 6 – 2017 – Score 4.00 – 2017/t000586.txt

A confession : I am an archaeologist and a museum curator, but a paradoxical one. For my museum, I collect things, but I also return things back to where they came from. I love museums because they ’ re social and educational, but I ’ m most drawn to them because of the magic of objects : a one-million-year-old hand axe, a totem **pole**, an impressionist painting all take us beyond our own imaginations. In museums, we pause to muse, to gaze upon our human empire of things in meditation and wonder. I understand why US museums alone host more than 850 million visits each year. Yet, in recent years, museums have become a battleground. Communities around the world don ’ t want to see their culture in distant institutions which they have no control over. They want to see their cultural treasures repatriated, returned to their places of origin. Greece seeks the return of the Parthenon Marbles, a collection of classical sculptures held by the British Museum. Egypt demands antiquities from Germany. New Zealand ’ s Maori want to see returned ancestral tattooed heads from museums everywhere. Yet these claims pale in comparison to those made by Native Americans. Already, US museums have returned more than one million **artifacts** and 50, 000 sets of Native American skeletons. To **illustrate** what ’ s at stake, let ’ s start with the War Gods. This is a wood carving made by members of the Zuni tribe in New Mexico. In the 1880s, anthropologists began to collect them as evidence of American Indian religion. They came to be seen as beautiful, the precursor to the stark sculptures of Picasso and Paul Klee, helping to usher in the modern art movement. From one viewpoint, the museum did exactly as it ’ s supposed to with the War God. It helped introduce a little-known art form for the world to appreciate. But from another point of view, the museum had committed a terrible crime of cultural violence. For Zunis, the War God is not a piece of art, it is not even a thing. It is a being. For Zunis, every year, priests ritually carve new War Gods, the Ahayu : da, breathing life into them in a long ceremony. They are placed on sacred shrines where they live to protect the Zuni people and keep the universe in balance. No one can own or sell a War God. They belong only to the earth. And so Zunis want them back from museums so they can go to their shrine homes to fulfill their spiritual purpose. What is a curator to do? I believe that the War Gods should be returned. This might be a startling answer. After all, my conclusion contradicts the refrain of the world ' s most famous archaeologist :"That belongs in a museum!"is what Indiana Jones said, not just to drive **movie** plots, but to drive home the unquestionable good of museums for society. I did not come to my view easily. I grew up in Tucson, Arizona, and fell in love with the Sonoran Desert ' s past. I was amazed that beneath the city ' s bland strip malls was 12, 000 years of history just waiting to be discovered. When I was 16 years old, I started taking archaeology classes and going out on digs. A high school teacher of mine even helped me set up my own laboratory to study animal bones. But in college, I came to learn that my future career had a dark history. Starting in the 1860s, Native American skeletons became a tool for science, collected in the thousands to prove new theories of social and racial hierarchies. Native American human remains were plundered from graves, even taken fresh from battlefields. When archaeologists came across white graves, the skeleton was often quickly reburied, while Native bones were deposited as specimens on museum shelves. In the wake of war, stolen land, boarding schools, laws banning religion, anthropologists collected sacred objects in the belief that Native peoples were on the cusp of extinction. You can call it racism or colonialism, but the labels don ' t matter as much as the fact that over the last century, Native American rights and culture were taken from them. In 1990, after years of Native protests, the US government, through the US Congress, finally passed a law that allowed Native Americans to reclaim cultural items, sacred objects and human remains from museums. Many archaeologists were panicked. For scientists, it can be hard to fully grasp how a piece of wood can be a living god or how spirits surround bones. And they knew that modern science, especially with DNA, can provide luminous insights into the past. As the anthropologist Frank Norwick declared,"We are doing important work that benefits all of mankind. We are not returning anything to anyone."As a college student, all of this was an enigma that was hard to decipher. Why did Native Americans want their heritage back from the very places preserving it? And how could scientists spend their entire lives studying dead Indians but seem to care so little about living ones? I graduated but wasn ' t sure what to do next, so I traveled. One day, in South Africa, I visited Nelson Mandela ' s former prison cell on Robben Island. I had an epiphany. Here was a man who helped a country bridge vast divides to seek, however imperfectly, reconciliation. I ' m no Mandela, but I ask myself : Could I, too, plant seeds of hope in the ruins of the past? In 2007, I was hired as a curator at the Denver Museum of Nature and Science. Our team agreed that unlike many other institutions, we needed to proactively confront the legacy of museum collecting. We started with the skeletons in our closet, 100 of them. After months and then years, we met with dozens of tribes to figure out how to get these remains home. And this is hard work. It involves negotiating who will receive the remains, how to respectfully transfer them, where will they go. Native American leaders become undertakers, planning funerals for dead relatives they had never wanted unearthed. A decade later, the Denver Museum and our Native partners have reburied nearly all of the human remains in the collection. We have returned hundreds of sacred objects. But I ' ve come to see that these battles are endless. Repatriation is now a permanent feature of the museum world. Hundreds of tribes are waiting their turn. There are always more museums with more stuff. Every catalogued War God in an American public museum has now been returned — 106, so far — but there are more beyond the reach of US law, in private collections and outside our borders. In 2014, I had the chance to travel with a respected religious leader from the Zuni tribe named Octavius Seowtewa to visit five museums in Europe with War Gods. At the Ethnological Museum of Berlin, we saw a War God with a history of dubious care. An overly enthusiastic curator had added chicken feathers to it. Its necklace had once been stolen. At the Muse du quai Branly in Paris, an official told us that the War God there is now state property with no provisions for repatriation. He insisted that the War God no longer served Zunis but museum visitors. He said,"We give all of the objects to the world."At the British Museum, we were warned that the Zuni case would establish a dangerous precedent for bigger disputes, such as the Parthenon Marbles, claimed by Greece. After visiting the five museums, Octavius returned home to his people empty- handed. He later told me,"It hurts my heart to see the Ahayu:da so far away. They all belong together. It ' s like a family member that ' s missing from a family dinner. When one is gone, their strength is broken."I wish that my colleagues in Europe and beyond could see that the War Gods do not represent the end of museums but the chance for a new beginning. When you walk the halls of a museum, you ' re likely just seeing about one percent of the total collections. The rest is in storage. Even after returning 500 cultural items and skeletons, my museum still retains 99. 999 percent of its total collections. Though we no longer have War Gods, we have Zuni traditional pottery, jewelry, tools, clothing and arts. And even more precious than these objects are the relationships that we formed with Native Americans through the process of repatriation. Now, we can ask Zunis to share their culture with us. Not long ago, I had the chance to visit the returned War Gods. A shrine sits up high atop a mesa overlooking beautiful Zuni homeland. The shrine is enclosed by a roofless stone building threaded at the top with barbed wire to ensure that they ' re not stolen again. And there they are, inside, the Ahayu:da, 106 War Gods amid offerings of turquoise, cornmeal, shell, even T-shirts... a modern gift to ancient beings. And standing there, I got a glimpse at the War Gods ' true purpose in the world. And it occurred to me then that we do not get to choose the histories that we inherit. Museum curators today did not pillage ancient graves or steal spiritual objects, but we can accept responsibility for correcting past mistakes. We can help restore dignity, hope and humanity to Native Americans, the very people who were once the voiceless objects of our curiosity. And this doesn ' t even require us to fully understand others ' beliefs, only that we respect them. Museums are temples to things past. Now they must also become places for living cultures. As I turned to walk away from the shrine, I drank in the warm summer air, and I watched an eagle turn lazy circles high above. I thought of the Zunis, whose offerings ensure that their culture is not dead and gone but alive and well, and I could think of no better place for the War Gods to be. Thank you.

Text Sample 887: POS Dim 6 – 2018 – Score 4.00 – 2018/t000633.txt

It ' s like a dream. Imagine, in the empty desert, you come upon a huge wheel ringed in skeletons, and someone invites you to come pull a series of heavy ropes at its base, so you walk to one side, where a team is waiting, and you all throw your backs into it, and you pull in turn, and eventually, the wheel roars to life, lights begin to flicker, and the audience cheers, and you ' ve just activated Peter Hudson ' s"Charon,"one of the world ' s largest zoetropes. This is the farthest thing from marketable art. It ' s huge, it ' s dangerous, it takes a dozen people to run, and it doesn ' t go with the sofa. It ' s beautifully crafted and completely useless, and it ' s wonderful. You ' re unlikely to see works like"Charon"in the art-world headlines. These days, the buying and selling of artwork often gets more attention than the works themselves. In the last year, a Jean-Michel Basquiat sold for 110 million dollars, the highest price ever achieved for the work of an American artist, and a painting by Leonardo da Vinci sold for 450 million, setting a new auction record. Still, these are big, important artists, but still, when you look at these works and you look at the headlines, you have to ask yourself,"Do I care about these because they move me, or do I care about them because they ' re expensive and I think they ' re supposed to?"In our contemporary world, it can be hard to separate those two things. But what if we tried? What if we redefined art ' s value — not by its price tag, but by the emotional connection it creates between the artist and the audience, or the benefits it gives our society, or the fulfillment it gives the artists themselves? This is Nevada ' s Black Rock Desert, about as far as you can get from the galleries of New York and London and Hong Kong. And here, for just about 30 years, at Burning Man, a movement has been forming that does exactly that. Since its early anarchist years, Burning Man has grown up. Today, it ' s more of an experiment in collective dreaming. It ' s a year-round community, and every August, for a single week, 70, 000 people power down their technology and pilgrimage out into the desert to build an anti-consumerist society outside the bounds of their everyday lives. The conditions are brutal. Strangers will hug you, and every year, you will swear it was better the last, but it ' s still ridiculous and freeing and alive, and the art is one thing that thrives here. So this is me on the desert playa last year with my brother, obviously hard at work. I ' d been studying the art of Burning Man for several years, for an exhibition I curated at the Smithsonian ' s Renwick Gallery, and what fascinates me the most isn ' t the quality of the work here, which is actually rather high, it ' s why people come out here into the desert again and again to get their hands dirty and make in our increasingly digital age. Because it seems like this gets to something that ' s essentially human. Really, the entire encampment of Burning Man could be thought of as one giant interactive art installation driven by the participation of everyone in it. One thing that sets this work aside from the commercial art world is that anyone who makes work can show it. These days, around 300 art installations and countless artistic gestures go to the playa. None of them are sold there. At the end of the week, if the works aren ' t burned, artists have to cart them back out and store them. It ' s a tremendous labor of love. Though there is certainly a Burning Man aesthetic, pioneered by artists like Kate Raudenbush and Michael Christian, much of the distinctive character of the work here comes from the desert itself. For a work to succeed, it has to be portable enough to make the journey, rugged enough to withstand the wind and weather and participants, stimulating in daylight and darkness, and engaging without interpretation. Encounters with monumental and intimate works here feel surreal. Scale has a tendency to fool the eyes. What looked enormous in an artist ' s studio could get lost on the playa, but there are virtually no **spatial** limits, so artists can dream as big as they can build. Some pieces bowl you over by their grace and others by the sheer audacity it took to bring them here. Burning Man ' s irreverent humor comes out in pieces like Rebekah Waites '"Church Trap,"a tiny country chapel set precariously on a wooden beam, like a mousetrap, that lured participants in to find religion — it was built and burned in 2013... while other works, like Christopher Schardt ' s"Firmament,"aim for the sublime. Here, under a canopy of dancing lights set to classical music, participants could escape the thumping rave beats and chaos all around. At night, the city swarms with mutant vehicles, the only cars allowed to roam the playa. And if necessity is the mother of invention, here, absurdity is its father. They zigzag from artwork to artwork like some bizarre, random public transportation system, pulsing with light and sound. When artists stop worrying about critics and collectors and start making work for themselves, these are the kinds of marvelous toys they create. And what ' s amazing is that, by and large, when people first come to Burning Man, they don ' t know how to make this stuff. It ' s the active collaborative maker community there that makes this possible. Collectives like Five Ton Crane come together to share skills and take on complex projects a single artist would never even attempt, from a Gothic rocket ship that appears ready to take off at any moment to a fairytale home inside a giant boot complete with shelves full of artist-made books, a blackbird pie in the oven and a climbable beanstalk. Skilled or unskilled, all are welcome. In fact, part of the charm and the innovation of the work here is that so many makers aren ' t artists at all, but scientists or engineers or welders or garbage collectors, and their works cross disciplinary boundaries, from a grove of origami mushrooms that developed out of the design for a yurt to a tree that responds to the voices and biorhythms of all those around it through 175, 000 LEDs embedded in its leaves. In museums, a **typical** visitor spends less than 30 seconds with a work of art, and I often watch people wander from label to label, searching for information, as though the entire story of a work of art could be contained in that one 80- word text. But in the desert, there are no gatekeepers and no placards explaining the art, just natural curiosity. You see a work on the horizon, and you ride towards it. When you arrive, you walk all around it, you touch it, you test it. Is it sturdy enough to climb on? Will I be impaled by it? Art becomes a place for extended interaction, and although the **display** might be short-lived, the experience stays with you. Nowhere is that truer at Burning Man than at the Temple. In 2000, David Best and Jack Haye built the first Temple, and after a member of their team was killed tragically in an accident shortly before the event, the building became a makeshift memorial. By itself, it ' s a magnificent piece of architecture, but the structure is only a shell until it disappears under a thick blanket of messages."I miss you.""Please forgive me.""Even a broken crayon still colors."Intimate testaments to the most universal of human experiences, the experience of loss. The collective emotion in this place is overpowering and indescribable, before it ' s set afire on the last night of the event. Every year, something compels people from all different walks of life, from all over the world, to go out into the desert and make art when there is no money in it. The work ' s not always refined, it ' s not always viable, it ' s not even always good, but it ' s authentic and optimistic in a way we rarely see anywhere else. In these cynical times, it ' s comforting to know that we ' re still capable of great feats of imagination, and that when we search for connection, we come together and build cathedrals in the dust. Forget the price tags. Forget the big names. What is art for in our contemporary world if not this? Thank you.

Text Sample 888: POS Dim 6 – 2018 – Score 4.00 – 2018/t000502.txt
It ' s 4am, you ' ve been awake for forty hours, when you unlock a puzzle containing this video of some kind of dance-off between a chicken and a roller- skating beaver. The confusion and delight you ' re experiencing is a **typical** moment at the MIT Mystery Hunt, which is basically the Olympics meets Burning Man for a specific type of nerd. Today, I ' m going to take you inside this strange,

intellectually masochistic and incredibly joyful world. But first, I have to explain what I mean when I say"puzzle."A puzzle-hunt- style puzzle is a data set. It can be a grid of letters, a sudoku, a video, an audio — it can be anything that contains hidden information that can eventually resolve into an answer that is a word or a phrase. So, to give you an example, this is a puzzle called"Master Pieces."It consists of 10 images of LEGO people looking at piles of LEGOs. And to save us some time, I ' m going to explain what ' s going on here. Each of the piles of LEGOs is a deconstructed work of art in the style of a famous artist. So, does anybody recognize the artist on the left? They used a lot of red. I heard"Rothko,"yeah. The second one? Mondrian. Alex Rosenthal : Yeah, well done. And the third one? This is the hardest one — Yeah, Klimt, I heard it. Well done, the color is the biggest clue there. So the puzzle has **various** clues that tell you what matters here are the artists, not the specific works of art. And what you need to do is then look at what you haven ' t used yet, which is the number of LEGO people in each painting. And you can count them and then count into the artists ' last names by the same number of letters. So there ' s three people in front of the Rothko on the left, so you take the third letter, which is a T. There ' s only one in front of the Mondrian, so you take the first letter, M. And there ' s three again in front of Klimt, so you take the third letter, I. You do that for all 10 of the original artists and put them in the order, and you get the answer, which is"illuminate."Puzzles like this are about communicating an idea. But where I ' m trying to be as clear as possible for you now, puzzles have to navigate the line between abstraction and clarity. They have to be obtuse enough to make you work for it, but elegant enough so you can get to the aha moment, where everything clicks into place. Puzzle solvers are junkies for this aha moment — it feels like a brief high and an instant of pristine clarity. And there ' s also a deeper fulfillment at play here, which is that humans are innate problem-solvers. That ' s why we love crosswords and escape rooms and figuring out how to explore the bottom of the ocean. Solving deviously difficult puzzles expands our minds in new directions, and it also helps us come at problems from diverse perspectives. These puzzles come in **various** puzzle hunts, which come in **various** shapes and sizes. There ' s one-hour ones designed for novices, 24-hour road rallies, and the puzzle hunt of puzzle hunts, the MIT Mystery Hunt. This is an event that takes place once a year and has around 2, 000 people descending on MIT ' s campus and solving puzzles in teams that range from a single person to over 100. My team has 60 people on it — that includes a national crossword puzzle tournament champion, a particle physicist, a composer, an actual deep-sea explorer, and me, feeling like"Mr. Bean goes to Bletchley Park."That ' s actually an apt comparison, because one year involved a puzzle where you had to construct a working Enigma machine out of pieces of cardboard. Each Mystery Hunt has a theme. Past ones have included"The Matrix"and"Alice in Wonderland."It ' s often pop culture- and literary-based themes. And the goal is to find the coin that ' s been hidden somewhere on MIT ' s campus. And in order to get there, you have to solve around 150 puzzles and do **various** events and challenges. I had done this for about 10 years without ever dreaming of winning, until January of 2016, where 53 hours into a hunt whose theme is the **movie**"Inception,"we haven ' t slept in days, so everything is hilarious... The tables are covered in piles of papers, of our notes and completed puzzles. The whiteboards are an unintelligible mess of three days ' worth of insights. And we ' re stuck on two puzzles. If we could crack them, we would get into the endgame, and after hours of work, in a magical moment, they both fall within 10 seconds of each other, and soon, we ' re on the final runaround, a series of clues that will lead us to the coin, and we ' re racing through the halls of MIT, trying not to knock over or terrify tour groups, when we realize we ’ re not alone, there ’ s another team on the runaround as well, and we don ’ t know who ’ s ahead. So, we ’ re a mess of anxiety, anticipation, exhilaration and sleep deprivation, when we arrive at the Alchemist, a sculpture in which we find... this coin. Yeah. And in claiming it, we win the MIT Mystery Hunt by a tiny margin of five minutes. What I didn ’ t mention before is that the prize for winning is that you get to construct the whole hunt for the following year. The punishment for winning is that you have to construct the whole hunt for the following year. At the beginning of 2016, I had never constructed a puzzle before — I had solved plenty of puzzles, but constructing and solving are entirely different beasts. But once again, I was lucky to be on a team full of brilliant mentors and collaborators. So, from a constructor ’ s point of view, a puzzle is where I have an idea, and instead of telling you what it is, I ’ m going to leave a trail of breadcrumbs so you can figure it out for yourself, and have the joy and experience of the aha moment. This is another way of looking at the aha moment. And what ’ s incredible to me is that this experience, which is very emotional and kind of almost physical, is something that can be carefully designed. So, to show you what I mean, this is a puzzle I co-constructed with my friend Matt Gruskin. It ’ s a text adventure, which is the old-school adventure game format, where you ’ re exploring, going north, east, south and west, picking up items and using them. And you could get to the end of the game part, but you won ’ t have solved the puzzle. In order to do so, you have to recognize a hidden layer of information, and the easiest way of seeing it is by mapping the game out. That looks something like this. Does anybody recognize what this is? Yeah, exactly. This text adventure takes place within”Settlers of Catan.”Who here knows what”Settlers”is? Nerds. If you don ’ t know,”Settlers”is a board game where you ’ re competing against other people to collect resources and use them to build structures. And within the text adventure, we hid information in **various** ways, with which you could reconstruct an entire game. You could figure out the roads, the cities, the towns, the resources, the numbers on the tiles, even the dice rolls. You put all that information together and you could extract an answer in a way that ’ s too complicated to explain right now. But find me afterwards if you really want to know. But what this puzzle emphasized for me is the value of perspective shifts in inspiring an aha. So, in this puzzle, you go from experiencing the world on the ground, as a character, to looking down on it from above as if you ’ re playing a board game, and in that shift, you completely reframe all the information you ’ ve been given. The hardest part of construction for me is coming up with a great idea for an aha. Fortunately, the world is a torrent of ideas and information. I ’ ve seen fantastic puzzles constructed out of the waggle dances of bees, and the **remarkable** coincidence that the 88 keys of a piano can be perfectly mapped to the 88 constellations in the sky. Once you find that out, you can ’ t not construct the puzzle, and it ’ s going to be about having the solvers make that connection in their own minds. Whether you give them stars on a keyboard or play the celestial music of the cosmos, you ’ re getting them there, one way or another. Before long, you find yourself staring at a turtle, and asking yourself,”Is this a puzzle?”And also, staring at a turtle and saying,”I never appreciated what multitudes this contains in its shell alone.”This might be a familiar experience to you, if you ’ ve ever been watching a TED Talk and asked yourself,”Is this a puzzle?”I ’ m not telling. But what I will say is that puzzles can be found in the most unexpected of places. That brings us back to one of my favorite puzzles of all time, which was constructed by Trip Payne. And this time, I ’ m going to play it for you with the sound on, so get ready to name that tune. Who knows what that is? Yeah,”You Make Me Feel Like a Natural Woman.”So you can identify that and seven other songs and clips, and then look at the videos themselves for clues, where the way that they are filmed and edited together plus things like the cutaways to the panel of five people sitting at a table, which is reminiscent of a panel of judges, all of this can suggest”reality competition show.”And either through internet research, or from just recognizing this, you can get to the aha, which is that these clips are shot- for-shot recreations of lip-synch battles from”RuPaul ’ s Drag Race.”So, why do we do this? You tell me, I don ’ t know. So, first of all, it ’ s really fun. But I think it also improves our lives in **various** ways. Being able to solve puzzles, when I ’ m confronted with a challenge, has allowed me to explore it from multiple perspectives before I lock in an approach. Also, the process of solving is great training for working with a team, knowing when to listen, when to share, and how to recognize and celebrate insight and being able to construct ahas is a very powerful tool. Think of how powerful and exciting and convincing an idea is that comes from your own mind, where you make all the connections yourself. So in January of 2017, after tens of thousands of hours of work, we finally run our Mystery Hunt. And it ’ s a different sort of satisfaction than the quick high of an aha moment. Instead, it ’ s the slow burn of saying something through perplexing abstraction, yet being understood. And when it was all over, in our exhaustion, we turned to each other and the world, and we said,”We ’ re never doing this again. It ’ s too much work. It ’ s really fun, but no more winning.”One year later, in January of 2018, we won the MIT Mystery Hunt again. So, we ’ re currently I don ’ t know how many tens of thousands of hours of work in, and we ’ re two months out from the 2019 Hunt. So, thank you for listening, I have to go write a puzzle.

Text Sample 889: POS Dim 6 – 2018 – Score 4.00 – 2018/t000550.txt

Do you remember when you were a child, you probably had a favorite toy that was a constant companion, like Christopher Robin had Winnie the Pooh, and your imagination fueled endless adventures? What could be more innocent than that? Well, let me introduce you to my friend Cayla. Cayla was voted toy of the year in countries around the world. She connects to the internet and uses speech recognition technology to answer your child ’ s questions, respond just like a friend. But the power doesn ’ t lie with your child ’ s imagination. It actually lies with the company harvesting masses of personal information while your family is innocently chatting away in the safety of their home, a dangerously false sense of security. This case sounded alarm bells for me, as it is my job to protect consumers ’ rights in my country. And with billions of devices such as cars, energy meters and even vacuum cleaners expected to come online by 2020, we thought this was a case worth investigating further. Because what was Cayla doing with all the interesting things she was learning? Did she have another friend she was loyal to and shared her information with? Yes, you guessed right. She did. In order to play with Cayla, you need to download an app to access all her features. Parents must consent to the terms being changed without notice. The recordings of the child, her friends and family, can be used for targeted advertising. And all this information can be shared with unnamed third parties. Enough? Not quite. Anyone with a smartphone can connect to Cayla within a certain distance. When we confronted the company that made and programmed Cayla, they issued a series of statements that one had to be an IT expert in order to breach the security. Shall we fact-check that statement and live hack Cayla together? Here she is. Cayla is equipped with a Bluetooth device which can transmit up to 60 feet, a bit less if there ’ s a wall between. That means I, or any stranger, can connect to the doll while being outside the room where Cayla and her friends are. And to **illustrate** this, I ' m going to turn Cayla on now. Let ' s see, one, two, three. There. She ' s on. And I asked a colleague to stand outside with his smartphone, and he ' s connected, and to make this a bit creepier... let ' s see what kids could hear Cayla say in the safety of their room. Man : Hi. My name is Cayla. What is yours? Finn Myrstad : Uh, Finn. Man : Is your mom close by? FM : Uh, no, she ' s in the store. Man : Ah. Do you want to come out and play with me? FM : That ' s a great idea. Man : Ah, great. FM : I ' m going to turn Cayla off now. We needed no password or to circumvent any other type of security to do this. We published a report in 20 countries around the world, exposing this significant security flaw and many other problematic issues. So what happened? Cayla was banned in Germany, taken off the shelves by Amazon and Wal-Mart, and she ' s now peacefully resting at the German Spy Museum in Berlin. However, Cayla was also for sale in stores around the world for more than a year after we published our report. What we uncovered is that there are few rules to protect us and the ones we have are not being properly enforced. We need to get the security and privacy of these devices right before they enter the market, because what is the point of locking a house with a key if anyone can enter it through a connected device? You may well think,"This will not happen to me. I will just stay away from these flawed devices."But that won ' t keep you safe, because simply by connecting to the internet, you are put in an impossible take-it-or-leave-it position. Let me show you. Like most of you, I have dozens of apps on my phone, and used properly, they can make our lives easier, more convenient and maybe even healthier. But have we been lulled into a false sense of security? It starts simply by ticking a box. Yes, we say, I ' ve read the terms. But have you really read the terms? Are you sure they didn ' t look too long and your phone was running out of battery, and the last time you tried they were impossible to understand, and you needed to use the service now? And now, the power imbalance is established, because we have agreed to our personal information being gathered and used on a scale we could never imagine. This is why my colleagues and I decided to take a deeper look at this. We set out to read the terms of popular apps on an average phone. And to show the world how unrealistic it is to expect consumers to actually read the terms, we printed them, more than 900 pages, and sat down in our office and read them out loud ourselves, streaming the experiment live on our websites. As you can see, it took quite a long time. It took us 31 hours, 49 minutes and 11 seconds to read the terms on an average phone. That is longer than a **movie marathon** of the"Harry Potter"**movies** and the"Godfather"**movies** combined. And reading is one thing. Understanding is another story. That would have taken us much, much longer. And this is a real problem, because companies have argued for 20 to 30 years against regulating the internet better, because users have consented to the terms and conditions. As we ' ve shown with this experiment, achieving informed consent is close to impossible. Do you think it ' s fair to put the burden of responsibility on the consumer? I don ' t. I think we should demand less take-it-or-leave-it and more understandable terms before we agree to them. Thank you. Now, I would like to tell you a story about love. Some of the world ' s most popular apps are dating apps, an industry now worth more than, or close to, three billion dollars a year. And of course, we ' re OK sharing our intimate details with our other half. But who else is snooping, saving and sharing our information while we are baring our souls? My team and I decided to investigate this. And in order to understand the issue from all angles and to truly do a thorough job, I realized I had to download one of the world ' s most popular dating apps myself. So I went home to my wife... who I had just married."Is it OK if I establish a profile on a very popular dating app for purely scientific purposes?"This is what we found. Hidden behind the main menu was a preticked box that gave the dating company access to all my personal pictures on Facebook, in my case more than 2, 000 of them, and some were quite personal. And to make matters worse, when we read the terms and conditions, we discovered the following, and I ' m going to need to take out my reading glasses for this one. And I ' m going to read it for you, because this is complicated. All right.”By posting content”— and content refers to your pictures, chat and other interactions in the dating service —”as a part of the service, you automatically grant to the company, its affiliates, licensees and successors an irrevocable”— which means you can ' t change your mind —”perpetual”— which means forever —”nonexclusive, transferrable, sublicensable, fully paid-up, worldwide right and license to use, copy, store, perform, **display**, reproduce, record, play, adapt, modify and distribute the content, prepare derivative works of the content, or incorporate the content into other works and grant and authorize sublicenses of the foregoing in any media now known or hereafter created.”That basically means that all your dating history and everything related to it can be used for any purpose for all time. Just imagine your children seeing your sassy dating photos in a birth control ad 20 years from now. But seriously, though — what might these commercial practices mean to you? For example, financial loss : based on your web browsing history, algorithms might decide whether you will get a mortgage or not. Subconscious manipulation : companies can analyze your emotions based on your photos and chats, targeting you with ads when you are at your most vulnerable. Discrimination : a fitness app can sell your data to a health insurance company, preventing you from getting coverage in the future. All of this is happening in the world today. But of course, not all uses of data are malign. Some are just flawed or need more work, and some are truly great. And there is some good news as well. The dating companies changed their policies globally after we filed a legal complaint. But organizations such as mine that fight for consumers ' rights can ' t be everywhere. Nor can consumers fix this on their own, because if we know that something innocent we said will come back to haunt us, we will stop speaking. If we know that we are being watched and monitored, we will change our behavior. And if we can ' t control who has our data and how it is being used, we have lost the control of our lives. The stories I have told you today are not random examples. They are everywhere, and they are a sign that things need to change. And how can we achieve that change? Well, companies need to realize that by prioritizing privacy and security, they can build trust and loyalty to their users. Governments must create a safer internet by ensuring enforcement and up-to-date rules. And us, the citizens? We can use our voice to remind the world that technology can only truly benefit society if it respects basic rights. Thank you so much.

Text Sample 890: POS Dim 6 – 2019 – Score 4.00 – 2019/t000918.txt

As a roboticist, I get asked a lot of questions.”When we will they start serving me breakfast?”So I thought the future of robotics would be looking more like us. I thought they would look like me, so I built eyes that would simulate my eyes. I built fingers that are dextrous enough to serve me... baseballs. Classical robots like this are built and become functional based on the fixed number of joints and actuators. And this means their functionality and shape are already fixed at the moment of their conception. So even though this arm has a really nice throw — it even hit the tripod at the end — it ' s not meant for cooking you breakfast per se. It ' s not really suited for scrambled eggs. So this was when I was hit by a new vision of future robotics : the transformers. They drive, they run, they fly, all depending on the ever- changing, new environment and task at hand. To make this a reality, you really have to rethink how robots are designed. So, imagine a robotic module in a polygon shape and using that simple polygon shape to reconstruct multiple different forms to create a new form of robot for different tasks. In CG, computer graphics, it ' s not any news — it ' s been done for a while, and that ' s how most of the **movies** are made. But if you ' re trying to make a robot that ' s physically moving, it ' s a completely new story. It ' s a completely new paradigm. But you ' ve all done this. Who hasn ' t made a paper airplane, paper boat, paper crane? Origami is a versatile platform for designers. From a single sheet of paper, you can make multiple shapes, and if you don ' t like it, you unfold and fold back again. Any 3D form can be made from 2D surfaces by folding, and this is proven mathematically. And imagine if you were to have an intelligent sheet that can self-fold into any form it wants, anytime. And that ' s what I ' ve been working on. I call this robotic origami,"robogami."This is our first robogami transformation that was made by me about 10 years ago. From a flat-sheeted robot, it turns into a pyramid and back into a flat sheet and into a space shuttle. Quite cute. Ten years later, with my group of ninja origami robotic researchers — about 22 of them right now — we have a new generation of robogamis, and they ' re a little more effective and they do more than that. So the new generation of robogamis actually serve a purpose. For example, this one actually navigates through different terrains autonomously. So when it ' s a dry and flat land, it crawls. And if it meets sudden rough terrain, it starts rolling. It does this — it ' s the same robot — but depending on which terrain it meets, it activates a different sequence of actuators that ' s on board. And once it meets an obstacle, it jumps over it. It does this by storing energy in each of its legs and releasing it and catapulting like a slingshot. And it even does gymnastics. Yay. So I just showed you what a single robogami can do. Imagine what they can do as a group. They can join forces to tackle more complex tasks. Each module, either active or passive, we can assemble them to create different shapes. Not only that, by controlling the folding joints, we ' re able to create and attack different tasks. The form is making new task space. And this time, what ' s most important is the assembly. They need to autonomously find each other in a different space, attach and detach, depending on the environment and task. And we can do this now. So what ' s next? Our imagination. This is a simulation of what you can achieve with this type of module. We decided that we were going to have a four-legged crawler turn into a little dog and make small gaits. With the same module, we can actually make it do something else : a manipulator, a **typical**, classical robotic task. So with a manipulator, it can pick up an object. Of course, you can add more modules to make the manipulator legs longer to attack or pick up objects that are bigger or smaller, or even have a third arm. For robogamis, there ' s no one fixed shape nor task. They can transform into anything, anywhere, anytime. So how do you make them? The biggest technical challenge of robogami is keeping them super thin, flexible, but still remaining functional. They ' re composed of multiple layers of circuits, motors, microcontrollers and sensors, all in the single body, and when you control individual folding joints, you ' ll be able to achieve soft motions like that upon your command. Instead of being a single robot that is specifically made for a single task, robogamis are optimized to do multi-tasks. And this is quite important for the difficult and unique environments on the Earth as well as in space. Space is a perfect environment for robogamis. You cannot afford to have one robot for one task. Who knows how many tasks you will encounter in space? What you want is a single robotic platform that can transform to do multi-tasks. What we want is a deck of thin robogami modules that can transform to do multiples of performing tasks. And don ’ t take my word for it, because the European Space Agency and Swiss Space Center are sponsoring this exact concept. So here you see a couple of images of reconfiguration of robogamis, exploring the foreign land aboveground, on the surface, as well as digging into the surface. It ’ s not just exploration. For astronauts, they need additional help, because you cannot afford to bring interns up there, either. They have to do every tedious task. They may be simple, but super interactive. So you need robots to facilitate their experiments, assisting them with the communications and just docking onto surfaces to be their third arm holding different tools. But how will they be able to control robogamis, for example, outside the space station? In this case, I show a robogami that is holding space debris. You can work with your vision so that you can control them, but what would be better is having the sensation of touch directly transported onto the hands of the astronauts. And what you need is a haptic device, a haptic **interface** that recreates the sensation of touch. And using robogamis, we can do this. This is the world ’ s smallest haptic **interface** that can recreate a sensation of touch just underneath your fingertip. We do this by moving the robogami by microscopic and macroscopic movements at the stage. And by having this, not only will you be able to feel how big the object is, the roundness and the lines, but also the stiffness and the texture. Alex has this **interface** just underneath his thumb, and if he were to use this with VR goggles and hand controllers, now the virtual reality is no longer virtual. It becomes a tangible reality. The blue **ball**, red **ball** and black **ball** that he ’ s looking at is no longer differentiated by colors. Now it is a rubber blue **ball**, sponge red **ball** and billiard black **ball**. This is now possible. Let me show you. This is really the first time this is shown live in front of a public grand audience, so hopefully this works. So what you see here is an atlas of anatomy and the robogami haptic **interface**. So, like all the other reconfigurable robots, it multitasks. Not only is it going to serve as a mouse, but also a haptic **interface**. So for example, we have a white background where there is no object. That means there is nothing to feel, so we can have a very, very flexible **interface**. Now, I use this as a mouse to approach skin, a muscular arm, so now let ’ s feel his biceps, or shoulders. So now you see how much stiffer it becomes. Let ’ s explore even more. Let ’ s approach the ribcage. And as soon as I move on top of the ribcage and between the intercostal muscles, which is softer and harder, I can feel the difference of the stiffness. Take my word for it. So now you see, it ’ s much stiffer in terms of the force it ’ s giving back to my fingertip. So I showed you the surfaces that aren ’ t moving. How about if I were to approach something that moves, for example, like a beating heart? What would I feel? This can be your beating heart. This can actually be inside your pocket while you ’ re shopping online. Now you ’ ll be able to feel the difference of the sweater that you ’ re buying, how soft it is, if it ’ s actually cashmere or not, or the bagel that you ’ re trying to buy, how hard it is or how crispy it is. This is now possible. The robotics technology is advancing to be more personalized and adaptive, to adapt to our everyday needs. This unique specie of reconfigurable robotics is actually the platform to provide this invisible, intuitive **interface** to meet our exact needs. These robots will no longer look like the characters from the **movies**. Instead, they will be whatever you want them to be. Thank you.

Text Sample 891: POS Dim 6 – 2019 – Score 3.00 – 2019/t000021.txt

Every other night in Japan, I step out of my apartment, I climb up a hill for 15 minutes, and then I head into my local health club, where three **ping-pong** tables are set up in a studio. And space is limited, so at every table, one pair of players practices forehands, another practices backhands, and every now and then, the **balls** collide in midair and everybody says,”Wow!”Then, choosing lots, we select partners and play doubles. But I honestly couldn ’ t tell you who ’ s won, because we change partners every five minutes. And everybody is trying really hard to win points, but nobody is keeping track of who is winning games. And after an hour or so of furious exertion, I can honestly tell you that not knowing who has won feels like the ultimate victory. In Japan, it ’ s been said, they ’ ve created a competitive spirit without competition. Now, all of you know that geopolitics is best followed by watching **ping-pong**. The two strongest powers in the world were fiercest enemies until, in 1972, an American **ping-pong** team was allowed to visit Communist China. And as soon as the former adversaries were gathered around some small green tables, each of them could claim a victory, and the whole world could breathe more easily. China ’ s leader, Mao Zedong, wrote a whole manual on **ping-pong**, and he called the sport”a spiritual nuclear weapon.”And it ’ s been said that the only honorary lifelong member of the US Table Tennis Association is the then- President Richard Nixon, who helped to engineer this win-win situation through **ping-pong** diplomacy. But long before that, really, the history of the modern world was best told through the bouncing white **ball**.”**Ping-pong**”sounds like a cousin of”sing-song,”like something Eastern, but actually, it ’ s believed that it was invented by high-class Brits during Victorian times, who started hitting wine corks over walls of books after dinner. No exaggeration. And by the end of World War I, the sport was dominated by players from the former Austro-Hungarian Empire : eight out of nine early world championships were claimed by Hungary. And Eastern Europeans grew so adept at hitting back everything that was hit at them that they almost brought the whole sport to a standstill. In one championship match in Prague in 1936, the first point is said to have lasted two hours and 12 minutes. The first point! Longer than a”Mad Max”**movie**. And according to one of the players, the umpire had to retire with a sore neck before the point was concluded. That player started hitting the **ball** back with his left hand and dictating chess moves between shots. Many in the audience started, of course, filing out, as that single point lasted maybe 12, 000 strokes. And an emergency meeting of the International Table Tennis Association had to be held then and there, and soon the rules were changed so that no game could last longer than 20 minutes. Sixteen years later, Japan entered the picture, when a little-known watchmaker called Hiroji Satoh showed up at the world championships in Bombay in 1952. And Satoh was not very big, he wasn ’ t highly rated, he was wearing spectacles, but he was armed with a paddle that was not pimpled, as other paddles were, but covered by a thick spongy rubber foam. And thanks to this silencing secret weapon, the little-known Satoh won a gold medal. One million people came out into the streets of Tokyo to greet him upon his return, and really, Japan ’ s postwar resurgence was set into motion. What I learned, though, at my regular games in Japan, is more what could be called the inner sport of global domination, sometimes known as life. We never play singles in our club, only doubles, and because, as I say, we change partners every five minutes, if you do happen to lose, you ’ re very likely to win six minutes later. We also play best-of-two sets, so often, there ’ s no loser at all. **Ping-pong** diplomacy. And I always remember that as a boy growing up in England, I was taught that the point of a game was to win. But in Japan, I ’ m encouraged to believe that, really, the point of a game is to make as many people as possible around you feel that they are winners. So you ’ re not careening up and down as an individual might, but you ’ re part of a regular, steady chorus. The most skillful players in our club deploy their skills to turn a

9-1 lead for their team into a 9-9 game in which everybody is intensely involved. And my friend who hits these high, looping lobs that smaller players flail at and miss — well, he wins a lot of points, but I think he ' s thought of as a loser. In Japan, a game of **ping-pong** is really like an act of love. You ' re learning how to play with somebody, rather than against her. And I ' ll confess, at first, this seemed to me to take all the fun out of the sport. I couldn ' t exult after a tremendous upset victory against our strongest players, because six minutes later, with a new partner, I was falling behind again. On the other hand, I never felt disconsolate. And when I flew away from Japan and started playing singles again with my English archrival, I noticed that after every defeat, I was really brokenhearted. But after every victory, I couldn ' t sleep either, because I knew there was only one way to go, and that was down. Now, if I were trying to do business in Japan, this would lead to endless frustration. In Japan, unlike elsewhere, if the score is still level after four hours, a baseball game ends in a tie, and because the league standings are based on winning percentage, a team with quite a few ties can finish ahead of a team with more victories. One of the first times an American was ever brought over to Japan to lead a professional Japanese baseball team, Bobby Valentine, in 1995, he took this really mediocre squad, he lead them to a stunning second-place finish, and he was instantly fired. Why?"Well,"said the team spokesman,"because of his emphasis on winning."Official Japan can feel quite a lot like that point that was said to last two hours and 12 minutes, and playing not to lose can take all the imagination, the daring, the excitement, out of things. At the same time, playing **ping-pong** in Japan reminds me why choirs regularly enjoy more fun than soloists. In a choir, your only job is to play your small part perfectly, to hit your notes with feeling, and by so doing, to help to create a beautiful harmony that ' s much greater than the sum of its parts. Yes, every choir does need a conductor, but I think a choir releases you from a child ' s simple sense of either-ors. You come to see that the opposite of winning isn ' t losing — it ' s failing to see the larger picture. As my life goes on, I ' m really startled to see that no event can properly be assessed for years after it has unfolded. I once lost everything I owned in the world, every last thing, in a wildfire. But in time, I came to see that it was that seeming loss that allowed me to live on the earth more gently, to write without notes, and actually, to move to Japan and the inner health club known as the **ping-pong** table. Conversely, I once stumbled into the perfect job, and I came to see that seeming happiness can stand in the way of true joy even more than misery does. Playing doubles in Japan really relieves me of all my anxiety, and at the end of an evening, I notice everybody is filing out in a more or less equal state of delight. I ' m reminded every night that not getting ahead isn ' t the same thing as falling behind any more than not being lively is the same thing as being dead. And I ' ve come to understand why it is that Chinese universities are said to offer degrees in **ping-pong**, and why researchers have found that ping- pong can actually help a little with mild mental disorders and even autism. But as I watch the 2020 Olympics in Tokyo, I ' m going to be keenly aware that it won ' t be possible to tell who ' s won or who ' s lost for a very long time. You remember that point I mentioned that was said to last for two hours and 12 minutes? Well, one of the players from that game ended up, six years later, in the concentration camps of Auschwitz and Dachau. But he walked out alive. Why? Simply because a guard in the gas chamber recognized him from his **ping-pong** playing days. Had he been the winner of that epic match? It hardly mattered. As you recall, many people had filed out before even the first point was concluded. The only thing that saved him was the fact that he took part. The best way to win any game, Japan tells me every other night, is never, never to think about the score. Thank you.

Text Sample 892: POS Dim 6 – 2019 – Score 3.00 – 2019/t002071.txt

So my story starts on July 4, 1992, the day my mother followed her college sweetheart to New York City from Egypt. As fireworks exploded behind the skyline, my father looked at my mother jokingly and said,"Look, habibti, Americans are celebrating your arrival."Unfortunately, it didn ' t feel much like a celebration when, growing up, my mother and I would wander past Queens into New York City streets, and my mother with her hijab and long flowy dresses would tighten her hand around my small fingers as she stood up against weathered comments like,"Go back to where you came from,""Learn English,""Stupid immigrant."These words were meant to make us feel unsafe, insecure in our own neighborhoods, in our own skin. But it was these same streets that made me fall in love with New York. Queens is one of the most diverse places in the world, with immigrant parents holding stories that always start with something between three and 15 dollars in a pocket, a voyage across a vast sea and a cash-only hustle sheltering families in jam-packed, busted apartments. And it was these same families that worked so hard to make sure that we had safe micro-communities — we, as immigrant children, to feel affirmed and loved in our identities. But it was mostly the women. And these women are the reason why, regardless of these statements that my mom faced, she remained unapologetic. And these women were some of the most powerful women I have ever met in my entire life. I mean, they had networks for everything. They had rotations for who watched whose kids when, for saving extra cash, for throwing belly dance parties and memorizing Koran and learning English. And they would collect small gold tokens to fundraise for the local mosque. And it was these same women, when I decided to wear my hijab, who supported me through it. And when I was bullied for being Muslim, I always felt like I had an army of unapologetic North African aunties who had my back. And so every morning at 15, I would wake up and stand in front of a mirror, and wrap beautiful bright silk around my head the way my mother does and my grandmother did. And one day that summer 2009, I stepped out into the streets of New York City on my way to volunteer at a domestic violence organization that a woman in my neighborhood had started. And I remember at that moment I felt a yank at the back of my head. Then someone pulled and grabbed me, trying to remove my hijab from off of my head. I turned around to a tall, broad- shouldered man, pure hate in his eyes. I struggled and fought back, and finally was able to get away, hid myself in the bathroom of that organization and cried and cried. I kept thinking to myself,"Why does he hate me? He doesn ' t even know me."Hate crimes against Muslims in the US increased by 1, 600 percent post-9 / 11, and one in every four women in the US will suffer some form of gender violence. And it may not seem like it, but Islamophobia and anti-Muslim violence is a form of gender violence, given the visibility of Muslim women in our hijabs. And so I was not alone, and that horrified me. It made me want to do something. It made me want to go out there and make sure that no one I loved, that no woman would have to feel this insecure in her own skin. So I started to think about how the women in my own neighborhood were able to build community for themselves, and how they were able to use the very little resources they had to actually offer something. And I began to think about what I could potentially offer to build safety and power for women. And through this journey, I learned a couple of things, and this is what I want to share with you today, some of these lessons. So lesson number one : start with what you know. At the time, I had been doing Shotokan karate for as long as I could remember, and so I had a black belt. Yeah. And so, I thought — surprise. I thought that maybe I should go out into my neighborhood and teach self- defense to young girls. And so I actually went out and knocked on doors, spoke to community leaders, to parents, to young women, and finally was able to secure a free community center basement and convince enough young women that they should come to my class. And it actually all worked out, because when I pitched the idea, most of the responses were, like,"All right, cute, this 5 ' 1"hijabi girl who knows karate. How nice."But in reality, I became the Queens, New York version of Mr. Miyagi at 16 years old, and I started teaching 13 young women in that community center basement self-defense. And with every single self-defense move, for eight sessions over the course of that summer, we began to understand the power of our bodies, and we began to share our experiences about our identities. And sometimes there were shocking realizations, and other times there were tears, but mostly it was laughs. And I ended that summer with this incredible sisterhood, and I began to feel much safer in my own skin. And it was because of these women that we just kept teaching. I never thought that I would continue, but we just kept teaching. And today, nine years, 17 cities, 12 countries, 760 courses and thousands of women and girls later, I ' m still teaching. And what started as a self-defense course in the basement of a community center is now an international grassroots organization focused on building safety and power for women around the world : Malikah. Now, for lesson number two : start with who you know. Oftentimes, it could be quite exciting, especially if you ' re an expert in something and you want to have impact, to swoop into a community and think you have the magic recipe. But very early on I learned that, as esteemed philosopher Kendrick Lamar once said, it ' s really important to be humble and to sit down. So, basically, at 15 years old, the only community that I had any business doing work with were the 14-year-old girls in my neighborhood, and that ' s because I was friends with them. Other than that, I didn ' t know what it meant to be a child of Bengali immigrants in Brooklyn or to be Senegalese in the Bronx. But I did know young women who were connected to those communities, and it was quite **remarkable** how they already had these layers of trust and awareness and relationship with their communities. So like my mother and the women in her neighborhood, they had these really strong social networks, and it was about providing capacity and believing in other women ' s definition of safety. Even though I was a self-defense instructor, I couldn ' t come into a community and define safety for any other woman who was not part of my own community. And it was because, as our network expanded, I learned that self-defense is not just physical. It ' s actually really emotional work. I mean, we would do a 60- minute self-defense class, and then we ' d have 30 minutes reserved for just talking and healing. And in those 30 minutes, women would share what brought them to the class to begin with but also **various** other experiences with violence. And, as an example, one time in one of those classes, one woman actually started to talk about the fact that she had been in a domestic violence relationship for over 30 years, and it was her first time being able to **articulate** that because we had established that safe space for her. So it ' s powerful work, but it only happens when we believe in women ' s agency to define what safety and what power looks like for themselves. All right, for lesson number three — and this was the hardest thing for me — the most important thing about this work is to start with the joy. When I started doing this work, I was reacting to a hate-based attack, so I was feeling insecure and anxious and overwhelmed. I was really afraid. And it makes sense, because if you take a step back, and I can imagine that a lot of women in this room can probably relate to this, the feeling, an overwhelming feeling of insecurity, is oftentimes with us constantly. I mean, imagine this : walking home late at night, hearing footsteps behind you. You wonder if you should walk faster or if you should slow down. You keep your keys in your hand in case you need to use them. You say,”Text me when you get home. I want to make sure you are safe.”And we mean those words. We ’ re afraid to put down our drinks. We ’ re afraid to speak too much or too little in a meeting. And imagine being woman and black and trans and queer and Latinx and undocumented and poor and immigrant, and you could then only imagine how overwhelming this work can be, especially within the context of personal safety. However, when I took a step to reflect on what brought me to this work to begin with, I began to realize it was actually the love that I had for women in my community. It was the way I saw them gather, their ability to build for each other, that inspired me to keep doing this work day in and day out. So whether I was in a refugee camp in Jordan or a community center in Dallas, Texas or a corporate office in Silicon Valley, women gathered in beautifully magical ways and they built together and supported each other in ways that shifted culture to empower and build safety for women. And that is how the change happens. It was through those relationships we built together. That ’ s why we don ’ t just teach self-defense, but we also throw dance parties and host potlucks and write love notes to each other and sing songs together. And it ’ s really about the friendship, and it ’ s been so, so fun. So the last thing I want to leave you with is that the key takeaway for me in teaching self-defense all of these years is that I actually don ’ t want women, as cool as the self-defense moves are, to go out and use these self-defense techniques. I don ’ t want any woman to have to de-escalate any violent situation. But for that to happen, the violence shouldn ’ t happen, and for the violence not to happen, the systems and the cultures that allow for this violence to take place to begin with needs to stop. And for that to happen, we need all hands on deck. So I ’ ve given you my secret recipe, and now it ’ s up to you. To start with what you know, to start with who you know and to start with joy. But just start. Thank you so much.

Text Sample 893: POS Dim 6 – 2022 – Score 4.00 – 2022/t003655.txt

David Biello : In the wake of Dr. Anthony Fauci ’ s announcement that he will be retiring as the head of the National Institute of Allergy and Infectious Diseases at the end of the year, I ’ ve invited him to join me for a conversation on the future of the COVID-19 pandemic, reflections on his career and the future of public health. Please welcome Brooklyn ’ s own Dr. Fauci. Anthony Fauci : Hi, David, nice to see you, thank you for having me. DB : Thank you for joining us. So my first question is very simple. Is the COVID-19 pandemic over? AF : You know, David, there ’ s a lot of misinterpretation about what the meaning of the word”over”is, it means different things to different people. I ’ m sure you ’ re referring to the comment made by the President a day or two ago. If you ’ re talking about the fulminant phase of the outbreak, when we were having anywhere from 800, 000 to 900, 000 infections a day and 3, 000 to 4, 000 deaths per day, that was several months ago. We are much, much better off now than we were then. So in that case, that fulminant phase of the outbreak is behind us. But as the President made very clear on the second half of his sentence, is that, which they don ’ t seem to show, is that he actually said we still have a lot of work to do, there ’ s still a challenge ahead, we ’ ve got to get people vaccinated. We still have a number of cases, we have 400 deaths per day. That ’ s an unacceptably high level. So again, it depends on the semantics of what your definition is. We don ’ t want anyone to get the impression that we don ’ t have a lot of work to do. We ’ ve got to get the level of infection considerably lower than it is. And we ’ ve certainly got to get the level of deaths lower than it is. So again, it depends on the semantics of what you mean by"end."DB : Well, let ' s get into those semantics then. What conditions would need to exist for the pandemic to be over? AF : You know, that ' s a call that officially is made by the World Health Organization. You know, my colleagues and I oh, it must have been more than ten years ago, because of the lack of clarity on what a pandemic means to one person versus another, we wrote a paper in the Journal of Infectious Diseases in which we talked about all the different variations of interpretations. Is it a widespread phenomenon? Is it the widespread nature throughout the world that makes it officially a pandemic? Or is it widespread and accelerating? Or is it widespread causing serious disease? It means different things to different people. So rather than try and give a definition, that ' s my definition versus another, we should stick with what the WHO is saying. And as Dr. Tedros said, that we ' re seeing the light at the end of the tunnel on that. So, again, you might interpret that, well, if that ' s the case, is it over? Well, again, what does"over"mean? There ' s a lot of semantics there, David. The easiest way not to confuse people is to say we still have a lot of cases, we still have 400 deaths, we only have 67 percent of the population vaccinated, we ' ve got to do better than that. Of those, only one half have gotten a single boost. And as a nation, we lag behind other developed countries and even some low- and middle-income countries in the level of vaccinations that we ' ve been able to implement. So if you want to look it that way, we have a lot of work to do, and that ' s exactly what the President said. DB : So speaking of vaccines, what ' s your advice? I know we have the bivalent available now. What is your advice on which vaccines to get and when that ' s appropriate? AF : Well, first, you ' ve got to get your primary series. So, I mean, that ' s the one where I said only 67 percent of the country has gotten their primary series, which for the most part, with some exceptions, is an mRNA vaccine, either Moderna or Pfizer, given anywhere from three to four weeks apart as the primary. Then the issue is about giving people booster shots. So right now, if you ' re asking a clear question, of today, with the bivalent BA. 4-5 boosters or updated vaccines is a better terminology to use, they are available now throughout the country. We ' ve ordered 171 million doses. Who should get it? Anyone who is vaccinated with the primary series and has not received a shot longer than two months ago should get it. So if I got my last shot, let ' s say in July – August, September, two months later, you should get the bivalent. If you were infected three months or more ago, you should then get the updated vaccine. Let me give you an example for clarity for the audience. I was infected in the end of June of this year, even though I had been vaccinated. Fortunately, because I was vaccinated, I had a relatively mild illness. At my age, had I not been vaccinated, the chances are I could have had a real severe outcome because elderly are more prone to get severity of disease. So if you take the end of June, take the end of July, the end of August, the end of September. I plan to get my updated BA. 4-5 bivalent at the end of September, the first week in October. DB : I think that ' s a very useful advice, and actually I will be doing the same. How are you personally navigating this stage in the pandemic? What precautions are you taking, if any? AF : Well, I certainly continue to take precautions. And I think it ' s important that you ask me personally that question, because people have different levels of risk of severity of disease. I am a person who is relatively healthy, but I ' m at an elderly age. I ' m 81 years old. I ' m going to be 82 in December. So I, statistically, would have more of a risk. So the precautions I take, I stay up to date on my vaccinations, number one. And when I go to a place that ' s a congregate indoor setting where there are a lot of people and I don ' t know the status of their infection, their vaccination or what have you, I would, for the most part, wear a **mask**. When I ' m with people who I know what their status is, people who are recently vaccinated or people who come in and test before they come in, I could have a dinner in my home or in the home of a friend without any concern. But if I go to crowded places, certainly on an airplane, even though it isn ’ t required any more, if I go on a prolonged or even a short airplane trip, again, because of my increased risk as an elderly person, I wear a **mask** on the plane. DB : So switching gears a little bit, or taking a step back, are consecutive pandemics kind of our new reality? We obviously had the monkeypox outbreak. And if that is the case, how do we cope with consecutive pandemics? AF : Well, you know, we have had, probably without the general public noticing it much, we have had, in the history of our civilization, outbreaks of emerging infections, some of which turn into pandemics. We ’ ve had them before recorded history. We ’ ve had them in the lifetime of some of us, you and I. We ’ re going through one right now. So given the fact that most of the outbreaks of new infections come from the animal-human **interface**, which is sometimes intruded upon, as it were, where people, either by climate change or by intruding on forests,”uninhabited by human”places in the world, you ’ re going to get jumping of species. Or in markets where you put animals from the wild in contact with humans, which is exactly what happened with SARS-CoV-1 and highly likely happened with SARS-CoV-2. We will continue to get outbreaks. Pandemic flu generally comes from a situation where you have the animal species that harbor influenza, pigs, fowl, birds and humans together in that environment. That ’ s how you get the bird flu that tend to challenge us a fair amount or the swine flu. So the short answer to your question, David, is that we will continue to get outbreaks of new infections. The critical issue is how do you prevent them from becoming pandemics? And that ’ s what ’ s called pandemic preparedness, which is a combination of scientific preparedness, like we did with the rapid development of vaccines for COVID-19, which was a highly successful scientific endeavor, matched with a public health response, which we didn ’ t do as well, in the public health response, because we had spread of infection in a way that we could have done better in controlling it. DB : So speaking of that, this is not your first epidemic, but you ’ ve made historic contributions to the AIDS epidemic and now COVID-19. But is there anything you wish you had done differently in those cases? AF : Well, there always is. I mean, it ’ s a question of when you ’ re involved – Nobody is perfect, certainly not I or any of my colleagues. But when you ’ re dealing with an emerging, moving, dynamic target, which by definition is what a pandemic is, particularly if it ’ s with a pathogen that you ’ ve never had experience with, like HIV in the very early 1980s, or the COVID-19 pandemic in the first months of 2020, you could always say, if we knew then what we know now, and there was a lot of things that we didn ’ t know, we certainly would have done things differently. And that ’ s why you have to be humble and modest to realize if you are going to be following the science, the science which gives you data and information and evidence is going to change, particularly in the early phases of the outbreak. Did we know how easily it was spread from human to human? No. Did we know that it was aerosol spread? No. We thought in the beginning it was like influenza, mostly droplets from a sick person. Did we know that 50 to 60 percent of the transmissions were [from] someone who had no symptoms at all, which clearly impacts how you approach an outbreak? Did we realize that instead of the **typical** outbreak, where it goes up, it comes down and then you ’ re done with it, we had no concept that you ’ d be seeing different waves and different variants that came along. So the answer to your question, if we knew all of that from the beginning, we certainly would have done things differently. But unfortunately we didn ’ t. And you try to be flexible enough and humble enough to change and modify how you approach things based on the recent data. That

’ s not flip-flopping. That is truly following the evidence and following the data. DB : Right, that ’ s just how science works, it ’ s constantly updating and especially in a real-time situation like this pandemic. But are there any, I don ’ t know, specific regrets you have, something you would take back if you could? AF : Well, it depends. Yeah, I mean, obviously in the beginning when we were under the impression, or didn ’ t fully realize that there was aerosol spread, we were under the impression, which was true, because we were told that, that there weren ’ t enough **masks** for the health care providers. And if we started everybody hoarding **masks**, there wouldn ’ t be **masks** available to the health care providers. We didn ’ t realize that out of the health care setting, like the hospital setting, that **masks** were effective in preventing acquisition and transmission. We didn ’ t know that. We know it now for sure, but we didn ’ t know it then. We didn ’ t know that the silent spread from people who were without symptoms. And that meant we didn ’ t know that, while we were looking for sick people, there were many, many, many more people without symptoms that were in society spreading the infection in a way that was not detectable, below the radar screen. Had we known that, I absolutely would have said right from the beginning, everybody wear a **mask** all the time in an indoor setting. But we didn ’ t say that then. It was only when it became obvious. So if we had known that early on, we would have told people to wear a **mask**. However, I must say, David, given the reluctance of people to wear **masks**, even now that we know all that stuff, I ’ m wondering how well that would have been received if at a time when there were ten or so documented infections, if you told the country that everybody should wear a **mask** in an indoor setting, not so sure that would have been broadly accepted. DB : Yeah. And this seems to be something that, the United States anyway, has been through before, with the Spanish flu and **masking** and then anti-mask protests. And it seems to be in our, let ’ s say, societal immune response to these pandemics. Flipping it a little bit, how do we make sure we ’ re not caught so, sort of, unprepared next time? AF : Well, you know, it ’ s interesting, David, what you mean by unprepared, because the Johns Hopkins School of Public Health evaluated different countries ’ preparedness for a pandemic. And guess who was evaluated to be first in the world? The United States of America. Guess who has the most deaths per population? The United States of America. So, you know, there ’ s preparedness, and there ’ s response, there ’ s execution of your preparedness plans that we did not do so well for any of a number of complicated reasons, one of which was the fact, and is, that this outbreak occurred at a time of really profound and deep divisiveness in our own country. That where we had something we hadn ’ t seen before, where political ideology played a role in whether you did or did not accept the recommended public health countermeasures, be it wearing a **mask**, indoor settings, quarantining, taking a vaccine, getting a boost. If you look at the country and look at the demography of the country with regard to ideology, there should never be that red states vaccinate much less than blue states. There ’ s no reason for that at all because public health risks and public health implementation should be uniform throughout. And we didn ’ t see that. DB : Yeah, let ’ s talk about that a little bit more because obviously, you ’ ve had to cope with an incredibly polarized response to your work in which not just ordinary citizens, but even politicians have called for your resignation and **various** other things. How do you deal with that, how do you keep your cool? You ’ re quite cool about this. AF : Well, calling for your resignation is mild compared to having somebody arrested who was trying to kill you. So, I mean, there ’ s a big spectrum of pushing back against public health people. And that ’ s one of the really unfortunate things. I mean, I keep my cool because that ’ s just the nature of the kind of person that I am. When you ’ re dealing with a very, very difficult situation, you ’ ve got to keep your cool. I mean, I learned that in my early training in medicine. When you ’ re in the middle of emergency, somebody is dying in front of you, you ’ ve got to keep your cool all the time. And that ’ s something that ’ s just part of my inherent training as a physician and as a person. But what we faced was well beyond that. I mean, public health officials, not only myself, I ’ m a very visible one, but many of my colleagues are being threatened and hassled and harassed, themselves and their families, the way my family is being harassed, merely because of saying things that are purely public health, common sense, tried and true principles of how to keep people safe. That is really extraordinary that that ’ s going on in our country. DB : Yeah. I ’ m also going to ascribe it to the Brooklyn upbringing. That gave you some cool, too. But a quick follow up, how do you, like, with all that going on, with those horrifying threats, how do you unwind from all this? How do you, you know, keep it together and give yourself some space to breathe? AF : Well, I have an extraordinarily supportive family, my wife who ’ s with me, my children are grown and live in different parts of the country. But they are very supportive of me with texts and calls and knowing what I ’ m going through. So I have three daughters, which they, you know, they try to take care of their daddy, so it really helps. But my wife is extremely supportive, and we do things together. I mean, I work a preposterous amount of hours a day, but every day I try to get some exercise in and it ’ s usually a few mile walk with my wife, whether that ’ s on the weekends very early in the morning, or during the week late at night when I come home. We try to get some form of exercise in to diffuse the tension, hopefully every day. And we ’ re pretty successful at that. AF : Well, good for you. So you ’ re retiring after a very long and distinguished career. Congratulations. You will have successors. What lessons would you want to offer your successors based on your tenure? AF : First of all, I ’ m not retiring in the classic sense. As my wife, says, David, I ’ m”rewiring,”not retiring, because I do intend to be very active. And that was one of the reasons why I stepped down at this point in time. Because while I still have the enthusiasm, the energy and thank goodness, the good health to be able to do something else for the next few years, I want to use the benefit of my experience of being at the NIH for almost 60 years, for being the director of the Institute for 38 years, and for having the privilege of advising seven presidents of the United States on public health issues, to use that experience to hopefully inspire by writing, reading, traveling, lecturing, inspiring the younger generation of scientists and would-be scientists to at least consider a career in public service, particularly in the arena of public health, and science and medicine. Having said that, my advice to the person who will ultimately replace me would be to focus on the science and be consistent with the science and do not get distracted by a lot of the peripheral things, the disinformation, the misinformation, the attacks on medicine and science and public health. Focus like a laser beam on what your job is and don ’ t get distracted by all the other noise that ’ s out there because there is a lot more noise now than there was a few decades ago. And by noise, I mean misinformation and disinformation about science. DB : For sure. I ’ ve definitely noticed that as the science curator. But pivoting a bit, let ’ s talk about hope. What gives you hope about the future? Are there treatments or other things coming down the pipeline that you ’ re excited about? AF : Well, science is an absolutely phenomenal discipline. It ’ s discovery. It ’ s... brand new knowledge. Pushing back the frontiers of knowledge that we would not have imagined we would be in. If you look at medicine and science, how it ’ s changed in a very, very positive way from the time I stepped into medical school in 1962 to the time now of the things that are available to me as a physician and as a scientist. I have great hope that if we continue the investments in basic and applied science, that we will be able to accomplish things in the arena of health, individual health, and public health and global health that were really unimaginable just decades ago. So I have a great deal of optimism about what the future holds for science and medicine. And that ' s the reason why one of the things I ' m going to try and do in my rewired post-government life is to encourage young individuals to consider a career in medicine, science and public health, because the opportunities are really limitless. We are at a stage now the likes of which people who antidated us never would have imagined the opportunities in science that we have. So I ' m very optimistic about where we ' re going. DB : Me, too. But you mentioned earlier how climate change is affecting pandemics. What worries you about the future that we ' re facing? AF : Well, just some of the things that you mentioned. We do have a growing element of anti-science in society. Disturbingly, growing in the United States. And when I talk to my colleagues internationally, depending upon the country that they live in to a greater or lesser degree, there ' s some element of that. The thing that bothers me is a denial of science and what science is showing us. A denial of the issues of climate and the environment, a denial of scientific principles and conspiracy theories about things that push people away. I mean, some of them are laughable, but you would be astounded, David, at the number of people who believe it. That the vaccines were made by Bill Gates and I and we put a chip in it so that we could follow people around and know what they ' re doing and get into their head. The idea that there ' s a conspiracy, that we ' re making billions of dollars on vaccines, and that ' s why people are promoting vaccines, so that people like me and others who are public figures... I mean, based on no data. But once it gets into the social media, it explodes and conspiracy theories explode. Even though all of the evidence proves it wrong, proving something wrong today doesn ' t seem to matter much. That ' s really strange and weird, isn ' t it, David? DB : Yeah, yeah. Well, it ' s, I think, an old phenomenon. There ' s a famous saying that I ' m going to mangle about by the time the truth gets out of the barn, the lie is halfway around the world. And that ' s definitely the world we ' re living in. I do know you have to go and, you know, finish out your tenure and then get ready for your rewiring, which I ' m excited about. I ' m excited to see what ' s next. But do you have a last bit of advice for the public? One thing you hope everyone takes away from your time as director and your time in public health? AF : Yeah, David, there are so many things, but I think one thing that stands out, particularly in the climate and the environment that we ' re in right now, is that people need to get involved in a proactive way in spreading the truth about what scientific principles are and what they mean and how they can be of great benefit to society and to try and make our population more science literate than it is right now by talking about things. The truth, you know, the easiest way to counter misinformation and disinformation is to be enthusiastic about spreading correct information. A little bit about the metaphor that you said about the truth and lies. Be spreaders of facts and truth. Everybody ' s got to be a contributor to that. And that ' s one of the things I think we can do better. DB : Yeah, I would agree. And we ' re definitely trying here at TED. Well, Dr. Fauci, I know our members are incredibly thankful, I personally am incredibly thankful. So thank you so much for your work and for joining us today. AF : My pleasure, David, thank you so much for having me. [Want to join conversations like this live?] [Become a TED Member!] [Sign up at ted. com / membership]

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Bruno Giussani : It ’ s difficult to think clearly of the Russian invasion of Ukraine, because wars, while they unfold, they ’ re kind of shrouded in a sort of fog. Information is abundant : the millions of refugees, the shocking suffering and the destruction, the politics. But sense is lacking. And that ’ s going to be the focus of this Membership conversation as we enter the third week of the war. We won ’ t talk about the events of the day but try to project a longer arc, a broader context. Our guest is geopolitical analyst Ian Bremmer. He ’ s the founder and president of Eurasia Group, and we asked him to lay the scene by talking first about the geopolitical shifts that have already been brought by the war in Ukraine. And after, we ’ re going to have a conversation, including questions from TED Members who are participating in this call. Ian, welcome. Ian Bremmer : Thank you very much. I ’ ll start by saying that in my lifetime, the most important geopolitical **artifact** is the fall of the Berlin Wall. I mean, you see it if you go into the new NATO headquarters in Brussels, just built a few years ago. And anyone that has a piece, something they ’ re very proud of, they know it affected their entire lives. I think that in 30 years ’ time, and I fear that in 30 years ’ time, if we look back, a second most important geopolitical **artifact** will be a piece of the rubble of the Maidan in Kyiv. I believe that the war that we are seeing right now is no more and no less than the end of the peace dividend that we all thought we had when the wall came down in 1989. The idea that the world could focus more on globalization and goods and services and people and ideas going faster and faster across borders, leading to unprecedented growth in human development and a global middle class. I think that this is a tipping point. Won ’ t end globalization, but it does end the peace dividend. It does mean that the Europeans overnight will and must prioritize spending on defense policy, on national security, coordination, on NATO. And the speech that was given by Olaf Scholz, the new chancellor, two weeks ago, in my view, the most significant speech given by a European leader in the post-Cold War environment, precisely because it ’ s now the post-post-Cold War environment, sending weapons to the Ukrainians, committing to over two percent of GDP spend on defense, investing in a new fund for defense infrastructure. But also recognizing that the way that the Germans and the Europeans as a whole looked at the world and looked at themselves was, unfortunately for all of us, outdated. A few other points I ’ d like to raise, just to kick off this conversation. One : One of the reasons I ’ m pretty negative about this, and I ’ m not usually very negative, I ’ m usually an existential optimist, I ’ m someone that ’ s just happy there ’ s water in the glass. But when I look at this conflict, I ’ m much more concerned. And that is because I do not see a scenario, a plausible scenario, in the foreseeable future where Putin emerges from this war in anything less than a radically weakened position compared to where he was before he announced the invasion. And I believe that both in terms of his domestic political orientation, how stable he is in his own country, also, of course, Russia ’ s economic position, and finally, Russia ’ s position in terms of global security and European security : ostensibly, the very reason that Putin began the war to begin with. Second big point is that the decoupling that we are seeing from Europe and the United States with Russia is, in my view, permanent. And that would be true even if there were a negotiated settlement and all the Russian **troops** were to pull out of Ukraine and we had peace. I still think that a lot of those companies would not come back with Putin in power. I ’ m convinced that the decisions by the Europeans to ramp up their national security capabilities will be permanent. Permanent deployments coming in the Baltic states, for example, forward deployments in Poland and Bulgaria and Romania. And also an unwind of Europe ’ s massive energy dependence : coal, oil and most importantly, gas on Russia. That does not make Russia a global pariah because you can ' t be a global pariah if the soon-to-be most important economy in the world, China, is actually your bestie on the global stage, and that indeed continues to be true despite China ' s efforts to portray themselves, towards Europe at least, as more neutral. We are going to see the Russians as a supplicant economically, in terms of energy flows, financially, in terms of transactions, and technologically, perhaps most important, aligned with China. That has big geopolitical implications long-term. Also, a lot of other developing economies, like the Indians, like the Gulf states, like Brazil, are also not going to work with Russia as a pariah. They ' ll continue to engage. Are there any silver linings? And I think there are a few. Of course, there is a much greater renewed purpose and mission of NATO. I mean, this is an organization that just a couple of years ago, France President Emmanuel Macron referred to as"brain-dead."It was increasingly drifting in terms of its importance. The Americans were focusing much more on Asia, the pivot. Not today. Today, NATO is purposeful, it ' s aligned, it ' s consolidated. it ' s going to get more resources, not less. That ' s also true of the European Union as a whole, even when we talk about countries like Hungary and Poland that have been much less aligned in terms of rule of law, in terms of an independent judiciary, much more aligned in terms of the importance of common values of Europe compared to that of what we ' re seeing in Moscow. I mentioned the German security and policy shift. The UK-EU relationship is much smoother and more functional than at any point since Brexit process actually started. And even the United States. I mean, if you watched the State of the Union for a brief moment in time, five or 10 minutes, when all of the Democrats and Republicans were standing and applauding together, you could be forgiven for believing that the United States had a functional representative democracy. Now I ' m not sure how long this is going to last, but at least as of now, leaders of the Democratic and Republican Party see Putin as much more of a threat, an enemy, than they do their opponents across the aisle in domestic politics. And two weeks ago, that was not true. That ' s significant. Final silver lining, and I wish it was more of one, but the Chinese, as much as they are strategically aligned with Russia and with the person of President Putin, they do not want a second Cold War. And they would rather a negotiated settlement. They ' re not willing to push very hard for it. But they certainly do not see a radical decoupling of the Russian, and potentially the Chinese economy, from the rest of the world, from Europe, from the US, from the advanced industrial democracies, as in any way in their interest. And ultimately, that does create at least some buffer, some guardrail on how much this is likely to escalate as a conflict going forward. BG : I want to make a step back and unpack some of that, maybe starting with a question that relates to your last point and is probably on the mind of many. And it is : Is there still some real space for negotiation? Is there still a potential relationship between Russia and Ukraine? IB : The foreign ministers of Russia and Ukraine just met recently in Turkey. It was as much of a non-event as the three preceding negotiations of more junior representatives of their teams on the Belarus border. The one thing that has been accomplished to a small degree has been humanitarian corridors, extending out of a number of Ukrainian cities that are being pounded by Russian military. That ' s because the Ukrainians are interested in protecting their civilians, and the Russians are interested in taking a lot of territory without necessarily having to kill so many Ukrainians, that could cause problems for them internationally as well as domestically inside Russia. But that is nowhere close to a negotiated settlement. Now, I mean, everyone I know that ' s involved in the negotiations right now responds that the President Putin himself is hell-bent on taking Kyiv and on removing Zelenskyy from power. Now I think, and by the way, they ' re getting quite close to being able to accomplish that militarily on the ground. I think within the next couple of weeks, certainly, it looks very likely. A couple of points here. One, there is no reason to put any stock in anything that the Russians are saying publicly in terms of their diplomacy. They lied to the face of every world leader about the invasion that they said they were not going to do into Ukraine. And then just today, Foreign Minister Sergey Lavrov publicly said, well, the Russians didn ' t attack Ukraine. I mean, this is Orwellian stuff, right? So first of all, do not report on Russian public statements as if they bear any semblance to reality on the ground. Secondarily, this looks like a huge loss for Putin right now. He understands it, and I think he would have a hard time, even with his control of information, spinning this to his public without removing Zelenskyy, without the"de-Nazification,"as he calls it – which is an obscenity in an environment where the Ukrainian president is actually Jewish – the disarmament of Ukraine, and of course, the ability of the Russians to change how they feel about Ukraine as a threat to the Russian homeland. BG : What level of support can we give Ukraine militarily, intel, economic, before Putin considers taking a strike on a NATO country? IB : Well, it ' s interesting the way you framed that, Bruno. Because I mean, I think that Putin is already considering strikes on NATO countries. I mean, there were massive attacks, cyberattacks and disinformation attacks, by Russia against NATO countries with reckless abandon over the course of the past years. And in fact, when President Biden met with Putin in Geneva back in June, it seems like years and years ago at this point, Biden set the agenda. Ukraine was largely not discussed, but what was discussed was cyberattacks on critical infrastructure. Because you may remember Bruno, that meeting came right after the cyber attacks against the Colonial Pipeline. And the Russians after that indeed pulled back on supporting those attacks by their criminal cyber syndicates. I expect those attacks to restart in very short order against NATO countries. I also believe that the fact that the West is sending weapons to Ukraine and is providing real-time intelligence reports on the disposition of Russian **troops** on the ground in Ukraine to better allow the Ukrainians to defend themselves and blow the Russians up, that is considered by the Russians to be an act of war, and they will retaliate. And how they retaliate is the question. I don ' t think they ' re going to send **troops** into Poland. But you know, when the Americans under Reagan were providing that kind of support to the mujahideen to help them defeat the Soviets in Afghanistan, the Soviets saw that as an act of war, and they engaged in acts of terrorism against the mujahideen in Pakistan. And I absolutely think that that is on the table in terms of front line NATO countries especially, like Poland, like the Baltic states, like Bulgaria, Romania. Would that be considered an Article V attack? Would that force NATO countries to strike the Russians back? I ' m not sure it would. Not directly, not militarily. So I mean, I do think that the fact that the NATO countries see there is some sort of a big red line between sending **troops** in, for example, and doing a no-fly zone because that could cause World War III, but everything short of that is just a proxy war. The Russians don ' t see it that way. And that gives the Russians some advantage tactically in terms of their willingness to escalate. BG : You ' re describing a spiral of escalation here that will touch the globe and not only Ukraine, not only the eastern flank of Europe, which means that not only this war has ripple effects everywhere, but this is also starting a sort of realignment of the global geopolitical situation and context. To me, it has been very striking how Europe and the US have kind of moved very fast in a cohesive way. And it has chosen, after years of prioritizing the economics in their international and global dealings, it ' s chosen to put politics over markets. It has adopted sanctions that will hurt Russia, but will also hurt Western businesses. It ' s the discussion about decoupling that you put forward before, an active kind of fencing out of the Russian economy. Talk to us about how do you see this decoupling playing out. IB : Yeah, I mean, I do think that for the Europeans, this is a permanent move. I mean, I ’ ve spoken to top leaders in the German government who tell me that Nord Stream was a strategic mistake, and they understand it. Who say that, you know, Scholz making this speech from the Social Democratic Party on the center left is the equivalent of Nixon going to China. No one else could have made that move. But having made it, everyone is on board. The popularity in Germany, even given the massive economic consequences, is extraordinary and is across the board. And what they need to do now is ensure that the diversification of fossil fuels in the near term away from Russia, towards Qatar and Azerbaijan and even, you know, sort of the United States for LNG, can be done as fast as humanly possible. And that further, even though it ’ s going to cost a lot, some of it will be uneconomic, the move towards renewables actually picks up and is faster. The Italians, Mario Draghi, I think his shift in strategic orientation that they will do, this is his”whatever it takes”moment. He had that in response to the 2008 financial crisis as the head of the European Central Bank, and that made him”Super Mario.”This is making him Super Mario as the Italian prime minister. This is the”whatever it takes”moment for the Italians who never make public statements that undermine their economic, their commercial interests like this in such a strategic way. The French, of course, have less to be concerned about in the sense that most of their energy comes from nuclear power and is domestic. So they are not as affected directly by a cut-off from Russia. And also because Macron fancies himself, especially as the leader of the European Commission this year, the rotating leadership, occupying the presidency, but also with his elections coming up, and just given his personal belief that he can drive diplomacy, I believe that even after Kyiv falls and after Zelenskyy is either killed or forced out that the Americans will not want to engage in direct diplomacy, the Germans probably won ’ t. The French will. And by the way, the Chinese will too. And I do believe that there is a potential, and this is a danger for the cohesiveness of the West, that the Chinese and Macron end up being the post-Kyiv Normandy format of diplomacy. That ’ s something that the Americans and the Germans right now are starting to worry about quite a bit. Now that ’ s the European shift. And I think, as I said, I think it ’ s permanent. I believe the UK is in that camp as well. I ’ m not so sure the United States is going to be as committed for as long a term. It doesn ’ t affect the Americans as much economically, it doesn ’ t affect the Americans as much in terms of a direct security issue. None of those refugees are coming to the United States. But also American inequality, American political polarization and dysfunction is so much greater than what you experience on the continent in Europe. So the potential that in six months ’ time or in two years ’ time, as we ’ re thinking about the 2024 election, that the Americans have largely forgotten about this Russia issue instead, are focusing once again on domestic political opponents as principal adversaries, which deeply undermines NATO, much more than anything that would come from the Europeans, I think that is a real open question going forward that is perhaps as significant as the question of where the Chinese go. BG : Let me pick up on the point you made about energy, because somehow Putin ’ s calculus can really change if Russian oil and gas stops flowing to Europe, if it becomes part of the sanctions, right? And this war indeed can kind of be read as a war about energy. Selling energy funds it for Russia, being dependent on Russian energy makes the European response more constrained. Rising energy insecurity, rising energy cost may or probably will destabilize European politics and economy in the coming months. How would you look at this from the perspective of energy, and is there any likelihood that

Russian oil and gas is going to stop flowing, either because Putin cuts it or the Europeans sanction it? IB : Yeah, or because it ' s blown up in some of the transit in Ukraine? I mean, keep in mind, so much of the gas transit is going through large pipeline networks, which have some redundancy across all of Ukraine. But there ' s a big war that ' s going on right there, and lots of people that could have incentive to create problems. The Americans, of course, the Canadians, have said that they ' re cutting off oil import from Russia, but those are nominal numbers, so they don ' t matter very much to the markets. The Europeans, as I said, want to decouple themselves as quickly as possible, but they believe that doing that this year would be economic suicide. So there isn ' t, despite everything we see from Russia, they ' re using thermobaric weapons now against the Ukrainian people, the Americans are warning that they could use chemical, biological weapons against Ukraine. I mean, you know, you even have some people saying, what if they use a tactical nuclear weapon? I mean... God willing, none of these things come to pass. But it is very hard to see a military scenario in Ukraine that leads the Europeans to completely cut off their inbound gas from Russia this year. It ' s very hard to see. And also, I would say, it ' s very hard to see any level of economic sanction that would change the mind of the Russians in terms of their military decision making on the ground in Ukraine. Now, I think there are a lot of things that the West is doing in terms of providing weapons for the Ukrainians that are having an impact on the ground. A lot more Russians are getting killed. It won ' t prevent them from taking to Kyiv, again in my mind I feel quite confident about that. But it ' s quite possible, perhaps even likely, that the west of Ukraine will remain in Ukrainian hands, which means that, you know, after this fighting is"over,"that a rump Ukrainian state in exile exists in the West, run by Zelenskyy or someone that ' s aligned with him, and that they continue to get enormous economic and military support from all of the NATO countries. So even though I don ' t think that the energy situation will become so parlous that it would affect Putin ' s decision making, I do think that the West ' s response does matter on the ground. BG : The war is kind of having radiating economic shock waves around the world now, ripple effects on food markets, for example and food security. We talk a lot about energy security, what about food security? IB : Well, you have the largest grain producer in the world invading the fifth largest grain producer in the world on the back of a two-year pandemic that ' s still ongoing. We don ' t talk about it much anymore, but it ' s still there. And of course, this hit the poorest countries in the world the hardest. The level of indebtedness and the unsustainability of paying that debt off was already becoming a massive problem for so many of the developing countries in the world. And the IMF provided a lot of relief in special drawing rights and direct aid over the course of the past 12 months, but that money is now running to an end. And what happens when commodity prices spike up and we have severe supply chain challenges with energy and food, and those things are obviously very related. What happens is that a lot of people die. What happens is we see a lot more starvation. The World Food Organization says about 10 million people a year die of starvation. That number in the next 12 months is going to be a lot higher than it otherwise would have been. The number of people who are food stressed in the world is going to go way up in sub-Saharan Africa, in Yemen, in Afghanistan, in Bangladesh. It ' s going to go way up. And you know, it ' s horrible to think about, but the massive impact of this Russia crisis is going to be much more global inequality. And this is, of course, a direct consequence of the end of the peace dividend more structurally. That over the last 30 years of globalization, what did you have? A lot of people were left behind, but the biggest thing you had was the explosion of a single global middle class. On the back of the pandemic and now this Russia-Ukraine war and the decoupling of the Russian economy from the West, which doesn ' t matter so much in terms of the size of the Russian economy, but it matters immensely in terms of commodities globally and supply chain, those two things are going to seriously unwind the growth of this global middle class, and they ' re going to stress developing countries to a much greater degree. They will lead to financial crises in countries like Turkey, for example, that will no longer be able to service their debt. You ' ll see more Lebanons out there. You ' ll see some in Latin America, you ' ll see some in sub-Saharan Africa. Those are the knock-on effects and so, so many people that have been saying over the last few weeks,"Why are we paying so much attention to Ukraine? It ' s because they ' re white people, because they ' re European. We wouldn ' t pay that much attention if they were Afghans or if they were, you know, Afghanis or if they were Yemenis. We wouldn ' t."I mean, first of all, you ' ve got millions and millions of Ukrainian refugees, and we ' re not paying as much attention to them as we did to the Syrian refugees precisely because of race, precisely because the Europeans are more willing to integrate millions and millions of"fellow Europeans"into Europe. But we are paying much more attention to the Ukraine crisis and we should, because the impact on the poorest people around the world is vastly greater from this conflict than anything that we ' ve seen in any of those smaller economies with less impact, despite all of the human depredation that ' s happened over the past 30 years. BG : Ian, I want to talk for a second about climate because another crisis that has, kind of, disappeared from the headlines is the climate crisis, right? Ten days ago, the IPCC released a report that the secretary general of the UN described as an"atlas of human suffering,"if I remember correctly. And it has been basically ignored. Over the last several years, much of the world had started to embark, with more or less enthusiasm, on a process of transitioning away from oil and gas and into kind of a clean energy future. And now the war comes in and we look at what you just described, the unraveling of global supply chains, the dependency on energy and so on. And there are kind of two schools of thought here. One says this war is going to accelerate the adoption of clean energy because we need to diminish dependence from Russia and these fossil fuels. And the other says, the other school of thought says it ' s going to derail the transition to clean energy because suddenly the priority is no longer decarbonization, suddenly the priority is energy security, energy supply. IB : The Europeans are largely in the first camp, and they will move towards faster decoupling and investment accordingly. The Americans are largely in the second camp, and they will move towards"Let ' s focus more on fossil fuels and partisan divide on this issue,"accordingly. The Chinese, who are the largest carbon emitter in the world by a long margin, though not per capita and not historically, but still in terms of every year totals, they will continue on the same path they ' ve been on, which is a net-zero target but without yet a very strong plan on how to get there and not feeling a lot of pressure to provide that plan, because they think the Americans are completely incoherent and incapable of effectuating a strategic long-term plan on climate themselves. So I mean, what we have is a lot of progress on climate and, of course, technology around renewable energies and around electric batteries and supply chain continue to get cheaper and cheaper as more money is being invested in it. And that does make me long-term more optimistic that by 2045, a majority of the world ' s energy will probably be coming from renewables. And five years ago, I wouldn ' t have said that. But still, I mean, when the news today is that the Americans are sending a high-level delegation to Caracas to figure out if we can reopen relations with Venezuela to get them to produce more oil again. With the Iranians, let ' s do any deal possible to get back into the JCPOA, the nuclear deal, so that we can get that oil on the market. Calling the Saudis, calling the Emiratis, and they ’ re not willing to take Biden ’ s phone call on this issue while they ’ re talking to Putin. Those are warning signals that in the near term, we ’ ve got some big challenges and a lot of those challenges are going to be filled with fossil fuels and fossil fuel development. And so I do think that the fact that both of the answers to your question are true in different places on net-net is more negative for how quickly we can transition. BG : Let ’ s talk a bit about China. Brigid, I think, who ’ s listening in asks,”What do you believe Xi Jinping is learning from the world ’ s response to the crisis, to the Ukrainian war?”IB : Well, certainly learning that this was a red line for the West. And I think that this would have surprised, it obviously surprised Putin, I think it would have surprised Xi Jinping as well. Xi Jinping saw Afghanistan. He saw that Merkel was out. He saw that Macron is focused on strategic autonomy. He sees Biden as much more focused on China and Asia. I think that this is a surprise to Xi Jinping. But Xi Jinping also sees that a lot of the world is not with NATO on this issue. 141 countries, if I remember correctly, voted to censor the Russians for their invasion of Ukraine at the United Nations General Assembly. But very large numbers of that 141 are not on board with all of these sanctions against Russia. They ’ re happy with the diplomatic censure, but they need to continue to work with the Russians. The Chinese see that too. The Chinese see just how much more fragmented the global order is. I thought the most significant thing that we ’ ve seen from the Chinese so far two issues. The first is, of course, when Putin went to Beijing and Xi Jinping made the public announcement that”this is our best friend on the global stage, and we will work much more strategically with them economically, diplomatically and militarily going forward.”And Xi Jinping knew very well where Ukraine was heading at that point and also knew that the likelihood of an invasion was coming. Didn ’ t stop him from making that announcement in the slightest. And then after the invasion, and it ’ s going badly, I mean, if you watch Chinese social media, the fact is that the censorship is all about Ukraine. I mean, the Chinese media space is pursuing a relentlessly pro-Putin policy. They have media embedded with Russian **troops** on the ground in Ukraine. Now, publicly, the Chinese government wants to be seen as :”We ’ re neutral, we like the Russians, we like the Ukrainians, we still want to work with everybody.”But the fact is that China feels no problem being publicly completely aligned with Putin, despite the fact that they are invading a democratic government with 44 million people in the middle of Europe. That ’ s a pretty astonishing statement from the Chinese. And there ’ s no question that they have learned that they ’ re in a vastly better economic position than they used to be, and that gives them influence. They are a government who projects its power primarily through economic and technological means, as opposed to Russia that projects it primarily through military means. And the Chinese believe that there is a level of decoupling that is already going on as the Americans focus on more industrial policy, as they focus on America first for American workers. A US foreign policy for the American middle class, as Biden put it, is one that really pushes a lot of capital to leave a country like China, which had served as the factory for the world, but at the expense of a lot of labor coming out of advanced industrial economies. And now, yes, there are definitely some dangers that come from the Chinese being perceived as too close to Russia, and they won ’ t want that, and they ’ ll want to make sure that they ’ re engaging diplomatically with the Europeans to try to minimize that damage. But I thought it was very interesting, and I ’ m not sure this is public yet, that the Chinese ambassador to Russia recently, in the last few days, organized a meeting of a lot of the top investors Chinese investors in Russia, saying,”This is a unique opportunity, the West is leaving, we should be going in and doing more. Because they ’ re going to be completely reliant on us going forward.”That is not a message that the Chinese ambassador delivers unless he is told directly to from Beijing. BG : Ian, I ’ m going to jump from topic to topic because there are several questions in the chat. Nancy is asking about whether Putin can be removed from power. There ’ s been a lot of discussion lately about regime change in Russia, either endogenous, like a palace coup, or provoked by sanctions and other policies. And so she asks,”How likely is that Putin will face a challenge from inside Russia, whether a popular uprising, a coup or other?”IB : It ’ s very, very unlikely until it happens. I mean, in the sense that there is absolutely no purpose in trying to say, oh, I mean, you know, there are rumors that Defense Minister Shoigu is unhappy and, you know, he might be making a move. And I ’ ve seen these from relatively credible analysts, I ’ m like, no, no, if there are such rumors, then we know it ’ s not happening because that ’ s the end of Shoigu and his family. But it ’ s very clear that there is more pressure on Putin now than at any point since he ’ s been president. Domestic pressure on Putin. About 10, 000 Russians have been arrested so far, detained, most of them have been released, for nonviolent anti-war protests. The Russians have shut down all the Western media. They ’ ve shut down all the Russian opposition and independent media. So Putin has control of the space, though if you look at Russian conversations on Telegram, you ’ ll still see a bunch of people that are seriously, seriously anti-war. But, you know, once the economy starts truly imploding and you can ’ t find goods on shelves in Russia in major cities, and this is coming, you know, very soon, this is a matter of days, in many of these cities, those demonstrations will likely become greater, some of them can become violent. You know, that ’ ll increase the pressure. Then you have the issue of how the Russians are fighting on the ground. I mean, what happens if you get a lot of desertions? We haven ’ t seen that so far. What happens if you get orders to bomb Kyiv and a whole bunch of Russian fighter pilots, bomber pilots, decide not to and they defect to Poland, to Germany. That would have a big impact on morale. That has not happened so far. I mean, do be aware of the fact that the Ukrainians are winning the war on information, and that means that the information that you are getting in the West about the war is much more pro-Ukrainian – morale, enthusiasm, how well the military is doing – than what ’ s actually happening on the ground. And also be aware of the fact that the Russians completely control the war on information inside Russia. BG : Exactly. IB : They ’ re not getting a balanced view. They ’ re getting a completely pro-Putin view. And most of them actually believe it in the same way that most people that voted for Trump in the US believe that the election was stolen and Trump is still president. I mean, it ’ s much worse in Russia in that regard than it is in the United States, and I think that that ’ s underappreciated in the West. So even though I think there ’ s pressure, I really don ’ t think that it ’ s super likely that Putin is out anytime imminently. BG : Ed is asking whether you see any off-ramp for Putin. IB : I think that the most likely off-ramp for Putin is after Kyiv is taken and Zelenskyy is removed one way or the other, at that point, the possibility of the Russians accepting a frozen conflict or a cease fire that could lead to ongoing negotiations is a lot higher because Putin can sell that as a win back home much more easily. But also because further Russian attacks at that point serve much less purpose for the Russians, are much harder to bring about, and potentially have much more negative consequences. So for me, that would be the near-term potential break where we could at least freeze issues largely where they are. Now whether that could then eventually lead to a climbdown or not, I mean, the Russians have been very happy with frozen conflicts on their borders for years and years and years. I ’ m thinking about Nagorno-Karabakh between Armenia and Azerbaijan, which basically stayed in place until the Azeris, over the course of a decade got enough military capacity that they could forcibly change the situation on the ground. Which, by the way, the Ukrainians might also be eventually thinking about because the West will be supplying them with advanced weapons all the way through. I ' m thinking South Ossetia in Georgia. I ' m also, of course, thinking about the two pieces of Ukraine they took back in 2014. So be aware of the fact that a negotiation that creates a cease fire does not mean you ' re anywhere close to a resolution or an end of the fighting that we ' re seeing. BG : Someone else in the chat, who didn ' t sign by his or her name, is asking about the nuclear fear that hangs over the conflict. How should we think of that? IB : Yeah, we don ' t like it when Putin uses the N-word, and there ' s no question, I mean, he and his direct reports have rattled nuclear sabers on at least five times that I ' ve seen in the past few weeks. I think that... In 1962, I wasn ' t alive, we had the Cuban Missile Crisis. There was a real possibility of nuclear confrontation between the world ' s two superpowers. At least for the last 30 years, there ' s been no chance of that. Functionally, no chance of that. I think we ' re now back in a world where a Cuban Missile Crisis is again a reality. Now, that doesn ' t mean that I think nuclear war is likely or imminent. I don ' t. And in fact, there is active deconfliction going on even today : the Americans and Russians with a new hotline, the secretary general of the UN, with the Russian defense minister engaging in deconfliction measures with UN offices being invited to Moscow. So as bad as it is right now, people that have been doing this for a long time are trying to avoid nuclear war. But that ' s the point. Is we ' re now in a situation where the conflict that we ' re going to experience needs to be actively managed because of the danger of nuclear confrontation. So it now becomes a risk on the horizon that we must be continually aware of, even if only at a low level, as we take and consider further actions, as we consider diplomacy, as we consider escalation. It is now on the table in a way that frankly is so debilitating. I mean, as human beings all on this call, one of the most painful things to think about is the fact that we still have these 5, 000 nuclear warheads in Russia and 5, 000 the United States that are still pointed at each other, and they still have the potential to destroy the planet. And we haven ' t had any real lessons that we ' ve been able to learn institutionally from 1962. BG : 5, 000 being a generic figure, not the exact figure, but we are kind of in that order of magnitude. Then of course, there is the question of civilian nuclear, so the two power plants, nuclear power plants, that have been seized by the Russians. One has been slightly damaged by a bomb, the other has been turned off. But those are also potentially gigantic nuclear problems just waiting to happen. IB : Chemical weapons, biological weapons. I mean, look, we have had two million refugees from Ukraine in two weeks. As this continues, you ' re looking at five to 10 million refugees. I mean, it is hard – Just take a step for a moment just as a human being. Imagine what it would take for a quarter of your country ' s population to say :"I am not living here anymore. I am leaving everything because of the condition of the country, because of this unjust war that has been imposed upon you by your neighbor."That ' s what we ' re looking at. And again, it ' s important for us to, you know, not lose the humanity of this crisis and the extraordinary hardship that is being visited upon 44 million Ukrainians that have done nothing wrong, they have committed no sin other than their desire to have an independent country. BG : One other country that has not yet taken a very clear position is India. IB : Well, they ' re a member of the Quad, and their relationship with China is pretty bad, and that ' s mutual. But in terms of Russia, there ' s been a longstanding relationship, trade relationship, defense relationship between India and Russia that the Russians are not going to jettison, and they see no reason to jettison it. And as long as you ' ve got a whole bunch of other countries out there that are substantial, that are willing to say, we ’ re going to keep playing **ball** with the Russians then the Indians will too. And that ’ s why you ’ ve got the abstention in the United Nations vote. And that ’ s why you ’ ve had very careful comments as opposed to overt and strong condemnation coming from the Indian leadership. BG : Phil in the chat is asking,”Will this cause a fragmentation of the financial system with kind of a Western system and an Eastern system?”So two different SWIFT-like systems, two different credit card systems, crypto, what ’ s the role of crypto in all this? IB : I hope not. I mean, I will tell you that before the invasion started, if you talk to most Western CEOs, and I ’ m talking across the entire sweep of sectors, so it ’ s finance and it ’ s manufacturing and its services and it ’ s tech, most of them would have told you that they did not in any way plan on reducing their footprint in China, and a lot of them said that China was their most important growth market in the world. Not a surprise. China is going to be the largest economy in the world in 2030. So, you know, a world that you ’ re decoupling is not a good world when China is going to be number one economically. I mean, that obviously is going to hurt the West in a big way. So there are strong incentives against that, and there remain very strong and powerful entrenched interests in the United States and Europe that will resist direct decoupling. Despite the fact that there are these more incremental moves towards friendsourcing and insourcing because, you know, Chinese labor is more expensive, you don ’ t need as much labor to get capital moving, given robotics and big data, deep learning all of those things. But I do think that the Russia conflict risks a level of second-order decoupling. Because if the Russians end up financially integrated with China in their own, not-as-effective SWIFT system, and all of their energy ends up going to China and the Chinese build that infrastructure and they get a discount on it, and Russia ’ s technology and their military industrial complex gets serviced by Chinese semiconductors and Chinese componentry, well, I do think that there will be knock-on decoupling that will be longer term and more strategic from the United States, from the Europeans and even from Japan and South Korea. So that is a worry, and I think the Chinese are highly aware of that. And over the coming months, they will do everything they can, both with the Europeans in particular, but also, I expect at least with some of the Asian economies, to try to limit the impact of that. Now, keep in mind, we haven ’ t talked at all about Asia yet outside of China. The new Japanese Prime Minister Kishida is at least as hawkish in his orientation towards China and Russia as Abe was. He is providing support for the Ukrainians, including some military capacity – unheard of for the Japanese. He ’ s allowing Ukrainian refugees – unheard of for the Japanese. And yesterday, the South Koreans had a very, very tight election, and Yoon is now in charge. He is on the right, and he is the guy that is strongly anti-China, was talking about South Korea having nuclear capabilities, wants a new THAAD missile defense system for the South Koreans and wants to rebuild the relationship with Tokyo. That matters. And that ’ s a big strategic change in the geopolitical map that will look more problematic on the decoupling front from Beijing ’ s perspective. BG : Three final quick questions that all come from the chat, Ian. One is, because you mentioned the rest of Asia outside of China,”What about the rest of the world? What about Africa and Latin America? How do they factor into this conversation or don ’ t they?”IB : They factor in. I mean, those that have significant commodities do well because the prices are going to be so high. Those that don ’ t are going to be under massive pressure for reasons we already talked about, but they are not going to be forced to pick a side on this one. I just don ’ t see it. In the same way that if you were Colombia in the last couple of years, you know, you found, even though you ’ re working very closely with an American ally, you ’ re still dealing with Huawei and 5G. This is knock-on effects of all of this. These are countries that are not going to take on significant economic burden, given how much they ’ re suffering right now geopolitically. BG : Another one is about sanctions. How do we even know when and how, at what point we start rolling back sanctions? IB : I think that as long as Ukraine is occupied by the Russian government for the foreseeable future and Putin is there, I can ’ t see these sanctions getting unwound. Now, if a rump Ukrainian government that is democratically elected were prepared to sue for peace and retakes most of Ukraine, but they give away Crimea and they give away the Donbass, could you see in that environment some of these sanctions unwound? Sure. But I mean, I am suggesting that I think that many of these sanctions are functionally permanent. They reflect a new way of doing business. And when people ask me what ’ s going to happen when this is over, my response is, what do you mean over? What ’ s over is the peace dividend. We are now in a new environment. BG : And one of the figures of this new environment and I want to close with that, is President Zelenskyy of Ukraine, who was not taken very seriously when he was elected, he has come out as a significant figure in this war. What do you make of President Zelenskyy? How do you read this character? IB : He ’ s very courageous. I ’ m obviously inspired by his ability to communicate and rally his people and take personal risk in Kyiv while this invasion is going on. But I ’ m very conflicted because I think many of the steps that Zelenskyy took in the run-up to this conflict actually made the likelihood of conflict worse. He was unwilling to take the advice of the Americans and Europeans seriously in the months leading up to the conflict. He was unwilling to mobilize his people to ready them for the potential of conflict. He was certainly unwilling to give an inch in terms of Ukraine ’ s desire to be a member of NATO, even though he knew completely that no one in NATO was prepared to offer a membership action plan, let alone actually bring them in as members. And part of that is a lack of experience and lack of any business being in that position in the run-up to this crisis. So I ’ m very deeply conflicted in my personal views on Zelenskyy, given the way he behaved before the invasion, compared to the extraordinary leadership that he has **displayed** to all of us over the last two weeks. BG : Ian, thank you for taking the time, for sharing your knowledge, and your analysis with us. We deeply appreciate it. Thank you very much. IB : Good to see all of you. [Get access to thought-provoking events you won ’ t want to miss.] [Become a TED Member at ted. com / membership]

Text Sample 895: POS Dim 6 – 2022 – Score 4.00 – 2022/t003585.txt

Creating intelligence on a computer. This has been the Holy Grail for artificial intelligence for quite some time. But how do we get there? So we view ourselves as highly intelligent beings. So it ’ s logical to study our own brains, the **substrate** of our cognition, for creating artificial intelligence. Imagine if we could replicate how our own brains work on a computer. But now consider the journey that would be required. The human brain contains 86 billion neurons. Each is constantly communicating with thousands of others, and each has individual characteristics of its own. Capturing the human brain on a computer may simply be too big and too complex a problem to tackle with the technology and the knowledge that we have today. I believe that we can capture a brain on a computer, but we have to start smaller. Much smaller. These insects have three of the most fascinating brains in the world to me. While they do not possess human-level intelligence, each is **remarkable** at a particular task. Think of them as highly trained specialists. African dung beetles are really good at rolling large **balls** in straight lines. Now, if you ' ve ever made a snowman, you know that rolling a large **ball** is not easy. Now picture trying to make that snowman when the **ball** of snow is as big as you are and you ' re standing on your head. Sahara desert ants are navigation specialists. They might have to wander a considerable distance to forage for food. But once they do find sustenance, they know how to calculate the straightest path home. And the dragonfly is a hunting specialist. In the wild, dragonflies capture approximately 95 percent of the prey they choose to go after. These insects are so good at their specialties that neuroscientists such as myself study them as model systems to understand how animal nervous systems solve particular problems. And in my own research, I study brains to bring these solutions, the best that biology has to offer, to computers. So consider the dragonfly brain. It has only on the order of one million neurons. Now, it ' s still not easy to unravel a circuit of even one million neurons. But given the choice between trying to tease apart the one-million-neuron brain versus the 86-billion-neuron brain, which would you choose to try first? When studying these smaller insect brains, the immediate goal is not human intelligence. We study these brains for what the insects do well. And in the case of the dragonfly, that ' s interception. So when dragonflies are hunting, they do more than just fly straight at the prey. They fly in such a way that they will intercept it. They aim for where the prey is going to be. Much like a soccer player, running to intercept a pass. To do this correctly, dragonflies need to perform what is known as a coordinate transformation, going from the eye ' s frame of reference, or what the dragonfly sees, to the body ' s frame of reference, or how the dragonfly needs to turn its body to intercept. Coordinate transformations are a basic calculation that animals need to perform to interact with the world. We do them instinctively every time we reach for something. When I reach for an object straight in front of me, my arm takes a very different trajectory than if I turn my head, look at that same object when it is off to one side and reach for it there. In both cases, my eyes see the same image of that object, but my brain is sending my arm on a very different trajectory based on the position of my neck. And dragonflies are fast. This means they calculate fast. The latency, or the time it takes for a dragonfly to respond once it sees the prey turn, is about 50 milliseconds. This latency is **remarkable**. For one thing, it ' s only half the time of a human eye blink. But for another thing, it suggests that dragonflies capture how to intercept in only relatively or surprisingly few computational steps. So in the brain, a computational step is a single neuron or a layer of neurons working in parallel. It takes a single neuron about 10 milliseconds to add up all its inputs and respond. The 50-millisecond response time means that once the dragonfly sees its prey turn, there ' s only time for maybe four of these computational steps or four layers of neurons, working in sequence, one after the other, to calculate how the dragonfly needs to turn. In other words, if I want to study how the dragonfly does coordinate transformations, the neural circuit that I need to understand, the neural circuit that I need to study, can have at most four layers of neurons. Each layer may have many neurons, but this is a small neural circuit. Small enough that we can identify it and study it with the tools that are available today. And this is what I ' m trying to do. I have built a model of what I believe is the neural circuit that calculates how the dragonfly should turn. And here is the cool result. In the model, dragonflies do coordinate transformations in only one computational step, one layer of neurons. This is something we can test and understand. In a computer simulation, I can predict the activities of individual neurons while the dragonfly is hunting. For example, here I am predicting the action potentials, or the spikes, that are fired by one of these neurons when the dragonfly sees the prey move. To test the model, my collaborators and I are now comparing these predicted neural responses with responses of neurons recorded in living dragonfly brains. These are ongoing experiments in which we put living dragonflies in virtual reality. Now, it ' s not practical to put VR goggles on a dragonfly. So instead, we show **movies** of moving targets to the dragonfly, while an electrode records activity patterns of individual neurons in the brain. Yeah, he likes the **movies**. If the responses that we record in the brain match those predicted by the model, we will have identified which neurons are responsible for coordinate transformations. The next step will be to understand the specifics of how these neurons work together to do the calculation. But this is how we begin to understand how brains do basic or primitive calculations. Calculations that I regard as building blocks for more complex functions, not only for interception but also for cognition. The way that these neurons compute may be different from anything that exists on a computer today. And the goal of this work is to do more than just write code that replicates the activity patterns of neurons. We aim to build a computer chip that not only does the same things as biological brains but does them in the same way as biological brains. This could lead to drones driven by computers the same size of the dragonfly ' s brain that captures some targets and avoid others. Personally, I ' m hoping for a small army of these to defend my backyard from mosquitoes in the summer. The GPS on your phone could be replaced by a new navigation device based on dung beetles or ants that could guide you to the straight or the easy path home. And what would the power requirements of these devices be like? As small as it is – Or, sorry – as large as it is, the human brain is estimated to have the same power requirements as a 20-watt light bulb. Imagine if all brain-inspired computers had the same extremely low-power requirements. Your smartphone or your smartwatch probably needs charging every day. Your new brain-inspired device might only need charging every few months, or maybe even every few years. The famous physicist, Richard Feynman, once said,"What I cannot create, I do not understand."What I see in insect nervous systems is an opportunity to understand brains through the creation of computers that work as brains do. And creation of these computers will not just be for knowledge. There ' s potential for real impact on your devices, your vehicles, maybe even artificial intelligences. So next time you see an insect, consider that these tiny brains can lead to **remarkable** computers. And think of the potential that they offer us for the future. Thank you.

Text Sample 896: POS Dim 6 – 2023 – Score 4.00 – 2023/t003468.txt

I became obsessed with the relationship between the brain and the mind after suffering a series of concussions playing football and rugby in college. I felt my mind change for years after. I was studying computers at the time, and it felt as though I had damaged my hardware and that my software was running differently. Over the following years, a close friend suffered a serious neck injury and multiple friends and family members were struggling with crippling mental health issues. All around me, people that I loved dearly were being afflicted by ailments of the nervous system or the mind. I was grappling with all of this while pursuing an MFA in Design and Technology at Parsons when a friend and fellow student showed me an open-source tutorial on how to build a low-cost single-channel EEG system to detect brain activity. After a couple long nights of hacking and tinkering, I saw my brainwaves dancing across the screen for the very first time. And that moment changed my life. In that moment, I felt as though I had the possibility to help myself and the people I loved. And I also realized that I couldn ' t do it alone. I needed help. So in 2013, in Brooklyn, with some like-minded friends, I started OpenBCI, an open-source neurotechnology company. In the beginning, our goal was to build an inward-pointing telescope and to share the blueprints with the world so that anybody with a computer could begin peering into their own brain. At first, we were an EEG-only company. We sold brain sensors to measure brain activity. I thought that ' s what people wanted. But over time, we discovered people doing very strange things with our technology. Some people were connecting the equipment to the stomach to measure the neurons in the gut and study gut-brain connection and the microbiome. Others were using the tools to build new muscle sensors and controllers for prosthetics and robotics. And some were designing new devices and peripheral add-ons that could be connected to the platform to measure new types of data that I had never heard of before. What we learned from all of this is that the brain by itself is actually quite boring. Turns out brain data alone lacks context. And what we ultimately care about is not the brain, but the mind, consciousness, human cognition. When we have things like EMG sensors to measure muscle activity or ECG sensors to measure heart activity, eye trackers and even environmental sensors to measure the world around us, all of this makes the brain data much more useful. But the organs around our body, our sensory receptors, are actually much easier to collect data from than the brain, and also arguably much more important for determining the things that we actually care about : emotions, intentions and the mind overall. Additionally, we realized that people weren ' t just interested in reading from the brain and the body. They were also interested in modulating the mind through **various** types of sensory stimulation. Things like light, sound, haptics and electricity. It ' s one thing to record the mind, it ' s another to modulate it. The idea of a combined system that can both read from and write to the brain or body is referred to as a closed-loop system or bidirectional human **interface**. This concept is truly profound, and it will define the next major revolution in computing technology. When you have products that not just are designed for the average user but are designed to actually adapt to their user, that ' s something truly special. When we know what the data of an emotion or a feeling looks like and we know how to make that data go up or down, then using AI, we can build constructive or destructive interference patterns to either amplify or suppress those emotions or feelings. In the very near future, we will have computers that we are resonantly and subconsciously connected to, enabling empathetic computing for the very first time. In 2018, we put these learnings to work and began development of a new tool for cognitive exploration. Named after my friend Gael, who passed from ALS in 2016, we call it Galea. It ' s a multimodal bio-sensing headset, and it is absolutely packed with sensors. It can measure the user ' s heart, skin, muscles, eyes and brain, and it combines that capability with head-mounted **displays** or augmented and virtual reality headsets. Additionally, we ' re exploring the integration of non-invasive electrical neural stimulation as a feature. The Galea software suite can turn the raw sensor data into meaningful metrics. With some of the sensors, we ' re able to provide new forms of real-time interactivity and control. And with all of the sensors, we ' re able to make quantifiable inferences about high-level states of mind, things like stress, fatigue, cognitive workload and focus. In 2019, a legendary neurohacker by the name of Christian Bayerlein reached out to me. He was actually one of our very first Kickstarter backers when we got started, early on. Christian was a very smart, intelligent, happy-go-lucky and easygoing guy. And so I worked up the courage to ask him,"Hey, Christian, can we connect you to our sensors?"At which point he said,"I thought you would never ask."So after 20 minutes, we had him rigged up to a bunch of electrodes, and we provided him with four new inputs to a computer. Little digital buttons, that he could control voluntarily. This essentially doubled his number of inputs to a computer. Years later, after many setbacks due to COVID, we flew to Germany to work with Christian in person to implement the first prototype of what we ' re going to be demoing here today. Christian then spent months training with that prototype and sending his data across the Atlantic to us in Brooklyn from Germany and flying a virtual drone in our offices. The first thing that we did was scour Christian ' s body for residual motor function. We then connected electrodes to the four muscles that he had the most voluntary control over, and then we turned those muscles into digital buttons. We then applied some smart **filtering** and signal processing to adapt those buttons into something more like a slider or a digital potentiometer. After that, we turned those four sliders and mapped them to a new virtual joystick. Christian then combined that new joystick with the joystick that he uses with his lip to control his wheelchair, and with the two joysticks combined, Christian finally had control over all the manual controls of a drone. I ' m going to stop talking about it, and we ' re going to show you. Christian, welcome. At this point, I ' m going to ask everybody to turn off your Bluetooth and put your phones in airplane mode so that you don ' t get hit in the face with a drone. How are you feeling, Christian? Christian Bayerlein : Yeah, let ' s do it. Conor Russomanno : Awesome. This is a heads-up **display** that ' s showing all of Christian ' s biometric data, as well as some information about the drone. On the left here, we can see Christian ' s muscle data. Christian is now going to attempt to fly the drone. How are you feeling, Christian, feeling good? CB : Yes. CR : All right. Rock and roll. Let ' s take this up for a joyride. Whenever you ' re ready. CB : I ' m ready. CR : All right, take her up. And now let ' s do something we probably shouldn ' t do and fly it over the audience. Alright, actually, let ' s do this. I ' m going to ask for people to call out some commands in the audience. So how about you? Straight forward. Straight forward. Alright. How about you? Man : Up! CR : Not down. Oh, he ' s doing what he wants right now. Amazing. Alright, let ' s bring it back. And what I ' m going to do right now is take control of the controller so that you guys know that there isn ' t someone backstage flying this drone. All right, Christian, you ' re alright with that? CB : Yeah. CR : Unplug. Forward. And we ' re going to land this guy now. CB : I think I was better than you. CR : Amazing. Now I ' m going to unplug it so it doesn ' t turn on on its own. Perfect. Christian has repurposed dormant muscles from around his body for extended and augmented interactivity. We have turned those muscles into a generic controller that in this case we ' ve mapped into a drone, but what ' s really cool is that joystick can be applied to anything. Another thing that ' s really cool is that even in individuals who are not living with motor disabilities, there exist dozens of dormant muscles around the body that we can tap into for augmented and expanded control interactivity. And lastly, all the code related to that virtual joystick, we ' re going to open source so that you can implement it and improve upon it. There ' s three things that have stood out to me from working on this project and many others over the years. One, we cannot conflate the brain with the mind. In order to understand emotions and tensions and the mind overall, we have to measure data from all over the body, not just the brain. Two, open-source technology access and literacy is one way that we can combat the potential ethical challenges we face in introducing neural technology to society. But that ' s not enough. We have to do much, much more than that. It ' s very important, imperative, that we set up guardrails and design the future that we want to live in. Three. It ' s the courage and resilience of trailblazers like Christian who don ' t get bogged down by what they can ' t do, but instead strive to prove that the impossible is in fact possible. And since none of this would have been possible without you, Christian, the stage is yours. CB : Yeah, hi, everybody. Audience : Hi. CB : I ’ m excited to be here today. I was born with a genetic condition that affects my mobility and requires me to have assistance. Despite my disability, I ’ m a very happy and fulfilled person. What truly holds me back are not my physical limitations. It ’ s rather the barriers in the environment. I ’ m a tech nerd and political activist. I believe that technology can empower disabled people. It can help create a better, more inclusive and accessible world for everyone. This demonstration is a perfect example. We saw what ’ s possible when cutting edge technology is combined with human curiosity and creativity. So let ’ s build tools that empower people, applications that break down barriers and systems that unlock a world of possibilities. I think that ’ s an idea worth spreading. Thank you.

Text Sample 897: POS Dim 6 – 2023 – Score 3.00 – 2023/t003371.txt

I was supposed to go in the very first session. And I had a plan. And that plan was I ’ d do TED, I ’ d see Vancouver, I ’ d chill out, I ’ d get this thing done, then I ’ d relax. I ’ d listen to other people ’ s speeches, rob their best ideas, bring them home and sound incredibly, incredibly brainy. And then your man, Chris, rings me last weekend. He says,”You know that last thing, the first thing? Why don ’ t you go last?”I ’ m like,”Oh, man.”That brought to mind the words of the great American philosopher, Mike Tyson. Who said of plans,”Everybody has a plan until they get a punch in the face.”And of course, you know, doing the last gig at TED, it ’ s not about the pressure, that ’ s not the problem. The pressure isn ’ t the problem. The problem is the sobriety. But the speeches have been amazing, right? And if I could kind of encapsulate them all, it ’ s really been about, as far as I can see, the pace of change and what that ’ s doing to us demographically, politically, socially, economically. Extraordinary stuff. And I ’ d now like to quote from a great liberal, democrat, a friend of TED, Vladimir Lenin. Who said of change and crises that there are decades when nothing happens and there are weeks when decades happen. And we ’ re living through those weeks, you get that feeling, and it ’ s really difficult to know where to start to think about these... How do you analyze this stuff? Now, if I was an American economist or a Canadian economist or, God forbid, an English economist – Actually the nice thing is they ’ re not good at economics anymore. It ’ s so beautiful, isn ’ t it? Praise the Lord. Praise the Lord. But I ’ d come here armed with the tools of my trade. You know, the graphs and the charts and the maths and all that good stuff. But I ’ m an Irish economist, so I ’ m only going to come here armed with some lines and some verses of poetry. Yeah. Thank you very much, the poets in the corner. We ’ re a small minority. But perfectly formed. Now, I want to talk to you about a poem that was written in 1919 called”The Second Coming”by W. B. Yeats, our national poet. And the fascinating thing about crisis, so in 1919, 100-odd years ago – the fascinating thing about crisis is that every generation feels that their crisis is the big one. We ’ re kind of narcissistic about it, right? But every generation experiences crisis. Every generation, our parents, our grandparents, they all did. How we deal with the crisis is the definitive issue for our generation. So Yates is sitting in Dublin, 1919. He ’ s trying to make sense of the world, right? His world is changing rapidly. The German Empire is over. The Austrian Empire is over. The Ottoman Empire is over. Ireland has declared a war of independence against Britain, which, if you were a betting man, you wouldn ’ t really give us, not our side, you wouldn ’ t have given us good odds. And Yeats is trying to figure out not just what is happening, but what is likely to happen. And he writes these words, and just listen to them and imagine they were written now."The Second Coming.""Turning and turning in the widening gyre / The falcon cannot hear the falconer ; / Things fall apart ; the centre cannot hold ; / Mere anarchy is loosed on the world, / A blood-dimmed tide is loosed, and everywhere / The procession of innocence is drowned ; / The best lack all conviction, the worst / Are full of passionate intensity."Just let those words lie."Things fall apart, the best people lack conviction, the worst are full of passionate intensity."Now, this is Yeats writing 100 years ago, and the historical rhythm and repetition is clear, I think, to all of us. But what really interests me as an economist is the contrast between what Yeats, the poet, was saying back then and what all the economists, the people who were employed to think about the future, were saying. So, Yeats said,"The center will not hold."Three years after he wrote this, three years, three to four years, Mussolini was in power in Italy, Stalin was on the ascendancy in the Soviet Union and a little, small, unprepossessing man with a mustache called Adolf Hitler had just orchestrated a putsch in Munich. Yates was right. Yates was right. And what were all the economists saying back then? The people who were paid to think about the future? They were all saying,"Oh, don ' t worry, we ' ll go back to the gold –"Not all, the vast majority –"We will go back to the gold standard. We ' ll trade again. Germany will pay all its reparations and the First World War will have been the war to end all wars."How wrong could they be? So what ' s always bugged me is why did the poet get things so right at a tipping point, and the economists get things so wrong? And I believe it is because the poet, the artist, the musician, these types of people give themselves, at a tipping point, the permission to think unconventionally. They see the world differently. Do we value the unconventional thinker? Do we? And maybe the best way to answer that question is to go all the way back to school, to the place we begin to learn. Now just bring yourselves back to your school days. Remember yourself at 13. Just remember the person you were. I remember the classroom. I remember the teachers, I remember the building, I remember the break, I remember everything. I remember the sports day, I remember the whole thing. In fact, there ' s a gang of five of us who ' ve hung around since we were in school. And those five lads I was in school with spent the entire of our school years looking out the window They just didn ' t get it. They just didn ' t get school at all. And they ' ve gone on to have incredibly successful lives in their own individual ways. But their intelligence, their form of intelligence, was not recognized. And as we get older, we realize the world is full of **various** different intelligences. But back in school, we only really recognized one type of intelligence. And that was like the little kid who could come in – big, wide-eyed kid, right? – he ' d come in, could absorb all this information into his head or her head, opened some sort of weird compartment, stuff it all in, stuff, stuff, stuff it all in. And then, in early June, when the weather gets good in Ireland – and I presume it ' s the same all over the place except for Vancouver – write like bejesus, right? That was the type of intelligence we rewarded. It ' s like getting a prize for being a walking filing cabinet, right? That ' s one intelligence. It is an intelligence, but it ' s only one type of intelligence. And what about all the other kids over here, the kids who didn ' t get it?... They weren ' t not only recognized, their intelligence was besmirched, humiliated and belittled. My daughter is dyslexic. She hated every single day of school. And as a result of this system, we have a strange thing in our society where there are thousands of incredibly clever people who left school feeling stupid. But the corollary is also the case. There are many actually quite stupid people – You know where I ' m going – Who left school feeling very, very clever. And those kids, when they were really good kids, they used to get the best marks in school. In Ireland, we have a little star over the right. And of course, the teacher told them they were clever, and the priests, the nuns, and of course, their mummy told them they were clever, right? If you want to get a sense of an Irish mummy, an Irish mummy is the sort of mother that makes an American soccer mom look unambitious. I know, I have one, right? I am the son of a retired schoolteacher. If anybody else in the room is the son or daughter of a schoolteacher, we can form a self-help group later on to deal with our trauma, right? But those sort of kids, they do extremely, extremely well in school, then they do very, very well in college. And then they get on, because they do very, well, on to the graduate trainee programs and they join the big banks, and they join the big firms and they join insurance companies, they join the consultancies, and they ’ re on the fast track. They ’ re on the fast track career-wise. And something weird happens as their career progresses. We like to think that we surround ourselves with people who think differently, but we don ’ t. There ’ s something called confirmation bias. We actually surround ourselves with people who confirm our biases. And how this works in institutions is the following. We end up employing people who think like us, right? And like, if you imagine an interview process, so the person goes into the interview and there ’ s two clever persons saying,”No, no, you ’ re clever.””You ’ re clever, you ’ re amazing.””You ’ ve got that grade.””You went to that university.””I read that paper you wrote.””You ’ re clever, clever, amazing.””Feck it, just have the job.”So the interview becomes – By the way, I said”feck”there. Not the other word. Feck is a sort of a linguistic version of a white lie. OK? It ’ s used at home. Now, what actually happens is the interview process just becomes like a Tinder for people who can do algebra. OK. So they all go into the institution and then what you get, you get groupthink because everyone ’ s thinking the same way. And of course, because these people have always been the clever kids with the right answer, they defined themselves,”I ’ m the person with the right answer.”And what happens when you ’ re the person who always has the right answer? It ’ s very hard to be wrong. So it breeds a sort of an overconfidence. And we know that overconfidence and overconfident people can really overestimate their competence ’ s critical moments. The thing comes from a thing called Dunning-Kruger in psychology. It ’ s a great story. A fella goes in to rob a bank in America. Somewhere, I think Pittsburgh, down there somewhere. In the ’ 90s, a lad goes into rob a bank, OK? He runs into the bank... he ’ s got no **mask**, no balaclava, no nothing, right? The security camera looks at him. And he winks at it. He winks again, big smile,”Hi, how are you?”He goes in, holds up the bank, robs the thing. Goes off. Coppers, come in, look at the securities.”See this geezer, just waved to the camera. Does anyone know him?””Oh, yeah, he lives around the corner, around the second block, sixth house, up third floor.”So they arrive and **bang** on the door, the man ’ s there, I presume, kind of eating a takeaway or something. And copper said,”We want to **bang** you off for the bank robbery.”And he said,”How do you mean, the bank robbery?””We ’ ve got you on camera for the bank robbery... here we go.””How do you mean?””We ’ ve got you!””How could you possibly know? I wore the juice!”Copper looks at him, says,”What?”Do you remember when we were kids? Invisible ink? And you put lemon juice on the invisible ink – this is a true story – and you disappear. Your man covered his face with lemon juice and he thought he was invisible. The overconfident overestimating their competence. Now, when the psychologists thought about this, they actually did lots and lots of tests. And apparently it ’ s true, that this is a thing in society. And surprise, surprise, the Dunning-Kruger effect is much more prevalent in men than women. Who would have known? Shock horror. I can see it in my own family. Our son, for example, comes home, teenager, after doing an exam. Hasn ’ t done a tap of work, right? Comes in the door and everyone in the family ’ s in the kitchen saying,”How did you do in your exam?”And he says,”Oh, Jesus, I aced that, man, no problem.”And he fails all the time. All the time. So you see these things and that again it happens. And of course, how this happens in institutions is you get very overconfident people, very intelligent, can ’ t make mistakes. And we saw that in the 2008 financial crisis. 2008, the biggest financial crisis the world has ever seen. And the vast majority of economists and the Fed and the Bank of England and the European Central Bank and Wall Street missed the whole thing. In fact, the Queen of England went to the LSC about two or three months after the crash and she said,”If you chaps were so clever, how come none of you saw this coming?”And she was right. Because they all were wearing the juice. The economics juice. Now, JK Galbraith, very famous Canadian economist. But Galbraith said something fascinating about the conventional person. He said,”When faced with the choice between changing his mind and finding the proof not to do so, the conventional man always gets busy looking for the proof.”And that ’ s when we make big mistakes at these critical moments. Leonard Cohen, the Canadian poet, put it differently, the same idea. Cohen said,”There is a crack in everything. And that is how the light gets in.”What Cohen was saying to us was, look for the cracks, look into the cracks. That ’ s where we ’ ll see the big picture. So let ’ s just leave the Canadian poet, go back to an Irish poet. The words of”The Second Coming”again, our world right now.”Turning and turning in the widening gyre / The falcon cannot hear the falconer ; / Things fall apart ; the center cannot hold ; / Mere anarchy is loosed on the world, / A blood-dimmed tide is loosed, and everywhere, / The procession of innocence is drowned.”And why, says Yeats? Because the best lack all conviction and the worst are full of passionate intensity. Now apply that to our world. And if you, at this tipping point say,”Nah, you know, it ’ s not my problem, I lack all conviction.”Right?”I go and watch baseball,”or rounders or whatever that game is you guys watch, right? But if you lack all conviction, what actually happens is the worst people in our society, with their passionate intensity, their certitude, their simple ideas, they will win the day at this tipping point. And they will win the day because the best people back away from the responsibility. And what Yates was also saying about the best people was the following. He said, mandate the best. The poets, the artists, the musicians, because they are the people who see the possibilities. They see the possibilities because they see the world from a different angle. They have to be part of the solution. So my idea that I think is worth spreading is the following. If you want to understand the world a little bit more clearly, listen less to my tribe, the economists, and listen more to Yeats ’ s tribe, the poets. Thank you very much. Thank you very much.

Text Sample 898: POS Dim 6 – 2023 – Score 3.00 – 2023/t003280.txt

Every month I spend a week on the front line. For over a year now, I have been a member of Aerial Reconnaissance Unit. Before the invasion, I had only used a drone to scout locations for my architectural practice. Now I pilot a UAV to help Ukrainian artillery spot targets. This time two years ago, I would probably introduce myself as the CEO of an architectural studio. Maybe I would also brag about co-owning two cafes in Kyiv, which I also designed. To be clear, I still am an architect and an entrepreneur. It ’ s just that now I consider those roles supporting. As a military volunteer, I have monthly shifts. This means I still get to spend time at home, my safe haven, trying to make sense of this new life. I travel back and forth from Kyiv to the front line and back to Kyiv. Not your regular commute, I suppose. Even though Kyiv suffers constant missile attacks, the contrast in between the capital and the hotspots is quite stark.

Back home, I can walk my dog Eva, grab a coffee and just barricade myself in my office. Of course, our work is often interrupted by the air-raid alarms, but this is something we have grown to live with. Compared to dusty trenches near Kherson, our bureau ' s open-space office seems like a perfect work environment. But somehow, whenever I try to focus, I keep thinking back to those trenches. It is there, just a few miles from the battlefield, where I can feel the most productive. And it is then, in times of distress, when I can clearly see what the future holds. Of course, it hasn ' t always been this way. Before Russia launched its full-scale war, I had a pretty regular life. Well, maybe not that regular. My architectural studio was busy pushing the envelope. We renovated an old church in San Francisco, California, turned a former military arsenal into a teeming food hall. And even designed the art installation for the Ukrainian polar research base. Seriously, here ' s the proof. All the way from Antarctica. All I ' m saying, we were doing just fine. Great, even. But then came the sickest plot twist one could imagine. On February 24, 2022, I woke up to the sounds of explosions. Now, I don ' t know if you know many people from Ukraine, but our humor can be rather dark. But back then, we all went numb. Most of my teammates fled to safer areas, unsure whether they would have a home to return to. Some stayed in Kyiv, sheltering in basements and subway stations. The others decided to take up arms for the first time ever. After the shock came a revelation. We had to turn our fear into action. So on day two, I stepped into a new role, switching from the coffee co-owner to emergency kitchen cofounder. And since there was plenty of food left in storage, we decided to cook meals for territorial defense units. Partnering up with other restaurants, we founded an organization called Kyiv Volunteer. Many people came in to help – musicians and lawyers, drivers and architects – all united around a common goal. And to give you a sense of scale, on peak days, we served up to 15, 000 hot meals to the military and medics. While our pots were clanking, Russian **troops** were advancing. Ukraine ' s future remained uncertain, but we were too busy to dwell on that thought. Working underground allowed us to prioritize the needs and address them, one by one. First we focused on basics : shelter, food, medicine. Then we had to address the recovery, both physical and mental. Talking to people on our humanitarian mission, we got a simple yet fundamental notion. What we all craved most was the sense of home. Not four walls and roof per se, but the little things. The rose bush behind the kitchen window. The attic full of hidden treasures. That slightly crooked bench by the gate, always attracting the neighborhood cats. No matter where Ukrainians happen to spend the night, be it in an emergency shelter or some Good Samaritan ' s home, they ' re always trying to make it feel cozy. Sometimes we even better our temporary homes by, say, fixing a long-broken fence. This observation led us to another insight. Home is the temple. And architecture can be heailing. When done with empathy, architecture can help people to sustain the trauma and envision the brighter future. Even if this future is still up in the air. Two weeks into invasion, my team and I remembered : we were still architects and could use our expertise for a good cause. This was when our bureau started working on an idea we call RE:Ukraine Housing. Knowing that nothing is more permanent than temporary, we tasked ourselves with raising the standard of temporary housing. Being both architects and displaced people, we knew how to make a difference. So what is the first thing that comes to your mind when you think of temporary housing? Probably those **typical** refugee camps, built from shipping containers. So this is something our team sought to change. Dignity no matter what. These four words became a basis for our work and in a way, our mission. In merely two weeks, we developed a system of dignified, temporary accommodation. Our vision of what displaced people need to feel at ease. We want residents to feel equally comfortable, so we made our rooms fit different life scenarios, whether it ’ s an elderly couple or a single parent with a newborn. We also knew people would need a place to work, study and care for their kids, so we provided them the space to do this as well. And to avoid **spatial** segregation, we proposed to make public spaces, open to anyone, building bridges in between host communities and IDPs. Technology-wise, we didn ’ t reinvent the wheel. But this is what and how we suggest building that made people reconsider the very essence of temporary housing. This May, after a year of preparation and paperwork, we broke ground on our pilot settlement. Bucha District, once known as the site of Russian atrocities, became a home for our big idea and a symbolic place to create new, happier memories. Now, RE:Ukraine is not just a housing project but a pilot for a recovery program. Our goal is to help displaced people get back on their feet and reintegrate, leaving their trauma behind. And since many places in the world are devastated by wars and climate change, the displacement challenge is the global one. Though rooted in Ukraine, the model can be replicated, helping to accommodate the displaced people elsewhere. Watching Ukrainians flock to Bucha and other liberated areas, translated into another insight. Our folks don ’ t want to wait to fix what ’ s broken. Fixing is the first thing we do after every attack, despite the threat of anything being destroyed again. Public spaces like schools and hospitals restored by local authorities. But when it comes to private housing, people are often left with this task [on] their own. So to help smaller householders rebuild, we created RE:Ukraine Villages, a free online tool for rural housing restoration. With just a few clicks, the tool generates you a step-by-step manual to restore a house or to design a new one from scratch. Floor plans, façade scans, technical description, a list of materials. Fool-proof guides for homeowners and volunteers. To show you how it works, here ’ s the project generated with the tool. And the model of a house one may build based on this manual. To preserve the traditional image of Ukrainian village, we decided to base our builder on a local design code. And since every region has its own quirks, the tool had to reflect them. So the Kyiv region became the starting point for our research. Our architects traveled from village to village exploring different types of roofing, windows and cornices. Then came the developers who transformed the data in an accessible tool. Just imagine, that tool can generate you over 211 million unique house configurations for the Kyiv region alone. And each house, though designed with modern materials, looks super familiar, like the one Ukrainians spent their summer holidays at. A reminder of happier days and a wholesome image of home. The peak of our work fell on the end 2022. This was when Russia targetted the power plants, to plunge our cities in total darkness. Our team had to work around the power cut schedule, trying to get the work done during the four-hour spans. This was when we discovered how to charge our computers from the car inverter. Now, I hope none of you ever happen to face a blackout. But if you do, remember this life hack. And to give you some context, the minibus we used as the power bank is the same I use on my military missions. So this is what Ukrainians call war-work-life balance. The thing we mastered over the last year and a half. Now, I would be lying if I said I ’ m not proud of our accomplishments, I truly am. But just imagine how much more we could achieve if not for the war. Despair and uncertainty can fuel your ideas. But do we really need to wait for this kind of fuel? The best time is always now. So if you have the privilege of living in peace, use this time wisely. And don ’ t worry too much about the future. It ’ s not that scary if you embrace it with dignity. Thank you.

Text Sample 899: POS Dim 6 – 2024 – Score 6.00 – 2024/t003277.txt
Chris Anderson : We are 10 days away or so from the 40-year anniversary of TED ’ s founding. 40 years! Unbelievable. And today we get to listen to the man who really has been the driver of TED for the whole first part of that voyage, an amazing man. Please welcome the incomparable Richard Saul Wurman. Richard Saul Wurman : Thank you, Chris. That ’ s the first time I ever heard you say that. CA : Ricky, it ’ s great to have you here. Thank you so much, for so much. Gosh, where to start? I think I would love to hear a bit about your story before TED was even a glimmer in your mind. Who are you? How did you become this information architect that conceived of this conference? RSW : I had the realization that there was two kinds of people, vertical and horizontal people. And success, in terms of money, power, fame, getting better at a certain task – painting, sculpture, playing the cello, being a magician – comes from doing that one thing better and better and better through your life. And you are, if you have the right PR or you just stand out yourself, you gain success in that profession and in society. And my attention span points to horizontality. I ’ m just as interested when I walk down the street in what ’ s left and what ’ s right, and the sign over there. And that horizontality is a life devoted to seeing patterns. And every meeting that I went, every gathering I went to, if it was eye, ear, nose and throat specialists, it was on the road to becoming a nose specialist and then one nostril. The whole society was being focused and still is, in many ways, most gatherings are about one thing, and people talk to each other about the one thing they can talk to each other about. And part of it is getting a job or selling a paper or getting a grant within that one specialty. And that ’ s good. This is not a pejorative. This is an observation of one of the ways, the major way the world turns. But when I went to the University of Pennsylvania in architecture, I got a special deal with the dean that I could take as many courses as I wanted as long as I kept a very high average. And so I was in class, every day, every night, taking very odd courses, inside painting and snuff bottles and Japanese swords and integration technology and ethnology and things on illuminated manuscripts and painting and history of astronomy. And I realized that I didn ’ t take notes because I had no time to study, that I engineered reverse-engineering, that you take notes, not to take notes, you take notes so you can study them. To pass the test. That all the educational system was about taking a test. And my learning was about my memory. So I learned to listen. And I see and visualize patterns between things. CA : And so although Ricky, you qualified as an architect and worked as an architect, but how long was it before you really thought of yourself and described yourself as an”information architect?”RSW : I graduated in 1958 / 9 with a master ’ s degree. And I was an assistant professor of architecture. And the first book I did was when I was 26 years old, and it was a book of comparative maps of 50 cities in the world. And that was my first book, was information architecture. Because the comparative analysis of things was the systemic way of showing maps to the same scale, which people basically don ’ t ever do now. If you look at the road atlases, every map on every page is a different scale. So I would say when I was 26. CA : And even from those early days, it feels like you had this obsession with just what it is to explain something. You know, how you make information interesting and useful. And it sounds like that came precisely because you were willing to go broad. You were willing not just to look at a thing in itself but how it connected, how the dots connected. RSW : Well, I don ’ t like to fail, but I was willing to fail, and I embrace it so I see what doesn ’ t work and what I can ’ t understand. So yes, explaining is a key word, but we ’ ve never explained how do you explain things. And we don ’ t understand how we understand things. They ’ re very simple. And when we ask a question,

most of the word is”quest.”There ’ s no”quest”in the question. We ask lousy questions. And information, most of the word is”inform,”and most information is data, doesn ’ t inform. I ’ m talking dumb words now, Chris. These are simple words : memory, memorize, understand, understand, explain, explain. They ’ re all simple : quest – question inform – information. CA : Well I ’ m going to come back to some of those questions at the end because I would love to know what you would say now about understanding understanding. But before then, talk about, like, you started to get very involved in conferences. There was a big conference that you were involved with before TED. Talk about that. RSW : Yes, I was a little schlepper in Philadelphia, and I ’ d done, with my partner, Al Levy, I ’ d done a book called”Our Man-made Environment”for kids. And had a nonprofit called GEE, Group for Environmental Education, GEE! And TIME Magazine, which was important then, TIME Magazine was the record of the week in the world that you believed, if it got into TIME Magazine. They did a big story on this book. I mean, we were just schleppers, a little teeny office in Philadelphia, architectural practice, Murphy, Levi, Wurman, MLW. And it was picked up and the people from Aspen from the International Design Conference in Aspen, asked my partner and myself, Al and myself, to come out and speak. He got ill. And this was one of the big moments in my life. I went out and I worked my – I just, I gave a speech. I think it ’ s the last speech I worked out and wrote before I gave it. In fact, it ’ s the one and only, but I had it down and I gave it. I don ’ t know if the audience liked it so much, but some of the board liked it because it was complex. It was very dense because I was trying to – I was ambitious, I really was ambitious. And the board was the stars of the design world, the stars. And at the end of the speech, somebody came up to me from the board and said,”That was a very good speech,”and very soon, I was in my 30s, I was the youngest person, they put me on the board before the conference was over. And before the year was out, and this was in June, before the year was out, I was asked to do the conference after the next one. So in ’ 70 I was put on the board. ’ 72 – they already had the person chosen for ’ 71 – ’ 72, I did a conference there for 1, 200 people called the Invisible City, and when you did a conference there, it was not funded, hardly funded at all. And you were God, you ’ re, you know, it ’ s nice to be king. I was in charge of the whole thing. And at that conference, Lou Kahn came and talked, incredible people came, and I learned how to fail and how to have things work. And I learned that you can ’ t go to school for this, but that I felt really comfortable being on stage. In fact, I was more comfortable being on stage than in the audience. Because I was seeing things were going wrong when I was in the audience and onstage, it was my mistake, and I enjoyed that. I knew I could do it better. So that was it and it was the best conference in the world, not mine. The National Design Conference, which was in Aspen, and the whole town was, the Aspen Institute was connected to it, and it was, 1, 200 people came from around the world. My partner in my guidebook company, I had a guidebook company called Access Press, and it was on the process of doing guides to 22 cities around the world. So I had that. And around the corner from me was a gentleman by the name of Harry Marks. The CEO of CBS was Frank Stanton. So Harry had never met Frank, and his eyes were wide open because he was such a big deal. But we were in California, and Harry and I were talking, and Harry wanted to do something because he was tired of what he was doing in television, which was doing ads for television programs on television networks. He invented this idea. And he said he knew I did the Aspen conference. Why don ’ t I invent a conference, and we ’ ll go into business together. So I got Frank to give me 10, 000 dollars, Harry to give me 10, 000 dollars, they both had money. I put in 10, 000 dollars, I didn ’ t have any money. And we were going to do this conference. But we signed a paper, and this is how close TED came to not happening, we signed a paper, quite clear, because Frank did not want to be attached to any failure. He was failure adverse, as sometimes top executives are. And Harry was just a little nervous of that. And I was a loose cannon, as you know. And we signed a paper that if we didn ’ t have X number of people signed up by December, we ’ d give the money back, and we wouldn ’ t do it. And so I was unethical. I was a liar. I broke that commitment. Neither one of them ever forgave me. And they have a right not to forgive me because I was not – I broke my word. But by that time, my assistant Janet Smith and I went down all the numbers and it showed that with the number of people who were signed up, and if we sell the rest of the tickets at 100 dollars each – was 395 – we sold them for 100 bucks, we could not break even, but we would lose less than if we canceled it now. And we ’ d even have to shell out more money because we had rented the room and rented the hotels and committed to things. So I went ahead and basically, the two of them never talked to me again. Frank broke off his relationship as partner of my guidebook company, more or less. And Harry didn ’ t talk except six years later. The first one was so good. Because we did it ’ 84, this is now 1989, Harry comes and says so many people have told him to do it again. CA : Alright, so he was ready to talk to you then. But before we go there, I want to go back to this first one and where the insight came from that there might be synergy between technology, entertainment and design? I mean, look, this was the year – RSW : You ’ re going to make me say it : the conference is not about the audience. I don ’ t care about synergy or about transforming the audience. I don ’ t care about getting letters in, I don ’ t care about rah rah rah. What I cared about was pleasing myself. I invited the people I wanted to hear from. It worked. The whole measure is me. These were interesting people to me, that most of them I hadn ’ t met, but calling them on the phone and appealing to their ego and the fact that some of them I knew, and if they were coming, other people would come. CA : Right. Before you could invite them, you had to invite them to something. And what you invited them to was this weird conference that was these three industries coming together. Did that just emerge from the discussions between the three of you that you were all kind of, from those three industries in some way? And so you thought, you know what? We could pull these things together. Who thought TED, T-E-D? How did that happen? RSW : I did that. I think up names for things. The logo that you now have, I hand drew. That ’ s not a typeface. I drew that logo. CA : Well, thank you. RSW : You ’ re welcome. CA : But why, who thought that technology, entertainment and design, as opposed to, say, software, architecture, there are many other ways that you could have combined. Why these three industries as the heart of something special? RSW : Chris, you tell me a better three things, and I will do it next time. That turned out to be OK. That ’ s all. CA : I mean, it turned out to be amazing. 1984 was the year that the Apple Mac was created. It was the year you had – RSW : The Mac was shown there for the first time. You could touch it. It was announced a month before, but the real ones, the people in the audience could touch. Mickey Schulhof could give away shiny little mirrors, and nobody had a CD player. I just happened to – CA : Right. But so like, right at that time, so a CD, you know, it ’ s technology, it ’ s entertainment and design, that it must have felt like an aha moment to a bunch of people then that, gosh, there really is this connectivity. And I just think it ’ s beautiful how that happened. And tell me this, Richard, even from the start – RSW : The name, the E is the one that many people, you know, to you, say,”Oh, it ’ s technology, education, design.”And it ’ s my way of being for entertainment, being understandable, ways of pulling you into understanding as opposed to the educational system. Design is what I was, and technology was out there. It ’ s a pretty dumb thing

: technology, entertainment, design. CA : Sir Ken Robinson himself said that when you reveal that you ’ re in education, everyone runs away from you at parties. It ’ s OK. But I mean, it worked out incredibly well. And I ’ m curious about like, did it happen that from the first conference that because people weren ’ t just talking to their own industry, that they made an extra effort to have their words accessible to a general audience? Was that something you insisted on? Did it just happen? How did that happen? How did that turn out? RSW : Build the **ball** field, they ’ ll come. You set up a situation where you gave people permission to talk to other people, and they get in touch with their curiosity. The people I invited, they had a **filter** of people I knew who were curious. They were open. The people in the audience who came were curious because they heard about it, and they wanted to know what the hell I was doing. So I mean, I wasn ’ t invisible at that point, and Frank Stanton was not invisible at that point. So they had a little trust, that was all. It wasn ’ t so planned. It was just trying to do good work. That ’ s all, it was not, I don ’ t – I can ’ t write a doctoral dissertation on the planning of what I did. I just did something that felt good. And it was good. CA : If people do want to get a flavor of that first conference, there ’ s actually a talk online by Nicholas Negroponte, the founder of the MIT Media Lab. RSW : He announced it at that conference. He got up and he said,”I ’ m closing the architecture machine and opening, you know, the MIT Media Lab. And he was codirector with the president of MIT. CA : He gave quite a long talk, longer than most TED Talks would be now, and made some predictions that have actually held up pretty well. RSW : If somebody kept on talking, it was good, I let them. But basically, it was supposed to be under 20 minutes. CA : So that first one, only a few hundred people showed up, less than you hoped for. And you lost money. You each lost 7, 000 dollars, I think you told me, of your 10. Each. And so TED didn ’ t then happen for another five or six years until – RSW : I wasn ’ t going to do it again. I didn ’ t want to do it, that was it. I tried that and it was good, but I was on to other things, and I was trying to make a living because I was not in good shape, you know, financially. CA : So Harry Marks came to you in ’ 89 and said,”Actually, it was a commercial failure, but people loved it. How about it? Maybe the time is better now.”How did he persuade you? RSW : This time, I wrote a letter you couldn ’ t get out of, that said if we don ’ t have enough money by a certain date, I can ’ t afford to lose any money. So I did something which was radically different than the lie, breaking a contract the first time. But then it filled up, and it filled up from then on in. But Harry didn ’ t like working with me, and after that conference, he says,”I just want out. I can ’ t, I don ’ t want to do this.”And we didn ’ t argue, we just didn ’ t get along. We just didn ’ t get along. CA : You bought him out for a dollar, right? RSW : He wanted a dollar so it was legal. So he wrote the contract, and he asked for a dollar, and I kept it. And there was never any problems about that afterwards. CA : And so you then held TED every year in Monterey, California. And there was this just growing buzz. I mean, that was the ’ 90s when there was just this growing sense of optimism and excitement about technology and everything it was connected to. I mean, talk about some of those early years, Richard, was there a moment when you just,”Oh my goodness, this thing is going to be amazing. This is more amazing than I know.”What really got you excited in some of those early years? RSW : It just changed my life. It just absolutely changed my life. And it changed the life of many people who were there, and it created circles in their lives. And there was not one person but even today, people said that the friends they have now are dominated by the friends they met at TED. It changed people ’ s acceptance of things outside of their circle and changed their businesses and expanded their feelings that they touched other things. And I wasn ’ t trying to do that. It just did that. It just did that. CA : There ’ s a lot of people listening here who are interested in events, and I think would love to tap into your wisdom about what it was that made it special. You were a very unusual and **remarkable** host. You sat on the stage while the speaker was speaking. You were unafraid to cut them off if they were getting boring. Like your client, as it were, was the audience, not the speaker. Or maybe the client was just your own interest and that that was a proxy for audience interest. What was it that made this thing become so special? RSW : It was human, that ’ s all. There was no lectern. So you couldn ’ t read a speech. I curated it by asking – not always done, not every speech was wonderful, but the best were wonderful – and I asked people to say something they hadn ’ t said before. And... I wasn ’ t interested in good speakers, I was interested in good conversations. I was interested in seeing things before other people saw them. I would interrupt some speakers if I didn ’ t understand something. So I was, in that sense, I curated for the audience, I was their conscience. And I think it was joyful, between the animal acts. I had animal acts because I always wanted to have animal acts. I mean in that sense, I was a pig in shit, I loved being there. CA : You were the ringmaster. RSW : Well, I enjoyed it as much as I hope when you were there, and you said you started coming in ’ 98. Did you know I was having a good time there? It was not painful. I mean, there was attention to detail. I tried to make the details of how – you had your program in your badge. You just held up your badge, and it was the program. You didn ’ t have to carry anything. And then I gave away all those free things until it got – Wired Magazine did a story and said I invented the idea of swag at that time. And I didn ’ t even know I invented it. But we gave away, you know, huge amounts of stuff that people sent in. And nobody would sell anything from the stage. So it wasn ’ t commercial. I didn ’ t have a political point of view, and I didn ’ t have a financial point of view. I had just – wasn ’ t it fun to learn these things? And, you know, some of them, they were up there were maybe slightly boring, but something I was interested in. And then sometimes people other people were interested in. You were there, you could tell me what it was like being there. What was it like being there, Chris? CA : Well, it was overwhelming for the first day, and I didn ’ t get it actually, for the first day. Like, I was intrigued, but I didn ’ t understand why. Like most people, I was in my groove, focused on, you know, trying to make magazines and trying to figure out why exactly am I listening to a designer talk about a chair or an architect or this? You know, it wasn ’ t until day three that you started to realize that something that someone said is connected in a really surprising way with something someone else had said. RSW : Absolutely. CA : And you realize, you know, that all of the best ideas happen through a weird kind of serendipity of things bumping together from outside your normal frame of reference. That ’ s how innovation happens. And Ricky, the human element like – Aimee Mullins, you brought her onstage and you did something that few people would dare to do today, I think. Like, she had lost her legs. She had artificial legs that she had used as an athlete to win. And you invited her to take them off. RSW : But nobody knew that she had artificial legs. They were so good. CA : Right! So this is the showman – RSW : And then I said,”Today, Aimee, take off your legs.”CA : The showman in you, there is a big showman in you, and it was like, you know, how could we really surprise people? I know, let ’ s ask someone, let ’ s ask a speaker to take off her legs. That doesn ’ t happen every day at a conference. And the thing is, she was completely cool with it and so human and told her story of her own empowerment, of how, you know, this technology and help for other people and so forth, that she just felt strong and full of possibility. And I was, by that stage, in the back row of the auditorium, you know, weeping, like, tears rolling down my cheeks. So that was when, I think, I really knew that this was not just interesting but truly special, like, it was moving. And I spoke with other people there, and they said things like,”This is the first week I carve out of my calendar every year.”Well, that gets your attention. That ’ s pretty special. Tell me about – You had courage on stage to do things that, again, most people wouldn ’ t do, and you insisted on a certain kind of vibe from speakers and audience. So there was the time, famously, when Nicholas Negroponte came back, like in his first talk, he was wearing a sort of jacket and tie. And you weren ’ t happy about that. What happened next? RSW : Well, I mean... I ’ d said that the dress is casual and no ties, suits, please. Because that has an effect, it ’ s different. So I just got scissors and cut off his tie. And the audience gasped. And then it became a joke. But people remember that because it was something. I ’ ll tell you a speech that – the audience came up with names. These are not mine, I didn ’ t create these. The audience did somehow. If somebody was there for the first time and they came out in conversation, people in the audience would say to them,”Oh, you ’ re a TED Virgin.”They came up with those things. They came up with things, a”TED moment”when something happened, like cutting off a tie. Or if you remember, Sherwin Nuland, I don ’ t know if you were there. CA : I was there, that was an astonishing talk. RSW : That was one of the most moving things for me. I ’ ll tell you a story of what curation is. Sherwin called me on the phone. I did not know him well. He had been to a conference, and he trusted me for some reason. I think because I don ’ t lie. And he said he ’ s always wanted to tell a story, and he thought he would do it at TED, would it be OK with me? I said, I don ’ t even want to know what the story is. If you want to tell a story, that ’ s for you to do. He says, well, it ’ s, OK. And he got up – and he was well-known then as a doctor and I mean, quite well. I always felt very humbled by getting him to come because he was quite famous in his field. And he came on stage, and he started a talk regularly. And then I looked at him, with the thing of, well, what ’ s the story you ’ re going to tell? And he nodded, and I nodded. And then he told the story, which I then cried, of course I cry a lot, but I cried heavily for his talk. And he talked about being clinically depressed. And committed to treatment in a hospital and committed you know, maybe for the rest of his life. I mean, he was really bad. And he asked for electroshock therapy because he could, as a doctor, which you ’ re supposed to get maximum three times, but it was not thought of well at that time. And by the conference, he gave this talk, it was a horror to think of that, but he asked for it to be given to him ten times. More than three times what the limit was, and it basically cured him. And he told a story which was a shock. His wife didn ’ t know that story, his second wife didn ’ t know it. He had never told the story before. CA : Whoa. She found out when he was onstage? RSW : She heard it for the first time then. It was just astonishing. CA : He must have worried that she would have not let him tell it. You can watch that talk online now. There ’ s this incredible moment when he says, he tells the history of electroshock therapy and then says – RSW : Did I get it right because I haven ’ t seen it? CA : You ’ ve said it exactly right. He says,”And then you may ask, why am I telling you this? Well, it ’ s for a specific reason.”And when he revealed that he himself had been, this was his treatment, yeah, the shock in the room was unbelievable. And it ’ s just a brilliant talk. Wow. So, look, I ’ m going to, in about less than ten minutes, I ’ m going to bring in – RSW : I want to give a compliment to you. Because I ’ ve been working on this. And you see, you ’ ve just interviewed me. You ’ ve messed me up here. I ’ ve been thinking about what you ’ ve done. And, you know, when I did the last one, I was petulant, and I missed it. And then over the years, I saw you, and I wouldn ’ t do this, I would do this. And then I ’ ve been thinking lately what you have done. And in a different style, but amazing what you have been able to do. And I looked online, and I researched you, you have 25 programs that go on. 25 programs. In a recent correspondence with you, I talked about an orchard and apple trees, and I don ' t think you knew what I was getting at, and I didn ' t quite either. But I read a book on Johnny Appleseed, and it was somewhat nonsense because he did – It was a person, and he did carry seeds with him all the time that he got from cider factories. They gave him the free seeds, and he did take them around and he planted them. But you really can ' t get good apples from a seed. You can ' t plant a tree. So what ' s the relationship between you and I, Chris? I think I gave you a tree. But you, as you do to grow American Delicious, all the apples you can grow, you grafted them, and you have grafted a tree with 25 different apples. 25 branches. And that ' s where the apples have come from, from these. Because apple seeds don ' t grow apple trees, apples on apple trees. Little apples, but not big apples. And you have done – It was amazing. For a thing that almost didn ' t happen because of the fear of failure to something that then filled up a year in advance, to something that was the first person who signed up came each time, to selling it, to my petulance, to you doing things that, I think you didn ' t want to do TEDx in the beginning. And then you were convinced to do it. Lara Stein, you had some great people. June Cohen with TED Talks and Lara. And they convinced you to do it. I would have said no because that would have been TED Light. I would have thought, oh, you don ' t want to do that. And it ' s been wonderful. I ' ve spoken at a few TEDxs, and they have been really interesting. And you did 13, 000 of them! 13, 000 TEDxs! So my hat ' s off to you at this time. CA : Maybe even more now. Well, you ' re a kind man. Thank you. That ' s very kind. We should probably tell people just a bit about how the transition happened, because it was a very intense time, you know, like, it was the year 2000. So I ' d be coming to TED for two years when, I think, word got out that you were thinking that it was maybe time to sell, you ' d reached the grand old age of 65 or something like that. And of course, companies like Ziff Davis and Time Warner and so forth were in the hunt for this amazing media property. I had a small media company and had become convinced that this thing was so special. And there was almost like a two-part thing to this, like I came and saw you and your wife Gloria, and we spoke about dreams and values. And, you know, I think you had, your fear was this thing you ' d created would get eaten by some corporation and turned into a, you know, a money-making thing or whatever. It would lose its magic. You probably feared that a bit with me as well. But the one thing I held on to was that, you know, I ' m not a big company, you know, we ' re an entrepreneurial-driven company. At the time, I was still working for the company I ' d founded, Future, and we had this magazine, Business 2. 0, and that seemed like there were connections with a lot of the internet people at TED in that magazine. Somehow you agreed to sell it to me. I suspect you may have got more money elsewhere. I don ' t know, but you sold it to me for, I think it ' s public record, it was six million dollars of cash and six million dollars of stock, I think, there or thereabouts. And the six million dollars of stock disappeared basically, basically because my company blew up soon after that. I don ' t know whether you were able to exit any of that in time. I hope you were. RSW : I have to correct you because we have to get the story straight online. It was 14 million dollars, 12 million in cash and two million in stock. And the stock bankrupted. The stock disappeared. CA : OK, there you go. See, I put my rose-tinted glasses on there, hoping that we hadn ' t spent that much on it initially because I then bought it back from that same company. When the company was blowing up, and it was time for me to leave and I had no money, I had a foundation with a bit of money in it. And so that foundation bought TED off the company for six million dollars in cash. And like, I now, with the benefit of hindsight, that seems like one of the best philanthropic investments ever made. From your point of view, you must have, during that period, I think you felt angered about aspects of the sale, like, you almost had some form of seller ’ s remorse or felt misled or whatever, and we definitely went through a couple of years where things were hard between us. RSW : We had difficult years. And I will take half of the blame for that. And I was petulant because all of a sudden I wasn ’ t doing this every year, and it was my life, and I missed it. So three years passed for a non-compete, and I invented a new conference called EG. CA : Back in Monterey. RSW : And it was... It was not in Monterey, we did it in LA. CA : It moved to Monterey later, right, I think. Maybe. RSW : It went back to Monterey, but the first one was not in Monterey. Because I was going to show everybody and myself I could do it not in Monterey and do it. And it was petulance. It turned out well and then I gave it away, because I realized what a baby I was. And then, you know, it was difficult. It was difficult for you, difficult for me. And then, I would say in the last, you ’ ve been doing it for about 20 years, I did it for about 20 years in the 40 years, give or take a few years. You put together something **remarkable**. I think each of us put together something **remarkable**, different, and yet really kissing cousins. It ’ s the tree and the branches of what you ’ ve done that I think is terrific, just terrific. And I got to do the animal acts. You haven ’ t had any animal acts. CA : You know, we ought to do something about that. Just, you know, 40th anniversary coming up, if only for that. We ’ ve definitely had a lot of animals on screen, spectacular animals. And those are some of the best talks, honestly. RSW : I don ’ t know if you were there for when the bear came on stage? CA : I wasn ’ t there for the bear. RSW : They had a black bear. CA : Amazing. RSW : They had a bear that they walked down the aisle. Two people with chains walked down this big black bear down the aisle on the stage. And I was told by the animal trainer, you know,”Go up and kiss it,”it turns out he thought I would be scared and not do it. And I went up and kissed him, and he turned white and said,”Very quietly, back up very slowly, you could be dead.”And I wasn ’ t supposed to do that. And he could have just taken my belly out with a hand. And that happened. CA : So you told me that story on stage in Monterey when we had you back, and you actually gave me my best – when things were still a bit awkward with us – and you gave me my best-ever line on stage, because I asked you,”Well, did anyone warn the bear?”And people liked that. I mean, in context, it went down. You have this amazing courage and this amazing sense of showmanship that I think has helped. You know, the whole problem with interesting information is that it gets lost in the sea of just noise out there, and it needs all the help it can get in terms of drama, theatricality and so forth. And I think it is one of the pieces of your genius, Ricky, which we ’ ve tried to carry forward, probably have not done in the way that you could. And we miss that. One thing that did happen, though, which and I ’ m, you know, this is just serendipity. I mean, technology came along that allowed TED to be shared with the world. And that, of course, is what what changed everything. We possibly, like, if I ’ d owned it privately, might never have done it. I might have been too frightened to do it. But because it was owned by a nonprofit, we decided we had to do it. Thank you, June Cohen, thank you, Kelly Stoetzel. And, you know, there was an amazing team around at that time who were brave. Jason Wishnow, the video editor, played a role. But we went for it and everything changed, you know, TED went viral and demand for the conference, to our surprise, went up, not down. And most people in the community said,”This is really cool, I can share it now with my family. Thank you.”And you know, it took us on our journey. But I think one of the areas that was uncomfortable for some people in the community and for you, was this feeling that what had been a dinner party, it had been created for the interests of everyone there, had to some extent become an annoying sort of place of”do-goodery”and”let ’ s make the world a better place”and all the rest of it. I mean, how much do you think there is a fundamental conflict there between what is interesting and what is useful for the public good? This is the question I find myself asking. RSW : It ’ s a fine line between, I think it ’ s particularly difficult right now where the”do-goodery”thing and the”watching what you say”thing and all those things, I would have been tarred and feathered for what I said and what I did when I ran TED, because it ’ s not acceptable. I think it is difficult. Your job has become much more difficult in curating now. CA : Possibly. Ricky, I ’ ve got a question from someone in the audience who sat there in the front row for many years of your curation and for many of mine, the wonderful Jim Young. RSW : Oh! CA : He wants to know what was the most memorable moment of your TED experience. Give us one more. RSW : Well, Jim, since you sat in the front row, it was probably two or three times your neck was almost broken when I threw out hats into the audience, and Jeff Bezos leapt from the fourth row across your head and almost broke the necks of everybody in the front row. And then we saw how ambitious he was, was to get a hat, and then he turned out OK. CA : We have to assume he wasn ’ t quite as financially successful then, you know, a hat was meaningful. RSW : You remember those times when people used to jump for the hats? CA : Yeah, no. Absolutely. RSW : But it was nice having you in the front row. It was unlike any other conference where the front row was the place you really wanted to be. Those were the prized seats, not the back row, where a lot of people sit when they go to a conference. CA : So people got married at TED. Engaged or married. Do you remember? RSW : Yes, Chris Fralic. Fralic got engaged on stage, yes? CA : Yeah. Yep yep yep. Chris Fralic is actually in the audience, and so it ’ s very cool that you remember that. RSW : Hi, Chris, yes, yes, I remember that. So my memory hasn ’ t gone anyway, and I ’ m going to be 89 next month. CA : I mean, that ’ s amazing. There ’ s a question here from Todd.”How do you inspire lifelong learning and innovating in people who don ’ t care to learn nor understand, nor change anything?”RSW : I don ’ t think... I think all you can do is give people permission, and if they want to take it, they should be exposed to things that are interesting and available. But no, everybody doesn ’ t have to be like me or you, Chris, or anybody else. One doesn ’ t have to learn. One doesn ’ t have to do things. But it should be available in a form that ’ s honest and understandable. And data by itself is not information, and it ’ s not accessible, and it doesn ’ t inform. So I believe that I have a responsibility, and others do, to make things available. And that ’ s why I invented the term”information architects,”to see the systemic way, not just making things look good. There ’ s a lot of charts, graphs and information that looks good. A lot of people who speak well and pretty and give good **presentations**, but you can ’ t understand it. So understanding, the thing you said you were going to get into later on in this, and explaining, first you have to explain something so you can understand something so you can take action. But that action goes back to having it clearly explained in the beginning. CA : Yeah. So Manoush Zomorodi, who ’ s the host of TED Radio Hour now, has a question that, you know,”Information has become much more nicheified. People want to know exactly what they ’ re getting before they watch / listen. How can we get people to be more general and curious?”RSW : That ’ s why we have the word E for entertaining. You have to make it so – Not entertaining like some song. CA : Song and dance. RSW : Not that. Entertaining in that it feels warm and interesting. You have a warm place. And when something is explained to you that you didn ’ t understand, it feels warm. And you feel warm when you ’ re entertained well. You have to make that available, and then it ’ s up to the person.

It ’ s not something that you need to be tested on. CA : Right. RSW : It ’ s not a homogenous audience out there. It ’ s not our duty to make a homogenous audience. CA : So the question in the audience from Dave,”One of your iconic books is ’ Information Anxiety. ’ What do you see humans are anxious about? What should we be anxious or concerned about?”RSW : Well, I ’ ve done two books called”Information Anxiety”and”Information Anxiety 2,”and might do a book called”3,”and they have a place. The first was just to show the difference between data and things that inform you. And there ’ s a duty, if you ’ re going to have data, to make it understandable to a 12-year-old. To make things you do understandable. And it ’ s your duty that if you don ’ t understand and you ’ re interested, to ask a good question. A question that has a quest. And a good question is better than a brilliant answer. That changed,”Information Anxiety 2”changed because the internet changed our ability and the masses of available data. And so it became more of a crisis of how – and cartography comes in here – how you find your way, how you map your way through information. And that ’ s why the underpinning of all of our data is cartography, not necessarily a map of a city, but the map of understanding. And that ’ s why Esri and people who make cartographic things are so important, because they show the pattern of understanding, and it makes things available and reduces your anxiety because everything takes place someplace. CA : Adrian Neubauer would like to know,”How do you think TED has changed our conceptual understanding of the lecture and lecturing?”So can you talk about the convergence of storytelling and lecturing? RSW : Well, I ’ ve seen the conference world change after TED. Now, I can ’ t say it changed because of TED, but I have a belief that it did. It was bumped along the way. Maybe somebody else would have done a TED-like thing and bumped it away a year later. I don ’ t know, but I believe that TED existed and had no right to exist, not supported by an institution or a university or a company or anything else, it was an independent thing, not based on our society and our businesses in society. That it has changed how companies put on gatherings, how other people put on gatherings, certainly changed Davos, which had no entertainment or anything. But I mean, Davos is a conference made up of back rooms. I mean it ’ s a whole city. They ’ ve turned all the hotels into back rooms. So everything is a closet in there. But they put a patina of entertainment. But there ’ s some genuine understanding and genuine mixed conferences that a lot of people put on today, and some very good ones besides TED. There ’ s The Nantucket Project is one. There ’ s one that used to be put on from with a TEDster Tony Chan up in Boston. There ’ s a number of conferences that I think were directly affected and given permission to happen because a schlepper from Philadelphia could put on a thing, and it became OK that they can do it, too. So I think it ’ s had an effect. CA : I completely agree. I think one of the things we ’ ve really tried to hold on to is when talks become boring, is when it ’ s clear the speaker has an agenda to promote or it ’ s about, here ’ s a company, here is an organization, here is something that I need to promote as opposed to :”I ’ m here with other human beings. There is something really interesting to me in my mind, something that has really lit me up, and I want to share it.”And the fact that people can share it and others can feel that same thing and learn from it, and that it can change their life ten years later, that is what is so beautiful, Richard. And you know, when I came, the first job title I took at TED was TED Custodian. Now this was before I understood properly that sometimes in America that means, you know, bathroom cleaner. But I still like the name. And because what I was promising to do there was... The values that started with what matters, what is interesting, what lights people up, what is inspiring, what is human, what is important and what can cross boundaries out of one sphere into another, that seemed to me so special, and I was determined not to let that go for anything. And so, despite the fact that we ’ ve occasionally disagreed, I have tried to stay true to that. And I think overall, TED has still stayed true to that. And one of the reasons it ’ s special is because it is still a place where no matter what your start point is, so long as you come in with curiosity and an open heart, you will learn something that matters. And that all came from you, my friend. That all came from you. And you know, thank you so much for that. I ’ m going to ask this question from Katherine McCartney. RSW : I ’ m going to interrupt for one second. You said something earlier, that I sat on stage the whole time. Let me give you a hint, there ’ s two reasons why I did that. I was the only person that saw the audience, remember, I kept the house lights up so I could see the audience. I watched the audience. That was very important to me, to sense the audience. And two, getting up and down off the stage is about four minutes, two minutes up and two minutes down. So I gave people four times 50. I gave 200 minutes back to the audience for somebody just getting up and down off the stage to introduce the next person. So it was a way to give the audience more for their money, and I didn ’ t have to get up and down because I was really fat then. CA : That ’ s beautiful. I like that explanation. I ’ ve got good news that through technological advances, we have figured out how to get up on stage again and back off in less than four minutes. RSW : OK. But you know what I ’ m saying. CA : I know exactly what you ’ re saying. Katherine McCartney, who was with me during the transition here, from your TED to ours. RSW : I know who she is, yes. CA : Dear colleague. So she wants to know what moment in your history would you wish you could repeat, either to change the outcome or just to enjoy the moment again? RSW : I can ’ t say the things that come instantly in my head right now, because it wouldn ’ t be good online. CA : Or you could just say it, and we could love you for it. RSW : Oh yeah, edit it out. I will tell you in the after speak, after we speak afterwards. But I think that ’ s a very good question. And I think that question should be mulled around. And anybody who ’ s listening, I hope there ’ s a few people listening, mull around in your heads. How would you answer that question? What, if anything, would you want to redo or repeat or change or do again? And it ’ s not singular things that come up because your mind, at least my mind, goes to different subjects, different moments. CA : I ’ ll ask this question, then, a more specific version of that. I don ’ t know that this is what Katherine was aiming at, but do you... With all that you now know, do you regret the decision to sell TED to me? RSW : Huh. I think, if I look back on it, I probably should have waited three years or so to get some of the ideas out of my system that made me petulant after I sold it. But selling it and doing other things also, because when I was doing TED I was also doing guidebooks. For eight years, I was doing TEDMED. So I got into medicine. And maybe a couple of years, but not selling it. No, I think when you learn how to do something fairly well, you shouldn ’ t do it anymore. For me, I ’ m speaking for myself. And I ’ ve done that with different things in my life, painting, sculpture, different things. I did some of them OK. And then it ’ s time to do something else. So I knew there was a time to do something else. But when it came up, the reality of it caught me off guard. But I think absolutely, If I can take the longer view of an old fart, I gave it to the right person. I sold it to the right person, no doubt, because I can ’ t imagine another person doing anything near what you were doing or squeezing the life out of it. So I think I sold it to, what turns out to be, the best person. CA : Well, those are moving words, obviously, to me. And I will say from my part that I can ’ t imagine a different version of my life. I mean, I loved being an entrepreneur. It was fun building a company. I didn ’ t find out who I wanted to be, Ricky, until I had a chance to pick up this amazing thing that you created. And especially when we had a chance to start sharing it with the world, it suddenly felt, gosh, you know, the fact that we ’ re in a time when ideas can spread beyond a theater to millions of people was such an extraordinary thing. And, you know, just learning to this day of people who gave a talk in that theater that you identified, and by the way, what a special magical theater, that theater in Monterey was. RSW : It turned out to be perfect. Well, see, let me tell you, that wasn ’ t by chance, I had done three conferences in that theater before I did TED. And I learned the town and the theater. So I didn ’ t go there as an amateur, just choosing that place. That ’ s why that place was important. And when I did it in New York or did it in Toronto, it wasn ’ t as good. CA : It was magic. So I can ’ t imagine a different version of it. And I think there are probably many, many millions of people around the world who, if they only knew this story, your story, would want to right now take off the hat and nod at you and say, thank you, Richard Saul Wurman, you ’ ve made a difference to my life. RSW : Chris, that ’ s lovely. CA : On behalf of so many people, thank you. RSW : Thank you, Chris, for having me. [Want to support TED?] [Become a TED Member!] [Learn more at ted. com / membership]

Text Sample 900: POS Dim 6 – 2024 – Score 4.00 – 2024/t003212.txt

Let me show you something. To be precise, I ’ m going to show you nothing. This was the world 540 million years ago. Pure, endless darkness. It wasn ’ t dark due to a lack of light. It was dark because of a lack of sight. Although sunshine did **filter** 1, 000 meters beneath the surface of ocean, a light permeated from hydrothermal vents to seafloor, brimming with life, there was not a single eye to be found in these ancient waters. No retinas, no corneas, no lenses. So all this light, all this life went unseen. There was a time that the very idea of seeing didn ’ t exist. It [had] simply never been done before. Until it was. So for reasons we ’ re only beginning to understand, trilobites, the first organisms that could sense light, emerged. They ’ re the first inhabitants of this reality that we take for granted. First to discover that there is something other than oneself. A world of many selves. The ability to see is thought to have ushered in Cambrian explosion, a period in which a huge variety of animal species entered fossil records. What began as a passive experience, the simple act of letting light in, soon became far more active. The nervous system began to evolve. Sight turning to insight. Seeing became understanding. Understanding led to actions. And all these gave rise to intelligence. Today, we ’ re no longer satisfied with just nature ’ s gift of visual intelligence. Curiosity urges us to create machines to see just as intelligently as we can, if not better. Nine years ago, on this stage, I delivered an early progress report on computer vision, a subfield of artificial intelligence. Three powerful forces converged for the first time. Aa family of algorithms called neural networks. Fast, specialized hardware called graphic processing units, or GPUs. And big data. Like the 15 million images that my lab spent years curating called ImageNet. Together, they ushered in the age of modern AI. We ’ ve come a long way. Back then, just putting labels on images was a big breakthrough. But the speed and accuracy of these algorithms just improved rapidly. The annual ImageNet challenge, led by my lab, gauged the performance of this progress. And on this plot, you ’ re seeing the annual improvement and milestone models. We went a step further and created algorithms that can segment objects or predict the dynamic relationships among them in these works done by my students and collaborators. And there ’ s more. Recall last time I showed you the first computer-vision algorithm that can describe a photo in human natural language. That was work done with my brilliant former student, Andrej Karpathy. At that time, I pushed my luck and said,"Andrej, can we make computers to do the reverse?"And Andrej said,"Ha ha, that ' s impossible."Well, as you can see from this post, recently the impossible has become possible. That ' s thanks to a family of diffusion models that powers today ' s generative AI algorithm, which can take human-prompted sentences and turn them into photos and videos of something that ' s entirely new. Many of you have seen the recent impressive results of Sora by OpenAI. But even without the enormous number of GPUs, my student and our collaborators have developed a generative video model called Walt months before Sora. And you ' re seeing some of these results. There is room for improvement. I mean, look at that cat ' s eye and the way it goes under the wave without ever getting wet. What a cat-astrophe. And if past is prologue, we will learn from these mistakes and create a future we imagine. And in this future, we want AI to do everything it can for us, or to help us. For years I have been saying that taking a picture is not the same as seeing and understanding. Today, I would like to add to that. Simply seeing is not enough. Seeing is for doing and learning. When we act upon this world in 3D space and time, we learn, and we learn to see and do better. Nature has created this virtuous cycle of seeing and doing powered by"**spatial** intelligence."To **illustrate** to you what your **spatial** intelligence is doing constantly, look at this picture. Raise your hand if you feel like you want to do something. In the last split of a second, your brain looked at the geometry of this glass, its place in 3D space, its relationship with the table, the cat and everything else. And you can predict what ' s going to happen next. The urge to act is innate to all beings with **spatial** intelligence, which links perception with action. And if we want to advance AI beyond its current capabilities, we want more than AI that can see and talk. We want AI that can do. Indeed, we ' re making exciting progress. The recent milestones in **spatial** intelligence is teaching computers to see, learn, do and learn to see and do better. This is not easy. It took nature millions of years to evolve **spatial** intelligence, which depends on the eye taking light, project 2D images on the retina and the brain to translate these data into 3D information. Only recently, a group of researchers from Google are able to develop an algorithm to take a bunch of photos and translate that into 3D space, like the examples we ' re showing here. My student and our collaborators have taken a step further and created an algorithm that takes one input image and turn that into 3D shape. Here are more examples. Recall, we talked about computer programs that can take a human sentence and turn it into videos. A group of researchers in University of Michigan have figured out a way to translate that line of sentence into 3D room layout, like shown here. And my colleagues at Stanford and their students have developed an algorithm that takes one image and generates infinitely plausible spaces for viewers to explore. These are prototypes of the first budding signs of a future possibility. One in which the human race can take our entire world and translate it into digital forms and model the richness and nuances. What nature did to us implicitly in our individual minds, **spatial** intelligence technology can hope to do for our collective consciousness. As the progress of **spatial** intelligence accelerates, a new era in this virtuous cycle is taking place in front of our eyes. This back and forth is catalyzing robotic learning, a key component for any embodied intelligence system that needs to understand and interact with the 3D world. A decade ago, ImageNet from my lab enabled a database of millions of high-quality photos to help train computers to see. Today, we ' re doing the same with behaviors and actions to train computers and robots how to act in the 3D world. But instead of collecting static images, we develop simulation environments powered by 3D **spatial** models so that the computers can have infinite varieties of possibilities to learn to act. And you ’ re just seeing a small number of examples to teach our robots in a project led by my lab called Behavior. We ’ re also making exciting progress in robotic language intelligence. Using large language model-based input, my students and our collaborators are among the first teams that can show a robotic arm performing a variety of tasks based on verbal instructions, like opening this drawer or unplugging a charged phone. Or making sandwiches, using bread, lettuce, tomatoes and even putting a napkin for the user. Typically I would like a little more for my sandwich, but this is a good start. In that primordial ocean, in our ancient times, the ability to see and perceive one ’ s environment kicked off the Cambrian explosion of interactions with other life forms. Today, that light is reaching the digital minds. **Spatial** intelligence is allowing machines to interact not only with one another, but with humans, and with 3D worlds, real or virtual. And as that future is taking shape, it will have a profound impact to many lives. Let ’ s take health care as an example. For the past decade, my lab has been taking some of the first steps in applying AI to tackle challenges that impact patient outcome and medical staff burnout. Together with our collaborators from Stanford School of Medicine and partnering hospitals, we ’ re piloting smart sensors that can detect clinicians going into patient rooms without properly washing their hands. Or keep track of surgical instruments. Or alert care teams when a patient is at physical risk, such as falling. We consider these techniques a form of ambient intelligence, like extra pairs of eyes that do make a difference. But I would like more interactive help for our patients, clinicians and caretakers, who desperately also need an extra pair of hands. Imagine an autonomous robot transporting medical supplies while caretakers focus on our patients or augmented reality, guiding surgeons to do safer, faster and less invasive operations. Or imagine patients with severe paralysis controlling robots with their thoughts. That ’ s right, brainwaves, to perform everyday tasks that you and I take for granted. You ’ re seeing a glimpse of that future in this pilot study from my lab recently. In this video, the robotic arm is cooking a Japanese sukiyaki meal controlled only by the brain electrical signal, non-invasively collected through an EEG cap. Thank you. The emergence of vision half a billion years ago turned a world of darkness upside down. It set off the most profound evolutionary process : the development of intelligence in the animal world. AI ’ s breathtaking progress in the last decade is just as astounding. But I believe the full potential of this digital Cambrian explosion won ’ t be fully realized until we power our computers and robots with **spatial** intelligence, just like what nature did to all of us. It ’ s an exciting time to teach our digital companion to learn to reason and to interact with this beautiful 3D space we call home, and also create many more new worlds that we can all explore. To realize this future won ’ t be easy. It requires all of us to take thoughtful steps and develop technologies that always put humans in the center. But if we do this right, the computers and robots powered by **spatial** intelligence will not only be useful tools but also trusted partners to enhance and augment our productivity and humanity while respecting our individual dignity and lifting our collective prosperity. What excites me the most in the future is a future in which that AI grows more perceptive, insightful and spatially aware, and they join us on our quest to always pursue a better way to make a better world. Thank you.

Text Sample 901: POS Dim 6 – 2024 – Score 3.00 – 2024/t003046.txt

How often are you frustrated by the time it takes to accurately get things in your mind into a computer? It is even worse for people like me, whose first language is not based on letters. I live and work in Australia, but I am originally from Taiwan. I moved to Sydney eight years ago and now run a university research center there. Most of us use keyboards every day to get things in our minds into the computer. We have to learn to type. The fact that you have to learn to do some things shows how unnatural it is. The finger-driven touch screen has been around for 60 years. It ' s convenient, but it is also slow. There are other ways to control computers – joystick or gestures – but they are not very useful in capturing the words in your mind. And it is words – they are critical to communication for human beings. The problem is about to be over, because of AI. Today, I will show you how AI can turn the speech in your mind into words on screen. Getting from the brain to the computer efficiently is a real bottleneck for any computer application. It has been my passion for 25 years. Many of you, or most of you, have heard of"brain-computer **interface**,"BCI. I have been working on BCI, for the direct communication between the brain and machine, since 2004. I developed a series of EEG headsets that do this. But they are not new. What is new is an **interface** that works in a natural way, based on how our brain is working naturally. Imagine reading words when someone is thinking, translating the brain signals into words. Today, you will see this in action, and with no implants. We are using AI to decode the brain signals on the top of your head and identify the biomarkers of speaking. That means that you can send the words in your mind into the computer with wearable technology. It ' s exciting, and I believe it will open up the bottleneck of how we engage with computers. We are making exciting progress in decoding EEG to test this. It ' s natural. We have had very promising results in decoding EEG when someone is speaking aloud. The frontier we are working on now is to decode EEG when the speech is not spoken aloud, the words that flow in your mind when you are listening to others or when you are talking to yourself or thinking. We are well on the way to make it a reality. Who would like to see this in action? Great, we are ready to demonstrate it to you. I am going to invite two of my team, Charles and Daniel, to show it to us again. This is the first world premiere for us, so I hope you can be patient with us. We are getting around 50 percent accuracy... in decoding the brain signals into words when someone is speaking silently. Here shows how it will work. We have a collection of words that we have trained our technology with. They are combined into sentences. Charles will select one sentence, and Daniel will read the sentence word by word, silently, and produce the brain signals that will be picked up by our sensors. Our technology will decode the brain signals into words. Charles, Daniel, are you ready to go ahead? This is the sentence that Daniel is going to read silently. Sorry, please keep silent. Here are the – decoded words. They are likely the intended words. You can see the probability ranking of the decoded words by our technology. The pattern shows our predicted sentence... is not so correct. Sorry, you see the other 50 percent, the system doesn ' t work. But actually, you still can see some keywords where we got it. Let ' s invite Charles and Daniel to do it again, but please keep silent when he ' s reading silently. Here is the sentence, another sentence that Daniel will read word by word, silently. Again, here are the decoded words. They are likely the intended words. The pattern shows our predictive sentence is very close to the ground-truth sentence this time. Thank you, thank you. How does it work? We pick up the brain signals with sensors and amplify and **filter** them to reduce the noise and get the right biomarkers. We use AI for the task. We use deep learning to decode the brain signals into the intended words. And then we use the large language model to make the match of the decoded words and make up for the mistakes in EEG decoding. All of this is going on in the AI, but for the user, the interaction is natural, through thoughts and natural language. We are very excited about the advances that we are making in understanding words and sentences. Another thing that is very natural to people is looking at something that has their attention. Imagine if you could select an item just by looking at it, not by picking it off the shelf or punching a code into the vending machine. Two years ago, in a project about hands-free control of robots, we were very excited about robot control via visual identification of the fingers. We are now beyond that. We need not any fingers. The AI is making it natural. There are four objects on the table. Toy car, toy animal, plastic flower and some food, which is also plastic, not left over from the breakfast this morning. You can also see the four objects ' photo on the screen. Daniel is going to look at the photos, and select an item in his mind. If it is working as it should, you will see the selected item pop up on screen. We use photos for this because they are very controllable. To show that this is not all just built into my **presentation**, Charles will pick up one item for Daniel to select in mind. Please, Charles. It ' s a car. So, Daniel, select... the car in his mind. Hamburger. It ' s incorrect. It ' s unlucky that the 30-percent error rate came with us again. Let ' s invite Charles and Daniel to show it again. It ' s a duck, a lovely duck. OK. Good. Thank you. Thank you. Daniel did this for his PhD. It ' s very impressive, isn ' t it? When Daniel selects an item in his mind, his brain recognizes and identifies the object and triggers his EEGs. Our technology decodes the triggers. We are working on our way through the technical challenges. We will work on overcoming the interference issue. That ' s why I asked for the phones to be turned off. Different people have different neural signatures, which are important to decoding accuracy. One reason I brought Daniel along here is because he can give off great neural signatures. Yeah, he can give us a great neural signature, as far as our technology is concerned. They are still cables here as well. It is not yet very portable. Probably one biggest barrier to people using this will be :"How do I turn it off?"Any one of you will have had times when you are happy that the people you are with don ' t know what you are really thinking. There are serious privacy and ethical issues that will have to be dealt with. I am very passionate about how important this technology can be. One exciting point is linking the brain-computer **interface** to wearable computers. You already have a computer on your head. The brain will be a natural **interface**. It is not only about controlling a computer. The natural BCI also provides another way for people to communicate with people. For example, it allows people who are not able to speak to communicate with others, or such as when privacy or silence are required. If your idea of nature is a lovely forest, you could wonder how natural this could be. My answer is, it ' s natural language, it ' s the natural thought process that you are using. There are no unnatural implants in your body. I am challenging you to think about what you regard as natural communication. Turning the speech in your mind into words. There is a standard way to finish up when talking with people – you say :"Just think about it."I hope you are as excited as we are for the prospect of a future in which, when you just think about something, the words in your mind appear on screen. Thank you.

Text Sample 902: POS Dim 6 – 2021 – Score 5.00 – 2021/t004019.txt

Hello there, this is Chris Anderson, and I am hugely, hugely, tremendously excited to welcome you to a new series of the TED interview. Now, then, this season, we ' re trying something new. We ' re organising the whole season around a single theme, albeit a theme that some of you may consider inappropriate. But hear me out. The theme is the case for optimism. And yes, I know the world has been hit with extraordinarily ugly things in the last few years.

Political division, a racial reckoning, technology run amuck, not to mention a global pandemic and impending climate catastrophe. What on earth are we thinking in this context? Optimism just seems so naive and unwanted, almost annoying. So here ’ s my position. Don ’ t think of optimism as a feeling. It ’ s not just this sort of shallow feeling of hope. Optimism is a search. It ’ s a determination to look for a pathway forward somewhere out there. I believe I truly believe there are amazing people whose minds contain the ideas, the visions, the solutions that can actually create that pathway forward. If given the support and resources they need, they may very well light the path out of this dark place we ’ re in. So these are the people who can present not optimism, but a case for optimism. They ’ re the people I ’ m talking to this season. So let ’ s see if they can persuade us now. Then the place I want to start is with A. I. artificial intelligence. This, of course, is the next innovative technology that is going to change everything as we know it, for better or for worse. Today was painted not with the usual dystopian brush, but by someone who truly believes in its potential. Sam Altman is the former president of Y Combinator, the legendary startup accelerator. And in 2015, he and a team launched a company called Open Eye, dedicated to one noble purpose to develop A. I. so that it benefits humanity as a whole. You may have heard, by the way, recently a lot of buzz around in A. I. technology called T3 that was developed by open eye improve the quality of the amazing team of researchers and developers they have work in. There will be hearing a lot about three in the conversation ahead. But sticking to this lofty mission of developing A. I. for humanity and finding the resources to realize it haven ’ t been simple. Open A. I. is certainly not without its critics, but their goal couldn ’ t be more important. And honestly, I found it really quite exciting to hear Sam ’ s vision for where all this could lead. OK, let ’ s do this. So, Sam Altman, welcome. Thank you for having me. So, Sam, here we are in 2021. A lot of people are fearful of the future at this moment in world history. How would you describe your attitude to the future? I think that the combination of scientific and technological progress and better societal decision making, better societal governance is going to solve in the next couple of decades all of our current most pressing problems, there will be new ones. But I think we are going to get very safe, very inexpensive, carbon free nuclear energy to work. And I think we ’ re going to talk about that time that the climate disaster looks so bad and how lucky we are. We got saved by science and technology, I think. And we ’ ve already now seen this with the rapidity that we were able to get vaccines deployed. We are going to find that we are able to cure or at least treat a significant percentage of human disease, including I think we ’ ll just actually make progress in helping people have much longer decades, longer health spans. And I think in the next couple of decades, that will look pretty clear. I think we will build systems with AI and otherwise that make access to an incredibly high quality education more possible than ever before. I think the lives we look forward like one hundred years, fifty years, even the quality of life available to anyone then will be much better than the quality of life available in the very best case to anyone today, to any single person today. So, yeah, I ’ m super optimistic. I think, like, it ’ s always easy to do scroll and think about how bad are the bad things are, but the good things are really good and getting much better. Is it your sincere belief that artificial intelligence can actually make that future better? Certainly. How look, with any technology. I don ’ t think it will all be better. I think there are always positive and negative use cases of anything new, and it ’ s our job to maximize the positive ones, minimize the negative ones. But I truly, genuinely believe that the positive impacts will be orders of magnitude bigger than the negative ones. I think we ’ re seeing a glimpse of that now. Now that we have the first general purpose built out in the world and available via things like RPI, I think we are seeing evidence of just the breadth of services that we will be able to offer as the sort of technological revolution really takes hold. And we will have people interact with services that are smart, really smart, and it will feel like as strange as the world before mobile phones feels now to us. Hmm, yeah, you mentioned your API, I guess that stands for what, application programming **interface**? It ' s the technology that allows complex technology to be accessible to others. So give me a sense of a couple of things that have got you most excited that are already out there and then how that gives you visibility to a pathway forward that is even more exciting. So I think that the things that we ' re seeing now are very much glimpse of the future. We released three, which is a general-purpose natural language text model in the summer of twenty twenty. You know, there ' s hundreds of applications that are now using it in production that ' s ramping up all of the time. But there are things where people use three to really understand the intent behind the search query and deliver results and sort of understand not only intent, but all of the data and deliver the thing of what you want. So you can sort of describe a fuzzy thing and it ' ll understand documents. It can understand, you know, short documents, not full books yet, but bring you back to the context of what you want. There ' s been a lot of excitement about using the generative capabilities to create sort of games or sort of interactive stories or letting people develop characters or chat with a sort of virtual friend. There are applications that, for example, help a job seeker polish a tailored application for each individual company. There ' s the beginning of tutors that can sort of teach people about different concepts and take on different personas. And we can go on for a long time. But I think anything that you can imagine that you do today via computer that you would like to really understand and get to know you. And not only that, but understand all of the data and knowledge in the world and help you have the best experience that is is possible that that will happen. So what gets opened up? What new adjacent possible state is that as a result of these powers from this question, from the point of view of someone who ' s starting out on a career, for example, they ' re trying to figure out what would be a really interesting thing to do in the future that has only recently become possible. What are some new things that this opens up in a world where you can talk to a computer? And get. The output that would normally require you hiring the world experts back immediately for almost no money, I would say think about what ' s possible there. So that could be like, as you said, what can normally only the best programmer in the world or a really great programmer do for me. And can I now instead just ask in English and have that program written? So all these people that, you know, want to develop an app and they have an idea, but they don ' t know how to program. Now they can have it. You know, what is the service look like when anyone on Earth who wants really great medical advice? Can get better medical advice than any single doctor could ever get, because this has the total medical knowledge and reasoning ability that the some humanity has ever produced. When you want to learn something, you have sort of a tutor that understands your exact style, how you best learn everything you know, and custom teaches you whatever concept you want to learn someday. You can imagine that like. You have an eye that reads your email and your task list and your calendar and the documents you ' ve been sent and in any meeting maximally perfectly prepares you and has all of the information that you need in all the context of your entire career right there for you to go on for a long time. But I think this will just be powerful systems. So it ' s really fun playing around with Chapatti three, one compelling example of someone who ' s more tax base is try Googling The Guardian essay that was written entirely by different GP2 three queries and stitched together. It ' s an essay on why artificial intelligence isn ' t a threat to humanity. And that ' s impressive. It ' s very compelling. I actually tried inputting one of the three online uses. I asked the question what is interesting about some ollman? Oh no. Here ' s what it came back with. It was it was rather philosophical, actually. Came back with. I don ' t understand what you mean by interesting. It seems to me that this word has no unambiguous meaning in the context of human society beyond its literal definition as somewhat pleasing or entertaining. I do not believe there to be any such thing as INTERESTINGNESS except in the mind of a human or other sentient being that to my knowledge, this is an entirely subjective state that varies from person to person. However, I will grant that this is an interesting question to ask. This does not mean it has been answered. There is no answer to be found. Well, so you can agree that somewhere between profound and gibberish is that almost well, with the state of play is I mean, that ' s where we are today. I think somewhere between profound and jibberish is the right way to think about the current capabilities of CGP three. I think they would definitely had a bubble of hype about three last summer. But the thing about bubbles is the reason that smart people fall for them is there ' s a kernel of something really real and really interesting that people get overexcited about. And I think people definitely got and still are overexcited about 3:00 today, but still probably underestimated the potential of where these models will go in the future. And so maybe there ' s this like short term overhyped and long term under hype for the entire field, for tax models, for whatever you ' d like. It ' s going on. And as you said, there ' s clearly some gibberish in there. But on the other hand, those were like **well-formed** sentences. And there were a couple of ideas and there that I was like, oh, like they actually maybe that ' s right. And I think if artificial intelligence, even in its current very larval state, can make us confront new things and sort of inspire new ideas, that ' s already pretty impressive. Give us a sense of what ' s actually happening in the background there. I think it ' s hard to understand because you read these words seem like someone is trying to mean something. Obviously, I think you believe that there ' s whatever you ' ve built there, that there ' s a sort of thinking, sentient thing that ' s going, oh, I must answer this question. So so what how would you describe what ' s going on? You ' ve got something that has read the entire Internet, essentially all of Wikipedia, etc. We ' ve read something that ' s read like a small fraction of a random sampling of the Internet. We will eventually train something that has read as much of the Internet or more of the Internet than we ' ve done right now. But we have a very long way to go. I mean, we ' re still, I think, relative to what we will have operated at quite small scale with quite small eyes. But what is happening is there is a model that is ingesting lots of text and it is trying to predict the next word. So we use Transformer ' s they take in a context, which is a particular architecture of an A. I. model, they take in a context of a lot of words, let ' s say like a thousand or something like that. And they try to predict the word that comes next in the sequence. And there ' s like a lot of other things that happen, but fundamentally that ' s it, and I think this is interesting because in the process of playing that little game of trying to predict the next word, these models have to develop a representation and understanding of what is likely to come next and. I think it is maybe not perfectly accurate, but certainly worth considering to say that intelligence is very near the ability to make accurate predictions. What ' s confusing about this is that there are so many words on the Internet which are foolish as well as the words that are wise. And and how do you build a model that can distinguish between those two? And this is prompted actually by another example that I typed in. Like I asked, you know, what is a powerful idea, very interested in ideas. That was my question as a powerful idea. And it came back with several things, some of which seemed moderately pronouncements, which seemed moderately gibberish. But then he was he was one that it came back with the idea that the human race has, quote, evolved, unquote, is false evolution or adaptation within a species was abandoned by biology and genetics long ago. Wait a sec. That ' s news to me. What have you been reading? And I presume this has been pulled out of some recesses of the Internet, but how is it possible, even in theory, to imagine how a model can gravitate towards truth, wisdom, as opposed to just like majority views? Or how how how do you avoid something taking us further into the sort of the maze of errors and bad thinking and so forth that has already been a worrying feature for the last few years? It ' s a fantastic question, and I think it is the most interesting area of research that we need to pursue. Now, I think at this point, the questions of whether we can build really powerful general-purpose AI system, I won ' t say there in the rearview mirror. We still have a lot of hard engineering work to do, but I ' m pretty confident we ' re going to be able to. And now the questions are like, what should we build? And how and why and what data should we train on and how do we build systems not just that can do these like phenomenally impressive things, but that we can ensure do the things that we want and that understand the concepts of truth and falsehood and, you know, alignment with human values and misalignment with human values. One of the pieces of research that we put out last year that I was most proud of and most excited about is what we call reinforcement learning from human feedback. And we showed that we can take these giant models that are trained on a bunch of stuff, some of it good, some of the bad, and then with a really quite small amount of feedback from human judgment about, hey, this is good, this is bad, this is wrong, this is the behavior I want I don ' t want this behavior. We can feed that information from the human judges back into the model and we can teach the model, behave more like this and less like that. And it works better than I ever imagined it would. And that gives me a lot of hope that we can build an aligned system. We ' ll do other things, too, like I think curating data sets where there ' s just less sort of bad data to train on. It will go a very long way. And as these models get smarter, I think they inherently develop the ability to sort out bad data from good data. And as they get really smart, they ' ll even start to do something we call active learning, which is where they ask us for exactly the data they need when they ' re missing something, when they ' re unsure, when they don ' t understand. But I think as a result of simply scaling these models up, building better, I hate to use the word cognition because it sounds so anthropomorphic, but let ' s say building a better ability to reason into the models, to think, to challenge, to try to understand and combining that with this idea of online into human values via this technique we developed, that ' s going to go a very long way. Now, there ' s another question, which you sort of just kicked the **ball** down the field, too, which is how do we as a society decide to which set of human values do we align these powerful systems? Yeah, indeed. So if I if I understand rightly what you ' re saying, that you ' re saying that it ' s possible to look at the output at any one time of three. And if we don ' t like what it ' s coming up with, some ways human can say, no, that was off, don ' t do that. Whatever algorithm or process led you to that, undo it. Yeah. And that the system is that incredibly powerful at avoiding that same kind of mistake in future because it sort of replicates the instructions, correct? Yeah. And eventually and not much longer, I believe that we ' ll be able to not only say that was good, that was bad, but say that was bad for this reason. And also tell me how you got to that answer so I can make sure I understand. But at the end of the day, someone needs to decide who is the wise human or short humans who are looking at the results. So it ' s a big difference. Someone who who grew up with intelligent design world view could look at that and go, that ' s a brilliant outcome. Well, Goldstar done. And someone else would say something is done awfully wrong here. So how do you avoid and this is a version of the problem that a lot of the, I guess, Silicon Valley companies are facing right now in terms of the pushback they ' re getting on the output of social media and so forth. How do you assemble that pool of experts who stand for human values that we actually want? I mean, we talk about this all the time, I don ' t think this is like solely or even not even close to majorly up to opening night to decide, I think we need to begin a societal conversation now about how we ' re going to make those decisions, how we ' re going to make sure we have representational input in that, and how we sort of make these very difficult global governance systems. My personal belief is that we should have pretty broad rules about what these systems will never do and will always do. But then the individual user should get a system that kind of behaves like they want. And there will be people do have very different value systems. Some of them are just fundamentally incompatible. No one gets to use eye to, like, exploit other people, for example, and hopefully we can all agree on. But do you want the AI to like. You know, support you and your belief of intelligent design, like, do I think openly, I should say it can ' t, even though I disagree with that is like a scientific conclusion. No, I wouldn ' t take that stance. I think the thing to remember about all of this is that history is still quite extraordinarily weak. It ' s still has such big problems and it ' s still so unreliable that for most use cases it ' s still unsuitable. But when we think about a system that is like a thousand times more powerful and let ' s say a million times more reliable, it just doesn ' t it doesn ' t say gibberish very often. It doesn ' t totally lose the plot and get distracted or system like that is going to be one that a lot of the economic activity in the world comes to rely on. And I think it ' s very important that we don ' t have a small group of people sort of saying you can never use it for this thing that, like most of the world wants to use it for because it doesn ' t match our personal beliefs. Talk a bit more about some of the other uses of it, because one of the things that ' s most surprising is it ' s not just about sort of text responses. It ' s it can take generalized human instructions and build things up. For example, you can say to it, write a Python program that is designed to put a flashing **cursor** in one corner of the screen, in the Google logo in the other corner. And and it can go your way and do something like that. Shockingly, quite well, effectively. Yeah, I it can. That ' s amazing. I mean, this is amazing to me. That opens the door to. An entirely way to think about programers for the future, that you could you could have people who can program just in human natural language potentially and gain rapid efficiency. I do the engineering. We ' re not that far away from that world. We ' re not that far away from the world where you will write a spec in English. And for a simple enough program, I will just write the code for you. As you said, you can see glimpses of that even in this very week three which was not trained to code like. I think this is important to remember. We trained it on the language on the Internet very rarely, you know, Internet let language on the Internet also includes some code snippets. And that was enough, so if we really try to go train a model on code itself and that ' s where we decide to put the horsepower of the model into, just imagine what will be possible will be quite impressive. But I think what you ' re pointing to there is that because models like three to some degree or other, and it ' s like very hard to know exactly how much understand the underlying concepts of what ' s going on. And they ' re not just regurgitating things they found in a website, but they can really apply them and say, oh, yeah, I kind of like know about this word and this idea and code. And this is probably what you ' re trying to do. And I won ' t get it right always. But sometimes I will just generate this like a brand new program for nothing that anyone has ever asked before. And it will work. That ’ s pretty cool. And data is data. So it can do that from English to code. It can do that from English to French. Again, we never told it to learn about translation. We never told it about the concepts of English and French, but it learned them, even though we never said this is what English is and this is what French is and this is what it means to translate, it can still do it. Wow, I mean, for creative people, is there a world coming where the sort of the palette of possibility that they can be exposed to is just explodes? I mean, if you ’ re a musician, is there a near future where you can say to your eye, OK, I ’ m going to bed now, but in the morning I ’ d love you to present me with a thousand tuba jingles with words attached that you have of a sort of mean factor to the and you come down in the morning and the computer shows you the stuff. And one of them, you go, wow, that is it. That is a top 10 hit and you build a song from it. Or is that going to be released? Actually be the value add. We released something last year called Jukebox, which is very near what you described, where you can say I want music generated for me in this style or this kind of stuff, and it can come up with the words as well. And it ’ s like pretty cool. And I really enjoy listening to music that it creates. And I can sort of do four songs, two bars of a jingle, whatever you ’ d like. And one of my very favorite artists reached out, called to open it after we release this and said that he wanted to talk. And I was like, well, I like total fanboy here. I ’ d love to join that call. And I was so nervous that he was going to say, this is terrible. This is like a really sad thing for human creativity. Like, you know, why are you doing this? This is like whatever. And he was so excited. And he ’ s like, this has been so inspiring. I want to do a new album with this. You know, it ’ s like, give me all these new ideas. It ’ s making me much better at my job. I ’ m going to make better music because of this tool. And that was awesome. And I hope that ’ s how it all continues to go. And I think it is going to lead to this. We see a similar thing now with Dolly, where graphic designers sometimes tell us that they just they see this new set of possibilities because there ’ s new creative inspiration and they ’ re cycle time, like the amount of time it takes to just come up with an idea and be able to look at it and then decide whether to go down that path or head in a different direction goes down so much. And so I think it ’ s going to just be this like incredible creative explosion for humans. And how far away are we some before? And I it comes up with a genuinely powerful new idea, an idea that solves the problem that humans have been wrestling with. It doesn ’ t have to be as quite on the scale as of, OK, we ’ ve got a virus coming. Please describe to us what a what a national rational response should look like, but some kind of genuinely innovative idea or solution like one one internal question we ’ ve asked ourselves is, when will the first genuinely interesting, purely AI written TED talk show up? I think that ’ s a great milestone. I will say it ’ s always hard to guess timeline ’ s I ’ m sure I ’ ll be wrong on this, but I would guess the first genuinely interesting. Ted talk, thought of written delivered by an AIDS within the kind of the seven ish year time frame. Maybe a little bit less. And it feels like I mean, just reading that Guardian essay that was kind of it was a composite of several different GPG three responses to questions about, you know, the threats of robotics or whatever. If you throw in a human editor into the mix, you could probably imagine something much sooner. Indeed. Like tomorrow. Yeah. So the hybrid the hybrid version where it ’ s basically a tool assisted TED talk, but that it is better than any TED talk a human could generate in one hundred hours or whatever, if you can sort of combine human discretion with A. I. horsepower. I suspect that ’ s like our next year or two years from now kind of thing where it ’ s just really quite good. That ’ s that ’ s really interesting. How do you view the impact of A. I. on jobs? There ' s obviously been the familiar story is that every White-Collar job is now up for destruction. What ' s what ' s your view there? You know, it ' s I think it ' s always hard to make these predictions. That is definitely the familiar story now. Five years ago, it was every blue collar job is up for destruction, maybe like last year it was. Every creative job is up for destruction because of things like Jukebox I. I think there will be an enormous impact on. The job market, and I really hate it, I think it ' s kind of gross when people like working on I pretend like there ' s not going to be or sort of say, oh, don ' t worry about it. It ' ll just all obviously better. It doesn ' t always obviously get better. I think what is true is. Every technological revolution produces a change in jobs, we always find new ones, at least so far. It ' s difficult to predict from where we ' re sitting now what the new ones will be and this technological revolution is likely to be. Again, it ' s always tempting to say this time it ' s different. Maybe I ' ll be totally wrong. But from what I see now, this technological revolution is likely to be more. Dramatic. More of a staccato note than most, and I think we as a society need to figure out how we ' re going to cushion everybody through that. I ' ve got my own ideas about how to do that. I, I wouldn ' t say that I have any reason to believe they ' re the right ones, but doing nothing and not really engaging with the magnitude of what ' s about to happen, I think it ' s like not an acceptable answer. So there ' s going to be huge impact. It ' s difficult to predict where it shows up the most. I think previous predictions have mostly been wrong, but I I ' d like to see us all as a society, certainly as a field, engage in what what the shifts we want to make to the social contract are to kind of get through that in a way that is maximally beneficial to everybody. I mean, in every past revolution, there ' s always been a space for humans to move to. That is, if you like, moving up the food chain, it ' s sort of we ' ve retreated to the things that humans could uniquely do, think better, be more creative and so forth. I guess the worry about A. I. is that in principle, I believe this, that there is no human cognitive feat that won ' t ultimately be doable, probably better by artificial general touch, simply because of the extra firepower that ultimately they can have, the vast knowledge they bring to the table and so forth. Is that basically right, that there is ultimately no safe sort of space where we can say, oh, but that would never be able to do that on a very long time horizon? I agree with you, but that ' s such a long time horizon. I think that, you know, like maybe we ' ve merged by that point, like maybe we ' re all plugged in and then, like, we ' re this sort of symbiotic thing. Like, I think there ' s an interesting example, as we were talking about a few minutes ago, where right now we have these systems that have sort of enormous horsepower but no steering wheel. It ' s like, you know, incredible capabilities, but no judgment. And there ' s like these obvious ways in which today even a human plus three is far better than either on their own. Many people speak about a world where it ' s sort of A. I. as this external threat you speak about. At some point, we actually merge with eyes in some way. What do you mean by that? There ' s a lot of different versions of what I think is possible there, you know, in some sense, I ' d argue the merge has already like begun the human technology merge like we have this thing in our hands that sort of dictates a lot of what we think, but it gives us real superpowers and that can go much, much further. Maybe it goes all the way to like the Elon Musk vision of neuro link and having our brains plugged into computers and sort of like literally we have a computer on the back of our head or goes the other direction and we get uploaded into one. Or maybe it ' s just that we all have a chat bot that kind of constantly steers us and helps us make better decisions than we could. But in any case, I think the fundamental thing is it ' s not like the humans versus the eyes competing to be the. Smartest sentient thing on earth or beyond. But it ’ s that this idea of being on the same team. Hmm. I certainly get very excited by the sort of the medium term potential for creative people of all sorts if they ’ re willing to expand their palette of possibilities. But with the use of A. I. to be willing to. I mean, the one thing that the history of technology has shown again and again is that something this powerful and with this much benefit is unstoppable and you will get rewarded for embracing it the most and the earliest. So talk about what can go wrong with that, so let ’ s move away from just the sort of economic displacement factor. You were a co-founder of Open Eye because you saw existential risks to humanity from high today. What would you put as the sort of the most worrying of those risks? And how is open eye working to minimize? I still think all of the really horrifying risks exist. I am more confident, much more confident than I was five years ago when we started that there are technical things we can do about. How we build these systems and the research and the alignment that make us much more likely to end up in the kind of really wonderful camp, but, you know, like maybe open I fall behind and maybe somebody else feels ajai that thinks about it in a very different way or doesn ’ t care as much as we ’ d like about safety and the risks or how to strike a different trade off of how fast we should go with this and where we should sort of just say, like, you know, like let ’ s push on for the economic benefits. But I think all of this sort of like, you know, traditionally what ’ s been in the realm of sci fi risks are real and we should not ignore them. And I still lose sleep over them. And just to update people is artificial general intelligence. Right now, we have incredible examples of powerful AI operating on specific areas. Ajai is the ability of a computer mind to connect the dots and to make decisions at the same level of breadth that that humans have had. What ’ s your sort of elevator pitch on Ajai about how to identify and how to think of it? Yeah, I mean, the way that I would say it is that for a while we were in this world of like very narrow A. I., you know, that could like classify images of cats or whatever, more advanced stuff in that. But that kind of thing. We are now in the era of general purpose, AI, where you have these systems that are still very much imperfect tools, but that can generalize. And one thing like GPP three can write essays and translate between languages and write computer code and do very complicated search. It ’ s like a single model that understands enough of what ’ s really going on to do a broad array of tasks and learn new things quickly, sort of like people can. And then eventually we ’ ll get to this other realm. Some people call it ajai, some people call ostler things. But I think it implies that the systems are like to some degree self directed, have some intentionality of their own is a simple summary to say that, like the fundamental risk is that there ’ s the potential with general artificial intelligence of a sort of runaway effect of self-improvement that can happen far faster than any kind of humans can even keep up with, so that the day after you get to ajai, suddenly computers are thousands of times more advanced than us and we have no way of controlling what they do with that power. Yeah, and that is certainly in the risk space, which is that we build this thing and at some point somewhat suddenly, it ’ s much more powerful than we are, we haven ’ t really done the full merge yet. There ’ s an event horizon there and it ’ s sort of hard to see to the other side of it. Again, lots of reasons to think it will go OK. Lots of reasons to think we won ’ t even get to that scenario. But that is something that. I don ’ t think people should brush under the rug as much as they do, it ’ s in the possibility space for sure, and in the possibility subspace of that is one where, like, we didn ’ t actually do as good of a job on the alignment work as we thought. And this sort of child of humanity kind of acts in a very different way than we think. A framework that I find useful is to sort of think about like a two by two matrix, which is short timelines to ajai and long timelines to ajai and a slow take off and a fast take off on the other axis. And in the short timelines, fast take off quadrant, which is not where I think we ’ re going to be. But if we get there, I think there ’ s a lot of scenarios in the direction that you are describing that are worrisome. And we would want to spend a lot of effort planning for. I mean, the fact that a computer could start editing its own code and improving itself while we ’ re asleep and you wake up in the morning and it ’ s got smarter, that is the start of something super powerful and potentially scary. I have tremendous misgivings about letting my system, not one we have today, but one that we might not have and too many more years start editing its own code while we ’ re not paying attention. I think that ’ s the kind of thing that is worth a great deal of societal discussion about, you know, just because we can do that. Should we? Yes, because one of the things that ’ s that ’ s been most shocking to you about the last few years has been just the power of unintended consequences. It ’ s like you don ’ t have to have a belief that there ’ s some sort of waking up of of an alien intelligence that suddenly decided it wants to wreak havoc on humans. That may never happen. What you can have is just incredible power that goes amok. So a lot of people would argue that what ’ s happened in technology in the last few years is actually an example of that. You know, social media companies created these intelligences that were programmed to maximally harvest attention, for example, for sure. And they understand this from that turned out to be in some ways horrifying and extraordinarily damaging. Is that a meaningful sort of canary in the coal mine saying, look out, humanity, this could be really dangerous? And how how on earth do you protect against those kinds of unintended consequences? I think you raise a great point in general, which is these systems don ’ t have to wish ill to humanity to cause ill just when you have, like, very powerful systems. I mean, unintended consequences for sure. But another version of that is and I think this applies at the technical level, at the company level, at the societal level, incentives are superpower ’ s. Charlie Munger had this thing on, which is incentives are so powerful that if you can spend any time whatsoever working on the incentive system, that ’ s what you should do before you work on anything else. And I really believe that. And I think that applies to the individual models we build and what their reward functions look like. I think it applies to society in a big way, and I think it applies to our corporate structure at open. I you know, we sort of observe that if you have very well-meaning people, but they have this incentive to sort of maximize attention harvesting and profit forever through no one ’ s ill intentions, that leads to a quite undesirable outcome. And so we set up opening is this thing called a capped profit model specifically so that we don ’ t have the system incentive to just generate maximum value forever with an AGI that seems like obviously quite broken. But even though we knew that was bad and even though we all like to think of ourselves as good people, it took us a long time to figure out the right structure, to figure out a charter that ’ s going to govern us and a set of incentives that we believe will let us do our work. And kind of these we have these like three elements that we talk about a lot research sort of engineering, development and deployment policy and safety. Put those all together under a system where you don ’ t have to rely on. Anything but the natural incentives to push in a direction that we hope will minimize the sort of negative unintended consequences. So help me understand this, because this is I think this is confusing to some people. So you started opening. I initially I think Elon Musk, the co-founder, and there was a group of you and the argument was this technology is too powerful to be left, developed in secret and to be left developed purely by corporations who have whatever incentive they may have. We need a nonprofit that will develop and share knowledge openly. First of all, just even at that early stage, some people were confused about this. It was saying if this thing is so dangerous, why on earth would you want to make it secrets even more available? Well, maybe giving the tools to that sort of AI terrorist in his bedroom somewhere, I think I think we got misunderstood in the way we were talking about that. We certainly don ’ t think that the right thing to do is to, like, build this a super weapon and hand it to a terrorist. That ’ s obviously awful. One of the reasons that we like our API model is it lets us make the most powerful AI technology anyone in the world has, as far as we know, available to ever would like to use it, but to put some controls on its usage. And also, if we make a mistake, to be able to pull it back or change it or tweak it or improve it or whatever. But we do want to put and this is continued will continue to be true with appropriate restrictions and guardrails, very powerful technology in the hands of people. I think that is fair. I think that will lead to the best results for the society as a whole. And I think it will sort of maximize benefit. But that ’ s very different than sort of shipping the whole model and saying, here, do whatever you want with it. We ’ re able to enforce rules on it. We also think and this is part of the mission that like something the field was doing a lot of that we didn ’ t feel good about was sort of saying like, oh, we ’ re going to keep the pace of progress and capabilities secret. That doesn ’ t feel right, because I think we do need a societal conversation about what ’ s what ’ s going on here, what the impacts are going to be. And so we although we don ’ t always say, like, you know, here ’ s the super weapon, hopefully we do try to say, like, this is really serious. This is a big deal. This is going to affect all of us. We need to have a big conversation about what to do with it. Help me understand the structure a bit better, because you definitely surprised much people when you announced that Microsoft were putting a billion dollars into the organization and in return, I guess they get certain exclusive licensing rights. And so, for example, they are the exclusive licensee of CP3. So talk about that structure of how you win. Microsoft presumably have invested not purely for altruistic purposes. They think that they will make money on that billion dollars. I sure hope they do. I love capitalism, but I think that I really loved even more about Microsoft as a partner. And I ’ ll talk about the structure and the exclusive license in a minute is that we like went around to people that might find us. And we said one of the things here is that we ’ re going to try to make you some money. But like Adjei going well is more important. And we need you to sign this document that says if things don ’ t go the way we think and we can ’ t make you money like you just cheerfully walk away from it and we do the right thing for humanity. And they were like, yes, we are enthusiastic about that. We get that the mission comes first here. So again, I hope a phenomenal investment for them. But they were like they really pleasantly surprised us on the upside of how aligned they were with us, about how strange the world may get here and the need for us to have flexibility and put our mission first, even if that means they lose all their money, which I hope they don ’ t and don ’ t think they will. So the way it ’ s set up is that if at some point in the coming year or two, two years, Microsoft decide that there ’ s some incredible commercial opportunity that they could realize out of the eye that you ’ ve built and you feel actually, no, that ’ s that ’ s damaging. You can block it. You can veto it. Correct. So the four most powerful version of three and its successors are available via the API, and we intend for that to continue. What Microsoft has is the ability to sort of put that model directly into their own technology. If they want to do that. We don ’ t plan to do that with other people because we can ’ t have all these controls that we talked about earlier. But they ’ re like a close trusted partner and they really care about safety, too. But our goal is that anybody who wants to use the API can have the most powerful versions of what we ’ ve trained. And the structure of the API lets us continue to increase the safety and fix problems when we find them. But but the structure. So we start out as a non-profit, as you said, we realized pretty quickly that although we went into this thinking that the way to get to ajai would be about smarter and smarter algorithms, that we just needed bigger and bigger computers as well. And that was going to require a scale of capital that no one will, at least certainly not me, could figure out how to raise is a nonprofit. We also needed to sort of be able to compensate very highly compensated, talented individuals that do this, but are full for profit company had runaway incentives problem, among other things. Also just one about sort of fairness in society and wealth concentration that didn ’ t feel right to us either. And so we came up with this kind of hybrid where we have a nonprofit that governs what we do, and it has a subsidiary, LLC, that we structure in a way to make a fixed amount of profit so that all of our investors and employees, hopefully if things go how we like, if not no one gets any money, but hopefully they get to make this one time great return on their investment or the time that they spent it open their equity here. And then beyond that, all the value flows back to the nonprofit and we figure out how to share it as fairly as we can with the world. And I think that this structure and this nonprofit with this very strong charter in place and everybody who joins signing up for the mission come in first and the fact the world may get strange, I think that. That was at least the best idea we could come up with, and I think it feels so far like the incentive system is working, just as I sort of watch the way that we and our partners make decisions. But if I read it right, the cap on the gain that investors can make is 100 Axum. It ’ s a massive call that was for our very first round investors. It ’ s way, way lower. Like as we now take a bit of capital, it ’ s way, way lower. So your deal with Microsoft isn ’ t you can only make the first hundred billion dollars. I don ’ t know. It ’ s way lower than after that. We ’ re giving it to the world. It ’ s way lower than that. Have you disclosed what I don ’ t know if we have, so I won ’ t accidentally do it now. All right. OK, so explain a bit more about the charter and how it is that you. Hope to avoid or I guess help contribute to an eye that is safe for humanity. What do you see as the keys to us avoiding the worst mistakes and really holding on to something that ’ s that ’ s beneficial for humanity? My answer there is actually more about, like technical and societal issues than the charter. So if it ’ s OK for me to answer it from that perspective, sure. OK, I ’ m happy to talk about the charter to. I think this question of alignment that we talked about a little earlier is paramount, and then I think to understand that it ’ s useful to differentiate between accidental misuse of a system and intentional misuse of a system. So like intentional would be a bad actor saying, I ’ ve got this powerful system, I ’ m going to use it to like hack into all the computers in the world and wreak havoc on the power grids. And accidental would be kind of the Nick Bostrom make a lot of paper clips and view humans as collateral damage in both cases. But to varying degrees, if we can really, truly, technically solve the alignment problem and the societal problem of deciding to which set of human values do we align, then the systems understand right and wrong, and they understand probably better than we ever can, unintended consequences from complex actions and very complex systems. And, you know, if we can train a system which is like. Don ’ t harm humanity and the system can really understand what we mean when we say that, again, who is we and what does that have some asterisks on them? Sorry, go ahead. Well, that ’ s if they could understand what it means to not harm humanity, that there ’ s a lot wrapped up in that sentence. Because what ’ s been so striking to me about efforts so far is that they seem to have been based on a very naive view of human nature. Go back to the sort of Facebook and Twitter examples of, well, the engineers building some of the systems would say we ’ ve just designed them around what humans want to do. You said, well, if someone wants to click on something, we will give them more of that thing. And what could possibly be wrong with that? We ’ re just supporting human choice, ignoring the fact that humans are complicated, farshid animals for sure, who are constantly making choices, that a more effective version of themselves would agree is not in their long term interests. So that ’ s one part of it. And then you ’ ve got layered on top of that or the complications of systemic complexity where, you know, multiple choices by thousands of people end up creating a reality that possibly have designed for how how to cut through that. Like an AI has to make a decision based on a moment, on a specific data set. As those decisions get more powerful, how can we be confident that they don ’ t lead to this sort of system crashing basically in some way? I think that I ’ ve heard a lot of behavioral psychologists and other people that have studied this say in different ways, are that I hate to keep picking on Facebook, but we can do it one more time since we ’ re on the topic. Maybe you can ’ t in any given moment in night where you ’ re tired and you have a stressful day, stop yourself from the dopamine hit of scrolling and Instagram, even though you know that ’ s bad for you and it ’ s not leading to your best life. But if you were asked in a reflective moment where you were sort of fully alert and thoughtful, do you want to spend as much time as you do scrolling through Instagram? Does it make you happier or not? You would actually be able to give like the right long term answer? It ’ s sort of the spirit is willing, but the flesh is weak kind of moment. And one thing that I am hopeful is that humans do know what we want and what. On the whole, and presented with research or sort of an objective view about what makes us happy and doesn ’ t we ’ re pretty, what ’ s so great about it, they ’ re pretty good. But in any particular moment, we are subjected to our animal instincts and it is easy for the lower brain to take over the eye. Well, I think be an even higher brain. And as we can teach it, you know, here is what we really do value. Here ’ s what we really do want. It will help us make better decisions than we are capable of, even in our best moments. So is that being proposed and talked about as an actual rule? Because it strikes me that there is something potentially super profound here to introduce some kind of rule for development of AIDS that they have to tap into not. What humans one, which is an ill defined question, but as to what humans in reflective mode want. Yeah, we talk about this a lot. I mean, do you see a real chance where something like that could be incorporated as a sort of an absolute golden rule and and if you like, spread around the community so that it seeps into corporations and elsewhere? Because that I ’ ve seen no evidence that, well, a little corporation that was potentially a game changer. Corporations have this weird incentive problem. Right. What I was trying to speak about was something that I think should be technologically possible, and that ’ s something that we as a society should demand. And I think it is technically possible for this to be sort of like a layer above the neocortex that makes even better decisions for us and our welfare and our long term happiness and fulfillment than we could make on our own. And I think it is possible for us as a society to demand that. And if we can do like a pincer move between what the technology is capable of and what we what we as society demand, maybe we can make everybody in the middle that way. I mean, there are instances of even though companies have their incentives to make money and so forth, they also in the knowledge age. Can ’ t make money if they have pissed off too many of their employees and customers and investors by analogy of the climate space right now, you can see more and more companies, even those that are emitting huge amounts of carbon dioxide, saying, wait a sec, we ’ re struggling to recruit talented people because they don ’ t want to work for someone who

’ s evil. And their customers are saying, we don ’ t want to buy something that is evil. And so, you know, ultimately you can picture processes where they do better. And I I believe that most engineers, for example, work in Silicon Valley. Companies are actually good people who want to design great products for humanity. I think that the people who run these companies want to be a net contribution to humanity. It ’ s we ’ ve we ’ ve rushed really quickly and design stuff without thinking it through properly. And it ’ s led to a mess up. So it ’ s like, OK, don ’ t move fast, break things, slow down and build beautiful things that are built on a real version of human nature and on a real version of system complexity and the risks associated with systemic complexity. Is that the agenda that fundamentally you think that you can push somehow? Yes, but I think the way we can push it is by getting the incentive system right. I think most people are fundamentally extremely good. Very few people wake up in the morning thinking about how can I make the world a worse place? But the incentive systems that we ’ re in are so powerful. And even those engineers who join with the absolute best of intentions get sucked into this world where they ’ re like trying to go up from it all for and five or whatever Facebook calls those things and you like, it ’ s pretty exciting. You get caught up playing the game, you ’ re rewarded for kind of doing things that move the company ’ s key metrics. It ’ s like fun to get promoted. It feels good to make more money and the incentive systems of the company. And that ’ s what it rewards. An individual performance are maybe like not what we all want. And here I don ’ t want to pick on Facebook at all because I think there ’ s versions of this at play it like every big tech company, including in some ways I ’ m sure it open I but to the degree that we can better align the incentives of companies with the welfare of society and then the incentives of an individual at those companies within the now realign incentives for those companies, the more likely we are to be able to have things like ajai that. Follow an incentive system of. What we want in our most reflective best moments and are even better than what we could think of ourselves is is it still the vision for open eye that you will get to? Artificial general intelligence ahead of. The corporations, so that you can somehow put a stake in the ground and build it the right way. Is that really a realistic thing to to dream for? And if not, how do you live up to the mission and help ensure that this thing doesn ’ t go off the rails? I think it is. Look, I certainly don ’ t think we will be the only group to build an AGI, but I think we could be the first. And I think if you are the first, you have a lot of norms that empower. And I think you ’ ve already seen that. You know, we have released some of the most powerful systems to date. And I think the way that we have done that kind of in controlled release where we ’ ve released a bigger model than a bigger one than a bigger one, and we sort of try and talk about the potential misuse cases and we try to like talk about the importance of releasing this behind an API so that you can make changes. Other groups have followed suit in some of those directions, and I think that ’ s good. So, yes, I don ’ t think we can be the only I do think we can be ahead. And if we are ahead, I think we can use that leverage to hopefully push people in a better direction or maybe we ’ re wrong and somebody else has a better direction. We ’ re doing something about do you have a structural advantage in that your mission is to do this for everyone as opposed to for some corporate objective. And that that that allows you that. Why is it that we came out of open eye and not someone else? It ’ s like it ’ s surprising in some ways when you ’ re up against so much money and so much talent in these other companies that you came up with this platform ahead of. You know, in some sense it ’ s surprising and in some sense, like the startup wins most of the time, like I ’ m a huge believer in startups as the best force for innovation we have in the world today. I talked a little bit about how we combine these three. Different clans of research, engineering and sort of safety and policy that don ’ t normally combine well and I think we have an unusual strength, there were clearly like well funded. We have super talented people. But what we really have is like intense focus and self belief that what we ’ re doing is possible and good. And I appreciate the implied compliment. But, you know, we, like, work really hard. And if we stop doing that, I ’ m sure someone would run by us fast. Tell us a bit more about some of your prior life sentences. Yeah, for several years, you were running Y Combinator, which had incredible impact on some 70 companies. There are so many startup stories that began at Y Combinator. What were key drivers in your own life that took you on the path you ’ re on? And how did that path end up at Y Combinator? No exaggeration. I think I have back to back had the two jobs that are at least the most interesting to me in all of Silicon Valley. I, I was I went to college to study computer science. I was a major computer nerd growing up. I knew like a little bit about startups, but not very much. I started working on this project the same year I started working on that. This thing called Y Combinator started and funded me and my co-founders. And we dropped out of school and did this company, which I ran for like seven years. And then after that I got acquired. I had stayed close to my comment the whole time. I thought it was just this incredible group of people and spirit and set of incentives and just badly misunderstood by most of the world, but obvious to everyone within it that it was going to create huge amounts of value and do a lot of new things. My company had acquired PJI, who is the founder of ICI, and like truly one of the most incredible humans and business people. And Paul Burrell, Paul Graham asked me if I wanted to run it. And kind of like the central learning of my career, why I individual startups has been that if you really scale them up, **remarkable** things can happen. And I. I did it and I was like, one of the things that would make this exciting for me personally motivating would be if I could sort of push it in the direction of doing these hard tech companies, one of which became open. I describe actually what Y Combinator is, you know, how many people come through it to give us a couple of stories of its impact. Yeah. So you basically apply as a handful of people and an idea, maybe a prototype and say, I would like to start a company and will you please fund me? And we review those applications and we I shouldn ’ t say we anymore. I guess they fund four hundred companies a year. You get about one hundred and fifty thousand dollars while she takes about seven percent ownership and then gives you lots of advice and then networking and sort of this like fast track program for starting a startup. I haven ’ t looked at this in a while, but at one point a significant fraction of the billion dollar plus companies in the US that got started. It all came through the Wiki program, some recently in the news ones have been like Airbnb, Jordache, Coinbase, insta card stripe. And I think it ’ s just it has become an incredible way to help. People who understand technology get a three month course in business, but instead of like herding you with an MBA, we actually teach you the things that matter and kind of go on to do incredible, incredible work. What is it about entrepreneurs? Why do they matter? Some people just find them kind of annoying. But I think you would argue I think I would argue that they have done as much as anyone to shape the future. Why? What is it about them? I think it is the ability to take. And idea and by force of will to make it happen in the world and in an incentive system that rewards you for making the most impact on the most people like in our system. That ’ s how we get most of the things that that we use. That ’ s how we got the computer that I ’ m using, the software I ’ m using to talk to you on it. Like all of this, you know, everyone in life, everything has a balance sheet. There ’ s plenty of very annoying things about them. And there ’ s plenty of very annoying things about the system that sort of idolizes them. But we do get something really important in return. And I think that as a force for making things that make all of our lives better happen, it ' s very cool. Otherwise, you know, like if you have, like, a great idea, but you don ' t actually do anything useful with it for people, that ' s still cool. It ' s still intellectually interesting. But like, there ' s got to be something about the reward function in society that is like, did you actually do something useful? Did you create value? And I think entrepreneurship and startups are a wonderful way to do that. You know, we get all these great software companies. But I also think it ' s like how we ' re going to get ajai, how we ' re going to get nuclear fusion, how we ' re going to get life extension. And like on any of those topics are a long list of other things I could point to. There ' s like a number of startups that I think are doing incredible work, some of which will actually deliver. It is a truly amazing thing when you put the camera back and to believe that a human being could be lying awake at night and something pops inside their mind as a patterning of the neurons in their brain that is effectively them saying, aha, I can see a way where the future could be better and and they can actually picture it. And then they wake up and then they talk to other people and they persuade them and they persuade investors and so forth. And the fact that this this system can happen and that they can then actually change the history changes in some sense. It is mind boggling that that happens that way and it happens again and again. So you ' ve seen so many of these stories happen. What would you say? Is the is there a key thing that differentiates good entrepreneurs from others? If you could double down on one trait, what would it be? If I could pick only one, I would pick determination. I think that is the biggest predictor of success, the biggest differentiator predictor. And if you would allow a second, I would pick like communication skills or evangelism or something in that direction as well. There are all of the obvious ones that matter, like intelligence, but there ' s like a lot of smart people in the world. And when I look back at kind of the thousands of entrepreneurs I ' ve worked with, all of many of whom were like quite capable, I would say that ' s like one and two of the surprisingly differentiated characteristics. What it ' s it ' s what I look at, the different things that you ' ve built and you ' re working on. I mean, it could not be more foundational for the future. I mean, entrepreneurship. I know this is I agree that this is really what has driven the future. Do you see some people get really now they look at Silicon Valley and they look at this story and they worry about the culture. Right. That it ' s this is a bro culture. Do you see prospects of that changing anytime soon? And would you welcome it? Can we get better companies by really working to expand a group of people who can be entrepreneurs and who can contribute to aid, for example? For sure. And in fact, I think I ' m hopeful, since these are the two things I ' ve thought the most about. I ' m excited for the day when someone combines them and uses A. I. to better select who did more fairly, maybe even select who to fund and how to advise them and really kind of make entrepreneurship super widely available that will lead to like better outcomes and sort of more societal wealth for all of us. So are. So, yeah, I think. Broadening the set of people able to start companies and that sort of get the resources that you need, that is like an unequivocally good thing and it ' s something that I think Silicon Valley is making some progress in. But I hope we see a lot more. And I do really, truly think that the technology industry entrepreneurship is one of the greatest forces for self betterment. If we can just figure out how to be a little bit more inclusive in how we do things. My last question today is about ideas were spreading. If you could inject one idea into the mind of everyone listening, what what would the idea be? We ' ve touched on it a bunch, but the one idea would be the ajai really is going to happen. You have to engage with it seriously, and you shouldn ’ t just listen to this and then brush aside and go about life as if it ’ s not going to happen because it is going to affect everything. And we will all we all, I think, have an obligation, but also an opportunity to figure out what not means and how we want the world and this sort of one time shift to go on. I ’ m kind of awed by the breadth of things are engaged with. Thank you so much for spending so much time sharing your vision. Thanks so much for having me. OK, that ’ s it for today. You can read more about open eyes, vision and progress at open eye dotcom. If you want to try playing with yourself, it ’ s a little tricky. You have to find a website that has licensed the API. The one I went to was philosopher ehi dot com, where you just you pay a few dollars to get access to a very strange mind. That ’ s actually quite a lot of fun. The interview is part of the TED Audio Collective, a collection of podcasts dedicated to sparking curiosity and sharing ideas that matter. This show is produced by Kim Net2Phone Pittas and edited by Grace Rubenstein and Sheila Boffano, Sambor Islamic Sir. Fact Check is by Paul Durbin and special thanks to Michele Quent, Colin Helmes and Anna Felin. If you like the show, please write and review it. It helps other people find us. We read every review, so thanks so much for listening. See you next time.

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Exactly 12 years ago in a TEDx Talk, I introduced my seven species of robot. A unique three-legged robot called STriDER, a wheel-leg hybrid robot called IMPASS, an amoeba robot and many more. Now, 12 years is a pretty long time, especially in a cutting-edge field like robotics. How have our robots evolved during the past 12 years? What new species of robot have we built since then? Hello, my name is Dennis Hong, and today, I would like to show you our next seven species of robot. Since my last TEDx Talk, we ’ ve been focusing on the development of humanoid robots : robots that look like us, two legs, torso, two arms and a head. Let me show you some of the humanoid robots that we developed at our lab, RoMeLa. DARwIn is a miniature humanoid robot for research and education that we made fully open-source. CHARLI is considered the United States ’ very first full-sized humanoid robot. THOR-RD is for disaster relief applications and Shipboard Autonomous Fire fighting Robot, SAFFiR and THOR, which uses this unique biologically-inspired actuator. And many more. I think we built more than a dozen different types of humanoid robots. However, after more than a decade of humanoid robotics research, I ’ m starting to have different thoughts. Now humanoid robots are important and they ’ re great. However, they always fall down, they ’ re too unstable. And they ’ re painfully slow. Yes, these days we ’ re trying to develop technology that resolves these two problems, but still, they are too expensive and they ’ re too complicated. And most importantly, they ’ re too dangerous. You do not want to be next to these big, heavy moving robots. It might fall on top of you. So what do we do? Now, why is it so difficult to make robots walk with two feet? I mean, we do it all the time. So once I wanted to throw away everything I know about robotics for a moment and try to think fundamentally, why is it so difficult to make robots walk with two feet? And I found something fascinating. So check this out. Now, when your robot tries to move forward, one of the reasons why it constantly falls down is because the distance between your left and right leg. Because your legs move forward and backward, up and down, it creates this unwanted twisting forces, we call these moments, right? Now, this is a video called a RoboOne Competition. It ’ s a robot fighting competition. And people who participate in this competition are robot hobbyists. And if you look at the robots, they develop, it ’ s brilliant. It ’ s fast, it ’ s nimble, and it doesn ’ t fall down. And if you look carefully, they always walk sideways, Because if you walk sideways, your left and right leg line up, so your twisting moments disappear. I actually got this inspiration by watching sports, fencing or ballet. They always walk sideways, and I think this is the reason why. Now, one of the problems of trying to make robots move sideways is you cannot really use your knees. So what do we do? We rotate our legs. It looks kind of awkward, right? So actually, let ’ s change the entire torso like this and make it walk this way. Now, the problem of this kind of configuration is that you really cannot use your feet and ankles. So what do we do? We get rid of them. Now let ’ s make more changes, let ’ s change the shin. And while doing that, let ’ s change the head. And something strange starts to emerge. This is NABi, Non Anthropomorphic Biped. In other words, it ’ s a robot with two legs, but it does not look like a human. How does it walk? It walks like this. Simple, fast, easy, low cost. This is great. Now, I briefly mentioned about a robot called SAFFiR, Shipboard Autonomous Fire Fighting Robot that we built. This is a robot for Navy ships. And I don ’ t know if you ’ ve been on a Navy ship, but if you open this door, it has a really high door seal. We call that”knee-knockers.”Before we had this robot, it was very difficult for the robot to climb over it. But if a robot does not have to be like a human, we can do something like this. Make the knee rotate at 360 degrees. Brilliant! Isn ’ t this cool? Yeah. Our robots can climb stairs, but it ’ s rather difficult. But now we can climb stairs like this. Now, is this just simulation and animation? No. In two weeks, from the idea, analysis, design and fabrication, in two weeks, we built our first robot. And you have not seen anything like this. Man : You ’ re wild. And we use many different types of tricks. For example, utilize the springiness of the feet we can store the energy and make it hop and jump. When we first introduced NABi to the community, it had a pretty good impact on the community. However, this robot is great for walking forward and backward. But how does it change directions? It can ’ t. So we decided to work on the next version called ALPHRED. ALPHRED stands for Autonomous Leg Personal Helper Robot with Enhanced Dynamics. So think of this robot as two NABis stacked onto each other. So it looks like this. So this robot is axisymmetric, which means that there ’ s no forwards, backwards or sideways. Now, when it walks, it has a very wide stance, right? So it can walk like this. And this is great for unstructured outdoor environments. However, it ’ s pretty slow. If you want to make it go really fast, you can change your body configuration. And if you go like this, you can have your forward legs and backward legs, and then you can gallop like a horse. There you go. And when you lift two of the arms it can walk like NABi, then the two of the other legs can be used as arms to pick up boxes or press buttons. And we ’ re pretty excited about this new type of configuration. We call this multimodal locomotion, and this is ALPHRED. Now, while developing these robots, we ’ ve been also working on this new type of actuator. Now, when we say actuators, we ’ re talking about these devices that make the robots move just like muscles in humans. Now, most of the robots that exist today, electric robots, use electric motors with gears, right? And these are really strong, and they ’ re precise, they ’ re good for manufacturing. But for legs, we need something different. We need something like biological muscles that ’ s compliant, springy, and not only the position, but also change the force. And we developed a new type of actuator called the BEAR actuator. Now, using this BEAR actuator with the NABi, we developed a second version called NABi-2. Now, if you remember, NABi was our experiment for studying a new type of bipedal locomotion. But it turns out that this NABi with these new actuators doesn ’ t work very well in terms of walking. However, unexpectedly, we didn ’ t know this, but this robot is great for jumping and hopping. It can jump forward, backward and turn directions, it ’ s brilliant. But the really cool thing is, once it jumps and when it lands, the motors become generators, and the impact is changed to electricity. And it can charge its battery, so it ’ s very energy efficient. And NABi-2 is ready to go. Or not. But this is NABi-2. So BEAR actuator with NABi was NABi-2, then BEAR actuator with ALPHRED is – ALPHRED-2. And this is the next generation of ALPHRED. So this is the very first time we turned on the switch. Look carefully at the reaction of our students. Man : Whoa! Oh, my God! What the –? Dennis Hong : Excuse me. Oh, my! Man 2 : That was amazing! I think it jumped about 1. 4 meters. This is **remarkable**. Walking is also very fast and stable. So we wanted to do something more interesting. We wanted to make it do taekwondo. Whoa! DH : We ’ re just having too much fun. And ALPHRED-2 is ready to go. Now, ALPHRED-2 is actually a very practical robot. For example, this is an example of picking up a box, delivering it and putting it nicely on the ground or on the table. I sped up the video because it was pretty slow, but you can get an idea how we can use these new type of robots for different applications. And this is ALPHRED-2. Now, we looked at two legs. We looked at four legs. What about six legs? Of course, we have a hexapod robot. Now, this robot, HEX, is about the size of a small car. Now, the cool thing about six-legged robots is this. Now, to have static stability, the minimum number of contacts with the ground is three. Think of a camera tripod. If you have a hexapod robot, you can always have three-point contact with the ground. Three, three, three. And this makes it really great for outdoor environment, rough environment. This particular robot we ’ re developing for demining applications, getting rid of landmines underground autonomously. If you see it in person, it ’ s kind of scary, it ’ s big and goes like this. So this is HEX. We also have a smaller version, this is called SiLVIA : Six Legged Vehicle with Intelligent Articulation. And as I mentioned, these type of robots are really great for unstructured terrain such as this. It can also climb up and down stairs. And this robot might be the strongest six-legged robot out there. Man : Ah, 20 kilograms. DH : Now, the reason why we wanted to make it really strong is because we want this robot to do some very interesting new type of stuff. So this is the very first robot that can brace between walls and climb up. So it ’ s like Spiderman, isn ’ t that cool? And it has some other tricks, too. If you put suction cups on the feet, it can stick to the wall and climb. And if you give it an arm, it can do some fun stuff on the blackboard, too. And this is SiLVIA. So we talked about vertical mobility. Twelve years ago, in my last TEDx Talk, I introduced a robot called CLIMBeR. It ’ s a robot for climbing cliffs using cables like a human. But this year, we wanted to get rid of the cable. We wanted to make it do free climbing. And I ’ m very excited to introduce a brand new robot called SCALER, Spine Enhanced Climbing Autonomous Legged Exploration Robot. Now, when you first look at it, it looks like a quadruped robot, four-legged robot. It ’ s actually very strong, but strength is not all. We had to develop this new type of gripper, we call these spine grippers, that can really grab onto the surfaces. And this video, we just took it a few days ago. So literally, it ’ s taking its very first few steps. Now, why are we interested in cliff-climbing robots? Rescue missions, of course. But we ’ re really interested in the planetary applications. Now, as you know, on Mars, we had these Rovers discovering the new planet. However, the really science-rich sites are always at the cliffs, and these Rovers cannot get there. So something like this will enable those kind of exploration, in the science-rich sites on planetary surfaces. This is SCALER. Now, we talked about legs, but what about wheels? Of course, we have robots with wheels. This is called OmBURo, Omnidirectional Balancing Unicycle Robot. And as the name implies, it has one wheel, it ’ s omnidirectional, it can go forward, backwards, sideways and **diagonal**. But if we look at it, it doesn ’ t even need to change its orientation. When you see it, it ’ s almost like magic. It feels like it ’ s gliding over the surface. Isn ’ t that cool? It can also balance and climb very steep surfaces. Now it follows path, now, yes, it has one wheel, but actually has these smaller rollers in the rim to generate this kind of motion as well. So why do we want to build a robot like this? We like to have robots living with us in this environment, and this environment sometimes is crowded, a lot of people or obstacles. To go through people or to avoid obstacles, it ’ s very important to have a very small cross section, and probably a single wheel is the best way to have a very small cross section. And this is OmBURo. Now, I ’ ve shown you many different shapes, sizes, mechanism robots. And if you remember this, the whole thing started because we had problems with this bipedal locomotion. Is there a way to make these humanoid robots not fall at all? There is. And this robot is called BALLU. This is my son, Ethan, by the way. Ethan Man : Uh-oh. DH : What if robots never fell? What if we could change the direction of gravity? I ’ m very excited to show you our new robot, BALLU, Buoyancy Assisted Lightweight Leg Unit. This robot is a bipedal robot the size of a human being. But the body is a helium balloon. It ’ s a balloon with two legs. How does it walk? It walks like this. Very elegant, isn ’ t it? This robot might be the safest robot out there. It can walk forward, backwards, sideways. It can turn directions. We want this robot to be living with us. So it can climb upstairs, downstairs. Go over obstacles. It ’ s fun to play with. It can hop, now it can hop up to a height of a table top. Oh, this robot is a good dancer. If you turn on the music, it listens to beat and it dances to the beat. Now we do crazy things when we do experiments. Once, we opened the window and threw the robot out the window. Isn ’ t it great? It solves all the problems of bilateral locomotion. It ’ s cheap, it doesn ’ t fall down. However, it also creates new problems. This robot is horrible outdoors because if the wind blows, it floats away. Now, these are some of the images of our secret projects. I ’ m not going to explain it, but we ’ re using projection mapping, vortex cannons, many different things. Oh, this robot literally can walk on water. You put **ping-pong balls** on the foot, it can walk on water. It can also walk on a tightrope. Isn ’ t this cool? This is BALLU. By the way, all of our videos end with a **bang** like this. Isn ’ t that cool? This is BALLU. Thank you. Now all of these new inventions and creativity came from the problem of humanoid robots. However, secretly, we ’ ve still been working on humanoid robots. And this robot is called ARTEMIS, our latest one, Advanced Robotics Technology for Enhanced Mobility and Improved Stability. And this is the very first time I ’ m showing it to the general public. This is a sneak peek of what is about to come. Now, this robot, ARTEMIS, uses a new type of technology. It ’ s based on that BEAR actuator. And because it ’ s so new, you cannot buy any of these components. So we have to design and fabricate every single part of this robot. We have just finished assembly of the lower body, and I ’ m very excited to show you a sneak peek of robot ARTEMIS. And ARTEMIS is coming too, I cannot wait to show it to you, probably on stage in my next talk. I ’ ve shown you a lot of our newest creation robots from our lab RoMeLa. I would like to give credit to our brilliant, hardworking students at RoMeLa. But what is the source of this creativity? What ’ s the source of this energy? What is the secret to this crazy productivity? I think the next video will be able to answer that question. Man : Look at that. DH : You remember BALLU? We ’ re doing the experiment of dancing. By the way, this is Saturday, 2am in the morning. We ’ re having too much fun, so, yeah, let ’ s bring out all of our robots, one by one. You remember NABi? An early prototype of ALPHRED. Everybody ’ s favorite, DARwIn-OP. The rotating knee mechanism for NABi. The six-legged hexapod robot. This is RoMeLa, the Robotics and Mechanism Laboratory at UCLA. By the way, that ’ s me on the left, dancing. You probably haven ’ t seen a professor dance and party with their students while conducting serious experiments, right? Openness, freedom, trust, having fun and truly believing what you do can change the world. These are some of the secrets behind our next seven species of robot. Thank you very much.

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Hey, Mark Leifer ’ s, it ’ s Adam. Season four is launching this spring. In the meantime, some exciting news. Each season we ’ ve done a couple of bonus episodes where I have a discussion or a debate with a thought leader. You asked for more of those. So we ’ ve delivered. This is a new series we ’ re calling taken for granted. Get it? The gist, more unscripted conversations with fascinating people about rethinking things we ’ ve taken for granted. This is taken for granted by podcast with the TED Radio Collective, I ’ m an organizational psychologist. My job is to think again about how we work, lead and live. I ’ m delighted that we ’ re starting this series with someone who never fails to challenge how I think about the world. Renee Brown is a social work. Professor Bernie has spent two decades studying vulnerability, empathy and shame. She ’ s the host of two new podcasts, Unlocking US and Dare to Lead. Her TED talk is one of the most watched ever with more than 50 million views, one of the big reasons it was such a sensation is that she models what vulnerability looks like. I ’ m going in. I ’ m going to figure this stuff out. I ’ m going to spend a year. I ’ m going to understand how vulnerability works and I ’ m going to outsmart it. As you know, it ’ s not going to turn out well. I ’ ve long been fascinated with the power of vulnerability and impressed with the power of Bernier ’ s insights. I ’ m excited to learn from her about the nuances of vulnerability at work, especially in places where it seems risky. I will say from one interview to another, I cannot stand small talk. I love conversations that go deep right away, and you are the queen of going deep and being vulnerable. So I feel like I feel like I have extra permission to do that here. I ’ m the queen of vulnerability and I ’ m the assistant queen of, like, boundaries. So we ’ ll see. Let ’ s go. Let ’ s go as deep as we can go and then we ’ ll see. So I guess I guess the place I want to start today is to say that you have convinced millions and millions of people that vulnerability is not a sign of weakness. It can actually be a source of strength and of connection. And yet so many people struggle to be vulnerable at work and they decide to put on armor instead. And I wonder, just for starters, if you could talk a little bit about why people feel special pressure to put armor on at work and what kinds of armor they wear. Yeah, so I hope that the work does and I hope what I do is help people dispel the mythology about a vulnerability. I think that ’ s such an important place to start that there ’ s this idea that vulnerability is weakness. I think most of us were raised with that that kind of ethos like self protect. Be careful. Don ’ t put yourself out there. I mean, vulnerability is very simply defined as uncertainty, risk and emotional exposure. It ’ s the affect or emotion we feel in times of great uncertainty, risk. And when an emotional exposure just means I put myself out there. And so the first thing we have to do is dispel that notion. We are raised to believe it ’ s important to be brave, but then we ’ re taught not to be vulnerable. And they ’ re really. In my you know, based on my research and our data, there just is no courage without vulnerability. I tell the story of asking a group of soldiers a very simple question. Give me an example of courage in your life or an example of courage that you ’ ve observed in someone else that did not require

uncertainty, risk or emotional exposure. And I think I was at Fort Bragg and there was just absolute silence until one guy stood up and said, three tours, ma ’ am, there is no courage without vulnerability. Then a week later, I ’ m doing work with Pete Carroll and the Seattle Seahawks. We ’ re talking about vulnerability. I asked the same question to that group of players. Give me an example of courage on the field or off that doesn ’ t require vulnerability. And it was so funny to me because they they had to huddle for a minute and then they kind of came back and said, there is no courage without vulnerability, not on or off the field. If you ’ re not all in if you ’ re not putting yourself out there, you just can ’ t be brave. And so I think the job is dispelling the mythology about vulnerability is weakness at work. I think we armourer more because there ’ s less trust, there ’ s less confidence. And I think we slip into kind of who we think we ’ re supposed to be at work. And we armer up using things like cynicism, perfectionism, needing to be the knower and be right versus the learner and get it right. There are a lot of forms of armor that can sometimes be rewarded at work. I think that ’ s so true. I love your observation that that people will use expertise or performance as armor at work. And it it really struck a chord with me on a very deep level because I don ’ t want to I don ’ t want to get to psychoanalytic here. But I you know, I had this this defining experience, I guess, in many ways when I was 12, where all of my close friends dropped me because I wasn ’ t cool enough for them. And I never really realized until until you started talking about performance armor that my way of coping with with that experience was to try to be exceptionally good at whatever I did, because then people would respect me or look up to me here like me. And so first it was in sports saying, OK, if I can excel in diving, you know, then I ’ ll have I ’ ll have earned, you know, a badge or some admiration and and maybe I ’ ll be accepted as opposed to rejected. And then I got to college and started to redefine my identity around excellence in school, which which was something that then got reinforced. And then over time it felt like every single project I took on was was one where I knew I could excel. And that way I was just fortifying my armor more and more. And I started to feel like this is a mistake and I ’ m missing out on opportunities for learning and for challenging myself and stretching myself. And so I just I ’ d love to get your reactions on on this idea of for those of us who are so accustomed to to treating excellence or expertises as our armor, how do we let go of that? Well, first of all, I have to say that your story of when you are 12 is real trauma. I mean, that ’ s really that that ’ s trauma. And yes, there are different kinds of trauma and different sizes of the bucket. But when you ’ re 12, it ’ s all about belonging. And I think I have the same story. I was I was 13, not 12, but I have a very similar story. But I just got really good at, like smoking cigarettes and being wild, which was probably not the best thing to pick. But but we all have stories like that and we build our armor around those stories. And no one talks about the big developmental milestone of midlife, which can happen, I think, anywhere between our late 30s and probably mid 50s, which is the armor that we put on to protect us when we were children and have less agency and less control over what was happening. That armor no longer serves us and it is heavy. And what it actually does is prevent us from being seen and prevent us from growing into areas because that armor doesn ’ t grow with us. It stays kind of the same size, I think. And so I you know, for me, I kind of switched armor, you know, when I moved from being kind of the loud party girl to, OK, this is scary, but I ’ m going to try to be the smart kid. And that worked. And I was rewarded for it. Then I went I moved solidly into a life of. Proving, performing, perfecting until I actually physically and emotionally couldn ’ t do it anymore, and so I think, again, the big challenge of midlife is the armor that we ’ re carrying, their armor that we ’ re, you know, that ’ s got us locked in and the weapons that were carrying that kept us safe at some point. What is OK to let go of because it ’ s no longer serving and then how do we peel it off, which is the scary part. You know, I do think that as I peeled mine off and that was work I did with a therapist, and now I ’ m constantly trying new things that I don ’ t know whether I ’ m going to be good at or not. And I am failing, you know, on occasion, you know, but feels but now that I ’ ve changed the goal to stretching and learning instead of proving and perfecting. It feels so different. Does that make sense? It does it it resonates. And I ’ ve I guess I ’ ve I ’ ve gone through a similar shift in saying, look, I want to be the kind of person who takes on projects that matter. Yes. And where I have the potential to contribute something meaningful. And even if I fail, at least I didn ’ t fail to try. That ’ s right. Yes. And I have let me tell you, I have I have taken on a couple of things in the last several years where I have just failed. But I don ’ t regret it. I don ’ t regret it. It ’ s such a great place to be and part of I guess part of what I ’ ve been wondering is, as I ’ ve been applying applying your work to my own life and also talking in class about it quite a bit. So just just as a little bit of back story, every year, I have students at the undergrad level apply to my class. And one of the questions is, what ’ s your favorite TED talk? Yours is every year. One of the topics I think one of the things that jumped out at me after the first few years was it was mostly women who were naming you as their favorite talk. And and I ’ m wondering, I ’ m sure some of this, of course, is, you know, men tend to choose male role models. Women tend to choose female. But it also seems like vulnerability is easier for women to process and maybe adopt. And it ’ s something men struggle a lot more with. And I wondered if you could talk about what you found in your research there and a little bit of the psychology of why men struggle so much with vulnerability. I have moved from talking about men and women just to talking about folks who are, you know, trying to meet masculine norms and folks trying to meet feminine norms just to be more inclusive. But here ’ s the thing. In terms of masculine norms, the biggest shame trigger is do not be perceived as weak. So just think about that calculus for a second. I ’ m asking people to be vulnerable. Ninety nine percent of them are raised to believe that vulnerability is weakness, and they know in their heart that they ’ ll experience shame if they ’ re perceived as weak. And that ’ s so it becomes a very big ask for. People who really value complying with masculine norms in terms of women and feminine norms, the number one shame trigger is. Perfection do not be perceived as imperfect, do it all, do it. Well, never let them see you sweat. So now I ’ m saying, hey, I want you to be vulnerable about your setbacks and your failures and where you ’ re struggling and your emotions. So really the biggest barrier to vulnerability. Is shame the fear of shame? The fear that I ’ m going to put myself out there and I ’ m going to find myself experiencing, you know, the definition we use it for, shame is the incredibly painful experience of believing or feeling that we are unworthy, that we ’ re flawed and unworthy of love and belonging and connection. And so you can see how vulnerability is a really big ask. It ’ s a really big ask you. It ’ s so interesting. Your analysis reminds me of the Boston and Vandeleur work on precarious manhood, which it basically says, look, you know, the problem with most definitions of masculinity or masculine cultures is we have to demonstrate it over and over and over again. And so no matter how many times you ’ ve proven your strength, if you show just one tiny bit of weakness, then all of a sudden you ’ re no longer a man or you ’ re no longer strong. I don ’ t know. Have you been reading any of that work? Well, I. I found it by asking some folks on my research team what is an alternative to toxic masculinity. Interesting as a phrase, because I ’ m not sure that phrase is helpful. I am so with you. I hate the phrase. It ’ s it ’ s always bothered me just at a basic level, because it seems to imply that masculinity is inherently toxic as opposed to what you said, which is there ' s a there ' s a brand or a flavor of masculinity that can be toxic. And I guess I came to wondering, is that part of the reason why why you see so many men motivated to armor up at work because they have to keep proving their strength over and over again? This is super interesting. And this is I ' m like, this is me thinking out loud with you just spit **balling** right now. Why is vulnerability? So much. More difficult to talk about with men in suits. In organizations or hoodies and jeans, because I work in Silicon Valley a lot, right. Why is it more difficult to talk about with them than it is with. Trauma surgeons and firefighters and military and professional athletes, and I guess this hypothesis that ' s forming in my head right now, which, you know, who knows if it ' s right or not, would be. Where masculinity is, you know, where where that kind of strength is seen every day on the pitch or on the football field or in their jobs, they ' re not having to prove it so often because it ' s part of what they do. Wow. But they ' re more open to those conversations. Do you know what I ' m saying? That is so fascinating. Right. So if I ' m if I ' m a firefighter, everybody knows I ' m a badass. So if you if I if I break down outside of a burning building into tears, I ' ve I ' ve I ' ve established already that I ' m tough and I can I can handle extreme crises and difficult situations, whereas if I ' m a software engineer in Silicon Valley or I ' m a trader on Wall Street, I haven ' t been able to demonstrate my strength in the same way as that is. Is that what you ' re getting at? Yeah, I mean, it ' s what I ' m getting at because, I mean, it ' s totally what I get. I ' m getting that because I ' m thinking to this moment where I ' m on an Air Force base, talking to a general, doing a lot of work with and these are like serious, serious ass people like these are fighter pilot kind of folks. Right. And before we got started, he and I are talking about the work. And I said, you know, one of the things that ' s going to be really interesting is I want to talk about care and connection are irreducible needs when we lead, meaning we have to care for and be connected to the people we lead are we can ' t do it effectively. And up until that point, I had had so much pushback from people like, that ' s bullshit. I ' m not here to like you. I ' m here to lead you. And this general turned toward me and said from the highest ranks of the Air Force, we believe you cannot lead people that you do not feel affection toward. Which I think takes care in connection a little another step affections, a bigger word right in my book and I said, Huh? And he goes, Yeah, that ' s that ' s just common. If you cannot find a way to feel affection for people you lead, then we need to move them out of your direct report line. You know, as you as you walk through some of these stories and examples, what seems to set apart the contexts where even tough, masculine leaders appreciate the importance of vulnerability and affection, is there doing work with with heavy emotional demands? Yes, whereas I. I mean, I can I ' m sure you know, and I also know some some very successful people in knowledge work who have managed to pull off a, you know, a pretty extraordinary career without ever really having to to take off the **mask** and show what they ' re really feeling. And so is is there something also built into their work in addition to the fact that you get to prove your strength, that you can ' t you can ' t hide from the emotions altogether? Yeah, I think the thing that I would ask I would ask the great contributions that they made without taking off the **mask**, how much greater could they have been? What changes could they have made had they had that combination of intellect plus humanity and, you know, reading think again was a catalyst for me, re-examining my. Bad assumption that we ' re all hungry for brave leadership. Thank you, Renee. I ' m honored that you read my book, Think Again, let alone that it struck a chord. It kind of grabbed me by the shoulders and gave him gave me a little shake, like a big shake, actually. I don ’ t think we can make the mistake of assuming that everyone wants brave leadership. Yes, because brave leadership puts demands on people to also be courageous, to also be self-aware, to also put more importance on getting things right than being right. I don ’ t think brave leadership for the leader or for the people who are being led is for the faint of heart. So I ’ m not so sure that everyone wants courageous leadership. And that ’ s. That ’ s a tough thing, don ’ t you think? Yeah, I agree. I think the first thing that comes to mind is I read some research earlier this week showing that when leaders have employees who are courageous and daring, that sometimes they feel like that gives them a pass and they say, OK, you know, my people have got this. And so I don ’ t have to demonstrate the same integrity, which, of course, is the exact opposite of what you would want to see. And it goes to something I ’ ve been wondering about that your work has really pushed me to think a lot about, which is, you know, if if if I ’ m going to play who who believes in the value of courage and vulnerability. But I have a boss or a leader who doesn ’ t get it. Well, how do I manage up? I mean, obviously, in an ideal world, I would just leave. But if I don ’ t have that option, is there anything I can do to nudge a leader to take off the armor, to not be so ashamed to to be a little bit more real? So one thing is I never tell people to leave because I mean, leave if you can. But I just you know, I think you and I are both careful about that in our in our work. Right. Because the reality is you may have a sick kid and this is your insurance, you know, like or this is how you know, definitely when I tell people to leave in this job market right now. So we try to give people real tools. No, you can ’ t manage by teaching your boss this work and asking her to take off the armor. I mean, it just doesn ’ t work like that. What I would say is, let ’ s say you ’ re my boss and you ’ re you ’ re a pretty armored. You ’ re not your you don ’ t do vulnerability. And we just had a big disappointment setback in our team. And, you know, your style is to just move forward. And so what I might say, instead of saying, you know, look, I think we need to be vulnerable and take the armor off and, you know, really embrace this failure that ’ s going to get you fired or in trouble. So what I would say to this person and depending on how they are, is I would say, hey, Adam. I have a question for you. Do you think it could be helpful for us to dig in around what happened as a group? What I ’ m seeing, as I think people are making up different stories about how we ended up in this setback, and I wonder if it ’ d be helpful if we sat down and just talked through it so we could get on the same page and figure out just as a group what went wrong so we could embed that in our team and not repeat it. So I would just practice the work. Not try to influence people to do the work. Does that make sense? It does. It does. And to me, what ’ s so clever about that approach is it takes the leader out of the spotlight a little bit. Right. So I ’ m not attacking or criticizing my boss. I ’ m actually saying, hey, you know what a lot of people are? I love your phrase. You know, the story I ’ m making up is and here you ’ re saying a lot of people are making up different stories. And so, you know, maybe we need to actually get the either set the record straight or learn from something from this experience. And so you ’ re almost inviting the leader to be a problem solver as opposed to. Yes. Calling them the problem. That ’ s right. And so it is. We have this list of kind of we you know, we call hard conversations, rumbles. And in fact, I have one schedule today and it literally says in my calendar, ad sales rumball four o ’ clock. And so what I know what that means in our organization, our culture is we ’ re expecting a lot of different opinions. We ’ re expecting a lot of competing priorities. Bring up a point of view, bring a lot of curiosity, be prepared to listen more than you talk and be prepared from for some discomfort, because we ’ re going to stay in this until we understand each other better. And so it ’ s just an attention set up for us. We

’ re going to rumble. And so we have these rumble starters and language that we give people who go through our training where. If I again, if I ’ m talking to you and we have different takes on, you know, the perks of vulnerability, I ’ m just practicing my work with you. So I ’ m saying, you know, I hear you, Adam. That ’ s not my experience of how OP showed up in that meeting. Can you walk me through what you saw that that is leading you to this belief? There ’ s just language tools that we can use. Yeah, and, you know, I have to ask, because this term rumbles from is is such a great reframe of these hard conversations that many of us avoid. And there ’ s a part of me that thinks, OK, you ’ re you ’ re activating a new mindset. It ’ s we ’ re going into WWE. Let ’ s get ready to rumble. And I can ’ t wait this this could be fun, but there ’ s a part of me that also thinks, oh, do you do you actually do you ruminate even more going into them thinking, oh, God, I ’ ve got to rumble. This is going to be horrible. Does the framing help or hurt? It depends. I think it depends on the culture. Right. And it ’ s funny. And this is the age difference maybe because I ’ ve got I think I ’ ve probably got a decade on you. And so when I think of rumble, I think of West Side Story, I think musicals, you know. And so to me, that ’ s where it came from. It came from this idea of a rumble in West Side Story was part part argument, part dance, because, you know, as a musical. So in our culture, where my number one job is that as the kind of CEO of this organization that I lead is to normalize discomfort, we onboard for discomfort. We teach you how to fail and get back up when you are brand new and weak one like we normalize discomfort because I just can ’ t have people working for me who are only doing what they ’ re already good at doing. I just that ’ s not that ’ s not effective. Like, that ’ s not innovative. So for me, it just means like, I will definitely have people who are two levels, you know, apart from me, who, you know, their bosses boss report to me and they will say to me, hey, Brian, can I rumble with you on the social media strategy? I am not I ’ m not down with it yet. I don ’ t get it. And Michael Rumble what ’ s what ’ s what do you think? And as you and I both know, one way to really measure embedded change in in cultures, which is a hard thing to do. Right, is language like language is a big indicator. And so I think you can ’ t just start using the word rumble if it ’ s not paired with a deeper culture change. I think that makes a lot of sense, and it it leaves me wondering a little bit if if actually wondering a lot. I ’ m just going to wonder, no matter how much I ’ m wondering about it, I I love measuring culture change by looking at the vocabulary people use. I guess I guess maybe this is a little bit of a chicken in the egg. But is it possible that the vocabulary you ’ re introducing is is creating the change, not just reflecting it? Yeah, I think language is praksis. It is. It is the theory of change. It is the practice of change all wrapped up into one thing. And you know, the greatest compliment that I get, people would come up after a talk and say, I already knew everything you said. I had no words for it and I thought it was just me. And so that ’ s that ’ s the goal of the work to give people language, language is power. And and here ’ s the myth. Here ’ s what I tell people. We think. That giving language to hard emotions like shame, our grief are our hard experiences, gives those experiences or those emotions power, but giving language to hard things gives us power. Yeah, that ’ s you know, that ’ s so true when when we can name an emotion, it at least allows us to analyze it or to reflect on it and to control it a little bit, because we realize we have choices about what we call it. And when I do something wrong, as you pointed out so many times, I have a choice about whether I ’ m going to interpret what I ’ m feeling as shame. I ’ m a bad person or guilt. I did a bad thing. That ’ s right, you wanna hear something crazy? Yes, yes, yeah, let me tell you something crazy. So we have communities of facilitators and when we look at evaluation ’ s from that that kind of deeply therapeutic work where people will spend a facilitator will spend 12 weeks taking people through, you know, daring, greatly rising, strong curriculum in the end. And the evaluation, what we see over and over again, understanding the difference between shame and guilt was the most important part of this work. Just understanding that there ' s a difference between shame I am bad and guilt, I did something bad and that how we talk to ourselves or how we use shame or guilt to parent or lead has profoundly different outcomes. Right. Just understanding that I can look at my child and say, you know what? You ' re a wonderful kid. That was a really stupid choice and how different that is from your stupid kid. Yeah. Which changes who people are, you know. Welcome back to Taken for Granted and my conversation about vulnerability with Bernie Brown, one of the fears that a lot of people carry around is people are worried that if they ' re vulnerable at the wrong time or with the wrong person, especially if they ' re in a more performance oriented culture at work, that they might not be seen as competent. I ' d love to hear your latest thinking on, especially in a virtual world. How do I figure out what appropriate vulnerability is and how do I know whether it ' s safe to be vulnerable? So I would say you don ' t you ' ll never succeed in a performative culture if you don ' t have some of the things that really are vulnerable, like curiosity. If you pretend like you know everything in a performative culture and you ' re not a learner, that house of cards is going to collapse at some point. So what I think people are asking is how much is too much to share about my feelings? And that always leads me to this very simple sentence, vulnerability minus boundaries is not vulnerability. Are you sharing your your emotions, your experiences to move? Work connection, our relationship forward, or are you working your shit out with somebody and work is not a place to do that. So I ' ll give you an example that I think is. You know, in fact, what I tell the story, I tell the story a lot when I ' m talking, but I think about you sometimes, Adam, when I tell the story, because of things that I have read that you ' ve written. No, no, it ' s good, I think. But it ' s this is a nuanced life is nuance. Right. And so I was working with a group of newly funded CEOs from Silicon Valley. And after my talk, one of them came up to me and said, I ' m going to be vulnerable. I ' m going to tell my investors, I ' m going to tell my employees, look, we ' re in over our head. I don ' t know what I ' m doing. And we ' re bleeding money. Right, and he said, I ' m just going to be vulnerable. And I said. You must have stepped out to go to the bathroom and get a coffee during the part where I said vulnerability minus boundaries is not vulnerability. We always have to interrogate our intention around sharing and we have to question who we ' re sharing with and is that the right thing? And he ' s saying, so what do you think is going to happen? You don ' t think I should do it? I said, I think you ' ll never get funding again. And I think you will unfairly put the people who ' ve probably left great jobs to follow you over here into a terrible position of fear. And he said, so I don ' t understand. And I said, if you are literally in over your head, you don ' t know what to do next and you ' re bleeding money. You should absolutely share that with someone, but the question is, who is the appropriate person to tell? I so appreciate the nuance that you bring to this idea of vulnerability, to say, look, you know, just just because vulnerability helps to build trust doesn ' t mean you should share everything in all situations with other people now. And I think that ' s such a common misconception about the idea. I guess it comes up a lot in discussions about authenticity to you that people think that that, you know, OK, I ' m trying to be authentic and that means I don ' t need to have a **filter** or, you know, I can defend my actions by saying I was just being myself like, well, you were just being a jerk. That ' s that ' s not OK. Yes, that that ' s exactly right. Look, I know some of the most vulnerable and authentic leaders I have ever had the pleasure of working with. Truly authentic, truly vulnerable, personally disclose very little. And some of the leaders that I work with disclose everything are the least authentic and vulnerable people I ’ ve been around. OK, so this is totally fascinating. Are you saying that I don ’ t I can. No, no. I actually love this because I have been criticized on this before. And I think you just you just gave me a new way of thinking about this, which is you ’ re saying I can be vulnerable without disclosing a ton about my emotions or my life. Yes. How I guess. I think this is what I ’ ve been trying to do. I ’ ll just give you the back the background in case it ’ s helpful. I have feedback from a bunch of people I work with that when, you know, when there ’ s something difficult going on in my life, I don ’ t care much about it. And my fear has been that when, you know, when people know that there ’ s something difficult going on in my life and then I don ’ t open up about it, they ’ re going to think that I ’ m you know, I ’ m not being honest or authentic with them or I ’ m lacking vulnerability. And you ’ re saying there are ways that I can maintain my, I guess, my degree of privacy that I made up for naturally and still be vulnerable. So I ’ m excited about this. Tell me more. This is what vulnerability can look like. Hey, I ’ m I ’ m really struggling right now. I ’ ve got some stuff going on with my mom and it ’ s hard. And I want a job to know. And I want you to know what support looks like for me is I ’ ll check in with you if I need something from you. And I may take some time off, but I want to know for me, support looks like being able to share this with you and being able to bring it up with you when it ’ s helpful for me, but not having to field a lot of questions for it. So I appreciate being able to tell you. I appreciate that we all will need different things when we have hard things going on in our lives. This is what I need right now that is incredibly empowering because what so let me tell you what people are actually worried about, Adam, when you ’ ve got someone who compartmentalizes in segments and when they know something really difficult is going on, they ’ ve got they ’ ve got a parent in chemotherapy or they ’ ve got, you know, like some something ’ s really hard going on. They ’ re concerned about you for sure. But what they ’ re also concerned about is Bernie does not give me permission to be human. I am not safe here. Unless I to compartmentalize and bring do not bring my whole self here and what you ’ re doing when you say what I just said, which is I understand we all have different things and different needs, and I love being a part of team A team that can respect that. I do trust you and want you to know that these hard things are going on. And for me, support looks like this right now. That ’ s awesome. Does that make sense? It does, it ’ s a different interpretation than I had made, which is, you know, when when I ’ m the first or second time, I was like, oh, OK. You know, this is somebody who who wants, you know, who just expects me to open up more that idea. And we have different preferences. That ’ s OK. And as it happened a few more times, I felt like, OK, maybe what ’ s going on is people are worried that they ’ re not as close to me as they thought they were, because if they were, I would have shared this. And you ’ re saying, yes, maybe that, but also that they ’ re worried that if they want to share something with me, that I ’ m not going to be receptive to it because I don ’ t do it and return. If you ’ re my boss and you ’ re the person that really is very private and doesn ’ t like to disclose and and I know something ’ s going on. It creates eggshells for me. Because you ’ re not being explicit about what you want or don ’ t want, it ’ s just off limits. Yes, and if you said that to me, I ’ d say, man Adam. I really appreciate you normalizing that we have lives outside of work, and I really appreciate you normalizing that. We all need different things when we ’ re in a hard time. And what I really appreciate is you asking us, you trusting us enough to share that you ’ re having a hard time and ask for what you need, which is not to talk about it unless you bring it up. That safety, that psychological safety. It ’ s such a powerful conversation that I clearly need to have with a bunch of people having to add to my list now, but it also it also makes me think that this it might be helpful to clarify to people what closeness means to me. Yes. You know, when I think about closeness, it ’ s less about disclosing, you know, my deepest emotions or what ’ s going on in my personal life. It ’ s more about, number one, if I made time for you, if that means you matter to me first off. Yes. And number two, and maybe more importantly, I do not I do not see a connection between the frequency of communication and the depth of a friendship or collaboration. And I feel like the people I ’ m closest to, I sometimes go a year or more without talking to them and we can just pick right up where we left off. And I know that if I ever needed them, they would be there for me and vice versa. And I don ’ t think I ’ ve actually let anyone know that, God, it ’ s huge. It ’ s so important. And that that you know what that is? That ’ s authenticity. That ’ s authenticity because let me tell you something, you sitting down with this, folks are concerned because you ’ re not sharing enough and and pushing yourself to share with them is actually not authenticity that ’ s assessing and acclimating to what you think people need. It ’ s not being you. That ’ s why this sentence, what you know, what a support look like for me, what a support look like for you, and that we can build a team, that we can have different needs because we see each other and respect each other. That ’ s authenticity. It ’ s nothing more. Nothing less. So if I can quote you to you, I think I know well, I mean, I ’ m going to do it. You know, you said something really profound about this. You ’ ve said that I think tell me if I ’ m if I misapplying it here. But I think that another way of saying what you just said in your words is that if I were basically just sharing because people expected me to share, then I ’ m fitting in as opposed to belonging. That ’ s right. That ’ s right. That ’ s right. And and there is room for all of us, like I have pushed and pushed and pushed people to be more like at my level of sharing, which is kind of somewhere in the middle, like I have my own personal line I ’ ll share with vulnerable, but I never share with sentiment. I ’ m really clear on all that. I don ’ t my kids are not in my posts. I ’ m not you know, I just I ’ m a public person. But that doesn ’ t mean that I forfeited everything in my life. And so what true belonging is, if you ’ ve got a team of seven people. That have seven different ways of showing up and different levels of comfort with sharing and that all of them feel like they have a space there, because what ’ s not honored is a way of being was honored is actually authenticity. You know, and that ’ s what we ’ re looking for, but let me tell you, that ’ s a shit ton of work and it ’ s more work than saying authenticity is crying, three point five, which I have really had leaders come up to me and literally say, how many times do I have to cry in front of them to, like, really be considered vulnerable? And I said, yeah, like, you don ’ t get it. There ’ s no hack here. There ’ s no hard wiring here. It ’ s about understanding and seeing. People love that. Well, thank you. I ’ ve learned so much from your work. It ’ s been you know, it ’ s been eye opening. It ’ s been at **various** points uplifting and also a little bit arresting in the best ways. And it ’ s been just a real treat and delight to have you on work life. God, thank you so much. I have to say, this is just one of my favorite conversations ever. So thank you for inviting me. If you ’ re still hungry for more from Bernie and me, we just had another conversation for her podcast, Dare to Lead. It ’ s on Spotify, Renee, with Adam Grant on the power of knowing what you don ’ t know. Taken for granted is a member of the TED Audio Collective. The show is hosted by me, Adam Grant, and produced by TED with Transmitter Media. Our team includes Colin Helmes Credico and Dan O ’ Donnell, Joanne DeLuna, Grace Rubenstein, Michelle Quint and Ben Chang. And Add a few in this episode was produced by Constanza Gado and Jessica Glaser. Our show is mixed by Rick Original Music by Hartsdale Sue and Alison Layton Brown.

12 NEG Dim 6

Text Sample 905: NEG Dim 6 – 2021 – Score 1.00 – 2021/t004082.txt
If you have credit card debt, you are far from alone. More than half of Americans are currently carrying around some form of credit card debt. And if you ’ re part of that majority or just want to put some preventative measures in place, I have four simple changes that can help you work towards becoming debt free. [Your Money and Your Mind with Wendy De La Rosa] Number one : prioritize your payments and prioritize them in the right way. So let ’ s say you have two cards, one with a 200 dollar balance and the other with a 1, 000 dollar balance. Which one do you pay off first? Well, if you ’ re like most people, when people have debt across multiple credit cards, they tend to pay off the one with a lower balance first because it feels like an easy win. It feels good. We call this the”completion bias.”When you give in to completion bias, you feel more productive because you ’ re able to check off quick tasks from your to-do list. And you might even get a rush of dopamine while you ’ re doing it. But rather than looking at the balance – 200 dollars versus 1, 000 dollars – what you should be looking at is the interest rate instead. If you ’ re busy paying off a credit card with a low interest, while the larger-interest-rate card lingers on and on, you could be in debt for much, much longer. You can still work the completion bias to your advantage. Here ’ s a quick tip : think of that 1, 000 dollar debt as a series of smaller, achievable tasks. For instance, you could decide to pay 100 dollars towards that credit card balance each month. And each time you pay that 100 dollars, you ’ re crossing something off your to-do list and achieving that completion high. You ’ ll not only get that completion high but you ’ ll have something to celebrate. Number two : call your credit card company and ask them to lower your interest rate. Get somebody on the phone. Typically, when you apply for a credit card, it ’ s either when you ’ re young or you lack liquidity. And in both instances, your credit score is lower. Your credit score is what lenders and banks and credit card companies use to determine how much of a risk it would be to lend you money. Now, the higher the credit score, the lower the risk, and the lower the interest rate. Conversely, the lower the credit score, the higher your interest rate, and the more money you ’ ll have to pay in interest in the long run. Over time, with on-time payments, your credit score improves. But guess what? Sometimes credit card companies don ’ t lower your interest rate as a result. But what we do know is that if you call, oftentimes, credit card companies will loosen up on their interest rates, given your credit score improvement. We only have a small sample, but in my research, we invited people to come in, call their credit card company and make the case for a lower interest rate. Fifty percent of them were successful. This is a simple way of having a fifty-fifty shot to save yourself thousands of dollars on interest. Now, while you have them on the phone, here ’ s tip number three : request a change in your payment date. The payment due date is usually a random day that the credit card companies set that doesn ’ t align with your cash flow. Now, I want you to think about when you typically get paid and what time of the month you usually have the most cash on hand. Based on that, pick a payment date that works best for you. I ’ ve seen that it ’ s easier for people to pay down their debt when they have more money in their bank account. So

if you have credit card debt, choose a new payment date that falls after your usual payday. So if you typically get paid on the 15th, the 16th might be a great day. Here ' s my last tip. Number four : fundamentally change how often you pay your credit card. So for example, if you usually pay 100 dollars a month towards your credit card debt, there are 12 months in a year, so in a year, you ' ll pay 1, 200 dollars. Good for you! But if you decide to spread that 100 dollar a month payment into weekly payments of 25 dollars, you ' ll end up paying more in the long run, because there are 52 weeks in a year. That means you ' ll end up paying 1, 300 dollars instead of 1, 200 dollars. You will also end up paying less on interest. It ' s a natural calendar hack that you can work to your benefit. And there are even companies that can help you do this automatically. These changes may seem small and subtle, but you ' ll be pleasantly surprised by the huge difference they ' ll make, whether you ' re overcoming debt or just trying to avoid it all together. I ' ll see you next episode with some more tips. Your future self will thank you.

Text Sample 906: NEG Dim 6 – 2021 – Score 1.00 – 2021/t004069.txt

Christiana Figueres : Today, February 19, 2021, at the beginning of a crucial year and a crucial decade for confronting the climate crisis, the United States rejoins the Paris Climate Agreement after four years of absence. Unanimously adopted by 195 nations, the Paris Agreement came into force in 2016, establishing targets and mechanisms to lead the global economy to a zero-emissions future. It was one of the most extraordinary examples of multilateralism ever, and one which I had the privilege to coordinate. One year later, the United States withdrew. The Biden-Harris administration is now bringing the United States back and has expressed strong commitment to responsible climate action. The two men you are about to see both played essential roles in birthing the Paris Agreement in 2015. Former Vice President Al Gore, a lifelong climate expert, made key contributions to the diplomatic process. John Kerry was the US Secretary of State and head of the US delegation. With his granddaughter sitting on his lap, he signed the Paris Agreement on behalf of the United States. He is now the US Special Envoy for Climate. TED Countdown has invited Al Gore to interview John Kerry as he begins his new role. Over to both of them. Al Gore : Well, thank you, Christiana, and John Kerry, thank you so much for doing this interview. I have to say on a personal basis, I was just absolutely thrilled when President Biden, then president-elect, announced you were going to be taking on this incredibly important role. And thank you for doing it. Let me just start by welcoming you to TED Countdown and asking you, how are you feeling as you step back into the middle of this issue that has been close to your heart for so long? John Kerry : Well, I feel safer being here with you. I honestly, I feel very energized, very focused. I think it ' s a privilege to be able to take on this task. And as you know better than anybody, it ' s going to take everybody coming together. There ' s going to have to be a massive movement of people to do what we have to do. So I feel privileged to be part of it, and I ' m honored to be here with you on this important day. AG : Well, it ' s been a privilege to be able to work with a dear friend for so long on this crisis. And, of course, on this historic day, when the United States now formally and legally rejoins the Paris Agreement, we have to acknowledge that the world is lagging behind the pace of change needed to successfully confront the climate crisis, because even if all countries kept the commitments made under the Paris Agreement – and I watched you sign it, you had your grandchild with you – I was there at the U. N, that was an inspiring moment, you signed on behalf of the United States, but even if all of those pledges were kept they ’ re not strong enough to keep the global temperature increase well below two degrees or below 1. 5 degrees, and emissions are still rising. So what needs to happen here in the US and globally in order to accelerate the pace of change? JK : Well, Al, you ’ re absolutely correct. It ’ s a very significant day, a day that never had to happen, America returning to this agreement. It is so sad that our previous president, without any scientific basis, without any legitimate economic rationale, decided to pull America out. And it hurt us and it hurt the world. Now we have an opportunity to try to make that up. And I approach that job with a lot of humility for the agony of the last four years of not moving faster. But we have to simply up our ambition on a global basis. United States is 15 percent of all the emissions. China is 30 percent. EU is somewhere around 14, 11, depends who you talk to. And India is about seven. So you add all those together, just four entities, and you ’ ve got well over 60 percent of all the emissions in the world. And yet none of those nations are at this moment doing enough to be able to get done what has to be done, let alone many others, at lower levels of emission. It ’ s going to take all of us. Even if tomorrow China went to zero, or the United States went to zero, you know full well, Al, we ’ re still not going to get there. We all have to be reducing greenhouse gas emissions. We have to do it much more rapidly. So the meeting in Glasgow rises in its importance. You and I, we ’ ve been to these meetings since way back in the beginning of the ’ 90s with Rio and even before, some of them parliamentary meetings. And we ’ re at this most critical moment where we have a capacity to define the decade of the ’ 20s, which will really make or break us in our ability to get to a 2050 net zero carbon economy. And so we all have to raise our ambition. That means coal has got to phase down faster. It means we ’ ve got to deploy renewables, all forms of alternative, renewable, sustainable energy. We ’ ve got to push the curve of discovery intensely. Whether we get to hydrogen economy or battery storage or any number of technologies, we are going to have to have an all-of-the-above approach to getting where we need to go to meet the target in this next 10 years. And I think Glasgow has to not only have countries come and raise ambition, but those countries are going to have to define in real terms, what their road map is for the next 10 years, then the next 30 years, so that we ’ re really talking a reality that we ’ ve never been able to completely assemble at any of these meetings thus far. AG : Well, hearing you talk, John, just highlights how painful it ’ s been for the US to be absent from the international effort for the last four years, and again, it makes me so happy President Biden has brought us back into the Paris Agreement. After this four year hiatus, how are you personally, as our Climate Envoy, planning to approach re-entry into the conversation? I know you ’ ve already started it, but is there anything tricky about that? Or I guess everything is tricky about it, but how are you planning to do it? JK : Well, I ’ m planning, first of all, to do it with humility, because I think it ’ s not appropriate for the United States to leap back in and start telling everybody what has to happen. We have to listen. We have to work very, very closely with other countries, many of whom have been carrying the load for the last four years in the absence of the United States. I don ’ t think we come in, Al, I want to emphasize this – I don ’ t believe we come to the table with our heads hanging down on behalf of many of our own efforts, because, as you know, President Obama worked very hard and we all did, together with you and others, to get the Paris Agreement. And we also have 38 states in America that have passed renewable portfolio laws. And during the four years of Trump being out, the governors of those 38 states, Republican and Democrat alike, continued to push forward and we ’ re still in movement. And more than a thousand mayors, the mayors of our biggest cities in America, all have forged ahead. So it ’ s not a totally, abjectly miserable story by the United States. I think we can come back and earn our credibility by stepping up in the next month or two with a strong national determined contribution. We ’ re going to have a summit on April, 22. That summit will bring together the major emitting nations of the world again. And because, as you recall in Paris, a number of nations felt left out of the conversation. The island states, some of the poorer nations, Bangladesh, others. And so we ’ re going to bring those stakeholders to the table, as well as the big emitters and developed countries, so that they can be heard from the get-go. And as we head on into Glasgow, hopefully we ’ ll be building a bigger momentum and we ’ ll have a larger consensus. And that ’ s our goal – have the summit, raise ambition, announce our national determined contribution, begin to break ground on entirely new initiatives, build towards the biodiversity convention in China, even though we ’ re not a party, we want to be helpful, and then go into the G7, the G20, the UNGA, the meeting of the United Nations in the fall, reconvene and reenergize, going for the last six weeks into Glasgow. In my judgment, Glasgow, and you ’ d know this full well, I think Glasgow is the last, best hope we have for our nations to really set us on that path. And so, you know, one key is, as I said, raising ambition. The other is defining how you ’ re going to get there, and then the third is finance. We ’ ve got to bring an unprecedented global finance plan to the table. And I think we ’ re already working with private sector entities. I believe there ’ s a way to do that in a very exciting way. AG : Well, that ’ s encouraging, and I ’ m going to come back to that in just a moment. But I ’ m glad you made those points about state and local governments actually moving forward during the last four years. A lot of US private companies have as well. And already I ’ m extremely encouraged by the suite of executive actions that President Biden has already taken during his first weeks in office. And there ’ s more to come. There ’ s also a push for legislative action to invest in the fantastic new opportunities in clean energy, electric vehicles and more. Yet you and I have both seen the difficulties of this approach in the past. How can we use all of this activity to well and truly convince the world that America is genuinely back to being part of the solution? I know we are. You know we are, but we ’ ve got to really restore that confidence. I think your appointment went a long way to doing that. But what else can we do to gain back the world ’ s confidence? JK : Well, we have to be honest and forthright and direct about the things that we ’ re prepared to do. And they have to be things we ’ re really going to do. We just held a meeting a few days ago with all of the domestic entities that President Biden has ordered to come to the table and be part of this effort. This is an all-of-government effort now. So we will have the Energy Department, the Homeland Security Department, the Defense Department, the Treasury. I mean, Janet Yellen was there talking about how she ’ s going to work and we ’ re going to work together to try to mobilize some of the finance. So I think, you know, we ’ re not going to convince anybody by just saying it. Nor should we. We have to do it. And I think the actions that we put together shortly after President Biden achieves the COVID legislation here, he will almost immediately introduce the rebuild effort, the infrastructure components, and those will be very much engaged in building out America ’ s grid capacity, doing things that we should have done years ago to facilitate the transmission of electricity from one part of the country to another, whether it ’ s renewable or otherwise. We just don ’ t have that ability now. We have a queue of backed up projects sitting in one of our regulatory agencies which have got to be broken free. And by creating this all-of-government effort, Al, our hope is we ’ re really going to be able to do that. The other thing that we ’ re doing is I ’ m reaching out, very rapidly, to colleagues all around the world. We ’ ve had meetings already, discussions with India, with Latin American countries, with

European countries, with the European Commission and others. And we ’ re going to try to build as much energy and momentum as possible towards these various benchmarks that I ’ ve talked about. And I mean, the proof will be in the pudding. We ’ re going to have to show people that we ’ ve got a strong NDC, we ’ re actually implementing, we ’ re passing legislation, and we ’ re moving forward in a collegiate manner with other countries around the world. For instance, I ’ ve talked to Australia, we had a very good conversation. Australia has had some differences with us. We ’ ve not been able to get on the same page completely. That was one of the problems in Madrid, as you recall, together with Brazil. Well, I ’ ve reached out to Brazil already, we ’ re starting to work on that. My hope is that we can build some new coalitions and approach this, hopefully in a new way. AG : Well, that ’ s exciting, and I do agree with your statement earlier that the COP26 conference in Glasgow this fall may be the world ’ s last, best chance, I like your phrase there. From your perspective, what would you list as the priorities for ensuring that this Glasgow conference is a success? JK : I think that perhaps one of the single most important things, which is why we ’ re focused on this summit of ours, is to get the 17 nations, that produced the vast majority of emissions, on the same page of committing to 2050 net zero, committing to this decade, having a road map that is going to lay down how they are going to accelerate the reduction of emissions in a way that keeps 1. 5 degrees as a floor alive and also in a way that guarantees that we are seeing the road map to get to net zero. I will personally be dissatisfied, disappointed if for our children ’ s sake and our grandkids sake we can ’ t say that when these adults came together to make this kind of a decision, we didn ’ t actually make it. We ’ ve got to make it. And I think if we can show people we ’ re actually on the road, I think you believe this as much as I do, that – I mean, you ’ re far more knowledgeable than I am about some of the technologies and you ’ ve helped break ground on some of them. The pace at which we are now beginning to accelerate, I mean, the reduction in cost of solar, the movement in storage and other kinds of things, I ’ m convinced we ’ re going to find one breakthrough or another. I don ’ t know what it ’ s going to be, but I do know that when we push the curve and we put the resources to work, the innovative creative capacity of humankind is such that we have an ability to surprise ourselves. We ’ ve always done it. When we went to the Moon in this incredible backdrop behind you. And that ’ s exactly what we did. And people today use products in everyday household use that came out of that quest that you never would have anticipated. That ’ s what ’ s going to happen now. We can move faster to electric vehicles. No question in my mind, we could absolutely phase down coal-fired emissions faster than we are in a plan to do it. So the available choices are there. The test is going to be whether we create the energy and momentum necessary to actually get those choices made. AG : One of the big challenges is one you referred to earlier on finance. Wealthy countries have promised financial assistance to the less wealthy countries to help them out with cutting emissions and to help them cope with the impacts of the climate crisis. But of course, we need to continue to work to meet this commitment, especially as countries around the world rebuild their economies in the wake of this pandemic. What are some of the most effective ways in which the wealthier countries can help those that don ’ t have as many resources, and why is this so important for the world to move forward? JK : Let me answer the last part first. It ’ s so important because it ’ s the only way we ’ re going to get there. I don ’ t believe that any government has either the money or the inclination to be able to do what ’ s necessary here. I believe the private sector, particularly driven by venture capital investment, by the quest to be able to create a product that then can help create wealth and actually provide a benefit to humankind drives a lot of things that we ’ ve done all through history. And I don ’ t think it ’ ll be any different now. I think the question is, can we pull together enough nations to leverage a uniform approach to the judgment about the kinds of investments that are being made. And I believe that if we can standardize to some degree, with disclosure requirements, which Janet Yellen is now seized of that issue, and Europe, there are folks working on that and European Commission elsewhere, if we could actually find a way to come together and harmonize some of those definitions and the marketplace begins to make those judgments as they qualify risk, looking way out, risk, because of climate crisis for investing is very, very real. And we all understand that. We spent 265 billion dollars in America two years ago just cleaning up after three storms, Maria, Harvey and Irma. And it ’ s crazy. You spend 265 billion to clean up after the storms, but we can ’ t put 100 billion together for the Green Climate Fund. That ’ s what this year has to be about. We ’ ve got to break that cycle. And I think business, I ’ m convinced of this, a lot of people will doubt me and say, have I lost my mind, but I ’ m convinced the private sector is going to be critical, if not the key to helping to make this happen. And that will leverage other money. I ’ ve talked to the IMF, we ’ ll be talking with the World Bank, we ’ re going to try to bring our own Finance Development Corporation in America. All of these things can help leverage investment into the sectors that can make the greatest difference to the rapid reduction of greenhouse gases. And I think people are going to get very excited about where this money is going to go and how much it is going to be. And my hope is in a matter of weeks to be in a position to make a couple of announcements with respect to that that could be helpful in building some of this momentum. AG : Well, that ’ s great. It sounds like some major news coming in a couple of weeks and just one example you used, the point about businessmen. I have a friend in Australia, Mike Cannon-Brookes, building a long undersea cable from the northern territories of Australia to take renewable electricity to Singapore. You have made the point about the need for the US to approach this with humility a number of times. In that spirit, what lessons can a country like ours learn from some of the lower income nations that are already beginning to tackle climate change? JK : Well, I think one of the most important things, Al, is to make sure that central to this transformation, to this transition to the new energy economy, central to it is environmental justice, is that we don ’ t leave people behind, that we ’ re not making whole communities the recipients of the downside of some particular choice, that the diesel trucks, for instance, aren ’ t all being routed through a particular low income community that doesn ’ t have the ability to make a different political decision. I think it is vital for the developed world to recognize that there are nations, 138 nations or more, way below one percent in terms of emissions. And they ’ re looking around some of them, like Tommy Remengesau, the president of Palau, who no longer can consider adaptation, he ’ s got to figure out where his people are going to go live, as do other very low-lying areas in the ocean. So that impact on people is really not known by the vast majority of people who live pretty good lives in a lot of countries in the world. And we have a responsibility to make sure that we ’ re learning the lesson of their lives and of their hopes and aspirations here. AG : Couldn ’ t agree more. And here in the US, if we had paid more attention to the differential impact on Black, Brown and Indigenous communities, we would have had a better early warning of what the whole country was facing. But let me shift subjects and ask you about China. I know that you, as you are close friends with Xie Zhenhua, as I have been over the years, and I was very happy when he was brought out of retirement to play the lead role for them. But the US is now in the middle of a somewhat contentious relationship with China. But successfully solving the climate crisis is going to require collaboration between the US and China, we ’ re the two biggest emitters and the two biggest economies. How can this collaboration be shaped, in your view? I know you played a role, as Joe Biden did before the Paris Agreement, in getting our two countries together. Can we do that again? JK : I hope so. I really do hope so, Al. As you just said, if we can – if we don ’ t get China to be cooperating and partnering with the rest of the world on this, we don ’ t solve the problem. And we unfortunately, we see too much investment in China right now in coal still. We ’ ve had some conversations about it. I was on a panel with Xie Zhenhua several months before the election by the University of California, and we had a very constructive conversation. My hope is that that will continue and can continue and that China will be just as constructive, if not more so, in this endeavor than they were in 2013 as we began the process to build up to Paris. AG : Well, that relationship is absolutely crucial. But in order to cover all the ground I want to cover here, let me shift again and ask you, what role do you expect that big corporations and also smaller businesses will play in moving this green transition forward? JK : I think they ’ re the biggest single players in it. I mean, governments are important and governments can and have made a difference with tax credits. For instance, our solar tax credit made an enormous difference and it will make one going forward. And even in the middle of COVID, we ’ ve been able to hold on to that. But we need to grow those kinds of efforts. But in the end, it ’ s not going to be government cash that makes this happen. It ’ s going to be the private sector investment that is coming in because it ’ s the right thing to do, because it ’ s also smart investing. And the truth is, you can talk to many – and you have, you ’ re one of the investors actually, Al – you and others have proven that you can invest in this sector of dealing with climate or environment or sustainability, whether it ’ s ESG or it ’ s pure climate. There are ways to have a good return on money. And during the last couple of years, we had something like, 13 to 17 trillion dollars sitting in parked banking situations around the world in net negative interest. In other words, they were paying for the privilege of sitting there, not invested in something. And so I think there ’ s just a massive opportunity here. And most of the CEOs I am talking to, at least now, are increasingly aware of the potential of these alternatives. And you were in early, I don ’ t know if you invested in it or not, but I know you ’ re involved with Tesla or have been. Tesla is the most highly valued automobile company in the world. And it only makes one thing : electric car. If that isn ’ t a message to people, I don ’ t know what is. AG : I wish I had invested in Tesla, John, but I ’ m a huge fan of Elon Musk and what he ’ s doing. I ’ m also a huge fan of Greta Thunberg. And I ’ m just curious what you think in practical terms is the real impact for change coming from these youth movements like Fridays for the Future? JK : I think it ’ s been gigantic and spectacular and in the best traditions of what young people do and have done historically. I mean, as you recall, in America, at least in the 1960s, it was young people who drove the environment movement, the peace movement, the women ’ s movement, the civil rights movement, and they were willing to put their lives on the line. And Greta has been just unbelievable. And in the way in which she has held adults accountable and it has created this wonderful movement. I ’ ve met so many young people, many of whom have worked in one fashion or another with me in the last few years, who were brought to it from Fridays for the Future, from the Sunrise Movement, or, you know, it ’ s all that focused youthful idealism and energy and it demands to be heard. And I think all of us, I mean, we should be ashamed of ourselves that we have to have people who were then 16 or 15 not going to school to get our attention. I mean, what the hell is the matter with adult leadership? That ’ s not leadership at all. So I salute her and all the young people who put themselves on the line. But I invite them, you know, it ' s not enough. You ' ve got to then – and I said this during the course of the election where I hope we created a lot of new voters. And I think environment, specifically climate crisis, became a real voting issue this year, just as it was back in 1970 when we created the EPA and the Clean Air Act and a host of things. And it proves that that kind of activism is necessary. And I hope we ' re going to keep young people at the table here and finish the job, that ' s the key now. AG : Yeah, I couldn ' t agree more. And another big movement that ' s having an impact is the environmental justice movement. You referred to it earlier. And I ' m so glad that President Biden is putting environmental justice at the heart of his climate agenda. It might be good if I could ask you to just take a moment and tell people why that is such an important part of this issue. JK : Well, I think it ' s important part of this issue for many reasons, the most basic is just moral, you know, what is morally right. And how do you redress a wrong that has for too many years held people back, killed people by virtue of disease or other things, and resulted in a basic inequality and unfairness in society. And I think you share a feeling, as I do, Al, that the fabric of a nation is built around certain organizational principles. And if you ' re holding yourselves out as a nation to be one thing, i. e. equal opportunity and fairness and all people created equal, and equal rights and so forth, if that ' s what you hold out there and it isn ' t there, eventually you get such a cynicism and such a backlash built up into your society that it doesn ' t hold together. To some degree, that is what we ' re seeing around the world today, is this nationalistic populism that is driven by this heightened inequality that has come through globalization that has mostly enriched already fairly well-off folks. And so if it ' s the upper one percent that ' s getting all the benefits and the rest of the world struggling to survive and they also have COVID, and then you tell them we ' ve got to do this or that in terms of climate, you ' re walking on very thin ice in terms of that sacred relationship between government and the people who are governed. It ' s not just an American phenomenon. You see it in Europe. You see it in alternative movements in various countries. And I think it is the great task of our generation not only to deal with climate, but to restore a sense of fairness to our economies, to our societies, to our world. And that is part of this battle, I think. AG : Yeah, I agree. And another common source of opposition to what governments are doing now has to do with the fear, both in the US and elsewhere, on the part of some, that jobs might be lost in this transition toward a green economy. You and I both know that there are a lot of jobs that can be created. But let me put the question to you. How can we approach this green transition in a way that lifts everyone up? JK : That is one of the most important things that we need to do, Al. And we can ' t lie to people. We can ' t say that some of the dislocation doesn ' t mean that a job that exists today might not be the same job in the future and that that person has to go through a process of getting there. And we need to make certain that nobody ' s abandoned. We need to make certain that there are real mechanisms in place to help folks be able to transition. And I just spoke the other day with Richie Trumka, the head of the American Federation of Labor, and he ' s been very focused on this. And we agreed to try to work through how do we integrate that into this transitional process so that we ' re guaranteeing that you don ' t abandon people. Now, one of the things we need to do is go to the places where there have been changes and there will be change. Southeastern Ohio, Kentucky, West Virginia. You know, if the marketplace is making the decision and it ' s – by the way, it ' s not government policy, it is the marketplace that has decided, in America at least, not to be building a new coal-fired plant. So where does that miner, or where does that person who worked in that supply chain go? We have to make sure that the new companies, that the new jobs are actually going into those communities that the coal community, the coal country, as we call it, in America, is actually being immediately and directly and realistically addressed in this to make sure that people are not abandoned and left behind. That is possible. That is doable. Historically, unfortunately, there have been too many words and not enough actual – not enough actually implementation and process. I think that can change. And I ’ m going to do everything possible in my ability to make sure that we do change it. AG : Well, that ’ s great. And another part of the context within which you are taking on this enormous challenge is the COVID pandemic, which has exposed the cost of ignoring pre-existing systemic risks, inequalities and sustainability. And now as we start to come out of this pandemic, how can we avoid sleepwalking back into old habits? JK : Boy, that ’ s probably the toughest of all. I mean, there ’ s a natural proclivity for people sometimes to just choose the easiest way. And clearly, some people already have and will resort to that. I think the key will be in President Biden ’ s proposal for the build back, which will actually fight hard to direct funds to the investments and to the sectors where we want to see a responsible build back. There ’ s another aspect, and I think that can be done, Al, I really feel that. For instance, someone who ’ s making a car today in South Carolina, where BMW has plants, and just to pick one place or Detroit, GM is obviously going to make this shift, they just announced it. The people building the car today are still going to have to put wheels on a car, build the car, put the seats in, do everything else. It ’ s just that instead of an internal combustion engine, they can be quickly trained to be able to put the platform in for the batteries and the engines themselves, etc, that will drive the car, the motors, that ’ s one way of dealing. Others are that there ’ s new work in some ways. We have to lay transmission lines in America. We do not have a grid in the United States, as you know. We have at east coast, west coast. Texas has its own grid, north part of America, but there ’ s a huge hole in the country. You can ’ t send energy efficiently from one place to another. We could lower prices for people and create more jobs in the build-out of all of that kind of new infrastructure. Not to mention the things that you and I, you know, there are going to be things that we can ’ t name today, some negative-emissions technology that ’ s going to grab CO2 out of the atmosphere and do something with it, like in Iceland, where they put it into the rock, mix it with liquid and it turns into stone. I mean, there are all kinds of different things people are exploring. Those are new jobs. AG : I just want to say, since we ’ ve come to the end of our time for this conversation, thank you again for taking on this crucial challenge on behalf of the United States of America and enabling the US to restore its traditional role in trying to bring the world together. And I know that everybody watching and listening to this conversation sends you their best wishes and hopes for all the success possible in this new work, John. Thank you so much for joining TED Countdown, and we wish you the very best. JK : Can I reciprocate for a minute? First of all, I want to thank you for your extraordinary leadership for years, I can remember when you were leading us in the Senate on this, and you ’ ve done so much since. And I am personally delighted to be working with you on this again and look forward to the next months and together with a lot of other folks. Let ’ s get this done.

Text Sample 907: NEG Dim 6 – 2021 – Score 1.00 – 2021/t004053.txt

I ’ ve spent the last couple of years traveling around the world giving talks to big corporations and little bitty start-ups and lots of leadership teams and women ’ s groups, and what I ’ ve been talking to people about, I ’ ve been trying really hard to convince people that we can change the way we work. But every time I do a talk, somebody comes backstage or follows me offstage and says,”You know, I ’ m so inspired by what you say. It ’ s so great, it makes so much sense. But we can ’ t.””We can ’ t because we ’ re regulated.””We can ’ t because our CFO says we can ’ t do it.””We can ’ t because we ’ re in Europe.””We can ’ t because we ’ re a service industry.””We can ’ t because we ’ re a nonprofit.”And then last year came the pandemic. And the pandemic changed everything all over the world. Service people started realizing that they had to suit up and wear masks and take temperatures and wash their hands. We had to start standing six feet apart in lines. We started working from home. We started working virtually. And we started learning all kinds of things because we had to. All that muscle around innovation and flexibility and creativity that we didn ’ t think we had, we had all along. And we now have realized that we can. So what have we learned? I mean, what did we learn right away? First of all, we learned we ’ re not family. The family is the toddler walking around behind you in the Zoom call with the pet. The family is somebody needing their diaper changed. The family is making sure you ’ re taking care of your mom. That ’ s your family. This is your team. And we ’ ve also learned that that separation between family and work has become this balancing act. And that when we used to say,”Well, this is my work home and this is my family home, and those are two completely different things,”for many of us, it ’ s exactly the same thing. You ’ re no longer at home and at work. For many of us, work is at home and the home is – and it ’ s confusing, and it ’ s creating a whole different level of complexity and coordination so that we understand that it ’ s easier actually to work when we can separate the work that we do as a team from the work that we do in our family. Furthermore, in order to be able to do all that, we have to recognize that we ’ re all adults. And here ’ s the deal about adults. Adults have responsibilities, adults have obligations. Adults have things that they have to commit to. And do you know that every single person that works for you, from the shop floor to the executive suite, is a grown-up? But we have been operating as if they aren ’ t. We operate as if only the smart adults are the people who are at the C Suite. And as we move through the organization, everybody sort of gets a little dumbed down and the rules get a lot stricter and we have to have more control. And the truth is, everybody ’ s a grown-up, we can see it now. Everybody has all of these things to figure out and coordinate. And so now we ’ re expecting from people adult behavior. We ’ re now focusing on the results that matter, not the work. And the way we track it now is we don ’ t walk by and see who ’ s working. We pay attention to what people are doing. And I think that that ’ s always been the best metric. And you know what? For the first time in my life, the concept of best practices is out the window. And you know what? We don ’ t care what Google ’ s doing because we ’ re not Google. We don ’ t care what some other company is doing. Nobody ’ s doing it best. We ’ re all figuring it out as we go along and we ’ re figuring it out for our organizations in our teams at this time. So in order for people to deliver the right results, in order for people ’ s hard work to matter, it has to be in the context of what success looks like for your organization. So if we start to think about context, it ’ s really important that we think about how we teach that. If we can teach everybody in the company how to read a profit and loss statement, if we can teach them what the different teams do, and what they ’ re setting out to accomplish, then people within their own small teams, and within themselves, can figure out what excellence looks like for them. And so then we can start operating relatively independently as a whole organization because we ’ re all moving in the same direction, trying to do the same thing. And there ’ s a really critically important part of making that work, and that ’ s communication. And everything about communication has changed. We tend to think that communication is this waterfall from the top to the bottom. The executives would tell somebody and the next level would tell somebody and we ’ d go all the way down to the shop floor and everybody would understand what ’ s going on. Well, it may not have worked that well then, but it certainly doesn ’ t work that well now. So now we have to recognize it ’ s a different heartbeat. What has it been before and what should it be now? How do we make sure that the messages are clear and consistent? Because that ’ s how people operate. That ’ s how those adults who get the freedom and the responsibility to produce great results operate best is when they understand what they need to know in order to make the best decisions. So that communication, that skill around being a great communicator is something that each of us needs to get better at. One of the things we have to do is think about what the right discipline is for that. If you used to communicate to your team by walking by and asking how they ’ re doing or if they had heard something, you ’ re going to have to schedule that now, it ’ s going to have to have discipline. We ’ ve got to check in with the people on the shop floor to make sure they ’ re hearing what they need to hear because it ’ s not going to automatically happen. One of the ideas I have is just jot down at the end of every day a sentence of what worked and what didn ’ t work. And you don ’ t have to look at it for a month. But when you look back, over a month, you want to look for,”Wow, that was surprising. I didn ’ t really think that would be as effective as it is.”Or maybe it would be, like,”We keep trying to have this in-person meeting in Zoom, and it turns out that there ’ s 14 people on the call and only two of them are talking. Maybe it ’ s an email.”So we have to rethink all of the ways, not just the work we ’ re doing, but the ways we ’ re doing it. So now I ’ m starting to hear a lot of nostalgia around the way it used to be. There are things we aren ’ t doing now that don ’ t matter. Maybe we don ’ t need to go back for five levels of approval. Maybe we don ’ t need to go back and do that annual performance review. Maybe we don ’ t need to do a whole bunch of things that were part of the way we do business that just aren ’ t making a difference. You know what? The way we used to do it not only is not the way of the future, but we ’ re discovering so many wonderful things right now. Let ’ s not lose it. We want to create a new organization, new workforce, that ’ s excited about taking all of the things that we ’ ve learned using that muscle, going forward. One of the most important things that we can do is realize the things that we aren ’ t doing now. The stuff that we ’ ve stopped doing and not go back and do it again. What if we don ’ t go back? What if we go forward and rethink the way we work? Thank you.

Text Sample 908: NEG Dim 6 – 2024 – Score 1.00 – 2024/t003254.txt

OK, so my name is Amber Cabral, and I teach people how to be good humans. What that essentially comes down to is I work with a lot of well-recognized brands on something that is a pretty consistent challenge day-to-day. And what that really comes down to is : how do we help people who are different from one another experience a sense of belonging and support in the workplace? Now we ’ ve all had that moment. We ’ ve been at work and we ’ re like,”Ah, I just don ’ t feel like I fit here.”And so I help organizations kind of start to figure out how to make the connections necessary for that to work. Now support sounds easy. Like we think we can just do that, because we do it all the time. We support people we love and care about. We support people that we think we understand, even if we don ’ t know them. But when it comes to people that we, like, don ’ t know at all or we don ’ t have that connection with, well, then we kind of get a little weird about it. And what we really need to do is pay attention to those differences, because there ’ s magic in those differences, there ’ s something in there that we actually have an opportunity to learn and grow from. But we ’ ve kind of been taught to politely ignore when we ’ re different from one another. We don ’ t actually pay attention to it. And here ’ s the thing : differences are inevitable. Look around this room. Think about your travel into this room today. You encountered all kinds of differences. We have different backgrounds, different ideas, different perspectives. We have different identities, we have different abilities, all of it. And it ’ s going to continue to be that way forever. But we still have this moment where we want to pull back when differences happen. And what I want to encourage you to do is instead of pull back, maybe take a moment and notice those differences. And not so that it ’ s awkward or weird. You know we try to do that thing where we see a thing, but we try to act like we don ’ t see the thing. Like, don ’ t do that! But definitely please, let ’ s be aware of the differences, because the magic that ’ s in those will help us move forward. And I ’ m going to give you some techniques based on just my own study and my own learnings that I think will help you be able to bridge some of those connections as we move forward, alright? So the first step... The first step – there are three. Let me warn you because sometimes we need to know that. Alright. The first step is to acknowledge that we all have some privilege. Now that word might have landed on some of you, like, ich. Right? But really. And I want to demonstrate it for you for just a moment. Did you wake up this morning with hot, clean running water? Did you think about it? Did you wonder if your shower was going to be your ideal temperature, or if you ’ d have clean water for your morning coffee or tea? Likely the answer is no. For most of us in this room. But I think we can all agree that access to water is indeed a privilege. In fact, a little more than a quarter of the world ’ s population does not have access to safely managed drinking water at home. So while it might be our norm, it is indeed a privilege. And this is how privilege shows up. We don ’ t see it. It ’ s regular to us. It ’ s a part of who we are. We don ’ t even think about it. But if we take a moment and we actually pay attention, what being aware of our privilege does for us is it helps us to see where we have access. It helps us see where we have ease. And access and ease give us power. Something as simple as being able to speak the language of our communities gives us privilege, and that privilege gives us power. So you may be wondering : how do I figure out my privileges if they ’ re regular? They ’ re everywhere. OK, let me give you a simple question you can ask yourself. What about me or my experience might someone look at and consider to be typical? It could be my upbringing. It could be the language I speak. It could be the way my body works. It could be my style of dress. It could be any of those elements. But it ’ s hidden in the things that we think of as everyday and regular. That ’ s where we ’ re going to find our privilege. So that leads me to the second step. The second step is we need to be willing to recognize the differences in folks and want to get to know them across those differences. Let me learn about your experiences so that I can develop empathy for you. That learning about the experiences part, it ’ s not just so that we can be curious. We love stories, alright? That ’ s great. Keep loving the stories. But also getting to know about folks ’ experience gives us a bit of perspective that we might not otherwise have. And so what we want to think about is, you know, considering the impact that having those kinds of connections can show for us. Even as a simple example, I mentioned we struggle to connect across differences. But think about this : a couple days ago, maybe a month ago, I saw online where Snoop Dogg, yes, the rapper Snoop Dogg, posted on his LinkedIn – which why he ’ s on LinkedIn, I have no idea – That he really valued his relationship with Martha Stewart. Now most of us have seen that relationship, and if we ’ re being honest, we kind of notice that there ’ s some differences between them that make that connection kind of unlikely, right? OK. But you know what? He pointed out specifically,”I learn so much from this friendship. I get so much value out of this relationship.”And it was the willingness of being able to connect across differences that most of us would kind of politely ignore, or would assume were not available to us. So learning about the experiences of others is about hearing the stories. Yes, but it ’ s also about having empathy and connection, and being able to broaden our view. Now I have another example where this happens, and this is the workplace, because I care a lot about how this shows up in the workplace, because we all, you know, at least for now, until the AI gets us, have to go into the workplace. OK, so when we ’ re thinking about the workplace, one of the things that always happens is interviews. I ’ ve had this conversation with a lot of my clients because clients are always thinking about how are we attracting and retaining talent. They ’ re always thinking about specifically Black folks, brown folks, young folks. And we want to make sure that we ’ re bridging those connections. But we know that our workplaces aren ’ t always inviting to those diverse identities. Now what my clients hear is a pipeline problem. But what I hear is an empathy problem. I ’ ll give an example of what that looks like. Candidates go into an interview. One of the questions is,”I ’ d like you to tell me about how you navigate ambiguity.”Most of the candidates tell a story about working in a previous position, having to solve a problem, not having a lot of resources or information, making some educated guesses and ultimately solving said problem amazingly. Alright? Creating a fantastic new product. We ’ ve heard this story. We work with those people. OK, great. But there ’ s one candidate, and when that question hits them, what they do is they tell a story about a 1989 Honda Accord. It ’ s a car that their father gave them before they went away to college. And it ran great. But the gas gauge didn ’ t work. So they talk about – feels real, right? Y ’ all know this person, too. So the gas gauge didn ’ t work. So they had to develop a process to make sure that they were able to track their miles so that they would know when they needed to fill up again. Right? They had to navigate the ambiguity of having no idea what was in their gas tank and make sure that they didn ’ t run out of gas, and they managed to do that.”I track my miles so that I know exactly when I need to fill up, so that I don ’ t have to worry about running out of gas.”How often would we hear a story like that in the workplace and discount that person? How often would we dismiss or not even consider the candidate because they didn ’ t give us a business example, or we didn ’ t hear it as a culture fit. When if we ’ re really honest with ourselves, the candidate with the Honda Accord not only navigated the ambiguity of not having any gas in the car, they also did something in their day-to-day life. It wasn ’ t just for work. This is one of the things I had to figure out how to do to survive. It ’ s innate. It ’ s in my body. Think about if we considered it that way. Think about if we considered getting to know folks ’ stories and having empathy for them as a part of the way that we evaluate where they have an opportunity to thrive and grow, instead of just listening for the things that we always listen for. That leads me to the third step. OK? The third step is extending your privilege to others. So I have another story. I was on a call the other day with a client. She is the head of HR for a pretty big company. It was she and I and one of her colleagues on the call. And as we were having the discussion, about 10 minutes in, she chimes in and says,”Wait, I ’ m going to hop off for a moment. I need to go change my top. I ’ ll be back.”A few minutes pass by. She returns to the call, and she ’ s in a plain black shirt. She ’ s taken her earrings off, makeup ’ s gone. So I ask, like,”Why ’ d you change clothes?”She says,”Oh, I ’ m getting my home appraised today.

So I ’ m actually going to pretend to be the cleaning lady, and my neighbor across the street is going to come over and pretend to be me. She ’ s white.”I ’ m nodding my head because I know what ’ s going on. The colleague on the call says,”Wait, what do you mean? I don ’ t understand.”So my client proceeds to explain about the stories about Black folks ’ homes being appraised for less when there ’ s evidence that the homeowners are Black. The colleague on the call is, like, almost outraged. Like,”What? Is that a real thing? I can ’ t believe this is happening.”Meanwhile, I am feeling grateful to the neighbor for being able to identify her privilege, for being able to empathize with my client, and for being willing to extend the privilege of her race in that situation. The three steps that I just covered with you are what most of us want when we think about support. I want someone to identify their privilege. I want somebody to actually hear my circumstances and experience, and empathize with me. And then I want you to extend your privilege respectfully and impactfully in the ways that you ’ re able to in that relationship. Now I have a little secret. A confession, if you will. There is a word I ’ ve been avoiding in this conversation. So I mentioned at the top of this, you know, dialogue that I work with lots of well-recognized brands. And what I didn ’ t tell you is that the work I do is DEI, or diversity, equity and inclusion. And the word that I ’ ve been avoiding is”allyship.”The three steps that we just talked through is what allyship looks like in action. Now I ’ ll be honest, I avoided it because I just wanted you to hear my talk. I know right now the media is all the rage. DEI is dying, right? And allyship is on its way with it. But I want us to just for a moment consider that maybe allyship isn ’ t grand historical gestures or another big program, or huge financial commitments. I mean, we need those too, but – It doesn ’ t have to just be that. It doesn ’ t have to just be those really big, visible things. It can be the small things, the things that we see and encounter every day. Allyship is understanding that the system is such that the way your neighbor is experiencing a home appraisal and the way you might be could be different, and being willing to step in and interrupt that impact. It could also be recognizing that access to job opportunities isn ’ t necessarily fair just because you got in. It ’ s left-handed scissors. It ’ s considering food allergies. It ’ s being willing to accept correction when you mispronounce someone ’ s name or misgender them. It ’ s all of those things that you encounter every single day. And I think when we think of it as just the big things, we miss an opportunity to step into what we independently, individually have the ability to do to be allies. And I know that word may still not sit well with a lot of you. But whether you call it support, love, friendship, solidarity, whatever you ’ d like to call it, let ’ s start taking the kinds of everyday actions that really bring it to life. Thank you.

Text Sample 909: NEG Dim 6 – 2024 – Score 1.00 – 2024/t003234.txt

I want to tell you what I see coming. I ’ ve been lucky enough to be working on AI for almost 15 years now. Back when I started, to describe it as fringe would be an understatement. Researchers would say,”No, no, we ’ re only working on machine learning.”Because working on AI was seen as way too out there. In 2010, just the very mention of the phrase”AGI,”artificial general intelligence, would get you some seriously strange looks and even a cold shoulder.”You ’ re actually building AGI?”people would say.”Isn ’ t that something out of science fiction?”People thought it was 50 years away or 100 years away, if it was even possible at all. Talk of AI was, I guess, kind of embarrassing. People generally thought we were weird. And I guess in some ways we kind of were. It wasn ’ t long, though, before AI started beating humans at a whole range of tasks that people previously thought were way out of reach. Understanding images, translating languages, transcribing speech, playing Go and chess and even diagnosing diseases. People started waking up to the fact that AI was going to have an enormous impact, and they were rightly asking technologists like me some pretty tough questions. Is it true that AI is going to solve the climate crisis? Will it make personalized education available to everyone? Does it mean we ' ll all get universal basic income and we won ' t have to work anymore? Should I be afraid? What does it mean for weapons and war? And of course, will China win? Are we in a race? Are we headed for a mass misinformation apocalypse? All good questions. But it was actually a simpler and much more kind of fundamental question that left me puzzled. One that actually gets to the very heart of my work every day. One morning over breakfast, my six-year-old nephew Caspian was playing with Pi, the AI I created at my last company, Inflection. With a mouthful of scrambled eggs, he looked at me plain in the face and said,"But Mustafa, what is an AI anyway?"He ' s such a sincere and curious and optimistic little guy. He ' d been talking to Pi about how cool it would be if one day in the future, he could visit dinosaurs at the zoo. And how he could make infinite amounts of chocolate at home. And why Pi couldn ' t yet play I Spy."Well,"I said,"it ' s a clever piece of software that ' s read most of the text on the open internet, and it can talk to you about anything you want.""Right. So like a person then?"I was stumped. Genuinely left scratching my head. All my boring stock answers came rushing through my mind."No, but AI is just another general-purpose technology, like printing or steam."It will be a tool that will augment us and make us smarter and more productive. And when it gets better over time, it ' ll be like an all-knowing oracle that will help us solve grand scientific challenges."You know, all of these responses started to feel, I guess, a little bit defensive. And actually better suited to a policy seminar than breakfast with a no-nonsense six-year-old."Why am I hesitating?"I thought to myself. You know, let ' s be honest. My nephew was asking me a simple question that those of us in AI just don ' t confront often enough. What is it that we are actually creating? What does it mean to make something totally new, fundamentally different to any invention that we have known before? It is clear that we are at an inflection point in the history of humanity. On our current trajectory, we ' re headed towards the emergence of something that we are all struggling to describe, and yet we cannot control what we don ' t understand. And so the metaphors, the mental models, the names, these all matter if we ' re to get the most out of AI whilst limiting its potential downsides. As someone who embraces the possibilities of this technology, but who ' s also always cared deeply about its ethics, we should, I think, be able to easily describe what it is we are building. And that includes the six-year-olds. So it ' s in that spirit that I offer up today the following metaphor for helping us to try to grapple with what this moment really is. I think AI should best be understood as something like a new digital species. Now, don ' t take this too literally, but I predict that we ' ll come to see them as digital companions, new partners in the journeys of all our lives. Whether you think we ' re on a 10-, 20- or 30-year path here, this is, in my view, the most accurate and most fundamentally honest way of describing what ' s actually coming. And above all, it enables everybody to prepare for and shape what comes next. Now I totally get, this is a strong claim, and I ' m going to explain to everyone as best I can why I ' m making it. But first, let me just try to set the context. From the very first microscopic organisms, life on Earth stretches back billions of years. Over that time, life evolved and diversified. Then a few million years ago, something began to shift. After countless cycles of growth and adaptation, one of life ' s branches began using tools, and that branch grew into us. We went on to produce a mesmerizing variety of tools, at first slowly and then with astonishing speed, we went from stone axes and fire to language, writing and eventually industrial technologies. One invention unleashed a thousand more. And in time, we became homo technologicus. Around 80 years ago, another new branch of technology began. With the invention of computers, we quickly jumped from the first mainframes and transistors to today ' s smartphones and virtual-reality headsets. Information, knowledge, communication, computation. In this revolution, creation has exploded like never before. And now a new wave is upon us. Artificial intelligence. These waves of history are clearly speeding up, as each one is amplified and accelerated by the last. And if you look back, it ' s clear that we are in the fastest and most consequential wave ever. The journeys of humanity and technology are now deeply intertwined. In just 18 months, over a billion people have used large language models. We ' ve witnessed one landmark event after another. Just a few years ago, people said that AI would never be creative. And yet AI now feels like an endless river of creativity, making poetry and images and music and video that stretch the imagination. People said it would never be empathetic. And yet today, millions of people enjoy meaningful conversations with AIs, talking about their hopes and dreams and helping them work through difficult emotional challenges. AIs can now drive cars, manage energy grids and even invent new molecules. Just a few years ago, each of these was impossible. And all of this is turbocharged by spiraling exponentials of data and computation. Last year, Inflection 2. 5, our last model, used five billion times more computation than the DeepMind AI that beat the old-school Atari games just over 10 years ago. That ' s nine orders of magnitude more computation. 10x per year, every year for almost a decade. Over the same time, the size of these models has grown from first tens of millions of parameters to then billions of parameters, and very soon, tens of trillions of parameters. If someone did nothing but read 24 hours a day for their entire life, they ' d consume eight billion words. And of course, that ' s a lot of words. But today, the most advanced AIs consume more than eight trillion words in a single month of training. And all of this is set to continue. The long arc of technological history is now in an extraordinary new phase. So what does this mean in practice? Well, just as the internet gave us the browser and the smartphone gave us apps, the cloud-based supercomputer is ushering in a new era of ubiquitous AIs. Everything will soon be represented by a conversational interface. Or, to put it another way, a personal AI. And these AIs will be infinitely knowledgeable, and soon they ' ll be factually accurate and reliable. They ' ll have near-perfect IQ. They ' ll also have exceptional EQ. They ' ll be kind, supportive, empathetic. These elements on their own would be transformational. Just imagine if everybody had a personalized tutor in their pocket and access to low-cost medical advice. A lawyer and a doctor, a business strategist and coach – all in your pocket 24 hours a day. But things really start to change when they develop what I call AQ, their"actions quotient."This is their ability to actually get stuff done in the digital and physical world. And before long, it won ' t just be people that have AIs. Strange as it may sound, every organization, from small business to nonprofit to national government, each will have their own. Every town, building and object will be represented by a unique interactive persona. And these won ' t just be mechanistic assistants. They ' ll be companions, confidants, colleagues, friends and partners, as varied and unique as we all are. At this point, AIs will convincingly imitate humans at most tasks. And we ' ll feel this at the most intimate of scales. An AI organizing a community get-together for an elderly neighbor. A sympathetic expert helping you make sense of a difficult diagnosis. But we ' ll also feel it at the largest scales. Accelerating scientific discovery, autonomous cars on the roads, drones in the skies. They ' ll both order the takeout and run the power station. They ' ll interact with us and, of course, with each other. They ' ll speak every language, take in every pattern of sensor data, sights, sounds, streams and streams of information, far surpassing what any one of us could consume in a thousand lifetimes. So what is this? What are these AIs? If we are to prioritize safety above all else, to ensure that this new wave always serves and amplifies humanity, then we need to find the right metaphors for what this might become. For years, we in the AI community, and I specifically, have had a tendency to refer to this as just tools. But that doesn ' t really capture what ' s actually happening here. AIs are clearly more dynamic, more ambiguous, more integrated and more emergent than mere tools, which are entirely subject to human control. So to contain this wave, to put human agency at its center and to mitigate the inevitable unintended consequences that are likely to arise, we should start to think about them as we might a new kind of digital species. Now it ' s just an analogy, it ' s not a literal description, and it ' s not perfect. For a start, they clearly aren ' t biological in any traditional sense, but just pause for a moment and really think about what they already do. They communicate in our languages. They see what we see. They consume unimaginably large amounts of information. They have memory. They have personality. They have creativity. They can even reason to some extent and formulate rudimentary plans. They can act autonomously if we allow them. And they do all this at levels of sophistication that is far beyond anything that we ' ve ever known from a mere tool. And so saying AI is mainly about the math or the code is like saying we humans are mainly about carbon and water. It ' s true, but it completely misses the point. And yes, I get it, this is a super arresting thought but I honestly think this frame helps sharpen our focus on the critical issues. What are the risks? What are the boundaries that we need to impose? What kind of AI do we want to build or allow to be built? This is a story that ' s still unfolding. Nothing should be accepted as a given. We all must choose what we create. What AIs we bring into the world, or not. These are the questions for all of us here today, and all of us alive at this moment. For me, the benefits of this technology are stunningly obvious, and they inspire my life ' s work every single day. But quite frankly, they ' ll speak for themselves. Over the years, I ' ve never shied away from highlighting risks and talking about downsides. Thinking in this way helps us focus on the huge challenges that lie ahead for all of us. But let ' s be clear. There is no path to progress where we leave technology behind. The prize for all of civilization is immense. We need solutions in health care and education, to our climate crisis. And if AI delivers just a fraction of its potential, the next decade is going to be the most productive in human history. Here ' s another way to think about it. In the past, unlocking economic growth often came with huge downsides. The economy expanded as people discovered new continents and opened up new frontiers. But they colonized populations at the same time. We built factories, but they were grim and dangerous places to work. We struck oil, but we polluted the planet. Now because we are still designing and building AI, we have the potential and opportunity to do it better, radically better. And today, we ' re not discovering a new continent and plundering its resources. We ' re building one from scratch. Sometimes people say that data or chips are the 21st century ' s new oil, but that ' s totally the wrong image. AI is to the mind what nuclear fusion is to energy. Limitless, abundant, world-changing. And AI really is different, and that means we have to think about it creatively and honestly. We have to push our analogies and our metaphors to the very limits to be able to grapple with what ' s coming. Because this is not just another invention. AI is itself an infinite inventor. And yes, this is exciting and promising and concerning and intriguing all at once. To be quite honest, it ' s pretty surreal. But step back, see it on the long view of glacial time, and these really are the very most appropriate metaphors that we have today. Since the beginning of life on Earth, we ’ ve been evolving, changing and then creating everything around us in our human world today. And AI isn ’ t something outside of this story. In fact, it ’ s the very opposite. It ’ s the whole of everything that we have created, distilled down into something that we can all interact with and benefit from. It ’ s a reflection of humanity across time, and in this sense, it isn ’ t a new species at all. This is where the metaphors end. Here ’ s what I ’ ll tell Caspian next time he asks. AI isn ’ t separate. AI isn ’ t even in some senses, new. AI is us. It ’ s all of us. And this is perhaps the most promising and vital thing of all that even a six-year-old can get a sense for. As we build out AI, we can and must reflect all that is good, all that we love, all that is special about humanity : our empathy, our kindness, our curiosity and our creativity. This, I would argue, is the greatest challenge of the 21st century, but also the most wonderful, inspiring and hopeful opportunity for all of us. Thank you. Chris Anderson : Thank you Mustafa. It ’ s an amazing vision and a super powerful metaphor. You ’ re in an amazing position right now. I mean, you were connected at the hip to the amazing work happening at OpenAI. You ’ re going to have resources made available, there are reports of these giant new data centers, 100 billion dollars invested and so forth. And a new species can emerge from it. I mean, in your book, you did, as well as painting an incredible optimistic vision, you were super eloquent on the dangers of AI. And I ’ m just curious, from the view that you have now, what is it that most keeps you up at night? Mustafa Suleyman : I think the great risk is that we get stuck in what I call the pessimism aversion trap. You know, we have to have the courage to confront the potential of dark scenarios in order to get the most out of all the benefits that we see. So the good news is that if you look at the last two or three years, there have been very, very few downsides, right? It ’ s very hard to say explicitly what harm an LLM has caused. But that doesn ’ t mean that that ’ s what the trajectory is going to be over the next 10 years. So I think if you pay attention to a few specific capabilities, take for example, autonomy. Autonomy is very obviously a threshold over which we increase risk in our society. And it ’ s something that we should step towards very, very closely. The other would be something like recursive self-improvement. If you allow the model to independently self-improve, update its own code, explore an environment without oversight, and, you know, without a human in control to change how it operates, that would obviously be more dangerous. But I think that we ’ re still some way away from that. I think it ’ s still a good five to 10 years before we have to really confront that. But it ’ s time to start talking about it now. CA : A digital species, unlike any biological species, can replicate not in nine months, but in nine nanoseconds, and produce an indefinite number of copies of itself, all of which have more power than we have in many ways. I mean, the possibility for unintended consequences seems pretty immense. And isn ’ t it true that if a problem happens, it could happen in an hour? MS : No. That is really not true. I think there ’ s no evidence to suggest that. And I think that, you know, that ’ s often referred to as the”intelligence explosion.”And I think it is a theoretical, hypothetical maybe that we ’ re all kind of curious to explore, but there ’ s no evidence that we ’ re anywhere near anything like that. And I think it ’ s very important that we choose our words super carefully. Because you ’ re right, that ’ s one of the weaknesses of the species framing, that we will design the capability for self-replication into it if people choose to do that. And I would actually argue that we should not, that would be one of the dangerous capabilities that we should step back from, right? So there ’ s no chance that this will”emerge”accidentally. I really think that ’ s a very low probability. It will happen if engineers deliberately design those capabilities in. And if they don ’ t take enough efforts to deliberately design them out. And so this is the point of being explicit and transparent about trying to introduce safety by design very early on. CA : Thank you, your vision of humanity injecting into this new thing the best parts of ourselves, avoiding all those weird, biological, freaky, horrible tendencies that we can have in certain circumstances, I mean, that is a very inspiring vision. And thank you so much for coming here and sharing it at TED. Thank you, good luck.

Text Sample 910: NEG Dim 6 – 2024 – Score 1.00 – 2024/t003231.txt
Auntie 1 : Why are you still single? Auntie 2 : How much do you earn? Auntie 3 : You look fat. Niceaunties : I just love aunties. Some of my aunties, I only see during Chinese New Year. I used to brace myself for inquisition prior to family gatherings. They call you fat but present you with platters of delicious food, made with love and great skill. This is exasperating and endearing. I ’ m not alone in this. This behavior and line of questioning is so common we even have memes for it. I grew up in a big family, where the women, my late grandmother, my mother and some of my 11 aunties, were expected to take care of children. They were big influences in my life, and I witnessed great personalities, strength, eccentricity and humor. They held back from fully expressing themselves because of societal pressures, family expectations and lack of support. My grandma spent most of her life at home. She did not travel and had little contact with the outside world. In the last 20 years of her life, she had dementia and was bedridden. I often imagined her in an alternate reality, where she ’ s out and about, having fun, laughing, chatting, enjoying life. But aunties are not just about family. Anyone of any age, of any culture, can be labelled an auntie. It is not something you really want to be associated with. As it could mean you ’ re old-fashioned, rigid in your thinking or naggy. So it is rather negative to be called one. Women of my generation live in fear of being labeled an auntie. But what if you could change this perception, turning it from dread to something fun and positive? Thank you. I ’ m a designer from Singapore, and in the last year, also an artist. I came across some intriguing and beautiful images on social media and then found out they were made with AI. Curious about what AI could do, I found a program, I signed up, I keyed in some words, and within seconds, images appeared. I was amazed and mesmerized. Never before have I found a medium that felt like the shortest path between my ideas and the visual. It became addictive. Rather than just making images, I wanted to do more with this tool, to tell stories about auntie culture. My mother engages in many social activities. She is very active, a result of her mother ’ s immobility and limited life. She dances and performs just like the aunties do in Singapore ’ s parks every day. This was a huge inspiration for me. So”Niceaunties”was born, a project about freedom, exuberance, self-expression and fun. It is a lighthearted perspective on the joyful side of auntie life. Also a surreal narrative, loosely based on the women in my family, their social lives, their relationships, their quirks, humor, and, of course, food. In the last 20 years, I worked in architecture, a profession of discipline, rules and clients. I always worked in a team. It was about fulfilling the client ’ s brief and budgets. With AI, it ’ s just me. It ’ s a breath of fresh air. This lack of restraint has transported me to an imaginary world, where my aunties can live freely as themselves, where anything fun can and does happen. For instance, aunties in public places, taking photos with sculptures in absurd ways. This is a common sight. A typical holiday photo becomes aunties trying to kiss the moon, but failing badly. Over the course of a year – instead of designing buildings, I developed”Niceaunties”into

a world-building project : the”Auntieverse.”Just like in any world, there are cities, citizens, their lives and their culture. Auntieverse is partly inspired by Singapore, where acronyms are used for everything. A sentence would typically sound like :”I ’ m going to take the MRT to the NTUC near my mom ’ s HDB to avoid the ERP.”In the Auntieverse, we have Tofu-Engineered Sushi Luxury Autos, aka TESLA. It ’ s a transportation factory – where aunties assemble food and legs to create vehicles running on leg power. It ’ s very sustainable. In Chinese dialects,”legs”have the same sound as”cars,”so a common joke would be to say,”Let ’ s get around on our own cars.”Meaning we can move around on our own legs, implying self-reliance. So I imagine at TESLA, we have the usual car assembly, a car wash and, of course, a car junkyard. But legs don ’ t just move cars around in the Auntieverse. Buildings use them too, as do animals and vegetables. Another destination in the Auntieverse is MoMA, the Museum of Modern Aunties... Where modern auntie culture is celebrated. This is a social space where aunties gather, hang out, eat, chitchat, have fun, chillax. So one day, I said to my mother that she should show others how to make, and then sell, her delicious food. She said”No lah, too tired.”So I imagined NASA, Nice Auntie Sushi Academy – A culinary school on the moon, where aunties share their cooking skills. Here, amazing food is made available for all, 24-7. Aunties in training learn how to forage fresh ingredients on the moon, how to slice fish, and the cat will ensure all lessons go according to plan. But food isn ’ t just used for eating – In the Auntieverse. It ’ s used for all kinds of relaxing activities. I would love to soak in this ramen hot tub with my aunties. Well, sushi is used extensively in Auntie Spa for all kinds of beauty treatments and relaxing body wraps, a reference to many aunties ’ beauty tips, like putting cucumbers and other foods on their faces and bodies. Personally, I have been rubbed down in coffee grounds, and then wrapped in cling film. It ’ s very unsettling. Along with many beauty treatments, like lasers and acupuncture, I felt like I had to try, under the promise of youth and beauty. So many women undergo the pain of these beauty treatments to maintain body ideals and notions of beauty standards. Sometimes, it feels like having construction sites on our faces and bodies. Does it actually work? I don ’ t know. On the lighter end of the spectrum, Aunties ’ Nail Spa provides many beautiful options for our fingers, including very useful tools for everyday life. Cute and practical. Meanwhile, on Auntlantis, a parallel world inspired by Plato and the Great Pacific Garbage Patch, my aunties take a break on the beach, after a day ’ s work cleaning out the ocean floor, in a world filled with plastic waste as we know it. My Auntieverse is an ever-expanding narrative of food, culture, family, relationships and the world we live in. I could not have imagined having created this without the help of AI. Before AI, I have never edited a video. This technology lowered the barrier to learning and allowed me to bring my imagination to life in only 12 months. But this isn ’ t just about AI. The Auntieverse is about honoring the memory of extraordinary women who nurtured one another and their families. It is about celebrating personalities and bringing into the spotlight the mundane and the bizarre, like some aunties ’ big permed hair.”The taller the hair, the closer to God,”some said. [Auntie ’ s Hair Spray] [smells nice] [holds shape] [animal testing] [long lasting] [while stocks last!] Thank you.

Text Sample 911: NEG Dim 6 – 2023 – Score 1.00 – 2023/t003117.txt

When I first moved to Singapore, I thought it was magical. Here ’ s this clean, bright, pretty, well-run city with twisted, tall skyscrapers made of steel and glass that ’ s rising out of what ’ s left of a tropical rainforest. And it wasn ’ t just tropical rainforests, it was animals. Walk around and you see sunbirds flitting amongst the flowers. You see hornbills hopping from branch to branch on campus. You see colugos and Sumatran flying dragons and paradise tree snakes gliding from tree to tree. All of this was just so amazing to me. The first week I came to Singapore there was a fight during the day, out in the open between a king cobra and a reticulated python on the campus of NTU. That same first week there was video of – and I ’ m trying to think which way I ’ m looking. There was video of a pangolin walking down the stairway between Yale-NUS and RC4. I started to look for a place to live. And I went to a flat that I wanted to rent. And I walked out on the balcony and a parakeet flew, a wild parakeet flew and landed on my shoulder. And the property manager asked me,”Are you going to take the flat?”And I said,”Of course I am. A little bird told me to.”This was just amazing to me. Everything about Singapore was just incredible to me. When I first moved here, I kept wondering to myself, you know, what is this place with these tall, twisted skyscrapers? And at the same time, we have brightly-colored birds flying, hornbills that are eyeing people on campus, gliding lizards and colugos and snakes. It was just amazing to me. To me, it was utterly fantastical. And everything, everywhere, was otters. So the otters started to blow up on social media just before I came to Singapore in 2015, and one family in particular started to get a lot of attention, the Bishan family, because it lived downtown near all the famous landmarks of Singapore. And so they were getting a lot of attention, and this was really exciting to me. And this is a video from 2016. These are phone videos for the most part, but this is the Bishan mom, and these are her second brood of pups. Now we never get this close to otters anymore, so I deserve some scolding. Those are some Yale-NUS students and an otter watcher. And if you look upstream, you ’ ll see the Bishan dad with a fish. And those whining noises. Those are begging calls from the pups. That is a little bit of something that ’ s not actually in my script. You can see the otters are sprainting, they have a communal latrine site. And I didn ’ t know this because I just started. And so I was actually sitting in their latrine. These otters that are swimming upriver here, these are the prior year ’ s brood. So these are three-yearling otters, they ’ re are sexually mature otters and they stay with the family. All of this was just incredible to me. And part of the reason it was incredible to everybody is that the otters were returning after a long absence. So we know that there were otters in Singapore sometime before the mid-20th century. And we know partly because of individual accounts, but partly because of things like this. This is Haw Par Villa and these are statues of otters in Haw Par Villa. But this installation was moved in 1937. So before 1937, we know that otters weren ’ t just present in Singapore, but they were prominent in Singapore. And prominent enough that their statues were put with Chinese legends. But then Singapore started to change. It modernized, it started to industrialize, and all of a sudden the waterways got filthy. And what happened was they started to fill with sludge, industrial pollution and dead animals to the point where they stank. And otters live in water, they eat fish in water, and they couldn ’ t eat and live in waterways that were that dirty. So they left. But things changed again. Singapore enacted policies to clean up their waterways, and they were really, really successful. So all of a sudden, instead of having waterways that were filled with filth, we had waterways that were filled with fish. And from the otters ’ point of view, they were feeding troughs, so they came back. And now we have lots of otters all over Singapore. We have about 12 families. These are smooth-coated otters. They ’ re a pretty large otter. The adults get up to about 10 kilograms, which is larger than the European common otter. They ’ re a little bit unusual as mammals go, because the adult offspring stay with the family as helpers. So a family group might have two dominant breeding individuals and one or two sets of adult offspring that are staying as helpers, and then a brood of pups, which is pretty typical. The family sizes are quite large in Singapore, we have families that are over 20 individuals, and we have more than a dozen families in Singapore. And those families are watched by otter watchers, some of whom go out and watch the otters every day. My students worked closely with some of the otter watchers, and we got to find out all sorts of interesting things. So these are otters playing. Otters have a really tough life. They wake up in the morning, they fish, they roll around in the grass, they play, they go to sleep. And then a few hours later, they do it again. So my students wanted to know whether adults and pups played differently. And what they did to look at this is [they] found literally dozens of interactions like this and looked at the frequencies of role switching among the individuals who are playing. So the pups here, if you watch them long enough, they do something where one pup will chase the other, and then it ’ ll switch roles so the second pup chases the first. It ’ s very much like”Tag, you ’ re it.”It turns out that pups do this a lot, and adults don ’ t do this much at all. If an adult is in a dominant position, it stays in a dominant position. But this tells us something. It tells us something about how play has different functions for pups versus adults. For pups, play is a way that they ’ re figuring things out and they ’ re learning. But for adults, they ’ re jockeying for a position in the social group. And that ’ s why they don ’ t give up their position of dominance. I don ’ t know if you noticed this, it was early in the video, but there ’ s a monitor lizard that ’ s watching the otters play. And it turns out this is pretty common. Monitor lizards and otters live in the same environments. They both eat fish, and they bump into each other all the time. And otters sometimes attack the monitor lizards, and sometimes they kill the monitor lizards, but not all the time. So my students wanted to know when do otters attack? What conditions lead to the otters attacking sometimes but not other times. This is a pair of otters that are approaching a monitor lizard, and if you notice, the monitor lizard ’ s frill is open, its throat frill is open. It ’ s in an aggressive posture right here. And in fact, just seconds after this picture was taken, the monitor lizard whipped its tail at the otters. So monitor lizards can also be aggressive. But most of the time, including here, they ’ re reacting to the otters. What my students found after looking at dozens of these kinds of interactions, was that otters were more likely to be aggressive if there were pups around, and they were most likely to be aggressive if there were more pups in a group than there were adults. So the otters are only aggressive to defend the pups. The otters are big and they ’ re fast and smart, and they don ’ t really have to worry about the monitor lizards very much. But the pups do because they ’ re small and they ’ re kind of dumb. It turns out that pups drive a lot of the things that otters do. So watch what ’ s going on here. Do you see how they ’ re swimming in a line? By the way, we never get this close to otters anymore. I scold my students if I see this, and I get scolded if I ’ m this close. We try and give otters a pretty wide berth. What they ’ re doing here is catching fish. But what ’ s interesting is they ’ re catching really small fish. This is called herding or corralling, depending on the medium in which they ’ re doing it. And it turns out that otters do this only when they have pups. And when they do it, they typically catch very small fish. So this isn ’ t a way for them to catch more fish, and it isn ’ t a way for them to catch larger fish. What it is, is a way for them to teach their pups how to hunt. And it ’ s not just that the pups are learning how to hunt by being with the adults. It ’ s that the adults are actively changing their behavior so that they can teach the pups what to do. So all of these discoveries, and there are a few more of them, we couldn ’ t have done without the otter watchers. And the otter watchers are incredibly dedicated to watching otters. We really owe them a lot, and I really want to kind of voice that gratitude towards them. And a lot of things motivate otter watchers. So otter watchers might be curious about the otters, they might like the otters. But a lot of the otter watchers are photographers, and that ' s their primary hobby. And for them, this is a chance to get really excellent photographs. The same thing in Singapore is true with birders. If you ' ve ever gone birding in Singapore, it ' s a little bit surreal because rather than go out and look for people with big binoculars, you look for people with big cameras. But otter watching and bird watching are gateways. They ' re gateways for people to interact with nature. And so people who might start otter watching because they want to get photographs of cute pups, might continue to do other things because they ' ve formed a connection with nature. And we see this all the time : that people care about nature when they form some connection to nature. Whether that connection is to otters or to a pair of hornbills on campus or to a bird that visits them on their balcony, we need these personal connections, and we see them all over the place. So that ' s one thing to emphasize. But I ' m not the only one who ' s thinking about this. Singapore has enacted a lot of policies that make these kinds of connections a lot easier. So we talked about Singapore cleaning up its waterways in the ' 70s and ' 80s. They ' ve done a lot of other things, too, including plant over a million trees. There are over 300 parks and nature reserves. Singapore has a plan that no one should be more than 10 minutes away from some kind of park. There are a lot of reasons to do this, and some of these are public health reasons. But one of the effects of this is that people will have more chances to interact with nature, and they ' ll have more chances to care. So I ' m really excited about these possibilities, especially in Singapore, especially because Singapore is one of the most densely populated countries in the world. Recently, NParks changed its motto. NParks had the motto that Singapore is a city in a garden, and they changed it to Singapore is the city in nature. And I think this is a real effort on Singapore ' s part to shift how they view their relationship to nature. I think we ' re trying to get away from something where nature is over there on the other side of a fence or a wall or something like that. Nature is something that ' s around us and above us and beside us, and Singapore is acknowledging that. And that ' s true in lots of places, including in cities. So I think this also raises other questions, such as : Can cities be wildlife refuges? Is the biodiversity we see in Singapore, is this the last hot ember of the biodiversity that was here before? Or is this something that we can protect and maybe foster and grow? Lots of cities have wildlife, it ' s not just Singapore. Most North American cities have wildlife like raccoons and coyotes, and some have bears and bobcats. But sometimes there are even more impressive things. For example, Los Angeles has mountain lions, and Mumbai has leopards. And there ' s still one jaguar kicking around Tucson. So there ' s still these big cats, not just, you know, predators, but big cats that are in other cities. And Singapore, unlike some of those cities, is just incredibly, incredibly dense in terms of human population. And yet we ' re ripe with wildlife. So I think it ' s a real question. Can we make cities that are wildlife refuges? Can we make cities that foster some kind of productive relationship between wildlife and humans? Can we make cities where we exist in close proximity, side by side? I think we can. The founder of Singapore, famously, or at least as legend has it, saw a large creature moving along and he thought it was a lion, and so he named Singapore after it. Singa Pura, Lion City. And much later, Singapore adopted its mascot of a merlion, which is supposed to celebrate Singapore ' s history as a sea city and this legacy of it being a Lion City. Obviously, there are no lions in Singapore outside of the zoo, and there never were. But we do have a ferocious mammalian predator that hunts in groups, that fights giant lizards.

That protects and teaches its offspring. And that ’ s frequent in the waterways around Singapore. Isn ’ t that predator sort of half fish and half lion? Singapore is magic.

Text Sample 912: NEG Dim 6 – 2023 – Score 1.00 – 2023/t003113.txt
When I was asked to do this TED Talk, I sat down with my cat, as you do, and I thought, what ’ s going to make the biggest difference to as many cats as possible? What could it be? And I looked at my cat for inspiration and thought,”Yes, that ’ s it, yes, Kato,”– that ’ s his name –”Let ’ s tell everybody about the power of play.”If some people seem to be the cat people like me, hands up. Yes, welcome. Other people seem to be dog people, hands up. You ’ re also welcome. It doesn ’ t matter if you ’ re not really a cat person. If cats aren ’ t really your thing, that ’ s fine. Just take the word cat in this talk and replace it with dog – or horse, goat, child, whatever your species of choice is. And that ’ s because play is so fundamental to happiness that it affects so many different types of animals. In fact, there ’ s research to show that even fish like to play. They actually jump into streams of bubbles and ride them to the top and jump out and do it again. In case you were wondering. Now when I tell people that I ’ m a cat behaviorist, other than the confused looks I get, which you can imagine, and people saying,”A cat, a cat what?”Then – they really are quite confused by it. Then if they don ’ t have a cat themselves, then they ’ ll soon tell me that,”Oh, yes, my mother ’ s neighbor ’ s friend ’ s sister has a cat,”and they ’ ll often ask me questions about that cat ’ s behavior. It ’ s a bit like those of you that are nurses and you get everybody ’ s medical questions. Now invariably, I ’ m often recommending interactive play and feeding enrichment. And that ’ s because it has such a positive difference, such a big impact on their happiness. As you can see here from my cat Kato, play is not only fun, it gets the gray matter working. It ’ s great way to connect with your cat. And it has lots of physical benefits as well, because all that extra exercise is an excellent tool for helping overweight cats shed those extra pounds. This is the very gorgeous Moodles, and she came into Cats Protection ’ s care weighing 6. 7 kilograms. Now her very fantastic owner not only continued the diet that was already in place, but she also implemented a program of interactive play and feeding enrichment. It ’ s certainly made dieting a lot more fun. And this is Moodles today at her target weight of four kilograms, and she ’ s a much happier and healthier cat for it. Play is important for every cat, young or old, all around the world. Now I know what you might be thinking.”I know how to play with my cat.”That ’ s fine. But if your cat were to give you a review, what ’ s it going to say? Now, it might not say”Could try harder.”But it might say :”Could try smarter.”So let ’ s have a look at how to play the cat way. I often act as a bit of a translator between cats and their caregivers. This cat, for example, is saying,”What is this? This mouse is dead. Bring it to me alive.”And that ’ s where the interactive play comes. Because if you just put it there in front of the cat, it ’ s not interesting. You need to make it move, you need to bring it to life. And so you bring the mouse to the cat, and the cat goes,”No, no, no, no, not like that. Make it run away from me.”Because let ’ s face it, not many – You know, no self-respecting mouse would ever go up to a cat and go,”Hiya! Here I am!”So we need to be thinking about how prey species actually move. And then we need to be thinking about the fact that cats have got different play styles. Some cats prefer a game of rabbit, a bit like this cat. And that ’ s where they grab the toy with their front claws and they bunny kick with the back legs. Other cats prefer a game where you move the toy across the floor, and maybe it makes like a little, a little quiver

over here and a mad dash across the floor, just like a mouse might. And then others prefer the toy to be moved up in the air, like this. So this is a game like an erratic moth. Now cats haven ’ t evolved to show an awful lot in their faces. If you find them hard to read, that ’ s OK, they ’ ve evolved that way. But if you look at this cat, you can see that the claws are out, the whiskers have come forwards and the mouth ’ s open, and they ’ re so engaged and focused on that toy. This is a cat showing the face of joy. You can also make it a lot more fun like this by just having it disappear out of sight, like under a newspaper or magazine like this. And that makes it much more interesting. It ’ s really important that the cat is actually allowed to catch and kill the toy. It makes the cat feel good. It ’ s those happy hormones. So very, very important, catch and kill the toy. And that ’ s where, spoiler alert, the laser pens can be a bit of a problem, because it can cause frustration where the cat can ’ t ever catch the red dot. So if you do have a laser toy at home, fear not. You just need to end the game always on a physical toy like this so they can actually catch and and kill the toy. You may be concerned, perhaps that, you know, maybe all of this play is just training our cats to become more proficient hunters. But you ’ ll be very pleased and relieved to hear that the latest research shows that if you just play with your cat for a few minutes, several times a day, it actually reduces hunting behavior. OK, now some people might think,”It ’ s all very well and good, but my cat doesn ’ t really like to play. They ’ re too old. They ’ re too lazy. That won ’ t work for my cat.”And some people really think that their cat doesn ’ t play at all. I had one client who said,”No, my cat doesn ’ t play.”And yet, during the conversation, I found out that her cat does play, but only for five minutes. This is play. It ’ s just that the owner expected her cat to play for half an hour or more. Much like a dog would. But cats are not small dogs. Now as children, we were taught not to play with our food. And I ’ m here to ask all of you to not only play with your cat ’ s food but, more importantly, get your cat to play with their food. Feeding cats in a way that ’ s more interesting gets their minds working. It ’ s mentally stimulating. In short, for very little effort, you can actually change your cat ’ s happiness and you can change their life in just three minutes, because that ’ s how long it takes to show a cat feeding enrichment, also known as puzzle feeders. And it simply just means feeding a cat in a way that ’ s much more interesting than just in a bowl. Now some people may have dabbled in feeding enrichment, probably accidentally, when they ’ ve gone down the pet shop and picked up a puzzle ball like this. And they ’ ve given it to their cat, and then they ’ ve got frustrated and thought,”It doesn ’ t work.”Because the cat ’ s disinterest has left them disheartened. But fear not, because I have a few handy tips up my sleeve and that is to keep it simple. Now, as much as I love, I do love the puzzle balls, they are brilliant, but these are medium-level toys. We need to start off really simple with something like this, OK? You can find this online. And you want to start everybody off in an egg box. It ’ s brilliant because they ’ re free, and you just put in the biscuits where the eggs would normally sit. And you need to pour it out with your fingers and show the cats how to use it. And then for the next level up, you can cover the biscuits with a bit of paper. If you want it to be really difficult, you can then shut the lid. And it ’ s so much fun watching these cats hunt for their food. Once they ’ ve mastered the egg box, you can move on to the toilet roll pyramid. This one is fantastic for children, a great one I ’ d like to see done in schools, particularly because there ’ s no sharp scissors involved. And you just need to sellotape all the joins front and back, and then put the biscuits in where the toilet roll tubes are, and then allow the cat to get them out. So you can see, very simple ideas, they ’ re free and a lot of fun. So those are my two tips. Keep it simple and show the cats how to do it. Many years ago, I helped a caregiver and I was explaining how to introduce a puzzle ball to the cat. So we talked through it, and she rung me up a while later, quite exasperated, and said,”Nicky, I ’ ve done what you said. I got down on my hands and knees, and I ’ ve been batting around this puzzle ball now for two hours, and the cat still doesn ’ t understand. She just follows me and eats all the biscuits.”Yeah, that cat knows what she ’ s doing. And that cat knows how to use a puzzle ball but she ’ s getting the caregiver to do all the hard work. And you get pairs of cats like this as well, where one cat does all the hard work and the other cat swoops on in and eats the biscuits. I particularly like this puzzle board because it ’ s got, if you can just about see, it ’ s got about four different sections on there. So it ’ s different ways for the cat to think and use their brain and figure out how they ’ re going to get the food. The little bowls you can see near the beginning, near the cat, are also really good for wet food, so it ’ s quite a good one for both wet and dry food. This one is one of our cats in care, in Cats Protection ’ s care, because we do use the feeding enrichment quite a lot. Now this is one of those”do not try this at home,”unless your cat has been doing a lot of enrichment and knows exactly what they ’ re doing, because this cat has worked through from easy through to middle right through to sort of hard, master level. But you can see that it ’ s a lot of fun. And also, it really does slow down feeding. It ’ s particularly good if you ’ ve got cats that will just eat a lot of food and then regurgitate it up. Where ’ s the biscuit? It ’ s on her back. They do make cute videos as well. So you can see that really, we ’ re not teaching cats how to play. They ’ re teaching us. We just have to listen to them, and we need to learn their preferences. Because once you get the basic principles of feeding enrichment, the world is your oyster, and you ’ re only limited by your imagination. So you just need to take the feeding enrichment and build up really slowly in complexity over time to your own individual cat ’ s pace. I think that feeding enrichment is just as fun for us as it is for the cats. So go ahead, give it a go. It ’ s fun, it ’ s easy, and it takes very little effort. You ’ ve got three minutes, right? It may not change the world, but it will change the world for your cat. Thank you very much.

Text Sample 913: NEG Dim 6 – 2023 – Score 1.00 – 2023/t003210.txt

I was born in Terengganu, a beautiful coastal state in Peninsular Malaysia. Growing up in a superstitious, conservative community, my siblings and I always had to follow the strangest rules that usually didn ’ t make sense to me. Things like : don ’ t whistle in the house, no nail clipping at night. And whenever we went out to play, we had to return home before sunset or before it got dark. This particular rule made the night seem mysterious to me. I spent my school year admiring the dark, but never got around to really exploring it. As I got older, I was really drawn to beautiful nature. While in mangrove forests, a forestry officer told me about kelip-kelip. A group of insects that can produce light that are easiest to see at night. Given my limited experience with darkness – I decided to take a boat ride across the mangrove estuary one night just to see what they look like. Well, the first 15 minutes of the journey was frightening. It was pitch-black out there, and the river was choppy. All I could hear was a faint breeze. As the river began to narrow, it was then that I noticed a mesmerizing sight. Countless tiny flashes of light started to flicker on the trees, all flashing in almost perfect unison. It was as if they were dancing to their own beat. That is the moment I will never forget. The moment I officially fell in love with kelip-kelip, known as fireflies in English. The rest is history. Now, I have been researching fireflies for more than 17 years, and I plan to spend my life uncovering and supporting the worlds of these creatures. There are more than

2, 000 firefly species that we know of, and they live all over the world. They are found in every continent except for Antarctica. Most of them have wings, and most species can emit light. The light is produced by special organs under their abdomens, and each species has its own unique light pattern. Some glow continuously, while others emit discreet flashing patterns, almost like a secret code. Fireflies are so much more than just pretty lights. They are an essential part of a healthy ecosystem. The life cycle of fireflies keeps the ecosystem balanced. Each firefly species in its [respective] life stage has specific needs for a habitat to thrive. They act as bioindicator to ascertain whether a particular habitat is healthy or not. For example, in a mangrove forest, when you see a population of fireflies decreasing, that could be due to water quality degradation, which can be a sign of a collapsing food chain. Why? Because firefly larvae eat snails, and snails need good water quality to thrive. In terrestrial habitats, firefly population declines can be attributed to light pollution. They are extremely sensitive to artificial light because this type of light can disorient, repel or blind them. So most of the time when you see a decline in firefly population, you can bet other species that live in the same habitat in which fireflies are found are also declining. This is indicative of an unhealthy ecosystem. This is bad for us, humans, too. It is a sign of overdevelopment that can induce climate effects such as flood and drought. As I delve deeper into the world of fireflies, I can ’ t help but wonder the fascinating mysteries that unfolded before me. It amazed me to learn that some firefly species prefer to fly solo in forests, searching for light signals from their nonwinged mates, while others choose to gather amongst trees where the ocean meets the land. Equally intriguing is the fact that some firefly species have larvae that dwell in slow-moving rivers and ponds, while others have larvae that stay on land. I find it inspiring. I find it inspiring to know that there are so many undiscovered species hiding within them. Imagine how many more remarkable firefly species are waiting to be found. And I have always wanted to find them. But it can be hard. There are other dangerous animals, too, at night. On one occasion, when I was in a place called Linggi, I stopped at a tree with firefly colonies to collect a few specimens. When we turned on our headlamps, we saw a row of red eyes. And soon we realized that we were surrounded by crocodiles. Audience : Oh! Sometimes the discovery of new species can be prompted by examining collections in the Natural History Museum. In 2016, I went to Singapore and studied a firefly collection from Southeast Asia at Lee Kong Chian Natural History Museum. I saw a firefly specimen lying in the collection that did not match any of the descriptions that I knew of. There ’ s no name attached to it, and nobody knew what it was. In fact, the first specimen of this unknown species was collected in 1989. That is akin to a human adult living up to 30 years without a name and without a proper description of this person. So my colleagues and I in 2018 went on a mission. We coordinated the first night survey to the habitat of this unknown firefly, which happens to be the last remaining freshwater swamp in Singapore. We spent three hours exploring only to be greeted by the empty darkness. We went home disappointed and tired. Not to mention dirty and smelly from the mud. We went in again three months later. Still nothing. At this point we thought, this is it. The species is gone for good. However, the third time ’ s the charm. During the third time visit to this swamp, we saw a few tiny flashes start to show up and perch on ferns. It was such an amazing feeling despite having to wade through the muddy, swampy waters and being feasted on by mosquitoes. The itchiness was still worth it. After two years of additional data collection and analysis, we confirmed that this unknown firefly species was new to science. It was also the first time since over a century that a new species of firefly, a luminous firefly from Singapore, was found. And so we named it Luciola singapura. There is a Malay saying."If you are going to reach into the pickle jar, might as well put your whole arm in."The further I have gone into this work as researcher and educator in the field of biodiversity and conservation at Monash University Malaysia, the more passionate I have become. I have transformed from a girl, a curious girl, wanting to know what kelip-kelip is, to a driven scientist committed to safeguarding the fireflies of the world. The problem is that firefly habitats are disappearing fast. We are now in a race against time to name species before they disappear. I am a cochair of the IUCN Firefly Specialist Group, but it is also my personal mission to collaborate, to work together with firefly lovers and stakeholders to identify and conserve threatened firefly species everywhere. Here ' s a flash of hope. Join me on this journey if you are keen. Fireflies need your help before they flash that one last time. Thank you.

Text Sample 914: NEG Dim 6 – 2022 – Score 1.00 – 2022/t003578.txt
I wrote this journal entry in 2013."I ' ve been sleeping restlessly for months. There ' s a constant feeling of tension in my body. I love ministry as a Catholic priest, the opportunity to share my faith with people and support them as we journey through life together. But I have a desire to share my life with another person. I knew, in discerning ministry, that celibacy, not getting married, was part of the package. But I don ' t know if I can do this any longer. I ' m rarely alone, but I feel bitterly lonely."I wrote this when I was on the cusp of making one of the biggest and most significant decisions of my life : whether to leave ministry as a Catholic priest. I think my life has been a reasonably unique and unusual one. Now in my early forties, I ' ve had careers as an opera singer, a Catholic priest, a corporate lawyer, and now, a management consultant. And when people hear the careers that I ' ve had, the most common question I get asked – maybe the one that ' s on your mind – is,"What on earth is the connection amongst it all?"Through all of these changes, through all of these careers, one constant has been present : the practice of reflection. Now people may think that reflection is something that happens in a dark, private room by candlelight or flashlight, to purge one ' s deepest, darkest secrets. But I want to bring this practice out of the dark and into the light. I want to share how this practice can help our everyday lives, and especially our work lives. I want to share how this practice helps to improve performance, to make better decisions, both big and small, and to build better relationships. Interestingly, in a 2020 survey of some 4, 000 respondents, conducted by BCG and BVA, the question was asked,"What makes a good leader? What are their main qualities and skills?"Notably, possessing a good capacity for reflection was listed as one of the top five skills, the others being empathy, listening, consideration and team development. Reflection is about learning. It ' s about looking at the events of our lives without judgment, but with a critical lens. I really like the way leadership professors James Bailey and Scheherazade Rehman describe it :"It requires taking an honest moment to look at what transpired, what worked, what didn ' t, what can be done and what can ' t. Reflection requires courage. It ' s thoughtful, and it ' s deliberate."So how do we do this, then? Let ' s take a look at sports. I grew up in Australia, and one of the most popular games played was cricket. Consistent high performance from batters, bowlers and fielders is essential. But given it ' s a team sport, it ' s not just about how... the individual performs, but about how the team performs collectively together. Cricketers reflect during a game, after a game and over time. During a game, they might think about a missed catch or how they could bat or bowl better the next delivery. After a game, they

might come together to watch video replays, to look at what worked and what didn ’ t, which might differ from what they actually experienced during the game itself. And then, over time, they might look to the patterns of their wins and losses, to glean even more meaningful conclusions and insights. The same process can and should be used in the workplace, and I don ’ t think we need to watch video replays of our meetings to dissect what took place. I don ’ t know about you, but that might feel kind of creepy. Reflection can be done through a variety of different formats : purposeful thinking, written journal entries, audio notes, pitches, a discussion with a mentor or honest friend. The point is to find what works for you and to make a regular commitment. To examine the events and experiences of your lives – what worked, what didn ’ t, and why – and then, to think about what and how you would like to do things differently next time. When we practice reflection as a habit, we gain even more meaningful insight, because we see patterns that reflection on stand-alone events doesn ’ t provide. I truly think reflection can help everyone – people in every industry, at every stage in one ’ s career, and in every point in one ’ s life. Let me share with you an example that I think most people can connect with. Most of us have meetings. I was due for a daily catch-up with a colleague. He had just led a client meeting for the first time. It was to give a progress update on the work stream he was responsible for in this project. He said to me that the meeting had gone terribly. He didn ’ t get through the actions taken. He wasn ’ t able to discuss the obstacles faced or the decisions required. He didn ’ t get to the next steps or the responsible persons. He felt angst, uneasy, upset. He was concerned about what the client had thought of him and the meeting, but more importantly, what the client felt about the work that was going on in the work stream. It would have been so easy for him to have pushed past this, to try and suppress the emotions, but that would have missed a massive opportunity. We took a few moments to think and objectively reflect over what took place, and then to put in place some commitments as to what and how he could do things differently next time. He decided that in [the] future, he would start each meeting with an agenda alignment, to make sure there was clarity on what needed to be achieved. And then, he ’ d resolve to make sure that he would take greater control over the meeting so that if topics came up beyond the scope of the agenda, that he ’ d note that a separate discussion should be had. Reflection helps to improve performance. Reflection helps to make better decisions. Imagine, for a moment, you ’ ve been in your current role for five years. It ’ s a creative role, but you don ’ t quite feel you ’ ve got that zing, that energy for it, anymore. You ’ ve been offered another opportunity in the organization. It ’ s actually a promotion. It ’ s a more senior role, managerial. But you ’ ll have responsibility for looking after 12 direct reports. A competitor has also recently reached out to you. They ’ ve offered you an opportunity for the same kind of role that you ’ ve got currently, but it pays a higher salary. There ’ s a big difference between being a creative and being a manager. So what is it that really makes you happy? What is it that really makes you fulfilled? Reflection provides a treasure trove of data to help you work through this. Have your reflections mentioned being bored with projects, or do you just want to try something new? Would you like to be a manager? Would you like to see people grow and form and develop them? Do you think you could do a better job than your own manager? Reflection provides great insight. It ’ s easy to get lost when you ’ ve got an opportunity of a fancier job title and more money. But reflection enables you to focus on what really matters, and to make better choices. Let me share with you a final example. Most of us have relationships in our work – bosses, customers, clients, suppliers, whomever. And I think most of us try to have good relationships with these people. If I ’ m honest, while I strive for this ambition, I haven ’ t always succeeded, but reflection has helped me to build better relationships. A number of years ago, I was giving a feedback session with a colleague, and after having done so, I took a few moments to jot down some thoughts as to how it went. I realized I had been too clinical. In fact, if only you could have seen the expressions on my colleague ’ s face. I ’ d raced through their various strengths and moved onto spending more time in their areas for development. If I ’ d really thought about this person, I would have spent far greater time actually on their strengths, actually emphasizing why they were such a valuable member of our organization, and then, creating a space where they felt psychologically safe, to be able to go on and explore these areas for development. Reflection has helped me to improve this and to build better relationships. So this might all sound a little fine and dandy, and may be obvious or trivial, but the truth is, so many of us don ’ t take time out for regular reflection. This practice has helped me and I am so grateful. After I left being a priest, I took some time out for reflection to think about who I was and what I wanted my life to be about. Reflection helped me to grapple with this, and it continues to help me today as I grapple with this and other topics. So as we end this day, or tomorrow, before you begin the next, sit down, take a breath, and reflect. And you ’ ll see the power that this habit brings to your life. Thank you.

Text Sample 915: NEG Dim 6 – 2022 – Score 1.00 – 2022/t003577.txt

So we know forests play an essential role in regulating the Earth ’ s climate. However, most of what we know about those forests is actually based on things we can measure aboveground. So historically, ecologists like myself would come to this place, and we ’ d count the number of tree stems we ’ d find. We ’ d identify which species they are, and today we ’ d probably remotely sense features of this forest canopy from space. And all of this absolutely makes sense. Aboveground is where photosynthesis happens. Photosynthesis is how carbon and energy enter forests. Photosynthesis is how trees can remove carbon dioxide from the atmosphere. However, we also know most trees are limited in some way, by soil resources like water or nutrients. And to access those resources, trees have to build roots. And trees build an incredible amount of roots. So in some forests, there can be as much or more biomass belowground, in root structures, as aboveground, in stems and leaves. Decades of research have now made very clear that belowground ecology – so what ’ s going on in the soil – is really essential to understanding how these forest systems work. However, if you follow these root systems all the way out to their terminal ends, the finest tips in the root system, and you look closely – I mean super closely, like, you ’ re going to need a microscope closely – you discover a place where the tree stops being a plant, and starts becoming a fungus. So most trees on Earth form a partnership, or what scientists call symbiosis, with mycorrhizal fungi. So this, in my opinion, is one of the most remarkable images ever captured of these organisms. So in the background, at the top, you can see this dense network of fungal hyphae. These are essentially like roots, but for fungi, instead of plants. And in the foreground, you can see these incredible, multinucleated fungal spores, which look totally unreal, but absolutely are. These are the reproductive structures of the fungus. These have the potential to become entirely new fungal networks. Mycorrhizal fungi are essential to how basically all plants access limiting soil resources. There ’ s actually evidence that when plants first made the evolutionary transition from living in water to living on land, they evolved this symbiosis before they even evolved roots. And so this partnership between forests and their fungi is ancient, and it stretches back hundreds of millions of years. However, these roots don ’ t have to be just fungi. They can also be, for instance, bacteria. So these circular structures in this root network are called root nodules. They house symbiotic, nitrogen-fixing bacteria. And what these bacteria do is actually convert nitrogen gas in the atmosphere into plant-usable forms, and in turn, they nurture plant growth. And the complexity of soil biology just keeps going. So these root symbionts are embedded in an even more complex network of free-living bacterial and fungal decomposers, and archaea and protists, microscopic soil animals, viruses... The biodiversity of soil communities is astonishing. We now know a handful of soil can easily contain over 1, 000 coexisting microbial species. And so all of this, this is the soil microbiome. This is the forest microbiome, this is the ecosystem microbiome. So breakthroughs in DNA sequencing technology have finally turned the lights on belowground. DNA has allowed us to see these microbial communities in unprecedented detail, and, only recently, at unprecedented scales. Yet despite these breakthroughs, I ’ d argue we still don ’ t know the answers to seemingly simple questions, like this :”What does a healthy forest microbiome look like?”We ’ re far closer to answering a question like this for people than we are for plants. The Human Microbiome Project has really led in this area. So the human body is a microbial ecosystem. Each of us houses an incredibly biodiverse community of bacteria in our gut, and that has a profound impact on our health. This was discovered by medical microbiologists using DNA sequencing to characterize which bacteria live in hundreds of people ’ s bodies. And importantly, also noting health features of those same people. So, are they sick? And if so, with what? What ’ s their blood pressure, their digestive health, their mental health? And by combining all of that information, those microbiologists could begin to identify combinations of bacteria linked to health and disease. And these analyses became a road map for the development of human microbiome transplant therapies, which is essentially ecosystem restoration, but for your gut microbiome. And these therapies are now on the road to market to treat some of these diseases today. And so drawing from this work, my team asked,”What would it look like to take the Human Microbiome Project approach, but apply it to the forest?”What could we discover about the forest carbon cycle? Could we identify places where we could actually do belowground microbial restoration, and, in the process, combat climate change? Over the past three years, we ’ ve been working with forest scientists across Europe to do exactly that. In each of these locations, scientists have been documenting forest health for decades. And so, we asked our forest research partners to go out to each of these forests and collect a small sample of soil, which they then shipped back to our lab in Zurich so we could extract and sequence DNA, which allowed us to understand which microorganisms, and particularly fungi, live in each of these forests. And then finally, we used statistics and machine learning to relate which microorganisms live in a forest to a really important forest health metric : tree growth rate and carbon-capture rate aboveground. Now, once we controlled for the environmental drivers of tree growth – so how warm and wet each of these places is, as well as other variables we know control background site fertility – we discovered that particularly which fungi colonize the roots of these trees is linked to threefold variation in how fast these trees grow, how fast they remove carbon dioxide from the atmosphere. So put another way, these correlations imply that you could have two pine forests, sitting side by side, experiencing the same climate, growing in the same soils. But if one of them was colonized by the right community of fungi on its roots, it could be growing up to three times as fast as that adjacent forest. And furthermore, these patterns were not driven by the presence of particularly high-performing species or strains, but instead, they were driven by biodiverse and completely different communities of fungi. And so these fungal signatures are super exciting to us because they imply an opportunity to manage, and in many cases, actually rewild the forest fungal microbiome. So, for example, can we reintroduce fungal biodiversity into a managed timber forestry landscape? And in the process, can we make those trees grow faster? Can we make them capture more carbon in their tree stems and in their soils? Can we rewild the soil and combat climate change? And these aren ' t just rhetorical questions – we ' ve actually started doing this. So this is one of our field trials in Wales, in the United Kingdom. It ' s run in collaboration with the charity there called the Carbon Community. It ' s 28 acres, or 11 hectares, and it ' s set up as a block-randomized controlled trial. This is analogous to how you would run a drug trial, but in this case, it ' s for trees instead of people. And here, we do a pretty straightforward experiment. We either plant trees, business as usual – which is just direct planting of seedlings into the ground – or we plant trees, and at the moment of planting, we add a small handful of soil. But it ' s not just any soil. It ' s soil sourced from a forest our analyses have identified as harboring potentially high-performing fungi. So since we reintroduced microbial biodiversity into some of these sites, we ' ve observed that where we actually did that, we ' ve been able to accelerate tree growth and carbon capture in tree stems by 30 to 70 percent, depending on the tree species. Or put another way – where we manipulated and rewilded the invisible microbiology of this place, we ' ve begun to change how that entire place works. Now it ' s important to emphasize that we ' re really excited about these findings, but we also understand they ' re still early. We want to see many more large-scale field trials and many more locations with many more years of data. However, beyond just these carbon and climate outcomes, I think the most exciting thing here is that we can actually do this with wild and native and biodiverse combinations of microorganisms. And while we pointed this approach at forestry, in principle, this kind of science has the potential to generalize to all of our managed landscapes. We can begin asking questions like,"What does a healthy agricultural microbiome look like?"Thinking across both food and forest agriculture. And there ' s reason to think a biodiversity-first approach may be particularly powerful here. And that ' s because the history of agriculture has been an exercise in reductionism. We ' ve identified high-performing plant species, and then strains, and then we ' ve selectively bred them, and now we genetically modify them. And finally, we plant those organisms out in vast monocultures. So a single plant species as far as you can see. And to be clear, this has produced very productive agroecosystems. But it ' s also produced ecosystems we ' re coming to understand are remarkably fragile. Systems increasingly sensitive to extreme climate events, novel pathogens. Systems incredibly reliant on chemical inputs, we ' re coming to understand have really serious externalities. So we now have the data, computational tools and the ecological theory to start going the other way, to lean into biodiversity and complexity. And once we do, the question really becomes, by rewilding our soils, can we make our managed food and forest landscapes reservoirs of belowground biodiversity? And in the process, can we enhance yields and carbon capture and all the other services we ask of these ecosystems? I think there ' s a lot of reason to be incredibly hopeful here. And I think we also shouldn ' t be so surprised that these microscopic organisms have the potential for such enormous, ecosystem-scale effects. And that ' s because we ' ve known now, really for a long time, that forests are fungi. And they ' re incredibly biodiverse communities of bacteria and archaea and protists and microscopic soil animals and viruses. Soil is the literal foundation of terrestrial ecosystems, and the microbial life that inhabits soil represents some of the most complex and biodiverse communities of life on Earth. For the first time, DNA sequencing is turning the lights on belowground. It ’ s allowing us to see these organisms in unprecedented detail and at unprecedented scales. Imagine studying plant biology, but you never really knew if you ’ re looking at a sequoia tree or a sphagnum moss. And then, all of a sudden, you did. That ’ s what ’ s happening right now in global environmental microbiology. And so we should expect this revolution in our understanding of these microscopic organisms, and particularly fungi, to transform how we understand and how we manage our ecosystems in a foundational way. Thank you.

Text Sample 916: NEG Dim 6 – 2022 – Score 1.00 – 2022/t003576.txt

I am a planet hunter and keeper of the keys at NASA ’ s Exoplanet Archive. In March 2022, we reached a major milestone in space exploration : 5, 000 known exoplanets. For thousands of years, we ’ ve wondered about planets outside of our solar system, now called exoplanets. But our technology only recently caught up with our imaginations. And yes, 5, 000 planets is incredible. What ’ s even more incredible is how space research will change as a result. When I started grad school, there were about 100 known exoplanets, all radically different from the Earth and from each other. I was determined to find more. I spent four years looking at nearly 87, 000 stars, one by one. Now you might have this romantic idea that I was gazing intently through a telescope, pondering some gorgeous view of the universe. I was not. I was looking at data like this, measuring the brightness of each star over time. If the brightness dipped, just briefly, just a little bit, it could be because a planet had orbited in front of that star, blocking some of the light from reaching my telescope. So I spent four years looking for decimal-level changes in these data. And after four years, I ’ d found... nothing. Zero exoplanets. Thankfully, they still gave me the PhD, I think, for effort. Then I moved to Harvard, where I worked on my first NASA mission, called EPOXI. I still didn ’ t find any exoplanets. Then in March 2010, I joined the Kepler Mission, NASA ’ s grand experiment with putting one of our planet-hunting instruments into space. Monday was my first day on the base in Silicon Valley. It was mostly spent in HR. Tuesday, I sat down and looked at the data for the first time, and I found my first exoplanet. A few minutes later, I found another one. There ’ s a saying that we ’ re the generation that was born too late to explore Earth and too soon to explore space. That ’ s not true anymore. That day and every day since, I ’ ve gotten to explore space. Kepler made it possible for us to measure stellar brightness much more precisely than we had before. And eventually I helped find thousands of exoplanets. And we ’ ve really only searched our local corner of the galaxy to find those planets. That means there ’ s likely tens of billions of planets just in our Milky Way. Now with so much data, we can start sorting and grouping and categorizing these planets to find trends. Think of it this way : if you wanted to learn about dogs and you had five dogs in your study, well, you ’ d learn a lot about those five dogs. That they ’ re all good dogs. But maybe not about dogs in general. If you had 5, 000 dogs in your study, then you ’ d start to see that there were German Shepherds and Dobermanns and beagles, and that these different breeds have different features. With demographic-level data on exoplanets, we can start asking some of these big questions for the first time, like : Of those thousands and billions of planets in our galaxy, how many are like the Earth, or like Jupiter? How many planets does a typical star have? Can a planet orbit more than one star? Yes. Can a planet exist without any star at all? Also yes. One surprising result from the study of planet populations is that the most common kind of planet in our galaxy might be one we don ’ t have in our solar system : a super-Earth up to twice as big and ten times as heavy as our Earth. We ’ ve found evaporating planets, disintegrating planets, planets clustered together in a clockwork dance, ultra-puffy planets, ultra-dense planets. It ’ s truly a wild and wonderful menagerie that I get to corral at the NASA Exoplanet Archive. But it gets even more interesting than that. With so much data, we might finally be able to figure out how planets are made. We see baby stars being born in stellar nurseries surrounded by dust and gas. And we see all the stars surrounded by completed planetary systems. But we still don ’ t really know what happens in between. With more data, we might find planets at some middle stage or many middle stages. And from there, be able to map out a timeline of planetary development. What triggers these diffused clouds of dust and gas to collapse and transform? And how does the chaos and turmoil of dust become pebbles, and pebbles become boulders, and boulders become planetesimals? And from there, after an intense series of bombardments eventually settle into an ordered series of planets. How often is one of those planets solid and warm, with a water ocean lapping a sandy shore? Where do we come from, and how did we get here? The more we learn about exoplanets, the easier it is to target the ones we want. So far, we haven ’ t found any planets that are like the Earth. But I hope we will. NASA just spent the last few years studying the idea of a very large telescope in space with next-generation technology that would allow us to take an image, an actual photograph, of a planet like the Earth. With that photo, we could search for biomarkers, signatures of life. I ’ ll probably spend the rest of my career working on that mission. I hope I get to take that photo. Thank you.

Text Sample 917: NEG Dim 6 – 2019 – Score 1.00 – 2019/t000012.txt
Now, this is Joanna. Joanna works at a university in Poland. And one Saturday morning at 3am, she got up, packed her rucksack and traveled more than a thousand kilometers, only to have a political argument with a stranger. His name is Christof, and he ’ s a customer manager from Germany. And the two had never met before. They only knew that they were totally at odds over European politics, over migration, or the relationship to Russia or whatever. And they were arguing for almost one day. And after that, Joanna sent me a somewhat irritating email.”That was really cool, and I enjoyed every single minute of it!”So these are Tom from the UK and Nils from Germany. They also were strangers, and they are both supporters of their local football team, as you may imagine, Borussia Dortmund and Tottenham Hotspurs. And so they met on the very spot where football roots were invented, on some field in Cambridge. And they didn ’ t argue about football, but about Brexit. And after talking for many hours about this contentious topic, they also sent a rather unexpected email.”It was delightful, and we both enjoyed it very much.”So in spring 2019, more than 17, 000 Europeans from 33 countries signed up to have a political argument. Thousands crossed their borders to meet a stranger with a different opinion, and they were all part of a project called”Europe Talks.”Now, talking about politics amongst people with different opinions has become really difficult, not only in Europe. Families are splitting, friends no longer talk to each other. We stay in our bubbles. And these so-called filter bubbles are amplified by social media, but they are not, in the core, a digital product. The filter bubble has always been there. It ’ s in our minds. As many studies repeatedly have shown, we, for example, ignore effects that contradict our convictions. So

correcting fake news is definitely necessary, but it ’ s not sufficient to get a divided society to rethink itself. Fortunately, according to at least some research, there may be a simple way to get a new perspective : a personal one-on-one discussion with someone who doesn ’ t have your opinion. It enables you to see the world in a new way, through someone else ’ s eyes. Now, I ’ m the editor of”ZEIT ONLINE,”one of the major digital news organizations in Germany. And we started what became”Europe Talks”as a really modest editorial exercise. As many journalists, we were impressed by Trump and by Brexit, and Germany was getting divided, too, especially over the issue of migration. So the arrival of more than a million refugees in 2015 and 2016 dominated somewhat the debate. And when we were thinking about our own upcoming election in 2017, we definitely knew that we had to reinvent the way we were dealing with politics. So digital nerds that we are, we came up with obviously many very strange digital product ideas, one of them being a Tinder for politics — a dating platform for political opposites, a tool that could help get people together with different opinions. And we decided to test it and launched what techies would call a”minimum viable product.”So it was really simple. We called it”Deutschland spricht”—”Germany Talks”— and we started with that in May 2017. And it was really simple. We used mainly Google Forms, a tool that each and every one of us here can use to make surveys online. And everywhere in our content, we embedded simple questions like this :”Did Germany take in too many refugees?”You click yes or no. We asked you more questions, like,”Does the West treat Russia fairly?”or,”Should gay couples be allowed to marry?”And if you answered all these questions, we asked one more question :”Hey, would you like to meet a neighbor who totally disagrees with you?”So this was a really simple experiment with no budget whatsoever. We expected some hundred-ish people to register, and we planned to match them by hand, the pairs. And after one day, 1, 000 people had registered. And after some weeks, 12, 000 Germans had signed up to meet someone else with a different opinion. So we had a problem. We hacked a quick and dirty algorithm that would find the perfect Tinder matches, like people living as close as possible having answered the questions as differently as possible. We introduced them via email. And, as you may imagine, we had many concerns. Maybe no one would show up in real life. Maybe all the discussions in real life would be awful. Or maybe we had an axe murderer in our database. But then, on a Sunday in June 2017, something beautiful happened. Thousands of Germans met in pairs and talked about politics peacefully. Like Anno. He ’ s a former policeman who ’ s against — or was against — gay marriage, and Anne, she ’ s an engineer who lives in a domestic partnership with another woman. And they were talking for hours about all the topics where they had different opinions. At one point, Anno told us later, he realized that Anne was hurt by his statements about gay marriage, and he started to question his own assumptions. And after talking for three hours, Anne invited Anno to her summer party, and today, years later, they still meet from time to time and are friends. So our algorithm matched, for example, this court bailiff. He ’ s also a spokesperson of the right-wing populist party AfD in Germany, and this counselor for pregnant women. She used to be an active member of the Green Party. We even matched this professor and his student. It ’ s an algorithm. We also matched a father-in-law and his very own daughter-in-law, because, obviously, they live close by but have really different opinions. So as a general rule, we did not observe, record, document the discussions, because we didn ’ t want people to perform in any way. But I made an exception. I took part myself. And so I met in my trendy Berlin neighborhood called Prenzlauer Berg, I met Mirko. This is me talking to Mirko. Mirko didn ’ t want to be in the picture. He ’ s a young plant operator, and he looked like all the hipsters in our area, like with a beard and a beanie. We were talking for hours, and I found him to be a wonderful person. And despite the fact that we had really different opinions about most of the topics — maybe with the exception of women ' s rights, where I couldn ' t comprehend his thoughts — it was really nice. After our discussion, I Googled Mirko. And I found out that in his teenage years, he used to be a neo-Nazi. So I called him and asked,"Hey, why didn ' t you tell me?"And he said,"You know, I didn ' t tell you because I want to get over it. I just don ' t want to talk about it anymore."I thought that people with a history like that could never change, and I had to rethink my assumptions, as did many of the participants who sent us thousands of emails and also selfies. No violence was recorded whatsoever. And we just don ' t know if some of the pairs got married. But, at least, we were really excited and wanted to do it again, especially in version 2. 0, wanted to expand the diversity of the participants, because obviously in the first round, they were mainly our readers. And so we embraced our competition and asked other media outlets to join. We coordinated via Slack. And this live collaboration among 11 major German media houses was definitely a first in Germany. The numbers more than doubled : 28, 000 people applied this time. And the German president — you see him here in the center of the picture — became our patron. And so, thousands of Germans met again in the summer of 2018 to talk to someone else with a different opinion. Some of the pairs we invited to Berlin to a special event. And there, this picture was taken, until today my favorite symbol for"Germany Talks."You see Henrik, a bus driver and boxing trainer, and Engelbert, the director of a children ' s help center. They answered all of the seven questions we asked differently. They had never met before this day, and they had a really intensive discussion and seemed to get along anyway with each other. So this time we also wanted to know if the discussion would have any impact on the participants. So we asked researchers to survey the participants. And two-thirds of the participants said that they learned something about their partner ' s attitudes. Sixty percent agreed that their viewpoints converged. The level of trust in society seemed also higher after the event, according to the researchers. Ninety percent said that they enjoyed their discussion. Ten percent said they didn ' t enjoy their discussion, eight percent only because, simply, their partner didn ' t show up. After"Germany Talks,"we got approached by many international media outlets, and we decided this time to build a serious and secure platform. We called it"My Country Talks."And in this short period of time,"My Country Talks"has already been used for more than a dozen local and national events like"Het grote gelijk"in Belgium or"Suomi puhuu"in Finland or"Britain Talks"in the UK. And as I mentioned at the beginning, we also launched"Europe Talks,"together with 15 international media partners, from the"Financial Times"in the UK to"Helsingin Sanomat"in Finland. Thousands of Europeans met with a total stranger to argue about politics. So far, we have been approached by more than 150 global media outlets, and maybe someday there will be something like"The World Talks,"with hundreds of thousands of participants. But what matters here are not the numbers, obviously. What matters here is... Whenever two people meet to talk in person for hours without anyone else listening, they change. And so do our societies. They change little by little, discussion by discussion. What matters here is that we relearn how to have these face-to-face discussions, without anyone else listening, with a stranger. Not only with a stranger we are introduced to by a Tinder for politics, but also with a stranger in a pub or in a gym or at a conference. So please meet someone and have an argument and enjoy it very much. Thank you. Wow!

Text Sample 918: NEG Dim 6 – 2019 – Score 1.00 – 2019/t000009.txt

I am obsessed with forming healthy communities, and that ’ s why I started Twitch â to help people watch other people play video games on the internet. Thank you for coming to my TED talk. So in seriousness, video games and communities truly are quite related. From our early human history, we made our entertainment together in small tribes. We shared stories around the campfire, we sang together, we danced together. Our earliest entertainment was both shared and interactive. It wasn ’ t until pretty recently on the grand scale of human history that interactivity took a back seat and broadcast entertainment took over. Radio and records brought music into our vehicles, into our homes. TV and VHS brought sports and drama into our living rooms. This access to broadcast entertainment was unprecedented. It gave people amazing content around the globe. It created a shared culture for millions of people. And now, if you want to go watch or listen to Mozart, you don ’ t have to buy an incredibly expensive ticket and find an orchestra. And if you like to sing â I can show you the world â then you have something in common with people around the world. But with this amazing access, we allowed for a separation between creator and consumer, and the relationship between the two became much more one-way. We wound up in a world where we had a smaller class of professional creators and most of us became spectators, and as a result it became far easier for us to enjoy that content alone. There ’ s a trend counteracting this : scarcity. So, Vienna in the 1900s, was famous for its cafĂŠ culture. And one of the big drivers of that cafĂŠ culture was expensive newspapers that were hard to get, and as a result, people would go to the cafĂŠ and read the shared copy there. And once they ’ re in the cafe, they meet the other people also reading the same newspaper, they converse, they exchange ideas and they form a community. In a similar way, TV and cable used to be more expensive, and so you might not watch the game at home. Instead you ’ d go to the local bar and cheer along with your fellow sports fans there. But as the price of media continues to fall over time thanks to technology, this shared necessity that used to bring our communities together falls away. We have so many amazing options for our entertainment, and yet it ’ s easier than ever for us to wind up consuming those options alone. Our communities are bearing the consequences. For example, the number of people who report having at least two close friends is at an all-time low. I believe that one of the major contributing causes to this is that our entertainment today allows us to be separate. There is one trend reversing this atomization of our society : modern multiplayer video games. Games are like a shared campfire. They ’ re both interactive and connecting. Now these campfires may have beautiful animations, heroic quests, occasionally too many loot boxes, but games today are very different than the solitary activity of 20 years ago. They ’ re deeply complex, they ’ re more intellectually stimulating, and most of all, they ’ re intrinsically social. One of the recent breakout genres exemplifying this change is the battle royale. 100 people parachute onto an island in a last-man-standing competition. Think of it as being kind of like”American Idol,”but with a lot more fighting and a lot less Simon Cowell. You may have heard of”Fortnite,”which is a breakout example of the battle royale genre, which has been played by more than 250 million people around the world. It ’ s everyone from kids in your neighborhood to Drake and Ellen DeGeneres. 2. 3 billion people in the world play video games. Early games like”Tetris”and”Mario”may have been simple puzzles or quests, but with the rise of arcades and then internet play, and now massively multiplayer games of huge, thriving online communities, games have emerged as the one form of entertainment where consumption truly requires human connection. So this brings us to streaming. Why do people stream themselves playing video games? And why do hundreds of millions of people around the world congregate to watch them? I want you all the imagine for second â imagine you land on an alien planet, and on this planet, there ’ s a giant green rectangle. And in this green rectangle, aliens in matching outfits are trying to push a checkered sphere between two posts using only their feet. It ’ s pretty evenly matched, so the ball is just going back and forth, but there ’ s hundreds of millions of people watching from home anyway, and cheering and getting excited and engaged right along with them. Now I grew up watching sports with my dad, so I get why soccer is entertaining and engaging. But if you don ’ t watch sports, maybe you like watching”Dancing with the Stars”or you enjoy”Top Chef.”Regardless, the principle is the same. If there is an activity that you really enjoy, you ’ re probably going to like watching other people do it with skill and panache. It might be perplexing to an alien, but bonding over shared passion is a human universal. So gamers grew up expecting this live, interactive entertainment, and passive consumption just doesn ’ t feel as fulfilling. That ’ s why livestreaming has taken off with video games. Because livestreaming offers that same kind of interactive feeling. So when you imagine what ’ s happening on Twitch, I don ’ t want you to think of a million livestreams of video games. Instead, what I want you to picture is millions of campfires. Some of them are bonfires â huge, roaring bonfires with hundreds of thousands of people around them. Some of them are smaller, more intimate community gatherings where everyone knows your name. Let ’ s try taking a seat by one of those campfires right now. Hey Cohh, how ’ s it going? Cohh : Hey, how ’ s it going, Emmett? ES : So I ’ m here at TED with about 1, 000 of my closest friends, and we thought we ’ d come and join you guys for a little stream. Cohh : Awesome! It ’ s great to hear from you guys. ES : So Cohh, can you share with the TED audience here â what have you learned about your community on Twitch? Cohh : Ah, man, where to begin? I ’ ve been doing this for over five years now, and if there ’ s one thing that doesn ’ t cease to impress me on the daily, it ’ s just kind of how incredible this whole thing is for communication. I ’ ve been playing games for 20 years of my life, I ’ ve led online MMO guilds for over 10, and it ’ s the kind of thing where there ’ s very few places in life where you can go to meet so many people with similar interests. I was listening in a bit earlier ; I love the campfire analogy, I actually use a similar one. I see it all as a bunch of people on a big couch but only one person has the controller. So it ’ s kind of like a”Pass the snack!”situation, you know? 700 people that way â but it ’ s great and really it ’ s just â ES : So Cohh, what is going on in chat right now? Can you explain that a little bit to us? Because my eyesight isn ’ t that good but I see a lot of emotes. Cohh : So this is my community ; this is the Cohhilition. I stream every single day. I actually just wrapped up a 2, 000-day challenge, and as such, we have developed a pretty incredible community here in the channel. Right now we have about 6200 people with us. What you ’ re seeing is a spam of”Hello, TED”good- vibe emotes, love emotes,”this is awesome,””Hi, guys,””Hi, everyone.”Basically just a huge collection of people â huge collection of gamers that are all just experiencing a positive event together. ES : So is there anything that â can we poll chat? I want ask chat a question. Is there anything that chat would like the world, and particularly these people here with me at TED right now, to know about what they get out of playing video games and being part of this community? Cohh : Oh, wow. I am already starting to see a lot of answers here.”I like the good vibes.””Best communities are on Twitch.””They get us through the rough patches in life.”Oh, that ’ s a message I definitely see a lot on Twitch, which is very good.”A very positive community,””a lot of positivity,”which is pretty great. ES : So Cohh, before I get back to my TED talk, which I actually should probably get back to doing at some point â Do you have anything else that you want to share with me or any question you wanted to ask, you ’ ve always wanted to get out there before an audience? Cohh : Honestly, not too much. I mean, I absolutely love what you ’ re doing right now. I think that the interactive streaming is the big unexplored frontier of the future in entertainment, and thank you for doing everything you ’ re doing up there. The more people that hear about what you do, the better â for everyone on here. ES : Awesome, Cohh. Thanks so much. I ’ m going to get back to giving this talk now, but we should catch up later. Cohh : Sounds great! ES : So that was a new way to interact. We could influence what happened on the stream, we could cocreate the experience along with him, and we really had a multiplayer experience with chat and with Cohh. At Twitch, we ’ ve started calling this, as a result,”multiplayer entertainment.”Because going from watching a video alone to watching a live interactive stream is similar to the difference between going from playing a single-player game to playing a multiplayer game. Gamers are often as the forefront of exploration in new technology. Microcomputers, for example, were used early on for video games, and the very first handheld, digital mass-market devices weren ’ t cell phones, they were Gameboys... for video games. And as a result, one way that you can get a hint of what the future might hold is to look to this fun, interactive sandbox of video games and ask yourself,”what are these gamers doing today?”And that might give you a hint as to what the future is going to hold for all of us. One of the things we ’ re already seeing on Twitch is multiplayer entertainment coming to sports. So, Twitch and the NFL teamed up to offer livestreaming football, but instead of network announcers in suits streaming the game, we got Twitch users to come in and stream it themselves on their own channel and interact with their community and make it a real multiplayer experience. So I actually think that if you look out into the future â only hundreds of people today get to be sports announcers. It ’ s a tiny, tiny number of people who have that opportunity. But sports are about to go multiplayer, and that means that anyone who wants to around the world is going to get the opportunity to become a sports announcer, to give it a shot. And I think that ’ s going to unlock incredible amounts of new talent for all of us. And we ’ re not going to be asking,”Did you catch the game?”Instead, we ’ re going to be asking,”Whose channel did you catch the game on?”We already see this happening with cooking, with singing â we even see people streaming welding. And all of this stuff is going to happen around the metaphorical campfire. There ’ s going to be millions of these campfires lit over the next few years. And on every topic, you ’ re going to be able to find a campfire that will allow you to bond with your people around the world. For most of human history, entertainment was simply multiplayer. We sang together in person, we shared news together in the town square in person, and somewhere along the way, that two-way conversation turned into a one-way transmission. As someone who cares about communities, I am excited for a world where our entertainment could connect us instead of isolating us. A world where we can bond with each other over our shared interests and create real, strong communities. Games, streams and the interactions they encourage, are only just beginning to turn the wheel back to our interactive, community-rich, multiplayer past. Thank you all for sharing this experience here with me, and may you all find your best campfire.

Text Sample 919: NEG Dim 6 – 2019 – Score 1.00 – 2019/t000029.txt

Keith Kirkland : We believe in making the world a better place, and we ’ re trying to do that through design, but the world for us starts with our company.

Our first product, Wayband, is a wearable haptic navigation device for the blind and visually impaired. We had gotten an article published and it had made its way to this guy, Simon Wheatcroft. He reached out to us and he was like,”If you can have this technology ready by November, I ’ ll run the New York City Marathon with it.”We saw it and was kind of like, moonshot, this is probably not going to work, but let ’ s just do it and see how far we can get. [Dropbox is a collaborative platform whose mission is to design a more enlightened way of working. They are invested in understanding how teams like WearWorks achieve extraordinary outcomes.] Jennifer Brook : In Keith ’ s story, I see a microcosm that reflects a greater macrocosm in terms of an emerging set of values. What we found is that everyone has a set of tensions in their work and life, things like making a living and making a life, or fulfilling the organization ’ s needs and the needs of the individual. We navigate those tensions using our own unique strategies, for example, working remotely, learning new skills, or self care. Navigating these tensions is critical, because a lot is at stake for us when we show up to work, like our sense of purpose or our connection to our families and communities. And for me, in this organization, this is why we should care about designing a more enlightened way of working. KK : We see a company as a vehicle to design life with as opposed to something to conform to. JB : WearWorks is really thinking holistically about navigating their tensions. For example, when it comes to a tension like making a living and making a life, what kinds of cultural practices and policies need to be in place to support a strategy like flexible or remote work? KK : At WearWorks, we have this policy called WhereverWorks, and the basic idea is that we don ’ t really care where people are as long as the work gets done. And by doing so, we make our employees ridiculously happy. And by doing that, we get the best work from people. JB : We ’ re seeing that people everywhere are challenging these inherited, default ways of working. They want access to more choices, more agency, the freedom to decide how they want to work. This supports them to better navigate the tensions in their work and life for the sake of preserving what ’ s at stake for them. KK : We did a six-month sprint, and on November 5, 2017, Simon ran the first 15 miles of the New York City Marathon without any sight assistance. Working with the blind and visually impaired was a wonderful challenge. It ’ s a different feeling when you go to work every day and the thing that you do actually helped the needle of humanity, in some way, move forward. JB : WearWorks embodies what an enlightened way of working means for people out in the world today.

Text Sample 920: NEG Dim 6 – 2018 – Score 1.00 – 2018/t000026.txt

Every year, more than four to five million people die due to exposure to outdoor air pollution around the world. This petri dish that you are looking at contains approximately 20 minutes ’ worth of pollution captured off a pyrolysis plant. This is PM 2. 5. These particles â you can see it right now, but when they ’ re out there in the air, you won ’ t see them. These are so tiny that our lungs â our bodies cannot filter them, and they end up in our bodies â give us asthma and lung cancer if not treated in the right time. On a trip back to India, when I was a student in 2012, I took this picture. This picture stuck in my head. On one side, you see this exhaust of a diesel generator, the same generator which is a sign of human progress, which is a sign of rapid industrialization and what we have become as a society in the last 100 years, generating energy. But on the other side, you see this very interesting triangular, black-colored swatch, that is produced by the same residual particulate waste created by the emissions of the generator. Now, this picture gave me an idea and got me thinking about rethinking both pollution and inks, because it was making that black-colored mark. Now, the reality is that most of the black ink that we use conventionally is traditionally produced by conventionally burning fossil fuels in factories. There are factories around the world that are burning fossil fuels to produce carbon black, to make black inks that we use on an everyday basis. But given that millions of liters of fossil fuels are already being burned out there by our cars, our engines and our exhaust out there, what if you could capture that pollution and use it to recycle and make those inks? I decided to give this experiment a shot. I went back to my lab back in Boston and conducted a small experiment. In Boston, I couldn ’ t find much pollution to play with, so I resorted to using a candle. This was an experiment. I burnt a candle, built this contraption that would suck in that candle soot, mixed it with some vegetable oil and vodka, because to a DIY hacker, these were really easily available. And after mixing them, you could churn out a very rudimentary form of ink that would go into a cartridge, and now you could print with it. This was my”Hello, World!”of experimenting with printing with pollution. This is the same pollution that I showed you in the petri dish, which is the result of any fossil fuel that is being burned out there. In 2015, I decided to take this experimentation forward and set up a lab in India to work on the capture and recycling of air pollution. In the good times, the lab used to look something like this. But experimentations were not always controlled, and disasters happened. And while experimentation would happen, the lab would end up looking something like this. Well, we knew where we wanted to go, but we were not sure how exactly to reach there. The passersby who used to go by that lab through that building used to, at times, think,”These guys are making bombs in there,”because there was too much fire, wires and smoke in the same vicinity. We decided, let ’ s move to a garage and take experiments forward. We took a garage, and during the early stages, we were driving around Bangalore with contraptions like these. This is an early-stage prototype. Imagine the looks people gave us,”What are these cars driving around doing?”This is an early- stage prototype of our system that would capture pollution that is being released from a conventional diesel-based car. This is an early stage of the technology. We advanced the technology and created this into this version that would capture pollution from static sources of pollution, like a diesel generator. If you see, all the fumes disappear as soon as you turn this machine on. Without affecting the performance of the engine, we are able to capture 95 percent worth of pollution released from the diesel generator. This is the particulate matter that we are talking about that we capture, in this case, within three to four hours of operation of a generator. And while our experiments and our research was advancing, a very big company, a very big brand, approached us and said,”We want to take this idea further with you guys, and take this further in a very big celebrated form.”They said,”Let ’ s do a global art campaign with the inks that you are making off this pollution.”I ’ ll show you what the ink looks like. So, this pen is made by recycling 40 to 50 minutes of that car pollution that we are talking about, the same pollution that is in the petri dish. And it ’ s a very sharp black that you can write with. So I ’ m going to write... PM 2. 5, that ’ s incorrect. So this is a very sharp black that is generated by the same pollution. After much work on the lab-level research, we got an offer from a big corporation to do a very big trial of this idea. And it happened to be a brand, and we didn ’ t think twice. We said,”Let ’ s go ahead.”Inventing in the lab is one thing and taking ideas and deploying them in the real world is completely another. During early stages, we had to resort to using our own houses and own kitchens as our ink-making factories, and our own bedrooms and living rooms as the first assembly line for making these inks. This is my cofounder Nikhil ’ s own bedroom, that is being used to supply inks to artists all around the world, who would paint with AIR-INK. And that ’ s him, delivering AIR-INKs to the ports so that the artists around the world can use it. Soon, we started seeing that thousands of artists around the world started using AIR-INK, and artworks started emerging like this. Soon, thousands of black-and-white, pollution-made artworks started emerging on a global scale. And believe me, for a group of scientists and engineers and inventors, there was nothing more satisfying than that the product of their work is now being used by some of the finest artists around the world. This is the cover of”Contagious”magazine last year, that was done by using the same ink that we made back in our labs. This is a famous painting by the British artist, Christian Furr, who painted it for the song”Paint It Black”by The Rolling Stones. Now, there ’ s more to this pen and this ink than just the popular and pop-culture artworks. And now our goal is to create a company that can actually make some black money â I mean, just money â and high-quality printing processes and inks that can replace the conventional black inks that have been produced for the last thousands of years around the world. Soon after our growing popularity and artworks around the world, we started facing a very different kind of a problem. We started getting spammed by polluters, who would send us bags full of pollution to our office address, asking us,”What can we do with this pollution?”Our lab back in Bombay right now has pollution samples that have come from London, from India, from China, you name it. And this is just the beginning. This polluter sent us this specific image, asking us that these are all bags filled with PM 2. 5, and can we recycle it for him if we paid him some money. Well, what would he have done if we did not take that pollution? He would probably find a nearby river or a landfill and dump it over there. But now, because we had the economics of AIR- INK figured out on the other side, we could incentivize him to give us this pollution and make inks from it, and turn it into even more valuable products. Now, pollution, as we all know, is a global killer. We can ’ t claim that our ink will solve the world ’ s pollution problem. But it does show what can be done if you look at this problem slightly differently. Look at this T-shirt I ’ m holding right now. This is made from the same AIR-INK I ’ m talking about. It ’ s made from the same pollution that is inside this petri dish. And the same pollution we are all breathing in when we are walking outdoors. And we are on our way to do better than this. Thank you very much.

Text Sample 921: NEG Dim 6 – 2018 – Score 1.00 – 2018/t000025.txt

Aja Monet : Our story begins like all great, young love stories. Phillip Agnew : She slid in my DMs... AM : He liked about 50 of my photos, back-to-back, in the middle of the night â PA : What I saw was an artist committed to truth and justice â and she ’ s beautiful, but I digress. AM : Our story actually begins across many worlds, over maqluba and red wine in Palestine. But how did we get there? PA : Well, I was born in Chicago, the son of a preacher and a teacher. My ears first rung with church songs sung by my mother on Saturday mornings. My father ’ s South Side sermons summoned me. My first words were more notes than quotes. It was music that molded me. Later on, it was Florida A&M University that first introduced me to organizing. In 2012, a young black male named Trayvon Martin was murdered, and it changed my life and millions of others ’. We were a ragtag group of college kids and not-quite adults who had decided enough was enough. Art and organizing became our answer to anger and anxiety. We built a movement and it traveled around the world and to Palestine, in 2015. AM : I was born to a single mother in the Pink House projects of Brooklyn, New York. Maddened by survival, I gravitated inwards towards books, poems and my brother ' s hand-me-down Walkman. I saw train-station theater, subwoofing streets and hood murals. In high school, I found a community of metaphor magicians and truth-telling poets in an organization called Urban Word NYC. Adopted by the Black Arts movement, I won the legendary Nuyorican Poets Cafe Grand Slam title. At Sarah Lawrence College, I worked with artists to respond to Hurricane Katrina and the earthquake ; I discovered the impact of poetry and the ability to not just articulate our feelings, but to get us to work towards changing things and doing something about it, when a friend, Maytha Alhassen, invited me to Palestine... PA : We were a delegation of artists and organizers, and we immersed ourselves in Palestinian culture, music, their stories. Late into the night, we would have discussions about the role of art in politics and the role of politics in art. Aja and I disagree. AM : Oh, we disagree. PA : But we quite quickly and unsurprisingly fell in love. Exhibit A : me working my magic. AM : Obvious, isn ' t it? Four months later, this artist â PA : and this organizer â AM : moved into a little home with a big backyard, in Miami. PA : Listen, five months before this ever happened, I predicted it all. I ' m going to tell you â a friend sat me down and said,"You ' ve done so much for organizing, when are you going to settle down?"I looked him straight in the face and I said,"The only way that it would ever happen is if it is a collision. This woman would have to knock me completely off course."I didn ' t know how right I was. Our first few months were like any between young lovers : filled with hot, passionate, all-night... AM : nonstop... PA : discussions. PA : Aja challenged everything I knew and understood about the world. She forced me â AM : lovingly â PA : to see our organizing work with new eyes. She helped me see the unseen things and how artists illuminate our interior worlds. AM : There were many days I did not want to get up out of bed and face the exterior world. I was discouraged. There was so much loss and death and artists were being used to numb, lull and exploit. While winning awards, accolades and grants soothed so many egos, people were still dying and I was seeking community. Meeting Phillip brought so much joy, love, truth into my life, and it pulled me out of isolation. He showed me that community and relationships wasn ' t just about building great movements. It was integral in creating powerful, meaningful art, and neither could be done in solitude. PA : Yeah, we realized many of our artist and organizer friends were also lost in these cycles of sadness, and we were in movements that often found themselves at funerals. We asked ourselves what becomes of a generation all too familiar with the untimely ends of lives streamed daily on our Timelines? It was during one of our late-night discussions that we saw beyond art and organizing and began to see that art was organizing. AM : The idea was set : art was an anchor, not an accessory to movement. Our home was a home of radical imagination ; an instrument of our nurturing hearts ; a place of risk where were dared to laugh, love, cry, debate. Art, books, records and all this stuff decorated our walls, and there was lizards â walls of palm trees that guided our guests into our backyard, where our neighbors would come and feel right at home. The wind â the wind was an affirmation for the people who walked into the space. And we learned that in a world â a bewildering world of so much distraction â we were able to cultivate a space where people could come and be present, and artists and organizers could find refuge. PA : This became Smoke Signals Studio. AM : As we struggle to clothe, house, feed and educate our communities ; our spirits hunger for connection, joy and purpose ; and as our bodies are out on the front lines, our souls still need to be fed, or else we succumb to despair and depression. Our art possesses rhythmic communication, coded emotional cues, improvised feelings of critical thought. Our social movements should be like jazz : encouraging active participation, listening, spontaneity and freedom. What people see as a party... PA : is actually a movement meeting. See, we aren ’ t all protest and pain. Here ’ s a place to be loved, to be felt, to be heard, and where we prepare for the most pressing political issues in our neighborhoods. See, laws never change culture, but culture always changes laws. Art â Art as organizing is even changing and opening doors in places seen as the opposite of freedom. Our weekly poetry series is transforming the lives of men incarcerated at Dade Correctional, and we ’ re so excited to bring you all the published work of one of those men, Echo Martinez. In the intro, he says... AM :”Poetry for the people is a sick pen ’ s penicillin. It ’ s a cuff key to a prisoner ’ s dreams. The Molotov in the ink. It is knowledge, it is overstanding, it is tasting ingredients in everything you ’ ve been force-fed, but most of all, it ’ s a reminder that we all have voices, we all can be heard even if we have to scream.”In 2018, we created our first annual Maroon Poetry Festival at the TACOLCY Center in Liberty City. There, the Last Poets, Sonia Sanchez, Emory Douglas and the late, great Ntozake Shange, performed and met with local artists and organizers. We were able to honor them for their commitment to radical truth- telling. And in addition to that, we transformed a public park into the physical manifestation of the world we are organizing for. Everything that we put into poetry, we put into the art, into the creativity, into the curated kids ’ games and into the stunning stage design. PA : Our work is in a long line of cultural organizers that understood to use art to animate a radical future. Artists like June Jordan, Emory Douglas and Nina Simone. They understood what many of us are just now realizing â that to get people to build the ship, you ’ ve got to get them to long for the sea ; that data rarely moves people, but great art always does. This understanding â This understanding informed the thinking behind the Dream Defenders ’”Freedom Papers,”a radical political vision for the future of Florida that talked about people over profits. Now, we could have done a policy paper. Instead, artists and organizers came together in their poetry to create incredible murals and did the video that we see behind us. We joined the political precision of the Black Panther Party and the beautiful poetry of Puerto Rican poet MartĂn Espada to bring our political vision to life. AM : Now thousands of Floridians across age, race, gender and class see the”Freedom Papers”as a vision for the future of their lives. For decades, our artists and our art has been used to exploit, lull, numb, sell things to us and to displace our communities, but we believe that the personal is political and the heart is measured by what is done, not what one feels. And so art as organizing is not just concerned with artists ’ intentions, but their actual impact. Great art is not a monologue. Great art is a dialogue between the artist and the people. PA : Four years ago, this artist... AM : and this organizer... PA : found that we were not just a match. AM : We were a mirror. PA : Our worlds truly did collide, and in many ways... AM : they combined. PA : We learned so much about movement, about love and about art at its most impactful : when it articulates the impossible and when it erodes individualism, when it plays into the gray places of our black and white worlds, when it does what our democracy does not, when it reminds us that we are not islands, when it adorns every street but Wall Street and Madison Avenue, when it reminds us that we are not islands and refuses to succumb to the numbness, when it indicts empire and inspires each and every one of us to love, tell the truth and make revolution irresistible. AM : For the wizards â AM : For the wizards and ways of our defiance, love-riot visions of our rising, risen, raised selves. The overcoming grace â fires, bitter tongues, wise as rickety rocking chairs, suffering salt and sand skies. Memories unshackled and shining stitches on a stretch-marked heart. For the flowers that bloom in midnight scars. How we suffered and sought a North Star. When there was no light, we glowed. We sparked this rejoice, this righteous delight. We have a cause to take joy in. How we weathered and persisted, tenacious, no stone unturned. How we witnessed the horror of mankind and did not become that which horrified us. PA : Thank you. AM : Thank you.

Text Sample 922: NEG Dim 6 – 2018 – Score 1.00 – 2018/t000005.txt
So this is the first time I ’ ve told this story in public, the personal aspects of it. Yogi Berra was a world-famous baseball player who said,”If you come to a fork in the road, take it.”Researchers had been, for more than a century, studying the immune system as a way to fight cancer, and cancer vaccines have, unfortunately, been disappointing. They ’ ve only worked in cancers caused by viruses, like cervical cancer or liver cancer. So cancer researchers basically gave up on the idea of using the immune system to fight cancer. And the immune system, in any case, did not evolve to fight cancer ; it evolved to fight pathogens invading from the outside. So its job is to kill bacteria and viruses. And the reason the immune system has trouble with most cancers is that it doesn ’ t invade from the outside ; it evolves from its own cells. And so either the immune system does not recognize the cancer as a problem, or it attacks a cancer and also our normal cells, leading to autoimmune diseases like colitis or multiple sclerosis. So how do you get around that? Our answer turned out to be synthetic immune systems that are designed to recognize and kill cancer cells. That ’ s right — I said a synthetic immune system. You do that with genetic engineering and synthetic biology. We did it with the naturally occurring parts of the immune system, called B cells and T cells. These were our building blocks. T cells have evolved to kill cells infected with viruses, and B cells are the cells that make antibodies that are secreted and then bind to kill bacteria. Well, what if you combined these two functions in a way that was designed to repurpose them to fight cancer? We realized it would be possible to insert the genes for antibodies from B cells into T cells. So how do you do that? Well, we used an HIV virus as a Trojan horse to get past the T cells ’ immune system. The result is a chimera, a fantastic fire-breathing creature from Greek mythology, with a lion ’ s head, a goat ’ s body and a serpent ’ s tail. So we decided that the paradoxical thing that we had created with our B-cell antibodies, our T cells carrier and the HIV Trojan horse should be called”Chimeric Antigen Receptor T cells,”or CAR T cells. The virus also inserts genetic information to activate the T cells and program them into their killing mode. So when CAR T cells are injected into somebody with cancer, what happens when those CAR T cells see and bind to their tumor target? They act like supercharged killer T cells on steroids. They start this crash-defense buildup system in the body and literally divide and multiply by the millions, where they then attack and kill the tumor. All of this means that CAR T cells are the first living drug in medicine. CAR T cells break the mold. Unlike normal drugs that you take — they do their job and get metabolized, and then you have to take them again — CAR T cells stay alive and on the job for years. We have had CAR T cells stay in the bodies of our cancer patients now for more than eight years. And these designer cancer T cells, CAR T cells, have a calculated half-life of more than 17 years. So one infusion can do the job ; they stay on patrol for the rest of your life. This is the beginning of a new paradigm in medicine. Now, there was one major challenge to these T-cell infusions. The only source of T cells that will work in a patient are your own T cells, unless you happen to have an identical

twin. So for most of us, we ’ re out of luck. So what we did was to make CAR T cells. We had to learn to grow the patient ’ s own T cells. And we developed a robust platform for this in the 1990s. Then in 1997, we first tested CAR T cells in patients with advanced HIV-AIDS. And we found that those CAR T cells survived in the patients for more than a decade. And it improved their immune system and decreased their viruses, but it didn ’ t cure them. So we went back to the laboratory, and over the next decade made improvements to the CAR T cell design. And by 2010, we began treating leukemia patients. And our team treated three patients with advanced chronic lymphocytic leukemia in 2012. It ’ s a form of incurable leukemia that afflicts approximately 20, 000 adults every year in the United States. The first patient that we treated was a retired Marine sergeant and a prison corrections officer. He had only weeks to live and had, in fact, already paid for his funeral. The cells were infused, and within days, he had high fevers. He developed multiple organ failures, was transferred to the ICU and was comatose. We thought he would die, and, in fact, he was given last rites. But then, another fork in the road happened. So, around 28 days after the CAR T cell infusion, he woke up, and the physicians finally examined him, and the cancer was gone. The big masses that had been there had melted. Bone marrow biopsies found no evidence of leukemia, and that year, in our first three patients we treated, two of three have had durable remissions now for eight years, and one had a partial remission. The CAR T cells had attacked the leukemia in these patients and had dissolved between 2. 9 and 7. 7 pounds of tumor in each patient. Their bodies had become veritable bioreactors for these CAR T cells, producing millions and millions of CAR T cells in the bone marrow, blood and tumor masses. And we discovered that these CAR T cells can punch far above their weight class, to use a boxing analogy. Just one CAR T cell can kill 1, 000 tumor cells. That ’ s right — it ’ s a ratio of one to a thousand. The CAR T cell and its daughter progeny cells can divide and divide and divide in the body until the last tumor cell is gone. There ’ s no precedent for this in cancer medicine. The first two patients who had full remission remain today leukemia-free, and we think they are cured. These are people who had run out of options, and by all traditional methods they had, they were like modern-day Lazarus cases. All I can say is : thank goodness for those forks in the road. Our next step was to get permission to treat children with acute leukemia, the most common form of cancer in kids. The first patient we enrolled on the trial was Emily Whitehead, and at that time, she was six years old. She had gone through a series of chemotherapy and radiation treatments over several years, and her leukemia had always come back. In fact, it had come back three times. When we first saw her, Emily was very ill. Her official diagnosis was advanced, incurable leukemia. Cancer had invaded her bone marrow, her liver, her spleen. And when we infused her with the CAR T cells in the spring of April 2012, over the next few days, she did not get better. She got worse, and in fact, much worse. As our prison corrections officer had in 2010, she, in 2012, was admitted to the ICU, and this was the scariest fork in the whole road of this story. By day three, she was comatose and on life support for kidney failure, lung failure and coma. Her fever was as high as 106 degrees Fahrenheit for three days. And we didn ’ t know what was causing those fevers. We did all the standard blood tests for infections, and we could not find an infectious cause for her fever. But we did find something very unusual in her blood that had never been seen before in medicine. She had elevated levels of a protein called interleukin-6, or IL-6, in her blood. It was, in fact, more than a thousandfold above the normal levels. And here ’ s where yet another fork in the road came in. By sheer coincidence, one of my daughters has a form of pediatric arthritis. And as a result, I had been following as a cancer doc, experimental therapies for arthritis for my daughter, in case she would need them. And it so happened that just months before Emily was admitted to the hospital, a new therapy had been approved by the FDA to treat elevated levels of interleukin-6. And it was approved for the arthritis that my daughter had. It ' s called tocilizumab. And, in fact, it had just been added to the pharmacy at Emily ' s hospital, for arthritis. So when we found Emily had these very high levels of IL-6, I called her doctors in the ICU and said,"Why don ' t you treat her with this arthritis drug?"They said I was a cowboy for suggesting that. And since her fever and low blood pressure had not responded to any other therapy, her doctor quickly asked permission to the institutional review board, her parents, and everybody, of course, said yes. And they tried it, and the results were nothing short of striking. Within hours after treatment with tocilizumab, Emily began to improve very rapidly. Twenty-three days after her treatment, she was declared cancer-free. And today, she ' s 12 years old and still in remission. So we now call this violent reaction of the high fevers and coma, following CAR T cells, cytokine release syndrome, or CRS. We ' ve found that it occurs in nearly all patients who respond to the therapy. But it does not happen in those patients who fail to respond. So paradoxically, our patients now hope for these high fevers after therapy, which feels like"the worst flu in their life,"when they get CAR T-cell therapies. They hope for this reaction because they know it ' s part of the twisting and turning path back to health. Unfortunately, not every patient recovers. Patients who do not get CRS are often those who are not cured. So there ' s a strong link now between CRS and the ability of the immune system to eradicate leukemia. That ' s why last summer, when the FDA approved CAR T cells for leukemia, they also co-approved the use of tocilizumab to block the IL-6 effects and the accompanying CRS in these patients. That was a very unusual event in medical history. Emily ' s doctors have now completed further trials and reported that 27 out of 30 patients, the first 30 we treated, or 90 percent, had a complete remission after CAR T cells, within a month. A 90 percent complete remission rate in patients with advanced cancer is unheard of in more than 50 years of cancer research. In fact, companies often declare success in a cancer trial if 15 percent of the patients had a complete response rate. A remarkable study appeared in the"New England Journal of Medicine"in 2013. An international study has since confirmed those results. And that led to the approval by the FDA for pediatric and young adult leukemia in August of 2017. So as a first-ever approval of a cell and gene therapy, CAR T-cell therapy has also been tested now in adults with refractory lymphoma. This disease afflicts about 20, 000 a year in the United States. The results were equally impressive and have been durable to date. And six months ago, the FDA approved the therapy of this advanced lymphoma with CAR T cells. So now there are many labs and physicians and scientists around the world who have tested CAR T cells across many different diseases, and understandably, we ' re all thrilled with the rapid pace of advancement. We ' re so grateful to see patients who were formerly terminal return to healthy lives, as Emily has. We ' re thrilled to see long remissions that may, in fact, be a cure. At the same time, we ' re also concerned about the financial cost. It can cost up to 150, 000 dollars to make the CAR T cells for each patient. And when you add in the cost of treating CRS and other complications, the cost can reach one million dollars per patient. We must remember that the cost of failure, though, is even worse. The current noncurative therapies for cancer are also expensive and, in addition, the patient dies. So, of course, we ' d like to see research done now to make this more efficient and increase affordability to all patients. Fortunately, this is a new and evolving field, and as with many other new therapies and services, prices will come down as industry learns to do things more efficiently. When I think about all the forks in the road that have led to CAR T-cell therapy, there is one thing that strikes me as very important. We ’ re reminded that discoveries of this magnitude don ’ t happen overnight. CAR T-cell therapies came to us after a 30-year journey, along a road full of setbacks and surprises. In all this world of instant gratification and 24 / 7, on-demand results, scientists require persistence, vision and patience to rise above all that. They can see that the fork in the road is not always a dilemma or a detour ; sometimes, even though we may not know it at the time, the fork is the way home. Thank you very much.

Text Sample 923: NEG Dim 6 – 2017 – Score 1.00 – 2017/t000607.txt

I come from Egypt, which is also called Umm al-Dunya, the Mother of the World. It ’ s a rich country filled with stories of rebellion, stories of civilizational triumph and downfall and the rich, religious, ethnic, cultural and linguistic diversity. Growing up in such an environment, I became a strong believer in the power of storytelling. As I searched for the medium with which to tell my story, I stumbled upon graphic design. I would like to share with you a project of how graphic design can bring the Arabic language to life. But first, let me tell you why I want to do this. I believe that graphic design can change the world. At least in my very own city of Cairo, it helped overthrow two separate dictators. As you can see from those photos, the power and potential of graphic design as a tool for positive change is undeniably strong. Egypt ’ s 2011 revolution was also a grassroots design revolution. Everyone became a creator. People were the real designers and, just overnight, Cairo was flooded with posters, signage, graffiti. Visual communication was the medium that spoke far louder than words when the population of over 90 million voices were suppressed for almost 30 years. It was precisely this political and social suppression, coupled with decades of colonialism and miseducation that slowly eroded the significance of the Arabic script in the region. All of these countries once used Arabic. Now it ’ s just the green and the blue. To put it simply, the Arabic script is dying. In postcolonial Arab countries functioning in an increasingly globalized world, it is a growing alarm that less and less people are using the Arabic script to communicate. As I was studying my master ’ s in Italy, I noticed myself missing Arabic. I missed looking at the letters, digesting their meaning. So one day, I walked into one of the biggest libraries in Italy in search of an Arabic book. I was surprised to find that this is what they had under the category of”Arabic / Middle Eastern books.”Fear, terrorism and destruction. One word : ISIS. My heart ached that this is how we are portrayed to the world, even from a literary perspective. I asked myself : Whatever happened to the world-renowned writers like Naguib Mahfouz, Khalil Gibran, iconic poets like Mutanabbi, Nizar Qabbani? Think about this. The cultural products of an entire region of the world, as rich, as diverse, have been deemed redundant, if not ignored altogether. The cultural products of an entire region of the world have been barred from imparting any kind of real impact on global media productions and contemporary social discourse. And then I reminded myself of my number one belief : design can change the world. All you need is for someone to catch a glimpse of your work, feel, connect. And so I started. I thought about how can I stop the world from seeing us as evil, as terrorists of this planet, and start perceiving us as equals, fellow humans? How can I save and honor the Arabic script and share it with other people, other cultures? And then it hit me : What if I combined the two most significant symbols of innocence and Arab identity? Maybe then people could resonate. What ’ s more pure, innocent and fun as LEGO? It ’ s a universal child ’ s toy. You play with them, you build with them, and with them, you imagine endless possibilities. My eureka moment was to find a bilingual solution for Arabic education, because effective communication and education is the road to more tolerant communities. However, the Arabic and Latin scripts do not only represent different worlds but also create technical difficulties for both Eastern and Western communities on a daily basis. There are so many reasons why Arabic and Latin are different, but here are some of the main ones. Yes, both use upward and downward strokes, but have completely different baselines. Arabic tends to be more calligraphic and connectivity is important to the Arabic language, whose letters have to be mostly joined in order to articulate a given word. It also uses an entirely different system of punctuation and diacritics. But most importantly, Arabic has no capital letters. Instead it has four different letter forms : initial, medial, isolated and final. I want to introduce the Arabic language to young learners, foreign speakers, but most importantly help refugees integrate to their host societies through creating a bilingual learning system, a two-way flow of communication. And I called it"Let ' s Play."The idea is to simply create a fun and engaging way of learning Modern Standard Arabic through LEGO. These are the two words."Let ' s Play."Every colored bar marks an Arabic letter. As you can see, the letter is explained in form, sound and examples of words in function, in addition to the equivalent in Latin. Together, they form a fun pocket book with the 29 Arabic letters and the four different forms, plus a 400-word dictionary. So this is how the page looks like. You have the letter, the transliteration in Latin and the description underneath. I ' ll take you through the process. So first in my tiny studio in Florence, I built the letters. I photographed each letter separately, and then I retouched every letter and chose the correct color background and typefaces to use. Ultimately, I created the full letter set, which is 29 letters times four different forms. That ' s 116 letters build just in one week. I believe that information should and can be fun, portable. This book is the final product, which I would eventually like to publish and translate into as many languages in the world, so that Arabic teaching and learning becomes fun, easy and accessible globally. With this book, I hope to save my nation ' s beautiful script. Thank you. Working on this project was a form of visual meditation, like a Sufi dance, a prayer to a better planet. One set of building blocks made two languages. LEGO is just a metaphor. It ' s because we are all made of the same building unit, is that I can see a future where the barriers between people all come tumbling down. So no matter how ugly the world around us gets, or how many discouraging books on ISIS, the terrorist group, and not Isis, the ancient Egyptian goddess, continue to be published, I will keep building one colorful world. Shukran, which means"thank you."Thank you. Thank you so much. Thank you.

Text Sample 924: NEG Dim 6 – 2017 – Score 1.00 – 2017/t000603.txt

Hey look, if you guys are anything like me, you have found it harder and harder to turn around recently without seeing words like"free-range,""farm-to- table,""organically produced,"especially here in Colorado. Now, as we ' ve become more conscientious of the way that we eat in recent years, these once unfamiliar words have worked their way into our daily lexicon. When we started to pay more attention to the way that the food we were eating interacted with our bodies and with the earth, the food industry had to listen. And the results have been really powerful. Now, those of you out there from states like Washington and Oregon and, of course, my fellow Coloradans — y ' all know what I ' m talking about. Because this is not... Words like"all- natural"and"homegrown"are not just being used in our diets. There ' s this whole new industry using this language now. You guys know. It ' s weed, an industry that taxed a sale of about six billion dollars worth of product in 2016. So what if I were to propose to you that some of what you think you know about this legalized marijuana thing could be wrong? Listen, I get it — talking about issues with legal weed is a pretty quick way to get uninvited from the cool kids ' table. I know that better than most, but I intend to do it anyway. First, before I get started, let me be perfectly clear about one thing : my fight is not against the casual adult use of marijuana — I don ' t care about that. What I care deeply about is this new industry that is working to convince us that we are consuming something natural while fixing social ills, when we aren ' t. So let ' s start with a little bit of Weed 101. Cannabis is a plant that grows naturally and has been used within textiles and even traditional Chinese medicine for thousands of years. Genesis 1:12 even tells us :"I have given you all of the seed-bearing plants and herbs to use."It ' s the microphone — it ' s got a TV preacher sort of thing. Now, cannabis is made up of hundreds of different chemicals, but two of those chemicals are by far the most interesting. That ' s CBD and THC. CBD is where almost all of the medicinal properties lie. It ' s an incredibly fascinating part of the plant with real potential to help people. It also is totally nonintoxicating. You could take a bath in the stuff while vaping pure CBD and drinking a CBD smoothie, and you still couldn ' t get high. I ' ve tried. I haven ' t, I haven ' t. That ' d cost a lot of money. Now, for as interesting and remarkable a part of the plant as CBD is, it actually makes up a really tiny portion of the commercial market. The real money is being made in that other chemical — in THC. THC is the natural part of the plant that gets you high. And before the 1970s, cannabis contained less than half of a percent of THC. That ' s what ' s naturally occurring. Over the last 40 years, as we became better gardeners, that — that percentage of THC started to slowly but steadily rise, until recently, when the chemists started to get involved. So these guys moved grow cycles — sorry — these guys moved cultivation exclusively indoors, and they made grow cycles extremely and unnaturally short. They also started to use pesticides and fertilizers in some ways that we should be concerned with. In fact, I was recently talking to a buddy who had just left a job at a commercial grow operation because he was so concerned with the chemicals that he was being asked to interact with. Some of his fellow employees were actually encouraged to wear hazmat suits while they were spraying the chemical cocktails on the plants. With that kind of manipulation, the products that are being sold today can contain above 30 percent THC. And our concentrates — our concentrates can actually contain above 95 percent THC — a far cry from the natural plant. Listen, this isn ' t your grandpa ' s weed. This isn ' t your dad ' s weed. Like, this isn ' t even my weed. If you ' ve ever set foot inside one of the thousands of dispensaries that have sprung up in recent years, you know that what we ' re really selling in them is THC. All of the weed that you buy commercially lists exactly how much THC it contains, as do our other, much more popular products like vape pens, coffee, ice cream, condiments, granola, gum, candy, baked goods, suppositories. And, of course, lube. Pretty much — no, for real — pretty much anything that you can imagine introducing into the human body. The vast majority of cannabis that ' s being sold today — it isn ' t really cannabis. It ' s THC in either a pure form or in an extremely high and unnatural concentration. To say that we have legalized weed is subtly misleading. We have commercialized THC. And it ' s happened really quickly. Now, the reason why the commercial market has so rapidly exploded is because there is a hell of a lot of money to be made in satisfying and increasing our desire to get high. And that money is no longer really being made by the mom- and-pop shops. So industry groups and corporations —

groups like the Drug Policy Alliance, the Marijuana Policy Project, Arcview Investment, the Cannabis Industry Association — they ’ ve chased out and helped to chase out a lot of the small-time growers. So these cats know that the best way to continue to profit off of us is if they follow the alcohol industry ’ s 80 / 20 rule. It ’ s simple — it ’ s where 80 percent of the product is consumed by 20 percent of the consumers — the problemed users. The wealthy, white, weed lobbyists — and seriously, they are almost all rich, white men — they know that we will consume more of what they ’ re selling if they jack up the potency. They also know that we are more than twice as likely to consume THC regularly if we earn under 20, 000 dollars a year than those who earn over 50, 000 dollars a year. In other words, the poorer you are, the more likely you are to spend your money on their products. And in this country, income and race are highly correlated. One of the reasons we often hear cited for the legalization of marijuana is that it will help to stop the disproportionate incarceration rates among minorities, which is something everybody in this room should be extremely concerned with. Unfortunately, we don ’ t have to look any further than arrest rates for juveniles here in Colorado to counter that argument. According to the Colorado Department of Public Safety, since we opened retail in 2014 — almost all of which are in poor, minority neighborhoods — we saw an eight percent reduction in the arrest of white kids for all weed-related activity. Good on ’ em. During that same time period, there was a 29 percent increase in the arrest of Hispanic kids for weed-related activity and a 58 percent increase in the arrest of black kids for weed-related crimes. You guys heard that, right? We are actually arresting more people of color in Colorado than we were prior to commercialization. And you ’ re not reading that in the Post. Colorado Department of Safety. Legal marijuana coming into focus. Another big issue that we have is in school suspension rates. So, schools that are predominantly white — that is, they have a minority population of 25 percent or fewer — in the first full year of data collection following commercialization, these schools had a grand total of 190 drug-related suspensions, almost all of which are for THC. At the same time, schools with a minority population of 75 to 100 percent had 801 drug-related suspensions, almost all of which were for THC. When discussing minority populations, one that unfortunately often gets left out of the conversation is the LGBTQ community. Members of this community are more than twice as likely to consume THC than those who identify as heterosexual or cisgender. They also, unfortunately, have higher rates of mental illness and suicide. According to a study published in 2014 called”Going to Pot,”we see that the unnaturally high levels of THC found in today ’ s products actually compound those issues. They make them worse. Unfortunately, that seems to matter very little to the folks who are selling these products, because as you just saw, clearly, this is a good consumer base. Listen, man — I get it. In many circles, legalized marijuana is too much of a sacred cow to question. But we need to start this conversation, because what ’ s being sold today is not natural, and lobbyists and industry are using social justice as a smoke screen so that they can get richer. It ’ s been my own journey to sobriety that led me to begin questioning a lot of what I was seeing ; that ’ s kind of one of the things that we ’ re taught to do. When I left Boulder for the Washington, DC, area at 12 years old, I was transported into a world where the kind of shoes you wore mattered more than just about anything else. And my family was just too poor to help me play that game. So I was faced with a pretty real crisis of identity. In this new scene where there ’ s more blacktop than treetops, man, I just didn ’ t know who I was. So I smoked weed for the first time when I was 13. And I loved it. I instantly found this social group, and I also just really liked being high. I finally found a way to shut this up. I quickly turned to other drugs and alcohol, and something just woke up inside of my brain. I was a daily user within a couple of months. My addictive use mirrors many of the stories that I ' m sure you ' ve heard before. It started out as fun, it got scary, and then it was just necessary. Enough said. I got wasted for the last time on June 15 of 1996. And I — Thank you. And I ' ve spent the last 21 years trying to both put my life back in order as well as trying to find some peace in this world. And one of the ways I ' ve done that is by working inside of nonprofit drug and alcohol treatment for the last 10 years, with groups like Phoenix Multisport, the University of Colorado Hospital and NALGAP — the National Association for Lesbian, Gay, Transgender, Bisexual Treatment Providers and their Allies. Even after all of my work on the front lines and as a former consumer myself, I was shocked and pissed when I started to see what commercialization was doing to cannabis, because, you see, our hope for something pure and natural is making it hard for us to see what ' s really going on, and that is that the rich are getting richer on the backs of the poor and lying to our faces the entire time. Thank you. My friends, once again I fear that we are allowing industry to take advantage of the most challenged among us in order to turn a profit, much like we saw with tobacco and food in years past. So when we told the food industry that we understood the impact our choices were having, and that we demanded better for ourselves and our families, that industry got into line. So is there any reason why we couldn ' t demand the same thing from this and from future industries who are trying to get a piece of our paychecks? What if we made these guys answer some hard questions? What if we held them to a higher standard than we are right now? Because as it stands, for many in our community, the grass isn ' t greener on this side of commercialization. They ' ve just been sold a bag of goods. Thank you. Jeremy Duhon : I know this is a sensitive topic but a very important one, so thank you for bringing this up and helping us explore it. You know, a lot of folks are experiencing health benefits from marijuana and cannabis. What would you say to that part of the community? Ben Cort : I ' m actually glad you brought that up. I think one of the most important things that we can do right now is to separate out medicinal, and especially what ' s happening and some of the advances that are being made using parts of this plant and even some whole-plant medicines, from the commercial market for THC. That ' s, I think, crucial. We ' ve got to stop putting them together, and we ' ve got to say,"OK, here ' s the part about getting high, and here ' s the part about the medicine."JD : So it sounds like your talk is less about being anti-cannabis and more about raising awareness about aspects of commercialization. Is that a fair way to put it? BC : Yes. So, I am not the anti-weed guy. I ' m the pro-logic guy. For me to cast stones — listen, I ' m a drug addict. I don ' t get to do that, and I don ' t want to do that. But what ' s bothering me and what ' s so hard for me is to see the way that we are just embracing without asking the hard questions, when if this was another industry, we ' d be holding their feet to the fire on some stuff. And no, I ' m not the anti-weed guy, I ' m the pro-thought guy. So : think. I don ' t even care if you ' re smoking when you do it, just so long as you ' re an adult. So long as you ' re an adult, just think.

Text Sample 925: NEG Dim 6 – 2017 – Score 1.00 – 2017/t000576.txt

How many of us have ever seen something, thought that we should report it, but decided not to? And not that I need to see a show of hands, but I ' m sure this has happened to someone in this room before. In fact, when this question was asked to a group of employees, 46 percent of them responded by saying that they had seen something and decided not to report it. So if you raised your hand, or quietly raised your hand, don ' t feel bad, you ' re not alone. This message of if you see something to say something is really all around us. Even when driving down the highway, you see billboards like this, encouraging us to report crime without revealing ourselves. But I still feel like a lot of us are really uncomfortable coming forward in the name of the truth. I ' m an accounting professor, and I do fraud research. And in my class, I encourage my students to come forward with information if they see it. Or in other words, encouraging my students to become whistle-blowers. But if I ' m being completely honest with myself, I am really conflicted with this message that I ' m sending to my students. And here ' s why. Whistle-blowers are under attack. Headline after headline shows us this. Many people choose not to become whistle-blowers due to the fear of retaliation. From demotions to death threats, to job loss — perpetual job loss. Choosing to become a whistle- blower is an uphill battle. Their loyalty becomes into question. Their motives, their trustworthiness. So how can I, as a professor who really cares about her students encourage them to become whistle-blowers, when I know how the world truly feels about them? So, one day I was getting ready for my annual whistle-blower lecture with my students. And I was working on an article for"Forbes,"entitled"Wells Fargo and Millennial Whistle-blowing. What Do We Tell Them?"And as I was working on this piece and reading about the case, I became outraged. And what made me angry was when I came to the fact and realized that the employees that tried to whistle-blow were actually fired. And it really made me think about the message that I was sharing with my students. And it made me think : What if my students had been Wells Fargo employees? On the one hand, if they whistle-blew, they would have gotten fired. But on the other hand, if they didn ' t report the frauds that they knew, the way current regulation is written, employees are held responsible if they knew something and didn ' t report it. So criminal prosecution is a real option. What ' s a person supposed to do with those type of odds? I of all people know the valuable contributions that whistle-blowers make. In fact, most frauds are discovered by them. Forty two percent of frauds are discovered by a whistle-blower in comparison to other methods, like measurement review and external audit. And when you think about some of the more classic or historical fraud cases, it always is around a whistle-blower. Think Watergate — discovered by a whistle-blower. Think Enron — discovered by a whistle- blower. And who can forget about Bernard Madoff, discovered by a whistle- blower? It takes a tremendous amount of courage to come forward in the name of the truth. But when we think about the term whistle-blower, we often think of some very descriptive words : rat, snake, traitor, tattletale, weasel. And those are the nice words, the ones I can say from the stage. And so when I ' m not in class, I go around the country and I interview white- collar felons, whistle-blowers and victims of fraud. Because really I ' m trying to understand what makes them tick and to bring those experiences back into the classroom. But it ' s my interviews with whistle-blowers that really stick with me. And they stick with me, because they make me question my own courage. When given the opportunity, would I actually speak up? And so, this is a couple stories that I want to share with you. This is Mary. Mary Willingham is the whistle-blower from the University of North Carolina at Chapel Hill, academic fraud case. And Mary was a learning specialist at the university, and she worked with students, primarily student athletes. And what she noticed, when she was working with students, is they were turning in term papers that seemed well beyond their reading levels. She started to ask a couple of questions and she found out that there was a database where the student athletes could retrieve papers and turn them in. And then she found out that some of her colleagues were funneling students into fake classes, just to keep them eligible to play. Now, when Mary found this out, she was outraged. And so what she tried to do was go to her direct supervisor. But they didn ’ t do anything. And then Mary tried to go to some internal university administrators. And they didn ’ t do anything. So, what happens when nobody listens? You blog. So Mary decided to develop a blog. Her blog went viral within 24 hours, and she was contacted by a reporter. Now, when she was contacted by this reporter, her identity was known. She was exposed. And when she was exposed, she received a demotion, death threats, over collegiate sports. Mary didn ’ t do anything wrong. She didn ’ t participate in the fraud. She really thought that she was giving voice to students that were voiceless. But her loyalty was questioned. Her trustworthiness and her motives. Now, whistle-blowing doesn ’ t always have to end in demotions or death threats. Actually, in 2002, this was the cover of”Time”magazine, where we were actually honoring three brave whistle-blowers for their decision to come forward in the name of the truth. And when you look at the research, 22 percent of whistle-blowers actually report retaliation. So there is a huge population of people that report and are not retaliated against and that gives me hope. So this is Kathe. Kathe Swanson is a retired city clerk from the city of Dixon. And one day, Kathe was doing her job, just like she always did, and she stumbled upon a pretty interesting case. See, Kathe was at the end of the month, and she was doing her treasures report for the city, and typically, her boss, Rita Crundwell, gave her a list of accounts and said,”Kathe, call the bank and get these specific accounts.”And Kathe did her job. But this particular day, Rita was out of town, and Kathe was busy. She picks up the phone, she calls the bank and says,”Fax me all of the accounts.”And when she gets the fax, she sees that there is an account that has some withdrawals and deposits in it that she did not know about. It was an account controlled only by Rita. So Kathe looked at the information, she reported it to her direct supervisor, which was then-mayor Burke, and this led into a huge investigation, a six-month investigation. Come to find out, Kathe ’ s boss, Rita Crundwell, was embezzling money. Rita was embezzling 53 million dollars over a 20-year period, and Kathe just happened to stumble upon it. Kathe is a hero. And actually, I had the opportunity of interviewing Kathe for my documentary,”All the Queen ’ s Horses.”And Kathe wasn ’ t seeking fame. In fact, she really didn ’ t want to talk to me for a really long time, but through strategic stalking, she ended up doing the interview. But she was seeking fairness, not fame. And if it wasn ’ t for Kathe, who ’ s to say this fraud would have ever been discovered? So, remember that”Forbes”article I was talking about, that I was working on before my lecture? Well, I posted it and something really fantastic happened. I started receiving emails from whistle-blowers all over the world. And as I was receiving these emails and responding back to them, there was a common theme in the message that I received, and this is what it was : they all said this,”I blew the whistle, people really hate me now. I got fired, but guess what? I would do it all over again if I could.”And so as I kept reading this message, all these messages, I wanted to think, what could I share with my students? And so, I pulled it all together and this is what I learned. It ’ s important for us to cultivate hope. Whistle-blowers are hopeful. Despite popular belief, they ’ re not all disgruntled employees that have a beef with the company. Their hopefulness really is what drives them to come forward. We also have to cultivate commitment. Whistle-blowers are committed. And it ’ s that passion to their organization that makes them want to come forward. Whistle- blowers are humble. Again, they ’ re not seeking fame, but they are seeking fairness. And we need to continue to cultivate bravery. Whistle-blowers are brave. Often, they underestimated the impact whistle-blowing had on their family, but what they continue to comment on is how hard it is to withhold the truth. With that, I want to leave you with one additional name : Peter Buxtun. Peter Buxtun was a 27-year-old employee for the US Public Health Service. And he was hired to interview people that had sexually transmitted diseases. And through the course of his work, he noticed a clinical study that was going on within the organization. And it was a study that was looking at the progression of untreated syphilis. And so, there were 600 African American males that were in this study. They were enticed into the study through being given free medical exams, burial insurance. And so, what happened through the course of this study, is penicillin was discovered to help treat syphilis. And what Peter noticed was, the participants in this study were not given the penicillin to treat their syphilis. And the participants didn ' t know. So similar to Mary, Peter tried to report and talk to his internal supervisors, but no one listened. And so Peter thought this was completely unfair and he tried to report again, and finally talked to a reporter — very similar to Mary. And in 1972, this was the front page of the"New York Times":"Syphilis Victims in US Study Went Untreated for 40 Years."This is known to us today as the Tuskegee syphilis experiment. And Peter was the whistle-blower. What happened to the 600 men, you may wonder, the 600 original men? Twenty eight men died from syphilis. One hundred died from syphilis complications, forty wives were infected and 10 children were born with congenital syphilis. Who ' s to say what these numbers would be if it wasn ' t for the brave, courageous act of Peter? We ' re all connected to Peter, actually. If you know anybody that ' s in a clinical trial, the reason why we have informed consent today is because of Peter ' s courageous act. So let me ask you a question. That original question, a variation of the original question. How many of us have ever used the term snitch, rat tattletale, snake, weasel, leak? Anybody? Before you get the urge to do that again, I want you to think a little bit. It might be the Mary, the Peter, the Kathes of the world. You might be the person that could shape history, or they could be the person that shapes yours. Thank you.

Text Sample 926: NEG Dim 6 – 2025 – Score 1.00 – 2025/t003001.txt

- I don ' t remember exactly, but... I use the most expensive hair color in the world. It ' s not that I care about the money, it ' s that I care about my hair. It ' s not just the color. I expect great color. What ' s worth more to me is the way my hair feels. Feels good around my neck. I can ' t remember it. I can ' t finish it all. If only we had more time. - Why don ' t we have more time? - I ' m dying. I ' m croaking! - It ' s tough, coming home. But this is what you do for a parent that you love. But, you know? She ' s not my mother. Technically. In my own family, there ' s a lot of dysfunction, a lot of mental illness, a lot of substance abuse. Every generation. My parents divorced when I was about five. My mother moved us out of state, so we had to fly to New York to see my father ' cause he was already working in advertising, and New York was the only place to be. - Many of the commercials you see on television begin life here. - Advertising is one of the main sources of messaging about who you should be. - Nowadays, most of you women lead two lives, combining homemaking with the role of glamorous hostess. - And in the ' 60s, the ads were often made from a male point of view. - In the pre-production meeting, everybody who has anything to do with the commercial is there. - It was mostly men. These superstar art directors and writers. Like my father, Gene Case."Mad Men"was like home movies, and he was Don Draper. He was very witty, but I don ' t think my father realized how powerful his words were. He knew it professionally. But with his kids, he could be incredibly cutting. Flay you just with a phrase. But it ' s the time with my mother that was the nightmare. She had a horrible childhood, but she sure as hell didn ’ t make any effort to overcome that in her relations with us. You know, it was really horrifying to my mother that I was not popular. But I was also really awkward because she had been undermining my confidence for all my life. She would pick apart your personality in detail. I was lazy, I looked bad, I acted bad, I was awkward. Emotionally, just out for blood. At a certain point in the night, I would know,”OK, the situation is dangerous. I need to get outta here.”I had this fantasy when I was in fourth grade that I vividly remember that some kind of outside agency was gonna come in and determine from all the salt on my pillow that I was crying myself to sleep every night and do something about it. No, it was awful. It was awful. And that was the point where Ilon came into my life. - Ilon, do you need anything to eat? - I don ’ t think so, but look up this wallpaper, Waverly Fabrics, and see if they have a rose pattern. - OK. Oh, what about this one here? - That ’ s not it. - No? - No, first of all, I was an art dir... I was a creative director at an advertising agency. The first thing I know is what I ’ m looking at. Ilon the girl. I don ’ t remember particularly being into advertising, but somehow I found myself in it. And I seemed to be very good at it. First of all, I ’ d always listened to everybody. And I noticed that I saw things differently than other people. - Women could get somewhere in advertising, but they were marginalized. - Women were given a lesser role. All the men were always arguing with you and always taking credit. - And one of the things that was coming through loud and clear is that women are for men. - Glory, glory, hallelujah Glory, glory, hallelujah Glory, glory, hallelujah His truth is marching on - Men were always the ones who everyone was pleasing. - What will your eyes say to him tonight? Wait till the fellas take a look at this plunging backline. - You look at ads from that period, and it ’ s jaw-dropping. - This is one of our nation ’ s most beautiful sights, the kind of girl who keeps her figure. - But that ’ s what feminism was pushing back against. - I was a feminist. I knew women were equal if not superior to men from the time I was a little girl. - And in advertising, she wanted to be a trailblazer that way. At one point, she worked in the same agency with my father, and that ’ s how they met. - He was a good writer, and I was a good writer. I liked the fact that he was good at what he did and cared about it. We got engaged, and then we got married. - I ’ m sitting there in the background. I ’ m grinning. I ’ m so thrilled because I adored her instantly. I thought she was super cool. And she treated me like I mattered. - She was a sweet little girl, and she deserved some caretaking. My mother was kinda terrible, so I knew that she needed support, and I gave her support. And that was what we did. - She was interested in me. She listened to me. Treated me like I was lovable. - She didn ’ t have anyone to recognize her being, to support her, to see who she could be, who she was. - She was very interested in what I would do with my life, and the assumption was that it would be something worthwhile. She just went way, way beyond the call of duty to make me feel like I was worth it. And she did that for a lot of other women, too. - I was working at McCann Erickson. We were given an assignment. L ’ Oréal, they wanted a campaign for Preference, it ’ s a hair color. - She ’ s in this room full of men brainstorming for this ad. - The men were busy flirting with the women, telling us we all looked like models and we didn ’ t look like advertising people, you know? - They ’ re like,”Oh, this woman is gonna be standing by an open window, and the wind will be blowing the curtains.”And she ’ s thinking, you know, this is just making the woman totally an object. - I was feeling angry. I ’ m not interested in writing anything about looking good for men. Fuck ’ em. Fuck you too. - Why fuck me? - Well, you ’ re a man. - Yeah, but I ’ m here trying to help you tell your story. - Yeah, that ’ s good. I like that part. I was pissed. We weren ’ t there to just dance for the men. - It is what she said. She just thought,”Fuck you.”And wrote this. - I just opened my eyes and knew. Because I ’ m worth it. The campaign got approved, but all the men were always telling you that you were doing something wrong or they knew how to do it better. - The agency insisted on a male voiceover. - She uses the most expensive hair color in the world, Preference by L ’ Oréal. - You know, it was just totally different. - Smooth and silky, but with body. It feels good against her neck. Actually, she doesn ’ t mind spending more for L ’ Oréal. Because she ’ s worth it. - It was wrong. This was not for men but for women and for other human beings. - I use the most expensive hair color in the world, Preference by L ’ Oréal. It ’ s not that I care about the money, it ’ s that I care about my hair. It ’ s not just the color, I expect great color. What ’ s worth more to me is the way my hair feels. Smooth and silky, but with body. It feels good against my neck. Actually, I don ’ t mind spending more for L ’ Oréal. Because I ’ m worth it. - I wrote that. - It was everywhere. It was a national campaign. - Yes, it ’ s expensive, but I ’ m worth it. - What they realized after they started running it was that that was only the tip of the iceberg. - It was supporting women, finally. - And I ’ m worth it. And I ’ m worth it. I ’ m worth it. - It ’ s a four-word feminist manifesto, defying this social pressure to think about yourself in relation to men. - You ’ re worth it. I ’ m worth it. I ’ m worth it. - It was a very big success. - At a certain point, L ’ Oréal adopted it as the tagline for their entire business. - Parce que je le vaux bien. - Even today, that ’ s their motto. - Because I ’ m worth it. - ’ Cause you ’ re worth it. - You ’ re worth it. - We ’ re worth it. - And I ’ m worth it. - She wrote the most influential tagline in women ’ s beauty advertising. - I ’ m worth it. - It ’ s magic, that phrase. - Those words just get better with age, don ’ t they? - I ’ m worth it. Because you are and always will be. - She came up with that line because she believed it. She believed it about herself, and she believed that women should believe it. She did that for me. You know, even after they split up, she was still a big part of my life. I ’ ve always kind of felt like I dodged a bullet that I have watched take down other members of my family. She saved me. And today, I am a professor of English at Williams College. I teach Victorian literature, and I also teach women ’ s, gender and sexuality studies. She was my mother. She also taught me how to be a mother. I had been very uncertain about motherhood for myself because my role model for that was so nightmarish. But Ilon really taught me, I can do this. I can be a good mother. She made me believe that I was worth it. And I was not either the first or the last. - I have to go away now. Go to my room. - It ’ s been lovely. - It ’ s been lovely. Oh... I ’ m not interested in advertising. I don ’ t give a shit. It wasn ’ t like it was my whole life. It was just a portion. It ’ s about humans ; it ’ s not about advertising. It ’ s about caring for people because... we ’ re all worth it, or no one is worth it.

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HW : Hi everyone, I am Helen Walters. I am head of Media and Curation here at TED. Welcome to another episode of TED Explains the World with the one and only, Ian Bremmer. It is Monday, February 24, a little more than a month since President Trump was inaugurated once more, and safe to say, a lot has been going on. So we figured we ’ d check in with Ian to determine what we should really be paying attention to amid the extreme noise. Ian, hi. Ian Bremmer : Helen, great to be back with you. HW : So we asked the TED community to share their questions for you, and we wanted them to share what they ’ re most curious to know. And we really got some amazing questions that I plan to pepper this conversation with. So let ’ s start with one right now. Two months into 2025, where does the US stand? IB : Very powerfully, in the sense that the US economy, of course, is performing considerably better than any of the other G7 economies coming still out of the pandemic. Technologically, only close competitor is China, ahead of the US in some areas, behind the US and other critical areas. But compared to every other country in the world, the US is head and shoulders and neck or waist above them. Militarily, of course, the US is the only country with global ability to project power. No one else is close. And the US dollar is still the global reserve currency, no one is approximate challenger. I say all of those things because those are the things that haven ’ t really changed over the course of the last few months, but I suspect that isn ’ t what the questioner was asking. What they were really asking is what ’ s happening politically. And politically, the United States is unwinding its own global order. It is no longer particularly interested in promoting collective security or NATO. It ’ s no longer really interested in promoting involvement and leadership of multilateral institutions, of consistent rule of law and free trade that ’ s well regulated, certainly not very interested in promoting democracy around the world. I mean, this is an environment where other countries around the world have to figure out how to adapt to the United States, and that the US has become the principal driver of geopolitical risk and uncertainty in the world today, unlike any other time in my lifetime and yours, that ’ s perhaps the biggest change. HW : That is quite a statement. Alright, you are a geopolitical expert. Let ’ s talk about Europe. So Defense Secretary Pete Hegseth stated that America ’ s foreign policy focus no longer lies in Europe. Vice President JD Vance caused some raised eyebrows, dropped jaws and, I suspect, some choice expletives after his recent speech at the Munich Security Conference, in which he accused European leaders of suppressing free speech and warned of the threat from within. So you were in Munich. Was it as dramatic as the stories we read had it? And what happens next? IB : It was. I was in the room. I was maybe 30 feet away from him when he gave that speech. Standing right in the front, right next to me was the president of Czechia, the president of Finland, the prime minister of Sweden, watching all of them and their reactions. What he said, let ’ s keep in mind, this is the Munich Security Conference. It ’ s been going on now for something like 61 years. And he was the head of the US delegation. And every year the head of the US delegation gives a big speech on the state of the transatlantic alliance and on global security. And he didn ’ t do that. I mean, the people in the audience were expecting a challenging speech. They were expecting that the Americans would be less committed to the alignment with the Europeans on Ukraine. The US had just had that direct Trump phone call with Vladimir Putin, 90 minutes long. Hadn ’ t coordinated that with the Europeans, never mind the Ukrainians, in advance. So, I mean, they were definitely prepared for a very challenging plenary. But Vance did none of that. Vance said,”I ’ m not going to talk about Ukraine or Russia or China, because the biggest problem is actually what ’ s happening inside Europe. The biggest problem, essentially, are you guys sitting in front of me, that I ’ m speaking to, because you ’ re suppressing free speech. You ’ re suppressing the far right. You ’ ve been infected by the woke mind virus, and you aren ’ t real democracies as a consequence.”And specifically he said, and this is something that I think is underappreciated in the United States, he attacked the German firewall. And said that – And let me explain what the German firewall is. That is a principle by the leaders of all of the mainstream German parties and their supporters that they will not, under any circumstances, work with, enter into coalition with the Alternative for Deutschland party, the AfD. Because the AfD, under surveillance from German intelligence agencies, is considered to be a neo-Nazi party. And the United States, of course, after World War II, led de-nazification of Germany. The day before, Vance had actually gone to Dachau and visited a concentration camp, and then came to speak in Germany about the firewall needing to be ended. And at that point someone yelled out from the crowd,"This is unacceptable!"And it was right in the front. And people in the room, about 1, 500 people in the room, standing room only, couldn ' t see who it was. They ' d just see the back of his head. I saw who it was. It was the German Defense Minister, Pistorius. And in 15 years of me going to the Munich Security Conference, I have never seen anything remotely like that. And the reason I mention all of this, and of course, on the back of that, let me be clear, Vance refused to meet with the German chancellor because he ' s basically, well, he ' s not relevant. He ' s only going to be there for a couple more months until a new government, so why should I bother? But does meet with the leader of this AfD, goes and meets with her directly, does a bilateral that evening. So that is a very important backdrop for this last weekend ' s German elections, where the victor, Friedrich Mertz, who is going to be the next chancellor, actually referred specifically to America ' s intervention in German democracy and support of the AfD being every bit as bad as what the Kremlin is doing inside Germany, and also said that the Europeans need to develop a defense policy which is independent of the United States, in other words, essentially saying that it will be the end of NATO. Again, staggering comments, unheard of, impossible to imagine even two weeks ago. And that ' s why it ' s very critical that you have the context of what happened between Vance, Elon Musk over the past couple of months, Trump, with the Germans, with the Europeans. So you can understand why the incoming German chancellor would make a statement like that. HW : What should we make of all of the conversation about the rise of Nazism, the return of Nazism? Obviously, the AfD just won 20 percent in the German election. We ' ve also seen people allegedly giving Nazi salutes at various US conventions and speeches. Do you think this is overblown? Is this a fear that people have rationally? And what should we make of that? IB : In Germany, the Alternative for Deutschland, are winning all of former East Germany. This is a group that increasingly understands that they ' re never going to really catch up to the rest of Germany. Remember, Germany hasn ' t really grown in five years now. They ' ve been in recession for the last two. And the average citizen, the average voter, living in former East Germany no longer are just angry about not catching up. They ' re basically saying,"I ' m done with this system."So that ' s why you ' re getting this very strident, revanchist nationalism that is, you know, it ' s not just taking a hard line on illegal immigrants. The AfD even supports removing citizens of Germany that they claim have not adequately integrated into German culture and nationality. And so, I mean, this is some very, very strong stuff, stuff that really does feel like what they were doing back in Nazi Germany. But in former East Germany, they win. They get the largest number of votes, where, in much of the wealthier part of West Germany, including, you know, where I was, in Munich, the AfD can ' t break out of single digits. So it ' s a very, very divided country. And also the AfD is doing very, very well among young men. There ' s a clear gender divide here as well, just as there is in the United States. In the US, I mean, I, of course, saw Elon Musk and that, you know, sort of grabbing his heart, saying,"My heart goes out to you"and then doing what appeared to be mimicking a Nazi salute. And then Steve Bannon doing the same without the initial heartfelt comment. It clearly was trolling. Clearly was trying to get a reaction from folks. It ' s very performative. And it ' s meant to also, you know, attack and attach, make the left lose their minds on something that isn ' t very critical to what the Trump administration is trying to do compared to the revolutionary changes inside the US government that they are very much prioritizing right now. So I ' m very concerned about the rise of neo-Nazism in Germany. I am very much not concerned about that in the US. I

’ m concerned about other things I ’ m thinking is being used as gaslighting for those that can use distractions or can be distracted by them. And I would also say that this election in Germany wasn ’ t surprising. It was exactly as the polls have expected for several months now. But the key elections in Germany and the key elections in Europe are the next cycle. I ’ ve spoken with a lot of people around the Trump administration that believe that the Europeans are one electoral cycle behind the United States. So in other words, Trumpism is coming, just not quite yet. And so in the UK, you ’ ve got Labour for a full term. But the Reform Party is increasingly the most popular. And just with a little nudge, maybe a little external money, and Elon said he ’ s already thinking about giving them 100 million dollars, which is a massive amount in UK politics, not so much in the US, that the Conservatives would fragment, a rump group would join with Reform, and they could win the next elections. In France, you ’ ve had all of these governments continue to collapse. Macron, very unpopular. Marine Le Pen and the National Rally party could easily win upcoming elections in 2027. In 2029, in Germany, if the Germans are incapable of turning their economy around, incapable of unwinding their debt break and spending some of the capital that ’ s just sitting there and as their debt to GDP goes down, but their economy contracts at the same time, then AfD could easily win in 2029, and then the firewall is no more. And then, of course, in the European Union elections overall, in five years ’ time, now four and a half, you could easily see the Patriots front, which is the equivalent to the far right grouping that these parties are a part of, they could all come together and be the dominant party. And then the EU as we know it is really a thing of the past. So these are existential changes that are happening not just for NATO right now, but also for the EU and quite plausibly, for the future of democracy as we know it. HW : Alright, let ’ s turn to Ukraine. So President Trump accused President Zelenskyy of starting the war with Russia and called him”a dictator without elections.”Last week, senior American and Russian officials met in Riyadh to discuss a potential cease-fire with no Ukrainians present. Now Zelenskyy has offered to step down in return for Ukrainian admission to NATO, although apparently he doesn ’ t really mean this. So what happens to Ukraine now? What should we be paying attention to? IB : Well, I mean, I think he would mean it if NATO membership happened. I ’ d love to see Trump call his bluff on that. But of course it ’ s not going to. And he wasn ’ t saying that six months or a year ago. He ’ s saying it now that NATO is truly being pulled off the table. I think it ’ s the right thing for him to do. It is, again, performative and helps to remind those around the world that Ukraine is fighting for self-determination. They ’ re fighting for the ability to join their own alliance, as any country in the world should be able to do. But that is not Trump ’ s belief. Trump believes that what you get to do is determined by how powerful you are, not by the fact that you happen to run a country. And indeed, the territorial integrity is something that is afforded to powerful countries, but not so much for weak countries. And that ’ s the way he feels about Panama, and that ’ s the way he feels about Denmark and Greenland. And that ’ s certainly the way he feels about Ukraine. So right now, as you and I are speaking, the United States is putting forward a new UN resolution at the General Assembly that says that the war has to end as fast as possible but does not recognize Ukraine ’ s territorial integrity, which has been a core precept of the international order that the United States has supported since creating the United Nations and since the end of World War II in 1945. The Americans are now throwing that out. And the Russians, of course, are fully supportive of that. America ’ s allies in Europe are not. But the US has gotten the Saudis to agree, and they ’ re getting a unified Arab block. The Americans have privately worked with a lot of poorer countries in Africa, in the Asia Pacific and others, smaller countries, and saying,"You ' re not going to get any aid from the US unless you vote in favor of this new resolution."I fully expect the resolution is going to pass. But of course, what that means is that the Americans are preparing to cut a deal on Ukraine over the heads of the Ukrainians. In other words, the US and Russia together will figure out what a ceasefire should and should not entail as part of a broader US-Russia rapprochement. This is obviously a disaster for the Ukrainians and the Ukrainians who initially refuse to accept this so-called deal, on Ukrainian natural resources that would grant the Americans $500 billion in access to such resources in return for"paying off"aid that the Americans had granted to the Ukrainians without conditions or strings, but apparently no longer. And so, you know, what does this all mean? Well it means it ' s yet another case of why people shouldn ' t trust the United States from one administration to the next, because foreign policy will change completely. And what you thought was an American commitment won ' t stand. It shows that there is a fundamental rift between the Americans and the Europeans and indeed, Emmanuel Macron in the United States right now, as you and I speak, hoping that he can do something that will keep Trump from, as Macron believes, essentially surrendering to, conceding to, the Russian position and maintaining some level of coordination with the Europeans. But he doesn ' t have a lot to offer. And the Ukrainians, you know, now in a position of desperation where maybe they will end up signing over a lot of their natural resources to the Americans. But what, if anything, will they get in return for that? And how empowered will the Russians be in any deal that is cut? And will an independent Ukraine even be able to survive, and what will the terms that will be required of them? Of course, Putin is saying things like, no European troops of any sort, whether in a NATO formulation or not, would be allowed as peacekeepers in Ukraine. Well, then, how would you guarantee, what kind of security commitments would you give to the Ukrainians? These are all open questions, but apparently questions that President Trump is prepared to largely resolve on Russia ' s terms. And again, this is such a dramatic change from where the United States and where the Western order was just a month or two ago. HW : So let ' s take some quick-fire questions from our community related to this topic. And I think the first one is particularly pertinent to what you just said, so it ' s simple. What does Putin have over Trump? IB : I don ' t think that Putin"has"anything over Trump. If he did, then why wouldn ' t Trump have granted Russia more in his first term? In his first term he gave the Ukrainians those javelin anti-tank missiles that Obama refused to give. And, you know, he thought they were too risky. He increased sanctions against Russia. Now that was his administration with a lot of hawkish in orientation advisors around him that he doesn ' t have this time around. They ' re all directly loyal to him. But, I mean, if it was a priority for him, he could have stopped it. And if the Russians had something over him, then why wouldn ' t they have used it in that case? It feels like a conspiracy theory. I don ' t buy it. I think very differently. We can explain what ' s happening because number one, Trump wants to end wars. It ' s very popular with his base. He ' s said this on Gaza. He ' s said this on Iran, where Israel wants him to support attacks on Iran ' s nuclear capabilities, and Trump has said no. And he ' s showing this on Ukraine. He doesn ' t want to spend money on the Ukrainians going forward after spending over 100 billion dollars under the last three years from the Biden administration. He doesn ' t want to spend that money going forward. And he also doesn ' t value a strong Europe. He actually thinks that a weak Europe is in America ' s interests, where you have more Brexits. He used to always talk to the French about that during his first term,"Why won ' t you do Brexit?"He wants all of these MAGA-type populist parties, including the AfD, to win in Europe because that legitimizes his own popularity. Just like he loves Milei in Argentina or Bukele in El Salvador. And so I think if you put those things together, you actually get to why Trump is doing what he ’ s doing with the Russians without needing to create a story about Trump somehow being threatened by some secret, shadowy information that the Russians have on Trump that they ’ ve let him know about but nobody else knows. I just don ’ t buy that. HW : Great. OK, another community question. If the US hands Ukraine to Russia, will Russia keep going? Will they attempt to take over Europe? IB : It ’ s interesting that Trump just met with the Polish president, President Duda. It was supposed to be an hour meeting, it was only 10 minutes. So he kind of embarrassed him at home in Poland. But he did say that the US is still committed to maintaining a troop presence on the ground in Poland. And so it ’ s very hard to imagine the Russians rolling through Ukraine and then into Poland, even though there ’ s been lots of asymmetric warfare, for example, and even there have been, you know, Russian missiles launched into Lviv, Ukrainian air defense, missile defense, and with Polish villagers getting killed because of the fighting. So, I mean, there ’ s been spillover. But America is still going to be in Poland. Why is that? They get along pretty well, and the Poles are heading towards five percent of GDP defense spend this year, which has been the high water mark of American demands of defense spend. So he ’ s, you know, been consistent on that over the past months. Baltic states. I mean, they ’ re all spending a lot more money on their defense. And so, I mean, in principle, you could imagine the United States maintaining these rotational deployments in the Baltics, which would prevent the Russians from invading. But might Trump say, as part of a deal,”Why do we have all those troops there?”You mentioned Pete Hegseth saying the Americans need to get troops out of Europe and get them to Asia, which is where their principal competitive, you know, sort of, strategy is going to be oriented. And the US hasn ’ t been pressing the Japanese or the Indians, the principal partners, allies in Asia vis-à-vis China, the way they ’ ve been pressing the French, the Brits, the Germans in Europe. So I think it is certainly plausible that the Russians will do more, especially with those countries that refuse to align with a Trump rapprochement towards Russia. The countries I would be most worried about are Moldova, are Georgia. I ’ d be most worried about the countries that are in the so-called”near abroad”that aren ’ t a part of NATO, and aren ’ t going to become a part of NATO, and therefore are much more vulnerable. But certainly in terms of Russia ’ s willingness to engage in espionage, to fund actors on the ground for arson attacks, assassinations, critical infrastructure attacks on fiber networks, on pipelines, I think all of those things are likely to increase against European states across the board, on the back of this US-Russia rapprochement. HW : OK, final community question for this segment. What happens if the US no longer supports NATO? IB : It ’ s possible that the United States is moving towards a similar posture in Europe that they presently have in Asia, which is strong individual bilateral deals with a number of countries : South Korea, Japan, Australia, but not a collective security agreement, where those countries together have more of a call on the United States and there ’ s a lot more free riding. So I could imagine that NATO falls apart and becomes that. The danger is that even though the Europeans now understand the nature of the threat, that it is very plausible and maybe even baseline, that the Americans are going to leave Europe collectively on its own. That that seriousness does not equate to the ability to increase spending adequately in the near term. The Germans did manage to pull together an electoral outcome that will allow for a so-called grand coalition, just barely. So you ’ ll have a two-party government in all likelihood instead of a three. But you also have a blocking capacity on the part of the far right and the far left that will prevent the Germans from really blowing out their spending, including their defense spending, unless they can pull something off before that new government is formed. Which is possible, but it ' s unlikely to be very big. The Brits are now talking about maybe moving the 2. 5 percent of GDP defense spend over years with a very challenging fiscal environment. The Spaniards, not even close, the Italians, not even close. So what does a European military capability really look like? And without the Americans, the answer is it ' s not there. It ' s not there. So I think that if NATO falls apart, the reality is that most of Europe will be incredibly vulnerable to external attack, especially Russian attack, And there won ' t be a way for them to defend themselves, and some of them will break and therefore want to work more directly and individually with the United States. And that could be the end, not only of NATO, but it could be the end of the European Union. I think this is, the stakes are incredibly high, this is an existential challenge for European governments that needed to take this seriously 30 years ago, 20 years ago, certainly in 2014, when the Russians invaded Ukraine, certainly, certainly in 2016 when Trump was elected, and certainly, certainly, certainly in 2022 when the Russians then invaded all of Ukraine. And at every point the Europeans were basically saying,"We ' re still fine with the Americans basically doing this."And now they are in very serious trouble. HW : No time like the present, I suppose. Alright, let ' s move on. So I mentioned the discussions in Saudi and obviously also on the agenda there with the discussions about Gaza and Trump ' s proposal to build a Gaza Riviera. What should we make of that? Or as one of our community members asked, what is the actual solution for Gaza? And surely it isn ' t what the US is proposing. IB : Well, look, here ' s what ' s interesting. You know, Trump says a lot of things, and when they don ' t work [out], he backs out fairly quickly. And you know, I thought – just stick with Ukraine for one second on this. It was interesting to me that Trump has said that he doesn ' t want to talk to Zelenskyy because he has no cards, he has no leverage. And so why should he let him at the table? And so Trump ' s going to cut the deal. Very interesting that if that is true, then why is it that the Secretary of Treasury brings this deal and says,"You ' ve got to sign this right now, Zelenskyy"And Zelenskyy refuses. And then a few days later, the Americans come back with a new deal with altered terms that are not quite as predatory vis-à-vis Ukraine. Why would the Americans change their tune for someone that has no leverage? Because, I don ' t think it ' s because Trump is a nice guy. He ' s not going soft at 78. I think it ' s because he overstated how little leverage Ukraine has. Ukraine is very popular among the GOP in the House and Senate, far more than Putin. Zelenskyy is far more popular in the United States among the voting population than Putin. And of course, he ' s also much more popular among the Europeans and even globally than Putin. And so it turns out that maybe there is a little bit of leverage that Zelenskyy has, and maybe Trump does have to give a little bit more. So this is relevant in the context of Gaza, because Trump was very annoyed that, you know, despite his ability to get a cease fire between Hamas and Israel over the goal line, that his friends in the Middle East were not coming up with a plan for Gaza. Where they were supposed to provide security and provide investment and lead reconstruction and deal with the Palestinian populations in the interim. And that wasn ' t happening. And so Trump then meeting with Prime Minister Netanyahu from Israel in Washington, then after that, summoning the Jordanian king, who was the ally that is most reliant on the United States and so, therefore, most needing to say yes to whatever Trump orders of him. Basically says, as you mentioned, Helen, that Gaza is going to be a Riviera, that the US is going to turn this into an incredible place. They ' re going to build wonderful homes for the Palestinians someplace else. And the Palestinians will all volunteer to leave Gaza and go to those homes, and Gaza will be depopulated and new people will come in and live there. That was the plan. And when he was asked, standing there with King Abdullah from Jordan, under whose authority? Trump ' s response was :"Under my authority."And that went over like a lead balloon among America ' s allies in the Middle East. They all said,"We ' re not up for this. We ' re not resettling Palestinian populations. They won ' t actually leave voluntarily. This will would actually be a huge problem for Israel ' s own national security long-term. And we ' re going to come up with another plan. And our plan will keep the Palestinians on the ground in Gaza."At which point Trump pivots to work with the new plan. And his advisers say that Trump was only trying to start the conversation. And what he said was his plan is not actually what he ' s going to do. So, what ' s sustainable? What would be sustainable would be a lot of Gulf money going in and maybe some European money too, though they ' re going to be really pressed, given everything you and I just talked about, to try to reconstruct a completely devastated Gaza that eventually some two million-plus Palestinians will be able to live normal lives in, and the governance will be provided by some technocratic group of a Palestinian Authority that will be working with, selected by, those that are involved in the reconstruction and the security. Namely the Egyptians, the Jordanians, the Saudis and the Emiratis. That is the best possible deal that they ' re going to get. But that ' s happening as Israel has just evacuated forcibly another 40, 000 Palestinians from refugee camps in the West Bank, sending tanks in for the first time in decades and probably precipitating more Israeli settlers taking more land on the ground in the West Bank. And so, as we are possibly taking a small step towards a sustainable solution in Gaza, we are taking a slightly larger step away from a sustainable solution for the Palestinians in the West Bank, twas ever thus over the past decades in this incredibly horrible problem and conflict. HW : And why is Israel doing that? I mean, are they feeling emboldened by the US, or why the incursions in the West Bank now? IB : Well, I think they ' re feeling emboldened by their own successes. They have shown that they are the dominant military power in the Middle East, and they have the ability to determine the level of escalation unilaterally, and their adversaries and enemies can ' t really do any damage to them. They proved that after October 7, in their response to Hamas. They proved that in decapitating and taking out the military capabilities of Hezbollah. And they ' ve proved that also with overturning – they didn ' t do it, but the fact that the Assad regime in Syria is gone, which means that the Iranians no longer have a land bridge to get additional weapons to what had been the strongest adversary of Israel militarily, Hezbollah. So all of that has shown that Israel can decide outcomes. They ' re the ones that have all the power here. And also that depopulation of Gaza is very popular among Israelis. There was a poll recently in the Jerusalem Post, which is pretty mainstream in Israel. 80 percent of Israeli respondents said that they wanted to depopulate Gaza of all Palestinians. Taking more land is very popular, especially in the West Bank, especially among Netanyahu ' s present right-wing coalition, that he would like to keep together, to continue to govern. And this is a sop to them as he is looking to potentially extend the ceasefire in Gaza and move from phase one to phase two, which is facing some delays and open questions right now. So, yes, certainly, the United States has been underpinning that military capability of Israel. And I think it is instructive to remember that not only does the US provide more military aid in peacetime to any country in the world, to Israel, but that unlike in Ukraine, where the Trump administration is saying,"No, I ' m not happy with the grants, you ' ve got to pay that back,"The billions and billions that the United States sends to Israel, nobody is suggesting that that should be alone or that the Americans should get technology rights or mineral rights or anything else from Israel, despite the fact that Trump ' s policy is America First. Israel is a very clear exception to America First in Trump ' s visioning of it. HW : Alright, so we talked about the fact that the foreign policy focus of America is shifting to Asia. So what are you hearing about how China is feeling in all of this? And what should we be paying attention to on that front? IB : China has had a couple of good weeks economically and specifically on the back of that huge announcement of DeepSeek, which has, you know, far less money behind it than the comparable American chatbots but is performing at an extremely high level. And, you know, given the fact that there are all of these export controls on semiconductors and other parts of advanced technology, that this is giving people more of a belief that the Chinese really are capable of driving innovation even in that space. And they ' re, of course, dominating the post-carbon energy space. We see that with electric vehicles and batteries. But we ' re also seeing that with massive investment now in green hydrogen, for example, we ' re seeing it in advanced nuclear capabilities, all of these things. And so there is more investment in that space that the Chinese are not only driving as a state, but that they ' re also increasingly able to raise and move in their own private sector. So the story that you and I have been discussing for the last couple of years is the Chinese economy is radically underperforming. That is still true, but there is an emerging counter-narrative here that I would be remiss not to mention. Now more broadly, what the Chinese here are seeing is that Trump ' s direct policies are going to hurt them economically. His tariffs that he ' s already announced on China, the 10 percent, on a whole bunch of Chinese goods for export, the reciprocal tariffs that are global, that will clearly hurt China and that they ' ll have a hard time negotiating out of, and also the willingness of the United States to expand their export controls, their sanctions on China, especially in areas that are considered relevant, even loosely relevant to American national security. All of that is going to constrain Chinese growth. All of that is going to negatively impact the Chinese economy. That ' s the downside. And the Americans are doing that not only in bilateral relations with China but also pressuring other countries to get Chinese tech out of their supply chain. And you see that with Americans pushing other countries that produce semiconductors to take the Chinese out. They ' re pushing American allies and trade partners to get China out of the export chain. So Mexico being pushed very hard by Trump to stop allowing China to pass goods through Mexico into the United States. India, same conversation. Vietnam, same conversation. A lot of that going on. So baseline, the US-China relationship is getting worse. But the Chinese see huge advantages in America ' s reversion to unilateralism that we haven ' t really seen since the original America First movement in the run up to World War II. Charles Lindbergh saying,"Don ' t get into the war, don ' t fight the Nazis. The US is far away. This isn ' t our our problem."And now you have the Americans not only doing that with Russia and Ukraine with the new America First, but you also see that happening with the Americans pulling out of the Paris Climate Accord, again, pulling out of the World Health Organization. You see it with the Americans shutting down USAID. And America has historically been responsible for about 40 percent of global humanitarian support and aid. And absolutely, some of that is corrupt. Absolutely, some of that is on woke programs that the vast majority of American taxpayers would never support. But the majority of that money is actually spent on programs that are really important in advancing American soft power and influence over countries in the Global South, all over the world. The Chinese know that. That ' s why they ' ve started humanitarian programs. It ' s not out of some great love for, you know, providing foreign aid for these countries. It ' s more because they see it creates influence. Now the Americans are leaving a vacuum that the Chinese can absolutely take advantage of. And we ' ll see this across, you know, sort of, microstates in the Pacific. We ' ll see this across sub-Saharan Africa. We ' ll see this across South America. The Chinese see huge opportunities from the United States creating a leadership vacuum. And the place they ' re most capable of doing that is if the US stops paying dues in the United Nations, where China is number two, and they ' ll suddenly have the most influence over all the key positions, which will give them a role in global governance that they ' ve really been constrained from having over the past decades. HW : What do you think are the chances that the US stops paying its dues to the UN? IB : You know, if it were up to Congress and only Congress and Trump doesn ' t lean on them, I ' d say fairly low. I ' d say that they ' d probably, you know, cut back on some programs. But in terms of baseline dues, they ' d keep paying it. But Trump personally, I think increasingly sees the UN as hostile to the United States. That ' s particularly true in terms of Israel policy, which Trump is leaning in on. And, you know – let me put it this way. Tulsi Gabbard did not have the votes to get confirmed. There were Republican senators that were going to vote against her. And Elon called those individual senators and threatened them. And this was, you know, fully aligned with what Trump wanted Elon to do. And those votes went away. And the only person that ended up voting against Tulsi was McConnell. And as a consequence, it went through. So look, I think that we should not underestimate the level of power and control that Trump and Elon have over the Republican Party this time around, compared to 2017, where it was really the Republican Party that was much stronger than Trump, both in his administration creating a lot of guardrails, but also in Congress. This time around, Trump has all the chips and he ' s using them. So if he decides he wants to stop paying dues, I think that ' s what ' s going to happen. This is a much more revolutionary presidency, domestically, where last time around it was much more transactional. HW : Alright. Let ' s talk a little bit more about what ' s actually happening inside the US. You bring up Elon, who obviously with DOGE has been sending out emails and asking for information from people and kind of trying to do a lot very quickly. Some of these high-profile moves are – I think what ' s interesting is that no one would really argue that there isn ' t inefficiency in government, that that ' s an understood reality. But the way that some of these moves are being made are maybe going to cause irreparable damage. And that ' s to people from red states, that ' s to scientists, that ' s to veterans, that ' s to people who voted for Trump and that ' s to people around the world. You talk about soft power and the United States, but what about actually what ' s happening inside the United States? And why do you think that the administration is willing to go so quickly, to go so fast and to cause such damage? IB : Well, and you saw Trump this weekend saying that he wants Elon to move faster, to be more aggressive. And I don ' t think that was just a troll. I think that ' s true. Trump is 78 years old, and with his friends he talks about the fact that he doesn ' t have much time. That in two years ' time, the Democrats could take the House in four years ' time, you know, that ' s it. He ' s not talking privately of"I ' m going to be president for life."He also knows he was almost killed. He was shot in the head. And do I think that changes somebody? Absolutely. I think that he believes that he was saved by God, and that his level of confidence that what he is doing is fundamentally right and necessary now is so much higher than the insecurity that I frequently saw in him in his first presidency. So I think his willingness to push, push, push, and his view that Elon is a great activator and executor on that intention, and someone that he had nowhere close to someone with that kind of capability and energy and execution ability in his first term. And now he does. So first of all, I think that relationship between Trump and Elon isn ' t going anywhere. I think it ' s actually very stable. People that say it is are people that want it to go away. But hope is not a strategy. And I also think that the intention of revolutionary politics domestically is to shoot first and ask questions later. It ' s to break things. It ' s a view that the bureaucracy has been weaponized against Trump first and foremost, and also against the will of the American people. And therefore it must be broken. And you ' re going to break eggs when you want to make that omelet. Now your point is that there are a lot of Republicans that work in government. There are a lot of Republicans and Trump voters that are losing their jobs and that are not being treated with a level of care and compassion as a result of what Trump and Elon are doing. And I think there will be a backlash. But I ' m not so sure that a one-term, 78-year-old-Trump presidency cares all that much about that. And, you know, if that means that his popularity slips to like, the high 30s, frankly, you know, in a very, very polarized country, high 30s isn ' t that bad. It still means 80, 90 percent of the people that voted for him still support him, and I think he ' s probably more comfortable with that this time around. So I think a lot more has to happen before we can start talking about whether there ' s really going to be significant pushback against what Trump is trying to accomplish at home. HW : Do you expect to see any of Congress standing up to Trump? We saw the governor of Maine recently kind of take him on. Do you expect to see more of that, or do you feel like for the next two years at least, people will just go where Trump takes them? IB : Well, sure, but, you know, in the case of Maine, that ' s a Democrat. So I don ' t know how much that means. I don ' t expect to see a lot of Republicans standing up against him. And of course, since the Republicans control the House and the Senate, that ' s the only thing that really matters. I mean, you know, you saw Kash Patel got through, Pete Hegseth got through. Now look, I actually think that a lot of the members of cabinet are very capable. I mean, they ' ve been picked for their loyalty and they will be loyal to Trump, but they ' re very capable. I think JD Vance is very capable. Mike Waltz, I know quite well. Marco Rubio, over the years. These are people that are very well respected, have been by their colleagues and could have served under any Republican administration. But there are some people that are completely incapable for the roles that they have been selected, completely incapable. They would not be selected in a role in any cabinet, in any democracy. And yet here they are, serving in critical roles in the most powerful country in the world. And yes, I ' m talking about the Secretary of Defense. I ' m talking about the Secretary of Health and Human Services. I ' m talking about the Director of National Intelligence. I ' m talking about the Director of FBI. These are important roles. And they were all confirmed. And if you ' re a Republican senator – and senator – I ' m not talking about the House here where, you know, it ' s a lot more diffuse and there are a lot more wackos and it ' s easier to get in. But Senate has always been like, responsible and more oriented towards bipartisanship and the rest, and they are utterly petrified of what it means if they publicly get crosswise with President Trump and number two, Elon. And they ' re not willing to do it. So I think that there ' s very little belief that you are going to see Congress step up. I think that you will see justices step up because those justices are independent. The problem is that many of those justices are progressive and have a known progressive lean. And so when cases come up and justices that have that political lean oppose something Trump wants, his willingness, his administration ' s willingness to attack them and refuse to execute on that judicial decision, unconstitutionally, is real. Now the case will of course go then up to the Supreme Court eventually. And I don ' t feel the same way about Trump refusing to listen to adhere to the rulings of the Supreme Court. But if there ' s 100 rulings like that and you overwhelm the judicial process, you quickly see how you might erode a core check on executive power in the United States. And Trump and Elon clearly intend to every day test those checks and balances and break them if they can. I mean, that is the intent, that is the intent. HW : So you mentioned democracy. When we last spoke, you said very clearly that the US is not Hungary, meaning that democracy is not threatened and that the US is not in danger of turning into an autocracy. Do you stand by that? IB : Yes, I do. I mean, in the sense that I think in two years we ’ re going to have midterm elections, and they will be largely free and fair. And in four years we ’ ll have presidential elections. And US elections are held at the federal level, right? They ’ re held state by state. They make the rules. It ’ s very easy to make people believe that the election has been stolen. It ’ s extremely hard to actually rig an American election. And I don ’ t believe that that is very likely at all. It ’ s very unlikely. The American military, despite the firings that we ’ ve seen just last Friday, is still an independent and professional military that I believe will carry out its orders and its duties professionally. I see the judiciary in the United States as independent and a clear check on the executive power in the United States. Where I see a problem, is that the US is becoming more kleptocratic every day. And the US was already much more of a kleptocracy over the past 10 years under Democrats and Republicans than any other advanced industrial democracy in the world. Clearly, money buys power in America in a way that it doesn ’ t in any other major democracy, wealthy democracy. That is getting dramatically worse. I saw specifically that Elon Musk, who is a walking conflict of interest at the highest level, met with Narendra Modi, the Prime Minister of India, in Blair House, so part of the official White House complex, right after Trump met with him. And Trump was asked later in the day, was Modi meeting with Elon in Elon ’ s private-sector capacity to advance his business interests or his official capacity to advance the policies of the United States? And Trump answered honestly. He said,”I don ’ t know.”And how could he know, right? Because, I mean, those two things are completely intertwined. But that is not remotely acceptable for rule of law in a functional democracy, and yet nothing is being done about it. In fact, it ’ s getting more and more entrenched every day. So the single place where the United States was least functional as an advanced democracy is now getting far, far worse. And I don ’ t see any checks and balances on that. I think that ’ s a very, very serious problem. HW : Alright, Ian, thank you, as always for all of your insights. Let ’ s close out with a couple more questions from our community. And this one indeed relates to the kleptocratic conversation that you were just raising. So why are the billionaires enabling fascism? What do they get out of it? IB : Fascism is a very politically loaded argument. That ’ s a political system where the cruelty against a subgroup or a perceived subgroup in the society is the intention, the expressed intention of the policy. And I certainly wouldn ’ t define the United States as a fascist system today. But I do believe that billionaires in the United States, at least a significant subset of them, are really under-interested in the well-being of their fellow countrymen and women. And this is a country that really makes heroic the individual. And it undermines the community. We ’ ve seen a significant erosion of civic engagement in the United States and civic associations. The church is not what it was in the US. Fewer people go, fewer people trust the church. Public education is not what it was. And wealthy people increasingly opt out. The family is not what it was. It ’ s increasingly atomized and people spend much more, much more of their time being, you know, isolated and atomized and intermediated by algorithms. And you and I have spoken about that problem in the past. We are increasingly a society of individuals as opposed to a nation together. I will tell you that that is a real problem when you have large numbers of billionaires that believe that they built it themselves, that they ’ re not responsible to a collective whole for any of their success and therefore they ’ re not accountable for their fellow citizens. And that ’ s what I think we have. I think we have a lot of winners in the United States and not a lot of leaders. And when you have winners, you have losers. When you have winners, you don ’ t have a collectivity. I will say that in this anti-DEI thing, I have a lot of sympathy for the people that don ’ t like DEI. I thought it was rammed down the throats of a lot of people And I ’ ve said this before, it was well beyond what the average American was willing to tolerate, and also it came with a level of high-handedness that that basically said, if you ’ re not with us in supporting these programs, you ’ re evil, you ’ re stupid, you ’ re uneducated. That ’ s unacceptable, too. It was a completely, you know, sort of oppositional way to try to have what ’ s a very challenging conversation. But I ’ ve never been a champion of diversity for diversity ’ s sake. What ’ s amazing about the history of the United States is despite all of our diversity, we find commonalities with each other. We find connectedness. That ’ s what ’ s amazing, is the connectedness, is that, you know, on this world, we ’ ve got eight billion people. And despite how different we all are, we all actually connect as human beings. That is what we need to focus on, not the diversity, the connectedness. And I think that ’ s what the billionaires today in the United States are losing. They ’ re doing incredibly well themselves. They ’ ve got theirs, they ’ ve got access to power, they ’ re paying good money for it, they ’ re going to get even more. But they ’ re forgetting that for us to succeed as a society and as a planet, we have to be connected to each other. You can ’ t throw out diversity and forget connectedness, and that ’ s what we ’ re forgetting right now. That ’ s why the country feels so much trauma and pain right now. And I think why the United States has given up on a lot of its really aspirational values that made such a difference to the world, that almost lost World War II for many generations. And now we ’ re giving it all away. HW : I mean, I think that ’ s why you find that diversity and inclusion go hand in hand, that that ’ s actually speaking to that. But in some of the moves that are being made to kind of throw out any type of diversity, to scan documents, to find words in order to highlight programs that are wasteful, that we ’ re actually going to lose what many, many studies actually highlight as being the benefits of diversity and the fact that you need diversity in order to be actually successful. IB : Shooting first and asking questions later, you know, moving fast and breaking things, those things make sense in a turbocharged venture capital technology environment. They do not work in government. They don ’ t work in government at all. In government, we have something called checks and balances, and we need them. And moving fast and breaking things undermines those checks and balances. So we ’ re going to do a lot of damage unintentionally. A lot of people are going to get hurt. A lot of folks don ’ t care about that in power right now, but they should because it would be much, much better. I know people want to move fast, but actually to think a little bit, do a little bit of your own research before you decide to kill that program, before you decide to fire that person. We ’ d be in a lot better shape as a government right now. HW : OK, the final question that I have for you actually comes from my 10-year-old son, and it ’ s a bit of a miserable question. We have a very lovely house, I promise. But his question nonetheless is are we heading into World War III? So, Ian, what should I tell him? IB : No, no, I don ’ t think we ’ re heading into World War III but we are absolutely heading into a world of much greater conflict. I think it ’ s much more likely, for example, in the next five, 10 years, that we ’ ll have a nuclear weapon go off. And I thought that was pretty unlikely since the Cuban Missile Crisis. In other words, over the course of my life. I think we ’ re going to see a lot more terrorism at scale, because a lot more people are going to feel angry and disaffected, I think. You know, you saw that assassination of the CEO of UnitedHealthcare just a few dozen blocks from my house. I think you ’ re going to see more things like that. And so, no, not World War III because the fact is that outside of the US and China, all the other countries know that they need to work with everybody. And there are even forces inside US-China that are trying to make sure that you don ’ t throw the baby out with the bathwater. But underneath that World War III risk, there ’ s an awful lot of damage that ’ s going to get done, an awful lot of crisis that I think we ’ re going to have to live through before we start to see what we create on the other side. A lot of defense as well, that we ’ re going to be playing in this environment. HW : Well, Ian, it is always a pleasure to talk to you, and we ’ re so glad that you are scanning the world to understand what is happening and then you come here and explain it to us all. Couldn ’ t be more grateful and thank you for everything. IB : Thank you.

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Hi there, my name is Sean Sherman, I am a chef. Unfortunately, I don ’ t have food for you guys tonight. Food for thought, I guess, maybe. I ’ m here to talk about Native American food. I was born and raised in Pine Ridge in South Dakota, and our focus are on Indigenous foods. And, you know, it ’ s been a really interesting journey so far. I started my company called The Sioux Chef – S-I-O-U-X, a little play on words – back in 2014. But it had come from quite a few years of trying to research and understand because I kind of grew up in restaurants. I grew up in Pine Ridge. I grew up in Spearfish and in South Dakota in the Black Hills. And I started working a lot of touristy restaurants. And, you know, I had just a long career. All through high school and college, I worked restaurants. After college, I moved to Minneapolis. I became a chef at a young age in the city. And I ’ d just been cheffing for a long time. And a few years into my chef career, I realized the complete absence of Indigenous foods. And even for myself, I realized that I couldn ’ t even name – I could name less than a handful of Lakota recipes that were truly Lakota, things without cream of mushroom soup in it, right? So I was really trying – It, you know, put me on a path to try and understand what happened, like where are all the Native American foods at, you know? And so it ’ s been really interesting. So Indigenous foods, that shouldn ’ t be – there shouldn ’ t be a big question mark, you know, we should know about it, because no matter where we are in North America, we ’ re – you know, North America obviously begins, all of its history begins with Indigenous history, right? And no matter where we are, we ’ re standing on indigenous land. And so we should have a really good, strong sense of Native American food because it ’ s just the land that we ’ re on. It ’ s just the history of the land that we ’ re on. So for us, it became more than just serving foods. It really became talking about it and talking about why it isn ’ t here. And I think it ’ s a really important story for us to know. And it ’ s also really important to see the benefit of why understanding Indigenous foods could really help all of us in the future. So, you know, but where are all the Native American restaurants? We live in a world today, you know, where we have – as the US, we ’ re like food capitals of the world, right? We have some of the best restaurants in New York City, in Chicago and LA, and zero Indigenous restaurants that are focused on the land that they ’ re sitting on, which is kind of insane. You can have every other restaurants – and Indian restaurants don ’ t count, because that was my only choice on Facebook, because when I was trying to decide how to describe our restaurant – is it Indian or is it new American or old American? But anyways, so what we ' ve done is like we tried to focus on, first off, just understanding what were precontact foods, precolonial foods. And I realized that that term didn ' t even really make a lot of sense to people. So I think it ' s really important to go through the storyline because to understand colonial or what is a precolonial food, you have to understand colonialism itself. And to understand colonialism, the easiest way is just to Google it. So if you Google the word"colonialism,"you ' ll get a definition,"it ' s a policy or practice of acquiring full or partial political control over another country, occupying it with settlers and exploiting it economically."And this is something that ' s happened not uniquely here. It ' s happened all across the globe. So all over the Americas, North and South, all over Africa, all over India, all over Southeast Asia, Australia, New Zealand, Hawaii, you name it, like this has been a very common history for a lot of areas around the globe. For the US, which is our focus, because we ' re right here smack dab in the middle of the United States, it ' s really important to understand the history because the US did a really good job of smudging its history a little bit. So if you ' re going through high school, the history you get on Indigenous peoples probably isn ' t the best history. So you really should read a little bit more about what really happened. So let ' s start with Manifest Destiny, which is really kind of something that was born from the idea of what was originally doctrine of discovery, which basically gave European powers their own rights to say, if we discover it, then we own it. Right? But that policy doesn ' t really work that well, because if you go into an Apple Store and you discover a brand new MacBook, most likely you ' re not going to have the rights to walk out the door with it. But a lot of our policies and a lot of – like, our country was built on this notion that we just have this right to everything, right? And people have to remember how young our history is. We ' re such a young country, you know? There ' s like, barely any time has passed. So just go back a couple hundred years and, like, start with 1800s. So in 1800, the United States is still not much more than just the 13 colonies at that point in history. And it ' s the 1800s that are the most deadly century for Indigenous peoples. So a lot of really bad things happened during this time period, because in 1800, in reality, almost all of what is the US is still completely occupied by Indigenous peoples and communities and a huge diversity of them across the board. Even despite European powers having big land claims, you know, France has a big section and Spain ' s got big chunks and England is holding on to chunks and Russia is coming in and there ' s all sorts of just big land grabs happening. But in reality, it ' s the Indigenous communities that have always been there. But this century is a mass century of change, you know. So during this time period, things move really fast. So this is just a really tough time. And for me, this is like my great-grandfather ' s era because my great-grandfather was born in the late 1850s and during his lifetime, he sees so much change so quickly, he sees so many battles between the Lakota and the US government. He sees the Battle of Little Bighorn when he ' s 18 years old, during the battle on the Lakota side. He sees his kids having to go to boarding school, cut their hair, learn to speak English, learn Christianity. He sees his children – some of his children even grow up to fight for the US government. So it ' s such a crazy amount of change to see in one single lifetime, right? And during this time period, people are getting pushed around. At the beginning of that century, over 80 percent of that landmass was under Indigenous control. And by the end of the century less than two percent, only because of the reservation systems. And this is all just part of the story of why there aren ' t Native American restaurants, because we just went through a really traumatic time in history where we ' re still – we haven ' t even had the time to heal yet, let alone evolve, right, when it comes down to all this. So the US history, you know, there ' s a lot of these big movements like the Indian Removal Act of 1830, the Homestead Act of 1862, the Indian Appropriation Act that basically said we ' re wards the states, that we ' re not our own entities anymore, the Dawes Act of 1887. And all these pieces were very focused and the government was really, really good at what they did, you know. And it all starts with taking our food away from us. So the loss of Indigenous food is something that starts from the very beginning. George Washington, one of his very first things that he does is send General Sullivan out to push all the native people outside of the US. He wanted them captured. He wanted them brought back. And they went on this march that lasts a single summer and does just that. So after a single summer, there ' s no more native people in all of that New York area, from D. C. all the way up, basically. And they named George Washington the president. They gave the name for a US president : Town Destroyer, which is still the name that they use today because he just devastated a whole area. And this is the precedent that gets set for how the US government treats the Indigenous peoples throughout the next century, basically. So here, in our area, the very systematic destruction of bison, which they knew would hurt a lot of people, and it did. And by the end of the century, there was less than 500 on the planet. And it was very purposeful. So... But I think what ' s most damaging for us and why we don ' t have a lot of Indigenous restaurants out there was the loss of our education, because this whole generation, like my great-grandfather ' s generation and my grandfather ' s generation especially, like, those generations should have been getting the full extent of Indigenous education. They should have been learning everything their ancestors intended them to learn. How to fish, how to hunt, how to gather, how to identify plants, how to live sustainably, utilising plants and animals around us. But instead, we went through a really intense assimilation period where we basically, you know, the boarding school systems stripped this whole generation of all that knowledge and education. And it became very traumatic because this was not a fun situation for these kids to go through. This was a military-style school and they popped up all over the US, all over Canada. These kids being again forced to speak different languages, forced to learn new religions, forced to learn skills that had nothing to do with them. And being forced to is the situation. You know, a lot of these kids perished. We shouldn ' t have to worry about sending kids to school to see if they ' ll survive or not. But this was a very harsh situation for kids to go through. And they went through physical abuse, sexual abuse. They went through mental abuse. And we ' re still reeling from that in our communities today because of this direct link to the trauma that happened there. And being Indigenous in the 1900s wasn ' t much better. My grandparents were born before they were even citizens, which doesn ' t happen until 1924. And then in the 40s and 60s, the US government started dismantling a lot of tribes. So over 100 tribes got dismantled so they could continue to take over more land spaces. We couldn ' t vote until 1965. We couldn ' t celebrate religions until ' 78, you know. So what does it look like for me growing up in this? I was born in the mid-70s and growing up in postcolonial America. Like, what kind of foods was I eating? And I get asked that a lot because people in the media are always like,"You ' re native, like what kind of foods did you grow up with?"Because they want to hear a cool story like,"I ' d get up in the morning, take down an elk with a slingshot, we ' d have a big family feast."But that wasn ' t the reality, because like I grew up with the Commodity Food Program because we were poor, like a lot of people on the reservation. And we didn ' t even have the pretty cans when I was growing up. We just had, you know, these black and white cans, beef with juices. And that ' s dinner, you know, and that sucks. So... And Indian tacos, you know, even when I was a kid, I was like, why does our Lakota food taste like Mexican food? It didn ’ t even make sense to me at the time. Because we could do better than this. There ’ s so much more to learn and more to offer with indigenous foods. So it ’ s really important to understand what Indigenous foods are. But first, you have to understand just like how diverse our nation is. We ’ re so diverse, there ’ s all sorts of plants and animals out there. And when you layer Indigenous peoples on it, you can see so much amazing diversity, you know? This is a language map. So just look at all those huge color blocks and within those color blocks there ’ s all sorts of diversity within those two, right? Still today, we have 634 tribes in Canada, 573 in the US and 20 percent of Mexico identifies as Indigenous. So there ’ s an immense amount of indigeneity out there today and we should be celebrating that diversity because it ’ s awesome. You know, just compare colonial settler states to Indigenous territories and you can see that diversity. It should change everywhere we go. You know, the US, the food system shouldn ’ t just be hamburgers across the board, or in Canada shouldn ’ t just be poutine. We could do so much better describing our foods, right? And so we have to really focus on Indigenous education because it ’ s important for us to learn. So when we ’ re looking at Indigenous education, it ’ s a study of all these pieces, wild food, permaculture, native agriculture, seed saving, seasonal lifestyles, ethno-oceanography, hunting, fishing, whole animal butchery, mycology, salt, sugar and fat productions, crafting, land stewardship, cooking, metallurgy, Indigenous history, traditional medicines, food preservation, fermentation, nutrition, health, spirituality, gender roles, sustainability – all of that stuff is this really important education that we need to learn, you know. So let ’ s just break down some foods real quick. Proteins are easy. We learn about how natives were able to use every single part of a bison. But that ’ s just because we didn ’ t have the privilege to be wasteful. We figured out how to be resourceful with everything that we had and we treated everything like that. But basically, anything moving around is literally game. And we cut out beef, pork and chicken because those animals didn ’ t exist here. And there are other animals to eat out there that aren ’ t those three. So there ’ s just a ton of stuff out there. And you shouldn ’ t be afraid of something if it ’ s not a cow, a pig or a chicken because there ’ s a lot of cool foods out there, and even insects, it ’ s so normal in so many parts of the world and it was normal here, too. But for us, our biggest love is plant knowledge because you start to learn the plants around us, you just see food and medicine everywhere. The Western diet has never really taken the time to learn this amazing biology that surround us and all these plants all around us. Because there ’ s so much to learn. There ’ s all sorts of staples out there, like the timpsula, which is the prairie turnip which grows around these plains. Camas root from the Pacific Northwest, wild rice from the Great Lakes, even just seaweed out there in the oceans, which a lot of families were utilizing, or in the deserts where all the plants look like they want to hurt you or maim you. The Indigenous peoples knew how to live with them. And another piece like the domesticated piece, with all the agriculture, it ’ s really important, because we think of this as agriculture but we know how damaging this is. And it ’ s scary when you see headlines like,”What should we do if glyphosate was found in our Cheerios?”You guys should be really scared about that. That stuff ’ s really nasty, you know. But it ’ s just amazing to learn about Indigenous agriculture because it goes back so far and people figured out all sorts of ways to farm and build sustained, huge civilisations, whether they ’ re in the middle of the desert, whether they ’ re on the coastal regions, or way up here in the Dakotas. People were able to farm amazing things that had an amazing amount of diversity that we need to protect. We are the stewards of what ’ s left of this diversity. And a lot of it got wiped off the map in the 1800s with all that colonialism that was going on. So we have to be understanding so we can protect these for the next generation because these could disappear if we don ' t do anything about it. So it ' s really important to understand that. So to use Indigenous knowledge in today ' s world, it ' s just important to open up your eyes, you know, stop calling everything a weed because that just means you don ' t know what it is. You know, our kids can name more K-Pop bands than they can trees and that ' s your fault, you know? We need to teach them things that are important. Because, like, just look around. There ' s food everywhere and we should be making pantries, like our grandparents did, and our great-grandparents. They just used the food that was around us. So we should just be making our own pantries that tastes like where we are, what makes us unique in our own region. And that ' s why we should have Native American food restaurants all over the nation, run by Indigenous peoples. There ' s so much to explore. There ' s so much flavor. There ' s so much health. And it ' s just super healthy, you know, and it ' s fun for chefs to create and play with all these flavors. Chefs should be really excited about getting to learn all of these plants that aren ' t in their diet because they ' re just going out of a French cookbook. And for us, we just want to get this food back into tribal communities especially, and make people healthy and happy and break a lot of the cycle of, you know, government reliance on food and huge rates of type 2 diabetes and obesity and heart disease because of this low nutritional food base that the government ' s been feeding us for too long. And we just need to think about how we can adjust and make a better lifestyle. We need to use our land spaces better. Lawns are fucking stupid. We need to really do something better. We could just be growing food out there, you know? We could just be putting food plants everywhere. We need more community gardens, more permacultural landscapes. It ' s that easy. If we can grow 30 golf courses in Palm Springs in the middle of the desert, just think what we could do if we just did that for good and just put food everywhere, you know? An organic food, food that wants to grow in that certain region. So, you know, Indigenous diet is really the most ideal diet. It ' s healthy fats. It ' s diverse proteins, it ' s low carbs, it ' s low salt. It ' s a ton of plant diversity. It ' s organic agriculture. It ' s celebrating cultural and regional diversity. And it ' s seasonal. It ' s just really good. It ' s like what the paleo diet wishes it was, when it comes down to it, because that just makes sense, you know, and we need to protect this. We need to get this out there. And again, it ' s not unique here. There ' s Indigenous peoples all around the world and there ' s an Indigenous knowledge base that ' s basically untapped because of the colonial structure that ' s been put everywhere. We need to be protecting people in Africa and India and Southeast Asia and Australia, New Zealand, Hawaii, South America, North America. We need to protect those. We need to be celebrating diversity instead of trying to build stupid walls to keep people out. We need to have, you know, healthy food access, cultural food producers, regional food systems, local control of food systems, not governmental control, access to Indigenous education and environmental protections to protect a lot of this natural food that surround us. We need to be better connected to our nature around us and really, truly understand how it ' s a symbiotic relationship. We ' re not above it, right? If we can control our food, we can control our future. And for us, it ' s an exciting time to be Indigenous because we are taking all of these lessons from our ancestors that should have been passed down to us, relearning them and utilizing the world today with everything it has to offer and becoming something different. We ' re at the stage where we ' re ready to evolve. This is an Indigenous evolution and revolution at the same time. So I hope someday that you can drive across this nation, stop at Indigenous-run food businesses and see this amazing amount of diversity out there and just think about it, you know.

Text Sample 929: NEG Dim 6 – 2020 – Score 1.00 – 2020/t003955.txt

When I was six years old, growing up in Medellín, Colombia, I made one of the most impactful decisions of my life. I asked my mother to change my school, to the school where she was teaching. To my surprise, she said yes. So I switched from a rich, private Catholic school to a public school where 99 percent of the students live in a condition of extreme poverty. The only meal some of my friends ate a day was the one that was given in school. My friends and I lived close to each other, but worlds apart. I lived in a neighborhood with a museum, a library, parks, and they lived in a neighborhood with the lack of the most basic necessities, such as potable water or electricity. More importantly, they lived in a place surrounded by danger, from guns to landslides. Their suffering was not unique. Up in the mountains, in Medellín informal settlements, thousands of families were having the same problems [as] my friends and their families, fearing that the police or the rains would take their homes away. I learned so much from my friends, but what continued to surprise me the most is their resilience and optimism in the face of adversity. Growing up with people that I care [about] is what had led me to the story of informal settlements. I teach now at the University of Colorado, Boulder, in the program of environmental design. I study informal settlements because even if they are invisible to most of us, they represent one of humanity ' s biggest challenges. And yet they provide great insight in how cities develop and innovate. There are three crucial things that I have learned about informal settlements that I want to share with you today. The first one is that informal settlements are a widespread form of city making. The second one is that by making visible populations in informal settlements, we can save their lives. The third one is that we pay more attention to the creativity of people who live in these places, we could be aware of innovations that can save the planet. Informal settlements can be broadly described as self-built neighborhoods outside of city regulations in conditions of extreme poverty. Nowadays, more than a billion people live in informal settlements all around the world. By the year 2050, one in three people on the planet will live in one of these places without potable water, adequate sanitation and in condition of extreme poverty. This makes informal settlements, what some call the slums, the most common form of urbanization of the planet. The paradox of informal settlements is that they are vast and common. However, the people and the places in which they live are the most invisible. There is much that we don ' t know about these places and that ignorance creates barriers to develop tools to help them. A first step to make visible these populations is to record the conditions in which they live. However, many countries where the informal settlements are do not have the resources to map these populations. And the countries who have the resources sometimes have legal restrictions that impede the state organizations to support the work on informal settlements. These unknowns create vacuums to understand informality and support the dissemination of misconceptions about the real challenges and opportunities of informality. As I started to learn more about the informal settlements, I realized the scarcity of data available. Most of our understanding about informality comes from separate and unreliable sources. There is not a single database that contains all the informal settlements in the world. To try to aid in such a puzzle, I created alongside hundreds of collaborators the Atlas of Informality. The Atlas is a creative attempt to visualize these invisible populations in an effort to understand the unique process of informal city making. A crucial question that we wanted to resolve here was how these places evolve over time. This was important not only to understand the past, but more importantly, the future of informal settlements and the future of all world cities. We at Environmental Design Program created a protocol with open-access software, remote sensing tools and direct mapping to identify and map the change of informal settlements over the last 15 years all over the world. The key was to develop a tool that was simple to use and that allows us to reach most of the planet. A tool that allows to compare these places at the same level. We have now mapped more than 400 informal settlements all over the world, and we have realized how each one of them is changing and expanding as a result of the arriving populations. We discovered things expected. Regions are expanding at different rates. Informal settlements in Latin America and Africa are expanding more rapidly than those in Asia. More importantly, we discovered that the entire sample continues expanding at a rate of 9. 85 percent. But what [does] this obscure number mean? It means that every year 2, 300 square kilometers of informal settlements are created out of the expansion of existing ones. This expansion means that every year, at the informal settlement, a slum, a city larger that some of the largest cities on the planet, such as Moscow, Houston or Tokyo, is created out of the expansion of existing settlements. As these places continue to grow in darkness, we are blinded to what happens in the cities emerging every day. This is why I have dedicated my life to the co-production with communities that live in informal settlements. Not only to try to improve their conditions of living, but to learn from them about the unique process of informal city making. Working with families and community members over the last 10 years, I have learned that to solve the informal settlements most challenging problems new cutting-edge strategies are needed. And that the source of that innovation resides already within the knowledge of these communities. I have learned that for each problem there is a community-based solution spearheaded by the people living there. For example, we learn fascinating things from communities like Carpinelo or Manantiales de Paz in Colombia, who organized themselves to build infrastructure improvements. They call these”combites”. These infrastructure improvements go from the creation of water systems to stairways, to roads. At the family level, we find incredible financing mechanisms. Like they ’ re renting of rooms to pay for home expansions or the creation of micro businesses tailored to the surrounding populations. One of my goals now is the emulation of those strategies at larger scales. Creative informal solutions follow a disruptive process that breaks away with traditional ways in which we think about cities. Planners, city officials and architects tend to operate in cities in similar ways as those set up at the beginning of the 20th century. What forced them and us to think about the informal settlements as a pathology, as a disease, as something that needs to be eradicated. This old-fashioned way of looking at slums forces the use of obsolete strategies. As a result, slum eradication programs have left millions homeless and have only displayed the problem to other places. In unbelievable contrast, the resources of informal dwellers for these populations to find unconventional ways to solve the same problems. Their solutions are less environmentally impactful and rely less on the need of big infrastructure improvements. These solutions could be as physical as the creation of pedestrian-friendly compact neighborhoods, or as strategic as the setup of community-based banking systems. These solutions could work both for informal settlements with less resources and to cities in the search of more sustainable development. Making these places visible is not only essential to help impoverished communities, it ’ s also vital for the rest of us. These populations living in scarcity are forced to innovate and create these disruptive urban products. Informal communities have always thrived, finding new opportunities out of necessity, from unofficial moto-taxis, private vehicles that serve the public, a response for the need for affordable transportation systems, or like the renting of rooms to pay for home expansions, what makes homes in informal settlements a self-sustainable urban model. Think about how radical this idea is. That instead of getting a loan to pay for your home, your home is the business that pays for the place that you live in. Of course, I don ’ t want to romanticize these solutions, as they are the result of innovation out of dramatic suffering. But what I want to say is that there is much that we could learn from them. In fact, I think there are some that are already learning. I argue that today, some of the most disruptive urban products, such as the ride apps, similar to the moto-taxis, or the home-sharing economy, similar to the self-financing urban model in informal settlements, started decades ago in the confines of informal settlements. If we pay more attention to visibilizing these invisible populations, we will not only have the opportunity to support the effort of billions, but we could learn from them how we can change the planet. Now, thinking back about my schoolmates, the communities which I collaborate with and the billions living in informal settlements, there are three things that we all need to do. The first one is that we need to make these communities more visible. They are part of our cities, they deserve to be respected and accounted. Second is that we need to pay more attention to the creativity and innovation that happen in these places. The next billion-dollar business, the next urban sustainable solution has already been invented in one of the thousands informal settlements around the world. And finally, we need to apply what we learned there. For the future one third of the planet and for our cities that need to be safe. Thank you.

Text Sample 930: NEG Dim 6 – 2020 – Score 1.00 – 2020/t004038.txt

First, a warning. As far as offensive words go, you are now entering a hard-hat area. We ’ re going to be unabashed in this, I am talking to you about a very particular word, a very powerful word, a very”see you next Tuesday”word. A word that is still so offensive that the funders of this event would only let me talk about it if we censored it on the slides, which rather proves my point, don ’ t you think? I love this word. Oh, my God, I love everything about this word, not just what it signifies, but the actual sound of it, the fact that the C and the T just cushion that”nnn”sound into this monosyllabic that you can just spit like a bullet or you can extend it out and roll it round your mouth,”cuuunt.”I love its dexterity. I love the fact that in Scotland it ’ s a term of endearment, but in America it ’ s horrendously offensive. I love it means something different with your friends than it does if you said it to your boss, it would probably cost you your job. I do not recommend it. I love this word. I love the fact that the first three letters are still the same chalice shape, all rolling through the word until they ’ re stopped in that plosive T at the end. I think the thing that I love most about it is its status as the nastiest of all the nasty words. Although that title is under some contention now. There are other obvious heavyweight contenders for the most offensive word. The N-word, for example. But here ’ s what I would say to you. I know why that word is offensive. I can look at the history, that word enabled the brutalization and racial genocide of an entire group of people. It played its part in dehumanizing Black people. What did”cunt”do? Does it not strike anyone else as odd that a word that just means the vulva could even be regarded in the same league of offense as the N-word? Are we saying that vulvas are that offensive? Surely not. But what I want to talk to you today about is how did we get here? Has it always been this offensive and how did it come to be so? The answer is no, it was not. But let ' s look at the history of it first of all. Where in the cunt does”cunt”come from? It ' s one of those words that ' s so old, etymologists and linguists, they lose sight of it eventually. It ' s the oldest word for the vulva that we have in the English language. It might even be the oldest in the world. There are some theories. There are also similar cognitions in Germanic languages all across Europe. The Vikings would be talking about”kuntas,”the Germans had”kuntō,”Dutch,”kont,”Germanic”kott,”and I think at one point we had”kott,”which I think may be due for a revival. After that, it gets a bit confusing as to what this word actually means. One of the leading theories is that it shares this root, this Proto-Indo-European root with this”gen”sound, which you also see in”genetics,””gene,”and that means”to create.”Another theory is that it comes from this sound,”gune,”which gives us”woman,”gynecology.”Create,””woman.”But what really fascinates linguists is this sound, the”cuu”sound, because that gave us”cunt”also it gave us”cunning.””Cunning”originally didn ' t mean”sneaky.”It meant you knew something. Cunning folk, cunning women were wise women. And in Scotland still today, if you can something, it means you know something.”I ken this.”It also gave us”queen”and”cow,”slightly bizarre, which is slightly less highbrow, but – It turns up again in the Middle Ages in”quaint,”which means”knowledge”and also means”cunt.”It has a Latin variation as well,”cunnis,”which also means”cunt,”which turns up all over the Roman world, including in graffiti in Pompeii. Some of my favorite Roman graffiti from the city of Pompeii I won ' t try and do the Latin, but it ' s translated to be”A hairy cunt is better fucked than a smooth one.””It wants cock and holds in steam.”There you go. However, I put it to you that the word”cunt,”as offensive as it may be today, stems from a root that means”woman,””knowledge,””create,””cow.”Has it always been this offensive? No. But we ' ll talk about this. So when we talk about these words,”vulva,””vagina,”trying to offer more palatable alternatives to”cunt,””vagina,”the word turns up in the 17th century. It ' s taken directly from Latin and it means a scabbard. It means something that a sword goes into.”Vulva”doesn ' t do much better. That appears in the 14th century, and it means”womb,”but some people suggest it comes from the French and means”wrapper.”Both these words derive their meaning and their import from the penis, basically. That ' s what a vagina is. It ' s something a sword goes into. I say that these words aren ' t as feminist as”cunt,”which comes from a word that means”queen,””create,””wisdom,””cow.”But when did it first start being used in English as we recognize it today? Gropecunte Lane, this is the first recorded incident in the Oxford English Dictionary, it turns up in 1230, a street name in London called Gropecunte Lane, which was exactly what it sounds like, this was in the red-light district of Southwark, it was a lane for groping cunts. And there wasn ' t just one in London, there was one in Bristol, there was one in York. It appears all over the British Isles. Here it is, the one in Bristol. Sorry, Oxford, there it is just in blue. But whereas Glaswegians might be calling each other and their friends”cunts,”it seems that medieval people were calling their children”cunts”because it turns up in a number of names, bizarrely enough. Godwin Clawecunte is recorded in 1066, Gunoka Cuntles in 1219, John Fillecunt in 1246, Robert Clevecunt, 1302, and a Miss Bele Wydecunthe turns up in the Norfolk subsidiary role. We don ' t know if these are aliases or if they ' re jokes, but we do have a lot of fun with medieval names. In fact, originally the word”fuck”did not mean what it means today. It means to strike something to hit, which gives us the fabulous name of a dairy farmer in 1290 who ' s known as Simon Fuckebotere. So was it this offensive to medieval people? No, it wasn ' t.”Cunts”turn up all over medieval culture and medieval literature. And they are certainly not offensive, it ' s just a descriptive term. Here ' s some examples. The”Proverbs of Hendyng”from 1325 advises women to”give your cunt cunningly and make your demands later,”i. e. get a ring on it first before you give it up. There ’ s a Welsh poet called Gwerful Mechain from the 15th century, and she advises male poets to celebrate the fine bright curtain of a cunt that flaps in place of greeting. It might surprise us that medieval culture was this open about cunts, but the truth was, they were more sexually liberated than we actually give them credit for. This idea of them being in a tower with a chastity belt on is largely a hatchet job on their reputation done by the Victorians. Now, it wasn ’ t a sexually liberated utopia. They had their own hang-ups, but they weren ’ t that offended by sex. What will get you in trouble, swearwords in the Middle Ages, was religious ones, blasphemous ones. If you said something like,”God ’ s wounds”or”God ’ s teeth,”that ’ s what you ’ d say if you caught your soft and danglies in your fly. One medieval poet who dropped the C-bomb with the precision of a military drone is this chap, Geoffrey Chaucer, who turns up in GCSEs and A-levels syllabuses, although his cunt jokes are generally not dwelled upon. He doesn ’ t use the word”cunt,”he uses the word”queynte”here, which again means”knowledge”and it means”cunt.”So this is his joke,”As the clerkes ben ful subtile and ful queynte, And prively he caughte hire by the queynte.”A rough translation means”the clerk was really cunning and he caught her by the cunt.”Shakespeare. It ’ s been suggestion that he uses that play, a quaint - queynte - cunt, in his Sonnet number 20. Here he is. It certainly turns up in a lot of his work. It ’ s a lot ruder than we often give him credit for. In Hamlet, act three, scene two, Hamlet says to Ophelia, he says,”Shall I lie in your lap?”And she says,”Oh, no, my Lord.”And then he says,”Do you think I meant country matters?”When David Tennant played that part, he paused”Do you think I meant count-ry matters?”to try and really drive it home. Another one,”Twelfth Night,”Malvolio says of his mistress ’ s handwriting,”There be her Cs, her Us, her Ts and thus she makes the very great Ps,”punning on”cunt”and”piss”simultaneously. The immortal Bard ’ s status as a smut peddler is often swept under the cultural rug. In 1807, Thomas Bowdler published”The Family Shakespeare,”where he edited out all of these jokes, all of the rude bits, and made it a completely cunt-free affair. It ’ s no surprise that about this time we start to get the first libel laws in Britain, the first banning of seditious and offensive pamphlets with the rise of Puritanism. For Shakespeare to be veiling his cunt jokes in kind of cheeky double entendres suggests that it ’ s not quite as free and open as Gunoka Cuntles and Gropecunte Lane would once have had. The Puritans repressed sexuality, we know this, and language is extremely important battleground for sexual liberation. How do you talk about your bodies if the very words you ’ re trying to use are considered to be offensive? How do you do that? And by the time we get to the Restoration period, the early modern period,”cunt”is most certainly offensive. And this chap here, John Wilmot Earl of Rochester, is the absolute poster boy of”fuck you.”If the Puritans tried to dam up sexuality, this guy surfed to notoriety on a wave of sexual repression that was unleashed when the plug was pulled on the Puritan rule. He uses”cunt”a lot and he ’ s very naughty about it. He wrote this poem about his mistress and how jealous he was of her other lovers.”When your lewd cunt came spewing home Drenched with the seed of half the town, My dram of sperm was supped up after For the digestive surfeit water. Full gorged at another time With a vast meal of slime Which your devouring cunt had drawn From porters ’ backs and footmen ’ s brawn.”Sorry, everyone. He uses that word to shock, and it ’ s easy to look at his work and think that he ’ s sexually liberated, but he ’ s actually quite angry at cunts and their owners and that goes all the way through it. From here on out, cunt is an offensive, naughty word. Georgian cunts, here we go. I ’ ll just let that settle.

So what happens about the 18th century is the print industry really explodes. And of course, we being humans, we didn ’ t just want to publish nice books. We published porn, yay. There ’ s a huge proliferation of porn that comes out of the 18th century. But oddly enough, most of it shies away from using that word”cunt.”In 1785, Francis Grose published his book,”A Dictionary of the Vulgar Tongue,”which is basically a dictionary of slang. And he defined”cunt”as a”nasty name for a nasty thing.”Such modesty from someone who also uses the word”buccaneer ’ s boot,””lobster pot,””skut”and”Mrs. Frub ’ s parlor”for the vulva. This book here,”Harris ’ s List,”this is an almanac, it ’ s a directory of sex workers in London at the time, who were selling sex. And it lists not only their address and their prices, but very, very intimate descriptions of what they do and their vulvas. But it doesn ’ t use”cunt”very much. This one here, this is an illustration, fabulous illustration from”Fanny Hill,”what ’ s often called the first pornographic novel, which was published in 1748 by John Cleland, who famously boasted that he did it without writing any rude words at all. These texts tend to use expressions like”mossy grot,””cupid ’ s coal hole,””Venus ’ s mound,”but we shy away from”cunt.”Victorians ; so despite their reputation for being sexually prudish, pornography flowed underneath Victoria upper-crust society like a river of slime in”Ghostbusters II.”They had pornography all over the place, visual and literary, and they had a lot of fun with”cunt.”One of their pornographic magazines”The Pearl”was published from 1879 to 1880. and it published in it”nursery rhymes”every month. I ’ ve got some here for you to have a look at.”There was a young lady of Hitchin, Who was scrotching her cunt in the kitchen ; Her father said, ’ Rose, It ’ s the crabs, I suppose. ’ ’ You ’ re right, pa, the buggers are itching. ’””There was a young man of Bombay Who fashioned a cunt out of clay ; But the heat of his prick Turned it into a brick, And chafed all his foreskin away.”Yeah, well done, Victorians, well done. Interestingly, it ’ s also in the 19th century that we get the first recorded use of”cunt”being used as an insult. As an actual,”You are a cunt.”That ’ s the first time that it ’ s used in the 19th century. In the 17th century, it started being used as a kind of a derogatory collective noun for women. Samuel Pepys writes about this aphrodisiac that ’ s going to make all the”cunts”chase after him. Charming. That ’ s when they weren ’ t stabbing him with pins for being too sexually aggressive. Anyway, the Victorians liked a well-placed”cunt.”One of the most important”cunt”moments in history is this. Is the publication and the subsequent obscenity trial of”Lady Chatterley ’ s Lover.”This book contained 14”cunts”and 40”fucks,”and it was banned and it had to go on trial in order to be published. And it was shocking, not just because of the graphic scenes of sex and the language used, but because it smashes down class boundaries. If you ’ re not familiar with this, it ’ s about Lady Constance Chatterley, a married woman who embarks on affair with with Sean Bean here but with Mellors the gamekeeper. And the idea is that it doesn ’ t matter all her heirs and graces and titles, she ’ s got a cunt, she ’ s a sexual woman and that levels them. But one of the pivotal scenes is where Mellors tries to tell her what”cunt”means. I won ’ t do the accent.”Nay nay! Fuck ’ s only what animals do, but cunt ’ s a lot more than that. It ’ s thee, dost see : there ’ s a lot more beside an animal, aren ’ t ter? Even ter fuck? Cunt! Eh, that ’ s the beauty of thee lass!””Cunt. That ’ s the beauty of thee, lass!”I love that. Now, despite a jury that agreed a work stuffed full of cunts does have artistic merit and they allowed it to be published, and you can see the pictures of the people queuing around the streets to get their hands on this book once it was,”cunt”never really made it back into the mainstream. Feminists have maintained a rather uneasy relationship with”cunt.”This is Judy Chicago. She led what was called The Cunt Art Movement of the 1970s. It first turned up in a film, a mainstream cinema in 1971 in”Carnal Knowledge”with Jack Nicholson, who screamed at a woman that she is a ball-busting son of a cunt bitch, or words to that effect. And in”The Exorcist”as well. It appears in”The Vagina Monologues,”1996, I think it was, with Eve Ensler when she talks about reclaiming”cunt.”But it ’ s still not off the linguistic naughty step, despite all of this work. Cunts today. It was it was finally admitted to the Oxford English Dictionary, despite having been around for thousands of years, in the ’ 70s. And then in 2014, they relented a little bit more and they added”cunty,””cuntish,””cunted”and”cunting.”So we all know exactly what that means. The Ofcom, the regulator for UK TV censorship in 2016 released a poll of what they regarded to be the most offensive words and”cunt”was bang up there. It was on top. It is still regarded as a horrendously offensive word. But here ’ s what I want to leave you with. What do you call yours? Because as far as I can see, words for vulva or cunts fall into a few categories. We ’ ve got child-like : a tuppence, a Twinkie, a foof, a minky, a Mary. Very medical : a vulva, pudendum, vagina. Slightly detached : down there, it ’ s down there. Bits, special area. Violent : axe wound, penis flytrap, gash or a growler. The taxi driver on my way in told me that Glaswegian slang for cunnilingus is”growling at the badger,”which – I ’ ll leave that with you. Or they just tend to be unpleasant, horrible images of fish and meat and general putrescence, fish taco, bacon sandwich, badly stuffed kebab, bearded clam, etc. Are these better alternatives to”cunt?”But I think the reason that we ’ re not prepared and we can ’ t handle”cunt”is because we can ’ t handle cunts generally. While it ’ s been linguistically sanitized, culturally, the only cunts we seem to be OK with are the ones that have been plucked and buffed and waxed and glued and covered in glitter. So, vajazzled by the way. The vaginoplasty business is booming. You can have your labia cut off, you can have your hymen rebuilt, you can have your pelvic floor resprung. Are we this uncomfortable with the cunt actually as it is? It ’ s a seat of enormous and awesome power. It can eat a penis and push out a baby, it ’ s not a twinkle. It is an old word. It ’ s an offensive word. But it ’ s an ancient and honest one, and this is the thing. This is the original word, everything else came after. So welcome to Team Cunt.

Text Sample 931: NEG Dim 6 – 2015 – Score 1.00 – 2015/t002700.txt

So, I ’ m here to recruit you. But not in the sense that you ’ re thinking. I know I ’ m a politician. I ’ ll save that for another day. I ’ m here to try and encourage you to take up a leadership role in public service in your country and on your continent. I ’ m here to convince you that your country and your continent need you - not later, not when you ’ re older and more experienced, but now - and that whether you realize it or not, your country ’ s politics are going to be doomed to fail unless you ’ re willing to get involved right now. So my recruitment pitch comes with a single disclaimer : I resigned from public office 18 months ago. I did it in order to take stock of my time in office, to think about the work that I had done, to capacitate myself with skills, knowledge, contacts, allies and experiences, and to find a little bit of personal and professional perspective. It ’ s one of the best decisions I think I ’ ve ever made. I imagine that some time during the next 18 minutes while I ’ m pitching you, you ’ re going to think,”Yeah, it ’ s easy for you to say I should go into public service. You ’ ve already done it and you ’ ve left.”But I hope I ’ ll be able to convince you that, in fact, we all find ourselves in exactly the same boat right now. Because being outside of politics for 18 months has reminded me just how important it is and just how much the political landscapes in my country and in your countries and on our continent are truly lacking in good leadership and political talent. So, I want to make a deal with you. I ’ m not going to return to active politics unless you come with me. I ’ m not going to do it alone. I won ’ t go back unless I can convince smart, entrepreneurial, highly skilled, talented, experienced young Africans like yourselves and millions more like you across the continent, that the best chance that our countries have, not just for survival but for lasting prosperity, is if our most talented citizens step forward and make themselves available, either for political party, leadership or for public service and government. So over the next 16-or-so minutes that are remaining, I ’ m going to alternately flatter you, as I just have, I ’ m going to challenge you, I ’ m going to talk to you about my experiences, about a couple of facts and figures ; I may even frighten you a little bit. And it ’ ll be entirely worth it if that fear convinces you of the urgency of the point in history that we find ourselves in today. Everything I say today will be in service of a single objective : convincing you, showing you, that your countries need you ; that Africa ’ s prosperity may depend on many things - entrepreneurialism, industrial development, health reform, social upliftment - but that all of these hinge upon the success of politics and government in our countries. I can ’ t begin a talk about public service, of course, without honoring my former president Nelson Mandela, the father of democratic South Africa. President Mandela passed away on this day in 2013. I really believe that when the people of my country look back on the day that he passed away, it ’ ll be seen as an inflection point in South Africa ’ s history. The day we decided whether we could, indeed, go it alone without him. What ’ s written in those history books will depend entirely on whether this generation, which includes all of you sitting in this room, recognizes that the time has come for us to take up the work that President Mandela left for us, before that work is captured by people who would use power and politics for empty vanity and personal gain. I ’ m referring, of course, to the young man who was here in London this very past week. Defiling the name of the visionary leader, the intellectual and political strategist, the formidable athlete, the Prince of the Abathembu nation who served as a South Africa ’ s first democratic president. The young man who tried to taint President Mandela ’ s legacy with a few throwaway lines, all in service of getting cheap headlines, which he got. People like this, who we leave public service to when we stay out of the fray of public service, are the reason your country and my country needs you and needs us. So let us begin. I want to first talk to you about the African diaspora. You may have heard about a study in 2013 that revealed that cash transfers from Africans living outside of the continent have now begun to exceed donor aid from foreign countries into Africa. In 2012, total remittances to Africa stood at 60 billion dollars while in the same year, official development aid to Sub-Saharan Africa totalled 44. 6 billion by comparison. Now, this got me thinking. If we can do such great work with our money from outside of Africa, what can we do with our skills, our talent, our experiences, our education and our passion for our countries and for our continent? I ’ ve spent the past semester at the Harvard Kennedy School as a fellow at the Institute of Politics. I ran a seminar which was called”How to build a democracy? Lessons from South Africa.”It was also about Zimbabwe and Malawi. And it wasn ’ t intended to make it seem like we got everything right in South Africa, but it was asking the critical question : Now that we have this legacy of peaceful transition, of constitutionalism, of difficult negotiations, which were very, very difficultly gotten, are we going to be successful in entrenching that democracy and making it last into the future? Now, one of the benefits of being an African in an academic setting like New England is that other African students reach out to you, they want to talk to you, and many of them express to you their desire to enter public service. So

I had students knocking down my door, wanting to talk to me in office hours about the fact that they have Ghanean parents but they were born in Texas. They really wanted to give back to Ghana, but they ' re afraid that if they go home, nobody will take them seriously as real Africans. I had students who said they had families, wives, children, husbands, partners to take care of, perhaps they were better off staying in the United States and providing for their families back home rather than going back and getting into public service. This got me thinking about the question of skills remittance, of talent remittance, of social and political remittance. If these young people have the passion to give back to their communities monetarily, imagine how different our politics would be if those same skills, influence, leadership, talent were put at work in service of the public good. And that includes all of you in this room because many of you are also part of the diaspora. I ' m here to recruit you. I ' m here to make a deal with you. I ' m not going back unless I take you with me. Now, I know that most of you, if not the vast majority of you, are completely fed up, turned off, discouraged, disgusted by politics, either in your country, in this country, all over the world. Perhaps you are discouraged by the fact that governments are slow to deliver. Perhaps they ' re inefficient. Perhaps they are thoroughly corrupt and rotten to the core. Perhaps they ' re responsible for conflicts that have claimed lives and livelihoods in the countries from which you come. So why would you sink your time and your energies into such a compromised system? One of the most powerful analyses of conflict, inefficiency, corruption, stagnation which I ' ve encountered in recent months is the question of a political economy. There is a reason that our governments are not performing as they should. It ' s not just because of a failure within the system. Consider the political economy of conflict and corruption in your own country. Why is it so difficult to overcome? Who is making money or amassing power because things don ' t work the way they should? Where does the back stop? Who has an incentive to keep the system dysfunctional? And how can we work together to overcome their total infection of the system, to ensure that we don ' t lose our grip on the very principle of democratic governance? The answer, I ' m afraid, because you were born into this political time, is simply by taking over - you have to get involved. There ' s no way around it. You have to join political organizations in numbers large enough to influence change from within. You have to actively seek to take up a leadership role in government, in the state, in the public service and deftly but decisively move its priorities to where they should be : not in the service of people who want to amass power and money for themselves, but to better the lives of the highest number of people. There will always be government, whether we like it or not, whether we find it palatable or not. But there won ' t always be democracy. If we ignore politics, the people who have been quietly lobbying our governments to prioritize development ahead of democracy, these are the people who will have their way, and the systems that we now take for granted will dissolve before our eyes. When I was campaigning in South Africa last year for the 2014 general election, the voter registration numbers looked a little bit like this, six months before the election : 23 % of potential voters in the 18-to-19-year-old age group were registered to vote. In the age group 20 to 29 years old, 55 % were registered. And from 30 upwards, the number varied from 79 to 100 % ; in fact, there were more people aged 80 and over who were registered than were in the census numbers in South Africa. Imagine that. Fully 100 % of people over a certain age consider voting to be an indispensable right, 21 years into democracy, and do not shirk their responsibility to register and turn out at the polls. But in the 18-to-19-year-old age group - and we must remember 19 is the average age on our continent ; 26 is the average age in South Africa - the number is 23 % to 55

%. What ’ s the political economy of voter apathy? Who benefits when we stay out of the system? Who gets to keep the status quo and empower themselves and enrich themselves and continue to infect our political system like a cancer. Who banks by us continuing with the status quo? Now even as I say all of this to you, that your country and your continent need you to enter public service, I know that if you take up my challenge, you ’ re going to face huge amounts of resistance - all because of these political economies that I have just described. I did. I was told that I was too young. I was too female. I didn ’ t have enough experience though no one could define what experience was enough. I had too much of a white accent ; I wasn ’ t a real African. I straightened my hair and wore weaves ; I wasn ’ t a real African. We should be honest about the things that hold people back from entering public service - humiliation, degradation ; it ’ s not an easy road - but all of these things should illustrate to you the extent to which the status quo is designed to enrich and empower a few at the expense of the many, and it should impart to you the urgency of you, as a generation, of now getting involved in public service to change that very culture. And if you decide to enter public service, you may even be tempted to believe some of these criticisms. They ’ re designed to keep you out ; that ’ s how gatekeeping works. Somebody is benefiting from the absence of excellence and disruption in politics and government. But these are challenges that have to be faced on. There is no other route ; there is no wishing this away. They are the reason that your country and your continent need you. We have this thing in politics in Africa ; it ’ s called the”big man.”The cult of personality - we ’ ve all heard different terminologies for it. In South Africa, in particular, this entails waiting for a great person to come and save us from ourselves. Currently, we ’ re waiting for Cyril Ramaphosa or Nkosazana Dlamini-Zuma or [inaudible] to come and save South Africa from itself, to save us from the mess that we find ourselves in that perhaps another big man put us in. But how can a single personality be held responsible for building or for running a whole nation? And where do we turn when they fail? If we haven ’ t cultivated any kind of pipeline of energetic, young people who wanted to enter public service now or in the future and, critically, who can do the job better, are we doomed to always have to choose between mediocrity and ego, and mediocrity and ego? Is that it? Is that all our government will ever be? Or worse : Are we going to stand by while presidents change constitutions so they can serve a third term and a fourth term and a fifth term, claiming that three million people signed a petition stating that they are the only person who can do the job? Is that what we ’ ll do? Now, there ’ s a new energy around entrepreneurism and innovation and growth in Africa today. But that energy isn ’ t going to translate into lasting prosperity unless we get our politics right. Political leaders who are gatekeepers of the status quo will claim that any success is their success. They ’ ll centralize power, and they ’ ll demand that we all be grateful for those little green shoots of achievement, and then they ’ ll claim that nobody else can do the job. They ’ ll argue that development must come first, freedom can come later, and that they are the best benevolent dictator to do the job. They ’ ll take your political voice from you when times are a little bit good, and when times go bad, they will refuse to give it back. There is no prosperity for our continent without a vibrant, diverse, and truly competitive politics, founded upon excellence, transparency and commitment to the public good. Our politics will not have any of these qualities unless talented, young people, the best people, step forward at this moment in Africa ’ s history, when we ’ re emerging from that stereotype of the dark continent, the hopeless continent, and commit themselves to public service. We must run for office. We must work in the civil service. We must disrupt the political status quo. We must prevent the rush to the bottom. You really are the ones that you have been waiting for. There are no great saviors waiting somewhere in the wings to save us from future problems. There ’ s nobody who is waiting in the wings to come and save us from ourselves ; there ’ s just us. And I ’ m not going back without you. So, will you take up the challenge? Thank you.

Text Sample 932: NEG Dim 6 – 2015 – Score 1.00 – 2015/t001620.txt

We are built out of very small stuff, and we are embedded in a very large cosmos, and the fact is that we are not very good at understanding reality at either of those scales, and that ’ s because our brains haven ’ t evolved to understand the world at that scale. Instead, we ’ re trapped on this very thin slice of perception right in the middle. But it gets strange, because even at that slice of reality that we call home, we ’ re not seeing most of the action that ’ s going on. So take the colors of our world. This is light waves, electromagnetic radiation that bounces off objects and it hits specialized receptors in the back of our eyes. But we ’ re not seeing all the waves out there. In fact, what we see is less than a 10 trillionth of what ’ s out there. So you have radio waves and microwaves and X- rays and gamma rays passing through your body right now and you ’ re completely unaware of it, because you don ’ t come with the proper biological receptors for picking it up. There are thousands of cell phone conversations passing through you right now, and you ’ re utterly blind to it. Now, it ’ s not that these things are inherently unseeable. Snakes include some infrared in their reality, and honeybees include ultraviolet in their view of the world, and of course we build machines in the dashboards of our cars to pick up on signals in the radio frequency range, and we built machines in hospitals to pick up on the X-ray range. But you can ’ t sense any of those by yourself, at least not yet, because you don ’ t come equipped with the proper sensors. Now, what this means is that our experience of reality is constrained by our biology, and that goes against the common sense notion that our eyes and our ears and our fingertips are just picking up the objective reality that ’ s out there. Instead, our brains are sampling just a little bit of the world. Now, across the animal kingdom, different animals pick up on different parts of reality. So in the blind and deaf world of the tick, the important signals are temperature and butyric acid ; in the world of the black ghost knifefish, its sensory world is lavishly colored by electrical fields ; and for the echolocating bat, its reality is constructed out of air compression waves. That ’ s the slice of their ecosystem that they can pick up on, and we have a word for this in science. It ’ s called the umwelt, which is the German word for the surrounding world. Now, presumably, every animal assumes that its umwelt is the entire objective reality out there, because why would you ever stop to imagine that there ’ s something beyond what we can sense. Instead, what we all do is we accept reality as it ’ s presented to us. Let ’ s do a consciousness-raiser on this. Imagine that you are a bloodhound dog. Your whole world is about smelling. You ’ ve got a long snout that has 200 million scent receptors in it, and you have wet nostrils that attract and trap scent molecules, and your nostrils even have slits so you can take big nosefuls of air. Everything is about smell for you. So one day, you stop in your tracks with a revelation. You look at your human owner and you think,”What is it like to have the pitiful, impoverished nose of a human? What is it like when you take a feeble little noseful of air? How can you not know that there ’ s a cat 100 yards away, or that your neighbor was on this very spot six hours ago?”So because we ’ re humans, we ’ ve never experienced that world of smell, so we don ’ t miss it, because we are firmly settled into our umwelt. But the question is, do we have to be stuck there? So as a neuroscientist, I ’ m interested in the way that technology might expand our umwelt, and how that ’ s going to change the experience of being human. So we already know that we can marry our technology to our biology, because there are hundreds of thousands of people walking around with artificial hearing and artificial vision. So the way this works is, you take a microphone and you digitize the signal, and you put an electrode strip directly into the inner ear. Or, with the retinal implant, you take a camera and you digitize the signal, and then you plug an electrode grid directly into the optic nerve. And as recently as 15 years ago, there were a lot of scientists who thought these technologies wouldn ’ t work. Why? It ’ s because these technologies speak the language of Silicon Valley, and it ’ s not exactly the same dialect as our natural biological sense organs. But the fact is that it works ; the brain figures out how to use the signals just fine. Now, how do we understand that? Well, here ’ s the big secret : Your brain is not hearing or seeing any of this. Your brain is locked in a vault of silence and darkness inside your skull. All it ever sees are electrochemical signals that come in along different data cables, and this is all it has to work with, and nothing more. Now, amazingly, the brain is really good at taking in these signals and extracting patterns and assigning meaning, so that it takes this inner cosmos and puts together a story of this, your subjective world. But here ’ s the key point : Your brain doesn ’ t know, and it doesn ’ t care, where it gets the data from. Whatever information comes in, it just figures out what to do with it. And this is a very efficient kind of machine. It ’ s essentially a general purpose computing device, and it just takes in everything and figures out what it ’ s going to do with it, and that, I think, frees up Mother Nature to tinker around with different sorts of input channels. So I call this the P. H. model of evolution, and I don ’ t want to get too technical here, but P. H. stands for Potato Head, and I use this name to emphasize that all these sensors that we know and love, like our eyes and our ears and our fingertips, these are merely peripheral plug-and-play devices : You stick them in, and you ’ re good to go. The brain figures out what to do with the data that comes in. And when you look across the animal kingdom, you find lots of peripheral devices. So snakes have heat pits with which to detect infrared, and the ghost knifefish has electroreceptors, and the star-nosed mole has this appendage with 22 fingers on it with which it feels around and constructs a 3D model of the world, and many birds have magnetite so they can orient to the magnetic field of the planet. So what this means is that nature doesn ’ t have to continually redesign the brain. Instead, with the principles of brain operation established, all nature has to worry about is designing new peripherals. Okay. So what this means is this : The lesson that surfaces is that there ’ s nothing really special or fundamental about the biology that we come to the table with. It ’ s just what we have inherited from a complex road of evolution. But it ’ s not what we have to stick with, and our best proof of principle of this comes from what ’ s called sensory substitution. And that refers to feeding information into the brain via unusual sensory channels, and the brain just figures out what to do with it. Now, that might sound speculative, but the first paper demonstrating this was published in the journal Nature in 1969. So a scientist named Paul Bach-y-Rita put blind people in a modified dental chair, and he set up a video feed, and he put something in front of the camera, and then you would feel that poked into your back with a grid of solenoids. So if you wiggle a coffee cup in front of the camera, you ’ re feeling that in your back, and amazingly, blind people got pretty good at being able to determine what was in front of the camera just by feeling it in the small of their back. Now, there have been many modern incarnations of this. The sonic glasses take a video feed right in front of you and turn that into a sonic landscape, so as things move around, and get closer and farther, it sounds like"Bzz, bzz, bzz."It sounds like a cacophony, but after several weeks, blind people start getting pretty good at understanding what ' s in front of them just based on what they ' re hearing. And it doesn ' t have to be through the ears : this system uses an electrotactile grid on the forehead, so whatever ' s in front of the video feed, you ' re feeling it on your forehead. Why the forehead? Because you ' re not using it for much else. The most modern incarnation is called the brainport, and this is a little electrogrid that sits on your tongue, and the video feed gets turned into these little electrotactile signals, and blind people get so good at using this that they can throw a ball into a basket, or they can navigate complex obstacle courses. They can come to see through their tongue. Now, that sounds completely insane, right? But remember, all vision ever is is electrochemical signals coursing around in your brain. Your brain doesn ' t know where the signals come from. It just figures out what to do with them. So my interest in my lab is sensory substitution for the deaf, and this is a project I ' ve undertaken with a graduate student in my lab, Scott Novich, who is spearheading this for his thesis. And here is what we wanted to do : we wanted to make it so that sound from the world gets converted in some way so that a deaf person can understand what is being said. And we wanted to do this, given the power and ubiquity of portable computing, we wanted to make sure that this would run on cell phones and tablets, and also we wanted to make this a wearable, something that you could wear under your clothing. So here ' s the concept. So as I ' m speaking, my sound is getting captured by the tablet, and then it ' s getting mapped onto a vest that ' s covered in vibratory motors, just like the motors in your cell phone. So as I ' m speaking, the sound is getting translated to a pattern of vibration on the vest. Now, this is not just conceptual : this tablet is transmitting Bluetooth, and I ' m wearing the vest right now. So as I ' m speaking — — the sound is getting translated into dynamic patterns of vibration. I ' m feeling the sonic world around me. So, we ' ve been testing this with deaf people now, and it turns out that after just a little bit of time, people can start feeling, they can start understanding the language of the vest. So this is Jonathan. He ' s 37 years old. He has a master ' s degree. He was born profoundly deaf, which means that there ' s a part of his umwelt that ' s unavailable to him. So we had Jonathan train with the vest for four days, two hours a day, and here he is on the fifth day. Scott Novich : You. David Eagleman : So Scott says a word, Jonathan feels it on the vest, and he writes it on the board. SN : Where. Where. DE : Jonathan is able to translate this complicated pattern of vibrations into an understanding of what ' s being said. SN : Touch. Touch. DE : Now, he ' s not doing this — — Jonathan is not doing this consciously, because the patterns are too complicated, but his brain is starting to unlock the pattern that allows it to figure out what the data mean, and our expectation is that, after wearing this for about three months, he will have a direct perceptual experience of hearing in the same way that when a blind person passes a finger over braille, the meaning comes directly off the page without any conscious intervention at all. Now, this technology has the potential to be a game-changer, because the only other solution for deafness is a cochlear implant, and that requires an invasive surgery. And this can be built for 40 times cheaper than a cochlear implant, which opens up this technology globally, even for the poorest countries. Now, we ' ve been very encouraged by our results with sensory substitution, but what we ' ve been thinking a lot about is sensory addition. How could we use a technology like this to add a completely new kind of sense, to expand the human umvelt? For example, could we feed real-time data from the Internet directly into somebody ' s brain, and can they develop a direct perceptual experience? So here ' s an experiment we ' re doing in the lab. A subject is feeling a real- time streaming feed from the Net of data for five seconds. Then, two buttons appear, and he has to make a choice. He doesn ’ t know what ’ s going on. He makes a choice, and he gets feedback after one second. Now, here ’ s the thing : The subject has no idea what all the patterns mean, but we ’ re seeing if he gets better at figuring out which button to press. He doesn ’ t know that what we ’ re feeding is real-time data from the stock market, and he ’ s making buy and sell decisions. And the feedback is telling him whether he did the right thing or not. And what we ’ re seeing is, can we expand the human umvelt so that he comes to have, after several weeks, a direct perceptual experience of the economic movements of the planet. So we ’ ll report on that later to see how well this goes. Here ’ s another thing we ’ re doing : During the talks this morning, we ’ ve been automatically scraping Twitter for the TED2015 hashtag, and we ’ ve been doing an automated sentiment analysis, which means, are people using positive words or negative words or neutral? And while this has been going on, I have been feeling this, and so I am plugged in to the aggregate emotion of thousands of people in real time, and that ’ s a new kind of human experience, because now I can know how everyone ’ s doing and how much you ’ re loving this. It ’ s a bigger experience than a human can normally have. We ’ re also expanding the umvelt of pilots. So in this case, the vest is streaming nine different measures from this quadcopter, so pitch and yaw and roll and orientation and heading, and that improves this pilot ’ s ability to fly it. It ’ s essentially like he ’ s extending his skin up there, far away. And that ’ s just the beginning. What we ’ re envisioning is taking a modern cockpit full of gauges and instead of trying to read the whole thing, you feel it. We live in a world of information now, and there is a difference between accessing big data and experiencing it. So I think there ’ s really no end to the possibilities on the horizon for human expansion. Just imagine an astronaut being able to feel the overall health of the International Space Station, or, for that matter, having you feel the invisible states of your own health, like your blood sugar and the state of your microbiome, or having 360-degree vision or seeing in infrared or ultraviolet. So the key is this : As we move into the future, we ’ re going to increasingly be able to choose our own peripheral devices. We no longer have to wait for Mother Nature ’ s sensory gifts on her timescales, but instead, like any good parent, she ’ s given us the tools that we need to go out and define our own trajectory. So the question now is, how do you want to go out and experience your universe? Thank you. Chris Anderson : Can you feel it? DE : Yeah. Actually, this was the first time I felt applause on the vest. It ’ s nice. It ’ s like a massage. CA : Twitter ’ s going crazy. Twitter ’ s going mad. So that stock market experiment. This could be the first experiment that secures its funding forevermore, right, if successful? DE : Well, that ’ s right, I wouldn ’ t have to write to NIH anymore. CA : Well look, just to be skeptical for a minute, I mean, this is amazing, but isn ’ t most of the evidence so far that sensory substitution works, not necessarily that sensory addition works? I mean, isn ’ t it possible that the blind person can see through their tongue because the visual cortex is still there, ready to process, and that that is needed as part of it? DE : That ’ s a great question. We actually have no idea what the theoretical limits are of what kind of data the brain can take in. The general story, though, is that it ’ s extraordinarily flexible. So when a person goes blind, what we used to call their visual cortex gets taken over by other things, by touch, by hearing, by vocabulary. So what that tells us is that the cortex is kind of a one-trick pony. It just runs certain kinds of computations on things. And when we look around at things like braille, for example, people are getting information through bumps on their fingers. So I don ’ t think we have any reason to think there ’ s a theoretical limit that we know the edge of. CA

: If this checks out, you ’ re going to be deluged. There are so many possible applications for this. Are you ready for this? What are you most excited about, the direction it might go? DE : I mean, I think there ’ s a lot of applications here. In terms of beyond sensory substitution, the things I started mentioning about astronauts on the space station, they spend a lot of their time monitoring things, and they could instead just get what ’ s going on, because what this is really good for is multidimensional data. The key is this : Our visual systems are good at detecting blobs and edges, but they ’ re really bad at what our world has become, which is screens with lots and lots of data. We have to crawl that with our attentional systems. So this is a way of just feeling the state of something, just like the way you know the state of your body as you ’ re standing around. So I think heavy machinery, safety, feeling the state of a factory, of your equipment, that ’ s one place it ’ ll go right away. CA : David Eagleman, that was one mind-blowing talk. Thank you very much. DE : Thank you, Chris.

Text Sample 933: NEG Dim 6 – 2015 – Score 1.00 – 2015/t001615.txt

I ’ m a potter, which seems like a fairly humble vocation. I know a lot about pots. I ’ ve spent about 15 years making them. One of the things that really excites me in my artistic practice and being trained as a potter is that you very quickly learn how to make great things out of nothing ; that I spent a lot of time at my wheel with mounds of clay trying stuff ; and that the limitations of my capacity, my ability, was based on my hands and my imagination ; that if I wanted to make a really nice bowl and I didn ’ t know how to make a foot yet, I would have to learn how to make a foot ; that that process of learning has been very, very helpful to my life. I feel like, as a potter, you also start to learn how to shape the world. There have been times in my artistic capacity that I wanted to reflect on other really important moments in the history of the U. S., the history of the world where tough things happened, but how do you talk about tough ideas without separating people from that content? Could I use art like these old, discontinued firehoses from Alabama, to talk about the complexities of a moment of civil rights in the ’ 60s? Is it possible to talk about my father and I doing labor projects? My dad was a roofer, construction guy, he owned small businesses, and at 80, he was ready to retire and his tar kettle was my inheritance. Now, a tar kettle doesn ’ t sound like much of an inheritance. It wasn ’ t. It was stinky and it took up a lot of space in my studio, but I asked my dad if he would be willing to make some art with me, if we could reimagine this kind of nothing material as something very special. And by elevating the material and my dad ’ s skill, could we start to think about tar just like clay, in a new way, shaping it differently, helping us to imagine what was possible? After clay, I was then kind of turned on to lots of different kinds of materials, and my studio grew a lot because I thought, well, it ’ s not really about the material, it ’ s about our capacity to shape things. I became more and more interested in ideas and more and more things that were happening just outside my studio. Just to give you a little bit of context, I live in Chicago. I live on the South Side now. I ’ m a West Sider. For those of you who are not Chicagoans, that won ’ t mean anything, but if I didn ’ t mention that I was a West Sider, there would be a lot of people in the city that would be very upset. The neighborhood that I live in is Grand Crossing. It ’ s a neighborhood that has seen better days. It is not a gated community by far. There is lots of abandonment in my neighborhood, and while I was kind of busy making pots and busy making art and having a good art career, there was all of this stuff that was happening just outside my studio. All of us know about failing housing markets and the challenges of blight, and I feel like we talk about it with some of our cities more than others, but I think a lot of our U. S. cities and beyond have the challenge of blight, abandoned buildings that people no longer know what to do anything with. And so I thought, is there a way that I could start to think about these buildings as an extension or an expansion of my artistic practice? And that if I was thinking along with other creatives — architects, engineers, real estate finance people — that us together might be able to kind of think in more complicated ways about the reshaping of cities. And so I bought a building. The building was really affordable. We tricked it out. We made it as beautiful as we could to try to just get some activity happening on my block. Once I bought the building for about 18, 000 dollars, I didn ’ t have any money left. So I started sweeping the building as a kind of performance. This is performance art, and people would come over, and I would start sweeping. Because the broom was free and sweeping was free. It worked out. But we would use the building, then, to stage exhibitions, small dinners, and we found that that building on my block, Dorchester — we now referred to the block as Dorchester projects — that in a way that building became a kind of gathering site for lots of different kinds of activity. We turned the building into what we called now the Archive House. The Archive House would do all of these amazing things. Very significant people in the city and beyond would find themselves in the middle of the hood. And that ’ s when I felt like maybe there was a relationship between my history with clay and this new thing that was starting to develop, that we were slowly starting to reshape how people imagined the South Side of the city. One house turned into a few houses, and we always tried to suggest that not only is creating a beautiful vessel important, but the contents of what happens in those buildings is also very important. So we were not only thinking about development, but we were thinking about the program, thinking about the kind of connections that could happen between one house and another, between one neighbor and another. This building became what we call the Listening House, and it has a collection of discarded books from the Johnson Publishing Corporation, and other books from an old bookstore that was going out of business. I was actually just wanting to activate these buildings as much as I could with whatever and whoever would join me. In Chicago, there ’ s amazing building stock. This building, which had been the former crack house on the block, and when the building became abandoned, it became a great opportunity to really imagine what else could happen there. So this space we converted into what we call Black Cinema House. Black Cinema House was an opportunity in the hood to screen films that were important and relevant to the folk who lived around me, that if we wanted to show an old Melvin Van Peebles film, we could. If we wanted to show”Car Wash,”we could. That would be awesome. The building we soon outgrew, and we had to move to a larger space. Black Cinema House, which was made from just a small piece of clay, had to grow into a much larger piece of clay, which is now my studio. What I realized was that for those of you who are zoning junkies, that some of the things that I was doing in these buildings that had been left behind, they were not the uses by which the buildings were built, and that there are city policies that say,”Hey, a house that is residential needs to stay residential.”But what do you do in neighborhoods when ain ’ t nobody interested in living there? That the people who have the means to leave have already left? What do we do with these abandoned buildings? And so I was trying to wake them up using culture. We found that that was so exciting for folk, and people were so responsive to the work, that we had to then find bigger buildings. By the time we found bigger buildings, there was, in part, the resources necessary to think about those things. In this bank that we called the Arts Bank, it was in pretty bad shape. There was about six feet of standing water. It was a difficult project to finance, because banks weren ’ t interested in the neighborhood because people weren ’ t interested in the neighborhood because nothing had happened there. It was dirt. It was nothing. It was nowhere. And so we just started imagining, what else could happen in this building? And so now that the rumor of my block has spread, and lots of people are starting to visit, we ’ ve found that the bank can now be a center for exhibition, archives, music performance, and that there are people who are now interested in being adjacent to those buildings because we brought some heat, that we kind of made a fire. One of the archives that we ’ ll have there is this Johnson Publishing Corporation. We ’ ve also started to collect memorabilia from American history, from people who live or have lived in that neighborhood. Some of these images are degraded images of black people, kind of histories of very challenging content, and where better than a neighborhood with young people who are constantly asking themselves about their identity to talk about some of the complexities of race and class? In some ways, the bank represents a hub, that we ’ re trying to create a pretty hardcore node of cultural activity, and that if we could start to make multiple hubs and connect some cool green stuff around there, that the buildings that we ’ ve purchased and rehabbed, which is now around 60 or 70 units, that if we could land miniature Versailles on top of that, and connect these buildings by a beautiful greenbelt — — that this place where people never wanted to be would become an important destination for folk from all over the country and world. In some ways, it feels very much like I ’ m a potter, that we tackle the things that are at our wheel, we try with the skill that we have to think about this next bowl that I want to make. And it went from a bowl to a singular house to a block to a neighborhood to a cultural district to thinking about the city, and at every point, there were things that I didn ’ t know that I had to learn. I ’ ve never learned so much about zoning law in my life. I never thought I ’ d have to. But as a result of that, I ’ m finding that there ’ s not just room for my own artistic practice, there ’ s room for a lot of other artistic practices. So people started asking us,”Well, Theaster, how are you going to go to scale?”and,”What ’ s your sustainability plan?”And what I found was that I couldn ’ t export myself, that what seems necessary in cities like Akron, Ohio, and Detroit, Michigan, and Gary, Indiana, is that there are people in those places who already believe in those places, that are already dying to make those places beautiful, and that often, those people who are passionate about a place are disconnected from the resources necessary to make cool things happen, or disconnected from a contingency of people that could help make things happen. So now, we ’ re starting to give advice around the country on how to start with what you got, how to start with the things that are in front of you, how to make something out of nothing, how to reshape your world at a wheel or at your block or at the scale of the city. Thank you so much. June Cohen : Thank you. So I think many people watching this will be asking themselves the question you just raised at the end : How can they do this in their own city? You can ’ t export yourself. Give us a few pages out of your playbook about what someone who is inspired about their city can do to take on projects like yours? Theaster Gates : One of the things I ’ ve found that ’ s really important is giving thought to not just the kind of individual project, like an old house, but what ’ s the relationship between an old house, a local school, a small bodega, and is there some kind of synergy between those things? Can you get those folk talking? I ’ ve found that in cases where neighborhoods have failed, they still often have a pulse. How do you identify the pulse in that place, the passionate people, and then how do you get folk who have been fighting, slogging for 20 years, reenergized about the place that they live? And so someone has to do that work. If I were a traditional developer, I would be talking about buildings alone, and then putting a"For Lease"sign in the window. I think that you actually have to curate more than that, that there ' s a way in which you have to be mindful about, what are the businesses that I want to grow here? And then, are there people who live in this place who want to grow those businesses with me? Because I think it ' s not just a cultural space or housing ; there has to be the recreation of an economic core. So thinking about those things together feels right. JC : It ' s hard to get people to create the spark again when people have been slogging for 20 years. Are there any methods you ' ve found that have helped break through? TG : Yeah, I think that now there are lots of examples of folk who are doing amazing work, but those methods are sometimes like, when the media is constantly saying that only violent things happen in a place, then based on your skill set and the particular context, what are the things that you can do in your neighborhood to kind of fight some of that? So I ' ve found that if you ' re a theater person, you have outdoor street theater festivals. In some cases, we don ' t have the resources in certain neighborhoods to do things that are a certain kind of splashy, but if we can then find ways of making sure that people who are local to a place, plus people who could be supportive of the things that are happening locally, when those people get together, I think really amazing things can happen. JC : So interesting. And how can you make sure that the projects you ' re creating are actually for the disadvantaged and not just for the sort of vegetarian indie movie crowd that might move in to take advantage of them. TG : Right on. So I think this is where it starts to get into the thick weeds. JC : Let ' s go there. TG : Right now, Grand Crossing is 99 percent black, or at least living, and we know that maybe who owns property in a place is different from who walks the streets every day. So it ' s reasonable to say that Grand Crossing is already in the process of being something different than it is today. But are there ways to think about housing trusts or land trusts or a mission-based development that starts to protect some of the space that happens, because when you have 7, 500 empty lots in a city, you want something to happen there, but you need entities that are not just interested in the development piece, but entities that are interested in the stabilization piece, and I feel like often the developer piece is really motivated, but the other work of a kind of neighborhood consciousness, that part doesn ' t live anymore. So how do you start to grow up important watchdogs that ensure that the resources that are made available to new folk that are coming in are also distributed to folk who have lived in a place for a long time. JC : That makes so much sense. One more question : You make such a compelling case for beauty and the importance of beauty and the arts. There would be others who would argue that funds would be better spent on basic services for the disadvantaged. How do you combat that viewpoint, or come against it? TG : I believe that beauty is a basic service. Often what I have found is that when there are resources that have not been made available to certain under-resourced cities or neighborhoods or communities, that sometimes culture is the thing that helps to ignite, and that I can ' t do everything, but I think that there ' s a way in which if you can start with culture and get people kind of reinvested in their place, other kinds of adjacent amenities start to grow, and then people can make a demand that ' s a poetic demand, and the political demands that are necessary to wake up our cities, they also become very poetic. JC : It makes perfect sense to me. Theaster, thank you so much for being here with us today. Thank you. Theaster Gates.

Text Sample 934: NEG Dim 6 – 2014 – Score 1.00 – 2014/t002699.txt

Hello everybody. Audience : Hello. Keith Lowe : Fantastic! This is like a schoolroom or something. My name is Keith Lowe. I am an historian of the Second World War and its aftermath, and even I have to admit that I ' ve chosen a pretty crowded field to study. I went into my local bookshop recently, and this is what I saw. Thousands of books about the Second World War are published every year, and, actually, to tell you the truth, this is only a very tiny selection of what ' s on offer. We in the West, and, actually, increasingly people in other parts of the world too, we are just a little bit obsessed by the Second World War. We have whole TV stations which seem devoted to it. We write books about it, we write novels about it, we make films about it. We have university courses which are devoted to the Second World War. Whole museums are built to house World War II collections. Even our politicians like to get in on this act. Whenever there is an important anniversary of the war, they tend to gather and commemorate it, and make speeches. So, for example, at the 70th anniversary of the D-Day landings, June 2014, 17 heads of state took time out of their schedules to come and spend the day on the Normandy beaches. Seventeen! This included people like Barack Obama, Vladimir Putin, the chancellor of Germany, Angela Merkel, and so on. From my own country, we sent not only our Prime Minister, but also Queen Elisabeth II, who is - I mean, she is now in her nineties and largely retired from public life. What other international event can do all this? Even international summits struggle to get so many world leaders into one place at the same time. My question is : Why? What is it that all these world leaders, in fact what is it that all of us, think it is that we ' re remembering? Why are we all so obsessed by the the Second World War? You might think that somebody like me would be pretty pleased with this situation. As long as the World War II industry is booming, I am always going to have a job, right? But actually there is something about it that I find really a little bit disturbing, and I don ' t know whether that is just because I have an inherent distrust of a lot of politicians, or whether it ' s because I ' ve been trained to always question everything. But it strikes me that a lot of the rhetoric that gets thrown around about the Second World War, particularly by people like politicians and journalists and diplomats and so on, a lot of it doesn ' t seem to be about the Second World War at all ; it seems to be about something else. I ' m not sure if I ' ve exactly put my finger on precisely what that thing is, but it seems to be something like a way of fostering national pride, or just trying to get people to feel good about themselves. Along the way, it seems to me that the Second World War has been turned into a little bit of a cartoon, where everybody knows who the good guys were, and everybody knows who the bad guys were. There is precious little space left anymore for any of that difficult grey area in between. To give you some kind of an idea about what on earth it is I am going on about, let me tell you a story from my own country, from Britain. In Britain, we like to think we are the real heroes of the Second World War. We tell stories about how we stood alone against the Nazis, about how we endured the bombing of the Blitz ; how we kept calm and carried on, and eventually fought our way back into Europe and liberated it. We still call the Second World War"Our Finest Hour", as if it is some kind of golden age in our history. So whenever there is any kind of anniversary, or important event based around the Second World War, we Brits really go for it. One of these events happened quite recently, in the summer of 2012, when we opened up a brand new war memorial right in the middle of central London. It was a memorial to the men of Bomber Command, the men who flew the planes over Germany, dropped bombs and so on. This is what it looks like. As you can see, it ' s not exactly a shy and retiring piece of architecture, it ' s actually quite huge. It ' s by far the largest war memorial that we have in London, and I can tell you there are a lot of war memorials in London. As you walk into this thing, there is an inscription which tells you that it ' s dedicated to the 55, 000 men of Bomber Command who lost their lives during World War II. Now, when this first opened, in the summer of 2012, I went along to have a look, see what I thought of it, have a walk around. There is something really quite moving about it actually. You step in through these great big pillars, and, up on the wall, you can see carved into the stone, there is a quotation from Winston Churchill saying exactly how much we owe to these men who lost their lives. Parts of the memorial are built out of an actual World War II aircraft that was shot down during the war. So they put a lot of thought into this. It ' s actually really quite inspiring. So as I was walking around this monument, I couldn ' t help but feel this real surge of pride. I felt proud of these men who had given their lives for something that I hold dear. I felt proud of my country, I felt proud of the British way of life which had produced heroes like this. And yet, there was this little voice in the back of my head which just wouldn ' t go away because I know that 55, 000 men of Bomber Command died during World War II, but I also know that 500, 000 Germans died beneath the bombs that these men dropped. A lot of those Germans were Nazis, and I dare say that a lot of them probably deserved it. But the vast majority of these people were just ordinary men, women and children, just like you and me. Are any of these people at all mentioned on this memorial? Of course they are not. If you would suggest such a thing in the summer of 2012, you probably would have been lynched. The Germans were our enemies during the Second World War. You can ' t mention the Germans on a British national monument. And yet not to mention them, to pretend that somehow this didn ' t happen, or even worse, that somehow it doesn ' t matter, that too makes me feel a little bit uncomfortable. Okay, the Germans are a difficult problem, so let ' s just put them to one side for a minute, and let ' s think instead about the other nationalities. And here is where the story starts getting interesting. Because if there is one thing that we Brits always forget about the bomber war, it ' s the fact that we didn ' t only bomb Germany. More than a third of British and American bombs dropped on Europe during the war were dropped not on Germany but on those countries we were supposed to be liberating. As a consequence, 50, 000 French civilians were killed by our bombs. 10, 000 Dutch civilians were killed by our bombs. Are any of these people mentioned on this memorial? Of course they ' re not. And it was while I was thinking about that particular group, that it finally dawned on me - actually something I probably should have realized right from the start - which is that memorials like this aren ' t designed to tell the whole story ; they are only designed to tell those parts of the story that make British people feel good about themselves. That ' s all very well and good, but it does come at a cost. And I couldn ' t help thinking when I was walking around this thing : This was the summer of 2012, this was the summer when the Olympic Games was coming to London. So at exactly the moment when the entire world was arriving in our city, the message that we were advertising was that we will remember our wartime dead, but we won ' t remember yours. It ' s like the exact opposite of the Olympic spirit. It ' s not only the British who do this, of course it ' s not. Every nation does it, the Americans, for example. The Americans like to call their wartime veterans"the greatest generation that any society has ever produced", as if they have some kind of monopoly on heroism or something. They backed it up with a thousand Hollywood movies full of square- jawed American heroes defeating evil in the name of truth and freedom. The Chinese are the same. You know that in 2013 alone, Chinese TV companies produced over 200 TV dramatizations about the Second World War, each of them telling almost an identical story. Only, of course, this time, it ' s the Japanese who are all the evil monsters, and the Chinese who are all selfless heroes. And, of course, I could say the same thing about the French or the Koreans or the Norwegians or the Greeks. We all do this. We all like to think that we were the heroes. We all like to think that we were the victims. But what we don ’ t like to remember is those grey areas. And it ’ s that which I find most uncomfortable about this, because, as far as I am concerned, it ’ s the grey areas that make history interesting. In a sense, all good history is about the grey areas. But a lot of people don ’ t seem to have time for complicated stories. They don ’ t have time for difficult and uncomfortable emotions. In fact, I ’ m quickly coming to the conclusion that a lot of people don ’ t really have time for history at all. What they really want is a myth. Now, you might ask yourselves : What does any of this matter? I mean, we all like a good story, don ’ t we? If that story makes us feel good about ourselves, then so much the better. It ’ s all in the past anyway, so what does it matter? But that ’ s just the problem, isn ’ t it? Because it ’ s not all in the past. And there is a dark side to all of these stories and myths that can be really damaging. When the former Yugoslavia tore itself apart in the 1990s, it did so with World War II songs on its lips, and World War II atrocities in its heart. When the Ukraine crisis broke out in 2014, Ukrainians and Russians accused one another of acting like Nazis. And then, of course, Hillary Clinton weighed in and started comparing Vladimir Putin to Hitler. These sorts of comparisons don ’ t do anything to foster rational debate. If you were in an argument with someone, the last thing that is going to calm things down is that you start accusing them of being a Nazi. If you don ’ t believe me, next time you ’ re in an argument with your wife or your husband, give it a try and see what kind of reaction you get. I can see some of you seem to have tried it. My point is that as soon as we start bringing the Second World War into any of our arguments, we get so sort of carried away with our own national myths that all we actually end up doing is stirring things back up again. Let me give you a couple of examples. Take this economic crisis which has rocked the world since 2008. Here we are in Athens, and you all know about the economic crisis. All across Southern Europe, people have been suffering a real austerity, and largely this has been imposed by the European Union. But it ’ s not always the European Union that gets the blame for this. Quite often, as the largest and most powerful country in the union, it ’ s Germany that gets the blame. Now, how has this been portrayed in the press? Have we had a calm, rational economic debate about it? This is the way that the Italian press portrayed the situation in 2012 :”Quarto Reich.””The Fourth Reich.”This is the Italian way of saying that modern-day Germany is no better than the Nazis, as if there is a direct link between World War II and today. And take a look at that picture. They ’ ve managed to dig something out that makes it look like Angela Merkel is making a Nazi salute. How about the Greek press? How have the Greeks portrayed it? Well, here is a Greek newspaper. from the same year, 2012, and you will notice a photograph of Angela Merkel once again, this time in a Nazi uniform. Obviously, it has been photoshopped. But what about that headline in red?”Memorandum macht frei.”This is a direct reference to the motto that was written above the gates of the concentration camps in places like Auschwitz and Dachau. The implication here is that the whole of Greece is going to become like one giant German concentration camp as a consequence of the economic deal they ’ ve had to do. Now, this is the sort of thing that makes an historian like me want to just give up and go and become a window cleaner or something. I mean, it ’ s historical nonsense. None of these headlines have anything to do with the Second World War at all. They are about a modern-day problem, a modern situation. The only reason to mention the Second World War is to provoke an emotional response. If I have one message that I want you to take away with you today, it is this : Whenever you hear a politician, or a journalist, or a diplomat, mention the Second World War, I want alarm bells to ring. Because when public figures speak about the Second World War, they are not talking about what actually happened, they ' re invoking a myth. So whenever you hear a politician mention the war, I want you to ask yourselves what it is he is really trying to do. Is he trying to inspire you? - In which case that is relatively harmless. Or is he trying to fill you with fear? Is he trying to draw people together? In which case, again, that is relatively harmless. Or is he really trying to drive people apart? And above all, I want you to remind yourselves, and to remind everybody you know, that the Second World War is over. We live in a different world, with different values and different problems. These problems will never be solved, and they certainly will never be solved peacefully, if all we can think to do is to resurrect the Second World War. You know, history is a messy business, particularly the history of the Second World War. It ' s not there to make us feel good about ourselves ; it ' s often ugly and uncomfortable, and desperately complicated. It ' s full of those grey areas. If we could all just learn to accept that, then this world, our world, would be a much more peaceful place. Thank you.

Text Sample 935: NEG Dim 6 – 2014 – Score 1.00 – 2014/t001905.txt

Two years ago, I have to say there was no problem. Two years ago, I knew exactly what an icon looked like. It looks like this. Everybody ' s icon, but also the default position of a curator of Italian Renaissance paintings, which I was then. And in a way, this is also another default selection. Leonardo da Vinci ' s exquisitely soulful image of the"Lady with an Ermine."And I use that word, soulful, deliberately. Or then there ' s this, or rather these : the two versions of Leonardo ' s"Virgin of the Rocks"that were about to come together in London for the very first time. In the exhibition that I was then in the absolute throes of organizing. I was literally up to my eyes in Leonardo, and I had been for three years. So, he was occupying every part of my brain. Leonardo had taught me, during that three years, about what a picture can do. About taking you from your own material world into a spiritual world. He said, actually, that he believed the job of the painter was to paint everything that was visible and invisible in the universe. That ' s a huge task. And yet, somehow he achieves it. He shows us, I think, the human soul. He shows us the capacity of ourselves to move into a spiritual realm. To see a vision of the universe that ' s more perfect than our own. To see God ' s own plan, in some sense. So this, in a sense, was really what I believed an icon was. At about that time, I started talking to Tom Campbell, director here of the Metropolitan Museum, about what my next move might be. The move, in fact, back to an earlier life, one I ' d begun at the British Museum, back to the world of three dimensions — of sculpture and of decorative arts — to take over the department of European sculpture and decorative arts, here at the Met. But it was an incredibly busy time. All the conversations were done at very peculiar times of the day — over the phone. In the end, I accepted the job without actually having been here. Again, I ' d been there a couple of years before, but on that particular visit. So, it was just before the time that the Leonardo show was due to open when I finally made it back to the Met, to New York, to see my new domain. To see what European sculpture and decorative arts looked like, beyond those Renaissance collections with which I was so already familiar. And I thought, on that very first day, I better tour the galleries. Fifty-seven of these galleries — like 57 varieties of baked beans, I believe. I walked through and I started in my comfort zone in the Italian Renaissance. And then I moved gradually around, feeling a little lost sometimes. My head, also still full of the Leonardo exhibition that was about to open, and I came across this. And I thought to myself : What the hell have I done? There was absolutely no connection in my mind at all and, in fact, if there was any emotion going on, it was a kind of repulsion. This object felt utterly and completely alien. Silly at a level that I hadn ’ t yet understood silliness to be. And then it was made worse — there were two of them. So, I started thinking about why it was, in fact, that I disliked this object so much. What was the anatomy of my distaste? Well, so much gold, so vulgar. You know, so nouveau riche, frankly. Leonardo himself had preached against the use of gold, so it was absolutely anathema at that moment. And then there ’ s little pretty sprigs of flowers everywhere. And finally, that pink. That damned pink. It ’ s such an extraordinarily artificial color. I mean, it ’ s a color that I can ’ t think of anything that you actually see in nature, that looks that shade. The object even has its own tutu. This little flouncy, spangly, bottomy bit that sits at the bottom of the vase. It reminded me, in an odd kind of way, of my niece ’ s fifth birthday party. Where all the little girls would come either as a princess or a fairy. There was one who would come as a fairy princess. You should have seen the looks. And I realize that this object was in my mind, born from the same mind, from the same womb, practically, as Barbie Ballerina. And then there ’ s the elephants. Those extraordinary elephants with their little, sort of strange, sinister expressions and Greta Garbo eyelashes, with these golden tusks and so on. I realized this was an elephant that had absolutely nothing to do with a majestic march across the Serengeti. It was a Dumbo nightmare. But something more profound was happening as well. These objects, it seemed to me, were quintessentially the kind that I and my liberal left friends in London had always seen as summing up something deplorable about the French aristocracy in the 18th century. The label had told me that these pieces were made by the Svres Manufactory, made of porcelain in the late 1750s, and designed by a designer called Jean-Claude Duplessis, actually somebody of extraordinary distinction as I later learned. But for me, they summed up a kind of, that sort of sheer uselessness of the aristocracy in the 18th century. I and my colleagues had always thought that these objects, in way, summed up the idea of, you know — no wonder there was a revolution. Or, indeed, thank God there was a revolution. There was a sort of idea really, that, if you owned a vase like this, then there was really only one fate possible. So, there I was — in a sort of paroxysm of horror. But I took the job and I went on looking at these vases. I sort of had to because they ’ re on a through route in the Met. So, almost anywhere I went, there they were. They had this kind of odd sort of fascination, like a car accident. Where I couldn ’ t stop looking. And as I did so, I started thinking : Well, what are we actually looking at here? And what I started with was understanding this as really a supreme piece of design. It took me a little time. But, that tutu for example — actually, this is a piece that does dance in its own way. It has an extraordinary lightness and yet, it is also amazing balanced. It has these kinds of sculptural ingredients. And then the play between — actually really quite carefully disposed color and gilding, and the sculptural surface, is really rather remarkable. And then I realized that this piece went into the kiln four times, at least four times in order to arrive at this. How many moments for accident can you think of that could have happened to this piece? And then remember, not just one, but two. So he ’ s having to arrive at two exactly matched vases of this kind. And then this question of uselessness. Well actually, the end of the trunks were originally candle holders. So what you would have had were candles on either side. Imagine that effect of candlelight on that surface. On the slightly uneven pink, on the beautiful gold. It would have glittered in an interior, a little like a little firework. And at that point, actually, a firework went off in my brain. Somebody reminded me that, that word ’ fancy ’ — which in a sense for me, encapsulated this object — actually comes from the same root as the word ’ fantasy. ’ And that what this object was just as much in a way, in its own way, as a Leonardo da Vinci painting, is a portal to somewhere else. This is an object of the imagination. If you think about the mad 18th-century operas of the time — set in the Orient. If you think about divans and perhaps even opium-induced visions of pink elephants, then at that point, this object starts to make sense. This is an object which is all about escapism. It ’ s about an escapism that happens — that the aristocracy in France sought very deliberately to distinguish themselves from ordinary people. It ’ s not an escapism that we feel particularly happy with today, however. And again, going on thinking about this, I realize that in a way we ’ re all victims of a certain kind of tyranny of the triumph of modernism whereby form and function in an object have to follow one another, or are deemed to do so. And the extraneous ornament is seen as really, essentially, criminal. It ’ s a triumph, in a way, of bourgeois values rather than aristocratic ones. And that seems fine. Except for the fact that it becomes a kind of sequestration of imagination. So just as in the 20th century, so many people had the idea that their faith took place on the Sabbath day, and the rest of their lives — their lives of washing machines and orthodontics — took place on another day. Then, I think we ’ ve started doing the same. We ’ ve allowed ourselves to lead our fantasy lives in front of screens. In the dark of the cinema, with the television in the corner of the room. We ’ ve eliminated, in a sense, that constant of the imagination that these vases represented in people ’ s lives. So maybe it ’ s time we got this back a little. I think it ’ s beginning to happen. In London, for example, with these extraordinary buildings that have been appearing over the last few years. Redolent, in a sense, of science fiction, turning London into a kind of fantasy playground. It ’ s actually amazing to look out of a high building nowadays there. But even then, there ’ s a resistance. London has called these buildings the Gherkin, the Shard, the Walkie Talkie — bringing these soaring buildings down to Earth. There ’ s an idea that we don ’ t want these anxious-making, imaginative journeys to happen in our daily lives. I feel lucky in a way, I ’ ve encountered this object. I found him on the Internet when I was looking up a reference. And there he was. And unlike the pink elephant vase, this was a kind of love at first sight. In fact, reader, I married him. I bought him. And he now adorns my office. He ’ s a Staffordshire figure made in the middle of the 19th century. He represents the actor, Edmund Kean, playing Shakespeare ’ s Richard III. And it ’ s based, actually, on a more elevated piece of porcelain. So I loved, on an art historical level, I loved that layered quality that he has. But more than that, I love him. In a way that I think would have been impossible without the pink Svres vase in my Leonardo days. I love his orange and pink breeches. I love the fact that he seems to be going off to war, having just finished the washing up. He seems also to have forgotten his sword. I love his pink little cheeks, his munchkin energy. In a way, he ’ s become my sort of alter ego. He ’ s, I hope, a little bit dignified, but mostly rather vulgar. And energetic, I hope, too. I let him into my life because the Svres pink elephant vase allowed me to do so. And before that Leonardo, I understood that this object could become part of a journey for me every day, sitting in my office. I really hope that others, all of you, visiting objects in the museum, and taking them home and finding them for yourselves, will allow those objects to flourish in your imaginative lives. Thank you very much.

Text Sample 936: NEG Dim 6 – 2014 – Score 1.00 – 2014/t001889.txt

Chris Anderson : We had Edward Snowden here a couple days ago, and this is response time. And several of you have written to me with questions to ask our guest here from the NSA. So Richard Ledgett is the 15th deputy director of the National Security Agency, and he ’ s a senior civilian officer there, acts as its chief operating officer, guiding strategies, setting internal policies, and serving as the principal advisor to the director. And all being well, welcome, Rick Ledgett, to TED. Richard Ledgett : I ’ m really thankful for the opportunity to talk to folks here. I look forward to the conversation, so thanks for arranging for that. CA : Thank you, Rick. We appreciate you joining us. It ’ s certainly quite a strong statement that the NSA is willing to reach out and show a more open face here. You saw, I think, the talk and interview that Edward Snowden gave here a couple days ago. What did you make of it? RL : So I think it was interesting. We didn ’ t realize that he was going to show up there, so kudos to you guys for arranging a nice surprise like that. I think that, like a lot of the things that have come out since Mr. Snowden started disclosing classified information, there were some kernels of truth in there, but a lot of extrapolations and half-truths in there, and I ’ m interested in helping to address those. I think this is a really important conversation that we ’ re having in the United States and internationally, and I think it is important and of import, and so given that, we need to have that be a fact-based conversation, and we want to help make that happen. CA : So the question that a lot of people have here is, what do you make of Snowden ’ s motivations for doing what he did, and did he have an alternative way that he could have gone? RL : He absolutely did have alternative ways that he could have gone, and I actually think that characterizing him as a whistleblower actually hurts legitimate whistleblowing activities. So what if somebody who works in the NSA — and there are over 35, 000 people who do. They ’ re all great citizens. They ’ re just like your husbands, fathers, sisters, brothers, neighbors, nephews, friends and relatives, all of whom are interested in doing the right thing for their country and for our allies internationally, and so there are a variety of venues to address if folks have a concern. First off, there ’ s their supervisor, and up through the supervisory chain within their organization. If folks aren ’ t comfortable with that, there are a number of inspectors general. In the case of Mr. Snowden, he had the option of the NSA inspector general, the Navy inspector general, the Pacific Command inspector general, the Department of Defense inspector general, and the intelligence community inspector general, any of whom would have both kept his concerns in classified channels and been happy to address them. He had the option to go to congressional committees, and there are mechanisms to do that that are in place, and so he didn ’ t do any of those things. CA : Now, you had said that Ed Snowden had other avenues for raising his concerns. The comeback on that is a couple of things : one, that he certainly believes that as a contractor, the avenues that would have been available to him as an employee weren ’ t available, two, there ’ s a track record of other whistleblowers, like [Thomas Andrews Drake] being treated pretty harshly, by some views, and thirdly, what he was taking on was not one specific flaw that he ’ d discovered, but programs that had been approved by all three branches of government. I mean, in that circumstance, couldn ’ t you argue that what he did was reasonable? RL : No, I don ’ t agree with that. I think that the — sorry, I ’ m getting feedback through the microphone there — the actions that he took were inappropriate because of the fact that he put people ’ s lives at risk, basically, in the long run, and I know there ’ s been a lot of talk in public by Mr. Snowden and some of the journalists that say that the things that have been disclosed have not put national security and people at risk, and that is categorically not true. They actually do. I think there ’ s also an amazing arrogance to the idea that he knows better than the framers of the Constitution in how the government should be designed and work for separation of powers and the fact that the executive and the legislative branch have to work together and they have checks and balances on each other, and then the judicial branch, which oversees the entire process. I think that ’ s extremely arrogant on his part. CA : Can you give a specific example of how he put people ’ s lives at risk? RL : Yeah, sure. So the things that he ’ s disclosed, the capabilities, and the NSA is a capabilities-based organization, so when we have foreign intelligence targets, legitimate things of interest — like, terrorists is the iconic example, but it includes things like human traffickers, drug traffickers, people who are trying to build advanced weaponry, nuclear weapons, and build delivery systems for those, and nation-states who might be executing aggression against their immediate neighbors, which you may have some visibility into some of that that ’ s going on right now, the capabilities are applied in very discrete and measured and controlled ways. So the unconstrained disclosure of those capabilities means that as adversaries see them and recognize,”Hey, I might be vulnerable to this,”they move away from that, and we have seen targets in terrorism, in the nation-state area, in smugglers of various types, and other folks who have, because of the disclosures, moved away from our ability to have insight into what they ’ re doing. The net effect of that is that our people who are overseas in dangerous places, whether they ’ re diplomats or military, and our allies who are in similar situations, are at greater risk because we don ’ t see the threats that are coming their way. CA : So that ’ s a general response saying that because of his revelations, access that you had to certain types of information has been shut down, has been closed down. But the concern is that the nature of that access was not necessarily legitimate in the first place. I mean, describe to us this Bullrun program where it ’ s alleged that the NSA specifically weakened security in order to get the type of access that you ’ ve spoken of. RL : So there are, when our legitimate foreign intelligence targets of the type that I described before, use the global telecommunications system as their communications methodology, and they do, because it ’ s a great system, it ’ s the most complex system ever devised by man, and it is a wonder, and lots of folks in the room there are responsible for the creation and enhancement of that, and it ’ s just a wonderful thing. But it ’ s also used by people who are working against us and our allies. And so if I ’ m going to pursue them, I need to have the capability to go after them, and again, the controls are in how I apply that capability, not that I have the capability itself. Otherwise, if we could make it so that all the bad guys used one corner of the Internet, we could have a domain, badguy. com. That would be awesome, and we could just concentrate all our efforts there. That ’ s not how it works. They ’ re trying to hide from the government ’ s ability to isolate and interdict their actions, and so we have to swim in that same space. But I will tell you this. So NSA has two missions. One is the Signals Intelligence mission that we ’ ve unfortunately read so much about in the press. The other one is the Information Assurance mission, which is to protect the national security systems of the United States, and by that, that ’ s things like the communications that the president uses, the communications that control our nuclear weapons, the communications that our military uses around the world, and the communications that we use with our allies, and that some of our allies themselves use. And so we make recommendations on standards to use, and we use those same standards, and so we are invested in making sure that those communications are secure for their intended purposes. CA : But it sounds like what you ’ re saying is that when it comes to the Internet at large, any strategy is fair game if it improves America ’ s safety. And I think this is partly where there is such a divide of opinion, that there ’ s a lot of people in this room and around the world who think very differently about the Internet. They think of it as a momentous invention of humanity, kind of on a par with the Gutenberg press, for example. It ’ s the bringer of knowledge to all. It ’ s the connector of all. And it ’ s viewed in those sort of idealistic terms. And from that lens, what the NSA has done is equivalent to the authorities back in Germany inserting some device into every printing press that would reveal which books people bought and what they read. Can you understand that from that viewpoint, it feels outrageous? RL : I do understand that, and I actually share the view of the utility of the Internet, and I would argue it ’ s bigger than the Internet. It is a global telecommunications system. The Internet is a big chunk of that, but there is a lot more. And I think that people have legitimate concerns about the balance between transparency and secrecy. That ’ s sort of been couched as a balance between privacy and national security. I don ’ t think that ’ s the right framing. I think it really is transparency and secrecy. And so that ’ s the national and international conversation that we ’ re having, and we want to participate in that, and want people to participate in it in an informed way. So there are things, let me talk there a little bit more, there are things that we need to be transparent about : our authorities, our processes, our oversight, who we are. We, NSA, have not done a good job of that, and I think that ’ s part of the reason that this has been so revelational and so sensational in the media. Nobody knew who we were. We were the No Such Agency, the Never Say Anything. There ’ s takeoffs of our logo of an eagle with headphones on around it. And so that ’ s the public characterization. And so we need to be more transparent about those things. What we don ’ t need to be transparent about, because it ’ s bad for the U. S., it ’ s bad for all those other countries that we work with and that we help provide information that helps them secure themselves and their people, it ’ s bad to expose operations and capabilities in a way that allows the people that we ’ re all working against, the generally recognized bad guys, to counter those. CA : But isn ’ t it also bad to deal a kind of body blow to the American companies that have essentially given the world most of the Internet services that matter? RL : It is. It ’ s really the companies are in a tough position, as are we, because the companies, we compel them to provide information, just like every other nation in the world does. Every industrialized nation in the world has a lawful intercept program where they are requiring companies to provide them with information that they need for their security, and the companies that are involved have complied with those programs in the same way that they have to do when they ’ re operating in Russia or the U. K. or China or India or France, any country that you choose to name. And so the fact that these revelations have been broadly characterized as”you can ’ t trust company A because your privacy is suspect with them”is actually only accurate in the sense that it ’ s accurate with every other company in the world that deals with any of those countries in the world. And so it ’ s being picked up by people as a marketing advantage, and it ’ s being marketed that way by several countries, including some of our allied countries, where they are saying,”Hey, you can ’ t trust the U. S., but you can trust our telecom company, because we ’ re safe.”And they ’ re actually using that to counter the very large technological edge that U. S. companies have in areas like the cloud and Internet-based technologies. CA : You ’ re sitting there with the American flag, and the American Constitution guarantees freedom from unreasonable search and seizure. How do you characterize the American citizen ’ s right to privacy? Is there such a right? RL : Yeah, of course there is. And we devote an inordinate amount of time and pressure, inordinate and appropriate, actually I should say, amount of time and effort in order to ensure that we protect that privacy. and beyond that, the privacy of citizens around the world, it ' s not just Americans. Several things come into play here. First, we ' re all in the same network. My communications, I ' m a user of a particular Internet email service that is the number one email service of choice by terrorists around the world, number one. So I ' m there right beside them in email space in the Internet. And so we need to be able to pick that apart and find the information that ' s relevant. In doing so, we ' re going to necessarily encounter Americans and innocent foreign citizens who are just going about their business, and so we have procedures in place that shreds that out, that says, when you find that, not if you find it, when you find it, because you ' re certain to find it, here ' s how you protect that. These are called minimization procedures. They ' re approved by the attorney general and constitutionally based. And so we protect those. And then, for people, citizens of the world who are going about their lawful business on a day-to-day basis, the president on his January 17 speech, laid out some additional protections that we are providing to them. So I think absolutely, folks do have a right to privacy, and that we work very hard to make sure that that right to privacy is protected. CA : What about foreigners using American companies ' Internet services? Do they have any privacy rights? RL : They do. They do, in the sense of, the only way that we are able to compel one of those companies to provide us information is when it falls into one of three categories : We can identify that this particular person, identified by a selector of some kind, is associated with counterterrorist or proliferation or other foreign intelligence target. CA : Much has been made of the fact that a lot of the information that you ' ve obtained through these programs is essentially metadata. It ' s not necessarily the actual words that someone has written in an email or given on a phone call. It ' s who they wrote to and when, and so forth. But it ' s been argued, and someone here in the audience has talked to a former NSA analyst who said metadata is actually much more invasive than the core data, because in the core data you present yourself as you want to be presented. With metadata, who knows what the conclusions are that are drawn? Is there anything to that? RL : I don ' t really understand that argument. I think that metadata ' s important for a couple of reasons. Metadata is the information that lets you find connections that people are trying to hide. So when a terrorist is corresponding with somebody else who ' s not known to us but is engaged in doing or supporting terrorist activity, or someone who ' s violating international sanctions by providing nuclear weapons-related material to a country like Iran or North Korea, is trying to hide that activity because it ' s illicit activity. What metadata lets you do is connect that. The alternative to that is one that ' s much less efficient and much more invasive of privacy, which is gigantic amounts of content collection. So metadata, in that sense, actually is privacy- enhancing. And we don ' t, contrary to some of the stuff that ' s been printed, we don ' t sit there and grind out metadata profiles of average people. If you ' re not connected to one of those valid intelligence targets, you are not of interest to us. CA : So in terms of the threats that face America overall, where would you place terrorism? RL : I think terrorism is still number one. I think that we have never been in a time where there are more places where things are going badly and forming the petri dish in which terrorists take advantage of the lack of governance. An old boss of mine, Tom Fargo, Admiral Fargo, used to describe it as arcs of instability. And so you have a lot of those arcs of instability in the world right now, in places like Syria, where there ' s a civil war going on and you have massive numbers, thousands and thousands of foreign fighters who are coming into Syria to learn how to be terrorists and practice that activity, and lots of those people are Westerners who hold passports to European countries or in some cases the United States, and so they are basically learning how to do jihad and have expressed intent to go out and do that later on in their home countries. You ’ ve got places like Iraq, which is suffering from a high level of sectarian violence, again a breeding ground for terrorism. And you have the activity in the Horn of Africa and the Sahel area of Africa. Again, lots of weak governance which forms a breeding ground for terrorist activity. So I think it ’ s very serious. I think it ’ s number one. I think number two is cyber threat. I think cyber is a threat in three ways : One way, and probably the most common way that people have heard about it, is due to the theft of intellectual property, so basically, foreign countries going in, stealing companies ’ secrets, and then providing that information to state-owned enterprises or companies connected to the government to help them leapfrog technology or to gain business intelligence that ’ s then used to win contracts overseas. That is a hugely costly set of activities that ’ s going on right now. Several nation- states are doing it. Second is the denial-of-service attacks. You ’ re probably aware that there have been a spate of those directed against the U. S. financial sector since 2012. Again, that ’ s a nation-state who is executing those attacks, and they ’ re doing that as a semi-anonymous way of reprisal. And the last one is destructive attacks, and those are the ones that concern me the most. Those are on the rise. You have the attack against Saudi Aramco in 2012, August of 2012. It took down about 35, 000 of their computers with a Wiper-style virus. You had a follow-on a week later to a Qatari company. You had March of 2013, you had a South Korean attack that was attributed in the press to North Korea that took out thousands of computers. Those are on the rise, and we see people expressing interest in those capabilities and a desire to employ them. CA : Okay, so a couple of things here, because this is really the core of this, almost. I mean, first of all, a lot of people who look at risk and look at the numbers don ’ t understand this belief that terrorism is still the number one threat. Apart from September 11, I think the numbers are that in the last 30 or 40 years about 500 Americans have died from terrorism, mostly from homegrown terrorists. The chance in the last few years of being killed by terrorism is far less than the chance of being killed by lightning. I guess you would say that a single nuclear incident or bioterrorism act or something like that would change those numbers. Would that be the point of view? RL : Well, I ’ d say two things. One is, the reason that there hasn ’ t been a major attack in the United States since 9 / 11, that is not an accident. That ’ s a lot of hard work that we have done, that other folks in the intelligence community have done, that the military has done, and that our allies around the globe have done. You ’ ve heard the numbers about the tip of the iceberg in terms of numbers of terrorist attacks that NSA programs contributed to stopping was 54, 25 of those in Europe, and of those 25, 18 of them occurred in three countries, some of which are our allies, and some of which are beating the heck out of us over the NSA programs, by the way. So that ’ s not an accident that those things happen. That ’ s hard work. That ’ s us finding intelligence on terrorist activities and interdicting them through one way or another, through law enforcement, through cooperative activities with other countries and sometimes through military action. The other thing I would say is that your idea of nuclear or chem-bio-threat is not at all far-fetched and in fact there are a number of groups who have for several years expressed interest and desire in obtaining those capabilities and work towards that. CA : It ’ s also been said that, of those 54 alleged incidents, that as few as zero of them were actually anything to do with these controversial programs that Mr. Snowden revealed, that it was basically through other forms of intelligence, that you ’ re looking for a needle in a haystack, and the effects of these programs, these controversial programs, is just to add hay to the stack, not to really find the needle. The needle was found by other methods. Isn ’ t there something to that? RL : No, there ’ s actually two programs that are typically implicated in that discussion. One is the section 215 program, the U. S. telephony metadata program, and the other one is popularly called the PRISM program, and it ’ s actually section 702 of the FISA Amendment Act. But the 215 program is only relevant to threats that are directed against the United States, and there have been a dozen threats where that was implicated. Now what you ’ ll see people say publicly is there is no”but for”case, and so there is no case where, but for that, the threat would have happened. But that actually indicates a lack of understanding of how terrorist investigations actually work. You think about on television, you watch a murder mystery. What do you start with? You start with a body, and then they work their way from there to solve the crime. We ’ re actually starting well before that, hopefully before there are any bodies, and we ’ re trying to build the case for who the people are, what they ’ re trying to do, and that involves massive amounts of information. Think of it is as mosaic, and it ’ s hard to say that any one piece of a mosaic was necessary to building the mosaic, but to build the complete picture, you need to have all the pieces of information. On the other, the non-U. S. -related threats out of those 54, the other 42 of them, the PRISM program was hugely relevant to that, and in fact was material in contributing to stopping those attacks. CA : Snowden said two days ago that terrorism has always been what is called in the intelligence world”a cover for action,”that it ’ s something that, because it invokes such a powerful emotional response in people, it allows the initiation of these programs to achieve powers that an organization like yours couldn ’ t otherwise have. Is there any internal debate about that? RL : Yeah. I mean, we debate these things all the time, and there is discussion that goes on in the executive branch and within NSA itself and the intelligence community about what ’ s right, what ’ s proportionate, what ’ s the correct thing to do. And it ’ s important to note that the programs that we ’ re talking about were all authorized by two different presidents, two different political parties, by Congress twice, and by federal judges 16 different times, and so this is not NSA running off and doing its own thing. This is a legitimate activity of the United States foreign government that was agreed to by all the branches of the United States government, and President Madison would have been proud. CA : And yet, when congressmen discovered what was actually being done with that authorization, many of them were completely shocked. Or do you think that is not a legitimate reaction, that it ’ s only because it ’ s now come out publicly, that they really knew exactly what you were doing with the powers they had granted you? RL : Congress is a big body. There ’ s 535 of them, and they change out frequently, in the case of the House, every two years, and I think that the NSA provided all the relevant information to our oversight committees, and then the dissemination of that information by the oversight committees throughout Congress is something that they manage. I think I would say that Congress members had the opportunity to make themselves aware, and in fact a significant number of them, the ones who are assigned oversight responsibility, did have the ability to do that. And you ’ ve actually had the chairs of those committees say that in public. CA : Now, you mentioned the threat of cyberattacks, and I don ’ t think anyone in this room would disagree that that is a huge concern, but do you accept that there ’ s a tradeoff between offensive and defensive strategies, and that it ’ s possible that the very measures taken to,”weaken encryption,”and allow yourself to find the bad guys, might also open the door to forms of cyberattack? RL : So I think two things. One is, you said weaken encryption. I didn ’ t. And the other one is that the NSA has both of those missions, and we are heavily biased towards defense, and, actually, the vulnerabilities that we find in the overwhelming majority of cases, we disclose to the people who are responsible for manufacturing or developing those products. We have a great track record of that, and we ’ re actually working on a proposal right now to be transparent and to publish transparency reports in the same way that the Internet companies are being allowed to publish transparency reports for them. We want to be more transparent about that. So again, we eat our own dog food. We use the standards, we use the products that we recommend, and so it ’ s in our interest to keep our communications protected in the same way that other people ’ s need to be. CA : Edward Snowden, when, after his talk, was wandering the halls here in the bot, and I heard him say to a couple of people, they asked him about what he thought of the NSA overall, and he was very complimentary about the people who work with you, said that it ’ s a really impassioned group of employees who are seeking to do the right thing, and that the problems have come from just some badly conceived policies. He came over certainly very reasonably and calmly. He didn ’ t come over like a crazy man. Would you accept that at least, even if you disagree with how he did it, that he has opened a debate that matters? RL : So I think that the discussion is an important one to have. I do not like the way that he did it. I think there were a number of other ways that he could have done that that would have not endangered our people and the people of other nations through losing visibility into what our adversaries are doing. But I do think it ’ s an important conversation. CA : It ’ s been reported that there ’ s almost a difference of opinion with you and your colleagues over any scenario in which he might be offered an amnesty deal. I think your boss, General Keith Alexander, has said that that would be a terrible example for others ; you can ’ t negotiate with someone who ’ s broken the law in that way. But you ’ ve been quoted as saying that, if Snowden could prove that he was surrendering all undisclosed documents, that a deal maybe should be considered. Do you still think that? RL : Yeah, so actually, this is my favorite thing about that”60 Minutes”interview was all the misquotes that came from that. What I actually said, in response to a question about, would you entertain any discussions of mitigating action against Snowden, I said, yeah, it ’ s worth a conversation. This is something that the attorney general of the United States and the president also actually have both talked about this, and I defer to the attorney general, because this is his lane. But there is a strong tradition in American jurisprudence of having discussions with people who have been charged with crimes in order to, if it benefits the government, to get something out of that, that there ’ s always room for that kind of discussion. So I ’ m not presupposing any outcome, but there is always room for discussion. CA : To a lay person it seems like he has certain things to offer the U. S., the government, you, others, in terms of putting things right and helping figure out a smarter policy, a smarter way forward for the future. Do you see, has that kind of possibility been entertained at all? RL : So that ’ s out of my lane. That ’ s not an NSA thing. That would be a Department of Justice sort of discussion. I ’ ll defer to them. CA : Rick, when Ed Snowden ended his talk, I offered him the chance to share an idea worth spreading. What would be your idea worth spreading for this group? RL : So I think, learn the facts. This is a really important conversation, and it impacts, it ’ s not just NSA, it ’ s not just the government, it ’ s you, it ’ s the Internet companies. The issue of privacy and personal data is much bigger than just the government, and so learn the facts. Don ’ t rely on headlines, don ’ t rely on sound bites, don ’ t rely on one-sided conversations. So that ’ s the idea, I think, worth spreading. We have a sign, a badge tab, we wear badges at work with lanyards, and if I could make a plug, my badge lanyard at work says,”Dallas Cowboys.”Go Dallas. I ’ ve just alienated half the audience, I know. So the lanyard that our people who work in the organization that does our crypto- analytic work have a tab that says,”Look at the data.”So that ’ s the idea worth spreading. Look at the data. CA : Rick, it took a certain amount of courage, I think, actually, to come and speak openly to this group. It ’ s not something the NSA has done a lot of in the past, and plus the technology has been challenging. We truly appreciate you doing that and sharing in this very important conversation. Thank you so much. RL : Thanks, Chris.

Text Sample 937: NEG Dim 6 – 2016 – Score 1.00 – 2016/t000605.txt
Thank you. You ’ re nice to invite me. What I want to do is talk a little bit about Watergate - which was a long time ago - what we ’ ve learned about corruption, what we ’ ve learned about Nixon since then - because of his secret taping system, there ’ s so much that comes out ; it ’ s almost as if every season, there is a new batch of Nixon tapes - and then try to address the question a little bit, What does that mean? What does Watergate, what does the kind of corruption that leads to the resignation of a president entail, and what happened? The beginning of Watergate was five burglars caught in the Democratic headquarters at the Watergate, and it seemed bizarre. I remember that morning being called in to the Post, and there ’ s, you know, burglars - I was sent down to the courthouse where the burglars were arraigned. They all had business suits. Now, there are lots of burglars in Washington ; none to my knowledge ever had business suits. And they were there before the judge, and the judge asked the leader where he worked, and the leader said, and the judge said,”Speak up!”And so it was James McCord, he said,”CIA.”And the judge then said,”No, speak up so we can hear.”And so, McCord said,”CIA.”I was in the front row, listening - and I ’ m going to speak English here because that ’ s what happened - I said, not in a whisper but kind of blurted it out,”Holy shit.”So it was 26 months of Watergate and the revelations that sort of tumbled out, and as you look at it now, you know, What really was Watergate? Wasn ’ t just that burglary ; it was the series of illegal activities designed to subvert the process of electing a president in this country and nominating - in the case in 1972 - of the Democratic opponent to Nixon. Sam Ervin, who led the Senate Watergate Committee, addressed the question,”What was Watergate?”and he said,”It was this subversion, but why? Why Watergate?”And his answer, which I think is right, is it was”a lust for political power,”retaining that political power that Nixon had. So as you look at”What was Watergate?”it was really five wars conducted by Nixon, led by Nixon. The first war was, interestingly enough, the war against the anti-war movement, the anti-Vietnam war movement, which was growing. Nixon had inherited the Vietnam War from President Johnson. He wanted to do it his way, which was”We ’ re going to withdraw troops, but we ’ re going to bomb and bomb and bomb,”which of course is what they did, but he hated the anti-war movement so strongly that he had one of his aides in the White House, Thomas Charles Huston, draw up a top-secret plan of how to deal with the domestic opposition, people who were thought to be radicals or were radicals. This Huston plan - Huston, in proposing it to Nixon, in writing said,”Oh, this is illegal,”because it involved break-ins, wiretapping, surveillance - everything necessary to bring the anti-war movement, or the violent parts of the anti-war movement, to toe. And Nixon approved it. J. Edgar Hoover, the FBI Director, objected, not because he thought, gee, these are illegal things, but he thought and he knew that break-ins, wiretapping, surveillance - that ’ s the job of the FBI, that ’ s our turf. We don ’ t want to share it. And so Nixon rescinded the Huston plan. But then, things in 1971 got so intense, and there ’ s a tape of a meeting in the Oval Office. It was June 17th, 1971. Now, this is a year before the Watergate

burglary, and it ' s Nixon, his Chief of Staff, Haldeman, and Henry Kissinger, the National Security Advisor, and there is a document that they want to get out of the Brookings Institution that ' s going to help in criticizing President Johnson and his conduct of the war, and they ' re talking about it, and, literally, Bob Haldeman said,"Oh, if we get this document, we ' ll be able to blackmail Lyndon Johnson."Think about it. Can you imagine John Adams, the second president, sitting around and saying,"Let ' s get something so we can blackmail George Washington"? But here, that was the proposal, and Nixon said,"Well, I want to burglarize the Brookings Institution to get this document."Listen to the tape. It is chilling - President of the United States saying,"Let ' s blow the safe,""I want to do this on a thievery basis."You know, it ' s that anybody would talk this way ; that your President would talk this way is stunning. Now, the second war of Nixon was against the news media ; set up an operation in the White House called the"Plumbers,"led by Howard Hunt and Gordon Liddy. They were to find out leaks to the news media. There were 17 wiretaps of reporters, White House officials to gain information. They burglarized the psychiatrist ' s office of Daniel Ellsberg, who had leaked the Pentagon Papers to The New York Times and The Washington Post. Now think of that. The government has decided - and Ellsberg is under criminal indictment also - they decide they ' re going to get him, so let ' s break into his psychiatrist ' s office to get dirt on him. The third war was to take this apparatus of Hunt and Liddy in the Plumbers, and the wiretapping and the break- ins and bring it into the White House and to turn it against the Democrats - and that was the Watergate operation - and sabotage and espionage that accompanied that. The fourth war was, significantly, the war against justice, which is the cover-up :"Oh, my God, we ' ve been caught. Now we have to lie."And if you listen to some of these tapes - here is the President of the United States proposing to his counsel, John Dean,"Well, we have to pay the Watergate burglars for their silence,"and John Dean, the counsel, says on this tape, said,"Well, it ' s going to be expensive."And Nixon said,"Well, how much?"And Dean said,"Well, a million dollars."Nixon :"Oh, I know where I can get that."In the White House. Level of criminality which is astonishing. Then the whole series of investigations, special prosecutor of Senate Watergate Committee, the impeachment investigation of Nixon, and he resigns in August, 1974. And what happened there, he lived for 20 more years, and the fifth war was the war against history, to say,"Oh, no, Watergate was really not that important. I did not commit impeachable offenses,"having surrogates go out and say,"Well, there are unanswered questions about Watergate"- which there are. But I think in a very important way, you look at Nixon and this record on the tapes - what he did is - the presidency is supposed to be something that can do good for a majority of people in the country. Listen to those tapes ; time and time again, it ' s"Let ' s use the power of the Presidency as an instrument of personal revenge. Let ' s get the IRS, the FBI, the CIA on Nixon ' s real or perceived opponents."The dog - and this is ultimately, I think, the final corruption of Nixon in Watergate - the dog that doesn ' t bark on the tapes. No one, to my knowledge, ever says, What does the country need? What would be good for the people? It ' s all in this vicious way of let ' s screw somebody, let ' s help Nixon ' s career, let ' s prosecute the Vietnam War so he ' ll get reelected even though - in one memo in my latest book, Nixon actually wrote out, which we didn ' t know at the time, just found this out last year, he said,"Well, the bombing has achieved zilch, nothing."And what did he do the year he was running for reelection? He increased the bombing, intensified it : 1. 1 million tons of bombs in Southeast Asia. So the day Nixon resigned, he called his aides, friends, cabinet officers to the East Room of the White House and had his wife, two daughters, son-in-law standing there, and there was no script - it was Nixon raw - and he talked about his mother and his father, and he was sweating. It was just a lot of people worried that he was going to come unglued, because it was televised nationally. Then at the end, Nixon kind of waved his hand,"Ah, here is why I called you all here."And he said the following :"Always remember, others may hate you, but those who hate you don ’ t win unless you hate them, and then you destroy yourself."Now think of the wisdom in that : hate was the poison, hate was what drove him and his presidency, and at that moment he had to leave office, he understood the poison and how hate had destroyed him, had put the country on edge, and if there is anything to reflect on now if you look back, as we ’ ve seen the current presidential election, on the surface and underneath, there is too much contempt, too much hate, and that, if you think about it, we need to know what ’ s really going on. We don ’ t know enough, and if we don ’ t know what ’ s going on, the judge who said it got it right :"Democracies die in darkness."And history proves that. And we think we ’ re resilient - I think we are. We think we ’ re strong. We think that we have a process that will protect us from that. But if we are infected with hate and infected with the lack of knowledge - who these people really are, what ’ s inside, what ’ s driving them - we could partake of losing our wonderful democracy. Thank you very much.

Text Sample 938: NEG Dim 6 – 2016 – Score 1.00 – 2016/t000580.txt

I love football. I started playing when I was 12 years old. It has been a dominant presence in my life ever since. From youth football, to high-school, and college, and now as a coach. Football has played a huge role in shaping me as a person. Recently, I achieved my dream job of becoming varsity line coach at my high-school alma mater. Last March I was at a coaching clinic. It was late in the afternoon and I was tired. These coaching clinics can tend to be a bit of a drag at times. Hours and hours, and days and days of the presentations on the x ’ s and o ’ s of football. There was one presentation, though, that looked interesting. It was titled"Disneyland."This presentation changed my life. This talk was not about the x ’ s and o ’ s of football. But rather about the emotional truth of why we coaches coach. We are coaches so that we can have a meaningful impact on young people ’ s lives, and help them become better people. As coaches, we want to have the kind of impact that assure that one day our players can hold hands with their children and walk into Disneyland. The moral of this incredible message was this : as coaches, we can preach to our players that it is more important for them be better people than it is great football players. But if we are not honest with them and do not practice what we preach, it will never work. In other words, teenagers are adept at sensing bullshit. As coaches, we must be willing to share our truth. And sharing our truth is the only way we can have the kind of impact we truly want to have. The coach who gave this presentation, his truth was that his son became a drug addict due to the bullying and pressure he was subjected to from his dad being the head football coach at his school in Eagle, Idaho, a program that at the time was not very succesful. I went from a presentation on how to run the power against a 3-4 defense to the most honest, emotional and inspirational talks on coaching philosophy I ’ d ever heard. So it got me thinking. What ’ s my bullshit? What is unique about me that I bring to the table that will have a meaningful impact? What is my truth? My truth is this. I ’ m a former college football player. I ’ m a current high-school football coach. And I am gay. I battled with this truth for a long time. Personally, professionally and certainly in my coaching and athletic career. It was an identity crisis that had a tremendous impact on my life. Long before I heard this presentation on Disneyland I thought about coming out in football. In the beginning, I told myself”You ’ re gay and you will take it to the grave.”Now, I ’ ve come a long way since that day. Thank you. Thank you. I ’ ve come a long way since that day, but the process almost cost me my life. I chose to come out to family and friends starting in August of 2014. I told my immediate family first, and then I carefully picked my way through family and friends, trying to choose the right time and place. Only being part way out of the closet meant that I had to be two different people and always be very aware of where I was and whom I was with. It caused a constant state of panic and anxiety. So the question became,”Is it important for me to come out publicly?”Yes it is, and here is why. My time in-and-out of the closet helped my realize that there are other people in my situation. And it is not a pretty one. It also helped me realize that coming out is not about sharing my bedroom habits, but about giving myself permission to live my life in its entirety. Let me explain. After talking to one of my fellow coaches, he told me :”I don ’ t talk about my sex life with players, why would you?”And it ’ s a legitimate question, but a common misconception about coming out. Again, I ’ m not talking about my sex life. I ’ m talking about my life. It ’ s often said that your private life is your private life. But imagine you are put in a situation where going virtually anywhere and doing very normal, healthy, human activities with your significant other could have substantial consequences on your life, reputation and career. You gain a very new perspective on what your private life actually is. That ’ s how being in the closet affected my life. It was quite literally a closet. No space, no freedom and no comfort. It ’ s a suffocating lifestyle with measurable effects. It wore me down until eventually I was abusing alcohol and prescription medications to keep my anxiety in check. It was a horrible time in my life. When you are put in a situation of having to live a double life, it strips you off dignity and normal coping mechanisms. At that time the only place I felt somewhat safe was at home. I was living with my parents, who had been nothing but supportive, but I did not feel comfortable bringing guys around them yet. So there I was. No safe place, no place to be myself, facing a new lifestyle I did not know how to navigate, the anxiety and depression were inescapable. And I responded the only way I felt I could at the time. Drugs and alcohol. Football teaches so many great life lessons to those who play or coach. Perseverance, toughness, respect, self-esteem. But the one negative thing that it does teach is that being gay is not OK. To be frank, the word”faggot”is used almost as much as the word”football.”There is a misconception about the prominence of gay men in football, and it has serious consequences. I ’ m the perfect example of this issue. An all-state player in high-school. A two-year varsity captain among a select few in school history to go play for a top tier division 1 football program. The youngest line coach in the history of the school. And I was ready to kill myself. Because I thought that even though this program was like a second family to me, I feared they would shun me. It was a crushing weight that I was carrying with me all the time. Months and months of sleep deprivation, severe anxiety and depression. And honestly, a lack of the will to live began to catch up with me. It was too much. I was desperate for a way out. Any way out. One night I reached for a bottle of vodka and a couple of pills. I didn ’ t see a way out. I just wanted it to be over. If I couldn ’ t be me and still live my life, what was the point? I passed out on the bathroom floor, and my mom found me. I was fortunate to wake up the next day. I escaped an overdose. It was the scariest moment of my life. When I woke up the next day, I knew I really wasn ’ t ready to give up. And when I heard that talk on Disneyland, I knew how, when and why it was important for me to share my story. I worried for so long about how the football community would react. And while at this point only time will tell, my experience has given me a theory. It ’ s simple. One day, being gay in football will be normal. But in order for that to happen, those of us who are gay need to stand up and own it. The coaches I know are perfect examples of this. I have been met with nothing but love and support from my fellow coaching friends. But now the challenge is to change this. And not just on the private level. The odds of a gay teen or young adult abusing drugs, alcohol, or experiencing anxiety, depression, or even attempting suicide are drastically high. When it comes to football, the social norm that we ’ ve created leaves many without hope. We ’ ve made incredible progress in the equal rights movement under the law. But now we must tackle a different problem. Social equality and the message we send to young people who play football. Continue to use sports to inspire, connect and to share a positive message. A healthy person cannot live life in the dark. And if you are out there, stand up and own it. Thank you.

Text Sample 939: NEG Dim 6 – 2016 – Score 1.00 – 2016/t000984.txt

”Judge, I want to tell you something. I want to tell you something. I been watching you and you ’ re not two-faced. You treat everybody the same.”That was said to me by a transgender prostitute who before I had gotten on the bench had fired her public defender, insulted the court officer and yelled at the person sitting next to her,”I don ’ t know what you ’ re looking at. I look better than the girl you ’ re with.”She said this to me after I said her male name low enough so that it could be picked up by the record, but I said her female name loud enough so that she could walk down the aisle towards counselor ’ s table with dignity. This is procedural justice, also known as procedural fairness, at its best. You see, I am the daughter of an African-American garbageman who was born in Harlem and spent his summers in the segregated South. Soy la hija de una peluquera dominicana. I do that to make sure you ’ re still paying attention. I ’ m the daughter of a Dominican beautician who came to this country for a better life for her unborn children. My parents taught me, you treat everyone you meet with dignity and respect, no matter how they look, no matter how they dress, no matter how they spoke. You see, the principles of fairness were taught to me at an early age, and unbeknownst to me, it would be the most important lesson that I carried with me to the Newark Municipal Court bench. And because I was dragged off the playground at the early age of 10 to translate for family members as they began to migrate to the United States, I understand how daunting it can be for a person, a novice, to navigate any government system. Every day across America and around the globe, people encounter our courts, and it is a place that is foreign, intimidating and often hostile towards them. They are confused about the nature of their charges, annoyed about their encounters with the police and facing consequences that might impact their relationships, their finances and even their liberty. Let me paint a picture for you of what it ’ s like for the average person who encounters our courts. First, they ’ re annoyed as they ’ re probed going through court security. They finally get through court security, they walk around the building, they ask different people the same question and get different answers. When they finally get to where they ’ re supposed to be, it gets really bad when they encounter the courts. What would you think if I told you that you could improve people ’ s court experience, increase their compliance with the law and court orders, all the while increasing the public ’ s trust in the justice system with a simple idea? Well, that simple idea is procedural justice and it ’ s a concept that says that if people perceive they are treated fairly and with dignity and respect, they ’ ll obey the law. Well, that ’ s what Yale professor Tom Tyler found when he began to study as far back in the ’ 70s why people obey the law. He found that if people see the justice system as a legitimate authority to impose rules and regulations, they would follow them. His research concluded that people would be satisfied with the judge ’ s rulings, even when the judge ruled against them, if they perceived that they were treated fairly and with dignity and respect. And that perception of fairness begins with what? Begins with how judges speak to court participants. Now, being a judge is sometimes like having a reserve seat to a tragic reality show that has no commercial interruptions and no season finale. It ’ s true. People come before me handcuffed, drug-sick, depressed, hungry and mentally ill. When I saw that their need for help was greater than my fear of appearing vulnerable on the bench, I realized that not only did I need to do something, but that in fact I could do something. The good news is is that the principles of procedural justice are easy and can be implemented as quickly as tomorrow. The even better news, that it can be done for free. The first principle is voice. Give people an opportunity to speak, even when you ’ re not going to let them speak. Explain it.”Sir, I ’ m not letting you speak right now. You don ’ t have an attorney. I don ’ t want you to say anything that ’ s going to hurt your case.”For me, assigning essays to defendants has been a tremendous way of giving them voice. I recently gave an 18-year-old college student an essay. He lamented his underage drinking charge. As he stood before me reading his essay, his voice cracking and his hands trembling, he said that he worried that he had become an alcoholic like his mom, who had died a couple of months prior due to alcohol- related liver disease. You see, assigning a letter to my father, a letter to my son,”If I knew then what I know now...””If I believed one positive thing about myself, how would my life be different?”gives the person an opportunity to be introspective, go on the inside, which is where all the answers are anyway. But it also gives them an opportunity to share something with the court that goes beyond their criminal record and their charges. The next principle is neutrality. When increasing public trust in the justice system, neutrality is paramount. The judge cannot be perceived to be favoring one side over the other. The judge has to make a conscious decision not to say things like,”my officer,””my prosecutor,””my defense attorney.”And this is challenging when we work in environments where you have people assigned to your courts, the same people coming in and out of your courts as well. When I think of neutrality, I ’ m reminded of when I was a new Rutgers Law grad and freshly minted attorney, and I entered an arbitration and I was greeted by two grey-haired men who were joking about the last game of golf they played together and planning future social outings. I knew my client couldn ’ t get a fair shot in that forum. The next principle is understand. It is critical that court participants understand the process, the consequences of the process and what ’ s expected of them. I like to say that legalese is the language we use to confuse. I am keenly aware that the people who appear before me, many of them have very little education and English is often their second language. So I speak plain English in court. A great example of this was when I was a young judge — oh no, I mean younger judge. When I was a younger judge, a senior judge comes to me, gives me a script and says,”If you think somebody has mental health issues, ask them these questions and you can get your evaluation.”So the first time I saw someone who had what I thought was a mental health issue, I went for my script and I started to ask questions.”Um, sir, do you take psycho — um, psychotrop — psychotropic medication?””Nope.””Uh, sir, have you treated with a psychiatrist before?””Nope.”But it was obvious that the person was suffering from mental illness. One day, in my frustration, I decided to scrap the script and ask one question.”Ma ’ am, do you take medication to clear your mind?””Yeah, judge, I take Haldol for my schizophrenia, Xanax for my anxiety.”The question works even when it doesn ’ t.”Mr. L, do you take medication to clear your mind?””No, judge, I don ’ t take no medication to clear my mind. I take medication to stop the voices in my head, but my mind is fine.”You see, once people understand the question, they can give you valuable information that allows the court to make meaningful decisions about the cases that are before them. The last principle is respect, that without it none of the other principles can work. Now, respect can be as simple as,”Good afternoon, sir.””Good morning, ma ’ am.”It ’ s looking the person in the eye who is standing before you, especially when you ’ re sentencing them. It ’ s when I say,”Um, how are you doing today? And what ’ s going on with you?”And not as a greeting, but as someone who is actually interested in the response. Respect is the difference between saying,”Ma ’ am, are you having difficulty understanding the information in the paperwork?”versus,”You can read and write, can ’ t you?”when you ’ ve realized there ’ s a literacy issue. And the good thing about respect is that it ’ s contagious. People see you being respectful to other folks and they impute that respect to themselves. You see, that ’ s what the transgender prostitute was telling me. I ’ m judging you just as much as you think you may be judging me. Now, I am not telling you what I think, I am telling you what I have lived, using procedural justice to change the culture at my courthouse and in the courtroom. After sitting comfortably for seven months as a traffic court judge, I was advised that I was being moved to the criminal court, Part Two, criminal courtroom. Now, I need you to understand, this was not good news. It was not. Part Two was known as the worst courtroom in the city, some folks would even say in the state. It was your typical urban courtroom with revolving door justice, you know, your regular lineup of low-level offenders — you know, the low-hanging fruit, the drug-addicted prostitute, the mentally ill homeless person with quality-of-life tickets, the high school dropout petty drug dealer and the misguided young people — you know, those folks doing a life sentence 30 days at a time. Fortunately, the City of Newark decided that Newarkers deserved better, and they partnered with the Center for Court Innovation and the New Jersey Judiciary to create Newark Community Solutions, a community court program that provided alternative sanctions. This means now a judge can sentence a defendant to punishment with assistance. So a defendant who would otherwise get a jail sentence would now be able to get individual counseling sessions, group counseling sessions as well as community giveback, which is what we call community service. The only problem is that this wonderful program was now coming to Newark and was going to be housed where? Part Two criminal courtroom. And the attitudes there were terrible. And the reason that the attitudes were terrible there was because everyone who was sent there understood they were being sent there as punishment. The officers who were facing disciplinary actions at times, the public defender and prosecutor felt like they were doing a 30-day jail sentence on their rotation, the judges understood they were being hazed just like a college sorority or fraternity. I was once told that an attorney who worked there referred to the defendants as”the scum of the earth”and then had to represent them. I would hear things from folks like,”Oh, how could you work with those people? They ’ re so nasty. You ’ re a judge, not a social worker.”But the reality is that as a society, we criminalize social ills, then sent people to a judge and say,”Do something.”I decided that I was going to lead by example. So my first foray into the approach came when a 60-something-year- old man appeared before me handcuffed. His head was lowered and his body was showing the signs of drug withdrawal. I asked him how long he had been addicted, and he said,”30 years.”And I asked him,”Do you have any kids?”And he said,”Yeah, I have a 32-year-old son.”And I said,”Oh, so you ’ ve never had the opportunity to be a father to your son because of your addiction.”He began to cry. I said,”You know what, I ’ m going to let you go home, and you ’ ll come back in two weeks, and when you come back, we ’ ll give you some assistance for your addiction.”Surprisingly, two weeks passed and he was sitting the courtroom. When he came up, he said,”Judge, I came back to court because you showed me more love than I had for myself.”And I thought, my God, he heard love from the bench? I could do this all day. Because the reality is that when the court behaves differently, then naturally people respond differently. The court becomes a place you can go to for assistance, like the 60-something-year-old schizophrenic homeless woman who was in distress and fighting with the voices in her head, and barges into court, and screams,”Judge! I just came by to see how you were doing.”I had been monitoring her case for a couple of months, her compliance with her medication, and had just closed out her case a couple of weeks ago. On this day she needed help, and she came to court. And after four hours of coaxing by the judge, the police officers and the staff, she is convinced to get into the ambulance that will take her to crisis unit so that she can get her medication. People become connected to their community when the court changes, like the 50-something-year-old man who told me,”Community service was terrible, Judge. I had to clean the park, and it was full of empty heroin envelopes, and the kids had to play there.”As he wrung his hands, he confessed,”Judge, I realized that it was my fault, because I used that same park to get high, and before you sent me there to do community service, I had never gone to the park when I wasn ’ t high, so I never noticed the children playing there.”Every addict in the courtroom lowered their head. Who better to teach that lesson? It helps the court reset its relationship with the community, like with the 20- something-year-old guy who gets a job interview through the court program. He gets a job interview at an office cleaning company, and he comes back to court to proudly say,”Judge, I even worked in my suit after the interview, because I wanted the guy to see how bad I wanted the job.”It ’ s what happens when a person in authority treats you with dignity and respect, like the 40-something-year-old guy who struts down the aisle and says,”Judge, do you notice anything different?”And when I look up, he ’ s pointing at his new teeth that he was able to get after getting a referral from the program, but he was able to get them to replace the old teeth that he lost as a result of years of heroin addiction. When he looks in the mirror, now he sees somebody who is worth saving. You see, I have a dream and that dream is that judges will use these tools to revolutionize the communities that they serve. Now, these tools are not miracle cure-alls, but they get us light-years closer to where we want to be, and where we want to be is a place that people enter our halls of justice and believe they will be treated with dignity and respect and know that justice will be served there. Imagine that, a simple idea. Thank you.

Text Sample 940: NEG Dim 6 – 2012 – Score 1.00 – 2012/t002459.txt

I ’ m going to start here. This is a hand-lettered sign that appeared in a mom and pop bakery in my old neighborhood in Brooklyn a few years ago. The store owned one of those machines that can print on plates of sugar. And kids could bring in drawings and have the store print a sugar plate for the top of their birthday cake. But unfortunately, one of the things kids liked to draw was cartoon characters. They liked to draw the Little Mermaid, they ’ d like to draw a smurf, they ’ d like to draw Micky Mouse. But it turns out to be illegal to print a child ’ s drawing of Micky Mouse onto a plate of sugar. And it ’ s a copyright violation. And policing copyright violations for children ’ s birthday cakes was such a hassle that the College Bakery said,"You know what, we ’ re getting out of that business. If you ’ re an amateur, you don ’ t have access to our machine anymore. If you want a printed sugar birthday cake, you have to use one of our prefab images — only for professionals."So there ’ s two bills in Congress right now. One is called SOPA, the other is called PIPA. SOPA stands for the Stop Online Piracy Act. It ’ s from the Senate. PIPA is short for PROTECTIP, which is itself short for Preventing Real Online Threats to Economic Creativity and Theft of Intellectual Property — because the congressional aides who name these things have a lot of time on their hands. And what SOPA and PIPA want to do is they want to do this. They want to raise the cost of copyright compliance to the point where people simply get out of the business of offering it as a capability to amateurs. Now the way they propose to do this is to identify sites that are substantially infringing on copyright — although how those sites are identified is never fully specified in the bills — and then they want to remove them from the domain name system. They want to take them out of the domain name system. Now the domain name system is the thing that turns human-readable names, like Google. com, into the kinds of addresses machines expect — 74. 125. 226. 212. Now the problem with this model of censorship, of identifying a site and then trying to remove it from the domain name system, is that it won ’ t work. And you ’ d think that would be a pretty big problem for a law, but Congress seems not to have let that bother them too much. Now the reason it won ’ t work is that you can still type 74. 125. 226. 212 into the browser or you can make it a clickable link and you ’ ll still go to Google. So the policing layer around the problem becomes the real threat of the act. Now to understand how Congress came to write a bill that won ’ t accomplish its stated goals, but will produce a lot of pernicious side effects, you have to understand a little bit about the back story. And the back story is this : SOPA and PIPA, as legislation, were drafted largely by media companies that were founded in the 20th century. The 20th century was a great time to be a media company, because the thing you really had on your side was scarcity. If you were making a TV show, it didn ’ t have to be better than all other TV shows ever made ; it only had to be better than the two other shows that were on at the same time — which is a very low threshold of competitive difficulty. Which meant that if you fielded average content, you got a third of the U. S. public for free — tens of millions of users for simply doing something that wasn ’ t too terrible. This is like having a license to print money and a barrel of free ink. But technology moved on, as technology is wont to do. And slowly, slowly, at the end of the 20th century, that scarcity started to get eroded — and I don ’ t mean by digital technology ; I mean by analog technology. Cassette tapes, video cassette recorders, even the humble Xerox machine created new opportunities for us to behave in ways that astonished the media business. Because it turned out we ’ re not really couch potatoes. We don ’ t really like to only consume. We do like to consume, but every time one of these new tools came along, it turned out we also like to produce and we like to share. And this freaked the media businesses out — it freaked them out every time. Jack Valenti, who was the head lobbyist for the Motion Picture Association of America, once likened the ferocious video cassette recorder to Jack the Ripper and poor, helpless Hollywood to a woman at home alone. That was the level of rhetoric. And so the media industries begged, insisted, demanded that Congress do something. And Congress did something. By the early 90s, Congress passed the law that changed everything. And that law was called the Audio Home Recording Act of 1992. What the Audio Home Recording Act of 1992 said was, look, if people are taping stuff off the radio and then making mixtapes for their friends, that is not a crime. That ’ s okay. Taping and remixing and sharing with your friends is okay. If you make lots and lots of high quality copies and you sell them, that ’ s not okay. But this taping business, fine, let it go. And they thought that they clarified the issue, because they ’ d set out a clear distinction between legal and illegal copying. But that wasn ’ t what the media businesses wanted. They had wanted Congress to outlaw copying full-stop. So when the Audio Home Recording Act of 1992 was passed, the media businesses gave up on the idea of legal versus illegal distinctions for copying because it was clear that if Congress was acting in their framework, they might actually increase the rights of citizens to participate in our own media environment. So they went for plan B. It took them a while to formulate plan B. Plan B appeared in its first full-blown form in 1998 — something called the Digital Millennium Copyright Act. It was a complicated piece of legislation, a lot of moving parts. But the main thrust of the DMCA was that it was legal to sell you uncopyable digital material — except that there ’ s no such things as uncopyable digital material. It would be, as Ed Felton once famously said,”Like handing out water that wasn ’ t wet.”Bits are copyable. That ’ s what computers do. That is a side effect of their ordinary operation. So in order to fake the ability to sell uncopyable bits, the DMCA also made it legal to force you to use systems that broke the copying function of your devices. Every DVD player and game player and television and computer you brought home — no matter what you thought you were getting when you bought it — could be broken by the content industries, if they wanted to set that as a condition of selling you the content. And to make sure you didn ’ t realize, or didn ’ t enact their capabilities as general purpose computing devices, they also made it illegal for you to try to reset the copyability of that content. The DMCA marks the moment when the media industries gave up on the legal system of distinguishing between legal and illegal copying and simply tried to prevent copying through technical means. Now the DMCA had, and is continuing to have, a lot of complicated effects, but in this one domain, limiting sharing, it has mostly not worked. And the main reason it hasn ’ t worked is the Internet has turned out to be far more popular and far more powerful than anyone imagined. The mixtape, the fanzine, that was nothing compared to what we ’ re seeing now with the Internet. We are in a world where most American citizens over the age of 12 share things with each other online. We share written things, we share images, we share audio, we share video. Some of the stuff we share is stuff we ’ ve made. Some of the stuff we share is stuff we ’ ve found. Some of the stuff we share is stuff we ’ ve made out of what we ’ ve found, and all of it horrifies those industries. So PIPA and SOPA are round two. But where the DMCA was surgical — we want to go down into your computer, we want to go down into your television set, down into your game machine, and prevent it from doing what they said it would do at the store — PIPA and SOPA are nuclear and they ’ re saying, we want to go anywhere in the world and censor content. Now the mechanism, as I said, for doing this, is you need to take out anybody pointing to those IP addresses. You need to take them out of search engines, you need to take them out of online directories, you need to take them out of user lists. And because the biggest producers of content on the Internet are not Google and Yahoo, they ’ re us, we ’ re the people getting policed. Because in the end, the real threat to the enactment of PIPA and SOPA is our ability to share things with one another. So what PIPA and SOPA risk doing is taking a centuries-old legal concept, innocent until proven guilty, and reversing it — guilty until proven innocent. You can ’ t share until you show us that you ’ re not sharing something we don ’ t like. Suddenly, the burden of proof for legal versus illegal falls affirmatively on us and on the services that might be offering us any new capabilities. And if it costs even a dime to police a user, that will crush a service with a hundred million users. So this is the Internet they have in mind. Imagine this sign everywhere — except imagine it doesn ’ t say College Bakery, imagine it says YouTube and Facebook and Twitter. Imagine it says TED, because the comments can ’ t be policed at any acceptable cost. The real effects of SOPA and PIPA are going to be different than the proposed effects. The threat, in fact, is this inversion of the burden of proof, where we suddenly are all treated like thieves at every moment we ’ re given the freedom to create, to produce or to share. And the people who provide those capabilities to us — the YouTubes, the Facebooks, the Twitters and TEDs — are in the business of having to police us, or being on the hook for contributory infringement. There ’ s two things you can do to help stop this — a simple thing and a complicated thing, an easy thing and a hard thing. The simple thing, the easy thing, is this : if you ’ re an American citizen, call your representative, call your senator. When you look at the people who co-signed on the SOPA bill, people who ’ ve co-signed on PIPA, what you see is that they have cumulatively received millions and millions of dollars from the traditional media industries. You don ’ t have millions and millions of dollars, but you can call your representatives, and you can remind them that you vote, and you can ask not to be treated like a thief, and you can suggest that you would prefer that the Internet not be broken. And if you ’ re not an American citizen, you can contact American citizens that you know and encourage them to do the same. Because this seems like a national issue, but it is not. These industries will not be content with breaking our Internet. If they break it, they will break it for everybody. That ’ s the easy thing. That ’ s the simple thing. The hard thing is this : get ready, because more is coming. SOPA is simply a reversion of COICA, which was purposed last year, which did not pass. And all of this goes back to the failure of the DMCA to disallow sharing as a technical means. And the DMCA goes back to the Audio Home Recording Act, which horrified those industries. Because the whole business of actually suggesting that someone is breaking the law and then gathering evidence and proving that, that turns out to be really inconvenient.”We ’ d prefer not to do that,”says the content industries. And what they want is not to have to do that. They don ’ t want legal distinctions between legal and illegal sharing. They just want the sharing to go away. PIPA and SOPA are not oddities, they ’ re not anomalies, they ’ re not events. They ’ re the next turn of this particular screw, which has been going on 20 years now. And if we defeat these, as I hope we do, more is coming. Because until we convince Congress that the way to deal with copyright violation is the way copyright violation was dealt with with Napster, with YouTube, which is to have a trial with all the presentation of evidence and the hashing out of facts and the assessment of remedies that goes on in democratic societies. That ’ s the way to handle this. In the meantime, the hard thing to do is to be ready. Because that ’ s the real message of PIPA and SOPA. Time Warner has called and they want us all back on the couch, just consuming — not producing, not sharing — and we should say,”No.”Thank you.

Text Sample 941: NEG Dim 6 – 2012 – Score 1.00 – 2012/t002449.txt

Good morning. I ’ m here today to talk about autonomous flying beach balls. No, agile aerial robots like this one. I ’ d like to tell you a little bit about the challenges in building these, and some of the terrific opportunities for applying this technology. So these robots are related to unmanned aerial vehicles. However, the vehicles you see here are big. They weigh thousands of pounds, are not by any means agile. They ’ re not even autonomous. In fact, many of these vehicles are operated by flight crews that can include multiple pilots, operators of sensors, and mission coordinators. What we ’ re interested in is developing robots like this — and here are two other pictures — of robots that you can buy off the shelf. So these are helicopters with four rotors, and they ' re roughly a meter or so in scale, and weigh several pounds. And so we retrofit these with sensors and processors, and these robots can fly indoors. Without GPS. The robot I ' m holding in my hand is this one, and it ' s been created by two students, Alex and Daniel. So this weighs a little more than a tenth of a pound. It consumes about 15 watts of power. And as you can see, it ' s about eight inches in diameter. So let me give you just a very quick tutorial on how these robots work. So it has four rotors. If you spin these rotors at the same speed, the robot hovers. If you increase the speed of each of these rotors, then the robot flies up, it accelerates up. Of course, if the robot were tilted, inclined to the horizontal, then it would accelerate in this direction. So to get it to tilt, there ' s one of two ways of doing it. So in this picture, you see that rotor four is spinning faster and rotor two is spinning slower. And when that happens, there ' s a moment that causes this robot to roll. And the other way around, if you increase the speed of rotor three and decrease the speed of rotor one, then the robot pitches forward. And then finally, if you spin opposite pairs of rotors faster than the other pair, then the robot yaws about the vertical axis. So an on-board processor essentially looks at what motions need to be executed and combines these motions, and figures out what commands to send to the motors — 600 times a second. That ' s basically how this thing operates. So one of the advantages of this design is when you scale things down, the robot naturally becomes agile. So here, R is the characteristic length of the robot. It ' s actually half the diameter. And there are lots of physical parameters that change as you reduce R. The one that ' s most important is the inertia, or the resistance to motion. So it turns out the inertia, which governs angular motion, scales as a fifth power of R. So the smaller you make R, the more dramatically the inertia reduces. So as a result, the angular acceleration, denoted by the Greek letter alpha here, goes as 1 over R. It ' s inversely proportional to R. The smaller you make it, the more quickly you can turn. So this should be clear in these videos. On the bottom right, you see a robot performing a 360-degree flip in less than half a second. Multiple flips, a little more time. So here the processes on board are getting feedback from accelerometers and gyros on board, and calculating, like I said before, commands at 600 times a second, to stabilize this robot. So on the left, you see Daniel throwing this robot up into the air, and it shows you how robust the control is. No matter how you throw it, the robot recovers and comes back to him. So why build robots like this? Well, robots like this have many applications. You can send them inside buildings like this, as first responders to look for intruders, maybe look for biochemical leaks, gaseous leaks. You can also use them for applications like construction. So here are robots carrying beams, columns and assembling cube-like structures. I ' ll tell you a little bit more about this. The robots can be used for transporting cargo. So one of the problems with these small robots is their payload-carrying capacity. So you might want to have multiple robots carry payloads. This is a picture of a recent experiment we did — actually not so recent anymore — in Sendai, shortly after the earthquake. So robots like this could be sent into collapsed buildings, to assess the damage after natural disasters, or sent into reactor buildings, to map radiation levels. So one fundamental problem that the robots have to solve if they are to be autonomous, is essentially figuring out how to get from point A to point B. So this gets a little challenging, because the dynamics of this robot are quite complicated. In fact, they live in a 12-dimensional space. So we use a little trick. We take this curved 12-dimensional space, and transform it into a flat, four-dimensional space. And that four-dimensional space consists of X, Y, Z, and then the yaw angle. And so what the robot does, is it plans what we call a minimum-snap trajectory. So to remind you of physics : You have position, derivative, velocity ; then acceleration ; and then comes jerk, and then comes snap. So this robot minimizes snap. So what that effectively does, is produce a smooth and graceful motion. And it does that avoiding obstacles. So these minimum-snap trajectories in this flat space are then transformed back into this complicated 12-dimensional space, which the robot must do for control and then execution. So let me show you some examples of what these minimum-snap trajectories look like. And in the first video, you ’ ll see the robot going from point A to point B, through an intermediate point. So the robot is obviously capable of executing any curve trajectory. So these are circular trajectories, where the robot pulls about two G ’ s. Here you have overhead motion capture cameras on the top that tell the robot where it is 100 times a second. It also tells the robot where these obstacles are. And the obstacles can be moving. And here, you ’ ll see Daniel throw this hoop into the air, while the robot is calculating the position of the hoop, and trying to figure out how to best go through the hoop. So as an academic, we ’ re always trained to be able to jump through hoops to raise funding for our labs, and we get our robots to do that. So another thing the robot can do is it remembers pieces of trajectory that it learns or is pre-programmed. So here, you see the robot combining a motion that builds up momentum, and then changes its orientation and then recovers. So it has to do this because this gap in the window is only slightly larger than the width of the robot. So just like a diver stands on a springboard and then jumps off it to gain momentum, and then does this pirouette, this two and a half somersault through and then gracefully recovers, this robot is basically doing that. So it knows how to combine little bits and pieces of trajectories to do these fairly difficult tasks. So I want change gears. So one of the disadvantages of these small robots is its size. And I told you earlier that we may want to employ lots and lots of robots to overcome the limitations of size. So one difficulty is : How do you coordinate lots of these robots? And so here, we looked to nature. So I want to show you a clip of Aphaenogaster desert ants, in Professor Stephen Pratt ’ s lab, carrying an object. So this is actually a piece of fig. Actually you take any object coated with fig juice, and the ants will carry it back to the nest. So these ants don ’ t have any central coordinator. They sense their neighbors. There ’ s no explicit communication. But because they sense the neighbors and because they sense the object, they have implicit coordination across the group. So this is the kind of coordination we want our robots to have. So when we have a robot which is surrounded by neighbors — and let ’ s look at robot I and robot J — what we want the robots to do, is to monitor the separation between them, as they fly in formation. And then you want to make sure that this separation is within acceptable levels. So again, the robots monitor this error and calculate the control commands 100 times a second, which then translates into motor commands, 600 times a second. So this also has to be done in a decentralized way. Again, if you have lots and lots of robots, it ’ s impossible to coordinate all this information centrally fast enough in order for the robots to accomplish the task. Plus, the robots have to base their actions only on local information — what they sense from their neighbors. And then finally, we insist that the robots be agnostic to who their neighbors are. So this is what we call anonymity. So what I want to show you next is a video of 20 of these little robots, flying in formation. They ’ re monitoring their neighbors ’ positions. They ’ re maintaining formation. The formations can change. They can be planar formations, they can be three-dimensional formations. As you can see here, they collapse from a three-dimensional formation into planar formation. And to fly through obstacles, they can adapt the formations on the fly. So again, these robots come really close together. As you can see in this figure-eight flight, they come within inches of each other. And despite the aerodynamic interactions with these propeller blades, they ’ re able to maintain stable flight. So once you know how to fly in formation, you can actually pick up objects cooperatively. So this just shows that we can double, triple, quadruple the robots ’ strength, by just getting them to team with neighbors, as you can see here. One of the disadvantages of doing that is, as you scale things up — so if you have lots of robots carrying the same thing, you ’ re essentially increasing the inertia, and therefore you pay a price ; they ’ re not as agile. But you do gain in terms of payload-carrying capacity. Another application I want to show you — again, this is in our lab. This is work done by Quentin Lindsey, who ’ s a graduate student. So his algorithm essentially tells these robots how to autonomously build cubic structures from truss-like elements. So his algorithm tells the robot what part to pick up, when, and where to place it. So in this video you see — and it ’ s sped up 10, 14 times — you see three different structures being built by these robots. And again, everything is autonomous, and all Quentin has to do is to give them a blueprint of the design that he wants to build. So all these experiments you ’ ve seen thus far, all these demonstrations, have been done with the help of motion-capture systems. So what happens when you leave your lab, and you go outside into the real world? And what if there ’ s no GPS? So this robot is actually equipped with a camera, and a laser rangefinder, laser scanner. And it uses these sensors to build a map of the environment. What that map consists of are features — like doorways, windows, people, furniture — and it then figures out where its position is, with respect to the features. So there is no global coordinate system. The coordinate system is defined based on the robot, where it is and what it ’ s looking at. And it navigates with respect to those features. So I want to show you a clip of algorithms developed by Frank Shen and Professor Nathan Michael, that shows this robot entering a building for the very first time, and creating this map on the fly. So the robot then figures out what the features are, it builds the map, it figures out where it is with respect to the features, and then estimates its position 100 times a second, allowing us to use the control algorithms that I described to you earlier. So this robot is actually being commanded remotely by Frank, but the robot can also figure out where to go on its own. So suppose I were to send this into a building, and I had no idea what this building looked like. I can ask this robot to go in, create a map, and then come back and tell me what the building looks like. So here, the robot is not only solving the problem of how to go from point A to point B in this map, but it ’ s figuring out what the best point B is at every time. So essentially it knows where to go to look for places that have the least information, and that ’ s how it populates this map. So I want to leave you with one last application. And there are many applications of this technology. I ’ m a professor, and we ’ re passionate about education. Robots like this can really change the way we do K-12 education. But we ’ re in Southern California, close to Los Angeles, so I have to conclude with something focused on entertainment. I want to conclude with a music video. I want to introduce the creators, Alex and Daniel, who created this video. So before I play this video, I want to tell you that they created it in the last three days, after getting a call from Chris. And the robots that play in the video are completely autonomous. You will see nine robots play six different instruments. And of course, it ’ s made exclusively for TED 2012. Let ’ s watch.

Text Sample 942: NEG Dim 6 – 2012 – Score 1.00 – 2012/t002446.txt

When I was nine years old, I went off to summer camp for the first time. And my mother packed me a suitcase full of books, which to me seemed like a perfectly natural thing to do. Because in my family, reading was the primary group activity. And this might sound antisocial to you, but for us it was really just a different way of being social. You have the animal warmth of your family sitting right next to you, but you are also free to go roaming around the adventureland inside your own mind. And I had this idea that camp was going to be just like this, but better. I had a vision of 10 girls sitting in a cabin cozily reading books in their matching nightgowns. Camp was more like a keg party without any alcohol. And on the very first day, our counselor gathered us all together and she taught us a cheer that she said we would be doing every day for the rest of the summer to instill camp spirit. And it went like this :"R-O-W-D-I-E, that ' s the way we spell rowdie. Rowdie, rowdie, let ' s get rowdie."Yeah. So I couldn ' t figure out for the life of me why we were supposed to be so rowdy, or why we had to spell this word incorrectly. But I recited a cheer. I recited a cheer along with everybody else. I did my best. And I just waited for the time that I could go off and read my books. But the first time that I took my book out of my suitcase, the coolest girl in the bunk came up to me and she asked me,"Why are you being so mellow?"— mellow, of course, being the exact opposite of R-O-W-D-I-E. And then the second time I tried it, the counselor came up to me with a concerned expression on her face and she repeated the point about camp spirit and said we should all work very hard to be outgoing. And so I put my books away, back in their suitcase, and I put them under my bed, and there they stayed for the rest of the summer. And I felt kind of guilty about this. I felt as if the books needed me somehow, and they were calling out to me and I was forsaking them. But I did forsake them and I didn ' t open that suitcase again until I was back home with my family at the end of the summer. Now, I tell you this story about summer camp. I could have told you 50 others just like it — all the times that I got the message that somehow my quiet and introverted style of being was not necessarily the right way to go, that I should be trying to pass as more of an extrovert. And I always sensed deep down that this was wrong and that introverts were pretty excellent just as they were. But for years I denied this intuition, and so I became a Wall Street lawyer, of all things, instead of the writer that I had always longed to be — partly because I needed to prove to myself that I could be bold and assertive too. And I was always going off to crowded bars when I really would have preferred to just have a nice dinner with friends. And I made these self- negating choices so reflexively, that I wasn ' t even aware that I was making them. Now this is what many introverts do, and it ' s our loss for sure, but it is also our colleagues ' loss and our communities ' loss. And at the risk of sounding grandiose, it is the world ' s loss. Because when it comes to creativity and to leadership, we need introverts doing what they do best. A third to a half of the population are introverts — a third to a half. So that ' s one out of every two or three people you know. So even if you ' re an extrovert yourself, I ' m talking about your coworkers and your spouses and your children and the person sitting next to you right now — all of them subject to this bias that is pretty deep and real in our society. We all internalize it from a very early age without even having a language for what we ' re doing. Now, to see the bias clearly, you need to understand what introversion is. It ' s different from being shy. Shyness is about fear of social judgment. Introversion is more about, how do you respond to stimulation, including social stimulation. So extroverts really crave large amounts of stimulation, whereas introverts feel at their most alive and their most switched-on and their most capable when they ' re in quieter, more low-key environments. Not all the time — these things aren ' t absolute — but a lot of the time. So the key then to maximizing our talents is for us all to put ourselves in the zone of stimulation that is right for us. But now here ' s where the bias comes in. Our most important institutions, our schools and our workplaces, they are designed mostly for extroverts and for extroverts ' need for lots of stimulation. And also we have this belief system right now that I call the new groupthink, which holds that all creativity and all productivity comes from a very oddly gregarious place. So if you picture the typical classroom nowadays : When I was going to school, we sat in rows. We sat in rows of desks like this, and we did most of our work pretty autonomously. But nowadays, your typical classroom has pods of desks — four or five or six or seven kids all facing each other. And kids are working in countless group assignments. Even in subjects like math and creative writing, which you think would depend on solo flights of thought, kids are now expected to act as committee members. And for the kids who prefer to go off by themselves or just to work alone, those kids are seen as outliers often or, worse, as problem cases. And the vast majority of teachers reports believing that the ideal student is an extrovert as opposed to an introvert, even though introverts actually get better grades and are more knowledgeable, according to research. Okay, same thing is true in our workplaces. Now, most of us work in open plan offices, without walls, where we are subject to the constant noise and gaze of our coworkers. And when it comes to leadership, introverts are routinely passed over for leadership positions, even though introverts tend to be very careful, much less likely to take outsize risks — which is something we might all favor nowadays. And interesting research by Adam Grant at the Wharton School has found that introverted leaders often deliver better outcomes than extroverts do, because when they are managing proactive employees, they ' re much more likely to let those employees run with their ideas, whereas an extrovert can, quite unwittingly, get so excited about things that they ' re putting their own stamp on things, and other people ' s ideas might not as easily then bubble up to the surface. Now in fact, some of our transformative leaders in history have been introverts. I ' ll give you some examples. Eleanor Roosevelt, Rosa Parks, Gandhi — all these people described themselves as quiet and soft-spoken and even shy. And they all took the spotlight, even though every bone in their bodies was telling them not to. And this turns out to have a special power all its own, because people could feel that these leaders were at the helm not because they enjoyed directing others and not out of the pleasure of being looked at ; they were there because they had no choice, because they were driven to do what they thought was right. Now I think at this point it ' s important for me to say that I actually love extroverts. I always like to say some of my best friends are extroverts, including my beloved husband. And we all fall at different points, of course, along the introvert / extrovert spectrum. Even Carl Jung, the psychologist who first popularized these terms, said that there ' s no such thing as a pure introvert or a pure extrovert. He said that such a man would be in a lunatic asylum, if he existed at all. And some people fall smack in the middle of the introvert / extrovert spectrum, and we call these people ambiverts. And I often think that they have the best of all worlds. But many of us do recognize ourselves as one type or the other. And what I ' m saying is that culturally, we need a much better balance. We need more of a yin and yang between these two types. This is especially important when it comes to creativity and to productivity, because when psychologists look at the lives of the most creative people, what they find are people who are very good at exchanging ideas and advancing ideas, but who also have a serious streak of introversion in them. And this is because solitude is a crucial ingredient often to creativity. So Darwin, he took long walks alone in the woods and emphatically turned down dinner-party invitations. Theodor Geisel, better known as Dr. Seuss, he dreamed up many of his amazing creations in a lonely bell tower office that he had in the back of his house in La Jolla, California. And he was actually afraid to meet the young children who read his books for fear that they were expecting him this kind of jolly Santa Claus-like figure and would be disappointed with his more reserved persona. Steve Wozniak invented the first Apple computer sitting alone in his cubicle in Hewlett-Packard where he was working at the time. And he says that he never would have become such an expert in the first place had he not been too introverted to leave the house when he was growing up. Now, of course, this does not mean that we should all stop collaborating — and case in point, is Steve Wozniak famously coming together with Steve Jobs to start Apple Computer — but it does mean that solitude matters and that for some people it is the air that they breathe. And in fact, we have known for centuries about the transcendent power of solitude. It ' s only recently that we ' ve strangely begun to forget it. If you look at most of the world ' s major religions, you will find seekers — Moses, Jesus, Buddha, Muhammad — seekers who are going off by themselves alone to the wilderness, where they then have profound epiphanies and revelations that they then bring back to the rest of the community. So, no wilderness, no revelations. This is no surprise, though, if you look at the insights of contemporary psychology. It turns out that we can ' t even be in a group of people without instinctively mirroring, mimicking their opinions. Even about seemingly personal and visceral things like who you ' re attracted to, you will start aping the beliefs of the people around you without even realizing that that ' s what you ' re doing. And groups famously follow the opinions of the most dominant or charismatic person in the room, even though there ' s zero correlation between being the best talker and having the best ideas — I mean zero. So — You might be following the person with the best ideas, but you might not. And do you really want to leave it up to chance? Much better for everybody to go off by themselves, generate their own ideas freed from the distortions of group dynamics, and then come together as a team to talk them through in a well- managed environment and take it from there. Now if all this is true, then why are we getting it so wrong? Why are we setting up our schools this way, and our workplaces? And why are we making these introverts feel so guilty about wanting to just go off by themselves some of the time? One answer lies deep in our cultural history. Western societies, and in particular the U. S., have always favored the man of action over the"man"of contemplation. But in America ' s early days, we lived in what historians call a culture of character, where we still, at that point, valued people for their inner selves and their moral rectitude. And if you look at the self-help books from this era, they all had titles with things like"Character, the Grandest Thing in the World."And they featured role models like Abraham Lincoln, who was praised for being modest and unassuming. Ralph Waldo Emerson called him"A man who does not offend by superiority."But then we hit the 20th century, and we entered a new culture that historians call the culture of personality. What happened is we had evolved an agricultural economy to a world of big business. And so suddenly people are moving from small towns to the cities. And instead of working alongside people they ' ve known all their lives, now they are having to prove themselves in a crowd of strangers. So, quite understandably, qualities like magnetism and charisma suddenly come to seem really important. And sure enough, the self-help books change to meet these new needs and they start to have names like"How to Win Friends and Influence People."And they feature as their role models really great salesmen. So that ' s the world we ' re living in today. That ' s our cultural inheritance. Now none of this is to say that social skills are unimportant, and I ' m also not calling for the abolishing of teamwork at all. The same religions who send their sages off to lonely mountain tops also teach us love and trust. And the problems that we are facing today in fields like science and in economics are so vast and so complex that we are going to need armies of people coming together to solve them working together. But I am saying that the more freedom that we give introverts to be themselves, the more likely that they are to come up with their own unique solutions to these problems. So now I ' d like to share with you what ' s in my suitcase today. Guess what? Books. I have a suitcase full of books. Here ' s Margaret Atwood,"Cat ' s Eye."Here ' s a novel by Milan Kundera. And here ' s"The Guide for the Perplexed"by Maimonides. But these are not exactly my books. I brought these books with me because they were written by my grandfather ' s favorite authors. My grandfather was a rabbi and he was a widower who lived alone in a small apartment in Brooklyn that was my favorite place in the world when I was growing up, partly because it was filled with his very gentle, very courtly presence and partly because it was filled with books. I mean literally every table, every chair in this apartment had yielded its original function to now serve as a surface for swaying stacks of books. Just like the rest of my family, my grandfather ' s favorite thing to do in the whole world was to read. But he also loved his congregation, and you could feel this love in the sermons that he gave every week for the 62 years that he was a rabbi. He would takes the fruits of each week ' s reading and he would weave these intricate tapestries of ancient and humanist thought. And people would come from all over to hear him speak. But here ' s the thing about my grandfather. Underneath this ceremonial role, he was really modest and really introverted — so much so that when he delivered these sermons, he had trouble making eye contact with the very same congregation that he had been speaking to for 62 years. And even away from the podium, when you called him to say hello, he would often end the conversation prematurely for fear that he was taking up too much of your time. But when he died at the age of 94, the police had to close down the streets of his neighborhood to accommodate the crowd of people who came out to mourn him. And so these days I try to learn from my grandfather ' s example in my own way. So I just published a book about introversion, and it took me about seven years to write. And for me, that seven years was like total bliss, because I was reading, I was writing, I was thinking, I was researching. It was my version of my grandfather ' s hours of the day alone in his library. But now all of a sudden my job is very different, and my job is to be out here talking about it, talking about introversion. And that ' s a lot harder for me, because as honored as I am to be here with all of you right now, this is not my natural milieu. So I prepared for moments like these as best I could. I spent the last year practicing public speaking every chance I could get. And I call this my"year of speaking dangerously."And that actually helped a lot. But I ' ll tell you, what helps even more is my sense, my belief, my hope that when it comes to our attitudes to introversion and to quiet and to solitude, we truly are poised on the brink on dramatic change. I mean, we are. And so I am going to leave you now with three calls for action for those who share this vision. Number one : Stop the madness for constant group work. Just stop it. Thank you. And I want to be clear about what I ' m saying, because I deeply believe our offices should be encouraging casual, chatty cafe-style types of interactions — you know, the kind where people come together and serendipitously have an exchange of ideas. That is great. It ' s great for introverts and it ' s great for extroverts. But we need much more privacy and much more freedom and much more autonomy at work. School, same thing. We need to be teaching kids to work together, for sure, but we also need to be teaching them how to work on their own. This is especially important for extroverted children too. They need to work on their own because that is where deep thought comes from in part. Okay, number two : Go to the wilderness. Be like Buddha, have your own revelations. I ’ m not saying that we all have to now go off and build our own cabins in the woods and never talk to each other again, but I am saying that we could all stand to unplug and get inside our own heads a little more often. Number three : Take a good look at what ’ s inside your own suitcase and why you put it there. So extroverts, maybe your suitcases are also full of books. Or maybe they ’ re full of champagne glasses or skydiving equipment. Whatever it is, I hope you take these things out every chance you get and grace us with your energy and your joy. But introverts, you being you, you probably have the impulse to guard very carefully what ’ s inside your own suitcase. And that ’ s okay. But occasionally, just occasionally, I hope you will open up your suitcases for other people to see, because the world needs you and it needs the things you carry. So I wish you the best of all possible journeys and the courage to speak softly. Thank you very much. Thank you. Thank you.

Text Sample 943: NEG Dim 6 – 2010 – Score 1.00 – 2010/t002906.txt

I think I ’ ll start out and just talk a little bit about what exactly autism is. Autism is a very big continuum that goes from very severe â the child remains nonverbal â all the way up to brilliant scientists and engineers. And I actually feel at home here, because there ’ s a lot of autism genetics here. You wouldn ’ t have any â It ’ s a continuum of traits. When does a nerd turn into Asperger, which is just mild autism? I mean, Einstein and Mozart and Tesla would all be probably diagnosed as autistic spectrum today. And one of the things that is really going to concern me is getting these kids to be the ones that are going to invent the next energy things that Bill Gates talked about this morning. OK, now, if you want to understand autism : animals. I want to talk to you now about different ways of thinking. You have to get away from verbal language. I think in pictures. I don ’ t think in language. Now, the thing about the autistic mind is it attends to details. This is a test where you either have to pick out the big letters or the little letters, and the autistic mind picks out the little letters more quickly. And the thing is, the normal brain ignores the details. Well, if you ’ re building a bridge, details are pretty important because it ’ ll fall down if you ignore the details. And one of my big concerns with a lot of policy things today is things are getting too abstract. People are getting away from doing hands-on stuff. I ’ m really concerned that a lot of the schools have taken out the hands-on classes, because art, and classes like that â those are the classes where I excelled. In my work with cattle, I noticed a lot of little things that most people don ’ t notice would make the cattle balk. For example, this flag waving right in front of the veterinary facility. This feed yard was going to tear down their whole veterinary facility ; all they needed to do was move the flag. Rapid movement, contrast. In the early ’ 70s when I started, I got right down in the chutes to see what cattle were seeing. People thought that was crazy. A coat on a fence would make them balk, shadows would make them balk, a hose on the floor â people weren ’ t noticing these things. A chain hanging down... And that ’ s shown very, very nicely in the movie. In fact, I loved the movie, how they duplicated all my projects. That ’ s the geek side. My drawings got to star in the movie, too. And, actually, it ’ s called”Temple Grandin,”not”Thinking in Pictures.”So what is thinking in pictures? It ’ s literally movies in your head. My mind works like Google for images. When I was a young kid, I didn ’ t know my thinking was different. I thought everybody thought in pictures. Then when I did my book,”Thinking in Pictures,”I started interviewing people about how they think. And I was shocked to find out that my thinking was quite different. Like if I say,”Think about a church steeple,”most people get this sort of generalized generic one. Now, maybe that ’ s not true in this room, but it ’ s going to be true in a lot of different places. I see only specific pictures. They flash up into my memory, just like Google for pictures. And in the movie, they ’ ve got a great scene in there, where the word”shoe”is said, and a whole bunch of ’ 50s and ’ 60s shoes pop into my imagination. OK, there ’ s my childhood church ; that ’ s specific. There ’ s some more, Fort Collins. OK, how about famous ones? And they just kind of come up, kind of like this. Just really quickly, like Google for pictures. And they come up one at a time, and then I think,”OK, well, maybe we can have it snow, or we can have a thunderstorm,”and I can hold it there and turn them into videos. Now, visual thinking was a tremendous asset in my work designing cattle- handling facilities. And I ’ ve worked really hard on improving how cattle are treated at the slaughter plant. I ’ m not going to go into any gucky slaughter slides. I ’ ve got that stuff up on YouTube, if you want to look at it. But one of the things that I was able to do in my design work is I could test- run a piece of equipment in my mind, just like a virtual reality computer system. And this is an aerial view of a recreation of one of my projects that was used in the movie. That was like just so super cool. And there were a lot of, kind of, Asperger types and autism types working out there on the movie set, too. But one of the things that really worries me is : Where ’ s the younger version of those kids going today? They ’ re not ending up in Silicon Valley, where they belong. One of the things I learned very early on because I wasn ’ t that social, is I had to sell my work, and not myself. And the way I sold livestock jobs is I showed off my drawings, I showed off pictures of things. Another thing that helped me as a little kid is, boy, in the ’ 50s, you were taught manners. You were taught you can ’ t pull the merchandise off the shelves in the store and throw it around. When kids get to be in third or fourth grade, you might see that this kid ’ s going to be a visual thinker, drawing in perspective. Now, I want to emphasize that not every autistic kid is going to be a visual thinker. Now, I had this brain scan done several years ago, and I used to joke around about having a gigantic Internet trunk line going deep into my visual cortex. This is tensor imaging. And my great big Internet trunk line is twice as big as the control ’ s. The red lines there are me, and the blue lines are the sex and age-matched control. And there I got a gigantic one, and the control over there, the blue one, has got a really small one. And some of the research now is showing that people on the spectrum actually think with the primary visual cortex. Now, the thing is, the visual thinker is just one kind of mind. You see, the autistic mind tends to be a specialist mind â good at one thing, bad at something else. And where I was bad was algebra. And I was never allowed to take geometry or trig. Gigantic mistake. I ’ m finding a lot of kids who need to skip algebra, go right to geometry and trig. Now, another kind of mind is the pattern thinker. More abstract. These are your engineers, your computer programmers. This is pattern thinking. That praying mantis is made from a single sheet of paper â no scotch tape, no cuts. And there in the background is the pattern for folding it. Here are the types of thinking : photo-realistic visual thinkers, like me ; pattern thinkers, music and math minds. Some of these oftentimes have problems with reading. You also will see these kind of problems with kids that are dyslexic. You ’ ll see these different kinds of minds. And then there ’ s a verbal mind, they know every fact about everything. Now, another thing is the sensory issues. I was really concerned about having to wear this gadget on my face. And I came in half an hour beforehand so I could have it put on and kind of get used to it, and they got it bent so it ’ s not hitting my chin. But sensory is an issue. Some kids are bothered by fluorescent lights ; others have problems with sound sensitivity. You know, it ' s going to be variable. Now, visual thinking gave me a whole lot of insight into the animal mind. Because think about it : an animal is a sensory-based thinker, not verbal â thinks in pictures, thinks in sounds, thinks in smells. Think about how much information there is on the local fire hydrant. He knows who ' s been there â When they were there. Are they friend or foe? Is there anybody he can go mate with? There ' s a ton of information on that fire hydrant. It ' s all very detailed information. And looking at these kind of details gave me a lot of insight into animals. Now, the animal mind, and also my mind, puts sensory-based information into categories. Man on a horse, and a man on the ground â that is viewed as two totally different things. You could have a horse that ' s been abused by a rider. They ' ll be absolutely fine with the veterinarian and with the horseshoer, but you can ' t ride him. You have another horse, where maybe the horseshoer beat him up, and he ' ll be terrible for anything on the ground with the veterinarian, but a person can ride him. Cattle are the same way. Man on a horse, a man on foot â they ' re two different things. You see, it ' s a different picture. See, I want you to think about just how specific this is. Now, this ability to put information into categories, I find a lot of people are not very good at this. When I ' m out troubleshooting equipment or problems with something in a plant, they don ' t seem to be able to figure out :"Do I have a training-people issue? Or do I have something wrong with the equipment?"In other words, categorize equipment problem from a people problem. I find a lot of people have difficulty doing that. Now, let ' s say I figure out it ' s an equipment problem. Is it a minor problem, with something simple I can fix? Or is the whole design of the system wrong? People have a hard time figuring that out. Let ' s just look at something like, you know, solving problems with making airlines safer. Yeah, I ' m a million-mile flier. I do lots and lots of flying, and if I was at the FAA, what would I be doing a lot of direct observation of? It would be their airplane tails. You know, five fatal wrecks in the last 20 years, the tail either came off, or steering stuff inside the tail broke in some way. It ' s tails, pure and simple. And when the pilots walk around the plane, guess what? They can ' t see that stuff inside the tail. Now as I think about that, I ' m pulling up all of that specific information. It ' s specific. See, my thinking ' s bottom-up. I take all the little pieces and I put the pieces together like a puzzle. Now, here is a horse that was deathly afraid of black cowboy hats. He ' d been abused by somebody with a black cowboy hat. White cowboy hats, that was absolutely fine. Now, the thing is, the world is going to need all of the different kinds of minds to work together. We ' ve got to work on developing all these different kinds of minds. And one of the things that is driving me really crazy as I travel around and I do autism meetings, is I ' m seeing a lot of smart, geeky, nerdy kids, and they just aren ' t very social, and nobody ' s working on developing their interest in something like science. And this brings up the whole thing of my science teacher. My science teacher is shown absolutely beautifully in the movie. I was a goofball student when I was in high school. I just didn ' t care at all about studying, until I had Mr. Carlock ' s science class. He was now Dr. Carlock in the movie. And he got me challenged to figure out an optical illusion room. This brings up the whole thing of you ' ve got to show kids interesting stuff. You know, one of the things that I think maybe TED ought to do is tell all the schools about all the great lectures that are on TED, and there ' s all kinds of great stuff on the Internet to get these kids turned on. Because I ' m seeing a lot of these geeky, nerdy kids, and the teachers out in the Midwest and other parts of the country when you get away from these tech areas, they don ' t know what to do with these kids. And they ' re not going down the right path. The thing is, you can make a mind to be more of a thinking and cognitive mind, or your mind can be wired to be more social. And what some of the research now has shown in autism is there may by extra wiring back here in the really brilliant mind, and we lose a few social circuits here. It ' s kind of a trade- off between thinking and social. And then you can get to the point where it ' s so severe, you ' re going to have a person that ' s going to be non-verbal. In the normal human mind, language covers up the visual thinking we share with animals. This is the work of Dr. Bruce Miller. He studied Alzheimer ' s patients that had frontal temporal lobe dementia. And the dementia ate out the language parts of the brain. And then this artwork came out of somebody who used to install stereos in cars. Now, Van Gogh doesn ' t know anything about physics, but I think it ' s very interesting that there was some work done to show that this eddy pattern in this painting followed a statistical model of turbulence, which brings up the whole interesting idea of maybe some of this mathematical patterns is in our own head. And the Wolfram stuff â I was taking notes and writing down all the search words I could use, because I think that ' s going to go on in my autism lectures. We ' ve got to show these kids interesting stuff. And they ' ve taken out the auto- shop class and the drafting class and the art class. I mean, art was my best subject in school. We ' ve got to think about all these different kinds of minds, and we ' ve got to absolutely work with these kind of minds, because we absolutely are going to need these kinds of people in the future. And let ' s talk about jobs. OK, my science teacher got me studying, because I was a goofball that didn ' t want to study. But you know what? I was getting work experience. I ' m seeing too many of these smart kids who haven ' t learned basic things, like how to be on time â I was taught that when I was eight years old. How to have table manners at granny ' s Sunday party. I was taught that when I was very, very young. And when I was 13, I had a job at a dressmaker ' s shop sewing clothes. I did internships in college, I was building things, and I also had to learn how to do assignments. You know, all I wanted to do was draw pictures of horses when I was little. My mother said,"Well let ' s do a picture of something else."They ' ve got to learn how to do something else. Let ' s say the kid is fixated on Legos. Let ' s get him working on building different things. The thing about the autistic mind is it tends to be fixated. Like if the kid loves race cars, let ' s use race cars for math. Let ' s figure out how long it takes a race car to go a certain distance. In other words, use that fixation in order to motivate that kid, that ' s one of the things we need to do. I really get fed up when the teachers, especially when you get away from this part of the country, they don ' t know what to do with these smart kids. It just drives me crazy. What can visual thinkers do when they grow up? They can do graphic design, all kinds of stuff with computers, photography, industrial design. The pattern thinkers â they ' re the ones that are going to be your mathematicians, your software engineers, your computer programmers, all of those kinds of jobs. And then you ' ve got the word minds ; they make great journalists, and they also make really, really good stage actors. Because the thing about being autistic is, I had to learn social skills like being in a play. You just kind of... you just have to learn it. And we need to be working with these students. And this brings up mentors. You know, my science teacher was not an accredited teacher. He was a NASA space scientist. Some states now are getting it to where, if you have a degree in biology or in chemistry, you can come into the school and teach biology or chemistry. We need to be doing that. Because what I ' m observing is, the good teachers, for a lot of these kids, are out in the community colleges. But we need to be getting some of these good teachers into the high schools. Another thing that can be very, very, very successful is : there ' s a lot of people that may have retired from working in the software industry, and they can teach your kid. And it doesn ' t matter if what they teach them is old, because what you ’ re doing is you ’ re lighting the spark. You ’ re getting that kid turned on. And you get him turned on, then you ’ ll learn all the new stuff. Mentors are just essential. I cannot emphasize enough what my science teacher did for me. And we ’ ve got to mentor them, hire them. And if you bring them in for internships in your companies, the thing about the autism, Asperger-y kind of mind, you ’ ve got to give them a specific task. Don ’ t just say,”Design new software.”You ’ ve got to tell them something more specific :”We ’ re designing software for a phone and it has to do some specific thing, and it can only use so much memory.”That ’ s the kind of specificity you need. Well, that ’ s the end of my talk. And I just want to thank everybody for coming. It was great to be here. Oh â you have a question for me? OK. Chris Anderson : Thank you so much for that. You know, you once wrote â I like this quote :”If by some magic, autism had been eradicated from the face of the Earth, then men would still be socializing in front of a wood fire at the entrance to a cave.”Temple Grandin : Because who do you think made the first stone spear? It was the Asperger guy, and if you were to get rid of all the autism genetics, there ’ d be no more Silicon Valley, and the energy crisis would not be solved. CA : I want to ask you a couple other questions, and if any of these feel inappropriate, it ’ s OK just to say,”Next question.”But if there is someone here who has an autistic child, or knows an autistic child and feels kind of cut off from them, what advice would you give them? TG : Well, first of all, we ’ ve got to look at age. If you have a two, three or four-year-old, no speech, no social interaction, I can ’ t emphasize enough : Don ’ t wait. You need at least 20 hours a week of one-to-one teaching. The thing is, autism comes in different degrees. About half of the people on the spectrum are not going to learn to talk, and they won ’ t be working in Silicon Valley. That would not be a reasonable thing for them to do. But then you get these smart, geeky kids with a touch of autism, and that ’ s where you ’ ve got to get them turned on with doing interesting things. I got social interaction through shared interests â I rode horses with other kids, I made model rockets with other kids, did electronics lab with other kids. And in the ’ 60s, it was gluing mirrors onto a rubber membrane on a speaker to make a light show. That was, like, we considered that super cool. CA : Is it unrealistic for them to hope or think that that child loves them, as some might, as most, wish? TG : Well, I tell you, that child will be loyal, and if your house is burning down, they ’ re going to get you out of it. CA : Wow. So most people, if you ask them what they ’ re most passionate about, they ’ d say things like,”My kids”or”My lover.”What are you most passionate about? TG : I ’ m passionate about that the things I do are going to make the world a better place. When I have a mother of an autistic child say,”My kid went to college because of your book or one of your lectures,”that makes me happy. You know, the slaughter plants I worked with in the ’ 80s ; they were absolutely awful. I developed a really simple scoring system for slaughter plants, where you just measure outcomes : How many cattle fell down? How many got poked with the prodder? How many cattle are mooing their heads off? And it ’ s very, very simple. You directly observe a few simple things. It ’ s worked really well. I get satisfaction out of seeing stuff that makes real change in the real world. We need a lot more of that, and a lot less abstract stuff. CA : Totally. CA : When we were talking on the phone, one of the things you said that really astonished me was that one thing you were passionate about was server farms. Tell me about that. TG : Well, the reason why I got really excited when I read about that, it contains knowledge. It ’ s libraries. And to me, knowledge is something that is extremely valuable. So, maybe over 10 years ago now, our library got flooded. This is before the Internet got really big. And I was really upset about all the books being wrecked, because it was knowledge being destroyed. And server farms, or data centers, are great libraries of knowledge. CA : Temple, can I just say, it ’ s an absolute delight to have you at TED. Thank you so much. TG : Well, thank you so much. Thank you.

Text Sample 944: NEG Dim 6 – 2010 – Score 1.00 – 2010/t002904.txt

I think it was in my second grade that I was caught drawing the bust of a nude by Michelangelo. I was sent straight away to my school principal, and my school principal, a sweet nun, looked at my book with disgust, flipped through the pages, saw all the nudes â you know, I ’ d been seeing my mother draw nudes and I ’ d copy her â and the nun slapped me on my face and said,”Sweet Jesus, this kid has already begun.”I had no clue what she was talking about, but it was convincing enough for me never to draw again until the ninth grade. Thanks to a really boring lecture, I started caricaturing my teachers in school. And, you know, I got a lot of popularity. I don ’ t play sports. I ’ m really bad at sports. I don ’ t have the fanciest gadgets at home. I ’ m not on top of the class. So for me, cartooning gave me a sense of identity. I got popular, but I was scared I ’ d get caught again. So what I did was I quickly put together a collage of all the teachers I had drawn, glorified my school principal, put him right on top, and gifted it to him. He had a good laugh at the other teachers and put it up on the notice board. This is a part of that. And I became a school hero. All my seniors knew me. I felt really special. I have to tell you a little bit about my family. That ’ s my mother. I love her to bits. She ’ s the one who taught me how to draw and, more importantly, how to love. She ’ s a bit of a hippie. She said,”Don ’ t say that,”but I ’ m saying it anyway. The rest of my family are boring academics, busy collecting Ivy League decals for our classic Ambassador car. My father ’ s a little different. My father believed in a holistic approach to living, and, you know, every time he taught us, he ’ d say,”I hate these books, because these books are hijacked by Industrial Revolution.”While he still held that worldview, I was 16, I got the best lawyer in town, my older brother Karthik, and I sat him down, and I said,”Pa, from today onwards I ’ ve decided I ’ m going to be disciplined, I ’ m going to be curious, I ’ m going to learn something new every day, I ’ m going to be very hard working, and I ’ m not going to depend on you emotionally or financially.”And he was very impressed. He was all tearing up. Ready to hug me. And I said,”Hold that thought.”I said,”Can I quit school then?”But, to cut a long story short, I quit school to pursue a career as a cartoonist. I must have done about 30, 000 caricatures. I would do birthday parties, weddings, divorces, anything for anyone who wanted to use my services. But, most importantly, while I was traveling, I taught children cartooning, and in exchange, I learned how to be spontaneous. And mad and crazy and fun. When I started teaching them, I said let me start doing this professionally. When I was 18 I started my own school. However, an 18 year-old trying to start a school is not easy unless you have a big patron or a big supporter. So I was flipping through the pages of the Times of India when I saw that the Prime Minister of India was visiting my home town, Bangalore. And, you know, just like how every cartoonist knows Bush here, and if you had to meet Bush, it would be the funnest thing because his face was a cartoonist ’ s delight. I had to meet my Prime Minister. I went to the place where his helicopter was about to land. I saw layers of security. I caricatured my way through three layers by just impressing the guards, but I got stuck. I got stuck at the third. And what happened was, to my luck, I saw a nuclear scientist at whose party I had done cartoons. I ran up to him, and said,”Hello, sir. How do you do?”He said,”What are you doing here, Raghava?”I said,”I ’ m here to meet the Prime Minister.”He said,”Oh, so am I.”I hopped into his car, and off we went through the remaining layers of security. Thank you. I sat him down, I caricatured him, and since then I ’ ve caricatured hundreds of celebrities. This is one I remember fondly. Salman Rushdie was pissed-off I think because I altered the map of New York, if you notice. Anyway, the next slide I ’ m about to show you â Should I just turn that off? The next slide I ’ m about to show you, is a little more serious. I was hesitant to include this in my presentation because this cartoon was published soon after 9 / 11. What was, for me, a very naive observation, turned out to be a disaster. That evening, I came home to hundreds of hate mails, hundreds of people telling me how they could have lived another day without seeing this. I was also asked to leave the organization, a cartoonists ’ organization in America, that for me was my lifeline. That ’ s when I realized, you know, cartoons are really powerful, art comes with responsibility. Anyway, what I did was I decided that I need to take a break. I quit my job at the papers, I closed my school, and I wrapped up my pencils and my brushes and inks, and I decided to go traveling. When I went traveling, I remember, I met this fabulous old man, who I met when I was caricaturing, who turned out to be an artist, in Italy. He invited me to his studio. He said,”Come and visit.”When I went, I saw the ghastliest thing ever. I saw this dead, naked effigy of himself hanging from the ceiling. I said,”Oh, my God. What is that?”And I asked him, and he said,”Oh, that thing? In the night, I die. In the morning, I am born again.”I thought he was koo koo, but something about that really stuck. I loved it. I thought there was something really beautiful about that. So I said,”I am dead, so I need to be born again.”So, I wanted to be a painter like him, except, I don ’ t know how to paint. So, I tried going to the art store. You know, there are a hundred types of brushes. Forget it, they will confuse you even if you know how to draw. So I decided, I ’ m going to learn to paint by myself. I ’ m going to show you a very quick clip to show you how I painted and a little bit about my city, Bangalore. They had to be larger than life. Everything had to be larger. The next painting was even bigger. And even bigger. And for me it was, I had to dance while I painted. It was so exciting. Except, I even started painting dancers. Here for example is a Flamenco dancer, except there was one problem. I didn ’ t know the dance form, so I started following them, and I made some money, sold my paintings and would rush off to France or Spain and work with them. That ’ s Pepe Linares, the renowned Flamenco singer. But I had one problem, my paintings never danced. As much energy as I put into them while making them, they never danced. So I decided â I had this crazy epiphany at two in the morning. I called my friends, painted on their bodies, and had them dance in front of a painting. And, all of a sudden, my paintings came alive. And then I was fortunate enough to actually perform this in California with Velocity Circus. And I sat like you guys there in the audience. And I saw my work come alive. You know, normally you work in isolation, and you show at a gallery, but here, the work was coming alive, and it had some other artists working with me. The collaborative effort was fabulous. I said, I ’ m going to collaborate with anybody and everybody I meet. I started doing fashion. This is a fashion show we held in London. The best collaboration, of course, is with children. They are ruthless, they are honest, but they ’ re full of energy and fun. This is a work, a library I designed for the Robin Hood Foundation. And I must say, I spent time in the Bronx working with these kids. And, in exchange for me working with them, they taught me how to be cool. I don ’ t think I ’ ve succeeded, but they ’ ve taught me. They said,”Stop saying sorry. Say, my bad.”Then I said, all this is good, but I want to paint like a real painter. American education is so expensive. I was in India, and I was walking down the streets, and I saw a billboard painter. And these guys paint humongous paintings, and they look really good. And I wondered how they did it from so close. So, one day I had the opportunity to meet one of these guys, and I said,”How do you paint like that? Who taught you?”And he said,”Oh, it ’ s very easy. I can teach you, but we ’ re leaving the city, because billboard painters are a dying, extinct bunch of artists, because digital printing has totally replaced them and hijacked them.”I said, in exchange for education in how to paint, I will support them, and I started a company. And since then, I ’ ve been painting all over the place. This is a painting I did of my wife in my apartment. This is another painting. And, in fact, I started painting on anything, and started sending them around town. Since I mentioned my wife, the most important collaboration has been with her, Netra. Netra and I met when she was 18. I must have been 19 and a half then, and it was love at first sight. I lived in India. She lived in America. She ’ d come every two months to visit me, and then I said I ’ m the man, I ’ m the man, and I have to reciprocate. I have to travel seven oceans, and I have to come and see you. I did that twice, and I went broke. So then I said,”Nets, what do I do?”She said,”Why don ’ t you send me your paintings? My dad knows a bunch of rich guys. We ’ ll try and con them into buying it, and then...”But it turned out, after I sent the works to her, that her dad ’ s friends, like most of you, are geeks. I ’ m joking. No, they were really big geeks, and they didn ’ t know much about art. So Netra was stuck with 30 paintings of mine. So what we did was we rented a little van and we drove all over the east coast trying to sell it. She contacted anyone and everyone who was willing to buy my work. She made enough money, she sold off the whole collection and made enough money to move me for four years with lawyers, a company, everything, and she became my manager. That ’ s us in New York. Notice one thing, we ’ re equal here. Something happened along the line. But this brought me â with Netra managing my career â it brought me a lot of success. I was really happy. I thought of myself as a bit of a rockstar. I loved the attention. This is all the press we got, and we said, it ’ s time to celebrate. And I said that the best way to celebrate is to marry Netra. I said,”Let ’ s get married.”And I said,”Not just married. Let ’ s invite everyone who ’ s helped us, all the people who bought our work.”And you won ’ t believe it, we put together a list of 7, 000 people, who had made a difference â a ridiculous list, but I was determined to bring them to India, so â a lot of them were in India. 150 artists volunteered to help me with my wedding. We had fashion designers, installation artists, models, we had makeup artists, jewelry designers, all kinds of people working with me to make my wedding an art installation. And I had a special installation in tribute to my in-laws. I had the vegetable carvers work on that for me. But all this excitement led to the press writing about us. We were in the papers, we ’ re still in the news three years later, but, unfortunately, something tragic happened right after. My mother fell very ill. I love my mother and I was told all of a sudden that she was going to die. And they said you have to say bye to her, you have to do what you have to do. And I was devastated. I had shows booked up for another year. I was on a high. And I couldn ’ t. I could not. My life was not exuberant. I could not live this larger than life person. I started exploring the darker abscesses of the human mind. Of course, my work turned ugly, but another thing happened. I lost all my audiences. The Bollywood stars who I would party with and buy my work disappeared. The collectors, the friends, the press, everyone said,”Nice, but thank you.””No thank you,”was more like it. But I wanted people to actually feel my work from their gut, because I was painting it from my gut. If they wanted beauty, I said, this is the beauty I ’ m willing to give you. It ’ s politicized. Of course, none of them liked it. My works also turned autobiographical. At this point, something else happened. A very, very dear friend of mine came out of the closet, and in India at that time, it was illegal to be gay, and it ’ s disgusting to see how people respond to a gay person. I was very upset. I remember the time when my mother used to dress me up as a little girl â that ’ s me there â because she wanted a girl, and she has only boys. Anyway, I don ’ t know what my friends are going to say after this talk. It ’ s a secret. So, after this, my works turned a little violent. I talked about this masculinity that one need not perform. And I talked about the weakness of male sexuality. This time, not only did my collectors disappear, the political activists decided to ban me and to threaten me and to forbid me from showing. It turned nasty, and I ’ m a bit of a chicken. I can ’ t deal with any threat. This was a big threat. So, I decided it was time to end and go back home. This time I said let ’ s try something different. I need to be reborn again. And I thought the best way, as most of you know who have children, the best way to have a new lease on life, is to have a child. I decided to have a child, and before I did that, I quickly studied what can go wrong. How can a family get dysfunctional? And Rudra was born. That ’ s my little son. And two magical things happened after he was born. My mother miraculously recovered after a serious operation, and this man was elected president of this country. You know I sat at home and I watched. I teared up and I said that ’ s where I want to be. So Netra and I wound up our life, closed up everything we had, and we decided to move to New York. And this was just eight months ago. I moved back to New York, my work has changed. Everything about my work has become more whimsical. This one is called”What the Fuck Was I Thinking?”It talks about mental incest. You know, I may appear to be a very nice, clean, sweet boy. But I ’ m not. I ’ m capable of thinking anything. But I ’ m very civil in my action, I assure you. These are just different cartoons. And, before I go, I want to tell you a little story. I was talking to mother and father this morning, and my dad said,”I know you have so much you want to say, but you have to talk about your work with children.”So I said, okay. I work with children all over the world, and that ’ s an entirely different talk, but I want to leave you with one story that really, really inspired me. I met Belinda when she was 16. I was 17. I was in Australia, and Belinda had cancer, and I was told she ’ s not going to live very long. They, in fact, told me three weeks. I walk into her room, and there was a shy girl, and she was bald, and she was trying to hide her baldness. I whipped out my pen, and I started drawing on her head and I drew a crown for her. And then, we started talking, and we spent a lovely time â I told her how I ended up in Australia, how I backpacked and who I conned, and how I got a ticket, and all the stories. And I drew it out for her. And then I left. Belinda died and within a few days of her death, they published a book for her, and she used my cartoon on the cover. And she wrote a little note, she said,”Hey Rags, thank you for the magic carpet ride around the world.”For me, my art is my magic carpet ride. I hope you will join me in this magic carpet ride, and touch children and be honest. Thank you so much.

Text Sample 945: NEG Dim 6 – 2010 – Score 1.00 – 2010/t002884.txt

Last year here at TED I asked you to give me your data, to put your data on the web, on the basis that if people put data onto the web â government data, scientific data, community data, whatever it is â it will be used by other people to do wonderful things, in ways that they never could have imagined. So, today I ’ m back just to show you a few things, to show you, in fact, there is an open data movement afoot, now, around the world. The cry of”Raw data now!”which I made people make in the auditorium, was heard around the world. So, let ’ s roll the video. A classic story, the first one which lots of people picked up, was when in March â on March 10th in fact, soon after TED â Paul Clarke, in the U. K. government, blogged,"Oh, I ' ve just got some raw data. Here it is, it ' s about bicycle accidents."Two days it took the Times Online to make a map, a mashable map â we call these things mash-ups â a mashed-up user interface that allows you to go in there and have a look and find out whether your bicycle route to work was affected. Here ' s more data, traffic survey data, again, put out by the U. K. government, and because they put it up using the Linked Data standards, then a user could just make a map, just by clicking. Does this data affect things? Well, let ' s get back to 2008. Look at Zanesville, Ohio. Here ' s a map a lawyer made. He put on it the water plant, and which houses are there, which houses have been connected to the water. And he got, from other data sources, information to show which houses are occupied by white people. Well, there was too much of a correlation, he felt, between which houses were occupied by white people and which houses had water, and the judge was not impressed either. The judge was not impressed to the tune of 10. 9 million dollars. That ' s the power of taking one piece of data, another piece of data, putting it together, and showing the result. Let ' s look at some data from the U. K. now. This is U. K. government data, a completely independent site, Where Does My Money Go. It allows anybody to go there and burrow down. You can burrow down by a particular type of spending, or you can go through all the different regions and compare them. So, that ' s happening in the U. K. with U. K. government data. Yes, certainly you can do it over here. Here ' s a site which allows you to look at recovery spending in California. Take an arbitrary example, Long Beach, California, you can go and have a look at what recovery money they ' ve been spending on different things such as energy. In fact, this is the graph of the number of data sets in the repositories of data. gov, and data. gov. uk. And I ' m delighted to see a great competition between the U. K. in blue, and the U. S. in red. How can you use this stuff? Well, for example, if you have lots of data about places you can take, from a postcode â which is like a zip plus four â for a specific group of houses, you can make paper, print off a paper which has got very, very specific things about the bus stops, the things specifically near you. On a larger scale, this is a mash-up of the data which was released about the Afghan elections. It allows you to set your own criteria for what sort of things you want to look at. The red circles are polling stations, selected by your criteria. And then you can select also other things on the map to see what other factors, like the threat level. So, that was government data. I also talked about community-generated data â in fact I edited some. This is the wiki map, this is the Open Street Map."Terrace Theater"I actually put on the map because it wasn ' t on the map before TED last year. I was not the only person editing the open street map. Each flash on this visualization â put together by ITO World â shows an edit in 2009 made to the Open Street Map. Let ' s now spin the world during the same year. Every flash is an edit. Somebody somewhere looking at the Open Street Map, and realizing it could be better. You can see Europe is ablaze with updates. Some places, perhaps not as much as they should be. Here focusing in on Haiti. The map of Port au-Prince at the end of 2009 was not all it could be, not as good as the map of California. Fortunately, just after the earthquake, GeoEye, a commercial company, released satellite imagery with a license, which allowed the open-source community to use it. This is January, in time lapse, of people editing... that ' s the earthquake. After the earthquake, immediately, people all over the world, mappers who wanted to help, and could, looked at that imagery, built the map, quickly building it up. We ' re focusing now on Port-au-Prince. The light blue is refugee camps these volunteers had spotted from the [satellite images]. So, now we have, immediately, a real-time map showing where there are refugee camps â rapidly became the best map to use if you ' re doing relief work in Port-au-Prince. Witness the fact that it ' s here on this Garmin device being used by rescue team in Haiti. There ' s the map showing, on the left-hand side, that hospital â actually that ' s a hospital ship. This is a real-time map that shows blocked roads, damaged buildings, refugee camps â it shows things that are needed [for rescue and relief work]. So, if you ' ve been involved in that at all, I just wanted to say : Whatever you ' ve been doing, whether you ' ve just been chanting,"Raw data now!"or you ' ve been putting government or scientific data online, I just wanted to take this opportunity to say : Thank you very much, and we have only just started!

Text Sample 946: NEG Dim 6 – 2013 – Score 1.00 – 2013/t002408.txt

Thank y ' all! This is going to be a motivational speech. Because â imagine my motivation standing between this strong, healthy crowd... and lunch. So... I ' m @Falkvinge on Twitter. Feel free to quote me if I say something memorable, stupid, funny, whatever. I love seeing my name on Twitter. So... Hi! I ' m Rick. I ' m a politician. I ' m sorry. How many in here have heard of the Swedish Pirate Party before? Let ' s see a show of hands. OK, that ' s practically everybody. Probably due to the fact that we are Sweden ' s neighbor. I frequently ask how many have heard of any other political party and there ' s always just scattered hands in the audience compared to this first question which is one-half to two-thirds. This is actually the first time ever that does not match. It was practically everybody. So, for those who haven ' t heard of us : well, the Pirate Party, we love the net. We love copying and sharing, and we love civil liberties. For that, some people call us pirates. Probably in an attempt to make us bow our heads and feel shame. That didn ' t work very well. We decided to stand tall about it instead. And so in 2006, I founded a new political party. I led it for its first five years. And the European elections, the last European elections, we became the largest party and the most coveted youth demographic, sub-30. And what ' s interesting is we did that on less than one percent of the competition ' s budget. We had a campaign budget total of 50, 000 euros. They had six million between them â and we beat them. That gave us a cost efficiency advantage of over two orders of magnitude. And I ' m gonna share the secret recipe of how we did that. We developed swarm methodologies. And they can be applied to any business or social cause. Well, almost any â there ' s a small asterisk by the end, and I ' ll get to that in just a minute. But applying these â and we ' ve done this dozens of times, we know that this works. We ' ve put two people in the European Parliament, we have 45 people in various German state parliaments, we ' re in the Icelandic parliament, the Czech senate, many, many, many more, local councils â and, as said, we ' ve spread to 70 countries. And that ' s not bad for a political movement that hasn ' t even been around for a decade. So today we ' re going to talk a bit about â how people are motivated to be part of change, to be part of something bigger than themselves. And how you can channel this into an organization that harnesses this great power of wanting to make the world a better place. And in the end, come out a little on the better. When I speak to businesspeople, I frequently make them very upset when I contradict them and say that no, your employees are not your most valuable asset. Your most valuable asset is the thousands of people who want to work for you for free. And you don ' t let them. They get very upset about that. A swarm is a congregation of tens of thousands of volunteers that have chosen of their own will to converge on a common goal. There ' s this"Futurama"quote :"When push comes to shove, you gotta do what you love â even if it ’ s not a good idea.”I mean, seriously, what kind of idiot thinks they can change the world by starting a political party? This kind of idiot, apparently. But it works! What you need to do is to put a stake in the ground. You need to announce your goal. Just say,”I want to accomplish this.”I ’ m going to do this. And it doesn ’ t need to be very costly. My announcement was just two lines in a chat channel.”Hey, look, the Pirate Party has its website up now after New Year ’ s.”And the address. That was all the advertising I ever did. The next time I had several hundred activists wanting to work with us. When you provide such a focus point, a swarm intelligence emerges. When people can rally to a flag. And that ’ s what gives you this two orders of magnitude of cost efficiency. It ’ s a huge advantage â you ’ re running circles around all the legacy organizations. And there are four goals that need to be fulfilled in your goal in order for this to work. These four criteria are that your goal must be : tangible, credible, inclusive and epic. Let ’ s take a look at them : It needs to be tangible. A lot of people say,”Well, you know, we should make the world a better place,”or,”Yeah, we should all feel good now.”Not going to work. You need a binary. Are we there yet, or are we not there yet? It needs to be credible. Somebody seeing the project plan that you ’ re posting needs to see that, yes, this project plan will take us from where we are to where we want to be. You need to break it down into subgoals that each by themselves are seen as doable, and when you add the subgoals together, we ’ ve gone to where we want. It needs to be â and this is where it gets exciting in terms of working swarmwise â it needs to be inclusive. Anybody who sees this project plan needs to immediately say,”I want to do this â and there ’ s my spot!”And they will be able to jump right into the project and start working on it without asking anybody ’ s permission. And that is exactly what ’ ll happen. And, last but not least, it needs to be epic. It needs to energize people. It needs to electrify people. Shoot for the moon! On second thought, don ’ t shoot for the moon, we ’ ve already been there â shoot for Mars! In contrast, you will never be able to get a volunteer swarm forming around making the most correct tax audit ever. Doesn ’ t electrify people. Go to Mars. A lot of people kind of balk at the obstacles. We ’ re going to climb a huge mountain. So how do you motivate people to do that? Well, it turns out that obstacles are not the problem. Not knowing the obstacles is the problem. If you know how high the mountain is, you know exactly what it takes to scale it. We know exactly how far away Mars is and what it takes to get there. If you can plan it like a project, you can plan what resources you need and you can execute it, exactly like a project. Let ’ s see : we ’ re going to Mars, we need two dozen volunteer rocket scientists, one dozen volunteer metallurgists, some crazy dude who will mix rocket fuel in his backyard and so on. When you can list the resources, you know what you need to get there. When you know what you need to get there, you can go there. And the next thing is to encourage this development of a swarm intelligence, which is where the cost efficiency comes in. There ’ s a TED Talk on motivation that debunks that we work for money, and it presents science on how we ’ re really motivated by three things, in terms of larger creative tasks, when we work for something bigger than ourselves. We work for autonomy, mastery and purpose. We ’ ve covered purpose already. As in, working for something bigger, tangible, credible, inclusive and epic. So, where that motivation talk ends, what it doesn ’ t answer is, how do you build an organization that harnesses this motivational power. And this is where working swarmwise comes in, this is where swarm intelligence comes in. Turns out that there are three factors that you optimize for â and each of these are in complete opposite to what you learn at a business school. But it works. We know it works. We have people in many, many parliaments to prove it. Those three factors are : speed, trust and scalability.

We optimize for speed by cutting bottlenecks out of the loop, cutting them out of the decision loop. That means cutting yourself out of the decision loop, which can be hard. But you ’ ve got to communicate your vision so passionately, so strongly, that everybody knows what the goal is and can find something, some step that takes the movement just a little closer to that goal. And when tens of thousands of people do that on a weekly basis, you become an unstoppable force. We had a three-person rule in our organization, saying that if three self-identified volunteers in the movement were in agreement that something was good for the movement, they had the green light from the highest office to go ahead and act in the name of the organization, including spending resources. When you talk about this kind of empowerment to traditional businesspeople, they think you belong in a zoo. But you know what? I led this organization for five years, there were 50, 000 registered members and many, many more anonymous activists. It was not abused once. Everybody had the key to the treasure chest. It was not abused one single time. Turns out when you give people the keys to the castle, and look them in the eye and say,”I trust you,”they step up to the plate. And that ’ s a beautiful thing to see happen. Obviously, not everything went according to plan, but that ’ s a different thing. We made mistakes. We should expect mistakes. If you ’ re pioneering something, that means you must, by definition, venture into the unknown. When you ’ re trying the unknown, some things won ’ t go as planned. That ’ s part of the definition of venturing into the unknown. To find the great, you must allow mistakes to happen. So you must communicate that we expect some things to go wrong to create a risk-positive environment. Therefore we optimize for iteration speed. Meaning that we try, we fail, we try again, we fail faster, we fail better, we try again, we fail better again. Maybe after we ’ ve tried 15 times, we ’ ve mastered some specific subject, so you want to minimize the time it takes to try those 15 times. We optimize on trust. We encourage diversity. You need to communicate your vision so strongly so that everybody can translate it into their own context because language is an incredibly strong inclusionary and exclusionary social marker. This one-brand-fits-all message â forget it! That ’ s what they teach you at business school â it doesn ’ t work. Or at least, it doesn ’ t give you the cost-efficiency advantage of working swarmwise. This leads to a lot of different approaches tried in parallel in different social groups who try out different methods of working toward the goal. Some of them will work but in order to find the great ones, you need this diversity. And you need to communicate that we need that diversity. If somebody on this side does not understand what those guys are doing, that ’ s OK because we all trust each other to work for the better of the movement. And it ’ s OK that I don ’ t understand their social context. I ’ m not expected to â I understand my social context. I contribute with something I know. Make people aware of this diversity. Finally, scalability. Get feet on the ground. Again, in business school, they teach you to use a lean organization. Forget that. Just scale up the organization from the get-go. Start with 10, 000 empty boxes and an org chart covering down to every minor city. When you have lots and lots of small responsibilities in such a scaffolding that supports the swarm, supports the activists, you ’ ll find that these boxes in the org charts are getting filled in quite rapidly, and they start to get filled in beyond your horizon with people you ’ ve never heard of. And so, this swarm keeps growing to tens of thousands of people, each taking on something small with very, very decentralized mandate to act on the organization. And this is when a swarm intelligence emerges. This is when you have this beehive logic where everybody knows what ’ s to be done. Everybody is taking their own small steps towards it. So the swarm starts to act as a coherent organism. And it ’ s amazing to watch. This is when you ’ re awarded by the cost-efficiency advantage over your competitors by two orders of magnitude. Two orders of magnitude. This is not just a silver bullet. So we ’ ve been talking a lot about the big picture today. You can use these swarm methods for a lot of stuff. Do you want to change the world? Do you want to bring clean water to a billion people? Teach three billion people to read? Maybe you ’ re into social change ; you want to introduce unconditional basic income. Or maybe you want to take humanity to Mars. You can do this using these methods. You can do this. It ’ s about leadership. It ’ s about deciding what you want to do and telling it to the world. Because no matter whether you think you can or cannot change the world, no matter whether you think you can or cannot change the world, you are probably right. So one question I want everybody here to ask themselves today is the observation that change doesn ’ t just happen, somebody makes it happen â do you want to be that person? Do you want to be that person? And then one last thing : There ’ s one component more that ’ s required to work swarmwise that I haven ’ t mentioned yet. And that is fun. This goes beyond just enjoying your job, this goes beyond having a pinball machine in the office. Because this is actually required to succeed in a swarmwise scenario. This is required to succeed to get that cost-efficiency advantage of two orders of magnitude. For the reason that you need to attract volunteers. And people, in this aspect, are rather predictable. People will go to other people who are having fun. In contrast, they will walk an extra mile to avoid people who are not having fun. So, having fun is more than just having a pinball machine in the office. It ’ s an absolute and unavoidable requirement for organizational and operational success when you ’ re working swarmwise. So, in summary â a recipe for a swarm organization using these motivational methods to a huge competitive advantage. Your goal : it needs to be tangible, credible, inclusive and epic. Your organization needs to be optimized for speed, trust and scalability. You need to enjoy yourselves. And that will reward you with two orders of magnitude of cost-efficiency advantage. Thank you.

Text Sample 947: NEG Dim 6 – 2013 – Score 1.00 – 2013/t002141.txt

There ’ s so many of you. When I was a kid, I hid my heart under the bed, because my mother said,”If you ’ re not careful, someday someone ’ s going to break it.”Take it from me : Under the bed is not a good hiding spot. I know because I ’ ve been shot down so many times, I get altitude sickness just from standing up for myself. But that ’ s what we were told.”Stand up for yourself.”And that ’ s hard to do if you don ’ t know who you are. We were expected to define ourselves at such an early age, and if we didn ’ t do it, others did it for us. Geek. Fatty. Slut. Fag. And at the same time we were being told what we were, we were being asked,”What do you want to be when you grow up?”I always thought that was an unfair question. It presupposes that we can ’ t be what we already are. We were kids. When I was a kid, I wanted to be a man. I wanted a registered retirement savings plan that would keep me in candy long enough to make old age sweet. When I was a kid, I wanted to shave. Now, not so much. When I was eight, I wanted to be a marine biologist. When I was nine, I saw the movie”Jaws,”and thought to myself,”No, thank you.”And when I was 10, I was told that my parents left because they didn ’ t want me. When I was 11, I wanted to be left alone. When I was 12, I wanted to die. When I was 13, I wanted to kill a kid. When I was 14, I was asked to seriously consider a career path. I said,”I ’ d like to be a writer.”And they said,”Choose something realistic.”So I said,”Professional wrestler.”And they said,”Don ’ t be stupid.”See, they asked me what I wanted to be, then told me what not to be. And I wasn

’ t the only one. We were being told that we somehow must become what we are not, sacrificing what we are to inherit the masquerade of what we will be. I was being told to accept the identity that others will give me. And I wondered, what made my dreams so easy to dismiss? Granted, my dreams are shy, because they ’ re Canadian. My dreams are self-conscious and overly apologetic. They ’ re standing alone at the high school dance, and they ’ ve never been kissed. See, my dreams got called names too. Silly. Foolish. Impossible. But I kept dreaming. I was going to be a wrestler. I had it all figured out. I was going to be The Garbage Man. My finishing move was going to be The Trash Compactor. My saying was going to be,”I ’ m taking out the trash!”And then this guy, Duke”The Dumpster”Droese, stole my entire shtick. I was crushed, as if by a trash compactor. I thought to myself,”What now? Where do I turn?”Poetry. Like a boomerang, the thing I loved came back to me. One of the first lines of poetry I can remember writing was in response to a world that demanded I hate myself. From age 15 to 18, I hated myself for becoming the thing that I loathed : a bully. When I was 19, I wrote,”I will love myself despite the ease with which I lean toward the opposite.”Standing up for yourself doesn ’ t have to mean embracing violence. When I was a kid, I traded in homework assignments for friendship, then gave each friend a late slip for never showing up on time, and in most cases, not at all. I gave myself a hall pass to get through each broken promise. And I remember this plan, born out of frustration from a kid who kept calling me”Yogi,”then pointed at my tummy and said,”Too many picnic baskets.”Turns out it ’ s not that hard to trick someone, and one day before class, I said,”Yeah, you can copy my homework,”and I gave him all the wrong answers that I ’ d written down the night before. He got his paper back expecting a near-perfect score, and couldn ’ t believe it when he looked across the room at me and held up a zero. I knew I didn ’ t have to hold up my paper of 28 out of 30, but my satisfaction was complete when he looked at me, puzzled, and I thought to myself,”Smarter than the average bear, motherfucker.”This is who I am. This is how I stand up for myself. When I was a kid, I used to think that pork chops and karate chops were the same thing. I thought they were both pork chops. My grandmother thought it was cute, and because they were my favorite, she let me keep doing it. Not really a big deal. One day, before I realized fat kids are not designed to climb trees, I fell out of a tree and bruised the right side of my body. I didn ’ t want to tell my grandmother because I was scared I ’ d get in trouble for playing somewhere I shouldn ’ t have been. The gym teacher noticed the bruise, and I got sent to the principal ’ s office. From there, I was sent to another small room with a really nice lady who asked me all kinds of questions about my life at home. I saw no reason to lie. As far as I was concerned, life was pretty good. I told her, whenever I ’ m sad, my grandmother gives me karate chops. This led to a full-scale investigation, and I was removed from the house for three days, until they finally decided to ask how I got the bruises. News of this silly little story quickly spread through the school, and I earned my first nickname : Porkchop. To this day, I hate pork chops. I ’ m not the only kid who grew up this way, surrounded by people who used to say that rhyme about sticks and stones, as if broken bones hurt more than the names we got called, and we got called them all. So we grew up believing no one would ever fall in love with us, that we ’ d be lonely forever, that we ’ d never meet someone to make us feel like the sun was something they built for us in their toolshed. So broken heartstrings bled the blues, and we tried to empty ourselves so we ’ d feel nothing. Don ’ t tell me that hurts less than a broken bone, that an ingrown life is something surgeons can cut away, that there ’ s no way for it to metastasize ; it does. She was eight years old, our first day of grade three when she got called ugly. We both got moved to the back of class so we would stop getting bombarded by spitballs. But the school halls were a battleground. We found ourselves outnumbered day after wretched day. We used to stay inside for recess, because outside was worse. Outside, we ’ d have to rehearse running away, or learn to stay still like statues, giving no clues that we were there. In grade five, they taped a sign to the front of her desk that read,”Beware of dog.”To this day, despite a loving husband, she doesn ’ t think she ’ s beautiful, because of a birthmark that takes up a little less than half her face. Kids used to say,”She looks like a wrong answer that someone tried to erase, but couldn ’ t quite get the job done.”And they ’ ll never understand that she ’ s raising two kids whose definition of beauty begins with the word”Mom,”because they see her heart before they see her skin, because she ’ s only ever always been amazing. He was a broken branch grafted onto a different family tree, adopted, not because his parents opted for a different destiny. He was three when he became a mixed drink of one part left alone and two parts tragedy, started therapy in eighth grade, had a personality made up of tests and pills, lived like the uphills were mountains and the downhills were cliffs, four-fifths suicidal, a tidal wave of antidepressants, and an adolescent being called”Popper,”one part because of the pills, 99 parts because of the cruelty. He tried to kill himself in grade 10 when a kid who could still go home to Mom and Dad had the audacity to tell him,”Get over it.”As if depression is something that could be remedied by any of the contents found in a first-aid kit. To this day, he is a stick of TNT lit from both ends, could describe to you in detail the way the sky bends in the moment before it ’ s about to fall, and despite an army of friends who all call him an inspiration, he remains a conversation piece between people who can ’ t understand sometimes being drug- free has less to do with addiction and more to do with sanity. We weren ’ t the only kids who grew up this way. To this day, kids are still being called names. The classics were”Hey, stupid,””Hey, spaz.”Seems like every school has an arsenal of names getting updated every year. And if a kid breaks in a school and no one around chooses to hear, do they make a sound? Are they just background noise from a soundtrack stuck on repeat, when people say things like,”Kids can be cruel.”Every school was a big top circus tent, and the pecking order went from acrobats to lion tamers, from clowns to carnies, all of these miles ahead of who we were. We were freaks — lobster-claw boys and bearded ladies, oddities juggling depression and loneliness, playing solitaire, spin the bottle, trying to kiss the wounded parts of ourselves and heal, but at night, while the others slept, we kept walking the tightrope. It was practice, and yes, some of us fell. But I want to tell them that all of this is just debris left over when we finally decide to smash all the things we thought we used to be, and if you can ’ t see anything beautiful about yourself, get a better mirror, look a little closer, stare a little longer, because there ’ s something inside you that made you keep trying despite everyone who told you to quit. You built a cast around your broken heart and signed it yourself,”They were wrong.”Because maybe you didn ’ t belong to a group or a clique. Maybe they decided to pick you last for basketball or everything. Maybe you used to bring bruises and broken teeth to show-and-tell, but never told, because how can you hold your ground if everyone around you wants to bury you beneath it? You have to believe that they were wrong. They have to be wrong. Why else would we still be here? We grew up learning to cheer on the underdog because we see ourselves in them. We stem from a root planted in the belief that we are not what we were called. We are not abandoned cars stalled out and sitting empty on some highway, and if in some way we are, don ’ t worry. We only got out to walk and get gas. We are graduating members from the class of We Made It, not the faded echoes of voices crying out,”Names will never hurt me.”Of course they did. But our lives will only ever always continue to be a balancing act that has less to do with pain and more to do with beauty.

Text Sample 948: NEG Dim 6 – 2013 – Score 1.00 – 2013/t002127.txt

So raise your hand if you know someone in your immediate family or circle of friends who suffers from some form of mental illness. Yeah. I thought so. Not surprised. And raise your hand if you think that basic research on fruit flies has anything to do with understanding mental illness in humans. Yeah. I thought so. I ’ m also not surprised. I can see I ’ ve got my work cut out for me here. As we heard from Dr. Insel this morning, psychiatric disorders like autism, depression and schizophrenia take a terrible toll on human suffering. We know much less about their treatment and the understanding of their basic mechanisms than we do about diseases of the body. Think about it : In 2013, the second decade of the millennium, if you ’ re concerned about a cancer diagnosis and you go to your doctor, you get bone scans, biopsies and blood tests. In 2013, if you ’ re concerned about a depression diagnosis, you go to your doctor, and what do you get? A questionnaire. Now, part of the reason for this is that we have an oversimplified and increasingly outmoded view of the biological basis of psychiatric disorders. We tend to view them — and the popular press aids and abets this view — as chemical imbalances in the brain, as if the brain were some kind of bag of chemical soup full of dopamine, serotonin and norepinephrine. This view is conditioned by the fact that many of the drugs that are prescribed to treat these disorders, like Prozac, act by globally changing brain chemistry, as if the brain were indeed a bag of chemical soup. But that can ’ t be the answer, because these drugs actually don ’ t work all that well. A lot of people won ’ t take them, or stop taking them, because of their unpleasant side effects. These drugs have so many side effects because using them to treat a complex psychiatric disorder is a bit like trying to change your engine oil by opening a can and pouring it all over the engine block. Some of it will dribble into the right place, but a lot of it will do more harm than good. Now, an emerging view that you also heard about from Dr. Insel this morning, is that psychiatric disorders are actually disturbances of neural circuits that mediate emotion, mood and affect. When we think about cognition, we analogize the brain to a computer. That ’ s no problem. Well it turns out that the computer analogy is just as valid for emotion. It ’ s just that we don ’ t tend to think about it that way. But we know much less about the circuit basis of psychiatric disorders because of the overwhelming dominance of this chemical imbalance hypothesis. Now, it ’ s not that chemicals are not important in psychiatric disorders. It ’ s just that they don ’ t bathe the brain like soup. Rather, they ’ re released in very specific locations and they act on specific synapses to change the flow of information in the brain. So if we ever really want to understand the biological basis of psychiatric disorders, we need to pinpoint these locations in the brain where these chemicals act. Otherwise, we ’ re going to keep pouring oil all over our mental engines and suffering the consequences. Now to begin to overcome our ignorance of the role of brain chemistry in brain circuitry, it ’ s helpful to work on what we biologists call”model organisms,”animals like fruit flies and laboratory mice, in which we can apply powerful genetic techniques to molecularly identify and pinpoint specific classes of neurons, as you heard about in Allan Jones ’ s talk this morning. Moreover, once we can do that, we can actually activate specific neurons or we can destroy or inhibit the activity of those neurons. So if we inhibit a particular type of neuron, and we find that a behavior is blocked, we can conclude that those neurons are necessary for that behavior. On the other hand, if we activate a group of neurons and we find that that produces the behavior, we can conclude that those neurons are sufficient for the behavior. So in this way, by doing this kind of test, we can draw cause and effect relationships between the activity of specific neurons in particular circuits and particular behaviors, something that is extremely difficult, if not impossible, to do right now in humans. But can an organism like a fruit fly, which is — it ' s a great model organism because it ' s got a small brain, it ' s capable of complex and sophisticated behaviors, it breeds quickly, and it ' s cheap. But can an organism like this teach us anything about emotion-like states? Do these organisms even have emotion-like states, or are they just little digital robots? Charles Darwin believed that insects have emotion and express them in their behaviors, as he wrote in his 1872 monograph on the expression of the emotions in man and animals. And my eponymous colleague, Seymour Benzer, believed it as well. Seymour is the man that introduced the use of drosophila here at CalTech in the ' 60s as a model organism to study the connection between genes and behavior. Seymour recruited me to CalTech in the late 1980s. He was my Jedi and my rabbi while he was here, and Seymour taught me both to love flies and also to play with science. So how do we ask this question? It ' s one thing to believe that flies have emotion-like states, but how do we actually find out whether that ' s true or not? Now, in humans we often infer emotional states, as you ' ll hear later today, from facial expressions. However, it ' s a little difficult to do that in fruit flies. It ' s kind of like landing on Mars and looking out the window of your spaceship at all the little green men who are surrounding it and trying to figure out,"How do I find out if they have emotions or not?"What can we do? It ' s not so easy. Well, one of the ways that we can start is to try to come up with some general characteristics or properties of emotion-like states such as arousal, and see if we can identify any fly behaviors that might exhibit some of those properties. So three important ones that I can think of are persistence, gradations in intensity, and valence. Persistence means long-lasting. We all know that the stimulus that triggers an emotion causes that emotion to last long after the stimulus is gone. Gradations of intensity means what it sounds like. You can dial up the intensity or dial down the intensity of an emotion. If you ' re a little bit unhappy, the corners of your mouth turn down and you sniffle, and if you ' re very unhappy, tears pour down your face and you might sob. Valence means good or bad, positive or negative. So we decided to see if flies could be provoked into showing the kind of behavior that you see by the proverbial wasp at the picnic table, you know, the one that keeps coming back to your hamburger the more vigorously you try to swat it away, and it seems to keep getting irritated. So we built a device, which we call a puff-o-mat, in which we could deliver little brief air puffs to fruit flies in these plastic tubes in our laboratory bench and blow them away. And what we found is that if we gave these flies in the puff-o-mat several puffs in a row, they became somewhat hyperactive and continued to run around for some time after the air puffs actually stopped and took a while to calm down. So we quantified this behavior using custom locomotor tracking software developed with my collaborator Pietro Perona, who ' s in the electrical engineering division here at CalTech. And what this quantification showed us is that, upon experiencing a train of these air puffs, the flies appear to enter a kind of state of hyperactivity which is persistent, long-lasting, and also appears to be graded. More puffs, or more intense puffs, make the state last for a longer period of time. So now we wanted to try to understand something about what controls the duration of this state. So we decided to use our puff-o-mat and our automated tracking software to screen through hundreds of lines of mutant fruit flies to see if we could find any that showed abnormal responses to the air puffs. And this is one of the great things about fruit flies. There are repositories where you can just pick up the phone and order hundreds of vials of flies of different mutants and screen them in your assay and then find out what gene is affected in the mutation. So doing the screen, we discovered one mutant that took much longer than normal to calm down after the air puffs, and when we examined the gene that was affected in this mutation, it turned out to encode a dopamine receptor. That ’ s right — flies, like people, have dopamine, and it acts on their brains and on their synapses through the same dopamine receptor molecules that you and I have. Dopamine plays a number of important functions in the brain, including in attention, arousal, reward, and disorders of the dopamine system have been linked to a number of mental disorders including drug abuse, Parkinson ’ s disease, and ADHD. Now, in genetics, it ’ s a little counterintuitive. We tend to infer the normal function of something by what doesn ’ t happen when we take it away, by the opposite of what we see when we take it away. So when we take away the dopamine receptor and the flies take longer to calm down, from that we infer that the normal function of this receptor and dopamine is to cause the flies to calm down faster after the puff. And that ’ s a bit reminiscent of ADHD, which has been linked to disorders of the dopamine system in humans. Indeed, if we increase the levels of dopamine in normal flies by feeding them cocaine after getting the appropriate DEA license — oh my God — — we find indeed that these cocaine-fed flies calm down faster than normal flies do, and that ’ s also reminiscent of ADHD, which is often treated with drugs like Ritalin that act similarly to cocaine. So slowly I began to realize that what started out as a rather playful attempt to try to annoy fruit flies might actually have some relevance to a human psychiatric disorder. Now, how far does this analogy go? As many of you know, individuals afflicted with ADHD also have learning disabilities. Is that true of our dopamine receptor mutant flies? Remarkably, the answer is yes. As Seymour showed back in the 1970s, flies, like songbirds, as you just heard, are capable of learning. You can train a fly to avoid an odor, shown here in blue, if you pair that odor with a shock. Then when you give those trained flies the chance to choose between a tube with the shock-paired odor and another odor, it avoids the tube containing the blue odor that was paired with shock. Well, if you do this test on dopamine receptor mutant flies, they don ’ t learn. Their learning score is zero. They flunk out of CalTech. So that means that these flies have two abnormalities, or phenotypes, as we geneticists call them, that one finds in ADHD : hyperactivity and learning disability. Now what ’ s the causal relationship, if anything, between these phenotypes? In ADHD, it ’ s often assumed that the hyperactivity causes the learning disability. The kids can ’ t sit still long enough to focus, so they don ’ t learn. But it could equally be the case that it ’ s the learning disabilities that cause the hyperactivity. Because the kids can ’ t learn, they look for other things to distract their attention. And a final possibility is that there ’ s no relationship at all between learning disabilities and hyperactivity, but that they are caused by a common underlying mechanism in ADHD. Now people have been wondering about this for a long time in humans, but in flies we can actually test this. And the way that we do this is to delve deeply into the mind of the fly and begin to untangle its circuitry using genetics. We take our dopamine receptor mutant flies and we genetically restore, or cure, the dopamine receptor by putting a good copy of the dopamine receptor gene back into the fly brain. But in each fly, we put it back only into certain neurons and not in others, and then we test each of these flies for their ability to learn and for hyperactivity. Remarkably, we find we can completely dissociate these two abnormalities. If we put a good copy of the dopamine receptor back in this elliptical structure called the central complex, the flies are no longer hyperactive, but they still can ’ t learn. On the other hand, if we put the receptor back in a different structure called the mushroom body, the learning deficit is rescued, the flies learn well, but they ’ re still hyperactive. What that tells us is that dopamine is not bathing the brain of these flies like soup. Rather, it ’ s acting to control two different functions on two different circuits, so the reason there are two things wrong with our dopamine receptor flies is that the same receptor is controlling two different functions in two different regions of the brain. Whether the same thing is true in ADHD in humans we don ’ t know, but these kinds of results should at least cause us to consider that possibility. So these results make me and my colleagues more convinced than ever that the brain is not a bag of chemical soup, and it ’ s a mistake to try to treat complex psychiatric disorders just by changing the flavor of the soup. What we need to do is to use our ingenuity and our scientific knowledge to try to design a new generation of treatments that are targeted to specific neurons and specific regions of the brain that are affected in particular psychiatric disorders. If we can do that, we may be able to cure these disorders without the unpleasant side effects, putting the oil back in our mental engines, just where it ’ s needed. Thank you very much.

Text Sample 949: NEG Dim 6 – 2011 – Score 1.00 – 2011/t002711.txt

The first thing I want to do is say thank you to all of you. The second thing I want to do is introduce my co-author and dear friend and co-teacher. Ken and I have been working together for almost 40 years. That ’ s Ken Sharpe over there. So there is among many people â certainly me and most of the people I talk to â a kind of collective dissatisfaction with the way things are working, with the way our institutions run. Our kids ’ teachers seem to be failing them. Our doctors don ’ t know who the hell we are, and they don ’ t have enough time for us. We certainly can ’ t trust the bankers, and we certainly can ’ t trust the brokers. They almost brought the entire financial system down. And even as we do our own work, all too often, we find ourselves having to choose between doing what we think is the right thing and doing the expected thing, or the required thing, or the profitable thing. So everywhere we look, pretty much across the board, we worry that the people we depend on don ’ t really have our interests at heart. Or if they do have our interests at heart, we worry that they don ’ t know us well enough to figure out what they need to do in order to allow us to secure those interests. They don ’ t understand us. They don ’ t have the time to get to know us. There are two kinds of responses that we make to this sort of general dissatisfaction. If things aren ’ t going right, the first response is : let ’ s make more rules, let ’ s set up a set of detailed procedures to make sure that people will do the right thing. Give teachers scripts to follow in the classroom, so even if they don ’ t know what they ’ re doing and don ’ t care about the welfare of our kids, as long as they follow the scripts, our kids will get educated. Give judges a list of mandatory sentences to impose for crimes, so that you don ’ t need to rely on judges using their judgment. Instead, all they have to do is look up on the list what kind of sentence goes with what kind of crime. Impose limits on what credit card companies can charge in interest and on what they can charge in fees. More and more rules to protect us against an indifferent, uncaring set of institutions we have to deal with. Or â or maybe and â in addition to rules, let ’ s see if we can come up with some really clever incentives so that, even if the people we deal with don ’ t particularly want to serve our interests, it is in their interest to serve our interest â the magic incentives that will get people to do the right thing even out of pure selfishness. So we offer teachers bonuses if the kids they teach score passing grades on these big test scores that are used to evaluate the quality of school systems. Rules and incentives â”sticks”and”carrots.”We passed a bunch of rules to regulate the financial industry in response to the recent collapse. There ’ s the Dodd-Frank Act, there ’ s the new Consumer Financial Protection Agency that is temporarily being headed through the backdoor by Elizabeth Warren. Maybe these rules will actually improve the way these financial services companies behave. We ’ ll see. In addition, we are struggling to find some way to create incentives for people in the financial services industry that will have them more interested in serving the long-term interests even of their own companies, rather than securing short-term profits. So if we find just the right incentives, they ’ ll do the right thing â as I said â selfishly, and if we come up with the right rules and regulations, they won ’ t drive us all over a cliff. And Ken [Sharpe] and I certainly know that you need to reign in the bankers. If there is a lesson to be learned from the financial collapse it is that. But what we believe, and what we argue in the book, is that there is no set of rules, no matter how detailed, no matter how specific, no matter how carefully monitored and enforced, there is no set of rules that will get us what we need. Why? Because bankers are smart people. And, like water, they will find cracks in any set of rules. You design a set of rules that will make sure that the particular reason why the financial system”almost-collapse”can ’ t happen again. It is naive beyond description to think that having blocked this source of financial collapse, you have blocked all possible sources of financial collapse. So it ’ s just a question of waiting for the next one and then marveling at how we could have been so stupid as not to protect ourselves against that. What we desperately need, beyond, or along with, better rules and reasonably smart incentives, is we need virtue. We need character. We need people who want to do the right thing. And in particular, the virtue that we need most of all is the virtue that Aristotle called”practical wisdom.”Practical wisdom is the moral will to do the right thing and the moral skill to figure out what the right thing is. So Aristotle was very interested in watching how the craftsmen around him worked. And he was impressed at how they would improvise novel solutions to novel problems â problems that they hadn ’ t anticipated. So one example is he sees these stonemasons working on the Isle of Lesbos, and they need to measure out round columns. Well if you think about it, it ’ s really hard to measure out round columns using a ruler. So what do they do? They fashion a novel solution to the problem. They created a ruler that bends, what we would call these days a tape measure â a flexible rule, a rule that bends. And Aristotle said,”Hah, they appreciated that sometimes to design rounded columns, you need to bend the rule.”And Aristotle said often in dealing with other people, we need to bend the rules. Dealing with other people demands a kind of flexibility that no set of rules can encompass. Wise people know when and how to bend the rules. Wise people know how to improvise. The way my co-author, Ken, and I talk about it, they are kind of like jazz musicians. The rules are like the notes on the page, and that gets you started, but then you dance around the notes on the page, coming up with just the right combination for this particular moment with this particular set of fellow players. So for Aristotle, the kind of rule-bending, rule exception-finding and improvisation that you see in skilled craftsmen is exactly what you need to be a skilled moral craftsman. And in interactions with people, almost all the time, it is this kind of flexibility that is required. A wise person knows when to bend the rules. A wise person knows when to improvise. And most important, a wise person does this improvising and rule- bending in the service of the right aims. If you are a rule-bender and an improviser mostly to serve yourself, what you get is ruthless manipulation of other people. So it matters that you do this wise practice in the service of others and not in the service of yourself. And so the will to do the right thing is just as important as the moral skill of improvisation and exception- finding. Together they comprise practical wisdom, which Aristotle thought was the master virtue. So I ’ ll give you an example of wise practice in action. It ’ s the case of Michael. Michael ’ s a young guy. He had a pretty low-wage job. He was supporting his wife and a child, and the child was going to parochial school. Then he lost his job. He panicked about being able to support his family. One night, he drank a little too much, and he robbed a cab driver â stole 50 dollars. He robbed him at gunpoint. It was a toy gun. He got caught. He got tried. He got convicted. The Pennsylvania sentencing guidelines required a minimum sentence for a crime like this of two years, 24 months. The judge on the case, Judge Lois Forer thought that this made no sense. He had never committed a crime before. He was a responsible husband and father. He had been faced with desperate circumstances. All this would do is wreck a family. And so she improvised a sentence â 11 months, and not only that, but release every day to go to work. Spend your night in jail, spend your day holding down a job. He did. He served out his sentence. He made restitution and found himself a new job. And the family was united. And it seemed on the road to some sort of a decent life â a happy ending to a story involving wise improvisation from a wise judge. But it turned out the prosecutor was not happy that Judge Forer ignored the sentencing guidelines and sort of invented her own, and so he appealed. And he asked for the mandatory minimum sentence for armed robbery. He did after all have a toy gun. The mandatory minimum sentence for armed robbery is five years. He won the appeal. Michael was sentenced to five years in prison. Judge Forer had to follow the law. And by the way, this appeal went through after he had finished serving his sentence, so he was out and working at a job and taking care of his family and he had to go back into jail. Judge Forer did what she was required to do, and then she quit the bench. And Michael disappeared. So that is an example, both of wisdom in practice and the subversion of wisdom by rules that are meant, of course, to make things better. Now consider Ms. Dewey. Ms. Dewey ’ s a teacher in a Texas elementary school. She found herself listening to a consultant one day who was trying to help teachers boost the test scores of the kids, so that the school would reach the elite category in percentage of kids passing big tests. All these schools in Texas compete with one another to achieve these milestones, and there are bonuses and various other treats that come if you beat the other schools. So here was the consultant ’ s advice : first, don ’ t waste your time on kids who are going to pass the test no matter what you do. Second, don ’ t waste your time on kids who can ’ t pass the test no matter what you do. Third, don ’ t waste your time on kids who moved into the district too late for their scores to be counted. Focus all of your time and attention on the kids who are on the bubble, the so-called”bubble kids”â kids where your intervention can get them just maybe over the line from failing to passing. So Ms. Dewey heard this, and she shook her head in despair while fellow teachers were sort of cheering each other on and nodding approvingly. It ’ s like they were about to go play a football game. For Ms. Dewey, this isn ’ t why she became a teacher. Now Ken and I are not naive, and we understand that you need to have rules. You need to have incentives. People have to make a living. But the problem with relying on rules and incentives is that they demoralize professional activity, and they demoralize professional activity in two senses. First, they demoralize the people who are engaged in the activity. Judge Forer quits, and Ms. Dewey in completely disheartened. And second, they demoralize the activity itself. The very practice is demoralized, and the practitioners are demoralized. It creates people â when you manipulate incentives to get people to do the right thing â it creates people who are addicted to incentives. That is to say, it creates people who only do things for incentives. Now the striking thing about this is that psychologists have known this for 30 years. Psychologists have known about the negative consequences of incentivizing everything for 30 years. We know that if you reward kids for drawing pictures, they stop caring about the drawing and care only about the reward. If you reward kids for reading books, they stop caring about what ’ s in the books and only care about how long they are. If you reward teachers for kids ’ test scores, they stop caring about educating and only care about test preparation. If you were to reward doctors for doing more procedures â which is the current system â they would do more. If instead you reward doctors for doing fewer procedures, they will do fewer. What we want, of course, is doctors who do just the right amount of procedures and do the right amount for the right reason â namely, to serve the welfare of their patients. Psychologists have known this for decades, and it ’ s time for policymakers to start paying attention and listen to psychologists a little bit, instead of economists. And it doesn ’ t have to be this way. We think, Ken and I, that there are real sources of hope. We identify one set of people in all of these practices who we call canny outlaws. These are people who, being forced to operate in a system that demands rule-following and creates incentives, find away around the rules, find a way to subvert the rules. So there are teachers who have these scripts to follow, and they know that if they follow these scripts, the kids will learn nothing. And so what they do is they follow the scripts, but they follow the scripts at double-time and squirrel away little bits of extra time during which they teach in the way that they actually know is effective. So these are little ordinary, everyday heroes, and they ’ re incredibly admirable, but there ’ s no way that they can sustain this kind of activity in the face of a system that either roots them out or grinds them down. So canny outlaws are better than nothing, but it ’ s hard to imagine any canny outlaw sustaining that for an indefinite period of time. More hopeful are people we call system-changers. These are people who are looking not to dodge the system ’ s rules and regulations, but to transform the system, and we talk about several. One in particular is a judge named Robert Russell. And one day he was faced with the case of Gary Pettengill. Pettengill was a 23-year-old vet who had planned to make the army a career, but then he got a severe back injury in Iraq, and that forced him to take a medical discharge. He was married, he had a third kid on the way, he suffered from PTSD, in addition to the bad back, and recurrent nightmares, and he had started using marijuana to ease some of the symptoms. He was only able to get part-time work because of his back, and so he was unable to earn enough to put food on the table and take care of his family. So he started selling marijuana. He was busted in a drug sweep. His family was kicked out of their apartment, and the welfare system was threatening to take away his kids. Under normal sentencing procedures, Judge Russell would have had little choice but to sentence Pettengill to serious jail-time as a drug felon. But Judge Russell did have an alternative. And that ’ s because he was in a special court. He was in a court called the Veterans ’ Court. In the Veterans ’ Court â this was the first of its kind in the United States. Judge Russell created the Veterans ’ Court. It was a court only for veterans who had broken the law. And he had created it exactly because mandatory sentencing laws were taking the judgment out of judging. No one wanted non-violent offenders â and especially non-violent offenders who were veterans to boot â to be thrown into prison. They wanted to do something about what we all know, namely the revolving door of the criminal justice system. And what the Veterans ’ Court did, was it treated each criminal as an individual, tried to get inside their problems, tried to fashion responses to their crimes that helped them to rehabilitate themselves, and didn ’ t forget about them once the judgment was made. Stayed with them, followed up on them, made sure that they were sticking to whatever plan had been jointly developed to get them over the hump. There are now 22 cities that have Veterans ’ Courts like this. Why has the idea spread? Well, one reason is that Judge Russell has now seen 108 vets in his Veterans ’ Court as of February of this year, and out of 108, guess how many have gone back through the revolving door of justice into prison. None. None. Anyone would glom onto a criminal justice system that has this kind of a record. So here ’ s is a system-changer, and it seems to be catching. There ’ s a banker who created a for-profit community bank that encouraged bankers â□□ I know this is hard to believe â□□ encouraged bankers who worked there to do well by doing good for their low-income clients. The bank helped finance the rebuilding of what was otherwise a dying community. Though their loan recipients were high-risk by ordinary standards, the default rate was extremely low. The bank was profitable. The bankers stayed with their loan recipients. They didn ’ t make loans and then sell the loans. They serviced the loans. They made sure that their loan recipients were staying up with their payments. Banking hasn ’ t always been the way we read about it now in the newspapers. Even Goldman Sachs once used to serve clients, before it turned into an institution that serves only itself. Banking wasn ’ t always this way, and it doesn ’ t have to be this way. So there are examples like this in medicine â□□ doctors at Harvard who are trying to transform medical education, so that you don ’ t get a kind of ethical erosion and loss of empathy, which characterizes most medical students in the course of their medical training. And the way they do it is to give third-year medical students patients who they follow for an entire year. So the patients are not organ systems, and they ’ re not diseases ; they ’ re people, people with lives. And in order to be an effective doctor, you need to treat people who have lives and not just disease. In addition to which there ’ s an enormous amount of back and forth, mentoring of one student by another, of all the students by the doctors, and the result is a generation â□□ we hope â□□ of doctors who do have time for the people they treat. We ’ ll see. So there are lots of examples like this that we talk about. Each of them shows that it is possible to build on and nurture character and keep a profession true to its proper mission â□□ what Aristotle would have called its proper telos. And Ken and I believe that this is what practitioners actually want. People want to be allowed to be virtuous. They want to have permission to do the right thing. They don ’ t want to feel like they need to take a shower to get the moral grime off their bodies everyday when they come home from work. Aristotle thought that practical wisdom was the key to happiness, and he was right. There ’ s now a lot of research being done in psychology on what makes people happy, and the two things that jump out in study after study â□□ I know this will come as a shock to all of you â□□ the two things that matter most to happiness are love and work. Love : managing successfully relations with the people who are close to you and with the communities of which you are a part. Work : engaging in activities that are meaningful and satisfying. If you have that, good close relations with other people, work that ’ s meaningful and fulfilling, you don ’ t much need anything else. Well, to love well and to work well, you need wisdom. Rules and incentives don ’ t tell you how to be a good friend, how to be a good parent, how to be a good spouse, or how to be a good doctor or a good lawyer or a good teacher. Rules and incentives are no substitutes for wisdom. Indeed, we argue, there is no substitute for wisdom. And so practical wisdom does not require heroic acts of self-sacrifice on the part of practitioners. In giving us the will and the skill to do the right thing â□□ to do right by others â□□ practical wisdom also gives us the will and the skill to do right by ourselves. Thanks.

Text Sample 950: NEG Dim 6 – 2011 – Score 1.00 – 2011/t002665.txt

I ’ m going to have a pretty simple idea that I ’ m just going to tell you over and over until I get you to believe it, and that is all of us are makers. I really believe that. All of us are makers. We ’ re born makers. We have this ability to make things, to grasp things with our hands. We use words like”grasp”metaphorically to also think about understanding things. We don ’ t just live, but we make. We create things. Well I ’ m going to show you a group of makers from Maker Faire and various places. It doesn ’ t come out particularly well, but that ’ s a particularly tall bicycle. It ’ s a scraper bike ; it ’ s called — from Oakland. And this is a particularly small scooter for a gentleman of this size. But he ’ s trying to power it, or motorize it, with a drill. And the question he had is,”Can I do it? Can it be done?”Apparently it can. So makers are enthusiasts ; they ’ re amateurs ; they ’ re people who love doing what they do. They don ’ t always even know why they ’ re doing it. We have begun organizing makers at our Maker Faire. There was one held in Detroit here last summer, and it will be held again next summer, at the Henry Ford. But we hold them in San Francisco — — and in New York. And it ’ s a fabulous event to just meet and talk to these people who make things and are there to just show them to you and talk about them and have a great conversation. Guy : I might get one of those. Dale Dougherty : These are electric muffins. Guy : Where did you guys get those? Muffin : Will you glide with us? DD : I know Ford has new electric vehicles coming out. We got there first. Lady : Will you glide with us? DD : This is something I call”swinging in the rain.”And you can barely see it, but it ’ s — a controller at top cycles the water to fall just before and after you pass through the bottom of the arc. So imagine a kid :”Am I going to get wet? Am I going to get wet? No, I didn ’ t get wet. Am I going to get wet? Am I going to get wet?”That ’ s the experience of a clever ride. And of course, we have fashion. People are remaking things into fashion. I don ’ t know if this is called a basket-bra, but it ought to be something like that. We have art students getting together, taking old radiator parts and doing an iron-pour to make something new out of it. They did that in the summer, and it was very warm. Now this one takes a little bit of explaining. You know what those are, right? Billy-Bob, or Billy Bass, or something like that. Now the background is — the guy who did this is a physicist. And here he ’ ll explain a little bit about what it does. Richard Carter : I ’ m Richard Carter, and this is the Sashimi Tabernacle Choir. Choir : When you hold me in your arms DD : This is all computer-controlled in an old Volvo. Choir : I ’ m hooked on a feelin ’ I ’ m high on believin ’ That you ’ re in love with me DD : So Richard came up from Houston last year to visit us in Detroit here and show the wonderful Sashimi Tabernacle Choir. So, are you a maker? How many people here would say you ’ re a maker, if you raise your hand? That ’ s a pretty good — but there ’ s some of you out there that won ’ t admit that you ’ re makers. And again, think about it. You ’ re makers of food ; you ’ re makers of shelter ; you ’ re makers of lots of different things, and partly what interests me today is you ’ re makers of your own world, and particularly the role that technology has in your life. You ’ re really a driver or a passenger — to use a Volkswagen phrase. Makers are in control. That ’ s what fascinates them. That ’ s why they do what they do. They want to figure out how things work ; they want to get access to it ; and they want to control it. They want to use it to their own purpose. Makers today, to some degree, are out on the edge. They ’ re not mainstream. They ’ re a little bit radical. They ’ re a bit subversive in what they do. But at one time, it was fairly commonplace to think of yourself as a maker. It was not something you ’ d even remark upon.

And I found this old video. And I ’ ll tell you more about it, but just... Narrator : Of all things Americans are, we are makers. With our strengths and our minds and spirit, we gather, we form, and we fashion. Makers and shapers and put-it-togetherers. DD : So it goes on to show you people making things out of wood, a grandfather making a ship in a bottle, a woman making a pie — somewhat standard fare of the day. But it was a sense of pride that we made things, that the world around us was made by us. It didn ’ t just exist. We made it, and we were connected to it that way. And I think that ’ s tremendously important. Now I ’ m going to tell you one funny thing about this. This particular reel — it ’ s an industrial video — but it was shown in drive-in theaters in 1961 — in the Detroit area, in fact — and it preceded Alfred Hitchcock ’ s”Psycho.”So I like to think there was something going on there of the new generation of makers coming out of this, plus”Psycho.”This is Andrew Archer. I met Andrew at one of our community meetings putting together Maker Faire. Andrew had moved to Detroit from Duluth, Minnesota. And I talked to his mom, and I ended up doing a story on him for a magazine called Kidrobot. He ’ s just a kid that grew up playing with tools instead of toys. He liked to take things apart. His mother gave him a part of the garage, and he collected things from yard sales, and he made stuff. And then he didn ’ t particularly like school that much, but he got involved in robotics competitions, and he realized he had a talent, and, more importantly, he had a real passion for it. And he began building robots. And when I sat down next to him, he was telling me about a company he formed, and he was building some robots for automobile factories to move things around on the factory floor. And that ’ s why he moved to Michigan. But he also moved here to meet other people doing what he ’ s doing. And this kind of gets to this important idea today. This is Jeff and Bilal and several others here in a hackerspace. And there ’ s about three hackerspaces or more in Detroit. And there ’ s probably even some new ones since I ’ ve been here last. But these are like clubs — they ’ re sharing tools, sharing space, sharing expertise in what to make. And so it ’ s a very interesting phenomenon that ’ s going across the world. But essentially these are people that are playing with technology. Let me say that again : playing. They don ’ t necessarily know what they ’ re doing or why they ’ re doing it. They ’ re playing to discover what the technology can do, and probably to discover what they can do themselves, what their own capabilities are. Now the other thing that I think is taking off, another reason making is taking off today, is there ’ s some great new tools out there. And you can ’ t see this very well on the screen, but Arduino — Arduino is an open-source hardware platform. It ’ s a micro-controller. If you don ’ t know what those are, they ’ re just the”brains.”So they ’ re the brains of maker projects, and here ’ s an example of one. And I don ’ t know if you can see it that well, but that ’ s a mailbox — so an ordinary mailbox and an Arduino. So you figure out how to program this, and you put this in your mailbox. And when someone opens your mailbox, you get a notification, an alert message goes to your iPhone. Now that could be a dog door, it could be someone going somewhere where they shouldn ’ t, like a little brother into a little sister ’ s room. There ’ s all kinds of different things that you can imagine for that. Now here ’ s something — a 3D printer. That ’ s another tool that ’ s really taken off — really, really interesting. This is Makerbot. And there are industrial versions of this — about 20, 000 dollars. These guys came up with a kit version for 750 dollars, and that means that hobbyists and ordinary folks can get a hold of this and begin playing with 3D printers. Now they don ’ t know what they want to do with it, but they ’ re going to figure it out. They will only figure it out by getting their hands on it and playing with it. One of the coolest things is, Makerbot sent out an upgrade, some new brackets for the box. Well you printed out the brackets and then replaced the old brackets with the new ones. Isn ’ t that cool? So makers harvest technology from all the places around us. This is a radar speed detector that was developed from a Hot Wheels toy. And they do interesting things. They ’ re really creating new areas and exploring areas that you might only think — the military is doing drones — well, there is a whole community of people building autonomous airplanes, or vehicles — something that you could program to fly on its own, without a stick or anything, to figure out what path it ’ s going. Fascinating work they ’ re doing. We just had an issue on space exploration, DIY space exploration. This is probably the best time in the history of mankind to love space. You could build your own satellite and get it into space for like 8, 000 dollars. Think how much money and how many years it took NASA to get satellites into space. In fact, these guys actually work for NASA, and they ’ re trying to pioneer using off-the- shelf components, cheap things that aren ’ t specialized that they can combine and send up into space. Makers are a source of innovation, and I think it relates back to something like the birth of the personal computer industry. This is Steve Wozniak. Where does he learn about computers? It ’ s the Homebrew Computer Club — just like a hackerspace. And he says,”I could go there all day long and talk to people and share ideas for free.”Well he did a little bit better than free. But it ’ s important to understand that a lot of the origins of our industries — even like Henry Ford — come from this idea of playing and figuring things out in groups. Well, if I haven ’ t convinced you that you ’ re a maker, I hope I could convince you that our next generation should be makers, that kids are particularly interested in this, in this ability to control the physical world and be able to use things like micro-controllers and build robots. And we ’ ve got to get this into schools, or into communities in many, many ways — the ability to tinker, to shape and reshape the world around us. There ’ s a great opportunity today — and that ’ s what I really care about the most. An the answer to the question : what will America make? It ’ s more makers. Thank you very much.

Text Sample 951: NEG Dim 6 – 2011 – Score 1.00 – 2011/t002660.txt

The maxim,”Know thyself”has been around since the ancient Greeks. Some attribute this golden world knowledge to Plato, others to Pythagoras. But the truth is it doesn ’ t really matter which sage said it first, because it ’ s still sage advice, even today.”Know thyself.”It ’ s pithy almost to the point of being meaningless, but it rings familiar and true, doesn ’ t it?”Know thyself.”I understand this timeless dictum as a statement about the problems, or more exactly, the confusions, of consciousness. I ’ ve always been fascinated with knowing the self. This fascination led me to submerge myself in art, study neuroscience, and later, to become a psychotherapist. Today I combine all my passions as the CEO of InteraXon, a thought-controlled computing company. My goal, quite simply, is to help people become more in tune with themselves. I take it from this little dictum,”Know thyself.”If you think about it, this imperative is kind of the defining characteristic of our species, isn ’ t it? I mean, it ’ s self-awareness that separates Homo sapiens from earlier instances of our mankind. Today we ’ re often too busy tending to our iPhones and iPods to really stop and get to know ourselves. Under the deluge of minute-to-minute text conversations, e-mails, relentless exchange of media channels and passwords and apps and reminders and Tweets and tags, we lose sight of what all this fuss is supposed to be about in the first place : Ourselves. Much of the time we ’ re transfixed by all of the ways we can reflect ourselves out into the world. And we can barely find the time to reflect deeply back in on our own selves. We ’ ve cluttered ourselves up with all this. And we feel like we have to get far, far away to a secluded retreat, leaving it all behind. So we go far away to the top of a mountain, assuming that perching ourselves on a piece is bound to give us the respite we need to sort the clutter, the chaotic everyday, and find ourselves again. But on that mountain where we gain that beautiful peace of mind, what are we really achieving? It ’ s really only a successful escape. Think of the term we use,”Retreat.”This is the term that armies use when they ’ ve lost a battle. It means we ’ ve got to get out of here. Is this how we feel about the pressures of our world, that in order to get inside ourselves, you have to run for the hills? And the problem with escaping your day-to-day life is that you have to come home, eventually. So when you think about it, we ’ re almost like a tourist visiting ourselves over there. And eventually, that vacation ’ s got to come to an end. So my question to you is, can we find ways to know ourselves without the escape? Can we redefine our relationship with the technologized world in order to have the heightened sense of self-awareness that we seek? Can we live here and now in our wired web and still follow those ancient instructions,”Know thyself?”I say the answer is yes. And I ’ m here today to share a new way that we ’ re working with technology to this end, to get familiar with our inner self like never before — humanizing technology and furthering that age-old quest of ours to more fully know the self. It ’ s called thought-controlled computing. You may or may not have noticed that I ’ m wearing a tiny electrode on my forehead. This is actually a brainwave sensor that ’ s reading the electrical activity of my brain as I give this talk. These brainwaves are being analyzed and we can see them as a graph. Let me show you what it looks like. That blue line there is my brainwave. It ’ s the direct signal being recorded from my head, rendered in real time. The green and red bars show that same signal displayed by frequency, with lower frequencies here and higher frequencies up here. You ’ re actually looking inside my head as I speak. These graphs are compelling, they ’ re undulating, but from a human ’ s perspective, they ’ re actually not very useful. That ’ s why we ’ ve spent a lot of time thinking about how to make this data meaningful to the people who use it. For instance, what if I could use this data to find out how relaxed I am at any moment? Or what if I can take that information and put it into an organic shape up on the screen? The shape on the right over here has become an indicator of what ’ s going on in my head. The more relaxed I am, the more the energy ’ s going to fall through it. I may also be interested in knowing how focused I am, so I can put my level of attention into the circuit board on the other side. And the more focused my brain is, the more the circuit board is going to surge with energy. Ordinarily, I would have no way of knowing how focused or relaxed I was in any tangible way. As we know, our feelings about how we ’ re feeling are notoriously unreliable. We ’ ve all had stress creep up on us without even noticing it until we lost it on someone who didn ’ t deserve it, and then we realize that we probably should have checked in with ourselves a little earlier. This new awareness opens up vast possibilities for applications that help improve our lives and ourselves. We ’ re trying to create technology that uses the insights to make our work more efficient, our breaks more relaxing and our connections deeper and more fulfilling than ever. I ’ m going to share some of these visions with you in a bit, but first I want to take a look at how we got here. By the way, feel free to check in on my head at any time. My team at InteraXon and I have been developing thought-controlled application for almost a decade now. In the first phase of development, we were really enthused by all the things we could control with our mind. We were making things activate, light up and work just by thinking. We were transcending the space between the mind and the device. We brought to life a vast array of prototypes and products that you could control with your mind, like thought- controlled home appliances or slot-car games or video games or a levitating chair. We created technology and applications that engaged people ’ s imaginations, and it was really exciting. And then we were asked to do something really big for the Olympics. We were invited to create a massive installation at the Vancouver 2010 winter Olympics, were used in Vancouver, got to control the lighting on the CN Tower, the Canadian Parliament buildings and Niagara Falls from all the way across the country using their minds. Over 17 days at the Olympics, 7, 000 visitors from all over the world actually got to individually control the light from the CN Tower, parliament and Niagara in real time with their minds from across the country, 3, 000 km away. So controlling stuff with your mind is pretty cool. But we ’ re always interested in multitiered levels of human interaction. And so we began looking into inventing thought-controlled applications in a more complex frame than just control. And that was responsiveness. We realized that we had a system that allowed technology to know something about you. And it could join into the relationship with you. We created the responsive room where the lights, music and blinds adjusted to your state. They followed these little shifts in your mental activity. So as you settled into relaxation at the end of a hard day, on the couch in our office, the music would mellow with you. When you read, the desk lamp would get brighter. If you nod off, the system would know, dimming to darkness as you do. We then realized that if technology could know something about you and use it to help you, there ’ s an even more valuable application than that. That you could know something about yourself. We could know sides of ourselves that were all but invisible and come to see things that were previously hidden. Let me show you an example of what I ’ m talking about here. Here ’ s an application that I created for the iPad. So the goal of the original game Zen Bound is to wrap a rope around a wooden form. So you use it with your headset. The headset connects wirelessly to an iPad or a smartphone. In that headset, you have fabric sensors on your forehead and above the ear. In the original Zen Bound game, you play it by scrolling your fingers over the pad. In the game that we created, of course, you control the wooden form that ’ s on the screen there with your mind. As you focus on the wooden form, it rotates. The more you focus, the faster the rotation. This is for real. This is not a fake. What ’ s really interesting to me though is at the end of the game, you get stats and feedback about how you did. You have graphs and charts that tell you how your brain was doing — not just how much rope you used or what your high score is, but what was going on inside of your mind. And this is valuable feedback that we can use to understand what ’ s going on inside of ourselves. I like to call this”intra-active.”Normally, we think about technology as interactive. This technology is intra-active. It understands what ’ s inside of you and builds a sort of responsive relationship between you and your technology so that you can use this information to move you forward. So you can use this information to understand you in a responsive loop. At InteraXon — intra-active technology is one of our really defining mandates. It ’ s how we understand the world inside and reflect it outside into this tight loop. For example, thought-controlled computing can teach children with ADD how to improve their focus. With ADD, children have a low proportion of beta waves for focus states and a high proportion of theta states. So you can create applications that reward focused brain states. So you can imagine kids playing video games with their brain waves and improving their ADD symptoms as they do it. This can be as effective as Ritalin. Perhaps even more importantly, thought-controlled computing can give children with ADD insights into their own fluctuating mental states, so they can better understand themselves and their learning needs. The way these children will be able to use their new awareness to improve themselves will upend many of the damaging and widespread social stigmas that people who are diagnosed as different are challenged with. We can peer inside our heads and interact with what was once locked away from us, what once mystified and separated us. Brainwave technology can understand us, anticipate our emotions and find the best solutions for our needs. Imagine this collected awareness of the individual computed and reflected across an entire lifespan. Imagine the insights that you can gain from this kind of second sight. It would be like plugging into your own personal Google. On the subject of Google, today you can search and tag images based on the thoughts and feelings you had while you watched them. You can tag pictures of baby animals as happy, or whatever baby animals are to you, and then you can search that database, navigating with your feelings, rather than the keywords that just hint at them. Or you could tag Facebook photos with the emotions that you had associated with those memories and then instantly prioritize the streams that catch your attention, just like this. Humanizing technology is about taking what ' s already natural about the human-tech experience and building technology seamlessly in tandem with it. As it aligns with our human behaviors, it can allow us to make better sense of what we do and, more importantly, why. Creating a big picture out of all the important little details that make up who we are. With humanized technology we can monitor the quality of your sleep cycles. When our productivity starts to slacken, we can go back to that data and see how we can make more effective balance between work and play. Do you know what causes fatigue in you or what brings out your energetic self, what triggers cause you to be depressed or what fun things are going to bring you out of that funk? Imagine if you had access to data that allowed you to rank on a scale of overall happiness which people in your life made you the happiest, or what activities brought you joy. Would you make more time for those people? Would you prioritize? Would you get a divorce? What thought-controlled computing can allow you to do is build colorful layered pictures of our lives. And with this, we can get the skinny on our psychological happenings and build a story of our behaviors over time. We can begin to see the underlying narratives that propel us forward and tell us about what ' s going on. And from this, we can learn how to change the plot, the outcome and the character of our personal stories. Two millennia ago, those Greeks had some powerful insights. They knew that a fundamental piece falls into place when you start to live out their little phrase, when you come into contact with yourself. They understood the power of human narrative and the value that we place on humans as changing, evolving and growing. But they understood something more fundamental — the sheer joy in discovery, the delight and fascination that we get from the world and being ourselves in it ; the richness that we get from seeing, feeling and knowing the lives that we are. My mom ' s an artist, and as a child, I ' d often see her bring things to life with the stroke of a brush. One moment, it was all white space, pure possibility. The next, it was alive with her colorful ideas and expressions. As I sat easel- side, watching her transform canvas after canvas, I learned that you could create your own world. I learned that our own inner worlds — our ideas, emotions and imaginations — were, in fact, not bound by our brains and bodies. If you could think it, if you could discover it, you could bring it to life. To me, thought-controlled computing is as simple and powerful as a paintbrush — one more tool to unlock and enliven the hidden worlds within us. I look forward to the day that I can sit beside you, easel-side, watching the world that we can create with our new toolboxes and the discoveries that we can make about ourselves. Thank you.

Text Sample 952: NEG Dim 6 – 2009 – Score 1.00 – 2009/t002995.txt

Today I ’ m going to talk to you about the problem of other minds. And the problem I ’ m going to talk about is not the familiar one from philosophy, which is,”How can we know whether other people have minds?”That is, maybe you have a mind, and everyone else is just a really convincing robot. So that ’ s a problem in philosophy, but for today ’ s purposes I ’ m going to assume that many people in this audience have a mind, and that I don ’ t have to worry about this. There is a second problem that is maybe even more familiar to us as parents and teachers and spouses and novelists, which is,”Why is it so hard to know what somebody else wants or believes?”Or perhaps, more relevantly,”Why is it so hard to change what somebody else wants or believes?”I think novelists put this best. Like Philip Roth, who said,”And yet, what are we to do about this terribly significant business of other people? So ill equipped are we all, to envision one another ’ s interior workings and invisible aims.”So as a teacher and as a spouse, this is, of course, a problem I confront every day. But as a scientist, I ’ m interested in a different problem of other minds, and that is the one I ’ m going to introduce to you today. And that problem is,”How is it so easy to know other minds?”So to start with an illustration, you need almost no information, one snapshot of a stranger, to guess what this woman is thinking, or what this man is. And put another way, the crux of the problem is the machine that we use for thinking about other minds, our brain, is made up of pieces, brain cells, that we share with all other animals, with monkeys and mice and even sea slugs. And yet, you put them together in a particular network, and what you get is the capacity to write Romeo and Juliet. Or to say, as Alan Greenspan did,”I know you think you understand what you thought I said, but I ’ m not sure you realize that what you heard is not what I meant.”So, the job of my field of cognitive neuroscience is to stand with these ideas, one in each hand. And to try to understand how you can put together simple units, simple messages over space and time, in a network, and get this amazing human capacity to think about minds. So I ’ m going to tell you three things about this today. Obviously the whole project here is huge. And I ’ m going to tell you just our first few steps about the discovery of a special brain region for thinking about other people ’ s thoughts. Some observations on the slow development of this system as we learn how to do this difficult job. And then finally, to show that some of the differences between people, in how we judge others, can be explained by differences in this brain system. So first, the first thing I want to tell you is that there is a brain region in the human brain, in your brains, whose job it is to think about other people ’ s thoughts. This is a picture of it. It ’ s called the Right Temporo-Parietal Junction. It ’ s above and behind your right ear. And this is the brain region you used when you saw the pictures I showed you, or when you read Romeo and Juliet or when you tried to understand Alan Greenspan. And you don ’ t use it for solving any other kinds of logical problems. So this brain region is called the Right TPJ. And this picture shows the average activation in a group of what we call typical human adults. They ’ re MIT undergraduates. The second thing I want to say about this brain system is that although we human adults are really good at understanding other minds, we weren ’ t always that way. It takes children a long time to break into the system. I ’ m going to show you a little bit of that long, extended process. The first thing I ’ m going to show you is a change between age three and five, as kids learn to understand that somebody else can have beliefs that are different from their own. So I ’ m going to show you a five-year-old who is getting a standard kind of puzzle that we call the false belief task. Rebecca Saxe : This is the first pirate. His name is Ivan. And you know what pirates really like? Child : What? RS : Pirates really like cheese sandwiches. Child : Cheese? I love cheese! RS : Yeah. So Ivan has this cheese sandwich, and he says,”Yum yum yum yum yum! I really love cheese sandwiches.”And Ivan puts his sandwich over here, on top of the pirate chest. And Ivan says,”You know what? I need a drink with my lunch.”And so Ivan goes to get a drink. And while Ivan is away the wind comes, and it blows the sandwich down onto the grass. And now, here comes the other pirate. This pirate is called Joshua. And Joshua also really loves cheese sandwiches. So Joshua has a cheese sandwich and he says,”Yum yum yum yum yum! I love cheese sandwiches.”And he puts his cheese sandwich over here on top of the pirate chest. Child : So, that one is his. RS : That one is Joshua ’ s. That ’ s right. Child : And then his went on the ground. RS : That ’ s exactly right. Child : So he won ’ t know which one is his. RS : Oh. So now Joshua goes off to get a drink. Ivan comes back and he says,”I want my cheese sandwich.”So which one do you think Ivan is going to take? Child : I think he is going to take that one. RS : Yeah, you think he ’ s going to take that one? All right. Let ’ s see. Oh yeah, you were right. He took that one. So that ’ s a five-year-old who clearly understands that other people can have false beliefs and what the consequences are for their actions. Now I ’ m going to show you a three-year-old who got the same puzzle. RS : And Ivan says,”I want my cheese sandwich.”Which sandwich is he going to take? Do you think he ’ s going to take that one? Let ’ s see what happens. Let ’ s see what he does. Here comes Ivan. And he says,”I want my cheese sandwich.”And he takes this one. Uh-oh. Why did he take that one? Child : His was on the grass. So the three-year-old does two things differently. First, he predicts Ivan will take the sandwich that ’ s really his. And second, when he sees Ivan taking the sandwich where he left his, where we would say he ’ s taking that one because he thinks it ’ s his, the three-year-old comes up with another explanation : He ’ s not taking his own sandwich because he doesn ’ t want it, because now it ’ s dirty, on the ground. So that ’ s why he ’ s taking the other sandwich. Now of course, development doesn ’ t end at five. And we can see the continuation of this process of learning to think about other people ’ s thoughts by upping the ante and asking children now, not for an action prediction, but for a moral judgment. So first I ’ m going to show you the three-year-old again. RS. : So is Ivan being mean and naughty for taking Joshua ’ s sandwich? Child : Yeah. RS : Should Ivan get in trouble for taking Joshua ’ s sandwich? Child : Yeah. So it ’ s maybe not surprising he thinks it was mean of Ivan to take Joshua ’ s sandwich, since he thinks Ivan only took Joshua ’ s sandwich to avoid having to eat his own dirty sandwich. But now I ’ m going to show you the five-year-old. Remember the five-year-old completely understood why Ivan took Joshua ’ s sandwich. RS : Was Ivan being mean and naughty for taking Joshua ’ s sandwich? Child : Um, yeah. And so, it is not until age seven that we get what looks more like an adult response. RS : Should Ivan get in trouble for taking Joshua ’ s sandwich? Child : No, because the wind should get in trouble. He says the wind should get in trouble for switching the sandwiches. And now what we ’ ve started to do in my lab is to put children into the brain scanner and ask what ’ s going on in their brain as they develop this ability to think about other people ’ s thoughts. So the first thing is that in children we see this same brain region, the Right TPJ, being used while children are thinking about other people. But it ’ s not quite like the adult brain. So whereas in the adults, as I told you, this brain region is almost completely specialized — it does almost nothing else except for thinking about other people ’ s thoughts — in children it ’ s much less so, when they are age five to eight, the age range of the children I just showed you. And actually if we even look at eight to 11-year-olds, getting into early adolescence, they still don ’ t have quite an adult-like brain region. And so, what we can see is that over the course of childhood and even into adolescence, both the cognitive system, our mind ’ s ability to think about other minds, and the brain system that supports it are continuing, slowly, to develop. But of course, as you ’ re probably aware, even in adulthood, people differ from one another in how good they are at thinking of other minds, how often they do it and how accurately. And so what we wanted to know was, could differences among adults in how they think about other people ’ s thoughts be explained in terms of differences in this brain region? So, the first thing that we did is we gave adults a version of the pirate problem that we gave to the kids. And I ’ m going to give that to you now. So Grace and her friend are on a tour of a chemical factory, and they take a break for coffee. And Grace ’ s friend asks for some sugar in her coffee. Grace goes to make the coffee and finds by the coffee a pot containing a white powder, which is sugar. But the powder is labeled”Deadly Poison,”so Grace thinks that the powder is a deadly poison. And she puts it in her friend ’ s coffee. And her friend drinks the coffee, and is fine. How many people think it was morally permissible for Grace to put the powder in the coffee? Okay. Good. So we ask people, how much should Grace be blamed in this case, which we call a failed attempt to harm? And we can compare that to another case, where everything in the real world is the same. The powder is still sugar, but what ’ s different is what Grace thinks. Now she thinks the powder is sugar. And perhaps unsurprisingly, if Grace thinks the powder is sugar and puts it in her friend ’ s coffee, people say she deserves no blame at all. Whereas if she thinks the powder was poison, even though it ’ s really sugar, now people say she deserves a lot of blame, even though what happened in the real world was exactly the same. And in fact, they say she deserves more blame in this case, the failed attempt to harm, than in another case, which we call an accident. Where Grace thought the powder was sugar, because it was labeled”sugar”and by the coffee machine, but actually the powder was poison. So even though when the powder was poison, the friend drank the coffee and died, people say Grace deserves less blame in that case, when she innocently thought it was sugar, than in the other case, where she thought it was poison and no harm occurred. People, though, disagree a little bit about exactly how much blame Grace should get in the accident case. Some people think she should deserve more blame, and other people less. And what I ’ m going to show you is what happened when we look inside the brains of people while they ’ re making that judgment. So what I ’ m showing you, from left to right, is how much activity there was in this brain region, and from top to bottom, how much blame people said that Grace deserved. And what you can see is, on the left when there was very little activity in this brain region, people paid little attention to her innocent belief and said she deserved a lot of blame for the accident. Whereas on the right, where there was a lot of activity, people paid a lot more attention to her innocent belief, and said she deserved a lot less blame for causing the accident. So that ’ s good, but of course what we ’ d rather is have a way to interfere with function in this brain region, and see if we could change people ’ s moral judgment. And we do have such a tool. It ’ s called Trans-Cranial Magnetic Stimulation, or TMS. This is a tool that lets us pass a magnetic pulse through somebody ’ s skull, into a small region of their brain, and temporarily disorganize the function of the neurons in that region. So I ’ m going to show you a demo of this. First, I ’ m going to show you that this is a magnetic pulse. I ’ m going to show you what happens when you put a quarter on the machine. When you hear clicks, we ’ re turning the machine on. So now I ’ m going to apply that same pulse to my brain, to the part of my brain that controls my hand. So there is no physical force, just a magnetic pulse. Woman : Ready, Rebecca? RS : Yes. Okay, so it causes a small involuntary contraction in my hand by putting a magnetic pulse in my brain. And we can use that same pulse, now applied to the RTPJ, to ask if we can change people ’ s moral judgments. So these are the judgments I showed you before, people ’ s normal moral judgments. And then we can apply TMS to the RTPJ and ask how people ’ s judgments change. And the first thing is, people can still do this task overall. So their judgments of the case when everything was fine remain the same. They say she deserves no blame. But in the case of a failed attempt to harm, where Grace thought that it was poison, although it was really sugar, people now say it was more okay, she deserves less blame for putting the powder in the coffee. And in the case of the accident, where she thought that it was sugar, but it was really poison and so she caused a death, people say that it was less okay, she deserves more blame. So what I ’ ve told you today is that people come, actually, especially well equipped to think about other people ’ s thoughts. We have a special brain system that lets us think about what other people are thinking. This system takes a long time to develop, slowly throughout the course of childhood and into early adolescence. And even in adulthood, differences in this brain region can explain differences among adults in how we think about and judge other people. But I want to give the last word back to the novelists, and to Philip Roth, who ended by saying,”The fact remains that getting people right is not what living is all about anyway. It ’ s getting them wrong that is living. Getting them wrong and wrong and wrong, and then on careful reconsideration, getting them wrong again.”Thank you. Chris Anderson : So, I have a question. When you start talking about using magnetic pulses to change people ’ s moral judgments, that sounds alarming. Please tell me that you ’ re not taking phone calls from the Pentagon, say. RS : I ’ m not. I mean, they ’ re calling, but I ’ m not taking the call. CA : They really are calling? So then seriously, you must lie awake at night sometimes wondering where this work leads. I mean, you ’ re clearly an incredible human being, but someone could take this knowledge and in some future not- torture chamber, do acts that people here might be worried about. RS : Yeah, we worry about this. So, there ’ s a couple of things to say about TMS. One is that you can ’ t be TMSed without knowing it. So it ’ s not a surreptitious technology. It ’ s quite hard, actually, to get those very small changes. The changes I showed you are impressive to me because of what they tell us about the function of the brain, but they ’ re small on the scale of the moral judgments that we actually make. And what we changed was not people ’ s moral judgments when they ’ re deciding what to do, when they ’ re making action choices. We changed their ability to judge other people ’ s actions. And so, I think of what I ’ m doing not so much as studying the defendant in a criminal trial, but studying the jury. CA : Is your work going to lead to any recommendations in education, to perhaps bring up a generation of kids able to make fairer moral judgments? RS : That ’ s one of the idealistic hopes. The whole research program here of studying the distinctive parts of the human brain is brand new. Until recently, what we knew about the brain were the things that any other animal ’ s brain could do too, so we could study it in animal models. We knew how brains see, and how they control the body and how they hear and sense. And the whole project of understanding how brains do the uniquely human things — learn language and abstract concepts, and thinking about other people ’ s thoughts — that ’ s brand new. And we don ’ t know yet what the implications will be of understanding it. CA : So I ’ ve got one last question. There is this thing called the hard problem of consciousness, that puzzles a lot of people. The notion that you can understand why a brain works, perhaps. But why does anyone have to feel anything? Why does it seem to require these beings who sense things for us to operate? You ’ re a brilliant young neuroscientist. I mean, what chances do you think there are that at some time in your career, someone, you or someone else, is going to come up with some paradigm shift in understanding what seems an impossible problem? RS : I hope they do. And I think they probably won ’ t. CA : Why? RS : It ’ s not called the hard problem of consciousness for nothing. CA : That ’ s a great answer. Rebecca Saxe, thank you very much. That was fantastic.

Text Sample 953: NEG Dim 6 – 2009 – Score 1.00 – 2009/t002989.txt

Good morning everybody. I ’ d like to talk about a couple of things today. The first thing is water. Now I see you ’ ve all been enjoying the water that ’ s been provided for you here at the conference, over the past couple of days. And I ’ m sure you ’ ll feel that it ’ s from a safe source. But what if it wasn ’ t? What if it was from a source like this? Then statistics would actually say that half of you would now be suffering with diarrhea. I talked a lot in the past about statistics, and the provision of safe drinking water for all. But they just don ’ t seem to get through. And I think I ’ ve worked out why. It ’ s because, using current thinking, the scale of the problem just seems too huge to contemplate solving. So we just switch off : us, governments and aid agencies. Well, today, I ’ d like to show you that through thinking differently, the problem has been solved. By the way, since I ’ ve been speaking, another 13, 000 people around the world are suffering now with diarrhea. And four children have just died. I invented Lifesaver bottle because I got angry. I, like most of you, was sitting down, the day after Christmas in 2004, when I was watching the devastating news of the Asian tsunami as it rolled in, playing out on TV. The days and weeks that followed, people fleeing to the hills, being forced to drink contaminated water or face death. That really stuck with me. Then, a few months later, Hurricane Katrina slammed into the side of America.”Okay,”I thought,”here ’ s a First World country, let ’ s see what they can do.”Day one : nothing. Day two : nothing. Do you know it took five days to get water to the Superdome? People were shooting each other on the streets for TV sets and water. That ’ s when I decided I had to do something. Now I spent a lot of time in my garage, over the next weeks and months, and also in my kitchen — much to the dismay of my wife. However, after a few failed prototypes, I finally came up with this, the Lifesaver bottle. Okay, now for the science bit. Before Lifesaver, the best hand filters were only capable of filtering down to about 200 nanometers. The smallest bacteria is about 200 nanometers. So a 200-nanometer bacteria is going to get through a 200-nanometer hole. The smallest virus, on the other hand, is about 25 nanometers. So that ’ s definitely going to get through those 200 nanometer holes. Lifesaver pores are 15 nanometers. So nothing is getting through. Okay, I ’ m going to give you a bit of a demonstration. Would you like to see that? I spent all the time setting this up, so I guess I should. We ’ re in the fine city of Oxford. So — someone ’ s done that up. Fine city of Oxford, so what I ’ ve done is I ’ ve gone and got some water from the River Cherwell, and the River Thames, that flow through here. And this is the water. But I got to thinking, you know, if we were in the middle of a flood zone in Bangladesh, the water wouldn ’ t look like this. So I ’ ve gone and got some stuff to add into it. And this is from my pond. Have a smell of that, mister cameraman. Okay. Right. We ’ re just going to pour that in there. Audience : Ugh! Michael Pritchard : Okay. We ’ ve got some runoff from a sewage plant farm. So I ’ m just going to put that in there. Put that in there. There we go. And some other bits and pieces, chuck that in there. And I ’ ve got a gift here from a friend of mine ’ s rabbit. So we ’ re just going to put that in there as well. Okay. Now. The Lifesaver bottle works really simply. You just scoop the water up. Today I

’ m going to use a jug just to show you all. Let ’ s get a bit of that poo in there. That ’ s not dirty enough. Let ’ s just stir that up a little bit. Okay, so I ’ m going to take this really filthy water, and put it in here. Do you want a drink yet? Okay. There we go. Replace the top. Give it a few pumps. Okay? That ’ s all that ’ s necessary. Now as soon as I pop the teat, sterile drinking water is going to come out. I ’ ve got to be quick. Okay, ready? There we go. Mind the electrics. That is safe, sterile drinking water. Cheers. There you go Chris. What ’ s it taste of? Chris Anderson : Delicious. Michael Pritchard : Okay. Let ’ s see Chris ’ s program throughout the rest of the show. Okay? Okay. Lifesaver bottle is used by thousands of people around the world. It ’ ll last for 6, 000 liters. And when it ’ s expired, using failsafe technology, the system will shut off, protecting the user. Pop the cartridge out. Pop a new one in. It ’ s good for another 6, 000 liters. So let ’ s look at the applications. Traditionally, in a crisis, what do we do? We ship water. Then, after a few weeks, we set up camps. And people are forced to come into the camps to get their safe drinking water. What happens when 20, 000 people congregate in a camp? Diseases spread. More resources are required. The problem just becomes self-perpetuating. But by thinking differently, and shipping these, people can stay put. They can make their own sterile drinking water, and start to get on with rebuilding their homes and their lives. Now, it doesn ’ t require a natural disaster for this to work. Using the old thinking, of national infrastructure and pipe work, is too expensive. When you run the numbers on a calculator, you run out of noughts. So here is the”thinking different”bit. Instead of shipping water, and using man-made processes to do it, let ’ s use Mother Nature. She ’ s got a fantastic system. She picks the water up from there, desalinates it, for free, transports it over there, and dumps it onto the mountains, rivers, and streams. And where do people live? Near water. All we ’ ve go to do is make it sterile. How do we do that? Well, we could use the Lifesaver bottle. Or we could use one of these. The same technology, in a jerry can. This will process 25, 000 liters of water ; that ’ s good enough for a family of four, for three years. And how much does it cost? About half a cent a day to run. Thank you. So, by thinking differently, and processing water at the point of use, mothers and children no longer have to walk four hours a day to collect their water. They can get it from a source nearby. So with just eight billion dollars, we can hit the millennium goal ’ s target of halving the number of people without access to safe drinking water. To put that into context, The U. K. government spends about 12 billion pounds a year on foreign aid. But why stop there? With 20 billion dollars, everyone can have access to safe drinking water. So the three-and-a-half billion people that suffer every year as a result, and the two million kids that die every year, will live. Thank you.

Text Sample 954: NEG Dim 6 – 2009 – Score 1.00 – 2009/t002988.txt

Today I want to talk to you about swimming across the North Pole, across the most northern place in the whole world. And perhaps the best place to start is with my late father. He was a great storyteller. He could tell a story about an event, and so you felt you were absolutely there at the moment. And one of the stories he told me so often when I was a young boy was of the first British atomic bomb test. He had been there and watched it go off. And he said that the explosion was so loud and the light was so intense, that he actually had to put his hands in front of his face to protect his eyes. And he said that he could actually see an x-ray of his fingers, because the light was so bright. And I know that watching that atomic bomb going off had a very, very big impact on my late father. Every holiday I had as a young boy was in a national park. What he was trying to do with me was to inspire me to protect the world, and show me just how fragile the world is. He also told me about the great explorers. He loved history. He would tell me about Captain Scott walking all the way to the South Pole and Sir Edmund Hillary climbing up Mount Everest. And so ever since I think I was just six years old, I dreamed of going to the polar regions. I really, really wanted to go to the Arctic. There was something about that place which drew me to it. And, well, sometimes it takes a long time for a dream to come true. But seven years ago, I went to the Arctic for the first time. And it was so beautiful that I ' ve been back there ever since, for the last seven years. I love the place. But I have seen that place change beyond all description, just in that short period of time. I have seen polar bears walking across very, very thin ice in search of food. I have swum in front of glaciers which have retreated so much. And I have also, every year, seen less and less sea ice. And I wanted the world to know what was happening up there. In the two years before my swim, 23 percent of the arctic sea ice cover just melted away. And I wanted to really shake the lapels of world leaders to get them to understand what is happening. So I decided to do this symbolic swim at the top of the world, in a place which should be frozen over, but which now is rapidly unfreezing. And the message was very clear : Climate change is for real, and we need to do something about it. And we need to do something about it right now. Well, swimming across the North Pole, it ' s not an ordinary thing to do. I mean, just to put it in perspective, 27 degrees is the temperature of a normal indoor swimming pool. This morning, the temperature of the English Channel was 18 degrees. The passengers who fell off the Titanic fell into water of just five degrees centigrade. Fresh water freezes at zero. And the water at the North Pole is minus 1. 7. It ' s fucking freezing. I ' m sorry, but there is no other way to describe it. And so I had to assemble an incredible team around me to help me with this task. I assembled this team of 29 people from 10 nations. Some people think that swimming is a very solo sport, you just dive into the sea and off you go. It couldn ' t be further from the truth for me. And I then went and did a huge amount of training, swimming in icy water, backwards and forwards. But the most important thing was to train my mind to prepare myself for what was going to happen. And I had to visualize the swim. I had to see it from the beginning all the way to the end. I had to taste the salt water in my mouth. I had to see my coach screaming for me,"Come on Lewis! Come on! Go! Go! Go! Don ' t slow down!"And so I literally swam across the North Pole hundreds and hundreds of times in my mind. And then, after a year of training, I felt ready. I felt confident that I could actually do this swim. So myself and the five members of the team, we hitched a ride on an icebreaker which was going to the North Pole. And on day four, we decided to just do a quick five minute test swim. I had never swum in water of minus 1. 7 degrees before, because it ' s just impossible to train in those types of conditions. So we stopped the ship, as you do. We all got down onto the ice, and I then got into my swimming costume and I dived into the sea. I have never in my life felt anything like that moment. I could barely breathe. I was gasping for air. I was hyperventilating so much, and within seconds my hands were numb. And it was — the paradox is that you ' re in freezing cold water, but actually you ' re on fire. I swam as hard as I could for five minutes. I remember just trying to get out of the water. I climbed out of the ice. And I remember taking the goggles off my face and looking down at my hands in sheer shock, because my fingers had swollen so much that they were like sausages. And they were swollen so much, I couldn ' t even close them. What had happened is that we are made partially of water, and when water freezes it expands. And so what had actually happened is that the cells in my fingers had frozen and expanded. And they had burst. And I was in so much agony. I immediately got rushed onto the ship and into a hot shower. And I remember standing underneath the hot shower and trying to defrost my fingers. And I thought, in two days ’ time, I was going to do this swim across the North Pole. I was going to try and do a 20-minute swim, for one kilometer across the North Pole. And this dream which I had had ever since I was a young boy with my father, was just going out the window. There is no possibility that this was going to happen. And I remember then getting out of the shower and realizing I couldn ’ t even feel my hands. And for a swimmer, you need to feel your hands because you need to be able to grab the water and pull it through with you. The next morning, I woke up and I was in such a state of depression, and all I could think about was Sir Ranulph Fiennes. For those of you who don ’ t know him, he ’ s the great British explorer. A number of years ago, he tried to ski all the way to the North Pole. He accidentally fell through the ice into the sea. And after just three minutes in that water, he was able to get himself out. And his hands were so badly frostbitten that he had to return to England. He went to a local hospital and there they said,”Ran, there is no possibility of us being able to save these fingers. We are going to actually have to take them off.”And Ran decided to go into his tool shed and take out a saw and do it himself. And all I could think of was, if that happened to Ran after three minutes, and I can ’ t feel my hands after five minutes, what on earth is going to happen if I try 20 minutes? At the very best, I ’ m going to end up losing some fingers. And at worst, I didn ’ t even want to think about it. We carried on sailing through the ice packs towards the North Pole. And my close friend David, he saw the way I was thinking, and he came up to me and he said,”Lewis, I ’ ve known you since you were 18 years old. I ’ ve known you, and I know, Lewis, deep down, right deep down here, that you are going to make this swim. I so believe in you Lewis. I ’ ve seen the way you ’ ve been training. And I realize the reason why you ’ re going to do this. This is such an important swim. We stand at a very, very important moment in this history, and you ’ re going to make a symbolic swim here to try to shake the lapels of world leaders. Lewis, have the courage to go in there, because we are going to look after you every moment of it.”And I just, I got so much confidence from him saying that, because he knew me so well. So we carried on sailing and we arrived at the North Pole. And we stopped the ship, and it was just as the scientists had predicted. There were open patches of sea everywhere. And I went down into my cabin and I put on my swimming costume. And then the doctor strapped on a chest monitor, which measures my core body temperature and my heart rate. And then we walked out onto the ice. And I remember looking into the ice, and there were big chunks of white ice in there, and the water was completely black. I had never seen black water before. And it is 4, 200 meters deep. And I said to myself,”Lewis, don ’ t look left, don ’ t look right. Just scuttle forward and go for it.”And so I now want to show you a short video of what happened there on the ice. Narrator : We ’ re just sailing out of harbor now, and it ’ s at this stage when one can have a bit of a wobble mentally. Everything just looks so gray around here, and looks so cold. We ’ ve just seen our first polar bears. It was absolutely magical. A mother and a cub, such a beautiful sight. And to think that in 30, 40 years they could become extinct. It ’ s a very frightening, very, very frightening thought. We ’ re finally at the North Pole. This is months and months and months of dreaming to get here, years of training and planning and preparation. Ooh. In a couple of hours ’ time I ’ m going to get in here and do my swim. It ’ s all a little bit frightening, and emotional. Amundson, you ready? Amudson : Ready. Lewis Pugh : Ten seconds to swim. Ten seconds to swim. Take the goggles off. Take the goggles off! Man : Take the shoes. Take the shoes. Well done lad! You did it! You did it Lewis! You did it! You did it man! LP : How on earth did we do that? Man : Against the current! You did it against the current! LP : Thank you very much. Thank you very much. Thank you so much. Audience : Encore! LP : I ’ d just like to end off by just saying this : It took me four months again to feel my hands. But was it worth it? Yes, absolutely it was. There are very, very few people who don ’ t know now about what is happening in the Arctic. And people ask me,”Lewis, what can we do about climate change?”And I say to them, I think we need to do three things. The first thing we need to do is we need to break this problem down into manageable chunks. You saw during that video all those flags. Those flags represented the countries from which my team came from. And equally, when it comes to climate change, every single country is going to have to make cuts. Britain, America, Japan, South Africa, the Congo. All of us together, we ’ re all on the same ship together. The second thing we need to do is we need to just look back at how far we have come in such a short period of time. I remember, just a few years ago, speaking about climate change, and people heckling me in the back and saying it doesn ’ t even exist. I ’ ve just come back from giving a series of speeches in some of the poorest townships in South Africa to young children as young as 10 years old. Four or five children sitting behind a desk, and even in those poorest conditions, they all have a very, very good grasp of climate change. We need to believe in ourselves. Now is the time to believe. We ’ ve come a long way. We ’ re doing good. But the most important thing we must do is, I think, we must all walk to the end of our lives and turn around, and ask ourselves a most fundamental question. And that is,”What type of world do we want to live in, and what decision are we going to make today to ensure that we all live in a sustainable world?”Ladies and gentlemen, thank you very, very much.

Text Sample 955: NEG Dim 6 – 2004 – Score 1.00 – 2004/t000487.txt
You know, when Chris first approached me to speak at TED, I said no, because I felt like I wasn ’ t going to be able to make that personal connection, you know, that I wanted to. It ’ s such a large conference. But he explained to me that he was in a bind, and that he was having trouble finding the kind of sex appeal and star power that the conference was known for. So I said fine, Ted — I mean Chris. I ’ ll come on two conditions. One : I want to speak as early in the morning as possible. And two : I want to pick the theme for TED 2006. And luckily he agreed. And the theme, in two years, is going to be”Cute Pictures Of Puppies.”[How to Dance Properly BASIC TWIRL] [NEW SCHOOL] [OLD SCHOOL] [WHO ’ S YOUR DADDY?] [”RIDE THE PONY”] [MAKE LOVE TO THE CROWD] [SMACKING THAT ASS] [STIR THE POT OF LOVE] [HANGING OUT... CASUAL] [WORD.] I invented the Placebo Camera. It doesn ’ t actually take pictures, but it ’ s a hell of a lot cheaper, and you still feel like you were there.”Dear Sir, good day, compliments of the day, and my best wishes to you and family. I know this letter will come to you surprisingly, but let it not be a surprise to you, for nature has a way of arriving unannounced, and, as an adage says, originals are very hard to find, but their echoes sound ouder. So I decided to contact you myself, for you to assure me of safety and honesty, if I have to entrust any amount of money under your custody. I am Mr Micheal Bangura, the son of late Mr Thaimu Bangura who was the Minister of Finance in Sierra Leone but was killed during the civil war. Knowing your country to be economical conducive for investment, and your people as transparent and trustworthy to engage in business, on which premise I write you. Before my father death, he had the sum of 23 million

United States dollars, which he kept away from the rebel leaders during the course of the war. This fund was supposed to be used for the rehabilitation of water reserves all over the country, before the outbreak of war. When the war broke out, the rebel leader demanded the fund be given to him, my father insisted it was not in his possession, and he was killed because of his refusal to release the fund. Meanwhile, my mother and I is the only person who knows about it because my father always confide in me. I made an arrangement with a Red Cross relief worker, who used his official van to transport the money to Lungi Airport, Freetown, although he did not know the real contents of the box. The fund was deposited as a family reasure, in a safe, reliable security company in Dakar, Senegal, where I was only given temporary asylum. I do not wish to invest the money in Senegal due to unfavorable economic climate, and so close to my country. The only assistance I need from you, which I know you would do for me, are the following : one, be a silent partner and receive the funds in your account in trust ; two, provide a bank account under your control to which the funds will be remitted ; three, receive the funds into your account in trust ; take out your commission ; and leave the rest of the money until I arrive, after the transfer is complete. Sincerely, Mr Micheal Bangura.”This is really embarrassing. I was told backstage that I have 18 minutes. I only prepared 15. So if it ’ s cool, I ’ d like to just wait for three. I ’ m really sorry. What ’ s your name? Mark Surfas. It ’ s pretty cool, huh? Pursuing happiness. Are you a virgin? Virgin? I mean — no, I mean like in the TED sense? Are you? Oh, yeah? So what are you, like, a thousand, two thousand, somewhere in there? Huh? Oh? You don ’ t know what I ’ m talking about? Ah, Mark — Surfas. 1, 860 — am I good? And that ’ s nothing to be ashamed of. That ’ s nothing to be ashamed of. Yeah, I was hanging out with some Google guys last night. Really cool, we were getting wasted. And they were telling me that Google software has gotten so advanced that, based on your interaction with Google over your lifetime, they can actually predict what you are going to say — next. And I was like,”Get the fuck out of here. That ’ s crazy.”But they said,”No, but don ’ t show anyone.”But they slipped up. And they said that I could just type in”What was I going to say next?”and my name, and it would tell me. And I have to tell you, this is an unadulterated piece of software, this is a real Internet browser and this is the actual Google site, and we ’ re going to test it out live today. What was I going to say next? And”Ze Frank”— that ’ s me. Am I feeling lucky? Am I feeling lucky? Audience : Yes! Yeah! Ze Frank : Oh! Amazing. In March of 2001 — I filmed myself dancing to Madonna ’ s”Justify My Love.”On a Thursday, I sent out a link to a website that featured those clips to 17 of my closest friends, as part of an invitation to my — an invitation to my th — th — 26th birthday party. By Monday, over a million people were coming to this site a day. Within a week, I received a call from Earthlink that said, due to a 10 cents per megabyte overage charge, I owed them 30, 000 dollars. Needless to say, I was able to leave my job. [WAS LAID OFF] And, finally, you know, become freelance. [UNEMPLOYED] But some people refer to me more as, like, an Internet guru or — [JACKASS] swami. I knew I had something. I ’ d basically distilled a very difficult-to-explain and complex philosophy, which I won ’ t get into here, because it ’ s a little too deep for all of you, but — It ’ s about what makes websites popular, and, you know, it ’ s — [DANCE LIKE AN IDIOT AND DON ’ T SELL ANYTHING] It ’ s unfortunate that I don ’ t have more time. Maybe I can come back next year, or something like that. I ’ m obsessed with email. I get a lot of it. Four years later, I still get probably two or three hundred emails a day from people I don ’ t know, and it ’ s been an amazing opportunity to kind of get to know different cultures, you know? It ’ s like a microscope to the rest of the world. You can kind of peer into other people

’ s lives. And I also feel like I get a lot of inspiration from the average user. For example, somebody wrote,”Hey Ze, if you ever come to Boulder, you should rock out with us,”and I said,”Why wait?”[rocking out] And they said,”Hey Ze, thanks for rocking out, but I meant the kind of rocking out where we ’ d be naked.”And that was embarrassing. But you know, it ’ s kind of a collaboration between me and the fans, so I said,”Sure.”[rocking out naked] I hear a lot of you whispering. And I know what you ’ re saying,”Holy crap! How is his presentation so smooth?”And I have to say that it ’ s not all me this year. I guess Chris has to take some credit here, because in years past, I guess there ’ s been some sort of subpar speakers at TED. I don ’ t know. And so, this year, Chris sent us a TED conference simulator. Which really allowed us as speakers to get there, in the trenches, and practice at home so that we would be ready for this experience. And I ’ ve got to say that, you know, it ’ s really, really great to be here. I ’ d like to tell all of you a little joke. Not just the good stuff, though. You can do heckler mode. Voice : Hey, moron, get off the stage! ZF : You get off the stage. Voice : We want Malcolm Gladwell. In case you run over time. Just one last thing I ’ d like to say, I ’ d, really — I ’ d like to thank all of you for being here. And frog mode.”Ah, the first time that I made love to a rock shrimp —”[Spam jokes are the new airplane jokes] It ’ s true. Some people say to me,”Ze, you ’ re doing all this stuff, this Internet stuff, and you ’ re not making any money.””Why?”And I say,”Mom, Dad — I ’ m trying.”I don ’ t know if you ’ re all aware of this, but the video game market, kids are playing these video games, but, supposedly, there ’ s tons of money. I mean, like, I think, 100, 000 dollars or so a year is being spent on these things. So I decided to try my hand. I came up with a few games. This is called”Atheist.”I figured it would be popular with the young kids. OK. Look, I ’ ll move around and say some things. [Game over. There is no replay.] So that didn ’ t go over so well. I don ’ t really understand why you ’ re laughing. Should have done this before I tried to pitch it.”Buddhist,”of course, looks very, very similar to”Atheist.”But you come back as a duck. And this is great because, you know, for a quarter, you can play this for a long time. And Chris had said in an email that we should really bring something new to TED, something that we haven ’ t shown anyone. So, I made this for TED. It ’ s”Christian.”It ’ s the third in the series. I ’ m hoping it ’ s going to do well this year. Do you have a preference? Good choice. So you can wait for the Second Coming, which is a random number between one and 500 million. So really, what are we talking about here? Oh, tech joy. Tech joy, to me, means something, because I get a lot of joy out of tech. And in fact, making things using technology — and I ’ m being serious here, even though I ’ m using my sarcastic voice — I won ’ t — hold on. Making things, you know — making things actually does give me a lot of joy. It ’ s the process of creation that keeps me sort of a bubble and a half above perpetual anxiety in my life, and it ’ s that feeling of being about 80 percent complete on a project — where you know you still have something to do, but it ’ s not finished, and you ’ re not starting something — that really fills my entire life. And so, what I ’ ve done is, I started getting interested in creating online social spaces to share that feeling with people who don ’ t consider themselves artists. We ’ re in a culture of guru-ship. It ’ s so hard to use some software because, you know, it ’ s unapproachable, people feel like they have to read the manual. So I try to create these very minimal activities that allow people to express themselves, and, hopefully — Whoa! I ’ m like — on the page, but it doesn ’ t exist. It ’ s, like — seriously, though — I try to create meaningful environments for people to express themselves. Here I created a contest called,”When Office Supplies Attack,”which, I think, really resonated with the working population. Over 500 entries in three weeks. Toilet paper fashion. Again, people from all over the country. The watch is particularly incredible. Online drawing tools — you ' ve probably seen a lot of them. I think they ' re wonderful. It ' s a chance for people to get to play with crayons and all that kind of stuff. But I ' m interested in the process of creating, as the real event that I ' m interested in. And the problem is that a lot of people suck at drawing, and they get bummed out at this, sort of, you know, stick figure, awful little thing that they created. And eventually, it just makes them stop playing with it, or they draw penises and things like that. So, the Scribbler is an attempt to create a generative tool. In other words, it ' s a helping tool. You can draw your simple stick figure, and it collaborates with you to create sort of a post-war German etching. In fact, it ' s tuned to be better at drawing things that look worse. So, we go ahead, and we start scribbling. So the idea is that you can really, you know, partake in this process, but watch something really crappy look beautiful. And here are some of my favorites. This is the little trap marionette that was submitted to me. Very cool. Darling. Beautiful stuff. I mean this is incredible. An 11-year-old girl drew this and submitted it. It ' s just gorgeous. I ' m dead serious here. This is not a joke. But, I think it ' s a really fun and wonderful thing. So this is called the"Fiction Project."This is an online space, which is basically a refurbished message board that encourages collaborative fiction writing. These are haikus. None of the haikus were written by the same person. In fact, each line is contributed by a different person at a different time. I think that the"now tied up, tied down, mistress cruel approaches me, now tied down, it ' s up."It ' s an amazing way, and I ' ll tell you, if you come home, and your spouse, or whoever it is, says,"Let ' s talk"— That, like, chills you to the very core. But it ' s peripheral activities like these that allow people to get together, doing fun things. They actually get to know each other, and it ' s sort of like low-threshold peripheral activities that I think are the key to bringing up some of our bonding social capital that we ' re lacking. And very, very quickly — I love puppets. Here ' s a puppet. It dances to music. Lotte Reiniger, an amazing shadow puppeteer in the 20s, that started doing more elaborate things. I became interested in puppets, and I just want to show one last thing to you. Oh, this is how you make puppets. Chris Anderson : Ladies and gentlemen, Mr. Ze Frank.

Text Sample 956: NEG Dim 6 – 2004 – Score 1.00 – 2004/t000459.txt

When you have 21 minutes to speak, two million years seems like a really long time. But evolutionarily, two million years is nothing. And yet, in two million years, the human brain has nearly tripled in mass, going from the one-and-a-quarter-pound brain of our ancestor here, Habilis, to the almost three-pound meatloaf everybody here has between their ears. What is it about a big brain that nature was so eager for every one of us to have one? Well, it turns out when brains triple in size, they don ' t just get three times bigger ; they gain new structures. And one of the main reasons our brain got so big is because it got a new part, called the"frontal lobe,"particularly, a part called the"pre-frontal cortex."What does a prefrontal cortex do for you that should justify the entire architectural overhaul of the human skull in the blink of evolutionary time? Well, it turns out the prefrontal cortex does lots of things, but one of the most important things it does is it ' s an experience simulator. Pilots practice in flight simulators so that they don ' t make real mistakes in planes. Human beings have this marvelous adaptation that they can actually have experiences in their heads before they try them out in real life. This is a trick that none of our ancestors could do, and that no other animal can do quite like we can. It ' s a marvelous adaptation. It ' s up there with opposable thumbs and standing upright and language as one of the things that got our species out of the trees and into the shopping mall. All of you have done this. Ben and Jerry ' s doesn ' t have"liver and onion"ice cream, and it ' s not because they whipped some up, tried it and went,"Yuck!"It ' s because, without leaving your armchair, you can simulate that flavor and say"yuck"before you make it. Let ' s see how your experience simulators are working. Let ' s just run a quick diagnostic before I proceed with the rest of the talk. Here ' s two different futures that I invite you to contemplate. You can try to simulate them and tell me which one you think you might prefer. One of them is winning the lottery. This is about 314 million dollars. And the other is becoming paraplegic. Just give it a moment of thought. You probably don ' t feel like you need a moment of thought. Interestingly, there are data on these two groups of people, data on how happy they are. And this is exactly what you expected, isn ' t it? But these aren ' t the data. I made these up! These are the data. You failed the pop quiz, and you ' re hardly five minutes into the lecture. Because the fact is that a year after losing the use of their legs and a year after winning the lotto, lottery winners and paraplegics are equally happy with their lives. Don ' t feel too bad about failing the first pop quiz, because everybody fails all of the pop quizzes all of the time. The research that my laboratory has been doing, that economists and psychologists around the country have been doing, has revealed something really quite startling to us, something we call the"impact bias,"which is the tendency for the simulator to work badly, for the simulator to make you believe that different outcomes are more different than, in fact, they really are. From field studies to laboratory studies, we see that winning or losing an election, gaining or losing a romantic partner, getting or not getting a promotion, passing or not passing a college test, on and on, have far less impact, less intensity and much less duration than people expect them to have. A recent study — this almost floors me — a recent study showing how major life traumas affect people suggests that if it happened over three months ago, with only a few exceptions, it has no impact whatsoever on your happiness. Why? Because happiness can be synthesized. Sir Thomas Brown wrote in 1642,"I am the happiest man alive. I have that in me that can convert poverty to riches, adversity to prosperity. I am more invulnerable than Achilles ; fortune hath not one place to hit me."What kind of remarkable machinery does this guy have in his head? Well, it turns out it ' s precisely the same remarkable machinery that all of us have. Human beings have something that we might think of as a"psychological immune system,"a system of cognitive processes, largely nonconscious cognitive processes, that help them change their views of the world, so that they can feel better about the worlds in which they find themselves. Like Sir Thomas, you have this machine. Unlike Sir Thomas, you seem not to know it. We synthesize happiness, but we think happiness is a thing to be found. Now, you don ' t need me to give you too many examples of people synthesizing happiness, I suspect, though I ' m going to show you some experimental evidence. You don ' t have to look very far for evidence. I took a copy of the"New York Times"and tried to find some instances of people synthesizing happiness. Here are three guys synthesizing happiness."I ' m better off physically, financially, mentally...""I don ' t have one minute ' s regret. It was a glorious experience.""I believe it turned out for the best."Who are these characters who are so damn happy? The first one is Jim Wright. Some of you are old enough to remember : he was the chairman of the House of Representatives, and he resigned in disgrace when this young Republican named Newt Gingrich found out about a shady book deal that he had done. He lost everything. The most powerful Democrat in the country lost everything : he lost his money, he lost his power. What does he have to say all these years later about it?"I am so much better off physically, financially, mentally and in almost every other way.”What other way would there be to be better off? Vegetably? Minerally? Animally? He ’ s pretty much covered them there. Moreese Bickham is somebody you ’ ve never heard of. Moreese Bickham uttered these words upon being released. He was 78 years old. He ’ d spent 37 years in Louisiana State Penitentiary for a crime he didn ’ t commit. He was ultimately [released for good behavior halfway through his sentence.] What did he have to say about his experience?”I don ’ t have one minute ’ s regret. It was a glorious experience.”Glorious! This guy ’ s not saying,”There were some nice guys. They had a gym.””Glorious”— a word we usually reserve for something like a religious experience. Harry S. Langerman uttered these words. He ’ s somebody you might have known but didn ’ t, because in 1949, he read a little article in the paper about a hamburger stand owned by these two brothers named McDonald. And he thought,”That ’ s a really neat idea!”So he went to find them. They said,”We can give you a franchise on this for 3, 000 bucks.”Harry went back to New York, asked his brother, an investment banker, to loan him 3, 000 dollars, and his brother ’ s immortal words were,”You idiot, nobody eats hamburgers.”He wouldn ’ t lend him the money. Of course, six months later, Ray Kroc had exactly the same idea. It turns out, people do eat hamburgers, and Ray Kroc, for a while, became the richest man in America. And then, finally, some of you recognize this young photo of Pete Best, who was the original drummer for the Beatles, until they, you know, sent him out on an errand and snuck away and picked up Ringo on a tour. Well, in 1994, when Pete Best was interviewed — yes, he ’ s still a drummer ; yes, he ’ s a studio musician — he had this to say :”I ’ m happier than I would have been with the Beatles.”OK, there ’ s something important to be learned from these people, and it is the secret of happiness. Here it is, finally to be revealed. First : accrue wealth, power and prestige, then lose it. Second : spend as much of your life in prison as you possibly can. Third : make somebody else really, really rich. And finally : never, ever join the Beatles. Yeah, right. Because when people synthesize happiness, as these gentlemen seem to have done, we all smile at them, but we kind of roll our eyes and say,”Yeah, right, you never really wanted the job.””Oh yeah, right — you really didn ’ t have that much in common with her, and you figured that out just about the time she threw the engagement ring in your face.”We smirk, because we believe that synthetic happiness is not of the same quality as what we might call”natural happiness.”What are these terms? Natural happiness is what we get when we get what we wanted, and synthetic happiness is what we make when we don ’ t get what we wanted. And in our society, we have a strong belief that synthetic happiness is of an inferior kind. Why do we have that belief? Well, it ’ s very simple. What kind of economic engine would keep churning if we believed that not getting what we want could make us just as happy as getting it? With all apologies to my friend Matthieu Ricard, a shopping mall full of Zen monks is not going to be particularly profitable, because they don ’ t want stuff enough. I want to suggest to you that synthetic happiness is every bit as real and enduring as the kind of happiness you stumble upon when you get exactly what you were aiming for. Now, I ’ m a scientist, so I ’ m going to do this not with rhetoric, but by marinating you in a little bit of data. Let me first show you an experimental paradigm that ’ s used to demonstrate the synthesis of happiness among regular old folks. This isn ’ t mine, it ’ s a 50- year-old paradigm called the”free choice paradigm.”It ’ s very simple. You bring in, say, six objects, and you ask a subject to rank them from the most to the least liked. In this case, because this experiment uses them, these are Monet prints. Everybody ranks these Monet prints from the one they like the most to the one they like the least. Now we give you a choice :”We happen to have some extra prints in the closet. We ’ re going to give you one as your prize to take home. We happen to have number three and number four,”we tell the subject. This is a bit of a difficult choice, because neither one is preferred strongly to the other, but naturally, people tend to pick number three, because they liked it a little better than number four. Sometime later — it could be 15 minutes, it could be 15 days — the same stimuli are put before the subject, and the subject is asked to re-rank the stimuli.”Tell us how much you like them now.”What happens? Watch as happiness is synthesized. This is the result that ’ s been replicated over and over again. You ’ re watching happiness be synthesized. Would you like to see it again? Happiness!”The one I got is really better than I thought! That other one I didn ’ t get sucks!”That ’ s the synthesis of happiness. Now, what ’ s the right response to that?”Yeah, right!”Now, here ’ s the experiment we did, and I hope this is going to convince you that”Yeah, right!”was not the right response. We did this experiment with a group of patients who had anterograde amnesia. These are hospitalized patients. Most of them have Korsakoff syndrome, a polyneuritic psychosis. They drank way too much, and they can ’ t make new memories. They remember their childhood, but if you walk in and introduce yourself and then leave the room, when you come back, they don ’ t know who you are. We took our Monet prints to the hospital. And we asked these patients to rank them from the one they liked the most to the one they liked the least. We then gave them the choice between number three and number four. Like everybody else, they said,”Gee, thanks Doc! That ’ s great! I could use a new print. I ’ ll take number three.”We explained we would have number three mailed to them. We gathered up our materials, and we went out of the room and counted to a half hour. Back into the room, we say,”Hi, we ’ re back.”The patients, bless them, say,”Ah, Doc, I ’ m sorry, I ’ ve got a memory problem ; that ’ s why I ’ m here. If I ’ ve met you before, I don ’ t remember.””Really, Jim, you don ’ t remember? I was just here with the Monet prints?””Sorry, Doc, I just don ’ t have a clue.””No problem, Jim. All I want you to do is rank these for me, from the one you like the most to the one you like the least.”What do they do? Well, let ’ s first check and make sure they ’ re really amnesiac. We ask these amnesiac patients to tell us which one they own, which one they chose last time, which one is theirs. And what we find is, amnesiac patients just guess. These are normal controls, where if I did this with you, all of you would know which print you chose. But if I do this with amnesiac patients, they don ’ t have a clue. They can ’ t pick their print out of a lineup. Here ’ s what normal controls do : they synthesize happiness. Right? This is the change in liking score, the change from the first time they ranked to the second time they ranked. Normal controls show — that was the magic I showed you ; now I ’ m showing it to you in graphical form —”The one I own is better than I thought. The one I didn ’ t own, the one I left behind, is not as good as I thought.”Amnesiacs do exactly the same thing. Think about this result. These people like better the one they own, but they don ’ t know they own it.”Yeah, right”is not the right response! What these people did when they synthesized happiness is they really, truly changed their affective, hedonic, aesthetic reactions to that poster. They ’ re not just saying it because they own it, because they don ’ t know they own it. When psychologists show you bars, you know that they are showing you averages of lots of people. And yet, all of us have this psychological immune system, this capacity to synthesize happiness, but some of us do this trick better than others. And some situations allow anybody to do it more effectively than other situations do. It turns out that freedom, the ability to make up your mind and change your mind, is the friend of natural happiness, because it allows you to choose among all those delicious futures and find the one that you would most enjoy. But freedom to choose, to change and make up your mind, is the enemy of synthetic happiness, and I ’ m going to show you why. Dilbert already knows, of course.”Dogbert ’ s tech support. How may I abuse you?””My printer prints a blank page after every document.””Why complain about getting free paper?””Free? Aren ’ t you just giving me my own paper?””Look at the quality of the free paper compared to your lousy regular paper! Only a fool or a liar would say that they look the same!””Now that you mention it, it does seem a little silkier!””What are you doing?””I ’ m helping people accept the things they cannot change.”Indeed. The psychological immune system works best when we are totally stuck, when we are trapped. This is the difference between dating and marriage. You go out on a date with a guy, and he picks his nose ; you don ’ t go out on another date. You ’ re married to a guy and he picks his nose? He has a heart of gold. Don ’ t touch the fruitcake! You find a way to be happy with what ’ s happened. Now, what I want to show you is that people don ’ t know this about themselves, and not knowing this can work to our supreme disadvantage. Here ’ s an experiment we did at Harvard. We created a black-and-white photography course, and we allowed students to come in and learn how to use a darkroom. So we gave them cameras, they went around campus, they took 12 pictures of their favorite professors and their dorm room and their dog, and all the other things they wanted to have Harvard memories of. They bring us the camera, we make up a contact sheet, they figure out which are the two best pictures. We now spend six hours teaching them about darkrooms, and they blow two of them up. They have two gorgeous 8 x 10 glossies of meaningful things to them, and we say,”Which one would you like to give up?””I have to give one up?””Yes, we need one as evidence of the class project. So you have to give me one. You have to make a choice. You get to keep one, and I get to keep one.”Now, there are two conditions in this experiment. In one case, the students are told,”But you know, if you want to change your mind, I ’ ll always have the other one here, and in the next four days, before I actually mail it to headquarters”— yeah,”headquarters”—”I ’ ll be glad to swap it out with you. In fact, I ’ ll come to your dorm room, just give me an email. Better yet, I ’ ll check with you. You ever want to change your mind, it ’ s totally returnable.”The other half of the students are told exactly the opposite :”Make your choice, and by the way, the mail is going out, gosh, in two minutes, to England. Your picture will be winging its way over the Atlantic. You will never see it again.”Half of the students in each of these conditions are asked to make predictions about how much they ’ re going to come to like the picture that they keep and the picture they leave behind. Other students are just sent back to their little dorm rooms and they are measured over the next three to six days on their satisfaction with the pictures. Look at what we find. First of all, here ’ s what students think is going to happen. They think they ’ re going to maybe come to like the picture they chose a little more than the one they left behind. But these are not statistically significant differences. It ’ s a very small increase, and it doesn ’ t much matter whether they were in the reversible or irreversible condition. Wrong-o. Bad simulators. Because here ’ s what ’ s really happening. Both right before the swap and five days later, people who are stuck with that picture, who have no choice, who can never change their mind, like it a lot. And people who are deliberating —”Should I return it? Have I gotten the right one? Maybe this isn ’ t the good one. Maybe I left the good one?”— have killed themselves. They don ’ t like their picture. In fact, even after the opportunity to swap has expired, they still don ’ t like their picture. Why? Because the [reversible] condition is not conducive to the synthesis of happiness. So here ’ s the final piece of this experiment. We bring in a whole new group of naive Harvard students and we say,”You know, we ’ re doing a photography course, and we can do it one of two ways. We could do it so that when you take the two pictures, you ’ d have four days to change your mind, or we ’ re doing another course where you take the two pictures and you make up your mind right away and you can never change it. Which course would you like to be in?”Duh! Sixty-six percent of the students, two-thirds, prefer to be in the course where they have the opportunity to change their mind. Hello? Sixty-six percent of the students choose to be in the course in which they will ultimately be deeply dissatisfied with the picture — because they do not know the conditions under which synthetic happiness grows. The Bard said everything best, of course, and he ’ s making my point here but he ’ s making it hyperbolically :”’ Tis nothing good or bad But thinking makes it so.”It ’ s nice poetry, but that can ’ t exactly be right. Is there really nothing good or bad? Is it really the case that gall bladder surgery and a trip to Paris are just the same thing? That seems like a one-question IQ test. They can ’ t be exactly the same. In more turgid prose, but closer to the truth, was the father of modern capitalism, Adam Smith, and he said this. This is worth contemplating :”The great source of both the misery and disorders of human life seems to arise from overrating the difference between one permanent situation and another. Some of these situations may, no doubt, deserve to be preferred to others, but none of them can deserve to be pursued with that passionate ardor which drives us to violate the rules either of prudence or of justice, or to corrupt the future tranquility of our minds, either by shame from the remembrance of our own folly, or by remorse for the horror of our own injustice.”In other words : yes, some things are better than others. We should have preferences that lead us into one future over another. But when those preferences drive us too hard and too fast because we have overrated the difference between these futures, we are at risk. When our ambition is bounded, it leads us to work joyfully. When our ambition is unbounded, it leads us to lie, to cheat, to steal, to hurt others, to sacrifice things of real value. When our fears are bounded, we ’ re prudent, we ’ re cautious, we ’ re thoughtful. When our fears are unbounded and overblown, we ’ re reckless, and we ’ re cowardly. The lesson I want to leave you with, from these data, is that our longings and our worries are both to some degree overblown, because we have within us the capacity to manufacture the very commodity we are constantly chasing when we choose experience. Thank you.

Text Sample 957: NEG Dim 6 – 2004 – Score 1.00 – 2004/t000446.txt

About 15 years ago, I went to visit a friend in Hong Kong. And at the time I was very superstitious. So, upon landing â this was still at the old Hong Kong airport that ’ s Kai Tak, when it was smack in the middle of the city â I thought,”If I see something good, I ’ m going to have a great time here in my two weeks. And if I see something negative, I ’ m going to be miserable, indeed.”So the plane landed in between the buildings and got to a full stop in front of this little billboard. And I actually went to see some of the design companies in Hong Kong in my stay there. And it turned out that â I just went to see, you know, what they are doing in Hong Kong. But I actually walked away with a great job offer. And I flew back to Austria, packed my bags, and, another week later, I was again on my way to Hong Kong still superstitions and thinking,”Well, if that ’ Winner ’ billboard is still up, I ’ m going to have a good time working here. But if it ’ s gone, it ’ s going to be really miserable and stressful.”So it turned out that not only was the billboard still up but they had put this one right next to it. On the other hand, it also taught me where superstition gets me because I really had a terrible time in Hong Kong. However, I did have a number of real moments of happiness in my life â of, you know, I think what the conference brochure refers to as”moments that take your breath away.”And since I ’ m a big list maker, I actually listed them all. Now, you don ’ t have to go through the trouble of reading them and I won ’ t read them for you. I know that it ’ s incredibly boring to hear about other people ’ s happinesses. What I did do, though is, I actually looked at them from a design standpoint and just eliminated all the ones that had nothing to do with design. And, very surprisingly, over half of them had, actually, something to do with design. So there are, of course, two different possibilities. There ’ s one from a consumer ’ s point of view â where I was happy while experiencing design. And I ’ ll just give you one example. I had gotten my first Walkman. This is 1983. My brother had this great Yamaha motorcycle that he was willing to borrow to me freely. And The Police ’ s”Synchronicity”cassette had just been released and there was no helmet law in my hometown of Bregenz. So you could drive up into the mountains freely blasting The Police on the new Sony Walkman. And I remember it as a true moment of happiness. You know, of course, they are related to this combination of at least two of them being, you know, design objects. And, you know, there ’ s a scale of happiness when you talk about in design but the motorcycle incident would definitely be, you know, situated somewhere here â right in there between Delight and Bliss. Now, there is the other part, from a designer ’ s standpoint â if you ’ re happy while actually doing it. And one way to see how happy designers are when they ’ re designing could be to look at the authors ’ photos on the back of their monographs? So, according to this, the Australians and the Japanese as well as the Mexicans are very happy. While, somewhat, the Spaniards... and, I think, particularly, the Swiss, don ’ t seem to be doing all that well. Last November, a museum opened in Tokyo called The Mori Museum, in a skyscraper, up on the 56th floor. And their inaugural exhibit was called”Happiness.”And I went, very eagerly, to see it, because â well, also, with an eye on this conference. And they interestingly sectioned the exhibit off into four different areas. Under”Arcadia,”they showed things like this, from the Edo period â a hundred ways to write”happiness”in different forms. Or they had this apple by Yoko Ono â that, of course, later on was, you know, made into the label for The Beatles. Under”Nirvana”they showed this Constable painting. And there was a little â an interesting theory about abstraction. This is a blue field â it ’ s actually an Yves Klein painting. And the theory was that if you abstract an image, you really, you know open as much room for the un-representable â and, therefore, you know, are able to involve the viewer more. Then, under”Desire,”they showed these Shunsho paintings â also from the Edo period â ink on silk. And, lastly, under”Harmony,”they had this 13th- century mandala from Tibet. Now, what I took away from the exhibit was that maybe with the exception of the mandala most of the pieces in there were actually about the visualization of happiness and not about happiness. And I felt a little bit cheated, because the visualization â that ’ s a really easy thing to do. And, you know, my studio â we ’ ve done it all the time. This is, you know, a book. A happy dog â and you take it out, it ’ s an aggressive dog. It ’ s a happy David Byrne and an angry David Byrne. Or a jazz poster with a happy face and a more aggressive face. You know, that ’ s not a big deal to accomplish. It has gotten to the point where, you know, within advertising or within the movie industry,”happy”has gotten such a bad reputation that if you actually want to do something with the subject and still appear authentic, you almost would have to, you know, do it from a cynical point of view. This is, you know, the movie poster. Or we, a couple of weeks ago, designed a box set for The Talking Heads where the happiness visualized on the cover definitely has, very much, a dark side to it. Much, much more difficult is this, where the designs actually can evoke happiness â and I ’ m going to just show you three that actually did this for me. This is a campaign done by a young artist in New York, who calls himself”True.”Everybody who has ridden the New York subway system will be familiar with these signs? True printed his own version of these signs. Met every Wednesday at a subway stop with 20 of his friends. They divided up the different subway lines and added their own version. So this is one. Now, the way this works in the system is that nobody ever looks at these signs. So you ’ re you ’ re really bored in the subway, and you kind of stare at something. And it takes you a while until it actually â you realize that this says something different than what it normally says. I mean, that ’ s, at least, how it made me happy. Now, True is a real humanitarian. He didn ’ t want any of his friends to be arrested, so he supplied everybody with this fake volunteer card. And also gave this fake letter from the MTA to everybody â sort of like pretending that it ’ s an art project financed by The Metropolitan Transit Authority. Another New York project. This is at P. S. 1 â a sculpture that ’ s basically a square room by James Turrell, that has a retractable ceiling. Opens up at dusk and dawn every day. You don ’ t see the horizon. You ’ re just in there, watching the incredible, subtle changes of color in the sky. And the room is truly something to be seen. People ’ s demeanor changes when they go in there. And, for sure, I haven ’ t looked at the sky in the same way after spending an hour in there. There are, of course, more than those three projects that I ’ m showing here. I would definitely say that observing Vik Muniz ’”Cloud”a couple of years ago in Manhattan for sure made me happy, as well. But my last project is, again, from a young designer in New York. He ’ s from Korea originally. And he took it upon himself to print 55, 000 speech bubbles â empty speech bubbles stickers, large ones and small ones. And he goes around New York and just puts them, empty as they are, on posters. And other people go and fill them in. This one says,”Please let me die in peace.”I think that was â the most surprising to myself was that the writing was actually so good. This is on a musician poster, that says :”I am concerned that my CD will not sell more than 200, 000 units and that, as a result, my recoupable advance from my label will be taken from me, after which, my contract will be cancelled, and I ’ ll be back doing Journey covers on Bleecker Street.”I think the reason this works so well is because everybody involved wins. Jee gets to have his project ; the public gets a sweeter environment ; and different public gets a place to express itself ; and the advertisers finally get somebody to look at their ads. Well, there was a question, of course, that was on my mind for a while : You know, can I do more of the things that I like doing in design and less of the ones that I don ’ t like to be doing? Which brought me back to my list making â you know, just to see what I actually like about my job. You know, one is : just working without pressure. Then : working concentrated, without being frazzled. Or, as Nancy said before, like really immerse oneself into it. Try not to get stuck doing the same thing â or try not get stuck behind the computer all day. This is, you know, related to it : getting out of the studio. Then, of course, trying to, you know, work on things where the content is actually important for me. And being able to enjoy the end results. And then I found another list in one of my diaries that actually contained all the things that I thought I learned in my life so far. And, just about at that time, an Austrian magazine called and asked if we would want to do six spreads â design six spreads that work like dividing pages between the different chapters in the magazine? And the whole thing just fell together. So I just picked one of the things that I thought I learned â in this case,”Everything I do always comes back to me”â and we made these spreads right out of this. So it was :”Everything I do always comes back to me.”A couple of weeks ago, a French company asked us to design five billboards for them. Again, we could supply the content for it. So I just picked another one. And this was two weeks ago. We flew to Arizona â the designer who works with me, and myself â and photographed this one. So it ' s :"Trying to look good limits my life."And then we did one more of these. This is, again, for a magazine, dividing pages. This is :"Having"â this is the same thing ; it ' s just, you know, photographed from the side. This is from the front. Then it ' s :"guts."Again, it ' s the same thing â"guts"is just the same room, reworked. Then it ' s :"always works out."Then it ' s"for,"with the light on. And it ' s"me."Thank you so much.

Text Sample 958: NEG Dim 6 – 2001 – Score 1.00 – 2001/t000154.txt

So I understand that this meeting was planned, and the slogan was From Was to Still. And I am illustrating Still. Which, of course, I am not agreeing with because, although I am 94, I am not still working. And anybody who asks me,"Are you still doing this or that?"I don ' t answer because I ' m not doing things still, I ' m doing it like I always did. I still have — or did I use the word still? I didn ' t mean that. I have my file which is called To Do. I have my plans. I have my clients. I am doing my work like I always did. So this takes care of my age. I want to show you my work so you know what I am doing and why I am here. This was about 1925. All of these things were made during the last 75 years. But, of course, I ' m working since 25, doing more or less what you see here. This is Castleton China. This was an exhibition at the Museum of Modern Art. This is now for sale at the Metropolitan Museum. This is still at the Metropolitan Museum now for sale. This is a portrait of my daughter and myself. These were just some of the things I ' ve made. I made hundreds of them for the last 75 years. I call myself a maker of things. I don ' t call myself an industrial designer because I ' m other things. Industrial designers want to make novel things. Novelty is a concept of commerce, not an aesthetic concept. The industrial design magazine, I believe, is called"Innovation."Innovation is not part of the aim of my work. Well, makers of things : they make things more beautiful, more elegant, more comfortable than just the craftsmen do. I have so much to say. I have to think what I am going to say. Well, to describe our profession otherwise, we are actually concerned with the playful search for beauty. That means the playful search for beauty was called the first activity of Man. Sarah Smith, who was a mathematics professor at MIT, wrote,"The playful search for beauty was Man ' s first activity — that all useful qualities and all material qualities were developed from the playful search for beauty."These are tiles. The word,"playful"is a necessary aspect of our work because, actually, one of our problems is that we have to make, produce, lovely things throughout all of life, and this for me is now 75 years. So how can you, without drying up, make things with the same pleasure, as a gift to others, for so long? The playful is therefore an important part of our quality as designer. Let me tell you some about my life. As I said, I started to do these things 75 years ago. My first exhibition in the United States was at the Sesquicentennial exhibition in 1926 — that the Hungarian government sent one of my hand-drawn pieces as part of the exhibit. My work actually took me through many countries, and showed me a great part of the world. This is not that they took me — the work didn ' t take me — I made the things particularly because I wanted to use them to see the world. I was incredibly curious to see the world, and I made all these things, which then finally did take me to see many countries and many cultures. I started as an apprentice to a Hungarian craftsman, and this taught me what the guild system was in Middle Ages. The guild system : that means when I was an apprentice, I had to apprentice myself in order to become a pottery master.

In my shop where I studied, or learned, there was a traditional hierarchy of master, journeyman and learned worker, and apprentice, and I worked as the apprentice. The work as an apprentice was very primitive. That means I had to actually learn every aspect of making pottery by hand. We mashed the clay with our feet when it came from the hillside. After that, it had to be kneaded. It had to then go in, kind of, a mangle. And then finally it was prepared for the throwing. And there I really worked as an apprentice. My master took me to set ovens because this was part of oven-making, oven-setting, in the time. And finally, I had received a document that I had accomplished my apprenticeship successfully, that I had behaved morally, and this document was given to me by the Guild of Roof- Coverers, Rail-Diggers, Oven-Setters, Chimney Sweeps and Potters. I also got at the time a workbook which explained my rights and my working conditions, and I still have that workbook. First I set up a shop in my own garden, and made pottery which I sold on the marketplace in Budapest. And there I was sitting, and my then-boyfriend — I didn ' t mean it was a boyfriend like it is meant today — but my boyfriend and I sat at the market and sold the pots. My mother thought that this was not very proper, so she sat with us to add propriety to this activity. However, after a while there was a new factory being built in Budapest, a pottery factory, a large one. And I visited it with several ladies, and asked all sorts of questions of the director. Then the director asked me, why do you ask all these questions? I said, I also have a pottery. So he asked me, could he please visit me, and then finally he did, and explained to me that what I did now in my shop was an anachronism, that the industrial revolution had broken out, and that I rather should join the factory. There he made an art department for me where I worked for several months. However, everybody in the factory spent his time at the art department. The director there said there were several women casting and producing my designs now in molds, and this was sold also to America. I remember that it was quite successful. However, the director, the chemist, model maker — everybody — concerned himself much more with the art department — that means, with my work — than making toilets, so finally they got a letter from the center, from the bank who owned the factory, saying, make toilet-setting behind the art department, and that was my end. So this gave me the possibility because now I was a journeyman, and journeymen also take their satchel and go to see the world. So as a journeyman, I put an ad into the paper that I had studied, that I was a down-to-earth potter ' s journeyman and I was looking for a job as a journeyman. And I got several answers, and I accepted the one which was farthest from home and practically, I thought, halfway to America. And that was in Hamburg. Then I first took this job in Hamburg, at an art pottery where everything was done on the wheel, and so I worked in a shop where there were several potters. And the first day, I was coming to take my place at the turntable — there were three or four turntables — and one of them, behind where I was sitting, was a hunchback, a deaf-mute hunchback, who smelled very bad. So I doused him in cologne every day, which he thought was very nice, and therefore he brought bread and butter every day, which I had to eat out of courtesy. The first day I came to work in this shop there was on my wheel a surprise for me. My colleagues had thoughtfully put on the wheel where I was supposed to work a very nicely modeled natural man ' s organs. After I brushed them off with a hand motion, they were very — I finally was now accepted, and worked there for some six months. This was my first job. If I go on like this, you will be here till midnight. So I will try speed it up a little Moderator : Eva, we have about five minutes. Eva Zeisel : Are you sure? Moderator : Yes, I am sure. EZ : Well, if you are sure, I have to tell you that within five minutes I will talk very fast. And actually, my work took me to many countries because I used my work to fill my curiosity. And among other things, other countries I worked, was in the Soviet Union, where I worked from ’ 32 to ’ 37 — actually, to ’ 36. I was finally there, although I had nothing to do — I was a foreign expert. I became art director of the china and glass industry, and eventually under Stalin ’ s purges — at the beginning of Stalin ’ s purges, I didn ’ t know that hundreds of thousands of innocent people were arrested. So I was arrested quite early in Stalin ’ s purges, and spent 16 months in a Russian prison. The accusation was that I had successfully prepared an Attentat on Stalin ’ s life. This was a very dangerous accusation. And if this is the end of my five minutes, I want to tell you that I actually did survive, which was a surprise. But since I survived and I ’ m here, and since this is the end of the five minutes, I will — Moderator : Tell me when your last trip to Russia was. Weren ’ t you there recently? EZ : Oh, this summer, in fact, the Lomonosov factory was bought by an American company, invited me. They found out that I had worked in ’ 33 at this factory, and they came to my studio in Rockland County, and brought the 15 of their artists to visit me here. And they invited myself to come to the Russian factory last summer, in July, to make some dishes, design some dishes. And since I don ’ t like to travel alone, they also invited my daughter, son-in-law and granddaughter, so we had a lovely trip to see Russia today, which is not a very pleasant and happy view. Here I am now, if this is the end? Thank you.

Text Sample 959: NEG Dim 6 – 2001 – Score 1.00 – 2001/t000092.txt

I coined my own definition of success in 1934, when I was teaching at a high school in South Bend, Indiana, being a little bit disappointed, and [disillusioned] perhaps, by the way parents of the youngsters in my English classes expected their youngsters to get an A or a B. They thought a C was all right for the neighbors ’ children, because they were all average. But they weren ’ t satisfied when their own — it would make the teacher feel that they had failed, or the youngster had failed. And that ’ s not right. The good Lord in his infinite wisdom didn ’ t create us all equal as far as intelligence is concerned, any more than we ’ re equal for size, appearance. Not everybody could earn an A or a B, and I didn ’ t like that way of judging, and I did know how the alumni of various schools back in the ’ 30s judged coaches and athletic teams. If you won them all, you were considered to be reasonably successful — not completely. Because I found out — we had a number of years at UCLA where we didn ’ t lose a game. But it seemed that we didn ’ t win each individual game by the margin that some of our alumni had predicted — And quite frequently I really felt that they had backed up their predictions in a more materialistic manner. But that was true back in the 30s, so I understood that. But I didn ’ t like it, I didn ’ t agree with it. I wanted to come up with something I hoped could make me a better teacher, and give the youngsters under my supervision, be it in athletics or the English classroom, something to which to aspire, other than just a higher mark in the classroom, or more points in some athletic contest. I thought about that for quite a spell, and I wanted to come up with my own definition. I thought that might help. And I knew how Mr. Webster defined it, as the accumulation of material possessions or the attainment of a position of power or prestige, or something of that sort, worthy accomplishments perhaps, but in my opinion, not necessarily indicative of success. So I wanted to come up with something of my own. And I recalled — I was raised on a small farm in Southern Indiana, and Dad tried to teach me and my brothers that you should never try to be better than someone else. I ’ m sure at the time he did that, I didn ’ t — it didn ’ t — well, somewhere, I guess in the hidden recesses of the mind, it popped out years later. Never try to be better than someone else, always learn from others. Never cease trying to be the best you can be — that ’ s under your control. If you get too engrossed and involved and concerned in regard to the things over which you have no control, it will adversely affect the things over which you have control. Then I ran across this simple verse that said,”At God ’ s footstool to confess, a poor soul knelt, and bowed his head. ’ I failed! ’ he cried. The Master said, ’ Thou didst thy best, that is success. ’”From those things, and one other perhaps, I coined my own definition of success, which is : Peace of mind attained only through self-satisfaction in knowing you made the effort to do the best of which you ’ re capable. I believe that ’ s true. If you make the effort to do the best of which you ’ re capable, trying to improve the situation that exists for you, I think that ’ s success, and I don ’ t think others can judge that ; it ’ s like character and reputation — your reputation is what you ’ re perceived to be ; your character is what you really are. And I think that character is much more important than what you are perceived to be. You ’ d hope they ’ d both be good, but they won ’ t necessarily be the same. Well, that was my idea that I was going to try to get across to the youngsters. I ran across other things. I love to teach, and it was mentioned by the previous speaker that I enjoy poetry, and I dabble in it a bit, and love it. There are some things that helped me, I think, be better than I would have been. I know I ’ m not what I ought to be, what I should be, but I think I ’ m better than I would have been if I hadn ’ t run across certain things. One was just a little verse that said,”No written word, no spoken plea can teach our youth what they should be ; nor all the books on all the shelves — it ’ s what the teachers are themselves.”That made an impression on me in the 1930s. And I tried to use that more or less in my teaching, whether it be in sports, or whether it be in the English classroom. I love poetry and always had an interest in that somehow. Maybe it ’ s because Dad used to read to us at night, by coal oil lamp — we didn ’ t have electricity in our farm home. And Dad would read poetry to us. So I always liked it. And about the same time I ran across this one verse, I ran across another one. Someone asked a lady teacher why she taught, and after some time, she said she wanted to think about that. Then she came up and said,”They ask me why I teach, and I reply, ’ Where could I find such splendid company? ’ There sits a statesman, strong, unbiased, wise ; another Daniel Webster, silver-tongued. A doctor sits beside him, whose quick, steady hand may mend a bone, or stem the life-blood ’ s flow. And there a builder ; upward rise the arch of a church he builds, wherein that minister may speak the word of God, and lead a stumbling soul to touch the Christ. And all about, a gathering of teachers, farmers, merchants, laborers — those who work and vote and build and plan and pray into a great tomorrow. And I may say, I may not see the church, or hear the word, or eat the food their hands may grow, but yet again I may ; And later I may say, I knew him once, and he was weak, or strong, or bold or proud or gay. I knew him once, but then he was a boy. They ask me why I teach and I reply, ’ Where could I find such splendid company? ’”And I believe the teaching profession — it ’ s true, you have so many youngsters, and I ’ ve got to think of my youngsters at UCLA — 30-some attorneys, 11 dentists and doctors, many, many teachers and other professions. And that gives you a great deal of pleasure, to see them go on. I always tried to make the youngsters feel that they ’ re there to get an education, number one ; basketball was second, because it was paying their way, and they do need a little time for social activities, but you let social activities take a little precedence over the other two, and you ’ re not going to have any very long. So that was the idea that I tried to get across to the youngsters under my supervision. I had three rules, pretty much, that I stuck with practically all the time. I ’ d learned these prior to coming to UCLA, and I decided they were very important. One was"Never be late."Later on I said certain things — the players, if we were leaving for somewhere, had to be neat and clean. There was a time when I made them wear jackets and shirts and ties. Then I saw our chancellor coming to school in denims and turtlenecks, and thought, it ' s not right for me to keep this other [rule] so I let them just — they had to be neat and clean. I had one of my greatest players that you probably heard of, Bill Walton. He came to catch the bus ; we were leaving for somewhere to play. And he wasn ' t clean and neat, so I wouldn ' t let him go. He couldn ' t get on the bus, he had to go home and get cleaned up to get to the airport. So I was a stickler for that. I believed in that. I believe in time ; very important. I believe you should be on time, but I felt at practice, for example — we start on time, we close on time. The youngsters didn ' t have to feel that we were going to keep them over. When I speak at coaching clinics, I often tell young coaches — and at coaching clinics, more or less, they ' ll be the younger coaches getting in the profession. Most of them are young, you know, and probably newly-married. And I tell them,"Don ' t run practices late, because you ' ll go home in a bad mood, and that ' s not good, for a young married man to go home in a bad mood. When you get older, it doesn ' t make any difference, but —"So I did believe : on time. I believe starting on time, and I believe closing on time. And another one I had was, not one word of profanity. One word of profanity, and you are out of here for the day. If I see it in a game, you ' re going to come out and sit on the bench. And the third one was, never criticize a teammate. I didn ' t want that. I used to tell them I was paid to do that. That ' s my job. I ' m paid to do it. Pitifully poor, but I am paid to do it. Not like the coaches today, for gracious sakes, no. It ' s a little different than it was in my day. Those were three things that I stuck with pretty closely all the time. And those actually came from my dad. That ' s what he tried to teach me and my brothers at one time. I came up with a pyramid eventually, that I don ' t have the time to go on that. But that helped me, I think, become a better teacher. It ' s something like this : And I had blocks in the pyramid, and the cornerstones being industriousness and enthusiasm, working hard and enjoying what you ' re doing, coming up to the apex, according to my definition of success. And right at the top, faith and patience. And I say to you, in whatever you ' re doing, you must be patient. You have to have patience to — we want things to happen. We talk about our youth being impatient a lot, and they are. They want to change everything. They think all change is progress. And we get a little older — we sort of let things go. And we forget there is no progress without change. So you must have patience, and I believe that we must have faith. I believe that we must believe, truly believe. Not just give it word service, believe that things will work out as they should, providing we do what we should. I think our tendency is to hope things will turn out the way we want them to much of the time, but we don ' t do the things that are necessary to make those things become reality. I worked on this for some 14 years, and I think it helped me become a better teacher. But it all revolved around that original definition of success. You know, a number of years ago, there was a Major League Baseball umpire by the name of George Moriarty. He spelled Moriarty with only one ' i '. I ' d never seen that before, but he did. Big league baseball players — they ' re very perceptive about those things, and they noticed he had only one ' i ' in his name. You ' d be surprised how many also told him that that was one more than he had in his head at various times. But he wrote something where I think he did what I tried to do in this pyramid. He called it"The Road Ahead, or the Road Behind."He said,"Sometimes I think the Fates must grin as we denounce them and insist the only reason we can ' t win, is the Fates themselves have missed. Yet there lives on the ancient claim : we win or lose within ourselves. The shining trophies on our shelves can never win tomorrow ’ s game. You and I know deeper down, there ’ s always a chance to win the crown. But when we fail to give our best, we simply haven ’ t met the test, of giving all and saving none until the game is really won ; of showing what is meant by grit ; of playing through when others quit ; of playing through, not letting up. It ’ s bearing down that wins the cup. Of dreaming there ’ s a goal ahead ; of hoping when our dreams are dead ; of praying when our hopes have fled ; yet losing, not afraid to fall, if, bravely, we have given all. For who can ask more of a man than giving all within his span. Giving all, it seems to me, is not so far from victory. And so the Fates are seldom wrong, no matter how they twist and wind. It ’ s you and I who make our fates — we open up or close the gates on the road ahead or the road behind.”Reminds me of another set of threes that my dad tried to get across to us : Don ’ t whine. Don ’ t complain. Don ’ t make excuses. Just get out there, and whatever you ’ re doing, do it to the best of your ability. And no one can do more than that. I tried to get across, too, that — my opponents will tell you — you never heard me mention winning. Never mention winning. My idea is that you can lose when you outscore somebody in a game, and you can win when you ’ re outscored. I ’ ve felt that way on certain occasions, at various times. And I just wanted them to be able to hold their head up after a game. I used to say that when a game is over, and you see somebody that didn ’ t know the outcome, I hope they couldn ’ t tell by your actions whether you outscored an opponent or the opponent outscored you. That ’ s what really matters : if you make an effort to do the best you can regularly, the results will be about what they should be. Not necessarily what you ’ d want them to be but they ’ ll be about what they should ; only you will know whether you can do that. And that ’ s what I wanted from them more than anything else. And as time went by, and I learned more about other things, I think it worked a little better, as far as the results. But I wanted the score of a game to be the byproduct of these other things, and not the end itself. I believe it was one great philosopher who said — no, no — Cervantes. Cervantes said,”The journey is better than the end.”And I like that. I think that it is — it ’ s getting there. Sometimes when you get there, there ’ s almost a let down. But it ’ s the getting there that ’ s the fun. As a basketball coach at UCLA, I liked our practices to be the journey, and the game would be the end, the end result. I liked to go up and sit in the stands and watch the players play, and see whether I ’ d done a decent job during the week. There again, it ’ s getting the players to get that self-satisfaction, in knowing that they ’ d made the effort to do the best of which they are capable. Sometimes I ’ m asked who was the best player I had, or the best teams. I can never answer that. As far as the individuals are concerned — I was asked one time about that, and they said,”Suppose that you, in some way, could make the perfect player. What would you want?”And I said,”Well, I ’ d want one that knew why he was at UCLA : to get an education, he was a good student, really knew why he was there in the first place. But I ’ d want one that could play, too. I ’ d want one to realize that defense usually wins championships, and who would work hard on defense. But I ’ d want one who would play offense, too. I ’ d want him to be unselfish, and look for the pass first and not shoot all the time. And I ’ d want one that could pass and would pass. I ’ ve had some that could and wouldn ’ t, and I ’ ve had some that would and could. So, yeah, I ’ d want that. And I wanted them to be able to shoot from the outside. I wanted them to be good inside too. I ’ d want them to be able to rebound well at both ends, too. Why not just take someone like Keith Wilkes and let it go at that. He had the qualifications. Not the only one, but he was one that I used in that particular category, because I think he made the effort to become the best. There was a couple. I mention in my book,”They Call Me Coach,”two players that gave me great satisfaction, that came as close as I think anyone I ever had to reach their full potential : one was Conrad Burke, and one was Doug McIntosh. When I saw them as freshmen, on our freshmen team — freshmen couldn ’ t play varsity when I taught. I thought,”Oh gracious, if these two players, either one of them”— they were different years, but I thought about each one at the time he was there —”Oh, if he ever makes the varsity, our varsity must be pretty miserable, if he ’ s good enough to make it.”And you know, one of them was a starting player for a season and a half. The other one, his next year, played 32 minutes in a national championship game, Did a tremendous job for us. The next year, he was a starting player on the national championship team, and here I thought he ’ d never play a minute, when he was — so those are the things that give you great joy, and great satisfaction to see. Neither one of those youngsters could shoot very well. But they had outstanding shooting percentages, because they didn ’ t force it. And neither one could jump very well, but they kept good position, and so they did well rebounding. They remembered that every shot that ’ s taken, they assumed would be missed. I ’ ve had too many stand around and wait to see if it ’ s missed, then they go and it ’ s too late, somebody else is in there ahead of them. They weren ’ t very quick, but they played good position, kept in good balance. And so they played pretty good defense for us. So they had qualities that — they came close to — as close to reaching possibly their full potential as any players I ever had. So I consider them to be as successful as Lewis Alcindor or Bill Walton, or many of the others that we had ; there were some outstanding players. Have I rambled enough? I was told that when he makes his appearance, I was supposed to shut up.

Text Sample 960: NEG Dim 6 – 2001 – Score 3.00 – 2001/t000083.txt

I thought I would read poems I have that relate to the subject of youth and age. I was sort of astonished to find out how many I have actually. The first one is dedicated to Spencer, and his grandmother, who was shocked by his work. My poem is called”Dirt.”My grandmother is washing my mouth out with soap ; half a long century gone and still she comes at me with that thick cruel yellow bar. All because of a word I said, not even said really, only repeated. But”Open,”she says,”open up!”her hand clawing at my head. I know now her life was hard ; she lost three children as babies, then her husband died too, leaving young sons, and no money. She ’ d stand me in the sink to pee because there was never room in the toilet. But oh, her soap! Might its bitter burning have been what made me a poet? The street she lived on was unpaved, her flat, two cramped rooms and a fetid kitchen where she stalked and caught me. Dare I admit that after she did it I never really loved her again? She lived to a hundred, even then. All along it was the sadness, the squalor, but I never, until now loved her again. When that was published in a magazine I got an irate letter from my uncle.”You have maligned a great woman.”It took some diplomacy. This is called”The Dress.”It ’ s a longer poem. In those days, those days which exist for me only as the most elusive memory now, when often the first sound you ’ d hear in the morning would be a storm of birdsong, then the soft clop of the hooves of the horse hauling a milk wagon down your block, and the last sound at night as likely as not would be your father pulling up in his car, having worked late again, always late, and going heavily down to the cellar, to the furnace, to shake out the ashes and damp the draft before he came upstairs to fall into bed — in those long-ago days, women, my mother, my friends ’ mothers, our neighbors, all the women I knew — wore, often much of the day, what were called house-dresses, cheap, printed, pulpy, seemingly purposefully shapeless light cotton shifts that you wore over your nightgown and, when you had to go look for a child, hang wash on the line, or run down to the grocery store on the corner, under a coat, the twisted hem of the nightgown always lank and yellowed, dangling beneath. More than the curlers some of the women seemed constantly to have in their hair in preparation for some great event — a ball, one would think — that never came to pass ; more than the way most women ' s faces not only were never made up during the day, but seemed scraped, bleached, and, with their plucked eyebrows, scarily masklike ; more than all that it was those dresses that made women so unknowable and forbidding, adepts of enigmas to which men could have no access, and boys no conception. Only later would I see the dresses also as a proclamation : that in your dim kitchen, your laundry, your bleak concrete yard, what you revealed of yourself was a fabulation ; your real sensual nature, veiled in those sexless vestments, was utterly your dominion. In those days, one hid much else as well : grown men didn ' t embrace one another, unless someone had died, and not always then ; you shook hands or, at a ball game, thumped your friend ' s back and exchanged blows meant to be codes for affection ; once out of childhood you ' d never again know the shock of your father ' s whiskers on your cheek, not until mores at last had evolved, and you could hug another man, then hold on for a moment, then even kiss. What release finally, the embrace : though we were wary — it seemed so audacious — how much unspoken joy there was in that affirmation of equality and communion, no matter how much misunderstanding and pain had passed between you by then. We knew so little in those days, as little as now, I suppose about healing those hurts : even the women, in their best dresses, with beads and sequins sewn on the bodices, even in lipstick and mascara, their hair aflow, could only stand wringing their hands, begging for peace, while father and son, like thugs, like thieves, like Romans, simmered and hissed and hated, inflicting sorrows that endured, the worst anyway, through the kiss and embrace, bleeding from brother to brother, into the generations. In those days there was still countryside close to the city, farms, cornfields, cows ; even not far from our building with its blurred brick and long shadowy hallway you could find tracts with hills and trees you could pretend were mountains and forests. Or you could go out by yourself even to a half-block-long empty lot, into the bushes : like a creature of leaves you ' d lurk, crouched, crawling, simplified, savage, alone ; already there was wanting to be simpler, wanting, when they called you, never to go back. This is another longish one, about the old and the young. It actually happened right at the time we met. Part of the poem takes place in space we shared and time we shared. It ' s called"The Neighbor."Her five horrid, deformed little dogs who incessantly yap on the roof under my window. Her cats, God knows how many, who must piss on her rugs — her landing ' s a sickening reek. Her shadow once, fumbling the chain on her door, then the door slamming fearfully shut, only the barking and the music — jazz — filtering as it does, day and night into the hall. The time it was Chris Connor singing"Lush Life"— how it brought back my college sweetheart, my first real love, who — till I left her — played the same record. And head on my shoulder, hand on my thigh, sang sweetly along, of regrets and depletions she was too young for, as I was too young, later, to believe in her pain. It startled, then bored, then repelled me. My starting to fancy she ' d ended up in this fire-trap in the Village, that my neighbor was her. My thinking we ' d meet, recognize one another, become friends, that I ' d accomplish a penance. My seeing her, it wasn ' t her, at the mailbox. Gray-yellow hair, army pants under a nightgown, her turning away, hiding her ravaged face in her hands, muttering an inappropriate"Hi."Sometimes there are frightening goings-on in the stairwell. A man shouting,"Shut up!"The dogs frantically snarling, claws scrabbling, then her — her voice hoarse, harsh, hollow, almost only a tone, incoherent, a note, a squawk, bone on metal, metal gone molten, calling them back,"Come back darlings, come back dear ones. My sweet angels, come back."Medea she was, next time I saw her. Sorceress, tranced, ecstatic, stock-still on the sidewalk ragged coat hanging agape, passersby flowing around her, her mouth torn suddenly open as though in a scream, silently though, as though only in her brain or breast had it erupted. A cry so pure, practiced, detached, it had no need of a voice, or could no longer bear one. These invisible links that allure, these transfigurations, even of anguish, that hold us. The girl, my old love, the last lost time I saw her when she came to find me at a party, her drunkenly stumbling, falling, sprawling, skirt hiked, eyes veined red, swollen with tears, her shame, her dishonor. My ignorant, arrogant coarseness, my secret pride, my turning away. Still life on a rooftop, dead trees in barrels, a bench broken, dogs, excrement, sky. What pathways through pain, what junctures of vulnerability, what crossings and counterings? Too many lives in our lives already, too many chances for sorrow, too many unaccounted-for pasts."Behold me,"the god of frenzied, inexhaustible love says, rising in bloody splendor,"Behold me."Her making her way down the littered vestibule stairs, one agonized step at a time. My holding the door. Her crossing the fragmented tiles, faltering at the step to the street, droning, not looking at me,"Can you help me?"Taking my arm, leaning lightly against me. Her wavering step into the world. Her whispering,"Thanks love."Lightly, lightly against me. I think I ' ll lighten up a little. Another, different kind of poem of youth and age. It ' s called"Gas."Wouldn ' t it be nice, I think, when the blue-haired lady in the doctor ' s waiting room bends over the magazine table and farts, just a little, and violently blushes. Wouldn ' t it be nice if intestinal gas came embodied in visible clouds, so she could see that her really quite inoffensive pop had only barely grazed my face before it drifted away. Besides, for this to have happened now is a nice coincidence. Because not an hour ago, while we were on our walk, my dog was startled by a backfire and jumped straight up like a horse bucking. And that brought back to me the stable I worked on weekends when I was 12, and a splendid piebald stallion, who whenever he was mounted would buck just like that, though more hugely of course, enormous, gleaming, resplendent. And the woman, her face abashedly buried in her"Elle"now, reminded me — I ' d forgotten that not the least part of my awe consisted of the fact that with every jump he took the horse would powerfully fart. Phwap! Phwap! Phwap! Something never mentioned in the dozens of books about horses and their riders I devoured in those days. All that savage grandeur, the steely glinting hooves, the eruptions driven from the creature ' s mighty innards, breath stopped, heart stopped, nostrils madly flared, I didn ' t know if I wanted to break him, or be him. This is called"Thirst."Many — most of my poems actually are urban poems. I happen to be reading a bunch that aren ' t."Thirst."Here was my relation with the woman who lived all last autumn and winter, day and night, on a bench in the 103rd Street subway station, until finally one day she vanished. We regarded each other, scrutinized one another. Me shyly, obliquely, trying not to be furtive. She boldly, unblinkingly, even pugnaciously, wrathfully even, when her bottle was empty. I was frightened of her. I felt like a child. I was afraid some repressed part of myself would go out of control, and I ' d be forever entrapped in the shocking seethe of her stench. Not excrement merely, not merely surface and orifice going unwashed, rediffusion of rum, there was will in it, and intention, power and purpose — a social, ethical rage and rebellion — despair too, though, grief, loss. Sometimes I ' d think I should take her home with me, bathe her, comfort her, dress her. She wouldn ' t have wanted me to, I would think. Instead, I ' d step into my train. How rich I would think, is the lexicon of our self-absolving. How enduring, our bland fatal assurance that reflection is righteousness being accomplished. The dance of our glances, the clash, pulling each other through our perceptual punctures, then holocaust, holocaust, host on host of ill, injured presences, squandered, consumed. Her vigil somewhere I know continues. Her occupancy, her absolute, faithful attendance. The dance of our glances, challenge, abdication, effacement, the perfume of our consternation. This is a newer poem, a brand new poem. The title is"This Happened."A student, a young woman in a fourth-floor hallway of her lycee, perched on the ledge of an open window chatting with friends between classes ; a teacher passes and chides her,"Be careful, you might fall,"almost banteringly chides her,"You might fall,"and the young woman, 18, a girl really, though she wouldn ' t think that, as brilliant as she is, first in her class, and"Beautiful, too,"she ' s often told, smiles back, and leans into the open window, which wouldn ' t even be open if it were winter — if it were winter someone would have closed it — leans into the window, farther, still smiling, farther and farther, though it takes less time than this, really an instant, and lets herself fall. Herself fall. A casual impulse, a fancy, never thought of until now, hardly thought of even now... No, more than impulse or fancy, the girl knows what she ' s doing, the girl means something, the girl means to mean, because it occurs to her in that instant, that beautiful or not, bright yes or no, she ' s not who she is, she ' s not the person she is, and the reason, she suddenly knows, is that there ' s been so much premeditation where she is, so much plotting and planning, there ' s hardly a person where she is, or if there is, it ' s not her, or not wholly her, it ' s a self inhabited, lived in by her, and seemingly even as she thinks it she knows what ' s been missing : grace, not premeditation but grace, a kind of being in the world spontaneously, with grace. Weightfully upon me was the world. Weightfully this self which graced the world yet never wholly itself. Weightfully this self which weighed upon me, the release from which is what I desire and what I achieve. And the girl remembers, in this infinite instant already now so many times divided, the sadness she felt once, hardly knowing she felt it, to merely inhabit herself. Yes, the girl falls, absurd to fall, even the earth with its compulsion to take unto itself all that falls must know that falling is absurd, yet the girl falling isn ' t myself, or she is myself, but a self I took of my own volition unto myself. Forever. With grace. This happened. I ' ll read just one more. I don ' t usually say that. I like to just end. But I ' m afraid that Ricky will come out here and shake his fist at me. This is called"Old Man,"appropriately enough."Special : big tits,"Says the advertisement for a soft-core magazine on our neighborhood newsstand. But forget her breasts. A lush, fresh-lipped blond, skin glowing gold, sprawls there, resplendent. 60 nearly, yet these hardly tangible, hardly better than harlots, can still stir me. Maybe a coming of age in the American sensual darkness, never seeing an unsmudged nipple, an uncensored vagina, has left me forever infected with an unquenchable lust of the eye. Always that erotic murmur, I ' m hardly myself if I ' m not in a state of incipient desire. God knows though, there are worse twists your obsessions can take. Last year in Israel, a young ultra-orthodox Rabbi guiding some teenage girls through the Shrine of the Shoah forbade them to look in one room. Because there were images in it he said were licentious. The display was a photo. Men and women stripped naked, some trying to cover their genitals, others too frightened to bother, lined up in snow waiting to be shot and thrown into a ditch. The girls, to my horror, averted their gaze. What carnal mistrust had their teacher taught them. Even that though. Another confession : Once in a book on pre-war Poland, a studio portrait, an absolute angel, an absolute angel with tormented, tormenting eyes. I kept finding myself at her page. That she died in the camps made her — I didn ' t dare wonder why —

more present, more precious. Died in the camps, that too people — or Jews anyway — kept from their children back then. But it was like sex, you didn ’ t have to be told. Sex and death, how close they can seem. So constantly conscious now of death moving towards me, sometimes I think I confound them. My wife ’ s loveliness almost consumes me. My passion for her goes beyond reasonable bounds. When we make love, her holding me everywhere all around me, I ’ m there and not there. My mind teems, jumbles of faces, voices, impressions, I live my life over, as though I were drowning. Then I am drowning, in despair at having to leave her, this, everything, all, unbearable, awful. Still, to be able to die with no special contrition, not having been slaughtered, or enslaved. And not having to know history ’ s next mad rage or regression, it might be a relief. No. Again, no. I don ’ t mean that for a moment. What I mean is the world holds me so tightly — the good and the bad — my own follies and weakness that even this counterfeit Venus with her sham heat, and her bosom probably plumped with gel, so moves me my breath catches. Vamp. Siren. Seductress. How much more she reveals in her glare of ink than she knows. How she incarnates our desperate human need for regard, our passion to live in beauty, to be beauty, to be cherished by glances, if by no more, of something like love, or love. Thank you.

Text Sample 961: NEG Dim 6 – 2003 – Score 1.00 – 2003/t000451.txt

You know, one of the intense pleasures of travel and one of the delights of ethnographic research is the opportunity to live amongst those who have not forgotten the old ways, who still feel their past in the wind, touch it in stones polished by rain, taste it in the bitter leaves of plants. Just to know that Jaguar shamans still journey beyond the Milky Way, or the myths of the Inuit elders still resonate with meaning, or that in the Himalaya, the Buddhists still pursue the breath of the Dharma, is to really remember the central revelation of anthropology, and that is the idea that the world in which we live does not exist in some absolute sense, but is just one model of reality, the consequence of one particular set of adaptive choices that our lineage made, albeit successfully, many generations ago. And of course, we all share the same adaptive imperatives. We ’ re all born. We all bring our children into the world. We go through initiation rites. We have to deal with the inexorable separation of death, so it shouldn ’ t surprise us that we all sing, we all dance, we all have art. But what ’ s interesting is the unique cadence of the song, the rhythm of the dance in every culture. And whether it is the Penan in the forests of Borneo, or the Voodoo acolytes in Haiti, or the warriors in the Kaisut desert of Northern Kenya, the Curandero in the mountains of the Andes, or a caravanserai in the middle of the Sahara — this is incidentally the fellow that I traveled into the desert with a month ago — or indeed a yak herder in the slopes of Qomolangma, Everest, the goddess mother of the world. All of these peoples teach us that there are other ways of being, other ways of thinking, other ways of orienting yourself in the Earth. And this is an idea, if you think about it, can only fill you with hope. Now, together the myriad cultures of the world make up a web of spiritual life and cultural life that envelops the planet, and is as important to the well-being of the planet as indeed is the biological web of life that you know as a biosphere. And you might think of this cultural web of life as being an ethnosphere, and you might define the ethnosphere as being the sum total of all thoughts and dreams, myths, ideas, inspirations, intuitions brought into being by the human imagination since the dawn of consciousness. The ethnosphere is humanity ’ s great legacy. It ’ s the symbol of all that we are and all that we can be as an astonishingly inquisitive species. And just as the biosphere has been severely eroded, so too is the ethnosphere — and, if anything, at a far greater rate. No biologists, for example, would dare suggest that 50 percent of all species or more have been or are on the brink of extinction because it simply is not true, and yet that — the most apocalyptic scenario in the realm of biological diversity — scarcely approaches what we know to be the most optimistic scenario in the realm of cultural diversity. And the great indicator of that, of course, is language loss. When each of you in this room were born, there were 6, 000 languages spoken on the planet. Now, a language is not just a body of vocabulary or a set of grammatical rules. A language is a flash of the human spirit. It ’ s a vehicle through which the soul of each particular culture comes into the material world. Every language is an old-growth forest of the mind, a watershed, a thought, an ecosystem of spiritual possibilities. And of those 6, 000 languages, as we sit here today in Monterey, fully half are no longer being whispered into the ears of children. They ’ re no longer being taught to babies, which means, effectively, unless something changes, they ’ re already dead. What could be more lonely than to be enveloped in silence, to be the last of your people to speak your language, to have no way to pass on the wisdom of the ancestors or anticipate the promise of the children? And yet, that dreadful fate is indeed the plight of somebody somewhere on Earth roughly every two weeks, because every two weeks, some elder dies and carries with him into the grave the last syllables of an ancient tongue. And I know there ’ s some of you who say,”Well, wouldn ’ t it be better, wouldn ’ t the world be a better place if we all just spoke one language?”And I say,”Great, let ’ s make that language Yoruba. Let ’ s make it Cantonese. Let ’ s make it Kogi.”And you ’ ll suddenly discover what it would be like to be unable to speak your own language. And so, what I ’ d like to do with you today is sort of take you on a journey through the ethnosphere, a brief journey through the ethnosphere, to try to begin to give you a sense of what in fact is being lost. Now, there are many of us who sort of forget that when I say”different ways of being,”I really do mean different ways of being. Take, for example, this child of a Barasana in the Northwest Amazon, the people of the anaconda who believe that mythologically they came up the milk river from the east in the belly of sacred snakes. Now, this is a people who cognitively do not distinguish the color blue from the color green because the canopy of the heavens is equated to the canopy of the forest upon which the people depend. They have a curious language and marriage rule which is called”linguistic exogamy :”you must marry someone who speaks a different language. And this is all rooted in the mythological past, yet the curious thing is in these long houses, where there are six or seven languages spoken because of intermarriage, you never hear anyone practicing a language. They simply listen and then begin to speak. Or, one of the most fascinating tribes I ever lived with, the Waorani of northeastern Ecuador, an astonishing people first contacted peacefully in 1958. In 1957, five missionaries attempted contact and made a critical mistake. They dropped from the air 8 x 10 glossy photographs of themselves in what we would say to be friendly gestures, forgetting that these people of the rainforest had never seen anything two-dimensional in their lives. They picked up these photographs from the forest floor, tried to look behind the face to find the form or the figure, found nothing, and concluded that these were calling cards from the devil, so they speared the five missionaries to death. But the Waorani didn ’ t just spear outsiders. They speared each other. 54 percent of their mortality was due to them spearing each other. We traced genealogies back eight generations, and we found two instances of natural death and when we pressured the people a little bit about it, they admitted that one of the fellows had gotten so old that he died getting old, so we speared him anyway. But at the same time they had a perspicacious knowledge of the forest that was astonishing. Their hunters could smell animal urine at 40 paces and tell you what species left it behind. In the early ’ 80s, I had a really astonishing assignment when I was asked by my professor at Harvard if I was interested in going down to Haiti, infiltrating the secret societies which were the foundation of Duvalier ’ s strength and Tonton Macoutes, and securing the poison used to make zombies. In order to make sense out of sensation, of course, I had to understand something about this remarkable faith of Vodoun. And Voodoo is not a black magic cult. On the contrary, it ’ s a complex metaphysical worldview. It ’ s interesting. If I asked you to name the great religions of the world, what would you say? Christianity, Islam, Buddhism, Judaism, whatever. There ’ s always one continent left out, the assumption being that sub-Saharan Africa had no religious beliefs. Well, of course, they did and Voodoo is simply the distillation of these very profound religious ideas that came over during the tragic Diaspora of the slavery era. But, what makes Voodoo so interesting is that it ’ s this living relationship between the living and the dead. So, the living give birth to the spirits. The spirits can be invoked from beneath the Great Water, responding to the rhythm of the dance to momentarily displace the soul of the living, so that for that brief shining moment, the acolyte becomes the god. That ’ s why the Voodooists like to say that”You white people go to church and speak about God. We dance in the temple and become God.”And because you are possessed, you are taken by the spirit — how can you be harmed? So you see these astonishing demonstrations : Voodoo acolytes in a state of trance handling burning embers with impunity, a rather astonishing demonstration of the ability of the mind to affect the body that bears it when catalyzed in the state of extreme excitation. Now, of all the peoples that I ’ ve ever been with, the most extraordinary are the Kogi of the Sierra Nevada de Santa Marta in northern Colombia. Descendants of the ancient Tairona civilization which once carpeted the Caribbean coastal plain of Colombia, in the wake of the conquest, these people retreated into an isolated volcanic massif that soars above the Caribbean coastal plain. In a bloodstained continent, these people alone were never conquered by the Spanish. To this day, they remain ruled by a ritual priesthood but the training for the priesthood is rather extraordinary. The young acolytes are taken away from their families at the age of three and four, sequestered in a shadowy world of darkness in stone huts at the base of glaciers for 18 years : two nine-year periods deliberately chosen to mimic the nine months of gestation they spend in their natural mother ’ s womb ; now they are metaphorically in the womb of the great mother. And for this entire time, they are inculturated into the values of their society, values that maintain the proposition that their prayers and their prayers alone maintain the cosmic — or we might say the ecological — balance. And at the end of this amazing initiation, one day they ’ re suddenly taken out and for the first time in their lives, at the age of 18, they see a sunrise. And in that crystal moment of awareness of first light as the Sun begins to bathe the slopes of the stunningly beautiful landscape, suddenly everything they have learned in the abstract is affirmed in stunning glory. And the priest steps back and says,”You see? It ’ s really as I ’ ve told you. It is that beautiful. It is yours to protect.”They call themselves the”elder brothers”and they say we, who are the younger brothers, are the ones responsible for destroying the world. Now, this level of intuition becomes very important. Whenever we think of indigenous people and landscape, we either invoke Rousseau and the old canard of the”noble savage,”which is an idea racist in its simplicity, or alternatively, we invoke Thoreau and say these people are closer to the Earth than we are. Well, indigenous people are neither sentimental nor weakened by nostalgia. There ’ s not a lot of room for either in the malarial swamps of the

Asmat or in the chilling winds of Tibet, but they have, nevertheless, through time and ritual, forged a traditional mystique of the Earth that is based not on the idea of being self-consciously close to it, but on a far subtler intuition : the idea that the Earth itself can only exist because it is breathed into being by human consciousness. Now, what does that mean? It means that a young kid from the Andes who ’ s raised to believe that that mountain is an Apu spirit that will direct his or her destiny will be a profoundly different human being and have a different relationship to that resource or that place than a young kid from Montana raised to believe that a mountain is a pile of rock ready to be mined. Whether it ’ s the abode of a spirit or a pile of ore is irrelevant. What ’ s interesting is the metaphor that defines the relationship between the individual and the natural world. I was raised in the forests of British Columbia to believe those forests existed to be cut. That made me a different human being than my friends amongst the Kwagiulth who believe that those forests were the abode of Huxwhukw and the Crooked Beak of Heaven and the cannibal spirits that dwelled at the north end of the world, spirits they would have to engage during their Hamatsa initiation. Now, if you begin to look at the idea that these cultures could create different realities, you could begin to understand some of their extraordinary discoveries. Take this plant here. It ’ s a photograph I took in the Northwest Amazon just last April. This is ayahuasca, which many of you have heard about, the most powerful psychoactive preparation of the shaman ’ s repertoire. What makes ayahuasca fascinating is not the sheer pharmacological potential of this preparation, but the elaboration of it. It ’ s made really of two different sources : on the one hand, this woody liana which has in it a series of beta- carbolines, harmine, harmaline, mildly hallucinogenic — to take the vine alone is rather to have sort of blue hazy smoke drift across your consciousness — but it ’ s mixed with the leaves of a shrub in the coffee family called Psychotria viridis. This plant had in it some very powerful tryptamines, very close to brain serotonin, dimethyltryptamine, 5-methoxydimethyltryptamine. If you ’ ve ever seen the Yanomami blowing that snuff up their noses, that substance they make from a different set of species also contains methoxydimethyltryptamine. To have that powder blown up your nose is rather like being shot out of a rifle barrel lined with baroque paintings and landing on a sea of electricity. It doesn ’ t create the distortion of reality ; it creates the dissolution of reality. In fact, I used to argue with my professor, Richard Evan Shultes — who is a man who sparked the psychedelic era with his discovery of the magic mushrooms in Mexico in the 1930s — I used to argue that you couldn ’ t classify these tryptamines as hallucinogenic because by the time you ’ re under the effects there ’ s no one home anymore to experience a hallucination. But the thing about tryptamines is they cannot be taken orally because they ’ re denatured by an enzyme found naturally in the human gut called monoamine oxidase. They can only be taken orally if taken in conjunction with some other chemical that denatures the MAO. Now, the fascinating things are that the beta- carbolines found within that liana are MAO inhibitors of the precise sort necessary to potentiate the tryptamine. So you ask yourself a question. How, in a flora of 80, 000 species of vascular plants, do these people find these two morphologically unrelated plants that when combined in this way, created a kind of biochemical version of the whole being greater than the sum of the parts? Well, we use that great euphemism,”trial and error,”which is exposed to be meaningless. But you ask the Indians, and they say,”The plants talk to us.”Well, what does that mean? This tribe, the Cofan, has 17 varieties of ayahuasca, all of which they distinguish a great distance in the forest, all of which are referable to our eye as one species. And then you ask them how they establish their taxonomy and they say,”I thought you knew something about plants. I mean, don ’ t you know anything?”And I said,”No.”Well, it turns out you take each of the 17 varieties in the night of a full moon, and it sings to you in a different key. Now, that ’ s not going to get you a Ph. D. at Harvard, but it ’ s a lot more interesting than counting stamens. Now — — the problem — the problem is that even those of us sympathetic with the plight of indigenous people view them as quaint and colorful but somehow reduced to the margins of history as the real world, meaning our world, moves on. Well, the truth is the 20th century, 300 years from now, is not going to be remembered for its wars or its technological innovations, but rather as the era in which we stood by and either actively endorsed or passively accepted the massive destruction of both biological and cultural diversity on the planet. Now, the problem isn ’ t change. All cultures through all time have constantly been engaged in a dance with new possibilities of life. And the problem is not technology itself. The Sioux Indians did not stop being Sioux when they gave up the bow and arrow any more than an American stopped being an American when he gave up the horse and buggy. It ’ s not change or technology that threatens the integrity of the ethnosphere. It is power, the crude face of domination. Wherever you look around the world, you discover that these are not cultures destined to fade away ; these are dynamic living peoples being driven out of existence by identifiable forces that are beyond their capacity to adapt to : whether it ’ s the egregious deforestation in the homeland of the Penan — a nomadic people from Southeast Asia, from Sarawak — a people who lived free in the forest until a generation ago, and now have all been reduced to servitude and prostitution on the banks of the rivers, where you can see the river itself is soiled with the silt that seems to be carrying half of Borneo away to the South China Sea, where the Japanese freighters hang light in the horizon ready to fill their holds with raw logs ripped from the forest — or, in the case of the Yanomami, it ’ s the disease entities that have come in, in the wake of the discovery of gold. Or if we go into the mountains of Tibet, where I ’ m doing a lot of research recently, you ’ ll see it ’ s a crude face of political domination. You know, genocide, the physical extinction of a people is universally condemned, but ethnocide, the destruction of people ’ s way of life, is not only not condemned, it ’ s universally, in many quarters, celebrated as part of a development strategy. And you cannot understand the pain of Tibet until you move through it at the ground level. I once travelled 6, 000 miles from Chengdu in Western China overland through southeastern Tibet to Lhasa with a young colleague, and it was only when I got to Lhasa that I understood the face behind the statistics you hear about : 6, 000 sacred monuments torn apart to dust and ashes, 1. 2 million people killed by the cadres during the Cultural Revolution. This young man ’ s father had been ascribed to the Panchen Lama. That meant he was instantly killed at the time of the Chinese invasion. His uncle fled with His Holiness in the Diaspora that took the people to Nepal. His mother was incarcerated for the crime of being wealthy. He was smuggled into the jail at the age of two to hide beneath her skirt tails because she couldn ’ t bear to be without him. The sister who had done that brave deed was put into an education camp. One day she inadvertently stepped on an armband of Mao, and for that transgression, she was given seven years of hard labor. The pain of Tibet can be impossible to bear, but the redemptive spirit of the people is something to behold. And in the end, then, it really comes down to a choice : do we want to live in a monochromatic world of monotony or do we want to embrace a polychromatic world of diversity? Margaret Mead, the great anthropologist, said, before she died, that her greatest fear was that as we drifted towards this blandly amorphous generic world view not only would we see the entire range of the human imagination reduced to a more narrow modality of thought, but that we would wake from a dream one day having forgotten there were even other possibilities. And it ' s humbling to remember that our species has, perhaps, been around for [150, 000] years. The Neolithic Revolution — which gave us agriculture, at which time we succumbed to the cult of the seed ; the poetry of the shaman was displaced by the prose of the priesthood ; we created hierarchy specialization surplus — is only 10, 000 years ago. The modern industrial world as we know it is barely 300 years old. Now, that shallow history doesn ' t suggest to me that we have all the answers for all of the challenges that will confront us in the ensuing millennia. When these myriad cultures of the world are asked the meaning of being human, they respond with 10, 000 different voices. And it ' s within that song that we will all rediscover the possibility of being what we are : a fully conscious species, fully aware of ensuring that all peoples and all gardens find a way to flourish. And there are great moments of optimism. This is a photograph I took at the northern tip of Baffin Island when I went narwhal hunting with some Inuit people, and this man, Olayuk, told me a marvelous story of his grandfather. The Canadian government has not always been kind to the Inuit people, and during the 1950s, to establish our sovereignty, we forced them into settlements. This old man ' s grandfather refused to go. The family, fearful for his life, took away all of his weapons, all of his tools. Now, you must understand that the Inuit did not fear the cold ; they took advantage of it. The runners of their sleds were originally made of fish wrapped in caribou hide. So, this man ' s grandfather was not intimidated by the Arctic night or the blizzard that was blowing. He simply slipped outside, pulled down his sealskin trousers and defecated into his hand. And as the feces began to freeze, he shaped it into the form of a blade. He put a spray of saliva on the edge of the shit knife and as it finally froze solid, he butchered a dog with it. He skinned the dog and improvised a harness, took the ribcage of the dog and improvised a sled, harnessed up an adjacent dog, and disappeared over the ice floes, shit knife in belt. Talk about getting by with nothing. And this, in many ways — — is a symbol of the resilience of the Inuit people and of all indigenous people around the world. The Canadian government in April of 1999 gave back to total control of the Inuit an area of land larger than California and Texas put together. It ' s our new homeland. It ' s called Nunavut. It ' s an independent territory. They control all mineral resources. An amazing example of how a nation-state can seek restitution with its people. And finally, in the end, I think it ' s pretty obvious at least to all of all us who ' ve traveled in these remote reaches of the planet, to realize that they ' re not remote at all. They ' re homelands of somebody. They represent branches of the human imagination that go back to the dawn of time. And for all of us, the dreams of these children, like the dreams of our own children, become part of the naked geography of hope. So, what we ' re trying to do at the National Geographic, finally, is, we believe that politicians will never accomplish anything. We think that polemics — — we think that polemics are not persuasive, but we think that storytelling can change the world, and so we are probably the best storytelling institution in the world. We get 35 million hits on our website every month. 156 nations carry our television channel. Our magazines are read by millions. And what we ' re doing is a series of journeys to the ethnosphere where we ' re going to take our audience to places of such cultural wonder that they cannot help but come away dazzled by what they have seen, and hopefully, therefore, embrace gradually, one by one, the central revelation of anthropology : that this world deserves to exist in a diverse way, that we can find a way to live in a truly multicultural, pluralistic world where all of the wisdom of all peoples can contribute to our collective well-being. Thank you very much.

Text Sample 962: NEG Dim 6 – 2003 – Score 1.00 – 2003/t000450.txt

I was asked to come here and speak about creation. And I only have 15 minutes, and I see they ’ re counting already. And I can â in 15 minutes, I think I can touch only a very rather janitorial branch of creation, which I call”creativity.”Creativity is how we cope with creation. While creation sometimes seems a bit un-graspable, or even pointless, creativity is always meaningful. See, for instance, in this picture. You know, creation is what put that dog in that picture, and creativity is what makes us see a chicken on his hindquarters. When you think about â you know, creativity has a lot to do with causality too. You know, when I was a teenager, I was a creator. I just did things. Then I became an adult and started knowing who I was, and tried to maintain that persona â I became creative. It wasn ’ t until I actually did a book and a retrospective exhibition, that I could track exactly â looks like all the craziest things that I had done, all my drinking, all my parties â they followed a straight line that brings me to the point that actually I ’ m talking to you at this moment. Though it ’ s actually true, you know, the reason I ’ m talking to you right now is because I was born in Brazil. If I was born in Monterey, probably would be in Brazil. You know, I was born in Brazil and grew up in the ’ 70s under a climate of political distress, and I was forced to learn to communicate in a very specific way â in a sort of a semiotic black market. You couldn ’ t really say what you wanted to say ; you had to invent ways of doing it. You didn ’ t trust information very much. That led me to another step of why I ’ m here today, is because I really liked media of all kinds. I was a media junkie, and eventually got involved with advertising. My first job in Brazil was actually to develop a way to improve the readability of billboards, and based on speed, angle of approach and actually blocks of text. It was very â actually, it was a very good study, and got me a job in an ad agency. And they also decided that I had to â to give me a very ugly Plexiglas trophy for it. And another point â why I ’ m here â is that the day I went to pick up the Plexiglas trophy, I rented a tuxedo for the first time in my life, picked the thing â didn ’ t have any friends. On my way out, I had to break a fight apart. Somebody was hitting somebody else with brass knuckles. They were in tuxedos, and fighting. It was very ugly. And also â advertising people do that all the time â â and I â well, what happened is when I went back, it was on the way back to my car, the guy who got hit decided to grab a gun â I don ’ t know why he had a gun â and shoot the first person he decided to be his aggressor. The first person was wearing a black tie, a tuxedo. It was me. Luckily, it wasn ’ t fatal, as you can all see. And, even more luckily, the guy said that he was sorry and I bribed him for compensation money, otherwise I press charges. And that ’ s how â with this money I paid for a ticket to come to the United States in 1983, and that ’ s very â the basic reason I ’ m talking to you here today : because I got shot. Well, when I started working with my own work, I decided that I shouldn ’ t do images. You know, I became â I took this very iconoclastic approach. Because when I decided to go into advertising, I wanted to do â I wanted to airbrush naked people on ice, for whiskey commercials, that ’ s what I really wanted to do. But I â they didn ’ t let me do it, so I just â you know, they would only let me do other things. But I wasn ’ t into selling whiskey ; I was into selling ice. The first works were actually objects. It was kind of a mixture of found object, product design and advertising. And I called them relics. They were displayed first at Stux Gallery in 1983. This is the clown skull. Is a remnant of a race of â a very evolved race of entertainers. They lived in Brazil, long time ago. This is the Ashanti joystick. Unfortunately, it has become obsolete because it was designed for Atari platform. A Playstation II is in the works, maybe for the next TED I ’ ll bring it. The rocking podium. This is the pre-Columbian coffeemaker. Actually, the idea came out of an argument that I had at Starbucks, that I insisted that I wasn ’ t having Colombian coffee ; the coffee was actually pre-Columbian. The Bonsai table. The entire Encyclopedia Britannica bound in a single volume, for travel purposes. And the half tombstone, for people who are not dead yet. I wanted to take that into the realm of images, and I decided to make things that had the same identity conflicts. So I decided to do work with clouds. Because clouds can mean anything you want. But now I wanted to work in a very low-tech way, so something that would mean at the same time a lump of cotton, a cloud and Durer ’ s praying hands â although this looks a lot more like Mickey Mouse ’ s praying hands. But I was still, you know â this is a kitty cloud. They ’ re called”Equivalents,”after Alfred Stieglitz ’ s work.”The Snail.”But I was still working with sculpture, and I was really trying to go flatter and flatter.”The Teapot.”I had a chance to go to Florence, in â I think it was ’ 94, and I saw Ghiberti ’ s”Door of Paradise.”And he did something that was very tricky. He put together two different media from different periods of time. First, he got an age-old way of making it, which was relief, and he worked this with three- point perspective, which was brand-new technology at the time. And it ’ s totally overkill. And your eye doesn ’ t know which level to read. And you become trapped into this kind of representation. So I decided to make these very simple renderings, that at first they are taken as a line drawing â you know, something that ’ s very â and then I did it with wire. The idea was to â because everybody overlooks white â like pencil drawings, you know? And they would look at it â”Ah, it ’ s a pencil drawing.”Then you have this double take and see that it ’ s actually something that existed in time. It had a physicality, and you start going deeper and deeper into sort of narrative that goes this way, towards the image. So this is”Monkey with Leica.””Relaxation.””Fiat Lux.”And the same way the history of representation evolved from line drawings to shaded drawings. And I wanted to deal with other subjects. I started taking that into the realm of landscape, which is something that ’ s almost a picture of nothing. I made these pictures called”Pictures of Thread,”and I named them after the amount of yards that I used to represent each picture. These always end up being a photograph at the end, or more like an etching in this case. So this is a lighthouse. This is”6, 500 Yards,”after Corot.”9, 000 Yards,”after Gerhard Richter. And I don ’ t know how many yards, after John Constable. Departing from the lines, I decided to tackle the idea of points, like which is more similar to the type of representation that we find in photographs themselves. I had met a group of children in the Caribbean island of Saint Kitts, and I did work and play with them. I got some photographs from them. Upon my arrival in New York, I decided â they were children of sugar plantation workers. And by manipulating sugar over a black paper, I made portraits of them. These are â â Thank you. This is”Valentina, the Fastest.”It was just the name of the child, with the little thing you get to know of somebody that you meet very briefly.”Valicia.””Jacynthe.”But another layer of representation was still introduced. Because I was doing this while I was making these pictures, I realized that I could add still another thing I was trying to make a subject â something that would interfere with the themes, so chocolate is very good, because it has â it brings to mind ideas that go from scatology to romance. And so I decided to make these pictures, and they were very large, so you had to walk away from it to be able to see them. So they ’ re called”Pictures of Chocolate.”Freud probably could explain chocolate better than I. He was the first subject. And Jackson Pollock also. Pictures of crowds are particularly interesting, because, you know, you go to that â you try to figure out the threshold with something you can define very easily, like a face, goes into becoming just a texture.”Paparazzi.”I used the dust at the Whitney Museum to render some pieces of their collection. And I picked minimalist pieces because they ’ re about specificity. And you render this with the most non-specific material, which is dust itself. Like, you know, you have the skin particles of every single museum visitor. They do a DNA scan of this, they will come up with a great mailing list. This is Richard Serra. I bought a computer, and [they] told me it had millions of colors in it. You know an artist ’ s first response to this is, who counted it? You know? And I realized that I never worked with color, because I had a hard time controlling the idea of single colors. But once they ’ re applied to numeric structure, then you can feel more comfortable. So the first time I worked with colors was by making these mosaics of Pantone swatches. They end up being very large pictures, and I photographed with a very large camera â an 8x10 camera. So you can see the surface of every single swatch â like in this picture of Chuck Close. And you have to walk very far to be able to see it. Also, the reference to Gerhard Richter ’ s use of color charts â and the idea also entering another realm of representation that ’ s very common to us today, which is the bit map. I ended up narrowing the subject to Monet ’ s”Haystacks.”This is something I used to do as a joke â you know, make â the same like â Robert Smithson ’ s”Spiral Jetty”â and then leaving traces, as if it was done on a tabletop. I tried to prove that he didn ’ t do that thing in the Salt Lake. But then, just doing the models, I was trying to explore the relationship between the model and the original. And I felt that I would have to actually go there and make some earthworks myself. I opt for very simple line drawings â kind of stupid looking. And at the same time, I was doing these very large constructions, being 150 meters away. Now I would do very small ones, which would be like â but under the same light, and I would show them together, so the viewer would have to really figure it out what one he was looking. I wasn ’ t interested in the very large things, or in the small things. I was more interested in the things in between, you know, because you can leave an enormous range for ambiguity there. This is like you see â the size of a person over there. This is a pipe. A hanger. And this is another thing that I did â you know working â everybody loves to watch somebody draw, but not many people have a chance to watch somebody draw in â a lot of people at the same time, to evidence a single drawing. And I love this work, because I did these cartoonish clouds over Manhattan for a period of two months. And it was quite wonderful, because I had an interest â an early interest â in theater, that ’ s justified on this thing. In theater, you have the character and the actor in the same place, trying to negotiate each other in front of an audience. And in this, you ’ d have like a â something that looks like a cloud, and it is a cloud at the same time. So they ’ re like perfect actors. My interest in acting, especially bad acting, goes a long way. Actually, I once paid like 60 dollars to see a very great actor to do a version of”King Lear,”and I felt really robbed, because by the time the actor started being King Lear, he stopped being the great actor that I had paid money to see. On the other hand, you know, I paid like three dollars, I think â and I went to a warehouse in Queens to see a version of”Othello”by an amateur group. And it was quite fascinating, because you know the guy â his name was Joey Grimaldi â he impersonated the Moorish general â you know, for the first three minutes he was really that general, and then he went back into plumber, he worked as a plumber, so â plumber, general, plumber, general â so for three dollars, I saw two tragedies for the price of one. See, I think it ’ s not really about impression, making people fall for a really perfect illusion, as much as it is to make â I usually work at the lowest threshold of visual illusion. Because it ’ s not about fooling somebody, it ’ s actually giving somebody a measure of their own belief : how much you want to be fooled. That ' s why we pay to go to magic shows and things like that. Well, I think that ' s it. My time is nearly up. Thank you very much.

Text Sample 963: NEG Dim 6 – 2003 – Score 1.00 – 2003/t000436.txt

I ' m supposed to scare you, because it ' s about fear, right? And you should be really afraid, but not for the reasons why you think you should be. You should be really afraid that — if we stick up the first slide on this thing — there we go — that you ' re missing out. Because if you spend this week thinking about Iraq and thinking about Bush and thinking about the stock market, you ' re going to miss one of the greatest adventures that we ' ve ever been on. And this is what this adventure ' s really about. This is crystallized DNA. Every life form on this planet — every insect, every bacteria, every plant, every animal, every human, every politician — is coded in that stuff. And if you want to take a single crystal of DNA, it looks like that. And we ' re just beginning to understand this stuff. And this is the single most exciting adventure that we have ever been on. It ' s the single greatest mapping project we ' ve ever been on. If you think that the mapping of America ' s made a difference, or landing on the moon, or this other stuff, it ' s the map of ourselves and the map of every plant and every insect and every bacteria that really makes a difference. And it ' s beginning to tell us a lot about evolution. It turns out that what this stuff is — and Richard Dawkins has written about this — is, this is really a river out of Eden. So, the 3. 2 billion base pairs inside each of your cells is really a history of where you ' ve been for the past billion years. And we could start dating things, and we could start changing medicine and archeology. It turns out that if you take the human species about 700 years ago, white Europeans diverged from black Africans in a very significant way. White Europeans were subject to the plague. And when they were subject to the plague, most people didn ' t survive, but those who survived had a mutation on the CCR5 receptor. And that mutation was passed on to their kids because they ' re the ones that survived, so there was a great deal of population pressure. In Africa, because you didn ' t have these cities, you didn ' t have that CCR5 population pressure mutation. We can date it to 700 years ago. That is one of the reasons why AIDS is raging across Africa as fast as it is, and not as fast across Europe. And we ' re beginning to find these little things for malaria, for sickle cell, for cancers. And in the measure that we map ourselves, this is the single greatest adventure that we ' ll ever be on. And this Friday, I want you to pull out a really good bottle of wine, and I want you to toast these two people. Because this Friday, 50 years ago, Watson and Crick found the structure of DNA, and that is almost as important a date as the 12th of February when we first mapped ourselves, but anyway, we ' ll get to that. I thought we ' d talk about the new zoo. So, all you guys have heard about DNA, all the stuff that DNA does, but some of the stuff we ' re discovering is kind of nifty because this turns out to be the single most abundant species on the planet. If you think you ' re successful or cockroaches are successful, it turns out that there ' s ten trillion trillion Pleurococcus sitting out there. And we didn ' t know that Pleurococcus was out there, which is part of the reason why this whole species-mapping project is so important. Because we ' re just beginning to learn where we came from and what we are. And we ' re finding amoebas like this. This is the amoeba dubia. And the amoeba dubia doesn ' t look like much, except that each of you has about 3. 2 billion letters, which is what makes you you, as far as gene code inside each of your cells, and this little amoeba which, you know, sits in water in hundreds and millions and billions, turns out to have

620 billion base pairs of gene code inside. So, this little thingamajig has a genome that ’ s 200 times the size of yours. And if you ’ re thinking of efficient information storage mechanisms, it may not turn out to be chips. It may turn out to be something that looks a little like that amoeba. And, again, we ’ re learning from life and how life works. This funky little thing : people didn ’ t used to think that it was worth taking samples out of nuclear reactors because it was dangerous and, of course, nothing lived there. And then finally somebody picked up a microscope and looked at the water that was sitting next to the cores. And sitting next to that water in the cores was this little Deinococcus radiodurans, doing a backstroke, having its chromosomes blown apart every day, six, seven times, restitching them, living in about 200 times the radiation that would kill you. And by now you should be getting a hint as to how diverse and how important and how interesting this journey into life is, and how many different life forms there are, and how there can be different life forms living in very different places, maybe even outside of this planet. Because if you can live in radiation that looks like this, that brings up a whole series of interesting questions. This little thingamajig : we didn ’ t know this thingamajig existed. We should have known that this existed because this is the only bacteria that you can see to the naked eye. So, this thing is 0. 75 millimeters. It lives in a deep trench off the coast of Namibia. And what you ’ re looking at with this namibiensis is the biggest bacteria we ’ ve ever seen. So, it ’ s about the size of a little period on a sentence. Again, we didn ’ t know this thing was there three years ago. We ’ re just beginning this journey of life in the new zoo. This is a really odd one. This is Ferroplasma. The reason why Ferroplasma is interesting is because it eats iron, lives inside the equivalent of battery acid, and excretes sulfuric acid. So, when you think of odd life forms, when you think of what it takes to live, it turns out this is a very efficient life form, and they call it an archaea. Archaea means”the ancient ones.”And the reason why they ’ re ancient is because this thing came up when this planet was covered by things like sulfuric acid in batteries, and it was eating iron when the earth was part of a melted core. So, it ’ s not just dogs and cats and whales and dolphins that you should be aware of and interested in on this little journey. Your fear should be that you are not, that you ’ re paying attention to stuff which is temporal. I mean, George Bush — he ’ s going to be gone, alright? Life isn ’ t. Whether the humans survive or don ’ t survive, these things are going to be living on this planet or other planets. And it ’ s just beginning to understand this code of DNA that ’ s really the most exciting intellectual adventure that we ’ ve ever been on. And you can do strange things with this stuff. This is a baby gaur. Conservation group gets together, tries to figure out how to breed an animal that ’ s almost extinct. They can ’ t do it naturally, so what they do with this thing is they take a spoon, take some cells out of an adult gaur ’ s mouth, code, take the cells from that and insert it into a fertilized cow ’ s egg, reprogram cow ’ s egg — different gene code. When you do that, the cow gives birth to a gaur. We are now experimenting with bongos, pandas, elands, Sumatran tigers, and the Australians — bless their hearts — are playing with these things. Now, the last of these things died in September 1936. These are Tasmanian tigers. The last known one died at the Hobart Zoo. But it turns out that as we learn more about gene code and how to reprogram species, we may be able to close the gene gaps in deteriorate DNA. And when we learn how to close the gene gaps, then we can put a full string of DNA together. And if we do that, and insert this into a fertilized wolf ’ s egg, we may give birth to an animal that hasn ’ t walked the earth since 1936. And then you can start going back further, and you can start thinking about dodos, and you can think about other species. And in other places, like Maryland, they ’ re trying to figure out what the primordial

ancestor is. Because each of us contains our entire gene code of where we ’ ve been for the past billion years, because we ’ ve evolved from that stuff, you can take that tree of life and collapse it back, and in the measure that you learn to reprogram, maybe we ’ ll give birth to something that is very close to the first primordial ooze. And it ’ s all coming out of things that look like this. These are companies that didn ’ t exist five years ago. Huge gene sequencing facilities the size of football fields. Some are public. Some are private. It takes about 5 billion dollars to sequence a human being the first time. Takes about 3 million dollars the second time. We will have a 1, 000-dollar genome within the next five to eight years. That means each of you will contain on a CD your entire gene code. And it will be really boring. It will read like this. The really neat thing about this stuff is that ’ s life. And Laurie ’ s going to talk about this one a little bit. Because if you happen to find this one inside your body, you ’ re in big trouble, because that ’ s the source code for Ebola. That ’ s one of the deadliest diseases known to humans. But plants work the same way and insects work the same way, and this apple works the same way. This apple is the same thing as this floppy disk. Because this thing codes ones and zeros, and this thing codes A, T, C, Gs, and it sits up there, absorbing energy on a tree, and one fine day it has enough energy to say, execute, and it goes [thump]. Right? And when it does that, pushes a. EXE, what it does is, it executes the first line of code, which reads just like that, AATCAGGGACCC, and that means : make a root. Next line of code : make a stem. Next line of code, TACGGGG : make a flower that ’ s white, that blooms in the spring, that smells like this. In the measure that you have the code and the measure that you read it — and, by the way, the first plant was read two years ago ; the first human was read two years ago ; the first insect was read two years ago. The first thing that we ever read was in 1995 : a little bacteria called Haemophilus influenzae. In the measure that you have the source code, as all of you know, you can change the source code, and you can reprogram life forms so that this little thingy becomes a vaccine, or this little thingy starts producing biomaterials, which is why DuPont is now growing a form of polyester that feels like silk in corn. This changes all rules. This is life, but we ’ re reprogramming it. This is what you look like. This is one of your chromosomes. And what you can do now is, you can outlay exactly what your chromosome is, and what the gene code on that chromosome is right here, and what those genes code for, and what animals they code against, and then you can tie it to the literature. And in the measure that you can do that, you can go home today, and get on the Internet, and access the world ’ s biggest public library, which is a library of life. And you can do some pretty strange things because in the same way as you can reprogram this apple, if you go to Cliff Tabin ’ s lab at the Harvard Medical School, he ’ s reprogramming chicken embryos to grow more wings. Why would Cliff be doing that? He doesn ’ t have a restaurant. The reason why he ’ s reprogramming that animal to have more wings is because when you used to play with lizards as a little child, and you picked up the lizard, sometimes the tail fell off, but it regrew. Not so in human beings : you cut off an arm, you cut off a leg — it doesn ’ t regrow. But because each of your cells contains your entire gene code, each cell can be reprogrammed, if we don ’ t stop stem cell research and if we don ’ t stop genomic research, to express different body functions. And in the measure that we learn how chickens grow wings, and what the program is for those cells to differentiate, one of the things we ’ re going to be able to do is to stop undifferentiated cells, which you know as cancer, and one of the things we ’ re going to learn how to do is how to reprogram cells like stem cells in such a way that they express bone, stomach, skin, pancreas. And you are likely to be wandering around — and your children — on regrown body parts in a reasonable period of time, in some places in the world where they don ’ t stop the research. How ’ s this stuff work? If each of you differs from the person next to you by one in a thousand, but only three percent codes, which means it ’ s only one in a thousand times three percent, very small differences in expression and punctuation can make a significant difference. Take a simple declarative sentence. Right? That ’ s perfectly clear. So, men read that sentence, and they look at that sentence, and they read this. Okay? Now, women look at that sentence and they say, uh-uh, wrong. This is the way it should be seen. That ’ s what your genes are doing. That ’ s why you differ from this person over here by one in a thousand. Right? But, you know, he ’ s reasonably good looking, but... I won ’ t go there. You can do this stuff even without changing the punctuation. You can look at this, right? And they look at the world a little differently. They look at the same world and they say... That ’ s how the same gene code — that ’ s why you have 30, 000 genes, mice have 30, 000 genes, husbands have 30, 000 genes. Mice and men are the same. Wives know that, but anyway. You can make very small changes in gene code and get really different outcomes, even with the same string of letters. That ’ s what your genes are doing every day. That ’ s why sometimes a person ’ s genes don ’ t have to change a lot to get cancer. These little chippies, these things are the size of a credit card. They will test any one of you for 60, 000 genetic conditions. That brings up questions of privacy and insurability and all kinds of stuff, but it also allows us to start going after diseases, because if you run a person who has leukemia through something like this, it turns out that three diseases with completely similar clinical syndromes are completely different diseases. Because in ALL leukemia, that set of genes over there over-expresses. In MLL, it ’ s the middle set of genes, and in AML, it ’ s the bottom set of genes. And if one of those particular things is expressing in your body, then you take Gleevec and you ’ re cured. If it is not expressing in your body, if you don ’ t have one of those types — a particular one of those types — don ’ t take Gleevec. It won ’ t do anything for you. Same thing with Receptin if you ’ ve got breast cancer. Don ’ t have an HER- 2 receptor? Don ’ t take Receptin. Changes the nature of medicine. Changes the predictions of medicine. Changes the way medicine works. The greatest repository of knowledge when most of us went to college was this thing, and it turns out that this is not so important any more. The U. S. Library of Congress, in terms of its printed volume of data, contains less data than is coming out of a good genomics company every month on a compound basis. Let me say that again : A single genomics company generates more data in a month, on a compound basis, than is in the printed collections of the Library of Congress. This is what ’ s been powering the U. S. economy. It ’ s Moore ’ s Law. So, all of you know that the price of computers halves every 18 months and the power doubles, right? Except that when you lay that side by side with the speed with which gene data ’ s being deposited in GenBank, Moore ’ s Law is right here : it ’ s the blue line. This is on a log scale, and that ’ s what superexponential growth means. This is going to push computers to have to grow faster than they ’ ve been growing, because so far, there haven ’ t been applications that have been required that need to go faster than Moore ’ s Law. This stuff does. And here ’ s an interesting map. This is a map which was finished at the Harvard Business School. One of the really interesting questions is, if all this data ’ s free, who ’ s using it? This is the greatest public library in the world. Well, it turns out that there ’ s about 27 trillion bits moving inside from the United States to the United States ; about 4. 6 trillion is going over to those European countries ; about 5. 5 ’ s going to Japan ; there ’ s almost no communication between Japan, and nobody else is literate in this stuff. It ’ s free. No one ’ s reading it. They ’ re focusing on the war ; they ’ re focusing on

Bush ; they ’ re not interested in life. So, this is what a new map of the world looks like. That is the genomically literate world. And that is a problem. In fact, it ’ s not a genomically literate world. You can break this out by states. And you can watch states rise and fall depending on their ability to speak a language of life, and you can watch New York fall off a cliff, and you can watch New Jersey fall off a cliff, and you can watch the rise of the new empires of intelligence. And you can break it out by counties, because it ’ s specific counties. And if you want to get more specific, it ’ s actually specific zip codes. So, you want to know where life is happening? Well, in Southern California it ’ s happening in 92121. And that ’ s it. And that ’ s the triangle between Salk, Scripps, UCSD, and it ’ s called Torrey Pines Road. That means you don ’ t need to be a big nation to be successful ; it means you don ’ t need a lot of people to be successful ; and it means you can move most of the wealth of a country in about three or four carefully picked 747s. Same thing in Massachusetts. Looks more spread out but — oh, by the way, the ones that are the same color are contiguous. What ’ s the net effect of this? In an agricultural society, the difference between the richest and the poorest, the most productive and the least productive, was five to one. Why? Because in agriculture, if you had 10 kids and you grow up a little bit earlier and you work a little bit harder, you could produce about five times more wealth, on average, than your neighbor. In a knowledge society, that number is now 427 to 1. It really matters if you ’ re literate, not just in reading and writing in English and French and German, but in Microsoft and Linux and Apple. And very soon it ’ s going to matter if you ’ re literate in life code. So, if there is something you should fear, it ’ s that you ’ re not keeping your eye on the ball. Because it really matters who speaks life. That ’ s why nations rise and fall. And it turns out that if you went back to the 1870s, the most productive nation on earth was Australia, per person. And New Zealand was way up there. And then the U. S. came in about 1950, and then Switzerland about 1973, and then the U. S. got back on top — beat up their chocolates and cuckoo clocks. And today, of course, you all know that the most productive nation on earth is Luxembourg, producing about one third more wealth per person per year than America. Tiny landlocked state. No oil. No diamonds. No natural resources. Just smart people moving bits. Different rules. Here ’ s differential productivity rates. Here ’ s how many people it takes to produce a single U. S. patent. So, about 3, 000 Americans, 6, 000 Koreans, 14, 000 Brits, 790, 000 Argentines. You want to know why Argentina ’ s crashing? It ’ s got nothing to do with inflation. It ’ s got nothing to do with privatization. You can take a Harvard-educated Ivy League economist, stick him in charge of Argentina. He still crashes the country because he doesn ’ t understand how the rules have changed. Oh, yeah, and it takes about 5. 6 million Indians. Well, watch what happens to India. India and China used to be 40 percent of the global economy just at the Industrial Revolution, and they are now about 4. 8 percent. Two billion people. One third of the global population producing 5 percent of the wealth because they didn ’ t get this change, because they kept treating their people like serfs instead of like shareholders of a common project. They didn ’ t keep the people who were educated. They didn ’ t foment the businesses. They didn ’ t do the IPOs. Silicon Valley did. And that ’ s why they say that Silicon Valley has been powered by ICs. Not integrated circuits : Indians and Chinese. Here ’ s what ’ s happening in the world. It turns out that if you ’ d gone to the U. N. in 1950, when it was founded, there were 50 countries in this world. It turns out there ’ s now about 192. Country after country is splitting, seceding, succeeding, failing — and it ’ s all getting very fragmented. And this has not stopped. In the 1990s, these are sovereign states that did not exist before 1990. And this doesn ’ t include fusions or name changes or changes in flags. We ’ re generating about 3. 12 states per year. People are taking control of their own states, sometimes for the better and sometimes for the worse. And the really interesting thing is, you and your kids are empowered to build great empires, and you don ’ t need a lot to do it. And, given that the music is over, I was going to talk about how you can use this to generate a lot of wealth, and how code works. Moderator : Two minutes. Juan Enriquez : No, I ’ m going to stop there and we ’ ll do it next year because I don ’ t want to take any of Laurie ’ s time. But thank you very much.

Text Sample 964: NEG Dim 6 – 2007 – Score 1.00 – 2007/t000422.txt

I thought in getting up to my TED wish I would try to begin by putting in perspective what I try to do and how it fits with what they try to do. We live in a world that everyone knows is interdependent, but insufficient in three major ways. It is, first of all, profoundly unequal : half the world ’ s people still living on less than two dollars a day ; a billion people with no access to clean water ; two and a half billion no access to sanitation ; a billion going to bed hungry every night ; one in four deaths every year from AIDS, TB, malaria and the variety of infections associated with dirty water — 80 percent of them under five years of age. Even in wealthy countries it is common now to see inequality growing. In the United States, since 2001 we ’ ve had five years of economic growth, five years of productivity growth in the workplace, but median wages are stagnant and the percentage of working families dropping below the poverty line is up by four percent. The percentage of working families without health care up by four percent. So this interdependent world which has been pretty good to most of us — which is why we ’ re all here in Northern California doing what we do for a living, enjoying this evening — is profoundly unequal. It is also unstable. Unstable because of the threats of terror, weapons of mass destruction, the spread of global disease and a sense that we are vulnerable to it in a way that we weren ’ t not so many years ago. And perhaps most important of all, it is unsustainable because of climate change, resource depletion and species destruction. When I think about the world I would like to leave to my daughter and the grandchildren I hope to have, it is a world that moves away from unequal, unstable, unsustainable interdependence to integrated communities — locally, nationally and globally — that share the characteristics of all successful communities : a broadly shared, accessible set of opportunities, a shared sense of responsibility for the success of the common enterprise and a genuine sense of belonging. All easier said than done. When the terrorist incidents occurred in the United Kingdom a couple of years ago, I think even though they didn ’ t claim as many lives as we lost in the United States on 9 / 11, I think the thing that troubled the British most was that the perpetrators were not invaders, but homegrown citizens whose religious and political identities were more important to them than the people they grew up with, went to school with, worked with, shared weekends with, shared meals with. In other words, they thought their differences were more important than their common humanity. It is the central psychological plague of humankind in the 21st century. Into this mix, people like us, who are not in public office, have more power to do good than at any time in history, because more than half the world ’ s people live under governments they voted in and can vote out. And even non-democratic governments are more sensitive to public opinion. Because primarily of the power of the Internet, people of modest means can band together and amass vast sums of money that can change the world for some public good if they all agree. When the tsunami hit South Asia, the United States contributed 1. 2 billion dollars.

30 percent of our households gave. Half of them gave over the Internet. The median contribution was somewhere around 57 dollars. And thirdly, because of the rise of non-governmental organizations. They, businesses, other citizens ’ groups, have enormous power to affect the lives of our fellow human beings. When I became president in 1993, there were none of these organizations in Russia. There are now a couple of hundred thousand. None in India. There are now at least a half a million active. None in China. There are now 250, 000 registered with the government, probably twice again that many who are not registered for political reasons. When I organized my foundation, and I thought about the world as it is and the world that I hope to leave to the next generation, and I tried to be realistic about what I had cared about all my life that I could still have an impact on. I wanted to focus on activities that would help to alleviate poverty, fight disease, combat climate change, bridge the religious, racial and other divides that torment the world, but to do it in a way that would either use whatever particular skills we could put together in our group to change the way some public good function was performed so that it would sweep across the world more. You saw one reference to that in what we were able to do with AIDS drugs. And I want to say that the head of our AIDS effort, and the person who also is primarily active in the wish I ’ ll make tonight, Ira Magaziner, is here with me and I want to thank him for everything he ’ s done. He ’ s over there. When I got out of office and was asked to work, first in the Caribbean, to try to help deal with the AIDS crisis, generic drugs were available for about 500 dollars a person a year. If you bought them in vast bulks, you could get them at a little under 400 dollars. The first country we went to work in, the Bahamas, was paying 3, 500 dollars for these drugs. The market was so terribly disorganized that they were buying this medicine through two agents who were gigging them sevenfold. So the very first week we were working, we got the price down to 500 dollars. And all of a sudden, they could save seven times as many lives for the same amount of money. Then we went to work with the manufacturers of AIDS medicines, one of whom was cited in the film, and negotiated a whole different change in business strategy, because even at 500 dollars, these drugs were being sold on a high- margin, low-volume, uncertain-payment basis. So we worked on improving the productivity of the operations and the supply chain, and went to a low-margin, high-volume, absolutely certain-payment business. I joked that the main contribution we made to the battle against AIDS was to get the manufacturers to change from a jewelry store to a grocery store strategy. But the price went to 140 dollars from 500. And pretty soon, the average price was 192 dollars. Now we can get it for about 100 dollars. Children ’ s medicine was 600 dollars, because nobody could afford to buy any of it. We negotiated it down to 190. Then, the French imposed their brilliantly conceived airline tax to create a something called UNITAID, got a bunch of other countries to help. That children ’ s medicine is now 60 dollars a person a year. The only thing that is keeping us from basically saving the lives of everybody who needs the medicine to stay alive are the absence of systems necessary to diagnose, treat and care for people and deliver this medicine. We started a childhood obesity initiative with the Heart Association in America. We tried to do the same thing by negotiating industry-right deals with the soft drink and the snack food industry to cut the caloric and other dangerous content of food going to our children in the schools. We just reorganized the markets. And it occurred to me that in this whole non-governmental world, somebody needs to be thinking about organizing public goods markets. And that is now what we ’ re trying to do, and working with this large cities group to fight climate change, to negotiate huge, big, volume deals that will enable cities which generate 75 percent of the world ’ s greenhouse gases, to

drastically and quickly reduce greenhouse gas emissions in a way that is good economics. And this whole discussion as if it ’ s some sort of economic burden, is a mystery to me. I think it ’ s a bird ’ s nest on the ground. When Al Gore won his well-deserved Oscar for the”Inconvenient Truth”movie, I was thrilled, but I had urged him to make a second movie quickly. For those of you who saw”An Inconvenient Truth,”the most important slide in the Gore lecture is the last one, which shows here ’ s where greenhouse gases are going if we don ’ t do anything, here ’ s where they could go. And then there are six different categories of things we can do to change the trajectory. We need a movie on those six categories. And all of you need to have it embedded in your brains and to organize yourselves around it. So we ’ re trying to do that. So organizing these markets is one thing we try to do. Now we have taken on a second thing, and this gets to my wish. It has been my experience in working in developing countries that while the headlines may all be — the pessimistic headlines may say, well, we can ’ t do this, that or the other thing because of corruption — I think incapacity is a far bigger problem in poor countries than corruption, and feeds corruption. We now have the money, given these low prices, to distribute AIDS drugs all over the world to people we cannot presently reach. Today these low prices are available in the 25 countries where we work, and in a total of 62 countries, and about 550, 000 people are getting the benefits of them. But the money is there to reach others. The systems are not there to reach the people. So what we have been trying to do, working first in Rwanda and then in Malawi and other places — but I want to talk about Rwanda tonight — is to develop a model for rural health care in a very poor area that can be used to deal with AIDS, TB, malaria, other infectious diseases, maternal and child health, and a whole range of health issues poor people are grappling with in the developing world, that can first be scaled for the whole nation of Rwanda, and then will be a model that could literally be implemented in any other poor country in the world. And the test is : one, will it do the job? Will it provide high quality care? And two, will it do it at a price that will enable the country to sustain a health care system without foreign donors after five to 10 years? Because the longer I deal with these problems, the more convinced I am that we have to — whether it ’ s economics, health, education, whatever — we have to build systems. And the absence of systems that function break the connection which got you all in this seat tonight. You think about whatever your life has been, however many obstacles you have faced in your life, at critical junctures you always knew there was a predictable connection between the effort you exerted and the result you achieved. In a world with no systems, with chaos, everything becomes a guerilla struggle, and this predictability is not there. And it becomes almost impossible to save lives, educate kids, develop economies, whatever. The person, in my view, who has done the best job of this in the health care area, of building a system in a very poor area, is Dr. Paul Farmer, who, many of you know, has worked for now 20 years with his group, Partners in Health, primarily in Haiti where he started, but they ’ ve also worked in Russia, in Peru and other places around the world. As poor as Haiti is, in the area where Farmer ’ s clinic is active — and they serve a catchment area far greater than the medical professionals they have would indicate they could serve — since 1988, they have not lost one person to tuberculosis, not one. And they ’ ve achieved a lot of other amazing health results. So when we decided to work in Rwanda on trying to dramatically increase the income of the country and fight the AIDS problem, we wanted to build a healthcare network, because it had been totally destroyed during the genocide in 1994, and the per capita income was still under a dollar a day. So I rang up, asked Paul Farmer if he would help. Because it seemed to me if we could prove there was a model in Haiti and a model in Rwanda that we could then take all over the country, number one, it would be a wonderful thing for a country that has suffered as much as any on Earth in the last 15 years, and number two, we would have something that could then be adapted to any other poor country anywhere in the world. And so we have set about doing that. Now, we started working together 18 months ago. And we ’ re working in an area called Southern Kayonza, which is one of the poorest areas in Rwanda, with a group that originally includes about 400, 000 people. We ’ re essentially implementing what Paul Farmer did in Haiti : he develops and trains paid community health workers who are able to identify health problems, ensure that people who have AIDS or TB are properly diagnosed and take their medicine regularly, who work on bringing about health education, clean water and sanitation, providing nutritional supplements and moving people up the chain of health care if they have problems of the severity that require it. The procedures that make this work have been perfected, as I said, by Paul Farmer and his team in their work in rural Haiti over the last 20 years. Recently we did an evaluation of the first 18 months of our efforts in Rwanda. And the results were so good that the Rwandan government has now agreed to adopt the model for the entire country, and has strongly supported and put the full resources of the government behind it. I ’ ll tell you a little bit about our team because it ’ s indicative of what we do. We have about 500 people around the world working in our AIDS program, some of them for nothing — just for transportation, room and board. And then we have others working in these other related programs. Our business plan in Rwanda was put together under the leadership of Diana Noble, who is an unusually gifted woman, but not unusual in the type of people who have been willing to do this kind of work. She was the youngest partner at Schroder Ventures in London in her 20s. She was CEO of a successful e-venture — she started and built Reed Elsevier Ventures — and at 45 she decided she wanted to do something different with her life. So she now works full-time on this for very little pay. She and her team of former business people have created a business plan that will enable us to scale this health system up for the whole country. And it would be worthy of the kind of private equity work she used to do when she was making a lot more money for it. When we came to this rural area, 45 percent of the children under the age of five had stunted growth due to malnutrition. 23 percent of them died before they reached the age of five. Mortality at birth was over two-and-a-half percent. Over 15 percent of the deaths among adults and children occurred because of intestinal parasites and diarrhea from dirty water and inadequate sanitation — all entirely preventable and treatable. Over 13 percent of the deaths were from respiratory illnesses — again, all preventable and treatable. And not a single soul in this area was being treated for AIDS or tuberculosis. Within the first 18 months, the following things happened : we went from zero to about 2, 000 people being treated for AIDS. That ’ s 80 percent of the people who need treatment in this area. Listen to this : less than four-tenths of one percent of those being treated stopped taking their medicine or otherwise defaulted on treatment. That ’ s lower than the figure in the United States. Less than three-tenths of one percent had to transfer to the more expensive second- line drugs. 400, 000 pregnant women were brought into counseling and will give birth for the first time within an organized healthcare system. That ’ s about 43 percent of all the pregnancies. About 40 percent of all the people — I said 400, 000. I meant 40, 000. About 40 percent of all the people who need TB treatment are now getting it — in just 18 months, up from zero when we started. 43 percent of the children in need of an infant feeding program to prevent malnutrition and early death are now getting the food supplements they need to stay alive and to grow. We ’ ve started the first malaria treatment programs they ’ ve ever had there. Patients admitted to a hospital that was destroyed during the genocide that we have renovated along with four other clinics, complete with solar power generators, good lab technology. We now are treating 325 people a month, despite the fact that almost 100 percent of the AIDS patients are now treated at home. And the most important thing is because we ’ ve implemented Paul Farmer ’ s model, using community health workers, we estimate that this system could be put into place for all of Rwanda for between five and six percent of GDP, and that the government could sustain that without depending on foreign aid after five or six years. And for those of you who understand healthcare economics you know that all wealthy countries spend between nine and 11 percent of GDP on health care, except for the United States, we spend 16 — but that ’ s a story for another day. We ’ re now working with Partners in Health and the Ministry of Health in Rwanda and our Foundation folks to scale this system up. We ’ re also beginning to do this in Malawi and Lesotho. And we have similar projects in Tanzania, Mozambique, Kenya and Ethiopia with other partners trying to achieve the same thing : to save as many lives as quickly as we can, but to do it in a systematic way that can be implemented nationwide and then with a model that can be implemented in any country in the world. We need initial upfront investment to train doctors, nurses, health administration and community health workers throughout the country, to set up the information technology, the solar energy, the water and sanitation, the transportation infrastructure. But over a five- to 10-year period, we will take down the need for outside assistance and eventually it will be phased out. My wish is that TED assist us in our work and help us to build a high-quality rural health system in a poor country, Rwanda, that can be a model for Africa, and indeed, for any poor country anywhere in the world. My belief is that this will help us to build a more integrated world with more partners and fewer terrorists, with more productive citizens and fewer haters, a place we ’ d all want our kids and our grandchildren to grow up in. It has been an honor for me, particularly, to work in Rwanda where we also have a major economic development project in partnership with Sir Tom Hunter, the Scottish philanthropist, where last year we, using the same thing with AIDS drugs, cut the cost of fertilizer and the interest rates on microcredit loans by 30 percent and achieved three- to four-hundred percent increases in crop yields with the farmers. These people have been through a lot and none of us, most of all me, helped them when they were on the verge of destroying each other. We ’ re undoing that now, and they are so over it and so into their future. We ’ re doing this in an environmentally responsible way. I ’ m doing my best to convince them not to run the electric grid to the 35 percent of the people that have no access, but to do it with clean energy. To have responsible reforestation projects, the Rwandans, interestingly enough, have been quite good, Mr. Wilson, in preserving their topsoil. There ’ s a couple of guys from southern farming families — the first thing I did when I went out to this place was to get down on my hands and knees and dig in the dirt and see what they ’ d done with it. We have a chance here to prove that a country that almost slaughtered itself out of existence can practice reconciliation, reorganize itself, focus on tomorrow and provide comprehensive, quality health care with minimal outside help. I am grateful for this prize, and I will use it to that end. We could use some more help to do this, but think of what it would mean if we could have a world-class health system in Rwanda — in a country with a less-than-one- dollar-a-day-per-capita income, one that could save hundreds of millions of lives over the next decade if applied to every similarly situated country on Earth. It ’ s worth a try and I believe it would succeed. Thank you and God bless you.

Text Sample 965: NEG Dim 6 – 2007 – Score 1.00 – 2007/t000401.txt

So I want to talk to you today about AIDS in sub-Saharan Africa. And this is a pretty well-educated audience, so I imagine you all know something about AIDS. You probably know that roughly 25 million people in Africa are infected with the virus, that AIDS is a disease of poverty, and that if we can bring Africa out of poverty, we would decrease AIDS as well. If you know something more, you probably know that Uganda, to date, is the only country in sub-Saharan Africa that has had success in combating the epidemic. Using a campaign that encouraged people to abstain, be faithful, and use condoms — the ABC campaign — they decreased their prevalence in the 1990s from about 15 percent to 6 percent over just a few years. If you follow policy, you probably know that a few years ago the president pledged 15 billion dollars to fight the epidemic over five years, and a lot of that money is going to go to programs that try to replicate Uganda and use behavior change to encourage people and decrease the epidemic. So today I ’ m going to talk about some things that you might not know about the epidemic, and I ’ m actually also going to challenge some of these things that you think that you do know. To do that I ’ m going to talk about my research as an economist on the epidemic. And I ’ m not really going to talk much about the economy. I ’ m not going to tell you about exports and prices. But I ’ m going to use tools and ideas that are familiar to economists to think about a problem that ’ s more traditionally part of public health and epidemiology. And I think in that sense, this fits really nicely with this lateral thinking idea. Here I ’ m really using the tools of one academic discipline to think about problems of another. So we think, first and foremost, AIDS is a policy issue. And probably for most people in this room, that ’ s how you think about it. But this talk is going to be about understanding facts about the epidemic. It ’ s going to be about thinking about how it evolves, and how people respond to it. I think it may seem like I ’ m ignoring the policy stuff, which is really the most important, but I ’ m hoping that at the end of this talk you will conclude that we actually cannot develop effective policy unless we really understand how the epidemic works. And the first thing that I want to talk about, the first thing I think we need to understand is : how do people respond to the epidemic? So AIDS is a sexually transmitted infection, and it kills you. So this means that in a place with a lot of AIDS, there ’ s a really significant cost of sex. If you ’ re an uninfected man living in Botswana, where the HIV rate is 30 percent, if you have one more partner this year — a long-term partner, girlfriend, mistress — your chance of dying in 10 years increases by three percentage points. That is a huge effect. And so I think that we really feel like then people should have less sex. And in fact among gay men in the US we did see that kind of change in the 1980s. So if we look in this particularly high-risk sample, they ’ re being asked,”Did you have more than one unprotected sexual partner in the last two months?”Over a period from ’ 84 to ’ 88, that share drops from about 85 percent to 55 percent. It ’ s a huge change in a very short period of time. We didn ’ t see anything like that in Africa. So we don ’ t have quite as good data, but you can see here the share of single men having pre-marital sex, or married men having extra-marital sex, and how that changes from the early ’ 90s to late ’ 90s, and late ’ 90s to early 2000s. The epidemic is getting worse. People are learning more things about it. We see almost no change in sexual behavior. These are just tiny decreases — two percentage points — not significant. This seems puzzling. But I ’ m going to argue that you shouldn ’ t be surprised by this, and that to understand this you need to think about health the way than an economist does — as an investment. So if you ’ re a software engineer and you ’ re trying to think about whether to add some new functionality to your program, it ’ s important to think about how much it costs. It ’ s also important to think about what the benefit is. And one part of that benefit is how much longer you think this program is going to be active. If version 10 is coming out next week, there ’ s no point in adding more functionality into version nine. But your health decisions are the same. Every time you have a carrot instead of a cookie, every time you go to the gym instead of going to the movies, that ’ s a costly investment in your health. But how much you want to invest is going to depend on how much longer you expect to live in the future, even if you don ’ t make those investments. AIDS is the same kind of thing. It ’ s costly to avoid AIDS. People really like to have sex. But, you know, it has a benefit in terms of future longevity. But life expectancy in Africa, even without AIDS, is really, really low : 40 or 50 years in a lot of places. I think it ’ s possible, if we think about that intuition, and think about that fact, that maybe that explains some of this low behavior change. But we really need to test that. And a great way to test that is to look across areas in Africa and see : do people with more life expectancy change their sexual behavior more? And the way that I ’ m going to do that is, I ’ m going to look across areas with different levels of malaria. So malaria is a disease that kills you. It ’ s a disease that kills a lot of adults in Africa, in addition to a lot of children. And so people who live in areas with a lot of malaria are going to have lower life expectancy than people who live in areas with limited malaria. So one way to test to see whether we can explain some of this behavior change by differences in life expectancy is to look and see is there more behavior change in areas where there ’ s less malaria. So that ’ s what this figure shows you. This shows you — in areas with low malaria, medium malaria, high malaria — what happens to the number of sexual partners as you increase HIV prevalence. If you look at the blue line, the areas with low levels of malaria, you can see in those areas, actually, the number of sexual partners is decreasing a lot as HIV prevalence goes up. Areas with medium levels of malaria it decreases some — it doesn ’ t decrease as much. And areas with high levels of malaria — actually, it ’ s increasing a little bit, although that ’ s not significant. This is not just through malaria. Young women who live in areas with high maternal mortality change their behavior less in response to HIV than young women who live in areas with low maternal mortality. There ’ s another risk, and they respond less to this existing risk. So by itself, I think this tells a lot about how people behave. It tells us something about why we see limited behavior change in Africa. But it also tells us something about policy. Even if you only cared about AIDS in Africa, it might still be a good idea to invest in malaria, in combating poor indoor air quality, in improving maternal mortality rates. Because if you improve those things, then people are going to have an incentive to avoid AIDS on their own. But it also tells us something about one of these facts that we talked about before. Education campaigns, like the one that the president is focusing on in his funding, may not be enough, at least not alone. If people have no incentive to avoid AIDS on their own, even if they know everything about the disease, they still may not change their behavior. So the other thing that I think we learn here is that AIDS is not going to fix itself. People aren ’ t changing their behavior enough to decrease the growth in the epidemic. So we ’ re going to need to think about policy and what kind of policies might be effective. And a great way to learn about policy is to look at what worked in the past. The reason that we know that the ABC campaign was effective in Uganda is we have good data on prevalence over time. In Uganda we see the prevalence went down. We know they had this campaign. That ’ s how we learn about what works. It ’ s not the only place we had any interventions. Other places have tried things, so why don ’ t we look at those places and see what happened to their prevalence? Unfortunately, there ' s almost no good data on HIV prevalence in the general population in Africa until about 2003. So if I asked you,"Why don ' t you go and find me the prevalence in Burkina Faso in 1991?"You get on Google, you Google, and you find, actually the only people tested in Burkina Faso in 1991 are STD patients and pregnant women, which is not a terribly representative group of people. Then if you poked a little more, you looked a little more at what was going on, you ' d find that actually that was a pretty good year, because in some years the only people tested are IV drug users. But even worse — some years it ' s only IV drug users, some years it ' s only pregnant women. We have no way to figure out what happened over time. We have no consistent testing. Now in the last few years, we actually have done some good testing. In Kenya, in Zambia, and a bunch of countries, there ' s been testing in random samples of the population. But this leaves us with a big gap in our knowledge. So I can tell you what the prevalence was in Kenya in 2003, but I can ' t tell you anything about 1993 or 1983. So this is a problem for policy. It was a problem for my research. And I started thinking about how else might we figure out what the prevalence of HIV was in Africa in the past. And I think that the answer is, we can look at mortality data, and we can use mortality data to figure out what the prevalence was in the past. To do this, we ' re going to have to rely on the fact that AIDS is a very specific kind of disease. It kills people in the prime of their lives. Not a lot of other diseases have that profile. And you can see here — this is a graph of death rates by age in Botswana and Egypt. Botswana is a place with a lot of AIDS, Egypt is a place without a lot of AIDS. And you see they have pretty similar death rates among young kids and old people. That suggests it ' s pretty similar levels of development. But in this middle region, between 20 and 45, the death rates in Botswana are much, much, much higher than in Egypt. But since there are very few other diseases that kill people, we can really attribute that mortality to HIV. But because people who died this year of AIDS got it a few years ago, we can use this data on mortality to figure out what HIV prevalence was in the past. So it turns out, if you use this technique, actually your estimates of prevalence are very close to what we get from testing random samples in the population, but they ' re very, very different than what UNAIDS tells us the prevalences are. So this is a graph of prevalence estimated by UNAIDS, and prevalence based on the mortality data for the years in the late 1990s in nine countries in Africa. You can see, almost without exception, the UNAIDS estimates are much higher than the mortality-based estimates. UNAIDS tell us that the HIV rate in Zambia is 20 percent, and mortality estimates suggest it ' s only about 5 percent. And these are not trivial differences in mortality rates. So this is another way to see this. You can see that for the prevalence to be as high as UNAIDS says, we have to really see 60 deaths per 10, 000 rather than 20 deaths per 10, 000 in this age group. I ' m going to talk a little bit in a minute about how we can use this kind of information to learn something that ' s going to help us think about the world. But this also tells us that one of these facts that I mentioned in the beginning may not be quite right. If you think that 25 million people are infected, if you think that the UNAIDS numbers are much too high, maybe that ' s more like 10 or 15 million. It doesn ' t mean that AIDS isn ' t a problem. It ' s a gigantic problem. But it does suggest that that number might be a little big. What I really want to do, is I want to use this new data to try to figure out what makes the HIV epidemic grow faster or slower. And I said in the beginning, I wasn ' t going to tell you about exports. When I started working on these projects, I was not thinking at all about economics, but eventually it kind of sucks you back in. So I am going to talk about exports and prices. And I want to talk about the relationship between economic activity, in particular export volume, and HIV infections. So obviously, as an economist, I ' m deeply familiar with the fact that development, that openness to trade, is really good for developing countries. It ' s good for improving people ' s lives. But openness and inter-connectedness, it comes with a cost when we think about disease. I don ' t think this should be a surprise. On Wednesday, I learned from Laurie Garrett that I ' m definitely going to get the bird flu, and I wouldn ' t be at all worried about that if we never had any contact with Asia. And HIV is actually particularly closely linked to transit. The epidemic was introduced to the US by actually one male steward on an airline flight, who got the disease in Africa and brought it back. And that was the genesis of the entire epidemic in the US. In Africa, epidemiologists have noted for a long time that truck drivers and migrants are more likely to be infected than other people. Areas with a lot of economic activity — with a lot of roads, with a lot of urbanization — those areas have higher prevalence than others. But that actually doesn ' t mean at all that if we gave people more exports, more trade, that that would increase prevalence. By using this new data, using this information about prevalence over time, we can actually test that. And so it seems to be — fortunately, I think — it seems to be the case that these things are positively related. More exports means more AIDS. And that effect is really big. So the data that I have suggests that if you double export volume, it will lead to a quadrupling of new HIV infections. So this has important implications both for forecasting and for policy. From a forecasting perspective, if we know where trade is likely to change, for example, because of the African Growth and Opportunities Act or other policies that encourage trade, we can actually think about which areas are likely to be heavily infected with HIV. And we can go and we can try to have pre-emptive preventive measures there. Likewise, as we ' re developing policies to try to encourage exports, if we know there ' s this externality — this extra thing that ' s going to happen as we increase exports — we can think about what the right kinds of policies are. But it also tells us something about one of these things that we think that we know. Even though it is the case that poverty is linked to AIDS, in the sense that Africa is poor and they have a lot of AIDS, it ' s not necessarily the case that improving poverty — at least in the short run, that improving exports and improving development — it ' s not necessarily the case that that ' s going to lead to a decline in HIV prevalence. So throughout this talk I ' ve mentioned a few times the special case of Uganda, and the fact that it ' s the only country in sub-Saharan Africa with successful prevention. It ' s been widely heralded. It ' s been replicated in Kenya, and Tanzania, and South Africa and many other places. But now I want to actually also question that. Because it is true that there was a decline in prevalence in Uganda in the 1990s. It ' s true that they had an education campaign. But there was actually something else that happened in Uganda in this period. There was a big decline in coffee prices. Coffee is Uganda ' s major export. Their exports went down a lot in the early 1990s — and actually that decline lines up really, really closely with this decline in new HIV infections. So you can see that both of these series — the black line is export value, the red line is new HIV infections — you can see they ' re both increasing. Starting about 1987 they ' re both going down a lot. And then actually they track each other a little bit on the increase later in the decade. So if you combine the intuition in this figure with some of the data that I talked about before, it suggests that somewhere between 25 percent and 50 percent of the decline in prevalence in Uganda actually would have happened even without any education campaign. But that ' s enormously important for policy. We ' re spending so much money to try to replicate this campaign. And if it was only 50 percent as effective as we think that it was, then there are all sorts of other things maybe we should be spending our money on instead.

Trying to change transmission rates by treating other sexually transmitted diseases. Trying to change them by engaging in male circumcision. There are tons of other things that we should think about doing. And maybe this tells us that we should be thinking more about those things. I hope that in the last 16 minutes I ’ ve told you something that you didn ’ t know about AIDS, and I hope that I ’ ve gotten you questioning a little bit some of the things that you did know. And I hope that I ’ ve convinced you maybe that it ’ s important to understand things about the epidemic in order to think about policy. But more than anything, you know, I ’ m an academic. And when I leave here, I ’ m going to go back and sit in my tiny office, and my computer, and my data. And the thing that ’ s most exciting about that is every time I think about research, there are more questions. There are more things that I think that I want to do. And what ’ s really, really great about being here is I ’ m sure that the questions that you guys have are very, very different than the questions that I think up myself. And I can ’ t wait to hear about what they are. So thank you very much.

Text Sample 966: NEG Dim 6 – 2007 – Score 1.00 – 2007/t000392.txt

So, I kind of believe that we ’ re in like the”cave-painting”era of computer interfaces. Like, they ’ re very kind of â they don ’ t go as deep or as emotionally engaging as they possibly could be and I ’ d like to change all that. Hit me. OK. So I mean, this is the kind of status quo interface, right? It ’ s very flat, kind of rigid. And OK, so you could sex it up and like go to a much more lickable Mac, you know, but really it ’ s the kind of same old crap we ’ ve had for the last, you know, 30 years. Like I think we really put up with a lot of crap with our computers. I mean it ’ s point and click, it ’ s like the menus, icons, it ’ s all the kind of same thing. And so one kind of information space that I take inspiration from is my real desk. It ’ s so much more subtle, so much more visceral â you know, what ’ s visible, what ’ s not. And I ’ d like to bring that experience to the desktop. So I kind of have a â this is BumpTop. It ’ s kind of like a new approach to desktop computing. So you can bump things â they ’ re all physically, you know, manipulable and stuff. And instead of that point and click, it ’ s like a push and pull, things collide as you ’ d expect them. Just like on my real desk, I can â let me just grab these guys â I can turn things into piles instead of just the folders that we have. And once things are in a pile I can browse them by throwing them into a grid, or you know, flip through them like a book or I can lay them out like a deck of cards. When they ’ re laid out, I can pull things to new locations or delete things or just quickly sort a whole pile, you know, just immediately, right? And then, it ’ s all smoothly animated, instead of these jarring changes you see in today ’ s interfaces. Also, if I want to add something to a pile, well, how do I do that? I just toss it to the pile, and it ’ s added right to the top. It ’ s a kind of nice way. Also some of the stuff we can do is, for these individual icons we thought â I mean, how can we play with the idea of an icon, and push that further? And one of the things I can do is make it bigger if I want to emphasize it and make it more important. But what ’ s really cool is that since there ’ s a physics simulation running under this, it ’ s actually heavier. So the lighter stuff doesn ’ t really move but if I throw it at the lighter guys, right? So it ’ s cute, but it ’ s also like a subtle channel of conveying information, right? This is heavy so it feels more important. So it ’ s kind of cool. Despite computers everywhere paper really hasn ’ t disappeared, because it has a lot of, I think, valuable properties. And some of those we wanted to transfer to the icons in our system. So one of the things you can do to our icons, just like paper, is crease them and fold them, just like paper. Remember, you know, something for later. Or if you want to be destructive, you can just crumple it up and, you know, toss it to the corner. Also just like paper, around our workspace we ’ ll pin things up to the wall to remember them later, and I can do the same thing here, and you know, you ’ ll see post-it notes and things like that around people ’ s offices. And I can pull them off when I want to work with them. So, one of the criticisms of this kind of approach to organization is that, you know,”Okay, well my real desk is really messy. I don ’ t want that mess on my computer.”So one thing we have for that is like a grid align, kind of â so you get that more traditional desktop. Things are kind of grid aligned. More boring, but you still have that kind of colliding and bumping. And you can still do fun things like make shelves on your desktop. Let ’ s just break this shelf. Okay, that shelf broke. I think beyond the icons, I think another really cool domain for this software â I think it applies to more than just icons and your desktop â but browsing photographs. I think you can really enrich the way we browse our photographs and bring it to that kind of shoebox of, you know, photos with your family on the kitchen table kind of thing. I can toss these things around. They ’ re so much more tangible and touchable â and you know I can double-click on something to take a look at it. And I can do all that kind of same stuff I showed you before. So I can pile things up, I can flip through it, I can, you know â okay, let ’ s move this photo to the back, let ’ s delete this guy here, and I think it ’ s just a much more rich kind of way of interacting with your information. And that ’ s BumpTop. Thanks!

Text Sample 967: NEG Dim 6 – 1998 – Score 1.00 – 1998/t000113.txt

Sheryl Shade : Hi, Aimee. Aimee Mullins : Hi. SS : Aimee and I thought we ’ d just talk a little bit, and I wanted her to tell all of you what makes her a distinctive athlete. AM : Well, for those of you who have seen the picture in the little bio â it might have given it away â I ’ m a double amputee, and I was born without fibulas in both legs. I was amputated at age one, and I ’ ve been running like hell ever since, all over the place. SS : Well, why don ’ t you tell them how you got to Georgetown â why don ’ t we start there? Why don ’ t we start there? AM : I ’ m a senior in Georgetown in the Foreign Service program. I won a full academic scholarship out of high school. They pick three students out of the nation every year to get involved in international affairs, and so I won a full ride to Georgetown and I ’ ve been there for four years. Love it. SS : When Aimee got there, she decided that she ’ s, kind of, curious about track and field, so she decided to call someone and start asking about it. So, why don ’ t you tell that story? AM : Yeah. Well, I guess I ’ ve always been involved in sports. I played softball for five years growing up. I skied competitively throughout high school, and I got a little restless in college because I wasn ’ t doing anything for about a year or two sports-wise. And I ’ d never competed on a disabled level, you know â I ’ d always competed against other able-bodied athletes. That ’ s all I ’ d ever known. In fact, I ’ d never even met another amputee until I was 17. And I heard that they do these track meets with all disabled runners, and I figured,”Oh, I don ’ t know about this, but before I judge it, let me go see what it ’ s all about.”So, I booked myself a flight to Boston in ’ 95, 19 years old and definitely the dark horse candidate at this race. I ’ d never done it before. I went out on a gravel track a couple of weeks before this meet to see how far I could run, and about 50 meters was enough for me, panting and heaving. And I had these legs that were made of a wood and plastic compound, attached with Velcro straps â big, thick, five-ply wool socks on â you know, not the most comfortable things, but all I ’ d ever known. And I ’ m up there in Boston against people wearing legs made of all things â carbon graphite and, you know, shock absorbers in them and all sorts of things â and they ’ re all looking at me like, OK, we know who ’ s not going to win this race. And, I mean, I went up there expecting â I don ’ t know what I was expecting â but, you know, when I saw a man who was missing an entire leg go up to the high jump, hop on one leg to the high jump and clear it at six feet, two inches... Dan O ’ Brien jumped 5 ’ 11”in ’ 96 in Atlanta, I mean, if it just gives you a comparison of â these are truly accomplished athletes, without qualifying that word”athlete.”And so I decided to give this a shot : heart pounding, I ran my first race and I beat the national record-holder by three hundredths of a second, and became the new national record-holder on my first try out. And, you know, people said,”Aimee, you know, you ’ ve got speed â you ’ ve got natural speed â but you don ’ t have any skill or finesse going down that track. You were all over the place. We all saw how hard you were working.”And so I decided to call the track coach at Georgetown. And I thank god I didn ’ t know just how huge this man is in the track and field world. He ’ s coached five Olympians, and the man ’ s office is lined from floor to ceiling with All America certificates of all these athletes he ’ s coached. He ’ s just a rather intimidating figure. And I called him up and said,”Listen, I ran one race and I won...””I want to see if I can, you know â I need to just see if I can sit in on some of your practices, see what drills you do and whatever.”That ’ s all I wanted â just two practices.”Can I just sit in and see what you do?”And he said,”Well, we should meet first, before we decide anything.”You know, he ’ s thinking,”What am I getting myself into?”So, I met the man, walked in his office, and saw these posters and magazine covers of people he has coached. And we got to talking, and it turned out to be a great partnership because he ’ d never coached a disabled athlete, so therefore he had no preconceived notions of what I was or wasn ’ t capable of, and I ’ d never been coached before. So this was like,”Here we go â let ’ s start on this trip.”So he started giving me four days a week of his lunch break, his free time, and I would come up to the track and train with him. So that ’ s how I met Frank. That was fall of ’ 95. But then, by the time that winter was rolling around, he said,”You know, you ’ re good enough. You can run on our women ’ s track team here.”And I said,”No, come on.”And he said,”No, no, really. You can. You can run with our women ’ s track team.”In the spring of 1996, with my goal of making the U. S. Paralympic team that May coming up full speed, I joined the women ’ s track team. And no disabled person had ever done that â run at a collegiate level. So I don ’ t know, it started to become an interesting mix. SS : Well, on your way to the Olympics, a couple of memorable events happened at Georgetown. Why don ’ t you just tell them? AM : Yes, well, you know, I ’ d won everything as far as the disabled meets â everything I competed in â and, you know, training in Georgetown and knowing that I was going to have to get used to seeing the backs of all these women ’ s shirts â you know, I ’ m running against the next Flo-Jo â and they ’ re all looking at me like,”Hmm, what ’ s, you know, what ’ s going on here?”And putting on my Georgetown uniform and going out there and knowing that, you know, in order to become better â and I ’ m already the best in the country â you know, you have to train with people who are inherently better than you. And I went out there and made it to the Big East, which was sort of the championship race at the end of the season. It was really, really hot. And it ’ s the first â I had just gotten these new sprinting legs that you see in that bio, and I didn ’ t realize at that time that the amount of sweating I would be doing in the sock â it actually acted like a lubricant and I ’ d be, kind of, pistoning in the socket. And at about 85 meters of my 100 meters sprint, in all my glory, I came out of my leg. Like, I almost came out of it, in front of, like, 5, 000 people. And I, I mean, was just mortified â because I was signed up for the 200, you know, which went off in a half hour. I went to my coach :”Please, don ’ t make me do this.”I can ’ t do this in front of all those people. My legs will come off. And if it came off at 85 there ’ s no way I ’ m going 200 meters. And he just sat there like this. My pleas fell on deaf ears, thank god. Because you know, the man is from Brooklyn ; he ’ s a big man. He says,”Aimee, so what if your leg falls off? You pick it up, you put the damn thing back on, and finish the goddamn race!”And I did. So, he kept me in line. He kept me on the right track. SS : So, then Aimee makes it to the 1996 Paralympics, and she ’ s all excited. Her family ’ s coming down â it ’ s a big deal. It ’ s now two years that you ’ ve been running? AM : No, a year. SS : A year. And why don ’ t you tell them what happened right before you go run your race? AM : Okay, well, Atlanta. The Paralympics, just for a little bit of clarification, are the Olympics for people with physical disabilities â amputees, persons with cerebral palsy, and wheelchair athletes â as opposed to the Special Olympics, which deals with people with mental disabilities. So, here we are, a week after the Olympics and down at Atlanta, and I ’ m just blown away by the fact that just a year ago, I got out on a gravel track and couldn ’ t run 50 meters. And so, here I am â never lost. I set new records at the U. S. Nationals â the Olympic trials â that May, and was sure that I was coming home with the gold. I was also the only, what they call”bilateral BK”â below the knee. I was the only woman who would be doing the long jump. I had just done the long jump, and a guy who was missing two legs came up to me and says,”How do you do that? You know, we ’ re supposed to have a planar foot, so we can ’ t get off on the springboard.”I said,”Well, I just did it. No one told me that.”So, it ’ s funny â I ’ m three inches within the world record â and kept on from that point, you know, so I ’ m signed up in the long jump â signed up? No, I made it for the long jump and the 100-meter. And I ’ m sure of it, you know? I made the front page of my hometown paper that I delivered for six years, you know? It was, like, this is my time for shine. And we ’ re at the trainee warm-up track, which is a few blocks away from the Olympic stadium. These legs that I was on, which I ’ ll take out right now â I was the first person in the world on these legs. I was the guinea pig., I ’ m telling you, this was, like â talk about a tourist attraction. Everyone was taking pictures â”What is this girl running on?”And I ’ m always looking around, like, where is my competition? It ’ s my first international meet. I tried to get it out of anybody I could, you know,”Who am I running against here?””Oh, Aimee, we ’ ll have to get back to you on that one.”I wanted to find out times.”Don ’ t worry, you ’ re doing great.”This is 20 minutes before my race in the Olympic stadium, and they post the heat sheets. And I go over and look. And my fastest time, which was the world record, was 15. 77. Then I ’ m looking : the next lane, lane two, is 12. 8. Lane three is 12. 5. Lane four is 12. 2. I said,”What ’ s going on?”And they shove us all into the shuttle bus, and all the women there are missing a hand. So, I ’ m just, like â they ’ re all looking at me like ’ which one of these is not like the other, ’ you know? I ’ m sitting there, like,”Oh, my god. Oh, my god.”You know, I ’ d never lost anything, like, whether it would be the scholarship or, you know, I ’ d won five golds when I skied. In everything, I came in first. And Georgetown â that was great. I was losing, but it was the best training because this was Atlanta. Here we are, like, crĂ¨me de la crĂ¨me, and there is no doubt about it, that I ’ m going to lose big. And, you know, I ’ m just thinking,”Oh, my god, my whole family got in a van and drove down here from Pennsylvania.”And, you know, I was the only female U. S. sprinter. So they call us out and, you know â”Ladies, you have one minute.”And I remember putting my blocks in and just feeling horrified because there was just this murmur coming over the crowd, like, the ones who are close enough to the starting line to see. And I ’ m like,”I know! Look! This isn ’ t right.”And I ’ m thinking that ’ s my last card to play here ; if I ’ m not going to beat these girls, I ’ m going to mess their heads a little, you know? I mean, it was definitely the”Rocky IV”sensation of me versus Germany, and everyone else â Estonia and Poland â was in this heat. And the gun went off, and all I remember was finishing last and fighting back tears of frustration and incredible â incredible â this feeling of just being overwhelmed. And I had to think,”Why did I do this?”If I had won everything â but it was like, what was the point? All this training â I had transformed my life. I became a collegiate athlete, you know. I became an Olympic athlete. And it made me really think about how the achievement was getting there. I mean, the fact that I set my sights, just a year and three months before, on becoming an Olympic athlete and saying,”Here ’ s my life going in this direction â and I want to take it here for a while, and just seeing how far I could push it.”And the fact that I asked for help â how many people jumped on board? How many people gave of their time and their expertise, and their patience, to deal with me? And that was this collective glory â that there was, you know, 50 people behind me that had joined in this incredible experience of going to Atlanta. So, I apply this sort of philosophy now to everything I do : sitting back and realizing the progression, how far you ’ ve come at this day to this goal, you know. It ’ s important to focus on a goal, I think, but also recognize the progression on the way there and how you ’ ve grown as a person. That ’ s the achievement, I think. That ’ s the real achievement. SS : Why don ’ t you show them your legs? AM : Oh, sure. SS : You know, show us more than one set of legs. AM : Well, these are my pretty legs. No, these are my cosmetic legs, actually, and they ’ re absolutely beautiful. You ’ ve got to come up and see them. There are hair follicles on them, and I can paint my toenails. And, seriously, like, I can wear heels. Like, you guys don ’ t understand what that ’ s like to be able to just go into a shoe store and buy whatever you want. SS : You got to pick your height? AM : I got to pick my height, exactly. Patrick Ewing, who played for Georgetown in the ’ 80s, comes back every summer. And I had incessant fun making fun of him in the training room because he ’ d come in with foot injuries. I ’ m like,”Get it off! Don ’ t worry about it, you know. You can be eight feet tall. Just take them off.”He didn ’ t find it as humorous as I did, anyway. OK, now, these are my sprinting legs, made of carbon graphite, like I said, and I ’ ve got to make sure I ’ ve got the right socket. No, I ’ ve got so many legs in here. These are â do you want to hold that actually? That ’ s another leg I have for, like, tennis and softball. It has a shock absorber in it so it, like,”Shhhh,”makes this neat sound when you jump around on it. All right. And then this is the silicon sheath I roll over, to keep it on. Which, when I sweat, you know, I ’ m pistoning out of it. SS : Are you a different height? AM : In these? SS : In these. AM : I don ’ t know. I don ’ t think so. I may be a little taller. I actually can put both of them on. SS : She can ’ t really stand on these legs. She has to be moving, so... AM : Yeah, I definitely have to be moving, and balance is a little bit of an art in them. But without having the silicon sock, I ’ m just going to try slip in it. And so, I run on these, and have shocked half the world on these. These are supposed to simulate the actual form of a sprinter when they run. If you ever watch a sprinter, the ball of their foot is the only thing that ever hits the track. So when I stand in these legs, my hamstring and my glutes are contracted, as they would be had I had feet and were standing on the ball of my feet. AM : It ’ s a company in San Diego called Flex-Foot. And I was a guinea pig, as I hope to continue to be in every new form of prosthetic limbs that come out. But actually these, like I said, are still the actual prototype. I need to get some new ones because the last meet

I was at, they were everywhere. You know, it ’ s like a big â it ’ s come full circle. Moderator : Aimee and the designer of them will be at TEDMED 2, and we ’ ll talk about the design of them. AM : Yes, we ’ ll do that. SS : Yes, there you go. AM : So, these are the sprint legs, and I can put my other... SS : Can you tell about who designed your other legs? AM : Yes. These I got in a place called Bournemouth, England, about two hours south of London, and I ’ m the only person in the United States with these, which is a crime because they are so beautiful. And I don ’ t even mean, like, because of the toes and everything. For me, while I ’ m such a serious athlete on the track, I want to be feminine off the track, and I think it ’ s so important not to be limited in any capacity, whether it ’ s, you know, your mobility or even fashion. I mean, I love the fact that I can go in anywhere and pick out what I want â the shoes I want, the skirts I want â and I ’ m hoping to try to bring these over here and make them accessible to a lot of people. They ’ re also silicon. This is a really basic, basic prosthetic limb under here. It ’ s like a Barbie foot under this. It is. It ’ s just stuck in this position, so I have to wear a two-inch heel. And, I mean, it ’ s really â let me take this off so you can see it. I don ’ t know how good you can see it, but, like, it really is. There ’ re veins on the feet, and then my heel is pink, and my Achilles ’ tendon â that moves a little bit. And it ’ s really an amazing store. I got them a year and two weeks ago. And this is just a silicon piece of skin. I mean, what happened was, two years ago this man in Belgium was saying,”God, if I can go to Madame Tussauds ’ wax museum and see Jerry Hall replicated down to the color of her eyes, looking so real as if she breathed, why can ’ t they build a limb for someone that looks like a leg, or an arm, or a hand?”I mean, they make ears for burn victims. They do amazing stuff with silicon. SS : Two weeks ago, Aimee was up for the Arthur Ashe award at the ESPYs. And she came into town and she rushed around and she said,”I have to buy some new shoes!”We ’ re an hour before the ESPYs, and she thought she ’ d gotten a two-inch heel but she ’ d actually bought a three-inch heel. AM : And this poses a problem for me, because it means I ’ m walking like that all night long. SS : For 45 minutes. Luckily, the hotel was terrific. They got someone to come in and saw off the shoes. AM : I said to the receptionist â I mean, I am just harried, and Sheryl ’ s at my side â I said,”Look, do you have anybody here who could help me? Because I have this problem...”You know, at first they were just going to write me off, like,”If you don ’ t like your shoes, sorry. It ’ s too late.””No, no, no, no. I ’ ve got these special feet that need a two-inch heel. I have a three-inch heel. I need a little bit off.”They didn ’ t even want to go there. They didn ’ t even want to touch that one. They just did it. No, these legs are great. I ’ m actually going back in a couple of weeks to get some improvements. I want to get legs like these made for flat feet so I can wear sneakers, because I can ’ t with these ones. So... Moderator : That ’ s it. SS : That ’ s Aimee Mullins.

Text Sample 968: NEG Dim 6 – 1998 – Score 1.00 – 1998/t000250.txt

As a clergyman, you can imagine how out of place I feel. I feel like a fish out of water, or maybe an owl out of the air. I was preaching in San Jose some time ago, and my friend Mark Kvamme, who helped introduce me to this conference, brought several CEOs and leaders of some of the companies here in the Silicon Valley to have breakfast with me, or I with them. And I was so stimulated. And had such — it was an eye-opening experience to hear them talk about the world that is yet to come through technology and science. I know that we ’ re near the end of this conference, and some of you may be wondering why they have a speaker from the field of religion. Richard can answer that, because he made that decision. But some years ago I was on an elevator in Philadelphia, coming down. I was to address a conference at a hotel. And on that elevator a man said,"I hear Billy Graham is staying in this hotel."And another man looked in my direction and said,"Yes, there he is. He ' s on this elevator with us."And this man looked me up and down for about 10 seconds, and he said,"My, what an anticlimax!"I hope that you won ' t feel that these few moments with me is not a — is an anticlimax, after all these tremendous talks that you ' ve heard, and addresses, which I intend to listen to every one of them. But I was on an airplane in the east some years ago, and the man sitting across the aisle from me was the mayor of Charlotte, North Carolina. His name was John Belk. Some of you will probably know him. And there was a drunk man on there, and he got up out of his seat two or three times, and he was making everybody upset by what he was trying to do. And he was slapping the stewardess and pinching her as she went by, and everybody was upset with him. And finally, John Belk said,"Do you know who ' s sitting here?"And the man said,"No, who?"He said,"It ' s Billy Graham, the preacher."He said,"You don ' t say!"And he turned to me, and he said,"Put her there!"He said,"Your sermons have certainly helped me."And I suppose that that ' s true with thousands of people. I know that as you have been peering into the future, and as we ' ve heard some of it here tonight, I would like to live in that age and see what is going to be. But I won ' t, because I ' m 80 years old. This is my eightieth year, and I know that my time is brief. I have phlebitis at the moment, in both legs, and that ' s the reason that I had to have a little help in getting up here, because I have Parkinson ' s disease in addition to that, and some other problems that I won ' t talk about. But this is not the first time that we ' ve had a technological revolution. We ' ve had others. And there ' s one that I want to talk about. In one generation, the nation of the people of Israel had a tremendous and dramatic change that made them a great power in the Near East. A man by the name of David came to the throne, and King David became one of the great leaders of his generation. He was a man of tremendous leadership. He had the favor of God with him. He was a brilliant poet, philosopher, writer, soldier — with strategies in battle and conflict that people study even today. But about two centuries before David, the Hittites had discovered the secret of smelting and processing of iron, and, slowly, that skill spread. But they wouldn ' t allow the Israelis to look into it, or to have any. But David changed all of that, and he introduced the Iron Age to Israel. And the Bible says that David laid up great stores of iron, and which archaeologists have found, that in present-day Palestine, there are evidences of that generation. Now, instead of crude tools made of sticks and stones, Israel now had iron plows, and sickles, and hoes and military weapons. And in the course of one generation, Israel was completely changed. The introduction of iron, in some ways, had an impact a little bit like the microchip has had on our generation. And David found that there were many problems that technology could not solve. There were many problems still left. And they ' re still with us, and you haven ' t solved them, and I haven ' t heard anybody here speak to that. How do we solve these three problems that I ' d like to mention? The first one that David saw was human evil. Where does it come from? How do we solve it? Over again and again in the Psalms, which Gladstone said was the greatest book in the world, David describes the evils of the human race. And yet he says,"He restores my soul."Have you ever thought about what a contradiction we are? On one hand, we can probe the deepest secrets of the universe and dramatically push back the frontiers of technology, as this conference vividly demonstrates. We ' ve seen under the sea, three miles down, or galaxies hundreds of billions of years out in the future. But on the other hand, something is wrong. Our battleships, our soldiers, are on a frontier now, almost ready to go to war with Iraq. Now, what causes this? Why do we have these wars in every generation, and in every part of the world? And revolutions? We can ’ t get along with other people, even in our own families. We find ourselves in the paralyzing grip of self-destructive habits we can ’ t break. Racism and injustice and violence sweep our world, bringing a tragic harvest of heartache and death. Even the most sophisticated among us seem powerless to break this cycle. I would like to see Oracle take up that, or some other technological geniuses work on this. How do we change man, so that he doesn ’ t lie and cheat, and our newspapers are not filled with stories of fraud in business or labor or athletics or wherever? The Bible says the problem is within us, within our hearts and our souls. Our problem is that we are separated from our Creator, which we call God, and we need to have our souls restored, something only God can do. Jesus said,”For out of the heart come evil thoughts : murders, sexual immorality, theft, false testimonies, slander.”The British philosopher Bertrand Russell was not a religious man, but he said,”It ’ s in our hearts that the evil lies, and it ’ s from our hearts that it must be plucked out.”Albert Einstein — I was just talking to someone, when I was speaking at Princeton, and I met Mr. Einstein. He didn ’ t have a doctor ’ s degree, because he said nobody was qualified to give him one. But he made this statement. He said,”It ’ s easier to denature plutonium than to denature the evil spirit of man.”And many of you, I ’ m sure, have thought about that and puzzled over it. You ’ ve seen people take beneficial technological advances, such as the Internet we ’ ve heard about tonight, and twist them into something corrupting. You ’ ve seen brilliant people devise computer viruses that bring down whole systems. The Oklahoma City bombing was simple technology, horribly used. The problem is not technology. The problem is the person or persons using it. King David said that he knew the depths of his own soul. He couldn ’ t free himself from personal problems and personal evils that included murder and adultery. Yet King David sought God ’ s forgiveness, and said,”You can restore my soul.”You see, the Bible teaches that we ’ re more than a body and a mind. We are a soul. And there ’ s something inside of us that is beyond our understanding. That ’ s the part of us that yearns for God, or something more than we find in technology. Your soul is that part of you that yearns for meaning in life, and which seeks for something beyond this life. It ’ s the part of you that yearns, really, for God. I find [that] young people all over the world are searching for something. They don ’ t know what it is. I speak at many universities, and I have many questions and answer periods, and whether it ’ s Cambridge, or Harvard, or Oxford — I ’ ve spoken at all of those universities. I ’ m going to Harvard in about three or four — no, it ’ s about two months from now — to give a lecture. And I ’ ll be asked the same questions that I was asked the last few times I ’ ve been there. And it ’ ll be on these questions : where did I come from? Why am I here? Where am I going? What ’ s life all about? Why am I here? Even if you have no religious belief, there are times when you wonder that there ’ s something else. Thomas Edison also said,”When you see everything that happens in the world of science, and in the working of the universe, you cannot deny that there ’ s a captain on the bridge.”I remember once, I sat beside Mrs. Gorbachev at a White House dinner. I went to Ambassador Dobrynin, whom I knew very well. And I ’ d been to Russia several times under the Communists, and they ’ d given me marvelous freedom that I didn ’ t expect. And I knew Mr. Dobrynin very well, and I said,”I ’ m going to sit beside Mrs. Gorbachev tonight. What shall I talk to her about?”And he surprised me with the answer. He said,”Talk to her about religion and philosophy. That ’ s what she ’ s really interested in.”I was a little bit surprised, but that evening that ’ s what we talked about, and it was a stimulating conversation. And afterward, she said,”You know, I ’ m an atheist, but I know that there ’ s something up there higher than we are.”The second problem that King David realized he could not solve was the problem of human suffering. Writing the oldest book in the world was Job, and he said,”Man is born unto trouble as the sparks fly upward.”Yes, to be sure, science has done much to push back certain types of human suffering. But I ’ m — in a few months, I ’ ll be 80 years of age. I admit that I ’ m very grateful for all the medical advances that have kept me in relatively good health all these years. My doctors at the Mayo Clinic urged me not to take this trip out here to this — to be here. I haven ’ t given a talk in nearly four months. And when you speak as much as I do, three or four times a day, you get rusty. That ’ s the reason I ’ m using this podium and using these notes. Every time you ever hear me on the television or somewhere, I ’ m ad-libbing. I ’ m not reading. I never read an address. I never read a speech or a talk or a lecture. I talk ad lib. But tonight, I ’ ve got some notes here so that if I begin to forget, which I do sometimes, I ’ ve got something I can turn to. But even here among us, most — in the most advanced society in the world, we have poverty. We have families that self-destruct, friends that betray us. Unbearable psychological pressures bear down on us. I ’ ve never met a person in the world that didn ’ t have a problem or a worry. Why do we suffer? It ’ s an age- old question that we haven ’ t answered. Yet David again and again said that he would turn to God. He said,”The Lord is my shepherd.”The final problem that David knew he could not solve was death. Many commentators have said that death is the forbidden subject of our generation. Most people live as if they ’ re never going to die. Technology projects the myth of control over our mortality. We see people on our screens. Marilyn Monroe is just as beautiful on the screen as she was in person, and our — many young people think she ’ s still alive. They don ’ t know that she ’ s dead. Or Clark Gable, or whoever it is. The old stars, they come to life. And they ’ re — they ’ re just as great on that screen as they were in person. But death is inevitable. I spoke some time ago to a joint session of Congress, last year. And we were meeting in that room, the statue room. About 300 of them were there. And I said,”There ’ s one thing that we have in common in this room, all of us together, whether Republican or Democrat, or whoever.”I said,”We ’ re all going to die. And we have that in common with all these great men of the past that are staring down at us.”And it ’ s often difficult for young people to understand that. It ’ s difficult for them to understand that they ’ re going to die. As the ancient writer of Ecclesiastes wrote, he said, there ’ s every activity under heaven. There ’ s a time to be born, and there ’ s a time to die. I ’ ve stood at the deathbed of several famous people, whom you would know. I ’ ve talked to them. I ’ ve seen them in those agonizing moments when they were scared to death. And yet, a few years earlier, death never crossed their mind. I talked to a woman this past week whose father was a famous doctor. She said he never thought of God, never talked about God, didn ’ t believe in God. He was an atheist. But she said, as he came to die, he sat up on the side of the bed one day, and he asked the nurse if he could see the chaplain. And he said, for the first time in his life he ’ d thought about the inevitable, and about God. Was there a God? A few years ago, a university student asked me,”What is the greatest surprise in your life?”And I said,”The greatest surprise in my life is the brevity of life. It passes so fast.”But it does not need to have to be that way. Wernher von Braun, in the aftermath of World War II concluded, quote :”science and religion are not antagonists. On the contrary, they ’ re sisters.”He put it on a personal basis. I knew Dr. von Braun very well. And he said,”Speaking for myself, I can only say that the grandeur of the cosmos serves only to confirm a belief in the certainty of a creator.”He also said,”In our search to know God, I ’ ve come to believe that the life of Jesus Christ should be the focus of our efforts and inspiration. The reality of this life and His resurrection is the hope of mankind.”I ’ ve done a lot of speaking in Germany and in France, and in different parts of the world — 105 countries it ’ s been my privilege to speak in. And I was invited one day to visit Chancellor Adenauer, who was looked upon as sort of the founder of modern Germany, since the war. And he once — and he said to me, he said,”Young man.”He said,”Do you believe in the resurrection of Jesus Christ?”And I said,”Sir, I do.”He said,”So do I.”He said,”When I leave office, I ’ m going to spend my time writing a book on why Jesus Christ rose again, and why it ’ s so important to believe that.”In one of his plays, Alexander Solzhenitsyn depicts a man dying, who says to those gathered around his bed,”The moment when it ’ s terrible to feel regret is when one is dying.”How should one live in order not to feel regret when one is dying? Blaise Pascal asked exactly that question in seventeenth-century France. Pascal has been called the architect of modern civilization. He was a brilliant scientist at the frontiers of mathematics, even as a teenager. He is viewed by many as the founder of the probability theory, and a creator of the first model of a computer. And of course, you are all familiar with the computer language named for him. Pascal explored in depth our human dilemmas of evil, suffering and death. He was astounded at the phenomenon we ’ ve been considering : that people can achieve extraordinary heights in science, the arts and human enterprise, yet they also are full of anger, hypocrisy and have — and self- hatreds. Pascal saw us as a remarkable mixture of genius and self-delusion. On November 23, 1654, Pascal had a profound religious experience. He wrote in his journal these words :”I submit myself, absolutely, to Jesus Christ, my redeemer.”A French historian said, two centuries later,”Seldom has so mighty an intellect submitted with such humility to the authority of Jesus Christ.”Pascal came to believe not only the love and the grace of God could bring us back into harmony, but he believed that his own sins and failures could be forgiven, and that when he died he would go to a place called heaven. He experienced it in a way that went beyond scientific observation and reason. It was he who penned the well-known words,”The heart has its reasons, which reason knows not of.”Equally well known is Pascal ’ s Wager. Essentially, he said this :”if you bet on God, and open yourself to his love, you lose nothing, even if you ’ re wrong. But if instead you bet that there is no God, then you can lose it all, in this life and the life to come.”For Pascal, scientific knowledge paled beside the knowledge of God. The knowledge of God was far beyond anything that ever crossed his mind. He was ready to face him when he died at the age of 39. King David lived to be 70, a long time in his era. Yet he too had to face death, and he wrote these words :”even though I walk through the valley of the shadow of death, I will fear no evil, for you are with me.”This was David ’ s answer to three dilemmas of evil, suffering and death. It can be yours, as well, as you seek the living God and allow him to fill your life and give you hope for the future. When I was 17 years of age, I was born and reared on a farm in North Carolina. I milked cows every morning, and I had to milk the same cows every evening when I came home from school. And there were 20 of them that I had — that I was responsible for, and I worked on the farm and tried to keep up with my studies. I didn ’ t make good grades in high school. I didn ’ t make them in college, until something happened in my heart. One day, I was faced face-to-face with Christ. He said,”I am the way, the truth and the life.”Can you imagine that?”I am the truth. I ’ m the embodiment of all truth.”He was a liar. Or he was insane. Or he was what he claimed to be. Which was he? I had to make that decision. I couldn ’ t prove it. I couldn ’ t take it to a laboratory and experiment with it. But by faith I said, I believe him, and he came into my heart and changed my life. And now I ’ m ready, when I hear that call, to go into the presence of God. Thank you, and God bless all of you. Thank you for the privilege. It was great. Richard Wurman : You did it. Thanks.

Text Sample 969: NEG Dim 6 – 1998 – Score 2.00 – 1998/t000134.txt

’ Theme and variations ’ is one of those forms that require a certain kind of intellectual activity, because you are always comparing the variation with the theme that you hold in your mind. You might say that the theme is nature and everything that follows is a variation on the subject. I was asked, I guess about six years ago, to do a series of paintings that in some way would celebrate the birth of Piero della Francesca. And it was very difficult for me to imagine how to paint pictures that were based on Piero until I realized that I could look at Piero as nature — that I would have the same attitude towards looking at Piero della Francesca as I would if I were looking out a window at a tree. And that was enormously liberating to me. Perhaps it ’ s not a very insightful observation, but that really started me on a path to be able to do a kind of theme and variations based on a work by Piero, in this case that remarkable painting that ’ s in the Uffizi,”The Duke of Montefeltro,”who faces his consort, Battista. Once I realized that I could take some liberties with the subject, I did the following series of drawings. That ’ s the real Piero della Francesca — one of the greatest portraits in human history. And these, I ’ ll just show these without comment. It ’ s just a series of variations on the head of the Duke of Montefeltro, who ’ s a great, great figure in the Renaissance, and probably the basis for Machiavelli ’ s”The Prince.”He apparently lost an eye in battle, which is why he is always shown in profile. And this is Battista. And then I decided I could move them around a little bit — so that for the first time in history, they ’ re facing the same direction. Whoops! Passed each other. And then a visitor from another painting by Piero, this is from”The Resurrection of Christ”— as though the cast had just gotten of the set to have a chat. And now, four large panels : this is upper left ; upper right ; lower left ; lower right. Incidentally, I ’ ve never understood the conflict between abstraction and naturalism. Since all paintings are inherently abstract to begin with there doesn ’ t seem to be an argument there. On another subject — — I was driving in the country one day with my wife, and I saw this sign, and I said,”That is a fabulous piece of design.”And she said,”What are you talking about?”I said,”Well, it ’ s so persuasive, because the purpose of that sign is to get you into the garage, and since most people are so suspicious of garages, and know that they ’ re going to be ripped off, they use the word ’ reliable. ’ But everybody says they ’ re reliable. But, reliable Dutchman”— —”Fantastic!”Because as soon as you hear the word Dutchman — which is an archaic word, nobody calls Dutch people”Dutchmen”anymore — but as soon as you hear Dutchman, you get this picture of the kid with his finger in the dike, preventing the thing from falling and flooding Holland, and so on. And so the entire issue is detoxified by the use of”Dutchman.”Now, if you think I ’ m exaggerating at all in this, all you have to do is substitute something else, like”Indonesian.”Or even”French.”Now,”Swiss”works, but you know it ’ s going to cost a lot of money. I ’ m going to take you quickly through the actual process of doing a poster. I do a lot of work for the School of Visual Arts, where I teach, and the director of this school — a remarkable man named Silas Rhodes — often gives you a piece of text and he says,”Do something with this.”And so he did. And this was the text —”In words as fashion the same rule will hold / Alike fantastic if too new or old / Be not the first by whom the new are tried / Nor yet the last to lay the old aside.”I could make nothing of that. And I really struggled with this one. And the first thing I did, which was sort of in the absence of another idea, was say I ’ ll sort of write it out and make some words big, and I ’ ll have some kind of design on the back somehow, and I was hoping — as one often does — to stumble into something. So I took another crack at it — you ’ ve got to keep it moving — and I Xeroxed some words on pieces of colored paper and I pasted them down on an ugly board. I thought that something would come out of it, like”Words rule fantastic new old first last Pope”because it ’ s by Alexander Pope, but I sort of made a mess out of it, and then I thought I ’ d repeat it in some way so it was legible. So, it was going nowhere. Sometimes, in the middle of a resistant problem, I write down things that I know about it. But you can see the beginning of an idea there, because you can see the word”new”emerging from the”old.”That ’ s what happens. There ’ s a relationship between the old and the new ; the new emerges from the context of the old. And then I did some variations of it, but it still wasn ’ t coalescing graphically at all. I had this other version which had something interesting about it in terms of being able to put it together in your mind from clues. The W was clearly a W, the N was clearly an N, even though they were very fragmentary and there wasn ’ t a lot of information in it. Then I got the words”new”and”old”and now I had regressed back to a point where there seemed to be no return. I was really desperate at this point. And so, I do something I ’ m truthfully ashamed of, which is that I took two drawings I had made for another purpose and I put them together. It says”dreams”at the top. And I was going to do a thing, I say,”Well, change the copy. Let it say something about dreams, and come to SVA and you ’ ll sort of fulfill your dreams.”But, to my credit, I was so embarrassed about doing that that I never submitted this sketch. And, finally, I arrived at the following solution. Now, it doesn ’ t look terribly interesting, but it does have something that distinguishes it from a lot of other posters. For one thing, it transgresses the idea of what a poster ’ s supposed to be, which is to be understood and seen immediately, and not explained. I remember hearing all of you in the graphic arts —”If you have to explain it, it ain ’ t working.”And one day I woke up and I said,”Well, suppose that ’ s not true?”So here ’ s what it says in my explanation at the bottom left. It says,”Thoughts : This poem is impossible. Silas usually has a better touch with his choice of quotations. This one generates no imagery at all.”I am now exposing myself to my audience, right? Which is something you never want to do professionally.”Maybe the words can make the image without anything else happening. What ’ s the heart of this poem? Don ’ t be trendy if you want to be serious. Is doing the poster this way trendy in itself? I guess one could reduce the idea further by suggesting that the new emerges behind and through the old, like this.”And then I show you a little drawing — you see, you remember that old thing I discarded? Well, I found a way to use it. So, there ’ s that little alternative over there, and I say,”Not bad,”— criticizing myself —”but more didactic than visual. Maybe what wants to be said is that old and the new are locked in a dialectical embrace, a kind of dance where each defines the other.”And then more self-questioning —”Am I being simple-minded? Is this the kind of simple that looks obvious, or the kind that looks profound? There ’ s a significant difference. This could be embarrassing. Actually, I realize fear of embarrassment drives me as much as any ambition. Do you think this sort of thing could really attract a student to the school?”Well, I think there are two fresh things here — two fresh things. One is the sort of willingness to expose myself to a critical audience, and not to suggest that I am confident about what I ’ m doing. And as you know, you have to have a front. I mean — you ’ ve got to be confident ; if you don ’ t believe in your work, who else is going to believe in it? So that ’ s one thing, to introduce the idea of doubt into graphics.

That can be a big contribution. The other thing is to actually give you two solutions for the price of one ; you get the big one and if you don ’ t like that, how about the little one? And that too is a relatively new idea. And here ’ s just a series of experiments where I ask the question of — does a poster have to be square? Now, this is a little illusion. That poster is not folded. It ’ s not folded, that ’ s a photograph and it ’ s cut on the diagonal. Same cheap trick in the upper left- hand corner. And here, a very peculiar poster because, simply because of using the isometric perspective in the computer, it won ’ t sit still in the space. At times, it seems to be wider at the back than the front, and then it shifts. And if you sit here long enough, it ’ ll float off the page into the audience. But, we don ’ t have time. And then an experiment — a little bit about the nature of perspective, where the outside shape is determined by the peculiarity of perspective, but the shape of the bottle — which is identical to the outside shape — is seen frontally. And another piece for the art directors ’ club is”Anna Rees”casting long shadows. This is another poster from the School of Visual Arts. There were 10 artists invited to participate in it, and it was one of those things where it was extremely competitive and I didn ’ t want to be embarrassed, so I worked very hard on this. The idea was — and it was a brilliant idea — was to have 10 posters distributed throughout the city ’ s subway system so every time you got on the subway you ’ d be passing a different poster, all of which had a different idea of what art is. But I was absolutely stuck on the idea of”art is”and trying to determine what art was. But then I gave up and I said,”Well, art is whatever.”And as soon as I said that, I discovered that the word”hat”was hidden in the word”whatever,”and that led me to the inevitable conclusion. But then again, it ’ s on my list of didactic posters. My intent is to have a literary accompaniment that explains the poster, in case you don ’ t get it. Now this says,”Note to the viewer : I thought I might use a visual cliche of our time — Magritte ’ s everyman — to express the idea that art is mystery, continuity and history. I ’ m also convinced that, in an age of computer manipulation, surrealism has become banal, a shadow of its former self. The phrase ’ Art is whatever ’ expresses the current inclusiveness that surrounds art-making — a sort of ’ it ain ’ t what you do, it ’ s the way that you do it ’ notion. The shadow of Magritte falls across the central part of the poster a poetic event that occurs as the shadow man isolates the word ’ hat, ’ hidden in the word ’ whatever. ’ The four hats shown in the poster suggests how art might be defined : as a thing itself, the worth of the thing, the shadow of the thing, and the shape of the thing. Whatever.”OK. And the one that I did not submit, which I still like, I wanted to use the same phrase. There were wonderful experiments by Bruno Munari on letterforms some years ago — sort of, see how far you could go and still be able to read them. And that idea stuck in my head. But then I took the pieces that I had taken off and put them at the bottom. And, of course those are the remains, and they ’ re so labeled. But what really happens is that you read it as :”Art is whatever remains.”Thank you.

Text Sample 970: NEG Dim 6 – 2008 – Score 1.00 – 2008/t002921.txt
The National Portrait Gallery is the place dedicated to presenting great American lives, amazing people. And that ’ s what it ’ s about. We use portraiture as a way to deliver those lives, but that ’ s it. And so I ’ m not going to talk about the painted portrait today. I ’ m going to talk about a program I started there, which, from my point of view, is the proudest thing I did. I started to worry about the fact that a lot of people don ’ t get their portraits painted anymore,

and they ’ re amazing people, and we want to deliver them to future generations. So, how do we do that? And so I came up with the idea of the living self-portrait series. And the living self-portrait series was the idea of basically my being a brush in the hand of amazing people who would come and I would interview. And so what I ’ m going to do is, not so much give you the great hits of that program, as to give you this whole notion of how you encounter people in that kind of situation, what you try to find out about them, and when people deliver and when they don ’ t and why. Now, I had two preconditions. One was that they be American. That ’ s just because, in the nature of the National Portrait Gallery, it ’ s created to look at American lives. That was easy, but then I made the decision, maybe arbitrary, that they needed to be people of a certain age, which at that point, when I created this program, seemed really old. Sixties, seventies, eighties and nineties. For obvious reasons, it doesn ’ t seem that old anymore to me. And why did I do that? Well, for one thing, we ’ re a youth-obsessed culture. And I thought really what we need is an elders program to just sit at the feet of amazing people and hear them talk. But the second part of it — and the older I get, the more convinced I am that that ’ s true. It ’ s amazing what people will say when they know how the story turned out. That ’ s the one advantage that older people have. Well, they have other, little bit of advantage, but they also have some disadvantages, but the one thing they or we have is that we ’ ve reached the point in life where we know how the story turned out. So, we can then go back in our lives, if we ’ ve got an interviewer who gets that, and begin to reflect on how we got there. All of those accidents that wound up creating the life narrative that we inherited. So, I thought okay, now, what is it going to take to make this work? There are many kinds of interviews. We know them. There are the journalist interviews, which are the interrogation that is expected. This is somewhat against resistance and caginess on the part of the interviewee. Then there ’ s the celebrity interview, where it ’ s more important who ’ s asking the question than who answers. That ’ s Barbara Walters and others like that, and we like that. That ’ s Frost-Nixon, where Frost seems to be as important as Nixon in that process. Fair enough. But I wanted interviews that were different. I wanted to be, as I later thought of it, empathic, which is to say, to feel what they wanted to say and to be an agent of their self-revelation. By the way, this was always done in public. This was not an oral history program. This was all about 300 people sitting at the feet of this individual, and having me be the brush in their self-portrait. Now, it turns out that I was pretty good at that. I didn ’ t know it coming into it. And the only reason I really know that is because of one interview I did with Senator William Fulbright, and that was six months after he ’ d had a stroke. And he had never appeared in public since that point. This was not a devastating stroke, but it did affect his speaking and so forth. And I thought it was worth a chance, he thought it was worth a chance, and so we got up on the stage, and we had an hour conversation about his life, and after that a woman rushed up to me, essentially did, and she said,”Where did you train as a doctor?”And I said,”I have no training as a doctor. I never claimed that.”And she said,”Well, something very weird was happening. When he started a sentence, particularly in the early parts of the interview, and paused, you gave him the word, the bridge to get to the end of the sentence, and by the end of it, he was speaking complete sentences on his own.”I didn ’ t know what was going on, but I was so part of the process of getting that out. So I thought, okay, fine, I ’ ve got empathy, or empathy, at any rate, is what ’ s critical to this kind of interview. But then I began to think of other things. Who makes a great interview in this context? It had nothing to do with their intellect, the quality of their intellect. Some of them were very brilliant, some of them were, you know, ordinary people who would never claim to be intellectuals, but it was never about that. It was about their energy. It ’ s energy that creates extraordinary interviews and extraordinary lives. I ’ m convinced of it. And it had nothing to do with the energy of being young. These were people through their 90s. In fact, the first person I interviewed was George Abbott, who was 97, and Abbott was filled with the life force — I guess that ’ s the way I think about it — filled with it. And so he filled the room, and we had an extraordinary conversation. He was supposed to be the toughest interview that anybody would ever do because he was famous for being silent, for never ever saying anything except maybe a word or two. And, in fact, he did wind up opening up — by the way, his energy is evidenced in other ways. He subsequently got married again at 102, so he, you know, he had a lot of the life force in him. But after the interview, I got a call, very gruff voice, from a woman. I didn ’ t know who she was, and she said,”Did you get George Abbott to talk?”And I said,”Yeah. Apparently I did.”And she said,”I ’ m his old girlfriend, Maureen Stapleton, and I could never do it.”And then she made me go up with the tape of it and prove that George Abbott actually could talk. So, you know, you want energy, you want the life force, but you really want them also to think that they have a story worth sharing. The worst interviews that you can ever have are with people who are modest. Never ever get up on a stage with somebody who ’ s modest, because all of these people have been assembled to listen to them, and they sit there and they say,”Aw, shucks, it was an accident.”There ’ s nothing that ever happens that justifies people taking good hours of the day to be with them. The worst interview I ever did : William L. Shirer. The journalist who did”The Rise and Fall of the Third Reich.”This guy had met Hitler and Gandhi within six months, and every time I ’ d ask him about it, he ’ d say,”Oh, I just happened to be there. Didn ’ t matter.”Whatever. Awful. I never would ever agree to interview a modest person. They have to think that they did something and that they want to share it with you. But it comes down, in the end, to how do you get through all the barriers we have. All of us are public and private beings, and if all you ’ re going to get from the interviewee is their public self, there ’ s no point in it. It ’ s pre- programmed. It ’ s infomercial, and we all have infomercials about our lives. We know the great lines, we know the great moments, we know what we ’ re not going to share, and the point of this was not to embarrass anybody. This wasn ’ t — and some of you will remember Mike Wallace ’ s old interviews — tough, aggressive and so forth. They have their place. I was trying to get them to say what they probably wanted to say, to break out of their own cocoon of the public self, and the more public they had been, the more entrenched that person, that outer person was. And let me tell you at once the worse moment and the best moment that happened in this interview series. It all has to do with that shell that most of us have, and particularly certain people. There ’ s an extraordinary woman named Clare Boothe Luce. It ’ ll be your generational determinant as to whether her name means much to you. She did so much. She was a playwright. She did an extraordinary play called”The Women.”She was a congresswoman when there weren ’ t very many congresswomen. She was editor of Vanity Fair, one of the great phenomenal women of her day. And, incidentally, I call her the Eleanor Roosevelt of the Right. She was sort of adored on the Right the way Eleanor Roosevelt was on the Left. And, in fact, when we did the interview — I did the living self-portrait with her — there were three former directors of the CIA basically sitting at her feet, just enjoying her presence. And I thought, this is going to be a piece of cake, because I always have preliminary talks with these people for just maybe 10 or 15 minutes. We never talk before that because if you talk before, you don ’ t get it on the stage. So she and I had a delightful conversation. We were on the stage and then — by the way, spectacular. It was all part of Clare Boothe Luce ' s look. She was in a great evening gown. She was 80, almost that day of the interview, and there she was and there I was, and I just proceeded into the questions. And she stonewalled me. It was unbelievable. Anything that I would ask, she would turn around, dismiss, and I was basically up there — any of you in the moderate-to-full entertainment world know what it is to die onstage. And I was dying. She was absolutely not giving me a thing. And I began to wonder what was going on, and you think while you talk, and basically, I thought, I got it. When we were alone, I was her audience. Now I ' m her competitor for the audience. That ' s the problem here, and she ' s fighting me for that, and so then I asked her a question — I didn ' t know how I was going to get out of it — I asked her a question about her days as a playwright, and again, characteristically, instead of saying,"Oh yes, I was a playwright, and this is what blah blah blah,"she said,"Oh, playwright. Everybody knows I was a playwright. Most people think that I was an actress. I was never an actress."But I hadn ' t asked that, and then she went off on a tear, and she said,"Oh, well, there was that one time that I was an actress. It was for a charity in Connecticut when I was a congresswoman, and I got up there,"and she went on and on,"And then I got on the stage."And then she turned to me and said,"And you know what those young actors did? They upstaged me."And she said,"Do you know what that is?"Just withering in her contempt. And I said,"I ' m learning."And she looked at me, and it was like the successful arm-wrestle, and then, after that, she delivered an extraordinary account of what her life really was like. I have to end that one. This is my tribute to Clare Boothe Luce. Again, a remarkable person. I ' m not politically attracted to her, but through her life force, I ' m attracted to her. And the way she died — she had, toward the end, a brain tumor. That ' s probably as terrible a way to die as you can imagine, and very few of us were invited to a dinner party. And she was in horrible pain. We all knew that. She stayed in her room. Everybody came. The butler passed around canapes. The usual sort of thing. Then at a certain moment, the door opened and she walked out perfectly dressed, completely composed. The public self, the beauty, the intellect, and she walked around and talked to every person there and then went back into the room and was never seen again. She wanted the control of her final moment, and she did it amazingly. Now, there are other ways that you get somebody to open up, and this is just a brief reference. It wasn ' t this arm-wrestle, but it was a little surprising for the person involved. I interviewed Steve Martin. It wasn ' t all that long ago. And we were sitting there, and almost toward the beginning of the interview, I turned to him and I said,"Steve,"or"Mr. Martin, it is said that all comedians have unhappy childhoods. Was yours unhappy?"And he looked at me, you know, as if to say,"This is how you ' re going to start this thing, right off?"And then he turned to me, not stupidly, and he said,"What was your childhood like?"And I said — these are all arm wrestles, but they ' re affectionate — and I said,"My father was loving and supportive, which is why I ' m not funny."And he looked at me, and then we heard the big sad story. His father was an SOB, and, in fact, he was another comedian with an unhappy childhood, but then we were off and running. So the question is : What is the key that ' s going to allow this to proceed? Now, these are arm wrestle questions, but I want to tell you about questions that are more related to empathy and that really, very often, are the questions that people have been waiting their whole lives to be asked. And I ' ll just give you two examples of this because of the time constraints. One was an interview I did with one of the great American biographers. Again, some of you will know him, most of you won ' t, Dumas Malone. He did a five- volume biography of Thomas Jefferson, spent virtually his whole life with Thomas Jefferson, and by the way, at one point I asked him,"Would you like to have met him?”And he said,”Well, of course, but actually, I know him better than anyone who ever met him, because I got to read all of his letters.”So, he was very satisfied with the kind of relationship they had over 50 years. And I asked him one question. I said,”Did Jefferson ever disappoint you?”And here is this man who had given his whole life to uncovering Jefferson and connecting with him, and he said,”Well...”— I ’ m going to do a bad southern accent. Dumas Malone was from Mississippi originally. But he said,”Well,”he said,”I ’ m afraid so.”He said,”You know, I ’ ve read everything, and sometimes Mr. Jefferson would smooth the truth a bit.”And he basically was saying that this was a man who lied more than he wished he had, because he saw the letters. He said,”But I understand that.”He said,”I understand that.”He said,”We southerners do like a smooth surface, so that there were times when he just didn ’ t want the confrontation.”And he said,”Now, John Adams was too honest.”And he started to talk about that, and later on he invited me to his house, and I met his wife who was from Massachusetts, and he and she had exactly the relationship of Thomas Jefferson and John Adams. She was the New Englander and abrasive, and he was this courtly fellow. But really the most important question I ever asked, and most of the times when I talk about it, people kind of suck in their breath at my audacity, or cruelty, but I promise you it was the right question. This was to Agnes de Mille. Agnes de Mille is one of the great choreographers in our history. She basically created the dances in”Oklahoma,”transforming the American theater. An amazing woman. At the time that I proposed to her that — by the way, I would have proposed to her ; she was extraordinary — but proposed to her that she come on. She said,”Come to my apartment.”She lived in New York.”Come to my apartment and we ’ ll talk for those 15 minutes, and then we ’ ll decide whether we proceed.”And so I showed up in this dark, rambling New York apartment, and she called out to me, and she was in bed. I had known that she had had a stroke, and that was some 10 years before. And so she spent almost all of her life in bed, but — I speak of the life force — her hair was askew. She wasn ’ t about to make up for this occasion. And she was sitting there surrounded by books, and her most interesting possession she felt at that moment was her will, which she had by her side. She wasn ’ t unhappy about this. She was resigned. She said,”I keep this will by my bed, memento mori, and I change it all the time just because I want to.”And she was loving the prospect of death as much as she had loved life. I thought, this is somebody I ’ ve got to get in this series. She agreed. She came on. Of course she was wheelchaired on. Half of her body was stricken, the other half not. She was, of course, done up for the occasion, but this was a woman in great physical distress. And we had a conversation, and then I asked her this unthinkable question. I said,”Was it a problem for you in your life that you were not beautiful?”And the audience just — you know, they ’ re always on the side of the interviewee, and they felt that this was a kind of assault, but this was the question she had wanted somebody to ask her whole life. And she began to talk about her childhood, when she was beautiful, and she literally turned — here she was, in this broken body — and she turned to the audience and described herself as the fair demoiselle with her red hair and her light steps and so forth, and then she said,”And then puberty hit.”And she began to talk about things that had happened to her body and her face, and how she could no longer count on her beauty, and her family then treated her like the ugly sister of the beautiful one for whom all the ballet lessons were given. And she had to go along just to be with her sister for company, and in that process, she made a number of decisions. First of all, was that dance, even though it hadn ’ t been offered to her, was her life. And secondly, she had better be, although she did dance for a while, a choreographer because then her looks didn ’ t matter. But she was thrilled to get that out as a real, real fact in her life. It was an amazing privilege to do this series. There were other moments like that, very few moments of silence. The key point was empathy because everybody in their lives is really waiting for people to ask them questions, so that they can be truthful about who they are and how they became what they are, and I commend that to you, even if you ’ re not doing interviews. Just be that way with your friends and particularly the older members of your family. Thank you very much.

Text Sample 971: NEG Dim 6 – 2008 – Score 1.00 – 2008/t000293.txt
Well, this is such an honor. And it ’ s wonderful to be in the presence of an organization that is really making a difference in the world. And I ’ m intensely grateful for the opportunity to speak to you today. And I ’ m also rather surprised, because when I look back on my life the last thing I ever wanted to do was write, or be in any way involved in religion. After I left my convent, I ’ d finished with religion, frankly. I thought that was it. And for 13 years I kept clear of it. I wanted to be an English literature professor. And I certainly didn ’ t even want to be a writer, particularly. But then I suffered a series of career catastrophes, one after the other, and finally found myself in television. I said that to Bill Moyers, and he said,”Oh, we take anybody.”And I was doing some rather controversial religious programs. This went down very well in the U. K., where religion is extremely unpopular. And so, for once, for the only time in my life, I was finally in the mainstream. But I got sent to Jerusalem to make a film about early Christianity. And there, for the first time, I encountered the other religious traditions : Judaism and Islam, the sister religions of Christianity. And while I found I knew nothing about these faiths at all — despite my own intensely religious background, I ’ d seen Judaism only as a kind of prelude to Christianity, and I knew nothing about Islam at all. But in that city, that tortured city, where you see the three faiths jostling so uneasily together, you also become aware of the profound connection between them. And it has been the study of other religious traditions that brought me back to a sense of what religion can be, and actually enabled me to look at my own faith in a different light. And I found some astonishing things in the course of my study that had never occurred to me. Frankly, in the days when I thought I ’ d had it with religion, I just found the whole thing absolutely incredible. These doctrines seemed unproven, abstract. And to my astonishment, when I began seriously studying other traditions, I began to realize that belief — which we make such a fuss about today — is only a very recent religious enthusiasm that surfaced only in the West, in about the 17th century. The word”belief”itself originally meant to love, to prize, to hold dear. In the 17th century, it narrowed its focus, for reasons that I ’ m exploring in a book I ’ m writing at the moment, to include — to mean an intellectual assent to a set of propositions, a credo.”I believe :”it did not mean,”I accept certain creedal articles of faith.”It meant :”I commit myself. I engage myself.”Indeed, some of the world traditions think very little of religious orthodoxy. In the Quran, religious opinion — religious orthodoxy — is dismissed as”zanna :”self-indulgent guesswork about matters that nobody can be certain of one way or the other, but which makes people quarrelsome and stupidly sectarian. So if religion is not about believing things, what is it about? What I ’ ve found, across the board, is that religion is about behaving differently. Instead of deciding whether or not you believe in God, first you do something. You behave in a committed way, and then you begin to understand the truths of religion. And religious doctrines are meant to be summons to action ; you only understand them when you put them into practice. Now, pride of place in this

practice is given to compassion. And it is an arresting fact that right across the board, in every single one of the major world faiths, compassion — the ability to feel with the other in the way we ’ ve been thinking about this evening — is not only the test of any true religiosity, it is also what will bring us into the presence of what Jews, Christians and Muslims call”God”or the”Divine.”It is compassion, says the Buddha, which brings you to Nirvana. Why? Because in compassion, when we feel with the other, we dethrone ourselves from the center of our world and we put another person there. And once we get rid of ego, then we ’ re ready to see the Divine. And in particular, every single one of the major world traditions has highlighted — has said — and put at the core of their tradition what ’ s become known as the Golden Rule. First propounded by Confucius five centuries before Christ :”Do not do to others what you would not like them to do to you.”That, he said, was the central thread which ran through all his teaching and that his disciples should put into practice all day and every day. And it was — the Golden Rule would bring them to the transcendent value that he called”ren,”human-heartedness, which was a transcendent experience in itself. And this is absolutely crucial to the monotheisms, too. There ’ s a famous story about the great rabbi, Hillel, the older contemporary of Jesus. A pagan came to him and offered to convert to Judaism if the rabbi could recite the whole of Jewish teaching while he stood on one leg. Hillel stood on one leg and said,”That which is hateful to you, do not do to your neighbor. That is the Torah. The rest is commentary. Go and study it.”And”go and study it”was what he meant. He said,”In your exegesis, you must make it clear that every single verse of the Torah is a commentary, a gloss upon the Golden Rule.”The great Rabbi Meir said that any interpretation of Scripture which led to hatred and disdain, or contempt of other people — any people whatsoever — was illegitimate. Saint Augustine made exactly the same point. Scripture, he says,”teaches nothing but charity, and we must not leave an interpretation of Scripture until we have found a compassionate interpretation of it.”And this struggle to find compassion in some of these rather rebarbative texts is a good dress rehearsal for doing the same in ordinary life. But now look at our world. And we are living in a world that is — where religion has been hijacked. Where terrorists cite Quranic verses to justify their atrocities. Where instead of taking Jesus ’ words,”Love your enemies. Don ’ t judge others,”we have the spectacle of Christians endlessly judging other people, endlessly using Scripture as a way of arguing with other people, putting other people down. Throughout the ages, religion has been used to oppress others, and this is because of human ego, human greed. We have a talent as a species for messing up wonderful things. So the traditions also insisted — and this is an important point, I think — that you could not and must not confine your compassion to your own group : your own nation, your own co-religionists, your own fellow countrymen. You must have what one of the Chinese sages called”jian ai”: concern for everybody. Love your enemies. Honor the stranger. We formed you, says the Quran, into tribes and nations so that you may know one another. And this, again — this universal outreach — is getting subdued in the strident use of religion — abuse of religion — for nefarious gains. Now, I ’ ve lost count of the number of taxi drivers who, when I say to them what I do for a living, inform me that religion has been the cause of all the major world wars in history. Wrong. The causes of our present woes are political. But, make no mistake about it, religion is a kind of fault line, and when a conflict gets ingrained in a region, religion can get sucked in and become part of the problem. Our modernity has been exceedingly violent. Between 1914 and 1945, 70 million people died in Europe alone as a result of armed conflict. And so many of our institutions, even football, which used to be a pleasant pastime, now causes riots where people even die. And it ’ s not surprising that religion, too, has been affected by this violent ethos. There ’ s also a great deal, I think, of religious illiteracy around. People seem to think, now equate religious faith with believing things. As though that — we call religious people often believers, as though that were the main thing that they do. And very often, secondary goals get pushed into the first place, in place of compassion and the Golden Rule. Because the Golden Rule is difficult. I sometimes — when I ’ m speaking to congregations about compassion, I sometimes see a mutinous expression crossing some of their faces because a lot of religious people prefer to be right, rather than compassionate. Now — but that ’ s not the whole story. Since September the 11th, when my work on Islam suddenly propelled me into public life, in a way that I ’ d never imagined, I ’ ve been able to sort of go all over the world, and finding, everywhere I go, a yearning for change. I ’ ve just come back from Pakistan, where literally thousands of people came to my lectures, because they were yearning, first of all, to hear a friendly Western voice. And especially the young people were coming. And were asking me — the young people were saying,”What can we do? What can we do to change things?”And my hosts in Pakistan said,”Look, don ’ t be too polite to us. Tell us where we ’ re going wrong. Let ’ s talk together about where religion is failing.”Because it seems to me that with — our current situation is so serious at the moment that any ideology that doesn ’ t promote a sense of global understanding and global appreciation of each other is failing the test of the time. And religion, with its wide following... Here in the United States, people may be being religious in a different way, as a report has just shown — but they still want to be religious. It ’ s only Western Europe that has retained its secularism, which is now beginning to look rather endearingly old-fashioned. But people want to be religious, and religion should be made to be a force for harmony in the world, which it can and should be — because of the Golden Rule.”Do not do to others what you would not have them do to you”: an ethos that should now be applied globally. We should not treat other nations as we would not wish to be treated ourselves. And these — whatever our wretched beliefs — is a religious matter, it ’ s a spiritual matter. It ’ s a profound moral matter that engages and should engage us all. And as I say, there is a hunger for change out there. Here in the United States, I think you see it in this election campaign : a longing for change. And people in churches all over and mosques all over this continent after September the 11th, coming together locally to create networks of understanding. With the mosque, with the synagogue, saying,”We must start to speak to one another.”I think it ’ s time that we moved beyond the idea of toleration and move toward appreciation of the other. I ’ d — there ’ s one story I ’ d just like to mention. This comes from”The Iliad.”But it tells you what this spirituality should be. You know the story of”The Iliad,”the 10-year war between Greece and Troy. In one incident, Achilles, the famous warrior of Greece, takes his troops out of the war, and the whole war effort suffers. And in the course of the ensuing muddle, his beloved friend, Patroclus, is killed — and killed in single combat by one of the Trojan princes, Hector. And Achilles goes mad with grief and rage and revenge, and he mutilates the body. He kills Hector, he mutilates his body and then he refuses to give the body back for burial to the family, which means that, in Greek ethos, Hector ’ s soul will wander eternally, lost. And then one night, Priam, king of Troy, an old man, comes into the Greek camp incognito, makes his way to Achilles ’ tent to ask for the body of his son. And everybody is shocked when the old man takes off his head covering and shows himself. And Achilles looks at him and thinks of his father. And he starts to weep. And Priam looks at the man who has murdered so many of his sons, and he, too, starts to weep. And the sound of their weeping filled the house. The Greeks believed that weeping together created a bond between people. And then Achilles takes the body of Hector, he hands it very tenderly to the father, and the two men look at each other, and see each other as divine. That is the ethos found, too, in all the religions. It ’ s what is meant by overcoming the horror that we feel when we are under threat of our enemies, and beginning to appreciate the other. It ’ s of great importance that the word for”holy”in Hebrew, applied to God, is”Kadosh”: separate, other. And it is often, perhaps, the very otherness of our enemies which can give us intimations of that utterly mysterious transcendence which is God. And now, here ’ s my wish : I wish that you would help with the creation, launch and propagation of a Charter for Compassion, crafted by a group of inspirational thinkers from the three Abrahamic traditions of Judaism, Christianity and Islam, and based on the fundamental principle of the Golden Rule. We need to create a movement among all these people that I meet in my travels — you probably meet, too — who want to join up, in some way, and reclaim their faith, which they feel, as I say, has been hijacked. We need to empower people to remember the compassionate ethos, and to give guidelines. This Charter would not be a massive document. I ’ d like to see it — to give guidelines as to how to interpret the Scriptures, these texts that are being abused. Remember what the rabbis and what Augustine said about how Scripture should be governed by the principle of charity. Let ’ s get back to that. And the idea, too, of Jews, Christians and Muslims — these traditions now so often at loggerheads — working together to create a document which we hope will be signed by a thousand, at least, of major religious leaders from all the traditions of the world. And you are the people. I ’ m just a solitary scholar. Despite the idea that I love a good time, which I was rather amazed to see coming up on me — I actually spend a great deal of time alone, studying, and I ’ m not very — you ’ re the people with media knowledge to explain to me how we can get this to everybody, everybody on the planet. I ’ ve had some preliminary talks, and Archbishop Desmond Tutu, for example, is very happy to give his name to this, as is Imam Feisal Rauf, the Imam in New York City. Also, I would be working with the Alliance of Civilizations at the United Nations. I was part of that United Nations initiative called the Alliance of Civilizations, which was asked by Kofi Annan to diagnose the causes of extremism, and to give practical guidelines to member states about how to avoid the escalation of further extremism. And the Alliance has told me that they are very happy to work with it. The importance of this is that this is — I can see some of you starting to look worried, because you think it ’ s a slow and cumbersome body — but what the United Nations can do is give us some neutrality, so that this isn ’ t seen as a Western or a Christian initiative, but that it ’ s coming, as it were, from the United Nations, from the world — who would help with the sort of bureaucracy of this. And so I do urge you to join me in making — in this charter — to building this charter, launching it and propagating it so that it becomes — I ’ d like to see it in every college, every church, every mosque, every synagogue in the world, so that people can look at their tradition, reclaim it, and make religion a source of peace in the world, which it can and should be. Thank you very much.

Text Sample 972: NEG Dim 6 – 2008 – Score 1.00 – 2008/t000290.txt

So, as researchers, something that we often do is use immense resources to achieve certain capabilities, or achieve certain goals. And this is essential to the progress of science, or exploration of what is possible. But it creates this unfortunate situation where a tiny, tiny fraction of the world can actually participate in this exploration or can benefit from that technology. Something that motivates me, and gets me really excited about my research, is when I see simple opportunities to drastically change that distribution and make the technology accessible to a much wider percentage of the population. I ’ m going to show you two videos that have gotten a lot of attention that I think embody this philosophy. And they actually use the Nintendo Wii Remote. For those of you who aren ’ t familiar with this device, it ’ s a $40 video game controller. And it ’ s mostly advertised for its motion-sensing capabilities : so you can swing a tennis racket, or hit a baseball bat. But what actually interests me a lot more is the fact that in the tip of each controller is a relatively high-performing infrared camera. And I ’ m going to show you two demos of why this is useful. So here, I have my computer set up with the projector, and I have a Wii Remote sitting on top of it. And, for example, if you ’ re in a school that doesn ’ t have a lot money, probably a lot of schools, or if you ’ re in an office environment, and you want an interactive whiteboard, normally these cost about two to three thousand dollars. So I ’ m going to show you how to create one with a Wii Remote. Now, this requires another piece of hardware, which is this infrared pen. You can probably make this yourself for about five dollars with a quick trip to the Radio Shack. It ’ s got a battery, a button and an infrared LED — you guys can ’ t see it — but it turns on whenever I push the button. Now, what this means is that if I run this piece of software, the camera sees the infrared dot, and I can register the location of the camera pixels to the projector pixels. And now this is like an interactive whiteboard surface. So for about $50 of hardware, you can have your own whiteboard. This is Adobe Photoshop. Thank you. The software for this I ’ ve actually put on my website and have let people download it for free. In the three months this project has been public, it ’ s been downloaded over half a million times. So teachers and students all around the world are already using this. Although it does do it for 50 dollars, there are some limitations of this approach. You get about 80 percent of the way there, for one percent of the cost. Another nice thing is that a camera can see multiple dots, so this is actually a multi-touch, interactive whiteboard system as well. For the second demo, I have this Wii Remote that ’ s actually next to the TV. So it ’ s pointing away from the display, rather than pointing at the display. And why this is interesting is that if you put on, say, a pair of safety glasses, that have two infrared dots in them, they are going to give the computer an approximation of your head location. And why this is interesting is I have this sort of application running on the computer monitor, which has a 3D room, with some targets floating in it. And you can see that it looks like a 3D room. kind of like a video game, it sort of looks 3D, but for the most part, the image looks pretty flat, and bound to the surface of the screen. But if we turn on head tracking — the computer can change the image that ’ s on the screen and make it respond to the head movements. So let ’ s switch back to that. So this has actually been a little bit startling to the game-development community. Because this is about 10 dollars of additional hardware if you already have a Nintendo Wii. So I ’ m looking forward to seeing some games, and actually Louis Castle, that ’ s him down there, last week announced that Electronic Arts, one of the largest game publishers, is releasing a game in May that has a little Easter egg feature for supporting this type of head tracking. And that ’ s from less than five months from a prototype in my lab to a major commercial product. Thank you. But actually, to me, what ’ s almost more interesting than either of these two projects is how people actually found out about them. YouTube has really changed the way, or changed the speed, in which a single individual can actually spread an idea around the world. I ’ m doing some research in my lab with a video camera, and within the first week, a million people had seen this work, and literally within days, engineers, teachers and students from around the world were already posting their own YouTube videos of them using my system or derivatives of this work. So I hope to see more of that in the future, and hope online video distribution to be embraced by the research community. So thank you very much.

Text Sample 973: NEG Dim 6 – 2006 – Score 1.00 – 2006/t000497.txt

Good morning. How are you? Good. It ’ s been great, hasn ’ t it? I ’ ve been blown away by the whole thing. In fact, I ’ m leaving. There have been three themes running through the conference, which are relevant to what I want to talk about. One is the extraordinary evidence of human creativity in all of the presentations that we ’ ve had and in all of the people here ; just the variety of it and the range of it. The second is that it ’ s put us in a place where we have no idea what ’ s going to happen in terms of the future. No idea how this may play out. I have an interest in education. Actually, what I find is, everybody has an interest in education. Don ’ t you? I find this very interesting. If you ’ re at a dinner party, and you say you work in education — actually, you ’ re not often at dinner parties, frankly. If you work in education, you ’ re not asked. And you ’ re never asked back, curiously. That ’ s strange to me. But if you are, and you say to somebody, you know, they say,”What do you do?”and you say you work in education, you can see the blood run from their face. They ’ re like,”Oh my God. Why me?””My one night out all week.”But if you ask about their education, they pin you to the wall, because it ’ s one of those things that goes deep with people, am I right? Like religion and money and other things. So I have a big interest in education, and I think we all do. We have a huge vested interest in it, partly because it ’ s education that ’ s meant to take us into this future that we can ’ t grasp. If you think of it, children starting school this year will be retiring in 2065. Nobody has a clue, despite all the expertise that ’ s been on parade for the past four days, what the world will look like in five years ’ time. And yet, we ’ re meant to be educating them for it. So the unpredictability, I think, is extraordinary. And the third part of this is that we ’ ve all agreed, nonetheless, on the really extraordinary capacities that children have — their capacities for innovation. I mean, Sirena last night was a marvel, wasn ’ t she? Just seeing what she could do. And she ’ s exceptional, but I think she ’ s not, so to speak, exceptional in the whole of childhood. What you have there is a person of extraordinary dedication who found a talent. And my contention is, all kids have tremendous talents, and we squander them, pretty ruthlessly. So I want to talk about education, and I want to talk about creativity. My contention is that creativity now is as important in education as literacy, and we should treat it with the same status. Thank you. That was it, by the way. Thank you very much. So, 15 minutes left.”Well, I was born...”I heard a great story recently — I love telling it — of a little girl who was in a drawing lesson. She was six, and she was at the back, drawing, and the teacher said this girl hardly ever paid attention, and in this drawing lesson, she did. The teacher was fascinated. She went over to her, and she said,”What are you drawing?”And the girl said,”I ’ m drawing a picture of God.”And the teacher said,”But nobody knows what God looks like.”And the girl said,”They will in a minute.”When my son was four in England — actually, he was four everywhere, to be honest. If we ’ re being strict about it, wherever he went, he was four that year. He was in the Nativity play. Do you remember the story? No, it was big, it was a big story. Mel Gibson did the sequel, you may have seen it.”Nativity II.”But James got the part of Joseph, which we were thrilled about. We considered this to be one of the lead parts. We had the place crammed full of agents in T-shirts :”James

Robinson IS Joseph!”He didn ’ t have to speak, but you know the bit where the three kings come in? They come in bearing gifts, gold, frankincense and myrrh. This really happened. We were sitting there, and I think they just went out of sequence, because we talked to the little boy afterward and said,”You OK with that?”They said,”Yeah, why? Was that wrong?”They just switched. The three boys came in, four-year-olds with tea towels on their heads. They put these boxes down, and the first boy said,”I bring you gold.”And the second boy said,”I bring you myrrh.”And the third boy said,”Frank sent this.”What these things have in common is that kids will take a chance. If they don ’ t know, they ’ ll have a go. Am I right? They ’ re not frightened of being wrong. I don ’ t mean to say that being wrong is the same thing as being creative. What we do know is, if you ’ re not prepared to be wrong, you ’ ll never come up with anything original — if you ’ re not prepared to be wrong. And by the time they get to be adults, most kids have lost that capacity. They have become frightened of being wrong. And we run our companies like this. We stigmatize mistakes. And we ’ re now running national education systems where mistakes are the worst thing you can make. And the result is that we are educating people out of their creative capacities. Picasso once said this, he said that all children are born artists. The problem is to remain an artist as we grow up. I believe this passionately, that we don ’ t grow into creativity, we grow out of it. Or rather, we get educated out of it. So why is this? I lived in Stratford-on-Avon until about five years ago. In fact, we moved from Stratford to Los Angeles. So you can imagine what a seamless transition this was. Actually, we lived in a place called Snitterfield, just outside Stratford, which is where Shakespeare ’ s father was born. Are you struck by a new thought? I was. You don ’ t think of Shakespeare having a father, do you? Do you? Because you don ’ t think of Shakespeare being a child, do you? Shakespeare being seven? I never thought of it. I mean, he was seven at some point. He was in somebody ’ s English class, wasn ’ t he? How annoying would that be?”Must try harder.”Being sent to bed by his dad, to Shakespeare,”Go to bed, now!”To William Shakespeare.”And put the pencil down!””And stop speaking like that.””It ’ s confusing everybody.”Anyway, we moved from Stratford to Los Angeles, and I just want to say a word about the transition. Actually, my son didn ’ t want to come. I ’ ve got two kids ; he ’ s 21 now, my daughter ’ s 16. He didn ’ t want to come to Los Angeles. He loved it, but he had a girlfriend in England. This was the love of his life, Sarah. He ’ d known her for a month. Mind you, they ’ d had their fourth anniversary, because it ’ s a long time when you ’ re 16. He was really upset on the plane. He said,”I ’ ll never find another girl like Sarah.”And we were rather pleased about that, frankly — because she was the main reason we were leaving the country. But something strikes you when you move to America and travel around the world : every education system on earth has the same hierarchy of subjects. Every one. Doesn ’ t matter where you go. You ’ d think it would be otherwise, but it isn ’ t. At the top are mathematics and languages, then the humanities. At the bottom are the arts. Everywhere on earth. And in pretty much every system, too, there ’ s a hierarchy within the arts. Art and music are normally given a higher status in schools than drama and dance. There isn ’ t an education system on the planet that teaches dance every day to children the way we teach them mathematics. Why? Why not? I think this is rather important. I think math is very important, but so is dance. Children dance all the time if they ’ re allowed to, we all do. We all have bodies, don ’ t we? Did I miss a meeting? Truthfully, what happens is, as children grow up, we start to educate them progressively from the waist up. And then we focus on their heads. And slightly to one side. If you were to visit education as an alien and say”What ’ s it for, public education?”I think you ’ d have to conclude, if you look at the output, who really succeeds by this, who does everything they should, who gets all the brownie points, who are the winners — I think you ’ d have to conclude the whole purpose of public education throughout the world is to produce university professors. Isn ’ t it? They ’ re the people who come out the top. And I used to be one, so there. And I like university professors, but, you know, we shouldn ’ t hold them up as the high-water mark of all human achievement. They ’ re just a form of life. Another form of life. But they ’ re rather curious. And I say this out of affection for them : there ’ s something curious about professors. In my experience — not all of them, but typically — they live in their heads. They live up there and slightly to one side. They ’ re disembodied, you know, in a kind of literal way. They look upon their body as a form of transport for their heads. Don ’ t they? It ’ s a way of getting their head to meetings. If you want real evidence of out-of-body experiences, by the way, get yourself along to a residential conference of senior academics and pop into the discotheque on the final night. And there, you will see it. Grown men and women writhing uncontrollably, off the beat. Waiting until it ends, so they can go home and write a paper about it. Our education system is predicated on the idea of academic ability. And there ’ s a reason. Around the world, there were no public systems of education, really, before the 19th century. They all came into being to meet the needs of industrialism. So the hierarchy is rooted on two ideas. Number one, that the most useful subjects for work are at the top. So you were probably steered benignly away from things at school when you were a kid, things you liked, on the grounds you would never get a job doing that. Is that right?”Don ’ t do music, you ’ re not going to be a musician ; don ’ t do art, you won ’ t be an artist.”Benign advice — now, profoundly mistaken. The whole world is engulfed in a revolution. And the second is academic ability, which has really come to dominate our view of intelligence, because the universities design the system in their image. If you think of it, the whole system of public education around the world is a protracted process of university entrance. And the consequence is that many highly talented, brilliant, creative people think they ’ re not, because the thing they were good at at school wasn ’ t valued, or was actually stigmatized. And I think we can ’ t afford to go on that way. In the next 30 years, according to UNESCO, more people worldwide will be graduating through education than since the beginning of history. More people. And it ’ s the combination of all the things we ’ ve talked about : technology and its transformational effect on work, and demography and the huge explosion in population. Suddenly, degrees aren ’ t worth anything. Isn ’ t that true? When I was a student, if you had a degree, you had a job. If you didn ’ t have a job, it ’ s because you didn ’ t want one. And I didn ’ t want one, frankly. But now kids with degrees are often heading home to carry on playing video games, because you need an MA where the previous job required a BA, and now you need a PhD for the other. It ’ s a process of academic inflation. And it indicates the whole structure of education is shifting beneath our feet. We need to radically rethink our view of intelligence. We know three things about intelligence. One, it ’ s diverse. We think about the world in all the ways that we experience it. We think visually, we think in sound, we think kinesthetically. We think in abstract terms, we think in movement. Secondly, intelligence is dynamic. If you look at the interactions of a human brain, as we heard yesterday from a number of presentations, intelligence is wonderfully interactive. The brain isn ’ t divided into compartments. In fact, creativity — which I define as the process of having original ideas that have value — more often than not comes about through the interaction of different disciplinary ways of seeing things. By the way, there ’ s a shaft of nerves that joins the two halves of the brain, called the corpus callosum. It ’ s thicker in women. Following off from Helen yesterday, this is probably why women are better at multitasking. Because you are, aren ’ t you? There ’ s a raft of research, but I know it from my personal life. If my wife is cooking a meal at home, which is not often... thankfully. No, she ’ s good at some things. But if she ’ s cooking, she ’ s dealing with people on the phone, she ’ s talking to the kids, she ’ s painting the ceiling — she ’ s doing open-heart surgery over here. If I ’ m cooking, the door is shut, the kids are out, the phone ’ s on the hook, if she comes in, I get annoyed. I say,”Terry, please, I ’ m trying to fry an egg in here.””Give me a break.”Actually, do you know that old philosophical thing,”If a tree falls in a forest, and nobody hears it, did it happen?”Remember that old chestnut? I saw a great T-shirt recently, which said,”If a man speaks his mind in a forest, and no woman hears him, is he still wrong?”And the third thing about intelligence is, it ’ s distinct. I ’ m doing a new book at the moment called”Epiphany,”which is based on a series of interviews with people about how they discovered their talent. I ’ m fascinated by how people got to be there. It ’ s really prompted by a conversation I had with a wonderful woman who maybe most people have never heard of, Gillian Lynne. Have you heard of her? Some have. She ’ s a choreographer, and everybody knows her work. She did”Cats”and”Phantom of the Opera.”She ’ s wonderful. I used to be on the board of The Royal Ballet, as you can see. Gillian and I had lunch one day. I said,”How did you get to be a dancer?”It was interesting. When she was at school, she was really hopeless. And the school, in the ’ 30s, wrote to her parents and said,”We think Gillian has a learning disorder.”She couldn ’ t concentrate ; she was fidgeting. I think now they ’ d say she had ADHD. Wouldn ’ t you? But this was the 1930s, and ADHD hadn ’ t been invented at this point. It wasn ’ t an available condition. People weren ’ t aware they could have that. Anyway, she went to see this specialist. So, this oak-paneled room, and she was there with her mother, and she was led and sat on this chair at the end, and she sat on her hands for 20 minutes, while this man talked to her mother about all the problems Gillian was having at school, because she was disturbing people, her homework was always late, and so on. Little kid of eight. In the end, the doctor went and sat next to Gillian and said,”I ’ ve listened to all these things your mother ’ s told me. I need to speak to her privately. Wait here. We ’ ll be back. We won ’ t be very long,”and they went and left her. But as they went out of the room, he turned on the radio that was sitting on his desk. And when they got out of the room, he said to her mother,”Just stand and watch her.”And the minute they left the room, she was on her feet, moving to the music. And they watched for a few minutes, and he turned to her mother and said,”Mrs. Lynne, Gillian isn ’ t sick. She ’ s a dancer. Take her to a dance school.”I said,”What happened?”She said,”She did. I can ’ t tell you how wonderful it was. We walked in this room, and it was full of people like me — people who couldn ’ t sit still, people who had to move to think.”Who had to move to think. They did ballet, they did tap, jazz ; they did modern ; they did contemporary. She was eventually auditioned for the Royal Ballet School. She became a soloist ; she had a wonderful career at the Royal Ballet. She eventually graduated from the Royal Ballet School, founded the Gillian Lynne Dance Company, met Andrew Lloyd Webber. She ’ s been responsible for some of the most successful musical theater productions in history, she ’ s given pleasure to millions, and she ’ s a multimillionaire. Somebody else might have put her on medication and told her to calm down. What I think it comes to is this : Al Gore spoke the other night about ecology and the revolution that was triggered by Rachel Carson. I believe our only hope for the future is to adopt a new conception of human ecology, one in which we start to reconstitute our conception of the richness of human capacity. Our education system has mined our minds in the way that we strip-mine the earth for a particular commodity. And for the future, it won ’ t serve us. We have to rethink the fundamental principles on which we ’ re educating our children. There was a wonderful quote by Jonas Salk, who said,”If all the insects were to disappear from the Earth, within 50 years, all life on Earth would end. If all human beings disappeared from the Earth, within 50 years, all forms of life would flourish.”And he ’ s right. What TED celebrates is the gift of the human imagination. We have to be careful now that we use this gift wisely, and that we avert some of the scenarios that we ’ ve talked about. And the only way we ’ ll do it is by seeing our creative capacities for the richness they are and seeing our children for the hope that they are. And our task is to educate their whole being, so they can face this future. By the way — we may not see this future, but they will. And our job is to help them make something of it. Thank you very much.

Text Sample 974: NEG Dim 6 – 2006 – Score 1.00 – 2006/t000496.txt

On September 10, the morning of my seventh birthday, I came downstairs to the kitchen, where my mother was washing the dishes and my father was reading the paper or something, and I sort of presented myself to them in the doorway, and they said,”Hey, happy birthday!”And I said,”I ’ m seven.”And my father smiled and said,”Well, you know what that means, don ’ t you?”And I said,”Yeah, that I ’ m going to have a party and a cake and get a lot of presents?”And my dad said,”Well, yes. But more importantly, being seven means that you ’ ve reached the age of reason, and you ’ re now capable of committing any and all sins against God and man.”Now, I had heard this phrase,”age of reason,”before. Sister Mary Kevin had been bandying it about my second-grade class at school. But when she said it, the phrase seemed all caught up in the excitement of preparations for our first communion and our first confession, and everybody knew that was really all about the white dress and the white veil. And anyway, I hadn ’ t really paid all that much attention to that phrase,”age of reason.”So, I said,”Yeah, yeah, age of reason. What does that mean again?”And my dad said,”Well, we believe, in the Catholic Church, that God knows that little kids don ’ t know the difference between right and wrong, but when you ’ re seven, you ’ re old enough to know better. So, you ’ ve grown up and reached the age of reason, and now God will start keeping notes on you, and begin your permanent record.”And I said,”Oh... Wait a minute. You mean all that time, up till today, all that time I was so good, God didn ’ t notice it?”And my mom said,”Well, I noticed it.”And I thought,”How could I not have known this before? How could it not have sunk in when they ’ d been telling me? All that being good and no real credit for it. And worst of all, how could I not have realized this very important information until the very day that it was basically useless to me?”So I said,”Well, Mom and Dad, what about Santa Claus? I mean, Santa Claus knows if you ’ re naughty or nice, right?”And my dad said,”Yeah, but, honey, I think that ’ s technically just between Thanksgiving and Christmas.”And my mother said,”Oh, Bob, stop it. Let ’ s just tell her. I mean, she ’ s seven. Julie, there is no Santa Claus.”Now, this was actually not that upsetting to me. My parents had this whole elaborate story about Santa Claus : how they had talked to Santa Claus himself and agreed that instead of Santa delivering our presents over the night of Christmas Eve, like he did for every other family who got to open their surprises first thing Christmas morning, our family would give Santa more time. Santa would come to our house while we were at nine o ’ clock high mass on Christmas morning, but only if all of us kids did not make a fuss. Which made me very suspicious. It was pretty obvious that it was really our parents giving us the presents. I mean, my dad had a very distinctive wrapping style, and my mother ' s handwriting was so close to Santa ' s. Plus, why would Santa save time by having to loop back to our house after he ' d gone to everybody else ' s? There was only one obvious conclusion to reach from this mountain of evidence : our family was too strange and weird for even Santa Claus to come visit, and my poor parents were trying to protect us from the embarrassment, this humiliation of rejection by Santa, who was jolly — but let ' s face it, he was also very judgmental. So to find out that there was no Santa Claus at all was actually sort of a relief. I left the kitchen not really in shock about Santa, but rather, I was just dumbfounded about how I could have missed this whole age of reason thing. It was too late for me, but maybe I could help someone else, someone who could use the information. They had to fit two criteria : they had to be old enough to be able to understand the whole concept of the age of reason, and not yet seven. The answer was clear : my brother Bill. He was six. Well, I finally found Bill about a block away from our house at this public school playground. It was a Saturday, and he was all by himself, just kicking a ball against the side of a wall. I ran up to him and said,"Bill! I just realized that the age of reason starts when you turn seven, and then you ' re capable of committing any and all sins against God and man."And Bill said,"So?"And I said,"So, you ' re six. You have a whole year to do anything you want to and God won ' t notice it."And he said,"So?"And I said,"So? So everything!"And I turned to run. I was so angry with him. But when I got to the top of the steps, I turned around dramatically and said,"Oh, by the way, Bill — there is no Santa Claus."Now, I didn ' t know it at the time, but I really wasn ' t turning seven on September 10th. For my 13th birthday, I planned a slumber party with all of my girlfriends, but a couple of weeks beforehand my mother took me aside and said,"I need to speak to you privately. September 10th is not your birthday. It ' s October 10th."And I said,"What?"And she said..."Listen. The cut-off date to start kindergarten was September 15th.""So I told them that your birthday was September 10th, and then I wasn ' t sure that you weren ' t just going to go blab it all over the place, so I started to tell you your birthday was September 10th. But, Julie, you were so ready to start school, honey. You were so ready."I thought about it, and when I was four, I was already the oldest of four children, and my mother even had another child to come, so what I think she — understandably — really meant was that she was so ready, she was so ready. Then she said,"Don ' t worry, Julie. Every year on October 10th, when it was your birthday but you didn ' t realize it, I made sure that you ate a piece of cake that day."Which was comforting, but troubling. My mother had been celebrating my birthday with me, without me. What was so upsetting about this new piece of information was not that I had to change the date of my slumber party with all of my girlfriends. What was most upsetting was that this meant I was not a Virgo. I had a huge Virgo poster in my bedroom. And I read my horoscope every single day, and it was so totally me. And this meant that I was a Libra? So, I took the bus downtown to get the new Libra poster. The Virgo poster is a picture of a beautiful woman with long hair, sort of lounging by some water, but the Libra poster is just a huge scale. This was around the time that I started filling out physically, and I was filling out a lot more than a lot of the other girls, and frankly, the whole idea that my astrological sign was a scale just seemed ominous and depressing. But I got the new Libra poster, and I started to read my new Libra horoscope, and I was astonished to find that it was also totally me. It wasn ' t until years later, looking back on this whole age-of-reason, change- of-birthday thing, that it dawned on me : I wasn ' t turning seven when I thought I turned seven. I had a whole other month to do anything I wanted to before God started keeping tabs on me. Oh, life can be so cruel. One day, two Mormon missionaries came to my door. Now, I just live off a main thoroughfare in Los Angeles, and my block is — well, it ’ s a natural beginning for people who are peddling things door to door. Sometimes I get little old ladies from the Seventh Day Adventist Church showing me these cartoon pictures of heaven. And sometimes I get teenagers who promise me that they won ’ t join a gang and just start robbing people, if I only buy some magazine subscriptions from them. So normally, I just ignore the doorbell, but on this day, I answered. And there stood two boys, each about 19, in white, starched short-sleeved shirts, and they had little name tags that identified them as official representatives of the Church of Jesus Christ of Latter-day Saints, and they said they had a message for me, from God. I said,”A message for me? From God?”And they said,”Yes.”Now, I was raised in the Pacific Northwest, around a lot of Church of Latter-day Saints people and, you know, I ’ ve worked with them and even dated them, but I never really knew the doctrine, or what they said to people when they were out on a mission, and I guess I was sort of curious, so I said,”Well, please, come in.”And they looked really happy, because I don ’ t think this happens to them all that often. And I sat them down, and I got them glasses of water — Ok, I got it, I got it. I got them glasses of water. Don ’ t touch my hair, that ’ s the thing. You can ’ t put a video of myself in front of me and expect me not to fix my hair. Ok. So I sat them down and I got them glasses of water, and after niceties, they said,”Do you believe that God loves you with all his heart?”And I thought,”Well, of course I believe in God, but you know, I don ’ t like that word ’ heart, ’ because it anthropomorphizes God, and I don ’ t like the word, ’ his, ’ either, because that sexualizes God.”But I didn ’ t want to argue semantics with these boys, so after a very long, uncomfortable pause, I said,”Yes, yes, I do. I feel very loved.”And they looked at each other and smiled, like that was the right answer. And then they said,”Do you believe that we ’ re all brothers and sisters on this planet?”And I said,”Yes, I do.”And I was so relieved that it was a question I could answer so quickly. And they said,”Well, then we have a story to tell you.”And they told me this story all about this guy named Lehi, who lived in Jerusalem in 600 BC. Now, apparently in Jerusalem in 600 BC, everyone was completely bad and evil. Every single one of them : man, woman, child, infant, fetus. And God came to Lehi and said to him,”Put your family on a boat and I will lead you out of here.”And God did lead them. He led them to America. I said,”America? From Jerusalem to America by boat in 600 BC?”And they said,”Yes.”Then they told me how Lehi and his descendants reproduced and reproduced, and over the course of 600 years, there were two great races of them, the Nephites and the Lamanites, and the Nephites were totally good — each and every one of them — and the Lamanites were totally bad and evil — every single one of them just bad to the bone. Then, after Jesus died on the cross for our sins, on his way up to heaven, he stopped by America and visited the Nephites. And he told them that if they all remained totally, totally good — each and every one of them — they would win the war against the evil Lamanites. But apparently somebody blew it, because the Lamanites were able to kill all the Nephites. All but one guy, this guy named Mormon, who managed to survive by hiding in the woods. And he made sure this whole story was written down in reformed Egyptian hieroglyphics chiseled onto gold plates, which he then buried near Palmyra, New York. Well, I was just on the edge of my seat. I said,”What happened to the Lamanites?”And they said,”Well, they became our Native Americans, here in the U. S.”And I said,”So, you believe the Native Americans are descended from a people who were totally evil?”And they said,”Yes.”Then they told me how this guy named Joseph Smith found those buried gold plates right in his backyard, and he also found this magic stone back there that he put into his hat and then buried his face into, and this allowed him to translate the gold plates from the reformed Egyptian into English. Well, at this point I just wanted to give these two boys some advice about their pitch. I wanted to say —"Ok, don ' t start with this story."I mean, even the Scientologists know to start with a personality test before they start — telling people all about Xenu, the evil intergalactic overlord. Then, they said,"Do you believe that God speaks to us through his righteous prophets?"And I said,"No, I don ' t,"because I was sort of upset about this Lamanite story and this crazy gold plate story, but the truth was, I hadn ' t really thought this through, so I backpedaled a little and I said,"Well, what exactly do you mean by ' righteous '? And what do you mean by prophets? Like, could the prophets be women?"And they said,"No."And I said,"Why?"And they said,"Well, it ' s because God gave women a gift that is so spectacular, it is so wonderful, that the only gift he had left over to give men was the gift of prophecy."What is this wonderful gift God gave women, I wondered? Maybe their greater ability to cooperate and adapt? Women ' s longer lifespan? The fact that women tend to be much less violent than men? But no — it wasn ' t any of these gifts. They said,"Well, it ' s her ability to bear children."I said,"Oh, come on. I mean, even if women tried to have a baby every single year from the time they were 15 to the time they were 45, assuming they didn ' t die from exhaustion, it still seems like some women would have some time left over to hear the word of God."And they said,"No."Well, then they didn ' t look so fresh-faced and cute to me any more, but they had more to say. They said,"Well, we also believe that if you ' re a Mormon, and if you ' re in good standing with the church, when you die, you get to go to heaven and be with your family for all eternity."And I said,"Oh, dear. That wouldn ' t be such a good incentive for me."And they said,"Oh. Hey! Well, we also believe that when you go to heaven, you get your body restored to you in its best original state. Like, if you ' d lost a leg, well, you get it back. Or, if you ' d gone blind, you could see."I said,"Oh. Now, I don ' t have a uterus, because I had cancer a few years ago. So does this mean that if I went to heaven, I would get my old uterus back?"And they said,"Sure."And I said,"I don ' t want it back. I ' m happy without it."Gosh. What if you had a nose job and you liked it? Would God force you to get your old nose back? Then they gave me this Book of Mormon, told me to read this chapter and that chapter, and said they ' d come back and check in on me, and I think I said something like,"Please don ' t hurry,"or maybe just,"Please don ' t,"and they were gone. Ok, so I initially felt really superior to these boys, and smug in my more conventional faith. But then the more I thought about it, the more I had to be honest with myself. If someone came to my door and I was hearing Catholic theology and dogma for the very first time, and they said,"We believe that God impregnated a very young girl without the use of intercourse, and the fact that she was a virgin is maniacally important to us.""And she had a baby, and that ' s the son of God,"I mean, I would think that ' s equally ridiculous. I ' m just so used to that story. So, I couldn ' t let myself feel condescending towards these boys. But the question they asked me when they first arrived really stuck in my head : Did I believe that God loved me with all his heart? Because I wasn ' t exactly sure how I felt about that question. Now, if they had asked me,"Do you feel that God loves you with all his heart?"Well, that would have been much different, I think I would have instantly answered,"Yes, yes, I feel it all the time. I feel God ' s love when I ' m hurt and confused, and I feel consoled and cared for. I take shelter in God ' s love when I don ' t understand why tragedy hits, and I feel God ' s love when I look with gratitude at all the beauty I see."But since they asked me that question with the word"believe"in it, somehow it was all different, because I wasn ' t exactly sure if I believed what I so clearly felt.

Text Sample 975: NEG Dim 6 – 2006 – Score 1.00 – 2006/t000492.txt

About 10 years ago, I took on the task to teach global development to Swedish undergraduate students. That was after having spent about 20 years, together with African institutions, studying hunger in Africa. So I was sort of expected to know a little about the world. And I started, in our medical university, Karolinska Institute, an undergraduate course called Global Health. But when you get that opportunity, you get a little nervous. I thought, these students coming to us actually have the highest grade you can get in the Swedish college system, so I thought, maybe they know everything I ' m going to teach them about. So I did a pretest when they came. And one of the questions from which I learned a lot was this one :"Which country has the highest child mortality of these five pairs?"And I put them together so that in each pair of countries, one has twice the child mortality of the other. And this means that it ' s much bigger, the difference, than the uncertainty of the data. I won ' t put you at a test here, but it ' s Turkey, which is highest there, Poland, Russia, Pakistan and South Africa. And these were the results of the Swedish students. I did it so I got the confidence interval, which is pretty narrow. And I got happy, of course — a 1. 8 right answer out of five possible. That means there was a place for a professor of international health and for my course. But one late night, when I was compiling the report, I really realized my discovery. I have shown that Swedish top students know, statistically, significantly less about the world than the chimpanzees. Because the chimpanzee would score half right if I gave them two bananas with Sri Lanka and Turkey. They would be right half of the cases. But the students are not there. The problem for me was not ignorance ; it was preconceived ideas. I did also an unethical study of the professors of the Karolinska Institute, which hands out the Nobel Prize in Medicine, and they are on par with the chimpanzee there. This is where I realized that there was really a need to communicate, because the data of what ' s happening in the world and the child health of every country is very well aware. So we did this software, which displays it like this. Every bubble here is a country. This country over here is China. This is India. The size of the bubble is the population, and on this axis here, I put fertility rate. Because my students, what they said when they looked upon the world, and I asked them,"What do you really think about the world?"Well, I first discovered that the textbook was Tintin, mainly. And they said,"The world is still ' we ' and ' them. ' And ' we ' is the Western world and ' them ' is the Third World.""And what do you mean with ' Western world? '"I said."Well, that ' s long life and small family. And ' Third World ' is short life and large family."So this is what I could display here. I put fertility rate here — number of children per woman : one, two, three, four, up to about eight children per woman. We have very good data since 1962, 1960, about, on the size of families in all countries. The error margin is narrow. Here, I put life expectancy at birth, from 30 years in some countries, up to about 70 years. And in 1962, there was really a group of countries here that were industrialized countries, and they had small families and long lives. And these were the developing countries. They had large families and they had relatively short lives. Now, what has happened since 1962? We want to see the change. Are the students right? It ' s still two types of countries? Or have these developing countries got smaller families and they live here? Or have they got longer lives and live up there? Let ' s see. We stopped the world then. This is all UN statistics that have been available. Here we go. Can you see there? It ' s China there, moving against better health there, improving there. All the green Latin American countries are moving towards smaller families. Your yellow ones here are the Arabic countries, and they get longer life, but not larger families. The Africans are the green here. They still remain here. This is India ;

Indonesia is moving on pretty fast. In the ’ 80s here, you have Bangladesh still among the African countries. But now, Bangladesh — it ’ s a miracle that happens in the ’ 80s — the imams start to promote family planning, and they move up into that corner. And in the ’ 90s, we have the terrible HIV epidemic that takes down the life expectancy of the African countries. And the rest of them all move up into the corner, where we have long lives and small family, and we have a completely new world. Let me make a comparison directly between the United States of America and Vietnam. 1964 : America had small families and long life ; Vietnam had large families and short lives. And this is what happens. The data during the war indicate that even with all the death, there was an improvement of life expectancy. By the end of the year, family planning started in Vietnam, and they went for smaller families. And the United States up there is getting longer life, keeping family size. And in the ’ 80s now, they give up Communist planning and they go for market economy, and it moves faster even than social life. And today, we have in Vietnam the same life expectancy and the same family size here in Vietnam, 2003, as in United States, 1974, by the end of the war. I think we all, if we don ’ t look at the data, we underestimate the tremendous change in Asia, which was in social change before we saw the economic change. So let ’ s move over to another way here in which we could display the distribution in the world of income. This is the world distribution of income of people. One dollar, 10 dollars or 100 dollars per day. There ’ s no gap between rich and poor any longer. This is a myth. There ’ s a little hump here. But there are people all the way. And if we look where the income ends up, this is 100 percent of the world ’ s annual income. And the richest 20 percent, they take out of that about 74 percent. And the poorest 20 percent, they take about two percent. And this shows that the concept of developing countries is extremely doubtful. We think about aid, like these people here giving aid to these people here. But in the middle, we have most of the world population, and they have now 24 percent of the income. We heard it in other forms. And who are these? Where are the different countries? I can show you Africa. This is Africa. Ten percent of the world population, most in poverty. This is OECD — the rich countries, the country club of the UN. And they are over here on this side. Quite an overlap between Africa and OECD. And this is Latin America. It has everything on this earth, from the poorest to the richest in Latin America. And on top of that, we can put East Europe, we can put East Asia, and we put South Asia. And what did it look like if we go back in time, to about 1970? Then, there was more of a hump. And most who lived in absolute poverty were Asians. The problem in the world was the poverty in Asia. And if I now let the world move forward, you will see that while population increases, there are hundreds of millions in Asia getting out of poverty, and some others getting into poverty, and this is the pattern we have today. And the best projection from the World Bank is that this will happen, and we will not have a divided world. We ’ ll have most people in the middle. Of course it ’ s a logarithmic scale here, but our concept of economy is growth with percent. We look upon it as a possibility of percentile increase. If I change this and take GDP per capita instead of family income, and I turn these individual data into regional data of gross domestic product, and I take the regions down here, the size of the bubble is still the population. And you have the OECD there, and you have sub-Saharan Africa there, and we take off the Arab states there, coming both from Africa and from Asia, and we put them separately, and we can expand this axis, and I can give it a new dimension here, by adding the social values there, child survival. Now I have money on that axis, and I have the possibility of children to survive there. In some countries, 99. 7 % of children survive to five years of age ; others, only 70. And here, it seems, there is a gap between OECD, Latin America, East Europe, East Asia, Arab states, South Asia and sub-Saharan Africa. The linearity is very strong between child survival and money. But let me split sub-Saharan Africa. Health is there and better health is up there. I can go here, and I can split sub-Saharan Africa into its countries. And when it bursts, the size of each country bubble is the size of the population. Sierra Leone down there, Mauritius is up there. Mauritius was the first country to get away with trade barriers, and they could sell their sugar, they could sell their textiles, on equal terms as the people in Europe and North America. There ' s a huge difference [within] Africa. And Ghana is here in the middle. In Sierra Leone, humanitarian aid. Here in Uganda, development aid. Here, time to invest ; there, you can go for a holiday. There ' s tremendous variation within Africa, which we very often make that it ' s equal everything. I can split South Asia here. India ' s the big bubble in the middle. But there ' s a huge difference between Afghanistan and Sri Lanka. I can split Arab states. How are they? Same climate, same culture, same religion — huge difference. Even between neighbors — Yemen, civil war ; United Arab Emirates, money, which was quite equally and well-used. Not as the myth is. And that includes all the children of the foreign workers who are in the country. Data is often better than you think. Many people say data is bad. There is an uncertainty margin, but we can see the difference here : Cambodia, Singapore. The differences are much bigger than the weakness of the data. East Europe : Soviet economy for a long time, but they come out after 10 years very, very differently. And there is Latin America. Today, we don ' t have to go to Cuba to find a healthy country in Latin America. Chile will have a lower child mortality than Cuba within some few years from now. Here, we have high-income countries in the OECD. And we get the whole pattern here of the world, which is more or less like this. And if we look at it, how the world looks, in 1960, it starts to move. This is Mao Zedong. He brought health to China. And then he died. And then Deng Xiaoping came and brought money to China, and brought them into the mainstream again. And we have seen how countries move in different directions like this, so it ' s sort of difficult to get an example country which shows the pattern of the world. But I would like to bring you back to about here, at 1960. I would like to compare South Korea, which is this one, with Brazil, which is this one. The label went away for me here. And I would like to compare Uganda, which is there. I can run it forward, like this. And you can see how South Korea is making a very, very fast advancement, whereas Brazil is much slower. And if we move back again, here, and we put trails on them, like this, you can see again that the speed of development is very, very different, and the countries are moving more or less at the same rate as money and health, but it seems you can move much faster if you are healthy first than if you are wealthy first. And to show that, you can put on the way of United Arab Emirates. They came from here, a mineral country. They cached all the oil ; they got all the money ; but health cannot be bought at the supermarket. You have to invest in health. You have to get kids into schooling. You have to train health staff. You have to educate the population. And Sheikh Zayed did that in a fairly good way. In spite of falling oil prices, he brought this country up here. So we ' ve got a much more mainstream appearance of the world, where all countries tend to use their money better than they used it in the past. Now, this is, more or less, if you look at the average data of the countries — they are like this. That ' s dangerous, to use average data, because there is such a lot of difference within countries. So if I go and look here, we can see that Uganda today is where South Korea was in 1960. If I split Uganda, there ' s quite a difference within Uganda. These are the quintiles of Uganda. The richest 20 percent of Ugandans are there. The poorest are down there. If I split South Africa, it ' s like this. And if I go down and look at Niger, where there was such a terrible famine [recently], it ’ s like this. The 20 percent poorest of Niger is out here, and the 20 percent richest of South Africa is there, and yet we tend to discuss what solutions there should be in Africa. Everything in this world exists in Africa. And you can ’ t discuss universal access to HIV [treatment] for that quintile up here with the same strategy as down here. The improvement of the world must be highly contextualized, and it ’ s not relevant to have it on a regional level. We must be much more detailed. We find that students get very excited when they can use this. And even more, policy makers and the corporate sectors would like to see how the world is changing. Now, why doesn ’ t this take place? Why are we not using the data we have? We have data in the United Nations, in the national statistical agencies and in universities and other nongovernmental organizations. Because the data is hidden down in the databases. And the public is there, and the internet is there, but we have still not used it effectively. All that information we saw changing in the world does not include publicly funded statistics. There are some web pages like this, you know, but they take some nourishment down from the databases, but people put prices on them, stupid passwords and boring statistics. And this won ’ t work. So what is needed? We have the databases. It ’ s not a new database that you need. We have wonderful design tools and more and more are added up here. So we started a nonprofit venture linking data to design, we called”Gapminder,”from the London Underground, where they warn you,”Mind the gap.”So we thought Gapminder was appropriate. And we started to write software which could link the data like this. And it wasn ’ t that difficult. It took some person years, and we have produced animations. You can take a data set and put it there. We are liberating UN data, some few UN organization. Some countries accept that their databases can go out on the world. But what we really need is, of course, a search function, a search function where we can copy the data up to a searchable format and get it out in the world. And what do we hear when we go around? I ’ ve done anthropology on the main statistical units. Everyone says,”It ’ s impossible. This can ’ t be done. Our information is so peculiar in detail, so that cannot be searched as others can be searched. We cannot give the data free to the students, free to the entrepreneurs of the world.”But this is what we would like to see, isn ’ t it? The publicly funded data is down here. And we would like flowers to grow out on the net. One of the crucial points is to make them searchable, and then people can use the different design tools to animate it there. And I have pretty good news for you. I have good news that the [current], new head of UN statistics doesn ’ t say it ’ s impossible. He only says,”We can ’ t do it.”And that ’ s a quite clever guy, huh? So we can see a lot happening in data in the coming years. We will be able to look at income distributions in completely new ways. This is the income distribution of China, 1970. This is the income distribution of the United States, 1970. Almost no overlap. Almost no overlap. And what has happened? What has happened is this : that China is growing, it ’ s not so equal any longer, and it ’ s appearing here, overlooking the United States, almost like a ghost, isn ’ t it? It ’ s pretty scary. But I think it ’ s very important to have all this information. We need really to see it. And instead of looking at this, I would like to end up by showing the internet users per 1, 000. In this software, we access about 500 variables from all the countries quite easily. It takes some time to change for this, but on the axes, you can quite easily get any variable you would like to have. And the thing would be to get up the databases free, to get them searchable, and with a second click, to get them into the graphic formats, where you can instantly understand them. Now, statisticians don ’ t like it, because they say that this will not show the reality ; we have to have statistical, analytical methods. But this is hypothesis-generating. I end now with the world. There, the internet is coming. The number of internet users are going up like this. This is the GDP per capita. And it ' s a new technology coming in, but then amazingly, how well it fits to the economy of the countries. That ' s why the $100 computer will be so important. But it ' s a nice tendency. It ' s as if the world is flattening off, isn ' t it? These countries are lifting more than the economy, and it will be very interesting to follow this over the year, as I would like you to be able to do with all the publicly funded data. Thank you very much.

Text Sample 976: NEG Dim 6 – 2002 – Score 1.00 – 2002/t000102.txt

When I was five years old I fell in love with airplanes. Now I ' m talking about the ' 30s. In the ' 30s an airplane had two wings and a round motor, and was always flown by a guy who looked like Cary Grant. He had high leather boots, jodhpurs, an old leather jacket, a wonderful helmet and those marvelous goggles — and, inevitably, a white scarf, to flow in the wind. He ' d always walk up to his airplane in a kind of saunter, devil-may-care saunter, flick the cigarette away, grab the girl waiting here, give her a kiss. And then mount his airplane, maybe for the last time. Of course I always wondered what would happen if he ' d kissed the airplane first. But this was real romance to me. Everything about flying in those years, which was — you have to stop and think for a moment — was probably the most advanced technological thing going on at the time. So as a youngster, I tried to get close to this by drawing airplanes, constantly drawing airplanes. It ' s the way I got a part of this romance. And of course, in a way, when I say romance, I mean in part the aesthetics of that whole situation. I think the word is the holistic experience revolving around a product. The product was that airplane. But it built a romance. Even the parts of the airplane had French names. Ze fuselage, ze empanage, ze nessal. You know, from a romance language. So that it was something that just got into your spirit. It did mine. And I decided I had to get closer than just drawing fantasy airplanes. I wanted to build airplanes. So I built model airplanes. And I found that in doing the model airplanes the appearance drawings were not enough. You couldn ' t transfer those to the model itself. If you wanted it to fly you had to learn the discipline of flying. You had to learn about aeronautics. You had to learn what made an airplane stay in the air. And of course, as a model in those years, you couldn ' t control it. So it had to be self-righting, and stay up without crashing. So I had to give up the approach of drawing the fantasy shapes and convert it to technical drawings — the shape of the wing, the shape of the fuselage and so on — and build an airplane over these drawings that I knew followed some of the principles of flying. And in so doing, I could produce a model that would fly, stay in the air. And it had, once it was in the air, some of this romance that I was in love with. Well the act of drawing airplanes led me to, when I had the opportunity to choose a course in school, led me to sign up for aeronautical engineering. And when I was sitting in classes — in which no one asked me to draw an airplane — to my surprise. I had to learn mathematics and mechanics and all this sort of thing. I ' d wile away my time drawing airplanes in the class. One day a young man looked over my shoulder, he said,"You draw very well. You should be in the art department."And I said,"Why?"And he said,"Well for one thing, there are more girls there."So my romance was temporarily shifted. And I went into art because they appreciated drawing. Studied painting ; didn ' t do very well at that. Went through design, some architecture. Eventually hired myself out as a designer. And for the following 25 years, living in Italy, living in America, I doled out a piece of this romance to anybody who ' d pay for it — this sense, this aesthetic feeling, for the experience revolving around a designed object. And it exists. Any of you who rode the automobiles — was it yesterday? — at the track, you know the romance revolving around those high performance cars. Well in 25 years I was mostly putting out pieces of this romance and not getting a lot back in because design on call doesn ’ t always connect you with a circumstance in which you can produce things of this nature. So after 25 years I began to feel as though I was running dry. And I quit. And I started up a very small operation — went from 40 people to one, in an effort to rediscover my innocence. I wanted to get back where the romance was. And I couldn ’ t choose airplanes because they had gotten sort of unromantic at that point, even though I ’ d done a lot of airplane work, on the interiors. So I chose furniture. And I chose chairs specifically because I knew something about them. I ’ d designed a lot of chairs, over the years for tractors and trucks and submarines — all kinds of things. But not office chairs. So I started doing that. And I found that there were ways to duplicate the same approach that I used to use on the airplane. Only this time, instead of the product being shaped by the wind, it was shaped by the human body. So the discipline was — as in the airplane you learn a lot about how to deal with the air, for a chair you have to learn a lot about how to deal with the body, and what the body needs, wants, indicates it needs. And that ’ s the way, ultimately after some ups and downs, I ended up designing the chair I ’ m going to show you. I should say one more thing. When I was doing those model airplanes, I did everything. I conceived the kind of airplane. I basically engineered it. I built it. And I flew it. And that ’ s the way I work now. When I started this chair it was not a preconceived notion. Design nowadays, if you mean it, you don ’ t start with styling sketches. I started with a lot of loose ideas, roughly eight or nine years ago. And the loose ideas had something to do with what I knew happened with people in the office, at the work place — people who worked, and used task seating, a great many of them sitting in front of a computer all day long. And I felt, the one thing they don ’ t need, is a chair that interferes with their main reason for sitting there. So I took the approach that the chair should do as much for them as humanly possible or as mechanistically possible so that they didn ’ t have to fuss with it. So my idea was that, instead of sitting down and reaching for a lot of controls, that you would sit on the chair, and it would automatically balance your weight against the force required to recline. Now that may not mean a lot to some of you. But you know most good chairs do recline because it ’ s beneficial to open up this joint between your legs and your upper body for better breathing and better flow. So that if you sit down on my chair, whether you ’ re five feet tall or six foot six, it always deals with your weight and transfers the amount of force required to recline in a way that you don ’ t have to look for something to adjust. I ’ ll tell you right up front, this is a trade off. There are drawbacks to this. One is : you can ’ t accommodate everybody. There are some very light people, some extremely heavy people, maybe people with a lot of bulk up top. They begin to fall off the end of your chart. But the compromise, I felt, was in my favor because most people don ’ t adjust their chairs. They will sit in them forever. I had somebody on the bus out to the racetrack tell me about his sister calling him. He said she had one of the new, better chairs. She said,”Oh I love it.”She said,”But it ’ s too high.”So he said,”Well I ’ ll come over and look at it.”He came over and looked at it. He reached down. He pulled a lever. And the chair sank down. She said,”Oh it ’ s wonderful. How did you do that?”And he showed her the lever. Well, that ’ s typical of a lot of us working in chairs. And why should you get a 20-page manual about how to run a chair? I had one for a wristwatch once. 20 pages. Anyway, I felt that it was important that you didn ’ t have to make an adjustment in order to get this kind of action. The other thing I felt was that armrests had never really been properly approached from the standpoint of how much of an aid they could be to your work life. But I felt it was too much to ask to have to adjust each individual armrest in order to get it where you wanted. So I spent a long time. I said I worked eight or nine years on it. And each of these things went along sort of in parallel but incrementally were a problem of their own. I worked a long time on figuring out how to move the arms over a much greater arc — that is up and down — and make them a lot easier, so that you didn ' t have to use a button. And so after many trials, many failures, we came up with a very simple arrangement in which we could just move one arm or the other. And they go up easily. And stop where you want. You can put them down, essentially out of the way. No arms at all. Or you can pull them up where you want them. And this was another thing that I felt, while not nearly as romantic as Cary Grant, nevertheless begins to grab a little bit of aesthetic operation, aesthetic performance into a product. The next area that was of interest to me was the fact that reclining was a very important factor. And the more you can recline, in a way, the better it is. The more the angle between here and here opens up — and nowadays, with a screen in front of you, you don ' t want to have your eye drop too far in the recline, so we keep it at more or less the same level — but you transfer weight off your tailbones. Would everybody put their hand under their bottom and feel their tailbone? You feel that bone under there? Just your own. There ' s two of them, one on either side. All the weight of your upper torso — your arms, your head — goes right down through your back, your spine, into those bones when you sit. And that ' s a lot of load. Just relieving your arms with armrests takes 20 percent of that load off. Now that, if your spine is not held in a good position, will help bend your spine the wrong way, and so on. So to unload that great weight — if that indeed exists — you can recline. When you recline you take away a lot of that load off your bottom end, and transfer it to your back. At the same time, as I say, you open up this joint. And breathability is good. But to do that, if you have any amount of recline, it gets to the point where you need a headrest because nearly always, automatically hold your head in a vertical position, see? As I recline, my head says more or less vertical. Well if you ' re reclined a great deal, you have to use muscle force to hold your head there. So that ' s where a headrest comes in. Now headrest is a challenge because you want it to adjust enough so that it ' ll fit, you know, a tall guy and a short girl. So here we are. I ' ve got five inches of adjustment here in order to get the headrest in the right place. But then I knew from experience and looking around in offices where there were chairs with headrests that nobody would ever bother to reach back and turn a knob and adjust the headrest to put it in position. And you need it in a different position when you ' re upright, then when you ' re reclined. So I knew that had to be solved, and had to be automatic. So if you watch this chair as I recline, the headrest comes up to meet my neck. Ideally you want to put the head support in the cranial area, right there. So that part of it took a long time to work out. And there is a variety of other things : the shape of the cushions, the gel we put. We stole the idea from bicycle seats, and put gel in the cushions and in the armrests to absorb point load — distributes the loading so you don ' t get hard spots. You cant hit your elbow on bottom. And I did want to demonstrate the fact that the chair can accommodate people. While you ' re sitting in it you can adjust it down for the five-footer, or you can adjust it for the six-foot- six guy — all within the scope of a few simple adjustments.

Text Sample 977: NEG Dim 6 – 2002 – Score 1.00 – 2002/t000101.txt

One thing I wanted to say about film making is â about this film â in thinking about some of the wonderful talks we ’ ve heard here, Michael Moschen, and some of the talks about music, this idea that there is a narrative line, and that music exists in time. A film also exists in time ; it ’ s an experience that you should go through emotionally. And in making this film I felt that so many of the documentaries I ’ ve seen were all about learning something, or knowledge, or driven by talking heads, and driven by ideas. And I wanted this film to be driven by emotions, and really to follow my journey. So instead of doing the talking head thing, instead it ’ s composed of scenes, and we meet people along the way. We only meet them once. They don ’ t come back several times, so it really chronicles a journey. It ’ s something like life, that once you get in it you can ’ t get out. There are two clips I want to show you, the first one is a kind of hodgepodge, its just three little moments, four little moments with three of the people who are here tonight. It ’ s not the way they occur in the film, because they are part of much larger scenes. They play off each other in a wonderful way. And that ends with a little clip of my father, of Lou, talking about something that is very dear to him, which is the accidents of life. I think he felt that the greatest things in life were accidental, and perhaps not planned at all. And those three clips will be followed by a scene of perhaps what, to me, is really his greatest building which is a building in Dhaka, Bangladesh. He built the capital over there. And I think you ’ ll enjoy this building, it ’ s never been seen â it ’ s been still photographed, but never photographed by a film crew. We were the first film crew in there. So you ’ ll see images of this remarkable building. A couple of things to keep in mind when you see it, it was built entirely by hand, I think they got a crane the last year. It was built entirely by hand off bamboo scaffolding, people carrying these baskets of concrete on their heads, dumping them in the forms. It is the capital of the country, and it took 23 years to build, which is something they seem to be very proud of over there. It took as long as the Taj Mahal. Unfortunately it took so long that Lou never saw it finished. He died in 1974. The building was finished in 1983. So it continued on for many years after he died. Think about that when you see that building, that sometimes the things we strive for so hard in life we never get to see finished. And that really struck me about my father, in the sense that he had such belief that somehow, doing these things giving in the way that he gave, that something good would come out of it, even in the middle of a war, there was a war with Pakistan at one point, and the construction stopped totally and he kept working, because he felt,”Well when the war is done they ’ ll need this building.”So, those are the two clips I ’ m going to show. Roll that tape. Richard Saul Wurman : I remember hearing him talk at Penn. And I came home and I said to my father and mother,”I just met this man : doesn ’ t have much work, and he ’ s sort of ugly, funny voice, and he ’ s a teacher at school. I know you ’ ve never heard of him, but just mark this day that someday you will hear of him, because he ’ s really an amazing man.”Frank Gehry : I heard he had some kind of a fling with Ingrid Bergman. Is that true? Nathaniel Kahn : If he did he was a very lucky man. NK : Did you hear that, really? FG : Yeah, when he was in Rome. Moshe Safdie : He was a real nomad. And you know, when I knew him when I was in the office, he would come in from a trip, and he would be in the office for two or three days intensely, and he would pack up and go. You know he ’ d be in the office till three in the morning working with us and there was this kind of sense of the nomad in him. I mean as tragic as his death was in a railway station, it was so consistent with his life, you know? I mean I often think I ’ m going to die in a plane, or I ’ m going to die in an airport, or die jogging without an identification on me. I don ’ t know why I sort of carry that from that memory of the way he died. But he was a sort of a nomad at heart.

Louis Kahn : How accidental our existences are really and how full of influence by circumstance. Man : We are the morning workers who come, all the time, here and enjoy the walking, city ’ s beauty and the atmosphere and this is the nicest place of Bangladesh. We are proud of it. NK : You ’ re proud of it? Man : Yes, it is the national image of Bangladesh. NK : Do you know anything about the architect? Man : Architect? I ’ ve heard about him ; he ’ s a top-ranking architect. NK : Well actually I ’ m here because I ’ m the architect ’ s son, he was my father. Man : Oh! Dad is Louis Farrakhan? NK : Yeah. No not Louis Farrakhan, Louis Kahn. Man : Louis Kahn, yes! Man : Your father, is he alive? NK : No, he ’ s been dead for 25 years. Man : Very pleased to welcome you back. NK : Thank you. NK : He never saw it finished, Pop. No, he never saw this. Shamsul Wares : It was almost impossible, building for a country like ours. In 30, 50 years back, it was nothing, only paddy fields, and since we invited him here, he felt that he has got a responsibility. He wanted to be a Moses here, he gave us democracy. He is not a political man, but in this guise he has given us the institution for democracy, from where we can rise. In that way it is so relevant. He didn ’ t care for how much money this country has, or whether he would be able to ever finish this building, but somehow he has been able to do it, build it, here. And this is the largest project he has got in here, the poorest country in the world. NK : It cost him his life. SW : Yeah, he paid. He paid his life for this, and that is why he is great and we ’ ll remember him. But he was also human. Now his failure to satisfy the family life, is an inevitable association of great people. But I think his son will understand this, and will have no sense of grudge, or sense of being neglected, I think. He cared in a very different manner, but it takes a lot of time to understand that. In social aspect of his life he was just like a child, he was not at all matured. He could not say no to anything, and that is why, that he cannot say no to things, we got this building today. You see, only that way you can be able to understand him. There is no other shortcut, no other way to really understand him. But I think he has given us this building and we feel all the time for him, that ’ s why, he has given love for us. He could not probably give the right kind of love for you, but for us, he has given the people the right kind of love, that is important. You have to understand that. He had an enormous amount of love, he loved everybody. To love everybody, he sometimes did not see the very closest ones, and that is inevitable for men of his stature.

Text Sample 978: NEG Dim 6 – 2002 – Score 1.00 – 2002/t000333.txt

Thank you. Ooh, I ’ m like,”Phew, phew, calm down. Get back into my body now.”Usually when I play out, the first thing that happens is people scream out,”What ’ s she doing?!”I ’ ll play at these rock shows, be on stage standing completely still, and they ’ re like,”What ’ s she doing?! What ’ s she doing?!”And then I ’ ll kind of be like — — and then they ’ re like,”Whoa!”I ’ m sure you ’ re trying to figure out,”Well, how does this thing work?”Well, what I ’ m doing is controlling the pitch with my left hand. See, the closer I get to this antenna, the higher the note gets — — and you can get it really low. And with this hand I ’ m controlling the volume, so the further away my right hand gets, the louder it gets. So basically, with both of your hands you ’ re controlling pitch and volume and kind of trying to create the illusion that you ’ re doing separate notes, when really it ’ s continuously going... Sometimes I startle myself : I ’ ll forget that I have it on, and I ’ ll lean over to pick up something, and then it goes like — —”Oh!”And it ’ s like a funny sound effect that follows you around if you don ’ t turn the thing off. Maybe we ’ ll go into the next tune, because I totally lost where this is going. We ’ re going to do a song by David Mash called”Listen : the Words Are Gone,”and maybe I ’ ll have words come back into me afterwards if I can relax. So, I ’ m trying to think of some of the questions that are commonly asked ; there are so many. And... Well, I guess I could tell you a little of the history of the theremin. It was invented around the 1920s, and the inventor, Lon Theremin — he also was a musician besides an inventor — he came up with the idea for making the theremin, I think, when he was working on some shortwave radios. And there ’ d be that sound in the signal — it ’ s like — and he thought,”Oh, what if I could control that sound and turn it into an instrument, because there are pitches in it.”And so somehow through developing that, he eventually came to make the theremin the way it is now. And a lot of times, even kids nowadays, they ’ ll make reference to a theremin by going,”Whoo-hoo-hoo-hoo,”because in the ’ 50s it was used in the sci-fi horror movies, that sound that ’ s like... It ’ s kind of a funny, goofy sound to do. And sometimes if I have too much coffee, then my vibrato gets out of hand. You ’ re really sensitive to your body and its functions when you ’ re behind this thing. You have to stay so still if you want to have the most control. It reminds me of the balancing act earlier on — what Michael was doing — because you ’ re fighting so hard to keep the balance with what you ’ re playing with and stay in tune, and at the same time you don ’ t want to focus so much on being in tune all the time ; you want to be feeling the music. And then also, you ’ re trying to stay very, very, very still because little movements with other parts of your body will affect the pitch, or sometimes if you ’ re holding a low note — — and breathing will make it... If I pass out on the next song... I think of it almost like like a yoga instrument because it makes you so aware of every little crazy thing your body is doing, or just aware of what you don ’ t want it to be doing while you ’ re playing ; you don ’ t want to have any sudden movements. And if I go to a club and play a gig, people are like,”Here, have some drinks on us!”And it ’ s like,”Well, I ’ m about to go on soon ; I don ’ t want to be like — — you know?”It really does reflect the mood that you ’ re in also, if you ’ re... it ’ s similar to being a vocalist, except instead of it coming out of your throat, you ’ re controlling it just in the air and you don ’ t really have a point of reference ; you ’ re always relying on your ears and adjusting constantly. You just have to always adjust to what ’ s happening and realize you ’ ll have bummer notes come here and there and listen to it, adjust it, and just move on, or else you ’ ll get too tied up and go crazy. Like me. I think we will play another tune now. I ’ m going to do”Lush Life.”It ’ s one of my favorite tunes to play.

Text Sample 979: NEG Dim 6 – 2005 – Score 1.00 – 2005/t000484.txt

Video : Narrator : An event seen from one point of view gives one impression. Seen from another point of view, it gives quite a different impression. But it ’ s only when you get the whole picture you can fully understand what ’ s going on. Sasha Vucinic : It ’ s a great clip, isn ’ t it? And I found that in 29 seconds, it tells more about the power of, and importance of, independent media than I could say in an hour. So I thought that it will be good to start with it. And also start with a little bit of statistics. According to relevant researchers, 83 percent of the population of this planet lives in the societies without independent press. Think about that number : 83 percent of the population on the whole planet does not really know what is going on in their countries. The information they get gets filtered through somebody who either twists that information, or colors that information, does something with it. So they ’ re deprived of understanding their reality. That is just to understand how big and important this problem is.

Now those of you who are lucky enough to live in those societies that represent 17 percent, I think should enjoy it until it lasts. You know, Sunday morning, you flick the paper, get your cappuccino. Enjoy it while it lasts. Because as we heard yesterday, countries can lose stars from their flags, but they can also lose press freedom, as I guess Americans among us can tell us more about. But that ’ s totally another and separate topic. So I can go back to my story. My story starts — the story I want to share — starts in 1991. At that time I was running B92, the only independent, for that matter the only electronic media, in the country. And I guess we were sharing — we had that regular life of the only independent media in the country, operating in hostile environment, where government really wants to make your life miserable. And there are different ways. Yeah, it was the usual cocktail : a little bit of threats, a little bit of friendly advice, a little bit of financial police, a little bit of text control, so you always have somebody who never leaves your office. But what they really do, which is very powerful, and that is what governments in the late ’ 90s started doing if they don ’ t like independent media companies — you know, they threaten your advertisers. Once they threaten your advertisers, market forces are actually, you know, destroyed, and the advertisers do not want to come — no matter how much does it make sense for them — do not want to come and advertise. And you have a problem making ends meet. At that time at the beginning of the ’ 90s, we had that problem, which was, you know, survival below one side, but what was really painful for me was, remember, the beginning of the ’ 90s, Yugoslavia is falling apart. We were sitting over there with a country in a downfall, in a slow-motion downfall. And we all had all of that on tapes. We had the ability to understand what was going on. We were actually recording history. The problem was that we had to re-tape that history a week later ; because if we did not, we could not afford enough tapes to keep archives of that history. So if I gave you that picture, I don ’ t want to go too long on that. In that context a gentleman came to my office at that time. It was still 1991. He was running a media systems organization which is still in business, the gentleman is still in business. And what did I know at that time about media systems? I would think media systems were organizations, which means they should help you. So I prepared two plans for that meeting, two strategic plans : the small one and the big one. The small one was, I just wanted him to help us get those damn tapes, so we can keep that archive for the next 50 years. The big plan was to ask him for a 1, 000, 000-dollar loan. Because I thought, I still maintain, that serious and independent media companies are great business. And I thought that B92 will survive and be a great company once Milosevic is gone, which turned out to be true. It ’ s now probably either the biggest or the second biggest media company in the country. And I thought that the only thing that we needed at that time was 1, 000, 000-dollar loan to take us through those hard times. To make a long story short, the gentleman comes into the office, great suit and tie. I gave him what I thought was a brilliant explanation of the political situation and explained how hard and difficult the war will be. Actually, I underestimated the atrocities, I have to admit. Anyway, after that whole, big, long explanation, the only question he had for me — and this is not a joke — is, are we paying royalties after we broadcast music of Michael Jackson? That was really the only question he had. He left, and I remember being actually very angry at myself because I thought there must be an institution in the world that is providing loans to media companies. It ’ s so obvious, straight in your face, and somebody must have thought of it. Somebody must have started something like that. And I thought, I ’ m just dumb and I cannot find it. You know, in my defense, there was no Google at that time ; you could not just Google in ’ 91. So I thought that that ’ s actually my problem. Now we go from here, fast forward to 1995. I have — I left the country, I have a meeting with George Soros, trying for the third time to convince him that his foundation should invest in something that should operate like a media bank. And basically what I was saying is very simple. You know, forget about charity ; it doesn ’ t work. Forget about handouts ; 20, 000 dollars do not help anybody. What you should do is you should treat media companies as a business. It ’ s business anywhere. Media business, or any other business, it needs to be capitalized. And what these guys need, actually, is access to capital. So third meeting, arguments are pretty well exercised. At the end of the meeting he says, look, it is not going to work ; you will never see your money back ; but my foundations will put 500, 000 dollars so you can test the idea. See that it will not work. He said, I ’ ll give you a rope to hang yourself. I knew two things after that meeting. First, under no circumstances I want to hang myself. And second, that I have no idea how to make it work. You see, at the level of a concept, it was a great concept. But it ’ s one thing to have a concept ; it ’ s a totally separate thing to actually make it work. So I had absolutely no idea how that could actually work. Had the wrong idea ; I thought that we can be a bank. You see banks — I don ’ t know if there are any bankers over here ; I apologize in advance — but it ’ s the best job in the world. You know, you find somebody who is respectable and has a lot of money. You give them more money ; they repay you that over a time. You collect interest and do nothing in between. So I thought, why don ’ t we get into that business? So here we are having our first client, brilliant. First independent newspaper in Slovakia. The government cutting them off from all the printing facilities in Bratislava. So here ’ s the daily newspaper that has to be printed 400 kilometers away from the capital. It ’ s a daily newspaper with a deadline of 4 p. m. That means that they have no sports ; they have no latest news ; circulation goes down. It ’ s a kind of very nice, sophisticated way how to economically strangle a daily newspaper. They come to us with a request for a loan. They want to — the only way for them to survive is to get a printing press. And we said, that ’ s fine ; let ’ s meet ; you ’ ll bring us your business plan, which eventually they did. We start the meeting. I get these two pieces of paper, not like this, A4 format, so it ’ s much bigger. A lot of numbers there. A lot of numbers. But however you put it, you know, the numbers do not make any sense. And that ’ s the best they could do. We were the best that they could do. So that is how we understood what our method is. It ’ s not a bank. We had to actually go into these companies and earn our return by fixing them — by establishing management systems, by providing all that knowledge, how do you run a business on one side — while they all know how to run, how to create content. Just quickly on the results. Over these 10 years, 40 million dollars in affordable financing, average interest rate five percent to six percent. Lately we are going wild, charging seven percent from time to time. We do it in 17 countries of the developing world. And here is the most stunning number. Return rate — the one that Soros was so worried about — 97 percent. 97 percent of all the scheduled repayments came back to us on time. What do we typically finance? We finance anything that a media company would need, from printing presses to transmitters. What is most important is we do it either in form of loans, equities, lease — whatever is appropriate for, you know, supporting anybody. But what is most important here is, who do we finance? We believe that in the last 10 years companies that we ’ ve financed are actually the best media companies in the developing world. That is a”Who is Who”list. And I could spend hours talking about them, because they ’ re all kind of heroes. And I can, but I ’ ll give you just, maybe one, and depending on time I may give you two examples who we work with. You see we started working in Eastern and Central Europe, and moved to Russia. Our first loan in Russia was in Chelyabinsk. I ' ll bet half of you have never heard of that place. In the south of Russia there ' s a guy called Boris Nikolayevich Kirshin, who is running an independent newspaper there. The city was closed until early ' 90s because, of all things, they were producing glass for Tupolev planes. Anyway, he ' s running independent newspaper there. After two years working with us, he becomes the most respected newspaper in that small place. Governor comes to him one day, actually invites him to come to his office. He goes and sees the governor. The governor says, Boris Nikolayevich, I understand you are doing a great job, and you are the most respected newspaper in our district. And I want to offer you a deal. Can you please give me your newspaper for the next nine months, because I have elections — there are elections coming up in nine months. I will not run, but it ' s very important for me who is going to succeed me. So give me the paper for nine months. I ' ll give it back to you. I have no interest in being in media business. How much would that cost? Boris Nikolayevich says,"It ' s not for sale."The governor says,"We will close you."Boris Nikolayevich says,"No, you cannot do it."Six months later the newspaper was closed. Luckily, we had enough time to help Boris Nikolayevich take all the assets out of that company and bring him into a new one, to get all the subscription lists, rehire staff. So what the governor got was an empty shell. But that is what happens if you ' re in business of independent media, and if you are a banker for independent media. So it sounds like a great story. Somewhere down the road we opened a media management center. We started our media lab, sounds like a real great story. But there is a second angle to that. The second angle, like in this clip. If you take the camera above, you start thinking about these numbers again. 40 million dollars over 10 years spread over 17 countries. That is not too much, is it? It ' s actually just a drop in the sea. Because when you think about the importance, some of the issues that we were talking about last night — this last session we had about Africa and his hypothetical 50 billion dollars destined for Africa. All of those, not all, half of those problems mentioned last night — government accountability, corruption, how do you fight corruption, giving voice to unheard, to poor — it ' s why independent media is in business. And it ' s why it was invented. So from that perspective, what we did is just really one drop in the sea of that need that we can identify. Now ours is just one story. I ' m sure that in this room there are, like, 15 other wonderful stories of nonprofits doing spectacular work. Here is where the problem is, and I ' ll explain to you as well as I can what the problem is. And it ' s called fundraising. Imagine that this third of this room is filled with people who represent different foundations. Imagine two thirds over here running excellent organizations, doing very important work. Now imagine that every second person over here is deaf, does not hear, and switch the lights off. Now that is how difficult it is to match people from this side of the room with people of that side of the room. So we thought that some kind of a big idea is needed to reform, to totally rethink fundraising. You know, instead of people running in this dark, trying to find their own match, who will be willing, who has the same goals. Instead of all of that we thought there is — something new needs to be invented. And we came up with this idea of issuing bonds, press freedom bonds. If there are investors willing to finance U. S. government budget deficit, why wouldn ' t we find investors willing to finance press freedom deficit? We ' ve decided to do it this fall ; we will issue them, probably in denominations of 1, 000 dollars. I don ' t want to advertise them too much ; that ' s not the point. But the point is, if we ever survive to actually issue them, find enough investors that this can be considered a success, there ' s nothing stopping the next organization to start to issue bonds next spring. And those can be environmental bonds. And then two weeks later, Iqbal Quadir can issue his electricity in Bangladesh bonds. And before you know it, any social cause can be actually financed in this way. Now we do daydreaming in 11:30 with 55 seconds left. But let ’ s take the idea further. You do it, you start it in the States, because it ’ s, you know, concepts are very, very close to American minds. But you can actually bring it to Europe, too. You can bring it to Asia. You can, once you have all of those different points, you can make it easy for investors. Put all of those bonds at one place and they sit down and click. Once you have more than 10 of them you have to develop some kind of a matrix. What do investors get? On one side financial, on the other side social. So that brings the idea of some kind of rating agency, Morningstar type. It says, you know, social impact over here is spectacular, five stars. Financial, they give you one percent, only one star. Now take it to the last step. Once you have all of that put together, there ’ s not one reason why you couldn ’ t actually have a marketplace for all of that, where you cannot dispose of all of those bonds in a pretty quick way. And in that way you organize the financing so there are no dark rooms, no blind people running around to find each other. Thank you.

Text Sample 980: NEG Dim 6 – 2005 – Score 1.00 – 2005/t000479.txt
This is really a two-hour presentation I give to high school students, cut down to three minutes. And it all started one day on a plane, on my way to TED, seven years ago. And in the seat next to me was a high school student, a teenager, and she came from a really poor family. And she wanted to make something of her life, and she asked me a simple little question. She said,”What leads to success?”And I felt really badly, because I couldn ’ t give her a good answer. So I get off the plane, and I come to TED. And I think, jeez, I ’ m in the middle of a room of successful people! So why don ’ t I ask them what helped them succeed, and pass it on to kids? So here we are, seven years, 500 interviews later, and I ’ m going to tell you what really leads to success and makes TEDsters tick. And the first thing is passion. Freeman Thomas says,”I ’ m driven by my passion.”TEDsters do it for love ; they don ’ t do it for money. Carol Coletta says,”I would pay someone to do what I do.”And the interesting thing is : if you do it for love, the money comes anyway. Work! Rupert Murdoch said to me,”It ’ s all hard work. Nothing comes easily. But I have a lot of fun.”Did he say fun? Rupert? Yes! TEDsters do have fun working. And they work hard. I figured, they ’ re not workaholics. They ’ re workafrolics. Good! Alex Garden says,”To be successful, put your nose down in something and get damn good at it.”There ’ s no magic ; it ’ s practice, practice, practice. And it ’ s focus. Norman Jewison said to me,”I think it all has to do with focusing yourself on one thing.”And push! David Gallo says,”Push yourself. Physically, mentally, you ’ ve got to push, push, push.”You ’ ve got to push through shyness and self-doubt. Goldie Hawn says,”I always had self-doubts. I wasn ’ t good enough ; I wasn ’ t smart enough. I didn ’ t think I ’ d make it.”Now it ’ s not always easy to push yourself, and that ’ s why they invented mothers. Frank Gehry said to me,”My mother pushed me.”Serve! Sherwin Nuland says,”It was a privilege to serve as a doctor.”A lot of kids want to be millionaires. The first thing I say is :”OK, well you can ’ t serve yourself ; you ’ ve got to serve others something of value. Because that ’ s the way people really get rich.”Ideas! TEDster Bill Gates says,”I had an idea : founding the first micro- computer software company.”I ’ d say it was a pretty good idea. And there ’ s no magic to creativity in coming up with ideas — it ’ s just doing some very simple things. And I give lots of evidence. Persist! Joe Kraus says,”Persistence is the number one reason for our success.”You ’ ve got to persist through failure. You ’ ve got to persist through crap! Which of

course means”Criticism, Rejection, Assholes and Pressure.”So, the answer to this question is simple : Pay 4, 000 bucks and come to TED. Or failing that, do the eight things — and trust me, these are the big eight things that lead to success. Thank you TEDsters for all your interviews!

Text Sample 981: NEG Dim 6 – 2005 – Score 1.00 – 2005/t000470.txt
I don ’ t know about you, but I haven ’ t quite figured out exactly what technology means in my life. I ’ ve spent the past year thinking about what it really should be about. Should I be pro-technology? Should I embrace it full arms? Should I be wary? Like you, I ’ m very tempted by the latest thing. But at the other hand, a couple of years ago I gave up all of my possessions, sold all my technology — except for a bicycle — and rode across 3, 000 miles on the U. S. back roads under the power of my one body, fuelled mostly by Twinkies and junk food. And I ’ ve since then tried to keep technology at arm ’ s length in many ways, so it doesn ’ t master my life. At the same time, I run a website on cool tools, where I issue a daily obsession of the latest things in technology. So I ’ m still perplexed about what the true meaning of technology is as it relates to humanity, as it relates to nature, as it relates to the spiritual. And I ’ m not even sure we know what technology is. And one definition of technology is that which is first recorded. This is the first example of the modern use of technology that I can find. It was the suggested syllabus for dealing with the Applied Arts and Science at Cambridge University in 1829. Before that, obviously, technology didn ’ t exist. But obviously it did. I like one of the definitions that Alan Kay has for technology. He says technology is anything that was invented after you were born. So it sums up a lot of what we ’ re talking about. Danny Hillis actually has an update on that — he says technology is anything that doesn ’ t quite work yet. Which also, I think, gets into a little bit of our current idea. But I was interested in another definition of technology. Something, again, that went back to something more fundamental. Something that was deeper. And as I struggled to understand that, I came up with a way of framing the question that seemed to work for me in my investigations. And I ’ m, this morning, going to talk about this for the first time. So this is a very rough attempt to think out loud. The question that I came up with was this question : what does technology want? And by that, I don ’ t mean, does it want chocolate or vanilla? By what it wants, I mean, what are its inherent trends and biases? What are its tendencies over time? One way to think about this is thinking about biological organisms, which we ’ ve heard a lot about. And the trick that Richard Dawkins does, which is to say, to look at them as simply as genes, as vehicles for genes. So he ’ s saying, what do genes want? The selfish gene. And I ’ m applying a similar trick to say, what if we looked at the universe in our culture through the eyes of technology? What does technology want? Obviously, this in an incomplete question, just as looking at an organism as only a gene is an incomplete way of looking at it. But it ’ s still very, very productive. So I ’ m attempting to say, if we take technology ’ s view of the world, what does it want? And I think once we ask that question we have to go back, actually, to life. Because obviously, if we keep extending the origins of technology far back, I think we come back to life at some point. So that ’ s where I want to begin my little exploration, is in life. And like you heard from the previous speakers, we don ’ t really know what life there is on Earth right now. We have really no idea. Craig Venter ’ s tremendous and brilliant attempt to DNA sequence things in the ocean is great. Brian Farrell ’ s work is all part of this agenda to try and actually discover all the species on Earth. And one of the things that we should

do is just make a grid of the globe and randomly go and inspect all the places that the grid intersects, just to see what ’ s on life. And if we did that with our little Martian probe, which we have not done on Earth, we would begin to see some incredible species. This is not on another planet. These are things that are hidden away on our planet. This is an ant that stores its colleagues ’ honey in its abdomen. Each one of these organisms that we ’ ve described — that you ’ ve seen from Jamie and others, these magnificent things — what they ’ re doing, each one of them, is they ’ re hacking the rules of life. I can ’ t think of a single general principle of biology that does not have an exception somewhere by some organism. Every single thing that we can think of — and if you heard Olivia ’ s talk about the sexual habits, you ’ ll realize that there isn ’ t anything we can say that ’ s true for all life, because every single one of them is hacking something about it. This is a solar-powered sea slug. It ’ s a nudibranch that has incorporated chloroplast inside it to drive its energy. This is another version of that. This is a sea dragon, and the one on the bottom, the blue one, is a juvenile that has not yet swallowed the acid, has not yet taken in the brown-green algae pond scum into its body to give it energy. These are hacks, and if we looked at the general shape of the approaches to hacking life there are, current consensus, six kingdoms. Six different broad approaches : the plants, the animals, the fungi, the protests — the little things — the bacteria and the Archaea bacteria. The Archaeas. Those are the general approaches to life. That ’ s one way to look at life on Earth today. But a more interesting way, the current way to take the long view, is to look at it in an evolutionary perspective. And here we have a view of evolution where rather than having evolution go over the linear time, we have it coming out from the center. So in the center is the most primitive, and this is a genealogical chart of all life on earth. This is all the same six kingdoms. You see 4, 000 representative species, and you can see where we are. But what I like about this is it shows that every living organism on Earth today is equally evolved. Those fungi and bacteria are as highly evolved as humans. They ’ ve been around just as long and gone through just the same kind of trial and error to get here. But we see that each one of these is actually hacking, and has a different way of finding out how to do life. And if we take the long-term trends of life, if we begin to say, what does evolution want? There ’ s several things that we see. One of the things about evolution is that nowhere on Earth have we ever been where we don ’ t find life. We find life at the bottom of every long-term, long-distance drilling core into the center of rock that we bring up — and there ’ s bacteria in the pores of that rock. And wherever life is, it never retreats. It ’ s ubiquitous and it wants to be more. More and more of the inert matter of the globe is being touched and animated by life. The second thing is is we see diversity. We also see specialization. We see the movement from a general-purpose cell to the more specific and specialized. And we see a drift towards complexity that ’ s very intuitive. And actually, we have current data that does show that there is an actual drift towards complexity over time. And the last thing, I bring back this nudibranch. One of the things we see about life is that it moves from the inner to increasing sociability. And by that it means that there is more and more of life whose entire environment is other life. Like those chloroplast cells — they ’ re completely surrounded by other life. They never touch the inner matter. There is more and more co-evolution. And so the general, long-term trends of evolution are roughly these five : ubiquity, diversity, specialization, complexity and socialization. Now, I took that and said, OK, what are the long-term trends in technology? And again, my question is, what does technology want? And so, remarkably, I discovered that there ’ s also a drift toward specialization. That we see there ’ s a general hammer, and hammers become more and more specific over time. There ’ s obviously diversity. Huge numbers of things. This is all the contents of a Japanese home. I actually had my daughter — gave her a tally counter, and I gave her an assignment last summer to go around and count the number of species of technology in our household. And it came up with 6, 000 different species of products. I did some research and found out that the King of England, Henry VIII, had only about 7, 000 items in his household. And he was the King of England, and that was the entire wealth of England at the time. So we ’ re seeing huge numbers of diversity in the kinds of things. This is a scene from Star Wars where the 3PO comes out and he sees machines making machines. How depraved! Well, this is actually what we ’ re headed towards : world machines. And the technology is only being thrown out by other technologies. Most machines will only ever be in contact with other technology and not non-technology, or even life. And thirdly, the idea that machines are becoming biological and complex is at this point a cliche. And I ’ m happy to say, I was partly responsible for that cliche that machines are becoming biological, but that ’ s pretty evident. So the major trends in technology evolution actually are the same as in biological evolution. The same drives that we see towards ubiquity, towards diversity, towards socialization, towards complexity. That is maybe not a big surprise because if we map out, say, the evolution of armor, you can actually follow a sort of an evolutionary-type cladistic tree. I suggest that, in fact, technology is the seventh kingdom of life. That its operations and how it works is so similar that we can think of it as the seventh kingdom. And so it would be sort of approximately up there, coming out of the animal kingdom. And if we were to do that, we would find out — we could actually approach technology in this way. This is Niles Eldredge. He was the co-developer with Stephen Jay Gould of the theory of punctuated equilibrium. But as a sideline, he happens to collect cornets. He has one of the world ’ s largest collections — about 500 of them. And he has decided to treat them as if they were trilobites, or snails, and to do a morphological analysis, and try to derive their genealogical history over time. This is his chart, which is not quite published yet. But the most interesting aspect about this is that if you look at those red lines at the bottom, those indicate basically a parentage of a type of cornet that was no longer made. That does not happen in biology. When something is extinct, you can ’ t have it as your parent. But that does happen in technology. And it turns out that that ’ s so distinctive that you can actually look at this tree, and you can actually use it to determine that this is a technological system versus a biological system. In fact, this idea of resurrecting the whole idea is so important that I began to think about what happens with old technology. And it turns out that, in fact, technologies don ’ t die. So I suggested this to an historian of science, and he said,”Well, what about, you know, come on, what about steam cars? They ’ re not around anymore.”Well actually, they are. In fact, they ’ re so around that you can buy new parts for a Stanley steam automobile. And this is a website of a guy who ’ s selling brand new parts for the Stanley automobile. And the thing that I liked is sort of this one-click, add-to-your-cart button — — for buying steam valves. I mean, it was just — it was really there. And so, I began to think about, well, maybe that ’ s just a random sample. Maybe I should do this sort of in a more conservative way. So I took the great big 1895 Montgomery Ward ’ s catalog and I randomly went through it. And I took a page — not quite a random page — I took a page that was actually more difficult than others because lots of the pages are filled with things that are still being made. But I took this page and I said, how many of these things are still being made? And not antiques. I want to know how many of these things are still in production. And the answer is : all of them. All of them are still being produced. So you ’ ve got corn shellers. I don ’ t know who needs a corn sheller. Be it corn shellers — you ’ ve got ploughs ; you ’ ve got fan mills ; all these things — and these are not, again, antiques. These are — you can order these. You can go to the web and you can buy them now, brand-new made. So in a certain sense, technologies don ’ t die. In fact, you can buy, for 50 bucks, a stone-age knife made exactly the same way that they were made 10, 000 years ago. It ’ s short, bone handle, 50 bucks. And in fact, what ’ s important is that this information actually never died out. It ’ s not just that it was resurrected. It ’ s continued all along. And in Papua New Guinea, they were making stone axes until two decades ago, just as a course of practical matters. Even when we try to get rid of a technology, it ’ s actually very hard. So we ’ ve all heard about the Amish giving up cars. We ’ ve heard about the Japanese giving up guns. We ’ ve heard about this and that. But I actually went back and took what I could find, the examples in history where there have been prohibitions against technology, and then I tried to find out when they came back in, because they always came back in. And it turns out that the time, the duration of when they were outlawed and prohibited, is decreasing over time. And that basically, you can delay technology, but you can ’ t kill it. So this makes sense, because in a certain sense what culture is, is the accumulation of ideas. That ’ s what it ’ s for. It ’ s so that ideas don ’ t die out. And when we take that, we take this idea of what culture is doing and add it to what the long- term trajectory — again, in life ’ s evolution — we find that each case — each of the major transitions in life — what they ’ re really about is accelerating and changing the way in which evolution happens. They ’ re actually changing the way in which ideas are generated. So all these steps in evolution are increasing, basically, the evolution of evolvability. So what ’ s happening over time in life is that the ways in which you generate these new ideas, these new hacks, are increasing. And the real tricks are ways in which you kind of explore the way of exploring. And then what we see in the singularity, that prophesized by Kurzweil and others — his idea that technology is accelerating evolution. It ’ s accelerating the way in which we search for ideas. So if you have life hacking — life means hacking, the game of survival — then evolution is a way to extend the game by changing the rules of the game. And what technology is really about is better ways to evolve. That is what we call an”infinite game.”That ’ s the definition of”infinite game.”A finite game is play to win, and an infinite game is played to keep playing. And I believe that technology is actually a cosmic force. The origins of technology was not in 1829, but was actually at the beginning of the Big Bang, and at that moment the entire huge billions of stars in the universe were compressed. The entire universe was compressed into a little quantum dot, and it was so tight in there, there was no room for any difference at all. That ’ s the definition. There was no temperature. There was no difference whatsoever. And at the Big Bang, what it expanded was the potential for difference. So as it expands and as things expand what we have is the potential for differences, diversity, options, choices, opportunities, possibilities and freedoms. Those are all basically the same thing. And those are the things that technology brings us. That ’ s what technology is bringing us : choices, possibilities, freedoms. That ’ s what it ’ s about. It ’ s this expansion of room to make differences. And so a hammer, when we grab a hammer, that ’ s what we ’ re grabbing. And that ’ s why we continue to grab technology — because we want those things. Those things are good. Differences, freedom, choices, possibilities. And each time we make a new opportunity place, we ’ re allowing a platform to make new ones. And I think it ’ s really important. Because if you can imagine Mozart before the technology of the piano was invented — what a loss to society there would be. Imagine Van Gogh being born before the technologies of cheap oil paints. Imagine Hitchcock before the technologies of film. Somewhere, today, there are millions of young children being born whose technology of self-expression has not yet been invented. We have a moral obligation to invent technology so that every person on the globe has the potential to realize their true difference. We want a trillion zillion species of one individuals. That ’ s what technology really wants. I ’ m going to skip through some of the objections because I don ’ t have answers to why there ’ s deforestation. I don ’ t have an answer to the fact that there seem to be bad technologies. I don ’ t have an answer to how this impacts on our dignity, other than to suggest that maybe the seventh kingdom, because it ’ s so close to what life is about, maybe we can bring it back and have it help us monitor life. Maybe in some ways the fact that what we ’ re trying to do with technology is find a good home for it. It ’ s a terrible thing to spray DDT on cotton fields, but it ’ s a really good thing to use to eliminate millions of cases of death due to malaria in a small village. Our humanity is actually defined by technology. All the things that we think that we really like about humanity is being driven by technology. This is the infinite game. That ’ s what we ’ re talking about. You see, technology is a way to evolve the evolution. It ’ s a way to explore possibilities and opportunities and create more. And it ’ s actually a way of playing the game, of playing all the games. That ’ s what technology wants. And so when I think about what technology wants, I think that it has to do with the fact that every person here — and I really believe this — every person here has an assignment. And your assignment is to spend your life discovering what your assignment is. That recursive nature is the infinite game. And if you play that well, you ’ ll have other people involved, so even that game extends and continues even when you ’ re gone. That is the infinite game. And what technology is is the medium in which we play that infinite game. And so I think that we should embrace technology because it is an essential part of our journey in finding out who we are. Thank you.

Text Sample 982: NEG Dim 6 – 1984 – Score 58.00 – 1984/t000297.txt

In this rather long sort of marathon presentation, I ’ ve tried to break it up into three parts : the first being a whole lot of examples on how it can be a little bit more pleasurable to deal with a computer and really address the qualities of the human interface. And these will be some simple design qualities and they will also be some qualities of, if you will, the intelligence of interaction. Then the second part will really just be examples of new technologies — new media falling very much into that mold. Again, I will go through them as fast as possible. And then the last one will be some examples I ’ ve been able to collect, which I think illustrate this at least as best I can, in the world of entertainment. People have this belief — and I share most of it — that we will be using the TV screens or their equivalents for electronic books of the future. But then you think,”My God! What a terrible image you get when you look at still pictures on TV.”Well, it doesn ’ t have to be terrible. And that is a slide taken from a TV set and it was pre- processed to be very sympathetic to the TV medium, and it absolutely looks beautiful. Well, what ’ s happened? How did people get into this mess? Where you are now, all of a sudden, sitting in front of personal computers and video text — teletext systems, and somewhat horrified by what you see on the screen? Well, you have to remember that TV was designed to be looked at eight times the distance of the diagonal. So you get a 13-inch, 19-inch, whatever, TV, and then you should multiply that by eight and that ’ s the distance you should sit away from the TV set. Now we ’ ve put people 18 inches in front of a TV, and all the artifacts that none of the original designers expected to be seen, all of a sudden, are staring you in the face : the shadow mask, the scan lines, all of that. And they can be treated very easily ; there are actually ways of getting rid of them, there are actually ways of just making absolutely beautiful pictures. I ’ m talking here a little bit about display technologies. Let me talk about how you might input information. And my favorite example is always fingers. I ’ m very interested in touch-sensitive displays. High-tech, high-touch. Isn ’ t that what some of you said? It ’ s certainly a very important medium for input, and a lot of people think that fingers are a very low- resolution sort of stylus for inputting to a display. In fact, they ’ re not : it ’ s really a very, very high-resolution input medium — you have to just do it twice, you have to touch the screen and then rotate your finger slightly — and you can move a cursor with great accuracy. And so when you see on the market these systems that have just a few light emitting diodes on the side and are very low resolution, it ’ s nice that they exist because it still is better than nothing. But it, in some sense, misses the point : namely, that fingers are a very, very high-resolution input medium. Now, what are some of the other advantages? Well, the one advantage is that you don ’ t have to pick them up, and people don ’ t realize how important that is — not having to pick up your fingers to use them. When you think for a second of the mouse on Macintosh — and I will not criticize the mouse too much — when you ’ re typing — what you have — you want to now put something — first of all, you ’ ve got to find the mouse. You have to probably stop. Maybe not come to a grinding halt, but you ’ ve got to sort of find that mouse. Then you find the mouse, and you ’ re going to have to wiggle it a little bit to see where the cursor is on the screen. And then when you finally see where it is, then you ’ ve got to move it to get the cursor over there, and then —”Bang”— you ’ ve got to hit a button or do whatever. That ’ s four separate steps versus typing and then touching and typing and just doing it all in one motion — or one-and-a-half, depending on how you want to count. Again, what I ’ m trying to do is just illustrate the kinds of problems that I think face the designers of new computer systems and entertainment systems and educational systems from the perspective of the quality of that interface. And another advantage, of course, of using fingers is you have 10 of them. And we have never known how to do this technically, so this slide is a fake slide. We never succeeded in using ten fingers, but there are certain things you can do, obviously, with more than one-finger input, which is rather fascinating. What we did stumble across was something... Again, which is typical of the computer field, is when you have a bug that you can ’ t get rid of you turn it into a feature. And maybe... maybe a mouse is a new kind of bug. But the bug in our case was in touch-sensitive displays : we wanted to be able to draw — you know, rub your finger across the screen to input continuous points — and there was just too much friction created between your finger and the glass — if glass was the substrate, which it usually is. So we found that that actually was a feature in the sense you could build a pressure-sensitive display. And when you touch it with your finger, you can actually, then, introduce all the forces on the face of that screen, and that actually has a certain amount of value. Let me see if I can load another disc and show you, quickly, an example. Now, imagine a screen, which is not only touch-sensitive now, it ’ s pressure-sensitive. And it ’ s pressure-sensitive to the forces both in the plane of the screen — X, Y, and Z at least in one direction ; we couldn ’ t figure out how to come in the other direction. But let me get rid of the slide, and let ’ s see if this comes on. OK. So there is the pressure-sensitive display in operation. The person ’ s just, if you will, pushing on the screen to make a curve. But this is the interesting part. I want to stop it for a second because the movie is very badly made. And the particular display was built about six years ago, and when we moved from one room to another room, a rather large person sat on it and it got destroyed. So all we have is this record. But imagine that screen having lots of objects on it and the person has touched an object — one of N — like he did there, and then pushed on it. Now, imagine a program where some of those objects are physically heavy and some are light : one is an anvil on a fuzzy rug and the other one is a ping-pong ball on a sheet of glass. And when you touch it, you have to really push very hard to move that anvil across the screen, and yet you touch the ping-pong ball very lightly and it just scoots across the screen. And what you can do — oops, I didn ’ t mean to do that — what you can do is actually feed back to the user the feeling of the physical properties. So again, they don ’ t have to be weight ; they could be a general trying to move troops, and he ’ s got to move an aircraft carrier versus a little boat. In fact, they funded it for that very reason. The whole notion, then, is one that at the interface there are physical properties in that transducer — in this case it ’ s pressure and touches — that allow you to present things to the user that you could never present before. So it ’ s not simply looking at the quality or, if you will, the luxury of that interface, but it ’ s actually looking at the idea of presenting things that previously couldn ’ t be presented before. I want to move on to another example, which is one of a different sort, where we ’ re trying to use computer and video disc technology now to come up with a new kind of book. Here, the idea is that you ’ re going to take this book, if you will, and it ’ s going to come alive. You ’ re going to sort of breathe life into it. We are so used to doing monologues. Filmmakers, for example, are the experts in monologue making : you make a film and it has a well-formed beginning, middle and end, and in some sense the art of it is that. And you then say,”There ’ s an opportunity for making conversational movies.”Well, what does that mean? And it sort of nibbles at the core of the whole profession and all the assumptions of that medium. So, book writing is the same thing. What I ’ ll show you very quickly is a new kind of book where it is mixed now with... all sorts of things live in there, but you have to keep a few things in mind. One is that this book knows about itself. Each frame of the movie has information about itself. So it knows, or at least there is computer-readable information in the medium itself. It ’ s just not a static movie frame. That ’ s one thing. The other is that you have to realize that it is a random access medium, and you can, in fact, branch and expand and elaborate and shrink. And here — again, my favorite example — is the cookbook, the”Larousse Gastronomique.”And I think I use the example all too often, but it ’ s a great one because there is a classic ending in that little encyclopedia-style cookbook that tells you how to do something like penguin, and you get to the end of the recipe and it says,”Cook until done.”Now, that would be, if you will, the top green track, which doesn ’ t mean too much. But you might have to elaborate for me or for somebody who isn ’ t an expert, and say,”Cook at 380 degrees for 45 minutes.”And then for a real beginner, you would go down even further and elaborate more — say,”Open the oven, preheat, wait for the light to go out, open the door, don ’ t leave it open too long, put the penguin in and shut the door...”whatever. And that ’ s a much more elaborate one than you dribble back. That ’ s one kind of use of random access. And the other is where you want to explain the same thing in different ways. If you ’ re in a classroom situation and somebody asks a question, the last thing you do is repeat what you just said. You try and think of a different way of saying the same thing, or if you know the particular student and that student ’ s cognitive style, then you might say it in a way that you think would have a good impedance match with that student. There are all sorts of techniques you will use — and again, this is a different kind of branching. So, what I will show you is... it ’ s a rather boring book, but I ’ m afraid sometimes you have to do boring books because your sponsors aren ’ t necessarily interested in fiction and entertainment. And this is a book on how to repair a transmission. Now, I don ' t even know what vintage the transmission is, but let me just show you very quickly some of it, and we ' ll move on. Narrator : And continue to get descriptions for each of these chapters. Nicholas Negroponte : Now, this is his table of contents. Just a picture of the transmission, and as you rub your finger across the transmission it highlights the various parts. Narrator : When I find a chapter that I want to see, I just touch the text and the system will format pages for me to read. The words or phrases that are lit up in red are glossary words, so I can get a different definition by just touching the word, and the definition appears, superimposed over the illustration. NN : This is about the oil pan, or the oil filter and all that. This is relatively important because it sets the page... Narrator : This is another example of a page with glossary words highlighted in red. I can get a definition of these words just by touching them, and the definition will appear in the illustration corner. I can get back to the illustration, but in this case it ' s not a single frame, but it ' s actually a movie of someone coming into the frame and doing the repair that ' s described in the text. The two-headed slider is a speed control that allows me to watch the movie at various speeds, in forward or reverse. And the movie is displayed as a full frame movie. I can go back to the beginning... and play the movie at full speed. Here ' s another step-by-step procedure, only in this case — NN : Okay, this movie is... Everybody ' s heard of sound-sync movies — this is text-sync movies, so as the movie plays, the text gets highlighted. We highlight the text as we go through the movie. Repairman :... Not too far out. Front poles, preferably. Don ' t loosen them too far. If you loosen them too far, you ' ll have a big mess. NN : I suspect that some of you might not even understand that language. OK. I ' m at the third and last part of this, which I said I would make an attempt to at least give you some examples that may be more directly related to the world of entertainment. And of course, good education has got to be good entertainment, so my first example will be drawn from a very recent experiment that we ' ve been doing — in this case, in Senegal — where we have tried to use personal computers as a pedagogical medium. But not as teaching machines at all ; the whole notion is to use this as an instrument where there is a complete reversal of roles — the child is, if you will, the teacher and the machine is the student — and the art of computer programming is a vehicle that sort of approximates thinking about thinking. But teaching kids programming per se is utterly irrelevant. And there are just a few slides I want to go through, but there ' s a story I ' d like to tell. And that was when, before we did this in any developing countries — we ' re doing it, in fact, in three developing countries right now : Pakistan, Colombia and Senegal — we did it in some pretty rough areas of New York City. And one child, whose name I ' ve forgotten, was about seven or eight years old, absolutely considered mentally handicapped — couldn ' t read, didn ' t even make it in the lowest section of the school ' s classes — and was pretty much not in school, though physically there. But did hang around the, quote,"computer room,"where there were quite a few computers, and learned this particular language called Logo — and learned it with great ease and found it a lot of fun, it was very interesting. And one day, by chance, some visitors from the NIE came by in their double-breasted suits looking at this setup, and none of the children who were normally there, except for this one child, were there. He was, and he said,"Let me show you how this works,"and they got an absolutely ingenuous, wonderful description of Logo. And the child was just zipping right through it, showing them all sorts of things until they asked him how to do something which he couldn ' t explain and so he flipped through the manual, found the explanation and typed the command and got it to do what they asked. They were delighted, and by the time it was time to go see the principal, whom they ’ d actually come to see — not the computer room — they went upstairs and they said,”This is absolutely remarkable! That child was very articulate and showed us and even dealt with the things he couldn ’ t do automatically with that manual. It was just absolutely fantastic.”The principal said,”There ’ s a dreadful mistake, because that child can ’ t read. And you obviously have been hoodwinked or you ’ ve talked about somebody else.”And they all got up and they all went downstairs and the child was still there. And they did something very intelligent : they asked the child,”Can you read?”And the child said,”No, I can ’ t.”And then they said,”But wait a minute. You just looked through that manual and you found...”and he said,”Oh, but that ’ s not reading.”And so they said,”Well, what ’ s reading then?”He says,”Well, reading is this junk they give me in little books to read. It ’ s absolutely irrelevant, and I get nothing for it. But here, with a little bit of effort I get a lot of return.”And it really meant something to the child. The child read beautifully, it turned out, and was really very competent. So it actually meant something. And that story has many other anecdotes that are similar, but wow. The key to the future of computers in education is right there, and it is : when does it mean something to a child? There is a myth, and it truly is a myth : we believe — and I ’ m sure a lot of you believe in this room — that it is harder to read and write than it is to learn how to speak. And it ’ s not, but we think speech —”My God, little children pick it up somehow, and by the age of two they ’ re doing a mediocre job, and by three and four they ’ re speaking reasonably well. And yet you ’ ve got to go to school to learn how to read, and you have to sit in a classroom and somebody has to teach you. Hence, it must be harder.”Well, it ’ s not harder. What the truth is is that speaking has great value to a child ; the child can get a great deal by talking to you. Reading and writing is utterly useless. There is no reason for a child to read and write except blind faith, and that it ’ s going to help you. So what happens is you go to school and people say,”Just believe me, you ’ re going to like it. You ’ re going to like reading,”and just read and read. On the other hand, you give a kid — a three-year-old kid — a computer and they type a little command and — Poof! — something happens. And all of a sudden... You may not call that reading and writing, but a certain bit of typing and reading stuff on the screen has a huge payoff, and it ’ s a lot of fun. And in fact, it ’ s a powerful educational instrument. Well, in Senegal we found that this was the traditional classroom : 120 kids — three per desk — one teacher, a little bit of chalk. This student was one of our first students, and it ’ s the girl on the left leaning with her chalkboard, and she came... within two days — I want to show you the program she wrote, and remember her hairstyle. And that is the program she made. That ’ s what meant something to her, is doing the hair pattern, and actually did it within two days — an hour each day — and found it was, to her, absolutely the most meaningful piece... But rooted in that, little did she know how much knowledge she was acquiring about geometry and just math and logic and all the rest. And again, I could talk for three hours about this subject. I will come to my last example and then quit. And my last example — as some of my former colleagues, whom I see in the room, can imagine what it will be. Yes, it is. It ’ s our work — that was a while ago, and it still is my favorite project — of teleconferencing. And the reason it remains a favorite project is that we were asked to do a teleconferencing system where you had the following situation : you had five people at five different sites — they were known people — and you had to have these people in teleconference, such that each one was utterly convinced that the other four were physically present. Now, that is sufficiently zany that we would, obviously, jump to the bait, and we did. And the fact that we knew the people — we had to take a page out of the history of Walt Disney — we actually went so far as to build CRTs in the shapes of the people ’ s faces. So if I wanted to call my friend Peter Sprague on the phone, my secretary would get his head out and bring it and set it on the desk, and that would be the TV used for the occasion. And it ’ s uncanny : there ’ s no way I can explain to you the amount of eye contact you get with that physical face projected on a 3D CRT of that sort. The next thing that we had to do is to persuade them that there needed to be spatial correspondence, which is straightforward, but again, it ’ s something that didn ’ t fall naturally out of a telecommunications or computing style of thinking ; it was a very, if you will, architectural or spatial concept. And that was to recognize that when you sit around the table, the actual location of the people becomes rather important. And when somebody gets up, in fact, to go answer a phone or use a bathroom or something, the empty seat becomes, if you will, that person. And you point frequently to the empty seat and you say,”He or she wouldn ’ t agree,”and the empty chair is that person and the spatiality is crucial. So we said,”Well, these will be on round tables and the order around the table had to be the same, so that at my site, I would be, if you will, real and then at each other ’ s site you ’ d have these plastic heads. And the plastic heads, sometimes you want to project them. And there are a number of schemes, which I don ’ t want to dwell on, but this is the one that we finally used where we projected onto rear screen material that was molded in the face — literally in the face of the person. And I ’ ll show you one more slide, where this is actually made from something called a solid photograph and is the screen. Now, we track, on the person ’ s head, the head motions — so we transmit with a video the head positions — and so this head moves in about two axes. So if I, all of a sudden, turn to the person to my left and start talking to that person, then at the person to my right ’ s site, he ’ ll see these two plastic heads talking to each other. And then if that person interrupts, then those two heads may turn. And it really is reconstructing, quite accurately, teleconferencing.